Vulkan® 1.4.366 - A Specification (with all ratified extensions)

The Khronos® Vulkan Working Group

Version 1.4.366, 2026-10-23 00:57:43Z: from git branch: github-main commit: a8b088a3fae12c8575407ea90c7c09bd653db923

Table of Contents

1. Preamble . . . 1
2. Introduction . . . 3
   2.1. Document Conventions . . . 3
3. Fundamentals . . . 7
   3.1. Host and Device Environment . . . 7
   3.2. Execution Model . . . 7
   3.3. Object Model . . . 9
   3.4. Application Binary Interface . . . 11
   3.5. Command Syntax and Duration . . . 12
   3.6. Threading Behavior . . . 16
   3.7. Valid Usage . . . 37
   3.8. `VkResult` Return Codes . . . 44
   3.9. Numeric Representation and Computation . . . 50
   3.10. Fixed-Point Data Conversions . . . 52
   3.11. String Representation . . . 53
   3.12. Common Object Types . . . 53
   3.13. API Name Aliases . . . 106
4. Initialization . . . 107
   4.1. Command Function Pointers . . . 107
   4.2. Instances . . . 110
5. Devices and Queues . . . 124
   5.1. Physical Devices . . . 124
   5.2. Devices . . . 177
   5.3. Queues . . . 199
6. Command Buffers . . . 210
   6.1. Command Buffer Lifecycle . . . 210
   6.2. Command Pools . . . 212
   6.3. Command Buffer Allocation and Management . . . 220
   6.4. Command Buffer Recording . . . 225
   6.5. Command Buffer Submission . . . 237
   6.6. Queue Forward Progress . . . 263
   6.7. Secondary Command Buffer Execution . . . 264
   6.8. Nested Command Buffers . . . 273
   6.9. Command Buffer Device Mask . . . 274
7. Synchronization and Cache Control . . . 278
   7.1. Execution and Memory Dependencies . . . 278
   7.2. Implicit Synchronization Guarantees . . . 320
   7.3. Fences . . . 323

7.4. Semaphores . . . 349
7.5. Events . . . 379
7.6. Pipeline Barriers . . . 411
7.7. Memory Barriers . . . 424
7.8. Wait Idle Operations . . . 509
7.9. Host Write Ordering Guarantees . . . 511
7.10. Synchronization and Multiple Physical Devices . . . 511
7.11. Calibrated Timestamps . . . 512
8. Render Pass . . . 517
8.1. Render Pass Objects . . . 553
8.2. Render Pass Creation . . . 555
8.3. Render Pass Compatibility . . . 625
8.4. Framebuffers . . . 626
8.5. Render Pass Load Operations . . . 639
8.6. Render Pass Store Operations . . . 641
8.7. Render Pass Multisample Resolve Operations . . . 642
8.8. Render Pass Commands . . . 644
8.9. Render Pass Creation Feedback . . . 677
8.10. Common Render Pass Data Races (Informative) . . . 682
9. Shaders . . . 685
9.1. Shader Objects . . . 685
9.2. Shader Modules . . . 711
9.3. Shader Module Identifiers . . . 716
9.4. Binding Shaders . . . 719
9.5. Shader Execution . . . 720
9.6. Shader Out-of-Bounds Memory Access . . . 722
9.7. Shader Memory Access Ordering . . . 727
9.8. Shader Inputs and Outputs . . . 728
9.9. Task Shaders . . . 728
9.10. Mesh Shaders . . . 728
9.11. Vertex Shaders . . . 729
9.12. Tessellation Control Shaders . . . 730
9.13. Tessellation Evaluation Shaders . . . 732
9.14. Geometry Shaders . . . 732
9.15. Fragment Shaders . . . 732
9.16. Compute Shaders . . . 732
9.17. Ray Generation Shaders . . . 733
9.18. Intersection Shaders . . . 733
9.19. Any-Hit Shaders . . . 733
9.20. Closest Hit Shaders . . . 734
9.21. Miss Shaders . . . 734

9.22. Callable Shaders . . . 734
9.23. Interpolation Decorations . . . 734
9.24. Static Use . . . 735
9.25. Scope . . . 735
9.26. Group Operations . . . 740
9.27. Quad Group Operations . . . 742
9.28. Derivative Operations . . . 742
9.29. Helper Invocations . . . 744
9.30. Cooperative Matrices . . . 745
9.31. Validation Cache . . . 758
10. Pipelines . . . 767
10.1. Multiple Pipeline Creation . . . 768
10.2. Pipeline Creation Flags . . . 769
10.3. Compute Pipelines . . . 781
10.4. Graphics Pipelines . . . 803
10.5. Ray Tracing Pipelines . . . 864
10.6. Pipeline Destruction . . . 886
10.7. Pipeline Derivatives . . . 887
10.8. Pipeline Cache . . . 887
10.9. Pipeline Binaries . . . 896
10.10. Specialization Constants . . . 909
10.11. Pipeline Libraries . . . 913
10.12. Pipeline Binding . . . 914
10.13. Dynamic State . . . 918
10.14. Pipeline Properties and Shader Information . . . 926
10.15. Pipeline Creation Feedback . . . 938
11. Memory Allocation . . . 941
11.1. Host Memory . . . 941
11.2. Device Memory . . . 948
12. Resource Creation . . . 1035
12.1. Buffers . . . 1035
12.2. Buffer Views . . . 1055
12.3. Buffer Device Addresses . . . 1060
12.4. Images . . . 1064
12.5. Image Layouts . . . 1131
12.6. Image Views . . . 1139
12.7. Acceleration Structures . . . 1164
12.8. Micromaps . . . 1187
12.9. Resource Memory Association . . . 1195
12.10. Resource Sharing Mode . . . 1232
12.11. Memory Aliasing . . . 1235

13. Images . . . 1239
13.1. Image Coordinate Validation . . . 1240
13.2. Image Reads . . . 1240
13.3. Image Writes . . . 1241
13.4. SPIR-V Image Access . . . 1241
14. Samplers . . . 1244
14.1. Sampler $Y'C_BC_R$ Conversion . . . 1257
14.2. Custom Border Colors . . . 1267
15. Resource Descriptors . . . 1274
15.1. Descriptor Types . . . 1274
15.2. Physical Storage Buffer Access . . . 1280
16. Descriptor Heaps . . . 1281
16.1. Writing Descriptors . . . 1281
16.2. Using Heaps . . . 1292
16.3. Shader Bindings . . . 1300
16.4. Packing descriptors more tightly . . . 1323
17. Descriptor Sets . . . 1325
17.1. Descriptor Set Layout . . . 1325
17.2. Pipeline Layouts . . . 1346
17.3. Allocation of Descriptor Sets . . . 1362
17.4. Descriptor Set Updates . . . 1377
17.5. Descriptor Update Templates . . . 1394
17.6. Descriptor Set Updates With Templates . . . 1395
17.7. Descriptor Set Binding . . . 1404
17.8. Push Descriptor Updates . . . 1413
17.9. Push Descriptor Updates With Descriptor Update Templates . . . 1420
17.10. Push Constant Updates . . . 1426
18. Descriptor Buffers . . . 1433
18.1. Putting Descriptors in Memory . . . 1433
18.2. Binding Descriptor Buffers . . . 1445
18.3. Updating Descriptor Buffers . . . 1463
18.4. Push Descriptors With Descriptor Buffers . . . 1463
18.5. Capture and Replay . . . 1464
19. Shader Interfaces . . . 1477
19.1. Shader Input and Output Interfaces . . . 1477
19.2. Vertex Input Interface . . . 1480
19.3. Fragment Output Interface . . . 1481
19.4. Legacy Dithering . . . 1486
19.5. Fragment Tile Image Interface . . . 1486
19.6. Fragment Input Attachment Interface . . . 1486
19.7. Ray Tracing Pipeline Interface . . . 1492

19.8. Shader Resource Interface . . . 1493
19.9. Built-In Variables . . . 1505
20. Sampling . . . 1546
20.1. Sampling Coordinate Systems . . . 1546
20.2. Sampling Operations . . . 1550
20.3. Normalized Texel Coordinate Operations . . . 1566
20.4. Unnormalized Texel Coordinate Operations . . . 1573
20.5. Integer Texel Coordinate Operations . . . 1575
20.6. Image Sample Operations . . . 1575
20.7. Image Operation Steps . . . 1580
20.8. LOD Query . . . 1581
21. Fragment Density Map Operations . . . 1582
21.1. Fragment Density Map Operations Overview . . . 1582
21.2. Fetch Density Value . . . 1582
21.3. Fragment Area Conversion . . . 1584
22. Queries . . . 1585
22.1. Query Pools . . . 1585
22.2. Query Operation . . . 1594
22.3. Occlusion Queries . . . 1632
22.4. Pipeline Statistics Queries . . . 1632
22.5. Timestamp Queries . . . 1635
22.6. Performance Queries . . . 1643
22.7. Transform Feedback Queries . . . 1647
22.8. Primitives Generated Queries . . . 1647
22.9. Mesh Shader Queries . . . 1647
22.10. Result Status Queries . . . 1648
22.11. Video Encode Feedback Queries . . . 1648
23. Clear Commands . . . 1655
23.1. Clearing Images Outside a Render Pass Instance . . . 1655
23.2. Clearing Images Inside a Render Pass Instance . . . 1661
23.3. Clear Values . . . 1665
23.4. Filling Memory . . . 1667
23.5. Updating Memory . . . 1670
23.6. Filling Buffers . . . 1673
23.7. Updating Buffers . . . 1675
24. Copy Commands . . . 1679
24.1. Copying Data Between Memory Ranges . . . 1679
24.2. Copying Data Between Buffers . . . 1685
24.3. Copying Data Between Images . . . 1692
24.4. Copying Data Between Memory Ranges and Images . . . 1718
24.5. Copying Data Between Buffers and Images . . . 1728

24.6. Indirect Copies . . . 1779
24.7. Image Copies With Scaling . . . 1790
24.8. Resolving Multisample Images . . . 1808
25. Drawing Commands . . . 1825
25.1. Primitive Topologies . . . 1829
25.2. Effective Primitive Topology . . . 1841
25.3. Primitive Order . . . 1843
25.4. Programmable Primitive Shading . . . 1843
25.5. Conditional Rendering . . . 2449
25.6. Programmable Mesh Shading . . . 2458
26. Fixed-Function Vertex Processing . . . 2656
26.1. Vertex Attributes . . . 2656
26.2. Vertex Input Description . . . 2661
26.3. Vertex Attribute Divisor in Instanced Rendering . . . 2679
26.4. Vertex Input Address Calculation . . . 2681
27. Tessellation . . . 2683
27.1. Tessellator . . . 2683
27.2. Tessellator Patch Discard . . . 2686
27.3. Tessellator Spacing . . . 2687
27.4. Tessellation Primitive Ordering . . . 2687
27.5. Tessellator Vertex Winding Order . . . 2688
27.6. Triangle Tessellation . . . 2688
27.7. Quad Tessellation . . . 2690
27.8. Isoline Tessellation . . . 2692
27.9. Tessellation Point Mode . . . 2692
27.10. Tessellation Pipeline State . . . 2692
28. Geometry Shading . . . 2697
28.1. Geometry Shader Input Primitives . . . 2697
28.2. Geometry Shader Output Primitives . . . 2698
28.3. Multiple Invocations of Geometry Shaders . . . 2698
28.4. Geometry Shader Primitive Ordering . . . 2698
29. Mesh Shading . . . 2699
29.1. Task Shader Input . . . 2699
29.2. Task Shader Output . . . 2699
29.3. Mesh Generation . . . 2699
29.4. Mesh Shader Input . . . 2699
29.5. Mesh Shader Output . . . 2700
29.6. Mesh Shader Primitive Ordering . . . 2701
30. Fixed-Function Vertex Post-Processing . . . 2702
30.1. Transform Feedback . . . 2702
30.2. Flat Shading . . . 2719

30.3. Primitive Clipping . . . 2723
30.4. Clipping Shader Outputs . . . 2730
30.5. Coordinate Transformations . . . 2731
30.6. Controlling the Viewport . . . 2731
31. Rasterization . . . 2742
31.1. Discarding Primitives Before Rasterization . . . 2747
31.2. Controlling the Vertex Stream Used for Rasterization . . . 2748
31.3. Rasterization Order . . . 2752
31.4. Multisampling . . . 2752
31.5. Custom Sample Locations . . . 2758
31.6. Fragment Shading Rates . . . 2763
31.7. Sample Shading . . . 2772
31.8. Barycentric Interpolation . . . 2772
31.9. Points . . . 2775
31.10. Line Segments . . . 2776
31.11. Polygons . . . 2787
32. Fragment Operations . . . 2810
32.1. Discard Rectangles Test . . . 2811
32.2. Scissor Test . . . 2818
32.3. Sample Mask Test . . . 2820
32.4. Fragment Shading . . . 2822
32.5. Multisample Coverage . . . 2826
32.6. Depth and Stencil Operations . . . 2830
32.7. Depth Bounds Test . . . 2833
32.8. Stencil Test . . . 2836
32.9. Depth Test . . . 2847
32.10. Sample Counting . . . 2856
32.11. Coverage Reduction . . . 2856
33. The Framebuffer . . . 2858
33.1. Blending . . . 2858
33.2. Logical Operations . . . 2889
33.3. Color Write Mask . . . 2894
33.4. Color Write Enable . . . 2895
34. Dispatching Commands . . . 2899
35. Device-Generated Commands . . . 2950
35.1. Indirect Commands Layout . . . 2950
35.2. Indirect Commands Generation and Execution . . . 2972
36. Sparse Resources . . . 3039
36.1. Sparse Resource Features . . . 3039
36.2. Sparse Buffers and Fully-Resident Images . . . 3040
36.3. Sparse Partially-Resident Buffers . . . 3041

36.4. Sparse Partially-Resident Images . . . 3041
36.5. Sparse Memory Aliasing . . . 3049
36.6. Sparse Resource Implementation Guidelines (Informative) . . . 3050
36.7. Sparse Resource API . . . 3051
37. Window System Integration (WSI) . . . 3080
37.1. WSI Platform . . . 3080
37.2. WSI Surface . . . 3080
37.3. Presenting Directly to Display Devices . . . 3099
37.4. Querying for WSI Support . . . 3129
37.5. Surface Queries . . . 3134
37.6. Full Screen Exclusive Control . . . 3173
37.7. Device Group Queries . . . 3175
37.8. Present Timing Queries . . . 3182
37.9. Present Wait . . . 3195
37.10. WSI Swapchain . . . 3196
37.11. HDR Metadata . . . 3261
38. Deferred Host Operations . . . 3264
38.1. Requesting Deferral . . . 3264
38.2. Deferred Host Operations API . . . 3265
39. Private Data . . . 3271
40. Acceleration Structures . . . 3277
40.1. Acceleration Structures . . . 3277
40.2. Host Acceleration Structure Operations . . . 3344
41. Micromap . . . 3362
41.1. Micromaps . . . 3362
41.2. Host Micromap Operations . . . 3385
42. Ray Traversal . . . 3396
42.1. Ray Intersection Candidate Determination . . . 3397
42.2. Ray Intersection Culling . . . 3400
42.3. Ray Intersection Confirmation . . . 3405
42.4. Ray Closest Hit Determination . . . 3407
42.5. Ray Result Determination . . . 3408
43. Ray Tracing . . . 3409
43.1. Shader Call Instructions . . . 3409
43.2. Ray Tracing Commands . . . 3411
43.3. Shader Binding Table . . . 3458
43.4. Ray Tracing Pipeline Stack . . . 3460
43.5. Ray Tracing Capture Replay . . . 3461
44. Memory Decompression . . . 3462
45. Video Coding . . . 3470
45.1. Video Picture Resources . . . 3470

45.2. Decoded Picture Buffer . . . 3472
45.3. Video Profiles . . . 3474
45.4. Video Capabilities . . . 3485
45.5. Video Sessions . . . 3494
45.6. Video Profile Compatibility . . . 3509
45.7. Video Session Parameters . . . 3511
45.8. Video Coding Scope . . . 3535
45.9. Video Coding Control . . . 3546
45.10. Inline Queries . . . 3549
45.11. Video Decode Operations . . . 3551
45.12. H.264 Decode Operations . . . 3572
45.13. H.265 Decode Operations . . . 3588
45.14. VP9 Decode Operations . . . 3604
45.15. AV1 Decode Operations . . . 3611
45.16. Video Encode Operations . . . 3625
45.17. Video Encode Intra Refresh . . . 3668
45.18. Video Encode Rate Control . . . 3676
45.19. Video Encode Quantization Maps . . . 3686
45.20. H.264 Encode Operations . . . 3695
45.21. H.265 Encode Operations . . . 3735
45.22. AV1 Encode Operations . . . 3782
46. Extending Vulkan . . . 3826
46.1. Functionality Levels . . . 3826
46.2. Instance and Device Versions . . . 3826
46.3. Core Versions . . . 3826
46.4. Layers . . . 3830
46.5. Extensions . . . 3834
46.6. Extension Dependencies . . . 3838
46.7. Compatibility Guarantees (Informative) . . . 3838
47. Features . . . 3843
47.1. Feature Requirements . . . 4048
48. Limits . . . 4059
48.1. Limit Requirements . . . 4157
48.2. Additional Multisampling Capabilities . . . 4189
48.3. Milestone Limits . . . 4190
49. Formats . . . 4194
49.1. Format Definition . . . 4194
49.2. Format Properties . . . 4250
49.3. Required Format Support . . . 4273
50. Additional Capabilities . . . 4298
50.1. Additional Image Capabilities . . . 4298

50.2. Additional Buffer Capabilities . . . 4324
50.3. Optional Semaphore Capabilities . . . 4326
50.4. Optional Fence Capabilities . . . 4333
50.5. Timestamp Calibration Capabilities . . . 4339
51. Debugging . . . 4341
51.1. Debug Utilities . . . 4344
51.2. Device Loss Debugging . . . 4369
51.3. Active Tooling Information . . . 4385
51.4. Frame Boundary . . . 4387
Appendix A: Vulkan Environment for SPIR-V . . . 4391
Versions and Formats . . . 4391
Capabilities . . . 4391
Validation Rules Within a Module . . . 4405
Precision and Operation of SPIR-V Instructions . . . 4444
Buffer Indexing Calculations . . . 4452
Signedness of SPIR-V Image Accesses . . . 4453
Image Format and Type Matching . . . 4453
Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types . . . 4455
Compatibility Between SPIR-V Image Formats and Vulkan Formats . . . 4456
Ray Query Precision and Operation . . . 4457
Appendix B: Memory Model . . . 4458
Agent . . . 4458
Memory Location . . . 4458
Allocation . . . 4458
Memory Operation . . . 4459
Reference . . . 4459
Program-Order . . . 4459
Shader Call Related . . . 4460
Shader Call Order . . . 4460
Scope . . . 4461
Atomic Operation . . . 4461
Scoped Modification Order . . . 4461
Memory Semantics . . . 4462
Release Sequence . . . 4463
Synchronizes-With . . . 4464
System-Synchronizes-With . . . 4466
Private vs. Non-Private . . . 4466
Inter-Thread-Happens-Before . . . 4467
Happens-Before . . . 4467
Availability and Visibility . . . 4467
Availability, Visibility, and Domain Operations . . . 4470

Availability and Visibility Semantics . . . 4470
Per-Instruction Availability and Visibility Semantics . . . 4471
Location-Ordered . . . 4471
Data Race . . . 4472
Visible-To . . . 4473
Acyclicity . . . 4473
Shader I/O . . . 4474
Deallocation . . . 4474
Descriptions (Informative) . . . 4474
Tessellation Output Ordering . . . 4475
Cooperative Matrix Memory Access . . . 4475
Appendix C: Compressed Image Formats . . . 4476
Block-Compressed Image Formats . . . 4477
ETC Compressed Image Formats . . . 4478
ASTC Compressed Image Formats . . . 4479
Appendix D: Core Revisions (Informative) . . . 4482
Vulkan Version 1.4 . . . 4482
Vulkan Version 1.3 . . . 4491
Vulkan Version 1.2 . . . 4500
Vulkan Version 1.1 . . . 4508
Vulkan Version 1.0 . . . 4520
Appendix E: Layers & Extensions (Informative) . . . 4534
Extension Dependencies . . . 4534
Extension Interactions . . . 4534
List of Current Extensions . . . 4535
List of Provisional Extensions . . . 5144
List of Deprecated Extensions . . . 5148
Appendix F: Vulkan Roadmap Milestones . . . 5392
Roadmap 2022 . . . 5392
Roadmap 2024 . . . 5396
Roadmap 2026 . . . 5398
Appendix G: Legacy and Superseded Functionality . . . 5401
List of Legacy Functionality . . . 5402
Appendix H: API Boilerplate . . . 5406
Vulkan Header Files . . . 5406
Window System-Specific Header Control (Informative) . . . 5410
Provisional Extension Header Control (Informative) . . . 5412
Video Std Headers . . . 5412
Appendix I: Invariance . . . 5414
Repeatability . . . 5414
Multi-Pass Algorithms . . . 5414

Invariance Rules 5414
Tessellation Invariance 5416
Appendix J: Lexicon 5418
Glossary 5418
Common Abbreviations 5447
Video-Specific Abbreviations 5448
Prefixes 5450
Appendix K: Credits (Informative) 5452
Working Group Contributors to Vulkan 5452
Other Credits 5464

Chapter 1. Preamble

Copyright 2014-2026 The Khronos Group Inc.

This Specification is protected by copyright laws and contains material proprietary to Khronos. Except as described by these terms, it or any components may not be reproduced, republished, distributed, transmitted, displayed, broadcast or otherwise exploited in any manner without the express prior written permission of Khronos.

Khronos grants a conditional copyright license to use and reproduce the unmodified Specification for any purpose, without fee or royalty, EXCEPT no licenses to any patent, trademark or other intellectual property rights are granted under these terms.

Khronos makes no, and expressly disclaims any, representations or warranties, express or implied, regarding this Specification, including, without limitation: merchantability, fitness for a particular purpose, non-infringement of any intellectual property, correctness, accuracy, completeness, timeliness, and reliability. Under no circumstances will Khronos, or any of its Promoters, Contributors or Members, or their respective partners, officers, directors, employees, agents or representatives be liable for any damages, whether direct, indirect, special or consequential damages for lost revenues, lost profits, or otherwise, arising from or in connection with these materials.

This Specification has been created under the Khronos Intellectual Property Rights Policy, which is Attachment A of the Khronos Group Membership Agreement available at https://www.khronos.org/files/member_agreement.pdf. Parties desiring to implement the Specification and make use of Khronos trademarks in relation to that implementation, and receive reciprocal patent license protection under the Khronos Intellectual Property Rights Policy must become Adopters and confirm the implementation as conformant under the process defined by Khronos for this Specification; see https://www.khronos.org/adopters.

This Specification contains substantially unmodified functionality from, and is a successor to, Khronos specifications including OpenGL, OpenGL ES and OpenCL.

The Khronos Intellectual Property Rights Policy defines the terms 'Scope', 'Compliant Portion', and 'Necessary Patent Claims'.

Some parts of this Specification are purely informative and so are EXCLUDED the Scope of this Specification. The Document Conventions section of the Introduction defines how these parts of the Specification are identified.

Where this Specification uses technical terminology, defined in the Glossary or otherwise, that refer to enabling technologies that are not expressly set forth in this Specification, those enabling technologies are EXCLUDED from the Scope of this Specification. For clarity, enabling technologies not disclosed with particularity in this Specification (e.g. semiconductor manufacturing technology, hardware architecture, processor architecture or microarchitecture, memory architecture, compiler technology, object oriented technology, basic operating system technology, compression technology, algorithms, and so on) are NOT to be considered expressly set forth; only those application program interfaces and data structures disclosed with particularity are included in the Scope of this Specification.

For purposes of the Khronos Intellectual Property Rights Policy as it relates to the definition of Necessary Patent Claims, all recommended or optional features, behaviors and functionality set forth in this Specification, if implemented, are considered to be included as Compliant Portions.

Where this Specification identifies specific sections of external references, only those specifically identified sections define normative functionality. The Khronos Intellectual Property Rights Policy excludes external references to materials and associated enabling technology not created by Khronos from the Scope of this Specification, and any licenses that may be required to implement such referenced materials and associated technologies must be obtained separately and may involve royalty payments.

Khronos and Vulkan are registered trademarks, and SPIR-V is a trademark of The Khronos Group Inc. OpenCL is a trademark of Apple Inc., used under license by Khronos. OpenGL is a registered trademark and the OpenGL ES logo is a trademark of Hewlett Packard Enterprise, used under license by Khronos. ASTC is a trademark of ARM Holdings PLC. All other product names, trademarks, and/or company names are used solely for identification and belong to their respective owners.

Chapter 2. Introduction

This document, referred to as the “Vulkan Specification” or just the “Specification” hereafter, describes the Vulkan Application Programming Interface (API). Vulkan is a C99 API designed for explicit control of low-level graphics and compute functionality.

The canonical version of the Specification is available in the official Vulkan Registry (https://registry.khronos.org/vulkan/). The source files used to generate the Vulkan specification are stored in the Vulkan Documentation Repository (https://github.com/KhronosGroup/Vulkan-Docs).

The source repository additionally has a public issue tracker and allows the submission of pull requests that improve the specification.

2.1. Document Conventions

The Vulkan specification is intended for use by both implementors of the API and application developers seeking to make use of the API, forming a contract between these parties. Specification text may address either party; typically the intended audience can be inferred from context, though some sections are defined to address only one of these parties. (For example, Valid Usage sections only address application developers). Any requirements, prohibitions, recommendations, or options defined in specification text are imposed only on the audience of that text.

2.1.1. Normative Requirements

The Vulkan Specification uses a combination of normative terminology and normative descriptions to express the requirements that it imposes on applications and implementations. An application which complies with all normative requirements imposed on applications is said to make **valid use** of the API; failing to comply with such requirements results in undefined behavior, as discussed in the Valid Usage section below. In the context of this document, an implementation which complies with all normative requirements imposed on implementations is said to be **conformant**.

NOTE	The Khronos Group imposes additional requirements on implementors who wish to make public statements describing their Vulkan implementations as conformant. These include signing the Vulkan Adopter’s Agreement, paying the associated fee, and making a successful conformance test submission to the Khronos Conformance Process. For details see the Khronos Trademark Guidelines (https://www.khronos.org/legal/khronos-trademark-guidelines).

Normative Terminology

Within this specification, the key words **must**, **required**, **should**, **may**, and **optional** are to be interpreted as described in RFC 2119 - Key words for use in RFCs to Indicate Requirement Levels (https://www.ietf.org/rfc/rfc2119.txt). The additional key word **optionally** is an alternate form of **optional**, for use where grammatically appropriate. These key words are highlighted in the specification to indicate that they are being used in a specific technical sense.

The additional key words **can** and **cannot** are to be interpreted as describing the capabilities of an application, as follows:

can

This word means that the application is able to perform the action described.

cannot

This word means that the API and/or the execution environment provide no mechanism through which the application can express or accomplish the action described.

These key words are never used in text addressing implementors.

NOTE	There is an important distinction between **cannot** and **must not**, as used in this Specification. **Cannot** refers to something the API provides no way for the application to express or accomplish. **Must not** describes something that the application is able to express, but that is not valid use of the API, and will have undefined and potentially unrecoverable consequences.

Normative Descriptions

In the Vulkan Specification, the normative term **must** is primarily used to describe **application** behavior, and in particular to constrain what inputs or commands issued by the application to the implementation are considered valid.

To constrain **implementation** behavior, the specification sometimes uses **must**, but more often simply describes the behavior of the implementation in response to specified commands and inputs. Unless explicitly stated otherwise, such references to implementation behavior describe the behavior of **conformant** implementations, and express normative requirements which an implementation must satisfy in order to conform to the specification. For example, if the specification says “Under **specified condition**, the error code VK_ERROR_FEATURE_NOT_PRESENT is returned”, that behavior is a requirement of the specification, and an implementation which does not return that error code under that condition is not conformant.

When the normative terms **may**, **should**, or **optional** are used to describe implementation behavior, they define alternative or optional behaviors which a conformant implementation may or may not exhibit. Such statements are also normative. For example, if the specification says “Under **specified condition**, the implementation **should** return A but **may** instead return B”, then an implementation that returns either A or B under that condition is conformant (assuming it does not violate other normative requirements), while an implementation that returns anything else is not.

2.1.2. Normative References

References to external documents are considered normative references if the Specification uses normative terminology or normative descriptions to refer to them or their requirements, either as a whole or in part.

The following documents are referenced by normative sections of the specification:

IEEE. August, 2008. *IEEE Standard for Floating-Point Arithmetic*. IEEE Std 754-2008. https://dx.doi.org/10.1109/IEEESTD.2008.4610935 .

Andrew Garrard. *Khronos Data Format Specification, version 1.3*. https://registry.khronos.org/DataFormat/specs/1.3/dataformat.1.3.html .

John Kessenich. *SPIR-V Extended Instructions for GLSL, Version 1.00* (February 10, 2016). https://registry.khronos.org/spir-v/ .

John Kessenich, Boaz Ouriel, Raun Krisch, and Diego Novillo. *SPIR-V Specification, Version 1.6.7, Revision 1, Unified* (March, 2026). https://registry.khronos.org/spir-v/ .

ITU-T. *H.264 Advanced Video Coding for Generic Audiovisual Services* (August, 2021). https://www.itu.int/rec/T-REC-H.264-202108-S/ .

ITU-T. *H.265 High Efficiency Video Coding* (August, 2021). https://www.itu.int/rec/T-REC-H.265-202108-S/ .

Google. *VP9 Bitstream & Decoding Process Specification* (February 22, 2017). https://storage.googleapis.com/downloads.webmproject.org/docs/vp9/vp9-bitstream-specification-v0.7-20170222-draft.pdf .

Alliance for Open Media. *AV1 Bitstream & Decoding Process Specification* (January 8, 2019). https://aomediacodec.github.io/av1-spec/av1-spec.pdf .

Jon Leech. *The Khronos Vulkan API Registry* (February 26, 2023). https://registry.khronos.org/vulkan/specs/latest/registry.html .

Jon Leech and Tobias Hector. *Vulkan Documentation and Extensions: Procedures and Conventions* (February 26, 2023). https://registry.khronos.org/vulkan/specs/latest/styleguide.html .

Architecture of the Vulkan Loader Interfaces (October, 2021). https://github.com/KhronosGroup/Vulkan-Loader/blob/main/docs/LoaderInterfaceArchitecture.md .

2.1.3. Informative Language

Some language in the specification is purely informative, intended to provide background information or make suggestions to implementors or developers. Such language does not impose normative requirements on implementations or applications.

All NOTEs are implicitly informative.

If an entire chapter, section, or appendix contains only informative language, its title will be suffixed with “(Informative)”. Unless so noted in the title, all chapters, sections, and appendices in this document are normative.

2.1.4. Technical Terminology

The Vulkan Specification makes use of common engineering and graphics terms such as **Pipeline**, **Shader**, and **Host** to identify and describe Vulkan API constructs and their attributes, states, and behaviors. The Glossary defines the basic meanings of these terms in the context of the Specification. The Specification text provides fuller definitions of the terms and may elaborate, extend, or clarify the Glossary definitions. When a term defined in the Glossary is used in

normative language within the Specification, the definitions within the Specification govern and supersede any meanings the terms may have in other technical contexts (i.e. outside the Specification).

2.1.5. Ratification

Ratification of a Vulkan core version or extension is a status conferred by vote of the Khronos Board of Promoters, bringing that core version or extension under the umbrella of the Khronos IP Rights Policy.

All Vulkan core versions and KHR extensions (including provisional specifications) are ratified, as are some multi-vendor EXT extensions. Ratification status of extensions is described in the Layers & Extensions (Informative) appendix.

NOTE

Ratification status is primarily of interest to IHVs developing GPU hardware and Vulkan implementations. For developers, ratification does not necessarily mean that an extension is “better”, has a more stable API, or is more widely supported than alternative ways of achieving that functionality.

Interactions between ratified and non-ratified extensions are not themselves ratified.

Chapter 3. Fundamentals

This chapter introduces fundamental concepts including the Vulkan architecture and execution model, API syntax, queues, pipeline configurations, numeric representation, state and state queries, and the different types of objects and shaders. It provides a framework for interpreting more specific descriptions of commands and behavior in the remainder of the Specification.

3.1. Host and Device Environment

The Vulkan Specification assumes and requires: the following properties of the host environment with respect to Vulkan implementations:

- The host **must** have runtime support for 8-, 16-, 32-, and 64-bit signed and unsigned twos-complement integers, all addressable at the granularity of their size in bytes.
- The host **must** have runtime support for 32- and 64-bit floating-point types satisfying the range and precision constraints in the Floating-Point Computation section.
- The representation and endianness of these types on the host **must** match the representation and endianness of the same types on every physical device supported.

NOTE Since a variety of data types and structures in Vulkan **may** be accessible by both host and physical device operations, the implementation **should** be able to access such data efficiently in both paths in order to facilitate writing portable and performant applications.

3.2. Execution Model

This section outlines the execution model of a Vulkan system.

Vulkan exposes one or more *devices*, each of which exposes one or more *queues* which **may** process work asynchronously to one another. The set of queues supported by a device is partitioned into *families*. Each family supports one or more types of functionality and **may** contain multiple queues with similar characteristics. Queues within a single family are considered *compatible* with one another, and work produced for a family of queues **can** be executed on any queue within that family. This specification defines the following types of functionality that queues **may** support: graphics, compute, video decode, video encode, protected memory management, sparse memory management, and transfer.

NOTE A single device **may** report multiple similar queue families rather than, or as well as, reporting multiple members of one or more of those families. This indicates that while members of those families have similar capabilities, they are *not* directly compatible with one another.

Device memory is explicitly managed by the application. Each device **may** advertise one or more heaps, representing different areas of memory. Memory heaps are either device-local or host-local, but are always visible to the device. Further detail about memory heaps is exposed via memory types available on that heap. Examples of memory areas that **may** be available on an

implementation include:

- *device-local* is memory that is physically connected to the device.
- *device-local, host visible* is device-local memory that is visible to the host.
- *host-local, host visible* is memory that is local to the host and visible to the device and host.

On other architectures, there **may** only be a single heap that **can** be used for any purpose.

3.2.1. Queue Operation

Each device supports a number of queues, which provide an interface for submitting work for execution on the device.

Queue submission commands are used to submit work, along with a set of synchronization primitives used to constrain the order of submitted operations. Queues are intended for asynchronous execution of submitted workloads, and queue submission commands **should** return as soon as the work has been submitted, without waiting for the work to complete. Once submitted to a queue, work will begin and complete execution without further application intervention.

There are no implicit ordering constraints between queue operations on different queues, or between queues and the host, so these **may** operate in any order with respect to each other. Explicit ordering constraints between different queues or with the host **can** be expressed with semaphores and fences.

Many commands for queues are recorded into *command buffers* first, before the command buffers are then submitted to a queue for execution. Command buffer submissions to a single queue respect submission order and other implicit ordering guarantees, but otherwise **may** overlap or execute out of order. Other types of batches and queue submissions against a single queue (e.g. sparse memory binding) have no implicit ordering constraints with any other queue submission or batch. Additional explicit ordering constraints between queue submissions and individual batches can be expressed with semaphores and fences.

Before a fence or semaphore is signaled, it is guaranteed that any previously submitted queue operations have completed execution, and that memory writes from those queue operations are available to future queue operations. Waiting on a signaled semaphore or fence guarantees that previous writes that are available are also visible to subsequent commands.

Command buffer boundaries, both between primary command buffers of the same or different batches or submissions as well as between primary and secondary command buffers, do not introduce any additional ordering constraints. In other words, submitting the set of command buffers (which **can** include executing secondary command buffers) between any semaphore or fence operations executes the recorded commands as if they had all been recorded into a single primary command buffer, except that the current state is reset on each boundary. Explicit ordering constraints **can** be expressed with explicit synchronization primitives.

There are a few implicit ordering guarantees between commands within a command buffer, but only covering a subset of execution. Additional explicit ordering constraints can be expressed with the various explicit synchronization primitives.

NOTE Implementations have significant freedom to overlap execution of work submitted to a queue, and this is common due to deep pipelining and parallelism in Vulkan devices.

Commands recorded in command buffers can perform actions, set state that persists across commands, synchronize other commands, or indirectly launch other commands, with some commands fulfilling several of these roles. The “Command Properties” section for each such command lists which of these roles the command takes:

Action

Action commands perform operations that can update values in memory. E.g. draw commands, dispatch commands.

State

State setting commands update the current state of a command buffer, affecting the operation of future action commands.

Synchronization

Synchronization commands impose ordering constraints on action commands, by introducing explicit execution and memory dependencies.

Indirection

Indirection commands execute other commands which were not directly recorded in the same command buffer.

NOTE In the absence of explicit synchronization or implicit ordering guarantees, action commands may overlap execution or execute out of order, potentially leading to data races. However, such reordering does not affect the current state observed by any action command. Each action command uses the state in effect at the point where the command occurs in the command buffer, regardless of when it is executed.

3.3. Object Model

The devices, queues, and other entities in Vulkan are represented by Vulkan objects. At the API level, all objects are referred to by handles. There are two classes of handles: dispatchable and non-dispatchable. *Dispatchable* handle types are a pointer to an opaque type. This pointer **may** be used by layers as part of intercepting API commands, and thus each API command takes a dispatchable type as its first parameter. Each object of a dispatchable type **must** have a unique handle value during its lifetime.

Non-dispatchable handle types are a 64-bit integer type whose meaning is implementation-dependent. If the `privateData` feature is enabled for a VkDevice, each object of a non-dispatchable type created on that device **must** have a handle value that is unique among objects created on that device, for the duration of the object’s lifetime. Otherwise, non-dispatchable handles **may** encode object information directly in the handle rather than acting as a reference to an underlying object, and thus **may** not have unique handle values. If handle values are not unique, then destroying one

such handle **must** not cause identical handles of other types to become invalid, and **must** not cause identical handles of the same type to become invalid if that handle value has been created more times than it has been destroyed.

All objects created or allocated from a `VkDevice` (i.e. with a `VkDevice` as the first parameter) are private to that device, and **must** not be used on other devices.

3.3.1. Object Lifetime

Objects are created or allocated by `vkCreate*` and `vkAllocate*` commands, respectively. Once an object is created or allocated, its "structure" is considered to be immutable, though the content of certain object types is still free to change. When an object is passed to another command, it **may** be accessed by the implementation, which **may** include both read and write access unless explicitly stated otherwise. Objects are destroyed or freed by `vkDestroy*` and `vkFree*` commands, respectively.

Objects that are allocated (rather than created) take resources from an existing pool object or memory heap, and when freed return resources to that pool or heap. While object creation and destruction are generally expected to be low-frequency occurrences during runtime, allocating and freeing objects **can** occur at high frequency. Pool objects help accommodate improved performance of the allocations and frees.

Applications are responsible for managing the lifetimes of Vulkan objects and memory passed into the Vulkan API. The access semantics of different functions in the API follow a typical pattern as laid out below, with any exceptions listed with the commands or objects that have them.

Application-owned memory and Vulkan objects **may** be accessed at any time during the execution of a command they are passed to. Vulkan objects that device addresses are retrieved from **may** be accessed by the implementation any time that memory backing the device address is accessed. Device addresses and Vulkan objects passed in during the creation or allocation of another object **may** be accessed by the implementation any time that the created/allocated object is accessed unless explicitly stated otherwise. Device addresses and Vulkan objects passed to a recording command (`vkCmd*`) **may** be accessed at any time during the execution of the command, when the command buffer is subsequently recorded into another command buffer, during any subsequent command that is recorded to either the command buffer or one it is recorded into, or while the command buffer is in the *pending state*, unless explicitly stated otherwise. If an application is using deferred host operations in a command, and that operation is successfully deferred, objects and memory passed to that command **may** be accessed at any time until the deferred operation is complete. Some additional operations hold references to other objects or application-owned memory beyond the duration of the command; in which case the access semantics and lifetime of those references are described by that command.

When destroying or freeing an object, implementations **must** not access any memory or other objects that **may** otherwise be accessed when the object is accessed. Applications **can** free or destroy objects in any order, except that parent objects **must** be freed only after all child objects are freed. An object is the parent of another child object if the parent was used as the first object parameter in the creation of the child. Once an object is freed or destroyed it **must** not be accessed again, either directly or via access through another object. Applications **must** not free or destroy any object while it is being accessed.

3.3.2. External Object Handles

As defined above, the scope of object handles created or allocated from a `VkDevice` is limited to that logical device. Objects which are not in scope are said to be external. To bring an external object into scope, an external handle **must** be exported from the object in the source scope and imported into the destination scope.

NOTE The scope of external handles and their associated resources **may** vary according to their type, but they **can** generally be shared across process and API boundaries.

3.4. Application Binary Interface

The mechanism by which Vulkan is made available to applications is platform- or implementation-defined. On many platforms the C interface described in this Specification is provided by a shared library. Since shared libraries can be changed independently of the applications that use them, they present particular compatibility challenges, and this Specification places some requirements on them.

Shared library implementations **must** use the default Application Binary Interface (ABI) of the standard C compiler for the platform, or provide customized API headers that cause application code to use the implementation's non-default ABI. An ABI in this context means the size, alignment, and layout of C data types; the procedure calling convention; and the naming convention for shared library symbols corresponding to C functions. Customizing the calling convention for a platform is usually accomplished by defining calling convention macros appropriately in `vk_platform.h`.

On platforms where Vulkan is provided as a shared library, library symbols beginning with "vk" and followed by a digit or uppercase letter are reserved for use by the implementation. Applications which use Vulkan **must** not provide definitions of these symbols. This allows the Vulkan shared library to be updated with additional symbols for new API versions or extensions without causing symbol conflicts with existing applications.

Shared library implementations **should** provide library symbols for commands in the highest version of this Specification they support, and for Window System Integration extensions relevant to the platform. They **may** also provide library symbols for commands defined by additional extensions.

NOTE

These requirements and recommendations are intended to allow implementors to take advantage of platform-specific conventions for SDKs, ABIs, library versioning mechanisms, etc. while still minimizing the code changes necessary to port applications or libraries between platforms. Platform vendors, or providers of the *de facto* standard Vulkan shared library for a platform, are encouraged to document what symbols the shared library provides and how it will be versioned when new symbols are added.

Applications **should** only rely on shared library symbols for commands in the minimum core version required by the application. vkGetInstanceProcAddr and vkGetDeviceProcAddr **should** be used to obtain function pointers for commands in core versions beyond the application's minimum required version.

3.5. Command Syntax and Duration

The Specification describes Vulkan commands as functions or procedures using C99 syntax. Language bindings for other languages such as C++ and JavaScript **may** allow for stricter parameter passing, or object-oriented interfaces.

Vulkan uses the standard C types for the base type of scalar parameters (e.g. types from `<stdint.h>`), with exceptions described below, or elsewhere in the text when appropriate:

VkBool32 represents boolean `True` and `False` values, since C does not have a sufficiently portable built-in boolean type:

```
// Provided by VK_VERSION_1_0
typedef uint32_t VkBool32;
```

VK_TRUE represents a boolean **True** (unsigned integer 1) value, and VK_FALSE a boolean **False** (unsigned integer 0) value.

All values returned from a Vulkan implementation in a VkBool32 will be either VK_TRUE or VK_FALSE.

Applications **must** not pass any other values than VK_TRUE or VK_FALSE into a Vulkan implementation where a VkBool32 is expected.

VK_TRUE is a constant representing a VkBool32 **True** value.

```
#define VK_TRUE                         1U
```

VK_FALSE is a constant representing a VkBool32 **False** value.

```
#define VK_FALSE                        0U
```

VkDeviceSize represents device memory size and offset values:

```
// Provided by VK_VERSION_1_0
typedef uint64_t VkDeviceSize;
```

VkDeviceAddress represents device buffer address values:

```
// Provided by VK_VERSION_1_0
typedef uint64_t VkDeviceAddress;
```

Valid Usage

- VUID-VkDeviceAddress-size-11364
  A valid VkDeviceAddress **must** be equal to the sum of an address retrieved from a VkBuffer via vkGetBufferDeviceAddress, and any offset in the range [0, `size`), where `size` is the value of VkBufferCreateInfo::`size` used to create that VkBuffer
- VUID-VkDeviceAddress-None-10894
  If a VkDeviceAddress was retrieved from a non-sparse buffer, that buffer **must** be bound completely and contiguously to a single `VkDeviceMemory` object

Commands that create Vulkan objects are of the form `vkCreate*` and take `Vk*CreateInfo` structures with the parameters needed to create the object. These Vulkan objects are destroyed with commands of the form `vkDestroy*`.

The last in-parameter to each command that creates or destroys a Vulkan object is `pAllocator`. The `pAllocator` parameter **can** be a non-`NULL` value, in which case allocations for the given object are delegated to an application provided callback. Refer to the Memory Allocation chapter for further details.

Commands that allocate Vulkan objects owned by pool objects are of the form `vkAllocate*`, and take `Vk*AllocateInfo` structures. These Vulkan objects are freed with commands of the form `vkFree*`. These objects do not take allocators; if host memory is needed, they will use the allocator that was specified when their parent pool was created.

Commands are recorded into a command buffer by calling API commands of the form `vkCmd*`. Each such command **may** have different restrictions on where it **can** be used: in a primary and/or secondary command buffer, inside and/or outside a render pass, and in one or more of the supported queue types. These restrictions are documented together with the definition of each such command.

The *duration* of a Vulkan command refers to the interval between calling the command and its return to the caller.

3.5.1. Lifetime of Retrieved Results

Information is retrieved from the implementation with commands of the form `vkGet*` and `vkEnumerate*`.

Unless otherwise specified for an individual command, the results are *invariant*; that is, they will remain unchanged when retrieved again by calling the same command with the same parameters, so long as those parameters themselves all remain valid.

3.5.2. Array Results

Some query commands of the form `vkGet*` and `vkEnumerate*` enable retrieving multiple results in the form of a return array. Such commands typically have two pointer arguments as follows:

- An element count pointer pointing to an integer variable, conventionally named as `p*Count`

where * is the capitalized singular form of the name of the retrieved values.

- A pointer to an array where the result array is retrieved, conventionally named as p* where * is the capitalized plural form of the name of the retrieved values.

If such commands are called with the array pointer set to NULL, then the number of retrievable elements is returned in the variable pointed to by the element count pointer. Otherwise, the element count pointer **must** point to a variable set by the application to the number of elements in the return array, and on return the variable is overwritten with the number of elements actually written to the return array. If the input element count is less than the number of retrievable array elements, the query will write only as many elements to the return array as specified by the element count variable set by the application, and the command will return VK_INCOMPLETE instead of VK_SUCCESS, to indicate that not all retrievable array elements were returned.

NOTE

In practice, this means that applications will typically call such query commands twice:

- First, with the array pointer set to NULL, to retrieve the number of retrievable elements.
- Second, with the array pointer pointing to an application allocated storage for at least as many elements as indicated by the variable pointed to by the element count pointer, to retrieve at most as many of the retrievable elements.

Query commands that return one or more structures, regardless of whether they return a single or an array of structures with or without a pNext chain, **may** also contain arrays within those structures. Such return arrays are typically defined in the form of two members as follows:

- An integer value specifying the element count, conventionally named as *Count where * is the singular form of the name of the retrieved values.
- A pointer to an array where the result array is retrieved, conventionally named as p* where * is the capitalized plural form of the name of the retrieved values.

Analogously to query commands that return multiple results, if the command is called with the array pointer member of the output structure in question set to NULL, then the number of retrievable elements is returned in the element count member of that output structure. Otherwise, the element count **must** specify the number of elements in the return array, and on return the element count member is overwritten with the number of elements actually written to the return array. If the input element count is less than the number of retrievable array elements, the query will write only as many elements to the return array as specified by the input element count, and the command will return VK_INCOMPLETE instead of VK_SUCCESS, if the query command has a VkResult return type, to indicate that not all retrievable array elements were returned.

NOTE

Applications need to separately track the value they provided as the input element count member for such arrays and compare those with the returned element counts in order to determine whether the actually returned element count is smaller than the size of the return array. Another side effect of this is that it is impossible for the application to determine if the number of retrievable elements has increased beyond the provided input element count so using return arrays in output

structures **should** be limited to *invariant* array results. In practice, this means that applications will typically call such query commands multiple times:

- First, with the array pointer member(s) set to `NULL`, to retrieve the number(s) of retrievable elements.
- Second, with the array pointer(s) pointing to an application allocated storage for at least as many elements as indicated by the element count member(s), to retrieve at most as many of the retrievable elements.
- Then the process **may** need to be repeated for all other newly introduced return arrays in any nested output structures indirectly specified through the previously retrieved result arrays.

Regardless of the type of query command, any array pointer member of an output structure **must** either be `NULL`, or point to an application-allocated array. Query commands **must** not return a pointer to implementation allocated storage in any output structure.

3.5.3. Opaque Binary Data Results

Some query commands of the form `vkGet*` retrieve opaque binary data in the form of a byte array and have a possible result code of VK_ERROR_NOT_ENOUGH_SPACE_KHR. Such commands always have two pointer arguments as follows:

- A binary data size pointer pointing to a `size_t` variable, conventionally named as `p*Size` where `*` is the capitalized form of the name of the retrieved binary data.
- A pointer to a byte array where the binary data is retrieved, conventionally named as `p*` where `*` is the capitalized form of the name of the retrieved binary data.

If such commands are called with the binary pointer not set to `NULL`, the binary size pointer **must** point to a variable set by the application to the allocated size of the binary pointer. These arguments **may** also be placed in an extensible structure, in which case the binary data size argument is not a pointer.

If the input binary size is less than the total retrievable binary size, the query will not write any data to the location pointed to by the binary pointer, and the command will return VK_ERROR_NOT_ENOUGH_SPACE_KHR instead of VK_SUCCESS.

If the return code is VK_SUCCESS or VK_ERROR_NOT_ENOUGH_SPACE_KHR, the total size of the binary data that **can** be retrieved is returned in the variable pointed to by the binary size pointer.

If multiple binaries are being retrieved, VK_ERROR_NOT_ENOUGH_SPACE_KHR will be returned if any input binary sizes are less than their respective total retrievable binary sizes. Unless otherwise specified, this command will determine writing data to each binary individually based on if their input binary sizes are sufficiently sized, following the behavior for single binary retrieval.

For all other error codes, the contents of the return structures are undefined.

NOTE

If VK_ERROR_NOT_ENOUGH_SPACE_KHR is returned with a command that returns multiple binaries, the application **can** determine which binaries are undersized by

comparing the total binary size that is returned for each binary against the allocated size that was provided to the command.

NOTE

Some binary queries do not behave consistently with this pattern for historical reasons, primarily that the VK_ERROR_NOT_ENOUGH_SPACE_KHR error code was not defined until after those queries were written.

A NOTE is added to each such query, describing such inconsistent behavior.

3.6. Threading Behavior

Vulkan is intended to provide scalable performance when used on multiple host threads. All commands support being called concurrently from multiple threads, but certain parameters, or components of parameters are defined to be *externally synchronized*. This means that the caller **must** guarantee that no more than one thread is using such a parameter at a given time.

More precisely, Vulkan commands use simple stores to update the state of Vulkan objects. The implementation **may** not synchronize accesses to memory parameters or object parameters declared as externally synchronized with other accesses. If two commands access the same object or memory and at least one of the commands declares the object to be externally synchronized, then the caller **must** guarantee not only that the commands do not execute simultaneously, but also that the two commands are separated by an appropriate memory barrier (if needed). Similarly, if a Vulkan command accesses a non-const memory parameter and the application also accesses that memory, or if the application writes to that memory and the command accesses it as a const memory parameter, the application **must** ensure the accesses are properly synchronized with a memory barrier if needed.

NOTE

Memory barriers are particularly relevant for hosts based on the ARM CPU architecture, which is more weakly ordered than many developers are accustomed to from x86/x64 programming. Fortunately, most higher-level synchronization primitives (like the pthread library) perform memory barriers as a part of mutual exclusion, so mutexing Vulkan objects via these primitives will have the desired effect.

Any object parameters that are not labeled as externally synchronized are either not mutated by the command or are internally synchronized. Additionally, certain objects related to a command's parameters (e.g. command pools and descriptor pools) **may** be affected by a command, and **must** also be externally synchronized. These implicit parameters are documented as described below.

Parameters of commands that are externally synchronized are listed below.

Externally Synchronized Parameters and Members

- The `instance` parameter in vkDestroyInstance
- The `device` parameter in vkDestroyDevice
- The `queue` parameter in vkQueueSubmit, if it was not created with

VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR

- The `fence` parameter in vkQueueSubmit
- The `queue` parameter in vkQueueWaitIdle, if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR
- The `memory` parameter in vkFreeMemory
- The `memory` parameter in vkMapMemory
- The `memory` parameter in vkUnmapMemory
- The `buffer` parameter in vkBindBufferMemory
- The `image` parameter in vkBindImageMemory
- The `queue` parameter in vkQueueBindSparse, if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR
- The `fence` parameter in vkQueueBindSparse
- The `fence` parameter in vkDestroyFence
- The `semaphore` parameter in vkDestroySemaphore
- The `queryPool` parameter in vkDestroyQueryPool
- The `buffer` parameter in vkDestroyBuffer
- The `image` parameter in vkDestroyImage
- The `imageView` parameter in vkDestroyImageView
- The `commandPool` parameter in vkDestroyCommandPool
- The `commandPool` parameter in vkResetCommandPool
- The `commandPool` parameter in vkFreeCommandBuffers
- The `commandBuffer` parameter in vkBeginCommandBuffer
- The `commandBuffer` parameter in vkEndCommandBuffer
- The `commandBuffer` parameter in vkResetCommandBuffer
- The `commandBuffer` parameter in vkCmdCopyBuffer
- The `commandBuffer` parameter in vkCmdCopyImage
- The `commandBuffer` parameter in vkCmdCopyBufferToImage
- The `commandBuffer` parameter in vkCmdCopyImageToBuffer
- The `commandBuffer` parameter in vkCmdUpdateBuffer
- The `commandBuffer` parameter in vkCmdFillBuffer
- The `commandBuffer` parameter in vkCmdPipelineBarrier
- The `commandBuffer` parameter in vkCmdBeginQuery
- The `commandBuffer` parameter in vkCmdEndQuery
- The `commandBuffer` parameter in vkCmdResetQueryPool
- The `commandBuffer` parameter in vkCmdWriteTimestamp

- The `commandBuffer` parameter in vkCmdCopyQueryPoolResults
- The `commandBuffer` parameter in vkCmdExecuteCommands
- The `event` parameter in vkDestroyEvent
- The `event` parameter in vkSetEvent
- The `event` parameter in vkResetEvent
- The `bufferView` parameter in vkDestroyBufferView
- The `shaderModule` parameter in vkDestroyShaderModule
- The `pipelineCache` parameter in vkDestroyPipelineCache
- The `dstCache` parameter in vkMergePipelineCaches, conditionally[1]
- The `pipelineCache` parameter in vkCreateComputePipelines, conditionally[1]
- The `pipeline` parameter in vkDestroyPipeline
- The `pipelineLayout` parameter in vkDestroyPipelineLayout
- The `sampler` parameter in vkDestroySampler
- The `descriptorSetLayout` parameter in vkDestroyDescriptorSetLayout
- The `descriptorPool` parameter in vkDestroyDescriptorPool
- The `descriptorPool` parameter in vkResetDescriptorPool
- The `descriptorPool` parameter in vkFreeDescriptorSets
- The `commandBuffer` parameter in vkCmdBindPipeline
- The `commandBuffer` parameter in vkCmdBindDescriptorSets
- The `commandBuffer` parameter in vkCmdClearColorImage
- The `commandBuffer` parameter in vkCmdDispatch
- The `commandBuffer` parameter in vkCmdDispatchIndirect
- The `commandBuffer` parameter in vkCmdSetEvent
- The `commandBuffer` parameter in vkCmdResetEvent
- The `commandBuffer` parameter in vkCmdWaitEvents
- The `commandBuffer` parameter in vkCmdPushConstants
- The `pipelineCache` parameter in vkCreateGraphicsPipelines, conditionally[1]
- The `framebuffer` parameter in vkDestroyFramebuffer
- The `renderPass` parameter in vkDestroyRenderPass
- The `commandBuffer` parameter in vkCmdSetViewport
- The `commandBuffer` parameter in vkCmdSetScissor
- The `commandBuffer` parameter in vkCmdSetLineWidth
- The `commandBuffer` parameter in vkCmdSetDepthBias
- The `commandBuffer` parameter in vkCmdSetBlendConstants
- The `commandBuffer` parameter in vkCmdSetDepthBounds

- The `commandBuffer` parameter in vkCmdSetStencilCompareMask
- The `commandBuffer` parameter in vkCmdSetStencilWriteMask
- The `commandBuffer` parameter in vkCmdSetStencilReference
- The `commandBuffer` parameter in vkCmdBindIndexBuffer
- The `commandBuffer` parameter in vkCmdBindVertexBuffers
- The `commandBuffer` parameter in vkCmdDraw
- The `commandBuffer` parameter in vkCmdDrawIndexed
- The `commandBuffer` parameter in vkCmdDrawIndirect
- The `commandBuffer` parameter in vkCmdDrawIndexedIndirect
- The `commandBuffer` parameter in vkCmdBlitImage
- The `commandBuffer` parameter in vkCmdClearDepthStencilImage
- The `commandBuffer` parameter in vkCmdClearAttachments
- The `commandBuffer` parameter in vkCmdResolveImage
- The `commandBuffer` parameter in vkCmdBeginRenderPass
- The `commandBuffer` parameter in vkCmdNextSubpass
- The `commandBuffer` parameter in vkCmdEndRenderPass
- The `commandBuffer` parameter in vkCmdSetDeviceMask
- The `commandPool` parameter in vkTrimCommandPool
- The `commandBuffer` parameter in vkCmdDispatchBase
- The `descriptorUpdateTemplate` parameter in vkDestroyDescriptorUpdateTemplate
- The `descriptorSet` parameter in vkUpdateDescriptorSetWithTemplate, conditionally[1]
- The `ycbcrConversion` parameter in vkDestroySamplerYcbcrConversion
- The `commandBuffer` parameter in vkCmdDrawIndirectCount
- The `commandBuffer` parameter in vkCmdDrawIndexedIndirectCount
- The `commandBuffer` parameter in vkCmdBeginRenderPass2
- The `commandBuffer` parameter in vkCmdNextSubpass2
- The `commandBuffer` parameter in vkCmdEndRenderPass2
- The `privateDataSlot` parameter in vkDestroyPrivateDataSlot
- The `commandBuffer` parameter in vkCmdPipelineBarrier2
- The `commandBuffer` parameter in vkCmdWriteTimestamp2
- The `queue` parameter in vkQueueSubmit2, if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR
- The `fence` parameter in vkQueueSubmit2
- The `commandBuffer` parameter in vkCmdCopyBuffer2
- The `commandBuffer` parameter in vkCmdCopyImage2

- The `commandBuffer` parameter in vkCmdCopyBufferToImage2
- The `commandBuffer` parameter in vkCmdCopyImageToBuffer2
- The `commandBuffer` parameter in vkCmdSetEvent2
- The `commandBuffer` parameter in vkCmdResetEvent2
- The `commandBuffer` parameter in vkCmdWaitEvents2
- The `commandBuffer` parameter in vkCmdBlitImage2
- The `commandBuffer` parameter in vkCmdResolveImage2
- The `commandBuffer` parameter in vkCmdBeginRendering
- The `commandBuffer` parameter in vkCmdEndRendering
- The `commandBuffer` parameter in vkCmdSetCullMode
- The `commandBuffer` parameter in vkCmdSetFrontFace
- The `commandBuffer` parameter in vkCmdSetPrimitiveTopology
- The `commandBuffer` parameter in vkCmdSetViewportWithCount
- The `commandBuffer` parameter in vkCmdSetScissorWithCount
- The `commandBuffer` parameter in vkCmdBindVertexBuffers2
- The `commandBuffer` parameter in vkCmdSetDepthTestEnable
- The `commandBuffer` parameter in vkCmdSetDepthWriteEnable
- The `commandBuffer` parameter in vkCmdSetDepthCompareOp
- The `commandBuffer` parameter in vkCmdSetDepthBoundsTestEnable
- The `commandBuffer` parameter in vkCmdSetStencilTestEnable
- The `commandBuffer` parameter in vkCmdSetStencilOp
- The `commandBuffer` parameter in vkCmdSetRasterizerDiscardEnable
- The `commandBuffer` parameter in vkCmdSetDepthBiasEnable
- The `commandBuffer` parameter in vkCmdSetPrimitiveRestartEnable
- The `commandBuffer` parameter in vkCmdPushDescriptorSet
- The `commandBuffer` parameter in vkCmdPushDescriptorSetWithTemplate
- The `commandBuffer` parameter in vkCmdBindDescriptorSets2
- The `commandBuffer` parameter in vkCmdPushConstants2
- The `commandBuffer` parameter in vkCmdPushDescriptorSet2
- The `commandBuffer` parameter in vkCmdPushDescriptorSetWithTemplate2
- The `commandBuffer` parameter in vkCmdSetLineStipple
- The `commandBuffer` parameter in vkCmdBindIndexBuffer2
- The `commandBuffer` parameter in vkCmdSetRenderingAttachmentLocations
- The `commandBuffer` parameter in vkCmdSetRenderingInputAttachmentIndices
- The `surface` parameter in vkDestroySurfaceKHR

- The `swapchain` parameter in vkDestroySwapchainKHR
- The `swapchain` parameter in vkAcquireNextImageKHR
- The `semaphore` parameter in vkAcquireNextImageKHR
- The `fence` parameter in vkAcquireNextImageKHR
- The `queue` parameter in vkQueuePresentKHR, if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR
- The `surface` parameter in vkGetDeviceGroupSurfacePresentModesKHR
- The `surface` parameter in vkGetPhysicalDevicePresentRectanglesKHR
- The `display` parameter in vkCreateDisplayModeKHR
- The `mode` parameter in vkGetDisplayPlaneCapabilitiesKHR
- The `videoSession` parameter in vkDestroyVideoSessionKHR
- The `videoSession` parameter in vkBindVideoSessionMemoryKHR
- The `videoSessionParameters` parameter in vkDestroyVideoSessionParametersKHR
- The `commandBuffer` parameter in vkCmdBeginVideoCodingKHR
- The `commandBuffer` parameter in vkCmdEndVideoCodingKHR
- The `commandBuffer` parameter in vkCmdControlVideoCodingKHR
- The `commandBuffer` parameter in vkCmdDecodeVideoKHR
- The `commandBuffer` parameter in vkCmdBeginRenderingKHR
- The `commandBuffer` parameter in vkCmdEndRenderingKHR
- The `commandBuffer` parameter in vkCmdSetDeviceMaskKHR
- The `commandBuffer` parameter in vkCmdDispatchBaseKHR
- The `commandPool` parameter in vkTrimCommandPoolKHR
- The `commandBuffer` parameter in vkCmdPushDescriptorSetKHR
- The `commandBuffer` parameter in vkCmdPushDescriptorSetWithTemplateKHR
- The `descriptorUpdateTemplate` parameter in vkDestroyDescriptorUpdateTemplateKHR
- The `descriptorSet` parameter in vkUpdateDescriptorSetWithTemplateKHR, conditionally[1]
- The `commandBuffer` parameter in vkCmdBeginRenderPass2KHR
- The `commandBuffer` parameter in vkCmdNextSubpass2KHR
- The `commandBuffer` parameter in vkCmdEndRenderPass2KHR
- The `swapchain` parameter in vkGetSwapchainStatusKHR
- The `ycbcrConversion` parameter in vkDestroySamplerYcbcrConversionKHR
- The `commandBuffer` parameter in vkCmdDrawIndirectCountKHR
- The `commandBuffer` parameter in vkCmdDrawIndexedIndirectCountKHR
- The `commandBuffer` parameter in vkCmdSetFragmentShadingRateKHR
- The `commandBuffer` parameter in vkCmdSetRenderingAttachmentLocationsKHR

- The `commandBuffer` parameter in vkCmdSetRenderingInputAttachmentIndicesKHR
- The `swapchain` parameter in vkWaitForPresentKHR
- The `operation` parameter in vkDestroyDeferredOperationKHR
- The `commandBuffer` parameter in vkCmdEncodeVideoKHR
- The `commandBuffer` parameter in vkCmdSetEvent2KHR
- The `commandBuffer` parameter in vkCmdResetEvent2KHR
- The `commandBuffer` parameter in vkCmdWaitEvents2KHR
- The `commandBuffer` parameter in vkCmdPipelineBarrier2KHR
- The `commandBuffer` parameter in vkCmdWriteTimestamp2KHR
- The `queue` parameter in vkQueueSubmit2KHR, if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR
- The `fence` parameter in vkQueueSubmit2KHR
- The `commandBuffer` parameter in vkCmdBindIndexBuffer3KHR
- The `commandBuffer` parameter in vkCmdBindVertexBuffers3KHR
- The `commandBuffer` parameter in vkCmdDrawIndirect2KHR
- The `commandBuffer` parameter in vkCmdDrawIndexedIndirect2KHR
- The `commandBuffer` parameter in vkCmdDispatchIndirect2KHR
- The `commandBuffer` parameter in vkCmdCopyMemoryKHR
- The `commandBuffer` parameter in vkCmdCopyMemoryToImageKHR
- The `commandBuffer` parameter in vkCmdCopyImageToMemoryKHR
- The `commandBuffer` parameter in vkCmdUpdateMemoryKHR
- The `commandBuffer` parameter in vkCmdFillMemoryKHR
- The `commandBuffer` parameter in vkCmdCopyQueryPoolResultsToMemoryKHR
- The `commandBuffer` parameter in vkCmdDrawIndirectCount2KHR
- The `commandBuffer` parameter in vkCmdDrawIndexedIndirectCount2KHR
- The `commandBuffer` parameter in vkCmdBeginConditionalRendering2EXT
- The `commandBuffer` parameter in vkCmdBindTransformFeedbackBuffers2EXT
- The `commandBuffer` parameter in vkCmdBeginTransformFeedback2EXT
- The `commandBuffer` parameter in vkCmdEndTransformFeedback2EXT
- The `commandBuffer` parameter in vkCmdDrawIndirectByteCount2EXT
- The `commandBuffer` parameter in vkCmdDrawMeshTasksIndirect2EXT
- The `commandBuffer` parameter in vkCmdDrawMeshTasksIndirectCount2EXT
- The `commandBuffer` parameter in vkCmdCopyBuffer2KHR
- The `commandBuffer` parameter in vkCmdCopyImage2KHR
- The `commandBuffer` parameter in vkCmdCopyBufferToImage2KHR

- The `commandBuffer` parameter in vkCmdCopyImageToBuffer2KHR
- The `commandBuffer` parameter in vkCmdBlitImage2KHR
- The `commandBuffer` parameter in vkCmdResolveImage2KHR
- The `commandBuffer` parameter in vkCmdTraceRaysIndirect2KHR
- The `commandBuffer` parameter in vkCmdBindIndexBuffer2KHR
- The `swapchain` parameter in vkWaitForPresent2KHR
- The `pipelineBinary` parameter in vkDestroyPipelineBinaryKHR
- The `commandBuffer` parameter in vkCmdSetLineStippleKHR
- The `commandBuffer` parameter in vkCmdBindDescriptorSets2KHR
- The `commandBuffer` parameter in vkCmdPushConstants2KHR
- The `commandBuffer` parameter in vkCmdPushDescriptorSet2KHR
- The `commandBuffer` parameter in vkCmdPushDescriptorSetWithTemplate2KHR
- The `commandBuffer` parameter in vkCmdSetDescriptorBufferOffsets2EXT
- The `commandBuffer` parameter in vkCmdBindDescriptorBufferEmbeddedSamplers2EXT
- The `commandBuffer` parameter in vkCmdCopyMemoryIndirectKHR
- The `commandBuffer` parameter in vkCmdCopyMemoryToImageIndirectKHR
- The `commandBuffer` parameter in vkCmdEndRendering2KHR
- The `commandBuffer` parameter in vkCmdBindTransformFeedbackBuffersEXT
- The `commandBuffer` parameter in vkCmdBeginTransformFeedbackEXT
- The `commandBuffer` parameter in vkCmdEndTransformFeedbackEXT
- The `commandBuffer` parameter in vkCmdBeginQueryIndexedEXT
- The `commandBuffer` parameter in vkCmdEndQueryIndexedEXT
- The `commandBuffer` parameter in vkCmdDrawIndirectByteCountEXT
- The `commandBuffer` parameter in vkCmdBeginConditionalRenderingEXT
- The `commandBuffer` parameter in vkCmdEndConditionalRenderingEXT
- The `commandBuffer` parameter in vkCmdSetDiscardRectangleEXT
- The `commandBuffer` parameter in vkCmdSetDiscardRectangleEnableEXT
- The `commandBuffer` parameter in vkCmdSetDiscardRectangleModeEXT
- The `objectHandle` member of the `pNameInfo` parameter in vkSetDebugUtilsObjectNameEXT
- The `queue` parameter in vkQueueBeginDebugUtilsLabelEXT, if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR
- The `queue` parameter in vkQueueEndDebugUtilsLabelEXT, if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR
- The `queue` parameter in vkQueueInsertDebugUtilsLabelEXT, if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR

- The `commandBuffer` parameter in vkCmdBeginDebugUtilsLabelEXT
- The `commandBuffer` parameter in vkCmdEndDebugUtilsLabelEXT
- The `commandBuffer` parameter in vkCmdInsertDebugUtilsLabelEXT
- The `messenger` parameter in vkDestroyDebugUtilsMessengerEXT
- The `commandBuffer` parameter in vkCmdBindSamplerHeapEXT
- The `commandBuffer` parameter in vkCmdBindResourceHeapEXT
- The `commandBuffer` parameter in vkCmdPushDataEXT
- The `commandBuffer` parameter in vkCmdSetSampleLocationsEXT
- The `validationCache` parameter in vkDestroyValidationCacheEXT
- The `dstCache` parameter in vkMergeValidationCachesEXT
- The `swapchain` parameter in vkSetSwapchainPresentTimingQueueSizeEXT
- The `swapchain` parameter in vkGetSwapchainTimingPropertiesEXT
- The `swapchain` parameter in vkGetSwapchainTimeDomainPropertiesEXT
- The `commandBuffer` parameter in vkCmdSetCullModeEXT
- The `commandBuffer` parameter in vkCmdSetFrontFaceEXT
- The `commandBuffer` parameter in vkCmdSetPrimitiveTopologyEXT
- The `commandBuffer` parameter in vkCmdSetViewportWithCountEXT
- The `commandBuffer` parameter in vkCmdSetScissorWithCountEXT
- The `commandBuffer` parameter in vkCmdBindVertexBuffers2EXT
- The `commandBuffer` parameter in vkCmdSetDepthTestEnableEXT
- The `commandBuffer` parameter in vkCmdSetDepthWriteEnableEXT
- The `commandBuffer` parameter in vkCmdSetDepthCompareOpEXT
- The `commandBuffer` parameter in vkCmdSetDepthBoundsTestEnableEXT
- The `commandBuffer` parameter in vkCmdSetStencilTestEnableEXT
- The `commandBuffer` parameter in vkCmdSetStencilOpEXT
- The `commandBuffer` parameter in vkCmdSetDepthBias2EXT
- The `commandBuffer` parameter in vkCmdBindDescriptorBuffersEXT
- The `commandBuffer` parameter in vkCmdSetDescriptorBufferOffsetsEXT
- The `commandBuffer` parameter in vkCmdBindDescriptorBufferEmbeddedSamplersEXT
- The `commandBuffer` parameter in vkCmdSetVertexInputEXT
- The `commandBuffer` parameter in vkCmdSetPatchControlPointsEXT
- The `commandBuffer` parameter in vkCmdSetRasterizerDiscardEnableEXT
- The `commandBuffer` parameter in vkCmdSetDepthBiasEnableEXT
- The `commandBuffer` parameter in vkCmdSetLogicOpEXT
- The `commandBuffer` parameter in vkCmdSetPrimitiveRestartEnableEXT

- The `commandBuffer` parameter in vkCmdSetColorWriteEnableEXT
- The `commandBuffer` parameter in vkCmdDrawMultiEXT
- The `commandBuffer` parameter in vkCmdDrawMultiIndexedEXT
- The `micromap` parameter in vkDestroyMicromapEXT
- The `commandBuffer` parameter in vkCmdBuildMicromapsEXT
- The `commandBuffer` parameter in vkCmdCopyMicromapEXT
- The `commandBuffer` parameter in vkCmdCopyMicromapToMemoryEXT
- The `commandBuffer` parameter in vkCmdCopyMemoryToMicromapEXT
- The `commandBuffer` parameter in vkCmdWriteMicromapsPropertiesEXT
- The `commandBuffer` parameter in vkCmdSetDepthClampEnableEXT
- The `commandBuffer` parameter in vkCmdSetPolygonModeEXT
- The `commandBuffer` parameter in vkCmdSetRasterizationSamplesEXT
- The `commandBuffer` parameter in vkCmdSetSampleMaskEXT
- The `commandBuffer` parameter in vkCmdSetAlphaToCoverageEnableEXT
- The `commandBuffer` parameter in vkCmdSetAlphaToOneEnableEXT
- The `commandBuffer` parameter in vkCmdSetLogicOpEnableEXT
- The `commandBuffer` parameter in vkCmdSetColorBlendEnableEXT
- The `commandBuffer` parameter in vkCmdSetColorBlendEquationEXT
- The `commandBuffer` parameter in vkCmdSetColorWriteMaskEXT
- The `commandBuffer` parameter in vkCmdSetTessellationDomainOriginEXT
- The `commandBuffer` parameter in vkCmdSetRasterizationStreamEXT
- The `commandBuffer` parameter in vkCmdSetConservativeRasterizationModeEXT
- The `commandBuffer` parameter in vkCmdSetExtraPrimitiveOverestimationSizeEXT
- The `commandBuffer` parameter in vkCmdSetDepthClipEnableEXT
- The `commandBuffer` parameter in vkCmdSetSampleLocationsEnableEXT
- The `commandBuffer` parameter in vkCmdSetColorBlendAdvancedEXT
- The `commandBuffer` parameter in vkCmdSetProvokingVertexModeEXT
- The `commandBuffer` parameter in vkCmdSetLineRasterizationModeEXT
- The `commandBuffer` parameter in vkCmdSetLineStippleEnableEXT
- The `commandBuffer` parameter in vkCmdSetDepthClipNegativeOneToOneEXT
- The `shader` parameter in vkDestroyShaderEXT
- The `commandBuffer` parameter in vkCmdBindShadersEXT
- The `commandBuffer` parameter in vkCmdSetDepthClampRangeEXT
- The `commandBuffer` parameter in vkCmdSetAttachmentFeedbackLoopEnableEXT
- The `commandBuffer` parameter in vkCmdDecompressMemoryEXT

- The `commandBuffer` parameter in vkCmdDecompressMemoryIndirectCountEXT
- The `commandBuffer` parameter in vkCmdPreprocessGeneratedCommandsEXT
- The `stateCommandBuffer` parameter in vkCmdPreprocessGeneratedCommandsEXT
- The `commandBuffer` parameter in vkCmdExecuteGeneratedCommandsEXT
- The `indirectCommandsLayout` parameter in vkDestroyIndirectCommandsLayoutEXT
- The `indirectExecutionSet` parameter in vkDestroyIndirectExecutionSetEXT
- The `indirectExecutionSet` parameter in vkUpdateIndirectExecutionSetPipelineEXT
- The `indirectExecutionSet` parameter in vkUpdateIndirectExecutionSetShaderEXT
- The `commandBuffer` parameter in vkCmdEndRendering2EXT
- The `commandBuffer` parameter in vkCmdBeginCustomResolveEXT
- The `commandBuffer` parameter in vkCmdSetPrimitiveRestartIndexEXT
- The `accelerationStructure` parameter in vkDestroyAccelerationStructureKHR
- The `commandBuffer` parameter in vkCmdBuildAccelerationStructuresKHR
- The `commandBuffer` parameter in vkCmdBuildAccelerationStructuresIndirectKHR
- The `commandBuffer` parameter in vkCmdCopyAccelerationStructureKHR
- The `commandBuffer` parameter in vkCmdCopyAccelerationStructureToMemoryKHR
- The `commandBuffer` parameter in vkCmdCopyMemoryToAccelerationStructureKHR
- The `commandBuffer` parameter in vkCmdWriteAccelerationStructuresPropertiesKHR
- The `commandBuffer` parameter in vkCmdTraceRaysKHR
- The `pipelineCache` parameter in vkCreateRayTracingPipelinesKHR, conditionally[1]
- The `commandBuffer` parameter in vkCmdTraceRaysIndirectKHR
- The `commandBuffer` parameter in vkCmdSetRayTracingPipelineStackSizeKHR
- The `commandBuffer` parameter in vkCmdDrawMeshTasksEXT
- The `commandBuffer` parameter in vkCmdDrawMeshTasksIndirectEXT
- The `commandBuffer` parameter in vkCmdDrawMeshTasksIndirectCountEXT
- The `commandPool` member of VkCommandBufferAllocateInfo
- The `dstSet` member of VkCopyDescriptorSet, conditionally[1]
- The `descriptorPool` member of VkDescriptorSetAllocateInfo
- The `buffer` member of VkBindBufferMemoryInfo
- The `image` member of VkBindImageMemoryInfo
- The `memory` member of VkMemoryMapInfo
- The `memory` member of VkMemoryUnmapInfo
- The `surface` member of VkSwapchainCreateInfoKHR
- The `oldSwapchain` member of VkSwapchainCreateInfoKHR
- The `swapchain` member of VkBindImageMemorySwapchainInfoKHR

- The `swapchain` member of VkAcquireNextImageInfoKHR
- The `semaphore` member of VkAcquireNextImageInfoKHR
- The `fence` member of VkAcquireNextImageInfoKHR
- The `semaphore` member of VkImportSemaphoreWin32HandleInfoKHR
- The `semaphore` member of VkImportSemaphoreFdInfoKHR
- The `fence` member of VkImportFenceWin32HandleInfoKHR
- The `fence` member of VkImportFenceFdInfoKHR
- The `mode` member of VkDisplayPlaneInfo2KHR
- The `pipeline` member of VkReleaseCapturedPipelineDataInfoKHR
- The `swapchain` member of VkReleaseSwapchainImagesInfoKHR
- The `objectHandle` member of VkDebugUtilsObjectTagInfoEXT
- The `swapchain` member of VkPastPresentationTimingInfoEXT

[1] See Valid Usage language for this token for details.

There are also a few instances where a command **can** take in an application-allocated list whose contents are externally synchronized parameters. In these cases, the caller **must** guarantee that at most one thread is using a given element within the list at a given time. These parameters are listed below.

Externally Synchronized Parameter and Member Lists

- Each element of the `pFences` parameter in vkResetFences
- Each element of the `pCommandBuffers` parameter in vkFreeCommandBuffers
- Each element of the `pDescriptorSets` parameter in vkFreeDescriptorSets
- The `dstSet` member of each element of the `pDescriptorWrites` parameter in vkUpdateDescriptorSets, conditionally[1]
- Each element of the `pSwapchains` parameter in vkSetHdrMetadataEXT
- Each element of the `pWaitSemaphores` member of VkPresentInfoKHR
- Each element of the `pSwapchains` member of VkPresentInfoKHR

[1] See Valid Usage language for this token for details.

In addition, there are some implicit parameters that need to be externally synchronized. For example, when a `commandBuffer` parameter needs to be externally synchronized, it implies that the `commandPool` from which that command buffer was allocated also needs to be externally synchronized. The implicit parameters and their associated object are listed below.

Implicit Externally Synchronized Parameters

- All VkPhysicalDevice objects enumerated from instance in vkDestroyInstance
- All VkQueue objects created from device in vkDestroyDevice
- All VkQueue objects created from device that are not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR in vkDeviceWaitIdle
- The VkCommandPool that commandBuffer was allocated from in vkBeginCommandBuffer
- The VkCommandPool that commandBuffer was allocated from in vkEndCommandBuffer
- The VkCommandPool that commandBuffer was allocated from in vkResetCommandBuffer
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyBuffer
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyImage
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyBufferToImage
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyImageToBuffer
- The VkCommandPool that commandBuffer was allocated from, in vkCmdUpdateBuffer
- The VkCommandPool that commandBuffer was allocated from, in vkCmdFillBuffer
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPipelineBarrier
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBeginQuery
- The VkCommandPool that commandBuffer was allocated from, in vkCmdEndQuery
- The VkCommandPool that commandBuffer was allocated from, in vkCmdResetQueryPool
- The VkCommandPool that commandBuffer was allocated from, in vkCmdWriteTimestamp
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyQueryPoolResults
- The VkCommandPool that commandBuffer was allocated from, in vkCmdExecuteCommands
- Any VkDescriptorSet objects allocated from descriptorPool in vkResetDescriptorPool
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindPipeline
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindDescriptorSets
- The VkCommandPool that commandBuffer was allocated from, in vkCmdClearColorImage
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDispatch
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDispatchIndirect
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetEvent
- The VkCommandPool that commandBuffer was allocated from, in vkCmdResetEvent
- The VkCommandPool that commandBuffer was allocated from, in vkCmdWaitEvents
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPushConstants
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetViewport
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetScissor

- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetLineWidth
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthBias
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetBlendConstants
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthBounds
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetStencilCompareMask
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetStencilWriteMask
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetStencilReference
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindIndexBuffer
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindVertexBuffers
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDraw
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawIndexed
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawIndirect
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawIndexedIndirect
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBlitImage
- The VkCommandPool that commandBuffer was allocated from, in vkCmdClearDepthStencilImage
- The VkCommandPool that commandBuffer was allocated from, in vkCmdClearAttachments
- The VkCommandPool that commandBuffer was allocated from, in vkCmdResolveImage
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBeginRenderPass
- The VkCommandPool that commandBuffer was allocated from, in vkCmdNextSubpass
- The VkCommandPool that commandBuffer was allocated from, in vkCmdEndRenderPass
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDeviceMask
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDispatchBase
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawIndirectCount
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawIndexedIndirectCount
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBeginRenderPass2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdNextSubpass2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdEndRenderPass2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPipelineBarrier2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdWriteTimestamp2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyBuffer2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyImage2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyBufferToImage2

- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyImageToBuffer2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetEvent2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdResetEvent2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdWaitEvents2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBlitImage2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdResolveImage2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBeginRendering
- The VkCommandPool that commandBuffer was allocated from, in vkCmdEndRendering
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetCullMode
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetFrontFace
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetPrimitiveTopology
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetViewportWithCount
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetScissorWithCount
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindVertexBuffers2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthTestEnable
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthWriteEnable
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthCompareOp
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthBoundsTestEnable
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetStencilTestEnable
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetStencilOp
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetRasterizerDiscardEnable
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthBiasEnable
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetPrimitiveRestartEnable
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPushDescriptorSet
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPushDescriptorSetWithTemplate
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindDescriptorSets2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPushConstants2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPushDescriptorSet2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPushDescriptorSetWithTemplate2
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetLineStipple
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindIndexBuffer2

- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetRenderingAttachmentLocations
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetRenderingInputAttachmentIndices
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBeginVideoCodingKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdEndVideoCodingKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdControlVideoCodingKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDecodeVideoKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBeginRenderingKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdEndRenderingKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDeviceMaskKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDispatchBaseKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPushDescriptorSetKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPushDescriptorSetWithTemplateKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBeginRenderPass2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdNextSubpass2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdEndRenderPass2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawIndirectCountKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawIndexedIndirectCountKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetFragmentShadingRateKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetRenderingAttachmentLocationsKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetRenderingInputAttachmentIndicesKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdEncodeVideoKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetEvent2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdResetEvent2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdWaitEvents2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPipelineBarrier2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdWriteTimestamp2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindIndexBuffer3KHR
- The VkCommandPool that commandBuffer was allocated from, in

vkCmdBindVertexBuffers3KHR

- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawIndirect2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawIndexedIndirect2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDispatchIndirect2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyMemoryKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyMemoryToImageKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyImageToMemoryKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdUpdateMemoryKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdFillMemoryKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyQueryPoolResultsToMemoryKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawIndirectCount2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawIndexedIndirectCount2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBeginConditionalRendering2EXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindTransformFeedbackBuffers2EXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBeginTransformFeedback2EXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdEndTransformFeedback2EXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawIndirectByteCount2EXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawMeshTasksIndirect2EXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawMeshTasksIndirectCount2EXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyBuffer2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyImage2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyBufferToImage2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyImageToBuffer2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBlitImage2KHR

- The VkCommandPool that commandBuffer was allocated from, in vkCmdResolveImage2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdTraceRaysIndirect2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindIndexBuffer2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetLineStippleKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindDescriptorSets2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPushConstants2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPushDescriptorSet2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPushDescriptorSetWithTemplate2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDescriptorBufferOffsets2EXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindDescriptorBufferEmbeddedSamplers2EXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyMemoryIndirectKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyMemoryToImageIndirectKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdEndRendering2KHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindTransformFeedbackBuffersEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBeginTransformFeedbackEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdEndTransformFeedbackEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBeginQueryIndexedEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdEndQueryIndexedEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawIndirectByteCountEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBeginConditionalRenderingEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdEndConditionalRenderingEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDiscardRectangleEXT
- The VkCommandPool that commandBuffer was allocated from, in

vkCmdSetDiscardRectangleEnableEXT

- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDiscardRectangleModeEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBeginDebugUtilsLabelEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdEndDebugUtilsLabelEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdInsertDebugUtilsLabelEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindSamplerHeapEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindResourceHeapEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPushDataEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetSampleLocationsEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetCullModeEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetFrontFaceEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetPrimitiveTopologyEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetViewportWithCountEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetScissorWithCountEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindVertexBuffers2EXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthTestEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthWriteEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthCompareOpEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthBoundsTestEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetStencilTestEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetStencilOpEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthBias2EXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindDescriptorBuffersEXT
- The VkCommandPool that commandBuffer was allocated from, in

vkCmdSetDescriptorBufferOffsetsEXT

- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindDescriptorBufferEmbeddedSamplersEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetVertexInputEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetPatchControlPointsEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetRasterizerDiscardEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthBiasEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetLogicOpEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetPrimitiveRestartEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetColorWriteEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawMultiEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawMultiIndexedEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBuildMicromapsEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyMicromapEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyMicromapToMemoryEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyMemoryToMicromapEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdWriteMicromapsPropertiesEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthClampEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetPolygonModeEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetRasterizationSamplesEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetSampleMaskEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetAlphaToCoverageEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetAlphaToOneEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetLogicOpEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetColorBlendEnableEXT

- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetColorBlendEquationEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetColorWriteMaskEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetTessellationDomainOriginEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetRasterizationStreamEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetConservativeRasterizationModeEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetExtraPrimitiveOverestimationSizeEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthClipEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetSampleLocationsEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetColorBlendAdvancedEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetProvokingVertexModeEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetLineRasterizationModeEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetLineStippleEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthClipNegativeOneToOneEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBindShadersEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetDepthClampRangeEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetAttachmentFeedbackLoopEnableEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDecompressMemoryEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDecompressMemoryIndirectCountEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdPreprocessGeneratedCommandsEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdExecuteGeneratedCommandsEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdEndRendering2EXT

- The VkCommandPool that commandBuffer was allocated from, in vkCmdBeginCustomResolveEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetPrimitiveRestartIndexEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBuildAccelerationStructuresKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdBuildAccelerationStructuresIndirectKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyAccelerationStructureKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyAccelerationStructureToMemoryKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdCopyMemoryToAccelerationStructureKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdWriteAccelerationStructuresPropertiesKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdTraceRaysKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdTraceRaysIndirectKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdSetRayTracingPipelineStackSizeKHR
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawMeshTasksEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawMeshTasksIndirectEXT
- The VkCommandPool that commandBuffer was allocated from, in vkCmdDrawMeshTasksIndirectCountEXT

3.7. Valid Usage

Valid usage defines a set of conditions which **must** be met in order to achieve well-defined runtime behavior in an application. These conditions depend only on Vulkan state, and the parameters or objects whose usage is constrained by the condition.

The core layer assumes applications are using the API correctly. Except as documented elsewhere in the Specification, the behavior of the core layer to an application using the API incorrectly is undefined, and **may** include program termination. However, implementations **must** ensure that incorrect usage by an application does not affect the integrity of the operating system, the Vulkan implementation, or other applications in the system using Vulkan. In particular, any guarantees made by an operating system about whether memory from one process **can** be visible to another process or not **must** not be violated by a Vulkan implementation for **any memory allocation**. Vulkan implementations are not **required** to make additional security or integrity guarantees beyond those provided by the OS unless explicitly directed by the application's use of a particular feature or extension.

NOTE

For instance, if an operating system guarantees that data in all its memory allocations are set to zero when newly allocated, the Vulkan implementation **must** make the same guarantees for any allocations it controls (e.g. VkDeviceMemory).

Similarly, if an operating system guarantees that use-after-free of host allocations will not result in values written by another process becoming visible, the same guarantees **must** be made by the Vulkan implementation for device memory.

If the `protectedMemory` feature is supported, the implementation provides additional guarantees when invalid usage occurs to prevent values in protected memory from being accessed or inferred outside of protected operations, as described in Protected Memory Access Rules.

Some valid usage conditions have dependencies on runtime limits or feature availability. It is possible to validate these conditions against Vulkan's minimum supported values for these limits and features, or some subset of other known values.

Valid usage conditions do not cover conditions where well-defined behavior (including returning an error code) exists.

Valid usage conditions **should** apply to the command or structure where complete information about the condition would be known during execution of an application. This is such that a validation layer or linter **can** be written directly against these statements at the point they are specified.

NOTE

This does lead to some non-obvious places for valid usage statements. For instance, the valid values for a structure might depend on a separate value in the calling command. In this case, the structure itself will not reference this valid usage as it is impossible to determine validity from the structure that it is invalid - instead this valid usage would be attached to the calling command.

Another example is draw state - the state setters are independent, and can cause a legitimately invalid state configuration between draw calls; so the valid usage statements are attached to the place where all state needs to be valid - at the drawing command.

Valid usage conditions are described in a block labeled "Valid Usage" following each command or structure they apply to.

3.7.1. Usage Validation

Vulkan is a layered API. The lowest layer is the core Vulkan layer, as defined by this Specification. The application **can** use additional layers above the core for debugging, validation, and other purposes.

One of the core principles of Vulkan is that building and submitting command buffers **should** be highly efficient. Thus error checking and validation of state in the core layer is minimal, although more rigorous validation **can** be enabled through the use of layers.

Validation of correct API usage is left to validation layers. Applications **should** be developed with

validation layers enabled, to help catch and eliminate errors. Once validated, released applications **should** not enable validation layers by default.

3.7.2. Implicit Valid Usage

Some valid usage conditions apply to all commands and structures in the API, unless explicitly denoted otherwise for a specific command or structure. These conditions are considered *implicit*, and are described in a block labeled "Valid Usage (Implicit)" following each command or structure they apply to. Implicit valid usage conditions are described in detail below.

Valid Usage for Object Handles

Any input parameter to a command that is an object handle **must** be a valid object handle, unless otherwise specified. An object handle is valid if:

- It has been created or allocated by a previous, successful call to the API. Such calls are noted in the Specification.
- It has not been deleted or freed by a previous call to the API. Such calls are noted in the Specification.
- Any objects used by that object, either as part of creation or execution, **must** also be valid.

The reserved values VK_NULL_HANDLE and `NULL` **can** be used in place of valid non-dispatchable handles and dispatchable handles, respectively, when *explicitly called out in the Specification*. Any command that creates an object successfully **must** not return these values. It is valid to pass these values to `vkDestroy*` or `vkFree*` commands, which will silently ignore these values.

Valid Usage for Pointers

Any parameter that is a pointer **must** be a *valid pointer* only if it is explicitly called out by a Valid Usage statement.

A pointer is "valid" if it points at memory containing values of the number and type(s) expected by the command, and all fundamental types accessed through the pointer (e.g. as elements of an array or as members of a structure) satisfy the alignment requirements of the host processor.

Valid Usage for Strings

Any parameter that is a pointer to `char` **must** be a finite sequence of values terminated by a null character, or if *explicitly called out in the Specification*, **can** be `NULL`.

Strings specified as UTF-8 encoded **must** not contain invalid UTF-8 sequences. See String Representation for additional information about strings.

Valid Usage for Enumerated Types

Any parameter of an enumerated type **must** be a valid enumerant for that type. Use of an enumerant is valid if the following conditions are true:

- The enumerant is defined as part of the enumerated type.

- The enumerant is not a value suffixed with `_MAX_ENUM`.
  - This value exists only to ensure that C `enum` types are 32 bits in size and **must** not be used by applications.
- If the enumerant is used in a function that has a VkInstance as its first parameter and either:
  - it was added by a core version that is supported (as reported by vkEnumerateInstanceVersion) and the value of VkApplicationInfo::`apiVersion` is greater than or equal to the version that added it; or
  - it was added by an instance extension that was enabled for the instance.
- If the enumerant is used in a function that has a VkPhysicalDevice object as its first parameter and either:
  - it was added by a core version that is supported by that device (as reported by VkPhysicalDeviceProperties::`apiVersion`);
  - it was added by an instance extension that was enabled for the instance; or
  - it was added by a device extension that is supported by that device.
- If the enumerant is used in a function that has any other dispatchable object as its first parameter and either:
  - it was added by a core version that is supported for the device (as reported by VkPhysicalDeviceProperties::`apiVersion`); or
  - it was added by a device extension that was enabled for the device.

Additionally, if the `maintenance5` feature is supported, any integer value representable in the range valid for the defined type is valid when used in a function that has a VkPhysicalDevice object as its first parameter. Physical device queries will either return results indicating lack of support, or ignore unsupported values when used as a bit flag in a `Vk*Flags*` parameter.

Any enumerated type returned from a query command or otherwise output from Vulkan to the application **must** not have a reserved value. Reserved values are values not defined by any extension for that enumerated type.

NOTE In some special cases, an enumerant is only meaningful if a feature defined by an extension is also enabled, as well as the extension itself. The global “valid enumerant” rule described here does not address such cases.

NOTE This language is intended to accommodate cases such as “hidden” extensions known only to driver internals, or layers enabling extensions without knowledge of the application, without allowing return of values not defined by any extension.

NOTE Application developers are encouraged to be careful when using `switch` statements with Vulkan API enums. This is because new extensions can add new values to existing enums. Using a `default:` statement within a `switch` may avoid future compilation issues.

This is particularly true for enums such as VkDriverId, which may have values

added that do not belong to a corresponding new extension.

Valid Usage for Flags

A collection of flags is represented by a bitmask using the type VkFlags:

```
// Provided by VK_VERSION_1_0
typedef uint32_t VkFlags;
```

Bitmasks are passed to many commands and structures to compactly represent options, but VkFlags is not used directly in the API. Instead, a `Vk*Flags` type which is an alias of VkFlags, and whose name matches the corresponding `Vk*FlagBits` that are valid for that type, is used.

Any `Vk*Flags` member or parameter used in the API as an input **must** be a valid combination of bit flags. A valid combination is the bitwise OR of valid bit flags.

An individual bit flag is valid for a `Vk*Flags` type if it would be a valid enumerant when used with the equivalent `Vk*FlagBits` type, where the bits type is obtained by taking the flag type and replacing the trailing `Flags` with `FlagBits`. For example, a flag value of type VkColorComponentFlags **must** contain only bit flags defined by VkColorComponentFlagBits.

Any `Vk*Flags` member or parameter returned from a query command or otherwise output from Vulkan to the application **may** contain bit flags undefined in its corresponding `Vk*FlagBits` type. An application **cannot** rely on the state of these unspecified bits.

Only the low-order 31 bits (bit positions zero through 30) are available for use as flag bits.

NOTE This restriction is due to poorly defined behavior by C compilers given a C enumerant value of `0x80000000`. In some cases adding this enumerant value may increase the size of the underlying `Vk*FlagBits` type, breaking the ABI.

A collection of 64-bit flags is represented by a bitmask using the type VkFlags64:

```
// Provided by VK_VERSION_1_3, VK_KHR_synchronization2
typedef uint64_t VkFlags64;
```

When the 31 bits available in VkFlags are insufficient, the VkFlags64 type can be passed to commands and structures to represent up to 64 options. VkFlags64 is not used directly in the API. Instead, a `Vk*Flags2` type which is an alias of VkFlags64, and whose name matches the corresponding `Vk*FlagBits2` that are valid for that type, is used.

Any `Vk*Flags2` member or parameter used in the API as an input **must** be a valid combination of bit flags. A valid combination is either zero or the bitwise OR of valid bit flags.

An individual bit flag is valid for a `Vk*Flags2` type if it would be a valid enumerant when used with the equivalent `Vk*FlagBits2` type, where the bits type is obtained by taking the flag type and replacing the trailing `Flags2` with `FlagBits2`. For example, a flag value of type VkAccessFlags2 **must**

contain only bit flags defined by VkAccessFlagBits2.

Any Vk*Flags2 member or parameter returned from a query command or otherwise output from Vulkan to the application **may** contain bit flags undefined in its corresponding Vk*FlagBits2 type. An application **cannot** rely on the state of these unspecified bits.

NOTE	Both the Vk*FlagBits2 type, and the individual bits defined for that type, are defined as uint64_t integers in the C API. This is in contrast to the 32-bit types, where the Vk*FlagBits type is defined as a C enum and the individual bits as enumerants belonging to that enum. As a result, there is less compile time type checking possible for the 64-bit types. This is unavoidable since there is no sufficiently portable way to define a 64-bit enum type in C99.

Effective Flags

When the effective value of a bitmask command parameter or structure member is determined by an extending structure providing a 64-bit version of the bitflags, that effective value is used in all specification language, including valid usage statements, referring to the original parameter.

Examples of such extending structures include (this list is not complete):

- VkBufferUsageFlags2CreateInfo
- VkImageUsageFlags2CreateInfoKHR

NOTE	When a 64-bit version of a flags type is defined, the implementation **must** always represent those flags internally as 64 bits, whether they are specified through the 32-bit or 64-bit APIs.

Valid Usage for Structure Types

Any parameter that is a structure containing a sType member **must** have a value of sType which is a valid VkStructureType value matching the type of the structure.

Valid Usage for Structure Pointer Chains

Any parameter that is a structure containing a void* pNext member **must** have a value of pNext that is either NULL, or is a pointer to a valid *extending structure*, containing sType and pNext members as described in the Vulkan Documentation and Extensions document in the section “Extending Structures”. The set of structures connected by pNext pointers is referred to as a pNext *chain*.

Each structure included in the pNext chain **must** be defined at runtime by either:

- a core version which is supported
- an extension which is enabled
- a supported device extension in the case of physical-device-level functionality added by the device extension

Each type of extending structure **must** not appear more than once in a pNext chain, including any

aliases. This general rule may be explicitly overridden for specific structures.

Any component of the implementation (the loader, any enabled layers, and drivers) **must** skip over, without processing (other than reading the `sType` and `pNext` members) any extending structures in the chain not defined by core versions or extensions supported by that component.

As a convenience to implementations and layers needing to iterate through a structure pointer chain, the Vulkan API provides two *base structures*. These structures allow for some type safety, and can be used by Vulkan API functions that operate on generic inputs and outputs.

The `VkBaseInStructure` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBaseInStructure {
    VkStructureType                    sType;
    const struct VkBaseInStructure*    pNext;
} VkBaseInStructure;
```

- `sType` is the structure type of the structure being iterated through.
- `pNext` is `NULL` or a pointer to the next structure in a structure chain.

`VkBaseInStructure` can be used to facilitate iterating through a read-only structure pointer chain.

The `VkBaseOutStructure` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBaseOutStructure {
    VkStructureType               sType;
    struct VkBaseOutStructure*    pNext;
} VkBaseOutStructure;
```

- `sType` is the structure type of the structure being iterated through.
- `pNext` is `NULL` or a pointer to the next structure in a structure chain.

`VkBaseOutStructure` can be used to facilitate iterating through a structure pointer chain that returns data back to the application.

Valid Usage for Nested Structures

The above conditions also apply recursively to members of structures provided as input to a command, either as a direct argument to the command, or themselves a member of another structure.

Specifics on valid usage of each command are covered in their individual sections.

Valid Usage for Extensions

Instance-level functionality or behavior added by an instance extension to the API **must** not be

used unless that extension is supported by the instance as determined by vkEnumerateInstanceExtensionProperties, and that extension is enabled in VkInstanceCreateInfo.

Physical-device-level functionality or behavior added by an instance extension to the API **must** not be used unless that extension is supported by the instance as determined by vkEnumerateInstanceExtensionProperties, and that extension is enabled in VkInstanceCreateInfo.

Physical-device-level functionality or behavior added by a device extension to the API **must** not be used unless the conditions described in Extending Physical Device From Device Extensions are met.

Device-level functionality added by a device extension that is dispatched from a VkDevice, or from a child object of a VkDevice **must** not be used unless that extension is supported by the device as determined by vkEnumerateDeviceExtensionProperties, and that extension is enabled in VkDeviceCreateInfo.

Valid Usage for Newer Core Versions

Instance-level functionality or behavior added by a new core version of the API **must** not be used unless it is supported by the instance as determined by vkEnumerateInstanceVersion and the specified version of VkApplicationInfo::`apiVersion`.

Physical-device-level functionality or behavior added by a new core version of the API **must** not be used unless it is supported by the physical device as determined by VkPhysicalDeviceProperties::`apiVersion` and the specified version of VkApplicationInfo::`apiVersion`.

Device-level functionality or behavior added by a new core version of the API **must** not be used unless it is supported by the device as determined by VkPhysicalDeviceProperties::`apiVersion` and the specified version of VkApplicationInfo::`apiVersion`.

3.8. `VkResult` Return Codes

While the core Vulkan API is not designed to capture incorrect usage, some circumstances still require return codes. Commands in Vulkan return their status via return codes that are in one of two categories:

- Successful completion codes are returned when a command needs to communicate success or status information. All successful completion codes are non-negative values.
- Runtime error codes are returned when a command needs to communicate a failure that could only be detected at runtime. All runtime error codes are negative values.

All return codes in Vulkan are reported via VkResult return values. The possible codes are:

```
// Provided by VK_VERSION_1_0
typedef enum VkResult {
    VK_SUCCESS = 0,
    VK_NOT_READY = 1,
    VK_TIMEOUT = 2,
    VK_EVENT_SET = 3,
    VK_EVENT_RESET = 4,
```

```
    VK_INCOMPLETE = 5,
    VK_ERROR_OUT_OF_HOST_MEMORY = -1,
    VK_ERROR_OUT_OF_DEVICE_MEMORY = -2,
    VK_ERROR_INITIALIZATION_FAILED = -3,
    VK_ERROR_DEVICE_LOST = -4,
    VK_ERROR_MEMORY_MAP_FAILED = -5,
    VK_ERROR_LAYER_NOT_PRESENT = -6,
    VK_ERROR_EXTENSION_NOT_PRESENT = -7,
    VK_ERROR_FEATURE_NOT_PRESENT = -8,
    VK_ERROR_INCOMPATIBLE_DRIVER = -9,
    VK_ERROR_TOO_MANY_OBJECTS = -10,
    VK_ERROR_FORMAT_NOT_SUPPORTED = -11,
    VK_ERROR_FRAGMENTED_POOL = -12,
    VK_ERROR_UNKNOWN = -13,
  // Provided by VK_VERSION_1_0
    VK_ERROR_VALIDATION_FAILED = -1000011001,
  // Provided by VK_VERSION_1_1
    VK_ERROR_OUT_OF_POOL_MEMORY = -1000069000,
  // Provided by VK_VERSION_1_1
    VK_ERROR_INVALID_EXTERNAL_HANDLE = -1000072003,
  // Provided by VK_VERSION_1_2
    VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS = -1000257000,
  // Provided by VK_VERSION_1_2
    VK_ERROR_FRAGMENTATION = -1000161000,
  // Provided by VK_VERSION_1_3
    VK_PIPELINE_COMPILE_REQUIRED = 1000297000,
  // Provided by VK_VERSION_1_4
    VK_ERROR_NOT_PERMITTED = -1000174001,
  // Provided by VK_KHR_surface
    VK_ERROR_SURFACE_LOST_KHR = -1000000000,
  // Provided by VK_KHR_surface
    VK_ERROR_NATIVE_WINDOW_IN_USE_KHR = -1000000001,
  // Provided by VK_KHR_swapchain
    VK_SUBOPTIMAL_KHR = 1000001003,
  // Provided by VK_KHR_swapchain
    VK_ERROR_OUT_OF_DATE_KHR = -1000001004,
  // Provided by VK_KHR_display_swapchain
    VK_ERROR_INCOMPATIBLE_DISPLAY_KHR = -1000003001,
  // Provided by VK_KHR_video_queue
    VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR = -1000023000,
  // Provided by VK_KHR_video_queue
    VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR = -1000023001,
  // Provided by VK_KHR_video_queue
    VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR = -1000023002,
  // Provided by VK_KHR_video_queue
    VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR = -1000023003,
  // Provided by VK_KHR_video_queue
    VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR = -1000023004,
  // Provided by VK_KHR_video_queue
    VK_ERROR_VIDEO_STD_VERSION_NOT_SUPPORTED_KHR = -1000023005,
  // Provided by VK_EXT_image_drm_format_modifier
```

```
    VK_ERROR_INVALID_DRM_FORMAT_MODIFIER_PLANE_LAYOUT_EXT = -1000158000,
  // Provided by VK_EXT_present_timing
    VK_ERROR_PRESENT_TIMING_QUEUE_FULL_EXT = -1000208000,
  // Provided by VK_EXT_full_screen_exclusive
    VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT = -1000255000,
  // Provided by VK_KHR_deferred_host_operations
    VK_THREAD_IDLE_KHR = 1000268000,
  // Provided by VK_KHR_deferred_host_operations
    VK_THREAD_DONE_KHR = 1000268001,
  // Provided by VK_KHR_deferred_host_operations
    VK_OPERATION_DEFERRED_KHR = 1000268002,
  // Provided by VK_KHR_deferred_host_operations
    VK_OPERATION_NOT_DEFERRED_KHR = 1000268003,
  // Provided by VK_KHR_video_encode_queue
    VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR = -1000299000,
  // Provided by VK_EXT_image_compression_control
    VK_ERROR_COMPRESSION_EXHAUSTED_EXT = -1000338000,
  // Provided by VK_EXT_shader_object
    VK_INCOMPATIBLE_SHADER_BINARY_EXT = 1000482000,
  // Provided by VK_KHR_pipeline_binary
    VK_PIPELINE_BINARY_MISSING_KHR = 1000483000,
  // Provided by VK_KHR_pipeline_binary
    VK_ERROR_NOT_ENOUGH_SPACE_KHR = -1000483000,
  // Provided by VK_KHR_maintenance1
    VK_ERROR_OUT_OF_POOL_MEMORY_KHR = VK_ERROR_OUT_OF_POOL_MEMORY,
  // Provided by VK_KHR_external_memory
    VK_ERROR_INVALID_EXTERNAL_HANDLE_KHR = VK_ERROR_INVALID_EXTERNAL_HANDLE,
  // Provided by VK_EXT_descriptor_indexing
    VK_ERROR_FRAGMENTATION_EXT = VK_ERROR_FRAGMENTATION,
  // Provided by VK_KHR_global_priority
    VK_ERROR_NOT_PERMITTED_KHR = VK_ERROR_NOT_PERMITTED,
  // Provided by VK_KHR_buffer_device_address
    VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR =
VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS,
  // Provided by VK_EXT_shader_object
  // VK_ERROR_INCOMPATIBLE_SHADER_BINARY_EXT is a legacy alias
    VK_ERROR_INCOMPATIBLE_SHADER_BINARY_EXT = VK_INCOMPATIBLE_SHADER_BINARY_EXT,
} VkResult;
```

Success Codes

- VK_SUCCESS Command successfully completed
- VK_NOT_READY A fence or query has not yet completed
- VK_TIMEOUT A wait operation has not completed in the specified time
- VK_EVENT_SET An event is signaled
- VK_EVENT_RESET An event is unsignaled
- VK_INCOMPLETE A return array was too small for the result

- VK_SUBOPTIMAL_KHR A swapchain no longer matches the surface properties exactly, but **can** still be used to present to the surface successfully.
- VK_THREAD_IDLE_KHR A deferred operation is not complete but there is currently no work for this thread to do at the time of this call.
- VK_THREAD_DONE_KHR A deferred operation is not complete but there is no work remaining to assign to additional threads.
- VK_OPERATION_DEFERRED_KHR A deferred operation was requested and at least some of the work was deferred.
- VK_OPERATION_NOT_DEFERRED_KHR A deferred operation was requested and no operations were deferred.
- VK_PIPELINE_COMPILE_REQUIRED A requested pipeline creation would have required compilation, but the application requested compilation to not be performed.
- VK_PIPELINE_BINARY_MISSING_KHR The application attempted to create a pipeline binary by querying an internal cache, but the internal cache entry did not exist.
- VK_INCOMPATIBLE_SHADER_BINARY_EXT The provided binary shader code is not compatible with this device.

NOTE	In the initial version of the `VK_EXT_shader_object` extension, this return code was named VK_ERROR_INCOMPATIBLE_SHADER_BINARY_EXT and improperly described as an error code. The name has been changed, but the old name is retained as an alias for compatibility with old code.

Error Codes

- VK_ERROR_OUT_OF_HOST_MEMORY A host memory allocation has failed.
- VK_ERROR_OUT_OF_DEVICE_MEMORY A device memory allocation has failed.
- VK_ERROR_INITIALIZATION_FAILED Initialization of an object could not be completed for implementation-specific reasons.
- VK_ERROR_DEVICE_LOST The logical or physical device has been lost. See Lost Device.
- VK_ERROR_MEMORY_MAP_FAILED Mapping of a memory object has failed.
- VK_ERROR_LAYER_NOT_PRESENT A requested layer is not present or could not be loaded.
- VK_ERROR_EXTENSION_NOT_PRESENT A requested extension is not supported.
- VK_ERROR_FEATURE_NOT_PRESENT A requested feature is not supported.
- VK_ERROR_INCOMPATIBLE_DRIVER The requested version of Vulkan is not supported by the driver or is otherwise incompatible for implementation-specific reasons.
- VK_ERROR_TOO_MANY_OBJECTS Too many objects of the type have already been created.
- VK_ERROR_FORMAT_NOT_SUPPORTED A requested format is not supported on this device.
- VK_ERROR_FRAGMENTED_POOL A pool allocation has failed due to fragmentation of the pool's memory. This **must** only be returned if no attempt to allocate host or device memory was made to accommodate the new allocation. This **should** be returned in preference to VK_ERROR_OUT_OF_POOL_MEMORY, but only if the implementation is certain that the pool

allocation failure was due to fragmentation.

- VK_ERROR_SURFACE_LOST_KHR A surface is no longer available.
- VK_ERROR_NATIVE_WINDOW_IN_USE_KHR The requested window is already in use by Vulkan or another API in a manner which prevents it from being used again.
- VK_ERROR_OUT_OF_DATE_KHR A surface has changed in such a way that it is no longer compatible with the swapchain, and further presentation requests using the swapchain will fail. Applications **must** query the new surface properties and recreate their swapchain if they wish to continue presenting to the surface.
- VK_ERROR_INCOMPATIBLE_DISPLAY_KHR The display used by a swapchain does not use the same presentable image layout, or is incompatible in a way that prevents sharing an image.
- VK_ERROR_OUT_OF_POOL_MEMORY A pool memory allocation has failed. This **must** only be returned if no attempt to allocate host or device memory was made to accommodate the new allocation. If the failure was definitely due to fragmentation of the pool, VK_ERROR_FRAGMENTED_POOL **should** be returned instead.
- VK_ERROR_INVALID_EXTERNAL_HANDLE An external handle is not a valid handle of the specified type.
- VK_ERROR_FRAGMENTATION A descriptor pool creation has failed due to fragmentation.
- VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS A buffer creation or memory allocation failed because the requested address is not available. A shader group handle assignment failed because the requested shader group handle information is no longer valid.
- VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT An operation on a swapchain created with VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT failed as it did not have exclusive full-screen access. This **may** occur due to implementation-dependent reasons, outside of the application's control.
- VK_ERROR_PRESENT_TIMING_QUEUE_FULL_EXT A present operation on a swapchain created with VK_SWAPCHAIN_CREATE_PRESENT_TIMING_BIT_EXT failed due to insufficient space in the swapchain's internal results queue to hold requested present timing data.
- VK_ERROR_VALIDATION_FAILED A command failed because invalid usage was detected by the implementation or a validation layer. This **may** result in the command not being dispatched to the ICD.
- VK_ERROR_COMPRESSION_EXHAUSTED_EXT An image creation failed because internal resources required for compression are exhausted. This **must** only be returned when fixed-rate compression is requested.
- VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR The requested VkImageUsageFlags are not supported.
- VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR The requested video picture layout is not supported.
- VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR A video profile operation specified via VkVideoProfileInfoKHR::`videoCodecOperation` is not supported.
- VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR Format parameters in a requested VkVideoProfileInfoKHR chain are not supported.

- VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR Codec-specific parameters in a requested VkVideoProfileInfoKHR chain are not supported.
- VK_ERROR_VIDEO_STD_VERSION_NOT_SUPPORTED_KHR The specified video Std header version is not supported.
- VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR The specified Video Std parameters do not adhere to the syntactic or semantic requirements of the used video compression standard, or values derived from parameters according to the rules defined by the used video compression standard do not adhere to the capabilities of the video compression standard or the implementation.
- VK_ERROR_NOT_PERMITTED The driver implementation has denied a request to acquire a priority above the default priority (VK_QUEUE_GLOBAL_PRIORITY_MEDIUM_EXT) because the application does not have sufficient privileges.
- VK_ERROR_NOT_ENOUGH_SPACE_KHR The application did not provide enough space to return all the required data.
- VK_ERROR_UNKNOWN An unknown error has occurred; either the application has provided invalid input, or an implementation failure has occurred.

If a command returns a runtime error, unless otherwise specified any output parameters will have undefined contents, except that if the output parameter is a structure with `sType` and `pNext` fields, those fields will be unmodified. Any structures chained from `pNext` will also have undefined contents, except that `sType` and `pNext` will be unmodified.

`VK_ERROR_OUT_OF_*_MEMORY` errors do not modify any currently existing Vulkan objects. Objects that have already been successfully created **can** still be used by the application.

NOTE As a general rule, `Free`, `Release`, and `Reset` commands do not return VK_ERROR_OUT_OF_HOST_MEMORY, while any other command with a return code **may** return it. Any exceptions from this rule are described for those commands.

VK_ERROR_UNKNOWN will be returned by an implementation when an unexpected error occurs that cannot be attributed to valid behavior of the application and implementation. Under these conditions, it **may** be returned from any command returning a VkResult.

NOTE VK_ERROR_UNKNOWN is not expected to ever be returned if the application behavior is valid, and if the implementation is bug-free. If VK_ERROR_UNKNOWN is returned, the application should be checked against the latest validation layers to verify correct behavior as much as possible. If no issues are identified it could be an implementation issue, and the implementor should be contacted for support.

Any command returning a VkResult **may** return VK_ERROR_VALIDATION_FAILED if a violation of valid usage is detected.

Unless otherwise noted, if an implementation encounters conflicting conditions such that both a runtime error and a status success code such as VK_INCOMPLETE would be returned, reporting of the runtime error **should** take precedence.

Performance-critical commands generally do not have return codes. If a runtime error occurs in

such commands, the implementation will defer reporting the error until a specified point. For commands that record into command buffers (`vkCmd*`) runtime errors are reported by `vkEndCommandBuffer`.

3.9. Numeric Representation and Computation

Implementations normally perform computations in floating-point, and **must** meet the range and precision requirements defined under “Floating-Point Computation” below.

These requirements only apply to computations performed in Vulkan operations outside of shader execution, such as texture image specification and sampling, and per-fragment operations. Range and precision requirements during shader execution differ, and are specified by the Precision and Operation of SPIR-V Instructions section.

In some cases, the representation and/or precision of operations is implicitly limited by the specified format of vertex or texel data consumed by Vulkan. Specific floating-point formats are described later in this section.

3.9.1. Floating-Point Computation

Most floating-point computation is performed in SPIR-V shader modules. The properties of computation within shaders are constrained as defined by the Precision and Operation of SPIR-V Instructions section.

Some floating-point computation is performed outside of shaders, such as viewport and depth range calculations. For these computations, we do not specify how floating-point numbers are to be represented, or the details of how operations on them are performed, but only place minimal requirements on representation and precision as described in the remainder of this section.

We require simply that numbers' floating-point parts contain enough bits and that their exponent fields are large enough so that individual results of floating-point operations are accurate to about 1 part in 10^5. The maximum representable magnitude for all floating-point values **must** be at least 2^{32}.

$x \times 0 = 0 \times x = 0$ for any non-infinite and non-NaN x.

$1 \times x = x \times 1 = x$.

$x + 0 = 0 + x = x$.

$0^0 = 1$.

Occasionally, further requirements will be specified. Most single-precision floating-point formats meet these requirements.

The special values Inf and -Inf encode values with magnitudes too large to be represented; the

special value NaN encodes “Not A Number” values resulting from undefined arithmetic operations such as 0 / 0. Implementations **may** support Inf and NaN in their floating-point computations. Any computation which does not support either Inf or NaN, for which that value is an input or output will yield an undefined value.

3.9.2. Floating-Point Format Conversions

When a value is converted to a defined floating-point representation, finite values falling between two representable finite values are rounded to one or the other. The rounding mode is not defined. Finite values whose magnitude is larger than that of any representable finite value may be rounded either to the closest representable finite value or to the appropriately signed infinity. For unsigned destination formats any negative values are converted to zero. Positive infinity is converted to positive infinity; negative infinity is converted to negative infinity in signed formats and to zero in unsigned formats; and any NaN is converted to a NaN.

3.9.3. 16-Bit Floating-Point Numbers

16-bit floating-point numbers are defined in the “16-bit floating-point numbers” section of the Khronos Data Format Specification.

3.9.4. Unsigned 11-Bit Floating-Point Numbers

Unsigned 11-bit floating-point numbers are defined in the “Unsigned 11-bit floating-point numbers” section of the Khronos Data Format Specification.

3.9.5. Unsigned 10-Bit Floating-Point Numbers

Unsigned 10-bit floating-point numbers are defined in the “Unsigned 10-bit floating-point numbers” section of the Khronos Data Format Specification.

3.9.6. 8-bit booleans

An 8-bit boolean uses the following representation for `true` and `false`:

- Zero to represent `false`
- Any non-zero value to represent `true`

3.9.7. General Requirements

Any representable floating-point value in the appropriate format is legal as input to a Vulkan command that requires floating-point data. The result of providing a value that is not a floating-point number to such a command is unspecified, but **must** not lead to Vulkan interruption or termination. For example, providing a negative zero (where applicable) or a denormalized number to a Vulkan command **must** yield deterministic results, while providing a NaN or Inf yields unspecified results.

Some calculations require division. In such cases (including implied divisions performed by vector normalization), division by zero produces an unspecified result but **must** not lead to Vulkan

interruption or termination.

3.10. Fixed-Point Data Conversions

When generic vertex attributes and pixel color or depth *components* are represented as integers, they are often (but not always) considered to be *normalized*. Normalized integer values are treated specially when being converted to and from floating-point values, and are usually referred to as *normalized fixed-point*.

In the remainder of this section, b denotes the bit width of the fixed-point integer representation. When the integer is one of the types defined by the API, b is the bit width of that type. When the integer comes from an image containing color or depth component texels, b is the number of bits allocated to that component in its specified image format.

The signed and unsigned fixed-point representations are assumed to be b-bit binary two's-complement integers and binary unsigned integers, respectively.

3.10.1. Conversion From Normalized Fixed-Point to Floating-Point

Unsigned normalized fixed-point integers represent numbers in the range [0,1]. The conversion from an unsigned normalized fixed-point value c to the corresponding floating-point value f is defined as

$$f = \frac{c}{2^b - 1}$$

Signed normalized fixed-point integers represent numbers in the range [-1,1]. The conversion from a signed normalized fixed-point value c to the corresponding floating-point value f is performed using

$$f = \max\left(\frac{c}{2^{b-1} - 1}, -1.0\right)$$

Only the range [$-2^{b-1} + 1$, $2^{b-1} - 1$] is used to represent signed fixed-point values in the range [-1,1]. For example, if b = 8, then the integer value -127 corresponds to -1.0 and the value 127 corresponds to 1.0. This equation is used everywhere that signed normalized fixed-point values are converted to floating-point.

Note that while zero is exactly expressible in this representation, one value (-128 in the example) is outside the representable range, and implementations **must** clamp it to -1.0. Where the value is subject to further processing by the implementation, e.g. during texture filtering, values less than -1.0 **may** be used but the result **must** be clamped before the value is returned to shaders.

3.10.2. Conversion From Floating-Point to Normalized Fixed-Point

The conversion from a floating-point value f to the corresponding unsigned normalized fixed-point value c is defined by first clamping f to the range [0,1], then computing

c = convertFloatToUint(f × (2^b - 1), b)

where convertFloatToUint(r,b) returns one of the two unsigned binary integer values with exactly b bits which are closest to the floating-point value r. Implementations **should** round to nearest. If r is equal to an integer, then that integer value **must** be returned. In particular, if f is equal to 0.0 or 1.0, then c **must** be assigned 0 or 2^b - 1, respectively.

The conversion from a floating-point value f to the corresponding signed normalized fixed-point value c is performed by clamping f to the range [-1,1], then computing

$$c = \text{convertFloatToInt}(f \times (2^{b-1} - 1), b)$$

where convertFloatToInt(r,b) returns one of the two signed two's-complement binary integer values with exactly b bits which are closest to the floating-point value r. Implementations **should** round to nearest. If r is equal to an integer, then that integer value **must** be returned. In particular, if f is equal to -1.0, 0.0, or 1.0, then c **must** be assigned $-(2^{b-1} - 1)$, 0, or $2^{b-1} - 1$, respectively.

This equation is used everywhere that floating-point values are converted to signed normalized fixed-point.

3.11. String Representation

Strings passed into and returned from Vulkan API commands are usually defined to be null-terminated and UTF-8 encoded.

NOTE	Exceptions to this rule exist only when strings are defined or used by operating system APIs where that OS has a different convention. For example, VkExportMemoryWin32HandleInfoKHR::`name` is a null-terminated UTF-16 encoded string used in conjunction with Windows handles.

When a UTF-8 string is **returned from** a Vulkan API query, it is returned in a fixed-length buffer of C `char`. For example, a string returned in VkPhysicalDeviceProperties::`deviceName` has maximum length VK_MAX_PHYSICAL_DEVICE_NAME_SIZE, and a string returned in VkExtensionProperties ::`extensionName` has maximum length VK_MAX_EXTENSION_NAME_SIZE. The string, **including** its null terminator, will always fit completely within this buffer. If the string is shorter than the buffer size, the contents of `char` in the buffer following the null terminator are undefined.

When a UTF-8 string is **passed into** a Vulkan API, such as VkDeviceCreateInfo ::`ppEnabledExtensionNames`, there is no explicit limit on the length of that string. However, the string **must** contain a valid UTF-8 encoded string and **must** be null-terminated.

3.12. Common Object Types

Some types of Vulkan objects are used in many different structures and command parameters, and are described here. These types include *offsets*, *extents*, and *rectangles*.

3.12.1. Offsets

Offsets are used to describe a pixel location within an image or framebuffer, as an (x,y) location for

two-dimensional images, or an (x,y,z) location for three-dimensional images.

A two-dimensional offset is defined by the structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkOffset2D {
    int32_t    x;
    int32_t    y;
} VkOffset2D;
```

- `x` is the x offset.
- `y` is the y offset.

A three-dimensional offset is defined by the structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkOffset3D {
    int32_t    x;
    int32_t    y;
    int32_t    z;
} VkOffset3D;
```

- `x` is the x offset.
- `y` is the y offset.
- `z` is the z offset.

3.12.2. Extents

Extents are used to describe the size of a rectangular region of pixels within an image or framebuffer, as (width,height) for two-dimensional images, or as (width,height,depth) for three-dimensional images.

A two-dimensional extent is defined by the structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkExtent2D {
    uint32_t    width;
    uint32_t    height;
} VkExtent2D;
```

- `width` is the width of the extent.
- `height` is the height of the extent.

A three-dimensional extent is defined by the structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkExtent3D {
    uint32_t    width;
    uint32_t    height;
    uint32_t    depth;
} VkExtent3D;
```

- `width` is the width of the extent.
- `height` is the height of the extent.
- `depth` is the depth of the extent.

3.12.3. Rectangles

Rectangles are used to describe a specified rectangular region of pixels within an image or framebuffer. Rectangles include both an offset and an extent of the same dimensionality, as described above. Two-dimensional rectangles are defined by the structure

```
// Provided by VK_VERSION_1_0
typedef struct VkRect2D {
    VkOffset2D    offset;
    VkExtent2D    extent;
} VkRect2D;
```

- `offset` is a VkOffset2D specifying the rectangle offset.
- `extent` is a VkExtent2D specifying the rectangle extent.

3.12.4. Host Address Ranges

A host address range indicates a range of host memory.

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkHostAddressRangeEXT {
    void*     address;
    size_t    size;
} VkHostAddressRangeEXT;
```

- `address` is a host memory address.
- `size` is the size of the range.

Valid Usage (Implicit)

- VUID-VkHostAddressRangeEXT-address-parameter
  `address` **must** be a valid pointer to an array of `size` bytes
- VUID-VkHostAddressRangeEXT-size-arraylength

`size` **must** be greater than `0`

A constant host address range indicates a range of host memory that cannot be altered.

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkHostAddressRangeConstEXT {
    const void*    address;
    size_t         size;
} VkHostAddressRangeConstEXT;
```

- `address` is a read-only host memory address.
- `size` is the size of the range.

Valid Usage (Implicit)

- VUID-VkHostAddressRangeConstEXT-address-parameter
  `address` **must** be a valid pointer to an array of `size` bytes
- VUID-VkHostAddressRangeConstEXT-size-arraylength
  `size` **must** be greater than `0`

3.12.5. Device Address Ranges

A device address range indicates a sized range of device memory.

```
// Provided by VK_KHR_device_address_commands
typedef struct VkDeviceAddressRangeKHR {
    VkDeviceAddress    address;
    VkDeviceSize       size;
} VkDeviceAddressRangeKHR;
```

```
// Provided by VK_EXT_descriptor_heap
// Equivalent to VkDeviceAddressRangeKHR
typedef VkDeviceAddressRangeKHR VkDeviceAddressRangeEXT;
```

- `address` is 0 or a VkDeviceAddress specifying the start of the range.
- `size` is a VkDeviceSize specifying the size of the range.

Valid Usage

- VUID-VkDeviceAddressRangeKHR-size-11411
  If `size` is not 0, `address` **must** not be 0
- VUID-VkDeviceAddressRangeKHR-address-11365

The sum of `address` and `size` **must** be less than or equal to the sum of an address retrieved from a VkBuffer and the value of VkBufferCreateInfo::`size` used to create that VkBuffer

Valid Usage (Implicit)

- VUID-VkDeviceAddressRangeKHR-address-parameter
  If `address` is not `0`, `address` **must** be a valid VkDeviceAddress value

A strided device address range is defined by the structure:

```
// Provided by VK_KHR_copy_memory_indirect, VK_KHR_device_address_commands
typedef struct VkStridedDeviceAddressRangeKHR {
    VkDeviceAddress    address;
    VkDeviceSize       size;
    VkDeviceSize       stride;
} VkStridedDeviceAddressRangeKHR;
```

- `address` is a VkDeviceAddress specifying the start of the range.
- `size` is a VkDeviceSize specifying the size of the range.
- `stride` is a VkDeviceSize specifying the stride of elements over the range.

Valid Usage

- VUID-VkStridedDeviceAddressRangeKHR-size-11411
  If `size` is not 0, `address` **must** not be 0
- VUID-VkStridedDeviceAddressRangeKHR-address-11365
  The sum of `address` and `size` **must** be less than or equal to the sum of an address retrieved from a VkBuffer and the value of VkBufferCreateInfo::`size` used to create that VkBuffer
- VUID-VkStridedDeviceAddressRangeKHR-stride-10957
  `stride` **must** be less than or equal to `size`

Valid Usage (Implicit)

- VUID-VkStridedDeviceAddressRangeKHR-address-parameter
  If `address` is not `0`, `address` **must** be a valid VkDeviceAddress value

Many Vulkan commands need additional information about an address range; this **may** be provided by the following flags.

Bits which **can** be set in a VkAddressCommandFlagsKHR mask are:

```
// Provided by VK_KHR_device_address_commands
typedef enum VkAddressCommandFlagBitsKHR {
    VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR = 0x00000001,
    VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR = 0x00000002,
    VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR = 0x00000004,
    VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR = 0x00000008,
  // Provided by VK_KHR_device_address_commands with VK_EXT_transform_feedback
    VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR = 0x00000010,
  // Provided by VK_KHR_device_address_commands with VK_EXT_transform_feedback
    VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR = 0x00000020,
} VkAddressCommandFlagBitsKHR;
```

- VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR specifies that an address range is allocated from protected memory.
- VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR specifies that an address range will be fully bound to physical memory when accessed.
- VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR specifies that all buffers containing any part of an address range were created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage.
- VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR specifies that whether buffers containing any part of an address range were created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage or not is unknown.
- VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR specifies that all buffers containing any part of an address range are allocated with the VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT usage.
- VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR specifies that whether buffers containing any part of an address range were created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT usage or not is unknown.

If neither VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR nor VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR are specified, the address range **must** not be aliased with any buffer allocated with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT. If neither VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR nor VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR are specified, the address range **must** not be aliased with any buffer allocated with the VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT.

NOTE

Buffers can return overlapping address ranges if they are bound to overlapping ranges of a VkDeviceMemory object. Applications should ensure that if they do this kind of aliasing, that they consistently either include or do not include VK_BUFFER_USAGE_STORAGE_BUFFER_BIT and VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT on aliased buffers when considering these flags.

```
// Provided by VK_KHR_device_address_commands
typedef VkFlags VkAddressCommandFlagsKHR;
```

VkAddressCommandFlagsKHR is a bitmask type for setting a mask of zero or more VkAddressCommandFlagBitsKHR.

3.12.6. Structure Types

Each value corresponds to a particular structure with a sType member with a matching name. As a general rule, the name of each VkStructureType value is obtained by taking the name of the structure, stripping the leading Vk, prefixing each capital letter except the first with _, converting the entire resulting string to upper case, and prefixing it with VK_STRUCTURE_TYPE_. For example, structures of type VkImageCreateInfo correspond to a VkStructureType value of VK_STRUCTURE_TYPE_IMAGE_CREATE_INFO, and thus a structure of this type **must** have its sType member set to this value before it is passed to the API.

The values VK_STRUCTURE_TYPE_LOADER_INSTANCE_CREATE_INFO and VK_STRUCTURE_TYPE_LOADER_DEVICE_CREATE_INFO are reserved for internal use by the loader, and do not have corresponding Vulkan structures in this Specification.

Structure types supported by the Vulkan API include:

```
// Provided by VK_VERSION_1_0
typedef enum VkStructureType {
    VK_STRUCTURE_TYPE_APPLICATION_INFO = 0,
    VK_STRUCTURE_TYPE_INSTANCE_CREATE_INFO = 1,
    VK_STRUCTURE_TYPE_DEVICE_QUEUE_CREATE_INFO = 2,
    VK_STRUCTURE_TYPE_DEVICE_CREATE_INFO = 3,
    VK_STRUCTURE_TYPE_SUBMIT_INFO = 4,
    VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_INFO = 5,
    VK_STRUCTURE_TYPE_MAPPED_MEMORY_RANGE = 6,
    VK_STRUCTURE_TYPE_BIND_SPARSE_INFO = 7,
    VK_STRUCTURE_TYPE_FENCE_CREATE_INFO = 8,
    VK_STRUCTURE_TYPE_SEMAPHORE_CREATE_INFO = 9,
    VK_STRUCTURE_TYPE_EVENT_CREATE_INFO = 10,
    VK_STRUCTURE_TYPE_QUERY_POOL_CREATE_INFO = 11,
    VK_STRUCTURE_TYPE_BUFFER_CREATE_INFO = 12,
    VK_STRUCTURE_TYPE_BUFFER_VIEW_CREATE_INFO = 13,
    VK_STRUCTURE_TYPE_IMAGE_CREATE_INFO = 14,
    VK_STRUCTURE_TYPE_IMAGE_VIEW_CREATE_INFO = 15,
    VK_STRUCTURE_TYPE_SHADER_MODULE_CREATE_INFO = 16,
    VK_STRUCTURE_TYPE_PIPELINE_CACHE_CREATE_INFO = 17,
    VK_STRUCTURE_TYPE_PIPELINE_SHADER_STAGE_CREATE_INFO = 18,
    VK_STRUCTURE_TYPE_PIPELINE_VERTEX_INPUT_STATE_CREATE_INFO = 19,
    VK_STRUCTURE_TYPE_PIPELINE_INPUT_ASSEMBLY_STATE_CREATE_INFO = 20,
    VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_STATE_CREATE_INFO = 21,
    VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_STATE_CREATE_INFO = 22,
    VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_STATE_CREATE_INFO = 23,
    VK_STRUCTURE_TYPE_PIPELINE_MULTISAMPLE_STATE_CREATE_INFO = 24,
```

```
    VK_STRUCTURE_TYPE_PIPELINE_DEPTH_STENCIL_STATE_CREATE_INFO = 25,
    VK_STRUCTURE_TYPE_PIPELINE_COLOR_BLEND_STATE_CREATE_INFO = 26,
    VK_STRUCTURE_TYPE_PIPELINE_DYNAMIC_STATE_CREATE_INFO = 27,
    VK_STRUCTURE_TYPE_GRAPHICS_PIPELINE_CREATE_INFO = 28,
    VK_STRUCTURE_TYPE_COMPUTE_PIPELINE_CREATE_INFO = 29,
    VK_STRUCTURE_TYPE_PIPELINE_LAYOUT_CREATE_INFO = 30,
    VK_STRUCTURE_TYPE_SAMPLER_CREATE_INFO = 31,
    VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_CREATE_INFO = 32,
    VK_STRUCTURE_TYPE_DESCRIPTOR_POOL_CREATE_INFO = 33,
    VK_STRUCTURE_TYPE_DESCRIPTOR_SET_ALLOCATE_INFO = 34,
    VK_STRUCTURE_TYPE_WRITE_DESCRIPTOR_SET = 35,
    VK_STRUCTURE_TYPE_COPY_DESCRIPTOR_SET = 36,
    VK_STRUCTURE_TYPE_FRAMEBUFFER_CREATE_INFO = 37,
    VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO = 38,
    VK_STRUCTURE_TYPE_COMMAND_POOL_CREATE_INFO = 39,
    VK_STRUCTURE_TYPE_COMMAND_BUFFER_ALLOCATE_INFO = 40,
    VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_INFO = 41,
    VK_STRUCTURE_TYPE_COMMAND_BUFFER_BEGIN_INFO = 42,
    VK_STRUCTURE_TYPE_RENDER_PASS_BEGIN_INFO = 43,
    VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER = 44,
    VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER = 45,
    VK_STRUCTURE_TYPE_MEMORY_BARRIER = 46,
    VK_STRUCTURE_TYPE_LOADER_INSTANCE_CREATE_INFO = 47,
    VK_STRUCTURE_TYPE_LOADER_DEVICE_CREATE_INFO = 48,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_INFO = 1000157000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_INFO = 1000157001,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_MEMORY_DEDICATED_REQUIREMENTS = 1000127000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_MEMORY_DEDICATED_ALLOCATE_INFO = 1000127001,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_FLAGS_INFO = 1000060000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_DEVICE_GROUP_COMMAND_BUFFER_BEGIN_INFO = 1000060004,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_DEVICE_GROUP_SUBMIT_INFO = 1000060005,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_DEVICE_GROUP_BIND_SPARSE_INFO = 1000060006,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_DEVICE_GROUP_INFO = 1000060013,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_DEVICE_GROUP_INFO = 1000060014,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GROUP_PROPERTIES = 1000070000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_DEVICE_GROUP_DEVICE_CREATE_INFO = 1000070001,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_BUFFER_MEMORY_REQUIREMENTS_INFO_2 = 1000146000,
  // Provided by VK_VERSION_1_1
```

```
    VK_STRUCTURE_TYPE_IMAGE_MEMORY_REQUIREMENTS_INFO_2 = 1000146001,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_IMAGE_SPARSE_MEMORY_REQUIREMENTS_INFO_2 = 1000146002,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_MEMORY_REQUIREMENTS_2 = 1000146003,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_SPARSE_IMAGE_MEMORY_REQUIREMENTS_2 = 1000146004,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FEATURES_2 = 1000059000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROPERTIES_2 = 1000059001,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_2 = 1000059002,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_IMAGE_FORMAT_PROPERTIES_2 = 1000059003,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_FORMAT_INFO_2 = 1000059004,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_QUEUE_FAMILY_PROPERTIES_2 = 1000059005,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PROPERTIES_2 = 1000059006,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_SPARSE_IMAGE_FORMAT_PROPERTIES_2 = 1000059007,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SPARSE_IMAGE_FORMAT_INFO_2 = 1000059008,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_IMAGE_VIEW_USAGE_CREATE_INFO = 1000117002,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PROTECTED_SUBMIT_INFO = 1000145000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROTECTED_MEMORY_FEATURES = 1000145001,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROTECTED_MEMORY_PROPERTIES = 1000145002,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_DEVICE_QUEUE_INFO_2 = 1000145003,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_IMAGE_FORMAT_INFO = 1000071000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_EXTERNAL_IMAGE_FORMAT_PROPERTIES = 1000071001,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_BUFFER_INFO = 1000071002,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_EXTERNAL_BUFFER_PROPERTIES = 1000071003,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ID_PROPERTIES = 1000071004,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_BUFFER_CREATE_INFO = 1000072000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_IMAGE_CREATE_INFO = 1000072001,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_EXPORT_MEMORY_ALLOCATE_INFO = 1000072002,
```

```
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_FENCE_INFO = 1000112000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_EXTERNAL_FENCE_PROPERTIES = 1000112001,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_EXPORT_FENCE_CREATE_INFO = 1000113000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_CREATE_INFO = 1000077000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_SEMAPHORE_INFO = 1000076000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_EXTERNAL_SEMAPHORE_PROPERTIES = 1000076001,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SUBGROUP_PROPERTIES = 1000094000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_16BIT_STORAGE_FEATURES = 1000083000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTERS_FEATURES = 1000120000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_CREATE_INFO = 1000085000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_3_PROPERTIES = 1000168000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_SUPPORT = 1000168001,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_CREATE_INFO = 1000156000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_INFO = 1000156001,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_BIND_IMAGE_PLANE_MEMORY_INFO = 1000156002,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_IMAGE_PLANE_MEMORY_REQUIREMENTS_INFO = 1000156003,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLER_YCBCR_CONVERSION_FEATURES = 1000156004,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_IMAGE_FORMAT_PROPERTIES = 1000156005,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_DEVICE_GROUP_RENDER_PASS_BEGIN_INFO = 1000060003,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_POINT_CLIPPING_PROPERTIES = 1000117000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_RENDER_PASS_INPUT_ATTACHMENT_ASPECT_CREATE_INFO = 1000117001,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_DOMAIN_ORIGIN_STATE_CREATE_INFO =
1000117003,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_RENDER_PASS_MULTIVIEW_CREATE_INFO = 1000053000,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_FEATURES = 1000053001,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_PROPERTIES = 1000053002,
```

```
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_DRAW_PARAMETERS_FEATURES = 1000063000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRIVER_PROPERTIES = 1000196000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_1_FEATURES = 49,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_1_PROPERTIES = 50,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_2_FEATURES = 51,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_2_PROPERTIES = 52,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_IMAGE_FORMAT_LIST_CREATE_INFO = 1000147000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_MEMORY_MODEL_FEATURES = 1000211000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_QUERY_RESET_FEATURES = 1000261000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_FEATURES = 1000207000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_PROPERTIES = 1000207001,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_SEMAPHORE_TYPE_CREATE_INFO = 1000207002,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_TIMELINE_SEMAPHORE_SUBMIT_INFO = 1000207003,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_SEMAPHORE_WAIT_INFO = 1000207004,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_SEMAPHORE_SIGNAL_INFO = 1000207005,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BUFFER_DEVICE_ADDRESS_FEATURES = 1000257000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_BUFFER_DEVICE_ADDRESS_INFO = 1000244001,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_BUFFER_OPAQUE_CAPTURE_ADDRESS_CREATE_INFO = 1000257002,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_MEMORY_OPAQUE_CAPTURE_ADDRESS_ALLOCATE_INFO = 1000257003,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_DEVICE_MEMORY_OPAQUE_CAPTURE_ADDRESS_INFO = 1000257004,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_8BIT_STORAGE_FEATURES = 1000177000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_INT64_FEATURES = 1000180000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT16_INT8_FEATURES = 1000082000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FLOAT_CONTROLS_PROPERTIES = 1000197000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_BINDING_FLAGS_CREATE_INFO = 1000161000,
  // Provided by VK_VERSION_1_2
```

```
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_INDEXING_FEATURES = 1000161001,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_INDEXING_PROPERTIES = 1000161002,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_DESCRIPTOR_SET_VARIABLE_DESCRIPTOR_COUNT_ALLOCATE_INFO =
1000161003,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_DESCRIPTOR_SET_VARIABLE_DESCRIPTOR_COUNT_LAYOUT_SUPPORT =
1000161004,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SCALAR_BLOCK_LAYOUT_FEATURES = 1000221000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLER_FILTER_MINMAX_PROPERTIES = 1000130000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_SAMPLER_REDUCTION_MODE_CREATE_INFO = 1000130001,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_UNIFORM_BUFFER_STANDARD_LAYOUT_FEATURES =
1000253000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_EXTENDED_TYPES_FEATURES =
1000175000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_2 = 1000109000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_2 = 1000109001,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_2 = 1000109002,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_SUBPASS_DEPENDENCY_2 = 1000109003,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO_2 = 1000109004,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_SUBPASS_BEGIN_INFO = 1000109005,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_SUBPASS_END_INFO = 1000109006,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_STENCIL_RESOLVE_PROPERTIES = 1000199000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_DEPTH_STENCIL_RESOLVE = 1000199001,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_IMAGE_STENCIL_USAGE_CREATE_INFO = 1000246000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGELESS_FRAMEBUFFER_FEATURES = 1000108000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENTS_CREATE_INFO = 1000108001,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENT_IMAGE_INFO = 1000108002,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_RENDER_PASS_ATTACHMENT_BEGIN_INFO = 1000108003,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SEPARATE_DEPTH_STENCIL_LAYOUTS_FEATURES =
```

```
1000241000,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_STENCIL_LAYOUT = 1000241001,
  // Provided by VK_VERSION_1_2
    VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_STENCIL_LAYOUT = 1000241002,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_3_FEATURES = 53,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_3_PROPERTIES = 54,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TOOL_PROPERTIES = 1000245000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIVATE_DATA_FEATURES = 1000295000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_DEVICE_PRIVATE_DATA_CREATE_INFO = 1000295001,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PRIVATE_DATA_SLOT_CREATE_INFO = 1000295002,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_MEMORY_BARRIER_2 = 1000314000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER_2 = 1000314001,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER_2 = 1000314002,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_DEPENDENCY_INFO = 1000314003,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_SUBMIT_INFO_2 = 1000314004,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_SEMAPHORE_SUBMIT_INFO = 1000314005,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_COMMAND_BUFFER_SUBMIT_INFO = 1000314006,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SYNCHRONIZATION_2_FEATURES = 1000314007,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_COPY_BUFFER_INFO_2 = 1000337000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_COPY_IMAGE_INFO_2 = 1000337001,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_COPY_BUFFER_TO_IMAGE_INFO_2 = 1000337002,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_COPY_IMAGE_TO_BUFFER_INFO_2 = 1000337003,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_BUFFER_COPY_2 = 1000337006,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_IMAGE_COPY_2 = 1000337007,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_BUFFER_IMAGE_COPY_2 = 1000337009,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TEXTURE_COMPRESSION_ASTC_HDR_FEATURES =
1000066000,
  // Provided by VK_VERSION_1_3
```

```
    VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_3 = 1000360000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_FEATURES = 1000413000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_PROPERTIES = 1000413001,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_DEVICE_BUFFER_MEMORY_REQUIREMENTS = 1000413002,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_DEVICE_IMAGE_MEMORY_REQUIREMENTS = 1000413003,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PIPELINE_CREATION_FEEDBACK_CREATE_INFO = 1000192000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TERMINATE_INVOCATION_FEATURES =
1000215000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_DEMOTE_TO_HELPER_INVOCATION_FEATURES =
1000276000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_CREATION_CACHE_CONTROL_FEATURES =
1000297000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ZERO_INITIALIZE_WORKGROUP_MEMORY_FEATURES =
1000325000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_ROBUSTNESS_FEATURES = 1000335000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SUBGROUP_SIZE_CONTROL_PROPERTIES = 1000225000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PIPELINE_SHADER_STAGE_REQUIRED_SUBGROUP_SIZE_CREATE_INFO =
1000225001,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SUBGROUP_SIZE_CONTROL_FEATURES = 1000225002,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INLINE_UNIFORM_BLOCK_FEATURES = 1000138000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INLINE_UNIFORM_BLOCK_PROPERTIES = 1000138001,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_WRITE_DESCRIPTOR_SET_INLINE_UNIFORM_BLOCK = 1000138002,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_DESCRIPTOR_POOL_INLINE_UNIFORM_BLOCK_CREATE_INFO = 1000138003,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_FEATURES =
1000280000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_PROPERTIES =
1000280001,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TEXEL_BUFFER_ALIGNMENT_PROPERTIES = 1000281001,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_BLIT_IMAGE_INFO_2 = 1000337004,
  // Provided by VK_VERSION_1_3
```

```
    VK_STRUCTURE_TYPE_RESOLVE_IMAGE_INFO_2 = 1000337005,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_IMAGE_BLIT_2 = 1000337008,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_IMAGE_RESOLVE_2 = 1000337010,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_RENDERING_INFO = 1000044000,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_INFO = 1000044001,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PIPELINE_RENDERING_CREATE_INFO = 1000044002,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_FEATURES = 1000044003,
  // Provided by VK_VERSION_1_3
    VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_RENDERING_INFO = 1000044004,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_4_FEATURES = 55,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_4_PROPERTIES = 56,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_DEVICE_QUEUE_GLOBAL_PRIORITY_CREATE_INFO = 1000174000,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GLOBAL_PRIORITY_QUERY_FEATURES = 1000388000,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_QUEUE_FAMILY_GLOBAL_PRIORITY_PROPERTIES = 1000388001,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INDEX_TYPE_UINT8_FEATURES = 1000265000,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_MEMORY_MAP_INFO = 1000271000,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_MEMORY_UNMAP_INFO = 1000271001,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_FEATURES = 1000470000,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_PROPERTIES = 1000470001,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_DEVICE_IMAGE_SUBRESOURCE_INFO = 1000470004,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_SUBRESOURCE_LAYOUT_2 = 1000338002,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_IMAGE_SUBRESOURCE_2 = 1000338003,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_BUFFER_USAGE_FLAGS_2_CREATE_INFO = 1000470006,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_FEATURES = 1000545000,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_PROPERTIES = 1000545001,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_BIND_MEMORY_STATUS = 1000545002,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_FEATURES = 1000270000,
```

```
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_PROPERTIES = 1000270001,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_MEMORY_TO_IMAGE_COPY = 1000270002,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_IMAGE_TO_MEMORY_COPY = 1000270003,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_COPY_IMAGE_TO_MEMORY_INFO = 1000270004,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_COPY_MEMORY_TO_IMAGE_INFO = 1000270005,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_HOST_IMAGE_LAYOUT_TRANSITION_INFO = 1000270006,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_COPY_IMAGE_TO_IMAGE_INFO = 1000270007,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_SUBRESOURCE_HOST_MEMCPY_SIZE = 1000270008,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_HOST_IMAGE_COPY_DEVICE_PERFORMANCE_QUERY = 1000270009,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_ROTATE_FEATURES = 1000416000,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT_CONTROLS_2_FEATURES = 1000528000,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_EXPECT_ASSUME_FEATURES = 1000544000,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PIPELINE_CREATE_FLAGS_2_CREATE_INFO = 1000470005,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PUSH_DESCRIPTOR_PROPERTIES = 1000080000,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_BIND_DESCRIPTOR_SETS_INFO = 1000545003,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PUSH_CONSTANTS_INFO = 1000545004,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_INFO = 1000545005,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_WITH_TEMPLATE_INFO = 1000545006,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_PROTECTED_ACCESS_FEATURES = 1000466000,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PIPELINE_ROBUSTNESS_CREATE_INFO = 1000068000,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_ROBUSTNESS_FEATURES = 1000068001,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_ROBUSTNESS_PROPERTIES = 1000068002,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_FEATURES = 1000259000,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_LINE_STATE_CREATE_INFO = 1000259001,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_PROPERTIES = 1000259002,
  // Provided by VK_VERSION_1_4
```

```
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_PROPERTIES =
1000525000,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PIPELINE_VERTEX_INPUT_DIVISOR_STATE_CREATE_INFO = 1000190001,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_FEATURES = 1000190002,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_RENDERING_AREA_INFO = 1000470003,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_LOCAL_READ_FEATURES =
1000232000,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_LOCATION_INFO = 1000232001,
  // Provided by VK_VERSION_1_4
    VK_STRUCTURE_TYPE_RENDERING_INPUT_ATTACHMENT_INDEX_INFO = 1000232002,
  // Provided by VK_KHR_swapchain
    VK_STRUCTURE_TYPE_SWAPCHAIN_CREATE_INFO_KHR = 1000001000,
  // Provided by VK_KHR_swapchain
    VK_STRUCTURE_TYPE_PRESENT_INFO_KHR = 1000001001,
  // Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_surface
    VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_CAPABILITIES_KHR = 1000060007,
  // Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
    VK_STRUCTURE_TYPE_IMAGE_SWAPCHAIN_CREATE_INFO_KHR = 1000060008,
  // Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
    VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_SWAPCHAIN_INFO_KHR = 1000060009,
  // Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
    VK_STRUCTURE_TYPE_ACQUIRE_NEXT_IMAGE_INFO_KHR = 1000060010,
  // Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
    VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_INFO_KHR = 1000060011,
  // Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
    VK_STRUCTURE_TYPE_DEVICE_GROUP_SWAPCHAIN_CREATE_INFO_KHR = 1000060012,
  // Provided by VK_KHR_display
    VK_STRUCTURE_TYPE_DISPLAY_MODE_CREATE_INFO_KHR = 1000002000,
  // Provided by VK_KHR_display
    VK_STRUCTURE_TYPE_DISPLAY_SURFACE_CREATE_INFO_KHR = 1000002001,
  // Provided by VK_KHR_display_swapchain
    VK_STRUCTURE_TYPE_DISPLAY_PRESENT_INFO_KHR = 1000003000,
  // Provided by VK_KHR_xlib_surface
    VK_STRUCTURE_TYPE_XLIB_SURFACE_CREATE_INFO_KHR = 1000004000,
  // Provided by VK_KHR_xcb_surface
    VK_STRUCTURE_TYPE_XCB_SURFACE_CREATE_INFO_KHR = 1000005000,
  // Provided by VK_KHR_wayland_surface
    VK_STRUCTURE_TYPE_WAYLAND_SURFACE_CREATE_INFO_KHR = 1000006000,
  // Provided by VK_KHR_android_surface
    VK_STRUCTURE_TYPE_ANDROID_SURFACE_CREATE_INFO_KHR = 1000008000,
```

```
  // Provided by VK_KHR_win32_surface
    VK_STRUCTURE_TYPE_WIN32_SURFACE_CREATE_INFO_KHR = 1000009000,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_VIDEO_PROFILE_INFO_KHR = 1000023000,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_VIDEO_CAPABILITIES_KHR = 1000023001,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_VIDEO_PICTURE_RESOURCE_INFO_KHR = 1000023002,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_VIDEO_SESSION_MEMORY_REQUIREMENTS_KHR = 1000023003,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_BIND_VIDEO_SESSION_MEMORY_INFO_KHR = 1000023004,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_VIDEO_SESSION_CREATE_INFO_KHR = 1000023005,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_VIDEO_SESSION_PARAMETERS_CREATE_INFO_KHR = 1000023006,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_VIDEO_SESSION_PARAMETERS_UPDATE_INFO_KHR = 1000023007,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_VIDEO_BEGIN_CODING_INFO_KHR = 1000023008,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_VIDEO_END_CODING_INFO_KHR = 1000023009,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_VIDEO_CODING_CONTROL_INFO_KHR = 1000023010,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_VIDEO_REFERENCE_SLOT_INFO_KHR = 1000023011,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_QUEUE_FAMILY_VIDEO_PROPERTIES_KHR = 1000023012,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_VIDEO_PROFILE_LIST_INFO_KHR = 1000023013,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_FORMAT_INFO_KHR = 1000023014,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_VIDEO_FORMAT_PROPERTIES_KHR = 1000023015,
  // Provided by VK_KHR_video_queue
    VK_STRUCTURE_TYPE_QUEUE_FAMILY_QUERY_RESULT_STATUS_PROPERTIES_KHR = 1000023016,
  // Provided by VK_KHR_video_decode_queue
    VK_STRUCTURE_TYPE_VIDEO_DECODE_INFO_KHR = 1000024000,
  // Provided by VK_KHR_video_decode_queue
    VK_STRUCTURE_TYPE_VIDEO_DECODE_CAPABILITIES_KHR = 1000024001,
  // Provided by VK_KHR_video_decode_queue
    VK_STRUCTURE_TYPE_VIDEO_DECODE_USAGE_INFO_KHR = 1000024002,
  // Provided by VK_EXT_transform_feedback
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TRANSFORM_FEEDBACK_FEATURES_EXT = 1000028000,
  // Provided by VK_EXT_transform_feedback
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TRANSFORM_FEEDBACK_PROPERTIES_EXT = 1000028001,
  // Provided by VK_EXT_transform_feedback
    VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_STATE_STREAM_CREATE_INFO_EXT =
1000028002,
  // Provided by VK_KHR_video_encode_h264
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_CAPABILITIES_KHR = 1000038000,
```

```
  // Provided by VK_KHR_video_encode_h264
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_CREATE_INFO_KHR =
1000038001,
  // Provided by VK_KHR_video_encode_h264
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_ADD_INFO_KHR = 1000038002,
  // Provided by VK_KHR_video_encode_h264
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_PICTURE_INFO_KHR = 1000038003,
  // Provided by VK_KHR_video_encode_h264
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_DPB_SLOT_INFO_KHR = 1000038004,
  // Provided by VK_KHR_video_encode_h264
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_NALU_SLICE_INFO_KHR = 1000038005,
  // Provided by VK_KHR_video_encode_h264
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_GOP_REMAINING_FRAME_INFO_KHR = 1000038006,
  // Provided by VK_KHR_video_encode_h264
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_PROFILE_INFO_KHR = 1000038007,
  // Provided by VK_KHR_video_encode_h264
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_RATE_CONTROL_INFO_KHR = 1000038008,
  // Provided by VK_KHR_video_encode_h264
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_RATE_CONTROL_LAYER_INFO_KHR = 1000038009,
  // Provided by VK_KHR_video_encode_h264
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_CREATE_INFO_KHR = 1000038010,
  // Provided by VK_KHR_video_encode_h264
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_QUALITY_LEVEL_PROPERTIES_KHR = 1000038011,
  // Provided by VK_KHR_video_encode_h264
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_GET_INFO_KHR = 1000038012,
  // Provided by VK_KHR_video_encode_h264
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_FEEDBACK_INFO_KHR =
1000038013,
  // Provided by VK_KHR_video_encode_h265
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_CAPABILITIES_KHR = 1000039000,
  // Provided by VK_KHR_video_encode_h265
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_CREATE_INFO_KHR =
1000039001,
  // Provided by VK_KHR_video_encode_h265
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_ADD_INFO_KHR = 1000039002,
  // Provided by VK_KHR_video_encode_h265
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_PICTURE_INFO_KHR = 1000039003,
  // Provided by VK_KHR_video_encode_h265
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_DPB_SLOT_INFO_KHR = 1000039004,
  // Provided by VK_KHR_video_encode_h265
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_NALU_SLICE_SEGMENT_INFO_KHR = 1000039005,
  // Provided by VK_KHR_video_encode_h265
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_GOP_REMAINING_FRAME_INFO_KHR = 1000039006,
  // Provided by VK_KHR_video_encode_h265
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_PROFILE_INFO_KHR = 1000039007,
  // Provided by VK_KHR_video_encode_h265
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_RATE_CONTROL_INFO_KHR = 1000039009,
  // Provided by VK_KHR_video_encode_h265
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_RATE_CONTROL_LAYER_INFO_KHR = 1000039010,
  // Provided by VK_KHR_video_encode_h265
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_CREATE_INFO_KHR = 1000039011,
```

```
  // Provided by VK_KHR_video_encode_h265
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_QUALITY_LEVEL_PROPERTIES_KHR = 1000039012,
  // Provided by VK_KHR_video_encode_h265
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_GET_INFO_KHR = 1000039013,
  // Provided by VK_KHR_video_encode_h265
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_FEEDBACK_INFO_KHR =
1000039014,
  // Provided by VK_KHR_video_decode_h264
    VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_CAPABILITIES_KHR = 1000040000,
  // Provided by VK_KHR_video_decode_h264
    VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_PICTURE_INFO_KHR = 1000040001,
  // Provided by VK_KHR_video_decode_h264
    VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_PROFILE_INFO_KHR = 1000040003,
  // Provided by VK_KHR_video_decode_h264
    VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_SESSION_PARAMETERS_CREATE_INFO_KHR =
1000040004,
  // Provided by VK_KHR_video_decode_h264
    VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_SESSION_PARAMETERS_ADD_INFO_KHR = 1000040005,
  // Provided by VK_KHR_video_decode_h264
    VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_DPB_SLOT_INFO_KHR = 1000040006,
  // Provided by VK_EXT_astc_decode_mode
    VK_STRUCTURE_TYPE_IMAGE_VIEW_ASTC_DECODE_MODE_EXT = 1000067000,
  // Provided by VK_EXT_astc_decode_mode
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ASTC_DECODE_FEATURES_EXT = 1000067001,
  // Provided by VK_KHR_external_memory_win32
    VK_STRUCTURE_TYPE_IMPORT_MEMORY_WIN32_HANDLE_INFO_KHR = 1000073000,
  // Provided by VK_KHR_external_memory_win32
    VK_STRUCTURE_TYPE_EXPORT_MEMORY_WIN32_HANDLE_INFO_KHR = 1000073001,
  // Provided by VK_KHR_external_memory_win32
    VK_STRUCTURE_TYPE_MEMORY_WIN32_HANDLE_PROPERTIES_KHR = 1000073002,
  // Provided by VK_KHR_external_memory_win32
    VK_STRUCTURE_TYPE_MEMORY_GET_WIN32_HANDLE_INFO_KHR = 1000073003,
  // Provided by VK_KHR_external_memory_fd
    VK_STRUCTURE_TYPE_IMPORT_MEMORY_FD_INFO_KHR = 1000074000,
  // Provided by VK_KHR_external_memory_fd
    VK_STRUCTURE_TYPE_MEMORY_FD_PROPERTIES_KHR = 1000074001,
  // Provided by VK_KHR_external_memory_fd
    VK_STRUCTURE_TYPE_MEMORY_GET_FD_INFO_KHR = 1000074002,
  // Provided by VK_KHR_win32_keyed_mutex
    VK_STRUCTURE_TYPE_WIN32_KEYED_MUTEX_ACQUIRE_RELEASE_INFO_KHR = 1000075000,
  // Provided by VK_KHR_external_semaphore_win32
    VK_STRUCTURE_TYPE_IMPORT_SEMAPHORE_WIN32_HANDLE_INFO_KHR = 1000078000,
  // Provided by VK_KHR_external_semaphore_win32
    VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_WIN32_HANDLE_INFO_KHR = 1000078001,
  // Provided by VK_KHR_external_semaphore_win32
    VK_STRUCTURE_TYPE_D3D12_FENCE_SUBMIT_INFO_KHR = 1000078002,
  // Provided by VK_KHR_external_semaphore_win32
    VK_STRUCTURE_TYPE_SEMAPHORE_GET_WIN32_HANDLE_INFO_KHR = 1000078003,
  // Provided by VK_KHR_external_semaphore_fd
    VK_STRUCTURE_TYPE_IMPORT_SEMAPHORE_FD_INFO_KHR = 1000079000,
  // Provided by VK_KHR_external_semaphore_fd
```

```
    VK_STRUCTURE_TYPE_SEMAPHORE_GET_FD_INFO_KHR = 1000079001,
  // Provided by VK_EXT_conditional_rendering
    VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_CONDITIONAL_RENDERING_INFO_EXT =
1000081000,
  // Provided by VK_EXT_conditional_rendering
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CONDITIONAL_RENDERING_FEATURES_EXT = 1000081001,
  // Provided by VK_EXT_conditional_rendering
    VK_STRUCTURE_TYPE_CONDITIONAL_RENDERING_BEGIN_INFO_EXT = 1000081002,
  // Provided by VK_KHR_incremental_present
    VK_STRUCTURE_TYPE_PRESENT_REGIONS_KHR = 1000084000,
  // Provided by VK_EXT_display_surface_counter
    VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_2_EXT = 1000090000,
  // Provided by VK_EXT_display_control
    VK_STRUCTURE_TYPE_DISPLAY_POWER_INFO_EXT = 1000091000,
  // Provided by VK_EXT_display_control
    VK_STRUCTURE_TYPE_DEVICE_EVENT_INFO_EXT = 1000091001,
  // Provided by VK_EXT_display_control
    VK_STRUCTURE_TYPE_DISPLAY_EVENT_INFO_EXT = 1000091002,
  // Provided by VK_EXT_display_control
    VK_STRUCTURE_TYPE_SWAPCHAIN_COUNTER_CREATE_INFO_EXT = 1000091003,
  // Provided by VK_EXT_discard_rectangles
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DISCARD_RECTANGLE_PROPERTIES_EXT = 1000099000,
  // Provided by VK_EXT_discard_rectangles
    VK_STRUCTURE_TYPE_PIPELINE_DISCARD_RECTANGLE_STATE_CREATE_INFO_EXT = 1000099001,
  // Provided by VK_EXT_conservative_rasterization
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CONSERVATIVE_RASTERIZATION_PROPERTIES_EXT =
1000101000,
  // Provided by VK_EXT_conservative_rasterization
    VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_CONSERVATIVE_STATE_CREATE_INFO_EXT =
1000101001,
  // Provided by VK_EXT_depth_clip_enable
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLIP_ENABLE_FEATURES_EXT = 1000102000,
  // Provided by VK_EXT_depth_clip_enable
    VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_DEPTH_CLIP_STATE_CREATE_INFO_EXT =
1000102001,
  // Provided by VK_EXT_hdr_metadata
    VK_STRUCTURE_TYPE_HDR_METADATA_EXT = 1000105000,
  // Provided by VK_KHR_shared_presentable_image
    VK_STRUCTURE_TYPE_SHARED_PRESENT_SURFACE_CAPABILITIES_KHR = 1000111000,
  // Provided by VK_KHR_external_fence_win32
    VK_STRUCTURE_TYPE_IMPORT_FENCE_WIN32_HANDLE_INFO_KHR = 1000114000,
  // Provided by VK_KHR_external_fence_win32
    VK_STRUCTURE_TYPE_EXPORT_FENCE_WIN32_HANDLE_INFO_KHR = 1000114001,
  // Provided by VK_KHR_external_fence_win32
    VK_STRUCTURE_TYPE_FENCE_GET_WIN32_HANDLE_INFO_KHR = 1000114002,
  // Provided by VK_KHR_external_fence_fd
    VK_STRUCTURE_TYPE_IMPORT_FENCE_FD_INFO_KHR = 1000115000,
  // Provided by VK_KHR_external_fence_fd
    VK_STRUCTURE_TYPE_FENCE_GET_FD_INFO_KHR = 1000115001,
  // Provided by VK_KHR_performance_query
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PERFORMANCE_QUERY_FEATURES_KHR = 1000116000,
```

```
  // Provided by VK_KHR_performance_query
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PERFORMANCE_QUERY_PROPERTIES_KHR = 1000116001,
  // Provided by VK_KHR_performance_query
    VK_STRUCTURE_TYPE_QUERY_POOL_PERFORMANCE_CREATE_INFO_KHR = 1000116002,
  // Provided by VK_KHR_performance_query
    VK_STRUCTURE_TYPE_PERFORMANCE_QUERY_SUBMIT_INFO_KHR = 1000116003,
  // Provided by VK_KHR_performance_query
    VK_STRUCTURE_TYPE_ACQUIRE_PROFILING_LOCK_INFO_KHR = 1000116004,
  // Provided by VK_KHR_performance_query
    VK_STRUCTURE_TYPE_PERFORMANCE_COUNTER_KHR = 1000116005,
  // Provided by VK_KHR_performance_query
    VK_STRUCTURE_TYPE_PERFORMANCE_COUNTER_DESCRIPTION_KHR = 1000116006,
  // Provided by VK_KHR_get_surface_capabilities2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SURFACE_INFO_2_KHR = 1000119000,
  // Provided by VK_KHR_get_surface_capabilities2
    VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_2_KHR = 1000119001,
  // Provided by VK_KHR_get_surface_capabilities2
    VK_STRUCTURE_TYPE_SURFACE_FORMAT_2_KHR = 1000119002,
  // Provided by VK_KHR_get_display_properties2
    VK_STRUCTURE_TYPE_DISPLAY_PROPERTIES_2_KHR = 1000121000,
  // Provided by VK_KHR_get_display_properties2
    VK_STRUCTURE_TYPE_DISPLAY_PLANE_PROPERTIES_2_KHR = 1000121001,
  // Provided by VK_KHR_get_display_properties2
    VK_STRUCTURE_TYPE_DISPLAY_MODE_PROPERTIES_2_KHR = 1000121002,
  // Provided by VK_KHR_get_display_properties2
    VK_STRUCTURE_TYPE_DISPLAY_PLANE_INFO_2_KHR = 1000121003,
  // Provided by VK_KHR_get_display_properties2
    VK_STRUCTURE_TYPE_DISPLAY_PLANE_CAPABILITIES_2_KHR = 1000121004,
  // Provided by VK_EXT_debug_utils
    VK_STRUCTURE_TYPE_DEBUG_UTILS_OBJECT_NAME_INFO_EXT = 1000128000,
  // Provided by VK_EXT_debug_utils
    VK_STRUCTURE_TYPE_DEBUG_UTILS_OBJECT_TAG_INFO_EXT = 1000128001,
  // Provided by VK_EXT_debug_utils
    VK_STRUCTURE_TYPE_DEBUG_UTILS_LABEL_EXT = 1000128002,
  // Provided by VK_EXT_debug_utils
    VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CALLBACK_DATA_EXT = 1000128003,
  // Provided by VK_EXT_debug_utils
    VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CREATE_INFO_EXT = 1000128004,
  // Provided by VK_EXT_descriptor_heap
    VK_STRUCTURE_TYPE_TEXEL_BUFFER_DESCRIPTOR_INFO_EXT = 1000135000,
  // Provided by VK_EXT_descriptor_heap
    VK_STRUCTURE_TYPE_IMAGE_DESCRIPTOR_INFO_EXT = 1000135001,
  // Provided by VK_EXT_descriptor_heap
    VK_STRUCTURE_TYPE_RESOURCE_DESCRIPTOR_INFO_EXT = 1000135002,
  // Provided by VK_EXT_descriptor_heap
    VK_STRUCTURE_TYPE_BIND_HEAP_INFO_EXT = 1000135003,
  // Provided by VK_EXT_descriptor_heap
    VK_STRUCTURE_TYPE_PUSH_DATA_INFO_EXT = 1000135004,
  // Provided by VK_EXT_descriptor_heap
    VK_STRUCTURE_TYPE_DESCRIPTOR_SET_AND_BINDING_MAPPING_EXT = 1000135005,
  // Provided by VK_EXT_descriptor_heap
```

```
    VK_STRUCTURE_TYPE_SHADER_DESCRIPTOR_SET_AND_BINDING_MAPPING_INFO_EXT = 1000135006,
  // Provided by VK_EXT_descriptor_heap
    VK_STRUCTURE_TYPE_OPAQUE_CAPTURE_DATA_CREATE_INFO_EXT = 1000135007,
  // Provided by VK_EXT_descriptor_heap
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_HEAP_PROPERTIES_EXT = 1000135008,
  // Provided by VK_EXT_descriptor_heap
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_HEAP_FEATURES_EXT = 1000135009,
  // Provided by VK_EXT_descriptor_heap
    VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_DESCRIPTOR_HEAP_INFO_EXT =
1000135010,
  // Provided by VK_EXT_custom_border_color with VK_EXT_descriptor_heap
    VK_STRUCTURE_TYPE_SAMPLER_CUSTOM_BORDER_COLOR_INDEX_CREATE_INFO_EXT = 1000135011,
  // Provided by VK_EXT_descriptor_heap with VK_EXT_fragment_density_map
    VK_STRUCTURE_TYPE_SUBSAMPLED_IMAGE_FORMAT_PROPERTIES_EXT = 1000135013,
  // Provided by VK_KHR_shader_bfloat16
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_BFLOAT16_FEATURES_KHR = 1000141000,
  // Provided by VK_EXT_sample_locations
    VK_STRUCTURE_TYPE_SAMPLE_LOCATIONS_INFO_EXT = 1000143000,
  // Provided by VK_EXT_sample_locations
    VK_STRUCTURE_TYPE_RENDER_PASS_SAMPLE_LOCATIONS_BEGIN_INFO_EXT = 1000143001,
  // Provided by VK_EXT_sample_locations
    VK_STRUCTURE_TYPE_PIPELINE_SAMPLE_LOCATIONS_STATE_CREATE_INFO_EXT = 1000143002,
  // Provided by VK_EXT_sample_locations
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLE_LOCATIONS_PROPERTIES_EXT = 1000143003,
  // Provided by VK_EXT_sample_locations
    VK_STRUCTURE_TYPE_MULTISAMPLE_PROPERTIES_EXT = 1000143004,
  // Provided by VK_EXT_blend_operation_advanced
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BLEND_OPERATION_ADVANCED_FEATURES_EXT =
1000148000,
  // Provided by VK_EXT_blend_operation_advanced
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BLEND_OPERATION_ADVANCED_PROPERTIES_EXT =
1000148001,
  // Provided by VK_EXT_blend_operation_advanced
    VK_STRUCTURE_TYPE_PIPELINE_COLOR_BLEND_ADVANCED_STATE_CREATE_INFO_EXT =
1000148002,
  // Provided by VK_KHR_acceleration_structure
    VK_STRUCTURE_TYPE_WRITE_DESCRIPTOR_SET_ACCELERATION_STRUCTURE_KHR = 1000150007,
  // Provided by VK_KHR_acceleration_structure
    VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_BUILD_GEOMETRY_INFO_KHR = 1000150000,
  // Provided by VK_KHR_acceleration_structure
    VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_DEVICE_ADDRESS_INFO_KHR = 1000150002,
  // Provided by VK_KHR_acceleration_structure
    VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_AABBS_DATA_KHR = 1000150003,
  // Provided by VK_KHR_acceleration_structure
    VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_INSTANCES_DATA_KHR = 1000150004,
  // Provided by VK_KHR_acceleration_structure
    VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_TRIANGLES_DATA_KHR = 1000150005,
  // Provided by VK_KHR_acceleration_structure
    VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_KHR = 1000150006,
  // Provided by VK_KHR_acceleration_structure
    VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_VERSION_INFO_KHR = 1000150009,
```

```
  // Provided by VK_KHR_acceleration_structure
    VK_STRUCTURE_TYPE_COPY_ACCELERATION_STRUCTURE_INFO_KHR = 1000150010,
  // Provided by VK_KHR_acceleration_structure
    VK_STRUCTURE_TYPE_COPY_ACCELERATION_STRUCTURE_TO_MEMORY_INFO_KHR = 1000150011,
  // Provided by VK_KHR_acceleration_structure
    VK_STRUCTURE_TYPE_COPY_MEMORY_TO_ACCELERATION_STRUCTURE_INFO_KHR = 1000150012,
  // Provided by VK_KHR_acceleration_structure
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ACCELERATION_STRUCTURE_FEATURES_KHR =
1000150013,
  // Provided by VK_KHR_acceleration_structure
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ACCELERATION_STRUCTURE_PROPERTIES_KHR =
1000150014,
  // Provided by VK_KHR_acceleration_structure
    VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CREATE_INFO_KHR = 1000150017,
  // Provided by VK_KHR_acceleration_structure
    VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_BUILD_SIZES_INFO_KHR = 1000150020,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_PIPELINE_FEATURES_KHR = 1000347000,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_PIPELINE_PROPERTIES_KHR =
1000347001,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_STRUCTURE_TYPE_RAY_TRACING_PIPELINE_CREATE_INFO_KHR = 1000150015,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_STRUCTURE_TYPE_RAY_TRACING_SHADER_GROUP_CREATE_INFO_KHR = 1000150016,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_STRUCTURE_TYPE_RAY_TRACING_PIPELINE_INTERFACE_CREATE_INFO_KHR = 1000150018,
  // Provided by VK_KHR_ray_query
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_QUERY_FEATURES_KHR = 1000348013,
  // Provided by VK_EXT_image_drm_format_modifier
    VK_STRUCTURE_TYPE_DRM_FORMAT_MODIFIER_PROPERTIES_LIST_EXT = 1000158000,
  // Provided by VK_EXT_image_drm_format_modifier
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_DRM_FORMAT_MODIFIER_INFO_EXT = 1000158002,
  // Provided by VK_EXT_image_drm_format_modifier
    VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_LIST_CREATE_INFO_EXT = 1000158003,
  // Provided by VK_EXT_image_drm_format_modifier
    VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_EXPLICIT_CREATE_INFO_EXT = 1000158004,
  // Provided by VK_EXT_image_drm_format_modifier
    VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_PROPERTIES_EXT = 1000158005,
  // Provided by VK_EXT_image_drm_format_modifier with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
    VK_STRUCTURE_TYPE_DRM_FORMAT_MODIFIER_PROPERTIES_LIST_2_EXT = 1000158006,
  // Provided by VK_EXT_validation_cache
    VK_STRUCTURE_TYPE_VALIDATION_CACHE_CREATE_INFO_EXT = 1000160000,
  // Provided by VK_EXT_validation_cache
    VK_STRUCTURE_TYPE_SHADER_MODULE_VALIDATION_CACHE_CREATE_INFO_EXT = 1000160001,
#ifdef VK_ENABLE_BETA_EXTENSIONS
  // Provided by VK_KHR_portability_subset
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PORTABILITY_SUBSET_FEATURES_KHR = 1000163000,
#endif
#ifdef VK_ENABLE_BETA_EXTENSIONS
```

```
  // Provided by VK_KHR_portability_subset
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PORTABILITY_SUBSET_PROPERTIES_KHR = 1000163001,
#endif
  // Provided by VK_EXT_filter_cubic
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_VIEW_IMAGE_FORMAT_INFO_EXT = 1000170000,
  // Provided by VK_EXT_filter_cubic
    VK_STRUCTURE_TYPE_FILTER_CUBIC_IMAGE_VIEW_IMAGE_FORMAT_PROPERTIES_EXT =
1000170001,
  // Provided by VK_EXT_external_memory_host
    VK_STRUCTURE_TYPE_IMPORT_MEMORY_HOST_POINTER_INFO_EXT = 1000178000,
  // Provided by VK_EXT_external_memory_host
    VK_STRUCTURE_TYPE_MEMORY_HOST_POINTER_PROPERTIES_EXT = 1000178001,
  // Provided by VK_EXT_external_memory_host
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_MEMORY_HOST_PROPERTIES_EXT =
1000178002,
  // Provided by VK_KHR_shader_clock
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_CLOCK_FEATURES_KHR = 1000181000,
  // Provided by VK_KHR_video_decode_h265
    VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_CAPABILITIES_KHR = 1000187000,
  // Provided by VK_KHR_video_decode_h265
    VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_SESSION_PARAMETERS_CREATE_INFO_KHR =
1000187001,
  // Provided by VK_KHR_video_decode_h265
    VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_SESSION_PARAMETERS_ADD_INFO_KHR = 1000187002,
  // Provided by VK_KHR_video_decode_h265
    VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_PROFILE_INFO_KHR = 1000187003,
  // Provided by VK_KHR_video_decode_h265
    VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_PICTURE_INFO_KHR = 1000187004,
  // Provided by VK_KHR_video_decode_h265
    VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_DPB_SLOT_INFO_KHR = 1000187005,
  // Provided by VK_EXT_present_timing
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_TIMING_FEATURES_EXT = 1000208000,
  // Provided by VK_EXT_present_timing
    VK_STRUCTURE_TYPE_SWAPCHAIN_TIMING_PROPERTIES_EXT = 1000208001,
  // Provided by VK_EXT_present_timing
    VK_STRUCTURE_TYPE_SWAPCHAIN_TIME_DOMAIN_PROPERTIES_EXT = 1000208002,
  // Provided by VK_EXT_present_timing
    VK_STRUCTURE_TYPE_PRESENT_TIMINGS_INFO_EXT = 1000208003,
  // Provided by VK_EXT_present_timing
    VK_STRUCTURE_TYPE_PRESENT_TIMING_INFO_EXT = 1000208004,
  // Provided by VK_EXT_present_timing
    VK_STRUCTURE_TYPE_PAST_PRESENTATION_TIMING_INFO_EXT = 1000208005,
  // Provided by VK_EXT_present_timing
    VK_STRUCTURE_TYPE_PAST_PRESENTATION_TIMING_PROPERTIES_EXT = 1000208006,
  // Provided by VK_EXT_present_timing
    VK_STRUCTURE_TYPE_PAST_PRESENTATION_TIMING_EXT = 1000208007,
  // Provided by VK_EXT_present_timing
    VK_STRUCTURE_TYPE_PRESENT_TIMING_SURFACE_CAPABILITIES_EXT = 1000208008,
  // Provided by VK_EXT_present_timing
    VK_STRUCTURE_TYPE_SWAPCHAIN_CALIBRATED_TIMESTAMP_INFO_EXT = 1000208009,
  // Provided by VK_EXT_pci_bus_info
```

```
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PCI_BUS_INFO_PROPERTIES_EXT = 1000212000,
  // Provided by VK_EXT_metal_surface
    VK_STRUCTURE_TYPE_METAL_SURFACE_CREATE_INFO_EXT = 1000217000,
  // Provided by VK_EXT_fragment_density_map
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_FEATURES_EXT = 1000218000,
  // Provided by VK_EXT_fragment_density_map
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_PROPERTIES_EXT =
1000218001,
  // Provided by VK_EXT_fragment_density_map
    VK_STRUCTURE_TYPE_RENDER_PASS_FRAGMENT_DENSITY_MAP_CREATE_INFO_EXT = 1000218002,
  // Provided by VK_EXT_fragment_density_map with VK_VERSION_1_3 or
VK_KHR_dynamic_rendering
    VK_STRUCTURE_TYPE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_INFO_EXT = 1000044007,
  // Provided by VK_KHR_fragment_shading_rate
    VK_STRUCTURE_TYPE_FRAGMENT_SHADING_RATE_ATTACHMENT_INFO_KHR = 1000226000,
  // Provided by VK_KHR_fragment_shading_rate
    VK_STRUCTURE_TYPE_PIPELINE_FRAGMENT_SHADING_RATE_STATE_CREATE_INFO_KHR =
1000226001,
  // Provided by VK_KHR_fragment_shading_rate
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_PROPERTIES_KHR =
1000226002,
  // Provided by VK_KHR_fragment_shading_rate
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_FEATURES_KHR = 1000226003,
  // Provided by VK_KHR_fragment_shading_rate
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_KHR = 1000226004,
  // Provided by VK_KHR_fragment_shading_rate with VK_VERSION_1_3 or
VK_KHR_dynamic_rendering
    VK_STRUCTURE_TYPE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_INFO_KHR =
1000044006,
  // Provided by VK_KHR_shader_constant_data
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_CONSTANT_DATA_FEATURES_KHR = 1000231000,
  // Provided by VK_KHR_shader_abort
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ABORT_FEATURES_KHR = 1000233000,
  // Provided by VK_KHR_shader_abort
    VK_STRUCTURE_TYPE_DEVICE_FAULT_SHADER_ABORT_MESSAGE_INFO_KHR = 1000233001,
  // Provided by VK_KHR_shader_abort
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ABORT_PROPERTIES_KHR = 1000233002,
  // Provided by VK_EXT_shader_image_atomic_int64
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_IMAGE_ATOMIC_INT64_FEATURES_EXT =
1000234000,
  // Provided by VK_KHR_shader_quad_control
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_QUAD_CONTROL_FEATURES_KHR = 1000235000,
  // Provided by VK_EXT_memory_budget
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_BUDGET_PROPERTIES_EXT = 1000237000,
  // Provided by VK_EXT_memory_priority
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PRIORITY_FEATURES_EXT = 1000238000,
  // Provided by VK_EXT_memory_priority
    VK_STRUCTURE_TYPE_MEMORY_PRIORITY_ALLOCATE_INFO_EXT = 1000238001,
  // Provided by VK_KHR_surface_protected_capabilities
    VK_STRUCTURE_TYPE_SURFACE_PROTECTED_CAPABILITIES_KHR = 1000239000,
  // Provided by VK_EXT_validation_features
```

```
    VK_STRUCTURE_TYPE_VALIDATION_FEATURES_EXT = 1000247000,
  // Provided by VK_KHR_present_wait
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_WAIT_FEATURES_KHR = 1000248000,
  // Provided by VK_EXT_fragment_shader_interlock
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_INTERLOCK_FEATURES_EXT =
1000251000,
  // Provided by VK_EXT_ycbcr_image_arrays
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_YCBCR_IMAGE_ARRAYS_FEATURES_EXT = 1000252000,
  // Provided by VK_EXT_provoking_vertex
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROVOKING_VERTEX_FEATURES_EXT = 1000254000,
  // Provided by VK_EXT_provoking_vertex
    VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_PROVOKING_VERTEX_STATE_CREATE_INFO_EXT =
1000254001,
  // Provided by VK_EXT_provoking_vertex
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROVOKING_VERTEX_PROPERTIES_EXT = 1000254002,
  // Provided by VK_EXT_full_screen_exclusive
    VK_STRUCTURE_TYPE_SURFACE_FULL_SCREEN_EXCLUSIVE_INFO_EXT = 1000255000,
  // Provided by VK_EXT_full_screen_exclusive
    VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_FULL_SCREEN_EXCLUSIVE_EXT = 1000255002,
  // Provided by VK_KHR_win32_surface with VK_EXT_full_screen_exclusive
    VK_STRUCTURE_TYPE_SURFACE_FULL_SCREEN_EXCLUSIVE_WIN32_INFO_EXT = 1000255001,
  // Provided by VK_EXT_headless_surface
    VK_STRUCTURE_TYPE_HEADLESS_SURFACE_CREATE_INFO_EXT = 1000256000,
  // Provided by VK_EXT_shader_atomic_float
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_FLOAT_FEATURES_EXT = 1000260000,
  // Provided by VK_EXT_extended_dynamic_state
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_FEATURES_EXT =
1000267000,
  // Provided by VK_KHR_pipeline_executable_properties
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_EXECUTABLE_PROPERTIES_FEATURES_KHR =
1000269000,
  // Provided by VK_KHR_pipeline_executable_properties
    VK_STRUCTURE_TYPE_PIPELINE_INFO_KHR = 1000269001,
  // Provided by VK_KHR_pipeline_executable_properties
    VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_PROPERTIES_KHR = 1000269002,
  // Provided by VK_KHR_pipeline_executable_properties
    VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_INFO_KHR = 1000269003,
  // Provided by VK_KHR_pipeline_executable_properties
    VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_STATISTIC_KHR = 1000269004,
  // Provided by VK_KHR_pipeline_executable_properties
    VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_INTERNAL_REPRESENTATION_KHR = 1000269005,
  // Provided by VK_EXT_map_memory_placed
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAP_MEMORY_PLACED_FEATURES_EXT = 1000272000,
  // Provided by VK_EXT_map_memory_placed
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAP_MEMORY_PLACED_PROPERTIES_EXT = 1000272001,
  // Provided by VK_EXT_map_memory_placed
    VK_STRUCTURE_TYPE_MEMORY_MAP_PLACED_INFO_EXT = 1000272002,
  // Provided by VK_EXT_shader_atomic_float2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_FLOAT_2_FEATURES_EXT = 1000273000,
  // Provided by VK_EXT_depth_bias_control
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_BIAS_CONTROL_FEATURES_EXT = 1000283000,
```

```
  // Provided by VK_EXT_depth_bias_control
    VK_STRUCTURE_TYPE_DEPTH_BIAS_INFO_EXT = 1000283001,
  // Provided by VK_EXT_depth_bias_control
    VK_STRUCTURE_TYPE_DEPTH_BIAS_REPRESENTATION_INFO_EXT = 1000283002,
  // Provided by VK_EXT_device_memory_report
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_MEMORY_REPORT_FEATURES_EXT = 1000284000,
  // Provided by VK_EXT_device_memory_report
    VK_STRUCTURE_TYPE_DEVICE_DEVICE_MEMORY_REPORT_CREATE_INFO_EXT = 1000284001,
  // Provided by VK_EXT_device_memory_report
    VK_STRUCTURE_TYPE_DEVICE_MEMORY_REPORT_CALLBACK_DATA_EXT = 1000284002,
  // Provided by VK_EXT_custom_border_color
    VK_STRUCTURE_TYPE_SAMPLER_CUSTOM_BORDER_COLOR_CREATE_INFO_EXT = 1000287000,
  // Provided by VK_EXT_custom_border_color
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CUSTOM_BORDER_COLOR_PROPERTIES_EXT = 1000287001,
  // Provided by VK_EXT_custom_border_color
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CUSTOM_BORDER_COLOR_FEATURES_EXT = 1000287002,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TEXTURE_COMPRESSION_ASTC_3D_FEATURES_EXT =
1000288000,
  // Provided by VK_KHR_pipeline_library
    VK_STRUCTURE_TYPE_PIPELINE_LIBRARY_CREATE_INFO_KHR = 1000290000,
  // Provided by VK_KHR_present_id
    VK_STRUCTURE_TYPE_PRESENT_ID_KHR = 1000294000,
  // Provided by VK_KHR_present_id
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_ID_FEATURES_KHR = 1000294001,
  // Provided by VK_KHR_video_encode_queue
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_INFO_KHR = 1000299000,
  // Provided by VK_KHR_video_encode_queue
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_RATE_CONTROL_INFO_KHR = 1000299001,
  // Provided by VK_KHR_video_encode_queue
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_RATE_CONTROL_LAYER_INFO_KHR = 1000299002,
  // Provided by VK_KHR_video_encode_queue
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_CAPABILITIES_KHR = 1000299003,
  // Provided by VK_KHR_video_encode_queue
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_USAGE_INFO_KHR = 1000299004,
  // Provided by VK_KHR_video_encode_queue
    VK_STRUCTURE_TYPE_QUERY_POOL_VIDEO_ENCODE_FEEDBACK_CREATE_INFO_KHR = 1000299005,
  // Provided by VK_KHR_video_encode_queue
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_QUALITY_LEVEL_INFO_KHR =
1000299006,
  // Provided by VK_KHR_video_encode_queue
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUALITY_LEVEL_PROPERTIES_KHR = 1000299007,
  // Provided by VK_KHR_video_encode_queue
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUALITY_LEVEL_INFO_KHR = 1000299008,
  // Provided by VK_KHR_video_encode_queue
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_SESSION_PARAMETERS_GET_INFO_KHR = 1000299009,
  // Provided by VK_KHR_video_encode_queue
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_SESSION_PARAMETERS_FEEDBACK_INFO_KHR = 1000299010,
  // Provided by VK_EXT_shader_split_barrier
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SPLIT_BARRIER_FEATURES_EXT = 1000305000,
  // Provided by VK_EXT_shader_split_barrier
```

```
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SPLIT_BARRIER_PROPERTIES_EXT =
1000305001,
  // Provided by VK_EXT_metal_objects
    VK_STRUCTURE_TYPE_EXPORT_METAL_OBJECT_CREATE_INFO_EXT = 1000311000,
  // Provided by VK_EXT_metal_objects
    VK_STRUCTURE_TYPE_EXPORT_METAL_OBJECTS_INFO_EXT = 1000311001,
  // Provided by VK_EXT_metal_objects
    VK_STRUCTURE_TYPE_EXPORT_METAL_DEVICE_INFO_EXT = 1000311002,
  // Provided by VK_EXT_metal_objects
    VK_STRUCTURE_TYPE_EXPORT_METAL_COMMAND_QUEUE_INFO_EXT = 1000311003,
  // Provided by VK_EXT_metal_objects
    VK_STRUCTURE_TYPE_EXPORT_METAL_BUFFER_INFO_EXT = 1000311004,
  // Provided by VK_EXT_metal_objects
    VK_STRUCTURE_TYPE_IMPORT_METAL_BUFFER_INFO_EXT = 1000311005,
  // Provided by VK_EXT_metal_objects
    VK_STRUCTURE_TYPE_EXPORT_METAL_TEXTURE_INFO_EXT = 1000311006,
  // Provided by VK_EXT_metal_objects
    VK_STRUCTURE_TYPE_IMPORT_METAL_TEXTURE_INFO_EXT = 1000311007,
  // Provided by VK_EXT_metal_objects
    VK_STRUCTURE_TYPE_EXPORT_METAL_IO_SURFACE_INFO_EXT = 1000311008,
  // Provided by VK_EXT_metal_objects
    VK_STRUCTURE_TYPE_IMPORT_METAL_IO_SURFACE_INFO_EXT = 1000311009,
  // Provided by VK_EXT_metal_objects
    VK_STRUCTURE_TYPE_EXPORT_METAL_SHARED_EVENT_INFO_EXT = 1000311010,
  // Provided by VK_EXT_metal_objects
    VK_STRUCTURE_TYPE_IMPORT_METAL_SHARED_EVENT_INFO_EXT = 1000311011,
  // Provided by VK_EXT_descriptor_buffer
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_BUFFER_PROPERTIES_EXT = 1000316000,
  // Provided by VK_EXT_descriptor_buffer
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_BUFFER_DENSITY_MAP_PROPERTIES_EXT =
1000316001,
  // Provided by VK_EXT_descriptor_buffer
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_BUFFER_FEATURES_EXT = 1000316002,
  // Provided by VK_EXT_descriptor_buffer
    VK_STRUCTURE_TYPE_DESCRIPTOR_ADDRESS_INFO_EXT = 1000316003,
  // Provided by VK_EXT_descriptor_buffer
    VK_STRUCTURE_TYPE_DESCRIPTOR_GET_INFO_EXT = 1000316004,
  // Provided by VK_EXT_descriptor_buffer
    VK_STRUCTURE_TYPE_BUFFER_CAPTURE_DESCRIPTOR_DATA_INFO_EXT = 1000316005,
  // Provided by VK_EXT_descriptor_buffer
    VK_STRUCTURE_TYPE_IMAGE_CAPTURE_DESCRIPTOR_DATA_INFO_EXT = 1000316006,
  // Provided by VK_EXT_descriptor_buffer
    VK_STRUCTURE_TYPE_IMAGE_VIEW_CAPTURE_DESCRIPTOR_DATA_INFO_EXT = 1000316007,
  // Provided by VK_EXT_descriptor_buffer
    VK_STRUCTURE_TYPE_SAMPLER_CAPTURE_DESCRIPTOR_DATA_INFO_EXT = 1000316008,
  // Provided by VK_EXT_descriptor_buffer
    VK_STRUCTURE_TYPE_OPAQUE_CAPTURE_DESCRIPTOR_DATA_CREATE_INFO_EXT = 1000316010,
  // Provided by VK_EXT_descriptor_buffer
    VK_STRUCTURE_TYPE_DESCRIPTOR_BUFFER_BINDING_INFO_EXT = 1000316011,
  // Provided by VK_EXT_descriptor_buffer
    VK_STRUCTURE_TYPE_DESCRIPTOR_BUFFER_BINDING_PUSH_DESCRIPTOR_BUFFER_HANDLE_EXT =
```

```
1000316012,
  // Provided by VK_EXT_descriptor_buffer with VK_KHR_acceleration_structure or
VK_NV_ray_tracing
    VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CAPTURE_DESCRIPTOR_DATA_INFO_EXT =
1000316009,
  // Provided by VK_KHR_device_address_commands
    VK_STRUCTURE_TYPE_DEVICE_MEMORY_COPY_KHR = 1000318000,
  // Provided by VK_KHR_device_address_commands
    VK_STRUCTURE_TYPE_COPY_DEVICE_MEMORY_INFO_KHR = 1000318001,
  // Provided by VK_KHR_device_address_commands
    VK_STRUCTURE_TYPE_DEVICE_MEMORY_IMAGE_COPY_KHR = 1000318002,
  // Provided by VK_KHR_device_address_commands
    VK_STRUCTURE_TYPE_COPY_DEVICE_MEMORY_IMAGE_INFO_KHR = 1000318003,
  // Provided by VK_KHR_device_address_commands
    VK_STRUCTURE_TYPE_MEMORY_RANGE_BARRIERS_INFO_KHR = 1000318004,
  // Provided by VK_KHR_device_address_commands
    VK_STRUCTURE_TYPE_MEMORY_RANGE_BARRIER_KHR = 1000318005,
  // Provided by VK_KHR_device_address_commands
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_ADDRESS_COMMANDS_FEATURES_KHR =
1000318006,
  // Provided by VK_KHR_device_address_commands
    VK_STRUCTURE_TYPE_BIND_INDEX_BUFFER_3_INFO_KHR = 1000318007,
  // Provided by VK_KHR_device_address_commands
    VK_STRUCTURE_TYPE_BIND_VERTEX_BUFFER_3_INFO_KHR = 1000318008,
  // Provided by VK_KHR_device_address_commands
    VK_STRUCTURE_TYPE_DRAW_INDIRECT_2_INFO_KHR = 1000318009,
  // Provided by VK_KHR_device_address_commands
    VK_STRUCTURE_TYPE_DRAW_INDIRECT_COUNT_2_INFO_KHR = 1000318010,
  // Provided by VK_KHR_device_address_commands
    VK_STRUCTURE_TYPE_DISPATCH_INDIRECT_2_INFO_KHR = 1000318011,
  // Provided by VK_KHR_device_address_commands with VK_EXT_conditional_rendering
    VK_STRUCTURE_TYPE_CONDITIONAL_RENDERING_BEGIN_INFO_2_EXT = 1000318012,
  // Provided by VK_KHR_device_address_commands with VK_EXT_transform_feedback
    VK_STRUCTURE_TYPE_BIND_TRANSFORM_FEEDBACK_BUFFER_2_INFO_EXT = 1000318013,
  // Provided by VK_KHR_acceleration_structure with VK_KHR_device_address_commands
    VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CREATE_INFO_2_KHR = 1000318015,
  // Provided by VK_EXT_graphics_pipeline_library
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GRAPHICS_PIPELINE_LIBRARY_FEATURES_EXT =
1000320000,
  // Provided by VK_EXT_graphics_pipeline_library
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GRAPHICS_PIPELINE_LIBRARY_PROPERTIES_EXT =
1000320001,
  // Provided by VK_EXT_graphics_pipeline_library
    VK_STRUCTURE_TYPE_GRAPHICS_PIPELINE_LIBRARY_CREATE_INFO_EXT = 1000320002,
  // Provided by VK_KHR_fragment_shader_barycentric
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_BARYCENTRIC_FEATURES_KHR =
1000203000,
  // Provided by VK_KHR_fragment_shader_barycentric with VK_EXT_provoking_vertex
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_BARYCENTRIC_PROPERTIES_KHR =
1000322000,
  // Provided by VK_KHR_shader_subgroup_uniform_control_flow
```

```
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_UNIFORM_CONTROL_FLOW_FEATURES_KHR =
1000323000,
  // Provided by VK_EXT_mesh_shader
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MESH_SHADER_FEATURES_EXT = 1000328000,
  // Provided by VK_EXT_mesh_shader
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MESH_SHADER_PROPERTIES_EXT = 1000328001,
  // Provided by VK_EXT_fragment_density_map2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_2_FEATURES_EXT =
1000332000,
  // Provided by VK_EXT_fragment_density_map2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_2_PROPERTIES_EXT =
1000332001,
  // Provided by VK_KHR_workgroup_memory_explicit_layout
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_WORKGROUP_MEMORY_EXPLICIT_LAYOUT_FEATURES_KHR =
1000336000,
  // Provided by VK_EXT_image_compression_control
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_COMPRESSION_CONTROL_FEATURES_EXT =
1000338000,
  // Provided by VK_EXT_image_compression_control
    VK_STRUCTURE_TYPE_IMAGE_COMPRESSION_CONTROL_EXT = 1000338001,
  // Provided by VK_EXT_image_compression_control
    VK_STRUCTURE_TYPE_IMAGE_COMPRESSION_PROPERTIES_EXT = 1000338004,
  // Provided by VK_EXT_attachment_feedback_loop_layout
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ATTACHMENT_FEEDBACK_LOOP_LAYOUT_FEATURES_EXT =
1000339000,
  // Provided by VK_EXT_device_fault
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FAULT_FEATURES_EXT = 1000341000,
  // Provided by VK_EXT_device_fault
    VK_STRUCTURE_TYPE_DEVICE_FAULT_COUNTS_EXT = 1000341001,
  // Provided by VK_EXT_device_fault
    VK_STRUCTURE_TYPE_DEVICE_FAULT_INFO_EXT = 1000341002,
  // Provided by VK_EXT_rgba10x6_formats
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RGBA10X6_FORMATS_FEATURES_EXT = 1000344000,
  // Provided by VK_EXT_directfb_surface
    VK_STRUCTURE_TYPE_DIRECTFB_SURFACE_CREATE_INFO_EXT = 1000346000,
  // Provided by VK_EXT_vertex_input_dynamic_state
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_INPUT_DYNAMIC_STATE_FEATURES_EXT =
1000352000,
  // Provided by VK_EXT_shader_object, VK_EXT_vertex_input_dynamic_state
    VK_STRUCTURE_TYPE_VERTEX_INPUT_BINDING_DESCRIPTION_2_EXT = 1000352001,
  // Provided by VK_EXT_shader_object, VK_EXT_vertex_input_dynamic_state
    VK_STRUCTURE_TYPE_VERTEX_INPUT_ATTRIBUTE_DESCRIPTION_2_EXT = 1000352002,
  // Provided by VK_EXT_physical_device_drm
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRM_PROPERTIES_EXT = 1000353000,
  // Provided by VK_EXT_device_address_binding_report
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ADDRESS_BINDING_REPORT_FEATURES_EXT =
1000354000,
  // Provided by VK_EXT_device_address_binding_report
    VK_STRUCTURE_TYPE_DEVICE_ADDRESS_BINDING_CALLBACK_DATA_EXT = 1000354001,
  // Provided by VK_EXT_depth_clip_control
```

```
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLIP_CONTROL_FEATURES_EXT = 1000355000,
  // Provided by VK_EXT_depth_clip_control
    VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_DEPTH_CLIP_CONTROL_CREATE_INFO_EXT =
1000355001,
  // Provided by VK_EXT_primitive_topology_list_restart
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIMITIVE_TOPOLOGY_LIST_RESTART_FEATURES_EXT =
1000356000,
  // Provided by VK_EXT_pipeline_properties
    VK_STRUCTURE_TYPE_PIPELINE_PROPERTIES_IDENTIFIER_EXT = 1000372000,
  // Provided by VK_EXT_pipeline_properties
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_PROPERTIES_FEATURES_EXT = 1000372001,
  // Provided by VK_EXT_frame_boundary
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAME_BOUNDARY_FEATURES_EXT = 1000375000,
  // Provided by VK_EXT_frame_boundary
    VK_STRUCTURE_TYPE_FRAME_BOUNDARY_EXT = 1000375001,
  // Provided by VK_EXT_multisampled_render_to_single_sampled

VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_FEATURES_EXT =
1000376000,
  // Provided by VK_EXT_multisampled_render_to_single_sampled
    VK_STRUCTURE_TYPE_SUBPASS_RESOLVE_PERFORMANCE_QUERY_EXT = 1000376001,
  // Provided by VK_EXT_multisampled_render_to_single_sampled
    VK_STRUCTURE_TYPE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_INFO_EXT = 1000376002,
  // Provided by VK_EXT_extended_dynamic_state2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_2_FEATURES_EXT =
1000377000,
  // Provided by VK_EXT_color_write_enable
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COLOR_WRITE_ENABLE_FEATURES_EXT = 1000381000,
  // Provided by VK_EXT_color_write_enable
    VK_STRUCTURE_TYPE_PIPELINE_COLOR_WRITE_CREATE_INFO_EXT = 1000381001,
  // Provided by VK_EXT_primitives_generated_query
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIMITIVES_GENERATED_QUERY_FEATURES_EXT =
1000382000,
  // Provided by VK_KHR_ray_tracing_maintenance1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_MAINTENANCE_1_FEATURES_KHR =
1000386000,
  // Provided by VK_KHR_shader_untyped_pointers
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_UNTYPED_POINTERS_FEATURES_KHR =
1000387000,
  // Provided by VK_EXT_image_view_min_lod
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_VIEW_MIN_LOD_FEATURES_EXT = 1000391000,
  // Provided by VK_EXT_image_view_min_lod
    VK_STRUCTURE_TYPE_IMAGE_VIEW_MIN_LOD_CREATE_INFO_EXT = 1000391001,
  // Provided by VK_EXT_multi_draw
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTI_DRAW_FEATURES_EXT = 1000392000,
  // Provided by VK_EXT_multi_draw
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTI_DRAW_PROPERTIES_EXT = 1000392001,
  // Provided by VK_EXT_image_2d_view_of_3d
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_2D_VIEW_OF_3D_FEATURES_EXT = 1000393000,
  // Provided by VK_EXT_shader_tile_image
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TILE_IMAGE_FEATURES_EXT = 1000395000,
```

```
  // Provided by VK_EXT_shader_tile_image
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TILE_IMAGE_PROPERTIES_EXT = 1000395001,
  // Provided by VK_EXT_opacity_micromap
    VK_STRUCTURE_TYPE_MICROMAP_BUILD_INFO_EXT = 1000396000,
  // Provided by VK_EXT_opacity_micromap
    VK_STRUCTURE_TYPE_MICROMAP_VERSION_INFO_EXT = 1000396001,
  // Provided by VK_EXT_opacity_micromap
    VK_STRUCTURE_TYPE_COPY_MICROMAP_INFO_EXT = 1000396002,
  // Provided by VK_EXT_opacity_micromap
    VK_STRUCTURE_TYPE_COPY_MICROMAP_TO_MEMORY_INFO_EXT = 1000396003,
  // Provided by VK_EXT_opacity_micromap
    VK_STRUCTURE_TYPE_COPY_MEMORY_TO_MICROMAP_INFO_EXT = 1000396004,
  // Provided by VK_EXT_opacity_micromap
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_FEATURES_EXT = 1000396005,
  // Provided by VK_EXT_opacity_micromap
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_PROPERTIES_EXT = 1000396006,
  // Provided by VK_EXT_opacity_micromap
    VK_STRUCTURE_TYPE_MICROMAP_CREATE_INFO_EXT = 1000396007,
  // Provided by VK_EXT_opacity_micromap
    VK_STRUCTURE_TYPE_MICROMAP_BUILD_SIZES_INFO_EXT = 1000396008,
  // Provided by VK_EXT_opacity_micromap
    VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_TRIANGLES_OPACITY_MICROMAP_EXT =
1000396009,
  // Provided by VK_EXT_border_color_swizzle
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BORDER_COLOR_SWIZZLE_FEATURES_EXT = 1000411000,
  // Provided by VK_EXT_border_color_swizzle
    VK_STRUCTURE_TYPE_SAMPLER_BORDER_COLOR_COMPONENT_MAPPING_CREATE_INFO_EXT =
1000411001,
  // Provided by VK_EXT_pageable_device_local_memory
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PAGEABLE_DEVICE_LOCAL_MEMORY_FEATURES_EXT =
1000412000,
  // Provided by VK_EXT_image_sliced_view_of_3d
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_SLICED_VIEW_OF_3D_FEATURES_EXT =
1000418000,
  // Provided by VK_EXT_image_sliced_view_of_3d
    VK_STRUCTURE_TYPE_IMAGE_VIEW_SLICED_CREATE_INFO_EXT = 1000418001,
  // Provided by VK_EXT_non_seamless_cube_map
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NON_SEAMLESS_CUBE_MAP_FEATURES_EXT = 1000422000,
  // Provided by VK_KHR_shader_maximal_reconvergence
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_MAXIMAL_RECONVERGENCE_FEATURES_KHR =
1000434000,
  // Provided by VK_EXT_image_compression_control_swapchain
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_COMPRESSION_CONTROL_SWAPCHAIN_FEATURES_EXT
= 1000437000,
  // Provided by VK_EXT_nested_command_buffer
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NESTED_COMMAND_BUFFER_FEATURES_EXT = 1000451000,
  // Provided by VK_EXT_nested_command_buffer
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NESTED_COMMAND_BUFFER_PROPERTIES_EXT =
1000451001,
  // Provided by VK_EXT_external_memory_acquire_unmodified
    VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_ACQUIRE_UNMODIFIED_EXT = 1000453000,
```

```
  // Provided by VK_EXT_extended_dynamic_state3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_3_FEATURES_EXT =
1000455000,
  // Provided by VK_EXT_extended_dynamic_state3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_3_PROPERTIES_EXT =
1000455001,
  // Provided by VK_EXT_subpass_merge_feedback
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SUBPASS_MERGE_FEEDBACK_FEATURES_EXT =
1000458000,
  // Provided by VK_EXT_subpass_merge_feedback
    VK_STRUCTURE_TYPE_RENDER_PASS_CREATION_CONTROL_EXT = 1000458001,
  // Provided by VK_EXT_subpass_merge_feedback
    VK_STRUCTURE_TYPE_RENDER_PASS_CREATION_FEEDBACK_CREATE_INFO_EXT = 1000458002,
  // Provided by VK_EXT_subpass_merge_feedback
    VK_STRUCTURE_TYPE_RENDER_PASS_SUBPASS_FEEDBACK_CREATE_INFO_EXT = 1000458003,
  // Provided by VK_EXT_shader_module_identifier
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_MODULE_IDENTIFIER_FEATURES_EXT =
1000462000,
  // Provided by VK_EXT_shader_module_identifier
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_MODULE_IDENTIFIER_PROPERTIES_EXT =
1000462001,
  // Provided by VK_EXT_shader_module_identifier
    VK_STRUCTURE_TYPE_PIPELINE_SHADER_STAGE_MODULE_IDENTIFIER_CREATE_INFO_EXT =
1000462002,
  // Provided by VK_EXT_shader_module_identifier
    VK_STRUCTURE_TYPE_SHADER_MODULE_IDENTIFIER_EXT = 1000462003,
  // Provided by VK_EXT_rasterization_order_attachment_access

VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RASTERIZATION_ORDER_ATTACHMENT_ACCESS_FEATURES_EXT =
1000342000,
  // Provided by VK_EXT_legacy_dithering
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LEGACY_DITHERING_FEATURES_EXT = 1000465000,
  // Provided by VK_KHR_present_id2
    VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_PRESENT_ID_2_KHR = 1000479000,
  // Provided by VK_KHR_present_id2
    VK_STRUCTURE_TYPE_PRESENT_ID_2_KHR = 1000479001,
  // Provided by VK_KHR_present_id2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_ID_2_FEATURES_KHR = 1000479002,
  // Provided by VK_KHR_present_wait2
    VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_PRESENT_WAIT_2_KHR = 1000480000,
  // Provided by VK_KHR_present_wait2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_WAIT_2_FEATURES_KHR = 1000480001,
  // Provided by VK_KHR_present_wait2
    VK_STRUCTURE_TYPE_PRESENT_WAIT_2_INFO_KHR = 1000480002,
  // Provided by VK_KHR_ray_tracing_position_fetch
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_POSITION_FETCH_FEATURES_KHR =
1000481000,
  // Provided by VK_EXT_shader_object
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_OBJECT_FEATURES_EXT = 1000482000,
  // Provided by VK_EXT_shader_object
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_OBJECT_PROPERTIES_EXT = 1000482001,
```

```
  // Provided by VK_EXT_shader_object
    VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT = 1000482002,
  // Provided by VK_KHR_pipeline_binary
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_BINARY_FEATURES_KHR = 1000483000,
  // Provided by VK_KHR_pipeline_binary
    VK_STRUCTURE_TYPE_PIPELINE_BINARY_CREATE_INFO_KHR = 1000483001,
  // Provided by VK_KHR_pipeline_binary
    VK_STRUCTURE_TYPE_PIPELINE_BINARY_INFO_KHR = 1000483002,
  // Provided by VK_KHR_pipeline_binary
    VK_STRUCTURE_TYPE_PIPELINE_BINARY_KEY_KHR = 1000483003,
  // Provided by VK_KHR_pipeline_binary
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_BINARY_PROPERTIES_KHR = 1000483004,
  // Provided by VK_KHR_pipeline_binary
    VK_STRUCTURE_TYPE_RELEASE_CAPTURED_PIPELINE_DATA_INFO_KHR = 1000483005,
  // Provided by VK_KHR_pipeline_binary
    VK_STRUCTURE_TYPE_PIPELINE_BINARY_DATA_INFO_KHR = 1000483006,
  // Provided by VK_KHR_pipeline_binary
    VK_STRUCTURE_TYPE_PIPELINE_CREATE_INFO_KHR = 1000483007,
  // Provided by VK_KHR_pipeline_binary
    VK_STRUCTURE_TYPE_DEVICE_PIPELINE_BINARY_INTERNAL_CACHE_CONTROL_KHR = 1000483008,
  // Provided by VK_KHR_pipeline_binary
    VK_STRUCTURE_TYPE_PIPELINE_BINARY_HANDLES_INFO_KHR = 1000483009,
  // Provided by VK_KHR_surface_maintenance1
    VK_STRUCTURE_TYPE_SURFACE_PRESENT_MODE_KHR = 1000274000,
  // Provided by VK_KHR_surface_maintenance1
    VK_STRUCTURE_TYPE_SURFACE_PRESENT_SCALING_CAPABILITIES_KHR = 1000274001,
  // Provided by VK_KHR_surface_maintenance1
    VK_STRUCTURE_TYPE_SURFACE_PRESENT_MODE_COMPATIBILITY_KHR = 1000274002,
  // Provided by VK_KHR_swapchain_maintenance1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SWAPCHAIN_MAINTENANCE_1_FEATURES_KHR =
1000275000,
  // Provided by VK_KHR_swapchain_maintenance1
    VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_FENCE_INFO_KHR = 1000275001,
  // Provided by VK_KHR_swapchain_maintenance1
    VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_MODES_CREATE_INFO_KHR = 1000275002,
  // Provided by VK_KHR_swapchain_maintenance1
    VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_MODE_INFO_KHR = 1000275003,
  // Provided by VK_KHR_swapchain_maintenance1
    VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_SCALING_CREATE_INFO_KHR = 1000275004,
  // Provided by VK_KHR_swapchain_maintenance1
    VK_STRUCTURE_TYPE_RELEASE_SWAPCHAIN_IMAGES_INFO_KHR = 1000275005,
  // Provided by VK_EXT_mutable_descriptor_type
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MUTABLE_DESCRIPTOR_TYPE_FEATURES_EXT =
1000351000,
  // Provided by VK_EXT_mutable_descriptor_type
    VK_STRUCTURE_TYPE_MUTABLE_DESCRIPTOR_TYPE_CREATE_INFO_EXT = 1000351002,
  // Provided by VK_EXT_legacy_vertex_attributes
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LEGACY_VERTEX_ATTRIBUTES_FEATURES_EXT =
1000495000,
  // Provided by VK_EXT_legacy_vertex_attributes
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LEGACY_VERTEX_ATTRIBUTES_PROPERTIES_EXT =
```

```
1000495001,
  // Provided by VK_EXT_layer_settings
    VK_STRUCTURE_TYPE_LAYER_SETTINGS_CREATE_INFO_EXT = 1000496000,
  // Provided by VK_EXT_dynamic_rendering_unused_attachments

VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_UNUSED_ATTACHMENTS_FEATURES_EXT =
1000499000,
  // Provided by VK_KHR_internally_synchronized_queues
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INTERNALLY_SYNCHRONIZED_QUEUES_FEATURES_KHR =
1000504000,
  // Provided by VK_KHR_cooperative_matrix
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_FEATURES_KHR = 1000506000,
  // Provided by VK_KHR_cooperative_matrix
    VK_STRUCTURE_TYPE_COOPERATIVE_MATRIX_PROPERTIES_KHR = 1000506001,
  // Provided by VK_KHR_cooperative_matrix
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_PROPERTIES_KHR = 1000506002,
  // Provided by VK_KHR_compute_shader_derivatives
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COMPUTE_SHADER_DERIVATIVES_FEATURES_KHR =
1000201000,
  // Provided by VK_KHR_compute_shader_derivatives
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COMPUTE_SHADER_DERIVATIVES_PROPERTIES_KHR =
1000511000,
  // Provided by VK_KHR_video_decode_av1
    VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_CAPABILITIES_KHR = 1000512000,
  // Provided by VK_KHR_video_decode_av1
    VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_PICTURE_INFO_KHR = 1000512001,
  // Provided by VK_KHR_video_decode_av1
    VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_PROFILE_INFO_KHR = 1000512003,
  // Provided by VK_KHR_video_decode_av1
    VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_SESSION_PARAMETERS_CREATE_INFO_KHR =
1000512004,
  // Provided by VK_KHR_video_decode_av1
    VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_DPB_SLOT_INFO_KHR = 1000512005,
  // Provided by VK_KHR_video_encode_av1
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_CAPABILITIES_KHR = 1000513000,
  // Provided by VK_KHR_video_encode_av1
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_SESSION_PARAMETERS_CREATE_INFO_KHR =
1000513001,
  // Provided by VK_KHR_video_encode_av1
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_PICTURE_INFO_KHR = 1000513002,
  // Provided by VK_KHR_video_encode_av1
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_DPB_SLOT_INFO_KHR = 1000513003,
  // Provided by VK_KHR_video_encode_av1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_AV1_FEATURES_KHR = 1000513004,
  // Provided by VK_KHR_video_encode_av1
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_PROFILE_INFO_KHR = 1000513005,
  // Provided by VK_KHR_video_encode_av1
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_RATE_CONTROL_INFO_KHR = 1000513006,
  // Provided by VK_KHR_video_encode_av1
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_RATE_CONTROL_LAYER_INFO_KHR = 1000513007,
  // Provided by VK_KHR_video_encode_av1
```

```
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_QUALITY_LEVEL_PROPERTIES_KHR = 1000513008,
  // Provided by VK_KHR_video_encode_av1
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_SESSION_CREATE_INFO_KHR = 1000513009,
  // Provided by VK_KHR_video_encode_av1
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_GOP_REMAINING_FRAME_INFO_KHR = 1000513010,
  // Provided by VK_KHR_video_decode_vp9
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_DECODE_VP9_FEATURES_KHR = 1000514000,
  // Provided by VK_KHR_video_decode_vp9
    VK_STRUCTURE_TYPE_VIDEO_DECODE_VP9_CAPABILITIES_KHR = 1000514001,
  // Provided by VK_KHR_video_decode_vp9
    VK_STRUCTURE_TYPE_VIDEO_DECODE_VP9_PICTURE_INFO_KHR = 1000514002,
  // Provided by VK_KHR_video_decode_vp9
    VK_STRUCTURE_TYPE_VIDEO_DECODE_VP9_PROFILE_INFO_KHR = 1000514003,
  // Provided by VK_KHR_video_maintenance1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_MAINTENANCE_1_FEATURES_KHR = 1000515000,
  // Provided by VK_KHR_video_maintenance1
    VK_STRUCTURE_TYPE_VIDEO_INLINE_QUERY_INFO_KHR = 1000515001,
  // Provided by VK_EXT_attachment_feedback_loop_dynamic_state

VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ATTACHMENT_FEEDBACK_LOOP_DYNAMIC_STATE_FEATURES_EXT
= 1000524000,
  // Provided by VK_KHR_unified_image_layouts
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_UNIFIED_IMAGE_LAYOUTS_FEATURES_KHR = 1000527000,
  // Provided by VK_KHR_unified_image_layouts with
VK_EXT_attachment_feedback_loop_layout and (VK_VERSION_1_3 or
VK_KHR_dynamic_rendering)
    VK_STRUCTURE_TYPE_ATTACHMENT_FEEDBACK_LOOP_INFO_EXT = 1000527001,
  // Provided by VK_KHR_calibrated_timestamps
    VK_STRUCTURE_TYPE_CALIBRATED_TIMESTAMP_INFO_KHR = 1000184000,
  // Provided by VK_KHR_maintenance6 with VK_EXT_descriptor_buffer
    VK_STRUCTURE_TYPE_SET_DESCRIPTOR_BUFFER_OFFSETS_INFO_EXT = 1000545007,
  // Provided by VK_KHR_maintenance6 with VK_EXT_descriptor_buffer
    VK_STRUCTURE_TYPE_BIND_DESCRIPTOR_BUFFER_EMBEDDED_SAMPLERS_INFO_EXT = 1000545008,
  // Provided by VK_KHR_copy_memory_indirect
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COPY_MEMORY_INDIRECT_FEATURES_KHR = 1000549000,
  // Provided by VK_KHR_copy_memory_indirect
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COPY_MEMORY_INDIRECT_PROPERTIES_KHR =
1000426001,
  // Provided by VK_KHR_copy_memory_indirect
    VK_STRUCTURE_TYPE_COPY_MEMORY_INDIRECT_INFO_KHR = 1000549002,
  // Provided by VK_KHR_copy_memory_indirect
    VK_STRUCTURE_TYPE_COPY_MEMORY_TO_IMAGE_INDIRECT_INFO_KHR = 1000549003,
  // Provided by VK_EXT_memory_decompression
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_DECOMPRESSION_FEATURES_EXT = 1000427000,
  // Provided by VK_EXT_memory_decompression
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_DECOMPRESSION_PROPERTIES_EXT =
1000427001,
  // Provided by VK_EXT_memory_decompression
    VK_STRUCTURE_TYPE_DECOMPRESS_MEMORY_INFO_EXT = 1000550002,
  // Provided by VK_KHR_video_encode_intra_refresh
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_INTRA_REFRESH_CAPABILITIES_KHR = 1000552000,
```

```
  // Provided by VK_KHR_video_encode_intra_refresh
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_SESSION_INTRA_REFRESH_CREATE_INFO_KHR = 1000552001,
  // Provided by VK_KHR_video_encode_intra_refresh
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_INTRA_REFRESH_INFO_KHR = 1000552002,
  // Provided by VK_KHR_video_encode_intra_refresh
    VK_STRUCTURE_TYPE_VIDEO_REFERENCE_INTRA_REFRESH_INFO_KHR = 1000552003,
  // Provided by VK_KHR_video_encode_intra_refresh
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_INTRA_REFRESH_FEATURES_KHR =
1000552004,
  // Provided by VK_KHR_video_encode_quantization_map
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUANTIZATION_MAP_CAPABILITIES_KHR = 1000553000,
  // Provided by VK_KHR_video_encode_quantization_map
    VK_STRUCTURE_TYPE_VIDEO_FORMAT_QUANTIZATION_MAP_PROPERTIES_KHR = 1000553001,
  // Provided by VK_KHR_video_encode_quantization_map
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUANTIZATION_MAP_INFO_KHR = 1000553002,
  // Provided by VK_KHR_video_encode_quantization_map
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUANTIZATION_MAP_SESSION_PARAMETERS_CREATE_INFO_KHR
= 1000553005,
  // Provided by VK_KHR_video_encode_quantization_map
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_QUANTIZATION_MAP_FEATURES_KHR =
1000553009,
  // Provided by VK_KHR_video_encode_h264 with VK_KHR_video_encode_quantization_map
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_QUANTIZATION_MAP_CAPABILITIES_KHR =
1000553003,
  // Provided by VK_KHR_video_encode_h265 with VK_KHR_video_encode_quantization_map
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_QUANTIZATION_MAP_CAPABILITIES_KHR =
1000553004,
  // Provided by VK_KHR_video_encode_h265 with VK_KHR_video_encode_quantization_map
    VK_STRUCTURE_TYPE_VIDEO_FORMAT_H265_QUANTIZATION_MAP_PROPERTIES_KHR = 1000553006,
  // Provided by VK_KHR_video_encode_av1 with VK_KHR_video_encode_quantization_map
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_QUANTIZATION_MAP_CAPABILITIES_KHR = 1000553007,
  // Provided by VK_KHR_video_encode_av1 with VK_KHR_video_encode_quantization_map
    VK_STRUCTURE_TYPE_VIDEO_FORMAT_AV1_QUANTIZATION_MAP_PROPERTIES_KHR = 1000553008,
  // Provided by VK_KHR_shader_relaxed_extended_instruction
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_RELAXED_EXTENDED_INSTRUCTION_FEATURES_KHR
= 1000558000,
  // Provided by VK_KHR_maintenance7
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_7_FEATURES_KHR = 1000562000,
  // Provided by VK_KHR_maintenance7
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_7_PROPERTIES_KHR = 1000562001,
  // Provided by VK_KHR_maintenance7
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_PROPERTIES_LIST_KHR = 1000562002,
  // Provided by VK_KHR_maintenance7
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_PROPERTIES_KHR = 1000562003,
  // Provided by VK_KHR_maintenance7
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_VULKAN_PROPERTIES_KHR = 1000562004,
  // Provided by VK_EXT_shader_replicated_composites
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_REPLICATED_COMPOSITES_FEATURES_EXT =
1000564000,
  // Provided by VK_EXT_shader_float8
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT8_FEATURES_EXT = 1000567000,
```

```
  // Provided by VK_EXT_device_generated_commands
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_GENERATED_COMMANDS_FEATURES_EXT =
1000572000,
  // Provided by VK_EXT_device_generated_commands
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_GENERATED_COMMANDS_PROPERTIES_EXT =
1000572001,
  // Provided by VK_EXT_device_generated_commands
    VK_STRUCTURE_TYPE_GENERATED_COMMANDS_MEMORY_REQUIREMENTS_INFO_EXT = 1000572002,
  // Provided by VK_EXT_device_generated_commands
    VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_CREATE_INFO_EXT = 1000572003,
  // Provided by VK_EXT_device_generated_commands
    VK_STRUCTURE_TYPE_GENERATED_COMMANDS_INFO_EXT = 1000572004,
  // Provided by VK_EXT_device_generated_commands
    VK_STRUCTURE_TYPE_INDIRECT_COMMANDS_LAYOUT_CREATE_INFO_EXT = 1000572006,
  // Provided by VK_EXT_device_generated_commands
    VK_STRUCTURE_TYPE_INDIRECT_COMMANDS_LAYOUT_TOKEN_EXT = 1000572007,
  // Provided by VK_EXT_device_generated_commands
    VK_STRUCTURE_TYPE_WRITE_INDIRECT_EXECUTION_SET_PIPELINE_EXT = 1000572008,
  // Provided by VK_EXT_device_generated_commands
    VK_STRUCTURE_TYPE_WRITE_INDIRECT_EXECUTION_SET_SHADER_EXT = 1000572009,
  // Provided by VK_EXT_device_generated_commands
    VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_PIPELINE_INFO_EXT = 1000572010,
  // Provided by VK_EXT_device_generated_commands
    VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_SHADER_INFO_EXT = 1000572011,
  // Provided by VK_EXT_device_generated_commands
    VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_SHADER_LAYOUT_INFO_EXT = 1000572012,
  // Provided by VK_EXT_device_generated_commands
    VK_STRUCTURE_TYPE_GENERATED_COMMANDS_PIPELINE_INFO_EXT = 1000572013,
  // Provided by VK_EXT_device_generated_commands
    VK_STRUCTURE_TYPE_GENERATED_COMMANDS_SHADER_INFO_EXT = 1000572014,
  // Provided by VK_KHR_device_fault
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FAULT_FEATURES_KHR = 1000573000,
  // Provided by VK_KHR_device_fault
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FAULT_PROPERTIES_KHR = 1000573001,
  // Provided by VK_KHR_device_fault
    VK_STRUCTURE_TYPE_DEVICE_FAULT_INFO_KHR = 1000573002,
  // Provided by VK_KHR_device_fault
    VK_STRUCTURE_TYPE_DEVICE_FAULT_DEBUG_INFO_KHR = 1000573003,
  // Provided by VK_KHR_maintenance8
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_8_FEATURES_KHR = 1000574000,
  // Provided by VK_KHR_maintenance8
    VK_STRUCTURE_TYPE_MEMORY_BARRIER_ACCESS_FLAGS_3_KHR = 1000574002,
  // Provided by VK_KHR_shader_fma
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FMA_FEATURES_KHR = 1000579000,
  // Provided by VK_EXT_ray_tracing_invocation_reorder
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_INVOCATION_REORDER_FEATURES_EXT =
1000581000,
  // Provided by VK_EXT_ray_tracing_invocation_reorder
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_INVOCATION_REORDER_PROPERTIES_EXT =
1000581001,
  // Provided by VK_EXT_depth_clamp_control
```

```
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLAMP_CONTROL_FEATURES_EXT = 1000582000,
  // Provided by VK_EXT_depth_clamp_control
    VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_DEPTH_CLAMP_CONTROL_CREATE_INFO_EXT =
1000582001,
  // Provided by VK_KHR_maintenance9
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_9_FEATURES_KHR = 1000584000,
  // Provided by VK_KHR_maintenance9
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_9_PROPERTIES_KHR = 1000584001,
  // Provided by VK_KHR_maintenance9
    VK_STRUCTURE_TYPE_QUEUE_FAMILY_OWNERSHIP_TRANSFER_PROPERTIES_KHR = 1000584002,
  // Provided by VK_KHR_video_maintenance2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_MAINTENANCE_2_FEATURES_KHR = 1000586000,
  // Provided by VK_KHR_video_decode_h264 with VK_KHR_video_maintenance2
    VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_INLINE_SESSION_PARAMETERS_INFO_KHR =
1000586001,
  // Provided by VK_KHR_video_decode_h265 with VK_KHR_video_maintenance2
    VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_INLINE_SESSION_PARAMETERS_INFO_KHR =
1000586002,
  // Provided by VK_KHR_video_decode_av1 with VK_KHR_video_maintenance2
    VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_INLINE_SESSION_PARAMETERS_INFO_KHR =
1000586003,
  // Provided by VK_KHR_video_encode_feedback2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_FEEDBACK_2_FEATURES_KHR =
1000598000,
  // Provided by VK_KHR_video_encode_feedback2
    VK_STRUCTURE_TYPE_VIDEO_ENCODE_FEEDBACK_2_CAPABILITIES_KHR = 1000598001,
  // Provided by VK_KHR_video_encode_feedback2
    VK_STRUCTURE_TYPE_QUERY_POOL_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_CREATE_INFO_KHR =
1000598002,
  // Provided by VK_EXT_external_memory_metal
    VK_STRUCTURE_TYPE_IMPORT_MEMORY_METAL_HANDLE_INFO_EXT = 1000602000,
  // Provided by VK_EXT_external_memory_metal
    VK_STRUCTURE_TYPE_MEMORY_METAL_HANDLE_PROPERTIES_EXT = 1000602001,
  // Provided by VK_EXT_external_memory_metal
    VK_STRUCTURE_TYPE_MEMORY_GET_METAL_HANDLE_INFO_EXT = 1000602002,
  // Provided by VK_KHR_depth_clamp_zero_one
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLAMP_ZERO_ONE_FEATURES_KHR = 1000421000,
  // Provided by VK_KHR_robustness2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_FEATURES_KHR = 1000286000,
  // Provided by VK_KHR_robustness2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_PROPERTIES_KHR = 1000286001,
  // Provided by VK_EXT_multisampled_render_to_swapchain
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTISAMPLED_RENDER_TO_SWAPCHAIN_FEATURES_EXT =
1000616000,
  // Provided by VK_EXT_multisampled_render_to_swapchain
    VK_STRUCTURE_TYPE_SWAPCHAIN_FLAGS_SURFACE_CAPABILITIES_EXT = 1000616001,
  // Provided by VK_EXT_fragment_density_map_offset
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_OFFSET_FEATURES_EXT =
1000425000,
  // Provided by VK_EXT_fragment_density_map_offset
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_OFFSET_PROPERTIES_EXT =
```

```
1000425001,
  // Provided by VK_EXT_fragment_density_map_offset
    VK_STRUCTURE_TYPE_RENDER_PASS_FRAGMENT_DENSITY_MAP_OFFSET_END_INFO_EXT =
1000425002,
  // Provided by VK_EXT_zero_initialize_device_memory
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ZERO_INITIALIZE_DEVICE_MEMORY_FEATURES_EXT =
1000620000,
  // Provided by VK_KHR_present_mode_fifo_latest_ready
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_MODE_FIFO_LATEST_READY_FEATURES_KHR =
1000361000,
  // Provided by VK_KHR_opacity_micromap
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_FEATURES_KHR = 1000623000,
  // Provided by VK_KHR_opacity_micromap
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_PROPERTIES_KHR = 1000623001,
  // Provided by VK_KHR_opacity_micromap
    VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_MICROMAP_DATA_KHR = 1000623002,
  // Provided by VK_KHR_opacity_micromap
    VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_TRIANGLES_OPACITY_MICROMAP_KHR =
1000623003,
  // Provided by VK_EXT_shader_64bit_indexing
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_64_BIT_INDEXING_FEATURES_EXT =
1000627000,
  // Provided by VK_EXT_custom_resolve
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CUSTOM_RESOLVE_FEATURES_EXT = 1000628000,
  // Provided by VK_EXT_custom_resolve with VK_KHR_dynamic_rendering or VK_VERSION_1_3
    VK_STRUCTURE_TYPE_BEGIN_CUSTOM_RESOLVE_INFO_EXT = 1000628001,
  // Provided by VK_EXT_custom_resolve with VK_KHR_dynamic_rendering or VK_VERSION_1_3
    VK_STRUCTURE_TYPE_CUSTOM_RESOLVE_CREATE_INFO_EXT = 1000628002,
  // Provided by VK_KHR_maintenance10
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_10_FEATURES_KHR = 1000630000,
  // Provided by VK_KHR_maintenance10
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_10_PROPERTIES_KHR = 1000630001,
  // Provided by VK_KHR_maintenance10
    VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_FLAGS_INFO_KHR = 1000630002,
  // Provided by VK_KHR_maintenance10
    VK_STRUCTURE_TYPE_RENDERING_END_INFO_KHR = 1000619003,
  // Provided by VK_KHR_maintenance10
    VK_STRUCTURE_TYPE_RESOLVE_IMAGE_MODE_INFO_KHR = 1000630004,
  // Provided by VK_EXT_shader_long_vector
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_LONG_VECTOR_FEATURES_EXT = 1000635000,
  // Provided by VK_EXT_shader_long_vector
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_LONG_VECTOR_PROPERTIES_EXT = 1000635001,
  // Provided by VK_EXT_shader_uniform_buffer_unsized_array
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_UNIFORM_BUFFER_UNSIZED_ARRAY_FEATURES_EXT
= 1000642000,
  // Provided by VK_KHR_pipeline_library_group_handles
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_LIBRARY_GROUP_HANDLES_FEATURES_KHR =
1000498000,
  // Provided by VK_KHR_maintenance11
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_11_FEATURES_KHR = 1000657000,
  // Provided by VK_KHR_maintenance11
```

```
    VK_STRUCTURE_TYPE_QUEUE_FAMILY_OPTIMAL_IMAGE_TRANSFER_GRANULARITY_PROPERTIES_KHR =
1000657001,
  // Provided by VK_EXT_cooperative_matrix_maintenance1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_MAINTENANCE_1_FEATURES_EXT =
1000659000,
  // Provided by VK_EXT_cooperative_matrix_maintenance1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_INFO_2_EXT = 1000659001,
  // Provided by VK_EXT_cooperative_matrix_maintenance1
    VK_STRUCTURE_TYPE_COOPERATIVE_MATRIX_PROPERTIES_2_EXT = 1000659002,
  // Provided by VK_EXT_shader_subgroup_partitioned
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_PARTITIONED_FEATURES_EXT =
1000662000,
  // Provided by VK_KHR_extended_flags
    VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_4_KHR = 1000668000,
  // Provided by VK_KHR_extended_flags
    VK_STRUCTURE_TYPE_IMAGE_CREATE_FLAGS_2_CREATE_INFO_KHR = 1000668001,
  // Provided by VK_KHR_extended_flags
    VK_STRUCTURE_TYPE_IMAGE_USAGE_FLAGS_2_CREATE_INFO_KHR = 1000668002,
  // Provided by VK_KHR_extended_flags
    VK_STRUCTURE_TYPE_IMAGE_VIEW_USAGE_2_CREATE_INFO_KHR = 1000668003,
  // Provided by VK_KHR_extended_flags
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_FLAGS_FEATURES_KHR = 1000668004,
  // Provided by VK_KHR_extended_flags with VK_VERSION_1_2 or
VK_EXT_separate_stencil_usage
    VK_STRUCTURE_TYPE_IMAGE_STENCIL_USAGE_2_CREATE_INFO_KHR = 1000668005,
  // Provided by VK_KHR_extended_flags with VK_KHR_shared_presentable_image
    VK_STRUCTURE_TYPE_SHARED_PRESENT_SURFACE_CAPABILITIES_2_KHR = 1000668006,
  // Provided by VK_EXT_shader_ocp_microscaling_types
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_OCP_MICROSCALING_TYPES_FEATURES_EXT =
1000672000,
  // Provided by VK_EXT_primitive_restart_index
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIMITIVE_RESTART_INDEX_FEATURES_EXT =
1000678000,
  // Provided by VK_EXT_image_tiling_control
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_TILING_CONTROL_FEATURES_EXT = 1000687000,
  // Provided by VK_EXT_image_tiling_control
    VK_STRUCTURE_TYPE_IMAGE_TILING_CONTROL_CREATE_INFO_EXT = 1000687001,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTER_FEATURES =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTERS_FEATURES,
  // Provided by VK_VERSION_1_1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_DRAW_PARAMETER_FEATURES =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_DRAW_PARAMETERS_FEATURES,
  // Provided by VK_KHR_dynamic_rendering
    VK_STRUCTURE_TYPE_RENDERING_INFO_KHR = VK_STRUCTURE_TYPE_RENDERING_INFO,
  // Provided by VK_KHR_dynamic_rendering
    VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_INFO_KHR =
VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_INFO,
  // Provided by VK_KHR_dynamic_rendering
    VK_STRUCTURE_TYPE_PIPELINE_RENDERING_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_PIPELINE_RENDERING_CREATE_INFO,
```

```
  // Provided by VK_KHR_dynamic_rendering
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_FEATURES,
  // Provided by VK_KHR_dynamic_rendering
    VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_RENDERING_INFO_KHR =
VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_RENDERING_INFO,
  // Provided by VK_KHR_multiview
    VK_STRUCTURE_TYPE_RENDER_PASS_MULTIVIEW_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_RENDER_PASS_MULTIVIEW_CREATE_INFO,
  // Provided by VK_KHR_multiview
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_FEATURES,
  // Provided by VK_KHR_multiview
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_PROPERTIES,
  // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FEATURES_2_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FEATURES_2,
  // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROPERTIES_2_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROPERTIES_2,
  // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_2_KHR = VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_2,
  // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_IMAGE_FORMAT_PROPERTIES_2_KHR =
VK_STRUCTURE_TYPE_IMAGE_FORMAT_PROPERTIES_2,
  // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_FORMAT_INFO_2_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_FORMAT_INFO_2,
  // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_QUEUE_FAMILY_PROPERTIES_2_KHR =
VK_STRUCTURE_TYPE_QUEUE_FAMILY_PROPERTIES_2,
  // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PROPERTIES_2_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PROPERTIES_2,
  // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_SPARSE_IMAGE_FORMAT_PROPERTIES_2_KHR =
VK_STRUCTURE_TYPE_SPARSE_IMAGE_FORMAT_PROPERTIES_2,
  // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SPARSE_IMAGE_FORMAT_INFO_2_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SPARSE_IMAGE_FORMAT_INFO_2,
  // Provided by VK_KHR_device_group
    VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_FLAGS_INFO_KHR =
VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_FLAGS_INFO,
  // Provided by VK_KHR_device_group
    VK_STRUCTURE_TYPE_DEVICE_GROUP_RENDER_PASS_BEGIN_INFO_KHR =
VK_STRUCTURE_TYPE_DEVICE_GROUP_RENDER_PASS_BEGIN_INFO,
  // Provided by VK_KHR_device_group
    VK_STRUCTURE_TYPE_DEVICE_GROUP_COMMAND_BUFFER_BEGIN_INFO_KHR =
VK_STRUCTURE_TYPE_DEVICE_GROUP_COMMAND_BUFFER_BEGIN_INFO,
  // Provided by VK_KHR_device_group
```

```
    VK_STRUCTURE_TYPE_DEVICE_GROUP_SUBMIT_INFO_KHR =
VK_STRUCTURE_TYPE_DEVICE_GROUP_SUBMIT_INFO,
  // Provided by VK_KHR_device_group
    VK_STRUCTURE_TYPE_DEVICE_GROUP_BIND_SPARSE_INFO_KHR =
VK_STRUCTURE_TYPE_DEVICE_GROUP_BIND_SPARSE_INFO,
  // Provided by VK_KHR_bind_memory2 with VK_KHR_device_group
    VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_DEVICE_GROUP_INFO_KHR =
VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_DEVICE_GROUP_INFO,
  // Provided by VK_KHR_bind_memory2 with VK_KHR_device_group
    VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_DEVICE_GROUP_INFO_KHR =
VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_DEVICE_GROUP_INFO,
  // Provided by VK_EXT_pipeline_robustness
    VK_STRUCTURE_TYPE_PIPELINE_ROBUSTNESS_CREATE_INFO_EXT =
VK_STRUCTURE_TYPE_PIPELINE_ROBUSTNESS_CREATE_INFO,
  // Provided by VK_EXT_pipeline_robustness
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_ROBUSTNESS_FEATURES_EXT =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_ROBUSTNESS_FEATURES,
  // Provided by VK_EXT_pipeline_robustness
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_ROBUSTNESS_PROPERTIES_EXT =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_ROBUSTNESS_PROPERTIES,
  // Provided by VK_KHR_device_group_creation
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GROUP_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GROUP_PROPERTIES,
  // Provided by VK_KHR_device_group_creation
    VK_STRUCTURE_TYPE_DEVICE_GROUP_DEVICE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_DEVICE_GROUP_DEVICE_CREATE_INFO,
  // Provided by VK_KHR_external_memory_capabilities
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_IMAGE_FORMAT_INFO_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_IMAGE_FORMAT_INFO,
  // Provided by VK_KHR_external_memory_capabilities
    VK_STRUCTURE_TYPE_EXTERNAL_IMAGE_FORMAT_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_EXTERNAL_IMAGE_FORMAT_PROPERTIES,
  // Provided by VK_KHR_external_memory_capabilities
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_BUFFER_INFO_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_BUFFER_INFO,
  // Provided by VK_KHR_external_memory_capabilities
    VK_STRUCTURE_TYPE_EXTERNAL_BUFFER_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_EXTERNAL_BUFFER_PROPERTIES,
  // Provided by VK_KHR_external_fence_capabilities,
VK_KHR_external_memory_capabilities, VK_KHR_external_semaphore_capabilities
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ID_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ID_PROPERTIES,
  // Provided by VK_KHR_external_memory
    VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_BUFFER_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_BUFFER_CREATE_INFO,
  // Provided by VK_KHR_external_memory
    VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_IMAGE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_IMAGE_CREATE_INFO,
  // Provided by VK_KHR_external_memory
    VK_STRUCTURE_TYPE_EXPORT_MEMORY_ALLOCATE_INFO_KHR =
VK_STRUCTURE_TYPE_EXPORT_MEMORY_ALLOCATE_INFO,
```

```
  // Provided by VK_KHR_external_semaphore_capabilities
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_SEMAPHORE_INFO_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_SEMAPHORE_INFO,
  // Provided by VK_KHR_external_semaphore_capabilities
    VK_STRUCTURE_TYPE_EXTERNAL_SEMAPHORE_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_EXTERNAL_SEMAPHORE_PROPERTIES,
  // Provided by VK_KHR_external_semaphore
    VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_CREATE_INFO,
  // Provided by VK_KHR_push_descriptor
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PUSH_DESCRIPTOR_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PUSH_DESCRIPTOR_PROPERTIES,
  // Provided by VK_KHR_shader_float16_int8
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT16_INT8_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT16_INT8_FEATURES,
  // Provided by VK_KHR_shader_float16_int8
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FLOAT16_INT8_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT16_INT8_FEATURES,
  // Provided by VK_KHR_16bit_storage
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_16BIT_STORAGE_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_16BIT_STORAGE_FEATURES,
  // Provided by VK_KHR_descriptor_update_template
    VK_STRUCTURE_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_CREATE_INFO,
  // Provided by VK_EXT_display_surface_counter
  // VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES2_EXT is a legacy alias
    VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES2_EXT =
VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_2_EXT,
  // Provided by VK_KHR_imageless_framebuffer
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGELESS_FRAMEBUFFER_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGELESS_FRAMEBUFFER_FEATURES,
  // Provided by VK_KHR_imageless_framebuffer
    VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENTS_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENTS_CREATE_INFO,
  // Provided by VK_KHR_imageless_framebuffer
    VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENT_IMAGE_INFO_KHR =
VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENT_IMAGE_INFO,
  // Provided by VK_KHR_imageless_framebuffer
    VK_STRUCTURE_TYPE_RENDER_PASS_ATTACHMENT_BEGIN_INFO_KHR =
VK_STRUCTURE_TYPE_RENDER_PASS_ATTACHMENT_BEGIN_INFO,
  // Provided by VK_KHR_create_renderpass2
    VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_2_KHR =
VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_2,
  // Provided by VK_KHR_create_renderpass2
    VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_2_KHR =
VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_2,
  // Provided by VK_KHR_create_renderpass2
    VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_2_KHR =
VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_2,
  // Provided by VK_KHR_create_renderpass2
    VK_STRUCTURE_TYPE_SUBPASS_DEPENDENCY_2_KHR =
```

```
VK_STRUCTURE_TYPE_SUBPASS_DEPENDENCY_2,
  // Provided by VK_KHR_create_renderpass2
    VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO_2_KHR =
VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO_2,
  // Provided by VK_KHR_create_renderpass2
    VK_STRUCTURE_TYPE_SUBPASS_BEGIN_INFO_KHR = VK_STRUCTURE_TYPE_SUBPASS_BEGIN_INFO,
  // Provided by VK_KHR_create_renderpass2
    VK_STRUCTURE_TYPE_SUBPASS_END_INFO_KHR = VK_STRUCTURE_TYPE_SUBPASS_END_INFO,
  // Provided by VK_KHR_external_fence_capabilities
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_FENCE_INFO_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_FENCE_INFO,
  // Provided by VK_KHR_external_fence_capabilities
    VK_STRUCTURE_TYPE_EXTERNAL_FENCE_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_EXTERNAL_FENCE_PROPERTIES,
  // Provided by VK_KHR_external_fence
    VK_STRUCTURE_TYPE_EXPORT_FENCE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_EXPORT_FENCE_CREATE_INFO,
  // Provided by VK_KHR_maintenance2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_POINT_CLIPPING_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_POINT_CLIPPING_PROPERTIES,
  // Provided by VK_KHR_maintenance2
    VK_STRUCTURE_TYPE_RENDER_PASS_INPUT_ATTACHMENT_ASPECT_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_RENDER_PASS_INPUT_ATTACHMENT_ASPECT_CREATE_INFO,
  // Provided by VK_KHR_maintenance2
    VK_STRUCTURE_TYPE_IMAGE_VIEW_USAGE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_IMAGE_VIEW_USAGE_CREATE_INFO,
  // Provided by VK_KHR_maintenance2
    VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_DOMAIN_ORIGIN_STATE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_DOMAIN_ORIGIN_STATE_CREATE_INFO,
  // Provided by VK_KHR_variable_pointers
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTERS_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTERS_FEATURES,
  // Provided by VK_KHR_variable_pointers
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTER_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTERS_FEATURES_KHR,
  // Provided by VK_KHR_dedicated_allocation
    VK_STRUCTURE_TYPE_MEMORY_DEDICATED_REQUIREMENTS_KHR =
VK_STRUCTURE_TYPE_MEMORY_DEDICATED_REQUIREMENTS,
  // Provided by VK_KHR_dedicated_allocation
    VK_STRUCTURE_TYPE_MEMORY_DEDICATED_ALLOCATE_INFO_KHR =
VK_STRUCTURE_TYPE_MEMORY_DEDICATED_ALLOCATE_INFO,
  // Provided by VK_KHR_get_memory_requirements2
    VK_STRUCTURE_TYPE_BUFFER_MEMORY_REQUIREMENTS_INFO_2_KHR =
VK_STRUCTURE_TYPE_BUFFER_MEMORY_REQUIREMENTS_INFO_2,
  // Provided by VK_KHR_get_memory_requirements2
    VK_STRUCTURE_TYPE_IMAGE_MEMORY_REQUIREMENTS_INFO_2_KHR =
VK_STRUCTURE_TYPE_IMAGE_MEMORY_REQUIREMENTS_INFO_2,
  // Provided by VK_KHR_get_memory_requirements2
    VK_STRUCTURE_TYPE_IMAGE_SPARSE_MEMORY_REQUIREMENTS_INFO_2_KHR =
VK_STRUCTURE_TYPE_IMAGE_SPARSE_MEMORY_REQUIREMENTS_INFO_2,
  // Provided by VK_KHR_get_memory_requirements2
```

```
    VK_STRUCTURE_TYPE_MEMORY_REQUIREMENTS_2_KHR =
VK_STRUCTURE_TYPE_MEMORY_REQUIREMENTS_2,
  // Provided by VK_KHR_get_memory_requirements2
    VK_STRUCTURE_TYPE_SPARSE_IMAGE_MEMORY_REQUIREMENTS_2_KHR =
VK_STRUCTURE_TYPE_SPARSE_IMAGE_MEMORY_REQUIREMENTS_2,
  // Provided by VK_KHR_image_format_list
    VK_STRUCTURE_TYPE_IMAGE_FORMAT_LIST_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_IMAGE_FORMAT_LIST_CREATE_INFO,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_CREATE_INFO,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_INFO_KHR =
VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_INFO,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_STRUCTURE_TYPE_BIND_IMAGE_PLANE_MEMORY_INFO_KHR =
VK_STRUCTURE_TYPE_BIND_IMAGE_PLANE_MEMORY_INFO,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_STRUCTURE_TYPE_IMAGE_PLANE_MEMORY_REQUIREMENTS_INFO_KHR =
VK_STRUCTURE_TYPE_IMAGE_PLANE_MEMORY_REQUIREMENTS_INFO,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLER_YCBCR_CONVERSION_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLER_YCBCR_CONVERSION_FEATURES,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_IMAGE_FORMAT_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_IMAGE_FORMAT_PROPERTIES,
  // Provided by VK_KHR_bind_memory2
    VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_INFO_KHR =
VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_INFO,
  // Provided by VK_KHR_bind_memory2
    VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_INFO_KHR =
VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_INFO,
  // Provided by VK_EXT_descriptor_indexing
    VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_BINDING_FLAGS_CREATE_INFO_EXT =
VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_BINDING_FLAGS_CREATE_INFO,
  // Provided by VK_EXT_descriptor_indexing
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_INDEXING_FEATURES_EXT =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_INDEXING_FEATURES,
  // Provided by VK_EXT_descriptor_indexing
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_INDEXING_PROPERTIES_EXT =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_INDEXING_PROPERTIES,
  // Provided by VK_EXT_descriptor_indexing
    VK_STRUCTURE_TYPE_DESCRIPTOR_SET_VARIABLE_DESCRIPTOR_COUNT_ALLOCATE_INFO_EXT =
VK_STRUCTURE_TYPE_DESCRIPTOR_SET_VARIABLE_DESCRIPTOR_COUNT_ALLOCATE_INFO,
  // Provided by VK_EXT_descriptor_indexing
    VK_STRUCTURE_TYPE_DESCRIPTOR_SET_VARIABLE_DESCRIPTOR_COUNT_LAYOUT_SUPPORT_EXT =
VK_STRUCTURE_TYPE_DESCRIPTOR_SET_VARIABLE_DESCRIPTOR_COUNT_LAYOUT_SUPPORT,
  // Provided by VK_KHR_maintenance3
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_3_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_3_PROPERTIES,
  // Provided by VK_KHR_maintenance3
```

```
    VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_SUPPORT_KHR =
VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_SUPPORT,
  // Provided by VK_KHR_shader_subgroup_extended_types
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_EXTENDED_TYPES_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_EXTENDED_TYPES_FEATURES,
  // Provided by VK_KHR_8bit_storage
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_8BIT_STORAGE_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_8BIT_STORAGE_FEATURES,
  // Provided by VK_KHR_shader_atomic_int64
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_INT64_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_INT64_FEATURES,
  // Provided by VK_KHR_global_priority
    VK_STRUCTURE_TYPE_DEVICE_QUEUE_GLOBAL_PRIORITY_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_DEVICE_QUEUE_GLOBAL_PRIORITY_CREATE_INFO,
  // Provided by VK_KHR_global_priority
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GLOBAL_PRIORITY_QUERY_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GLOBAL_PRIORITY_QUERY_FEATURES,
  // Provided by VK_KHR_global_priority
    VK_STRUCTURE_TYPE_QUEUE_FAMILY_GLOBAL_PRIORITY_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_QUEUE_FAMILY_GLOBAL_PRIORITY_PROPERTIES,
  // Provided by VK_KHR_driver_properties
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRIVER_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRIVER_PROPERTIES,
  // Provided by VK_KHR_shader_float_controls
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FLOAT_CONTROLS_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FLOAT_CONTROLS_PROPERTIES,
  // Provided by VK_KHR_depth_stencil_resolve
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_STENCIL_RESOLVE_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_STENCIL_RESOLVE_PROPERTIES,
  // Provided by VK_KHR_depth_stencil_resolve
    VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_DEPTH_STENCIL_RESOLVE_KHR =
VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_DEPTH_STENCIL_RESOLVE,
  // Provided by VK_KHR_timeline_semaphore
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_FEATURES,
  // Provided by VK_KHR_timeline_semaphore
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_PROPERTIES,
  // Provided by VK_KHR_timeline_semaphore
    VK_STRUCTURE_TYPE_SEMAPHORE_TYPE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_SEMAPHORE_TYPE_CREATE_INFO,
  // Provided by VK_KHR_timeline_semaphore
    VK_STRUCTURE_TYPE_TIMELINE_SEMAPHORE_SUBMIT_INFO_KHR =
VK_STRUCTURE_TYPE_TIMELINE_SEMAPHORE_SUBMIT_INFO,
  // Provided by VK_KHR_timeline_semaphore
    VK_STRUCTURE_TYPE_SEMAPHORE_WAIT_INFO_KHR = VK_STRUCTURE_TYPE_SEMAPHORE_WAIT_INFO,
  // Provided by VK_KHR_timeline_semaphore
    VK_STRUCTURE_TYPE_SEMAPHORE_SIGNAL_INFO_KHR =
VK_STRUCTURE_TYPE_SEMAPHORE_SIGNAL_INFO,
  // Provided by VK_KHR_vulkan_memory_model
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_MEMORY_MODEL_FEATURES_KHR =
```

```
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_MEMORY_MODEL_FEATURES,
  // Provided by VK_KHR_shader_terminate_invocation
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TERMINATE_INVOCATION_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TERMINATE_INVOCATION_FEATURES,
  // Provided by VK_KHR_dynamic_rendering_local_read
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_LOCAL_READ_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_LOCAL_READ_FEATURES,
  // Provided by VK_KHR_dynamic_rendering_local_read
    VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_LOCATION_INFO_KHR =
VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_LOCATION_INFO,
  // Provided by VK_KHR_dynamic_rendering_local_read
    VK_STRUCTURE_TYPE_RENDERING_INPUT_ATTACHMENT_INDEX_INFO_KHR =
VK_STRUCTURE_TYPE_RENDERING_INPUT_ATTACHMENT_INDEX_INFO,
  // Provided by VK_KHR_separate_depth_stencil_layouts
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SEPARATE_DEPTH_STENCIL_LAYOUTS_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SEPARATE_DEPTH_STENCIL_LAYOUTS_FEATURES,
  // Provided by VK_KHR_separate_depth_stencil_layouts
    VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_STENCIL_LAYOUT_KHR =
VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_STENCIL_LAYOUT,
  // Provided by VK_KHR_separate_depth_stencil_layouts
    VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_STENCIL_LAYOUT_KHR =
VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_STENCIL_LAYOUT,
  // Provided by VK_KHR_uniform_buffer_standard_layout
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_UNIFORM_BUFFER_STANDARD_LAYOUT_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_UNIFORM_BUFFER_STANDARD_LAYOUT_FEATURES,
  // Provided by VK_KHR_buffer_device_address
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BUFFER_DEVICE_ADDRESS_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BUFFER_DEVICE_ADDRESS_FEATURES,
  // Provided by VK_KHR_buffer_device_address
    VK_STRUCTURE_TYPE_BUFFER_DEVICE_ADDRESS_INFO_KHR =
VK_STRUCTURE_TYPE_BUFFER_DEVICE_ADDRESS_INFO,
  // Provided by VK_KHR_buffer_device_address
    VK_STRUCTURE_TYPE_BUFFER_OPAQUE_CAPTURE_ADDRESS_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_BUFFER_OPAQUE_CAPTURE_ADDRESS_CREATE_INFO,
  // Provided by VK_KHR_buffer_device_address
    VK_STRUCTURE_TYPE_MEMORY_OPAQUE_CAPTURE_ADDRESS_ALLOCATE_INFO_KHR =
VK_STRUCTURE_TYPE_MEMORY_OPAQUE_CAPTURE_ADDRESS_ALLOCATE_INFO,
  // Provided by VK_KHR_buffer_device_address
    VK_STRUCTURE_TYPE_DEVICE_MEMORY_OPAQUE_CAPTURE_ADDRESS_INFO_KHR =
VK_STRUCTURE_TYPE_DEVICE_MEMORY_OPAQUE_CAPTURE_ADDRESS_INFO,
  // Provided by VK_EXT_host_image_copy
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_FEATURES_EXT =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_FEATURES,
  // Provided by VK_EXT_host_image_copy
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_PROPERTIES_EXT =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_PROPERTIES,
  // Provided by VK_EXT_host_image_copy
    VK_STRUCTURE_TYPE_MEMORY_TO_IMAGE_COPY_EXT =
VK_STRUCTURE_TYPE_MEMORY_TO_IMAGE_COPY,
  // Provided by VK_EXT_host_image_copy
    VK_STRUCTURE_TYPE_IMAGE_TO_MEMORY_COPY_EXT =
```

```
VK_STRUCTURE_TYPE_IMAGE_TO_MEMORY_COPY,
  // Provided by VK_EXT_host_image_copy
    VK_STRUCTURE_TYPE_COPY_IMAGE_TO_MEMORY_INFO_EXT =
VK_STRUCTURE_TYPE_COPY_IMAGE_TO_MEMORY_INFO,
  // Provided by VK_EXT_host_image_copy
    VK_STRUCTURE_TYPE_COPY_MEMORY_TO_IMAGE_INFO_EXT =
VK_STRUCTURE_TYPE_COPY_MEMORY_TO_IMAGE_INFO,
  // Provided by VK_EXT_host_image_copy
    VK_STRUCTURE_TYPE_HOST_IMAGE_LAYOUT_TRANSITION_INFO_EXT =
VK_STRUCTURE_TYPE_HOST_IMAGE_LAYOUT_TRANSITION_INFO,
  // Provided by VK_EXT_host_image_copy
    VK_STRUCTURE_TYPE_COPY_IMAGE_TO_IMAGE_INFO_EXT =
VK_STRUCTURE_TYPE_COPY_IMAGE_TO_IMAGE_INFO,
  // Provided by VK_EXT_host_image_copy
    VK_STRUCTURE_TYPE_SUBRESOURCE_HOST_MEMCPY_SIZE_EXT =
VK_STRUCTURE_TYPE_SUBRESOURCE_HOST_MEMCPY_SIZE,
  // Provided by VK_EXT_host_image_copy
    VK_STRUCTURE_TYPE_HOST_IMAGE_COPY_DEVICE_PERFORMANCE_QUERY_EXT =
VK_STRUCTURE_TYPE_HOST_IMAGE_COPY_DEVICE_PERFORMANCE_QUERY,
  // Provided by VK_KHR_map_memory2
    VK_STRUCTURE_TYPE_MEMORY_MAP_INFO_KHR = VK_STRUCTURE_TYPE_MEMORY_MAP_INFO,
  // Provided by VK_KHR_map_memory2
    VK_STRUCTURE_TYPE_MEMORY_UNMAP_INFO_KHR = VK_STRUCTURE_TYPE_MEMORY_UNMAP_INFO,
  // Provided by VK_EXT_surface_maintenance1
    VK_STRUCTURE_TYPE_SURFACE_PRESENT_MODE_EXT =
VK_STRUCTURE_TYPE_SURFACE_PRESENT_MODE_KHR,
  // Provided by VK_EXT_surface_maintenance1
    VK_STRUCTURE_TYPE_SURFACE_PRESENT_SCALING_CAPABILITIES_EXT =
VK_STRUCTURE_TYPE_SURFACE_PRESENT_SCALING_CAPABILITIES_KHR,
  // Provided by VK_EXT_surface_maintenance1
    VK_STRUCTURE_TYPE_SURFACE_PRESENT_MODE_COMPATIBILITY_EXT =
VK_STRUCTURE_TYPE_SURFACE_PRESENT_MODE_COMPATIBILITY_KHR,
  // Provided by VK_EXT_swapchain_maintenance1
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SWAPCHAIN_MAINTENANCE_1_FEATURES_EXT =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SWAPCHAIN_MAINTENANCE_1_FEATURES_KHR,
  // Provided by VK_EXT_swapchain_maintenance1
    VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_FENCE_INFO_EXT =
VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_FENCE_INFO_KHR,
  // Provided by VK_EXT_swapchain_maintenance1
    VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_MODES_CREATE_INFO_EXT =
VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_MODES_CREATE_INFO_KHR,
  // Provided by VK_EXT_swapchain_maintenance1
    VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_MODE_INFO_EXT =
VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_MODE_INFO_KHR,
  // Provided by VK_EXT_swapchain_maintenance1
    VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_SCALING_CREATE_INFO_EXT =
VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_SCALING_CREATE_INFO_KHR,
  // Provided by VK_EXT_swapchain_maintenance1
    VK_STRUCTURE_TYPE_RELEASE_SWAPCHAIN_IMAGES_INFO_EXT =
VK_STRUCTURE_TYPE_RELEASE_SWAPCHAIN_IMAGES_INFO_KHR,
  // Provided by VK_KHR_shader_integer_dot_product
```

```
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_FEATURES,
  // Provided by VK_KHR_shader_integer_dot_product
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_PROPERTIES,
  // Provided by VK_EXT_robustness2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_FEATURES_EXT =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_FEATURES_KHR,
  // Provided by VK_EXT_robustness2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_PROPERTIES_EXT =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_PROPERTIES_KHR,
  // Provided by VK_KHR_synchronization2
    VK_STRUCTURE_TYPE_MEMORY_BARRIER_2_KHR = VK_STRUCTURE_TYPE_MEMORY_BARRIER_2,
  // Provided by VK_KHR_synchronization2
    VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER_2_KHR =
VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER_2,
  // Provided by VK_KHR_synchronization2
    VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER_2_KHR =
VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER_2,
  // Provided by VK_KHR_synchronization2
    VK_STRUCTURE_TYPE_DEPENDENCY_INFO_KHR = VK_STRUCTURE_TYPE_DEPENDENCY_INFO,
  // Provided by VK_KHR_synchronization2
    VK_STRUCTURE_TYPE_SUBMIT_INFO_2_KHR = VK_STRUCTURE_TYPE_SUBMIT_INFO_2,
  // Provided by VK_KHR_synchronization2
    VK_STRUCTURE_TYPE_SEMAPHORE_SUBMIT_INFO_KHR =
VK_STRUCTURE_TYPE_SEMAPHORE_SUBMIT_INFO,
  // Provided by VK_KHR_synchronization2
    VK_STRUCTURE_TYPE_COMMAND_BUFFER_SUBMIT_INFO_KHR =
VK_STRUCTURE_TYPE_COMMAND_BUFFER_SUBMIT_INFO,
  // Provided by VK_KHR_synchronization2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SYNCHRONIZATION_2_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SYNCHRONIZATION_2_FEATURES,
  // Provided by VK_KHR_zero_initialize_workgroup_memory
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ZERO_INITIALIZE_WORKGROUP_MEMORY_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ZERO_INITIALIZE_WORKGROUP_MEMORY_FEATURES,
  // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_COPY_BUFFER_INFO_2_KHR = VK_STRUCTURE_TYPE_COPY_BUFFER_INFO_2,
  // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_COPY_IMAGE_INFO_2_KHR = VK_STRUCTURE_TYPE_COPY_IMAGE_INFO_2,
  // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_COPY_BUFFER_TO_IMAGE_INFO_2_KHR =
VK_STRUCTURE_TYPE_COPY_BUFFER_TO_IMAGE_INFO_2,
  // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_COPY_IMAGE_TO_BUFFER_INFO_2_KHR =
VK_STRUCTURE_TYPE_COPY_IMAGE_TO_BUFFER_INFO_2,
  // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_BLIT_IMAGE_INFO_2_KHR = VK_STRUCTURE_TYPE_BLIT_IMAGE_INFO_2,
  // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_RESOLVE_IMAGE_INFO_2_KHR =
VK_STRUCTURE_TYPE_RESOLVE_IMAGE_INFO_2,
  // Provided by VK_KHR_copy_commands2
```

```
    VK_STRUCTURE_TYPE_BUFFER_COPY_2_KHR = VK_STRUCTURE_TYPE_BUFFER_COPY_2,
  // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_IMAGE_COPY_2_KHR = VK_STRUCTURE_TYPE_IMAGE_COPY_2,
  // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_IMAGE_BLIT_2_KHR = VK_STRUCTURE_TYPE_IMAGE_BLIT_2,
  // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_BUFFER_IMAGE_COPY_2_KHR = VK_STRUCTURE_TYPE_BUFFER_IMAGE_COPY_2,
  // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_IMAGE_RESOLVE_2_KHR = VK_STRUCTURE_TYPE_IMAGE_RESOLVE_2,
  // Provided by VK_EXT_image_compression_control
    VK_STRUCTURE_TYPE_SUBRESOURCE_LAYOUT_2_EXT =
VK_STRUCTURE_TYPE_SUBRESOURCE_LAYOUT_2,
  // Provided by VK_EXT_image_compression_control
    VK_STRUCTURE_TYPE_IMAGE_SUBRESOURCE_2_EXT = VK_STRUCTURE_TYPE_IMAGE_SUBRESOURCE_2,
  // Provided by VK_KHR_format_feature_flags2
    VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_3_KHR = VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_3,
  // Provided by VK_EXT_present_mode_fifo_latest_ready
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_MODE_FIFO_LATEST_READY_FEATURES_EXT =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_MODE_FIFO_LATEST_READY_FEATURES_KHR,
  // Provided by VK_EXT_pipeline_properties
    VK_STRUCTURE_TYPE_PIPELINE_INFO_EXT = VK_STRUCTURE_TYPE_PIPELINE_INFO_KHR,
  // Provided by VK_KHR_maintenance4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_FEATURES,
  // Provided by VK_KHR_maintenance4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_PROPERTIES,
  // Provided by VK_KHR_maintenance4
    VK_STRUCTURE_TYPE_DEVICE_BUFFER_MEMORY_REQUIREMENTS_KHR =
VK_STRUCTURE_TYPE_DEVICE_BUFFER_MEMORY_REQUIREMENTS,
  // Provided by VK_KHR_maintenance4
    VK_STRUCTURE_TYPE_DEVICE_IMAGE_MEMORY_REQUIREMENTS_KHR =
VK_STRUCTURE_TYPE_DEVICE_IMAGE_MEMORY_REQUIREMENTS,
  // Provided by VK_KHR_shader_subgroup_rotate
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_ROTATE_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_ROTATE_FEATURES,
  // Provided by VK_EXT_depth_clamp_zero_one
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLAMP_ZERO_ONE_FEATURES_EXT =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLAMP_ZERO_ONE_FEATURES_KHR,
  // Provided by VK_EXT_pipeline_protected_access
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_PROTECTED_ACCESS_FEATURES_EXT =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_PROTECTED_ACCESS_FEATURES,
  // Provided by VK_KHR_maintenance5
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_FEATURES,
  // Provided by VK_KHR_maintenance5
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_PROPERTIES,
  // Provided by VK_KHR_maintenance5
    VK_STRUCTURE_TYPE_RENDERING_AREA_INFO_KHR = VK_STRUCTURE_TYPE_RENDERING_AREA_INFO,
  // Provided by VK_KHR_maintenance5
```

```
    VK_STRUCTURE_TYPE_DEVICE_IMAGE_SUBRESOURCE_INFO_KHR =
VK_STRUCTURE_TYPE_DEVICE_IMAGE_SUBRESOURCE_INFO,
  // Provided by VK_KHR_maintenance5
    VK_STRUCTURE_TYPE_SUBRESOURCE_LAYOUT_2_KHR =
VK_STRUCTURE_TYPE_SUBRESOURCE_LAYOUT_2,
  // Provided by VK_KHR_maintenance5
    VK_STRUCTURE_TYPE_IMAGE_SUBRESOURCE_2_KHR = VK_STRUCTURE_TYPE_IMAGE_SUBRESOURCE_2,
  // Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
    VK_STRUCTURE_TYPE_PIPELINE_CREATE_FLAGS_2_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_PIPELINE_CREATE_FLAGS_2_CREATE_INFO,
  // Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
    VK_STRUCTURE_TYPE_BUFFER_USAGE_FLAGS_2_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_BUFFER_USAGE_FLAGS_2_CREATE_INFO,
  // Provided by VK_EXT_shader_object
    VK_STRUCTURE_TYPE_SHADER_REQUIRED_SUBGROUP_SIZE_CREATE_INFO_EXT =
VK_STRUCTURE_TYPE_PIPELINE_SHADER_STAGE_REQUIRED_SUBGROUP_SIZE_CREATE_INFO,
  // Provided by VK_EXT_pipeline_library_group_handles
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_LIBRARY_GROUP_HANDLES_FEATURES_EXT =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_LIBRARY_GROUP_HANDLES_FEATURES_KHR,
  // Provided by VK_KHR_vertex_attribute_divisor
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_PROPERTIES,
  // Provided by VK_KHR_vertex_attribute_divisor
    VK_STRUCTURE_TYPE_PIPELINE_VERTEX_INPUT_DIVISOR_STATE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_PIPELINE_VERTEX_INPUT_DIVISOR_STATE_CREATE_INFO,
  // Provided by VK_KHR_vertex_attribute_divisor
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_FEATURES,
  // Provided by VK_KHR_shader_float_controls2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT_CONTROLS_2_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT_CONTROLS_2_FEATURES,
  // Provided by VK_KHR_index_type_uint8
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INDEX_TYPE_UINT8_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INDEX_TYPE_UINT8_FEATURES,
  // Provided by VK_KHR_line_rasterization
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_FEATURES,
  // Provided by VK_KHR_line_rasterization
    VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_LINE_STATE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_LINE_STATE_CREATE_INFO,
  // Provided by VK_KHR_line_rasterization
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_PROPERTIES,
  // Provided by VK_KHR_shader_expect_assume
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_EXPECT_ASSUME_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_EXPECT_ASSUME_FEATURES,
  // Provided by VK_KHR_maintenance6
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_FEATURES,
  // Provided by VK_KHR_maintenance6
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_PROPERTIES_KHR =
```

```
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_PROPERTIES,
  // Provided by VK_KHR_maintenance6
    VK_STRUCTURE_TYPE_BIND_MEMORY_STATUS_KHR = VK_STRUCTURE_TYPE_BIND_MEMORY_STATUS,
  // Provided by VK_KHR_maintenance6
    VK_STRUCTURE_TYPE_BIND_DESCRIPTOR_SETS_INFO_KHR =
VK_STRUCTURE_TYPE_BIND_DESCRIPTOR_SETS_INFO,
  // Provided by VK_KHR_maintenance6
    VK_STRUCTURE_TYPE_PUSH_CONSTANTS_INFO_KHR = VK_STRUCTURE_TYPE_PUSH_CONSTANTS_INFO,
  // Provided by VK_KHR_maintenance6 with VK_KHR_push_descriptor
    VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_INFO_KHR =
VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_INFO,
  // Provided by VK_KHR_maintenance6 with VK_KHR_push_descriptor
    VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_WITH_TEMPLATE_INFO_KHR =
VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_WITH_TEMPLATE_INFO,
  // Provided by VK_EXT_fragment_density_map_offset
    VK_STRUCTURE_TYPE_RENDERING_END_INFO_EXT =
VK_STRUCTURE_TYPE_RENDERING_END_INFO_KHR,
} VkStructureType;
```

3.13. API Name Aliases

A small number of APIs did not follow the naming conventions when initially defined. For consistency, when we discover an API name that violates the naming conventions, we rename it in the Specification, XML, and header files. For backwards compatibility, the original (incorrect) name is retained as a “typo alias”. The alias is legacy and should not be used, but will be retained indefinitely.

NOTE VK_STENCIL_FRONT_AND_BACK is an example of a *typo alias*. It was initially defined as part of VkStencilFaceFlagBits. Once the naming inconsistency was noticed, it was renamed to VK_STENCIL_FACE_FRONT_AND_BACK, and the old name was aliased to the correct name.

Chapter 4. Initialization

Before using Vulkan, an application **must** initialize it by loading the Vulkan commands, and creating a `VkInstance` object.

4.1. Command Function Pointers

Vulkan commands are not necessarily exposed by static linking on a platform. Commands to query function pointers for Vulkan commands are described below.

NOTE

When extensions are promoted or otherwise incorporated into another extension or Vulkan core version, command aliases may be included. Whilst the behavior of each command alias is identical, the behavior of retrieving each alias's function pointer is not. A function pointer for a given alias can only be retrieved if the extension or version that introduced that alias is supported and enabled, irrespective of whether any other alias is available.

Function pointers for all Vulkan commands **can** be obtained by calling:

```
// Provided by VK_VERSION_1_0
PFN_vkVoidFunction vkGetInstanceProcAddr(
    VkInstance                                  instance,
    const char*                                 pName);
```

- `instance` is the instance that the function pointer will be compatible with, or `NULL` for commands not dependent on any instance.
- `pName` is the name of the command to obtain.

`vkGetInstanceProcAddr` itself is obtained in a platform- and loader- specific manner. Typically, the loader library will export this command as a function symbol, so applications **can** link against the loader library, or load it dynamically and look up the symbol using platform-specific APIs.

The table below defines the various use cases for `vkGetInstanceProcAddr` and expected return value ("fp" is "function pointer") for each case. A valid returned function pointer ("fp") **must** not be `NULL`.

The returned function pointer is of type PFN_vkVoidFunction, and **must** be cast to the type of the command being queried before use.

Table 1. `vkGetInstanceProcAddr` *behavior*

`instance`	`pName`	**return value**
*[1]	`NULL`	undefined
invalid non-`NULL` instance	*[1]	undefined
`NULL`	*global command*[2]	fp
`NULL`	vkGetInstanceProcAddr	fp[5]

instance	pName	return value
instance	vkGetInstanceProcAddr	fp
instance	core *dispatchable command*	fp[3]
instance	enabled instance extension dispatchable command for instance	fp[3]
instance	available device extension[4] dispatchable command for instance	fp[3]
any other case, not covered above		NULL

1

"*" means any representable value for the parameter (including valid values, invalid values, and NULL).

2

The global commands are: vkEnumerateInstanceVersion, vkEnumerateInstanceExtensionProperties, vkEnumerateInstanceLayerProperties, and vkCreateInstance. Dispatchable commands are all other commands which are not global.

3

The returned function pointer **must** only be called with a dispatchable object (the first parameter) that is instance or a child of instance, e.g. VkInstance, VkPhysicalDevice, VkDevice, VkQueue, or VkCommandBuffer.

4

An “available device extension” is a device extension supported by any physical device enumerated by instance.

5

Starting with Vulkan 1.2, vkGetInstanceProcAddr can resolve itself with a NULL instance pointer.

Valid Usage (Implicit)

- VUID-vkGetInstanceProcAddr-instance-parameter
  If instance is not NULL, instance **must** be a valid VkInstance handle
- VUID-vkGetInstanceProcAddr-pName-parameter
  pName **must** be a null-terminated UTF-8 string

In order to support systems with multiple Vulkan implementations, the function pointers returned by vkGetInstanceProcAddr **may** point to dispatch code that calls a different real implementation for different VkDevice objects or their child objects. The overhead of the internal dispatch for VkDevice objects can be avoided by obtaining device-specific function pointers for any commands that use a

device or device-child object as their dispatchable object. Such function pointers **can** be obtained by calling:

```
// Provided by VK_VERSION_1_0
PFN_vkVoidFunction vkGetDeviceProcAddr(
    VkDevice                                    device,
    const char*                                 pName);
```

The table below defines the various use cases for `vkGetDeviceProcAddr` and expected return value ("fp" is "function pointer") for each case. A valid returned function pointer ("fp") **must** not be `NULL`.

The returned function pointer is of type PFN_vkVoidFunction, and **must** be cast to the type of the command being queried before use. The function pointer **must** only be called with a dispatchable object (the first parameter) that is `device` or a child of `device`.

Table 2. `vkGetDeviceProcAddr` behavior

`device`	`pName`	**return value**
`NULL`	*[1]	undefined
invalid device	*[1]	undefined
device	`NULL`	undefined
device	requested core version[2] device-level dispatchable command[3]	fp[4]
device	enabled extension device-level dispatchable command[3]	fp[4]
any other case, not covered above		`NULL`

1

"*" means any representable value for the parameter (including valid values, invalid values, and `NULL`).

2

Device-level commands which are part of the core version specified by VkApplicationInfo ::`apiVersion` when creating the instance will always return a valid function pointer. If the `maintenance5` feature is enabled, core commands beyond that version which are supported by the implementation will return `NULL`, otherwise the implementation **may** either return `NULL` or a function pointer. If a function pointer is returned, it **must** not be called.

3

In this function, device-level excludes all physical-device-level commands.

4

The returned function pointer **must** only be called with a dispatchable object (the first parameter) that is `device` or a child of `device` e.g. VkDevice, VkQueue, or VkCommandBuffer.

Valid Usage (Implicit)

- VUID-vkGetDeviceProcAddr-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDeviceProcAddr-pName-parameter
  `pName` **must** be a null-terminated UTF-8 string

The definition of PFN_vkVoidFunction is:

```
// Provided by VK_VERSION_1_0
typedef void (*PFN_vkVoidFunction)(void);
```

This type is returned from command function pointer queries, and **must** be cast to an actual command function pointer before use.

4.1.1. Extending Physical Device Core Functionality

New core physical-device-level functionality **can** be used when both VkPhysicalDeviceProperties::`apiVersion` and VkApplicationInfo::`apiVersion` are greater than or equal to the version of Vulkan that added the new functionality. The Vulkan version supported by a physical device **can** be obtained by calling vkGetPhysicalDeviceProperties.

4.1.2. Extending Physical Device From Device Extensions

When the `VK_KHR_get_physical_device_properties2` extension is enabled, or when both the instance and the physical-device versions are at least 1.1, physical-device-level functionality of a device extension **can** be used with a physical device if the corresponding extension is enumerated by vkEnumerateDeviceExtensionProperties for that physical device, even before a logical device has been created.

To obtain a function pointer for a physical-device-level command from a device extension, an application **can** use vkGetInstanceProcAddr. This function pointer **may** point to dispatch code, which calls a different real implementation for different `VkPhysicalDevice` objects. Applications **must** not use a VkPhysicalDevice in any command added by an extension or core version that is not supported by that physical device.

Device extensions **may** define structures that **can** be added to the `pNext` chain of physical-device-level commands.

4.2. Instances

There is no global state in Vulkan and all per-application state is stored in a `VkInstance` object. Creating a `VkInstance` object initializes the Vulkan library and allows the application to pass information about itself to the implementation.

Instances are represented by `VkInstance` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_HANDLE(VkInstance)
```

To query the version of instance-level functionality supported by the implementation, call:

```
// Provided by VK_VERSION_1_1
VkResult vkEnumerateInstanceVersion(
    uint32_t*                                   pApiVersion);
```

- `pApiVersion` is a pointer to a `uint32_t`, which is the version of Vulkan supported by instance-level functionality, encoded as described in Version Numbers.

NOTE

The intended behavior of vkEnumerateInstanceVersion is that an implementation **should** not need to perform memory allocations and **should** unconditionally return VK_SUCCESS. The loader, and any enabled layers, **may** return VK_ERROR_OUT_OF_HOST_MEMORY in the case of a failed memory allocation.

Valid Usage (Implicit)

- VUID-vkEnumerateInstanceVersion-pApiVersion-parameter
  `pApiVersion` **must** be a valid pointer to a `uint32_t` value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To create an instance object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateInstance(
    const VkInstanceCreateInfo*                 pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkInstance*                                 pInstance);
```

- `pCreateInfo` is a pointer to a VkInstanceCreateInfo structure controlling creation of the instance.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

- `pInstance` is a pointer to a VkInstance handle in which the resulting instance is returned.

`vkCreateInstance` verifies that the requested layers exist. If not, `vkCreateInstance` will return VK_ERROR_LAYER_NOT_PRESENT. Next `vkCreateInstance` verifies that the requested extensions are supported (e.g. in the implementation or in any enabled instance layer) and if any requested extension is not supported, `vkCreateInstance` **must** return VK_ERROR_EXTENSION_NOT_PRESENT. After verifying and enabling the instance layers and extensions the `VkInstance` object is created and returned to the application. If a requested extension is only supported by a layer, both the layer and the extension need to be specified at `vkCreateInstance` time for the creation to succeed.

Valid Usage

- VUID-vkCreateInstance-ppEnabledExtensionNames-01388
  All required extensions for each extension in the VkInstanceCreateInfo ::`ppEnabledExtensionNames` list **must** also be present in that list

Valid Usage (Implicit)

- VUID-vkCreateInstance-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkInstanceCreateInfo structure
- VUID-vkCreateInstance-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateInstance-pInstance-parameter
  `pInstance` **must** be a valid pointer to a VkInstance handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_EXTENSION_NOT_PRESENT
- VK_ERROR_INCOMPATIBLE_DRIVER
- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_LAYER_NOT_PRESENT
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkInstanceCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkInstanceCreateInfo {
    VkStructureType             sType;
    const void*                 pNext;
    VkInstanceCreateFlags       flags;
    const VkApplicationInfo*    pApplicationInfo;
    uint32_t                    enabledLayerCount;
    const char* const*          ppEnabledLayerNames;
    uint32_t                    enabledExtensionCount;
    const char* const*          ppEnabledExtensionNames;
} VkInstanceCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkInstanceCreateFlagBits indicating the behavior of the instance.
- `pApplicationInfo` is `NULL` or a pointer to a `VkApplicationInfo` structure. If not `NULL`, this information helps implementations recognize behavior inherent to classes of applications. VkApplicationInfo is defined in detail below.
- `enabledLayerCount` is the number of global layers to enable.
- `ppEnabledLayerNames` is a pointer to an array of `enabledLayerCount` null-terminated UTF-8 strings containing the names of layers to enable for the created instance. The layers are loaded in the order they are listed in this array, with the first array element being the closest to the application, and the last array element being the closest to the driver. See the Layers section for further details.
- `enabledExtensionCount` is the number of global extensions to enable.
- `ppEnabledExtensionNames` is a pointer to an array of `enabledExtensionCount` null-terminated UTF-8 strings containing the names of extensions to enable.

To capture events that occur while creating or destroying an instance, an application **can** link a VkDebugUtilsMessengerCreateInfoEXT structure to the `pNext` chain of the `VkInstanceCreateInfo` structure passed to `vkCreateInstance`. This callback is only valid for the duration of the vkCreateInstance and the vkDestroyInstance call. Use vkCreateDebugUtilsMessengerEXT to create persistent callback objects.

Valid Usage

- VUID-VkInstanceCreateInfo-pNext-04926
  If the `pNext` chain of `VkInstanceCreateInfo` includes a VkDebugUtilsMessengerCreateInfoEXT structure, the list of enabled extensions in `ppEnabledExtensionNames` **must** contain `VK_EXT_debug_utils`
- VUID-VkInstanceCreateInfo-pNext-06779
  If the `pNext` chain includes a VkExportMetalObjectCreateInfoEXT structure, its `exportObjectType` member **must** be either VK_EXPORT_METAL_OBJECT_TYPE_METAL_DEVICE_BIT_EXT or

VK_EXPORT_METAL_OBJECT_TYPE_METAL_COMMAND_QUEUE_BIT_EXT

- VUID-VkInstanceCreateInfo-flags-06559
  If `flags` has the VK_INSTANCE_CREATE_ENUMERATE_PORTABILITY_BIT_KHR bit set, the list of enabled extensions in `ppEnabledExtensionNames` **must** contain `VK_KHR_portability_enumeration`
- VUID-VkInstanceCreateInfo-pNext-10242
  If the `pNext` chain of `VkInstanceCreateInfo` includes a VkLayerSettingsCreateInfoEXT structure, the list of enabled extensions in `ppEnabledExtensionNames` **must** contain VK_EXT_layer_settings
- VUID-VkInstanceCreateInfo-pNext-10243
  If the `pNext` chain of `VkInstanceCreateInfo` includes a VkValidationFeaturesEXT structure, the list of enabled extensions in `ppEnabledExtensionNames` **must** contain VK_EXT_validation_features

Valid Usage (Implicit)

- VUID-VkInstanceCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_INSTANCE_CREATE_INFO
- VUID-VkInstanceCreateInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkDebugUtilsMessengerCreateInfoEXT, VkExportMetalObjectCreateInfoEXT, VkLayerSettingsCreateInfoEXT, or VkValidationFeaturesEXT
- VUID-VkInstanceCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique, with the exception of structures of type VkDebugUtilsMessengerCreateInfoEXT, VkExportMetalObjectCreateInfoEXT, or VkLayerSettingsCreateInfoEXT
- VUID-VkInstanceCreateInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkInstanceCreateFlagBits values
- VUID-VkInstanceCreateInfo-pApplicationInfo-parameter
  If `pApplicationInfo` is not `NULL`, `pApplicationInfo` **must** be a valid pointer to a valid VkApplicationInfo structure
- VUID-VkInstanceCreateInfo-ppEnabledLayerNames-parameter
  If `enabledLayerCount` is not `0`, `ppEnabledLayerNames` **must** be a valid pointer to an array of `enabledLayerCount` null-terminated UTF-8 strings
- VUID-VkInstanceCreateInfo-ppEnabledExtensionNames-parameter
  If `enabledExtensionCount` is not `0`, `ppEnabledExtensionNames` **must** be a valid pointer to an array of `enabledExtensionCount` null-terminated UTF-8 strings

```
// Provided by VK_VERSION_1_0
typedef enum VkInstanceCreateFlagBits {
  // Provided by VK_KHR_portability_enumeration
    VK_INSTANCE_CREATE_ENUMERATE_PORTABILITY_BIT_KHR = 0x00000001,
} VkInstanceCreateFlagBits;
```

- VK_INSTANCE_CREATE_ENUMERATE_PORTABILITY_BIT_KHR specifies that the instance will enumerate available Vulkan Portability-compliant physical devices and groups in addition to the Vulkan physical devices and groups that are enumerated by default.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkInstanceCreateFlags;
```

VkInstanceCreateFlags is a bitmask type for setting a mask of zero or more VkInstanceCreateFlagBits.

When creating a Vulkan instance for which you wish to enable or disable specific validation features, add a VkValidationFeaturesEXT structure to the pNext chain of the VkInstanceCreateInfo structure, specifying the features to be enabled or disabled.

```
// Provided by VK_EXT_validation_features
typedef struct VkValidationFeaturesEXT {
    VkStructureType                      sType;
    const void*                          pNext;
    uint32_t                             enabledValidationFeatureCount;
    const VkValidationFeatureEnableEXT*  pEnabledValidationFeatures;
    uint32_t                             disabledValidationFeatureCount;
    const VkValidationFeatureDisableEXT* pDisabledValidationFeatures;
} VkValidationFeaturesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- enabledValidationFeatureCount is the number of features to enable.
- pEnabledValidationFeatures is a pointer to an array of VkValidationFeatureEnableEXT values specifying the validation features to be enabled.
- disabledValidationFeatureCount is the number of features to disable.
- pDisabledValidationFeatures is a pointer to an array of VkValidationFeatureDisableEXT values specifying the validation features to be disabled.

Valid Usage

- VUID-VkValidationFeaturesEXT-pEnabledValidationFeatures-02967
  If the pEnabledValidationFeatures array contains VK_VALIDATION_FEATURE_ENABLE_GPU_ASSISTED_RESERVE_BINDING_SLOT_EXT, then

it **must** also contain VK_VALIDATION_FEATURE_ENABLE_GPU_ASSISTED_EXT or VK_VALIDATION_FEATURE_ENABLE_DEBUG_PRINTF_EXT

Valid Usage (Implicit)

- VUID-VkValidationFeaturesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VALIDATION_FEATURES_EXT
- VUID-VkValidationFeaturesEXT-pEnabledValidationFeatures-parameter
  If `enabledValidationFeatureCount` is not `0`, `pEnabledValidationFeatures` **must** be a valid pointer to an array of `enabledValidationFeatureCount` valid VkValidationFeatureEnableEXT values
- VUID-VkValidationFeaturesEXT-pDisabledValidationFeatures-parameter
  If `disabledValidationFeatureCount` is not `0`, `pDisabledValidationFeatures` **must** be a valid pointer to an array of `disabledValidationFeatureCount` valid VkValidationFeatureDisableEXT values

Structure Chaining

Extends the structures

- VkComputePipelineCreateInfo
- VkGraphicsPipelineCreateInfo
- VkInstanceCreateInfo
- VkPipelineShaderStageCreateInfo
- VkRayTracingPipelineCreateInfoKHR
- VkShaderCreateInfoEXT
- VkShaderModuleCreateInfo

Possible values of elements of the VkValidationFeaturesEXT::`pEnabledValidationFeatures` array, specifying validation features to be enabled, are:

```
// Provided by VK_EXT_validation_features
typedef enum VkValidationFeatureEnableEXT {
    VK_VALIDATION_FEATURE_ENABLE_GPU_ASSISTED_EXT = 0,
    VK_VALIDATION_FEATURE_ENABLE_GPU_ASSISTED_RESERVE_BINDING_SLOT_EXT = 1,
    VK_VALIDATION_FEATURE_ENABLE_BEST_PRACTICES_EXT = 2,
    VK_VALIDATION_FEATURE_ENABLE_DEBUG_PRINTF_EXT = 3,
    VK_VALIDATION_FEATURE_ENABLE_SYNCHRONIZATION_VALIDATION_EXT = 4,
} VkValidationFeatureEnableEXT;
```

- VK_VALIDATION_FEATURE_ENABLE_GPU_ASSISTED_EXT specifies that GPU-assisted validation is enabled. Activating this feature instruments shader programs to generate additional

diagnostic data. This feature is disabled by default.

- VK_VALIDATION_FEATURE_ENABLE_GPU_ASSISTED_RESERVE_BINDING_SLOT_EXT specifies that the validation layers reserve a descriptor set binding slot for their own use. The layer reports a value for VkPhysicalDeviceLimits::maxBoundDescriptorSets that is one less than the value reported by the device. If the device supports the binding of only one descriptor set, the validation layer does not perform GPU-assisted validation. This feature is disabled by default.
- VK_VALIDATION_FEATURE_ENABLE_BEST_PRACTICES_EXT specifies that Vulkan best-practices validation is enabled. Activating this feature enables the output of warnings related to common misuse of the API, but which are not explicitly prohibited by the specification. This feature is disabled by default.
- VK_VALIDATION_FEATURE_ENABLE_DEBUG_PRINTF_EXT specifies that the layers will process debugPrintfEXT operations in shaders and send the resulting output to the debug callback. This feature is disabled by default.
- VK_VALIDATION_FEATURE_ENABLE_SYNCHRONIZATION_VALIDATION_EXT specifies that Vulkan synchronization validation is enabled. This feature reports resource access conflicts due to missing or incorrect synchronization operations between actions (Draw, Copy, Dispatch, Blit) reading or writing the same regions of memory. This feature is disabled by default.

Possible values of elements of the VkValidationFeaturesEXT::pDisabledValidationFeatures array, specifying validation features to be disabled, are:

```
// Provided by VK_EXT_validation_features
typedef enum VkValidationFeatureDisableEXT {
    VK_VALIDATION_FEATURE_DISABLE_ALL_EXT = 0,
    VK_VALIDATION_FEATURE_DISABLE_SHADERS_EXT = 1,
    VK_VALIDATION_FEATURE_DISABLE_THREAD_SAFETY_EXT = 2,
    VK_VALIDATION_FEATURE_DISABLE_API_PARAMETERS_EXT = 3,
    VK_VALIDATION_FEATURE_DISABLE_OBJECT_LIFETIMES_EXT = 4,
    VK_VALIDATION_FEATURE_DISABLE_CORE_CHECKS_EXT = 5,
    VK_VALIDATION_FEATURE_DISABLE_UNIQUE_HANDLES_EXT = 6,
    VK_VALIDATION_FEATURE_DISABLE_SHADER_VALIDATION_CACHE_EXT = 7,
} VkValidationFeatureDisableEXT;
```

- VK_VALIDATION_FEATURE_DISABLE_ALL_EXT specifies that all validation checks are disabled.
- VK_VALIDATION_FEATURE_DISABLE_SHADERS_EXT specifies that shader validation, both runtime and standalone, is disabled. This validation occurs inside VkShaderCreateInfoEXT and VkShaderModuleCreateInfo. This feature is enabled by default.
- VK_VALIDATION_FEATURE_DISABLE_THREAD_SAFETY_EXT specifies that thread safety validation is disabled. This feature is enabled by default.
- VK_VALIDATION_FEATURE_DISABLE_API_PARAMETERS_EXT specifies that stateless parameter validation is disabled. This feature is enabled by default.
- VK_VALIDATION_FEATURE_DISABLE_OBJECT_LIFETIMES_EXT specifies that object lifetime validation is disabled. This feature is enabled by default.
- VK_VALIDATION_FEATURE_DISABLE_CORE_CHECKS_EXT specifies that core validation checks

are disabled. This feature is enabled by default. If this feature is disabled, VK_VALIDATION_FEATURE_DISABLE_SHADERS_EXT is implied.

- VK_VALIDATION_FEATURE_DISABLE_UNIQUE_HANDLES_EXT specifies that protection against duplicate non-dispatchable object handles is disabled. This feature is enabled by default.
- VK_VALIDATION_FEATURE_DISABLE_SHADER_VALIDATION_CACHE_EXT specifies that there will be no caching of shader validation results and every shader will be validated on every application execution. Shader validation caching is enabled by default.

NOTE	Disabling checks such as parameter validation and object lifetime validation prevents the reporting of error conditions that can cause other validation checks to behave incorrectly or crash. Some validation checks assume that their inputs are already valid and do not always revalidate them.

To create a Vulkan instance with a specific configuration of layer settings, add VkLayerSettingsCreateInfoEXT structures to the pNext chain of the VkInstanceCreateInfo structure, specifying the settings to be configured.

```
// Provided by VK_EXT_layer_settings
typedef struct VkLayerSettingsCreateInfoEXT {
    VkStructureType             sType;
    const void*                 pNext;
    uint32_t                    settingCount;
    const VkLayerSettingEXT*    pSettings;
} VkLayerSettingsCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- settingCount is the number of settings to configure.
- pSettings is a pointer to an array of settingCount VkLayerSettingEXT values specifying the settings to be configured.

Valid Usage (Implicit)

- VUID-VkLayerSettingsCreateInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_LAYER_SETTINGS_CREATE_INFO_EXT
- VUID-VkLayerSettingsCreateInfoEXT-pSettings-parameter
  If settingCount is not 0, pSettings **must** be a valid pointer to an array of settingCount valid VkLayerSettingEXT structures

Structure Chaining

Extends the structure

- VkInstanceCreateInfo

The values of elements of the VkLayerSettingsCreateInfoEXT::pSettings array, specifying layer settings to be configured, are:

```
// Provided by VK_EXT_layer_settings
typedef struct VkLayerSettingEXT {
    const char*              pLayerName;
    const char*              pSettingName;
    VkLayerSettingTypeEXT    type;
    uint32_t                 valueCount;
    const void*              pValues;
} VkLayerSettingEXT;
```

- pLayerName is a pointer to a null-terminated UTF-8 string naming the layer to configure the setting from.
- pSettingName is a pointer to a null-terminated UTF-8 string naming the setting to configure. Values of pSettingName that are unknown to the layer are ignored.
- type is a VkLayerSettingTypeEXT value specifying the type of the pValues values.
- valueCount is the number of values used to configure the layer setting.
- pValues is a pointer to an array of valueCount values of the type indicated by type to configure the layer setting.

When multiple VkLayerSettingsCreateInfoEXT structures are chained and the same pSettingName is referenced for the same pLayerName, the value of the first reference of the layer setting is used.

Valid Usage

- VUID-VkLayerSettingEXT-valueCount-10070
  If valueCount is not 0, pValues **must** be a valid pointer to an array of valueCount values of the type indicated by type

Valid Usage (Implicit)

- VUID-VkLayerSettingEXT-pLayerName-parameter
  pLayerName **must** be a null-terminated UTF-8 string
- VUID-VkLayerSettingEXT-pSettingName-parameter
  pSettingName **must** be a null-terminated UTF-8 string
- VUID-VkLayerSettingEXT-type-parameter
  type **must** be a valid VkLayerSettingTypeEXT value

Possible values of VkLayerSettingEXT::type, specifying the type of the data returned in VkLayerSettingEXT::pValues, are:

```
// Provided by VK_EXT_layer_settings
typedef enum VkLayerSettingTypeEXT {
    VK_LAYER_SETTING_TYPE_BOOL32_EXT = 0,
    VK_LAYER_SETTING_TYPE_INT32_EXT = 1,
    VK_LAYER_SETTING_TYPE_INT64_EXT = 2,
    VK_LAYER_SETTING_TYPE_UINT32_EXT = 3,
    VK_LAYER_SETTING_TYPE_UINT64_EXT = 4,
    VK_LAYER_SETTING_TYPE_FLOAT32_EXT = 5,
    VK_LAYER_SETTING_TYPE_FLOAT64_EXT = 6,
    VK_LAYER_SETTING_TYPE_STRING_EXT = 7,
} VkLayerSettingTypeEXT;
```

- VK_LAYER_SETTING_TYPE_BOOL32_EXT specifies that the layer setting's type is VkBool32.
- VK_LAYER_SETTING_TYPE_INT32_EXT specifies that the layer setting's type is signed 32-bit integer.
- VK_LAYER_SETTING_TYPE_INT64_EXT specifies that the layer setting's type is signed 64-bit integer.
- VK_LAYER_SETTING_TYPE_UINT32_EXT specifies that the layer setting's type is unsigned 32-bit integer.
- VK_LAYER_SETTING_TYPE_UINT64_EXT specifies that the layer setting's type is unsigned 64-bit integer.
- VK_LAYER_SETTING_TYPE_FLOAT32_EXT specifies that the layer setting's type is 32-bit floating-point.
- VK_LAYER_SETTING_TYPE_FLOAT64_EXT specifies that the layer setting's type is 64-bit floating-point.
- VK_LAYER_SETTING_TYPE_STRING_EXT specifies that the layer setting's type is a pointer to a null-terminated UTF-8 string.

The VkApplicationInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkApplicationInfo {
    VkStructureType    sType;
    const void*        pNext;
    const char*        pApplicationName;
    uint32_t           applicationVersion;
    const char*        pEngineName;
    uint32_t           engineVersion;
    uint32_t           apiVersion;
} VkApplicationInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pApplicationName is NULL or is a pointer to a null-terminated UTF-8 string containing the name of

the application.

- `applicationVersion` is an unsigned integer variable containing the developer-supplied version number of the application.
- `pEngineName` is `NULL` or is a pointer to a null-terminated UTF-8 string containing the name of the engine (if any) used to create the application.
- `engineVersion` is an unsigned integer variable containing the developer-supplied version number of the engine used to create the application.
- `apiVersion` **must** be the highest version of Vulkan that the application is designed to use, encoded as described in Version Numbers. The patch version number specified in `apiVersion` is ignored when creating an instance object. The variant version of the instance **must** match that requested in `apiVersion`.

Vulkan 1.0 implementations were required to return VK_ERROR_INCOMPATIBLE_DRIVER if `apiVersion` was larger than 1.0. Implementations that support Vulkan 1.1 or later **must** not return VK_ERROR_INCOMPATIBLE_DRIVER for any value of `apiVersion` .

NOTE

Because Vulkan 1.0 implementations **may** fail with VK_ERROR_INCOMPATIBLE_DRIVER, applications **should** determine the version of Vulkan available before calling vkCreateInstance. If vkGetInstanceProcAddr returns `NULL` for vkEnumerateInstanceVersion, it is a Vulkan 1.0 implementation. Otherwise, the application **can** call vkEnumerateInstanceVersion to determine the version of Vulkan.

As long as the instance supports at least Vulkan 1.1, an application **can** use different versions of Vulkan with an instance than it does with a device or physical device.

NOTE

The Khronos validation layers will treat `apiVersion` as the highest API version the application targets, and will validate API usage against the minimum of that version and the implementation version (instance or device, depending on context). If an application tries to use functionality from a greater version than this, a validation error will be triggered.

For example, if the instance supports Vulkan 1.1 and three physical devices support Vulkan 1.0, Vulkan 1.1, and Vulkan 1.2, respectively, and if the application sets `apiVersion` to 1.2, the application **can** use the following versions of Vulkan:

- Vulkan 1.0 **can** be used with the instance and with all physical devices.
- Vulkan 1.1 **can** be used with the instance and with the physical devices that support Vulkan 1.1 and Vulkan 1.2.
- Vulkan 1.2 **can** be used with the physical device that supports Vulkan 1.2.

If we modify the above example so that the application sets `apiVersion` to 1.1, then the application **must** not use Vulkan 1.2 functionality on the physical device that supports Vulkan 1.2.

NOTE

Providing a `NULL` VkInstanceCreateInfo::`pApplicationInfo` or providing an `apiVersion`

> of 0 is equivalent to providing an `apiVersion` of `VK_MAKE_API_VERSION(0,1,0,0)`.

Valid Usage

- VUID-VkApplicationInfo-apiVersion-04010
  If `apiVersion` is not `0`, then it **must** be greater than or equal to VK_API_VERSION_1_0

Valid Usage (Implicit)

- VUID-VkApplicationInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_APPLICATION_INFO
- VUID-VkApplicationInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkApplicationInfo-pApplicationName-parameter
  If `pApplicationName` is not `NULL`, `pApplicationName` **must** be a null-terminated UTF-8 string
- VUID-VkApplicationInfo-pEngineName-parameter
  If `pEngineName` is not `NULL`, `pEngineName` **must** be a null-terminated UTF-8 string

To destroy an instance, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyInstance(
    VkInstance                                  instance,
    const VkAllocationCallbacks*                pAllocator);
```

- `instance` is the handle of the instance to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Prior to destroying an instance, an application is responsible for destroying/freeing any Vulkan objects with explicit `vkDestroy*` or `vkFree*` commands that were created using that instance, or any VkPhysicalDevice object retrieved from it, as the first parameter of the corresponding `vkCreate*` or `vkAllocate*` command.

Valid Usage

- VUID-vkDestroyInstance-instance-00629
  All child objects that were created with `instance` or with a VkPhysicalDevice retrieved from it, and that **can** be destroyed or freed, **must** have been destroyed or freed prior to destroying `instance`
- VUID-vkDestroyInstance-instance-00630
  If `VkAllocationCallbacks` were provided when `instance` was created, a compatible set of callbacks **must** be provided here

- VUID-vkDestroyInstance-instance-00631
  If no `VkAllocationCallbacks` were provided when `instance` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyInstance-instance-parameter
  If `instance` is not `NULL`, `instance` **must** be a valid VkInstance handle
- VUID-vkDestroyInstance-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure

Host Synchronization

- Host access to `instance` **must** be externally synchronized
- Host access to all `VkPhysicalDevice` objects enumerated from `instance` **must** be externally synchronized

Chapter 5. Devices and Queues

Once Vulkan is initialized, devices and queues are the primary objects used to interact with a Vulkan implementation.

Vulkan separates the concept of *physical* and *logical* devices. A physical device usually represents a single complete implementation of Vulkan (excluding instance-level functionality) available to the host, of which there are a finite number. A logical device represents an instance of that implementation with its own state and resources independent of other logical devices.

Physical devices **cannot** be independently destroyed, and are instead destroyed with the VkInstance that they were retrieved from.

Physical devices are represented by VkPhysicalDevice handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_HANDLE(VkPhysicalDevice)
```

5.1. Physical Devices

To retrieve a list of physical device objects representing the physical devices installed in the system, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEnumeratePhysicalDevices(
    VkInstance                                  instance,
    uint32_t*                                   pPhysicalDeviceCount,
    VkPhysicalDevice*                           pPhysicalDevices);
```

- instance is a handle to a Vulkan instance previously created with vkCreateInstance.
- pPhysicalDeviceCount is a pointer to an integer related to the number of physical devices available or queried, as described below.
- pPhysicalDevices is either NULL or a pointer to an array of VkPhysicalDevice handles.

If pPhysicalDevices is NULL, then the number of physical devices available is returned in pPhysicalDeviceCount. Otherwise, pPhysicalDeviceCount **must** point to a variable set by the application to the number of elements in the pPhysicalDevices array, and on return the variable is overwritten with the number of handles actually written to pPhysicalDevices. If pPhysicalDeviceCount is less than the number of physical devices available, at most pPhysicalDeviceCount structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available physical devices were returned.

Valid Usage (Implicit)

- VUID-vkEnumeratePhysicalDevices-instance-parameter

`instance` **must** be a valid VkInstance handle

- VUID-vkEnumeratePhysicalDevices-pPhysicalDeviceCount-parameter
  `pPhysicalDeviceCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkEnumeratePhysicalDevices-pPhysicalDevices-parameter
  If the value referenced by `pPhysicalDeviceCount` is not `0`, and `pPhysicalDevices` is not `NULL`, `pPhysicalDevices` **must** be a valid pointer to an array of `pPhysicalDeviceCount` VkPhysicalDevice handles

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To query general properties of physical devices once enumerated, call:

WARNING This functionality is superseded by vkGetPhysicalDeviceProperties2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceProperties(
    VkPhysicalDevice                            physicalDevice,
    VkPhysicalDeviceProperties*                 pProperties);
```

- `physicalDevice` is the handle to the physical device whose properties will be queried.
- `pProperties` is a pointer to a VkPhysicalDeviceProperties structure in which properties are returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceProperties-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceProperties-pProperties-parameter
  `pProperties` **must** be a valid pointer to a VkPhysicalDeviceProperties structure

The VkPhysicalDeviceProperties structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPhysicalDeviceProperties {
    uint32_t                            apiVersion;
    uint32_t                            driverVersion;
    uint32_t                            vendorID;
    uint32_t                            deviceID;
    VkPhysicalDeviceType                deviceType;
    char                                deviceName[VK_MAX_PHYSICAL_DEVICE_NAME_SIZE];
    uint8_t                             pipelineCacheUUID[VK_UUID_SIZE];
    VkPhysicalDeviceLimits              limits;
    VkPhysicalDeviceSparseProperties    sparseProperties;
} VkPhysicalDeviceProperties;
```

- apiVersion is the version of Vulkan supported by the device, encoded as described in Version Numbers.
- driverVersion is the vendor-specified version of the driver.
- vendorID is a unique identifier for the *vendor* (see below) of the physical device.
- deviceID is a unique identifier for the physical device among devices available from the vendor.
- deviceType is a VkPhysicalDeviceType specifying the type of device.
- deviceName is an array of VK_MAX_PHYSICAL_DEVICE_NAME_SIZE char containing a null-terminated UTF-8 string which is the name of the device.
- pipelineCacheUUID is an array of VK_UUID_SIZE uint8_t values representing a universally unique identifier for the device.
- limits is the VkPhysicalDeviceLimits structure specifying device-specific limits of the physical device. See Limits for details.
- sparseProperties is the VkPhysicalDeviceSparseProperties structure specifying various sparse related properties of the physical device. See Sparse Properties for details.

NOTE

The value of apiVersion **may** be different than the version returned by vkEnumerateInstanceVersion; either higher or lower. In such cases, the application **must** not use functionality that exceeds the version of Vulkan associated with a given object. The pApiVersion parameter returned by vkEnumerateInstanceVersion is the version associated with a VkInstance and its children, except for a VkPhysicalDevice and its children. VkPhysicalDeviceProperties::apiVersion is the version associated with a VkPhysicalDevice and its children.

NOTE

The encoding of driverVersion is implementation-defined. It **may** not use the same encoding as apiVersion. Applications should follow information from the *vendor* on how to extract the version information from driverVersion.

On implementations that claim support for the Roadmap 2022 profile, the major and minor version expressed by apiVersion **must** be at least Vulkan 1.3.

The `vendorID` and `deviceID` fields are provided to allow applications to adapt to device characteristics that are not adequately exposed by other Vulkan queries.

NOTE	These **may** include performance profiles, hardware errata, or other characteristics.

The *vendor* identified by `vendorID` is the entity responsible for the most salient characteristics of the underlying implementation of the VkPhysicalDevice being queried.

NOTE	For example, in the case of a discrete GPU implementation, this **should** be the GPU chipset vendor. In the case of a hardware accelerator integrated into a system-on-chip (SoC), this **should** be the supplier of the silicon IP used to create the accelerator.

If the vendor has a PCI vendor ID, the low 16 bits of `vendorID` **must** contain that PCI vendor ID, and the remaining bits **must** be zero. Otherwise, the value returned **must** be a valid Khronos vendor ID, obtained as described in the Vulkan Documentation and Extensions: Procedures and Conventions document in the section “Registering a Vendor ID with Khronos”. Khronos vendor IDs are allocated starting at 0x10000, to distinguish them from the PCI vendor ID namespace. Khronos vendor IDs are symbolically defined in the VkVendorId type.

The vendor is also responsible for the value returned in `deviceID`. If the implementation is driven primarily by a PCI device with a PCI device ID, the low 16 bits of `deviceID` **must** contain that PCI device ID, and the remaining bits **must** be zero. Otherwise, the choice of what values to return **may** be dictated by operating system or platform policies - but **should** uniquely identify both the device version and any major configuration options (for example, core count in the case of multicore devices).

NOTE	The same device ID **should** be used for all physical implementations of that device version and configuration. For example, all uses of a specific silicon IP GPU version and configuration **should** use the same device ID, even if those uses occur in different SoCs.

Khronos vendor IDs which **may** be returned in VkPhysicalDeviceProperties::`vendorID` are:

```
// Provided by VK_VERSION_1_0
typedef enum VkVendorId {
    VK_VENDOR_ID_KHRONOS = 0x10000,
    VK_VENDOR_ID_VIV = 0x10001,
    VK_VENDOR_ID_VSI = 0x10002,
    VK_VENDOR_ID_KAZAN = 0x10003,
    VK_VENDOR_ID_CODEPLAY = 0x10004,
    VK_VENDOR_ID_MESA = 0x10005,
    VK_VENDOR_ID_POCL = 0x10006,
    VK_VENDOR_ID_MOBILEYE = 0x10007,
    VK_VENDOR_ID_APE = 0x10008,
    VK_VENDOR_ID_APEX = 0x10009,
} VkVendorId;
```

NOTE

Khronos vendor IDs may be allocated by vendors at any time. Only the latest canonical versions of this Specification, of the corresponding `vk.xml` API Registry, and of the corresponding `vulkan_core.h` header file **must** contain all reserved Khronos vendor IDs.

Only Khronos vendor IDs are given symbolic names at present. PCI vendor IDs returned by the implementation can be looked up in the PCI-SIG database.

VK_MAX_PHYSICAL_DEVICE_NAME_SIZE is the length in `char` values of an array containing a physical device name string, as returned in VkPhysicalDeviceProperties::`deviceName`.

```
#define VK_MAX_PHYSICAL_DEVICE_NAME_SIZE  256U
```

The physical device types which **may** be returned in VkPhysicalDeviceProperties::`deviceType` are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPhysicalDeviceType {
    VK_PHYSICAL_DEVICE_TYPE_OTHER = 0,
    VK_PHYSICAL_DEVICE_TYPE_INTEGRATED_GPU = 1,
    VK_PHYSICAL_DEVICE_TYPE_DISCRETE_GPU = 2,
    VK_PHYSICAL_DEVICE_TYPE_VIRTUAL_GPU = 3,
    VK_PHYSICAL_DEVICE_TYPE_CPU = 4,
} VkPhysicalDeviceType;
```

- VK_PHYSICAL_DEVICE_TYPE_OTHER - the device does not match any other available types.
- VK_PHYSICAL_DEVICE_TYPE_INTEGRATED_GPU - the device is typically one embedded in or tightly coupled with the host.
- VK_PHYSICAL_DEVICE_TYPE_DISCRETE_GPU - the device is typically a separate processor connected to the host via an interlink.
- VK_PHYSICAL_DEVICE_TYPE_VIRTUAL_GPU - the device is typically a virtual node in a virtualization environment.
- VK_PHYSICAL_DEVICE_TYPE_CPU - the device is typically running on the same processors as the host.

The physical device type is advertised for informational purposes only, and does not directly affect the operation of the system. However, the device type **may** correlate with other advertised properties or capabilities of the system, such as how many memory heaps there are.

To query general properties of physical devices once enumerated, call:

```
// Provided by VK_VERSION_1_1
void vkGetPhysicalDeviceProperties2(
    VkPhysicalDevice                            physicalDevice,
    VkPhysicalDeviceProperties2*                pProperties);
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to vkGetPhysicalDeviceProperties2
void vkGetPhysicalDeviceProperties2KHR(
    VkPhysicalDevice                            physicalDevice,
    VkPhysicalDeviceProperties2*                pProperties);
```

- `physicalDevice` is the handle to the physical device whose properties will be queried.
- `pProperties` is a pointer to a VkPhysicalDeviceProperties2 structure in which properties are returned.

Each structure in `pProperties` and its `pNext` chain contains members corresponding to implementation-dependent properties, behaviors, or limits. `vkGetPhysicalDeviceProperties2` fills in each member to specify the corresponding value for the implementation.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceProperties2-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceProperties2-pProperties-parameter
  `pProperties` **must** be a valid pointer to a VkPhysicalDeviceProperties2 structure

The `VkPhysicalDeviceProperties2` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceProperties2 {
    VkStructureType              sType;
    void*                        pNext;
    VkPhysicalDeviceProperties   properties;
} VkPhysicalDeviceProperties2;
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to VkPhysicalDeviceProperties2
typedef VkPhysicalDeviceProperties2 VkPhysicalDeviceProperties2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `properties` is a VkPhysicalDeviceProperties structure describing properties of the physical device. This structure is written with the same values as if it were written by vkGetPhysicalDeviceProperties.

The `pNext` chain of this structure is used to extend the structure with properties defined by extensions.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceProperties2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROPERTIES_2
- VUID-VkPhysicalDeviceProperties2-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkPhysicalDeviceAccelerationStructurePropertiesKHR, VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT, VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR, VkPhysicalDeviceConservativeRasterizationPropertiesEXT, VkPhysicalDeviceCooperativeMatrixPropertiesKHR, VkPhysicalDeviceCopyMemoryIndirectPropertiesKHR, VkPhysicalDeviceCustomBorderColorPropertiesEXT, VkPhysicalDeviceDepthStencilResolveProperties, VkPhysicalDeviceDescriptorBufferDensityMapPropertiesEXT, VkPhysicalDeviceDescriptorBufferPropertiesEXT, VkPhysicalDeviceDescriptorHeapPropertiesEXT, VkPhysicalDeviceDescriptorIndexingProperties, VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT, VkPhysicalDeviceDiscardRectanglePropertiesEXT, VkPhysicalDeviceDriverProperties, VkPhysicalDeviceDrmPropertiesEXT, VkPhysicalDeviceExtendedDynamicState3PropertiesEXT, VkPhysicalDeviceExternalMemoryHostPropertiesEXT, VkPhysicalDeviceFaultPropertiesKHR, VkPhysicalDeviceFloatControlsProperties, VkPhysicalDeviceFragmentDensityMap2PropertiesEXT, VkPhysicalDeviceFragmentDensityMapOffsetPropertiesEXT, VkPhysicalDeviceFragmentDensityMapPropertiesEXT, VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR, VkPhysicalDeviceFragmentShadingRatePropertiesKHR, VkPhysicalDeviceGraphicsPipelineLibraryPropertiesEXT, VkPhysicalDeviceHostImageCopyProperties, VkPhysicalDeviceIDProperties, VkPhysicalDeviceInlineUniformBlockProperties, VkPhysicalDeviceLayeredApiPropertiesListKHR, VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT, VkPhysicalDeviceLineRasterizationProperties, VkPhysicalDeviceMaintenance10PropertiesKHR, VkPhysicalDeviceMaintenance3Properties, VkPhysicalDeviceMaintenance4Properties, VkPhysicalDeviceMaintenance5Properties, VkPhysicalDeviceMaintenance6Properties, VkPhysicalDeviceMaintenance7PropertiesKHR, VkPhysicalDeviceMaintenance9PropertiesKHR, VkPhysicalDeviceMapMemoryPlacedPropertiesEXT, VkPhysicalDeviceMemoryDecompressionPropertiesEXT, VkPhysicalDeviceMeshShaderPropertiesEXT, VkPhysicalDeviceMultiDrawPropertiesEXT, VkPhysicalDeviceMultiviewProperties, VkPhysicalDeviceNestedCommandBufferPropertiesEXT, VkPhysicalDeviceOpacityMicromapPropertiesEXT,

VkPhysicalDeviceOpacityMicromapPropertiesKHR,
VkPhysicalDevicePCIBusInfoPropertiesEXT,
VkPhysicalDevicePerformanceQueryPropertiesKHR,
VkPhysicalDevicePipelineBinaryPropertiesKHR,
VkPhysicalDevicePipelineRobustnessProperties,
VkPhysicalDevicePointClippingProperties,
VkPhysicalDevicePortabilitySubsetPropertiesKHR,
VkPhysicalDeviceProtectedMemoryProperties,
VkPhysicalDeviceProvokingVertexPropertiesEXT,
VkPhysicalDevicePushDescriptorProperties,
VkPhysicalDeviceRayTracingInvocationReorderPropertiesEXT,
VkPhysicalDeviceRayTracingPipelinePropertiesKHR,
VkPhysicalDeviceRobustness2PropertiesKHR,
VkPhysicalDeviceSampleLocationsPropertiesEXT,
VkPhysicalDeviceSamplerFilterMinmaxProperties,
VkPhysicalDeviceShaderAbortPropertiesKHR,
VkPhysicalDeviceShaderIntegerDotProductProperties,
VkPhysicalDeviceShaderLongVectorPropertiesEXT,
VkPhysicalDeviceShaderModuleIdentifierPropertiesEXT,
VkPhysicalDeviceShaderObjectPropertiesEXT,
VkPhysicalDeviceShaderSplitBarrierPropertiesEXT,
VkPhysicalDeviceShaderTileImagePropertiesEXT, VkPhysicalDeviceSubgroupProperties,
VkPhysicalDeviceSubgroupSizeControlProperties,
VkPhysicalDeviceTexelBufferAlignmentProperties,
VkPhysicalDeviceTimelineSemaphoreProperties,
VkPhysicalDeviceTransformFeedbackPropertiesEXT,
VkPhysicalDeviceVertexAttributeDivisorProperties, VkPhysicalDeviceVulkan11Properties,
VkPhysicalDeviceVulkan12Properties, VkPhysicalDeviceVulkan13Properties, or
VkPhysicalDeviceVulkan14Properties

- VUID-VkPhysicalDeviceProperties2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

The `VkPhysicalDeviceVulkan11Properties` structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceVulkan11Properties {
    VkStructureType            sType;
    void*                      pNext;
    uint8_t                    deviceUUID[VK_UUID_SIZE];
    uint8_t                    driverUUID[VK_UUID_SIZE];
    uint8_t                    deviceLUID[VK_LUID_SIZE];
    uint32_t                   deviceNodeMask;
    VkBool32                   deviceLUIDValid;
    uint32_t                   subgroupSize;
    VkShaderStageFlags         subgroupSupportedStages;
    VkSubgroupFeatureFlags     subgroupSupportedOperations;
    VkBool32                   subgroupQuadOperationsInAllStages;
    VkPointClippingBehavior    pointClippingBehavior;
    uint32_t                   maxMultiviewViewCount;
    uint32_t                   maxMultiviewInstanceIndex;
    VkBool32                   protectedNoFault;
    uint32_t                   maxPerSetDescriptors;
    VkDeviceSize               maxMemoryAllocationSize;
} VkPhysicalDeviceVulkan11Properties;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `deviceUUID` is an array of VK_UUID_SIZE `uint8_t` values representing a universally unique identifier for the device.
- `driverUUID` is an array of VK_UUID_SIZE `uint8_t` values representing a universally unique identifier for the driver build in use by the device.
- `deviceLUID` is an array of VK_LUID_SIZE `uint8_t` values representing a locally unique identifier for the device.
- `deviceNodeMask` is a `uint32_t` bitfield identifying the node within a linked device adapter corresponding to the device.
- `deviceLUIDValid` is a boolean value that will be VK_TRUE if `deviceLUID` contains a valid LUID and `deviceNodeMask` contains a valid node mask, and VK_FALSE if they do not.
- `subgroupSize` is the default number of invocations in each subgroup. `subgroupSize` is at least 1 if any of the physical device's queues support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT. `subgroupSize` is a power-of-two.
- `subgroupSupportedStages` is a bitfield of VkShaderStageFlagBits describing the shader stages that group operations with subgroup scope are supported in. `subgroupSupportedStages` will have the VK_SHADER_STAGE_COMPUTE_BIT bit set if any of the physical device's queues support VK_QUEUE_COMPUTE_BIT.
- `subgroupSupportedOperations` is a bitmask of VkSubgroupFeatureFlagBits specifying the sets of group operations with subgroup scope supported on this device. `subgroupSupportedOperations` will have the VK_SUBGROUP_FEATURE_BASIC_BIT bit set if any of the physical device's queues support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT.

- `subgroupQuadOperationsInAllStages` is a boolean specifying whether quad group operations are available in all stages, or are restricted to fragment and compute stages.
- `pointClippingBehavior` is a VkPointClippingBehavior value specifying the point clipping behavior supported by the implementation.
- `maxMultiviewViewCount` is one greater than the maximum view index that **can** be used in a subpass.
- `maxMultiviewInstanceIndex` is the maximum valid value of instance index allowed to be generated by a drawing command recorded within a subpass of a multiview render pass instance.
- `protectedNoFault` specifies how an implementation behaves when an application attempts to write to unprotected memory in a protected queue operation, read from protected memory in an unprotected queue operation, or perform a query in a protected queue operation. If this limit is VK_TRUE, such writes will be discarded or have undefined values written; reads and queries will return poison. If this limit is VK_FALSE, applications **must** not perform these operations. See Protected Memory Access Rules for more information.
- `maxPerSetDescriptors` is a maximum number of descriptors (summed over all descriptor types) in a single descriptor set that is guaranteed to satisfy any implementation-dependent constraints on the size of a descriptor set itself. Applications **can** query whether a descriptor set that goes beyond this limit is supported using vkGetDescriptorSetLayoutSupport.
- `maxMemoryAllocationSize` is the maximum size of a memory allocation that **can** be created, even if there is more space available in the heap. If VkMemoryAllocateInfo::`allocationSize` is larger the error VK_ERROR_OUT_OF_DEVICE_MEMORY **may** be returned.

If the `VkPhysicalDeviceVulkan11Properties` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

These properties correspond to Vulkan 1.1 functionality.

The members of `VkPhysicalDeviceVulkan11Properties` have the same values as the corresponding members of VkPhysicalDeviceIDProperties, VkPhysicalDeviceSubgroupProperties, VkPhysicalDevicePointClippingProperties, VkPhysicalDeviceMultiviewProperties, VkPhysicalDeviceProtectedMemoryProperties, and VkPhysicalDeviceMaintenance3Properties.

NOTE

The `subgroupSupportedStages`, `subgroupSupportedOperations`, and `subgroupQuadOperationsInAllStages` members of this structure correspond respectively to the VkPhysicalDeviceSubgroupProperties::`supportedStages`, VkPhysicalDeviceSubgroupProperties::`supportedOperations`, and VkPhysicalDeviceSubgroupProperties::`quadOperationsInAllStages` members, but add the `subgroup` prefix to the member name.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVulkan11Properties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_1_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceVulkan12Properties structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceVulkan12Properties {
    VkStructureType                      sType;
    void*                                pNext;
    VkDriverId                           driverID;
    char                                 driverName[VK_MAX_DRIVER_NAME_SIZE];
    char                                 driverInfo[VK_MAX_DRIVER_INFO_SIZE];
    VkConformanceVersion                 conformanceVersion;
    VkShaderFloatControlsIndependence    denormBehaviorIndependence;
    VkShaderFloatControlsIndependence    roundingModeIndependence;
    VkBool32                             shaderSignedZeroInfNanPreserveFloat16;
    VkBool32                             shaderSignedZeroInfNanPreserveFloat32;
    VkBool32                             shaderSignedZeroInfNanPreserveFloat64;
    VkBool32                             shaderDenormPreserveFloat16;
    VkBool32                             shaderDenormPreserveFloat32;
    VkBool32                             shaderDenormPreserveFloat64;
    VkBool32                             shaderDenormFlushToZeroFloat16;
    VkBool32                             shaderDenormFlushToZeroFloat32;
    VkBool32                             shaderDenormFlushToZeroFloat64;
    VkBool32                             shaderRoundingModeRTEFloat16;
    VkBool32                             shaderRoundingModeRTEFloat32;
    VkBool32                             shaderRoundingModeRTEFloat64;
    VkBool32                             shaderRoundingModeRTZFloat16;
    VkBool32                             shaderRoundingModeRTZFloat32;
    VkBool32                             shaderRoundingModeRTZFloat64;
    uint32_t                             maxUpdateAfterBindDescriptorsInAllPools;
    VkBool32
shaderUniformBufferArrayNonUniformIndexingNative;
    VkBool32
shaderSampledImageArrayNonUniformIndexingNative;
    VkBool32
shaderStorageBufferArrayNonUniformIndexingNative;
    VkBool32
shaderStorageImageArrayNonUniformIndexingNative;
    VkBool32
shaderInputAttachmentArrayNonUniformIndexingNative;
    VkBool32                             robustBufferAccessUpdateAfterBind;
    VkBool32                             quadDivergentImplicitLod;
    uint32_t                             maxPerStageDescriptorUpdateAfterBindSamplers;
    uint32_t
maxPerStageDescriptorUpdateAfterBindUniformBuffers;
    uint32_t
```

```
maxPerStageDescriptorUpdateAfterBindStorageBuffers;
    uint32_t
maxPerStageDescriptorUpdateAfterBindSampledImages;
    uint32_t
maxPerStageDescriptorUpdateAfterBindStorageImages;
    uint32_t
maxPerStageDescriptorUpdateAfterBindInputAttachments;
    uint32_t                            maxPerStageUpdateAfterBindResources;
    uint32_t                            maxDescriptorSetUpdateAfterBindSamplers;
    uint32_t
maxDescriptorSetUpdateAfterBindUniformBuffers;
    uint32_t
maxDescriptorSetUpdateAfterBindUniformBuffersDynamic;
    uint32_t
maxDescriptorSetUpdateAfterBindStorageBuffers;
    uint32_t
maxDescriptorSetUpdateAfterBindStorageBuffersDynamic;
    uint32_t                            maxDescriptorSetUpdateAfterBindSampledImages;
    uint32_t                            maxDescriptorSetUpdateAfterBindStorageImages;
    uint32_t
maxDescriptorSetUpdateAfterBindInputAttachments;
    VkResolveModeFlags                  supportedDepthResolveModes;
    VkResolveModeFlags                  supportedStencilResolveModes;
    VkBool32                            independentResolveNone;
    VkBool32                            independentResolve;
    VkBool32                            filterMinmaxSingleComponentFormats;
    VkBool32                            filterMinmaxImageComponentMapping;
    uint64_t                            maxTimelineSemaphoreValueDifference;
    VkSampleCountFlags                  framebufferIntegerColorSampleCounts;
} VkPhysicalDeviceVulkan12Properties;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `driverID` is a unique identifier for the driver of the physical device.
- `driverName` is an array of VK_MAX_DRIVER_NAME_SIZE `char` containing a null-terminated UTF-8 string which is the name of the driver.
- `driverInfo` is an array of VK_MAX_DRIVER_INFO_SIZE `char` containing a null-terminated UTF-8 string with additional information about the driver.
- `conformanceVersion` is the latest version of the Vulkan conformance test that the implementor has successfully tested this driver against prior to release (see VkConformanceVersion).
- `denormBehaviorIndependence` is a VkShaderFloatControlsIndependence value indicating whether, and how, denorm behavior can be set independently for different bit widths.
- `roundingModeIndependence` is a VkShaderFloatControlsIndependence value indicating whether, and how, rounding modes can be set independently for different bit widths.
- `shaderSignedZeroInfNanPreserveFloat16` is a boolean value indicating whether sign of a zero, Nans and ±∞ **can** be preserved in 16-bit floating-point computations. It also indicates whether

the SignedZeroInfNanPreserve execution mode **can** be used for 16-bit floating-point types.

- shaderSignedZeroInfNanPreserveFloat32 is a boolean value indicating whether sign of a zero, Nans and ±∞ **can** be preserved in 32-bit floating-point computations. It also indicates whether the SignedZeroInfNanPreserve execution mode **can** be used for 32-bit floating-point types.
- shaderSignedZeroInfNanPreserveFloat64 is a boolean value indicating whether sign of a zero, Nans and ±∞ **can** be preserved in 64-bit floating-point computations. It also indicates whether the SignedZeroInfNanPreserve execution mode **can** be used for 64-bit floating-point types.
- shaderDenormPreserveFloat16 is a boolean value indicating whether denormals **can** be preserved in 16-bit floating-point computations. It also indicates whether the DenormPreserve execution mode **can** be used for 16-bit floating-point types.
- shaderDenormPreserveFloat32 is a boolean value indicating whether denormals **can** be preserved in 32-bit floating-point computations. It also indicates whether the DenormPreserve execution mode **can** be used for 32-bit floating-point types.
- shaderDenormPreserveFloat64 is a boolean value indicating whether denormals **can** be preserved in 64-bit floating-point computations. It also indicates whether the DenormPreserve execution mode **can** be used for 64-bit floating-point types.
- shaderDenormFlushToZeroFloat16 is a boolean value indicating whether denormals **can** be flushed to zero in 16-bit floating-point computations. It also indicates whether the DenormFlushToZero execution mode **can** be used for 16-bit floating-point types.
- shaderDenormFlushToZeroFloat32 is a boolean value indicating whether denormals **can** be flushed to zero in 32-bit floating-point computations. It also indicates whether the DenormFlushToZero execution mode **can** be used for 32-bit floating-point types.
- shaderDenormFlushToZeroFloat64 is a boolean value indicating whether denormals **can** be flushed to zero in 64-bit floating-point computations. It also indicates whether the DenormFlushToZero execution mode **can** be used for 64-bit floating-point types.
- shaderRoundingModeRTEFloat16 is a boolean value indicating whether an implementation supports the round-to-nearest-even rounding mode for 16-bit floating-point arithmetic and conversion instructions. It also indicates whether the RoundingModeRTE execution mode **can** be used for 16-bit floating-point types.
- shaderRoundingModeRTEFloat32 is a boolean value indicating whether an implementation supports the round-to-nearest-even rounding mode for 32-bit floating-point arithmetic and conversion instructions. It also indicates whether the RoundingModeRTE execution mode **can** be used for 32-bit floating-point types.
- shaderRoundingModeRTEFloat64 is a boolean value indicating whether an implementation supports the round-to-nearest-even rounding mode for 64-bit floating-point arithmetic and conversion instructions. It also indicates whether the RoundingModeRTE execution mode **can** be used for 64-bit floating-point types.
- shaderRoundingModeRTZFloat16 is a boolean value indicating whether an implementation supports the round-towards-zero rounding mode for 16-bit floating-point arithmetic and conversion instructions. It also indicates whether the RoundingModeRTZ execution mode **can** be used for 16-bit floating-point types.
- shaderRoundingModeRTZFloat32 is a boolean value indicating whether an implementation supports the round-towards-zero rounding mode for 32-bit floating-point arithmetic and

conversion instructions. It also indicates whether the RoundingModeRTZ execution mode **can** be used for 32-bit floating-point types.

- shaderRoundingModeRTZFloat64 is a boolean value indicating whether an implementation supports the round-towards-zero rounding mode for 64-bit floating-point arithmetic and conversion instructions. It also indicates whether the RoundingModeRTZ execution mode **can** be used for 64-bit floating-point types.
- maxUpdateAfterBindDescriptorsInAllPools is the maximum number of descriptors (summed over all descriptor types) that **can** be created across all pools that are created with the VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT bit set. Pool creation **may** fail when this limit is exceeded, or when the space this limit represents is unable to satisfy a pool creation due to fragmentation.
- shaderUniformBufferArrayNonUniformIndexingNative is a boolean value indicating whether uniform buffer descriptors natively support non-uniform indexing. If this is VK_FALSE, then a single dynamic instance of an instruction that non-uniformly indexes an array of uniform buffers **may** execute multiple times in order to access all the descriptors.
- shaderSampledImageArrayNonUniformIndexingNative is a boolean value indicating whether sampler and image descriptors natively support non-uniform indexing. If this is VK_FALSE, then a single dynamic instance of an instruction that non-uniformly indexes an array of samplers or images **may** execute multiple times in order to access all the descriptors.
- shaderStorageBufferArrayNonUniformIndexingNative is a boolean value indicating whether storage buffer descriptors natively support non-uniform indexing. If this is VK_FALSE, then a single dynamic instance of an instruction that non-uniformly indexes an array of storage buffers **may** execute multiple times in order to access all the descriptors.
- shaderStorageImageArrayNonUniformIndexingNative is a boolean value indicating whether storage image descriptors natively support non-uniform indexing. If this is VK_FALSE, then a single dynamic instance of an instruction that non-uniformly indexes an array of storage images **may** execute multiple times in order to access all the descriptors.
- shaderInputAttachmentArrayNonUniformIndexingNative is a boolean value indicating whether input attachment descriptors natively support non-uniform indexing. If this is VK_FALSE, then a single dynamic instance of an instruction that non-uniformly indexes an array of input attachments **may** execute multiple times in order to access all the descriptors.
- robustBufferAccessUpdateAfterBind is a boolean value indicating whether robustBufferAccess **can** be enabled on a device simultaneously with descriptorBindingUniformBufferUpdateAfterBind, descriptorBindingStorageBufferUpdateAfterBind, descriptorBindingUniformTexelBufferUpdateAfterBind, and/or descriptorBindingStorageTexelBufferUpdateAfterBind. If this is VK_FALSE, then either robustBufferAccess **must** be disabled or all of these update-after-bind features **must** be disabled. Similarly, if this property is VK_FALSE, robustness **must** not be enabled through the VkPipelineRobustnessCreateInfo mechanism.
- quadDivergentImplicitLod is a boolean value indicating whether implicit LOD calculations for image operations have well-defined results when the image and/or sampler objects used for the instruction are not uniform within a quad. See Derivative Image Operations.
- maxPerStageDescriptorUpdateAfterBindSamplers is similar to maxPerStageDescriptorSamplers but counts descriptors from descriptor sets created with or without the

VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.

- maxPerStageDescriptorUpdateAfterBindUniformBuffers is similar to maxPerStageDescriptorUniformBuffers but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- maxPerStageDescriptorUpdateAfterBindStorageBuffers is similar to maxPerStageDescriptorStorageBuffers but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- maxPerStageDescriptorUpdateAfterBindSampledImages is similar to maxPerStageDescriptorSampledImages but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- maxPerStageDescriptorUpdateAfterBindStorageImages is similar to maxPerStageDescriptorStorageImages but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- maxPerStageDescriptorUpdateAfterBindInputAttachments is similar to maxPerStageDescriptorInputAttachments but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- maxPerStageUpdateAfterBindResources is similar to maxPerStageResources but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- maxDescriptorSetUpdateAfterBindSamplers is similar to maxDescriptorSetSamplers but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- maxDescriptorSetUpdateAfterBindUniformBuffers is similar to maxDescriptorSetUniformBuffers but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- maxDescriptorSetUpdateAfterBindUniformBuffersDynamic is similar to maxDescriptorSetUniformBuffersDynamic but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set. While an application **can** allocate dynamic uniform buffer descriptors from a pool created with VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT, and **can** include dynamic uniform buffer descriptors in descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT, bindings for these descriptors **must** not be present in any descriptor set layout that includes bindings created with VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT. Therefore this limit has no practical effect and **may** be zero.
- maxDescriptorSetUpdateAfterBindStorageBuffers is similar to maxDescriptorSetStorageBuffers but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- maxDescriptorSetUpdateAfterBindStorageBuffersDynamic is similar to maxDescriptorSetStorageBuffersDynamic but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set. While an application **can** allocate dynamic storage buffer descriptors from a pool created with VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT, and **can** include dynamic storage

buffer descriptors in descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT, bindings for these descriptors **must** not be present in any descriptor set layout that includes bindings created with VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT. Therefore this limit has no practical effect and **may** be zero.

- `maxDescriptorSetUpdateAfterBindSampledImages` is similar to `maxDescriptorSetSampledImages` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxDescriptorSetUpdateAfterBindStorageImages` is similar to `maxDescriptorSetStorageImages` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxDescriptorSetUpdateAfterBindInputAttachments` is similar to `maxDescriptorSetInputAttachments` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set. While an application **can** allocate input attachment descriptors from a pool created with VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT, and **can** include input attachment descriptors in descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT, bindings for these descriptors **must** not be created with VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT. Therefore this limit only exists to allow implementations exposing a higher limit for input attachment descriptors in a pipeline layout including any descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT than in a pipeline layout not including any descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT and **may** be zero to indicate that only the `maxDescriptorSetInputAttachments` applies.
- `supportedDepthResolveModes` is a bitmask of VkResolveModeFlagBits indicating the set of supported depth resolve modes. VK_RESOLVE_MODE_SAMPLE_ZERO_BIT **must** be included in the set but implementations **may** support additional modes.
- `supportedStencilResolveModes` is a bitmask of VkResolveModeFlagBits indicating the set of supported stencil resolve modes. VK_RESOLVE_MODE_SAMPLE_ZERO_BIT **must** be included in the set but implementations **may** support additional modes. VK_RESOLVE_MODE_AVERAGE_BIT **must** not be included in the set.
- `independentResolveNone` is VK_TRUE if the implementation supports setting the depth and stencil resolve modes to different values when one of those modes is VK_RESOLVE_MODE_NONE. Otherwise the implementation only supports setting both modes to the same value.
- `independentResolve` is VK_TRUE if the implementation supports all combinations of the supported depth and stencil resolve modes, including setting either depth or stencil resolve mode to VK_RESOLVE_MODE_NONE. An implementation that supports `independentResolve` **must** also support `independentResolveNone`.
- `filterMinmaxSingleComponentFormats` is a boolean value indicating whether a minimum set of required formats support min/max filtering.
- `filterMinmaxImageComponentMapping` is a boolean value indicating whether the implementation supports non-identity component mapping of the image when doing min/max filtering.
- `maxTimelineSemaphoreValueDifference` indicates the maximum difference allowed by the

implementation between the current value of a timeline semaphore and any pending signal or wait operations.

- `framebufferIntegerColorSampleCounts` is a bitmask of VkSampleCountFlagBits indicating the color sample counts that are supported for all framebuffer color attachments with integer formats.

If the `VkPhysicalDeviceVulkan12Properties` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

These properties correspond to Vulkan 1.2 functionality.

The members of `VkPhysicalDeviceVulkan12Properties` **must** have the same values as the corresponding members of VkPhysicalDeviceDriverProperties, VkPhysicalDeviceFloatControlsProperties, VkPhysicalDeviceDescriptorIndexingProperties, VkPhysicalDeviceDepthStencilResolveProperties, VkPhysicalDeviceSamplerFilterMinmaxProperties, and VkPhysicalDeviceTimelineSemaphoreProperties.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVulkan12Properties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_2_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceVulkan13Properties` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceVulkan13Properties {
    VkStructureType       sType;
    void*                 pNext;
    uint32_t              minSubgroupSize;
    uint32_t              maxSubgroupSize;
    uint32_t              maxComputeWorkgroupSubgroups;
    VkShaderStageFlags    requiredSubgroupSizeStages;
    uint32_t              maxInlineUniformBlockSize;
    uint32_t              maxPerStageDescriptorInlineUniformBlocks;
    uint32_t              maxPerStageDescriptorUpdateAfterBindInlineUniformBlocks;
    uint32_t              maxDescriptorSetInlineUniformBlocks;
    uint32_t              maxDescriptorSetUpdateAfterBindInlineUniformBlocks;
    uint32_t              maxInlineUniformTotalSize;
    VkBool32              integerDotProduct8BitUnsignedAccelerated;
    VkBool32              integerDotProduct8BitSignedAccelerated;
```

```
    VkBool32               integerDotProduct8BitMixedSignednessAccelerated;
    VkBool32               integerDotProduct4x8BitPackedUnsignedAccelerated;
    VkBool32               integerDotProduct4x8BitPackedSignedAccelerated;
    VkBool32               integerDotProduct4x8BitPackedMixedSignednessAccelerated;
    VkBool32               integerDotProduct16BitUnsignedAccelerated;
    VkBool32               integerDotProduct16BitSignedAccelerated;
    VkBool32               integerDotProduct16BitMixedSignednessAccelerated;
    VkBool32               integerDotProduct32BitUnsignedAccelerated;
    VkBool32               integerDotProduct32BitSignedAccelerated;
    VkBool32               integerDotProduct32BitMixedSignednessAccelerated;
    VkBool32               integerDotProduct64BitUnsignedAccelerated;
    VkBool32               integerDotProduct64BitSignedAccelerated;
    VkBool32               integerDotProduct64BitMixedSignednessAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating8BitUnsignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating8BitSignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating8BitMixedSignednessAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating4x8BitPackedUnsignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating4x8BitPackedSignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating4x8BitPackedMixedSignednessAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating16BitUnsignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating16BitSignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating16BitMixedSignednessAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating32BitUnsignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating32BitSignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating32BitMixedSignednessAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating64BitUnsignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating64BitSignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating64BitMixedSignednessAccelerated;
    VkDeviceSize           storageTexelBufferOffsetAlignmentBytes;
    VkBool32               storageTexelBufferOffsetSingleTexelAlignment;
    VkDeviceSize           uniformTexelBufferOffsetAlignmentBytes;
    VkBool32               uniformTexelBufferOffsetSingleTexelAlignment;
    VkDeviceSize           maxBufferSize;
} VkPhysicalDeviceVulkan13Properties;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `minSubgroupSize` is the minimum subgroup size supported by this device. `minSubgroupSize` is at least one if any of the physical device's queues support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT. `minSubgroupSize` is a power-of-two. `minSubgroupSize` is less than or equal to `maxSubgroupSize`. `minSubgroupSize` is less than or equal to `subgroupSize`.
- `maxSubgroupSize` is the maximum subgroup size supported by this device. `maxSubgroupSize` is at least one if any of the physical device's queues support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT. `maxSubgroupSize` is a power-of-two. `maxSubgroupSize` is greater than or equal to `minSubgroupSize`. `maxSubgroupSize` is greater than or equal to `subgroupSize`.
- `maxComputeWorkgroupSubgroups` is the maximum number of subgroups supported by the implementation within a workgroup.
- `requiredSubgroupSizeStages` is a bitfield of what shader stages support having a required subgroup size specified.
- `maxInlineUniformBlockSize` is the maximum size in bytes of an inline uniform block binding.
- `maxPerStageDescriptorInlineUniformBlocks` is the maximum number of inline uniform block bindings that **can** be accessible to a single shader stage in a pipeline layout. Descriptor bindings with a descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK count against this limit. Only descriptor bindings in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit.
- `maxPerStageDescriptorUpdateAfterBindInlineUniformBlocks` is similar to `maxPerStageDescriptorInlineUniformBlocks` but counts descriptor bindings from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxDescriptorSetInlineUniformBlocks` is the maximum number of inline uniform block bindings that **can** be included in descriptor bindings in a pipeline layout across all pipeline shader stages and descriptor set numbers. Descriptor bindings with a descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK count against this limit. Only descriptor bindings in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit.
- `maxDescriptorSetUpdateAfterBindInlineUniformBlocks` is similar to `maxDescriptorSetInlineUniformBlocks` but counts descriptor bindings from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxInlineUniformTotalSize` is the maximum total size in bytes of all inline uniform block bindings, across all pipeline shader stages and descriptor set numbers, that **can** be included in a pipeline layout. Descriptor bindings with a descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK count against this limit.
- `integerDotProduct8BitUnsignedAccelerated` is a boolean that will be VK_TRUE if the support for 8-bit unsigned dot product operations using the `OpUDotKHR` SPIR-V instruction is accelerated as defined below.

- integerDotProduct8BitSignedAccelerated is a boolean that will be VK_TRUE if the support for 8-bit signed dot product operations using the OpSDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct8BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 8-bit mixed signedness dot product operations using the OpSUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct4x8BitPackedUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 8-bit unsigned dot product operations from operands packed into 32-bit integers using the OpUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct4x8BitPackedSignedAccelerated is a boolean that will be VK_TRUE if the support for 8-bit signed dot product operations from operands packed into 32-bit integers using the OpSDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct4x8BitPackedMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 8-bit mixed signedness dot product operations from operands packed into 32-bit integers using the OpSUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct16BitUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 16-bit unsigned dot product operations using the OpUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct16BitSignedAccelerated is a boolean that will be VK_TRUE if the support for 16-bit signed dot product operations using the OpSDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct16BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 16-bit mixed signedness dot product operations using the OpSUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct32BitUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 32-bit unsigned dot product operations using the OpUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct32BitSignedAccelerated is a boolean that will be VK_TRUE if the support for 32-bit signed dot product operations using the OpSDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct32BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 32-bit mixed signedness dot product operations using the OpSUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct64BitUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 64-bit unsigned dot product operations using the OpUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct64BitSignedAccelerated is a boolean that will be VK_TRUE if the support for 64-bit signed dot product operations using the OpSDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct64BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 64-bit mixed signedness dot product operations using the OpSUDotKHR SPIR-V instruction is accelerated as defined below.

- integerDotProductAccumulatingSaturating8BitUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 8-bit unsigned accumulating saturating dot product operations using the OpUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating8BitSignedAccelerated is a boolean that will be VK_TRUE if the support for 8-bit signed accumulating saturating dot product operations using the OpSDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating8BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 8-bit mixed signedness accumulating saturating dot product operations using the OpSUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating4x8BitPackedUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 8-bit unsigned accumulating saturating dot product operations from operands packed into 32-bit integers using the OpUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating4x8BitPackedSignedAccelerated is a boolean that will be VK_TRUE if the support for 8-bit signed accumulating saturating dot product operations from operands packed into 32-bit integers using the OpSDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating4x8BitPackedMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 8-bit mixed signedness accumulating saturating dot product operations from operands packed into 32-bit integers using the OpSUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating16BitUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 16-bit unsigned accumulating saturating dot product operations using the OpUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating16BitSignedAccelerated is a boolean that will be VK_TRUE if the support for 16-bit signed accumulating saturating dot product operations using the OpSDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating16BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 16-bit mixed signedness accumulating saturating dot product operations using the OpSUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating32BitUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 32-bit unsigned accumulating saturating dot product operations using the OpUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating32BitSignedAccelerated is a boolean that will be VK_TRUE if the support for 32-bit signed accumulating saturating dot product operations using the OpSDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating32BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 32-bit mixed signedness accumulating saturating dot product operations using the OpSUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating64BitUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 64-bit unsigned accumulating saturating dot product operations using the OpUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating64BitSignedAccelerated is a boolean that will be

VK_TRUE if the support for 64-bit signed accumulating saturating dot product operations using the OpSDotAccSatKHR SPIR-V instruction is accelerated as defined below.

- integerDotProductAccumulatingSaturating64BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 64-bit mixed signedness accumulating saturating dot product operations using the OpSUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- storageTexelBufferOffsetAlignmentBytes is a byte alignment that is sufficient for a storage texel buffer of any format. The value **must** be a power of two.
- storageTexelBufferOffsetSingleTexelAlignment indicates whether single texel alignment is sufficient for a storage texel buffer of any format.
- uniformTexelBufferOffsetAlignmentBytes is a byte alignment that is sufficient for a uniform texel buffer of any format. The value **must** be a power of two.
- uniformTexelBufferOffsetSingleTexelAlignment indicates whether single texel alignment is sufficient for a uniform texel buffer of any format.
- maxBufferSize is the maximum size of a VkBuffer that **can** be created.

If the VkPhysicalDeviceVulkan13Properties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

These properties correspond to Vulkan 1.3 functionality.

The members of VkPhysicalDeviceVulkan13Properties **must** have the same values as the corresponding members of VkPhysicalDeviceInlineUniformBlockProperties and VkPhysicalDeviceSubgroupSizeControlProperties.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVulkan13Properties-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_3_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceVulkan14Properties structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceVulkan14Properties {
    VkStructureType                       sType;
    void*                                 pNext;
    uint32_t                              lineSubPixelPrecisionBits;
    uint32_t                              maxVertexAttribDivisor;
    VkBool32                              supportsNonZeroFirstInstance;
    uint32_t                              maxPushDescriptors;
    VkBool32
dynamicRenderingLocalReadDepthStencilAttachments;
    VkBool32
dynamicRenderingLocalReadMultisampledAttachments;
    VkBool32
earlyFragmentMultisampleCoverageAfterSampleCounting;
    VkBool32
earlyFragmentSampleMaskTestBeforeSampleCounting;
    VkBool32                              depthStencilSwizzleOneSupport;
    VkBool32                              polygonModePointSize;
    VkBool32
nonStrictSinglePixelWideLinesUseParallelogram;
    VkBool32                              nonStrictWideLinesUseParallelogram;
    VkBool32                              blockTexelViewCompatibleMultipleLayers;
    uint32_t                              maxCombinedImageSamplerDescriptorCount;
    VkBool32                              fragmentShadingRateClampCombinerInputs;
    VkPipelineRobustnessBufferBehavior    defaultRobustnessStorageBuffers;
    VkPipelineRobustnessBufferBehavior    defaultRobustnessUniformBuffers;
    VkPipelineRobustnessBufferBehavior    defaultRobustnessVertexInputs;
    VkPipelineRobustnessImageBehavior     defaultRobustnessImages;
    uint32_t                              copySrcLayoutCount;
    VkImageLayout*                        pCopySrcLayouts;
    uint32_t                              copyDstLayoutCount;
    VkImageLayout*                        pCopyDstLayouts;
    uint8_t                               optimalTilingLayoutUUID[VK_UUID_SIZE];
    VkBool32                              identicalMemoryTypeRequirements;
} VkPhysicalDeviceVulkan14Properties;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `lineSubPixelPrecisionBits` is the number of bits of subpixel precision in framebuffer coordinates x_f and y_f when rasterizing line segments.
- `maxVertexAttribDivisor` is the maximum value of the number of instances that will repeat the value of vertex attribute data when instanced rendering is enabled.
- `supportsNonZeroFirstInstance` specifies whether a non-zero value for the `firstInstance` parameter of drawing commands is supported when VkVertexInputBindingDivisorDescription ::`divisor` is not `1`.
- `maxPushDescriptors` is the maximum number of descriptors that **can** be used in a descriptor set

layout created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT set.

- `dynamicRenderingLocalReadDepthStencilAttachments` is VK_TRUE if the implementation supports local reads of depth/stencil attachments, VK_FALSE otherwise.
- `dynamicRenderingLocalReadMultisampledAttachments` is VK_TRUE if the implementation supports local reads of multisampled attachments, VK_FALSE otherwise.
- `earlyFragmentMultisampleCoverageAfterSampleCounting` is a boolean value indicating whether the fragment shading and multisample coverage operations are performed after sample counting for fragment shaders with `EarlyFragmentTests` execution mode.
- `earlyFragmentSampleMaskTestBeforeSampleCounting` is a boolean value indicating whether the sample mask test operation is performed before sample counting for fragment shaders using the `EarlyFragmentTests` execution mode.
- `depthStencilSwizzleOneSupport` is a boolean indicating that depth/stencil texturing operations with VK_COMPONENT_SWIZZLE_ONE have defined behavior.
- `polygonModePointSize` is a boolean value indicating whether the point size of the final rasterization of polygons with VK_POLYGON_MODE_POINT is controlled by `PointSize`.
- `nonStrictSinglePixelWideLinesUseParallelogram` is a boolean value indicating whether non-strict lines with a width of 1.0 are rasterized as parallelograms or using Bresenham's algorithm.
- `nonStrictWideLinesUseParallelogram` is a boolean value indicating whether non-strict lines with a width greater than 1.0 are rasterized as parallelograms or using Bresenham's algorithm.
- `blockTexelViewCompatibleMultipleLayers` is a boolean value indicating that an implementation supports creating image views with VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT where the `layerCount` member of `subresourceRange` is greater than 1.
- `maxCombinedImageSamplerDescriptorCount` is the maximum number of descriptors that the implementation uses for sampled images with $Y'C_BC_R$ conversion.
- `fragmentShadingRateClampCombinerInputs` is a boolean value indicating that an implementation clamps the inputs to combiner operations.
- `defaultRobustnessStorageBuffers` describes the behavior of out of bounds accesses made to storage buffers when no robustness features are enabled.
- `defaultRobustnessUniformBuffers` describes the behavior of out of bounds accesses made to uniform buffers when no robustness features are enabled.
- `defaultRobustnessVertexInputs` describes the behavior of out of bounds accesses made to vertex input attributes when no robustness features are enabled.
- `defaultRobustnessImages` describes the behavior of out of bounds accesses made to images when no robustness features are enabled.
- `copySrcLayoutCount` is an integer related to the number of image layouts for host copies from images available or queried, as described below.
- `pCopySrcLayouts` is a pointer to an array of VkImageLayout in which supported image layouts for use with host copy operations from images are returned.
- `copyDstLayoutCount` is an integer related to the number of image layouts for host copies to images available or queried, as described below.

- `pCopyDstLayouts` is a pointer to an array of VkImageLayout in which supported image layouts for use with host copy operations to images are returned.
- `optimalTilingLayoutUUID` is an array of VK_UUID_SIZE `uint8_t` values representing a universally unique identifier for the implementation's swizzling layout of images created with VK_IMAGE_TILING_OPTIMAL.
- `identicalMemoryTypeRequirements` indicates that specifying the VK_IMAGE_USAGE_HOST_TRANSFER_BIT flag in VkImageCreateInfo::`usage` does not affect the memory type requirements of the image.

If the `VkPhysicalDeviceVulkan14Properties` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

These properties correspond to Vulkan 1.4 functionality.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVulkan14Properties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_4_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceIDProperties` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceIDProperties {
    VkStructureType    sType;
    void*              pNext;
    uint8_t            deviceUUID[VK_UUID_SIZE];
    uint8_t            driverUUID[VK_UUID_SIZE];
    uint8_t            deviceLUID[VK_LUID_SIZE];
    uint32_t           deviceNodeMask;
    VkBool32           deviceLUIDValid;
} VkPhysicalDeviceIDProperties;
```

```
// Provided by VK_KHR_external_fence_capabilities,
VK_KHR_external_memory_capabilities, VK_KHR_external_semaphore_capabilities
// Equivalent to VkPhysicalDeviceIDProperties
typedef VkPhysicalDeviceIDProperties VkPhysicalDeviceIDPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.

- pNext is NULL or a pointer to a structure extending this structure.
- deviceUUID is an array of VK_UUID_SIZE uint8_t values representing a universally unique identifier for the device.
- driverUUID is an array of VK_UUID_SIZE uint8_t values representing a universally unique identifier for the driver build in use by the device.
- deviceLUID is an array of VK_LUID_SIZE uint8_t values representing a locally unique identifier for the device.
- deviceNodeMask is a uint32_t bitfield identifying the node within a linked device adapter corresponding to the device.
- deviceLUIDValid is a boolean value that will be VK_TRUE if deviceLUID contains a valid LUID and deviceNodeMask contains a valid node mask, and VK_FALSE if they do not.

If the VkPhysicalDeviceIDProperties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

deviceUUID **must** be immutable for a given device across instances, processes, driver APIs, driver versions, and system reboots.

Applications **can** compare the driverUUID value across instance and process boundaries, and **can** make similar queries in external APIs to determine whether they are capable of sharing memory objects and resources using them with the device.

deviceUUID and/or driverUUID **must** be used to determine whether a particular external object can be shared between driver components, where such a restriction exists as defined in the compatibility table for the particular object type:

- External memory handle types compatibility
- External semaphore handle types compatibility
- External fence handle types compatibility

If deviceLUIDValid is VK_FALSE, the values of deviceLUID and deviceNodeMask are undefined. If deviceLUIDValid is VK_TRUE and Vulkan is running on the Windows operating system, the contents of deviceLUID **can** be cast to an LUID object and **must** be equal to the locally unique identifier of a IDXGIAdapter1 object that corresponds to physicalDevice. If deviceLUIDValid is VK_TRUE, deviceNodeMask **must** contain exactly one bit. If Vulkan is running on an operating system that supports the Direct3D 12 API and physicalDevice corresponds to an individual device in a linked device adapter, deviceNodeMask identifies the Direct3D 12 node corresponding to physicalDevice. Otherwise, deviceNodeMask **must** be 1.

NOTE

Although they have identical descriptions, VkPhysicalDeviceIDProperties ::deviceUUID may differ from VkPhysicalDeviceProperties2 ::properties.pipelineCacheUUID. The former is intended to identify and correlate devices across API and driver boundaries, while the latter is used to identify a compatible device and driver combination to use when serializing and de-serializing pipeline state.

Implementations **should** return `deviceUUID` values which are likely to be unique even in the presence of multiple Vulkan implementations (such as a GPU driver and a software renderer; two drivers for different GPUs; or the same Vulkan driver running on two logically different devices).

Khronos' conformance testing is unable to guarantee that `deviceUUID` values are actually unique, so implementors **should** make their own best efforts to ensure this. In particular, hard-coded `deviceUUID` values, especially all-`0` bits, **should** never be used.

A combination of values unique to the vendor, the driver, and the hardware environment can be used to provide a `deviceUUID` which is unique to a high degree of certainty. Some possible inputs to such a computation are:

- Information reported by vkGetPhysicalDeviceProperties
- PCI device ID (if defined)
- PCI bus ID, or similar system configuration information.
- Driver binary checksums.

NOTE

While VkPhysicalDeviceIDProperties::`deviceUUID` is specified to remain consistent across driver versions and system reboots, it is not intended to be usable as a serializable persistent identifier for a device. It may change when a device is physically added to, removed from, or moved to a different connector in a system while that system is powered down. Further, there is no reasonable way to verify with conformance testing that a given device retains the same UUID in a given system across all driver versions supported in that system. While implementations should make every effort to report consistent device UUIDs across driver versions, applications should avoid relying on the persistence of this value for uses other than identifying compatible devices for external object sharing purposes.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceIDProperties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ID_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

VK_UUID_SIZE is the length in `uint8_t` values of an array containing a universally unique device or driver build identifier, as returned in VkPhysicalDeviceIDProperties::`deviceUUID` and VkPhysicalDeviceIDProperties::`driverUUID`.

```
#define VK_UUID_SIZE                      16U
```

VK_LUID_SIZE is the length in uint8_t values of an array containing a locally unique device identifier, as returned in VkPhysicalDeviceIDProperties::deviceLUID.

```
#define VK_LUID_SIZE                      8U
```

```
#define VK_LUID_SIZE_KHR                  VK_LUID_SIZE
```

The VkPhysicalDeviceDriverProperties structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceDriverProperties {
    VkStructureType         sType;
    void*                   pNext;
    VkDriverId              driverID;
    char                    driverName[VK_MAX_DRIVER_NAME_SIZE];
    char                    driverInfo[VK_MAX_DRIVER_INFO_SIZE];
    VkConformanceVersion    conformanceVersion;
} VkPhysicalDeviceDriverProperties;
```

```
// Provided by VK_KHR_driver_properties
// Equivalent to VkPhysicalDeviceDriverProperties
typedef VkPhysicalDeviceDriverProperties VkPhysicalDeviceDriverPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- driverID is a unique identifier for the driver of the physical device.
- driverName is an array of VK_MAX_DRIVER_NAME_SIZE char containing a null-terminated UTF-8 string which is the name of the driver.
- driverInfo is an array of VK_MAX_DRIVER_INFO_SIZE char containing a null-terminated UTF-8 string with additional information about the driver.
- conformanceVersion is the latest version of the Vulkan conformance test that the implementor has successfully tested this driver against prior to release (see VkConformanceVersion).

If the VkPhysicalDeviceDriverProperties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

These are properties of the driver corresponding to a physical device.

driverID **must** be immutable for a given driver across instances, processes, driver versions, and

system reboots.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDriverProperties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRIVER_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

Khronos driver IDs which **may** be returned in VkPhysicalDeviceDriverProperties::`driverID` are:

```
// Provided by VK_VERSION_1_2
typedef enum VkDriverId {
    VK_DRIVER_ID_AMD_PROPRIETARY = 1,
    VK_DRIVER_ID_AMD_OPEN_SOURCE = 2,
    VK_DRIVER_ID_MESA_RADV = 3,
    VK_DRIVER_ID_NVIDIA_PROPRIETARY = 4,
    VK_DRIVER_ID_INTEL_PROPRIETARY_WINDOWS = 5,
    VK_DRIVER_ID_INTEL_OPEN_SOURCE_MESA = 6,
    VK_DRIVER_ID_IMAGINATION_PROPRIETARY = 7,
    VK_DRIVER_ID_QUALCOMM_PROPRIETARY = 8,
    VK_DRIVER_ID_ARM_PROPRIETARY = 9,
    VK_DRIVER_ID_GOOGLE_SWIFTSHADER = 10,
    VK_DRIVER_ID_GGP_PROPRIETARY = 11,
    VK_DRIVER_ID_BROADCOM_PROPRIETARY = 12,
    VK_DRIVER_ID_MESA_LLVMPIPE = 13,
    VK_DRIVER_ID_MOLTENVK = 14,
    VK_DRIVER_ID_COREAVI_PROPRIETARY = 15,
    VK_DRIVER_ID_JUICE_PROPRIETARY = 16,
    VK_DRIVER_ID_VERISILICON_PROPRIETARY = 17,
    VK_DRIVER_ID_MESA_TURNIP = 18,
    VK_DRIVER_ID_MESA_V3DV = 19,
    VK_DRIVER_ID_MESA_PANVK = 20,
    VK_DRIVER_ID_SAMSUNG_PROPRIETARY = 21,
    VK_DRIVER_ID_MESA_VENUS = 22,
    VK_DRIVER_ID_MESA_DOZEN = 23,
    VK_DRIVER_ID_MESA_NVK = 24,
    VK_DRIVER_ID_IMAGINATION_OPEN_SOURCE_MESA = 25,
    VK_DRIVER_ID_MESA_HONEYKRISP = 26,
    VK_DRIVER_ID_VULKAN_SC_EMULATION_ON_VULKAN = 27,
    VK_DRIVER_ID_MESA_KOSMICKRISP = 28,
    VK_DRIVER_ID_MESA_GFXSTREAM = 29,
    VK_DRIVER_ID_APE_SOFT = 30,
    VK_DRIVER_ID_RESERVED_31 = 31,
```

```
    VK_DRIVER_ID_APEX = 32,
    VK_DRIVER_ID_NIKKI = 33,
    VK_DRIVER_ID_MOMOTEN = 34,
  // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_AMD_PROPRIETARY_KHR = VK_DRIVER_ID_AMD_PROPRIETARY,
  // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_AMD_OPEN_SOURCE_KHR = VK_DRIVER_ID_AMD_OPEN_SOURCE,
  // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_MESA_RADV_KHR = VK_DRIVER_ID_MESA_RADV,
  // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_NVIDIA_PROPRIETARY_KHR = VK_DRIVER_ID_NVIDIA_PROPRIETARY,
  // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_INTEL_PROPRIETARY_WINDOWS_KHR =
VK_DRIVER_ID_INTEL_PROPRIETARY_WINDOWS,
  // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_INTEL_OPEN_SOURCE_MESA_KHR = VK_DRIVER_ID_INTEL_OPEN_SOURCE_MESA,
  // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_IMAGINATION_PROPRIETARY_KHR = VK_DRIVER_ID_IMAGINATION_PROPRIETARY,
  // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_QUALCOMM_PROPRIETARY_KHR = VK_DRIVER_ID_QUALCOMM_PROPRIETARY,
  // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_ARM_PROPRIETARY_KHR = VK_DRIVER_ID_ARM_PROPRIETARY,
  // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_GOOGLE_SWIFTSHADER_KHR = VK_DRIVER_ID_GOOGLE_SWIFTSHADER,
  // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_GGP_PROPRIETARY_KHR = VK_DRIVER_ID_GGP_PROPRIETARY,
  // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_BROADCOM_PROPRIETARY_KHR = VK_DRIVER_ID_BROADCOM_PROPRIETARY,
} VkDriverId;
```

```
// Provided by VK_KHR_driver_properties
// Equivalent to VkDriverId
typedef VkDriverId VkDriverIdKHR;
```

NOTE

Khronos driver IDs may be allocated by vendors at any time. There may be multiple driver IDs for the same vendor, representing different drivers (for e.g. different platforms, proprietary or open source, etc.). Only the latest canonical versions of this Specification, of the corresponding `vk.xml` API Registry, and of the corresponding `vulkan_core.h` header file **must** contain all reserved Khronos driver IDs.

Only driver IDs registered with Khronos are given symbolic names. There **may** be unregistered driver IDs returned.

VK_MAX_DRIVER_NAME_SIZE is the length in `char` values of an array containing a driver name string, as returned in VkPhysicalDeviceDriverProperties::`driverName`.

```
#define VK_MAX_DRIVER_NAME_SIZE           256U
```

```
#define VK_MAX_DRIVER_NAME_SIZE_KHR       VK_MAX_DRIVER_NAME_SIZE
```

VK_MAX_DRIVER_INFO_SIZE is the length in `char` values of an array containing a driver information string, as returned in VkPhysicalDeviceDriverProperties::`driverInfo`.

```
#define VK_MAX_DRIVER_INFO_SIZE           256U
```

```
#define VK_MAX_DRIVER_INFO_SIZE_KHR       VK_MAX_DRIVER_INFO_SIZE
```

The conformance test suite version an implementation is compliant with is described with the `VkConformanceVersion` structure:

```
// Provided by VK_VERSION_1_2
typedef struct VkConformanceVersion {
    uint8_t    major;
    uint8_t    minor;
    uint8_t    subminor;
    uint8_t    patch;
} VkConformanceVersion;
```

```
// Provided by VK_KHR_driver_properties
// Equivalent to VkConformanceVersion
typedef VkConformanceVersion VkConformanceVersionKHR;
```

- `major` is the major version number of the conformance test suite.
- `minor` is the minor version number of the conformance test suite.
- `subminor` is the subminor version number of the conformance test suite.
- `patch` is the patch version number of the conformance test suite.

The `VkPhysicalDevicePCIBusInfoPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_pci_bus_info
typedef struct VkPhysicalDevicePCIBusInfoPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           pciDomain;
    uint32_t           pciBus;
    uint32_t           pciDevice;
    uint32_t           pciFunction;
} VkPhysicalDevicePCIBusInfoPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pciDomain` is the PCI bus domain.
- `pciBus` is the PCI bus identifier.
- `pciDevice` is the PCI device identifier.
- `pciFunction` is the PCI device function identifier.

If the `VkPhysicalDevicePCIBusInfoPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

These are properties of the PCI bus information of a physical device.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePCIBusInfoPropertiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PCI_BUS_INFO_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceDrmPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_physical_device_drm
typedef struct VkPhysicalDeviceDrmPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           hasPrimary;
    VkBool32           hasRender;
    int64_t            primaryMajor;
    int64_t            primaryMinor;
    int64_t            renderMajor;
    int64_t            renderMinor;
} VkPhysicalDeviceDrmPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `hasPrimary` is a boolean indicating whether the physical device has a DRM primary node.
- `hasRender` is a boolean indicating whether the physical device has a DRM render node.
- `primaryMajor` is the DRM primary node major number, if any.
- `primaryMinor` is the DRM primary node minor number, if any.
- `renderMajor` is the DRM render node major number, if any.
- `renderMinor` is the DRM render node minor number, if any.

If the `VkPhysicalDeviceDrmPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

These are properties of the DRM information of a physical device.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDrmPropertiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRM_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceShaderIntegerDotProductProperties` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceShaderIntegerDotProductProperties {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           integerDotProduct8BitUnsignedAccelerated;
    VkBool32           integerDotProduct8BitSignedAccelerated;
    VkBool32           integerDotProduct8BitMixedSignednessAccelerated;
    VkBool32           integerDotProduct4x8BitPackedUnsignedAccelerated;
    VkBool32           integerDotProduct4x8BitPackedSignedAccelerated;
    VkBool32           integerDotProduct4x8BitPackedMixedSignednessAccelerated;
    VkBool32           integerDotProduct16BitUnsignedAccelerated;
    VkBool32           integerDotProduct16BitSignedAccelerated;
    VkBool32           integerDotProduct16BitMixedSignednessAccelerated;
    VkBool32           integerDotProduct32BitUnsignedAccelerated;
    VkBool32           integerDotProduct32BitSignedAccelerated;
    VkBool32           integerDotProduct32BitMixedSignednessAccelerated;
    VkBool32           integerDotProduct64BitUnsignedAccelerated;
    VkBool32           integerDotProduct64BitSignedAccelerated;
    VkBool32           integerDotProduct64BitMixedSignednessAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating8BitUnsignedAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating8BitSignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating8BitMixedSignednessAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating4x8BitPackedUnsignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating4x8BitPackedSignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating4x8BitPackedMixedSignednessAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating16BitUnsignedAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating16BitSignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating16BitMixedSignednessAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating32BitUnsignedAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating32BitSignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating32BitMixedSignednessAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating64BitUnsignedAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating64BitSignedAccelerated;
    VkBool32
integerDotProductAccumulatingSaturating64BitMixedSignednessAccelerated;
} VkPhysicalDeviceShaderIntegerDotProductProperties;
```

```
// Provided by VK_KHR_shader_integer_dot_product
// Equivalent to VkPhysicalDeviceShaderIntegerDotProductProperties
typedef VkPhysicalDeviceShaderIntegerDotProductProperties
VkPhysicalDeviceShaderIntegerDotProductPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- integerDotProduct8BitUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 8-bit unsigned dot product operations using the OpUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct8BitSignedAccelerated is a boolean that will be VK_TRUE if the support for 8-bit signed dot product operations using the OpSDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct8BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 8-bit mixed signedness dot product operations using the OpSUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct4x8BitPackedUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 8-bit unsigned dot product operations from operands packed into 32-bit integers using the OpUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct4x8BitPackedSignedAccelerated is a boolean that will be VK_TRUE if the support for 8-bit signed dot product operations from operands packed into 32-bit integers using the OpSDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct4x8BitPackedMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 8-bit mixed signedness dot product operations from operands packed into 32-bit integers using the OpSUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct16BitUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 16-bit unsigned dot product operations using the OpUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct16BitSignedAccelerated is a boolean that will be VK_TRUE if the support for 16-bit signed dot product operations using the OpSDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct16BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 16-bit mixed signedness dot product operations using the OpSUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct32BitUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 32-bit unsigned dot product operations using the OpUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct32BitSignedAccelerated is a boolean that will be VK_TRUE if the support for 32-bit signed dot product operations using the OpSDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct32BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the

support for 32-bit mixed signedness dot product operations using the OpSUDotKHR SPIR-V instruction is accelerated as defined below.

- integerDotProduct64BitUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 64-bit unsigned dot product operations using the OpUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct64BitSignedAccelerated is a boolean that will be VK_TRUE if the support for 64-bit signed dot product operations using the OpSDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProduct64BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 64-bit mixed signedness dot product operations using the OpSUDotKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating8BitUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 8-bit unsigned accumulating saturating dot product operations using the OpUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating8BitSignedAccelerated is a boolean that will be VK_TRUE if the support for 8-bit signed accumulating saturating dot product operations using the OpSDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating8BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 8-bit mixed signedness accumulating saturating dot product operations using the OpSUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating4x8BitPackedUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 8-bit unsigned accumulating saturating dot product operations from operands packed into 32-bit integers using the OpUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating4x8BitPackedSignedAccelerated is a boolean that will be VK_TRUE if the support for 8-bit signed accumulating saturating dot product operations from operands packed into 32-bit integers using the OpSDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating4x8BitPackedMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 8-bit mixed signedness accumulating saturating dot product operations from operands packed into 32-bit integers using the OpSUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating16BitUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 16-bit unsigned accumulating saturating dot product operations using the OpUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating16BitSignedAccelerated is a boolean that will be VK_TRUE if the support for 16-bit signed accumulating saturating dot product operations using the OpSDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating16BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 16-bit mixed signedness accumulating saturating dot product operations using the OpSUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating32BitUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 32-bit unsigned accumulating saturating dot product operations

using the OpUDotAccSatKHR SPIR-V instruction is accelerated as defined below.

- integerDotProductAccumulatingSaturating32BitSignedAccelerated is a boolean that will be VK_TRUE if the support for 32-bit signed accumulating saturating dot product operations using the OpSDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating32BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 32-bit mixed signedness accumulating saturating dot product operations using the OpSUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating64BitUnsignedAccelerated is a boolean that will be VK_TRUE if the support for 64-bit unsigned accumulating saturating dot product operations using the OpUDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating64BitSignedAccelerated is a boolean that will be VK_TRUE if the support for 64-bit signed accumulating saturating dot product operations using the OpSDotAccSatKHR SPIR-V instruction is accelerated as defined below.
- integerDotProductAccumulatingSaturating64BitMixedSignednessAccelerated is a boolean that will be VK_TRUE if the support for 64-bit mixed signedness accumulating saturating dot product operations using the OpSUDotAccSatKHR SPIR-V instruction is accelerated as defined below.

If the VkPhysicalDeviceShaderIntegerDotProductProperties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

These are properties of the integer dot product acceleration information of a physical device.

NOTE

A dot product operation is deemed accelerated if its implementation provides a performance advantage over application-provided code composed from elementary instructions and/or other dot product instructions, either because the implementation uses optimized machine code sequences whose generation from application-provided code cannot be guaranteed or because it uses hardware features that cannot otherwise be targeted from application-provided code.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderIntegerDotProductProperties-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceShaderTileImagePropertiesEXT structure is defined as:

```
// Provided by VK_EXT_shader_tile_image
typedef struct VkPhysicalDeviceShaderTileImagePropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderTileImageCoherentReadAccelerated;
    VkBool32           shaderTileImageReadSampleFromPixelRateInvocation;
    VkBool32           shaderTileImageReadFromHelperInvocation;
} VkPhysicalDeviceShaderTileImagePropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderTileImageCoherentReadAccelerated` is a boolean that will be VK_TRUE if coherent reads of tile image data are accelerated.
- `shaderTileImageReadSampleFromPixelRateInvocation` is a boolean that will be VK_TRUE if reading from samples from a pixel rate fragment invocation is supported when VkPipelineMultisampleStateCreateInfo::`rasterizationSamples` > 1.
- `shaderTileImageReadFromHelperInvocation` is a boolean that will be VK_TRUE if reads of tile image data from helper fragment invocations result in valid values.

If the `VkPhysicalDeviceShaderTileImagePropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

These are properties of the tile image information of a physical device.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderTileImagePropertiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TILE_IMAGE_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

To query properties of queues available on a physical device, call:

WARNING	This functionality is superseded by vkGetPhysicalDeviceQueueFamilyProperties2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceQueueFamilyProperties(
    VkPhysicalDevice                            physicalDevice,
    uint32_t*                                   pQueueFamilyPropertyCount,
    VkQueueFamilyProperties*                    pQueueFamilyProperties);
```

- `physicalDevice` is the handle to the physical device whose properties will be queried.
- `pQueueFamilyPropertyCount` is a pointer to an integer related to the number of queue families available or queried, as described below.
- `pQueueFamilyProperties` is either `NULL` or a pointer to an array of VkQueueFamilyProperties structures.

If `pQueueFamilyProperties` is `NULL`, then the number of queue families available is returned in `pQueueFamilyPropertyCount`. Implementations **must** support at least one queue family. Otherwise, `pQueueFamilyPropertyCount` **must** point to a variable set by the application to the number of elements in the `pQueueFamilyProperties` array, and on return the variable is overwritten with the number of structures actually written to `pQueueFamilyProperties`. If `pQueueFamilyPropertyCount` is less than the number of queue families available, at most `pQueueFamilyPropertyCount` structures will be written.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceQueueFamilyProperties-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceQueueFamilyProperties-pQueueFamilyPropertyCount-parameter
  `pQueueFamilyPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceQueueFamilyProperties-pQueueFamilyProperties-parameter
  If the value referenced by `pQueueFamilyPropertyCount` is not `0`, and `pQueueFamilyProperties` is not `NULL`, `pQueueFamilyProperties` **must** be a valid pointer to an array of `pQueueFamilyPropertyCount` VkQueueFamilyProperties structures

The `VkQueueFamilyProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkQueueFamilyProperties {
    VkQueueFlags    queueFlags;
    uint32_t        queueCount;
    uint32_t        timestampValidBits;
    VkExtent3D      minImageTransferGranularity;
} VkQueueFamilyProperties;
```

- `queueFlags` is a bitmask of VkQueueFlagBits indicating capabilities of the queues in this queue family.

- `queueCount` is the unsigned integer count of queues in this queue family. Each queue family **must** support at least one queue.
- `timestampValidBits` is the unsigned integer count of meaningful bits in the timestamps written via vkCmdWriteTimestamp2 or vkCmdWriteTimestamp. The valid range for the count is 36 to 64 bits, or a value of 0, indicating no support for timestamps. Bits outside the valid range are guaranteed to be zeros.
- `minImageTransferGranularity` is the minimum granularity supported for image transfer operations on the queues in this queue family.

The value returned in `minImageTransferGranularity` has a unit of compressed texel blocks for images having a block-compressed format, and a unit of texels otherwise.

Possible values of `minImageTransferGranularity` are:

- (0,0,0) specifies that only whole mip levels **must** be transferred using the image transfer operations on the corresponding queues. In this case, the following restrictions apply to all offset and extent parameters of image transfer operations:
  - The `x`, `y`, and `z` members of a VkOffset3D parameter **must** always be zero.
  - The `width`, `height`, and `depth` members of a VkExtent3D parameter **must** always match the width, height, and depth of the image subresource corresponding to the parameter, respectively.
- (A_x, A_y, A_z) where A_x, A_y, and A_z are all integer powers of two. In this case the following restrictions apply to all image transfer operations:
  - `x`, `y`, and `z` of a VkOffset3D parameter **must** be integer multiples of A_x, A_y, and A_z, respectively.
  - `width` of a VkExtent3D parameter **must** be an integer multiple of A_x, or else `x` + `width` **must** equal the width of the image subresource corresponding to the parameter.
  - `height` of a VkExtent3D parameter **must** be an integer multiple of A_y, or else `y` + `height` **must** equal the height of the image subresource corresponding to the parameter.
  - `depth` of a VkExtent3D parameter **must** be an integer multiple of A_z, or else `z` + `depth` **must** equal the depth of the image subresource corresponding to the parameter.
  - If the format of the image corresponding to the parameters is one of the block-compressed formats then for the purposes of the above calculations the granularity **must** be scaled up by the compressed texel block dimensions.

Queues supporting graphics and/or compute and/or, if the `maintenance11` feature is supported, transfer operations, **must** report (1,1,1) in `minImageTransferGranularity`, meaning that there are no additional restrictions on the granularity of image transfer operations for these queues. Other queues supporting image transfer operations are only **required** to support whole mip level transfers, thus `minImageTransferGranularity` for queues belonging to such queue families **may** be (0,0,0).

If the `maintenance11` feature is supported, VkQueueFamilyOptimalImageTransferGranularityPropertiesKHR::`optimalImageTransferGranularity` **can** be used to determine the optimal image transfer granularity for a particular queue family.

The Device Memory section describes memory properties queried from the physical device.

For physical device feature queries see the Features chapter.

Bits which **may** be set in VkQueueFamilyProperties::`queueFlags`, indicating capabilities of queues in a queue family are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueueFlagBits {
    VK_QUEUE_GRAPHICS_BIT = 0x00000001,
    VK_QUEUE_COMPUTE_BIT = 0x00000002,
    VK_QUEUE_TRANSFER_BIT = 0x00000004,
    VK_QUEUE_SPARSE_BINDING_BIT = 0x00000008,
  // Provided by VK_VERSION_1_1
    VK_QUEUE_PROTECTED_BIT = 0x00000010,
  // Provided by VK_KHR_video_decode_queue
    VK_QUEUE_VIDEO_DECODE_BIT_KHR = 0x00000020,
  // Provided by VK_KHR_video_encode_queue
    VK_QUEUE_VIDEO_ENCODE_BIT_KHR = 0x00000040,
} VkQueueFlagBits;
```

- VK_QUEUE_GRAPHICS_BIT specifies that queues in this queue family support graphics operations.
- VK_QUEUE_COMPUTE_BIT specifies that queues in this queue family support compute operations.
- VK_QUEUE_TRANSFER_BIT specifies that queues in this queue family support transfer operations.
- VK_QUEUE_SPARSE_BINDING_BIT specifies that queues in this queue family support sparse memory management operations (see Sparse Resources). If any of the sparse resource features are supported, then at least one queue family **must** support this bit.
- VK_QUEUE_VIDEO_DECODE_BIT_KHR specifies that queues in this queue family support video decode operations.
- VK_QUEUE_VIDEO_ENCODE_BIT_KHR specifies that queues in this queue family support video encode operations.
- VK_QUEUE_PROTECTED_BIT specifies that queues in this queue family support the VK_DEVICE_QUEUE_CREATE_PROTECTED_BIT bit (see Protected Memory). If the physical device supports the `protectedMemory` feature, at least one of its queue families **must** support this bit.

At least one queue family of at least one physical device exposed by the implementation **must** support at least one of the following sets of operations:

- graphics operations
- compute operations
- video encode operations
- video decode operations

If an implementation exposes any queue family that supports graphics operations, at least one queue family of at least one physical device exposed by the implementation **must** support both graphics and compute operations.

Furthermore, if the `protectedMemory` physical device feature is supported, then at least one queue family of at least one physical device exposed by the implementation **must** support graphics operations, compute operations, and protected memory operations.

NOTE All commands that are allowed on a queue that supports transfer operations are also allowed on a queue that supports either graphics or compute operations. Thus, if the capabilities of a queue family include VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT, then reporting the VK_QUEUE_TRANSFER_BIT capability separately for that queue family is **optional**.

For further details see Queues.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkQueueFlags;
```

`VkQueueFlags` is a bitmask type for setting a mask of zero or more VkQueueFlagBits.

To query properties of queues available on a physical device, call:

```
// Provided by VK_VERSION_1_1
void vkGetPhysicalDeviceQueueFamilyProperties2(
    VkPhysicalDevice                            physicalDevice,
    uint32_t*                                   pQueueFamilyPropertyCount,
    VkQueueFamilyProperties2*                   pQueueFamilyProperties);
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to vkGetPhysicalDeviceQueueFamilyProperties2
void vkGetPhysicalDeviceQueueFamilyProperties2KHR(
    VkPhysicalDevice                            physicalDevice,
    uint32_t*                                   pQueueFamilyPropertyCount,
    VkQueueFamilyProperties2*                   pQueueFamilyProperties);
```

- `physicalDevice` is the handle to the physical device whose properties will be queried.
- `pQueueFamilyPropertyCount` is a pointer to an integer related to the number of queue families available or queried, as described in vkGetPhysicalDeviceQueueFamilyProperties.
- `pQueueFamilyProperties` is either `NULL` or a pointer to an array of VkQueueFamilyProperties2 structures.

`vkGetPhysicalDeviceQueueFamilyProperties2` behaves similarly to vkGetPhysicalDeviceQueueFamilyProperties, with the ability to return extended information in a `pNext` chain of output structures.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceQueueFamilyProperties2-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceQueueFamilyProperties2-pQueueFamilyPropertyCount-parameter
  `pQueueFamilyPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceQueueFamilyProperties2-pQueueFamilyProperties-parameter
  If the value referenced by `pQueueFamilyPropertyCount` is not `0`, and `pQueueFamilyProperties` is not `NULL`, `pQueueFamilyProperties` **must** be a valid pointer to an array of `pQueueFamilyPropertyCount` VkQueueFamilyProperties2 structures

The `VkQueueFamilyProperties2` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkQueueFamilyProperties2 {
    VkStructureType            sType;
    void*                      pNext;
    VkQueueFamilyProperties    queueFamilyProperties;
} VkQueueFamilyProperties2;
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to VkQueueFamilyProperties2
typedef VkQueueFamilyProperties2 VkQueueFamilyProperties2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `queueFamilyProperties` is a VkQueueFamilyProperties structure which is populated with the same values as in vkGetPhysicalDeviceQueueFamilyProperties.

Valid Usage (Implicit)

- VUID-VkQueueFamilyProperties2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_QUEUE_FAMILY_PROPERTIES_2
- VUID-VkQueueFamilyProperties2-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkQueueFamilyGlobalPriorityProperties, VkQueueFamilyOptimalImageTransferGranularityPropertiesKHR, VkQueueFamilyOwnershipTransferPropertiesKHR, VkQueueFamilyQueryResultStatusPropertiesKHR, or VkQueueFamilyVideoPropertiesKHR
- VUID-VkQueueFamilyProperties2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

The VkQueueFamilyGlobalPriorityProperties structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkQueueFamilyGlobalPriorityProperties {
    VkStructureType          sType;
    void*                    pNext;
    uint32_t                 priorityCount;
    VkQueueGlobalPriority    priorities[VK_MAX_GLOBAL_PRIORITY_SIZE];
} VkQueueFamilyGlobalPriorityProperties;
```

```
// Provided by VK_KHR_global_priority
// Equivalent to VkQueueFamilyGlobalPriorityProperties
typedef VkQueueFamilyGlobalPriorityProperties
VkQueueFamilyGlobalPriorityPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- priorityCount is the number of supported global queue priorities in this queue family, and it **must** be greater than 0.
- priorities is an array of VK_MAX_GLOBAL_PRIORITY_SIZE VkQueueGlobalPriority enums representing all supported global queue priorities in this queue family. The first priorityCount elements of the array will be valid.

If the VkQueueFamilyGlobalPriorityProperties structure is included in the pNext chain of the VkQueueFamilyProperties2 structure passed to vkGetPhysicalDeviceQueueFamilyProperties2, it is filled in with the list of supported global queue priorities for the indicated family.

The valid elements of priorities **must** not contain any duplicate values.

The valid elements of priorities **must** be a continuous sequence of VkQueueGlobalPriority enums in ascending order.

NOTE For example, returning priorityCount as 3 with supported priorities as VK_QUEUE_GLOBAL_PRIORITY_LOW, VK_QUEUE_GLOBAL_PRIORITY_MEDIUM and VK_QUEUE_GLOBAL_PRIORITY_REALTIME is not allowed.

Valid Usage (Implicit)

- VUID-VkQueueFamilyGlobalPriorityProperties-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_QUEUE_FAMILY_GLOBAL_PRIORITY_PROPERTIES

Structure Chaining

Extends the structure

- VkQueueFamilyProperties2

VK_MAX_GLOBAL_PRIORITY_SIZE is the length of an array of VkQueueGlobalPriority enumerants representing supported queue priorities, as returned in VkQueueFamilyGlobalPriorityProperties ::priorities.

```
#define VK_MAX_GLOBAL_PRIORITY_SIZE       16U
```

```
#define VK_MAX_GLOBAL_PRIORITY_SIZE_KHR   VK_MAX_GLOBAL_PRIORITY_SIZE
```

The VkQueueFamilyVideoPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkQueueFamilyVideoPropertiesKHR {
    VkStructureType                 sType;
    void*                           pNext;
    VkVideoCodecOperationFlagsKHR   videoCodecOperations;
} VkQueueFamilyVideoPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- videoCodecOperations is a bitmask of VkVideoCodecOperationFlagBitsKHR that indicates the set of video codec operations supported by the queue family.

If this structure is included in the pNext chain of the VkQueueFamilyProperties2 structure passed to vkGetPhysicalDeviceQueueFamilyProperties2, then it is filled with the set of video codec operations supported by the specified queue family.

Valid Usage (Implicit)

- VUID-VkQueueFamilyVideoPropertiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_QUEUE_FAMILY_VIDEO_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkQueueFamilyProperties2

The VkQueueFamilyQueryResultStatusPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkQueueFamilyQueryResultStatusPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           queryResultStatusSupport;
} VkQueueFamilyQueryResultStatusPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- queryResultStatusSupport reports VK_TRUE if query type VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR and use of VK_QUERY_RESULT_WITH_STATUS_BIT_KHR are supported.

If this structure is included in the pNext chain of the VkQueueFamilyProperties2 structure passed to vkGetPhysicalDeviceQueueFamilyProperties2, then it is filled with information about whether result status queries are supported by the specified queue family.

Valid Usage (Implicit)

- VUID-VkQueueFamilyQueryResultStatusPropertiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_QUEUE_FAMILY_QUERY_RESULT_STATUS_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkQueueFamilyProperties2

To enumerate the performance query counters available on a queue family of a physical device, call:

```
// Provided by VK_KHR_performance_query
VkResult vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR(
    VkPhysicalDevice                            physicalDevice,
    uint32_t                                    queueFamilyIndex,
    uint32_t*                                   pCounterCount,
    VkPerformanceCounterKHR*                    pCounters,
    VkPerformanceCounterDescriptionKHR*         pCounterDescriptions);
```

- physicalDevice is the handle to the physical device whose queue family performance query counter properties will be queried.
- queueFamilyIndex is the index into the queue family of the physical device we want to get

properties for.

- `pCounterCount` is a pointer to an integer related to the number of counters available or queried, as described below.
- `pCounters` is either `NULL` or a pointer to an array of VkPerformanceCounterKHR structures.
- `pCounterDescriptions` is either `NULL` or a pointer to an array of VkPerformanceCounterDescriptionKHR structures.

If `pCounters` is `NULL` and `pCounterDescriptions` is `NULL`, then the number of counters available is returned in `pCounterCount`. Otherwise, `pCounterCount` **must** point to a variable set by the application to the number of elements in the `pCounters`, `pCounterDescriptions`, or both arrays and on return the variable is overwritten with the number of structures actually written out. If `pCounterCount` is less than the number of counters available, at most `pCounterCount` structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available counters were returned.

Valid Usage (Implicit)

- VUID-vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR-pCounterCount-parameter
  `pCounterCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR-pCounters-parameter
  If the value referenced by `pCounterCount` is not `0`, and `pCounters` is not `NULL`, `pCounters` **must** be a valid pointer to an array of `pCounterCount` VkPerformanceCounterKHR structures
- VUID-vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR-pCounterDescriptions-parameter
  If the value referenced by `pCounterCount` is not `0`, and `pCounterDescriptions` is not `NULL`, `pCounterDescriptions` **must** be a valid pointer to an array of `pCounterCount` VkPerformanceCounterDescriptionKHR structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY

- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkPerformanceCounterKHR structure is defined as:

```
// Provided by VK_KHR_performance_query
typedef struct VkPerformanceCounterKHR {
    VkStructureType                  sType;
    void*                            pNext;
    VkPerformanceCounterUnitKHR      unit;
    VkPerformanceCounterScopeKHR     scope;
    VkPerformanceCounterStorageKHR   storage;
    uint8_t                          uuid[VK_UUID_SIZE];
} VkPerformanceCounterKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- unit is a VkPerformanceCounterUnitKHR specifying the unit that the counter data will record.
- scope is a VkPerformanceCounterScopeKHR specifying the scope that the counter belongs to.
- storage is a VkPerformanceCounterStorageKHR specifying the storage type that the counter's data uses.
- uuid is an array of size VK_UUID_SIZE, containing 8-bit values that represent a universally unique identifier for the counter of the physical device.

Valid Usage (Implicit)

- VUID-VkPerformanceCounterKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PERFORMANCE_COUNTER_KHR
- VUID-VkPerformanceCounterKHR-pNext-pNext
  pNext **must** be NULL

Performance counters have an associated unit. This unit describes how to interpret the performance counter result.

The performance counter unit types which **may** be returned in VkPerformanceCounterKHR::unit are:

```
// Provided by VK_KHR_performance_query
typedef enum VkPerformanceCounterUnitKHR {
    VK_PERFORMANCE_COUNTER_UNIT_GENERIC_KHR = 0,
    VK_PERFORMANCE_COUNTER_UNIT_PERCENTAGE_KHR = 1,
    VK_PERFORMANCE_COUNTER_UNIT_NANOSECONDS_KHR = 2,
    VK_PERFORMANCE_COUNTER_UNIT_BYTES_KHR = 3,
    VK_PERFORMANCE_COUNTER_UNIT_BYTES_PER_SECOND_KHR = 4,
    VK_PERFORMANCE_COUNTER_UNIT_KELVIN_KHR = 5,
    VK_PERFORMANCE_COUNTER_UNIT_WATTS_KHR = 6,
    VK_PERFORMANCE_COUNTER_UNIT_VOLTS_KHR = 7,
    VK_PERFORMANCE_COUNTER_UNIT_AMPS_KHR = 8,
    VK_PERFORMANCE_COUNTER_UNIT_HERTZ_KHR = 9,
    VK_PERFORMANCE_COUNTER_UNIT_CYCLES_KHR = 10,
} VkPerformanceCounterUnitKHR;
```

- VK_PERFORMANCE_COUNTER_UNIT_GENERIC_KHR - the performance counter unit is a generic data point.
- VK_PERFORMANCE_COUNTER_UNIT_PERCENTAGE_KHR - the performance counter unit is a percentage (%).
- VK_PERFORMANCE_COUNTER_UNIT_NANOSECONDS_KHR - the performance counter unit is a value of nanoseconds (ns).
- VK_PERFORMANCE_COUNTER_UNIT_BYTES_KHR - the performance counter unit is a value of bytes.
- VK_PERFORMANCE_COUNTER_UNIT_BYTES_PER_SECOND_KHR - the performance counter unit is a value of bytes/s.
- VK_PERFORMANCE_COUNTER_UNIT_KELVIN_KHR - the performance counter unit is a temperature reported in Kelvin.
- VK_PERFORMANCE_COUNTER_UNIT_WATTS_KHR - the performance counter unit is a value of watts (W).
- VK_PERFORMANCE_COUNTER_UNIT_VOLTS_KHR - the performance counter unit is a value of volts (V).
- VK_PERFORMANCE_COUNTER_UNIT_AMPS_KHR - the performance counter unit is a value of amps (A).
- VK_PERFORMANCE_COUNTER_UNIT_HERTZ_KHR - the performance counter unit is a value of hertz (Hz).
- VK_PERFORMANCE_COUNTER_UNIT_CYCLES_KHR - the performance counter unit is a value of cycles.

Performance counters have an associated scope. This scope describes the granularity of a performance counter.

The performance counter scope types which **may** be returned in VkPerformanceCounterKHR::scope are:

```
// Provided by VK_KHR_performance_query
typedef enum VkPerformanceCounterScopeKHR {
    VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_BUFFER_KHR = 0,
    VK_PERFORMANCE_COUNTER_SCOPE_RENDER_PASS_KHR = 1,
    VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_KHR = 2,
  // VK_QUERY_SCOPE_COMMAND_BUFFER_KHR is a legacy alias
    VK_QUERY_SCOPE_COMMAND_BUFFER_KHR =
VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_BUFFER_KHR,
  // VK_QUERY_SCOPE_RENDER_PASS_KHR is a legacy alias
    VK_QUERY_SCOPE_RENDER_PASS_KHR = VK_PERFORMANCE_COUNTER_SCOPE_RENDER_PASS_KHR,
  // VK_QUERY_SCOPE_COMMAND_KHR is a legacy alias
    VK_QUERY_SCOPE_COMMAND_KHR = VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_KHR,
} VkPerformanceCounterScopeKHR;
```

- VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_BUFFER_KHR - the performance counter scope is a single complete command buffer.
- VK_PERFORMANCE_COUNTER_SCOPE_RENDER_PASS_KHR - the performance counter scope is zero or more complete render passes. The performance query containing the performance counter **must** begin and end outside a render pass instance.
- VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_KHR - the performance counter scope is zero or more commands.

Performance counters have an associated storage. This storage describes the payload of a counter result.

The performance counter storage types which **may** be returned in VkPerformanceCounterKHR ::`storage` are:

```
// Provided by VK_KHR_performance_query
typedef enum VkPerformanceCounterStorageKHR {
    VK_PERFORMANCE_COUNTER_STORAGE_INT32_KHR = 0,
    VK_PERFORMANCE_COUNTER_STORAGE_INT64_KHR = 1,
    VK_PERFORMANCE_COUNTER_STORAGE_UINT32_KHR = 2,
    VK_PERFORMANCE_COUNTER_STORAGE_UINT64_KHR = 3,
    VK_PERFORMANCE_COUNTER_STORAGE_FLOAT32_KHR = 4,
    VK_PERFORMANCE_COUNTER_STORAGE_FLOAT64_KHR = 5,
} VkPerformanceCounterStorageKHR;
```

- VK_PERFORMANCE_COUNTER_STORAGE_INT32_KHR - the performance counter storage is a 32-bit signed integer.
- VK_PERFORMANCE_COUNTER_STORAGE_INT64_KHR - the performance counter storage is a 64-bit signed integer.
- VK_PERFORMANCE_COUNTER_STORAGE_UINT32_KHR - the performance counter storage is a 32-bit unsigned integer.
- VK_PERFORMANCE_COUNTER_STORAGE_UINT64_KHR - the performance counter storage is a 64-bit unsigned integer.

- VK_PERFORMANCE_COUNTER_STORAGE_FLOAT32_KHR - the performance counter storage is a 32-bit floating-point.
- VK_PERFORMANCE_COUNTER_STORAGE_FLOAT64_KHR - the performance counter storage is a 64-bit floating-point.

The VkPerformanceCounterDescriptionKHR structure is defined as:

```
// Provided by VK_KHR_performance_query
typedef struct VkPerformanceCounterDescriptionKHR {
    VkStructureType                            sType;
    void*                                      pNext;
    VkPerformanceCounterDescriptionFlagsKHR    flags;
    char                                       name[VK_MAX_DESCRIPTION_SIZE];
    char                                       category[VK_MAX_DESCRIPTION_SIZE];
    char                                       description[VK_MAX_DESCRIPTION_SIZE];
} VkPerformanceCounterDescriptionKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask of VkPerformanceCounterDescriptionFlagBitsKHR indicating the usage behavior for the counter.
- name is an array of size VK_MAX_DESCRIPTION_SIZE, containing a null-terminated UTF-8 string specifying the name of the counter.
- category is an array of size VK_MAX_DESCRIPTION_SIZE, containing a null-terminated UTF-8 string specifying the category of the counter.
- description is an array of size VK_MAX_DESCRIPTION_SIZE, containing a null-terminated UTF-8 string specifying the description of the counter.

Valid Usage (Implicit)

- VUID-VkPerformanceCounterDescriptionKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PERFORMANCE_COUNTER_DESCRIPTION_KHR
- VUID-VkPerformanceCounterDescriptionKHR-pNext-pNext
  pNext **must** be NULL

Bits which **can** be set in VkPerformanceCounterDescriptionKHR::flags, specifying usage behavior of a performance counter, are:

```
// Provided by VK_KHR_performance_query
typedef enum VkPerformanceCounterDescriptionFlagBitsKHR {
    VK_PERFORMANCE_COUNTER_DESCRIPTION_PERFORMANCE_IMPACTING_BIT_KHR = 0x00000001,
    VK_PERFORMANCE_COUNTER_DESCRIPTION_CONCURRENTLY_IMPACTED_BIT_KHR = 0x00000002,
  // VK_PERFORMANCE_COUNTER_DESCRIPTION_PERFORMANCE_IMPACTING_KHR is a legacy alias
    VK_PERFORMANCE_COUNTER_DESCRIPTION_PERFORMANCE_IMPACTING_KHR =
VK_PERFORMANCE_COUNTER_DESCRIPTION_PERFORMANCE_IMPACTING_BIT_KHR,
  // VK_PERFORMANCE_COUNTER_DESCRIPTION_CONCURRENTLY_IMPACTED_KHR is a legacy alias
    VK_PERFORMANCE_COUNTER_DESCRIPTION_CONCURRENTLY_IMPACTED_KHR =
VK_PERFORMANCE_COUNTER_DESCRIPTION_CONCURRENTLY_IMPACTED_BIT_KHR,
} VkPerformanceCounterDescriptionFlagBitsKHR;
```

- VK_PERFORMANCE_COUNTER_DESCRIPTION_PERFORMANCE_IMPACTING_BIT_KHR specifies that recording the counter **may** have a noticeable performance impact.
- VK_PERFORMANCE_COUNTER_DESCRIPTION_CONCURRENTLY_IMPACTED_BIT_KHR specifies that concurrently recording the counter while other submitted command buffers are running **may** impact the accuracy of the recording.

```
// Provided by VK_KHR_performance_query
typedef VkFlags VkPerformanceCounterDescriptionFlagsKHR;
```

VkPerformanceCounterDescriptionFlagsKHR is a bitmask type for setting a mask of zero or more VkPerformanceCounterDescriptionFlagBitsKHR.

The VkQueueFamilyOwnershipTransferPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_maintenance9
typedef struct VkQueueFamilyOwnershipTransferPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           optimalImageTransferToQueueFamilies;
} VkQueueFamilyOwnershipTransferPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- optimalImageTransferToQueueFamilies is a bitmask of queue family indices that indicates which queue families belonging to the same logical device support implicitly acquiring optimal image resources owned by this queue family, without the resources' contents becoming undefined.

If this structure is included in the pNext chain of the VkQueueFamilyProperties2 structure passed to vkGetPhysicalDeviceQueueFamilyProperties2, then it is filled with the queue family ownership properties for the specified queue family.

Valid Usage (Implicit)

- VUID-VkQueueFamilyOwnershipTransferPropertiesKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_QUEUE_FAMILY_OWNERSHIP_TRANSFER_PROPERTIES_KHR`

Structure Chaining

Extends the structure

- VkQueueFamilyProperties2

The VkQueueFamilyOptimalImageTransferGranularityPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_maintenance11
typedef struct VkQueueFamilyOptimalImageTransferGranularityPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkExtent3D         optimalImageTransferGranularity;
} VkQueueFamilyOptimalImageTransferGranularityPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `optimalImageTransferGranularity` is the optimal granularity for image copy operations in this queue family.

If this structure is included in the `pNext` chain of the VkQueueFamilyProperties2 structure passed to vkGetPhysicalDeviceQueueFamilyProperties2, then it is filled with the optimal image transfer granularity for the specified queue family.

The value returned in `optimalImageTransferGranularity` has a unit of compressed texel blocks for images having a block-compressed format, and a unit of texels otherwise.

Possible values of `optimalImageTransferGranularity` are:

- (0,0,0) specifies that an image copy operation is optimal only when copying whole mip levels, i.e. all of the following conditions are met:
  - The `x`, `y`, and `z` members of a VkOffset3D are zero.
  - The `width`, `height`, and `depth` members of a VkExtent3D parameter match the width, height, and depth of the image subresource corresponding to the parameter, respectively.
- (A_x, A_y, A_z) where A_x, A_y, and A_z are all integer powers of two. An image copy operation is optimal when all of the following conditions are met:
  - `width` of a VkExtent3D parameter is an integer multiple of A_x, or else `x` + `width` equals the width of the image subresource corresponding to the parameter.

- `height` of a VkExtent3D parameter is an integer multiple of A_y, or else `y` + `height` equals the height of the image subresource corresponding to the parameter.
- `depth` of a VkExtent3D parameter is an integer multiple of A_z, or else `z` + `depth` equals the depth of the image subresource corresponding to the parameter.
- If the format of the image corresponding to the parameters is one of the block-compressed formats then for the purposes of the above calculations the granularity **must** be scaled up by the compressed texel block dimensions.

NOTE While it is not required that copies are optimal, there may be a performance cost for copies not aligned to `optimalImageTransferGranularity`.

Valid Usage (Implicit)

- VUID-VkQueueFamilyOptimalImageTransferGranularityPropertiesKHR-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_QUEUE_FAMILY_OPTIMAL_IMAGE_TRANSFER_GRANULARITY_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkQueueFamilyProperties2

5.2. Devices

Device objects represent logical connections to physical devices. Each device exposes a number of *queue families* each having one or more *queues*. All queues in a queue family support the same operations.

As described in Physical Devices, a Vulkan application will first query for all physical devices in a system. Each physical device **can** then be queried for its capabilities, including its queue and queue family properties. Once an acceptable physical device is identified, an application will create a corresponding logical device. The created logical device is then the primary interface to the physical device.

How to enumerate the physical devices in a system and query those physical devices for their queue family properties is described in the Physical Device Enumeration section above.

A single logical device **can** be created from multiple physical devices, if those physical devices belong to the same device group. A *device group* is a set of physical devices that support accessing each other's memory and recording a single command buffer that **can** be executed on all the physical devices. Device groups are enumerated by calling vkEnumeratePhysicalDeviceGroups, and a logical device is created from a subset of the physical devices in a device group by passing the physical devices through VkDeviceGroupDeviceCreateInfo. For two physical devices to be in the same device group, they **must** support identical extensions, features, and properties.

NOTE

Physical devices in the same device group **must** be so similar because there are no rules for how different features/properties would interact. They **must** return the same values for nearly every invariant `vkGetPhysicalDevice*` feature, property, capability, etc., but could potentially differ for certain queries based on things like having a different display connected, or a different compositor. The specification does not attempt to enumerate which state is in each category, because such a list would quickly become out of date.

To retrieve a list of the device groups present in the system, call:

```
// Provided by VK_VERSION_1_1
VkResult vkEnumeratePhysicalDeviceGroups(
    VkInstance                                  instance,
    uint32_t*                                   pPhysicalDeviceGroupCount,
    VkPhysicalDeviceGroupProperties*            pPhysicalDeviceGroupProperties);
```

```
// Provided by VK_KHR_device_group_creation
// Equivalent to vkEnumeratePhysicalDeviceGroups
VkResult vkEnumeratePhysicalDeviceGroupsKHR(
    VkInstance                                  instance,
    uint32_t*                                   pPhysicalDeviceGroupCount,
    VkPhysicalDeviceGroupProperties*            pPhysicalDeviceGroupProperties);
```

- `instance` is a handle to a Vulkan instance previously created with vkCreateInstance.
- `pPhysicalDeviceGroupCount` is a pointer to an integer related to the number of device groups available or queried, as described below.
- `pPhysicalDeviceGroupProperties` is either `NULL` or a pointer to an array of VkPhysicalDeviceGroupProperties structures.

If `pPhysicalDeviceGroupProperties` is `NULL`, then the number of device groups available is returned in `pPhysicalDeviceGroupCount`. Otherwise, `pPhysicalDeviceGroupCount` **must** point to a variable set by the application to the number of elements in the `pPhysicalDeviceGroupProperties` array, and on return the variable is overwritten with the number of structures actually written to `pPhysicalDeviceGroupProperties`. If `pPhysicalDeviceGroupCount` is less than the number of device groups available, at most `pPhysicalDeviceGroupCount` structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available device groups were returned.

Every physical device **must** be in exactly one device group.

Valid Usage (Implicit)

- VUID-vkEnumeratePhysicalDeviceGroups-instance-parameter
  `instance` **must** be a valid VkInstance handle
- VUID-vkEnumeratePhysicalDeviceGroups-pPhysicalDeviceGroupCount-parameter

pPhysicalDeviceGroupCount **must** be a valid pointer to a uint32_t value

- VUID-vkEnumeratePhysicalDeviceGroups-pPhysicalDeviceGroupProperties-parameter
  If the value referenced by pPhysicalDeviceGroupCount is not 0, and pPhysicalDeviceGroupProperties is not NULL, pPhysicalDeviceGroupProperties **must** be a valid pointer to an array of pPhysicalDeviceGroupCount VkPhysicalDeviceGroupProperties structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkPhysicalDeviceGroupProperties structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceGroupProperties {
    VkStructureType     sType;
    void*               pNext;
    uint32_t            physicalDeviceCount;
    VkPhysicalDevice    physicalDevices[VK_MAX_DEVICE_GROUP_SIZE];
    VkBool32            subsetAllocation;
} VkPhysicalDeviceGroupProperties;
```

```
// Provided by VK_KHR_device_group_creation
// Equivalent to VkPhysicalDeviceGroupProperties
typedef VkPhysicalDeviceGroupProperties VkPhysicalDeviceGroupPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- physicalDeviceCount is the number of physical devices in the group.
- physicalDevices is an array of VK_MAX_DEVICE_GROUP_SIZE VkPhysicalDevice handles representing all physical devices in the group. The first physicalDeviceCount elements of the array will be valid.

- `subsetAllocation` specifies whether logical devices created from the group support allocating device memory on a subset of devices, via the `deviceMask` member of the VkMemoryAllocateFlagsInfo. If this is VK_FALSE, then all device memory allocations are made across all physical devices in the group. If `physicalDeviceCount` is `1`, then `subsetAllocation` **must** be VK_FALSE.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceGroupProperties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GROUP_PROPERTIES
- VUID-VkPhysicalDeviceGroupProperties-pNext-pNext
  `pNext` **must** be `NULL`

VK_MAX_DEVICE_GROUP_SIZE is the length of an array containing VkPhysicalDevice handle values representing all physical devices in a group, as returned in VkPhysicalDeviceGroupProperties ::`physicalDevices`.

```
#define VK_MAX_DEVICE_GROUP_SIZE          32U
```

```
#define VK_MAX_DEVICE_GROUP_SIZE_KHR      VK_MAX_DEVICE_GROUP_SIZE
```

5.2.1. Device Creation

Logical devices are represented by `VkDevice` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_HANDLE(VkDevice)
```

A logical device is created as a *connection* to a physical device. To create a logical device, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateDevice(
    VkPhysicalDevice                            physicalDevice,
    const VkDeviceCreateInfo*                   pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkDevice*                                   pDevice);
```

- `physicalDevice` **must** be one of the device handles returned from a call to `vkEnumeratePhysicalDevices` (see Physical Device Enumeration).
- `pCreateInfo` is a pointer to a VkDeviceCreateInfo structure containing information about how to create the device.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

- `pDevice` is a pointer to a handle in which the created VkDevice is returned.

`vkCreateDevice` verifies that extensions and features requested in the `ppEnabledExtensionNames` and `pEnabledFeatures` members of `pCreateInfo`, respectively, are supported by the implementation. If any requested extension is not supported, `vkCreateDevice` **must** return VK_ERROR_EXTENSION_NOT_PRESENT. If any requested feature is not supported, `vkCreateDevice` **must** return VK_ERROR_FEATURE_NOT_PRESENT. Support for extensions **can** be checked before creating a device by querying vkEnumerateDeviceExtensionProperties. Support for features **can** similarly be checked by querying vkGetPhysicalDeviceFeatures.

After verifying and enabling the extensions the `VkDevice` object is created and returned to the application.

Multiple logical devices **can** be created from the same physical device. Logical device creation **may** fail due to lack of device-specific resources (in addition to other errors). If that occurs, `vkCreateDevice` will return VK_ERROR_TOO_MANY_OBJECTS.

Valid Usage

- VUID-vkCreateDevice-ppEnabledExtensionNames-01387
  All required device extensions for each extension in the VkDeviceCreateInfo ::`ppEnabledExtensionNames` list **must** also be present in that list

Valid Usage (Implicit)

- VUID-vkCreateDevice-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkCreateDevice-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkDeviceCreateInfo structure
- VUID-vkCreateDevice-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateDevice-pDevice-parameter
  `pDevice` **must** be a valid pointer to a VkDevice handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_EXTENSION_NOT_PRESENT
- VK_ERROR_FEATURE_NOT_PRESENT

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_TOO_MANY_OBJECTS
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkDeviceCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDeviceCreateInfo {
    VkStructureType                    sType;
    const void*                        pNext;
    VkDeviceCreateFlags                flags;
    uint32_t                           queueCreateInfoCount;
    const VkDeviceQueueCreateInfo*     pQueueCreateInfos;
    // enabledLayerCount is legacy and not used
    uint32_t                           enabledLayerCount;
    // ppEnabledLayerNames is legacy and not used
    const char* const*                 ppEnabledLayerNames;
    uint32_t                           enabledExtensionCount;
    const char* const*                 ppEnabledExtensionNames;
    const VkPhysicalDeviceFeatures*    pEnabledFeatures;
} VkDeviceCreateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is reserved for future use.
- queueCreateInfoCount is the unsigned integer size of the pQueueCreateInfos array. Refer to the Queue Creation section below for further details.
- pQueueCreateInfos is a pointer to an array of VkDeviceQueueCreateInfo structures describing the queues that are requested to be created along with the logical device. Refer to the Queue Creation section below for further details.
- enabledLayerCount is legacy and not used. Implementations **must** not access this parameter. It **must** be 0. See Device Layers: Superseded via instance layers.
- ppEnabledLayerNames is legacy and not used. Implementations **must** not access this parameter. It **must** be NULL. See Device Layers: Superseded via instance layers.
- enabledExtensionCount is the number of device extensions to enable.
- ppEnabledExtensionNames is a pointer to an array of enabledExtensionCount null-terminated UTF-8 strings containing the names of extensions to enable for the created device. See the Extensions section for further details.
- pEnabledFeatures is NULL or a pointer to a VkPhysicalDeviceFeatures structure containing

boolean indicators of all the features to be enabled. Refer to the Features section for further details. This field is legacy. See Physical Device Queries: Superseded via version 2.

Valid Usage

- VUID-VkDeviceCreateInfo-queueFamilyIndex-02802
  The combination of the values in the `queueFamilyIndex` and `flags` members of each element of `pQueueCreateInfos` **must** be unique within `pQueueCreateInfos`
- VUID-VkDeviceCreateInfo-pQueueCreateInfos-06755
  If multiple elements of `pQueueCreateInfos` share the same `queueFamilyIndex`, the sum of their `queueCount` members **must** be less than or equal to the `queueCount` member of the `VkQueueFamilyProperties` structure, as returned by `vkGetPhysicalDeviceQueueFamilyProperties` in the `pQueueFamilyProperties`[queueFamilyIndex]
- VUID-VkDeviceCreateInfo-pQueueCreateInfos-06654
  If multiple elements of `pQueueCreateInfos` share the same `queueFamilyIndex`, then all of such elements **must** have the same global priority level, which **can** be specified explicitly by including a VkDeviceQueueGlobalPriorityCreateInfo structure in the `pNext` chain, or by the implicit default value
- VUID-VkDeviceCreateInfo-pNext-00373
  If the `pNext` chain includes a VkPhysicalDeviceFeatures2 structure, then `pEnabledFeatures` **must** be `NULL`
- VUID-VkDeviceCreateInfo-pNext-02829
  If the `pNext` chain includes a VkPhysicalDeviceVulkan11Features structure, then it **must** not include a VkPhysicalDevice16BitStorageFeatures, VkPhysicalDeviceMultiviewFeatures, VkPhysicalDeviceVariablePointersFeatures, VkPhysicalDeviceProtectedMemoryFeatures, VkPhysicalDeviceSamplerYcbcrConversionFeatures, or VkPhysicalDeviceShaderDrawParametersFeatures structure
- VUID-VkDeviceCreateInfo-pNext-02830
  If the `pNext` chain includes a VkPhysicalDeviceVulkan12Features structure, then it **must** not include a VkPhysicalDevice8BitStorageFeatures, VkPhysicalDeviceShaderAtomicInt64Features, VkPhysicalDeviceShaderFloat16Int8Features, VkPhysicalDeviceDescriptorIndexingFeatures, VkPhysicalDeviceScalarBlockLayoutFeatures, VkPhysicalDeviceImagelessFramebufferFeatures, VkPhysicalDeviceUniformBufferStandardLayoutFeatures, VkPhysicalDeviceShaderSubgroupExtendedTypesFeatures, VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures, VkPhysicalDeviceHostQueryResetFeatures, VkPhysicalDeviceTimelineSemaphoreFeatures, VkPhysicalDeviceBufferDeviceAddressFeatures, or VkPhysicalDeviceVulkanMemoryModelFeatures structure
- VUID-VkDeviceCreateInfo-ppEnabledExtensionNames-04476

If ppEnabledExtensionNames contains "VK_KHR_shader_draw_parameters" and the pNext chain includes a VkPhysicalDeviceVulkan11Features structure, then VkPhysicalDeviceVulkan11Features::shaderDrawParameters **must** be VK_TRUE

- VUID-VkDeviceCreateInfo-ppEnabledExtensionNames-02831
  If ppEnabledExtensionNames contains "VK_KHR_draw_indirect_count" and the pNext chain includes a VkPhysicalDeviceVulkan12Features structure, then VkPhysicalDeviceVulkan12Features::drawIndirectCount **must** be VK_TRUE
- VUID-VkDeviceCreateInfo-ppEnabledExtensionNames-02832
  If ppEnabledExtensionNames contains "VK_KHR_sampler_mirror_clamp_to_edge" and the pNext chain includes a VkPhysicalDeviceVulkan12Features structure, then VkPhysicalDeviceVulkan12Features::samplerMirrorClampToEdge **must** be VK_TRUE
- VUID-VkDeviceCreateInfo-ppEnabledExtensionNames-02833
  If ppEnabledExtensionNames contains "VK_EXT_descriptor_indexing" and the pNext chain includes a VkPhysicalDeviceVulkan12Features structure, then VkPhysicalDeviceVulkan12Features::descriptorIndexing **must** be VK_TRUE
- VUID-VkDeviceCreateInfo-pNext-06532
  If the pNext chain includes a VkPhysicalDeviceVulkan13Features structure, then it **must** not include a VkPhysicalDeviceDynamicRenderingFeatures, VkPhysicalDeviceImageRobustnessFeatures, VkPhysicalDeviceInlineUniformBlockFeatures, VkPhysicalDeviceMaintenance4Features, VkPhysicalDevicePipelineCreationCacheControlFeatures, VkPhysicalDevicePrivateDataFeatures, VkPhysicalDeviceShaderDemoteToHelperInvocationFeatures, VkPhysicalDeviceShaderIntegerDotProductFeatures, VkPhysicalDeviceShaderTerminateInvocationFeatures, VkPhysicalDeviceSubgroupSizeControlFeatures, VkPhysicalDeviceSynchronization2Features, VkPhysicalDeviceTextureCompressionASTCHDRFeatures, or VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures structure
- VUID-VkDeviceCreateInfo-pNext-10360
  If the pNext chain includes a VkPhysicalDeviceVulkan14Features structure, then it **must** not include a VkPhysicalDeviceGlobalPriorityQueryFeatures, VkPhysicalDeviceShaderSubgroupRotateFeatures, VkPhysicalDeviceShaderFloatControls2Features, VkPhysicalDeviceShaderExpectAssumeFeatures, VkPhysicalDeviceLineRasterizationFeatures, VkPhysicalDeviceVertexAttributeDivisorFeatures, VkPhysicalDeviceIndexTypeUint8Features, VkPhysicalDeviceDynamicRenderingLocalReadFeatures, VkPhysicalDeviceMaintenance5Features, VkPhysicalDeviceMaintenance6Features, VkPhysicalDevicePipelineProtectedAccessFeatures, VkPhysicalDevicePipelineRobustnessFeatures, or VkPhysicalDeviceHostImageCopyFeatures structure
- VUID-VkDeviceCreateInfo-ppEnabledExtensionNames-10858
  If ppEnabledExtensionNames contains "VK_KHR_push_descriptor" and the pNext chain includes

a VkPhysicalDeviceVulkan14Features structure, then VkPhysicalDeviceVulkan14Features ::pushDescriptor **must** be VK_TRUE

- VUID-VkDeviceCreateInfo-pProperties-04451
  If the VK_KHR_portability_subset extension is included in pProperties of vkEnumerateDeviceExtensionProperties, ppEnabledExtensionNames **must** include "VK_KHR_portability_subset"
- VUID-VkDeviceCreateInfo-fragmentDensityMap-04481
  If the fragmentDensityMap feature is enabled, the pipelineFragmentShadingRate feature **must** not be enabled
- VUID-VkDeviceCreateInfo-fragmentDensityMap-04482
  If the fragmentDensityMap feature is enabled, the primitiveFragmentShadingRate feature **must** not be enabled
- VUID-VkDeviceCreateInfo-fragmentDensityMap-04483
  If the fragmentDensityMap feature is enabled, the attachmentFragmentShadingRate feature **must** not be enabled
- VUID-VkDeviceCreateInfo-None-04896
  If the sparseImageInt64Atomics feature is enabled, shaderImageInt64Atomics **must** be enabled
- VUID-VkDeviceCreateInfo-None-04897
  If the sparseImageFloat32Atomics feature is enabled, shaderImageFloat32Atomics **must** be enabled
- VUID-VkDeviceCreateInfo-None-04898
  If the sparseImageFloat32AtomicAdd feature is enabled, shaderImageFloat32AtomicAdd **must** be enabled
- VUID-VkDeviceCreateInfo-sparseImageFloat32AtomicMinMax-04975
  If the sparseImageFloat32AtomicMinMax feature is enabled, shaderImageFloat32AtomicMinMax **must** be enabled
- VUID-VkDeviceCreateInfo-robustBufferAccess-10247
  If the robustBufferAccess feature is enabled, and robustBufferAccessUpdateAfterBind is VK_FALSE, then descriptorBindingUniformBufferUpdateAfterBind, descriptorBindingStorageBufferUpdateAfterBind, descriptorBindingUniformTexelBufferUpdateAfterBind, and descriptorBindingStorageTexelBufferUpdateAfterBind **must** not be enabled
- VUID-VkDeviceCreateInfo-None-10778
  If the maintenance9 feature is not supported, queueCreateInfoCount **must** be greater than 0
- VUID-VkDeviceCreateInfo-enabledLayerCount-12384
  enabledLayerCount **must** be 0
- VUID-VkDeviceCreateInfo-ppEnabledLayerNames-12385
  ppEnabledLayerNames **must** be NULL

Valid Usage (Implicit)

- VUID-VkDeviceCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_CREATE_INFO
- VUID-VkDeviceCreateInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkDeviceDeviceMemoryReportCreateInfoEXT, VkDeviceGroupDeviceCreateInfo, VkDevicePipelineBinaryInternalCacheControlKHR, VkDevicePrivateDataCreateInfo, VkPhysicalDevice16BitStorageFeatures, VkPhysicalDevice8BitStorageFeatures, VkPhysicalDeviceASTCDecodeFeaturesEXT, VkPhysicalDeviceAccelerationStructureFeaturesKHR, VkPhysicalDeviceAddressBindingReportFeaturesEXT, VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT, VkPhysicalDeviceAttachmentFeedbackLoopLayoutFeaturesEXT, VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT, VkPhysicalDeviceBorderColorSwizzleFeaturesEXT, VkPhysicalDeviceBufferDeviceAddressFeatures, VkPhysicalDeviceColorWriteEnableFeaturesEXT, VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR, VkPhysicalDeviceConditionalRenderingFeaturesEXT, VkPhysicalDeviceCooperativeMatrixFeaturesKHR, VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT, VkPhysicalDeviceCopyMemoryIndirectFeaturesKHR, VkPhysicalDeviceCustomBorderColorFeaturesEXT, VkPhysicalDeviceCustomResolveFeaturesEXT, VkPhysicalDeviceDepthBiasControlFeaturesEXT, VkPhysicalDeviceDepthClampControlFeaturesEXT, VkPhysicalDeviceDepthClampZeroOneFeaturesKHR, VkPhysicalDeviceDepthClipControlFeaturesEXT, VkPhysicalDeviceDepthClipEnableFeaturesEXT, VkPhysicalDeviceDescriptorBufferFeaturesEXT, VkPhysicalDeviceDescriptorHeapFeaturesEXT, VkPhysicalDeviceDescriptorIndexingFeatures, VkPhysicalDeviceDeviceAddressCommandsFeaturesKHR, VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT, VkPhysicalDeviceDeviceMemoryReportFeaturesEXT, VkPhysicalDeviceDynamicRenderingFeatures, VkPhysicalDeviceDynamicRenderingLocalReadFeatures, VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT, VkPhysicalDeviceExtendedDynamicState2FeaturesEXT, VkPhysicalDeviceExtendedDynamicState3FeaturesEXT, VkPhysicalDeviceExtendedDynamicStateFeaturesEXT, VkPhysicalDeviceExtendedFlagsFeaturesKHR, VkPhysicalDeviceFaultFeaturesEXT, VkPhysicalDeviceFaultFeaturesKHR, VkPhysicalDeviceFeatures2, VkPhysicalDeviceFragmentDensityMap2FeaturesEXT, VkPhysicalDeviceFragmentDensityMapFeaturesEXT, VkPhysicalDeviceFragmentDensityMapOffsetFeaturesEXT,

VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR,
VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT,
VkPhysicalDeviceFragmentShadingRateFeaturesKHR,
VkPhysicalDeviceFrameBoundaryFeaturesEXT,
VkPhysicalDeviceGlobalPriorityQueryFeatures,
VkPhysicalDeviceGraphicsPipelineLibraryFeaturesEXT,
VkPhysicalDeviceHostImageCopyFeatures, VkPhysicalDeviceHostQueryResetFeatures,
VkPhysicalDeviceImage2DViewOf3DFeaturesEXT,
VkPhysicalDeviceImageCompressionControlFeaturesEXT,
VkPhysicalDeviceImageCompressionControlSwapchainFeaturesEXT,
VkPhysicalDeviceImageRobustnessFeatures,
VkPhysicalDeviceImageSlicedViewOf3DFeaturesEXT,
VkPhysicalDeviceImageTilingControlFeaturesEXT,
VkPhysicalDeviceImageViewMinLodFeaturesEXT,
VkPhysicalDeviceImagelessFramebufferFeatures,
VkPhysicalDeviceIndexTypeUint8Features, VkPhysicalDeviceInlineUniformBlockFeatures,
VkPhysicalDeviceInternallySynchronizedQueuesFeaturesKHR,
VkPhysicalDeviceLegacyDitheringFeaturesEXT,
VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT,
VkPhysicalDeviceLineRasterizationFeatures,
VkPhysicalDeviceMaintenance10FeaturesKHR,
VkPhysicalDeviceMaintenance11FeaturesKHR, VkPhysicalDeviceMaintenance4Features,
VkPhysicalDeviceMaintenance5Features, VkPhysicalDeviceMaintenance6Features,
VkPhysicalDeviceMaintenance7FeaturesKHR,
VkPhysicalDeviceMaintenance8FeaturesKHR,
VkPhysicalDeviceMaintenance9FeaturesKHR,
VkPhysicalDeviceMapMemoryPlacedFeaturesEXT,
VkPhysicalDeviceMemoryDecompressionFeaturesEXT,
VkPhysicalDeviceMemoryPriorityFeaturesEXT,
VkPhysicalDeviceMeshShaderFeaturesEXT, VkPhysicalDeviceMultiDrawFeaturesEXT,
VkPhysicalDeviceMultisampledRenderToSingleSampledFeaturesEXT,
VkPhysicalDeviceMultisampledRenderToSwapchainFeaturesEXT,
VkPhysicalDeviceMultiviewFeatures,
VkPhysicalDeviceMutableDescriptorTypeFeaturesEXT,
VkPhysicalDeviceNestedCommandBufferFeaturesEXT,
VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT,
VkPhysicalDeviceOpacityMicromapFeaturesEXT,
VkPhysicalDeviceOpacityMicromapFeaturesKHR,
VkPhysicalDevicePageableDeviceLocalMemoryFeaturesEXT,
VkPhysicalDevicePerformanceQueryFeaturesKHR,
VkPhysicalDevicePipelineBinaryFeaturesKHR,
VkPhysicalDevicePipelineCreationCacheControlFeatures,
VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR,
VkPhysicalDevicePipelineLibraryGroupHandlesFeaturesKHR,
VkPhysicalDevicePipelinePropertiesFeaturesEXT,
VkPhysicalDevicePipelineProtectedAccessFeatures,
VkPhysicalDevicePipelineRobustnessFeatures,
VkPhysicalDevicePortabilitySubsetFeaturesKHR,

VkPhysicalDevicePresentId2FeaturesKHR, VkPhysicalDevicePresentIdFeaturesKHR,
VkPhysicalDevicePresentModeFifoLatestReadyFeaturesKHR,
VkPhysicalDevicePresentTimingFeaturesEXT,
VkPhysicalDevicePresentWait2FeaturesKHR, VkPhysicalDevicePresentWaitFeaturesKHR,
VkPhysicalDevicePrimitiveRestartIndexFeaturesEXT,
VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT,
VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT,
VkPhysicalDevicePrivateDataFeatures, VkPhysicalDeviceProtectedMemoryFeatures,
VkPhysicalDeviceProvokingVertexFeaturesEXT,
VkPhysicalDeviceRGBA10X6FormatsFeaturesEXT,
VkPhysicalDeviceRasterizationOrderAttachmentAccessFeaturesEXT,
VkPhysicalDeviceRayQueryFeaturesKHR,
VkPhysicalDeviceRayTracingInvocationReorderFeaturesEXT,
VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR,
VkPhysicalDeviceRayTracingPipelineFeaturesKHR,
VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR,
VkPhysicalDeviceRobustness2FeaturesKHR,
VkPhysicalDeviceSamplerYcbcrConversionFeatures,
VkPhysicalDeviceScalarBlockLayoutFeatures,
VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures,
VkPhysicalDeviceShader64BitIndexingFeaturesEXT,
VkPhysicalDeviceShaderAbortFeaturesKHR,
VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT,
VkPhysicalDeviceShaderAtomicFloatFeaturesEXT,
VkPhysicalDeviceShaderAtomicInt64Features,
VkPhysicalDeviceShaderBfloat16FeaturesKHR,
VkPhysicalDeviceShaderClockFeaturesKHR,
VkPhysicalDeviceShaderConstantDataFeaturesKHR,
VkPhysicalDeviceShaderDemoteToHelperInvocationFeatures,
VkPhysicalDeviceShaderDrawParametersFeatures,
VkPhysicalDeviceShaderExpectAssumeFeatures,
VkPhysicalDeviceShaderFloat16Int8Features, VkPhysicalDeviceShaderFloat8FeaturesEXT,
VkPhysicalDeviceShaderFloatControls2Features,
VkPhysicalDeviceShaderFmaFeaturesKHR,
VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT,
VkPhysicalDeviceShaderIntegerDotProductFeatures,
VkPhysicalDeviceShaderLongVectorFeaturesEXT,
VkPhysicalDeviceShaderMaximalReconvergenceFeaturesKHR,
VkPhysicalDeviceShaderModuleIdentifierFeaturesEXT,
VkPhysicalDeviceShaderOCPMicroscalingTypesFeaturesEXT,
VkPhysicalDeviceShaderObjectFeaturesEXT,
VkPhysicalDeviceShaderQuadControlFeaturesKHR,
VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR,
VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT,
VkPhysicalDeviceShaderSplitBarrierFeaturesEXT,
VkPhysicalDeviceShaderSubgroupExtendedTypesFeatures,
VkPhysicalDeviceShaderSubgroupPartitionedFeaturesEXT,
VkPhysicalDeviceShaderSubgroupRotateFeatures,

VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR, VkPhysicalDeviceShaderTerminateInvocationFeatures, VkPhysicalDeviceShaderTileImageFeaturesEXT, VkPhysicalDeviceShaderUniformBufferUnsizedArrayFeaturesEXT, VkPhysicalDeviceShaderUntypedPointersFeaturesKHR, VkPhysicalDeviceSubgroupSizeControlFeatures, VkPhysicalDeviceSubpassMergeFeedbackFeaturesEXT, VkPhysicalDeviceSwapchainMaintenance1FeaturesKHR, VkPhysicalDeviceSynchronization2Features, VkPhysicalDeviceTextureCompressionASTC3DFeaturesEXT, VkPhysicalDeviceTextureCompressionASTCHDRFeatures, VkPhysicalDeviceTimelineSemaphoreFeatures, VkPhysicalDeviceTransformFeedbackFeaturesEXT, VkPhysicalDeviceUnifiedImageLayoutsFeaturesKHR, VkPhysicalDeviceUniformBufferStandardLayoutFeatures, VkPhysicalDeviceVariablePointersFeatures, VkPhysicalDeviceVertexAttributeDivisorFeatures, VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT, VkPhysicalDeviceVideoDecodeVP9FeaturesKHR, VkPhysicalDeviceVideoEncodeAV1FeaturesKHR, VkPhysicalDeviceVideoEncodeFeedback2FeaturesKHR, VkPhysicalDeviceVideoEncodeIntraRefreshFeaturesKHR, VkPhysicalDeviceVideoEncodeQuantizationMapFeaturesKHR, VkPhysicalDeviceVideoMaintenance1FeaturesKHR, VkPhysicalDeviceVideoMaintenance2FeaturesKHR, VkPhysicalDeviceVulkan11Features, VkPhysicalDeviceVulkan12Features, VkPhysicalDeviceVulkan13Features, VkPhysicalDeviceVulkan14Features, VkPhysicalDeviceVulkanMemoryModelFeatures, VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR, VkPhysicalDeviceYcbcrImageArraysFeaturesEXT, VkPhysicalDeviceZeroInitializeDeviceMemoryFeaturesEXT, or VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures

- VUID-VkDeviceCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique, with the exception of structures of type VkDeviceDeviceMemoryReportCreateInfoEXT or VkDevicePrivateDataCreateInfo
- VUID-VkDeviceCreateInfo-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkDeviceCreateInfo-pQueueCreateInfos-parameter
  If `queueCreateInfoCount` is not `0`, `pQueueCreateInfos` **must** be a valid pointer to an array of `queueCreateInfoCount` valid VkDeviceQueueCreateInfo structures
- VUID-VkDeviceCreateInfo-ppEnabledExtensionNames-parameter
  If `enabledExtensionCount` is not `0`, `ppEnabledExtensionNames` **must** be a valid pointer to an array of `enabledExtensionCount` null-terminated UTF-8 strings
- VUID-VkDeviceCreateInfo-pEnabledFeatures-parameter
  If `pEnabledFeatures` is not `NULL`, `pEnabledFeatures` **must** be a valid pointer to a valid VkPhysicalDeviceFeatures structure

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDeviceCreateFlags;
```

VkDeviceCreateFlags is a bitmask type for setting a mask, but is currently reserved for future use.

A logical device **can** be created that connects to one or more physical devices by adding a VkDeviceGroupDeviceCreateInfo structure to the pNext chain of VkDeviceCreateInfo. The VkDeviceGroupDeviceCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkDeviceGroupDeviceCreateInfo {
    VkStructureType            sType;
    const void*                pNext;
    uint32_t                   physicalDeviceCount;
    const VkPhysicalDevice*    pPhysicalDevices;
} VkDeviceGroupDeviceCreateInfo;
```

```
// Provided by VK_KHR_device_group_creation
// Equivalent to VkDeviceGroupDeviceCreateInfo
typedef VkDeviceGroupDeviceCreateInfo VkDeviceGroupDeviceCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- physicalDeviceCount is the number of elements in the pPhysicalDevices array.
- pPhysicalDevices is a pointer to an array of physical device handles belonging to the same device group.

The elements of the pPhysicalDevices array are an ordered list of the physical devices that the logical device represents. These **must** be a subset of a single device group, and need not be in the same order as they were enumerated. The order of the physical devices in the pPhysicalDevices array determines the *device index* of each physical device, with element i being assigned a device index of i. Certain commands and structures refer to one or more physical devices by using device indices or *device masks* formed using device indices.

A logical device created without using VkDeviceGroupDeviceCreateInfo, or with physicalDeviceCount equal to zero, is equivalent to a physicalDeviceCount of one and pPhysicalDevices pointing to the physicalDevice parameter to vkCreateDevice. In particular, the device index of that physical device is zero.

Valid Usage

- VUID-VkDeviceGroupDeviceCreateInfo-pPhysicalDevices-00375
  Each element of pPhysicalDevices **must** be unique
- VUID-VkDeviceGroupDeviceCreateInfo-pPhysicalDevices-00376

All elements of pPhysicalDevices **must** be in the same device group as enumerated by vkEnumeratePhysicalDeviceGroups

- VUID-VkDeviceGroupDeviceCreateInfo-physicalDeviceCount-00377
  If physicalDeviceCount is not 0, the physicalDevice parameter of vkCreateDevice **must** be an element of pPhysicalDevices

Valid Usage (Implicit)

- VUID-VkDeviceGroupDeviceCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DEVICE_GROUP_DEVICE_CREATE_INFO
- VUID-VkDeviceGroupDeviceCreateInfo-pPhysicalDevices-parameter
  If physicalDeviceCount is not 0, pPhysicalDevices **must** be a valid pointer to an array of physicalDeviceCount valid VkPhysicalDevice handles

Structure Chaining

Extends the structure

- VkDeviceCreateInfo

To register callbacks for underlying device memory events of type VkDeviceMemoryReportEventTypeEXT, add one or multiple VkDeviceDeviceMemoryReportCreateInfoEXT structures to the pNext chain of the VkDeviceCreateInfo structure.

```
// Provided by VK_EXT_device_memory_report
typedef struct VkDeviceDeviceMemoryReportCreateInfoEXT {
    VkStructureType                         sType;
    const void*                             pNext;
    VkDeviceMemoryReportFlagsEXT            flags;
    PFN_vkDeviceMemoryReportCallbackEXT     pfnUserCallback;
    void*                                   pUserData;
} VkDeviceDeviceMemoryReportCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is 0 and reserved for future use.
- pfnUserCallback is the application callback function to call.
- pUserData is NULL or an application-defined user data pointer to be passed to the callback.

The callback **may** be called from multiple threads simultaneously.

The callback **must** be called only once by the implementation when a

VkDeviceMemoryReportEventTypeEXT event occurs.

NOTE	The callback could be called from a background thread other than the thread calling the Vulkan commands.

Valid Usage (Implicit)

- VUID-VkDeviceDeviceMemoryReportCreateInfoEXT-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_DEVICE_MEMORY_REPORT_CREATE_INFO_EXT`
- VUID-VkDeviceDeviceMemoryReportCreateInfoEXT-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkDeviceDeviceMemoryReportCreateInfoEXT-pfnUserCallback-parameter
  `pfnUserCallback` **must** be a valid PFN_vkDeviceMemoryReportCallbackEXT value

Structure Chaining

Extends the structure

- VkDeviceCreateInfo

The prototype for the VkDeviceDeviceMemoryReportCreateInfoEXT::`pfnUserCallback` function implemented by the application is:

```
// Provided by VK_EXT_device_memory_report
typedef void (*PFN_vkDeviceMemoryReportCallbackEXT)(
    const VkDeviceMemoryReportCallbackDataEXT*  pCallbackData,
    void*                                       pUserData);
```

- `pCallbackData` contains all the callback related data in the VkDeviceMemoryReportCallbackDataEXT structure.
- `pUserData` is the application-defined user data pointer, equal to the value of VkDeviceDeviceMemoryReportCreateInfoEXT::`pUserData` specified when the VkDevice object was created.

The callback **must** not make calls to any Vulkan commands.

The definition of `VkDeviceMemoryReportCallbackDataEXT` is:

```
// Provided by VK_EXT_device_memory_report
typedef struct VkDeviceMemoryReportCallbackDataEXT {
    VkStructureType                      sType;
    void*                                pNext;
    VkDeviceMemoryReportFlagsEXT         flags;
    VkDeviceMemoryReportEventTypeEXT     type;
    uint64_t                             memoryObjectId;
    VkDeviceSize                         size;
    VkObjectType                         objectType;
    uint64_t                             objectHandle;
    uint32_t                             heapIndex;
} VkDeviceMemoryReportCallbackDataEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is 0 and reserved for future use.
- `type` is a VkDeviceMemoryReportEventTypeEXT type specifying the type of event reported in this `VkDeviceMemoryReportCallbackDataEXT` structure.
- `memoryObjectId` is the unique id for the underlying memory object as described below.
- `size` is the size of the memory object in bytes. If `type` is VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATE_EXT, VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_IMPORT_EXT or VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATION_FAILED_EXT, `size` is a valid VkDeviceSize value. Otherwise, `size` is undefined.
- `objectType` is a VkObjectType value specifying the type of the object associated with this device memory report event. If `type` is VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATE_EXT, VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_FREE_EXT, VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_IMPORT_EXT, VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_UNIMPORT_EXT or VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATION_FAILED_EXT, `objectType` is a valid VkObjectType enum. Otherwise, `objectType` is undefined.
- `objectHandle` is the object this device memory report event is attributed to. If `type` is VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATE_EXT, VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_FREE_EXT, VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_IMPORT_EXT or VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_UNIMPORT_EXT, `objectHandle` is a valid Vulkan handle of the type associated with `objectType` as defined in the `VkObjectType` and Vulkan Handle Relationship table. Otherwise, `objectHandle` is undefined.
- `heapIndex` describes which memory heap this device memory allocation is made from. If `type` is VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATE_EXT or VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATION_FAILED_EXT, `heapIndex` corresponds to one of the valid heaps from the VkPhysicalDeviceMemoryProperties structure. Otherwise, `heapIndex` is undefined.

`memoryObjectId` is used to avoid double-counting on the same memory object.

If an internally-allocated device memory object or a VkDeviceMemory **cannot** be exported, `memoryObjectId` **must** be unique in the VkDevice.

Implementor's Note

If the heap backing an internally-allocated device memory **cannot** be used to back VkDeviceMemory, implementations **can** advertise that heap with no types.

NOTE

This structure should only be considered valid during the lifetime of the triggered callback.

For VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATE_EXT and VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_IMPORT_EXT events, `objectHandle` usually will not yet exist when the application or tool receives the callback. `objectHandle` will only exist when the create or allocate call that triggered the event returns, and if the allocation or import ends up failing `objectHandle` will not ever exist.

Valid Usage (Implicit)

- VUID-VkDeviceMemoryReportCallbackDataEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_MEMORY_REPORT_CALLBACK_DATA_EXT
- VUID-VkDeviceMemoryReportCallbackDataEXT-pNext-pNext
  `pNext` **must** be `NULL`

```
// Provided by VK_EXT_device_memory_report
typedef VkFlags VkDeviceMemoryReportFlagsEXT;
```

`VkDeviceMemoryReportFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

Possible values of VkDeviceMemoryReportCallbackDataEXT::`type`, specifying event types which cause the device driver to call the callback, are:

```
// Provided by VK_EXT_device_memory_report
typedef enum VkDeviceMemoryReportEventTypeEXT {
    VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATE_EXT = 0,
    VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_FREE_EXT = 1,
    VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_IMPORT_EXT = 2,
    VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_UNIMPORT_EXT = 3,
    VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATION_FAILED_EXT = 4,
} VkDeviceMemoryReportEventTypeEXT;
```

- VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATE_EXT specifies this event corresponds to the allocation of an internal device memory object or a VkDeviceMemory.
- VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_FREE_EXT specifies this event corresponds to the deallocation of an internally-allocated device memory object or a VkDeviceMemory.
- VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_IMPORT_EXT specifies this event corresponds to the import of an external memory object.
- VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_UNIMPORT_EXT specifies this event is the release of an imported external memory object.
- VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATION_FAILED_EXT specifies this event corresponds to the failed allocation of an internal device memory object or a VkDeviceMemory.

To reserve private data storage slots, add a VkDevicePrivateDataCreateInfo structure to the pNext chain of the VkDeviceCreateInfo structure. Reserving slots in this manner is not strictly necessary, but doing so **may** improve performance.

```
// Provided by VK_VERSION_1_3
typedef struct VkDevicePrivateDataCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           privateDataSlotRequestCount;
} VkDevicePrivateDataCreateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- privateDataSlotRequestCount is the number of slots to reserve.

Valid Usage (Implicit)

- VUID-VkDevicePrivateDataCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DEVICE_PRIVATE_DATA_CREATE_INFO

Structure Chaining

Extends the structure

- VkDeviceCreateInfo

To disable the implementation's internal pipeline cache, add a VkDevicePipelineBinaryInternalCacheControlKHR structure to the pNext chain of the VkDeviceCreateInfo structure.

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkDevicePipelineBinaryInternalCacheControlKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           disableInternalCache;
} VkDevicePipelineBinaryInternalCacheControlKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `disableInternalCache` specifies whether or not to disable the implementation's internal pipeline cache.

If the `VkDeviceCreateInfo`::`pNext` chain does not include this structure, then `disableInternalCache` defaults to VK_FALSE.

Valid Usage

- VUID-VkDevicePipelineBinaryInternalCacheControlKHR-disableInternalCache-09602
  If VkPhysicalDevicePipelineBinaryPropertiesKHR::`pipelineBinaryInternalCacheControl` is VK_FALSE, `disableInternalCache` **must** be VK_FALSE

Valid Usage (Implicit)

- VUID-VkDevicePipelineBinaryInternalCacheControlKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_PIPELINE_BINARY_INTERNAL_CACHE_CONTROL_KHR

Structure Chaining

Extends the structure

- VkDeviceCreateInfo

5.2.2. Device Use

The following is a high-level list of `VkDevice` uses along with references on where to find more information:

- Creation of queues. See the Queues section below for further details.
- Creation and tracking of various synchronization constructs. See Synchronization and Cache Control for further details.
- Allocating, freeing, and managing memory. See Memory Allocation and Resource Creation for further details.

- Creation and destruction of command buffers and command buffer pools. See Command Buffers for further details.
- Creation, destruction, and management of graphics state. See Pipelines and Resource Descriptors, among others, for further details.

5.2.3. Lost Device

A logical device **may** become *lost* for a number of implementation-specific reasons, indicating that pending and future command execution **may** fail and cause resources and backing memory to become undefined.

NOTE

Typical reasons for device loss will include things like execution timing out (to prevent denial of service), power management events, platform resource management, and implementation errors.

Applications not adhering to valid usage may also result in device loss being reported, however this is not guaranteed. Even if device loss is reported, the system may be in an unrecoverable state, and further usage of the API is still considered invalid.

When this happens, certain commands will return VK_ERROR_DEVICE_LOST. After any such event, the logical device is considered *lost*. It is not possible to reset the logical device to a non-lost state, however the lost state is specific to a logical device (`VkDevice`), and the corresponding physical device (`VkPhysicalDevice`) **may** be otherwise unaffected.

In some cases, the physical device **may** also be lost, and attempting to create a new logical device will fail, returning VK_ERROR_DEVICE_LOST. This is usually indicative of a problem with the underlying implementation, or its connection to the host. If the physical device has not been lost, and a new logical device is successfully created from that physical device, it **must** be in the non-lost state.

NOTE

Whilst logical device loss **may** be recoverable, in the case of physical device loss, it is unlikely that an application will be able to recover unless additional, unaffected physical devices exist on the system. The error is largely informational and intended only to inform the application that a platform issue has occurred, and **should** be investigated further. For example, underlying hardware **may** have developed a fault or become physically disconnected from the rest of the system. In many cases, physical device loss **may** cause other more serious issues such as the operating system crashing; in which case it **may** not be reported via the Vulkan API.

When a device is lost, its child objects are not implicitly destroyed and their handles are still valid. Those objects **must** still be destroyed before their parents or the device **can** be destroyed (see the Object Lifetime section). The host address space corresponding to device memory mapped using vkMapMemory is still valid, and host memory accesses to these mapped regions are still valid, but the contents are undefined. It is still legal to call any API command on the device and child objects.

Once a device is lost, command execution **may** fail, and certain commands that return a VkResult **may** return VK_ERROR_DEVICE_LOST. These commands can be identified by the inclusion of

VK_ERROR_DEVICE_LOST in the Return Codes section for each command. Commands that do not allow runtime errors **must** still operate correctly for valid usage and, if applicable, return valid data.

Commands that wait indefinitely for device execution (namely vkDeviceWaitIdle, vkQueueWaitIdle, vkWaitForFences or vkAcquireNextImageKHR with a maximum `timeout`, and vkGetQueryPoolResults with the VK_QUERY_RESULT_WAIT_BIT bit set in `flags`) **must** return in finite time even in the case of a lost device, and return either VK_SUCCESS or VK_ERROR_DEVICE_LOST. For any command that **may** return VK_ERROR_DEVICE_LOST, for the purpose of determining whether a command buffer is in the pending state, or whether resources are considered in-use by the device, a return value of VK_ERROR_DEVICE_LOST is equivalent to VK_SUCCESS.

If a device was created with the `maintenance5` feature enabled, and any device command returns VK_ERROR_DEVICE_LOST, then all device commands for which VK_ERROR_DEVICE_LOST is a valid return value and which happen-after it on the same host thread **must** return VK_ERROR_DEVICE_LOST.

Device commands executing on other threads **must** begin returning VK_ERROR_DEVICE_LOST within finite time.

The content of any external memory objects that have been exported from or imported to a lost device becomes undefined. Objects on other logical devices or in other APIs which are associated with the same underlying memory resource as the external memory objects on the lost device are unaffected other than their content becoming undefined. The layout of subresources of images on other logical devices that are bound to `VkDeviceMemory` objects associated with the same underlying memory resources as external memory objects on the lost device becomes VK_IMAGE_LAYOUT_UNDEFINED.

The state of `VkSemaphore` objects on other logical devices created by importing a semaphore payload with temporary permanence which was exported from the lost device is undefined. The state of `VkSemaphore` objects on other logical devices that permanently share a semaphore payload with a `VkSemaphore` object on the lost device is undefined, and remains undefined following any subsequent signal operations. Implementations **must** ensure pending and subsequently submitted wait operations on such semaphores behave as defined in Semaphore State Requirements For Wait Operations for external semaphores not in a valid state for a wait operation.

5.2.4. Device Destruction

To destroy a device, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyDevice(
    VkDevice                                    device,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

To ensure that no work is active on the device, vkDeviceWaitIdle **can** be used to gate the destruction of the device. Prior to destroying a device, an application is responsible for destroying/freeing any Vulkan objects with explicit `vkDestroy*` or `vkFree*` commands that were created using that device as the first parameter of the corresponding `vkCreate*` or `vkAllocate*` command.

NOTE	The lifetime of each of these objects is bound by the lifetime of the `VkDevice` object. Therefore, to avoid resource leaks, it is critical that an application explicitly free all of these resources prior to calling `vkDestroyDevice`.

Valid Usage

- VUID-vkDestroyDevice-device-05137
  All child objects created on `device` that can be destroyed or freed **must** have been destroyed or freed prior to destroying `device`
- VUID-vkDestroyDevice-device-00379
  If `VkAllocationCallbacks` were provided when `device` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyDevice-device-00380
  If no `VkAllocationCallbacks` were provided when `device` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyDevice-device-parameter
  If `device` is not `NULL`, `device` **must** be a valid VkDevice handle
- VUID-vkDestroyDevice-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure

Host Synchronization

- Host access to `device` **must** be externally synchronized
- Host access to all `VkQueue` objects created from `device` **must** be externally synchronized

5.3. Queues

5.3.1. Queue Family Properties

As discussed in the Physical Device Enumeration section above, the vkGetPhysicalDeviceQueueFamilyProperties command is used to retrieve details about the queue families and queues supported by a device.

Each index in the pQueueFamilyProperties array returned by vkGetPhysicalDeviceQueueFamilyProperties describes a unique queue family on that physical device. These indices are used when creating queues, and they correspond directly with the queueFamilyIndex that is passed to the vkCreateDevice command via the VkDeviceQueueCreateInfo structure as described in the Queue Creation section below.

Grouping of queue families within a physical device is implementation-dependent.

NOTE	The general expectation is that a physical device groups all queues of matching capabilities into a single family. However, while implementations **should** do this, it is possible that a physical device **may** return two separate queue families with the same capabilities.

Once an application has identified a physical device with the queue(s) that it desires to use, it will create those queues in conjunction with a logical device. This is described in the following section.

5.3.2. Queue Creation

Creating a logical device also creates the queues associated with that device. The queues to create are described by a set of VkDeviceQueueCreateInfo structures that are passed to vkCreateDevice in pQueueCreateInfos. Queues **cannot** be independently destroyed, and are instead destroyed with the VkDevice that they were created from.

Queues are represented by VkQueue handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_HANDLE(VkQueue)
```

The VkDeviceQueueCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDeviceQueueCreateInfo {
    VkStructureType             sType;
    const void*                 pNext;
    VkDeviceQueueCreateFlags    flags;
    uint32_t                    queueFamilyIndex;
    uint32_t                    queueCount;
    const float*                pQueuePriorities;
} VkDeviceQueueCreateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask indicating behavior of the queues.
- queueFamilyIndex is an unsigned integer indicating the index of the queue family in which to create the queues on this device. This index corresponds to the index of an element of the pQueueFamilyProperties array that was returned by vkGetPhysicalDeviceQueueFamilyProperties.

- `queueCount` is an unsigned integer specifying the number of queues to create in the queue family indicated by `queueFamilyIndex`, and with the behavior specified by `flags`.
- `pQueuePriorities` is a pointer to an array of `queueCount` normalized floating-point values, specifying priorities of work that will be submitted to each created queue. See Queue Priority for more information.

Some queue families support functionality which requires a device feature or extension to be enabled, as indicated by the VkQueueFamilyProperties::`queueFlags` member. Applications **may** create such queues and submit queue submission commands to them without enabling the corresponding feature or extension, but **must** not utilize the specific functionality that they did not enable.

For example, sparse memory management operations **can** be performed on queues from queue families exposing the VK_QUEUE_SPARSE_BINDING_BIT bit, provided the sparseBinding feature is enabled. If a queue family supports both the VK_QUEUE_SPARSE_BINDING_BIT and VK_QUEUE_TRANSFER_BIT bits, applications **may** create a queue from this family and issue transfer operations without enabling the sparseBinding feature.

Valid Usage

- VUID-VkDeviceQueueCreateInfo-queueFamilyIndex-00381
  `queueFamilyIndex` **must** be less than `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties`
- VUID-VkDeviceQueueCreateInfo-queueCount-00382
  `queueCount` **must** be less than or equal to the `queueCount` member of the `VkQueueFamilyProperties` structure, as returned by `vkGetPhysicalDeviceQueueFamilyProperties` in the `pQueueFamilyProperties`[queueFamilyIndex]
- VUID-VkDeviceQueueCreateInfo-pQueuePriorities-00383
  Each element of `pQueuePriorities` **must** be between `0.0` and `1.0` inclusive
- VUID-VkDeviceQueueCreateInfo-flags-02861
  If the `protectedMemory` feature is not enabled, the VK_DEVICE_QUEUE_CREATE_PROTECTED_BIT bit of `flags` **must** not be set
- VUID-VkDeviceQueueCreateInfo-flags-06449
  If `flags` includes VK_DEVICE_QUEUE_CREATE_PROTECTED_BIT, `queueFamilyIndex` **must** be the index of a queue family that includes the VK_QUEUE_PROTECTED_BIT capability
- VUID-VkDeviceQueueCreateInfo-internallySynchronizedQueues-12348
  If the `internallySynchronizedQueues` feature is not enabled, `flags` **must** not include VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR

Valid Usage (Implicit)

- VUID-VkDeviceQueueCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_QUEUE_CREATE_INFO

- VUID-VkDeviceQueueCreateInfo-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkDeviceQueueGlobalPriorityCreateInfo
- VUID-VkDeviceQueueCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkDeviceQueueCreateInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkDeviceQueueCreateFlagBits values
- VUID-VkDeviceQueueCreateInfo-pQueuePriorities-parameter
  `pQueuePriorities` **must** be a valid pointer to an array of `queueCount` `float` values
- VUID-VkDeviceQueueCreateInfo-queueCount-arraylength
  `queueCount` **must** be greater than `0`

Bits which **can** be set in VkDeviceQueueCreateInfo::`flags`, specifying usage behavior of a queue, are:

```
// Provided by VK_VERSION_1_1
typedef enum VkDeviceQueueCreateFlagBits {
  // Provided by VK_VERSION_1_1
    VK_DEVICE_QUEUE_CREATE_PROTECTED_BIT = 0x00000001,
  // Provided by VK_KHR_internally_synchronized_queues
    VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR = 0x00000004,
} VkDeviceQueueCreateFlagBits;
```

- VK_DEVICE_QUEUE_CREATE_PROTECTED_BIT specifies that the device queue is a protected-capable queue.
- VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR specifies that the device queue is internally synchronized and does not require external synchronization.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDeviceQueueCreateFlags;
```

`VkDeviceQueueCreateFlags` is a bitmask type for setting a mask of zero or more VkDeviceQueueCreateFlagBits.

Queues **can** be created with a system-wide priority by adding a `VkDeviceQueueGlobalPriorityCreateInfo` structure to the `pNext` chain of VkDeviceQueueCreateInfo.

The `VkDeviceQueueGlobalPriorityCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkDeviceQueueGlobalPriorityCreateInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkQueueGlobalPriority    globalPriority;
} VkDeviceQueueGlobalPriorityCreateInfo;
```

```
// Provided by VK_KHR_global_priority
// Equivalent to VkDeviceQueueGlobalPriorityCreateInfo
typedef VkDeviceQueueGlobalPriorityCreateInfo
VkDeviceQueueGlobalPriorityCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `globalPriority` is the system-wide priority associated to these queues as specified by VkQueueGlobalPriority.

Queues created without specifying `VkDeviceQueueGlobalPriorityCreateInfo` will default to VK_QUEUE_GLOBAL_PRIORITY_MEDIUM.

Valid Usage (Implicit)

- VUID-VkDeviceQueueGlobalPriorityCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_QUEUE_GLOBAL_PRIORITY_CREATE_INFO
- VUID-VkDeviceQueueGlobalPriorityCreateInfo-globalPriority-parameter
  `globalPriority` **must** be a valid VkQueueGlobalPriority value

Structure Chaining

Extends the structure

- VkDeviceQueueCreateInfo

Possible values of VkDeviceQueueGlobalPriorityCreateInfo::`globalPriority`, specifying a system-wide priority level are:

```
// Provided by VK_VERSION_1_4
typedef enum VkQueueGlobalPriority {
    VK_QUEUE_GLOBAL_PRIORITY_LOW = 128,
    VK_QUEUE_GLOBAL_PRIORITY_MEDIUM = 256,
    VK_QUEUE_GLOBAL_PRIORITY_HIGH = 512,
    VK_QUEUE_GLOBAL_PRIORITY_REALTIME = 1024,
  // Provided by VK_KHR_global_priority
    VK_QUEUE_GLOBAL_PRIORITY_LOW_KHR = VK_QUEUE_GLOBAL_PRIORITY_LOW,
  // Provided by VK_KHR_global_priority
    VK_QUEUE_GLOBAL_PRIORITY_MEDIUM_KHR = VK_QUEUE_GLOBAL_PRIORITY_MEDIUM,
  // Provided by VK_KHR_global_priority
    VK_QUEUE_GLOBAL_PRIORITY_HIGH_KHR = VK_QUEUE_GLOBAL_PRIORITY_HIGH,
  // Provided by VK_KHR_global_priority
    VK_QUEUE_GLOBAL_PRIORITY_REALTIME_KHR = VK_QUEUE_GLOBAL_PRIORITY_REALTIME,
} VkQueueGlobalPriority;
```

```
// Provided by VK_KHR_global_priority
// Equivalent to VkQueueGlobalPriority
typedef VkQueueGlobalPriority VkQueueGlobalPriorityKHR;
```

Priority values are sorted in ascending order. A comparison operation on the enum values can be used to determine the priority order.

- VK_QUEUE_GLOBAL_PRIORITY_LOW is below the system default. Useful for non-interactive tasks.
- VK_QUEUE_GLOBAL_PRIORITY_MEDIUM is the system default priority.
- VK_QUEUE_GLOBAL_PRIORITY_HIGH is above the system default.
- VK_QUEUE_GLOBAL_PRIORITY_REALTIME is the highest priority. Useful for critical tasks.

Queues with higher system priority **may** be allotted more processing time than queues with lower priority. An implementation **may** allow a higher-priority queue to starve a lower-priority queue until the higher-priority queue has no further commands to execute.

Priorities imply no ordering or scheduling constraints.

No specific guarantees are made about higher priority queues receiving more processing time or better quality of service than lower priority queues.

The global priority level of a queue takes precedence over the per-process queue priority (VkDeviceQueueCreateInfo::pQueuePriorities).

Abuse of this feature **may** result in starving the rest of the system of implementation resources. Therefore, the driver implementation **may** deny requests to acquire a priority above the default priority (VK_QUEUE_GLOBAL_PRIORITY_MEDIUM) if the caller does not have sufficient privileges. In this scenario VK_ERROR_NOT_PERMITTED is returned.

The driver implementation **may** fail the queue allocation request if resources required to complete

the operation have been exhausted (either by the same process or a different process). In this scenario VK_ERROR_INITIALIZATION_FAILED is returned.

If the globalPriorityQuery feature is enabled and the requested global priority is not reported via VkQueueFamilyGlobalPriorityProperties, the driver implementation **must** fail the queue creation. In this scenario, VK_ERROR_INITIALIZATION_FAILED is returned.

To retrieve a handle to a VkQueue object, call:

```
// Provided by VK_VERSION_1_0
void vkGetDeviceQueue(
    VkDevice                                    device,
    uint32_t                                    queueFamilyIndex,
    uint32_t                                    queueIndex,
    VkQueue*                                    pQueue);
```

- device is the logical device that owns the queue.
- queueFamilyIndex is the index of the queue family to which the queue belongs.
- queueIndex is the index within this queue family of the queue to retrieve.
- pQueue is a pointer to a VkQueue object that will be filled with the handle for the requested queue.

vkGetDeviceQueue **must** only be used to get queues that were created with the flags parameter of VkDeviceQueueCreateInfo set to zero. To get queues that were created with a non-zero flags parameter use vkGetDeviceQueue2.

Valid Usage

- VUID-vkGetDeviceQueue-queueFamilyIndex-00384
  queueFamilyIndex **must** be one of the queue family indices specified when device was created, via the VkDeviceQueueCreateInfo structure
- VUID-vkGetDeviceQueue-queueIndex-00385
  queueIndex **must** be less than the value of VkDeviceQueueCreateInfo::queueCount for the queue family indicated by queueFamilyIndex when device was created
- VUID-vkGetDeviceQueue-flags-01841
  VkDeviceQueueCreateInfo::flags **must** have been zero when device was created

Valid Usage (Implicit)

- VUID-vkGetDeviceQueue-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkGetDeviceQueue-pQueue-parameter
  pQueue **must** be a valid pointer to a VkQueue handle

To retrieve a handle to a VkQueue object with specific VkDeviceQueueCreateFlags creation flags, call:

```
// Provided by VK_VERSION_1_1
void vkGetDeviceQueue2(
    VkDevice                                    device,
    const VkDeviceQueueInfo2*                   pQueueInfo,
    VkQueue*                                    pQueue);
```

- device is the logical device that owns the queue.
- pQueueInfo is a pointer to a VkDeviceQueueInfo2 structure, describing parameters of the device queue to be retrieved.
- pQueue is a pointer to a VkQueue object that will be filled with the handle for the requested queue.

Valid Usage (Implicit)

- VUID-vkGetDeviceQueue2-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkGetDeviceQueue2-pQueueInfo-parameter
  pQueueInfo **must** be a valid pointer to a valid VkDeviceQueueInfo2 structure
- VUID-vkGetDeviceQueue2-pQueue-parameter
  pQueue **must** be a valid pointer to a VkQueue handle

The VkDeviceQueueInfo2 structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkDeviceQueueInfo2 {
    VkStructureType             sType;
    const void*                 pNext;
    VkDeviceQueueCreateFlags    flags;
    uint32_t                    queueFamilyIndex;
    uint32_t                    queueIndex;
} VkDeviceQueueInfo2;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure. The pNext chain of VkDeviceQueueInfo2 **can** be used to provide additional device queue parameters to vkGetDeviceQueue2.
- flags is a VkDeviceQueueCreateFlags value indicating the flags used to create the device queue.
- queueFamilyIndex is the index of the queue family to which the queue belongs.
- queueIndex is the index of the queue to retrieve from within the set of queues that share both the queue family and flags specified.

The queue returned by vkGetDeviceQueue2 **must** have the same flags value from this structure as that used at device creation time in a VkDeviceQueueCreateInfo structure.

NOTE

Normally, if you create both protected-capable and non-protected-capable queues with the same family, they are treated as separate lists of queues and queueIndex is relative to the start of the list of queues specified by both queueFamilyIndex and flags. However, for historical reasons, some implementations may exhibit different behavior. These divergent implementations instead concatenate the lists of queues and treat queueIndex as relative to the start of the first list of queues with the given queueFamilyIndex. This only matters in cases where an application has created both protected-capable and non-protected-capable queues from the same queue family.

For such divergent implementations, the maximum value of queueIndex is equal to the sum of VkDeviceQueueCreateInfo::queueCount minus one, for all VkDeviceQueueCreateInfo structures that share a common queueFamilyIndex.

Such implementations will return NULL for either the protected or unprotected queues when calling vkGetDeviceQueue2 with queueIndex in the range zero to VkDeviceQueueCreateInfo::queueCount minus one. In cases where these implementations returned NULL, the corresponding queues are instead located in the extended range described in the preceding two paragraphs.

This behavior will not be observed on any driver that has passed Vulkan conformance test suite version 1.3.3.0, or any subsequent version. This information can be found by querying VkPhysicalDeviceDriverProperties::conformanceVersion.

Valid Usage

- VUID-VkDeviceQueueInfo2-queueFamilyIndex-01842
  queueFamilyIndex **must** be one of the queue family indices specified when device was created, via the VkDeviceQueueCreateInfo structure
- VUID-VkDeviceQueueInfo2-flags-06225
  flags **must** be equal to VkDeviceQueueCreateInfo::flags for a VkDeviceQueueCreateInfo structure for the queue family indicated by queueFamilyIndex when device was created
- VUID-VkDeviceQueueInfo2-queueIndex-01843
  queueIndex **must** be less than VkDeviceQueueCreateInfo::queueCount for the corresponding queue family and flags indicated by queueFamilyIndex and flags when device was created

Valid Usage (Implicit)

- VUID-VkDeviceQueueInfo2-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DEVICE_QUEUE_INFO_2
- VUID-VkDeviceQueueInfo2-pNext-pNext
  pNext **must** be NULL
- VUID-VkDeviceQueueInfo2-flags-parameter

If `flags` is not `0`, `flags` **must** be a valid combination of VkDeviceQueueCreateFlagBits values

5.3.3. Queue Family Index

The queue family index is used in multiple places in Vulkan in order to tie operations to a specific family of queues.

When retrieving a handle to the queue via `vkGetDeviceQueue`, the queue family index is used to select which queue family to retrieve the `VkQueue` handle from as described in the previous section.

When creating a `VkCommandPool` object (see Command Pools), a queue family index is specified in the VkCommandPoolCreateInfo structure. Command buffers from this pool **can** only be submitted on queues corresponding to this queue family.

When creating `VkImage` (see Images) and `VkBuffer` (see Buffers) resources, a set of queue families is included in the VkImageCreateInfo and VkBufferCreateInfo structures to specify the queue families that **can** access the resource.

When inserting a VkBufferMemoryBarrier or VkImageMemoryBarrier (see Pipeline Barriers), a source and destination queue family index is specified to allow the ownership of a buffer or image to be transferred from one queue family to another. See the Resource Sharing section for details.

5.3.4. Queue Priority

Each queue is assigned a priority, as set in the VkDeviceQueueCreateInfo structures when creating the device. The priority of each queue is a normalized floating-point value between 0.0 and 1.0, which is then translated to a discrete priority level by the implementation. Higher values indicate a higher priority, with 0.0 being the lowest priority and 1.0 being the highest.

Within the same device, queues with higher priority **may** be allotted more processing time than queues with lower priority. The implementation makes no guarantees with regards to ordering or scheduling among queues with the same priority, other than the constraints defined by any explicit synchronization primitives. The implementation makes no guarantees with regards to queues across different devices.

An implementation **may** allow a higher-priority queue to starve a lower-priority queue on the same `VkDevice` until the higher-priority queue has no further commands to execute. The relationship of queue priorities **must** not cause queues on one `VkDevice` to starve queues on another `VkDevice`.

No specific guarantees are made about higher priority queues receiving more processing time or better quality of service than lower priority queues.

5.3.5. Queue Submission

Work is submitted to a queue via *queue submission* commands such as vkQueueSubmit2 or vkQueuePresentKHR or vkQueueSubmit. Queue submission commands define a set of *queue operations* to be executed by the underlying physical device, including synchronization with semaphores and fences.

Submission commands take as parameters a target queue, zero or more *batches* of work, and an **optional** fence to signal upon completion. Each batch consists of three distinct parts:

1. Zero or more semaphores to wait on before execution of the rest of the batch.
    - If present, these describe a semaphore wait operation.
2. Zero or more work items to execute.
    - If present, these describe a *queue operation* matching the work described.
3. Zero or more semaphores to signal upon completion of the work items.
    - If present, these describe a semaphore signal operation.

If a fence is present in a queue submission, it describes a fence signal operation.

All work described by a queue submission command **must** be submitted to the queue before the command returns.

Sparse Memory Binding

In Vulkan it is possible to sparsely bind memory to buffers and images as described in the Sparse Resource chapter. Sparse memory binding is a queue operation. A queue whose flags include the VK_QUEUE_SPARSE_BINDING_BIT **must** be able to support the mapping of a virtual address to a physical address on the device. This causes an update to the page table mappings on the device. This update **must** be synchronized on a queue to avoid corrupting page table mappings during execution of graphics commands. By binding the sparse memory resources on queues, all commands that are dependent on the updated bindings are synchronized to only execute after the binding is updated. See the Synchronization and Cache Control chapter for how this synchronization is accomplished.

5.3.6. Queue Destruction

Queues are created along with a logical device during `vkCreateDevice`. All queues associated with a logical device are destroyed when `vkDestroyDevice` is called on that device.

Chapter 6. Command Buffers

Command buffers are objects used to record commands which **can** be subsequently submitted to a device queue for execution. There are two levels of command buffers - *primary command buffers*, which **can** execute secondary command buffers, and which are submitted to queues, and *secondary command buffers*, which **can** be executed by primary command buffers, and which are not directly submitted to queues.

Command buffers are represented by `VkCommandBuffer` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_HANDLE(VkCommandBuffer)
```

Recorded commands include commands to bind pipelines and descriptor sets to the command buffer, commands to modify dynamic state, commands to draw (for graphics rendering), commands to dispatch (for compute), commands to execute secondary command buffers (for primary command buffers only), commands to copy buffers and images, and other commands.

Each command buffer manages state independently of other command buffers. There is no inheritance of state across primary and secondary command buffers, or between secondary command buffers. When a command buffer begins recording, all state in that command buffer is undefined. When secondary command buffer(s) are recorded to execute on a primary command buffer, the secondary command buffer inherits no state from the primary command buffer, and all state of the primary command buffer is undefined after an execute secondary command buffer command is recorded, with the following exception(s):

- If the primary command buffer is inside a render pass instance, then the render pass and subpass state is not disturbed by executing secondary command buffers.
- If the primary command buffer has a descriptor heap bound, and the address of that descriptor heap is specified in VkCommandBufferInheritanceDescriptorHeapInfoEXT for every secondary command buffer, that heap binding is not disturbed by executing secondary command buffers.

For state dependent commands (such as draws and dispatches), any state consumed by those commands **must** not be undefined.

Unless otherwise specified, and without explicit synchronization, the various commands submitted to a queue via command buffers **may** execute in arbitrary order relative to each other, and/or concurrently. Also, the memory side effects of those commands **may** not be directly visible to other commands without explicit memory dependencies. This is true within a command buffer, and across command buffers submitted to a given queue. See the synchronization chapter for information on implicit and explicit synchronization between commands.

6.1. Command Buffer Lifecycle

Each command buffer is always in one of the following states:

Initial

Command buffers in the *initial state* **can** only be transitioned to the recording state, or freed. When a command buffer is allocated, it is in the *initial state*. Command buffers in any state other than the pending state **can** be transitioned to the initial state by calling vkResetCommandPool on the pool they were allocated from. Command buffers allocated from a command pool created with VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT and in any state other than the pending state **can** also be transitioned to the initial state by calling vkResetCommandBuffer.

Recording

Command buffers in the *recording state* **can** be used to record commands via `vkCmd*` commands, be reset, or be freed. Command buffers in the initial state **can** be transitioned to the recording state by vkBeginCommandBuffer. Command buffers allocated from a command pool created with VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT in the invalid, recording, or executable state **can** also be transitioned to the recording state by vkBeginCommandBuffer.

Executable

A command buffer in the *executable state* **can** be submitted for execution, reset, freed, or recorded to another command buffer. Command buffers in the recording state are transitioned to the executable state by vkEndCommandBuffer. Command buffers in the pending state that were recorded without the VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT flag immediately return to the executable state as the final command in the command buffer completes all execution, which can be observed via synchronization commands.

Pending

A command buffer in the *pending state* **must** not be modified by the application, as it **may** be executing on the device. Command buffers in the pending state that were recorded without the VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT flag, or with the VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT flag, **must** not be submitted to the device for execution. Command buffers in the executable state **can** be transitioned to the pending state by queue submission commands. Once commands in the command buffer have completed all execution, the command buffer is immediately in either the executable state, or in the invalid state if it was recorded with the VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT flag. A synchronization command **can** be used to detect when command execution is complete.

Invalid

A command buffer in the *invalid* state **can** be reset or freed. Command buffers in any state other than the pending state will transition to the invalid state if any memory or object, other than VkCommandPool, that **may** be accessed when the command buffer is accessed (e.g. an object bound to the command buffer) is destroyed or freed. Command buffers in any state other than the pending state will transition to the invalid state if any command buffer executed within it via vkCmdExecuteCommands transitions to any state other than the pending or executable state. Command buffers in the pending state that were recorded with the VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT flag are immediately invalid as the final command in the command buffer completes all execution, which can be observed via synchronization commands.

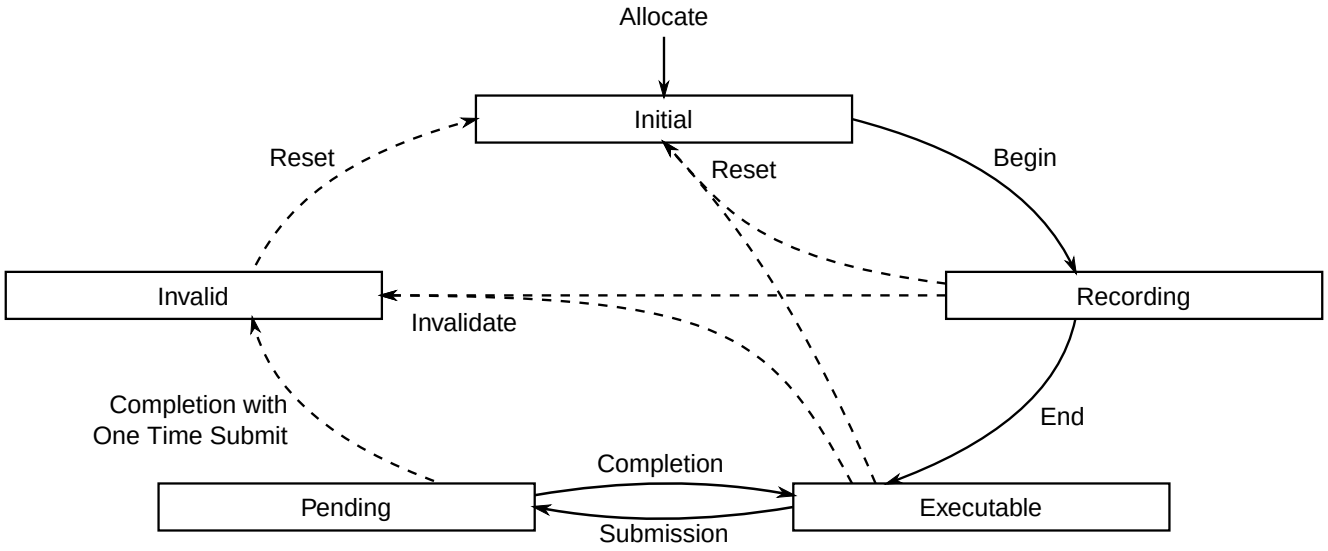


Figure 1. Lifecycle of a command buffer

NOTE

The pending state is tied entirely to the status of execution of a command buffer's commands; once all commands are finished executing, the command buffer is in the subsequent state (executable or invalid) with no further intervention.

Detection of the final command in a command buffer completing all stages of execution (i.e. VK_PIPELINE_STAGE_ALL_COMMANDS_BIT) with a synchronization command is sufficient to ensure the command buffer is in the expected state.

Secondary command buffers **can** be recorded to a primary command buffer via vkCmdExecuteCommands. This partially ties the lifecycle of the two command buffers together - if the primary is submitted to a queue, both the primary and any secondaries recorded to it move to the *pending state*. Similarly, once execution of the primary completes, it completes for any command buffer recorded within it.

NOTE

Resetting or freeing a primary command buffer removes the lifecycle linkage to all secondary command buffers that were recorded into it.

6.2. Command Pools

Command pools are opaque objects that command buffer memory is allocated from, and which allow the implementation to amortize the cost of resource creation across multiple command buffers. Command pools are externally synchronized, meaning that a command pool **must** not be used concurrently in multiple threads. That includes use via recording commands on any command buffers allocated from the pool, as well as operations that allocate, free, and reset command buffers or the pool itself.

Command pools are represented by `VkCommandPool` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkCommandPool)
```

To create a command pool, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateCommandPool(
    VkDevice                                    device,
    const VkCommandPoolCreateInfo*              pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkCommandPool*                              pCommandPool);
```

- `device` is the logical device that creates the command pool.
- `pCreateInfo` is a pointer to a VkCommandPoolCreateInfo structure specifying the state of the command pool object.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pCommandPool` is a pointer to a VkCommandPool handle in which the created pool is returned.

Valid Usage

- VUID-vkCreateCommandPool-queueFamilyIndex-01937
  `pCreateInfo->queueFamilyIndex` **must** be the index of a queue family available in the logical device `device`

Valid Usage (Implicit)

- VUID-vkCreateCommandPool-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateCommandPool-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkCommandPoolCreateInfo structure
- VUID-vkCreateCommandPool-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateCommandPool-pCommandPool-parameter
  `pCommandPool` **must** be a valid pointer to a VkCommandPool handle
- VUID-vkCreateCommandPool-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkCommandPoolCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkCommandPoolCreateInfo {
    VkStructureType             sType;
    const void*                 pNext;
    VkCommandPoolCreateFlags    flags;
    uint32_t                    queueFamilyIndex;
} VkCommandPoolCreateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask of VkCommandPoolCreateFlagBits indicating usage behavior for the pool and command buffers allocated from it.
- queueFamilyIndex designates a queue family as described in section Queue Family Properties. All command buffers allocated from this command pool **must** be submitted on queues from the same queue family.

Valid Usage

- VUID-VkCommandPoolCreateInfo-flags-02860
  If the protectedMemory feature is not enabled, the VK_COMMAND_POOL_CREATE_PROTECTED_BIT bit of flags **must** not be set

Valid Usage (Implicit)

- VUID-VkCommandPoolCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_COMMAND_POOL_CREATE_INFO
- VUID-VkCommandPoolCreateInfo-pNext-pNext
  pNext **must** be NULL
- VUID-VkCommandPoolCreateInfo-flags-parameter
  If flags is not 0, flags **must** be a valid combination of VkCommandPoolCreateFlagBits values

Bits which **can** be set in VkCommandPoolCreateInfo::flags, specifying usage behavior for a command pool, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCommandPoolCreateFlagBits {
    VK_COMMAND_POOL_CREATE_TRANSIENT_BIT = 0x00000001,
    VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT = 0x00000002,
  // Provided by VK_VERSION_1_1
    VK_COMMAND_POOL_CREATE_PROTECTED_BIT = 0x00000004,
} VkCommandPoolCreateFlagBits;
```

- VK_COMMAND_POOL_CREATE_TRANSIENT_BIT specifies that command buffers allocated from the pool will be short-lived, meaning that they will be reset or freed in a relatively short timeframe. This flag **may** be used by the implementation to control memory allocation behavior within the pool.
- VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT allows any command buffer allocated from a pool to be individually reset to the initial state; either by calling vkResetCommandBuffer, or via the implicit reset when calling vkBeginCommandBuffer. If this flag is not set on a pool, then `vkResetCommandBuffer` **must** not be called for any command buffer allocated from that pool.
- VK_COMMAND_POOL_CREATE_PROTECTED_BIT specifies that command buffers allocated from the pool are protected command buffers.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkCommandPoolCreateFlags;
```

`VkCommandPoolCreateFlags` is a bitmask type for setting a mask of zero or more VkCommandPoolCreateFlagBits.

To trim a command pool, call:

```
// Provided by VK_VERSION_1_1
void vkTrimCommandPool(
    VkDevice                                    device,
    VkCommandPool                               commandPool,
    VkCommandPoolTrimFlags                      flags);
```

```
// Provided by VK_KHR_maintenance1
// Equivalent to vkTrimCommandPool
void vkTrimCommandPoolKHR(
    VkDevice                                    device,
    VkCommandPool                               commandPool,
    VkCommandPoolTrimFlags                      flags);
```

- `device` is the logical device that owns the command pool.
- `commandPool` is the command pool to trim.

- `flags` is reserved for future use.

Trimming a command pool recycles unused memory from the command pool back to the system. Command buffers allocated from the pool are not affected by the command.

NOTE

This command provides applications with some control over the internal memory allocations used by command pools.

Unused memory normally arises from command buffers that have been recorded and later reset, such that they are no longer using the memory. On reset, a command buffer can return memory to its command pool, but the only way to release memory from a command pool to the system requires calling vkResetCommandPool, which cannot be executed while any command buffers from that pool are still in use. Subsequent recording operations into command buffers will reuse this memory but since total memory requirements fluctuate over time, unused memory can accumulate.

In this situation, trimming a command pool **may** be useful to return unused memory back to the system, returning the total outstanding memory allocated by the pool back to a more "average" value.

Implementations utilize many internal allocation strategies that make it impossible to guarantee that all unused memory is released back to the system. For instance, an implementation of a command pool **may** involve allocating memory in bulk from the system and sub-allocating from that memory. In such an implementation any live command buffer that holds a reference to a bulk allocation would prevent that allocation from being freed, even if only a small proportion of the bulk allocation is in use.

In most cases trimming will result in a reduction in allocated but unused memory, but it does not guarantee the "ideal" behavior.

Trimming **may** be an expensive operation, and **should** not be called frequently. Trimming **should** be treated as a way to relieve memory pressure after application-known points when there exists enough unused memory that the cost of trimming is "worth" it.

Valid Usage (Implicit)

- VUID-vkTrimCommandPool-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkTrimCommandPool-commandPool-parameter
  `commandPool` **must** be a valid VkCommandPool handle
- VUID-vkTrimCommandPool-flags-zerobitmask
  `flags` **must** be `0`
- VUID-vkTrimCommandPool-commandPool-parent
  `commandPool` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `commandPool` **must** be externally synchronized

```
// Provided by VK_VERSION_1_1
typedef VkFlags VkCommandPoolTrimFlags;
```

```
// Provided by VK_KHR_maintenance1
// Equivalent to VkCommandPoolTrimFlags
typedef VkCommandPoolTrimFlags VkCommandPoolTrimFlagsKHR;
```

`VkCommandPoolTrimFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

To reset a command pool, call:

```
// Provided by VK_VERSION_1_0
VkResult vkResetCommandPool(
    VkDevice                                    device,
    VkCommandPool                               commandPool,
    VkCommandPoolResetFlags                     flags);
```

- `device` is the logical device that owns the command pool.
- `commandPool` is the command pool to reset.
- `flags` is a bitmask of VkCommandPoolResetFlagBits controlling the reset operation.

Resetting a command pool recycles all of the resources from all of the command buffers allocated from the command pool back to the command pool. All command buffers that have been allocated from the command pool are put in the initial state.

Any primary command buffer allocated from another VkCommandPool that is in the recording or executable state and has a secondary command buffer allocated from `commandPool` recorded into it, becomes invalid.

Valid Usage

- VUID-vkResetCommandPool-commandPool-00040
  All `VkCommandBuffer` objects allocated from `commandPool` **must** not be in the pending state

Valid Usage (Implicit)

- VUID-vkResetCommandPool-device-parameter
  `device` **must** be a valid VkDevice handle

- VUID-vkResetCommandPool-commandPool-parameter
  `commandPool` **must** be a valid VkCommandPool handle
- VUID-vkResetCommandPool-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkCommandPoolResetFlagBits values
- VUID-vkResetCommandPool-commandPool-parent
  `commandPool` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `commandPool` **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Bits which **can** be set in vkResetCommandPool::`flags`, controlling the reset operation, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCommandPoolResetFlagBits {
    VK_COMMAND_POOL_RESET_RELEASE_RESOURCES_BIT = 0x00000001,
} VkCommandPoolResetFlagBits;
```

- VK_COMMAND_POOL_RESET_RELEASE_RESOURCES_BIT specifies that resetting a command pool recycles all of the resources from the command pool back to the system.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkCommandPoolResetFlags;
```

`VkCommandPoolResetFlags` is a bitmask type for setting a mask of zero or more VkCommandPoolResetFlagBits.

To destroy a command pool, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyCommandPool(
    VkDevice                                    device,
    VkCommandPool                               commandPool,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the command pool.
- `commandPool` is the handle of the command pool to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

When a pool is destroyed, all command buffers allocated from the pool are freed.

Any primary command buffer allocated from another VkCommandPool that is in the recording or executable state and has a secondary command buffer allocated from `commandPool` recorded into it, becomes invalid.

Valid Usage

- VUID-vkDestroyCommandPool-commandPool-00041
  All `VkCommandBuffer` objects allocated from `commandPool` **must** not be in the pending state
- VUID-vkDestroyCommandPool-commandPool-00042
  If `VkAllocationCallbacks` were provided when `commandPool` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyCommandPool-commandPool-00043
  If no `VkAllocationCallbacks` were provided when `commandPool` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyCommandPool-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyCommandPool-commandPool-parameter
  If `commandPool` is not VK_NULL_HANDLE, `commandPool` **must** be a valid VkCommandPool handle
- VUID-vkDestroyCommandPool-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyCommandPool-commandPool-parent
  If `commandPool` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `commandPool` **must** be externally synchronized

6.3. Command Buffer Allocation and Management

To allocate command buffers, call:

```
// Provided by VK_VERSION_1_0
VkResult vkAllocateCommandBuffers(
    VkDevice                                    device,
    const VkCommandBufferAllocateInfo*          pAllocateInfo,
    VkCommandBuffer*                            pCommandBuffers);
```

- `device` is the logical device that owns the command pool.
- `pAllocateInfo` is a pointer to a VkCommandBufferAllocateInfo structure describing parameters of the allocation. `commandPool` **may** be accessed any time one of the resulting command buffers is accessed.
- `pCommandBuffers` is a pointer to an array of VkCommandBuffer handles in which the resulting command buffer objects are returned. The array **must** be at least the length specified by the `commandBufferCount` member of `pAllocateInfo`. Each allocated command buffer begins in the initial state.

`vkAllocateCommandBuffers` **can** be used to allocate multiple command buffers. If the allocation of any of those command buffers fails, the implementation **must** free all successfully allocated command buffer objects from this command, set all entries of the `pCommandBuffers` array to `NULL` and return the error.

NOTE Filling `pCommandBuffers` with `NULL` values on failure is an exception to the default error behavior that output parameters will have undefined contents.

When command buffers are first allocated, they are in the initial state.

Valid Usage (Implicit)

- VUID-vkAllocateCommandBuffers-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkAllocateCommandBuffers-pAllocateInfo-parameter
  `pAllocateInfo` **must** be a valid pointer to a valid VkCommandBufferAllocateInfo structure
- VUID-vkAllocateCommandBuffers-pCommandBuffers-parameter
  `pCommandBuffers` **must** be a valid pointer to an array of `pAllocateInfo->commandBufferCount` VkCommandBuffer handles
- VUID-vkAllocateCommandBuffers-device-queuecount

The device **must** have been created with at least 1 queue

- VUID-vkAllocateCommandBuffers-pAllocateInfo::commandBufferCount-arraylength
  `pAllocateInfo->commandBufferCount` **must** be greater than `0`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkCommandBufferAllocateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkCommandBufferAllocateInfo {
    VkStructureType         sType;
    const void*             pNext;
    VkCommandPool           commandPool;
    VkCommandBufferLevel    level;
    uint32_t                commandBufferCount;
} VkCommandBufferAllocateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `commandPool` is the command pool from which the command buffers are allocated.
- `level` is a VkCommandBufferLevel value specifying the command buffer level.
- `commandBufferCount` is the number of command buffers to allocate from the pool.

Valid Usage (Implicit)

- VUID-VkCommandBufferAllocateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COMMAND_BUFFER_ALLOCATE_INFO
- VUID-VkCommandBufferAllocateInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkCommandBufferAllocateInfo-commandPool-parameter
  `commandPool` **must** be a valid VkCommandPool handle
- VUID-VkCommandBufferAllocateInfo-level-parameter

`level` **must** be a valid VkCommandBufferLevel value

Host Synchronization

- Host access to `commandPool` **must** be externally synchronized

Possible values of VkCommandBufferAllocateInfo::`level`, specifying the command buffer level, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCommandBufferLevel {
    VK_COMMAND_BUFFER_LEVEL_PRIMARY = 0,
    VK_COMMAND_BUFFER_LEVEL_SECONDARY = 1,
} VkCommandBufferLevel;
```

- VK_COMMAND_BUFFER_LEVEL_PRIMARY specifies a primary command buffer.
- VK_COMMAND_BUFFER_LEVEL_SECONDARY specifies a secondary command buffer.

To reset a command buffer, call:

```
// Provided by VK_VERSION_1_0
VkResult vkResetCommandBuffer(
    VkCommandBuffer                             commandBuffer,
    VkCommandBufferResetFlags                   flags);
```

- `commandBuffer` is the command buffer to reset. The command buffer **can** be in any state other than pending, and is moved into the initial state.
- `flags` is a bitmask of VkCommandBufferResetFlagBits controlling the reset operation.

Any primary command buffer that is in the recording or executable state and has `commandBuffer` recorded into it, becomes invalid.

After a command buffer is reset, any objects or memory specified by commands recorded into the command buffer **must** no longer be accessed when the command buffer is accessed by the implementation.

Valid Usage

- VUID-vkResetCommandBuffer-commandBuffer-00045
  `commandBuffer` **must** not be in the pending state
- VUID-vkResetCommandBuffer-commandBuffer-00046
  `commandBuffer` **must** have been allocated from a pool that was created with the VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT

Valid Usage (Implicit)

- VUID-vkResetCommandBuffer-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkResetCommandBuffer-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkCommandBufferResetFlagBits values

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Bits which **can** be set in vkResetCommandBuffer::`flags`, controlling the reset operation, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCommandBufferResetFlagBits {
    VK_COMMAND_BUFFER_RESET_RELEASE_RESOURCES_BIT = 0x00000001,
} VkCommandBufferResetFlagBits;
```

- VK_COMMAND_BUFFER_RESET_RELEASE_RESOURCES_BIT specifies that most or all memory resources currently owned by the command buffer **should** be returned to the parent command pool. If this flag is not set, then the command buffer **may** hold onto memory resources and reuse them when recording commands. `commandBuffer` is moved to the initial state.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkCommandBufferResetFlags;
```

`VkCommandBufferResetFlags` is a bitmask type for setting a mask of zero or more VkCommandBufferResetFlagBits.

To free command buffers, call:

```
// Provided by VK_VERSION_1_0
void vkFreeCommandBuffers(
    VkDevice                                    device,
    VkCommandPool                               commandPool,
    uint32_t                                    commandBufferCount,
    const VkCommandBuffer*                      pCommandBuffers);
```

- `device` is the logical device that owns the command pool.
- `commandPool` is the command pool from which the command buffers were allocated.
- `commandBufferCount` is the length of the `pCommandBuffers` array.
- `pCommandBuffers` is a pointer to an array of handles of command buffers to free.

Any primary command buffer that is in the recording or executable state and has any element of `pCommandBuffers` recorded into it, becomes invalid.

Valid Usage

- VUID-vkFreeCommandBuffers-pCommandBuffers-00047
  All elements of `pCommandBuffers` **must** not be in the pending state
- VUID-vkFreeCommandBuffers-pCommandBuffers-00048
  `pCommandBuffers` **must** be a valid pointer to an array of `commandBufferCount` `VkCommandBuffer` handles, each element of which **must** either be a valid handle or `NULL`

Valid Usage (Implicit)

- VUID-vkFreeCommandBuffers-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkFreeCommandBuffers-commandPool-parameter
  `commandPool` **must** be a valid VkCommandPool handle
- VUID-vkFreeCommandBuffers-commandBufferCount-arraylength
  `commandBufferCount` **must** be greater than `0`
- VUID-vkFreeCommandBuffers-commandPool-parent
  `commandPool` **must** have been created, allocated, or retrieved from `device`
- VUID-vkFreeCommandBuffers-pCommandBuffers-parent
  Each element of `pCommandBuffers` that is a valid handle **must** have been created, allocated, or retrieved from `commandPool`

Host Synchronization

- Host access to `commandPool` **must** be externally synchronized

- Host access to each member of `pCommandBuffers` **must** be externally synchronized

6.4. Command Buffer Recording

To begin recording a command buffer, call:

```
// Provided by VK_VERSION_1_0
VkResult vkBeginCommandBuffer(
    VkCommandBuffer                             commandBuffer,
    const VkCommandBufferBeginInfo*             pBeginInfo);
```

- `commandBuffer` is the handle of the command buffer which is to be put in the recording state.
- `pBeginInfo` is a pointer to a VkCommandBufferBeginInfo structure defining additional information about how the command buffer begins recording.

Valid Usage

- VUID-vkBeginCommandBuffer-commandBuffer-00049
  `commandBuffer` **must** not be in the pending state
- VUID-vkBeginCommandBuffer-commandBuffer-00050
  If `commandBuffer` was allocated from a VkCommandPool which did not have the VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT flag set, `commandBuffer` **must** be in the initial state
- VUID-vkBeginCommandBuffer-commandBuffer-00051
  If `commandBuffer` is a secondary command buffer, the `pInheritanceInfo` member of `pBeginInfo` **must** be a valid `VkCommandBufferInheritanceInfo` structure
- VUID-vkBeginCommandBuffer-commandBuffer-00052
  If `commandBuffer` is a secondary command buffer and either the `occlusionQueryEnable` member of the `pInheritanceInfo` member of `pBeginInfo` is VK_FALSE, or the `occlusionQueryPrecise` feature is not enabled, then `pBeginInfo->pInheritanceInfo->queryFlags` **must** not contain VK_QUERY_CONTROL_PRECISE_BIT
- VUID-vkBeginCommandBuffer-commandBuffer-02840
  If `commandBuffer` is a primary command buffer, then `pBeginInfo->flags` **must** not set both the VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT and the VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT flags

Valid Usage (Implicit)

- VUID-vkBeginCommandBuffer-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkBeginCommandBuffer-pBeginInfo-parameter
  `pBeginInfo` **must** be a valid pointer to a valid VkCommandBufferBeginInfo structure

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkCommandBufferBeginInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkCommandBufferBeginInfo {
    VkStructureType                          sType;
    const void*                              pNext;
    VkCommandBufferUsageFlags                flags;
    const VkCommandBufferInheritanceInfo*    pInheritanceInfo;
} VkCommandBufferBeginInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkCommandBufferUsageFlagBits specifying usage behavior for the command buffer.
- `pInheritanceInfo` is a pointer to a VkCommandBufferInheritanceInfo structure, used if `commandBuffer` is a secondary command buffer. If this is a primary command buffer, then this value is ignored.

Valid Usage

- VUID-VkCommandBufferBeginInfo-flags-09123
  If `flags` contains VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT, the VkCommandPool that `commandBuffer` was allocated from **must** support graphics operations
- VUID-VkCommandBufferBeginInfo-flags-00055
  If `flags` contains VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT, the

framebuffer member of pInheritanceInfo **must** be either VK_NULL_HANDLE, or a valid VkFramebuffer that is compatible with the renderPass member of pInheritanceInfo

- VUID-VkCommandBufferBeginInfo-flags-09240
  If flags contains VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT and the dynamicRendering feature is not enabled, the renderPass member of pInheritanceInfo **must** not be VK_NULL_HANDLE
- VUID-VkCommandBufferBeginInfo-flags-06002
  If flags contains VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT and the renderPass member of pInheritanceInfo is VK_NULL_HANDLE, the pNext chain of pInheritanceInfo **must** include a VkCommandBufferInheritanceRenderingInfo structure
- VUID-VkCommandBufferBeginInfo-flags-06000
  If flags contains VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT and the renderPass member of pInheritanceInfo is not VK_NULL_HANDLE, the renderPass member of pInheritanceInfo **must** be a valid VkRenderPass
- VUID-VkCommandBufferBeginInfo-flags-06001
  If flags contains VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT and the renderPass member of pInheritanceInfo is not VK_NULL_HANDLE, the subpass member of pInheritanceInfo **must** be a valid subpass index within the renderPass member of pInheritanceInfo

Valid Usage (Implicit)

- VUID-VkCommandBufferBeginInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_COMMAND_BUFFER_BEGIN_INFO
- VUID-VkCommandBufferBeginInfo-pNext-pNext
  pNext **must** be NULL or a pointer to a valid instance of VkDeviceGroupCommandBufferBeginInfo
- VUID-VkCommandBufferBeginInfo-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkCommandBufferBeginInfo-flags-parameter
  If flags is not 0, flags **must** be a valid combination of VkCommandBufferUsageFlagBits values

Bits which **can** be set in VkCommandBufferBeginInfo::flags, specifying usage behavior for a command buffer, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCommandBufferUsageFlagBits {
    VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT = 0x00000001,
    VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT = 0x00000002,
    VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT = 0x00000004,
} VkCommandBufferUsageFlagBits;
```

- VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT specifies that each recording of the command buffer will only be submitted once, and the command buffer will be reset and recorded again between each submission.
- VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT specifies that a secondary command buffer is considered to be entirely inside a render pass. If this is a primary command buffer, then this bit is ignored.
- VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT specifies that a command buffer **can** be resubmitted to any queue of the same queue family while it is in the *pending state*, and recorded into multiple primary command buffers.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkCommandBufferUsageFlags;
```

VkCommandBufferUsageFlags is a bitmask type for setting a mask of zero or more VkCommandBufferUsageFlagBits.

If the command buffer is a secondary command buffer, then the VkCommandBufferInheritanceInfo structure defines any state that will be inherited from the primary command buffer:

```
// Provided by VK_VERSION_1_0
typedef struct VkCommandBufferInheritanceInfo {
    VkStructureType                  sType;
    const void*                      pNext;
    VkRenderPass                     renderPass;
    uint32_t                         subpass;
    VkFramebuffer                    framebuffer;
    VkBool32                         occlusionQueryEnable;
    VkQueryControlFlags              queryFlags;
    VkQueryPipelineStatisticFlags    pipelineStatistics;
} VkCommandBufferInheritanceInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- renderPass is a VkRenderPass object defining which render passes the VkCommandBuffer will be compatible with and **can** be executed within.
- subpass is the index of the subpass within the render pass instance that the VkCommandBuffer will be executed within.
- framebuffer **can** refer to the VkFramebuffer object that the VkCommandBuffer will be rendering to if it is executed within a render pass instance. It **can** be VK_NULL_HANDLE if the framebuffer is not known.

NOTE	Specifying the exact framebuffer that the secondary command buffer will be executed with **may** result in better performance at command buffer execution time.

- `occlusionQueryEnable` specifies whether the command buffer **can** be executed while an occlusion query is active in the primary command buffer. If this is VK_TRUE, then this command buffer **can** be executed whether the primary command buffer has an occlusion query active or not. If this is VK_FALSE, then the primary command buffer **must** not have an occlusion query active.
- `queryFlags` specifies the query flags that **can** be used by an active occlusion query in the primary command buffer when this secondary command buffer is executed. If this value includes the VK_QUERY_CONTROL_PRECISE_BIT bit, then the active query **can** return boolean results or actual sample counts. If this bit is not set, then the active query **must** not use the VK_QUERY_CONTROL_PRECISE_BIT bit.
- `pipelineStatistics` is a bitmask of VkQueryPipelineStatisticFlagBits specifying the set of pipeline statistics that **can** be counted by an active query in the primary command buffer when this secondary command buffer is executed. If this value includes a given bit, then this command buffer **can** be executed whether the primary command buffer has a pipeline statistics query active that includes this bit or not. If this value excludes a given bit, then the active pipeline statistics query **must** not be from a query pool that counts that statistic.

If the VkCommandBuffer will not be executed within a render pass instance, or if the render pass instance was begun with vkCmdBeginRendering, `renderPass`, `subpass`, and `framebuffer` are ignored.

Valid Usage

- VUID-VkCommandBufferInheritanceInfo-occlusionQueryEnable-00056
  If the `inheritedQueries` feature is not enabled, `occlusionQueryEnable` **must** be VK_FALSE
- VUID-VkCommandBufferInheritanceInfo-queryFlags-00057
  If the `inheritedQueries` feature is enabled, `queryFlags` **must** be a valid combination of VkQueryControlFlagBits values
- VUID-VkCommandBufferInheritanceInfo-queryFlags-02788
  If the `inheritedQueries` feature is not enabled, `queryFlags` **must** be `0`
- VUID-VkCommandBufferInheritanceInfo-pipelineStatistics-02789
  If the `pipelineStatisticsQuery` feature is enabled, `pipelineStatistics` **must** be a valid combination of VkQueryPipelineStatisticFlagBits values
- VUID-VkCommandBufferInheritanceInfo-pipelineStatistics-00058
  If the `pipelineStatisticsQuery` feature is not enabled, `pipelineStatistics` **must** be `0`

Valid Usage (Implicit)

- VUID-VkCommandBufferInheritanceInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_INFO
- VUID-VkCommandBufferInheritanceInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkCommandBufferInheritanceConditionalRenderingInfoEXT, VkCommandBufferInheritanceDescriptorHeapInfoEXT,

VkCommandBufferInheritanceRenderingInfo, VkCustomResolveCreateInfoEXT, VkRenderingAttachmentLocationInfo, or VkRenderingInputAttachmentIndexInfo

- VUID-VkCommandBufferInheritanceInfo-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkCommandBufferInheritanceInfo-commonparent
  Both of framebuffer, and renderPass that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

NOTE

On some implementations, not using the VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT bit enables command buffers to be patched in-place if needed, rather than creating a copy of the command buffer.

If a command buffer is in the invalid, or executable state, and the command buffer was allocated from a command pool with the VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT flag set, then vkBeginCommandBuffer implicitly resets the command buffer, behaving as if vkResetCommandBuffer had been called with VK_COMMAND_BUFFER_RESET_RELEASE_RESOURCES_BIT not set. After the implicit reset, commandBuffer is moved to the recording state.

If the pNext chain of VkCommandBufferInheritanceInfo includes a VkCommandBufferInheritanceConditionalRenderingInfoEXT structure, then that structure controls whether a command buffer **can** be executed while conditional rendering is active in the primary command buffer.

The VkCommandBufferInheritanceConditionalRenderingInfoEXT structure is defined as:

```
// Provided by VK_EXT_conditional_rendering
typedef struct VkCommandBufferInheritanceConditionalRenderingInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           conditionalRenderingEnable;
} VkCommandBufferInheritanceConditionalRenderingInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- conditionalRenderingEnable specifies whether the command buffer **can** be executed while conditional rendering is active in the primary command buffer. If this is VK_TRUE, then this command buffer **can** be executed whether the primary command buffer has active conditional rendering or not. If this is VK_FALSE, then the primary command buffer **must** not have conditional rendering active.

If this structure is not present, the behavior is as if conditionalRenderingEnable is VK_FALSE.

Valid Usage

- VUID-VkCommandBufferInheritanceConditionalRenderingInfoEXT-conditionalRenderingEnable-01977
  If the `inheritedConditionalRendering` feature is not enabled, `conditionalRenderingEnable` **must** be VK_FALSE

Valid Usage (Implicit)

- VUID-VkCommandBufferInheritanceConditionalRenderingInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_CONDITIONAL_RENDERING_INFO_EXT

Structure Chaining

Extends the structure

- VkCommandBufferInheritanceInfo

The `VkCommandBufferInheritanceRenderingInfo` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkCommandBufferInheritanceRenderingInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkRenderingFlags         flags;
    uint32_t                 viewMask;
    uint32_t                 colorAttachmentCount;
    const VkFormat*          pColorAttachmentFormats;
    VkFormat                 depthAttachmentFormat;
    VkFormat                 stencilAttachmentFormat;
    VkSampleCountFlagBits    rasterizationSamples;
} VkCommandBufferInheritanceRenderingInfo;
```

```
// Provided by VK_KHR_dynamic_rendering
// Equivalent to VkCommandBufferInheritanceRenderingInfo
typedef VkCommandBufferInheritanceRenderingInfo
VkCommandBufferInheritanceRenderingInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkRenderingFlagBits used by the render pass instance.

- `viewMask` is the view mask used for rendering.
- `colorAttachmentCount` is the number of color attachments specified in the render pass instance.
- `pColorAttachmentFormats` is a pointer to an array of VkFormat values defining the format of color attachments.
- `depthAttachmentFormat` is a VkFormat value defining the format of the depth attachment.
- `stencilAttachmentFormat` is a VkFormat value defining the format of the stencil attachment.
- `rasterizationSamples` is a VkSampleCountFlagBits specifying the number of samples used in rasterization.

If the `pNext` chain of VkCommandBufferInheritanceInfo includes a `VkCommandBufferInheritanceRenderingInfo` structure, then that structure controls parameters of dynamic render pass instances that the VkCommandBuffer **can** be executed within. If VkCommandBufferInheritanceInfo::`renderPass` is not VK_NULL_HANDLE, or VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT is not specified in VkCommandBufferBeginInfo::`flags`, parameters of this structure are ignored.

If `colorAttachmentCount` is `0` and the `variableMultisampleRate` feature is enabled, `rasterizationSamples` is ignored.

If `depthAttachmentFormat`, `stencilAttachmentFormat`, or any element of `pColorAttachmentFormats` is VK_FORMAT_UNDEFINED, it indicates that the corresponding attachment is unused within the render pass and writes to those attachments are discarded.

Valid Usage

- VUID-VkCommandBufferInheritanceRenderingInfo-colorAttachmentCount-06004
  If `colorAttachmentCount` is not `0`, `rasterizationSamples` **must** be a valid VkSampleCountFlagBits value
- VUID-VkCommandBufferInheritanceRenderingInfo-variableMultisampleRate-06005
  If the `variableMultisampleRate` feature is not enabled, `rasterizationSamples` **must** be a valid VkSampleCountFlagBits value
- VUID-VkCommandBufferInheritanceRenderingInfo-depthAttachmentFormat-06540
  If `depthAttachmentFormat` is not VK_FORMAT_UNDEFINED, it **must** be a format that includes a depth component
- VUID-VkCommandBufferInheritanceRenderingInfo-depthAttachmentFormat-06007
  If `depthAttachmentFormat` is not VK_FORMAT_UNDEFINED, it **must** be a format with potential format features that include VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkCommandBufferInheritanceRenderingInfo-pColorAttachmentFormats-06492
  If any element of `pColorAttachmentFormats` is not VK_FORMAT_UNDEFINED, it **must** be a format with potential format features that include VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT
- VUID-VkCommandBufferInheritanceRenderingInfo-stencilAttachmentFormat-06541
  If `stencilAttachmentFormat` is not VK_FORMAT_UNDEFINED, it **must** be a format that

includes a stencil aspect

- VUID-VkCommandBufferInheritanceRenderingInfo-stencilAttachmentFormat-06199
  If `stencilAttachmentFormat` is not VK_FORMAT_UNDEFINED, it **must** be a format with potential format features that include VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkCommandBufferInheritanceRenderingInfo-depthAttachmentFormat-06200
  If `depthAttachmentFormat` is not VK_FORMAT_UNDEFINED and `stencilAttachmentFormat` is not VK_FORMAT_UNDEFINED, `depthAttachmentFormat` **must** equal `stencilAttachmentFormat`
- VUID-VkCommandBufferInheritanceRenderingInfo-multiview-06008
  If the `multiview` feature is not enabled, `viewMask` **must** be `0`
- VUID-VkCommandBufferInheritanceRenderingInfo-viewMask-06009
  The index of the most significant bit in `viewMask` **must** be less than `maxMultiviewViewCount`

Valid Usage (Implicit)

- VUID-VkCommandBufferInheritanceRenderingInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_RENDERING_INFO
- VUID-VkCommandBufferInheritanceRenderingInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkRenderingFlagBits values
- VUID-VkCommandBufferInheritanceRenderingInfo-pColorAttachmentFormats-parameter
  If `colorAttachmentCount` is not `0`, `pColorAttachmentFormats` **must** be a valid pointer to an array of `colorAttachmentCount` valid VkFormat values
- VUID-VkCommandBufferInheritanceRenderingInfo-depthAttachmentFormat-parameter
  `depthAttachmentFormat` **must** be a valid VkFormat value
- VUID-VkCommandBufferInheritanceRenderingInfo-stencilAttachmentFormat-parameter
  `stencilAttachmentFormat` **must** be a valid VkFormat value
- VUID-VkCommandBufferInheritanceRenderingInfo-rasterizationSamples-parameter
  If `rasterizationSamples` is not `0`, `rasterizationSamples` **must** be a valid VkSampleCountFlagBits value

Structure Chaining

Extends the structure

- VkCommandBufferInheritanceInfo

If the `pNext` chain of VkCommandBufferInheritanceInfo includes a `VkCommandBufferInheritanceDescriptorHeapInfoEXT` structure, then that structure indicates that the secondary will use the same descriptor heaps as the primary command buffer.

The `VkCommandBufferInheritanceDescriptorHeapInfoEXT` structure is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkCommandBufferInheritanceDescriptorHeapInfoEXT {
    VkStructureType             sType;
    const void*                 pNext;
    const VkBindHeapInfoEXT*    pSamplerHeapBindInfo;
    const VkBindHeapInfoEXT*    pResourceHeapBindInfo;
} VkCommandBufferInheritanceDescriptorHeapInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pSamplerHeapBindInfo` specifies the VkBindHeapInfoEXT of the sampler heap bound using vkCmdBindSamplerHeapEXT in the primary. If this is `NULL`, it indicates that no sampler heap is bound.
- `pResourceHeapBindInfo` specifies the VkBindHeapInfoEXT of the resource heap bound using vkCmdBindResourceHeapEXT in the primary. If this is `NULL`, it indicates that no resource heap is bound.

If this structure is not present, the behavior is as if `pSamplerHeapBindInfo` and `pResourceHeapBindInfo` were both `NULL`.

Valid Usage

- VUID-VkCommandBufferInheritanceDescriptorHeapInfoEXT-descriptorHeap-11200
  If the `descriptorHeap` feature is not enabled, `pSamplerHeapBindInfo` **must** be `NULL`
- VUID-VkCommandBufferInheritanceDescriptorHeapInfoEXT-descriptorHeap-11201
  If the `descriptorHeap` feature is not enabled, `pResourceHeapBindInfo` **must** be `NULL`
- VUID-VkCommandBufferInheritanceDescriptorHeapInfoEXT-pSamplerHeapBindInfo-11470
  If `pSamplerHeapBindInfo` is not `NULL`, `pSamplerHeapBindInfo->heapRange` **must** be a device address range allocated to the application from a buffer created with the VK_BUFFER_USAGE_DESCRIPTOR_HEAP_BIT_EXT usage flag set
- VUID-VkCommandBufferInheritanceDescriptorHeapInfoEXT-pResourceHeapBindInfo-11471
  If `pResourceHeapBindInfo` is not `NULL`, `pResourceHeapBindInfo->heapRange` **must** be a device address range allocated to the application from a buffer created with the VK_BUFFER_USAGE_DESCRIPTOR_HEAP_BIT_EXT usage flag set

Valid Usage (Implicit)

- VUID-VkCommandBufferInheritanceDescriptorHeapInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_DESCRIPTOR_HEAP_INFO_EXT

- VUID-VkCommandBufferInheritanceDescriptorHeapInfoEXT-pSamplerHeapBindInfo-parameter
  If `pSamplerHeapBindInfo` is not `NULL`, `pSamplerHeapBindInfo` **must** be a valid pointer to a valid VkBindHeapInfoEXT structure
- VUID-VkCommandBufferInheritanceDescriptorHeapInfoEXT-pResourceHeapBindInfo-parameter
  If `pResourceHeapBindInfo` is not `NULL`, `pResourceHeapBindInfo` **must** be a valid pointer to a valid VkBindHeapInfoEXT structure

Structure Chaining

Extends the structure

- VkCommandBufferInheritanceInfo

Once recording starts, an application records a sequence of commands (`vkCmd*`) to set state in the command buffer, draw, dispatch, and other commands.

Several commands can also be recorded indirectly from `VkBuffer` content, see Device-Generated Commands.

To complete recording of a command buffer, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEndCommandBuffer(
    VkCommandBuffer                             commandBuffer);
```

- `commandBuffer` is the command buffer to complete recording.

The command buffer **must** have been in the recording state, and, if successful, is moved to the executable state.

If there was an error during recording, the application will be notified by an unsuccessful return code returned by `vkEndCommandBuffer`, and the command buffer will be moved to the invalid state.

In case the application recorded one or more video encode operations into the command buffer, implementations **may** return the VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR error if any of the specified Video Std parameters do not adhere to the syntactic or semantic requirements of the used video compression standard, or if values derived from parameters according to the rules defined by the used video compression standard do not adhere to the capabilities of the video compression standard or the implementation.

NOTE

Applications **should** not rely on the VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR error being returned by any command as a means to verify Video Std parameters, as implementations are not required to report the error in any specific set of cases.

Valid Usage

- VUID-vkEndCommandBuffer-commandBuffer-00059
  `commandBuffer` **must** be in the recording state
- VUID-vkEndCommandBuffer-commandBuffer-00060
  If `commandBuffer` is a primary command buffer, there **must** not be an active render pass instance
- VUID-vkEndCommandBuffer-commandBuffer-00061
  All queries made active during the recording of `commandBuffer` **must** have been made inactive
- VUID-vkEndCommandBuffer-None-01978
  Conditional rendering **must** not be active
- VUID-vkEndCommandBuffer-None-06991
  There **must** be no video session object bound
- VUID-vkEndCommandBuffer-commandBuffer-01815
  If `commandBuffer` is a secondary command buffer, there **must** not be an outstanding vkCmdBeginDebugUtilsLabelEXT command recorded to `commandBuffer` that has not previously been ended by a call to vkCmdEndDebugUtilsLabelEXT

Valid Usage (Implicit)

- VUID-vkEndCommandBuffer-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN

- VK_ERROR_VALIDATION_FAILED

When a command buffer is in the executable state, it **can** be submitted to a queue for execution.

6.5. Command Buffer Submission

NOTE	Submission can be a high overhead operation, and applications **should** attempt to batch work together into as few calls to vkQueueSubmit or vkQueueSubmit2 as possible.

To submit command buffers to a queue, call:

```
// Provided by VK_VERSION_1_3
VkResult vkQueueSubmit2(
    VkQueue                                     queue,
    uint32_t                                    submitCount,
    const VkSubmitInfo2*                        pSubmits,
    VkFence                                     fence);
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to vkQueueSubmit2
VkResult vkQueueSubmit2KHR(
    VkQueue                                     queue,
    uint32_t                                    submitCount,
    const VkSubmitInfo2*                        pSubmits,
    VkFence                                     fence);
```

- queue is the queue that the command buffers will be submitted to.
- submitCount is the number of elements in the pSubmits array.
- pSubmits is a pointer to an array of VkSubmitInfo2 structures, each specifying a command buffer submission batch. Command buffers and semaphores specified in this array **may** be accessed at any point until the queue operations they define complete execution on the device.
- fence is an **optional** handle to a fence to be signaled once all submitted command buffers have completed execution. If fence is not VK_NULL_HANDLE, it defines a fence signal operation. If it is not VK_NULL_HANDLE, fence **may** be accessed at any point until this command completes on the device.

vkQueueSubmit2 is a queue submission command, with each batch defined by an element of pSubmits.

The first synchronization scope of each semaphore signal operation defined by this command includes every command in the same batch that the signal operation is defined in, and all commands that occur earlier in submission order. The scope is limited by the stageMask member of the VkSemaphoreSubmitInfo used to define each such operation.

The second synchronization scope of each semaphore wait operation defined by this command includes every command in the same batch that the wait operation is defined in, and all commands

that occur later in submission order. The scope is limited by the `stageMask` member of the VkSemaphoreSubmitInfo used to define each such operation.

If any command buffer submitted to this queue is in the executable state, it is moved to the pending state. Once execution of all submissions of a command buffer complete, it moves from the pending state, back to the executable state. If a command buffer was recorded with the VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT flag, it instead moves back to the invalid state.

If `vkQueueSubmit2` fails, it **may** return VK_ERROR_OUT_OF_HOST_MEMORY or VK_ERROR_OUT_OF_DEVICE_MEMORY. If it does, the implementation **must** ensure that the state and contents of any resources or synchronization primitives referenced by the submitted command buffers and any semaphores referenced by `pSubmits` is unaffected by the call or its failure. If `vkQueueSubmit2` fails in such a way that the implementation is unable to make that guarantee, the implementation **must** return VK_ERROR_DEVICE_LOST. See Lost Device.

Valid Usage

- VUID-vkQueueSubmit2-fence-04894
  If `fence` is not VK_NULL_HANDLE, `fence` **must** be unsignaled
- VUID-vkQueueSubmit2-fence-04895
  If `fence` is not VK_NULL_HANDLE, `fence` **must** not be associated with any other queue command that has not yet completed execution on that queue
- VUID-vkQueueSubmit2-synchronization2-03866
  The `synchronization2` feature **must** be enabled
- VUID-vkQueueSubmit2-commandBuffer-03867
  If a command recorded into the `commandBuffer` member of any element of the `pCommandBufferInfos` member of any element of `pSubmits` referenced a VkEvent, that event **must** not be referenced by a command that has been submitted to another queue and is still in the *pending state*
- VUID-vkQueueSubmit2-semaphore-03868
  The `semaphore` member of any binary semaphore element of the `pSignalSemaphoreInfos` member of any element of `pSubmits` **must** be unsignaled when the semaphore signal operation it defines is executed on the device
- VUID-vkQueueSubmit2-stageMask-03869
  The `stageMask` member of any element of the `pSignalSemaphoreInfos` member of any element of `pSubmits` **must** only include pipeline stages that are supported by the queue family which `queue` belongs to
- VUID-vkQueueSubmit2-stageMask-03870
  The `stageMask` member of any element of the `pWaitSemaphoreInfos` member of any element of `pSubmits` **must** only include pipeline stages that are supported by the queue family which `queue` belongs to
- VUID-vkQueueSubmit2-semaphore-03871
  When a semaphore wait operation for a binary semaphore is executed, as defined by the `semaphore` member of any element of the `pWaitSemaphoreInfos` member of any element of

pSubmits, there **must** be no other queues waiting on the same semaphore

- VUID-vkQueueSubmit2-semaphore-03873
  The semaphore member of any element of the pWaitSemaphoreInfos member of any element of pSubmits that was created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_BINARY **must** reference a semaphore signal operation that has been submitted for execution and any semaphore signal operations on which it depends **must** have also been submitted for execution
- VUID-vkQueueSubmit2-commandBuffer-03874
  The commandBuffer member of any element of the pCommandBufferInfos member of any element of pSubmits **must** be in the pending or executable state
- VUID-vkQueueSubmit2-commandBuffer-03875
  If a command recorded into the commandBuffer member of any element of the pCommandBufferInfos member of any element of pSubmits was not recorded with the VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT, it **must** not be in the pending state
- VUID-vkQueueSubmit2-commandBuffer-03876
  Any secondary command buffers recorded into the commandBuffer member of any element of the pCommandBufferInfos member of any element of pSubmits **must** be in the pending or executable state
- VUID-vkQueueSubmit2-commandBuffer-03877
  If any secondary command buffers recorded into the commandBuffer member of any element of the pCommandBufferInfos member of any element of pSubmits was not recorded with the VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT, it **must** not be in the pending state
- VUID-vkQueueSubmit2-commandBuffer-03878
  The commandBuffer member of any element of the pCommandBufferInfos member of any element of pSubmits **must** have been allocated from a VkCommandPool that was created for the same queue family queue belongs to
- VUID-vkQueueSubmit2-commandBuffer-03879
  If a command recorded into the commandBuffer member of any element of the pCommandBufferInfos member of any element of pSubmits includes a Queue Family Ownership Transfer Acquire Operation, there **must** exist a previously submitted Queue Family Ownership Transfer Release Operation on a queue in the queue family identified by the acquire operation, with parameters matching the acquire operation as defined in the definition of such acquire operations, and which happens before the acquire operation
- VUID-vkQueueSubmit2-commandBuffer-10910
  If a command recorded into the commandBuffer member of any element of the pCommandBufferInfos member of any element of pSubmits includes a Queue Family Ownership Transfer Acquire Operation, the affected resource **must** not be modified in any way between the last matching release operation and the acquire operation
- VUID-vkQueueSubmit2-commandBuffer-03880
  If a command recorded into the commandBuffer member of any element of the pCommandBufferInfos member of any element of pSubmits was a vkCmdBeginQuery whose

queryPool was created with a queryType of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, the profiling lock **must** have been held continuously on the VkDevice that queue was retrieved from, throughout recording of those command buffers

- VUID-vkQueueSubmit2-queue-06447
  If queue was not created with VK_DEVICE_QUEUE_CREATE_PROTECTED_BIT, the flags member of any element of pSubmits **must** not include VK_SUBMIT_PROTECTED_BIT_KHR

Valid Usage (Implicit)

- VUID-vkQueueSubmit2-queue-parameter
  queue **must** be a valid VkQueue handle
- VUID-vkQueueSubmit2-pSubmits-parameter
  If submitCount is not 0, pSubmits **must** be a valid pointer to an array of submitCount valid VkSubmitInfo2 structures
- VUID-vkQueueSubmit2-fence-parameter
  If fence is not VK_NULL_HANDLE, fence **must** be a valid VkFence handle
- VUID-vkQueueSubmit2-commonparent
  Both of fence, and queue that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to queue **must** be externally synchronized if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR
- Host access to fence **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	Any	-

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_OUT_OF_DEVICE_MEMORY

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkSubmitInfo2 structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkSubmitInfo2 {
    VkStructureType                       sType;
    const void*                           pNext;
    VkSubmitFlags                         flags;
    uint32_t                              waitSemaphoreInfoCount;
    const VkSemaphoreSubmitInfo*          pWaitSemaphoreInfos;
    uint32_t                              commandBufferInfoCount;
    const VkCommandBufferSubmitInfo*      pCommandBufferInfos;
    uint32_t                              signalSemaphoreInfoCount;
    const VkSemaphoreSubmitInfo*          pSignalSemaphoreInfos;
} VkSubmitInfo2;
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to VkSubmitInfo2
typedef VkSubmitInfo2 VkSubmitInfo2KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask of VkSubmitFlagBits.
- waitSemaphoreInfoCount is the number of elements in pWaitSemaphoreInfos.
- pWaitSemaphoreInfos is a pointer to an array of VkSemaphoreSubmitInfo structures defining semaphore wait operations.
- commandBufferInfoCount is the number of elements in pCommandBufferInfos and the number of command buffers to execute in the batch.
- pCommandBufferInfos is a pointer to an array of VkCommandBufferSubmitInfo structures describing command buffers to execute in the batch.
- signalSemaphoreInfoCount is the number of elements in pSignalSemaphoreInfos.
- pSignalSemaphoreInfos is a pointer to an array of VkSemaphoreSubmitInfo describing semaphore signal operations.

Valid Usage

- VUID-VkSubmitInfo2-semaphore-03881
  If the same semaphore is used as the semaphore member of both an element of pSignalSemaphoreInfos and pWaitSemaphoreInfos, and that semaphore is a timeline

semaphore, the value member of the pSignalSemaphoreInfos element **must** be greater than the value member of the pWaitSemaphoreInfos element

- VUID-VkSubmitInfo2-semaphore-03882
  If the semaphore member of any element of pSignalSemaphoreInfos is a timeline semaphore, the value member of that element **must** have a value greater than the current value of the semaphore when the semaphore signal operation is executed
- VUID-VkSubmitInfo2-semaphore-03883
  If the semaphore member of any element of pSignalSemaphoreInfos is a timeline semaphore, the value member of that element **must** have a value which does not differ from the current value of the semaphore or the value of any outstanding semaphore wait or signal operation on that semaphore by more than maxTimelineSemaphoreValueDifference
- VUID-VkSubmitInfo2-semaphore-03884
  If the semaphore member of any element of pWaitSemaphoreInfos is a timeline semaphore, the value member of that element **must** have a value which does not differ from the current value of the semaphore or the value of any outstanding semaphore wait or signal operation on that semaphore by more than maxTimelineSemaphoreValueDifference
- VUID-VkSubmitInfo2-flags-03886
  If flags includes VK_SUBMIT_PROTECTED_BIT, all elements of pCommandBuffers **must** be protected command buffers
- VUID-VkSubmitInfo2-flags-03887
  If flags does not include VK_SUBMIT_PROTECTED_BIT, each element of pCommandBuffers **must** not be a protected command buffer
- VUID-VkSubmitInfo2-commandBuffer-06192
  If any commandBuffer member of an element of pCommandBufferInfos contains any resumed render pass instances, they **must** be suspended by a render pass instance earlier in submission order within pCommandBufferInfos
- VUID-VkSubmitInfo2-commandBuffer-06010
  If any commandBuffer member of an element of pCommandBufferInfos contains any suspended render pass instances, they **must** be resumed by a render pass instance later in submission order within pCommandBufferInfos
- VUID-VkSubmitInfo2-commandBuffer-06011
  If any commandBuffer member of an element of pCommandBufferInfos contains any suspended render pass instances, there **must** be no action or synchronization commands between that render pass instance and the render pass instance that resumes it
- VUID-VkSubmitInfo2-commandBuffer-06012
  If any commandBuffer member of an element of pCommandBufferInfos contains any suspended render pass instances, there **must** be no render pass instances between that render pass instance and the render pass instance that resumes it
- VUID-VkSubmitInfo2-variableSampleLocations-06013
  If the variableSampleLocations limit is not supported, and any commandBuffer member of an element of pCommandBufferInfos contains any suspended render pass instances, where a graphics pipeline has been bound, any pipelines bound in the render pass instance that resumes it, or any subsequent render pass instances that resume from that one and so on, **must** use the same sample locations

Valid Usage (Implicit)

- VUID-VkSubmitInfo2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SUBMIT_INFO_2
- VUID-VkSubmitInfo2-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkFrameBoundaryEXT, VkPerformanceQuerySubmitInfoKHR, or VkWin32KeyedMutexAcquireReleaseInfoKHR
- VUID-VkSubmitInfo2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkSubmitInfo2-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkSubmitFlagBits values
- VUID-VkSubmitInfo2-pWaitSemaphoreInfos-parameter
  If `waitSemaphoreInfoCount` is not `0`, `pWaitSemaphoreInfos` **must** be a valid pointer to an array of `waitSemaphoreInfoCount` valid VkSemaphoreSubmitInfo structures
- VUID-VkSubmitInfo2-pCommandBufferInfos-parameter
  If `commandBufferInfoCount` is not `0`, `pCommandBufferInfos` **must** be a valid pointer to an array of `commandBufferInfoCount` valid VkCommandBufferSubmitInfo structures
- VUID-VkSubmitInfo2-pSignalSemaphoreInfos-parameter
  If `signalSemaphoreInfoCount` is not `0`, `pSignalSemaphoreInfos` **must** be a valid pointer to an array of `signalSemaphoreInfoCount` valid VkSemaphoreSubmitInfo structures

Bits which **can** be set in VkSubmitInfo2::`flags`, specifying submission behavior, are:

```
// Provided by VK_VERSION_1_3
typedef enum VkSubmitFlagBits {
    VK_SUBMIT_PROTECTED_BIT = 0x00000001,
  // Provided by VK_KHR_synchronization2
    VK_SUBMIT_PROTECTED_BIT_KHR = VK_SUBMIT_PROTECTED_BIT,
} VkSubmitFlagBits;
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to VkSubmitFlagBits
typedef VkSubmitFlagBits VkSubmitFlagBitsKHR;
```

- VK_SUBMIT_PROTECTED_BIT specifies that this batch is a protected submission.

```
// Provided by VK_VERSION_1_3
typedef VkFlags VkSubmitFlags;
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to VkSubmitFlags
typedef VkSubmitFlags VkSubmitFlagsKHR;
```

VkSubmitFlags is a bitmask type for setting a mask of zero or more VkSubmitFlagBits.

The VkSemaphoreSubmitInfo structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkSemaphoreSubmitInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkSemaphore              semaphore;
    uint64_t                 value;
    VkPipelineStageFlags2    stageMask;
    uint32_t                 deviceIndex;
} VkSemaphoreSubmitInfo;
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to VkSemaphoreSubmitInfo
typedef VkSemaphoreSubmitInfo VkSemaphoreSubmitInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- semaphore is a VkSemaphore affected by this operation.
- value is either the value used to signal semaphore or the value waited on by semaphore, if semaphore is a timeline semaphore. Otherwise it is ignored.
- stageMask is a VkPipelineStageFlags2 mask of pipeline stages which limit the first synchronization scope of a semaphore signal operation, or second synchronization scope of a semaphore wait operation as described in the semaphore wait operation and semaphore signal operation sections of the synchronization chapter.
- deviceIndex is the index of the device within a device group that executes the semaphore wait or signal operation.

Whether this structure defines a semaphore wait or signal operation is defined by how it is used. The first synchronization scope of a semaphore signal operation or the second synchronization scope of a semaphore wait operation defined by this structure are limited to operations in stages indicated by stageMask.

Valid Usage

- VUID-VkSemaphoreSubmitInfo-stageMask-03929
  If the geometryShader feature is not enabled, stageMask **must** not contain VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT

- VUID-VkSemaphoreSubmitInfo-stageMask-03930
  If the `tessellationShader` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-VkSemaphoreSubmitInfo-stageMask-03931
  If the `conditionalRendering` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT
- VUID-VkSemaphoreSubmitInfo-stageMask-03932
  If the `fragmentDensityMap` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-VkSemaphoreSubmitInfo-stageMask-03933
  If the `transformFeedback` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-VkSemaphoreSubmitInfo-stageMask-03934
  If the `meshShader` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT
- VUID-VkSemaphoreSubmitInfo-stageMask-03935
  If the `taskShader` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT
- VUID-VkSemaphoreSubmitInfo-stageMask-07317
  If the `attachmentFragmentShadingRate` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkSemaphoreSubmitInfo-stageMask-07947
  If the `rayTracingPipeline` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkSemaphoreSubmitInfo-stageMask-10751
  If the `accelerationStructure` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkSemaphoreSubmitInfo-stageMask-10752
  If the `rayTracingMaintenance1` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR
- VUID-VkSemaphoreSubmitInfo-stageMask-11541
  If the `VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkSemaphoreSubmitInfo-device-03888
  If the `device` that `semaphore` was created on is not a device group, `deviceIndex` **must** be `0`
- VUID-VkSemaphoreSubmitInfo-device-03889
  If the `device` that `semaphore` was created on is a device group, `deviceIndex` **must** be a valid device index

Valid Usage (Implicit)

- VUID-VkSemaphoreSubmitInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SEMAPHORE_SUBMIT_INFO
- VUID-VkSemaphoreSubmitInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkSemaphoreSubmitInfo-semaphore-parameter
  `semaphore` **must** be a valid VkSemaphore handle
- VUID-VkSemaphoreSubmitInfo-stageMask-parameter
  If `stageMask` is not `0`, `stageMask` **must** be a valid combination of VkPipelineStageFlagBits2 values

The `VkCommandBufferSubmitInfo` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkCommandBufferSubmitInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkCommandBuffer    commandBuffer;
    uint32_t           deviceMask;
} VkCommandBufferSubmitInfo;
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to VkCommandBufferSubmitInfo
typedef VkCommandBufferSubmitInfo VkCommandBufferSubmitInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `commandBuffer` is a VkCommandBuffer to be submitted for execution.
- `deviceMask` is a bitmask indicating which devices in a device group execute the command buffer. A `deviceMask` of `0` is equivalent to setting all bits corresponding to valid devices in the group to `1`.

Valid Usage

- VUID-VkCommandBufferSubmitInfo-commandBuffer-03890
  `commandBuffer` **must** not have been allocated with VK_COMMAND_BUFFER_LEVEL_SECONDARY
- VUID-VkCommandBufferSubmitInfo-deviceMask-03891
  If `deviceMask` is not `0`, it **must** be a valid device mask

Valid Usage (Implicit)

- VUID-VkCommandBufferSubmitInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COMMAND_BUFFER_SUBMIT_INFO
- VUID-VkCommandBufferSubmitInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkCommandBufferSubmitInfo-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle

To submit command buffers to a queue, call:

WARNING	This functionality is superseded by vkQueueSubmit2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
VkResult vkQueueSubmit(
    VkQueue                                     queue,
    uint32_t                                    submitCount,
    const VkSubmitInfo*                         pSubmits,
    VkFence                                     fence);
```

- `queue` is the queue that the command buffers will be submitted to.
- `submitCount` is the number of elements in the `pSubmits` array.
- `pSubmits` is a pointer to an array of VkSubmitInfo structures, each specifying a command buffer submission batch. Command buffers and semaphores specified in this array **may** be accessed at any point until the queue operations they define complete execution on the device.
- `fence` is an **optional** handle to a fence to be signaled once all submitted command buffers have completed execution. If `fence` is not VK_NULL_HANDLE, it defines a fence signal operation. If it is not VK_NULL_HANDLE, `fence` **may** be accessed at any point until this command completes on the device.

`vkQueueSubmit` is a queue submission command, with each batch defined by an element of `pSubmits`. Batches begin execution in the order they appear in `pSubmits`, but **may** complete out of order.

The order that batches appear in `pSubmits` is used to determine submission order, and thus all the implicit ordering guarantees that respect it. Other than these implicit ordering guarantees and any explicit synchronization primitives, these batches **may** overlap or otherwise execute out of order.

Fence operations submitted with vkQueueSubmit have additional ordering constraints compared to other submission commands, with dependencies involving previous and subsequent queue operations. Information about these additional constraints can be found in the fence sections of the synchronization chapter.

The first synchronization scope of each semaphore signal operation defined by this command includes every command in the same batch that the signal operation is defined in, and all

commands that occur earlier in submission order.

The second synchronization scope of each semaphore wait operation defined by this command includes every command in the same batch that the wait operation is defined in, and all commands that occur later in submission order. The scope is limited by the `pWaitDstStageMask` for each batch, as described in VkSubmitInfo.

If any command buffer submitted to this queue is in the executable state, it is moved to the pending state. Once execution of all submissions of a command buffer complete, it moves from the pending state, back to the executable state. If a command buffer was recorded with the VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT flag, it instead moves to the invalid state.

If `vkQueueSubmit` fails, it **may** return VK_ERROR_OUT_OF_HOST_MEMORY or VK_ERROR_OUT_OF_DEVICE_MEMORY. If it does, the implementation **must** ensure that the state and contents of any resources or synchronization primitives referenced by the submitted command buffers and any semaphores referenced by `pSubmits` is unaffected by the call or its failure. If `vkQueueSubmit` fails in such a way that the implementation is unable to make that guarantee, the implementation **must** return VK_ERROR_DEVICE_LOST. See Lost Device.

Valid Usage

- VUID-vkQueueSubmit-fence-00063
  If `fence` is not VK_NULL_HANDLE, `fence` **must** be unsignaled
- VUID-vkQueueSubmit-fence-00064
  If `fence` is not VK_NULL_HANDLE, `fence` **must** not be associated with any other queue command that has not yet completed execution on that queue
- VUID-vkQueueSubmit-pCommandBuffers-00065
  Any calls to vkCmdSetEvent, vkCmdResetEvent or vkCmdWaitEvents that have been recorded into any of the command buffer elements of the `pCommandBuffers` member of any element of `pSubmits`, **must** not reference any VkEvent that is referenced by any of those commands in a command buffer that has been submitted to another queue and is still in the *pending state*
- VUID-vkQueueSubmit-pWaitDstStageMask-00066
  Any stage flag included in any element of the `pWaitDstStageMask` member of any element of `pSubmits` **must** be a pipeline stage supported by one of the capabilities of `queue`, as specified in the table of supported pipeline stages
- VUID-vkQueueSubmit-pSignalSemaphores-00067
  Each binary semaphore element of the `pSignalSemaphores` member of any element of `pSubmits` **must** be unsignaled when the semaphore signal operation it defines is executed on the device
- VUID-vkQueueSubmit-pWaitSemaphores-00068
  When a semaphore wait operation referring to a binary semaphore defined by any element of the `pWaitSemaphores` member of any element of `pSubmits` executes on `queue`, there **must** be no other queues waiting on the same semaphore
- VUID-vkQueueSubmit-pWaitSemaphores-03238
  All elements of the `pWaitSemaphores` member of all elements of `pSubmits` created with a

VkSemaphoreType of VK_SEMAPHORE_TYPE_BINARY **must** reference a semaphore signal operation that has been submitted for execution and any semaphore signal operations on which it depends **must** have also been submitted for execution

- VUID-vkQueueSubmit-pCommandBuffers-00070
  Each element of the pCommandBuffers member of each element of pSubmits **must** be in the pending or executable state
- VUID-vkQueueSubmit-pCommandBuffers-00071
  If any element of the pCommandBuffers member of any element of pSubmits was not recorded with the VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT, it **must** not be in the pending state
- VUID-vkQueueSubmit-pCommandBuffers-00072
  Any secondary command buffers recorded into any element of the pCommandBuffers member of any element of pSubmits **must** be in the pending or executable state
- VUID-vkQueueSubmit-pCommandBuffers-00073
  If any secondary command buffers recorded into any element of the pCommandBuffers member of any element of pSubmits was not recorded with the VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT, it **must** not be in the pending state
- VUID-vkQueueSubmit-pCommandBuffers-00074
  Each element of the pCommandBuffers member of each element of pSubmits **must** have been allocated from a VkCommandPool that was created for the same queue family queue belongs to
- VUID-vkQueueSubmit-pSubmits-02207
  If any element of pSubmits->pCommandBuffers includes a Queue Family Ownership Transfer Acquire Operation, there **must** exist a previously submitted Queue Family Ownership Transfer Release Operation on a queue in the queue family identified by the acquire operation, with parameters matching the acquire operation as defined in the definition of such acquire operations, and which happens-before the acquire operation
- VUID-vkQueueSubmit-pSubmits-10911
  If any element of pSubmits->pCommandBuffers includes a Queue Family Ownership Transfer Acquire Operation, the affected resource **must** not be modified in any way between the last matching release operation and the acquire operation
- VUID-vkQueueSubmit-pCommandBuffers-03220
  If a command recorded into any element of pCommandBuffers was a vkCmdBeginQuery whose queryPool was created with a queryType of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, the profiling lock **must** have been held continuously on the VkDevice that queue was retrieved from, throughout recording of those command buffers
- VUID-vkQueueSubmit-pSubmits-02808
  Any resource created with VK_SHARING_MODE_EXCLUSIVE that is read by an operation specified by pSubmits **must** not be owned by any queue family other than the one which queue belongs to, at the time it is executed
- VUID-vkQueueSubmit-pSubmits-04626
  Any resource created with VK_SHARING_MODE_CONCURRENT that is accessed by an

operation specified by `pSubmits` **must** have included the queue family of `queue` at resource creation time

- VUID-vkQueueSubmit-queue-06448
  If `queue` was not created with VK_DEVICE_QUEUE_CREATE_PROTECTED_BIT, there **must** be no element of `pSubmits` that includes a VkProtectedSubmitInfo structure in its `pNext` chain with `protectedSubmit` equal to VK_TRUE

Valid Usage (Implicit)

- VUID-vkQueueSubmit-queue-parameter
  `queue` **must** be a valid VkQueue handle
- VUID-vkQueueSubmit-pSubmits-parameter
  If `submitCount` is not `0`, `pSubmits` **must** be a valid pointer to an array of `submitCount` valid VkSubmitInfo structures
- VUID-vkQueueSubmit-fence-parameter
  If `fence` is not VK_NULL_HANDLE, `fence` **must** be a valid VkFence handle
- VUID-vkQueueSubmit-commonparent
  Both of `fence`, and `queue` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `queue` **must** be externally synchronized if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR
- Host access to `fence` **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	Any	-

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_OUT_OF_DEVICE_MEMORY

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkSubmitInfo structure is defined as:

WARNING	This functionality is superseded by VkSubmitInfo2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef struct VkSubmitInfo {
    VkStructureType                sType;
    const void*                    pNext;
    uint32_t                       waitSemaphoreCount;
    const VkSemaphore*             pWaitSemaphores;
    const VkPipelineStageFlags*    pWaitDstStageMask;
    uint32_t                       commandBufferCount;
    const VkCommandBuffer*         pCommandBuffers;
    uint32_t                       signalSemaphoreCount;
    const VkSemaphore*             pSignalSemaphores;
} VkSubmitInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- waitSemaphoreCount is the number of semaphores upon which to wait before executing the command buffers for the batch.
- pWaitSemaphores is a pointer to an array of VkSemaphore handles upon which to wait before the command buffers for this batch begin execution. If semaphores to wait on are provided, they define a semaphore wait operation.
- pWaitDstStageMask is a pointer to an array of pipeline stages at which each corresponding semaphore wait will occur.
- commandBufferCount is the number of command buffers to execute in the batch.
- pCommandBuffers is a pointer to an array of VkCommandBuffer handles to execute in the batch.
- signalSemaphoreCount is the number of semaphores to be signaled once the commands specified in pCommandBuffers have completed execution.
- pSignalSemaphores is a pointer to an array of VkSemaphore handles which will be signaled when the command buffers for this batch have completed execution. If semaphores to be signaled are provided, they define a semaphore signal operation.

The order that command buffers appear in pCommandBuffers is used to determine submission order, and thus all the implicit ordering guarantees that respect it. Other than these implicit ordering guarantees and any explicit synchronization primitives, these command buffers **may** overlap or otherwise execute out of order.

The second synchronization scope of each semaphore wait operation defined by this structure is limited to operations in stages indicated by the corresponding element of pWaitDstStageMask.

NOTE

A common scenario for using pWaitDstStageMask with values other than VK_PIPELINE_STAGE_ALL_COMMANDS_BIT is when synchronizing a window system presentation operation against subsequent command buffers which render the next frame. In this case, a presentation image **must** not be overwritten until the presentation operation completes, but other pipeline stages **can** execute without waiting. A mask of VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT prevents subsequent color attachment writes from executing until the semaphore signals. Some implementations **may** be able to execute transfer operations and/or pre-rasterization work before the semaphore is signaled.

If an image layout transition needs to be performed on a presentable image before it is used in a framebuffer, that **can** be performed as the first operation submitted to the queue after acquiring the image, and **should** not prevent other work from overlapping with the presentation operation. For example, a VkImageMemoryBarrier could use:

- srcStageMask = VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT
- srcAccessMask = 0
- dstStageMask = VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT
- dstAccessMask = VK_ACCESS_COLOR_ATTACHMENT_READ_BIT | VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT.
- oldLayout = VK_IMAGE_LAYOUT_PRESENT_SRC_KHR
- newLayout = VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL

Alternatively, oldLayout **can** be VK_IMAGE_LAYOUT_UNDEFINED, if the image's contents need not be preserved.

This barrier accomplishes a dependency chain between previous presentation operations and subsequent color attachment output operations, with the layout transition performed in between, and does not introduce a dependency between previous work and any pre-rasterization shader stages. More precisely, the semaphore signals after the presentation operation completes, the semaphore wait stalls the VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT stage, and there is a dependency from that same stage to itself with the layout transition performed in between.

Valid Usage

- VUID-VkSubmitInfo-pWaitDstStageMask-04090
  If the geometryShader feature is not enabled, pWaitDstStageMask **must** not contain VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT
- VUID-VkSubmitInfo-pWaitDstStageMask-04091
  If the tessellationShader feature is not enabled, pWaitDstStageMask **must** not contain

VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT

- VUID-VkSubmitInfo-pWaitDstStageMask-04092
  If the `conditionalRendering` feature is not enabled, `pWaitDstStageMask` **must** not contain VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT
- VUID-VkSubmitInfo-pWaitDstStageMask-04093
  If the `fragmentDensityMap` feature is not enabled, `pWaitDstStageMask` **must** not contain VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-VkSubmitInfo-pWaitDstStageMask-04094
  If the `transformFeedback` feature is not enabled, `pWaitDstStageMask` **must** not contain VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-VkSubmitInfo-pWaitDstStageMask-04095
  If the `meshShader` feature is not enabled, `pWaitDstStageMask` **must** not contain VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT
- VUID-VkSubmitInfo-pWaitDstStageMask-04096
  If the `taskShader` feature is not enabled, `pWaitDstStageMask` **must** not contain VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT
- VUID-VkSubmitInfo-pWaitDstStageMask-07319
  If the `attachmentFragmentShadingRate` feature is not enabled, `pWaitDstStageMask` **must** not contain VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkSubmitInfo-pWaitDstStageMask-03937
  If the `synchronization2` feature is not enabled, `pWaitDstStageMask` **must** not be `0`
- VUID-VkSubmitInfo-pWaitDstStageMask-07950
  If the `rayTracingPipeline` feature is not enabled, `pWaitDstStageMask` **must** not contain VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkSubmitInfo-pWaitDstStageMask-10754
  If the `accelerationStructure` feature is not enabled, `pWaitDstStageMask` **must** not contain VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkSubmitInfo-pCommandBuffers-00075
  Each element of `pCommandBuffers` **must** not have been allocated with VK_COMMAND_BUFFER_LEVEL_SECONDARY
- VUID-VkSubmitInfo-pWaitDstStageMask-00078
  Each element of `pWaitDstStageMask` **must** not include VK_PIPELINE_STAGE_HOST_BIT
- VUID-VkSubmitInfo-pWaitSemaphores-03239
  If any element of `pWaitSemaphores` or `pSignalSemaphores` was created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE, then the `pNext` chain **must** include a VkTimelineSemaphoreSubmitInfo structure
- VUID-VkSubmitInfo-pNext-03240
  If the `pNext` chain of this structure includes a VkTimelineSemaphoreSubmitInfo structure and any element of `pWaitSemaphores` was created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE, then its `waitSemaphoreValueCount` member **must** equal `waitSemaphoreCount`

- VUID-VkSubmitInfo-pNext-03241
  If the pNext chain of this structure includes a VkTimelineSemaphoreSubmitInfo structure and any element of pSignalSemaphores was created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE, then its signalSemaphoreValueCount member **must** equal signalSemaphoreCount
- VUID-VkSubmitInfo-pSignalSemaphores-03242
  For each element of pSignalSemaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE the corresponding element of VkTimelineSemaphoreSubmitInfo::pSignalSemaphoreValues **must** have a value greater than the current value of the semaphore when the semaphore signal operation is executed
- VUID-VkSubmitInfo-pWaitSemaphores-03243
  For each element of pWaitSemaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE the corresponding element of VkTimelineSemaphoreSubmitInfo::pWaitSemaphoreValues **must** have a value which does not differ from the current value of the semaphore or the value of any outstanding semaphore wait or signal operation on that semaphore by more than maxTimelineSemaphoreValueDifference
- VUID-VkSubmitInfo-pSignalSemaphores-03244
  For each element of pSignalSemaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE the corresponding element of VkTimelineSemaphoreSubmitInfo::pSignalSemaphoreValues **must** have a value which does not differ from the current value of the semaphore or the value of any outstanding semaphore wait or signal operation on that semaphore by more than maxTimelineSemaphoreValueDifference
- VUID-VkSubmitInfo-pNext-04120
  If the pNext chain of this structure does not include a VkProtectedSubmitInfo structure with protectedSubmit set to VK_TRUE, then each element of the pCommandBuffers array **must** be an unprotected command buffer
- VUID-VkSubmitInfo-pNext-04148
  If the pNext chain of this structure includes a VkProtectedSubmitInfo structure with protectedSubmit set to VK_TRUE, then each element of the pCommandBuffers array **must** be a protected command buffer
- VUID-VkSubmitInfo-pCommandBuffers-06193
  If pCommandBuffers contains any resumed render pass instances, they **must** be suspended by a render pass instance earlier in submission order within pCommandBuffers
- VUID-VkSubmitInfo-pCommandBuffers-06014
  If pCommandBuffers contains any suspended render pass instances, they **must** be resumed by a render pass instance later in submission order within pCommandBuffers
- VUID-VkSubmitInfo-pCommandBuffers-06015
  If pCommandBuffers contains any suspended render pass instances, there **must** be no action or synchronization commands executed in a primary or secondary command buffer between that render pass instance and the render pass instance that resumes it
- VUID-VkSubmitInfo-pCommandBuffers-06016

If pCommandBuffers contains any suspended render pass instances, there **must** be no render pass instances between that render pass instance and the render pass instance that resumes it

- VUID-VkSubmitInfo-variableSampleLocations-06017
  If the variableSampleLocations limit is not supported, and any element of pCommandBuffers contains any suspended render pass instances, where a graphics pipeline has been bound, any pipelines bound in the render pass instance that resumes it, or any subsequent render pass instances that resume from that one and so on, **must** use the same sample locations

Valid Usage (Implicit)

- VUID-VkSubmitInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_SUBMIT_INFO
- VUID-VkSubmitInfo-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkD3D12FenceSubmitInfoKHR, VkDeviceGroupSubmitInfo, VkFrameBoundaryEXT, VkPerformanceQuerySubmitInfoKHR, VkProtectedSubmitInfo, VkTimelineSemaphoreSubmitInfo, or VkWin32KeyedMutexAcquireReleaseInfoKHR
- VUID-VkSubmitInfo-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkSubmitInfo-pWaitSemaphores-parameter
  If waitSemaphoreCount is not 0, pWaitSemaphores **must** be a valid pointer to an array of waitSemaphoreCount valid VkSemaphore handles
- VUID-VkSubmitInfo-pWaitDstStageMask-parameter
  If waitSemaphoreCount is not 0, pWaitDstStageMask **must** be a valid pointer to an array of waitSemaphoreCount valid combinations of VkPipelineStageFlagBits values
- VUID-VkSubmitInfo-pCommandBuffers-parameter
  If commandBufferCount is not 0, pCommandBuffers **must** be a valid pointer to an array of commandBufferCount valid VkCommandBuffer handles
- VUID-VkSubmitInfo-pSignalSemaphores-parameter
  If signalSemaphoreCount is not 0, pSignalSemaphores **must** be a valid pointer to an array of signalSemaphoreCount valid VkSemaphore handles
- VUID-VkSubmitInfo-commonparent
  Each of the elements of pCommandBuffers, the elements of pSignalSemaphores, and the elements of pWaitSemaphores that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

To specify the values to use when waiting for and signaling semaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE, add a VkTimelineSemaphoreSubmitInfo structure to the pNext chain of the VkSubmitInfo structure when using vkQueueSubmit or the VkBindSparseInfo structure when using vkQueueBindSparse . The VkTimelineSemaphoreSubmitInfo

structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkTimelineSemaphoreSubmitInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           waitSemaphoreValueCount;
    const uint64_t*    pWaitSemaphoreValues;
    uint32_t           signalSemaphoreValueCount;
    const uint64_t*    pSignalSemaphoreValues;
} VkTimelineSemaphoreSubmitInfo;
```

```
// Provided by VK_KHR_timeline_semaphore
// Equivalent to VkTimelineSemaphoreSubmitInfo
typedef VkTimelineSemaphoreSubmitInfo VkTimelineSemaphoreSubmitInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `waitSemaphoreValueCount` is the number of semaphore wait values specified in `pWaitSemaphoreValues`.
- `pWaitSemaphoreValues` is a pointer to an array of `waitSemaphoreValueCount` values for the corresponding semaphores in VkSubmitInfo::`pWaitSemaphores` to wait for.
- `signalSemaphoreValueCount` is the number of semaphore signal values specified in `pSignalSemaphoreValues`.
- `pSignalSemaphoreValues` is a pointer to an array of `signalSemaphoreValueCount` values for the corresponding semaphores in VkSubmitInfo::`pSignalSemaphores` to set when signaled.

If the semaphore in VkSubmitInfo::`pWaitSemaphores` or VkSubmitInfo::`pSignalSemaphores` corresponding to an entry in `pWaitSemaphoreValues` or `pSignalSemaphoreValues` respectively was not created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE, the implementation **must** ignore the value in the `pWaitSemaphoreValues` or `pSignalSemaphoreValues` entry.

Valid Usage (Implicit)

- VUID-VkTimelineSemaphoreSubmitInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_TIMELINE_SEMAPHORE_SUBMIT_INFO
- VUID-VkTimelineSemaphoreSubmitInfo-pWaitSemaphoreValues-parameter
  If `waitSemaphoreValueCount` is not `0`, `pWaitSemaphoreValues` **must** be a valid pointer to an array of `waitSemaphoreValueCount` `uint64_t` values
- VUID-VkTimelineSemaphoreSubmitInfo-pSignalSemaphoreValues-parameter
  If `signalSemaphoreValueCount` is not `0`, `pSignalSemaphoreValues` **must** be a valid pointer to an array of `signalSemaphoreValueCount` `uint64_t` values

Structure Chaining

Extends the structures

- VkBindSparseInfo
- VkSubmitInfo

To specify the values to use when waiting for and signaling semaphores whose current payload refers to a Direct3D 12 fence, add a VkD3D12FenceSubmitInfoKHR structure to the pNext chain of the VkSubmitInfo structure. The VkD3D12FenceSubmitInfoKHR structure is defined as:

```
// Provided by VK_KHR_external_semaphore_win32
typedef struct VkD3D12FenceSubmitInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           waitSemaphoreValuesCount;
    const uint64_t*    pWaitSemaphoreValues;
    uint32_t           signalSemaphoreValuesCount;
    const uint64_t*    pSignalSemaphoreValues;
} VkD3D12FenceSubmitInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- waitSemaphoreValuesCount is the number of semaphore wait values specified in pWaitSemaphoreValues.
- pWaitSemaphoreValues is a pointer to an array of waitSemaphoreValuesCount values for the corresponding semaphores in VkSubmitInfo::pWaitSemaphores to wait for.
- signalSemaphoreValuesCount is the number of semaphore signal values specified in pSignalSemaphoreValues.
- pSignalSemaphoreValues is a pointer to an array of signalSemaphoreValuesCount values for the corresponding semaphores in VkSubmitInfo::pSignalSemaphores to set when signaled.

If the semaphore in VkSubmitInfo::pWaitSemaphores or VkSubmitInfo::pSignalSemaphores corresponding to an entry in pWaitSemaphoreValues or pSignalSemaphoreValues respectively does not currently have a payload referring to a Direct3D 12 fence, the implementation **must** ignore the value in the pWaitSemaphoreValues or pSignalSemaphoreValues entry.

NOTE

As the introduction of the external semaphore handle type VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT predates that of timeline semaphores, support for importing semaphore payloads from external handles of that type into semaphores created (implicitly or explicitly) with a VkSemaphoreType of VK_SEMAPHORE_TYPE_BINARY is preserved for backwards compatibility. However, applications **should** prefer importing such handle types into semaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE, and use the VkTimelineSemaphoreSubmitInfo

structure instead of the VkD3D12FenceSubmitInfoKHR structure to specify the values to use when waiting for and signaling such semaphores.

Valid Usage

- VUID-VkD3D12FenceSubmitInfoKHR-waitSemaphoreValuesCount-00079
  waitSemaphoreValuesCount **must** be the same value as VkSubmitInfo::waitSemaphoreCount, where this structure is in the pNext chain of a VkSubmitInfo structure
- VUID-VkD3D12FenceSubmitInfoKHR-signalSemaphoreValuesCount-00080
  signalSemaphoreValuesCount **must** be the same value as VkSubmitInfo::signalSemaphoreCount, where this structure is in the pNext chain of a VkSubmitInfo structure

Valid Usage (Implicit)

- VUID-VkD3D12FenceSubmitInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_D3D12_FENCE_SUBMIT_INFO_KHR
- VUID-VkD3D12FenceSubmitInfoKHR-pWaitSemaphoreValues-parameter
  If waitSemaphoreValuesCount is not 0, and pWaitSemaphoreValues is not NULL, pWaitSemaphoreValues **must** be a valid pointer to an array of waitSemaphoreValuesCount uint64_t values
- VUID-VkD3D12FenceSubmitInfoKHR-pSignalSemaphoreValues-parameter
  If signalSemaphoreValuesCount is not 0, and pSignalSemaphoreValues is not NULL, pSignalSemaphoreValues **must** be a valid pointer to an array of signalSemaphoreValuesCount uint64_t values

Structure Chaining

Extends the structure

- VkSubmitInfo

When submitting work that operates on memory imported from a Direct3D 11 resource to a queue, the keyed mutex mechanism **may** be used in addition to Vulkan semaphores to synchronize the work. Keyed mutexes are a property of a properly created shareable Direct3D 11 resource. They **can** only be used if the imported resource was created with the D3D11_RESOURCE_MISC_SHARED_KEYEDMUTEX flag.

To acquire keyed mutexes before submitted work and/or release them after, add a VkWin32KeyedMutexAcquireReleaseInfoKHR structure to the pNext chain of the VkSubmitInfo structure.

The VkWin32KeyedMutexAcquireReleaseInfoKHR structure is defined as:

```
// Provided by VK_KHR_win32_keyed_mutex
typedef struct VkWin32KeyedMutexAcquireReleaseInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    uint32_t                 acquireCount;
    const VkDeviceMemory*    pAcquireSyncs;
    const uint64_t*          pAcquireKeys;
    const uint32_t*          pAcquireTimeouts;
    uint32_t                 releaseCount;
    const VkDeviceMemory*    pReleaseSyncs;
    const uint64_t*          pReleaseKeys;
} VkWin32KeyedMutexAcquireReleaseInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `acquireCount` is the number of entries in the `pAcquireSyncs`, `pAcquireKeys`, and `pAcquireTimeouts` arrays.
- `pAcquireSyncs` is a pointer to an array of VkDeviceMemory objects which were imported from Direct3D 11 resources.
- `pAcquireKeys` is a pointer to an array of mutex key values to wait for prior to beginning the submitted work. Entries refer to the keyed mutex associated with the corresponding entries in `pAcquireSyncs`.
- `pAcquireTimeouts` is a pointer to an array of timeout values, in millisecond units, for each acquire specified in `pAcquireKeys`.
- `releaseCount` is the number of entries in the `pReleaseSyncs` and `pReleaseKeys` arrays.
- `pReleaseSyncs` is a pointer to an array of VkDeviceMemory objects which were imported from Direct3D 11 resources.
- `pReleaseKeys` is a pointer to an array of mutex key values to set when the submitted work has completed. Entries refer to the keyed mutex associated with the corresponding entries in `pReleaseSyncs`.

Valid Usage

- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-pAcquireSyncs-00081
  Each member of `pAcquireSyncs` and `pReleaseSyncs` **must** be a device memory object imported by setting VkImportMemoryWin32HandleInfoKHR::`handleType` to VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT or VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT

Valid Usage (Implicit)

- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-sType-sType
  `sType` **must** be

VK_STRUCTURE_TYPE_WIN32_KEYED_MUTEX_ACQUIRE_RELEASE_INFO_KHR

- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-pAcquireSyncs-parameter
  If `acquireCount` is not `0`, `pAcquireSyncs` **must** be a valid pointer to an array of `acquireCount` valid VkDeviceMemory handles
- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-pAcquireKeys-parameter
  If `acquireCount` is not `0`, `pAcquireKeys` **must** be a valid pointer to an array of `acquireCount` `uint64_t` values
- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-pAcquireTimeouts-parameter
  If `acquireCount` is not `0`, `pAcquireTimeouts` **must** be a valid pointer to an array of `acquireCount` `uint32_t` values
- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-pReleaseSyncs-parameter
  If `releaseCount` is not `0`, `pReleaseSyncs` **must** be a valid pointer to an array of `releaseCount` valid VkDeviceMemory handles
- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-pReleaseKeys-parameter
  If `releaseCount` is not `0`, `pReleaseKeys` **must** be a valid pointer to an array of `releaseCount` `uint64_t` values
- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-commonparent
  Both of the elements of `pAcquireSyncs`, and the elements of `pReleaseSyncs` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Structure Chaining

Extends the structures

- VkSubmitInfo
- VkSubmitInfo2

If the `pNext` chain of VkSubmitInfo includes a `VkProtectedSubmitInfo` structure, then the structure indicates whether the batch is protected. The `VkProtectedSubmitInfo` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkProtectedSubmitInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           protectedSubmit;
} VkProtectedSubmitInfo;
```

- `protectedSubmit` specifies whether the batch is protected. If `protectedSubmit` is VK_TRUE, the batch is protected. If `protectedSubmit` is VK_FALSE, the batch is unprotected. If the `VkSubmitInfo` ::`pNext` chain does not include this structure, the batch is unprotected.

Valid Usage (Implicit)

- VUID-VkProtectedSubmitInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PROTECTED_SUBMIT_INFO

Structure Chaining

Extends the structure

- VkSubmitInfo

If the `pNext` chain of VkSubmitInfo includes a `VkDeviceGroupSubmitInfo` structure, then that structure includes device indices and masks specifying which physical devices execute semaphore operations and command buffers.

The `VkDeviceGroupSubmitInfo` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkDeviceGroupSubmitInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           waitSemaphoreCount;
    const uint32_t*    pWaitSemaphoreDeviceIndices;
    uint32_t           commandBufferCount;
    const uint32_t*    pCommandBufferDeviceMasks;
    uint32_t           signalSemaphoreCount;
    const uint32_t*    pSignalSemaphoreDeviceIndices;
} VkDeviceGroupSubmitInfo;
```

```
// Provided by VK_KHR_device_group
// Equivalent to VkDeviceGroupSubmitInfo
typedef VkDeviceGroupSubmitInfo VkDeviceGroupSubmitInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `waitSemaphoreCount` is the number of elements in the `pWaitSemaphoreDeviceIndices` array.
- `pWaitSemaphoreDeviceIndices` is a pointer to an array of `waitSemaphoreCount` device indices indicating which physical device executes the semaphore wait operation in the corresponding element of VkSubmitInfo::`pWaitSemaphores`.
- `commandBufferCount` is the number of elements in the `pCommandBufferDeviceMasks` array.
- `pCommandBufferDeviceMasks` is a pointer to an array of `commandBufferCount` device masks indicating which physical devices execute the command buffer in the corresponding element of VkSubmitInfo::`pCommandBuffers`. A physical device executes the command buffer if the

corresponding bit is set in the mask.

- `signalSemaphoreCount` is the number of elements in the `pSignalSemaphoreDeviceIndices` array.
- `pSignalSemaphoreDeviceIndices` is a pointer to an array of `signalSemaphoreCount` device indices indicating which physical device executes the semaphore signal operation in the corresponding element of VkSubmitInfo::`pSignalSemaphores`.

If this structure is not present, semaphore operations and command buffers execute on device index zero.

Valid Usage

- VUID-VkDeviceGroupSubmitInfo-waitSemaphoreCount-00082
  `waitSemaphoreCount` **must** equal VkSubmitInfo::`waitSemaphoreCount`
- VUID-VkDeviceGroupSubmitInfo-commandBufferCount-00083
  `commandBufferCount` **must** equal VkSubmitInfo::`commandBufferCount`
- VUID-VkDeviceGroupSubmitInfo-signalSemaphoreCount-00084
  `signalSemaphoreCount` **must** equal VkSubmitInfo::`signalSemaphoreCount`
- VUID-VkDeviceGroupSubmitInfo-pWaitSemaphoreDeviceIndices-00085
  All elements of `pWaitSemaphoreDeviceIndices` and `pSignalSemaphoreDeviceIndices` **must** be valid device indices
- VUID-VkDeviceGroupSubmitInfo-pCommandBufferDeviceMasks-00086
  All elements of `pCommandBufferDeviceMasks` **must** be valid device masks

Valid Usage (Implicit)

- VUID-VkDeviceGroupSubmitInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_GROUP_SUBMIT_INFO
- VUID-VkDeviceGroupSubmitInfo-pWaitSemaphoreDeviceIndices-parameter
  If `waitSemaphoreCount` is not `0`, `pWaitSemaphoreDeviceIndices` **must** be a valid pointer to an array of `waitSemaphoreCount` `uint32_t` values
- VUID-VkDeviceGroupSubmitInfo-pCommandBufferDeviceMasks-parameter
  If `commandBufferCount` is not `0`, `pCommandBufferDeviceMasks` **must** be a valid pointer to an array of `commandBufferCount` `uint32_t` values
- VUID-VkDeviceGroupSubmitInfo-pSignalSemaphoreDeviceIndices-parameter
  If `signalSemaphoreCount` is not `0`, `pSignalSemaphoreDeviceIndices` **must** be a valid pointer to an array of `signalSemaphoreCount` `uint32_t` values

Structure Chaining

Extends the structure

- VkSubmitInfo

If the pNext chain of VkSubmitInfo includes a VkPerformanceQuerySubmitInfoKHR structure, then the structure indicates which counter pass is active for the batch in that submit.

The VkPerformanceQuerySubmitInfoKHR structure is defined as:

```
// Provided by VK_KHR_performance_query
typedef struct VkPerformanceQuerySubmitInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           counterPassIndex;
} VkPerformanceQuerySubmitInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- counterPassIndex specifies which counter pass index is active.

If the VkSubmitInfo::pNext chain does not include this structure, the batch defaults to use counter pass index 0.

Valid Usage

- VUID-VkPerformanceQuerySubmitInfoKHR-counterPassIndex-03221
  counterPassIndex **must** be less than the number of counter passes required by any queries within the batch. The required number of counter passes for a performance query is obtained by calling vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR

Valid Usage (Implicit)

- VUID-VkPerformanceQuerySubmitInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PERFORMANCE_QUERY_SUBMIT_INFO_KHR

Structure Chaining

Extends the structures

- VkSubmitInfo
- VkSubmitInfo2

6.6. Queue Forward Progress

When using binary semaphores, the application **must** ensure that command buffer submissions will be able to complete without any subsequent operations by the application on any queue. After any call to vkQueueSubmit (or other queue operation), for every queued wait on a semaphore created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_BINARY there **must** be a prior signal of that

semaphore that will not be consumed by a different wait on the semaphore.

When using timeline semaphores, wait-before-signal behavior is well-defined and applications **can** submit work via `vkQueueSubmit` defining a timeline semaphore wait operation before submitting a corresponding semaphore signal operation. For each timeline semaphore wait operation defined by a call to `vkQueueSubmit`, the application **must** ensure that a corresponding semaphore signal operation is executed before forward progress can be made.

If a command buffer submission waits for any events to be signaled, the application **must** ensure that command buffer submissions will be able to complete without any subsequent operations by the application. Events signaled by the host **must** be signaled before the command buffer waits on those events.

NOTE

The ability for commands to wait on the host to set events was originally added to allow low-latency updates to resources between host and device. However, to ensure quality of service, implementations would necessarily detect extended stalls in execution and timeout after a short period. As this period is not defined in the Vulkan specification, it is impossible to correctly validate any application with any wait period. Since the original users of this functionality were highly limited and platform-specific, this functionality is now considered defunct and should not be used.

6.7. Secondary Command Buffer Execution

Secondary command buffers **must** not be directly submitted to a queue. To record a secondary command buffer to execute as part of a primary command buffer, call:

```
// Provided by VK_VERSION_1_0
void vkCmdExecuteCommands(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    commandBufferCount,
    const VkCommandBuffer*                      pCommandBuffers);
```

- `commandBuffer` is a handle to a primary command buffer that the secondary command buffers are executed in.
- `commandBufferCount` is the length of the `pCommandBuffers` array.
- `pCommandBuffers` is a pointer to an array of `commandBufferCount` secondary command buffer handles, which are recorded to execute in the primary command buffer in the order they are listed in the array.

If any element of `pCommandBuffers` was not recorded with the VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT flag, and it was recorded into any other primary command buffer which is currently in the executable or recording state, that primary command buffer becomes invalid.

If the `nestedCommandBuffer` feature is enabled it is valid usage for `vkCmdExecuteCommands` to also be recorded to a secondary command buffer.

Valid Usage

- VUID-vkCmdExecuteCommands-pCommandBuffers-00088
  Each element of pCommandBuffers **must** have been allocated with a level of VK_COMMAND_BUFFER_LEVEL_SECONDARY
- VUID-vkCmdExecuteCommands-pCommandBuffers-00089
  Each element of pCommandBuffers **must** be in the pending or executable state
- VUID-vkCmdExecuteCommands-pCommandBuffers-00091
  If any element of pCommandBuffers was not recorded with the VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT flag, it **must** not be in the pending state
- VUID-vkCmdExecuteCommands-pCommandBuffers-00092
  If any element of pCommandBuffers was not recorded with the VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT flag, it **must** not have already been recorded to commandBuffer
- VUID-vkCmdExecuteCommands-pCommandBuffers-00093
  If any element of pCommandBuffers was not recorded with the VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT flag, it **must** not appear more than once in pCommandBuffers
- VUID-vkCmdExecuteCommands-pCommandBuffers-00094
  Each element of pCommandBuffers **must** have been allocated from a VkCommandPool that was created for the same queue family as the VkCommandPool from which commandBuffer was allocated
- VUID-vkCmdExecuteCommands-pCommandBuffers-00096
  If this command is called within a render pass instance, each element of pCommandBuffers **must** have been recorded with the VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT
- VUID-vkCmdExecuteCommands-pCommandBuffers-00099
  If this command is called within a render pass instance, and any element of pCommandBuffers was recorded with VkCommandBufferInheritanceInfo::framebuffer not equal to VK_NULL_HANDLE, that VkFramebuffer **must** match the VkFramebuffer used in the current render pass instance
- VUID-vkCmdExecuteCommands-contents-09680
  If this command is called within a render pass instance begun with vkCmdBeginRenderPass, and vkCmdNextSubpass has not been called in the current render pass instance, the contents parameter of vkCmdBeginRenderPass **must** have been VK_SUBPASS_CONTENTS_SECONDARY_COMMAND_BUFFERS , or VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_EXT
- VUID-vkCmdExecuteCommands-None-09681
  If this command is called within a render pass instance begun with vkCmdBeginRenderPass, and vkCmdNextSubpass has been called in the current render pass instance, the contents parameter of the last call to vkCmdNextSubpass **must** have been VK_SUBPASS_CONTENTS_SECONDARY_COMMAND_BUFFERS , or VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR

- VUID-vkCmdExecuteCommands-pCommandBuffers-06019
  If this command is called within a render pass instance begun with vkCmdBeginRenderPass, each element of pCommandBuffers **must** have been recorded with VkCommandBufferInheritanceInfo::subpass set to the index of the subpass which the given command buffer will be executed in
- VUID-vkCmdExecuteCommands-pBeginInfo-06020
  If this command is called within a render pass instance begun with vkCmdBeginRenderPass, the render passes specified in the pBeginInfo->pInheritanceInfo->renderPass members of the vkBeginCommandBuffer commands used to begin recording each element of pCommandBuffers **must** be compatible with the current render pass
- VUID-vkCmdExecuteCommands-pCommandBuffers-00100
  If vkCmdExecuteCommands is not being called within a render pass instance, each element of pCommandBuffers **must** not have been recorded with the VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT
- VUID-vkCmdExecuteCommands-commandBuffer-00101
  If the inheritedQueries feature is not enabled, commandBuffer **must** not have any queries active
- VUID-vkCmdExecuteCommands-commandBuffer-00102
  If commandBuffer has a VK_QUERY_TYPE_OCCLUSION query active, then each element of pCommandBuffers **must** have been recorded with VkCommandBufferInheritanceInfo::occlusionQueryEnable set to VK_TRUE
- VUID-vkCmdExecuteCommands-commandBuffer-00103
  If commandBuffer has a VK_QUERY_TYPE_OCCLUSION query active, then each element of pCommandBuffers **must** have been recorded with VkCommandBufferInheritanceInfo::queryFlags having all bits set that are set for the query
- VUID-vkCmdExecuteCommands-commandBuffer-00104
  If commandBuffer has a VK_QUERY_TYPE_PIPELINE_STATISTICS query active, then each element of pCommandBuffers **must** have been recorded with VkCommandBufferInheritanceInfo::pipelineStatistics having all bits set that are set in the VkQueryPool the query uses
- VUID-vkCmdExecuteCommands-pCommandBuffers-00105
  Each element of pCommandBuffers **must** not begin any query types that are active in commandBuffer
- VUID-vkCmdExecuteCommands-commandBuffer-07594
  commandBuffer **must** not have any queries other than VK_QUERY_TYPE_OCCLUSION and VK_QUERY_TYPE_PIPELINE_STATISTICS active
- VUID-vkCmdExecuteCommands-commandBuffer-01820
  If commandBuffer is a protected command buffer and protectedNoFault is not supported, each element of pCommandBuffers **must** be a protected command buffer
- VUID-vkCmdExecuteCommands-commandBuffer-01821
  If commandBuffer is an unprotected command buffer and protectedNoFault is not supported, each element of pCommandBuffers **must** be an unprotected command buffer
- VUID-vkCmdExecuteCommands-None-02286

This command **must** not be recorded when transform feedback is active

- VUID-vkCmdExecuteCommands-commandBuffer-06533
  If this command is called within a render pass instance and any recorded command in `commandBuffer` in the current subpass will write to an image subresource as an attachment, commands recorded in elements of `pCommandBuffers` **must** not read from the memory backing that image subresource in any other way
- VUID-vkCmdExecuteCommands-commandBuffer-06534
  If this command is called within a render pass instance and any recorded command in `commandBuffer` in the current subpass will read from an image subresource used as an attachment in any way other than as an attachment, commands recorded in elements of `pCommandBuffers` **must** not write to that image subresource as an attachment
- VUID-vkCmdExecuteCommands-pCommandBuffers-06535
  If this command is called within a render pass instance and any recorded command in a given element of `pCommandBuffers` will write to an image subresource as an attachment, commands recorded in elements of `pCommandBuffers` at a higher index **must** not read from the memory backing that image subresource in any other way
- VUID-vkCmdExecuteCommands-pCommandBuffers-06536
  If this command is called within a render pass instance and any recorded command in a given element of `pCommandBuffers` will read from an image subresource used as an attachment in any way other than as an attachment, commands recorded in elements of `pCommandBuffers` at a higher index **must** not write to that image subresource as an attachment
- VUID-vkCmdExecuteCommands-pCommandBuffers-06021
  If `pCommandBuffers` contains any suspended render pass instances, there **must** be no action or synchronization commands between that render pass instance and any render pass instance that resumes it
- VUID-vkCmdExecuteCommands-pCommandBuffers-06022
  If `pCommandBuffers` contains any suspended render pass instances, there **must** be no render pass instances between that render pass instance and any render pass instance that resumes it
- VUID-vkCmdExecuteCommands-variableSampleLocations-06023
  If the `variableSampleLocations` limit is not supported, and any element of `pCommandBuffers` contains any suspended render pass instances, where a graphics pipeline has been bound, any pipelines bound in the render pass instance that resumes it, or any subsequent render pass instances that resume from that one and so on, **must** use the same sample locations
- VUID-vkCmdExecuteCommands-flags-06024
  If this command is called within a render pass instance begun with vkCmdBeginRendering, its VkRenderingInfo::`flags` parameter **must** have included VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT
- VUID-vkCmdExecuteCommands-pBeginInfo-06025
  If this command is called within a render pass instance begun with vkCmdBeginRendering, the render passes specified in the `pBeginInfo->pInheritanceInfo->renderPass` members of the vkBeginCommandBuffer commands used to begin recording

each element of pCommandBuffers **must** be VK_NULL_HANDLE

- VUID-vkCmdExecuteCommands-flags-06026
  If this command is called within a render pass instance begun with vkCmdBeginRendering, the flags member of the VkCommandBufferInheritanceRenderingInfo structure included in the pNext chain of VkCommandBufferBeginInfo::pInheritanceInfo used to begin recording each element of pCommandBuffers **must** be equal to the VkRenderingInfo::flags parameter to vkCmdBeginRendering, excluding VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT
- VUID-vkCmdExecuteCommands-colorAttachmentCount-06027
  If this command is called within a render pass instance begun with vkCmdBeginRendering, the colorAttachmentCount member of the VkCommandBufferInheritanceRenderingInfo structure included in the pNext chain of VkCommandBufferBeginInfo::pInheritanceInfo used to begin recording each element of pCommandBuffers **must** be equal to the VkRenderingInfo::colorAttachmentCount parameter to vkCmdBeginRendering
- VUID-vkCmdExecuteCommands-imageView-06028
  If this command is called within a render pass instance begun with vkCmdBeginRendering, if the imageView member of an element of the VkRenderingInfo::pColorAttachments parameter to vkCmdBeginRendering is not VK_NULL_HANDLE, the corresponding element of the pColorAttachmentFormats member of the VkCommandBufferInheritanceRenderingInfo structure included in the pNext chain of VkCommandBufferBeginInfo::pInheritanceInfo used to begin recording each element of pCommandBuffers **must** be equal to the format used to create that image view
- VUID-vkCmdExecuteCommands-imageView-07606
  If this command is called within a render pass instance begun with vkCmdBeginRendering, if the imageView member of an element of the VkRenderingInfo::pColorAttachments parameter to vkCmdBeginRendering is VK_NULL_HANDLE, the corresponding element of the pColorAttachmentFormats member of the VkCommandBufferInheritanceRenderingInfo structure included in the pNext chain of VkCommandBufferBeginInfo::pInheritanceInfo used to begin recording each element of pCommandBuffers **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteCommands-pDepthAttachment-06029
  If this command is called within a render pass instance begun with vkCmdBeginRendering, if the VkRenderingInfo::pDepthAttachment->imageView parameter to vkCmdBeginRendering is not VK_NULL_HANDLE, the value of the depthAttachmentFormat member of the VkCommandBufferInheritanceRenderingInfo structure included in the pNext chain of VkCommandBufferBeginInfo::pInheritanceInfo used to begin recording each element of pCommandBuffers **must** be equal to the format used to create that image view
- VUID-vkCmdExecuteCommands-pStencilAttachment-06030
  If this command is called within a render pass instance begun with vkCmdBeginRendering, if the VkRenderingInfo::pStencilAttachment->imageView parameter to vkCmdBeginRendering is not VK_NULL_HANDLE, the value of the stencilAttachmentFormat member of the VkCommandBufferInheritanceRenderingInfo

structure included in the pNext chain of VkCommandBufferBeginInfo::pInheritanceInfo used to begin recording each element of pCommandBuffers **must** be equal to the format used to create that image view

- VUID-vkCmdExecuteCommands-pDepthAttachment-06774
  If this command is called within a render pass instance begun with vkCmdBeginRendering and the VkRenderingInfo::pDepthAttachment->imageView parameter to vkCmdBeginRendering was VK_NULL_HANDLE, the value of the depthAttachmentFormat member of the VkCommandBufferInheritanceRenderingInfo structure included in the pNext chain of VkCommandBufferBeginInfo::pInheritanceInfo used to begin recording each element of pCommandBuffers **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteCommands-pStencilAttachment-06775
  If this command is called within a render pass instance begun with vkCmdBeginRendering and the VkRenderingInfo::pStencilAttachment->imageView parameter to vkCmdBeginRendering was VK_NULL_HANDLE, the value of the stencilAttachmentFormat member of the VkCommandBufferInheritanceRenderingInfo structure included in the pNext chain of VkCommandBufferBeginInfo::pInheritanceInfo used to begin recording each element of pCommandBuffers **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteCommands-pCommandBuffers-11500
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, then each element of pCommandBuffers **must** have been recorded with a VkCustomResolveCreateInfoEXT struct chained to its VkCommandBufferInheritanceInfo
- VUID-vkCmdExecuteCommands-pCommandBuffers-11501
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, then each element of pCommandBuffers **must** not have been recorded with a VkCustomResolveCreateInfoEXT struct chained to its VkCommandBufferInheritanceInfo
- VUID-vkCmdExecuteCommands-pCommandBuffers-11502
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then each element of pCommandBuffers **must** have been recorded with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdExecuteCommands-pCommandBuffers-11503
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, then each element of pCommandBuffers **must** have been recorded with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdExecuteCommands-colorAttachmentCount-11532
  If this command is called within a render pass instance begun with vkCmdBeginRendering that contains a custom resolve, the colorAttachmentCount member of the VkCustomResolveCreateInfoEXT structure included in the pNext chain of VkCommandBufferBeginInfo::pInheritanceInfo used to begin recording each element of pCommandBuffers **must** be equal to the VkRenderingInfo::colorAttachmentCount parameter to vkCmdBeginRendering
- VUID-vkCmdExecuteCommands-resolveImageView-11533

If this command is called within a render pass instance begun with vkCmdBeginRendering that contains a custom resolve, if the `resolveImageView` member of an element of the VkRenderingInfo::`pColorAttachments` parameter to vkCmdBeginRendering is not VK_NULL_HANDLE, the corresponding element of the `pColorAttachmentFormats` member of the VkCustomResolveCreateInfoEXT structure included in the `pNext` chain of VkCommandBufferBeginInfo::`pInheritanceInfo` used to begin recording each element of `pCommandBuffers` **must** be equal to the format used to create that image view

- VUID-vkCmdExecuteCommands-resolveImageView-11534
  If this command is called within a render pass instance begun with vkCmdBeginRendering that contains a custom resolve, if the `resolveImageView` member of an element of the VkRenderingInfo::`pColorAttachments` parameter to vkCmdBeginRendering is VK_NULL_HANDLE, the corresponding element of the `pColorAttachmentFormats` member of the VkCustomResolveCreateInfoEXT structure included in the `pNext` chain of VkCommandBufferBeginInfo::`pInheritanceInfo` used to begin recording each element of `pCommandBuffers` **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteCommands-pDepthAttachment-11535
  If this command is called within a render pass instance begun with vkCmdBeginRendering, if the VkRenderingInfo::`pDepthAttachment->imageView` parameter to vkCmdBeginRendering is not VK_NULL_HANDLE, the value of the `depthAttachmentFormat` member of the VkCommandBufferInheritanceRenderingInfo structure included in the `pNext` chain of VkCommandBufferBeginInfo::`pInheritanceInfo` used to begin recording each element of `pCommandBuffers` **must** be equal to the format used to create that image view
- VUID-vkCmdExecuteCommands-pStencilAttachment-11536
  If this command is called within a render pass instance begun with vkCmdBeginRendering that contains a custom resolve, if the VkRenderingInfo::`pStencilAttachment->resolveImageView` parameter to vkCmdBeginRendering is not VK_NULL_HANDLE, the value of the `stencilAttachmentFormat` member of the VkCustomResolveCreateInfoEXT structure included in the `pNext` chain of VkCommandBufferBeginInfo::`pInheritanceInfo` used to begin recording each element of `pCommandBuffers` **must** be equal to the format used to create that image view
- VUID-vkCmdExecuteCommands-pDepthAttachment-11537
  If this command is called within a render pass instance begun with vkCmdBeginRendering that contains a custom resolve and the VkRenderingInfo::`pDepthAttachment->resolveImageView` parameter to vkCmdBeginRendering was VK_NULL_HANDLE, the value of the `depthAttachmentFormat` member of the VkCustomResolveCreateInfoEXT structure included in the `pNext` chain of VkCommandBufferBeginInfo::`pInheritanceInfo` used to begin recording each element of `pCommandBuffers` **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteCommands-pStencilAttachment-11538
  If this command is called within a render pass instance begun with vkCmdBeginRendering that contains a custom resolve and the VkRenderingInfo::`pStencilAttachment->resolveImageView` parameter to vkCmdBeginRendering was VK_NULL_HANDLE, the value of the `stencilAttachmentFormat` member of the VkCustomResolveCreateInfoEXT structure included in the `pNext` chain of

VkCommandBufferBeginInfo::pInheritanceInfo used to begin recording each element of pCommandBuffers **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdExecuteCommands-resolveImageView-11526
  If this command is called within a render pass instance begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and the resolveImageView member of an element of the VkRenderingInfo::pColorAttachments parameter to vkCmdBeginRendering is not VK_NULL_HANDLE, the value of VkCommandBufferInheritanceRenderingInfo::rasterizationSamples **must** be equal to the sample count used to create that resolveImageView
- VUID-vkCmdExecuteCommands-pDepthAttachment-11527
  If this command is called within a render pass instance begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and the VkRenderingInfo::pDepthAttachment->resolveImageView parameter to vkCmdBeginRendering is not VK_NULL_HANDLE, the value of VkCommandBufferInheritanceRenderingInfo::rasterizationSamples **must** be equal to the sample count used to create that image view
- VUID-vkCmdExecuteCommands-pStencilAttachment-11528
  If this command is called within a render pass instance begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and the VkRenderingInfo::pStencilAttachment->resolveImageView parameter to vkCmdBeginRendering is not VK_NULL_HANDLE, the value of VkCommandBufferInheritanceRenderingInfo::rasterizationSamples **must** be equal to the sample count used to create that image view
- VUID-vkCmdExecuteCommands-viewMask-06031
  If this command is called within a render pass instance begun with vkCmdBeginRendering, the viewMask member of the VkCommandBufferInheritanceRenderingInfo structure included in the pNext chain of VkCommandBufferBeginInfo::pInheritanceInfo used to begin recording each element of pCommandBuffers **must** be equal to the VkRenderingInfo::viewMask parameter to vkCmdBeginRendering
- VUID-vkCmdExecuteCommands-commandBuffer-09375
  commandBuffer **must** not be a secondary command buffer unless the nestedCommandBuffer feature is enabled
- VUID-vkCmdExecuteCommands-nestedCommandBuffer-09376
  If the nestedCommandBuffer feature is enabled, and commandBuffer is a secondary command buffer, the command buffer nesting level of each element of pCommandBuffers **must** be less than maxCommandBufferNestingLevel
- VUID-vkCmdExecuteCommands-nestedCommandBufferRendering-09377
  If the nestedCommandBufferRendering feature is not enabled, and commandBuffer is a secondary command buffer, commandBuffer **must** not have been recorded with VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT
- VUID-vkCmdExecuteCommands-nestedCommandBufferSimultaneousUse-09378
  If the nestedCommandBufferSimultaneousUse feature is not enabled, and commandBuffer is a secondary command buffer, each element of pCommandBuffers **must** not have been recorded with VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT

- VUID-vkCmdExecuteCommands-pCommandBuffers-09504
  If this command is called within a render pass instance begun with vkCmdBeginRendering, the color attachment mapping state specified by VkRenderingAttachmentLocationInfo in the inheritance info of each element of pCommandBuffers and in the current state of commandBuffer **must** match
- VUID-vkCmdExecuteCommands-pCommandBuffers-09505
  If this command is called within a render pass instance begun with vkCmdBeginRendering, the input attachment mapping state specified by VkRenderingInputAttachmentIndexInfo in the inheritance info of each element of pCommandBuffers and in the current state of commandBuffer **must** match
- VUID-vkCmdExecuteCommands-commandBuffer-11351
  If there is a sampler descriptor heap bound to commandBuffer, each element of pCommandBuffers **must** have been recorded with a value of VkCommandBufferInheritanceDescriptorHeapInfoEXT::pSamplerHeapBindInfo that is either NULL or a pointer to a bind info that is identical to that set via the last call to vkCmdBindSamplerHeapEXT
- VUID-vkCmdExecuteCommands-commandBuffer-11352
  If there is a resource descriptor heap bound to commandBuffer, each element of pCommandBuffers **must** have been recorded with a value of VkCommandBufferInheritanceDescriptorHeapInfoEXT::pResourceHeapBindInfo that is either NULL or a pointer to a bind info that is identical to that set via the last call to vkCmdBindResourceHeapEXT
- VUID-vkCmdExecuteCommands-commandBuffer-11473
  If there is no sampler descriptor heap bound to commandBuffer, each element of pCommandBuffers **must** have been recorded with a value of VkCommandBufferInheritanceDescriptorHeapInfoEXT::pSamplerHeapBindInfo set to NULL
- VUID-vkCmdExecuteCommands-commandBuffer-11474
  If there is no resource descriptor heap bound to commandBuffer, each element of pCommandBuffers **must** have been recorded with a value of VkCommandBufferInheritanceDescriptorHeapInfoEXT::pResourceHeapBindInfo set to NULL

Valid Usage (Implicit)

- VUID-vkCmdExecuteCommands-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdExecuteCommands-pCommandBuffers-parameter
  pCommandBuffers **must** be a valid pointer to an array of commandBufferCount valid VkCommandBuffer handles
- VUID-vkCmdExecuteCommands-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdExecuteCommands-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, or VK_QUEUE_TRANSFER_BIT

operations

- VUID-vkCmdExecuteCommands-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdExecuteCommands-commandBufferCount-arraylength
  `commandBufferCount` **must** be greater than `0`
- VUID-vkCmdExecuteCommands-commonparent
  Both of `commandBuffer`, and the elements of `pCommandBuffers` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT	Indirection

Conditional Rendering

vkCmdExecuteCommands is not affected by conditional rendering

6.8. Nested Command Buffers

In addition to secondary command buffer execution from primary command buffers, an implementation **may** support nested command buffers, which enable secondary command buffers to be executed from other secondary command buffers. If the `nestedCommandBuffer` feature is enabled, the implementation supports nested command buffers.

Nested command buffer execution works the same as primary-to-secondary execution, except that it is subject to some additional implementation-defined limits.

Each secondary command buffer has a command buffer nesting level, which is determined at

vkEndCommandBuffer time and evaluated at vkCmdExecuteCommands time. A secondary command buffer that executes no other secondary command buffers has a command buffer nesting level of zero. Otherwise, the command buffer nesting level of a secondary command buffer is equal to the maximum nesting level of all secondary command buffers executed by that command buffer plus one. Some implementations **may** have a limit on the maximum nesting level of secondary command buffers that **can** be recorded. This limit is advertised in `maxCommandBufferNestingLevel`.

If the `nestedCommandBufferRendering` feature is enabled, the implementation supports calling vkCmdExecuteCommands inside secondary command buffers recorded with VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT. If the `nestedCommandBufferSimultaneousUse` feature is enabled, the implementation supports calling vkCmdExecuteCommands with secondary command buffers recorded with VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT.

Whenever vkCmdExecuteCommands is recorded inside a secondary command buffer recorded with VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT, each member of `pCommandBuffers` **must** have been recorded with a VkCommandBufferBeginInfo with VkCommandBufferInheritanceInfo compatible with the VkCommandBufferInheritanceInfo of the command buffer into which the vkCmdExecuteCommands call is being recorded. The VkCommandBufferInheritanceRenderingInfo structures are compatible when the `VkCommandBufferInheritanceRenderingInfo::renderpass` are compatible, or if they are VK_NULL_HANDLE then the VkCommandBufferInheritanceRenderingInfo members match, and all other members of `VkCommandBufferInheritanceRenderingInfo` match. This requirement applies recursively, down to the most nested command buffer and up to the command buffer where the render pass was originally begun.

6.9. Command Buffer Device Mask

Each command buffer has a piece of state storing the current device mask of the command buffer. This mask controls which physical devices within the logical device all subsequent commands will execute on, including state-setting commands, action commands, and synchronization commands.

Scissor and viewport state (excluding the count of each) **can** be different values on each physical device (only when set as dynamic state), and each physical device will render using its local copy of the state. Other state is shared between physical devices, such that all physical devices use the most recently set values for the state. However, when recording an action command that uses a piece of state, the most recent command that set that state **must** have included all physical devices that execute the action command in its current device mask.

The command buffer's device mask is orthogonal to the `pCommandBufferDeviceMasks` member of VkDeviceGroupSubmitInfo. Commands only execute on a physical device if the device index is set in both device masks.

If the `pNext` chain of VkCommandBufferBeginInfo includes a `VkDeviceGroupCommandBufferBeginInfo` structure, then that structure includes an initial device mask for the command buffer.

The `VkDeviceGroupCommandBufferBeginInfo` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkDeviceGroupCommandBufferBeginInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           deviceMask;
} VkDeviceGroupCommandBufferBeginInfo;
```

```
// Provided by VK_KHR_device_group
// Equivalent to VkDeviceGroupCommandBufferBeginInfo
typedef VkDeviceGroupCommandBufferBeginInfo VkDeviceGroupCommandBufferBeginInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `deviceMask` is the initial value of the command buffer's device mask.

The initial device mask also acts as an upper bound on the set of devices that **can** ever be in the device mask in the command buffer.

If this structure is not present, the initial value of a command buffer's device mask includes all physical devices in the logical device when the command buffer begins recording.

Valid Usage

- VUID-VkDeviceGroupCommandBufferBeginInfo-deviceMask-00106
  `deviceMask` **must** be a valid device mask value
- VUID-VkDeviceGroupCommandBufferBeginInfo-deviceMask-00107
  `deviceMask` **must** not be zero

Valid Usage (Implicit)

- VUID-VkDeviceGroupCommandBufferBeginInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_GROUP_COMMAND_BUFFER_BEGIN_INFO

Structure Chaining

Extends the structure

- VkCommandBufferBeginInfo

To update the current device mask of a command buffer, call:

```
// Provided by VK_VERSION_1_1
void vkCmdSetDeviceMask(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    deviceMask);
```

```
// Provided by VK_KHR_device_group
// Equivalent to vkCmdSetDeviceMask
void vkCmdSetDeviceMaskKHR(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    deviceMask);
```

- `commandBuffer` is the command buffer whose current device mask is modified.
- `deviceMask` is the new value of the current device mask.

`deviceMask` is used to filter out subsequent commands from executing on all physical devices whose bit indices are not set in the mask, except commands beginning a render pass instance, commands transitioning to the next subpass in the render pass instance, and commands ending a render pass instance, which always execute on the set of physical devices whose bit indices are included in the `deviceMask` member of the VkDeviceGroupRenderPassBeginInfo structure passed to the command beginning the corresponding render pass instance.

Valid Usage

- VUID-vkCmdSetDeviceMask-deviceMask-00108
  `deviceMask` **must** be a valid device mask value
- VUID-vkCmdSetDeviceMask-deviceMask-00109
  `deviceMask` **must** not be zero
- VUID-vkCmdSetDeviceMask-deviceMask-00110
  `deviceMask` **must** not include any set bits that were not in the VkDeviceGroupCommandBufferBeginInfo::`deviceMask` value when the command buffer began recording
- VUID-vkCmdSetDeviceMask-deviceMask-00111
  If `vkCmdSetDeviceMask` is called inside a render pass instance, `deviceMask` **must** not include any set bits that were not in the VkDeviceGroupRenderPassBeginInfo::`deviceMask` value when the render pass instance began recording

Valid Usage (Implicit)

- VUID-vkCmdSetDeviceMask-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDeviceMask-commandBuffer-recording
  `commandBuffer` **must** be in the recording state

- VUID-vkCmdSetDeviceMask-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, or VK_QUEUE_TRANSFER_BIT operations

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT	State

Conditional Rendering

vkCmdSetDeviceMask is not affected by conditional rendering

Chapter 7. Synchronization and Cache Control

Synchronization of access to resources is primarily the responsibility of the application in Vulkan. The order of execution of commands with respect to the host and other commands on the device has few implicit guarantees, and needs to be explicitly specified. Memory caches and other optimizations are also explicitly managed, requiring that the flow of data through the system is largely under application control.

Whilst some implicit guarantees exist between commands, five explicit synchronization mechanisms are exposed by Vulkan:

Fences

Fences **can** be used to communicate to the host that execution of some task on the device has completed, controlling resource access between host and device.

Semaphores

Semaphores **can** be used to control resource access across multiple queues.

Events

Events provide a fine-grained synchronization primitive which **can** be signaled either within a command buffer or by the host, and **can** be waited upon within a command buffer or queried on the host. Events **can** be used to control resource access within a single queue.

Pipeline Barriers

Pipeline barriers also provide synchronization control within a command buffer, but at a single point, rather than with separate signal and wait operations. Pipeline barriers **can** be used to control resource access within a single queue.

Render Pass Objects

Render pass objects provide a synchronization framework for rendering tasks, built upon the concepts in this chapter. Many cases that would otherwise need an application to use other synchronization primitives **can** be expressed more efficiently as part of a render pass. Render pass objects **can** be used to control resource access within a single queue.

7.1. Execution and Memory Dependencies

An *operation* is an arbitrary amount of work to be executed on the host, a device, or an external entity such as a presentation engine. Synchronization commands introduce explicit *execution dependencies*, and *memory dependencies* between two sets of operations defined by the command's two *synchronization scopes*.

The synchronization scopes define which other operations a synchronization command is able to create execution dependencies with. Any type of operation that is not in a synchronization command's synchronization scopes will not be included in the resulting dependency. For example, for many synchronization commands, the synchronization scopes **can** be limited to just operations executing in specific pipeline stages, which allows other pipeline stages to be excluded from a

dependency. Other scoping options are possible, depending on the particular command.

An *execution dependency* is a guarantee that for two sets of operations, the first set **must** *happen-before* the second set. If an operation happens-before another operation, then the first operation **must** complete before the second operation is initiated. More precisely:

- Let $\mathbf{Ops}_1$ and $\mathbf{Ops}_2$ be separate sets of operations.
- Let **Sync** be a synchronization command.
- Let $\mathbf{Scope}_{1st}$ and $\mathbf{Scope}_{2nd}$ be the synchronization scopes of **Sync**.
- Let $\mathbf{ScopedOps}_1$ be the intersection of sets $\mathbf{Ops}_1$ and $\mathbf{Scope}_{1st}$.
- Let $\mathbf{ScopedOps}_2$ be the intersection of sets $\mathbf{Ops}_2$ and $\mathbf{Scope}_{2nd}$.
- Submitting $\mathbf{Ops}_1$, **Sync** and $\mathbf{Ops}_2$ for execution, in that order, will result in execution dependency **ExeDep** between $\mathbf{ScopedOps}_1$ and $\mathbf{ScopedOps}_2$.
- Execution dependency **ExeDep** guarantees that $\mathbf{ScopedOps}_1$ happen-before $\mathbf{ScopedOps}_2$.

For any pair of execution dependencies, $\mathbf{ExeDep}_A$ and $\mathbf{ExeDep}_B$, an *execution dependency chain* exists if the intersection of $\mathbf{ScopedOps}_2$ in $\mathbf{ExeDep}_A$ and $\mathbf{ScopedOps}_1$ in $\mathbf{ExeDep}_B$ is not an empty set. The formation of a single execution dependency from an execution dependency chain can be described by substituting the following in the description of execution dependencies:

- Let **Sync** be a set of synchronization commands that form an execution dependency chain.
- Let $\mathbf{Scope}_{1st}$ be the first synchronization scope of the first command in **Sync**.
- Let $\mathbf{Scope}_{2nd}$ be the second synchronization scope of the last command in **Sync**.

Execution dependencies formed by execution dependency chains recursively form execution dependency chains, building up a total ordering of execution.

Execution dependencies alone are not sufficient to guarantee that values resulting from writes in one set of operations **can** be read from another set of operations.

Three additional types of operations are used to control memory access. *Availability operations* cause the values generated by specified memory write accesses to become *available* to a memory domain for future access. Any available value remains available until a subsequent write to the same memory location occurs (whether it is made available or not) or the memory is freed. *Memory domain operations* cause writes that are available to a source memory domain to become available to a destination memory domain (an example of this is making writes available to the host domain available to the device domain). *Visibility operations* cause values available to a memory domain to become *visible* to specified memory accesses.

Availability, visibility, memory domains, and memory domain operations are formally defined in the Availability and Visibility section of the Memory Model chapter. Which API operations perform each of these operations is defined in Availability, Visibility, and Domain Operations.

A *memory dependency* is an execution dependency which includes availability and visibility operations such that:

- The first set of operations happens-before the availability operation.

- The availability operation happens-before the visibility operation.
- The visibility operation happens-before the second set of operations.

Once written values are made visible to a particular type of memory access, they **can** be read or written by that type of memory access. Most synchronization commands in Vulkan define a memory dependency.

The specific memory accesses that are made available and visible are defined by the *access scopes* of a memory dependency. Any type of access that is in a memory dependency's first access scope and occurs in $\textbf{ScopedOps}_1$ is made available. Any type of access that is in a memory dependency's second access scope and occurs in $\textbf{ScopedOps}_2$ has any available writes made visible to it. Any type of operation that is not in a synchronization command's access scopes will not be included in the resulting dependency.

A memory dependency enforces availability and visibility of memory accesses and execution order between two sets of operations. Adding to the description of execution dependency chains:

- Let $\textbf{MemOps}_1$ be the set of memory accesses performed by $\textbf{ScopedOps}_1$.
- Let $\textbf{MemOps}_2$ be the set of memory accesses performed by $\textbf{ScopedOps}_2$.
- Let $\textbf{AccessScope}_{1st}$ be the first access scope of the first command in the **Sync** chain.
- Let $\textbf{AccessScope}_{2nd}$ be the second access scope of the last command in the **Sync** chain.
- Let $\textbf{ScopedMemOps}_1$ be the intersection of sets $\textbf{MemOps}_1$ and $\textbf{AccessScope}_{1st}$.
- Let $\textbf{ScopedMemOps}_2$ be the intersection of sets $\textbf{MemOps}_2$ and $\textbf{AccessScope}_{2nd}$.
- Submitting $\textbf{Ops}_1$, **Sync**, and $\textbf{Ops}_2$ for execution, in that order, will result in a memory dependency **MemDep** between $\textbf{ScopedOps}_1$ and $\textbf{ScopedOps}_2$.
- Memory dependency **MemDep** guarantees that:
  - Memory writes in $\textbf{ScopedMemOps}_1$ are made available.
  - Available memory writes, including those from $\textbf{ScopedMemOps}_1$, are made visible to $\textbf{ScopedMemOps}_2$.

NOTE

Execution and memory dependencies are used to solve data hazards, i.e. to ensure that read and write operations occur in a well-defined order. Write-after-read hazards can be solved with just an execution dependency, but read-after-write and write-after-write hazards need appropriate memory dependencies to be included between them. If an application does not include dependencies to solve these hazards, it results in a data race.

7.1.1. Image Layout Transitions

Image subresources **can** be transitioned from one layout to another as part of a memory dependency (e.g. by using an image memory barrier). When a layout transition is specified in a memory dependency, it happens-after the availability operations in the memory dependency, and happens-before the visibility operations. Image layout transitions **may** perform read and write accesses on all memory bound to the image subresource range, so applications **must** ensure that all memory writes have been made available before a layout transition is executed. Available memory

is automatically made visible to a layout transition, and writes performed by a layout transition are automatically made available.

Layout transitions always apply to a particular image subresource range, and specify both an old layout and new layout. The old layout **must** either be VK_IMAGE_LAYOUT_UNDEFINED, or match the current layout of the image subresource range. If the old layout matches the current layout of the image subresource range, the transition preserves the contents of that range. If the old layout is VK_IMAGE_LAYOUT_UNDEFINED, the contents of that range **may** be discarded.

NOTE

Image layout transitions with VK_IMAGE_LAYOUT_UNDEFINED allow the implementation to discard the image subresource range, which can provide performance or power benefits. Tile-based architectures may be able to avoid flushing tile data to memory, and immediate style renderers may be able to achieve fast metadata clears to reinitialize frame buffer compression state, or similar.

If the contents of an attachment are not needed after a render pass completes, then applications **should** use VK_ATTACHMENT_STORE_OP_DONT_CARE.

As image layout transitions **may** perform read and write accesses on the memory bound to the image, if the image subresource affected by the layout transition is bound to peer memory for any device in the current device mask then the memory heap the bound memory comes from **must** support the VK_PEER_MEMORY_FEATURE_GENERIC_SRC_BIT and VK_PEER_MEMORY_FEATURE_GENERIC_DST_BIT capabilities as returned by vkGetDeviceGroupPeerMemoryFeatures.

NOTE

Applications **must** ensure that layout transitions happen-after all operations accessing the image with the old layout, and happen-before any operations that will access the image with the new layout. Layout transitions are potentially read/write operations, so not defining appropriate memory dependencies to guarantee this will result in a data race.

Image layout transitions interact with memory aliasing.

Layout transitions that are performed via image memory barriers execute in their entirety in submission order, relative to other image layout transitions submitted to the same queue, including those performed by render passes. This ordering of image layout transitions only applies if the implementation performs actual read/write operations during the transition. An application **must** not rely on ordering of image layout transitions to influence ordering of other commands.

The image layout of each image subresource of a depth/stencil image created with VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT is dependent on the last sample locations used to render to the image subresource as a depth/stencil attachment, thus when the `image` member of an image memory barrier is an image created with this flag the application **can** chain a VkSampleLocationsInfoEXT structure to the `pNext` chain of VkImageMemoryBarrier2 or VkImageMemoryBarrier to specify the sample locations to use during any image layout transition.

If the `VkSampleLocationsInfoEXT` structure does not match the sample location state last used to render to the image subresource range specified by `subresourceRange`, or if no `VkSampleLocationsInfoEXT` structure is present, then the contents of the given image subresource

range becomes undefined as if `oldLayout` would equal VK_IMAGE_LAYOUT_UNDEFINED.

7.1.2. Pipeline Stages

The work performed by an action command consists of multiple operations, which are performed as a sequence of logically independent steps known as *pipeline stages*. The exact pipeline stages executed depend on the particular command that is used, and current command buffer state when the command was recorded.

NOTE

Operations performed by synchronization commands (e.g. availability and visibility operations) are not executed by a defined pipeline stage. However other commands can still synchronize with them by using the synchronization scopes to create a dependency chain.

Execution of operations across pipeline stages **must** adhere to implicit ordering guarantees, particularly including pipeline stage order. Otherwise, execution across pipeline stages **may** overlap or execute out of order with regards to other stages, unless otherwise enforced by an execution dependency.

Several of the synchronization commands include pipeline stage parameters, restricting the synchronization scopes for that command to just those stages. This allows fine grained control over the exact execution dependencies and accesses performed by action commands. Implementations **should** use these pipeline stages to avoid unnecessary stalls or cache flushing.

Bits which **can** be set in a VkPipelineStageFlags2 mask, specifying stages of execution, are:

```
// Provided by VK_VERSION_1_3
// Flag bits for VkPipelineStageFlagBits2
typedef VkFlags64 VkPipelineStageFlagBits2;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_NONE = 0ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_TOP_OF_PIPE_BIT =
0x00000001ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT =
0x00000002ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT =
0x00000004ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT =
0x00000008ULL;
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT = 0x00000010ULL;
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT = 0x00000020ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT =
0x00000040ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT =
0x00000080ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT =
0x00000100ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT =
```

```
0x00000200ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT
= 0x00000400ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT =
0x00000800ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT =
0x00001000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_TRANSFER_BIT =
0x00001000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_BOTTOM_OF_PIPE_BIT =
0x00002000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_HOST_BIT = 0x00004000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT =
0x00008000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT =
0x00010000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_COPY_BIT = 0x100000000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_RESOLVE_BIT =
0x200000000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_BLIT_BIT = 0x400000000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_CLEAR_BIT = 0x800000000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT =
0x1000000000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT =
0x2000000000ULL;
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_PRE_RASTERIZATION_SHADERS_BIT = 0x4000000000ULL;
// Provided by VK_KHR_video_decode_queue
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR =
0x04000000ULL;
// Provided by VK_KHR_video_encode_queue
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR =
0x08000000ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_NONE_KHR = 0ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_TOP_OF_PIPE_BIT_KHR =
0x00000001ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT_KHR =
0x00000002ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT_KHR =
0x00000004ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT_KHR =
0x00000008ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT_KHR = 0x00000010ULL;
// Provided by VK_KHR_synchronization2
```

```
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT_KHR = 0x00000020ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT_KHR =
0x00000040ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT_KHR =
0x00000080ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT_KHR
= 0x00000100ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT_KHR
= 0x00000200ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT_KHR = 0x00000400ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT_KHR =
0x00000800ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT_KHR =
0x00001000ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_TRANSFER_BIT_KHR =
0x00001000ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_BOTTOM_OF_PIPE_BIT_KHR =
0x00002000ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_HOST_BIT_KHR =
0x00004000ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT_KHR =
0x00008000ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT_KHR =
0x00010000ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_COPY_BIT_KHR =
0x100000000ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_RESOLVE_BIT_KHR =
0x200000000ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_BLIT_BIT_KHR =
0x400000000ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_CLEAR_BIT_KHR =
0x800000000ULL;
// Provided by VK_KHR_synchronization2
```

```
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT_KHR =
0x1000000000ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT_KHR = 0x2000000000ULL;
// Provided by VK_KHR_synchronization2
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_PRE_RASTERIZATION_SHADERS_BIT_KHR = 0x4000000000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_transform_feedback
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT =
0x01000000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_conditional_rendering
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT = 0x00040000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_device_generated_commands
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT =
0x00020000ULL;
// Provided by VK_KHR_fragment_shading_rate with VK_KHR_synchronization2
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR = 0x00400000ULL;
// Provided by VK_KHR_acceleration_structure with VK_KHR_synchronization2
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR = 0x02000000ULL;
// Provided by VK_KHR_ray_tracing_pipeline with VK_KHR_synchronization2
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR =
0x00200000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_fragment_density_map
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT = 0x00800000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_mesh_shader
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT =
0x00080000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_mesh_shader
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT =
0x00100000ULL;
// Provided by VK_KHR_ray_tracing_maintenance1 with VK_KHR_synchronization2 or
VK_VERSION_1_3
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR = 0x10000000ULL;
// Provided by VK_EXT_opacity_micromap
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT =
0x40000000ULL;
// Provided by VK_KHR_copy_memory_indirect
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_COPY_INDIRECT_BIT_KHR =
0x400000000000ULL;
// Provided by VK_EXT_memory_decompression
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
= 0x200000000000ULL;
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to VkPipelineStageFlagBits2
typedef VkPipelineStageFlagBits2 VkPipelineStageFlagBits2KHR;
```

- VK_PIPELINE_STAGE_2_NONE specifies no stages of execution.
- VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT specifies the stage of the pipeline where indirect command parameters are consumed. This stage also includes reading commands written by vkCmdPreprocessGeneratedCommandsEXT.
- VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT specifies the task shader stage.
- VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT specifies the mesh shader stage.
- VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT specifies the stage of the pipeline where index buffers are consumed.
- VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT specifies the stage of the pipeline where vertex buffers are consumed.
- VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT is equivalent to the logical OR of:
  - VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT
  - VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT
- VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT specifies the vertex shader stage.
- VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT specifies the tessellation control shader stage.
- VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT specifies the tessellation evaluation shader stage.
- VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT specifies the geometry shader stage.
- VK_PIPELINE_STAGE_2_PRE_RASTERIZATION_SHADERS_BIT is equivalent to specifying all supported pre-rasterization shader stages:
  - VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT
  - VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT
  - VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT
  - VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT
  - VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT
  - VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT
- VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT specifies the fragment shader stage.
- VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT specifies the stage of the pipeline where early fragment tests (depth and stencil tests before fragment shading) are performed. This stage also includes render pass load operations for framebuffer attachments with a depth/stencil format.
- VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT specifies the stage of the pipeline where late fragment tests (depth and stencil tests after fragment shading) are performed. This stage

also includes render pass store operations for framebuffer attachments with a depth/stencil format.

- VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT specifies the stage of the pipeline where final color values are output from the pipeline. This stage includes blending, logic operations, render pass load and store operations for color attachments, render pass multisample resolve operations, and vkCmdClearAttachments.
- VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT specifies the compute shader stage.
- VK_PIPELINE_STAGE_2_HOST_BIT specifies a pseudo-stage indicating execution on the host of reads/writes of device memory. This stage is not invoked by any commands recorded in a command buffer.
- VK_PIPELINE_STAGE_2_COPY_BIT specifies the execution of all copy commands, including vkCmdCopyQueryPoolResults, with the exception of vkCmdBlitImage2 and vkCmdBlitImage.
- VK_PIPELINE_STAGE_2_BLIT_BIT specifies the execution of vkCmdBlitImage2 and vkCmdBlitImage.
- VK_PIPELINE_STAGE_2_RESOLVE_BIT specifies the execution of vkCmdResolveImage2 and vkCmdResolveImage.
- VK_PIPELINE_STAGE_2_CLEAR_BIT specifies the execution of clear commands, with the exception of vkCmdClearAttachments.
- VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT is equivalent to specifying all of:
  - VK_PIPELINE_STAGE_2_COPY_BIT
  - VK_PIPELINE_STAGE_2_BLIT_BIT
  - VK_PIPELINE_STAGE_2_RESOLVE_BIT
  - VK_PIPELINE_STAGE_2_CLEAR_BIT
  - VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR
- VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR specifies the execution of the ray tracing shader stages.
- VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR specifies the execution of acceleration structure commands or acceleration structure copy commands.
- VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR specifies the execution of acceleration structure copy commands.
- VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT specifies the execution of all graphics pipeline stages, and is equivalent to the logical OR of:
  - VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT
  - VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT
  - VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT
  - VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT
  - VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT
  - VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT

    - VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT
    - VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT
    - VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT
    - VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT
    - VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT
    - VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT
    - VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT
    - VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
    - VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
    - VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT

- VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT specifies all operations performed by all commands supported on the queue it is used with.
- VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT specifies the stage of the pipeline where the predicate of conditional rendering is consumed.
- VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT specifies the stage of the pipeline where vertex attribute output values are written to the transform feedback buffers.
- VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT specifies the stage of the pipeline where device-side generation of commands via vkCmdPreprocessGeneratedCommandsEXT is handled.
- VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR specifies the stage of the pipeline where the fragment shading rate attachment is read to determine the fragment shading rate for portions of a rasterized primitive.
- VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT specifies the stage of the pipeline where the fragment density map is read to generate the fragment areas.
- VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR specifies the execution of video decode operations.
- VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR specifies the execution of video encode operations.
- VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT specifies the execution of micromap commands for VkMicromapEXT objects.
- VK_PIPELINE_STAGE_2_COPY_INDIRECT_BIT_KHR specifies the stage of the pipeline where indirect copy commands (vkCmdCopyMemoryIndirect* and vkCmdCopyMemoryToImageIndirect*) parameters are consumed.
- VK_PIPELINE_STAGE_2_TOP_OF_PIPE_BIT is equivalent to VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT with VkAccessFlags2 set to `0` when specified in the second synchronization scope, but equivalent to VK_PIPELINE_STAGE_2_NONE in the first scope.
- VK_PIPELINE_STAGE_2_BOTTOM_OF_PIPE_BIT is equivalent to VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT with VkAccessFlags2 set to `0` when specified in the first synchronization scope, but equivalent to VK_PIPELINE_STAGE_2_NONE in the second

scope.

NOTE	The TOP and BOTTOM pipeline stages are legacy, and applications should prefer VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT and VK_PIPELINE_STAGE_2_NONE.

NOTE	The VkPipelineStageFlags2 bitmask goes beyond the 31 individual bit flags allowable within a C99 enum, which is how VkPipelineStageFlagBits is defined. The first 31 values are common to both, and are interchangeable.

VkPipelineStageFlags2 is a bitmask type for setting a mask of zero or more VkPipelineStageFlagBits2 flags:

```
// Provided by VK_VERSION_1_3
typedef VkFlags64 VkPipelineStageFlags2;
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to VkPipelineStageFlags2
typedef VkPipelineStageFlags2 VkPipelineStageFlags2KHR;
```

Bits which **can** be set in a VkPipelineStageFlags mask, specifying stages of execution, are:

WARNING	This functionality is superseded by VkPipelineStageFlagBits2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef enum VkPipelineStageFlagBits {
    VK_PIPELINE_STAGE_TOP_OF_PIPE_BIT = 0x00000001,
    VK_PIPELINE_STAGE_DRAW_INDIRECT_BIT = 0x00000002,
    VK_PIPELINE_STAGE_VERTEX_INPUT_BIT = 0x00000004,
    VK_PIPELINE_STAGE_VERTEX_SHADER_BIT = 0x00000008,
    VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT = 0x00000010,
    VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT = 0x00000020,
    VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT = 0x00000040,
    VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT = 0x00000080,
    VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT = 0x00000100,
    VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT = 0x00000200,
    VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT = 0x00000400,
    VK_PIPELINE_STAGE_COMPUTE_SHADER_BIT = 0x00000800,
    VK_PIPELINE_STAGE_TRANSFER_BIT = 0x00001000,
    VK_PIPELINE_STAGE_BOTTOM_OF_PIPE_BIT = 0x00002000,
    VK_PIPELINE_STAGE_HOST_BIT = 0x00004000,
    VK_PIPELINE_STAGE_ALL_GRAPHICS_BIT = 0x00008000,
    VK_PIPELINE_STAGE_ALL_COMMANDS_BIT = 0x00010000,
  // Provided by VK_VERSION_1_3
    VK_PIPELINE_STAGE_NONE = 0,
  // Provided by VK_EXT_transform_feedback
    VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT = 0x01000000,
  // Provided by VK_EXT_conditional_rendering
    VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT = 0x00040000,
  // Provided by VK_KHR_acceleration_structure
    VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR = 0x02000000,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR = 0x00200000,
  // Provided by VK_EXT_fragment_density_map
    VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT = 0x00800000,
  // Provided by VK_KHR_fragment_shading_rate
    VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR = 0x00400000,
  // Provided by VK_EXT_mesh_shader
    VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT = 0x00080000,
  // Provided by VK_EXT_mesh_shader
    VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT = 0x00100000,
  // Provided by VK_EXT_device_generated_commands
    VK_PIPELINE_STAGE_COMMAND_PREPROCESS_BIT_EXT = 0x00020000,
  // Provided by VK_KHR_synchronization2
    VK_PIPELINE_STAGE_NONE_KHR = VK_PIPELINE_STAGE_NONE,
} VkPipelineStageFlagBits;
```

These values all have the same meaning as the equivalently named values for VkPipelineStageFlags2.

- VK_PIPELINE_STAGE_NONE specifies no stages of execution.
- VK_PIPELINE_STAGE_DRAW_INDIRECT_BIT specifies the stage of the pipeline where VkDrawIndirect* / VkDispatchIndirect* / VkTraceRaysIndirect* data structures are consumed. This

stage also includes reading commands written by vkCmdPreprocessGeneratedCommandsEXT.

- VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT specifies the task shader stage.
- VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT specifies the mesh shader stage.
- VK_PIPELINE_STAGE_VERTEX_INPUT_BIT specifies the stage of the pipeline where vertex and index buffers are consumed.
- VK_PIPELINE_STAGE_VERTEX_SHADER_BIT specifies the vertex shader stage.
- VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT specifies the tessellation control shader stage.
- VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT specifies the tessellation evaluation shader stage.
- VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT specifies the geometry shader stage.
- VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT specifies the fragment shader stage.
- VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT specifies the stage of the pipeline where early fragment tests (depth and stencil tests before fragment shading) are performed. This stage also includes render pass load operations for framebuffer attachments with a depth/stencil format.
- VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT specifies the stage of the pipeline where late fragment tests (depth and stencil tests after fragment shading) are performed. This stage also includes render pass store operations for framebuffer attachments with a depth/stencil format.
- VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT specifies the stage of the pipeline after blending where the final color values are output from the pipeline. This stage includes blending, logic operations, render pass load and store operations for color attachments, render pass multisample resolve operations, and vkCmdClearAttachments.
- VK_PIPELINE_STAGE_COMPUTE_SHADER_BIT specifies the execution of a compute shader.
- VK_PIPELINE_STAGE_TRANSFER_BIT specifies the following commands:
  - All copy commands, including vkCmdCopyQueryPoolResults
  - vkCmdBlitImage2 and vkCmdBlitImage
  - vkCmdResolveImage2 and vkCmdResolveImage
  - All clear commands, with the exception of vkCmdClearAttachments
- VK_PIPELINE_STAGE_HOST_BIT specifies a pseudo-stage indicating execution on the host of reads/writes of device memory. This stage is not invoked by any commands recorded in a command buffer.
- VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR specifies the execution of vkCmdBuildAccelerationStructuresKHR, vkCmdBuildAccelerationStructuresIndirectKHR, vkCmdCopyAccelerationStructureKHR, vkCmdCopyAccelerationStructureToMemoryKHR, vkCmdCopyMemoryToAccelerationStructureKHR, and vkCmdWriteAccelerationStructuresPropertiesKHR.
- VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR specifies the execution of the ray tracing shader stages, via vkCmdTraceRaysKHR, or vkCmdTraceRaysIndirectKHR
- VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT specifies the execution of

decompression commands with vkCmdDecompressMemoryEXT and vkCmdDecompressMemoryIndirectCountEXT.

- VK_PIPELINE_STAGE_ALL_GRAPHICS_BIT specifies the execution of all graphics pipeline stages, and is equivalent to the logical OR of:
  - VK_PIPELINE_STAGE_DRAW_INDIRECT_BIT
  - VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT
  - VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT
  - VK_PIPELINE_STAGE_VERTEX_INPUT_BIT
  - VK_PIPELINE_STAGE_VERTEX_SHADER_BIT
  - VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT
  - VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT
  - VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT
  - VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT
  - VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT
  - VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT
  - VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT
  - VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT
  - VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT
  - VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
  - VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VK_PIPELINE_STAGE_ALL_COMMANDS_BIT specifies all operations performed by all commands supported on the queue it is used with.
- VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT specifies the stage of the pipeline where the predicate of conditional rendering is consumed.
- VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT specifies the stage of the pipeline where vertex attribute output values are written to the transform feedback buffers.
- VK_PIPELINE_STAGE_COMMAND_PREPROCESS_BIT_EXT specifies the stage of the pipeline where device-side preprocessing for generated commands via vkCmdPreprocessGeneratedCommandsEXT is handled.
- VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR specifies the stage of the pipeline where the fragment shading rate attachment is read to determine the fragment shading rate for portions of a rasterized primitive.
- VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT specifies the stage of the pipeline where the fragment density map is read to generate the fragment areas.
- VK_PIPELINE_STAGE_TOP_OF_PIPE_BIT is equivalent to VK_PIPELINE_STAGE_ALL_COMMANDS_BIT with VkAccessFlags set to `0` when specified in the second synchronization scope, but specifies no stage of execution when specified in the first scope.

- VK_PIPELINE_STAGE_BOTTOM_OF_PIPE_BIT is equivalent to VK_PIPELINE_STAGE_ALL_COMMANDS_BIT with VkAccessFlags set to `0` when specified in the first synchronization scope, but specifies no stage of execution when specified in the second scope.

WARNING	This functionality is superseded by VkPipelineStageFlags2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineStageFlags;
```

VkPipelineStageFlags is a bitmask type for setting a mask of zero or more VkPipelineStageFlagBits.

If a synchronization command includes a source stage mask, its first synchronization scope only includes execution of the pipeline stages specified in that mask and any logically earlier stages. Its first access scope only includes memory accesses performed by pipeline stages explicitly specified in the source stage mask.

If a synchronization command includes a destination stage mask, its second synchronization scope only includes execution of the pipeline stages specified in that mask and any logically later stages. Its second access scope only includes memory accesses performed by pipeline stages explicitly specified in the destination stage mask.

NOTE	Note that access scopes do not interact with the logically earlier or later stages for either scope - only the stages the application specifies are considered part of each access scope.

Certain pipeline stages are only available on queues that support a particular set of operations. The following table lists, for each pipeline stage flag, which queue capability flag **must** be supported by the queue. When multiple flags are enumerated in the second column of the table, it means that the pipeline stage is supported on the queue if it supports any of the listed capability flags. For further details on queue capabilities see Physical Device Enumeration and Queues.

Table 3. Supported Pipeline Stage Flags

Pipeline stage flag	**Required queue capability flag**
VK_PIPELINE_STAGE_2_NONE	None required
VK_PIPELINE_STAGE_2_TOP_OF_PIPE_BIT	None required
VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT

Pipeline stage flag	Required queue capability flag
VK_PIPELINE_STAGE_2_COPY_INDIRECT_BIT_KHR	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT	VK_QUEUE_COMPUTE_BIT
VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_BOTTOM_OF_PIPE_BIT	None required
VK_PIPELINE_STAGE_2_HOST_BIT	None required
VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT	None required
VK_PIPELINE_STAGE_2_COPY_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_RESOLVE_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT

Pipeline stage flag	Required queue capability flag
VK_PIPELINE_STAGE_2_BLIT_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_CLEAR_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_PRE_RASTERIZATION_SHADERS_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR	VK_QUEUE_VIDEO_DECODE_BIT_KHR
VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR	VK_QUEUE_VIDEO_ENCODE_BIT_KHR
VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT
VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT
VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT
VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR	VK_QUEUE_COMPUTE_BIT
VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR	VK_QUEUE_COMPUTE_BIT
VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT	VK_QUEUE_GRAPHICS_BIT

Pipeline stage flag	Required queue capability flag
VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT	VK_QUEUE_COMPUTE_BIT

Pipeline stages that execute as a result of a command logically complete execution in a specific order, such that completion of a logically later pipeline stage **must** not happen-before completion of a logically earlier stage. This means that including any stage in the source stage mask for a particular synchronization command also implies that any logically earlier stages are included in $\textbf{Scope}_{1st}$ for that command.

Similarly, initiation of a logically earlier pipeline stage **must** not happen-after initiation of a logically later pipeline stage. Including any given stage in the destination stage mask for a particular synchronization command also implies that any logically later stages are included in $\textbf{Scope}_{2nd}$ for that command.

NOTE

Implementations **may** not support synchronization at every pipeline stage for every synchronization operation. If a pipeline stage that an implementation does not support synchronization for appears in a source stage mask, it **may** substitute any logically later stage in its place for the first synchronization scope. If a pipeline stage that an implementation does not support synchronization for appears in a destination stage mask, it **may** substitute any logically earlier stage in its place for the second synchronization scope.

For example, if an implementation is unable to signal an event immediately after vertex shader execution is complete, it **may** instead signal the event after color attachment output has completed.

If an implementation makes such a substitution, it **must** not affect the semantics of execution or memory dependencies or image and buffer memory barriers.

Graphics pipelines are executable on queues supporting VK_QUEUE_GRAPHICS_BIT. Stages executed by graphics pipelines **can** only be specified in commands recorded for queues supporting VK_QUEUE_GRAPHICS_BIT.

The graphics primitive pipeline executes the following stages, with the logical ordering of the stages matching the order specified here:

- VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT
- VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT
- VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT
- VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT

- VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT
- VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT
- VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT
- VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
- VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT
- VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT
- VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT
- VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT

The graphics mesh pipeline executes the following stages, with the logical ordering of the stages matching the order specified here:

- VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT
- VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT
- VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT
- VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT
- VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT
- VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT
- VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT

For the compute pipeline, the following stages occur in this order:

- VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT
- VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT

For graphics pipeline commands executing in a render pass with a fragment density map attachment, the following pipeline stage where the fragment density map read happens has no particular order relative to the other stages, except that it is logically earlier than VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT:

- VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT

The conditional rendering stage is formally part of both the graphics, and the compute pipeline. The pipeline stage where the predicate read happens has unspecified order relative to other stages of these pipelines:

- VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT

For the transfer pipeline, the following stages occur in this order:

- VK_PIPELINE_STAGE_2_COPY_INDIRECT_BIT_KHR
- VK_PIPELINE_STAGE_2_TRANSFER_BIT

For host operations, only one pipeline stage occurs, so no order is guaranteed:

- VK_PIPELINE_STAGE_2_HOST_BIT

For the command preprocessing pipeline, the following stages occur in this order:

- VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT

For the decompression pipeline, the following stages occur in this order:

- VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT

For acceleration structure build operations, only one pipeline stage occurs, so no order is guaranteed:

- VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR

For acceleration structure copy operations, only one pipeline stage occurs, so no order is guaranteed:

- VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR

For opacity micromap build operations, only one pipeline stage occurs, so no order is guaranteed:

- VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT

For the ray tracing pipeline, the following stages occur in this order:

- VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT
- VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR

For the video decode pipeline, the following stages occur in this order:

- VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR

For the video encode pipeline, the following stages occur in this order:

- VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR

7.1.3. Access Types

Memory in Vulkan **can** be accessed from within shader invocations and via some fixed-function stages of the pipeline. The *access type* is a function of the descriptor type used, or how a fixed-function stage accesses memory.

Some synchronization commands take sets of access types as parameters to define the access scopes of a memory dependency. If a synchronization command includes a *source access mask*, its first access scope only includes accesses via the access types specified in that mask. Similarly, if a

synchronization command includes a *destination access mask*, its second access scope only includes accesses via the access types specified in that mask.

Bits which **can** be set in the `srcAccessMask` and `dstAccessMask` members of VkMemoryBarrier2, VkImageMemoryBarrier2, and VkBufferMemoryBarrier2, specifying access behavior, are:

```
// Provided by VK_VERSION_1_3
// Flag bits for VkAccessFlagBits2
typedef VkFlags64 VkAccessFlagBits2;
static const VkAccessFlagBits2 VK_ACCESS_2_NONE = 0ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT = 0x00000001ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_INDEX_READ_BIT = 0x00000002ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT = 0x00000004ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_UNIFORM_READ_BIT = 0x00000008ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT = 0x00000010ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_READ_BIT = 0x00000020ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_WRITE_BIT = 0x00000040ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT = 0x00000080ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT = 0x00000100ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT =
0x00000200ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT =
0x00000400ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_TRANSFER_READ_BIT = 0x00000800ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_TRANSFER_WRITE_BIT = 0x00001000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_HOST_READ_BIT = 0x00002000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_HOST_WRITE_BIT = 0x00004000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_MEMORY_READ_BIT = 0x00008000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_MEMORY_WRITE_BIT = 0x00010000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_SAMPLED_READ_BIT = 0x100000000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_STORAGE_READ_BIT = 0x200000000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT = 0x400000000ULL;
// Provided by VK_KHR_video_decode_queue
static const VkAccessFlagBits2 VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR = 0x800000000ULL;
// Provided by VK_KHR_video_decode_queue
static const VkAccessFlagBits2 VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR =
0x1000000000ULL;
// Provided by VK_EXT_descriptor_heap
static const VkAccessFlagBits2 VK_ACCESS_2_SAMPLER_HEAP_READ_BIT_EXT =
0x20000000000000ULL;
// Provided by VK_EXT_descriptor_heap
static const VkAccessFlagBits2 VK_ACCESS_2_RESOURCE_HEAP_READ_BIT_EXT =
0x40000000000000ULL;
// Provided by VK_KHR_video_encode_queue
static const VkAccessFlagBits2 VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR =
0x2000000000ULL;
// Provided by VK_KHR_video_encode_queue
static const VkAccessFlagBits2 VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR =
0x4000000000ULL;
// Provided by VK_KHR_synchronization2
```

```
static const VkAccessFlagBits2 VK_ACCESS_2_NONE_KHR = 0ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT_KHR =
0x00000001ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_INDEX_READ_BIT_KHR = 0x00000002ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT_KHR =
0x00000004ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_UNIFORM_READ_BIT_KHR = 0x00000008ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT_KHR =
0x00000010ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_READ_BIT_KHR = 0x00000020ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_WRITE_BIT_KHR = 0x00000040ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT_KHR =
0x00000080ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT_KHR =
0x00000100ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT_KHR =
0x00000200ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT_KHR =
0x00000400ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_TRANSFER_READ_BIT_KHR = 0x00000800ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_TRANSFER_WRITE_BIT_KHR = 0x00001000ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_HOST_READ_BIT_KHR = 0x00002000ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_HOST_WRITE_BIT_KHR = 0x00004000ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_MEMORY_READ_BIT_KHR = 0x00008000ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_MEMORY_WRITE_BIT_KHR = 0x00010000ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_SAMPLED_READ_BIT_KHR =
0x100000000ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_STORAGE_READ_BIT_KHR =
0x200000000ULL;
// Provided by VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT_KHR =
0x400000000ULL;
```

```
// Provided by VK_KHR_synchronization2 with VK_EXT_transform_feedback
static const VkAccessFlagBits2 VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT =
0x02000000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_transform_feedback
static const VkAccessFlagBits2 VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT =
0x04000000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_transform_feedback
static const VkAccessFlagBits2 VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT =
0x08000000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_conditional_rendering
static const VkAccessFlagBits2 VK_ACCESS_2_CONDITIONAL_RENDERING_READ_BIT_EXT =
0x00100000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_device_generated_commands
static const VkAccessFlagBits2 VK_ACCESS_2_COMMAND_PREPROCESS_READ_BIT_EXT =
0x00020000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_device_generated_commands
static const VkAccessFlagBits2 VK_ACCESS_2_COMMAND_PREPROCESS_WRITE_BIT_EXT =
0x00040000ULL;
// Provided by VK_KHR_fragment_shading_rate with VK_KHR_synchronization2
static const VkAccessFlagBits2
VK_ACCESS_2_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR = 0x00800000ULL;
// Provided by VK_KHR_acceleration_structure with VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR =
0x00200000ULL;
// Provided by VK_KHR_acceleration_structure with VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR =
0x00400000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_fragment_density_map
static const VkAccessFlagBits2 VK_ACCESS_2_FRAGMENT_DENSITY_MAP_READ_BIT_EXT =
0x01000000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_blend_operation_advanced
static const VkAccessFlagBits2 VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT =
0x00080000ULL;
// Provided by VK_EXT_descriptor_buffer
static const VkAccessFlagBits2 VK_ACCESS_2_DESCRIPTOR_BUFFER_READ_BIT_EXT =
0x20000000000ULL;
// Provided by VK_KHR_ray_tracing_maintenance1 with (VK_KHR_synchronization2 or
VK_VERSION_1_3) and VK_KHR_ray_tracing_pipeline
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR =
0x10000000000ULL;
// Provided by VK_EXT_opacity_micromap
static const VkAccessFlagBits2 VK_ACCESS_2_MICROMAP_READ_BIT_EXT = 0x100000000000ULL;
// Provided by VK_EXT_opacity_micromap
static const VkAccessFlagBits2 VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT = 0x200000000000ULL;
// Provided by VK_EXT_memory_decompression
static const VkAccessFlagBits2 VK_ACCESS_2_MEMORY_DECOMPRESSION_READ_BIT_EXT =
0x80000000000000ULL;
// Provided by VK_EXT_memory_decompression
static const VkAccessFlagBits2 VK_ACCESS_2_MEMORY_DECOMPRESSION_WRITE_BIT_EXT =
0x100000000000000ULL;
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to VkAccessFlagBits2
typedef VkAccessFlagBits2 VkAccessFlagBits2KHR;
```

- VK_ACCESS_2_NONE specifies no accesses.
- VK_ACCESS_2_MEMORY_READ_BIT specifies all read accesses. It is always valid in any access mask, and is treated as equivalent to setting all READ access flags that are valid where it is used.
- VK_ACCESS_2_MEMORY_WRITE_BIT specifies all write accesses. It is always valid in any access mask, and is treated as equivalent to setting all WRITE access flags that are valid where it is used.
- VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT specifies read access to command data read from indirect buffers as part of an indirect build, trace, drawing or dispatch command. Such access occurs in the VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT pipeline stage. It also specifies read access to command data read from indirect buffers as part of a copy command with access occurring in the VK_PIPELINE_STAGE_2_COPY_INDIRECT_BIT_KHR pipeline stage.
- VK_ACCESS_2_INDEX_READ_BIT specifies read access to an index buffer as part of an indexed drawing command, bound by vkCmdBindIndexBuffer2 and vkCmdBindIndexBuffer. Such access occurs in the VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT pipeline stage.
- VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT specifies read access to a vertex buffer as part of a drawing command, bound by vkCmdBindVertexBuffers. Such access occurs in the VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT pipeline stage.
- VK_ACCESS_2_UNIFORM_READ_BIT specifies read access to a uniform buffer in any shader pipeline stage.
- VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT specifies read access to an input attachment within a render pass during fragment shading. Such access occurs in the VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT pipeline stage.
- VK_ACCESS_2_SHADER_SAMPLED_READ_BIT specifies read access to a uniform texel buffer or sampled image in any shader pipeline stage.
- VK_ACCESS_2_SHADER_STORAGE_READ_BIT specifies read access to a storage buffer, physical storage buffer, storage texel buffer, or storage image in any shader pipeline stage.
- VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR specifies read access to a shader binding table in any shader pipeline stage.
- VK_ACCESS_2_SHADER_READ_BIT is equivalent to the logical OR of:
  - VK_ACCESS_2_SHADER_SAMPLED_READ_BIT
  - VK_ACCESS_2_SHADER_STORAGE_READ_BIT
  - VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR
- VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT specifies write access to a storage buffer, physical storage buffer, storage texel buffer, or storage image in any shader pipeline stage.
- VK_ACCESS_2_SHADER_WRITE_BIT is equivalent to VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT.
- VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT specifies read access to a color attachment,

such as via blending (other than advanced blend operations), logic operations or certain render pass load operations in the VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT pipeline stage or via fragment shader tile image reads in the VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT pipeline stage.

- VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT specifies write access to a color attachment during a render pass or via certain render pass load, store, and multisample resolve operations. This includes multisample resolve operations for depth/stencil resolve attachments. Such access occurs in the VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT pipeline stage.
- VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT specifies read access to a depth/stencil attachment, via depth or stencil operations or certain render pass load operations in the VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT or VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT pipeline stages or via fragment shader tile image reads in the VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT pipeline stage.
- VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT specifies write access to a depth/stencil attachment, via depth or stencil operations or certain render pass load and store operations. Such access occurs in the VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT or VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT pipeline stages.
- VK_ACCESS_2_TRANSFER_READ_BIT specifies read access to an image or buffer in a copy operation. Such access occurs in the VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, or VK_PIPELINE_STAGE_2_RESOLVE_BIT pipeline stages.
- VK_ACCESS_2_TRANSFER_WRITE_BIT specifies write access to an image or buffer in a clear or copy operation. Such access occurs in the VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_CLEAR_BIT, or VK_PIPELINE_STAGE_2_RESOLVE_BIT pipeline stages.
- VK_ACCESS_2_HOST_READ_BIT specifies read access by a host operation. Accesses of this type are not performed through a resource, but directly on memory. Such access occurs in the VK_PIPELINE_STAGE_2_HOST_BIT pipeline stage.
- VK_ACCESS_2_HOST_WRITE_BIT specifies write access by a host operation. Accesses of this type are not performed through a resource, but directly on memory. Such access occurs in the VK_PIPELINE_STAGE_2_HOST_BIT pipeline stage.
- VK_ACCESS_2_CONDITIONAL_RENDERING_READ_BIT_EXT specifies read access to a predicate as part of conditional rendering. Such access occurs in the VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT pipeline stage.
- VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT specifies write access to a transform feedback buffer made when transform feedback is active. Such access occurs in the VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT pipeline stage.
- VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT specifies read access to a transform feedback counter buffer which is read when vkCmdBeginTransformFeedbackEXT executes. Such access occurs in the VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT pipeline stage.
- VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT specifies write access to a transform feedback counter buffer which is written when vkCmdEndTransformFeedbackEXT executes. Such access occurs in the VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT

pipeline stage.

- VK_ACCESS_2_COMMAND_PREPROCESS_READ_BIT_EXT specifies reads from buffer inputs to vkCmdPreprocessGeneratedCommandsEXT. Such access occurs in the VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT pipeline stage.
- VK_ACCESS_2_COMMAND_PREPROCESS_WRITE_BIT_EXT specifies writes to the target command buffer preprocess outputs. Such access occurs in the VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT pipeline stage.
- VK_ACCESS_2_MEMORY_DECOMPRESSION_READ_BIT_EXT specifies read access to memory in decompression commands vkCmdDecompressMemoryEXT and vkCmdDecompressMemoryIndirectCountEXT. Such access occurs in the VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT pipeline stage.
- VK_ACCESS_2_MEMORY_DECOMPRESSION_WRITE_BIT_EXT specifies write access to memory in decompression commands vkCmdDecompressMemoryEXT and vkCmdDecompressMemoryIndirectCountEXT. Such access occurs in the VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT pipeline stage.
- VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT specifies read access to color attachments, including advanced blend operations. Such access occurs in the VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT pipeline stage.
- VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR specifies read access to an acceleration structure as part of a trace, build, or copy command, or to an acceleration structure scratch buffer as part of a build command. Such access occurs in the VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR pipeline stage or VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR pipeline stage.
- VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR specifies write access to an acceleration structure or acceleration structure scratch buffer as part of a build or copy command. Such access occurs in the VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR pipeline stage.
- VK_ACCESS_2_FRAGMENT_DENSITY_MAP_READ_BIT_EXT specifies read access to a fragment density map attachment during dynamic fragment density map operations. Such access occurs in the VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT pipeline stage.
- VK_ACCESS_2_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR specifies read access to a fragment shading rate attachment during rasterization. Such access occurs in the VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR pipeline stage.
- VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR specifies read access to an image or buffer resource in a video decode operation. Such access occurs in the VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR pipeline stage.
- VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR specifies write access to an image or buffer resource in a video decode operation. Such access occurs in the VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR pipeline stage.
- VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR specifies read access to an image or buffer resource in a video encode operation. Such access occurs in the VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR pipeline stage.
- VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR specifies write access to an image or buffer

resource in a video encode operation. Such access occurs in the VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR pipeline stage.

- VK_ACCESS_2_DESCRIPTOR_BUFFER_READ_BIT_EXT specifies read access to a descriptor buffer in any shader pipeline stage.
- VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT specifies write access to a micromap object. Such access occurs in the VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT pipeline stage.
- VK_ACCESS_2_MICROMAP_READ_BIT_EXT specifies read access to a micromap object. Such access occurs in the VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT and VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR pipeline stages.
- VK_ACCESS_2_SAMPLER_HEAP_READ_BIT_EXT specifies read access to a sampler heap in any shader pipeline stage.
- VK_ACCESS_2_RESOURCE_HEAP_READ_BIT_EXT specifies read access to a resource heap in any shader pipeline stage.

Certain access types are only performed by a subset of pipeline stages, as described in more detail for VkAccessFlagBits. The Supported Access Types table lists, for each access flag, which pipeline stages **can** perform that type of access.

NOTE	In situations where an application wishes to select all access types for a given set of pipeline stages, VK_ACCESS_2_MEMORY_READ_BIT or VK_ACCESS_2_MEMORY_WRITE_BIT can be used. This is particularly useful when specifying stages that only have a single access type.

NOTE	The VkAccessFlags2 bitmask goes beyond the 31 individual bit flags allowable within a C99 enum, which is how VkAccessFlagBits is defined. The first 31 values are common to both, and are interchangeable.

VkAccessFlags2 is a bitmask type for setting a mask of zero or more VkAccessFlagBits2:

```
// Provided by VK_VERSION_1_3
typedef VkFlags64 VkAccessFlags2;
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to VkAccessFlags2
typedef VkAccessFlags2 VkAccessFlags2KHR;
```

An application **can** link a VkMemoryBarrierAccessFlags3KHR structure in the pNext chain of VkMemoryRangeBarrierKHR, VkMemoryBarrier2, VkBufferMemoryBarrier2, or VkImageMemoryBarrier2 to provide additional access flags beyond those available in VkAccessFlagBits2.

The VkMemoryBarrierAccessFlags3KHR structure is defined as:

```
// Provided by VK_KHR_maintenance8
typedef struct VkMemoryBarrierAccessFlags3KHR {
    VkStructureType      sType;
    const void*          pNext;
    VkAccessFlags3KHR    srcAccessMask3;
    VkAccessFlags3KHR    dstAccessMask3;
} VkMemoryBarrierAccessFlags3KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcAccessMask3` is a VkAccessFlags3KHR mask of access flags to be included in the first access scope.
- `dstAccessMask3` is a VkAccessFlags3KHR mask of access flags to be included in the second access scope.

Valid Usage (Implicit)

- VUID-VkMemoryBarrierAccessFlags3KHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_BARRIER_ACCESS_FLAGS_3_KHR
- VUID-VkMemoryBarrierAccessFlags3KHR-srcAccessMask3-parameter
  If `srcAccessMask3` is not `0`, `srcAccessMask3` **must** be a valid combination of VkAccessFlagBits3KHR values
- VUID-VkMemoryBarrierAccessFlags3KHR-dstAccessMask3-parameter
  If `dstAccessMask3` is not `0`, `dstAccessMask3` **must** be a valid combination of VkAccessFlagBits3KHR values

Structure Chaining

Extends the structures

- VkBufferMemoryBarrier2
- VkImageMemoryBarrier2
- VkMemoryBarrier2
- VkMemoryRangeBarrierKHR
- VkSubpassDependency2

Bits which **can** be set in the `srcAccessMask3` and `dstAccessMask3` members of VkMemoryBarrierAccessFlags3KHR, specifying access behavior, are:

```
// Provided by VK_KHR_maintenance8
// Flag bits for VkAccessFlagBits3KHR
typedef VkFlags64 VkAccessFlagBits3KHR;
static const VkAccessFlagBits3KHR VK_ACCESS_3_NONE_KHR = 0ULL;
```

- VK_ACCESS_3_NONE_KHR specifies no additional accesses.

VkAccessFlags3KHR is a bitmask type for setting a mask of zero or more VkAccessFlagBits3KHR:

```
// Provided by VK_KHR_maintenance8
typedef VkFlags64 VkAccessFlags3KHR;
```

When a VkMemoryBarrierAccessFlags3KHR structure is linked in the pNext field of VkMemoryBarrier2, VkBufferMemoryBarrier2, or VkImageMemoryBarrier2, the flags specified in the srcAccessMask3 and dstAccessMask3 fields are considered in addition to the flags in the srcAccessMask and dstAccessMask fields, respectively, to allow up to 128 total access types to be specified for the first or second access scope.

NOTE

When VkAccessFlagBits3KHR and VkAccessFlagBits2 are used together, the two sets of 64 flags bits are combined together into 128 flag bits (effectively OR'ing them together). This is different from VkAccessFlagBits2 and VkAccessFlagBits, where the 64 bit VkAccessFlagBits2 extends and *replaces* the 32 bit VkAccessFlagBits.

Bits which **can** be set in the srcAccessMask and dstAccessMask members of VkSubpassDependency, VkSubpassDependency2, VkMemoryBarrier, VkBufferMemoryBarrier, and VkImageMemoryBarrier, specifying access behavior, are:

WARNING

This functionality is superseded by VkAccessFlagBits2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef enum VkAccessFlagBits {
    VK_ACCESS_INDIRECT_COMMAND_READ_BIT = 0x00000001,
    VK_ACCESS_INDEX_READ_BIT = 0x00000002,
    VK_ACCESS_VERTEX_ATTRIBUTE_READ_BIT = 0x00000004,
    VK_ACCESS_UNIFORM_READ_BIT = 0x00000008,
    VK_ACCESS_INPUT_ATTACHMENT_READ_BIT = 0x00000010,
    VK_ACCESS_SHADER_READ_BIT = 0x00000020,
    VK_ACCESS_SHADER_WRITE_BIT = 0x00000040,
    VK_ACCESS_COLOR_ATTACHMENT_READ_BIT = 0x00000080,
    VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT = 0x00000100,
    VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT = 0x00000200,
    VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT = 0x00000400,
    VK_ACCESS_TRANSFER_READ_BIT = 0x00000800,
    VK_ACCESS_TRANSFER_WRITE_BIT = 0x00001000,
    VK_ACCESS_HOST_READ_BIT = 0x00002000,
    VK_ACCESS_HOST_WRITE_BIT = 0x00004000,
    VK_ACCESS_MEMORY_READ_BIT = 0x00008000,
    VK_ACCESS_MEMORY_WRITE_BIT = 0x00010000,
  // Provided by VK_VERSION_1_3
    VK_ACCESS_NONE = 0,
  // Provided by VK_EXT_transform_feedback
    VK_ACCESS_TRANSFORM_FEEDBACK_WRITE_BIT_EXT = 0x02000000,
  // Provided by VK_EXT_transform_feedback
    VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT = 0x04000000,
  // Provided by VK_EXT_transform_feedback
    VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT = 0x08000000,
  // Provided by VK_EXT_conditional_rendering
    VK_ACCESS_CONDITIONAL_RENDERING_READ_BIT_EXT = 0x00100000,
  // Provided by VK_EXT_blend_operation_advanced
    VK_ACCESS_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT = 0x00080000,
  // Provided by VK_KHR_acceleration_structure
    VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR = 0x00200000,
  // Provided by VK_KHR_acceleration_structure
    VK_ACCESS_ACCELERATION_STRUCTURE_WRITE_BIT_KHR = 0x00400000,
  // Provided by VK_EXT_fragment_density_map
    VK_ACCESS_FRAGMENT_DENSITY_MAP_READ_BIT_EXT = 0x01000000,
  // Provided by VK_KHR_fragment_shading_rate
    VK_ACCESS_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR = 0x00800000,
  // Provided by VK_EXT_device_generated_commands
    VK_ACCESS_COMMAND_PREPROCESS_READ_BIT_EXT = 0x00020000,
  // Provided by VK_EXT_device_generated_commands
    VK_ACCESS_COMMAND_PREPROCESS_WRITE_BIT_EXT = 0x00040000,
  // Provided by VK_KHR_synchronization2
    VK_ACCESS_NONE_KHR = VK_ACCESS_NONE,
} VkAccessFlagBits;
```

These values all have the same meaning as the equivalently named values for VkAccessFlags2.

- VK_ACCESS_NONE specifies no accesses.

- VK_ACCESS_MEMORY_READ_BIT specifies all read accesses. It is always valid in any access mask, and is treated as equivalent to setting all READ access flags that are valid where it is used.
- VK_ACCESS_MEMORY_WRITE_BIT specifies all write accesses. It is always valid in any access mask, and is treated as equivalent to setting all WRITE access flags that are valid where it is used.
- VK_ACCESS_INDIRECT_COMMAND_READ_BIT specifies read access to indirect command data read as part of an indirect build, trace, drawing or dispatching command. Such access occurs in the VK_PIPELINE_STAGE_DRAW_INDIRECT_BIT pipeline stage.
- VK_ACCESS_INDEX_READ_BIT specifies read access to an index buffer as part of an indexed drawing command, bound by vkCmdBindIndexBuffer2 and vkCmdBindIndexBuffer. Such access occurs in the VK_PIPELINE_STAGE_VERTEX_INPUT_BIT pipeline stage.
- VK_ACCESS_VERTEX_ATTRIBUTE_READ_BIT specifies read access to a vertex buffer as part of a drawing command, bound by vkCmdBindVertexBuffers. Such access occurs in the VK_PIPELINE_STAGE_VERTEX_INPUT_BIT pipeline stage.
- VK_ACCESS_UNIFORM_READ_BIT specifies read access to a uniform buffer in any shader pipeline stage.
- VK_ACCESS_INPUT_ATTACHMENT_READ_BIT specifies read access to an input attachment within a render pass during fragment shading. Such access occurs in the VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT pipeline stage.
- VK_ACCESS_SHADER_READ_BIT specifies read access to a uniform texel buffer, sampled image, storage buffer, physical storage buffer, shader binding table, storage texel buffer, or storage image in any shader pipeline stage.
- VK_ACCESS_SHADER_WRITE_BIT specifies write access to a storage buffer, physical storage buffer, storage texel buffer, or storage image in any shader pipeline stage.
- VK_ACCESS_COLOR_ATTACHMENT_READ_BIT specifies read access to a color attachment, such as via blending (other than advanced blend operations), logic operations or certain render pass load operations in the VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT pipeline stage or via fragment shader tile image reads in the VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT pipeline stage.
- VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT specifies write access to a color attachment during a render pass or via certain render pass load, store, and multisample resolve operations. This includes multisample resolve operations for depth/stencil resolve attachments. Such access occurs in the VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT pipeline stage.
- VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT specifies read access to a depth/stencil attachment, via depth or stencil operations or certain render pass load operations in the VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT or VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT pipeline stages or via fragment shader tile image reads in the VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT pipeline stage.
- VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT specifies write access to a depth/stencil attachment, via depth or stencil operations or certain render pass load and store operations. Such access occurs in the VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT or VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT pipeline stages.
- VK_ACCESS_TRANSFER_READ_BIT specifies read access to an image or buffer in a copy operation. Such access occurs in the VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT pipeline stage.

- VK_ACCESS_TRANSFER_WRITE_BIT specifies write access to an image or buffer in a clear or copy operation. Such access occurs in the VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT pipeline stage.
- VK_ACCESS_HOST_READ_BIT specifies read access by a host operation. Accesses of this type are not performed through a resource, but directly on memory. Such access occurs in the VK_PIPELINE_STAGE_HOST_BIT pipeline stage.
- VK_ACCESS_HOST_WRITE_BIT specifies write access by a host operation. Accesses of this type are not performed through a resource, but directly on memory. Such access occurs in the VK_PIPELINE_STAGE_HOST_BIT pipeline stage.
- VK_ACCESS_CONDITIONAL_RENDERING_READ_BIT_EXT specifies read access to a predicate as part of conditional rendering. Such access occurs in the VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT pipeline stage.
- VK_ACCESS_TRANSFORM_FEEDBACK_WRITE_BIT_EXT specifies write access to a transform feedback buffer made when transform feedback is active. Such access occurs in the VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT pipeline stage.
- VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT specifies read access to a transform feedback counter buffer which is read when vkCmdBeginTransformFeedbackEXT executes. Such access occurs in the VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT pipeline stage.
- VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT specifies write access to a transform feedback counter buffer which is written when vkCmdEndTransformFeedbackEXT executes. Such access occurs in the VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT pipeline stage.
- VK_ACCESS_COMMAND_PREPROCESS_READ_BIT_EXT specifies reads from buffer inputs to vkCmdPreprocessGeneratedCommandsEXT. Such access occurs in the VK_PIPELINE_STAGE_COMMAND_PREPROCESS_BIT_EXT pipeline stage.
- VK_ACCESS_COMMAND_PREPROCESS_WRITE_BIT_EXT specifies writes to the target command buffer preprocess outputs in vkCmdPreprocessGeneratedCommandsEXT. Such access occurs in the VK_PIPELINE_STAGE_COMMAND_PREPROCESS_BIT_EXT pipeline stage.
- VK_ACCESS_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT specifies read access to color attachments, including advanced blend operations. Such access occurs in the VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT pipeline stage.
- VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR specifies read access to an acceleration structure as part of a trace, build, or copy command, or to an acceleration structure scratch buffer as part of a build command. Such access occurs in the VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR pipeline stage or VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR pipeline stage.
- VK_ACCESS_ACCELERATION_STRUCTURE_WRITE_BIT_KHR specifies write access to an acceleration structure or acceleration structure scratch buffer as part of a build or copy command. Such access occurs in the VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR pipeline stage.
- VK_ACCESS_FRAGMENT_DENSITY_MAP_READ_BIT_EXT specifies read access to a fragment density map attachment during dynamic fragment density map operations. Such access occurs

in the VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT pipeline stage.

- VK_ACCESS_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR specifies read access to a fragment shading rate attachment during rasterization. Such access occurs in the VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR pipeline stage.

Certain access types are only performed by a subset of pipeline stages. Any synchronization command that takes both stage masks and access masks uses both to define the access scopes - only the specified access types performed by the specified stages are included in the access scope. An application **must** not specify an access flag in a synchronization command if it does not include a pipeline stage in the corresponding stage mask that is able to perform accesses of that type. The following table lists, for each access flag, which pipeline stages **can** perform that type of access.

Table 4. Supported Access Types

Access flag	Supported pipeline stages
VK_ACCESS_2_NONE, VK_ACCESS_NONE	Any
VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT, VK_ACCESS_INDIRECT_COMMAND_READ_BIT	VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_COPY_INDIRECT_BIT_KHR
VK_ACCESS_2_INDEX_READ_BIT, VK_ACCESS_INDEX_READ_BIT	VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT
VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT, VK_ACCESS_VERTEX_ATTRIBUTE_READ_BIT	VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT
VK_ACCESS_2_UNIFORM_READ_BIT, VK_ACCESS_UNIFORM_READ_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT, VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT,
VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT, VK_ACCESS_INPUT_ATTACHMENT_READ_BIT	VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT,

Access flag	Supported pipeline stages
VK_ACCESS_2_SHADER_READ_BIT, VK_ACCESS_SHADER_READ_BIT	VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT, VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT, VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT ,
VK_ACCESS_2_SHADER_WRITE_BIT, VK_ACCESS_SHADER_WRITE_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT, VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT ,
VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT, VK_ACCESS_COLOR_ATTACHMENT_READ_BIT	VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT
VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT, VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT	VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT

Access flag	Supported pipeline stages
VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT, VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT	VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT
VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT, VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT	VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT
VK_ACCESS_2_TRANSFER_READ_BIT, VK_ACCESS_TRANSFER_READ_BIT	VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT,
VK_ACCESS_2_TRANSFER_WRITE_BIT, VK_ACCESS_TRANSFER_WRITE_BIT	VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_CLEAR_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT,
VK_ACCESS_2_HOST_READ_BIT, VK_ACCESS_HOST_READ_BIT	VK_PIPELINE_STAGE_2_HOST_BIT
VK_ACCESS_2_HOST_WRITE_BIT, VK_ACCESS_HOST_WRITE_BIT	VK_PIPELINE_STAGE_2_HOST_BIT
VK_ACCESS_2_MEMORY_READ_BIT, VK_ACCESS_MEMORY_READ_BIT	Any
VK_ACCESS_2_MEMORY_WRITE_BIT, VK_ACCESS_MEMORY_WRITE_BIT	Any

Access flag	Supported pipeline stages
VK_ACCESS_2_SHADER_SAMPLED_READ_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT, VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT,
VK_ACCESS_2_SHADER_STORAGE_READ_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT, VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT,
VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT, VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT,

Access flag	Supported pipeline stages
VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR	VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR	VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR	VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR	VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT, VK_ACCESS_TRANSFORM_FEEDBACK_WRITE_BIT_EXT	VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT, VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT	VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT, VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT	VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
VK_ACCESS_2_CONDITIONAL_RENDERING_READ_BIT_EXT, VK_ACCESS_CONDITIONAL_RENDERING_READ_BIT_EXT	VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT
VK_ACCESS_2_COMMAND_PREPROCESS_READ_BIT_EXT, VK_ACCESS_COMMAND_PREPROCESS_READ_BIT_EXT	VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT
VK_ACCESS_2_COMMAND_PREPROCESS_WRITE_BIT_EXT, VK_ACCESS_COMMAND_PREPROCESS_WRITE_BIT_EXT	VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT
VK_ACCESS_2_MEMORY_DECOMPRESSION_READ_BIT_EXT	VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
VK_ACCESS_2_MEMORY_DECOMPRESSION_WRITE_BIT_EXT	VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
VK_ACCESS_2_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR, VK_ACCESS_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR	VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR

Access flag	Supported pipeline stages
VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT, VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR,
VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR, VK_ACCESS_ACCELERATION_STRUCTURE_WRITE_BIT_KHR	VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR
VK_ACCESS_2_FRAGMENT_DENSITY_MAP_READ_BIT_EXT, VK_ACCESS_FRAGMENT_DENSITY_MAP_READ_BIT_EXT	VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT
VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT, VK_ACCESS_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT	VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT

Access flag	Supported pipeline stages
VK_ACCESS_2_DESCRIPTOR_BUFFER_READ_BIT_EXT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT, VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT ,
VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT, VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT ,
VK_ACCESS_2_MICROMAP_READ_BIT_EXT	VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT	VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT

Access flag	Supported pipeline stages
VK_ACCESS_2_SAMPLER_HEAP_READ_BIT_EXT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT, VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT ,
VK_ACCESS_2_RESOURCE_HEAP_READ_BIT_EXT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT, VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT ,

WARNING This functionality is superseded by VkAccessFlags2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkAccessFlags;
```

`VkAccessFlags` is a bitmask type for setting a mask of zero or more VkAccessFlagBits.

If a memory object does not have the VK_MEMORY_PROPERTY_HOST_COHERENT_BIT property, then vkFlushMappedMemoryRanges **must** be called in order to guarantee that writes to the memory object from the host are made available to the host domain, where they **can** be further made available to the device domain via a domain operation. Similarly, vkInvalidateMappedMemoryRanges **must** be called to guarantee that writes which are available to the host domain are made visible to host operations.

If the memory object does have the VK_MEMORY_PROPERTY_HOST_COHERENT_BIT property flag, writes to the memory object from the host are automatically made available to the host domain. Similarly, writes made available to the host domain are automatically made visible to the host.

NOTE	Queue submission commands automatically perform a domain operation from host to device for all writes performed before the command executes, so in most cases an explicit memory barrier is not needed for this case. In the few circumstances where a submit does not occur between the host write and the device read access, writes **can** be made available by using an explicit memory barrier.

7.1.4. Framebuffer Region Dependencies

Pipeline stages that operate on, or with respect to, the framebuffer are collectively the *framebuffer-space* pipeline stages. These stages are:

- VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT
- VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT
- VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT
- VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT

For these pipeline stages, an execution or memory dependency from the first set of operations to the second set **can** either be a single *framebuffer-global* dependency, or split into multiple *framebuffer-local* dependencies. A dependency with non-framebuffer-space pipeline stages is neither framebuffer-global nor framebuffer-local.

Framebuffer-global dependencies require synchronization from all framebuffer operations in the first synchronization scope to all framebuffer operations in the second synchronization scope.

Framebuffer-local dependencies instead partition the framebuffer operations into *framebuffer regions*, and dependencies are only formed between framebuffer regions with overlapping (x, y, layer, view, sample) coordinates. How a framebuffer-local dependency is partitioned depends on the nature of the dependency and what is being synchronized, as outlined below.

- If the VK_DEPENDENCY_BY_REGION_BIT flag is specified in a synchronization command, each region in that dependency covers a single (x,y,layer) position.
  - If the dependency is between two operations where the sample counts match, VK_RENDERING_FRAGMENT_REGION_BIT_EXT is not also specified, and the operations are not both in subpasses specified with VK_SUBPASS_DESCRIPTION_FRAGMENT_REGION_BIT_EXT, the regions are further split to only cover a single sample.
- If the VK_DEPENDENCY_VIEW_LOCAL_BIT flag is specified in a synchronization command, each region covers a single (view).
- Each region in an automatic framebuffer-local dependency inserted between framebuffer-space operations performed by pipelines created with VK_PIPELINE_COLOR_BLEND_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_ACCESS_BIT_EXT,

VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT_EXT, or VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_BIT_EXT, covers a single (x,y,layer,sample) position.

If none of the above conditions partitions the regions across one of the x, y, layer, view, or sample dimensions, all coordinates for that dimension are included in each region. If this is true for all of those dimensions, the dependency is framebuffer-global.

NOTE	Framebuffer-local dependencies are more efficient for most architectures; particularly tile-based architectures - which can keep framebuffer-regions entirely in on-chip registers and thus avoid external bandwidth across such a dependency. Including a framebuffer-global dependency in your rendering will usually force all implementations to flush data to memory, or to a higher level cache, breaking any potential locality optimizations.
NOTE	It does not matter whether a framebuffer local dependency is within a single render pass or between two different render passes - the dependencies are always between regions with overlapping framebuffer coordinates.

7.1.5. Device-Local Dependencies

Dependencies **can** be either device-local or non-device-local. A device-local dependency acts as multiple separate dependencies, one for each physical device that executes the synchronization command, where each dependency only includes operations from that physical device in both synchronization scopes. A non-device-local dependency is a single dependency where both synchronization scopes include operations from all physical devices that participate in the synchronization command. For subpass dependencies, all physical devices in the VkDeviceGroupRenderPassBeginInfo::`deviceMask` participate in the dependency, and for pipeline barriers all physical devices that are set in the command buffer's current device mask participate in the dependency.

If a synchronization command includes a `dependencyFlags` parameter and specifies the VK_DEPENDENCY_DEVICE_GROUP_BIT flag, then it defines a non-device-local dependency for that synchronization command. If no `dependencyFlags` parameter is included or the VK_DEPENDENCY_DEVICE_GROUP_BIT flag is not specified, then it defines device-local dependencies for that synchronization command, for all participating physical devices.

Semaphore and event dependencies are device-local and only execute on the one physical device that performs the dependency.

7.2. Implicit Synchronization Guarantees

A small number of implicit ordering guarantees are provided by Vulkan, ensuring that the order in which commands are submitted is meaningful, and avoiding unnecessary complexity in common operations.

Submission order is a fundamental ordering in Vulkan, giving meaning to the order in which action

and synchronization commands are recorded and submitted to a single queue. Explicit and implicit ordering guarantees between commands in Vulkan all work on the premise that this ordering is meaningful. This order does not itself define any execution or memory dependencies; synchronization commands and other orderings within the API use this ordering to define their scopes.

Submission order for any given set of commands is based on the order in which they were recorded to command buffers and then submitted. This order is determined as follows:

1. The initial order is determined by the order in which vkQueueSubmit and vkQueueSubmit2 commands are executed on the host, for a single queue, from first to last.
2. The order in which VkSubmitInfo structures are specified in the `pSubmits` parameter of vkQueueSubmit, or in which VkSubmitInfo2 structures are specified in the `pSubmits` parameter of vkQueueSubmit2, from lowest index to highest.
3. The order in which command buffers are specified in the `pCommandBuffers` member of VkSubmitInfo or the `pCommandBufferInfos` member of VkSubmitInfo2 from lowest index to highest.
4. The order in which commands outside of a render pass were recorded to a command buffer on the host, from first to last.
5. The order in which commands inside a single subpass were recorded to a command buffer on the host, from first to last.

NOTE

When using a render pass object with multiple subpasses, commands in different subpasses have no defined submission order relative to each other, regardless of the order in which the subpasses were recorded. Commands within a subpass are still ordered relative to other commands in the same subpass, and those outside of the render pass.

State commands do not execute any operations on the device, instead they set the state of the command buffer when they execute on the host, in the order that they are recorded. Action commands consume the current state of the command buffer when they are recorded, and will execute state changes on the device as required to match the recorded state.

The order of primitives passing through the graphics pipeline and image layout transitions as part of an image memory barrier provide additional guarantees based on submission order.

Execution of pipeline stages within a given command also has a loose ordering, dependent only on a single command.

Signal operation order is a fundamental ordering in Vulkan, giving meaning to the order in which semaphore and fence signal operations occur when submitted to a single queue. The signal operation order for queue operations is determined as follows:

1. The initial order is determined by the order in which vkQueueSubmit and vkQueueSubmit2 commands are executed on the host, for a single queue, from first to last.
2. The order in which VkSubmitInfo structures are specified in the `pSubmits` parameter of vkQueueSubmit, or in which VkSubmitInfo2 structures are specified in the `pSubmits` parameter

of vkQueueSubmit2, from lowest index to highest.

3. The fence signal operation defined by the `fence` parameter of the following commands and the VkAcquireNextImageInfoKHR::`fence` member of the variable referred to by the `pAcquireInfo` parameter of vkAcquireNextImage2KHR is ordered after all semaphore signal operations defined by that command:
   - vkQueueSubmit
   - vkQueueSubmit2
   - vkQueueBindSparse
   - vkAcquireNextImageKHR

Semaphore signal operations defined by a single VkSubmitInfo or VkSubmitInfo2 or VkBindSparseInfo structure are unordered with respect to other semaphore signal operations defined within the same structure.

The vkSignalSemaphore command does not execute on a queue but instead performs the signal operation from the host. The semaphore signal operation defined by executing a vkSignalSemaphore command happens-after the vkSignalSemaphore command is invoked and happens-before the command returns.

NOTE

When signaling timeline semaphores, it is the responsibility of the application to ensure that they are ordered such that the semaphore value is strictly increasing. Because the first synchronization scope for a semaphore signal operation contains all semaphore signal operations which occur earlier in submission order, all semaphore signal operations contained in any given batch are guaranteed to happen-after all semaphore signal operations contained in any previous batches. However, no ordering guarantee is provided between the semaphore signal operations defined within a single batch. This, combined with the requirement that timeline semaphore values strictly increase, means that it is invalid to signal the same timeline semaphore twice within a single batch.

If an application wishes to ensure that some semaphore signal operation happens-after some other semaphore signal operation, it can submit a separate batch containing only semaphore signal operations, which will happen-after the semaphore signal operations in any earlier batches.

When signaling a semaphore from the host, the only ordering guarantee is that the signal operation happens-after when vkSignalSemaphore is called and happens-before it returns. Therefore, it is invalid to call `vkSignalSemaphore` while there are any outstanding signal operations on that semaphore from any queue submissions unless those queue submissions have some dependency which ensures that they happen-after the host signal operation. One example of this would be if the pending signal operation is, itself, waiting on the same semaphore at a lower value and the call to `vkSignalSemaphore` signals that lower value. Furthermore, if there are two or more processes or threads signaling the same timeline semaphore from the host, the application must ensure that the `vkSignalSemaphore` with the lower semaphore value returns before `vkSignalSemaphore` is called with the higher value.

7.3. Fences

Fences are a synchronization primitive that **can** be used to insert a dependency from a queue to the host. Fences have two states - signaled and unsignaled. A fence **can** be signaled as part of the execution of a queue submission command. Fences **can** be unsignaled on the host with vkResetFences. Fences **can** be waited on by the host with the vkWaitForFences command, and the current state **can** be queried with vkGetFenceStatus.

The internal data of a fence **may** include a reference to any resources and pending work associated with signal or unsignal operations performed on that fence object, collectively referred to as the fence's *payload*. Mechanisms to import and export that internal data to and from fences are provided below. These mechanisms indirectly enable applications to share fence state between two or more fences and other synchronization primitives across process and API boundaries.

Fences are represented by VkFence handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkFence)
```

To create a fence, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateFence(
    VkDevice                                    device,
    const VkFenceCreateInfo*                    pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkFence*                                    pFence);
```

- `device` is the logical device that creates the fence.
- `pCreateInfo` is a pointer to a VkFenceCreateInfo structure containing information about how the fence is to be created.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pFence` is a pointer to a handle in which the resulting fence object is returned.

Valid Usage (Implicit)

- VUID-vkCreateFence-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateFence-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkFenceCreateInfo structure
- VUID-vkCreateFence-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateFence-pFence-parameter

`pFence` **must** be a valid pointer to a VkFence handle

- VUID-vkCreateFence-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkFenceCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkFenceCreateInfo {
    VkStructureType       sType;
    const void*           pNext;
    VkFenceCreateFlags    flags;
} VkFenceCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkFenceCreateFlagBits specifying the initial state and behavior of the fence.

Valid Usage (Implicit)

- VUID-VkFenceCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_FENCE_CREATE_INFO
- VUID-VkFenceCreateInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkExportFenceCreateInfo or VkExportFenceWin32HandleInfoKHR
- VUID-VkFenceCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkFenceCreateInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkFenceCreateFlagBits values

```
// Provided by VK_VERSION_1_0
typedef enum VkFenceCreateFlagBits {
    VK_FENCE_CREATE_SIGNALED_BIT = 0x00000001,
} VkFenceCreateFlagBits;
```

- VK_FENCE_CREATE_SIGNALED_BIT specifies that the fence object is created in the signaled state. Otherwise, it is created in the unsignaled state.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkFenceCreateFlags;
```

VkFenceCreateFlags is a bitmask type for setting a mask of zero or more VkFenceCreateFlagBits.

To create a fence whose payload **can** be exported to external handles, add a VkExportFenceCreateInfo structure to the pNext chain of the VkFenceCreateInfo structure. The VkExportFenceCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkExportFenceCreateInfo {
    VkStructureType                   sType;
    const void*                       pNext;
    VkExternalFenceHandleTypeFlags    handleTypes;
} VkExportFenceCreateInfo;
```

```
// Provided by VK_KHR_external_fence
// Equivalent to VkExportFenceCreateInfo
typedef VkExportFenceCreateInfo VkExportFenceCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- handleTypes is a bitmask of VkExternalFenceHandleTypeFlagBits specifying one or more fence handle types the application **can** export from the resulting fence. The application **can** request multiple handle types for the same fence.

Valid Usage

- VUID-VkExportFenceCreateInfo-handleTypes-01446
  The bits in handleTypes **must** be supported and compatible, as reported by VkExternalFenceProperties

Valid Usage (Implicit)

- VUID-VkExportFenceCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_EXPORT_FENCE_CREATE_INFO
- VUID-VkExportFenceCreateInfo-handleTypes-parameter
  If `handleTypes` is not `0`, `handleTypes` **must** be a valid combination of VkExternalFenceHandleTypeFlagBits values

Structure Chaining

Extends the structure

- VkFenceCreateInfo

To specify additional attributes of NT handles exported from a fence, add a VkExportFenceWin32HandleInfoKHR structure to the `pNext` chain of the VkFenceCreateInfo structure. The `VkExportFenceWin32HandleInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_fence_win32
typedef struct VkExportFenceWin32HandleInfoKHR {
    VkStructureType               sType;
    const void*                   pNext;
    const SECURITY_ATTRIBUTES*    pAttributes;
    DWORD                         dwAccess;
    LPCWSTR                       name;
} VkExportFenceWin32HandleInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pAttributes` is a pointer to a Windows `SECURITY_ATTRIBUTES` structure specifying security attributes of the handle.
- `dwAccess` is a `DWORD` specifying access rights of the handle.
- `name` is a null-terminated UTF-16 string to associate with the underlying synchronization primitive referenced by NT handles exported from the created fence.

If VkExportFenceCreateInfo is not included in the same `pNext` chain, this structure is ignored.

If VkExportFenceCreateInfo is included in the `pNext` chain of VkFenceCreateInfo with a Windows `handleType`, but either `VkExportFenceWin32HandleInfoKHR` is not included in the `pNext` chain, or it is included but `pAttributes` is `NULL`, default security descriptor values will be used, and child processes created by the application will not inherit the handle, as described in the MSDN documentation for "Synchronization Object Security and Access Rights"[1]. Further, if the structure is not present, the access rights will be

`DXGI_SHARED_RESOURCE_READ` | `DXGI_SHARED_RESOURCE_WRITE`

for handles of the following types:

VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT

1

https://docs.microsoft.com/en-us/windows/win32/sync/synchronization-object-security-and-access-rights

Valid Usage

- VUID-VkExportFenceWin32HandleInfoKHR-handleTypes-01447
  If VkExportFenceCreateInfo::handleTypes does not include VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT, a VkExportFenceWin32HandleInfoKHR structure **must** not be included in the pNext chain of VkFenceCreateInfo

Valid Usage (Implicit)

- VUID-VkExportFenceWin32HandleInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_EXPORT_FENCE_WIN32_HANDLE_INFO_KHR
- VUID-VkExportFenceWin32HandleInfoKHR-pAttributes-parameter
  If pAttributes is not NULL, pAttributes **must** be a valid pointer to a valid SECURITY_ATTRIBUTES value

Structure Chaining

Extends the structure

- VkFenceCreateInfo

To export a Windows handle representing the state of a fence, call:

```
// Provided by VK_KHR_external_fence_win32
VkResult vkGetFenceWin32HandleKHR(
    VkDevice                                    device,
    const VkFenceGetWin32HandleInfoKHR*         pGetWin32HandleInfo,
    HANDLE*                                     pHandle);
```

- device is the logical device that created the fence being exported.
- pGetWin32HandleInfo is a pointer to a VkFenceGetWin32HandleInfoKHR structure containing parameters of the export operation.
- pHandle will return the Windows handle representing the fence state.

For handle types defined as NT handles, the handles returned by vkGetFenceWin32HandleKHR are owned by the application. To avoid leaking resources, the application **must** release ownership of them using the CloseHandle system call when they are no longer needed.

Exporting a Windows handle from a fence **may** have side effects depending on the transference of the specified handle type, as described in Importing Fence Payloads.

Valid Usage (Implicit)

- VUID-vkGetFenceWin32HandleKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetFenceWin32HandleKHR-pGetWin32HandleInfo-parameter
  `pGetWin32HandleInfo` **must** be a valid pointer to a valid VkFenceGetWin32HandleInfoKHR structure
- VUID-vkGetFenceWin32HandleKHR-pHandle-parameter
  `pHandle` **must** be a valid pointer to a `HANDLE` value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_TOO_MANY_OBJECTS
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkFenceGetWin32HandleInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_fence_win32
typedef struct VkFenceGetWin32HandleInfoKHR {
    VkStructureType                       sType;
    const void*                           pNext;
    VkFence                               fence;
    VkExternalFenceHandleTypeFlagBits     handleType;
} VkFenceGetWin32HandleInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `fence` is the fence from which state will be exported.
- `handleType` is a VkExternalFenceHandleTypeFlagBits value specifying the type of handle requested.

The properties of the handle returned depend on the value of `handleType`. See VkExternalFenceHandleTypeFlagBits for a description of the properties of the defined external fence handle types.

Valid Usage

- VUID-VkFenceGetWin32HandleInfoKHR-handleType-01448
  `handleType` **must** have been included in VkExportFenceCreateInfo::`handleTypes` when the `fence`'s current payload was created
- VUID-VkFenceGetWin32HandleInfoKHR-handleType-01449
  If `handleType` is defined as an NT handle, vkGetFenceWin32HandleKHR **must** be called no more than once for each valid unique combination of `fence` and `handleType`
- VUID-VkFenceGetWin32HandleInfoKHR-fence-01450
  `fence` **must** not currently have its payload replaced by an imported payload as described below in Importing Fence Payloads unless that imported payload's handle type was included in VkExternalFenceProperties::`exportFromImportedHandleTypes` for `handleType`
- VUID-VkFenceGetWin32HandleInfoKHR-handleType-01451
  If `handleType` refers to a handle type with copy payload transference semantics, `fence` **must** be signaled, or have an associated fence signal operation pending execution
- VUID-VkFenceGetWin32HandleInfoKHR-handleType-01452
  `handleType` **must** be defined as an NT handle or a global share handle

Valid Usage (Implicit)

- VUID-VkFenceGetWin32HandleInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_FENCE_GET_WIN32_HANDLE_INFO_KHR
- VUID-VkFenceGetWin32HandleInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkFenceGetWin32HandleInfoKHR-fence-parameter
  `fence` **must** be a valid VkFence handle
- VUID-VkFenceGetWin32HandleInfoKHR-handleType-parameter
  `handleType` **must** be a valid VkExternalFenceHandleTypeFlagBits value

To export a POSIX file descriptor representing the payload of a fence, call:

```
// Provided by VK_KHR_external_fence_fd
VkResult vkGetFenceFdKHR(
    VkDevice                                    device,
    const VkFenceGetFdInfoKHR*                  pGetFdInfo,
    int*                                        pFd);
```

- `device` is the logical device that created the fence being exported.
- `pGetFdInfo` is a pointer to a VkFenceGetFdInfoKHR structure containing parameters of the export operation.
- `pFd` will return the file descriptor representing the fence payload.

Each call to vkGetFenceFdKHR **must** create a new file descriptor and transfer ownership of it to the application. To avoid leaking resources, the application **must** release ownership of the file descriptor when it is no longer needed.

NOTE	Ownership can be released in many ways. For example, the application can call close() on the file descriptor, or transfer ownership back to Vulkan by using the file descriptor to import a fence payload.

If pGetFdInfo->handleType is VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT and the fence is signaled at the time vkGetFenceFdKHR is called, pFd **may** return the value -1 instead of a valid file descriptor.

Where supported by the operating system, the implementation **must** set the file descriptor to be closed automatically when an execve system call is made.

Exporting a file descriptor from a fence **may** have side effects depending on the transference of the specified handle type, as described in Importing Fence State.

Valid Usage (Implicit)

- VUID-vkGetFenceFdKHR-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkGetFenceFdKHR-pGetFdInfo-parameter
  pGetFdInfo **must** be a valid pointer to a valid VkFenceGetFdInfoKHR structure
- VUID-vkGetFenceFdKHR-pFd-parameter
  pFd **must** be a valid pointer to an int value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_TOO_MANY_OBJECTS
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkFenceGetFdInfoKHR structure is defined as:

```
// Provided by VK_KHR_external_fence_fd
typedef struct VkFenceGetFdInfoKHR {
    VkStructureType                       sType;
    const void*                           pNext;
    VkFence                               fence;
    VkExternalFenceHandleTypeFlagBits     handleType;
} VkFenceGetFdInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `fence` is the fence from which state will be exported.
- `handleType` is a VkExternalFenceHandleTypeFlagBits value specifying the type of handle requested.

The properties of the file descriptor returned depend on the value of `handleType`. See VkExternalFenceHandleTypeFlagBits for a description of the properties of the defined external fence handle types.

Valid Usage

- VUID-VkFenceGetFdInfoKHR-handleType-01453
  `handleType` **must** have been included in VkExportFenceCreateInfo::`handleTypes` when `fence`'s current payload was created
- VUID-VkFenceGetFdInfoKHR-handleType-01454
  If `handleType` refers to a handle type with copy payload transference semantics, `fence` **must** be signaled, or have an associated fence signal operation pending execution
- VUID-VkFenceGetFdInfoKHR-fence-01455
  `fence` **must** not currently have its payload replaced by an imported payload as described below in Importing Fence Payloads unless that imported payload's handle type was included in VkExternalFenceProperties::`exportFromImportedHandleTypes` for `handleType`
- VUID-VkFenceGetFdInfoKHR-handleType-01456
  `handleType` **must** be defined as a POSIX file descriptor handle

Valid Usage (Implicit)

- VUID-VkFenceGetFdInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_FENCE_GET_FD_INFO_KHR
- VUID-VkFenceGetFdInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkFenceGetFdInfoKHR-fence-parameter
  `fence` **must** be a valid VkFence handle
- VUID-VkFenceGetFdInfoKHR-handleType-parameter

handleType **must** be a valid VkExternalFenceHandleTypeFlagBits value

To destroy a fence, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyFence(
    VkDevice                                    device,
    VkFence                                     fence,
    const VkAllocationCallbacks*                pAllocator);
```

- device is the logical device that destroys the fence.
- fence is the handle of the fence to destroy.
- pAllocator controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyFence-fence-01120
  All queue submission commands that refer to fence **must** have completed execution
- VUID-vkDestroyFence-fence-01121
  If VkAllocationCallbacks were provided when fence was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyFence-fence-01122
  If no VkAllocationCallbacks were provided when fence was created, pAllocator **must** be NULL

Valid Usage (Implicit)

- VUID-vkDestroyFence-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkDestroyFence-fence-parameter
  If fence is not VK_NULL_HANDLE, fence **must** be a valid VkFence handle
- VUID-vkDestroyFence-pAllocator-parameter
  If pAllocator is not NULL, pAllocator **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyFence-fence-parent
  If fence is a valid handle, it **must** have been created, allocated, or retrieved from device

Host Synchronization

- Host access to fence **must** be externally synchronized

To query the status of a fence from the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkGetFenceStatus(
    VkDevice                                    device,
    VkFence                                     fence);
```

- `device` is the logical device that owns the fence.
- `fence` is the handle of the fence to query.

Upon success, `vkGetFenceStatus` returns the status of the fence object, with the following return codes:

Table 5. Fence Object Status Codes

Status	Meaning
VK_SUCCESS	The fence specified by `fence` is signaled.
VK_NOT_READY	The fence specified by `fence` is unsignaled.
VK_ERROR_DEVICE_LOST	The device has been lost. See Lost Device.

If a queue submission command is pending execution, then the value returned by this command **may** immediately be out of date.

If the device has been lost (see Lost Device), `vkGetFenceStatus` **may** return any of the above status codes. If the device has been lost and `vkGetFenceStatus` is called repeatedly, it will eventually return either VK_SUCCESS or VK_ERROR_DEVICE_LOST.

Valid Usage (Implicit)

- VUID-vkGetFenceStatus-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetFenceStatus-fence-parameter
  `fence` **must** be a valid VkFence handle
- VUID-vkGetFenceStatus-fence-parent
  `fence` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_NOT_READY
- VK_SUCCESS

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To set the state of fences to unsignaled from the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkResetFences(
    VkDevice                                    device,
    uint32_t                                    fenceCount,
    const VkFence*                              pFences);
```

- `device` is the logical device that owns the fences.
- `fenceCount` is the number of fences to reset.
- `pFences` is a pointer to an array of fence handles to reset.

If any member of `pFences` currently has its payload imported with temporary permanence, that fence's prior permanent payload is first restored. The remaining operations described therefore operate on the restored payload.

When vkResetFences is executed on the host, it defines a *fence unsignal operation* for each fence, which resets the fence to the unsignaled state.

If any member of `pFences` is already in the unsignaled state when vkResetFences is executed, then vkResetFences has no effect on that fence.

Valid Usage

- VUID-vkResetFences-pFences-01123
  Each element of `pFences` **must** not be currently associated with any queue command that has not yet completed execution on that queue

Valid Usage (Implicit)

- VUID-vkResetFences-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkResetFences-pFences-parameter
  `pFences` **must** be a valid pointer to an array of `fenceCount` valid VkFence handles
- VUID-vkResetFences-fenceCount-arraylength
  `fenceCount` **must** be greater than `0`

- VUID-vkResetFences-pFences-parent
  Each element of `pFences` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to each member of `pFences` **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

When a fence is submitted to a queue as part of a queue submission command, it defines a memory dependency on the batches that were submitted as part of that command, and defines a *fence signal operation* which sets the fence to the signaled state.

The first synchronization scope includes every batch submitted in the same queue submission command. Fence signal operations that are defined by vkQueueSubmit or vkQueueSubmit2 additionally include in the first synchronization scope all commands that occur earlier in submission order. Fence signal operations that are defined by vkQueueSubmit or vkQueueSubmit2 or vkQueueBindSparse additionally include in the first synchronization scope any semaphore and fence signal operations that occur earlier in signal operation order.

The second synchronization scope only includes the fence signal operation.

The first access scope includes all memory access performed by the device.

The second access scope is empty.

To wait for one or more fences to enter the signaled state on the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkWaitForFences(
    VkDevice                                    device,
    uint32_t                                    fenceCount,
    const VkFence*                              pFences,
    VkBool32                                    waitAll,
    uint64_t                                    timeout);
```

- `device` is the logical device that owns the fences.

- `fenceCount` is the number of fences to wait on.
- `pFences` is a pointer to an array of `fenceCount` fence handles.
- `waitAll` is the condition that **must** be satisfied to successfully unblock the wait. If `waitAll` is VK_TRUE, then the condition is that all fences in `pFences` are signaled. Otherwise, the condition is that at least one fence in `pFences` is signaled.
- `timeout` is the timeout period in units of nanoseconds. `timeout` is adjusted to the closest value allowed by the implementation-dependent timeout accuracy, which **may** be substantially longer than one nanosecond, and **may** be longer than the requested period.

If the condition is satisfied when `vkWaitForFences` is called, then `vkWaitForFences` returns immediately. If the condition is not satisfied at the time `vkWaitForFences` is called, then `vkWaitForFences` will block and wait until the condition is satisfied or the `timeout` has expired, whichever is sooner.

If `timeout` is zero, then `vkWaitForFences` does not wait, but simply returns the current state of the fences. VK_TIMEOUT will be returned in this case if the condition is not satisfied, even though no actual wait was performed.

If the condition is satisfied before the `timeout` has expired, `vkWaitForFences` returns VK_SUCCESS. Otherwise, `vkWaitForFences` returns VK_TIMEOUT after the `timeout` has expired.

If device loss occurs (see Lost Device) before the timeout has expired, `vkWaitForFences` **must** return in finite time with either VK_SUCCESS or VK_ERROR_DEVICE_LOST.

NOTE	While we guarantee that `vkWaitForFences` **must** return in finite time, no guarantees are made that it returns immediately upon device loss. However, the application can reasonably expect that the delay will be on the order of seconds and that calling `vkWaitForFences` will not result in a permanently (or seemingly permanently) dead process.

Valid Usage (Implicit)

- VUID-vkWaitForFences-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkWaitForFences-pFences-parameter
  `pFences` **must** be a valid pointer to an array of `fenceCount` valid VkFence handles
- VUID-vkWaitForFences-fenceCount-arraylength
  `fenceCount` **must** be greater than `0`
- VUID-vkWaitForFences-pFences-parent
  Each element of `pFences` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS
- VK_TIMEOUT

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

An execution dependency is defined by waiting for a fence to become signaled, either via vkWaitForFences or by polling on vkGetFenceStatus.

The first synchronization scope includes only the fence signal operation.

The second synchronization scope includes the host operations of vkWaitForFences or vkGetFenceStatus indicating that the fence has become signaled.

NOTE Signaling a fence and waiting on the host does not guarantee that the results of memory accesses will be visible to the host, as the access scope of a memory dependency defined by a fence only includes device access. A memory barrier or other memory dependency **must** be used to guarantee this. See the description of host access types for more information.

7.3.1. Alternate Methods to Signal Fences

Besides submitting a fence to a queue as part of a queue submission command, a fence **may** also be signaled when a particular event occurs on a device or display.

To create a fence that will be signaled when an event occurs on a device, call:

```
// Provided by VK_EXT_display_control
VkResult vkRegisterDeviceEventEXT(
    VkDevice                                    device,
    const VkDeviceEventInfoEXT*                 pDeviceEventInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkFence*                                    pFence);
```

- `device` is a logical device on which the event **may** occur.
- `pDeviceEventInfo` is a pointer to a VkDeviceEventInfoEXT structure describing the event of interest to the application.

- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pFence` is a pointer to a handle in which the resulting fence object is returned.

Valid Usage (Implicit)

- VUID-vkRegisterDeviceEventEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkRegisterDeviceEventEXT-pDeviceEventInfo-parameter
  `pDeviceEventInfo` **must** be a valid pointer to a valid VkDeviceEventInfoEXT structure
- VUID-vkRegisterDeviceEventEXT-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkRegisterDeviceEventEXT-pFence-parameter
  `pFence` **must** be a valid pointer to a VkFence handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkDeviceEventInfoEXT` structure is defined as:

```
// Provided by VK_EXT_display_control
typedef struct VkDeviceEventInfoEXT {
    VkStructureType         sType;
    const void*             pNext;
    VkDeviceEventTypeEXT    deviceEvent;
} VkDeviceEventInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `deviceEvent` is a VkDeviceEventTypeEXT value specifying when the fence will be signaled.

Valid Usage (Implicit)

- VUID-VkDeviceEventInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_EVENT_INFO_EXT

- VUID-VkDeviceEventInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDeviceEventInfoEXT-deviceEvent-parameter
  `deviceEvent` **must** be a valid VkDeviceEventTypeEXT value

Possible values of VkDeviceEventInfoEXT::`deviceEvent`, specifying when a fence will be signaled, are:

```
// Provided by VK_EXT_display_control
typedef enum VkDeviceEventTypeEXT {
    VK_DEVICE_EVENT_TYPE_DISPLAY_HOTPLUG_EXT = 0,
} VkDeviceEventTypeEXT;
```

- VK_DEVICE_EVENT_TYPE_DISPLAY_HOTPLUG_EXT specifies that the fence is signaled when a display is plugged into or unplugged from the specified device. Applications **can** use this notification to determine when they need to re-enumerate the available displays on a device.

To create a fence that will be signaled when an event occurs on a VkDisplayKHR object, call:

```
// Provided by VK_EXT_display_control
VkResult vkRegisterDisplayEventEXT(
    VkDevice                                    device,
    VkDisplayKHR                                display,
    const VkDisplayEventInfoEXT*                pDisplayEventInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkFence*                                    pFence);
```

- `device` is a logical device associated with `display`.
- `display` is the display on which the event **may** occur.
- `pDisplayEventInfo` is a pointer to a VkDisplayEventInfoEXT structure describing the event of interest to the application.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pFence` is a pointer to a handle in which the resulting fence object is returned.

Valid Usage (Implicit)

- VUID-vkRegisterDisplayEventEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkRegisterDisplayEventEXT-display-parameter
  `display` **must** be a valid VkDisplayKHR handle
- VUID-vkRegisterDisplayEventEXT-pDisplayEventInfo-parameter
  `pDisplayEventInfo` **must** be a valid pointer to a valid VkDisplayEventInfoEXT structure
- VUID-vkRegisterDisplayEventEXT-pAllocator-parameter

If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure

- VUID-vkRegisterDisplayEventEXT-pFence-parameter
  `pFence` **must** be a valid pointer to a VkFence handle
- VUID-vkRegisterDisplayEventEXT-commonparent
  Both of `device`, and `display` **must** have been created, allocated, or retrieved from the same VkPhysicalDevice

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkDisplayEventInfoEXT` structure is defined as:

```
// Provided by VK_EXT_display_control
typedef struct VkDisplayEventInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkDisplayEventTypeEXT    displayEvent;
} VkDisplayEventInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `displayEvent` is a VkDisplayEventTypeEXT specifying when the fence will be signaled.

Valid Usage (Implicit)

- VUID-VkDisplayEventInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DISPLAY_EVENT_INFO_EXT
- VUID-VkDisplayEventInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDisplayEventInfoEXT-displayEvent-parameter
  `displayEvent` **must** be a valid VkDisplayEventTypeEXT value

Possible values of VkDisplayEventInfoEXT::`displayEvent`, specifying when a fence will be signaled, are:

```
// Provided by VK_EXT_display_control
typedef enum VkDisplayEventTypeEXT {
    VK_DISPLAY_EVENT_TYPE_FIRST_PIXEL_OUT_EXT = 0,
} VkDisplayEventTypeEXT;
```

- VK_DISPLAY_EVENT_TYPE_FIRST_PIXEL_OUT_EXT specifies that the fence is signaled when the first pixel of the next display refresh cycle leaves the display engine for the display.

7.3.2. Importing Fence Payloads

Applications **can** import a fence payload into an existing fence using an external fence handle. The effects of the import operation will be either temporary or permanent, as specified by the application. If the import is temporary, the fence will be *restored* to its permanent state the next time that fence is passed to vkResetFences.

NOTE Restoring a fence to its prior permanent payload is a distinct operation from resetting a fence payload. See vkResetFences for more detail.

Performing a subsequent temporary import on a fence before resetting it has no effect on this requirement; the next unsignal of the fence **must** still restore its last permanent state. A permanent payload import behaves as if the target fence was destroyed, and a new fence was created with the same handle but the imported payload. Because importing a fence payload temporarily or permanently detaches the existing payload from a fence, similar usage restrictions to those applied to `vkDestroyFence` are applied to any command that imports a fence payload. Which of these import types is used is referred to as the import operation's *permanence*. Each handle type supports either one or both types of permanence.

The implementation **must** perform the import operation by either referencing or copying the payload referred to by the specified external fence handle, depending on the handle's type. The import method used is referred to as the handle type's *transference*. When using handle types with reference transference, importing a payload to a fence adds the fence to the set of all fences sharing that payload. This set includes the fence from which the payload was exported. Fence signaling, waiting, and resetting operations performed on any fence in the set **must** behave as if the set were a single fence. Importing a payload using handle types with copy transference creates a duplicate copy of the payload at the time of import, but makes no further reference to it. Fence signaling, waiting, and resetting operations performed on the target of copy imports **must** not affect any other fence or payload.

Export operations have the same transference as the specified handle type's import operations. Additionally, exporting a fence payload to a handle with copy transference has the same side effects on the source fence's payload as executing a fence reset operation. If the fence was using a temporarily imported payload, the fence's prior permanent payload will be restored.

NOTE The tables Handle Types Supported by `VkImportFenceWin32HandleInfoKHR` and Handle Types Supported by `VkImportFenceFdInfoKHR` define the permanence and transference of each handle type.

External synchronization allows implementations to modify an object's internal state, i.e. payload, without internal synchronization. However, for fences sharing a payload across processes, satisfying the external synchronization requirements of `VkFence` parameters as if all fences in the set were the same object is sometimes infeasible. Satisfying valid usage constraints on the state of a fence would similarly require impractical coordination or levels of trust between processes. Therefore, these constraints only apply to a specific fence handle, not to its payload. For distinct fence objects which share a payload:

- If multiple commands which queue a signal operation, or which unsignal a fence, are called concurrently, behavior will be as if the commands were called in an arbitrary sequential order.
- If a queue submission command is called with a fence that is sharing a payload, and the payload is already associated with another queue command that has not yet completed execution, either one or both of the commands will cause the fence to become signaled when they complete execution.
- If a fence payload is reset while it is associated with a queue command that has not yet completed execution, the payload will become unsignaled, but **may** become signaled again when the command completes execution.
- In the preceding cases, any of the devices associated with the fences sharing the payload **may** be lost, or any of the queue submission or fence reset commands **may** return VK_ERROR_INITIALIZATION_FAILED.

Other than these non-deterministic results, behavior is well defined. In particular:

- The implementation **must** not crash or enter an internally inconsistent state where future valid Vulkan commands might cause undefined results,
- Timeouts on future wait commands on fences sharing the payload **must** be effective.

NOTE

These rules allow processes to synchronize access to shared memory without trusting each other. However, such processes must still be cautious not to use the shared fence for more than synchronizing access to the shared memory. For example, a process should not use a fence with shared payload to tell when commands it submitted to a queue have completed and objects used by those commands may be destroyed, since the other process can accidentally or maliciously cause the fence to signal before the commands actually complete.

When a fence is using an imported payload, its VkExportFenceCreateInfo::`handleTypes` value is specified when creating the fence from which the payload was exported, rather than specified when creating the fence. Additionally, VkExternalFenceProperties::`exportFromImportedHandleTypes` restricts which handle types **can** be exported from such a fence based on the specific handle type used to import the current payload. Passing a fence to vkAcquireNextImageKHR is equivalent to temporarily importing a fence payload to that fence.

NOTE

Because the exportable handle types of an imported fence correspond to its current imported payload, and vkAcquireNextImageKHR behaves the same as a temporary import operation for which the source fence is opaque to the application, applications have no way of determining whether any external handle types **can** be exported from a fence in this state. Therefore, applications **must** not attempt to

> export handles from fences using a temporarily imported payload from vkAcquireNextImageKHR.

When importing a fence payload, it is the responsibility of the application to ensure the external handles meet all valid usage requirements. However, implementations **must** perform sufficient validation of external handles to ensure that the operation results in a valid fence which will not cause program termination, device loss, queue stalls, host thread stalls, or corruption of other resources when used as allowed according to its import parameters. If the external handle provided does not meet these requirements, the implementation **must** fail the fence payload import operation with the error code VK_ERROR_INVALID_EXTERNAL_HANDLE.

To import a fence payload from a Windows handle, call:

```
// Provided by VK_KHR_external_fence_win32
VkResult vkImportFenceWin32HandleKHR(
    VkDevice                                    device,
    const VkImportFenceWin32HandleInfoKHR*      pImportFenceWin32HandleInfo);
```

- `device` is the logical device that created the fence.
- `pImportFenceWin32HandleInfo` is a pointer to a VkImportFenceWin32HandleInfoKHR structure specifying the fence and import parameters.

Importing a fence payload from Windows handles does not transfer ownership of the handle to the Vulkan implementation. For handle types defined as NT handles, the application **must** release ownership using the `CloseHandle` system call when the handle is no longer needed.

Applications **can** import the same fence payload into multiple instances of Vulkan, into the same instance from which it was exported, and multiple times into a given Vulkan instance.

Valid Usage

- VUID-vkImportFenceWin32HandleKHR-fence-04448
  `fence` **must** not be associated with any queue command that has not yet completed execution on that queue

Valid Usage (Implicit)

- VUID-vkImportFenceWin32HandleKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkImportFenceWin32HandleKHR-pImportFenceWin32HandleInfo-parameter
  `pImportFenceWin32HandleInfo` **must** be a valid pointer to a valid VkImportFenceWin32HandleInfoKHR structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_EXTERNAL_HANDLE
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkImportFenceWin32HandleInfoKHR structure is defined as:

```
// Provided by VK_KHR_external_fence_win32
typedef struct VkImportFenceWin32HandleInfoKHR {
    VkStructureType                       sType;
    const void*                           pNext;
    VkFence                               fence;
    VkFenceImportFlags                    flags;
    VkExternalFenceHandleTypeFlagBits     handleType;
    HANDLE                                handle;
    LPCWSTR                               name;
} VkImportFenceWin32HandleInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- fence is the fence into which the state will be imported.
- flags is a bitmask of VkFenceImportFlagBits specifying additional parameters for the fence payload import operation.
- handleType is a VkExternalFenceHandleTypeFlagBits value specifying the type of handle.
- handle is NULL or the external handle to import.
- name is NULL or a null-terminated UTF-16 string naming the underlying synchronization primitive to import.

The handle types supported by handleType are:

Table 6. Handle Types Supported by VkImportFenceWin32HandleInfoKHR

Handle Type	Transference	Permanence Supported
VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT	Reference	Temporary,Permanent

Handle Type	Transference	Permanence Supported
VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT	Reference	Temporary,Permanent

Valid Usage

- VUID-VkImportFenceWin32HandleInfoKHR-handleType-01457
  `handleType` **must** be a value included in the Handle Types Supported by `VkImportFenceWin32HandleInfoKHR` table
- VUID-VkImportFenceWin32HandleInfoKHR-handleType-01459
  If `handleType` is not VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT, `name` **must** be `NULL`
- VUID-VkImportFenceWin32HandleInfoKHR-handleType-01460
  If `handle` is `NULL`, `name` **must** name a valid synchronization primitive of the type specified by `handleType`
- VUID-VkImportFenceWin32HandleInfoKHR-handleType-01461
  If `name` is `NULL`, `handle` **must** be a valid handle of the type specified by `handleType`
- VUID-VkImportFenceWin32HandleInfoKHR-handle-01462
  If `handle` is not `NULL`, `name` **must** be `NULL`
- VUID-VkImportFenceWin32HandleInfoKHR-handle-01539
  If `handle` is not `NULL`, it **must** obey any requirements listed for `handleType` in external fence handle types compatibility
- VUID-VkImportFenceWin32HandleInfoKHR-name-01540
  If `name` is not `NULL`, it **must** obey any requirements listed for `handleType` in external fence handle types compatibility

Valid Usage (Implicit)

- VUID-VkImportFenceWin32HandleInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMPORT_FENCE_WIN32_HANDLE_INFO_KHR
- VUID-VkImportFenceWin32HandleInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkImportFenceWin32HandleInfoKHR-fence-parameter
  `fence` **must** be a valid VkFence handle
- VUID-VkImportFenceWin32HandleInfoKHR-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkFenceImportFlagBits values

Host Synchronization

- Host access to `fence` **must** be externally synchronized

To import a fence payload from a POSIX file descriptor, call:

```
// Provided by VK_KHR_external_fence_fd
VkResult vkImportFenceFdKHR(
    VkDevice                                    device,
    const VkImportFenceFdInfoKHR*               pImportFenceFdInfo);
```

- `device` is the logical device that created the fence.
- `pImportFenceFdInfo` is a pointer to a VkImportFenceFdInfoKHR structure specifying the fence and import parameters.

Importing a fence payload from a file descriptor transfers ownership of the file descriptor from the application to the Vulkan implementation. The application **must** not perform any operations on the file descriptor after a successful import.

Applications **can** import the same fence payload into multiple instances of Vulkan, into the same instance from which it was exported, and multiple times into a given Vulkan instance.

Valid Usage

- VUID-vkImportFenceFdKHR-fence-01463
  `fence` **must** not be associated with any queue command that has not yet completed execution on that queue

Valid Usage (Implicit)

- VUID-vkImportFenceFdKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkImportFenceFdKHR-pImportFenceFdInfo-parameter
  `pImportFenceFdInfo` **must** be a valid pointer to a valid VkImportFenceFdInfoKHR structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_EXTERNAL_HANDLE
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkImportFenceFdInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_fence_fd
typedef struct VkImportFenceFdInfoKHR {
    VkStructureType                       sType;
    const void*                           pNext;
    VkFence                               fence;
    VkFenceImportFlags                    flags;
    VkExternalFenceHandleTypeFlagBits     handleType;
    int                                   fd;
} VkImportFenceFdInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `fence` is the fence into which the payload will be imported.
- `flags` is a bitmask of VkFenceImportFlagBits specifying additional parameters for the fence payload import operation.
- `handleType` is a VkExternalFenceHandleTypeFlagBits value specifying the type of `fd`.
- `fd` is the external handle to import.

The handle types supported by `handleType` are:

Table 7. Handle Types Supported by `VkImportFenceFdInfoKHR`

Handle Type	Transference	Permanence Supported
VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_FD_BIT	Reference	Temporary,Permanent
VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT	Copy	Temporary

Valid Usage

- VUID-VkImportFenceFdInfoKHR-handleType-01464
  `handleType` **must** be a value included in the Handle Types Supported by `VkImportFenceFdInfoKHR` table
- VUID-VkImportFenceFdInfoKHR-fd-01541
  `fd` **must** obey any requirements listed for `handleType` in external fence handle types compatibility
- VUID-VkImportFenceFdInfoKHR-handleType-07306
  If `handleType` refers to a handle type with copy payload transference semantics, `flags` **must** contain VK_FENCE_IMPORT_TEMPORARY_BIT

If `handleType` is VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT, the special value `-1` for `fd` is treated like a valid sync file descriptor referring to an object that has already signaled. The import

operation will succeed and the VkFence will have a temporarily imported payload as if a valid file descriptor had been provided.

NOTE This special behavior for importing an invalid sync file descriptor allows easier interoperability with other system APIs which use the convention that an invalid sync file descriptor represents work that has already completed and does not need to be waited for. It is consistent with the option for implementations to return a -1 file descriptor when exporting a VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT from a VkFence which is signaled.

Valid Usage (Implicit)

- VUID-VkImportFenceFdInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_IMPORT_FENCE_FD_INFO_KHR
- VUID-VkImportFenceFdInfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkImportFenceFdInfoKHR-fence-parameter
  fence **must** be a valid VkFence handle
- VUID-VkImportFenceFdInfoKHR-flags-parameter
  If flags is not 0, flags **must** be a valid combination of VkFenceImportFlagBits values
- VUID-VkImportFenceFdInfoKHR-handleType-parameter
  handleType **must** be a valid VkExternalFenceHandleTypeFlagBits value

Host Synchronization

- Host access to fence **must** be externally synchronized

Bits which **can** be set in

- VkImportFenceWin32HandleInfoKHR::flags
- VkImportFenceFdInfoKHR::flags

specifying additional parameters of a fence import operation are:

```
// Provided by VK_VERSION_1_1
typedef enum VkFenceImportFlagBits {
    VK_FENCE_IMPORT_TEMPORARY_BIT = 0x00000001,
  // Provided by VK_KHR_external_fence
    VK_FENCE_IMPORT_TEMPORARY_BIT_KHR = VK_FENCE_IMPORT_TEMPORARY_BIT,
} VkFenceImportFlagBits;
```

```
// Provided by VK_KHR_external_fence
// Equivalent to VkFenceImportFlagBits
typedef VkFenceImportFlagBits VkFenceImportFlagBitsKHR;
```

- VK_FENCE_IMPORT_TEMPORARY_BIT specifies that the fence payload will be imported only temporarily, as described in Importing Fence Payloads, regardless of the permanence of handleType.

```
// Provided by VK_VERSION_1_1
typedef VkFlags VkFenceImportFlags;
```

```
// Provided by VK_KHR_external_fence
// Equivalent to VkFenceImportFlags
typedef VkFenceImportFlags VkFenceImportFlagsKHR;
```

VkFenceImportFlags is a bitmask type for setting a mask of zero or more VkFenceImportFlagBits.

7.4. Semaphores

Semaphores are a synchronization primitive that **can** be used to insert a dependency between queue operations or between a queue operation and the host. Binary semaphores have two states - signaled and unsignaled. Timeline semaphores have a strictly increasing 64-bit unsigned integer payload and are signaled with respect to a particular reference value. A semaphore **can** be signaled after execution of a queue operation is completed, and a queue operation **can** wait for a semaphore to become signaled before it begins execution. A timeline semaphore **can** additionally be signaled from the host with the vkSignalSemaphore command and waited on from the host with the vkWaitSemaphores command.

The internal data of a semaphore **may** include a reference to any resources and pending work associated with signal or unsignal operations performed on that semaphore object, collectively referred to as the semaphore's *payload*. Mechanisms to import and export that internal data to and from semaphores are provided below. These mechanisms indirectly enable applications to share semaphore state between two or more semaphores and other synchronization primitives across process and API boundaries.

Semaphores are represented by VkSemaphore handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkSemaphore)
```

To create a semaphore, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateSemaphore(
    VkDevice                                    device,
    const VkSemaphoreCreateInfo*                pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkSemaphore*                                pSemaphore);
```

- `device` is the logical device that creates the semaphore.
- `pCreateInfo` is a pointer to a VkSemaphoreCreateInfo structure containing information about how the semaphore is to be created.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pSemaphore` is a pointer to a handle in which the resulting semaphore object is returned.

Valid Usage (Implicit)

- VUID-vkCreateSemaphore-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateSemaphore-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkSemaphoreCreateInfo structure
- VUID-vkCreateSemaphore-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateSemaphore-pSemaphore-parameter
  `pSemaphore` **must** be a valid pointer to a VkSemaphore handle
- VUID-vkCreateSemaphore-device-queuecount
  The device **must** have been created with at least `1` queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkSemaphoreCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSemaphoreCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkSemaphoreCreateFlags    flags;
} VkSemaphoreCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.

Valid Usage

- VUID-VkSemaphoreCreateInfo-pNext-06789
  If the `pNext` chain includes a VkExportMetalObjectCreateInfoEXT structure, its `exportObjectType` member **must** be VK_EXPORT_METAL_OBJECT_TYPE_METAL_SHARED_EVENT_BIT_EXT

Valid Usage (Implicit)

- VUID-VkSemaphoreCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SEMAPHORE_CREATE_INFO
- VUID-VkSemaphoreCreateInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkExportMetalObjectCreateInfoEXT, VkExportSemaphoreCreateInfo, VkExportSemaphoreWin32HandleInfoKHR, VkImportMetalSharedEventInfoEXT, or VkSemaphoreTypeCreateInfo
- VUID-VkSemaphoreCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique, with the exception of structures of type VkExportMetalObjectCreateInfoEXT
- VUID-VkSemaphoreCreateInfo-flags-zerobitmask
  `flags` **must** be `0`

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSemaphoreCreateFlags;
```

`VkSemaphoreCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

The `VkSemaphoreTypeCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkSemaphoreTypeCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkSemaphoreType    semaphoreType;
    uint64_t           initialValue;
} VkSemaphoreTypeCreateInfo;
```

```
// Provided by VK_KHR_timeline_semaphore
// Equivalent to VkSemaphoreTypeCreateInfo
typedef VkSemaphoreTypeCreateInfo VkSemaphoreTypeCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `semaphoreType` is a VkSemaphoreType value specifying the type of the semaphore.
- `initialValue` is the initial payload value if `semaphoreType` is VK_SEMAPHORE_TYPE_TIMELINE.

To create a semaphore of a specific type, add a `VkSemaphoreTypeCreateInfo` structure to the VkSemaphoreCreateInfo::`pNext` chain.

If no `VkSemaphoreTypeCreateInfo` structure is included in the `pNext` chain of VkSemaphoreCreateInfo, then the created semaphore will have a default VkSemaphoreType of VK_SEMAPHORE_TYPE_BINARY.

Valid Usage

- VUID-VkSemaphoreTypeCreateInfo-timelineSemaphore-03252
  If the `timelineSemaphore` feature is not enabled, `semaphoreType` **must** not equal VK_SEMAPHORE_TYPE_TIMELINE
- VUID-VkSemaphoreTypeCreateInfo-semaphoreType-03279
  If `semaphoreType` is VK_SEMAPHORE_TYPE_BINARY, `initialValue` **must** be zero

Valid Usage (Implicit)

- VUID-VkSemaphoreTypeCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SEMAPHORE_TYPE_CREATE_INFO
- VUID-VkSemaphoreTypeCreateInfo-semaphoreType-parameter
  `semaphoreType` **must** be a valid VkSemaphoreType value

Structure Chaining

Extends the structures

- VkPhysicalDeviceExternalSemaphoreInfo
- VkSemaphoreCreateInfo

Possible values of VkSemaphoreTypeCreateInfo::`semaphoreType`, specifying the type of a semaphore, are:

```
// Provided by VK_VERSION_1_2
typedef enum VkSemaphoreType {
    VK_SEMAPHORE_TYPE_BINARY = 0,
    VK_SEMAPHORE_TYPE_TIMELINE = 1,
  // Provided by VK_KHR_timeline_semaphore
    VK_SEMAPHORE_TYPE_BINARY_KHR = VK_SEMAPHORE_TYPE_BINARY,
  // Provided by VK_KHR_timeline_semaphore
    VK_SEMAPHORE_TYPE_TIMELINE_KHR = VK_SEMAPHORE_TYPE_TIMELINE,
} VkSemaphoreType;
```

```
// Provided by VK_KHR_timeline_semaphore
// Equivalent to VkSemaphoreType
typedef VkSemaphoreType VkSemaphoreTypeKHR;
```

- VK_SEMAPHORE_TYPE_BINARY specifies a *binary semaphore* type that has a boolean payload indicating whether the semaphore is currently signaled or unsignaled. When created, the semaphore is in the unsignaled state.
- VK_SEMAPHORE_TYPE_TIMELINE specifies a *timeline semaphore* type that has a strictly increasing 64-bit unsigned integer payload indicating whether the semaphore is signaled with respect to a particular reference value. When created, the semaphore payload has the value given by the `initialValue` field of VkSemaphoreTypeCreateInfo.

To create a semaphore whose payload **can** be exported to external handles, add a VkExportSemaphoreCreateInfo structure to the `pNext` chain of the VkSemaphoreCreateInfo structure. The `VkExportSemaphoreCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkExportSemaphoreCreateInfo {
    VkStructureType                       sType;
    const void*                           pNext;
    VkExternalSemaphoreHandleTypeFlags    handleTypes;
} VkExportSemaphoreCreateInfo;
```

```
// Provided by VK_KHR_external_semaphore
// Equivalent to VkExportSemaphoreCreateInfo
typedef VkExportSemaphoreCreateInfo VkExportSemaphoreCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `handleTypes` is a bitmask of VkExternalSemaphoreHandleTypeFlagBits specifying one or more semaphore handle types the application **can** export from the resulting semaphore. The application **can** request multiple handle types for the same semaphore.

Valid Usage

- VUID-VkExportSemaphoreCreateInfo-handleTypes-01124
  The bits in `handleTypes` **must** be supported and compatible, as reported by VkExternalSemaphoreProperties

Valid Usage (Implicit)

- VUID-VkExportSemaphoreCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_CREATE_INFO
- VUID-VkExportSemaphoreCreateInfo-handleTypes-parameter
  If `handleTypes` is not `0`, `handleTypes` **must** be a valid combination of VkExternalSemaphoreHandleTypeFlagBits values

Structure Chaining

Extends the structure

- VkSemaphoreCreateInfo

To specify additional attributes of NT handles exported from a semaphore, add a `VkExportSemaphoreWin32HandleInfoKHR` structure to the `pNext` chain of the VkSemaphoreCreateInfo structure. The `VkExportSemaphoreWin32HandleInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_semaphore_win32
typedef struct VkExportSemaphoreWin32HandleInfoKHR {
    VkStructureType               sType;
    const void*                   pNext;
    const SECURITY_ATTRIBUTES*    pAttributes;
    DWORD                         dwAccess;
    LPCWSTR                       name;
} VkExportSemaphoreWin32HandleInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pAttributes` is a pointer to a Windows `SECURITY_ATTRIBUTES` structure specifying security attributes of the handle.
- `dwAccess` is a `DWORD` specifying access rights of the handle.
- `name` is a null-terminated UTF-16 string to associate with the underlying synchronization primitive referenced by NT handles exported from the created semaphore.

If VkExportSemaphoreCreateInfo is not included in the same `pNext` chain, this structure is ignored.

If VkExportSemaphoreCreateInfo is included in the `pNext` chain of VkSemaphoreCreateInfo with a Windows `handleType`, but either `VkExportSemaphoreWin32HandleInfoKHR` is not included in the `pNext` chain, or it is included but `pAttributes` is `NULL`, default security descriptor values will be used, and child processes created by the application will not inherit the handle, as described in the MSDN documentation for "Synchronization Object Security and Access Rights"[1]. Further, if the structure is not present, the access rights used depend on the handle type.

For handles of the following types:

VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT

The implementation **must** ensure the access rights allow both signal and wait operations on the semaphore.

For handles of the following types:

VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT

The access rights **must** be:

`GENERIC_ALL`

1

https://docs.microsoft.com/en-us/windows/win32/sync/synchronization-object-security-and-access-rights

Valid Usage

- VUID-VkExportSemaphoreWin32HandleInfoKHR-handleTypes-01125
  If VkExportSemaphoreCreateInfo::`handleTypes` does not include VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT or VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT, `VkExportSemaphoreWin32HandleInfoKHR` **must** not be included in the `pNext` chain of VkSemaphoreCreateInfo

Valid Usage (Implicit)

- VUID-VkExportSemaphoreWin32HandleInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_WIN32_HANDLE_INFO_KHR
- VUID-VkExportSemaphoreWin32HandleInfoKHR-pAttributes-parameter
  If `pAttributes` is not `NULL`, `pAttributes` **must** be a valid pointer to a valid `SECURITY_ATTRIBUTES` value

Structure Chaining

Extends the structure

- VkSemaphoreCreateInfo

To export a Windows handle representing the payload of a semaphore, call:

```
// Provided by VK_KHR_external_semaphore_win32
VkResult vkGetSemaphoreWin32HandleKHR(
    VkDevice                                    device,
    const VkSemaphoreGetWin32HandleInfoKHR*     pGetWin32HandleInfo,
    HANDLE*                                     pHandle);
```

- `device` is the logical device that created the semaphore being exported.
- `pGetWin32HandleInfo` is a pointer to a VkSemaphoreGetWin32HandleInfoKHR structure containing parameters of the export operation.
- `pHandle` will return the Windows handle representing the semaphore state.

For handle types defined as NT handles, the handles returned by `vkGetSemaphoreWin32HandleKHR` are owned by the application. To avoid leaking resources, the application **must** release ownership of them using the `CloseHandle` system call when they are no longer needed.

Exporting a Windows handle from a semaphore **may** have side effects depending on the transference of the specified handle type, as described in Importing Semaphore Payloads.

Valid Usage (Implicit)

- VUID-vkGetSemaphoreWin32HandleKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetSemaphoreWin32HandleKHR-pGetWin32HandleInfo-parameter
  `pGetWin32HandleInfo` **must** be a valid pointer to a valid VkSemaphoreGetWin32HandleInfoKHR structure
- VUID-vkGetSemaphoreWin32HandleKHR-pHandle-parameter
  `pHandle` **must** be a valid pointer to a `HANDLE` value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_TOO_MANY_OBJECTS
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkSemaphoreGetWin32HandleInfoKHR structure is defined as:

```
// Provided by VK_KHR_external_semaphore_win32
typedef struct VkSemaphoreGetWin32HandleInfoKHR {
    VkStructureType                       sType;
    const void*                           pNext;
    VkSemaphore                           semaphore;
    VkExternalSemaphoreHandleTypeFlagBits    handleType;
} VkSemaphoreGetWin32HandleInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- semaphore is the semaphore from which state will be exported.
- handleType is a VkExternalSemaphoreHandleTypeFlagBits value specifying the type of handle requested.

The properties of the handle returned depend on the value of handleType. See VkExternalSemaphoreHandleTypeFlagBits for a description of the properties of the defined external semaphore handle types.

Valid Usage

- VUID-VkSemaphoreGetWin32HandleInfoKHR-handleType-01126
  handleType **must** have been included in VkExportSemaphoreCreateInfo::handleTypes when the semaphore's current payload was created
- VUID-VkSemaphoreGetWin32HandleInfoKHR-handleType-01127
  If handleType is defined as an NT handle, vkGetSemaphoreWin32HandleKHR **must** be called no more than once for each valid unique combination of semaphore and handleType
- VUID-VkSemaphoreGetWin32HandleInfoKHR-semaphore-01128
  semaphore **must** not currently have its payload replaced by an imported payload as described below in Importing Semaphore Payloads unless that imported payload's handle type was included in VkExternalSemaphoreProperties::exportFromImportedHandleTypes for

handleType

- VUID-VkSemaphoreGetWin32HandleInfoKHR-handleType-01129
  If handleType refers to a handle type with copy payload transference semantics, as defined below in Importing Semaphore Payloads, there **must** be no queue waiting on semaphore
- VUID-VkSemaphoreGetWin32HandleInfoKHR-handleType-01130
  If handleType refers to a handle type with copy payload transference semantics, semaphore **must** be signaled, or have an associated semaphore signal operation pending execution
- VUID-VkSemaphoreGetWin32HandleInfoKHR-handleType-01131
  handleType **must** be defined as an NT handle or a global share handle

Valid Usage (Implicit)

- VUID-VkSemaphoreGetWin32HandleInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_SEMAPHORE_GET_WIN32_HANDLE_INFO_KHR
- VUID-VkSemaphoreGetWin32HandleInfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkSemaphoreGetWin32HandleInfoKHR-semaphore-parameter
  semaphore **must** be a valid VkSemaphore handle
- VUID-VkSemaphoreGetWin32HandleInfoKHR-handleType-parameter
  handleType **must** be a valid VkExternalSemaphoreHandleTypeFlagBits value

To export a POSIX file descriptor representing the payload of a semaphore, call:

```
// Provided by VK_KHR_external_semaphore_fd
VkResult vkGetSemaphoreFdKHR(
    VkDevice                                    device,
    const VkSemaphoreGetFdInfoKHR*              pGetFdInfo,
    int*                                        pFd);
```

- device is the logical device that created the semaphore being exported.
- pGetFdInfo is a pointer to a VkSemaphoreGetFdInfoKHR structure containing parameters of the export operation.
- pFd will return the file descriptor representing the semaphore payload.

Each call to vkGetSemaphoreFdKHR **must** create a new file descriptor and transfer ownership of it to the application. To avoid leaking resources, the application **must** release ownership of the file descriptor when it is no longer needed.

NOTE Ownership can be released in many ways. For example, the application can call close() on the file descriptor, or transfer ownership back to Vulkan by using the file descriptor to import a semaphore payload.

Where supported by the operating system, the implementation **must** set the file descriptor to be

closed automatically when an execve system call is made.

Exporting a file descriptor from a semaphore **may** have side effects depending on the transference of the specified handle type, as described in Importing Semaphore State.

Valid Usage (Implicit)

- VUID-vkGetSemaphoreFdKHR-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkGetSemaphoreFdKHR-pGetFdInfo-parameter
  pGetFdInfo **must** be a valid pointer to a valid VkSemaphoreGetFdInfoKHR structure
- VUID-vkGetSemaphoreFdKHR-pFd-parameter
  pFd **must** be a valid pointer to an int value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_TOO_MANY_OBJECTS
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkSemaphoreGetFdInfoKHR structure is defined as:

```
// Provided by VK_KHR_external_semaphore_fd
typedef struct VkSemaphoreGetFdInfoKHR {
    VkStructureType                           sType;
    const void*                               pNext;
    VkSemaphore                               semaphore;
    VkExternalSemaphoreHandleTypeFlagBits     handleType;
} VkSemaphoreGetFdInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- semaphore is the semaphore from which state will be exported.
- handleType is a VkExternalSemaphoreHandleTypeFlagBits value specifying the type of handle requested.

The properties of the file descriptor returned depend on the value of handleType. See VkExternalSemaphoreHandleTypeFlagBits for a description of the properties of the defined

external semaphore handle types.

Valid Usage

- VUID-VkSemaphoreGetFdInfoKHR-handleType-01132
  `handleType` **must** have been included in VkExportSemaphoreCreateInfo::`handleTypes` when `semaphore`'s current payload was created
- VUID-VkSemaphoreGetFdInfoKHR-semaphore-01133
  `semaphore` **must** not currently have its payload replaced by an imported payload as described below in Importing Semaphore Payloads unless that imported payload's handle type was included in VkExternalSemaphoreProperties::`exportFromImportedHandleTypes` for `handleType`
- VUID-VkSemaphoreGetFdInfoKHR-handleType-01134
  If `handleType` refers to a handle type with copy payload transference semantics, as defined below in Importing Semaphore Payloads, there **must** be no queue waiting on `semaphore`
- VUID-VkSemaphoreGetFdInfoKHR-handleType-01135
  If `handleType` refers to a handle type with copy payload transference semantics, `semaphore` **must** be signaled, or have an associated semaphore signal operation pending execution
- VUID-VkSemaphoreGetFdInfoKHR-handleType-01136
  `handleType` **must** be defined as a POSIX file descriptor handle
- VUID-VkSemaphoreGetFdInfoKHR-handleType-03253
  If `handleType` refers to a handle type with copy payload transference semantics, `semaphore` **must** have been created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_BINARY
- VUID-VkSemaphoreGetFdInfoKHR-handleType-03254
  If `handleType` refers to a handle type with copy payload transference semantics, `semaphore` **must** have an associated semaphore signal operation that has been submitted for execution and any semaphore signal operations on which it depends **must** have also been submitted for execution

Valid Usage (Implicit)

- VUID-VkSemaphoreGetFdInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SEMAPHORE_GET_FD_INFO_KHR
- VUID-VkSemaphoreGetFdInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkSemaphoreGetFdInfoKHR-semaphore-parameter
  `semaphore` **must** be a valid VkSemaphore handle
- VUID-VkSemaphoreGetFdInfoKHR-handleType-parameter
  `handleType` **must** be a valid VkExternalSemaphoreHandleTypeFlagBits value

To destroy a semaphore, call:

```
// Provided by VK_VERSION_1_0
void vkDestroySemaphore(
    VkDevice                                    device,
    VkSemaphore                                 semaphore,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the semaphore.
- `semaphore` is the handle of the semaphore to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroySemaphore-semaphore-05149
  All submitted batches that refer to `semaphore` **must** have completed execution
- VUID-vkDestroySemaphore-semaphore-01138
  If `VkAllocationCallbacks` were provided when `semaphore` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroySemaphore-semaphore-01139
  If no `VkAllocationCallbacks` were provided when `semaphore` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroySemaphore-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroySemaphore-semaphore-parameter
  If `semaphore` is not VK_NULL_HANDLE, `semaphore` **must** be a valid VkSemaphore handle
- VUID-vkDestroySemaphore-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroySemaphore-semaphore-parent
  If `semaphore` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `semaphore` **must** be externally synchronized

7.4.1. Semaphore Signaling

When a batch is submitted to a queue via a queue submission, and it includes semaphores to be

signaled, it defines a memory dependency on the batch, and defines *semaphore signal operations* which set the semaphores to the signaled state.

In case of semaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE the semaphore is considered signaled with respect to the counter value set to be signaled as specified in VkTimelineSemaphoreSubmitInfo or VkSemaphoreSignalInfo.

The first synchronization scope of a semaphore signal operation is defined by the command which defines it.

The second synchronization scope includes only the semaphore signal operation.

The first access scope includes all memory access performed by the device.

The second access scope is empty.

7.4.2. Semaphore Waiting

When a batch is submitted to a queue via a queue submission, and it includes semaphores to be waited on, it defines a memory dependency between prior semaphore signal operations and the batch, and defines *semaphore wait operations*.

Such semaphore wait operations set the semaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_BINARY to the unsignaled state. In case of semaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE a prior semaphore signal operation defines a memory dependency with a semaphore wait operation if the value the semaphore is signaled with is greater than or equal to the value the semaphore is waited with, thus the semaphore will continue to be considered signaled with respect to the counter value waited on as specified in VkTimelineSemaphoreSubmitInfo.

The first synchronization scope of any wait semaphore operation includes one semaphore signal operation for each semaphore waited on by this batch. The specific signal operation waited on for each semaphore **must** meet the following criteria:

- for binary semaphores, the signal operation is either earlier in submission order on the same queue, or is submitted by a command whose host operation happens-before this batch is submitted on the host
- for binary semaphores, no wait operation exists that happens-after the signal operation and happens-before this wait operation
- the signal operation is not guaranteed to happen-after the semaphore wait operation in this batch
- for timeline semaphores, the signal value is greater than or equal to the wait value

If multiple semaphore signal operations meet these criteria, any of those operations **may** be included in the first synchronization scope. When waiting on a binary semaphore, applications **must** ensure that exactly one semaphore signal operation meets these criteria.

The second synchronization scope of a wait operation is defined by each command.

The first access scope is empty.

The second access scope includes all memory access performed by the device.

The semaphore wait operation happens-after the first set of operations in the execution dependency, and happens-before the second set of operations in the execution dependency.

NOTE	Unlike timeline semaphores, fences or events, waiting for a binary semaphore also unsignals that semaphore when the wait completes. Applications **must** ensure that between two such wait operations, the semaphore is signaled again, with execution dependencies used to ensure these occur in order. Binary semaphore waits and signals should thus occur in discrete 1:1 pairs.

7.4.3. Semaphore State Requirements for Wait Operations

Before waiting on a semaphore, the application **must** ensure the semaphore is in a valid state for a wait operation. Specifically, when a semaphore wait operation is submitted to a queue:

- A binary semaphore **must** be signaled, or have an associated semaphore signal operation that is pending execution.
- Any semaphore signal operations on which the pending binary semaphore signal operation depends **must** also be completed or pending execution.
- There **must** be no other queue waiting on the same binary semaphore when the operation executes.

7.4.4. Host Operations on Semaphores

In addition to semaphore signal operations and semaphore wait operations submitted to device queues, timeline semaphores support the following host operations:

- Query the current counter value of the semaphore using the vkGetSemaphoreCounterValue command.
- Wait for a set of semaphores to reach particular counter values using the vkWaitSemaphores command.
- Signal the semaphore with a particular counter value from the host using the vkSignalSemaphore command.

To query the current counter value of a semaphore created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE from the host, call:

```
// Provided by VK_VERSION_1_2
VkResult vkGetSemaphoreCounterValue(
    VkDevice                                    device,
    VkSemaphore                                 semaphore,
    uint64_t*                                   pValue);
```

```
// Provided by VK_KHR_timeline_semaphore
// Equivalent to vkGetSemaphoreCounterValue
VkResult vkGetSemaphoreCounterValueKHR(
    VkDevice                                    device,
    VkSemaphore                                 semaphore,
    uint64_t*                                   pValue);
```

- `device` is the logical device that owns the semaphore.
- `semaphore` is the handle of the semaphore to query.
- `pValue` is a pointer to a 64-bit integer value in which the current counter value of the semaphore is returned.

NOTE If a queue submission command is pending execution, then the value returned by this command **may** immediately be out of date.

Valid Usage

- VUID-vkGetSemaphoreCounterValue-semaphore-03255
  `semaphore` **must** have been created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE

Valid Usage (Implicit)

- VUID-vkGetSemaphoreCounterValue-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetSemaphoreCounterValue-semaphore-parameter
  `semaphore` **must** be a valid VkSemaphore handle
- VUID-vkGetSemaphoreCounterValue-pValue-parameter
  `pValue` **must** be a valid pointer to a `uint64_t` value
- VUID-vkGetSemaphoreCounterValue-semaphore-parent
  `semaphore` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY

- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To wait for a set of semaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE to reach particular counter values on the host, call:

```
// Provided by VK_VERSION_1_2
VkResult vkWaitSemaphores(
    VkDevice                                    device,
    const VkSemaphoreWaitInfo*                  pWaitInfo,
    uint64_t                                    timeout);
```

```
// Provided by VK_KHR_timeline_semaphore
// Equivalent to vkWaitSemaphores
VkResult vkWaitSemaphoresKHR(
    VkDevice                                    device,
    const VkSemaphoreWaitInfo*                  pWaitInfo,
    uint64_t                                    timeout);
```

- device is the logical device that owns the semaphores.
- pWaitInfo is a pointer to a VkSemaphoreWaitInfo structure containing information about the wait condition.
- timeout is the timeout period in units of nanoseconds. timeout is adjusted to the closest value allowed by the implementation-dependent timeout accuracy, which **may** be substantially longer than one nanosecond, and **may** be longer than the requested period.

If the condition is satisfied when vkWaitSemaphores is called, then vkWaitSemaphores returns immediately. If the condition is not satisfied at the time vkWaitSemaphores is called, then vkWaitSemaphores will block and wait until the condition is satisfied or the timeout has expired, whichever is sooner.

If timeout is zero, then vkWaitSemaphores does not wait, but simply returns information about the current state of the semaphores. VK_TIMEOUT will be returned in this case if the condition is not satisfied, even though no actual wait was performed.

If the condition is satisfied before the timeout has expired, vkWaitSemaphores returns VK_SUCCESS. Otherwise, vkWaitSemaphores returns VK_TIMEOUT after the timeout has expired.

If device loss occurs (see Lost Device) before the timeout has expired, vkWaitSemaphores **must** return in finite time with either VK_SUCCESS or VK_ERROR_DEVICE_LOST.

Valid Usage (Implicit)

- VUID-vkWaitSemaphores-device-parameter
  device **must** be a valid VkDevice handle

- VUID-vkWaitSemaphores-pWaitInfo-parameter
  `pWaitInfo` **must** be a valid pointer to a valid VkSemaphoreWaitInfo structure

Return Codes

Success

- VK_SUCCESS
- VK_TIMEOUT

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkSemaphoreWaitInfo` structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkSemaphoreWaitInfo {
    VkStructureType         sType;
    const void*             pNext;
    VkSemaphoreWaitFlags    flags;
    uint32_t                semaphoreCount;
    const VkSemaphore*      pSemaphores;
    const uint64_t*         pValues;
} VkSemaphoreWaitInfo;
```

```
// Provided by VK_KHR_timeline_semaphore
// Equivalent to VkSemaphoreWaitInfo
typedef VkSemaphoreWaitInfo VkSemaphoreWaitInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkSemaphoreWaitFlagBits specifying additional parameters for the semaphore wait operation.
- `semaphoreCount` is the number of semaphores to wait on.
- `pSemaphores` is a pointer to an array of `semaphoreCount` semaphore handles to wait on.
- `pValues` is a pointer to an array of `semaphoreCount` timeline semaphore values.

Valid Usage

- VUID-VkSemaphoreWaitInfo-pSemaphores-03256
  All of the elements of `pSemaphores` **must** reference a semaphore that was created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE

Valid Usage (Implicit)

- VUID-VkSemaphoreWaitInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SEMAPHORE_WAIT_INFO
- VUID-VkSemaphoreWaitInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkSemaphoreWaitInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkSemaphoreWaitFlagBits values
- VUID-VkSemaphoreWaitInfo-pSemaphores-parameter
  `pSemaphores` **must** be a valid pointer to an array of `semaphoreCount` valid VkSemaphore handles
- VUID-VkSemaphoreWaitInfo-pValues-parameter
  `pValues` **must** be a valid pointer to an array of `semaphoreCount` `uint64_t` values
- VUID-VkSemaphoreWaitInfo-semaphoreCount-arraylength
  `semaphoreCount` **must** be greater than `0`

Bits which **can** be set in VkSemaphoreWaitInfo::`flags`, specifying additional parameters of a semaphore wait operation, are:

```
// Provided by VK_VERSION_1_2
typedef enum VkSemaphoreWaitFlagBits {
    VK_SEMAPHORE_WAIT_ANY_BIT = 0x00000001,
  // Provided by VK_KHR_timeline_semaphore
    VK_SEMAPHORE_WAIT_ANY_BIT_KHR = VK_SEMAPHORE_WAIT_ANY_BIT,
} VkSemaphoreWaitFlagBits;
```

```
// Provided by VK_KHR_timeline_semaphore
// Equivalent to VkSemaphoreWaitFlagBits
typedef VkSemaphoreWaitFlagBits VkSemaphoreWaitFlagBitsKHR;
```

- VK_SEMAPHORE_WAIT_ANY_BIT specifies that the semaphore wait condition is that at least one of the semaphores in `VkSemaphoreWaitInfo`::`pSemaphores` has reached the value specified by the corresponding element of `VkSemaphoreWaitInfo`::`pValues`. If VK_SEMAPHORE_WAIT_ANY_BIT is not set, the semaphore wait condition is that all of the semaphores in `VkSemaphoreWaitInfo`::`pSemaphores` have reached the value specified by the corresponding element of `VkSemaphoreWaitInfo`::`pValues`.

```
// Provided by VK_VERSION_1_2
typedef VkFlags VkSemaphoreWaitFlags;
```

```
// Provided by VK_KHR_timeline_semaphore
// Equivalent to VkSemaphoreWaitFlags
typedef VkSemaphoreWaitFlags VkSemaphoreWaitFlagsKHR;
```

VkSemaphoreWaitFlags is a bitmask type for setting a mask of zero or more VkSemaphoreWaitFlagBits.

To signal a semaphore created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE with a particular counter value, on the host, call:

```
// Provided by VK_VERSION_1_2
VkResult vkSignalSemaphore(
    VkDevice                                    device,
    const VkSemaphoreSignalInfo*                pSignalInfo);
```

```
// Provided by VK_KHR_timeline_semaphore
// Equivalent to vkSignalSemaphore
VkResult vkSignalSemaphoreKHR(
    VkDevice                                    device,
    const VkSemaphoreSignalInfo*                pSignalInfo);
```

- device is the logical device that owns the semaphore.
- pSignalInfo is a pointer to a VkSemaphoreSignalInfo structure containing information about the signal operation.

When vkSignalSemaphore is executed on the host, it defines and immediately executes a *semaphore signal operation* which sets the timeline semaphore to the given value.

The first synchronization scope is defined by the host execution model, but includes execution of vkSignalSemaphore on the host and anything that happened-before it.

The second synchronization scope is empty.

Valid Usage (Implicit)

- VUID-vkSignalSemaphore-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkSignalSemaphore-pSignalInfo-parameter
  pSignalInfo **must** be a valid pointer to a valid VkSemaphoreSignalInfo structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkSemaphoreSignalInfo` structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkSemaphoreSignalInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkSemaphore        semaphore;
    uint64_t           value;
} VkSemaphoreSignalInfo;
```

```
// Provided by VK_KHR_timeline_semaphore
// Equivalent to VkSemaphoreSignalInfo
typedef VkSemaphoreSignalInfo VkSemaphoreSignalInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `semaphore` is the handle of the semaphore to signal.
- `value` is the value to signal.

Valid Usage

- VUID-VkSemaphoreSignalInfo-semaphore-03257
  `semaphore` **must** have been created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE
- VUID-VkSemaphoreSignalInfo-value-03258
  `value` **must** have a value greater than the current value of the semaphore
- VUID-VkSemaphoreSignalInfo-value-03259
  `value` **must** be less than the value of any pending semaphore signal operations
- VUID-VkSemaphoreSignalInfo-value-03260
  `value` **must** have a value which does not differ from the current value of the semaphore or the value of any outstanding semaphore wait or signal operation on `semaphore` by more

than `maxTimelineSemaphoreValueDifference`

Valid Usage (Implicit)

- VUID-VkSemaphoreSignalInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SEMAPHORE_SIGNAL_INFO
- VUID-VkSemaphoreSignalInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkSemaphoreSignalInfo-semaphore-parameter
  `semaphore` **must** be a valid VkSemaphore handle

7.4.5. Importing Semaphore Payloads

Applications **can** import a semaphore payload into an existing semaphore using an external semaphore handle. The effects of the import operation will be either temporary or permanent, as specified by the application. If the import is temporary, the implementation **must** restore the semaphore to its prior permanent state after submitting the next semaphore wait operation. Performing a subsequent temporary import on a semaphore before performing a semaphore wait has no effect on this requirement; the next wait submitted on the semaphore **must** still restore its last permanent state. A permanent payload import behaves as if the target semaphore was destroyed, and a new semaphore was created with the same handle but the imported payload. Because importing a semaphore payload temporarily or permanently detaches the existing payload from a semaphore, similar usage restrictions to those applied to `vkDestroySemaphore` are applied to any command that imports a semaphore payload. Which of these import types is used is referred to as the import operation's *permanence*. Each handle type supports either one or both types of permanence.

The implementation **must** perform the import operation by either referencing or copying the payload referred to by the specified external semaphore handle, depending on the handle's type. The import method used is referred to as the handle type's *transference*. When using handle types with reference transference, importing a payload to a semaphore adds the semaphore to the set of all semaphores sharing that payload. This set includes the semaphore from which the payload was exported. Semaphore signaling and waiting operations performed on any semaphore in the set **must** behave as if the set were a single semaphore. Importing a payload using handle types with copy transference creates a duplicate copy of the payload at the time of import, but makes no further reference to it. Semaphore signaling and waiting operations performed on the target of copy imports **must** not affect any other semaphore or payload.

Export operations have the same transference as the specified handle type's import operations. Additionally, exporting a semaphore payload to a handle with copy transference has the same side effects on the source semaphore's payload as executing a semaphore wait operation. If the semaphore was using a temporarily imported payload, the semaphore's prior permanent payload will be restored.

NOTE The permanence and transference of handle types can be found in:

- Handle Types Supported by VkImportSemaphoreWin32HandleInfoKHR
- Handle Types Supported by VkImportSemaphoreFdInfoKHR

External synchronization allows implementations to modify an object's internal state, i.e. payload, without internal synchronization. However, for semaphores sharing a payload across processes, satisfying the external synchronization requirements of VkSemaphore parameters as if all semaphores in the set were the same object is sometimes infeasible. Satisfying the wait operation state requirements would similarly require impractical coordination or levels of trust between processes. Therefore, these constraints only apply to a specific semaphore handle, not to its payload. For distinct semaphore objects which share a payload, if the semaphores are passed to separate queue submission commands concurrently, behavior will be as if the commands were called in an arbitrary sequential order. If the wait operation state requirements are violated for the shared payload by a queue submission command, or if a signal operation is queued for a shared payload that is already signaled or has a pending signal operation, effects **must** be limited to one or more of the following:

- Returning VK_ERROR_INITIALIZATION_FAILED from the command which resulted in the violation.
- Losing the logical device on which the violation occurred immediately or at a future time, resulting in a VK_ERROR_DEVICE_LOST error from subsequent commands, including the one causing the violation.
- Continuing execution of the violating command or operation as if the semaphore wait completed successfully after an implementation-dependent timeout. In this case, the state of the payload becomes undefined, and future operations on semaphores sharing the payload will be subject to these same rules. The semaphore **must** be destroyed or have its payload replaced by an import operation to again have a well-defined state.

NOTE

These rules allow processes to synchronize access to shared memory without trusting each other. However, such processes must still be cautious not to use the shared semaphore for more than synchronizing access to the shared memory. For example, a process should not use a shared semaphore as part of an execution dependency chain that, when complete, leads to objects being destroyed, if it does not trust other processes sharing the semaphore payload.

When a semaphore is using an imported payload, its VkExportSemaphoreCreateInfo::handleTypes value is specified when creating the semaphore from which the payload was exported, rather than specified when creating the semaphore. Additionally, VkExternalSemaphoreProperties ::exportFromImportedHandleTypes restricts which handle types **can** be exported from such a semaphore based on the specific handle type used to import the current payload. Passing a semaphore to vkAcquireNextImageKHR is equivalent to temporarily importing a semaphore payload to that semaphore.

NOTE

Because the exportable handle types of an imported semaphore correspond to its current imported payload, and vkAcquireNextImageKHR behaves the same as a temporary import operation for which the source semaphore is opaque to the application, applications have no way of determining whether any external handle types **can** be exported from a semaphore in this state. Therefore, applications **must**

not attempt to export external handles from semaphores using a temporarily imported payload from vkAcquireNextImageKHR.

When importing a semaphore payload, it is the responsibility of the application to ensure the external handles meet all valid usage requirements. However, implementations **must** perform sufficient validation of external handles to ensure that the operation results in a valid semaphore which will not cause program termination, device loss, queue stalls, or corruption of other resources when used as allowed according to its import parameters, and excepting those side effects allowed for violations of the valid semaphore state for wait operations rules. If the external handle provided does not meet these requirements, the implementation **must** fail the semaphore payload import operation with the error code VK_ERROR_INVALID_EXTERNAL_HANDLE.

In addition, when importing a semaphore payload that is not compatible with the payload type corresponding to the VkSemaphoreType the semaphore was created with, the implementation **may** fail the semaphore payload import operation with the error code VK_ERROR_INVALID_EXTERNAL_HANDLE.

NOTE

As the introduction of the external semaphore handle type VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT predates that of timeline semaphores, support for importing semaphore payloads from external handles of that type into semaphores created (implicitly or explicitly) with a VkSemaphoreType of VK_SEMAPHORE_TYPE_BINARY is preserved for backwards compatibility. However, applications **should** prefer importing such handle types into semaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE.

To import a semaphore payload from a Windows handle, call:

```
// Provided by VK_KHR_external_semaphore_win32
VkResult vkImportSemaphoreWin32HandleKHR(
    VkDevice                                    device,
    const VkImportSemaphoreWin32HandleInfoKHR*  pImportSemaphoreWin32HandleInfo);
```

- `device` is the logical device that created the semaphore.
- `pImportSemaphoreWin32HandleInfo` is a pointer to a VkImportSemaphoreWin32HandleInfoKHR structure specifying the semaphore and import parameters.

Importing a semaphore payload from Windows handles does not transfer ownership of the handle to the Vulkan implementation. For handle types defined as NT handles, the application **must** release ownership using the `CloseHandle` system call when the handle is no longer needed.

Applications **can** import the same semaphore payload into multiple instances of Vulkan, into the same instance from which it was exported, and multiple times into a given Vulkan instance.

Valid Usage (Implicit)

- VUID-vkImportSemaphoreWin32HandleKHR-device-parameter

device **must** be a valid VkDevice handle

- VUID-vkImportSemaphoreWin32HandleKHR-pImportSemaphoreWin32HandleInfo-parameter
  pImportSemaphoreWin32HandleInfo **must** be a valid pointer to a valid VkImportSemaphoreWin32HandleInfoKHR structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_EXTERNAL_HANDLE
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkImportSemaphoreWin32HandleInfoKHR structure is defined as:

```
// Provided by VK_KHR_external_semaphore_win32
typedef struct VkImportSemaphoreWin32HandleInfoKHR {
    VkStructureType                          sType;
    const void*                              pNext;
    VkSemaphore                              semaphore;
    VkSemaphoreImportFlags                   flags;
    VkExternalSemaphoreHandleTypeFlagBits    handleType;
    HANDLE                                   handle;
    LPCWSTR                                  name;
} VkImportSemaphoreWin32HandleInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- semaphore is the semaphore into which the payload will be imported.
- flags is a bitmask of VkSemaphoreImportFlagBits specifying additional parameters for the semaphore payload import operation.
- handleType is a VkExternalSemaphoreHandleTypeFlagBits value specifying the type of handle.
- handle is NULL or the external handle to import.
- name is NULL or a null-terminated UTF-16 string naming the underlying synchronization primitive to import.

The handle types supported by handleType are:

Table 8. Handle Types Supported by VkImportSemaphoreWin32HandleInfoKHR

Handle Type	Transference	Permanence Supported
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT	Reference	Temporary,Permanent
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT	Reference	Temporary,Permanent
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT	Reference	Temporary,Permanent

Valid Usage

- VUID-VkImportSemaphoreWin32HandleInfoKHR-handleType-01140
  `handleType` **must** be a value included in the Handle Types Supported by `VkImportSemaphoreWin32HandleInfoKHR` table
- VUID-VkImportSemaphoreWin32HandleInfoKHR-handleType-01466
  If `handleType` is not VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT or VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT, `name` **must** be `NULL`
- VUID-VkImportSemaphoreWin32HandleInfoKHR-handleType-01467
  If `handle` is `NULL`, `name` **must** name a valid synchronization primitive of the type specified by `handleType`
- VUID-VkImportSemaphoreWin32HandleInfoKHR-handleType-01468
  If `name` is `NULL`, `handle` **must** be a valid handle of the type specified by `handleType`
- VUID-VkImportSemaphoreWin32HandleInfoKHR-handle-01469
  If `handle` is not `NULL`, `name` **must** be `NULL`
- VUID-VkImportSemaphoreWin32HandleInfoKHR-handle-01542
  If `handle` is not `NULL`, it **must** obey any requirements listed for `handleType` in external semaphore handle types compatibility
- VUID-VkImportSemaphoreWin32HandleInfoKHR-name-01543
  If `name` is not `NULL`, it **must** obey any requirements listed for `handleType` in external semaphore handle types compatibility
- VUID-VkImportSemaphoreWin32HandleInfoKHR-handleType-03261
  If `handleType` is VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT or VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT, the VkSemaphoreCreateInfo::`flags` field **must** match that of the semaphore from which `handle` or `name` was exported
- VUID-VkImportSemaphoreWin32HandleInfoKHR-handleType-03262
  If `handleType` is VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT or VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT, the VkSemaphoreTypeCreateInfo::`semaphoreType` field **must** match that of the semaphore from which `handle` or `name` was exported

- VUID-VkImportSemaphoreWin32HandleInfoKHR-flags-03322
  If `flags` contains VK_SEMAPHORE_IMPORT_TEMPORARY_BIT, the VkSemaphoreTypeCreateInfo::`semaphoreType` field of the semaphore from which `handle` or `name` was exported **must** not be VK_SEMAPHORE_TYPE_TIMELINE

Valid Usage (Implicit)

- VUID-VkImportSemaphoreWin32HandleInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMPORT_SEMAPHORE_WIN32_HANDLE_INFO_KHR
- VUID-VkImportSemaphoreWin32HandleInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkImportSemaphoreWin32HandleInfoKHR-semaphore-parameter
  `semaphore` **must** be a valid VkSemaphore handle
- VUID-VkImportSemaphoreWin32HandleInfoKHR-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkSemaphoreImportFlagBits values

Host Synchronization

- Host access to `semaphore` **must** be externally synchronized

To import a semaphore payload from a POSIX file descriptor, call:

```
// Provided by VK_KHR_external_semaphore_fd
VkResult vkImportSemaphoreFdKHR(
    VkDevice                                    device,
    const VkImportSemaphoreFdInfoKHR*           pImportSemaphoreFdInfo);
```

- `device` is the logical device that created the semaphore.
- `pImportSemaphoreFdInfo` is a pointer to a VkImportSemaphoreFdInfoKHR structure specifying the semaphore and import parameters.

Importing a semaphore payload from a file descriptor transfers ownership of the file descriptor from the application to the Vulkan implementation. The application **must** not perform any operations on the file descriptor after a successful import.

Applications **can** import the same semaphore payload into multiple instances of Vulkan, into the same instance from which it was exported, and multiple times into a given Vulkan instance.

Valid Usage

- VUID-vkImportSemaphoreFdKHR-semaphore-01142
  `semaphore` **must** not be associated with any queue command that has not yet completed execution on that queue

Valid Usage (Implicit)

- VUID-vkImportSemaphoreFdKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkImportSemaphoreFdKHR-pImportSemaphoreFdInfo-parameter
  `pImportSemaphoreFdInfo` **must** be a valid pointer to a valid VkImportSemaphoreFdInfoKHR structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_EXTERNAL_HANDLE
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkImportSemaphoreFdInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_semaphore_fd
typedef struct VkImportSemaphoreFdInfoKHR {
    VkStructureType                           sType;
    const void*                               pNext;
    VkSemaphore                               semaphore;
    VkSemaphoreImportFlags                    flags;
    VkExternalSemaphoreHandleTypeFlagBits     handleType;
    int                                       fd;
} VkImportSemaphoreFdInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `semaphore` is the semaphore into which the payload will be imported.
- `flags` is a bitmask of VkSemaphoreImportFlagBits specifying additional parameters for the semaphore payload import operation.
- `handleType` is a VkExternalSemaphoreHandleTypeFlagBits value specifying the type of `fd`.
- `fd` is the external handle to import.

The handle types supported by `handleType` are:

Table 9. Handle Types Supported by `VkImportSemaphoreFdInfoKHR`

Handle Type	Transference	Permanence Supported
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT	Reference	Temporary,Permanent
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT	Copy	Temporary

Valid Usage

- VUID-VkImportSemaphoreFdInfoKHR-handleType-01143
  `handleType` **must** be a value included in the Handle Types Supported by `VkImportSemaphoreFdInfoKHR` table
- VUID-VkImportSemaphoreFdInfoKHR-fd-01544
  `fd` **must** obey any requirements listed for `handleType` in external semaphore handle types compatibility
- VUID-VkImportSemaphoreFdInfoKHR-handleType-03263
  If `handleType` is VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT, the VkSemaphoreCreateInfo::`flags` field **must** match that of the semaphore from which `fd` was exported
- VUID-VkImportSemaphoreFdInfoKHR-handleType-07307
  If `handleType` refers to a handle type with copy payload transference semantics, `flags` **must** contain VK_SEMAPHORE_IMPORT_TEMPORARY_BIT
- VUID-VkImportSemaphoreFdInfoKHR-handleType-03264
  If `handleType` is VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT, the VkSemaphoreTypeCreateInfo::`semaphoreType` field **must** match that of the semaphore from which `fd` was exported
- VUID-VkImportSemaphoreFdInfoKHR-flags-03323
  If `flags` contains VK_SEMAPHORE_IMPORT_TEMPORARY_BIT, the VkSemaphoreTypeCreateInfo::`semaphoreType` field of the semaphore from which `fd` was exported **must** not be VK_SEMAPHORE_TYPE_TIMELINE

If `handleType` is VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT, the special value `-1` for `fd` is treated like a valid sync file descriptor referring to an object that has already signaled. The import operation will succeed and the `VkSemaphore` will have a temporarily imported payload as if a valid file descriptor had been provided.

NOTE

This special behavior for importing an invalid sync file descriptor allows easier interoperability with other system APIs which use the convention that an invalid sync file descriptor represents work that has already completed and does not need to be waited for. It is consistent with the option for implementations to return a `-1` file descriptor when exporting a VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT from a `VkSemaphore`

which is signaled.

Valid Usage (Implicit)

- VUID-VkImportSemaphoreFdInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMPORT_SEMAPHORE_FD_INFO_KHR
- VUID-VkImportSemaphoreFdInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkImportSemaphoreFdInfoKHR-semaphore-parameter
  `semaphore` **must** be a valid VkSemaphore handle
- VUID-VkImportSemaphoreFdInfoKHR-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkSemaphoreImportFlagBits values
- VUID-VkImportSemaphoreFdInfoKHR-handleType-parameter
  `handleType` **must** be a valid VkExternalSemaphoreHandleTypeFlagBits value

Host Synchronization

- Host access to `semaphore` **must** be externally synchronized

Bits which **can** be set in

- VkImportSemaphoreWin32HandleInfoKHR::`flags`
- VkImportSemaphoreFdInfoKHR::`flags`

specifying additional parameters of a semaphore import operation are:

```
// Provided by VK_VERSION_1_1
typedef enum VkSemaphoreImportFlagBits {
    VK_SEMAPHORE_IMPORT_TEMPORARY_BIT = 0x00000001,
  // Provided by VK_KHR_external_semaphore
    VK_SEMAPHORE_IMPORT_TEMPORARY_BIT_KHR = VK_SEMAPHORE_IMPORT_TEMPORARY_BIT,
} VkSemaphoreImportFlagBits;
```

```
// Provided by VK_KHR_external_semaphore
// Equivalent to VkSemaphoreImportFlagBits
typedef VkSemaphoreImportFlagBits VkSemaphoreImportFlagBitsKHR;
```

These bits have the following meanings:

- VK_SEMAPHORE_IMPORT_TEMPORARY_BIT specifies that the semaphore payload will be imported only temporarily, as described in Importing Semaphore Payloads, regardless of the permanence of `handleType`.

```
// Provided by VK_VERSION_1_1
typedef VkFlags VkSemaphoreImportFlags;
```

```
// Provided by VK_KHR_external_semaphore
// Equivalent to VkSemaphoreImportFlags
typedef VkSemaphoreImportFlags VkSemaphoreImportFlagsKHR;
```

VkSemaphoreImportFlags is a bitmask type for setting a mask of zero or more VkSemaphoreImportFlagBits.

7.5. Events

Events are a synchronization primitive that **can** be used to insert a fine-grained dependency between commands submitted to the same queue, or between the host and a queue. Events **must** not be used to insert a dependency between commands submitted to different queues. Events have two states - signaled and unsignaled. An application **can** signal or unsignal an event either on the host or on the device. A device **can** be made to wait for an event to become signaled before executing further operations. No command exists to wait for an event to become signaled on the host, but the current state of an event **can** be queried.

Events are represented by VkEvent handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkEvent)
```

To create an event, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateEvent(
    VkDevice                                    device,
    const VkEventCreateInfo*                    pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkEvent*                                    pEvent);
```

- device is the logical device that creates the event.
- pCreateInfo is a pointer to a VkEventCreateInfo structure containing information about how the event is to be created.
- pAllocator controls host memory allocation as described in the Memory Allocation chapter.
- pEvent is a pointer to a handle in which the resulting event object is returned.

When created, the event object is in the unsignaled state.

Valid Usage

- VUID-vkCreateEvent-device-09672
  `device` **must** support at least one queue family with one of the VK_QUEUE_VIDEO_ENCODE_BIT_KHR, VK_QUEUE_VIDEO_DECODE_BIT_KHR, VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT capabilities
- VUID-vkCreateEvent-events-04468
  If the `VK_KHR_portability_subset` extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::`events` is VK_FALSE, then the implementation does not support events, and vkCreateEvent **must** not be used

Valid Usage (Implicit)

- VUID-vkCreateEvent-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateEvent-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkEventCreateInfo structure
- VUID-vkCreateEvent-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateEvent-pEvent-parameter
  `pEvent` **must** be a valid pointer to a VkEvent handle
- VUID-vkCreateEvent-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkEventCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkEventCreateInfo {
    VkStructureType       sType;
    const void*           pNext;
    VkEventCreateFlags    flags;
} VkEventCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkEventCreateFlagBits defining additional creation parameters.

Valid Usage

- VUID-VkEventCreateInfo-pNext-06790
  If the `pNext` chain includes a VkExportMetalObjectCreateInfoEXT structure, its `exportObjectType` member **must** be VK_EXPORT_METAL_OBJECT_TYPE_METAL_SHARED_EVENT_BIT_EXT

Valid Usage (Implicit)

- VUID-VkEventCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_EVENT_CREATE_INFO
- VUID-VkEventCreateInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkExportMetalObjectCreateInfoEXT or VkImportMetalSharedEventInfoEXT
- VUID-VkEventCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique, with the exception of structures of type VkExportMetalObjectCreateInfoEXT
- VUID-VkEventCreateInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkEventCreateFlagBits values

```
// Provided by VK_VERSION_1_0
typedef enum VkEventCreateFlagBits {
  // Provided by VK_VERSION_1_3
    VK_EVENT_CREATE_DEVICE_ONLY_BIT = 0x00000001,
  // Provided by VK_KHR_synchronization2
    VK_EVENT_CREATE_DEVICE_ONLY_BIT_KHR = VK_EVENT_CREATE_DEVICE_ONLY_BIT,
} VkEventCreateFlagBits;
```

- VK_EVENT_CREATE_DEVICE_ONLY_BIT specifies that host event commands will not be used with this event.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkEventCreateFlags;
```

VkEventCreateFlags is a bitmask type for setting a mask of VkEventCreateFlagBits.

To destroy an event, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyEvent(
    VkDevice                                    device,
    VkEvent                                     event,
    const VkAllocationCallbacks*                pAllocator);
```

- device is the logical device that destroys the event.
- event is the handle of the event to destroy.
- pAllocator controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyEvent-event-01145
  All submitted commands that refer to event **must** have completed execution
- VUID-vkDestroyEvent-event-01146
  If VkAllocationCallbacks were provided when event was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyEvent-event-01147
  If no VkAllocationCallbacks were provided when event was created, pAllocator **must** be NULL

Valid Usage (Implicit)

- VUID-vkDestroyEvent-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkDestroyEvent-event-parameter
  If event is not VK_NULL_HANDLE, event **must** be a valid VkEvent handle
- VUID-vkDestroyEvent-pAllocator-parameter
  If pAllocator is not NULL, pAllocator **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyEvent-event-parent
  If event is a valid handle, it **must** have been created, allocated, or retrieved from device

Host Synchronization

- Host access to `event` **must** be externally synchronized

To query the state of an event from the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkGetEventStatus(
    VkDevice                                    device,
    VkEvent                                     event);
```

- `device` is the logical device that owns the event.
- `event` is the handle of the event to query.

Upon success, `vkGetEventStatus` returns the state of the event object with the following return codes:

Table 10. Event Object Status Codes

Status	Meaning
VK_EVENT_SET	The event specified by `event` is signaled.
VK_EVENT_RESET	The event specified by `event` is unsignaled.

If a `vkCmdSetEvent` or `vkCmdResetEvent` command is in a command buffer that is in the pending state, then the value returned by this command **may** immediately be out of date.

The state of an event **can** be updated by the host. The state of the event is immediately changed, and subsequent calls to `vkGetEventStatus` will return the new state. If an event is already in the requested state, then updating it to the same state has no effect.

Valid Usage

- VUID-vkGetEventStatus-event-03940
  `event` **must** not have been created with VK_EVENT_CREATE_DEVICE_ONLY_BIT

Valid Usage (Implicit)

- VUID-vkGetEventStatus-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetEventStatus-event-parameter
  `event` **must** be a valid VkEvent handle
- VUID-vkGetEventStatus-event-parent
  `event` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_EVENT_RESET
- VK_EVENT_SET

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To set the state of an event to signaled from the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkSetEvent(
    VkDevice                                    device,
    VkEvent                                     event);
```

- `device` is the logical device that owns the event.
- `event` is the event to set.

When vkSetEvent is executed on the host, it defines an *event signal operation* which sets the event to the signaled state.

If `event` is already in the signaled state when vkSetEvent is executed, then vkSetEvent has no effect, and no event signal operation occurs.

NOTE If a command buffer is waiting for an event to be signaled from the host, the application must signal the event before submitting the command buffer, as described in the queue forward progress section.

Valid Usage

- VUID-vkSetEvent-event-03941
  `event` **must** not have been created with VK_EVENT_CREATE_DEVICE_ONLY_BIT
- VUID-vkSetEvent-event-09543
  `event` **must** not be waited on by a command buffer in the pending state

Valid Usage (Implicit)

- VUID-vkSetEvent-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkSetEvent-event-parameter
  `event` **must** be a valid VkEvent handle
- VUID-vkSetEvent-event-parent
  `event` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `event` **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To set the state of an event to unsignaled from the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkResetEvent(
    VkDevice                                    device,
    VkEvent                                     event);
```

- `device` is the logical device that owns the event.
- `event` is the event to reset.

When vkResetEvent is executed on the host, it defines an *event unsignal operation* which resets the event to the unsignaled state.

If `event` is already in the unsignaled state when vkResetEvent is executed, then vkResetEvent has no effect, and no event unsignal operation occurs.

Valid Usage

- VUID-vkResetEvent-event-03821
  There **must** be an execution dependency between `vkResetEvent` and the execution of any vkCmdWaitEvents that includes `event` in its `pEvents` parameter
- VUID-vkResetEvent-event-03822
  There **must** be an execution dependency between `vkResetEvent` and the execution of any vkCmdWaitEvents2 that includes `event` in its `pEvents` parameter
- VUID-vkResetEvent-event-03823
  `event` **must** not have been created with VK_EVENT_CREATE_DEVICE_ONLY_BIT

Valid Usage (Implicit)

- VUID-vkResetEvent-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkResetEvent-event-parameter
  `event` **must** be a valid VkEvent handle
- VUID-vkResetEvent-event-parent
  `event` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `event` **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The state of an event **can** also be updated on the device by commands inserted in command buffers.

To signal an event from a device, call:

```
// Provided by VK_VERSION_1_3
void vkCmdSetEvent2(
    VkCommandBuffer                             commandBuffer,
    VkEvent                                     event,
    const VkDependencyInfo*                     pDependencyInfo);
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to vkCmdSetEvent2
void vkCmdSetEvent2KHR(
    VkCommandBuffer                             commandBuffer,
    VkEvent                                     event,
    const VkDependencyInfo*                     pDependencyInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `event` is the event that will be signaled.
- `pDependencyInfo` is a pointer to a VkDependencyInfo structure defining the first scopes of this operation.

When vkCmdSetEvent2 is submitted to a queue, it defines the first half of memory dependencies defined by `pDependencyInfo`, as well as an event signal operation which sets the event to the signaled state. A memory dependency is defined between the event signal operation and commands that occur earlier in submission order.

The first synchronization scope and access scope are defined by the union of all the memory dependencies defined by `pDependencyInfo`, and are applied to all operations that occur earlier in submission order. Queue family ownership transfers and image layout transitions defined by `pDependencyInfo` are also included in the first scopes.

The second synchronization scope includes only the event signal operation, and any queue family ownership transfers and image layout transitions defined by `pDependencyInfo`.

The second access scope includes only queue family ownership transfers and image layout transitions.

If VK_DEPENDENCY_ASYMMETRIC_EVENT_BIT_KHR is not set in `pDependencyInfo->dependencyFlags`, future vkCmdWaitEvents2 commands rely on all values of each element in `pDependencyInfo` matching exactly with those used to signal the corresponding event. If VK_DEPENDENCY_ASYMMETRIC_EVENT_BIT_KHR is set, `vkCmdSetEvent2` **must** only include the source stage mask of the first synchronization scope in `pDependencyInfo->pMemoryBarriers`[0].`srcStageMask`. vkCmdWaitEvents **must** not be used to wait on the result of a signal operation defined by `vkCmdSetEvent2`.

NOTE

The extra information provided by vkCmdSetEvent2 compared to vkCmdSetEvent allows implementations to more efficiently schedule the operations required to satisfy the requested dependencies. With vkCmdSetEvent, the full dependency information is not known until vkCmdWaitEvents is recorded, forcing

> implementations to insert the required operations at that point and not before.

If `event` is already in the signaled state when vkCmdSetEvent2 is executed on the device, then vkCmdSetEvent2 has no effect, no event signal operation occurs, and no dependency is generated.

Valid Usage

- VUID-vkCmdSetEvent2-synchronization2-03824
  The `synchronization2` feature **must** be enabled
- VUID-vkCmdSetEvent2-dependencyFlags-03825
  The `dependencyFlags` member of `pDependencyInfo` **must** be `0` or VK_DEPENDENCY_ASYMMETRIC_EVENT_BIT_KHR
- VUID-vkCmdSetEvent2-srcStageMask-09391
  The `srcStageMask` member of any element of the `pMemoryBarriers`, `pBufferMemoryBarriers`, or `pImageMemoryBarriers` members of `pDependencyInfo` **must** not include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-vkCmdSetEvent2-dstStageMask-09392
  The `dstStageMask` member of any element of the `pMemoryBarriers`, `pBufferMemoryBarriers`, or `pImageMemoryBarriers` members of `pDependencyInfo` **must** not include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-vkCmdSetEvent2-commandBuffer-03826
  The current device mask of `commandBuffer` **must** include exactly one physical device
- VUID-vkCmdSetEvent2-srcStageMask-03827
  The `srcStageMask` member of any element of the `pMemoryBarriers`, `pBufferMemoryBarriers`, or `pImageMemoryBarriers` members of `pDependencyInfo` **must** only include pipeline stages valid for the queue family that was used to create the command pool that `commandBuffer` was allocated from
- VUID-vkCmdSetEvent2-dstStageMask-03828
  The `dstStageMask` member of any element of the `pMemoryBarriers`, `pBufferMemoryBarriers`, or `pImageMemoryBarriers` members of `pDependencyInfo` **must** only include pipeline stages valid for the queue family that was used to create the command pool that `commandBuffer` was allocated from
- VUID-vkCmdSetEvent2-dependencyFlags-10785
  If the `dependencyFlags` member of `pDependencyInfo` includes VK_DEPENDENCY_ASYMMETRIC_EVENT_BIT_KHR, the `bufferMemoryBarrierCount` and `imageMemoryBarrierCount` members of `pDependencyInfo` **must** be `0`
- VUID-vkCmdSetEvent2-dependencyFlags-10786
  If the `dependencyFlags` member of `pDependencyInfo` includes VK_DEPENDENCY_ASYMMETRIC_EVENT_BIT_KHR, the `memoryBarrierCount` member of `pDependencyInfo` **must** be `1`
- VUID-vkCmdSetEvent2-dependencyFlags-10787
  If the `dependencyFlags` member of `pDependencyInfo` includes VK_DEPENDENCY_ASYMMETRIC_EVENT_BIT_KHR, the `srcAccessMask`, `dstStageMask`, and `dstAccessMask` members of `pDependencyInfo->pMemoryBarriers`[0] **must** be `0`

Valid Usage (Implicit)

- VUID-vkCmdSetEvent2-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetEvent2-event-parameter
  `event` **must** be a valid VkEvent handle
- VUID-vkCmdSetEvent2-pDependencyInfo-parameter
  `pDependencyInfo` **must** be a valid pointer to a valid VkDependencyInfo structure
- VUID-vkCmdSetEvent2-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetEvent2-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdSetEvent2-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdSetEvent2-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdSetEvent2-commonparent
  Both of `commandBuffer`, and `event` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Synchronization

Conditional Rendering

vkCmdSetEvent2 is not affected by conditional rendering

The VkDependencyInfo structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkDependencyInfo {
    VkStructureType                  sType;
    const void*                      pNext;
    VkDependencyFlags                dependencyFlags;
    uint32_t                         memoryBarrierCount;
    const VkMemoryBarrier2*          pMemoryBarriers;
    uint32_t                         bufferMemoryBarrierCount;
    const VkBufferMemoryBarrier2*    pBufferMemoryBarriers;
    uint32_t                         imageMemoryBarrierCount;
    const VkImageMemoryBarrier2*     pImageMemoryBarriers;
} VkDependencyInfo;
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to VkDependencyInfo
typedef VkDependencyInfo VkDependencyInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- dependencyFlags is a bitmask of VkDependencyFlagBits specifying how execution and memory dependencies are formed.

- `memoryBarrierCount` is the length of the `pMemoryBarriers` array.
- `pMemoryBarriers` is a pointer to an array of VkMemoryBarrier2 structures defining memory dependencies between any memory accesses.
- `bufferMemoryBarrierCount` is the length of the `pBufferMemoryBarriers` array.
- `pBufferMemoryBarriers` is a pointer to an array of VkBufferMemoryBarrier2 structures defining memory dependencies between buffer ranges.
- `imageMemoryBarrierCount` is the length of the `pImageMemoryBarriers` array.
- `pImageMemoryBarriers` is a pointer to an array of VkImageMemoryBarrier2 structures defining memory dependencies between image subresources.

This structure defines a set of memory dependencies, as well as queue family ownership transfer operations and image layout transitions.

Each member of `pMemoryBarriers`, `pBufferMemoryBarriers`, and `pImageMemoryBarriers` defines a separate memory dependency.

Valid Usage

- VUID-VkDependencyInfo-pMemoryBarriers-10605
  For each element of `pMemoryBarriers`, the `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkDependencyInfo-pMemoryBarriers-10606
  For each element of `pMemoryBarriers`, `pNext` **must** be either `NULL` or a pointer to a valid instance of VkMemoryBarrierAccessFlags3KHR

Valid Usage (Implicit)

- VUID-VkDependencyInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEPENDENCY_INFO
- VUID-VkDependencyInfo-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkMemoryRangeBarriersInfoKHR
- VUID-VkDependencyInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkDependencyInfo-dependencyFlags-parameter
  If `dependencyFlags` is not `0`, `dependencyFlags` **must** be a valid combination of VkDependencyFlagBits values
- VUID-VkDependencyInfo-pMemoryBarriers-parameter
  If `memoryBarrierCount` is not `0`, `pMemoryBarriers` **must** be a valid pointer to an array of `memoryBarrierCount` valid VkMemoryBarrier2 structures
- VUID-VkDependencyInfo-pBufferMemoryBarriers-parameter
  If `bufferMemoryBarrierCount` is not `0`, `pBufferMemoryBarriers` **must** be a valid pointer to an array of `bufferMemoryBarrierCount` valid VkBufferMemoryBarrier2 structures

- VUID-VkDependencyInfo-pImageMemoryBarriers-parameter
  If `imageMemoryBarrierCount` is not `0`, `pImageMemoryBarriers` **must** be a valid pointer to an array of `imageMemoryBarrierCount` valid VkImageMemoryBarrier2 structures

To set the state of an event to signaled from a device, call:

WARNING	This functionality is superseded by vkCmdSetEvent2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkCmdSetEvent(
    VkCommandBuffer                             commandBuffer,
    VkEvent                                     event,
    VkPipelineStageFlags                        stageMask);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `event` is the event that will be signaled.
- `stageMask` specifies the source stage mask used to determine the first synchronization scope.

`vkCmdSetEvent` behaves identically to vkCmdSetEvent2, except that it does not define an access scope, and **must** only be used with vkCmdWaitEvents, not vkCmdWaitEvents2.

Valid Usage

- VUID-vkCmdSetEvent-stageMask-04090
  If the `geometryShader` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT
- VUID-vkCmdSetEvent-stageMask-04091
  If the `tessellationShader` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-vkCmdSetEvent-stageMask-04092
  If the `conditionalRendering` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT
- VUID-vkCmdSetEvent-stageMask-04093
  If the `fragmentDensityMap` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-vkCmdSetEvent-stageMask-04094
  If the `transformFeedback` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-vkCmdSetEvent-stageMask-04095
  If the `meshShader` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT

- VUID-vkCmdSetEvent-stageMask-04096
  If the taskShader feature is not enabled, stageMask **must** not contain VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT
- VUID-vkCmdSetEvent-stageMask-07319
  If the attachmentFragmentShadingRate feature is not enabled, stageMask **must** not contain VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdSetEvent-stageMask-03937
  If the synchronization2 feature is not enabled, stageMask **must** not be 0
- VUID-vkCmdSetEvent-stageMask-07950
  If the rayTracingPipeline feature is not enabled, stageMask **must** not contain VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR
- VUID-vkCmdSetEvent-stageMask-10754
  If the accelerationStructure feature is not enabled, stageMask **must** not contain VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-vkCmdSetEvent-stageMask-06457
  Any pipeline stage included in stageMask **must** be supported by the capabilities of the queue family specified by the queueFamilyIndex member of the VkCommandPoolCreateInfo structure that was used to create the VkCommandPool that commandBuffer was allocated from, as specified in the table of supported pipeline stages
- VUID-vkCmdSetEvent-stageMask-01149
  stageMask **must** not include VK_PIPELINE_STAGE_HOST_BIT
- VUID-vkCmdSetEvent-commandBuffer-01152
  The current device mask of commandBuffer **must** include exactly one physical device

Valid Usage (Implicit)

- VUID-vkCmdSetEvent-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetEvent-event-parameter
  event **must** be a valid VkEvent handle
- VUID-vkCmdSetEvent-stageMask-parameter
  If stageMask is not 0, stageMask **must** be a valid combination of VkPipelineStageFlagBits values
- VUID-vkCmdSetEvent-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdSetEvent-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdSetEvent-renderpass
  This command **must** only be called outside of a render pass instance

- VUID-vkCmdSetEvent-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdSetEvent-commonparent
  Both of `commandBuffer`, and `event` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Synchronization

Conditional Rendering

vkCmdSetEvent is not affected by conditional rendering

To unsignal the event from a device, call:

```
// Provided by VK_VERSION_1_3
void vkCmdResetEvent2(
    VkCommandBuffer                             commandBuffer,
    VkEvent                                     event,
    VkPipelineStageFlags2                       stageMask);
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to vkCmdResetEvent2
void vkCmdResetEvent2KHR(
    VkCommandBuffer                             commandBuffer,
    VkEvent                                     event,
    VkPipelineStageFlags2                       stageMask);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `event` is the event that will be unsignaled.
- `stageMask` is a VkPipelineStageFlags2 mask of pipeline stages used to determine the first synchronization scope.

When vkCmdResetEvent2 is submitted to a queue, it defines an execution dependency on commands that were submitted before it, and defines an event unsignal operation which resets the event to the unsignaled state.

The first synchronization scope includes all commands that occur earlier in submission order. The synchronization scope is limited to operations by `stageMask` or stages that are logically earlier than `stageMask`.

The second synchronization scope includes only the event unsignal operation.

If `event` is already in the unsignaled state when vkCmdResetEvent2 is executed on the device, then this command has no effect, no event unsignal operation occurs, and no execution dependency is generated.

Valid Usage

- VUID-vkCmdResetEvent2-stageMask-03929
  If the `geometryShader` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT
- VUID-vkCmdResetEvent2-stageMask-03930
  If the `tessellationShader` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-vkCmdResetEvent2-stageMask-03931
  If the `conditionalRendering` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT
- VUID-vkCmdResetEvent2-stageMask-03932
  If the `fragmentDensityMap` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-vkCmdResetEvent2-stageMask-03933
  If the `transformFeedback` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-vkCmdResetEvent2-stageMask-03934

If the meshShader feature is not enabled, stageMask **must** not contain VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT

- VUID-vkCmdResetEvent2-stageMask-03935
  If the taskShader feature is not enabled, stageMask **must** not contain VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT
- VUID-vkCmdResetEvent2-stageMask-07317
  If the attachmentFragmentShadingRate feature is not enabled, stageMask **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdResetEvent2-stageMask-07947
  If the rayTracingPipeline feature is not enabled, stageMask **must** not contain VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-vkCmdResetEvent2-stageMask-10751
  If the accelerationStructure feature is not enabled, stageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-vkCmdResetEvent2-stageMask-10752
  If the rayTracingMaintenance1 feature is not enabled, stageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR
- VUID-vkCmdResetEvent2-stageMask-11541
  If the VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap feature is not enabled, stageMask **must** not contain VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-vkCmdResetEvent2-synchronization2-03829
  The synchronization2 feature **must** be enabled
- VUID-vkCmdResetEvent2-stageMask-03830
  stageMask **must** not include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-vkCmdResetEvent2-event-03831
  There **must** be an execution dependency between vkCmdResetEvent2 and the execution of any earlier vkCmdWaitEvents in submission order that includes event in its pEvents parameter
- VUID-vkCmdResetEvent2-event-03832
  There **must** be an execution dependency between vkCmdResetEvent2 and the execution of any earlier vkCmdWaitEvents2 in submission order that includes event in its pEvents parameter
- VUID-vkCmdResetEvent2-commandBuffer-03833
  commandBuffer's current device mask **must** include exactly one physical device

Valid Usage (Implicit)

- VUID-vkCmdResetEvent2-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdResetEvent2-event-parameter
  event **must** be a valid VkEvent handle
- VUID-vkCmdResetEvent2-stageMask-parameter

If `stageMask` is not `0`, `stageMask` **must** be a valid combination of VkPipelineStageFlagBits2 values

- VUID-vkCmdResetEvent2-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdResetEvent2-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdResetEvent2-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdResetEvent2-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdResetEvent2-commonparent
  Both of `commandBuffer`, and `event` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Synchronization

Conditional Rendering

vkCmdResetEvent2 is not affected by conditional rendering

To set the state of an event to unsignaled from a device, call:

WARNING	This functionality is superseded by vkCmdResetEvent2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkCmdResetEvent(
    VkCommandBuffer                             commandBuffer,
    VkEvent                                     event,
    VkPipelineStageFlags                        stageMask);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `event` is the event that will be unsignaled.
- `stageMask` is a bitmask of VkPipelineStageFlagBits specifying the source stage mask used to determine when the `event` is unsignaled.

`vkCmdResetEvent` behaves identically to vkCmdResetEvent2.

Valid Usage

- VUID-vkCmdResetEvent-stageMask-04090
  If the `geometryShader` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT
- VUID-vkCmdResetEvent-stageMask-04091
  If the `tessellationShader` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-vkCmdResetEvent-stageMask-04092
  If the `conditionalRendering` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT
- VUID-vkCmdResetEvent-stageMask-04093
  If the `fragmentDensityMap` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-vkCmdResetEvent-stageMask-04094
  If the `transformFeedback` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-vkCmdResetEvent-stageMask-04095
  If the `meshShader` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT
- VUID-vkCmdResetEvent-stageMask-04096
  If the `taskShader` feature is not enabled, `stageMask` **must** not contain VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT
- VUID-vkCmdResetEvent-stageMask-07319

If the attachmentFragmentShadingRate feature is not enabled, stageMask **must** not contain VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR

- VUID-vkCmdResetEvent-stageMask-03937
  If the synchronization2 feature is not enabled, stageMask **must** not be 0
- VUID-vkCmdResetEvent-stageMask-07950
  If the rayTracingPipeline feature is not enabled, stageMask **must** not contain VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR
- VUID-vkCmdResetEvent-stageMask-10754
  If the accelerationStructure feature is not enabled, stageMask **must** not contain VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-vkCmdResetEvent-stageMask-06458
  Any pipeline stage included in stageMask **must** be supported by the capabilities of the queue family specified by the queueFamilyIndex member of the VkCommandPoolCreateInfo structure that was used to create the VkCommandPool that commandBuffer was allocated from, as specified in the table of supported pipeline stages
- VUID-vkCmdResetEvent-stageMask-01153
  stageMask **must** not include VK_PIPELINE_STAGE_HOST_BIT
- VUID-vkCmdResetEvent-event-03834
  There **must** be an execution dependency between vkCmdResetEvent and the execution of any earlier vkCmdWaitEvents in submission order that includes event in its pEvents parameter
- VUID-vkCmdResetEvent-event-03835
  There **must** be an execution dependency between vkCmdResetEvent and the execution of any earlier vkCmdWaitEvents2 in submission order that includes event in its pEvents parameter
- VUID-vkCmdResetEvent-commandBuffer-01157
  commandBuffer's current device mask **must** include exactly one physical device

Valid Usage (Implicit)

- VUID-vkCmdResetEvent-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdResetEvent-event-parameter
  event **must** be a valid VkEvent handle
- VUID-vkCmdResetEvent-stageMask-parameter
  If stageMask is not 0, stageMask **must** be a valid combination of VkPipelineStageFlagBits values
- VUID-vkCmdResetEvent-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdResetEvent-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT,

VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations

- VUID-vkCmdResetEvent-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdResetEvent-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdResetEvent-commonparent
  Both of `commandBuffer`, and `event` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Synchronization

Conditional Rendering

vkCmdResetEvent is not affected by conditional rendering

To wait for one or more events to enter the signaled state on a device, call:

```
// Provided by VK_VERSION_1_3
void vkCmdWaitEvents2(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    eventCount,
    const VkEvent*                              pEvents,
    const VkDependencyInfo*                     pDependencyInfos);
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to vkCmdWaitEvents2
void vkCmdWaitEvents2KHR(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    eventCount,
    const VkEvent*                              pEvents,
    const VkDependencyInfo*                     pDependencyInfos);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `eventCount` is the length of the `pEvents` array.
- `pEvents` is a pointer to an array of `eventCount` events to wait on.
- `pDependencyInfos` is a pointer to an array of `eventCount` VkDependencyInfo structures, defining the second synchronization scope.

When `vkCmdWaitEvents2` is submitted to a queue, it inserts memory dependencies according to the elements of `pDependencyInfos` and each corresponding element of `pEvents`. `vkCmdWaitEvents2` **must** not be used to wait on event signal operations occurring on other queues, or signal operations executed by vkCmdSetEvent.

The first synchronization scope and access scope of each memory dependency defined by any element i of `pDependencyInfos` are applied to operations that occurred earlier in submission order than the last event signal operation on element i of `pEvents`.

Signal operations for an event at index i are only included if:

- The event was signaled by a vkCmdSetEvent2 command that occurred earlier in submission order with a `dependencyInfo` parameter exactly equal to the element of `pDependencyInfos` at index i; or
- The event was created without VK_EVENT_CREATE_DEVICE_ONLY_BIT, and the first synchronization scope defined by the element of `pDependencyInfos` at index i only includes host operations (VK_PIPELINE_STAGE_2_HOST_BIT).

The second synchronization scope and access scope of each memory dependency defined by any element i of `pDependencyInfos` are applied to operations that occurred later in submission order than `vkCmdWaitEvents2`.

NOTE vkCmdWaitEvents2 is used with vkCmdSetEvent2 to define a memory dependency between two sets of action commands, roughly in the same way as pipeline barriers, but split into two commands such that work between the two **may** execute

unhindered.

NOTE Applications should be careful to avoid race conditions when using events. There is no direct ordering guarantee between `vkCmdSetEvent2` and vkCmdResetEvent2, vkCmdResetEvent, or vkCmdSetEvent. Another execution dependency (e.g. a pipeline barrier or semaphore with VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT) is needed to prevent such a race condition.

Valid Usage

- VUID-vkCmdWaitEvents2-oldLayout-01181
  If `vkCmdWaitEvents2` is called within a render pass instance, the `oldLayout` and `newLayout` members of any image memory barrier included in this command **must** be equal
- VUID-vkCmdWaitEvents2-srcQueueFamilyIndex-01182
  If `vkCmdWaitEvents2` is called within a render pass instance, the `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members of any memory barrier included in this command **must** be equal
- VUID-vkCmdWaitEvents2-synchronization2-03836
  The `synchronization2` feature **must** be enabled
- VUID-vkCmdWaitEvents2-pEvents-03837
  Members of `pEvents` **must** not have been signaled by vkCmdSetEvent
- VUID-vkCmdWaitEvents2-pEvents-10788
  For each element i of `pEvents`, if the `dependencyFlags` member of the ith element of `pDependencyInfos` does not include VK_DEPENDENCY_ASYMMETRIC_EVENT_BIT_KHR, and if that event is signaled by vkCmdSetEvent2, that command's `dependencyInfo` parameter **must** be exactly equal to the ith element of `pDependencyInfos`
- VUID-vkCmdWaitEvents2-pEvents-10789
  For each element i of `pEvents`, if the `dependencyFlags` member of the ith element of `pDependencyInfos` includes VK_DEPENDENCY_ASYMMETRIC_EVENT_BIT_KHR, that event **must** be signaled by vkCmdSetEvent2 with VK_DEPENDENCY_ASYMMETRIC_EVENT_BIT_KHR
- VUID-vkCmdWaitEvents2-pEvents-10790
  For each element i of `pEvents`, if the `dependencyFlags` member of the ith element of `pDependencyInfos` includes VK_DEPENDENCY_ASYMMETRIC_EVENT_BIT_KHR, the union of `srcStageMask` members of all elements of `pMemoryBarriers`, `pBufferMemoryBarriers`, and `pImageMemoryBarriers` of the ith element of `pDependencyInfos` **must** equal `pDependencyInfos->pMemoryBarriers`[0].`srcStageMask` in the vkCmdSetEvent2 command
- VUID-vkCmdWaitEvents2-pEvents-03839
  For each element i of `pEvents`, if that event is signaled by vkSetEvent, barriers in the ith element of `pDependencyInfos` **must** include only host operations in their first synchronization scope
- VUID-vkCmdWaitEvents2-pEvents-03840
  For each element i of `pEvents`, if barriers in the ith element of `pDependencyInfos` include only host operations, the ith element of `pEvents` **must** be signaled before

vkCmdWaitEvents2 is executed

- VUID-vkCmdWaitEvents2-pEvents-03841
  For each element i of pEvents, if barriers in the ith element of pDependencyInfos do not include host operations, the ith element of pEvents **must** be signaled by a corresponding vkCmdSetEvent2 that occurred earlier in submission order
- VUID-vkCmdWaitEvents2-srcStageMask-03842
  The srcStageMask member of any element of the pMemoryBarriers, pBufferMemoryBarriers, or pImageMemoryBarriers members of pDependencyInfos **must** only include pipeline stages valid for the queue family that was used to create the command pool that commandBuffer was allocated from
- VUID-vkCmdWaitEvents2-dstStageMask-03843
  The dstStageMask member of any element of the pMemoryBarriers, pBufferMemoryBarriers, or pImageMemoryBarriers members of pDependencyInfos **must** only include pipeline stages valid for the queue family that was used to create the command pool that commandBuffer was allocated from
- VUID-vkCmdWaitEvents2-dependencyFlags-10394
  The dependencyFlags member of any element of pDependencyInfos **must** not include any of the following bits:
  - VK_DEPENDENCY_BY_REGION_BIT
  - VK_DEPENDENCY_DEVICE_GROUP_BIT
  - VK_DEPENDENCY_VIEW_LOCAL_BIT
  - VK_DEPENDENCY_FEEDBACK_LOOP_BIT_EXT
- VUID-vkCmdWaitEvents2-maintenance8-10205
  If the maintenance8 feature is not enabled, the dependencyFlags members of any element of pDependencyInfos **must** not include VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR
- VUID-vkCmdWaitEvents2-dependencyFlags-03844
  If this command is called inside a render pass instance, the srcStageMask member of any element of the pMemoryBarriers, pBufferMemoryBarriers, or pImageMemoryBarriers members of pDependencyInfos **must** not include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-vkCmdWaitEvents2-commandBuffer-03846
  commandBuffer's current device mask **must** include exactly one physical device

Valid Usage (Implicit)

- VUID-vkCmdWaitEvents2-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdWaitEvents2-pEvents-parameter
  pEvents **must** be a valid pointer to an array of eventCount valid VkEvent handles
- VUID-vkCmdWaitEvents2-pDependencyInfos-parameter
  pDependencyInfos **must** be a valid pointer to an array of eventCount valid VkDependencyInfo structures

- VUID-vkCmdWaitEvents2-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdWaitEvents2-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdWaitEvents2-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdWaitEvents2-eventCount-arraylength
  `eventCount` **must** be greater than `0`
- VUID-vkCmdWaitEvents2-commonparent
  Both of `commandBuffer`, and the elements of `pEvents` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Synchronization

Conditional Rendering

vkCmdWaitEvents2 is not affected by conditional rendering

To wait for one or more events to enter the signaled state on a device, call:

WARNING	This functionality is superseded by vkCmdWaitEvents2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkCmdWaitEvents(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    eventCount,
    const VkEvent*                              pEvents,
    VkPipelineStageFlags                        srcStageMask,
    VkPipelineStageFlags                        dstStageMask,
    uint32_t                                    memoryBarrierCount,
    const VkMemoryBarrier*                      pMemoryBarriers,
    uint32_t                                    bufferMemoryBarrierCount,
    const VkBufferMemoryBarrier*                pBufferMemoryBarriers,
    uint32_t                                    imageMemoryBarrierCount,
    const VkImageMemoryBarrier*                 pImageMemoryBarriers);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `eventCount` is the length of the `pEvents` array.
- `pEvents` is a pointer to an array of event object handles to wait on.
- `srcStageMask` is a bitmask of VkPipelineStageFlagBits specifying the source stage mask.
- `dstStageMask` is a bitmask of VkPipelineStageFlagBits specifying the destination stage mask.
- `memoryBarrierCount` is the length of the `pMemoryBarriers` array.
- `pMemoryBarriers` is a pointer to an array of VkMemoryBarrier structures.
- `bufferMemoryBarrierCount` is the length of the `pBufferMemoryBarriers` array.
- `pBufferMemoryBarriers` is a pointer to an array of VkBufferMemoryBarrier structures.
- `imageMemoryBarrierCount` is the length of the `pImageMemoryBarriers` array.
- `pImageMemoryBarriers` is a pointer to an array of VkImageMemoryBarrier structures.

`vkCmdWaitEvents` is largely similar to vkCmdWaitEvents2, but **can** only wait on signal operations defined by vkCmdSetEvent. As vkCmdSetEvent does not define any access scopes, `vkCmdWaitEvents` defines the first access scope for each event signal operation in addition to its own access scopes.

NOTE	Since vkCmdSetEvent does not have any dependency information beyond a stage mask, implementations do not have the same opportunity to perform availability and visibility operations or image layout transitions in advance as they do with vkCmdSetEvent2 and vkCmdWaitEvents2.

When `vkCmdWaitEvents` is submitted to a queue, it defines a memory dependency between prior event signal operations on the same queue or the host, and subsequent commands. `vkCmdWaitEvents` **must** not be used to wait on event signal operations occurring on other queues.

The first synchronization scope only includes event signal operations that operate on members of `pEvents`, and the operations that happened-before the event signal operations. Event signal

operations performed by vkCmdSetEvent that occur earlier in submission order are included in the first synchronization scope, if the logically latest pipeline stage in their stageMask parameter is logically earlier than or equal to the logically latest pipeline stage in srcStageMask. Event signal operations performed by vkSetEvent are only included in the first synchronization scope if VK_PIPELINE_STAGE_HOST_BIT is included in srcStageMask.

The second synchronization scope includes all commands that occur later in submission order. The second synchronization scope is limited to operations on the pipeline stages determined by the destination stage mask specified by dstStageMask.

The first access scope is limited to accesses in the pipeline stages determined by the source stage mask specified by srcStageMask. Within that, the first access scope only includes the first access scopes defined by elements of the pMemoryBarriers, pBufferMemoryBarriers and pImageMemoryBarriers arrays, which each define a set of memory barriers. If no memory barriers are specified, then the first access scope includes no accesses.

The second access scope is limited to accesses in the pipeline stages determined by the destination stage mask specified by dstStageMask. Within that, the second access scope only includes the second access scopes defined by elements of the pMemoryBarriers, pBufferMemoryBarriers and pImageMemoryBarriers arrays, which each define a set of memory barriers. If no memory barriers are specified, then the second access scope includes no accesses.

Valid Usage

- VUID-vkCmdWaitEvents-srcStageMask-04090
  If the geometryShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT
- VUID-vkCmdWaitEvents-srcStageMask-04091
  If the tessellationShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-vkCmdWaitEvents-srcStageMask-04092
  If the conditionalRendering feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT
- VUID-vkCmdWaitEvents-srcStageMask-04093
  If the fragmentDensityMap feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-vkCmdWaitEvents-srcStageMask-04094
  If the transformFeedback feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-vkCmdWaitEvents-srcStageMask-04095
  If the meshShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT
- VUID-vkCmdWaitEvents-srcStageMask-04096
  If the taskShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT

- VUID-vkCmdWaitEvents-srcStageMask-07319
  If the attachmentFragmentShadingRate feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdWaitEvents-srcStageMask-03937
  If the synchronization2 feature is not enabled, srcStageMask **must** not be 0
- VUID-vkCmdWaitEvents-srcStageMask-07950
  If the rayTracingPipeline feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR
- VUID-vkCmdWaitEvents-srcStageMask-10754
  If the accelerationStructure feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-vkCmdWaitEvents-srcAccessMask-06257
  If the rayQuery feature is not enabled and a memory barrier srcAccessMask includes VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, or VK_PIPELINE_STAGE_ALL_COMMANDS_BIT
- VUID-vkCmdWaitEvents-dstStageMask-04090
  If the geometryShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT
- VUID-vkCmdWaitEvents-dstStageMask-04091
  If the tessellationShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-vkCmdWaitEvents-dstStageMask-04092
  If the conditionalRendering feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT
- VUID-vkCmdWaitEvents-dstStageMask-04093
  If the fragmentDensityMap feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-vkCmdWaitEvents-dstStageMask-04094
  If the transformFeedback feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-vkCmdWaitEvents-dstStageMask-04095
  If the meshShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT
- VUID-vkCmdWaitEvents-dstStageMask-04096
  If the taskShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT
- VUID-vkCmdWaitEvents-dstStageMask-07319
  If the attachmentFragmentShadingRate feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR

- VUID-vkCmdWaitEvents-dstStageMask-03937
  If the synchronization2 feature is not enabled, dstStageMask **must** not be 0
- VUID-vkCmdWaitEvents-dstStageMask-07950
  If the rayTracingPipeline feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR
- VUID-vkCmdWaitEvents-dstStageMask-10754
  If the accelerationStructure feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-vkCmdWaitEvents-dstAccessMask-06257
  If the rayQuery feature is not enabled and a memory barrier dstAccessMask includes VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, or VK_PIPELINE_STAGE_ALL_COMMANDS_BIT
- VUID-vkCmdWaitEvents-srcAccessMask-02815
  The srcAccessMask member of each element of pMemoryBarriers **must** only include access flags that are supported by one or more of the pipeline stages in srcStageMask, as specified in the table of supported access types
- VUID-vkCmdWaitEvents-dstAccessMask-02816
  The dstAccessMask member of each element of pMemoryBarriers **must** only include access flags that are supported by one or more of the pipeline stages in dstStageMask, as specified in the table of supported access types
- VUID-vkCmdWaitEvents-pBufferMemoryBarriers-02817
  For each element of pBufferMemoryBarriers, if its srcQueueFamilyIndex and dstQueueFamilyIndex members are equal, or if its srcQueueFamilyIndex is the queue family index that was used to create the command pool that commandBuffer was allocated from, then its srcAccessMask member **must** only contain access flags that are supported by one or more of the pipeline stages in srcStageMask, as specified in the table of supported access types
- VUID-vkCmdWaitEvents-pBufferMemoryBarriers-02818
  For each element of pBufferMemoryBarriers, if its srcQueueFamilyIndex and dstQueueFamilyIndex members are equal, or if its dstQueueFamilyIndex is the queue family index that was used to create the command pool that commandBuffer was allocated from, then its dstAccessMask member **must** only contain access flags that are supported by one or more of the pipeline stages in dstStageMask, as specified in the table of supported access types
- VUID-vkCmdWaitEvents-pImageMemoryBarriers-02819
  For each element of pImageMemoryBarriers, if its srcQueueFamilyIndex and dstQueueFamilyIndex members are equal, or if its srcQueueFamilyIndex is the queue family index that was used to create the command pool that commandBuffer was allocated from, then its srcAccessMask member **must** only contain access flags that are supported by one or more of the pipeline stages in srcStageMask, as specified in the table of supported access types

- VUID-vkCmdWaitEvents-pImageMemoryBarriers-02820
  For each element of pImageMemoryBarriers, if its srcQueueFamilyIndex and dstQueueFamilyIndex members are equal, or if its dstQueueFamilyIndex is the queue family index that was used to create the command pool that commandBuffer was allocated from, then its dstAccessMask member **must** only contain access flags that are supported by one or more of the pipeline stages in dstStageMask, as specified in the table of supported access types
- VUID-vkCmdWaitEvents-oldLayout-01181
  If vkCmdWaitEvents is called within a render pass instance, the oldLayout and newLayout members of any image memory barrier included in this command **must** be equal
- VUID-vkCmdWaitEvents-srcQueueFamilyIndex-01182
  If vkCmdWaitEvents is called within a render pass instance, the srcQueueFamilyIndex and dstQueueFamilyIndex members of any memory barrier included in this command **must** be equal
- VUID-vkCmdWaitEvents-srcStageMask-06459
  Any pipeline stage included in srcStageMask **must** be supported by the capabilities of the queue family specified by the queueFamilyIndex member of the VkCommandPoolCreateInfo structure that was used to create the VkCommandPool that commandBuffer was allocated from, as specified in the table of supported pipeline stages
- VUID-vkCmdWaitEvents-dstStageMask-06460
  Any pipeline stage included in dstStageMask **must** be supported by the capabilities of the queue family specified by the queueFamilyIndex member of the VkCommandPoolCreateInfo structure that was used to create the VkCommandPool that commandBuffer was allocated from, as specified in the table of supported pipeline stages
- VUID-vkCmdWaitEvents-srcStageMask-01158
  srcStageMask **must** be the bitwise OR of the stageMask parameter used in previous calls to vkCmdSetEvent with any of the elements of pEvents and VK_PIPELINE_STAGE_HOST_BIT if any of the elements of pEvents was set using vkSetEvent
- VUID-vkCmdWaitEvents-srcStageMask-07308
  If this command is called inside a render pass instance, srcStageMask **must** not include VK_PIPELINE_STAGE_HOST_BIT
- VUID-vkCmdWaitEvents-srcQueueFamilyIndex-02803
  The srcQueueFamilyIndex and dstQueueFamilyIndex members of any element of pBufferMemoryBarriers or pImageMemoryBarriers **must** be equal
- VUID-vkCmdWaitEvents-commandBuffer-01167
  commandBuffer's current device mask **must** include exactly one physical device
- VUID-vkCmdWaitEvents-pEvents-03847
  Elements of pEvents **must** not have been signaled by vkCmdSetEvent2

Valid Usage (Implicit)

- VUID-vkCmdWaitEvents-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle

- VUID-vkCmdWaitEvents-pEvents-parameter
  `pEvents` **must** be a valid pointer to an array of `eventCount` valid VkEvent handles
- VUID-vkCmdWaitEvents-srcStageMask-parameter
  If `srcStageMask` is not `0`, `srcStageMask` **must** be a valid combination of VkPipelineStageFlagBits values
- VUID-vkCmdWaitEvents-dstStageMask-parameter
  If `dstStageMask` is not `0`, `dstStageMask` **must** be a valid combination of VkPipelineStageFlagBits values
- VUID-vkCmdWaitEvents-pMemoryBarriers-parameter
  If `memoryBarrierCount` is not `0`, `pMemoryBarriers` **must** be a valid pointer to an array of `memoryBarrierCount` valid VkMemoryBarrier structures
- VUID-vkCmdWaitEvents-pBufferMemoryBarriers-parameter
  If `bufferMemoryBarrierCount` is not `0`, `pBufferMemoryBarriers` **must** be a valid pointer to an array of `bufferMemoryBarrierCount` valid VkBufferMemoryBarrier structures
- VUID-vkCmdWaitEvents-pImageMemoryBarriers-parameter
  If `imageMemoryBarrierCount` is not `0`, `pImageMemoryBarriers` **must** be a valid pointer to an array of `imageMemoryBarrierCount` valid VkImageMemoryBarrier structures
- VUID-vkCmdWaitEvents-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdWaitEvents-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdWaitEvents-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdWaitEvents-eventCount-arraylength
  `eventCount` **must** be greater than `0`
- VUID-vkCmdWaitEvents-commonparent
  Both of `commandBuffer`, and the elements of `pEvents` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Synchronization

Conditional Rendering

vkCmdWaitEvents is not affected by conditional rendering

7.6. Pipeline Barriers

To record a pipeline barrier, call:

```
// Provided by VK_VERSION_1_3
void vkCmdPipelineBarrier2(
    VkCommandBuffer                             commandBuffer,
    const VkDependencyInfo*                     pDependencyInfo);
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to vkCmdPipelineBarrier2
void vkCmdPipelineBarrier2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkDependencyInfo*                     pDependencyInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `pDependencyInfo` is a pointer to a VkDependencyInfo structure defining the scopes of this operation.

When vkCmdPipelineBarrier2 is submitted to a queue, it defines memory dependencies between commands that were submitted to the same queue before it, and those submitted to the same queue after it.

The first synchronization scope and access scope of each memory dependency defined by

pDependencyInfo are applied to operations that occurred earlier in submission order.

The second synchronization scope and access scope of each memory dependency defined by pDependencyInfo are applied to operations that occurred later in submission order.

If vkCmdPipelineBarrier2 is recorded within a render pass instance, the synchronization scopes are limited to a subset of operations within the same subpass or render pass instance.

Valid Usage

- VUID-vkCmdPipelineBarrier2-None-07889
  If vkCmdPipelineBarrier2 is called within a render pass instance using a VkRenderPass object, the render pass **must** have been created with at least one subpass dependency that expresses a dependency from the current subpass to itself, does not include VK_DEPENDENCY_BY_REGION_BIT if this command does not, does not include VK_DEPENDENCY_VIEW_LOCAL_BIT if this command does not, and has synchronization scopes and access scopes that are all supersets of the scopes defined in this command
- VUID-vkCmdPipelineBarrier2-bufferMemoryBarrierCount-01178
  If vkCmdPipelineBarrier2 is called within a render pass instance using a VkRenderPass object, it **must** not include any buffer memory barriers
- VUID-vkCmdPipelineBarrier2-image-04073
  If vkCmdPipelineBarrier2 is called within a render pass instance using a VkRenderPass object, the image member of any image memory barrier included in this command **must** be an attachment used in the current subpass both as an input attachment, and as either a color, or depth/stencil attachment
- VUID-vkCmdPipelineBarrier2-None-07890
  If vkCmdPipelineBarrier2 is called within a render pass instance, and the source stage masks of any memory barriers include framebuffer-space stages, destination stage masks of all memory barriers **must** only include framebuffer-space stages
- VUID-vkCmdPipelineBarrier2-dependencyFlags-07891
  If vkCmdPipelineBarrier2 is called within a render pass instance, and the source stage masks of any memory barriers include framebuffer-space stages, then dependencyFlags **must** include VK_DEPENDENCY_BY_REGION_BIT
- VUID-vkCmdPipelineBarrier2-None-07892
  If vkCmdPipelineBarrier2 is called within a render pass instance, the source and destination stage masks of any memory barriers **must** only include graphics pipeline stages
- VUID-vkCmdPipelineBarrier2-dependencyFlags-01186
  If vkCmdPipelineBarrier2 is called outside of a render pass instance, the dependency flags **must** not include VK_DEPENDENCY_VIEW_LOCAL_BIT
- VUID-vkCmdPipelineBarrier2-None-07893
  If vkCmdPipelineBarrier2 is called inside a render pass instance, and there is more than one view in the current subpass, dependency flags **must** include VK_DEPENDENCY_VIEW_LOCAL_BIT
- VUID-vkCmdPipelineBarrier2-None-09553

If none of the shaderTileImageColorReadAccess, shaderTileImageStencilReadAccess, or shaderTileImageDepthReadAccess features are enabled, and the dynamicRenderingLocalRead feature is not enabled, vkCmdPipelineBarrier2 **must** not be called within a render pass instance started with vkCmdBeginRendering

- VUID-vkCmdPipelineBarrier2-None-09554
  If the dynamicRenderingLocalRead feature is not enabled, and vkCmdPipelineBarrier2 is called within a render pass instance started with vkCmdBeginRendering, there **must** be no buffer or image memory barriers specified by this command
- VUID-vkCmdPipelineBarrier2-None-09586
  If the dynamicRenderingLocalRead feature is not enabled, and vkCmdPipelineBarrier2 is called within a render pass instance started with vkCmdBeginRendering, memory barriers specified by this command **must** only include VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT, VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT, VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT, or VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT in their access masks
- VUID-vkCmdPipelineBarrier2-image-09555
  If vkCmdPipelineBarrier2 is called within a render pass instance started with vkCmdBeginRendering, and the image member of any image memory barrier is used as an attachment in the current render pass instance, it **must** be in the VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ or VK_IMAGE_LAYOUT_GENERAL layout
- VUID-vkCmdPipelineBarrier2-srcStageMask-09556
  If vkCmdPipelineBarrier2 is called within a render pass instance started with vkCmdBeginRendering, this command **must** only specify framebuffer-space stages in srcStageMask and dstStageMask
- VUID-vkCmdPipelineBarrier2-oldLayout-01181
  If vkCmdPipelineBarrier2 is called within a render pass instance, the oldLayout and newLayout members of any image memory barrier included in this command **must** be equal
- VUID-vkCmdPipelineBarrier2-srcQueueFamilyIndex-01182
  If vkCmdPipelineBarrier2 is called within a render pass instance, the srcQueueFamilyIndex and dstQueueFamilyIndex members of any memory barrier included in this command **must** be equal
- VUID-vkCmdPipelineBarrier2-synchronization2-03848
  The synchronization2 feature **must** be enabled
- VUID-vkCmdPipelineBarrier2-srcStageMask-09673
  The srcStageMask member of any element of the pMemoryBarriers member of pDependencyInfo **must** only include pipeline stages valid for the queue family that was used to create the command pool that commandBuffer was allocated from
- VUID-vkCmdPipelineBarrier2-dstStageMask-09674
  The dstStageMask member of any element of the pMemoryBarriers member of pDependencyInfo **must** only include pipeline stages valid for the queue family that was used to create the command pool that commandBuffer was allocated from

- VUID-vkCmdPipelineBarrier2-srcStageMask-09675
  If a buffer or image memory barrier does not specify an acquire operation, or if it does but `pDependencyInfo->dependencyFlags` includes VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR, the respective `srcStageMask` member of the element of the `pBufferMemoryBarriers` or `pImageMemoryBarriers` members of `pDependencyInfo` **must** only include pipeline stages valid for the queue family that was used to create the command pool that `commandBuffer` was allocated from
- VUID-vkCmdPipelineBarrier2-dstStageMask-09676
  If a buffer or image memory barrier does not specify an release operation, or if it does but `pDependencyInfo->dependencyFlags` includes VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR, the respective `dstStageMask` member of the element of the `pBufferMemoryBarriers` or `pImageMemoryBarriers` members of `pDependencyInfo` **must** only include pipeline stages valid for the queue family that was used to create the command pool that `commandBuffer` was allocated from
- VUID-vkCmdPipelineBarrier2-srcQueueFamilyIndex-10387
  If a buffer or image memory barrier specifies a queue family ownership transfer operation, either the `srcQueueFamilyIndex` or `dstQueueFamilyIndex` member of the element of the `pBufferMemoryBarriers` or `pImageMemoryBarriers` members of `pDependencyInfo` and the queue family index that was used to create the command pool that `commandBuffer` was allocated from **must** be equal

Valid Usage (Implicit)

- VUID-vkCmdPipelineBarrier2-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdPipelineBarrier2-pDependencyInfo-parameter
  `pDependencyInfo` **must** be a valid pointer to a valid VkDependencyInfo structure
- VUID-vkCmdPipelineBarrier2-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdPipelineBarrier2-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_TRANSFER_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdPipelineBarrier2-suspended
  This command **must** not be called between suspended render pass instances

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally

synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Synchronization

Conditional Rendering

vkCmdPipelineBarrier2 is not affected by conditional rendering

To record a pipeline barrier, call:

WARNING This functionality is superseded by vkCmdPipelineBarrier2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkCmdPipelineBarrier(
    VkCommandBuffer                             commandBuffer,
    VkPipelineStageFlags                        srcStageMask,
    VkPipelineStageFlags                        dstStageMask,
    VkDependencyFlags                           dependencyFlags,
    uint32_t                                    memoryBarrierCount,
    const VkMemoryBarrier*                      pMemoryBarriers,
    uint32_t                                    bufferMemoryBarrierCount,
    const VkBufferMemoryBarrier*                pBufferMemoryBarriers,
    uint32_t                                    imageMemoryBarrierCount,
    const VkImageMemoryBarrier*                 pImageMemoryBarriers);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `srcStageMask` is a bitmask of VkPipelineStageFlagBits specifying the source stages.

- `dstStageMask` is a bitmask of VkPipelineStageFlagBits specifying the destination stages.
- `dependencyFlags` is a bitmask of VkDependencyFlagBits specifying how execution and memory dependencies are formed.
- `memoryBarrierCount` is the length of the `pMemoryBarriers` array.
- `pMemoryBarriers` is a pointer to an array of VkMemoryBarrier structures.
- `bufferMemoryBarrierCount` is the length of the `pBufferMemoryBarriers` array.
- `pBufferMemoryBarriers` is a pointer to an array of VkBufferMemoryBarrier structures.
- `imageMemoryBarrierCount` is the length of the `pImageMemoryBarriers` array.
- `pImageMemoryBarriers` is a pointer to an array of VkImageMemoryBarrier structures.

`vkCmdPipelineBarrier` operates almost identically to vkCmdPipelineBarrier2, except that the scopes and barriers are defined as direct parameters rather than being defined by a VkDependencyInfo.

When vkCmdPipelineBarrier is submitted to a queue, it defines a memory dependency between commands that were submitted to the same queue before it, and those submitted to the same queue after it.

If vkCmdPipelineBarrier was recorded outside a render pass instance, the first synchronization scope includes all commands that occur earlier in submission order. If vkCmdPipelineBarrier was recorded inside a render pass instance, the first synchronization scope includes only commands that occur earlier in submission order within the same subpass. In either case, the first synchronization scope is limited to operations on the pipeline stages determined by the source stage mask specified by `srcStageMask`.

If vkCmdPipelineBarrier was recorded outside a render pass instance, the second synchronization scope includes all commands that occur later in submission order. If vkCmdPipelineBarrier was recorded inside a render pass instance, the second synchronization scope includes only commands that occur later in submission order within the same subpass. In either case, the second synchronization scope is limited to operations on the pipeline stages determined by the destination stage mask specified by `dstStageMask`.

The first access scope is limited to accesses in the pipeline stages determined by the source stage mask specified by `srcStageMask`. Within that, the first access scope only includes the first access scopes defined by elements of the `pMemoryBarriers`, `pBufferMemoryBarriers` and `pImageMemoryBarriers` arrays, which each define a set of memory barriers. If no memory barriers are specified, then the first access scope includes no accesses.

The second access scope is limited to accesses in the pipeline stages determined by the destination stage mask specified by `dstStageMask`. Within that, the second access scope only includes the second access scopes defined by elements of the `pMemoryBarriers`, `pBufferMemoryBarriers` and `pImageMemoryBarriers` arrays, which each define a set of memory barriers. If no memory barriers are specified, then the second access scope includes no accesses.

If `dependencyFlags` includes VK_DEPENDENCY_BY_REGION_BIT, then any dependency between framebuffer-space pipeline stages is framebuffer-local - otherwise it is framebuffer-global.

Valid Usage

- VUID-vkCmdPipelineBarrier-srcStageMask-04090
  If the `geometryShader` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT
- VUID-vkCmdPipelineBarrier-srcStageMask-04091
  If the `tessellationShader` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-vkCmdPipelineBarrier-srcStageMask-04092
  If the `conditionalRendering` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT
- VUID-vkCmdPipelineBarrier-srcStageMask-04093
  If the `fragmentDensityMap` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-vkCmdPipelineBarrier-srcStageMask-04094
  If the `transformFeedback` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-vkCmdPipelineBarrier-srcStageMask-04095
  If the `meshShader` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT
- VUID-vkCmdPipelineBarrier-srcStageMask-04096
  If the `taskShader` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT
- VUID-vkCmdPipelineBarrier-srcStageMask-07319
  If the `attachmentFragmentShadingRate` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdPipelineBarrier-srcStageMask-03937
  If the `synchronization2` feature is not enabled, `srcStageMask` **must** not be `0`
- VUID-vkCmdPipelineBarrier-srcStageMask-07950
  If the `rayTracingPipeline` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR
- VUID-vkCmdPipelineBarrier-srcStageMask-10754
  If the `accelerationStructure` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-vkCmdPipelineBarrier-srcAccessMask-06257
  If the `rayQuery` feature is not enabled and a memory barrier `srcAccessMask` includes VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR, `srcStageMask` **must** include VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, or VK_PIPELINE_STAGE_ALL_COMMANDS_BIT
- VUID-vkCmdPipelineBarrier-dstStageMask-04090

If the geometryShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT

- VUID-vkCmdPipelineBarrier-dstStageMask-04091
  If the tessellationShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-vkCmdPipelineBarrier-dstStageMask-04092
  If the conditionalRendering feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT
- VUID-vkCmdPipelineBarrier-dstStageMask-04093
  If the fragmentDensityMap feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-vkCmdPipelineBarrier-dstStageMask-04094
  If the transformFeedback feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-vkCmdPipelineBarrier-dstStageMask-04095
  If the meshShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT
- VUID-vkCmdPipelineBarrier-dstStageMask-04096
  If the taskShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT
- VUID-vkCmdPipelineBarrier-dstStageMask-07319
  If the attachmentFragmentShadingRate feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdPipelineBarrier-dstStageMask-03937
  If the synchronization2 feature is not enabled, dstStageMask **must** not be 0
- VUID-vkCmdPipelineBarrier-dstStageMask-07950
  If the rayTracingPipeline feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR
- VUID-vkCmdPipelineBarrier-dstStageMask-10754
  If the accelerationStructure feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-vkCmdPipelineBarrier-dstAccessMask-06257
  If the rayQuery feature is not enabled and a memory barrier dstAccessMask includes VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, or VK_PIPELINE_STAGE_ALL_COMMANDS_BIT
- VUID-vkCmdPipelineBarrier-srcAccessMask-02815
  The srcAccessMask member of each element of pMemoryBarriers **must** only include access flags that are supported by one or more of the pipeline stages in srcStageMask, as specified in the table of supported access types

- VUID-vkCmdPipelineBarrier-dstAccessMask-02816
  The `dstAccessMask` member of each element of `pMemoryBarriers` **must** only include access flags that are supported by one or more of the pipeline stages in `dstStageMask`, as specified in the table of supported access types
- VUID-vkCmdPipelineBarrier-pBufferMemoryBarriers-02817
  For each element of `pBufferMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `srcQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `srcAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `srcStageMask`, as specified in the table of supported access types
- VUID-vkCmdPipelineBarrier-pBufferMemoryBarriers-02818
  For each element of `pBufferMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `dstQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `dstAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `dstStageMask`, as specified in the table of supported access types
- VUID-vkCmdPipelineBarrier-pImageMemoryBarriers-02819
  For each element of `pImageMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `srcQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `srcAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `srcStageMask`, as specified in the table of supported access types
- VUID-vkCmdPipelineBarrier-pImageMemoryBarriers-02820
  For each element of `pImageMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `dstQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `dstAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `dstStageMask`, as specified in the table of supported access types
- VUID-vkCmdPipelineBarrier-oldLayout-01181
  If `vkCmdPipelineBarrier` is called within a render pass instance, the `oldLayout` and `newLayout` members of any image memory barrier included in this command **must** be equal
- VUID-vkCmdPipelineBarrier-srcQueueFamilyIndex-01182
  If `vkCmdPipelineBarrier` is called within a render pass instance, the `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members of any memory barrier included in this command **must** be equal
- VUID-vkCmdPipelineBarrier-None-07889
  If `vkCmdPipelineBarrier` is called within a render pass instance using a VkRenderPass object, the render pass **must** have been created with at least one subpass dependency that expresses a dependency from the current subpass to itself, does not include

VK_DEPENDENCY_BY_REGION_BIT if this command does not, does not include VK_DEPENDENCY_VIEW_LOCAL_BIT if this command does not, and has synchronization scopes and access scopes that are all supersets of the scopes defined in this command

- VUID-vkCmdPipelineBarrier-bufferMemoryBarrierCount-01178
  If vkCmdPipelineBarrier is called within a render pass instance using a VkRenderPass object, it **must** not include any buffer memory barriers
- VUID-vkCmdPipelineBarrier-image-04073
  If vkCmdPipelineBarrier is called within a render pass instance using a VkRenderPass object, the image member of any image memory barrier included in this command **must** be an attachment used in the current subpass both as an input attachment, and as either a color, or depth/stencil attachment
- VUID-vkCmdPipelineBarrier-None-07890
  If vkCmdPipelineBarrier is called within a render pass instance, and the source stage masks of any memory barriers include framebuffer-space stages, destination stage masks of all memory barriers **must** only include framebuffer-space stages
- VUID-vkCmdPipelineBarrier-dependencyFlags-07891
  If vkCmdPipelineBarrier is called within a render pass instance, and the source stage masks of any memory barriers include framebuffer-space stages, then dependencyFlags **must** include VK_DEPENDENCY_BY_REGION_BIT
- VUID-vkCmdPipelineBarrier-None-07892
  If vkCmdPipelineBarrier is called within a render pass instance, the source and destination stage masks of any memory barriers **must** only include graphics pipeline stages
- VUID-vkCmdPipelineBarrier-dependencyFlags-01186
  If vkCmdPipelineBarrier is called outside of a render pass instance, the dependency flags **must** not include VK_DEPENDENCY_VIEW_LOCAL_BIT
- VUID-vkCmdPipelineBarrier-None-07893
  If vkCmdPipelineBarrier is called inside a render pass instance, and there is more than one view in the current subpass, dependency flags **must** include VK_DEPENDENCY_VIEW_LOCAL_BIT
- VUID-vkCmdPipelineBarrier-None-09553
  If none of the shaderTileImageColorReadAccess, shaderTileImageStencilReadAccess, or shaderTileImageDepthReadAccess features are enabled, and the dynamicRenderingLocalRead feature is not enabled, vkCmdPipelineBarrier **must** not be called within a render pass instance started with vkCmdBeginRendering
- VUID-vkCmdPipelineBarrier-None-09554
  If the dynamicRenderingLocalRead feature is not enabled, and vkCmdPipelineBarrier is called within a render pass instance started with vkCmdBeginRendering, there **must** be no buffer or image memory barriers specified by this command
- VUID-vkCmdPipelineBarrier-None-09586
  If the dynamicRenderingLocalRead feature is not enabled, and vkCmdPipelineBarrier is called within a render pass instance started with vkCmdBeginRendering, memory barriers specified by this command **must** only include VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT, VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT,

VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT, or VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT in their access masks

- VUID-vkCmdPipelineBarrier-image-09555
  If `vkCmdPipelineBarrier` is called within a render pass instance started with vkCmdBeginRendering, and the `image` member of any image memory barrier is used as an attachment in the current render pass instance, it **must** be in the VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ or VK_IMAGE_LAYOUT_GENERAL layout
- VUID-vkCmdPipelineBarrier-srcStageMask-09556
  If `vkCmdPipelineBarrier` is called within a render pass instance started with vkCmdBeginRendering, this command **must** only specify framebuffer-space stages in `srcStageMask` and `dstStageMask`
- VUID-vkCmdPipelineBarrier-oldLayout-10758
  If called within a render pass instance using a VkRenderPass object, the `oldLayout` member of any image memory barrier included in this command **must** be equal to the layout that the corresponding attachment uses during the subpass
- VUID-vkCmdPipelineBarrier-oldLayout-10759
  If called within a render pass instance started with vkCmdBeginRendering, the `oldLayout` member of any image memory barrier included in this command **must** be equal to the layout that the corresponding attachment uses during the render pass instance
- VUID-vkCmdPipelineBarrier-srcStageMask-06461
  Any pipeline stage included in `srcStageMask` **must** be supported by the capabilities of the queue family specified by the `queueFamilyIndex` member of the VkCommandPoolCreateInfo structure that was used to create the `VkCommandPool` that `commandBuffer` was allocated from, as specified in the table of supported pipeline stages
- VUID-vkCmdPipelineBarrier-dstStageMask-06462
  Any pipeline stage included in `dstStageMask` **must** be supported by the capabilities of the queue family specified by the `queueFamilyIndex` member of the VkCommandPoolCreateInfo structure that was used to create the `VkCommandPool` that `commandBuffer` was allocated from, as specified in the table of supported pipeline stages
- VUID-vkCmdPipelineBarrier-srcStageMask-09633
  If either `srcStageMask` or `dstStageMask` includes VK_PIPELINE_STAGE_HOST_BIT, for each element of `pImageMemoryBarriers`, `srcQueueFamilyIndex` and `dstQueueFamilyIndex` **must** be equal
- VUID-vkCmdPipelineBarrier-srcStageMask-09634
  If either `srcStageMask` or `dstStageMask` includes VK_PIPELINE_STAGE_HOST_BIT, for each element of `pBufferMemoryBarriers`, `srcQueueFamilyIndex` and `dstQueueFamilyIndex` **must** be equal
- VUID-vkCmdPipelineBarrier-srcQueueFamilyIndex-10388
  If a buffer or image memory barrier specifies a queue family ownership transfer operation, either the `srcQueueFamilyIndex` or `dstQueueFamilyIndex` member and the queue family index that was used to create the command pool that `commandBuffer` was allocated from **must** be equal
- VUID-vkCmdPipelineBarrier-maintenance8-10206
  If the `maintenance8` feature is not enabled, `dependencyFlags` **must** not include

VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR

Valid Usage (Implicit)

- VUID-vkCmdPipelineBarrier-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdPipelineBarrier-srcStageMask-parameter
  If `srcStageMask` is not `0`, `srcStageMask` **must** be a valid combination of VkPipelineStageFlagBits values
- VUID-vkCmdPipelineBarrier-dstStageMask-parameter
  If `dstStageMask` is not `0`, `dstStageMask` **must** be a valid combination of VkPipelineStageFlagBits values
- VUID-vkCmdPipelineBarrier-dependencyFlags-parameter
  If `dependencyFlags` is not `0`, `dependencyFlags` **must** be a valid combination of VkDependencyFlagBits values
- VUID-vkCmdPipelineBarrier-pMemoryBarriers-parameter
  If `memoryBarrierCount` is not `0`, `pMemoryBarriers` **must** be a valid pointer to an array of `memoryBarrierCount` valid VkMemoryBarrier structures
- VUID-vkCmdPipelineBarrier-pBufferMemoryBarriers-parameter
  If `bufferMemoryBarrierCount` is not `0`, `pBufferMemoryBarriers` **must** be a valid pointer to an array of `bufferMemoryBarrierCount` valid VkBufferMemoryBarrier structures
- VUID-vkCmdPipelineBarrier-pImageMemoryBarriers-parameter
  If `imageMemoryBarrierCount` is not `0`, `pImageMemoryBarriers` **must** be a valid pointer to an array of `imageMemoryBarrierCount` valid VkImageMemoryBarrier structures
- VUID-vkCmdPipelineBarrier-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdPipelineBarrier-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_TRANSFER_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdPipelineBarrier-suspended
  This command **must** not be called between suspended render pass instances

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Synchronization

Conditional Rendering

vkCmdPipelineBarrier is not affected by conditional rendering

Bits which **can** be set in vkCmdPipelineBarrier::dependencyFlags, specifying how execution and memory dependencies are formed, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkDependencyFlagBits {
    VK_DEPENDENCY_BY_REGION_BIT = 0x00000001,
  // Provided by VK_VERSION_1_1
    VK_DEPENDENCY_DEVICE_GROUP_BIT = 0x00000004,
  // Provided by VK_VERSION_1_1
    VK_DEPENDENCY_VIEW_LOCAL_BIT = 0x00000002,
  // Provided by VK_EXT_attachment_feedback_loop_layout
    VK_DEPENDENCY_FEEDBACK_LOOP_BIT_EXT = 0x00000008,
  // Provided by VK_KHR_maintenance8
    VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR = 0x00000020,
  // Provided by VK_KHR_maintenance9
    VK_DEPENDENCY_ASYMMETRIC_EVENT_BIT_KHR = 0x00000040,
  // Provided by VK_KHR_multiview
    VK_DEPENDENCY_VIEW_LOCAL_BIT_KHR = VK_DEPENDENCY_VIEW_LOCAL_BIT,
  // Provided by VK_KHR_device_group
    VK_DEPENDENCY_DEVICE_GROUP_BIT_KHR = VK_DEPENDENCY_DEVICE_GROUP_BIT,
} VkDependencyFlagBits;
```

- VK_DEPENDENCY_BY_REGION_BIT specifies that dependencies will be split into multiple framebuffer-local regions according to the (x,y,layer,sample) coordinates.

- VK_DEPENDENCY_VIEW_LOCAL_BIT specifies that dependencies will be split into multiple framebuffer-local regions according to the view.
- VK_DEPENDENCY_DEVICE_GROUP_BIT specifies that dependencies are non-device-local.
- VK_DEPENDENCY_FEEDBACK_LOOP_BIT_EXT specifies that the render pass will write to and read from the same image with feedback loop enabled.
- VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR specifies that source and destination stages are not ignored when performing a queue family ownership transfer.
- VK_DEPENDENCY_ASYMMETRIC_EVENT_BIT_KHR specifies that the access scopes of vkCmdSetEvent2 and vkCmdWaitEvents2 do not need to match for a given event when it is specified in both commands, and the access scope of vkCmdSetEvent2 is empty.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDependencyFlags;
```

VkDependencyFlags is a bitmask type for setting a mask of zero or more VkDependencyFlagBits.

7.7. Memory Barriers

Memory barriers are used to explicitly control access to buffer and image subresource ranges. Memory barriers are used to transfer ownership between queue families, change image layouts, and define availability and visibility operations. They explicitly define the access types and buffer and image subresource ranges that are included in the access scopes of a memory dependency that is created by a synchronization command that includes them.

7.7.1. Global Memory Barriers

Global memory barriers apply to memory accesses involving all memory objects that exist at the time of its execution.

The VkMemoryBarrier2 structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkMemoryBarrier2 {
    VkStructureType          sType;
    const void*              pNext;
    VkPipelineStageFlags2    srcStageMask;
    VkAccessFlags2           srcAccessMask;
    VkPipelineStageFlags2    dstStageMask;
    VkAccessFlags2           dstAccessMask;
} VkMemoryBarrier2;
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to VkMemoryBarrier2
typedef VkMemoryBarrier2 VkMemoryBarrier2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcStageMask` is a VkPipelineStageFlags2 mask of pipeline stages to be included in the first synchronization scope.
- `srcAccessMask` is a VkAccessFlags2 mask of access flags to be included in the first access scope.
- `dstStageMask` is a VkPipelineStageFlags2 mask of pipeline stages to be included in the second synchronization scope.
- `dstAccessMask` is a VkAccessFlags2 mask of access flags to be included in the second access scope.

This structure defines a memory dependency affecting all device memory.

The first synchronization scope and access scope described by this structure include only operations and memory accesses specified by the source stage mask and source access mask.

The second synchronization scope and access scope described by this structure include only operations and memory accesses specified by the destination stage mask and destination access mask.

Valid Usage

- VUID-VkMemoryBarrier2-srcStageMask-03929
  If the `geometryShader` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT
- VUID-VkMemoryBarrier2-srcStageMask-03930
  If the `tessellationShader` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-VkMemoryBarrier2-srcStageMask-03931
  If the `conditionalRendering` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT
- VUID-VkMemoryBarrier2-srcStageMask-03932
  If the `fragmentDensityMap` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-VkMemoryBarrier2-srcStageMask-03933
  If the `transformFeedback` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-VkMemoryBarrier2-srcStageMask-03934
  If the `meshShader` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT

- VUID-VkMemoryBarrier2-srcStageMask-03935
  If the taskShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT
- VUID-VkMemoryBarrier2-srcStageMask-07317
  If the attachmentFragmentShadingRate feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkMemoryBarrier2-srcStageMask-07947
  If the rayTracingPipeline feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkMemoryBarrier2-srcStageMask-10751
  If the accelerationStructure feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkMemoryBarrier2-srcStageMask-10752
  If the rayTracingMaintenance1 feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR
- VUID-VkMemoryBarrier2-srcStageMask-11541
  If the VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkMemoryBarrier2-srcAccessMask-03900
  If srcAccessMask includes VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_COPY_INDIRECT_BIT_KHR, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03901
  If srcAccessMask includes VK_ACCESS_2_INDEX_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03902
  If srcAccessMask includes VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03903
  If srcAccessMask includes VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_SUBPASS_SHADER_BIT_HUAWEI, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03904
  If srcAccessMask includes VK_ACCESS_2_UNIFORM_READ_BIT, srcStageMask **must** include

VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages

- VUID-VkMemoryBarrier2-srcAccessMask-03905
  If srcAccessMask includes VK_ACCESS_2_SHADER_SAMPLED_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkMemoryBarrier2-srcAccessMask-03906
  If srcAccessMask includes VK_ACCESS_2_SHADER_STORAGE_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkMemoryBarrier2-srcAccessMask-03907
  If srcAccessMask includes VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkMemoryBarrier2-srcAccessMask-07454
  If srcAccessMask includes VK_ACCESS_2_SHADER_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkMemoryBarrier2-srcAccessMask-03909
  If srcAccessMask includes VK_ACCESS_2_SHADER_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkMemoryBarrier2-srcAccessMask-03910
  If srcAccessMask includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03911
  If srcAccessMask includes VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03912
  If srcAccessMask includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or

VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkMemoryBarrier2-srcAccessMask-03913
  If srcAccessMask includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03914
  If srcAccessMask includes VK_ACCESS_2_TRANSFER_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03915
  If srcAccessMask includes VK_ACCESS_2_TRANSFER_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_CLEAR_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03916
  If srcAccessMask includes VK_ACCESS_2_HOST_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03917
  If srcAccessMask includes VK_ACCESS_2_HOST_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03918
  If srcAccessMask includes VK_ACCESS_2_CONDITIONAL_RENDERING_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03919
  If srcAccessMask includes VK_ACCESS_2_FRAGMENT_DENSITY_MAP_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03920
  If srcAccessMask includes VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkMemoryBarrier2-srcAccessMask-04747
  If `srcAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03922
  If `srcAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03926
  If `srcAccessMask` includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03927
  If `srcAccessMask` includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the `rayQuery` feature is enabled, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-03928
  If `srcAccessMask` includes VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-06256
  If `srcAccessMask` includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the `rayQuery` feature is not enabled, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-srcAccessMask-07272
  If `srcAccessMask` includes VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT or VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkMemoryBarrier2-srcAccessMask-04858

If srcAccessMask includes VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR

- VUID-VkMemoryBarrier2-srcAccessMask-04859
  If srcAccessMask includes VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
- VUID-VkMemoryBarrier2-srcAccessMask-04860
  If srcAccessMask includes VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- VUID-VkMemoryBarrier2-srcAccessMask-04861
  If srcAccessMask includes VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- VUID-VkMemoryBarrier2-srcAccessMask-07457
  If srcAccessMask includes VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkMemoryBarrier2-srcAccessMask-07458
  If srcAccessMask includes VK_ACCESS_2_MICROMAP_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT or VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkMemoryBarrier2-srcAccessMask-08118
  If srcAccessMask includes VK_ACCESS_2_DESCRIPTOR_BUFFER_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkMemoryBarrier2-srcAccessMask-11771
  If srcAccessMask includes VK_ACCESS_2_MEMORY_DECOMPRESSION_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- VUID-VkMemoryBarrier2-srcAccessMask-11772
  If srcAccessMask includes VK_ACCESS_2_MEMORY_DECOMPRESSION_WRITE_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- VUID-VkMemoryBarrier2-srcAccessMask-11294
  If srcAccessMask includes VK_ACCESS_2_SAMPLER_HEAP_READ_BIT_EXT or VK_ACCESS_2_RESOURCE_HEAP_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkMemoryBarrier2-dstStageMask-03929
  If the geometryShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT
- VUID-VkMemoryBarrier2-dstStageMask-03930
  If the tessellationShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-VkMemoryBarrier2-dstStageMask-03931

If the conditionalRendering feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT

- VUID-VkMemoryBarrier2-dstStageMask-03932
  If the fragmentDensityMap feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-VkMemoryBarrier2-dstStageMask-03933
  If the transformFeedback feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-VkMemoryBarrier2-dstStageMask-03934
  If the meshShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT
- VUID-VkMemoryBarrier2-dstStageMask-03935
  If the taskShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT
- VUID-VkMemoryBarrier2-dstStageMask-07317
  If the attachmentFragmentShadingRate feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkMemoryBarrier2-dstStageMask-07947
  If the rayTracingPipeline feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkMemoryBarrier2-dstStageMask-10751
  If the accelerationStructure feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkMemoryBarrier2-dstStageMask-10752
  If the rayTracingMaintenance1 feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR
- VUID-VkMemoryBarrier2-dstStageMask-11541
  If the VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkMemoryBarrier2-dstAccessMask-03900
  If dstAccessMask includes VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_COPY_INDIRECT_BIT_KHR, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-03901
  If dstAccessMask includes VK_ACCESS_2_INDEX_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-03902
  If dstAccessMask includes VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT, dstStageMask

must include VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkMemoryBarrier2-dstAccessMask-03903
  If `dstAccessMask` includes VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_SUBPASS_SHADER_BIT_HUAWEI, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-03904
  If `dstAccessMask` includes VK_ACCESS_2_UNIFORM_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkMemoryBarrier2-dstAccessMask-03905
  If `dstAccessMask` includes VK_ACCESS_2_SHADER_SAMPLED_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkMemoryBarrier2-dstAccessMask-03906
  If `dstAccessMask` includes VK_ACCESS_2_SHADER_STORAGE_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkMemoryBarrier2-dstAccessMask-03907
  If `dstAccessMask` includes VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkMemoryBarrier2-dstAccessMask-07454
  If `dstAccessMask` includes VK_ACCESS_2_SHADER_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkMemoryBarrier2-dstAccessMask-03909
  If `dstAccessMask` includes VK_ACCESS_2_SHADER_WRITE_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkMemoryBarrier2-dstAccessMask-03910
  If `dstAccessMask` includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or

VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkMemoryBarrier2-dstAccessMask-03911
  If dstAccessMask includes VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-03912
  If dstAccessMask includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-03913
  If dstAccessMask includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-03914
  If dstAccessMask includes VK_ACCESS_2_TRANSFER_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-03915
  If dstAccessMask includes VK_ACCESS_2_TRANSFER_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_CLEAR_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-03916
  If dstAccessMask includes VK_ACCESS_2_HOST_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-03917
  If dstAccessMask includes VK_ACCESS_2_HOST_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-03918
  If dstAccessMask includes VK_ACCESS_2_CONDITIONAL_RENDERING_READ_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkMemoryBarrier2-dstAccessMask-03919
  If `dstAccessMask` includes VK_ACCESS_2_FRAGMENT_DENSITY_MAP_READ_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-03920
  If `dstAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-04747
  If `dstAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-03922
  If `dstAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-03926
  If `dstAccessMask` includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-03927
  If `dstAccessMask` includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the `rayQuery` feature is enabled, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-03928
  If `dstAccessMask` includes VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryBarrier2-dstAccessMask-06256

If dstAccessMask includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the rayQuery feature is not enabled, dstStageMask **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkMemoryBarrier2-dstAccessMask-07272
  If dstAccessMask includes VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT or VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkMemoryBarrier2-dstAccessMask-04858
  If dstAccessMask includes VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
- VUID-VkMemoryBarrier2-dstAccessMask-04859
  If dstAccessMask includes VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
- VUID-VkMemoryBarrier2-dstAccessMask-04860
  If dstAccessMask includes VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- VUID-VkMemoryBarrier2-dstAccessMask-04861
  If dstAccessMask includes VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- VUID-VkMemoryBarrier2-dstAccessMask-07457
  If dstAccessMask includes VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkMemoryBarrier2-dstAccessMask-07458
  If dstAccessMask includes VK_ACCESS_2_MICROMAP_READ_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT or VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkMemoryBarrier2-dstAccessMask-08118
  If dstAccessMask includes VK_ACCESS_2_DESCRIPTOR_BUFFER_READ_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkMemoryBarrier2-dstAccessMask-11771
  If dstAccessMask includes VK_ACCESS_2_MEMORY_DECOMPRESSION_READ_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- VUID-VkMemoryBarrier2-dstAccessMask-11772
  If dstAccessMask includes VK_ACCESS_2_MEMORY_DECOMPRESSION_WRITE_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- VUID-VkMemoryBarrier2-dstAccessMask-11294
  If dstAccessMask includes VK_ACCESS_2_SAMPLER_HEAP_READ_BIT_EXT or VK_ACCESS_2_RESOURCE_HEAP_READ_BIT_EXT, dstStageMask **must** include

VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages

Valid Usage (Implicit)

- VUID-VkMemoryBarrier2-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_MEMORY_BARRIER_2
- VUID-VkMemoryBarrier2-srcStageMask-parameter
  If srcStageMask is not 0, srcStageMask **must** be a valid combination of VkPipelineStageFlagBits2 values
- VUID-VkMemoryBarrier2-srcAccessMask-parameter
  If srcAccessMask is not 0, srcAccessMask **must** be a valid combination of VkAccessFlagBits2 values
- VUID-VkMemoryBarrier2-dstStageMask-parameter
  If dstStageMask is not 0, dstStageMask **must** be a valid combination of VkPipelineStageFlagBits2 values
- VUID-VkMemoryBarrier2-dstAccessMask-parameter
  If dstAccessMask is not 0, dstAccessMask **must** be a valid combination of VkAccessFlagBits2 values

Structure Chaining

Extends the structure

- VkSubpassDependency2

The VkMemoryBarrier structure is defined as:

WARNING This functionality is superseded by VkMemoryBarrier2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef struct VkMemoryBarrier {
    VkStructureType    sType;
    const void*        pNext;
    VkAccessFlags      srcAccessMask;
    VkAccessFlags      dstAccessMask;
} VkMemoryBarrier;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- srcAccessMask is a bitmask of VkAccessFlagBits specifying a source access mask.

- `dstAccessMask` is a bitmask of VkAccessFlagBits specifying a destination access mask.

The first access scope is limited to access types in the source access mask specified by `srcAccessMask` and, if a VkMemoryBarrierAccessFlags3KHR is passed in `pNext`, `srcAccessMask3`.

The second access scope is limited to access types in the destination access mask specified by `dstAccessMask` and, if a VkMemoryBarrierAccessFlags3KHR is passed in `pNext`, `dstAccessMask3`.

Valid Usage (Implicit)

- VUID-VkMemoryBarrier-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_BARRIER
- VUID-VkMemoryBarrier-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkMemoryBarrier-srcAccessMask-parameter
  If `srcAccessMask` is not `0`, `srcAccessMask` **must** be a valid combination of VkAccessFlagBits values
- VUID-VkMemoryBarrier-dstAccessMask-parameter
  If `dstAccessMask` is not `0`, `dstAccessMask` **must** be a valid combination of VkAccessFlagBits values

7.7.2. Memory Range Barriers

Memory range barriers only apply to memory accesses involving a specific device address range. That is, a memory dependency formed from a memory range barrier is scoped to access via the specified device address range. Memory range barriers **can** also be used to define a queue family ownership transfer for the specified device address range.

The `VkMemoryRangeBarriersInfoKHR` structure is defined as:

```
// Provided by VK_KHR_device_address_commands
typedef struct VkMemoryRangeBarriersInfoKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    uint32_t                        memoryRangeBarrierCount;
    const VkMemoryRangeBarrierKHR*  pMemoryRangeBarriers;
} VkMemoryRangeBarriersInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryRangeBarrierCount` is the length of the `pMemoryRangeBarriers` array.
- `pMemoryRangeBarriers` is a pointer to an array of VkMemoryRangeBarrierKHR structures defining memory dependencies between accesses to specified memory ranges.

When this structure is included in the `pNext` chain of VkDependencyInfo, it defines a set of memory

dependencies, as well as queue family ownership transfer operations, for a specified set of memory ranges.

If this structure is not included in the pNext chain of VkDependencyInfo, it is equivalent to specifying it with a memoryRangeBarrierCount of 0.

Each member of pMemoryRangeBarriers defines a separate memory dependency.

Valid Usage (Implicit)

- VUID-VkMemoryRangeBarriersInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_MEMORY_RANGE_BARRIERS_INFO_KHR
- VUID-VkMemoryRangeBarriersInfoKHR-pMemoryRangeBarriers-parameter
  If memoryRangeBarrierCount is not 0, pMemoryRangeBarriers **must** be a valid pointer to an array of memoryRangeBarrierCount valid VkMemoryRangeBarrierKHR structures

Structure Chaining

Extends the structure

- VkDependencyInfo

The VkMemoryRangeBarrierKHR structure is defined as:

```
// Provided by VK_KHR_device_address_commands
typedef struct VkMemoryRangeBarrierKHR {
    VkStructureType             sType;
    const void*                 pNext;
    VkPipelineStageFlags2       srcStageMask;
    VkAccessFlags2              srcAccessMask;
    VkPipelineStageFlags2       dstStageMask;
    VkAccessFlags2              dstAccessMask;
    uint32_t                    srcQueueFamilyIndex;
    uint32_t                    dstQueueFamilyIndex;
    VkDeviceAddressRangeKHR     addressRange;
    VkAddressCommandFlagsKHR    addressFlags;
} VkMemoryRangeBarrierKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- srcStageMask is a VkPipelineStageFlags2 mask of pipeline stages to be included in the first synchronization scope.
- srcAccessMask is a VkAccessFlags2 mask of access flags to be included in the first access scope.
- dstStageMask is a VkPipelineStageFlags2 mask of pipeline stages to be included in the second synchronization scope.

- `dstAccessMask` is a VkAccessFlags2 mask of access flags to be included in the second access scope.
- `srcQueueFamilyIndex` is the source queue family for a queue family ownership transfer.
- `dstQueueFamilyIndex` is the destination queue family for a queue family ownership transfer.
- `addressRange` is a VkDeviceAddressRangeKHR structure specifying the address range affected by the barrier.
- `addressFlags` is a VkAddressCommandFlagsKHR value defining the flags for the address range.

This structure defines a memory dependency limited to an address range, and **can** define a queue family ownership transfer operation for that range.

The first synchronization scope and access scope described by this structure include only operations and memory accesses specified by `srcStageMask` and `srcAccessMask`.

The second synchronization scope and access scope described by this structure include only operations and memory accesses specified by `dstStageMask` and `dstAccessMask`.

Both access scopes are limited to only memory accesses to memory in `addressRange`.

If the buffer from which `addressRange` was queried was created with a sharing mode of VK_SHARING_MODE_EXCLUSIVE, and `srcQueueFamilyIndex` is not equal to `dstQueueFamilyIndex`, this memory range barrier defines a queue family ownership transfer operation. When executed on a queue in the family identified by `srcQueueFamilyIndex`, this barrier defines a queue family release operation for the specified address range, and the second synchronization scope does not apply to this operation. When executed on a queue in the family identified by `dstQueueFamilyIndex`, this barrier defines a queue family acquire operation for the specified address range, and the first synchronization scope does not apply to this operation.

A queue family ownership transfer operation is also defined if the values are not equal, and either is one of the special queue family values reserved for external memory ownership transfers, as described in Queue Family Ownership Transfer. A queue family release operation is defined when `dstQueueFamilyIndex` is one of those values, and a queue family acquire operation is defined when `srcQueueFamilyIndex` is one of those values.

Valid Usage

- VUID-VkMemoryRangeBarrierKHR-addressRange-13097
  If the range specified by `addressRange` is not bound completely to memory when accessed, `addressFlags` **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-addressRange-13098
  If the buffer from which the range specified by `addressRange` was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `addressFlags` **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-addressRange-13099
  If the buffer from which the range specified by `addressRange` was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `addressFlags` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR

- VUID-VkMemoryRangeBarrierKHR-addressFlags-13100
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-addressRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-addressRange-13123
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-addressFlags-13101
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-addressRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `addressFlags` **must** include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-addressRange-13125
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created without VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `addressFlags` **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-srcStageMask-03929
  If the `geometryShader` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT
- VUID-VkMemoryRangeBarrierKHR-srcStageMask-03930
  If the `tessellationShader` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-VkMemoryRangeBarrierKHR-srcStageMask-03931
  If the `conditionalRendering` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-srcStageMask-03932
  If the `fragmentDensityMap` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT

- VUID-VkMemoryRangeBarrierKHR-srcStageMask-03933
  If the transformFeedback feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-srcStageMask-03934
  If the meshShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-srcStageMask-03935
  If the taskShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-srcStageMask-07317
  If the attachmentFragmentShadingRate feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-srcStageMask-07947
  If the rayTracingPipeline feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-srcStageMask-10751
  If the accelerationStructure feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-srcStageMask-10752
  If the rayTracingMaintenance1 feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-srcStageMask-11541
  If the VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03900
  If srcAccessMask includes VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_COPY_INDIRECT_BIT_KHR, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03901
  If srcAccessMask includes VK_ACCESS_2_INDEX_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03902
  If srcAccessMask includes VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03903
  If srcAccessMask includes VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT, srcStageMask

must include VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_SUBPASS_SHADER_BIT_HUAWEI, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03904
  If `srcAccessMask` includes VK_ACCESS_2_UNIFORM_READ_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03905
  If `srcAccessMask` includes VK_ACCESS_2_SHADER_SAMPLED_READ_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03906
  If `srcAccessMask` includes VK_ACCESS_2_SHADER_STORAGE_READ_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03907
  If `srcAccessMask` includes VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-07454
  If `srcAccessMask` includes VK_ACCESS_2_SHADER_READ_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03909
  If `srcAccessMask` includes VK_ACCESS_2_SHADER_WRITE_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03910
  If `srcAccessMask` includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03911
  If `srcAccessMask` includes VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or

VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03912
  If `srcAccessMask` includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03913
  If `srcAccessMask` includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03914
  If `srcAccessMask` includes VK_ACCESS_2_TRANSFER_READ_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03915
  If `srcAccessMask` includes VK_ACCESS_2_TRANSFER_WRITE_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_CLEAR_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03916
  If `srcAccessMask` includes VK_ACCESS_2_HOST_READ_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03917
  If `srcAccessMask` includes VK_ACCESS_2_HOST_WRITE_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03918
  If `srcAccessMask` includes VK_ACCESS_2_CONDITIONAL_RENDERING_READ_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03919
  If `srcAccessMask` includes VK_ACCESS_2_FRAGMENT_DENSITY_MAP_READ_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03920
  If srcAccessMask includes VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-04747
  If srcAccessMask includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03922
  If srcAccessMask includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03926
  If srcAccessMask includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03927
  If srcAccessMask includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the rayQuery feature is enabled, srcStageMask **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-03928
  If srcAccessMask includes VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-06256
  If srcAccessMask includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the rayQuery feature is not enabled, srcStageMask **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-07272
  If `srcAccessMask` includes VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT or VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-04858
  If `srcAccessMask` includes VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-04859
  If `srcAccessMask` includes VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-04860
  If `srcAccessMask` includes VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-04861
  If `srcAccessMask` includes VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-07457
  If `srcAccessMask` includes VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-07458
  If `srcAccessMask` includes VK_ACCESS_2_MICROMAP_READ_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT or VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-08118
  If `srcAccessMask` includes VK_ACCESS_2_DESCRIPTOR_BUFFER_READ_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-11771
  If `srcAccessMask` includes VK_ACCESS_2_MEMORY_DECOMPRESSION_READ_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-11772
  If `srcAccessMask` includes VK_ACCESS_2_MEMORY_DECOMPRESSION_WRITE_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-11294
  If `srcAccessMask` includes VK_ACCESS_2_SAMPLER_HEAP_READ_BIT_EXT or VK_ACCESS_2_RESOURCE_HEAP_READ_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkMemoryRangeBarrierKHR-dstStageMask-03929
  If the `geometryShader` feature is not enabled, `dstStageMask` **must** not contain VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT

- VUID-VkMemoryRangeBarrierKHR-dstStageMask-03930
  If the tessellationShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-VkMemoryRangeBarrierKHR-dstStageMask-03931
  If the conditionalRendering feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-dstStageMask-03932
  If the fragmentDensityMap feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-dstStageMask-03933
  If the transformFeedback feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-dstStageMask-03934
  If the meshShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-dstStageMask-03935
  If the taskShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-dstStageMask-07317
  If the attachmentFragmentShadingRate feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-dstStageMask-07947
  If the rayTracingPipeline feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-dstStageMask-10751
  If the accelerationStructure feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-dstStageMask-10752
  If the rayTracingMaintenance1 feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-dstStageMask-11541
  If the VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03900
  If dstAccessMask includes VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_COPY_INDIRECT_BIT_KHR, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03901
  If dstAccessMask includes VK_ACCESS_2_INDEX_READ_BIT, dstStageMask **must** include

VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03902
  If dstAccessMask includes VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03903
  If dstAccessMask includes VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_SUBPASS_SHADER_BIT_HUAWEI, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03904
  If dstAccessMask includes VK_ACCESS_2_UNIFORM_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03905
  If dstAccessMask includes VK_ACCESS_2_SHADER_SAMPLED_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03906
  If dstAccessMask includes VK_ACCESS_2_SHADER_STORAGE_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03907
  If dstAccessMask includes VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-07454
  If dstAccessMask includes VK_ACCESS_2_SHADER_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03909
  If dstAccessMask includes VK_ACCESS_2_SHADER_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the

VK_PIPELINE_STAGE_*_SHADER_BIT stages

- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03910
  If dstAccessMask includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03911
  If dstAccessMask includes VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03912
  If dstAccessMask includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03913
  If dstAccessMask includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03914
  If dstAccessMask includes VK_ACCESS_2_TRANSFER_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03915
  If dstAccessMask includes VK_ACCESS_2_TRANSFER_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_CLEAR_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03916
  If dstAccessMask includes VK_ACCESS_2_HOST_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03917
  If dstAccessMask includes VK_ACCESS_2_HOST_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_HOST_BIT

- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03918
  If dstAccessMask includes VK_ACCESS_2_CONDITIONAL_RENDERING_READ_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03919
  If dstAccessMask includes VK_ACCESS_2_FRAGMENT_DENSITY_MAP_READ_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03920
  If dstAccessMask includes VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-04747
  If dstAccessMask includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03922
  If dstAccessMask includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03926
  If dstAccessMask includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03927
  If dstAccessMask includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the rayQuery feature is enabled, dstStageMask **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-03928
  If dstAccessMask includes VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR,

dstStageMask **must** include VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-06256
  If dstAccessMask includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the rayQuery feature is not enabled, dstStageMask **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-07272
  If dstAccessMask includes VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT or VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-04858
  If dstAccessMask includes VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-04859
  If dstAccessMask includes VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-04860
  If dstAccessMask includes VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-04861
  If dstAccessMask includes VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-07457
  If dstAccessMask includes VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-07458
  If dstAccessMask includes VK_ACCESS_2_MICROMAP_READ_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT or VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-08118
  If dstAccessMask includes VK_ACCESS_2_DESCRIPTOR_BUFFER_READ_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-11771
  If dstAccessMask includes VK_ACCESS_2_MEMORY_DECOMPRESSION_READ_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-11772

If dstAccessMask includes VK_ACCESS_2_MEMORY_DECOMPRESSION_WRITE_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT

- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-11294
  If dstAccessMask includes VK_ACCESS_2_SAMPLER_HEAP_READ_BIT_EXT or VK_ACCESS_2_RESOURCE_HEAP_READ_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkMemoryRangeBarrierKHR-None-09097
  If the VK_KHR_external_memory extension is not enabled, and the value of VkApplicationInfo::apiVersion used to create the VkInstance is not greater than or equal to Version 1.1, srcQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_EXTERNAL
- VUID-VkMemoryRangeBarrierKHR-None-09098
  If the VK_KHR_external_memory extension is not enabled, and the value of VkApplicationInfo::apiVersion used to create the VkInstance is not greater than or equal to Version 1.1, dstQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_EXTERNAL
- VUID-VkMemoryRangeBarrierKHR-srcQueueFamilyIndex-09099
  If the VK_EXT_queue_family_foreign extension is not enabled, srcQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_FOREIGN_EXT
- VUID-VkMemoryRangeBarrierKHR-dstQueueFamilyIndex-09100
  If the VK_EXT_queue_family_foreign extension is not enabled, dstQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_FOREIGN_EXT
- VUID-VkMemoryRangeBarrierKHR-address-13087
  If the buffer from which addressRange was queried was created with a sharing mode of VK_SHARING_MODE_EXCLUSIVE, and srcQueueFamilyIndex and dstQueueFamilyIndex are not equal, srcQueueFamilyIndex **must** be VK_QUEUE_FAMILY_EXTERNAL, VK_QUEUE_FAMILY_FOREIGN_EXT, or a valid queue family
- VUID-VkMemoryRangeBarrierKHR-address-13088
  If the buffer from which addressRange was queried was created with a sharing mode of VK_SHARING_MODE_EXCLUSIVE, and srcQueueFamilyIndex and dstQueueFamilyIndex are not equal, dstQueueFamilyIndex **must** be VK_QUEUE_FAMILY_EXTERNAL, VK_QUEUE_FAMILY_FOREIGN_EXT, or a valid queue family
- VUID-VkMemoryRangeBarrierKHR-srcStageMask-13089
  If either srcStageMask or dstStageMask includes VK_PIPELINE_STAGE_2_HOST_BIT, srcQueueFamilyIndex and dstQueueFamilyIndex **must** be equal

Valid Usage (Implicit)

- VUID-VkMemoryRangeBarrierKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_MEMORY_RANGE_BARRIER_KHR
- VUID-VkMemoryRangeBarrierKHR-pNext-pNext
  pNext **must** be NULL or a pointer to a valid instance of VkMemoryBarrierAccessFlags3KHR

- VUID-VkMemoryRangeBarrierKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkMemoryRangeBarrierKHR-srcStageMask-parameter
  If `srcStageMask` is not `0`, `srcStageMask` **must** be a valid combination of VkPipelineStageFlagBits2 values
- VUID-VkMemoryRangeBarrierKHR-srcAccessMask-parameter
  If `srcAccessMask` is not `0`, `srcAccessMask` **must** be a valid combination of VkAccessFlagBits2 values
- VUID-VkMemoryRangeBarrierKHR-dstStageMask-parameter
  If `dstStageMask` is not `0`, `dstStageMask` **must** be a valid combination of VkPipelineStageFlagBits2 values
- VUID-VkMemoryRangeBarrierKHR-dstAccessMask-parameter
  If `dstAccessMask` is not `0`, `dstAccessMask` **must** be a valid combination of VkAccessFlagBits2 values
- VUID-VkMemoryRangeBarrierKHR-addressFlags-parameter
  If `addressFlags` is not `0`, `addressFlags` **must** be a valid combination of VkAddressCommandFlagBitsKHR values

7.7.3. Buffer Memory Barriers

Buffer memory barriers only apply to memory accesses involving a specific buffer range. That is, a memory dependency formed from a buffer memory barrier is scoped to access via the specified buffer range. Buffer memory barriers **can** also be used to define a queue family ownership transfer for the specified buffer range.

The `VkBufferMemoryBarrier2` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkBufferMemoryBarrier2 {
    VkStructureType          sType;
    const void*              pNext;
    VkPipelineStageFlags2    srcStageMask;
    VkAccessFlags2           srcAccessMask;
    VkPipelineStageFlags2    dstStageMask;
    VkAccessFlags2           dstAccessMask;
    uint32_t                 srcQueueFamilyIndex;
    uint32_t                 dstQueueFamilyIndex;
    VkBuffer                 buffer;
    VkDeviceSize             offset;
    VkDeviceSize             size;
} VkBufferMemoryBarrier2;
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to VkBufferMemoryBarrier2
typedef VkBufferMemoryBarrier2 VkBufferMemoryBarrier2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcStageMask` is a VkPipelineStageFlags2 mask of pipeline stages to be included in the first synchronization scope.
- `srcAccessMask` is a VkAccessFlags2 mask of access flags to be included in the first access scope.
- `dstStageMask` is a VkPipelineStageFlags2 mask of pipeline stages to be included in the second synchronization scope.
- `dstAccessMask` is a VkAccessFlags2 mask of access flags to be included in the second access scope.
- `srcQueueFamilyIndex` is the source queue family for a queue family ownership transfer.
- `dstQueueFamilyIndex` is the destination queue family for a queue family ownership transfer.
- `buffer` is a handle to the buffer whose backing memory is affected by the barrier.
- `offset` is an offset in bytes into the backing memory for `buffer`; this is relative to the base offset as bound to the buffer (see vkBindBufferMemory).
- `size` is a size in bytes of the affected area of backing memory for `buffer`, or VK_WHOLE_SIZE to use the range from `offset` to the end of the buffer.

This structure defines a memory dependency limited to a range of a buffer, and **can** define a queue family ownership transfer operation for that range.

The first synchronization scope and access scope described by this structure include only operations and memory accesses specified by the source stage mask and the source access mask.

The second synchronization scope and access scope described by this structure include only operations and memory accesses specified by the destination stage mask and the destination access mask.

Both access scopes are limited to only memory accesses to `buffer` in the range defined by `offset` and `size`.

If `buffer` was created with VK_SHARING_MODE_EXCLUSIVE, and `srcQueueFamilyIndex` is not equal to `dstQueueFamilyIndex`, this memory barrier defines a queue family ownership transfer operation. When executed on a queue in the family identified by `srcQueueFamilyIndex`, this barrier defines a queue family release operation for the specified buffer range, and if VkDependencyInfoKHR ::`dependencyFlags` did not include VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR, the second synchronization scope does not apply to this operation. When executed on a queue in the family identified by `dstQueueFamilyIndex`, this barrier defines a queue family acquire operation for the specified buffer range, and if VkDependencyInfoKHR::`dependencyFlags` did not include VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR, the first synchronization scope does not apply to this operation.

A queue family ownership transfer operation is also defined if the values are not equal, and either is one of the special queue family values reserved for external memory ownership transfers, as described in Queue Family Ownership Transfer. A queue family release operation is defined when `dstQueueFamilyIndex` is one of those values, and a queue family acquire operation is defined when `srcQueueFamilyIndex` is one of those values.

Valid Usage

- VUID-VkBufferMemoryBarrier2-srcStageMask-03929
  If the `geometryShader` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT
- VUID-VkBufferMemoryBarrier2-srcStageMask-03930
  If the `tessellationShader` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-VkBufferMemoryBarrier2-srcStageMask-03931
  If the `conditionalRendering` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT
- VUID-VkBufferMemoryBarrier2-srcStageMask-03932
  If the `fragmentDensityMap` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-VkBufferMemoryBarrier2-srcStageMask-03933
  If the `transformFeedback` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-VkBufferMemoryBarrier2-srcStageMask-03934
  If the `meshShader` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT
- VUID-VkBufferMemoryBarrier2-srcStageMask-03935
  If the `taskShader` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT
- VUID-VkBufferMemoryBarrier2-srcStageMask-07317
  If the `attachmentFragmentShadingRate` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkBufferMemoryBarrier2-srcStageMask-07947
  If the `rayTracingPipeline` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkBufferMemoryBarrier2-srcStageMask-10751
  If the `accelerationStructure` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkBufferMemoryBarrier2-srcStageMask-10752
  If the `rayTracingMaintenance1` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR
- VUID-VkBufferMemoryBarrier2-srcStageMask-11541

If the VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT

- VUID-VkBufferMemoryBarrier2-srcAccessMask-03900
  If srcAccessMask includes VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_COPY_INDIRECT_BIT_KHR, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03901
  If srcAccessMask includes VK_ACCESS_2_INDEX_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03902
  If srcAccessMask includes VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03903
  If srcAccessMask includes VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_SUBPASS_SHADER_BIT_HUAWEI, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03904
  If srcAccessMask includes VK_ACCESS_2_UNIFORM_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03905
  If srcAccessMask includes VK_ACCESS_2_SHADER_SAMPLED_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03906
  If srcAccessMask includes VK_ACCESS_2_SHADER_STORAGE_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03907
  If srcAccessMask includes VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the

VK_PIPELINE_STAGE_*_SHADER_BIT stages

- VUID-VkBufferMemoryBarrier2-srcAccessMask-07454
  If srcAccessMask includes VK_ACCESS_2_SHADER_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03909
  If srcAccessMask includes VK_ACCESS_2_SHADER_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03910
  If srcAccessMask includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03911
  If srcAccessMask includes VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03912
  If srcAccessMask includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03913
  If srcAccessMask includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03914
  If srcAccessMask includes VK_ACCESS_2_TRANSFER_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03915
  If srcAccessMask includes VK_ACCESS_2_TRANSFER_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT,

VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_CLEAR_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkBufferMemoryBarrier2-srcAccessMask-03916
  If `srcAccessMask` includes VK_ACCESS_2_HOST_READ_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03917
  If `srcAccessMask` includes VK_ACCESS_2_HOST_WRITE_BIT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03918
  If `srcAccessMask` includes VK_ACCESS_2_CONDITIONAL_RENDERING_READ_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03919
  If `srcAccessMask` includes VK_ACCESS_2_FRAGMENT_DENSITY_MAP_READ_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03920
  If `srcAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-04747
  If `srcAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03922
  If `srcAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03926
  If `srcAccessMask` includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or

VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkBufferMemoryBarrier2-srcAccessMask-03927
  If srcAccessMask includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the rayQuery feature is enabled, srcStageMask **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03928
  If srcAccessMask includes VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-06256
  If srcAccessMask includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the rayQuery feature is not enabled, srcStageMask **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-07272
  If srcAccessMask includes VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT or VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkBufferMemoryBarrier2-srcAccessMask-04858
  If srcAccessMask includes VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
- VUID-VkBufferMemoryBarrier2-srcAccessMask-04859
  If srcAccessMask includes VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
- VUID-VkBufferMemoryBarrier2-srcAccessMask-04860
  If srcAccessMask includes VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- VUID-VkBufferMemoryBarrier2-srcAccessMask-04861
  If srcAccessMask includes VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- VUID-VkBufferMemoryBarrier2-srcAccessMask-07457
  If srcAccessMask includes VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-07458
  If srcAccessMask includes VK_ACCESS_2_MICROMAP_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT or

VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR

- VUID-VkBufferMemoryBarrier2-srcAccessMask-08118
  If srcAccessMask includes VK_ACCESS_2_DESCRIPTOR_BUFFER_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkBufferMemoryBarrier2-srcAccessMask-11771
  If srcAccessMask includes VK_ACCESS_2_MEMORY_DECOMPRESSION_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-11772
  If srcAccessMask includes VK_ACCESS_2_MEMORY_DECOMPRESSION_WRITE_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- VUID-VkBufferMemoryBarrier2-srcAccessMask-11294
  If srcAccessMask includes VK_ACCESS_2_SAMPLER_HEAP_READ_BIT_EXT or VK_ACCESS_2_RESOURCE_HEAP_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkBufferMemoryBarrier2-dstStageMask-03929
  If the geometryShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT
- VUID-VkBufferMemoryBarrier2-dstStageMask-03930
  If the tessellationShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-VkBufferMemoryBarrier2-dstStageMask-03931
  If the conditionalRendering feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT
- VUID-VkBufferMemoryBarrier2-dstStageMask-03932
  If the fragmentDensityMap feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-VkBufferMemoryBarrier2-dstStageMask-03933
  If the transformFeedback feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-VkBufferMemoryBarrier2-dstStageMask-03934
  If the meshShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT
- VUID-VkBufferMemoryBarrier2-dstStageMask-03935
  If the taskShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT
- VUID-VkBufferMemoryBarrier2-dstStageMask-07317
  If the attachmentFragmentShadingRate feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR

- VUID-VkBufferMemoryBarrier2-dstStageMask-07947
  If the rayTracingPipeline feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkBufferMemoryBarrier2-dstStageMask-10751
  If the accelerationStructure feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkBufferMemoryBarrier2-dstStageMask-10752
  If the rayTracingMaintenance1 feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR
- VUID-VkBufferMemoryBarrier2-dstStageMask-11541
  If the VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03900
  If dstAccessMask includes VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_COPY_INDIRECT_BIT_KHR, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03901
  If dstAccessMask includes VK_ACCESS_2_INDEX_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03902
  If dstAccessMask includes VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03903
  If dstAccessMask includes VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_SUBPASS_SHADER_BIT_HUAWEI, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03904
  If dstAccessMask includes VK_ACCESS_2_UNIFORM_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03905
  If dstAccessMask includes VK_ACCESS_2_SHADER_SAMPLED_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the

VK_PIPELINE_STAGE_*_SHADER_BIT stages

- VUID-VkBufferMemoryBarrier2-dstAccessMask-03906
  If dstAccessMask includes VK_ACCESS_2_SHADER_STORAGE_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03907
  If dstAccessMask includes VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkBufferMemoryBarrier2-dstAccessMask-07454
  If dstAccessMask includes VK_ACCESS_2_SHADER_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03909
  If dstAccessMask includes VK_ACCESS_2_SHADER_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03910
  If dstAccessMask includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03911
  If dstAccessMask includes VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03912
  If dstAccessMask includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03913
  If dstAccessMask includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkBufferMemoryBarrier2-dstAccessMask-03914
  If `dstAccessMask` includes VK_ACCESS_2_TRANSFER_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03915
  If `dstAccessMask` includes VK_ACCESS_2_TRANSFER_WRITE_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_CLEAR_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03916
  If `dstAccessMask` includes VK_ACCESS_2_HOST_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03917
  If `dstAccessMask` includes VK_ACCESS_2_HOST_WRITE_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03918
  If `dstAccessMask` includes VK_ACCESS_2_CONDITIONAL_RENDERING_READ_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03919
  If `dstAccessMask` includes VK_ACCESS_2_FRAGMENT_DENSITY_MAP_READ_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03920
  If `dstAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-04747
  If `dstAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03922

If dstAccessMask includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkBufferMemoryBarrier2-dstAccessMask-03926
  If dstAccessMask includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03927
  If dstAccessMask includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the rayQuery feature is enabled, dstStageMask **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03928
  If dstAccessMask includes VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-06256
  If dstAccessMask includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the rayQuery feature is not enabled, dstStageMask **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-07272
  If dstAccessMask includes VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT or VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkBufferMemoryBarrier2-dstAccessMask-04858
  If dstAccessMask includes VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
- VUID-VkBufferMemoryBarrier2-dstAccessMask-04859
  If dstAccessMask includes VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
- VUID-VkBufferMemoryBarrier2-dstAccessMask-04860
  If dstAccessMask includes VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR

- VUID-VkBufferMemoryBarrier2-dstAccessMask-04861
  If `dstAccessMask` includes VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- VUID-VkBufferMemoryBarrier2-dstAccessMask-07457
  If `dstAccessMask` includes VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-07458
  If `dstAccessMask` includes VK_ACCESS_2_MICROMAP_READ_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT or VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkBufferMemoryBarrier2-dstAccessMask-08118
  If `dstAccessMask` includes VK_ACCESS_2_DESCRIPTOR_BUFFER_READ_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkBufferMemoryBarrier2-dstAccessMask-11771
  If `dstAccessMask` includes VK_ACCESS_2_MEMORY_DECOMPRESSION_READ_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-11772
  If `dstAccessMask` includes VK_ACCESS_2_MEMORY_DECOMPRESSION_WRITE_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- VUID-VkBufferMemoryBarrier2-dstAccessMask-11294
  If `dstAccessMask` includes VK_ACCESS_2_SAMPLER_HEAP_READ_BIT_EXT or VK_ACCESS_2_RESOURCE_HEAP_READ_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkBufferMemoryBarrier2-offset-01187
  `offset` **must** be less than the size of `buffer`
- VUID-VkBufferMemoryBarrier2-size-01188
  If `size` is not equal to VK_WHOLE_SIZE, `size` **must** be greater than `0`
- VUID-VkBufferMemoryBarrier2-size-01189
  If `size` is not equal to VK_WHOLE_SIZE, `size` **must** be less than or equal to the size of `buffer` minus `offset`
- VUID-VkBufferMemoryBarrier2-buffer-01931
  If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkBufferMemoryBarrier2-buffer-09095
  If `buffer` was created with a sharing mode of VK_SHARING_MODE_EXCLUSIVE, and `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are not equal, `srcQueueFamilyIndex` **must** be VK_QUEUE_FAMILY_EXTERNAL, VK_QUEUE_FAMILY_FOREIGN_EXT, or a valid queue family
- VUID-VkBufferMemoryBarrier2-buffer-09096

If `buffer` was created with a sharing mode of VK_SHARING_MODE_EXCLUSIVE, and `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are not equal, `dstQueueFamilyIndex` **must** be VK_QUEUE_FAMILY_EXTERNAL, VK_QUEUE_FAMILY_FOREIGN_EXT, or a valid queue family

- VUID-VkBufferMemoryBarrier2-None-09097
  If the VK_KHR_external_memory extension is not enabled, and the value of VkApplicationInfo::`apiVersion` used to create the VkInstance is not greater than or equal to Version 1.1, `srcQueueFamilyIndex` **must** not be VK_QUEUE_FAMILY_EXTERNAL
- VUID-VkBufferMemoryBarrier2-None-09098
  If the VK_KHR_external_memory extension is not enabled, and the value of VkApplicationInfo::`apiVersion` used to create the VkInstance is not greater than or equal to Version 1.1, `dstQueueFamilyIndex` **must** not be VK_QUEUE_FAMILY_EXTERNAL
- VUID-VkBufferMemoryBarrier2-srcQueueFamilyIndex-09099
  If the VK_EXT_queue_family_foreign extension is not enabled, `srcQueueFamilyIndex` **must** not be VK_QUEUE_FAMILY_FOREIGN_EXT
- VUID-VkBufferMemoryBarrier2-dstQueueFamilyIndex-09100
  If the VK_EXT_queue_family_foreign extension is not enabled, `dstQueueFamilyIndex` **must** not be VK_QUEUE_FAMILY_FOREIGN_EXT
- VUID-VkBufferMemoryBarrier2-srcStageMask-03851
  If either `srcStageMask` or `dstStageMask` includes VK_PIPELINE_STAGE_2_HOST_BIT, `srcQueueFamilyIndex` and `dstQueueFamilyIndex` **must** be equal

Valid Usage (Implicit)

- VUID-VkBufferMemoryBarrier2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER_2
- VUID-VkBufferMemoryBarrier2-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkExternalMemoryAcquireUnmodifiedEXT or VkMemoryBarrierAccessFlags3KHR
- VUID-VkBufferMemoryBarrier2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkBufferMemoryBarrier2-srcStageMask-parameter
  If `srcStageMask` is not `0`, `srcStageMask` **must** be a valid combination of VkPipelineStageFlagBits2 values
- VUID-VkBufferMemoryBarrier2-srcAccessMask-parameter
  If `srcAccessMask` is not `0`, `srcAccessMask` **must** be a valid combination of VkAccessFlagBits2 values
- VUID-VkBufferMemoryBarrier2-dstStageMask-parameter
  If `dstStageMask` is not `0`, `dstStageMask` **must** be a valid combination of VkPipelineStageFlagBits2 values
- VUID-VkBufferMemoryBarrier2-dstAccessMask-parameter

If `dstAccessMask` is not `0`, `dstAccessMask` **must** be a valid combination of VkAccessFlagBits2 values

- VUID-VkBufferMemoryBarrier2-buffer-parameter
  `buffer` **must** be a valid VkBuffer handle

The `VkBufferMemoryBarrier` structure is defined as:

WARNING This functionality is superseded by VkBufferMemoryBarrier2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef struct VkBufferMemoryBarrier {
    VkStructureType    sType;
    const void*        pNext;
    VkAccessFlags      srcAccessMask;
    VkAccessFlags      dstAccessMask;
    uint32_t           srcQueueFamilyIndex;
    uint32_t           dstQueueFamilyIndex;
    VkBuffer           buffer;
    VkDeviceSize       offset;
    VkDeviceSize       size;
} VkBufferMemoryBarrier;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcAccessMask` is a bitmask of VkAccessFlagBits specifying a source access mask.
- `dstAccessMask` is a bitmask of VkAccessFlagBits specifying a destination access mask.
- `srcQueueFamilyIndex` is the source queue family for a queue family ownership transfer.
- `dstQueueFamilyIndex` is the destination queue family for a queue family ownership transfer.
- `buffer` is a handle to the buffer whose backing memory is affected by the barrier.
- `offset` is an offset in bytes into the backing memory for `buffer`; this is relative to the base offset as bound to the buffer (see vkBindBufferMemory).
- `size` is a size in bytes of the affected area of backing memory for `buffer`, or VK_WHOLE_SIZE to use the range from `offset` to the end of the buffer.

The first access scope is limited to access to memory through the specified buffer range, via access types in the source access mask specified by `srcAccessMask` and, if a VkMemoryBarrierAccessFlags3KHR is passed in `pNext`, `srcAccessMask3`. If the source access mask includes VK_ACCESS_HOST_WRITE_BIT, a memory domain operation is performed where available memory in the host domain is also made available to the device domain.

The second access scope is limited to access to memory through the specified buffer range, via access types in the destination access mask specified by `dstAccessMask` and, if a VkMemoryBarrierAccessFlags3KHR is passed in `pNext`, `dstAccessMask3`. If the destination access

mask includes VK_ACCESS_HOST_WRITE_BIT or VK_ACCESS_HOST_READ_BIT, a memory domain operation is performed where available memory in the device domain is also made available to the host domain.

NOTE

Host writes to device memory that was allocated without VK_MEMORY_PROPERTY_HOST_COHERENT_BIT have to be flushed with vkFlushMappedMemoryRanges before they can be accessed safely on the device. Similarly, device writes to such memory have to be invalidated with vkInvalidateMappedMemoryRanges before they can be accessed safely on the host.

Memory allocated with VK_MEMORY_PROPERTY_HOST_COHERENT_BIT does not need to have these additional operations performed.

If `srcQueueFamilyIndex` is not equal to `dstQueueFamilyIndex`, and `srcQueueFamilyIndex` is equal to the current queue family, then the memory barrier defines a queue family release operation for the specified buffer range, and if `dependencyFlags` did not include VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR, the second synchronization scope of the calling command does not apply to this operation.

If `dstQueueFamilyIndex` is not equal to `srcQueueFamilyIndex`, and `dstQueueFamilyIndex` is equal to the current queue family, then the memory barrier defines a queue family acquire operation for the specified buffer range, and if `dependencyFlags` did not include VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR, the first synchronization scope of the calling command does not apply to this operation.

Valid Usage

- VUID-VkBufferMemoryBarrier-offset-01187
  `offset` **must** be less than the size of `buffer`
- VUID-VkBufferMemoryBarrier-size-01188
  If `size` is not equal to VK_WHOLE_SIZE, `size` **must** be greater than `0`
- VUID-VkBufferMemoryBarrier-size-01189
  If `size` is not equal to VK_WHOLE_SIZE, `size` **must** be less than or equal to the size of `buffer` minus `offset`
- VUID-VkBufferMemoryBarrier-buffer-01931
  If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkBufferMemoryBarrier-buffer-09095
  If `buffer` was created with a sharing mode of VK_SHARING_MODE_EXCLUSIVE, and `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are not equal, `srcQueueFamilyIndex` **must** be VK_QUEUE_FAMILY_EXTERNAL, VK_QUEUE_FAMILY_FOREIGN_EXT, or a valid queue family
- VUID-VkBufferMemoryBarrier-buffer-09096
  If `buffer` was created with a sharing mode of VK_SHARING_MODE_EXCLUSIVE, and `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are not equal, `dstQueueFamilyIndex` **must** be VK_QUEUE_FAMILY_EXTERNAL, VK_QUEUE_FAMILY_FOREIGN_EXT, or a valid queue

family

- VUID-VkBufferMemoryBarrier-None-09097
  If the VK_KHR_external_memory extension is not enabled, and the value of VkApplicationInfo::apiVersion used to create the VkInstance is not greater than or equal to Version 1.1, srcQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_EXTERNAL
- VUID-VkBufferMemoryBarrier-None-09098
  If the VK_KHR_external_memory extension is not enabled, and the value of VkApplicationInfo::apiVersion used to create the VkInstance is not greater than or equal to Version 1.1, dstQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_EXTERNAL
- VUID-VkBufferMemoryBarrier-srcQueueFamilyIndex-09099
  If the VK_EXT_queue_family_foreign extension is not enabled, srcQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_FOREIGN_EXT
- VUID-VkBufferMemoryBarrier-dstQueueFamilyIndex-09100
  If the VK_EXT_queue_family_foreign extension is not enabled, dstQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_FOREIGN_EXT
- VUID-VkBufferMemoryBarrier-None-09049
  If the synchronization2 feature is not enabled, and buffer was created with a sharing mode of VK_SHARING_MODE_CONCURRENT, at least one of srcQueueFamilyIndex and dstQueueFamilyIndex **must** be VK_QUEUE_FAMILY_IGNORED
- VUID-VkBufferMemoryBarrier-None-09050
  If the synchronization2 feature is not enabled, and buffer was created with a sharing mode of VK_SHARING_MODE_CONCURRENT, srcQueueFamilyIndex **must** be VK_QUEUE_FAMILY_IGNORED or VK_QUEUE_FAMILY_EXTERNAL
- VUID-VkBufferMemoryBarrier-None-09051
  If the synchronization2 feature is not enabled, and buffer was created with a sharing mode of VK_SHARING_MODE_CONCURRENT, dstQueueFamilyIndex **must** be VK_QUEUE_FAMILY_IGNORED or VK_QUEUE_FAMILY_EXTERNAL

Valid Usage (Implicit)

- VUID-VkBufferMemoryBarrier-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER
- VUID-VkBufferMemoryBarrier-pNext-pNext
  pNext **must** be NULL or a pointer to a valid instance of VkExternalMemoryAcquireUnmodifiedEXT
- VUID-VkBufferMemoryBarrier-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkBufferMemoryBarrier-buffer-parameter
  buffer **must** be a valid VkBuffer handle

VK_WHOLE_SIZE is a special value indicating that the entire remaining length of a buffer or allocation following a given offset should be used. It **can** be specified for VkBufferMemoryBarrier

::size, vkMapMemory::size, and other similar structures.

```
#define VK_WHOLE_SIZE                     (~0ULL)
```

7.7.4. Image Memory Barriers

Image memory barriers only apply to memory accesses involving a specific image subresource range. That is, a memory dependency formed from an image memory barrier is scoped to access via the specified image subresource range. Image memory barriers **can** also be used to define image layout transitions or a queue family ownership transfer for the specified image subresource range.

The VkImageMemoryBarrier2 structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkImageMemoryBarrier2 {
    VkStructureType            sType;
    const void*                pNext;
    VkPipelineStageFlags2      srcStageMask;
    VkAccessFlags2             srcAccessMask;
    VkPipelineStageFlags2      dstStageMask;
    VkAccessFlags2             dstAccessMask;
    VkImageLayout              oldLayout;
    VkImageLayout              newLayout;
    uint32_t                   srcQueueFamilyIndex;
    uint32_t                   dstQueueFamilyIndex;
    VkImage                    image;
    VkImageSubresourceRange    subresourceRange;
} VkImageMemoryBarrier2;
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to VkImageMemoryBarrier2
typedef VkImageMemoryBarrier2 VkImageMemoryBarrier2KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- srcStageMask is a VkPipelineStageFlags2 mask of pipeline stages to be included in the first synchronization scope.
- srcAccessMask is a VkAccessFlags2 mask of access flags to be included in the first access scope.
- dstStageMask is a VkPipelineStageFlags2 mask of pipeline stages to be included in the second synchronization scope.
- dstAccessMask is a VkAccessFlags2 mask of access flags to be included in the second access scope.
- oldLayout is the old layout in an image layout transition.

- `newLayout` is the new layout in an image layout transition.
- `srcQueueFamilyIndex` is the source queue family for a queue family ownership transfer.
- `dstQueueFamilyIndex` is the destination queue family for a queue family ownership transfer.
- `image` is a handle to the image affected by this barrier.
- `subresourceRange` describes the image subresource range within `image` that is affected by this barrier.

This structure defines a memory dependency limited to an image subresource range, and **can** define a queue family ownership transfer operation and image layout transition for that subresource range.

The first synchronization scope and access scope described by this structure include only operations and memory accesses specified by the source stage mask and the source access mask.

The second synchronization scope and access scope described by this structure include only operations and memory accesses specified by the destination stage mask and the destination access mask.

Both access scopes are limited to only memory accesses to `image` in the subresource range defined by `subresourceRange`.

If `image` was created with VK_SHARING_MODE_EXCLUSIVE, and `srcQueueFamilyIndex` is not equal to `dstQueueFamilyIndex`, this memory barrier defines a queue family ownership transfer operation. When executed on a queue in the family identified by `srcQueueFamilyIndex`, this barrier defines a queue family release operation for the specified image subresource range, and if VkDependencyInfoKHR::`dependencyFlags` did not include VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR, the second synchronization scope does not apply to this operation. When executed on a queue in the family identified by `dstQueueFamilyIndex`, this barrier defines a queue family acquire operation for the specified image subresource range, and if VkDependencyInfoKHR::`dependencyFlags` did not include VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR, the first synchronization scope does not apply to this operation.

A queue family ownership transfer operation is also defined if the values are not equal, and either is one of the special queue family values reserved for external memory ownership transfers, as described in Queue Family Ownership Transfer. A queue family release operation is defined when `dstQueueFamilyIndex` is one of those values, and a queue family acquire operation is defined when `srcQueueFamilyIndex` is one of those values.

If `oldLayout` is not equal to `newLayout`, then the memory barrier defines an image layout transition for the specified image subresource range. If this memory barrier defines a queue family ownership transfer operation, the layout transition is only executed once between the queues.

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are equal and `oldLayout` and `newLayout` are also equal, the layout values are ignored and the image contents are preserved regardless of the values of `oldLayout`, `newLayout`, and the current layout of the image.

If `image` is a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT and the

maintenance9 feature is enabled, the baseArrayLayer and layerCount members of subresourceRange specify the subset of slices of the 3D image affected by the memory barrier, including the layout transition. Any slices of a 3D image not included in subresourceRange are not affected by the memory barrier and remain in their existing layout.

NOTE

Enabling the maintenance9 feature modifies the behavior of image barriers targeting 3D images created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT. Previously, a layerCount equal to 1 would cover the entire 3D image, but this has a different meaning when the maintenance9 feature is enabled. Linking this behavioral change solely to the maintenance9 feature caused an unintended break in forward-compatibility. Validation layers are expected to flag a warning for the scenario where the maintenance9 feature is not enabled, and the application uses layerCount equal to 1 on this kind of 3D image. layerCount can be set to VK_REMAINING_ARRAY_LAYERS instead, which has the same semantics with or without the extension. This validation check should make it feasible for software to avoid any breaking changes should the maintenance9 feature be enabled in the future, either explicitly by the application or by a layer outside the control of the application.

If image has a multi-planar format and the image is *disjoint*, then including VK_IMAGE_ASPECT_COLOR_BIT in the aspectMask member of subresourceRange is equivalent to including VK_IMAGE_ASPECT_PLANE_0_BIT, VK_IMAGE_ASPECT_PLANE_1_BIT, and (for three-plane formats only) VK_IMAGE_ASPECT_PLANE_2_BIT.

Valid Usage

- VUID-VkImageMemoryBarrier2-srcStageMask-03929
  If the geometryShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT
- VUID-VkImageMemoryBarrier2-srcStageMask-03930
  If the tessellationShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-VkImageMemoryBarrier2-srcStageMask-03931
  If the conditionalRendering feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT
- VUID-VkImageMemoryBarrier2-srcStageMask-03932
  If the fragmentDensityMap feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-VkImageMemoryBarrier2-srcStageMask-03933
  If the transformFeedback feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-VkImageMemoryBarrier2-srcStageMask-03934
  If the meshShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT

- VUID-VkImageMemoryBarrier2-srcStageMask-03935
  If the taskShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT
- VUID-VkImageMemoryBarrier2-srcStageMask-07317
  If the attachmentFragmentShadingRate feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkImageMemoryBarrier2-srcStageMask-07947
  If the rayTracingPipeline feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkImageMemoryBarrier2-srcStageMask-10751
  If the accelerationStructure feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkImageMemoryBarrier2-srcStageMask-10752
  If the rayTracingMaintenance1 feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR
- VUID-VkImageMemoryBarrier2-srcStageMask-11541
  If the VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03900
  If srcAccessMask includes VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_COPY_INDIRECT_BIT_KHR, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03901
  If srcAccessMask includes VK_ACCESS_2_INDEX_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03902
  If srcAccessMask includes VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03903
  If srcAccessMask includes VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_SUBPASS_SHADER_BIT_HUAWEI, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03904
  If srcAccessMask includes VK_ACCESS_2_UNIFORM_READ_BIT, srcStageMask **must** include

VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages

- VUID-VkImageMemoryBarrier2-srcAccessMask-03905
  If srcAccessMask includes VK_ACCESS_2_SHADER_SAMPLED_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkImageMemoryBarrier2-srcAccessMask-03906
  If srcAccessMask includes VK_ACCESS_2_SHADER_STORAGE_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkImageMemoryBarrier2-srcAccessMask-03907
  If srcAccessMask includes VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkImageMemoryBarrier2-srcAccessMask-07454
  If srcAccessMask includes VK_ACCESS_2_SHADER_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkImageMemoryBarrier2-srcAccessMask-03909
  If srcAccessMask includes VK_ACCESS_2_SHADER_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkImageMemoryBarrier2-srcAccessMask-03910
  If srcAccessMask includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03911
  If srcAccessMask includes VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03912
  If srcAccessMask includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or

VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-srcAccessMask-03913
  If srcAccessMask includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03914
  If srcAccessMask includes VK_ACCESS_2_TRANSFER_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03915
  If srcAccessMask includes VK_ACCESS_2_TRANSFER_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_CLEAR_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03916
  If srcAccessMask includes VK_ACCESS_2_HOST_READ_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03917
  If srcAccessMask includes VK_ACCESS_2_HOST_WRITE_BIT, srcStageMask **must** include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03918
  If srcAccessMask includes VK_ACCESS_2_CONDITIONAL_RENDERING_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03919
  If srcAccessMask includes VK_ACCESS_2_FRAGMENT_DENSITY_MAP_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03920
  If srcAccessMask includes VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-srcAccessMask-04747
  If `srcAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03922
  If `srcAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03926
  If `srcAccessMask` includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03927
  If `srcAccessMask` includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the `rayQuery` feature is enabled, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-03928
  If `srcAccessMask` includes VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-06256
  If `srcAccessMask` includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the `rayQuery` feature is not enabled, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-srcAccessMask-07272
  If `srcAccessMask` includes VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR, `srcStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT or VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkImageMemoryBarrier2-srcAccessMask-04858

If srcAccessMask includes VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR

- VUID-VkImageMemoryBarrier2-srcAccessMask-04859
  If srcAccessMask includes VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
- VUID-VkImageMemoryBarrier2-srcAccessMask-04860
  If srcAccessMask includes VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- VUID-VkImageMemoryBarrier2-srcAccessMask-04861
  If srcAccessMask includes VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR, srcStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- VUID-VkImageMemoryBarrier2-srcAccessMask-07457
  If srcAccessMask includes VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkImageMemoryBarrier2-srcAccessMask-07458
  If srcAccessMask includes VK_ACCESS_2_MICROMAP_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT or VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkImageMemoryBarrier2-srcAccessMask-08118
  If srcAccessMask includes VK_ACCESS_2_DESCRIPTOR_BUFFER_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkImageMemoryBarrier2-srcAccessMask-11771
  If srcAccessMask includes VK_ACCESS_2_MEMORY_DECOMPRESSION_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- VUID-VkImageMemoryBarrier2-srcAccessMask-11772
  If srcAccessMask includes VK_ACCESS_2_MEMORY_DECOMPRESSION_WRITE_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- VUID-VkImageMemoryBarrier2-srcAccessMask-11294
  If srcAccessMask includes VK_ACCESS_2_SAMPLER_HEAP_READ_BIT_EXT or VK_ACCESS_2_RESOURCE_HEAP_READ_BIT_EXT, srcStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkImageMemoryBarrier2-dstStageMask-03929
  If the geometryShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT
- VUID-VkImageMemoryBarrier2-dstStageMask-03930
  If the tessellationShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-VkImageMemoryBarrier2-dstStageMask-03931

If the conditionalRendering feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT

- VUID-VkImageMemoryBarrier2-dstStageMask-03932
  If the fragmentDensityMap feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-VkImageMemoryBarrier2-dstStageMask-03933
  If the transformFeedback feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-VkImageMemoryBarrier2-dstStageMask-03934
  If the meshShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT
- VUID-VkImageMemoryBarrier2-dstStageMask-03935
  If the taskShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT
- VUID-VkImageMemoryBarrier2-dstStageMask-07317
  If the attachmentFragmentShadingRate feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkImageMemoryBarrier2-dstStageMask-07947
  If the rayTracingPipeline feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkImageMemoryBarrier2-dstStageMask-10751
  If the accelerationStructure feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkImageMemoryBarrier2-dstStageMask-10752
  If the rayTracingMaintenance1 feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR
- VUID-VkImageMemoryBarrier2-dstStageMask-11541
  If the VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03900
  If dstAccessMask includes VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_COPY_INDIRECT_BIT_KHR, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03901
  If dstAccessMask includes VK_ACCESS_2_INDEX_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03902
  If dstAccessMask includes VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT, dstStageMask

must include VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-03903
  If `dstAccessMask` includes VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_SUBPASS_SHADER_BIT_HUAWEI, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03904
  If `dstAccessMask` includes VK_ACCESS_2_UNIFORM_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkImageMemoryBarrier2-dstAccessMask-03905
  If `dstAccessMask` includes VK_ACCESS_2_SHADER_SAMPLED_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkImageMemoryBarrier2-dstAccessMask-03906
  If `dstAccessMask` includes VK_ACCESS_2_SHADER_STORAGE_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkImageMemoryBarrier2-dstAccessMask-03907
  If `dstAccessMask` includes VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkImageMemoryBarrier2-dstAccessMask-07454
  If `dstAccessMask` includes VK_ACCESS_2_SHADER_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkImageMemoryBarrier2-dstAccessMask-03909
  If `dstAccessMask` includes VK_ACCESS_2_SHADER_WRITE_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkImageMemoryBarrier2-dstAccessMask-03910
  If `dstAccessMask` includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or

VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-03911
  If dstAccessMask includes VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03912
  If dstAccessMask includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03913
  If dstAccessMask includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03914
  If dstAccessMask includes VK_ACCESS_2_TRANSFER_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03915
  If dstAccessMask includes VK_ACCESS_2_TRANSFER_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_CLEAR_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03916
  If dstAccessMask includes VK_ACCESS_2_HOST_READ_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03917
  If dstAccessMask includes VK_ACCESS_2_HOST_WRITE_BIT, dstStageMask **must** include VK_PIPELINE_STAGE_2_HOST_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03918
  If dstAccessMask includes VK_ACCESS_2_CONDITIONAL_RENDERING_READ_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-03919
  If `dstAccessMask` includes VK_ACCESS_2_FRAGMENT_DENSITY_MAP_READ_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03920
  If `dstAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-04747
  If `dstAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03922
  If `dstAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03926
  If `dstAccessMask` includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03927
  If `dstAccessMask` includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the `rayQuery` feature is enabled, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-03928
  If `dstAccessMask` includes VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT
- VUID-VkImageMemoryBarrier2-dstAccessMask-06256

If dstAccessMask includes VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR, and the rayQuery feature is not enabled, dstStageMask **must** include VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-07272
  If dstAccessMask includes VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT or VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkImageMemoryBarrier2-dstAccessMask-04858
  If dstAccessMask includes VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
- VUID-VkImageMemoryBarrier2-dstAccessMask-04859
  If dstAccessMask includes VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
- VUID-VkImageMemoryBarrier2-dstAccessMask-04860
  If dstAccessMask includes VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- VUID-VkImageMemoryBarrier2-dstAccessMask-04861
  If dstAccessMask includes VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR, dstStageMask **must** include VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- VUID-VkImageMemoryBarrier2-dstAccessMask-07457
  If dstAccessMask includes VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-VkImageMemoryBarrier2-dstAccessMask-07458
  If dstAccessMask includes VK_ACCESS_2_MICROMAP_READ_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT or VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkImageMemoryBarrier2-dstAccessMask-08118
  If dstAccessMask includes VK_ACCESS_2_DESCRIPTOR_BUFFER_READ_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages
- VUID-VkImageMemoryBarrier2-dstAccessMask-11771
  If dstAccessMask includes VK_ACCESS_2_MEMORY_DECOMPRESSION_READ_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- VUID-VkImageMemoryBarrier2-dstAccessMask-11772
  If dstAccessMask includes VK_ACCESS_2_MEMORY_DECOMPRESSION_WRITE_BIT_EXT, dstStageMask **must** include VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- VUID-VkImageMemoryBarrier2-dstAccessMask-11294
  If dstAccessMask includes VK_ACCESS_2_SAMPLER_HEAP_READ_BIT_EXT or VK_ACCESS_2_RESOURCE_HEAP_READ_BIT_EXT, dstStageMask **must** include

VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages

- VUID-VkImageMemoryBarrier2-oldLayout-01197
  If layouts are not ignored, oldLayout **must** be VK_IMAGE_LAYOUT_UNDEFINED or the current layout of the image subresources affected by the barrier
- VUID-VkImageMemoryBarrier2-newLayout-01198
  If layouts are not ignored, newLayout **must** not be VK_IMAGE_LAYOUT_UNDEFINED or VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT or VK_IMAGE_LAYOUT_PREINITIALIZED
- VUID-VkImageMemoryBarrier2-image-09117
  If image was created with a sharing mode of VK_SHARING_MODE_EXCLUSIVE, and srcQueueFamilyIndex and dstQueueFamilyIndex are not equal, srcQueueFamilyIndex **must** be VK_QUEUE_FAMILY_EXTERNAL, VK_QUEUE_FAMILY_FOREIGN_EXT, or a valid queue family
- VUID-VkImageMemoryBarrier2-image-09118
  If image was created with a sharing mode of VK_SHARING_MODE_EXCLUSIVE, and srcQueueFamilyIndex and dstQueueFamilyIndex are not equal, dstQueueFamilyIndex **must** be VK_QUEUE_FAMILY_EXTERNAL, VK_QUEUE_FAMILY_FOREIGN_EXT, or a valid queue family
- VUID-VkImageMemoryBarrier2-None-09097
  If the VK_KHR_external_memory extension is not enabled, and the value of VkApplicationInfo::apiVersion used to create the VkInstance is not greater than or equal to Version 1.1, srcQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_EXTERNAL
- VUID-VkImageMemoryBarrier2-None-09098
  If the VK_KHR_external_memory extension is not enabled, and the value of VkApplicationInfo::apiVersion used to create the VkInstance is not greater than or equal to Version 1.1, dstQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_EXTERNAL
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-09099
  If the VK_EXT_queue_family_foreign extension is not enabled, srcQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_FOREIGN_EXT
- VUID-VkImageMemoryBarrier2-dstQueueFamilyIndex-09100
  If the VK_EXT_queue_family_foreign extension is not enabled, dstQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_FOREIGN_EXT
- VUID-VkImageMemoryBarrier2-subresourceRange-01486
  subresourceRange.baseMipLevel **must** be less than the mipLevels specified in VkImageCreateInfo when image was created
- VUID-VkImageMemoryBarrier2-subresourceRange-01724
  If subresourceRange.levelCount is not VK_REMAINING_MIP_LEVELS, subresourceRange.baseMipLevel + subresourceRange.levelCount **must** be less than or equal to the mipLevels specified in VkImageCreateInfo when image was created
- VUID-VkImageMemoryBarrier2-subresourceRange-01488
  If image is not a 3D image or was created without

VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set, or the maintenance9 feature is not enabled, subresourceRange.baseArrayLayer **must** be less than the arrayLayers specified in VkImageCreateInfo when image was created

- VUID-VkImageMemoryBarrier2-maintenance9-10798
  If the maintenance9 feature is enabled and image is a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set, subresourceRange.baseArrayLayer **must** be less than the depth computed from baseMipLevel and extent.depth specified in VkImageCreateInfo when image was created, according to the formula defined in Image Mip Level Sizing
- VUID-VkImageMemoryBarrier2-maintenance9-10799
  If the maintenance9 feature is enabled and image is a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set and either subresourceRange.baseArrayLayer is not equal to 0 or subresourceRange.layerCount is not equal to VK_REMAINING_ARRAY_LAYERS, subresourceRange.levelCount **must** be 1
- VUID-VkImageMemoryBarrier2-subresourceRange-01725
  If image is not a 3D image or was created without VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set, or the maintenance9 feature is not enabled, and subresourceRange.layerCount is not VK_REMAINING_ARRAY_LAYERS, subresourceRange.baseArrayLayer + subresourceRange.layerCount **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when image was created
- VUID-VkImageMemoryBarrier2-maintenance9-10800
  If the maintenance9 feature is enabled, subresourceRange.layerCount is not VK_REMAINING_ARRAY_LAYERS, and image is a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set, subresourceRange.baseArrayLayer + subresourceRange.layerCount **must** be less than or equal to the depth computed from baseMipLevel and extent.depth specified in VkImageCreateInfo when image was created, according to the formula defined in Image Mip Level Sizing
- VUID-VkImageMemoryBarrier2-image-01932
  If image is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-VkImageMemoryBarrier2-image-09241
  If image has a color format that is single-plane, then the aspectMask member of subresourceRange **must** be VK_IMAGE_ASPECT_COLOR_BIT
- VUID-VkImageMemoryBarrier2-image-09242
  If image has a color format and is not *disjoint*, then the aspectMask member of subresourceRange **must** be VK_IMAGE_ASPECT_COLOR_BIT
- VUID-VkImageMemoryBarrier2-image-01672
  If image has a multi-planar format and the image is *disjoint*, then the aspectMask member of subresourceRange **must** include at least one multi-planar aspect mask bit or VK_IMAGE_ASPECT_COLOR_BIT
- VUID-VkImageMemoryBarrier2-image-03320
  If image has a depth/stencil format with both depth and stencil and the separateDepthStencilLayouts feature is not enabled, then the aspectMask member of subresourceRange **must** include both VK_IMAGE_ASPECT_DEPTH_BIT and

VK_IMAGE_ASPECT_STENCIL_BIT

- VUID-VkImageMemoryBarrier2-image-03319
  If `image` has a depth/stencil format with both depth and stencil and the `separateDepthStencilLayouts` feature is enabled, then the `aspectMask` member of `subresourceRange` **must** include either or both VK_IMAGE_ASPECT_DEPTH_BIT and VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkImageMemoryBarrier2-image-10749
  If `image` has a depth-only format then the `aspectMask` member of `subresourceRange` **must** be VK_IMAGE_ASPECT_DEPTH_BIT
- VUID-VkImageMemoryBarrier2-image-10750
  If `image` has a stencil-only format then the `aspectMask` member of `subresourceRange` **must** be VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkImageMemoryBarrier2-aspectMask-08702
  If the `aspectMask` member of `subresourceRange` includes VK_IMAGE_ASPECT_DEPTH_BIT, `oldLayout` and `newLayout` **must** not be one of VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkImageMemoryBarrier2-aspectMask-08703
  If the `aspectMask` member of `subresourceRange` includes VK_IMAGE_ASPECT_STENCIL_BIT, `oldLayout` and `newLayout` **must** not be one of VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL
- VUID-VkImageMemoryBarrier2-subresourceRange-09601
  `subresourceRange.aspectMask` **must** be valid for the `format` the `image` was created with
- VUID-VkImageMemoryBarrier2-oldLayout-01208
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier2-oldLayout-01209
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier2-oldLayout-01210
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier2-oldLayout-01211
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_SAMPLED_BIT or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier2-oldLayout-01212

If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set

- VUID-VkImageMemoryBarrier2-oldLayout-01213
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-VkImageMemoryBarrier2-oldLayout-10767
  If the zeroInitializeDeviceMemory feature is not enabled, `oldLayout` **must** not be VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT
- VUID-VkImageMemoryBarrier2-oldLayout-10768
  If `oldLayout` is VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT, then all subresources **must** be included in the barrier
- VUID-VkImageMemoryBarrier2-oldLayout-01658
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier2-oldLayout-01659
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-04065
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL then `image` **must** have been created with at least one of the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, VK_IMAGE_USAGE_SAMPLED_BIT, or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flags set
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-04066
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-04067
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL then `image` **must** have been created with at least one of the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, VK_IMAGE_USAGE_SAMPLED_BIT, or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flags set
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-04068
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier2-synchronization2-07793

If the synchronization2 feature is not enabled, oldLayout **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR or VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR

- VUID-VkImageMemoryBarrier2-synchronization2-07794
  If the synchronization2 feature is not enabled, newLayout **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR or VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-03938
  If layouts are not ignored, oldLayout or newLayout is VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL, image **must** have been created with the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-03939
  If layouts are not ignored, oldLayout or newLayout is VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL, image **must** have been created with at least one of the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, VK_IMAGE_USAGE_SAMPLED_BIT, or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flags set
- VUID-VkImageMemoryBarrier2-oldLayout-02088
  If layouts are not ignored, oldLayout or newLayout is VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR then image **must** have been created with the VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR usage flag set
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-07120
  If layouts are not ignored, oldLayout or newLayout is VK_IMAGE_LAYOUT_VIDEO_DECODE_SRC_KHR then image **must** have been created with the VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR usage flag set
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-07121
  If layouts are not ignored, oldLayout or newLayout is VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR then image **must** have been created with the VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR usage flag set
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-07122
  If layouts are not ignored, oldLayout or newLayout is VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR then image **must** have been created with the VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR usage flag set
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-07123
  If layouts are not ignored, oldLayout or newLayout is VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR then image **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR usage flag set
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-07124
  If layouts are not ignored, oldLayout or newLayout is VK_IMAGE_LAYOUT_VIDEO_ENCODE_DST_KHR then image **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR usage flag set
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-07125

If layouts are not ignored, oldLayout or newLayout is VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR then image **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR usage flag set

- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-10287
  If layouts are not ignored, oldLayout or newLayout is VK_IMAGE_LAYOUT_VIDEO_ENCODE_QUANTIZATION_MAP_KHR then image **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR usage flags set
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-07006
  If layouts are not ignored, oldLayout or newLayout is VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT then image **must** have been created with either the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flags set, and the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT or VK_IMAGE_USAGE_SAMPLED_BIT usage flags set, and the VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT usage flag set
- VUID-VkImageMemoryBarrier2-attachmentFeedbackLoopLayout-07313
  If the attachmentFeedbackLoopLayout feature is not enabled, newLayout **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-09550
  If layouts are not ignored, oldLayout or newLayout is VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ then image **must** have been created with either the VK_IMAGE_USAGE_STORAGE_BIT usage flag set, or with both the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flag and either of the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flags set
- VUID-VkImageMemoryBarrier2-dynamicRenderingLocalRead-09551
  If the dynamicRenderingLocalRead feature is not enabled, oldLayout **must** not be VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ
- VUID-VkImageMemoryBarrier2-dynamicRenderingLocalRead-09552
  If the dynamicRenderingLocalRead feature is not enabled, newLayout **must** not be VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ
- VUID-VkImageMemoryBarrier2-srcStageMask-03854
  If either srcStageMask or dstStageMask includes VK_PIPELINE_STAGE_2_HOST_BIT, srcQueueFamilyIndex and dstQueueFamilyIndex **must** be equal
- VUID-VkImageMemoryBarrier2-srcStageMask-03855
  If srcStageMask includes VK_PIPELINE_STAGE_2_HOST_BIT, and srcQueueFamilyIndex and dstQueueFamilyIndex define a queue family ownership transfer or oldLayout and newLayout define an image layout transition, oldLayout **must** be one of VK_IMAGE_LAYOUT_PREINITIALIZED, VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT, VK_IMAGE_LAYOUT_UNDEFINED, or VK_IMAGE_LAYOUT_GENERAL

Valid Usage (Implicit)

- VUID-VkImageMemoryBarrier2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER_2
- VUID-VkImageMemoryBarrier2-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkExternalMemoryAcquireUnmodifiedEXT, VkMemoryBarrierAccessFlags3KHR, or VkSampleLocationsInfoEXT
- VUID-VkImageMemoryBarrier2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkImageMemoryBarrier2-srcStageMask-parameter
  If `srcStageMask` is not `0`, `srcStageMask` **must** be a valid combination of VkPipelineStageFlagBits2 values
- VUID-VkImageMemoryBarrier2-srcAccessMask-parameter
  If `srcAccessMask` is not `0`, `srcAccessMask` **must** be a valid combination of VkAccessFlagBits2 values
- VUID-VkImageMemoryBarrier2-dstStageMask-parameter
  If `dstStageMask` is not `0`, `dstStageMask` **must** be a valid combination of VkPipelineStageFlagBits2 values
- VUID-VkImageMemoryBarrier2-dstAccessMask-parameter
  If `dstAccessMask` is not `0`, `dstAccessMask` **must** be a valid combination of VkAccessFlagBits2 values
- VUID-VkImageMemoryBarrier2-oldLayout-parameter
  `oldLayout` **must** be a valid VkImageLayout value
- VUID-VkImageMemoryBarrier2-newLayout-parameter
  `newLayout` **must** be a valid VkImageLayout value
- VUID-VkImageMemoryBarrier2-image-parameter
  `image` **must** be a valid VkImage handle
- VUID-VkImageMemoryBarrier2-subresourceRange-parameter
  `subresourceRange` **must** be a valid VkImageSubresourceRange structure

The `VkImageMemoryBarrier` structure is defined as:

WARNING	This functionality is superseded by VkImageMemoryBarrier2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef struct VkImageMemoryBarrier {
    VkStructureType            sType;
    const void*                pNext;
    VkAccessFlags              srcAccessMask;
    VkAccessFlags              dstAccessMask;
    VkImageLayout              oldLayout;
    VkImageLayout              newLayout;
    uint32_t                   srcQueueFamilyIndex;
    uint32_t                   dstQueueFamilyIndex;
    VkImage                    image;
    VkImageSubresourceRange    subresourceRange;
} VkImageMemoryBarrier;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcAccessMask` is a bitmask of VkAccessFlagBits specifying a source access mask.
- `dstAccessMask` is a bitmask of VkAccessFlagBits specifying a destination access mask.
- `oldLayout` is the old layout in an image layout transition.
- `newLayout` is the new layout in an image layout transition.
- `srcQueueFamilyIndex` is the source queue family for a queue family ownership transfer.
- `dstQueueFamilyIndex` is the destination queue family for a queue family ownership transfer.
- `image` is a handle to the image affected by this barrier.
- `subresourceRange` describes the image subresource range within `image` that is affected by this barrier.

The first access scope is limited to access to memory through the specified image subresource range, via access types in the source access mask specified by `srcAccessMask`. If `srcAccessMask` includes VK_ACCESS_HOST_WRITE_BIT, memory writes performed by that access type are also made visible, as that access type is not performed through a resource.

The second access scope is limited to access to memory through the specified image subresource range, via access types in the destination access mask specified by `dstAccessMask`. If `dstAccessMask` includes VK_ACCESS_HOST_WRITE_BIT or VK_ACCESS_HOST_READ_BIT, available memory writes are also made visible to accesses of those types, as those access types are not performed through a resource.

If `srcQueueFamilyIndex` is not equal to `dstQueueFamilyIndex`, and `srcQueueFamilyIndex` is equal to the current queue family, then the memory barrier defines a queue family release operation for the specified image subresource range, and if `dependencyFlags` did not include VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR, the second synchronization scope of the calling command does not apply to this operation.

If `dstQueueFamilyIndex` is not equal to `srcQueueFamilyIndex`, and `dstQueueFamilyIndex` is equal to the current queue family, then the memory barrier defines a queue family acquire operation for the

specified image subresource range, and if dependencyFlags did not include VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR, the first synchronization scope of the calling command does not apply to this operation.

If the synchronization2 feature is not enabled or oldLayout is not equal to newLayout, oldLayout and newLayout define an image layout transition for the specified image subresource range.

If the synchronization2 feature is enabled, srcQueueFamilyIndex and dstQueueFamilyIndex are equal, and oldLayout and newLayout are also equal, the layout values are ignored and the image contents are preserved regardless of the values of oldLayout, newLayout, and the current layout of the image.

If image is a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT and the maintenance9 feature is enabled, the baseArrayLayer and layerCount members of subresourceRange specify the subset of slices of the 3D image affected by the memory barrier, including the layout transition. Any slices of a 3D image not included in subresourceRange are not affected by the memory barrier and remain in their existing layout.

If image has a multi-planar format and the image is *disjoint*, then including VK_IMAGE_ASPECT_COLOR_BIT in the aspectMask member of subresourceRange is equivalent to including VK_IMAGE_ASPECT_PLANE_0_BIT, VK_IMAGE_ASPECT_PLANE_1_BIT, and (for three-plane formats only) VK_IMAGE_ASPECT_PLANE_2_BIT.

Valid Usage

- VUID-VkImageMemoryBarrier-oldLayout-01197
  If layouts are not ignored, oldLayout **must** be VK_IMAGE_LAYOUT_UNDEFINED or the current layout of the image subresources affected by the barrier
- VUID-VkImageMemoryBarrier-newLayout-01198
  If layouts are not ignored, newLayout **must** not be VK_IMAGE_LAYOUT_UNDEFINED or VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT or VK_IMAGE_LAYOUT_PREINITIALIZED
- VUID-VkImageMemoryBarrier-image-09117
  If image was created with a sharing mode of VK_SHARING_MODE_EXCLUSIVE, and srcQueueFamilyIndex and dstQueueFamilyIndex are not equal, srcQueueFamilyIndex **must** be VK_QUEUE_FAMILY_EXTERNAL, VK_QUEUE_FAMILY_FOREIGN_EXT, or a valid queue family
- VUID-VkImageMemoryBarrier-image-09118
  If image was created with a sharing mode of VK_SHARING_MODE_EXCLUSIVE, and srcQueueFamilyIndex and dstQueueFamilyIndex are not equal, dstQueueFamilyIndex **must** be VK_QUEUE_FAMILY_EXTERNAL, VK_QUEUE_FAMILY_FOREIGN_EXT, or a valid queue family
- VUID-VkImageMemoryBarrier-None-09097
  If the VK_KHR_external_memory extension is not enabled, and the value of VkApplicationInfo::apiVersion used to create the VkInstance is not greater than or equal to Version 1.1, srcQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_EXTERNAL
- VUID-VkImageMemoryBarrier-None-09098
  If the VK_KHR_external_memory extension is not enabled, and the value of

VkApplicationInfo::apiVersion used to create the VkInstance is not greater than or equal to Version 1.1, dstQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_EXTERNAL

- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-09099
  If the VK_EXT_queue_family_foreign extension is not enabled, srcQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_FOREIGN_EXT
- VUID-VkImageMemoryBarrier-dstQueueFamilyIndex-09100
  If the VK_EXT_queue_family_foreign extension is not enabled, dstQueueFamilyIndex **must** not be VK_QUEUE_FAMILY_FOREIGN_EXT
- VUID-VkImageMemoryBarrier-subresourceRange-01486
  subresourceRange.baseMipLevel **must** be less than the mipLevels specified in VkImageCreateInfo when image was created
- VUID-VkImageMemoryBarrier-subresourceRange-01724
  If subresourceRange.levelCount is not VK_REMAINING_MIP_LEVELS, subresourceRange.baseMipLevel + subresourceRange.levelCount **must** be less than or equal to the mipLevels specified in VkImageCreateInfo when image was created
- VUID-VkImageMemoryBarrier-subresourceRange-01488
  If image is not a 3D image or was created without VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set, or the maintenance9 feature is not enabled, subresourceRange.baseArrayLayer **must** be less than the arrayLayers specified in VkImageCreateInfo when image was created
- VUID-VkImageMemoryBarrier-maintenance9-10798
  If the maintenance9 feature is enabled and image is a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set, subresourceRange.baseArrayLayer **must** be less than the depth computed from baseMipLevel and extent.depth specified in VkImageCreateInfo when image was created, according to the formula defined in Image Mip Level Sizing
- VUID-VkImageMemoryBarrier-maintenance9-10799
  If the maintenance9 feature is enabled and image is a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set and either subresourceRange.baseArrayLayer is not equal to 0 or subresourceRange.layerCount is not equal to VK_REMAINING_ARRAY_LAYERS, subresourceRange.levelCount **must** be 1
- VUID-VkImageMemoryBarrier-subresourceRange-01725
  If image is not a 3D image or was created without VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set, or the maintenance9 feature is not enabled, and subresourceRange.layerCount is not VK_REMAINING_ARRAY_LAYERS, subresourceRange.baseArrayLayer + subresourceRange.layerCount **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when image was created
- VUID-VkImageMemoryBarrier-maintenance9-10800
  If the maintenance9 feature is enabled, subresourceRange.layerCount is not VK_REMAINING_ARRAY_LAYERS, and image is a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set, subresourceRange.baseArrayLayer + subresourceRange.layerCount **must** be less than or equal to the depth computed from baseMipLevel and extent.depth specified in VkImageCreateInfo when image was created, according to the formula defined in Image Mip Level Sizing

- VUID-VkImageMemoryBarrier-image-01932
  If `image` is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkImageMemoryBarrier-image-09241
  If `image` has a color format that is single-plane, then the `aspectMask` member of `subresourceRange` **must** be VK_IMAGE_ASPECT_COLOR_BIT
- VUID-VkImageMemoryBarrier-image-09242
  If `image` has a color format and is not *disjoint*, then the `aspectMask` member of `subresourceRange` **must** be VK_IMAGE_ASPECT_COLOR_BIT
- VUID-VkImageMemoryBarrier-image-01672
  If `image` has a multi-planar format and the image is *disjoint*, then the `aspectMask` member of `subresourceRange` **must** include at least one multi-planar aspect mask bit or VK_IMAGE_ASPECT_COLOR_BIT
- VUID-VkImageMemoryBarrier-image-03320
  If `image` has a depth/stencil format with both depth and stencil and the `separateDepthStencilLayouts` feature is not enabled, then the `aspectMask` member of `subresourceRange` **must** include both VK_IMAGE_ASPECT_DEPTH_BIT and VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkImageMemoryBarrier-image-03319
  If `image` has a depth/stencil format with both depth and stencil and the `separateDepthStencilLayouts` feature is enabled, then the `aspectMask` member of `subresourceRange` **must** include either or both VK_IMAGE_ASPECT_DEPTH_BIT and VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkImageMemoryBarrier-image-10749
  If `image` has a depth-only format then the `aspectMask` member of `subresourceRange` **must** be VK_IMAGE_ASPECT_DEPTH_BIT
- VUID-VkImageMemoryBarrier-image-10750
  If `image` has a stencil-only format then the `aspectMask` member of `subresourceRange` **must** be VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkImageMemoryBarrier-aspectMask-08702
  If the `aspectMask` member of `subresourceRange` includes VK_IMAGE_ASPECT_DEPTH_BIT, `oldLayout` and `newLayout` **must** not be one of VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkImageMemoryBarrier-aspectMask-08703
  If the `aspectMask` member of `subresourceRange` includes VK_IMAGE_ASPECT_STENCIL_BIT, `oldLayout` and `newLayout` **must** not be one of VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL
- VUID-VkImageMemoryBarrier-subresourceRange-09601
  `subresourceRange.aspectMask` **must** be valid for the `format` the `image` was created with
- VUID-VkImageMemoryBarrier-oldLayout-01208
  If layouts are not ignored, `oldLayout` or `newLayout` is

VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT usage flag set

- VUID-VkImageMemoryBarrier-oldLayout-01209
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier-oldLayout-01210
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier-oldLayout-01211
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_SAMPLED_BIT or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier-oldLayout-01212
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-VkImageMemoryBarrier-oldLayout-01213
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-VkImageMemoryBarrier-oldLayout-10767
  If the zeroInitializeDeviceMemory feature is not enabled, `oldLayout` **must** not be VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT
- VUID-VkImageMemoryBarrier-oldLayout-10768
  If `oldLayout` is VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT, then all subresources **must** be included in the barrier
- VUID-VkImageMemoryBarrier-oldLayout-01658
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier-oldLayout-01659
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-04065
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL then `image` **must** have been created with at least one of the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT,

VK_IMAGE_USAGE_SAMPLED_BIT, or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flags set

- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-04066
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-04067
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL then `image` **must** have been created with at least one of the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, VK_IMAGE_USAGE_SAMPLED_BIT, or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flags set
- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-04068
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier-synchronization2-07793
  If the `synchronization2` feature is not enabled, `oldLayout` **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR or VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VUID-VkImageMemoryBarrier-synchronization2-07794
  If the `synchronization2` feature is not enabled, `newLayout` **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR or VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-03938
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL, `image` **must** have been created with the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-03939
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL, `image` **must** have been created with at least one of the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, VK_IMAGE_USAGE_SAMPLED_BIT, or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flags set
- VUID-VkImageMemoryBarrier-oldLayout-02088
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR usage flag set
- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-07120
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_VIDEO_DECODE_SRC_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR usage flag set

- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-07121
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR usage flag set
- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-07122
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR usage flag set
- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-07123
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR usage flag set
- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-07124
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_VIDEO_ENCODE_DST_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR usage flag set
- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-07125
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR usage flag set
- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-10287
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_VIDEO_ENCODE_QUANTIZATION_MAP_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR usage flags set
- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-07006
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT then `image` **must** have been created with either the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flags set, and the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT or VK_IMAGE_USAGE_SAMPLED_BIT usage flags set, and the VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT usage flag set
- VUID-VkImageMemoryBarrier-attachmentFeedbackLoopLayout-07313
  If the `attachmentFeedbackLoopLayout` feature is not enabled, `newLayout` **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT
- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-09550
  If layouts are not ignored, `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ then `image` **must** have been created with either the VK_IMAGE_USAGE_STORAGE_BIT usage flag set, or with both the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flag and either of the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flags set

- VUID-VkImageMemoryBarrier-dynamicRenderingLocalRead-09551
  If the dynamicRenderingLocalRead feature is not enabled, oldLayout **must** not be VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ
- VUID-VkImageMemoryBarrier-dynamicRenderingLocalRead-09552
  If the dynamicRenderingLocalRead feature is not enabled, newLayout **must** not be VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ
- VUID-VkImageMemoryBarrier-None-09052
  If the synchronization2 feature is not enabled, and image was created with a sharing mode of VK_SHARING_MODE_CONCURRENT, at least one of srcQueueFamilyIndex and dstQueueFamilyIndex **must** be VK_QUEUE_FAMILY_IGNORED
- VUID-VkImageMemoryBarrier-None-09053
  If the synchronization2 feature is not enabled, and image was created with a sharing mode of VK_SHARING_MODE_CONCURRENT, srcQueueFamilyIndex **must** be VK_QUEUE_FAMILY_IGNORED or VK_QUEUE_FAMILY_EXTERNAL
- VUID-VkImageMemoryBarrier-None-09054
  If the synchronization2 feature is not enabled, and image was created with a sharing mode of VK_SHARING_MODE_CONCURRENT, dstQueueFamilyIndex **must** be VK_QUEUE_FAMILY_IGNORED or VK_QUEUE_FAMILY_EXTERNAL
- VUID-VkImageMemoryBarrier-dstQueueFamilyIndex-12331
  If dstQueueFamilyIndex is VK_QUEUE_FAMILY_EXTERNAL and image was created with VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT or VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT in VkExternalMemoryImageCreateInfo::handleTypes, newLayout **must** be VK_IMAGE_LAYOUT_GENERAL
- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-12332
  If srcQueueFamilyIndex is VK_QUEUE_FAMILY_EXTERNAL and image was created with VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT or VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT in VkExternalMemoryImageCreateInfo::handleTypes, oldLayout **must** be VK_IMAGE_LAYOUT_GENERAL or VK_IMAGE_LAYOUT_UNDEFINED

Valid Usage (Implicit)

- VUID-VkImageMemoryBarrier-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER
- VUID-VkImageMemoryBarrier-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkExternalMemoryAcquireUnmodifiedEXT or VkSampleLocationsInfoEXT
- VUID-VkImageMemoryBarrier-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkImageMemoryBarrier-oldLayout-parameter
  oldLayout **must** be a valid VkImageLayout value

- VUID-VkImageMemoryBarrier-newLayout-parameter
  `newLayout` **must** be a valid VkImageLayout value
- VUID-VkImageMemoryBarrier-image-parameter
  `image` **must** be a valid VkImage handle
- VUID-VkImageMemoryBarrier-subresourceRange-parameter
  `subresourceRange` **must** be a valid VkImageSubresourceRange structure

To facilitate usage of images whose memory is initialized on the host, Vulkan allows image layout transitions to be performed by the host as well, albeit supporting limited layouts.

To perform an image layout transition on the host, call:

```
// Provided by VK_VERSION_1_4
VkResult vkTransitionImageLayout(
    VkDevice                                    device,
    uint32_t                                    transitionCount,
    const VkHostImageLayoutTransitionInfo*      pTransitions);
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to vkTransitionImageLayout
VkResult vkTransitionImageLayoutEXT(
    VkDevice                                    device,
    uint32_t                                    transitionCount,
    const VkHostImageLayoutTransitionInfo*      pTransitions);
```

- `device` is the device which owns `pTransitions`[i].`image`.
- `transitionCount` is the number of image layout transitions to perform.
- `pTransitions` is a pointer to an array of VkHostImageLayoutTransitionInfo structures specifying the image and subresource ranges within them to transition.

Valid Usage (Implicit)

- VUID-vkTransitionImageLayout-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkTransitionImageLayout-pTransitions-parameter
  `pTransitions` **must** be a valid pointer to an array of `transitionCount` valid VkHostImageLayoutTransitionInfo structures
- VUID-vkTransitionImageLayout-transitionCount-arraylength
  `transitionCount` **must** be greater than `0`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_MEMORY_MAP_FAILED
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkHostImageLayoutTransitionInfo` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkHostImageLayoutTransitionInfo {
    VkStructureType            sType;
    const void*                pNext;
    VkImage                    image;
    VkImageLayout              oldLayout;
    VkImageLayout              newLayout;
    VkImageSubresourceRange    subresourceRange;
} VkHostImageLayoutTransitionInfo;
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to VkHostImageLayoutTransitionInfo
typedef VkHostImageLayoutTransitionInfo VkHostImageLayoutTransitionInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `image` is a handle to the image affected by this layout transition.
- `oldLayout` is the old layout in an image layout transition.
- `newLayout` is the new layout in an image layout transition.
- `subresourceRange` describes the image subresource range within `image` that is affected by this layout transition.

`vkTransitionImageLayout` does not check whether the device memory associated with an image is currently in use before performing the layout transition. The application **must** guarantee that any previously submitted command that reads from or writes to this subresource has completed before the host performs the layout transition. The memory of `image` is accessed by the host as if coherent.

If `image` is a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT and the `maintenance9` feature is enabled, the `baseArrayLayer` and `layerCount` members of `subresourceRange` specify the subset of slices of the 3D image affected by the memory barrier, including the layout transition. Any slices of a 3D image not included in `subresourceRange` are not affected by the memory barrier and remain in their existing layout.

NOTE

Image layout transitions performed on the host do not require queue family ownership transfers as the physical layout of the image will not vary between queue families for the layouts supported by this function.

NOTE

If the device has written to the image memory, it is not automatically made available to the host. Before this command can be called, a memory barrier for this image **must** have been issued on the device with the second synchronization scope including VK_PIPELINE_STAGE_HOST_BIT and VK_ACCESS_HOST_READ_BIT.

Because queue submissions automatically make host memory visible to the device, there would not be a need for a memory barrier before using the results of this layout transition on the device.

Valid Usage

- VUID-VkHostImageLayoutTransitionInfo-image-09055
  `image` **must** have been created with the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag set
- VUID-VkHostImageLayoutTransitionInfo-subresourceRange-01486
  `subresourceRange.baseMipLevel` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `image` was created
- VUID-VkHostImageLayoutTransitionInfo-subresourceRange-01724
  If `subresourceRange.levelCount` is not VK_REMAINING_MIP_LEVELS, `subresourceRange.baseMipLevel` + `subresourceRange.levelCount` **must** be less than or equal to the `mipLevels` specified in VkImageCreateInfo when `image` was created
- VUID-VkHostImageLayoutTransitionInfo-subresourceRange-01488
  If `image` is not a 3D image or was created without VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set, or the `maintenance9` feature is not enabled, `subresourceRange.baseArrayLayer` **must** be less than the `arrayLayers` specified in VkImageCreateInfo when `image` was created
- VUID-VkHostImageLayoutTransitionInfo-maintenance9-10798
  If the `maintenance9` feature is enabled and `image` is a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set, `subresourceRange.baseArrayLayer` **must** be less than the depth computed from `baseMipLevel` and `extent.depth` specified in VkImageCreateInfo when `image` was created, according to the formula defined in Image Mip Level Sizing
- VUID-VkHostImageLayoutTransitionInfo-maintenance9-10799
  If the `maintenance9` feature is enabled and `image` is a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set and either

subresourceRange.baseArrayLayer is not equal to 0 or subresourceRange.layerCount is not equal to VK_REMAINING_ARRAY_LAYERS, subresourceRange.levelCount **must** be 1

- VUID-VkHostImageLayoutTransitionInfo-subresourceRange-01725
  If image is not a 3D image or was created without VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set, or the maintenance9 feature is not enabled, and subresourceRange.layerCount is not VK_REMAINING_ARRAY_LAYERS, subresourceRange.baseArrayLayer + subresourceRange.layerCount **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when image was created
- VUID-VkHostImageLayoutTransitionInfo-maintenance9-10800
  If the maintenance9 feature is enabled, subresourceRange.layerCount is not VK_REMAINING_ARRAY_LAYERS, and image is a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set, subresourceRange.baseArrayLayer + subresourceRange.layerCount **must** be less than or equal to the depth computed from baseMipLevel and extent.depth specified in VkImageCreateInfo when image was created, according to the formula defined in Image Mip Level Sizing
- VUID-VkHostImageLayoutTransitionInfo-image-01932
  If image is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-VkHostImageLayoutTransitionInfo-image-09241
  If image has a color format that is single-plane, then the aspectMask member of subresourceRange **must** be VK_IMAGE_ASPECT_COLOR_BIT
- VUID-VkHostImageLayoutTransitionInfo-image-09242
  If image has a color format and is not *disjoint*, then the aspectMask member of subresourceRange **must** be VK_IMAGE_ASPECT_COLOR_BIT
- VUID-VkHostImageLayoutTransitionInfo-image-01672
  If image has a multi-planar format and the image is *disjoint*, then the aspectMask member of subresourceRange **must** include at least one multi-planar aspect mask bit or VK_IMAGE_ASPECT_COLOR_BIT
- VUID-VkHostImageLayoutTransitionInfo-image-03320
  If image has a depth/stencil format with both depth and stencil and the separateDepthStencilLayouts feature is not enabled, then the aspectMask member of subresourceRange **must** include both VK_IMAGE_ASPECT_DEPTH_BIT and VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkHostImageLayoutTransitionInfo-image-03319
  If image has a depth/stencil format with both depth and stencil and the separateDepthStencilLayouts feature is enabled, then the aspectMask member of subresourceRange **must** include either or both VK_IMAGE_ASPECT_DEPTH_BIT and VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkHostImageLayoutTransitionInfo-image-10749
  If image has a depth-only format then the aspectMask member of subresourceRange **must** be VK_IMAGE_ASPECT_DEPTH_BIT
- VUID-VkHostImageLayoutTransitionInfo-image-10750
  If image has a stencil-only format then the aspectMask member of subresourceRange **must** be VK_IMAGE_ASPECT_STENCIL_BIT

- VUID-VkHostImageLayoutTransitionInfo-aspectMask-08702
  If the `aspectMask` member of `subresourceRange` includes VK_IMAGE_ASPECT_DEPTH_BIT, `oldLayout` and `newLayout` **must** not be one of VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkHostImageLayoutTransitionInfo-aspectMask-08703
  If the `aspectMask` member of `subresourceRange` includes VK_IMAGE_ASPECT_STENCIL_BIT, `oldLayout` and `newLayout` **must** not be one of VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL
- VUID-VkHostImageLayoutTransitionInfo-subresourceRange-09601
  `subresourceRange.aspectMask` **must** be valid for the `format` the `image` was created with
- VUID-VkHostImageLayoutTransitionInfo-oldLayout-01208
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT usage flag set
- VUID-VkHostImageLayoutTransitionInfo-oldLayout-01209
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkHostImageLayoutTransitionInfo-oldLayout-01210
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkHostImageLayoutTransitionInfo-oldLayout-01211
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_SAMPLED_BIT or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flag set
- VUID-VkHostImageLayoutTransitionInfo-oldLayout-01212
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-VkHostImageLayoutTransitionInfo-oldLayout-01213
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL then `image` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-VkHostImageLayoutTransitionInfo-oldLayout-10767
  If the zeroInitializeDeviceMemory feature is not enabled, `oldLayout` **must** not be VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT
- VUID-VkHostImageLayoutTransitionInfo-oldLayout-10768
  If `oldLayout` is VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT, then all subresources **must** be included in the barrier
- VUID-VkHostImageLayoutTransitionInfo-oldLayout-01658
  If `oldLayout` or `newLayout` is

VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL then image **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set

- VUID-VkHostImageLayoutTransitionInfo-oldLayout-01659
  If oldLayout or newLayout is VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL then image **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkHostImageLayoutTransitionInfo-srcQueueFamilyIndex-04065
  If oldLayout or newLayout is VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL then image **must** have been created with at least one of the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, VK_IMAGE_USAGE_SAMPLED_BIT, or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flags set
- VUID-VkHostImageLayoutTransitionInfo-srcQueueFamilyIndex-04066
  If oldLayout or newLayout is VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL then image **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkHostImageLayoutTransitionInfo-srcQueueFamilyIndex-04067
  If oldLayout or newLayout is VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL then image **must** have been created with at least one of the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, VK_IMAGE_USAGE_SAMPLED_BIT, or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flags set
- VUID-VkHostImageLayoutTransitionInfo-srcQueueFamilyIndex-04068
  If oldLayout or newLayout is VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL then image **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkHostImageLayoutTransitionInfo-synchronization2-07793
  If the synchronization2 feature is not enabled, oldLayout **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR or VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VUID-VkHostImageLayoutTransitionInfo-synchronization2-07794
  If the synchronization2 feature is not enabled, newLayout **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR or VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VUID-VkHostImageLayoutTransitionInfo-srcQueueFamilyIndex-03938
  If oldLayout or newLayout is VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL, image **must** have been created with the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkHostImageLayoutTransitionInfo-srcQueueFamilyIndex-03939
  If oldLayout or newLayout is VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL, image **must** have been created with at least one of the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT,

VK_IMAGE_USAGE_SAMPLED_BIT, or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flags set

- VUID-VkHostImageLayoutTransitionInfo-oldLayout-02088
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR usage flag set
- VUID-VkHostImageLayoutTransitionInfo-srcQueueFamilyIndex-07120
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_VIDEO_DECODE_SRC_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR usage flag set
- VUID-VkHostImageLayoutTransitionInfo-srcQueueFamilyIndex-07121
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR usage flag set
- VUID-VkHostImageLayoutTransitionInfo-srcQueueFamilyIndex-07122
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR usage flag set
- VUID-VkHostImageLayoutTransitionInfo-srcQueueFamilyIndex-07123
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR usage flag set
- VUID-VkHostImageLayoutTransitionInfo-srcQueueFamilyIndex-07124
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_VIDEO_ENCODE_DST_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR usage flag set
- VUID-VkHostImageLayoutTransitionInfo-srcQueueFamilyIndex-07125
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR usage flag set
- VUID-VkHostImageLayoutTransitionInfo-srcQueueFamilyIndex-10287
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_VIDEO_ENCODE_QUANTIZATION_MAP_KHR then `image` **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR usage flags set
- VUID-VkHostImageLayoutTransitionInfo-srcQueueFamilyIndex-07006
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT then `image` **must** have been created with either the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flags set, and the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT or VK_IMAGE_USAGE_SAMPLED_BIT usage flags set, and the VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT usage flag

set

- VUID-VkHostImageLayoutTransitionInfo-attachmentFeedbackLoopLayout-07313
  If the `attachmentFeedbackLoopLayout` feature is not enabled, `newLayout` **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT
- VUID-VkHostImageLayoutTransitionInfo-srcQueueFamilyIndex-09550
  If `oldLayout` or `newLayout` is VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ then `image` **must** have been created with either the VK_IMAGE_USAGE_STORAGE_BIT usage flag set, or with both the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flag and either of the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flags set
- VUID-VkHostImageLayoutTransitionInfo-dynamicRenderingLocalRead-09551
  If the `dynamicRenderingLocalRead` feature is not enabled, `oldLayout` **must** not be VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ
- VUID-VkHostImageLayoutTransitionInfo-dynamicRenderingLocalRead-09552
  If the `dynamicRenderingLocalRead` feature is not enabled, `newLayout` **must** not be VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ
- VUID-VkHostImageLayoutTransitionInfo-oldLayout-09229
  `oldLayout` **must** be either VK_IMAGE_LAYOUT_UNDEFINED or the current layout of the image subresources as specified in `subresourceRange`
- VUID-VkHostImageLayoutTransitionInfo-oldLayout-09230
  If `oldLayout` is not VK_IMAGE_LAYOUT_UNDEFINED, VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT, or VK_IMAGE_LAYOUT_PREINITIALIZED, it **must** be one of the layouts in VkPhysicalDeviceHostImageCopyProperties::`pCopySrcLayouts`
- VUID-VkHostImageLayoutTransitionInfo-newLayout-09057
  `newLayout` **must** be one of the layouts in VkPhysicalDeviceHostImageCopyProperties::`pCopyDstLayouts`

Valid Usage (Implicit)

- VUID-VkHostImageLayoutTransitionInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_HOST_IMAGE_LAYOUT_TRANSITION_INFO
- VUID-VkHostImageLayoutTransitionInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkHostImageLayoutTransitionInfo-image-parameter
  `image` **must** be a valid VkImage handle
- VUID-VkHostImageLayoutTransitionInfo-oldLayout-parameter
  `oldLayout` **must** be a valid VkImageLayout value
- VUID-VkHostImageLayoutTransitionInfo-newLayout-parameter
  `newLayout` **must** be a valid VkImageLayout value
- VUID-VkHostImageLayoutTransitionInfo-subresourceRange-parameter
  `subresourceRange` **must** be a valid VkImageSubresourceRange structure

7.7.5. Queue Family Ownership Transfer

Resources created with a VkSharingMode of VK_SHARING_MODE_EXCLUSIVE **must** have their ownership explicitly transferred from one queue family to another in order to access their content in a well-defined manner on a queue in a different queue family.

The special queue family index VK_QUEUE_FAMILY_IGNORED indicates that a queue family parameter or member is ignored.

```
#define VK_QUEUE_FAMILY_IGNORED           (~0U)
```

Resources shared with external APIs or instances using external memory **must** also explicitly manage ownership transfers between local and external queues (or equivalent constructs in external APIs) regardless of the VkSharingMode specified when creating them.

The special queue family index VK_QUEUE_FAMILY_EXTERNAL represents any queue external to the resource's current Vulkan instance, as long as the queue uses the same underlying device group or physical device, and the same driver version as the resource's VkDevice, as indicated by VkPhysicalDeviceIDProperties::`deviceUUID` and VkPhysicalDeviceIDProperties::`driverUUID`.

```
#define VK_QUEUE_FAMILY_EXTERNAL          (~1U)
```

```
#define VK_QUEUE_FAMILY_EXTERNAL_KHR      VK_QUEUE_FAMILY_EXTERNAL
```

The special queue family index VK_QUEUE_FAMILY_FOREIGN_EXT represents any queue external to the resource's current Vulkan instance, regardless of the queue's underlying physical device or driver version. This includes, for example, queues for fixed-function image processing devices, media codec devices, and display devices, as well as all queues that use the same underlying device group or physical device, and the same driver version as the resource's VkDevice.

```
#define VK_QUEUE_FAMILY_FOREIGN_EXT       (~2U)
```

If memory dependencies are correctly expressed between uses of such a resource between two queues in different families, but no ownership transfer is defined, the contents of that resource are undefined for any read accesses performed by the second queue family.

NOTE If an application does not need the contents of a resource to remain valid when transferring from one queue family to another, then the ownership transfer **should** be skipped.

NOTE Applications should expect transfers to/from VK_QUEUE_FAMILY_FOREIGN_EXT to be more expensive than transfers to/from VK_QUEUE_FAMILY_EXTERNAL.

A queue family ownership transfer consists of two distinct parts:

1. Release exclusive ownership from the source queue family
2. Acquire exclusive ownership for the destination queue family

An application **must** ensure that these operations occur in the correct order by defining an execution dependency between them, e.g. using a semaphore.

A *release operation* is used to release exclusive ownership of a range of a buffer or image subresource range. A release operation is defined by executing a buffer memory barrier (for a buffer range) or an image memory barrier (for an image subresource range) using a pipeline barrier command, on a queue from the source queue family. The `srcQueueFamilyIndex` parameter of the barrier **must** be the source queue family index, and the `dstQueueFamilyIndex` parameter to the destination queue family index. The destination access mask is ignored for such a barrier, such that no visibility operation is executed - the value of this mask does not affect the validity of the barrier. The release operation happens-after the availability operation. If `dependencyFlags` does not include VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR, `dstStageMask` is also ignored for such a barrier as defined by buffer memory ownership transfer and image memory ownership transfer. After a release operation is performed, the contents and image layout (if applicable) of the released resource are undefined until a matching acquire operation is performed.

NOTE It is valid to never call the acquire operation after a release, and instead simply start using the resource on any queue (even the releasing queue), but the contents should be reinitialized before being read. In the case of images, an image layout transition away from VK_IMAGE_LAYOUT_UNDEFINED is also required. Executing an acquire operation after this without another release is invalid.

An *acquire operation* is used to acquire exclusive ownership of a range of a buffer or image subresource range. An acquire operation is defined by executing a buffer memory barrier (for a buffer range) or an image memory barrier (for an image subresource range) using a pipeline barrier command, on a queue from the destination queue family. The buffer range or image subresource range specified in an acquire operation **must** match exactly that of a previous release operation. The `srcQueueFamilyIndex` parameter of the barrier **must** be the source queue family index, and the `dstQueueFamilyIndex` parameter to the destination queue family index. The source access mask is ignored for such a barrier, such that no availability operation is executed - the value of this mask does not affect the validity of the barrier. The acquire operation happens-before the visibility operation. If `dependencyFlags` does not include VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR, `srcStageMask` is also ignored for such a barrier as defined by buffer memory ownership transfer and image memory ownership transfer. Applications **must** not execute an acquire operation on any resource where it has been modified in any way other than a matching release operation.

NOTE Whilst it is not invalid to provide destination or source access masks for memory barriers used for release or acquire operations, respectively, they have no practical effect. Access after a release operation has undefined results, and so visibility for those accesses has no practical effect. Similarly, write access before an acquire operation will produce undefined results for future access, so availability of those writes has no practical use. In an earlier version of the specification, these were required to match on both sides - but this was subsequently relaxed. These masks

should be set to 0.

NOTE

To ensure that an acquire and release operation are valid, the release operation must happen-before the acquire operation. Often, semaphores are used for this directly, with the semaphore signaling after a release and then waiting before an acquire. Prior to the introduction of VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR, VK_PIPELINE_STAGE_ALL_COMMANDS_BIT is the only valid stage to wait on or wait for these operations, as the acquire and release operations do not occur in a defined stage. When VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR is specified however, these can be synchronized with the stages which would otherwise be ignored, as these stages now synchronize the acquire and release operations, providing a way to avoid full pipeline stalls.

If the transfer is via an image memory barrier, and an image layout transition is desired, then the values of `oldLayout` and `newLayout` in the *release operation*'s memory barrier **must** be equal to values of `oldLayout` and `newLayout` in the *acquire operation*'s memory barrier. Although the image layout transition is submitted twice, it will only be executed once. A layout transition specified in this way happens-after the *release operation* and happens-before the *acquire operation*.

If the values of `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are equal, no ownership transfer is performed, and the barrier operates as if they were both set to VK_QUEUE_FAMILY_IGNORED.

Queue family ownership transfers **may** perform read and write accesses on all memory bound to the image subresource or buffer range, so applications **must** ensure that all memory writes have been made available before a queue family ownership transfer is executed. Available memory is automatically made visible to queue family release and acquire operations, and writes performed by those operations are automatically made available.

Once a queue family has acquired ownership of a buffer range or image subresource range of a VK_SHARING_MODE_EXCLUSIVE resource, its contents are undefined to other queue families unless ownership is transferred. The contents of any portion of another resource which aliases memory that is bound to the transferred buffer or image subresource range are undefined after a release or acquire operation.

NOTE

Because events **cannot** be used directly for inter-queue synchronization, and because vkCmdSetEvent does not have the queue family index or memory barrier parameters needed by a *release operation*, the release and acquire operations of a queue family ownership transfer **can** only be performed using vkCmdPipelineBarrier.

An *acquire operation* **may** have a performance penalty when acquiring ownership of a subresource range from one of the special queue families reserved for external memory ownership transfers described above. The application **can** reduce the performance penalty in some cases by adding a VkExternalMemoryAcquireUnmodifiedEXT structure to the `pNext` chain of the *acquire operation*'s memory barrier structure.

The VkExternalMemoryAcquireUnmodifiedEXT structure is defined as:

```
// Provided by VK_EXT_external_memory_acquire_unmodified
typedef struct VkExternalMemoryAcquireUnmodifiedEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           acquireUnmodifiedMemory;
} VkExternalMemoryAcquireUnmodifiedEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- acquireUnmodifiedMemory specifies, if VK_TRUE, that no range of VkDeviceMemory bound to the resource of the memory barrier's subresource range was modified at any time since the resource's most recent release of ownership to the queue family specified by the memory barrier's srcQueueFamilyIndex. If VK_FALSE, it specifies nothing.

If the application releases ownership of the subresource range to one of the special queue families reserved for external memory ownership transfers with a memory barrier structure, and later re-acquires ownership from the same queue family with a memory barrier structure, and if no range of VkDeviceMemory bound to the resource was modified at any time between the *release operation* and the *acquire operation*, then the application **should** add a VkExternalMemoryAcquireUnmodifiedEXT structure to the pNext chain of the *acquire operation*'s memory barrier structure because this **may** reduce the performance penalty.

This structure is ignored if acquireUnmodifiedMemory is VK_FALSE. In particular, VK_FALSE does *not* specify that memory was modified.

This structure is ignored if the memory barrier's srcQueueFamilyIndex is not a special queue family reserved for external memory ownership transfers.

NOTE

The method by which the application determines whether memory was modified between the *release operation* and *acquire operation* is outside the scope of Vulkan.

For any Vulkan operation that accesses a resource, the application **must** not assume the implementation accesses the resource's memory as read-only, even for *apparently* read-only operations such as transfer commands and shader reads.

The validity of VkExternalMemoryAcquireUnmodifiedEXT::acquireUnmodifiedMemory is independent of memory ranges outside the ranges of VkDeviceMemory bound to the resource. In particular, it is independent of any implementation-private memory associated with the resource.

Valid Usage

- VUID-VkExternalMemoryAcquireUnmodifiedEXT-acquireUnmodifiedMemory-08922
  If acquireUnmodifiedMemory is VK_TRUE, and the memory barrier's srcQueueFamilyIndex is a special queue family reserved for external memory ownership transfers (as described in

Queue Family Ownership Transfer), then each range of VkDeviceMemory bound to the resource **must** have remained unmodified during all time since the resource's most recent release of ownership to the queue family

Valid Usage (Implicit)

- VUID-VkExternalMemoryAcquireUnmodifiedEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_ACQUIRE_UNMODIFIED_EXT

Structure Chaining

Extends the structures

- VkBufferMemoryBarrier
- VkBufferMemoryBarrier2
- VkImageMemoryBarrier
- VkImageMemoryBarrier2

7.8. Wait Idle Operations

To wait on the host for the completion of outstanding queue operations for a given queue, call:

```
// Provided by VK_VERSION_1_0
VkResult vkQueueWaitIdle(
    VkQueue                                     queue);
```

- `queue` is the queue on which to wait.

`vkQueueWaitIdle` is equivalent to having submitted a valid fence to every previously executed queue submission command that accepts a fence, then waiting for all of those fences to signal using vkWaitForFences with an infinite timeout and `waitAll` set to VK_TRUE.

NOTE Even though vkQueuePresentKHR does not have a fence parameter, it does accept a fence through VkSwapchainPresentFenceInfoEXT.

Valid Usage (Implicit)

- VUID-vkQueueWaitIdle-queue-parameter
  `queue` **must** be a valid VkQueue handle

Host Synchronization

- Host access to `queue` **must** be externally synchronized if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	Any	-

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To wait on the host for the completion of outstanding queue operations for all queues on a given logical device, call:

```
// Provided by VK_VERSION_1_0
VkResult vkDeviceWaitIdle(
    VkDevice                                    device);
```

- `device` is the logical device to idle.

`vkDeviceWaitIdle` is equivalent to calling `vkQueueWaitIdle` for all queues owned by `device`.

Valid Usage (Implicit)

- VUID-vkDeviceWaitIdle-device-parameter
  `device` **must** be a valid VkDevice handle

Host Synchronization

- Host access to all `VkQueue` objects created from `device` that are not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

7.9. Host Write Ordering Guarantees

When batches of command buffers are submitted to a queue via a queue submission command, it defines a memory dependency with prior host operations, and execution of command buffers submitted to the queue.

The first synchronization scope includes execution of vkQueueSubmit on the host and anything that happened-before it, as defined by the host memory model.

NOTE Some systems allow writes that do not directly integrate with the host memory model; these have to be synchronized by the application manually. One example of this is non-temporal store instructions on x86; to ensure these happen-before submission, applications should call `_mm_sfence()`.

The second synchronization scope includes all commands submitted in the same queue submission, and all commands that occur later in submission order.

The first access scope includes all host writes to mappable device memory that are available to the host memory domain.

The second access scope includes all memory access performed by the device.

7.10. Synchronization and Multiple Physical Devices

If a logical device includes more than one physical device, then fences, semaphores, and events all still have a single instance of the signaled state.

A fence becomes signaled when all physical devices complete the necessary queue operations.

Semaphore wait and signal operations all include a device index that is the sole physical device that performs the operation. These indices are provided in the VkDeviceGroupSubmitInfo and VkDeviceGroupBindSparseInfo structures. Semaphores are not exclusively owned by any physical device. For example, a semaphore can be signaled by one physical device and then waited on by a different physical device.

An event **can** only be waited on by the same physical device that signaled it (or the host).

7.11. Calibrated Timestamps

In order to be able to correlate the time a particular operation took place at on timelines of different time domains (e.g. a device operation vs. a host operation), Vulkan allows querying calibrated timestamps from multiple time domains.

To query calibrated timestamps from a set of time domains, call:

```
// Provided by VK_KHR_calibrated_timestamps
VkResult vkGetCalibratedTimestampsKHR(
    VkDevice                                    device,
    uint32_t                                    timestampCount,
    const VkCalibratedTimestampInfoKHR*         pTimestampInfos,
    uint64_t*                                   pTimestamps,
    uint64_t*                                   pMaxDeviation);
```

- `device` is the logical device used to perform the query.
- `timestampCount` is the number of timestamps to query.
- `pTimestampInfos` is a pointer to an array of `timestampCount` VkCalibratedTimestampInfoKHR structures, describing the time domains the calibrated timestamps should be captured from.
- `pTimestamps` is a pointer to an array of `timestampCount` 64-bit unsigned integer values in which the requested calibrated timestamp values are returned.
- `pMaxDeviation` is a pointer to a 64-bit unsigned integer value in which the strictly positive maximum deviation, in nanoseconds, of the calibrated timestamp values is returned.

NOTE

The maximum deviation **may** vary between calls to `vkGetCalibratedTimestampsKHR` even for the same set of time domains due to implementation and platform specific reasons. It is the application's responsibility to assess whether the returned maximum deviation makes the timestamp values suitable for any particular purpose and **can** choose to re-issue the timestamp calibration call pursuing a lower deviation value.

Calibrated timestamp values **can** be extrapolated to estimate future coinciding timestamp values, however, depending on the nature of the time domains and other properties of the platform extrapolating values over a sufficiently long period of time **may** no longer be accurate enough to fit any particular purpose, so applications are expected to re-calibrate the timestamps on a regular

basis.

Valid Usage

- VUID-vkGetCalibratedTimestampsKHR-timeDomain-09246
  The `timeDomain` value of each VkCalibratedTimestampInfoKHR in `pTimestampInfos` **must** be unique except for those with a value of VK_TIME_DOMAIN_PRESENT_STAGE_LOCAL_EXT

Valid Usage (Implicit)

- VUID-vkGetCalibratedTimestampsKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetCalibratedTimestampsKHR-pTimestampInfos-parameter
  `pTimestampInfos` **must** be a valid pointer to an array of `timestampCount` valid VkCalibratedTimestampInfoKHR structures
- VUID-vkGetCalibratedTimestampsKHR-pTimestamps-parameter
  `pTimestamps` **must** be a valid pointer to an array of `timestampCount` `uint64_t` values
- VUID-vkGetCalibratedTimestampsKHR-pMaxDeviation-parameter
  `pMaxDeviation` **must** be a valid pointer to a `uint64_t` value
- VUID-vkGetCalibratedTimestampsKHR-timestampCount-arraylength
  `timestampCount` **must** be greater than `0`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkCalibratedTimestampInfoKHR` structure is defined as:

```
// Provided by VK_KHR_calibrated_timestamps
typedef struct VkCalibratedTimestampInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkTimeDomainKHR    timeDomain;
} VkCalibratedTimestampInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `timeDomain` is a VkTimeDomainKHR value specifying the time domain from which the calibrated timestamp value should be returned.

Valid Usage

- VUID-VkCalibratedTimestampInfoKHR-timeDomain-02354
  `timeDomain` **must** be one of the VkTimeDomainKHR values returned by vkGetPhysicalDeviceCalibrateableTimeDomainsKHR
- VUID-VkCalibratedTimestampInfoKHR-timeDomain-12227
  If `timeDomain` is VK_TIME_DOMAIN_SWAPCHAIN_LOCAL_EXT or VK_TIME_DOMAIN_PRESENT_STAGE_LOCAL_EXT, the `pNext` chain **must** include a VkSwapchainCalibratedTimestampInfoEXT structure

Valid Usage (Implicit)

- VUID-VkCalibratedTimestampInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_CALIBRATED_TIMESTAMP_INFO_KHR
- VUID-VkCalibratedTimestampInfoKHR-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkSwapchainCalibratedTimestampInfoEXT
- VUID-VkCalibratedTimestampInfoKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkCalibratedTimestampInfoKHR-timeDomain-parameter
  `timeDomain` **must** be a valid VkTimeDomainKHR value

The set of supported time domains consists of:

```
// Provided by VK_KHR_calibrated_timestamps
typedef enum VkTimeDomainKHR {
    VK_TIME_DOMAIN_DEVICE_KHR = 0,
    VK_TIME_DOMAIN_CLOCK_MONOTONIC_KHR = 1,
    VK_TIME_DOMAIN_CLOCK_MONOTONIC_RAW_KHR = 2,
    VK_TIME_DOMAIN_QUERY_PERFORMANCE_COUNTER_KHR = 3,
  // Provided by VK_EXT_present_timing
    VK_TIME_DOMAIN_PRESENT_STAGE_LOCAL_EXT = 1000208000,
  // Provided by VK_EXT_present_timing
    VK_TIME_DOMAIN_SWAPCHAIN_LOCAL_EXT = 1000208001,
} VkTimeDomainKHR;
```

- VK_TIME_DOMAIN_DEVICE_KHR specifies the device time domain. Timestamp values in this time domain use the same units and are comparable with device timestamp values captured

using vkCmdWriteTimestamp or vkCmdWriteTimestamp2 and are defined to be incrementing according to the timestampPeriod of the device.

- VK_TIME_DOMAIN_PRESENT_STAGE_LOCAL_EXT specifies a time domain unique to a particular swapchain and a specific present stage. Timestamp values in this time domain are in units of nanoseconds and are comparable only with other values from the same swapchain and present stage.
- VK_TIME_DOMAIN_SWAPCHAIN_LOCAL_EXT specifies a time domain unique to a particular swapchain. Timestamp values in this time domain are in units of nanoseconds and are comparable only with other values from the same swapchain.
- VK_TIME_DOMAIN_CLOCK_MONOTONIC_KHR specifies the CLOCK_MONOTONIC time domain available on POSIX platforms. Timestamp values in this time domain are in units of nanoseconds and are comparable with platform timestamp values captured using the POSIX clock_gettime API as computed by this example:

NOTE An implementation supporting VK_KHR_calibrated_timestamps will use the same time domain for all its VkQueue so that timestamp values reported for VK_TIME_DOMAIN_DEVICE_KHR can be matched to any timestamp captured through vkCmdWriteTimestamp or vkCmdWriteTimestamp2 .

```
struct timespec tv;
clock_gettime(CLOCK_MONOTONIC, &tv);
return tv.tv_nsec + tv.tv_sec*1000000000ull;
```

- VK_TIME_DOMAIN_CLOCK_MONOTONIC_RAW_KHR specifies the CLOCK_MONOTONIC_RAW time domain available on POSIX platforms. Timestamp values in this time domain are in units of nanoseconds and are comparable with platform timestamp values captured using the POSIX clock_gettime API as computed by this example:

```
struct timespec tv;
clock_gettime(CLOCK_MONOTONIC_RAW, &tv);
return tv.tv_nsec + tv.tv_sec*1000000000ull;
```

- VK_TIME_DOMAIN_QUERY_PERFORMANCE_COUNTER_KHR specifies the performance counter (QPC) time domain available on Windows. Timestamp values in this time domain are in the same units as those provided by the Windows QueryPerformanceCounter API and are comparable with platform timestamp values captured using that API as computed by this example:

```
LARGE_INTEGER counter;
QueryPerformanceCounter(&counter);
return counter.QuadPart;
```

The VkSwapchainCalibratedTimestampInfoEXT structure is defined as:

```
// Provided by VK_EXT_present_timing
typedef struct VkSwapchainCalibratedTimestampInfoEXT {
    VkStructureType           sType;
    const void*               pNext;
    VkSwapchainKHR            swapchain;
    VkPresentStageFlagsEXT    presentStage;
    uint64_t                  timeDomainId;
} VkSwapchainCalibratedTimestampInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `swapchain` is the swapchain to retrieve the swapchain-local timestamp from.
- `presentStage` is zero or a VkPresentStageFlagsEXT value used to identify a single present stage when calibrating a timestamp in the VK_TIME_DOMAIN_PRESENT_STAGE_LOCAL_EXT time domain.
- `timeDomainId` is the id for the opaque time domain being calibrated.

`timeDomainId` **must** be an id previously reported by vkGetSwapchainTimeDomainPropertiesEXT for `swapchain`. If the `timeDomainId` is no longer supported by the `swapchain`, implementations **may** report zero as the calibrated timestamp value.

Valid Usage

- VUID-VkSwapchainCalibratedTimestampInfoEXT-timeDomain-12228
  If the `timeDomain` member of the VkCalibratedTimestampInfoKHR structure in this structure's `pNext` chain is VK_TIME_DOMAIN_PRESENT_STAGE_LOCAL_EXT, `presentStage` **must** specify one and only one present stage

Valid Usage (Implicit)

- VUID-VkSwapchainCalibratedTimestampInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SWAPCHAIN_CALIBRATED_TIMESTAMP_INFO_EXT
- VUID-VkSwapchainCalibratedTimestampInfoEXT-swapchain-parameter
  `swapchain` **must** be a valid VkSwapchainKHR handle
- VUID-VkSwapchainCalibratedTimestampInfoEXT-presentStage-parameter
  If `presentStage` is not `0`, `presentStage` **must** be a valid combination of VkPresentStageFlagBitsEXT values

Structure Chaining

Extends the structure

- VkCalibratedTimestampInfoKHR

Chapter 8. Render Pass

Draw commands **must** be recorded within a *render pass instance*. Each render pass instance defines a set of image resources, referred to as *attachments*, used during rendering.

To begin a render pass instance, call:

```
// Provided by VK_VERSION_1_3
void vkCmdBeginRendering(
    VkCommandBuffer                             commandBuffer,
    const VkRenderingInfo*                      pRenderingInfo);
```

```
// Provided by VK_KHR_dynamic_rendering
// Equivalent to vkCmdBeginRendering
void vkCmdBeginRenderingKHR(
    VkCommandBuffer                             commandBuffer,
    const VkRenderingInfo*                      pRenderingInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pRenderingInfo` is a pointer to a VkRenderingInfo structure specifying details of the render pass instance to begin.

After beginning a render pass instance, the command buffer is ready to record draw commands.

If `pRenderingInfo->flags` includes VK_RENDERING_RESUMING_BIT then this render pass is resumed from a render pass instance that has been suspended earlier in submission order.

Valid Usage

- VUID-vkCmdBeginRendering-dynamicRendering-06446
  The `dynamicRendering` feature **must** be enabled
- VUID-vkCmdBeginRendering-commandBuffer-06068
  If `commandBuffer` is a secondary command buffer, and the `nestedCommandBuffer` feature is not enabled, `pRenderingInfo->flags` **must** not include VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT
- VUID-vkCmdBeginRendering-commandBuffer-10914
  If `commandBuffer` is a secondary command buffer, VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT **must** not have been set in VkCommandBufferBeginInfo::`flags` when `commandBuffer` began
- VUID-vkCmdBeginRendering-pRenderingInfo-09588
  If `pRenderingInfo->pDepthAttachment` is not `NULL` and `pRenderingInfo->pDepthAttachment->imageView` is not VK_NULL_HANDLE, when `pRenderingInfo->pDepthAttachment->imageView` is accessed it **must** be in the layout specified by `pRenderingInfo->pDepthAttachment->imageLayout`

- VUID-vkCmdBeginRendering-pRenderingInfo-09589
  If pRenderingInfo->pDepthAttachment is not NULL, pRenderingInfo->pDepthAttachment->imageView is not VK_NULL_HANDLE, pRenderingInfo->pDepthAttachment->resolveMode is not VK_RESOLVE_MODE_NONE, and pRenderingInfo->pDepthAttachment->resolveImageView is not VK_NULL_HANDLE, pRenderingInfo->pDepthAttachment->resolveImageView **must** be in the layout specified by pRenderingInfo->pDepthAttachment->resolveImageLayout
- VUID-vkCmdBeginRendering-pRenderingInfo-09590
  If pRenderingInfo->pStencilAttachment is not NULL and pRenderingInfo->pStencilAttachment->imageView is not VK_NULL_HANDLE, when pRenderingInfo->pStencilAttachment->imageView is accessed it **must** be in the layout specified by pRenderingInfo->pStencilAttachment->imageLayout
- VUID-vkCmdBeginRendering-pRenderingInfo-09591
  If pRenderingInfo->pStencilAttachment is not NULL, pRenderingInfo->pStencilAttachment->imageView is not VK_NULL_HANDLE, pRenderingInfo->pStencilAttachment->resolveMode is not VK_RESOLVE_MODE_NONE, and pRenderingInfo->pStencilAttachment->resolveImageView is not VK_NULL_HANDLE, pRenderingInfo->pStencilAttachment->resolveImageView **must** be in the layout specified by pRenderingInfo->pStencilAttachment->resolveImageLayout
- VUID-vkCmdBeginRendering-pRenderingInfo-09592
  For each element of pRenderingInfo->pColorAttachments, if imageView is not VK_NULL_HANDLE, when that image view is accessed it **must** be in the layout specified by the imageLayout member of that same element of pRenderingInfo->pColorAttachments
- VUID-vkCmdBeginRendering-pRenderingInfo-09593
  For each element of pRenderingInfo->pColorAttachments, if imageView is not VK_NULL_HANDLE and resolveMode is not VK_RESOLVE_MODE_NONE, and resolveImageView is not VK_NULL_HANDLE, resolveImageView **must** be in the layout specified by resolveImageLayout
- VUID-vkCmdBeginRendering-pRenderingInfo-11750
  If pRenderingInfo->flags contains VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR, maintenance10 **must** be enabled
- VUID-vkCmdBeginRendering-pRenderingInfo-11751
  If pRenderingInfo->flags does not contain VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR, attachments **must** not specify VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdBeginRendering-imageView-12276
  If VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView is not equal to VK_NULL_HANDLE, when imageView is accessed it **must** be in the layout specified by VkRenderingFragmentDensityMapAttachmentInfoEXT::imageLayout
- VUID-vkCmdBeginRendering-imageView-12277
  If VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView is not equal to VK_NULL_HANDLE, when imageView is accessed it **must** be in the layout specified by VkRenderingFragmentShadingRateAttachmentInfoKHR::imageLayout

Valid Usage (Implicit)

- VUID-vkCmdBeginRendering-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBeginRendering-pRenderingInfo-parameter
  `pRenderingInfo` **must** be a valid pointer to a valid VkRenderingInfo structure
- VUID-vkCmdBeginRendering-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBeginRendering-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBeginRendering-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdBeginRendering-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdBeginRendering-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_GRAPHICS_BIT	Action State

Conditional Rendering

vkCmdBeginRendering is not affected by conditional rendering

The `VkRenderingInfo` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkRenderingInfo {
    VkStructureType                     sType;
    const void*                         pNext;
    VkRenderingFlags                    flags;
    VkRect2D                            renderArea;
    uint32_t                            layerCount;
    uint32_t                            viewMask;
    uint32_t                            colorAttachmentCount;
    const VkRenderingAttachmentInfo*    pColorAttachments;
    const VkRenderingAttachmentInfo*    pDepthAttachment;
    const VkRenderingAttachmentInfo*    pStencilAttachment;
} VkRenderingInfo;
```

```
// Provided by VK_KHR_dynamic_rendering
// Equivalent to VkRenderingInfo
typedef VkRenderingInfo VkRenderingInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkRenderingFlagBits.
- `renderArea` is the render area that is affected by the render pass instance.
- `layerCount` is the number of layers rendered to in each attachment when `viewMask` is `0`.
- `viewMask` is a bitfield of view indices describing which views are active during rendering, when it is not `0`.
- `colorAttachmentCount` is the number of elements in `pColorAttachments`.
- `pColorAttachments` is a pointer to an array of `colorAttachmentCount` VkRenderingAttachmentInfo structures describing any color attachments used.
- `pDepthAttachment` is a pointer to a VkRenderingAttachmentInfo structure describing a depth attachment.
- `pStencilAttachment` is a pointer to a VkRenderingAttachmentInfo structure describing a stencil attachment.

If `viewMask` is not `0`, multiview is enabled.

If there is an instance of VkDeviceGroupRenderPassBeginInfo included in the `pNext` chain and its `deviceRenderAreaCount` member is not `0`, then `renderArea` is ignored, and the render area is defined per-device by that structure.

Each element of the `pColorAttachments` array corresponds to an output location in the shader, i.e. if the shader declares an output variable decorated with a `Location` value of **X**, then it uses the attachment provided in `pColorAttachments`[**X**]. If the `imageView` member of any element of `pColorAttachments` is VK_NULL_HANDLE, writes to the corresponding location by a fragment are discarded.

The aspectMask of any image view specified for pDepthAttachment or pStencilAttachment is ignored. Instead, depth attachments are automatically treated as if VK_IMAGE_ASPECT_DEPTH_BIT was specified for their aspect masks, and stencil attachments are automatically treated as if VK_IMAGE_ASPECT_STENCIL_BIT was specified for their aspect masks.

Valid Usage

- VUID-VkRenderingInfo-viewMask-06069
  If viewMask is 0, layerCount **must** not be 0
- VUID-VkRenderingInfo-multisampledRenderToSingleSampled-06857
  If none of the following are enabled:
  - The multisampledRenderToSingleSampled feature

  imageView members of pDepthAttachment, pStencilAttachment, and elements of pColorAttachments that are not VK_NULL_HANDLE **must** have been created with the same sampleCount
- VUID-VkRenderingInfo-imageView-09429
  imageView members of elements of pColorAttachments that are not VK_NULL_HANDLE **must** have been created with the same sampleCount , if the multisampledRenderToSingleSampled feature is not enabled
- VUID-VkRenderingInfo-None-08994
  If VkDeviceGroupRenderPassBeginInfo::deviceRenderAreaCount is 0, renderArea.extent.width **must** be greater than 0
- VUID-VkRenderingInfo-None-08995
  If VkDeviceGroupRenderPassBeginInfo::deviceRenderAreaCount is 0, renderArea.extent.height **must** be greater than 0
- VUID-VkRenderingInfo-imageView-06858
  If multisampled-render-to-single-sampled is enabled, then all attachments referenced by imageView members of pDepthAttachment, pStencilAttachment, and elements of pColorAttachments that are not VK_NULL_HANDLE **must** have a sample count that is either VK_SAMPLE_COUNT_1_BIT or equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-VkRenderingInfo-imageView-06859
  If multisampled-render-to-single-sampled is enabled, then all attachments referenced by imageView members of pDepthAttachment, pStencilAttachment, and elements of pColorAttachments that are not VK_NULL_HANDLE and have a sample count of VK_SAMPLE_COUNT_1_BIT **must** have been created with VK_IMAGE_CREATE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT in their VkImageCreateInfo::flags
- VUID-VkRenderingInfo-pNext-06077
  If the pNext chain does not contain VkDeviceGroupRenderPassBeginInfo or its deviceRenderAreaCount member is equal to 0, renderArea.offset.x **must** be greater than or equal to 0
- VUID-VkRenderingInfo-pNext-06078

If the pNext chain does not contain VkDeviceGroupRenderPassBeginInfo or its deviceRenderAreaCount member is equal to 0, renderArea.offset.y **must** be greater than or equal to 0

- VUID-VkRenderingInfo-pNext-07815
  If the pNext chain does not contain VkDeviceGroupRenderPassBeginInfo or its deviceRenderAreaCount member is equal to 0, the sum of renderArea.extent.width and renderArea.offset.x **must** be less than or equal to maxFramebufferWidth
- VUID-VkRenderingInfo-pNext-07816
  If the pNext chain does not contain VkDeviceGroupRenderPassBeginInfo or its deviceRenderAreaCount member is equal to 0, the sum of renderArea.extent.height and renderArea.offset.y **must** be less than or equal to maxFramebufferHeight
- VUID-VkRenderingInfo-pNext-06079
  If the pNext chain does not contain VkDeviceGroupRenderPassBeginInfo or its deviceRenderAreaCount member is equal to 0, the width of the imageView member of each element of pColorAttachments, pDepthAttachment, or pStencilAttachment that is not VK_NULL_HANDLE **must** be greater than or equal to renderArea.offset.x + renderArea.extent.width
- VUID-VkRenderingInfo-pNext-06080
  If the pNext chain does not contain VkDeviceGroupRenderPassBeginInfo or its deviceRenderAreaCount member is equal to 0, the height of the imageView member of each element of pColorAttachments, pDepthAttachment, or pStencilAttachment that is not VK_NULL_HANDLE **must** be greater than or equal to renderArea.offset.y + renderArea.extent.height
- VUID-VkRenderingInfo-pNext-06083
  If the pNext chain contains VkDeviceGroupRenderPassBeginInfo, the width of the imageView member of any element of pColorAttachments, pDepthAttachment, or pStencilAttachment that is not VK_NULL_HANDLE **must** be greater than or equal to the sum of the offset.x and extent.width members of each element of pDeviceRenderAreas
- VUID-VkRenderingInfo-pNext-06084
  If the pNext chain contains VkDeviceGroupRenderPassBeginInfo, the height of the imageView member of any element of pColorAttachments, pDepthAttachment, or pStencilAttachment that is not VK_NULL_HANDLE **must** be greater than or equal to the sum of the offset.y and extent.height members of each element of pDeviceRenderAreas
- VUID-VkRenderingInfo-pDepthAttachment-06085
  If neither pDepthAttachment nor pStencilAttachment are NULL and the imageView member of either structure is not VK_NULL_HANDLE, the imageView member of each structure **must** be the same
- VUID-VkRenderingInfo-pDepthAttachment-06086
  If neither pDepthAttachment nor pStencilAttachment are NULL, and the resolveMode member of each is not VK_RESOLVE_MODE_NONE, the resolveImageView member of each structure **must** be the same
- VUID-VkRenderingInfo-colorAttachmentCount-06087
  If colorAttachmentCount is not 0 and the imageView member of an element of pColorAttachments is not VK_NULL_HANDLE, that imageView **must** have been created with

the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT usage flag set

- VUID-VkRenderingInfo-colorAttachmentCount-09476
  If `colorAttachmentCount` is not `0` and there is an element of `pColorAttachments` with its `imageView` member not VK_NULL_HANDLE, its `resolveMode` member not set to VK_RESOLVE_MODE_NONE, and `resolveImageView` is not VK_NULL_HANDLE, then the `resolveImageView` member of that element of `pColorAttachments` **must** have been created with the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT usage flag set
- VUID-VkRenderingInfo-pDepthAttachment-06547
  If `pDepthAttachment` is not `NULL` and `pDepthAttachment->imageView` is not VK_NULL_HANDLE, `pDepthAttachment->imageView` **must** have been created with a format that includes a depth component
- VUID-VkRenderingInfo-pDepthAttachment-06088
  If `pDepthAttachment` is not `NULL` and `pDepthAttachment->imageView` is not VK_NULL_HANDLE, `pDepthAttachment->imageView` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkRenderingInfo-pDepthAttachment-09477
  If `pDepthAttachment` is not `NULL`, `pDepthAttachment->resolveMode` is not VK_RESOLVE_MODE_NONE, and `pDepthAttachment->resolveImageView` is not VK_NULL_HANDLE, then `pDepthAttachment->resolveImageView` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkRenderingInfo-pStencilAttachment-06548
  If `pStencilAttachment` is not `NULL` and `pStencilAttachment->imageView` is not VK_NULL_HANDLE, `pStencilAttachment->imageView` **must** have been created with a format that includes a stencil aspect
- VUID-VkRenderingInfo-pStencilAttachment-06089
  If `pStencilAttachment` is not `NULL` and `pStencilAttachment->imageView` is not VK_NULL_HANDLE, `pStencilAttachment->imageView` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkRenderingInfo-pStencilAttachment-09478
  If `pStencilAttachment` is not `NULL`, `pStencilAttachment->resolveMode` is not VK_RESOLVE_MODE_NONE, and `pStencilAttachment->resolveImageView` is not VK_NULL_HANDLE, then `pStencilAttachment->resolveImageView` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkRenderingInfo-colorAttachmentCount-06090
  If `colorAttachmentCount` is not `0` and the `imageView` member of an element of `pColorAttachments` is not VK_NULL_HANDLE, the `imageLayout` member of that element of `pColorAttachments` **must** not be VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkRenderingInfo-colorAttachmentCount-06091
  If `colorAttachmentCount` is not `0` and the `imageView` member of an element of `pColorAttachments` is not VK_NULL_HANDLE, if the `resolveMode` member of that element of `pColorAttachments` is not VK_RESOLVE_MODE_NONE, its `resolveImageLayout` member **must** not be VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL or

VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkRenderingInfo-pDepthAttachment-06092
  If pDepthAttachment is not NULL and pDepthAttachment->imageView is not VK_NULL_HANDLE, pDepthAttachment->imageLayout **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL
- VUID-VkRenderingInfo-pDepthAttachment-06093
  If pDepthAttachment is not NULL, pDepthAttachment->imageView is not VK_NULL_HANDLE, and pDepthAttachment->resolveMode is not VK_RESOLVE_MODE_NONE, pDepthAttachment->resolveImageLayout **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL
- VUID-VkRenderingInfo-pStencilAttachment-06094
  If pStencilAttachment is not NULL and pStencilAttachment->imageView is not VK_NULL_HANDLE, pStencilAttachment->imageLayout **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL
- VUID-VkRenderingInfo-flags-11514
  If flags contains VK_RENDERING_CUSTOM_RESOLVE_BIT_EXT or VK_RENDERING_FRAGMENT_REGION_BIT_EXT, then the customResolve feature **must** be enabled
- VUID-VkRenderingInfo-pColorAttachments-11515
  For any element of pColorAttachments, pDepthAttachment, or pStencilAttachment, if resolveMode contains VK_RESOLVE_MODE_CUSTOM_BIT_EXT, then flags **must** contain VK_RENDERING_CUSTOM_RESOLVE_BIT_EXT
- VUID-VkRenderingInfo-flags-11516
  If flags contains VK_RENDERING_CUSTOM_RESOLVE_BIT_EXT, then for any element of pColorAttachments, pDepthAttachment, or pStencilAttachment, resolveMode **must** be VK_RESOLVE_MODE_CUSTOM_BIT_EXT or VK_RESOLVE_MODE_NONE
- VUID-VkRenderingInfo-pStencilAttachment-06095
  If pStencilAttachment is not NULL, pStencilAttachment->imageView is not VK_NULL_HANDLE, and pStencilAttachment->resolveMode is not VK_RESOLVE_MODE_NONE, pStencilAttachment->resolveImageLayout **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL
- VUID-VkRenderingInfo-layerCount-07817
  layerCount **must** be less than or equal to maxFramebufferLayers
- VUID-VkRenderingInfo-viewMask-10859
  If viewMask is 0, each pColorAttachment->imageView and pColorAttachment->resolveImageView that is not VK_NULL_HANDLE **must** have a layerCount that is greater than or equal to VkRenderingInfo::layerCount
- VUID-VkRenderingInfo-viewMask-10860
  If viewMask is 0, each pDepthAttachment->imageView and pDepthAttachment->resolveImageView that is not VK_NULL_HANDLE **must** have a layerCount that is greater than or equal to VkRenderingInfo::layerCount
- VUID-VkRenderingInfo-viewMask-10861
  If viewMask is 0, each pStencilAttachment->imageView and pStencilAttachment->resolveImageView that is not VK_NULL_HANDLE **must** have a layerCount that is greater

than or equal to VkRenderingInfo::layerCount

- VUID-VkRenderingInfo-viewMask-12403
  If viewMask is not 0, each pColorAttachment->imageView and pColorAttachment->resolveImageView that is not VK_NULL_HANDLE **must** have a layerCount that is greater than the index of the most significant bit in viewMask
- VUID-VkRenderingInfo-viewMask-12404
  If viewMask is not 0, each pDepthAttachment->imageView and pDepthAttachment->resolveImageView that is not VK_NULL_HANDLE **must** have a layerCount that is greater than the index of the most significant bit in viewMask
- VUID-VkRenderingInfo-viewMask-12405
  If viewMask is not 0, each pStencilAttachment->imageView and pStencilAttachment->resolveImageView that is not VK_NULL_HANDLE **must** have a layerCount that is greater than the index of the most significant bit in viewMask
- VUID-VkRenderingInfo-colorAttachmentCount-06096
  If colorAttachmentCount is not 0 and the imageView member of an element of pColorAttachments is not VK_NULL_HANDLE, the imageLayout member of that element of pColorAttachments **must** not be VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkRenderingInfo-colorAttachmentCount-06097
  If colorAttachmentCount is not 0 and the imageView member of an element of pColorAttachments is not VK_NULL_HANDLE, if the resolveMode member of that element of pColorAttachments is not VK_RESOLVE_MODE_NONE, its resolveImageLayout member **must** not be VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkRenderingInfo-pDepthAttachment-06098
  If pDepthAttachment is not NULL, pDepthAttachment->imageView is not VK_NULL_HANDLE, and pDepthAttachment->resolveMode is not VK_RESOLVE_MODE_NONE, pDepthAttachment->resolveImageLayout **must** not be VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL
- VUID-VkRenderingInfo-pStencilAttachment-06099
  If pStencilAttachment is not NULL, pStencilAttachment->imageView is not VK_NULL_HANDLE, and pStencilAttachment->resolveMode is not VK_RESOLVE_MODE_NONE, pStencilAttachment->resolveImageLayout **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkRenderingInfo-colorAttachmentCount-06100
  If colorAttachmentCount is not 0 and the imageView member of an element of pColorAttachments is not VK_NULL_HANDLE, the imageLayout member of that element of pColorAttachments **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkRenderingInfo-colorAttachmentCount-06101
  If colorAttachmentCount is not 0 and the imageView member of an element of

pColorAttachments is not VK_NULL_HANDLE, if the resolveMode member of that element of pColorAttachments is not VK_RESOLVE_MODE_NONE, its resolveImageLayout member **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkRenderingInfo-pDepthAttachment-07732
  If pDepthAttachment is not NULL and pDepthAttachment->imageView is not VK_NULL_HANDLE, pDepthAttachment->imageLayout **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkRenderingInfo-pDepthAttachment-07733
  If pDepthAttachment is not NULL, pDepthAttachment->imageView is not VK_NULL_HANDLE, and pDepthAttachment->resolveMode is not VK_RESOLVE_MODE_NONE, pDepthAttachment->resolveImageLayout **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkRenderingInfo-pStencilAttachment-07734
  If pStencilAttachment is not NULL and pStencilAttachment->imageView is not VK_NULL_HANDLE, pStencilAttachment->imageLayout **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL
- VUID-VkRenderingInfo-pStencilAttachment-07735
  If pStencilAttachment is not NULL, pStencilAttachment->imageView is not VK_NULL_HANDLE, and pStencilAttachment->resolveMode is not VK_RESOLVE_MODE_NONE, pStencilAttachment->resolveImageLayout **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL
- VUID-VkRenderingInfo-pDepthAttachment-06102
  If pDepthAttachment is not NULL and pDepthAttachment->imageView is not VK_NULL_HANDLE, pDepthAttachment->resolveMode **must** be one of the bits set in VkPhysicalDeviceDepthStencilResolveProperties::supportedDepthResolveModes
- VUID-VkRenderingInfo-pStencilAttachment-06103
  If pStencilAttachment is not NULL and pStencilAttachment->imageView is not VK_NULL_HANDLE, pStencilAttachment->resolveMode **must** be one of the bits set in VkPhysicalDeviceDepthStencilResolveProperties::supportedStencilResolveModes
- VUID-VkRenderingInfo-pDepthAttachment-06104
  If pDepthAttachment and pStencilAttachment are both not NULL, pDepthAttachment->imageView and pStencilAttachment->imageView are both not VK_NULL_HANDLE, and VkPhysicalDeviceDepthStencilResolveProperties::independentResolveNone is VK_FALSE, the resolveMode of both structures **must** be the same value
- VUID-VkRenderingInfo-pDepthAttachment-06105
  If pDepthAttachment and pStencilAttachment are both not NULL, pDepthAttachment->imageView and pStencilAttachment->imageView are both not VK_NULL_HANDLE, VkPhysicalDeviceDepthStencilResolveProperties::independentResolve is VK_FALSE, and the resolveMode of neither structure is VK_RESOLVE_MODE_NONE, the resolveMode of both

structures **must** be the same value

- VUID-VkRenderingInfo-colorAttachmentCount-06106
  `colorAttachmentCount` **must** be less than or equal to VkPhysicalDeviceLimits ::`maxColorAttachments`
- VUID-VkRenderingInfo-imageView-06107
  If the `imageView` member of a VkRenderingFragmentDensityMapAttachmentInfoEXT structure included in the `pNext` chain is not VK_NULL_HANDLE, and the `fragmentDensityMapNonSubsampledImages` feature is not enabled, valid `imageView` and `resolveImageView` members of `pDepthAttachment`, `pStencilAttachment`, and each element of `pColorAttachments` **must** be a VkImageView created with VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-VkRenderingInfo-imageView-06108
  If the `imageView` member of a VkRenderingFragmentDensityMapAttachmentInfoEXT structure included in the `pNext` chain is not VK_NULL_HANDLE, and `viewMask` is not `0`, `imageView` **must** have a `layerCount` greater than the index of the most significant bit in `viewMask`
- VUID-VkRenderingInfo-imageView-06109
  If the `imageView` member of a VkRenderingFragmentDensityMapAttachmentInfoEXT structure included in the `pNext` chain is not VK_NULL_HANDLE, and `viewMask` is `0`, `imageView` **must** have a `layerCount` equal to `1`
- VUID-VkRenderingInfo-pNext-06112
  If the `pNext` chain does not contain VkDeviceGroupRenderPassBeginInfo or its `deviceRenderAreaCount` member is equal to 0 and the `imageView` member of a VkRenderingFragmentDensityMapAttachmentInfoEXT structure included in the `pNext` chain is not VK_NULL_HANDLE, `imageView` **must** have a width greater than or equal to $\lceil \frac{renderArea_x + renderArea_{width}}{maxFragmentDensityTexelSize_{width}} \rceil$
- VUID-VkRenderingInfo-pNext-06114
  If the `pNext` chain does not contain VkDeviceGroupRenderPassBeginInfo or its `deviceRenderAreaCount` member is equal to 0 and the `imageView` member of a VkRenderingFragmentDensityMapAttachmentInfoEXT structure included in the `pNext` chain is not VK_NULL_HANDLE, `imageView` **must** have a height greater than or equal to $\lceil \frac{renderArea_y + renderArea_{height}}{maxFragmentDensityTexelSize_{height}} \rceil$
- VUID-VkRenderingInfo-pNext-06113
  If the `pNext` chain contains a VkDeviceGroupRenderPassBeginInfo structure, its `deviceRenderAreaCount` member is not 0, and the `imageView` member of a VkRenderingFragmentDensityMapAttachmentInfoEXT structure included in the `pNext` chain is not VK_NULL_HANDLE, `imageView` **must** have a width greater than or equal to $\lceil \frac{pDeviceRenderAreas_x + pDeviceRenderAreas_{width}}{maxFragmentDensityTexelSize_{width}} \rceil$ for each element of `pDeviceRenderAreas`
- VUID-VkRenderingInfo-pNext-06115
  If the `pNext` chain contains a VkDeviceGroupRenderPassBeginInfo structure, its `deviceRenderAreaCount` member is not 0, and the `imageView` member of a VkRenderingFragmentDensityMapAttachmentInfoEXT structure included in the `pNext` chain is not VK_NULL_HANDLE, `imageView` **must** have a height greater than or equal to $\lceil \frac{pDeviceRenderAreas_y + pDeviceRenderAreas_{height}}{maxFragmentDensityTexelSize_{height}} \rceil$ for each element of `pDeviceRenderAreas`

- VUID-VkRenderingInfo-imageView-06116
  If the imageView member of a VkRenderingFragmentDensityMapAttachmentInfoEXT structure included in the pNext chain is not VK_NULL_HANDLE, it **must** not be equal to the imageView or resolveImageView member of pDepthAttachment, pStencilAttachment, or any element of pColorAttachments
- VUID-VkRenderingInfo-pNext-06119
  If the maintenance7 feature is not enabled or the robustFragmentShadingRateAttachmentAccess limit is VK_FALSE or the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure was created with VkImageSubresourceRange::baseMipLevel greater than 0, the pNext chain does not contain VkDeviceGroupRenderPassBeginInfo or its deviceRenderAreaCount member is equal to 0, and the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure included in the pNext chain is not VK_NULL_HANDLE, imageView **must** have a width greater than or equal to $\lceil \frac{renderArea_x + renderArea_{width}}{shadingRateAttachmentTexelSize_{width}} \rceil$
- VUID-VkRenderingInfo-pNext-06121
  If the maintenance7 feature is not enabled or the robustFragmentShadingRateAttachmentAccess limit is VK_FALSE or the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure was created with VkImageSubresourceRange::baseMipLevel greater than 0, the pNext chain does not contain VkDeviceGroupRenderPassBeginInfo or its deviceRenderAreaCount member is equal to 0 and the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure included in the pNext chain is not VK_NULL_HANDLE, imageView **must** have a height greater than or equal to $\lceil \frac{renderArea_y + renderArea_{height}}{shadingRateAttachmentTexelSize_{height}} \rceil$
- VUID-VkRenderingInfo-pNext-06120
  If the maintenance7 feature is not enabled or the robustFragmentShadingRateAttachmentAccess limit is VK_FALSE or the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure was created with VkImageSubresourceRange::baseMipLevel greater than 0, the pNext chain contains a VkDeviceGroupRenderPassBeginInfo structure, its deviceRenderAreaCount member is not 0, and the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure included in the pNext chain is not VK_NULL_HANDLE, imageView **must** have a width greater than or equal to $\lceil \frac{pDeviceRenderAreas_x + pDeviceRenderAreas_{width}}{shadingRateAttachmentTexelSize_{width}} \rceil$ for each element of pDeviceRenderAreas
- VUID-VkRenderingInfo-pNext-06122
  If the maintenance7 feature is not enabled or the robustFragmentShadingRateAttachmentAccess limit is VK_FALSE or the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure was created with VkImageSubresourceRange::baseMipLevel greater than 0, the pNext chain contains a VkDeviceGroupRenderPassBeginInfo structure, its deviceRenderAreaCount member is not 0, and the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure included in the pNext chain is not VK_NULL_HANDLE, imageView **must** have a height greater than or equal to $\lceil \frac{pDeviceRenderAreas_y + pDeviceRenderAreas_{height}}{shadingRateAttachmentTexelSize_{height}} \rceil$ for each element of pDeviceRenderAreas
- VUID-VkRenderingInfo-imageView-06123
  If the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure included in the pNext chain is not VK_NULL_HANDLE, and viewMask is 0,

imageView **must** have a layerCount that is either equal to 1 or greater than or equal to VkRenderingInfo::layerCount

- VUID-VkRenderingInfo-imageView-06124
  If the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure included in the pNext chain is not VK_NULL_HANDLE, and viewMask is not 0, imageView **must** have a layerCount that is either equal to 1 or greater than the index of the most significant bit in viewMask
- VUID-VkRenderingInfo-imageView-06125
  If the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure included in the pNext chain is not VK_NULL_HANDLE, it **must** not be equal to the imageView or resolveImageView member of pDepthAttachment, pStencilAttachment, or any element of pColorAttachments
- VUID-VkRenderingInfo-imageView-06126
  If the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure included in the pNext chain is not VK_NULL_HANDLE, it **must** not be equal to the imageView member of a VkRenderingFragmentDensityMapAttachmentInfoEXT structure included in the pNext chain
- VUID-VkRenderingInfo-multiview-06127
  If the multiview feature is not enabled, viewMask **must** be 0
- VUID-VkRenderingInfo-viewMask-06128
  The index of the most significant bit in viewMask **must** be less than maxMultiviewViewCount
- VUID-VkRenderingInfo-None-09044
  Valid attachments specified by this structure **must** not be bound to memory locations that are bound to any other valid attachments specified by this structure
- VUID-VkRenderingInfo-flags-10012
  If flags includes VK_RENDERING_CONTENTS_INLINE_BIT_KHR then at least one of the following features **must** be enabled
  - maintenance7
  - nestedCommandBuffer
- VUID-VkRenderingInfo-colorAttachmentCount-09479
  If colorAttachmentCount is not 0 and the imageView member of an element of pColorAttachments is not VK_NULL_HANDLE, that imageView **must** have been created with the identity swizzle
- VUID-VkRenderingInfo-colorAttachmentCount-09480
  If colorAttachmentCount is not 0, and there is an element of pColorAttachments with its imageView member not set to VK_NULL_HANDLE, its resolveMode member not set to VK_RESOLVE_MODE_NONE, and resolveImageView is not VK_NULL_HANDLE, then the resolveImageView member of that element of pColorAttachments **must** have been created with the identity swizzle
- VUID-VkRenderingInfo-pDepthAttachment-09481
  If pDepthAttachment is not NULL and pDepthAttachment->imageView is not VK_NULL_HANDLE, pDepthAttachment->imageView **must** have been created with the identity swizzle
- VUID-VkRenderingInfo-pDepthAttachment-09482

If pDepthAttachment is not NULL, pDepthAttachment->imageView is not VK_NULL_HANDLE, pDepthAttachment->resolveMode is not VK_RESOLVE_MODE_NONE, and pDepthAttachment->resolveImageView is not VK_NULL_HANDLE, pDepthAttachment->resolveImageView **must** have been created with the identity swizzle

- VUID-VkRenderingInfo-pStencilAttachment-09483
  If pStencilAttachment is not NULL and pStencilAttachment->imageView is not VK_NULL_HANDLE, pStencilAttachment->imageView **must** have been created with the identity swizzle
- VUID-VkRenderingInfo-pStencilAttachment-09484
  If pStencilAttachment is not NULL, pStencilAttachment->imageView is not VK_NULL_HANDLE, pStencilAttachment->resolveMode is not VK_RESOLVE_MODE_NONE, and pStencilAttachment->resolveImageView is not VK_NULL_HANDLE, pStencilAttachment->resolveImageView **must** have been created with the identity swizzle
- VUID-VkRenderingInfo-imageView-09485
  If the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure included in the pNext chain is not VK_NULL_HANDLE, it **must** have been created with the identity swizzle
- VUID-VkRenderingInfo-imageView-09486
  If the imageView member of a VkRenderingFragmentDensityMapAttachmentInfoEXT structure included in the pNext chain is not VK_NULL_HANDLE, it **must** have been created with the identity swizzle

Valid Usage (Implicit)

- VUID-VkRenderingInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_RENDERING_INFO
- VUID-VkRenderingInfo-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkDeviceGroupRenderPassBeginInfo, VkMultisampledRenderToSingleSampledInfoEXT, VkRenderingFragmentDensityMapAttachmentInfoEXT, or VkRenderingFragmentShadingRateAttachmentInfoKHR
- VUID-VkRenderingInfo-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkRenderingInfo-flags-parameter
  If flags is not 0, flags **must** be a valid combination of VkRenderingFlagBits values
- VUID-VkRenderingInfo-pColorAttachments-parameter
  If colorAttachmentCount is not 0, pColorAttachments **must** be a valid pointer to an array of colorAttachmentCount valid VkRenderingAttachmentInfo structures
- VUID-VkRenderingInfo-pDepthAttachment-parameter
  If pDepthAttachment is not NULL, pDepthAttachment **must** be a valid pointer to a valid VkRenderingAttachmentInfo structure
- VUID-VkRenderingInfo-pStencilAttachment-parameter

If `pStencilAttachment` is not `NULL`, `pStencilAttachment` **must** be a valid pointer to a valid VkRenderingAttachmentInfo structure

Bits which **can** be set in VkRenderingInfo::`flags` describing additional properties of the render pass are:

```
// Provided by VK_VERSION_1_3
typedef enum VkRenderingFlagBits {
    VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT = 0x00000001,
    VK_RENDERING_SUSPENDING_BIT = 0x00000002,
    VK_RENDERING_RESUMING_BIT = 0x00000004,
  // Provided by VK_EXT_legacy_dithering with (VK_KHR_dynamic_rendering or
VK_VERSION_1_3) and (VK_VERSION_1_4 or VK_KHR_extended_flags or VK_KHR_maintenance5)
    VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT = 0x00000008,
  // Provided by VK_KHR_maintenance7
    VK_RENDERING_CONTENTS_INLINE_BIT_KHR = 0x00000010,
  // Provided by VK_EXT_custom_resolve with VK_KHR_dynamic_rendering or VK_VERSION_1_3
    VK_RENDERING_FRAGMENT_REGION_BIT_EXT = 0x00000040,
  // Provided by VK_EXT_custom_resolve with VK_KHR_dynamic_rendering or VK_VERSION_1_3
    VK_RENDERING_CUSTOM_RESOLVE_BIT_EXT = 0x00000080,
  // Provided by VK_KHR_maintenance10 with (VK_VERSION_1_4 or
VK_KHR_dynamic_rendering_local_read) and (VK_VERSION_1_3 or VK_KHR_dynamic_rendering)
    VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR = 0x00000100,
  // Provided by VK_KHR_dynamic_rendering
    VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT_KHR =
VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT,
  // Provided by VK_KHR_dynamic_rendering
    VK_RENDERING_SUSPENDING_BIT_KHR = VK_RENDERING_SUSPENDING_BIT,
  // Provided by VK_KHR_dynamic_rendering
    VK_RENDERING_RESUMING_BIT_KHR = VK_RENDERING_RESUMING_BIT,
  // Provided by VK_EXT_nested_command_buffer
    VK_RENDERING_CONTENTS_INLINE_BIT_EXT = VK_RENDERING_CONTENTS_INLINE_BIT_KHR,
} VkRenderingFlagBits;
```

```
// Provided by VK_KHR_dynamic_rendering
// Equivalent to VkRenderingFlagBits
typedef VkRenderingFlagBits VkRenderingFlagBitsKHR;
```

- VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT specifies that contents for the render pass instance will be recorded in secondary command buffers, and vkCmdExecuteCommands is the only valid command in the command buffer until vkCmdEndRendering2KHR or vkCmdEndRendering. If the `nestedCommandBuffer` feature is enabled, the contents **can** come from both inline and vkCmdExecuteCommands.
- VK_RENDERING_RESUMING_BIT specifies that the render pass instance is resuming an earlier suspended render pass instance.
- VK_RENDERING_SUSPENDING_BIT specifies that the render pass instance will be suspended.

- VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT specifies that Legacy Dithering is enabled for the render pass instance.
- VK_RENDERING_CONTENTS_INLINE_BIT_KHR specifies that contents for the render pass instance **can** be recorded inline within the current command buffer. This **can** be combined with the VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT bit to allow contents to be recorded both inline and in secondary command buffers.
- VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR specifies that VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR will always be specified for any attachment which invokes the behavior described by that flag.
- VK_RENDERING_FRAGMENT_REGION_BIT_EXT specifies that the render pass **can** access samples which are not covered in its SampleMask.
- VK_RENDERING_CUSTOM_RESOLVE_BIT_EXT specifies that the render pass contains a custom resolve. When this bit is set, vkCmdBeginCustomResolveEXT **can** be called.

The contents of pRenderingInfo **must** match between suspended render pass instances and the render pass instances that resume them, other than the presence or absence of the VK_RENDERING_RESUMING_BIT, VK_RENDERING_SUSPENDING_BIT, and VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT flags. No action or synchronization commands, or other render pass instances, are allowed between suspending and resuming render pass instances.

```
// Provided by VK_VERSION_1_3
typedef VkFlags VkRenderingFlags;
```

```
// Provided by VK_KHR_dynamic_rendering
// Equivalent to VkRenderingFlags
typedef VkRenderingFlags VkRenderingFlagsKHR;
```

VkRenderingFlags is a bitmask type for setting a mask of zero or more VkRenderingFlagBits.

The VkRenderingAttachmentInfo structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkRenderingAttachmentInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkImageView              imageView;
    VkImageLayout            imageLayout;
    VkResolveModeFlagBits    resolveMode;
    VkImageView              resolveImageView;
    VkImageLayout            resolveImageLayout;
    VkAttachmentLoadOp       loadOp;
    VkAttachmentStoreOp      storeOp;
    VkClearValue             clearValue;
} VkRenderingAttachmentInfo;
```

```
// Provided by VK_KHR_dynamic_rendering
// Equivalent to VkRenderingAttachmentInfo
typedef VkRenderingAttachmentInfo VkRenderingAttachmentInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `imageView` is the image view that will be used for rendering.
- `imageLayout` is the layout that `imageView` will be in during rendering.
- `resolveMode` is a VkResolveModeFlagBits value defining how data written to `imageView` will be resolved into `resolveImageView`.
- `resolveImageView` is an image view used to write resolved data at the end of rendering.
- `resolveImageLayout` is the layout that `resolveImageView` will be in during rendering.
- `loadOp` is a VkAttachmentLoadOp value defining the load operation for the attachment.
- `storeOp` is a VkAttachmentStoreOp value defining the store operation for the attachment.
- `clearValue` is a VkClearValue structure defining values used to clear `imageView` when `loadOp` is VK_ATTACHMENT_LOAD_OP_CLEAR.

Values in `imageView` are loaded and stored according to the values of `loadOp` and `storeOp`, within the render area for each device specified in VkRenderingInfo. If `imageView` is VK_NULL_HANDLE, other members of this structure are ignored; writes to this attachment will be discarded, and no load, store, or multisample resolve operations will be performed.

If `resolveMode` is VK_RESOLVE_MODE_NONE, then `resolveImageView` is ignored. If `resolveMode` is not VK_RESOLVE_MODE_NONE, multisampled rendering to single-sampled attachments is enabled, and `imageView` has a sample count of VK_SAMPLE_COUNT_1_BIT, `resolveImageView` **must** be VK_NULL_HANDLE. Otherwise, `resolveImageView` **must** not be VK_NULL_HANDLE.

If `resolveMode` is not VK_RESOLVE_MODE_NONE, a render pass multisample resolve operation is defined for the attachment subresource.

The contents of a resolve attachment within the render area become undefined at the time vkCmdBeginCustomResolveEXT is called if all of the following conditions are true:

- VK_RENDERING_CUSTOM_RESOLVE_BIT_EXT is set.
- The attachment sets `resolveMode` to VK_RESOLVE_MODE_CUSTOM_BIT_EXT.

This affects color, depth, and stencil attachments. In addition, there is an implicit store operation of VK_ATTACHMENT_STORE_OP_STORE for these attachments.

NOTE	The resolve mode and store operation are independent; it is valid to write both resolved and unresolved values, and equally valid to discard the unresolved values while writing the resolved ones.

Store and resolve operations are only performed at the end of a render pass instance that does not specify the VK_RENDERING_SUSPENDING_BIT_KHR flag. If the VK_RENDERING_CUSTOM_RESOLVE_BIT_EXT is specified and an attachment uses the VK_RESOLVE_MODE_CUSTOM_BIT_EXT resolve mode, the resolve attachment will only be written by draws recorded following a call to vkCmdBeginCustomResolveEXT.

Load operations are only performed at the beginning of a render pass instance that does not specify the VK_RENDERING_RESUMING_BIT_KHR flag.

Image contents at the end of a suspended render pass instance remain defined for access by a resuming render pass instance.

Valid Usage

- VUID-VkRenderingAttachmentInfo-imageView-06129
  If `imageView` is not VK_NULL_HANDLE and has a non-integer color format, `resolveMode` **must** be VK_RESOLVE_MODE_NONE or VK_RESOLVE_MODE_CUSTOM_BIT_EXT or VK_RESOLVE_MODE_AVERAGE_BIT
- VUID-VkRenderingAttachmentInfo-imageView-06130
  If `imageView` is not VK_NULL_HANDLE and has an integer color format, `resolveMode` **must** be VK_RESOLVE_MODE_NONE or VK_RESOLVE_MODE_CUSTOM_BIT_EXT or VK_RESOLVE_MODE_SAMPLE_ZERO_BIT
- VUID-VkRenderingAttachmentInfo-imageView-06861
  If all of the following are true:
  - `imageView` is not VK_NULL_HANDLE
  - `resolveMode` is not VK_RESOLVE_MODE_NONE
  - the `pNext` chain of VkRenderingInfo does not include a VkMultisampledRenderToSingleSampledInfoEXT structure with the `multisampledRenderToSingleSampledEnable` field equal to VK_TRUE

  `imageView` **must** not have a sample count of VK_SAMPLE_COUNT_1_BIT
- VUID-VkRenderingAttachmentInfo-imageView-06862

If all of the following are true:

- `imageView` is not VK_NULL_HANDLE
- `resolveMode` is not VK_RESOLVE_MODE_NONE
- the `pNext` chain of VkRenderingInfo does not include a VkMultisampledRenderToSingleSampledInfoEXT structure with the `multisampledRenderToSingleSampledEnable` field equal to VK_TRUE

`resolveImageView` **must** not be VK_NULL_HANDLE

- VUID-VkRenderingAttachmentInfo-imageView-06863
  If `imageView` is not VK_NULL_HANDLE, `resolveMode` is not VK_RESOLVE_MODE_NONE, the `pNext` chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with the `multisampledRenderToSingleSampledEnable` field equal to VK_TRUE, and `imageView` has a sample count of VK_SAMPLE_COUNT_1_BIT, `resolveImageView` **must** be VK_NULL_HANDLE
- VUID-VkRenderingAttachmentInfo-None-12256
  If all of the following are true:

  - `imageView` is not VK_NULL_HANDLE
  - `imageView` has a sample count of VK_SAMPLE_COUNT_1_BIT
  - the `pNext` chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with the `multisampledRenderToSingleSampledEnable` field equal to VK_TRUE
  - `storeOp` is VK_ATTACHMENT_STORE_OP_STORE

  then `resolveMode` **must** not be VK_RESOLVE_MODE_NONE
- VUID-VkRenderingAttachmentInfo-imageView-06864
  If `imageView` is not VK_NULL_HANDLE, `resolveImageView` is not VK_NULL_HANDLE, and `resolveMode` is not VK_RESOLVE_MODE_NONE, `resolveImageView` **must** have a sample count of VK_SAMPLE_COUNT_1_BIT
- VUID-VkRenderingAttachmentInfo-imageView-06865
  If all of the following are true:

  - `imageView` is not VK_NULL_HANDLE
  - `resolveImageView` is not VK_NULL_HANDLE
  - `resolveMode` is not VK_RESOLVE_MODE_NONE
  - `resolveMode` is not VK_RESOLVE_MODE_CUSTOM_BIT_EXT

  `imageView` and `resolveImageView` **must** have the same VkFormat
- VUID-VkRenderingAttachmentInfo-imageView-06135
  If `imageView` is not VK_NULL_HANDLE, `imageLayout` **must** not be VK_IMAGE_LAYOUT_UNDEFINED, VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL, VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT,

VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL, or VK_IMAGE_LAYOUT_PREINITIALIZED

- VUID-VkRenderingAttachmentInfo-imageView-06136
  If `resolveImageView` is not VK_NULL_HANDLE, `resolveImageLayout` **must** not be VK_IMAGE_LAYOUT_UNDEFINED, VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL, VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT, VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL, or VK_IMAGE_LAYOUT_PREINITIALIZED
- VUID-VkRenderingAttachmentInfo-imageView-06137
  If `resolveImageView` is not VK_NULL_HANDLE, `resolveImageLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkRenderingAttachmentInfo-imageView-06140
  If `imageView` is not VK_NULL_HANDLE, `imageLayout` **must** not be VK_IMAGE_LAYOUT_FRAGMENT_DENSITY_MAP_OPTIMAL_EXT
- VUID-VkRenderingAttachmentInfo-imageView-06141
  If `resolveImageView` is not VK_NULL_HANDLE, `resolveImageLayout` **must** not be VK_IMAGE_LAYOUT_FRAGMENT_DENSITY_MAP_OPTIMAL_EXT
- VUID-VkRenderingAttachmentInfo-imageView-06142
  If `resolveImageView` is not VK_NULL_HANDLE, `resolveImageLayout` **must** not be VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VUID-VkRenderingAttachmentInfo-imageView-06143
  If `imageView` is not VK_NULL_HANDLE, `imageLayout` **must** not be VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR
- VUID-VkRenderingAttachmentInfo-imageView-06144
  If `resolveImageView` is not VK_NULL_HANDLE, `resolveImageLayout` **must** not be VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR
- VUID-VkRenderingAttachmentInfo-imageView-10780
  If feedback loop is enabled for the attachment identified by `imageView`, then `imageView` **must** have been created with a `usage` value including VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT, either VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, and either VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT or VK_IMAGE_USAGE_SAMPLED_BIT
- VUID-VkRenderingAttachmentInfo-imageView-06145
  If `imageView` is not VK_NULL_HANDLE, `imageLayout` **must** not be VK_IMAGE_LAYOUT_PRESENT_SRC_KHR
- VUID-VkRenderingAttachmentInfo-imageView-06146
  If `resolveImageView` is not VK_NULL_HANDLE, `resolveImageLayout` **must** not be VK_IMAGE_LAYOUT_PRESENT_SRC_KHR
- VUID-VkRenderingAttachmentInfo-pNext-11752
  If the `pNext` chain includes a VkRenderingAttachmentFlagsInfoKHR structure, and `flags`

includes VK_RENDERING_ATTACHMENT_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR or VK_RENDERING_ATTACHMENT_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR, imageView **must** have a format using sRGB encoding

- VUID-VkRenderingAttachmentInfo-pNext-11753
  If the pNext chain includes a VkRenderingAttachmentFlagsInfoKHR structure, and flags includes VK_RENDERING_ATTACHMENT_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR or VK_RENDERING_ATTACHMENT_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR, resolveMode **must** be equal to VK_RESOLVE_MODE_AVERAGE_BIT
- VUID-VkRenderingAttachmentInfo-pNext-11754
  If the pNext chain includes a VkRenderingAttachmentFlagsInfoKHR structure, and flags includes VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, imageView **must** have an image that was created with the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flag set
- VUID-VkRenderingAttachmentInfo-imageView-12470
  If imageView is not VK_NULL_HANDLE, it **must** have been created with a VkImageViewCreateInfo::viewType not equal to VK_IMAGE_VIEW_TYPE_3D
- VUID-VkRenderingAttachmentInfo-resolveImageView-12471
  If resolveImageView is not VK_NULL_HANDLE, it **must** have been created with a VkImageViewCreateInfo::viewType not equal to VK_IMAGE_VIEW_TYPE_3D
- VUID-VkRenderingAttachmentInfo-imageView-12486
  If imageView is not VK_NULL_HANDLE it **must** only specify a single mip level
- VUID-VkRenderingAttachmentInfo-resolveImageView-12487
  If resolveImageView is not VK_NULL_HANDLE it **must** only specify a single mip level

Valid Usage (Implicit)

- VUID-VkRenderingAttachmentInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_INFO
- VUID-VkRenderingAttachmentInfo-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkAttachmentFeedbackLoopInfoEXT or VkRenderingAttachmentFlagsInfoKHR
- VUID-VkRenderingAttachmentInfo-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkRenderingAttachmentInfo-imageView-parameter
  If imageView is not VK_NULL_HANDLE, imageView **must** be a valid VkImageView handle
- VUID-VkRenderingAttachmentInfo-imageLayout-parameter
  imageLayout **must** be a valid VkImageLayout value
- VUID-VkRenderingAttachmentInfo-resolveMode-parameter
  If resolveMode is not 0, resolveMode **must** be a valid VkResolveModeFlagBits value
- VUID-VkRenderingAttachmentInfo-resolveImageView-parameter
  If resolveImageView is not VK_NULL_HANDLE, resolveImageView **must** be a valid

VkImageView handle

- VUID-VkRenderingAttachmentInfo-resolveImageLayout-parameter
  `resolveImageLayout` **must** be a valid VkImageLayout value
- VUID-VkRenderingAttachmentInfo-loadOp-parameter
  `loadOp` **must** be a valid VkAttachmentLoadOp value
- VUID-VkRenderingAttachmentInfo-storeOp-parameter
  `storeOp` **must** be a valid VkAttachmentStoreOp value
- VUID-VkRenderingAttachmentInfo-clearValue-parameter
  `clearValue` **must** be a valid VkClearValue union
- VUID-VkRenderingAttachmentInfo-commonparent
  Both of `imageView`, and `resolveImageView` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

To specify an attachment as an *input attachment* or to specify resolve operation flags, the `VkRenderingAttachmentFlagsInfoKHR` structure **can** be added to the `pNext` chain of VkRenderingAttachmentInfo.

The `VkRenderingAttachmentFlagsInfoKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance10
typedef struct VkRenderingAttachmentFlagsInfoKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    VkRenderingAttachmentFlagsKHR   flags;
} VkRenderingAttachmentFlagsInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkRenderingAttachmentFlagsKHR.

Valid Usage

- VUID-VkRenderingAttachmentFlagsInfoKHR-flags-11755
  `flags` **must** not include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR if the `dynamicRenderingLocalRead` feature is not enabled
- VUID-VkRenderingAttachmentFlagsInfoKHR-flags-11756
  If `flags` includes VK_RENDERING_ATTACHMENT_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR, `flags` **must** not include VK_RENDERING_ATTACHMENT_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR
- VUID-VkRenderingAttachmentFlagsInfoKHR-flags-11757
  If `flags` includes

VK_RENDERING_ATTACHMENT_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR or VK_RENDERING_ATTACHMENT_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR, `resolveSrgbFormatSupportsTransferFunctionControl` **must** be VK_TRUE

Valid Usage (Implicit)

- VUID-VkRenderingAttachmentFlagsInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_FLAGS_INFO_KHR
- VUID-VkRenderingAttachmentFlagsInfoKHR-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkRenderingAttachmentFlagBitsKHR values

Structure Chaining

Extends the structure

- VkRenderingAttachmentInfo

Bits which **can** be set in VkRenderingAttachmentFlagsInfoKHR::`flags`, describing additional properties of a rendering attachment, are:

```
// Provided by VK_KHR_maintenance10
typedef enum VkRenderingAttachmentFlagBitsKHR {
  // Provided by VK_KHR_maintenance10 with (VK_VERSION_1_4 or
VK_KHR_dynamic_rendering_local_read) and (VK_VERSION_1_3 or VK_KHR_dynamic_rendering)
    VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR = 0x00000001,
  // Provided by VK_KHR_maintenance10 with VK_VERSION_1_3 or VK_KHR_dynamic_rendering
    VK_RENDERING_ATTACHMENT_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR = 0x00000002,
  // Provided by VK_KHR_maintenance10 with VK_VERSION_1_3 or VK_KHR_dynamic_rendering
    VK_RENDERING_ATTACHMENT_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR = 0x00000004,
} VkRenderingAttachmentFlagBitsKHR;
```

- VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR specifies that the attachment **can** be used concurrently as both an input attachment and a write-only attachment during the render pass, creating a feedback loop while processing a fragment, and without a VK_DEPENDENCY_BY_REGION_BIT barrier separating the write attachment and input attachment usage. Using this flag does not remove the general requirement to use a VK_DEPENDENCY_BY_REGION_BIT barrier to resolve hazards when two different fragments access a particular attachment region, where one of them performs an attachment write, and a subsequent fragment performs an input attachment read. If VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR is specified in the rendering info, this flag **must** be set for an attachment to be used concurrently as an input attachment and a write attachment in this manner. If VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR is not specified in the rendering info, this flag is implied to be set for any attachment which has a combination of

image layouts and image view usage flags which support input attachment usage.

- VK_RENDERING_ATTACHMENT_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR specifies that resolve operations happening to an sRGB encoded attachment **must** not convert samples from nonlinear to linear before averaging.
- VK_RENDERING_ATTACHMENT_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR specifies that resolve operations happening to an sRGB encoded attachment **must** convert samples from nonlinear to linear before averaging.

NOTE

VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR is intended to give implementations similar information as a subpass where an attachment could be used as both a color attachment and input attachment. Some implementations require extra work to make this scenario work beyond just considering the image layouts. Implementations which have no such considerations may treat this flag as a noop. The primary use case for this flag is to enable feedback loops inside a single shader.

Applications are encouraged to use VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR if `maintenance10` is available and they use feedback loops with VK_KHR_dynamic_rendering_local_read. Feedback loops are still allowed when not using the rendering flag, but the performance implication was an oversight in the original definition of VK_KHR_dynamic_rendering_local_read.

NOTE

In some scenarios, resolving sRGB in nonlinear space instead of the expected linear space can improve perceptual aliasing at the cost of inaccurate color blending.

```
// Provided by VK_KHR_maintenance10
typedef VkFlags VkRenderingAttachmentFlagsKHR;
```

`VkRenderingAttachmentFlagsKHR` is a bitmask type for setting a mask of zero or more VkRenderingAttachmentFlagBitsKHR.

To enable feedback loop for an attachment, the `VkAttachmentFeedbackLoopInfoEXT` structure **can** be added to the `pNext` chain of VkRenderingAttachmentInfo.

The `VkAttachmentFeedbackLoopInfoEXT` structure is defined as:

```
// Provided by VK_KHR_unified_image_layouts with
VK_EXT_attachment_feedback_loop_layout and (VK_VERSION_1_3 or
VK_KHR_dynamic_rendering)
typedef struct VkAttachmentFeedbackLoopInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           feedbackLoopEnable;
} VkAttachmentFeedbackLoopInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `feedbackLoopEnable` specifies that feedback loop is enabled for the attachment identified by VkRenderingAttachmentInfo::`imageView`.

Valid Usage

- VUID-VkAttachmentFeedbackLoopInfoEXT-unifiedImageLayouts-10782
  If the `unifiedImageLayouts` feature is not enabled, `feedbackLoopEnable` **must** be VK_FALSE

Valid Usage (Implicit)

- VUID-VkAttachmentFeedbackLoopInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_ATTACHMENT_FEEDBACK_LOOP_INFO_EXT

Structure Chaining

Extends the structure

- VkRenderingAttachmentInfo

The `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure is defined as:

```
// Provided by VK_KHR_fragment_shading_rate with VK_VERSION_1_3 or
VK_KHR_dynamic_rendering
typedef struct VkRenderingFragmentShadingRateAttachmentInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkImageView        imageView;
    VkImageLayout      imageLayout;
    VkExtent2D         shadingRateAttachmentTexelSize;
} VkRenderingFragmentShadingRateAttachmentInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `imageView` is the image view that will be used as a fragment shading rate attachment.
- `imageLayout` is the layout that `imageView` will be in during rendering.
- `shadingRateAttachmentTexelSize` specifies the number of pixels corresponding to each texel in `imageView`.

This structure can be included in the `pNext` chain of VkRenderingInfo to define a fragment shading rate attachment. If `imageView` is VK_NULL_HANDLE, or if this structure is not specified, the implementation behaves as if a valid shading rate attachment was specified with all texels

specifying a single pixel per fragment.

Valid Usage

- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06147
  If `imageView` is not VK_NULL_HANDLE, `imageLayout` **must** be VK_IMAGE_LAYOUT_GENERAL or VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06148
  If `imageView` is not VK_NULL_HANDLE, it **must** have been created with the VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR usage flag set
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06149
  If `imageView` is not VK_NULL_HANDLE, `shadingRateAttachmentTexelSize.width` **must** be a power of two value
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06150
  If `imageView` is not VK_NULL_HANDLE, `shadingRateAttachmentTexelSize.width` **must** be less than or equal to `maxFragmentShadingRateAttachmentTexelSize.width`
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06151
  If `imageView` is not VK_NULL_HANDLE, `shadingRateAttachmentTexelSize.width` **must** be greater than or equal to `minFragmentShadingRateAttachmentTexelSize.width`
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06152
  If `imageView` is not VK_NULL_HANDLE, `shadingRateAttachmentTexelSize.height` **must** be a power of two value
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06153
  If `imageView` is not VK_NULL_HANDLE, `shadingRateAttachmentTexelSize.height` **must** be less than or equal to `maxFragmentShadingRateAttachmentTexelSize.height`
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06154
  If `imageView` is not VK_NULL_HANDLE, `shadingRateAttachmentTexelSize.height` **must** be greater than or equal to `minFragmentShadingRateAttachmentTexelSize.height`
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06155
  If `imageView` is not VK_NULL_HANDLE, the quotient of `shadingRateAttachmentTexelSize.width` and `shadingRateAttachmentTexelSize.height` **must** be less than or equal to `maxFragmentShadingRateAttachmentTexelSizeAspectRatio`
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06156
  If `imageView` is not VK_NULL_HANDLE, the quotient of `shadingRateAttachmentTexelSize.height` and `shadingRateAttachmentTexelSize.width` **must** be less than or equal to `maxFragmentShadingRateAttachmentTexelSizeAspectRatio`

Valid Usage (Implicit)

- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_INFO_K

HR

- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-parameter
  If `imageView` is not VK_NULL_HANDLE, `imageView` **must** be a valid VkImageView handle
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageLayout-parameter
  `imageLayout` **must** be a valid VkImageLayout value

Structure Chaining

Extends the structure

- VkRenderingInfo

The `VkRenderingFragmentDensityMapAttachmentInfoEXT` structure is defined as:

```
// Provided by VK_EXT_fragment_density_map with VK_VERSION_1_3 or
VK_KHR_dynamic_rendering
typedef struct VkRenderingFragmentDensityMapAttachmentInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkImageView        imageView;
    VkImageLayout      imageLayout;
} VkRenderingFragmentDensityMapAttachmentInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `imageView` is the image view that will be used as a fragment density map attachment.
- `imageLayout` is the layout that `imageView` will be in during rendering.

This structure can be included in the `pNext` chain of VkRenderingInfo to define a fragment density map. If this structure is not included in the `pNext` chain, `imageView` is treated as VK_NULL_HANDLE.

Valid Usage

- VUID-VkRenderingFragmentDensityMapAttachmentInfoEXT-imageView-06157
  If `imageView` is not VK_NULL_HANDLE, `imageLayout` **must** be VK_IMAGE_LAYOUT_GENERAL or VK_IMAGE_LAYOUT_FRAGMENT_DENSITY_MAP_OPTIMAL_EXT
- VUID-VkRenderingFragmentDensityMapAttachmentInfoEXT-imageView-06158
  If `imageView` is not VK_NULL_HANDLE, it **must** have been created with the VK_IMAGE_USAGE_FRAGMENT_DENSITY_MAP_BIT_EXT usage flag set
- VUID-VkRenderingFragmentDensityMapAttachmentInfoEXT-imageView-06159
  If `imageView` is not VK_NULL_HANDLE, it **must** not have been created with VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT

- VUID-VkRenderingFragmentDensityMapAttachmentInfoEXT-apiVersion-07908
  If the `multiview` feature is not enabled, VkPhysicalDeviceProperties::`apiVersion` is less than Vulkan 1.1, and `imageView` is not VK_NULL_HANDLE, it **must** have a `layerCount` equal to `1`

Valid Usage (Implicit)

- VUID-VkRenderingFragmentDensityMapAttachmentInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_INFO_EXT
- VUID-VkRenderingFragmentDensityMapAttachmentInfoEXT-imageView-parameter
  `imageView` **must** be a valid VkImageView handle
- VUID-VkRenderingFragmentDensityMapAttachmentInfoEXT-imageLayout-parameter
  `imageLayout` **must** be a valid VkImageLayout value

Structure Chaining

Extends the structure

- VkRenderingInfo

To query the render area granularity for a render pass instance, call:

```
// Provided by VK_VERSION_1_4
void vkGetRenderingAreaGranularity(
    VkDevice                                    device,
    const VkRenderingAreaInfo*                  pRenderingAreaInfo,
    VkExtent2D*                                 pGranularity);
```

```
// Provided by VK_KHR_maintenance5
// Equivalent to vkGetRenderingAreaGranularity
void vkGetRenderingAreaGranularityKHR(
    VkDevice                                    device,
    const VkRenderingAreaInfo*                  pRenderingAreaInfo,
    VkExtent2D*                                 pGranularity);
```

- `device` is the logical device that owns the render pass instance.
- `pRenderingAreaInfo` is a pointer to a VkRenderingAreaInfo structure specifying details of the render pass instance to query the render area granularity for.
- `pGranularity` is a pointer to a VkExtent2D structure in which the granularity is returned.

The conditions leading to an optimal `renderArea` are:

- the `offset.x` member in `renderArea` is a multiple of the `width` member of the returned

VkExtent2D (the horizontal granularity).

- the `offset.y` member in `renderArea` is a multiple of the `height` member of the returned VkExtent2D (the vertical granularity).
- either the `extent.width` member in `renderArea` is a multiple of the horizontal granularity or `offset.x`+`extent.width` is equal to the `width` of each attachment used in the render pass instance.
- either the `extent.height` member in `renderArea` is a multiple of the vertical granularity or `offset.y`+`extent.height` is equal to the `height` of each attachment used in the render pass instance.

Valid Usage (Implicit)

- VUID-vkGetRenderingAreaGranularity-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetRenderingAreaGranularity-pRenderingAreaInfo-parameter
  `pRenderingAreaInfo` **must** be a valid pointer to a valid VkRenderingAreaInfo structure
- VUID-vkGetRenderingAreaGranularity-pGranularity-parameter
  `pGranularity` **must** be a valid pointer to a VkExtent2D structure

The `VkRenderingAreaInfo` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkRenderingAreaInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           viewMask;
    uint32_t           colorAttachmentCount;
    const VkFormat*    pColorAttachmentFormats;
    VkFormat           depthAttachmentFormat;
    VkFormat           stencilAttachmentFormat;
} VkRenderingAreaInfo;
```

```
// Provided by VK_KHR_maintenance5
// Equivalent to VkRenderingAreaInfo
typedef VkRenderingAreaInfo VkRenderingAreaInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `viewMask` is the viewMask used for rendering.
- `colorAttachmentCount` is the number of entries in `pColorAttachmentFormats`.
- `pColorAttachmentFormats` is a pointer to an array of VkFormat values defining the format of color attachments used in the render pass instance.
- `depthAttachmentFormat` is a VkFormat value defining the format of the depth attachment used in

the render pass instance.

- `stencilAttachmentFormat` is a VkFormat value defining the format of the stencil attachment used in the render pass instance.

Valid Usage (Implicit)

- VUID-VkRenderingAreaInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RENDERING_AREA_INFO
- VUID-VkRenderingAreaInfo-pNext-pNext
  `pNext` **must** be `NULL`

To begin resolving attachments using render pass draws, call:

```
// Provided by VK_EXT_custom_resolve with VK_KHR_dynamic_rendering or VK_VERSION_1_3
void vkCmdBeginCustomResolveEXT(
    VkCommandBuffer                             commandBuffer,
    const VkBeginCustomResolveInfoEXT*          pBeginCustomResolveInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pBeginCustomResolveInfo` is an optional struct with which to extend functionality.

Following this call, any `resolveImageView` with `resolveMode` set to VK_RESOLVE_MODE_CUSTOM_BIT_EXT will be written by outputs which would otherwise have written to the `imageView` image until the end of the current render pass instance.

Following this call, the fragment area **may** be reduced to (1,1) if a fragment density map is attached. If this occurs, reads of input attachments mapped to a color, depth, or stencil attachment return the value for the original larger fragment containing the smaller fragment. Reads of input attachments not mapped to a color, depth, or stencil attachment use the new fragment area.

NOTE Because the content of any depth/stencil resolve attachment as well as any color resolve attachment is undefined at the beginning of a resolve operation, any depth testing, stencil testing, or blending operation which sources these undefined values also has undefined result value.

During a custom resolve pass, multiple fragment invocations writing to the same (x, y, layer, view, sample) coordinate, i.e. overdraw, will produce undefined behavior.

NOTE Implementations are allowed to implement custom resolve attachment writes through other mechanisms than framebuffer attachment writes, which would normally obey rules of rasterization order.

Valid Usage

- VUID-vkCmdBeginCustomResolveEXT-commandBuffer-11517

The current render pass instance **must** have been started or resumed by vkCmdBeginRendering in this `commandBuffer`

- VUID-vkCmdBeginCustomResolveEXT-None-11518
  vkCmdBeginCustomResolveEXT **must** not have already been recorded in the current render pass instance
- VUID-vkCmdBeginCustomResolveEXT-None-11519
  The current render pass instance **must** have specified VK_RENDERING_CUSTOM_RESOLVE_BIT_EXT
- VUID-vkCmdBeginCustomResolveEXT-None-11520
  The current render pass instance **must** not have specified VK_RENDERING_SUSPENDING_BIT

Valid Usage (Implicit)

- VUID-vkCmdBeginCustomResolveEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBeginCustomResolveEXT-pBeginCustomResolveInfo-parameter
  If `pBeginCustomResolveInfo` is not `NULL`, `pBeginCustomResolveInfo` **must** be a valid pointer to a valid VkBeginCustomResolveInfoEXT structure
- VUID-vkCmdBeginCustomResolveEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBeginCustomResolveEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBeginCustomResolveEXT-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdBeginCustomResolveEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdBeginCustomResolveEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdBeginCustomResolveEXT is affected by conditional rendering

The VkBeginCustomResolveInfoEXT structure is defined as:

```
// Provided by VK_EXT_custom_resolve with VK_KHR_dynamic_rendering or VK_VERSION_1_3
typedef struct VkBeginCustomResolveInfoEXT {
    VkStructureType    sType;
    void*              pNext;
} VkBeginCustomResolveInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.

Valid Usage (Implicit)

- VUID-VkBeginCustomResolveInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_BEGIN_CUSTOM_RESOLVE_INFO_EXT

To end a render pass instance, call:

```
// Provided by VK_VERSION_1_3
void vkCmdEndRendering(
    VkCommandBuffer                             commandBuffer);
```

```
// Provided by VK_KHR_dynamic_rendering
// Equivalent to vkCmdEndRendering
void vkCmdEndRenderingKHR(
    VkCommandBuffer                             commandBuffer);
```

- commandBuffer is the command buffer in which to record the command.

If the value of pRenderingInfo->flags used to begin this render pass instance included

VK_RENDERING_SUSPENDING_BIT, then this render pass is suspended and will be resumed later in submission order.

There is no implicit ordering between separate render passes, even in the same command buffer, and even when the attachments match. Some applications rely on the continuation of rasterization order between multiple render passes with attachments defined in the same way, in order to perform non-rendering operations (such as copies or compute operations) between draw calls, but this has never been required by the specification. There is also no explicit barrier currently in the API that provides the guarantee that applications rely on without additional performance penalties.

NOTE

New applications should avoid relying on this ordering until an appropriate barrier is added to the API.

Implementations where applications are performing this splitting are encouraged to continue supporting this guarantee until a suitable barrier is added to the API.

Existing applications relying on this ordering should expect that it will continue working on platforms where it currently does. Once a new extension adds support for a new barrier, developers are encouraged to adapt their applications to use this when available.

Valid Usage

- VUID-vkCmdEndRendering-None-06161
  The current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdEndRendering-commandBuffer-06162
  The current render pass instance **must** have been begun in `commandBuffer`
- VUID-vkCmdEndRendering-None-06781
  This command **must** not be recorded when transform feedback is active
- VUID-vkCmdEndRendering-None-06999
  If `vkCmdBeginQuery*` was called within the render pass, the corresponding `vkCmdEndQuery*` **must** have been called subsequently within the same subpass

Valid Usage (Implicit)

- VUID-vkCmdEndRendering-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdEndRendering-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdEndRendering-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations

- VUID-vkCmdEndRendering-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdEndRendering-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdEndRendering-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action State

Conditional Rendering

vkCmdEndRendering is not affected by conditional rendering

To end a render pass instance, call:

```
// Provided by VK_KHR_maintenance10
void vkCmdEndRendering2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkRenderingEndInfoKHR*                pRenderingEndInfo);
```

```
// Provided by VK_EXT_fragment_density_map_offset
// Equivalent to vkCmdEndRendering2KHR
void vkCmdEndRendering2EXT(
    VkCommandBuffer                             commandBuffer,
    const VkRenderingEndInfoKHR*                pRenderingEndInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pRenderingEndInfo` is `NULL` or a pointer to a VkRenderingEndInfoKHR structure containing information about how the render pass will be ended.

If the value of pRenderingInfo->flags used to begin this render pass instance included VK_RENDERING_SUSPENDING_BIT, then this render pass is suspended and will be resumed later in submission order.

There is no implicit ordering between separate render passes, even in the same command buffer, and even when the attachments match. Some applications rely on the continuation of rasterization order between multiple render passes with attachments defined in the same way, in order to perform non-rendering operations (such as copies or compute operations) between draw calls, but this has never been required by the specification. There is also no explicit barrier currently in the API that provides the guarantee that applications rely on without additional performance penalties.

NOTE

New applications should avoid relying on this ordering until an appropriate barrier is added to the API.

Implementations where applications are performing this splitting are encouraged to continue supporting this guarantee until a suitable barrier is added to the API.

Existing applications relying on this ordering should expect that it will continue working on platforms where it currently does. Once a new extension adds support for a new barrier, developers are encouraged to adapt their applications to use this when available.

Valid Usage

- VUID-vkCmdEndRendering2KHR-None-10610
  The current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdEndRendering2KHR-commandBuffer-10611
  The current render pass instance **must** have been begun in commandBuffer
- VUID-vkCmdEndRendering2KHR-None-10612
  This command **must** not be recorded when transform feedback is active
- VUID-vkCmdEndRendering2KHR-None-10613
  If vkCmdBeginQuery* was called within the render pass, the corresponding vkCmdEndQuery* **must** have been called subsequently within the same subpass

Valid Usage (Implicit)

- VUID-vkCmdEndRendering2KHR-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdEndRendering2KHR-pRenderingEndInfo-parameter
  If pRenderingEndInfo is not NULL, pRenderingEndInfo **must** be a valid pointer to a valid VkRenderingEndInfoKHR structure
- VUID-vkCmdEndRendering2KHR-commandBuffer-recording
  commandBuffer **must** be in the recording state

- VUID-vkCmdEndRendering2KHR-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdEndRendering2KHR-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdEndRendering2KHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdEndRendering2KHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action State

Conditional Rendering

vkCmdEndRendering2KHR is not affected by conditional rendering

The VkRenderingEndInfoKHR structure is defined as:

```
// Provided by VK_KHR_maintenance10
typedef struct VkRenderingEndInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
} VkRenderingEndInfoKHR;
```

```
// Provided by VK_EXT_fragment_density_map_offset
// Equivalent to VkRenderingEndInfoKHR
typedef VkRenderingEndInfoKHR VkRenderingEndInfoEXT;
```

- sType is a VkStructureType value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.

Valid Usage (Implicit)

- VUID-VkRenderingEndInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RENDERING_END_INFO_KHR
- VUID-VkRenderingEndInfoKHR-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkRenderPassFragmentDensityMapOffsetEndInfoEXT
- VUID-VkRenderingEndInfoKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

NOTE For more complex rendering graphs, it is possible to pre-define a static *render pass* object, which as well as allowing draw commands, allows the definition of framebuffer-local dependencies between multiple subpasses. These objects have a lot of setup cost compared to vkCmdBeginRendering, but use of subpass dependencies can confer important performance benefits on some devices.

8.1. Render Pass Objects

A render pass object represents a collection of attachments, subpasses, and dependencies between the subpasses, and describes how the attachments are used over the course of the subpasses.

Render passes are represented by `VkRenderPass` handles:

WARNING This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkRenderPass)
```

An *attachment description* describes the properties of an attachment including its format, sample count, and how its contents are treated at the beginning and end of each render pass instance.

A *subpass* represents a phase of rendering that reads and writes a subset of the attachments in a render pass. Rendering commands are recorded into a particular subpass of a render pass instance.

A *subpass description* describes the subset of attachments that is involved in the execution of a subpass. Each subpass **can** read from some attachments as *input attachments*, write to some as *color attachments* or *depth/stencil attachments*, perform *shader resolve operations* to *color_attachments* or *depth/stencil_attachments*, and perform *multisample resolve operations* to *resolve attachments*. A subpass description **can** also include a set of *preserve attachments*, which are attachments that are not read or written by the subpass but whose contents **must** be preserved throughout the subpass.

A subpass *uses an attachment* if the attachment is a color, depth/stencil, resolve, depth/stencil resolve, fragment shading rate, or input attachment for that subpass (as determined by the `pColorAttachments`, `pDepthStencilAttachment`, `pResolveAttachments`, VkSubpassDescriptionDepthStencilResolve::`pDepthStencilResolveAttachment`, VkFragmentShadingRateAttachmentInfoKHR::`pFragmentShadingRateAttachment->attachment`, and `pInputAttachments` members of VkSubpassDescription, respectively). A subpass does not use an attachment if that attachment is preserved by the subpass. The *first use of an attachment* is in the lowest numbered subpass that uses that attachment. Similarly, the *last use of an attachment* is in the highest numbered subpass that uses that attachment.

The subpasses in a render pass all render to the same dimensions, and fragments for pixel (x,y,layer) in one subpass **can** only read attachment contents written by previous subpasses at that same (x,y,layer) location. For multi-pixel fragments, the pixel read from an input attachment is selected from the pixels covered by that fragment in an implementation-dependent manner. However, this selection **must** be made consistently for any fragment with the same shading rate for the lifetime of the VkDevice.

NOTE

By describing a complete set of subpasses in advance, render passes provide the implementation an opportunity to optimize the storage and transfer of attachment data between subpasses.

In practice, this means that subpasses with a simple framebuffer-space dependency **may** be merged into a single tiled rendering pass, keeping the attachment data on-chip for the duration of a render pass instance. However, it is also quite common for a render pass to only contain a single subpass.

Subpass dependencies describe execution and memory dependencies between subpasses.

A *subpass dependency chain* is a sequence of subpass dependencies in a render pass, where the source subpass of each subpass dependency (after the first) equals the destination subpass of the previous dependency.

Execution of subpasses **may** overlap or execute out of order with regards to other subpasses, unless otherwise enforced by an execution dependency. Each subpass only respects submission order for commands recorded in the same subpass, and the vkCmdBeginRenderPass and vkCmdEndRenderPass commands that delimit the render pass - commands within other subpasses are not included. This affects most other implicit ordering guarantees.

A render pass describes the structure of subpasses and attachments independent of any specific image views for the attachments. The specific image views that will be used for the attachments, and their dimensions, are specified in `VkFramebuffer` objects. Framebuffers are created with respect to a specific render pass that the framebuffer is compatible with (see Render Pass Compatibility). Collectively, a render pass and a framebuffer define the complete render target state for one or more subpasses as well as the algorithmic dependencies between the subpasses.

The various pipeline stages of the drawing commands for a given subpass **may** execute concurrently and/or out of order, both within and across drawing commands, whilst still respecting pipeline order. However for a given (x,y,layer,sample) sample location, certain per-sample operations are performed in rasterization order.

VK_ATTACHMENT_UNUSED is a constant indicating that a render pass attachment is not used.

```
#define VK_ATTACHMENT_UNUSED              (~0U)
```

8.2. Render Pass Creation

To create a render pass, call:

WARNING	This functionality is superseded by vkCreateRenderPass2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
VkResult vkCreateRenderPass(
    VkDevice                                    device,
    const VkRenderPassCreateInfo*               pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkRenderPass*                               pRenderPass);
```

- `device` is the logical device that creates the render pass.
- `pCreateInfo` is a pointer to a VkRenderPassCreateInfo structure describing the parameters of the render pass.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pRenderPass` is a pointer to a VkRenderPass handle in which the resulting render pass object is returned.

Valid Usage

- VUID-vkCreateRenderPass-device-10000
  `device` **must** support at least one queue family with the VK_QUEUE_GRAPHICS_BIT capability

Valid Usage (Implicit)

- VUID-vkCreateRenderPass-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateRenderPass-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkRenderPassCreateInfo structure
- VUID-vkCreateRenderPass-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateRenderPass-pRenderPass-parameter
  `pRenderPass` **must** be a valid pointer to a VkRenderPass handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkRenderPassCreateInfo structure is defined as:

WARNING	This functionality is superseded by VkRenderPassCreateInfo2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef struct VkRenderPassCreateInfo {
    VkStructureType                   sType;
    const void*                       pNext;
    VkRenderPassCreateFlags           flags;
    uint32_t                          attachmentCount;
    const VkAttachmentDescription*    pAttachments;
    uint32_t                          subpassCount;
    const VkSubpassDescription*       pSubpasses;
    uint32_t                          dependencyCount;
    const VkSubpassDependency*        pDependencies;
} VkRenderPassCreateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is reserved for future use.
- attachmentCount is the number of attachments used by this render pass.
- pAttachments is a pointer to an array of attachmentCount VkAttachmentDescription structures describing the attachments used by the render pass.
- subpassCount is the number of subpasses to create.
- pSubpasses is a pointer to an array of subpassCount VkSubpassDescription structures describing each subpass.
- dependencyCount is the number of memory dependencies between pairs of subpasses.
- pDependencies is a pointer to an array of dependencyCount VkSubpassDependency structures describing dependencies between pairs of subpasses.

NOTE Care should be taken to avoid a data race here; if any subpasses access attachments with overlapping memory locations, and one of those accesses is a write, a subpass dependency needs to be included between them.

Valid Usage

- VUID-VkRenderPassCreateInfo-attachment-00834
  If the `attachment` member of any element of `pInputAttachments`, `pColorAttachments`, `pResolveAttachments` or `pDepthStencilAttachment`, or any element of `pPreserveAttachments` in any element of `pSubpasses` is not VK_ATTACHMENT_UNUSED, then it **must** be less than `attachmentCount`
- VUID-VkRenderPassCreateInfo-fragmentDensityMapAttachment-06471
  If the `pNext` chain includes a VkRenderPassFragmentDensityMapCreateInfoEXT structure and the `fragmentDensityMapAttachment` member is not VK_ATTACHMENT_UNUSED, then `attachment` **must** be less than `attachmentCount`
- VUID-VkRenderPassCreateInfo-pAttachments-00836
  For any member of `pAttachments` with a `loadOp` equal to VK_ATTACHMENT_LOAD_OP_CLEAR, the first use of that attachment **must** not specify a `layout` equal to VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkRenderPassCreateInfo-pAttachments-02511
  For any member of `pAttachments` with a `stencilLoadOp` equal to VK_ATTACHMENT_LOAD_OP_CLEAR, the first use of that attachment **must** not specify a `layout` equal to VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkRenderPassCreateInfo-pAttachments-01566
  For any member of `pAttachments` with a `loadOp` equal to VK_ATTACHMENT_LOAD_OP_CLEAR, the first use of that attachment **must** not specify a `layout` equal to VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL
- VUID-VkRenderPassCreateInfo-pAttachments-01567
  For any member of `pAttachments` with a `stencilLoadOp` equal to VK_ATTACHMENT_LOAD_OP_CLEAR, the first use of that attachment **must** not specify a `layout` equal to VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkRenderPassCreateInfo-pNext-01926
  If the `pNext` chain includes a VkRenderPassInputAttachmentAspectCreateInfo structure, the `subpass` member of each element of its `pAspectReferences` member **must** be less than `subpassCount`
- VUID-VkRenderPassCreateInfo-pNext-01927
  If the `pNext` chain includes a VkRenderPassInputAttachmentAspectCreateInfo structure, the `inputAttachmentIndex` member of each element of its `pAspectReferences` member **must** be less than the value of `inputAttachmentCount` in the element of `pSubpasses` identified by its `subpass` member

- VUID-VkRenderPassCreateInfo-pNext-01963
  If the pNext chain includes a VkRenderPassInputAttachmentAspectCreateInfo structure, for any element of the pInputAttachments member of any element of pSubpasses where the attachment member is not VK_ATTACHMENT_UNUSED, the aspectMask member of the corresponding element of VkRenderPassInputAttachmentAspectCreateInfo ::pAspectReferences **must** only include aspects that are present in images of the format specified by the element of pAttachments at attachment
- VUID-VkRenderPassCreateInfo-pNext-01928
  If the pNext chain includes a VkRenderPassMultiviewCreateInfo structure, and its subpassCount member is not zero, that member **must** be equal to the value of subpassCount
- VUID-VkRenderPassCreateInfo-pNext-01929
  If the pNext chain includes a VkRenderPassMultiviewCreateInfo structure, if its dependencyCount member is not zero, it **must** be equal to dependencyCount
- VUID-VkRenderPassCreateInfo-pNext-01930
  If the pNext chain includes a VkRenderPassMultiviewCreateInfo structure, for each non-zero element of pViewOffsets, the srcSubpass and dstSubpass members of pDependencies at the same index **must** not be equal
- VUID-VkRenderPassCreateInfo-pNext-02512
  If the pNext chain includes a VkRenderPassMultiviewCreateInfo structure, for each element of pDependencies with a dependencyFlags member that does not include VK_DEPENDENCY_VIEW_LOCAL_BIT, the corresponding element of the pViewOffsets member of that VkRenderPassMultiviewCreateInfo instance **must** be 0
- VUID-VkRenderPassCreateInfo-pNext-02513
  If the pNext chain includes a VkRenderPassMultiviewCreateInfo structure, elements of its pViewMasks member **must** either all be 0, or all not be 0
- VUID-VkRenderPassCreateInfo-pNext-02514
  If the pNext chain includes a VkRenderPassMultiviewCreateInfo structure, and each element of its pViewMasks member is 0, the dependencyFlags member of each element of pDependencies **must** not include VK_DEPENDENCY_VIEW_LOCAL_BIT
- VUID-VkRenderPassCreateInfo-pNext-02515
  If the pNext chain includes a VkRenderPassMultiviewCreateInfo structure, and each element of its pViewMasks member is 0, its correlationMaskCount member **must** be 0
- VUID-VkRenderPassCreateInfo-pDependencies-00837
  For each element of pDependencies, if the srcSubpass is not VK_SUBPASS_EXTERNAL, all stage flags included in the srcStageMask member of that dependency **must** be VK_PIPELINE_STAGE_ALL_COMMANDS_BIT or a pipeline stage supported by the pipeline identified by the pipelineBindPoint member of the source subpass
- VUID-VkRenderPassCreateInfo-pDependencies-00838
  For each element of pDependencies, if the dstSubpass is not VK_SUBPASS_EXTERNAL, all stage flags included in the dstStageMask member of that dependency **must** be VK_PIPELINE_STAGE_ALL_COMMANDS_BIT or a pipeline stage supported by the pipeline identified by the pipelineBindPoint member of the destination subpass
- VUID-VkRenderPassCreateInfo-pDependencies-06866
  For each element of pDependencies, if its srcSubpass is not VK_SUBPASS_EXTERNAL, it **must**

be less than subpassCount

- VUID-VkRenderPassCreateInfo-pDependencies-06867
  For each element of pDependencies, if its dstSubpass is not VK_SUBPASS_EXTERNAL, it **must** be less than subpassCount
- VUID-VkRenderPassCreateInfo-None-10915
  If any subpass preserves an attachment, there **must** be a subpass dependency from a prior subpass which uses or preserves that attachment

Valid Usage (Implicit)

- VUID-VkRenderPassCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO
- VUID-VkRenderPassCreateInfo-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkRenderPassFragmentDensityMapCreateInfoEXT, VkRenderPassInputAttachmentAspectCreateInfo, or VkRenderPassMultiviewCreateInfo
- VUID-VkRenderPassCreateInfo-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkRenderPassCreateInfo-flags-zerobitmask
  flags **must** be 0
- VUID-VkRenderPassCreateInfo-pAttachments-parameter
  If attachmentCount is not 0, pAttachments **must** be a valid pointer to an array of attachmentCount valid VkAttachmentDescription structures
- VUID-VkRenderPassCreateInfo-pSubpasses-parameter
  pSubpasses **must** be a valid pointer to an array of subpassCount valid VkSubpassDescription structures
- VUID-VkRenderPassCreateInfo-pDependencies-parameter
  If dependencyCount is not 0, pDependencies **must** be a valid pointer to an array of dependencyCount valid VkSubpassDependency structures
- VUID-VkRenderPassCreateInfo-subpassCount-arraylength
  subpassCount **must** be greater than 0

Bits which **can** be set in VkRenderPassCreateInfo::flags, describing additional properties of the render pass, are:

WARNING	This functionality is superseded by Vulkan Version 1.2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef enum VkRenderPassCreateFlagBits {
} VkRenderPassCreateFlagBits;
```

NOTE All bits for this type are defined by extensions, and none of those extensions are enabled in this build of the specification.

WARNING This functionality is superseded by Vulkan Version 1.2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkRenderPassCreateFlags;
```

VkRenderPassCreateFlags is a bitmask type for setting a mask of zero or more VkRenderPassCreateFlagBits.

If the VkRenderPassCreateInfo::pNext chain includes a VkRenderPassMultiviewCreateInfo structure, then that structure includes an array of view masks, view offsets, and correlation masks for the render pass.

The VkRenderPassMultiviewCreateInfo structure is defined as:

WARNING This functionality is superseded by Vulkan Version 1.2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_1
typedef struct VkRenderPassMultiviewCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           subpassCount;
    const uint32_t*    pViewMasks;
    uint32_t           dependencyCount;
    const int32_t*     pViewOffsets;
    uint32_t           correlationMaskCount;
    const uint32_t*    pCorrelationMasks;
} VkRenderPassMultiviewCreateInfo;
```

```
// Provided by VK_KHR_multiview
// Equivalent to VkRenderPassMultiviewCreateInfo
typedef VkRenderPassMultiviewCreateInfo VkRenderPassMultiviewCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- subpassCount is zero or the number of subpasses in the render pass.
- pViewMasks is a pointer to an array of subpassCount view masks, where each mask is a bitfield of view indices describing which views rendering is broadcast to in each subpass, when multiview is enabled. If subpassCount is zero, each view mask is treated as zero.
- dependencyCount is zero or the number of dependencies in the render pass.

- `pViewOffsets` is a pointer to an array of `dependencyCount` view offsets, one for each dependency. If `dependencyCount` is zero, each dependency's view offset is treated as zero. Each view offset controls which views in the source subpass the views in the destination subpass depend on.
- `correlationMaskCount` is zero or the number of correlation masks.
- `pCorrelationMasks` is a pointer to an array of `correlationMaskCount` view masks indicating sets of views that **may** be more efficient to render concurrently.

When a subpass uses a non-zero view mask, *multiview* functionality is considered to be enabled. Multiview is all-or-nothing for a render pass - that is, either all subpasses **must** have a non-zero view mask (though some subpasses **may** have only one view) or all **must** be zero. Multiview causes all drawing and clear commands in the subpass to behave as if they were broadcast to each view, where a view is represented by one layer of the framebuffer attachments. All draws and clears are broadcast to each *view index* whose bit is set in the view mask. The view index is provided in the `ViewIndex` shader input variable, and color, depth/stencil, and input attachments all read/write the layer of the framebuffer corresponding to the view index.

If the view mask is zero for all subpasses, multiview is considered to be disabled and all drawing commands execute normally, without this additional broadcasting.

Some implementations **may** not support multiview in conjunction with mesh shaders, geometry shaders or tessellation shaders.

When multiview is enabled, the VK_DEPENDENCY_VIEW_LOCAL_BIT bit in a dependency **can** be used to express a view-local dependency, meaning that each view in the destination subpass depends on a single view in the source subpass. Unlike pipeline barriers, a subpass dependency **can** potentially have a different view mask in the source subpass and the destination subpass. If the dependency is view-local, then each view (dstView) in the destination subpass depends on the view dstView + `pViewOffsets`[dependency] in the source subpass. If there is not such a view in the source subpass, then this dependency does not affect that view in the destination subpass. If the dependency is not view-local, then all views in the destination subpass depend on all views in the source subpass, and the view offset is ignored. A non-zero view offset is not allowed in a self-dependency.

The elements of `pCorrelationMasks` are a set of masks of views indicating that views in the same mask **may** exhibit spatial coherency between the views, making it more efficient to render them concurrently. Correlation masks **must** not have a functional effect on the results of the multiview rendering.

When multiview is enabled, at the beginning of each subpass all non-render pass state is undefined. In particular, each time vkCmdBeginRenderPass or vkCmdNextSubpass is called the graphics pipeline **must** be bound, any relevant descriptor sets or vertex/index buffers **must** be bound, and any relevant dynamic state or push constants **must** be set before they are used.

Valid Usage

- VUID-VkRenderPassMultiviewCreateInfo-pCorrelationMasks-00841
  Each view index **must** not be set in more than one element of `pCorrelationMasks`
- VUID-VkRenderPassMultiviewCreateInfo-multiview-06555

If the multiview feature is not enabled, each element of pViewMasks **must** be 0

- VUID-VkRenderPassMultiviewCreateInfo-pViewMasks-06697
  The index of the most significant bit in each element of pViewMasks **must** be less than maxMultiviewViewCount

Valid Usage (Implicit)

- VUID-VkRenderPassMultiviewCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_RENDER_PASS_MULTIVIEW_CREATE_INFO
- VUID-VkRenderPassMultiviewCreateInfo-pViewMasks-parameter
  If subpassCount is not 0, pViewMasks **must** be a valid pointer to an array of subpassCount uint32_t values
- VUID-VkRenderPassMultiviewCreateInfo-pViewOffsets-parameter
  If dependencyCount is not 0, pViewOffsets **must** be a valid pointer to an array of dependencyCount int32_t values
- VUID-VkRenderPassMultiviewCreateInfo-pCorrelationMasks-parameter
  If correlationMaskCount is not 0, pCorrelationMasks **must** be a valid pointer to an array of correlationMaskCount uint32_t values

Structure Chaining

Extends the structure

- VkRenderPassCreateInfo

If the VkRenderPassCreateInfo::pNext chain includes a VkRenderPassFragmentDensityMapCreateInfoEXT structure, then that structure includes a fragment density map attachment for the render pass.

The VkRenderPassFragmentDensityMapCreateInfoEXT structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4 and VK_KHR_dynamic_rendering_local_read. See Legacy Functionality for more information.

```
// Provided by VK_EXT_fragment_density_map
typedef struct VkRenderPassFragmentDensityMapCreateInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkAttachmentReference    fragmentDensityMapAttachment;
} VkRenderPassFragmentDensityMapCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.

- `fragmentDensityMapAttachment` is the fragment density map to use for the render pass.

The fragment density map is read at an implementation-dependent time with the following constraints determined by the attachment's image view `flags`:

- VK_IMAGE_VIEW_CREATE_FRAGMENT_DENSITY_MAP_DYNAMIC_BIT_EXT specifies that the fragment density map will be read by the device during VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VK_IMAGE_VIEW_CREATE_FRAGMENT_DENSITY_MAP_DEFERRED_BIT_EXT specifies that the fragment density map will be read by the host during vkEndCommandBuffer of the primary command buffer that the render pass is recorded into
- Otherwise the fragment density map will be read by the host during vkCmdBeginRenderPass

The fragment density map **may** additionally be read by the device during VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT for any mode.

If this structure is not present, it is as if `fragmentDensityMapAttachment` was given as VK_ATTACHMENT_UNUSED.

Valid Usage

- VUID-VkRenderPassFragmentDensityMapCreateInfoEXT-fragmentDensityMapAttachment-02548
  If `fragmentDensityMapAttachment` is not VK_ATTACHMENT_UNUSED, `fragmentDensityMapAttachment` **must** not be an element of `VkSubpassDescription::pInputAttachments`, `VkSubpassDescription::pColorAttachments`, `VkSubpassDescription::pResolveAttachments`, `VkSubpassDescription::pDepthStencilAttachment`, or `VkSubpassDescription::pPreserveAttachments` for any subpass
- VUID-VkRenderPassFragmentDensityMapCreateInfoEXT-fragmentDensityMapAttachment-02549
  If `fragmentDensityMapAttachment` is not VK_ATTACHMENT_UNUSED, `layout` **must** be equal to VK_IMAGE_LAYOUT_FRAGMENT_DENSITY_MAP_OPTIMAL_EXT, or VK_IMAGE_LAYOUT_GENERAL
- VUID-VkRenderPassFragmentDensityMapCreateInfoEXT-fragmentDensityMapAttachment-02550
  If `fragmentDensityMapAttachment` is not VK_ATTACHMENT_UNUSED, `fragmentDensityMapAttachment` **must** reference an attachment with a `loadOp` equal to VK_ATTACHMENT_LOAD_OP_LOAD or VK_ATTACHMENT_LOAD_OP_DONT_CARE
- VUID-VkRenderPassFragmentDensityMapCreateInfoEXT-fragmentDensityMapAttachment-02551
  If `fragmentDensityMapAttachment` is not VK_ATTACHMENT_UNUSED, `fragmentDensityMapAttachment` **must** reference an attachment with a `storeOp` equal to VK_ATTACHMENT_STORE_OP_DONT_CARE

Valid Usage (Implicit)

- VUID-VkRenderPassFragmentDensityMapCreateInfoEXT-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_RENDER_PASS_FRAGMENT_DENSITY_MAP_CREATE_INFO_EXT`
- VUID-VkRenderPassFragmentDensityMapCreateInfoEXT-fragmentDensityMapAttachment-parameter
  `fragmentDensityMapAttachment` **must** be a valid VkAttachmentReference structure

Structure Chaining

Extends the structures

- VkRenderPassCreateInfo
- VkRenderPassCreateInfo2

The `VkAttachmentDescription` structure is defined as:

WARNING This functionality is superseded by VkAttachmentDescription2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef struct VkAttachmentDescription {
    VkAttachmentDescriptionFlags    flags;
    VkFormat                        format;
    VkSampleCountFlagBits           samples;
    VkAttachmentLoadOp              loadOp;
    VkAttachmentStoreOp             storeOp;
    VkAttachmentLoadOp              stencilLoadOp;
    VkAttachmentStoreOp             stencilStoreOp;
    VkImageLayout                   initialLayout;
    VkImageLayout                   finalLayout;
} VkAttachmentDescription;
```

- `flags` is a bitmask of VkAttachmentDescriptionFlagBits specifying additional properties of the attachment.
- `format` is a VkFormat value specifying the format of the image view that will be used for the attachment.
- `samples` is a VkSampleCountFlagBits value specifying the number of samples of the image.
- `loadOp` is a VkAttachmentLoadOp value specifying how the contents of color and depth components of the attachment are treated at the beginning of the subpass where it is first used.
- `storeOp` is a VkAttachmentStoreOp value specifying how the contents of color and depth components of the attachment are treated at the end of the subpass where it is last used.

- `stencilLoadOp` is a VkAttachmentLoadOp value specifying how the contents of stencil components of the attachment are treated at the beginning of the subpass where it is first used.
- `stencilStoreOp` is a VkAttachmentStoreOp value specifying how the contents of stencil components of the attachment are treated at the end of the last subpass where it is used.
- `initialLayout` is the layout the attachment image subresource will be in when a render pass instance begins.
- `finalLayout` is the layout the attachment image subresource will be transitioned to when a render pass instance ends.

If the attachment uses a color format, then `loadOp` and `storeOp` are used, and `stencilLoadOp` and `stencilStoreOp` are ignored. If the format has depth and/or stencil components, `loadOp` and `storeOp` apply only to the depth data, while `stencilLoadOp` and `stencilStoreOp` define how the stencil data is handled. `loadOp` and `stencilLoadOp` define the load operations for the attachment. `storeOp` and `stencilStoreOp` define the store operations for the attachment. If an attachment is not used by any subpass, `loadOp`, `storeOp`, `stencilStoreOp`, and `stencilLoadOp` will be ignored for that attachment, and no load or store ops will be performed. However, any transition specified by `initialLayout` and `finalLayout` will still be executed.

If `flags` includes VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT, then the attachment is treated as if it shares physical memory with another attachment in the same render pass. This information limits the ability of the implementation to reorder certain operations (like layout transitions and the `loadOp`) such that it is not improperly reordered against other uses of the same physical memory via a different attachment. This is described in more detail below.

If a render pass uses multiple attachments that alias the same device memory, those attachments **must** each include the VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT bit in their attachment description flags. Attachments aliasing the same memory occurs in multiple ways:

- Multiple attachments being assigned the same image view as part of framebuffer creation.
- Attachments using distinct image views that correspond to the same image subresource of an image.
- Attachments using views of distinct image subresources which are bound to overlapping memory ranges.

NOTE

Render passes **must** include subpass dependencies (either directly or via a subpass dependency chain) between any two subpasses that operate on the same attachment or aliasing attachments and those subpass dependencies **must** include execution and memory dependencies separating uses of the aliases, if at least one of those subpasses writes to one of the aliases. These dependencies **must** not include the VK_DEPENDENCY_BY_REGION_BIT if the aliases are views of distinct image subresources which overlap in memory.

Multiple attachments that alias the same memory **must** not be used in a single subpass. A given attachment index **must** not be used multiple times in a single subpass, with one exception: two subpass attachments **can** use the same attachment index if at least one use is as an input attachment and neither use is as a resolve or preserve attachment. In other words, the same view **can** be used simultaneously as an input and color or depth/stencil attachment, but **must** not be

used as multiple color or depth/stencil attachments nor as resolve or preserve attachments.

If a set of attachments alias each other, then all except the first to be used in the render pass **must** use an `initialLayout` of VK_IMAGE_LAYOUT_UNDEFINED, since the earlier uses of the other aliases make their contents undefined. Once an alias has been used and a different alias has been used after it, the first alias **must** not be used in any later subpasses. However, an application **can** assign the same image view to multiple aliasing attachment indices, which allows that image view to be used multiple times even if other aliases are used in between.

NOTE

Once an attachment needs the VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT bit, there **should** be no additional cost of introducing additional aliases, and using these additional aliases **may** allow more efficient clearing of the attachments on multiple uses via VK_ATTACHMENT_LOAD_OP_CLEAR.

Valid Usage

- VUID-VkAttachmentDescription-format-06699
  If `format` includes a color or depth component and `loadOp` is VK_ATTACHMENT_LOAD_OP_LOAD, then `initialLayout` **must** not be VK_IMAGE_LAYOUT_UNDEFINED
- VUID-VkAttachmentDescription-finalLayout-00843
  `finalLayout` **must** not be VK_IMAGE_LAYOUT_UNDEFINED or VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT or VK_IMAGE_LAYOUT_PREINITIALIZED
- VUID-VkAttachmentDescription-format-03280
  If `format` is a color format, `initialLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription-format-03281
  If `format` is a depth/stencil format, `initialLayout` **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL
- VUID-VkAttachmentDescription-format-03282
  If `format` is a color format, `finalLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription-format-03283
  If `format` is a depth/stencil format, `finalLayout` **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL
- VUID-VkAttachmentDescription-format-06487
  If `format` is a color format, `initialLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL
- VUID-VkAttachmentDescription-format-06488
  If `format` is a color format, `finalLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL

- VUID-VkAttachmentDescription-separateDepthStencilLayouts-03284
  If the `separateDepthStencilLayouts` feature is not enabled, `initialLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription-separateDepthStencilLayouts-03285
  If the `separateDepthStencilLayouts` feature is not enabled, `finalLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription-format-03286
  If `format` is a color format, `initialLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription-format-03287
  If `format` is a color format, `finalLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription-format-06906
  If `format` is a depth/stencil format which includes both depth and stencil components, `initialLayout` **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription-format-06907
  If `format` is a depth/stencil format which includes both depth and stencil components, `finalLayout` **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription-format-03290
  If `format` is a depth/stencil format which includes only the depth component, `initialLayout` **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription-format-03291
  If `format` is a depth/stencil format which includes only the depth component, `finalLayout` **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription-synchronization2-06908
  If the `synchronization2` feature is not enabled, `initialLayout` **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR or VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR

- VUID-VkAttachmentDescription-synchronization2-06909
  If the `synchronization2` feature is not enabled, `finalLayout` **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR or VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VUID-VkAttachmentDescription-attachmentFeedbackLoopLayout-07309
  If the `attachmentFeedbackLoopLayout` feature is not enabled, `initialLayout` **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT
- VUID-VkAttachmentDescription-attachmentFeedbackLoopLayout-07310
  If the `attachmentFeedbackLoopLayout` feature is not enabled, `finalLayout` **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT
- VUID-VkAttachmentDescription-dynamicRenderingLocalRead-09544
  If the `dynamicRenderingLocalRead` feature is not enabled, `initialLayout` **must** not be VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ
- VUID-VkAttachmentDescription-dynamicRenderingLocalRead-09545
  If the `dynamicRenderingLocalRead` feature is not enabled, `finalLayout` **must** not be VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ
- VUID-VkAttachmentDescription-flags-11773
  If `flags` includes VK_ATTACHMENT_DESCRIPTION_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR, `flags` **must** not include VK_ATTACHMENT_DESCRIPTION_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR
- VUID-VkAttachmentDescription-flags-11774
  If `flags` includes VK_ATTACHMENT_DESCRIPTION_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR or VK_ATTACHMENT_DESCRIPTION_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR, `resolveSrgbFormatSupportsTransferFunctionControl` **must** be VK_TRUE
- VUID-VkAttachmentDescription-flags-11775
  If `flags` includes VK_ATTACHMENT_DESCRIPTION_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR or VK_ATTACHMENT_DESCRIPTION_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR, `maintenance10` **must** be enabled
- VUID-VkAttachmentDescription-flags-11776
  If `flags` includes VK_ATTACHMENT_DESCRIPTION_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR or VK_ATTACHMENT_DESCRIPTION_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR, `format` **must** use sRGB encoding
- VUID-VkAttachmentDescription-flags-11777
  If `flags` includes VK_ATTACHMENT_DESCRIPTION_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR or VK_ATTACHMENT_DESCRIPTION_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR, `samples` **must** be VK_SAMPLE_COUNT_1_BIT
- VUID-VkAttachmentDescription-format-06698
  `format` **must** not be VK_FORMAT_UNDEFINED

- VUID-VkAttachmentDescription-format-06700
  If `format` includes a stencil component and `stencilLoadOp` is VK_ATTACHMENT_LOAD_OP_LOAD, then `initialLayout` **must** not be VK_IMAGE_LAYOUT_UNDEFINED
- VUID-VkAttachmentDescription-format-03292
  If `format` is a depth/stencil format which includes only the stencil component, `initialLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription-format-03293
  If `format` is a depth/stencil format which includes only the stencil component, `finalLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription-format-06242
  If `format` is a depth/stencil format which includes both depth and stencil components, `initialLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription-format-06243
  If `format` is a depth/stencil format which includes both depth and stencil components, `finalLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL

Valid Usage (Implicit)

- VUID-VkAttachmentDescription-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkAttachmentDescriptionFlagBits values
- VUID-VkAttachmentDescription-format-parameter
  `format` **must** be a valid VkFormat value
- VUID-VkAttachmentDescription-samples-parameter
  `samples` **must** be a valid VkSampleCountFlagBits value
- VUID-VkAttachmentDescription-loadOp-parameter
  `loadOp` **must** be a valid VkAttachmentLoadOp value
- VUID-VkAttachmentDescription-storeOp-parameter
  `storeOp` **must** be a valid VkAttachmentStoreOp value
- VUID-VkAttachmentDescription-stencilLoadOp-parameter
  `stencilLoadOp` **must** be a valid VkAttachmentLoadOp value
- VUID-VkAttachmentDescription-stencilStoreOp-parameter
  `stencilStoreOp` **must** be a valid VkAttachmentStoreOp value
- VUID-VkAttachmentDescription-initialLayout-parameter
  `initialLayout` **must** be a valid VkImageLayout value
- VUID-VkAttachmentDescription-finalLayout-parameter
  `finalLayout` **must** be a valid VkImageLayout value

Bits which **can** be set in VkAttachmentDescription::flags, describing additional properties of the attachment, are:

WARNING	This functionality is superseded by Vulkan Version 1.2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef enum VkAttachmentDescriptionFlagBits {
    VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT = 0x00000001,
  // Provided by VK_KHR_maintenance10
    VK_ATTACHMENT_DESCRIPTION_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR = 0x00000002,
  // Provided by VK_KHR_maintenance10
    VK_ATTACHMENT_DESCRIPTION_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR = 0x00000004,
} VkAttachmentDescriptionFlagBits;
```

- VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT specifies that the attachment aliases the same device memory as other attachments.
- VK_ATTACHMENT_DESCRIPTION_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR specifies that resolve operations happening to an sRGB encoded attachment **must** not convert samples from nonlinear to linear before averaging.
- VK_ATTACHMENT_DESCRIPTION_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR specifies that resolve operations happening to an sRGB encoded attachment **must** convert samples from nonlinear to linear before averaging.

WARNING	This functionality is superseded by Vulkan Version 1.2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkAttachmentDescriptionFlags;
```

VkAttachmentDescriptionFlags is a bitmask type for setting a mask of zero or more VkAttachmentDescriptionFlagBits.

The VkRenderPassInputAttachmentAspectCreateInfo structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_1
typedef struct VkRenderPassInputAttachmentAspectCreateInfo {
    VkStructureType                           sType;
    const void*                               pNext;
    uint32_t                                  aspectReferenceCount;
    const VkInputAttachmentAspectReference*   pAspectReferences;
} VkRenderPassInputAttachmentAspectCreateInfo;
```

```
// Provided by VK_KHR_maintenance2
// Equivalent to VkRenderPassInputAttachmentAspectCreateInfo
typedef VkRenderPassInputAttachmentAspectCreateInfo
VkRenderPassInputAttachmentAspectCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `aspectReferenceCount` is the number of elements in the `pAspectReferences` array.
- `pAspectReferences` is a pointer to an array of `aspectReferenceCount` VkInputAttachmentAspectReference structures containing a mask describing which aspect(s) **can** be accessed for a given input attachment within a given subpass.

To specify which aspects of an input attachment **can** be read, add a VkRenderPassInputAttachmentAspectCreateInfo structure to the `pNext` chain of the VkRenderPassCreateInfo structure:

An application **can** access any aspect of an input attachment that does not have a specified aspect mask in the `pAspectReferences` array. Otherwise, an application **must** not access aspect(s) of an input attachment other than those in its specified aspect mask.

Valid Usage (Implicit)

- VUID-VkRenderPassInputAttachmentAspectCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RENDER_PASS_INPUT_ATTACHMENT_ASPECT_CREATE_INFO
- VUID-VkRenderPassInputAttachmentAspectCreateInfo-pAspectReferences-parameter
  `pAspectReferences` **must** be a valid pointer to an array of `aspectReferenceCount` valid VkInputAttachmentAspectReference structures
- VUID-VkRenderPassInputAttachmentAspectCreateInfo-aspectReferenceCount-arraylength
  `aspectReferenceCount` **must** be greater than `0`

Structure Chaining

Extends the structure

- VkRenderPassCreateInfo

The VkInputAttachmentAspectReference structure is defined as:

WARNING This functionality is superseded by Vulkan Version 1.2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_1
typedef struct VkInputAttachmentAspectReference {
    uint32_t              subpass;
    uint32_t              inputAttachmentIndex;
    VkImageAspectFlags    aspectMask;
} VkInputAttachmentAspectReference;
```

```
// Provided by VK_KHR_maintenance2
// Equivalent to VkInputAttachmentAspectReference
typedef VkInputAttachmentAspectReference VkInputAttachmentAspectReferenceKHR;
```

- subpass is an index into the pSubpasses array of the parent VkRenderPassCreateInfo structure.
- inputAttachmentIndex is an index into the pInputAttachments of the specified subpass.
- aspectMask is a mask of which aspect(s) **can** be accessed within the specified subpass.

This structure specifies an aspect mask for a specific input attachment of a specific subpass in the render pass.

subpass and inputAttachmentIndex index into the render pass as:

```
pCreateInfo->pSubpasses[subpass].pInputAttachments[inputAttachmentIndex]
```

Valid Usage

- VUID-VkInputAttachmentAspectReference-aspectMask-01964
  aspectMask **must** not include VK_IMAGE_ASPECT_METADATA_BIT
- VUID-VkInputAttachmentAspectReference-aspectMask-02250
  aspectMask **must** not include VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT for any index *i*

Valid Usage (Implicit)

- VUID-VkInputAttachmentAspectReference-aspectMask-parameter
  aspectMask **must** be a valid combination of VkImageAspectFlagBits values

The VkSubpassDescription structure is defined as:

WARNING This functionality is superseded by VkSubpassDescription2. See Legacy

> Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef struct VkSubpassDescription {
    VkSubpassDescriptionFlags       flags;
    VkPipelineBindPoint             pipelineBindPoint;
    uint32_t                        inputAttachmentCount;
    const VkAttachmentReference*    pInputAttachments;
    uint32_t                        colorAttachmentCount;
    const VkAttachmentReference*    pColorAttachments;
    const VkAttachmentReference*    pResolveAttachments;
    const VkAttachmentReference*    pDepthStencilAttachment;
    uint32_t                        preserveAttachmentCount;
    const uint32_t*                 pPreserveAttachments;
} VkSubpassDescription;
```

- `flags` is a bitmask of VkSubpassDescriptionFlagBits specifying usage of the subpass.
- `pipelineBindPoint` is a VkPipelineBindPoint value specifying the pipeline type supported for this subpass.
- `inputAttachmentCount` is the number of input attachments.
- `pInputAttachments` is a pointer to an array of VkAttachmentReference structures defining the input attachments for this subpass and their layouts.
- `colorAttachmentCount` is the number of color attachments.
- `pColorAttachments` is a pointer to an array of `colorAttachmentCount` VkAttachmentReference structures defining the color attachments for this subpass and their layouts.
- `pResolveAttachments` is `NULL` or a pointer to an array of `colorAttachmentCount` VkAttachmentReference structures defining the resolve attachments for this subpass and their layouts.
- `pDepthStencilAttachment` is a pointer to a VkAttachmentReference structure specifying the depth/stencil attachment for this subpass and its layout.
- `preserveAttachmentCount` is the number of preserved attachments.
- `pPreserveAttachments` is a pointer to an array of `preserveAttachmentCount` render pass attachment indices identifying attachments that are not used by this subpass, but whose contents **must** be preserved throughout the subpass.

Each element of the `pInputAttachments` array corresponds to an input attachment index in a fragment shader, i.e. if a shader declares an image variable decorated with a `InputAttachmentIndex` value of **X**, then it uses the attachment provided in `pInputAttachments`[**X**]. Input attachments **must** also be bound to the pipeline in a descriptor set. If the `attachment` member of any element of `pInputAttachments` is VK_ATTACHMENT_UNUSED, the application **must** not read from the corresponding input attachment index. Fragment shaders **can** use subpass input variables to access the contents of an input attachment at the fragment's (x_f, y_f) framebuffer coordinates and layer.

Each element of the `pColorAttachments` array corresponds to an output location in the shader, i.e. if

the shader declares an output variable decorated with a `Location` value of **X**, then it uses the attachment provided in `pColorAttachments`[**X**]. If the `attachment` member of any element of `pColorAttachments` is VK_ATTACHMENT_UNUSED, or if Color Write Enable has been disabled for the corresponding attachment index, then writes to the corresponding location by a fragment shader are discarded.

If `flags` does not include VK_SUBPASS_DESCRIPTION_CUSTOM_RESOLVE_BIT_EXT, and if `pResolveAttachments` is not `NULL`, each of its elements corresponds to a color attachment (the element in `pColorAttachments` at the same index), and a multisample resolve operation is defined for each attachment unless the resolve attachment index is VK_ATTACHMENT_UNUSED.

Similarly, if `flags` does not include VK_SUBPASS_DESCRIPTION_CUSTOM_RESOLVE_BIT_EXT, and VkSubpassDescriptionDepthStencilResolve::`pDepthStencilResolveAttachment` is not `NULL` and does not have the value VK_ATTACHMENT_UNUSED, it corresponds to the depth/stencil attachment in `pDepthStencilAttachment`, and multisample resolve operations for depth and stencil are defined by VkSubpassDescriptionDepthStencilResolve::`depthResolveMode` and VkSubpassDescriptionDepthStencilResolve::`stencilResolveMode`, respectively. If VkSubpassDescriptionDepthStencilResolve::`depthResolveMode` is VK_RESOLVE_MODE_NONE or the `pDepthStencilResolveAttachment` does not have a depth aspect, no resolve operation is performed for the depth attachment. If VkSubpassDescriptionDepthStencilResolve::`stencilResolveMode` is VK_RESOLVE_MODE_NONE or the `pDepthStencilResolveAttachment` does not have a stencil aspect, no resolve operation is performed for the stencil attachment.

If the image subresource range referenced by the depth/stencil attachment is created with VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT, then the multisample resolve operation uses the sample locations state specified in the `sampleLocationsInfo` member of the element of the `VkRenderPassSampleLocationsBeginInfoEXT`::`pPostSubpassSampleLocations` for the subpass.

If `pDepthStencilAttachment` is `NULL`, or if its attachment index is VK_ATTACHMENT_UNUSED, it indicates that no depth/stencil attachment will be used in the subpass.

Following this call, if the `customResolve` feature is enabled the fragment area **may** be reduced to (1,1) if a fragment density map is attached to the render pass. If this occurs, reads of input attachments **may** return the value for the original larger fragment containing the smaller fragment or use the new fragment area.

The contents of an attachment within the render area become undefined at the start of a subpass **S** if all of the following conditions are true:

- The attachment is used as a color, depth/stencil, or resolve attachment in any subpass in the render pass.
- There is a subpass $\mathbf{S}_1$ that uses or preserves the attachment, and a subpass dependency from $\mathbf{S}_1$ to **S**.
- The attachment is not used or preserved in subpass **S**.

In addition, the contents of an attachment within the render area become undefined at the start of a subpass **S** if all of the following conditions are true:

- VK_SUBPASS_DESCRIPTION_CUSTOM_RESOLVE_BIT_EXT is set.
- The attachment is used as a color or depth/stencil attachment in the subpass.

For color attachments, this operation will be performed in the VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT pipeline stage, with any image accesses performed via VK_ACCESS_INPUT_ATTACHMENT_READ_BIT, VK_ACCESS_COLOR_ATTACHMENT_READ_BIT, and VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT. For depth/stencil attachments, this operation **may** be performed in either the VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT or VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT pipeline stage, with any image accesses performed via VK_ACCESS_INPUT_ATTACHMENT_READ_BIT, VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT, and VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT.

Once the contents of an attachment become undefined in subpass **S**, they remain undefined for subpasses in subpass dependency chains starting with subpass **S** until they are written again. However, they remain valid for subpasses in other subpass dependency chains starting with subpass $\mathbf{S}_1$ if all subpasses in each chain use or preserve the attachment.

NOTE If a subpass has multiple dependency chains where some of the chains preserve the attachment, and others do not, the contents of the attachment are undefined for that subpass.

Valid Usage

- VUID-VkSubpassDescription-attachment-06912
  If the `attachment` member of an element of `pInputAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL
- VUID-VkSubpassDescription-attachment-06913
  If the `attachment` member of an element of `pColorAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL
- VUID-VkSubpassDescription-attachment-06914
  If the `attachment` member of an element of `pResolveAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL
- VUID-VkSubpassDescription-attachment-06915
  If the `attachment` member of `pDepthStencilAttachment` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL
- VUID-VkSubpassDescription-attachment-06916
  If the `attachment` member of an element of `pColorAttachments` is not

VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL

- VUID-VkSubpassDescription-attachment-06917
  If the `attachment` member of an element of `pResolveAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL
- VUID-VkSubpassDescription-attachment-06918
  If the `attachment` member of an element of `pInputAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL
- VUID-VkSubpassDescription-attachment-06919
  If the `attachment` member of an element of `pColorAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkSubpassDescription-attachment-06920
  If the `attachment` member of an element of `pResolveAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkSubpassDescription-attachment-06921
  If the `attachment` member of an element of `pInputAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR
- VUID-VkSubpassDescription-attachment-06922
  If the `attachment` member of an element of `pColorAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VUID-VkSubpassDescription-attachment-06923
  If the `attachment` member of an element of `pResolveAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VUID-VkSubpassDescription-inputAttachmentCount-12293
  `inputAttachmentCount` **must** be less than or equal to `maxPerStageDescriptorInputAttachments`
- VUID-VkSubpassDescription-colorAttachmentCount-00845
  `colorAttachmentCount` **must** be less than or equal to `maxColorAttachments`
- VUID-VkSubpassDescription-loadOp-00846

If the first use of an attachment in this render pass is as an input attachment, and the attachment is not also used as a color or depth/stencil attachment in the same subpass, then `loadOp` **must** not be VK_ATTACHMENT_LOAD_OP_CLEAR

- VUID-VkSubpassDescription-pipelineBindPoint-04952
  `pipelineBindPoint` **must** be VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-VkSubpassDescription-pResolveAttachments-00847
  If `pResolveAttachments` is not `NULL`, for each resolve attachment that is not VK_ATTACHMENT_UNUSED, the corresponding color attachment **must** not be VK_ATTACHMENT_UNUSED
- VUID-VkSubpassDescription-pResolveAttachments-00848
  If `pResolveAttachments` is not `NULL`, for each resolve attachment that is not VK_ATTACHMENT_UNUSED, the corresponding color attachment **must** not have a sample count of VK_SAMPLE_COUNT_1_BIT
- VUID-VkSubpassDescription-pResolveAttachments-00849
  If `pResolveAttachments` is not `NULL`, each resolve attachment that is not VK_ATTACHMENT_UNUSED **must** have a sample count of VK_SAMPLE_COUNT_1_BIT
- VUID-VkSubpassDescription-pResolveAttachments-00850
  If `pResolveAttachments` is not `NULL`, each resolve attachment that is not VK_ATTACHMENT_UNUSED **must** have the same VkFormat as its corresponding color attachment
- VUID-VkSubpassDescription-pColorAttachments-09430
  All attachments in `pColorAttachments` that are not VK_ATTACHMENT_UNUSED **must** have the same sample count
- VUID-VkSubpassDescription-pInputAttachments-02647
  All attachments in `pInputAttachments` that are not VK_ATTACHMENT_UNUSED **must** have image formats whose potential format features contain at least VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT or VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkSubpassDescription-pColorAttachments-02648
  All attachments in `pColorAttachments` that are not VK_ATTACHMENT_UNUSED **must** have image formats whose potential format features contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT
- VUID-VkSubpassDescription-pResolveAttachments-02649
  All attachments in `pResolveAttachments` that are not VK_ATTACHMENT_UNUSED **must** have image formats whose potential format features contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT
- VUID-VkSubpassDescription-pDepthStencilAttachment-02650
  If `pDepthStencilAttachment` is not `NULL` and the attachment is not VK_ATTACHMENT_UNUSED then it **must** have an image format whose potential format features contain VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkSubpassDescription-pDepthStencilAttachment-01418

  all attachments in `pDepthStencilAttachment` and `pColorAttachments` that are not

VK_ATTACHMENT_UNUSED **must** have the same sample count

- VUID-VkSubpassDescription-attachment-00853
  Each element of `pPreserveAttachments` **must** not be VK_ATTACHMENT_UNUSED
- VUID-VkSubpassDescription-pPreserveAttachments-00854
  Each element of `pPreserveAttachments` **must** not also be an element of any other member of the subpass description
- VUID-VkSubpassDescription-layout-02519
  If any attachment is used by more than one VkAttachmentReference member, then each use **must** use the same `layout`
- VUID-VkSubpassDescription-flags-03341
  If `flags` includes VK_SUBPASS_DESCRIPTION_CUSTOM_RESOLVE_BIT_EXT, and if `pResolveAttachments` is not `NULL`, then each resolve attachment **must** be VK_ATTACHMENT_UNUSED
- VUID-VkSubpassDescription-flags-03343
  If `flags` includes VK_SUBPASS_DESCRIPTION_CUSTOM_RESOLVE_BIT_EXT, then the subpass **must** be the last subpass in a subpass dependency chain
- VUID-VkSubpassDescription-pDepthStencilAttachment-04438
  `pDepthStencilAttachment` and `pColorAttachments` **must** not contain references to the same attachment

Valid Usage (Implicit)

- VUID-VkSubpassDescription-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkSubpassDescriptionFlagBits values
- VUID-VkSubpassDescription-pipelineBindPoint-parameter
  `pipelineBindPoint` **must** be a valid VkPipelineBindPoint value
- VUID-VkSubpassDescription-pInputAttachments-parameter
  If `inputAttachmentCount` is not `0`, `pInputAttachments` **must** be a valid pointer to an array of `inputAttachmentCount` valid VkAttachmentReference structures
- VUID-VkSubpassDescription-pColorAttachments-parameter
  If `colorAttachmentCount` is not `0`, `pColorAttachments` **must** be a valid pointer to an array of `colorAttachmentCount` valid VkAttachmentReference structures
- VUID-VkSubpassDescription-pResolveAttachments-parameter
  If `colorAttachmentCount` is not `0`, and `pResolveAttachments` is not `NULL`, `pResolveAttachments` **must** be a valid pointer to an array of `colorAttachmentCount` valid VkAttachmentReference structures
- VUID-VkSubpassDescription-pDepthStencilAttachment-parameter
  If `pDepthStencilAttachment` is not `NULL`, `pDepthStencilAttachment` **must** be a valid pointer to a valid VkAttachmentReference structure
- VUID-VkSubpassDescription-pPreserveAttachments-parameter
  If `preserveAttachmentCount` is not `0`, `pPreserveAttachments` **must** be a valid pointer to an

array of `preserveAttachmentCount` `uint32_t` values

Bits which **can** be set in VkSubpassDescription::`flags`, specifying usage of the subpass, are:

WARNING	This functionality is superseded by Vulkan Version 1.2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef enum VkSubpassDescriptionFlagBits {
  // Provided by VK_EXT_rasterization_order_attachment_access
    VK_SUBPASS_DESCRIPTION_RASTERIZATION_ORDER_ATTACHMENT_COLOR_ACCESS_BIT_EXT =
0x00000010,
  // Provided by VK_EXT_rasterization_order_attachment_access
    VK_SUBPASS_DESCRIPTION_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT_EXT =
0x00000020,
  // Provided by VK_EXT_rasterization_order_attachment_access
    VK_SUBPASS_DESCRIPTION_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_BIT_EXT =
0x00000040,
  // Provided by VK_EXT_legacy_dithering
    VK_SUBPASS_DESCRIPTION_ENABLE_LEGACY_DITHERING_BIT_EXT = 0x00000080,
  // Provided by VK_EXT_custom_resolve
    VK_SUBPASS_DESCRIPTION_FRAGMENT_REGION_BIT_EXT = 0x00000004,
  // Provided by VK_EXT_custom_resolve
    VK_SUBPASS_DESCRIPTION_CUSTOM_RESOLVE_BIT_EXT = 0x00000008,
} VkSubpassDescriptionFlagBits;
```

- VK_SUBPASS_DESCRIPTION_FRAGMENT_REGION_BIT_EXT specifies that the framebuffer region is the fragment region, that is, the minimum region dependencies are by pixel rather than by sample, such that any fragment shader invocation **can** access any sample associated with that fragment shader invocation.
- VK_SUBPASS_DESCRIPTION_CUSTOM_RESOLVE_BIT_EXT specifies that the subpass performs shader resolve operations.
- VK_SUBPASS_DESCRIPTION_RASTERIZATION_ORDER_ATTACHMENT_COLOR_ACCESS_BIT_EXT specifies that this subpass supports pipelines created with VK_PIPELINE_COLOR_BLEND_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_ACCESS_BIT_EXT.
- VK_SUBPASS_DESCRIPTION_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT_EXT specifies that this subpass supports pipelines created with VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT_EXT.
- VK_SUBPASS_DESCRIPTION_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_BIT_EXT specifies that this subpass supports pipelines created with VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_BIT_EXT.
- VK_SUBPASS_DESCRIPTION_ENABLE_LEGACY_DITHERING_BIT_EXT specifies that Legacy

Dithering is enabled for this subpass.

NOTE Shader resolve operations allow for custom resolve operations, but overdrawing pixels **may** have a performance and/or power cost. Furthermore, since the content of any depth stencil attachment or color attachment is undefined at the beginning of a shader resolve subpass, any depth testing, stencil testing, or blending operation which sources these undefined values also has undefined result value.

NOTE There is no equivalent to VK_SUBPASS_DESCRIPTION_RASTERIZATION_ORDER_ATTACHMENT_COLOR_ACCESS_BIT_EXT, VK_SUBPASS_DESCRIPTION_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT_EXT, or VK_SUBPASS_DESCRIPTION_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_BIT_EXT for render pass instances begun with vkCmdBeginRendering. For such render passes, only the corresponding pipeline flags are specified.

WARNING This functionality is superseded by Vulkan Version 1.2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSubpassDescriptionFlags;
```

VkSubpassDescriptionFlags is a bitmask type for setting a mask of zero or more VkSubpassDescriptionFlagBits.

The VkAttachmentReference structure is defined as:

WARNING This functionality is superseded by VkAttachmentReference2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef struct VkAttachmentReference {
    uint32_t         attachment;
    VkImageLayout    layout;
} VkAttachmentReference;
```

- attachment is either an integer value identifying an attachment at the corresponding index in VkRenderPassCreateInfo::pAttachments, or VK_ATTACHMENT_UNUSED to signify that this attachment is not used.
- layout is a VkImageLayout value specifying the layout the attachment uses during the subpass.

Valid Usage

- VUID-VkAttachmentReference-layout-03077

If attachment is not VK_ATTACHMENT_UNUSED, layout **must** not be VK_IMAGE_LAYOUT_UNDEFINED, VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT, VK_IMAGE_LAYOUT_PREINITIALIZED, or VK_IMAGE_LAYOUT_PRESENT_SRC_KHR

- VUID-VkAttachmentReference-separateDepthStencilLayouts-03313
  If the separateDepthStencilLayouts feature is not enabled, and attachment is not VK_ATTACHMENT_UNUSED, layout **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentReference-synchronization2-06910
  If the synchronization2 feature is not enabled, layout **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR or VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VUID-VkAttachmentReference-attachmentFeedbackLoopLayout-07311
  If the attachmentFeedbackLoopLayout feature is not enabled, layout **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT
- VUID-VkAttachmentReference-dynamicRenderingLocalRead-09546
  If the dynamicRenderingLocalRead feature is not enabled, layout **must** not be VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ

Valid Usage (Implicit)

- VUID-VkAttachmentReference-layout-parameter
  layout **must** be a valid VkImageLayout value

VK_SUBPASS_EXTERNAL is a special subpass index value expanding synchronization scope outside a subpass. It is described in more detail by VkSubpassDependency.

```
#define VK_SUBPASS_EXTERNAL             (~0U)
```

The VkSubpassDependency structure is defined as:

WARNING This functionality is superseded by VkSubpassDependency2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef struct VkSubpassDependency {
    uint32_t                srcSubpass;
    uint32_t                dstSubpass;
    VkPipelineStageFlags    srcStageMask;
    VkPipelineStageFlags    dstStageMask;
    VkAccessFlags           srcAccessMask;
    VkAccessFlags           dstAccessMask;
    VkDependencyFlags       dependencyFlags;
} VkSubpassDependency;
```

- `srcSubpass` is the subpass index of the first subpass in the dependency, or VK_SUBPASS_EXTERNAL.
- `dstSubpass` is the subpass index of the second subpass in the dependency, or VK_SUBPASS_EXTERNAL.
- `srcStageMask` is a bitmask of VkPipelineStageFlagBits specifying the source stage mask. If set to VK_PIPELINE_STAGE_ALL_COMMANDS_BIT, it is equivalent to setting it to VK_PIPELINE_STAGE_ALL_GRAPHICS_BIT.
- `dstStageMask` is a bitmask of VkPipelineStageFlagBits specifying the destination stage mask. If set to VK_PIPELINE_STAGE_ALL_COMMANDS_BIT, it is equivalent to setting it to VK_PIPELINE_STAGE_ALL_GRAPHICS_BIT.
- `srcAccessMask` is a bitmask of VkAccessFlagBits specifying a source access mask.
- `dstAccessMask` is a bitmask of VkAccessFlagBits specifying a destination access mask.
- `dependencyFlags` is a bitmask of VkDependencyFlagBits.

If `srcSubpass` is equal to `dstSubpass` then the VkSubpassDependency does not directly define a dependency. Instead, it enables pipeline barriers to be used in a render pass instance within the identified subpass, where the scopes of one pipeline barrier **must** be a subset of those described by one subpass dependency. Subpass dependencies specified in this way that include framebuffer-space stages in the `srcStageMask` **must** only include framebuffer-space stages in `dstStageMask`, and **must** include VK_DEPENDENCY_BY_REGION_BIT. When a subpass dependency is specified in this way for a subpass that has more than one view in its view mask, its `dependencyFlags` **must** include VK_DEPENDENCY_VIEW_LOCAL_BIT.

If `srcSubpass` and `dstSubpass` are not equal, when a render pass instance which includes a subpass dependency is submitted to a queue, it defines a dependency between the subpasses identified by `srcSubpass` and `dstSubpass`.

If `srcSubpass` is equal to VK_SUBPASS_EXTERNAL, the first synchronization scope includes commands that occur earlier in submission order than the vkCmdBeginRenderPass used to begin the render pass instance. Otherwise, the first set of commands includes all commands submitted as part of the subpass instance identified by `srcSubpass` and any load, store, or multisample resolve operations on attachments used in `srcSubpass`. In either case, the first synchronization scope is limited to operations on the pipeline stages determined by the source stage mask specified by `srcStageMask`.

If dstSubpass is equal to VK_SUBPASS_EXTERNAL, the second synchronization scope includes commands that occur later in submission order than the vkCmdEndRenderPass used to end the render pass instance. Otherwise, the second set of commands includes all commands submitted as part of the subpass instance identified by dstSubpass and any load, store, and multisample resolve operations on attachments used in dstSubpass. In either case, the second synchronization scope is limited to operations on the pipeline stages determined by the destination stage mask specified by dstStageMask.

The first access scope is limited to accesses in the pipeline stages determined by the source stage mask specified by srcStageMask. It is also limited to access types in the source access mask specified by srcAccessMask.

The second access scope is limited to accesses in the pipeline stages determined by the destination stage mask specified by dstStageMask. It is also limited to access types in the destination access mask specified by dstAccessMask.

The availability and visibility operations defined by a subpass dependency affect the execution of image layout transitions within the render pass.

NOTE

For non-attachment resources, the memory dependency expressed by subpass dependency is nearly identical to that of a VkMemoryBarrier (with matching srcAccessMask and dstAccessMask parameters) submitted as a part of a vkCmdPipelineBarrier (with matching srcStageMask and dstStageMask parameters). The only difference being that its scopes are limited to the identified subpasses rather than potentially affecting everything before and after.

For attachments however, subpass dependencies work more like a VkImageMemoryBarrier defined similarly to the VkMemoryBarrier above, the queue family indices set to VK_QUEUE_FAMILY_IGNORED, and layouts as follows:

- The equivalent to oldLayout is the attachment's layout according to the subpass description for srcSubpass.
- The equivalent to newLayout is the attachment's layout according to the subpass description for dstSubpass.

Valid Usage

- VUID-VkSubpassDependency-srcStageMask-04090
  If the geometryShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT
- VUID-VkSubpassDependency-srcStageMask-04091
  If the tessellationShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-VkSubpassDependency-srcStageMask-04092
  If the conditionalRendering feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT

- VUID-VkSubpassDependency-srcStageMask-04093
  If the fragmentDensityMap feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-VkSubpassDependency-srcStageMask-04094
  If the transformFeedback feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-VkSubpassDependency-srcStageMask-04095
  If the meshShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT
- VUID-VkSubpassDependency-srcStageMask-04096
  If the taskShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT
- VUID-VkSubpassDependency-srcStageMask-07319
  If the attachmentFragmentShadingRate feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkSubpassDependency-srcStageMask-03937
  If the synchronization2 feature is not enabled, srcStageMask **must** not be 0
- VUID-VkSubpassDependency-srcStageMask-07950
  If the rayTracingPipeline feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkSubpassDependency-srcStageMask-10754
  If the accelerationStructure feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkSubpassDependency-dstStageMask-04090
  If the geometryShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT
- VUID-VkSubpassDependency-dstStageMask-04091
  If the tessellationShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-VkSubpassDependency-dstStageMask-04092
  If the conditionalRendering feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT
- VUID-VkSubpassDependency-dstStageMask-04093
  If the fragmentDensityMap feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-VkSubpassDependency-dstStageMask-04094
  If the transformFeedback feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-VkSubpassDependency-dstStageMask-04095
  If the meshShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT

- VUID-VkSubpassDependency-dstStageMask-04096
  If the `taskShader` feature is not enabled, `dstStageMask` **must** not contain VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT
- VUID-VkSubpassDependency-dstStageMask-07319
  If the `attachmentFragmentShadingRate` feature is not enabled, `dstStageMask` **must** not contain VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkSubpassDependency-dstStageMask-03937
  If the `synchronization2` feature is not enabled, `dstStageMask` **must** not be `0`
- VUID-VkSubpassDependency-dstStageMask-07950
  If the `rayTracingPipeline` feature is not enabled, `dstStageMask` **must** not contain VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkSubpassDependency-dstStageMask-10754
  If the `accelerationStructure` feature is not enabled, `dstStageMask` **must** not contain VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkSubpassDependency-srcSubpass-00864
  `srcSubpass` **must** be less than or equal to `dstSubpass`, unless one of them is VK_SUBPASS_EXTERNAL, to avoid cyclic dependencies and ensure a valid execution order
- VUID-VkSubpassDependency-srcSubpass-00865
  `srcSubpass` and `dstSubpass` **must** not both be equal to VK_SUBPASS_EXTERNAL
- VUID-VkSubpassDependency-srcSubpass-06809
  If `srcSubpass` is equal to `dstSubpass` and `srcStageMask` includes a framebuffer-space stage, `dstStageMask` **must** only contain framebuffer-space stages
- VUID-VkSubpassDependency-srcAccessMask-00868
  Any access flag included in `srcAccessMask` **must** be supported by one of the pipeline stages in `srcStageMask`, as specified in the table of supported access types
- VUID-VkSubpassDependency-dstAccessMask-00869
  Any access flag included in `dstAccessMask` **must** be supported by one of the pipeline stages in `dstStageMask`, as specified in the table of supported access types
- VUID-VkSubpassDependency-srcSubpass-02243
  If `srcSubpass` equals `dstSubpass`, and `srcStageMask` and `dstStageMask` both include a framebuffer-space stage, then `dependencyFlags` **must** include VK_DEPENDENCY_BY_REGION_BIT
- VUID-VkSubpassDependency-dependencyFlags-02520
  If `dependencyFlags` includes VK_DEPENDENCY_VIEW_LOCAL_BIT, `srcSubpass` **must** not be equal to VK_SUBPASS_EXTERNAL
- VUID-VkSubpassDependency-dependencyFlags-02521
  If `dependencyFlags` includes VK_DEPENDENCY_VIEW_LOCAL_BIT, `dstSubpass` **must** not be equal to VK_SUBPASS_EXTERNAL
- VUID-VkSubpassDependency-srcSubpass-00872
  If `srcSubpass` equals `dstSubpass` and that subpass has more than one bit set in the view mask, then `dependencyFlags` **must** include VK_DEPENDENCY_VIEW_LOCAL_BIT
- VUID-VkSubpassDependency-dependencyFlags-10203

`dependencyFlags` **must** not include VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR

Valid Usage (Implicit)

- VUID-VkSubpassDependency-srcStageMask-parameter
  If `srcStageMask` is not `0`, `srcStageMask` **must** be a valid combination of VkPipelineStageFlagBits values
- VUID-VkSubpassDependency-dstStageMask-parameter
  If `dstStageMask` is not `0`, `dstStageMask` **must** be a valid combination of VkPipelineStageFlagBits values
- VUID-VkSubpassDependency-srcAccessMask-parameter
  If `srcAccessMask` is not `0`, `srcAccessMask` **must** be a valid combination of VkAccessFlagBits values
- VUID-VkSubpassDependency-dstAccessMask-parameter
  If `dstAccessMask` is not `0`, `dstAccessMask` **must** be a valid combination of VkAccessFlagBits values
- VUID-VkSubpassDependency-dependencyFlags-parameter
  If `dependencyFlags` is not `0`, `dependencyFlags` **must** be a valid combination of VkDependencyFlagBits values

When multiview is enabled, the execution of the multiple views of one subpass **may** not occur simultaneously or even back-to-back, and rather **may** be interleaved with the execution of other subpasses. The load and store operations apply to attachments on a per-view basis. For example, an attachment using VK_ATTACHMENT_LOAD_OP_CLEAR will have each view cleared on first use, but the first use of one view may be temporally distant from the first use of another view.

NOTE

A good mental model for multiview is to think of a multiview subpass as if it were a collection of individual (per-view) subpasses that are logically grouped together and described as a single multiview subpass in the API. Similarly, a multiview attachment can be thought of like several individual attachments that happen to be layers in a single image. A view-local dependency between two multiview subpasses acts like a set of one-to-one dependencies between corresponding pairs of per-view subpasses. A view-global dependency between two multiview subpasses acts like a set of N × M dependencies between all pairs of per-view subpasses in the source and destination. Thus, it is a more compact representation which also makes clear the commonality and reuse that is present between views in a subpass. This interpretation motivates the answers to questions like “when does the load op apply” - it is on the first use of each view of an attachment, as if each view was a separate attachment.

The content of each view follows the description in attachment content behavior. In particular, if an attachment is preserved, all views within the attachment are preserved.

If any two subpasses of a render pass activate transform feedback to the same bound transform feedback buffers, a subpass dependency **must** be included (either directly or via some intermediate subpasses) between them.

If there is no subpass dependency from VK_SUBPASS_EXTERNAL to the first subpass that uses an attachment, then an implicit subpass dependency exists from VK_SUBPASS_EXTERNAL to the first subpass it is used in. The implicit subpass dependency only exists if there exists an automatic layout transition away from `initialLayout`. The subpass dependency operates as if defined with the following parameters:

```
VkSubpassDependency implicitDependency = {
    .srcSubpass = VK_SUBPASS_EXTERNAL,
    .dstSubpass = firstSubpass, // First subpass attachment is used in
    .srcStageMask = VK_PIPELINE_STAGE_NONE,
    .dstStageMask = VK_PIPELINE_STAGE_ALL_COMMANDS_BIT,
    .srcAccessMask = 0,
    .dstAccessMask = VK_ACCESS_INPUT_ATTACHMENT_READ_BIT |
                     VK_ACCESS_COLOR_ATTACHMENT_READ_BIT |
                     VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT |
                     VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT |
                     VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT,
    .dependencyFlags = 0
};
```

Similarly, if there is no subpass dependency from the last subpass that uses an attachment to VK_SUBPASS_EXTERNAL, then an implicit subpass dependency exists from the last subpass it is used in to VK_SUBPASS_EXTERNAL. The implicit subpass dependency only exists if there exists an automatic layout transition into `finalLayout`. The subpass dependency operates as if defined with the following parameters:

```
VkSubpassDependency implicitDependency = {
    .srcSubpass = lastSubpass, // Last subpass attachment is used in
    .dstSubpass = VK_SUBPASS_EXTERNAL,
    .srcStageMask = VK_PIPELINE_STAGE_ALL_COMMANDS_BIT,
    .dstStageMask = VK_PIPELINE_STAGE_NONE,
    .srcAccessMask = VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT |
                     VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT,
    .dstAccessMask = 0,
    .dependencyFlags = 0
};
```

As subpasses **may** overlap or execute out of order with regards to other subpasses unless a subpass dependency chain describes otherwise, the layout transitions required between subpasses **cannot** be known to an application. Instead, an application provides the layout that each attachment **must** be in at the start and end of a render pass, and the layout it **must** be in during each subpass it is used in. The implementation then **must** execute layout transitions between subpasses in order to guarantee that the images are in the layouts required by each subpass, and in the final layout at the

end of the render pass.

Automatic layout transitions apply to the entire image subresource attached to the framebuffer. If multiview is not enabled and the attachment is a view of a 1D or 2D image, the automatic layout transitions apply to the number of layers specified by VkFramebufferCreateInfo::layers. If multiview is enabled and the attachment is a view of a 1D or 2D image, the automatic layout transitions apply to the layers corresponding to views which are used by some subpass in the render pass, even if that subpass does not reference the given attachment. If the maintenance9 feature is not enabled, and the attachment view is a 2D or 2D array view of a 3D image, even if the attachment view only refers to a subset of the slices of the selected mip level of the 3D image, automatic layout transitions apply to the entire subresource referenced which is the entire mip level in this case. If the attachment view is a 2D or 2D array view of a 3D image and the maintenance9 feature is enabled, layout transitions apply only to the subresource the attachment view refers to.

Automatic layout transitions away from the layout used in a subpass happen-after the availability operations for all dependencies with that subpass as the srcSubpass.

Automatic layout transitions into the layout used in a subpass happen-before the visibility operations for all dependencies with that subpass as the dstSubpass.

Automatic layout transitions away from initialLayout happen-after the availability operations for all dependencies with a srcSubpass equal to VK_SUBPASS_EXTERNAL, where dstSubpass uses the attachment that will be transitioned. For attachments created with VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT, automatic layout transitions away from initialLayout happen-after the availability operations for all dependencies with a srcSubpass equal to VK_SUBPASS_EXTERNAL, where dstSubpass uses any aliased attachment.

Automatic layout transitions into finalLayout happen-before the visibility operations for all dependencies with a dstSubpass equal to VK_SUBPASS_EXTERNAL, where srcSubpass uses the attachment that will be transitioned. For attachments created with VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT, automatic layout transitions into finalLayout happen-before the visibility operations for all dependencies with a dstSubpass equal to VK_SUBPASS_EXTERNAL, where srcSubpass uses any aliased attachment.

The image layout of the depth aspect of a depth/stencil attachment referring to an image created with VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT is dependent on the last sample locations used to render to the attachment, thus automatic layout transitions use the sample locations state specified in VkRenderPassSampleLocationsBeginInfoEXT.

Automatic layout transitions of an attachment referring to a depth/stencil image created with VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT use the sample locations the image subresource range referenced by the attachment was last rendered with. If the current render pass does not use the attachment as a depth/stencil attachment in any subpass that happens-before, the automatic layout transition uses the sample locations state specified in the sampleLocationsInfo member of the element of the VkRenderPassSampleLocationsBeginInfoEXT::pAttachmentInitialSampleLocations array for which the attachmentIndex member equals the attachment index of the attachment, if one is specified. Otherwise, the automatic layout transition uses the sample locations state specified in the sampleLocationsInfo member of the element of the

VkRenderPassSampleLocationsBeginInfoEXT::pPostSubpassSampleLocations array for which the subpassIndex member equals the index of the subpass that last used the attachment as a depth/stencil attachment, if one is specified.

If no sample locations state has been specified for an automatic layout transition performed on an attachment referring to a depth/stencil image created with VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT the contents of the depth aspect of the depth/stencil attachment become undefined as if the layout of the attachment was transitioned from the VK_IMAGE_LAYOUT_UNDEFINED layout.

If two subpasses use the same attachment, and both subpasses use the attachment in a read-only layout, no subpass dependency needs to be specified between those subpasses. If an implementation treats those layouts separately, it **must** insert an implicit subpass dependency between those subpasses to separate the uses in each layout. The subpass dependency operates as if defined with the following parameters:

```
// Used for input attachments
VkPipelineStageFlags inputAttachmentStages = VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT;
VkAccessFlags inputAttachmentDstAccess = VK_ACCESS_INPUT_ATTACHMENT_READ_BIT;

// Used for depth/stencil attachments
VkPipelineStageFlags depthStencilAttachmentStages =
VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT |
VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT;
VkAccessFlags depthStencilAttachmentDstAccess =
VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT;

VkSubpassDependency implicitDependency = {
    .srcSubpass = firstSubpass;
    .dstSubpass = secondSubpass;
    .srcStageMask = inputAttachmentStages | depthStencilAttachmentStages;
    .dstStageMask = inputAttachmentStages | depthStencilAttachmentStages;
    .srcAccessMask = 0;
    .dstAccessMask = inputAttachmentDstAccess | depthStencilAttachmentDstAccess;
    .dependencyFlags = 0;
};
```

When drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT set in VkPipelineDynamicStateCreateInfo::pDynamicStates, the application **must** specify which types of attachments that are written to during a render pass will also be accessed as non-attachments in the render pass.

To dynamically set whether a pipeline **can** access a resource as a non-attachment while it is also used as an attachment that is written to, call:

```
// Provided by VK_EXT_attachment_feedback_loop_dynamic_state
void vkCmdSetAttachmentFeedbackLoopEnableEXT(
    VkCommandBuffer                             commandBuffer,
    VkImageAspectFlags                          aspectMask);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `aspectMask` specifies the types of attachments for which feedback loops will be enabled. Attachment types whose aspects are not included in `aspectMask` will have feedback loops disabled.

For attachments that are written to in a render pass, only attachments with the aspects specified in `aspectMask` **can** be accessed as non-attachments by subsequent drawing commands.

Valid Usage

- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-attachmentFeedbackLoopDynamicState-08862
  The `attachmentFeedbackLoopDynamicState` feature **must** be enabled
- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-aspectMask-08863
  `aspectMask` **must** only include VK_IMAGE_ASPECT_NONE, VK_IMAGE_ASPECT_COLOR_BIT, VK_IMAGE_ASPECT_DEPTH_BIT, and VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-attachmentFeedbackLoopLayout-08864
  If the `attachmentFeedbackLoopLayout` feature is not enabled, `aspectMask` **must** be VK_IMAGE_ASPECT_NONE

Valid Usage (Implicit)

- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-aspectMask-parameter
  If `aspectMask` is not `0`, `aspectMask` **must** be a valid combination of VkImageAspectFlagBits values
- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetAttachmentFeedbackLoopEnableEXT is not affected by conditional rendering

A more extensible version of render pass creation is also defined below.

To create a render pass, call:

WARNING This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
VkResult vkCreateRenderPass2(
    VkDevice                                    device,
    const VkRenderPassCreateInfo2*              pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkRenderPass*                               pRenderPass);
```

```
// Provided by VK_KHR_create_renderpass2
// Equivalent to vkCreateRenderPass2
VkResult vkCreateRenderPass2KHR(
    VkDevice                                    device,
    const VkRenderPassCreateInfo2*              pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkRenderPass*                               pRenderPass);
```

- `device` is the logical device that creates the render pass.
- `pCreateInfo` is a pointer to a VkRenderPassCreateInfo2 structure describing the parameters of the render pass.

- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pRenderPass` is a pointer to a VkRenderPass handle in which the resulting render pass object is returned.

This command is functionally identical to vkCreateRenderPass, but includes extensible sub-structures that include `sType` and `pNext` parameters, allowing them to be more easily extended.

Valid Usage

- VUID-vkCreateRenderPass2-device-10001
  `device` **must** support at least one queue family with the VK_QUEUE_GRAPHICS_BIT capability

Valid Usage (Implicit)

- VUID-vkCreateRenderPass2-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateRenderPass2-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkRenderPassCreateInfo2 structure
- VUID-vkCreateRenderPass2-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateRenderPass2-pRenderPass-parameter
  `pRenderPass` **must** be a valid pointer to a VkRenderPass handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkRenderPassCreateInfo2` structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
typedef struct VkRenderPassCreateInfo2 {
    VkStructureType                   sType;
    const void*                       pNext;
    VkRenderPassCreateFlags           flags;
    uint32_t                          attachmentCount;
    const VkAttachmentDescription2*   pAttachments;
    uint32_t                          subpassCount;
    const VkSubpassDescription2*      pSubpasses;
    uint32_t                          dependencyCount;
    const VkSubpassDependency2*       pDependencies;
    uint32_t                          correlatedViewMaskCount;
    const uint32_t*                   pCorrelatedViewMasks;
} VkRenderPassCreateInfo2;
```

```
// Provided by VK_KHR_create_renderpass2
// Equivalent to VkRenderPassCreateInfo2
typedef VkRenderPassCreateInfo2 VkRenderPassCreateInfo2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `attachmentCount` is the number of attachments used by this render pass.
- `pAttachments` is a pointer to an array of `attachmentCount` VkAttachmentDescription2 structures describing the attachments used by the render pass.
- `subpassCount` is the number of subpasses to create.
- `pSubpasses` is a pointer to an array of `subpassCount` VkSubpassDescription2 structures describing each subpass.
- `dependencyCount` is the number of dependencies between pairs of subpasses.
- `pDependencies` is a pointer to an array of `dependencyCount` VkSubpassDependency2 structures describing dependencies between pairs of subpasses.
- `correlatedViewMaskCount` is the number of correlation masks.
- `pCorrelatedViewMasks` is a pointer to an array of view masks indicating sets of views that **may** be more efficient to render concurrently.

Parameters defined by this structure with the same name as those in VkRenderPassCreateInfo have the identical effect to those parameters; the child structures are variants of those used in VkRenderPassCreateInfo which add `sType` and `pNext` parameters, allowing them to be extended.

If the VkSubpassDescription2::`viewMask` member of any element of `pSubpasses` is not zero, *multiview* functionality is considered to be enabled for this render pass.

`correlatedViewMaskCount` and `pCorrelatedViewMasks` have the same effect as

VkRenderPassMultiviewCreateInfo::correlationMaskCount and VkRenderPassMultiviewCreateInfo::pCorrelationMasks, respectively.

Valid Usage

- VUID-VkRenderPassCreateInfo2-None-03049
  If any two subpasses operate on attachments with overlapping ranges of the same VkDeviceMemory object, and at least one subpass writes to that area of VkDeviceMemory, a subpass dependency **must** be included (either directly or via some intermediate subpasses) between them
- VUID-VkRenderPassCreateInfo2-attachment-03050
  If the attachment member of any element of pInputAttachments, pColorAttachments, pResolveAttachments or pDepthStencilAttachment, or the attachment indexed by any element of pPreserveAttachments in any element of pSubpasses is bound to a range of a VkDeviceMemory object that overlaps with any other attachment in any subpass (including the same subpass), the VkAttachmentDescription2 structures describing them **must** include VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT in flags
- VUID-VkRenderPassCreateInfo2-attachment-03051
  If the attachment member of any element of pInputAttachments, pColorAttachments, pResolveAttachments or pDepthStencilAttachment, or any element of pPreserveAttachments in any element of pSubpasses is not VK_ATTACHMENT_UNUSED, then it **must** be less than attachmentCount
- VUID-VkRenderPassCreateInfo2-fragmentDensityMapAttachment-06472
  If the pNext chain includes a VkRenderPassFragmentDensityMapCreateInfoEXT structure and the fragmentDensityMapAttachment member is not VK_ATTACHMENT_UNUSED, then attachment **must** be less than attachmentCount
- VUID-VkRenderPassCreateInfo2-pSubpasses-06473
  If the pSubpasses pNext chain includes a VkSubpassDescriptionDepthStencilResolve structure and the pDepthStencilResolveAttachment member is not NULL and does not have the value VK_ATTACHMENT_UNUSED, then attachment **must** be less than attachmentCount
- VUID-VkRenderPassCreateInfo2-pSubpasses-12529
  If the pSubpasses pNext chain includes a VkFragmentShadingRateAttachmentInfoKHR structure and the pFragmentShadingRateAttachment member is not NULL and does not have the value VK_ATTACHMENT_UNUSED, then attachment **must** be less than attachmentCount
- VUID-VkRenderPassCreateInfo2-pAttachments-02522
  For any member of pAttachments with a loadOp equal to VK_ATTACHMENT_LOAD_OP_CLEAR, the first use of that attachment **must** not specify a layout equal to VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL, or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL
- VUID-VkRenderPassCreateInfo2-pAttachments-02523
  For any member of pAttachments with a stencilLoadOp equal to VK_ATTACHMENT_LOAD_OP_CLEAR, the first use of that attachment **must** not specify a layout equal to VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL, or

VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkRenderPassCreateInfo2-pDependencies-03054
  For each element of pDependencies, if the srcSubpass is not VK_SUBPASS_EXTERNAL, all stage flags included in the srcStageMask member of that dependency **must** be VK_PIPELINE_STAGE_ALL_COMMANDS_BIT or a pipeline stage supported by the pipeline identified by the pipelineBindPoint member of the source subpass
- VUID-VkRenderPassCreateInfo2-pDependencies-03055
  For each element of pDependencies, if the dstSubpass is not VK_SUBPASS_EXTERNAL, all stage flags included in the dstStageMask member of that dependency **must** be VK_PIPELINE_STAGE_ALL_COMMANDS_BIT or a pipeline stage supported by the pipeline identified by the pipelineBindPoint member of the destination subpass
- VUID-VkRenderPassCreateInfo2-pCorrelatedViewMasks-03056
  The set of bits included in any element of pCorrelatedViewMasks **must** not overlap with the set of bits included in any other element of pCorrelatedViewMasks
- VUID-VkRenderPassCreateInfo2-viewMask-03057
  If the VkSubpassDescription2::viewMask member of all elements of pSubpasses is 0, correlatedViewMaskCount **must** be 0
- VUID-VkRenderPassCreateInfo2-viewMask-03058
  The VkSubpassDescription2::viewMask member of all elements of pSubpasses **must** either all be 0, or all not be 0
- VUID-VkRenderPassCreateInfo2-viewMask-03059
  If the VkSubpassDescription2::viewMask member of all elements of pSubpasses is 0, the dependencyFlags member of any element of pDependencies **must** not include VK_DEPENDENCY_VIEW_LOCAL_BIT
- VUID-VkRenderPassCreateInfo2-pDependencies-03060
  For each element of pDependencies where its srcSubpass member equals its dstSubpass member, if the viewMask member of the corresponding element of pSubpasses includes more than one bit, its dependencyFlags member **must** include VK_DEPENDENCY_VIEW_LOCAL_BIT
- VUID-VkRenderPassCreateInfo2-attachment-02525
  If the attachment member of any element of the pInputAttachments member of any element of pSubpasses is not VK_ATTACHMENT_UNUSED, the aspectMask member of that element of pInputAttachments **must** only include aspects that are present in images of the format specified by the element of pAttachments specified by attachment
- VUID-VkRenderPassCreateInfo2-srcSubpass-02526
  The srcSubpass member of each element of pDependencies **must** be less than subpassCount
- VUID-VkRenderPassCreateInfo2-dstSubpass-02527
  The dstSubpass member of each element of pDependencies **must** be less than subpassCount
- VUID-VkRenderPassCreateInfo2-pAttachments-04585
  If any element of pAttachments is used as a fragment shading rate attachment in any subpass, it **must** not be used as any other attachment in the render pass
- VUID-VkRenderPassCreateInfo2-pAttachments-09387
  If any element of pAttachments is used as a fragment shading rate attachment, the loadOp

for that attachment **must** not be VK_ATTACHMENT_LOAD_OP_CLEAR

- VUID-VkRenderPassCreateInfo2-pAttachments-04586
  If any element of `pAttachments` is used as a fragment shading rate attachment in any subpass, it **must** have an image format whose potential format features contain VK_FORMAT_FEATURE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkRenderPassCreateInfo2-attachment-06244
  If the `attachment` member of the `pDepthStencilAttachment` member of an element of `pSubpasses` is not VK_ATTACHMENT_UNUSED, the `layout` member of that same structure is either VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, and the `pNext` chain of that structure does not include a VkAttachmentReferenceStencilLayout structure, then the element of `pAttachments` with an index equal to `attachment` **must** not have a `format` that includes both depth and stencil components
- VUID-VkRenderPassCreateInfo2-attachment-06245
  If the `attachment` member of the `pDepthStencilAttachment` member of an element of `pSubpasses` is not VK_ATTACHMENT_UNUSED and the `layout` member of that same structure is either VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL, then the element of `pAttachments` with an index equal to `attachment` **must** have a `format` that includes only a stencil component
- VUID-VkRenderPassCreateInfo2-attachment-06246
  If the `attachment` member of the `pDepthStencilAttachment` member of an element of `pSubpasses` is not VK_ATTACHMENT_UNUSED and the `layout` member of that same structure is either VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, then the element of `pAttachments` with an index equal to `attachment` **must** not have a `format` that includes only a stencil component
- VUID-VkRenderPassCreateInfo2-None-10916
  If any subpass preserves an attachment, there **must** be a subpass dependency from a prior subpass which uses or preserves that attachment

Valid Usage (Implicit)

- VUID-VkRenderPassCreateInfo2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO_2
- VUID-VkRenderPassCreateInfo2-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkRenderPassCreationControlEXT, VkRenderPassCreationFeedbackCreateInfoEXT, or VkRenderPassFragmentDensityMapCreateInfoEXT
- VUID-VkRenderPassCreateInfo2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkRenderPassCreateInfo2-flags-zerobitmask

flags must be 0

- VUID-VkRenderPassCreateInfo2-pAttachments-parameter
  If attachmentCount is not 0, pAttachments must be a valid pointer to an array of attachmentCount valid VkAttachmentDescription2 structures
- VUID-VkRenderPassCreateInfo2-pSubpasses-parameter
  pSubpasses must be a valid pointer to an array of subpassCount valid VkSubpassDescription2 structures
- VUID-VkRenderPassCreateInfo2-pDependencies-parameter
  If dependencyCount is not 0, pDependencies must be a valid pointer to an array of dependencyCount valid VkSubpassDependency2 structures
- VUID-VkRenderPassCreateInfo2-pCorrelatedViewMasks-parameter
  If correlatedViewMaskCount is not 0, pCorrelatedViewMasks must be a valid pointer to an array of correlatedViewMaskCount uint32_t values
- VUID-VkRenderPassCreateInfo2-subpassCount-arraylength
  subpassCount must be greater than 0

The VkAttachmentDescription2 structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
typedef struct VkAttachmentDescription2 {
    VkStructureType                 sType;
    const void*                     pNext;
    VkAttachmentDescriptionFlags    flags;
    VkFormat                        format;
    VkSampleCountFlagBits           samples;
    VkAttachmentLoadOp              loadOp;
    VkAttachmentStoreOp             storeOp;
    VkAttachmentLoadOp              stencilLoadOp;
    VkAttachmentStoreOp             stencilStoreOp;
    VkImageLayout                   initialLayout;
    VkImageLayout                   finalLayout;
} VkAttachmentDescription2;
```

```
// Provided by VK_KHR_create_renderpass2
// Equivalent to VkAttachmentDescription2
typedef VkAttachmentDescription2 VkAttachmentDescription2KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask of VkAttachmentDescriptionFlagBits specifying additional properties of the

attachment.

- `format` is a VkFormat value specifying the format of the image that will be used for the attachment.
- `samples` is a VkSampleCountFlagBits value specifying the number of samples of the image.
- `loadOp` is a VkAttachmentLoadOp value specifying how the contents of color and depth components of the attachment are treated at the beginning of the subpass where it is first used.
- `storeOp` is a VkAttachmentStoreOp value specifying how the contents of color and depth components of the attachment are treated at the end of the subpass where it is last used.
- `stencilLoadOp` is a VkAttachmentLoadOp value specifying how the contents of stencil components of the attachment are treated at the beginning of the subpass where it is first used.
- `stencilStoreOp` is a VkAttachmentStoreOp value specifying how the contents of stencil components of the attachment are treated at the end of the last subpass where it is used.
- `initialLayout` is the layout the attachment image subresource will be in when a render pass instance begins.
- `finalLayout` is the layout the attachment image subresource will be transitioned to when a render pass instance ends.

Parameters defined by this structure with the same name as those in VkAttachmentDescription have the identical effect to those parameters.

If the `separateDepthStencilLayouts` feature is enabled, and `format` is a depth/stencil format, `initialLayout` and `finalLayout` **can** be set to a layout that only specifies the layout of the depth aspect.

If the `pNext` chain includes a VkAttachmentDescriptionStencilLayout structure, then the `stencilInitialLayout` and `stencilFinalLayout` members specify the initial and final layouts of the stencil aspect of a depth/stencil format, and `initialLayout` and `finalLayout` only apply to the depth aspect. For depth-only formats, the VkAttachmentDescriptionStencilLayout structure is ignored. For stencil-only formats, the initial and final layouts of the stencil aspect are taken from the VkAttachmentDescriptionStencilLayout structure if present, or `initialLayout` and `finalLayout` if not present.

If `format` is a depth/stencil format, and either `initialLayout` or `finalLayout` does not specify a layout for the stencil aspect, then the application **must** specify the initial and final layouts of the stencil aspect by including a VkAttachmentDescriptionStencilLayout structure in the `pNext` chain.

`loadOp` and `storeOp` are ignored for fragment shading rate attachments. No access to the shading rate attachment is performed in `loadOp` and `storeOp`. Instead, access to VK_ACCESS_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR is performed as fragments are rasterized.

Valid Usage

- VUID-VkAttachmentDescription2-format-06699
  If `format` includes a color or depth component and `loadOp` is

VK_ATTACHMENT_LOAD_OP_LOAD, then initialLayout **must** not be VK_IMAGE_LAYOUT_UNDEFINED

- VUID-VkAttachmentDescription2-finalLayout-00843
  finalLayout **must** not be VK_IMAGE_LAYOUT_UNDEFINED or VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT or VK_IMAGE_LAYOUT_PREINITIALIZED
- VUID-VkAttachmentDescription2-format-03280
  If format is a color format, initialLayout **must** not be VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription2-format-03281
  If format is a depth/stencil format, initialLayout **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL
- VUID-VkAttachmentDescription2-format-03282
  If format is a color format, finalLayout **must** not be VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription2-format-03283
  If format is a depth/stencil format, finalLayout **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL
- VUID-VkAttachmentDescription2-format-06487
  If format is a color format, initialLayout **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL
- VUID-VkAttachmentDescription2-format-06488
  If format is a color format, finalLayout **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL
- VUID-VkAttachmentDescription2-separateDepthStencilLayouts-03284
  If the separateDepthStencilLayouts feature is not enabled, initialLayout **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription2-separateDepthStencilLayouts-03285
  If the separateDepthStencilLayouts feature is not enabled, finalLayout **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription2-format-03286
  If format is a color format, initialLayout **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or

VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkAttachmentDescription2-format-03287
  If `format` is a color format, `finalLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription2-format-06906
  If `format` is a depth/stencil format which includes both depth and stencil components, `initialLayout` **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription2-format-06907
  If `format` is a depth/stencil format which includes both depth and stencil components, `finalLayout` **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription2-format-03290
  If `format` is a depth/stencil format which includes only the depth component, `initialLayout` **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription2-format-03291
  If `format` is a depth/stencil format which includes only the depth component, `finalLayout` **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription2-synchronization2-06908
  If the `synchronization2` feature is not enabled, `initialLayout` **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR or VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VUID-VkAttachmentDescription2-synchronization2-06909
  If the `synchronization2` feature is not enabled, `finalLayout` **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR or VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VUID-VkAttachmentDescription2-attachmentFeedbackLoopLayout-07309
  If the `attachmentFeedbackLoopLayout` feature is not enabled, `initialLayout` **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT
- VUID-VkAttachmentDescription2-attachmentFeedbackLoopLayout-07310
  If the `attachmentFeedbackLoopLayout` feature is not enabled, `finalLayout` **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT
- VUID-VkAttachmentDescription2-dynamicRenderingLocalRead-09544
  If the `dynamicRenderingLocalRead` feature is not enabled, `initialLayout` **must** not be VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ
- VUID-VkAttachmentDescription2-dynamicRenderingLocalRead-09545
  If the `dynamicRenderingLocalRead` feature is not enabled, `finalLayout` **must** not be VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ

- VUID-VkAttachmentDescription2-flags-11773
  If `flags` includes VK_ATTACHMENT_DESCRIPTION_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR, `flags` **must** not include VK_ATTACHMENT_DESCRIPTION_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR
- VUID-VkAttachmentDescription2-flags-11774
  If `flags` includes VK_ATTACHMENT_DESCRIPTION_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR or VK_ATTACHMENT_DESCRIPTION_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR, `resolveSrgbFormatSupportsTransferFunctionControl` **must** be VK_TRUE
- VUID-VkAttachmentDescription2-flags-11775
  If `flags` includes VK_ATTACHMENT_DESCRIPTION_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR or VK_ATTACHMENT_DESCRIPTION_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR, `maintenance10` **must** be enabled
- VUID-VkAttachmentDescription2-flags-11776
  If `flags` includes VK_ATTACHMENT_DESCRIPTION_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR or VK_ATTACHMENT_DESCRIPTION_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR, `format` **must** use sRGB encoding
- VUID-VkAttachmentDescription2-flags-11777
  If `flags` includes VK_ATTACHMENT_DESCRIPTION_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR or VK_ATTACHMENT_DESCRIPTION_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR, `samples` **must** be VK_SAMPLE_COUNT_1_BIT
- VUID-VkAttachmentDescription2-pNext-06704
  If the `pNext` chain does not include a VkAttachmentDescriptionStencilLayout structure, `format` includes a stencil component, and `stencilLoadOp` is VK_ATTACHMENT_LOAD_OP_LOAD, then `initialLayout` **must** not be VK_IMAGE_LAYOUT_UNDEFINED
- VUID-VkAttachmentDescription2-pNext-06705
  If the `pNext` chain includes a VkAttachmentDescriptionStencilLayout structure, `format` includes a stencil component, and `stencilLoadOp` is VK_ATTACHMENT_LOAD_OP_LOAD, then VkAttachmentDescriptionStencilLayout::`stencilInitialLayout` **must** not be VK_IMAGE_LAYOUT_UNDEFINED
- VUID-VkAttachmentDescription2-format-06249
  If `format` is a depth/stencil format which includes both depth and stencil components, and `initialLayout` is VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, the `pNext` chain **must** include a VkAttachmentDescriptionStencilLayout structure
- VUID-VkAttachmentDescription2-format-06250
  If `format` is a depth/stencil format which includes both depth and stencil components, and `finalLayout` is VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, the `pNext` chain **must** include a VkAttachmentDescriptionStencilLayout structure

- VUID-VkAttachmentDescription2-format-06247
  If the `pNext` chain does not include a VkAttachmentDescriptionStencilLayout structure and `format` only includes a stencil component, `initialLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription2-format-06248
  If the `pNext` chain does not include a VkAttachmentDescriptionStencilLayout structure and `format` only includes a stencil component, `finalLayout` **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL
- VUID-VkAttachmentDescription2-format-09332
  `format` **must** not be VK_FORMAT_UNDEFINED

Valid Usage (Implicit)

- VUID-VkAttachmentDescription2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_2
- VUID-VkAttachmentDescription2-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkAttachmentDescriptionStencilLayout
- VUID-VkAttachmentDescription2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkAttachmentDescription2-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkAttachmentDescriptionFlagBits values
- VUID-VkAttachmentDescription2-format-parameter
  `format` **must** be a valid VkFormat value
- VUID-VkAttachmentDescription2-samples-parameter
  `samples` **must** be a valid VkSampleCountFlagBits value
- VUID-VkAttachmentDescription2-loadOp-parameter
  `loadOp` **must** be a valid VkAttachmentLoadOp value
- VUID-VkAttachmentDescription2-storeOp-parameter
  `storeOp` **must** be a valid VkAttachmentStoreOp value
- VUID-VkAttachmentDescription2-stencilLoadOp-parameter
  `stencilLoadOp` **must** be a valid VkAttachmentLoadOp value
- VUID-VkAttachmentDescription2-stencilStoreOp-parameter
  `stencilStoreOp` **must** be a valid VkAttachmentStoreOp value
- VUID-VkAttachmentDescription2-initialLayout-parameter
  `initialLayout` **must** be a valid VkImageLayout value
- VUID-VkAttachmentDescription2-finalLayout-parameter
  `finalLayout` **must** be a valid VkImageLayout value

The VkAttachmentDescriptionStencilLayout structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
typedef struct VkAttachmentDescriptionStencilLayout {
    VkStructureType    sType;
    void*              pNext;
    VkImageLayout      stencilInitialLayout;
    VkImageLayout      stencilFinalLayout;
} VkAttachmentDescriptionStencilLayout;
```

```
// Provided by VK_KHR_separate_depth_stencil_layouts
// Equivalent to VkAttachmentDescriptionStencilLayout
typedef VkAttachmentDescriptionStencilLayout VkAttachmentDescriptionStencilLayoutKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- stencilInitialLayout is the layout the stencil aspect of the attachment image subresource will be in when a render pass instance begins.
- stencilFinalLayout is the layout the stencil aspect of the attachment image subresource will be transitioned to when a render pass instance ends.

Valid Usage

- VUID-VkAttachmentDescriptionStencilLayout-stencilInitialLayout-03308
  stencilInitialLayout **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL, or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL
- VUID-VkAttachmentDescriptionStencilLayout-stencilFinalLayout-03309
  stencilFinalLayout **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL, or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL
- VUID-VkAttachmentDescriptionStencilLayout-stencilFinalLayout-03310
  stencilFinalLayout **must** not be VK_IMAGE_LAYOUT_UNDEFINED or

VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT or VK_IMAGE_LAYOUT_PREINITIALIZED

Valid Usage (Implicit)

- VUID-VkAttachmentDescriptionStencilLayout-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_STENCIL_LAYOUT
- VUID-VkAttachmentDescriptionStencilLayout-stencilInitialLayout-parameter
  `stencilInitialLayout` **must** be a valid VkImageLayout value
- VUID-VkAttachmentDescriptionStencilLayout-stencilFinalLayout-parameter
  `stencilFinalLayout` **must** be a valid VkImageLayout value

Structure Chaining

Extends the structure

- VkAttachmentDescription2

The `VkSubpassDescription2` structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
typedef struct VkSubpassDescription2 {
    VkStructureType                 sType;
    const void*                     pNext;
    VkSubpassDescriptionFlags       flags;
    VkPipelineBindPoint             pipelineBindPoint;
    uint32_t                        viewMask;
    uint32_t                        inputAttachmentCount;
    const VkAttachmentReference2*   pInputAttachments;
    uint32_t                        colorAttachmentCount;
    const VkAttachmentReference2*   pColorAttachments;
    const VkAttachmentReference2*   pResolveAttachments;
    const VkAttachmentReference2*   pDepthStencilAttachment;
    uint32_t                        preserveAttachmentCount;
    const uint32_t*                 pPreserveAttachments;
} VkSubpassDescription2;
```

```
// Provided by VK_KHR_create_renderpass2
// Equivalent to VkSubpassDescription2
typedef VkSubpassDescription2 VkSubpassDescription2KHR;
```

- `sType` is a VkStructureType value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkSubpassDescriptionFlagBits specifying usage of the subpass.
- `pipelineBindPoint` is a VkPipelineBindPoint value specifying the pipeline type supported for this subpass.
- `viewMask` is a bitfield of view indices describing which views rendering is broadcast to in this subpass, when multiview is enabled.
- `inputAttachmentCount` is the number of input attachments.
- `pInputAttachments` is a pointer to an array of VkAttachmentReference2 structures defining the input attachments for this subpass and their layouts.
- `colorAttachmentCount` is the number of color attachments.
- `pColorAttachments` is a pointer to an array of `colorAttachmentCount` VkAttachmentReference2 structures defining the color attachments for this subpass and their layouts.
- `pResolveAttachments` is `NULL` or a pointer to an array of `colorAttachmentCount` VkAttachmentReference2 structures defining the resolve attachments for this subpass and their layouts.
- `pDepthStencilAttachment` is a pointer to a VkAttachmentReference2 structure specifying the depth/stencil attachment for this subpass and its layout.
- `preserveAttachmentCount` is the number of preserved attachments.
- `pPreserveAttachments` is a pointer to an array of `preserveAttachmentCount` render pass attachment indices identifying attachments that are not used by this subpass, but whose contents **must** be preserved throughout the subpass.

Parameters defined by this structure with the same name as those in VkSubpassDescription have the identical effect to those parameters.

`viewMask` has the same effect for the described subpass as VkRenderPassMultiviewCreateInfo ::`pViewMasks` has on each corresponding subpass.

If a VkFragmentShadingRateAttachmentInfoKHR structure is included in the `pNext` chain, `pFragmentShadingRateAttachment` is not `NULL`, and its `attachment` member is not VK_ATTACHMENT_UNUSED, the identified attachment defines a fragment shading rate attachment for that subpass.

Valid Usage

- VUID-VkSubpassDescription2-attachment-06912
  If the `attachment` member of an element of `pInputAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL
- VUID-VkSubpassDescription2-attachment-06913
  If the `attachment` member of an element of `pColorAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be

VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL

- VUID-VkSubpassDescription2-attachment-06914
  If the `attachment` member of an element of `pResolveAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL
- VUID-VkSubpassDescription2-attachment-06915
  If the `attachment` member of `pDepthStencilAttachment` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL
- VUID-VkSubpassDescription2-attachment-06916
  If the `attachment` member of an element of `pColorAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL
- VUID-VkSubpassDescription2-attachment-06917
  If the `attachment` member of an element of `pResolveAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL
- VUID-VkSubpassDescription2-attachment-06918
  If the `attachment` member of an element of `pInputAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL
- VUID-VkSubpassDescription2-attachment-06919
  If the `attachment` member of an element of `pColorAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkSubpassDescription2-attachment-06920
  If the `attachment` member of an element of `pResolveAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkSubpassDescription2-attachment-06921
  If the `attachment` member of an element of `pInputAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR

- VUID-VkSubpassDescription2-attachment-06922
  If the `attachment` member of an element of `pColorAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VUID-VkSubpassDescription2-attachment-06923
  If the `attachment` member of an element of `pResolveAttachments` is not VK_ATTACHMENT_UNUSED, its `layout` member **must** not be VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VUID-VkSubpassDescription2-inputAttachmentCount-12293
  `inputAttachmentCount` **must** be less than or equal to `maxPerStageDescriptorInputAttachments`
- VUID-VkSubpassDescription2-colorAttachmentCount-00845
  `colorAttachmentCount` **must** be less than or equal to `maxColorAttachments`
- VUID-VkSubpassDescription2-loadOp-00846
  If the first use of an attachment in this render pass is as an input attachment, and the attachment is not also used as a color or depth/stencil attachment in the same subpass, then `loadOp` **must** not be VK_ATTACHMENT_LOAD_OP_CLEAR
- VUID-VkSubpassDescription2-attachment-06251
  If the `attachment` member of `pDepthStencilAttachment` is not VK_ATTACHMENT_UNUSED and its `pNext` chain includes a VkAttachmentReferenceStencilLayout structure, the `layout` member of `pDepthStencilAttachment` **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkSubpassDescription2-pipelineBindPoint-04953
  `pipelineBindPoint` **must** be VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-VkSubpassDescription2-pResolveAttachments-03065
  If `pResolveAttachments` is not `NULL`, for each resolve attachment that does not have the value VK_ATTACHMENT_UNUSED, the corresponding color attachment **must** not have the value VK_ATTACHMENT_UNUSED
- VUID-VkSubpassDescription2-pResolveAttachments-03066
  If `pResolveAttachments` is not `NULL`, for each resolve attachment that is not VK_ATTACHMENT_UNUSED, the corresponding color attachment **must** not have a sample count of VK_SAMPLE_COUNT_1_BIT
- VUID-VkSubpassDescription2-pResolveAttachments-03068
  Each element of `pResolveAttachments` **must** have the same VkFormat as its corresponding color attachment
- VUID-VkSubpassDescription2-pResolveAttachments-03067
  If `pResolveAttachments` is not `NULL`, each resolve attachment that is not VK_ATTACHMENT_UNUSED **must** have a sample count of VK_SAMPLE_COUNT_1_BIT
- VUID-VkSubpassDescription2-multisampledRenderToSingleSampled-06869
  If the `multisampledRenderToSingleSampled` feature is not enabled, all attachments in `pColorAttachments` that are not VK_ATTACHMENT_UNUSED **must** have the same sample count
- VUID-VkSubpassDescription2-pInputAttachments-02897

All attachments in pInputAttachments that are not VK_ATTACHMENT_UNUSED

must have image formats whose potential format features contain at least VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT or VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT

- VUID-VkSubpassDescription2-pColorAttachments-02898
  All attachments in pColorAttachments that are not VK_ATTACHMENT_UNUSED **must** have image formats whose potential format features contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT
- VUID-VkSubpassDescription2-pResolveAttachments-02899
  All attachments in pResolveAttachments that are not VK_ATTACHMENT_UNUSED **must** have image formats whose potential format features contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT
- VUID-VkSubpassDescription2-pDepthStencilAttachment-02900
  If pDepthStencilAttachment is not NULL and the attachment is not VK_ATTACHMENT_UNUSED then it **must** have an image format whose potential format features contain VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkSubpassDescription2-pNext-06870
  If the pNext chain includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then all attachments in pColorAttachments and pDepthStencilAttachment that are not VK_ATTACHMENT_UNUSED **must** have a sample count that is either VK_SAMPLE_COUNT_1_BIT or equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-VkSubpassDescription2-pNext-06871
  If the pNext chain includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, and pDepthStencilAttachment is not NULL, does not have the value VK_ATTACHMENT_UNUSED, and has a sample count of VK_SAMPLE_COUNT_1_BIT, the pNext chain **must** also include a VkSubpassDescriptionDepthStencilResolve structure with pDepthStencilResolveAttachment that is either NULL or has the value VK_ATTACHMENT_UNUSED
- VUID-VkSubpassDescription2-multisampledRenderToSingleSampled-06872
  If none of the following are enabled:
  - The multisampledRenderToSingleSampled feature

  all attachments in pDepthStencilAttachment and pColorAttachments that are not VK_ATTACHMENT_UNUSED **must** have the same sample count
- VUID-VkSubpassDescription2-attachment-03073
  Each element of pPreserveAttachments **must** not be VK_ATTACHMENT_UNUSED
- VUID-VkSubpassDescription2-pPreserveAttachments-03074
  Each element of pPreserveAttachments **must** not also be an element of any other member of the subpass description
- VUID-VkSubpassDescription2-layout-02528
  If any attachment is used by more than one VkAttachmentReference2 member, then each

use **must** use the same `layout`

- VUID-VkSubpassDescription2-attachment-02799
  If the `attachment` member of any element of `pInputAttachments` is not VK_ATTACHMENT_UNUSED, then the `aspectMask` member **must** be a valid combination of VkImageAspectFlagBits
- VUID-VkSubpassDescription2-attachment-02800
  If the `attachment` member of any element of `pInputAttachments` is not VK_ATTACHMENT_UNUSED, then the `aspectMask` member **must** not be `0`
- VUID-VkSubpassDescription2-attachment-02801
  If the `attachment` member of any element of `pInputAttachments` is not VK_ATTACHMENT_UNUSED, then the `aspectMask` member **must** not include VK_IMAGE_ASPECT_METADATA_BIT
- VUID-VkSubpassDescription2-attachment-04563
  If the `attachment` member of any element of `pInputAttachments` is not VK_ATTACHMENT_UNUSED, then the `aspectMask` member **must** not include `VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT` for any index *i*
- VUID-VkSubpassDescription2-pDepthStencilAttachment-04440
  An attachment **must** not be used in both `pDepthStencilAttachment` and `pColorAttachments`
- VUID-VkSubpassDescription2-multiview-06558
  If the `multiview` feature is not enabled, `viewMask` **must** be `0`
- VUID-VkSubpassDescription2-viewMask-06706
  The index of the most significant bit in `viewMask` **must** be less than `maxMultiviewViewCount`
- VUID-VkSubpassDescription2-flags-04907
  If `flags` includes VK_SUBPASS_DESCRIPTION_CUSTOM_RESOLVE_BIT_EXT, and if `pResolveAttachments` is not `NULL`, then each resolve attachment **must** be VK_ATTACHMENT_UNUSED
- VUID-VkSubpassDescription2-flags-04908
  If `flags` includes VK_SUBPASS_DESCRIPTION_CUSTOM_RESOLVE_BIT_EXT, and if `pDepthStencilResolveAttachment` is not `NULL`, then the depth/stencil resolve attachment **must** be VK_ATTACHMENT_UNUSED
- VUID-VkSubpassDescription2-flags-04909
  If `flags` includes VK_SUBPASS_DESCRIPTION_CUSTOM_RESOLVE_BIT_EXT, then the subpass **must** be the last subpass in a subpass dependency chain

Valid Usage (Implicit)

- VUID-VkSubpassDescription2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_2
- VUID-VkSubpassDescription2-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkFragmentShadingRateAttachmentInfoKHR, VkMultisampledRenderToSingleSampledInfoEXT, VkRenderPassCreationControlEXT,

VkRenderPassSubpassFeedbackCreateInfoEXT, or VkSubpassDescriptionDepthStencilResolve

- VUID-VkSubpassDescription2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkSubpassDescription2-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkSubpassDescriptionFlagBits values
- VUID-VkSubpassDescription2-pipelineBindPoint-parameter
  `pipelineBindPoint` **must** be a valid VkPipelineBindPoint value
- VUID-VkSubpassDescription2-pInputAttachments-parameter
  If `inputAttachmentCount` is not `0`, `pInputAttachments` **must** be a valid pointer to an array of `inputAttachmentCount` valid VkAttachmentReference2 structures
- VUID-VkSubpassDescription2-pColorAttachments-parameter
  If `colorAttachmentCount` is not `0`, `pColorAttachments` **must** be a valid pointer to an array of `colorAttachmentCount` valid VkAttachmentReference2 structures
- VUID-VkSubpassDescription2-pResolveAttachments-parameter
  If `colorAttachmentCount` is not `0`, and `pResolveAttachments` is not `NULL`, `pResolveAttachments` **must** be a valid pointer to an array of `colorAttachmentCount` valid VkAttachmentReference2 structures
- VUID-VkSubpassDescription2-pDepthStencilAttachment-parameter
  If `pDepthStencilAttachment` is not `NULL`, `pDepthStencilAttachment` **must** be a valid pointer to a valid VkAttachmentReference2 structure
- VUID-VkSubpassDescription2-pPreserveAttachments-parameter
  If `preserveAttachmentCount` is not `0`, `pPreserveAttachments` **must** be a valid pointer to an array of `preserveAttachmentCount` `uint32_t` values

The `VkSubpassDescriptionDepthStencilResolve` structure is defined as:

WARNING This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
typedef struct VkSubpassDescriptionDepthStencilResolve {
    VkStructureType                  sType;
    const void*                      pNext;
    VkResolveModeFlagBits            depthResolveMode;
    VkResolveModeFlagBits            stencilResolveMode;
    const VkAttachmentReference2*    pDepthStencilResolveAttachment;
} VkSubpassDescriptionDepthStencilResolve;
```

```
// Provided by VK_KHR_depth_stencil_resolve
// Equivalent to VkSubpassDescriptionDepthStencilResolve
typedef VkSubpassDescriptionDepthStencilResolve
VkSubpassDescriptionDepthStencilResolveKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `depthResolveMode` is a VkResolveModeFlagBits value describing the depth resolve mode.
- `stencilResolveMode` is a VkResolveModeFlagBits value describing the stencil resolve mode.
- `pDepthStencilResolveAttachment` is `NULL` or a pointer to a VkAttachmentReference2 structure defining the depth/stencil resolve attachment for this subpass and its layout.

If the `pNext` chain of VkSubpassDescription2 includes a `VkSubpassDescriptionDepthStencilResolve` structure, then that structure describes multisample resolve operations for the depth/stencil attachment in a subpass. If this structure is not included in the `pNext` chain of VkSubpassDescription2, or if it is and either `pDepthStencilResolveAttachment` is `NULL` or its attachment index is VK_ATTACHMENT_UNUSED, it indicates that no depth/stencil resolve attachment will be used in the subpass.

Valid Usage

- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03177
  If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value VK_ATTACHMENT_UNUSED, `pDepthStencilAttachment` **must** not be `NULL` or have the value VK_ATTACHMENT_UNUSED
- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03179
  If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value VK_ATTACHMENT_UNUSED, `pDepthStencilAttachment` **must** not have a sample count of VK_SAMPLE_COUNT_1_BIT
- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03180
  If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value VK_ATTACHMENT_UNUSED, `pDepthStencilResolveAttachment` **must** have a sample count of VK_SAMPLE_COUNT_1_BIT
- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-02651
  If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value VK_ATTACHMENT_UNUSED then it **must** have an image format whose potential format features contain VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03181
  If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value VK_ATTACHMENT_UNUSED and VkFormat of `pDepthStencilResolveAttachment` has a depth component, then the VkFormat of `pDepthStencilAttachment` **must** have a depth component with the same number of bits and numeric format
- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03182

If pDepthStencilResolveAttachment is not NULL and does not have the value VK_ATTACHMENT_UNUSED, and VkFormat of pDepthStencilResolveAttachment has a stencil component, then the VkFormat of pDepthStencilAttachment **must** have a stencil component with the same number of bits and numeric format

- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03178
  If pDepthStencilResolveAttachment is not NULL and does not have the value VK_ATTACHMENT_UNUSED, depthResolveMode and stencilResolveMode **must** not both be VK_RESOLVE_MODE_NONE
- VUID-VkSubpassDescriptionDepthStencilResolve-depthResolveMode-03183
  If pDepthStencilResolveAttachment is not NULL and does not have the value VK_ATTACHMENT_UNUSED and the VkFormat of pDepthStencilResolveAttachment has a depth component, then the value of depthResolveMode **must** be one of the bits set in VkPhysicalDeviceDepthStencilResolveProperties::supportedDepthResolveModes or VK_RESOLVE_MODE_NONE
- VUID-VkSubpassDescriptionDepthStencilResolve-stencilResolveMode-03184
  If pDepthStencilResolveAttachment is not NULL and does not have the value VK_ATTACHMENT_UNUSED and the VkFormat of pDepthStencilResolveAttachment has a stencil component, then the value of stencilResolveMode **must** be one of the bits set in VkPhysicalDeviceDepthStencilResolveProperties::supportedStencilResolveModes or VK_RESOLVE_MODE_NONE
- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03185
  If pDepthStencilResolveAttachment is not NULL and does not have the value VK_ATTACHMENT_UNUSED, the VkFormat of pDepthStencilResolveAttachment has both depth and stencil components, VkPhysicalDeviceDepthStencilResolveProperties::independentResolve is VK_FALSE, and VkPhysicalDeviceDepthStencilResolveProperties::independentResolveNone is VK_FALSE, then the values of depthResolveMode and stencilResolveMode **must** be identical
- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03186
  If pDepthStencilResolveAttachment is not NULL and does not have the value VK_ATTACHMENT_UNUSED, the VkFormat of pDepthStencilResolveAttachment has both depth and stencil components, VkPhysicalDeviceDepthStencilResolveProperties::independentResolve is VK_FALSE and VkPhysicalDeviceDepthStencilResolveProperties::independentResolveNone is VK_TRUE, then the values of depthResolveMode and stencilResolveMode **must** be identical or one of them **must** be VK_RESOLVE_MODE_NONE
- VUID-VkSubpassDescriptionDepthStencilResolve-pNext-06873
  If the pNext chain of VkSubpassDescription2 includes a VkMultisampledRenderToSingleSampledInfoEXT structure, the multisampledRenderToSingleSampledEnable field is VK_TRUE, and pDepthStencilAttachment is not NULL and does not have the value VK_ATTACHMENT_UNUSED, depthResolveMode and stencilResolveMode **must** not both be VK_RESOLVE_MODE_NONE
- VUID-VkSubpassDescriptionDepthStencilResolve-pNext-06874
  If the pNext chain of VkSubpassDescription2 includes a VkMultisampledRenderToSingleSampledInfoEXT structure whose multisampledRenderToSingleSampledEnable field is VK_TRUE, and pDepthStencilAttachment is not NULL, does not have the value VK_ATTACHMENT_UNUSED, and has a VkFormat that

has a depth component, then the value of depthResolveMode **must** be one of the bits set in VkPhysicalDeviceDepthStencilResolveProperties::supportedDepthResolveModes or VK_RESOLVE_MODE_NONE

- VUID-VkSubpassDescriptionDepthStencilResolve-pNext-06875
  If the pNext chain of VkSubpassDescription2 includes a VkMultisampledRenderToSingleSampledInfoEXT structure whose multisampledRenderToSingleSampledEnable field is VK_TRUE, and pDepthStencilAttachment is not NULL, does not have the value VK_ATTACHMENT_UNUSED, and has a VkFormat with a stencil component, then the value of stencilResolveMode **must** be one of the bits set in VkPhysicalDeviceDepthStencilResolveProperties::supportedStencilResolveModes or VK_RESOLVE_MODE_NONE
- VUID-VkSubpassDescriptionDepthStencilResolve-pNext-06876
  If the pNext chain of VkSubpassDescription2 includes a VkMultisampledRenderToSingleSampledInfoEXT structure whose multisampledRenderToSingleSampledEnable field is VK_TRUE, pDepthStencilAttachment is not NULL, does not have the value VK_ATTACHMENT_UNUSED, and has a VkFormat with both depth and stencil components, and both VkPhysicalDeviceDepthStencilResolveProperties::independentResolve and VkPhysicalDeviceDepthStencilResolveProperties::independentResolveNone are VK_FALSE, then the values of depthResolveMode and stencilResolveMode **must** be identical
- VUID-VkSubpassDescriptionDepthStencilResolve-pNext-06877
  If the pNext chain of VkSubpassDescription2 includes a VkMultisampledRenderToSingleSampledInfoEXT structure whose multisampledRenderToSingleSampledEnable field is VK_TRUE, pDepthStencilAttachment is not NULL, does not have the value VK_ATTACHMENT_UNUSED, and has a VkFormat with both depth and stencil components, VkPhysicalDeviceDepthStencilResolveProperties::independentResolve is VK_FALSE, and VkPhysicalDeviceDepthStencilResolveProperties::independentResolveNone is VK_TRUE, then the values of depthResolveMode and stencilResolveMode **must** be identical or one of them **must** be VK_RESOLVE_MODE_NONE

Valid Usage (Implicit)

- VUID-VkSubpassDescriptionDepthStencilResolve-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_DEPTH_STENCIL_RESOLVE
- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-parameter
  If pDepthStencilResolveAttachment is not NULL, pDepthStencilResolveAttachment **must** be a valid pointer to a valid VkAttachmentReference2 structure

Structure Chaining

Extends the structure

- VkSubpassDescription2

The VkFragmentShadingRateAttachmentInfoKHR structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4 and VK_KHR_dynamic_rendering_local_read. See Legacy Functionality for more information.

```
// Provided by VK_KHR_fragment_shading_rate
typedef struct VkFragmentShadingRateAttachmentInfoKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    const VkAttachmentReference2*   pFragmentShadingRateAttachment;
    VkExtent2D                      shadingRateAttachmentTexelSize;
} VkFragmentShadingRateAttachmentInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pFragmentShadingRateAttachment is NULL or a pointer to a VkAttachmentReference2 structure defining the fragment shading rate attachment for this subpass.
- shadingRateAttachmentTexelSize specifies the size of the portion of the framebuffer corresponding to each texel in pFragmentShadingRateAttachment.

If no shading rate attachment is specified, or if this structure is not specified, the implementation behaves as if a valid shading rate attachment was specified with all texels specifying a single pixel per fragment.

Valid Usage

- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04524
  If pFragmentShadingRateAttachment is not NULL and its attachment member is not VK_ATTACHMENT_UNUSED, its layout member **must** be equal to VK_IMAGE_LAYOUT_GENERAL or VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR
- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04525
  If pFragmentShadingRateAttachment is not NULL and its attachment member is not VK_ATTACHMENT_UNUSED, shadingRateAttachmentTexelSize.width **must** be a power of two value
- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04526
  If pFragmentShadingRateAttachment is not NULL and its attachment member is not VK_ATTACHMENT_UNUSED, shadingRateAttachmentTexelSize.width **must** be less than or equal to maxFragmentShadingRateAttachmentTexelSize.width
- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-

04527
If pFragmentShadingRateAttachment is not NULL and its attachment member is not VK_ATTACHMENT_UNUSED, shadingRateAttachmentTexelSize.width **must** be greater than or equal to minFragmentShadingRateAttachmentTexelSize.width

- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04528
  If pFragmentShadingRateAttachment is not NULL and its attachment member is not VK_ATTACHMENT_UNUSED, shadingRateAttachmentTexelSize.height **must** be a power of two value
- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04529
  If pFragmentShadingRateAttachment is not NULL and its attachment member is not VK_ATTACHMENT_UNUSED, shadingRateAttachmentTexelSize.height **must** be less than or equal to maxFragmentShadingRateAttachmentTexelSize.height
- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04530
  If pFragmentShadingRateAttachment is not NULL and its attachment member is not VK_ATTACHMENT_UNUSED, shadingRateAttachmentTexelSize.height **must** be greater than or equal to minFragmentShadingRateAttachmentTexelSize.height
- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04531
  If pFragmentShadingRateAttachment is not NULL and its attachment member is not VK_ATTACHMENT_UNUSED, the quotient of shadingRateAttachmentTexelSize.width and shadingRateAttachmentTexelSize.height **must** be less than or equal to maxFragmentShadingRateAttachmentTexelSizeAspectRatio
- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04532
  If pFragmentShadingRateAttachment is not NULL and its attachment member is not VK_ATTACHMENT_UNUSED, the quotient of shadingRateAttachmentTexelSize.height and shadingRateAttachmentTexelSize.width **must** be less than or equal to maxFragmentShadingRateAttachmentTexelSizeAspectRatio

Valid Usage (Implicit)

- VUID-VkFragmentShadingRateAttachmentInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_FRAGMENT_SHADING_RATE_ATTACHMENT_INFO_KHR
- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-parameter
  If pFragmentShadingRateAttachment is not NULL, pFragmentShadingRateAttachment **must** be a valid pointer to a valid VkAttachmentReference2 structure

Structure Chaining

Extends the structure

- VkSubpassDescription2

If the pNext chain of VkSubpassDescription2 or VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure, then that structure describes how multisampled rendering is performed on single sampled attachments in that subpass.

The VkMultisampledRenderToSingleSampledInfoEXT structure is defined as:

```
// Provided by VK_EXT_multisampled_render_to_single_sampled
typedef struct VkMultisampledRenderToSingleSampledInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkBool32                 multisampledRenderToSingleSampledEnable;
    VkSampleCountFlagBits    rasterizationSamples;
} VkMultisampledRenderToSingleSampledInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- multisampledRenderToSingleSampledEnable controls whether multisampled rendering to single-sampled attachments is performed as described below.
- rasterizationSamples is a VkSampleCountFlagBits specifying the number of samples used in rasterization.

Valid Usage

- VUID-VkMultisampledRenderToSingleSampledInfoEXT-rasterizationSamples-06878
  The value of rasterizationSamples **must** not be VK_SAMPLE_COUNT_1_BIT
- VUID-VkMultisampledRenderToSingleSampledInfoEXT-pNext-06880
  If added to the pNext chain of VkRenderingInfo, each imageView member of any element of VkRenderingInfo::pColorAttachments, VkRenderingInfo::pDepthAttachment, or VkRenderingInfo::pStencilAttachment that is not VK_NULL_HANDLE **must** have a format that supports the sample count specified in rasterizationSamples

Valid Usage (Implicit)

- VUID-VkMultisampledRenderToSingleSampledInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_INFO_EXT
- VUID-VkMultisampledRenderToSingleSampledInfoEXT-rasterizationSamples-parameter
  rasterizationSamples **must** be a valid VkSampleCountFlagBits value

Structure Chaining

Extends the structures

- VkRenderingInfo
- VkSubpassDescription2

If the pNext chain of VkSubpassDescription2 or VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure whose multisampledRenderToSingleSampledEnable field is VK_TRUE, the graphics pipelines **must** have VkGraphicsPipelineCreateInfo::rasterizationSamples equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples, and the subpass attachments **can** have a sample count of VK_SAMPLE_COUNT_1_BIT. For attachments with a sample count of VK_SAMPLE_COUNT_1_BIT, multisampled rendering is performed to an intermediate multisampled image with VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples samples, implicitly allocated by the implementation for the duration of the subpass. For such attachments:

- If loadOp is equal to VK_ATTACHMENT_LOAD_OP_LOAD, samples of the implicit image are initialized by replicating the value from the corresponding pixel in the attachment.
- If storeOp or stencilStoreOp is equal to VK_ATTACHMENT_STORE_OP_STORE, the implicit image is implicitly resolved prior to storage in the attachment.

Memory constraints due to high primitive counts **may** result in an implicit split of the subpass. This is the equivalent of partial rasterization of geometry in a render pass that ends in storeOp and stencilStoreOp equal to VK_ATTACHMENT_STORE_OP_STORE, followed by another render pass with loadOp and stencilLoadOp equal to VK_ATTACHMENT_LOAD_OP_LOAD with appropriate barriers in between. When VkMultisampledRenderToSingleSampledInfoEXT is used, the implementation is allowed to resolve attachments with a sample count of VK_SAMPLE_COUNT_1_BIT and lose multisampled data on such splits. The implementation **may** similarly split the render pass at subpass boundaries even if they use the same value for VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples.

The VkAttachmentReference2 structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
typedef struct VkAttachmentReference2 {
    VkStructureType       sType;
    const void*           pNext;
    uint32_t              attachment;
    VkImageLayout         layout;
    VkImageAspectFlags    aspectMask;
} VkAttachmentReference2;
```

```
// Provided by VK_KHR_create_renderpass2
// Equivalent to VkAttachmentReference2
typedef VkAttachmentReference2 VkAttachmentReference2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `attachment` is either an integer value identifying an attachment at the corresponding index in VkRenderPassCreateInfo2::`pAttachments`, or VK_ATTACHMENT_UNUSED to signify that this attachment is not used.
- `layout` is a VkImageLayout value specifying the layout the attachment uses during the subpass.
- `aspectMask` is a mask of which aspect(s) **can** be accessed within the specified subpass as an input attachment.

Parameters defined by this structure with the same name as those in VkAttachmentReference have the identical effect to those parameters.

`aspectMask` is ignored when this structure is used to describe anything other than an input attachment reference.

If the `separateDepthStencilLayouts` feature is enabled, and `attachment` has a depth/stencil format, `layout` **can** be set to a layout that only specifies the layout of the depth aspect.

If `layout` only specifies the layout of the depth aspect of the attachment, the layout of the stencil aspect is specified by the `stencilLayout` member of a VkAttachmentReferenceStencilLayout structure included in the `pNext` chain. Otherwise, `layout` describes the layout for all relevant image aspects.

Valid Usage

- VUID-VkAttachmentReference2-layout-03077
  If `attachment` is not VK_ATTACHMENT_UNUSED, `layout` **must** not be VK_IMAGE_LAYOUT_UNDEFINED, VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT, VK_IMAGE_LAYOUT_PREINITIALIZED, or VK_IMAGE_LAYOUT_PRESENT_SRC_KHR
- VUID-VkAttachmentReference2-separateDepthStencilLayouts-03313
  If the `separateDepthStencilLayouts` feature is not enabled, and `attachment` is not VK_ATTACHMENT_UNUSED, `layout` **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VUID-VkAttachmentReference2-synchronization2-06910
  If the `synchronization2` feature is not enabled, `layout` **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR or VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VUID-VkAttachmentReference2-attachmentFeedbackLoopLayout-07311

If the attachmentFeedbackLoopLayout feature is not enabled, layout **must** not be VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT

- VUID-VkAttachmentReference2-dynamicRenderingLocalRead-09546
  If the dynamicRenderingLocalRead feature is not enabled, layout **must** not be VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ

Valid Usage (Implicit)

- VUID-VkAttachmentReference2-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_2
- VUID-VkAttachmentReference2-pNext-pNext
  pNext **must** be NULL or a pointer to a valid instance of VkAttachmentReferenceStencilLayout
- VUID-VkAttachmentReference2-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkAttachmentReference2-layout-parameter
  layout **must** be a valid VkImageLayout value

The VkAttachmentReferenceStencilLayout structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkAttachmentReferenceStencilLayout {
    VkStructureType    sType;
    void*              pNext;
    VkImageLayout      stencilLayout;
} VkAttachmentReferenceStencilLayout;
```

```
// Provided by VK_KHR_separate_depth_stencil_layouts
// Equivalent to VkAttachmentReferenceStencilLayout
typedef VkAttachmentReferenceStencilLayout VkAttachmentReferenceStencilLayoutKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- stencilLayout is a VkImageLayout value specifying the layout the stencil aspect of the attachment uses during the subpass.

Valid Usage

- VUID-VkAttachmentReferenceStencilLayout-stencilLayout-03318
  stencilLayout **must** not be VK_IMAGE_LAYOUT_UNDEFINED, VK_IMAGE_LAYOUT_PREINITIALIZED, VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL,

VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL,
VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL,
VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL,
VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL,
VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL,
VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL, or
VK_IMAGE_LAYOUT_PRESENT_SRC_KHR

Valid Usage (Implicit)

- VUID-VkAttachmentReferenceStencilLayout-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_STENCIL_LAYOUT
- VUID-VkAttachmentReferenceStencilLayout-stencilLayout-parameter
  `stencilLayout` **must** be a valid VkImageLayout value

Structure Chaining

Extends the structure

- VkAttachmentReference2

The `VkSubpassDependency2` structure is defined as:

WARNING This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
typedef struct VkSubpassDependency2 {
    VkStructureType         sType;
    const void*             pNext;
    uint32_t                srcSubpass;
    uint32_t                dstSubpass;
    VkPipelineStageFlags    srcStageMask;
    VkPipelineStageFlags    dstStageMask;
    VkAccessFlags           srcAccessMask;
    VkAccessFlags           dstAccessMask;
    VkDependencyFlags       dependencyFlags;
    int32_t                 viewOffset;
} VkSubpassDependency2;
```

```
// Provided by VK_KHR_create_renderpass2
// Equivalent to VkSubpassDependency2
typedef VkSubpassDependency2 VkSubpassDependency2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcSubpass` is the subpass index of the first subpass in the dependency, or VK_SUBPASS_EXTERNAL.
- `dstSubpass` is the subpass index of the second subpass in the dependency, or VK_SUBPASS_EXTERNAL.
- `srcStageMask` is a bitmask of VkPipelineStageFlagBits specifying the source stage mask. If set to VK_PIPELINE_STAGE_ALL_COMMANDS_BIT, it is equivalent to setting it to VK_PIPELINE_STAGE_ALL_GRAPHICS_BIT.
- `dstStageMask` is a bitmask of VkPipelineStageFlagBits specifying the destination stage mask. If set to VK_PIPELINE_STAGE_ALL_COMMANDS_BIT, it is equivalent to setting it to VK_PIPELINE_STAGE_ALL_GRAPHICS_BIT.
- `srcAccessMask` is a bitmask of VkAccessFlagBits specifying a source access mask.
- `dstAccessMask` is a bitmask of VkAccessFlagBits specifying a destination access mask.
- `dependencyFlags` is a bitmask of VkDependencyFlagBits.
- `viewOffset` controls which views in the source subpass the views in the destination subpass depend on.

Parameters defined by this structure with the same name as those in VkSubpassDependency have the identical effect to those parameters.

`viewOffset` has the same effect for the described subpass dependency as VkRenderPassMultiviewCreateInfo::`pViewOffsets` has on each corresponding subpass dependency.

If a VkMemoryBarrier2 is included in the `pNext` chain, `srcStageMask`, `dstStageMask`, `srcAccessMask`, and `dstAccessMask` parameters are ignored. The synchronization and access scopes instead are defined by the parameters of VkMemoryBarrier2. If either `srcStageMask` or `dstStageMask` are set to VK_PIPELINE_STAGE_ALL_COMMANDS_BIT, it is equivalent to setting VK_PIPELINE_STAGE_ALL_GRAPHICS_BIT.

Valid Usage

- VUID-VkSubpassDependency2-srcStageMask-04090
  If the `geometryShader` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT
- VUID-VkSubpassDependency2-srcStageMask-04091
  If the `tessellationShader` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-VkSubpassDependency2-srcStageMask-04092
  If the `conditionalRendering` feature is not enabled, `srcStageMask` **must** not contain VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT
- VUID-VkSubpassDependency2-srcStageMask-04093

If the fragmentDensityMap feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT

- VUID-VkSubpassDependency2-srcStageMask-04094
  If the transformFeedback feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-VkSubpassDependency2-srcStageMask-04095
  If the meshShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT
- VUID-VkSubpassDependency2-srcStageMask-04096
  If the taskShader feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT
- VUID-VkSubpassDependency2-srcStageMask-07319
  If the attachmentFragmentShadingRate feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkSubpassDependency2-srcStageMask-03937
  If the synchronization2 feature is not enabled, srcStageMask **must** not be 0
- VUID-VkSubpassDependency2-srcStageMask-07950
  If the rayTracingPipeline feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkSubpassDependency2-srcStageMask-10754
  If the accelerationStructure feature is not enabled, srcStageMask **must** not contain VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkSubpassDependency2-dstStageMask-04090
  If the geometryShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT
- VUID-VkSubpassDependency2-dstStageMask-04091
  If the tessellationShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-VkSubpassDependency2-dstStageMask-04092
  If the conditionalRendering feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT
- VUID-VkSubpassDependency2-dstStageMask-04093
  If the fragmentDensityMap feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-VkSubpassDependency2-dstStageMask-04094
  If the transformFeedback feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-VkSubpassDependency2-dstStageMask-04095
  If the meshShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT
- VUID-VkSubpassDependency2-dstStageMask-04096

If the taskShader feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT

- VUID-VkSubpassDependency2-dstStageMask-07319
  If the attachmentFragmentShadingRate feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkSubpassDependency2-dstStageMask-03937
  If the synchronization2 feature is not enabled, dstStageMask **must** not be 0
- VUID-VkSubpassDependency2-dstStageMask-07950
  If the rayTracingPipeline feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR
- VUID-VkSubpassDependency2-dstStageMask-10754
  If the accelerationStructure feature is not enabled, dstStageMask **must** not contain VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-VkSubpassDependency2-srcSubpass-03084
  srcSubpass **must** be less than or equal to dstSubpass, unless one of them is VK_SUBPASS_EXTERNAL, to avoid cyclic dependencies and ensure a valid execution order
- VUID-VkSubpassDependency2-srcSubpass-03085
  srcSubpass and dstSubpass **must** not both be equal to VK_SUBPASS_EXTERNAL
- VUID-VkSubpassDependency2-srcSubpass-06810
  If srcSubpass is equal to dstSubpass and srcStageMask includes a framebuffer-space stage, dstStageMask **must** only contain framebuffer-space stages
- VUID-VkSubpassDependency2-srcAccessMask-03088
  Any access flag included in srcAccessMask **must** be supported by one of the pipeline stages in srcStageMask, as specified in the table of supported access types
- VUID-VkSubpassDependency2-dstAccessMask-03089
  Any access flag included in dstAccessMask **must** be supported by one of the pipeline stages in dstStageMask, as specified in the table of supported access types
- VUID-VkSubpassDependency2-dependencyFlags-03090
  If dependencyFlags includes VK_DEPENDENCY_VIEW_LOCAL_BIT, srcSubpass **must** not be equal to VK_SUBPASS_EXTERNAL
- VUID-VkSubpassDependency2-dependencyFlags-03091
  If dependencyFlags includes VK_DEPENDENCY_VIEW_LOCAL_BIT, dstSubpass **must** not be equal to VK_SUBPASS_EXTERNAL
- VUID-VkSubpassDependency2-srcSubpass-02245
  If srcSubpass equals dstSubpass, and srcStageMask and dstStageMask both include a framebuffer-space stage, then dependencyFlags **must** include VK_DEPENDENCY_BY_REGION_BIT
- VUID-VkSubpassDependency2-viewOffset-02530
  If viewOffset is not equal to 0, srcSubpass **must** not be equal to dstSubpass
- VUID-VkSubpassDependency2-dependencyFlags-03092
  If dependencyFlags does not include VK_DEPENDENCY_VIEW_LOCAL_BIT, viewOffset **must** be 0

- VUID-VkSubpassDependency2-dependencyFlags-10204
  `dependencyFlags` **must** not include `VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR`

Valid Usage (Implicit)

- VUID-VkSubpassDependency2-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_SUBPASS_DEPENDENCY_2`
- VUID-VkSubpassDependency2-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkMemoryBarrier2` or `VkMemoryBarrierAccessFlags3KHR`
- VUID-VkSubpassDependency2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkSubpassDependency2-srcStageMask-parameter
  If `srcStageMask` is not `0`, `srcStageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values
- VUID-VkSubpassDependency2-dstStageMask-parameter
  If `dstStageMask` is not `0`, `dstStageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values
- VUID-VkSubpassDependency2-srcAccessMask-parameter
  If `srcAccessMask` is not `0`, `srcAccessMask` **must** be a valid combination of `VkAccessFlagBits` values
- VUID-VkSubpassDependency2-dstAccessMask-parameter
  If `dstAccessMask` is not `0`, `dstAccessMask` **must** be a valid combination of `VkAccessFlagBits` values
- VUID-VkSubpassDependency2-dependencyFlags-parameter
  If `dependencyFlags` is not `0`, `dependencyFlags` **must** be a valid combination of `VkDependencyFlagBits` values

To destroy a render pass, call:

WARNING	This functionality is superseded by `Vulkan Version 1.4`. See `Legacy Functionality` for more information.

```
// Provided by VK_VERSION_1_0
void vkDestroyRenderPass(
    VkDevice                                    device,
    VkRenderPass                                renderPass,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the render pass.

- `renderPass` is the handle of the render pass to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyRenderPass-renderPass-00873
  All submitted commands that refer to `renderPass` **must** have completed execution
- VUID-vkDestroyRenderPass-renderPass-00874
  If `VkAllocationCallbacks` were provided when `renderPass` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyRenderPass-renderPass-00875
  If no `VkAllocationCallbacks` were provided when `renderPass` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyRenderPass-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyRenderPass-renderPass-parameter
  If `renderPass` is not VK_NULL_HANDLE, `renderPass` **must** be a valid VkRenderPass handle
- VUID-vkDestroyRenderPass-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyRenderPass-renderPass-parent
  If `renderPass` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `renderPass` **must** be externally synchronized

8.3. Render Pass Compatibility

Framebuffers and graphics pipelines are created based on a specific render pass object. They **must** only be used with that render pass object, or one compatible with it.

Two attachment references are compatible if they have matching format and sample count, or are both VK_ATTACHMENT_UNUSED.

Two arrays of attachment references are compatible if all corresponding pairs of attachments are compatible. If the arrays are of different lengths, attachment references not present in the smaller array are treated as VK_ATTACHMENT_UNUSED. If the pointer that would contain the reference is

NULL, the attachment reference is treated as VK_ATTACHMENT_UNUSED.

Two render passes are compatible if their corresponding color, input, resolve, and depth/stencil attachment references are compatible and if they are otherwise identical except for:

- Initial and final image layout in attachment descriptions
- Load and store operations in attachment descriptions
- Image layout in attachment references
- The inclusion of VK_ATTACHMENT_DESCRIPTION_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR
- The inclusion of VK_ATTACHMENT_DESCRIPTION_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR

As an additional special case, if two render passes have a single subpass, the resolve attachment reference and depth/stencil resolve mode compatibility requirements are ignored.

A framebuffer is compatible with a render pass if it was created using the same render pass or a compatible render pass.

8.4. Framebuffers

Render passes operate in conjunction with *framebuffers*. Framebuffers represent a collection of specific memory attachments that a render pass instance uses.

Framebuffers are represented by VkFramebuffer handles:

> **WARNING** This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkFramebuffer)
```

To create a framebuffer, call:

> **WARNING** This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
VkResult vkCreateFramebuffer(
    VkDevice                                    device,
    const VkFramebufferCreateInfo*              pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkFramebuffer*                              pFramebuffer);
```

- device is the logical device that creates the framebuffer.

- `pCreateInfo` is a pointer to a VkFramebufferCreateInfo structure describing additional information about framebuffer creation.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pFramebuffer` is a pointer to a VkFramebuffer handle in which the resulting framebuffer object is returned.

Valid Usage

- VUID-vkCreateFramebuffer-device-10002
  `device` **must** support at least one queue family with the VK_QUEUE_GRAPHICS_BIT capability
- VUID-vkCreateFramebuffer-pCreateInfo-02777
  If `pCreateInfo->flags` does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, and `attachmentCount` is not `0`, each element of `pCreateInfo->pAttachments` **must** have been created on `device`

Valid Usage (Implicit)

- VUID-vkCreateFramebuffer-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateFramebuffer-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkFramebufferCreateInfo structure
- VUID-vkCreateFramebuffer-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateFramebuffer-pFramebuffer-parameter
  `pFramebuffer` **must** be a valid pointer to a VkFramebuffer handle
- VUID-vkCreateFramebuffer-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkFramebufferCreateInfo structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef struct VkFramebufferCreateInfo {
    VkStructureType             sType;
    const void*                 pNext;
    VkFramebufferCreateFlags    flags;
    VkRenderPass                renderPass;
    uint32_t                    attachmentCount;
    const VkImageView*          pAttachments;
    uint32_t                    width;
    uint32_t                    height;
    uint32_t                    layers;
} VkFramebufferCreateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask of VkFramebufferCreateFlagBits.
- renderPass is a render pass defining what render passes the framebuffer will be compatible with. See Render Pass Compatibility for details. The implementation **must** not access this object outside of the duration of the command this structure is passed to.
- attachmentCount is the number of attachments.
- pAttachments is a pointer to an array of VkImageView handles, each of which will be used as the corresponding attachment in a render pass instance. If flags includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, this parameter is ignored.
- width, height and layers define the dimensions of the framebuffer. If the render pass uses multiview, then layers **must** be one and each attachment requires a number of layers that is greater than the maximum bit index set in the view mask in the subpasses in which it is used.

For any depth/stencil attachments used by this framebuffer in pAttachments, or set later through VkRenderPassAttachmentBeginInfoKHR::pAttachments, the aspectMask is ignored.

It is legal for a subpass to use no color or depth/stencil attachments, either because it has no attachment references or because all of them are VK_ATTACHMENT_UNUSED. This kind of subpass **can** use shader side effects such as image stores and atomics to produce an output. In this case, the subpass continues to use the width, height, and layers of the framebuffer to define the dimensions of the rendering area, and the rasterizationSamples from each pipeline's VkPipelineMultisampleStateCreateInfo to define the number of samples used in rasterization; however, if VkPhysicalDeviceFeatures::variableMultisampleRate is VK_FALSE, then all pipelines to be bound with the subpass **must** have the same value for VkPipelineMultisampleStateCreateInfo::rasterizationSamples. In all such cases, rasterizationSamples **must** be a valid VkSampleCountFlagBits value that is set in VkPhysicalDeviceLimits

::framebufferNoAttachmentsSampleCounts.

Valid Usage

- VUID-VkFramebufferCreateInfo-attachmentCount-00876
  attachmentCount **must** be equal to the attachment count specified in renderPass
- VUID-VkFramebufferCreateInfo-flags-02778
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT and attachmentCount is not 0, pAttachments **must** be a valid pointer to an array of attachmentCount valid VkImageView handles
- VUID-VkFramebufferCreateInfo-pAttachments-00877
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of pAttachments that is used as a color attachment or resolve attachment by renderPass **must** have been created with the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT usage flag set
- VUID-VkFramebufferCreateInfo-pAttachments-02633
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of pAttachments that is used as a depth/stencil attachment by renderPass **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkFramebufferCreateInfo-pAttachments-02634
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of pAttachments that is used as a depth/stencil resolve attachment by renderPass **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-VkFramebufferCreateInfo-pAttachments-00879
  If renderpass is not VK_NULL_HANDLE, flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of pAttachments that is used as an input attachment by renderPass **must** have been created with the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flag set
- VUID-VkFramebufferCreateInfo-pAttachments-02552
  Each element of pAttachments that is used as a fragment density map attachment by renderPass **must** not have been created with a flags value including VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-VkFramebufferCreateInfo-renderPass-02553
  If renderPass has a fragment density map attachment and the fragmentDensityMapNonSubsampledImages feature is not enabled, each element of pAttachments **must** have been created with a flags value including VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT unless that element is the fragment density map attachment
- VUID-VkFramebufferCreateInfo-pAttachments-00880
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of pAttachments **must** have been created with a VkFormat value that matches the VkFormat specified by the corresponding VkAttachmentDescription in renderPass
- VUID-VkFramebufferCreateInfo-pAttachments-00881
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of

pAttachments **must** have been created with a samples value that matches the samples value specified by the corresponding VkAttachmentDescription in renderPass

- VUID-VkFramebufferCreateInfo-flags-04533
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of pAttachments that is used as an input, color, resolve, or depth/stencil attachment by renderPass **must** have been created with a VkImageCreateInfo::extent.width greater than or equal to width
- VUID-VkFramebufferCreateInfo-flags-04534
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of pAttachments that is used as an input, color, resolve, or depth/stencil attachment by renderPass **must** have been created with a VkImageCreateInfo::extent.height greater than or equal to height
- VUID-VkFramebufferCreateInfo-flags-04535
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of pAttachments that is used as an input, color, resolve, or depth/stencil attachment by renderPass **must** have been created with a VkImageViewCreateInfo ::subresourceRange.layerCount greater than or equal to layers
- VUID-VkFramebufferCreateInfo-renderPass-04536
  If renderPass was specified with non-zero view masks, each element of pAttachments that is used as an input, color, resolve, or depth/stencil attachment by renderPass **must** have a layerCount greater than the index of the most significant bit set in any of those view masks
- VUID-VkFramebufferCreateInfo-renderPass-02746
  Each element of pAttachments that is referenced by fragmentDensityMapAttachment **must** have a layerCount equal to 1 or if renderPass was specified with non-zero view masks, greater than the index of the most significant bit set in any of those view masks
- VUID-VkFramebufferCreateInfo-pAttachments-02555
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, an element of pAttachments that is referenced by fragmentDensityMapAttachment **must** have a width at least as large as $\lceil \frac{width}{maxFragmentDensityTexelSize_{width}} \rceil$
- VUID-VkFramebufferCreateInfo-pAttachments-02556
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, an element of pAttachments that is referenced by fragmentDensityMapAttachment **must** have a height at least as large as $\lceil \frac{height}{maxFragmentDensityTexelSize_{height}} \rceil$
- VUID-VkFramebufferCreateInfo-flags-04537
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, and renderPass was specified with non-zero view masks, each element of pAttachments that is used as a fragment shading rate attachment by renderPass **must** have a layerCount that is either 1, or greater than the index of the most significant bit set in any of those view masks
- VUID-VkFramebufferCreateInfo-flags-04538
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, and renderPass was not specified with non-zero view masks, each element of pAttachments that is used as a fragment shading rate attachment by renderPass **must** have a layerCount that is either 1, or greater than layers
- VUID-VkFramebufferCreateInfo-flags-04539

If the maintenance7 feature is not enabled or the robustFragmentShadingRateAttachmentAccess limit is VK_FALSE or the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure was created with VkImageSubresourceRange::baseMipLevel greater than 0, flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, an element of pAttachments that is used as a fragment shading rate attachment **must** have a width at least as large as ⌈width / texelWidth⌉, where texelWidth is the largest value of shadingRateAttachmentTexelSize.width in a VkFragmentShadingRateAttachmentInfoKHR which references that attachment

- VUID-VkFramebufferCreateInfo-flags-04540
  If the maintenance7 feature is not enabled or the robustFragmentShadingRateAttachmentAccess limit is VK_FALSE or the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure was created with VkImageSubresourceRange::baseMipLevel greater than 0, flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, an element of pAttachments that is used as a fragment shading rate attachment **must** have a height at least as large as ⌈height / texelHeight⌉, where texelHeight is the largest value of shadingRateAttachmentTexelSize.height in a VkFragmentShadingRateAttachmentInfoKHR which references that attachment
- VUID-VkFramebufferCreateInfo-pAttachments-00883
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of pAttachments **must** only specify a single mip level
- VUID-VkFramebufferCreateInfo-pAttachments-00884
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of pAttachments **must** have been created with the identity swizzle
- VUID-VkFramebufferCreateInfo-width-00885
  width **must** be greater than 0
- VUID-VkFramebufferCreateInfo-width-00886
  width **must** be less than or equal to maxFramebufferWidth
- VUID-VkFramebufferCreateInfo-height-00887
  height **must** be greater than 0
- VUID-VkFramebufferCreateInfo-height-00888
  height **must** be less than or equal to maxFramebufferHeight
- VUID-VkFramebufferCreateInfo-layers-00889
  layers **must** be greater than 0
- VUID-VkFramebufferCreateInfo-layers-00890
  layers **must** be less than or equal to maxFramebufferLayers
- VUID-VkFramebufferCreateInfo-renderPass-02531
  If renderPass was specified with non-zero view masks, layers **must** be 1
- VUID-VkFramebufferCreateInfo-pAttachments-00891
  If flags does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of pAttachments that is a 2D or 2D array image view taken from a 3D image **must** not be a depth/stencil format

- VUID-VkFramebufferCreateInfo-flags-03189
  If the imagelessFramebuffer feature is not enabled, flags **must** not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT
- VUID-VkFramebufferCreateInfo-flags-03190
  If flags includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the pNext chain **must** include a VkFramebufferAttachmentsCreateInfo structure
- VUID-VkFramebufferCreateInfo-flags-03191
  If flags includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the attachmentImageInfoCount member of a VkFramebufferAttachmentsCreateInfo structure in the pNext chain **must** be equal to either zero or attachmentCount
- VUID-VkFramebufferCreateInfo-flags-04541
  If flags includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the width member of any element of the pAttachmentImageInfos member of a VkFramebufferAttachmentsCreateInfo structure in the pNext chain that is used as an input, color, resolve, or depth/stencil attachment in renderPass **must** be greater than or equal to width
- VUID-VkFramebufferCreateInfo-flags-04542
  If flags includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the height member of any element of the pAttachmentImageInfos member of a VkFramebufferAttachmentsCreateInfo structure in the pNext chain that is used as an input, color, resolve, or depth/stencil attachment in renderPass **must** be greater than or equal to height
- VUID-VkFramebufferCreateInfo-flags-03196
  If flags includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the width member of any element of the pAttachmentImageInfos member of a VkFramebufferAttachmentsCreateInfo structure in the pNext chain that is referenced by VkRenderPassFragmentDensityMapCreateInfoEXT::fragmentDensityMapAttachment in renderPass **must** be greater than or equal to $\lceil \frac{width}{maxFragmentDensityTexelSize_{width}} \rceil$
- VUID-VkFramebufferCreateInfo-flags-03197
  If flags includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the height member of any element of the pAttachmentImageInfos member of a VkFramebufferAttachmentsCreateInfo structure included in the pNext chain that is referenced by VkRenderPassFragmentDensityMapCreateInfoEXT::fragmentDensityMapAttachment in renderPass **must** be greater than or equal to $\lceil \frac{height}{maxFragmentDensityTexelSize_{height}} \rceil$
- VUID-VkFramebufferCreateInfo-flags-04543
  If the maintenance7 feature is not enabled or the robustFragmentShadingRateAttachmentAccess limit is VK_FALSE or the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure was created with VkImageSubresourceRange::baseMipLevel greater than 0, and flags includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the width member of any element of the pAttachmentImageInfos member of a VkFramebufferAttachmentsCreateInfo structure in the pNext chain that is used as a fragment shading rate attachment **must** be greater than or equal to ⌈width / texelWidth⌉, where texelWidth is the largest value of shadingRateAttachmentTexelSize.width in a VkFragmentShadingRateAttachmentInfoKHR which references that attachment
- VUID-VkFramebufferCreateInfo-flags-04544
  If the maintenance7 feature is not enabled or the

robustFragmentShadingRateAttachmentAccess limit is VK_FALSE or the imageView member of a VkRenderingFragmentShadingRateAttachmentInfoKHR structure was created with VkImageSubresourceRange::baseMipLevel greater than 0, and flags includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the height member of any element of the pAttachmentImageInfos member of a VkFramebufferAttachmentsCreateInfo structure in the pNext chain that is used as a fragment shading rate attachment **must** be greater than or equal to ⌈height / texelHeight⌉, where texelHeight is the largest value of shadingRateAttachmentTexelSize.height in a VkFragmentShadingRateAttachmentInfoKHR which references that attachment

- VUID-VkFramebufferCreateInfo-flags-04545
  If flags includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the layerCount member of any element of the pAttachmentImageInfos member of a VkFramebufferAttachmentsCreateInfo structure in the pNext chain that is used as a fragment shading rate attachment **must** be either 1, or greater than or equal to layers
- VUID-VkFramebufferCreateInfo-flags-04587
  If flags includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT and renderPass was specified with non-zero view masks, the layerCount member of any element of the pAttachmentImageInfos member of a VkFramebufferAttachmentsCreateInfo structure in the pNext chain that is used as a fragment shading rate attachment **must** be either 1, or greater than the index of the most significant bit set in any of those view masks
- VUID-VkFramebufferCreateInfo-renderPass-03198
  If multiview is enabled for renderPass and flags includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the layerCount member of any element of the pAttachmentImageInfos member of a VkFramebufferAttachmentsCreateInfo structure included in the pNext chain used as an input, color, resolve, or depth/stencil attachment in renderPass **must** be greater than the maximum bit index set in the view mask in the subpasses in which it is used in renderPass
- VUID-VkFramebufferCreateInfo-renderPass-04546
  If multiview is not enabled for renderPass and flags includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the layerCount member of any element of the pAttachmentImageInfos member of a VkFramebufferAttachmentsCreateInfo structure included in the pNext chain used as an input, color, resolve, or depth/stencil attachment in renderPass **must** be greater than or equal to layers
- VUID-VkFramebufferCreateInfo-flags-03201
  If flags includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the usage member of any element of the pAttachmentImageInfos member of a VkFramebufferAttachmentsCreateInfo structure included in the pNext chain that refers to an attachment used as a color attachment or resolve attachment by renderPass **must** include VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT
- VUID-VkFramebufferCreateInfo-flags-03202
  If flags includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the usage member of any element of the pAttachmentImageInfos member of a VkFramebufferAttachmentsCreateInfo structure included in the pNext chain that refers to an attachment used as a depth/stencil attachment by renderPass **must** include VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT

- VUID-VkFramebufferCreateInfo-flags-03203
  If `flags` includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the `usage` member of any element of the `pAttachmentImageInfos` member of a VkFramebufferAttachmentsCreateInfo structure included in the `pNext` chain that refers to an attachment used as a depth/stencil resolve attachment by `renderPass` **must** include VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkFramebufferCreateInfo-flags-03204
  If `flags` includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the `usage` member of any element of the `pAttachmentImageInfos` member of a VkFramebufferAttachmentsCreateInfo structure included in the `pNext` chain that refers to an attachment used as an input attachment by `renderPass` **must** include VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT
- VUID-VkFramebufferCreateInfo-flags-03205
  If `flags` includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, at least one element of the `pViewFormats` member of any element of the `pAttachmentImageInfos` member of a VkFramebufferAttachmentsCreateInfo structure included in the `pNext` chain **must** be equal to the corresponding value of VkAttachmentDescription::`format` used to create `renderPass`
- VUID-VkFramebufferCreateInfo-flags-04113
  If `flags` does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of `pAttachments` **must** have been created with VkImageViewCreateInfo::`viewType` not equal to VK_IMAGE_VIEW_TYPE_3D
- VUID-VkFramebufferCreateInfo-flags-04548
  If `flags` does not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of `pAttachments` that is used as a fragment shading rate attachment by `renderPass` **must** have been created with the VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR usage flag set
- VUID-VkFramebufferCreateInfo-flags-04549
  If `flags` includes VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the `usage` member of any element of the `pAttachmentImageInfos` member of a VkFramebufferAttachmentsCreateInfo structure included in the `pNext` chain that refers to an attachment used as a fragment shading rate attachment by `renderPass` **must** include VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkFramebufferCreateInfo-samples-06881
  If multisampled-render-to-single-sampled is enabled for any subpass, all color, depth/stencil, and input attachments used in that subpass which have `VkAttachmentDescription`::`samples` or `VkAttachmentDescription2`::`samples` equal to VK_SAMPLE_COUNT_1_BIT **must** have been created with VK_IMAGE_CREATE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT in their VkImageCreateInfo::`flags`
- VUID-VkFramebufferCreateInfo-samples-07009
  If multisampled-render-to-single-sampled is enabled for any subpass, all color, depth/stencil, and input attachments used in that subpass which have `VkAttachmentDescription`::`samples` or `VkAttachmentDescription2`::`samples` equal to VK_SAMPLE_COUNT_1_BIT **must** have a format that supports the sample count specified in VkMultisampledRenderToSingleSampledInfoEXT::`rasterizationSamples`

Valid Usage (Implicit)

- VUID-VkFramebufferCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_FRAMEBUFFER_CREATE_INFO
- VUID-VkFramebufferCreateInfo-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkFramebufferAttachmentsCreateInfo
- VUID-VkFramebufferCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkFramebufferCreateInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkFramebufferCreateFlagBits values
- VUID-VkFramebufferCreateInfo-renderPass-parameter
  `renderPass` **must** be a valid VkRenderPass handle
- VUID-VkFramebufferCreateInfo-commonparent
  Both of `renderPass`, and the elements of `pAttachments` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

The `VkFramebufferAttachmentsCreateInfo` structure is defined as:

WARNING This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
typedef struct VkFramebufferAttachmentsCreateInfo {
    VkStructureType                              sType;
    const void*                                  pNext;
    uint32_t                                     attachmentImageInfoCount;
    const VkFramebufferAttachmentImageInfo*      pAttachmentImageInfos;
} VkFramebufferAttachmentsCreateInfo;
```

```
// Provided by VK_KHR_imageless_framebuffer
// Equivalent to VkFramebufferAttachmentsCreateInfo
typedef VkFramebufferAttachmentsCreateInfo VkFramebufferAttachmentsCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `attachmentImageInfoCount` is the number of attachments being described.
- `pAttachmentImageInfos` is a pointer to an array of VkFramebufferAttachmentImageInfo structures, each structure describing a number of parameters of the corresponding attachment in a render pass instance.

Valid Usage (Implicit)

- VUID-VkFramebufferAttachmentsCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENTS_CREATE_INFO
- VUID-VkFramebufferAttachmentsCreateInfo-pAttachmentImageInfos-parameter
  If `attachmentImageInfoCount` is not `0`, `pAttachmentImageInfos` **must** be a valid pointer to an array of `attachmentImageInfoCount` valid VkFramebufferAttachmentImageInfo structures

Structure Chaining

Extends the structure

- VkFramebufferCreateInfo

The `VkFramebufferAttachmentImageInfo` structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
typedef struct VkFramebufferAttachmentImageInfo {
    VkStructureType       sType;
    const void*           pNext;
    VkImageCreateFlags    flags;
    VkImageUsageFlags     usage;
    uint32_t              width;
    uint32_t              height;
    uint32_t              layerCount;
    uint32_t              viewFormatCount;
    const VkFormat*       pViewFormats;
} VkFramebufferAttachmentImageInfo;
```

```
// Provided by VK_KHR_imageless_framebuffer
// Equivalent to VkFramebufferAttachmentImageInfo
typedef VkFramebufferAttachmentImageInfo VkFramebufferAttachmentImageInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkImageCreateFlagBits, matching the value of VkImageCreateInfo::`flags` used to create an image that will be used with this framebuffer.
- `usage` is a bitmask of VkImageUsageFlagBits, matching the value of VkImageCreateInfo::`usage` used to create an image used with this framebuffer.
- `width` is the width of the image view used for rendering.

- `height` is the height of the image view used for rendering.
- `layerCount` is the number of array layers of the image view used for rendering.
- `viewFormatCount` is the number of entries in the `pViewFormats` array, matching the value of VkImageFormatListCreateInfo::`viewFormatCount` used to create an image used with this framebuffer.
- `pViewFormats` is a pointer to an array of VkFormat values specifying all of the formats which **can** be used when creating views of the image, matching the value of VkImageFormatListCreateInfo::`pViewFormats` used to create an image used with this framebuffer.

`flags` defines the effective create flags for the image. If the `pNext` chain includes a VkImageCreateFlags2CreateInfoKHR structure, `flags` is ignored, and the effective create flags are defined by VkImageCreateFlags2CreateInfoKHR::`flags`.

`usage` defines the effective usage flags for the image. If the `pNext` chain includes a VkImageUsageFlags2CreateInfoKHR structure, `usage` is ignored, and the effective usage flags are defined by VkImageUsageFlags2CreateInfoKHR::`flags`.

Images that **can** be used with the framebuffer when beginning a render pass, as specified by VkRenderPassAttachmentBeginInfo, **must** be created with parameters that are identical to those specified here.

Valid Usage

- VUID-VkFramebufferAttachmentImageInfo-viewFormatCount-09536
  If `viewFormatCount` is not 0, each element of `pViewFormats` **must** not be VK_FORMAT_UNDEFINED

Valid Usage (Implicit)

- VUID-VkFramebufferAttachmentImageInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENT_IMAGE_INFO
- VUID-VkFramebufferAttachmentImageInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkImageCreateFlags2CreateInfoKHR or VkImageUsageFlags2CreateInfoKHR
- VUID-VkFramebufferAttachmentImageInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkFramebufferAttachmentImageInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkImageCreateFlagBits values
- VUID-VkFramebufferAttachmentImageInfo-usage-parameter
  `usage` **must** be a valid combination of VkImageUsageFlagBits values
- VUID-VkFramebufferAttachmentImageInfo-pViewFormats-parameter
  If `viewFormatCount` is not `0`, `pViewFormats` **must** be a valid pointer to an array of

`viewFormatCount` valid VkFormat values

Bits which **can** be set in VkFramebufferCreateInfo::`flags`, specifying options for framebuffers, are:

WARNING This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef enum VkFramebufferCreateFlagBits {
  // Provided by VK_VERSION_1_2
    VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT = 0x00000001,
  // Provided by VK_KHR_imageless_framebuffer
    VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT_KHR = VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT,
} VkFramebufferCreateFlagBits;
```

- VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT specifies that image views are not specified, and only attachment compatibility information will be provided via a VkFramebufferAttachmentImageInfo structure.

WARNING This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkFramebufferCreateFlags;
```

`VkFramebufferCreateFlags` is a bitmask type for setting a mask of zero or more VkFramebufferCreateFlagBits.

To destroy a framebuffer, call:

WARNING This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkDestroyFramebuffer(
    VkDevice                                    device,
    VkFramebuffer                               framebuffer,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the framebuffer.
- `framebuffer` is the handle of the framebuffer to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyFramebuffer-framebuffer-00892
  All submitted commands that refer to `framebuffer` **must** have completed execution
- VUID-vkDestroyFramebuffer-framebuffer-00893
  If `VkAllocationCallbacks` were provided when `framebuffer` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyFramebuffer-framebuffer-00894
  If no `VkAllocationCallbacks` were provided when `framebuffer` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyFramebuffer-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyFramebuffer-framebuffer-parameter
  If `framebuffer` is not VK_NULL_HANDLE, `framebuffer` **must** be a valid VkFramebuffer handle
- VUID-vkDestroyFramebuffer-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyFramebuffer-framebuffer-parent
  If `framebuffer` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `framebuffer` **must** be externally synchronized

8.5. Render Pass Load Operations

Render pass load operations define the initial values of an attachment during a render pass instance.

Load operations for attachments with a depth/stencil format execute in the VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT pipeline stage. Load operations for attachments with a color format execute in the VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT pipeline stage. The load operation for each sample in an attachment happens-before any recorded command which accesses the sample in that render pass instance via that attachment or an alias. In a render pass object with multiple subpasses, load operations are performed in the first subpass which uses an attachment.

NOTE

Because load operations always happen first, external synchronization with attachment access only needs to synchronize the load operations with previous commands; not the operations within the render pass instance. This does not apply when using VK_ATTACHMENT_LOAD_OP_NONE.

Load operations only update values within the defined render area for the render pass instance. However, any writes performed by a load operation (as defined by its access masks) to a given attachment **may** read and write back any memory locations within the image subresource bound for that attachment. For depth/stencil images, if the maintenance7 feature is not enabled or `separateDepthStencilAttachmentAccess` is VK_FALSE, writes to one aspect **may** also result in read-modify-write operations for the other aspect. If the subresource is bound to an attachment with feedback loop enabled, implementations **must** not access pixels outside of the render area.

NOTE

As entire subresources could be accessed by load operations, applications cannot safely access values outside of the render area during a render pass instance when a load operation that modifies values is used.

Load operations that **can** be used for a render pass are:

```
// Provided by VK_VERSION_1_0
typedef enum VkAttachmentLoadOp {
    VK_ATTACHMENT_LOAD_OP_LOAD = 0,
    VK_ATTACHMENT_LOAD_OP_CLEAR = 1,
    VK_ATTACHMENT_LOAD_OP_DONT_CARE = 2,
  // Provided by VK_VERSION_1_4
    VK_ATTACHMENT_LOAD_OP_NONE = 1000400000,
  // Provided by VK_EXT_load_store_op_none
    VK_ATTACHMENT_LOAD_OP_NONE_EXT = VK_ATTACHMENT_LOAD_OP_NONE,
  // Provided by VK_KHR_load_store_op_none
    VK_ATTACHMENT_LOAD_OP_NONE_KHR = VK_ATTACHMENT_LOAD_OP_NONE,
} VkAttachmentLoadOp;
```

- VK_ATTACHMENT_LOAD_OP_LOAD specifies that the previous contents of the image within the render area will be preserved as the initial values. For attachments with a depth/stencil format, this uses the access type VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT. For attachments with a color format, this uses the access type VK_ACCESS_COLOR_ATTACHMENT_READ_BIT.
- VK_ATTACHMENT_LOAD_OP_CLEAR specifies that the contents within the render area will be cleared to a uniform value, which is specified when a render pass instance is begun. For attachments with a depth/stencil format, this uses the access type VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT. For attachments with a color format, this uses the access type VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT.
- VK_ATTACHMENT_LOAD_OP_DONT_CARE specifies that the previous contents within the area need not be preserved; the contents of the attachment will be undefined inside the render area. For attachments with a depth/stencil format, this uses the access type VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT. For attachments with a color format,

this uses the access type VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT.

- VK_ATTACHMENT_LOAD_OP_NONE specifies that the previous contents of the image will be undefined inside the render pass. No access type is used as the image is not accessed.

During a render pass instance, input and color attachments with color formats that have a component size of 8, 16, or 32 bits **must** be represented in the attachment's format throughout the instance. Attachments with other floating- or fixed-point color formats, or with depth components **may** be represented in a format with a precision higher than the attachment format, but **must** be represented with the same range. When such a component is loaded via the `loadOp`, it will be converted into an implementation-dependent format used by the render pass. Such components **must** be converted from the render pass format, to the format of the attachment, before they are resolved or stored at the end of a render pass instance via `storeOp`. Conversions occur as described in Numeric Representation and Computation and Fixed-Point Data Conversions.

8.6. Render Pass Store Operations

Render pass store operations define how values written to an attachment during a render pass instance are stored to memory.

Store operations for attachments with a depth/stencil format execute in the VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT pipeline stage. Store operations for attachments with a color format execute in the VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT pipeline stage. The store operation for each sample in an attachment happens-after any recorded command which accesses the sample via that attachment or an alias. In a render pass object with multiple subpasses, store operations are performed in the last subpass which uses an attachment.

NOTE Because store operations always happen after other accesses in a render pass instance, external synchronization with attachment access in an earlier render pass only needs to synchronize with the store operations; not the operations within the render pass instance. This does not apply when using VK_ATTACHMENT_STORE_OP_NONE.

Store operations only update values within the defined render area for the render pass instance. However, any writes performed by a store operation (as defined by its access masks) to a given attachment **may** read and write back any memory locations within the image subresource bound for that attachment. For depth/stencil images, if `separateDepthStencilAttachmentAccess` is VK_FALSE, writes to one aspect **may** also result in read-modify-write operations for the other aspect. If the subresource is bound to an attachment with feedback loop enabled, implementations **must** not access pixels outside of the render area.

NOTE As entire subresources could be accessed by store operations, applications cannot safely access values outside of the render area via aliased resources during a render pass instance when a store operation that modifies values is used.

Possible values of VkAttachmentDescription::`storeOp` and `stencilStoreOp`, specifying how the contents of the attachment are treated, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkAttachmentStoreOp {
    VK_ATTACHMENT_STORE_OP_STORE = 0,
    VK_ATTACHMENT_STORE_OP_DONT_CARE = 1,
  // Provided by VK_VERSION_1_3
    VK_ATTACHMENT_STORE_OP_NONE = 1000301000,
  // Provided by VK_KHR_dynamic_rendering, VK_KHR_load_store_op_none
    VK_ATTACHMENT_STORE_OP_NONE_KHR = VK_ATTACHMENT_STORE_OP_NONE,
  // Provided by VK_EXT_load_store_op_none
    VK_ATTACHMENT_STORE_OP_NONE_EXT = VK_ATTACHMENT_STORE_OP_NONE,
} VkAttachmentStoreOp;
```

- VK_ATTACHMENT_STORE_OP_STORE specifies the contents generated during the render pass and within the render area are written to memory. For attachments with a depth/stencil format, this uses the access type VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT. For attachments with a color format, this uses the access type VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT.
- VK_ATTACHMENT_STORE_OP_DONT_CARE specifies the contents within the render area are not needed after rendering, and **may** be discarded; the contents of the attachment will be undefined inside the render area. For attachments with a depth/stencil format, this uses the access type VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT. For attachments with a color format, this uses the access type VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT.
- VK_ATTACHMENT_STORE_OP_NONE specifies the contents within the render area are not accessed by the store operation as long as no values are written to the attachment during the render pass. If values are written during the render pass, this behaves identically to VK_ATTACHMENT_STORE_OP_DONT_CARE and with matching access semantics.

NOTE	VK_ATTACHMENT_STORE_OP_DONT_CARE **can** cause contents generated during previous render passes to be discarded before reaching memory, even if no write to the attachment occurs during the current render pass.

8.7. Render Pass Multisample Resolve Operations

Fixed-function render pass multisample resolve operations combine sample values from a single pixel in a multisample attachment and store the result to the corresponding pixel in a single sample attachment.

NOTE	Because VK_RESOLVE_MODE_CUSTOM_BIT_EXT specifies that the attachment will be resolved by shaders in the render pass instead of fixed-function operations, resolve operations with VK_RESOLVE_MODE_CUSTOM_BIT_EXT do not count as fixed-function multisample resolve operations.

Fixed-function multisample resolve operations for attachments execute in the VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT pipeline stage. A final resolve operation for all pixels in the render area happens-after any recorded command which writes a pixel via the multisample attachment to be resolved or an explicit alias of it in the subpass that it is specified.

Any single sample attachment specified for use in a fixed-function multisample resolve operation **may** have its contents modified at any point once rendering begins for the render pass instance. Reads from the multisample attachment can be synchronized with VK_ACCESS_COLOR_ATTACHMENT_READ_BIT. Access to the single sample attachment can be synchronized with VK_ACCESS_COLOR_ATTACHMENT_READ_BIT and VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT. These pipeline stage and access types are used whether the attachments are color or depth/stencil attachments.

When using render pass objects, a subpass dependency specified with the above pipeline stages and access flags will ensure synchronization with fixed-function multisample resolve operations for any attachments that were last accessed by that subpass. This allows later subpasses to read resolved values as input attachments.

Fixed-function resolve operations only update values within the defined render area for the render pass instance. However, any writes performed by a resolve operation (as defined by its access masks) to a given attachment **may** read and write back any memory locations within the image subresource bound for that attachment. For depth/stencil images, if `separateDepthStencilAttachmentAccess` is VK_FALSE, writes to one aspect **may** also result in read-modify-write operations for the other aspect. If the subresource is bound to an attachment with feedback loop enabled, implementations **must** not access pixels outside of the render area.

NOTE As entire subresources could be accessed by fixed-function multisample resolve operations, applications cannot safely access values outside of the render area via aliased resources during a render pass instance when a multisample resolve operation is performed.

Multisample values in a multisample attachment are combined according to the resolve mode used:

```
// Provided by VK_VERSION_1_2
typedef enum VkResolveModeFlagBits {
    VK_RESOLVE_MODE_NONE = 0,
    VK_RESOLVE_MODE_SAMPLE_ZERO_BIT = 0x00000001,
    VK_RESOLVE_MODE_AVERAGE_BIT = 0x00000002,
    VK_RESOLVE_MODE_MIN_BIT = 0x00000004,
    VK_RESOLVE_MODE_MAX_BIT = 0x00000008,
  // Provided by VK_EXT_custom_resolve with VK_KHR_dynamic_rendering or VK_VERSION_1_3
    VK_RESOLVE_MODE_CUSTOM_BIT_EXT = 0x00000020,
  // Provided by VK_KHR_depth_stencil_resolve
    VK_RESOLVE_MODE_NONE_KHR = VK_RESOLVE_MODE_NONE,
  // Provided by VK_KHR_depth_stencil_resolve
    VK_RESOLVE_MODE_SAMPLE_ZERO_BIT_KHR = VK_RESOLVE_MODE_SAMPLE_ZERO_BIT,
  // Provided by VK_KHR_depth_stencil_resolve
    VK_RESOLVE_MODE_AVERAGE_BIT_KHR = VK_RESOLVE_MODE_AVERAGE_BIT,
  // Provided by VK_KHR_depth_stencil_resolve
    VK_RESOLVE_MODE_MIN_BIT_KHR = VK_RESOLVE_MODE_MIN_BIT,
  // Provided by VK_KHR_depth_stencil_resolve
    VK_RESOLVE_MODE_MAX_BIT_KHR = VK_RESOLVE_MODE_MAX_BIT,
} VkResolveModeFlagBits;
```

```
// Provided by VK_KHR_depth_stencil_resolve
// Equivalent to VkResolveModeFlagBits
typedef VkResolveModeFlagBits VkResolveModeFlagBitsKHR;
```

- VK_RESOLVE_MODE_NONE specifies that no resolve operation is done.
- VK_RESOLVE_MODE_SAMPLE_ZERO_BIT specifies that the result of the resolve operation is equal to the value of sample 0.
- VK_RESOLVE_MODE_AVERAGE_BIT specifies that the result of the resolve operation is the average of the sample values.
- VK_RESOLVE_MODE_MIN_BIT specifies that the result of the resolve operation is the minimum of the sample values.
- VK_RESOLVE_MODE_MAX_BIT specifies that the result of the resolve operation is the maximum of the sample values.
- VK_RESOLVE_MODE_CUSTOM_BIT_EXT specifies that the attachment will be resolved by shaders in the render pass instead of fixed-function operations.

If no resolve mode is otherwise specified, VK_RESOLVE_MODE_AVERAGE_BIT is used.

If VK_RESOLVE_MODE_AVERAGE_BIT is used, and the source format is a floating-point or normalized type, the sample values for each pixel are resolved with implementation-defined numerical precision.

If the numeric format of the resolve attachment uses sRGB encoding, the implementation **should** convert samples from nonlinear to linear before averaging samples as described in the "sRGB EOTF" section of the Khronos Data Format Specification. In this case, the implementation **must** convert the linear averaged value to nonlinear before writing the resolved result to the resolve attachment. If the `maintenance10` feature is enabled, whether a nonlinear to linear conversion happens for sRGB resolve is defined by `resolveSrgbFormatAppliesTransferFunction`. This behavior **can** be overridden with appropriate `VK_*`*RESOLVE*{SKIP,ENABLE}_TRANSFER_FUNCTION_BIT_KHR flag usage.

```
// Provided by VK_VERSION_1_2
typedef VkFlags VkResolveModeFlags;
```

```
// Provided by VK_KHR_depth_stencil_resolve
// Equivalent to VkResolveModeFlags
typedef VkResolveModeFlags VkResolveModeFlagsKHR;
```

`VkResolveModeFlags` is a bitmask type for setting a mask of zero or more VkResolveModeFlagBits.

8.8. Render Pass Commands

An application records the commands for a render pass instance one subpass at a time, by

beginning a render pass instance, iterating over the subpasses to record commands for that subpass, and then ending the render pass instance.

To begin a render pass instance, call:

WARNING	This functionality is superseded by vkCmdBeginRenderPass2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkCmdBeginRenderPass(
    VkCommandBuffer                             commandBuffer,
    const VkRenderPassBeginInfo*                pRenderPassBegin,
    VkSubpassContents                           contents);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pRenderPassBegin` is a pointer to a VkRenderPassBeginInfo structure specifying the render pass to begin an instance of, and the framebuffer the instance uses.
- `contents` is a VkSubpassContents value specifying how the commands in the first subpass will be provided.

After beginning a render pass instance, the command buffer is ready to record the commands for the first subpass of that render pass.

Valid Usage

- VUID-vkCmdBeginRenderPass-initialLayout-00895
  If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT usage flag set
- VUID-vkCmdBeginRenderPass-initialLayout-01758
  If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-vkCmdBeginRenderPass-initialLayout-02842

If any of the initialLayout or finalLayout member of the VkAttachmentDescription structures or the layout member of the VkAttachmentReference structures specified when creating the render pass specified in the renderPass member of pRenderPassBegin is VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL then the corresponding attachment image view of the framebuffer specified in the framebuffer member of pRenderPassBegin **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set

- VUID-vkCmdBeginRenderPass-stencilInitialLayout-02843
  If any of the stencilInitialLayout or stencilFinalLayout member of the VkAttachmentDescriptionStencilLayout structures or the stencilLayout member of the VkAttachmentReferenceStencilLayout structures specified when creating the render pass specified in the renderPass member of pRenderPassBegin is VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL then the corresponding attachment image view of the framebuffer specified in the framebuffer member of pRenderPassBegin **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-vkCmdBeginRenderPass-initialLayout-00897
  If any of the initialLayout or finalLayout member of the VkAttachmentDescription structures or the layout member of the VkAttachmentReference structures specified when creating the render pass specified in the renderPass member of pRenderPassBegin is VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL then the corresponding attachment image view of the framebuffer specified in the framebuffer member of pRenderPassBegin **must** have been created with the VK_IMAGE_USAGE_SAMPLED_BIT or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flags set
- VUID-vkCmdBeginRenderPass-initialLayout-00898
  If any of the initialLayout or finalLayout member of the VkAttachmentDescription structures or the layout member of the VkAttachmentReference structures specified when creating the render pass specified in the renderPass member of pRenderPassBegin is VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL then the corresponding attachment image view of the framebuffer specified in the framebuffer member of pRenderPassBegin **must** have been created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-vkCmdBeginRenderPass-initialLayout-00899
  If any of the initialLayout or finalLayout member of the VkAttachmentDescription structures or the layout member of the VkAttachmentReference structures specified when creating the render pass specified in the renderPass member of pRenderPassBegin is VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL then the corresponding attachment image view of the framebuffer specified in the framebuffer member of pRenderPassBegin **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-vkCmdBeginRenderPass-initialLayout-00900
  If the initialLayout member of any of the VkAttachmentDescription structures specified when creating the render pass specified in the renderPass member of pRenderPassBegin is not VK_IMAGE_LAYOUT_UNDEFINED, then each such initialLayout **must** be equal to the

current layout of the corresponding attachment image subresource of the framebuffer specified in the framebuffer member of pRenderPassBegin

- VUID-vkCmdBeginRenderPass-srcStageMask-06451
  The srcStageMask members of any element of the pDependencies member of VkRenderPassCreateInfo used to create renderPass **must** be supported by the capabilities of the queue family identified by the queueFamilyIndex member of the VkCommandPoolCreateInfo used to create the command pool which commandBuffer was allocated from
- VUID-vkCmdBeginRenderPass-dstStageMask-06452
  The dstStageMask members of any element of the pDependencies member of VkRenderPassCreateInfo used to create renderPass **must** be supported by the capabilities of the queue family identified by the queueFamilyIndex member of the VkCommandPoolCreateInfo used to create the command pool which commandBuffer was allocated from
- VUID-vkCmdBeginRenderPass-framebuffer-02532
  For any attachment in framebuffer that is used by renderPass and is bound to memory locations that are also bound to another attachment used by renderPass, and if at least one of those uses causes either attachment to be written to, both attachments **must** have had the VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT set
- VUID-vkCmdBeginRenderPass-framebuffer-09045
  If any attachments specified in framebuffer are used by renderPass and are bound to overlapping memory locations, there **must** be only one that is used as a color attachment, depth/stencil, or resolve attachment in any subpass
- VUID-vkCmdBeginRenderPass-initialLayout-07000
  If any of the initialLayout or finalLayout member of the VkAttachmentDescription structures or the layout member of the VkAttachmentReference structures specified when creating the render pass specified in the renderPass member of pRenderPassBegin is VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT then the corresponding attachment image view of the framebuffer specified in the framebuffer member of pRenderPassBegin **must** have been created with either the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flags set, and either the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT or VK_IMAGE_USAGE_SAMPLED_BIT usage flags set
- VUID-vkCmdBeginRenderPass-initialLayout-07001
  If any of the initialLayout or finalLayout member of the VkAttachmentDescription structures or the layout member of the VkAttachmentReference structures specified when creating the render pass specified in the renderPass member of pRenderPassBegin is VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT then the corresponding attachment image view of the framebuffer specified in the framebuffer member of pRenderPassBegin **must** have been created with the VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT usage flag set
- VUID-vkCmdBeginRenderPass-initialLayout-09537
  If any of the initialLayout or finalLayout member of the VkAttachmentDescription structures or the layout member of the VkAttachmentReference structures specified when

creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with either the VK_IMAGE_USAGE_STORAGE_BIT usage flag set, or both the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT and either of VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flags set

- VUID-vkCmdBeginRenderPass-contents-09640
  If `contents` is VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR, then at least one of the following features **must** be enabled:
  - `maintenance7`
  - `nestedCommandBuffer`

Valid Usage (Implicit)

- VUID-vkCmdBeginRenderPass-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBeginRenderPass-pRenderPassBegin-parameter
  `pRenderPassBegin` **must** be a valid pointer to a valid VkRenderPassBeginInfo structure
- VUID-vkCmdBeginRenderPass-contents-parameter
  `contents` **must** be a valid VkSubpassContents value
- VUID-vkCmdBeginRenderPass-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBeginRenderPass-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBeginRenderPass-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdBeginRenderPass-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdBeginRenderPass-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBeginRenderPass-bufferlevel
  `commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Outside	VK_QUEUE_GRAPHICS_BIT	Action State Synchronization

Conditional Rendering

vkCmdBeginRenderPass is not affected by conditional rendering

To begin a render pass, call:

WARNING This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
void vkCmdBeginRenderPass2(
    VkCommandBuffer                             commandBuffer,
    const VkRenderPassBeginInfo*                pRenderPassBegin,
    const VkSubpassBeginInfo*                   pSubpassBeginInfo);
```

```
// Provided by VK_KHR_create_renderpass2
// Equivalent to vkCmdBeginRenderPass2
void vkCmdBeginRenderPass2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkRenderPassBeginInfo*                pRenderPassBegin,
    const VkSubpassBeginInfo*                   pSubpassBeginInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pRenderPassBegin` is a pointer to a VkRenderPassBeginInfo structure specifying the render pass to begin an instance of, and the framebuffer the instance uses. After recording this command, the render pass and framebuffer **may** be accessed at any point that `commandBuffer` is in the recording or pending state until it is reset.
- `pSubpassBeginInfo` is a pointer to a VkSubpassBeginInfo structure containing information about the subpass which is about to begin rendering.

After beginning a render pass instance, the command buffer is ready to record the commands for the first subpass of that render pass.

Valid Usage

- VUID-vkCmdBeginRenderPass2-framebuffer-02779
  Both the `framebuffer` and `renderPass` members of `pRenderPassBegin` **must** have been created on the same VkDevice that `commandBuffer` was allocated on
- VUID-vkCmdBeginRenderPass2-initialLayout-03094
  If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT usage flag set
- VUID-vkCmdBeginRenderPass2-initialLayout-03096
  If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-vkCmdBeginRenderPass2-initialLayout-02844
  If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set
- VUID-vkCmdBeginRenderPass2-stencilInitialLayout-02845
  If any of the `stencilInitialLayout` or `stencilFinalLayout` member of the `VkAttachmentDescriptionStencilLayout` structures or the `stencilLayout` member of the `VkAttachmentReferenceStencilLayout` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with the VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flag set

- VUID-vkCmdBeginRenderPass2-initialLayout-03097
  If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with the VK_IMAGE_USAGE_SAMPLED_BIT or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flags set
- VUID-vkCmdBeginRenderPass2-initialLayout-03098
  If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-vkCmdBeginRenderPass2-initialLayout-03099
  If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-vkCmdBeginRenderPass2-initialLayout-03100
  If the `initialLayout` member of any of the `VkAttachmentDescription` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is not VK_IMAGE_LAYOUT_UNDEFINED, then each such `initialLayout` **must** be equal to the current layout of the corresponding attachment image subresource of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin`
- VUID-vkCmdBeginRenderPass2-srcStageMask-06453
  The `srcStageMask` members of any element of the `pDependencies` member of VkRenderPassCreateInfo used to create `renderPass` **must** be supported by the capabilities of the queue family identified by the `queueFamilyIndex` member of the VkCommandPoolCreateInfo used to create the command pool which `commandBuffer` was allocated from
- VUID-vkCmdBeginRenderPass2-dstStageMask-06454
  The `dstStageMask` members of any element of the `pDependencies` member of VkRenderPassCreateInfo used to create `renderPass` **must** be supported by the capabilities of the queue family identified by the `queueFamilyIndex` member of the VkCommandPoolCreateInfo used to create the command pool which `commandBuffer` was allocated from
- VUID-vkCmdBeginRenderPass2-framebuffer-02533
  For any attachment in `framebuffer` that is used by `renderPass` and is bound to memory locations that are also bound to another attachment used by `renderPass`, and if at least one of those uses causes either attachment to be written to, both attachments **must** have had the VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT set

- VUID-vkCmdBeginRenderPass2-framebuffer-09046
  If any attachments specified in `framebuffer` are used by `renderPass` and are bound to overlapping memory locations, there **must** be only one that is used as a color attachment, depth/stencil, or resolve attachment in any subpass
- VUID-vkCmdBeginRenderPass2-initialLayout-07002
  If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with either the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flags set, and either the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT or VK_IMAGE_USAGE_SAMPLED_BIT usage flags set
- VUID-vkCmdBeginRenderPass2-initialLayout-07003
  If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with the VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT usage flag set
- VUID-vkCmdBeginRenderPass2-initialLayout-09538
  If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with either the VK_IMAGE_USAGE_STORAGE_BIT usage flag set, or both the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT and either of the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flags set

Valid Usage (Implicit)

- VUID-vkCmdBeginRenderPass2-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBeginRenderPass2-pRenderPassBegin-parameter
  `pRenderPassBegin` **must** be a valid pointer to a valid VkRenderPassBeginInfo structure
- VUID-vkCmdBeginRenderPass2-pSubpassBeginInfo-parameter
  `pSubpassBeginInfo` **must** be a valid pointer to a valid VkSubpassBeginInfo structure
- VUID-vkCmdBeginRenderPass2-commandBuffer-recording
  `commandBuffer` **must** be in the recording state

- VUID-vkCmdBeginRenderPass2-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBeginRenderPass2-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdBeginRenderPass2-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdBeginRenderPass2-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBeginRenderPass2-bufferlevel
  `commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Outside	VK_QUEUE_GRAPHICS_BIT	Action State Synchronization

Conditional Rendering

vkCmdBeginRenderPass2 is not affected by conditional rendering

The `VkRenderPassBeginInfo` structure is defined as:

WARNING This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef struct VkRenderPassBeginInfo {
    VkStructureType        sType;
    const void*            pNext;
    VkRenderPass           renderPass;
    VkFramebuffer          framebuffer;
    VkRect2D               renderArea;
    uint32_t               clearValueCount;
    const VkClearValue*    pClearValues;
} VkRenderPassBeginInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `renderPass` is the render pass to begin an instance of.
- `framebuffer` is the framebuffer containing the attachments that are used with the render pass.
- `renderArea` is the render area that is affected by the render pass instance, and is described in more detail below.
- `clearValueCount` is the number of elements in `pClearValues`.
- `pClearValues` is a pointer to an array of `clearValueCount` VkClearValue structures containing clear values for each attachment, if the attachment uses a `loadOp` value of VK_ATTACHMENT_LOAD_OP_CLEAR or if the attachment has a depth/stencil format and uses a `stencilLoadOp` value of VK_ATTACHMENT_LOAD_OP_CLEAR. The array is indexed by attachment number. Only elements corresponding to cleared attachments are used. Other elements of `pClearValues` are ignored.

`renderArea` is the render area that is affected by the render pass instance. The effects of attachment load, store, and multisample resolve operations are restricted to the pixels whose x and y coordinates fall within the render area on all attachments. The render area extends to all layers of `framebuffer`. The application **must** ensure (using scissor if necessary) that all rendering is contained within the render area. The render area **must** be contained within the framebuffer dimensions.

NOTE There **may** be a performance cost for using a render area smaller than the framebuffer, unless it matches the render area granularity for the render pass.

Valid Usage

- VUID-VkRenderPassBeginInfo-clearValueCount-00902
  `clearValueCount` **must** be greater than the largest attachment index in `renderPass` specifying a `loadOp` (or `stencilLoadOp`, if the attachment has a depth/stencil format) of VK_ATTACHMENT_LOAD_OP_CLEAR
- VUID-VkRenderPassBeginInfo-clearValueCount-04962
  If `clearValueCount` is not `0`, `pClearValues` **must** be a valid pointer to an array of `clearValueCount` VkClearValue unions
- VUID-VkRenderPassBeginInfo-renderPass-00904

`renderPass` **must** be compatible with the `renderPass` member of the VkFramebufferCreateInfo structure specified when creating `framebuffer`

- VUID-VkRenderPassBeginInfo-None-08996
  If the `pNext` chain does not contain VkDeviceGroupRenderPassBeginInfo or its `deviceRenderAreaCount` member is equal to 0, `renderArea.extent.width` **must** be greater than 0
- VUID-VkRenderPassBeginInfo-None-08997
  If the `pNext` chain does not contain VkDeviceGroupRenderPassBeginInfo or its `deviceRenderAreaCount` member is equal to 0, `renderArea.extent.height` **must** be greater than 0
- VUID-VkRenderPassBeginInfo-pNext-02850
  If the `pNext` chain does not contain VkDeviceGroupRenderPassBeginInfo or its `deviceRenderAreaCount` member is equal to 0, `renderArea.offset.x` **must** be greater than or equal to 0
- VUID-VkRenderPassBeginInfo-pNext-02851
  If the `pNext` chain does not contain VkDeviceGroupRenderPassBeginInfo or its `deviceRenderAreaCount` member is equal to 0, `renderArea.offset.y` **must** be greater than or equal to 0
- VUID-VkRenderPassBeginInfo-pNext-02852
  If the `pNext` chain does not contain VkDeviceGroupRenderPassBeginInfo or its `deviceRenderAreaCount` member is equal to 0, `renderArea.offset.x` + `renderArea.extent.width` **must** be less than or equal to VkFramebufferCreateInfo::`width` the `framebuffer` was created with
- VUID-VkRenderPassBeginInfo-pNext-02853
  If the `pNext` chain does not contain VkDeviceGroupRenderPassBeginInfo or its `deviceRenderAreaCount` member is equal to 0, `renderArea.offset.y` + `renderArea.extent.height` **must** be less than or equal to VkFramebufferCreateInfo::`height` the `framebuffer` was created with
- VUID-VkRenderPassBeginInfo-pNext-02856
  If the `pNext` chain contains VkDeviceGroupRenderPassBeginInfo, `offset.x` + `extent.width` of each element of `pDeviceRenderAreas` **must** be less than or equal to VkFramebufferCreateInfo::`width` the `framebuffer` was created with
- VUID-VkRenderPassBeginInfo-pNext-02857
  If the `pNext` chain contains VkDeviceGroupRenderPassBeginInfo, `offset.y` + `extent.height` of each element of `pDeviceRenderAreas` **must** be less than or equal to VkFramebufferCreateInfo::`height` the `framebuffer` was created with
- VUID-VkRenderPassBeginInfo-framebuffer-03207
  If `framebuffer` was created with a VkFramebufferCreateInfo::`flags` value that did not include VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, and the `pNext` chain includes a VkRenderPassAttachmentBeginInfo structure, its `attachmentCount` **must** be zero
- VUID-VkRenderPassBeginInfo-framebuffer-03208
  If `framebuffer` was created with a VkFramebufferCreateInfo::`flags` value that included VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, the `attachmentCount` of a VkRenderPassAttachmentBeginInfo structure included in the `pNext` chain **must** be equal

to the value of VkFramebufferAttachmentsCreateInfo::attachmentImageInfoCount used to create framebuffer

- VUID-VkRenderPassBeginInfo-framebuffer-02780
  If framebuffer was created with a VkFramebufferCreateInfo::flags value that included VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of the pAttachments member of a VkRenderPassAttachmentBeginInfo structure included in the pNext chain **must** have been created on the same VkDevice as framebuffer and renderPass
- VUID-VkRenderPassBeginInfo-framebuffer-03209
  If framebuffer was created with a VkFramebufferCreateInfo::flags value that included VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of the pAttachments member of a VkRenderPassAttachmentBeginInfo structure included in the pNext chain **must** be a VkImageView of an image created with a value of VkImageCreateInfo::flags equal to the flags member of the corresponding element of VkFramebufferAttachmentsCreateInfo ::pAttachmentImageInfos used to create framebuffer
- VUID-VkRenderPassBeginInfo-framebuffer-04627
  If framebuffer was created with a VkFramebufferCreateInfo::flags value that included VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of the pAttachments member of a VkRenderPassAttachmentBeginInfo structure included in the pNext chain **must** be a VkImageView with an inherited usage equal to the usage member of the corresponding element of VkFramebufferAttachmentsCreateInfo::pAttachmentImageInfos used to create framebuffer
- VUID-VkRenderPassBeginInfo-framebuffer-03211
  If framebuffer was created with a VkFramebufferCreateInfo::flags value that included VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of the pAttachments member of a VkRenderPassAttachmentBeginInfo structure included in the pNext chain **must** be a VkImageView with a width equal to the width member of the corresponding element of VkFramebufferAttachmentsCreateInfo::pAttachmentImageInfos used to create framebuffer
- VUID-VkRenderPassBeginInfo-framebuffer-03212
  If framebuffer was created with a VkFramebufferCreateInfo::flags value that included VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of the pAttachments member of a VkRenderPassAttachmentBeginInfo structure included in the pNext chain **must** be a VkImageView with a height equal to the height member of the corresponding element of VkFramebufferAttachmentsCreateInfo::pAttachmentImageInfos used to create framebuffer
- VUID-VkRenderPassBeginInfo-framebuffer-03213
  If framebuffer was created with a VkFramebufferCreateInfo::flags value that included VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of the pAttachments member of a VkRenderPassAttachmentBeginInfo structure included in the pNext chain **must** be a VkImageView of an image created with a value of VkImageViewCreateInfo ::subresourceRange.layerCount equal to the layerCount member of the corresponding element of VkFramebufferAttachmentsCreateInfo::pAttachmentImageInfos used to create framebuffer
- VUID-VkRenderPassBeginInfo-framebuffer-03214
  If framebuffer was created with a VkFramebufferCreateInfo::flags value that included VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of the pAttachments member of a VkRenderPassAttachmentBeginInfo structure included in the pNext chain **must** be a

VkImageView of an image created with a value of VkImageFormatListCreateInfo::viewFormatCount equal to the viewFormatCount member of the corresponding element of VkFramebufferAttachmentsCreateInfo::pAttachmentImageInfos used to create framebuffer

- VUID-VkRenderPassBeginInfo-framebuffer-03215
  If framebuffer was created with a VkFramebufferCreateInfo::flags value that included VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of the pAttachments member of a VkRenderPassAttachmentBeginInfo structure included in the pNext chain **must** be a VkImageView of an image created with a set of elements in VkImageFormatListCreateInfo::pViewFormats equal to the set of elements in the pViewFormats member of the corresponding element of VkFramebufferAttachmentsCreateInfo::pAttachmentImageInfos used to create framebuffer
- VUID-VkRenderPassBeginInfo-framebuffer-03216
  If framebuffer was created with a VkFramebufferCreateInfo::flags value that included VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of the pAttachments member of a VkRenderPassAttachmentBeginInfo structure included in the pNext chain **must** be a VkImageView of an image created with a value of VkImageViewCreateInfo::format equal to the corresponding value of VkAttachmentDescription::format in renderPass
- VUID-VkRenderPassBeginInfo-framebuffer-09047
  If framebuffer was created with a VkFramebufferCreateInfo::flags value that included VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT, each element of the pAttachments member of a VkRenderPassAttachmentBeginInfo structure included in the pNext chain **must** be a VkImageView of an image created with a value of VkImageCreateInfo::samples equal to the corresponding value of VkAttachmentDescription::samples in renderPass , or VK_SAMPLE_COUNT_1_BIT if renderPass was created with a VkMultisampledRenderToSingleSampledInfoEXT structure in the pNext chain with multisampledRenderToSingleSampledEnable equal to VK_TRUE

Valid Usage (Implicit)

- VUID-VkRenderPassBeginInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_RENDER_PASS_BEGIN_INFO
- VUID-VkRenderPassBeginInfo-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkDeviceGroupRenderPassBeginInfo, VkRenderPassAttachmentBeginInfo, or VkRenderPassSampleLocationsBeginInfoEXT
- VUID-VkRenderPassBeginInfo-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkRenderPassBeginInfo-renderPass-parameter
  renderPass **must** be a valid VkRenderPass handle
- VUID-VkRenderPassBeginInfo-framebuffer-parameter
  framebuffer **must** be a valid VkFramebuffer handle
- VUID-VkRenderPassBeginInfo-commonparent
  Both of framebuffer, and renderPass **must** have been created, allocated, or retrieved from

the same VkDevice

The image layout of the depth aspect of a depth/stencil attachment referring to an image created with VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT is dependent on the last sample locations used to render to the image subresource, thus preserving the contents of such depth/stencil attachments across subpass boundaries requires the application to specify these sample locations whenever a layout transition of the attachment **may** occur. This information **can** be provided by adding a VkRenderPassSampleLocationsBeginInfoEXT structure to the pNext chain of VkRenderPassBeginInfo.

The VkRenderPassSampleLocationsBeginInfoEXT structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4 and VK_KHR_dynamic_rendering_local_read. See Legacy Functionality for more information.

```
// Provided by VK_EXT_sample_locations
typedef struct VkRenderPassSampleLocationsBeginInfoEXT {
    VkStructureType                              sType;
    const void*                                  pNext;
    uint32_t                                     attachmentInitialSampleLocationsCount;
    const VkAttachmentSampleLocationsEXT*        pAttachmentInitialSampleLocations;
    uint32_t                                     postSubpassSampleLocationsCount;
    const VkSubpassSampleLocationsEXT*           pPostSubpassSampleLocations;
} VkRenderPassSampleLocationsBeginInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- attachmentInitialSampleLocationsCount is the number of elements in the pAttachmentInitialSampleLocations array.
- pAttachmentInitialSampleLocations is a pointer to an array of attachmentInitialSampleLocationsCount VkAttachmentSampleLocationsEXT structures specifying the attachment indices and their corresponding sample location state. Each element of pAttachmentInitialSampleLocations **can** specify the sample location state to use in the automatic layout transition performed to transition a depth/stencil attachment from the initial layout of the attachment to the image layout specified for the attachment in the first subpass using it.
- postSubpassSampleLocationsCount is the number of elements in the pPostSubpassSampleLocations array.
- pPostSubpassSampleLocations is a pointer to an array of postSubpassSampleLocationsCount VkSubpassSampleLocationsEXT structures specifying the subpass indices and their corresponding sample location state. Each element of pPostSubpassSampleLocations **can** specify the sample location state to use in the automatic layout transition performed to transition the depth/stencil attachment used by the specified subpass to the image layout specified in a dependent subpass or to the final layout of the attachment in case the specified subpass is the

last subpass using that attachment. In addition, if VkPhysicalDeviceSampleLocationsPropertiesEXT::variableSampleLocations is VK_FALSE, each element of pPostSubpassSampleLocations **must** specify the sample location state that matches the sample locations used by all pipelines that will be bound to a command buffer during the specified subpass. If variableSampleLocations is VK_TRUE, the sample locations used for rasterization do not depend on pPostSubpassSampleLocations.

Valid Usage (Implicit)

- VUID-VkRenderPassSampleLocationsBeginInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_RENDER_PASS_SAMPLE_LOCATIONS_BEGIN_INFO_EXT
- VUID-VkRenderPassSampleLocationsBeginInfoEXT-pAttachmentInitialSampleLocations-parameter
  If attachmentInitialSampleLocationsCount is not 0, pAttachmentInitialSampleLocations **must** be a valid pointer to an array of attachmentInitialSampleLocationsCount valid VkAttachmentSampleLocationsEXT structures
- VUID-VkRenderPassSampleLocationsBeginInfoEXT-pPostSubpassSampleLocations-parameter
  If postSubpassSampleLocationsCount is not 0, pPostSubpassSampleLocations **must** be a valid pointer to an array of postSubpassSampleLocationsCount valid VkSubpassSampleLocationsEXT structures

Structure Chaining

Extends the structure

- VkRenderPassBeginInfo

The VkAttachmentSampleLocationsEXT structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4 and VK_KHR_dynamic_rendering_local_read. See Legacy Functionality for more information.

```
// Provided by VK_EXT_sample_locations
typedef struct VkAttachmentSampleLocationsEXT {
    uint32_t                    attachmentIndex;
    VkSampleLocationsInfoEXT    sampleLocationsInfo;
} VkAttachmentSampleLocationsEXT;
```

- attachmentIndex is the index of the attachment for which the sample locations state is provided.
- sampleLocationsInfo is the sample locations state to use for the layout transition of the given attachment from the initial layout of the attachment to the image layout specified for the

attachment in the first subpass using it.

If the image referenced by the framebuffer attachment at index attachmentIndex was not created with VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT then the values specified in sampleLocationsInfo are ignored.

Valid Usage

- VUID-VkAttachmentSampleLocationsEXT-attachmentIndex-01531 attachmentIndex **must** be less than the attachmentCount specified in VkRenderPassCreateInfo the render pass specified by VkRenderPassBeginInfo::renderPass was created with

Valid Usage (Implicit)

- VUID-VkAttachmentSampleLocationsEXT-sampleLocationsInfo-parameter sampleLocationsInfo **must** be a valid VkSampleLocationsInfoEXT structure

The VkSubpassSampleLocationsEXT structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4 and VK_KHR_dynamic_rendering_local_read. See Legacy Functionality for more information.

```
// Provided by VK_EXT_sample_locations
typedef struct VkSubpassSampleLocationsEXT {
    uint32_t                    subpassIndex;
    VkSampleLocationsInfoEXT    sampleLocationsInfo;
} VkSubpassSampleLocationsEXT;
```

- subpassIndex is the index of the subpass for which the sample locations state is provided.
- sampleLocationsInfo is the sample locations state to use for the layout transition of the depth/stencil attachment away from the image layout the attachment is used with in the subpass specified in subpassIndex.

If the image referenced by the depth/stencil attachment used in the subpass identified by subpassIndex was not created with VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT or if the subpass does not use a depth/stencil attachment, and VkPhysicalDeviceSampleLocationsPropertiesEXT ::variableSampleLocations is VK_TRUE then the values specified in sampleLocationsInfo are ignored.

Valid Usage

- VUID-VkSubpassSampleLocationsEXT-subpassIndex-01532 subpassIndex **must** be less than the subpassCount specified in VkRenderPassCreateInfo the

render pass specified by VkRenderPassBeginInfo::renderPass was created with

Valid Usage (Implicit)

- VUID-VkSubpassSampleLocationsEXT-sampleLocationsInfo-parameter
  sampleLocationsInfo **must** be a valid VkSampleLocationsInfoEXT structure

The VkSubpassBeginInfo structure is defined as:

WARNING This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
typedef struct VkSubpassBeginInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkSubpassContents    contents;
} VkSubpassBeginInfo;
```

```
// Provided by VK_KHR_create_renderpass2
// Equivalent to VkSubpassBeginInfo
typedef VkSubpassBeginInfo VkSubpassBeginInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- contents is a VkSubpassContents value specifying how the commands in the next subpass will be provided.

Valid Usage

- VUID-VkSubpassBeginInfo-contents-09382
  If contents is VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR, then at least one of the following features **must** be enabled:
  - maintenance7
  - nestedCommandBuffer

Valid Usage (Implicit)

- VUID-VkSubpassBeginInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_SUBPASS_BEGIN_INFO

- VUID-VkSubpassBeginInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkSubpassBeginInfo-contents-parameter
  `contents` **must** be a valid VkSubpassContents value

Possible values of vkCmdBeginRenderPass::`contents`, specifying how the commands in the first subpass will be provided, are:

WARNING	This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef enum VkSubpassContents {
    VK_SUBPASS_CONTENTS_INLINE = 0,
    VK_SUBPASS_CONTENTS_SECONDARY_COMMAND_BUFFERS = 1,
  // Provided by VK_KHR_maintenance7
    VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR = 1000451000,
  // Provided by VK_EXT_nested_command_buffer
    VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_EXT =
VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR,
} VkSubpassContents;
```

- VK_SUBPASS_CONTENTS_INLINE specifies that the contents of the subpass will be recorded inline in the primary command buffer, and secondary command buffers **must** not be executed within the subpass.
- VK_SUBPASS_CONTENTS_SECONDARY_COMMAND_BUFFERS specifies that the contents are recorded in secondary command buffers that will be called from the primary command buffer, and vkCmdExecuteCommands is the only valid command in the command buffer until vkCmdNextSubpass or vkCmdEndRenderPass.
- VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR specifies that the contents of the subpass **can** be recorded both inline and in secondary command buffers executed from this command buffer with vkCmdExecuteCommands.

If the `pNext` chain of VkRenderPassBeginInfo or VkRenderingInfo includes a `VkDeviceGroupRenderPassBeginInfo` structure, then that structure includes a device mask and set of render areas for the render pass instance.

The `VkDeviceGroupRenderPassBeginInfo` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkDeviceGroupRenderPassBeginInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           deviceMask;
    uint32_t           deviceRenderAreaCount;
    const VkRect2D*    pDeviceRenderAreas;
} VkDeviceGroupRenderPassBeginInfo;
```

```
// Provided by VK_KHR_device_group
// Equivalent to VkDeviceGroupRenderPassBeginInfo
typedef VkDeviceGroupRenderPassBeginInfo VkDeviceGroupRenderPassBeginInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `deviceMask` is the device mask for the render pass instance.
- `deviceRenderAreaCount` is the number of elements in the `pDeviceRenderAreas` array.
- `pDeviceRenderAreas` is a pointer to an array of VkRect2D structures defining the render area for each physical device.

The `deviceMask` serves several purposes. It is an upper bound on the set of physical devices that **can** be used during the render pass instance, and the initial device mask when the render pass instance begins. In addition, commands transitioning to the next subpass in a render pass instance and commands ending the render pass instance, and, accordingly render pass load, store, and multisample resolve operations and subpass dependencies corresponding to the render pass instance, are executed on the physical devices included in the device mask provided here.

If `deviceRenderAreaCount` is not zero, then the elements of `pDeviceRenderAreas` override the value of VkRenderPassBeginInfo::`renderArea`, and provide a render area specific to each physical device. These render areas serve the same purpose as VkRenderPassBeginInfo::`renderArea`, including controlling the region of attachments that are cleared by VK_ATTACHMENT_LOAD_OP_CLEAR and that are resolved into resolve attachments.

If this structure is not present, the render pass instance's device mask is the value of VkDeviceGroupCommandBufferBeginInfo::`deviceMask`. If this structure is not present or if `deviceRenderAreaCount` is zero, VkRenderPassBeginInfo::`renderArea` is used for all physical devices.

Valid Usage

- VUID-VkDeviceGroupRenderPassBeginInfo-deviceMask-00905
  `deviceMask` **must** be a valid device mask value
- VUID-VkDeviceGroupRenderPassBeginInfo-deviceMask-00906
  `deviceMask` **must** not be zero
- VUID-VkDeviceGroupRenderPassBeginInfo-deviceMask-00907

`deviceMask` **must** be a subset of the command buffer's initial device mask

- VUID-VkDeviceGroupRenderPassBeginInfo-deviceRenderAreaCount-00908
  `deviceRenderAreaCount` **must** either be zero or equal to the number of physical devices in the logical device
- VUID-VkDeviceGroupRenderPassBeginInfo-offset-06166
  The `offset.x` member of any element of `pDeviceRenderAreas` **must** be greater than or equal to 0
- VUID-VkDeviceGroupRenderPassBeginInfo-offset-06167
  The `offset.y` member of any element of `pDeviceRenderAreas` **must** be greater than or equal to 0
- VUID-VkDeviceGroupRenderPassBeginInfo-offset-06168
  The sum of the `offset.x` and `extent.width` members of any element of `pDeviceRenderAreas` **must** be less than or equal to `maxFramebufferWidth`
- VUID-VkDeviceGroupRenderPassBeginInfo-offset-06169
  The sum of the `offset.y` and `extent.height` members of any element of `pDeviceRenderAreas` **must** be less than or equal to `maxFramebufferHeight`
- VUID-VkDeviceGroupRenderPassBeginInfo-extent-08998
  The `extent.width` member of any element of `pDeviceRenderAreas` **must** be greater than 0
- VUID-VkDeviceGroupRenderPassBeginInfo-extent-08999
  The `extent.height` member of any element of `pDeviceRenderAreas` **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkDeviceGroupRenderPassBeginInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_GROUP_RENDER_PASS_BEGIN_INFO
- VUID-VkDeviceGroupRenderPassBeginInfo-pDeviceRenderAreas-parameter
  If `deviceRenderAreaCount` is not `0`, `pDeviceRenderAreas` **must** be a valid pointer to an array of `deviceRenderAreaCount` VkRect2D structures

Structure Chaining

Extends the structures

- VkRenderPassBeginInfo
- VkRenderingInfo

The `VkRenderPassAttachmentBeginInfo` structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
typedef struct VkRenderPassAttachmentBeginInfo {
    VkStructureType       sType;
    const void*           pNext;
    uint32_t              attachmentCount;
    const VkImageView*    pAttachments;
} VkRenderPassAttachmentBeginInfo;
```

```
// Provided by VK_KHR_imageless_framebuffer
// Equivalent to VkRenderPassAttachmentBeginInfo
typedef VkRenderPassAttachmentBeginInfo VkRenderPassAttachmentBeginInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `attachmentCount` is the number of attachments.
- `pAttachments` is a pointer to an array of `VkImageView` handles, each of which will be used as the corresponding attachment in the render pass instance.

Valid Usage

- VUID-VkRenderPassAttachmentBeginInfo-pAttachments-03218
  Each element of `pAttachments` **must** only specify a single mip level
- VUID-VkRenderPassAttachmentBeginInfo-pAttachments-03219
  Each element of `pAttachments` **must** have been created with the identity swizzle
- VUID-VkRenderPassAttachmentBeginInfo-pAttachments-04114
  Each element of `pAttachments` **must** have been created with VkImageViewCreateInfo ::`viewType` not equal to VK_IMAGE_VIEW_TYPE_3D
- VUID-VkRenderPassAttachmentBeginInfo-pAttachments-07010
  If multisampled-render-to-single-sampled is enabled for any subpass, all elements of `pAttachments` which have a sample count equal to VK_SAMPLE_COUNT_1_BIT **must** have a format that supports the sample count specified in VkMultisampledRenderToSingleSampledInfoEXT::`rasterizationSamples`

Valid Usage (Implicit)

- VUID-VkRenderPassAttachmentBeginInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RENDER_PASS_ATTACHMENT_BEGIN_INFO
- VUID-VkRenderPassAttachmentBeginInfo-pAttachments-parameter
  If `attachmentCount` is not `0`, `pAttachments` **must** be a valid pointer to an array of `attachmentCount` valid VkImageView handles

Structure Chaining

Extends the structure

- VkRenderPassBeginInfo

To query the render area granularity, call:

WARNING	This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkGetRenderAreaGranularity(
    VkDevice                                    device,
    VkRenderPass                                renderPass,
    VkExtent2D*                                 pGranularity);
```

- `device` is the logical device that owns the render pass.
- `renderPass` is a handle to a render pass.
- `pGranularity` is a pointer to a VkExtent2D structure in which the granularity is returned.

The conditions leading to an optimal `renderArea` are:

- the `offset.x` member in `renderArea` is a multiple of the `width` member of the returned VkExtent2D (the horizontal granularity).
- the `offset.y` member in `renderArea` is a multiple of the `height` member of the returned VkExtent2D (the vertical granularity).
- either the `extent.width` member in `renderArea` is a multiple of the horizontal granularity or `offset.x`+`extent.width` is equal to the `width` of the `framebuffer` in the VkRenderPassBeginInfo.
- either the `extent.height` member in `renderArea` is a multiple of the vertical granularity or `offset.y`+`extent.height` is equal to the `height` of the `framebuffer` in the VkRenderPassBeginInfo.

Subpass dependencies are not affected by the render area, and apply to the entire image subresources attached to the framebuffer as specified in the description of automatic layout transitions. Similarly, pipeline barriers are valid even if their effect extends outside the render area.

Valid Usage (Implicit)

- VUID-vkGetRenderAreaGranularity-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetRenderAreaGranularity-renderPass-parameter
  `renderPass` **must** be a valid VkRenderPass handle
- VUID-vkGetRenderAreaGranularity-pGranularity-parameter

`pGranularity` **must** be a valid pointer to a VkExtent2D structure

- VUID-vkGetRenderAreaGranularity-renderPass-parent
  `renderPass` **must** have been created, allocated, or retrieved from `device`

To transition to the next subpass in the render pass instance after recording the commands for a subpass, call:

WARNING	This functionality is superseded by vkCmdNextSubpass2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkCmdNextSubpass(
    VkCommandBuffer                             commandBuffer,
    VkSubpassContents                           contents);
```

- `commandBuffer` is the command buffer in which to record the command.
- `contents` specifies how the commands in the next subpass will be provided, in the same fashion as the corresponding parameter of vkCmdBeginRenderPass.

The subpass index for a render pass begins at zero when `vkCmdBeginRenderPass` is recorded, and increments each time `vkCmdNextSubpass` is recorded.

After transitioning to the next subpass, the application **can** record the commands for that subpass.

Valid Usage

- VUID-vkCmdNextSubpass-None-00909
  The current subpass index **must** be less than the number of subpasses in the render pass minus one
- VUID-vkCmdNextSubpass-None-02349
  This command **must** not be recorded when transform feedback is active

Valid Usage (Implicit)

- VUID-vkCmdNextSubpass-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdNextSubpass-contents-parameter
  `contents` **must** be a valid VkSubpassContents value
- VUID-vkCmdNextSubpass-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdNextSubpass-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations

- VUID-vkCmdNextSubpass-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdNextSubpass-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdNextSubpass-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdNextSubpass-bufferlevel
  `commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action State Synchronization

Conditional Rendering

vkCmdNextSubpass is not affected by conditional rendering

To transition to the next subpass in the render pass instance after recording the commands for a subpass, call:

WARNING This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
void vkCmdNextSubpass2(
    VkCommandBuffer                             commandBuffer,
    const VkSubpassBeginInfo*                   pSubpassBeginInfo,
    const VkSubpassEndInfo*                     pSubpassEndInfo);
```

```
// Provided by VK_KHR_create_renderpass2
// Equivalent to vkCmdNextSubpass2
void vkCmdNextSubpass2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkSubpassBeginInfo*                   pSubpassBeginInfo,
    const VkSubpassEndInfo*                     pSubpassEndInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pSubpassBeginInfo` is a pointer to a VkSubpassBeginInfo structure containing information about the subpass which is about to begin rendering.
- `pSubpassEndInfo` is a pointer to a VkSubpassEndInfo structure containing information about how the previous subpass will be ended.

`vkCmdNextSubpass2` is semantically identical to vkCmdNextSubpass, except that it is extensible, and that `contents` is provided as part of an extensible structure instead of as a flat parameter.

Valid Usage

- VUID-vkCmdNextSubpass2-None-03102
  The current subpass index **must** be less than the number of subpasses in the render pass minus one
- VUID-vkCmdNextSubpass2-None-02350
  This command **must** not be recorded when transform feedback is active

Valid Usage (Implicit)

- VUID-vkCmdNextSubpass2-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdNextSubpass2-pSubpassBeginInfo-parameter
  `pSubpassBeginInfo` **must** be a valid pointer to a valid VkSubpassBeginInfo structure
- VUID-vkCmdNextSubpass2-pSubpassEndInfo-parameter
  `pSubpassEndInfo` **must** be a valid pointer to a valid VkSubpassEndInfo structure
- VUID-vkCmdNextSubpass2-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdNextSubpass2-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdNextSubpass2-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdNextSubpass2-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdNextSubpass2-videocoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdNextSubpass2-bufferlevel
  `commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action State Synchronization

Conditional Rendering

vkCmdNextSubpass2 is not affected by conditional rendering

To record a command to end a render pass instance after recording the commands for the last subpass, call:

WARNING This functionality is superseded by vkCmdEndRenderPass2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkCmdEndRenderPass(
    VkCommandBuffer                             commandBuffer);
```

- `commandBuffer` is the command buffer in which to end the current render pass instance.

Ending a render pass instance performs any multisample resolve operations on the final subpass.

NOTE There is no implicit ordering between separate render passes, even in the same command buffer, and even when the attachments match. Some applications rely on the continuation of rasterization order between multiple render passes with attachments defined in the same way, in order to perform non-rendering operations (such as copies or compute operations) between draw calls, but this has never been required by the specification. There is also no explicit barrier currently in the API

that provides the guarantee that applications rely on without additional performance penalties.

New applications should avoid relying on this ordering until an appropriate barrier is added to the API.

Implementations where applications are performing this splitting are encouraged to continue supporting this guarantee until a suitable barrier is added to the API.

Existing applications relying on this ordering should expect that it will continue working on platforms where it currently does. Once a new extension adds support for a new barrier, developers are encouraged to adapt their applications to use this when available.

Valid Usage

- VUID-vkCmdEndRenderPass-None-00910
  The current subpass index **must** be equal to the number of subpasses in the render pass minus one
- VUID-vkCmdEndRenderPass-None-02351
  This command **must** not be recorded when transform feedback is active
- VUID-vkCmdEndRenderPass-None-06170
  The current render pass instance **must** not have been begun with vkCmdBeginRendering
- VUID-vkCmdEndRenderPass-None-07004
  If `vkCmdBeginQuery*` was called within a subpass of the render pass, the corresponding `vkCmdEndQuery*` **must** have been called subsequently within the same subpass

Valid Usage (Implicit)

- VUID-vkCmdEndRenderPass-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdEndRenderPass-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdEndRenderPass-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdEndRenderPass-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdEndRenderPass-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdEndRenderPass-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdEndRenderPass-bufferlevel

`commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action State Synchronization

Conditional Rendering

vkCmdEndRenderPass is not affected by conditional rendering

To record a command to end a render pass instance after recording the commands for the last subpass, call:

WARNING This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
void vkCmdEndRenderPass2(
    VkCommandBuffer                             commandBuffer,
    const VkSubpassEndInfo*                     pSubpassEndInfo);
```

```
// Provided by VK_KHR_create_renderpass2
// Equivalent to vkCmdEndRenderPass2
void vkCmdEndRenderPass2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkSubpassEndInfo*                     pSubpassEndInfo);
```

- `commandBuffer` is the command buffer in which to end the current render pass instance.
- `pSubpassEndInfo` is a pointer to a VkSubpassEndInfo structure containing information about how the last subpass will be ended.

vkCmdEndRenderPass2 is semantically identical to vkCmdEndRenderPass, except that it is extensible.

There is no implicit ordering between separate render passes, even in the same command buffer, and even when the attachments match. Some applications rely on the continuation of rasterization order between multiple render passes with attachments defined in the same way, in order to perform non-rendering operations (such as copies or compute operations) between draw calls, but this has never been required by the specification. There is also no explicit barrier currently in the API that provides the guarantee that applications rely on without additional performance penalties.

NOTE

New applications should avoid relying on this ordering until an appropriate barrier is added to the API.

Implementations where applications are performing this splitting are encouraged to continue supporting this guarantee until a suitable barrier is added to the API.

Existing applications relying on this ordering should expect that it will continue working on platforms where it currently does. Once a new extension adds support for a new barrier, developers are encouraged to adapt their applications to use this when available.

Valid Usage

- VUID-vkCmdEndRenderPass2-None-03103
  The current subpass index **must** be equal to the number of subpasses in the render pass minus one
- VUID-vkCmdEndRenderPass2-None-02352
  This command **must** not be recorded when transform feedback is active
- VUID-vkCmdEndRenderPass2-None-06171
  The current render pass instance **must** not have been begun with vkCmdBeginRendering
- VUID-vkCmdEndRenderPass2-None-07005
  If vkCmdBeginQuery* was called within a subpass of the render pass, the corresponding vkCmdEndQuery* **must** have been called subsequently within the same subpass

Valid Usage (Implicit)

- VUID-vkCmdEndRenderPass2-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdEndRenderPass2-pSubpassEndInfo-parameter
  pSubpassEndInfo **must** be a valid pointer to a valid VkSubpassEndInfo structure
- VUID-vkCmdEndRenderPass2-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdEndRenderPass2-commandBuffer-cmdpool

The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations

- VUID-vkCmdEndRenderPass2-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdEndRenderPass2-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdEndRenderPass2-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdEndRenderPass2-bufferlevel
  commandBuffer **must** be a primary VkCommandBuffer

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action State Synchronization

Conditional Rendering

vkCmdEndRenderPass2 is not affected by conditional rendering

The VkSubpassEndInfo structure is defined as:

WARNING This functionality is superseded by Vulkan Version 1.4. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_2
typedef struct VkSubpassEndInfo {
    VkStructureType    sType;
    const void*        pNext;
} VkSubpassEndInfo;
```

```
// Provided by VK_KHR_create_renderpass2
// Equivalent to VkSubpassEndInfo
typedef VkSubpassEndInfo VkSubpassEndInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

Valid Usage (Implicit)

- VUID-VkSubpassEndInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SUBPASS_END_INFO
- VUID-VkSubpassEndInfo-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkRenderPassFragmentDensityMapOffsetEndInfoEXT
- VUID-VkSubpassEndInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

If the VkSubpassEndInfo::`pNext` chain or VkRenderingEndInfoEXT::`pNext` chain includes a `VkRenderPassFragmentDensityMapOffsetEndInfoEXT` structure, then that structure includes an array of fragment density map offsets per layer for the render pass.

The `VkRenderPassFragmentDensityMapOffsetEndInfoEXT` structure is defined as:

```
// Provided by VK_EXT_fragment_density_map_offset
typedef struct VkRenderPassFragmentDensityMapOffsetEndInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    uint32_t             fragmentDensityOffsetCount;
    const VkOffset2D*    pFragmentDensityOffsets;
} VkRenderPassFragmentDensityMapOffsetEndInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `fragmentDensityOffsetCount` is the number of offsets being specified.
- `pFragmentDensityOffsets` is a pointer to an array of VkOffset2D structs, each of which describes the offset per layer.

The array elements are given per `layer` as defined by Fetch Density Value, where index = layer. Each (x,y) offset is in framebuffer pixels and shifts the fetch of the fragment density map by that amount. Offsets can be positive or negative.

If neither the VkSubpassEndInfo::`pNext` chain for the last subpass of a render pass nor the VkRenderingEndInfoEXT::`pNext` chain of a dynamic render pass include `VkRenderPassFragmentDensityMapOffsetEndInfoEXT`, or if `fragmentDensityOffsetCount` is zero, then the

offset (0,0) is used for Fetch Density Value.

Valid Usage

- VUID-VkRenderPassFragmentDensityMapOffsetEndInfoEXT-fragmentDensityMapOffsets-06503
  If the `fragmentDensityMapOffset` feature is not enabled or fragment density map is not enabled in the render pass, `fragmentDensityOffsetCount` **must** equal `0`
- VUID-VkRenderPassFragmentDensityMapOffsetEndInfoEXT-fragmentDensityMapAttachment-06504
  If VkRenderPassFragmentDensityMapCreateInfoEXT::`fragmentDensityMapAttachment` is not VK_ATTACHMENT_UNUSED and was not created with VK_IMAGE_CREATE_FRAGMENT_DENSITY_MAP_OFFSET_BIT_EXT, `fragmentDensityOffsetCount` **must** equal `0`
- VUID-VkRenderPassFragmentDensityMapOffsetEndInfoEXT-pDepthStencilAttachment-06505
  If the depth or stencil attachments for the render pass are used and were not created with VK_IMAGE_CREATE_FRAGMENT_DENSITY_MAP_OFFSET_BIT_EXT, `fragmentDensityOffsetCount` **must** equal `0`
- VUID-VkRenderPassFragmentDensityMapOffsetEndInfoEXT-pInputAttachments-06506
  If any used input attachments for the render pass were not created with VK_IMAGE_CREATE_FRAGMENT_DENSITY_MAP_OFFSET_BIT_EXT, `fragmentDensityOffsetCount` **must** equal `0`
- VUID-VkRenderPassFragmentDensityMapOffsetEndInfoEXT-pColorAttachments-06507
  If any used color attachments for the render pass were not created with VK_IMAGE_CREATE_FRAGMENT_DENSITY_MAP_OFFSET_BIT_EXT, `fragmentDensityOffsetCount` **must** equal `0`
- VUID-VkRenderPassFragmentDensityMapOffsetEndInfoEXT-pResolveAttachments-06508
  If any used resolve attachments for the render pass were not created with VK_IMAGE_CREATE_FRAGMENT_DENSITY_MAP_OFFSET_BIT_EXT, `fragmentDensityOffsetCount` **must** equal `0`
- VUID-VkRenderPassFragmentDensityMapOffsetEndInfoEXT-pPreserveAttachments-06509
  If any used preserve attachments for the render pass were not created with VK_IMAGE_CREATE_FRAGMENT_DENSITY_MAP_OFFSET_BIT_EXT, `fragmentDensityOffsetCount` **must** equal `0`
- VUID-VkRenderPassFragmentDensityMapOffsetEndInfoEXT-fragmentDensityOffsetCount-06510
  If `fragmentDensityOffsetCount` is not `0` and multiview is enabled for the render pass, `fragmentDensityOffsetCount` **must** equal the `layerCount` that was specified in creating the fragment density map attachment view
- VUID-VkRenderPassFragmentDensityMapOffsetEndInfoEXT-fragmentDensityOffsetCount-06511
  If `fragmentDensityOffsetCount` is not `0` and multiview is not enabled for the render pass, `fragmentDensityOffsetCount` **must** equal `1`

- VUID-VkRenderPassFragmentDensityMapOffsetEndInfoEXT-x-06512
  The `x` component of each element of `pFragmentDensityOffsets` **must** be an integer multiple of `fragmentDensityOffsetGranularity.width`
- VUID-VkRenderPassFragmentDensityMapOffsetEndInfoEXT-y-06513
  The `y` component of each element of `pFragmentDensityOffsets` **must** be an integer multiple of `fragmentDensityOffsetGranularity.height`
- VUID-VkRenderPassFragmentDensityMapOffsetEndInfoEXT-pFragmentDensityOffsets-10730
  Each element of `pFragmentDensityOffsets` **must** be identical for every vkCmdEndRendering2KHR call made in a render pass

Valid Usage (Implicit)

- VUID-VkRenderPassFragmentDensityMapOffsetEndInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RENDER_PASS_FRAGMENT_DENSITY_MAP_OFFSET_END_INFO_EXT
- VUID-VkRenderPassFragmentDensityMapOffsetEndInfoEXT-pFragmentDensityOffsets-parameter
  If `fragmentDensityOffsetCount` is not `0`, `pFragmentDensityOffsets` **must** be a valid pointer to an array of `fragmentDensityOffsetCount` VkOffset2D structures

Structure Chaining

Extends the structures

- VkRenderingEndInfoKHR
- VkSubpassEndInfo

8.9. Render Pass Creation Feedback

A `VkRenderPassCreationControlEXT` structure **can** be included in the `pNext` chain of `VkRenderPassCreateInfo2` or `pNext` chain of VkSubpassDescription2. The `VkRenderPassCreationControlEXT` structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4 and VK_KHR_dynamic_rendering_local_read. See Legacy Functionality for more information.

```
// Provided by VK_EXT_subpass_merge_feedback
typedef struct VkRenderPassCreationControlEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           disallowMerging;
} VkRenderPassCreationControlEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `disallowMerging` is a boolean value indicating whether subpass merging will be disabled.

If a `VkRenderPassCreationControlEXT` structure is included in the `pNext` chain of VkRenderPassCreateInfo2 and its value of `disallowMerging` is VK_TRUE, the implementation will disable subpass merging for the entire render pass. If a `VkRenderPassCreationControlEXT` structure is included in the `pNext` chain of VkSubpassDescription2 and its value of `disallowMerging` is VK_TRUE, the implementation will disable merging the described subpass with previous subpasses in the render pass.

Valid Usage (Implicit)

- VUID-VkRenderPassCreationControlEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RENDER_PASS_CREATION_CONTROL_EXT

Structure Chaining

Extends the structures

- VkRenderPassCreateInfo2
- VkSubpassDescription2

To obtain feedback about the creation of a render pass, include a `VkRenderPassCreationFeedbackCreateInfoEXT` structure in the `pNext` chain of VkRenderPassCreateInfo2. The `VkRenderPassCreationFeedbackCreateInfoEXT` structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4 and VK_KHR_dynamic_rendering_local_read. See Legacy Functionality for more information.

```
// Provided by VK_EXT_subpass_merge_feedback
typedef struct VkRenderPassCreationFeedbackCreateInfoEXT {
    VkStructureType                         sType;
    const void*                             pNext;
    VkRenderPassCreationFeedbackInfoEXT*    pRenderPassFeedback;
} VkRenderPassCreationFeedbackCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pRenderPassFeedback` is a pointer to a VkRenderPassCreationFeedbackInfoEXT structure in which feedback is returned.

Valid Usage (Implicit)

- VUID-VkRenderPassCreationFeedbackCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RENDER_PASS_CREATION_FEEDBACK_CREATE_INFO_EXT
- VUID-VkRenderPassCreationFeedbackCreateInfoEXT-pRenderPassFeedback-parameter
  `pRenderPassFeedback` **must** be a valid pointer to a VkRenderPassCreationFeedbackInfoEXT structure

Structure Chaining

Extends the structure

- VkRenderPassCreateInfo2

The `VkRenderPassCreationFeedbackInfoEXT` structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4 and VK_KHR_dynamic_rendering_local_read. See Legacy Functionality for more information.

```
// Provided by VK_EXT_subpass_merge_feedback
typedef struct VkRenderPassCreationFeedbackInfoEXT {
    uint32_t    postMergeSubpassCount;
} VkRenderPassCreationFeedbackInfoEXT;
```

- `postMergeSubpassCount` is the subpass count after merge.

Feedback about the creation of a subpass **can** be obtained by including a `VkRenderPassSubpassFeedbackCreateInfoEXT` structure in the `pNext` chain of VkSubpassDescription2. The `VkRenderPassSubpassFeedbackCreateInfoEXT` structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4 and VK_KHR_dynamic_rendering_local_read. See Legacy Functionality for more information.

```
// Provided by VK_EXT_subpass_merge_feedback
typedef struct VkRenderPassSubpassFeedbackCreateInfoEXT {
    VkStructureType                         sType;
    const void*                             pNext;
    VkRenderPassSubpassFeedbackInfoEXT*     pSubpassFeedback;
} VkRenderPassSubpassFeedbackCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pSubpassFeedback` is a pointer to a VkRenderPassSubpassFeedbackInfoEXT structure in which feedback is returned.

Valid Usage (Implicit)

- VUID-VkRenderPassSubpassFeedbackCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RENDER_PASS_SUBPASS_FEEDBACK_CREATE_INFO_EXT
- VUID-VkRenderPassSubpassFeedbackCreateInfoEXT-pSubpassFeedback-parameter
  `pSubpassFeedback` **must** be a valid pointer to a VkRenderPassSubpassFeedbackInfoEXT structure

Structure Chaining

Extends the structure

- VkSubpassDescription2

The `VkRenderPassSubpassFeedbackInfoEXT` structure is defined as:

WARNING	This functionality is superseded by Vulkan Version 1.4 and VK_KHR_dynamic_rendering_local_read. See Legacy Functionality for more information.

```
// Provided by VK_EXT_subpass_merge_feedback
typedef struct VkRenderPassSubpassFeedbackInfoEXT {
    VkSubpassMergeStatusEXT    subpassMergeStatus;
    char                       description[VK_MAX_DESCRIPTION_SIZE];
    uint32_t                   postMergeIndex;
} VkRenderPassSubpassFeedbackInfoEXT;
```

- `subpassMergeStatus` is a VkSubpassMergeStatusEXT value specifying information about whether the subpass is merged with the previous subpass and the reason why it is not merged.
- `description` is an array of VK_MAX_DESCRIPTION_SIZE `char` containing a null-terminated UTF-8

string which provides additional details.

- `postMergeIndex` is the subpass index after the subpass merging.

Possible values of VkRenderPassSubpassFeedbackInfoEXT::`subpassMergeStatus` are:

```
// Provided by VK_EXT_subpass_merge_feedback
typedef enum VkSubpassMergeStatusEXT {
    VK_SUBPASS_MERGE_STATUS_MERGED_EXT = 0,
    VK_SUBPASS_MERGE_STATUS_DISALLOWED_EXT = 1,
    VK_SUBPASS_MERGE_STATUS_NOT_MERGED_SIDE_EFFECTS_EXT = 2,
    VK_SUBPASS_MERGE_STATUS_NOT_MERGED_SAMPLES_MISMATCH_EXT = 3,
    VK_SUBPASS_MERGE_STATUS_NOT_MERGED_VIEWS_MISMATCH_EXT = 4,
    VK_SUBPASS_MERGE_STATUS_NOT_MERGED_ALIASING_EXT = 5,
    VK_SUBPASS_MERGE_STATUS_NOT_MERGED_DEPENDENCIES_EXT = 6,
    VK_SUBPASS_MERGE_STATUS_NOT_MERGED_INCOMPATIBLE_INPUT_ATTACHMENT_EXT = 7,
    VK_SUBPASS_MERGE_STATUS_NOT_MERGED_TOO_MANY_ATTACHMENTS_EXT = 8,
    VK_SUBPASS_MERGE_STATUS_NOT_MERGED_INSUFFICIENT_STORAGE_EXT = 9,
    VK_SUBPASS_MERGE_STATUS_NOT_MERGED_DEPTH_STENCIL_COUNT_EXT = 10,
    VK_SUBPASS_MERGE_STATUS_NOT_MERGED_RESOLVE_ATTACHMENT_REUSE_EXT = 11,
    VK_SUBPASS_MERGE_STATUS_NOT_MERGED_SINGLE_SUBPASS_EXT = 12,
    VK_SUBPASS_MERGE_STATUS_NOT_MERGED_UNSPECIFIED_EXT = 13,
} VkSubpassMergeStatusEXT;
```

- VK_SUBPASS_MERGE_STATUS_MERGED_EXT specifies that the subpass is merged with a previous subpass.
- VK_SUBPASS_MERGE_STATUS_DISALLOWED_EXT specifies that the subpass is not merged because merging was disabled using VkRenderPassCreationControlEXT. If the render pass does not allow subpass merging, then all subpass statuses are set to this value. If a subpass description does not allow subpass merging, then only that subpass's status is set to this value.
- VK_SUBPASS_MERGE_STATUS_NOT_MERGED_SIDE_EFFECTS_EXT specifies that the subpass is not merged because it contains side effects.
- VK_SUBPASS_MERGE_STATUS_NOT_MERGED_SAMPLES_MISMATCH_EXT specifies that the subpass is not merged because sample count is not compatible with the previous subpass.
- VK_SUBPASS_MERGE_STATUS_NOT_MERGED_VIEWS_MISMATCH_EXT specifies that the subpass is not merged because view masks do not match with previous subpass.
- VK_SUBPASS_MERGE_STATUS_NOT_MERGED_ALIASING_EXT specifies that the subpass is not merged because of attachments aliasing between them.
- VK_SUBPASS_MERGE_STATUS_NOT_MERGED_DEPENDENCIES_EXT specifies that the subpass is not merged because subpass dependencies do not allow merging.
- VK_SUBPASS_MERGE_STATUS_NOT_MERGED_INCOMPATIBLE_INPUT_ATTACHMENT_EXT specifies that the subpass is not merged because input attachment is not a color attachment from previous subpass or the formats are incompatible.
- VK_SUBPASS_MERGE_STATUS_NOT_MERGED_TOO_MANY_ATTACHMENTS_EXT specifies that the subpass is not merged because of too many attachments.

- VK_SUBPASS_MERGE_STATUS_NOT_MERGED_INSUFFICIENT_STORAGE_EXT specifies that the subpass is not merged because of insufficient memory.
- VK_SUBPASS_MERGE_STATUS_NOT_MERGED_DEPTH_STENCIL_COUNT_EXT specifies that the subpass is not merged because of too many depth/stencil attachments.
- VK_SUBPASS_MERGE_STATUS_NOT_MERGED_RESOLVE_ATTACHMENT_REUSE_EXT specifies that the subpass is not merged because a resolve attachment is reused as an input attachment in a subsequent subpass.
- VK_SUBPASS_MERGE_STATUS_NOT_MERGED_SINGLE_SUBPASS_EXT specifies that the subpass is not merged because the render pass has only one subpass.
- VK_SUBPASS_MERGE_STATUS_NOT_MERGED_UNSPECIFIED_EXT specifies that the subpass is not merged for unspecified reasons. Implementations **should** return this value when no other VkSubpassMergeStatusEXT value is appropriate.

8.10. Common Render Pass Data Races (Informative)

Due to the complexity of how rendering is performed, there are several ways an application can accidentally introduce a data race, usually by doing something that may seem benign but actually cannot be supported. This section indicates a number of the more common cases as guidelines to help avoid them.

8.10.1. Sampling From a Read-Only Attachment

Vulkan includes read-only layouts for depth/stencil images, that allow the images to be both read during a render pass for the purposes of depth/stencil tests, and read as a non-attachment.

However, because VK_ATTACHMENT_STORE_OP_STORE and VK_ATTACHMENT_STORE_OP_DONT_CARE may perform write operations, even if no recorded command writes to an attachment, reading from an image while also using it as an attachment with these store operations can result in a data race. If the reads from the non-attachment are performed in a fragment shader where the accessed samples match those covered by the fragment shader, no data race will occur as store operations are guaranteed to operate after fragment shader execution for the set of samples the fragment covers. Notably, input attachments can also be used for this case. Reading other samples or in any other shader stage can result in unexpected behavior due to the potential for a data race, and validation errors should be generated for doing so. In practice, many applications have shipped reading samples outside of the covered fragment without any observable issue, but there is no guarantee that this will always work, and it is not advisable to rely on this in new or re-worked code bases. As VK_ATTACHMENT_STORE_OP_NONE is guaranteed to perform no writes, applications wishing to read an image as both an attachment and a non-attachment should make use of this store operation, coupled with a load operation that also performs no writes.

8.10.2. Non-Overlapping Access Between Resources

When relying on non-overlapping accesses between attachments and other resources, it is important to note that load and store operations have fairly wide alignment requirements - potentially affecting entire subresources and adjacent depth/stencil aspects. This makes it invalid to

access a non-attachment subresource that is simultaneously being used as an attachment where either access performs a write operation.

The only exception to this is if a subresource is explicitly specified for use as a feedback loop attachment. Feedback loop is enabled for an attachment if either of the following conditions are satisfied:

- The corresponding attachment is in the VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT layout
- For the attachment identified by VkRenderingAttachmentInfo::`imageView`, VkAttachmentFeedbackLoopInfoEXT::`feedbackLoopEnable` is VK_TRUE

In that case the overlap is defined to occur at a per-pixel granularity, and applications can read data from pixels outside the render area without introducing a data race.

In addition to the attachment layout requirements, extra state **must** be set for non-attachment uses to be valid. For every image aspect accessed with feedback loops, one of the following conditions **must** be satisfied:

- The feedback loop attachment is a color attachment, and the bound pipeline was created with VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT.
- The feedback loop attachment is a depth-stencil attachment, and the bound pipeline was created with VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT.
- There is no bound pipeline or the bound pipeline was created with VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT and the `aspectMask` state in vkCmdSetAttachmentFeedbackLoopEnableEXT has been set and not subsequently invalidated in the current command buffer, and `aspectMask` includes all aspects which require feedback loop.

8.10.3. Depth/Stencil and Input Attachments

When rendering to only the depth OR stencil aspect of an image, an input attachment accessing the other aspect will not cause a data race only under very specific conditions. To avoid a data race, the aspect not being written **must** be in a read-only layout, and writes to it **must** be disabled in the draw state. For example, to read from stencil while writing depth, the attachment **must** be in VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL (or equivalent), and the stencil write mask **must** be 0. Similarly to read from depth while writing stencil, the attachment **must** be in VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL (or equivalent), and depth write enable **must** be VK_FALSE.

8.10.4. Access Via Different Image Resources Types

If overlapping ranges of a non-attachment image resource are accessed via different resource types in the same render pass instance, and at least one of the accesses is a write operation, there is a data race.

For example, a write to a storage image **may** not be visible to a subsequent read from a sampled

image. This is true even if the resources use the same layout, such as VK_IMAGE_LAYOUT_GENERAL.

If the image resource is also used as an attachment in the render pass instance, the overlapping range is as described for feedback loops.

8.10.5. Synchronization Options

There are several synchronization options available to synchronize between accesses to resources within a render pass. Some of the options are outlined below:

- A VkSubpassDependency in a render pass object can synchronize attachment writes and multisample resolve operations from a prior subpass for subsequent input attachment reads.
- A vkCmdPipelineBarrier inside a subpass can synchronize prior attachment writes in the subpass with subsequent input attachment reads.
- A vkCmdPipelineBarrier inside a subpass can synchronize prior attachment writes in the subpass with subsequent non-attachment reads if feedback loop is enabled for the attachment.
- If a subresource is used as a color and input attachment, and the pipeline performing the read was created with VK_PIPELINE_COLOR_BLEND_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_ACCESS_BIT_EXT
- If a subresource is used as a depth and input attachment, and the pipeline performing the read was created with VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT_EXT
- If a subresource is used as a stencil and input attachment, and the pipeline performing the read was created with VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_BIT_EXT
- If a subresource is used as two separate non-attachment resources, writes to a pixel or individual sample in a fragment shader can be synchronized with access to the same pixel or sample in another fragment shader by using one of the fragment interlock execution modes.

Chapter 9. Shaders

A shader specifies programmable operations that execute for each vertex, control point, tessellated vertex, primitive, fragment, or workgroup in the corresponding stage(s) of the graphics and compute pipelines.

Graphics pipelines include vertex shader execution as a result of primitive assembly, followed, if enabled, by tessellation control and evaluation shaders operating on patches, geometry shaders, if enabled, operating on primitives, and fragment shaders, if present, operating on fragments generated by Rasterization. In this specification, vertex, tessellation control, tessellation evaluation and geometry shaders are collectively referred to as pre-rasterization shader stages and occur in the logical pipeline before rasterization. The fragment shader occurs logically after rasterization.

Only the compute shader stage is included in a compute pipeline. Compute shaders operate on compute invocations in a workgroup.

Shaders **can** read from input variables, and read from and write to output variables. Input and output variables **can** be used to transfer data between shader stages, or to allow the shader to interact with values that exist in the execution environment. Similarly, the execution environment provides constants describing capabilities.

Shader variables are associated with execution environment-provided inputs and outputs using *built-in* decorations in the shader. The available decorations for each stage are documented in the following subsections.

9.1. Shader Objects

Shaders **may** be compiled and linked into pipeline objects as described in the Pipelines chapter, or if the `shaderObject` feature is enabled they **may** be compiled into individual per-stage *shader objects* which **can** be bound on a command buffer independently from one another. Unlike pipelines, shader objects are not intrinsically tied to any specific set of state. Instead, state is specified dynamically in the command buffer.

Each shader object represents a single compiled shader stage, which **may optionally** be linked with one or more other stages.

Shader objects are represented by `VkShaderEXT` handles:

```
// Provided by VK_EXT_shader_object
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkShaderEXT)
```

9.1.1. Shader Object Creation

Shader objects **may** be created from shader code provided as SPIR-V, or in an opaque, implementation-defined binary format specific to the physical device.

To create one or more shader objects, call:

```
// Provided by VK_EXT_shader_object
VkResult vkCreateShadersEXT(
    VkDevice                                    device,
    uint32_t                                    createInfoCount,
    const VkShaderCreateInfoEXT*                pCreateInfos,
    const VkAllocationCallbacks*                pAllocator,
    VkShaderEXT*                                pShaders);
```

- `device` is the logical device that creates the shader objects.
- `createInfoCount` is the length of the `pCreateInfos` and `pShaders` arrays.
- `pCreateInfos` is a pointer to an array of VkShaderCreateInfoEXT structures.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pShaders` is a pointer to an array of VkShaderEXT handles in which the resulting shader objects are returned.

When this function returns, whether or not it succeeds, it is guaranteed that every element of `pShaders` will have been overwritten by either VK_NULL_HANDLE or a valid `VkShaderEXT` handle.

This means that whenever shader creation fails, the application **can** determine which shader the returned error pertains to by locating the first VK_NULL_HANDLE element in `pShaders`. It also means that an application **can** reliably clean up from a failed call by iterating over the `pShaders` array and destroying every element that is not VK_NULL_HANDLE.

Valid Usage

- VUID-vkCreateShadersEXT-stage-09670
  If the `stage` member of any element of `pCreateInfos` is VK_SHADER_STAGE_COMPUTE_BIT, `device` **must** support at least one queue family with the VK_QUEUE_COMPUTE_BIT capability
- VUID-vkCreateShadersEXT-stage-09671
  If the `stage` member of any element of `pCreateInfos` is VK_SHADER_STAGE_TASK_BIT_EXT, VK_SHADER_STAGE_MESH_BIT_EXT, VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, VK_SHADER_STAGE_GEOMETRY_BIT, or VK_SHADER_STAGE_FRAGMENT_BIT, `device` **must** support at least one queue family with the VK_QUEUE_GRAPHICS_BIT capability
- VUID-vkCreateShadersEXT-None-08400
  The `shaderObject` feature **must** be enabled
- VUID-vkCreateShadersEXT-pCreateInfos-08402
  If the `flags` member of any element of `pCreateInfos` includes VK_SHADER_CREATE_LINK_STAGE_BIT_EXT, the `flags` member of all other elements of `pCreateInfos` whose `stage` is VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT,

VK_SHADER_STAGE_GEOMETRY_BIT, or VK_SHADER_STAGE_FRAGMENT_BIT **must** also include VK_SHADER_CREATE_LINK_STAGE_BIT_EXT

- VUID-vkCreateShadersEXT-pCreateInfos-08403
  If the `flags` member of any element of `pCreateInfos` includes VK_SHADER_CREATE_LINK_STAGE_BIT_EXT, the `flags` member of all other elements of `pCreateInfos` whose `stage` is VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT **must** also include VK_SHADER_CREATE_LINK_STAGE_BIT_EXT
- VUID-vkCreateShadersEXT-pCreateInfos-08404
  If the `flags` member of any element of `pCreateInfos` whose `stage` is VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT includes VK_SHADER_CREATE_LINK_STAGE_BIT_EXT, there **must** be no member of `pCreateInfos` whose `flags` member includes VK_SHADER_CREATE_LINK_STAGE_BIT_EXT and whose `stage` is any of VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_GEOMETRY_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, or VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-vkCreateShadersEXT-pCreateInfos-08405
  If there is any element of `pCreateInfos` whose `stage` is VK_SHADER_STAGE_MESH_BIT_EXT and whose `flags` member includes both VK_SHADER_CREATE_LINK_STAGE_BIT_EXT and VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT, there **must** be no element of `pCreateInfos` whose `stage` is VK_SHADER_STAGE_TASK_BIT_EXT and whose `flags` member includes VK_SHADER_CREATE_LINK_STAGE_BIT_EXT
- VUID-vkCreateShadersEXT-pCreateInfos-08409
  For each element of `pCreateInfos` whose `flags` member includes VK_SHADER_CREATE_LINK_STAGE_BIT_EXT, if there is any other element of `pCreateInfos` whose `stage` is logically later than the `stage` of the former and whose `flags` member also includes VK_SHADER_CREATE_LINK_STAGE_BIT_EXT, the `nextStage` of the former **must** be equal to the `stage` of the element with the logically earliest `stage` following the `stage` of the former whose `flags` member also includes VK_SHADER_CREATE_LINK_STAGE_BIT_EXT
- VUID-vkCreateShadersEXT-pCreateInfos-08410
  The `stage` member of each element of `pCreateInfos` whose `flags` member includes VK_SHADER_CREATE_LINK_STAGE_BIT_EXT **must** be unique
- VUID-vkCreateShadersEXT-pCreateInfos-08411
  The `codeType` member of all elements of `pCreateInfos` whose `flags` member includes VK_SHADER_CREATE_LINK_STAGE_BIT_EXT **must** be the same
- VUID-vkCreateShadersEXT-pCreateInfos-12224
  If `pCreateInfos` contains elements with both VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, both elements' `flags` include VK_SHADER_CREATE_LINK_STAGE_BIT_EXT, both elements' `codeType` is VK_SHADER_CODE_TYPE_SPIRV_EXT, at least one **must** contain an `OpExecutionMode` instruction specifying the orientation of triangles generated by the tessellator
- VUID-vkCreateShadersEXT-pCreateInfos-12225
  If `pCreateInfos` contains elements with both VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and

VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, both elements' flags include VK_SHADER_CREATE_LINK_STAGE_BIT_EXT, both elements' codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, at least one **must** contain an OpExecutionMode instruction specifying the spacing of segments on the edges of tessellated primitives

- VUID-vkCreateShadersEXT-pCreateInfos-08867
  If pCreateInfos contains elements with both VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, both elements' flags include VK_SHADER_CREATE_LINK_STAGE_BIT_EXT, both elements' codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, and both stages contain an OpExecutionMode instruction specifying the type of subdivision, they **must** be the same
- VUID-vkCreateShadersEXT-pCreateInfos-08868
  If pCreateInfos contains elements with both VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, both elements' flags include VK_SHADER_CREATE_LINK_STAGE_BIT_EXT, both elements' codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, and both stages contain an OpExecutionMode instruction specifying the orientation of triangles, they **must** be the same
- VUID-vkCreateShadersEXT-pCreateInfos-08870
  If pCreateInfos contains elements with both VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, both elements' flags include VK_SHADER_CREATE_LINK_STAGE_BIT_EXT, both elements' codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, and both stages contain an OpExecutionMode instruction specifying the spacing of segments on the edges of tessellated primitives, they **must** be the same
- VUID-vkCreateShadersEXT-pCreateInfos-08871
  If pCreateInfos contains elements with both VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, both elements' flags include VK_SHADER_CREATE_LINK_STAGE_BIT_EXT, both elements' codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, and both stages contain an OpExecutionMode instruction specifying the output patch size, they **must** be the same
- VUID-vkCreateShadersEXT-pCreateInfos-09632
  If pCreateInfos contains a VK_SHADER_STAGE_MESH_BIT_EXT with codeType of VK_SHADER_CODE_TYPE_SPIRV_EXT and VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT is not set, then the mesh shader's entry point **must** not declare a variable with a DrawIndex BuiltIn decoration
- VUID-vkCreateShadersEXT-pCreateInfos-11413
  If any element of pCreateInfos sets VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT and includes embedded sampler mappings, there **must** be less than (maxSamplerAllocationCount - (minSamplerHeapReservedRangeWithEmbedded / samplerDescriptorSize)) VkSampler objects currently created on the device
- VUID-vkCreateShadersEXT-pCreateInfos-11428
  If any element of pCreateInfos sets VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT

and includes embedded sampler mappings, this command **must** not cause the total number of unique embedded samplers in pipelines and shaders on this device to exceed `maxDescriptorHeapEmbeddedSamplers`

- VUID-vkCreateShadersEXT-flags-11472
  If the `flags` member of any element of `pCreateInfos` includes VK_SHADER_CREATE_LINK_STAGE_BIT_EXT and VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT, the `flags` member of all other elements of `pCreateInfos` whose `stage` is VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, VK_SHADER_STAGE_GEOMETRY_BIT, VK_SHADER_STAGE_TASK_BIT_EXT, VK_SHADER_STAGE_MESH_BIT_EXT, or VK_SHADER_STAGE_FRAGMENT_BIT **must** also include VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT

Valid Usage (Implicit)

- VUID-vkCreateShadersEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateShadersEXT-pCreateInfos-parameter
  `pCreateInfos` **must** be a valid pointer to an array of `createInfoCount` valid VkShaderCreateInfoEXT structures
- VUID-vkCreateShadersEXT-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateShadersEXT-pShaders-parameter
  `pShaders` **must** be a valid pointer to an array of `createInfoCount` VkShaderEXT handles
- VUID-vkCreateShadersEXT-createInfoCount-arraylength
  `createInfoCount` **must** be greater than `0`

Return Codes

Success

- VK_INCOMPATIBLE_SHADER_BINARY_EXT
- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkShaderCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_shader_object
typedef struct VkShaderCreateInfoEXT {
    VkStructureType                 sType;
    const void*                     pNext;
    VkShaderCreateFlagsEXT          flags;
    VkShaderStageFlagBits           stage;
    VkShaderStageFlags              nextStage;
    VkShaderCodeTypeEXT             codeType;
    size_t                          codeSize;
    const void*                     pCode;
    const char*                     pName;
    uint32_t                        setLayoutCount;
    const VkDescriptorSetLayout*    pSetLayouts;
    uint32_t                        pushConstantRangeCount;
    const VkPushConstantRange*      pPushConstantRanges;
    const VkSpecializationInfo*     pSpecializationInfo;
} VkShaderCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask of VkShaderCreateFlagBitsEXT describing additional parameters of the shader.
- stage is a VkShaderStageFlagBits value specifying a single shader stage.
- nextStage is a bitmask of VkShaderStageFlagBits specifying which stages **can** be used as a logically next bound stage when drawing with the shader bound. A value of zero indicates this shader stage **must** be the last one.
- codeType is a VkShaderCodeTypeEXT value specifying the type of the shader code pointed to by pCode.
- codeSize is the size in bytes of the shader code pointed to by pCode.
- pCode is a pointer to the shader code to use to create the shader.
- pName is a pointer to a null-terminated UTF-8 string specifying the entry point name of the shader for this stage.
- setLayoutCount is the number of descriptor set layouts pointed to by pSetLayouts.
- pSetLayouts is a pointer to an array of VkDescriptorSetLayout objects used by the shader stage. The implementation **must** not access these objects outside of the duration of the command this structure is passed to.
- pushConstantRangeCount is the number of push constant ranges pointed to by pPushConstantRanges.
- pPushConstantRanges is a pointer to an array of VkPushConstantRange structures used by the shader stage.
- pSpecializationInfo is a pointer to a VkSpecializationInfo structure, as described in

Specialization Constants, or NULL.

When determining pipeline layout compatibility, compatibility is determined as if the shader was created with a pipeline layout object that was created with identical values of setLayoutCount, pSetLayouts, pushConstantRangeCount, and pPushConstantRanges, and with VkPipelineLayoutCreateInfo::flags set to 0. If flags includes VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT, the shader is not considered created with a pipeline layout.

When specifying descriptor heap mappings, only mappings corresponding to bindings that are actually present in the SPIR-V shader affect compilation. Mappings are ignored when codeType is VK_SHADER_CODE_TYPE_BINARY_EXT. The resulting compiled binary from two different SPIR-V shaders which would have identical bit patterns **must** remain identical even if the mapping entries vary in any of the following ways:

- Different numbers of unused mapping structures
- Different binding counts for unused bindings
- Unused parameters in mapping structures (e.g. sampler offsets)
- Ignored parameters in mapping structures
- Different order of mapping structures, used or unused

NOTE	Calculating the same offset in a mapping via different parameter values is not guaranteed to provide identical results.

Valid Usage

- VUID-VkShaderCreateInfoEXT-codeSize-08735
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, codeSize **must** be a multiple of 4
- VUID-VkShaderCreateInfoEXT-pCode-08736
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, pCode **must** point to valid SPIR-V code, formatted and packed as described by the Khronos SPIR-V Specification
- VUID-VkShaderCreateInfoEXT-pCode-08737
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, pCode **must** adhere to the validation rules described by the Validation Rules within a Module section of the SPIR-V Environment appendix
- VUID-VkShaderCreateInfoEXT-pCode-08738
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, pCode **must** declare the Shader capability for SPIR-V code
- VUID-VkShaderCreateInfoEXT-pCode-08739
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, pCode **must** not declare any capability that is not supported by the API, as described by the Capabilities section of the SPIR-V Environment appendix
- VUID-VkShaderCreateInfoEXT-pCode-08740
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, and pCode declares any of the

capabilities listed in the SPIR-V Environment appendix, one of the corresponding requirements **must** be satisfied

- VUID-VkShaderCreateInfoEXT-pCode-08741
  If `codeType` is VK_SHADER_CODE_TYPE_SPIRV_EXT, `pCode` **must** not declare any SPIR-V extension that is not supported by the API, as described by the Extension section of the SPIR-V Environment appendix
- VUID-VkShaderCreateInfoEXT-pCode-08742
  If `codeType` is VK_SHADER_CODE_TYPE_SPIRV_EXT, and `pCode` declares any of the SPIR-V extensions listed in the SPIR-V Environment appendix, one of the corresponding requirements **must** be satisfied
- VUID-VkShaderCreateInfoEXT-descriptorHeap-11314
  If the `descriptorHeap` feature is not enabled, VkShaderDescriptorSetAndBindingMappingInfoEXT::`mappingCount` **must** be 0
- VUID-VkShaderCreateInfoEXT-pNext-11315
  If the `pNext` chain specifies a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, the mapped resource in the shader **must** be a variable with a structure type decorated with `Block` in the `Uniform Storage Class`
- VUID-VkShaderCreateInfoEXT-pNext-11316
  If the `pNext` chain specifies a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, the mapped structure **must** not be larger than `maxPushDataSize` minus the `pushDataOffset` used in the mapping
- VUID-VkShaderCreateInfoEXT-pNext-12454
  If the `pNext` chain specifies a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, and there is a variable in the shader that matches the mapping, the `stage` **must** be a ray tracing stage
- VUID-VkShaderCreateInfoEXT-pNext-11317
  If the `pNext` chain specifies a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, the sum of mapped structure size and `shaderRecordDataOffset` used in the mapping **must** not be larger than `maxShaderGroupStride`
- VUID-VkShaderCreateInfoEXT-pNext-11318
  If the `pNext` chain specifies a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the mapped resource in the shader **must** be one of:
  - A variable with a structure type decorated with `Block` in the `Uniform Storage Class`
  - A variable with a structure type decorated with `BufferBlock` in the `Uniform Storage`

Class

- A variable with a structure type decorated with Block in the StorageBuffer Storage Class
- A OpTypeAccelerationStructureKHR variable

- VUID-VkShaderCreateInfoEXT-pNext-11378
  If the pNext chain specifies a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the OpArrayLength or OpUntypedArrayLengthKHR instruction **must** not be used on that resource
- VUID-VkShaderCreateInfoEXT-pNext-11399
  If the pNext chain specifies a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_CONSTANT_OFFSET_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, and the mapped resource declaration is an array, the pEmbeddedSampler member of the corresponding mapping structure **must** be NULL
- VUID-VkShaderCreateInfoEXT-flags-08412
  If stage is not VK_SHADER_STAGE_TASK_BIT_EXT, VK_SHADER_STAGE_MESH_BIT_EXT, VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, VK_SHADER_STAGE_GEOMETRY_BIT, or VK_SHADER_STAGE_FRAGMENT_BIT, flags **must** not include VK_SHADER_CREATE_LINK_STAGE_BIT_EXT
- VUID-VkShaderCreateInfoEXT-flags-08486
  If stage is not VK_SHADER_STAGE_FRAGMENT_BIT, flags **must** not include VK_SHADER_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_EXT
- VUID-VkShaderCreateInfoEXT-flags-08487
  If the attachmentFragmentShadingRate feature is not enabled, flags **must** not include VK_SHADER_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_EXT
- VUID-VkShaderCreateInfoEXT-flags-08488
  If stage is not VK_SHADER_STAGE_FRAGMENT_BIT, flags **must** not include VK_SHADER_CREATE_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-VkShaderCreateInfoEXT-flags-08489
  If the fragmentDensityMap feature is not enabled, flags **must** not include VK_SHADER_CREATE_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-VkShaderCreateInfoEXT-micromap-11623
  If flags includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT, the VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap feature **must** be enabled
- VUID-VkShaderCreateInfoEXT-flags-09404
  If flags includes VK_SHADER_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT_EXT, the

subgroupSizeControl feature **must** be enabled

- VUID-VkShaderCreateInfoEXT-flags-09405
  If flags includes VK_SHADER_CREATE_REQUIRE_FULL_SUBGROUPS_BIT_EXT, the computeFullSubgroups feature **must** be enabled
- VUID-VkShaderCreateInfoEXT-flags-11005
  If flags includes VK_SHADER_CREATE_INDIRECT_BINDABLE_BIT_EXT, then the VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT::deviceGeneratedCommands feature **must** be enabled
- VUID-VkShaderCreateInfoEXT-flags-11006
  If flags includes VK_SHADER_CREATE_INDIRECT_BINDABLE_BIT_EXT, then the identified entry point **must** not specify Xfb execution mode
- VUID-VkShaderCreateInfoEXT-flags-08992
  If flags includes VK_SHADER_CREATE_REQUIRE_FULL_SUBGROUPS_BIT_EXT, stage **must** be one of VK_SHADER_STAGE_MESH_BIT_EXT, VK_SHADER_STAGE_TASK_BIT_EXT, or VK_SHADER_STAGE_COMPUTE_BIT
- VUID-VkShaderCreateInfoEXT-flags-08485
  If stage is not VK_SHADER_STAGE_COMPUTE_BIT, flags **must** not include VK_SHADER_CREATE_DISPATCH_BASE_BIT_EXT
- VUID-VkShaderCreateInfoEXT-flags-08414
  If stage is not VK_SHADER_STAGE_MESH_BIT_EXT, flags **must** not include VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT
- VUID-VkShaderCreateInfoEXT-flags-08416
  If flags includes both VK_SHADER_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT_EXT and VK_SHADER_CREATE_REQUIRE_FULL_SUBGROUPS_BIT_EXT, the local workgroup size in the X dimension of the shader **must** be a multiple of maxSubgroupSize
- VUID-VkShaderCreateInfoEXT-flags-08417
  If flags includes VK_SHADER_CREATE_REQUIRE_FULL_SUBGROUPS_BIT_EXT but not VK_SHADER_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT_EXT and no VkShaderRequiredSubgroupSizeCreateInfoEXT structure is included in the pNext chain, the local workgroup size in the X dimension of the shader **must** be a multiple of subgroupSize
- VUID-VkShaderCreateInfoEXT-stage-08418
  stage **must** not be VK_SHADER_STAGE_ALL_GRAPHICS or VK_SHADER_STAGE_ALL
- VUID-VkShaderCreateInfoEXT-stage-08419
  If the tessellationShader feature is not enabled, stage **must** not be VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT or VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-VkShaderCreateInfoEXT-stage-08420
  If the geometryShader feature is not enabled, stage **must** not be VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-VkShaderCreateInfoEXT-stage-08421
  If the taskShader feature is not enabled, stage **must** not be VK_SHADER_STAGE_TASK_BIT_EXT

- VUID-VkShaderCreateInfoEXT-stage-08422
  If the meshShader feature is not enabled, stage **must** not be VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-VkShaderCreateInfoEXT-nextStage-08427
  If stage is VK_SHADER_STAGE_VERTEX_BIT, nextStage **must** not include any bits other than VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_GEOMETRY_BIT, and VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-VkShaderCreateInfoEXT-nextStage-08428
  If the tessellationShader feature is not enabled, nextStage **must** not include VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT or VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-VkShaderCreateInfoEXT-nextStage-08429
  If the geometryShader feature is not enabled, nextStage **must** not include VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-VkShaderCreateInfoEXT-nextStage-08430
  If stage is VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, nextStage **must** not include any bits other than VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-VkShaderCreateInfoEXT-nextStage-08431
  If stage is VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, nextStage **must** not include any bits other than VK_SHADER_STAGE_GEOMETRY_BIT and VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-VkShaderCreateInfoEXT-nextStage-08433
  If stage is VK_SHADER_STAGE_GEOMETRY_BIT, nextStage **must** not include any bits other than VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-VkShaderCreateInfoEXT-nextStage-08434
  If stage is VK_SHADER_STAGE_FRAGMENT_BIT or VK_SHADER_STAGE_COMPUTE_BIT, nextStage **must** be 0
- VUID-VkShaderCreateInfoEXT-nextStage-08435
  If stage is VK_SHADER_STAGE_TASK_BIT_EXT, nextStage **must** not include any bits other than VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-VkShaderCreateInfoEXT-nextStage-08436
  If stage is VK_SHADER_STAGE_MESH_BIT_EXT, nextStage **must** not include any bits other than VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-VkShaderCreateInfoEXT-pName-08440
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, pName **must** be the name of an OpEntryPoint in pCode with an execution model that matches stage
- VUID-VkShaderCreateInfoEXT-pCode-08492
  If codeType is VK_SHADER_CODE_TYPE_BINARY_EXT, pCode **must** be aligned to 16 bytes
- VUID-VkShaderCreateInfoEXT-pCode-08493
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, pCode **must** be aligned to 4 bytes
- VUID-VkShaderCreateInfoEXT-pCode-08451
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, and the identified entry point includes any variable in its interface that is declared with the SampleMask BuiltIn decoration, that

variable **must** not have an array size greater than VkPhysicalDeviceLimits ::maxSampleMaskWords

- VUID-VkShaderCreateInfoEXT-pCode-08453
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, and stage is VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT or VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, and the identified entry point has an OpExecutionMode instruction specifying a patch size with OutputVertices, the patch size **must** be greater than 0 and less than or equal to VkPhysicalDeviceLimits ::maxTessellationPatchSize
- VUID-VkShaderCreateInfoEXT-pCode-08454
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, and stage is VK_SHADER_STAGE_GEOMETRY_BIT, the identified entry point **must** have an OpExecutionMode instruction specifying a maximum output vertex count that is greater than 0 and less than or equal to VkPhysicalDeviceLimits::maxGeometryOutputVertices
- VUID-VkShaderCreateInfoEXT-pCode-08455
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, and stage is VK_SHADER_STAGE_GEOMETRY_BIT, the identified entry point **must** have an OpExecutionMode instruction specifying an invocation count that is greater than 0 and less than or equal to VkPhysicalDeviceLimits::maxGeometryShaderInvocations
- VUID-VkShaderCreateInfoEXT-pCode-08456
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, and stage is a pre-rasterization shader stage, and the identified entry point writes to Layer for any primitive, it **must** write the same value to Layer for all vertices of a given primitive
- VUID-VkShaderCreateInfoEXT-pCode-08457
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, and stage is a pre-rasterization shader stage, and the identified entry point writes to ViewportIndex for any primitive, it **must** write the same value to ViewportIndex for all vertices of a given primitive
- VUID-VkShaderCreateInfoEXT-pCode-08459
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, and stage is VK_SHADER_STAGE_FRAGMENT_BIT, and the identified entry point writes to FragDepth in any execution path, all execution paths that are not exclusive to helper invocations **must** either discard the fragment, or write or initialize the value of FragDepth
- VUID-VkShaderCreateInfoEXT-pCode-08460
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, the shader code in pCode **must** be valid as described by the Khronos SPIR-V Specification after applying the specializations provided in pSpecializationInfo, if any, and then converting all specialization constants into fixed constants
- VUID-VkShaderCreateInfoEXT-codeType-08872
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, and stage is VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, pCode **must** contain an OpExecutionMode instruction specifying the type of subdivision
- VUID-VkShaderCreateInfoEXT-codeType-12226
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, and stage is VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, pCode **must** contain an

OpExecutionMode instruction specifying the output patch size

- VUID-VkShaderCreateInfoEXT-pPushConstantRanges-10063
  Any two elements of pPushConstantRanges **must** not include the same stage in stageFlags
- VUID-VkShaderCreateInfoEXT-codeType-10064
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, flags does not include VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT and if a push constant block is declared in a shader, then an element of pPushConstantRanges->stageFlags **must** match stage
- VUID-VkShaderCreateInfoEXT-codeType-10065
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, flags does not include VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT and if a push constant block is declared in a shader, the block must be contained inside the element of pPushConstantRanges that matches the stage
- VUID-VkShaderCreateInfoEXT-codeType-10383
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, flags does not include VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT and a resource variable is declared in a shader, the corresponding descriptor set in pSetLayouts **must** match the shader stage
- VUID-VkShaderCreateInfoEXT-codeType-10384
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, flags does not include VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT and a resource variable is declared in a shader, and the descriptor type is not VK_DESCRIPTOR_TYPE_MUTABLE_EXT, the corresponding descriptor set in pSetLayouts **must** match the descriptor type
- VUID-VkShaderCreateInfoEXT-codeType-10385
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, flags does not include VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT and a resource variable is declared in a shader as an array, the corresponding descriptor set in pSetLayouts **must** match the descriptor count
- VUID-VkShaderCreateInfoEXT-setLayoutCount-13359
  If setLayoutCount is not 0, pSetLayouts is not NULL, and flags does not include VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR, then pSetLayouts must be a valid pointer to an array of setLayoutCount valid VkDescriptorSetLayout handles
- VUID-VkShaderCreateInfoEXT-setLayoutCount-13360
  If setLayoutCount is not 0, pSetLayouts is not NULL, and flags includes VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR, then pSetLayouts must be a valid pointer to an array of setLayoutCount VkDescriptorSetLayout or VK_NULL_HANDLE handles
- VUID-VkShaderCreateInfoEXT-stage-12428
  If stage is VK_SHADER_STAGE_COMPUTE_BIT, then flags **must** not include VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR
- VUID-VkShaderCreateInfoEXT-flags-12490
  If flags includes VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT, it **must** not include VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR
- VUID-VkShaderCreateInfoEXT-codeType-10386
  If codeType is VK_SHADER_CODE_TYPE_SPIRV_EXT, flags does not include

VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT and a resource variable is declared in a shader as an array of descriptors, then the descriptor type of that variable **must** not be VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK

- VUID-VkShaderCreateInfoEXT-flags-11758
  If the shader64BitIndexing feature is not enabled, `flags` **must** not contain VK_SHADER_CREATE_64_BIT_INDEXING_BIT_EXT
- VUID-VkShaderCreateInfoEXT-setLayoutCount-12257
  `setLayoutCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxBoundDescriptorSets`
- VUID-VkShaderCreateInfoEXT-flags-11290
  If `flags` includes VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT, `setLayoutCount` **must** be 0
- VUID-VkShaderCreateInfoEXT-flags-11291
  If `flags` includes VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT, `pSetLayouts` **must** be `NULL`
- VUID-VkShaderCreateInfoEXT-flags-11370
  If `flags` includes VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT, `pushConstantRangeCount` **must** be 0
- VUID-VkShaderCreateInfoEXT-flags-11371
  If `flags` includes VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT, `pPushConstantRanges` **must** be `NULL`
- VUID-VkShaderCreateInfoEXT-flags-11292
  If `flags` includes VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT, and `codeType` is VK_SHADER_CODE_TYPE_SPIRV_EXT, all shader variables in the shader resource interface with a `DescriptorSet` and `Binding` decoration **must** have a mapping declared in VkShaderDescriptorSetAndBindingMappingInfoEXT::`pMappings`
- VUID-VkShaderCreateInfoEXT-stage-12445
  `stage` **must** not contain VK_SHADER_STAGE_RAYGEN_BIT_KHR, VK_SHADER_STAGE_ANY_HIT_BIT_KHR, VK_SHADER_STAGE_CLOSEST_HIT_BIT_KHR, VK_SHADER_STAGE_MISS_BIT_KHR, VK_SHADER_STAGE_INTERSECTION_BIT_KHR, or VK_SHADER_STAGE_CALLABLE_BIT_KHR

Valid Usage (Implicit)

- VUID-VkShaderCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT
- VUID-VkShaderCreateInfoEXT-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkCustomResolveCreateInfoEXT, VkPipelineShaderStageRequiredSubgroupSizeCreateInfo, VkShaderDescriptorSetAndBindingMappingInfoEXT, or VkValidationFeaturesEXT
- VUID-VkShaderCreateInfoEXT-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

- VUID-VkShaderCreateInfoEXT-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkShaderCreateFlagBitsEXT values
- VUID-VkShaderCreateInfoEXT-stage-parameter
  `stage` **must** be a valid VkShaderStageFlagBits value
- VUID-VkShaderCreateInfoEXT-nextStage-parameter
  If `nextStage` is not `0`, `nextStage` **must** be a valid combination of VkShaderStageFlagBits values
- VUID-VkShaderCreateInfoEXT-codeType-parameter
  `codeType` **must** be a valid VkShaderCodeTypeEXT value
- VUID-VkShaderCreateInfoEXT-pCode-parameter
  `pCode` **must** be a valid pointer to an array of `codeSize` bytes
- VUID-VkShaderCreateInfoEXT-pName-parameter
  If `pName` is not `NULL`, `pName` **must** be a null-terminated UTF-8 string
- VUID-VkShaderCreateInfoEXT-pPushConstantRanges-parameter
  If `pushConstantRangeCount` is not `0`, `pPushConstantRanges` **must** be a valid pointer to an array of `pushConstantRangeCount` valid VkPushConstantRange structures
- VUID-VkShaderCreateInfoEXT-pSpecializationInfo-parameter
  If `pSpecializationInfo` is not `NULL`, `pSpecializationInfo` **must** be a valid pointer to a valid VkSpecializationInfo structure
- VUID-VkShaderCreateInfoEXT-codeSize-arraylength
  `codeSize` **must** be greater than `0`

```
// Provided by VK_EXT_shader_object
typedef VkFlags VkShaderCreateFlagsEXT;
```

`VkShaderCreateFlagsEXT` is a bitmask type for setting a mask of zero or more VkShaderCreateFlagBitsEXT.

Possible values of the `flags` member of VkShaderCreateInfoEXT specifying how a shader object is created, are:

```
// Provided by VK_EXT_shader_object
typedef enum VkShaderCreateFlagBitsEXT {
    VK_SHADER_CREATE_LINK_STAGE_BIT_EXT = 0x00000001,
  // Provided by VK_EXT_descriptor_heap with VK_EXT_shader_object
    VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT = 0x00000400,
  // Provided by VK_EXT_shader_object with VK_EXT_subgroup_size_control or
VK_VERSION_1_3
    VK_SHADER_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT_EXT = 0x00000002,
  // Provided by VK_EXT_shader_object with VK_EXT_subgroup_size_control or
VK_VERSION_1_3
    VK_SHADER_CREATE_REQUIRE_FULL_SUBGROUPS_BIT_EXT = 0x00000004,
  // Provided by VK_EXT_shader_object with VK_EXT_mesh_shader or VK_NV_mesh_shader
    VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT = 0x00000008,
  // Provided by VK_EXT_shader_object with VK_KHR_device_group or VK_VERSION_1_1
    VK_SHADER_CREATE_DISPATCH_BASE_BIT_EXT = 0x00000010,
  // Provided by VK_KHR_fragment_shading_rate with VK_EXT_shader_object
    VK_SHADER_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_EXT = 0x00000020,
  // Provided by VK_EXT_fragment_density_map with VK_EXT_shader_object
    VK_SHADER_CREATE_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT = 0x00000040,
  // Provided by VK_EXT_device_generated_commands
    VK_SHADER_CREATE_INDIRECT_BINDABLE_BIT_EXT = 0x00000080,
  // Provided by VK_KHR_opacity_micromap with VK_EXT_shader_object
    VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT =
0x00001000,
  // Provided by VK_EXT_shader_64bit_indexing
    VK_SHADER_CREATE_64_BIT_INDEXING_BIT_EXT = 0x00008000,
  // Provided by VK_KHR_maintenance11 with VK_EXT_shader_object
    VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR = 0x00040000,
} VkShaderCreateFlagBitsEXT;
```

- VK_SHADER_CREATE_LINK_STAGE_BIT_EXT specifies that a shader is linked to all other shaders created in the same vkCreateShadersEXT call whose VkShaderCreateInfoEXT structures' `flags` include VK_SHADER_CREATE_LINK_STAGE_BIT_EXT.
- VK_SHADER_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT_EXT specifies that the `SubgroupSize` **may** vary in a task, mesh, or compute shader.
- VK_SHADER_CREATE_REQUIRE_FULL_SUBGROUPS_BIT_EXT specifies that the subgroup sizes **must** be launched with all invocations active in a task, mesh, or compute shader.
- VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT specifies that a mesh shader **must** only be used without a task shader. Otherwise, the mesh shader **must** only be used with a task shader.
- VK_SHADER_CREATE_DISPATCH_BASE_BIT_EXT specifies that a compute shader **can** be used with vkCmdDispatchBase with a non-zero base workgroup.
- VK_SHADER_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_EXT specifies that a fragment shader **can** be used with a fragment shading rate attachment.
- VK_SHADER_CREATE_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT specifies that a fragment shader **can** be used with a fragment density map attachment.
- VK_SHADER_CREATE_INDIRECT_BINDABLE_BIT_EXT specifies that the shader **can** be used in

combination with Device-Generated Commands.

- VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT specifies that shader objects **cannot** be used with acceleration structures which are built with geometry that have an index buffer including both special indices and indices pointing to an associated micromap array. Geometry which has an index buffer using only special indices without an associated micromap array **can** be used with this flag.
- VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR specifies that implementations **must** ensure that the properties and/or absence of a particular descriptor set do not influence any other descriptor sets for the shader. This allows shader objects to be created with a subset of the total descriptor sets.
- VK_SHADER_CREATE_64_BIT_INDEXING_BIT_EXT specifies that the shader enables 64-bit indexing.
- VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT specifies that the shader will use descriptor heap mappings instead of descriptor set layouts.

NOTE

The behavior of VK_SHADER_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_EXT and VK_SHADER_CREATE_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT differs subtly from the behavior of VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR and VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT in that the shader bit allows, but does not require the shader to be used with that type of attachment. This means that the application need not create multiple shaders when it does not know in advance whether the shader will be used with or without the attachment type, or when it needs the same shader to be compatible with usage both with and without. This **may** come at some performance cost on some implementations, so applications **should** still only set bits that are actually necessary.

Shader objects **can** be created using different types of shader code. Possible values of VkShaderCreateInfoEXT::codeType, are:

```
// Provided by VK_EXT_shader_object
typedef enum VkShaderCodeTypeEXT {
    VK_SHADER_CODE_TYPE_BINARY_EXT = 0,
    VK_SHADER_CODE_TYPE_SPIRV_EXT = 1,
} VkShaderCodeTypeEXT;
```

- VK_SHADER_CODE_TYPE_BINARY_EXT specifies shader code in an opaque, implementation-defined binary format specific to the physical device.
- VK_SHADER_CODE_TYPE_SPIRV_EXT specifies shader code in SPIR-V format.

9.1.2. Binary Shader Code

Binary shader code **can** be retrieved from a shader object using the command:

```
// Provided by VK_EXT_shader_object
VkResult vkGetShaderBinaryDataEXT(
    VkDevice                                    device,
    VkShaderEXT                                 shader,
    size_t*                                     pDataSize,
    void*                                       pData);
```

- `device` is the logical device that the shader object was created from.
- `shader` is the shader object to retrieve binary shader code from.
- `pDataSize` is a pointer to a `size_t` value related to the size of the binary shader code, as described below.
- `pData` is either `NULL` or a pointer to a buffer.

If `pData` is `NULL`, then the size of the binary shader code of the shader object, in bytes, is returned in `pDataSize`. Otherwise, `pDataSize` **must** point to a variable set by the application to the size of the buffer, in bytes, pointed to by `pData`, and on return the variable is overwritten with the amount of data actually written to `pData`. If `pDataSize` is less than the size of the binary shader code, nothing is written to `pData`, and VK_INCOMPLETE will be returned instead of VK_SUCCESS.

NOTE

The behavior of this command when `pDataSize` is too small differs from how some other getter-type commands work in Vulkan. Because shader binary data is only usable in its entirety, it would never be useful for the implementation to return partial data. Because of this, nothing is written to `pData` unless `pDataSize` is large enough to fit the data in its entirety.

This behavior is not consistent with the behavior described in Opaque Binary Data Results, for historical reasons.

If the amount of data available is larger than the passed `pDataSize`, the query returns a VK_INCOMPLETE success status instead of a VK_ERROR_NOT_ENOUGH_SPACE_KHR error status.

Binary shader code retrieved using `vkGetShaderBinaryDataEXT` **can** be passed to a subsequent call to vkCreateShadersEXT on a compatible physical device by specifying VK_SHADER_CODE_TYPE_BINARY_EXT in the `codeType` member of `VkShaderCreateInfoEXT`.

The shader code returned by repeated calls to this function with the same `VkShaderEXT` is guaranteed to be invariant for the lifetime of the `VkShaderEXT` object.

Valid Usage

- VUID-vkGetShaderBinaryDataEXT-None-08461
  The `shaderObject` feature **must** be enabled

- VUID-vkGetShaderBinaryDataEXT-None-08499
  If `pData` is not `NULL`, it **must** be aligned to `16` bytes

Valid Usage (Implicit)

- VUID-vkGetShaderBinaryDataEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetShaderBinaryDataEXT-shader-parameter
  `shader` **must** be a valid VkShaderEXT handle
- VUID-vkGetShaderBinaryDataEXT-pDataSize-parameter
  `pDataSize` **must** be a valid pointer to a `size_t` value
- VUID-vkGetShaderBinaryDataEXT-pData-parameter
  If the value referenced by `pDataSize` is not `0`, and `pData` is not `NULL`, `pData` **must** be a valid pointer to an array of `pDataSize` bytes
- VUID-vkGetShaderBinaryDataEXT-shader-parent
  `shader` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

9.1.3. Binary Shader Compatibility

Binary shader compatibility means that binary shader code returned from a call to vkGetShaderBinaryDataEXT **can** be passed to a later call to vkCreateShadersEXT, potentially on a different logical and/or physical device, and that this will result in the successful creation of a shader object functionally equivalent to the shader object that the code was originally queried from.

Binary shader code queried from vkGetShaderBinaryDataEXT is not guaranteed to be compatible across all devices, but implementations are required to provide some compatibility guarantees. Applications **may** determine binary shader compatibility using either (or both) of two mechanisms.

Guaranteed compatibility of shader binaries is expressed through a combination of the

shaderBinaryUUID and shaderBinaryVersion members of the VkPhysicalDeviceShaderObjectPropertiesEXT structure queried from a physical device. Binary shaders retrieved from a physical device with a certain shaderBinaryUUID are guaranteed to be compatible with all other physical devices reporting the same shaderBinaryUUID and the same or higher shaderBinaryVersion.

Whenever a new version of an implementation incorporates any changes that affect the output of vkGetShaderBinaryDataEXT, the implementation **should** either increment shaderBinaryVersion if binary shader code retrieved from older versions remains compatible with the new implementation, or else replace shaderBinaryUUID with a new value if backward compatibility has been broken. Binary shader code queried from a device with a matching shaderBinaryUUID and lower shaderBinaryVersion relative to the device on which vkCreateShadersEXT is being called **may** be suboptimal for the new device in ways that do not change shader functionality, but it is still guaranteed to be usable to successfully create the shader object(s).

NOTE	Implementations are encouraged to share shaderBinaryUUID between devices and driver versions to the maximum extent their hardware naturally allows, and are **strongly** discouraged from ever changing the shaderBinaryUUID for the same hardware unless absolutely necessary.

In addition to the shader compatibility guarantees described above, it is valid for an application to call vkCreateShadersEXT with binary shader code created on a device with a different or unknown shaderBinaryUUID and/or higher shaderBinaryVersion. In this case, the implementation **may** use any unspecified means of its choosing to determine whether the provided binary shader code is usable. If it is, vkCreateShadersEXT **must** return VK_SUCCESS, and the created shader object is guaranteed to be valid. Otherwise, in the absence of some error, vkCreateShadersEXT **must** return VK_INCOMPATIBLE_SHADER_BINARY_EXT to indicate that the provided binary shader code is not compatible with the device.

9.1.4. Binding Shader Objects

Once shader objects have been created, they **can** be bound to the command buffer using the command:

```
// Provided by VK_EXT_shader_object
void vkCmdBindShadersEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    stageCount,
    const VkShaderStageFlagBits*                pStages,
    const VkShaderEXT*                          pShaders);
```

- commandBuffer is the command buffer that the shader object will be bound to.
- stageCount is the length of the pStages and pShaders arrays.
- pStages is a pointer to an array of VkShaderStageFlagBits values specifying one stage per array index that is affected by the corresponding value in the pShaders array.
- pShaders is a pointer to an array of VkShaderEXT handles and/or VK_NULL_HANDLE values

describing the shader binding operations to be performed on each stage in pStages.

When binding linked shaders, an application **may** bind them in any combination of one or more calls to vkCmdBindShadersEXT (i.e., shaders that were created linked together do not need to be bound in the same vkCmdBindShadersEXT call).

Any shader object bound to a particular stage **may** be unbound by setting its value in pShaders to VK_NULL_HANDLE. If pShaders is NULL, vkCmdBindShadersEXT behaves as if pShaders was an array of stageCount VK_NULL_HANDLE values (i.e., any shaders bound to the stages specified in pStages are unbound).

Valid Usage

- VUID-vkCmdBindShadersEXT-None-08462
  The shaderObject feature **must** be enabled
- VUID-vkCmdBindShadersEXT-pStages-08463
  Every element of pStages **must** be unique
- VUID-vkCmdBindShadersEXT-pStages-08464
  pStages **must** not contain VK_SHADER_STAGE_ALL_GRAPHICS or VK_SHADER_STAGE_ALL
- VUID-vkCmdBindShadersEXT-pStages-08465
  pStages **must** not contain VK_SHADER_STAGE_RAYGEN_BIT_KHR, VK_SHADER_STAGE_ANY_HIT_BIT_KHR, VK_SHADER_STAGE_CLOSEST_HIT_BIT_KHR, VK_SHADER_STAGE_MISS_BIT_KHR, VK_SHADER_STAGE_INTERSECTION_BIT_KHR, or VK_SHADER_STAGE_CALLABLE_BIT_KHR
- VUID-vkCmdBindShadersEXT-pShaders-08469
  For each element of pStages, if pShaders is not NULL, and the element of the pShaders array with the same index is not VK_NULL_HANDLE, it **must** have been created with a stage equal to the corresponding element of pStages
- VUID-vkCmdBindShadersEXT-pShaders-08470
  If pStages contains both VK_SHADER_STAGE_TASK_BIT_EXT and VK_SHADER_STAGE_VERTEX_BIT, and pShaders is not NULL, and the same index in pShaders as VK_SHADER_STAGE_TASK_BIT_EXT in pStages is not VK_NULL_HANDLE, the same index in pShaders as VK_SHADER_STAGE_VERTEX_BIT in pStages **must** be VK_NULL_HANDLE
- VUID-vkCmdBindShadersEXT-pShaders-08471
  If pStages contains both VK_SHADER_STAGE_MESH_BIT_EXT and VK_SHADER_STAGE_VERTEX_BIT, and pShaders is not NULL, and the same index in pShaders as VK_SHADER_STAGE_MESH_BIT_EXT in pStages is not VK_NULL_HANDLE, the same index in pShaders as VK_SHADER_STAGE_VERTEX_BIT in pStages **must** be VK_NULL_HANDLE
- VUID-vkCmdBindShadersEXT-pShaders-08476
  If pStages contains VK_SHADER_STAGE_COMPUTE_BIT, the VkCommandPool that commandBuffer was allocated from **must** support compute operations
- VUID-vkCmdBindShadersEXT-pShaders-08477

If `pStages` contains VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, VK_SHADER_STAGE_GEOMETRY_BIT, or VK_SHADER_STAGE_FRAGMENT_BIT, the `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

- VUID-vkCmdBindShadersEXT-pShaders-08478
  If `pStages` contains VK_SHADER_STAGE_MESH_BIT_EXT or VK_SHADER_STAGE_TASK_BIT_EXT, the `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

Valid Usage (Implicit)

- VUID-vkCmdBindShadersEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindShadersEXT-pStages-parameter
  `pStages` **must** be a valid pointer to an array of `stageCount` valid VkShaderStageFlagBits values
- VUID-vkCmdBindShadersEXT-pShaders-parameter
  If `pShaders` is not `NULL`, `pShaders` **must** be a valid pointer to an array of `stageCount` valid or VK_NULL_HANDLE VkShaderEXT handles
- VUID-vkCmdBindShadersEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBindShadersEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindShadersEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindShadersEXT-stageCount-arraylength
  `stageCount` **must** be greater than `0`
- VUID-vkCmdBindShadersEXT-commonparent
  Both of `commandBuffer`, and the elements of `pShaders` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindShadersEXT is not affected by conditional rendering

9.1.5. Setting State

Whenever shader objects are used to issue drawing commands, the appropriate dynamic state setting commands **must** have been called to set the relevant state in the command buffer prior to drawing:

- vkCmdSetViewportWithCount
- vkCmdSetScissorWithCount
- vkCmdSetRasterizerDiscardEnable

If a shader is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, the following commands **must** have been called in the command buffer prior to drawing:

- vkCmdSetVertexInputEXT
- vkCmdSetPrimitiveTopology
- vkCmdSetPrimitiveRestartEnable

If a shader is bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage, the following command **must** have been called in the command buffer prior to drawing:

- vkCmdSetPatchControlPointsEXT, if `primitiveTopology` is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST

If a shader is bound to the VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage, the following command **must** have been called in the command buffer prior to drawing:

- vkCmdSetTessellationDomainOriginEXT

If `rasterizerDiscardEnable` is VK_FALSE, the following commands **must** have been called in the command buffer prior to drawing:

- vkCmdSetRasterizationSamplesEXT

- vkCmdSetSampleMaskEXT
- vkCmdSetAlphaToCoverageEnableEXT
- vkCmdSetAlphaToOneEnableEXT, if the alphaToOne feature is enabled
- vkCmdSetPolygonModeEXT
- vkCmdSetLineWidth, if the effective rasterization input topology is in the line topology class
- vkCmdSetCullMode
- vkCmdSetFrontFace
- vkCmdSetDepthTestEnable
- vkCmdSetDepthWriteEnable, if `depthTestEnable` is VK_TRUE
- vkCmdSetDepthCompareOp, if `depthTestEnable` is VK_TRUE
- vkCmdSetDepthBoundsTestEnable, if the depthBounds feature is enabled
- vkCmdSetDepthBounds, if `depthBoundsTestEnable` is VK_TRUE
- vkCmdSetDepthBiasEnable
- vkCmdSetDepthBias or vkCmdSetDepthBias2EXT, if `depthBiasEnable` is VK_TRUE
- vkCmdSetDepthClampEnableEXT, if the depthClamp feature is enabled
- vkCmdSetStencilTestEnable
- vkCmdSetStencilOp, if `stencilTestEnable` is VK_TRUE
- vkCmdSetStencilCompareMask, if `stencilTestEnable` is VK_TRUE
- vkCmdSetStencilWriteMask, if `stencilTestEnable` is VK_TRUE
- vkCmdSetStencilReference, if `stencilTestEnable` is VK_TRUE

If a shader is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, and `rasterizerDiscardEnable` is VK_FALSE, the following commands **must** have been called in the command buffer prior to drawing:

- vkCmdSetLogicOpEnableEXT, if the `logicOp` feature is enabled
- vkCmdSetLogicOpEXT, if `logicOpEnable` is VK_TRUE
- vkCmdSetColorBlendEnableEXT and vkCmdSetColorWriteMaskEXT, if color attachments are bound, with values set for every color attachment in the render pass instance active at draw time
- vkCmdSetColorBlendEquationEXT or vkCmdSetColorBlendAdvancedEXT, if color attachments are bound, for every attachment whose index in `pColorBlendEnables` is a pointer to a value of VK_TRUE
- vkCmdSetBlendConstants, if any index in `pColorBlendEnables` is VK_TRUE, and the same index in `pColorBlendEquations` is a `VkColorBlendEquationEXT` structure with any VkBlendFactor member with a value of VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA

If the `pipelineFragmentShadingRate` feature is enabled, and `rasterizerDiscardEnable` is VK_FALSE, the

following command **must** have been called in the command buffer prior to drawing:

- vkCmdSetFragmentShadingRateKHR

If the `geometryStreams` feature is enabled, and a shader is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage, the following command **must** have been called in the command buffer prior to drawing:

- vkCmdSetRasterizationStreamEXT

If the `VK_EXT_discard_rectangles` extension is enabled, and `rasterizerDiscardEnable` is VK_FALSE, the following commands **must** have been called in the command buffer prior to drawing:

- vkCmdSetDiscardRectangleEnableEXT
- vkCmdSetDiscardRectangleModeEXT, if `discardRectangleEnable` is VK_TRUE
- vkCmdSetDiscardRectangleEXT, if `discardRectangleEnable` is VK_TRUE

If the `VK_EXT_conservative_rasterization` extension is enabled, and `rasterizerDiscardEnable` is VK_FALSE, the following commands **must** have been called in the command buffer prior to drawing:

- vkCmdSetConservativeRasterizationModeEXT
- vkCmdSetExtraPrimitiveOverestimationSizeEXT, if `conservativeRasterizationMode` is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT

If the `depthClipEnable` feature is enabled, the following command **must** have been called in the command buffer prior to drawing:

- vkCmdSetDepthClipEnableEXT

If the `VK_EXT_sample_locations` extension is enabled, and `rasterizerDiscardEnable` is VK_FALSE, the following commands **must** have been called in the command buffer prior to drawing:

- vkCmdSetSampleLocationsEnableEXT
- vkCmdSetSampleLocationsEXT, if `sampleLocationsEnable` is VK_TRUE

If the `VK_EXT_provoking_vertex` extension is enabled, and `rasterizerDiscardEnable` is VK_FALSE, and a shader is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, the following command **must** have been called in the command buffer prior to drawing:

- vkCmdSetProvokingVertexModeEXT

If any of the `stippledRectangularLines`, `stippledBresenhamLines`, or `stippledSmoothLines` features are enabled, and `rasterizerDiscardEnable` is VK_FALSE, and if the effective rasterization input topology is in the line topology class, the following commands **must** have been called in the command buffer prior to drawing:

- vkCmdSetLineStipple, if `stippledLineEnable` is VK_TRUE

If the `depthClipControl` feature is enabled, the following command **must** have been called in the

command buffer prior to drawing:

- vkCmdSetDepthClipNegativeOneToOneEXT

If the `colorWriteEnable` feature is enabled, and a shader is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, and `rasterizerDiscardEnable` is VK_FALSE, the following command **must** have been called in the command buffer prior to drawing:

- vkCmdSetColorWriteEnableEXT, with values set for every color attachment in the render pass instance active at draw time

If the attachmentFeedbackLoopDynamicState feature is enabled, and a shader is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, and `rasterizerDiscardEnable` is VK_FALSE, the following command **must** have been called in the command buffer prior to drawing:

- vkCmdSetAttachmentFeedbackLoopEnableEXT

If the depthClamp and `depthClampControl` features are enabled, and `depthClampEnable` is VK_TRUE, the following command **must** have been called in the command buffer prior to drawing:

- vkCmdSetDepthClampRangeEXT

State **can** be set either at any time before or after shader objects are bound, but all required state **must** be set prior to issuing drawing commands.

9.1.6. Interaction With Pipelines

Calling vkCmdBindShadersEXT causes the pipeline bind points corresponding to each stage in `pStages` to be disturbed, meaning that any pipelines that had previously been bound to those pipeline bind points are no longer bound.

If VK_PIPELINE_BIND_POINT_GRAPHICS is disturbed (i.e., if `pStages` contains any graphics stage), any graphics pipeline state that the previously bound pipeline did not specify as dynamic becomes undefined, and **must** be set in the command buffer before issuing drawing commands using shader objects.

Calls to vkCmdBindPipeline likewise disturb the shader stage(s) corresponding to `pipelineBindPoint`, meaning that any shaders that had previously been bound to any of those stages are no longer bound, even if the pipeline was created without shaders for some of those stages.

9.1.7. Shader Object Destruction

To destroy a shader object, call:

```
// Provided by VK_EXT_shader_object
void vkDestroyShaderEXT(
    VkDevice                                    device,
    VkShaderEXT                                 shader,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the shader object.
- `shader` is the handle of the shader object to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Destroying a shader object used by one or more command buffers in the recording or executable state causes those command buffers to move into the *invalid state*.

Valid Usage

- VUID-vkDestroyShaderEXT-None-08481
  The `shaderObject` feature **must** be enabled
- VUID-vkDestroyShaderEXT-shader-08482
  All submitted commands that refer to `shader` **must** have completed execution
- VUID-vkDestroyShaderEXT-pAllocator-08483
  If `VkAllocationCallbacks` were provided when `shader` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyShaderEXT-pAllocator-08484
  If no `VkAllocationCallbacks` were provided when `shader` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyShaderEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyShaderEXT-shader-parameter
  If `shader` is not VK_NULL_HANDLE, `shader` **must** be a valid VkShaderEXT handle
- VUID-vkDestroyShaderEXT-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyShaderEXT-shader-parent
  If `shader` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `shader` **must** be externally synchronized

9.2. Shader Modules

Shader modules contain *shader code* and one or more entry points. Shaders are selected from a shader module by specifying an entry point as part of pipeline creation. The stages of a pipeline **can** use shaders that come from different modules. The shader code defining a shader module **must** be

in the SPIR-V format, as described by the Vulkan Environment for SPIR-V appendix.

Shader modules are represented by VkShaderModule handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkShaderModule)
```

To create a shader module, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateShaderModule(
    VkDevice                                    device,
    const VkShaderModuleCreateInfo*             pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkShaderModule*                             pShaderModule);
```

- device is the logical device that creates the shader module.
- pCreateInfo is a pointer to a VkShaderModuleCreateInfo structure.
- pAllocator controls host memory allocation as described in the Memory Allocation chapter.
- pShaderModule is a pointer to a VkShaderModule handle in which the resulting shader module object is returned.

Once a shader module has been created, any entry points it contains **can** be used in pipeline shader stages as described in Compute Pipelines and Graphics Pipelines .

NOTE If the maintenance5 feature is enabled, shader module creation can be omitted entirely. Instead, applications should provide the VkShaderModuleCreateInfo structure directly in to pipeline creation by chaining it to VkPipelineShaderStageCreateInfo. This avoids the overhead of creating and managing an additional object.

Valid Usage

- VUID-vkCreateShaderModule-pCreateInfo-06904
  If pCreateInfo is not NULL, pCreateInfo->pNext **must** be NULL or a pointer to a valid instance of
  - VkShaderModuleValidationCacheCreateInfoEXT
  - VkValidationFeaturesEXT

Valid Usage (Implicit)

- VUID-vkCreateShaderModule-device-parameter
  device **must** be a valid VkDevice handle

- VUID-vkCreateShaderModule-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkShaderModuleCreateInfo structure
- VUID-vkCreateShaderModule-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateShaderModule-pShaderModule-parameter
  `pShaderModule` **must** be a valid pointer to a VkShaderModule handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkShaderModuleCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkShaderModuleCreateInfo {
    VkStructureType              sType;
    const void*                  pNext;
    VkShaderModuleCreateFlags    flags;
    size_t                       codeSize;
    const uint32_t*              pCode;
} VkShaderModuleCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `codeSize` is the size, in bytes, of the code pointed to by `pCode`.
- `pCode` is a pointer to code that is used to create the shader module. The type and format of the code is determined from the content of the memory addressed by `pCode`.

Valid Usage

- VUID-VkShaderModuleCreateInfo-codeSize-08735
  `codeSize` **must** be a multiple of 4
- VUID-VkShaderModuleCreateInfo-pCode-08736

`pCode` **must** point to valid SPIR-V code, formatted and packed as described by the Khronos SPIR-V Specification

- VUID-VkShaderModuleCreateInfo-pCode-08737
  `pCode` **must** adhere to the validation rules described by the Validation Rules within a Module section of the SPIR-V Environment appendix
- VUID-VkShaderModuleCreateInfo-pCode-08738
  `pCode` **must** declare the `Shader` capability for SPIR-V code
- VUID-VkShaderModuleCreateInfo-pCode-08739
  `pCode` **must** not declare any capability that is not supported by the API, as described by the Capabilities section of the SPIR-V Environment appendix
- VUID-VkShaderModuleCreateInfo-pCode-08740
  and `pCode` declares any of the capabilities listed in the SPIR-V Environment appendix, one of the corresponding requirements **must** be satisfied
- VUID-VkShaderModuleCreateInfo-pCode-08741
  `pCode` **must** not declare any SPIR-V extension that is not supported by the API, as described by the Extension section of the SPIR-V Environment appendix
- VUID-VkShaderModuleCreateInfo-pCode-08742
  and `pCode` declares any of the SPIR-V extensions listed in the SPIR-V Environment appendix, one of the corresponding requirements **must** be satisfied
- VUID-VkShaderModuleCreateInfo-codeSize-01085
  `codeSize` **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkShaderModuleCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SHADER_MODULE_CREATE_INFO
- VUID-VkShaderModuleCreateInfo-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkShaderModuleCreateInfo-pCode-parameter
  `pCode` **must** be a valid pointer to an array of $\frac{codeSize}{4}$ `uint32_t` values

Structure Chaining

Extends the structure

- VkPipelineShaderStageCreateInfo

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkShaderModuleCreateFlags;
```

`VkShaderModuleCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future

use.

To use a VkValidationCacheEXT to cache shader validation results, add a VkShaderModuleValidationCacheCreateInfoEXT structure to the pNext chain of the VkShaderModuleCreateInfo structure, specifying the cache object to use.

The VkShaderModuleValidationCacheCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_validation_cache
typedef struct VkShaderModuleValidationCacheCreateInfoEXT {
    VkStructureType         sType;
    const void*             pNext;
    VkValidationCacheEXT    validationCache;
} VkShaderModuleValidationCacheCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- validationCache is the validation cache object from which the results of prior validation attempts will be written, and to which new validation results for this VkShaderModule will be written (if not already present). The implementation **must** not access this object outside of the duration of the command this structure is passed to.

Valid Usage (Implicit)

- VUID-VkShaderModuleValidationCacheCreateInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_SHADER_MODULE_VALIDATION_CACHE_CREATE_INFO_EXT
- VUID-VkShaderModuleValidationCacheCreateInfoEXT-validationCache-parameter
  validationCache **must** be a valid VkValidationCacheEXT handle

Structure Chaining

Extends the structures

- VkPipelineShaderStageCreateInfo
- VkShaderModuleCreateInfo

To destroy a shader module, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyShaderModule(
    VkDevice                                    device,
    VkShaderModule                              shaderModule,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the shader module.
- `shaderModule` is the handle of the shader module to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

A shader module **can** be destroyed while pipelines created using its shaders are still in use.

Valid Usage

- VUID-vkDestroyShaderModule-shaderModule-01092
  If `VkAllocationCallbacks` were provided when `shaderModule` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyShaderModule-shaderModule-01093
  If no `VkAllocationCallbacks` were provided when `shaderModule` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyShaderModule-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyShaderModule-shaderModule-parameter
  If `shaderModule` is not VK_NULL_HANDLE, `shaderModule` **must** be a valid VkShaderModule handle
- VUID-vkDestroyShaderModule-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyShaderModule-shaderModule-parent
  If `shaderModule` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `shaderModule` **must** be externally synchronized

9.3. Shader Module Identifiers

Shader modules have unique identifiers associated with them. To query an implementation provided identifier, call:

```
// Provided by VK_EXT_shader_module_identifier
void vkGetShaderModuleIdentifierEXT(
    VkDevice                                    device,
    VkShaderModule                              shaderModule,
    VkShaderModuleIdentifierEXT*                pIdentifier);
```

- `device` is the logical device that created the shader module.
- `shaderModule` is the handle of the shader module.
- `pIdentifier` is a pointer to the returned VkShaderModuleIdentifierEXT.

The identifier returned by the implementation **must** only depend on `shaderModuleIdentifierAlgorithmUUID` and information provided in the VkShaderModuleCreateInfo which created `shaderModule`. The implementation **may** return equal identifiers for two different VkShaderModuleCreateInfo structures if the difference does not affect pipeline compilation. Identifiers are only meaningful on different VkDevice objects if the device the identifier was queried from had the same `shaderModuleIdentifierAlgorithmUUID` as the device consuming the identifier.

Valid Usage

- VUID-vkGetShaderModuleIdentifierEXT-shaderModuleIdentifier-06884
  The `shaderModuleIdentifier` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetShaderModuleIdentifierEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetShaderModuleIdentifierEXT-shaderModule-parameter
  `shaderModule` **must** be a valid VkShaderModule handle
- VUID-vkGetShaderModuleIdentifierEXT-pIdentifier-parameter
  `pIdentifier` **must** be a valid pointer to a VkShaderModuleIdentifierEXT structure
- VUID-vkGetShaderModuleIdentifierEXT-shaderModule-parent
  `shaderModule` **must** have been created, allocated, or retrieved from `device`

VkShaderModuleCreateInfo structures have unique identifiers associated with them. To query an implementation provided identifier, call:

```
// Provided by VK_EXT_shader_module_identifier
void vkGetShaderModuleCreateInfoIdentifierEXT(
    VkDevice                                    device,
    const VkShaderModuleCreateInfo*             pCreateInfo,
    VkShaderModuleIdentifierEXT*                pIdentifier);
```

- `device` is the logical device that **can** create a VkShaderModule from `pCreateInfo`.
- `pCreateInfo` is a pointer to a VkShaderModuleCreateInfo structure.
- `pIdentifier` is a pointer to the returned VkShaderModuleIdentifierEXT.

The identifier returned by the implementation **must** only depend on `shaderModuleIdentifierAlgorithmUUID` and information provided in the VkShaderModuleCreateInfo. The implementation **may** return equal identifiers for two different VkShaderModuleCreateInfo structures if the difference does not affect pipeline compilation. Identifiers are only meaningful on different VkDevice objects if the device the identifier was queried from had the same `shaderModuleIdentifierAlgorithmUUID` as the device consuming the identifier.

The identifier returned by the implementation in vkGetShaderModuleCreateInfoIdentifierEXT **must** be equal to the identifier returned by vkGetShaderModuleIdentifierEXT given equivalent definitions of VkShaderModuleCreateInfo and any chained `pNext` structures.

Valid Usage

- VUID-vkGetShaderModuleCreateInfoIdentifierEXT-shaderModuleIdentifier-06885
  The `shaderModuleIdentifier` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetShaderModuleCreateInfoIdentifierEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetShaderModuleCreateInfoIdentifierEXT-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkShaderModuleCreateInfo structure
- VUID-vkGetShaderModuleCreateInfoIdentifierEXT-pIdentifier-parameter
  `pIdentifier` **must** be a valid pointer to a VkShaderModuleIdentifierEXT structure

VkShaderModuleIdentifierEXT represents a shader module identifier returned by the implementation.

```
// Provided by VK_EXT_shader_module_identifier
typedef struct VkShaderModuleIdentifierEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           identifierSize;
    uint8_t            identifier[VK_MAX_SHADER_MODULE_IDENTIFIER_SIZE_EXT];
} VkShaderModuleIdentifierEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `identifierSize` is the size, in bytes, of valid data returned in `identifier`.

- `identifier` is a buffer of opaque data specifying an identifier.

Any returned values beyond the first `identifierSize` bytes are undefined. Implementations **must** return an `identifierSize` greater than 0, and less-or-equal to VK_MAX_SHADER_MODULE_IDENTIFIER_SIZE_EXT.

Two identifiers are considered equal if `identifierSize` is equal and the first `identifierSize` bytes of `identifier` compare equal.

Implementations **may** return a different `identifierSize` for different modules. Implementations **should** ensure that `identifierSize` is large enough to uniquely define a shader module.

Valid Usage (Implicit)

- VUID-VkShaderModuleIdentifierEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SHADER_MODULE_IDENTIFIER_EXT
- VUID-VkShaderModuleIdentifierEXT-pNext-pNext
  `pNext` **must** be `NULL`

VK_MAX_SHADER_MODULE_IDENTIFIER_SIZE_EXT is the length in bytes of a shader module identifier, as returned in VkShaderModuleIdentifierEXT::`identifierSize`.

```
#define VK_MAX_SHADER_MODULE_IDENTIFIER_SIZE_EXT 32U
```

9.4. Binding Shaders

Before a shader can be used it **must** be first bound to the command buffer.

Calling vkCmdBindPipeline binds all stages corresponding to the VkPipelineBindPoint. Calling vkCmdBindShadersEXT binds all stages in `pStages`.

The following table describes the relationship between shader stages and pipeline bind points:

Shader stage	Pipeline bind point	behavior controlled
• VK_SHADER_STAGE_VERTEX_BIT • VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT • VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT • VK_SHADER_STAGE_GEOMETRY_BIT • VK_SHADER_STAGE_FRAGMENT_BIT • VK_SHADER_STAGE_TASK_BIT_EXT • VK_SHADER_STAGE_MESH_BIT_EXT	VK_PIPELINE_BIND_POINT_GRAPHICS	all drawing commands
• VK_SHADER_STAGE_COMPUTE_BIT	VK_PIPELINE_BIND_POINT_COMPUTE	all dispatch commands
• VK_SHADER_STAGE_ANY_HIT_BIT_KHR • VK_SHADER_STAGE_CALLABLE_BIT_KHR • VK_SHADER_STAGE_CLOSEST_HIT_BIT_KHR • VK_SHADER_STAGE_INTERSECTION_BIT_KHR • VK_SHADER_STAGE_MISS_BIT_KHR • VK_SHADER_STAGE_RAYGEN_BIT_KHR	VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR	vkCmdTraceRaysKHR and vkCmdTraceRaysIndirectKHR

9.5. Shader Execution

At each stage of the pipeline, multiple invocations of a shader **may** execute simultaneously. Further, invocations of a single shader produced as the result of different commands **may** execute simultaneously. The relative execution order of invocations of the same shader type is undefined. Shader invocations **may** complete in a different order than that in which the primitives they originated from were drawn or dispatched by the application. However, fragment shader outputs are written to attachments in rasterization order.

The relative execution order of invocations of different shader types is largely undefined. However, when invoking a shader whose inputs are generated from a previous pipeline stage, the shader invocations from the previous stage are guaranteed to have executed far enough to generate input

values for all required shader inputs.

NOTE	This execution ordering guarantee is limited to variables in the shader input and output interface and payloads sent between stages. In particular, stores or atomic operations on storage buffers or images are not part of these interfaces and such operations in a previous pipeline stage are not guaranteed to be visible in the subsequent pipeline stage.

9.5.1. Shader Termination

A shader invocation that is *terminated* has finished executing instructions.

Executing `OpReturn` in the entry point, or executing `OpTerminateInvocation` in any function will terminate an invocation. Implementations **may** also terminate a shader invocation when `OpKill` is executed in any function; otherwise it becomes a helper invocation.

In addition to the above conditions, helper invocations **may** be terminated when all non-helper invocations in the same derivative group either terminate or become helper invocations.

A shader stage for a given command completes execution when all invocations for that stage have terminated.

NOTE	Depending on the implementation, `OpKill` will be functionally equivalent to either `OpTerminateInvocation` or `OpDemoteToHelperInvocation`. To obtain the most predictable behavior, shader authors should use `OpTerminateInvocation` or `OpDemoteToHelperInvocation` rather than `OpKill` wherever possible.

9.5.2. Varying Subgroup Size

Shaders **may** report a different value of `SubgroupSize` for different subgroups if varying subgroups are allowed.

If varying subgroups are not allowed, `SubgroupSize` will always be equal to [limits-subgroupSize]. If varying subgroups are allowed, `SubgroupSize` **must** be a power-of-two value between `minSubgroupSize` and `maxSubgroupSize` and **must** be uniform with subgroup scope. For task, mesh, or compute shaders, subgroups in the same command scope will have the same size. For other stages, every subgroup **may** have a different size even within a single command.

Varying subgroups are allowed if any of the following conditions are true:

- The shader was created from SPIR-V with version number of 1.6 or higher.
- The shader was created with vkCreateShadersEXT and the VK_SHADER_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT_EXT flag was set.
- The shader was created as part of a pipeline and the VK_PIPELINE_SHADER_STAGE_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT flag was set.

9.5.3. Full Subgroups

When full subgroups are **required**, subgroups **must** be launched with all invocations active, i.e., there is an active invocation with `SubgroupLocalInvocationId` for each value in range [0,`SubgroupSize` -1].

Full subgroups are **required** when the `X` dimension of the workgroup size is a multiple of the reported value of the `SubgroupSize` built-in, and one of the following is true:

- The shader was created from SPIR-V with version number of 1.6 or higher.
- The shader was created with vkCreateShadersEXT and the VK_SHADER_CREATE_REQUIRE_FULL_SUBGROUPS_BIT_EXT flag was set.
- The shader was created as part of a pipeline and the VK_PIPELINE_SHADER_STAGE_CREATE_REQUIRE_FULL_SUBGROUPS_BIT flag was set.

NOTE If varying subgroup sizes are allowed, this can only be statically guaranteed if the x dimension of the workgroup size is a multiple of the `maxSubgroupSize`. Otherwise, the subgroup size in a particular subgroup may be set to any of the supported sizes, making this a dynamic condition. Requiring a specific subgroup size will also ensure this is statically known.

NOTE Many algorithms cannot easily be implemented without full subgroups. For implementations, full subgroup guarantees can be provided more often with varying subgroup sizes. The behavior of this built-in without that guarantee is considered legacy, and implementations are encouraged to implement the tighter behavior unconditionally.

9.5.4. Required Subgroup Size

A specific subgroup size **can** be **required** for a shader between `minSubgroupSize` and `maxSubgroupSize`. The required size **can** be specified by:

- Adding an instance of VkPipelineShaderStageRequiredSubgroupSizeCreateInfo to the `pNext` chain of VkPipelineShaderStageCreateInfo with `requiredSize` set to the desired value.
- Adding an instance of VkShaderRequiredSubgroupSizeCreateInfoEXT to the `pNext` chain of VkShaderCreateInfoEXT with `requiredSize` set to the desired value.

When specified, the value of `SubgroupSize` **must** match the required value for all invocations of the shader.

9.6. Shader Out-of-Bounds Memory Access

Shader accesses to memory are not automatically bounds checked by the implementation. Applications **must** not execute operations that would access out of bounds memory locations unless some form of bounds checking is enabled or behavior is otherwise explicitly defined. An access is considered out of bounds if any part of the access is outside of any specified memory range, whether that is the array length specified in a shader or a range specified in the API (e.g. descriptor

size).

NOTE External tooling such as the Vulkan Validation Layers can be used to help validate that accesses are not out of bounds.

An access **can** be independently out of bounds for each range that applies; if one is bounds checked and the others are not, behavior is still undefined.

NOTE For example, given the following shader declaration

```
// Buffer type
struct MySSBO {
    uint32_t data[2];
};
```

accessing `data` at an index greater than 1 is undefined behavior, whether the underlying buffer is bigger than that or not.

Vulkan provides functionality that enables automatic bounds checking in some cases, as outlined below.

NOTE Automatic bounds checking can be used to ensure that accesses outside of certain bounds have predictable results, acting as a safety net for untrusted code, or simply as a way for applications to avoid their own bounds checks. While there may be a performance cost for enabling these features, they should not be slower than an application performing equivalent checks. Automatic checks do not necessarily account for all possible bounds - e.g. Robust Buffer Access will not prevent undefined behavior in the buffer access example in the prior note.

9.6.1. Robust Buffer Access

Robust buffer access **can** be enabled by specifying VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS in VkPipelineRobustnessCreateInfo, or specifying VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_DEVICE_DEFAULT and enabling the `robustBufferAccess` feature.

When robust buffer access is enabled, access to a buffer via a descriptor is bounds checked against the range specified for the descriptor, and access to vertex input data is bounds checked against the bound vertex buffer range. Reads from a vertex input **may** instead be bounds checked against a range rounded down to the nearest multiple of the stride of its binding.

NOTE The range of a descriptor is not necessarily equivalent to the size of the underlying resource; applications may suballocate descriptors from larger buffers, for instance. The APIs specifying the descriptor range vary between resource types and descriptor interfaces, but for example include the ranges specified by VkDescriptorBufferInfo or VkBufferViewCreateInfo.

If any vertex input read is outside of the checked range, all other vertex input reads through the same binding in the same shader invocation **may** behave as if they were outside of the checked range.

If any access to a uniform, storage, uniform texel, or storage texel buffer is outside of the checked range, any access of the same type (write, read-modify-write, or read) to the same buffer that is less than 16 bytes away from the first access **may** behave as if it is also outside of the checked range.

Any non-atomic access to a uniform, storage, uniform texel, or storage texel buffer wider than 32-bits **may** be treated as multiple 32-bit accesses that are separately bounds checked.

Writes to a storage or storage texel buffer outside of the checked range will either be discarded, or modify values within the memory range(s) bound to the underlying buffer (including outside of the checked range). They will not modify any other memory.

NOTE	Non-atomic writes outside of the checked range **can** lead to data races, as the application has no control over where the data will be written.

Atomic read-modify-write operations to a storage or storage texel buffer outside of the checked range will behave the same as a write outside of the checked range, but will return an undefined value.

Reading a uniform, storage, uniform texel, or storage texel buffer outside of the checked range will return one of the following values:

- Values from anywhere within the memory range(s) bound to the underlying buffer object, which **may** include bytes beyond the size of the buffer itself.
- Zero values
- For 4-component vectors, a value of (0,0,0,x), where x is any of
  - 0, 1, or the maximum positive integer value for integer components
  - 0.0 or 1.0 for floating-point components
- The value of the last store to the same out-of-bounds location in the same shader invocation.
  - Using the `Volatile`/`VolatileTexel` memory/image operand, the `Volatile` memory semantic, or the `Volatile` decoration to load the value will prevent prior stored values from being returned.

NOTE	Getting the value of the previous store is possible as implementations are free to optimize multiple accesses in the general case. There are several ways this **can** be prevented, but using volatile loads is by far the simplest.

Reads from a vertex input outside of the checked range will produce one of the following values:

- Values from anywhere within the memory range(s) bound to the underlying buffer object, which **may** include bytes beyond the size of the buffer itself, converted via input extraction.
- Zero values, converted via input extraction.
- Zero values

- For 4-component vectors, a value of (0,0,0,x), where x is any of
  - 0, 1, or the maximum positive integer value for integer components
  - 0.0 or 1.0 for floating-point components

Accesses via OpCooperativeMatrixLoadKHR and OpCooperativeMatrixStoreKHR are only bounds checked in the above manner if the VkPhysicalDeviceCooperativeMatrixFeaturesKHR::cooperativeMatrixRobustBufferAccess feature is enabled.

9.6.2. Robust Buffer Access 2

Robust buffer access 2 **can** be enabled by specifying VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 in VkPipelineRobustnessCreateInfo, or specifying VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_DEVICE_DEFAULT and enabling the robustBufferAccess2 feature.

When robust buffer access 2 is enabled, access to a buffer via a descriptor is bounds checked against the range specified for the descriptor, and access to vertex input data is bounds checked against the bound vertex buffer range, similarly to Robust Buffer Access, but with tighter bounds on the results.

Accesses to a uniform buffer **may** instead be bounds checked against a range rounded up to robustUniformBufferAccessSizeAlignment. Accesses inside the aligned range **may** behave as if they are in bounds, even if they are outside of the unaligned descriptor range, and access memory accordingly. The same is true for accesses to a storage buffer, using the robustStorageBufferAccessSizeAlignment limit instead.

NOTE	To avoid unexpected data races between neighboring descriptor ranges, applications may wish to ensure suballocated ranges of buffers are aligned to these limits.

Any access to a uniform, storage, uniform texel, or storage texel buffer wider than 32-bits **may** be treated as multiple 32-bit accesses that are separately bounds checked.

NOTE	Accesses to null descriptors are not considered out-of-bounds and have separate behavior controlled by the nullDescriptor feature.

Writes to a storage or storage texel buffer outside of the checked range will not modify any memory.

Atomic read-modify-write operations to a storage or storage texel buffer outside of the checked range will behave the same as a write outside of the checked range, but will return an undefined value.

Reads from a uniform or storage buffer outside of the checked range will return zero values. If a value was previously written to the same out of bounds location in the same shader invocation, that value **may** be returned instead; using the Volatile/VolatileTexel memory/image operand, the Volatile memory semantic, or the Volatile decoration to load the value will prevent prior stored

values from being returned.

Reading a uniform texel or storage texel buffer outside of the checked range will produce zero values, but component substitution will still be applied based on the buffer view's format, with the resulting value returned to the shader. If a value was previously written to the same out of bounds location in the same shader invocation, that value **may** be returned instead; using the `Volatile` /`VolatileTexel` memory/image operand, the `Volatile` memory semantic, or the `Volatile` decoration to load the value will prevent prior stored values from being returned.

Reads from a vertex input outside of the checked range will produce zero values, but input extraction will still be applied, filling missing G, B, or A components with (0,0,1).

Accesses via `OpCooperativeMatrixLoadKHR` and `OpCooperativeMatrixStoreKHR` are only bounds checked in the above manner if the `VkPhysicalDeviceCooperativeMatrixFeaturesKHR ::cooperativeMatrixRobustBufferAccess` feature is enabled.

9.6.3. Image Sampling

Sampling operations on an image descriptor are always well-defined when coordinates exceeding the dimensions specified for the descriptor are accessed, as described in the Wrapping Operation section.

9.6.4. Robust Image Access

Robust image access **can** be enabled by specifying VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS in VkPipelineRobustnessCreateInfo, or specifying VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_DEVICE_DEFAULT and enabling the `robustImageAccess` feature.

If robust image access is enabled, accesses to image descriptors are bounds checked against the image view dimensions specified for the descriptor.

Writes or atomic read-modify-write operations to a storage image outside of the checked dimensions will not modify any memory.

Reads, atomic read-modify-write operations, or fetches from images outside of the checked dimensions will return zero values, with (0,0,1) or (0,0,0) values inserted for missing G, B, or A components based on the format.

If a value was previously written to the same out of bounds location in the same shader invocation, that value **may** be returned instead; using the `VolatileTexel` image operand, the `Volatile` memory semantic, or the `Volatile` decoration to load the value will prevent prior stored values from being returned.

9.6.5. Robust Image Access 2

NOTE This is largely identical to Robust Image Access; the only difference being that the alpha channel must be replaced with 1, rather than 1 or 0, for out of bounds texel

access.

Robust image access 2 **can** be enabled by specifying VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS_2 in VkPipelineRobustnessCreateInfo, or specifying VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_DEVICE_DEFAULT and enabling the `robustImageAccess2` feature.

If robust image access 2 is enabled, accesses to image descriptors are bounds checked against the image view dimensions specified for the descriptor.

Writes or atomic read-modify-write operations to a storage image outside of the checked dimensions will not modify any memory.

Reads, atomic read-modify-write operations, or fetches from images outside of the checked dimensions will return zero values, with (0,0,1) values inserted for missing G, B, or A components based on the format.

If a value was previously written to the same out of bounds location in the same shader invocation, that value **may** be returned instead; using the `VolatileTexel` image operand, the `Volatile` memory semantic, or the `Volatile` decoration to load the value will prevent prior stored values from being returned.

9.7. Shader Memory Access Ordering

The order in which image or buffer memory is read or written by shaders is largely undefined. For some shader types (vertex, tessellation evaluation, and in some cases, fragment), even the number of shader invocations that **may** perform loads and stores is undefined.

In particular, the following rules apply:

- Vertex and tessellation evaluation shaders will be invoked at least once for each unique vertex, as defined in those sections.
- Fragment shaders will be invoked zero or more times, as defined in that section.
- The relative execution order of invocations of the same shader type is undefined. A store issued by a shader when working on primitive B might complete prior to a store for primitive A, even if primitive A is specified prior to primitive B. This applies even to fragment shaders; while fragment shader outputs are always written to the framebuffer in rasterization order, stores executed by fragment shader invocations are not.
- The relative execution order of invocations of different shader types is largely undefined.

NOTE The above limitations on shader invocation order make some forms of synchronization between shader invocations within a single set of primitives unimplementable. For example, having one invocation poll memory written by another invocation assumes that the other invocation has been launched and will complete its writes in finite time.

The Memory Model appendix defines the terminology and rules for how to correctly communicate

between shader invocations, such as when a write is Visible-To a read, and what constitutes a Data Race. Applications **must** not cause a data race.

9.8. Shader Inputs and Outputs

Data is passed into and out of shaders using variables with input or output storage class, respectively. User-defined inputs and outputs are connected between stages by matching their `Location` decorations. Additionally, data **can** be provided by or communicated to special functions provided by the execution environment using `BuiltIn` decorations.

In many cases, the same `BuiltIn` decoration **can** be used in multiple shader stages with similar meaning. The specific behavior of variables decorated as `BuiltIn` is documented in the following sections.

9.9. Task Shaders

Task shaders operate in conjunction with the mesh shaders to produce a collection of primitives that will be processed by subsequent stages of the graphics pipeline. Their primary purpose is to create a variable amount of subsequent mesh shader invocations.

Task shaders are invoked via the execution of the programmable mesh shading pipeline.

The task shader has no fixed-function inputs other than variables identifying the specific workgroup and invocation. In the `TaskEXT Execution Model` the number of mesh shader workgroups to create is specified via the `OpEmitMeshTasksEXT` instruction.

The task shader can write additional outputs to task memory, which can be read by all of the mesh shader workgroups it created.

9.9.1. Task Shader Execution

Task workloads are formed from groups of work items called workgroups and processed by the task shader in the current graphics pipeline. A workgroup is a collection of shader invocations that execute the same shader, potentially in parallel. Task shaders execute in *global workgroups* which are divided into a number of *local workgroups* with a size that **can** be set by assigning a value to the `LocalSize` or `LocalSizeId` execution mode or via an object decorated by the `WorkgroupSize` decoration. An invocation within a local workgroup **can** share data with other members of the local workgroup through shared variables and issue memory and control flow barriers to synchronize with other members of the local workgroup. If the subpass includes multiple views in its view mask, a Task shader using `TaskEXT Execution Model` **may** be invoked separately for each view.

9.10. Mesh Shaders

Mesh shaders operate in workgroups to produce a collection of primitives that will be processed by subsequent stages of the graphics pipeline. Each workgroup emits zero or more output primitives and the group of vertices and their associated data required for each output primitive.

Mesh shaders are invoked via the execution of the programmable mesh shading pipeline.

The only inputs available to the mesh shader are variables identifying the specific workgroup and invocation and, if applicable, any outputs written to task memory by the task shader that spawned the mesh shader's workgroup. The mesh shader can operate without a task shader as well.

The invocations of the mesh shader workgroup write an output mesh, comprising a set of primitives with per-primitive attributes, a set of vertices with per-vertex attributes, and an array of indices identifying the mesh vertices that belong to each primitive. The primitives of this mesh are then processed by subsequent graphics pipeline stages, where the outputs of the mesh shader form an interface with the fragment shader.

9.10.1. Mesh Shader Execution

Mesh workloads are formed from groups of work items called workgroups and processed by the mesh shader in the current graphics pipeline. A workgroup is a collection of shader invocations that execute the same shader, potentially in parallel. Mesh shaders execute in *global workgroups* which are divided into a number of *local workgroups* with a size that **can** be set by assigning a value to the `LocalSize` or `LocalSizeId` execution mode or via an object decorated by the `WorkgroupSize` decoration. An invocation within a local workgroup **can** share data with other members of the local workgroup through shared variables and issue memory and control flow barriers to synchronize with other members of the local workgroup.

The *global workgroups* may be generated explicitly via the API, or implicitly through the task shader's work creation mechanism. If the subpass includes multiple views in its view mask, a Mesh shader using `MeshEXT Execution Model` **may** be invoked separately for each view.

9.11. Vertex Shaders

Each vertex shader invocation operates on one vertex and its associated vertex attribute data, and outputs one vertex and associated data. Graphics pipelines using primitive shading **must** include a vertex shader, and the vertex shader stage is always the first shader stage in the graphics pipeline.

9.11.1. Vertex Shader Execution

A vertex shader **must** be executed at least once for each vertex specified by a drawing command. If the subpass includes multiple views in its view mask, the shader **may** be invoked separately for each view. During execution, the shader is presented with the index of the vertex and instance for which it has been invoked. Input variables declared in the vertex shader are filled by the implementation with the values of vertex attributes associated with the invocation being executed.

If the same vertex is specified multiple times in a drawing command (e.g. by including the same index value multiple times in an index buffer) the implementation **may** reuse the results of vertex shading if it can statically determine that the vertex shader invocations will produce identical results.

NOTE	It is implementation-dependent when and if results of vertex shading are reused, and thus how many times the vertex shader will be executed. This is true also if the vertex shader contains stores or atomic operations (see `vertexPipelineStoresAndAtomics`).

9.12. Tessellation Control Shaders

The tessellation control shader is used to read an input patch provided by the application and to produce an output patch. Each tessellation control shader invocation operates on an input patch (after all control points in the patch are processed by a vertex shader) and its associated data, and outputs a single control point of the output patch and its associated data, and **can** also output additional per-patch data. The input patch is sized according to the `patchControlPoints` member of VkPipelineTessellationStateCreateInfo, as part of input assembly.

The input patch can also be dynamically sized with the `patchControlPoints` parameter of vkCmdSetPatchControlPointsEXT.

To dynamically set the number of control points per patch, call:

```
// Provided by VK_EXT_extended_dynamic_state2, VK_EXT_shader_object
void vkCmdSetPatchControlPointsEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    patchControlPoints);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `patchControlPoints` specifies the number of control points per patch.

This command sets the number of control points per patch for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineTessellationStateCreateInfo ::`patchControlPoints` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetPatchControlPointsEXT-None-09422
  At least one of the following **must** be true:
  - The `extendedDynamicState2PatchControlPoints` feature is enabled
  - The `shaderObject` feature is enabled
- VUID-vkCmdSetPatchControlPointsEXT-patchControlPoints-04874
  `patchControlPoints` **must** be greater than zero and less than or equal to `VkPhysicalDeviceLimits`::`maxTessellationPatchSize`

Valid Usage (Implicit)

- VUID-vkCmdSetPatchControlPointsEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetPatchControlPointsEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state

- VUID-vkCmdSetPatchControlPointsEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetPatchControlPointsEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetPatchControlPointsEXT is not affected by conditional rendering

The size of the output patch is controlled by the `OpExecutionMode OutputVertices` specified in the tessellation control or tessellation evaluation shaders, which **must** be specified in at least one of the shaders. The size of the input and output patches **must** each be greater than zero and less than or equal to `VkPhysicalDeviceLimits`::`maxTessellationPatchSize`.

9.12.1. Tessellation Control Shader Execution

A tessellation control shader is invoked at least once for each *output* vertex in a patch. If the subpass includes multiple views in its view mask, the shader **may** be invoked separately for each view.

Inputs to the tessellation control shader are generated by the vertex shader. Each invocation of the tessellation control shader **can** read the attributes of any incoming vertices and their associated data. The invocations corresponding to a given patch execute logically in parallel, with undefined relative execution order. However, the `OpControlBarrier` instruction **can** be used to provide limited control of the execution order by synchronizing invocations within a patch, effectively dividing tessellation control shader execution into a set of phases. Tessellation control shaders will read undefined values if one invocation reads a per-vertex or per-patch output written by another invocation at any point during the same phase, or if two invocations attempt to write different values to the same per-patch output in a single phase.

9.13. Tessellation Evaluation Shaders

The Tessellation Evaluation Shader operates on an input patch of control points and their associated data, and a single input barycentric coordinate indicating the invocation's relative position within the subdivided patch, and outputs a single vertex and its associated data.

9.13.1. Tessellation Evaluation Shader Execution

A tessellation evaluation shader is invoked at least once for each unique vertex generated by the tessellator. If the subpass includes multiple views in its view mask, the shader **may** be invoked separately for each view.

9.14. Geometry Shaders

The geometry shader operates on a group of vertices and their associated data assembled from a single input primitive, and emits zero or more output primitives and the group of vertices and their associated data required for each output primitive.

9.14.1. Geometry Shader Execution

A geometry shader is invoked at least once for each primitive produced by the tessellation stages, or at least once for each primitive generated by primitive assembly when tessellation is not in use. A shader can request that the geometry shader runs multiple instances. A geometry shader is invoked at least once for each instance. If the subpass includes multiple views in its view mask, the shader **may** be invoked separately for each view.

9.15. Fragment Shaders

Fragment shaders are invoked as a fragment operation in a graphics pipeline. Each fragment shader invocation operates on a single fragment and its associated data. With few exceptions, fragment shaders do not have access to any data associated with other fragments and are considered to execute in isolation of fragment shader invocations associated with other fragments.

9.16. Compute Shaders

Compute shaders are invoked via dispatching commands. In general, they have access to similar resources as shader stages executing as part of a graphics pipeline.

Compute workloads are formed from groups of work items called workgroups and processed by the compute shader in the current compute pipeline. A workgroup is a collection of shader invocations that execute the same shader, potentially in parallel. Compute shaders execute in *global workgroups* which are divided into a number of *local workgroups* with a size that **can** be set by assigning a value to the `LocalSize` or `LocalSizeId` execution mode or via an object decorated by the `WorkgroupSize` decoration. An invocation within a local workgroup **can** share data with other members of the local workgroup through shared variables and issue memory and control flow barriers to synchronize with other members of the local workgroup.

9.17. Ray Generation Shaders

A ray generation shader is similar to a compute shader. Its main purpose is to execute ray tracing queries using pipeline trace ray instructions (such as `OpTraceRayKHR`) and process the results.

9.17.1. Ray Generation Shader Execution

One ray generation shader is executed per ray tracing dispatch. Its location in the shader binding table (see Shader Binding Table for details) is passed directly into vkCmdTraceRaysKHR using the `pRaygenShaderBindingTable` parameter or .

9.18. Intersection Shaders

Intersection shaders enable the implementation of arbitrary, application defined geometric primitives. An intersection shader for a primitive is executed whenever its axis-aligned bounding box is hit by a ray.

Like other ray tracing shader domains, an intersection shader operates on a single ray at a time. It also operates on a single primitive at a time. It is therefore the purpose of an intersection shader to compute the ray-primitive intersections and report them. To report an intersection, the shader calls the `OpReportIntersectionKHR` instruction.

An intersection shader communicates with any-hit and closest hit shaders by generating attribute values that they **can** read. Intersection shaders **cannot** read or modify the ray payload.

9.18.1. Intersection Shader Execution

The order in which intersections are found along a ray, and therefore the order in which intersection shaders are executed, is unspecified.

The intersection shader of the closest AABB which intersects the ray is guaranteed to be executed at some point during traversal, unless the ray is forcibly terminated.

9.19. Any-Hit Shaders

The any-hit shader is executed after the intersection shader reports an intersection that lies within the current $[t_{min},t_{max}]$ of the ray. The main use of any-hit shaders is to programmatically decide whether or not an intersection will be accepted. The intersection will be accepted unless the shader calls the `OpIgnoreIntersectionKHR` instruction. Any-hit shaders have read-only access to the attributes generated by the corresponding intersection shader, and **can** read or modify the ray payload.

9.19.1. Any-Hit Shader Execution

The order in which intersections are found along a ray, and therefore the order in which any-hit shaders are executed, is unspecified.

The any-hit shader of the closest hit is guaranteed to be executed at some point during traversal,

unless the ray is forcibly terminated.

9.20. Closest Hit Shaders

Closest hit shaders have read-only access to the attributes generated by the corresponding intersection shader, and **can** read or modify the ray payload. They also have access to a number of system-generated values. Closest hit shaders **can** call pipeline trace ray instructions to recursively trace rays.

9.20.1. Closest Hit Shader Execution

Exactly one closest hit shader is executed when traversal is finished and an intersection has been found and accepted.

9.21. Miss Shaders

Miss shaders **can** access the ray payload and **can** trace new rays through the pipeline trace ray instructions, but **cannot** access attributes since they are not associated with an intersection.

9.21.1. Miss Shader Execution

A miss shader is executed instead of a closest hit shader if no intersection was found during traversal.

9.22. Callable Shaders

Callable shaders **can** access a callable payload that works similarly to ray payloads to do subroutine work.

9.22.1. Callable Shader Execution

A callable shader is executed by calling `OpExecuteCallableKHR` from an allowed shader stage.

9.23. Interpolation Decorations

Variables in the `Input` storage class in a fragment shader's interface are interpolated from the values specified by the primitive being rasterized.

NOTE	Interpolation decorations can be present on input and output variables in pre-rasterization shaders but have no effect on the interpolation performed.

An undecorated input variable will be interpolated with perspective-correct interpolation according to the primitive type being rasterized. Lines and polygons are interpolated in the same way as the primitive's clip coordinates. If the `NoPerspective` decoration is present, linear interpolation is instead used for lines and polygons. For points, as there is only a single vertex, input values are never interpolated and instead take the value written for the single vertex.

If the Flat decoration is present on an input variable, the value is not interpolated, and instead takes its value directly from the provoking vertex. Fragment shader inputs that are signed or unsigned integers, integer vectors, or any double-precision floating-point type **must** be decorated with Flat.

Interpolation of input variables is performed at an implementation-defined position within the fragment area being shaded. The position is further constrained as follows:

- If the Centroid decoration is used, the interpolation position used for the variable **must** also fall within the bounds of the primitive being rasterized.
- If the Sample decoration is used, the interpolation position used for the variable **must** be at the position of the sample being shaded by the current fragment shader invocation.
- If a sample count of 1 is used, and the sampleLocationSampleCounts limit includes VK_SAMPLE_COUNT_1_BIT, the interpolation position **must** be at the position of the sample being shaded by the current fragment shader invocation.
- If a sample count of 1 is used and VK_SAMPLE_COUNT_1_BIT is not advertised in the sampleLocationSampleCounts limit, the interpolation position **must** be at the center of the fragment area.

NOTE

As Centroid constrains the interpolation position to lie within the covered area of the primitive, using it may cause the position to differ between neighboring fragments when it otherwise would not. Derivatives calculated based on these differing locations can produce inconsistent results compared to undecorated inputs. Thus using Centroid with input variables used in derivative calculations is not recommended.

If the PerVertexKHR decoration is present on an input variable, the value is not interpolated, and instead values from all input vertices are available in an array. Each index of the array corresponds to one of the vertices of the primitive that produced the fragment.

9.24. Static Use

A SPIR-V module declares a global object in memory using the OpVariable or OpUntypedVariableKHR instruction, which results in a pointer x to that object. A specific entry point in a SPIR-V module is said to *statically use* that object if that entry point's call tree contains a function containing an instruction with x as an id operand. A shader entry point also *statically uses* any variables explicitly declared in its interface.

9.25. Scope

A *scope* describes a set of shader invocations, where each such set is a *scope instance*. Each invocation belongs to one or more scope instances, but belongs to no more than one scope instance for each scope.

The operations available between invocations in a given scope instance vary, with smaller scopes generally able to perform more operations, and with greater efficiency.

9.25.1. Cross Device

All invocations executed in a Vulkan instance fall into a single *cross device scope instance*.

Whilst the `CrossDevice` scope is defined in SPIR-V, it is disallowed in Vulkan. API synchronization commands **can** be used to communicate between devices.

9.25.2. Device

All invocations executed on a single device form a *device scope instance*.

If the `vulkanMemoryModel` and `vulkanMemoryModelDeviceScope` features are enabled, this scope is represented in SPIR-V by the `Device Scope`, which **can** be used as a `Memory Scope` for barrier and atomic operations.

If both the `shaderDeviceClock` and `vulkanMemoryModelDeviceScope` features are enabled, using the `Device Scope` with the `OpReadClockKHR` instruction will read from a clock that is consistent across invocations in the same device scope instance.

There is no method to synchronize the execution of these invocations within SPIR-V, and this **can** only be done with API synchronization primitives.

Invocations executing on different devices in a device group operate in separate device scope instances.

9.25.3. Queue Family

Invocations executed by queues in a given queue family form a *queue family scope instance*.

This scope is identified in SPIR-V as the `QueueFamily Scope` if the `vulkanMemoryModel` feature is enabled, or if not, the `Device Scope`, which **can** be used as a `Memory Scope` for barrier and atomic operations.

If the `shaderDeviceClock` feature is enabled, but the `vulkanMemoryModelDeviceScope` feature is not enabled, using the `Device Scope` with the `OpReadClockKHR` instruction will read from a clock that is consistent across invocations in the same queue family scope instance.

There is no method to synchronize the execution of these invocations within SPIR-V, and this **can** only be done with API synchronization primitives.

Each invocation in a queue family scope instance **must** be in the same device scope instance.

9.25.4. Command

Any shader invocations executed as the result of a single command such as vkCmdDispatch or vkCmdDraw form a *command scope instance*. For indirect drawing commands with `drawCount` greater than one, invocations from separate draws are in separate command scope instances. For ray tracing shaders, an invocation group is an implementation-dependent subset of the set of shader invocations of a given shader stage which are produced by a single trace rays command.

There is no specific `Scope` for communication across invocations in a command scope instance. As

this has a clear boundary at the API level, coordination here **can** be performed in the API, rather than in SPIR-V.

Each invocation in a command scope instance **must** be in the same queue-family scope instance.

For shaders without defined workgroups, this set of invocations forms an *invocation group* as defined in the SPIR-V specification.

9.25.5. Primitive

Any fragment shader invocations executed as the result of rasterization of a single primitive form a *primitive scope instance*.

There is no specific `Scope` for communication across invocations in a primitive scope instance.

Any generated helper invocations are included in this scope instance.

Each invocation in a primitive scope instance **must** be in the same command scope instance.

Any input variables decorated with `Flat` are uniform within a primitive scope instance.

9.25.6. Shader Call

Any shader-call-related invocations that are executed in one or more ray tracing execution models form a *shader call scope instance*.

The `ShaderCallKHR Scope` can be used as `Memory Scope` for barrier and atomic operations.

Each invocation in a shader call scope instance **must** be in the same queue family scope instance.

9.25.7. Workgroup

A *local workgroup* is a set of invocations that can synchronize and share data with each other using memory in the `Workgroup` storage class.

The `Workgroup Scope` can be used as both an `Execution Scope` and `Memory Scope` for barrier and atomic operations.

Each invocation in a local workgroup **must** be in the same command scope instance.

Only task, mesh, and compute shaders have defined workgroups - other shader types **cannot** use workgroup functionality. For shaders that have defined workgroups, this set of invocations forms an *invocation group* as defined in the SPIR-V specification.

When variables declared with the `Workgroup` storage class are explicitly laid out (hence they are also decorated with `Block`), the amount of storage consumed is the size of the largest Block variable, not counting any padding at the end. The amount of storage consumed by the non-Block variables declared with the `Workgroup` storage class is implementation-dependent. However, the amount of storage consumed may not exceed the largest block size that would be obtained if all active non-Block variables declared with `Workgroup` storage class were assigned offsets in an arbitrary order by successively taking the smallest valid offset according to the Standard Storage Buffer Layout rules,

and with `Boolean` values considered as 32-bit integer values for the purpose of this calculation. (This is equivalent to using the GLSL std430 layout rules.)

9.25.8. Subgroup

A *subgroup* (see the subsection "Control Flow" of section 2 of the SPIR-V 1.3 Revision 1 specification) is a set of invocations that can synchronize and share data with each other efficiently.

The `Subgroup Scope` can be used as both an `Execution Scope` and `Memory Scope` for barrier and atomic operations. Other subgroup features allow the use of group operations with subgroup scope.

If the `shaderSubgroupClock` feature is enabled, using the `Subgroup Scope` with the `OpReadClockKHR` instruction will read from a clock that is consistent across invocations in the same subgroup.

For shaders that have defined workgroups, each invocation in a subgroup **must** be in the same local workgroup.

In other shader stages, each invocation in a subgroup **must** be in the same device scope instance.

Only shader stages that support subgroup operations have defined subgroups.

NOTE

Subgroups are not guaranteed to be a subset of a single command in shaders that do not have defined workgroups. Values that are guaranteed to be uniform for a given command or sub command may then not be uniform for the subgroup, and vice versa. As such, applications must take care when dealing with mixed uniformity.

A somewhat common example of this would be something like trying to optimize access to per-draw data using subgroup operations:

```
buffer { uint draw_data[]; };

flat in int vDrawID; // Passed through from vertex shader

void main()
{
    uint local_draw_data = subgroupBroadcastFirst(draw_data[vDrawID]);
}
```

This can be done in an attempt to optimize the shader to only perform the loads once per subgroup. However, if the implementation packs multiple draws into a single subgroup, invocations from draws with a different drawID are now receiving data from the wrong invocation. Applications should rely on implementations to do this kind of optimization automatically where the implementation can, rather than trying to force it.

9.25.9. Quad

A *quad scope instance* is formed of four shader invocations.

In a fragment shader, each invocation in a quad scope instance is formed of invocations in neighboring framebuffer locations (x_i, y_i), where:

- i is the index of the invocation within the scope instance.
- w and h are the number of pixels the fragment covers in the x and y axes.
- w and h are identical for all participating invocations.
- (x_0) = (x_1 - w) = (x_2) = (x_3 - w)
- (y_0) = (y_1) = (y_2 - h) = (y_3 - h)
- Each invocation has the same layer and sample indices.

In a mesh, task, or compute shader, if the `DerivativeGroupQuadsKHR` execution mode is specified, each invocation in a quad scope instance is formed of invocations with adjacent local invocation IDs (x_i, y_i), where:

- i is the index of the invocation within the quad scope instance.
- (x_0) = (x_1 - 1) = (x_2) = (x_3 - 1)
- (y_0) = (y_1) = (y_2 - 1) = (y_3 - 1)
- x_0 and y_0 are integer multiples of 2.
- Each invocation has the same z coordinate.

In a mesh, task, or compute shader, if the `DerivativeGroupLinearKHR` execution mode is specified, each invocation in a quad scope instance is formed of invocations with adjacent local invocation indices (l_i), where:

- i is the index of the invocation within the quad scope instance.
- (l_0) = (l_1 - 1) = (l_2 - 2) = (l_3 - 3)
- l_0 is an integer multiple of 4.

In all shaders, each invocation in a quad scope instance is formed of invocations in adjacent subgroup invocation indices (s_i), where:

- i is the index of the invocation within the quad scope instance.
- (s_0) = (s_1 - 1) = (s_2 - 2) = (s_3 - 3)
- s_0 is an integer multiple of 4.

Each invocation in a quad scope instance **must** be in the same subgroup.

In a fragment shader, each invocation in a quad scope instance **must** be in the same primitive scope instance.

Fragment , mesh, task, and compute shaders have defined quad scope instances. If the `quadOperationsInAllStages` limit is supported, any shader stages that support subgroup operations also have defined quad scope instances.

9.25.10. Fragment Interlock

A *fragment interlock scope instance* is formed of fragment shader invocations based on their framebuffer locations (x,y,layer,sample), executed by commands inside a single subpass.

The specific set of invocations included varies based on the execution mode as follows:

- If the `SampleInterlockOrderedEXT` or `SampleInterlockUnorderedEXT` execution modes are used, only invocations with identical framebuffer locations (x,y,layer,sample) are included.
- If the `PixelInterlockOrderedEXT` or `PixelInterlockUnorderedEXT` execution modes are used, fragments with different sample ids are also included.
- If the `ShadingRateInterlockOrderedEXT` or `ShadingRateInterlockUnorderedEXT` execution modes are used, fragments from neighboring framebuffer locations are also included. The fragment shading rate determines these fragments.

Only fragment shaders with one of the above execution modes have defined fragment interlock scope instances.

There is no specific `Scope` value for communication across invocations in a fragment interlock scope instance. However, this is implicitly used as a memory scope by `OpBeginInvocationInterlockEXT` and `OpEndInvocationInterlockEXT`.

Each invocation in a fragment interlock scope instance **must** be in the same queue family scope instance.

9.25.11. Invocation

The smallest *scope* is a single invocation; this is represented by the `Invocation Scope` in SPIR-V.

Fragment shader invocations **must** be in a primitive scope instance.

Invocations in fragment shaders that have a defined fragment interlock scope **must** be in a fragment interlock scope instance.

Invocations in shaders that have defined workgroups **must** be in a local workgroup.

Invocations in shaders that have a defined subgroup scope **must** be in a subgroup.

Invocations in shaders that have a defined quad scope **must** be in a quad scope instance.

All invocations in all stages **must** be in a command scope instance.

9.26. Group Operations

Group operations are executed by multiple invocations within a scope instance; with each invocation involved in calculating the result. This provides a mechanism for efficient communication between invocations in a particular scope instance.

Group operations all take a `Scope` defining the desired scope instance to operate within. Only the `Subgroup` scope **can** be used for these operations; the `subgroupSupportedOperations` limit defines

which types of operation **can** be used.

9.26.1. Basic Group Operations

Basic group operations include the use of `OpGroupNonUniformElect`, `OpControlBarrier`, `OpMemoryBarrier`, and atomic operations.

`OpGroupNonUniformElect` **can** be used to choose a single invocation to perform a task for the whole group. Only the invocation with the lowest id in the group will return `true`.

The Memory Model appendix defines the operation of barriers and atomics.

9.26.2. Vote Group Operations

The vote group operations allow invocations within a group to compare values across a group. The types of votes enabled are:

- Do all active group invocations agree that an expression is true?
- Do any active group invocations evaluate an expression to true?
- Do all active group invocations have the same value of an expression?

NOTE	These operations are useful in combination with control flow in that they allow for developers to check whether conditions match across the group and choose potentially faster code-paths in these cases.

9.26.3. Arithmetic Group Operations

The arithmetic group operations allow invocations to perform scans and reductions across a group. The operators supported are add, mul, min, max, and, or, xor.

For reductions, every invocation in a group will obtain the cumulative result of these operators applied to all values in the group. For exclusive scans, each invocation in a group will obtain the cumulative result of these operators applied to all values in invocations with a lower index in the group. Inclusive scans are identical to exclusive scans, except the cumulative result includes the operator applied to the value in the current invocation.

The order in which these operators are applied is implementation-dependent.

9.26.4. Ballot Group Operations

The ballot group operations allow invocations to perform more complex votes across the group. The ballot functionality allows all invocations within a group to provide a boolean value and get as a result what each invocation provided as their boolean value. The broadcast functionality allows values to be broadcast from an invocation to all other invocations within the group.

9.26.5. Shuffle Group Operations

The shuffle group operations allow invocations to read values from other invocations within a

group.

9.26.6. Shuffle Relative Group Operations

The shuffle relative group operations allow invocations to read values from other invocations within the group relative to the current invocation in the group. The relative operations supported allow data to be shifted up and down through the invocations within a group.

9.26.7. Clustered Group Operations

The clustered group operations allow invocations to perform an operation among partitions of a group, such that the operation is only performed within the group invocations within a partition. The partitions for clustered group operations are consecutive power-of-two size groups of invocations and the cluster size **must** be known at pipeline creation time. The operations supported are add, mul, min, max, and, or, xor.

9.26.8. Rotate Group Operations

The rotate group operations allow invocations to read values from other invocations within the group relative to the current invocation and modulo the size of the group. Clustered rotate group operations perform the same operation within individual partitions of a group.

The partitions for clustered rotate group operations are consecutive power-of-two size groups of invocations and the cluster size **must** be known at pipeline creation time.

9.27. Quad Group Operations

Quad group operations (`OpGroupNonUniformQuad*`) are a specialized type of group operations that only operate on quad scope instances. Whilst these instructions do include a `Scope` parameter, this scope is always overridden; only the quad scope instance is included in its execution scope.

Fragment shaders that statically execute either `OpGroupNonUniformQuadBroadcast` or `OpGroupNonUniformQuadSwap` **must** launch sufficient invocations to ensure their correct operation; additional helper invocations are launched for framebuffer locations not covered by rasterized fragments if necessary.

The index used to select participating invocations is i, as described for a quad scope instance, defined as the *quad index* in the SPIR-V specification.

For `OpGroupNonUniformQuadBroadcast` this value is equal to `Index`. For `OpGroupNonUniformQuadSwap`, it is equal to the implicit `Index` used by each participating invocation.

9.28. Derivative Operations

Derivative operations calculate the partial derivative for an expression P as a function of an invocation's x and y coordinates.

Derivative operations operate on a set of invocations known as a *derivative group* as defined in the SPIR-V specification.

A derivative group in a fragment shader is equivalent to the quad scope instance if the QuadDerivativesKHR execution mode is specified, otherwise it is equivalent to the primitive scope instance. A derivative group in a mesh, task, or compute shader is equivalent to the quad scope instance.

Derivatives are calculated assuming that P is piecewise linear and continuous within the derivative group.

The following control-flow restrictions apply to derivative operations:

- If the QuadDerivativesKHR execution mode is specified, dynamic instances of any derivative operations **must** be executed in control flow that is uniform within the current quad scope instance.
- If the QuadDerivativesKHR execution mode is not specified:
  - dynamic instances of explicit derivative instructions (OpDPdx*, OpDPdy*, and OpFwidth*) **must** be executed in control flow that is uniform within a derivative group.
  - dynamic instances of implicit derivative operations **can** be executed in control flow that is not uniform within the derivative group, but results are undefined.

Fragment shaders that statically execute derivative operations **must** launch sufficient invocations to ensure their correct operation; additional helper invocations are launched for framebuffer locations not covered by rasterized fragments if necessary.

NOTE	In a mesh, task, or compute shader, it is the application's responsibility to ensure that sufficient invocations are launched.

Derivative operations calculate their results as the difference between the result of P across invocations in the quad. For fine derivative operations (OpDPdxFine and OpDPdyFine), the values of DPdx(P_i) are calculated as

$$DPdx(P_0) = DPdx(P_1) = P_1 - P_0$$

$$DPdx(P_2) = DPdx(P_3) = P_3 - P_2$$

and the values of DPdy(P_i) are calculated as

$$DPdy(P_0) = DPdy(P_2) = P_2 - P_0$$

$$DPdy(P_1) = DPdy(P_3) = P_3 - P_1$$

where i is the index of each invocation as described in Quad.

Coarse derivative operations (OpDPdxCoarse and OpDPdyCoarse), calculate their results in roughly the same manner, but **may** only calculate two values instead of four (one for each of DPdx and DPdy),

reusing the same result no matter the originating invocation. If an implementation does this, it **should** use the fine derivative calculations described for P_0.

NOTE

Derivative values are calculated between fragments rather than pixels. If the fragment shader invocations involved in the calculation cover multiple pixels, these operations cover a wider area, resulting in larger derivative values. This in turn will result in a coarser LOD being selected for image sampling operations using derivatives.

Applications may want to account for this when using multi-pixel fragments; if pixel derivatives are desired, applications should use explicit derivative operations and divide the results by the size of the fragment in each dimension as follows:

$$DPdx(P_n)' = DPdx(P_n) / w$$

$$DPdy(P_n)' = DPdy(P_n) / h$$

where w and h are the size of the fragments in the quad, and $DPdx(P_n)'$ and $DPdy(P_n)'$ are the pixel derivatives.

The results for `OpDPdx` and `OpDPdy` **may** be calculated as either fine or coarse derivatives, with implementations favoring the most efficient approach. Implementations **must** choose coarse or fine consistently between the two.

Executing `OpFwidthFine`, `OpFwidthCoarse`, or `OpFwidth` is equivalent to executing the corresponding `OpDPdx*` and `OpDPdy*` instructions, taking the absolute value of the results, and summing them.

Executing an `OpImage*Sample*ImplicitLod` instruction is equivalent to executing `OpDPdx`(`Coordinate`) and `OpDPdy`(`Coordinate`), and passing the results as the `Grad` operands `dx` and `dy`.

NOTE

It is expected that using the `ImplicitLod` variants of sampling functions will be substantially more efficient than using the `ExplicitLod` variants with explicitly generated derivatives.

9.29. Helper Invocations

When performing derivative or quad group operations in a fragment shader, additional invocations **may** be spawned in order to ensure correct results. These additional invocations are known as *helper invocations* and **can** be identified by a non-zero value in the `HelperInvocation` built-in. Stores and atomics performed by helper invocations **must** not have any effect on memory except for the `Function`, `Private` and `Output` storage classes, and values returned by atomic instructions in helper invocations are undefined.

NOTE

While storage to `Output` storage class has an effect even in helper invocations, it does not mean that helper invocations have an effect on the framebuffer. `Output` variables in fragment shaders can be read from as well, and they behave more like

> `Private` variables for the duration of the shader invocation.

If the `MaximallyReconvergesKHR` execution mode is applied to the entry point, helper invocations **must** remain active for all instructions for the lifetime of the quad scope instance they are a part of. If the `MaximallyReconvergesKHR` execution mode is not applied to the entry point, helper invocations **may** be considered inactive for group operations other than derivative and quad group operations. All invocations in a quad scope instance **may** become permanently inactive at any point once the only remaining invocations in that quad scope instance are helper invocations.

9.30. Cooperative Matrices

A *cooperative matrix* type is a SPIR-V type where the storage for and computations performed on the matrix are spread across the invocations in a scope instance. These types give the implementation freedom in how to optimize matrix multiplies.

SPIR-V defines the types and instructions, but does not specify rules about what sizes/combinations are valid, and it is expected that different implementations **may** support different sizes.

To enumerate the supported cooperative matrix types and operations, call:

```
// Provided by VK_EXT_cooperative_matrix_maintenance1
VkResult vkGetPhysicalDeviceCooperativeMatrixProperties2EXT(
    VkPhysicalDevice                            physicalDevice,
    const VkPhysicalDeviceCooperativeMatrixInfo2EXT* pCooperativeMatrixInfo,
    uint32_t*                                   pPropertyCount,
    VkCooperativeMatrixProperties2EXT*          pProperties);
```

- `physicalDevice` is the physical device.
- `pCooperativeMatrixInfo` is a pointer to a VkPhysicalDeviceCooperativeMatrixInfo2EXT structure describing which cooperative matrix properties to query.
- `pPropertyCount` is a pointer to an integer related to the number of cooperative matrix properties available or queried.
- `pProperties` is either `NULL` or a pointer to an array of VkCooperativeMatrixProperties2EXT structures.

If `pProperties` is `NULL`, then the number of cooperative matrix properties available is returned in `pPropertyCount`. Otherwise, `pPropertyCount` **must** point to a variable set by the application to the number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If `pPropertyCount` is less than the number of cooperative matrix properties available, at most `pPropertyCount` structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available cooperative matrix properties were returned.

The implementation **must** not return `pPropertyCount` equal to `0` for the following parameters, when VkPhysicalDeviceCooperativeMatrixInfo2EXT::`flags` does not contain VK_COOPERATIVE_MATRIX_SATURATING_ACCUMULATION_BIT_EXT:

- `scope` equals VK_SCOPE_SUBGROUP_KHR and `subgroupSize` is `0`
- `scope` equals VK_SCOPE_SUBGROUP_KHR and `subgroupSize` is equal to VkPhysicalDeviceSubgroupProperties::`subgroupSize`

For each of the above required parameters, at least one property **must** be returned for each of the following combinations of component types:

- `AType` and `BType` are VK_COMPONENT_TYPE_FLOAT16_KHR, `CType` and `ResultType` are either both VK_COMPONENT_TYPE_FLOAT16_KHR or both VK_COMPONENT_TYPE_FLOAT32_KHR
- `AType` and `BType` are VK_COMPONENT_TYPE_UINT8_KHR, `CType` and `ResultType` are either both VK_COMPONENT_TYPE_UINT32_KHR or both VK_COMPONENT_TYPE_SINT32_KHR
- `AType` and `BType` are VK_COMPONENT_TYPE_SINT8_KHR, `CType` and `ResultType` are VK_COMPONENT_TYPE_SINT32_KHR

Valid Usage

- VUID-vkGetPhysicalDeviceCooperativeMatrixProperties2EXT-cooperativeMatrixProperties2-13371
  The `cooperativeMatrixProperties2` feature **must** be supported by `physicalDevice`

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceCooperativeMatrixProperties2EXT-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceCooperativeMatrixProperties2EXT-pCooperativeMatrixInfo-parameter
  `pCooperativeMatrixInfo` **must** be a valid pointer to a valid VkPhysicalDeviceCooperativeMatrixInfo2EXT structure
- VUID-vkGetPhysicalDeviceCooperativeMatrixProperties2EXT-pPropertyCount-parameter
  `pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceCooperativeMatrixProperties2EXT-pProperties-parameter
  If the value referenced by `pPropertyCount` is not `0`, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` VkCooperativeMatrixProperties2EXT structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkPhysicalDeviceCooperativeMatrixInfo2EXT structure is defined as:

```
// Provided by VK_EXT_cooperative_matrix_maintenance1
typedef struct VkPhysicalDeviceCooperativeMatrixInfo2EXT {
    VkStructureType                sType;
    const void*                    pNext;
    VkScopeKHR                     scope;
    uint32_t                       invocations;
    uint32_t                       subgroupSize;
    VkCooperativeMatrixFlagsEXT    flags;
} VkPhysicalDeviceCooperativeMatrixInfo2EXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- scope is the scope of the matrices, of type VkScopeKHR.
- invocations is the number of invocations within the local workgroup. invocations is ignored unless scope is VK_SCOPE_WORKGROUP_KHR.
- subgroupSize is the size of the subgroup.
- flags is a bitmask of VkCooperativeMatrixFlagBitsEXT selecting which class of matrix properties are being queried.

If subgroupSize is equal to 0, then the properties returned will be valid for use with shaders that execute with a varying subgroup size, and the returned properties are not guaranteed to be valid for any specific effective subgroup size.

NOTE

The intended usage of subgroupSize is:

- If the matrix will be used with a pipeline that enables VK_PIPELINE_SHADER_STAGE_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT or a shader that enables VK_SHADER_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT_EXT, or a SPIR-V module version 1.6 or later with no required subgroup size specified, the application should query the properties with subgroupSize equal to 0, since the effective subgroup size is not known at pipeline creation time.
- If the matrix will be used with a shader whose subgroup size is forced via VkPipelineShaderStageRequiredSubgroupSizeCreateInfo, or VkShaderRequiredSubgroupSizeCreateInfoEXT for shader objects, the application should query with subgroupSize equal to the required value.
- Otherwise (the SPIR-V module version is less than 1.6, and the shader neither requests a specific subgroup size nor allows it to vary), the effective subgroup

size is VkPhysicalDeviceSubgroupProperties::subgroupSize, and the application should query with that value.

Valid Usage

- VUID-VkPhysicalDeviceCooperativeMatrixInfo2EXT-scope-13372
  scope **must** equal VK_SCOPE_SUBGROUP_KHR
- VUID-VkPhysicalDeviceCooperativeMatrixInfo2EXT-cooperativeMatrixWorkgroupScope-13373
  scope **must** not equal VK_SCOPE_WORKGROUP_KHR
- VUID-VkPhysicalDeviceCooperativeMatrixInfo2EXT-subgroupSize-13374
  If subgroupSize does not equal 0, subgroupSize **must** be a power of two
- VUID-VkPhysicalDeviceCooperativeMatrixInfo2EXT-scope-13375
  If scope is VK_SCOPE_SUBGROUP_KHR, subgroupSize does not equal 0, and the subgroupSizeControl feature is supported, subgroupSize **must** be greater than or equal to minSubgroupSize
- VUID-VkPhysicalDeviceCooperativeMatrixInfo2EXT-scope-13376
  If scope is VK_SCOPE_SUBGROUP_KHR and the subgroupSizeControl feature is supported, subgroupSize **must** be less than or equal to maxSubgroupSize
- VUID-VkPhysicalDeviceCooperativeMatrixInfo2EXT-scope-13377
  If scope is VK_SCOPE_SUBGROUP_KHR, the subgroupSizeControl feature is not supported, and subgroupSize does not equal 0, subgroupSize **must** be equal to VkPhysicalDeviceSubgroupProperties::subgroupSize
- VUID-VkPhysicalDeviceCooperativeMatrixInfo2EXT-invocations-13378
  If scope is VK_SCOPE_WORKGROUP_KHR, invocations **must** be a power of two
- VUID-VkPhysicalDeviceCooperativeMatrixInfo2EXT-subgroupSize-13379
  If scope is VK_SCOPE_WORKGROUP_KHR, and subgroupSize does not equal 0, invocations **must** be a multiple of subgroupSize
- VUID-VkPhysicalDeviceCooperativeMatrixInfo2EXT-invocations-13381
  If scope is not VK_SCOPE_WORKGROUP_KHR, invocations **must** be equal to 0

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceCooperativeMatrixInfo2EXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_INFO_2_EXT
- VUID-VkPhysicalDeviceCooperativeMatrixInfo2EXT-pNext-pNext
  pNext **must** be NULL
- VUID-VkPhysicalDeviceCooperativeMatrixInfo2EXT-scope-parameter
  scope **must** be a valid VkScopeKHR value
- VUID-VkPhysicalDeviceCooperativeMatrixInfo2EXT-flags-parameter
  If flags is not 0, flags **must** be a valid combination of VkCooperativeMatrixFlagBitsEXT

values

Bits which **can** be set in VkPhysicalDeviceCooperativeMatrixInfo2EXT::flags to select the class of cooperative matrix properties being queried are:

```
// Provided by VK_EXT_cooperative_matrix_maintenance1
typedef enum VkCooperativeMatrixFlagBitsEXT {
  // Provided by VK_EXT_cooperative_matrix_maintenance1
    VK_COOPERATIVE_MATRIX_SATURATING_ACCUMULATION_BIT_EXT = 0x00000001,
} VkCooperativeMatrixFlagBitsEXT;
```

- VK_COOPERATIVE_MATRIX_SATURATING_ACCUMULATION_BIT_EXT specifies that the query returns properties for which the shader **must** use the SaturatingAccumulation operand on OpCooperativeMatrixMulAddKHR. If this bit is not set, the query returns properties for which the shader **must** not use the SaturatingAccumulation operand on OpCooperativeMatrixMulAddKHR.

```
// Provided by VK_EXT_cooperative_matrix_maintenance1
typedef VkFlags VkCooperativeMatrixFlagsEXT;
```

VkCooperativeMatrixFlagsEXT is a bitmask type for setting a mask of zero or more VkCooperativeMatrixFlagBitsEXT.

To enumerate the supported cooperative matrix types and operations, call:

```
// Provided by VK_KHR_cooperative_matrix
VkResult vkGetPhysicalDeviceCooperativeMatrixPropertiesKHR(
    VkPhysicalDevice                            physicalDevice,
    uint32_t*                                   pPropertyCount,
    VkCooperativeMatrixPropertiesKHR*           pProperties);
```

- physicalDevice is the physical device.
- pPropertyCount is a pointer to an integer related to the number of cooperative matrix properties available or queried.
- pProperties is either NULL or a pointer to an array of VkCooperativeMatrixPropertiesKHR structures.

If pProperties is NULL, then the number of cooperative matrix properties available is returned in pPropertyCount. Otherwise, pPropertyCount **must** point to a variable set by the application to the number of elements in the pProperties array, and on return the variable is overwritten with the number of structures actually written to pProperties. If pPropertyCount is less than the number of cooperative matrix properties available, at most pPropertyCount structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available cooperative matrix properties were returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceCooperativeMatrixPropertiesKHR-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceCooperativeMatrixPropertiesKHR-pPropertyCount-parameter
  `pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceCooperativeMatrixPropertiesKHR-pProperties-parameter
  If the value referenced by `pPropertyCount` is not `0`, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` VkCooperativeMatrixPropertiesKHR structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Each VkCooperativeMatrixPropertiesKHR structure describes a single supported combination of types for a matrix multiply/add operation (`OpCooperativeMatrixMulAddKHR`). The multiply **can** be described in terms of the following variables and types (in SPIR-V pseudocode):

```
    %A is of type OpTypeCooperativeMatrixKHR %AType %scope %MSize %KSize %MatrixAKHR
    %B is of type OpTypeCooperativeMatrixKHR %BType %scope %KSize %NSize %MatrixBKHR
    %C is of type OpTypeCooperativeMatrixKHR %CType %scope %MSize %NSize
%MatrixAccumulatorKHR
    %Result is of type OpTypeCooperativeMatrixKHR %ResultType %scope %MSize %NSize
%MatrixAccumulatorKHR

    %Result = %A * %B + %C // using OpCooperativeMatrixMulAddKHR
```

A matrix multiply with these dimensions is known as an *MxNxK* matrix multiply.

The `VkCooperativeMatrixProperties2EXT` structure is defined as:

```
// Provided by VK_EXT_cooperative_matrix_maintenance1
typedef struct VkCooperativeMatrixProperties2EXT {
    VkStructureType       sType;
    void*                 pNext;
    uint32_t              MGranularity;
    uint32_t              NGranularity;
    uint32_t              KGranularity;
    VkComponentTypeKHR    AType;
    VkComponentTypeKHR    BType;
    VkComponentTypeKHR    CType;
    VkComponentTypeKHR    ResultType;
} VkCooperativeMatrixProperties2EXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `MGranularity` is the granularity of the number of rows in matrices `A`, `C`, and `Result`. The rows **must** be an integer multiple of this value.
- `KGranularity` is the granularity of columns in matrix `A` and rows in matrix `B`. The columns/rows **must** be an integer multiple of this value.
- `NGranularity` is the granularity of columns in matrices `B`, `C`, `Result`. The columns **must** be an integer multiple of this value.
- `AType` is the component type of matrix `A`, of type VkComponentTypeKHR.
- `BType` is the component type of matrix `B`, of type VkComponentTypeKHR.
- `CType` is the component type of matrix `C`, of type VkComponentTypeKHR.
- `ResultType` is the component type of matrix `Result`, of type VkComponentTypeKHR.

If some types are preferred over other types (e.g. for performance), they **should** appear earlier in the list enumerated by vkGetPhysicalDeviceCooperativeMatrixProperties2EXT.

At least one entry in the list **must** have power of two values for all of `MGranularity`, `KGranularity`, and `NGranularity`.

Valid Usage (Implicit)

- VUID-VkCooperativeMatrixProperties2EXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COOPERATIVE_MATRIX_PROPERTIES_2_EXT
- VUID-VkCooperativeMatrixProperties2EXT-pNext-pNext
  `pNext` **must** be `NULL`

The `VkCooperativeMatrixPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_cooperative_matrix
typedef struct VkCooperativeMatrixPropertiesKHR {
    VkStructureType       sType;
    void*                 pNext;
    uint32_t              MSize;
    uint32_t              NSize;
    uint32_t              KSize;
    VkComponentTypeKHR    AType;
    VkComponentTypeKHR    BType;
    VkComponentTypeKHR    CType;
    VkComponentTypeKHR    ResultType;
    VkBool32              saturatingAccumulation;
    VkScopeKHR            scope;
} VkCooperativeMatrixPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `MSize` is the number of rows in matrices `A`, `C`, and `Result`.
- `KSize` is the number of columns in matrix `A` and rows in matrix `B`.
- `NSize` is the number of columns in matrices `B`, `C`, `Result`.
- `AType` is the component type of matrix `A`, of type VkComponentTypeKHR.
- `BType` is the component type of matrix `B`, of type VkComponentTypeKHR.
- `CType` is the component type of matrix `C`, of type VkComponentTypeKHR.
- `ResultType` is the component type of matrix `Result`, of type VkComponentTypeKHR.
- `saturatingAccumulation` indicates whether the `SaturatingAccumulation` operand to `OpCooperativeMatrixMulAddKHR` **must** be present or not. If it is VK_TRUE, the `SaturatingAccumulation` operand **must** be present. If it is VK_FALSE, the `SaturatingAccumulation` operand **must** not be present.
- `scope` is the scope of all the matrix types, of type VkScopeKHR.

If some types are preferred over other types (e.g. for performance), they **should** appear earlier in the list enumerated by vkGetPhysicalDeviceCooperativeMatrixPropertiesKHR.

At least one entry in the list **must** have power of two values for all of `MSize`, `KSize`, and `NSize`.

`scope` **must** be VK_SCOPE_SUBGROUP_KHR.

Valid Usage (Implicit)

- VUID-VkCooperativeMatrixPropertiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COOPERATIVE_MATRIX_PROPERTIES_KHR
- VUID-VkCooperativeMatrixPropertiesKHR-pNext-pNext
  `pNext` **must** be `NULL`

Possible values for VkScopeKHR include:

```
// Provided by VK_KHR_cooperative_matrix
typedef enum VkScopeKHR {
    VK_SCOPE_DEVICE_KHR = 1,
    VK_SCOPE_WORKGROUP_KHR = 2,
    VK_SCOPE_SUBGROUP_KHR = 3,
    VK_SCOPE_QUEUE_FAMILY_KHR = 5,
} VkScopeKHR;
```

- VK_SCOPE_DEVICE_KHR corresponds to SPIR-V `Device` scope.
- VK_SCOPE_WORKGROUP_KHR corresponds to SPIR-V `Workgroup` scope.
- VK_SCOPE_SUBGROUP_KHR corresponds to SPIR-V `Subgroup` scope.
- VK_SCOPE_QUEUE_FAMILY_KHR corresponds to SPIR-V `QueueFamily` scope.

All enum values match the corresponding SPIR-V value.

Possible values for VkComponentTypeKHR include:

```
// Provided by VK_KHR_cooperative_matrix
typedef enum VkComponentTypeKHR {
    VK_COMPONENT_TYPE_FLOAT16_KHR = 0,
    VK_COMPONENT_TYPE_FLOAT32_KHR = 1,
    VK_COMPONENT_TYPE_FLOAT64_KHR = 2,
    VK_COMPONENT_TYPE_SINT8_KHR = 3,
    VK_COMPONENT_TYPE_SINT16_KHR = 4,
    VK_COMPONENT_TYPE_SINT32_KHR = 5,
    VK_COMPONENT_TYPE_SINT64_KHR = 6,
    VK_COMPONENT_TYPE_UINT8_KHR = 7,
    VK_COMPONENT_TYPE_UINT16_KHR = 8,
    VK_COMPONENT_TYPE_UINT32_KHR = 9,
    VK_COMPONENT_TYPE_UINT64_KHR = 10,
  // Provided by VK_KHR_cooperative_matrix with VK_KHR_shader_bfloat16
    VK_COMPONENT_TYPE_BFLOAT16_KHR = 1000141000,
  // Provided by VK_KHR_cooperative_matrix with VK_EXT_shader_float8
    VK_COMPONENT_TYPE_FLOAT8_E4M3_EXT = 1000491002,
  // Provided by VK_KHR_cooperative_matrix with VK_EXT_shader_float8
    VK_COMPONENT_TYPE_FLOAT8_E5M2_EXT = 1000491003,
  // Provided by VK_KHR_cooperative_matrix with VK_EXT_shader_ocp_microscaling_types
    VK_COMPONENT_TYPE_FLOAT6_E2M3_EXT = 1000672000,
  // Provided by VK_KHR_cooperative_matrix with VK_EXT_shader_ocp_microscaling_types
    VK_COMPONENT_TYPE_FLOAT6_E3M2_EXT = 1000672001,
  // Provided by VK_KHR_cooperative_matrix with VK_EXT_shader_ocp_microscaling_types
    VK_COMPONENT_TYPE_FLOAT4_E2M1_EXT = 1000672002,
  // Provided by VK_KHR_cooperative_matrix with VK_EXT_shader_ocp_microscaling_types
    VK_COMPONENT_TYPE_FLOAT8_UNSIGNED_E8M0_EXT = 1000672003,
  // Provided by VK_KHR_cooperative_matrix with VK_EXT_shader_ocp_microscaling_types
    VK_COMPONENT_TYPE_MXINT8_EXT = 1000672004,
} VkComponentTypeKHR;
```

- VK_COMPONENT_TYPE_FLOAT16_KHR corresponds to SPIR-V OpTypeFloat 16.
- VK_COMPONENT_TYPE_FLOAT32_KHR corresponds to SPIR-V OpTypeFloat 32.
- VK_COMPONENT_TYPE_FLOAT64_KHR corresponds to SPIR-V OpTypeFloat 64.
- VK_COMPONENT_TYPE_SINT8_KHR corresponds to SPIR-V OpTypeInt 8 0/1.
- VK_COMPONENT_TYPE_SINT16_KHR corresponds to SPIR-V OpTypeInt 16 0/1.
- VK_COMPONENT_TYPE_SINT32_KHR corresponds to SPIR-V OpTypeInt 32 0/1.
- VK_COMPONENT_TYPE_SINT64_KHR corresponds to SPIR-V OpTypeInt 64 0/1.
- VK_COMPONENT_TYPE_UINT8_KHR corresponds to SPIR-V OpTypeInt 8 0/1.
- VK_COMPONENT_TYPE_UINT16_KHR corresponds to SPIR-V OpTypeInt 16 0/1.
- VK_COMPONENT_TYPE_UINT32_KHR corresponds to SPIR-V OpTypeInt 32 0/1.
- VK_COMPONENT_TYPE_UINT64_KHR corresponds to SPIR-V OpTypeInt 64 0/1.
- VK_COMPONENT_TYPE_BFLOAT16_KHR corresponds to SPIR-V OpTypeFloat 16 BFloat16KHR.
- VK_COMPONENT_TYPE_SINT8_PACKED_NV corresponds to four 8-bit signed integers packed in

a 32-bit unsigned integer.

- VK_COMPONENT_TYPE_UINT8_PACKED_NV corresponds to four 8-bit unsigned integers packed in a 32-bit unsigned integer.
- VK_COMPONENT_TYPE_FLOAT8_E4M3_EXT corresponds to SPIR-V `OpTypeFloat` 8 Float8E4M3EXT.
- VK_COMPONENT_TYPE_FLOAT8_E5M2_EXT corresponds to SPIR-V `OpTypeFloat` 8 Float8E5M2EXT.
- VK_COMPONENT_TYPE_FLOAT6_E2M3_EXT corresponds to SPIR-V `OpTypeFloat` 6 Float6E2M3EXT.
- VK_COMPONENT_TYPE_FLOAT6_E3M2_EXT corresponds to SPIR-V `OpTypeFloat` 6 Float6E3M2EXT.
- VK_COMPONENT_TYPE_FLOAT4_E2M1_EXT corresponds to SPIR-V `OpTypeFloat` 4 Float4E2M1EXT.
- VK_COMPONENT_TYPE_FLOAT8_UNSIGNED_E8M0_EXT corresponds to SPIR-V `OpTypeFloat` 8 Float8UnsignedE8M0EXT.
- VK_COMPONENT_TYPE_MXINT8_EXT corresponds to SPIR-V `OpTypeFloat` 8 MXInt8EXT.

9.30.1. Cooperative Matrix Property Matching

Cooperative matrix types and instructions match cooperative matrix properties as defined in this section.

VkCooperativeMatrixPropertiesKHR Matching

The following rules define how cooperative matrix types and instructions match a VkCooperativeMatrixPropertiesKHR structure returned by vkGetPhysicalDeviceCooperativeMatrixPropertiesKHR.

VkCooperativeMatrixPropertiesKHR Type Matching

An `OpTypeCooperativeMatrixKHR` type matches a VkCooperativeMatrixPropertiesKHR structure if all of the following conditions are satisfied:

- Its scope matches VkCooperativeMatrixPropertiesKHR::`scope`.
- If `Use` is `MatrixA`, its number of rows matches VkCooperativeMatrixPropertiesKHR::`MSize`, its number of columns matches VkCooperativeMatrixPropertiesKHR::`KSize`, and its component type matches VkCooperativeMatrixPropertiesKHR::`AType`.
- If `Use` is `MatrixB`, its number of rows matches VkCooperativeMatrixPropertiesKHR::`KSize`, its number of columns matches VkCooperativeMatrixPropertiesKHR::`NSize`, and its component type matches VkCooperativeMatrixPropertiesKHR::`BType`.
- If `Use` is `MatrixAccumulator`, its number of rows matches VkCooperativeMatrixPropertiesKHR::`MSize`, its number of columns matches VkCooperativeMatrixPropertiesKHR::`NSize`, and its component type matches either VkCooperativeMatrixPropertiesKHR::`CType` or VkCooperativeMatrixPropertiesKHR::`ResultType`.

VkCooperativeMatrixPropertiesKHR Multiply-Add Matching

The operands of an OpCooperativeMatrixMulAddKHR instruction match a VkCooperativeMatrixPropertiesKHR structure if all of the following conditions are satisfied:

- The type of A has Rows matching VkCooperativeMatrixPropertiesKHR::MSize, Columns matching VkCooperativeMatrixPropertiesKHR::KSize, Use equal to MatrixAKHR, and ComponentType matching VkCooperativeMatrixPropertiesKHR::AType.
- The type of B has Rows matching VkCooperativeMatrixPropertiesKHR::KSize, Columns matching VkCooperativeMatrixPropertiesKHR::NSize, Use equal to MatrixBKHR, and ComponentType matching VkCooperativeMatrixPropertiesKHR::BType.
- The type of C has Rows matching VkCooperativeMatrixPropertiesKHR::MSize, Columns matching VkCooperativeMatrixPropertiesKHR::NSize, Use equal to MatrixAccumulatorKHR, and ComponentType matching VkCooperativeMatrixPropertiesKHR::CType.
- The type of Result has Rows matching VkCooperativeMatrixPropertiesKHR::MSize, Columns matching VkCooperativeMatrixPropertiesKHR::NSize, Use equal to MatrixAccumulatorKHR, and ComponentType matching VkCooperativeMatrixPropertiesKHR::ResultType.
- MatrixASignedComponents is used if and only if VkCooperativeMatrixPropertiesKHR::AType is a signed integer type.
- MatrixBSignedComponents is used if and only if VkCooperativeMatrixPropertiesKHR::BType is a signed integer type.
- MatrixCSignedComponents is used if and only if VkCooperativeMatrixPropertiesKHR::CType is a signed integer type.
- MatrixResultSignedComponents is used if and only if VkCooperativeMatrixPropertiesKHR ::ResultType is a signed integer type.
- SaturatingAccumulationKHR is used if and only if VkCooperativeMatrixPropertiesKHR ::saturatingAccumulation is VK_TRUE.
- SaturatingAccumulationKHR is not used if and only if VkCooperativeMatrixPropertiesKHR ::saturatingAccumulation is VK_FALSE.
- The scope of all cooperative matrix operands matches VkCooperativeMatrixPropertiesKHR ::scope.

VkCooperativeMatrixProperties2EXT Matching

The following rules define how cooperative matrix types and instructions match a VkCooperativeMatrixProperties2EXT structure returned by vkGetPhysicalDeviceCooperativeMatrixProperties2EXT.

VkPhysicalDeviceCooperativeMatrixInfo2EXT Query Applicability

A query using a VkPhysicalDeviceCooperativeMatrixInfo2EXT structure is applicable to a cooperative matrix type in a shader if all of the following conditions are satisfied:

- VkPhysicalDeviceCooperativeMatrixInfo2EXT::scope matches the type's scope.
- If VkPhysicalDeviceCooperativeMatrixInfo2EXT::scope is VK_SCOPE_WORKGROUP_KHR,

VkPhysicalDeviceCooperativeMatrixInfo2EXT::invocations matches the number of invocations in the local workgroup.

- If the shader was created with VK_PIPELINE_SHADER_STAGE_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT or VK_SHADER_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT_EXT set, or if the SPIR-V module is version 1.6 or later and no structure specifying a required subgroup size was chained, VkPhysicalDeviceCooperativeMatrixInfo2EXT::subgroupSize is 0.
- If a structure specifying a required subgroup size was chained, or if the SPIR-V module is less than version 1.6 and the shader was not created with VK_PIPELINE_SHADER_STAGE_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT or VK_SHADER_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT_EXT set, VkPhysicalDeviceCooperativeMatrixInfo2EXT::subgroupSize equals the shader's effective subgroup size.

VkCooperativeMatrixProperties2EXT Type Matching

An OpTypeCooperativeMatrixKHR type in a shader matches a VkCooperativeMatrixProperties2EXT structure returned by a query if all of the following conditions are satisfied:

- The query is applicable to the type in the shader as defined in VkPhysicalDeviceCooperativeMatrixInfo2EXT Query Applicability.
- If Use is MatrixA, its number of rows is a multiple of VkCooperativeMatrixProperties2EXT::MGranularity, its number of columns is a multiple of VkCooperativeMatrixProperties2EXT::KGranularity, and its component type matches VkCooperativeMatrixProperties2EXT::AType.
- If Use is MatrixB, its number of rows is a multiple of VkCooperativeMatrixProperties2EXT::KGranularity, its number of columns is a multiple of VkCooperativeMatrixProperties2EXT::NGranularity, and its component type matches VkCooperativeMatrixProperties2EXT::BType.
- If Use is MatrixAccumulator, its number of rows is a multiple of VkCooperativeMatrixProperties2EXT::MGranularity, its number of columns is a multiple of VkCooperativeMatrixProperties2EXT::NGranularity, and its component type matches either VkCooperativeMatrixProperties2EXT::CType or VkCooperativeMatrixProperties2EXT::ResultType.
- The dimensions match the granularities exactly.

VkCooperativeMatrixProperties2EXT Multiply-Add Matching

The operands of an OpCooperativeMatrixMulAddKHR instruction in a shader match a VkCooperativeMatrixProperties2EXT structure returned by a query if all of the following conditions are satisfied:

- The query is applicable to every cooperative matrix operand type in the shader as defined in VkPhysicalDeviceCooperativeMatrixInfo2EXT Query Applicability.
- The type of A has Rows that are a multiple of VkCooperativeMatrixProperties2EXT::MGranularity, Columns that are a multiple of VkCooperativeMatrixProperties2EXT::KGranularity, Use equal to MatrixAKHR, and ComponentType matching VkCooperativeMatrixProperties2EXT::AType.
- The type of B has Rows that are a multiple of VkCooperativeMatrixProperties2EXT::KGranularity, Columns that are a multiple of VkCooperativeMatrixProperties2EXT::NGranularity, Use equal to

MatrixBKHR, and ComponentType matching VkCooperativeMatrixProperties2EXT::BType.

- The type of C has Rows that are a multiple of VkCooperativeMatrixProperties2EXT::MGranularity, Columns that are a multiple of VkCooperativeMatrixProperties2EXT::NGranularity, Use equal to MatrixAccumulatorKHR, and ComponentType matching VkCooperativeMatrixProperties2EXT::CType.
- The type of Result has Rows that are a multiple of VkCooperativeMatrixProperties2EXT::MGranularity, Columns that are a multiple of VkCooperativeMatrixProperties2EXT::NGranularity, Use equal to MatrixAccumulatorKHR, and ComponentType matching VkCooperativeMatrixProperties2EXT::ResultType.
- MatrixASignedComponents is used if and only if VkCooperativeMatrixProperties2EXT::AType is a signed integer type.
- MatrixBSignedComponents is used if and only if VkCooperativeMatrixProperties2EXT::BType is a signed integer type.
- MatrixCSignedComponents is used if and only if VkCooperativeMatrixProperties2EXT::CType is a signed integer type.
- MatrixResultSignedComponents is used if and only if VkCooperativeMatrixProperties2EXT::ResultType is a signed integer type.
- SaturatingAccumulationKHR is used if and only if VkPhysicalDeviceCooperativeMatrixInfo2EXT::flags contains VK_COOPERATIVE_MATRIX_SATURATING_ACCUMULATION_BIT_EXT.
- The dimensions match the granularities exactly.

9.31. Validation Cache

Validation cache objects allow the result of internal validation to be reused, both within a single application run and between multiple runs. Reuse within a single run is achieved by passing the same validation cache object when creating supported Vulkan objects. Reuse across runs of an application is achieved by retrieving validation cache contents in one run of an application, saving the contents, and using them to preinitialize a validation cache on a subsequent run. The contents of the validation cache objects are managed by the validation layers. Applications **can** manage the host memory consumed by a validation cache object and control the amount of data retrieved from a validation cache object.

Validation cache objects are represented by VkValidationCacheEXT handles:

```
// Provided by VK_EXT_validation_cache
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkValidationCacheEXT)
```

To create validation cache objects, call:

```
// Provided by VK_EXT_validation_cache
VkResult vkCreateValidationCacheEXT(
    VkDevice                                    device,
    const VkValidationCacheCreateInfoEXT*       pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkValidationCacheEXT*                       pValidationCache);
```

- `device` is the logical device that creates the validation cache object.
- `pCreateInfo` is a pointer to a VkValidationCacheCreateInfoEXT structure containing the initial parameters for the validation cache object.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pValidationCache` is a pointer to a VkValidationCacheEXT handle in which the resulting validation cache object is returned.

NOTE Applications **can** track and manage the total host memory size of a validation cache object using the `pAllocator`. Applications **can** limit the amount of data retrieved from a validation cache object in `vkGetValidationCacheDataEXT`. Implementations **should** not internally limit the total number of entries added to a validation cache object or the total host memory consumed.

Once created, a validation cache **can** be passed to the `vkCreateShaderModule` command by adding this object to the VkShaderModuleCreateInfo structure's `pNext` chain. If a VkShaderModuleValidationCacheCreateInfoEXT object is included in the VkShaderModuleCreateInfo::`pNext` chain, and its `validationCache` field is not VK_NULL_HANDLE, the implementation will query it for possible reuse opportunities and update it with new content. The use of the validation cache object in these commands is internally synchronized, and the same validation cache object **can** be used in multiple threads simultaneously.

NOTE Implementations **should** make every effort to limit any critical sections to the actual accesses to the cache, which is expected to be significantly shorter than the duration of the `vkCreateShaderModule` command.

Valid Usage (Implicit)

- VUID-vkCreateValidationCacheEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateValidationCacheEXT-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkValidationCacheCreateInfoEXT structure
- VUID-vkCreateValidationCacheEXT-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateValidationCacheEXT-pValidationCache-parameter
  `pValidationCache` **must** be a valid pointer to a VkValidationCacheEXT handle
- VUID-vkCreateValidationCacheEXT-device-queuecount

The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkValidationCacheCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_validation_cache
typedef struct VkValidationCacheCreateInfoEXT {
    VkStructureType                   sType;
    const void*                       pNext;
    VkValidationCacheCreateFlagsEXT   flags;
    size_t                            initialDataSize;
    const void*                       pInitialData;
} VkValidationCacheCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is reserved for future use.
- initialDataSize is the number of bytes in pInitialData. If initialDataSize is zero, the validation cache will initially be empty.
- pInitialData is a pointer to previously retrieved validation cache data. If the validation cache data is incompatible (as defined below) with the device, the validation cache will be initially empty. If initialDataSize is zero, pInitialData is ignored.

Valid Usage

- VUID-VkValidationCacheCreateInfoEXT-initialDataSize-01534
  If initialDataSize is not 0, it **must** be equal to the size of pInitialData, as returned by vkGetValidationCacheDataEXT when pInitialData was originally retrieved
- VUID-VkValidationCacheCreateInfoEXT-initialDataSize-01535
  If initialDataSize is not 0, pInitialData **must** have been retrieved from a previous call to vkGetValidationCacheDataEXT

Valid Usage (Implicit)

- VUID-VkValidationCacheCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VALIDATION_CACHE_CREATE_INFO_EXT
- VUID-VkValidationCacheCreateInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkValidationCacheCreateInfoEXT-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkValidationCacheCreateInfoEXT-pInitialData-parameter
  If `initialDataSize` is not `0`, `pInitialData` **must** be a valid pointer to an array of `initialDataSize` bytes

```
// Provided by VK_EXT_validation_cache
typedef VkFlags VkValidationCacheCreateFlagsEXT;
```

`VkValidationCacheCreateFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

Validation cache objects **can** be merged using the command:

```
// Provided by VK_EXT_validation_cache
VkResult vkMergeValidationCachesEXT(
    VkDevice                                    device,
    VkValidationCacheEXT                        dstCache,
    uint32_t                                    srcCacheCount,
    const VkValidationCacheEXT*                 pSrcCaches);
```

- `device` is the logical device that owns the validation cache objects.
- `dstCache` is the handle of the validation cache to merge results into.
- `srcCacheCount` is the length of the `pSrcCaches` array.
- `pSrcCaches` is a pointer to an array of validation cache handles, which will be merged into `dstCache`. The previous contents of `dstCache` are included after the merge.

NOTE The details of the merge operation are implementation-dependent, but implementations **should** merge the contents of the specified validation caches and prune duplicate entries.

Valid Usage

- VUID-vkMergeValidationCachesEXT-dstCache-01536
  `dstCache` **must** not appear in the list of source caches

Valid Usage (Implicit)

- VUID-vkMergeValidationCachesEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkMergeValidationCachesEXT-dstCache-parameter
  `dstCache` **must** be a valid VkValidationCacheEXT handle
- VUID-vkMergeValidationCachesEXT-pSrcCaches-parameter
  `pSrcCaches` **must** be a valid pointer to an array of `srcCacheCount` valid VkValidationCacheEXT handles
- VUID-vkMergeValidationCachesEXT-srcCacheCount-arraylength
  `srcCacheCount` **must** be greater than `0`
- VUID-vkMergeValidationCachesEXT-dstCache-parent
  `dstCache` **must** have been created, allocated, or retrieved from `device`
- VUID-vkMergeValidationCachesEXT-pSrcCaches-parent
  Each element of `pSrcCaches` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `dstCache` **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Data **can** be retrieved from a validation cache object using the command:

```
// Provided by VK_EXT_validation_cache
VkResult vkGetValidationCacheDataEXT(
    VkDevice                                    device,
    VkValidationCacheEXT                        validationCache,
    size_t*                                     pDataSize,
    void*                                       pData);
```

- `device` is the logical device that owns the validation cache.

- validationCache is the validation cache to retrieve data from.
- pDataSize is a pointer to a value related to the amount of data in the validation cache, as described below.
- pData is either NULL or a pointer to a buffer.

If pData is NULL, then the maximum size of the data that **can** be retrieved from the validation cache, in bytes, is returned in pDataSize. Otherwise, pDataSize **must** point to a variable set by the application to the size of the buffer, in bytes, pointed to by pData, and on return the variable is overwritten with the amount of data actually written to pData. If pDataSize is less than the maximum size that **can** be retrieved by the validation cache, at most pDataSize bytes will be written to pData, and vkGetValidationCacheDataEXT will return VK_INCOMPLETE instead of VK_SUCCESS, to indicate that not all of the validation cache was returned.

Any data written to pData is valid and **can** be provided as the pInitialData member of the VkValidationCacheCreateInfoEXT structure passed to vkCreateValidationCacheEXT.

Two calls to vkGetValidationCacheDataEXT with the same parameters **must** retrieve the same data unless a command that modifies the contents of the cache is called between them.

Applications **can** store the data retrieved from the validation cache, and use these data, possibly in a future run of the application, to populate new validation cache objects. The results of validation, however, **may** depend on the vendor ID, device ID, driver version, and other details of the device. To enable applications to detect when previously retrieved data is incompatible with the device, the initial bytes written to pData **must** be a header consisting of the following members:

Table 11. Layout for Validation Cache Header Version VK_VALIDATION_CACHE_HEADER_VERSION_ONE_EXT

Offset	Size	Meaning
0	4	length in bytes of the entire validation cache header written as a stream of bytes, with the least significant byte first
4	4	a VkValidationCacheHeaderVersionEXT value written as a stream of bytes, with the least significant byte first
8	VK_UUID_SIZE	a layer commit ID expressed as a UUID, which uniquely identifies the version of the validation layers used to generate these validation results

The first four bytes encode the length of the entire validation cache header, in bytes. This value includes all fields in the header including the validation cache version field and the size of the length field.

The next four bytes encode the validation cache version, as described for VkValidationCacheHeaderVersionEXT. A consumer of the validation cache **should** use the cache version to interpret the remainder of the cache header.

If pDataSize is less than what is necessary to store this header, nothing will be written to pData and

zero will be written to pDataSize.

NOTE

This query does not behave consistently with the behavior described in Opaque Binary Data Results, for historical reasons.

If the amount of data available is larger than the passed pDataSize, the query returns up to the size of the passed buffer, and signals overflow with a VK_INCOMPLETE success status instead of returning a VK_ERROR_NOT_ENOUGH_SPACE_KHR error status.

Valid Usage (Implicit)

- VUID-vkGetValidationCacheDataEXT-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkGetValidationCacheDataEXT-validationCache-parameter
  validationCache **must** be a valid VkValidationCacheEXT handle
- VUID-vkGetValidationCacheDataEXT-pDataSize-parameter
  pDataSize **must** be a valid pointer to a size_t value
- VUID-vkGetValidationCacheDataEXT-pData-parameter
  If the value referenced by pDataSize is not 0, and pData is not NULL, pData **must** be a valid pointer to an array of pDataSize bytes
- VUID-vkGetValidationCacheDataEXT-validationCache-parent
  validationCache **must** have been created, allocated, or retrieved from device

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Possible values of the second group of four bytes in the header returned by vkGetValidationCacheDataEXT, encoding the validation cache version, are:

```
// Provided by VK_EXT_validation_cache
typedef enum VkValidationCacheHeaderVersionEXT {
    VK_VALIDATION_CACHE_HEADER_VERSION_ONE_EXT = 1,
} VkValidationCacheHeaderVersionEXT;
```

- VK_VALIDATION_CACHE_HEADER_VERSION_ONE_EXT specifies version one of the validation cache.

To destroy a validation cache, call:

```
// Provided by VK_EXT_validation_cache
void vkDestroyValidationCacheEXT(
    VkDevice                                    device,
    VkValidationCacheEXT                        validationCache,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the validation cache object.
- `validationCache` is the handle of the validation cache to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyValidationCacheEXT-validationCache-01537
  If `VkAllocationCallbacks` were provided when `validationCache` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyValidationCacheEXT-validationCache-01538
  If no `VkAllocationCallbacks` were provided when `validationCache` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyValidationCacheEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyValidationCacheEXT-validationCache-parameter
  If `validationCache` is not VK_NULL_HANDLE, `validationCache` **must** be a valid VkValidationCacheEXT handle
- VUID-vkDestroyValidationCacheEXT-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyValidationCacheEXT-validationCache-parent
  If `validationCache` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `validationCache` **must** be externally synchronized

Chapter 10. Pipelines

The following figure shows a block diagram of the Vulkan pipelines. Some Vulkan commands specify geometric objects to be drawn or computational work to be performed, while others specify state controlling how objects are handled by the various pipeline stages, or control data transfer between memory organized as images and buffers. Commands are effectively sent through a processing pipeline, such as a *graphics pipeline*, a *ray tracing pipeline*, or a *compute pipeline*.

The graphics pipeline can be operated in two modes, as either *primitive shading* or *mesh shading* pipeline.

Primitive Shading

The first stage of the graphics pipeline (Input Assembler) assembles vertices to form geometric primitives such as points, lines, and triangles, based on a requested primitive topology. In the next stage (Vertex Shader) vertices **can** be transformed, computing positions and attributes for each vertex. If tessellation and/or geometry shaders are supported, they **can** then generate multiple primitives from a single input primitive, possibly changing the primitive topology or generating additional attribute data in the process.

Mesh Shading

When using the *mesh shading* pipeline input primitives are not assembled implicitly, but explicitly through the (Mesh Shader). The work on the mesh pipeline is initiated by the application drawing a set of mesh tasks.

If an optional (Task Shader) is active, each task triggers the execution of a task shader workgroup that will generate a new set of tasks upon completion. Each of these spawned tasks, or each of the original dispatched tasks if no task shader is present, triggers the execution of a mesh shader workgroup that produces an output mesh with a variable-sized number of primitives assembled from vertices stored in the output mesh.

Common

The final resulting primitives are clipped to a clip volume in preparation for the next stage, Rasterization. The rasterizer produces a series of *fragments* associated with a region of the framebuffer, from a two-dimensional description of a point, line segment, or triangle. These fragments are processed by fragment operations to determine whether generated values will be written to the framebuffer. Fragment shading determines the values to be written to the framebuffer attachments. Framebuffer operations then read and write the color and depth/stencil attachments of the framebuffer for a given subpass of a render pass instance. The attachments **can** be used as input attachments in the fragment shader in a later subpass of the same render pass.

The compute pipeline is a separate pipeline from the graphics pipeline, which operates on one-, two-, or three-dimensional workgroups which **can** read from and write to buffer and image memory.

This ordering is meant only as a tool for describing Vulkan, not as a strict rule of how Vulkan is implemented, and we present it only as a means to organize the various operations of the pipelines.

Actual ordering guarantees between pipeline stages are explained in detail in the synchronization chapter.

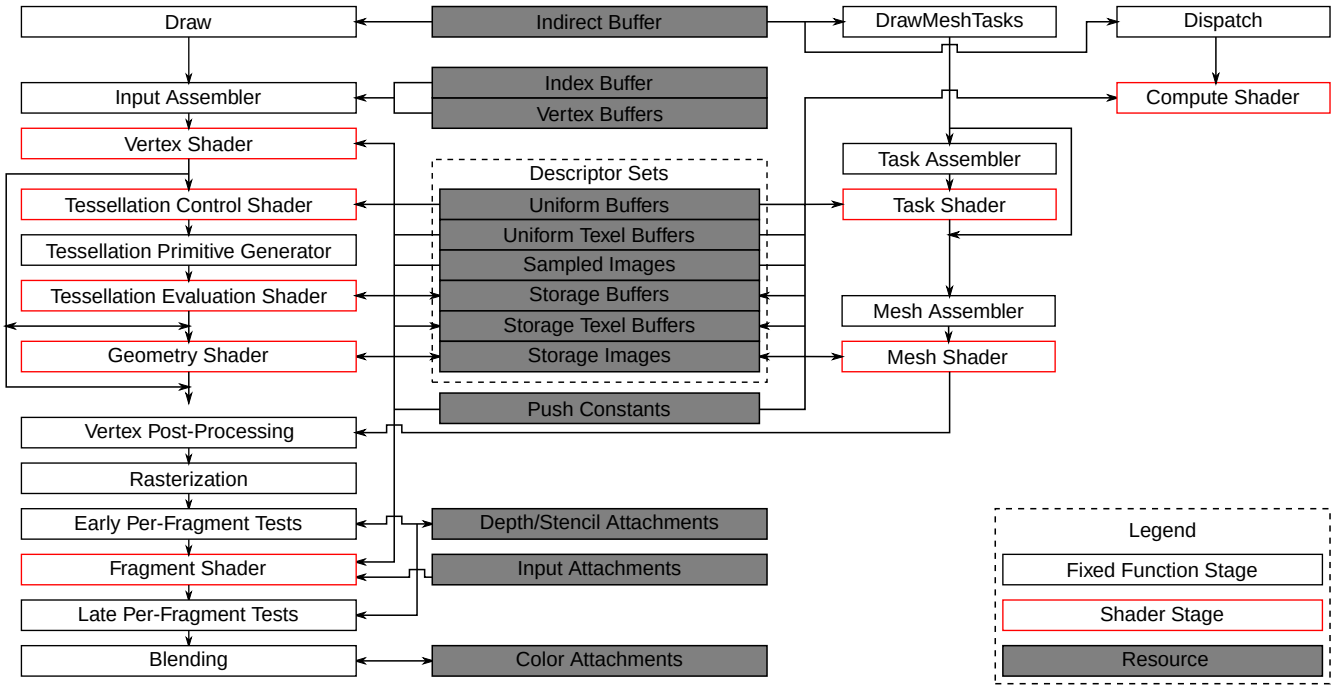


Figure 2. Block diagram of the Vulkan pipeline

Each pipeline is controlled by a monolithic object created from a description of all of the shader stages and any relevant fixed-function stages. Linking the whole pipeline together allows the optimization of shaders based on their input/outputs and eliminates expensive draw time state validation.

A pipeline object is bound to the current state using vkCmdBindPipeline. Any pipeline object state that is specified as dynamic is not applied to the current state when the pipeline object is bound, but is instead set by dynamic state setting commands.

No state, including dynamic state, is inherited from one command buffer to another.

Compute, ray tracing, and graphics pipelines are each represented by `VkPipeline` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkPipeline)
```

10.1. Multiple Pipeline Creation

Multiple pipelines **can** be created in a single call by commands such as:

- vkCreateRayTracingPipelinesKHR,
- vkCreateGraphicsPipelines,
- vkCreateComputePipelines.

The creation commands are passed an array `pCreateInfos` of `Vk*PipelineCreateInfo` structures specifying parameters of each pipeline to be created, and return a corresponding array of handles

in pPipelines. Each element index *i* of pPipelines is created based on the corresponding element *i* of pCreateInfos.

Applications **can** group together similar pipelines to be created in a single call, and implementations are encouraged to look for reuse opportunities when creating a group.

When attempting to create many pipelines in a single command, it is possible that creation **may** fail for a subset of them. In this case, the corresponding elements of pPipelines will be VK_NULL_HANDLE. If creation fails for a pipeline despite valid arguments (for example, due to out of memory errors), the VkResult code returned by the pipeline creation command will indicate why. The implementation will attempt to create all pipelines, and only return VK_NULL_HANDLE values for those that actually failed.

If creation fails for a pipeline that has the VK_PIPELINE_CREATE_EARLY_RETURN_ON_FAILURE_BIT set in its Vk*PipelineCreateInfo, pipelines at an index in the pPipelines array greater than or equal to that of the failing pipeline will be VK_NULL_HANDLE.

If creation fails for multiple pipelines, the returned VkResult **must** be the return value of any one of the pipelines which did not succeed. An application **can** reliably clean up from a failed call by iterating over the pPipelines array and destroying every element that is not VK_NULL_HANDLE.

If the entire command fails and no pipelines are created, all elements of pPipelines will be VK_NULL_HANDLE.

10.2. Pipeline Creation Flags

When creating a pipeline, the application **can** specify a set of flags that control how the pipeline is created and certain aspects of its behavior and capabilities. These flags are specified by chaining the VkPipelineCreateFlags2CreateInfo structure to the pipeline creation structure with a bitmask of VkPipelineCreateFlagBits2, or by setting the flags member of the pipeline creation structure to a bitmask of VkPipelineCreateFlagBits.

The VkPipelineCreateFlags2CreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPipelineCreateFlags2CreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineCreateFlags2    flags;
} VkPipelineCreateFlags2CreateInfo;
```

```
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
// Equivalent to VkPipelineCreateFlags2CreateInfo
typedef VkPipelineCreateFlags2CreateInfo VkPipelineCreateFlags2CreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.

- `flags` is a bitmask of VkPipelineCreateFlagBits2 specifying how a pipeline will be generated.

If this structure is included in the `pNext` chain of a pipeline creation structure, `flags` is used instead of the corresponding `flags` value passed in that creation structure, allowing additional creation flags to be specified.

Valid Usage (Implicit)

- VUID-VkPipelineCreateFlags2CreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_CREATE_FLAGS_2_CREATE_INFO
- VUID-VkPipelineCreateFlags2CreateInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkPipelineCreateFlagBits2 values

Structure Chaining

Extends the structures

- VkComputePipelineCreateInfo
- VkGraphicsPipelineCreateInfo
- VkRayTracingPipelineCreateInfoKHR

Bits which **can** be set in VkPipelineCreateFlags2CreateInfo::`flags`, specifying how a pipeline is created, are:

```
// Provided by VK_VERSION_1_4
// Flag bits for VkPipelineCreateFlagBits2
typedef VkFlags64 VkPipelineCreateFlagBits2;
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_DISABLE_OPTIMIZATION_BIT =
0x00000001ULL;
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_ALLOW_DERIVATIVES_BIT =
0x00000002ULL;
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_DERIVATIVE_BIT =
0x00000004ULL;
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_VIEW_INDEX_FROM_DEVICE_INDEX_BIT = 0x00000008ULL;
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_DISPATCH_BASE_BIT =
0x00000010ULL;
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT = 0x00000100ULL;
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_EARLY_RETURN_ON_FAILURE_BIT = 0x00000200ULL;
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_NO_PROTECTED_ACCESS_BIT =
0x08000000ULL;
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_PROTECTED_ACCESS_ONLY_BIT
= 0x40000000ULL;
// Provided by VK_EXT_descriptor_heap
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT =
```

```
0x1000000000ULL;
// Provided by VK_KHR_ray_tracing_pipeline
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_BUILT_IN_PRIMITIVES_BIT_KHR = 0x00001000ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_opacity_micromap,
VK_EXT_opacity_micromap with VK_KHR_maintenance5 or VK_VERSION_1_4
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_RAY_TRACING_OPACITY_MICROMAP_BIT_EXT = 0x01000000ULL;
// Provided by VK_EXT_legacy_dithering with (VK_KHR_dynamic_rendering or
VK_VERSION_1_3) and (VK_VERSION_1_4 or VK_KHR_extended_flags or VK_KHR_maintenance5)
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT = 0x400000000ULL;
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_DISABLE_OPTIMIZATION_BIT_KHR = 0x00000001ULL;
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_ALLOW_DERIVATIVES_BIT_KHR
= 0x00000002ULL;
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_DERIVATIVE_BIT_KHR =
0x00000004ULL;
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_VIEW_INDEX_FROM_DEVICE_INDEX_BIT_KHR = 0x00000008ULL;
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_DISPATCH_BASE_BIT_KHR =
0x00000010ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_pipeline_executable_properties,
VK_KHR_maintenance5 with VK_KHR_pipeline_executable_properties
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_CAPTURE_STATISTICS_BIT_KHR
= 0x00000040ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_pipeline_executable_properties,
VK_KHR_maintenance5 with VK_KHR_pipeline_executable_properties
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR = 0x00000080ULL;
// Provided by VK_KHR_extended_flags with VK_VERSION_1_3 or
VK_EXT_pipeline_creation_cache_control, VK_KHR_maintenance5 with VK_VERSION_1_3 or
VK_EXT_pipeline_creation_cache_control
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT_KHR = 0x00000100ULL;
// Provided by VK_KHR_extended_flags with VK_VERSION_1_3 or
VK_EXT_pipeline_creation_cache_control, VK_KHR_maintenance5 with VK_VERSION_1_3 or
VK_EXT_pipeline_creation_cache_control
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_EARLY_RETURN_ON_FAILURE_BIT_KHR = 0x00000200ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_graphics_pipeline_library,
VK_KHR_maintenance5 with VK_EXT_graphics_pipeline_library
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_LINK_TIME_OPTIMIZATION_BIT_EXT = 0x00000400ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_graphics_pipeline_library,
VK_KHR_maintenance5 with VK_EXT_graphics_pipeline_library
```

```
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_RETAIN_LINK_TIME_OPTIMIZATION_INFO_BIT_EXT = 0x00800000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_pipeline_library, VK_KHR_maintenance5
with VK_KHR_pipeline_library
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_LIBRARY_BIT_KHR =
0x00000800ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_ray_tracing_pipeline,
VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR = 0x00001000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_ray_tracing_pipeline,
VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_AABBS_BIT_KHR = 0x00002000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_ray_tracing_pipeline,
VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR = 0x00004000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_ray_tracing_pipeline,
VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR = 0x00008000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_ray_tracing_pipeline,
VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR = 0x00010000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_ray_tracing_pipeline,
VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR = 0x00020000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_ray_tracing_pipeline,
VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR =
0x00080000ULL;
// Provided by VK_KHR_extended_flags with (VK_KHR_dynamic_rendering or VK_VERSION_1_3)
and VK_KHR_fragment_shading_rate, VK_KHR_maintenance5 with (VK_KHR_dynamic_rendering
or VK_VERSION_1_3) and VK_KHR_fragment_shading_rate
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR =
0x00200000ULL;
// Provided by VK_KHR_extended_flags with (VK_KHR_dynamic_rendering or VK_VERSION_1_3)
and VK_EXT_fragment_density_map, VK_KHR_maintenance5 with (VK_KHR_dynamic_rendering or
VK_VERSION_1_3) and VK_EXT_fragment_density_map
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT =
0x00400000ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_attachment_feedback_loop_layout,
VK_KHR_maintenance5 with VK_EXT_attachment_feedback_loop_layout
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT = 0x02000000ULL;
```

```
// Provided by VK_KHR_extended_flags with VK_EXT_attachment_feedback_loop_layout,
VK_KHR_maintenance5 with VK_EXT_attachment_feedback_loop_layout
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT = 0x04000000ULL;
// Provided by VK_KHR_maintenance5 with VK_VERSION_1_4 or
VK_EXT_pipeline_protected_access
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_NO_PROTECTED_ACCESS_BIT_EXT = 0x08000000ULL;
// Provided by VK_KHR_maintenance5 with VK_VERSION_1_4 or
VK_EXT_pipeline_protected_access
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_PROTECTED_ACCESS_ONLY_BIT_EXT = 0x40000000ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_descriptor_buffer,
VK_KHR_maintenance5 with VK_EXT_descriptor_buffer
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_DESCRIPTOR_BUFFER_BIT_EXT
= 0x20000000ULL;
// Provided by VK_KHR_pipeline_binary
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR =
0x80000000ULL;
// Provided by VK_EXT_device_generated_commands
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT
= 0x4000000000ULL;
// Provided by VK_KHR_opacity_micromap with VK_VERSION_1_4 or VK_KHR_extended_flags or
VK_KHR_maintenance5
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR = 0x01000000ULL;
// Provided by VK_KHR_opacity_micromap with VK_VERSION_1_4 or VK_KHR_extended_flags or
VK_KHR_maintenance5
static const VkPipelineCreateFlagBits2
VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR =
0x20000000000ULL;
// Provided by VK_EXT_shader_64bit_indexing
static const VkPipelineCreateFlagBits2 VK_PIPELINE_CREATE_2_64_BIT_INDEXING_BIT_EXT =
0x800000000000ULL;
```

```
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
// Equivalent to VkPipelineCreateFlagBits2
typedef VkPipelineCreateFlagBits2 VkPipelineCreateFlagBits2KHR;
```

- VK_PIPELINE_CREATE_2_DISABLE_OPTIMIZATION_BIT specifies that the created pipeline will not be optimized. Using this flag **may** reduce the time taken to create the pipeline.
- VK_PIPELINE_CREATE_2_ALLOW_DERIVATIVES_BIT specifies that the pipeline to be created is allowed to be the parent of a pipeline that will be created in a subsequent pipeline creation call.
- VK_PIPELINE_CREATE_2_DERIVATIVE_BIT specifies that the pipeline to be created will be a child of a previously created parent pipeline.
- VK_PIPELINE_CREATE_2_VIEW_INDEX_FROM_DEVICE_INDEX_BIT specifies that any shader input variables decorated as `ViewIndex` will be assigned values as if they were decorated as

DeviceIndex.

- VK_PIPELINE_CREATE_2_DISPATCH_BASE_BIT specifies that a compute pipeline **can** be used with vkCmdDispatchBase with a non-zero base workgroup.
- VK_PIPELINE_CREATE_2_CAPTURE_STATISTICS_BIT_KHR specifies that the shader compiler should capture statistics for the pipeline executables produced by the compile process which **can** later be retrieved by calling vkGetPipelineExecutableStatisticsKHR. Enabling this flag **must** not affect the final compiled pipeline but **may** disable pipeline caching or otherwise affect pipeline creation time.
- VK_PIPELINE_CREATE_2_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR specifies that the shader compiler should capture the internal representations of pipeline executables produced by the compile process which **can** later be retrieved by calling vkGetPipelineExecutableInternalRepresentationsKHR. Enabling this flag **must** not affect the final compiled pipeline but **may** disable pipeline caching or otherwise affect pipeline creation time. When capturing IR from pipelines created with pipeline libraries, there is no guarantee that IR from libraries **can** be retrieved from the linked pipeline. Applications **should** retrieve IR from each library, and any linked pipelines, separately.
- VK_PIPELINE_CREATE_2_LIBRARY_BIT_KHR specifies that the pipeline **cannot** be used directly, and instead defines a *pipeline library* that **can** be combined with other pipelines using the VkPipelineLibraryCreateInfoKHR structure. This is available in ray tracing and graphics pipelines.
- VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR specifies that an any-hit shader will always be present when an any-hit shader would be executed. A NULL any-hit shader is an any-hit shader which is effectively VK_SHADER_UNUSED_KHR, such as from a shader group consisting entirely of zeros.
- VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR specifies that a closest hit shader will always be present when a closest hit shader would be executed. A NULL closest hit shader is a closest hit shader which is effectively VK_SHADER_UNUSED_KHR, such as from a shader group consisting entirely of zeros.
- VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR specifies that a miss shader will always be present when a miss shader would be executed. A NULL miss shader is a miss shader which is effectively VK_SHADER_UNUSED_KHR, such as from a shader group consisting entirely of zeros.
- VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR specifies that an intersection shader will always be present when an intersection shader would be executed. A NULL intersection shader is an intersection shader which is effectively VK_SHADER_UNUSED_KHR, such as from a shader group consisting entirely of zeros.
- VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR specifies that all built-in primitives will be skipped during traversal using pipeline trace ray instructions.
- VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_BUILT_IN_PRIMITIVES_BIT_KHR is an alias for VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR.
- VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_AABBS_BIT_KHR specifies that AABB primitives will be skipped during traversal using pipeline trace ray instructions.
- VK_PIPELINE_CREATE_2_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KH

R specifies that the shader group handles **can** be saved and reused on a subsequent run (e.g. for trace capture and replay).

- VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT specifies that the pipeline **can** be used in a `VkIndirectExecutionSetEXT`.
- VK_PIPELINE_CREATE_2_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT specifies that pipeline creation will fail if a compile is required for creation of a valid VkPipeline object; VK_PIPELINE_COMPILE_REQUIRED will be returned by pipeline creation, and the VkPipeline will be VK_NULL_HANDLE.
- When creating multiple pipelines, VK_PIPELINE_CREATE_2_EARLY_RETURN_ON_FAILURE_BIT specifies that control will be returned to the application if any individual pipeline returns a result which is not VK_SUCCESS rather than continuing to create additional pipelines.
- VK_PIPELINE_CREATE_2_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR specifies that the pipeline will be used with a fragment shading rate attachment.
- VK_PIPELINE_CREATE_2_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT specifies that the pipeline will be used with a fragment density map attachment.
- VK_PIPELINE_CREATE_2_LINK_TIME_OPTIMIZATION_BIT_EXT specifies that pipeline libraries being linked into this library **should** have link time optimizations applied. If this bit is omitted, implementations **should** instead perform linking as rapidly as possible.
- VK_PIPELINE_CREATE_2_RETAIN_LINK_TIME_OPTIMIZATION_INFO_BIT_EXT specifies that pipeline libraries should retain any information necessary to later perform an optimal link with VK_PIPELINE_CREATE_2_LINK_TIME_OPTIMIZATION_BIT_EXT.
- VK_PIPELINE_CREATE_2_DESCRIPTOR_BUFFER_BIT_EXT specifies that a pipeline will be used with descriptor buffers, rather than descriptor sets.
- VK_PIPELINE_CREATE_2_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT specifies that the pipeline **may** be used with an attachment feedback loop including color attachments.
- VK_PIPELINE_CREATE_2_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT specifies that the pipeline **may** be used with an attachment feedback loop including depth-stencil attachments.
- VK_PIPELINE_CREATE_2_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR specifies that the ray tracing pipeline **can** be used with acceleration structures which reference an opacity micromap array. This flag has no effect on using opacity micromaps with ray queries.
- VK_PIPELINE_CREATE_2_NO_PROTECTED_ACCESS_BIT specifies that the pipeline **must** not be bound to a protected command buffer.
- VK_PIPELINE_CREATE_2_PROTECTED_ACCESS_ONLY_BIT specifies that the pipeline **must** not be bound to an unprotected command buffer.
- VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR specifies that `VkPipelineBinaryKHR` objects **can** be created from the pipeline. If VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR is used, implementations **should** not store pipeline data to an internal cache, if such a cache exists as stated by `pipelineBinaryInternalCache`. If `pipelineBinaryPrefersInternalCache` is VK_TRUE, applications **should** not use VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR.
- VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT specifies that the pipeline will be used in a render pass that is begun with

VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT.

- VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR specifies that the pipeline **cannot** be used with acceleration structures which are built with geometry that have an index buffer including both special indices and indices pointing to an associated micromap array. Geometry which has an index buffer using only special indices without an associated micromap array **can** be used with this flag.
- VK_PIPELINE_CREATE_2_64_BIT_INDEXING_BIT_EXT specifies that the pipeline enables 64-bit indexing.
- VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT specifies that the pipeline will use descriptor heap mappings instead of descriptor set layouts.

It is valid to set both VK_PIPELINE_CREATE_2_ALLOW_DERIVATIVES_BIT and VK_PIPELINE_CREATE_2_DERIVATIVE_BIT. This allows a pipeline to be both a parent and possibly a child in a pipeline hierarchy. See Pipeline Derivatives for more information.

When an implementation is looking up a pipeline in a pipeline cache, if that pipeline is being created using linked libraries, implementations **should** always return an equivalent pipeline created with VK_PIPELINE_CREATE_2_LINK_TIME_OPTIMIZATION_BIT_EXT if available, whether or not that bit was specified.

NOTE

Using VK_PIPELINE_CREATE_2_LINK_TIME_OPTIMIZATION_BIT_EXT (or not) when linking pipeline libraries is intended as a performance tradeoff between host and device. If the bit is omitted, linking should be faster and produce a pipeline more rapidly, but performance of the pipeline on the target device may be reduced. If the bit is included, linking may be slower but should produce a pipeline with device performance comparable to a monolithically created pipeline. Using both options can allow latency-sensitive applications to generate a suboptimal but usable pipeline quickly, and then perform an optimal link in the background, substituting the result for the suboptimally linked pipeline as soon as it is available.

```
// Provided by VK_VERSION_1_4
typedef VkFlags64 VkPipelineCreateFlags2;
```

```
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
// Equivalent to VkPipelineCreateFlags2
typedef VkPipelineCreateFlags2 VkPipelineCreateFlags2KHR;
```

VkPipelineCreateFlags2 is a bitmask type for setting a mask of zero or more VkPipelineCreateFlagBits2.

Bits which **can** be set in

- VkGraphicsPipelineCreateInfo::flags
- VkComputePipelineCreateInfo::flags
- VkRayTracingPipelineCreateInfoKHR::flags

specify how a pipeline is created, and are:

WARNING	This functionality is superseded by VkPipelineCreateFlagBits2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef enum VkPipelineCreateFlagBits {
    VK_PIPELINE_CREATE_DISABLE_OPTIMIZATION_BIT = 0x00000001,
    VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT = 0x00000002,
    VK_PIPELINE_CREATE_DERIVATIVE_BIT = 0x00000004,
  // Provided by VK_VERSION_1_1
    VK_PIPELINE_CREATE_DISPATCH_BASE_BIT = 0x00000010,
  // Provided by VK_VERSION_1_1
    VK_PIPELINE_CREATE_VIEW_INDEX_FROM_DEVICE_INDEX_BIT = 0x00000008,
  // Provided by VK_VERSION_1_3
    VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT = 0x00000100,
  // Provided by VK_VERSION_1_3
    VK_PIPELINE_CREATE_EARLY_RETURN_ON_FAILURE_BIT = 0x00000200,
  // Provided by VK_VERSION_1_4
    VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT = 0x08000000,
  // Provided by VK_VERSION_1_4
    VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT = 0x40000000,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR = 0x00004000,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR = 0x00008000,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR = 0x00010000,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR = 0x00020000,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR = 0x00001000,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR = 0x00002000,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR =
0x00080000,
  // Provided by VK_EXT_fragment_density_map with VK_VERSION_1_3 or
VK_KHR_dynamic_rendering
    VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT = 0x00400000,
  // Provided by VK_KHR_fragment_shading_rate with VK_VERSION_1_3 or
VK_KHR_dynamic_rendering
    VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR =
0x00200000,
  // Provided by VK_KHR_pipeline_executable_properties
    VK_PIPELINE_CREATE_CAPTURE_STATISTICS_BIT_KHR = 0x00000040,
  // Provided by VK_KHR_pipeline_executable_properties
    VK_PIPELINE_CREATE_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR = 0x00000080,
  // Provided by VK_KHR_pipeline_library
    VK_PIPELINE_CREATE_LIBRARY_BIT_KHR = 0x00000800,
```

```
  // Provided by VK_EXT_descriptor_buffer
    VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT = 0x20000000,
  // Provided by VK_EXT_graphics_pipeline_library
    VK_PIPELINE_CREATE_RETAIN_LINK_TIME_OPTIMIZATION_INFO_BIT_EXT = 0x00800000,
  // Provided by VK_EXT_graphics_pipeline_library
    VK_PIPELINE_CREATE_LINK_TIME_OPTIMIZATION_BIT_EXT = 0x00000400,
  // Provided by VK_EXT_attachment_feedback_loop_layout
    VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT = 0x02000000,
  // Provided by VK_EXT_attachment_feedback_loop_layout
    VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT = 0x04000000,
  // Provided by VK_KHR_opacity_micromap
    VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR = 0x01000000,
  // Provided by VK_VERSION_1_1
  // VK_PIPELINE_CREATE_DISPATCH_BASE is a legacy alias
    VK_PIPELINE_CREATE_DISPATCH_BASE = VK_PIPELINE_CREATE_DISPATCH_BASE_BIT,
  // Provided by VK_KHR_device_group
    VK_PIPELINE_CREATE_VIEW_INDEX_FROM_DEVICE_INDEX_BIT_KHR =
VK_PIPELINE_CREATE_VIEW_INDEX_FROM_DEVICE_INDEX_BIT,
  // Provided by VK_KHR_device_group
    VK_PIPELINE_CREATE_DISPATCH_BASE_BIT_KHR = VK_PIPELINE_CREATE_DISPATCH_BASE_BIT,
  // Provided by VK_KHR_device_group
  // VK_PIPELINE_CREATE_DISPATCH_BASE_KHR is a legacy alias
    VK_PIPELINE_CREATE_DISPATCH_BASE_KHR = VK_PIPELINE_CREATE_DISPATCH_BASE_BIT,
  // Provided by VK_EXT_fragment_density_map with VK_VERSION_1_3 or
VK_KHR_dynamic_rendering
  // VK_PIPELINE_RASTERIZATION_STATE_CREATE_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT is
a legacy alias
    VK_PIPELINE_RASTERIZATION_STATE_CREATE_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT =
VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT,
  // Provided by VK_KHR_fragment_shading_rate with VK_VERSION_1_3 or
VK_KHR_dynamic_rendering
  // VK_PIPELINE_RASTERIZATION_STATE_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
is a legacy alias
    VK_PIPELINE_RASTERIZATION_STATE_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR =
VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR,
  // Provided by VK_EXT_opacity_micromap
    VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_EXT =
VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR,
  // Provided by VK_EXT_pipeline_protected_access
    VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT_EXT =
VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT,
  // Provided by VK_EXT_pipeline_protected_access
    VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT_EXT =
VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT,
} VkPipelineCreateFlagBits;
```

- VK_PIPELINE_CREATE_DISABLE_OPTIMIZATION_BIT specifies that the created pipeline will not be optimized. Using this flag **may** reduce the time taken to create the pipeline.
- VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT specifies that the pipeline to be created is allowed to be the parent of a pipeline that will be created in a subsequent pipeline creation call.

- VK_PIPELINE_CREATE_DERIVATIVE_BIT specifies that the pipeline to be created will be a child of a previously created parent pipeline.
- VK_PIPELINE_CREATE_VIEW_INDEX_FROM_DEVICE_INDEX_BIT specifies that any shader input variables decorated as `ViewIndex` will be assigned values as if they were decorated as `DeviceIndex`.
- VK_PIPELINE_CREATE_DISPATCH_BASE_BIT specifies that a compute pipeline **can** be used with vkCmdDispatchBase with a non-zero base workgroup.
- VK_PIPELINE_CREATE_CAPTURE_STATISTICS_BIT_KHR specifies that the shader compiler should capture statistics for the pipeline executables produced by the compile process which **can** later be retrieved by calling vkGetPipelineExecutableStatisticsKHR. Enabling this flag **must** not affect the final compiled pipeline but **may** disable pipeline caching or otherwise affect pipeline creation time.
- VK_PIPELINE_CREATE_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR specifies that the shader compiler should capture the internal representations of pipeline executables produced by the compile process which **can** later be retrieved by calling vkGetPipelineExecutableInternalRepresentationsKHR. Enabling this flag **must** not affect the final compiled pipeline but **may** disable pipeline caching or otherwise affect pipeline creation time. When capturing IR from pipelines created with pipeline libraries, there is no guarantee that IR from libraries **can** be retrieved from the linked pipeline. Applications **should** retrieve IR from each library, and any linked pipelines, separately.
- VK_PIPELINE_CREATE_LIBRARY_BIT_KHR specifies that the pipeline **cannot** be used directly, and instead defines a *pipeline library* that **can** be combined with other pipelines using the VkPipelineLibraryCreateInfoKHR structure. This is available in ray tracing and graphics pipelines.
- VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR specifies that an any-hit shader will always be present when an any-hit shader would be executed. A NULL any-hit shader is an any-hit shader which is effectively VK_SHADER_UNUSED_KHR, such as from a shader group consisting entirely of zeros.
- VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR specifies that a closest hit shader will always be present when a closest hit shader would be executed. A NULL closest hit shader is a closest hit shader which is effectively VK_SHADER_UNUSED_KHR, such as from a shader group consisting entirely of zeros.
- VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR specifies that a miss shader will always be present when a miss shader would be executed. A NULL miss shader is a miss shader which is effectively VK_SHADER_UNUSED_KHR, such as from a shader group consisting entirely of zeros.
- VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR specifies that an intersection shader will always be present when an intersection shader would be executed. A NULL intersection shader is an intersection shader which is effectively VK_SHADER_UNUSED_KHR, such as from a shader group consisting entirely of zeros.
- VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR specifies that triangle primitives will be skipped during traversal using pipeline trace ray instructions.
- VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR specifies that AABB primitives will

be skipped during traversal using pipeline trace ray instructions.

- VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR specifies that the shader group handles **can** be saved and reused on a subsequent run (e.g. for trace capture and replay).
- VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT specifies that pipeline creation will fail if a compile is required for creation of a valid VkPipeline object; VK_PIPELINE_COMPILE_REQUIRED will be returned by pipeline creation, and the VkPipeline will be VK_NULL_HANDLE.
- When creating multiple pipelines, VK_PIPELINE_CREATE_EARLY_RETURN_ON_FAILURE_BIT specifies that control will be returned to the application if any individual pipeline returns a result which is not VK_SUCCESS rather than continuing to create additional pipelines.
- VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR specifies that the pipeline will be used with a fragment shading rate attachment and dynamic rendering.
- VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT specifies that the pipeline will be used with a fragment density map attachment and dynamic rendering.
- VK_PIPELINE_CREATE_LINK_TIME_OPTIMIZATION_BIT_EXT specifies that pipeline libraries being linked into this library **should** have link time optimizations applied. If this bit is omitted, implementations **should** instead perform linking as rapidly as possible.
- VK_PIPELINE_CREATE_RETAIN_LINK_TIME_OPTIMIZATION_INFO_BIT_EXT specifies that pipeline libraries should retain any information necessary to later perform an optimal link with VK_PIPELINE_CREATE_LINK_TIME_OPTIMIZATION_BIT_EXT.
- VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT specifies that a pipeline will be used with descriptor buffers, rather than descriptor sets.
- VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT specifies that the pipeline **may** be used with an attachment feedback loop including color attachments. It is ignored if VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT is set in pDynamicStates.
- VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT specifies that the pipeline **may** be used with an attachment feedback loop including depth-stencil attachments. It is ignored if VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT is set in pDynamicStates.
- VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR specifies that the ray tracing pipeline **can** be used with acceleration structures which reference an opacity micromap array. This flag has no effect on using opacity micromaps with ray queries.
- VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT specifies that the pipeline **must** not be bound to a protected command buffer.
- VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT specifies that the pipeline **must** not be bound to an unprotected command buffer.

It is valid to set both VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT and VK_PIPELINE_CREATE_DERIVATIVE_BIT. This allows a pipeline to be both a parent and possibly a

child in a pipeline hierarchy. See Pipeline Derivatives for more information.

When an implementation is looking up a pipeline in a pipeline cache, if that pipeline is being created using linked libraries, implementations **should** always return an equivalent pipeline created with VK_PIPELINE_CREATE_LINK_TIME_OPTIMIZATION_BIT_EXT if available, whether or not that bit was specified.

NOTE

Using VK_PIPELINE_CREATE_LINK_TIME_OPTIMIZATION_BIT_EXT (or not) when linking pipeline libraries is intended as a performance tradeoff between host and device. If the bit is omitted, linking should be faster and produce a pipeline more rapidly, but performance of the pipeline on the target device may be reduced. If the bit is included, linking may be slower but should produce a pipeline with device performance comparable to a monolithically created pipeline. Using both options can allow latency-sensitive applications to generate a suboptimal but usable pipeline quickly, and then perform an optimal link in the background, substituting the result for the suboptimally linked pipeline as soon as it is available.

WARNING

This functionality is superseded by VkPipelineCreateFlags2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineCreateFlags;
```

VkPipelineCreateFlags is a bitmask type for setting a mask of zero or more VkPipelineCreateFlagBits.

10.3. Compute Pipelines

Compute pipelines consist of a single static compute shader stage and the pipeline layout.

The compute pipeline represents a compute shader and is created by calling vkCreateComputePipelines with module and pName selecting an entry point from a shader module, where that entry point defines a valid compute shader, in the VkPipelineShaderStageCreateInfo structure contained within the VkComputePipelineCreateInfo structure.

To create compute pipelines, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateComputePipelines(
    VkDevice                                    device,
    VkPipelineCache                             pipelineCache,
    uint32_t                                    createInfoCount,
    const VkComputePipelineCreateInfo*          pCreateInfos,
    const VkAllocationCallbacks*                pAllocator,
    VkPipeline*                                 pPipelines);
```

- `device` is the logical device that creates the compute pipelines.
- `pipelineCache` is either VK_NULL_HANDLE, indicating that pipeline caching is disabled, or to enable caching, the handle of a valid VkPipelineCache object. The implementation **must** not access this object outside of the duration of this command.
- `createInfoCount` is the length of the `pCreateInfos` and `pPipelines` arrays.
- `pCreateInfos` is a pointer to an array of VkComputePipelineCreateInfo structures.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pPipelines` is a pointer to an array of VkPipeline handles in which the resulting compute pipeline objects are returned.

Pipelines are created and returned as described for Multiple Pipeline Creation.

Valid Usage

- VUID-vkCreateComputePipelines-device-09661
  `device` **must** support at least one queue family with the VK_QUEUE_COMPUTE_BIT capability
- VUID-vkCreateComputePipelines-flags-00695
  If the `flags` member of any element of `pCreateInfos` contains the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag, and the `basePipelineIndex` member of that same element is not `-1`, `basePipelineIndex` **must** be less than the index into `pCreateInfos` that corresponds to that element
- VUID-vkCreateComputePipelines-flags-00696
  If the `flags` member of any element of `pCreateInfos` contains the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag, the base pipeline **must** have been created with the VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT flag set
- VUID-vkCreateComputePipelines-pipelineCache-02873
  If `pipelineCache` was created with VK_PIPELINE_CACHE_CREATE_EXTERNALLY_SYNCHRONIZED_BIT, host access to `pipelineCache` **must** be externally synchronized
- VUID-vkCreateComputePipelines-pNext-09616
  If VkPipelineBinaryInfoKHR::`binaryCount` is not `0` for any element of `pCreateInfos`, `pipelineCache` **must** be VK_NULL_HANDLE
- VUID-vkCreateComputePipelines-pNext-09617
  If a VkPipelineCreateFlags2CreateInfoKHR structure with the VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR flag set is included in the `pNext` chain of any element of `pCreateInfos`, `pipelineCache` **must** be VK_NULL_HANDLE
- VUID-vkCreateComputePipelines-binaryCount-09620
  If VkPipelineBinaryInfoKHR::`binaryCount` is not `0` for any element of `pCreateInfos`, VK_PIPELINE_CREATION_FEEDBACK_APPLICATION_PIPELINE_CACHE_HIT_BIT **must** not be set in the `flags` of that element
- VUID-vkCreateComputePipelines-binaryCount-09621
  If VkPipelineBinaryInfoKHR::`binaryCount` is not `0` for any element of `pCreateInfos`,

VK_PIPELINE_CREATION_FEEDBACK_BASE_PIPELINE_ACCELERATION_BIT **must** not be set in the `flags` of that element

- VUID-vkCreateComputePipelines-binaryCount-09622
  If VkPipelineBinaryInfoKHR::`binaryCount` is not `0` for any element of `pCreateInfos`, VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT_EXT **must** not be set in the `flags` of that element
- VUID-vkCreateComputePipelines-pCreateInfos-11414
  If any element of `pCreateInfos` sets VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT and includes embedded sampler mappings, there **must** be less than (`maxSamplerAllocationCount` - (`minSamplerHeapReservedRangeWithEmbedded` / `samplerDescriptorSize`)) VkSampler objects currently created on the device
- VUID-vkCreateComputePipelines-pCreateInfos-11429
  If any element of `pCreateInfos` sets VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT and includes embedded sampler mappings, this command **must** not cause the total number of unique embedded samplers in pipelines and shaders on this device to exceed `maxDescriptorHeapEmbeddedSamplers`

Valid Usage (Implicit)

- VUID-vkCreateComputePipelines-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateComputePipelines-pipelineCache-parameter
  If `pipelineCache` is not VK_NULL_HANDLE, `pipelineCache` **must** be a valid VkPipelineCache handle
- VUID-vkCreateComputePipelines-pCreateInfos-parameter
  `pCreateInfos` **must** be a valid pointer to an array of `createInfoCount` valid VkComputePipelineCreateInfo structures
- VUID-vkCreateComputePipelines-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateComputePipelines-pPipelines-parameter
  `pPipelines` **must** be a valid pointer to an array of `createInfoCount` VkPipeline handles
- VUID-vkCreateComputePipelines-createInfoCount-arraylength
  `createInfoCount` **must** be greater than `0`
- VUID-vkCreateComputePipelines-pipelineCache-parent
  If `pipelineCache` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkComputePipelineCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkComputePipelineCreateInfo {
    VkStructureType                    sType;
    const void*                        pNext;
    VkPipelineCreateFlags              flags;
    VkPipelineShaderStageCreateInfo    stage;
    VkPipelineLayout                   layout;
    VkPipeline                         basePipelineHandle;
    int32_t                            basePipelineIndex;
} VkComputePipelineCreateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask of VkPipelineCreateFlagBits specifying how the pipeline will be generated.
- stage is a VkPipelineShaderStageCreateInfo structure describing the compute shader.
- layout is the description of binding locations used by both the pipeline and descriptor sets used with the pipeline. If VkPhysicalDeviceProperties::apiVersion is greater than or equal to Vulkan 1.3 or VK_KHR_maintenance4 is enabled layout **must** not be accessed by the implementation outside of the duration of the command this structure is passed to.
- basePipelineHandle is a pipeline to derive from.
- basePipelineIndex is an index into the pCreateInfos parameter to use as a pipeline to derive from.

The parameters basePipelineHandle and basePipelineIndex are described in more detail in Pipeline Derivatives.

If the pNext chain includes a VkPipelineCreateFlags2CreateInfo structure, VkPipelineCreateFlags2CreateInfo::flags from that structure is used instead of flags from this structure.

Valid Usage

- VUID-VkComputePipelineCreateInfo-None-09497
  If the pNext chain does not include a VkPipelineCreateFlags2CreateInfo structure, flags **must** be a valid combination of VkPipelineCreateFlagBits values

- VUID-VkComputePipelineCreateInfo-flags-07984
  If `flags` contains the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag, and `basePipelineIndex` is -1, `basePipelineHandle` **must** be a valid compute `VkPipeline` handle
- VUID-VkComputePipelineCreateInfo-flags-07985
  If `flags` contains the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag, and `basePipelineHandle` is VK_NULL_HANDLE, `basePipelineIndex` **must** be a valid index into the calling command's `pCreateInfos` parameter
- VUID-VkComputePipelineCreateInfo-flags-07986
  If `flags` contains the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag, `basePipelineIndex` **must** be -1 or `basePipelineHandle` **must** be VK_NULL_HANDLE
- VUID-VkComputePipelineCreateInfo-layout-07987
  If a push constant block is declared in a shader and `layout` is not VK_NULL_HANDLE, a push constant range in `layout` **must** match the shader stage
- VUID-VkComputePipelineCreateInfo-layout-10069
  If a push constant block is declared in a shader and `layout` is not VK_NULL_HANDLE, the block must be contained inside the push constant range in `layout` that matches the stage
- VUID-VkComputePipelineCreateInfo-layout-07988
  If a resource variable is declared in a shader and `layout` is not VK_NULL_HANDLE, the corresponding descriptor set in `layout` **must** match the shader stage
- VUID-VkComputePipelineCreateInfo-layout-07990
  If a resource variable is declared in a shader, `layout` is not VK_NULL_HANDLE, and the descriptor type is not VK_DESCRIPTOR_TYPE_MUTABLE_EXT, the corresponding descriptor set in `layout` **must** match the descriptor type
- VUID-VkComputePipelineCreateInfo-layout-07991
  If a resource variable is declared in a shader as an array and `layout` is not VK_NULL_HANDLE, the corresponding descriptor binding used to create `layout` **must** have a `descriptorCount` that is greater than or equal to the length of the array
- VUID-VkComputePipelineCreateInfo-None-10391
  If a resource variable is declared in a shader as an array of descriptors, then the descriptor type of that variable **must** not be VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK
- VUID-VkComputePipelineCreateInfo-flags-11798
  If the shader64BitIndexing feature is not enabled, `flags` **must** not contain VK_PIPELINE_CREATE_2_64_BIT_INDEXING_BIT_EXT
- VUID-VkComputePipelineCreateInfo-pipelineCreationCacheControl-02878
  If the `pipelineCreationCacheControl` feature is not enabled, `flags` **must** not include VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT nor VK_PIPELINE_CREATE_EARLY_RETURN_ON_FAILURE_BIT
- VUID-VkComputePipelineCreateInfo-pipelineProtectedAccess-07368
  If the `pipelineProtectedAccess` feature is not enabled, `flags` **must** not include VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT nor VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT
- VUID-VkComputePipelineCreateInfo-flags-07369

flags **must** not include both VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT and VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT

- VUID-VkComputePipelineCreateInfo-flags-11311
  If VkPipelineCreateFlags2CreateInfoKHR::flags includes VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, layout **must** be VK_NULL_HANDLE
- VUID-VkComputePipelineCreateInfo-flags-11312
  If VkPipelineCreateFlags2CreateInfoKHR::flags includes VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, all shader variables in the shader resource interface with a DescriptorSet and Binding decoration **must** have a mapping declared in VkShaderDescriptorSetAndBindingMappingInfoEXT::pMappings
- VUID-VkComputePipelineCreateInfo-flags-03365
  flags **must** not include VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR
- VUID-VkComputePipelineCreateInfo-flags-03366
  flags **must** not include VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR
- VUID-VkComputePipelineCreateInfo-flags-03367
  flags **must** not include VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR
- VUID-VkComputePipelineCreateInfo-flags-03368
  flags **must** not include VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR
- VUID-VkComputePipelineCreateInfo-flags-03369
  flags **must** not include VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR
- VUID-VkComputePipelineCreateInfo-flags-03370
  flags **must** not include VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR
- VUID-VkComputePipelineCreateInfo-flags-03576
  flags **must** not include VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR
- VUID-VkComputePipelineCreateInfo-flags-11007
  If flags includes VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT, then the VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT::deviceGeneratedCommands feature **must** be enabled
- VUID-VkComputePipelineCreateInfo-stage-00701
  The stage member of stage **must** be VK_SHADER_STAGE_COMPUTE_BIT
- VUID-VkComputePipelineCreateInfo-stage-00702
  The shader code for the entry point identified by stage and the rest of the state identified by this structure **must** adhere to the pipeline linking rules described in the Shader Interfaces chapter
- VUID-VkComputePipelineCreateInfo-layout-01687

If `layout` is not VK_NULL_HANDLE, the number of resources in `layout` accessible to the compute shader stage **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageResources`

- VUID-VkComputePipelineCreateInfo-shaderEnqueue-09177
  `flags` **must** not include VK_PIPELINE_CREATE_LIBRARY_BIT_KHR
- VUID-VkComputePipelineCreateInfo-pipelineStageCreationFeedbackCount-06566
  If VkPipelineCreationFeedbackCreateInfo::`pipelineStageCreationFeedbackCount` is not `0`, it **must** be `1`
- VUID-VkComputePipelineCreateInfo-flags-07367
  `flags` **must** not include VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR
- VUID-VkComputePipelineCreateInfo-flags-11593
  If the `VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap` feature is not enabled, `flags` **must** not include VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-VkComputePipelineCreateInfo-layout-12396
  If `layout` is not VK_NULL_HANDLE, it **must** not have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT
- VUID-VkComputePipelineCreateInfo-None-11367
  If VkPipelineCreateFlags2CreateInfoKHR::`flags` does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, `layout` **must** not be VK_NULL_HANDLE

Valid Usage (Implicit)

- VUID-VkComputePipelineCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COMPUTE_PIPELINE_CREATE_INFO
- VUID-VkComputePipelineCreateInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkPipelineBinaryInfoKHR, VkPipelineCreateFlags2CreateInfo, VkPipelineCreationFeedbackCreateInfo, VkPipelineRobustnessCreateInfo, or VkValidationFeaturesEXT
- VUID-VkComputePipelineCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkComputePipelineCreateInfo-stage-parameter
  `stage` **must** be a valid VkPipelineShaderStageCreateInfo structure
- VUID-VkComputePipelineCreateInfo-layout-parameter
  If `layout` is not VK_NULL_HANDLE, `layout` **must** be a valid VkPipelineLayout handle
- VUID-VkComputePipelineCreateInfo-commonparent
  Both of `basePipelineHandle`, and `layout` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Structure Chaining

Extends the structure

- VkPipelineCreateInfoKHR

The `VkPipelineShaderStageCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineShaderStageCreateInfo {
    VkStructureType                     sType;
    const void*                         pNext;
    VkPipelineShaderStageCreateFlags    flags;
    VkShaderStageFlagBits               stage;
    VkShaderModule                      module;
    const char*                         pName;
    const VkSpecializationInfo*         pSpecializationInfo;
} VkPipelineShaderStageCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkPipelineShaderStageCreateFlagBits specifying how the pipeline shader stage will be generated.
- `stage` is a VkShaderStageFlagBits value specifying a single pipeline stage.
- `module` is optionally a VkShaderModule object containing the shader code for this stage. The implementation **must** not access this object outside of the duration of the command this structure is passed to.
- `pName` is a pointer to a null-terminated UTF-8 string specifying the entry point name of the shader for this stage.
- `pSpecializationInfo` is a pointer to a VkSpecializationInfo structure, as described in Specialization Constants, or `NULL`.

If `module` is not VK_NULL_HANDLE, the shader code used by the pipeline is defined by `module`. If `module` is VK_NULL_HANDLE, the shader code is defined by the chained VkShaderModuleCreateInfo if present.

If the `shaderModuleIdentifier` feature is enabled, applications **can** omit shader code for `stage` and instead provide a module identifier. This is done by including a VkPipelineShaderStageModuleIdentifierCreateInfoEXT structure with `identifierSize` not equal to 0 in the `pNext` chain. A shader stage created in this way is equivalent to one created using a shader module with the same identifier. The identifier allows an implementation to look up a pipeline without consuming a valid SPIR-V module. If a pipeline is not found, pipeline compilation is not possible and the implementation **must** fail as specified by VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT.

When an identifier is used in lieu of a shader module, implementations **may** fail pipeline compilation with VK_PIPELINE_COMPILE_REQUIRED for any reason.

NOTE The rationale for the relaxed requirement on implementations to return a pipeline with VkPipelineShaderStageModuleIdentifierCreateInfoEXT is that layers or tools may intercept pipeline creation calls and require the full SPIR-V context to operate correctly. ICDs are not expected to fail pipeline compilation if the pipeline exists in a cache somewhere.

Applications **can** use identifiers when creating pipelines with VK_PIPELINE_CREATE_LIBRARY_BIT_KHR. When creating such pipelines, VK_SUCCESS **may** be returned, but subsequently fail when referencing the pipeline in a VkPipelineLibraryCreateInfoKHR struct. Applications **must** allow pipeline compilation to fail during link steps with VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT as it **may** not be possible to determine if a pipeline **can** be created from identifiers until the link step.

Valid Usage

- VUID-VkPipelineShaderStageCreateInfo-descriptorHeap-11314
  If the `descriptorHeap` feature is not enabled, VkShaderDescriptorSetAndBindingMappingInfoEXT::`mappingCount` **must** be 0
- VUID-VkPipelineShaderStageCreateInfo-pNext-11315
  If the `pNext` chain specifies a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, the mapped resource in the shader **must** be a variable with a structure type decorated with `Block` in the `Uniform Storage Class`
- VUID-VkPipelineShaderStageCreateInfo-pNext-11316
  If the `pNext` chain specifies a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, the mapped structure **must** not be larger than `maxPushDataSize` minus the `pushDataOffset` used in the mapping
- VUID-VkPipelineShaderStageCreateInfo-pNext-12454
  If the `pNext` chain specifies a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, and there is a variable in the shader that matches the mapping, the `stage` **must** be a ray tracing stage
- VUID-VkPipelineShaderStageCreateInfo-pNext-11317
  If the `pNext` chain specifies a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, the sum of mapped structure size and `shaderRecordDataOffset` used in the mapping **must** not be larger than `maxShaderGroupStride`
- VUID-VkPipelineShaderStageCreateInfo-pNext-11318
  If the `pNext` chain specifies a descriptor mapping using

VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the mapped resource in the shader **must** be one of:

  - A variable with a structure type decorated with `Block` in the `Uniform Storage Class`
  - A variable with a structure type decorated with `BufferBlock` in the `Uniform Storage Class`
  - A variable with a structure type decorated with `Block` in the `StorageBuffer Storage Class`
  - A `OpTypeAccelerationStructureKHR` variable

- VUID-VkPipelineShaderStageCreateInfo-pNext-11378
  If the `pNext` chain specifies a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the `OpArrayLength` or `OpUntypedArrayLengthKHR` instruction **must** not be used on that resource
- VUID-VkPipelineShaderStageCreateInfo-pNext-11399
  If the `pNext` chain specifies a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_CONSTANT_OFFSET_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, and the mapped resource declaration is an array, the `pEmbeddedSampler` member of the corresponding mapping structure **must** be `NULL`
- VUID-VkPipelineShaderStageCreateInfo-stage-00704
  If the `geometryShader` feature is not enabled, `stage` **must** not be VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-VkPipelineShaderStageCreateInfo-stage-00705
  If the `tessellationShader` feature is not enabled, `stage` **must** not be VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT or VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-VkPipelineShaderStageCreateInfo-stage-02091
  If the `meshShader` feature is not enabled, `stage` **must** not be VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-VkPipelineShaderStageCreateInfo-stage-02092
  If the `taskShader` feature is not enabled, `stage` **must** not be VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-VkPipelineShaderStageCreateInfo-stage-00706
  `stage` **must** not be VK_SHADER_STAGE_ALL_GRAPHICS, or VK_SHADER_STAGE_ALL
- VUID-VkPipelineShaderStageCreateInfo-pName-00707
  `pName` **must** be the name of an `OpEntryPoint` in `module` with an execution model that matches `stage`

- VUID-VkPipelineShaderStageCreateInfo-maxSampleMaskWords-00711
  If the identified entry point includes any variable in its interface that is declared with the `SampleMask BuiltIn` decoration, that variable **must** not have an array size greater than `VkPhysicalDeviceLimits`::`maxSampleMaskWords`
- VUID-VkPipelineShaderStageCreateInfo-stage-00713
  If `stage` is VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT or VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, and the identified entry point has an `OpExecutionMode` instruction specifying a patch size with `OutputVertices`, the patch size **must** be greater than `0` and less than or equal to `VkPhysicalDeviceLimits`::`maxTessellationPatchSize`
- VUID-VkPipelineShaderStageCreateInfo-stage-00714
  If `stage` is VK_SHADER_STAGE_GEOMETRY_BIT, the identified entry point **must** have an `OpExecutionMode` instruction specifying a maximum output vertex count that is greater than `0` and less than or equal to `VkPhysicalDeviceLimits`::`maxGeometryOutputVertices`
- VUID-VkPipelineShaderStageCreateInfo-stage-00715
  If `stage` is VK_SHADER_STAGE_GEOMETRY_BIT, the identified entry point **must** have an `OpExecutionMode` instruction specifying an invocation count that is greater than `0` and less than or equal to `VkPhysicalDeviceLimits`::`maxGeometryShaderInvocations`
- VUID-VkPipelineShaderStageCreateInfo-stage-02596
  If `stage` is either VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, or VK_SHADER_STAGE_GEOMETRY_BIT, and the identified entry point writes to `Layer` for any primitive, it **must** write the same value to `Layer` for all vertices of a given primitive
- VUID-VkPipelineShaderStageCreateInfo-stage-02597
  If `stage` is either VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, or VK_SHADER_STAGE_GEOMETRY_BIT, and the identified entry point writes to `ViewportIndex` for any primitive, it **must** write the same value to `ViewportIndex` for all vertices of a given primitive
- VUID-VkPipelineShaderStageCreateInfo-stage-06685
  If `stage` is VK_SHADER_STAGE_FRAGMENT_BIT, and the identified entry point writes to `FragDepth` in any execution path, all execution paths that are not exclusive to helper invocations **must** either discard the fragment, or write or initialize the value of `FragDepth`
- VUID-VkPipelineShaderStageCreateInfo-stage-06686
  If `stage` is VK_SHADER_STAGE_FRAGMENT_BIT, and the identified entry point writes to `FragStencilRefEXT` in any execution path, all execution paths that are not exclusive to helper invocations **must** either discard the fragment, or write or initialize the value of `FragStencilRefEXT`
- VUID-VkPipelineShaderStageCreateInfo-flags-02784
  If `flags` has the VK_PIPELINE_SHADER_STAGE_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT flag set, the `subgroupSizeControl` feature **must** be enabled

- VUID-VkPipelineShaderStageCreateInfo-flags-02785
  If `flags` has the VK_PIPELINE_SHADER_STAGE_CREATE_REQUIRE_FULL_SUBGROUPS_BIT flag set, the `computeFullSubgroups` feature **must** be enabled
- VUID-VkPipelineShaderStageCreateInfo-flags-08988
  If `flags` includes VK_PIPELINE_SHADER_STAGE_CREATE_REQUIRE_FULL_SUBGROUPS_BIT, `stage` **must** be one of VK_SHADER_STAGE_MESH_BIT_EXT, VK_SHADER_STAGE_TASK_BIT_EXT, or VK_SHADER_STAGE_COMPUTE_BIT
- VUID-VkPipelineShaderStageCreateInfo-pNext-02754
  If a VkPipelineShaderStageRequiredSubgroupSizeCreateInfo structure is included in the `pNext` chain, `flags` **must** not have the VK_PIPELINE_SHADER_STAGE_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT flag set
- VUID-VkPipelineShaderStageCreateInfo-pNext-02755
  If a VkPipelineShaderStageRequiredSubgroupSizeCreateInfo structure is included in the `pNext` chain, the `subgroupSizeControl` feature **must** be enabled, and `stage` **must** be a valid bit specified in `requiredSubgroupSizeStages`
- VUID-VkPipelineShaderStageCreateInfo-pNext-02756
  If a VkPipelineShaderStageRequiredSubgroupSizeCreateInfo structure is included in the `pNext` chain and `stage` is VK_SHADER_STAGE_COMPUTE_BIT, VK_SHADER_STAGE_MESH_BIT_EXT, or VK_SHADER_STAGE_TASK_BIT_EXT, the local workgroup size of the shader **must** be less than or equal to the product of VkPipelineShaderStageRequiredSubgroupSizeCreateInfo::`requiredSubgroupSize` and `maxComputeWorkgroupSubgroups`
- VUID-VkPipelineShaderStageCreateInfo-pNext-02757
  If a VkPipelineShaderStageRequiredSubgroupSizeCreateInfo structure is included in the `pNext` chain, and `flags` has the VK_PIPELINE_SHADER_STAGE_CREATE_REQUIRE_FULL_SUBGROUPS_BIT flag set, the local workgroup size in the X dimension of the pipeline **must** be a multiple of VkPipelineShaderStageRequiredSubgroupSizeCreateInfo::`requiredSubgroupSize`
- VUID-VkPipelineShaderStageCreateInfo-flags-02758
  If `flags` has both the VK_PIPELINE_SHADER_STAGE_CREATE_REQUIRE_FULL_SUBGROUPS_BIT and VK_PIPELINE_SHADER_STAGE_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT flags set, the local workgroup size in the X dimension of the pipeline **must** be a multiple of `maxSubgroupSize`
- VUID-VkPipelineShaderStageCreateInfo-flags-02759
  If `flags` has the VK_PIPELINE_SHADER_STAGE_CREATE_REQUIRE_FULL_SUBGROUPS_BIT flag set and `flags` does not have the VK_PIPELINE_SHADER_STAGE_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT flag set and no VkPipelineShaderStageRequiredSubgroupSizeCreateInfo structure is included in the `pNext` chain, the local workgroup size in the X dimension of the pipeline **must** be a multiple of `subgroupSize`
- VUID-VkPipelineShaderStageCreateInfo-module-08987
  If `module` uses the `OpTypeCooperativeMatrixKHR` instruction with a `Scope` equal to `Subgroup`, then the local workgroup size in the X dimension of the pipeline **must** be a multiple of the

effective subgroup size

- VUID-VkPipelineShaderStageCreateInfo-stage-08771
  If a shader module identifier is not specified for this `stage`, `module` **must** be a valid VkShaderModule , or the `pNext` chain of the parent `Vk*CreateInfo` structure **must** set VkPipelineBinaryInfoKHR::`binaryCount` to a value greater than `0`, if none of the following features are enabled:
  - `graphicsPipelineLibrary`
  - `maintenance5`
- VUID-VkPipelineShaderStageCreateInfo-stage-06845
  If a shader module identifier is not specified for this `stage`, `module` **must** be a valid VkShaderModule, or there **must** be a valid VkShaderModuleCreateInfo structure in the `pNext` chain , or the `pNext` chain of the parent `Vk*CreateInfo` structure **must** set VkPipelineBinaryInfoKHR::`binaryCount` to a value greater than `0`,
- VUID-VkPipelineShaderStageCreateInfo-stage-06844
  If a shader module identifier is specified for this `stage`, the `pNext` chain **must** not include a VkShaderModuleCreateInfo structure
- VUID-VkPipelineShaderStageCreateInfo-stage-06848
  If a shader module identifier is specified for this `stage`, `module` **must** be VK_NULL_HANDLE
- VUID-VkPipelineShaderStageCreateInfo-pSpecializationInfo-06849
  If a shader module identifier is not specified, the shader code used by the pipeline **must** be valid as described by the Khronos SPIR-V Specification after applying the specializations provided in `pSpecializationInfo`, if any, and then converting all specialization constants into fixed constants

Valid Usage (Implicit)

- VUID-VkPipelineShaderStageCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_SHADER_STAGE_CREATE_INFO
- VUID-VkPipelineShaderStageCreateInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkDebugUtilsObjectNameInfoEXT, VkPipelineRobustnessCreateInfo, VkPipelineShaderStageModuleIdentifierCreateInfoEXT, VkPipelineShaderStageRequiredSubgroupSizeCreateInfo, VkShaderDescriptorSetAndBindingMappingInfoEXT, VkShaderModuleCreateInfo, VkShaderModuleValidationCacheCreateInfoEXT, or VkValidationFeaturesEXT
- VUID-VkPipelineShaderStageCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkPipelineShaderStageCreateInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkPipelineShaderStageCreateFlagBits values
- VUID-VkPipelineShaderStageCreateInfo-stage-parameter
  `stage` **must** be a valid VkShaderStageFlagBits value

- VUID-VkPipelineShaderStageCreateInfo-module-parameter
  If `module` is not VK_NULL_HANDLE, `module` **must** be a valid VkShaderModule handle
- VUID-VkPipelineShaderStageCreateInfo-pName-parameter
  `pName` **must** be a null-terminated UTF-8 string
- VUID-VkPipelineShaderStageCreateInfo-pSpecializationInfo-parameter
  If `pSpecializationInfo` is not `NULL`, `pSpecializationInfo` **must** be a valid pointer to a valid VkSpecializationInfo structure

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineShaderStageCreateFlags;
```

`VkPipelineShaderStageCreateFlags` is a bitmask type for setting a mask of zero or more VkPipelineShaderStageCreateFlagBits.

Possible values of the `flags` member of VkPipelineShaderStageCreateInfo specifying how a pipeline shader stage is created, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPipelineShaderStageCreateFlagBits {
  // Provided by VK_VERSION_1_3
    VK_PIPELINE_SHADER_STAGE_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT = 0x00000001,
  // Provided by VK_VERSION_1_3
    VK_PIPELINE_SHADER_STAGE_CREATE_REQUIRE_FULL_SUBGROUPS_BIT = 0x00000002,
} VkPipelineShaderStageCreateFlagBits;
```

- VK_PIPELINE_SHADER_STAGE_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT specifies that the `SubgroupSize` **may** vary in the shader stage.
- VK_PIPELINE_SHADER_STAGE_CREATE_REQUIRE_FULL_SUBGROUPS_BIT specifies that the subgroup sizes **must** be launched with all invocations active in the task, mesh, or compute stage.

NOTE

If VK_PIPELINE_SHADER_STAGE_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT_EXT and VK_PIPELINE_SHADER_STAGE_CREATE_REQUIRE_FULL_SUBGROUPS_BIT_EXT are specified and `minSubgroupSize` does not equal `maxSubgroupSize` and no required subgroup size is specified, then the only way to guarantee that the 'X' dimension of the local workgroup size is a multiple of `SubgroupSize` is to make it a multiple of `maxSubgroupSize`. Under these conditions, you are guaranteed full subgroups but not any particular subgroup size.

Bits which **can** be set by commands and structures, specifying one or more shader stages, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkShaderStageFlagBits {
    VK_SHADER_STAGE_VERTEX_BIT = 0x00000001,
    VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT = 0x00000002,
    VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT = 0x00000004,
    VK_SHADER_STAGE_GEOMETRY_BIT = 0x00000008,
    VK_SHADER_STAGE_FRAGMENT_BIT = 0x00000010,
    VK_SHADER_STAGE_COMPUTE_BIT = 0x00000020,
    VK_SHADER_STAGE_ALL_GRAPHICS = 0x0000001F,
    VK_SHADER_STAGE_ALL = 0x7FFFFFFF,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_SHADER_STAGE_RAYGEN_BIT_KHR = 0x00000100,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_SHADER_STAGE_ANY_HIT_BIT_KHR = 0x00000200,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_SHADER_STAGE_CLOSEST_HIT_BIT_KHR = 0x00000400,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_SHADER_STAGE_MISS_BIT_KHR = 0x00000800,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_SHADER_STAGE_INTERSECTION_BIT_KHR = 0x00001000,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_SHADER_STAGE_CALLABLE_BIT_KHR = 0x00002000,
  // Provided by VK_EXT_mesh_shader
    VK_SHADER_STAGE_TASK_BIT_EXT = 0x00000040,
  // Provided by VK_EXT_mesh_shader
    VK_SHADER_STAGE_MESH_BIT_EXT = 0x00000080,
} VkShaderStageFlagBits;
```

- VK_SHADER_STAGE_VERTEX_BIT specifies the vertex stage.
- VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT specifies the tessellation control stage.
- VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT specifies the tessellation evaluation stage.
- VK_SHADER_STAGE_GEOMETRY_BIT specifies the geometry stage.
- VK_SHADER_STAGE_FRAGMENT_BIT specifies the fragment stage.
- VK_SHADER_STAGE_COMPUTE_BIT specifies the compute stage.
- VK_SHADER_STAGE_ALL_GRAPHICS is a combination of bits used as shorthand to specify all graphics stages defined above (excluding the compute stage).
- VK_SHADER_STAGE_ALL is a combination of bits used as shorthand to specify all shader stages supported by the device, including all additional stages which are introduced by extensions.
- VK_SHADER_STAGE_TASK_BIT_EXT specifies the task stage.
- VK_SHADER_STAGE_MESH_BIT_EXT specifies the mesh stage.
- VK_SHADER_STAGE_RAYGEN_BIT_KHR specifies the ray generation stage.
- VK_SHADER_STAGE_ANY_HIT_BIT_KHR specifies the any-hit stage.
- VK_SHADER_STAGE_CLOSEST_HIT_BIT_KHR specifies the closest hit stage.

- VK_SHADER_STAGE_MISS_BIT_KHR specifies the miss stage.
- VK_SHADER_STAGE_INTERSECTION_BIT_KHR specifies the intersection stage.
- VK_SHADER_STAGE_CALLABLE_BIT_KHR specifies the callable stage.

NOTE	VK_SHADER_STAGE_ALL_GRAPHICS only includes the original five graphics stages included in Vulkan 1.0, and not any stages added by extensions. Thus, it may not have the desired effect in all cases.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkShaderStageFlags;
```

VkShaderStageFlags is a bitmask type for setting a mask of zero or more VkShaderStageFlagBits.

The VkPipelineShaderStageRequiredSubgroupSizeCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPipelineShaderStageRequiredSubgroupSizeCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           requiredSubgroupSize;
} VkPipelineShaderStageRequiredSubgroupSizeCreateInfo;
```

```
// Provided by VK_EXT_shader_object
// Equivalent to VkPipelineShaderStageRequiredSubgroupSizeCreateInfo
typedef VkPipelineShaderStageRequiredSubgroupSizeCreateInfo
VkShaderRequiredSubgroupSizeCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- requiredSubgroupSize is an unsigned integer value specifying the required subgroup size for the newly created pipeline shader stage.

If a VkPipelineShaderStageRequiredSubgroupSizeCreateInfo structure is included in the pNext chain of VkPipelineShaderStageCreateInfo, it specifies that the pipeline shader stage being compiled has a required subgroup size.

If a VkShaderRequiredSubgroupSizeCreateInfoEXT structure is included in the pNext chain of VkShaderCreateInfoEXT, it specifies that the shader being compiled has a required subgroup size.

Valid Usage

- VUID-VkPipelineShaderStageRequiredSubgroupSizeCreateInfo-requiredSubgroupSize-02760
  requiredSubgroupSize **must** be a power-of-two integer

- VUID-VkPipelineShaderStageRequiredSubgroupSizeCreateInfo-requiredSubgroupSize-02761
  `requiredSubgroupSize` **must** be greater than or equal to `minSubgroupSize`
- VUID-VkPipelineShaderStageRequiredSubgroupSizeCreateInfo-requiredSubgroupSize-02762
  `requiredSubgroupSize` **must** be less than or equal to `maxSubgroupSize`

Valid Usage (Implicit)

- VUID-VkPipelineShaderStageRequiredSubgroupSizeCreateInfo-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_SHADER_STAGE_REQUIRED_SUBGROUP_SIZE_CREATE_INFO`

Structure Chaining

Extends the structures

- VkPipelineShaderStageCreateInfo
- VkShaderCreateInfoEXT

The `VkPipelineRobustnessCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPipelineRobustnessCreateInfo {
    VkStructureType                         sType;
    const void*                             pNext;
    VkPipelineRobustnessBufferBehavior      storageBuffers;
    VkPipelineRobustnessBufferBehavior      uniformBuffers;
    VkPipelineRobustnessBufferBehavior      vertexInputs;
    VkPipelineRobustnessImageBehavior       images;
} VkPipelineRobustnessCreateInfo;
```

```
// Provided by VK_EXT_pipeline_robustness
// Equivalent to VkPipelineRobustnessCreateInfo
typedef VkPipelineRobustnessCreateInfo VkPipelineRobustnessCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `storageBuffers` sets the behavior of out of bounds accesses made to resources bound as:
  - VK_DESCRIPTOR_TYPE_STORAGE_BUFFER
  - VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER

    - VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC
- `uniformBuffers` describes the behavior of out of bounds accesses made to resources bound as:
    - VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER
    - VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER
    - VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC
    - VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK
- `vertexInputs` describes the behavior of out of bounds accesses made to vertex input attributes
- `images` describes the behavior of out of bounds accesses made to resources bound as:
    - VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE
    - VK_DESCRIPTOR_TYPE_STORAGE_IMAGE

Resources bound as VK_DESCRIPTOR_TYPE_MUTABLE_EXT will have the robustness behavior that covers its active descriptor type.

The scope of the effect of `VkPipelineRobustnessCreateInfo` depends on which structure's `pNext` chain it is included in.

- `VkGraphicsPipelineCreateInfo`, `VkRayTracingPipelineCreateInfoKHR`, `VkComputePipelineCreateInfo`:
  The robustness behavior described by `VkPipelineRobustnessCreateInfo` applies to all accesses through this pipeline
- `VkPipelineShaderStageCreateInfo`:
  The robustness behavior described by `VkPipelineRobustnessCreateInfo` applies to all accesses emanating from the shader code of this shader stage

If `VkPipelineRobustnessCreateInfo` is specified for both a pipeline and a pipeline stage, the `VkPipelineRobustnessCreateInfo` specified for the pipeline stage will take precedence.

When `VkPipelineRobustnessCreateInfo` is specified for a pipeline, it only affects the subset of the pipeline that is specified by the create info, as opposed to subsets linked from pipeline libraries. For VkGraphicsPipelineCreateInfo, that subset is specified by VkGraphicsPipelineLibraryCreateInfoEXT::`flags`. For VkRayTracingPipelineCreateInfoKHR, that subset is specified by the specific stages in VkRayTracingPipelineCreateInfoKHR::`pStages`.

Valid Usage

- VUID-VkPipelineRobustnessCreateInfo-pipelineRobustness-06926
  If the `pipelineRobustness` feature is not enabled, `storageBuffers` **must** be VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_DEVICE_DEFAULT
- VUID-VkPipelineRobustnessCreateInfo-pipelineRobustness-06927
  If the `pipelineRobustness` feature is not enabled, `uniformBuffers` **must** be VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_DEVICE_DEFAULT
- VUID-VkPipelineRobustnessCreateInfo-pipelineRobustness-06928
  If the `pipelineRobustness` feature is not enabled, `vertexInputs` **must** be VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_DEVICE_DEFAULT

- VUID-VkPipelineRobustnessCreateInfo-pipelineRobustness-06929
  If the `pipelineRobustness` feature is not enabled, `images` **must** be VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_DEVICE_DEFAULT
- VUID-VkPipelineRobustnessCreateInfo-robustImageAccess-06930
  If the `robustImageAccess` feature is not supported, `images` **must** not be VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS
- VUID-VkPipelineRobustnessCreateInfo-robustBufferAccess2-06931
  If the `robustBufferAccess2` feature is not supported, `storageBuffers` **must** not be VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2
- VUID-VkPipelineRobustnessCreateInfo-robustBufferAccess2-06932
  If the `robustBufferAccess2` feature is not supported, `uniformBuffers` **must** not be VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2
- VUID-VkPipelineRobustnessCreateInfo-robustBufferAccess2-06933
  If the `robustBufferAccess2` feature is not supported, `vertexInputs` **must** not be VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2
- VUID-VkPipelineRobustnessCreateInfo-robustImageAccess2-06934
  If the `robustImageAccess2` feature is not supported, `images` **must** not be VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS_2
- VUID-VkPipelineRobustnessCreateInfo-storageBuffers-10636
  If `storageBuffers` is VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2, and either the `descriptorBindingStorageBufferUpdateAfterBind` feature or the `descriptorBindingStorageTexelBufferUpdateAfterBind` feature is enabled on the device, `robustBufferAccessUpdateAfterBind` **must** be VK_TRUE
- VUID-VkPipelineRobustnessCreateInfo-uniformBuffers-10637
  If `uniformBuffers` is VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2, and either the `descriptorBindingInlineUniformBlockUpdateAfterBind` feature, the `descriptorBindingUniformBufferUpdateAfterBind` feature, or the `descriptorBindingUniformTexelBufferUpdateAfterBind` feature is enabled on the device, `robustBufferAccessUpdateAfterBind` **must** be VK_TRUE
- VUID-VkPipelineRobustnessCreateInfo-images-10638
  If `images` is VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS or VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS_2, and either the `descriptorBindingStorageImageUpdateAfterBind` feature or the `descriptorBindingSampledImageUpdateAfterBind` feature is enabled on the device, `robustBufferAccessUpdateAfterBind` **must** be VK_TRUE

Valid Usage (Implicit)

- VUID-VkPipelineRobustnessCreateInfo-sType-sType

`sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_ROBUSTNESS_CREATE_INFO

- VUID-VkPipelineRobustnessCreateInfo-storageBuffers-parameter
  `storageBuffers` **must** be a valid VkPipelineRobustnessBufferBehavior value
- VUID-VkPipelineRobustnessCreateInfo-uniformBuffers-parameter
  `uniformBuffers` **must** be a valid VkPipelineRobustnessBufferBehavior value
- VUID-VkPipelineRobustnessCreateInfo-vertexInputs-parameter
  `vertexInputs` **must** be a valid VkPipelineRobustnessBufferBehavior value
- VUID-VkPipelineRobustnessCreateInfo-images-parameter
  `images` **must** be a valid VkPipelineRobustnessImageBehavior value

Structure Chaining

Extends the structures

- VkComputePipelineCreateInfo
- VkGraphicsPipelineCreateInfo
- VkPipelineShaderStageCreateInfo
- VkRayTracingPipelineCreateInfoKHR

Possible values of the `storageBuffers`, `uniformBuffers`, and `vertexInputs` members of VkPipelineRobustnessCreateInfo are:

```
// Provided by VK_VERSION_1_4
typedef enum VkPipelineRobustnessBufferBehavior {
    VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_DEVICE_DEFAULT = 0,
    VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_DISABLED = 1,
    VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS = 2,
    VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 = 3,
  // Provided by VK_EXT_pipeline_robustness
    VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_DEVICE_DEFAULT_EXT =
VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_DEVICE_DEFAULT,
  // Provided by VK_EXT_pipeline_robustness
    VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_DISABLED_EXT =
VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_DISABLED,
  // Provided by VK_EXT_pipeline_robustness
    VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_EXT =
VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS,
  // Provided by VK_EXT_pipeline_robustness
    VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2_EXT =
VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2,
} VkPipelineRobustnessBufferBehavior;
```

```
// Provided by VK_EXT_pipeline_robustness
// Equivalent to VkPipelineRobustnessBufferBehavior
typedef VkPipelineRobustnessBufferBehavior VkPipelineRobustnessBufferBehaviorEXT;
```

- VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_DEVICE_DEFAULT specifies that out of bounds buffer accesses follow the behavior of robust buffer access features enabled for the device.
- VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_DISABLED specifies that buffer accesses **must** not be out of bounds.
- VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS specifies that buffer accesses conform to Robust Buffer Access guarantees.
- VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 specifies that buffer accesses conform to Robust Buffer Access 2 guarantees.

Possible values of the `images` member of VkPipelineRobustnessCreateInfo are:

```
// Provided by VK_VERSION_1_4
typedef enum VkPipelineRobustnessImageBehavior {
    VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_DEVICE_DEFAULT = 0,
    VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_DISABLED = 1,
    VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS = 2,
    VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS_2 = 3,
  // Provided by VK_EXT_pipeline_robustness
    VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_DEVICE_DEFAULT_EXT =
VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_DEVICE_DEFAULT,
  // Provided by VK_EXT_pipeline_robustness
    VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_DISABLED_EXT =
VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_DISABLED,
  // Provided by VK_EXT_pipeline_robustness
    VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS_EXT =
VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS,
  // Provided by VK_EXT_pipeline_robustness
    VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS_2_EXT =
VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS_2,
} VkPipelineRobustnessImageBehavior;
```

```
// Provided by VK_EXT_pipeline_robustness
// Equivalent to VkPipelineRobustnessImageBehavior
typedef VkPipelineRobustnessImageBehavior VkPipelineRobustnessImageBehaviorEXT;
```

- VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_DEVICE_DEFAULT specifies that out of bounds image accesses follow the behavior of robust image access features enabled for the device.
- VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_DISABLED specifies that image accesses **must** not be out of bounds.
- VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS specifies that image

accesses conform to Robust Image Access guarantees.

- VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS_2 specifies that image accesses conform to Robust Image Access 2 guarantees.

An identifier **can** be provided instead of shader code in an attempt to compile pipelines without providing complete SPIR-V to the implementation.

The VkPipelineShaderStageModuleIdentifierCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_shader_module_identifier
typedef struct VkPipelineShaderStageModuleIdentifierCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           identifierSize;
    const uint8_t*     pIdentifier;
} VkPipelineShaderStageModuleIdentifierCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- identifierSize is the size, in bytes, of the buffer pointed to by pIdentifier.
- pIdentifier is a pointer to a buffer of opaque data specifying an identifier.

Any identifier **can** be used. If the pipeline being created with identifier requires compilation to complete the pipeline creation call, pipeline compilation **must** fail as defined by VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT.

pIdentifier and identifierSize **can** be obtained from an VkShaderModuleIdentifierEXT queried earlier.

Valid Usage

- VUID-VkPipelineShaderStageModuleIdentifierCreateInfoEXT-pNext-06850
  If this structure is included in a pNext chain and identifierSize is not equal to 0, the shaderModuleIdentifier feature **must** be enabled
- VUID-VkPipelineShaderStageModuleIdentifierCreateInfoEXT-pNext-06851
  If this structure is included in a pNext chain of VkPipelineShaderStageCreateInfo and identifierSize is not equal to 0, the pipeline **must** be created with the VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT flag set
- VUID-VkPipelineShaderStageModuleIdentifierCreateInfoEXT-identifierSize-06852
  identifierSize **must** be less-or-equal to VK_MAX_SHADER_MODULE_IDENTIFIER_SIZE_EXT

Valid Usage (Implicit)

- VUID-VkPipelineShaderStageModuleIdentifierCreateInfoEXT-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_SHADER_STAGE_MODULE_IDENTIFIER_CREATE_INFO_EXT`
- VUID-VkPipelineShaderStageModuleIdentifierCreateInfoEXT-pIdentifier-parameter
  If `identifierSize` is not `0`, `pIdentifier` **must** be a valid pointer to an array of `identifierSize` `uint8_t` values

Structure Chaining

Extends the structure

- VkPipelineShaderStageCreateInfo

10.4. Graphics Pipelines

Graphics pipelines consist of multiple shader stages, multiple fixed-function pipeline stages, and a pipeline layout.

To create graphics pipelines, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateGraphicsPipelines(
    VkDevice                                    device,
    VkPipelineCache                             pipelineCache,
    uint32_t                                    createInfoCount,
    const VkGraphicsPipelineCreateInfo*         pCreateInfos,
    const VkAllocationCallbacks*                pAllocator,
    VkPipeline*                                 pPipelines);
```

- `device` is the logical device that creates the graphics pipelines.
- `pipelineCache` is either VK_NULL_HANDLE, indicating that pipeline caching is disabled, or to enable caching, the handle of a valid VkPipelineCache object. The implementation **must** not access this object outside of the duration of this command.
- `createInfoCount` is the length of the `pCreateInfos` and `pPipelines` arrays.
- `pCreateInfos` is a pointer to an array of VkGraphicsPipelineCreateInfo structures.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pPipelines` is a pointer to an array of VkPipeline handles in which the resulting graphics pipeline objects are returned.

The VkGraphicsPipelineCreateInfo structure includes an array of VkPipelineShaderStageCreateInfo

structures for each of the desired active shader stages, as well as creation information for all relevant fixed-function stages, and a pipeline layout.

Pipelines are created and returned as described for Multiple Pipeline Creation.

Valid Usage

- VUID-vkCreateGraphicsPipelines-device-09662
  `device` **must** support at least one queue family with the VK_QUEUE_GRAPHICS_BIT capability
- VUID-vkCreateGraphicsPipelines-flags-00720
  If the `flags` member of any element of `pCreateInfos` contains the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag, and the `basePipelineIndex` member of that same element is not `-1`, `basePipelineIndex` **must** be less than the index into `pCreateInfos` that corresponds to that element
- VUID-vkCreateGraphicsPipelines-flags-00721
  If the `flags` member of any element of `pCreateInfos` contains the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag, the base pipeline **must** have been created with the VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT flag set
- VUID-vkCreateGraphicsPipelines-pipelineCache-02876
  If `pipelineCache` was created with VK_PIPELINE_CACHE_CREATE_EXTERNALLY_SYNCHRONIZED_BIT, host access to `pipelineCache` **must** be externally synchronized
- VUID-vkCreateGraphicsPipelines-pNext-09616
  If VkPipelineBinaryInfoKHR::`binaryCount` is not `0` for any element of `pCreateInfos`, `pipelineCache` **must** be VK_NULL_HANDLE
- VUID-vkCreateGraphicsPipelines-pNext-09617
  If a VkPipelineCreateFlags2CreateInfoKHR structure with the VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR flag set is included in the `pNext` chain of any element of `pCreateInfos`, `pipelineCache` **must** be VK_NULL_HANDLE
- VUID-vkCreateGraphicsPipelines-binaryCount-09620
  If VkPipelineBinaryInfoKHR::`binaryCount` is not `0` for any element of `pCreateInfos`, VK_PIPELINE_CREATION_FEEDBACK_APPLICATION_PIPELINE_CACHE_HIT_BIT **must** not be set in the `flags` of that element
- VUID-vkCreateGraphicsPipelines-binaryCount-09621
  If VkPipelineBinaryInfoKHR::`binaryCount` is not `0` for any element of `pCreateInfos`, VK_PIPELINE_CREATION_FEEDBACK_BASE_PIPELINE_ACCELERATION_BIT **must** not be set in the `flags` of that element
- VUID-vkCreateGraphicsPipelines-binaryCount-09622
  If VkPipelineBinaryInfoKHR::`binaryCount` is not `0` for any element of `pCreateInfos`, VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT_EXT **must** not be set in the `flags` of that element
- VUID-vkCreateGraphicsPipelines-pCreateInfos-11414
  If any element of `pCreateInfos` sets VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT

and includes embedded sampler mappings, there **must** be less than (`maxSamplerAllocationCount` - (`minSamplerHeapReservedRangeWithEmbedded` / `samplerDescriptorSize`)) VkSampler objects currently created on the device

- VUID-vkCreateGraphicsPipelines-pCreateInfos-11429
  If any element of `pCreateInfos` sets VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT and includes embedded sampler mappings, this command **must** not cause the total number of unique embedded samplers in pipelines and shaders on this device to exceed `maxDescriptorHeapEmbeddedSamplers`

NOTE

An implicit cache may be provided by the implementation or a layer. For this reason, it is still valid to set VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT on `flags` for any element of `pCreateInfos` while passing VK_NULL_HANDLE for `pipelineCache`.

Valid Usage (Implicit)

- VUID-vkCreateGraphicsPipelines-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateGraphicsPipelines-pipelineCache-parameter
  If `pipelineCache` is not VK_NULL_HANDLE, `pipelineCache` **must** be a valid VkPipelineCache handle
- VUID-vkCreateGraphicsPipelines-pCreateInfos-parameter
  `pCreateInfos` **must** be a valid pointer to an array of `createInfoCount` valid VkGraphicsPipelineCreateInfo structures
- VUID-vkCreateGraphicsPipelines-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateGraphicsPipelines-pPipelines-parameter
  `pPipelines` **must** be a valid pointer to an array of `createInfoCount` VkPipeline handles
- VUID-vkCreateGraphicsPipelines-createInfoCount-arraylength
  `createInfoCount` **must** be greater than `0`
- VUID-vkCreateGraphicsPipelines-pipelineCache-parent
  If `pipelineCache` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkGraphicsPipelineCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkGraphicsPipelineCreateInfo {
    VkStructureType                                 sType;
    const void*                                     pNext;
    VkPipelineCreateFlags                           flags;
    uint32_t                                        stageCount;
    const VkPipelineShaderStageCreateInfo*          pStages;
    const VkPipelineVertexInputStateCreateInfo*     pVertexInputState;
    const VkPipelineInputAssemblyStateCreateInfo*   pInputAssemblyState;
    const VkPipelineTessellationStateCreateInfo*    pTessellationState;
    const VkPipelineViewportStateCreateInfo*        pViewportState;
    const VkPipelineRasterizationStateCreateInfo*   pRasterizationState;
    const VkPipelineMultisampleStateCreateInfo*     pMultisampleState;
    const VkPipelineDepthStencilStateCreateInfo*    pDepthStencilState;
    const VkPipelineColorBlendStateCreateInfo*      pColorBlendState;
    const VkPipelineDynamicStateCreateInfo*         pDynamicState;
    VkPipelineLayout                                layout;
    VkRenderPass                                    renderPass;
    uint32_t                                        subpass;
    VkPipeline                                      basePipelineHandle;
    int32_t                                         basePipelineIndex;
} VkGraphicsPipelineCreateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask of VkPipelineCreateFlagBits specifying how the pipeline will be generated.
- stageCount is the number of entries in the pStages array.
- pStages is a pointer to an array of stageCount VkPipelineShaderStageCreateInfo structures describing the set of the shader stages to be included in the graphics pipeline.
- pVertexInputState is a pointer to a VkPipelineVertexInputStateCreateInfo structure. It is ignored if the pipeline includes a mesh shader stage. It **can** be NULL if the pipeline is created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state set.
- pInputAssemblyState is a pointer to a VkPipelineInputAssemblyStateCreateInfo structure which determines input assembly behavior for vertex shading, as described in Drawing Commands. If the VK_EXT_extended_dynamic_state3 extension is enabled, it **can** be NULL if the pipeline is created with both VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE, and VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic states set and dynamicPrimitiveTopologyUnrestricted is VK_TRUE. It is ignored if the pipeline includes a mesh shader stage.

- pTessellationState is a pointer to a VkPipelineTessellationStateCreateInfo structure defining tessellation state used by tessellation shaders. It **can** be NULL if the pipeline is created with the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state set.
- pViewportState is a pointer to a VkPipelineViewportStateCreateInfo structure defining viewport state used when rasterization is enabled. If the VK_EXT_extended_dynamic_state3 extension is enabled, it **can** be NULL if the pipeline is created with both VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT, and VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic states set.
- pRasterizationState is a pointer to a VkPipelineRasterizationStateCreateInfo structure defining rasterization state. If the VK_EXT_extended_dynamic_state3 extension is enabled, it **can** be NULL if the pipeline is created with all of VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT, VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE, VK_DYNAMIC_STATE_POLYGON_MODE_EXT, VK_DYNAMIC_STATE_CULL_MODE, VK_DYNAMIC_STATE_FRONT_FACE, VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE, VK_DYNAMIC_STATE_DEPTH_BIAS, and VK_DYNAMIC_STATE_LINE_WIDTH dynamic states set.
- pMultisampleState is a pointer to a VkPipelineMultisampleStateCreateInfo structure defining multisample state used when rasterization is enabled. If the VK_EXT_extended_dynamic_state3 extension is enabled, it **can** be NULL if the pipeline is created with all of VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT, VK_DYNAMIC_STATE_SAMPLE_MASK_EXT, and VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic states set, and either the alphaToOne feature is not enabled or VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT is set, in which case VkPipelineMultisampleStateCreateInfo::sampleShadingEnable is assumed to be VK_FALSE.
- pDepthStencilState is a pointer to a VkPipelineDepthStencilStateCreateInfo structure defining depth/stencil state used when rasterization is enabled for depth or stencil attachments accessed during rendering. If the VK_EXT_extended_dynamic_state3 extension is enabled, it **can** be NULL if the pipeline is created with all of VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE, VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE, VK_DYNAMIC_STATE_DEPTH_COMPARE_OP, VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE, VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE, VK_DYNAMIC_STATE_STENCIL_OP, and VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic states set.
- pColorBlendState is a pointer to a VkPipelineColorBlendStateCreateInfo structure defining color blend state used when rasterization is enabled for any color attachments accessed during rendering. If the VK_EXT_extended_dynamic_state3 extension is enabled, it **can** be NULL if the pipeline is created with all of VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT, VK_DYNAMIC_STATE_LOGIC_OP_EXT, VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT, VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT, VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic states set.
- pDynamicState is a pointer to a VkPipelineDynamicStateCreateInfo structure defining which properties of the pipeline state object are dynamic and **can** be changed independently of the pipeline state. This **can** be NULL, which means no state in the pipeline is considered dynamic.
- layout is the description of binding locations used by both the pipeline and descriptor sets used with the pipeline. If VkPhysicalDeviceProperties::apiVersion is greater than or equal to Vulkan

1.3 or VK_KHR_maintenance4 is enabled `layout` **must** not be accessed by the implementation outside of the duration of the command this structure is passed to.

- `renderPass` is a handle to a render pass object describing the environment in which the pipeline will be used. The pipeline **must** only be used with a render pass instance compatible with the one provided. See Render Pass Compatibility for more information. The implementation **must** not access this object outside of the duration of the command this structure is passed to.
- `subpass` is the index of the subpass in the render pass where this pipeline will be used.
- `basePipelineHandle` is a pipeline to derive from.
- `basePipelineIndex` is an index into the `pCreateInfos` parameter to use as a pipeline to derive from.

The parameters `basePipelineHandle` and `basePipelineIndex` are described in more detail in Pipeline Derivatives.

NOTE	With `VK_EXT_extended_dynamic_state3`, it is possible that many of the `VkGraphicsPipelineCreateInfo` members above **can** be `NULL` because all their state is dynamic and therefore ignored. This is optional so the application **can** still use a valid pointer if it needs to set the `pNext` or `flags` fields to specify state for other extensions.

The state required for a graphics pipeline is divided into vertex input state, pre-rasterization shader state, fragment shader state, and fragment output state.

Vertex Input State

Vertex input state is defined by:

- VkPipelineVertexInputStateCreateInfo
- VkPipelineInputAssemblyStateCreateInfo

If this pipeline specifies pre-rasterization state either directly or by including it as a pipeline library and its `pStages` includes a vertex shader, this state **must** be specified to create a complete graphics pipeline.

If a pipeline includes VK_GRAPHICS_PIPELINE_LIBRARY_VERTEX_INPUT_INTERFACE_BIT_EXT in VkGraphicsPipelineLibraryCreateInfoEXT::`flags` either explicitly or as a default, and either the conditions requiring this state for a complete graphics pipeline are met or this pipeline does not specify pre-rasterization state in any way, that pipeline **must** specify this state directly.

Pre-Rasterization Shader State

Pre-rasterization shader state is defined by:

- VkPipelineShaderStageCreateInfo entries for:
  - Vertex shaders
  - Tessellation control shaders

    - Tessellation evaluation shaders
    - Geometry shaders
    - Task shaders
    - Mesh shaders
- Within the VkPipelineLayout, all descriptor sets with pre-rasterization shader bindings if VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT was specified.
    - If VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT was not specified, the full pipeline layout **must** be specified.
- VkPipelineViewportStateCreateInfo
- VkPipelineRasterizationStateCreateInfo
- VkPipelineTessellationStateCreateInfo
- VkRenderPass and `subpass` parameter
- The `viewMask` parameter of VkPipelineRenderingCreateInfo (formats are ignored)
- VkPipelineDiscardRectangleStateCreateInfoEXT
- VkPipelineFragmentShadingRateStateCreateInfoKHR

This state **must** be specified to create a complete graphics pipeline.

If either the `pNext` chain includes a VkGraphicsPipelineLibraryCreateInfoEXT structure with VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT included in `flags`, or it is not specified and would default to include that value, this state **must** be specified in the pipeline.

Fragment Shader State

Fragment shader state is defined by:

- A VkPipelineShaderStageCreateInfo entry for the fragment shader
- Within the VkPipelineLayout, all descriptor sets with fragment shader bindings if VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT was specified.
    - If VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT was not specified, the full pipeline layout **must** be specified.
- VkPipelineMultisampleStateCreateInfo if sample shading is enabled or `renderpass` is not VK_NULL_HANDLE
- VkPipelineDepthStencilStateCreateInfo
- VkRenderPass and `subpass` parameter
- The `viewMask` parameter of VkPipelineRenderingCreateInfo (formats are ignored)
- VkPipelineFragmentShadingRateStateCreateInfoKHR
- Inclusion/omission of the VK_PIPELINE_RASTERIZATION_STATE_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR flag

- Inclusion/omission of the VK_PIPELINE_RASTERIZATION_STATE_CREATE_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT flag
- VkRenderingInputAttachmentIndexInfo
- The `customResolve` parameter of VkCustomResolveCreateInfoEXT. Formats are ignored, and not including the structure behaves identically to setting `customResolve` to VK_FALSE, unlike in fragment output interface state.

If a pipeline specifies pre-rasterization state either directly or by including it as a pipeline library and `rasterizerDiscardEnable` is VK_FALSE or VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE is used, this state **must** be specified to create a complete graphics pipeline.

If a pipeline includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT in VkGraphicsPipelineLibraryCreateInfoEXT::`flags` either explicitly or as a default, and either the conditions requiring this state for a complete graphics pipeline are met or this pipeline does not specify pre-rasterization state in any way, that pipeline **must** specify this state directly.

Fragment Output State

Fragment output state is defined by:

- VkPipelineColorBlendStateCreateInfo
- VkRenderPass and `subpass` parameter
- VkPipelineMultisampleStateCreateInfo
- The format parameters of VkPipelineRenderingCreateInfo (`viewMask` is ignored)
- VkCustomResolveCreateInfoEXT
- Inclusion/omission of the VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT and VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT flags
- Inclusion/omission of the VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VkRenderingAttachmentLocationInfo

If a pipeline specifies pre-rasterization state either directly or by including it as a pipeline library and `rasterizerDiscardEnable` is VK_FALSE or VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE is used, this state **must** be specified to create a complete graphics pipeline.

If a pipeline includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT in VkGraphicsPipelineLibraryCreateInfoEXT::`flags` either explicitly or as a default, and either the conditions requiring this state for a complete graphics pipeline are met or this pipeline does not specify pre-rasterization state in any way, that pipeline **must** specify this state directly.

Dynamic State

Dynamic state values set via `pDynamicState` **must** be ignored if the state they correspond to is not otherwise statically set by one of the state subsets used to create the pipeline. Additionally, setting

dynamic state values **must** not modify whether state in a linked library is static or dynamic; this is set and unchangeable when the library is created. For example, if a pipeline only included pre-rasterization shader state, then any dynamic state value corresponding to depth or stencil testing has no effect. Any linked library that has dynamic state enabled that same dynamic state **must** also be enabled in all the other linked libraries to which that dynamic state applies.

Complete Graphics Pipelines

A complete graphics pipeline always includes pre-rasterization shader state, with other subsets included depending on that state as specified in the above sections.

Graphics Pipeline Library Layouts

If different subsets are linked together with pipeline layouts created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT, the final effective pipeline layout is effectively the union of the linked pipeline layouts. When binding descriptor sets for this pipeline, the pipeline layout used **must** be compatible with this union. This pipeline layout **can** be overridden when linking with VK_PIPELINE_CREATE_LINK_TIME_OPTIMIZATION_BIT_EXT by providing a VkPipelineLayout that is compatible with this union other than VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT, or when linking without VK_PIPELINE_CREATE_LINK_TIME_OPTIMIZATION_BIT_EXT by providing a VkPipelineLayout that is fully compatible with this union.

If the `pNext` chain includes a VkPipelineCreateFlags2CreateInfo structure, VkPipelineCreateFlags2CreateInfo::`flags` from that structure is used instead of `flags` from this structure.

Valid Usage

- VUID-VkGraphicsPipelineCreateInfo-None-09497
  If the `pNext` chain does not include a VkPipelineCreateFlags2CreateInfo structure, `flags` **must** be a valid combination of VkPipelineCreateFlagBits values
- VUID-VkGraphicsPipelineCreateInfo-flags-07984
  If `flags` contains the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag, and `basePipelineIndex` is -1, `basePipelineHandle` **must** be a valid graphics `VkPipeline` handle
- VUID-VkGraphicsPipelineCreateInfo-flags-07985
  If `flags` contains the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag, and `basePipelineHandle` is VK_NULL_HANDLE, `basePipelineIndex` **must** be a valid index into the calling command's `pCreateInfos` parameter
- VUID-VkGraphicsPipelineCreateInfo-flags-07986
  If `flags` contains the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag, `basePipelineIndex` **must** be -1 or `basePipelineHandle` **must** be VK_NULL_HANDLE
- VUID-VkGraphicsPipelineCreateInfo-layout-07987
  If a push constant block is declared in a shader and `layout` is not VK_NULL_HANDLE, a push constant range in `layout` **must** match the shader stage
- VUID-VkGraphicsPipelineCreateInfo-layout-10069

If a push constant block is declared in a shader and `layout` is not VK_NULL_HANDLE, the block must be contained inside the push constant range in `layout` that matches the stage

- VUID-VkGraphicsPipelineCreateInfo-layout-07988
  If a resource variable is declared in a shader and `layout` is not VK_NULL_HANDLE, the corresponding descriptor set in `layout` **must** match the shader stage
- VUID-VkGraphicsPipelineCreateInfo-layout-07990
  If a resource variable is declared in a shader, `layout` is not VK_NULL_HANDLE, and the descriptor type is not VK_DESCRIPTOR_TYPE_MUTABLE_EXT, the corresponding descriptor set in `layout` **must** match the descriptor type
- VUID-VkGraphicsPipelineCreateInfo-layout-07991
  If a resource variable is declared in a shader as an array and `layout` is not VK_NULL_HANDLE, the corresponding descriptor binding used to create `layout` **must** have a `descriptorCount` that is greater than or equal to the length of the array
- VUID-VkGraphicsPipelineCreateInfo-None-10391
  If a resource variable is declared in a shader as an array of descriptors, then the descriptor type of that variable **must** not be VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK
- VUID-VkGraphicsPipelineCreateInfo-flags-11798
  If the shader64BitIndexing feature is not enabled, `flags` **must** not contain VK_PIPELINE_CREATE_2_64_BIT_INDEXING_BIT_EXT
- VUID-VkGraphicsPipelineCreateInfo-pipelineCreationCacheControl-02878
  If the `pipelineCreationCacheControl` feature is not enabled, `flags` **must** not include VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT nor VK_PIPELINE_CREATE_EARLY_RETURN_ON_FAILURE_BIT
- VUID-VkGraphicsPipelineCreateInfo-pipelineProtectedAccess-07368
  If the `pipelineProtectedAccess` feature is not enabled, `flags` **must** not include VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT nor VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT
- VUID-VkGraphicsPipelineCreateInfo-flags-07369
  `flags` **must** not include both VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT and VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT
- VUID-VkGraphicsPipelineCreateInfo-flags-11311
  If VkPipelineCreateFlags2CreateInfoKHR::`flags` includes VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, `layout` **must** be VK_NULL_HANDLE
- VUID-VkGraphicsPipelineCreateInfo-flags-11312
  If VkPipelineCreateFlags2CreateInfoKHR::`flags` includes VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, all shader variables in the shader resource interface with a `DescriptorSet` and `Binding` decoration **must** have a mapping declared in VkShaderDescriptorSetAndBindingMappingInfoEXT::`pMappings`
- VUID-VkGraphicsPipelineCreateInfo-stage-02096
  If the pipeline requires pre-rasterization shader state the `stage` member of one element of `pStages` **must** be VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT

- VUID-VkGraphicsPipelineCreateInfo-pStages-02095
  If the pipeline requires pre-rasterization shader state the geometric shader stages provided in pStages **must** be either from the mesh shading pipeline (stage is VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT) or from the primitive shading pipeline (stage is VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, or VK_SHADER_STAGE_GEOMETRY_BIT)
- VUID-VkGraphicsPipelineCreateInfo-pStages-09631
  If the pipeline requires pre-rasterization shader state and pStages contains both VK_SHADER_STAGE_TASK_BIT_EXT and VK_SHADER_STAGE_MESH_BIT_EXT, then the mesh shader's entry point **must** not declare a variable with a DrawIndex BuiltIn decoration
- VUID-VkGraphicsPipelineCreateInfo-pStages-00729
  If the pipeline requires pre-rasterization shader state and pStages includes a tessellation control shader stage, it **must** include a tessellation evaluation shader stage
- VUID-VkGraphicsPipelineCreateInfo-pStages-00730
  If the pipeline requires pre-rasterization shader state and pStages includes a tessellation evaluation shader stage, it **must** include a tessellation control shader stage
- VUID-VkGraphicsPipelineCreateInfo-pStages-09022
  If the pipeline requires pre-rasterization shader state and pStages includes a tessellation control shader stage, and the VK_EXT_extended_dynamic_state3 extension is not enabled or the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state is not set, pTessellationState **must** be a valid pointer to a valid VkPipelineTessellationStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-pTessellationState-09023
  If pTessellationState is not NULL it **must** be a pointer to a valid VkPipelineTessellationStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-pStages-00732
  If the pipeline requires pre-rasterization shader state and pStages includes tessellation shader stages, the shader code of at least one stage **must** contain an OpExecutionMode instruction specifying the type of subdivision in the pipeline
- VUID-VkGraphicsPipelineCreateInfo-pStages-00733
  If the pipeline requires pre-rasterization shader state and pStages includes tessellation shader stages, and the shader code of both stages contain an OpExecutionMode instruction specifying the type of subdivision in the pipeline, they **must** both specify the same subdivision mode
- VUID-VkGraphicsPipelineCreateInfo-pStages-00734
  If the pipeline requires pre-rasterization shader state and pStages includes tessellation shader stages, the shader code of at least one stage **must** contain an OpExecutionMode instruction specifying the output patch size in the pipeline
- VUID-VkGraphicsPipelineCreateInfo-pStages-00735
  If the pipeline requires pre-rasterization shader state and pStages includes tessellation shader stages, and the shader code of both contain an OpExecutionMode instruction

specifying the output patch size in the pipeline, they **must** both specify the same patch size

- VUID-VkGraphicsPipelineCreateInfo-pStages-08888
  If the pipeline is being created with pre-rasterization shader state and vertex input state and `pStages` includes tessellation shader stages, and either VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state is not enabled or `dynamicPrimitiveTopologyUnrestricted` is VK_FALSE, the `topology` member of `pInputAssemblyState` **must** be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST
- VUID-VkGraphicsPipelineCreateInfo-topology-08889
  If the pipeline is being created with pre-rasterization shader state and vertex input state and the `topology` member of `pInputAssemblyState` is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, and either VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state is not enabled or `dynamicPrimitiveTopologyUnrestricted` is VK_FALSE, then `pStages` **must** include tessellation shader stages
- VUID-VkGraphicsPipelineCreateInfo-TessellationEvaluation-07723
  If the pipeline is being created with a `TessellationEvaluation` `Execution Model`, no `Geometry` `Execution Model`, uses the `PointMode` `Execution Mode`, and the `shaderTessellationAndGeometryPointSize` feature is enabled, a `PointSize` decorated variable **must** be written to if the `maintenance5` feature is not enabled
- VUID-VkGraphicsPipelineCreateInfo-topology-08773
  If the pipeline is being created with a `Vertex` `Execution Model` and no `TessellationEvaluation` or `Geometry` `Execution Model`, and the `topology` member of `pInputAssemblyState` is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, and either VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state is not enabled or `dynamicPrimitiveTopologyUnrestricted` is VK_FALSE, a `PointSize` decorated variable **must** be written to if the `maintenance5` feature is not enabled
- VUID-VkGraphicsPipelineCreateInfo-TessellationEvaluation-07724
  If the pipeline is being created with a `TessellationEvaluation` `Execution Model`, no `Geometry` `Execution Model`, uses the `PointMode` `Execution Mode`, and the `shaderTessellationAndGeometryPointSize` feature is not enabled, a `PointSize` decorated variable **must** not be written to
- VUID-VkGraphicsPipelineCreateInfo-shaderTessellationAndGeometryPointSize-08776
  If the pipeline is being created with a `Geometry` `Execution Model`, uses the `OutputPoints` `Execution Mode`, and the `shaderTessellationAndGeometryPointSize` feature is enabled, a `PointSize` decorated variable **must** be written to for every vertex emitted if the `maintenance5` feature is not enabled
- VUID-VkGraphicsPipelineCreateInfo-Geometry-07726
  If the pipeline is being created with a `Geometry` `Execution Model`, uses the `OutputPoints` `Execution Mode`, and the `shaderTessellationAndGeometryPointSize` feature is not enabled, a `PointSize` decorated variable **must** not be written to
- VUID-VkGraphicsPipelineCreateInfo-pStages-00738
  If the pipeline requires pre-rasterization shader state and `pStages` includes a geometry shader stage, and does not include any tessellation shader stages, its shader code **must** contain an `OpExecutionMode` instruction specifying an input primitive type that is

compatible with the primitive topology specified in pInputAssembly

- VUID-VkGraphicsPipelineCreateInfo-pStages-00739
  If the pipeline requires pre-rasterization shader state and pStages includes a geometry shader stage, and also includes tessellation shader stages, its shader code **must** contain an OpExecutionMode instruction specifying an input primitive type that is compatible with the primitive topology that is output by the tessellation stages
- VUID-VkGraphicsPipelineCreateInfo-pStages-00740
  If the pipeline requires pre-rasterization shader state and fragment shader state, it includes both a fragment shader and a geometry shader, and the fragment shader code reads from an input variable that is decorated with PrimitiveId, then the geometry shader code **must** write to a matching output variable, decorated with PrimitiveId, in all execution paths
- VUID-VkGraphicsPipelineCreateInfo-PrimitiveId-06264
  If the pipeline requires pre-rasterization shader state, it includes a mesh shader and the fragment shader code reads from an input variable that is decorated with PrimitiveId, then the mesh shader code **must** write to a matching output variable, decorated with PrimitiveId, in all execution paths
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06038
  If renderPass is not VK_NULL_HANDLE and the pipeline is being created with fragment shader state the fragment shader **must** not read from any input attachment that is defined as VK_ATTACHMENT_UNUSED in subpass
- VUID-VkGraphicsPipelineCreateInfo-pStages-00742
  If the pipeline requires pre-rasterization shader state and multiple pre-rasterization shader stages are included in pStages, the shader code for the entry points identified by those pStages and the rest of the state identified by this structure **must** adhere to the pipeline linking rules described in the Shader Interfaces chapter
- VUID-VkGraphicsPipelineCreateInfo-None-04889
  If the pipeline requires pre-rasterization shader state and fragment shader state, the fragment shader and last pre-rasterization shader stage and any relevant state **must** adhere to the pipeline linking rules described in the Shader Interfaces chapter
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06041
  If renderPass is not VK_NULL_HANDLE, and the pipeline is being created with fragment output interface state, then for each color attachment in the subpass, if the potential format features of the format of the corresponding attachment description do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the blendEnable member of the corresponding element of the pAttachments member of pColorBlendState **must** be VK_FALSE
- VUID-VkGraphicsPipelineCreateInfo-renderPass-07609
  If renderPass is not VK_NULL_HANDLE, the pipeline is being created with fragment output interface state, the pColorBlendState pointer is not NULL, the attachmentCount member of pColorBlendState is not ignored, and the subpass uses color attachments, the attachmentCount member of pColorBlendState **must** be equal to the colorAttachmentCount used to create subpass
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04130

If the pipeline requires pre-rasterization shader state, and pViewportState->pViewports is not dynamic, then pViewportState->pViewports **must** be a valid pointer to an array of pViewportState->viewportCount valid VkViewport structures

- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04131
  If the pipeline requires pre-rasterization shader state, and pViewportState->pScissors is not dynamic, then pViewportState->pScissors **must** be a valid pointer to an array of pViewportState->scissorCount VkRect2D structures
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-00749
  If the pipeline requires pre-rasterization shader state, and the wideLines feature is not enabled, and no element of the pDynamicStates member of pDynamicState is VK_DYNAMIC_STATE_LINE_WIDTH, the lineWidth member of pRasterizationState **must** be 1.0
- VUID-VkGraphicsPipelineCreateInfo-rasterizerDiscardEnable-09024
  If the pipeline requires pre-rasterization shader state, and the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state is enabled or the rasterizerDiscardEnable member of pRasterizationState is VK_FALSE, and related dynamic state is not set, pViewportState **must** be a valid pointer to a valid VkPipelineViewportStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-pViewportState-09025
  If pViewportState is not NULL it **must** be a valid pointer to a valid VkPipelineViewportStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-pMultisampleState-09026
  If the pipeline requires fragment output interface state, and the VK_EXT_extended_dynamic_state3 extension is not enabled or any of the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT, VK_DYNAMIC_STATE_SAMPLE_MASK_EXT, or VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic states is not set, or the alphaToOne feature is enabled and VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT is not set, pMultisampleState **must** be a valid pointer to a valid VkPipelineMultisampleStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-pMultisampleState-09027
  If pMultisampleState is not NULL it **must** be a valid pointer to a valid VkPipelineMultisampleStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-alphaToCoverageEnable-08891
  If the pipeline is being created with fragment shader state, the VkPipelineMultisampleStateCreateInfo::alphaToCoverageEnable is not ignored and is VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-VkGraphicsPipelineCreateInfo-renderPass-09028
  If renderPass is not VK_NULL_HANDLE, the pipeline is being created with fragment shader state, and subpass uses a depth/stencil attachment, and related dynamic state is not set, pDepthStencilState **must** be a valid pointer to a valid VkPipelineDepthStencilStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-pDepthStencilState-09029

If pDepthStencilState is not NULL it **must** be a valid pointer to a valid VkPipelineDepthStencilStateCreateInfo structure

- VUID-VkGraphicsPipelineCreateInfo-renderPass-09030
  If renderPass is not VK_NULL_HANDLE, the pipeline is being created with fragment output interface state, and subpass uses color attachments, and related dynamic state is not set, pColorBlendState **must** be a valid pointer to a valid VkPipelineColorBlendStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-00754
  If the pipeline requires pre-rasterization shader state, the depthBiasClamp feature is not enabled, no element of the pDynamicStates member of pDynamicState is VK_DYNAMIC_STATE_DEPTH_BIAS, and the depthBiasEnable member of pRasterizationState is VK_TRUE, the depthBiasClamp member of pRasterizationState **must** be 0.0
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-02510
  If the pipeline requires fragment shader state, the VK_EXT_depth_range_unrestricted extension is not enabled and no element of the pDynamicStates member of pDynamicState is VK_DYNAMIC_STATE_DEPTH_BOUNDS, and the depthBoundsTestEnable member of pDepthStencilState is VK_TRUE, the minDepthBounds and maxDepthBounds members of pDepthStencilState **must** be between 0.0 and 1.0, inclusive
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-10913
  If the pipeline requires fragment shader state, no element of the pDynamicStates member of pDynamicState is VK_DYNAMIC_STATE_DEPTH_BOUNDS, and pDynamicStates includes VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE or the depthBoundsTestEnable member of pDepthStencilState is VK_TRUE, minDepthBounds **must** be less than or equal to maxDepthBounds
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-07610
  If the pipeline requires fragment shader state or fragment output interface state, and rasterizationSamples and sampleLocationsInfo are not dynamic, and VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable included in the pNext chain of pMultisampleState is VK_TRUE, sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-07611
  If the pipeline requires fragment shader state or fragment output interface state, and rasterizationSamples and sampleLocationsInfo are not dynamic, and VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable included in the pNext chain of pMultisampleState is VK_TRUE or VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT is used, sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-07612

If the pipeline requires fragment shader state or fragment output interface state, and `rasterizationSamples` and `sampleLocationsInfo` are not dynamic, and VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` included in the `pNext` chain of `pMultisampleState` is VK_TRUE or VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT is used, `sampleLocationsInfo.sampleLocationsPerPixel` **must** equal `rasterizationSamples`

- VUID-VkGraphicsPipelineCreateInfo-sampleLocationsEnable-01524
  If the pipeline requires fragment shader state, and the `sampleLocationsEnable` member of a VkPipelineSampleLocationsStateCreateInfoEXT structure included in the `pNext` chain of `pMultisampleState` is VK_TRUE, the fragment shader code **must** not statically use the extended instruction `InterpolateAtSample`
- VUID-VkGraphicsPipelineCreateInfo-multisampledRenderToSingleSampled-06853
  If the pipeline requires fragment output interface state, the `multisampledRenderToSingleSampled` feature is not enabled, the `VK_AMD_mixed_attachment_samples` extension is not enabled, the `VK_NV_framebuffer_mixed_samples` extension is not enabled, `rasterizationSamples` is not dynamic, and `subpass` uses color and/or depth/stencil attachments, then the `rasterizationSamples` member of `pMultisampleState` **must** be the same as the sample count for those subpass attachments
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06854
  If `renderPass` is not VK_NULL_HANDLE, the `multisampledRenderToSingleSampled` feature is not enabled, `rasterizationSamples` is not dynamic, and `subpass` has a VkMultisampledRenderToSingleSampledInfoEXT structure included in the VkSubpassDescription2::`pNext` chain with `multisampledRenderToSingleSampledEnable` equal to VK_TRUE, then the `rasterizationSamples` member of `pMultisampleState` **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::`rasterizationSamples`
- VUID-VkGraphicsPipelineCreateInfo-subpass-00758
  If the pipeline requires fragment output interface state, `rasterizationSamples` is not dynamic, and `subpass` does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` member of `pMultisampleState` **must** follow the rules for a zero-attachment subpass
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06046
  If `renderPass` is not VK_NULL_HANDLE, `subpass` **must** be a valid subpass within `renderPass`
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06047
  If `renderPass` is not VK_NULL_HANDLE, the pipeline is being created with pre-rasterization shader state, `subpass` viewMask is not `0`, and `multiviewTessellationShader` is not enabled, then `pStages` **must** not include tessellation shaders
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06048
  If `renderPass` is not VK_NULL_HANDLE, the pipeline is being created with pre-rasterization shader state, `subpass` viewMask is not `0`, and `multiviewGeometryShader` is not enabled, then `pStages` **must** not include a geometry shader
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06050
  If `renderPass` is not VK_NULL_HANDLE and the pipeline is being created with pre-rasterization shader state, and `subpass` viewMask is not `0`, then all of the shaders in the pipeline **must** not include variables decorated with the `Layer` built-in decoration in their

interfaces

- VUID-VkGraphicsPipelineCreateInfo-renderPass-07064
  If renderPass is not VK_NULL_HANDLE, the pipeline is being created with pre-rasterization shader state, subpass viewMask is not 0, and multiviewMeshShader is not enabled, then pStages **must** not include a mesh shader
- VUID-VkGraphicsPipelineCreateInfo-renderPass-12325
  If renderPass is not VK_NULL_HANDLE, the pipeline is being created with pre-rasterization shader state, pStages includes a mesh shader, and subpass viewMask is not 0, then the index of the most significant bit in viewMask **must** be less than maxMeshMultiviewViewCount
- VUID-VkGraphicsPipelineCreateInfo-flags-00764
  flags **must** not contain the VK_PIPELINE_CREATE_DISPATCH_BASE_BIT flag
- VUID-VkGraphicsPipelineCreateInfo-pStages-01565
  If the pipeline requires fragment shader state and an input attachment was referenced by an aspectMask at renderPass creation time, the fragment shader **must** only read from the aspects that were specified for that input attachment
- VUID-VkGraphicsPipelineCreateInfo-layout-01688
  The number of resources in layout accessible to each shader stage that is used by the pipeline **must** be less than or equal to VkPhysicalDeviceLimits::maxPerStageResources
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04058
  If the pipeline requires pre-rasterization shader state, and no element of the pDynamicStates member of pDynamicState is VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT, and if pNext chain includes a VkPipelineDiscardRectangleStateCreateInfoEXT structure, and if its discardRectangleCount member is not 0, then its pDiscardRectangles member **must** be a valid pointer to an array of discardRectangleCount VkRect2D structures
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-07855
  If VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT is included in the pDynamicStates array then the implementation **must** support at least specVersion 2 of the VK_EXT_discard_rectangles extension
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-07856
  If VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT is included in the pDynamicStates array then the implementation **must** support at least specVersion 2 of the VK_EXT_discard_rectangles extension
- VUID-VkGraphicsPipelineCreateInfo-pStages-02097
  If the pipeline requires vertex input state, and pVertexInputState is not dynamic, then pVertexInputState **must** be a valid pointer to a valid VkPipelineVertexInputStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-Input-07904
  If the maintenance9 feature is not enabled, and the pipeline is being created with vertex input state and pVertexInputState is not dynamic, then all variables with the Input storage class decorated with Location in the Vertex Execution Model OpEntryPoint **must** contain a location in VkVertexInputAttributeDescription::location

- VUID-VkGraphicsPipelineCreateInfo-Input-08733
  If the pipeline requires vertex input state and pVertexInputState is not dynamic, then the numeric type associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be the same as VkVertexInputAttributeDescription::format
- VUID-VkGraphicsPipelineCreateInfo-pVertexInputState-08929
  If the pipeline is being created with vertex input state and pVertexInputState is not dynamic, and VkVertexInputAttributeDescription::format has a 64-bit component, then the scalar width associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be 64-bit
- VUID-VkGraphicsPipelineCreateInfo-pVertexInputState-08930
  If the pipeline is being created with vertex input state and pVertexInputState is not dynamic, and the scalar width associated with a Location decorated Input variable in the Vertex Execution Model OpEntryPoint is 64-bit, then the corresponding VkVertexInputAttributeDescription::format **must** have a 64-bit component
- VUID-VkGraphicsPipelineCreateInfo-pVertexInputState-09198
  If the pipeline is being created with vertex input state and pVertexInputState is not dynamic, and VkVertexInputAttributeDescription::format has a 64-bit component, then all Input variables at the corresponding Location in the Vertex Execution Model OpEntryPoint **must** not use components that are not present in the format
- VUID-VkGraphicsPipelineCreateInfo-dynamicPrimitiveTopologyUnrestricted-09031
  If the pipeline requires vertex input state, and related dynamic state is not set, pInputAssemblyState **must** be a valid pointer to a valid VkPipelineInputAssemblyStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-pInputAssemblyState-09032
  If pInputAssemblyState is not NULL it **must** be a valid pointer to a valid VkPipelineInputAssemblyStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-pStages-02317
  If the pipeline requires pre-rasterization shader state, the Xfb execution mode **can** be specified by no more than one shader stage in pStages
- VUID-VkGraphicsPipelineCreateInfo-pStages-02318
  If the pipeline requires pre-rasterization shader state, and any shader stage in pStages specifies Xfb execution mode it **must** be the last pre-rasterization shader stage
- VUID-VkGraphicsPipelineCreateInfo-rasterizationStream-02319
  If the pipeline requires pre-rasterization shader state, and a VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream value other than zero is specified, all variables in the output interface of the entry point being compiled decorated with Position, PointSize, ClipDistance, or CullDistance **must** be decorated with identical Stream values that match the rasterizationStream
- VUID-VkGraphicsPipelineCreateInfo-rasterizationStream-02320
  If the pipeline requires pre-rasterization shader state, and VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream is zero, or not specified, all variables in the output interface of the entry point being compiled decorated with Position, PointSize, ClipDistance, or CullDistance **must** be decorated with a Stream

value of zero, or **must** not specify the `Stream` decoration

- VUID-VkGraphicsPipelineCreateInfo-geometryStreams-02321
  If the pipeline requires pre-rasterization shader state, and the last pre-rasterization shader stage is a geometry shader, and that geometry shader uses the `GeometryStreams` capability, then `VkPhysicalDeviceTransformFeedbackFeaturesEXT`::`geometryStreams` feature **must** be enabled
- VUID-VkGraphicsPipelineCreateInfo-None-02322
  If the pipeline requires pre-rasterization shader state, and there are any mesh shader stages in the pipeline there **must** not be any shader stage in the pipeline with a `Xfb` execution mode
- VUID-VkGraphicsPipelineCreateInfo-lineRasterizationMode-02766
  If the pipeline requires pre-rasterization shader state and at least one of fragment output interface state or fragment shader state, and `pMultisampleState` is not `NULL`, the `lineRasterizationMode` member of a VkPipelineRasterizationLineStateCreateInfo structure included in the `pNext` chain of `pRasterizationState` is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the `alphaToCoverageEnable`, `alphaToOneEnable`, and `sampleShadingEnable` members of `pMultisampleState` **must** all be VK_FALSE
- VUID-VkGraphicsPipelineCreateInfo-stippledLineEnable-02767
  If the pipeline requires pre-rasterization shader state, the `stippledLineEnable` member of VkPipelineRasterizationLineStateCreateInfo is VK_TRUE, and no element of the `pDynamicStates` member of `pDynamicState` is VK_DYNAMIC_STATE_LINE_STIPPLE, then the `lineStippleFactor` member of VkPipelineRasterizationLineStateCreateInfo **must** be in the range [1,256]
- VUID-VkGraphicsPipelineCreateInfo-flags-03372
  `flags` **must** not include VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR
- VUID-VkGraphicsPipelineCreateInfo-flags-03373
  `flags` **must** not include VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR
- VUID-VkGraphicsPipelineCreateInfo-flags-03374
  `flags` **must** not include VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR
- VUID-VkGraphicsPipelineCreateInfo-flags-03375
  `flags` **must** not include VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR
- VUID-VkGraphicsPipelineCreateInfo-flags-03376
  `flags` **must** not include VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR
- VUID-VkGraphicsPipelineCreateInfo-flags-03377
  `flags` **must** not include VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR
- VUID-VkGraphicsPipelineCreateInfo-flags-03577
  `flags` **must** not include

VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR

- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-03378
  If the extendedDynamicState feature is not enabled, and the minimum value of VkApplicationInfo::apiVersion used to create the VkInstance and apiVersion supported by the physical device is less than Version 1.3 there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_CULL_MODE, VK_DYNAMIC_STATE_FRONT_FACE, VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY, VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT, VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT, VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE, VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE, VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE, VK_DYNAMIC_STATE_DEPTH_COMPARE_OP, VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE, VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE, or VK_DYNAMIC_STATE_STENCIL_OP
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-03379
  If the pipeline requires pre-rasterization shader state, and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT is included in the pDynamicStates array then viewportCount **must** be zero
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-03380
  If the pipeline requires pre-rasterization shader state, and VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT is included in the pDynamicStates array then scissorCount **must** be zero
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04132
  If the pipeline requires pre-rasterization shader state, and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT is included in the pDynamicStates array then VK_DYNAMIC_STATE_VIEWPORT **must** not be present
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04133
  If the pipeline requires pre-rasterization shader state, and VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT is included in the pDynamicStates array then VK_DYNAMIC_STATE_SCISSOR **must** not be present
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-07065
  If the pipeline requires pre-rasterization shader state, and includes a mesh shader, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY, or VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04868
  If the extendedDynamicState2 feature is not enabled, and the minimum value of VkApplicationInfo::apiVersion used to create the VkInstance and apiVersion supported by the physical device is less than Version 1.3 there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE, VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE, or VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04869

If the extendedDynamicState2LogicOp feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_LOGIC_OP_EXT

- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04870
  If the extendedDynamicState2PatchControlPoints feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-07066
  If the pipeline requires pre-rasterization shader state, and includes a mesh shader, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE, or VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT
- VUID-VkGraphicsPipelineCreateInfo-flags-11000
  If flags includes VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT, then the VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT::deviceGeneratedCommands feature **must** be enabled
- VUID-VkGraphicsPipelineCreateInfo-flags-11001
  If the pipeline requires pre-rasterization shader state and flags includes VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT, then all stages **must** not specify Xfb execution mode
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04494
  If the pipeline requires pre-rasterization shader state or fragment shader state and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.width **must** be greater than or equal to 1
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04495
  If the pipeline requires pre-rasterization shader state or fragment shader state and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.height **must** be greater than or equal to 1
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04496
  If the pipeline requires pre-rasterization shader state or fragment shader state and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.width **must** be a power-of-two value
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04497
  If the pipeline requires pre-rasterization shader state or fragment shader state and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.height **must** be a power-of-two value
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04498
  If the pipeline requires pre-rasterization shader state or fragment shader state and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.width **must** be less than or equal to 4

- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04499
  If the pipeline requires pre-rasterization shader state or fragment shader state and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.height **must** be less than or equal to 4
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04500
  If the pipeline requires pre-rasterization shader state or fragment shader state and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, and the pipelineFragmentShadingRate feature is not enabled, VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.width and VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.height **must** both be equal to 1
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-06567
  If the pipeline requires pre-rasterization shader state or fragment shader state and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, VkPipelineFragmentShadingRateStateCreateInfoKHR::combinerOps[0] **must** be a valid VkFragmentShadingRateCombinerOpKHR value
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-06568
  If the pipeline requires pre-rasterization shader state or fragment shader state and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, VkPipelineFragmentShadingRateStateCreateInfoKHR::combinerOps[1] **must** be a valid VkFragmentShadingRateCombinerOpKHR value
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04501
  If the pipeline requires pre-rasterization shader state or fragment shader state and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, and the primitiveFragmentShadingRate feature is not enabled, VkPipelineFragmentShadingRateStateCreateInfoKHR::combinerOps[0] **must** be VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04502
  If the pipeline requires pre-rasterization shader state or fragment shader state and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, and the attachmentFragmentShadingRate feature is not enabled, VkPipelineFragmentShadingRateStateCreateInfoKHR::combinerOps[1] **must** be VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR
- VUID-VkGraphicsPipelineCreateInfo-primitiveFragmentShadingRateWithMultipleViewports-04503
  If the pipeline requires pre-rasterization shader state and the primitiveFragmentShadingRateWithMultipleViewports limit is not supported, VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT is not included in pDynamicState->pDynamicStates, and VkPipelineViewportStateCreateInfo::viewportCount is greater than 1, entry points specified in pStages **must** not write to the PrimitiveShadingRateKHR built-in
- VUID-VkGraphicsPipelineCreateInfo-primitiveFragmentShadingRateWithMultipleViewports-04504
  If the pipeline requires pre-rasterization shader state and the primitiveFragmentShadingRateWithMultipleViewports limit is not supported, and entry

points specified in pStages write to the ViewportIndex built-in, they **must** not also write to the PrimitiveShadingRateKHR built-in

- VUID-VkGraphicsPipelineCreateInfo-fragmentShadingRateNonTrivialCombinerOps-04506
  If the pipeline requires pre-rasterization shader state or fragment shader state, the fragmentShadingRateNonTrivialCombinerOps limit is not supported, and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, elements of VkPipelineFragmentShadingRateStateCreateInfoKHR ::combinerOps **must** be VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR or VK_FRAGMENT_SHADING_RATE_COMBINER_OP_REPLACE_KHR
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-03578
  All elements of the pDynamicStates member of pDynamicState **must** not be VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04807
  If the pipeline requires pre-rasterization shader state and the vertexInputDynamicState feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_VERTEX_INPUT_EXT
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-07067
  If the pipeline requires pre-rasterization shader state, and includes a mesh shader, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_VERTEX_INPUT_EXT
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04800
  If the colorWriteEnable feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT
- VUID-VkGraphicsPipelineCreateInfo-rasterizationSamples-04899
  If the pipeline requires fragment shader state, and the customResolve feature is enabled, rasterizationSamples is not dynamic, and if subpass has any input attachments, and if the subpass description contains VK_SUBPASS_DESCRIPTION_FRAGMENT_REGION_BIT_EXT, then the sample count of the input attachments **must** equal rasterizationSamples
- VUID-VkGraphicsPipelineCreateInfo-sampleShadingEnable-04900
  If the pipeline requires fragment shader state, and the customResolve feature is enabled, and if the subpass description contains VK_SUBPASS_DESCRIPTION_FRAGMENT_REGION_BIT_EXT, then sampleShadingEnable **must** be false
- VUID-VkGraphicsPipelineCreateInfo-dynamicRendering-06576
  If the dynamicRendering feature is not enabled and the pipeline requires pre-rasterization shader state, fragment shader state, or fragment output interface state, renderPass **must** not be VK_NULL_HANDLE
- VUID-VkGraphicsPipelineCreateInfo-pNext-12427
  If VkPipelineRenderingCreateInfo is included in the pNext chain, then renderPass **must** be VK_NULL_HANDLE
- VUID-VkGraphicsPipelineCreateInfo-multiview-06577
  If the multiview feature is not enabled, the pipeline requires pre-rasterization shader state or fragment shader state, and renderPass is VK_NULL_HANDLE,

VkPipelineRenderingCreateInfo::viewMask **must** be 0

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06578
  If the pipeline requires pre-rasterization shader state or fragment shader state, and renderPass is VK_NULL_HANDLE, the index of the most significant bit in VkPipelineRenderingCreateInfo::viewMask **must** be less than maxMultiviewViewCount
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06579
  If the pipeline requires fragment output interface state, and renderPass is VK_NULL_HANDLE, and VkPipelineRenderingCreateInfo::colorAttachmentCount is not 0, VkPipelineRenderingCreateInfo::pColorAttachmentFormats **must** be a valid pointer to an array of colorAttachmentCount valid VkFormat values
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06580
  If the pipeline requires fragment output interface state, and renderPass is VK_NULL_HANDLE, each element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats **must** be a valid VkFormat value
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06582
  If the pipeline requires fragment output interface state, renderPass is VK_NULL_HANDLE, and any element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats is not VK_FORMAT_UNDEFINED, that format **must** be a format with potential format features that include VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06583
  If the pipeline requires fragment output interface state, and renderPass is VK_NULL_HANDLE, VkPipelineRenderingCreateInfo::depthAttachmentFormat **must** be a valid VkFormat value
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06584
  If the pipeline requires fragment output interface state, and renderPass is VK_NULL_HANDLE, VkPipelineRenderingCreateInfo::stencilAttachmentFormat **must** be a valid VkFormat value
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06585
  If the pipeline requires fragment output interface state, renderPass is VK_NULL_HANDLE, and VkPipelineRenderingCreateInfo::depthAttachmentFormat is not VK_FORMAT_UNDEFINED, it **must** be a format with potential format features that include VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06586
  If the pipeline requires fragment output interface state, renderPass is VK_NULL_HANDLE, and VkPipelineRenderingCreateInfo::stencilAttachmentFormat is not VK_FORMAT_UNDEFINED, it **must** be a format with potential format features that include VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06587
  If the pipeline requires fragment output interface state, renderPass is VK_NULL_HANDLE, and VkPipelineRenderingCreateInfo::depthAttachmentFormat is not VK_FORMAT_UNDEFINED, it **must** be a format that includes a depth component
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06588
  If the pipeline requires fragment output interface state, renderPass is VK_NULL_HANDLE, and VkPipelineRenderingCreateInfo::stencilAttachmentFormat is not

VK_FORMAT_UNDEFINED, it **must** be a format that includes a stencil component

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06589
  If the pipeline requires fragment output interface state, `renderPass` is VK_NULL_HANDLE, VkPipelineRenderingCreateInfo::`depthAttachmentFormat` is not VK_FORMAT_UNDEFINED, and VkPipelineRenderingCreateInfo::`stencilAttachmentFormat` is not VK_FORMAT_UNDEFINED, `depthAttachmentFormat` **must** equal `stencilAttachmentFormat`
- VUID-VkGraphicsPipelineCreateInfo-renderPass-09033
  If `renderPass` is VK_NULL_HANDLE, the pipeline is being created with fragment shader state and fragment output interface state, and either of VkPipelineRenderingCreateInfo::`depthAttachmentFormat` or VkPipelineRenderingCreateInfo::`stencilAttachmentFormat` are not VK_FORMAT_UNDEFINED, and the `VK_EXT_extended_dynamic_state3` extension is not enabled or any of the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE, VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE, VK_DYNAMIC_STATE_DEPTH_COMPARE_OP, VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE, VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE, VK_DYNAMIC_STATE_STENCIL_OP, or VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic states are not set, `pDepthStencilState` **must** be a valid pointer to a valid VkPipelineDepthStencilStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-pDepthStencilState-09034
  If `pDepthStencilState` is not `NULL` it **must** be a valid pointer to a valid VkPipelineDepthStencilStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-renderPass-09035
  If `renderPass` is VK_NULL_HANDLE and the pipeline is being created with fragment shader state but not fragment output interface state, and the `VK_EXT_extended_dynamic_state3` extension is not enabled, or any of the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE, VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE, VK_DYNAMIC_STATE_DEPTH_COMPARE_OP, VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE, VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE, VK_DYNAMIC_STATE_STENCIL_OP, or VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic states are not set, `pDepthStencilState` **must** be a valid pointer to a valid VkPipelineDepthStencilStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-pDepthStencilState-09036
  If `pDepthStencilState` is not `NULL` it **must** be a valid pointer to a valid VkPipelineDepthStencilStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-renderPass-09037
  If `renderPass` is VK_NULL_HANDLE, the pipeline is being created with fragment output interface state, and any element of VkPipelineRenderingCreateInfo::`pColorAttachmentFormats` is not VK_FORMAT_UNDEFINED, and the `VK_EXT_extended_dynamic_state3` extension is not enabled, or any of the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT, VK_DYNAMIC_STATE_LOGIC_OP_EXT, VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT, VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT, VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, or VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic states are not set, `pColorBlendState` **must** be a valid pointer to a valid VkPipelineColorBlendStateCreateInfo structure

- VUID-VkGraphicsPipelineCreateInfo-pColorBlendState-09038
  If pColorBlendState is not NULL it **must** be a valid pointer to a valid VkPipelineColorBlendStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06055
  If renderPass is VK_NULL_HANDLE, the attachmentCount member of pColorBlendState is not ignored, and the pipeline is being created with fragment output interface state, pColorBlendState->attachmentCount **must** be equal to VkPipelineRenderingCreateInfo ::colorAttachmentCount
- VUID-VkGraphicsPipelineCreateInfo-renderPass-11504
  If renderPass is VK_NULL_HANDLE, a VkCustomResolveCreateInfoEXT structure is in the pNext chain, and the pipeline is being created with fragment output interface state, VkPipelineRenderingCreateInfo::colorAttachmentCount **must** be equal to VkCustomResolveCreateInfoEXT::colorAttachmentCount
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06057
  If renderPass is VK_NULL_HANDLE, the pipeline is being created with pre-rasterization shader state, VkPipelineRenderingCreateInfo::viewMask is not 0, and the multiviewTessellationShader feature is not enabled, then pStages **must** not include tessellation shaders
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06058
  If renderPass is VK_NULL_HANDLE, the pipeline is being created with pre-rasterization shader state, VkPipelineRenderingCreateInfo::viewMask is not 0, and the multiviewGeometryShader feature is not enabled, then pStages **must** not include a geometry shader
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06059
  If renderPass is VK_NULL_HANDLE, the pipeline is being created with pre-rasterization shader state, and VkPipelineRenderingCreateInfo::viewMask is not 0, all of the shaders in the pipeline **must** not include variables decorated with the Layer built-in decoration in their interfaces
- VUID-VkGraphicsPipelineCreateInfo-renderPass-07720
  If renderPass is VK_NULL_HANDLE, the pipeline is being created with pre-rasterization shader state, and VkPipelineRenderingCreateInfo::viewMask is not 0, and multiviewMeshShader is not enabled, then pStages **must** not include a mesh shader
- VUID-VkGraphicsPipelineCreateInfo-renderPass-12326
  If renderPass is VK_NULL_HANDLE, the pipeline is being created with pre-rasterization shader state, pStages includes a mesh shader, and VkPipelineRenderingCreateInfo ::viewMask is not 0, then the index of the most significant bit in viewMask **must** be less than maxMeshMultiviewViewCount
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06061
  If the dynamicRenderingLocalRead feature is not enabled, the pipeline requires fragment shader state, and renderPass is VK_NULL_HANDLE, fragment shaders in pStages **must** not include the InputAttachment capability
- VUID-VkGraphicsPipelineCreateInfo-renderPass-08710
  If the pipeline requires fragment shader state and renderPass is not VK_NULL_HANDLE, fragment shaders in pStages **must** not include any of the TileImageColorReadAccessEXT,

TileImageDepthReadAccessEXT, or TileImageStencilReadAccessEXT capabilities

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06062
  If the pipeline requires fragment output interface state and renderPass is VK_NULL_HANDLE, for each color attachment format defined by the pColorAttachmentFormats member of VkPipelineRenderingCreateInfo, if its potential format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the blendEnable member of the corresponding element of the pAttachments member of pColorBlendState **must** be VK_FALSE
- VUID-VkGraphicsPipelineCreateInfo-flags-06591
  If pStages includes a fragment shader stage, and the fragment shader declares the EarlyFragmentTests execution mode, the flags member of VkPipelineDepthStencilStateCreateInfo **must** not include VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT_EXT or VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_BIT_EXT
- VUID-VkGraphicsPipelineCreateInfo-flags-06482
  If the dynamicRenderingLocalRead feature is not enabled, the pipeline requires fragment output interface state, and the flags member of VkPipelineColorBlendStateCreateInfo includes VK_PIPELINE_COLOR_BLEND_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_ACCESS_BIT_EXT, renderPass **must** not be VK_NULL_HANDLE
- VUID-VkGraphicsPipelineCreateInfo-None-09526
  If the dynamicRenderingLocalRead feature is not enabled, the pipeline requires fragment output interface state, and the flags member of VkPipelineDepthStencilStateCreateInfo includes VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT_EXT or VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_BIT_EXT, renderPass **must** not be VK_NULL_HANDLE
- VUID-VkGraphicsPipelineCreateInfo-renderPass-09527
  If the pipeline requires fragment output interface state, renderPass is not VK_NULL_HANDLE, and the flags member of VkPipelineColorBlendStateCreateInfo includes VK_PIPELINE_COLOR_BLEND_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_ACCESS_BIT_EXT subpass **must** have been created with VK_SUBPASS_DESCRIPTION_RASTERIZATION_ORDER_ATTACHMENT_COLOR_ACCESS_BIT_EXT
- VUID-VkGraphicsPipelineCreateInfo-renderPass-09528
  If the pipeline requires fragment shader state, renderPass is not VK_NULL_HANDLE, and the flags member of VkPipelineDepthStencilStateCreateInfo includes VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT_EXT, subpass **must** have been created with VK_SUBPASS_DESCRIPTION_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT

_EXT

- VUID-VkGraphicsPipelineCreateInfo-renderPass-09529
  If the pipeline requires fragment shader state, `renderPass` is not VK_NULL_HANDLE, and the `flags` member of VkPipelineDepthStencilStateCreateInfo includes VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_BIT_EXT, `subpass` **must** have been created with VK_SUBPASS_DESCRIPTION_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_BIT_EXT
- VUID-VkGraphicsPipelineCreateInfo-pipelineStageCreationFeedbackCount-06594
  If VkPipelineCreationFeedbackCreateInfo::`pipelineStageCreationFeedbackCount` is not `0`, it **must** be equal to `stageCount`
- VUID-VkGraphicsPipelineCreateInfo-pStages-06600
  If the pipeline requires pre-rasterization shader state or fragment shader state, `pStages` **must** be a valid pointer to an array of `stageCount` valid VkPipelineShaderStageCreateInfo structures
- VUID-VkGraphicsPipelineCreateInfo-stageCount-09587
  If the pipeline does not require pre-rasterization shader state or fragment shader state, `stageCount` **must** be zero
- VUID-VkGraphicsPipelineCreateInfo-pRasterizationState-06601
  If the pipeline requires pre-rasterization shader state, and related dynamic state is not set, `pRasterizationState` **must** be a valid pointer to a valid VkPipelineRasterizationStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-pRasterizationState-09039
  If VkGraphicsPipelineLibraryCreateInfoEXT::`flags` includes VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT, and related dynamic state is not set, then `pMultisampleState` **must** be a valid pointer to a valid VkPipelineMultisampleStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-pRasterizationState-09040
  If `pRasterizationState` is not `NULL` it **must** be a valid pointer to a valid VkPipelineRasterizationStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-layout-06602
  If the pipeline requires fragment shader state or pre-rasterization shader state, and VkPipelineCreateFlags2CreateInfoKHR::`flags` does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, `layout` **must** be a valid VkPipelineLayout handle
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06603
  If the pipeline requires pre-rasterization shader state, fragment shader state, or fragment output state, and `renderPass` is not VK_NULL_HANDLE, `renderPass` **must** be a valid VkRenderPass handle
- VUID-VkGraphicsPipelineCreateInfo-stageCount-09530
  If the pipeline requires pre-rasterization shader state, `stageCount` **must** be greater than `0`
- VUID-VkGraphicsPipelineCreateInfo-graphicsPipelineLibrary-06606
  If the `graphicsPipelineLibrary` feature is not enabled, `flags` **must** not include

VK_PIPELINE_CREATE_LIBRARY_BIT_KHR

- VUID-VkGraphicsPipelineCreateInfo-flags-06608
  If the pipeline is being created with all possible state subsets, `flags` **must** not include VK_PIPELINE_CREATE_LIBRARY_BIT_KHR
- VUID-VkGraphicsPipelineCreateInfo-flags-06609
  If `flags` includes VK_PIPELINE_CREATE_LINK_TIME_OPTIMIZATION_BIT_EXT, pipeline libraries included via VkPipelineLibraryCreateInfoKHR **must** have been created with VK_PIPELINE_CREATE_RETAIN_LINK_TIME_OPTIMIZATION_INFO_BIT_EXT
- VUID-VkGraphicsPipelineCreateInfo-flags-09245
  If `flags` includes VK_PIPELINE_CREATE_RETAIN_LINK_TIME_OPTIMIZATION_INFO_BIT_EXT, `flags` **must** also include VK_PIPELINE_CREATE_LIBRARY_BIT_KHR
- VUID-VkGraphicsPipelineCreateInfo-flags-06610
  If `flags` includes VK_PIPELINE_CREATE_RETAIN_LINK_TIME_OPTIMIZATION_INFO_BIT_EXT, pipeline libraries included via VkPipelineLibraryCreateInfoKHR **must** have been created with VK_PIPELINE_CREATE_RETAIN_LINK_TIME_OPTIMIZATION_INFO_BIT_EXT
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06611
  Any pipeline libraries included via VkPipelineLibraryCreateInfoKHR::`pLibraries` **must** not include any state subset already defined by this structure or defined by any other pipeline library in VkPipelineLibraryCreateInfoKHR::`pLibraries`
- VUID-VkGraphicsPipelineCreateInfo-flags-06612
  If VkPipelineCreateFlags2CreateInfoKHR::`flags` does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, VkGraphicsPipelineLibraryCreateInfoEXT::`flags` includes only one of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, and an element of VkPipelineLibraryCreateInfoKHR::`pLibraries` includes the other flag, and `layout` was not created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT, then the `layout` used by this pipeline and the library **must** be *identically defined*
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06613
  If VkPipelineCreateFlags2CreateInfoKHR::`flags` does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, one element of VkPipelineLibraryCreateInfoKHR::`pLibraries` includes VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT and another element includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, and the `layout` specified by either library was not created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT, then the `layout` used by each library **must** be identically defined
- VUID-VkGraphicsPipelineCreateInfo-flags-06614
  If VkPipelineCreateFlags2CreateInfoKHR::`flags` does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, VkGraphicsPipelineLibraryCreateInfoEXT::`flags` includes only one of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, an element of

VkPipelineLibraryCreateInfoKHR::pLibraries includes the other subset, and layout was created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT, then the layout used by the library **must** also have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT

- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06615
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, one element of VkPipelineLibraryCreateInfoKHR::pLibraries includes VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT and another element includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, and the layout specified by either library was created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT, then the layout used by both libraries **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT
- VUID-VkGraphicsPipelineCreateInfo-flags-06616
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, VkGraphicsPipelineLibraryCreateInfoEXT::flags includes only one of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, an element of VkPipelineLibraryCreateInfoKHR::pLibraries includes the other subset, and layout was created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT, elements of the pSetLayouts array which layout was created with that are not VK_NULL_HANDLE **must** be identically defined to the element at the same index of pSetLayouts used to create the library's layout
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06617
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, one element of VkPipelineLibraryCreateInfoKHR::pLibraries includes VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT and another element includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, and the layout specified by either library was created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT, elements of the pSetLayouts array which either layout was created with that are not VK_NULL_HANDLE **must** be identically defined to the element at the same index of pSetLayouts used to create the other library's layout
- VUID-VkGraphicsPipelineCreateInfo-flags-06618
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, VkGraphicsPipelineLibraryCreateInfoEXT::flags includes only one of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, and an element of VkPipelineLibraryCreateInfoKHR::pLibraries includes the other flag, any descriptor set layout *N* specified by layout in both this pipeline and the library which include bindings accessed by shader stages in each **must** be *identically defined*
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06619

If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, one element of VkPipelineLibraryCreateInfoKHR::pLibraries includes VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT and another element includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, any descriptor set layout *N* specified by layout in both libraries which include bindings accessed by shader stages in each **must** be identically defined

- VUID-VkGraphicsPipelineCreateInfo-flags-06620
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, VkGraphicsPipelineLibraryCreateInfoEXT::flags includes only one of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, and an element of VkPipelineLibraryCreateInfoKHR::pLibraries includes the other flag, push constants specified in layout in both this pipeline and the library which are available to shader stages in each **must** be identically defined
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06621
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, and one element of VkPipelineLibraryCreateInfoKHR::pLibraries includes VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT and another element includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, push constants specified in layout in both this pipeline and the library which are available to shader stages in each **must** be identically defined
- VUID-VkGraphicsPipelineCreateInfo-flags-06679
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, and VkGraphicsPipelineLibraryCreateInfoEXT::flags includes only one of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, an element of VkPipelineLibraryCreateInfoKHR::pLibraries includes the other subset, VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT was not used, and any element of the pSetLayouts array when layout was created and the corresponding element of the pSetLayouts array used to create the library's layout **must** not both be VK_NULL_HANDLE
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06681
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, one element of VkPipelineLibraryCreateInfoKHR::pLibraries includes VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT and another element includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT was not used, and any element of the pSetLayouts array used to create each library's layout was VK_NULL_HANDLE, then the corresponding element of the pSetLayouts array used to create the other library's layout **must** not be VK_NULL_HANDLE
- VUID-VkGraphicsPipelineCreateInfo-flags-06756

If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, VkGraphicsPipelineLibraryCreateInfoEXT::flags includes only one of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, an element of VkPipelineLibraryCreateInfoKHR::pLibraries includes the other subset, and any element of the pSetLayouts array which layout was created with was VK_NULL_HANDLE, then the corresponding element of the pSetLayouts array used to create the library's layout **must** not have shader bindings for shaders in the other subset

- VUID-VkGraphicsPipelineCreateInfo-flags-06757
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, VkGraphicsPipelineLibraryCreateInfoEXT::flags includes only one of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, an element of VkPipelineLibraryCreateInfoKHR::pLibraries includes the other subset, and any element of the pSetLayouts array used to create the library's layout was VK_NULL_HANDLE, then the corresponding element of the pSetLayouts array used to create this pipeline's layout **must** not have shader bindings for shaders in the other subset
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06758
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, one element of VkPipelineLibraryCreateInfoKHR::pLibraries includes VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT and another element includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, and any element of the pSetLayouts array used to create each library's layout was VK_NULL_HANDLE, then the corresponding element of the pSetLayouts array used to create the other library's layout **must** not have shader bindings for shaders in the other subset
- VUID-VkGraphicsPipelineCreateInfo-flags-06682
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, and VkGraphicsPipelineLibraryCreateInfoEXT::flags includes both VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT and VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, layout **must** have been created with no elements of the pSetLayouts array set to VK_NULL_HANDLE
- VUID-VkGraphicsPipelineCreateInfo-flags-06683
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, VkGraphicsPipelineLibraryCreateInfoEXT::flags includes VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT, and pRasterizationState->rasterizerDiscardEnable is VK_TRUE, layout **must** have been created with no elements of the pSetLayouts array set to VK_NULL_HANDLE
- VUID-VkGraphicsPipelineCreateInfo-flags-06684
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags includes at least one of and no more than two of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT,

VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT, and an element of VkPipelineLibraryCreateInfoKHR::pLibraries includes one of the other flags, the value of subpass **must** be equal to that used to create the library

- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06623
  If one element of VkPipelineLibraryCreateInfoKHR::pLibraries includes at least one of and no more than two of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT, VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT, and another element of VkPipelineLibraryCreateInfoKHR::pLibraries includes one of the other flags, the value of subpass used to create each library **must** be identical
- VUID-VkGraphicsPipelineCreateInfo-renderpass-06624
  If renderpass is not VK_NULL_HANDLE, VkGraphicsPipelineLibraryCreateInfoEXT::flags includes at least one of and no more than two of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT, VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT, and an element of VkPipelineLibraryCreateInfoKHR::pLibraries includes one of the other flags, renderPass **must** be compatible with that used to create the library
- VUID-VkGraphicsPipelineCreateInfo-renderpass-06625
  If renderpass is VK_NULL_HANDLE, VkGraphicsPipelineLibraryCreateInfoEXT::flags includes at least one of and no more than two of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT, VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT, and an element of VkPipelineLibraryCreateInfoKHR::pLibraries includes one of the other flags, the value of renderPass used to create that library **must** also be VK_NULL_HANDLE
- VUID-VkGraphicsPipelineCreateInfo-flags-06626
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags includes either VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, and the other element is included in VkPipelineLibraryCreateInfoKHR::pLibraries, and renderPass is VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::viewMask used by this pipeline and that specified by the library **must** be identical
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06627
  If VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT and VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT are included in two elements of VkPipelineLibraryCreateInfoKHR::pLibraries, and renderPass was VK_NULL_HANDLE for both libraries, the value of VkPipelineRenderingCreateInfo::viewMask set by each library **must** be identical
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06628
  If one element of VkPipelineLibraryCreateInfoKHR::pLibraries includes at least one of and no more than two of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT,

VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT, and another element of VkPipelineLibraryCreateInfoKHR::pLibraries includes one of the other flags, the renderPass objects used to create each library **must** be compatible or all equal to VK_NULL_HANDLE

- VUID-VkGraphicsPipelineCreateInfo-renderpass-06631
  If renderPass is not VK_NULL_HANDLE, the pipeline requires fragment shader state, and the VK_EXT_extended_dynamic_state3 extension is not enabled or any of the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT, VK_DYNAMIC_STATE_SAMPLE_MASK_EXT, or VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic states is not set, or the alphaToOne feature is enabled and VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT is not set, then pMultisampleState **must** be a valid pointer to a valid VkPipelineMultisampleStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-Input-06632
  If the pipeline requires fragment shader state with a fragment shader that either enables sample shading or decorates any variable in the Input storage class with Sample, and the VK_EXT_extended_dynamic_state3 extension is not enabled or any of the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT, VK_DYNAMIC_STATE_SAMPLE_MASK_EXT, or VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic states is not set, or the alphaToOne feature is enabled and VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT is not set, then pMultisampleState **must** be a valid pointer to a valid VkPipelineMultisampleStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-flags-11856
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags includes either VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, VkCustomResolveCreateInfoEXT is included and VkCustomResolveCreateInfoEXT::customResolve is VK_TRUE, and an element of VkPipelineLibraryCreateInfoKHR::pLibraries also includes either VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, the library **must** also include VkCustomResolveCreateInfoEXT and the VkCustomResolveCreateInfoEXT::customResolve specified by the library **must** be VK_TRUE
- VUID-VkGraphicsPipelineCreateInfo-flags-11857
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags includes either VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, VkCustomResolveCreateInfoEXT is not included or VkCustomResolveCreateInfoEXT::customResolve is VK_FALSE, and an element of VkPipelineLibraryCreateInfoKHR::pLibraries also includes either VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, either the library **must** not include VkCustomResolveCreateInfoEXT or the VkCustomResolveCreateInfoEXT::customResolve specified by the library **must** be

VK_FALSE

- VUID-VkGraphicsPipelineCreateInfo-customResolve-11858
  If one element of VkPipelineLibraryCreateInfoKHR includes either VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, VkCustomResolveCreateInfoEXT is included and VkCustomResolveCreateInfoEXT::customResolve is VK_TRUE, and another element of VkPipelineLibraryCreateInfoKHR::pLibraries also includes either VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, the other library **must** also include VkCustomResolveCreateInfoEXT and the VkCustomResolveCreateInfoEXT::customResolve specified by the library **must** be VK_TRUE
- VUID-VkGraphicsPipelineCreateInfo-customResolve-11859
  If one element of VkPipelineLibraryCreateInfoKHR includes either VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, VkCustomResolveCreateInfoEXT is not included or VkCustomResolveCreateInfoEXT::customResolve is VK_FALSE, and another element of VkPipelineLibraryCreateInfoKHR::pLibraries also includes either VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, either the other library **must** not include VkCustomResolveCreateInfoEXT or the VkCustomResolveCreateInfoEXT::customResolve specified by the library **must** be VK_FALSE
- VUID-VkGraphicsPipelineCreateInfo-flags-06633
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT with a pMultisampleState that was not NULL, and an element of VkPipelineLibraryCreateInfoKHR::pLibraries was created with VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT, pMultisampleState **must** be identically defined to that used to create the library
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06634
  If an element of VkPipelineLibraryCreateInfoKHR::pLibraries was created with VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT with a pMultisampleState that was not NULL, and if VkGraphicsPipelineLibraryCreateInfoEXT::flags includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT, pMultisampleState **must** be identically defined to that used to create the library
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06635
  If one element of VkPipelineLibraryCreateInfoKHR::pLibraries was created with VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT with a pMultisampleState that was not NULL, and if a different element of VkPipelineLibraryCreateInfoKHR::pLibraries was created with VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT, the pMultisampleState used to create each library **must** be identically defined

- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06636
  If one element of VkPipelineLibraryCreateInfoKHR::pLibraries was created with VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT and a value of pMultisampleState->sampleShadingEnable equal to VK_TRUE, and if a different element of VkPipelineLibraryCreateInfoKHR::pLibraries was created with VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, the pMultisampleState used to create each library **must** be identically defined
- VUID-VkGraphicsPipelineCreateInfo-flags-06637
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT, pMultisampleState->sampleShadingEnable is VK_TRUE, and an element of VkPipelineLibraryCreateInfoKHR::pLibraries was created with VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, pMultisampleState **must** be identically defined to that used to create the library
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-09567
  If one element of VkPipelineLibraryCreateInfoKHR::pLibraries was created with VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT and a value of pMultisampleState->sampleShadingEnable equal to VK_TRUE, and if VkGraphicsPipelineLibraryCreateInfoEXT::flags includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, pMultisampleState **must** be identically defined to that used to create the library
- VUID-VkGraphicsPipelineCreateInfo-flags-06638
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags includes only one of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, and an element of VkPipelineLibraryCreateInfoKHR::pLibraries includes the other flag, values specified in VkPipelineFragmentShadingRateStateCreateInfoKHR for both this pipeline and that library **must** be identical
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06639
  If one element of VkPipelineLibraryCreateInfoKHR::pLibraries includes VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT and another element includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, values specified in VkPipelineFragmentShadingRateStateCreateInfoKHR for both this pipeline and that library **must** be identical
- VUID-VkGraphicsPipelineCreateInfo-flags-06640
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags includes VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, pStages **must** be a valid pointer to an array of stageCount valid VkPipelineShaderStageCreateInfo structures
- VUID-VkGraphicsPipelineCreateInfo-flags-06642
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, VkGraphicsPipelineLibraryCreateInfoEXT::flags includes VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, layout **must** be a valid

VkPipelineLayout handle

- VUID-VkGraphicsPipelineCreateInfo-flags-06643
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags includes VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT, VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT, and renderPass is not VK_NULL_HANDLE, renderPass **must** be a valid VkRenderPass handle
- VUID-VkGraphicsPipelineCreateInfo-flags-06644
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags includes VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT or VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, stageCount **must** be greater than 0
- VUID-VkGraphicsPipelineCreateInfo-flags-06645
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags is non-zero, if flags includes VK_PIPELINE_CREATE_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR, any libraries **must** have also been created with VK_PIPELINE_CREATE_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06646
  If VkPipelineLibraryCreateInfoKHR::pLibraries includes more than one library, and any library was created with VK_PIPELINE_CREATE_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR, all libraries **must** have also been created with VK_PIPELINE_CREATE_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-06647
  If VkPipelineLibraryCreateInfoKHR::pLibraries includes at least one library, VkGraphicsPipelineLibraryCreateInfoEXT::flags is non-zero, and any library was created with VK_PIPELINE_CREATE_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR, flags **must** include VK_PIPELINE_CREATE_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR
- VUID-VkGraphicsPipelineCreateInfo-None-07826
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, the pipeline includes a complete set of state, and there are no libraries included in VkPipelineLibraryCreateInfoKHR::pLibraries, then VkPipelineLayout **must** be a valid pipeline layout
- VUID-VkGraphicsPipelineCreateInfo-layout-07827
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, the pipeline includes a complete set of state specified entirely by libraries, and each library was created with a VkPipelineLayout created without VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT, then layout **must** be compatible with the layouts in those libraries
- VUID-VkGraphicsPipelineCreateInfo-flags-06729
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, flags includes VK_PIPELINE_CREATE_LINK_TIME_OPTIMIZATION_BIT_EXT, the pipeline includes a complete set of state specified entirely by libraries, and each library was created with a

VkPipelineLayout created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT, then `layout` **must** be compatible with the union of the libraries' pipeline layouts other than the inclusion/exclusion of VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT

- VUID-VkGraphicsPipelineCreateInfo-flags-06730
  If VkPipelineCreateFlags2CreateInfoKHR::`flags` does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, `flags` does not include VK_PIPELINE_CREATE_LINK_TIME_OPTIMIZATION_BIT_EXT, the pipeline includes a complete set of state specified entirely by libraries, and each library was created with a VkPipelineLayout created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT, then `layout` **must** be compatible with the union of the libraries' pipeline layouts
- VUID-VkGraphicsPipelineCreateInfo-conservativePointAndLineRasterization-08892
  If `conservativePointAndLineRasterization` is not supported and the effective rasterization input topology is in line or point topology class, then VkPipelineRasterizationConservativeStateCreateInfoEXT::`conservativeRasterizationMode` **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-VkGraphicsPipelineCreateInfo-pStages-06894
  If the pipeline requires pre-rasterization shader state but not fragment shader state, elements of `pStages` **must** not have `stage` set to VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-VkGraphicsPipelineCreateInfo-pStages-06895
  If the pipeline requires fragment shader state but not pre-rasterization shader state, elements of `pStages` **must** not have `stage` set to a shader stage which participates in pre-rasterization
- VUID-VkGraphicsPipelineCreateInfo-pStages-06896
  If the pipeline requires pre-rasterization shader state, all elements of `pStages` **must** have a `stage` set to a shader stage which participates in fragment shader state or pre-rasterization shader state
- VUID-VkGraphicsPipelineCreateInfo-stage-06897
  If the pipeline requires fragment shader state and/or pre-rasterization shader state, any value of `stage` **must** not be set in more than one element of `pStages`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3TessellationDomainOrigin-07370
  If the `extendedDynamicState3TessellationDomainOrigin` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3DepthClampEnable-07371
  If the `extendedDynamicState3DepthClampEnable` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3PolygonMode-07372
  If the `extendedDynamicState3PolygonMode` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to VK_DYNAMIC_STATE_POLYGON_MODE_EXT

- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3RasterizationSamples-07373
  If the extendedDynamicState3RasterizationSamples feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3SampleMask-07374
  If the extendedDynamicState3SampleMask feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_SAMPLE_MASK_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3AlphaToCoverageEnable-07375
  If the extendedDynamicState3AlphaToCoverageEnable feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3AlphaToOneEnable-07376
  If the extendedDynamicState3AlphaToOneEnable feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3LogicOpEnable-07377
  If the extendedDynamicState3LogicOpEnable feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ColorBlendEnable-07378
  If the extendedDynamicState3ColorBlendEnable feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ColorBlendEquation-07379
  If the extendedDynamicState3ColorBlendEquation feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ColorWriteMask-07380
  If the extendedDynamicState3ColorWriteMask feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3RasterizationStream-07381
  If the extendedDynamicState3RasterizationStream feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ConservativeRasterizationMode-07382
  If the extendedDynamicState3ConservativeRasterizationMode feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ExtraPrimitiveOverestimationSize-07383

If the extendedDynamicState3ExtraPrimitiveOverestimationSize feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT

- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-09639
  If the pipeline requires pre-rasterization shader state, pDynamicState includes VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT, and pDynamicState does not include VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT, pRasterizationState **must** include a VkPipelineRasterizationConservativeStateCreateInfoEXT in its pNext chain
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3DepthClipEnable-07384
  If the extendedDynamicState3DepthClipEnable feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3SampleLocationsEnable-07385
  If the extendedDynamicState3SampleLocationsEnable feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ColorBlendAdvanced-07386
  If the extendedDynamicState3ColorBlendAdvanced feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ProvokingVertexMode-07387
  If the extendedDynamicState3ProvokingVertexMode feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3LineRasterizationMode-07388
  If the extendedDynamicState3LineRasterizationMode feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3LineStippleEnable-07389
  If the extendedDynamicState3LineStippleEnable feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3DepthClipNegativeOneToOne-07390
  If the extendedDynamicState3DepthClipNegativeOneToOne feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ViewportWScalingEnable-07391
  If the extendedDynamicState3ViewportWScalingEnable feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to

VK_DYNAMIC_STATE_VIEWPORT_W_SCALING_ENABLE_NV

- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ViewportSwizzle-07392
  If the extendedDynamicState3ViewportSwizzle feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_VIEWPORT_SWIZZLE_NV
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3CoverageToColorEnable-07393
  If the extendedDynamicState3CoverageToColorEnable feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_COVERAGE_TO_COLOR_ENABLE_NV
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3CoverageToColorLocation-07394
  If the extendedDynamicState3CoverageToColorLocation feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_COVERAGE_TO_COLOR_LOCATION_NV
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3CoverageModulationMode-07395
  If the extendedDynamicState3CoverageModulationMode feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_COVERAGE_MODULATION_MODE_NV
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3CoverageModulationTableEnable-07396
  If the extendedDynamicState3CoverageModulationTableEnable feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_COVERAGE_MODULATION_TABLE_ENABLE_NV
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3CoverageModulationTable-07397
  If the extendedDynamicState3CoverageModulationTable feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_COVERAGE_MODULATION_TABLE_NV
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3CoverageReductionMode-07398
  If the extendedDynamicState3CoverageReductionMode feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_COVERAGE_REDUCTION_MODE_NV
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3RepresentativeFragmentTestEnable-07399
  If the extendedDynamicState3RepresentativeFragmentTestEnable feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to VK_DYNAMIC_STATE_REPRESENTATIVE_FRAGMENT_TEST_ENABLE_NV
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ShadingRateImageEnable-07400
  If the extendedDynamicState3ShadingRateImageEnable feature is not enabled, there **must** be no element of the pDynamicStates member of pDynamicState set to

VK_DYNAMIC_STATE_SHADING_RATE_IMAGE_ENABLE_NV

- VUID-VkGraphicsPipelineCreateInfo-flags-07401
  `flags` **must** not include VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR
- VUID-VkGraphicsPipelineCreateInfo-pLibraries-11594
  If `flags` includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR, each element of VkPipelineLibraryCreateInfoKHR::`pLibraries` **must** have been created with the VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR bit set
- VUID-VkGraphicsPipelineCreateInfo-flags-11595
  If the `VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap` feature is not enabled, `flags` **must** not include VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-VkGraphicsPipelineCreateInfo-pStages-08711
  If `pStages` includes a fragment shader stage, VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE is not set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` member of VkPipelineDepthStencilStateCreateInfo **must** be VK_FALSE
- VUID-VkGraphicsPipelineCreateInfo-pStages-08712
  If `pStages` includes a fragment shader stage, VK_DYNAMIC_STATE_STENCIL_WRITE_MASK is not set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the value of VkStencilOpState::`writeMask` for both `front` and `back` in VkPipelineDepthStencilStateCreateInfo **must** be `0`
- VUID-VkGraphicsPipelineCreateInfo-flags-08897
  If VkGraphicsPipelineLibraryCreateInfoEXT::`flags` includes VK_GRAPHICS_PIPELINE_LIBRARY_VERTEX_INPUT_INTERFACE_BIT_EXT, pre-rasterization shader state is specified either in a library or by the inclusion of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT, and that state includes a vertex shader stage in `pStages`, the pipeline **must** define vertex input state
- VUID-VkGraphicsPipelineCreateInfo-flags-08898
  If VkGraphicsPipelineLibraryCreateInfoEXT::`flags` includes VK_GRAPHICS_PIPELINE_LIBRARY_VERTEX_INPUT_INTERFACE_BIT_EXT, and pre-rasterization shader state is not specified, the pipeline **must** define vertex input state
- VUID-VkGraphicsPipelineCreateInfo-flags-08899
  If `flags` does not include VK_PIPELINE_CREATE_LIBRARY_BIT_KHR, pre-rasterization shader state is specified either in a library or by the inclusion of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT, and that state includes a vertex shader stage in `pStages`, the pipeline **must** either define vertex input state or include that state in a linked pipeline library

- VUID-VkGraphicsPipelineCreateInfo-flags-08900
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags includes VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT the pipeline **must** define pre-rasterization shader state
- VUID-VkGraphicsPipelineCreateInfo-flags-08901
  If flags does not include VK_PIPELINE_CREATE_LIBRARY_BIT_KHR, the pipeline **must** either define pre-rasterization shader state or include that state in a linked pipeline library
- VUID-VkGraphicsPipelineCreateInfo-flags-08903
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, pre-rasterization shader state is specified either in a library or by the inclusion of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT, and that state either includes VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE or has pRasterizationState->rasterizerDiscardEnable set to VK_FALSE, the pipeline **must** define fragment shader state
- VUID-VkGraphicsPipelineCreateInfo-flags-08904
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT, and pre-rasterization shader state is not specified, the pipeline **must** define fragment shader state
- VUID-VkGraphicsPipelineCreateInfo-flags-08906
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT, pre-rasterization shader state is specified either in a library or by the inclusion of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT, and that state either includes VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE or has pRasterizationState->rasterizerDiscardEnable set to VK_FALSE, the pipeline **must** define fragment output interface state
- VUID-VkGraphicsPipelineCreateInfo-flags-08907
  If VkGraphicsPipelineLibraryCreateInfoEXT::flags includes VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT, and pre-rasterization shader state is not specified, the pipeline **must** define fragment output interface state
- VUID-VkGraphicsPipelineCreateInfo-flags-08909
  If flags does not include VK_PIPELINE_CREATE_LIBRARY_BIT_KHR, pre-rasterization shader state is specified either in a library or by the inclusion of VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT, and that state either includes VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE or has pRasterizationState->rasterizerDiscardEnable set to VK_FALSE, the pipeline **must** define fragment output interface state and fragment shader state or include those states in linked pipeline libraries
- VUID-VkGraphicsPipelineCreateInfo-None-09043
  If pDynamicState->pDynamicStates does not include VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the colorWriteMask member of the

corresponding element of pColorBlendState->pAttachments **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them

- VUID-VkGraphicsPipelineCreateInfo-renderPass-09531
  If the pipeline is being created with fragment shader state and fragment output state, the value of renderPass is VK_NULL_HANDLE, and VkRenderingInputAttachmentIndexInfo is included, VkRenderingInputAttachmentIndexInfo::colorAttachmentCount **must** be equal to VkPipelineRenderingCreateInfo::colorAttachmentCount
- VUID-VkGraphicsPipelineCreateInfo-renderPass-09652
  If the pipeline is being created with fragment shader state and fragment output state, the value of renderPass is VK_NULL_HANDLE, and VkRenderingInputAttachmentIndexInfo is not included, the fragment shader **must** not contain any input attachments with a InputAttachmentIndex greater than or equal to VkPipelineRenderingCreateInfo::colorAttachmentCount
- VUID-VkGraphicsPipelineCreateInfo-renderPass-09532
  If the pipeline is being created with fragment output state, and the value of renderPass is VK_NULL_HANDLE, VkRenderingAttachmentLocationInfo::colorAttachmentCount **must** be equal to VkPipelineRenderingCreateInfo::colorAttachmentCount
- VUID-VkGraphicsPipelineCreateInfo-flags-11273
  If VkPipelineCreateFlags2CreateInfoKHR::flags includes VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, all libraries linked to this pipeline **must** also have that flag set
- VUID-VkGraphicsPipelineCreateInfo-flags-11274
  If VkPipelineCreateFlags2CreateInfoKHR::flags does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, all libraries linked to this pipeline **must** also not have that flag set
- VUID-VkGraphicsPipelineCreateInfo-flags-12355
  If flags includes VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, all libraries linked to this pipeline **must** also have that flag set
- VUID-VkGraphicsPipelineCreateInfo-flags-12356
  If flags does not include VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, all libraries linked to this pipeline **must** also not have that flag set
- VUID-VkGraphicsPipelineCreateInfo-flags-12357
  If flags includes VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT, all libraries linked to this pipeline **must** also have that flag set
- VUID-VkGraphicsPipelineCreateInfo-flags-12358
  If flags does not include VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT, all libraries linked to this pipeline **must** also not have that flag set
- VUID-VkGraphicsPipelineCreateInfo-flags-12359
  If flags includes VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT, all libraries linked to this pipeline **must** also have that flag set
- VUID-VkGraphicsPipelineCreateInfo-flags-12360
  If flags does not include VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT, all

libraries linked to this pipeline **must** also not have that flag set

Valid Usage (Implicit)

- VUID-VkGraphicsPipelineCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_GRAPHICS_PIPELINE_CREATE_INFO
- VUID-VkGraphicsPipelineCreateInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkCustomResolveCreateInfoEXT, VkGraphicsPipelineLibraryCreateInfoEXT, VkPipelineBinaryInfoKHR, VkPipelineCreateFlags2CreateInfo, VkPipelineCreationFeedbackCreateInfo, VkPipelineDiscardRectangleStateCreateInfoEXT, VkPipelineFragmentShadingRateStateCreateInfoKHR, VkPipelineLibraryCreateInfoKHR, VkPipelineRenderingCreateInfo, VkPipelineRobustnessCreateInfo, VkRenderingAttachmentLocationInfo, VkRenderingInputAttachmentIndexInfo, or VkValidationFeaturesEXT
- VUID-VkGraphicsPipelineCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-parameter
  If `pDynamicState` is not `NULL`, `pDynamicState` **must** be a valid pointer to a valid VkPipelineDynamicStateCreateInfo structure
- VUID-VkGraphicsPipelineCreateInfo-commonparent
  Each of `basePipelineHandle`, `layout`, and `renderPass` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Structure Chaining

Extends the structure

- VkPipelineCreateInfoKHR

The `VkPipelineRenderingCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPipelineRenderingCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           viewMask;
    uint32_t           colorAttachmentCount;
    const VkFormat*    pColorAttachmentFormats;
    VkFormat           depthAttachmentFormat;
    VkFormat           stencilAttachmentFormat;
} VkPipelineRenderingCreateInfo;
```

```
// Provided by VK_KHR_dynamic_rendering
// Equivalent to VkPipelineRenderingCreateInfo
typedef VkPipelineRenderingCreateInfo VkPipelineRenderingCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `viewMask` is a bitfield of view indices describing which views are active during rendering. It **must** match VkRenderingInfo::`viewMask` when rendering.
- `colorAttachmentCount` is the number of entries in `pColorAttachmentFormats`.
- `pColorAttachmentFormats` is a pointer to an array of VkFormat values defining the format of color attachments used in this pipeline.
- `depthAttachmentFormat` is a VkFormat value defining the format of the depth attachment used in this pipeline.
- `stencilAttachmentFormat` is a VkFormat value defining the format of the stencil attachment used in this pipeline.

When a pipeline is created without a VkRenderPass, if the `pNext` chain of VkGraphicsPipelineCreateInfo includes this structure, it specifies the view mask and format of attachments used for rendering. If this structure is not specified, and the pipeline does not include a VkRenderPass, `viewMask` and `colorAttachmentCount` are `0`, and `depthAttachmentFormat` and `stencilAttachmentFormat` are VK_FORMAT_UNDEFINED. If a graphics pipeline is created with a valid VkRenderPass, parameters of this structure are ignored.

If `depthAttachmentFormat`, `stencilAttachmentFormat`, or any element of `pColorAttachmentFormats` is VK_FORMAT_UNDEFINED, it indicates that the corresponding attachment is unused within the render pass. Valid formats indicate that an attachment **can** be used - but it is still valid to set the attachment to `NULL` when beginning rendering.

Valid Usage

- VUID-VkPipelineRenderingCreateInfo-colorAttachmentCount-09533
  `colorAttachmentCount` **must** be less than or equal to `maxColorAttachments`

Valid Usage (Implicit)

- VUID-VkPipelineRenderingCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_RENDERING_CREATE_INFO

Structure Chaining

Extends the structure

- VkGraphicsPipelineCreateInfo

The VkCustomResolveCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_custom_resolve with VK_KHR_dynamic_rendering or VK_VERSION_1_3
typedef struct VkCustomResolveCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           customResolve;
    uint32_t           colorAttachmentCount;
    const VkFormat*    pColorAttachmentFormats;
    VkFormat           depthAttachmentFormat;
    VkFormat           stencilAttachmentFormat;
} VkCustomResolveCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- customResolve indicates whether this pipeline will be used for a resolve operation.
- colorAttachmentCount is the number of entries in pColorAttachmentFormats.
- pColorAttachmentFormats is a pointer to an array of VkFormat values defining the format of color resolve attachments used in custom resolves in the same render pass.
- depthAttachmentFormat is a VkFormat value defining the format of the depth resolve attachment used in custom resolves in the same render pass.
- stencilAttachmentFormat is a VkFormat value defining the format of the stencil resolve attachment used in custom resolves in the same render pass.

If the pNext chain includes this structure for one of:

- a VkGraphicsPipelineCreateInfo for a pipeline created without a VkRenderPass
- a VkCommandBufferInheritanceInfo for a secondary command buffer within a render pass instance begun with vkCmdBeginRendering.

it specifies the formats used in custom resolves within the same render pass. It also specifies that the corresponding object will be used in a render pass which contains a custom resolve operation.

If the pNext chain includes this structure for a VkShaderCreateInfoEXT for a fragment shader object, it only specifies that the fragment shader will be used in a custom resolve operation.

If a graphics pipeline is created with a valid VkRenderPass, parameters of this structure are ignored.

If customResolve is VK_FALSE, the pipeline **can** only be used outside the custom resolve section. If customResolve is VK_TRUE, the pipeline **can** only be used inside the custom resolve section.

When a dynamic render pass instance contains a custom resolve operation and the dynamicRenderingUnusedAttachments feature is not enabled , all pipelines used to draw in such a render pass **must** include this structure and have identical format information in it. When a dynamic render pass does not contain a custom resolve operation and the

dynamicRenderingUnusedAttachments feature is not enabled , all pipelines used to draw in such a render pass **must** not include this structure.

If the dynamicRenderingUnusedAttachments feature is enabled, then when this structure is not included in the pNext chain for VkGraphicsPipelineCreateInfo, customResolve is VK_FALSE, colorAttachmentCount is 0, and depthAttachmentFormat and stencilAttachmentFormat are VK_FORMAT_UNDEFINED.

If depthAttachmentFormat, stencilAttachmentFormat, or any element of pColorAttachmentFormats is VK_FORMAT_UNDEFINED, it indicates that the corresponding attachment is unused within the resolve portion of the render pass. Valid formats indicate that an attachment **can** be used - but it is still valid to set the attachment to NULL when beginning rendering.

When passed as a pNext member to a VkShaderCreateInfoEXT struct for use with fragment density maps, the colorAttachmentCount, pColorAttachmentFormats, depthAttachmentFormat, and stencilAttachmentFormat members of this struct are ignored. When not passed as a pNext member, customResolve is VK_FALSE.

Valid Usage

- VUID-VkCustomResolveCreateInfoEXT-colorAttachmentCount-11507
  colorAttachmentCount **must** be less than or equal to maxColorAttachments
- VUID-VkCustomResolveCreateInfoEXT-depthAttachmentFormat-11508
  If depthAttachmentFormat is not VK_FORMAT_UNDEFINED, it **must** be a format that includes a depth component
- VUID-VkCustomResolveCreateInfoEXT-depthAttachmentFormat-11509
  If depthAttachmentFormat is not VK_FORMAT_UNDEFINED, it **must** be a format with potential format features that include VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkCustomResolveCreateInfoEXT-pColorAttachmentFormats-11510
  If any element of pColorAttachmentFormats is not VK_FORMAT_UNDEFINED, it **must** be a format with potential format features that include VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT
- VUID-VkCustomResolveCreateInfoEXT-stencilAttachmentFormat-11511
  If stencilAttachmentFormat is not VK_FORMAT_UNDEFINED, it **must** be a format that includes a stencil aspect
- VUID-VkCustomResolveCreateInfoEXT-stencilAttachmentFormat-11512
  If stencilAttachmentFormat is not VK_FORMAT_UNDEFINED, it **must** be a format with potential format features that include VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkCustomResolveCreateInfoEXT-depthAttachmentFormat-11513
  If depthAttachmentFormat is not VK_FORMAT_UNDEFINED and stencilAttachmentFormat is not VK_FORMAT_UNDEFINED, depthAttachmentFormat **must** equal stencilAttachmentFormat

Valid Usage (Implicit)

- VUID-VkCustomResolveCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_CUSTOM_RESOLVE_CREATE_INFO_EXT

Structure Chaining

Extends the structures

- VkCommandBufferInheritanceInfo
- VkGraphicsPipelineCreateInfo
- VkShaderCreateInfoEXT

The `VkPipelineBinaryInfoKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineBinaryInfoKHR {
    VkStructureType               sType;
    const void*                   pNext;
    uint32_t                      binaryCount;
    const VkPipelineBinaryKHR*    pPipelineBinaries;
} VkPipelineBinaryInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `binaryCount` is the number of elements in the `pPipelineBinaries` array.
- `pPipelineBinaries` is a pointer to an array of VkPipelineBinaryKHR handles.

If a `VkPipelineBinaryInfoKHR` structure with a `binaryCount` greater than 0 is included in the `pNext` chain of any `Vk*PipelineCreateInfo` structure when creating a pipeline, implementations **must** use the data in `pPipelineBinaries` instead of recalculating it. Any shader module identifiers, shader modules, or chained VkShaderModuleCreateInfo structures declared in VkPipelineShaderStageCreateInfo instances are ignored. Any VkShaderDescriptorSetAndBindingMappingInfoEXT in the `pNext` chains of VkPipelineShaderStageCreateInfo instances are ignored.

If this structure is not included in the `pNext` chain, it is equivalent to specifying this structure with a `binaryCount` of `0`.

Valid Usage

- VUID-VkPipelineBinaryInfoKHR-binaryCount-09603
  `binaryCount` and the order of the elements in `pPipelineBinaries` **must** exactly match that returned by vkCreatePipelineBinariesKHR for the matching `Vk*PipelineCreateInfo`

structure and its pNext chain, ignoring the presence of the VkPipelineBinaryInfoKHR structure, the presence of the VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR flag, and absence of any shader module identifiers, shader modules, or VkShaderModuleCreateInfo structures, for the same global pipeline key, from either:

- VkPipelineBinaryCreateInfoKHR::pPipelineCreateInfo, or
- VkPipelineBinaryCreateInfoKHR::pipeline

Valid Usage (Implicit)

- VUID-VkPipelineBinaryInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PIPELINE_BINARY_INFO_KHR
- VUID-VkPipelineBinaryInfoKHR-pPipelineBinaries-parameter
  If binaryCount is not 0, pPipelineBinaries **must** be a valid pointer to an array of binaryCount valid VkPipelineBinaryKHR handles

Structure Chaining

Extends the structures

- VkComputePipelineCreateInfo
- VkGraphicsPipelineCreateInfo
- VkRayTracingPipelineCreateInfoKHR

The VkGraphicsPipelineLibraryCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_graphics_pipeline_library
typedef struct VkGraphicsPipelineLibraryCreateInfoEXT {
    VkStructureType                       sType;
    const void*                           pNext;
    VkGraphicsPipelineLibraryFlagsEXT     flags;
} VkGraphicsPipelineLibraryCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask of VkGraphicsPipelineLibraryFlagBitsEXT specifying the subsets of the graphics pipeline that are being compiled.

If a VkGraphicsPipelineLibraryCreateInfoEXT structure is included in the pNext chain of VkGraphicsPipelineCreateInfo, it specifies the subsets of the graphics pipeline being created, excluding any subsets from linked pipeline libraries. If the pipeline is created with pipeline libraries, state from those libraries is aggregated with said subset.

If this structure is omitted, and either VkGraphicsPipelineCreateInfo::flags includes

VK_PIPELINE_CREATE_LIBRARY_BIT_KHR or the VkGraphicsPipelineCreateInfo::pNext chain includes a VkPipelineLibraryCreateInfoKHR structure with a libraryCount greater than 0, it is as if flags is 0. Otherwise if this structure is omitted, it is as if flags includes all possible subsets of the graphics pipeline (i.e. a complete graphics pipeline).

Valid Usage (Implicit)

- VUID-VkGraphicsPipelineLibraryCreateInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_GRAPHICS_PIPELINE_LIBRARY_CREATE_INFO_EXT
- VUID-VkGraphicsPipelineLibraryCreateInfoEXT-flags-parameter
  flags **must** be a valid combination of VkGraphicsPipelineLibraryFlagBitsEXT values

Structure Chaining

Extends the structure

- VkGraphicsPipelineCreateInfo

```
// Provided by VK_EXT_graphics_pipeline_library
typedef VkFlags VkGraphicsPipelineLibraryFlagsEXT;
```

VkGraphicsPipelineLibraryFlagsEXT is a bitmask type for setting a mask of zero or more VkGraphicsPipelineLibraryFlagBitsEXT.

Possible values of the flags member of VkGraphicsPipelineLibraryCreateInfoEXT, specifying the subsets of a graphics pipeline to compile are:

```
// Provided by VK_EXT_graphics_pipeline_library
typedef enum VkGraphicsPipelineLibraryFlagBitsEXT {
    VK_GRAPHICS_PIPELINE_LIBRARY_VERTEX_INPUT_INTERFACE_BIT_EXT = 0x00000001,
    VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT = 0x00000002,
    VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT = 0x00000004,
    VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT = 0x00000008,
} VkGraphicsPipelineLibraryFlagBitsEXT;
```

- VK_GRAPHICS_PIPELINE_LIBRARY_VERTEX_INPUT_INTERFACE_BIT_EXT specifies that a pipeline will include vertex input interface state.
- VK_GRAPHICS_PIPELINE_LIBRARY_PRE_RASTERIZATION_SHADERS_BIT_EXT specifies that a pipeline will include pre-rasterization shader state.
- VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_SHADER_BIT_EXT specifies that a pipeline will include fragment shader state.
- VK_GRAPHICS_PIPELINE_LIBRARY_FRAGMENT_OUTPUT_INTERFACE_BIT_EXT specifies that a pipeline will include fragment output interface state.

The VkPipelineDynamicStateCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineDynamicStateCreateInfo {
    VkStructureType                      sType;
    const void*                          pNext;
    VkPipelineDynamicStateCreateFlags    flags;
    uint32_t                             dynamicStateCount;
    const VkDynamicState*                pDynamicStates;
} VkPipelineDynamicStateCreateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is reserved for future use.
- dynamicStateCount is the number of elements in the pDynamicStates array.
- pDynamicStates is a pointer to an array of VkDynamicState values specifying which pieces of pipeline state will use the values from dynamic state commands rather than from pipeline state creation information.

Valid Usage

- VUID-VkPipelineDynamicStateCreateInfo-pDynamicStates-01442
  Each element of pDynamicStates **must** be unique

Valid Usage (Implicit)

- VUID-VkPipelineDynamicStateCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PIPELINE_DYNAMIC_STATE_CREATE_INFO
- VUID-VkPipelineDynamicStateCreateInfo-pNext-pNext
  pNext **must** be NULL
- VUID-VkPipelineDynamicStateCreateInfo-flags-zerobitmask
  flags **must** be 0
- VUID-VkPipelineDynamicStateCreateInfo-pDynamicStates-parameter
  If dynamicStateCount is not 0, pDynamicStates **must** be a valid pointer to an array of dynamicStateCount valid VkDynamicState values

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineDynamicStateCreateFlags;
```

VkPipelineDynamicStateCreateFlags is a bitmask type for setting a mask, but is currently reserved for future use.

The source of different pieces of dynamic state is specified by the VkPipelineDynamicStateCreateInfo::pDynamicStates property of the currently active pipeline, each of whose elements **must** be one of the values:

```
// Provided by VK_VERSION_1_0
typedef enum VkDynamicState {
    VK_DYNAMIC_STATE_VIEWPORT = 0,
    VK_DYNAMIC_STATE_SCISSOR = 1,
    VK_DYNAMIC_STATE_LINE_WIDTH = 2,
    VK_DYNAMIC_STATE_DEPTH_BIAS = 3,
    VK_DYNAMIC_STATE_BLEND_CONSTANTS = 4,
    VK_DYNAMIC_STATE_DEPTH_BOUNDS = 5,
    VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK = 6,
    VK_DYNAMIC_STATE_STENCIL_WRITE_MASK = 7,
    VK_DYNAMIC_STATE_STENCIL_REFERENCE = 8,
  // Provided by VK_VERSION_1_3
    VK_DYNAMIC_STATE_CULL_MODE = 1000267000,
  // Provided by VK_VERSION_1_3
    VK_DYNAMIC_STATE_FRONT_FACE = 1000267001,
  // Provided by VK_VERSION_1_3
    VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY = 1000267002,
  // Provided by VK_VERSION_1_3
    VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT = 1000267003,
  // Provided by VK_VERSION_1_3
    VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT = 1000267004,
  // Provided by VK_VERSION_1_3
    VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE = 1000267005,
  // Provided by VK_VERSION_1_3
    VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE = 1000267006,
  // Provided by VK_VERSION_1_3
    VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE = 1000267007,
  // Provided by VK_VERSION_1_3
    VK_DYNAMIC_STATE_DEPTH_COMPARE_OP = 1000267008,
  // Provided by VK_VERSION_1_3
    VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE = 1000267009,
  // Provided by VK_VERSION_1_3
    VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE = 1000267010,
  // Provided by VK_VERSION_1_3
    VK_DYNAMIC_STATE_STENCIL_OP = 1000267011,
  // Provided by VK_VERSION_1_3
    VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE = 1000377001,
  // Provided by VK_VERSION_1_3
    VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE = 1000377002,
  // Provided by VK_VERSION_1_3
    VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE = 1000377004,
  // Provided by VK_VERSION_1_4
    VK_DYNAMIC_STATE_LINE_STIPPLE = 1000259000,
  // Provided by VK_EXT_discard_rectangles
    VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT = 1000099000,
  // Provided by VK_EXT_discard_rectangles
```

```
    VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT = 1000099001,
  // Provided by VK_EXT_discard_rectangles
    VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT = 1000099002,
  // Provided by VK_EXT_sample_locations
    VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT = 1000143000,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR = 1000347000,
  // Provided by VK_KHR_fragment_shading_rate
    VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR = 1000226000,
  // Provided by VK_EXT_vertex_input_dynamic_state
    VK_DYNAMIC_STATE_VERTEX_INPUT_EXT = 1000352000,
  // Provided by VK_EXT_extended_dynamic_state2
    VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT = 1000377000,
  // Provided by VK_EXT_extended_dynamic_state2
    VK_DYNAMIC_STATE_LOGIC_OP_EXT = 1000377003,
  // Provided by VK_EXT_color_write_enable
    VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT = 1000381000,
  // Provided by VK_EXT_extended_dynamic_state3
    VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT = 1000455003,
  // Provided by VK_EXT_extended_dynamic_state3
    VK_DYNAMIC_STATE_POLYGON_MODE_EXT = 1000455004,
  // Provided by VK_EXT_extended_dynamic_state3
    VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT = 1000455005,
  // Provided by VK_EXT_extended_dynamic_state3
    VK_DYNAMIC_STATE_SAMPLE_MASK_EXT = 1000455006,
  // Provided by VK_EXT_extended_dynamic_state3
    VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT = 1000455007,
  // Provided by VK_EXT_extended_dynamic_state3
    VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT = 1000455008,
  // Provided by VK_EXT_extended_dynamic_state3
    VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT = 1000455009,
  // Provided by VK_EXT_extended_dynamic_state3
    VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT = 1000455010,
  // Provided by VK_EXT_extended_dynamic_state3
    VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT = 1000455011,
  // Provided by VK_EXT_extended_dynamic_state3
    VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT = 1000455012,
  // Provided by VK_EXT_extended_dynamic_state3 with VK_KHR_maintenance2 or
VK_VERSION_1_1
    VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT = 1000455002,
  // Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_transform_feedback
    VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT = 1000455013,
  // Provided by VK_EXT_conservative_rasterization with VK_EXT_extended_dynamic_state3
    VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT = 1000455014,
  // Provided by VK_EXT_conservative_rasterization with VK_EXT_extended_dynamic_state3
    VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT = 1000455015,
  // Provided by VK_EXT_depth_clip_enable with VK_EXT_extended_dynamic_state3
    VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT = 1000455016,
  // Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_sample_locations
    VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT = 1000455017,
  // Provided by VK_EXT_blend_operation_advanced with VK_EXT_extended_dynamic_state3
```

```
    VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT = 1000455018,
  // Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_provoking_vertex
    VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT = 1000455019,
  // Provided by VK_EXT_extended_dynamic_state3 with VK_VERSION_1_4 or
VK_KHR_line_rasterization or VK_EXT_line_rasterization
    VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT = 1000455020,
  // Provided by VK_EXT_extended_dynamic_state3 with VK_VERSION_1_4 or
VK_KHR_line_rasterization or VK_EXT_line_rasterization
    VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT = 1000455021,
  // Provided by VK_EXT_depth_clip_control with VK_EXT_extended_dynamic_state3
    VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT = 1000455022,
  // Provided by VK_EXT_attachment_feedback_loop_dynamic_state
    VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT = 1000524000,
  // Provided by VK_EXT_depth_clamp_control
    VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT = 1000582000,
  // Provided by VK_EXT_extended_dynamic_state
    VK_DYNAMIC_STATE_CULL_MODE_EXT = VK_DYNAMIC_STATE_CULL_MODE,
  // Provided by VK_EXT_extended_dynamic_state
    VK_DYNAMIC_STATE_FRONT_FACE_EXT = VK_DYNAMIC_STATE_FRONT_FACE,
  // Provided by VK_EXT_extended_dynamic_state
    VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY_EXT = VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY,
  // Provided by VK_EXT_extended_dynamic_state
    VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT_EXT = VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT,
  // Provided by VK_EXT_extended_dynamic_state
    VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT_EXT = VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT,
  // Provided by VK_EXT_extended_dynamic_state
    VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE_EXT =
VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE,
  // Provided by VK_EXT_extended_dynamic_state
    VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE_EXT = VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE,
  // Provided by VK_EXT_extended_dynamic_state
    VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE_EXT = VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE,
  // Provided by VK_EXT_extended_dynamic_state
    VK_DYNAMIC_STATE_DEPTH_COMPARE_OP_EXT = VK_DYNAMIC_STATE_DEPTH_COMPARE_OP,
  // Provided by VK_EXT_extended_dynamic_state
    VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE_EXT =
VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE,
  // Provided by VK_EXT_extended_dynamic_state
    VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE_EXT = VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE,
  // Provided by VK_EXT_extended_dynamic_state
    VK_DYNAMIC_STATE_STENCIL_OP_EXT = VK_DYNAMIC_STATE_STENCIL_OP,
  // Provided by VK_EXT_extended_dynamic_state2
    VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE_EXT =
VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE,
  // Provided by VK_EXT_extended_dynamic_state2
    VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE_EXT = VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE,
  // Provided by VK_EXT_extended_dynamic_state2
    VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE_EXT =
VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE,
  // Provided by VK_KHR_line_rasterization
    VK_DYNAMIC_STATE_LINE_STIPPLE_KHR = VK_DYNAMIC_STATE_LINE_STIPPLE,
```

```
} VkDynamicState;
```

- VK_DYNAMIC_STATE_VIEWPORT specifies that the pViewports state in VkPipelineViewportStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetViewport before any drawing commands. The number of viewports used by a pipeline is still specified by the viewportCount member of VkPipelineViewportStateCreateInfo.
- VK_DYNAMIC_STATE_SCISSOR specifies that the pScissors state in VkPipelineViewportStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetScissor before any drawing commands. The number of scissor rectangles used by a pipeline is still specified by the scissorCount member of VkPipelineViewportStateCreateInfo.
- VK_DYNAMIC_STATE_LINE_WIDTH specifies that the lineWidth state in VkPipelineRasterizationStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetLineWidth before any drawing commands that generate line primitives for the rasterizer.
- VK_DYNAMIC_STATE_DEPTH_BIAS specifies that any instance of VkDepthBiasRepresentationInfoEXT included in the pNext chain of VkPipelineRasterizationStateCreateInfo as well as the depthBiasConstantFactor, depthBiasClamp and depthBiasSlopeFactor states in VkPipelineRasterizationStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetDepthBias or vkCmdSetDepthBias2EXT before any draws are performed with depth bias enabled.
- VK_DYNAMIC_STATE_BLEND_CONSTANTS specifies that the blendConstants state in VkPipelineColorBlendStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetBlendConstants before any draws are performed with a pipeline state with VkPipelineColorBlendAttachmentState member blendEnable set to VK_TRUE and any of the blend functions using a constant blend color.
- VK_DYNAMIC_STATE_DEPTH_BOUNDS specifies that the minDepthBounds and maxDepthBounds states of VkPipelineDepthStencilStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetDepthBounds before any draws are performed with a pipeline state with VkPipelineDepthStencilStateCreateInfo member depthBoundsTestEnable set to VK_TRUE.
- VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK specifies that the compareMask state in VkPipelineDepthStencilStateCreateInfo for both front and back will be ignored and **must** be set dynamically with vkCmdSetStencilCompareMask before any draws are performed with a pipeline state with VkPipelineDepthStencilStateCreateInfo member stencilTestEnable set to VK_TRUE.
- VK_DYNAMIC_STATE_STENCIL_WRITE_MASK specifies that the writeMask state in VkPipelineDepthStencilStateCreateInfo for both front and back will be ignored and **must** be set dynamically with vkCmdSetStencilWriteMask before any draws are performed with a pipeline state with VkPipelineDepthStencilStateCreateInfo member stencilTestEnable set to VK_TRUE.
- VK_DYNAMIC_STATE_STENCIL_REFERENCE specifies that the reference state in VkPipelineDepthStencilStateCreateInfo for both front and back will be ignored and **must** be set dynamically with vkCmdSetStencilReference before any draws are performed with a pipeline state with VkPipelineDepthStencilStateCreateInfo member stencilTestEnable set to VK_TRUE.
- VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT specifies that the pDiscardRectangles state in VkPipelineDiscardRectangleStateCreateInfoEXT will be ignored and **must** be set dynamically

with vkCmdSetDiscardRectangleEXT before any draw or clear commands.

- VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT specifies that the presence of the VkPipelineDiscardRectangleStateCreateInfoEXT structure in the VkGraphicsPipelineCreateInfo chain with a `discardRectangleCount` greater than zero does not implicitly enable discard rectangles and they **must** be enabled dynamically with vkCmdSetDiscardRectangleEnableEXT before any draw commands. This is available on implementations that support at least `specVersion` 2 of the `VK_EXT_discard_rectangles` extension.
- VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT specifies that the `discardRectangleMode` state in VkPipelineDiscardRectangleStateCreateInfoEXT will be ignored and **must** be set dynamically with vkCmdSetDiscardRectangleModeEXT before any draw commands. This is available on implementations that support at least `specVersion` 2 of the `VK_EXT_discard_rectangles` extension.
- VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT specifies that the `sampleLocationsInfo` state in VkPipelineSampleLocationsStateCreateInfoEXT will be ignored and **must** be set dynamically with vkCmdSetSampleLocationsEXT before any draw or clear commands. Enabling custom sample locations is still indicated by the `sampleLocationsEnable` member of VkPipelineSampleLocationsStateCreateInfoEXT.
- VK_DYNAMIC_STATE_LINE_STIPPLE specifies that the `lineStippleFactor` and `lineStipplePattern` state in VkPipelineRasterizationLineStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetLineStipple before any draws are performed with a pipeline state with VkPipelineRasterizationLineStateCreateInfo member `stippledLineEnable` set to VK_TRUE.
- VK_DYNAMIC_STATE_CULL_MODE specifies that the `cullMode` state in VkPipelineRasterizationStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetCullMode before any drawing commands.
- VK_DYNAMIC_STATE_FRONT_FACE specifies that the `frontFace` state in VkPipelineRasterizationStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetFrontFace before any drawing commands.
- VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY specifies that the `topology` state in VkPipelineInputAssemblyStateCreateInfo only specifies the topology class, and the specific topology order and adjacency **must** be set dynamically with vkCmdSetPrimitiveTopology before any drawing commands.
- VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT specifies that the `viewportCount` and `pViewports` state in VkPipelineViewportStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetViewportWithCount before any draw call.
- VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT specifies that the `scissorCount` and `pScissors` state in VkPipelineViewportStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetScissorWithCount before any draw call.
- VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE specifies that the `stride` state in VkVertexInputBindingDescription will be ignored and **must** be set dynamically with vkCmdBindVertexBuffers2 before any draw call.
- VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE specifies that the `depthTestEnable` state in VkPipelineDepthStencilStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetDepthTestEnable before any draw call.

- VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE specifies that the depthWriteEnable state in VkPipelineDepthStencilStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetDepthWriteEnable before any draw call.
- VK_DYNAMIC_STATE_DEPTH_COMPARE_OP specifies that the depthCompareOp state in VkPipelineDepthStencilStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetDepthCompareOp before any draw call.
- VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE specifies that the depthBoundsTestEnable state in VkPipelineDepthStencilStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetDepthBoundsTestEnable before any draw call.
- VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE specifies that the stencilTestEnable state in VkPipelineDepthStencilStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetStencilTestEnable before any draw call.
- VK_DYNAMIC_STATE_STENCIL_OP specifies that the failOp, passOp, depthFailOp, and compareOp states in VkPipelineDepthStencilStateCreateInfo for both front and back will be ignored and **must** be set dynamically with vkCmdSetStencilOp before any draws are performed with a pipeline state with VkPipelineDepthStencilStateCreateInfo member stencilTestEnable set to VK_TRUE.
- VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT specifies that the patchControlPoints state in VkPipelineTessellationStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetPatchControlPointsEXT before any drawing commands.
- VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE specifies that the rasterizerDiscardEnable state in VkPipelineRasterizationStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetRasterizerDiscardEnable before any drawing commands.
- VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE specifies that the depthBiasEnable state in VkPipelineRasterizationStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetDepthBiasEnable before any drawing commands.
- VK_DYNAMIC_STATE_LOGIC_OP_EXT specifies that the logicOp state in VkPipelineColorBlendStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetLogicOpEXT before any drawing commands.
- VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE specifies that the primitiveRestartEnable state in VkPipelineInputAssemblyStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetPrimitiveRestartEnable before any drawing commands.
- VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR specifies that state in VkPipelineFragmentShadingRateStateCreateInfoKHR will be ignored and **must** be set dynamically with vkCmdSetFragmentShadingRateKHR before any drawing commands.
- VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR specifies that the default stack size computation for the pipeline will be ignored and **must** be set dynamically with vkCmdSetRayTracingPipelineStackSizeKHR before any ray tracing calls are performed.
- VK_DYNAMIC_STATE_VERTEX_INPUT_EXT specifies that the pVertexInputState state will be ignored and **must** be set dynamically with vkCmdSetVertexInputEXT before any drawing commands.
- VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT specifies that the pColorWriteEnables state in VkPipelineColorWriteCreateInfoEXT will be ignored and **must** be set dynamically with

vkCmdSetColorWriteEnableEXT before any draw call.

- VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT specifies that the `domainOrigin` state in VkPipelineTessellationDomainOriginStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetTessellationDomainOriginEXT before any draw call.
- VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT specifies that the `depthClampEnable` state in VkPipelineRasterizationStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetDepthClampEnableEXT before any draw call.
- VK_DYNAMIC_STATE_POLYGON_MODE_EXT specifies that the `polygonMode` state in VkPipelineRasterizationStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetPolygonModeEXT before any draw call.
- VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT specifies that the `rasterizationSamples` state in VkPipelineMultisampleStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetRasterizationSamplesEXT before any draw call.
- VK_DYNAMIC_STATE_SAMPLE_MASK_EXT specifies that the `pSampleMask` state in VkPipelineMultisampleStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetSampleMaskEXT before any draw call.
- VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT specifies that the `alphaToCoverageEnable` state in VkPipelineMultisampleStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetAlphaToCoverageEnableEXT before any draw call.
- VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT specifies that the `alphaToOneEnable` state in VkPipelineMultisampleStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetAlphaToOneEnableEXT before any draw call.
- VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT specifies that the `logicOpEnable` state in VkPipelineColorBlendStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetLogicOpEnableEXT before any draw call.
- VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT specifies that the `blendEnable` state in VkPipelineColorBlendAttachmentState will be ignored and **must** be set dynamically with vkCmdSetColorBlendEnableEXT before any draw call.
- VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT specifies that the `srcColorBlendFactor`, `dstColorBlendFactor`, `colorBlendOp`, `srcAlphaBlendFactor`, `dstAlphaBlendFactor`, and `alphaBlendOp` states in VkPipelineColorBlendAttachmentState will be ignored and **must** be set dynamically with vkCmdSetColorBlendEquationEXT before any draw call.
- VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT specifies that the `colorWriteMask` state in VkPipelineColorBlendAttachmentState will be ignored and **must** be set dynamically with vkCmdSetColorWriteMaskEXT before any draw call.
- VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT specifies that the `rasterizationStream` state in VkPipelineRasterizationStateStreamCreateInfoEXT will be ignored and **must** be set dynamically with vkCmdSetRasterizationStreamEXT before any draw call.
- VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT specifies that the `conservativeRasterizationMode` state in VkPipelineRasterizationConservativeStateCreateInfoEXT will be ignored and **must** be set dynamically with vkCmdSetConservativeRasterizationModeEXT before any draw call.

- VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT specifies that the `extraPrimitiveOverestimationSize` state in VkPipelineRasterizationConservativeStateCreateInfoEXT will be ignored and **must** be set dynamically with vkCmdSetExtraPrimitiveOverestimationSizeEXT before any draw call.
- VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT specifies that the `depthClipEnable` state in VkPipelineRasterizationDepthClipStateCreateInfoEXT will be ignored and **must** be set dynamically with vkCmdSetDepthClipEnableEXT before any draw call.
- VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT specifies that the `sampleLocationsEnable` state in VkPipelineSampleLocationsStateCreateInfoEXT will be ignored and **must** be set dynamically with vkCmdSetSampleLocationsEnableEXT before any draw call.
- VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT specifies that the `colorBlendOp` state in VkPipelineColorBlendAttachmentState, and `srcPremultiplied`, `dstPremultiplied`, and `blendOverlap` states in VkPipelineColorBlendAdvancedStateCreateInfoEXT will be ignored and **must** be set dynamically with vkCmdSetColorBlendAdvancedEXT before any draw call.
- VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT specifies that the `provokingVertexMode` state in VkPipelineRasterizationProvokingVertexStateCreateInfoEXT will be ignored and **must** be set dynamically with vkCmdSetProvokingVertexModeEXT before any draw call.
- VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT specifies that the `lineRasterizationMode` state in VkPipelineRasterizationLineStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetLineRasterizationModeEXT before any draw call.
- VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT specifies that the `stippledLineEnable` state in VkPipelineRasterizationLineStateCreateInfo will be ignored and **must** be set dynamically with vkCmdSetLineStippleEnableEXT before any draw call.
- VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT specifies that the `negativeOneToOne` state in VkPipelineViewportDepthClipControlCreateInfoEXT will be ignored and **must** be set dynamically with vkCmdSetDepthClipNegativeOneToOneEXT before any draw call.
- VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT specifies that the `depthClampMode` and `pDepthClampRange` state in VkPipelineViewportDepthClampControlCreateInfoEXT will be ignored and **must** be set dynamically with vkCmdSetDepthClampRangeEXT before any draw call.
- VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT specifies that the VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT and VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT flags will be ignored and **must** be set dynamically with vkCmdSetAttachmentFeedbackLoopEnableEXT before any draw call.

10.4.1. Valid Combinations of Stages for Graphics Pipelines

Primitive processing can be handled either on a per primitive basis by the vertex, tessellation, and geometry shader stages, or on a per mesh basis using task and mesh shader stages. If the pipeline includes a mesh shader stage, it uses the mesh pipeline, otherwise it uses the primitive pipeline.

If a task shader is omitted, the task shading stage is skipped.

If tessellation shader stages are omitted, the tessellation shading and fixed-function stages of the

pipeline are skipped.

If a geometry shader is omitted, the geometry shading stage is skipped.

If a fragment shader is omitted, fragment color outputs have undefined values, and the fragment depth value is determined by Fragment Operations state. This **can** be useful for depth-only rendering.

Presence of a shader stage in a pipeline is indicated by including a valid VkPipelineShaderStageCreateInfo with `module` and `pName` selecting an entry point from a shader module, where that entry point is valid for the stage specified by `stage`.

Presence of some of the fixed-function stages in the pipeline is implicitly derived from enabled shaders and provided state. For example, the fixed-function tessellator is always present when the pipeline has valid Tessellation Control and Tessellation Evaluation shaders.

For example:

- Depth/stencil-only rendering in a subpass with no color attachments
  - Active Pipeline Shader Stages
    - Vertex Shader
  - Required: Fixed-Function Pipeline Stages
    - VkPipelineVertexInputStateCreateInfo
    - VkPipelineInputAssemblyStateCreateInfo
    - VkPipelineViewportStateCreateInfo
    - VkPipelineRasterizationStateCreateInfo
    - VkPipelineMultisampleStateCreateInfo
    - VkPipelineDepthStencilStateCreateInfo
- Color-only rendering in a subpass with no depth/stencil attachment
  - Active Pipeline Shader Stages
    - Vertex Shader
    - Fragment Shader
  - Required: Fixed-Function Pipeline Stages
    - VkPipelineVertexInputStateCreateInfo
    - VkPipelineInputAssemblyStateCreateInfo
    - VkPipelineViewportStateCreateInfo
    - VkPipelineRasterizationStateCreateInfo
    - VkPipelineMultisampleStateCreateInfo
    - VkPipelineColorBlendStateCreateInfo
- Rendering pipeline with tessellation and geometry shaders
  - Active Pipeline Shader Stages

        - Vertex Shader
        - Tessellation Control Shader
        - Tessellation Evaluation Shader
        - Geometry Shader
        - Fragment Shader
    - Required: Fixed-Function Pipeline Stages
        - VkPipelineVertexInputStateCreateInfo
        - VkPipelineInputAssemblyStateCreateInfo
        - VkPipelineTessellationStateCreateInfo
        - VkPipelineViewportStateCreateInfo
        - VkPipelineRasterizationStateCreateInfo
        - VkPipelineMultisampleStateCreateInfo
        - VkPipelineDepthStencilStateCreateInfo
        - VkPipelineColorBlendStateCreateInfo
- Rendering pipeline with task and mesh shaders
    - Active Pipeline Shader Stages
        - Task Shader
        - Mesh Shader
        - Fragment Shader
    - Required: Fixed-Function Pipeline Stages
        - VkPipelineViewportStateCreateInfo
        - VkPipelineRasterizationStateCreateInfo
        - VkPipelineMultisampleStateCreateInfo
        - VkPipelineDepthStencilStateCreateInfo
        - VkPipelineColorBlendStateCreateInfo

10.5. Ray Tracing Pipelines

Ray tracing pipelines consist of multiple shader stages, fixed-function traversal stages, and a pipeline layout.

10.5.1. Ray Tracing Pipeline Enums and Types

VK_SHADER_UNUSED_KHR is a special shader index used to indicate that a ray generation, miss, or callable shader member is not used.

```
#define VK_SHADER_UNUSED_KHR            (~0U)
```

The VkRayTracingShaderGroupTypeKHR enumeration is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef enum VkRayTracingShaderGroupTypeKHR {
    VK_RAY_TRACING_SHADER_GROUP_TYPE_GENERAL_KHR = 0,
    VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR = 1,
    VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR = 2,
} VkRayTracingShaderGroupTypeKHR;
```

- VK_RAY_TRACING_SHADER_GROUP_TYPE_GENERAL_KHR specifies a shader group with a single VK_SHADER_STAGE_RAYGEN_BIT_KHR, VK_SHADER_STAGE_MISS_BIT_KHR, or VK_SHADER_STAGE_CALLABLE_BIT_KHR shader in it.
- VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR specifies a shader group that only hits triangles and **must** not contain an intersection shader, only closest hit and any-hit shaders.
- VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR specifies a shader group that only intersects with custom geometry and **must** contain an intersection shader and **may** contain closest hit and any-hit shaders.

NOTE For current group types, the hit group type could be inferred from the presence or absence of the intersection shader, but we provide the type explicitly for future hit groups that do not have that property.

10.5.2. Ray Tracing Pipeline Creation

To create ray tracing pipelines, call:

```
// Provided by VK_KHR_ray_tracing_pipeline
VkResult vkCreateRayTracingPipelinesKHR(
    VkDevice                                    device,
    VkDeferredOperationKHR                      deferredOperation,
    VkPipelineCache                             pipelineCache,
    uint32_t                                    createInfoCount,
    const VkRayTracingPipelineCreateInfoKHR*    pCreateInfos,
    const VkAllocationCallbacks*                pAllocator,
    VkPipeline*                                 pPipelines);
```

- `device` is the logical device that creates the ray tracing pipelines.
- `deferredOperation` is VK_NULL_HANDLE or the handle of a valid VkDeferredOperationKHR request deferral object for this command.
- `pipelineCache` is either VK_NULL_HANDLE, indicating that pipeline caching is disabled, or to enable caching, the handle of a valid VkPipelineCache object. The implementation **must** not access this object outside of the duration of this command.
- `createInfoCount` is the length of the `pCreateInfos` and `pPipelines` arrays.

- `pCreateInfos` is a pointer to an array of VkRayTracingPipelineCreateInfoKHR structures.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pPipelines` is a pointer to an array in which the resulting ray tracing pipeline objects are returned.

The VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS error is returned if the implementation is unable to reuse the shader group handles provided in VkRayTracingShaderGroupCreateInfoKHR::`pShaderGroupCaptureReplayHandle` when VkPhysicalDeviceRayTracingPipelineFeaturesKHR::`rayTracingPipelineShaderGroupHandleCaptureReplay` is enabled.

Pipelines are created and returned as described for Multiple Pipeline Creation.

Valid Usage

- VUID-vkCreateRayTracingPipelinesKHR-device-09677
  `device` **must** support at least one queue family with the VK_QUEUE_COMPUTE_BIT capability
- VUID-vkCreateRayTracingPipelinesKHR-flags-03415
  If the `flags` member of any element of `pCreateInfos` contains the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag, and the `basePipelineIndex` member of that same element is not `-1`, `basePipelineIndex` **must** be less than the index into `pCreateInfos` that corresponds to that element
- VUID-vkCreateRayTracingPipelinesKHR-flags-03416
  If the `flags` member of any element of `pCreateInfos` contains the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag, the base pipeline **must** have been created with the VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT flag set
- VUID-vkCreateRayTracingPipelinesKHR-flags-03816
  `flags` **must** not contain the VK_PIPELINE_CREATE_DISPATCH_BASE_BIT flag
- VUID-vkCreateRayTracingPipelinesKHR-pipelineCache-02903
  If `pipelineCache` was created with VK_PIPELINE_CACHE_CREATE_EXTERNALLY_SYNCHRONIZED_BIT, host access to `pipelineCache` **must** be externally synchronized
- VUID-vkCreateRayTracingPipelinesKHR-deferredOperation-03678
  Any previous deferred operation that was associated with `deferredOperation` **must** be complete
- VUID-vkCreateRayTracingPipelinesKHR-pNext-09616
  If VkPipelineBinaryInfoKHR::`binaryCount` is not `0` for any element of `pCreateInfos`, `pipelineCache` **must** be VK_NULL_HANDLE
- VUID-vkCreateRayTracingPipelinesKHR-pNext-09617
  If a VkPipelineCreateFlags2CreateInfoKHR structure with the VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR flag set is included in the `pNext` chain of any element of `pCreateInfos`, `pipelineCache` **must** be VK_NULL_HANDLE
- VUID-vkCreateRayTracingPipelinesKHR-binaryCount-09620

If VkPipelineBinaryInfoKHR::binaryCount is not 0 for any element of pCreateInfos, VK_PIPELINE_CREATION_FEEDBACK_APPLICATION_PIPELINE_CACHE_HIT_BIT **must** not be set in the flags of that element

- VUID-vkCreateRayTracingPipelinesKHR-binaryCount-09621
  If VkPipelineBinaryInfoKHR::binaryCount is not 0 for any element of pCreateInfos, VK_PIPELINE_CREATION_FEEDBACK_BASE_PIPELINE_ACCELERATION_BIT **must** not be set in the flags of that element
- VUID-vkCreateRayTracingPipelinesKHR-binaryCount-09622
  If VkPipelineBinaryInfoKHR::binaryCount is not 0 for any element of pCreateInfos, VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT_EXT **must** not be set in the flags of that element
- VUID-vkCreateRayTracingPipelinesKHR-pCreateInfos-11414
  If any element of pCreateInfos sets VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT and includes embedded sampler mappings, there **must** be less than (maxSamplerAllocationCount - (minSamplerHeapReservedRangeWithEmbedded / samplerDescriptorSize)) VkSampler objects currently created on the device
- VUID-vkCreateRayTracingPipelinesKHR-pCreateInfos-11429
  If any element of pCreateInfos sets VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT and includes embedded sampler mappings, this command **must** not cause the total number of unique embedded samplers in pipelines and shaders on this device to exceed maxDescriptorHeapEmbeddedSamplers
- VUID-vkCreateRayTracingPipelinesKHR-rayTracingPipeline-03586
  The rayTracingPipeline feature **must** be enabled
- VUID-vkCreateRayTracingPipelinesKHR-deferredOperation-03587
  If deferredOperation is not VK_NULL_HANDLE, the flags member of elements of pCreateInfos **must** not include VK_PIPELINE_CREATE_EARLY_RETURN_ON_FAILURE_BIT

Valid Usage (Implicit)

- VUID-vkCreateRayTracingPipelinesKHR-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkCreateRayTracingPipelinesKHR-deferredOperation-parameter
  If deferredOperation is not VK_NULL_HANDLE, deferredOperation **must** be a valid VkDeferredOperationKHR handle
- VUID-vkCreateRayTracingPipelinesKHR-pipelineCache-parameter
  If pipelineCache is not VK_NULL_HANDLE, pipelineCache **must** be a valid VkPipelineCache handle
- VUID-vkCreateRayTracingPipelinesKHR-pCreateInfos-parameter
  pCreateInfos **must** be a valid pointer to an array of createInfoCount valid VkRayTracingPipelineCreateInfoKHR structures
- VUID-vkCreateRayTracingPipelinesKHR-pAllocator-parameter
  If pAllocator is not NULL, pAllocator **must** be a valid pointer to a valid VkAllocationCallbacks structure

- VUID-vkCreateRayTracingPipelinesKHR-pPipelines-parameter
  `pPipelines` **must** be a valid pointer to an array of `createInfoCount` VkPipeline handles
- VUID-vkCreateRayTracingPipelinesKHR-createInfoCount-arraylength
  `createInfoCount` **must** be greater than `0`
- VUID-vkCreateRayTracingPipelinesKHR-deferredOperation-parent
  If `deferredOperation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`
- VUID-vkCreateRayTracingPipelinesKHR-pipelineCache-parent
  If `pipelineCache` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_OPERATION_DEFERRED_KHR
- VK_OPERATION_NOT_DEFERRED_KHR
- VK_SUCCESS

Failure

- VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkRayTracingPipelineCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef struct VkRayTracingPipelineCreateInfoKHR {
    VkStructureType                                   sType;
    const void*                                       pNext;
    VkPipelineCreateFlags                             flags;
    uint32_t                                          stageCount;
    const VkPipelineShaderStageCreateInfo*            pStages;
    uint32_t                                          groupCount;
    const VkRayTracingShaderGroupCreateInfoKHR*       pGroups;
    uint32_t                                          maxPipelineRayRecursionDepth;
    const VkPipelineLibraryCreateInfoKHR*             pLibraryInfo;
    const VkRayTracingPipelineInterfaceCreateInfoKHR* pLibraryInterface;
    const VkPipelineDynamicStateCreateInfo*           pDynamicState;
    VkPipelineLayout                                  layout;
    VkPipeline                                        basePipelineHandle;
    int32_t                                           basePipelineIndex;
} VkRayTracingPipelineCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkPipelineCreateFlagBits specifying how the pipeline will be generated.
- `stageCount` is the number of entries in the `pStages` array.
- `pStages` is a pointer to an array of `stageCount` VkPipelineShaderStageCreateInfo structures describing the set of the shader stages to be included in the ray tracing pipeline.
- `groupCount` is the number of entries in the `pGroups` array.
- `pGroups` is a pointer to an array of `groupCount` VkRayTracingShaderGroupCreateInfoKHR structures describing the set of the shader stages to be included in each shader group in the ray tracing pipeline.
- `maxPipelineRayRecursionDepth` is the maximum recursion depth of shaders executed by this pipeline.
- `pLibraryInfo` is a pointer to a VkPipelineLibraryCreateInfoKHR structure defining pipeline libraries to include.
- `pLibraryInterface` is a pointer to a VkRayTracingPipelineInterfaceCreateInfoKHR structure defining additional information when using pipeline libraries.
- `pDynamicState` is a pointer to a VkPipelineDynamicStateCreateInfo structure, and is used to indicate which properties of the pipeline state object are dynamic and **can** be changed independently of the pipeline state. This **can** be `NULL`, which means no state in the pipeline is considered dynamic.
- `layout` is the description of binding locations used by both the pipeline and descriptor sets used with the pipeline. If VkPhysicalDeviceProperties::`apiVersion` is greater than or equal to Vulkan 1.3 or VK_KHR_maintenance4 is enabled `layout` **must** not be accessed by the implementation outside of the duration of the command this structure is passed to.
- `basePipelineHandle` is a pipeline to derive from.

- `basePipelineIndex` is an index into the `pCreateInfos` parameter to use as a pipeline to derive from.

The parameters `basePipelineHandle` and `basePipelineIndex` are described in more detail in Pipeline Derivatives.

When VK_PIPELINE_CREATE_LIBRARY_BIT_KHR is specified, this pipeline defines a *pipeline library* which **cannot** be bound as a ray tracing pipeline directly. Instead, pipeline libraries define common shaders and shader groups which **can** be included in future pipeline creation.

If pipeline libraries are included in `pLibraryInfo`, shaders defined in those libraries are treated as if they were defined as additional entries in `pStages`, appended in the order they appear in the `pLibraries` array and in the `pStages` array when those libraries were defined.

When referencing shader groups in order to obtain a shader group handle, groups defined in those libraries are treated as if they were defined as additional entries in `pGroups`, appended in the order they appear in the `pLibraries` array and in the `pGroups` array when those libraries were defined. The shaders these groups reference are set when the pipeline library is created, referencing those specified in the pipeline library, not in the pipeline that includes it.

The default stack size for a pipeline if VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR is not provided is computed as described in Ray Tracing Pipeline Stack.

If the `pNext` chain includes a VkPipelineCreateFlags2CreateInfo structure, VkPipelineCreateFlags2CreateInfo::`flags` from that structure is used instead of `flags` from this structure.

Valid Usage

- VUID-VkRayTracingPipelineCreateInfoKHR-None-09497
  If the `pNext` chain does not include a VkPipelineCreateFlags2CreateInfo structure, `flags` **must** be a valid combination of VkPipelineCreateFlagBits values
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-07984
  If `flags` contains the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag, and `basePipelineIndex` is -1, `basePipelineHandle` **must** be a valid ray tracing `VkPipeline` handle
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-07985
  If `flags` contains the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag, and `basePipelineHandle` is VK_NULL_HANDLE, `basePipelineIndex` **must** be a valid index into the calling command's `pCreateInfos` parameter
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-07986
  If `flags` contains the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag, `basePipelineIndex` **must** be -1 or `basePipelineHandle` **must** be VK_NULL_HANDLE
- VUID-VkRayTracingPipelineCreateInfoKHR-layout-07987
  If a push constant block is declared in a shader and `layout` is not VK_NULL_HANDLE, a push constant range in `layout` **must** match the shader stage
- VUID-VkRayTracingPipelineCreateInfoKHR-layout-10069

If a push constant block is declared in a shader and `layout` is not VK_NULL_HANDLE, the block must be contained inside the push constant range in `layout` that matches the stage

- VUID-VkRayTracingPipelineCreateInfoKHR-layout-07988
  If a resource variable is declared in a shader and `layout` is not VK_NULL_HANDLE, the corresponding descriptor set in `layout` **must** match the shader stage
- VUID-VkRayTracingPipelineCreateInfoKHR-layout-07990
  If a resource variable is declared in a shader, `layout` is not VK_NULL_HANDLE, and the descriptor type is not VK_DESCRIPTOR_TYPE_MUTABLE_EXT, the corresponding descriptor set in `layout` **must** match the descriptor type
- VUID-VkRayTracingPipelineCreateInfoKHR-layout-07991
  If a resource variable is declared in a shader as an array and `layout` is not VK_NULL_HANDLE, the corresponding descriptor binding used to create `layout` **must** have a `descriptorCount` that is greater than or equal to the length of the array
- VUID-VkRayTracingPipelineCreateInfoKHR-None-10391
  If a resource variable is declared in a shader as an array of descriptors, then the descriptor type of that variable **must** not be VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-11798
  If the shader64BitIndexing feature is not enabled, `flags` **must** not contain VK_PIPELINE_CREATE_2_64_BIT_INDEXING_BIT_EXT
- VUID-VkRayTracingPipelineCreateInfoKHR-pipelineCreationCacheControl-02878
  If the `pipelineCreationCacheControl` feature is not enabled, `flags` **must** not include VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT nor VK_PIPELINE_CREATE_EARLY_RETURN_ON_FAILURE_BIT
- VUID-VkRayTracingPipelineCreateInfoKHR-pipelineProtectedAccess-07368
  If the `pipelineProtectedAccess` feature is not enabled, `flags` **must** not include VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT nor VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-07369
  `flags` **must** not include both VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT and VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-11311
  If VkPipelineCreateFlags2CreateInfoKHR::`flags` includes VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, `layout` **must** be VK_NULL_HANDLE
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-11312
  If VkPipelineCreateFlags2CreateInfoKHR::`flags` includes VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, all shader variables in the shader resource interface with a `DescriptorSet` and `Binding` decoration **must** have a mapping declared in VkShaderDescriptorSetAndBindingMappingInfoEXT::`pMappings`
- VUID-VkRayTracingPipelineCreateInfoKHR-pStages-03426
  The shader code for the entry points identified by `pStages`, and the rest of the state identified by this structure **must** adhere to the pipeline linking rules described in the

Shader Interfaces chapter

- VUID-VkRayTracingPipelineCreateInfoKHR-layout-03428
  The number of resources in `layout` accessible to each shader stage that is used by the pipeline **must** be less than or equal to VkPhysicalDeviceLimits::`maxPerStageResources`
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-12341
  `flags` **must** not include VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT nor VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT
- VUID-VkRayTracingPipelineCreateInfoKHR-layout-12402
  If `layout` is not VK_NULL_HANDLE, it **must** not have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT
- VUID-VkRayTracingPipelineCreateInfoKHR-stage-03425
  If `flags` does not include VK_PIPELINE_CREATE_LIBRARY_BIT_KHR, the `stage` member of at least one element of `pStages`, including those implicitly added by `pLibraryInfo`, **must** be VK_SHADER_STAGE_RAYGEN_BIT_KHR
- VUID-VkRayTracingPipelineCreateInfoKHR-maxPipelineRayRecursionDepth-03589
  `maxPipelineRayRecursionDepth` **must** be less than or equal to VkPhysicalDeviceRayTracingPipelinePropertiesKHR::`maxRayRecursionDepth`
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-03465
  If `flags` includes VK_PIPELINE_CREATE_LIBRARY_BIT_KHR, `pLibraryInterface` **must** not be `NULL`
- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraryInfo-03590
  If `pLibraryInfo` is not `NULL` and its `libraryCount` member is greater than `0`, `pLibraryInterface` **must** not be `NULL`
- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraries-03591
  Each element of `pLibraryInfo->pLibraries` **must** have been created with the value of `maxPipelineRayRecursionDepth` equal to that in this pipeline
- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraryInfo-03592
  If `pLibraryInfo` is not `NULL` and `layout` is not VK_NULL_HANDLE, each element of `pLibraryInfo->pLibraries` **must** have been created with a `layout` that is compatible with the `layout` in this pipeline
- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraryInfo-03593
  If `pLibraryInfo` is not `NULL`, each element of its `pLibraries` member **must** have been created with values of the `maxPipelineRayPayloadSize` and `maxPipelineRayHitAttributeSize` members of `pLibraryInterface` equal to those in this pipeline
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-03594
  If `flags` includes VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR, each element of `pLibraryInfo->pLibraries` **must** have been created with the VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR bit set
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-04718
  If `flags` includes VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR, each element of `pLibraryInfo->pLibraries` **must** have been created with the

VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR bit set

- VUID-VkRayTracingPipelineCreateInfoKHR-flags-04719
  If flags includes VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR, each element of pLibraryInfo->pLibraries **must** have been created with the VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR bit set
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-04720
  If flags includes VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR, each element of pLibraryInfo->pLibraries **must** have been created with the VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR bit set
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-04721
  If flags includes VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR, each element of pLibraryInfo->pLibraries **must** have been created with the VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR bit set
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-04722
  If flags includes VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR, each element of pLibraryInfo->pLibraries **must** have been created with the VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR bit set
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-04723
  If flags includes VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR, each element of pLibraryInfo->pLibraries **must** have been created with the VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR bit set
- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraryInfo-03595
  If the VK_KHR_pipeline_library extension is not enabled, pLibraryInfo and pLibraryInterface **must** be NULL
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-03470
  If flags includes VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR, for each element of pGroups with a type of VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR or VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR, the anyHitShader of that element **must** not be VK_SHADER_UNUSED_KHR
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-03471
  If flags includes VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR, for each element of pGroups with a type of VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR or VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR, the closestHitShader of that element **must** not be VK_SHADER_UNUSED_KHR
- VUID-VkRayTracingPipelineCreateInfoKHR-rayTraversalPrimitiveCulling-03596

If the rayTraversalPrimitiveCulling feature is not enabled, flags **must** not include VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR

- VUID-VkRayTracingPipelineCreateInfoKHR-rayTraversalPrimitiveCulling-03597
  If the rayTraversalPrimitiveCulling feature is not enabled, flags **must** not include VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-06546
  flags **must** not include both VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR and VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-03598
  If flags includes VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR, rayTracingPipelineShaderGroupHandleCaptureReplay **must** be enabled
- VUID-VkRayTracingPipelineCreateInfoKHR-rayTracingPipelineShaderGroupHandleCaptureReplay-03599
  If VkPhysicalDeviceRayTracingPipelineFeaturesKHR::rayTracingPipelineShaderGroupHandleCaptureReplay is VK_TRUE and the pShaderGroupCaptureReplayHandle member of any element of pGroups is not NULL, flags **must** include VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR
- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraryInfo-07999
  If pLibraryInfo is NULL or its libraryCount is 0, stageCount **must** not be 0
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-08700
  If flags does not include VK_PIPELINE_CREATE_LIBRARY_BIT_KHR and either pLibraryInfo is NULL or its libraryCount is 0, groupCount **must** not be 0
- VUID-VkRayTracingPipelineCreateInfoKHR-pDynamicStates-03602
  Any element of the pDynamicStates member of pDynamicState **must** be VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR
- VUID-VkRayTracingPipelineCreateInfoKHR-pipelineStageCreationFeedbackCount-06652
  If VkPipelineCreationFeedbackCreateInfo::pipelineStageCreationFeedbackCount is not 0, it **must** be equal to stageCount
- VUID-VkRayTracingPipelineCreateInfoKHR-stage-06899
  The stage value in all pStages elements **must** be one of VK_SHADER_STAGE_RAYGEN_BIT_KHR, VK_SHADER_STAGE_ANY_HIT_BIT_KHR, VK_SHADER_STAGE_CLOSEST_HIT_BIT_KHR, VK_SHADER_STAGE_MISS_BIT_KHR, VK_SHADER_STAGE_INTERSECTION_BIT_KHR, or VK_SHADER_STAGE_CALLABLE_BIT_KHR
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-11597
  If flags includes VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR, at least one of the following features **must** be enabled
  - VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap
  - VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap

- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraries-11598
  If `flags` includes VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR, each element of `pLibraryInfo->pLibraries` **must** have been created with the VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR bit set
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-11599
  If the VkPhysicalDeviceOpacityMicromapFeaturesKHR::`micromap` feature is not enabled or `flags` does not include VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR, `flags` **must** not include VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-11275
  If VkPipelineCreateFlags2CreateInfoKHR::`flags` includes VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, all libraries linked to this pipeline **must** also have that flag set
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-11276
  If VkPipelineCreateFlags2CreateInfoKHR::`flags` does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, all libraries linked to this pipeline **must** also not have that flag set
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-12361
  If `flags` includes VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, all libraries linked to this pipeline **must** also have that flag set
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-12362
  If `flags` does not include VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, all libraries linked to this pipeline **must** also not have that flag set
- VUID-VkRayTracingPipelineCreateInfoKHR-None-11369
  If VkPipelineCreateFlags2CreateInfoKHR::`flags` does not include VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, `layout` **must** not be VK_NULL_HANDLE

Valid Usage (Implicit)

- VUID-VkRayTracingPipelineCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RAY_TRACING_PIPELINE_CREATE_INFO_KHR
- VUID-VkRayTracingPipelineCreateInfoKHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkPipelineBinaryInfoKHR, VkPipelineCreateFlags2CreateInfo, VkPipelineCreationFeedbackCreateInfo, VkPipelineRobustnessCreateInfo, or VkValidationFeaturesEXT
- VUID-VkRayTracingPipelineCreateInfoKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkRayTracingPipelineCreateInfoKHR-pStages-parameter
  If `stageCount` is not `0`, `pStages` **must** be a valid pointer to an array of `stageCount` valid

VkPipelineShaderStageCreateInfo structures

- VUID-VkRayTracingPipelineCreateInfoKHR-pGroups-parameter
  If `groupCount` is not `0`, `pGroups` **must** be a valid pointer to an array of `groupCount` valid VkRayTracingShaderGroupCreateInfoKHR structures
- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraryInfo-parameter
  If `pLibraryInfo` is not `NULL`, `pLibraryInfo` **must** be a valid pointer to a valid VkPipelineLibraryCreateInfoKHR structure
- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraryInterface-parameter
  If `pLibraryInterface` is not `NULL`, `pLibraryInterface` **must** be a valid pointer to a valid VkRayTracingPipelineInterfaceCreateInfoKHR structure
- VUID-VkRayTracingPipelineCreateInfoKHR-pDynamicState-parameter
  If `pDynamicState` is not `NULL`, `pDynamicState` **must** be a valid pointer to a valid VkPipelineDynamicStateCreateInfo structure
- VUID-VkRayTracingPipelineCreateInfoKHR-layout-parameter
  If `layout` is not VK_NULL_HANDLE, `layout` **must** be a valid VkPipelineLayout handle
- VUID-VkRayTracingPipelineCreateInfoKHR-commonparent
  Both of `basePipelineHandle`, and `layout` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Structure Chaining

Extends the structure

- VkPipelineCreateInfoKHR

The `VkRayTracingShaderGroupCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef struct VkRayTracingShaderGroupCreateInfoKHR {
    VkStructureType                   sType;
    const void*                       pNext;
    VkRayTracingShaderGroupTypeKHR    type;
    uint32_t                          generalShader;
    uint32_t                          closestHitShader;
    uint32_t                          anyHitShader;
    uint32_t                          intersectionShader;
    const void*                       pShaderGroupCaptureReplayHandle;
} VkRayTracingShaderGroupCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `type` is the type of hit group specified in this structure.
- `generalShader` is the index of the ray generation, miss, or callable shader from

VkRayTracingPipelineCreateInfoKHR::pStages in the group if the shader group has type of VK_RAY_TRACING_SHADER_GROUP_TYPE_GENERAL_KHR, and VK_SHADER_UNUSED_KHR otherwise.

- closestHitShader is the optional index of the closest hit shader from VkRayTracingPipelineCreateInfoKHR::pStages in the group if the shader group has type of VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR or VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR, and VK_SHADER_UNUSED_KHR otherwise.
- anyHitShader is the optional index of the any-hit shader from VkRayTracingPipelineCreateInfoKHR::pStages in the group if the shader group has type of VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR or VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR, and VK_SHADER_UNUSED_KHR otherwise.
- intersectionShader is the index of the intersection shader from VkRayTracingPipelineCreateInfoKHR::pStages in the group if the shader group has type of VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR, and VK_SHADER_UNUSED_KHR otherwise.
- pShaderGroupCaptureReplayHandle is NULL or a pointer to replay information for this shader group queried from vkGetRayTracingCaptureReplayShaderGroupHandlesKHR, as described in Ray Tracing Capture Replay. Ignored if VkPhysicalDeviceRayTracingPipelineFeaturesKHR::rayTracingPipelineShaderGroupHandleCaptureReplay is VK_FALSE.

If the pipeline is created with VK_PIPELINE_CREATE_LIBRARY_BIT_KHR and the pipelineLibraryGroupHandles feature is enabled, pShaderGroupCaptureReplayHandle is inherited by all pipelines which link against this pipeline and remains bitwise identical for any pipeline which references this pipeline library.

Valid Usage

- VUID-VkRayTracingShaderGroupCreateInfoKHR-type-03474
  If type is VK_RAY_TRACING_SHADER_GROUP_TYPE_GENERAL_KHR then generalShader **must** be a valid index into the list of shaders, formed by shaders in VkRayTracingPipelineCreateInfoKHR::pStages and the shaders imported from pipeline libraries, referring to a shader of VK_SHADER_STAGE_RAYGEN_BIT_KHR, VK_SHADER_STAGE_MISS_BIT_KHR, or VK_SHADER_STAGE_CALLABLE_BIT_KHR
- VUID-VkRayTracingShaderGroupCreateInfoKHR-type-03475
  If type is VK_RAY_TRACING_SHADER_GROUP_TYPE_GENERAL_KHR then closestHitShader, anyHitShader, and intersectionShader **must** be VK_SHADER_UNUSED_KHR
- VUID-VkRayTracingShaderGroupCreateInfoKHR-type-03476
  If type is VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR then intersectionShader **must** be a valid index into the list of shaders, formed by shaders in VkRayTracingPipelineCreateInfoKHR::pStages and the shaders imported from pipeline libraries, referring to a shader of VK_SHADER_STAGE_INTERSECTION_BIT_KHR
- VUID-VkRayTracingShaderGroupCreateInfoKHR-type-03477
  If type is VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR then

intersectionShader **must** be VK_SHADER_UNUSED_KHR

- VUID-VkRayTracingShaderGroupCreateInfoKHR-closestHitShader-03478
  closestHitShader **must** be either VK_SHADER_UNUSED_KHR or a valid index into the list of shaders, formed by shaders in VkRayTracingPipelineCreateInfoKHR::pStages and the shaders imported from pipeline libraries, referring to a shader of VK_SHADER_STAGE_CLOSEST_HIT_BIT_KHR
- VUID-VkRayTracingShaderGroupCreateInfoKHR-anyHitShader-03479
  anyHitShader **must** be either VK_SHADER_UNUSED_KHR or a valid index into the list of shaders, formed by shaders in VkRayTracingPipelineCreateInfoKHR::pStages and the shaders imported from pipeline libraries, referring to a shader of VK_SHADER_STAGE_ANY_HIT_BIT_KHR
- VUID-VkRayTracingShaderGroupCreateInfoKHR-rayTracingPipelineShaderGroupHandleCaptureReplayMixed-03603
  If VkPhysicalDeviceRayTracingPipelineFeaturesKHR::rayTracingPipelineShaderGroupHandleCaptureReplayMixed is VK_FALSE then pShaderGroupCaptureReplayHandle **must** not be provided if it has not been provided on a previous call to ray tracing pipeline creation
- VUID-VkRayTracingShaderGroupCreateInfoKHR-rayTracingPipelineShaderGroupHandleCaptureReplayMixed-03604
  If VkPhysicalDeviceRayTracingPipelineFeaturesKHR::rayTracingPipelineShaderGroupHandleCaptureReplayMixed is VK_FALSE then the caller **must** guarantee that no ray tracing pipeline creation commands with pShaderGroupCaptureReplayHandle provided execute simultaneously with ray tracing pipeline creation commands without pShaderGroupCaptureReplayHandle provided

Valid Usage (Implicit)

- VUID-VkRayTracingShaderGroupCreateInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_RAY_TRACING_SHADER_GROUP_CREATE_INFO_KHR
- VUID-VkRayTracingShaderGroupCreateInfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkRayTracingShaderGroupCreateInfoKHR-type-parameter
  type **must** be a valid VkRayTracingShaderGroupTypeKHR value

The VkRayTracingPipelineInterfaceCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef struct VkRayTracingPipelineInterfaceCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           maxPipelineRayPayloadSize;
    uint32_t           maxPipelineRayHitAttributeSize;
} VkRayTracingPipelineInterfaceCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxPipelineRayPayloadSize` is the maximum payload size in bytes used by any shader in the pipeline.
- `maxPipelineRayHitAttributeSize` is the maximum attribute structure size in bytes used by any shader in the pipeline.

`maxPipelineRayPayloadSize` is calculated as the maximum size of the block (in bytes) declared in the `RayPayloadKHR` or `IncomingRayPayloadKHR` storage classes. `maxPipelineRayHitAttributeSize` is calculated as the maximum size of any block (in bytes) declared in the `HitAttributeKHR` or `HitObjectAttributeEXT` storage class. As variables in these storage classes do not have explicit offsets, the size should be calculated as if each variable has a scalar alignment equal to the largest scalar alignment of any of the block's members.

NOTE There is no explicit upper limit for `maxPipelineRayPayloadSize`, but in practice it should be kept as small as possible. Similar to invocation local memory, it must be allocated for each shader invocation and for devices which support many simultaneous invocations, this storage can rapidly be exhausted, resulting in failure.

Valid Usage

- VUID-VkRayTracingPipelineInterfaceCreateInfoKHR-maxPipelineRayHitAttributeSize-03605
  `maxPipelineRayHitAttributeSize` **must** be less than or equal to VkPhysicalDeviceRayTracingPipelinePropertiesKHR::`maxRayHitAttributeSize`

Valid Usage (Implicit)

- VUID-VkRayTracingPipelineInterfaceCreateInfoKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_RAY_TRACING_PIPELINE_INTERFACE_CREATE_INFO_KHR`
- VUID-VkRayTracingPipelineInterfaceCreateInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`

10.5.3. Ray Tracing Pipeline Queries and Commands

To query the opaque handles of shaders in the ray tracing pipeline, call:

```
// Provided by VK_KHR_ray_tracing_pipeline
VkResult vkGetRayTracingShaderGroupHandlesKHR(
    VkDevice                                    device,
    VkPipeline                                  pipeline,
    uint32_t                                    firstGroup,
    uint32_t                                    groupCount,
    size_t                                      dataSize,
    void*                                       pData);
```

- `device` is the logical device containing the ray tracing pipeline.
- `pipeline` is the ray tracing pipeline object containing the shaders.
- `firstGroup` is the index of the first group to retrieve a handle for from the VkRayTracingPipelineCreateInfoKHR::`pGroups` array.
- `groupCount` is the number of shader handles to retrieve.
- `dataSize` is the size in bytes of the buffer pointed to by `pData`.
- `pData` is a pointer to an application-allocated buffer where the results will be written.

On success, an array of `groupCount` shader handles will be written to `pData`, with each element being of size VkPhysicalDeviceRayTracingPipelinePropertiesKHR::`shaderGroupHandleSize`. The first handle written corresponds to the group identified by `firstGroup`, with subsequent handles corresponding to consecutive shader groups in the order they appear in the VkRayTracingPipelineCreateInfoKHR ::`pGroups` array.

If `pipeline` was created with VK_PIPELINE_CREATE_LIBRARY_BIT_KHR and the pipelineLibraryGroupHandles feature is enabled applications **can** query group handles from that pipeline, even if the pipeline is a library and is never bound to a command buffer. These group handles remain bitwise identical for any `pipeline` which references the pipeline library. Group indices are assigned as-if the pipeline was created without VK_PIPELINE_CREATE_LIBRARY_BIT_KHR.

Valid Usage

- VUID-vkGetRayTracingShaderGroupHandlesKHR-pipeline-04619
  `pipeline` **must** be a ray tracing pipeline
- VUID-vkGetRayTracingShaderGroupHandlesKHR-firstGroup-04050
  `firstGroup` **must** be less than the number of shader groups in `pipeline`
- VUID-vkGetRayTracingShaderGroupHandlesKHR-firstGroup-02419
  The sum of `firstGroup` and `groupCount` **must** be less than or equal to the number of shader groups in `pipeline`
- VUID-vkGetRayTracingShaderGroupHandlesKHR-dataSize-02420
  `dataSize` **must** be at least VkPhysicalDeviceRayTracingPipelinePropertiesKHR ::`shaderGroupHandleSize` × `groupCount`
- VUID-vkGetRayTracingShaderGroupHandlesKHR-pipeline-07828
  If the pipelineLibraryGroupHandles feature is not enabled, `pipeline` **must** not have been

created with VK_PIPELINE_CREATE_LIBRARY_BIT_KHR

Valid Usage (Implicit)

- VUID-vkGetRayTracingShaderGroupHandlesKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetRayTracingShaderGroupHandlesKHR-pipeline-parameter
  `pipeline` **must** be a valid VkPipeline handle
- VUID-vkGetRayTracingShaderGroupHandlesKHR-pData-parameter
  `pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-vkGetRayTracingShaderGroupHandlesKHR-dataSize-arraylength
  `dataSize` **must** be greater than `0`
- VUID-vkGetRayTracingShaderGroupHandlesKHR-pipeline-parent
  `pipeline` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To query the opaque capture data of shader groups in a ray tracing pipeline, call:

```
// Provided by VK_KHR_ray_tracing_pipeline
VkResult vkGetRayTracingCaptureReplayShaderGroupHandlesKHR(
    VkDevice                                    device,
    VkPipeline                                  pipeline,
    uint32_t                                    firstGroup,
    uint32_t                                    groupCount,
    size_t                                      dataSize,
    void*                                       pData);
```

- `device` is the logical device containing the ray tracing pipeline.
- `pipeline` is the ray tracing pipeline object containing the shaders.
- `firstGroup` is the index of the first group to retrieve a handle for from the VkRayTracingPipelineCreateInfoKHR::`pGroups` array.

- `groupCount` is the number of shader handles to retrieve.
- `dataSize` is the size in bytes of the buffer pointed to by `pData`.
- `pData` is a pointer to an application-allocated buffer where the results will be written.

On success, an array of `groupCount` shader handles will be written to `pData`, with each element being of size VkPhysicalDeviceRayTracingPipelinePropertiesKHR::`shaderGroupHandleCaptureReplaySize`. The first handle written corresponds to the group identified by `firstGroup`, with subsequent handles corresponding to consecutive shader groups in the order they appear in the VkRayTracingPipelineCreateInfoKHR::`pGroups` array.

Once queried, this opaque data **can** be provided at pipeline creation time (in a subsequent execution), using VkRayTracingShaderGroupCreateInfoKHR::`pShaderGroupCaptureReplayHandle`, as described in Ray Tracing Capture Replay.

If `pipeline` was created with VK_PIPELINE_CREATE_LIBRARY_BIT_KHR and the pipelineLibraryGroupHandles feature is enabled applications **can** query capture replay group handles from that pipeline. The capture replay handle remains bitwise identical for any `pipeline` which references the pipeline library. Group indices are assigned as-if the pipeline was created without VK_PIPELINE_CREATE_LIBRARY_BIT_KHR.

Valid Usage

- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-pipeline-04620
  `pipeline` **must** be a ray tracing pipeline
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-firstGroup-04051
  `firstGroup` **must** be less than the number of shader groups in `pipeline`
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-firstGroup-03483
  The sum of `firstGroup` and `groupCount` **must** be less than or equal to the number of shader groups in `pipeline`
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-dataSize-03484
  `dataSize` **must** be at least VkPhysicalDeviceRayTracingPipelinePropertiesKHR::`shaderGroupHandleCaptureReplaySize` × `groupCount`
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-rayTracingPipelineShaderGroupHandleCaptureReplay-03606
  VkPhysicalDeviceRayTracingPipelineFeaturesKHR::`rayTracingPipelineShaderGroupHandleCaptureReplay` **must** be enabled to call this function
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-pipeline-03607
  `pipeline` **must** have been created with a `flags` that included VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-pipeline-07829
  If the pipelineLibraryGroupHandles feature is not enabled, `pipeline` **must** not have been created with VK_PIPELINE_CREATE_LIBRARY_BIT_KHR

Valid Usage (Implicit)

- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-device-parameter `device` **must** be a valid VkDevice handle
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-pipeline-parameter `pipeline` **must** be a valid VkPipeline handle
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-pData-parameter `pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-dataSize-arraylength `dataSize` **must** be greater than `0`
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-pipeline-parent `pipeline` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To query the pipeline stack size of shaders in a shader group in the ray tracing pipeline, call:

```
// Provided by VK_KHR_ray_tracing_pipeline
VkDeviceSize vkGetRayTracingShaderGroupStackSizeKHR(
    VkDevice                                    device,
    VkPipeline                                  pipeline,
    uint32_t                                    group,
    VkShaderGroupShaderKHR                      groupShader);
```

- `device` is the logical device containing the ray tracing pipeline.
- `pipeline` is the ray tracing pipeline object containing the shader groups.
- `group` is the index of the shader group to query.
- `groupShader` is the type of shader from the group to query.

The return value is the ray tracing pipeline stack size in bytes for the specified shader as called from the specified shader group.

Valid Usage

- VUID-vkGetRayTracingShaderGroupStackSizeKHR-pipeline-04622
  `pipeline` **must** be a ray tracing pipeline
- VUID-vkGetRayTracingShaderGroupStackSizeKHR-group-03608
  The value of `group` **must** be less than the number of shader groups in `pipeline`
- VUID-vkGetRayTracingShaderGroupStackSizeKHR-groupShader-03609
  The shader identified by `groupShader` in `group` **must** not be VK_SHADER_UNUSED_KHR

Valid Usage (Implicit)

- VUID-vkGetRayTracingShaderGroupStackSizeKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetRayTracingShaderGroupStackSizeKHR-pipeline-parameter
  `pipeline` **must** be a valid VkPipeline handle
- VUID-vkGetRayTracingShaderGroupStackSizeKHR-groupShader-parameter
  `groupShader` **must** be a valid VkShaderGroupShaderKHR value
- VUID-vkGetRayTracingShaderGroupStackSizeKHR-pipeline-parent
  `pipeline` **must** have been created, allocated, or retrieved from `device`

Possible values of `groupShader` in vkGetRayTracingShaderGroupStackSizeKHR are:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef enum VkShaderGroupShaderKHR {
    VK_SHADER_GROUP_SHADER_GENERAL_KHR = 0,
    VK_SHADER_GROUP_SHADER_CLOSEST_HIT_KHR = 1,
    VK_SHADER_GROUP_SHADER_ANY_HIT_KHR = 2,
    VK_SHADER_GROUP_SHADER_INTERSECTION_KHR = 3,
} VkShaderGroupShaderKHR;
```

- VK_SHADER_GROUP_SHADER_GENERAL_KHR uses the shader specified in the group with VkRayTracingShaderGroupCreateInfoKHR::`generalShader`
- VK_SHADER_GROUP_SHADER_CLOSEST_HIT_KHR uses the shader specified in the group with VkRayTracingShaderGroupCreateInfoKHR::`closestHitShader`
- VK_SHADER_GROUP_SHADER_ANY_HIT_KHR uses the shader specified in the group with VkRayTracingShaderGroupCreateInfoKHR::`anyHitShader`
- VK_SHADER_GROUP_SHADER_INTERSECTION_KHR uses the shader specified in the group with VkRayTracingShaderGroupCreateInfoKHR::`intersectionShader`

To dynamically set the stack size for a ray tracing pipeline, call:

```
// Provided by VK_KHR_ray_tracing_pipeline
void vkCmdSetRayTracingPipelineStackSizeKHR(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    pipelineStackSize);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pipelineStackSize` is the stack size to use for subsequent ray tracing trace commands.

This command sets the stack size for subsequent ray tracing commands when the ray tracing pipeline is created with VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, the stack size is computed as described in Ray Tracing Pipeline Stack.

Valid Usage

- VUID-vkCmdSetRayTracingPipelineStackSizeKHR-pipelineStackSize-03610
  `pipelineStackSize` **must** be large enough for any dynamic execution through the shaders in the ray tracing pipeline used by a subsequent trace call

Valid Usage (Implicit)

- VUID-vkCmdSetRayTracingPipelineStackSizeKHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetRayTracingPipelineStackSizeKHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetRayTracingPipelineStackSizeKHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdSetRayTracingPipelineStackSizeKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdSetRayTracingPipelineStackSizeKHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	State

Conditional Rendering

vkCmdSetRayTracingPipelineStackSizeKHR is not affected by conditional rendering

10.6. Pipeline Destruction

To destroy a pipeline, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyPipeline(
    VkDevice                                    device,
    VkPipeline                                  pipeline,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the pipeline.
- `pipeline` is the handle of the pipeline to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyPipeline-pipeline-00765
  All submitted commands that refer to `pipeline` **must** have completed execution
- VUID-vkDestroyPipeline-pipeline-00766
  If `VkAllocationCallbacks` were provided when `pipeline` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyPipeline-pipeline-00767
  If no `VkAllocationCallbacks` were provided when `pipeline` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyPipeline-device-parameter
  `device` **must** be a valid VkDevice handle

- VUID-vkDestroyPipeline-pipeline-parameter
  If `pipeline` is not VK_NULL_HANDLE, `pipeline` **must** be a valid VkPipeline handle
- VUID-vkDestroyPipeline-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyPipeline-pipeline-parent
  If `pipeline` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `pipeline` **must** be externally synchronized

10.7. Pipeline Derivatives

A pipeline derivative is a child pipeline created from a parent pipeline, where the child and parent are expected to have much commonality.

The goal of derivative pipelines is that they be cheaper to create using the parent as a starting point, and that it be more efficient (on either host or device) to switch/bind between children of the same parent.

A derivative pipeline is created by setting the VK_PIPELINE_CREATE_DERIVATIVE_BIT flag in the `Vk*PipelineCreateInfo` structure. If this is set, then exactly one of `basePipelineHandle` or `basePipelineIndex` members of the structure **must** have a valid handle/index, and specifies the parent pipeline. If `basePipelineHandle` is used, the parent pipeline **must** have already been created. If `basePipelineIndex` is used, then the parent is being created in the same command. VK_NULL_HANDLE acts as the invalid handle for `basePipelineHandle`, and -1 is the invalid index for `basePipelineIndex`. If `basePipelineIndex` is used, the base pipeline **must** appear earlier in the array. The base pipeline **must** have been created with the VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT flag set.

10.8. Pipeline Cache

Pipeline cache objects allow the result of pipeline construction to be reused between pipelines and between runs of an application. Reuse between pipelines is achieved by passing the same pipeline cache object when creating multiple related pipelines. Reuse across runs of an application is achieved by retrieving pipeline cache contents in one run of an application, saving the contents, and using them to preinitialize a pipeline cache on a subsequent run. The contents of the pipeline cache objects are managed by the implementation. Applications **can** manage the host memory consumed by a pipeline cache object and control the amount of data retrieved from a pipeline cache object.

Pipeline cache objects are represented by `VkPipelineCache` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkPipelineCache)
```

10.8.1. Creating a Pipeline Cache

To create pipeline cache objects, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreatePipelineCache(
    VkDevice                                    device,
    const VkPipelineCacheCreateInfo*            pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkPipelineCache*                            pPipelineCache);
```

- `device` is the logical device that creates the pipeline cache object.
- `pCreateInfo` is a pointer to a VkPipelineCacheCreateInfo structure containing initial parameters for the pipeline cache object.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pPipelineCache` is a pointer to a VkPipelineCache handle in which the resulting pipeline cache object is returned.

NOTE Applications **can** track and manage the total host memory size of a pipeline cache object using the `pAllocator`. Applications **can** limit the amount of data retrieved from a pipeline cache object in `vkGetPipelineCacheData`. Implementations **should** not internally limit the total number of entries added to a pipeline cache object or the total host memory consumed.

Once created, a pipeline cache **can** be passed to the vkCreateGraphicsPipelines, vkCreateRayTracingPipelinesKHR, and vkCreateComputePipelines commands. If the pipeline cache passed into these commands is not VK_NULL_HANDLE, the implementation will query it for possible reuse opportunities and update it with new content. The use of the pipeline cache object in these commands is internally synchronized, and the same pipeline cache object **can** be used in multiple threads simultaneously.

If `flags` of `pCreateInfo` includes VK_PIPELINE_CACHE_CREATE_EXTERNALLY_SYNCHRONIZED_BIT, all commands that modify the returned pipeline cache object **must** be externally synchronized.

NOTE Implementations **should** make every effort to limit any critical sections to the actual accesses to the cache, which is expected to be significantly shorter than the duration of the `vkCreate*Pipelines` commands.

Valid Usage (Implicit)

- VUID-vkCreatePipelineCache-device-parameter

`device` **must** be a valid VkDevice handle

- VUID-vkCreatePipelineCache-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkPipelineCacheCreateInfo structure
- VUID-vkCreatePipelineCache-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreatePipelineCache-pPipelineCache-parameter
  `pPipelineCache` **must** be a valid pointer to a VkPipelineCache handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkPipelineCacheCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineCacheCreateInfo {
    VkStructureType               sType;
    const void*                   pNext;
    VkPipelineCacheCreateFlags    flags;
    size_t                        initialDataSize;
    const void*                   pInitialData;
} VkPipelineCacheCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkPipelineCacheCreateFlagBits specifying the behavior of the pipeline cache.
- `initialDataSize` is the number of bytes in `pInitialData`. If `initialDataSize` is zero, the pipeline cache will initially be empty.
- `pInitialData` is a pointer to previously retrieved pipeline cache data. If the pipeline cache data is incompatible (as defined below) with the device, the pipeline cache will be initially empty. If `initialDataSize` is zero, `pInitialData` is ignored.

Valid Usage

- VUID-VkPipelineCacheCreateInfo-initialDataSize-00768
  If `initialDataSize` is not `0`, it **must** be equal to the size of `pInitialData`, as returned by `vkGetPipelineCacheData` when `pInitialData` was originally retrieved
- VUID-VkPipelineCacheCreateInfo-initialDataSize-00769
  If `initialDataSize` is not `0`, `pInitialData` **must** have been retrieved from a previous call to `vkGetPipelineCacheData`
- VUID-VkPipelineCacheCreateInfo-pipelineCreationCacheControl-02892
  If the `pipelineCreationCacheControl` feature is not enabled, `flags` **must** not include VK_PIPELINE_CACHE_CREATE_EXTERNALLY_SYNCHRONIZED_BIT
- VUID-VkPipelineCacheCreateInfo-maintenance8-10200
  If the `maintenance8` feature is not enabled, `flags` **must** not include VK_PIPELINE_CACHE_CREATE_INTERNALLY_SYNCHRONIZED_MERGE_BIT_KHR
- VUID-VkPipelineCacheCreateInfo-flags-10201
  If `flags` includes VK_PIPELINE_CACHE_CREATE_EXTERNALLY_SYNCHRONIZED_BIT, it **must** not include VK_PIPELINE_CACHE_CREATE_INTERNALLY_SYNCHRONIZED_MERGE_BIT_KHR

Valid Usage (Implicit)

- VUID-VkPipelineCacheCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_CACHE_CREATE_INFO
- VUID-VkPipelineCacheCreateInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkPipelineCacheCreateInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkPipelineCacheCreateFlagBits values
- VUID-VkPipelineCacheCreateInfo-pInitialData-parameter
  If `initialDataSize` is not `0`, `pInitialData` **must** be a valid pointer to an array of `initialDataSize` bytes

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineCacheCreateFlags;
```

`VkPipelineCacheCreateFlags` is a bitmask type for setting a mask of zero or more VkPipelineCacheCreateFlagBits.

Bits which **can** be set in VkPipelineCacheCreateInfo::`flags`, specifying behavior of the pipeline cache, are:

```
// Provided by VK_VERSION_1_0, VK_KHR_maintenance8
typedef enum VkPipelineCacheCreateFlagBits {
  // Provided by VK_VERSION_1_3
    VK_PIPELINE_CACHE_CREATE_EXTERNALLY_SYNCHRONIZED_BIT = 0x00000001,
  // Provided by VK_KHR_maintenance8
    VK_PIPELINE_CACHE_CREATE_INTERNALLY_SYNCHRONIZED_MERGE_BIT_KHR = 0x00000008,
} VkPipelineCacheCreateFlagBits;
```

- VK_PIPELINE_CACHE_CREATE_EXTERNALLY_SYNCHRONIZED_BIT specifies that all commands that modify the created VkPipelineCache will be externally synchronized. When set, the implementation **may** skip any unnecessary processing needed to support simultaneous modification from multiple threads where allowed.
- VK_PIPELINE_CACHE_CREATE_INTERNALLY_SYNCHRONIZED_MERGE_BIT_KHR specifies that when the created VkPipelineCache is used as the `dstCache` parameter of vkMergePipelineCaches, it does not need to be externally synchronized. This flag is mutually exclusive with VK_PIPELINE_CACHE_CREATE_EXTERNALLY_SYNCHRONIZED_BIT.

10.8.2. Merging Pipeline Caches

Pipeline cache objects **can** be merged using the command:

```
// Provided by VK_VERSION_1_0
VkResult vkMergePipelineCaches(
    VkDevice                                    device,
    VkPipelineCache                             dstCache,
    uint32_t                                    srcCacheCount,
    const VkPipelineCache*                      pSrcCaches);
```

- `device` is the logical device that owns the pipeline cache objects.
- `dstCache` is the handle of the pipeline cache to merge results into.
- `srcCacheCount` is the length of the `pSrcCaches` array.
- `pSrcCaches` is a pointer to an array of pipeline cache handles, which will be merged into `dstCache`. The previous contents of `dstCache` are included after the merge.

NOTE	The details of the merge operation are implementation-dependent, but implementations **should** merge the contents of the specified pipelines and prune duplicate entries.

Valid Usage

- VUID-vkMergePipelineCaches-dstCache-00770
  `dstCache` **must** not appear in the list of source caches
- VUID-vkMergePipelineCaches-dstCache-10202
  Host access to `dstCache` **must** be externally synchronized if it was not created with

VK_PIPELINE_CACHE_CREATE_INTERNALLY_SYNCHRONIZED_MERGE_BIT_KHR

Valid Usage (Implicit)

- VUID-vkMergePipelineCaches-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkMergePipelineCaches-dstCache-parameter
  `dstCache` **must** be a valid VkPipelineCache handle
- VUID-vkMergePipelineCaches-pSrcCaches-parameter
  `pSrcCaches` **must** be a valid pointer to an array of `srcCacheCount` valid VkPipelineCache handles
- VUID-vkMergePipelineCaches-srcCacheCount-arraylength
  `srcCacheCount` **must** be greater than `0`
- VUID-vkMergePipelineCaches-dstCache-parent
  `dstCache` **must** have been created, allocated, or retrieved from `device`
- VUID-vkMergePipelineCaches-pSrcCaches-parent
  Each element of `pSrcCaches` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

10.8.3. Retrieving Pipeline Cache Data

Data **can** be retrieved from a pipeline cache object using the command:

```
// Provided by VK_VERSION_1_0
VkResult vkGetPipelineCacheData(
    VkDevice                                    device,
    VkPipelineCache                             pipelineCache,
    size_t*                                     pDataSize,
    void*                                       pData);
```

- `device` is the logical device that owns the pipeline cache.

- `pipelineCache` is the pipeline cache to retrieve data from.
- `pDataSize` is a pointer to a `size_t` value related to the amount of data in the pipeline cache, as described below.
- `pData` is either `NULL` or a pointer to a buffer.

If `pData` is `NULL`, then the maximum size of the data that **can** be retrieved from the pipeline cache, in bytes, is returned in `pDataSize`. Otherwise, `pDataSize` **must** point to a variable set by the application to the size of the buffer, in bytes, pointed to by `pData`, and on return the variable is overwritten with the amount of data actually written to `pData`. If `pDataSize` is less than the maximum size that **can** be retrieved by the pipeline cache, at most `pDataSize` bytes will be written to `pData`, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all of the pipeline cache was returned.

Any data written to `pData` is valid and **can** be provided as the `pInitialData` member of the VkPipelineCacheCreateInfo structure passed to `vkCreatePipelineCache`.

Two calls to `vkGetPipelineCacheData` with the same parameters **must** retrieve the same data unless a command that modifies the contents of the cache is called between them.

The initial bytes written to `pData` **must** be a header as described in the Pipeline Cache Header section.

If `pDataSize` is less than what is necessary to store this header, nothing will be written to `pData` and zero will be written to `pDataSize`.

NOTE

This query does not behave consistently with the behavior described in Opaque Binary Data Results, for historical reasons.

If the amount of data available is larger than the passed `pDataSize`, the query returns up to the size of the passed buffer, and signals overflow with a VK_INCOMPLETE success status instead of returning a VK_ERROR_NOT_ENOUGH_SPACE_KHR error status.

Valid Usage (Implicit)

- VUID-vkGetPipelineCacheData-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetPipelineCacheData-pipelineCache-parameter
  `pipelineCache` **must** be a valid VkPipelineCache handle
- VUID-vkGetPipelineCacheData-pDataSize-parameter
  `pDataSize` **must** be a valid pointer to a `size_t` value
- VUID-vkGetPipelineCacheData-pData-parameter
  If the value referenced by `pDataSize` is not `0`, and `pData` is not `NULL`, `pData` **must** be a valid pointer to an array of `pDataSize` bytes
- VUID-vkGetPipelineCacheData-pipelineCache-parent
  `pipelineCache` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

10.8.4. Pipeline Cache Header

Applications **can** store the data retrieved from the pipeline cache, and use these data, possibly in a future run of the application, to populate new pipeline cache objects. The results of pipeline compiles, however, **may** depend on the vendor ID, device ID, driver version, and other details of the device. To enable applications to detect when previously retrieved data is incompatible with the device, the pipeline cache data **must** begin with a valid pipeline cache header.

NOTE Structures described in this section are not part of the Vulkan API and are only used to describe the representation of data elements in pipeline cache data. Accordingly, the valid usage clauses defined for structures defined in this section do not define valid usage conditions for APIs accepting pipeline cache data as input, as providing invalid pipeline cache data as input to any Vulkan API commands will result in the provided pipeline cache data being ignored.

Version one of the pipeline cache header is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineCacheHeaderVersionOne {
    uint32_t                        headerSize;
    VkPipelineCacheHeaderVersion    headerVersion;
    uint32_t                        vendorID;
    uint32_t                        deviceID;
    uint8_t                         pipelineCacheUUID[VK_UUID_SIZE];
} VkPipelineCacheHeaderVersionOne;
```

- `headerSize` is the length in bytes of the pipeline cache header.
- `headerVersion` is a VkPipelineCacheHeaderVersion value specifying the version of the header. A consumer of the pipeline cache **should** use the cache version to interpret the remainder of the cache header. `headerVersion` **must** be written as exactly 4 bytes.
- `vendorID` is the `VkPhysicalDeviceProperties`::`vendorID` of the implementation.
- `deviceID` is the `VkPhysicalDeviceProperties`::`deviceID` of the implementation.

- `pipelineCacheUUID` is the `VkPhysicalDeviceProperties`::`pipelineCacheUUID` of the implementation.

Unlike most structures declared by the Vulkan API, all fields of this structure are written with the least significant byte first, regardless of host byte-order.

The C language specification does not define the packing of structure members. This layout assumes tight structure member packing, with members laid out in the order listed in the structure, and the intended size of the structure is 32 bytes. If a compiler produces code that diverges from that pattern, applications **must** employ another method to set values at the correct offsets.

Valid Usage

- VUID-VkPipelineCacheHeaderVersionOne-headerSize-04967
  `headerSize` **must** be 32
- VUID-VkPipelineCacheHeaderVersionOne-headerVersion-04968
  `headerVersion` **must** be VK_PIPELINE_CACHE_HEADER_VERSION_ONE
- VUID-VkPipelineCacheHeaderVersionOne-headerSize-08990
  `headerSize` **must** not exceed the size of the pipeline cache

Valid Usage (Implicit)

- VUID-VkPipelineCacheHeaderVersionOne-headerVersion-parameter
  `headerVersion` **must** be a valid VkPipelineCacheHeaderVersion value

Possible values of the `headerVersion` value of the pipeline cache header are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPipelineCacheHeaderVersion {
    VK_PIPELINE_CACHE_HEADER_VERSION_ONE = 1,
} VkPipelineCacheHeaderVersion;
```

- VK_PIPELINE_CACHE_HEADER_VERSION_ONE specifies version one of the pipeline cache, described by VkPipelineCacheHeaderVersionOne.

10.8.5. Destroying a Pipeline Cache

To destroy a pipeline cache, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyPipelineCache(
    VkDevice                                    device,
    VkPipelineCache                             pipelineCache,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the pipeline cache object.
- `pipelineCache` is the handle of the pipeline cache to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyPipelineCache-pipelineCache-00771
  If `VkAllocationCallbacks` were provided when `pipelineCache` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyPipelineCache-pipelineCache-00772
  If no `VkAllocationCallbacks` were provided when `pipelineCache` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyPipelineCache-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyPipelineCache-pipelineCache-parameter
  If `pipelineCache` is not VK_NULL_HANDLE, `pipelineCache` **must** be a valid VkPipelineCache handle
- VUID-vkDestroyPipelineCache-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyPipelineCache-pipelineCache-parent
  If `pipelineCache` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `pipelineCache` **must** be externally synchronized

10.9. Pipeline Binaries

Pipeline binary objects allow the result of pipeline construction to be reused between pipelines and between runs of an application. Reuse is achieved by extracting pipeline binaries from a VkPipeline object, associating them with a corresponding VkPipelineBinaryKeyKHR and then adding a VkPipelineBinaryInfoKHR to the `pNext` chain of any `Vk*PipelineCreateInfo` when creating a pipeline. Pipeline binaries can be reused between runs by extracting `VkPipelineBinaryDataKHR` from `VkPipelineBinaryKHR` objects, saving the contents, and then using them to create a `VkPipelineBinaryKHR` object on subsequent runs.

When creating a pipeline that includes VkPipelineBinaryInfoKHR in the `pNext` chain, or has the

VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR flag set, the use of VkPipelineCache objects is not allowed.

Pipeline binary objects are represented by VkPipelineBinaryKHR handles:

```
// Provided by VK_KHR_pipeline_binary
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkPipelineBinaryKHR)
```

10.9.1. Generating the Pipeline Key

To generate the key for a particular pipeline creation info, call:

```
// Provided by VK_KHR_pipeline_binary
VkResult vkGetPipelineKeyKHR(
    VkDevice                                    device,
    const VkPipelineCreateInfoKHR*              pPipelineCreateInfo,
    VkPipelineBinaryKeyKHR*                     pPipelineKey);
```

- device is the logical device that creates the pipeline object.
- pPipelineCreateInfo is NULL or a pointer to a VkPipelineCreateInfoKHR structure.
- pPipelineKey is a pointer to a VkPipelineBinaryKeyKHR structure in which the resulting key is returned.

If pPipelineCreateInfo is NULL, then the implementation **must** return the global key that applies to all pipelines. If the key obtained in this way changes between saving and restoring data obtained from vkGetPipelineBinaryDataKHR in a different VkDevice, then the application **must** assume that the restored data is invalid and cannot be passed to vkCreatePipelineBinariesKHR. Otherwise the application **can** assume the data is still valid.

If pPipelineCreateInfo is not NULL, the key obtained functions as a method to compare two pipeline creation info structures. Implementations **may** not compare parts of a pipeline creation info which would not contribute to the final binary output. If a shader module identifier is used instead of a shader module, the pPipelineKey generated **must** be equal to the key generated when using the shader module from which the identifier was queried. If the content of two pPipelineKey are equal, pipelines created with the two pPipelineCreateInfo->pNext create infos **must** produce the same VkPipelineBinaryKHR contents.

The pipeline key is distinct from pipeline binary key. Pipeline binary keys **can** only be obtained after compilation. The pipeline key is intended to optionally allow associating pipeline create info with multiple pipeline binary keys.

Valid Usage

- VUID-vkGetPipelineKeyKHR-pNext-09605
  The pNext chain of pPipelineCreateInfo **must** not set VkPipelineBinaryInfoKHR ::binaryCount to a value greater than 0

Valid Usage (Implicit)

- VUID-vkGetPipelineKeyKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetPipelineKeyKHR-pPipelineCreateInfo-parameter
  If `pPipelineCreateInfo` is not `NULL`, `pPipelineCreateInfo` **must** be a valid pointer to a valid VkPipelineCreateInfoKHR structure
- VUID-vkGetPipelineKeyKHR-pPipelineKey-parameter
  `pPipelineKey` **must** be a valid pointer to a VkPipelineBinaryKeyKHR structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkPipelineCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineCreateInfoKHR {
    VkStructureType    sType;
    void*              pNext;
} VkPipelineCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is a pointer to a structure extending this structure.

Valid Usage

- VUID-VkPipelineCreateInfoKHR-pNext-09604
  `pNext` **must** be a pointer to a valid instance of VkGraphicsPipelineCreateInfo, VkRayTracingPipelineCreateInfoKHR, or VkComputePipelineCreateInfo

Valid Usage (Implicit)

- VUID-VkPipelineCreateInfoKHR-sType-sType

sType must be VK_STRUCTURE_TYPE_PIPELINE_CREATE_INFO_KHR

The VkPipelineBinaryKeyKHR structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineBinaryKeyKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           keySize;
    uint8_t            key[VK_MAX_PIPELINE_BINARY_KEY_SIZE_KHR];
} VkPipelineBinaryKeyKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- keySize is the size, in bytes, of valid data returned in key.
- key is a buffer of opaque data specifying a pipeline binary key.

Any returned values beyond the first keySize bytes are undefined. Implementations **must** return a keySize greater than 0, and less-or-equal to VK_MAX_PIPELINE_BINARY_KEY_SIZE_KHR.

Two keys are considered equal if keySize is equal and the first keySize bytes of key compare equal.

Implementations **may** return a different keySize for different binaries.

Implementations **should** ensure that keySize is large enough to uniquely identify a pipeline binary.

Valid Usage (Implicit)

- VUID-VkPipelineBinaryKeyKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PIPELINE_BINARY_KEY_KHR
- VUID-VkPipelineBinaryKeyKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkPipelineBinaryKeyKHR-key-parameter
  Each element of key **must** be a valid pointer to an array of keySize uint8_t values
- VUID-VkPipelineBinaryKeyKHR-keySize-arraylength
  keySize **must** be greater than 0

VK_MAX_PIPELINE_BINARY_KEY_SIZE_KHR is the length in bytes of a binary key, as returned in VkPipelineBinaryKeyKHR::keySize.

```
#define VK_MAX_PIPELINE_BINARY_KEY_SIZE_KHR 32U
```

10.9.2. Creating Pipeline Binaries

To create pipeline binary objects, call:

```
// Provided by VK_KHR_pipeline_binary
VkResult vkCreatePipelineBinariesKHR(
    VkDevice                                    device,
    const VkPipelineBinaryCreateInfoKHR*        pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkPipelineBinaryHandlesInfoKHR*             pBinaries);
```

- `device` is the logical device that creates the pipeline binary objects.
- `pCreateInfo` is a pointer to a VkPipelineBinaryCreateInfoKHR structure that contains the data to create the pipeline binaries from.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pBinaries` is a pointer to a VkPipelineBinaryHandlesInfoKHR structure in which the resulting pipeline binaries are returned.

The implementation will attempt to create all pipeline binaries. If creation fails for any pipeline binary, then:

- The corresponding entry in the `pPipelineBinaries` output array will be filled with VK_NULL_HANDLE.
- The VkResult returned by vkCreatePipelineBinariesKHR will contain the error value for the first entry in the output array in `pBinaries` containing VK_NULL_HANDLE.

Valid Usage (Implicit)

- VUID-vkCreatePipelineBinariesKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreatePipelineBinariesKHR-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkPipelineBinaryCreateInfoKHR structure
- VUID-vkCreatePipelineBinariesKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreatePipelineBinariesKHR-pBinaries-parameter
  `pBinaries` **must** be a valid pointer to a VkPipelineBinaryHandlesInfoKHR structure

Return Codes

Success

- VK_INCOMPLETE
- VK_PIPELINE_BINARY_MISSING_KHR

- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkPipelineBinaryHandlesInfoKHR structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineBinaryHandlesInfoKHR {
    VkStructureType         sType;
    const void*             pNext;
    uint32_t                pipelineBinaryCount;
    VkPipelineBinaryKHR*    pPipelineBinaries;
} VkPipelineBinaryHandlesInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pipelineBinaryCount is the number of binaries associated with this pipeline or the number of entries in the pPipelineBinaries array.
- pPipelineBinaries is NULL or a pointer to an array of VkPipelineBinaryKHR handles in which the resulting pipeline binaries are returned.

If pPipelineBinaries is NULL, the number of binaries that would be created is returned in pipelineBinaryCount. Otherwise, pipelineBinaryCount **must** be the number of entries in the pPipelineBinaries array, and on return from vkCreatePipelineBinariesKHR pipelineBinaryCount is overwritten with the number of handles actually written to pPipelineBinaries. If the value of pipelineBinaryCount is less than the number of binaries that would have been created, at most pipelineBinaryCount handles will be written to pPipelineBinaries and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that pPipelineBinaries was not large enough to create all the binaries.

Valid Usage (Implicit)

- VUID-VkPipelineBinaryHandlesInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PIPELINE_BINARY_HANDLES_INFO_KHR
- VUID-VkPipelineBinaryHandlesInfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkPipelineBinaryHandlesInfoKHR-pPipelineBinaries-parameter
  If pipelineBinaryCount is not 0, and pPipelineBinaries is not NULL, pPipelineBinaries **must** be a valid pointer to an array of pipelineBinaryCount VkPipelineBinaryKHR handles

The VkPipelineBinaryCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineBinaryCreateInfoKHR {
    VkStructureType                          sType;
    const void*                              pNext;
    const VkPipelineBinaryKeysAndDataKHR*    pKeysAndDataInfo;
    VkPipeline                               pipeline;
    const VkPipelineCreateInfoKHR*           pPipelineCreateInfo;
} VkPipelineBinaryCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pKeysAndDataInfo is NULL or a pointer to a VkPipelineBinaryKeysAndDataKHR structure that contains keys and data to create the pipeline binaries from.
- pipeline is VK_NULL_HANDLE or a VkPipeline that contains data to create the pipeline binaries from.
- pPipelineCreateInfo is NULL or a pointer to a VkPipelineCreateInfoKHR structure with the pipeline creation info. This is used to probe the implementation's internal cache for pipeline binaries.

When pPipelineCreateInfo is not NULL, an implementation will attempt to retrieve pipeline binary data from an internal cache external to the application if pipelineBinaryInternalCache is VK_TRUE. Applications **can** use this to determine if a pipeline **can** be created without compilation. If the implementation fails to create a pipeline binary due to missing an internal cache entry, VK_PIPELINE_BINARY_MISSING_KHR is returned. If creation succeeds, the resulting binary **can** be used to create a pipeline. VK_PIPELINE_BINARY_MISSING_KHR **may** be returned for any reason in this situation, even if creating a pipeline binary with the same parameters that succeeded earlier.

If pipelineBinaryPrecompiledInternalCache is VK_TRUE, the implementation **may** be able to create pipeline binaries even when pPipelineCreateInfo has not been used to create binaries before by the application.

NOTE	On some platforms, internal pipeline caches may be pre-populated before running the application.

Valid Usage

- VUID-VkPipelineBinaryCreateInfoKHR-pipeline-09607
  If pipeline is not VK_NULL_HANDLE, pipeline **must** have been created with VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR
- VUID-VkPipelineBinaryCreateInfoKHR-pipeline-09608
  If pipeline is not VK_NULL_HANDLE, vkReleaseCapturedPipelineDataKHR **must** not have been called on pipeline prior to this command
- VUID-VkPipelineBinaryCreateInfoKHR-pipelineBinaryInternalCache-09609

If pipelineBinaryInternalCache is VK_FALSE pPipelineCreateInfo **must** be NULL

- VUID-VkPipelineBinaryCreateInfoKHR-device-09610
  If device was created with VkDevicePipelineBinaryInternalCacheControlKHR ::disableInternalCache set to VK_TRUE, pPipelineCreateInfo **must** be NULL
- VUID-VkPipelineBinaryCreateInfoKHR-pKeysAndDataInfo-09619
  One and only one of pKeysAndDataInfo, pipeline, or pPipelineCreateInfo **must** be non-NULL
- VUID-VkPipelineBinaryCreateInfoKHR-pPipelineCreateInfo-09606
  If pPipelineCreateInfo is not NULL, the pNext chain of pPipelineCreateInfo **must** not set VkPipelineBinaryInfoKHR::binaryCount to a value greater than 0

Valid Usage (Implicit)

- VUID-VkPipelineBinaryCreateInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PIPELINE_BINARY_CREATE_INFO_KHR
- VUID-VkPipelineBinaryCreateInfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkPipelineBinaryCreateInfoKHR-pKeysAndDataInfo-parameter
  If pKeysAndDataInfo is not NULL, pKeysAndDataInfo **must** be a valid pointer to a valid VkPipelineBinaryKeysAndDataKHR structure
- VUID-VkPipelineBinaryCreateInfoKHR-pipeline-parameter
  If pipeline is not VK_NULL_HANDLE, pipeline **must** be a valid VkPipeline handle
- VUID-VkPipelineBinaryCreateInfoKHR-pPipelineCreateInfo-parameter
  If pPipelineCreateInfo is not NULL, pPipelineCreateInfo **must** be a valid pointer to a valid VkPipelineCreateInfoKHR structure

The VkPipelineBinaryKeysAndDataKHR structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineBinaryKeysAndDataKHR {
    uint32_t                            binaryCount;
    const VkPipelineBinaryKeyKHR*       pPipelineBinaryKeys;
    const VkPipelineBinaryDataKHR*      pPipelineBinaryData;
} VkPipelineBinaryKeysAndDataKHR;
```

- binaryCount is the size of the pPipelineBinaryKeys and pPipelineBinaryData arrays.
- pPipelineBinaryKeys is a pointer to an array of VkPipelineBinaryKeyKHR structures containing the pipeline binary keys.
- pPipelineBinaryData is a pointer to an array of VkPipelineBinaryDataKHR structures containing the pipeline binary data.

Valid Usage (Implicit)

- VUID-VkPipelineBinaryKeysAndDataKHR-pPipelineBinaryKeys-parameter
  `pPipelineBinaryKeys` **must** be a valid pointer to an array of `binaryCount` valid `VkPipelineBinaryKeyKHR` structures
- VUID-VkPipelineBinaryKeysAndDataKHR-pPipelineBinaryData-parameter
  `pPipelineBinaryData` **must** be a valid pointer to an array of `binaryCount` valid `VkPipelineBinaryDataKHR` structures
- VUID-VkPipelineBinaryKeysAndDataKHR-binaryCount-arraylength
  `binaryCount` **must** be greater than `0`

The `VkPipelineBinaryDataKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineBinaryDataKHR {
    size_t    dataSize;
    void*     pData;
} VkPipelineBinaryDataKHR;
```

- `dataSize` is the size of the `pData` buffer in bytes.
- `pData` is a pointer to a buffer of `dataSize` bytes that contains pipeline binary data obtained from `vkGetPipelineBinaryDataKHR`.

Valid Usage (Implicit)

- VUID-VkPipelineBinaryDataKHR-pData-parameter
  `pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-VkPipelineBinaryDataKHR-dataSize-arraylength
  `dataSize` **must** be greater than `0`

10.9.3. Retrieving Pipeline Binary Data

Data **can** be retrieved from a pipeline binary object using the command:

```
// Provided by VK_KHR_pipeline_binary
VkResult vkGetPipelineBinaryDataKHR(
    VkDevice                                    device,
    const VkPipelineBinaryDataInfoKHR*          pInfo,
    VkPipelineBinaryKeyKHR*                     pPipelineBinaryKey,
    size_t*                                     pPipelineBinaryDataSize,
    void*                                       pPipelineBinaryData);
```

- `device` is the logical device that created the pipeline binary.

- `pInfo` is a pointer to a VkPipelineBinaryDataInfoKHR structure which describes the pipeline binary to get data from.
- `pPipelineBinaryKey` is a pointer to a VkPipelineBinaryKeyKHR structure where the key for this binary will be written.
- `pPipelineBinaryDataSize` is a pointer to a `size_t` value related to the amount of data in the pipeline binary, as described below.
- `pPipelineBinaryData` is either `NULL` or a pointer to a buffer.

If `pPipelineBinaryData` is `NULL`, then the size of the data, in bytes, that is required to store the binary is returned in `pPipelineBinaryDataSize`. Otherwise, `pPipelineBinaryDataSize` **must** contain the size of the buffer, in bytes, pointed to by `pPipelineBinaryData`, and on return `pPipelineBinaryDataSize` is overwritten with the size of the data, in bytes, that is required to store the binary. If `pPipelineBinaryDataSize` is less than the size that is required to store the binary, nothing is written to `pPipelineBinaryData` and VK_ERROR_NOT_ENOUGH_SPACE_KHR will be returned, instead of VK_SUCCESS.

If the call returns one of the success return codes, the pipeline binary key is written to `pPipelineBinaryKey`, regardless of whether `pPipelineBinaryData` is `NULL` or not.

If pipelineBinaryCompressedData is VK_FALSE, implementations **should** not return compressed pipeline binary data to the application.

Valid Usage (Implicit)

- VUID-vkGetPipelineBinaryDataKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetPipelineBinaryDataKHR-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkPipelineBinaryDataInfoKHR structure
- VUID-vkGetPipelineBinaryDataKHR-pPipelineBinaryKey-parameter
  `pPipelineBinaryKey` **must** be a valid pointer to a VkPipelineBinaryKeyKHR structure
- VUID-vkGetPipelineBinaryDataKHR-pPipelineBinaryDataSize-parameter
  `pPipelineBinaryDataSize` **must** be a valid pointer to a `size_t` value
- VUID-vkGetPipelineBinaryDataKHR-pPipelineBinaryData-parameter
  If the value referenced by `pPipelineBinaryDataSize` is not `0`, and `pPipelineBinaryData` is not `NULL`, `pPipelineBinaryData` **must** be a valid pointer to an array of `pPipelineBinaryDataSize` bytes

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_NOT_ENOUGH_SPACE_KHR

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkPipelineBinaryDataInfoKHR structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineBinaryDataInfoKHR {
    VkStructureType        sType;
    void*                  pNext;
    VkPipelineBinaryKHR    pipelineBinary;
} VkPipelineBinaryDataInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pipelineBinary is the pipeline binary to get data from.

Valid Usage (Implicit)

- VUID-VkPipelineBinaryDataInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PIPELINE_BINARY_DATA_INFO_KHR
- VUID-VkPipelineBinaryDataInfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkPipelineBinaryDataInfoKHR-pipelineBinary-parameter
  pipelineBinary **must** be a valid VkPipelineBinaryKHR handle

10.9.4. Releasing Captured Pipeline Binary Data

To release pipeline resources captured with VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR, call:

```
// Provided by VK_KHR_pipeline_binary
VkResult vkReleaseCapturedPipelineDataKHR(
    VkDevice                                    device,
    const VkReleaseCapturedPipelineDataInfoKHR* pInfo,
    const VkAllocationCallbacks*                pAllocator);
```

- device is the logical device that created the pipeline object.
- pInfo is a pointer to a VkReleaseCapturedPipelineDataInfoKHR structure which describes the pipeline to release the data from.
- pAllocator controls host memory allocation as described in the Memory Allocation chapter.

The implementation **may** free any resources captured as a result of creating the pipeline with VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR and put the pipeline into a state as if VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR had not been provided at pipeline creation time.

NOTE	Any resources captured as a result of creating the pipeline with VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR are implicitly freed by vkDestroyPipeline.

Valid Usage

- VUID-vkReleaseCapturedPipelineDataKHR-pipeline-09611
  If `VkAllocationCallbacks` were provided when `pipeline` was created, a compatible set of callbacks **must** be provided in `pAllocator`
- VUID-vkReleaseCapturedPipelineDataKHR-pipeline-09612
  If no VkAllocationCallbacks were provided when `pipeline` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkReleaseCapturedPipelineDataKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkReleaseCapturedPipelineDataKHR-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkReleaseCapturedPipelineDataInfoKHR structure
- VUID-vkReleaseCapturedPipelineDataKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkReleaseCapturedPipelineDataInfoKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkReleaseCapturedPipelineDataInfoKHR {
    VkStructureType    sType;
    void*              pNext;
    VkPipeline         pipeline;
} VkReleaseCapturedPipelineDataInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pipeline` is the handle of the pipeline object to release the data from.

Valid Usage

- VUID-VkReleaseCapturedPipelineDataInfoKHR-pipeline-09613
  `pipeline` **must** have been created with VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR
- VUID-VkReleaseCapturedPipelineDataInfoKHR-pipeline-09618
  `pipeline` **must** not have been used in a previous call to `vkReleaseCapturedPipelineDataKHR`

Valid Usage (Implicit)

- VUID-VkReleaseCapturedPipelineDataInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RELEASE_CAPTURED_PIPELINE_DATA_INFO_KHR
- VUID-VkReleaseCapturedPipelineDataInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkReleaseCapturedPipelineDataInfoKHR-pipeline-parameter
  `pipeline` **must** be a valid VkPipeline handle

Host Synchronization

- Host access to `pipeline` **must** be externally synchronized

10.9.5. Destroying Pipeline Binaries

To destroy a VkPipelineBinaryKHR, call:

```
// Provided by VK_KHR_pipeline_binary
void vkDestroyPipelineBinaryKHR(
    VkDevice                                    device,
    VkPipelineBinaryKHR                         pipelineBinary,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that created the pipeline binary object.
- `pipelineBinary` is the handle of the pipeline binary object to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyPipelineBinaryKHR-pipelineBinary-09614
  If `VkAllocationCallbacks` were provided when `pipelineBinary` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyPipelineBinaryKHR-pipelineBinary-09615
  If no VkAllocationCallbacks were provided when `pipelineBinary` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyPipelineBinaryKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyPipelineBinaryKHR-pipelineBinary-parameter
  If `pipelineBinary` is not VK_NULL_HANDLE, `pipelineBinary` **must** be a valid VkPipelineBinaryKHR handle
- VUID-vkDestroyPipelineBinaryKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyPipelineBinaryKHR-pipelineBinary-parent
  If `pipelineBinary` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `pipelineBinary` **must** be externally synchronized

10.10. Specialization Constants

Specialization constants are a mechanism whereby constants in a SPIR-V module **can** have their constant value specified at the time the `VkPipeline` is created. This allows a SPIR-V module to have constants that **can** be modified while executing an application that uses the Vulkan API.

NOTE	Specialization constants are useful to allow a compute shader to have its local workgroup size changed at runtime by the user, for example.

Each VkPipelineShaderStageCreateInfo structure contains a `pSpecializationInfo` member, which **can** be `NULL` to indicate no specialization constants, or point to a `VkSpecializationInfo` structure.

The VkSpecializationInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSpecializationInfo {
    uint32_t                           mapEntryCount;
    const VkSpecializationMapEntry*    pMapEntries;
    size_t                             dataSize;
    const void*                        pData;
} VkSpecializationInfo;
```

- mapEntryCount is the number of entries in the pMapEntries array.
- pMapEntries is a pointer to an array of VkSpecializationMapEntry structures, which map constant IDs to offsets in pData.
- dataSize is the byte size of the pData buffer.
- pData contains the actual constant values to specialize with.

Valid Usage

- VUID-VkSpecializationInfo-offset-00773
  The offset member of each element of pMapEntries **must** be less than dataSize
- VUID-VkSpecializationInfo-pMapEntries-00774
  The size member of each element of pMapEntries **must** be less than or equal to dataSize minus offset
- VUID-VkSpecializationInfo-constantID-04911
  The constantID value of each element of pMapEntries **must** be unique within pMapEntries

Valid Usage (Implicit)

- VUID-VkSpecializationInfo-pMapEntries-parameter
  If mapEntryCount is not 0, pMapEntries **must** be a valid pointer to an array of mapEntryCount valid VkSpecializationMapEntry structures
- VUID-VkSpecializationInfo-pData-parameter
  If dataSize is not 0, pData **must** be a valid pointer to an array of dataSize bytes

The VkSpecializationMapEntry structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSpecializationMapEntry {
    uint32_t    constantID;
    uint32_t    offset;
    size_t      size;
} VkSpecializationMapEntry;
```

- `constantID` is the ID of the specialization constant in SPIR-V.
- `offset` is the byte offset of the specialization constant value within the supplied data buffer.
- `size` is the byte size of the specialization constant value within the supplied data buffer.

If a `constantID` value is not a specialization constant ID used in the shader, that map entry does not affect the behavior of the pipeline.

Valid Usage

- VUID-VkSpecializationMapEntry-constantID-00776
  For a `constantID` specialization constant declared in a shader, `size` **must** match the byte size of the `constantID`. If the specialization constant is of type `boolean`, `size` **must** be the byte size of VkBool32

In human readable SPIR-V:

```
OpDecorate %x SpecId 13 ; decorate .x component of WorkgroupSize with ID 13
OpDecorate %y SpecId 42 ; decorate .y component of WorkgroupSize with ID 42
OpDecorate %z SpecId 3  ; decorate .z component of WorkgroupSize with ID 3
OpDecorate %wgsize BuiltIn WorkgroupSize ; decorate WorkgroupSize onto constant
%i32 = OpTypeInt 32 0 ; declare an unsigned 32-bit type
%uvec3 = OpTypeVector %i32 3 ; declare a 3 element vector type of unsigned 32-bit
%x = OpSpecConstant %i32 1 ; declare the .x component of WorkgroupSize
%y = OpSpecConstant %i32 1 ; declare the .y component of WorkgroupSize
%z = OpSpecConstant %i32 1 ; declare the .z component of WorkgroupSize
%wgsize = OpSpecConstantComposite %uvec3 %x %y %z ; declare WorkgroupSize
```

From the above we have three specialization constants, one for each of the x, y, and z elements of the WorkgroupSize vector.

Now to specialize the above via the specialization constants mechanism:

```
const VkSpecializationMapEntry entries[] =
{
    {
        .constantID = 13,
        .offset = 0 * sizeof(uint32_t),
        .size = sizeof(uint32_t)
    },
    {
        .constantID = 42,
        .offset = 1 * sizeof(uint32_t),
        .size = sizeof(uint32_t)
    },
    {
        .constantID = 3,
        .offset = 2 * sizeof(uint32_t),
```

```
        .size = sizeof(uint32_t)
    }
};

const uint32_t data[] = { 16, 8, 4 }; // our workgroup size is 16x8x4

const VkSpecializationInfo info =
{
    .mapEntryCount = 3,
    .pMapEntries  = entries,
    .dataSize = 3 * sizeof(uint32_t),
    .pData = data,
};
```

Then when calling vkCreateComputePipelines, and passing the VkSpecializationInfo we defined as the pSpecializationInfo parameter of VkPipelineShaderStageCreateInfo, we will create a compute pipeline with the runtime specified local workgroup size.

Another example would be that an application has a SPIR-V module that has some platform-dependent constants they wish to use.

In human readable SPIR-V:

```
OpDecorate %1 SpecId 0  ; decorate our signed 32-bit integer constant
OpDecorate %2 SpecId 12 ; decorate our 32-bit floating-point constant
%i32 = OpTypeInt 32 1   ; declare a signed 32-bit type
%float = OpTypeFloat 32 ; declare a 32-bit floating-point type
%1 = OpSpecConstant %i32 -1 ; some signed 32-bit integer constant
%2 = OpSpecConstant %float 0.5 ; some 32-bit floating-point constant
```

From the above we have two specialization constants, one is a signed 32-bit integer and the second is a 32-bit floating-point value.

Now to specialize the above via the specialization constants mechanism:

```
struct SpecializationData {
    int32_t data0;
    float data1;
};

const VkSpecializationMapEntry entries[] =
{
    {
        .constantID = 0,
        .offset = offsetof(SpecializationData, data0),
        .size = sizeof(SpecializationData::data0)
    },
    {
        .constantID = 12,
```

```
        .offset = offsetof(SpecializationData, data1),
        .size = sizeof(SpecializationData::data1)
    }
};

SpecializationData data;
data.data0 = -42;    // set the data for the 32-bit integer
data.data1 = 42.0f;  // set the data for the 32-bit floating-point

const VkSpecializationInfo info =
{
    .mapEntryCount = 2,
    .pMapEntries = entries,
    .dataSize = sizeof(data),
    .pData = &data,
};
```

It is legal for a SPIR-V module with specializations to be compiled into a pipeline where no specialization information was provided. SPIR-V specialization constants contain default values such that if a specialization is not provided, the default value will be used. In the examples above, it would be valid for an application to only specialize some of the specialization constants within the SPIR-V module, and let the other constants use their default values encoded within the OpSpecConstant declarations.

10.11. Pipeline Libraries

A pipeline library is a special pipeline that was created using the VK_PIPELINE_CREATE_LIBRARY_BIT_KHR and cannot be bound, instead it defines a set of pipeline state which can be linked into other pipelines. For ray tracing pipelines this includes shaders and shader groups. For graphics pipelines this includes distinct library types defined by VkGraphicsPipelineLibraryFlagBitsEXT. The application **must** maintain the lifetime of a pipeline library based on the pipelines that link with it.

This linkage is achieved by using the following structure within the appropriate creation mechanisms:

The VkPipelineLibraryCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_pipeline_library
typedef struct VkPipelineLibraryCreateInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    uint32_t             libraryCount;
    const VkPipeline*    pLibraries;
} VkPipelineLibraryCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.

- `libraryCount` is the number of pipeline libraries in `pLibraries`.
- `pLibraries` is a pointer to an array of VkPipeline structures specifying pipeline libraries to use when creating a pipeline.

Valid Usage

- VUID-VkPipelineLibraryCreateInfoKHR-pLibraries-03381
  Each element of `pLibraries` **must** have been created with VK_PIPELINE_CREATE_LIBRARY_BIT_KHR
- VUID-VkPipelineLibraryCreateInfoKHR-pLibraries-06855
  If any library in `pLibraries` was created with a shader stage with VkPipelineShaderStageModuleIdentifierCreateInfoEXT and `identifierSize` not equal to 0, the pipeline **must** be created with the VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT flag set

Valid Usage (Implicit)

- VUID-VkPipelineLibraryCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_LIBRARY_CREATE_INFO_KHR
- VUID-VkPipelineLibraryCreateInfoKHR-pLibraries-parameter
  If `libraryCount` is not `0`, `pLibraries` **must** be a valid pointer to an array of `libraryCount` valid VkPipeline handles

Structure Chaining

Extends the structure

- VkGraphicsPipelineCreateInfo

Pipelines created with VK_PIPELINE_CREATE_LIBRARY_BIT_KHR libraries **can** depend on other pipeline libraries in VkPipelineLibraryCreateInfoKHR.

A pipeline library is considered in-use, as long as one of the linking pipelines is in-use. This applies recursively if a pipeline library includes other pipeline libraries.

10.12. Pipeline Binding

Once a pipeline has been created, it **can** be bound to the command buffer using the command:

```
// Provided by VK_VERSION_1_0
void vkCmdBindPipeline(
    VkCommandBuffer                             commandBuffer,
    VkPipelineBindPoint                         pipelineBindPoint,
    VkPipeline                                  pipeline);
```

- `commandBuffer` is the command buffer that the pipeline will be bound to.
- `pipelineBindPoint` is a VkPipelineBindPoint value specifying to which bind point the pipeline is bound. Binding one does not disturb the others.
- `pipeline` is the pipeline to be bound.

Once bound, a pipeline binding affects subsequent commands that interact with the given pipeline type in the command buffer until a different pipeline of the same type is bound to the bind point, or until the pipeline bind point is disturbed by binding a shader object as described in Interaction with Pipelines. Commands that do not interact with the given pipeline type **must** not be affected by the pipeline state.

Valid Usage

- VUID-vkCmdBindPipeline-pipelineBindPoint-00777
  If `pipelineBindPoint` is VK_PIPELINE_BIND_POINT_COMPUTE, the `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations
- VUID-vkCmdBindPipeline-pipelineBindPoint-00778
  If `pipelineBindPoint` is VK_PIPELINE_BIND_POINT_GRAPHICS, the `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdBindPipeline-pipelineBindPoint-00779
  If `pipelineBindPoint` is VK_PIPELINE_BIND_POINT_COMPUTE, `pipeline` **must** be a compute pipeline
- VUID-vkCmdBindPipeline-pipelineBindPoint-00780
  If `pipelineBindPoint` is VK_PIPELINE_BIND_POINT_GRAPHICS, `pipeline` **must** be a graphics pipeline
- VUID-vkCmdBindPipeline-pipeline-00781
  If the `variableMultisampleRate` feature is not supported, `pipeline` is a graphics pipeline, the current subpass uses no attachments, and this is not the first call to this function with a graphics pipeline after transitioning to the current subpass, then the sample count specified by this pipeline **must** match that set in the previous pipeline
- VUID-vkCmdBindPipeline-variableSampleLocations-01525
  If VkPhysicalDeviceSampleLocationsPropertiesEXT::`variableSampleLocations` is VK_FALSE, and `pipeline` is a graphics pipeline created with a `renderPass` that is not VK_NULL_HANDLE and with a VkPipelineSampleLocationsStateCreateInfoEXT structure having its `sampleLocationsEnable` member set to VK_TRUE but without VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT enabled then the current render pass instance **must** have been begun by specifying a VkRenderPassSampleLocationsBeginInfoEXT structure whose

pPostSubpassSampleLocations member contains an element with a subpassIndex matching the current subpass index and the sampleLocationsInfo member of that element **must** match the sampleLocationsInfo specified in VkPipelineSampleLocationsStateCreateInfoEXT when the pipeline was created

- VUID-vkCmdBindPipeline-None-02323
  This command **must** not be recorded when transform feedback is active
- VUID-vkCmdBindPipeline-pipelineBindPoint-02391
  If pipelineBindPoint is VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR, the VkCommandPool that commandBuffer was allocated from **must** support compute operations
- VUID-vkCmdBindPipeline-pipelineBindPoint-02392
  If pipelineBindPoint is VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR, pipeline **must** be a ray tracing pipeline
- VUID-vkCmdBindPipeline-pipelineBindPoint-06721
  If pipelineBindPoint is VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR, commandBuffer **must** not be a protected command buffer
- VUID-vkCmdBindPipeline-pipelineProtectedAccess-07408
  If the pipelineProtectedAccess feature is enabled, and commandBuffer is a protected command buffer, pipeline **must** have been created without VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT
- VUID-vkCmdBindPipeline-pipelineProtectedAccess-07409
  If the pipelineProtectedAccess feature is enabled, and commandBuffer is not a protected command buffer, pipeline **must** have been created without VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT
- VUID-vkCmdBindPipeline-pipeline-03382
  pipeline **must** not have been created with VK_PIPELINE_CREATE_LIBRARY_BIT_KHR set
- VUID-vkCmdBindPipeline-pipelineBindPoint-04881
  If pipelineBindPoint is VK_PIPELINE_BIND_POINT_GRAPHICS and the provokingVertexModePerPipeline limit is VK_FALSE, then the pipeline's VkPipelineRasterizationProvokingVertexStateCreateInfoEXT::provokingVertexMode **must** be the same as that of any other pipelines previously bound to this bind point within the current render pass instance, including any pipeline already bound when beginning the render pass instance

Valid Usage (Implicit)

- VUID-vkCmdBindPipeline-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindPipeline-pipelineBindPoint-parameter
  pipelineBindPoint **must** be a valid VkPipelineBindPoint value
- VUID-vkCmdBindPipeline-pipeline-parameter
  pipeline **must** be a valid VkPipeline handle
- VUID-vkCmdBindPipeline-commandBuffer-recording

commandBuffer **must** be in the recording state

- VUID-vkCmdBindPipeline-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindPipeline-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindPipeline-commonparent
  Both of commandBuffer, and pipeline **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindPipeline is not affected by conditional rendering

Possible values of vkCmdBindPipeline::pipelineBindPoint, specifying the bind point of a pipeline object, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPipelineBindPoint {
    VK_PIPELINE_BIND_POINT_GRAPHICS = 0,
    VK_PIPELINE_BIND_POINT_COMPUTE = 1,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR = 1000165000,
} VkPipelineBindPoint;
```

- VK_PIPELINE_BIND_POINT_COMPUTE specifies binding as a compute pipeline.

- VK_PIPELINE_BIND_POINT_GRAPHICS specifies binding as a graphics pipeline.
- VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR specifies binding as a ray tracing pipeline.

10.12.1. Interaction With Shader Objects

If the `shaderObject` feature is enabled, applications **can** use both pipelines and shader objects at the same time. The interaction between pipelines and shader objects is described in Interaction with Pipelines.

10.13. Dynamic State

When a pipeline object is bound, any pipeline object state that is not specified as dynamic is applied to the command buffer state. Pipeline object state that is specified as dynamic is not applied to the command buffer state at this time.

Instead, dynamic state **can** be modified at any time and persists for the lifetime of the command buffer, or until modified by another dynamic state setting command, or made invalid by binding a pipeline in which that state is statically specified.

When a pipeline object is bound, the following applies to each state parameter:

- If the state is not specified as dynamic in the new pipeline object, then that command buffer state is overwritten by the state in the new pipeline object. Before any draw or dispatch call with this pipeline there **must** not have been any calls to any of the corresponding dynamic state setting commands after this pipeline was bound.
- If the state is specified as dynamic in the new pipeline object, then that command buffer state is not disturbed. Before any draw or dispatch call with this pipeline there **must** have been at least one call to each of the corresponding dynamic state setting commands. The state-setting commands **must** be recorded after command buffer recording was begun, or after the last command binding a pipeline object with that state specified as static, whichever was the latter.
- If the state is not included (corresponding pointer in VkGraphicsPipelineCreateInfo was `NULL` or was ignored) in the new pipeline object, then that command buffer state is not disturbed. For example, mesh shading pipelines do not include vertex input state and therefore do not disturb any such command buffer state.

Dynamic state that does not affect the result of operations **can** be left undefined.

NOTE	For example, if blending is disabled by the pipeline object state then the dynamic color blend constants do not need to be specified in the command buffer, even if this state is specified as dynamic in the pipeline object.

NOTE	Applications running on Vulkan implementations advertising a VkPhysicalDeviceDriverProperties::`conformanceVersion` less than 1.3.8.0 should be aware that rebinding the bound pipeline object may not reapply static state.

The following lists all the dynamic state with their logical state name.

Some dynamic state is tied to a single shader stage and can be ignored if that stage is not being used.

Some dynamic state is tied to certain graphics pipeline state. The dynamic state in VkPipelineDynamicStateCreateInfo::pDynamicStates is ignored if the pipeline is being created without the corresponding pipeline state.

Some dynamic state is ignored if there is no rasterization stage. The rasterizerDiscardEnable state is used to control if the rasterization stage occurs.

Some dynamic state is ignored if other dynamic state is not enabled. Some conditions involve special conditions to be met.

Each dynamic state has at least one command that can be set while recording the command buffer. If using a pipeline, there is a VkDynamicState associated for each of these commands.

When using vkCreateShadersEXT to create a shader object, all dynamic state is dynamic, with no static equivalent.

Table 12. Dynamic State Table

Dynamic State	Shader Stage	Pipeline State	Requires Rasterization	Depends On
viewport	Any	Pre-Rasterization	No	None
	• vkCmdSetViewport (VK_DYNAMIC_STATE_VIEWPORT) - pipelines only • vkCmdSetViewportWithCount (VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT)			
scissor	Any	Pre-Rasterization	No	None
	• vkCmdSetScissor (VK_DYNAMIC_STATE_SCISSOR) - pipelines only • vkCmdSetScissorWithCount (VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT)			
rasterizerDiscardEnable	Any	Pre-Rasterization	No	None
	• vkCmdSetRasterizerDiscardEnable (VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE)			
vertexInput	Vertex	Vertex Input	No	None
	• vkCmdSetVertexInputEXT (VK_DYNAMIC_STATE_VERTEX_INPUT_EXT) • vkCmdBindVertexBuffers2 (VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE) - pipelines only			

Dynamic State	Shader Stage	Pipeline State	Requires Rasterization	Depends On
primitiveTopology	Vertex	Vertex Input	No	None
	• vkCmdSetPrimitiveTopology (VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY)			
primitiveRestartEnable	Vertex	Vertex Input	No	None
	• vkCmdSetPrimitiveRestartEnable (VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE)			
provokingVertexMode	Vertex	Pre-Rasterization	Yes	None
	• vkCmdSetProvokingVertexModeEXT (VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT) • Requires the VK_EXT_provoking_vertex extension			
blendConstants	Fragment	Fragment Output	Yes	colorBlendFactor
	• vkCmdSetBlendConstants (VK_DYNAMIC_STATE_BLEND_CONSTANTS)			
logicOpEnable	Fragment	Fragment Output	Yes	None
	• vkCmdSetLogicOpEnableEXT (VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT) • Requires the logicOp feature			
logicOp	Fragment	Fragment Output	Yes	logicOpEnable
	• vkCmdSetLogicOpEXT (VK_DYNAMIC_STATE_LOGIC_OP_EXT)			
colorWriteEnable	Fragment	Fragment Output	Yes	boundColorAttachment
	• vkCmdSetColorWriteEnableEXT (VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT) • Requires the colorWriteEnable feature			
colorBlendEnable	Fragment	Fragment Output	Yes	boundColorAttachment
	• vkCmdSetColorBlendEnableEXT (VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT)			
colorWriteMask	Fragment	Fragment Output	Yes	boundColorAttachment
	• vkCmdSetColorWriteMaskEXT (VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT)			

Dynamic State	Shader Stage	Pipeline State	Requires Rasterization	Depends On
colorBlendEquation	Fragment	Fragment Output	Yes	colorBlendEnableIndex
	• vkCmdSetColorBlendEquationEXT (VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT)			
colorBlendAdvanced	Fragment	Fragment Output	Yes	colorBlendEnableIndex
	• vkCmdSetColorBlendAdvancedEXT (VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT)			
attachmentFeedbackLoopEnable	Fragment	Pre-Rasterization, Fragment Shader, Fragment Output	Yes	None
	• vkCmdSetAttachmentFeedbackLoopEnableEXT (VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT) • Requires the attachmentFeedbackLoopDynamicState feature			
depthBiasEnable	Any	Pre-Rasterization	Yes	None
	• vkCmdSetDepthBiasEnable (VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE)			
depthBias	Any	Pre-Rasterization	Yes	depthBiasEnable
	• vkCmdSetDepthBias (VK_DYNAMIC_STATE_DEPTH_BIAS) • vkCmdSetDepthBias2EXT (VK_DYNAMIC_STATE_DEPTH_BIAS)			
depthBoundsTestEnable	Any	Fragment Shader	Yes	None
	• vkCmdSetDepthBoundsTestEnable (VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE) • Requires the depthBounds feature			
depthBounds	Any	Fragment Shader	Yes	depthBoundsTestEnable
	• vkCmdSetDepthBounds (VK_DYNAMIC_STATE_DEPTH_BOUNDS)			
depthTestEnable	Any	Fragment Shader	Yes	None
	• vkCmdSetDepthTestEnable (VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE)			

Dynamic State	Shader Stage	Pipeline State	Requires Rasterization	Depends On
depthWriteEnable	Any	Fragment Shader	Yes	depthTestEnable
	• vkCmdSetDepthWriteEnable (VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE)			
depthCompareOp	Any	Fragment Shader	Yes	depthTestEnable
	• vkCmdSetDepthCompareOp (VK_DYNAMIC_STATE_DEPTH_COMPARE_OP)			
depthClipEnable	Any	Pre-Rasterization	No	None
	• vkCmdSetDepthClipEnableEXT (VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT) • Requires the depthClipEnable feature			
depthClipNegativeOneToOne	Any	Pre-Rasterization	No	None
	• vkCmdSetDepthClipNegativeOneToOneEXT (VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT) • Requires the depthClipControl feature			
depthClampEnable	Any	Pre-Rasterization	Yes	None
	• vkCmdSetDepthClampEnableEXT (VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT) • Requires the depthClamp feature			
depthClampRange	Any	Pre-Rasterization	No	depthClampEnable
	• vkCmdSetDepthClampRangeEXT (VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT) • Requires the depthClampControl feature			
stencilTestEnable	Any	Fragment Shader	Yes	None
	• vkCmdSetStencilTestEnable (VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE)			
stencilCompareMask	Any	Fragment Shader	Yes	stencilTestEnable
	• vkCmdSetStencilCompareMask (VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK)			

Dynamic State	Shader Stage	Pipeline State	Requires Rasterization	Depends On
stencilWriteMask	Any	Fragment Shader	Yes	stencilTestEnable
	• vkCmdSetStencilWriteMask (VK_DYNAMIC_STATE_STENCIL_WRITE_MASK)			
stencilReference	Any	Fragment Shader	Yes	stencilTestEnable
	• vkCmdSetStencilReference (VK_DYNAMIC_STATE_STENCIL_REFERENCE)			
stencilOp	Any	Fragment Shader	Yes	stencilTestEnable
	• vkCmdSetStencilOp (VK_DYNAMIC_STATE_STENCIL_OP)			
cullMode	Any	Pre-Rasterization	Yes	None
	• vkCmdSetCullMode (VK_DYNAMIC_STATE_CULL_MODE)			
frontFace	Any	Pre-Rasterization	Yes	None
	• vkCmdSetFrontFace (VK_DYNAMIC_STATE_FRONT_FACE)			
discardRectangleEnable	Any	Pre-Rasterization	Yes	None
	• vkCmdSetDiscardRectangleEnableEXT (VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT) • Requires the VK_EXT_discard_rectangles extension			
discardRectangle	Any	Pre-Rasterization	Yes	discardRectangleEnable
	• vkCmdSetDiscardRectangleEXT (VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT) • Requires the VK_EXT_discard_rectangles extension			
discardRectangleMode	Any	Pre-Rasterization	Yes	discardRectangleEnable
	• vkCmdSetDiscardRectangleModeEXT (VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT) • Requires the VK_EXT_discard_rectangles extension			

Dynamic State	Shader Stage	Pipeline State	Requires Rasterization	Depends On
sampleLocationsEnable	Any	Fragment Shader, Fragment Output	Yes	None
	• vkCmdSetSampleLocationsEnableEXT (VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT) • Requires the VK_EXT_sample_locations extension			
sampleLocations	Any	Fragment Shader, Fragment Output	Yes	sampleLocationsEnable
	• vkCmdSetSampleLocationsEXT (VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT) • Requires the VK_EXT_sample_locations extension			
fragmentShadingRate	Any	Pre-Rasterization, Fragment Shader	Yes	None
	• vkCmdSetFragmentShadingRateKHR (VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR) • Requires the pipelineFragmentShadingRate feature			
lineWidth	Any	Pre-Rasterization	Yes	lineTopology
	• vkCmdSetLineWidth (VK_DYNAMIC_STATE_LINE_WIDTH)			
lineStippleEnable	Vertex	Pre-Rasterization	Yes	stippledLineTopology
	• vkCmdSetLineStippleEnableEXT (VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT) • Requires any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features			
lineStipple	Any	Pre-Rasterization	Yes	lineStippleEnable and stippledLineTopology
	• vkCmdSetLineStipple (VK_DYNAMIC_STATE_LINE_STIPPLE) • Requires any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features			

Dynamic State	Shader Stage	Pipeline State	Requires Rasterization	Depends On
`lineRasterizationMode`	Vertex	Pre-Rasterization	Yes	stippledLineTopology
	• vkCmdSetLineRasterizationModeEXT (VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT) • Requires any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features			
`rasterizationStream`	Geometry	Pre-Rasterization	No	None
	• vkCmdSetRasterizationStreamEXT (VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT) • Requires the `geometryStreams` feature			
`patchControlPoints`	Tessellation Control	Pre-Rasterization	No	patchListPrimitiveTopology
	• vkCmdSetPatchControlPointsEXT (VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT)			
`tessellationDomainOrigin`	Tessellation Evaluation	Pre-Rasterization	No	None
	• vkCmdSetTessellationDomainOriginEXT (VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT)			
`polygonMode`	Any	Pre-Rasterization	Yes	None
	• vkCmdSetPolygonModeEXT (VK_DYNAMIC_STATE_POLYGON_MODE_EXT)			
`rasterizationSamples`	Any	Fragment Shader, Fragment Output	Yes	None
	• vkCmdSetRasterizationSamplesEXT (VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT)			
`sampleMask`	Any	Fragment Shader, Fragment Output	Yes	None
	• vkCmdSetSampleMaskEXT (VK_DYNAMIC_STATE_SAMPLE_MASK_EXT)			

Dynamic State	Shader Stage	Pipeline State	Requires Rasterization	Depends On
alphaToCoverageEnable	Any	Fragment Shader, Fragment Output	Yes	None
	• vkCmdSetAlphaToCoverageEnableEXT (VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT)			
alphaToOneEnable	Any	Fragment Shader, Fragment Output	Yes	None
	• vkCmdSetAlphaToOneEnableEXT (VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT) • Requires the alphaToOne feature			
conservativeRasterizationMode	Any	Pre-Rasterization	Yes	None
	• vkCmdSetConservativeRasterizationModeEXT (VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT) • Requires the VK_EXT_conservative_rasterization extension			
extraPrimitiveOverestimationSize	Any	Pre-Rasterization	Yes	conservativeRasterizationMode
	• vkCmdSetExtraPrimitiveOverestimationSizeEXT (VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT) • Requires the VK_EXT_conservative_rasterization extension			
rayTracingPipelineStackSize	Any	None	No	None
	• vkCmdSetRayTracingPipelineStackSizeKHR (VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR) • Requires the VK_KHR_ray_tracing_pipeline extension			

10.14. Pipeline Properties and Shader Information

When a pipeline is created, its state and shaders are compiled into zero or more device-specific executables, which are used when executing commands against that pipeline. To query the properties of these pipeline executables, call:

```
// Provided by VK_KHR_pipeline_executable_properties
VkResult vkGetPipelineExecutablePropertiesKHR(
    VkDevice                                    device,
    const VkPipelineInfoKHR*                    pPipelineInfo,
    uint32_t*                                   pExecutableCount,
    VkPipelineExecutablePropertiesKHR*          pProperties);
```

- `device` is the device that created the pipeline.
- `pPipelineInfo` describes the pipeline being queried.
- `pExecutableCount` is a pointer to an integer related to the number of pipeline executables available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of VkPipelineExecutablePropertiesKHR structures.

If `pProperties` is `NULL`, then the number of pipeline executables associated with the pipeline is returned in `pExecutableCount`. Otherwise, `pExecutableCount` **must** point to a variable set by the application to the number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If `pExecutableCount` is less than the number of pipeline executables associated with the pipeline, at most `pExecutableCount` structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available properties were returned.

Valid Usage

- VUID-vkGetPipelineExecutablePropertiesKHR-pipelineExecutableInfo-03270
  The `pipelineExecutableInfo` feature **must** be enabled
- VUID-vkGetPipelineExecutablePropertiesKHR-pipeline-03271
  The `pipeline` member of `pPipelineInfo` **must** have been created with `device`

Valid Usage (Implicit)

- VUID-vkGetPipelineExecutablePropertiesKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetPipelineExecutablePropertiesKHR-pPipelineInfo-parameter
  `pPipelineInfo` **must** be a valid pointer to a valid VkPipelineInfoKHR structure
- VUID-vkGetPipelineExecutablePropertiesKHR-pExecutableCount-parameter
  `pExecutableCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPipelineExecutablePropertiesKHR-pProperties-parameter
  If the value referenced by `pExecutableCount` is not `0`, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pExecutableCount` VkPipelineExecutablePropertiesKHR structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkPipelineExecutablePropertiesKHR structure is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef struct VkPipelineExecutablePropertiesKHR {
    VkStructureType       sType;
    void*                 pNext;
    VkShaderStageFlags    stages;
    char                  name[VK_MAX_DESCRIPTION_SIZE];
    char                  description[VK_MAX_DESCRIPTION_SIZE];
    uint32_t              subgroupSize;
} VkPipelineExecutablePropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- stages is a bitmask of zero or more VkShaderStageFlagBits indicating which shader stages (if any) were principally used as inputs to compile this pipeline executable.
- name is an array of VK_MAX_DESCRIPTION_SIZE char containing a null-terminated UTF-8 string which is a short human readable name for this pipeline executable.
- description is an array of VK_MAX_DESCRIPTION_SIZE char containing a null-terminated UTF-8 string which is a human readable description for this pipeline executable.
- subgroupSize is the subgroup size with which this pipeline executable is dispatched.

Not all implementations have a 1:1 mapping between shader stages and pipeline executables and some implementations **may** reduce a given shader stage to fixed function hardware programming such that no pipeline executable is available. No guarantees are provided about the mapping between shader stages and pipeline executables and stages **should** be considered a best effort hint. Because the application **cannot** rely on the stages field to provide an exact description, name and description provide a human readable name and description which more accurately describes the given pipeline executable.

Valid Usage (Implicit)

- VUID-VkPipelineExecutablePropertiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_PROPERTIES_KHR
- VUID-VkPipelineExecutablePropertiesKHR-pNext-pNext
  `pNext` **must** be `NULL`

To query the pipeline properties call:

```
// Provided by VK_EXT_pipeline_properties
VkResult vkGetPipelinePropertiesEXT(
    VkDevice                                    device,
    const VkPipelineInfoKHR*                    pPipelineInfo,
    VkBaseOutStructure*                         pPipelineProperties);
```

- `device` is the logical device that created the pipeline.
- `pPipelineInfo` is a pointer to a VkPipelineInfoEXT structure which describes the pipeline being queried.
- `pPipelineProperties` is a pointer to a VkBaseOutStructure structure in which the pipeline properties will be written.

To query a pipeline's `pipelineIdentifier` pass a VkPipelinePropertiesIdentifierEXT structure in `pPipelineProperties`. Each pipeline is associated with a `pipelineIdentifier` and the identifier is implementation specific.

Valid Usage

- VUID-vkGetPipelinePropertiesEXT-pipeline-06738
  The `pipeline` member of `pPipelineInfo` **must** have been created with `device`
- VUID-vkGetPipelinePropertiesEXT-pPipelineProperties-06739
  `pPipelineProperties` **must** be a valid pointer to a VkPipelinePropertiesIdentifierEXT structure
- VUID-vkGetPipelinePropertiesEXT-None-06766
  The `pipelinePropertiesIdentifier` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetPipelinePropertiesEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetPipelinePropertiesEXT-pPipelineInfo-parameter
  `pPipelineInfo` **must** be a valid pointer to a valid VkPipelineInfoKHR structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkPipelinePropertiesIdentifierEXT structure is defined as:

```
// Provided by VK_EXT_pipeline_properties
typedef struct VkPipelinePropertiesIdentifierEXT {
    VkStructureType    sType;
    void*              pNext;
    uint8_t            pipelineIdentifier[VK_UUID_SIZE];
} VkPipelinePropertiesIdentifierEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pipelineIdentifier is an array of VK_UUID_SIZE uint8_t values into which the pipeline identifier will be written.

Valid Usage (Implicit)

- VUID-VkPipelinePropertiesIdentifierEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PIPELINE_PROPERTIES_IDENTIFIER_EXT
- VUID-VkPipelinePropertiesIdentifierEXT-pNext-pNext
  pNext **must** be NULL

The VkPipelineInfoKHR structure is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef struct VkPipelineInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkPipeline         pipeline;
} VkPipelineInfoKHR;
```

```
// Provided by VK_EXT_pipeline_properties
// Equivalent to VkPipelineInfoKHR
typedef VkPipelineInfoKHR VkPipelineInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pipeline` is a `VkPipeline` handle.

Valid Usage (Implicit)

- VUID-VkPipelineInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_INFO_KHR
- VUID-VkPipelineInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkPipelineInfoKHR-pipeline-parameter
  `pipeline` **must** be a valid VkPipeline handle

Each pipeline executable **may** have a set of statistics associated with it that are generated by the pipeline compilation process. These statistics **may** include things such as instruction counts, amount of spilling (if any), maximum number of simultaneous threads, or anything else which **may** aid developers in evaluating the expected performance of a shader. To query the compile time statistics associated with a pipeline executable, call:

```
// Provided by VK_KHR_pipeline_executable_properties
VkResult vkGetPipelineExecutableStatisticsKHR(
    VkDevice                                    device,
    const VkPipelineExecutableInfoKHR*          pExecutableInfo,
    uint32_t*                                   pStatisticCount,
    VkPipelineExecutableStatisticKHR*           pStatistics);
```

- `device` is the device that created the pipeline.
- `pExecutableInfo` describes the pipeline executable being queried.
- `pStatisticCount` is a pointer to an integer related to the number of statistics available or queried, as described below.
- `pStatistics` is either `NULL` or a pointer to an array of VkPipelineExecutableStatisticKHR structures.

If `pStatistics` is `NULL`, then the number of statistics associated with the pipeline executable is returned in `pStatisticCount`. Otherwise, `pStatisticCount` **must** point to a variable set by the application to the number of elements in the `pStatistics` array, and on return the variable is overwritten with the number of structures actually written to `pStatistics`. If `pStatisticCount` is less than the number of statistics associated with the pipeline executable, at most `pStatisticCount` structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to

indicate that not all the available statistics were returned.

Valid Usage

- VUID-vkGetPipelineExecutableStatisticsKHR-pipelineExecutableInfo-03272
  The `pipelineExecutableInfo` feature **must** be enabled
- VUID-vkGetPipelineExecutableStatisticsKHR-pipeline-03273
  The `pipeline` member of `pExecutableInfo` **must** have been created with `device`
- VUID-vkGetPipelineExecutableStatisticsKHR-pipeline-03274
  The `pipeline` member of `pExecutableInfo` **must** have been created with VK_PIPELINE_CREATE_CAPTURE_STATISTICS_BIT_KHR

Valid Usage (Implicit)

- VUID-vkGetPipelineExecutableStatisticsKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetPipelineExecutableStatisticsKHR-pExecutableInfo-parameter
  `pExecutableInfo` **must** be a valid pointer to a valid VkPipelineExecutableInfoKHR structure
- VUID-vkGetPipelineExecutableStatisticsKHR-pStatisticCount-parameter
  `pStatisticCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPipelineExecutableStatisticsKHR-pStatistics-parameter
  If the value referenced by `pStatisticCount` is not `0`, and `pStatistics` is not `NULL`, `pStatistics` **must** be a valid pointer to an array of `pStatisticCount` VkPipelineExecutableStatisticKHR structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkPipelineExecutableInfoKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef struct VkPipelineExecutableInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkPipeline         pipeline;
    uint32_t           executableIndex;
} VkPipelineExecutableInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pipeline` is the pipeline to query.
- `executableIndex` is the index of the pipeline executable to query in the array of executable properties returned by vkGetPipelineExecutablePropertiesKHR.

Valid Usage

- VUID-VkPipelineExecutableInfoKHR-executableIndex-03275
  `executableIndex` **must** be less than the number of pipeline executables associated with `pipeline` as returned in the `pExecutableCount` parameter of `vkGetPipelineExecutablePropertiesKHR`

Valid Usage (Implicit)

- VUID-VkPipelineExecutableInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_INFO_KHR
- VUID-VkPipelineExecutableInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkPipelineExecutableInfoKHR-pipeline-parameter
  `pipeline` **must** be a valid VkPipeline handle

The `VkPipelineExecutableStatisticKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef struct VkPipelineExecutableStatisticKHR {
    VkStructureType                           sType;
    void*                                     pNext;
    char                                      name[VK_MAX_DESCRIPTION_SIZE];
    char                                      description[VK_MAX_DESCRIPTION_SIZE];
    VkPipelineExecutableStatisticFormatKHR    format;
    VkPipelineExecutableStatisticValueKHR     value;
} VkPipelineExecutableStatisticKHR;
```

- `sType` is a VkStructureType value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `name` is an array of VK_MAX_DESCRIPTION_SIZE `char` containing a null-terminated UTF-8 string which is a short human readable name for this statistic.
- `description` is an array of VK_MAX_DESCRIPTION_SIZE `char` containing a null-terminated UTF-8 string which is a human readable description for this statistic.
- `format` is a VkPipelineExecutableStatisticFormatKHR value specifying the format of the data found in `value`.
- `value` is the value of this statistic.

Valid Usage (Implicit)

- VUID-VkPipelineExecutableStatisticKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_STATISTIC_KHR
- VUID-VkPipelineExecutableStatisticKHR-pNext-pNext
  `pNext` **must** be `NULL`

The VkPipelineExecutableStatisticFormatKHR enum is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef enum VkPipelineExecutableStatisticFormatKHR {
    VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_BOOL32_KHR = 0,
    VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_INT64_KHR = 1,
    VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_UINT64_KHR = 2,
    VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_FLOAT64_KHR = 3,
} VkPipelineExecutableStatisticFormatKHR;
```

- VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_BOOL32_KHR specifies that the statistic is returned as a 32-bit boolean value which **must** be either VK_TRUE or VK_FALSE and **should** be read from the `b32` field of `VkPipelineExecutableStatisticValueKHR`.
- VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_INT64_KHR specifies that the statistic is returned as a signed 64-bit integer and **should** be read from the `i64` field of `VkPipelineExecutableStatisticValueKHR`.
- VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_UINT64_KHR specifies that the statistic is returned as an unsigned 64-bit integer and **should** be read from the `u64` field of `VkPipelineExecutableStatisticValueKHR`.
- VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_FLOAT64_KHR specifies that the statistic is returned as a 64-bit floating-point value and **should** be read from the `f64` field of `VkPipelineExecutableStatisticValueKHR`.

The `VkPipelineExecutableStatisticValueKHR` union is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef union VkPipelineExecutableStatisticValueKHR {
    VkBool32    b32;
    int64_t     i64;
    uint64_t    u64;
    double      f64;
} VkPipelineExecutableStatisticValueKHR;
```

- `b32` is the 32-bit boolean value if the VkPipelineExecutableStatisticFormatKHR is VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_BOOL32_KHR.
- `i64` is the signed 64-bit integer value if the VkPipelineExecutableStatisticFormatKHR is VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_INT64_KHR.
- `u64` is the unsigned 64-bit integer value if the VkPipelineExecutableStatisticFormatKHR is VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_UINT64_KHR.
- `f64` is the 64-bit floating-point value if the VkPipelineExecutableStatisticFormatKHR is VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_FLOAT64_KHR.

Each pipeline executable **may** have one or more text or binary internal representations associated with it which are generated as part of the compile process. These **may** include the final shader assembly, a binary form of the compiled shader, or the shader compiler's internal representation at any number of intermediate compile steps. To query the internal representations associated with a pipeline executable, call:

```
// Provided by VK_KHR_pipeline_executable_properties
VkResult vkGetPipelineExecutableInternalRepresentationsKHR(
    VkDevice                                    device,
    const VkPipelineExecutableInfoKHR*          pExecutableInfo,
    uint32_t*                                   pInternalRepresentationCount,
    VkPipelineExecutableInternalRepresentationKHR* pInternalRepresentations);
```

- `device` is the device that created the pipeline.
- `pExecutableInfo` describes the pipeline executable being queried.
- `pInternalRepresentationCount` is a pointer to an integer related to the number of internal representations available or queried, as described below.
- `pInternalRepresentations` is either `NULL` or a pointer to an array of VkPipelineExecutableInternalRepresentationKHR structures.

If `pInternalRepresentations` is `NULL`, then the number of internal representations associated with the pipeline executable is returned in `pInternalRepresentationCount`. Otherwise, `pInternalRepresentationCount` **must** point to a variable set by the application to the number of elements in the `pInternalRepresentations` array, and on return the variable is overwritten with the number of structures actually written to `pInternalRepresentations`. If `pInternalRepresentationCount` is less than the number of internal representations associated with the pipeline executable, at most `pInternalRepresentationCount` structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available representations were returned.

While the details of the internal representations remain implementation-dependent, the implementation **should** order the internal representations in the order in which they occur in the compiled pipeline with the final shader assembly (if any) last.

Valid Usage

- VUID-vkGetPipelineExecutableInternalRepresentationsKHR-pipelineExecutableInfo-03276
  The `pipelineExecutableInfo` feature **must** be enabled
- VUID-vkGetPipelineExecutableInternalRepresentationsKHR-pipeline-03277
  The `pipeline` member of `pExecutableInfo` **must** have been created with `device`
- VUID-vkGetPipelineExecutableInternalRepresentationsKHR-pipeline-03278
  The `pipeline` member of `pExecutableInfo` **must** have been created with VK_PIPELINE_CREATE_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR

Valid Usage (Implicit)

- VUID-vkGetPipelineExecutableInternalRepresentationsKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetPipelineExecutableInternalRepresentationsKHR-pExecutableInfo-parameter
  `pExecutableInfo` **must** be a valid pointer to a valid VkPipelineExecutableInfoKHR structure
- VUID-vkGetPipelineExecutableInternalRepresentationsKHR-pInternalRepresentationCount-parameter
  `pInternalRepresentationCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPipelineExecutableInternalRepresentationsKHR-pInternalRepresentations-parameter
  If the value referenced by `pInternalRepresentationCount` is not `0`, and `pInternalRepresentations` is not `NULL`, `pInternalRepresentations` **must** be a valid pointer to an array of `pInternalRepresentationCount` VkPipelineExecutableInternalRepresentationKHR structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkPipelineExecutableInternalRepresentationKHR structure is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef struct VkPipelineExecutableInternalRepresentationKHR {
    VkStructureType    sType;
    void*              pNext;
    char               name[VK_MAX_DESCRIPTION_SIZE];
    char               description[VK_MAX_DESCRIPTION_SIZE];
    VkBool32           isText;
    size_t             dataSize;
    void*              pData;
} VkPipelineExecutableInternalRepresentationKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- name is an array of VK_MAX_DESCRIPTION_SIZE char containing a null-terminated UTF-8 string which is a short human readable name for this internal representation.
- description is an array of VK_MAX_DESCRIPTION_SIZE char containing a null-terminated UTF-8 string which is a human readable description for this internal representation.
- isText specifies whether the returned data is text or opaque data. If isText is VK_TRUE then the data returned in pData is text and is guaranteed to be a null-terminated UTF-8 string.
- dataSize is an integer related to the size, in bytes, of the internal representation's data, as described below.
- pData is either NULL or a pointer to a block of data into which the implementation will write the internal representation.

If pData is NULL, then the size, in bytes, of the internal representation data is returned in dataSize. Otherwise, dataSize **must** be the size of the buffer, in bytes, pointed to by pData and on return dataSize is overwritten with the number of bytes of data actually written to pData including any trailing null character. If dataSize is less than the size, in bytes, of the internal representation's data, at most dataSize bytes of data will be written to pData, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available representation was returned.

If isText is VK_TRUE and pData is not NULL and dataSize is not zero, the last byte written to pData will be a null character.

Valid Usage (Implicit)

- VUID-VkPipelineExecutableInternalRepresentationKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_INTERNAL_REPRESENTATION_KHR
- VUID-VkPipelineExecutableInternalRepresentationKHR-pNext-pNext
  pNext **must** be NULL

10.15. Pipeline Creation Feedback

Feedback about the creation of a particular pipeline object **can** be obtained by adding a VkPipelineCreationFeedbackCreateInfo structure to the pNext chain of VkGraphicsPipelineCreateInfo, VkRayTracingPipelineCreateInfoKHR, or VkComputePipelineCreateInfo. The VkPipelineCreationFeedbackCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPipelineCreationFeedbackCreateInfo {
    VkStructureType                 sType;
    const void*                     pNext;
    VkPipelineCreationFeedback*     pPipelineCreationFeedback;
    uint32_t                        pipelineStageCreationFeedbackCount;
    VkPipelineCreationFeedback*     pPipelineStageCreationFeedbacks;
} VkPipelineCreationFeedbackCreateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pPipelineCreationFeedback is a pointer to a VkPipelineCreationFeedback structure.
- pipelineStageCreationFeedbackCount is the number of elements in pPipelineStageCreationFeedbacks.
- pPipelineStageCreationFeedbacks is a pointer to an array of pipelineStageCreationFeedbackCount VkPipelineCreationFeedback structures.

An implementation **should** write pipeline creation feedback to pPipelineCreationFeedback and **may** write pipeline stage creation feedback to pPipelineStageCreationFeedbacks. An implementation **must** set or clear the VK_PIPELINE_CREATION_FEEDBACK_VALID_BIT in VkPipelineCreationFeedback::flags for pPipelineCreationFeedback and every element of pPipelineStageCreationFeedbacks.

NOTE One common scenario for an implementation to skip per-stage feedback is when VK_PIPELINE_CREATION_FEEDBACK_APPLICATION_PIPELINE_CACHE_HIT_BIT is set in pPipelineCreationFeedback.

When chained to VkRayTracingPipelineCreateInfoKHR, or VkGraphicsPipelineCreateInfo, the i element of pPipelineStageCreationFeedbacks corresponds to the i element of VkRayTracingPipelineCreateInfoKHR::pStages, or VkGraphicsPipelineCreateInfo::pStages. When chained to VkComputePipelineCreateInfo, the first element of pPipelineStageCreationFeedbacks corresponds to VkComputePipelineCreateInfo::stage.

Valid Usage (Implicit)

- VUID-VkPipelineCreationFeedbackCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PIPELINE_CREATION_FEEDBACK_CREATE_INFO
- VUID-VkPipelineCreationFeedbackCreateInfo-pPipelineCreationFeedback-parameter

pPipelineCreationFeedback **must** be a valid pointer to a VkPipelineCreationFeedback structure

- VUID-VkPipelineCreationFeedbackCreateInfo-pPipelineStageCreationFeedbacks-parameter
  If pipelineStageCreationFeedbackCount is not 0, pPipelineStageCreationFeedbacks **must** be a valid pointer to an array of pipelineStageCreationFeedbackCount VkPipelineCreationFeedback structures

Structure Chaining

Extends the structures

- VkComputePipelineCreateInfo
- VkGraphicsPipelineCreateInfo
- VkRayTracingPipelineCreateInfoKHR

The VkPipelineCreationFeedback structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPipelineCreationFeedback {
    VkPipelineCreationFeedbackFlags    flags;
    uint64_t                           duration;
} VkPipelineCreationFeedback;
```

- flags is a bitmask of VkPipelineCreationFeedbackFlagBits providing feedback about the creation of a pipeline or of a pipeline stage.
- duration is the duration spent creating a pipeline or pipeline stage in nanoseconds.

If the VK_PIPELINE_CREATION_FEEDBACK_VALID_BIT is not set in flags, an implementation **must** not set any other bits in flags, and the values of all other VkPipelineCreationFeedback data members are undefined.

Possible values of the flags member of VkPipelineCreationFeedback are:

```
// Provided by VK_VERSION_1_3
typedef enum VkPipelineCreationFeedbackFlagBits {
    VK_PIPELINE_CREATION_FEEDBACK_VALID_BIT = 0x00000001,
    VK_PIPELINE_CREATION_FEEDBACK_APPLICATION_PIPELINE_CACHE_HIT_BIT = 0x00000002,
    VK_PIPELINE_CREATION_FEEDBACK_BASE_PIPELINE_ACCELERATION_BIT = 0x00000004,
} VkPipelineCreationFeedbackFlagBits;
```

- VK_PIPELINE_CREATION_FEEDBACK_VALID_BIT specifies that the feedback information is valid.
- VK_PIPELINE_CREATION_FEEDBACK_APPLICATION_PIPELINE_CACHE_HIT_BIT specifies that a

readily usable pipeline or pipeline stage was found in the pipelineCache specified by the application in the pipeline creation command.

An implementation **should** set the VK_PIPELINE_CREATION_FEEDBACK_APPLICATION_PIPELINE_CACHE_HIT_BIT bit if it was able to avoid the large majority of pipeline or pipeline stage creation work by using the pipelineCache parameter of vkCreateGraphicsPipelines, vkCreateRayTracingPipelinesKHR, or vkCreateComputePipelines. When an implementation sets this bit for the entire pipeline, it **may** leave it unset for any stage.

> **NOTE** Implementations are encouraged to provide a meaningful signal to applications using this bit. The intention is to communicate to the application that the pipeline or pipeline stage was created "as fast as it gets" using the pipeline cache provided by the application. If an implementation uses an internal cache, it is discouraged from setting this bit as the feedback would be unactionable.

- VK_PIPELINE_CREATION_FEEDBACK_BASE_PIPELINE_ACCELERATION_BIT specifies that the base pipeline specified by the basePipelineHandle or basePipelineIndex member of the Vk*PipelineCreateInfo structure was used to accelerate the creation of the pipeline.

  An implementation **should** set the VK_PIPELINE_CREATION_FEEDBACK_BASE_PIPELINE_ACCELERATION_BIT bit if it was able to avoid a significant amount of work by using the base pipeline.

  > **NOTE** While "significant amount of work" is subjective, implementations are encouraged to provide a meaningful signal to applications using this bit. For example, a 1% reduction in duration may not warrant setting this bit, while a 50% reduction would.

```
// Provided by VK_VERSION_1_3
typedef VkFlags VkPipelineCreationFeedbackFlags;
```

VkPipelineCreationFeedbackFlags is a bitmask type for providing zero or more VkPipelineCreationFeedbackFlagBits.

Chapter 11. Memory Allocation

Vulkan memory is broken up into two categories, *host memory* and *device memory*.

11.1. Host Memory

Host memory is memory needed by the Vulkan implementation for non-device-visible storage.

> **NOTE** This memory **may** be used to store the implementation's representation and state of Vulkan objects.

Vulkan provides applications the opportunity to perform host memory allocations on behalf of the Vulkan implementation. If this feature is not used, the implementation will perform its own memory allocations. Since most memory allocations are off the critical path, this is not meant as a performance feature. Rather, this **can** be useful for certain embedded systems, for debugging purposes (e.g. putting a guard page after all host allocations), or for memory allocation logging.

Allocators are provided by the application as a pointer to a `VkAllocationCallbacks` structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkAllocationCallbacks {
    void*                                   pUserData;
    PFN_vkAllocationFunction                pfnAllocation;
    PFN_vkReallocationFunction              pfnReallocation;
    PFN_vkFreeFunction                      pfnFree;
    PFN_vkInternalAllocationNotification    pfnInternalAllocation;
    PFN_vkInternalFreeNotification          pfnInternalFree;
} VkAllocationCallbacks;
```

- `pUserData` is `NULL` or an application-defined user data pointer to be interpreted by the implementation of the callbacks. When any of the callbacks in `VkAllocationCallbacks` are called, the Vulkan implementation will pass this value as the first parameter to the callback. This value **can** vary each time an allocator is passed into a command, even when the same object takes an allocator in multiple commands.
- `pfnAllocation` is a PFN_vkAllocationFunction pointer to an application-defined memory allocation function.
- `pfnReallocation` is a PFN_vkReallocationFunction pointer to an application-defined memory reallocation function.
- `pfnFree` is a PFN_vkFreeFunction pointer to an application-defined memory free function.
- `pfnInternalAllocation` is a PFN_vkInternalAllocationNotification pointer to an application-defined function that is called by the implementation when the implementation makes internal allocations.
- `pfnInternalFree` is a PFN_vkInternalFreeNotification pointer to an application-defined function that is called by the implementation when the implementation frees internal allocations.

Valid Usage

- VUID-VkAllocationCallbacks-pfnAllocation-00632
  `pfnAllocation` **must** be a valid pointer to a valid application-defined PFN_vkAllocationFunction
- VUID-VkAllocationCallbacks-pfnReallocation-00633
  `pfnReallocation` **must** be a valid pointer to a valid application-defined PFN_vkReallocationFunction
- VUID-VkAllocationCallbacks-pfnFree-00634
  `pfnFree` **must** be a valid pointer to a valid application-defined PFN_vkFreeFunction
- VUID-VkAllocationCallbacks-pfnInternalAllocation-00635
  If either of `pfnInternalAllocation` or `pfnInternalFree` is not `NULL`, both **must** be valid callbacks

The type of `pfnAllocation` is:

```
// Provided by VK_VERSION_1_0
typedef void* (*PFN_vkAllocationFunction)(
    void*                                       pUserData,
    size_t                                      size,
    size_t                                      alignment,
    VkSystemAllocationScope                     allocationScope);
```

- `pUserData` is the value specified for VkAllocationCallbacks::`pUserData` in the allocator specified by the application.
- `size` is the size in bytes of the requested allocation.
- `alignment` is the requested alignment of the allocation in bytes and **must** be a power of two.
- `allocationScope` is a VkSystemAllocationScope value specifying the allocation scope of the lifetime of the allocation, as described here.

If `pfnAllocation` is unable to allocate the requested memory, it **must** return `NULL`. If the allocation was successful, it **must** return a valid pointer to memory allocation containing at least `size` bytes, and with the pointer value being a multiple of `alignment`.

NOTE

Correct Vulkan operation **cannot** be assumed if the application does not follow these rules.

For example, `pfnAllocation` (or `pfnReallocation`) could cause termination of running Vulkan instance(s) on a failed allocation for debugging purposes, either directly or indirectly. In these circumstances, it **cannot** be assumed that any part of any affected VkInstance objects are going to operate correctly (even vkDestroyInstance), and the application **must** ensure it cleans up properly via other means (e.g. process termination).

If pfnAllocation returns NULL, and if the implementation is unable to continue correct processing of the current command without the requested allocation, it **must** treat this as a runtime error, and generate VK_ERROR_OUT_OF_HOST_MEMORY at the appropriate time for the command in which the condition was detected, as described in Return Codes.

If the implementation is able to continue correct processing of the current command without the requested allocation, then it **may** do so, and **must** not generate VK_ERROR_OUT_OF_HOST_MEMORY as a result of this failed allocation.

The type of pfnReallocation is:

```
// Provided by VK_VERSION_1_0
typedef void* (*PFN_vkReallocationFunction)(
    void*                                       pUserData,
    void*                                       pOriginal,
    size_t                                      size,
    size_t                                      alignment,
    VkSystemAllocationScope                     allocationScope);
```

- pUserData is the value specified for VkAllocationCallbacks::pUserData in the allocator specified by the application.
- pOriginal **must** be either NULL or a pointer previously returned by pfnReallocation or pfnAllocation of a compatible allocator.
- size is the size in bytes of the requested allocation.
- alignment is the requested alignment of the allocation in bytes and **must** be a power of two.
- allocationScope is a VkSystemAllocationScope value specifying the allocation scope of the lifetime of the allocation, as described here.

If the reallocation was successful, pfnReallocation **must** return an allocation with enough space for size bytes, and the contents of the original allocation from bytes zero to min(original size, new size) - 1 **must** be preserved in the returned allocation. If size is larger than the old size, the contents of the additional space are undefined. If satisfying these requirements involves creating a new allocation, then the old allocation **should** be freed.

If pOriginal is NULL, then pfnReallocation **must** behave equivalently to a call to PFN_vkAllocationFunction with the same parameter values (without pOriginal).

If size is zero, then pfnReallocation **must** behave equivalently to a call to PFN_vkFreeFunction with the same pUserData parameter value, and pMemory equal to pOriginal.

If pOriginal is non-NULL, the implementation **must** ensure that alignment is equal to the alignment used to originally allocate pOriginal.

If this function fails and pOriginal is non-NULL the application **must** not free the old allocation.

pfnReallocation **must** follow the same rules for return values as PFN_vkAllocationFunction.

The type of pfnFree is:

```
// Provided by VK_VERSION_1_0
typedef void (*PFN_vkFreeFunction)(
    void*                                       pUserData,
    void*                                       pMemory);
```

- `pUserData` is the value specified for VkAllocationCallbacks::`pUserData` in the allocator specified by the application.
- `pMemory` is the allocation to be freed.

`pMemory` **may** be `NULL`, which the callback **must** handle safely. If `pMemory` is non-`NULL`, it **must** be a pointer previously allocated by `pfnAllocation` or `pfnReallocation`. The application **should** free this memory.

The type of `pfnInternalAllocation` is:

```
// Provided by VK_VERSION_1_0
typedef void (*PFN_vkInternalAllocationNotification)(
    void*                                       pUserData,
    size_t                                      size,
    VkInternalAllocationType                    allocationType,
    VkSystemAllocationScope                     allocationScope);
```

- `pUserData` is the value specified for VkAllocationCallbacks::`pUserData` in the allocator specified by the application.
- `size` is the requested size of an allocation.
- `allocationType` is a VkInternalAllocationType value specifying the requested type of an allocation.
- `allocationScope` is a VkSystemAllocationScope value specifying the allocation scope of the lifetime of the allocation, as described here.

This is a purely informational callback.

The type of `pfnInternalFree` is:

```
// Provided by VK_VERSION_1_0
typedef void (*PFN_vkInternalFreeNotification)(
    void*                                       pUserData,
    size_t                                      size,
    VkInternalAllocationType                    allocationType,
    VkSystemAllocationScope                     allocationScope);
```

- `pUserData` is the value specified for VkAllocationCallbacks::`pUserData` in the allocator specified by the application.
- `size` is the requested size of an allocation.

- `allocationType` is a VkInternalAllocationType value specifying the requested type of an allocation.
- `allocationScope` is a VkSystemAllocationScope value specifying the allocation scope of the lifetime of the allocation, as described here.

Each allocation has an *allocation scope* defining its lifetime and which object it is associated with. Possible values passed to the `allocationScope` parameter of the callback functions specified by VkAllocationCallbacks, indicating the allocation scope, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSystemAllocationScope {
    VK_SYSTEM_ALLOCATION_SCOPE_COMMAND = 0,
    VK_SYSTEM_ALLOCATION_SCOPE_OBJECT = 1,
    VK_SYSTEM_ALLOCATION_SCOPE_CACHE = 2,
    VK_SYSTEM_ALLOCATION_SCOPE_DEVICE = 3,
    VK_SYSTEM_ALLOCATION_SCOPE_INSTANCE = 4,
} VkSystemAllocationScope;
```

- VK_SYSTEM_ALLOCATION_SCOPE_COMMAND specifies that the allocation is scoped to the duration of the Vulkan command.
- VK_SYSTEM_ALLOCATION_SCOPE_OBJECT specifies that the allocation is scoped to the lifetime of the Vulkan object that is being created or used.
- VK_SYSTEM_ALLOCATION_SCOPE_CACHE specifies that the allocation is scoped to the lifetime of a `VkPipelineCache` or `VkValidationCacheEXT` object.
- VK_SYSTEM_ALLOCATION_SCOPE_DEVICE specifies that the allocation is scoped to the lifetime of the Vulkan device.
- VK_SYSTEM_ALLOCATION_SCOPE_INSTANCE specifies that the allocation is scoped to the lifetime of the Vulkan instance.

Most Vulkan commands operate on a single object, or there is a sole object that is being created or manipulated. When an allocation uses an allocation scope of VK_SYSTEM_ALLOCATION_SCOPE_OBJECT or VK_SYSTEM_ALLOCATION_SCOPE_CACHE, the allocation is scoped to the object being created or manipulated.

When an implementation requires host memory, it will make callbacks to the application using the most specific allocator and allocation scope available:

- If an allocation is scoped to the duration of a command, the allocator will use the VK_SYSTEM_ALLOCATION_SCOPE_COMMAND allocation scope. The most specific allocator available is used: if the object being created or manipulated has an allocator, that object's allocator will be used, else if the parent `VkDevice` has an allocator it will be used, else if the parent `VkInstance` has an allocator it will be used. Else,
- If an allocation is associated with a `VkValidationCacheEXT` or `VkPipelineCache` object, the allocator will use the VK_SYSTEM_ALLOCATION_SCOPE_CACHE allocation scope. The most specific allocator available is used (cache, else device, else instance). Else,

- If an allocation is scoped to the lifetime of an object, that object is being created or manipulated by the command, and that object's type is not `VkDevice` or `VkInstance`, the allocator will use an allocation scope of VK_SYSTEM_ALLOCATION_SCOPE_OBJECT. The most specific allocator available is used (object, else device, else instance). Else,
- If an allocation is scoped to the lifetime of a device, the allocator will use an allocation scope of VK_SYSTEM_ALLOCATION_SCOPE_DEVICE. The most specific allocator available is used (device, else instance). Else,
- If the allocation is scoped to the lifetime of an instance and the instance has an allocator, its allocator will be used with an allocation scope of VK_SYSTEM_ALLOCATION_SCOPE_INSTANCE.
- Otherwise an implementation will allocate memory through an alternative mechanism that is unspecified.

Objects that are allocated from pools do not specify their own allocator. When an implementation requires host memory for such an object, that memory is sourced from the object's parent pool's allocator.

The application is not expected to handle allocating memory that is intended for execution by the host due to the complexities of differing security implementations across multiple platforms. The implementation will allocate such memory internally and invoke an application provided informational callback when these *internal allocations* are allocated and freed. Upon allocation of executable memory, `pfnInternalAllocation` will be called. Upon freeing executable memory, `pfnInternalFree` will be called. An implementation will only call an informational callback for executable memory allocations and frees.

The `allocationType` parameter to the `pfnInternalAllocation` and `pfnInternalFree` functions **may** be one of the following values:

```
// Provided by VK_VERSION_1_0
typedef enum VkInternalAllocationType {
    VK_INTERNAL_ALLOCATION_TYPE_EXECUTABLE = 0,
} VkInternalAllocationType;
```

- VK_INTERNAL_ALLOCATION_TYPE_EXECUTABLE specifies that the allocation is intended for execution by the host.

An implementation **must** only make calls into an application-provided allocator during the execution of an API command. An implementation **must** only make calls into an application-provided allocator from the same thread that called the provoking API command. The implementation **should** not synchronize calls to any of the callbacks. If synchronization is needed, the callbacks **must** provide it themselves. The informational callbacks are subject to the same restrictions as the allocation callbacks.

If an implementation intends to make calls through a `VkAllocationCallbacks` structure between the time a `vkCreate*` command returns and the time a corresponding `vkDestroy*` command begins, that implementation **must** save a copy of the allocator before the `vkCreate*` command returns. The callback functions and any data structures they rely upon **must** remain valid for the lifetime of the object they are associated with.

If an allocator is provided to a vkCreate* command, a *compatible* allocator **must** be provided to the corresponding vkDestroy* command. Two VkAllocationCallbacks structures are compatible if memory allocated with pfnAllocation or pfnReallocation in each **can** be freed with pfnReallocation or pfnFree in the other. An allocator **must** not be provided to a vkDestroy* command if an allocator was not provided to the corresponding vkCreate* command.

If a non-NULL allocator is used, the pfnAllocation, pfnReallocation and pfnFree members **must** be non-NULL and point to valid implementations of the callbacks. An application **can** choose to not provide informational callbacks by setting both pfnInternalAllocation and pfnInternalFree to NULL. pfnInternalAllocation and pfnInternalFree **must** either both be NULL or both be non-NULL.

If pfnAllocation or pfnReallocation fail, the implementation **may** fail object creation and/or generate a VK_ERROR_OUT_OF_HOST_MEMORY error, as appropriate.

Allocation callbacks **must** not call any Vulkan commands.

The following sets of rules define when an implementation is permitted to call the allocator callbacks.

pfnAllocation or pfnReallocation **may** be called in the following situations:

- Allocations scoped to a VkDevice or VkInstance **may** be allocated from any API command.
- Allocations scoped to a command **may** be allocated from any API command.
- Allocations scoped to a VkPipelineCache **may** only be allocated from:
  - vkCreatePipelineCache
  - vkMergePipelineCaches for dstCache
  - vkCreateGraphicsPipelines for pipelineCache
  - vkCreateComputePipelines for pipelineCache
- Allocations scoped to a VkValidationCacheEXT **may** only be allocated from:
  - vkCreateValidationCacheEXT
  - vkMergeValidationCachesEXT for dstCache
  - vkCreateShaderModule for validationCache in VkShaderModuleValidationCacheCreateInfoEXT
- Allocations scoped to a VkDescriptorPool **may** only be allocated from:
  - any command that takes the pool as a direct argument
  - vkAllocateDescriptorSets for the descriptorPool member of its pAllocateInfo parameter
  - vkCreateDescriptorPool
- Allocations scoped to a VkCommandPool **may** only be allocated from:
  - any command that takes the pool as a direct argument
  - vkCreateCommandPool
  - vkAllocateCommandBuffers for the commandPool member of its pAllocateInfo parameter
  - any vkCmd* command whose commandBuffer was allocated from that VkCommandPool

- Allocations scoped to any other object **may** only be allocated in that object's `vkCreate*` command.

`pfnFree`, or `pfnReallocation` with zero `size`, **may** be called in the following situations:

- Allocations scoped to a `VkDevice` or `VkInstance` **may** be freed from any API command.
- Allocations scoped to a command **must** be freed by any API command which allocates such memory.
- Allocations scoped to a `VkPipelineCache` **may** be freed from `vkDestroyPipelineCache`.
- Allocations scoped to a `VkValidationCacheEXT` **may** be freed from `vkDestroyValidationCacheEXT`.
- Allocations scoped to a `VkDescriptorPool` **may** be freed from:
  - any command that takes the pool as a direct argument
- Allocations scoped to a `VkCommandPool` **may** be freed from:
  - any command that takes the pool as a direct argument
  - `vkResetCommandBuffer` whose `commandBuffer` was allocated from that `VkCommandPool`
- Allocations scoped to any other object **may** be freed in that object's `vkDestroy*` command.
- Any command that allocates host memory **may** also free host memory of the same scope.

11.2. Device Memory

Device memory is memory that is visible to the device — for example the contents of the image or buffer objects, which **can** be natively used by the device.

11.2.1. Device Memory Properties

Memory properties of a physical device describe the memory heaps and memory types available.

To query memory properties, call:

WARNING	This functionality is superseded by vkGetPhysicalDeviceMemoryProperties2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceMemoryProperties(
    VkPhysicalDevice                            physicalDevice,
    VkPhysicalDeviceMemoryProperties*           pMemoryProperties);
```

- `physicalDevice` is the handle to the device to query.
- `pMemoryProperties` is a pointer to a VkPhysicalDeviceMemoryProperties structure in which the properties are returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceMemoryProperties-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceMemoryProperties-pMemoryProperties-parameter
  `pMemoryProperties` **must** be a valid pointer to a VkPhysicalDeviceMemoryProperties structure

The `VkPhysicalDeviceMemoryProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPhysicalDeviceMemoryProperties {
    uint32_t        memoryTypeCount;
    VkMemoryType    memoryTypes[VK_MAX_MEMORY_TYPES];
    uint32_t        memoryHeapCount;
    VkMemoryHeap    memoryHeaps[VK_MAX_MEMORY_HEAPS];
} VkPhysicalDeviceMemoryProperties;
```

- `memoryTypeCount` is the number of valid elements in the `memoryTypes` array.
- `memoryTypes` is an array of VK_MAX_MEMORY_TYPES VkMemoryType structures describing the *memory types* that **can** be used to access memory allocated from the heaps specified by `memoryHeaps`.
- `memoryHeapCount` is the number of valid elements in the `memoryHeaps` array.
- `memoryHeaps` is an array of VK_MAX_MEMORY_HEAPS VkMemoryHeap structures describing the *memory heaps* from which memory **can** be allocated.

The `VkPhysicalDeviceMemoryProperties` structure describes a number of *memory heaps* as well as a number of *memory types* that **can** be used to access memory allocated in those heaps. Each heap describes a memory resource of a particular size, and each memory type describes a set of memory properties (e.g. host cached vs. uncached) that **can** be used with a given memory heap. Allocations using a particular memory type will consume resources from the heap indicated by that memory type's heap index. More than one memory type **may** share each heap, and the heaps and memory types provide a mechanism to advertise an accurate size of the physical memory resources while allowing the memory to be used with a variety of different properties.

The number of memory heaps is given by `memoryHeapCount` and is less than or equal to VK_MAX_MEMORY_HEAPS. Each heap is described by an element of the `memoryHeaps` array as a VkMemoryHeap structure. The number of memory types available across all memory heaps is given by `memoryTypeCount` and is less than or equal to VK_MAX_MEMORY_TYPES. Each memory type is described by an element of the `memoryTypes` array as a VkMemoryType structure.

At least one heap **must** include VK_MEMORY_HEAP_DEVICE_LOCAL_BIT in VkMemoryHeap::`flags`. If there are multiple heaps that all have similar performance characteristics, they **may** all include VK_MEMORY_HEAP_DEVICE_LOCAL_BIT. In a unified memory architecture (UMA) system there is often only a single memory heap which is considered to be equally "local" to the host and to the

device, and such an implementation **must** advertise the heap as device-local.

Each memory type returned by vkGetPhysicalDeviceMemoryProperties **must** have its `propertyFlags` set to one of the following values:

- 0
- VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT | VK_MEMORY_PROPERTY_HOST_COHERENT_BIT
- VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT | VK_MEMORY_PROPERTY_HOST_CACHED_BIT
- VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT | VK_MEMORY_PROPERTY_HOST_CACHED_BIT | VK_MEMORY_PROPERTY_HOST_COHERENT_BIT
- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT
- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT | VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT | VK_MEMORY_PROPERTY_HOST_COHERENT_BIT
- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT | VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT | VK_MEMORY_PROPERTY_HOST_CACHED_BIT
- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT | VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT | VK_MEMORY_PROPERTY_HOST_CACHED_BIT | VK_MEMORY_PROPERTY_HOST_COHERENT_BIT
- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT | VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT
- VK_MEMORY_PROPERTY_PROTECTED_BIT
- VK_MEMORY_PROPERTY_PROTECTED_BIT | VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT

There **must** be at least one memory type with both the VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT and VK_MEMORY_PROPERTY_HOST_COHERENT_BIT bits set in its `propertyFlags`. There **must** be at least one memory type with the VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT bit set in its `propertyFlags`.

For each pair of elements **X** and **Y** returned in `memoryTypes`, **X** **must** be placed at a lower index position than **Y** if:

- the set of bit flags returned in the `propertyFlags` member of **X** is a strict subset of the set of bit flags returned in the `propertyFlags` member of **Y**; or
- the `propertyFlags` members of **X** and **Y** are equal, and **X** belongs to a memory heap with greater performance (as determined in an implementation-specific manner)

NOTE

There is no ordering requirement between **X** and **Y** elements for the case their `propertyFlags` members are not in a subset relation. That potentially allows more than one possible way to order the same set of memory types. Notice that the list of

> all allowed memory property flag combinations is written in a valid order. But if instead VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT was before VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT | VK_MEMORY_PROPERTY_HOST_COHERENT_BIT, the list would still be in a valid order.

This ordering requirement enables applications to use a simple search loop to select the desired memory type along the lines of:

```
// Find a memory in `memoryTypeBitsRequirement` that includes all of
`requiredProperties`
int32_t findProperties(const VkPhysicalDeviceMemoryProperties* pMemoryProperties,
                       uint32_t memoryTypeBitsRequirement,
                       VkMemoryPropertyFlags requiredProperties) {
    const uint32_t memoryCount = pMemoryProperties->memoryTypeCount;
    for (uint32_t memoryIndex = 0; memoryIndex < memoryCount; ++memoryIndex) {
        const uint32_t memoryTypeBits = (1 << memoryIndex);
        const bool isRequiredMemoryType = memoryTypeBitsRequirement & memoryTypeBits;

        const VkMemoryPropertyFlags properties =
            pMemoryProperties->memoryTypes[memoryIndex].propertyFlags;
        const bool hasRequiredProperties =
            (properties & requiredProperties) == requiredProperties;

        if (isRequiredMemoryType && hasRequiredProperties)
            return static_cast<int32_t>(memoryIndex);
    }

    // failed to find memory type
    return -1;
}

// Try to find an optimal memory type, or if it does not exist try fallback memory
type
// `device` is the VkDevice
// `image` is the VkImage that requires memory to be bound
// `memoryProperties` properties as returned by vkGetPhysicalDeviceMemoryProperties
// `requiredProperties` are the property flags that must be present
// `optimalProperties` are the property flags that are preferred by the application
VkMemoryRequirements memoryRequirements;
vkGetImageMemoryRequirements(device, image, &memoryRequirements);
int32_t memoryType =
    findProperties(&memoryProperties, memoryRequirements.memoryTypeBits,
optimalProperties);
if (memoryType == -1) // not found; try fallback properties
    memoryType =
        findProperties(&memoryProperties, memoryRequirements.memoryTypeBits,
requiredProperties);
```

VK_MAX_MEMORY_TYPES is the length of an array of VkMemoryType structures describing memory types, as returned in VkPhysicalDeviceMemoryProperties::memoryTypes.

```
#define VK_MAX_MEMORY_TYPES             32U
```

VK_MAX_MEMORY_HEAPS is the length of an array of VkMemoryHeap structures describing memory heaps, as returned in VkPhysicalDeviceMemoryProperties::memoryHeaps.

```
#define VK_MAX_MEMORY_HEAPS             16U
```

To query memory properties, call:

```
// Provided by VK_VERSION_1_1
void vkGetPhysicalDeviceMemoryProperties2(
    VkPhysicalDevice                            physicalDevice,
    VkPhysicalDeviceMemoryProperties2*          pMemoryProperties);
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to vkGetPhysicalDeviceMemoryProperties2
void vkGetPhysicalDeviceMemoryProperties2KHR(
    VkPhysicalDevice                            physicalDevice,
    VkPhysicalDeviceMemoryProperties2*          pMemoryProperties);
```

- physicalDevice is the handle to the device to query.
- pMemoryProperties is a pointer to a VkPhysicalDeviceMemoryProperties2 structure in which the properties are returned.

vkGetPhysicalDeviceMemoryProperties2 behaves similarly to vkGetPhysicalDeviceMemoryProperties, with the ability to return extended information in a pNext chain of output structures.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceMemoryProperties2-physicalDevice-parameter
  physicalDevice **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceMemoryProperties2-pMemoryProperties-parameter
  pMemoryProperties **must** be a valid pointer to a VkPhysicalDeviceMemoryProperties2 structure

The VkPhysicalDeviceMemoryProperties2 structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceMemoryProperties2 {
    VkStructureType                     sType;
    void*                               pNext;
    VkPhysicalDeviceMemoryProperties    memoryProperties;
} VkPhysicalDeviceMemoryProperties2;
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to VkPhysicalDeviceMemoryProperties2
typedef VkPhysicalDeviceMemoryProperties2 VkPhysicalDeviceMemoryProperties2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryProperties` is a VkPhysicalDeviceMemoryProperties structure which is populated with the same values as in vkGetPhysicalDeviceMemoryProperties.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMemoryProperties2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PROPERTIES_2
- VUID-VkPhysicalDeviceMemoryProperties2-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkPhysicalDeviceMemoryBudgetPropertiesEXT
- VUID-VkPhysicalDeviceMemoryProperties2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

The `VkMemoryHeap` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMemoryHeap {
    VkDeviceSize         size;
    VkMemoryHeapFlags    flags;
} VkMemoryHeap;
```

- `size` is the total memory size in bytes in the heap.
- `flags` is a bitmask of VkMemoryHeapFlagBits specifying attribute flags for the heap.

Bits which **may** be set in VkMemoryHeap::`flags`, indicating attribute flags for the heap, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkMemoryHeapFlagBits {
    VK_MEMORY_HEAP_DEVICE_LOCAL_BIT = 0x00000001,
  // Provided by VK_VERSION_1_1
    VK_MEMORY_HEAP_MULTI_INSTANCE_BIT = 0x00000002,
  // Provided by VK_KHR_device_group_creation
    VK_MEMORY_HEAP_MULTI_INSTANCE_BIT_KHR = VK_MEMORY_HEAP_MULTI_INSTANCE_BIT,
} VkMemoryHeapFlagBits;
```

- VK_MEMORY_HEAP_DEVICE_LOCAL_BIT specifies that the heap corresponds to device-local memory. Device-local memory **may** have different performance characteristics than host-local memory, and **may** support different memory property flags.
- VK_MEMORY_HEAP_MULTI_INSTANCE_BIT specifies that in a logical device representing more than one physical device, there is a per-physical device instance of the heap memory. By default, an allocation from such a heap will be replicated to each physical device's instance of the heap.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkMemoryHeapFlags;
```

`VkMemoryHeapFlags` is a bitmask type for setting a mask of zero or more VkMemoryHeapFlagBits.

The `VkMemoryType` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMemoryType {
    VkMemoryPropertyFlags    propertyFlags;
    uint32_t                 heapIndex;
} VkMemoryType;
```

- `heapIndex` describes which memory heap this memory type corresponds to, and **must** be less than `memoryHeapCount` from the VkPhysicalDeviceMemoryProperties structure.
- `propertyFlags` is a bitmask of VkMemoryPropertyFlagBits of properties for this memory type.

Bits which **may** be set in VkMemoryType::`propertyFlags`, indicating properties of a memory type, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkMemoryPropertyFlagBits {
    VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT = 0x00000001,
    VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT = 0x00000002,
    VK_MEMORY_PROPERTY_HOST_COHERENT_BIT = 0x00000004,
    VK_MEMORY_PROPERTY_HOST_CACHED_BIT = 0x00000008,
    VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT = 0x00000010,
  // Provided by VK_VERSION_1_1
    VK_MEMORY_PROPERTY_PROTECTED_BIT = 0x00000020,
} VkMemoryPropertyFlagBits;
```

- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT bit specifies that memory allocated with this type is the most efficient for device access. This property will be set if and only if the memory type belongs to a heap with the VK_MEMORY_HEAP_DEVICE_LOCAL_BIT set.
- VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT bit specifies that memory allocated with this type **can** be mapped for host access using vkMapMemory.
- VK_MEMORY_PROPERTY_HOST_COHERENT_BIT bit specifies that the host cache management commands vkFlushMappedMemoryRanges and vkInvalidateMappedMemoryRanges are not needed to manage availability and visibility on the host.
- VK_MEMORY_PROPERTY_HOST_CACHED_BIT bit specifies that memory allocated with this type is cached on the host. Host memory read accesses to uncached memory are slower than to cached memory, however uncached memory is always host coherent.
- VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT bit specifies that the memory type only allows device access to the memory. Memory types **must** not have both VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT and VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT set. Additionally, the object's backing memory **may** be provided by the implementation lazily as specified in Lazily Allocated Memory.
- VK_MEMORY_PROPERTY_PROTECTED_BIT bit specifies that the memory type only allows device access to the memory, and allows protected queue operations to access the memory. Memory types **must** not have VK_MEMORY_PROPERTY_PROTECTED_BIT set and any of VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT set, or VK_MEMORY_PROPERTY_HOST_COHERENT_BIT set, or VK_MEMORY_PROPERTY_HOST_CACHED_BIT set.

NOTE

Between the memory types which support VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT on any particular implementation, there may be different properties which cater to different memory access patterns.

VK_MEMORY_PROPERTY_HOST_CACHED_BIT is expected to be used when the host performs any read or read-modify-write operations on the memory. The typical scenario for this is when data flows from device to host. A typical implementation of this is normal `malloc`-like memory.

VK_MEMORY_PROPERTY_HOST_COHERENT_BIT without VK_MEMORY_PROPERTY_HOST_CACHED_BIT is expected for upload operations where data flows from host to device, for purposes of initializing resources or

similar. A typical implementation of this memory type is write-combined memory which avoids polluting caches on the host, but has severe performance penalties if it is read from. The host is also expected to write complete cache lines for maximum performance.

Some implementations may not support uncached memory types. The expectation is that applications will end up using a memory type that supports VK_MEMORY_PROPERTY_HOST_COHERENT_BIT and VK_MEMORY_PROPERTY_HOST_CACHED_BIT together. Either way, applications are expected to look for the first memory type which supports VK_MEMORY_PROPERTY_HOST_COHERENT_BIT and use that. On VK_PHYSICAL_DEVICE_TYPE_INTEGRATED_GPU with a unified memory architecture, VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT is less of a concern since all memory is effectively device-local, but on VK_PHYSICAL_DEVICE_TYPE_DISCRETE_GPU, there may be a distinction between a VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT memory type which is additionally VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT or not.

On discrete GPUs, using VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT, VK_MEMORY_PROPERTY_HOST_COHERENT_BIT, and VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT together is intended for small uploads to device memory which is optimized for fast device access without the need to go through a separate copy operation. Also known as BAR on PCI-e platforms. Peak write performance may be lower than system memory, and is not recommended for large uploads to the device. Common use cases for this type may be updating descriptors in a descriptor heap with vkWriteResourceDescriptorsEXT and vkWriteSamplerDescriptorsEXT, updating descriptors in a descriptor buffer with vkGetDescriptorEXT, or writing uniform buffer data.

Some implementations may support host visible types which are VK_MEMORY_PROPERTY_HOST_CACHED_BIT, where some types support VK_MEMORY_PROPERTY_HOST_COHERENT_BIT alongside CACHED, and some types which do not support VK_MEMORY_PROPERTY_HOST_COHERENT_BIT in combination with CACHED. Unless coherency is required by the application's usage patterns, the VK_MEMORY_PROPERTY_HOST_COHERENT_BIT supporting type is likely to be less optimal. If an application only looks for the CACHED property, it must come before the CACHED | COHERENT memory type in the memory types list, so this suggestion will be followed by default.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkMemoryPropertyFlags;
```

VkMemoryPropertyFlags is a bitmask type for setting a mask of zero or more VkMemoryPropertyFlagBits.

If the VkPhysicalDeviceMemoryBudgetPropertiesEXT structure is included in the pNext chain of VkPhysicalDeviceMemoryProperties2, it is filled with the current memory budgets and usages.

The VkPhysicalDeviceMemoryBudgetPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_memory_budget
typedef struct VkPhysicalDeviceMemoryBudgetPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceSize       heapBudget[VK_MAX_MEMORY_HEAPS];
    VkDeviceSize       heapUsage[VK_MAX_MEMORY_HEAPS];
} VkPhysicalDeviceMemoryBudgetPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- heapBudget is an array of VK_MAX_MEMORY_HEAPS VkDeviceSize values in which memory budgets are returned, with one element for each memory heap. A heap's budget provides a guideline for how much total memory the current process **can** use at any given time before allocations **may** start failing or causing performance degradation. The values **may** change based on other activity in the system that is outside the scope and control of the Vulkan implementation.
- heapUsage is an array of VK_MAX_MEMORY_HEAPS VkDeviceSize values in which memory usages are returned, with one element for each memory heap. A heap's usage is an estimate of how much memory the current process has allocated from that heap.

The values returned in this structure are not invariant. The heapBudget and heapUsage values **must** be zero for array elements greater than or equal to VkPhysicalDeviceMemoryProperties::memoryHeapCount. The heapBudget value **must** be non-zero for array elements less than VkPhysicalDeviceMemoryProperties::memoryHeapCount. The heapBudget value **must** be less than or equal to VkMemoryHeap::size for each heap.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMemoryBudgetPropertiesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_BUDGET_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceMemoryProperties2

11.2.2. Device Memory Objects

A Vulkan device operates on data in device memory via memory objects that are represented in the API by a VkDeviceMemory handle:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDeviceMemory)
```

11.2.3. Device Memory Allocation

To allocate memory objects, call:

```
// Provided by VK_VERSION_1_0
VkResult vkAllocateMemory(
    VkDevice                                    device,
    const VkMemoryAllocateInfo*                 pAllocateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkDeviceMemory*                             pMemory);
```

- `device` is the logical device that owns the memory.
- `pAllocateInfo` is a pointer to a VkMemoryAllocateInfo structure describing parameters of the allocation. A successfully returned allocation **must** use the requested parameters — no substitution is permitted by the implementation.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pMemory` is a pointer to a VkDeviceMemory handle in which information about the allocated memory is returned.

Allocations returned by `vkAllocateMemory` are guaranteed to meet any alignment requirement of the implementation. For example, if an implementation requires 128 byte alignment for images and 64 byte alignment for buffers, the device memory returned through this mechanism would be 128-byte aligned. This ensures that applications **can** correctly suballocate objects of different types (with potentially different alignment requirements) in the same memory object.

When memory is allocated, its contents are undefined with the following constraint:

- The contents of unprotected memory **must** not be a function of the contents of data protected memory objects, even if those memory objects were previously freed.

NOTE The contents of memory allocated by one application **should** not be a function of data from protected memory objects of another application, even if those memory objects were previously freed.

The maximum number of valid memory allocations that **can** exist simultaneously within a VkDevice **may** be restricted by implementation- or platform-dependent limits. The `maxMemoryAllocationCount` feature describes the number of allocations that **can** exist simultaneously before encountering these internal limits.

NOTE For historical reasons, if `maxMemoryAllocationCount` is exceeded, some implementations may return VK_ERROR_TOO_MANY_OBJECTS. Exceeding this limit will result in undefined behavior, and an application should not rely on the use of

the returned error code in order to identify when the limit is reached.

NOTE

Many protected memory implementations involve complex hardware and system software support, and often have additional and much lower limits on the number of simultaneous protected memory allocations (from memory types with the VK_MEMORY_PROPERTY_PROTECTED_BIT property) than for non-protected memory allocations. These limits can be system-wide, and depend on a variety of factors outside of the Vulkan implementation, so they cannot be queried in Vulkan. Applications **should** use as few allocations as possible from such memory types by suballocating aggressively, and be prepared for allocation failure even when there is apparently plenty of capacity remaining in the memory heap. As a guideline, the Vulkan conformance test suite requires that at least 80 minimum-size allocations can exist concurrently when no other uses of protected memory are active in the system.

Some platforms **may** have a limit on the maximum size of a single allocation. For example, certain systems **may** fail to create allocations with a size greater than or equal to 4GB. Such a limit is implementation-dependent, and if such a failure occurs then the error VK_ERROR_OUT_OF_DEVICE_MEMORY **must** be returned. This limit is advertised in VkPhysicalDeviceMaintenance3Properties::`maxMemoryAllocationSize`.

If the VkPhysicalDevicePageableDeviceLocalMemoryFeaturesEXT::`pageableDeviceLocalMemory` feature is enabled, memory allocations made from a heap that includes VK_MEMORY_HEAP_DEVICE_LOCAL_BIT in VkMemoryHeap::`flags` **may** be transparently moved to host-local memory allowing multiple applications to share device-local memory. If there is no space left in device-local memory when this new allocation is made, other allocations **may** be moved out transparently to make room. The operating system will determine which allocations to move to device-local memory or host-local memory based on platform-specific criteria. To help the operating system make good choices, the application **should** set the appropriate memory priority with VkMemoryPriorityAllocateInfoEXT and adjust it as necessary with vkSetDeviceMemoryPriorityEXT. Higher priority allocations will be moved to device-local memory first.

Memory allocations made on heaps without the VK_MEMORY_HEAP_DEVICE_LOCAL_BIT property will not be transparently promoted to device-local memory by the operating system.

Valid Usage

- VUID-vkAllocateMemory-pAllocateInfo-01713
  `pAllocateInfo->allocationSize` **must** be less than or equal to VkPhysicalDeviceMemoryProperties::`memoryHeaps`[memindex].`size` where `memindex` = VkPhysicalDeviceMemoryProperties::`memoryTypes`[`pAllocateInfo->memoryTypeIndex`].`heapIndex` as returned by vkGetPhysicalDeviceMemoryProperties for the VkPhysicalDevice that `device` was created from
- VUID-vkAllocateMemory-pAllocateInfo-01714
  `pAllocateInfo->memoryTypeIndex` **must** be less than VkPhysicalDeviceMemoryProperties::`memoryTypeCount` as returned by vkGetPhysicalDeviceMemoryProperties for the

VkPhysicalDevice that device was created from

- VUID-vkAllocateMemory-maxMemoryAllocationCount-04101
  There **must** be less than VkPhysicalDeviceLimits::maxMemoryAllocationCount device memory allocations currently allocated on the device

Valid Usage (Implicit)

- VUID-vkAllocateMemory-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkAllocateMemory-pAllocateInfo-parameter
  pAllocateInfo **must** be a valid pointer to a valid VkMemoryAllocateInfo structure
- VUID-vkAllocateMemory-pAllocator-parameter
  If pAllocator is not NULL, pAllocator **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkAllocateMemory-pMemory-parameter
  pMemory **must** be a valid pointer to a VkDeviceMemory handle
- VUID-vkAllocateMemory-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_EXTERNAL_HANDLE
- VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkMemoryAllocateInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMemoryAllocateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceSize       allocationSize;
    uint32_t           memoryTypeIndex;
} VkMemoryAllocateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `allocationSize` is the size of the allocation in bytes.
- `memoryTypeIndex` is an index identifying a memory type from the `memoryTypes` array of the VkPhysicalDeviceMemoryProperties structure.

The internal data of an allocated device memory object **must** include a reference to implementation-specific resources, referred to as the memory object's *payload*. Applications **can** also import and export that internal data to and from device memory objects to share data between Vulkan instances and other compatible APIs. A `VkMemoryAllocateInfo` structure defines a memory import operation if its `pNext` chain includes one of the following structures:

- VkImportMemoryWin32HandleInfoKHR with a non-zero `handleType` value
- VkImportMemoryFdInfoKHR with a non-zero `handleType` value
- VkImportMemoryHostPointerInfoEXT with a non-zero `handleType` value
- VkImportMemoryMetalHandleInfoEXT with a non-zero `handleType` value

If the parameters define an import operation and the external handle type is VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT, VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT, or VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT, `allocationSize` is ignored. The implementation **must** query the size of these allocations from the OS.

If the parameters define an import operation and the external handle type is VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLTEXTURE_BIT_EXT, `allocationSize` is ignored. The implementation **must** query the size of these allocations from the OS.

Whether device memory objects constructed via a memory import operation hold a reference to their payload depends on the properties of the handle type used to perform the import, as defined below for each valid handle type. Importing memory **must** not modify the content of the memory. Implementations **must** ensure that importing memory does not enable the importing Vulkan instance to access any memory or resources in other Vulkan instances other than that corresponding to the memory object imported. Implementations **must** also ensure accessing imported memory which has not been initialized does not allow the importing Vulkan instance to obtain data from the exporting Vulkan instance or vice-versa.

NOTE How exported and imported memory is isolated is left to the implementation, but applications should be aware that such isolation **may** prevent implementations from placing multiple exportable memory objects in the same physical or virtual page. Hence, applications **should** avoid creating many small external memory objects whenever possible.

Importing memory **must** not increase overall heap usage within a system. However, it **must** affect the following per-process values:

- VkPhysicalDeviceLimits::`maxMemoryAllocationCount`

- VkPhysicalDeviceMemoryBudgetPropertiesEXT::heapUsage

When performing a memory import operation, it is the responsibility of the application to ensure the external handles and their associated payloads meet all valid usage requirements. However, implementations **must** perform sufficient validation of external handles and payloads to ensure that the operation results in a valid memory object which will not cause program termination, device loss, queue stalls, or corruption of other resources when used as allowed according to its allocation parameters. If the external handle provided does not meet these requirements, the implementation **must** fail the memory import operation with the error code VK_ERROR_INVALID_EXTERNAL_HANDLE.

Valid Usage

- VUID-VkMemoryAllocateInfo-allocationSize-07897
  If the parameters do not define an import or export operation, `allocationSize` **must** be greater than `0`
- VUID-VkMemoryAllocateInfo-None-06657
  The parameters **must** not define more than one import operation
- VUID-VkMemoryAllocateInfo-allocationSize-07899
  If the parameters define an export operation , `allocationSize` **must** be greater than `0`
- VUID-VkMemoryAllocateInfo-pNext-00639
  If the `pNext` chain includes a `VkExportMemoryAllocateInfo` structure, and any of the handle types specified in `VkExportMemoryAllocateInfo`::`handleTypes` require a dedicated allocation, as reported by vkGetPhysicalDeviceImageFormatProperties2 in `VkExternalImageFormatProperties`::`externalMemoryProperties.externalMemoryFeatures`, or by vkGetPhysicalDeviceExternalBufferProperties in `VkExternalBufferProperties`::`externalMemoryProperties.externalMemoryFeatures`, the `pNext` chain **must** include a `VkMemoryDedicatedAllocateInfo` or `VkDedicatedAllocationMemoryAllocateInfoNV` structure with either its `image` or `buffer` member set to a value other than VK_NULL_HANDLE
- VUID-VkMemoryAllocateInfo-allocationSize-01742
  If the parameters define an import operation, the external handle specified was created by the Vulkan API, and the external handle type is VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT, then the values of `allocationSize` and `memoryTypeIndex` **must** match those specified when the payload being imported was created
- VUID-VkMemoryAllocateInfo-None-00643
  If the parameters define an import operation and the external handle specified was created by the Vulkan API, the device mask specified by VkMemoryAllocateFlagsInfo **must** match the mask specified when the payload being imported was allocated
- VUID-VkMemoryAllocateInfo-None-00644
  If the parameters define an import operation and the external handle specified was created by the Vulkan API, the list of physical devices that comprise the logical device passed to vkAllocateMemory **must** match the list of physical devices that comprise the logical device on which the payload was originally allocated
- VUID-VkMemoryAllocateInfo-memoryTypeIndex-00645

If the parameters define an import operation and the external handle is an NT handle or a global share handle created outside of the Vulkan API, the value of `memoryTypeIndex` **must** be one of those returned by vkGetMemoryWin32HandlePropertiesKHR

- VUID-VkMemoryAllocateInfo-allocationSize-01743
  If the parameters define an import operation, the external handle was created by the Vulkan API, and the external handle type is VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT or VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT, then the values of `allocationSize` and `memoryTypeIndex` **must** match those specified when the payload being imported was created
- VUID-VkMemoryAllocateInfo-allocationSize-00647
  If the parameters define an import operation and the external handle type is VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT, `allocationSize` **must** match the size specified when creating the Direct3D 12 heap from which the payload was extracted
- VUID-VkMemoryAllocateInfo-memoryTypeIndex-00648
  If the parameters define an import operation and the external handle is a POSIX file descriptor created outside of the Vulkan API, the value of `memoryTypeIndex` **must** be one of those returned by vkGetMemoryFdPropertiesKHR
- VUID-VkMemoryAllocateInfo-None-12523
  If the parameters define an import operation and the external handle type is VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT, the number of instances allocated **must** be 1
- VUID-VkMemoryAllocateInfo-memoryTypeIndex-01872
  If the `protectedMemory` feature is not enabled, the `VkMemoryAllocateInfo`::`memoryTypeIndex` **must** not indicate a memory type that reports VK_MEMORY_PROPERTY_PROTECTED_BIT
- VUID-VkMemoryAllocateInfo-memoryTypeIndex-01744
  If the parameters define an import operation and the external handle is a host pointer, the value of `memoryTypeIndex` **must** be one of those returned by vkGetMemoryHostPointerPropertiesEXT
- VUID-VkMemoryAllocateInfo-allocationSize-01745
  If the parameters define an import operation and the external handle is a host pointer, `allocationSize` **must** be an integer multiple of `VkPhysicalDeviceExternalMemoryHostPropertiesEXT`::`minImportedHostPointerAlignment`
- VUID-VkMemoryAllocateInfo-pNext-02806
  If the parameters define an import operation and the external handle is a host pointer, the `pNext` chain **must** not include a VkMemoryDedicatedAllocateInfo structure with either its `image` or `buffer` field set to a value other than VK_NULL_HANDLE
- VUID-VkMemoryAllocateInfo-pNext-10395
  If the parameters define an import operation and the external handle is a VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLTEXTURE_BIT_EXT, then `pNext` **must** include a VkMemoryDedicatedAllocateInfo with `image` that is not VK_NULL_HANDLE
- VUID-VkMemoryAllocateInfo-pNext-10396
  If the parameters define an import operation, the external handle is a Metal MTLTexture,

and the pNext chain includes a VkMemoryDedicatedAllocateInfo structure with image that is not VK_NULL_HANDLE, the width, height, array layer dimensions, and mipmap levels of image and the Metal MTLTexture **must** be identical

- VUID-VkMemoryAllocateInfo-pNext-10397
  If the parameters define an import operation, the external handle is a Metal MTLTexture, and the pNext chain includes a VkMemoryDedicatedAllocateInfo structure with image that is not VK_NULL_HANDLE, allocationSize **must** be 0
- VUID-VkMemoryAllocateInfo-opaqueCaptureAddress-03329
  If VkMemoryOpaqueCaptureAddressAllocateInfo::opaqueCaptureAddress is not zero, VkMemoryAllocateFlagsInfo::flags **must** include VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT
- VUID-VkMemoryAllocateInfo-flags-03330
  If VkMemoryAllocateFlagsInfo::flags includes VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT, the bufferDeviceAddressCaptureReplay feature **must** be enabled
- VUID-VkMemoryAllocateInfo-flags-03331
  If VkMemoryAllocateFlagsInfo::flags includes VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT, the bufferDeviceAddress feature **must** be enabled
- VUID-VkMemoryAllocateInfo-pNext-03332
  If the pNext chain includes a VkImportMemoryHostPointerInfoEXT structure, VkMemoryOpaqueCaptureAddressAllocateInfo::opaqueCaptureAddress **must** be zero
- VUID-VkMemoryAllocateInfo-opaqueCaptureAddress-03333
  If the parameters define an import operation, VkMemoryOpaqueCaptureAddressAllocateInfo::opaqueCaptureAddress **must** be zero
- VUID-VkMemoryAllocateInfo-pNext-06780
  If the pNext chain includes a VkExportMetalObjectCreateInfoEXT structure, its exportObjectType member **must** be VK_EXPORT_METAL_OBJECT_TYPE_METAL_BUFFER_BIT_EXT

Valid Usage (Implicit)

- VUID-VkMemoryAllocateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_INFO
- VUID-VkMemoryAllocateInfo-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkExportMemoryAllocateInfo, VkExportMemoryWin32HandleInfoKHR, VkExportMetalObjectCreateInfoEXT, VkImportMemoryFdInfoKHR, VkImportMemoryHostPointerInfoEXT, VkImportMemoryMetalHandleInfoEXT, VkImportMemoryWin32HandleInfoKHR, VkImportMetalBufferInfoEXT, VkMemoryAllocateFlagsInfo, VkMemoryDedicatedAllocateInfo, VkMemoryOpaqueCaptureAddressAllocateInfo, or VkMemoryPriorityAllocateInfoEXT

- VUID-VkMemoryAllocateInfo-sType-unique
  The sType value of each structure in the pNext chain **must** be unique, with the exception of structures of type VkExportMetalObjectCreateInfoEXT

If the pNext chain includes a VkMemoryDedicatedAllocateInfo structure, then that structure includes a handle of the sole buffer or image resource that the memory **can** be bound to.

The VkMemoryDedicatedAllocateInfo structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkMemoryDedicatedAllocateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkImage            image;
    VkBuffer           buffer;
} VkMemoryDedicatedAllocateInfo;
```

```
// Provided by VK_KHR_dedicated_allocation
// Equivalent to VkMemoryDedicatedAllocateInfo
typedef VkMemoryDedicatedAllocateInfo VkMemoryDedicatedAllocateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- image is VK_NULL_HANDLE or a handle of an image which this memory will be bound to.
- buffer is VK_NULL_HANDLE or a handle of a buffer which this memory will be bound to.

Valid Usage

- VUID-VkMemoryDedicatedAllocateInfo-image-01432
  At least one of image and buffer **must** be VK_NULL_HANDLE
- VUID-VkMemoryDedicatedAllocateInfo-image-02964
  If image is not VK_NULL_HANDLE , VkMemoryAllocateInfo::allocationSize **must** be greater than or equal to the VkMemoryRequirements::size of the image
- VUID-VkMemoryDedicatedAllocateInfo-image-01434
  If image is not VK_NULL_HANDLE, image **must** have been created without VK_IMAGE_CREATE_SPARSE_BINDING_BIT set in VkImageCreateInfo::flags
- VUID-VkMemoryDedicatedAllocateInfo-buffer-02965
  If buffer is not VK_NULL_HANDLE , VkMemoryAllocateInfo::allocationSize **must** be greater than or equal to the VkMemoryRequirements::size of the buffer
- VUID-VkMemoryDedicatedAllocateInfo-buffer-01436
  If buffer is not VK_NULL_HANDLE, buffer **must** have been created without VK_BUFFER_CREATE_SPARSE_BINDING_BIT set in VkBufferCreateInfo::flags
- VUID-VkMemoryDedicatedAllocateInfo-image-01876

If `image` is not VK_NULL_HANDLE and VkMemoryAllocateInfo defines a memory import operation with handle type VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT, VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT, VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT, VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT, VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT, or VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT, and the external handle was created by the Vulkan API, then the memory being imported **must** also be a dedicated image allocation and `image` **must** be identical to the image associated with the imported memory

- VUID-VkMemoryDedicatedAllocateInfo-buffer-01877
  If `buffer` is not VK_NULL_HANDLE and VkMemoryAllocateInfo defines a memory import operation with handle type VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT, VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT, VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT, VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT, VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT, or VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT, and the external handle was created by the Vulkan API, then the memory being imported **must** also be a dedicated buffer allocation and `buffer` **must** be identical to the buffer associated with the imported memory
- VUID-VkMemoryDedicatedAllocateInfo-image-01878
  If `image` is not VK_NULL_HANDLE and VkMemoryAllocateInfo defines a memory import operation with handle type VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT, the memory being imported **must** also be a dedicated image allocation and `image` **must** be identical to the image associated with the imported memory
- VUID-VkMemoryDedicatedAllocateInfo-buffer-01879
  If `buffer` is not VK_NULL_HANDLE and VkMemoryAllocateInfo defines a memory import operation with handle type VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT, the memory being imported **must** also be a dedicated buffer allocation and `buffer` **must** be identical to the buffer associated with the imported memory
- VUID-VkMemoryDedicatedAllocateInfo-image-01797
  If `image` is not VK_NULL_HANDLE, `image` **must** not have been created with VK_IMAGE_CREATE_DISJOINT_BIT set in VkImageCreateInfo::`flags`

Valid Usage (Implicit)

- VUID-VkMemoryDedicatedAllocateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_DEDICATED_ALLOCATE_INFO
- VUID-VkMemoryDedicatedAllocateInfo-image-parameter
  If `image` is not VK_NULL_HANDLE, `image` **must** be a valid VkImage handle
- VUID-VkMemoryDedicatedAllocateInfo-buffer-parameter

If `buffer` is not VK_NULL_HANDLE, `buffer` **must** be a valid VkBuffer handle

- VUID-VkMemoryDedicatedAllocateInfo-commonparent
  Both of `buffer`, and `image` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Structure Chaining

Extends the structure

- VkMemoryAllocateInfo

If the `pNext` chain includes a `VkMemoryPriorityAllocateInfoEXT` structure, then that structure includes a priority for the memory.

The `VkMemoryPriorityAllocateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_memory_priority
typedef struct VkMemoryPriorityAllocateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    float              priority;
} VkMemoryPriorityAllocateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `priority` is a floating-point value between `0` and `1`, indicating the priority of the allocation relative to other memory allocations. Larger values are higher priority. The granularity of the priorities is implementation-dependent.

Memory allocations with higher priority **may** be more likely to stay in device-local memory when the system is under memory pressure.

If this structure is not included, it is as if the `priority` value were `0.5`.

Valid Usage

- VUID-VkMemoryPriorityAllocateInfoEXT-priority-02602
  `priority` **must** be between `0` and `1`, inclusive

Valid Usage (Implicit)

- VUID-VkMemoryPriorityAllocateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_PRIORITY_ALLOCATE_INFO_EXT

Structure Chaining

Extends the structure

- VkMemoryAllocateInfo

To modify the priority of an existing memory allocation, call:

```
// Provided by VK_EXT_pageable_device_local_memory
void vkSetDeviceMemoryPriorityEXT(
    VkDevice                                    device,
    VkDeviceMemory                              memory,
    float                                       priority);
```

- `device` is the logical device that owns the memory.
- `memory` is the VkDeviceMemory object to which the new priority will be applied.
- `priority` is a floating-point value between `0` and `1`, indicating the priority of the allocation relative to other memory allocations. Larger values are higher priority. The granularity of the priorities is implementation-dependent.

Memory allocations with higher priority **may** be more likely to stay in device-local memory when the system is under memory pressure.

Valid Usage

- VUID-vkSetDeviceMemoryPriorityEXT-priority-06258
  `priority` **must** be between `0` and `1`, inclusive

Valid Usage (Implicit)

- VUID-vkSetDeviceMemoryPriorityEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkSetDeviceMemoryPriorityEXT-memory-parameter
  `memory` **must** be a valid VkDeviceMemory handle
- VUID-vkSetDeviceMemoryPriorityEXT-memory-parent
  `memory` **must** have been created, allocated, or retrieved from `device`

When allocating memory whose payload **may** be exported to another process or Vulkan instance, add a VkExportMemoryAllocateInfo structure to the `pNext` chain of the VkMemoryAllocateInfo structure, specifying the handle types that **may** be exported.

The VkExportMemoryAllocateInfo structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkExportMemoryAllocateInfo {
    VkStructureType                    sType;
    const void*                        pNext;
    VkExternalMemoryHandleTypeFlags    handleTypes;
} VkExportMemoryAllocateInfo;
```

```
// Provided by VK_KHR_external_memory
// Equivalent to VkExportMemoryAllocateInfo
typedef VkExportMemoryAllocateInfo VkExportMemoryAllocateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `handleTypes` is zero or a bitmask of VkExternalMemoryHandleTypeFlagBits specifying one or more memory handle types the application **can** export from the resulting allocation. The application **can** request multiple handle types for the same allocation.

Valid Usage

- VUID-VkExportMemoryAllocateInfo-handleTypes-09860
  The bits in `handleTypes` **must** be supported and compatible, as reported by VkExternalImageFormatProperties, or VkExternalBufferProperties

Valid Usage (Implicit)

- VUID-VkExportMemoryAllocateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_EXPORT_MEMORY_ALLOCATE_INFO
- VUID-VkExportMemoryAllocateInfo-handleTypes-parameter
  If `handleTypes` is not `0`, `handleTypes` **must** be a valid combination of VkExternalMemoryHandleTypeFlagBits values

Structure Chaining

Extends the structure

- VkMemoryAllocateInfo

11.2.4. Win32 External Memory

To specify additional attributes of NT handles exported from a memory object, add a VkExportMemoryWin32HandleInfoKHR structure to the `pNext` chain of the VkMemoryAllocateInfo structure. The `VkExportMemoryWin32HandleInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_memory_win32
typedef struct VkExportMemoryWin32HandleInfoKHR {
    VkStructureType               sType;
    const void*                   pNext;
    const SECURITY_ATTRIBUTES*    pAttributes;
    DWORD                         dwAccess;
    LPCWSTR                       name;
} VkExportMemoryWin32HandleInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pAttributes` is a pointer to a Windows `SECURITY_ATTRIBUTES` structure specifying security attributes of the handle.
- `dwAccess` is a `DWORD` specifying access rights of the handle.
- `name` is a null-terminated UTF-16 string to associate with the payload referenced by NT handles exported from the created memory.

If VkExportMemoryAllocateInfo is not included in the same `pNext` chain, this structure is ignored.

If VkExportMemoryAllocateInfo is included in the `pNext` chain of VkMemoryAllocateInfo with a Windows `handleType`, but either `VkExportMemoryWin32HandleInfoKHR` is not included in the `pNext` chain, or it is included but `pAttributes` is set to `NULL`, default security descriptor values will be used, and child processes created by the application will not inherit the handle, as described in the MSDN documentation for "Synchronization Object Security and Access Rights"[1]. Further, if the structure is not present, the access rights used depend on the handle type.

For handles of the following types:

- VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT

The implementation **must** ensure the access rights allow read and write access to the memory.

1

https://docs.microsoft.com/en-us/windows/win32/sync/synchronization-object-security-and-access-rights

Valid Usage

- VUID-VkExportMemoryWin32HandleInfoKHR-handleTypes-00657
  If VkExportMemoryAllocateInfo::`handleTypes` does not include VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT, a `VkExportMemoryWin32HandleInfoKHR` structure **must** not be included in the `pNext` chain of VkMemoryAllocateInfo

Valid Usage (Implicit)

- VUID-VkExportMemoryWin32HandleInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_EXPORT_MEMORY_WIN32_HANDLE_INFO_KHR
- VUID-VkExportMemoryWin32HandleInfoKHR-pAttributes-parameter
  If `pAttributes` is not `NULL`, `pAttributes` **must** be a valid pointer to a valid `SECURITY_ATTRIBUTES` value

Structure Chaining

Extends the structure

- VkMemoryAllocateInfo

To import memory from a Windows handle, add a VkImportMemoryWin32HandleInfoKHR structure to the `pNext` chain of the VkMemoryAllocateInfo structure.

The `VkImportMemoryWin32HandleInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_memory_win32
typedef struct VkImportMemoryWin32HandleInfoKHR {
    VkStructureType                       sType;
    const void*                           pNext;
    VkExternalMemoryHandleTypeFlagBits    handleType;
    HANDLE                                handle;
    LPCWSTR                               name;
} VkImportMemoryWin32HandleInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `handleType` is a VkExternalMemoryHandleTypeFlagBits value specifying the type of `handle` or `name`.
- `handle` is `NULL` or the external handle to import.
- `name` is `NULL` or a null-terminated UTF-16 string naming the payload to import.

Importing memory object payloads from Windows handles does not transfer ownership of the handle to the Vulkan implementation. For handle types defined as NT handles, the application **must** release handle ownership using the `CloseHandle` system call when the handle is no longer needed. For handle types defined as NT handles, the imported memory object holds a reference to its payload.

NOTE	Non-NT handle import operations do not add a reference to their associated payload. If the original object owning the payload is destroyed, all resources and handles sharing that payload will become invalid.

Applications **can** import the same payload into multiple instances of Vulkan, into the same instance from which it was exported, and multiple times into a given Vulkan instance. In all cases, each import operation **must** create a distinct `VkDeviceMemory` object.

Valid Usage

- VUID-VkImportMemoryWin32HandleInfoKHR-handleType-09861
  If `handleType` is not `0`, it **must** be supported for import, as reported by VkExternalImageFormatProperties, or VkExternalBufferProperties
- VUID-VkImportMemoryWin32HandleInfoKHR-handle-00659
  The memory from which `handle` was exported, or the memory named by `name` **must** have been created on the same underlying physical device as `device`
- VUID-VkImportMemoryWin32HandleInfoKHR-handleType-00660
  If `handleType` is not `0`, it **must** be defined as an NT handle or a global share handle
- VUID-VkImportMemoryWin32HandleInfoKHR-handleType-01439
  If `handleType` is not VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT, VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT, VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT, or VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT, `name` **must** be `NULL`
- VUID-VkImportMemoryWin32HandleInfoKHR-handleType-01440
  If `handleType` is not `0` and `handle` is `NULL`, `name` **must** name a valid memory resource of the type specified by `handleType`
- VUID-VkImportMemoryWin32HandleInfoKHR-handleType-00661
  If `handleType` is not `0` and `name` is `NULL`, `handle` **must** be a valid handle of the type specified by `handleType`
- VUID-VkImportMemoryWin32HandleInfoKHR-handle-01441
  If `handle` is not `NULL`, `name` **must** be `NULL`
- VUID-VkImportMemoryWin32HandleInfoKHR-handle-01518
  If `handle` is not `NULL`, it **must** obey any requirements listed for `handleType` in external memory handle types compatibility
- VUID-VkImportMemoryWin32HandleInfoKHR-name-01519
  If `name` is not `NULL`, it **must** obey any requirements listed for `handleType` in external memory handle types compatibility

Valid Usage (Implicit)

- VUID-VkImportMemoryWin32HandleInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMPORT_MEMORY_WIN32_HANDLE_INFO_KHR
- VUID-VkImportMemoryWin32HandleInfoKHR-handleType-parameter
  If `handleType` is not `0`, `handleType` **must** be a valid VkExternalMemoryHandleTypeFlagBits value

Structure Chaining

Extends the structure

- VkMemoryAllocateInfo

To export a Windows handle representing the payload of a Vulkan device memory object, call:

```
// Provided by VK_KHR_external_memory_win32
VkResult vkGetMemoryWin32HandleKHR(
    VkDevice                                    device,
    const VkMemoryGetWin32HandleInfoKHR*        pGetWin32HandleInfo,
    HANDLE*                                     pHandle);
```

- `device` is the logical device that created the device memory being exported.
- `pGetWin32HandleInfo` is a pointer to a VkMemoryGetWin32HandleInfoKHR structure containing parameters of the export operation.
- `pHandle` will return the Windows handle representing the payload of the device memory object.

For handle types defined as NT handles, the handles returned by `vkGetMemoryWin32HandleKHR` are owned by the application and hold a reference to their payload. To avoid leaking resources, the application **must** release ownership of them using the `CloseHandle` system call when they are no longer needed.

NOTE	Non-NT handle types do not add a reference to their associated payload. If the original object owning the payload is destroyed, all resources and handles sharing that payload will become invalid.

Valid Usage (Implicit)

- VUID-vkGetMemoryWin32HandleKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetMemoryWin32HandleKHR-pGetWin32HandleInfo-parameter
  `pGetWin32HandleInfo` **must** be a valid pointer to a valid VkMemoryGetWin32HandleInfoKHR structure
- VUID-vkGetMemoryWin32HandleKHR-pHandle-parameter
  `pHandle` **must** be a valid pointer to a `HANDLE` value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_TOO_MANY_OBJECTS
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkMemoryGetWin32HandleInfoKHR structure is defined as:

```
// Provided by VK_KHR_external_memory_win32
typedef struct VkMemoryGetWin32HandleInfoKHR {
    VkStructureType                       sType;
    const void*                           pNext;
    VkDeviceMemory                        memory;
    VkExternalMemoryHandleTypeFlagBits    handleType;
} VkMemoryGetWin32HandleInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- memory is the memory object from which the handle will be exported.
- handleType is a VkExternalMemoryHandleTypeFlagBits value specifying the type of handle requested.

The properties of the handle returned depend on the value of handleType. See VkExternalMemoryHandleTypeFlagBits for a description of the properties of the defined external memory handle types.

Valid Usage

- VUID-VkMemoryGetWin32HandleInfoKHR-handleType-00662
  handleType **must** have been included in VkExportMemoryAllocateInfo::handleTypes when memory was created
- VUID-VkMemoryGetWin32HandleInfoKHR-handleType-00663
  If handleType is defined as an NT handle, vkGetMemoryWin32HandleKHR **must** be called no more than once for each valid unique combination of memory and handleType
- VUID-VkMemoryGetWin32HandleInfoKHR-handleType-00664
  handleType **must** be defined as an NT handle or a global share handle

Valid Usage (Implicit)

- VUID-VkMemoryGetWin32HandleInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_MEMORY_GET_WIN32_HANDLE_INFO_KHR
- VUID-VkMemoryGetWin32HandleInfoKHR-pNext-pNext

pNext must be NULL

- VUID-VkMemoryGetWin32HandleInfoKHR-memory-parameter
memory must be a valid VkDeviceMemory handle
- VUID-VkMemoryGetWin32HandleInfoKHR-handleType-parameter
handleType must be a valid VkExternalMemoryHandleTypeFlagBits value

Windows memory handles compatible with Vulkan **may** also be created by non-Vulkan APIs using methods beyond the scope of this specification. To determine the correct parameters to use when importing such handles, call:

```
// Provided by VK_KHR_external_memory_win32
VkResult vkGetMemoryWin32HandlePropertiesKHR(
    VkDevice                                    device,
    VkExternalMemoryHandleTypeFlagBits          handleType,
    HANDLE                                      handle,
    VkMemoryWin32HandlePropertiesKHR*           pMemoryWin32HandleProperties);
```

- device is the logical device that will be importing handle.
- handleType is a VkExternalMemoryHandleTypeFlagBits value specifying the type of the handle handle.
- handle is the handle which will be imported.
- pMemoryWin32HandleProperties is a pointer to a VkMemoryWin32HandlePropertiesKHR structure in which properties of handle are returned.

Valid Usage

- VUID-vkGetMemoryWin32HandlePropertiesKHR-handle-00665
handle must point to a valid Windows memory handle
- VUID-vkGetMemoryWin32HandlePropertiesKHR-handleType-00666
handleType must not be one of the handle types defined as opaque

Valid Usage (Implicit)

- VUID-vkGetMemoryWin32HandlePropertiesKHR-device-parameter
device must be a valid VkDevice handle
- VUID-vkGetMemoryWin32HandlePropertiesKHR-handleType-parameter
handleType must be a valid VkExternalMemoryHandleTypeFlagBits value
- VUID-vkGetMemoryWin32HandlePropertiesKHR-pMemoryWin32HandleProperties-parameter
pMemoryWin32HandleProperties must be a valid pointer to a VkMemoryWin32HandlePropertiesKHR structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_EXTERNAL_HANDLE
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkMemoryWin32HandlePropertiesKHR structure returned is defined as:

```
// Provided by VK_KHR_external_memory_win32
typedef struct VkMemoryWin32HandlePropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           memoryTypeBits;
} VkMemoryWin32HandlePropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- memoryTypeBits is a bitmask containing one bit set for every memory type which the specified Windows handle **can** be imported as.

Valid Usage (Implicit)

- VUID-VkMemoryWin32HandlePropertiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_MEMORY_WIN32_HANDLE_PROPERTIES_KHR
- VUID-VkMemoryWin32HandlePropertiesKHR-pNext-pNext
  pNext **must** be NULL

11.2.5. File Descriptor External Memory

To import memory from a POSIX file descriptor handle, add a VkImportMemoryFdInfoKHR structure to the pNext chain of the VkMemoryAllocateInfo structure. The VkImportMemoryFdInfoKHR structure is defined as:

```
// Provided by VK_KHR_external_memory_fd
typedef struct VkImportMemoryFdInfoKHR {
    VkStructureType                       sType;
    const void*                           pNext;
    VkExternalMemoryHandleTypeFlagBits    handleType;
    int                                   fd;
} VkImportMemoryFdInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `handleType` is a VkExternalMemoryHandleTypeFlagBits value specifying the handle type of `fd`.
- `fd` is the external handle to import.

Importing memory from a file descriptor transfers ownership of the file descriptor from the application to the Vulkan implementation. The application **must** not perform any operations on the file descriptor after a successful import. The imported memory object holds a reference to its payload.

Applications **can** import the same payload into multiple instances of Vulkan, into the same instance from which it was exported, and multiple times into a given Vulkan instance. In all cases, each import operation **must** create a distinct `VkDeviceMemory` object.

Valid Usage

- VUID-VkImportMemoryFdInfoKHR-handleType-09862
  If `handleType` is not `0`, it **must** be supported for import, as reported by VkExternalImageFormatProperties or VkExternalBufferProperties
- VUID-VkImportMemoryFdInfoKHR-fd-00668
  The memory from which `fd` was exported **must** have been created on the same underlying physical device as `device`
- VUID-VkImportMemoryFdInfoKHR-handleType-00669
  If `handleType` is not `0`, it **must** be VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT or VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT
- VUID-VkImportMemoryFdInfoKHR-handleType-00670
  If `handleType` is not `0`, `fd` **must** be a valid handle of the type specified by `handleType`
- VUID-VkImportMemoryFdInfoKHR-fd-01746
  The memory represented by `fd` **must** have been created from a physical device and driver that is compatible with `device` and `handleType`, as described in external memory handle types compatibility
- VUID-VkImportMemoryFdInfoKHR-fd-01520
  `fd` **must** obey any requirements listed for `handleType` in external memory handle types compatibility

Valid Usage (Implicit)

- VUID-VkImportMemoryFdInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMPORT_MEMORY_FD_INFO_KHR
- VUID-VkImportMemoryFdInfoKHR-handleType-parameter
  If `handleType` is not `0`, `handleType` **must** be a valid VkExternalMemoryHandleTypeFlagBits value

Structure Chaining

Extends the structure

- VkMemoryAllocateInfo

To export a POSIX file descriptor referencing the payload of a Vulkan device memory object, call:

```
// Provided by VK_KHR_external_memory_fd
VkResult vkGetMemoryFdKHR(
    VkDevice                                    device,
    const VkMemoryGetFdInfoKHR*                 pGetFdInfo,
    int*                                        pFd);
```

- `device` is the logical device that created the device memory being exported.
- `pGetFdInfo` is a pointer to a VkMemoryGetFdInfoKHR structure containing parameters of the export operation.
- `pFd` will return a file descriptor referencing the payload of the device memory object.

Each call to `vkGetMemoryFdKHR` **must** create a new file descriptor holding a reference to the memory object's payload and transfer ownership of the file descriptor to the application. To avoid leaking resources, the application **must** release ownership of the file descriptor using the `close` system call when it is no longer needed, or by importing a Vulkan memory object from it. Where supported by the operating system, the implementation **must** set the file descriptor to be closed automatically when an `execve` system call is made.

Valid Usage (Implicit)

- VUID-vkGetMemoryFdKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetMemoryFdKHR-pGetFdInfo-parameter
  `pGetFdInfo` **must** be a valid pointer to a valid VkMemoryGetFdInfoKHR structure
- VUID-vkGetMemoryFdKHR-pFd-parameter
  `pFd` **must** be a valid pointer to an `int` value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_TOO_MANY_OBJECTS
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkMemoryGetFdInfoKHR structure is defined as:

```
// Provided by VK_KHR_external_memory_fd
typedef struct VkMemoryGetFdInfoKHR {
    VkStructureType                       sType;
    const void*                           pNext;
    VkDeviceMemory                        memory;
    VkExternalMemoryHandleTypeFlagBits    handleType;
} VkMemoryGetFdInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- memory is the memory object from which the handle will be exported.
- handleType is a VkExternalMemoryHandleTypeFlagBits value specifying the type of handle requested.

The properties of the file descriptor exported depend on the value of handleType. See VkExternalMemoryHandleTypeFlagBits for a description of the properties of the defined external memory handle types.

NOTE The size of the exported file **may** be larger than the size requested by VkMemoryAllocateInfo::allocationSize. If handleType is VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT, then the application **can** query the file's actual size with lseek.

Valid Usage

- VUID-VkMemoryGetFdInfoKHR-handleType-00671
  handleType **must** have been included in VkExportMemoryAllocateInfo::handleTypes when memory was created
- VUID-VkMemoryGetFdInfoKHR-handleType-00672
  handleType **must** be VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT or

VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT

- VUID-VkMemoryGetFdInfoKHR-handleType-12524
  If `handleType` is VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT, `memory` **must** not have been allocated with multiple instances

Valid Usage (Implicit)

- VUID-VkMemoryGetFdInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_GET_FD_INFO_KHR
- VUID-VkMemoryGetFdInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkMemoryGetFdInfoKHR-memory-parameter
  `memory` **must** be a valid VkDeviceMemory handle
- VUID-VkMemoryGetFdInfoKHR-handleType-parameter
  `handleType` **must** be a valid VkExternalMemoryHandleTypeFlagBits value

POSIX file descriptor memory handles compatible with Vulkan **may** also be created by non-Vulkan APIs using methods beyond the scope of this specification. To determine the correct parameters to use when importing such handles, call:

```
// Provided by VK_KHR_external_memory_fd
VkResult vkGetMemoryFdPropertiesKHR(
    VkDevice                                    device,
    VkExternalMemoryHandleTypeFlagBits          handleType,
    int                                         fd,
    VkMemoryFdPropertiesKHR*                    pMemoryFdProperties);
```

- `device` is the logical device that will be importing `fd`.
- `handleType` is a VkExternalMemoryHandleTypeFlagBits value specifying the type of the handle `fd`.
- `fd` is the handle which will be imported.
- `pMemoryFdProperties` is a pointer to a VkMemoryFdPropertiesKHR structure in which the properties of the handle `fd` are returned.

Valid Usage

- VUID-vkGetMemoryFdPropertiesKHR-fd-00673
  `fd` **must** point to a valid POSIX file descriptor memory handle
- VUID-vkGetMemoryFdPropertiesKHR-handleType-00674
  `handleType` **must** not be VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT

Valid Usage (Implicit)

- VUID-vkGetMemoryFdPropertiesKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetMemoryFdPropertiesKHR-handleType-parameter
  `handleType` **must** be a valid VkExternalMemoryHandleTypeFlagBits value
- VUID-vkGetMemoryFdPropertiesKHR-pMemoryFdProperties-parameter
  `pMemoryFdProperties` **must** be a valid pointer to a VkMemoryFdPropertiesKHR structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_EXTERNAL_HANDLE
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkMemoryFdPropertiesKHR` structure returned is defined as:

```
// Provided by VK_KHR_external_memory_fd
typedef struct VkMemoryFdPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           memoryTypeBits;
} VkMemoryFdPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryTypeBits` is a bitmask containing one bit set for every memory type which the specified file descriptor **can** be imported as.

Valid Usage (Implicit)

- VUID-VkMemoryFdPropertiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_FD_PROPERTIES_KHR
- VUID-VkMemoryFdPropertiesKHR-pNext-pNext
  `pNext` **must** be `NULL`

11.2.6. Host External Memory

To import memory from a host pointer, add a VkImportMemoryHostPointerInfoEXT structure to the pNext chain of the VkMemoryAllocateInfo structure. The VkImportMemoryHostPointerInfoEXT structure is defined as:

```
// Provided by VK_EXT_external_memory_host
typedef struct VkImportMemoryHostPointerInfoEXT {
    VkStructureType                       sType;
    const void*                           pNext;
    VkExternalMemoryHandleTypeFlagBits    handleType;
    void*                                 pHostPointer;
} VkImportMemoryHostPointerInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- handleType is a VkExternalMemoryHandleTypeFlagBits value specifying the handle type.
- pHostPointer is the host pointer to import from.

Importing memory from a host pointer shares ownership of the memory between the host and the Vulkan implementation. The application **can** continue to access the memory through the host pointer but it is the application's responsibility to synchronize device and non-device access to the payload as defined in Host Access to Device Memory Objects.

Applications **can** import the same payload into multiple instances of Vulkan and multiple times into a given Vulkan instance. However, implementations **may** fail to import the same payload multiple times into a given physical device due to platform constraints.

Importing memory from a particular host pointer **may** not be possible due to additional platform-specific restrictions beyond the scope of this specification in which case the implementation **must** fail the memory import operation with the error code VK_ERROR_INVALID_EXTERNAL_HANDLE_KHR.

Whether device memory objects imported from a host pointer hold a reference to their payload is undefined. As such, the application **must** ensure that the imported memory range remains valid and accessible for the lifetime of the imported memory object.

Implementations **may** support importing host pointers for memory types which are not host-visible. In this case, after a successful call to vkAllocateMemory, the memory range imported from pHostPointer **must** not be accessed by the application until the VkDeviceMemory has been destroyed. Memory contents for the host memory become undefined on import, and are left undefined after the VkDeviceMemory has been destroyed. Applications **must** also not access host memory which is mapped to the same physical memory as pHostPointer, but mapped to a different host pointer while the VkDeviceMemory handle is valid. Implementations running on general-purpose operating systems **should** not support importing host pointers for memory types which are not host-visible.

NOTE

Using host pointers to back non-host visible allocations is a platform-specific use case, and applications should not attempt to do this unless instructed by the platform.

Valid Usage

- VUID-VkImportMemoryHostPointerInfoEXT-handleType-01747
  If `handleType` is not `0`, it **must** be supported for import, as reported in VkExternalMemoryProperties
- VUID-VkImportMemoryHostPointerInfoEXT-handleType-01748
  If `handleType` is not `0`, it **must** be VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT or VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT
- VUID-VkImportMemoryHostPointerInfoEXT-pHostPointer-01749
  `pHostPointer` **must** be a pointer aligned to an integer multiple of `VkPhysicalDeviceExternalMemoryHostPropertiesEXT`::`minImportedHostPointerAlignment`
- VUID-VkImportMemoryHostPointerInfoEXT-handleType-01750
  If `handleType` is VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT, `pHostPointer` **must** be a pointer to `allocationSize` number of bytes of host memory, where `allocationSize` is the member of the `VkMemoryAllocateInfo` structure this structure is chained to
- VUID-VkImportMemoryHostPointerInfoEXT-handleType-01751
  If `handleType` is VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT, `pHostPointer` **must** be a pointer to `allocationSize` number of bytes of host mapped foreign memory, where `allocationSize` is the member of the `VkMemoryAllocateInfo` structure this structure is chained to

Valid Usage (Implicit)

- VUID-VkImportMemoryHostPointerInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMPORT_MEMORY_HOST_POINTER_INFO_EXT
- VUID-VkImportMemoryHostPointerInfoEXT-handleType-parameter
  `handleType` **must** be a valid VkExternalMemoryHandleTypeFlagBits value
- VUID-VkImportMemoryHostPointerInfoEXT-pHostPointer-parameter
  `pHostPointer` **must** be a pointer value

Structure Chaining

Extends the structure

- VkMemoryAllocateInfo

To determine the correct parameters to use when importing host pointers, call:

```
// Provided by VK_EXT_external_memory_host
VkResult vkGetMemoryHostPointerPropertiesEXT(
    VkDevice                                    device,
    VkExternalMemoryHandleTypeFlagBits          handleType,
    const void*                                 pHostPointer,
    VkMemoryHostPointerPropertiesEXT*           pMemoryHostPointerProperties);
```

- `device` is the logical device that will be importing `pHostPointer`.
- `handleType` is a VkExternalMemoryHandleTypeFlagBits value specifying the type of the handle `pHostPointer`.
- `pHostPointer` is the host pointer to import from.
- `pMemoryHostPointerProperties` is a pointer to a VkMemoryHostPointerPropertiesEXT structure in which the host pointer properties are returned.

Valid Usage

- VUID-vkGetMemoryHostPointerPropertiesEXT-handleType-01752
  `handleType` **must** be VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT or VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT
- VUID-vkGetMemoryHostPointerPropertiesEXT-pHostPointer-01753
  `pHostPointer` **must** be a pointer aligned to an integer multiple of `VkPhysicalDeviceExternalMemoryHostPropertiesEXT`::`minImportedHostPointerAlignment`
- VUID-vkGetMemoryHostPointerPropertiesEXT-handleType-01754
  If `handleType` is VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT, `pHostPointer` **must** be a pointer to host memory
- VUID-vkGetMemoryHostPointerPropertiesEXT-handleType-01755
  If `handleType` is VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT, `pHostPointer` **must** be a pointer to host mapped foreign memory

Valid Usage (Implicit)

- VUID-vkGetMemoryHostPointerPropertiesEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetMemoryHostPointerPropertiesEXT-handleType-parameter
  `handleType` **must** be a valid VkExternalMemoryHandleTypeFlagBits value
- VUID-vkGetMemoryHostPointerPropertiesEXT-pHostPointer-parameter
  `pHostPointer` **must** be a pointer value
- VUID-vkGetMemoryHostPointerPropertiesEXT-pMemoryHostPointerProperties-

parameter `pMemoryHostPointerProperties` **must** be a valid pointer to a VkMemoryHostPointerPropertiesEXT structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_EXTERNAL_HANDLE
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkMemoryHostPointerPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_external_memory_host
typedef struct VkMemoryHostPointerPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           memoryTypeBits;
} VkMemoryHostPointerPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryTypeBits` is a bitmask containing one bit set for every memory type which the specified host pointer **can** be imported as.

The value returned by `memoryTypeBits` **should** only include bits that identify memory types which are host visible. Implementations **may** include bits that identify memory types which are not host visible. Behavior for imported pointers of such types is defined by VkImportMemoryHostPointerInfoEXT.

Valid Usage (Implicit)

- VUID-VkMemoryHostPointerPropertiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_HOST_POINTER_PROPERTIES_EXT
- VUID-VkMemoryHostPointerPropertiesEXT-pNext-pNext
  `pNext` **must** be `NULL`

11.2.7. Metal Objects

A Vulkan implementation that is layered on top of Metal on Apple device platform, and implements the VK_EXT_metal_objects extension, supports the ability to import and export the underlying Metal objects associated with specific Vulkan objects.

The underlying Metal objects associated with certain Vulkan objects can be exported from those Vulkan objects using the pNext chain of the VkExportMetalObjectsInfoEXT parameter of the vkExportMetalObjectsEXT command.

A VkDeviceMemory object can be allocated on an existing MTLBuffer object, by including the MTLBuffer object in a VkImportMetalBufferInfoEXT structure in the pNext chain of the VkMemoryAllocateInfo structure in the vkAllocateMemory command.

A new VkImage object can be created on an existing IOSurface object, or one or more existing Metal MTLTexture objects, by including those Metal objects in either VkImportMetalIOSurfaceInfoEXT or VkImportMetalTextureInfoEXT structures in the pNext chain of the VkImageCreateInfo structure in the vkCreateImage command.

To export Metal objects from Vulkan objects, the application **must** first indicate the intention to do so during the creation of the Vulkan object, by including one or more VkExportMetalObjectCreateInfoEXT structures in the pNext chain of the VkInstanceCreateInfo, VkMemoryAllocateInfo, VkImageCreateInfo, VkImageViewCreateInfo, VkBufferViewCreateInfo, VkSemaphoreCreateInfo, or VkEventCreateInfo, in the corresponding Vulkan object creation command.

The VkExportMetalObjectCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_metal_objects
typedef struct VkExportMetalObjectCreateInfoEXT {
    VkStructureType                       sType;
    const void*                           pNext;
    VkExportMetalObjectTypeFlagBitsEXT    exportObjectType;
} VkExportMetalObjectCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- exportObjectType is a VkExportMetalObjectTypeFlagBitsEXT indicating the type of Metal object that the application may request to be exported from the Vulkan object.

Valid Usage (Implicit)

- VUID-VkExportMetalObjectCreateInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_EXPORT_METAL_OBJECT_CREATE_INFO_EXT
- VUID-VkExportMetalObjectCreateInfoEXT-exportObjectType-parameter
  If exportObjectType is not 0, exportObjectType **must** be a valid VkExportMetalObjectTypeFlagBitsEXT value

Structure Chaining

Extends the structures

- VkBufferViewCreateInfo
- VkEventCreateInfo
- VkImageCreateInfo
- VkImageViewCreateInfo
- VkInstanceCreateInfo
- VkMemoryAllocateInfo
- VkSemaphoreCreateInfo

Bits which indicate the types of Metal objects that may be exported from a corresponding Vulkan object are:

```
// Provided by VK_EXT_metal_objects
typedef enum VkExportMetalObjectTypeFlagBitsEXT {
    VK_EXPORT_METAL_OBJECT_TYPE_METAL_DEVICE_BIT_EXT = 0x00000001,
    VK_EXPORT_METAL_OBJECT_TYPE_METAL_COMMAND_QUEUE_BIT_EXT = 0x00000002,
    VK_EXPORT_METAL_OBJECT_TYPE_METAL_BUFFER_BIT_EXT = 0x00000004,
    VK_EXPORT_METAL_OBJECT_TYPE_METAL_TEXTURE_BIT_EXT = 0x00000008,
    VK_EXPORT_METAL_OBJECT_TYPE_METAL_IOSURFACE_BIT_EXT = 0x00000010,
    VK_EXPORT_METAL_OBJECT_TYPE_METAL_SHARED_EVENT_BIT_EXT = 0x00000020,
} VkExportMetalObjectTypeFlagBitsEXT;
```

- VK_EXPORT_METAL_OBJECT_TYPE_METAL_DEVICE_BIT_EXT specifies that a Metal MTLDevice may be exported.
- VK_EXPORT_METAL_OBJECT_TYPE_METAL_COMMAND_QUEUE_BIT_EXT specifies that a Metal MTLCommandQueue may be exported.
- VK_EXPORT_METAL_OBJECT_TYPE_METAL_BUFFER_BIT_EXT specifies that a Metal MTLBuffer may be exported.
- VK_EXPORT_METAL_OBJECT_TYPE_METAL_TEXTURE_BIT_EXT specifies that a Metal MTLTexture may be exported.
- VK_EXPORT_METAL_OBJECT_TYPE_METAL_IOSURFACE_BIT_EXT specifies that a Metal IOSurface may be exported.
- VK_EXPORT_METAL_OBJECT_TYPE_METAL_SHARED_EVENT_BIT_EXT specifies that a Metal MTLSharedEvent may be exported.

```
// Provided by VK_EXT_metal_objects
typedef VkFlags VkExportMetalObjectTypeFlagsEXT;
```

VkExportMetalObjectTypeFlagsEXT is a bitmask type for setting a mask of zero or more

VkExportMetalObjectTypeFlagBitsEXT.

To export Metal objects that underlie Vulkan objects, call:

```
// Provided by VK_EXT_metal_objects
void vkExportMetalObjectsEXT(
    VkDevice                                    device,
    VkExportMetalObjectsInfoEXT*                pMetalObjectsInfo);
```

- `device` is the device that created the Vulkan objects.
- `pMetalObjectsInfo` is a pointer to a VkExportMetalObjectsInfoEXT structure whose `pNext` chain contains structures, each identifying a Vulkan object and providing a pointer through which the Metal object will be returned.

Valid Usage (Implicit)

- VUID-vkExportMetalObjectsEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkExportMetalObjectsEXT-pMetalObjectsInfo-parameter
  `pMetalObjectsInfo` **must** be a valid pointer to a VkExportMetalObjectsInfoEXT structure

The `VkExportMetalObjectsInfoEXT` structure is defined as:

```
// Provided by VK_EXT_metal_objects
typedef struct VkExportMetalObjectsInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
} VkExportMetalObjectsInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

Valid Usage

- VUID-VkExportMetalObjectsInfoEXT-pNext-06791
  If the `pNext` chain includes a VkExportMetalDeviceInfoEXT structure, the VkInstance **must** have been created with VK_EXPORT_METAL_OBJECT_TYPE_METAL_DEVICE_BIT_EXT in the `exportObjectType` member of a VkExportMetalObjectCreateInfoEXT structure in the `pNext` chain of the VkInstanceCreateInfo structure in the vkCreateInstance command
- VUID-VkExportMetalObjectsInfoEXT-pNext-06792
  If the `pNext` chain includes a VkExportMetalCommandQueueInfoEXT structure, the VkInstance **must** have been created with VK_EXPORT_METAL_OBJECT_TYPE_METAL_COMMAND_QUEUE_BIT_EXT in the `exportObjectType` member of a VkExportMetalObjectCreateInfoEXT structure in the `pNext`

chain of the VkInstanceCreateInfo structure in the vkCreateInstance command

- VUID-VkExportMetalObjectsInfoEXT-pNext-06793
  If the pNext chain includes a VkExportMetalBufferInfoEXT structure, the VkDeviceMemory in its memory member **must** have been allocated with VK_EXPORT_METAL_OBJECT_TYPE_METAL_BUFFER_BIT_EXT in the exportObjectType member of a VkExportMetalObjectCreateInfoEXT structure in the pNext chain of the VkMemoryAllocateInfo structure in the vkAllocateMemory command
- VUID-VkExportMetalObjectsInfoEXT-pNext-06794
  If the pNext chain includes a VkExportMetalTextureInfoEXT structure, exactly one of its image, imageView, or bufferView members **must** not be VK_NULL_HANDLE
- VUID-VkExportMetalObjectsInfoEXT-pNext-06795
  If the pNext chain includes a VkExportMetalTextureInfoEXT structure, and its image member is not VK_NULL_HANDLE, the VkImage in its image member **must** have been created with VK_EXPORT_METAL_OBJECT_TYPE_METAL_TEXTURE_BIT_EXT in the exportObjectType member of a VkExportMetalObjectCreateInfoEXT structure in the pNext chain of the VkImageCreateInfo structure in the vkCreateImage command
- VUID-VkExportMetalObjectsInfoEXT-pNext-06796
  If the pNext chain includes a VkExportMetalTextureInfoEXT structure, and its imageView member is not VK_NULL_HANDLE, the VkImageView in its imageView member **must** have been created with VK_EXPORT_METAL_OBJECT_TYPE_METAL_TEXTURE_BIT_EXT in the exportObjectType member of a VkExportMetalObjectCreateInfoEXT structure in the pNext chain of the VkImageViewCreateInfo structure in the vkCreateImageView command
- VUID-VkExportMetalObjectsInfoEXT-pNext-06797
  If the pNext chain includes a VkExportMetalTextureInfoEXT structure, and its bufferView member is not VK_NULL_HANDLE, the VkBufferView in its bufferView member **must** have been created with VK_EXPORT_METAL_OBJECT_TYPE_METAL_TEXTURE_BIT_EXT in the exportObjectType member of a VkExportMetalObjectCreateInfoEXT structure in the pNext chain of the VkBufferViewCreateInfo structure in the vkCreateBufferView command
- VUID-VkExportMetalObjectsInfoEXT-pNext-06798
  If the pNext chain includes a VkExportMetalTextureInfoEXT structure, and if either its image or imageView member is not VK_NULL_HANDLE, then plane **must** be VK_IMAGE_ASPECT_PLANE_0_BIT, VK_IMAGE_ASPECT_PLANE_1_BIT, or VK_IMAGE_ASPECT_PLANE_2_BIT
- VUID-VkExportMetalObjectsInfoEXT-pNext-06799
  If the pNext chain includes a VkExportMetalTextureInfoEXT structure, and if the VkImage in its image member does not have a multi-planar format, then its plane member **must** be VK_IMAGE_ASPECT_PLANE_0_BIT
- VUID-VkExportMetalObjectsInfoEXT-pNext-06800
  If the pNext chain includes a VkExportMetalTextureInfoEXT structure, and if the VkImage in its image member has a multi-planar format with only two planes, then its plane member **must** not be VK_IMAGE_ASPECT_PLANE_2_BIT
- VUID-VkExportMetalObjectsInfoEXT-pNext-06801
  If the pNext chain includes a VkExportMetalTextureInfoEXT structure, and if the VkImageView in its imageView member does not have a multi-planar format, then its plane

member **must** be VK_IMAGE_ASPECT_PLANE_0_BIT

- VUID-VkExportMetalObjectsInfoEXT-pNext-06802
  If the pNext chain includes a VkExportMetalTextureInfoEXT structure, and if the VkImageView in its imageView member has a multi-planar format with only two planes, then its plane member **must** not be VK_IMAGE_ASPECT_PLANE_2_BIT
- VUID-VkExportMetalObjectsInfoEXT-pNext-06803
  If the pNext chain includes a VkExportMetalIOSurfaceInfoEXT structure, the VkImage in its image member **must** have been created with VK_EXPORT_METAL_OBJECT_TYPE_METAL_IOSURFACE_BIT_EXT in the exportObjectType member of a VkExportMetalObjectCreateInfoEXT structure in the pNext chain of the VkImageCreateInfo structure in the vkCreateImage command
- VUID-VkExportMetalObjectsInfoEXT-pNext-06804
  If the pNext chain includes a VkExportMetalSharedEventInfoEXT structure, exactly one of its semaphore or event members **must** not be VK_NULL_HANDLE
- VUID-VkExportMetalObjectsInfoEXT-pNext-06805
  If the pNext chain includes a VkExportMetalSharedEventInfoEXT structure, and its semaphore member is not VK_NULL_HANDLE, the VkSemaphore in its semaphore member **must** have been created with VK_EXPORT_METAL_OBJECT_TYPE_METAL_SHARED_EVENT_BIT_EXT in the exportObjectType member of a VkExportMetalObjectCreateInfoEXT structure in the pNext chain of the VkSemaphoreCreateInfo structure in the vkCreateSemaphore command
- VUID-VkExportMetalObjectsInfoEXT-pNext-06806
  If the pNext chain includes a VkExportMetalSharedEventInfoEXT structure, and its event member is not VK_NULL_HANDLE, the VkEvent in its event member **must** have been created with VK_EXPORT_METAL_OBJECT_TYPE_METAL_SHARED_EVENT_BIT_EXT in the exportObjectType member of a VkExportMetalObjectCreateInfoEXT structure in the pNext chain of the VkEventCreateInfo structure in the vkCreateEvent command

Valid Usage (Implicit)

- VUID-VkExportMetalObjectsInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_EXPORT_METAL_OBJECTS_INFO_EXT
- VUID-VkExportMetalObjectsInfoEXT-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkExportMetalBufferInfoEXT, VkExportMetalCommandQueueInfoEXT, VkExportMetalDeviceInfoEXT, VkExportMetalIOSurfaceInfoEXT, VkExportMetalSharedEventInfoEXT, or VkExportMetalTextureInfoEXT
- VUID-VkExportMetalObjectsInfoEXT-sType-unique
  The sType value of each structure in the pNext chain **must** be unique, with the exception of structures of type VkExportMetalBufferInfoEXT, VkExportMetalCommandQueueInfoEXT, VkExportMetalIOSurfaceInfoEXT, VkExportMetalSharedEventInfoEXT, or VkExportMetalTextureInfoEXT

To export the Metal MTLDevice object underlying the VkPhysicalDevice associated with a VkDevice object, include a VkExportMetalDeviceInfoEXT structure in the pNext chain of the pMetalObjectsInfo parameter of a vkExportMetalObjectsEXT call.

The VkExportMetalDeviceInfoEXT structure is defined as:

```
// Provided by VK_EXT_metal_objects
typedef struct VkExportMetalDeviceInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    MTLDevice_id       mtlDevice;
} VkExportMetalDeviceInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- mtlDevice is the Metal id<MTLDevice> object underlying the VkPhysicalDevice associated with the VkDevice object identified in the call. The implementation will return the MTLDevice in this member, or it will return NULL if no MTLDevice could be found underlying the VkPhysicalDevice object.

Valid Usage (Implicit)

- VUID-VkExportMetalDeviceInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_EXPORT_METAL_DEVICE_INFO_EXT

Structure Chaining

Extends the structure

- VkExportMetalObjectsInfoEXT

The type id<MTLDevice> is defined in Apple's Metal framework, but to remove an unnecessary compile time dependency, an incomplete type definition of MTLDevice_id is provided in the Vulkan headers:

```
// Provided by VK_EXT_metal_objects
#ifdef __OBJC__
@protocol MTLDevice;
typedef __unsafe_unretained id<MTLDevice> MTLDevice_id;
#else
typedef void* MTLDevice_id;
#endif
```

To export the Metal MTLCommandQueue object underlying a VkQueue object, include a VkExportMetalCommandQueueInfoEXT structure in the pNext chain of the pMetalObjectsInfo parameter

of a vkExportMetalObjectsEXT call.

The VkExportMetalCommandQueueInfoEXT structure is defined as:

```
// Provided by VK_EXT_metal_objects
typedef struct VkExportMetalCommandQueueInfoEXT {
    VkStructureType       sType;
    const void*           pNext;
    VkQueue               queue;
    MTLCommandQueue_id    mtlCommandQueue;
} VkExportMetalCommandQueueInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- queue is a VkQueue.
- mtlCommandQueue is the Metal id<MTLCommandQueue> object underlying the VkQueue object in queue. The implementation will return the MTLCommandQueue in this member, or it will return NULL if no MTLCommandQueue could be found underlying the VkQueue object.

Valid Usage (Implicit)

- VUID-VkExportMetalCommandQueueInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_EXPORT_METAL_COMMAND_QUEUE_INFO_EXT
- VUID-VkExportMetalCommandQueueInfoEXT-queue-parameter
  queue **must** be a valid VkQueue handle

Structure Chaining

Extends the structure

- VkExportMetalObjectsInfoEXT

The type id<MTLCommandQueue> is defined in Apple's Metal framework, but to remove an unnecessary compile time dependency, an incomplete type definition of MTLCommandQueue_id is provided in the Vulkan headers:

```
// Provided by VK_EXT_metal_objects
#ifdef __OBJC__
@protocol MTLCommandQueue;
typedef __unsafe_unretained id<MTLCommandQueue> MTLCommandQueue_id;
#else
typedef void* MTLCommandQueue_id;
#endif
```

To export the Metal MTLBuffer object underlying a VkDeviceMemory object, include a VkExportMetalBufferInfoEXT structure in the pNext chain of the pMetalObjectsInfo parameter of a vkExportMetalObjectsEXT call.

The VkExportMetalBufferInfoEXT structure is defined as:

```
// Provided by VK_EXT_metal_objects
typedef struct VkExportMetalBufferInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceMemory     memory;
    MTLBuffer_id       mtlBuffer;
} VkExportMetalBufferInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- memory is a VkDeviceMemory.
- mtlBuffer is the Metal id<MTLBuffer> object underlying the VkDeviceMemory object in memory. The implementation will return the MTLBuffer in this member, or it will return NULL if no MTLBuffer could be found underlying the VkDeviceMemory object.

Valid Usage (Implicit)

- VUID-VkExportMetalBufferInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_EXPORT_METAL_BUFFER_INFO_EXT
- VUID-VkExportMetalBufferInfoEXT-memory-parameter
  memory **must** be a valid VkDeviceMemory handle

Structure Chaining

Extends the structure

- VkExportMetalObjectsInfoEXT

To import a Metal MTLBuffer object to underlie a VkDeviceMemory object, include a VkImportMetalBufferInfoEXT structure in the pNext chain of the VkMemoryAllocateInfo structure in a vkAllocateMemory command.

The VkImportMetalBufferInfoEXT structure is defined as:

```
// Provided by VK_EXT_metal_objects
typedef struct VkImportMetalBufferInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    MTLBuffer_id       mtlBuffer;
} VkImportMetalBufferInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `mtlBuffer` is the Metal `id<MTLBuffer>` object that is to underlie the VkDeviceMemory.

The application **must** ensure that the configuration of the `id<MTLBuffer>` object is compatible with the configuration of the VkDeviceMemory. Failure to do so results in undefined behavior.

Valid Usage (Implicit)

- VUID-VkImportMetalBufferInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMPORT_METAL_BUFFER_INFO_EXT

Structure Chaining

Extends the structure

- VkMemoryAllocateInfo

The type `id<MTLBuffer>` is defined in Apple's Metal framework, but to remove an unnecessary compile time dependency, an incomplete type definition of MTLBuffer_id is provided in the Vulkan headers:

```
// Provided by VK_EXT_metal_objects
#ifdef __OBJC__
@protocol MTLBuffer;
typedef __unsafe_unretained id<MTLBuffer> MTLBuffer_id;
#else
typedef void* MTLBuffer_id;
#endif
```

To export a Metal `MTLTexture` object underlying a VkImage, VkImageView, or VkBufferView object, include a `VkExportMetalTextureInfoEXT` structure in the `pNext` chain of the `pMetalObjectsInfo` parameter of a vkExportMetalObjectsEXT call.

The `VkExportMetalTextureInfoEXT` structure is defined as:

```
// Provided by VK_EXT_metal_objects
typedef struct VkExportMetalTextureInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkImage                  image;
    VkImageView              imageView;
    VkBufferView             bufferView;
    VkImageAspectFlagBits    plane;
    MTLTexture_id            mtlTexture;
} VkExportMetalTextureInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `image` is VK_NULL_HANDLE or a VkImage.
- `imageView` is VK_NULL_HANDLE or a VkImageView.
- `bufferView` is VK_NULL_HANDLE or a VkBufferView.
- `plane` specifies the plane of a multi-planar VkImage or VkImageView.
- `mtlTexture` is the Metal `id<MTLTexture>` object underlying the VkImage, VkImageView, or VkBufferView object in `image`, `imageView`, or `bufferView`, respectively, at the plane indicated in `plane`. The implementation will return the `MTLTexture` in this member, or it will return `NULL` if no `MTLTexture` could be found underlying the VkImage, VkImageView, or VkBufferView object, at the plane indicated in `plane`.

Valid Usage (Implicit)

- VUID-VkExportMetalTextureInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_EXPORT_METAL_TEXTURE_INFO_EXT
- VUID-VkExportMetalTextureInfoEXT-image-parameter
  If `image` is not VK_NULL_HANDLE, `image` **must** be a valid VkImage handle
- VUID-VkExportMetalTextureInfoEXT-imageView-parameter
  If `imageView` is not VK_NULL_HANDLE, `imageView` **must** be a valid VkImageView handle
- VUID-VkExportMetalTextureInfoEXT-bufferView-parameter
  If `bufferView` is not VK_NULL_HANDLE, `bufferView` **must** be a valid VkBufferView handle
- VUID-VkExportMetalTextureInfoEXT-plane-parameter
  `plane` **must** be a valid VkImageAspectFlagBits value
- VUID-VkExportMetalTextureInfoEXT-commonparent
  Each of `bufferView`, `image`, and `imageView` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Structure Chaining

Extends the structure

- VkExportMetalObjectsInfoEXT

To import one or more existing Metal `MTLTexture` objects to underlie a VkImage object, include one or more `VkImportMetalTextureInfoEXT` structures in the `pNext` chain of the VkImageCreateInfo structure in a vkCreateImage command.

The `VkImportMetalTextureInfoEXT` structure is defined as:

```
// Provided by VK_EXT_metal_objects
typedef struct VkImportMetalTextureInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkImageAspectFlagBits    plane;
    MTLTexture_id            mtlTexture;
} VkImportMetalTextureInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `plane` specifies the plane of the VkImage that the `id<MTLTexture>` object should be attached to.
- `mtlTexture` is the Metal `id<MTLTexture>` object that is to underlie the VkImage plane.

The `pNext` chain **must** include one `VkImportMetalTextureInfoEXT` structure for each plane in the VkImage. The application **must** ensure that the configuration of the Metal `id<MTLTexture>` objects is compatible with the configuration of the VkImage. Failure to do so results in undefined behavior.

NOTE Due to `id<MTLTexture>` already being backed by memory, images created with `VkImportMetalTextureInfoEXT` in the `pNext` of the VkImageCreateInfo will be treated as bound to a VkDeviceMemory.

Valid Usage (Implicit)

- VUID-VkImportMetalTextureInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMPORT_METAL_TEXTURE_INFO_EXT
- VUID-VkImportMetalTextureInfoEXT-plane-parameter
  `plane` **must** be a valid VkImageAspectFlagBits value

Structure Chaining

Extends the structure

- VkImageCreateInfo

The type id<MTLTexture> is defined in Apple's Metal framework, but to remove an unnecessary compile time dependency, an incomplete type definition of MTLTexture_id is provided in the Vulkan headers:

```
// Provided by VK_EXT_metal_objects
#ifdef __OBJC__
@protocol MTLTexture;
typedef __unsafe_unretained id<MTLTexture> MTLTexture_id;
#else
typedef void* MTLTexture_id;
#endif
```

To export the Metal IOSurfaceRef object underlying a VkImage object, include a VkExportMetalIOSurfaceInfoEXT structure in the pNext chain of the pMetalObjectsInfo parameter of a vkExportMetalObjectsEXT call.

The VkExportMetalIOSurfaceInfoEXT structure is defined as:

```
// Provided by VK_EXT_metal_objects
typedef struct VkExportMetalIOSurfaceInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkImage            image;
    IOSurfaceRef       ioSurface;
} VkExportMetalIOSurfaceInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- image is a VkImage.
- ioSurface is the Metal IOSurfaceRef object underlying the VkImage object in image. The implementation will return the IOSurfaceRef in this member, or it will return NULL if no IOSurfaceRef could be found underlying the VkImage object.

Valid Usage (Implicit)

- VUID-VkExportMetalIOSurfaceInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_EXPORT_METAL_IO_SURFACE_INFO_EXT
- VUID-VkExportMetalIOSurfaceInfoEXT-image-parameter
  image **must** be a valid VkImage handle

Structure Chaining

Extends the structure

- VkExportMetalObjectsInfoEXT

To import, or create, a Metal IOSurfaceRef object to underlie a VkImage object, include a VkImportMetalIOSurfaceInfoEXT structure in the pNext chain of the VkImageCreateInfo structure in a vkCreateImage command.

The VkImportMetalIOSurfaceInfoEXT structure is defined as:

```
// Provided by VK_EXT_metal_objects
typedef struct VkImportMetalIOSurfaceInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    IOSurfaceRef       ioSurface;
} VkImportMetalIOSurfaceInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- ioSurface is VK_NULL_HANDLE or the Metal IOSurfaceRef object that is to underlie the VkImage.

If ioSurface is not VK_NULL_HANDLE, it will be used to underlie the VkImage. If ioSurface is VK_NULL_HANDLE, the implementation will create a new IOSurface to underlie the VkImage.

If provided, the application **must** ensure that the configuration of the IOSurfaceRef object is compatible with the configuration of the VkImage. Failure to do so results in undefined behavior.

Valid Usage (Implicit)

- VUID-VkImportMetalIOSurfaceInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_IMPORT_METAL_IO_SURFACE_INFO_EXT

Structure Chaining

Extends the structure

- VkImageCreateInfo

The type IOSurfaceRef is defined in Apple's CoreGraphics framework, but to remove an unnecessary compile time dependency, an incomplete type definition of IOSurfaceRef is provided in the Vulkan headers:

```
// Provided by VK_EXT_metal_objects
typedef struct __IOSurface* IOSurfaceRef;
```

To export the Metal MTLSharedEvent object underlying a VkSemaphore or VkEvent object, include a VkExportMetalSharedEventInfoEXT structure in the pNext chain of the pMetalObjectsInfo parameter of a vkExportMetalObjectsEXT call.

The VkExportMetalSharedEventInfoEXT structure is defined as:

```
// Provided by VK_EXT_metal_objects
typedef struct VkExportMetalSharedEventInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkSemaphore          semaphore;
    VkEvent              event;
    MTLSharedEvent_id    mtlSharedEvent;
} VkExportMetalSharedEventInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- semaphore is VK_NULL_HANDLE or a VkSemaphore.
- event is VK_NULL_HANDLE or a VkEvent.
- mtlSharedEvent is the Metal id<MTLSharedEvent> object underlying the VkSemaphore or VkEvent object in semaphore or event, respectively. The implementation will return the MTLSharedEvent in this member, or it will return NULL if no MTLSharedEvent could be found underlying the VkSemaphore or VkEvent object.

Valid Usage (Implicit)

- VUID-VkExportMetalSharedEventInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_EXPORT_METAL_SHARED_EVENT_INFO_EXT
- VUID-VkExportMetalSharedEventInfoEXT-semaphore-parameter
  If semaphore is not VK_NULL_HANDLE, semaphore **must** be a valid VkSemaphore handle
- VUID-VkExportMetalSharedEventInfoEXT-event-parameter
  If event is not VK_NULL_HANDLE, event **must** be a valid VkEvent handle
- VUID-VkExportMetalSharedEventInfoEXT-commonparent
  Both of event, and semaphore that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Structure Chaining

Extends the structure

- VkExportMetalObjectsInfoEXT

To import a Metal id<MTLSharedEvent> object to underlie a VkSemaphore or VkEvent object, include a VkImportMetalSharedEventInfoEXT structure in the pNext chain of the VkSemaphoreCreateInfo or VkEventCreateInfo structure in a vkCreateSemaphore or vkCreateEvent command, respectively.

The VkImportMetalSharedEventInfoEXT structure is defined as:

```
// Provided by VK_EXT_metal_objects
typedef struct VkImportMetalSharedEventInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    MTLSharedEvent_id    mtlSharedEvent;
} VkImportMetalSharedEventInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `mtlSharedEvent` is the Metal `id<MTLSharedEvent>` object that is to underlie the VkSemaphore or VkEvent.

If the `pNext` chain of the VkSemaphoreCreateInfo structure includes both `VkImportMetalSharedEventInfoEXT` and VkSemaphoreTypeCreateInfo, the `signaledValue` property of the imported `id<MTLSharedEvent>` object will be set to VkSemaphoreTypeCreateInfo::`initialValue`.

Valid Usage (Implicit)

- VUID-VkImportMetalSharedEventInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMPORT_METAL_SHARED_EVENT_INFO_EXT

Structure Chaining

Extends the structures

- VkEventCreateInfo
- VkSemaphoreCreateInfo

The type `id<MTLSharedEvent>` is defined in Apple's Metal framework, but to remove an unnecessary compile time dependency, an incomplete type definition of MTLSharedEvent_id is provided in the Vulkan headers:

```
// Provided by VK_EXT_metal_objects
#ifdef __OBJC__
@protocol MTLSharedEvent;
typedef __unsafe_unretained id<MTLSharedEvent> MTLSharedEvent_id;
#else
typedef void* MTLSharedEvent_id;
#endif
```

11.2.8. Metal External Memory

To import memory from a Metal handle, add a VkImportMemoryMetalHandleInfoEXT structure to the `pNext` chain of the VkMemoryAllocateInfo structure.

The VkImportMemoryMetalHandleInfoEXT structure is defined as:

```
// Provided by VK_EXT_external_memory_metal
typedef struct VkImportMemoryMetalHandleInfoEXT {
    VkStructureType                       sType;
    const void*                           pNext;
    VkExternalMemoryHandleTypeFlagBits    handleType;
    void*                                 handle;
} VkImportMemoryMetalHandleInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- handleType is a VkExternalMemoryHandleTypeFlagBits value specifying the type of handle.
- handle is NULL or the external handle to import.

Importing memory object payloads from Metal handles shares the ownership of the handle to the Vulkan implementation.

Applications **can** import the same payload into multiple instances of Vulkan, into the same instance from which it was exported, and multiple times into a given Vulkan instance. In all cases, each import operation **must** create a distinct VkDeviceMemory object.

Valid Usage

- VUID-VkImportMemoryMetalHandleInfoEXT-handleType-10408
  If handleType is not 0, it **must** be supported for import, as reported by VkExternalImageFormatProperties or VkExternalBufferProperties
- VUID-VkImportMemoryMetalHandleInfoEXT-handle-10409
  The memory from which handle was exported **must** have been created on the same underlying physical device as device
- VUID-VkImportMemoryMetalHandleInfoEXT-handleType-10410
  If handleType is not 0, it **must** be VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLBUFFER_BIT_EXT, VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLTEXTURE_BIT_EXT or VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLHEAP_BIT_EXT
- VUID-VkImportMemoryMetalHandleInfoEXT-handleType-10411
  If handleType is not 0, handle **must** be a valid non-NULL handle of the type specified by handleType
- VUID-VkImportMemoryMetalHandleInfoEXT-handle-10412
  handle **must** obey any requirements listed for handleType in external memory handle types compatibility

Valid Usage (Implicit)

- VUID-VkImportMemoryMetalHandleInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMPORT_MEMORY_METAL_HANDLE_INFO_EXT
- VUID-VkImportMemoryMetalHandleInfoEXT-handleType-parameter
  If `handleType` is not `0`, `handleType` **must** be a valid VkExternalMemoryHandleTypeFlagBits value

Structure Chaining

Extends the structure

- VkMemoryAllocateInfo

To export a Metal handle representing the payload of a Vulkan device memory object, call:

```
// Provided by VK_EXT_external_memory_metal
VkResult vkGetMemoryMetalHandleEXT(
    VkDevice                                    device,
    const VkMemoryGetMetalHandleInfoEXT*        pGetMetalHandleInfo,
    void**                                      pHandle);
```

- `device` is the logical device that created the device memory being exported.
- `pGetMetalHandleInfo` is a pointer to a VkMemoryGetMetalHandleInfoEXT structure containing parameters of the export operation.
- `pHandle` will return the Metal handle representing the payload of the device memory object.

Unless the app retains the handle object returned by the call, the lifespan will be the same as the associated `VkDeviceMemory`.

Valid Usage (Implicit)

- VUID-vkGetMemoryMetalHandleEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetMemoryMetalHandleEXT-pGetMetalHandleInfo-parameter
  `pGetMetalHandleInfo` **must** be a valid pointer to a valid VkMemoryGetMetalHandleInfoEXT structure
- VUID-vkGetMemoryMetalHandleEXT-pHandle-parameter
  `pHandle` **must** be a valid pointer to a pointer value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_TOO_MANY_OBJECTS
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkMemoryGetMetalHandleInfoEXT structure is defined as:

```
// Provided by VK_EXT_external_memory_metal
typedef struct VkMemoryGetMetalHandleInfoEXT {
    VkStructureType                       sType;
    const void*                           pNext;
    VkDeviceMemory                        memory;
    VkExternalMemoryHandleTypeFlagBits    handleType;
} VkMemoryGetMetalHandleInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- memory is the memory object from which the handle will be exported.
- handleType is a VkExternalMemoryHandleTypeFlagBits value specifying the type of handle requested.

The properties of the handle returned depend on the value of handleType. See VkExternalMemoryHandleTypeFlagBits for a description of the properties of the defined external memory handle types.

Valid Usage

- VUID-VkMemoryGetMetalHandleInfoEXT-memory-10413
  memory **must** have been created with a valid VkExportMemoryAllocateInfo
- VUID-VkMemoryGetMetalHandleInfoEXT-handleType-10414
  handleType **must** have been included in VkExportMemoryAllocateInfo::handleTypes when memory was created
- VUID-VkMemoryGetMetalHandleInfoEXT-handleType-10415
  handleType **must** be VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLBUFFER_BIT_EXT, VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLTEXTURE_BIT_EXT or VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLHEAP_BIT_EXT

Valid Usage (Implicit)

- VUID-VkMemoryGetMetalHandleInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_GET_METAL_HANDLE_INFO_EXT
- VUID-VkMemoryGetMetalHandleInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkMemoryGetMetalHandleInfoEXT-memory-parameter
  `memory` **must** be a valid VkDeviceMemory handle
- VUID-VkMemoryGetMetalHandleInfoEXT-handleType-parameter
  `handleType` **must** be a valid VkExternalMemoryHandleTypeFlagBits value

Metal memory handles compatible with Vulkan **may** also be created by non-Vulkan APIs using methods beyond the scope of this specification. To determine the correct parameters to use when importing such handles, call:

```
// Provided by VK_EXT_external_memory_metal
VkResult vkGetMemoryMetalHandlePropertiesEXT(
    VkDevice                                    device,
    VkExternalMemoryHandleTypeFlagBits          handleType,
    const void*                                 pHandle,
    VkMemoryMetalHandlePropertiesEXT*           pMemoryMetalHandleProperties);
```

- `device` is the logical device that will be importing `pHandle`.
- `handleType` is a VkExternalMemoryHandleTypeFlagBits value specifying the type of the handle `pHandle`.
- `pHandle` is the handle which will be imported.
- `pMemoryMetalHandleProperties` is a pointer to a VkMemoryMetalHandlePropertiesEXT structure in which properties of `pHandle` are returned.

Valid Usage

- VUID-vkGetMemoryMetalHandlePropertiesEXT-handle-10416
  `pHandle` **must** point to a valid id<MTLBuffer>, id<MTLTexture> or id<MTLDevice>
- VUID-vkGetMemoryMetalHandlePropertiesEXT-handleType-10417
  `handleType` **must** be VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLBUFFER_BIT_EXT, VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLTEXTURE_BIT_EXT or VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLHEAP_BIT_EXT

Valid Usage (Implicit)

- VUID-vkGetMemoryMetalHandlePropertiesEXT-device-parameter
  `device` **must** be a valid VkDevice handle

- VUID-vkGetMemoryMetalHandlePropertiesEXT-handleType-parameter
  `handleType` **must** be a valid VkExternalMemoryHandleTypeFlagBits value
- VUID-vkGetMemoryMetalHandlePropertiesEXT-pHandle-parameter
  `pHandle` **must** be a pointer value
- VUID-vkGetMemoryMetalHandlePropertiesEXT-pMemoryMetalHandleProperties-parameter
  `pMemoryMetalHandleProperties` **must** be a valid pointer to a VkMemoryMetalHandlePropertiesEXT structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_EXTERNAL_HANDLE
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkMemoryMetalHandlePropertiesEXT` structure returned is defined as:

```
// Provided by VK_EXT_external_memory_metal
typedef struct VkMemoryMetalHandlePropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           memoryTypeBits;
} VkMemoryMetalHandlePropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryTypeBits` is a bitmask containing one bit set for every memory type which the specified Metal handle **can** be imported as.

Valid Usage (Implicit)

- VUID-VkMemoryMetalHandlePropertiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_METAL_HANDLE_PROPERTIES_EXT
- VUID-VkMemoryMetalHandlePropertiesEXT-pNext-pNext
  `pNext` **must** be `NULL`

11.2.9. Device Group Memory Allocations

If the pNext chain of VkMemoryAllocateInfo includes a VkMemoryAllocateFlagsInfo structure, then that structure includes flags and a device mask controlling how many instances of the memory will be allocated.

The VkMemoryAllocateFlagsInfo structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkMemoryAllocateFlagsInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkMemoryAllocateFlags    flags;
    uint32_t                 deviceMask;
} VkMemoryAllocateFlagsInfo;
```

```
// Provided by VK_KHR_device_group
// Equivalent to VkMemoryAllocateFlagsInfo
typedef VkMemoryAllocateFlagsInfo VkMemoryAllocateFlagsInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask of VkMemoryAllocateFlagBits controlling the allocation.
- deviceMask is a mask of physical devices in the logical device, indicating that memory **must** be allocated on each device in the mask, if VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT is set in flags.

If VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT is not set, the number of instances allocated depends on whether VK_MEMORY_HEAP_MULTI_INSTANCE_BIT is set in the memory heap. If VK_MEMORY_HEAP_MULTI_INSTANCE_BIT is set, then memory is allocated for every physical device in the logical device (as if deviceMask has bits set for all device indices). If VK_MEMORY_HEAP_MULTI_INSTANCE_BIT is not set, then a single instance of memory is allocated (as if deviceMask is set to one).

On some implementations, allocations from a multi-instance heap **may** consume memory on all physical devices even if the deviceMask excludes some devices. If VkPhysicalDeviceGroupProperties::subsetAllocation is VK_TRUE, then memory is only consumed for the devices in the device mask.

NOTE	In practice, most allocations on a multi-instance heap will be allocated across all physical devices. Unicast allocation support is an optional optimization for a minority of allocations.

Valid Usage

- VUID-VkMemoryAllocateFlagsInfo-deviceMask-00675
  If VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT is set, `deviceMask` **must** be a valid device mask
- VUID-VkMemoryAllocateFlagsInfo-deviceMask-00676
  If VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT is set, `deviceMask` **must** not be zero
- VUID-VkMemoryAllocateFlagsInfo-flags-10760
  If the allocation is performing a memory import operation, then `flags` **must** not contain VK_MEMORY_ALLOCATE_ZERO_INITIALIZE_BIT_EXT
- VUID-VkMemoryAllocateFlagsInfo-flags-10761
  If the allocation uses protected memory, then `flags` **must** not contain VK_MEMORY_ALLOCATE_ZERO_INITIALIZE_BIT_EXT

Valid Usage (Implicit)

- VUID-VkMemoryAllocateFlagsInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_FLAGS_INFO
- VUID-VkMemoryAllocateFlagsInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkMemoryAllocateFlagBits values

Structure Chaining

Extends the structure

- VkMemoryAllocateInfo

Bits which **can** be set in VkMemoryAllocateFlagsInfo::`flags`, controlling device memory allocation, are:

```
// Provided by VK_VERSION_1_1
typedef enum VkMemoryAllocateFlagBits {
    VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT = 0x00000001,
  // Provided by VK_VERSION_1_2
    VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT = 0x00000002,
  // Provided by VK_VERSION_1_2
    VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT = 0x00000004,
  // Provided by VK_EXT_zero_initialize_device_memory
    VK_MEMORY_ALLOCATE_ZERO_INITIALIZE_BIT_EXT = 0x00000008,
  // Provided by VK_KHR_device_group
    VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT_KHR = VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT,
  // Provided by VK_KHR_buffer_device_address
    VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT_KHR = VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT,
  // Provided by VK_KHR_buffer_device_address
    VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR =
VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT,
} VkMemoryAllocateFlagBits;
```

```
// Provided by VK_KHR_device_group
// Equivalent to VkMemoryAllocateFlagBits
typedef VkMemoryAllocateFlagBits VkMemoryAllocateFlagBitsKHR;
```

- VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT specifies that memory will be allocated for the devices in VkMemoryAllocateFlagsInfo::`deviceMask`.
- VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT specifies that the memory **can** be attached to a buffer object created with the VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT usage flag set.
- VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT specifies that the memory's address **can** be saved and reused on a subsequent run (e.g. for trace capture and replay), see VkBufferOpaqueCaptureAddressCreateInfo for more detail. If this bit is set, VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT **must** also be set.
- VK_MEMORY_ALLOCATE_ZERO_INITIALIZE_BIT_EXT specifies that the memory will be zeroed automatically by the implementation before the application is able to access it.

NOTE

Setting VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT is not expected to reduce performance on any known implementation. Applications should feel free to set it on every allocation, to help avoid needing to know in advance whether a device address will be needed for any buffer bound to the memory. Some development and/or debugging tools, such as a capture/replay layer, might have overhead for memory allocated with this bit, as they might need to track the device addresses of those allocations. However, this is not expected to be significant enough to be a reason for applications to avoid setting the bit unconditionally.

```
// Provided by VK_VERSION_1_1
typedef VkFlags VkMemoryAllocateFlags;
```

```
// Provided by VK_KHR_device_group
// Equivalent to VkMemoryAllocateFlags
typedef VkMemoryAllocateFlags VkMemoryAllocateFlagsKHR;
```

VkMemoryAllocateFlags is a bitmask type for setting a mask of zero or more VkMemoryAllocateFlagBits.

11.2.10. Opaque Capture Address Allocation

To request a specific device address for a memory allocation, add a VkMemoryOpaqueCaptureAddressAllocateInfo structure to the pNext chain of the VkMemoryAllocateInfo structure. The VkMemoryOpaqueCaptureAddressAllocateInfo structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkMemoryOpaqueCaptureAddressAllocateInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint64_t           opaqueCaptureAddress;
} VkMemoryOpaqueCaptureAddressAllocateInfo;
```

```
// Provided by VK_KHR_buffer_device_address
// Equivalent to VkMemoryOpaqueCaptureAddressAllocateInfo
typedef VkMemoryOpaqueCaptureAddressAllocateInfo
VkMemoryOpaqueCaptureAddressAllocateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- opaqueCaptureAddress is the opaque capture address requested for the memory allocation.

If opaqueCaptureAddress is zero, no specific address is requested.

If opaqueCaptureAddress is not zero, it **should** be an address retrieved from vkGetDeviceMemoryOpaqueCaptureAddress on an identically created memory allocation on the same implementation.

NOTE

In most cases, it is expected that a non-zero opaqueCaptureAddress is an address retrieved from vkGetDeviceMemoryOpaqueCaptureAddress on an identically created memory allocation. If this is not the case, it is likely that VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS errors will occur.

This is, however, not a strict requirement because trace capture/replay tools may need to adjust memory allocation parameters for imported memory.

If this structure is not present, it is as if `opaqueCaptureAddress` is zero.

Valid Usage (Implicit)

- VUID-VkMemoryOpaqueCaptureAddressAllocateInfo-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_OPAQUE_CAPTURE_ADDRESS_ALLOCATE_INFO`

Structure Chaining

Extends the structure

- VkMemoryAllocateInfo

11.2.11. Freeing Device Memory

To free a memory object, call:

```
// Provided by VK_VERSION_1_0
void vkFreeMemory(
    VkDevice                                    device,
    VkDeviceMemory                              memory,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that owns the memory.
- `memory` is the VkDeviceMemory object to be freed.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Before freeing a memory object, an application **must** ensure the memory object is no longer in use by the device — for example by command buffers in the *pending state*. Memory **can** be freed whilst still bound to resources, but those resources **must** not be used afterwards. Freeing a memory object releases the reference it held, if any, to its payload. If there are still any bound images or buffers, the memory object's payload **may** not be immediately released by the implementation, but **must** be released by the time all bound images and buffers have been destroyed. Once all references to a payload are released, it is returned to the heap from which it was allocated.

How memory objects are bound to Images and Buffers is described in detail in the Resource Memory Association section.

If a memory object is mapped at the time it is freed, it is implicitly unmapped.

NOTE As described below, host writes are not implicitly flushed when the memory object is unmapped, but the implementation **must** guarantee that writes that have not

> been flushed do not affect any other memory.

Valid Usage

- VUID-vkFreeMemory-memory-00677
  All submitted commands that refer to `memory` (via images or buffers) **must** have completed execution

Valid Usage (Implicit)

- VUID-vkFreeMemory-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkFreeMemory-memory-parameter
  If `memory` is not VK_NULL_HANDLE, `memory` **must** be a valid VkDeviceMemory handle
- VUID-vkFreeMemory-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkFreeMemory-memory-parent
  If `memory` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `memory` **must** be externally synchronized

11.2.12. Host Access to Device Memory Objects

Memory objects created with vkAllocateMemory are not directly host accessible.

Memory objects created with the memory property VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT are considered *mappable*. Memory objects **must** be mappable in order to be successfully mapped on the host.

To retrieve a host virtual address pointer to a region of a mappable memory object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkMapMemory(
    VkDevice                                    device,
    VkDeviceMemory                              memory,
    VkDeviceSize                                offset,
    VkDeviceSize                                size,
    VkMemoryMapFlags                            flags,
    void**                                      ppData);
```

- `device` is the logical device that owns the memory.
- `memory` is the VkDeviceMemory object to be mapped.
- `offset` is a zero-based byte offset from the beginning of the memory object.
- `size` is the size of the memory range to map, or VK_WHOLE_SIZE to map from `offset` to the end of the allocation.
- `flags` is a bitmask of VkMemoryMapFlagBits specifying additional parameters of the memory map operation.
- `ppData` is a pointer to a `void*` variable in which a host-accessible pointer to the beginning of the mapped range is returned. The value of the returned pointer minus `offset` **must** be aligned to VkPhysicalDeviceLimits::`minMemoryMapAlignment`.

After a successful call to `vkMapMemory` the memory object `memory` is considered to be currently *host mapped*.

NOTE It is an application error to call `vkMapMemory` on a memory object that is already *host mapped*.

NOTE `vkMapMemory` will fail if the implementation is unable to allocate an appropriately sized contiguous virtual address range, e.g. due to virtual address space fragmentation or platform limits. In such cases, `vkMapMemory` **must** return VK_ERROR_MEMORY_MAP_FAILED. The application **can** improve the likelihood of success by reducing the size of the mapped range and/or removing unneeded mappings using vkUnmapMemory.

`vkMapMemory` does not check whether the device memory is currently in use before returning the host-accessible pointer. The application **must** guarantee that any previously submitted command that writes to this range has completed before the host reads from or writes to that range, and that any previously submitted command that reads from that range has completed before the host writes to that region (see here for details on fulfilling such a guarantee). If the device memory was allocated without the VK_MEMORY_PROPERTY_HOST_COHERENT_BIT set, these guarantees **must** be made for an extended range: the application **must** round down the start of the range to the nearest multiple of VkPhysicalDeviceLimits::`nonCoherentAtomSize`, and round the end of the range up to the nearest multiple of VkPhysicalDeviceLimits::`nonCoherentAtomSize`.

While a range of device memory is host mapped, the application is responsible for synchronizing both device and host access to that memory range.

NOTE It is important for the application developer to become meticulously familiar with all of the mechanisms described in the chapter on Synchronization and Cache Control as they are crucial to maintaining memory access ordering.

Calling `vkMapMemory` is equivalent to calling vkMapMemory2 with an empty `pNext` chain.

Valid Usage

- VUID-vkMapMemory-memory-00678
  `memory` **must** not be currently host mapped
- VUID-vkMapMemory-offset-00679
  `offset` **must** be less than the size of `memory`
- VUID-vkMapMemory-size-00680
  If `size` is not equal to VK_WHOLE_SIZE, `size` **must** be greater than `0`
- VUID-vkMapMemory-size-00681
  If `size` is not equal to VK_WHOLE_SIZE, `size` **must** be less than or equal to the size of the `memory` minus `offset`
- VUID-vkMapMemory-memory-00682
  `memory` **must** have been created with a memory type that reports VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT
- VUID-vkMapMemory-memory-00683
  `memory` **must** not have been allocated with multiple instances
- VUID-vkMapMemory-flags-09568
  VK_MEMORY_MAP_PLACED_BIT_EXT **must** not be set in `flags`

Valid Usage (Implicit)

- VUID-vkMapMemory-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkMapMemory-memory-parameter
  `memory` **must** be a valid VkDeviceMemory handle
- VUID-vkMapMemory-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkMemoryMapFlagBits values
- VUID-vkMapMemory-ppData-parameter
  `ppData` **must** be a valid pointer to a pointer value
- VUID-vkMapMemory-memory-parent
  `memory` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `memory` **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_MEMORY_MAP_FAILED

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Bits which **can** be set in vkMapMemory::`flags` and VkMemoryMapInfo::`flags`, specifying additional properties of a memory map, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkMemoryMapFlagBits {
  // Provided by VK_EXT_map_memory_placed
    VK_MEMORY_MAP_PLACED_BIT_EXT = 0x00000001,
} VkMemoryMapFlagBits;
```

- VK_MEMORY_MAP_PLACED_BIT_EXT requests that the implementation place the memory map at the virtual address specified by the application via VkMemoryMapPlacedInfoEXT ::`pPlacedAddress`, replacing any existing mapping at that address. This flag **must** not be used with vkMapMemory as there is no way to specify the placement address.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkMemoryMapFlags;
```

`VkMemoryMapFlags` is a bitmask type for setting a mask of zero or more VkMemoryMapFlagBits.

To retrieve a host virtual address pointer to a region of a mappable memory object, call:

```
// Provided by VK_VERSION_1_4
VkResult vkMapMemory2(
    VkDevice                                    device,
    const VkMemoryMapInfo*                      pMemoryMapInfo,
    void**                                      ppData);
```

```
// Provided by VK_KHR_map_memory2
// Equivalent to vkMapMemory2
VkResult vkMapMemory2KHR(
    VkDevice                                    device,
    const VkMemoryMapInfo*                      pMemoryMapInfo,
    void**                                      ppData);
```

- `device` is the logical device that owns the memory.
- `pMemoryMapInfo` is a pointer to a VkMemoryMapInfo structure describing parameters of the map.
- `ppData` is a pointer to a `void *` variable in which is returned a host-accessible pointer to the beginning of the mapped range. This pointer minus VkMemoryMapInfo::`offset` **must** be aligned

to at least VkPhysicalDeviceLimits::minMemoryMapAlignment.

This function behaves identically to vkMapMemory except that it gets its parameters via an extensible structure pointer rather than directly as function arguments.

Valid Usage (Implicit)

- VUID-vkMapMemory2-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkMapMemory2-pMemoryMapInfo-parameter
  pMemoryMapInfo **must** be a valid pointer to a valid VkMemoryMapInfo structure
- VUID-vkMapMemory2-ppData-parameter
  ppData **must** be a valid pointer to a pointer value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_MEMORY_MAP_FAILED
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkMemoryMapInfo structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkMemoryMapInfo {
    VkStructureType     sType;
    const void*         pNext;
    VkMemoryMapFlags    flags;
    VkDeviceMemory      memory;
    VkDeviceSize        offset;
    VkDeviceSize        size;
} VkMemoryMapInfo;
```

```
// Provided by VK_KHR_map_memory2
// Equivalent to VkMemoryMapInfo
typedef VkMemoryMapInfo VkMemoryMapInfoKHR;
```

- sType is a VkStructureType value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkMemoryMapFlagBits specifying additional parameters of the memory map operation.
- `memory` is the VkDeviceMemory object to be mapped.
- `offset` is a zero-based byte offset from the beginning of the memory object.
- `size` is the size of the memory range to map, or VK_WHOLE_SIZE to map from `offset` to the end of the allocation.

Valid Usage

- VUID-VkMemoryMapInfo-memory-07958
  `memory` **must** not be currently host mapped
- VUID-VkMemoryMapInfo-offset-07959
  `offset` **must** be less than the size of `memory`
- VUID-VkMemoryMapInfo-size-07960
  If `size` is not equal to VK_WHOLE_SIZE, `size` **must** be greater than `0`
- VUID-VkMemoryMapInfo-size-07961
  If `size` is not equal to VK_WHOLE_SIZE, `size` **must** be less than or equal to the size of the `memory` minus `offset`
- VUID-VkMemoryMapInfo-memory-07962
  `memory` **must** have been created with a memory type that reports VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT
- VUID-VkMemoryMapInfo-memory-07963
  `memory` **must** not have been allocated with multiple instances
- VUID-VkMemoryMapInfo-flags-09569
  If VK_MEMORY_MAP_PLACED_BIT_EXT is set in `flags`, the `memoryMapPlaced` feature **must** be enabled
- VUID-VkMemoryMapInfo-flags-09570
  If VK_MEMORY_MAP_PLACED_BIT_EXT is set in `flags`, the `pNext` chain **must** include a VkMemoryMapPlacedInfoEXT structure and `VkMemoryMapPlacedInfoEXT`::`pPlacedAddress` **must** not be `NULL`
- VUID-VkMemoryMapInfo-flags-09571
  If VK_MEMORY_MAP_PLACED_BIT_EXT is set in `flags` and the `memoryMapRangePlaced` feature is not enabled, `offset` **must** be zero
- VUID-VkMemoryMapInfo-flags-09572
  If VK_MEMORY_MAP_PLACED_BIT_EXT is set in `flags` and the `memoryMapRangePlaced` feature is not enabled, `size` **must** be VK_WHOLE_SIZE or `VkMemoryAllocateInfo`::`allocationSize`
- VUID-VkMemoryMapInfo-flags-09573
  If VK_MEMORY_MAP_PLACED_BIT_EXT is set in `flags` and the `memoryMapRangePlaced` feature is enabled, `offset` **must** be aligned to an integer multiple of `VkPhysicalDeviceMapMemoryPlacedPropertiesEXT`::`minPlacedMemoryMapAlignment`

- VUID-VkMemoryMapInfo-flags-09574
  If VK_MEMORY_MAP_PLACED_BIT_EXT is set in `flags` and `size` is not VK_WHOLE_SIZE, `size` **must** be aligned to an integer multiple of `VkPhysicalDeviceMapMemoryPlacedPropertiesEXT`::`minPlacedMemoryMapAlignment`
- VUID-VkMemoryMapInfo-flags-09651
  If VK_MEMORY_MAP_PLACED_BIT_EXT is set in `flags` and `size` is VK_WHOLE_SIZE, `VkMemoryAllocateInfo`::`allocationSize` **must** be aligned to an integer multiple of `VkPhysicalDeviceMapMemoryPlacedPropertiesEXT`::`minPlacedMemoryMapAlignment`
- VUID-VkMemoryMapInfo-flags-09575
  If VK_MEMORY_MAP_PLACED_BIT_EXT is set in `flags`, the memory object **must** not have been imported from a handle type of VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT or VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT

Valid Usage (Implicit)

- VUID-VkMemoryMapInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_MAP_INFO
- VUID-VkMemoryMapInfo-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkMemoryMapPlacedInfoEXT
- VUID-VkMemoryMapInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkMemoryMapInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkMemoryMapFlagBits values
- VUID-VkMemoryMapInfo-memory-parameter
  `memory` **must** be a valid VkDeviceMemory handle

Host Synchronization

- Host access to `memory` **must** be externally synchronized

If VK_MEMORY_MAP_PLACED_BIT_EXT is set in `VkMemoryMapInfo`::`flags` and the `pNext` chain of VkMemoryMapInfo includes a `VkMemoryMapPlacedInfoEXT` structure, then that structure specifies the placement address of the memory map. The implementation will place the memory map at the specified address, replacing any existing maps in the specified memory range. Replacing memory maps in this way does not implicitly unmap Vulkan memory objects. Instead, the application **must** ensure no other Vulkan memory objects are mapped anywhere in the specified virtual address range. If successful, `ppData` will be set to the same value as `VkMemoryMapPlacedInfoEXT`::`pPlacedAddress` and `vkMapMemory2` will return VK_SUCCESS. If it cannot place the map at the requested address for any reason, the memory object is left unmapped and `vkMapMemory2` will return VK_ERROR_MEMORY_MAP_FAILED.

The `VkMemoryMapPlacedInfoEXT` structure is defined as:

```
// Provided by VK_EXT_map_memory_placed
typedef struct VkMemoryMapPlacedInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    void*              pPlacedAddress;
} VkMemoryMapPlacedInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pPlacedAddress` is the virtual address at which to place the address. If `VkMemoryMapInfo`::`flags` does not contain VK_MEMORY_MAP_PLACED_BIT_EXT, this value is ignored.

Valid Usage

- VUID-VkMemoryMapPlacedInfoEXT-pPlacedAddress-09577
  `pPlacedAddress` **must** be aligned to an integer multiple of `VkPhysicalDeviceMapMemoryPlacedPropertiesEXT`::`minPlacedMemoryMapAlignment`
- VUID-VkMemoryMapPlacedInfoEXT-pPlacedAddress-09578
  The address range specified by `pPlacedAddress` and `VkMemoryMapInfo`::`size` **must** not overlap any existing Vulkan memory object mapping

Valid Usage (Implicit)

- VUID-VkMemoryMapPlacedInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_MAP_PLACED_INFO_EXT

Structure Chaining

Extends the structure

- VkMemoryMapInfo

Two commands are provided to enable applications to work with non-coherent memory allocations: `vkFlushMappedMemoryRanges` and `vkInvalidateMappedMemoryRanges`.

NOTE	If the memory object was created with the VK_MEMORY_PROPERTY_HOST_COHERENT_BIT set, `vkFlushMappedMemoryRanges` and `vkInvalidateMappedMemoryRanges` are unnecessary and **may** have a performance cost. However, availability and visibility operations still need to be managed on the device. See the description of host access types for more information.
NOTE	While memory objects imported from a handle type of VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT or

> VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT are inherently mapped to host address space, they are not considered to be host mapped device memory unless they are explicitly host mapped using vkMapMemory. That means flushing or invalidating host caches with respect to host accesses performed on such memory through the original host pointer specified at import time is the responsibility of the application and **must** be performed with appropriate synchronization primitives provided by the platform which are outside the scope of Vulkan. vkFlushMappedMemoryRanges and vkInvalidateMappedMemoryRanges, however, **can** still be used on such memory objects to synchronize host accesses performed through the host pointer of the host mapped device memory range returned by vkMapMemory.

After a successful call to vkMapMemory or vkMapMemory2 the memory object memory is considered to be currently *host mapped*.

To flush ranges of non-coherent memory from the host caches, call:

```
// Provided by VK_VERSION_1_0
VkResult vkFlushMappedMemoryRanges(
    VkDevice                                    device,
    uint32_t                                    memoryRangeCount,
    const VkMappedMemoryRange*                  pMemoryRanges);
```

- device is the logical device that owns the memory ranges.
- memoryRangeCount is the length of the pMemoryRanges array.
- pMemoryRanges is a pointer to an array of VkMappedMemoryRange structures describing the memory ranges to flush.

vkFlushMappedMemoryRanges guarantees that host writes to the memory ranges described by pMemoryRanges are made available to the host memory domain, such that they **can** be made available to the device memory domain via memory domain operations using the VK_ACCESS_HOST_WRITE_BIT access type.

The first synchronization scope includes all host operations that happened-before it, as defined by the host memory model.

NOTE

> Some systems allow writes that do not directly integrate with the host memory model; these have to be synchronized by the application manually. One example of this is non-temporal store instructions on x86; to ensure these happen-before submission, applications should call _mm_sfence().

The second synchronization scope is empty.

The first access scope includes host writes to the specified memory ranges.

NOTE

> When a host write to a memory location is made available in this way, each whole aligned set of nonCoherentAtomSize bytes that the memory location exists in will also

be made available as if they were written by the host. For example, with a `nonCoherentAtomSize` of 128, if an application writes to the first byte of a memory object via a host mapping, the first 128 bytes of the memory object will be made available by this command. While the value of the following 127 bytes will be unchanged, this does count as an access for the purpose of synchronization, so care must be taken to avoid data races.

The second access scope is empty.

Unmapping non-coherent memory does not implicitly flush the host mapped memory, and host writes that have not been flushed **may** not ever be visible to the device. However, implementations **must** ensure that writes that have not been flushed do not become visible to any other memory.

NOTE The above guarantee avoids a potential memory corruption in scenarios where host writes to a mapped memory object have not been flushed before the memory is unmapped (or freed), and the virtual address range is subsequently reused for a different mapping (or memory allocation).

Valid Usage (Implicit)

- VUID-vkFlushMappedMemoryRanges-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkFlushMappedMemoryRanges-pMemoryRanges-parameter
  `pMemoryRanges` **must** be a valid pointer to an array of `memoryRangeCount` valid VkMappedMemoryRange structures
- VUID-vkFlushMappedMemoryRanges-memoryRangeCount-arraylength
  `memoryRangeCount` **must** be greater than `0`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To invalidate ranges of non-coherent memory from the host caches, call:

```
// Provided by VK_VERSION_1_0
VkResult vkInvalidateMappedMemoryRanges(
    VkDevice                                    device,
    uint32_t                                    memoryRangeCount,
    const VkMappedMemoryRange*                  pMemoryRanges);
```

- `device` is the logical device that owns the memory ranges.
- `memoryRangeCount` is the length of the `pMemoryRanges` array.
- `pMemoryRanges` is a pointer to an array of VkMappedMemoryRange structures describing the memory ranges to invalidate.

`vkInvalidateMappedMemoryRanges` guarantees that device writes to the memory ranges described by `pMemoryRanges`, which have been made available to the host memory domain using the VK_ACCESS_HOST_WRITE_BIT and VK_ACCESS_HOST_READ_BIT access types, are made visible to the host. If a range of non-coherent memory is written by the host and then invalidated without first being flushed, its contents are undefined.

The first synchronization scope includes all host operations that happened-before it, as defined by the host memory model.

NOTE

This function does not synchronize with device operations directly - other host synchronization operations that depend on device operations such as vkWaitForFences must be executed beforehand. So for any non-coherent device write to be made visible to the host, there has to be a dependency chain along the following lines:

1. Device write
2. Device memory barrier including host reads in its second scope
3. Signal on the device (e.g. a fence signal operation)
4. Wait on the host (e.g. vkWaitForFences)
5. vkInvalidateMappedMemoryRanges

The second synchronization scope includes all host operations that happen-after it, as defined by the host memory model.

The first access scope is empty.

The second access scope includes host reads to the specified memory ranges.

NOTE

When a device write to a memory location is made visible to the host in this way, each whole aligned set of `nonCoherentAtomSize` bytes that the memory location exists in will also be made visible as if they were written by the device. For example, with a `nonCoherentAtomSize` of 128, if an application writes to the first byte of a memory object on the device, the first 128 bytes of the memory object will be made visible by this command. While the value of the following 127 bytes will be unchanged, this does count as an access for the purpose of synchronization, so care must be taken to

avoid data races.

NOTE Mapping non-coherent memory does not implicitly invalidate that memory.

Valid Usage (Implicit)

- VUID-vkInvalidateMappedMemoryRanges-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkInvalidateMappedMemoryRanges-pMemoryRanges-parameter
  `pMemoryRanges` **must** be a valid pointer to an array of `memoryRangeCount` valid VkMappedMemoryRange structures
- VUID-vkInvalidateMappedMemoryRanges-memoryRangeCount-arraylength
  `memoryRangeCount` **must** be greater than `0`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkMappedMemoryRange` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMappedMemoryRange {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceMemory     memory;
    VkDeviceSize       offset;
    VkDeviceSize       size;
} VkMappedMemoryRange;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memory` is the memory object to which this range belongs.
- `offset` is the zero-based byte offset from the beginning of the memory object.
- `size` is either the size of the range, or VK_WHOLE_SIZE to affect the range from `offset` to the end of the current mapping of the allocation.

Valid Usage

- VUID-VkMappedMemoryRange-memory-00684
  `memory` **must** be currently host mapped
- VUID-VkMappedMemoryRange-size-00685
  If `size` is not equal to VK_WHOLE_SIZE, `offset` and `size` **must** specify a range contained within the currently mapped range of `memory`
- VUID-VkMappedMemoryRange-size-00686
  If `size` is equal to VK_WHOLE_SIZE, `offset` **must** be within the currently mapped range of `memory`
- VUID-VkMappedMemoryRange-offset-00687
  `offset` **must** be a multiple of VkPhysicalDeviceLimits::`nonCoherentAtomSize`
- VUID-VkMappedMemoryRange-size-01389
  If `size` is equal to VK_WHOLE_SIZE, the end of the current mapping of `memory` **must** either be a multiple of VkPhysicalDeviceLimits::`nonCoherentAtomSize` bytes from the beginning of the memory object, or be equal to the end of the memory object
- VUID-VkMappedMemoryRange-size-01390
  If `size` is not equal to VK_WHOLE_SIZE, `size` **must** either be a multiple of VkPhysicalDeviceLimits::`nonCoherentAtomSize`, or `offset` plus `size` **must** equal the size of `memory`

Valid Usage (Implicit)

- VUID-VkMappedMemoryRange-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MAPPED_MEMORY_RANGE
- VUID-VkMappedMemoryRange-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkMappedMemoryRange-memory-parameter
  `memory` **must** be a valid VkDeviceMemory handle

To unmap a memory object once host access to it is no longer needed by the application, call:

```
// Provided by VK_VERSION_1_0
void vkUnmapMemory(
    VkDevice                                    device,
    VkDeviceMemory                              memory);
```

- `device` is the logical device that owns the memory.
- `memory` is the memory object to be unmapped.

Calling `vkUnmapMemory` is equivalent to calling vkUnmapMemory2 with an empty `pNext` chain and `flags` set to zero.

Valid Usage

- VUID-vkUnmapMemory-memory-00689
  `memory` **must** be currently host mapped

Valid Usage (Implicit)

- VUID-vkUnmapMemory-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkUnmapMemory-memory-parameter
  `memory` **must** be a valid VkDeviceMemory handle
- VUID-vkUnmapMemory-memory-parent
  `memory` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `memory` **must** be externally synchronized

To unmap a memory object once host access to it is no longer needed by the application, call:

```
// Provided by VK_VERSION_1_4
VkResult vkUnmapMemory2(
    VkDevice                                    device,
    const VkMemoryUnmapInfo*                    pMemoryUnmapInfo);
```

```
// Provided by VK_KHR_map_memory2
// Equivalent to vkUnmapMemory2
VkResult vkUnmapMemory2KHR(
    VkDevice                                    device,
    const VkMemoryUnmapInfo*                    pMemoryUnmapInfo);
```

- `device` is the logical device that owns the memory.
- `pMemoryUnmapInfo` is a pointer to a VkMemoryUnmapInfo structure describing parameters of the unmap.

This function behaves identically to vkUnmapMemory except that it gets its parameters via an extensible structure pointer rather than directly as function arguments.

Valid Usage (Implicit)

- VUID-vkUnmapMemory2-device-parameter
  `device` **must** be a valid VkDevice handle

- VUID-vkUnmapMemory2-pMemoryUnmapInfo-parameter
  `pMemoryUnmapInfo` **must** be a valid pointer to a valid VkMemoryUnmapInfo structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_MEMORY_MAP_FAILED
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkMemoryUnmapInfo` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkMemoryUnmapInfo {
    VkStructureType       sType;
    const void*           pNext;
    VkMemoryUnmapFlags    flags;
    VkDeviceMemory        memory;
} VkMemoryUnmapInfo;
```

```
// Provided by VK_KHR_map_memory2
// Equivalent to VkMemoryUnmapInfo
typedef VkMemoryUnmapInfo VkMemoryUnmapInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkMemoryUnmapFlagBits specifying additional parameters of the memory unmap operation.
- `memory` is the VkDeviceMemory object to be unmapped.

Valid Usage

- VUID-VkMemoryUnmapInfo-memory-07964
  `memory` **must** be currently host mapped
- VUID-VkMemoryUnmapInfo-flags-09579
  If VK_MEMORY_UNMAP_RESERVE_BIT_EXT is set in `flags`, the `memoryUnmapReserve` feature **must** be enabled
- VUID-VkMemoryUnmapInfo-flags-09580
  If VK_MEMORY_UNMAP_RESERVE_BIT_EXT is set in `flags`, the memory object **must** not

have been imported from a handle type of VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT or VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT

Valid Usage (Implicit)

- VUID-VkMemoryUnmapInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_UNMAP_INFO
- VUID-VkMemoryUnmapInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkMemoryUnmapInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkMemoryUnmapFlagBits values
- VUID-VkMemoryUnmapInfo-memory-parameter
  `memory` **must** be a valid VkDeviceMemory handle

Host Synchronization

- Host access to `memory` **must** be externally synchronized

Bits which **can** be set in VkMemoryUnmapInfo::`flags`, specifying additional properties of a memory unmap, are:

```
// Provided by VK_VERSION_1_4
typedef enum VkMemoryUnmapFlagBits {
  // Provided by VK_EXT_map_memory_placed
    VK_MEMORY_UNMAP_RESERVE_BIT_EXT = 0x00000001,
} VkMemoryUnmapFlagBits;
```

```
// Provided by VK_KHR_map_memory2
// Equivalent to VkMemoryUnmapFlagBits
typedef VkMemoryUnmapFlagBits VkMemoryUnmapFlagBitsKHR;
```

- VK_MEMORY_UNMAP_RESERVE_BIT_EXT requests that the virtual address range currently occupied by the memory map remain reserved after the vkUnmapMemory2 call completes. Future system memory map operations or calls to vkMapMemory or vkMapMemory2 will not return addresses in that range unless the range has since been unreserved by the client or the mapping is explicitly placed in that range by calling vkMapMemory2 with VK_MEMORY_MAP_PLACED_BIT_EXT, or doing the system memory map equivalent. When VK_MEMORY_UNMAP_RESERVE_BIT_EXT is set, the memory unmap operation **may** fail, in which case the memory object will remain host mapped and vkUnmapMemory2 will return VK_ERROR_MEMORY_MAP_FAILED.

```
// Provided by VK_VERSION_1_4
typedef VkFlags VkMemoryUnmapFlags;
```

```
// Provided by VK_KHR_map_memory2
// Equivalent to VkMemoryUnmapFlags
typedef VkMemoryUnmapFlags VkMemoryUnmapFlagsKHR;
```

VkMemoryUnmapFlags is a bitmask type for setting a mask of zero or more VkMemoryUnmapFlagBits.

11.2.13. Lazily Allocated Memory

If the memory object is allocated from a heap with the VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT bit set, that object's backing memory **may** be provided by the implementation lazily. The actual committed size of the memory **may** initially be as small as zero (or as large as the requested size), and monotonically increases as additional memory is needed.

A memory type with this flag set is only allowed to be bound to a VkImage whose usage flags include VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT.

NOTE	Using lazily allocated memory objects for framebuffer attachments that are not needed once a render pass instance has completed **may** allow some implementations to never allocate memory for such attachments.

To determine the amount of lazily-allocated memory that is currently committed for a memory object, call:

```
// Provided by VK_VERSION_1_0
void vkGetDeviceMemoryCommitment(
    VkDevice                                    device,
    VkDeviceMemory                              memory,
    VkDeviceSize*                               pCommittedMemoryInBytes);
```

- device is the logical device that owns the memory.
- memory is the memory object being queried.
- pCommittedMemoryInBytes is a pointer to a VkDeviceSize value in which the number of bytes currently committed is returned, on success.

The implementation **may** update the commitment at any time, and the value returned by this query **may** be out of date.

The implementation guarantees to allocate any committed memory from the heapIndex indicated by the memory type that the memory object was created with.

Valid Usage

- VUID-vkGetDeviceMemoryCommitment-memory-00690
  `memory` **must** have been created with a memory type that reports VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT

Valid Usage (Implicit)

- VUID-vkGetDeviceMemoryCommitment-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDeviceMemoryCommitment-memory-parameter
  `memory` **must** be a valid VkDeviceMemory handle
- VUID-vkGetDeviceMemoryCommitment-pCommittedMemoryInBytes-parameter
  `pCommittedMemoryInBytes` **must** be a valid pointer to a VkDeviceSize value
- VUID-vkGetDeviceMemoryCommitment-memory-parent
  `memory` **must** have been created, allocated, or retrieved from `device`

11.2.14. Protected Memory

Protected memory divides device memory into protected device memory and unprotected device memory.

Protected memory adds the following concepts:

- Memory:
  - Unprotected device memory, which **can** be visible to the device and **can** be visible to the host
  - Protected device memory, which **can** be visible to the device but **must** not be visible to the host
- Resources:
  - Unprotected images and unprotected buffers, to which unprotected memory **can** be bound
  - Protected images and protected buffers, to which protected memory **can** be bound
- Command buffers:
  - Unprotected command buffers, which **can** be submitted to a device queue to execute unprotected queue operations
  - Protected command buffers, which **can** be submitted to a protected-capable device queue to execute protected queue operations
- Device queues:
  - Unprotected device queues, to which unprotected command buffers **can** be submitted
  - Protected-capable device queues, to which unprotected command buffers or protected

command buffers **can** be submitted

- Queue submissions
  - Unprotected queue submissions, through which unprotected command buffers **can** be submitted
  - Protected queue submissions, through which protected command buffers **can** be submitted
- Queue operations
  - Unprotected queue operations
  - Protected queue operations

NOTE

When the `protectedMemory` feature is enabled, all pipelines **may** be recorded in either protected or unprotected command buffers (or both), which may incur an extra cost on some implementations. This **can** be mitigated by enabling the `pipelineProtectedAccess` feature, in which case pipelines created with VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT may only be recorded in protected command buffers, and pipelines created with VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT may only be recorded in unprotected command buffers.

Protected Memory Access Rules

If VkPhysicalDeviceProtectedMemoryProperties::`protectedNoFault` is VK_FALSE, applications **must** not perform any of the following operations:

- Write to unprotected memory within protected queue operations.
- Access protected memory within protected queue operations other than in framebuffer-space pipeline stages, the compute shader stage, in video decode operations, in video encode operations, or the transfer stage.
- Perform a query within protected queue operations.

If VkPhysicalDeviceProtectedMemoryProperties::`protectedNoFault` is VK_TRUE, these operations are valid, but reads will return undefined values, and writes will either be dropped or store undefined values.

Additionally, indirect operations **must** not be performed within protected queue operations.

Whether these operations are valid or not, or if any other invalid usage is performed, the implementation **must** guarantee that:

- Protected device memory **must** never be visible to the host.
- Values written to unprotected device memory **must** not be a function of values from protected memory.

11.2.15. Peer Memory Features

Peer memory is memory that is allocated for a given physical device and then bound to a resource and accessed by a different physical device, in a logical device that represents multiple physical

devices. Some ways of reading and writing peer memory **may** not be supported by a device.

To determine how peer memory **can** be accessed, call:

```
// Provided by VK_VERSION_1_1
void vkGetDeviceGroupPeerMemoryFeatures(
    VkDevice                                    device,
    uint32_t                                    heapIndex,
    uint32_t                                    localDeviceIndex,
    uint32_t                                    remoteDeviceIndex,
    VkPeerMemoryFeatureFlags*                   pPeerMemoryFeatures);
```

```
// Provided by VK_KHR_device_group
// Equivalent to vkGetDeviceGroupPeerMemoryFeatures
void vkGetDeviceGroupPeerMemoryFeaturesKHR(
    VkDevice                                    device,
    uint32_t                                    heapIndex,
    uint32_t                                    localDeviceIndex,
    uint32_t                                    remoteDeviceIndex,
    VkPeerMemoryFeatureFlags*                   pPeerMemoryFeatures);
```

- `device` is the logical device that owns the memory.
- `heapIndex` is the index of the memory heap from which the memory is allocated.
- `localDeviceIndex` is the device index of the physical device that performs the memory access.
- `remoteDeviceIndex` is the device index of the physical device that the memory is allocated for.
- `pPeerMemoryFeatures` is a pointer to a VkPeerMemoryFeatureFlags bitmask indicating which types of memory accesses are supported for the combination of heap, local, and remote devices.

Valid Usage

- VUID-vkGetDeviceGroupPeerMemoryFeatures-heapIndex-00691
  `heapIndex` **must** be less than `memoryHeapCount`
- VUID-vkGetDeviceGroupPeerMemoryFeatures-localDeviceIndex-00692
  `localDeviceIndex` **must** be a valid device index
- VUID-vkGetDeviceGroupPeerMemoryFeatures-remoteDeviceIndex-00693
  `remoteDeviceIndex` **must** be a valid device index
- VUID-vkGetDeviceGroupPeerMemoryFeatures-localDeviceIndex-00694
  `localDeviceIndex` **must** not equal `remoteDeviceIndex`

Valid Usage (Implicit)

- VUID-vkGetDeviceGroupPeerMemoryFeatures-device-parameter
  `device` **must** be a valid VkDevice handle

- VUID-vkGetDeviceGroupPeerMemoryFeatures-pPeerMemoryFeatures-parameter `pPeerMemoryFeatures` **must** be a valid pointer to a VkPeerMemoryFeatureFlags value

Bits which **may** be set in vkGetDeviceGroupPeerMemoryFeatures::`pPeerMemoryFeatures`, indicating supported peer memory features, are:

```
// Provided by VK_VERSION_1_1
typedef enum VkPeerMemoryFeatureFlagBits {
    VK_PEER_MEMORY_FEATURE_COPY_SRC_BIT = 0x00000001,
    VK_PEER_MEMORY_FEATURE_COPY_DST_BIT = 0x00000002,
    VK_PEER_MEMORY_FEATURE_GENERIC_SRC_BIT = 0x00000004,
    VK_PEER_MEMORY_FEATURE_GENERIC_DST_BIT = 0x00000008,
  // Provided by VK_KHR_device_group
    VK_PEER_MEMORY_FEATURE_COPY_SRC_BIT_KHR = VK_PEER_MEMORY_FEATURE_COPY_SRC_BIT,
  // Provided by VK_KHR_device_group
    VK_PEER_MEMORY_FEATURE_COPY_DST_BIT_KHR = VK_PEER_MEMORY_FEATURE_COPY_DST_BIT,
  // Provided by VK_KHR_device_group
    VK_PEER_MEMORY_FEATURE_GENERIC_SRC_BIT_KHR =
VK_PEER_MEMORY_FEATURE_GENERIC_SRC_BIT,
  // Provided by VK_KHR_device_group
    VK_PEER_MEMORY_FEATURE_GENERIC_DST_BIT_KHR =
VK_PEER_MEMORY_FEATURE_GENERIC_DST_BIT,
} VkPeerMemoryFeatureFlagBits;
```

```
// Provided by VK_KHR_device_group
// Equivalent to VkPeerMemoryFeatureFlagBits
typedef VkPeerMemoryFeatureFlagBits VkPeerMemoryFeatureFlagBitsKHR;
```

- VK_PEER_MEMORY_FEATURE_COPY_SRC_BIT specifies that the memory **can** be accessed as the source of any `vkCmdCopy*` command.
- VK_PEER_MEMORY_FEATURE_COPY_DST_BIT specifies that the memory **can** be accessed as the destination of any `vkCmdCopy*` command.
- VK_PEER_MEMORY_FEATURE_GENERIC_SRC_BIT specifies that the memory **can** be read as any memory access type.
- VK_PEER_MEMORY_FEATURE_GENERIC_DST_BIT specifies that the memory **can** be written as any memory access type. Shader atomics are considered to be writes.

NOTE	The peer memory features of a memory heap also apply to any accesses that **may** be performed during image layout transitions.

VK_PEER_MEMORY_FEATURE_COPY_DST_BIT **must** be supported for all host local heaps and for at least one device-local memory heap.

If a device does not support a peer memory feature, it is still valid to use a resource that includes both local and peer memory bindings with the corresponding access type as long as only the local

bindings are actually accessed. For example, an application doing split-frame rendering would use framebuffer attachments that include both local and peer memory bindings, but would scissor the rendering to only update local memory.

```
// Provided by VK_VERSION_1_1
typedef VkFlags VkPeerMemoryFeatureFlags;
```

```
// Provided by VK_KHR_device_group
// Equivalent to VkPeerMemoryFeatureFlags
typedef VkPeerMemoryFeatureFlags VkPeerMemoryFeatureFlagsKHR;
```

VkPeerMemoryFeatureFlags is a bitmask type for setting a mask of zero or more VkPeerMemoryFeatureFlagBits.

11.2.16. Opaque Capture Address Query

To query a 64-bit opaque capture address value from a memory object, call:

```
// Provided by VK_VERSION_1_2
uint64_t vkGetDeviceMemoryOpaqueCaptureAddress(
    VkDevice                                    device,
    const VkDeviceMemoryOpaqueCaptureAddressInfo* pInfo);
```

```
// Provided by VK_KHR_buffer_device_address
// Equivalent to vkGetDeviceMemoryOpaqueCaptureAddress
uint64_t vkGetDeviceMemoryOpaqueCaptureAddressKHR(
    VkDevice                                    device,
    const VkDeviceMemoryOpaqueCaptureAddressInfo* pInfo);
```

- device is the logical device that the memory object was allocated on.
- pInfo is a pointer to a VkDeviceMemoryOpaqueCaptureAddressInfo structure specifying the memory object to retrieve an address for.

The 64-bit return value is an opaque address representing the start of pInfo->memory.

If the memory object was allocated with a non-zero value of VkMemoryOpaqueCaptureAddressAllocateInfo::opaqueCaptureAddress, the return value **must** be the same address.

NOTE	The expected usage for these opaque addresses is only for trace capture/replay tools to store these addresses in a trace and subsequently specify them during replay.

Valid Usage

- VUID-vkGetDeviceMemoryOpaqueCaptureAddress-None-03334
  The `bufferDeviceAddress` and `bufferDeviceAddressCaptureReplay` features **must** be enabled
- VUID-vkGetDeviceMemoryOpaqueCaptureAddress-pInfo-10727
  `pInfo->memory` **must** have been allocated using the VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT flag
- VUID-vkGetDeviceMemoryOpaqueCaptureAddress-device-03335
  If `device` was created with multiple physical devices, then the `bufferDeviceAddressMultiDevice` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetDeviceMemoryOpaqueCaptureAddress-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDeviceMemoryOpaqueCaptureAddress-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkDeviceMemoryOpaqueCaptureAddressInfo structure

The `VkDeviceMemoryOpaqueCaptureAddressInfo` structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkDeviceMemoryOpaqueCaptureAddressInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceMemory     memory;
} VkDeviceMemoryOpaqueCaptureAddressInfo;
```

```
// Provided by VK_KHR_buffer_device_address
// Equivalent to VkDeviceMemoryOpaqueCaptureAddressInfo
typedef VkDeviceMemoryOpaqueCaptureAddressInfo
VkDeviceMemoryOpaqueCaptureAddressInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memory` specifies the memory whose address is being queried.

Valid Usage

- VUID-VkDeviceMemoryOpaqueCaptureAddressInfo-memory-03336
  `memory` **must** have been allocated with VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT

Valid Usage (Implicit)

- VUID-VkDeviceMemoryOpaqueCaptureAddressInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_MEMORY_OPAQUE_CAPTURE_ADDRESS_INFO
- VUID-VkDeviceMemoryOpaqueCaptureAddressInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDeviceMemoryOpaqueCaptureAddressInfo-memory-parameter
  `memory` **must** be a valid VkDeviceMemory handle

Chapter 12. Resource Creation

Vulkan supports two primary resource types: *buffers* and *images*. Resources are views of memory with associated formatting and dimensionality. Buffers provide access to raw arrays of bytes, whereas images **can** be multidimensional and **may** have associated metadata.

Other resource types, such as acceleration structures and micromaps use buffers as the backing store for opaque data structures.

12.1. Buffers

Buffers represent linear arrays of data which are used for various purposes by binding them to a graphics or compute pipeline via descriptor sets or certain commands, or by directly specifying them as parameters to certain commands.

Buffers are represented by `VkBuffer` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkBuffer)
```

To create buffers, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateBuffer(
    VkDevice                                    device,
    const VkBufferCreateInfo*                   pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkBuffer*                                   pBuffer);
```

- `device` is the logical device that creates the buffer object.
- `pCreateInfo` is a pointer to a VkBufferCreateInfo structure containing parameters affecting creation of the buffer.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pBuffer` is a pointer to a VkBuffer handle in which the resulting buffer object is returned.

Implementations **may** fail to create a buffer if the effective usage includes the VK_BUFFER_USAGE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and `size` is greater than the maximum of `maxResourceHeapSize` and `maxSamplerHeapSize`. If this happens, VK_ERROR_OUT_OF_DEVICE_MEMORY will be returned.

NOTE

This is an issue identified with VK_EXT_descriptor_heap, which we plan to tighten up for the KHR version. Applications using VK_EXT_descriptor_heap may wish to avoid suballocating heaps from the same buffer, instead creating one buffer per heap, to avoid situations where this causes issues.

Valid Usage

- VUID-vkCreateBuffer-device-09664
  `device` **must** support at least one queue family with one of the VK_QUEUE_VIDEO_ENCODE_BIT_KHR, VK_QUEUE_VIDEO_DECODE_BIT_KHR, VK_QUEUE_SPARSE_BINDING_BIT, VK_QUEUE_TRANSFER_BIT, VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT capabilities
- VUID-vkCreateBuffer-flags-00911
  If the `flags` member of `pCreateInfo` includes VK_BUFFER_CREATE_SPARSE_BINDING_BIT, creating this `VkBuffer` **must** not cause the total required sparse memory for all currently valid sparse resources on the device to exceed `VkPhysicalDeviceLimits ::sparseAddressSpaceSize`

Valid Usage (Implicit)

- VUID-vkCreateBuffer-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateBuffer-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkBufferCreateInfo structure
- VUID-vkCreateBuffer-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateBuffer-pBuffer-parameter
  `pBuffer` **must** be a valid pointer to a VkBuffer handle
- VUID-vkCreateBuffer-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkBufferCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBufferCreateInfo {
    VkStructureType        sType;
    const void*            pNext;
    VkBufferCreateFlags    flags;
    VkDeviceSize           size;
    VkBufferUsageFlags     usage;
    VkSharingMode          sharingMode;
    uint32_t               queueFamilyIndexCount;
    const uint32_t*        pQueueFamilyIndices;
} VkBufferCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkBufferCreateFlagBits specifying additional parameters of the buffer.
- `size` is the size in bytes of the buffer to be created.
- `usage` is a bitmask of VkBufferUsageFlagBits specifying allowed usages of the buffer.
- `sharingMode` is a VkSharingMode value specifying the sharing mode of the buffer when it will be accessed by multiple queue families.
- `queueFamilyIndexCount` is the number of entries in the `pQueueFamilyIndices` array.
- `pQueueFamilyIndices` is a pointer to an array of queue families that will access this buffer. It is ignored if `sharingMode` is not VK_SHARING_MODE_CONCURRENT.

`usage` defines the effective usage flags for the buffer. If the `pNext` chain includes a VkBufferUsageFlags2CreateInfo structure, `usage` is ignored, and the effective usage flags are defined by VkBufferUsageFlags2CreateInfo::`usage`.

Valid Usage

- VUID-VkBufferCreateInfo-None-09499
  If the `pNext` chain does not include a VkBufferUsageFlags2CreateInfo structure, `usage` **must** be a valid combination of VkBufferUsageFlagBits values
- VUID-VkBufferCreateInfo-None-09500
  If the `pNext` chain does not include a VkBufferUsageFlags2CreateInfo structure, `usage` **must** not be 0
- VUID-VkBufferCreateInfo-size-00912
  `size` **must** be greater than `0`
- VUID-VkBufferCreateInfo-sharingMode-00913
  If `sharingMode` is VK_SHARING_MODE_CONCURRENT, `pQueueFamilyIndices` **must** be a valid pointer to an array of `queueFamilyIndexCount` `uint32_t` values
- VUID-VkBufferCreateInfo-maintenance11-13353
  If the `maintenance11` feature is enabled and `sharingMode` is VK_SHARING_MODE_CONCURRENT, then `queueFamilyIndexCount` **must** be greater than `0`

- VUID-VkBufferCreateInfo-sharingMode-00914
  If the maintenance11 feature is not enabled and sharingMode is VK_SHARING_MODE_CONCURRENT, then queueFamilyIndexCount **must** be greater than 1
- VUID-VkBufferCreateInfo-sharingMode-01419
  If sharingMode is VK_SHARING_MODE_CONCURRENT, each element of pQueueFamilyIndices **must** be unique and **must** be less than pQueueFamilyPropertyCount returned by either vkGetPhysicalDeviceQueueFamilyProperties2 or vkGetPhysicalDeviceQueueFamilyProperties for the physicalDevice that was used to create device
- VUID-VkBufferCreateInfo-flags-00915
  If the sparseBinding feature is not enabled, flags **must** not contain VK_BUFFER_CREATE_SPARSE_BINDING_BIT
- VUID-VkBufferCreateInfo-flags-00916
  If the sparseResidencyBuffer feature is not enabled, flags **must** not contain VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT
- VUID-VkBufferCreateInfo-flags-00917
  If the sparseResidencyAliased feature is not enabled, flags **must** not contain VK_BUFFER_CREATE_SPARSE_ALIASED_BIT
- VUID-VkBufferCreateInfo-flags-00918
  If flags contains VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT or VK_BUFFER_CREATE_SPARSE_ALIASED_BIT, it **must** also contain VK_BUFFER_CREATE_SPARSE_BINDING_BIT
- VUID-VkBufferCreateInfo-pNext-00920
  If the pNext chain includes a VkExternalMemoryBufferCreateInfo structure, its handleTypes member **must** only contain bits that are also in VkExternalBufferProperties::externalMemoryProperties.compatibleHandleTypes, as returned by vkGetPhysicalDeviceExternalBufferProperties with pExternalBufferInfo->handleType equal to any one of the handle types specified in VkExternalMemoryBufferCreateInfo::handleTypes
- VUID-VkBufferCreateInfo-flags-01887
  If the protectedMemory feature is not enabled, flags **must** not contain VK_BUFFER_CREATE_PROTECTED_BIT
- VUID-VkBufferCreateInfo-None-01888
  If any of the bits VK_BUFFER_CREATE_SPARSE_BINDING_BIT, VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT, or VK_BUFFER_CREATE_SPARSE_ALIASED_BIT are set, VK_BUFFER_CREATE_PROTECTED_BIT **must** not also be set
- VUID-VkBufferCreateInfo-opaqueCaptureAddress-03337
  If VkBufferOpaqueCaptureAddressCreateInfo::opaqueCaptureAddress is not zero, flags **must** include VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT
- VUID-VkBufferCreateInfo-flags-03338
  If flags includes VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT, the bufferDeviceAddressCaptureReplay feature **must** be enabled

- VUID-VkBufferCreateInfo-usage-04813
  If `usage` includes VK_BUFFER_USAGE_VIDEO_DECODE_SRC_BIT_KHR or VK_BUFFER_USAGE_VIDEO_DECODE_DST_BIT_KHR, and `flags` does not include VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR, then the `pNext` chain **must** include a VkVideoProfileListInfoKHR structure with `profileCount` greater than `0` and `pProfiles` including at least one VkVideoProfileInfoKHR structure with a `videoCodecOperation` member specifying a decode operation
- VUID-VkBufferCreateInfo-usage-04814
  If `usage` includes VK_BUFFER_USAGE_VIDEO_ENCODE_SRC_BIT_KHR or VK_BUFFER_USAGE_VIDEO_ENCODE_DST_BIT_KHR, and `flags` does not include VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR, then the `pNext` chain **must** include a VkVideoProfileListInfoKHR structure with `profileCount` greater than `0` and `pProfiles` including at least one VkVideoProfileInfoKHR structure with a `videoCodecOperation` member specifying an encode operation
- VUID-VkBufferCreateInfo-flags-08325
  If `flags` includes VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR, then `videoMaintenance1` **must** be enabled
- VUID-VkBufferCreateInfo-pNext-10783
  If the `pNext` chain includes a VkVideoProfileListInfoKHR structure and for any element of its `pProfiles` member `videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR, then the `videoDecodeVP9` feature **must** be enabled
- VUID-VkBufferCreateInfo-pNext-10249
  If the `pNext` chain includes a VkVideoProfileListInfoKHR structure and for any element of its `pProfiles` member `videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then the `videoEncodeAV1` feature **must** be enabled
- VUID-VkBufferCreateInfo-size-06409
  `size` **must** be less than or equal to VkPhysicalDeviceMaintenance4Properties::`maxBufferSize`
- VUID-VkBufferCreateInfo-usage-08097
  If `usage` includes VK_BUFFER_USAGE_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT, creating this `VkBuffer` **must** not cause the total required space for all currently valid buffers using this flag on the device to exceed VkPhysicalDeviceDescriptorBufferPropertiesEXT::`samplerDescriptorBufferAddressSpaceSize` or VkPhysicalDeviceDescriptorBufferPropertiesEXT::`descriptorBufferAddressSpaceSize`
- VUID-VkBufferCreateInfo-usage-08098
  If `usage` includes VK_BUFFER_USAGE_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT, creating this `VkBuffer` **must** not cause the total required space for all currently valid buffers using this flag on the device to exceed VkPhysicalDeviceDescriptorBufferPropertiesEXT::`resourceDescriptorBufferAddressSpaceSize` or VkPhysicalDeviceDescriptorBufferPropertiesEXT::`descriptorBufferAddressSpaceSize`
- VUID-VkBufferCreateInfo-flags-08099
  If `flags` includes VK_BUFFER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT, the `descriptorBufferCaptureReplay` feature **must** be enabled

- VUID-VkBufferCreateInfo-pNext-08100
  If the pNext chain includes a VkOpaqueCaptureDescriptorDataCreateInfoEXT structure, flags **must** contain VK_BUFFER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT
- VUID-VkBufferCreateInfo-usage-08101
  If usage includes VK_BUFFER_USAGE_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT, the descriptorBufferPushDescriptors feature **must** be enabled
- VUID-VkBufferCreateInfo-usage-08102
  If usage includes VK_BUFFER_USAGE_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT, VkPhysicalDeviceDescriptorBufferPropertiesEXT::bufferlessPushDescriptors **must** be VK_FALSE
- VUID-VkBufferCreateInfo-usage-08103
  If usage includes VK_BUFFER_USAGE_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT, usage **must** contain at least one of VK_BUFFER_USAGE_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT or VK_BUFFER_USAGE_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-VkBufferCreateInfo-flags-09641
  If flags includes VK_BUFFER_CREATE_PROTECTED_BIT, then the effective usage flags **must** not contain bits other than
  - VK_BUFFER_USAGE_TRANSFER_SRC_BIT
  - VK_BUFFER_USAGE_TRANSFER_DST_BIT
  - VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT
  - VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT
  - VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT
  - VK_BUFFER_USAGE_STORAGE_BUFFER_BIT
  - VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT
  - VK_BUFFER_USAGE_2_VIDEO_DECODE_SRC_BIT_KHR
  - VK_BUFFER_USAGE_2_VIDEO_ENCODE_DST_BIT_KHR
  - VK_BUFFER_USAGE_2_DESCRIPTOR_HEAP_BIT_EXT
- VUID-VkBufferCreateInfo-flags-11277
  If the protectedDescriptorHeaps property is not supported and the effective usage flags include the VK_BUFFER_USAGE_2_DESCRIPTOR_HEAP_BIT_EXT flag, flags **must** not include the VK_BUFFER_CREATE_PROTECTED_BIT flag
- VUID-VkBufferCreateInfo-flags-11279
  If the sparseDescriptorHeaps property is not supported and the effective usage flags include the VK_BUFFER_USAGE_2_DESCRIPTOR_HEAP_BIT_EXT flag, flags **must** not include any of the VK_BUFFER_CREATE_SPARSE_BINDING_BIT, VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT, or VK_BUFFER_CREATE_SPARSE_ALIASED_BIT flags

Valid Usage (Implicit)

- VUID-VkBufferCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BUFFER_CREATE_INFO
- VUID-VkBufferCreateInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkBufferOpaqueCaptureAddressCreateInfo, VkBufferUsageFlags2CreateInfo, VkExternalMemoryBufferCreateInfo, VkOpaqueCaptureDescriptorDataCreateInfoEXT, or VkVideoProfileListInfoKHR
- VUID-VkBufferCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkBufferCreateInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkBufferCreateFlagBits values
- VUID-VkBufferCreateInfo-sharingMode-parameter
  `sharingMode` **must** be a valid VkSharingMode value

The `VkBufferUsageFlags2CreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkBufferUsageFlags2CreateInfo {
    VkStructureType        sType;
    const void*            pNext;
    VkBufferUsageFlags2    usage;
} VkBufferUsageFlags2CreateInfo;
```

```
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
// Equivalent to VkBufferUsageFlags2CreateInfo
typedef VkBufferUsageFlags2CreateInfo VkBufferUsageFlags2CreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `usage` is a bitmask of VkBufferUsageFlagBits2 specifying allowed usages of the buffer.

If this structure is included in the `pNext` chain of a buffer creation structure, `usage` is used instead of the corresponding `usage` value passed in that creation structure, allowing additional usage flags to be specified.

If this structure is included in the `pNext` chain of a query structure including VkBufferUsageFlags, the usage flags of the buffer are returned in `flags` of this structure, allowing additional usage flags to be queried. The flags representable in VkBufferUsageFlags are still returned in the query structure in this case.

Valid Usage (Implicit)

- VUID-VkBufferUsageFlags2CreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BUFFER_USAGE_FLAGS_2_CREATE_INFO
- VUID-VkBufferUsageFlags2CreateInfo-usage-parameter
  `usage` **must** be a valid combination of VkBufferUsageFlagBits2 values

Structure Chaining

Extends the structures

- VkBufferCreateInfo
- VkBufferViewCreateInfo
- VkDescriptorBufferBindingInfoEXT
- VkPhysicalDeviceExternalBufferInfo

Bits which **can** be set in VkBufferUsageFlags2CreateInfo::`usage`, specifying usage behavior of a buffer, are:

```
// Provided by VK_VERSION_1_4
// Flag bits for VkBufferUsageFlagBits2
typedef VkFlags64 VkBufferUsageFlagBits2;
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_TRANSFER_SRC_BIT =
0x00000001ULL;
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_TRANSFER_DST_BIT =
0x00000002ULL;
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_UNIFORM_TEXEL_BUFFER_BIT =
0x00000004ULL;
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_STORAGE_TEXEL_BUFFER_BIT =
0x00000008ULL;
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_UNIFORM_BUFFER_BIT =
0x00000010ULL;
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_STORAGE_BUFFER_BIT =
0x00000020ULL;
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_INDEX_BUFFER_BIT =
0x00000040ULL;
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_VERTEX_BUFFER_BIT =
0x00000080ULL;
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_INDIRECT_BUFFER_BIT =
0x00000100ULL;
// Provided by VK_VERSION_1_4
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_SHADER_DEVICE_ADDRESS_BIT =
0x00020000ULL;
// Provided by VK_EXT_descriptor_heap
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_DESCRIPTOR_HEAP_BIT_EXT =
0x10000000ULL;
```

```
// Provided by VK_KHR_extended_flags with VK_EXT_opacity_micromap,
VK_EXT_opacity_micromap with VK_KHR_maintenance5 or VK_VERSION_1_4
static const VkBufferUsageFlagBits2
VK_BUFFER_USAGE_2_MICROMAP_BUILD_INPUT_READ_ONLY_BIT_EXT = 0x00800000ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_opacity_micromap,
VK_EXT_opacity_micromap with VK_KHR_maintenance5 or VK_VERSION_1_4
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_MICROMAP_STORAGE_BIT_EXT =
0x01000000ULL;
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_TRANSFER_SRC_BIT_KHR =
0x00000001ULL;
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_TRANSFER_DST_BIT_KHR =
0x00000002ULL;
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_UNIFORM_TEXEL_BUFFER_BIT_KHR =
0x00000004ULL;
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_STORAGE_TEXEL_BUFFER_BIT_KHR =
0x00000008ULL;
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_UNIFORM_BUFFER_BIT_KHR =
0x00000010ULL;
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_STORAGE_BUFFER_BIT_KHR =
0x00000020ULL;
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_INDEX_BUFFER_BIT_KHR =
0x00000040ULL;
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_VERTEX_BUFFER_BIT_KHR =
0x00000080ULL;
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_INDIRECT_BUFFER_BIT_KHR =
0x00000100ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_conditional_rendering,
VK_KHR_maintenance5 with VK_EXT_conditional_rendering
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_CONDITIONAL_RENDERING_BIT_EXT =
0x00000200ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_ray_tracing_pipeline,
VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_SHADER_BINDING_TABLE_BIT_KHR =
0x00000400ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_transform_feedback,
VK_KHR_maintenance5 with VK_EXT_transform_feedback
static const VkBufferUsageFlagBits2
VK_BUFFER_USAGE_2_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT = 0x00000800ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_transform_feedback,
VK_KHR_maintenance5 with VK_EXT_transform_feedback
static const VkBufferUsageFlagBits2
VK_BUFFER_USAGE_2_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT = 0x00001000ULL;
```

```
// Provided by VK_KHR_extended_flags with VK_KHR_video_decode_queue,
VK_KHR_maintenance5 with VK_KHR_video_decode_queue
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_VIDEO_DECODE_SRC_BIT_KHR =
0x00002000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_video_decode_queue,
VK_KHR_maintenance5 with VK_KHR_video_decode_queue
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_VIDEO_DECODE_DST_BIT_KHR =
0x00004000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_video_encode_queue,
VK_KHR_maintenance5 with VK_KHR_video_encode_queue
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_VIDEO_ENCODE_DST_BIT_KHR =
0x00008000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_video_encode_queue,
VK_KHR_maintenance5 with VK_KHR_video_encode_queue
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_VIDEO_ENCODE_SRC_BIT_KHR =
0x00010000ULL;
// Provided by VK_KHR_extended_flags with VK_VERSION_1_2 or
VK_KHR_buffer_device_address or VK_EXT_buffer_device_address, VK_KHR_maintenance5 with
VK_VERSION_1_2 or VK_KHR_buffer_device_address or VK_EXT_buffer_device_address
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_SHADER_DEVICE_ADDRESS_BIT_KHR =
0x00020000ULL;
// Provided by VK_KHR_acceleration_structure with VK_KHR_extended_flags,
VK_KHR_acceleration_structure with VK_KHR_maintenance5
static const VkBufferUsageFlagBits2
VK_BUFFER_USAGE_2_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR =
0x00080000ULL;
// Provided by VK_KHR_acceleration_structure with VK_KHR_extended_flags,
VK_KHR_acceleration_structure with VK_KHR_maintenance5
static const VkBufferUsageFlagBits2
VK_BUFFER_USAGE_2_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR = 0x00100000ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_descriptor_buffer,
VK_KHR_maintenance5 with VK_EXT_descriptor_buffer
static const VkBufferUsageFlagBits2
VK_BUFFER_USAGE_2_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT = 0x00200000ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_descriptor_buffer,
VK_KHR_maintenance5 with VK_EXT_descriptor_buffer
static const VkBufferUsageFlagBits2
VK_BUFFER_USAGE_2_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT = 0x00400000ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_descriptor_buffer,
VK_KHR_maintenance5 with VK_EXT_descriptor_buffer
static const VkBufferUsageFlagBits2
VK_BUFFER_USAGE_2_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT = 0x04000000ULL;
// Provided by VK_EXT_memory_decompression
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_MEMORY_DECOMPRESSION_BIT_EXT =
0x100000000ULL;
// Provided by VK_EXT_device_generated_commands
static const VkBufferUsageFlagBits2 VK_BUFFER_USAGE_2_PREPROCESS_BUFFER_BIT_EXT =
0x80000000ULL;
```

```
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
// Equivalent to VkBufferUsageFlagBits2
typedef VkBufferUsageFlagBits2 VkBufferUsageFlagBits2KHR;
```

- VK_BUFFER_USAGE_2_TRANSFER_SRC_BIT specifies that the buffer **can** be used as the source of a *transfer command* (see the definition of VK_PIPELINE_STAGE_TRANSFER_BIT).
- VK_BUFFER_USAGE_2_TRANSFER_DST_BIT specifies that the buffer **can** be used as the destination of a transfer command.
- VK_BUFFER_USAGE_2_UNIFORM_TEXEL_BUFFER_BIT specifies that the buffer **can** be used to create a VkBufferView suitable for occupying a VkDescriptorSet slot of type VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER.
- VK_BUFFER_USAGE_2_STORAGE_TEXEL_BUFFER_BIT specifies that the buffer **can** be used to create a VkBufferView suitable for occupying a VkDescriptorSet slot of type VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER.
- VK_BUFFER_USAGE_2_UNIFORM_BUFFER_BIT specifies that the buffer **can** be used in a VkDescriptorBufferInfo suitable for occupying a VkDescriptorSet slot either of type VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC.
- VK_BUFFER_USAGE_2_STORAGE_BUFFER_BIT specifies that the buffer **can** be used in a VkDescriptorBufferInfo suitable for occupying a VkDescriptorSet slot either of type VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC.
- VK_BUFFER_USAGE_2_INDEX_BUFFER_BIT specifies that the buffer is suitable for passing as the buffer parameter to vkCmdBindIndexBuffer2 and vkCmdBindIndexBuffer.
- VK_BUFFER_USAGE_2_VERTEX_BUFFER_BIT specifies that the buffer is suitable for passing as an element of the pBuffers array to vkCmdBindVertexBuffers.
- VK_BUFFER_USAGE_2_INDIRECT_BUFFER_BIT specifies that the buffer is suitable for passing as the buffer parameter to vkCmdDrawIndirect, vkCmdDrawIndexedIndirect, vkCmdDrawMeshTasksIndirectEXT, vkCmdDrawMeshTasksIndirectCountEXT, or vkCmdDispatchIndirect. It is also suitable for passing as the underlying buffer of either the preprocessAddress or sequenceCountAddress members of VkGeneratedCommandsInfoEXT.
- VK_BUFFER_USAGE_2_CONDITIONAL_RENDERING_BIT_EXT specifies that the buffer is suitable for passing as the buffer parameter to vkCmdBeginConditionalRenderingEXT.
- VK_BUFFER_USAGE_2_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT specifies that the buffer is suitable for using for binding as a transform feedback buffer with vkCmdBindTransformFeedbackBuffers2EXT or vkCmdBindTransformFeedbackBuffersEXT.
- VK_BUFFER_USAGE_2_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT specifies that the buffer is suitable for using as a counter buffer with vkCmdBeginTransformFeedback2EXT, vkCmdEndTransformFeedback2EXT, vkCmdBeginTransformFeedbackEXT, and vkCmdEndTransformFeedbackEXT.
- VK_BUFFER_USAGE_2_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT specifies that the buffer is suitable to contain sampler and combined image sampler descriptors when bound as a

descriptor buffer. Buffers containing combined image sampler descriptors **must** also specify VK_BUFFER_USAGE_2_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT.

- VK_BUFFER_USAGE_2_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT specifies that the buffer is suitable to contain resource descriptors when bound as a descriptor buffer.
- VK_BUFFER_USAGE_2_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT specifies that the buffer, when bound, **can** be used by the implementation to support push descriptors when using descriptor buffers.
- VK_BUFFER_USAGE_2_SHADER_BINDING_TABLE_BIT_KHR specifies that the buffer is suitable for use as a Shader Binding Table.
- VK_BUFFER_USAGE_2_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR specifies that the buffer is suitable for use as a read-only input to an acceleration structure build.
- VK_BUFFER_USAGE_2_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR specifies that the buffer is suitable for storage space for a VkAccelerationStructureKHR.
- VK_BUFFER_USAGE_2_SHADER_DEVICE_ADDRESS_BIT specifies that the buffer **can** be used to retrieve a buffer device address via vkGetBufferDeviceAddress and use that address to access the buffer's memory from a shader.
- VK_BUFFER_USAGE_2_VIDEO_DECODE_SRC_BIT_KHR specifies that the buffer **can** be used as the source video bitstream buffer in a video decode operation.
- VK_BUFFER_USAGE_2_VIDEO_DECODE_DST_BIT_KHR is reserved for future use.
- VK_BUFFER_USAGE_2_VIDEO_ENCODE_DST_BIT_KHR specifies that the buffer **can** be used as the destination video bitstream buffer in a video encode operation.
- VK_BUFFER_USAGE_2_VIDEO_ENCODE_SRC_BIT_KHR is reserved for future use.
- VK_BUFFER_USAGE_2_PREPROCESS_BUFFER_BIT_EXT specifies that the buffer **can** be used as a preprocess buffer for Device-Generated Commands.
- VK_BUFFER_USAGE_2_MICROMAP_BUILD_INPUT_READ_ONLY_BIT_EXT specifies that the buffer **can** be used as a read-only VkMicromapEXT build input, which includes the `data` and `triangleArray` parameters.
- VK_BUFFER_USAGE_2_MICROMAP_STORAGE_BIT_EXT specifies that the buffer **can** be used to create VkMicromapEXT objects.
- VK_BUFFER_USAGE_2_MEMORY_DECOMPRESSION_BIT_EXT specifies that the buffer **can** be used as a destination buffer in memory decompression.
- VK_BUFFER_USAGE_2_DESCRIPTOR_HEAP_BIT_EXT specifies that the buffer **can** be used as a descriptor heap.

```
// Provided by VK_VERSION_1_4
typedef VkFlags64 VkBufferUsageFlags2;
```

```
// Provided by VK_KHR_extended_flags, VK_KHR_maintenance5
// Equivalent to VkBufferUsageFlags2
typedef VkBufferUsageFlags2 VkBufferUsageFlags2KHR;
```

VkBufferUsageFlags2 is a bitmask type for setting a mask of zero or more VkBufferUsageFlagBits2.

Bits which **can** be set in VkBufferCreateInfo::usage, specifying usage behavior of a buffer, are:

WARNING	This functionality is superseded by VkBufferUsageFlagBits2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef enum VkBufferUsageFlagBits {
    VK_BUFFER_USAGE_TRANSFER_SRC_BIT = 0x00000001,
    VK_BUFFER_USAGE_TRANSFER_DST_BIT = 0x00000002,
    VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT = 0x00000004,
    VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT = 0x00000008,
    VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT = 0x00000010,
    VK_BUFFER_USAGE_STORAGE_BUFFER_BIT = 0x00000020,
    VK_BUFFER_USAGE_INDEX_BUFFER_BIT = 0x00000040,
    VK_BUFFER_USAGE_VERTEX_BUFFER_BIT = 0x00000080,
    VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT = 0x00000100,
  // Provided by VK_VERSION_1_2
    VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT = 0x00020000,
  // Provided by VK_KHR_video_decode_queue
    VK_BUFFER_USAGE_VIDEO_DECODE_SRC_BIT_KHR = 0x00002000,
  // Provided by VK_KHR_video_decode_queue
    VK_BUFFER_USAGE_VIDEO_DECODE_DST_BIT_KHR = 0x00004000,
  // Provided by VK_EXT_transform_feedback
    VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT = 0x00000800,
  // Provided by VK_EXT_transform_feedback
    VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT = 0x00001000,
  // Provided by VK_EXT_conditional_rendering
    VK_BUFFER_USAGE_CONDITIONAL_RENDERING_BIT_EXT = 0x00000200,
  // Provided by VK_EXT_descriptor_heap
    VK_BUFFER_USAGE_DESCRIPTOR_HEAP_BIT_EXT = 0x10000000,
  // Provided by VK_KHR_acceleration_structure
    VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR = 0x00080000,
  // Provided by VK_KHR_acceleration_structure
    VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR = 0x00100000,
  // Provided by VK_KHR_ray_tracing_pipeline
    VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR = 0x00000400,
  // Provided by VK_KHR_video_encode_queue
    VK_BUFFER_USAGE_VIDEO_ENCODE_DST_BIT_KHR = 0x00008000,
  // Provided by VK_KHR_video_encode_queue
    VK_BUFFER_USAGE_VIDEO_ENCODE_SRC_BIT_KHR = 0x00010000,
  // Provided by VK_EXT_descriptor_buffer
    VK_BUFFER_USAGE_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT = 0x00200000,
  // Provided by VK_EXT_descriptor_buffer
    VK_BUFFER_USAGE_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT = 0x00400000,
  // Provided by VK_EXT_descriptor_buffer
    VK_BUFFER_USAGE_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT = 0x04000000,
  // Provided by VK_EXT_opacity_micromap
    VK_BUFFER_USAGE_MICROMAP_BUILD_INPUT_READ_ONLY_BIT_EXT = 0x00800000,
  // Provided by VK_EXT_opacity_micromap
    VK_BUFFER_USAGE_MICROMAP_STORAGE_BIT_EXT = 0x01000000,
  // Provided by VK_KHR_buffer_device_address
    VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT_KHR =
VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT,
} VkBufferUsageFlagBits;
```

- VK_BUFFER_USAGE_TRANSFER_SRC_BIT specifies that the buffer **can** be used as the source of a *transfer command* (see the definition of VK_PIPELINE_STAGE_TRANSFER_BIT).
- VK_BUFFER_USAGE_TRANSFER_DST_BIT specifies that the buffer **can** be used as the destination of a transfer command.
- VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT specifies that the buffer **can** be used to create a VkBufferView suitable for occupying a VkDescriptorSet slot of type VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER.
- VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT specifies that the buffer **can** be used to create a VkBufferView suitable for occupying a VkDescriptorSet slot of type VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER.
- VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT specifies that the buffer **can** be used in a VkDescriptorBufferInfo suitable for occupying a VkDescriptorSet slot either of type VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC.
- VK_BUFFER_USAGE_STORAGE_BUFFER_BIT specifies that the buffer **can** be used in a VkDescriptorBufferInfo suitable for occupying a VkDescriptorSet slot either of type VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC.
- VK_BUFFER_USAGE_INDEX_BUFFER_BIT specifies that the buffer is suitable for passing as the buffer parameter to vkCmdBindIndexBuffer2 and vkCmdBindIndexBuffer.
- VK_BUFFER_USAGE_VERTEX_BUFFER_BIT specifies that the buffer is suitable for passing as an element of the pBuffers array to vkCmdBindVertexBuffers.
- VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT specifies that the buffer is suitable for passing as the buffer parameter to vkCmdDrawIndirect, vkCmdDrawIndexedIndirect, vkCmdDrawMeshTasksIndirectEXT, vkCmdDrawMeshTasksIndirectCountEXT, or vkCmdDispatchIndirect. It is also suitable for passing as the underlying buffer of either the preprocessAddress or sequenceCountAddress members of VkGeneratedCommandsInfoEXT.
- VK_BUFFER_USAGE_CONDITIONAL_RENDERING_BIT_EXT specifies that the buffer is suitable for passing as the buffer parameter to vkCmdBeginConditionalRenderingEXT.
- VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT specifies that the buffer is suitable for using for binding as a transform feedback buffer with vkCmdBindTransformFeedbackBuffers2EXT or vkCmdBindTransformFeedbackBuffersEXT.
- VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT specifies that the buffer is suitable for using as a counter buffer with vkCmdBeginTransformFeedback2EXT, vkCmdEndTransformFeedback2EXT, vkCmdBeginTransformFeedbackEXT, and vkCmdEndTransformFeedbackEXT.
- VK_BUFFER_USAGE_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT specifies that the buffer is suitable to contain sampler and combined image sampler descriptors when bound as a descriptor buffer. Buffers containing combined image sampler descriptors **must** also specify VK_BUFFER_USAGE_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT.
- VK_BUFFER_USAGE_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT specifies that the buffer is suitable to contain resource descriptors when bound as a descriptor buffer.

- VK_BUFFER_USAGE_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT specifies that the buffer, when bound, **can** be used by the implementation to support push descriptors when using descriptor buffers.
- VK_BUFFER_USAGE_MICROMAP_BUILD_INPUT_READ_ONLY_BIT_EXT specifies that when building a VkMicromapEXT, the buffer **can** be used as a read-only micromap build input, which includes the `data` and `triangleArray` parameters.
- VK_BUFFER_USAGE_MICROMAP_STORAGE_BIT_EXT specifies that the buffer **can** be used to create VkMicromapEXT objects.
- VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR specifies that the buffer is suitable for use as a Shader Binding Table.
- VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR specifies that the buffer is suitable for use as a read-only input to an acceleration structure build.
- VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR specifies that the buffer is suitable for storage space for a VkAccelerationStructureKHR.
- VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT specifies that the buffer **can** be used to retrieve a buffer device address via vkGetBufferDeviceAddress and use that address to access the buffer's memory from a shader.
- VK_BUFFER_USAGE_VIDEO_DECODE_SRC_BIT_KHR specifies that the buffer **can** be used as the source video bitstream buffer in a video decode operation.
- VK_BUFFER_USAGE_VIDEO_DECODE_DST_BIT_KHR is reserved for future use.
- VK_BUFFER_USAGE_VIDEO_ENCODE_DST_BIT_KHR specifies that the buffer **can** be used as the destination video bitstream buffer in a video encode operation.
- VK_BUFFER_USAGE_VIDEO_ENCODE_SRC_BIT_KHR is reserved for future use.
- VK_BUFFER_USAGE_DESCRIPTOR_HEAP_BIT_EXT specifies that the buffer **can** be used as a descriptor heap.

WARNING	This functionality is superseded by VkBufferUsageFlags2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkBufferUsageFlags;
```

`VkBufferUsageFlags` is a bitmask type for setting a mask of zero or more VkBufferUsageFlagBits.

Bits which **can** be set in VkBufferCreateInfo::`flags`, specifying additional parameters of a buffer, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkBufferCreateFlagBits {
    VK_BUFFER_CREATE_SPARSE_BINDING_BIT = 0x00000001,
    VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT = 0x00000002,
    VK_BUFFER_CREATE_SPARSE_ALIASED_BIT = 0x00000004,
  // Provided by VK_VERSION_1_1
    VK_BUFFER_CREATE_PROTECTED_BIT = 0x00000008,
  // Provided by VK_VERSION_1_2
    VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT = 0x00000010,
  // Provided by VK_EXT_descriptor_buffer
    VK_BUFFER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT = 0x00000020,
  // Provided by VK_KHR_video_maintenance1
    VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR = 0x00000040,
  // Provided by VK_KHR_buffer_device_address
    VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR =
VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT,
} VkBufferCreateFlagBits;
```

- VK_BUFFER_CREATE_SPARSE_BINDING_BIT specifies that the buffer will be backed using sparse memory binding.
- VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT specifies that the buffer **can** be partially backed using sparse memory binding. Buffers created with this flag **must** also be created with the VK_BUFFER_CREATE_SPARSE_BINDING_BIT flag.
- VK_BUFFER_CREATE_SPARSE_ALIASED_BIT specifies that the buffer will be backed using sparse memory binding with memory ranges that might also simultaneously be backing another buffer (or another portion of the same buffer). Buffers created with this flag **must** also be created with the VK_BUFFER_CREATE_SPARSE_BINDING_BIT flag.
- VK_BUFFER_CREATE_PROTECTED_BIT specifies that the buffer is a protected buffer.
- VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT specifies that the buffer's address **can** be saved and reused on a subsequent run (e.g. for trace capture and replay), see VkBufferOpaqueCaptureAddressCreateInfo for more detail.
- VK_BUFFER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT specifies that the buffer **can** be used with descriptor buffers when capturing and replaying (e.g. for trace capture and replay), see VkOpaqueCaptureDescriptorDataCreateInfoEXT for more detail.
- VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR specifies that the buffer **can** be used in video coding operations without having to specify at buffer creation time the set of video profiles the buffer will be used with.

See Sparse Resource Features and Physical Device Features for details of the sparse memory features supported on a device.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkBufferCreateFlags;
```

VkBufferCreateFlags is a bitmask type for setting a mask of zero or more VkBufferCreateFlagBits.

To define a set of external memory handle types that **may** be used as backing store for a buffer, add a VkExternalMemoryBufferCreateInfo structure to the pNext chain of the VkBufferCreateInfo structure. The VkExternalMemoryBufferCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkExternalMemoryBufferCreateInfo {
    VkStructureType                    sType;
    const void*                        pNext;
    VkExternalMemoryHandleTypeFlags    handleTypes;
} VkExternalMemoryBufferCreateInfo;
```

```
// Provided by VK_KHR_external_memory
// Equivalent to VkExternalMemoryBufferCreateInfo
typedef VkExternalMemoryBufferCreateInfo VkExternalMemoryBufferCreateInfoKHR;
```

NOTE A VkExternalMemoryBufferCreateInfo structure with a non-zero handleTypes field must be included in the creation parameters for a buffer that will be bound to memory that is either exported or imported.

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- handleTypes is zero or a bitmask of VkExternalMemoryHandleTypeFlagBits specifying one or more external memory handle types.

Valid Usage (Implicit)

- VUID-VkExternalMemoryBufferCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_BUFFER_CREATE_INFO
- VUID-VkExternalMemoryBufferCreateInfo-handleTypes-parameter
  If handleTypes is not 0, handleTypes **must** be a valid combination of VkExternalMemoryHandleTypeFlagBits values

Structure Chaining

Extends the structure

- VkBufferCreateInfo

To request a specific device address for a buffer, add a VkBufferOpaqueCaptureAddressCreateInfo structure to the pNext chain of the VkBufferCreateInfo structure. The VkBufferOpaqueCaptureAddressCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkBufferOpaqueCaptureAddressCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint64_t           opaqueCaptureAddress;
} VkBufferOpaqueCaptureAddressCreateInfo;
```

```
// Provided by VK_KHR_buffer_device_address
// Equivalent to VkBufferOpaqueCaptureAddressCreateInfo
typedef VkBufferOpaqueCaptureAddressCreateInfo
VkBufferOpaqueCaptureAddressCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `opaqueCaptureAddress` is the opaque capture address requested for the buffer.

If `opaqueCaptureAddress` is zero, no specific address is requested.

If `opaqueCaptureAddress` is not zero, then it **should** be an address retrieved from vkGetBufferOpaqueCaptureAddress for an identically created buffer on the same implementation.

If this structure is not present, it is as if `opaqueCaptureAddress` is zero.

Applications **should** avoid creating buffers with application-provided addresses and implementation-provided addresses in the same process, to reduce the likelihood of VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS errors.

NOTE

The expected usage for this is that a trace capture/replay tool will add the VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT flag to all buffers that use VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT, and during capture will save the queried opaque device addresses in the trace. During replay, the buffers will be created specifying the original address so any address values stored in the trace data will remain valid.

Implementations are expected to separate such buffers in the GPU address space so normal allocations will avoid using these addresses. Applications and tools should avoid mixing application-provided and implementation-provided addresses for buffers created with VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT, to avoid address space allocation conflicts.

Valid Usage (Implicit)

- VUID-VkBufferOpaqueCaptureAddressCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BUFFER_OPAQUE_CAPTURE_ADDRESS_CREATE_INFO

Structure Chaining

Extends the structure

- VkBufferCreateInfo

To destroy a buffer, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyBuffer(
    VkDevice                                    device,
    VkBuffer                                    buffer,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the buffer.
- `buffer` is the buffer to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyBuffer-buffer-00922
  All submitted commands that refer to `buffer`, either directly or via a `VkBufferView`, **must** have completed execution
- VUID-vkDestroyBuffer-buffer-00923
  If `VkAllocationCallbacks` were provided when `buffer` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyBuffer-buffer-00924
  If no `VkAllocationCallbacks` were provided when `buffer` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyBuffer-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyBuffer-buffer-parameter
  If `buffer` is not VK_NULL_HANDLE, `buffer` **must** be a valid VkBuffer handle
- VUID-vkDestroyBuffer-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyBuffer-buffer-parent
  If `buffer` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `buffer` **must** be externally synchronized

12.2. Buffer Views

A *buffer view* represents a contiguous range of a buffer and a specific format to be used to interpret the data. Buffer views are used to enable shaders to access buffer contents using image operations. In order to create a valid buffer view, the buffer **must** have been created with at least one of the following usage flags:

- VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT
- VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT

Buffer views are represented by `VkBufferView` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkBufferView)
```

To create a buffer view, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateBufferView(
    VkDevice                                    device,
    const VkBufferViewCreateInfo*               pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkBufferView*                               pView);
```

- `device` is the logical device that creates the buffer view.
- `pCreateInfo` is a pointer to a VkBufferViewCreateInfo structure containing parameters to be used to create the buffer view.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pView` is a pointer to a VkBufferView handle in which the resulting buffer view object is returned.

Valid Usage

- VUID-vkCreateBufferView-device-09665
  `device` **must** support at least one queue family with one of the VK_QUEUE_COMPUTE_BIT or VK_QUEUE_GRAPHICS_BIT capabilities

Valid Usage (Implicit)

- VUID-vkCreateBufferView-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateBufferView-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkBufferViewCreateInfo structure
- VUID-vkCreateBufferView-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateBufferView-pView-parameter
  `pView` **must** be a valid pointer to a VkBufferView handle
- VUID-vkCreateBufferView-device-queuecount
  The device **must** have been created with at least `1` queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkBufferViewCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBufferViewCreateInfo {
    VkStructureType            sType;
    const void*                pNext;
    VkBufferViewCreateFlags    flags;
    VkBuffer                   buffer;
    VkFormat                   format;
    VkDeviceSize               offset;
    VkDeviceSize               range;
} VkBufferViewCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `buffer` is a VkBuffer on which the view will be created.

- `format` is a VkFormat describing the format of the data elements in the buffer.
- `offset` is an offset in bytes from the base address of the buffer. Accesses to the buffer view from shaders use addressing that is relative to this starting offset.
- `range` is a size in bytes of the buffer view. If `range` is equal to VK_WHOLE_SIZE, the range from `offset` to the end of the buffer is used. If VK_WHOLE_SIZE is used and the remaining size of the buffer is not a multiple of the texel block size of `format`, the nearest smaller multiple is used.

The buffer view has a *buffer view usage* identifying which descriptor types can be created from it. This usage **can** be defined by including the VkBufferUsageFlags2CreateInfo structure in the `pNext` chain, and specifying the `usage` value there. If this structure is not included, it is equal to the VkBufferCreateInfo::`usage` value used to create `buffer`.

Valid Usage

- VUID-VkBufferViewCreateInfo-offset-00925
  `offset` **must** be less than the size of `buffer`
- VUID-VkBufferViewCreateInfo-range-00928
  If `range` is not equal to VK_WHOLE_SIZE, `range` **must** be greater than `0`
- VUID-VkBufferViewCreateInfo-range-00929
  If `range` is not equal to VK_WHOLE_SIZE, `range` **must** be an integer multiple of the texel block size of `format`
- VUID-VkBufferViewCreateInfo-range-00930
  If `range` is not equal to VK_WHOLE_SIZE, the number of texel buffer elements given by (⌊`range` / (texel block size)⌋ × (texels per block)) where texel block size and texels per block are as defined in the Compatible Formats table for `format`, **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxTexelBufferElements`
- VUID-VkBufferViewCreateInfo-offset-00931
  If `range` is not equal to VK_WHOLE_SIZE, the sum of `offset` and `range` **must** be less than or equal to the size of `buffer`
- VUID-VkBufferViewCreateInfo-range-04059
  If `range` is equal to VK_WHOLE_SIZE, the number of texel buffer elements given by (⌊(size - `offset`) / (texel block size)⌋ × (texels per block)) where size is the size of `buffer`, and texel block size and texels per block are as defined in the Compatible Formats table for `format`, **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxTexelBufferElements`
- VUID-VkBufferViewCreateInfo-buffer-00932
  `buffer` **must** have been created with at least one of the VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT or VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT usage flags set
- VUID-VkBufferViewCreateInfo-format-08778
  If the buffer view usage contains VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT, then format features of `format` **must** contain VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT
- VUID-VkBufferViewCreateInfo-format-08779
  If the buffer view usage contains VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT, then

format features of `format` **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT

- VUID-VkBufferViewCreateInfo-buffer-00935
  If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkBufferViewCreateInfo-buffer-02750
  If `buffer` was created with the VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT usage flag set, `offset` **must** be a multiple of the effective alignment requirement of `format` for VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER as defined by `minTexelBufferOffsetAlignment`
- VUID-VkBufferViewCreateInfo-buffer-02751
  If `buffer` was created with the VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT usage flag set, `offset` **must** be a multiple of the effective alignment requirement of `format` for VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER as defined by `minTexelBufferOffsetAlignment`
- VUID-VkBufferViewCreateInfo-pNext-06782
  If the `pNext` chain includes a VkExportMetalObjectCreateInfoEXT structure, its `exportObjectType` member **must** be VK_EXPORT_METAL_OBJECT_TYPE_METAL_TEXTURE_BIT_EXT
- VUID-VkBufferViewCreateInfo-pNext-08780
  If the `pNext` chain includes a VkBufferUsageFlags2CreateInfo, its `usage` **must** not contain any other bit than VK_BUFFER_USAGE_2_UNIFORM_TEXEL_BUFFER_BIT or VK_BUFFER_USAGE_2_STORAGE_TEXEL_BUFFER_BIT
- VUID-VkBufferViewCreateInfo-pNext-08781
  If the `pNext` chain includes a VkBufferUsageFlags2CreateInfo, its `usage` **must** be a subset of the VkBufferCreateInfo::`usage` specified or VkBufferUsageFlags2CreateInfo::`usage` from VkBufferCreateInfo::`pNext` when creating `buffer`

Valid Usage (Implicit)

- VUID-VkBufferViewCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BUFFER_VIEW_CREATE_INFO
- VUID-VkBufferViewCreateInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkBufferUsageFlags2CreateInfo or VkExportMetalObjectCreateInfoEXT
- VUID-VkBufferViewCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique, with the exception of structures of type VkExportMetalObjectCreateInfoEXT
- VUID-VkBufferViewCreateInfo-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkBufferViewCreateInfo-buffer-parameter
  `buffer` **must** be a valid VkBuffer handle

- VUID-VkBufferViewCreateInfo-format-parameter
  `format` **must** be a valid VkFormat value

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkBufferViewCreateFlags;
```

`VkBufferViewCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

To destroy a buffer view, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyBufferView(
    VkDevice                                    device,
    VkBufferView                                bufferView,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the buffer view.
- `bufferView` is the buffer view to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyBufferView-bufferView-00936
  All submitted commands that refer to `bufferView` **must** have completed execution
- VUID-vkDestroyBufferView-bufferView-00937
  If `VkAllocationCallbacks` were provided when `bufferView` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyBufferView-bufferView-00938
  If no `VkAllocationCallbacks` were provided when `bufferView` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyBufferView-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyBufferView-bufferView-parameter
  If `bufferView` is not VK_NULL_HANDLE, `bufferView` **must** be a valid VkBufferView handle
- VUID-vkDestroyBufferView-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyBufferView-bufferView-parent

If `bufferView` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `bufferView` **must** be externally synchronized

12.2.1. Buffer View Format Features

Valid uses of a VkBufferView **may** depend on the buffer view's *format features*, defined below. Such constraints are documented in the affected valid usage statement.

- If Vulkan 1.3 is supported or the `VK_KHR_format_feature_flags2` extension is supported, then the buffer view's set of *format features* is the value of VkFormatProperties3::`bufferFeatures` found by calling vkGetPhysicalDeviceFormatProperties2 on the same `format` as VkBufferViewCreateInfo::`format`.

12.3. Buffer Device Addresses

To query a 64-bit buffer device address value which can be used to identify a buffer to API commands or through which buffer memory **can** be accessed, call:

```
// Provided by VK_VERSION_1_2
VkDeviceAddress vkGetBufferDeviceAddress(
    VkDevice                                    device,
    const VkBufferDeviceAddressInfo*            pInfo);
```

```
// Provided by VK_KHR_buffer_device_address
// Equivalent to vkGetBufferDeviceAddress
VkDeviceAddress vkGetBufferDeviceAddressKHR(
    VkDevice                                    device,
    const VkBufferDeviceAddressInfo*            pInfo);
```

- `device` is the logical device that the buffer was created on.
- `pInfo` is a pointer to a VkBufferDeviceAddressInfo structure specifying the buffer to retrieve an address for.

The 64-bit return value, `bufferBaseAddress`, is an address of the start of `pInfo->buffer`. Addresses in the range [`bufferBaseAddress`, `bufferBaseAddress` + VkBufferCreateInfo::`size`) **can** be used to access the memory bound to this buffer on the device.

A value of zero is reserved as a "null" pointer and **must** not be returned as a valid buffer device address.

If the buffer was created with a non-zero value of VkBufferOpaqueCaptureAddressCreateInfo

::opaqueCaptureAddress, the return value will be the same address that was returned at capture time.

The returned address **must** satisfy alignment requirements if certain conditions are met, outlined below. The required alignment **must** be observed for either the returned address of the buffer, or, if the buffer was created without VK_BUFFER_CREATE_SPARSE_BINDING_BIT, the returned address of the buffer minus VkBindBufferMemoryInfo::memoryOffset or equivalent vkBindBufferMemory offset. The required alignment is the maximum of the following values:

- The value of VkMemoryRequirements::alignment queried via vkGetBufferMemoryRequirements for pInfo->buffer
- minStorageBufferOffsetAlignment if pInfo->buffer was created with VK_BUFFER_USAGE_2_STORAGE_BUFFER_BIT
- minUniformBufferOffsetAlignment if pInfo->buffer was created with VK_BUFFER_USAGE_2_UNIFORM_BUFFER_BIT
- minTexelBufferOffsetAlignment if pInfo->buffer was created with VK_BUFFER_USAGE_2_STORAGE_TEXEL_BUFFER_BIT or VK_BUFFER_USAGE_2_UNIFORM_TEXEL_BUFFER_BIT
- resourceHeapAlignment if pInfo->buffer was created with VK_BUFFER_USAGE_2_DESCRIPTOR_HEAP_BIT_EXT
- samplerHeapAlignment if pInfo->buffer was created with VK_BUFFER_USAGE_2_DESCRIPTOR_HEAP_BIT_EXT
- descriptorBufferOffsetAlignment if pInfo->buffer was created with VK_BUFFER_USAGE_2_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT or VK_BUFFER_USAGE_2_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT
- 256, if pInfo->buffer was created with VK_BUFFER_USAGE_2_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR, and VK_KHR_device_address_commands is supported

If multiple VkBuffer objects are bound to overlapping ranges of VkDeviceMemory, implementations **may** return address ranges which overlap. In this case, it is ambiguous which VkBuffer is associated with any given device address. For purposes of valid usage, if multiple VkBuffer objects **can** be attributed to a device address, a VkBuffer is selected such that valid usage passes, if it exists.

Valid Usage

- VUID-vkGetBufferDeviceAddress-bufferDeviceAddress-03324
  The bufferDeviceAddress feature **must** be enabled, and at least one of the following conditions **must** be met
  - buffer is sparse
  - buffer is bound completely and contiguously to a single VkDeviceMemory object
- VUID-vkGetBufferDeviceAddress-device-03325
  If device was created with multiple physical devices, then the bufferDeviceAddressMultiDevice feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetBufferDeviceAddress-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetBufferDeviceAddress-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkBufferDeviceAddressInfo structure

The `VkBufferDeviceAddressInfo` structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkBufferDeviceAddressInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkBuffer           buffer;
} VkBufferDeviceAddressInfo;
```

```
// Provided by VK_KHR_buffer_device_address
// Equivalent to VkBufferDeviceAddressInfo
typedef VkBufferDeviceAddressInfo VkBufferDeviceAddressInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `buffer` specifies the buffer whose address is being queried.

Valid Usage

- VUID-VkBufferDeviceAddressInfo-buffer-02601
  `buffer` **must** have been created with the VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT usage flag set

Valid Usage (Implicit)

- VUID-VkBufferDeviceAddressInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BUFFER_DEVICE_ADDRESS_INFO
- VUID-VkBufferDeviceAddressInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkBufferDeviceAddressInfo-buffer-parameter
  `buffer` **must** be a valid VkBuffer handle

To query a 64-bit buffer opaque capture address, call:

```
// Provided by VK_VERSION_1_2
uint64_t vkGetBufferOpaqueCaptureAddress(
    VkDevice                                    device,
    const VkBufferDeviceAddressInfo*            pInfo);
```

```
// Provided by VK_KHR_buffer_device_address
// Equivalent to vkGetBufferOpaqueCaptureAddress
uint64_t vkGetBufferOpaqueCaptureAddressKHR(
    VkDevice                                    device,
    const VkBufferDeviceAddressInfo*            pInfo);
```

- `device` is the logical device that the buffer was created on.
- `pInfo` is a pointer to a VkBufferDeviceAddressInfo structure specifying the buffer to retrieve an address for.

The 64-bit return value is an opaque capture address of the start of `pInfo->buffer`.

If the buffer was created with a non-zero value of VkBufferOpaqueCaptureAddressCreateInfo ::`opaqueCaptureAddress` the return value **must** be the same address.

Valid Usage

- VUID-vkGetBufferOpaqueCaptureAddress-None-03326
  The `bufferDeviceAddress` and `bufferDeviceAddressCaptureReplay` features **must** be enabled
- VUID-vkGetBufferOpaqueCaptureAddress-pInfo-10725
  `pInfo->buffer` **must** have been created with the VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT flag
- VUID-vkGetBufferOpaqueCaptureAddress-device-03327
  If `device` was created with multiple physical devices, then the `bufferDeviceAddressMultiDevice` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetBufferOpaqueCaptureAddress-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetBufferOpaqueCaptureAddress-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkBufferDeviceAddressInfo structure

The `VkStridedDeviceAddressRegionKHR` structure is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef struct VkStridedDeviceAddressRegionKHR {
    VkDeviceAddress    deviceAddress;
    VkDeviceSize       stride;
    VkDeviceSize       size;
} VkStridedDeviceAddressRegionKHR;
```

- `deviceAddress` is the device address (as returned by the vkGetBufferDeviceAddress command) at which the region starts, or zero if the region is unused.
- `stride` is the byte stride between consecutive elements.
- `size` is the size in bytes of the region starting at `deviceAddress`.

Valid Usage

- VUID-VkStridedDeviceAddressRegionKHR-size-04631
  If `size` is not zero, all addresses between `deviceAddress` and `deviceAddress` + `size` - 1 **must** be in the buffer device address range of the same buffer
- VUID-VkStridedDeviceAddressRegionKHR-size-04632
  If `size` is not zero, `stride` **must** be less than or equal to the size of the buffer from which `deviceAddress` was queried

Valid Usage (Implicit)

- VUID-VkStridedDeviceAddressRegionKHR-deviceAddress-parameter
  If `deviceAddress` is not `0`, `deviceAddress` **must** be a valid VkDeviceAddress value

12.4. Images

Images are specialized resources that have multi-dimensional access, as outlined in the Images chapter. Images **can** be used for various purposes, such as rendering attachments, for copy operations, or accessed through shaders via resource descriptors.

Images are represented by `VkImage` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkImage)
```

To create images, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateImage(
    VkDevice                                    device,
    const VkImageCreateInfo*                    pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkImage*                                    pImage);
```

- `device` is the logical device that creates the image.
- `pCreateInfo` is a pointer to a VkImageCreateInfo structure containing parameters to be used to create the image.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pImage` is a pointer to a VkImage handle in which the resulting image object is returned.

Valid Usage

- VUID-vkCreateImage-device-09666
  `device` **must** support at least one queue family with one of the VK_QUEUE_VIDEO_ENCODE_BIT_KHR, VK_QUEUE_VIDEO_DECODE_BIT_KHR, VK_QUEUE_SPARSE_BINDING_BIT, VK_QUEUE_TRANSFER_BIT, VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT capabilities
- VUID-vkCreateImage-flags-00939
  If the `flags` member of `pCreateInfo` includes VK_IMAGE_CREATE_SPARSE_BINDING_BIT, creating this `VkImage` **must** not cause the total required sparse memory for all currently valid sparse resources on the device to exceed `VkPhysicalDeviceLimits::sparseAddressSpaceSize`

Valid Usage (Implicit)

- VUID-vkCreateImage-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateImage-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkImageCreateInfo structure
- VUID-vkCreateImage-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateImage-pImage-parameter
  `pImage` **must** be a valid pointer to a VkImage handle
- VUID-vkCreateImage-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_COMPRESSION_EXHAUSTED_EXT
- VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkImageCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageCreateInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkImageCreateFlags       flags;
    VkImageType              imageType;
    VkFormat                 format;
    VkExtent3D               extent;
    uint32_t                 mipLevels;
    uint32_t                 arrayLayers;
    VkSampleCountFlagBits    samples;
    VkImageTiling            tiling;
    VkImageUsageFlags        usage;
    VkSharingMode            sharingMode;
    uint32_t                 queueFamilyIndexCount;
    const uint32_t*          pQueueFamilyIndices;
    VkImageLayout            initialLayout;
} VkImageCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkImageCreateFlagBits describing additional parameters of the image.
- `imageType` is a VkImageType value specifying the basic dimensionality of the image. Layers in array textures do not count as a dimension for the purposes of the image type.
- `format` is a VkFormat describing the format and type of the texel blocks that will be contained in the image.
- `extent` is a VkExtent3D describing the number of texels/pixels in each dimension of the base level.

- `mipLevels` describes the number of levels of detail available for minified sampling of the image.
- `arrayLayers` is the number of layers in the image.
- `samples` is a VkSampleCountFlagBits value specifying the number of samples per texel.
- `tiling` is a VkImageTiling value specifying the tiling arrangement of the texel blocks in memory.
- `usage` is a bitmask of VkImageUsageFlagBits describing the intended usage of the image.
- `sharingMode` is a VkSharingMode value specifying the sharing mode of the image when it will be accessed by multiple queue families.
- `queueFamilyIndexCount` is the number of entries in the `pQueueFamilyIndices` array.
- `pQueueFamilyIndices` is a pointer to an array of queue families that will access this image. It is ignored if `sharingMode` is not VK_SHARING_MODE_CONCURRENT.
- `initialLayout` is a VkImageLayout value specifying the initial VkImageLayout of all image subresources of the image. See Image Layouts.

`flags` defines the effective create flags for the image. If the `pNext` chain includes a VkImageCreateFlags2CreateInfoKHR structure, `flags` is ignored, and the effective create flags are defined by VkImageCreateFlags2CreateInfoKHR::`flags`.

`usage` defines the effective usage flags for the image. If the `pNext` chain includes a VkImageUsageFlags2CreateInfoKHR structure, `usage` is ignored, and the effective usage flags are defined by VkImageUsageFlags2CreateInfoKHR::`usage`.

Images created with `tiling` equal to VK_IMAGE_TILING_LINEAR have further restrictions on their limits and capabilities compared to images created with `tiling` equal to VK_IMAGE_TILING_OPTIMAL. Creation of images with tiling VK_IMAGE_TILING_LINEAR **may** not be supported unless other parameters meet all of the constraints:

- `imageType` is VK_IMAGE_TYPE_2D
- `format` is not a depth/stencil format
- `mipLevels` is 1
- `arrayLayers` is 1
- `samples` is VK_SAMPLE_COUNT_1_BIT
- `usage` only includes VK_IMAGE_USAGE_TRANSFER_SRC_BIT and/or VK_IMAGE_USAGE_TRANSFER_DST_BIT

Images created with one of the formats that require a sampler $Y'C_BC_R$ conversion have further restrictions on their limits and capabilities compared to images created with other formats. Creation of images with a format requiring $Y'C_BC_R$ conversion **may** not be supported unless other parameters meet all of the constraints:

- `imageType` is VK_IMAGE_TYPE_2D
- `mipLevels` is 1
- `arrayLayers` is 1, unless the `ycbcrImageArrays` feature is enabled, or otherwise indicated by VkImageFormatProperties::`maxArrayLayers`, as returned by

vkGetPhysicalDeviceImageFormatProperties

- `samples` is VK_SAMPLE_COUNT_1_BIT

Implementations **may** support additional limits and capabilities beyond those listed above.

To determine the set of valid `usage` bits for a given format, call vkGetPhysicalDeviceFormatProperties.

If the size of the resultant image would exceed `maxResourceSize`, then vkCreateImage **must** fail and return VK_ERROR_OUT_OF_DEVICE_MEMORY. This failure **may** occur even when all image creation parameters satisfy their valid usage requirements.

If the implementation reports VK_TRUE in VkPhysicalDeviceHostImageCopyProperties::`identicalMemoryTypeRequirements`, usage of VK_IMAGE_USAGE_HOST_TRANSFER_BIT **must** not affect the memory type requirements of the image as described in Sparse Resource Memory Requirements and Resource Memory Association.

NOTE

For images created without the VK_IMAGE_CREATE_EXTENDED_USAGE_BIT flag set, a `usage` bit is valid if it is supported for the format the image is created with.

For images created with VK_IMAGE_CREATE_EXTENDED_USAGE_BIT a `usage` bit is valid if it is supported for at least one of the formats a `VkImageView` created from the image **can** have (see Image Views for more detail).

Image Creation Limits

Valid values for some image creation parameters are limited by a numerical upper bound or by inclusion in a bitset. For example, VkImageCreateInfo::`arrayLayers` is limited by `imageCreateMaxArrayLayers`, defined below; and VkImageCreateInfo::`samples` is limited by `imageCreateSampleCounts`, also defined below.

Several limiting values are defined below, as well as assisting values from which the limiting values are derived. The limiting values are referenced by the relevant valid usage statements of VkImageCreateInfo.

- Let `uint64_t imageCreateDrmFormatModifiers[]` be the set of Linux DRM format modifiers that the resultant image **may** have.
  - If `tiling` is not VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then `imageCreateDrmFormatModifiers` is empty.
  - If VkImageCreateInfo::`pNext` contains VkImageDrmFormatModifierExplicitCreateInfoEXT, then `imageCreateDrmFormatModifiers` contains exactly one modifier, VkImageDrmFormatModifierExplicitCreateInfoEXT::`drmFormatModifier`.
  - If VkImageCreateInfo::`pNext` contains VkImageDrmFormatModifierListCreateInfoEXT, then `imageCreateDrmFormatModifiers` contains the entire array VkImageDrmFormatModifierListCreateInfoEXT::`pDrmFormatModifiers`.
- Let `VkBool32 imageCreateMaybeLinear` indicate if the resultant image may be linear.

- If `tiling` is VK_IMAGE_TILING_LINEAR, then `imageCreateMaybeLinear` is VK_TRUE.
- If `tiling` is VK_IMAGE_TILING_OPTIMAL, then `imageCreateMaybeLinear` is VK_FALSE.
- If `tiling` is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then `imageCreateMaybeLinear` is VK_TRUE if and only if `imageCreateDrmFormatModifiers` contains `DRM_FORMAT_MOD_LINEAR`.

- Let `VkFormatFeatureFlags imageCreateFormatFeatures` be the set of valid *format features* available during image creation.
  - If `tiling` is VK_IMAGE_TILING_LINEAR, then `imageCreateFormatFeatures` is the value of VkFormatProperties::`linearTilingFeatures` found by calling vkGetPhysicalDeviceFormatProperties with parameter `format` equal to VkImageCreateInfo::`format`.
  - If `tiling` is VK_IMAGE_TILING_OPTIMAL, then `imageCreateFormatFeatures` is the value of VkFormatProperties::`optimalTilingFeatures` found by calling vkGetPhysicalDeviceFormatProperties with parameter `format` equal to VkImageCreateInfo::`format`.
  - If `tiling` is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then the value of `imageCreateFormatFeatures` is found by calling vkGetPhysicalDeviceFormatProperties2 with `format` equal to VkImageCreateInfo::`format` and with VkDrmFormatModifierPropertiesListEXT chained into VkFormatProperties2; by collecting all members of the returned array VkDrmFormatModifierPropertiesListEXT::`pDrmFormatModifierProperties` whose `drmFormatModifier` belongs to `imageCreateDrmFormatModifiers`; and by taking the bitwise intersection, over the collected array members, of `drmFormatModifierTilingFeatures`. The resultant `imageCreateFormatFeatures` **may** be empty.
- Let `VkImageFormatProperties2 imageCreateImageFormatPropertiesList[]` be the list of structures obtained by calling vkGetPhysicalDeviceImageFormatProperties2, possibly multiple times, as follows:
  - The parameters VkPhysicalDeviceImageFormatInfo2::`format`, `imageType`, `tiling`, `usage`, and `flags` **must** be equal to those in VkImageCreateInfo.
  - If VkImageCreateInfo::`pNext` contains a VkExternalMemoryImageCreateInfo structure whose `handleTypes` is not `0`, then VkPhysicalDeviceImageFormatInfo2::`pNext` **must** contain a VkPhysicalDeviceExternalImageFormatInfo structure whose `handleType` is not `0`; and vkGetPhysicalDeviceImageFormatProperties2 **must** be called for each handle type in VkExternalMemoryImageCreateInfo::`handleTypes`, successively setting VkPhysicalDeviceExternalImageFormatInfo::`handleType` on each call.
  - If VkImageCreateInfo::`pNext` contains no VkExternalMemoryImageCreateInfo structure, or contains a structure whose `handleTypes` is `0`, then VkPhysicalDeviceImageFormatInfo2::`pNext` **must** either contain no VkPhysicalDeviceExternalImageFormatInfo structure, or contain a structure whose `handleType` is `0`.
  - If VkImageCreateInfo::`pNext` contains a VkVideoProfileListInfoKHR structure then VkPhysicalDeviceImageFormatInfo2::`pNext` **must** also contain the same VkVideoProfileListInfoKHR structure on each call.

- If `tiling` is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then:
  - VkPhysicalDeviceImageFormatInfo2::`pNext` **must** contain a VkPhysicalDeviceImageDrmFormatModifierInfoEXT structure where `sharingMode` is equal to VkImageCreateInfo::`sharingMode`;
  - if `sharingMode` is VK_SHARING_MODE_CONCURRENT, then `queueFamilyIndexCount` and `pQueueFamilyIndices` **must** be equal to those in VkImageCreateInfo;
  - if `flags` contains VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT, then the VkImageFormatListCreateInfo structure included in the `pNext` chain of VkPhysicalDeviceImageFormatInfo2 **must** be equivalent to the one included in the `pNext` chain of VkImageCreateInfo;
  - if VkImageCreateInfo::`pNext` contains a VkImageCompressionControlEXT structure, then the VkPhysicalDeviceImageFormatInfo2::`pNext` chain **must** contain an equivalent structure;
  - vkGetPhysicalDeviceImageFormatProperties2 **must** be called for each modifier in `imageCreateDrmFormatModifiers`, successively setting VkPhysicalDeviceImageDrmFormatModifierInfoEXT::`drmFormatModifier` on each call.
- If `tiling` is not VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then VkPhysicalDeviceImageFormatInfo2::`pNext` **must** contain no VkPhysicalDeviceImageDrmFormatModifierInfoEXT structure.
- If any call to vkGetPhysicalDeviceImageFormatProperties2 returns an error, then `imageCreateImageFormatPropertiesList` is defined to be the empty list.

- Let `uint32_t imageCreateMaxMipLevels` be the minimum value of VkImageFormatProperties::`maxMipLevels` in `imageCreateImageFormatPropertiesList`. The value is undefined if `imageCreateImageFormatPropertiesList` is empty.
- Let `uint32_t imageCreateMaxArrayLayers` be the minimum value of VkImageFormatProperties::`maxArrayLayers` in `imageCreateImageFormatPropertiesList`. The value is undefined if `imageCreateImageFormatPropertiesList` is empty.
- Let `VkExtent3D imageCreateMaxExtent` be the component-wise minimum over all VkImageFormatProperties::`maxExtent` values in `imageCreateImageFormatPropertiesList`. The value is undefined if `imageCreateImageFormatPropertiesList` is empty.
- Let `VkSampleCountFlags imageCreateSampleCounts` be the intersection of each VkImageFormatProperties::`sampleCounts` in `imageCreateImageFormatPropertiesList`. The value is undefined if `imageCreateImageFormatPropertiesList` is empty.
- Let `VkVideoFormatPropertiesKHR videoFormatProperties[]` be defined as follows.
  - If VkImageCreateInfo::`pNext` contains a VkVideoProfileListInfoKHR structure, then `videoFormatProperties` is the list of structures obtained by calling vkGetPhysicalDeviceVideoFormatPropertiesKHR with VkPhysicalDeviceVideoFormatInfoKHR::`imageUsage` equal to the `usage` member of VkImageCreateInfo and VkPhysicalDeviceVideoFormatInfoKHR::`pNext` containing the same VkVideoProfileListInfoKHR structure chained to VkImageCreateInfo.
  - If VkImageCreateInfo::`pNext` contains no VkVideoProfileListInfoKHR structure, then

videoFormatProperties is an empty list.

- Let VkBool32 supportedVideoFormat indicate if the image parameters are supported by the specified video profiles.
  - supportedVideoFormat is VK_TRUE if there exists an element in the videoFormatProperties list for which all of the following conditions are true:
    - VkImageCreateInfo::format equals VkVideoFormatPropertiesKHR::format.
    - VkImageCreateInfo::flags only contains VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR and/or bits also set in VkVideoFormatPropertiesKHR::imageCreateFlags.
    - VkImageCreateInfo::imageType equals VkVideoFormatPropertiesKHR::imageType.
    - VkImageCreateInfo::tiling equals VkVideoFormatPropertiesKHR::imageTiling.
    - VkImageCreateInfo::usage only contains bits also set in VkVideoFormatPropertiesKHR::imageUsageFlags, or VkImageCreateInfo::flags includes VK_IMAGE_CREATE_EXTENDED_USAGE_BIT.
  - Otherwise supportedVideoFormat is VK_FALSE.

Valid Usage

- VUID-VkImageCreateInfo-imageCreateMaxMipLevels-02251
  Each of the following values (as described in Image Creation Limits) **must** not be undefined : imageCreateMaxMipLevels, imageCreateMaxArrayLayers, imageCreateMaxExtent, and imageCreateSampleCounts
- VUID-VkImageCreateInfo-sharingMode-00941
  If sharingMode is VK_SHARING_MODE_CONCURRENT, pQueueFamilyIndices **must** be a valid pointer to an array of queueFamilyIndexCount uint32_t values
- VUID-VkImageCreateInfo-maintenance11-13354
  If the maintenance11 feature is enabled and sharingMode is VK_SHARING_MODE_CONCURRENT, then queueFamilyIndexCount **must** be greater than 0
- VUID-VkImageCreateInfo-sharingMode-00942
  If the maintenance11 feature is not enabled and sharingMode is VK_SHARING_MODE_CONCURRENT, then queueFamilyIndexCount **must** be greater than 1
- VUID-VkImageCreateInfo-sharingMode-01420
  If sharingMode is VK_SHARING_MODE_CONCURRENT, each element of pQueueFamilyIndices **must** be unique and **must** be less than pQueueFamilyPropertyCount returned by either vkGetPhysicalDeviceQueueFamilyProperties or vkGetPhysicalDeviceQueueFamilyProperties2 for the physicalDevice that was used to create device
- VUID-VkImageCreateInfo-format-00943
  format **must** not be VK_FORMAT_UNDEFINED
- VUID-VkImageCreateInfo-extent-00944
  extent.width **must** be greater than 0

- VUID-VkImageCreateInfo-extent-00945
  `extent.height` **must** be greater than `0`
- VUID-VkImageCreateInfo-extent-00946
  `extent.depth` **must** be greater than `0`
- VUID-VkImageCreateInfo-mipLevels-00947
  `mipLevels` **must** be greater than `0`
- VUID-VkImageCreateInfo-arrayLayers-00948
  `arrayLayers` **must** be greater than `0`
- VUID-VkImageCreateInfo-flags-00949
  If `flags` contains VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT, `imageType` **must** be VK_IMAGE_TYPE_2D
- VUID-VkImageCreateInfo-flags-08865
  If `flags` contains VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT, `extent.width` and `extent.height` **must** be equal
- VUID-VkImageCreateInfo-flags-08866
  If `flags` contains VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT, `arrayLayers` **must** be greater than or equal to 6
- VUID-VkImageCreateInfo-initialLayout-10765
  If the zeroInitializeDeviceMemory feature is not enabled, `initialLayout` **must** not be VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT
- VUID-VkImageCreateInfo-flags-02557
  If `usage` contains VK_IMAGE_USAGE_FRAGMENT_DENSITY_MAP_BIT_EXT, `imageType` **must** be VK_IMAGE_TYPE_2D
- VUID-VkImageCreateInfo-flags-00950
  If `flags` contains VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT, `imageType` **must** be VK_IMAGE_TYPE_3D
- VUID-VkImageCreateInfo-flags-09403
  If `flags` contains VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT, `flags` **must** not include VK_IMAGE_CREATE_SPARSE_ALIASED_BIT, VK_IMAGE_CREATE_SPARSE_BINDING_BIT, or VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT
- VUID-VkImageCreateInfo-flags-07755
  If `flags` contains VK_IMAGE_CREATE_2D_VIEW_COMPATIBLE_BIT_EXT, `imageType` **must** be VK_IMAGE_TYPE_3D
- VUID-VkImageCreateInfo-imageType-10197
  If `flags` contains VK_IMAGE_CREATE_2D_VIEW_COMPATIBLE_BIT_EXT and either the `maintenance9` feature is not enabled on the device or `image2DViewOf3DSparse` is VK_FALSE , `flags` **must** not include VK_IMAGE_CREATE_SPARSE_ALIASED_BIT, VK_IMAGE_CREATE_SPARSE_BINDING_BIT, or VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT
- VUID-VkImageCreateInfo-extent-02252
  `extent.width` **must** be less than or equal to `imageCreateMaxExtent.width` (as defined in Image Creation Limits)

- VUID-VkImageCreateInfo-extent-02253
  `extent.height` **must** be less than or equal to `imageCreateMaxExtent.height` (as defined in Image Creation Limits)
- VUID-VkImageCreateInfo-extent-02254
  `extent.depth` **must** be less than or equal to `imageCreateMaxExtent.depth` (as defined in Image Creation Limits)
- VUID-VkImageCreateInfo-imageType-00956
  If `imageType` is VK_IMAGE_TYPE_1D, both `extent.height` and `extent.depth` **must** be `1`
- VUID-VkImageCreateInfo-imageType-00957
  If `imageType` is VK_IMAGE_TYPE_2D, `extent.depth` **must** be `1`
- VUID-VkImageCreateInfo-mipLevels-00958
  `mipLevels` **must** be less than or equal to the number of levels in the complete mipmap chain based on `extent.width`, `extent.height`, and `extent.depth`
- VUID-VkImageCreateInfo-mipLevels-02255
  `mipLevels` **must** be less than or equal to `imageCreateMaxMipLevels` (as defined in Image Creation Limits)
- VUID-VkImageCreateInfo-arrayLayers-02256
  `arrayLayers` **must** be less than or equal to `imageCreateMaxArrayLayers` (as defined in Image Creation Limits)
- VUID-VkImageCreateInfo-imageType-00961
  If `imageType` is VK_IMAGE_TYPE_3D, `arrayLayers` **must** be `1`
- VUID-VkImageCreateInfo-samples-02257
  If `samples` is not VK_SAMPLE_COUNT_1_BIT, then `imageType` **must** be VK_IMAGE_TYPE_2D, `flags` **must** not contain VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT, `mipLevels` **must** be equal to `1`, and `imageCreateMaybeLinear` (as defined in Image Creation Limits) **must** be VK_FALSE
- VUID-VkImageCreateInfo-samples-02558
  If `samples` is not VK_SAMPLE_COUNT_1_BIT, `usage` **must** not contain VK_IMAGE_USAGE_FRAGMENT_DENSITY_MAP_BIT_EXT
- VUID-VkImageCreateInfo-usage-00963
  If `usage` includes VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT, then bits other than VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT, VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, and VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT **must** not be set
- VUID-VkImageCreateInfo-usage-00964
  If `usage` includes VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT, VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT, or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT, `extent.width` **must** be less than or equal to `VkPhysicalDeviceLimits::maxFramebufferWidth`
- VUID-VkImageCreateInfo-usage-00965
  If `usage` includes VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT, VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT,

VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT, or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT, extent.height **must** be less than or equal to VkPhysicalDeviceLimits::maxFramebufferHeight

- VUID-VkImageCreateInfo-usage-02559
  If usage includes VK_IMAGE_USAGE_FRAGMENT_DENSITY_MAP_BIT_EXT, extent.width **must** be less than or equal to $\lceil \frac{maxFramebufferWidth}{minFragmentDensityTexelSize_{width}} \rceil$
- VUID-VkImageCreateInfo-usage-02560
  If usage includes VK_IMAGE_USAGE_FRAGMENT_DENSITY_MAP_BIT_EXT, extent.height **must** be less than or equal to $\lceil \frac{maxFramebufferHeight}{minFragmentDensityTexelSize_{height}} \rceil$
- VUID-VkImageCreateInfo-usage-00966
  If usage includes VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT, usage **must** also contain at least one of VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT, VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT
- VUID-VkImageCreateInfo-samples-02258
  samples **must** be a valid VkSampleCountFlagBits value that is set in imageCreateSampleCounts (as defined in Image Creation Limits)
- VUID-VkImageCreateInfo-usage-00968
  If the shaderStorageImageMultisample feature is not enabled, and usage contains VK_IMAGE_USAGE_STORAGE_BIT, samples **must** be VK_SAMPLE_COUNT_1_BIT
- VUID-VkImageCreateInfo-flags-00969
  If the sparseBinding feature is not enabled, flags **must** not contain VK_IMAGE_CREATE_SPARSE_BINDING_BIT
- VUID-VkImageCreateInfo-flags-01924
  If the sparseResidencyAliased feature is not enabled, flags **must** not contain VK_IMAGE_CREATE_SPARSE_ALIASED_BIT
- VUID-VkImageCreateInfo-tiling-04121
  If tiling is VK_IMAGE_TILING_LINEAR, flags **must** not contain VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT
- VUID-VkImageCreateInfo-imageType-00970
  If imageType is VK_IMAGE_TYPE_1D, flags **must** not contain VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT
- VUID-VkImageCreateInfo-imageType-00971
  If the sparseResidencyImage2D feature is not enabled, and imageType is VK_IMAGE_TYPE_2D, flags **must** not contain VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT
- VUID-VkImageCreateInfo-imageType-00972
  If the sparseResidencyImage3D feature is not enabled, and imageType is VK_IMAGE_TYPE_3D, flags **must** not contain VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT
- VUID-VkImageCreateInfo-imageType-00973
  If the sparseResidency2Samples feature is not enabled, imageType is VK_IMAGE_TYPE_2D, and samples is VK_SAMPLE_COUNT_2_BIT, flags **must** not contain VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT
- VUID-VkImageCreateInfo-imageType-00974

If the sparseResidency4Samples feature is not enabled, imageType is VK_IMAGE_TYPE_2D, and samples is VK_SAMPLE_COUNT_4_BIT, flags **must** not contain VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT

- VUID-VkImageCreateInfo-imageType-00975
  If the sparseResidency8Samples feature is not enabled, imageType is VK_IMAGE_TYPE_2D, and samples is VK_SAMPLE_COUNT_8_BIT, flags **must** not contain VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT
- VUID-VkImageCreateInfo-imageType-00976
  If the sparseResidency16Samples feature is not enabled, imageType is VK_IMAGE_TYPE_2D, and samples is VK_SAMPLE_COUNT_16_BIT, flags **must** not contain VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT
- VUID-VkImageCreateInfo-flags-00987
  If flags contains VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT or VK_IMAGE_CREATE_SPARSE_ALIASED_BIT, it **must** also contain VK_IMAGE_CREATE_SPARSE_BINDING_BIT
- VUID-VkImageCreateInfo-None-01925
  If any of the bits VK_IMAGE_CREATE_SPARSE_BINDING_BIT, VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT, or VK_IMAGE_CREATE_SPARSE_ALIASED_BIT are set, VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT **must** not also be set
- VUID-VkImageCreateInfo-flags-01890
  If the protectedMemory feature is not enabled, flags **must** not contain VK_IMAGE_CREATE_PROTECTED_BIT
- VUID-VkImageCreateInfo-None-01891
  If any of the bits VK_IMAGE_CREATE_SPARSE_BINDING_BIT, VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT, or VK_IMAGE_CREATE_SPARSE_ALIASED_BIT are set, VK_IMAGE_CREATE_PROTECTED_BIT **must** not also be set
- VUID-VkImageCreateInfo-pNext-00990
  If the pNext chain includes a VkExternalMemoryImageCreateInfo structure, its handleTypes member **must** only contain bits that are also in VkExternalImageFormatProperties::externalMemoryProperties.compatibleHandleTypes, as returned by vkGetPhysicalDeviceImageFormatProperties2 with format, imageType, tiling, usage, and flags equal to those in this structure, and with a VkPhysicalDeviceExternalImageFormatInfo structure included in the pNext chain, with a handleType equal to any one of the handle types specified in VkExternalMemoryImageCreateInfo::handleTypes
- VUID-VkImageCreateInfo-physicalDeviceCount-01421
  If the logical device was created with VkDeviceGroupDeviceCreateInfo::physicalDeviceCount equal to 1, flags **must** not contain VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT
- VUID-VkImageCreateInfo-flags-02259
  If flags contains VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT, then mipLevels **must** be one, arrayLayers **must** be one, imageType **must** be VK_IMAGE_TYPE_2D,

and imageCreateMaybeLinear (as defined in Image Creation Limits) **must** be VK_FALSE

- VUID-VkImageCreateInfo-flags-01572
  If flags contains VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT, then format **must** be a compressed image format
- VUID-VkImageCreateInfo-flags-01573
  If flags contains VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT, then flags **must** also contain VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT
- VUID-VkImageCreateInfo-initialLayout-00993
  initialLayout **must** be VK_IMAGE_LAYOUT_UNDEFINED, VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT, or VK_IMAGE_LAYOUT_PREINITIALIZED
- VUID-VkImageCreateInfo-initialLayout-12478
  If initialLayout is VK_IMAGE_LAYOUT_PREINITIALIZED, tiling **must** be VK_IMAGE_TILING_LINEAR
- VUID-VkImageCreateInfo-pNext-01443
  If the pNext chain includes a VkExternalMemoryImageCreateInfo or VkExternalMemoryImageCreateInfoNV structure whose handleTypes member is not 0, initialLayout **must** be VK_IMAGE_LAYOUT_UNDEFINED
- VUID-VkImageCreateInfo-format-06410
  If the image format is one of the formats that require a sampler $Y'C_BC_R$ conversion, mipLevels **must** be 1
- VUID-VkImageCreateInfo-format-06411
  If the image format is one of the formats that require a sampler $Y'C_BC_R$ conversion, samples **must** be VK_SAMPLE_COUNT_1_BIT
- VUID-VkImageCreateInfo-format-06412
  If the image format is one of the formats that require a sampler $Y'C_BC_R$ conversion, imageType **must** be VK_IMAGE_TYPE_2D
- VUID-VkImageCreateInfo-imageCreateFormatFeatures-02260
  If format is a *multi-planar* format, and if imageCreateFormatFeatures (as defined in Image Creation Limits) does not contain VK_FORMAT_FEATURE_DISJOINT_BIT, then flags **must** not contain VK_IMAGE_CREATE_DISJOINT_BIT
- VUID-VkImageCreateInfo-format-01577
  If format is not a *multi-planar* format, and flags does not include VK_IMAGE_CREATE_ALIAS_BIT, flags **must** not contain VK_IMAGE_CREATE_DISJOINT_BIT
- VUID-VkImageCreateInfo-format-04712
  If format has a _422 or _420 suffix, extent.width **must** be a multiple of 2
- VUID-VkImageCreateInfo-format-04713
  If format has a _420 suffix, extent.height **must** be a multiple of 2
- VUID-VkImageCreateInfo-tiling-02261
  If tiling is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then the pNext chain **must** include exactly one of VkImageDrmFormatModifierListCreateInfoEXT or VkImageDrmFormatModifierExplicitCreateInfoEXT structures

- VUID-VkImageCreateInfo-pNext-02262
  If the pNext chain includes a VkImageDrmFormatModifierListCreateInfoEXT or VkImageDrmFormatModifierExplicitCreateInfoEXT structure, then tiling **must** be VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT
- VUID-VkImageCreateInfo-tiling-02353
  If tiling is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT and flags contains VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT, then the pNext chain **must** include a VkImageFormatListCreateInfo structure with non-zero viewFormatCount
- VUID-VkImageCreateInfo-flags-01533
  If flags contains VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT, format **must** be a depth or depth/stencil format
- VUID-VkImageCreateInfo-format-02795
  If format is a depth-stencil format, usage includes VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, and the pNext chain includes a VkImageStencilUsageCreateInfo structure, then its VkImageStencilUsageCreateInfo ::stencilUsage member **must** also include VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkImageCreateInfo-format-02796
  If format is a depth-stencil format, usage does not include VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, and the pNext chain includes a VkImageStencilUsageCreateInfo structure, then its VkImageStencilUsageCreateInfo ::stencilUsage member **must** also not include VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkImageCreateInfo-format-02797
  If format is a depth-stencil format, usage includes VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT, and the pNext chain includes a VkImageStencilUsageCreateInfo structure, then its VkImageStencilUsageCreateInfo ::stencilUsage member **must** also include VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT
- VUID-VkImageCreateInfo-format-02798
  If format is a depth-stencil format, usage does not include VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT, and the pNext chain includes a VkImageStencilUsageCreateInfo structure, then its VkImageStencilUsageCreateInfo ::stencilUsage member **must** also not include VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT
- VUID-VkImageCreateInfo-Format-02536
  If format is a depth-stencil format and the pNext chain includes a VkImageStencilUsageCreateInfo structure with its stencilUsage member including VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT, extent.width **must** be less than or equal to VkPhysicalDeviceLimits::maxFramebufferWidth
- VUID-VkImageCreateInfo-format-02537
  If format is a depth-stencil format and the pNext chain includes a VkImageStencilUsageCreateInfo structure with its stencilUsage member including VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT, extent.height **must** be less than or equal to

VkPhysicalDeviceLimits::maxFramebufferHeight

- VUID-VkImageCreateInfo-format-02538
  If the shaderStorageImageMultisample feature is not enabled, format is a depth-stencil format and the pNext chain includes a VkImageStencilUsageCreateInfo structure with its stencilUsage including VK_IMAGE_USAGE_STORAGE_BIT, samples **must** be VK_SAMPLE_COUNT_1_BIT
- VUID-VkImageCreateInfo-imageType-02082
  If usage includes VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR, imageType **must** be VK_IMAGE_TYPE_2D
- VUID-VkImageCreateInfo-samples-02083
  If usage includes VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR, samples **must** be VK_SAMPLE_COUNT_1_BIT
- VUID-VkImageCreateInfo-flags-02565
  If flags contains VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT, tiling **must** be VK_IMAGE_TILING_OPTIMAL
- VUID-VkImageCreateInfo-flags-02566
  If flags contains VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT, imageType **must** be VK_IMAGE_TYPE_2D
- VUID-VkImageCreateInfo-flags-02567
  If flags contains VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT, flags **must** not contain VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT
- VUID-VkImageCreateInfo-flags-02568
  If flags contains VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT, mipLevels **must** be 1
- VUID-VkImageCreateInfo-imageView2DOn3DImage-04459
  If the VK_KHR_portability_subset extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::imageView2DOn3DImage is VK_FALSE, flags **must** not contain VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT
- VUID-VkImageCreateInfo-multisampleArrayImage-04460
  If the VK_KHR_portability_subset extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::multisampleArrayImage is VK_FALSE, and samples is not VK_SAMPLE_COUNT_1_BIT, then arrayLayers **must** be 1
- VUID-VkImageCreateInfo-pNext-06722
  If a VkImageFormatListCreateInfo structure was included in the pNext chain and format is not a multi-planar format and VkImageFormatListCreateInfo::viewFormatCount is not zero, then each format in VkImageFormatListCreateInfo::pViewFormats **must** either be compatible with the format as described in the compatibility table or, if flags contains VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT, be an uncompressed format that is size-compatible with format
- VUID-VkImageCreateInfo-pNext-10062
  If a VkImageFormatListCreateInfo structure was included in the pNext chain and format is a multi-planar format and flags contains VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT

and VkImageFormatListCreateInfo::viewFormatCount is not zero, then each format in VkImageFormatListCreateInfo::pViewFormats **must** be compatible with the VkFormat for the plane of the image format

- VUID-VkImageCreateInfo-flags-04738
  If flags does not contain VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT and the pNext chain includes a VkImageFormatListCreateInfo structure, then VkImageFormatListCreateInfo ::viewFormatCount **must** be 0 or 1
- VUID-VkImageCreateInfo-usage-04815
  If usage includes VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR, VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR, or VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR, and flags does not include VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR, then the pNext chain **must** include a VkVideoProfileListInfoKHR structure with profileCount greater than 0 and pProfiles including at least one VkVideoProfileInfoKHR structure with a videoCodecOperation member specifying a decode operation
- VUID-VkImageCreateInfo-usage-04816
  If usage includes VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR, VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR, or VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR, and flags does not include VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR, then the pNext chain **must** include a VkVideoProfileListInfoKHR structure with profileCount greater than 0 and pProfiles including at least one VkVideoProfileInfoKHR structure with a videoCodecOperation member specifying an encode operation
- VUID-VkImageCreateInfo-flags-08328
  If flags includes VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR, then videoMaintenance1 **must** be enabled
- VUID-VkImageCreateInfo-flags-08329
  If flags includes VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR and usage does not include VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR, then usage **must** not include VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR
- VUID-VkImageCreateInfo-flags-08331
  If flags includes VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR, then usage **must** not include VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR, VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR, or VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR
- VUID-VkImageCreateInfo-pNext-06811
  If the pNext chain includes a VkVideoProfileListInfoKHR structure with profileCount greater than 0, then supportedVideoFormat **must** be VK_TRUE
- VUID-VkImageCreateInfo-pNext-10784
  If the pNext chain includes a VkVideoProfileListInfoKHR structure and for any element of its pProfiles member videoCodecOperation is VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR, then the videoDecodeVP9 feature **must** be enabled

- VUID-VkImageCreateInfo-pNext-10250
  If the pNext chain includes a VkVideoProfileListInfoKHR structure and for any element of its pProfiles member videoCodecOperation is VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then the videoEncodeAV1 feature **must** be enabled
- VUID-VkImageCreateInfo-usage-10251
  If usage includes VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR, then the videoEncodeQuantizationMap feature **must** be enabled
- VUID-VkImageCreateInfo-usage-10252
  If usage includes VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR, imageType **must** be VK_IMAGE_TYPE_2D
- VUID-VkImageCreateInfo-usage-10253
  If usage includes VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR, samples **must** be VK_SAMPLE_COUNT_1_BIT
- VUID-VkImageCreateInfo-usage-10254
  If usage includes VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR, then the pNext chain **must** include a VkVideoProfileListInfoKHR structure with profileCount equal to 1 and pProfiles pointing to a VkVideoProfileInfoKHR structure with a videoCodecOperation member specifying an encode operation
- VUID-VkImageCreateInfo-usage-10255
  If usage includes VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR, then VkVideoEncodeCapabilitiesKHR::flags **must** include VK_VIDEO_ENCODE_CAPABILITY_QUANTIZATION_DELTA_MAP_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified in the pProfiles member of the VkVideoProfileListInfoKHR structure included in the pNext chain
- VUID-VkImageCreateInfo-usage-10256
  If usage includes VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR, then VkVideoEncodeCapabilitiesKHR::flags **must** include VK_VIDEO_ENCODE_CAPABILITY_EMPHASIS_MAP_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified in the pProfiles member of the VkVideoProfileListInfoKHR structure included in the pNext chain
- VUID-VkImageCreateInfo-usage-10257
  If usage includes VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR, extent.width **must** be less than or equal to VkVideoEncodeQuantizationMapCapabilitiesKHR

::maxQuantizationMapExtent.width, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified in the pProfiles member of the VkVideoProfileListInfoKHR structure included in the pNext chain

- VUID-VkImageCreateInfo-usage-10258
  If usage includes VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR, extent.height **must** be less than or equal to VkVideoEncodeQuantizationMapCapabilitiesKHR::maxQuantizationMapExtent.height, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified in the pProfiles member of the VkVideoProfileListInfoKHR structure included in the pNext chain
- VUID-VkImageCreateInfo-multisampledRenderToSingleSampled-06882
  If the multisampledRenderToSingleSampled feature is not enabled, flags **must** not contain VK_IMAGE_CREATE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT
- VUID-VkImageCreateInfo-flags-06883
  If flags contains VK_IMAGE_CREATE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT, samples **must** be VK_SAMPLE_COUNT_1_BIT
- VUID-VkImageCreateInfo-pNext-06743
  If the pNext chain includes a VkImageCompressionControlEXT structure, format is a multi-planar format, and VkImageCompressionControlEXT::flags includes VK_IMAGE_COMPRESSION_FIXED_RATE_EXPLICIT_EXT, then VkImageCompressionControlEXT::compressionControlPlaneCount **must** be equal to the number of planes in format
- VUID-VkImageCreateInfo-pNext-06744
  If the pNext chain includes a VkImageCompressionControlEXT structure, format is not a multi-planar format, and VkImageCompressionControlEXT::flags includes VK_IMAGE_COMPRESSION_FIXED_RATE_EXPLICIT_EXT, then VkImageCompressionControlEXT::compressionControlPlaneCount **must** be 1
- VUID-VkImageCreateInfo-pNext-06746
  If the pNext chain includes a VkImageCompressionControlEXT structure, it **must** not contain a VkImageDrmFormatModifierExplicitCreateInfoEXT structure
- VUID-VkImageCreateInfo-flags-08104
  If flags includes VK_IMAGE_CREATE_DESCRIPTOR_HEAP_CAPTURE_REPLAY_BIT_EXT, the descriptorBufferCaptureReplay or descriptorHeapCaptureReplay feature **must** be enabled
- VUID-VkImageCreateInfo-pNext-08105
  If the pNext chain includes a VkOpaqueCaptureDescriptorDataCreateInfoEXT structure, flags **must** contain VK_IMAGE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT
- VUID-VkImageCreateInfo-pNext-06783
  If the pNext chain includes a VkExportMetalObjectCreateInfoEXT structure, its exportObjectType member **must** be either VK_EXPORT_METAL_OBJECT_TYPE_METAL_TEXTURE_BIT_EXT or VK_EXPORT_METAL_OBJECT_TYPE_METAL_IOSURFACE_BIT_EXT

- VUID-VkImageCreateInfo-pNext-06784
  If the pNext chain includes a VkImportMetalTextureInfoEXT structure, its plane member **must** be VK_IMAGE_ASPECT_PLANE_0_BIT, VK_IMAGE_ASPECT_PLANE_1_BIT, or VK_IMAGE_ASPECT_PLANE_2_BIT
- VUID-VkImageCreateInfo-pNext-06785
  If the pNext chain includes a VkImportMetalTextureInfoEXT structure and the image does not have a multi-planar format, then VkImportMetalTextureInfoEXT::plane **must** be VK_IMAGE_ASPECT_PLANE_0_BIT
- VUID-VkImageCreateInfo-pNext-06786
  If the pNext chain includes a VkImportMetalTextureInfoEXT structure and the image has a multi-planar format with only two planes, then VkImportMetalTextureInfoEXT::plane **must** not be VK_IMAGE_ASPECT_PLANE_2_BIT
- VUID-VkImageCreateInfo-imageCreateFormatFeatures-09048
  If imageCreateFormatFeatures (as defined in Image Creation Limits) does not contain VK_FORMAT_FEATURE_2_HOST_IMAGE_TRANSFER_BIT, then usage **must** not contain VK_IMAGE_USAGE_HOST_TRANSFER_BIT
- VUID-VkImageCreateInfo-usage-10245
  If usage includes VK_IMAGE_USAGE_HOST_TRANSFER_BIT, then the hostImageCopy feature **must** be enabled
- VUID-VkImageCreateInfo-flags-13355
  If flags contains VK_IMAGE_CREATE_ALIAS_SINGLE_LAYER_DESCRIPTOR_BIT_KHR, the maintenance11 feature **must** be enabled
- VUID-VkImageCreateInfo-flags-13356
  If flags contains VK_IMAGE_CREATE_ALIAS_SINGLE_LAYER_DESCRIPTOR_BIT_KHR, imageType **must** be VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D
- VUID-VkImageCreateInfo-flags-11281
  If VkOpaqueCaptureDataCreateInfoEXT::pData is not NULL, flags **must** contain VK_IMAGE_CREATE_DESCRIPTOR_HEAP_CAPTURE_REPLAY_BIT_EXT
- VUID-VkImageCreateInfo-pData-11286
  If flags contains VK_IMAGE_CREATE_DESCRIPTOR_HEAP_CAPTURE_REPLAY_BIT_EXT, and VkOpaqueCaptureDataCreateInfoEXT::pData is not NULL, VkOpaqueCaptureDataCreateInfoEXT::pData->size **must** be equal to imageCaptureReplayOpaqueDataSize
- VUID-VkImageCreateInfo-pNext-12480
  If the pNext chain includes a VkImageTilingControlCreateInfoEXT structure, then the pNext chain **must** not include a VkImageDrmFormatModifierExplicitCreateInfoEXT structure
- VUID-VkImageCreateInfo-pNext-12442
  pNext **must** contain at most one of the structures VkImageStencilUsageCreateInfo and VkImageStencilUsage2CreateInfoKHR

Valid Usage (Implicit)

- VUID-VkImageCreateInfo-sType-sType

sType **must** be VK_STRUCTURE_TYPE_IMAGE_CREATE_INFO

- VUID-VkImageCreateInfo-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkExportMetalObjectCreateInfoEXT, VkExternalMemoryImageCreateInfo, VkImageCompressionControlEXT, VkImageCreateFlags2CreateInfoKHR, VkImageDrmFormatModifierExplicitCreateInfoEXT, VkImageDrmFormatModifierListCreateInfoEXT, VkImageFormatListCreateInfo, VkImageStencilUsage2CreateInfoKHR, VkImageStencilUsageCreateInfo, VkImageSwapchainCreateInfoKHR, VkImageTilingControlCreateInfoEXT, VkImageUsageFlags2CreateInfoKHR, VkImportMetalIOSurfaceInfoEXT, VkImportMetalTextureInfoEXT, VkOpaqueCaptureDataCreateInfoEXT, VkOpaqueCaptureDescriptorDataCreateInfoEXT, or VkVideoProfileListInfoKHR
- VUID-VkImageCreateInfo-sType-unique
  The sType value of each structure in the pNext chain **must** be unique, with the exception of structures of type VkExportMetalObjectCreateInfoEXT or VkImportMetalTextureInfoEXT
- VUID-VkImageCreateInfo-flags-parameter
  If flags is not 0, flags **must** be a valid combination of VkImageCreateFlagBits values
- VUID-VkImageCreateInfo-imageType-parameter
  imageType **must** be a valid VkImageType value
- VUID-VkImageCreateInfo-format-parameter
  format **must** be a valid VkFormat value
- VUID-VkImageCreateInfo-samples-parameter
  samples **must** be a valid VkSampleCountFlagBits value
- VUID-VkImageCreateInfo-tiling-parameter
  tiling **must** be a valid VkImageTiling value
- VUID-VkImageCreateInfo-usage-parameter
  usage **must** be a valid combination of VkImageUsageFlagBits values
- VUID-VkImageCreateInfo-sharingMode-parameter
  sharingMode **must** be a valid VkSharingMode value
- VUID-VkImageCreateInfo-initialLayout-parameter
  initialLayout **must** be a valid VkImageLayout value

The VkImageCreateFlags2CreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_extended_flags
typedef struct VkImageCreateFlags2CreateInfoKHR {
    VkStructureType           sType;
    void*                     pNext;
    VkImageCreateFlags2KHR    flags;
} VkImageCreateFlags2CreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkImageCreateFlagBits2KHR specifying creation flags of the image.

If this structure is included in the `pNext` chain of an image creation structure, `flags` is used instead of the corresponding `flags` value passed in that creation structure, allowing additional create flags to be specified.

If this structure is included in the `pNext` chain of a query structure including VkImageCreateFlags, the create flags of the image are returned in `flags` of this structure, allowing additional create flags to be queried. The flags representable in VkImageCreateFlags are still returned in the query structure in this case.

Valid Usage (Implicit)

- VUID-VkImageCreateFlags2CreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_CREATE_FLAGS_2_CREATE_INFO_KHR
- VUID-VkImageCreateFlags2CreateInfoKHR-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkImageCreateFlagBits2KHR values

Structure Chaining

Extends the structures

- VkFramebufferAttachmentImageInfo
- VkImageCreateInfo
- VkPhysicalDeviceImageFormatInfo2
- VkVideoFormatPropertiesKHR

The `VkImageUsageFlags2CreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_extended_flags
typedef struct VkImageUsageFlags2CreateInfoKHR {
    VkStructureType          sType;
    void*                    pNext;
    VkImageUsageFlags2KHR    usage;
} VkImageUsageFlags2CreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `usage` is a bitmask of VkImageUsageFlagBits2KHR specifying allowed usages of the image.

If this structure is included in the `pNext` chain of an image creation structure, `usage` is used instead of the corresponding `usage` value passed in that creation structure, allowing additional usage flags to be specified.

If this structure is included in the pNext chain of a query structure including VkImageCreateFlags, the usage flags of the image are returned in flags of this structure, allowing additional usage flags to be queried. The flags representable in VkImageUsageFlags are still returned in the query structure in this case.

Valid Usage (Implicit)

- VUID-VkImageUsageFlags2CreateInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_IMAGE_USAGE_FLAGS_2_CREATE_INFO_KHR
- VUID-VkImageUsageFlags2CreateInfoKHR-usage-parameter
  usage **must** be a valid combination of VkImageUsageFlagBits2KHR values

Structure Chaining

Extends the structures

- VkFramebufferAttachmentImageInfo
- VkImageCreateInfo
- VkPhysicalDeviceImageFormatInfo2
- VkPhysicalDeviceSparseImageFormatInfo2
- VkPhysicalDeviceVideoFormatInfoKHR
- VkSurfaceCapabilities2KHR
- VkSwapchainCreateInfoKHR
- VkVideoFormatPropertiesKHR

The VkImageStencilUsageCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkImageStencilUsageCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkImageUsageFlags    stencilUsage;
} VkImageStencilUsageCreateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- stencilUsage is a bitmask of VkImageUsageFlagBits describing the intended usage of the stencil aspect of the image.

If the pNext chain of VkImageCreateInfo includes a VkImageStencilUsageCreateInfo structure, then that structure includes the usage flags specific to the stencil aspect of the image for an image with a depth-stencil format.

This structure specifies image usages which only apply to the stencil aspect of a depth/stencil format image. When this structure is included in the pNext chain of VkImageCreateInfo, the stencil aspect of the image **must** only be used as specified by stencilUsage. Use of other aspects of an image is unaffected by this structure.

This structure **can** also be included in the pNext chain of VkPhysicalDeviceImageFormatInfo2 to query additional capabilities specific to image creation parameter combinations including a separate set of usage flags for the stencil aspect of the image using vkGetPhysicalDeviceImageFormatProperties2. When this structure is not included in the pNext chain of VkPhysicalDeviceImageFormatInfo2 then the implicit value of stencilUsage matches that of VkPhysicalDeviceImageFormatInfo2::usage.

Valid Usage

- VUID-VkImageStencilUsageCreateInfo-stencilUsage-02539
  If stencilUsage includes VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT, it **must** not include bits other than VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT

Valid Usage (Implicit)

- VUID-VkImageStencilUsageCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_IMAGE_STENCIL_USAGE_CREATE_INFO
- VUID-VkImageStencilUsageCreateInfo-stencilUsage-parameter
  stencilUsage **must** be a valid combination of VkImageUsageFlagBits values

Structure Chaining

Extends the structures

- VkImageCreateInfo
- VkPhysicalDeviceImageFormatInfo2

The VkImageStencilUsage2CreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_extended_flags with VK_VERSION_1_2 or
VK_EXT_separate_stencil_usage
typedef struct VkImageStencilUsage2CreateInfoKHR {
    VkStructureType          sType;
    void*                    pNext;
    VkImageUsageFlags2KHR    stencilUsage;
} VkImageStencilUsage2CreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stencilUsage` is a bitmask of VkImageUsageFlagBits2KHR describing the intended usage of the stencil aspect of the image.

If the `pNext` chain of VkImageCreateInfo includes a `VkImageStencilUsage2CreateInfoKHR` structure, then that structure includes the usage flags specific to the stencil aspect of the image for an image with a depth-stencil format.

This structure specifies image usages which only apply to the stencil aspect of a depth/stencil format image. When this structure is included in the `pNext` chain of VkImageCreateInfo, the stencil aspect of the image **must** only be used as specified by `stencilUsage`. Use of other aspects of an image is unaffected by this structure.

This structure **can** also be included in the `pNext` chain of VkPhysicalDeviceImageFormatInfo2 to query additional capabilities specific to image creation parameter combinations including a separate set of usage flags for the stencil aspect of the image using vkGetPhysicalDeviceImageFormatProperties2. When this structure is not included in the `pNext` chain of `VkPhysicalDeviceImageFormatInfo2` then the implicit value of `stencilUsage` matches that of `VkPhysicalDeviceImageFormatInfo2`::`usage`.

Valid Usage

- VUID-VkImageStencilUsage2CreateInfoKHR-stencilUsage-12443
  If `stencilUsage` includes VK_IMAGE_USAGE_2_TRANSIENT_ATTACHMENT_BIT_KHR, it **must** not include bits other than VK_IMAGE_USAGE_2_DEPTH_STENCIL_ATTACHMENT_BIT_KHR or VK_IMAGE_USAGE_2_INPUT_ATTACHMENT_BIT_KHR

Valid Usage (Implicit)

- VUID-VkImageStencilUsage2CreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_STENCIL_USAGE_2_CREATE_INFO_KHR
- VUID-VkImageStencilUsage2CreateInfoKHR-stencilUsage-parameter
  `stencilUsage` **must** be a valid combination of VkImageUsageFlagBits2KHR values

Structure Chaining

Extends the structures

- VkImageCreateInfo
- VkPhysicalDeviceImageFormatInfo2

To define a set of external memory handle types that **may** be used as backing store for an image, add a VkExternalMemoryImageCreateInfo structure to the `pNext` chain of the VkImageCreateInfo structure. The `VkExternalMemoryImageCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkExternalMemoryImageCreateInfo {
    VkStructureType                       sType;
    const void*                           pNext;
    VkExternalMemoryHandleTypeFlags       handleTypes;
} VkExternalMemoryImageCreateInfo;
```

```
// Provided by VK_KHR_external_memory
// Equivalent to VkExternalMemoryImageCreateInfo
typedef VkExternalMemoryImageCreateInfo VkExternalMemoryImageCreateInfoKHR;
```

NOTE	A VkExternalMemoryImageCreateInfo structure with a non-zero handleTypes field must be included in the creation parameters for an image that will be bound to memory that is either exported or imported.

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- handleTypes is zero or a bitmask of VkExternalMemoryHandleTypeFlagBits specifying one or more external memory handle types.

Valid Usage (Implicit)

- VUID-VkExternalMemoryImageCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_IMAGE_CREATE_INFO
- VUID-VkExternalMemoryImageCreateInfo-handleTypes-parameter
  If handleTypes is not 0, handleTypes **must** be a valid combination of VkExternalMemoryHandleTypeFlagBits values

Structure Chaining

Extends the structure

- VkImageCreateInfo

If the pNext chain of VkImageCreateInfo includes a VkImageSwapchainCreateInfoKHR structure, then that structure includes a swapchain handle indicating that the image will be bound to memory from that swapchain.

The VkImageSwapchainCreateInfoKHR structure is defined as:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
typedef struct VkImageSwapchainCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkSwapchainKHR     swapchain;
} VkImageSwapchainCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `swapchain` is VK_NULL_HANDLE or a handle of a swapchain that the image will be bound to.

Valid Usage

- VUID-VkImageSwapchainCreateInfoKHR-swapchain-00995
  If `swapchain` is not VK_NULL_HANDLE, the fields of VkImageCreateInfo **must** match the implied image creation parameters of the swapchain

Valid Usage (Implicit)

- VUID-VkImageSwapchainCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_SWAPCHAIN_CREATE_INFO_KHR
- VUID-VkImageSwapchainCreateInfoKHR-swapchain-parameter
  If `swapchain` is not VK_NULL_HANDLE, `swapchain` **must** be a valid VkSwapchainKHR handle

Structure Chaining

Extends the structure

- VkImageCreateInfo

If the `pNext` chain of VkImageCreateInfo includes a `VkImageFormatListCreateInfo` structure, then that structure contains a list of all formats that **can** be used when creating views of this image.

The `VkImageFormatListCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkImageFormatListCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           viewFormatCount;
    const VkFormat*    pViewFormats;
} VkImageFormatListCreateInfo;
```

```
// Provided by VK_KHR_image_format_list
// Equivalent to VkImageFormatListCreateInfo
typedef VkImageFormatListCreateInfo VkImageFormatListCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `viewFormatCount` is the number of entries in the `pViewFormats` array.
- `pViewFormats` is a pointer to an array of VkFormat values specifying all formats which **can** be used when creating views of this image.

If `viewFormatCount` is zero, `pViewFormats` is ignored and the image is created as if the `VkImageFormatListCreateInfo` structure were not included in the `pNext` chain of VkImageCreateInfo.

Valid Usage

- VUID-VkImageFormatListCreateInfo-viewFormatCount-09540
  If `viewFormatCount` is not 0, each element of `pViewFormats` **must** not be VK_FORMAT_UNDEFINED

Valid Usage (Implicit)

- VUID-VkImageFormatListCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_FORMAT_LIST_CREATE_INFO
- VUID-VkImageFormatListCreateInfo-pViewFormats-parameter
  If `viewFormatCount` is not `0`, `pViewFormats` **must** be a valid pointer to an array of `viewFormatCount` valid VkFormat values

Structure Chaining

Extends the structures

- VkImageCreateInfo
- VkPhysicalDeviceImageFormatInfo2
- VkSwapchainCreateInfoKHR

If the pNext chain of VkImageCreateInfo includes a VkImageDrmFormatModifierListCreateInfoEXT structure, then the image will be created with one of the Linux DRM format modifiers listed in the structure. The choice of modifier **can** be influenced by also including a VkImageTilingControlCreateInfoEXT structure in the pNext chain, but otherwise, it is implementation-dependent.

The VkImageDrmFormatModifierListCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier
typedef struct VkImageDrmFormatModifierListCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           drmFormatModifierCount;
    const uint64_t*    pDrmFormatModifiers;
} VkImageDrmFormatModifierListCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- drmFormatModifierCount is the length of the pDrmFormatModifiers array.
- pDrmFormatModifiers is a pointer to an array of *Linux DRM format modifiers*.

Valid Usage

- VUID-VkImageDrmFormatModifierListCreateInfoEXT-pDrmFormatModifiers-02263
  Each *modifier* in pDrmFormatModifiers **must** be compatible with the parameters in VkImageCreateInfo and its pNext chain, as determined by querying VkPhysicalDeviceImageFormatInfo2 extended with VkPhysicalDeviceImageDrmFormatModifierInfoEXT

Valid Usage (Implicit)

- VUID-VkImageDrmFormatModifierListCreateInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_LIST_CREATE_INFO_EXT
- VUID-VkImageDrmFormatModifierListCreateInfoEXT-pDrmFormatModifiers-parameter
  pDrmFormatModifiers **must** be a valid pointer to an array of drmFormatModifierCount uint64_t values
- VUID-VkImageDrmFormatModifierListCreateInfoEXT-drmFormatModifierCount-arraylength
  drmFormatModifierCount **must** be greater than 0

Structure Chaining

Extends the structure

- VkImageCreateInfo

If the `pNext` chain of VkImageCreateInfo includes a VkImageDrmFormatModifierExplicitCreateInfoEXT structure, then the image will be created with the Linux DRM format modifier and memory layout defined by the structure.

The VkImageDrmFormatModifierExplicitCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier
typedef struct VkImageDrmFormatModifierExplicitCreateInfoEXT {
    VkStructureType                sType;
    const void*                    pNext;
    uint64_t                       drmFormatModifier;
    uint32_t                       drmFormatModifierPlaneCount;
    const VkSubresourceLayout*     pPlaneLayouts;
} VkImageDrmFormatModifierExplicitCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `drmFormatModifier` is the *Linux DRM format modifier* with which the image will be created.
- `drmFormatModifierPlaneCount` is the number of *memory planes* in the image (as reported by VkDrmFormatModifierPropertiesEXT) as well as the length of the `pPlaneLayouts` array.
- `pPlaneLayouts` is a pointer to an array of VkSubresourceLayout structures describing the image's *memory planes*.

The i^{th} member of `pPlaneLayouts` describes the layout of the image's i^{th} *memory plane* (that is, `VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT`). In each element of `pPlaneLayouts`, the implementation **must** ignore `size`. The implementation calculates the size of each plane, which the application **can** query with vkGetImageSubresourceLayout.

When creating an image with VkImageDrmFormatModifierExplicitCreateInfoEXT, it is the application's responsibility to satisfy all valid usage requirements. However, the implementation **must** validate that the provided `pPlaneLayouts`, when combined with the provided `drmFormatModifier` and other creation parameters in VkImageCreateInfo and its `pNext` chain, produce a valid image. (This validation is necessarily implementation-dependent and outside the scope of Vulkan, and therefore not described by valid usage requirements). If this validation fails, then vkCreateImage returns VK_ERROR_INVALID_DRM_FORMAT_MODIFIER_PLANE_LAYOUT_EXT.

Valid Usage

- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-drmFormatModifier-02264
  `drmFormatModifier` **must** be compatible with the parameters in VkImageCreateInfo and its

pNext chain, as determined by querying VkPhysicalDeviceImageFormatInfo2 extended with VkPhysicalDeviceImageDrmFormatModifierInfoEXT

- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-drmFormatModifierPlaneCount-02265
  drmFormatModifierPlaneCount **must** be equal to the VkDrmFormatModifierPropertiesEXT::drmFormatModifierPlaneCount associated with VkImageCreateInfo::format and drmFormatModifier, as found by querying VkDrmFormatModifierPropertiesListEXT
- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-size-02267
  For each element of pPlaneLayouts, size **must** be 0
- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-arrayPitch-02268
  For each element of pPlaneLayouts, arrayPitch **must** be 0 if VkImageCreateInfo::arrayLayers is 1
- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-depthPitch-02269
  For each element of pPlaneLayouts, depthPitch **must** be 0 if VkImageCreateInfo::extent.depth is 1

Valid Usage (Implicit)

- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_EXPLICIT_CREATE_INFO_EXT
- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-pPlaneLayouts-parameter
  pPlaneLayouts **must** be a valid pointer to an array of drmFormatModifierPlaneCount VkSubresourceLayout structures
- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-drmFormatModifierPlaneCount-arraylength
  drmFormatModifierPlaneCount **must** be greater than 0

Structure Chaining

Extends the structure

- VkImageCreateInfo

If the pNext list of VkImageCreateInfo includes a VkImageCompressionControlEXT structure, then that structure describes compression controls for this image.

The VkImageCompressionControlEXT structure is defined as:

```
// Provided by VK_EXT_image_compression_control
typedef struct VkImageCompressionControlEXT {
    VkStructureType                          sType;
    const void*                              pNext;
    VkImageCompressionFlagsEXT               flags;
    uint32_t                                 compressionControlPlaneCount;
    VkImageCompressionFixedRateFlagsEXT*     pFixedRateFlags;
} VkImageCompressionControlEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkImageCompressionFlagBitsEXT describing compression controls for the image.
- `compressionControlPlaneCount` is the number of entries in the `pFixedRateFlags` array.
- `pFixedRateFlags` is `NULL` or a pointer to an array of VkImageCompressionFixedRateFlagsEXT bitfields describing allowed fixed-rate compression rates of each image plane. It is ignored if `flags` does not include VK_IMAGE_COMPRESSION_FIXED_RATE_EXPLICIT_EXT.

If enabled, fixed-rate compression is done in an implementation-defined manner and **may** be applied at block granularity. In that case, a write to an individual texel **may** modify the value of other texels in the same block.

Valid Usage

- VUID-VkImageCompressionControlEXT-flags-06747
  `flags` **must** be one of VK_IMAGE_COMPRESSION_DEFAULT_EXT, VK_IMAGE_COMPRESSION_FIXED_RATE_DEFAULT_EXT, VK_IMAGE_COMPRESSION_FIXED_RATE_EXPLICIT_EXT, or VK_IMAGE_COMPRESSION_DISABLED_EXT
- VUID-VkImageCompressionControlEXT-flags-06748
  If `flags` includes VK_IMAGE_COMPRESSION_FIXED_RATE_EXPLICIT_EXT, `pFixedRateFlags` **must** not be `NULL`

Valid Usage (Implicit)

- VUID-VkImageCompressionControlEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_COMPRESSION_CONTROL_EXT

Structure Chaining

Extends the structures

- VkImageCreateInfo

- VkPhysicalDeviceImageFormatInfo2
- VkSwapchainCreateInfoKHR

NOTE Some combinations of compression properties may not be supported. For example, some implementations may not support different fixed-rate compression rates per plane of a multi-planar format and will not be able to enable fixed-rate compression for any plane if the requested rates differ.

Possible values of VkImageCompressionControlEXT::`flags`, specifying compression controls for an image, are:

```
// Provided by VK_EXT_image_compression_control
typedef enum VkImageCompressionFlagBitsEXT {
    VK_IMAGE_COMPRESSION_DEFAULT_EXT = 0,
    VK_IMAGE_COMPRESSION_FIXED_RATE_DEFAULT_EXT = 0x00000001,
    VK_IMAGE_COMPRESSION_FIXED_RATE_EXPLICIT_EXT = 0x00000002,
    VK_IMAGE_COMPRESSION_DISABLED_EXT = 0x00000004,
} VkImageCompressionFlagBitsEXT;
```

- VK_IMAGE_COMPRESSION_DEFAULT_EXT specifies that the default image compression setting is used. Implementations **must** not apply fixed-rate compression.
- VK_IMAGE_COMPRESSION_FIXED_RATE_DEFAULT_EXT specifies that the implementation **may** choose any supported fixed-rate compression setting in an implementation-defined manner based on the properties of the image.
- VK_IMAGE_COMPRESSION_FIXED_RATE_EXPLICIT_EXT specifies that fixed-rate compression **may** be used and that the allowed compression rates are specified by VkImageCompressionControlEXT::`pFixedRateFlags`.
- VK_IMAGE_COMPRESSION_DISABLED_EXT specifies that all lossless and fixed-rate compression **should** be disabled.

If VkImageCompressionControlEXT::`flags` is VK_IMAGE_COMPRESSION_FIXED_RATE_EXPLICIT_EXT, then the i^{th} member of the `pFixedRateFlags` array specifies the allowed compression rates for the image's i^{th} plane.

NOTE If VK_IMAGE_COMPRESSION_DISABLED_EXT is included in VkImageCompressionControlEXT::`flags`, both lossless and fixed-rate compression will be disabled. This is likely to have a negative impact on performance and is only intended to be used for debugging purposes.

```
// Provided by VK_EXT_image_compression_control
typedef VkFlags VkImageCompressionFlagsEXT;
```

`VkImageCompressionFlagsEXT` is a bitmask type for setting a mask of zero or more VkImageCompressionFlagBitsEXT.

```
// Provided by VK_EXT_image_compression_control
typedef VkFlags VkImageCompressionFixedRateFlagsEXT;
```

VkImageCompressionFixedRateFlagsEXT is a bitmask type for setting a mask of zero or more VkImageCompressionFixedRateFlagBitsEXT.

Bits which **can** be set in VkImageCompressionControlEXT::pFixedRateFlags, specifying allowed compression rates for an image plane, are:

```
// Provided by VK_EXT_image_compression_control
typedef enum VkImageCompressionFixedRateFlagBitsEXT {
    VK_IMAGE_COMPRESSION_FIXED_RATE_NONE_EXT = 0,
    VK_IMAGE_COMPRESSION_FIXED_RATE_1BPC_BIT_EXT = 0x00000001,
    VK_IMAGE_COMPRESSION_FIXED_RATE_2BPC_BIT_EXT = 0x00000002,
    VK_IMAGE_COMPRESSION_FIXED_RATE_3BPC_BIT_EXT = 0x00000004,
    VK_IMAGE_COMPRESSION_FIXED_RATE_4BPC_BIT_EXT = 0x00000008,
    VK_IMAGE_COMPRESSION_FIXED_RATE_5BPC_BIT_EXT = 0x00000010,
    VK_IMAGE_COMPRESSION_FIXED_RATE_6BPC_BIT_EXT = 0x00000020,
    VK_IMAGE_COMPRESSION_FIXED_RATE_7BPC_BIT_EXT = 0x00000040,
    VK_IMAGE_COMPRESSION_FIXED_RATE_8BPC_BIT_EXT = 0x00000080,
    VK_IMAGE_COMPRESSION_FIXED_RATE_9BPC_BIT_EXT = 0x00000100,
    VK_IMAGE_COMPRESSION_FIXED_RATE_10BPC_BIT_EXT = 0x00000200,
    VK_IMAGE_COMPRESSION_FIXED_RATE_11BPC_BIT_EXT = 0x00000400,
    VK_IMAGE_COMPRESSION_FIXED_RATE_12BPC_BIT_EXT = 0x00000800,
    VK_IMAGE_COMPRESSION_FIXED_RATE_13BPC_BIT_EXT = 0x00001000,
    VK_IMAGE_COMPRESSION_FIXED_RATE_14BPC_BIT_EXT = 0x00002000,
    VK_IMAGE_COMPRESSION_FIXED_RATE_15BPC_BIT_EXT = 0x00004000,
    VK_IMAGE_COMPRESSION_FIXED_RATE_16BPC_BIT_EXT = 0x00008000,
    VK_IMAGE_COMPRESSION_FIXED_RATE_17BPC_BIT_EXT = 0x00010000,
    VK_IMAGE_COMPRESSION_FIXED_RATE_18BPC_BIT_EXT = 0x00020000,
    VK_IMAGE_COMPRESSION_FIXED_RATE_19BPC_BIT_EXT = 0x00040000,
    VK_IMAGE_COMPRESSION_FIXED_RATE_20BPC_BIT_EXT = 0x00080000,
    VK_IMAGE_COMPRESSION_FIXED_RATE_21BPC_BIT_EXT = 0x00100000,
    VK_IMAGE_COMPRESSION_FIXED_RATE_22BPC_BIT_EXT = 0x00200000,
    VK_IMAGE_COMPRESSION_FIXED_RATE_23BPC_BIT_EXT = 0x00400000,
    VK_IMAGE_COMPRESSION_FIXED_RATE_24BPC_BIT_EXT = 0x00800000,
} VkImageCompressionFixedRateFlagBitsEXT;
```

- VK_IMAGE_COMPRESSION_FIXED_RATE_NONE_EXT specifies that fixed-rate compression **must** not be used.
- VK_IMAGE_COMPRESSION_FIXED_RATE_1BPC_BIT_EXT specifies that fixed-rate compression with a bitrate of [1,2) bits per component **may** be used.
- VK_IMAGE_COMPRESSION_FIXED_RATE_2BPC_BIT_EXT specifies that fixed-rate compression with a bitrate of [2,3) bits per component **may** be used.
- VK_IMAGE_COMPRESSION_FIXED_RATE_3BPC_BIT_EXT specifies that fixed-rate compression with a bitrate of [3,4) bits per component **may** be used.

- VK_IMAGE_COMPRESSION_FIXED_RATE_4BPC_BIT_EXT specifies that fixed-rate compression with a bitrate of [4,5) bits per component **may** be used.
- VK_IMAGE_COMPRESSION_FIXED_RATE_5BPC_BIT_EXT specifies that fixed-rate compression with a bitrate of [5,6) bits per component **may** be used.
- VK_IMAGE_COMPRESSION_FIXED_RATE_6BPC_BIT_EXT specifies that fixed-rate compression with a bitrate of [6,7) bits per component **may** be used.
- VK_IMAGE_COMPRESSION_FIXED_RATE_7BPC_BIT_EXT specifies that fixed-rate compression with a bitrate of [7,8) bits per component **may** be used.
- VK_IMAGE_COMPRESSION_FIXED_RATE_8BPC_BIT_EXT specifies that fixed-rate compression with a bitrate of [8,9) bits per component **may** be used.
- VK_IMAGE_COMPRESSION_FIXED_RATE_9BPC_BIT_EXT specifies that fixed-rate compression with a bitrate of [9,10) bits per component **may** be used.
- VK_IMAGE_COMPRESSION_FIXED_RATE_10BPC_BIT_EXT specifies that fixed-rate compression with a bitrate of [10,11) bits per component **may** be used.
- VK_IMAGE_COMPRESSION_FIXED_RATE_11BPC_BIT_EXT specifies that fixed-rate compression with a bitrate of [11,12) bits per component **may** be used.
- VK_IMAGE_COMPRESSION_FIXED_RATE_12BPC_BIT_EXT specifies that fixed-rate compression with a bitrate of at least 12 bits per component **may** be used.

If the format has a different bit rate for different components, VkImageCompressionControlEXT ::`pFixedRateFlags` describes the rate of the component with the largest number of bits assigned to it, scaled pro rata. For example, to request that a VK_FORMAT_A2R10G10B10_UNORM_PACK32 format be stored at a rate of 8 bits per pixel, use VK_IMAGE_COMPRESSION_FIXED_RATE_2BPC_BIT_EXT (10 bits for the largest component, stored at quarter the original size, 2.5 bits, rounded down).

If `flags` includes VK_IMAGE_COMPRESSION_FIXED_RATE_EXPLICIT_EXT, and multiple bits are set in VkImageCompressionControlEXT::`pFixedRateFlags` for a plane, implementations **should** apply the lowest allowed bitrate that is supported.

NOTE

The choice of “bits per component” terminology was chosen so that the same compression rate describes the same degree of compression applied to formats that differ only in the number of components. For example, VK_FORMAT_R8G8_UNORM compressed to half its original size is a rate of 4 bits per component, 8 bits per pixel. VK_FORMAT_R8G8B8A8_UNORM compressed to half *its* original size is 4 bits per component, 16 bits per pixel. Both of these cases can be requested with VK_IMAGE_COMPRESSION_FIXED_RATE_4BPC_BIT_EXT.

To query the compression properties of an image, add a VkImageCompressionPropertiesEXT structure to the `pNext` chain of the VkSubresourceLayout2 structure in a call to vkGetImageSubresourceLayout2.

To determine the compression rates that are supported for a given image format, add a VkImageCompressionPropertiesEXT structure to the `pNext` chain of the VkImageFormatProperties2 structure in a call to vkGetPhysicalDeviceImageFormatProperties2.

NOTE	Since fixed-rate compression is disabled by default, the VkImageCompressionPropertiesEXT structure passed to vkGetPhysicalDeviceImageFormatProperties2 will not indicate any fixed-rate compression support unless a VkImageCompressionControlEXT structure is also included in the pNext chain of the VkPhysicalDeviceImageFormatInfo2 structure passed to the same command.

The VkImageCompressionPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_image_compression_control
typedef struct VkImageCompressionPropertiesEXT {
    VkStructureType                         sType;
    void*                                   pNext;
    VkImageCompressionFlagsEXT              imageCompressionFlags;
    VkImageCompressionFixedRateFlagsEXT     imageCompressionFixedRateFlags;
} VkImageCompressionPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- imageCompressionFlags returns a value describing the compression controls that apply to the image. The value will be either VK_IMAGE_COMPRESSION_DEFAULT_EXT to indicate no fixed-rate compression, VK_IMAGE_COMPRESSION_FIXED_RATE_EXPLICIT_EXT to indicate fixed-rate compression, or VK_IMAGE_COMPRESSION_DISABLED_EXT to indicate no compression.
- imageCompressionFixedRateFlags returns a VkImageCompressionFixedRateFlagsEXT value describing the compression rates that apply to the specified aspect of the image.

Valid Usage (Implicit)

- VUID-VkImageCompressionPropertiesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_IMAGE_COMPRESSION_PROPERTIES_EXT

Structure Chaining

Extends the structures

- VkImageFormatProperties2
- VkSubresourceLayout2
- VkSurfaceFormat2KHR

If the pNext chain of VkImageCreateInfo includes a VkImageTilingControlCreateInfoEXT structure, then that structure specifies the optimal tiling arrangement for this image.

The VkImageTilingControlCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_image_tiling_control
typedef struct VkImageTilingControlCreateInfoEXT {
    VkStructureType            sType;
    const void*                pNext;
    VkImageTilingControlEXT    tilingControl;
} VkImageTilingControlCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `tilingControl` is a VkImageTilingControlEXT value specifying the optimal tiling arrangement for the image.

Valid Usage

- VUID-VkImageTilingControlCreateInfoEXT-tilingControl-12481
  If `VkImageTilingControlCreateInfoEXT::tilingControl` is not VK_IMAGE_TILING_CONTROL_DEFAULT_EXT, the `imageTilingControl` feature **must** be enabled
- VUID-VkImageTilingControlCreateInfoEXT-tilingControl-12479
  If `VkImageTilingControlCreateInfoEXT::tilingControl` is not VK_IMAGE_TILING_CONTROL_DEFAULT_EXT, VkImageCreateInfo::`tiling` **must** not be VK_IMAGE_TILING_LINEAR

Valid Usage (Implicit)

- VUID-VkImageTilingControlCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_TILING_CONTROL_CREATE_INFO_EXT
- VUID-VkImageTilingControlCreateInfoEXT-tilingControl-parameter
  `tilingControl` **must** be a valid VkImageTilingControlEXT value

Structure Chaining

Extends the structure

- VkImageCreateInfo

Possible values of VkImageTilingControlCreateInfoEXT::`tilingControl`, specifying image tiling arrangement, are:

```
// Provided by VK_EXT_image_tiling_control
typedef enum VkImageTilingControlEXT {
    VK_IMAGE_TILING_CONTROL_DEFAULT_EXT = 0,
    VK_IMAGE_TILING_CONTROL_MIN_SIZE_EXT = 1,
    VK_IMAGE_TILING_CONTROL_MAX_PERFORMANCE_EXT = 2,
} VkImageTilingControlEXT;
```

- VK_IMAGE_TILING_CONTROL_DEFAULT_EXT specifies that the image tiling arrangement balances memory size requirements and memory access performance. It is the same selection as with the VkImageTilingControlCreateInfoEXT structure omitted from the pNext chain of VkImageCreateInfo.
- VK_IMAGE_TILING_CONTROL_MIN_SIZE_EXT specifies that the image tiling arrangement minimizes memory size requirements with reduced consideration for memory access performance.
- VK_IMAGE_TILING_CONTROL_MAX_PERFORMANCE_EXT specifies that the image tiling arrangement maximizes memory access performance with reduced consideration for increases in memory size requirements.

These enumeration values only select the preferred tiling arrangement out of the subset that the implementation supports, which is dependent on the other image creation parameters.

There are cases where multiple enumeration values will result in the same tiling arrangement, especially for images with extents that are multiples of optimal tiling block sizes. For these cases, no tradeoff exists for the image.

The memory size requirements of the image are less than or equal when using VK_IMAGE_TILING_CONTROL_MIN_SIZE_EXT. However, there are no performance guarantees when using VK_IMAGE_TILING_CONTROL_MAX_PERFORMANCE_EXT as image access times are dependent on other variables such as the shader access pattern.

Bits which **can** be set in VkImageUsageFlags2CreateInfoKHR::usage, specifying intended usage of an image, are:

```
// Provided by VK_KHR_extended_flags
// Flag bits for VkImageUsageFlagBits2KHR
typedef VkFlags64 VkImageUsageFlagBits2KHR;
static const VkImageUsageFlagBits2KHR VK_IMAGE_USAGE_2_TRANSFER_SRC_BIT_KHR =
0x00000001ULL;
static const VkImageUsageFlagBits2KHR VK_IMAGE_USAGE_2_TRANSFER_DST_BIT_KHR =
0x00000002ULL;
static const VkImageUsageFlagBits2KHR VK_IMAGE_USAGE_2_SAMPLED_BIT_KHR =
0x00000004ULL;
static const VkImageUsageFlagBits2KHR VK_IMAGE_USAGE_2_STORAGE_BIT_KHR =
0x00000008ULL;
static const VkImageUsageFlagBits2KHR VK_IMAGE_USAGE_2_COLOR_ATTACHMENT_BIT_KHR =
0x00000010ULL;
static const VkImageUsageFlagBits2KHR
VK_IMAGE_USAGE_2_DEPTH_STENCIL_ATTACHMENT_BIT_KHR = 0x00000020ULL;
```

```
static const VkImageUsageFlagBits2KHR VK_IMAGE_USAGE_2_TRANSIENT_ATTACHMENT_BIT_KHR =
0x00000040ULL;
static const VkImageUsageFlagBits2KHR VK_IMAGE_USAGE_2_INPUT_ATTACHMENT_BIT_KHR =
0x00000080ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_fragment_shading_rate
static const VkImageUsageFlagBits2KHR
VK_IMAGE_USAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR = 0x00000100ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_fragment_density_map
static const VkImageUsageFlagBits2KHR VK_IMAGE_USAGE_2_FRAGMENT_DENSITY_MAP_BIT_EXT =
0x00000200ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_video_decode_queue
static const VkImageUsageFlagBits2KHR VK_IMAGE_USAGE_2_VIDEO_DECODE_DST_BIT_KHR =
0x00000400ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_video_decode_queue
static const VkImageUsageFlagBits2KHR VK_IMAGE_USAGE_2_VIDEO_DECODE_SRC_BIT_KHR =
0x00000800ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_video_decode_queue
static const VkImageUsageFlagBits2KHR VK_IMAGE_USAGE_2_VIDEO_DECODE_DPB_BIT_KHR =
0x00001000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_video_encode_queue
static const VkImageUsageFlagBits2KHR VK_IMAGE_USAGE_2_VIDEO_ENCODE_DST_BIT_KHR =
0x00002000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_video_encode_queue
static const VkImageUsageFlagBits2KHR VK_IMAGE_USAGE_2_VIDEO_ENCODE_SRC_BIT_KHR =
0x00004000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_video_encode_queue
static const VkImageUsageFlagBits2KHR VK_IMAGE_USAGE_2_VIDEO_ENCODE_DPB_BIT_KHR =
0x00008000ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_attachment_feedback_loop_layout
static const VkImageUsageFlagBits2KHR
VK_IMAGE_USAGE_2_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT = 0x00080000ULL;
// Provided by VK_VERSION_1_4 with VK_KHR_extended_flags
static const VkImageUsageFlagBits2KHR VK_IMAGE_USAGE_2_HOST_TRANSFER_BIT_KHR =
0x00400000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_video_encode_quantization_map
static const VkImageUsageFlagBits2KHR
VK_IMAGE_USAGE_2_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR = 0x02000000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_video_encode_quantization_map
static const VkImageUsageFlagBits2KHR
VK_IMAGE_USAGE_2_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR = 0x04000000ULL;
```

- VK_IMAGE_USAGE_2_TRANSFER_SRC_BIT_KHR specifies that the image **can** be used as the source of a transfer command.
- VK_IMAGE_USAGE_2_TRANSFER_DST_BIT_KHR specifies that the image **can** be used as the destination of a transfer command.
- VK_IMAGE_USAGE_2_SAMPLED_BIT_KHR specifies that the image **can** be used to create a VkImageView suitable for occupying a VkDescriptorSet slot either of type VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, and be sampled by a shader.

- VK_IMAGE_USAGE_2_STORAGE_BIT_KHR specifies that the image **can** be used to create a VkImageView suitable for occupying a VkDescriptorSet slot of type VK_DESCRIPTOR_TYPE_STORAGE_IMAGE.
- VK_IMAGE_USAGE_2_COLOR_ATTACHMENT_BIT_KHR specifies that the image **can** be used to create a VkImageView suitable for use as a color or resolve attachment in a VkFramebuffer.
- VK_IMAGE_USAGE_2_DEPTH_STENCIL_ATTACHMENT_BIT_KHR specifies that the image **can** be used to create a VkImageView suitable for use as a depth/stencil or depth/stencil resolve attachment in a VkFramebuffer.
- VK_IMAGE_USAGE_2_TRANSIENT_ATTACHMENT_BIT_KHR specifies that implementations **may** support using memory allocations with the VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT to back an image with this usage. This bit **can** be set for any image that **can** be used to create a VkImageView suitable for use as a color, resolve, depth/stencil, or input attachment.
- VK_IMAGE_USAGE_2_INPUT_ATTACHMENT_BIT_KHR specifies that the image **can** be used to create a VkImageView suitable for occupying a VkDescriptorSet slot of type VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT; be read from a shader as an input attachment; and be used as an input attachment in a framebuffer.
- VK_IMAGE_USAGE_2_FRAGMENT_DENSITY_MAP_BIT_EXT specifies that the image **can** be used to create a VkImageView suitable for use as a fragment density map image.
- VK_IMAGE_USAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR specifies that the image **can** be used to create a VkImageView suitable for use as a fragment shading rate attachment
- VK_IMAGE_USAGE_2_VIDEO_DECODE_DST_BIT_KHR specifies that the image **can** be used as a decode output picture in a video decode operation.
- VK_IMAGE_USAGE_2_VIDEO_DECODE_SRC_BIT_KHR is reserved for future use.
- VK_IMAGE_USAGE_2_VIDEO_DECODE_DPB_BIT_KHR specifies that the image **can** be used as an output reconstructed picture or an input reference picture in a video decode operation.
- VK_IMAGE_USAGE_2_VIDEO_ENCODE_DST_BIT_KHR is reserved for future use.
- VK_IMAGE_USAGE_2_VIDEO_ENCODE_SRC_BIT_KHR specifies that the image **can** be used as an encode input picture in a video encode operation.
- VK_IMAGE_USAGE_2_VIDEO_ENCODE_DPB_BIT_KHR specifies that the image **can** be used as an output reconstructed picture or an input reference picture in a video encode operation.
- VK_IMAGE_USAGE_2_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT specifies that the image **can** be used as a color or depth/stencil attachment with feedback loop enabled.
- VK_IMAGE_USAGE_2_HOST_TRANSFER_BIT_KHR specifies that the image **can** be used with host copy commands and host layout transitions.
- VK_IMAGE_USAGE_2_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR specifies that the image **can** be used as a quantization delta map in a video encode operation.
- VK_IMAGE_USAGE_2_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR specifies that the image **can** be used as an emphasis map in a video encode operation.

```
// Provided by VK_KHR_extended_flags
typedef VkFlags64 VkImageUsageFlags2KHR;
```

VkImageUsageFlags2KHR is a bitmask type for setting a mask of zero or more VkImageUsageFlagBits2KHR.

Bits which **can** be set in

- VkImageViewUsageCreateInfo::usage
- VkImageStencilUsageCreateInfo::stencilUsage
- VkImageCreateInfo::usage

specify intended usage of an image, and are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageUsageFlagBits {
    VK_IMAGE_USAGE_TRANSFER_SRC_BIT = 0x00000001,
    VK_IMAGE_USAGE_TRANSFER_DST_BIT = 0x00000002,
    VK_IMAGE_USAGE_SAMPLED_BIT = 0x00000004,
    VK_IMAGE_USAGE_STORAGE_BIT = 0x00000008,
    VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT = 0x00000010,
    VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT = 0x00000020,
    VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT = 0x00000040,
    VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT = 0x00000080,
  // Provided by VK_VERSION_1_4
    VK_IMAGE_USAGE_HOST_TRANSFER_BIT = 0x00400000,
  // Provided by VK_KHR_video_decode_queue
    VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR = 0x00000400,
  // Provided by VK_KHR_video_decode_queue
    VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR = 0x00000800,
  // Provided by VK_KHR_video_decode_queue
    VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR = 0x00001000,
  // Provided by VK_EXT_fragment_density_map
    VK_IMAGE_USAGE_FRAGMENT_DENSITY_MAP_BIT_EXT = 0x00000200,
  // Provided by VK_KHR_fragment_shading_rate
    VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR = 0x00000100,
  // Provided by VK_KHR_video_encode_queue
    VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR = 0x00002000,
  // Provided by VK_KHR_video_encode_queue
    VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR = 0x00004000,
  // Provided by VK_KHR_video_encode_queue
    VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR = 0x00008000,
  // Provided by VK_EXT_attachment_feedback_loop_layout
    VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT = 0x00080000,
  // Provided by VK_KHR_video_encode_quantization_map
    VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR = 0x02000000,
  // Provided by VK_KHR_video_encode_quantization_map
    VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR = 0x04000000,
  // Provided by VK_EXT_host_image_copy
    VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT = VK_IMAGE_USAGE_HOST_TRANSFER_BIT,
} VkImageUsageFlagBits;
```

- VK_IMAGE_USAGE_TRANSFER_SRC_BIT specifies that the image **can** be used as the source of a transfer command.
- VK_IMAGE_USAGE_TRANSFER_DST_BIT specifies that the image **can** be used as the destination of a transfer command.
- VK_IMAGE_USAGE_SAMPLED_BIT specifies that the image **can** be used to create a VkImageView suitable for occupying a VkDescriptorSet slot either of type VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, and be sampled by a shader.
- VK_IMAGE_USAGE_STORAGE_BIT specifies that the image **can** be used to create a VkImageView suitable for occupying a VkDescriptorSet slot of type VK_DESCRIPTOR_TYPE_STORAGE_IMAGE.

- VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT specifies that the image **can** be used to create a VkImageView suitable for use as a color or resolve attachment in a VkFramebuffer.
- VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT specifies that the image **can** be used to create a VkImageView suitable for use as a depth/stencil or depth/stencil resolve attachment in a VkFramebuffer.
- VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT specifies that implementations **may** support using memory allocations with the VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT to back an image with this usage. This bit **can** be set for any image that **can** be used to create a VkImageView suitable for use as a color, resolve, depth/stencil, or input attachment.
- VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT specifies that the image **can** be used to create a VkImageView suitable for occupying a VkDescriptorSet slot of type VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT; be read from a shader as an input attachment; and be used as an input attachment in a framebuffer.
- VK_IMAGE_USAGE_FRAGMENT_DENSITY_MAP_BIT_EXT specifies that the image **can** be used to create a VkImageView suitable for use as a fragment density map image.
- VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR specifies that the image **can** be used to create a VkImageView suitable for use as a fragment shading rate attachment
- VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR specifies that the image **can** be used as a decode output picture in a video decode operation.
- VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR is reserved for future use.
- VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR specifies that the image **can** be used as an output reconstructed picture or an input reference picture in a video decode operation.
- VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR is reserved for future use.
- VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR specifies that the image **can** be used as an encode input picture in a video encode operation.
- VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR specifies that the image **can** be used as an output reconstructed picture or an input reference picture in a video encode operation.
- VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT specifies that the image **can** be used as a color or depth/stencil attachment with feedback loop enabled.
- VK_IMAGE_USAGE_HOST_TRANSFER_BIT specifies that the image **can** be used with host copy commands and host layout transitions.
- VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR specifies that the image **can** be used as a quantization delta map in a video encode operation.
- VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR specifies that the image **can** be used as an emphasis map in a video encode operation.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkImageUsageFlags;
```

VkImageUsageFlags is a bitmask type for setting a mask of zero or more VkImageUsageFlagBits.

Bits which **can** be set in VkImageCreateFlags2CreateInfoKHR::flags, specifying additional parameters of an image, are:

```
// Provided by VK_KHR_extended_flags
// Flag bits for VkImageCreateFlagBits2KHR
typedef VkFlags64 VkImageCreateFlagBits2KHR;
static const VkImageCreateFlagBits2KHR VK_IMAGE_CREATE_2_SPARSE_BINDING_BIT_KHR =
0x00000001ULL;
static const VkImageCreateFlagBits2KHR VK_IMAGE_CREATE_2_SPARSE_RESIDENCY_BIT_KHR =
0x00000002ULL;
static const VkImageCreateFlagBits2KHR VK_IMAGE_CREATE_2_SPARSE_ALIASED_BIT_KHR =
0x00000004ULL;
static const VkImageCreateFlagBits2KHR VK_IMAGE_CREATE_2_MUTABLE_FORMAT_BIT_KHR =
0x00000008ULL;
static const VkImageCreateFlagBits2KHR VK_IMAGE_CREATE_2_CUBE_COMPATIBLE_BIT_KHR =
0x00000010ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_maintenance11
static const VkImageCreateFlagBits2KHR
VK_IMAGE_CREATE_2_ALIAS_SINGLE_LAYER_DESCRIPTOR_BIT_KHR = 0x00400000ULL;
// Provided by VK_VERSION_1_0 with VK_KHR_extended_flags
static const VkImageCreateFlagBits2KHR VK_IMAGE_CREATE_2_2D_ARRAY_COMPATIBLE_BIT_KHR =
0x00000020ULL;
// Provided by VK_VERSION_1_0 with VK_KHR_extended_flags
static const VkImageCreateFlagBits2KHR
VK_IMAGE_CREATE_2_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR = 0x00000040ULL;
// Provided by VK_VERSION_1_0 with VK_KHR_extended_flags
static const VkImageCreateFlagBits2KHR
VK_IMAGE_CREATE_2_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT_KHR = 0x00000080ULL;
// Provided by VK_VERSION_1_0 with VK_KHR_extended_flags
static const VkImageCreateFlagBits2KHR VK_IMAGE_CREATE_2_EXTENDED_USAGE_BIT_KHR =
0x00000100ULL;
// Provided by VK_VERSION_1_0 with VK_KHR_extended_flags
static const VkImageCreateFlagBits2KHR VK_IMAGE_CREATE_2_DISJOINT_BIT_KHR =
0x00000200ULL;
// Provided by VK_VERSION_1_0 with VK_KHR_extended_flags
static const VkImageCreateFlagBits2KHR VK_IMAGE_CREATE_2_ALIAS_BIT_KHR =
0x00000400ULL;
// Provided by VK_VERSION_1_0 with VK_KHR_extended_flags
static const VkImageCreateFlagBits2KHR VK_IMAGE_CREATE_2_PROTECTED_BIT_KHR =
0x00000800ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_sample_locations
static const VkImageCreateFlagBits2KHR
VK_IMAGE_CREATE_2_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT = 0x00001000ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_fragment_density_map
static const VkImageCreateFlagBits2KHR VK_IMAGE_CREATE_2_SUBSAMPLED_BIT_EXT =
0x00004000ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_fragment_density_map_offset
static const VkImageCreateFlagBits2KHR
VK_IMAGE_CREATE_2_FRAGMENT_DENSITY_MAP_OFFSET_BIT_EXT = 0x00008000ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_descriptor_buffer
```

```
static const VkImageCreateFlagBits2KHR
VK_IMAGE_CREATE_2_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT = 0x00010000ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_image_2d_view_of_3d
static const VkImageCreateFlagBits2KHR VK_IMAGE_CREATE_2_2D_VIEW_COMPATIBLE_BIT_EXT =
0x00020000ULL;
// Provided by VK_KHR_extended_flags with VK_EXT_multisampled_render_to_single_sampled
static const VkImageCreateFlagBits2KHR
VK_IMAGE_CREATE_2_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT = 0x00040000ULL;
// Provided by VK_KHR_extended_flags with VK_KHR_video_maintenance1
static const VkImageCreateFlagBits2KHR
VK_IMAGE_CREATE_2_VIDEO_PROFILE_INDEPENDENT_BIT_KHR = 0x00100000ULL;
```

- VK_IMAGE_CREATE_2_SPARSE_BINDING_BIT_KHR specifies that the image will be backed using sparse memory binding.
- VK_IMAGE_CREATE_2_SPARSE_RESIDENCY_BIT_KHR specifies that the image **can** be partially backed using sparse memory binding. Images created with this flag **must** also be created with the VK_IMAGE_CREATE_2_SPARSE_BINDING_BIT_KHR flag.
- VK_IMAGE_CREATE_2_SPARSE_ALIASED_BIT_KHR specifies that the image will be backed using sparse memory binding with memory ranges that might also simultaneously be backing another image (or another portion of the same image). Images created with this flag **must** also be created with the VK_IMAGE_CREATE_2_SPARSE_BINDING_BIT_KHR flag.
- VK_IMAGE_CREATE_2_MUTABLE_FORMAT_BIT_KHR specifies that the image **can** be used to create a VkImageView with a different format from the image. For multi-planar formats, VK_IMAGE_CREATE_2_MUTABLE_FORMAT_BIT_KHR specifies that a VkImageView can be created of a *plane* of the image.
- VK_IMAGE_CREATE_2_CUBE_COMPATIBLE_BIT_KHR specifies that the image **can** be used to create a VkImageView of type VK_IMAGE_VIEW_TYPE_CUBE or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY.
- VK_IMAGE_CREATE_2_2D_ARRAY_COMPATIBLE_BIT_KHR specifies that the image **can** be used to create a VkImageView of type VK_IMAGE_VIEW_TYPE_2D or VK_IMAGE_VIEW_TYPE_2D_ARRAY.
- VK_IMAGE_CREATE_2_2D_VIEW_COMPATIBLE_BIT_EXT specifies that the image **can** be used to create a VkImageView of type VK_IMAGE_VIEW_TYPE_2D.
- VK_IMAGE_CREATE_2_PROTECTED_BIT_KHR specifies that the image is a protected image.
- VK_IMAGE_CREATE_2_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR specifies that the image **can** be used with a non-zero value of the splitInstanceBindRegionCount member of a VkBindImageMemoryDeviceGroupInfo structure passed into vkBindImageMemory2. This flag also has the effect of making the image use the standard sparse image block dimensions.
- VK_IMAGE_CREATE_2_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT_KHR specifies that the image having a compressed format **can** be used to create a VkImageView with an uncompressed format where each texel in the image view corresponds to a compressed texel block of the image.
- VK_IMAGE_CREATE_2_EXTENDED_USAGE_BIT_KHR specifies that the image **can** be created with usage flags that are not supported for the format the image is created with but are supported for at least one format a VkImageView created from the image **can** have.

- VK_IMAGE_CREATE_2_DISJOINT_BIT_KHR specifies that an image with a multi-planar format **must** have each plane separately bound to memory, rather than having a single memory binding for the whole image; the presence of this bit distinguishes a *disjoint image* from an image without this bit set.
- VK_IMAGE_CREATE_2_ALIAS_BIT_KHR specifies that two images created with the same creation parameters and aliased to the same memory **can** interpret the contents of the memory consistently with each other, subject to the rules described in the Memory Aliasing section. This flag further specifies that each plane of a *disjoint* image **can** share an in-memory non-linear representation with single-plane images, and that a single-plane image **can** share an in-memory non-linear representation with a plane of a multi-planar disjoint image, according to the rules in Compatible Formats of Planes of Multi-Planar Formats. If the `pNext` chain includes a VkExternalMemoryImageCreateInfo structure whose `handleTypes` member is not `0`, it is as if VK_IMAGE_CREATE_2_ALIAS_BIT_KHR is set.
- VK_IMAGE_CREATE_2_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT specifies that an image with a depth or depth/stencil format **can** be used with custom sample locations when used as a depth/stencil attachment.
- VK_IMAGE_CREATE_2_SUBSAMPLED_BIT_EXT specifies that an image **can** be in a subsampled format which **may** be more optimal when written as an attachment by a render pass that has a fragment density map attachment. Accessing a subsampled image has additional considerations:
  - Image data read as an image sampler will have undefined values if the sampler was not created with `flags` containing VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT or was not sampled through the use of a combined image sampler with an immutable sampler in `VkDescriptorSetLayoutBinding`.
  - Image data read with an input attachment will have undefined values if the contents were not written as an attachment in an earlier subpass of the same render pass.
  - Image data read as an image sampler in the fragment shader will additionally be read by the device during VK_PIPELINE_STAGE_VERTEX_SHADER_BIT if `VkPhysicalDeviceFragmentDensityMap2PropertiesEXT::subsampledCoarseReconstructionEarlyAccess` is VK_TRUE and the sampler was created with `flags` containing VK_SAMPLER_CREATE_SUBSAMPLED_COARSE_RECONSTRUCTION_BIT_EXT.
  - Image data read with load operations are resampled to the fragment density of the render pass if `VkPhysicalDeviceFragmentDensityMap2PropertiesEXT::subsampledLoads` is VK_TRUE. Otherwise, values of image data are undefined.
  - Image contents outside of the render area take on undefined values if the image is stored as a render pass attachment.
- VK_IMAGE_CREATE_2_FRAGMENT_DENSITY_MAP_OFFSET_BIT_EXT specifies that an image **can** be used in a render pass with non-zero fragment density map offsets. In a render pass with non-zero offsets, fragment density map attachments, input attachments, color attachments, depth/stencil attachment, resolve attachments, and preserve attachments **must** be created with VK_IMAGE_CREATE_2_FRAGMENT_DENSITY_MAP_OFFSET_BIT_EXT.
- VK_IMAGE_CREATE_2_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT specifies that the image **can** be used with descriptor buffers when capturing and replaying (e.g. for trace capture and replay), see VkOpaqueCaptureDescriptorDataCreateInfoEXT for more detail.

- VK_IMAGE_CREATE_2_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT specifies that an image **can** be used with multisampled rendering as a single-sampled framebuffer attachment.
- VK_IMAGE_CREATE_2_VIDEO_PROFILE_INDEPENDENT_BIT_KHR specifies that the image **can** be used in video coding operations without having to specify at image creation time the set of video profiles the image will be used with, except for images used only as DPB pictures, as long as the image is otherwise compatible with the video profile in question.

  NOTE

  This enables exchanging video picture data without additional copies or conversions when used as:

  - Decode output pictures, regardless of the video profile used to produce them.
  - Encode input pictures, regardless of the video profile used to consume them.

  This includes images created with both VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR and VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR, which is necessary to use the same video picture as the reconstructed picture and decode output picture in a video decode operation on implementations supporting VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR.

  However, images with only DPB usage remain tied to the video profiles the image was created with, as the data layout of such DPB-only images **may** be implementation- and codec-dependent.

  If an application would like to share or reuse the device memory backing such images (e.g. for the purposes of temporal aliasing), then it **should** create separate image objects for each video profile and bind them to the same underlying device memory range, similar to how memory resources **can** be shared across separate video sessions or any other memory-backed resource.

See Sparse Resource Features and Sparse Physical Device Features for more details.

```
// Provided by VK_KHR_extended_flags
typedef VkFlags64 VkImageCreateFlags2KHR;
```

VkImageCreateFlags2KHR is a bitmask type for setting a mask of zero or more VkImageCreateFlagBits2KHR.

Bits which **can** be set in VkImageCreateInfo::flags, specifying additional parameters of an image, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageCreateFlagBits {
    VK_IMAGE_CREATE_SPARSE_BINDING_BIT = 0x00000001,
    VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT = 0x00000002,
    VK_IMAGE_CREATE_SPARSE_ALIASED_BIT = 0x00000004,
```

```
    VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT = 0x00000008,
    VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT = 0x00000010,
  // Provided by VK_VERSION_1_1
    VK_IMAGE_CREATE_ALIAS_BIT = 0x00000400,
  // Provided by VK_VERSION_1_1
    VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT = 0x00000040,
  // Provided by VK_VERSION_1_1
    VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT = 0x00000020,
  // Provided by VK_VERSION_1_1
    VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT = 0x00000080,
  // Provided by VK_VERSION_1_1
    VK_IMAGE_CREATE_EXTENDED_USAGE_BIT = 0x00000100,
  // Provided by VK_VERSION_1_1
    VK_IMAGE_CREATE_PROTECTED_BIT = 0x00000800,
  // Provided by VK_VERSION_1_1
    VK_IMAGE_CREATE_DISJOINT_BIT = 0x00000200,
  // Provided by VK_EXT_descriptor_heap
    VK_IMAGE_CREATE_DESCRIPTOR_HEAP_CAPTURE_REPLAY_BIT_EXT = 0x00010000,
  // Provided by VK_EXT_sample_locations
    VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT = 0x00001000,
  // Provided by VK_EXT_fragment_density_map
    VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT = 0x00004000,
  // Provided by VK_EXT_multisampled_render_to_single_sampled
    VK_IMAGE_CREATE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT = 0x00040000,
  // Provided by VK_EXT_image_2d_view_of_3d
    VK_IMAGE_CREATE_2D_VIEW_COMPATIBLE_BIT_EXT = 0x00020000,
  // Provided by VK_KHR_video_maintenance1
    VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR = 0x00100000,
  // Provided by VK_EXT_fragment_density_map_offset
    VK_IMAGE_CREATE_FRAGMENT_DENSITY_MAP_OFFSET_BIT_EXT = 0x00008000,
  // Provided by VK_KHR_maintenance11
    VK_IMAGE_CREATE_ALIAS_SINGLE_LAYER_DESCRIPTOR_BIT_KHR = 0x00400000,
  // Provided by VK_KHR_bind_memory2 with VK_KHR_device_group
    VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR =
VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT,
  // Provided by VK_KHR_maintenance1
    VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT_KHR =
VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT,
  // Provided by VK_KHR_maintenance2
    VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT_KHR =
VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT,
  // Provided by VK_KHR_maintenance2
    VK_IMAGE_CREATE_EXTENDED_USAGE_BIT_KHR = VK_IMAGE_CREATE_EXTENDED_USAGE_BIT,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_IMAGE_CREATE_DISJOINT_BIT_KHR = VK_IMAGE_CREATE_DISJOINT_BIT,
  // Provided by VK_KHR_bind_memory2
    VK_IMAGE_CREATE_ALIAS_BIT_KHR = VK_IMAGE_CREATE_ALIAS_BIT,
  // Provided by VK_EXT_descriptor_buffer
    VK_IMAGE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT =
VK_IMAGE_CREATE_DESCRIPTOR_HEAP_CAPTURE_REPLAY_BIT_EXT,
```

```
} VkImageCreateFlagBits;
```

- VK_IMAGE_CREATE_SPARSE_BINDING_BIT specifies that the image will be backed using sparse memory binding.
- VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT specifies that the image **can** be partially backed using sparse memory binding. Images created with this flag **must** also be created with the VK_IMAGE_CREATE_SPARSE_BINDING_BIT flag.
- VK_IMAGE_CREATE_SPARSE_ALIASED_BIT specifies that the image will be backed using sparse memory binding with memory ranges that might also simultaneously be backing another image (or another portion of the same image). Images created with this flag **must** also be created with the VK_IMAGE_CREATE_SPARSE_BINDING_BIT flag.
- VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT specifies that the image **can** be used to create a VkImageView with a different format from the image. For multi-planar formats, VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT specifies that a VkImageView can be created of a *plane* of the image.
- VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT specifies that the image **can** be used to create a VkImageView of type VK_IMAGE_VIEW_TYPE_CUBE or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY.
- VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT specifies that the image **can** be used to create a VkImageView of type VK_IMAGE_VIEW_TYPE_2D or VK_IMAGE_VIEW_TYPE_2D_ARRAY.
- VK_IMAGE_CREATE_2D_VIEW_COMPATIBLE_BIT_EXT specifies that the image **can** be used to create a VkImageView of type VK_IMAGE_VIEW_TYPE_2D.
- VK_IMAGE_CREATE_PROTECTED_BIT specifies that the image is a protected image.
- VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT specifies that the image **can** be used with a non-zero value of the splitInstanceBindRegionCount member of a VkBindImageMemoryDeviceGroupInfo structure passed into vkBindImageMemory2. This flag also has the effect of making the image use the standard sparse image block dimensions.
- VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT specifies that the image having a compressed format **can** be used to create a VkImageView with an uncompressed format where each texel in the image view corresponds to a compressed texel block of the image.
- VK_IMAGE_CREATE_EXTENDED_USAGE_BIT specifies that the image **can** be created with usage flags that are not supported for the format the image is created with but are supported for at least one format a VkImageView created from the image **can** have.
- VK_IMAGE_CREATE_DISJOINT_BIT specifies that an image with a multi-planar format **must** have each plane separately bound to memory, rather than having a single memory binding for the whole image; the presence of this bit distinguishes a *disjoint image* from an image without this bit set.
- VK_IMAGE_CREATE_ALIAS_BIT specifies that two images created with the same creation parameters and aliased to the same memory **can** interpret the contents of the memory consistently with each other, subject to the rules described in the Memory Aliasing section. This flag further specifies that each plane of a *disjoint* image **can** share an in-memory non-linear representation with single-plane images, and that a single-plane image **can** share an in-memory non-linear representation with a plane of a multi-planar disjoint image, according to the rules in Compatible Formats of Planes of Multi-Planar Formats. If the pNext chain includes a

VkExternalMemoryImageCreateInfo structure whose handleTypes member is not 0, it is as if VK_IMAGE_CREATE_ALIAS_BIT is set.

- VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT specifies that an image with a depth or depth/stencil format **can** be used with custom sample locations when used as a depth/stencil attachment.
- VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT specifies that an image **can** be in a subsampled format which **may** be more optimal when written as an attachment by a render pass that has a fragment density map attachment. Accessing a subsampled image has additional considerations:
  - Image data read as an image sampler will have undefined values if the sampler was not created with flags containing VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT or was not sampled through a combined embedded sampler and image mapping if using descriptor heaps, or the use of a combined image sampler with an immutable sampler in VkDescriptorSetLayoutBinding.
  - Image data read with an input attachment will have undefined values if the contents were not written as an attachment in an earlier subpass of the same render pass.
  - Image data read as an image sampler in the fragment shader will additionally be read by the device during VK_PIPELINE_STAGE_VERTEX_SHADER_BIT if VkPhysicalDeviceFragmentDensityMap2PropertiesEXT::subsampledCoarseReconstructionEarlyAccess is VK_TRUE and the sampler was created with flags containing VK_SAMPLER_CREATE_SUBSAMPLED_COARSE_RECONSTRUCTION_BIT_EXT.
  - Image data read with load operations are resampled to the fragment density of the render pass if VkPhysicalDeviceFragmentDensityMap2PropertiesEXT::subsampledLoads is VK_TRUE. Otherwise, values of image data are undefined.
  - Image contents outside of the render area take on undefined values if the image is stored as a render pass attachment.
- VK_IMAGE_CREATE_FRAGMENT_DENSITY_MAP_OFFSET_BIT_EXT specifies that an image **can** be used in a render pass with non-zero fragment density map offsets. In a render pass with non-zero offsets, fragment density map attachments, input attachments, color attachments, depth/stencil attachment, resolve attachments, and preserve attachments **must** be created with VK_IMAGE_CREATE_FRAGMENT_DENSITY_MAP_OFFSET_BIT_EXT.
- VK_IMAGE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT specifies that the image **can** be used with descriptor buffers when capturing and replaying (e.g. for trace capture and replay), see VkOpaqueCaptureDescriptorDataCreateInfoEXT for more detail.
- VK_IMAGE_CREATE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT specifies that an image **can** be used with multisampled rendering as a single-sampled render pass attachment.
- VK_IMAGE_CREATE_ALIAS_SINGLE_LAYER_DESCRIPTOR_BIT_KHR specifies that a single layer image view created from this image can be accessed with both Arrayed equal to 0 and Arrayed equal to 1 in a shader. If accessed with Arrayed equal to 1 in a shader, and VkImageViewCreateInfo::viewType is VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D, the image is treated as having an array size of 1, and subresourceRange.baseArrayLayer is accessed. If accessed with Arrayed equal to 0 in a shader, and VkImageViewCreateInfo::viewType is VK_IMAGE_VIEW_TYPE_1D_ARRAY or VK_IMAGE_VIEW_TYPE_2D_ARRAY, the descriptor is

accessed as-if the view was created with VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D respectively. Cube map view aliasing is not included by this flag bit. Also, image views created with a multi-planar format **cannot** alias like this.

- VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR specifies that the image **can** be used in video coding operations without having to specify at image creation time the set of video profiles the image will be used with, except for images used only as DPB pictures, as long as the image is otherwise compatible with the video profile in question.

  NOTE

  This enables exchanging video picture data without additional copies or conversions when used as:

  - Decode output pictures, regardless of the video profile used to produce them.
  - Encode input pictures, regardless of the video profile used to consume them.

  This includes images created with both VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR and VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR, which is necessary to use the same video picture as the reconstructed picture and decode output picture in a video decode operation on implementations supporting VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR.

  However, images with only DPB usage remain tied to the video profiles the image was created with, as the data layout of such DPB-only images **may** be implementation- and codec-dependent.

  If an application would like to share or reuse the device memory backing such images (e.g. for the purposes of temporal aliasing), then it **should** create separate image objects for each video profile and bind them to the same underlying device memory range, similar to how memory resources **can** be shared across separate video sessions or any other memory-backed resource.

See Sparse Resource Features and Sparse Physical Device Features for more details.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkImageCreateFlags;
```

`VkImageCreateFlags` is a bitmask type for setting a mask of zero or more VkImageCreateFlagBits.

Possible values of VkImageCreateInfo::`imageType`, specifying the basic dimensionality of an image, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageType {
    VK_IMAGE_TYPE_1D = 0,
    VK_IMAGE_TYPE_2D = 1,
    VK_IMAGE_TYPE_3D = 2,
} VkImageType;
```

- VK_IMAGE_TYPE_1D specifies a one-dimensional image.
- VK_IMAGE_TYPE_2D specifies a two-dimensional image.
- VK_IMAGE_TYPE_3D specifies a three-dimensional image.

Possible values of VkImageCreateInfo::`tiling`, specifying the tiling arrangement of texel blocks in an image, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageTiling {
    VK_IMAGE_TILING_OPTIMAL = 0,
    VK_IMAGE_TILING_LINEAR = 1,
  // Provided by VK_EXT_image_drm_format_modifier
    VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT = 1000158000,
} VkImageTiling;
```

- VK_IMAGE_TILING_OPTIMAL specifies optimal tiling (texels are laid out in an implementation-dependent arrangement, for more efficient memory access).
- VK_IMAGE_TILING_LINEAR specifies linear tiling (texels are laid out in memory in row-major order, possibly with some padding on each row).
- VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT specifies that the image's tiling is defined by a Linux DRM format modifier. The modifier is specified at image creation with VkImageDrmFormatModifierListCreateInfoEXT or VkImageDrmFormatModifierExplicitCreateInfoEXT, and **can** be queried with vkGetImageDrmFormatModifierPropertiesEXT.

To query the memory layout of an image subresource, call:

```
// Provided by VK_VERSION_1_0
void vkGetImageSubresourceLayout(
    VkDevice                                    device,
    VkImage                                     image,
    const VkImageSubresource*                   pSubresource,
    VkSubresourceLayout*                        pLayout);
```

- `device` is the logical device that owns the image.
- `image` is the image whose layout is being queried.
- `pSubresource` is a pointer to a VkImageSubresource structure selecting a specific image

subresource from the image.

- `pLayout` is a pointer to a VkSubresourceLayout structure in which the layout is returned.

If the image is linear, then the returned layout is valid for host access.

If the image's tiling is VK_IMAGE_TILING_LINEAR and its format is a multi-planar format, then `vkGetImageSubresourceLayout` describes one *format plane* of the image. If the image's tiling is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then `vkGetImageSubresourceLayout` describes one *memory plane* of the image. If the image's tiling is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT and the image is non-linear, then the returned layout has an implementation-dependent meaning; the vendor of the image's DRM format modifier **may** provide documentation that explains how to interpret the returned layout.

`vkGetImageSubresourceLayout` is invariant for the lifetime of a single image.

Valid Usage

- VUID-vkGetImageSubresourceLayout-image-07790
  `image` **must** have been created with `tiling` equal to VK_IMAGE_TILING_LINEAR or VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT
- VUID-vkGetImageSubresourceLayout-aspectMask-00997
  The `aspectMask` member of `pSubresource` **must** only have a single bit set
- VUID-vkGetImageSubresourceLayout-mipLevel-01716
  The `mipLevel` member of `pSubresource` **must** be less than the `mipLevels` specified in `image`
- VUID-vkGetImageSubresourceLayout-arrayLayer-01717
  The `arrayLayer` member of `pSubresource` **must** be less than the `arrayLayers` specified in `image`
- VUID-vkGetImageSubresourceLayout-format-08886
  If `format` of the `image` is a color format that is not a multi-planar format, and `tiling` of the `image` is VK_IMAGE_TILING_LINEAR or VK_IMAGE_TILING_OPTIMAL, the `aspectMask` member of `pSubresource` **must** be VK_IMAGE_ASPECT_COLOR_BIT
- VUID-vkGetImageSubresourceLayout-format-04462
  If `format` of the `image` has a depth component, the `aspectMask` member of `pSubresource` **must** contain VK_IMAGE_ASPECT_DEPTH_BIT
- VUID-vkGetImageSubresourceLayout-format-04463
  If `format` of the `image` has a stencil component, the `aspectMask` member of `pSubresource` **must** contain VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkGetImageSubresourceLayout-format-04464
  If `format` of the `image` does not contain a stencil or depth component, the `aspectMask` member of `pSubresource` **must** not contain VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkGetImageSubresourceLayout-tiling-08717
  If the `tiling` of the `image` is VK_IMAGE_TILING_LINEAR and has a multi-planar format, then the `aspectMask` member of `pSubresource` **must** be a single valid multi-planar aspect

mask bit

- VUID-vkGetImageSubresourceLayout-tiling-09433
  If the `tiling` of the `image` is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then the `aspectMask` member of `pSubresource` **must** be `VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT` and the index *i* **must** be less than the VkDrmFormatModifierPropertiesEXT::`drmFormatModifierPlaneCount` associated with the image's `format` and VkImageDrmFormatModifierPropertiesEXT::`drmFormatModifier`

Valid Usage (Implicit)

- VUID-vkGetImageSubresourceLayout-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetImageSubresourceLayout-image-parameter
  `image` **must** be a valid VkImage handle
- VUID-vkGetImageSubresourceLayout-pSubresource-parameter
  `pSubresource` **must** be a valid pointer to a valid VkImageSubresource structure
- VUID-vkGetImageSubresourceLayout-pLayout-parameter
  `pLayout` **must** be a valid pointer to a VkSubresourceLayout structure
- VUID-vkGetImageSubresourceLayout-image-parent
  `image` **must** have been created, allocated, or retrieved from `device`

The `VkImageSubresource` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageSubresource {
    VkImageAspectFlags    aspectMask;
    uint32_t              mipLevel;
    uint32_t              arrayLayer;
} VkImageSubresource;
```

- `aspectMask` is a VkImageAspectFlags value selecting the image *aspect*.
- `mipLevel` selects the mipmap level.
- `arrayLayer` selects the array layer.

Valid Usage (Implicit)

- VUID-VkImageSubresource-aspectMask-parameter
  `aspectMask` **must** be a valid combination of VkImageAspectFlagBits values

Information about the layout of the image subresource is returned in a `VkSubresourceLayout` structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkSubresourceLayout {
    VkDeviceSize    offset;
    VkDeviceSize    size;
    VkDeviceSize    rowPitch;
    VkDeviceSize    arrayPitch;
    VkDeviceSize    depthPitch;
} VkSubresourceLayout;
```

- `offset` is the byte offset from the start of the image or the plane where the image subresource begins.
- `size` is the size in bytes of the image subresource. `size` includes any extra memory that is required based on `rowPitch`.
- `rowPitch` describes the number of bytes between each row of texels in an image.
- `arrayPitch` describes the number of bytes between each array layer of an image.
- `depthPitch` describes the number of bytes between each slice of a 3D image.

If the image is linear, then `rowPitch`, `arrayPitch` and `depthPitch` describe the layout of the image subresource in linear memory. For uncompressed formats, `rowPitch` is the number of bytes between texels with the same x coordinate in adjacent rows (y coordinates differ by one). `arrayPitch` is the number of bytes between texels with the same x and y coordinate in adjacent array layers of the image (array layer values differ by one). `depthPitch` is the number of bytes between texels with the same x and y coordinate in adjacent slices of a 3D image (z coordinates differ by one). Expressed as an addressing formula, the starting byte of a texel in the image subresource has address:

```
// (x,y,z,layer) are in texel coordinates
address(x,y,z,layer) = layer*arrayPitch + z*depthPitch + y*rowPitch + x*elementSize +
offset
```

For compressed formats, the `rowPitch` is the number of bytes between compressed texel blocks in adjacent rows. `arrayPitch` is the number of bytes between compressed texel blocks in adjacent array layers. `depthPitch` is the number of bytes between compressed texel blocks in adjacent slices of a 3D image.

```
// (x,y,z,layer) are in compressed texel block coordinates
address(x,y,z,layer) = layer*arrayPitch + z*depthPitch + y*rowPitch + x
*compressedTexelBlockByteSize + offset;
```

The value of `arrayPitch` is undefined for images that were not created as arrays. `depthPitch` is defined only for 3D images.

If the image has a *single-plane* color format and its tiling is VK_IMAGE_TILING_LINEAR , then the `aspectMask` member of `VkImageSubresource` **must** be VK_IMAGE_ASPECT_COLOR_BIT.

If the image has a depth/stencil format and its tiling is VK_IMAGE_TILING_LINEAR , then `aspectMask`

must be either VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT. On implementations that store depth and stencil aspects separately, querying each of these image subresource layouts will return a different `offset` and `size` representing the region of memory used for that aspect. On implementations that store depth and stencil aspects interleaved, the same `offset` and `size` are returned and represent the interleaved memory allocation.

If the image has a multi-planar format and its tiling is VK_IMAGE_TILING_LINEAR , then the `aspectMask` member of `VkImageSubresource` **must** be VK_IMAGE_ASPECT_PLANE_0_BIT, VK_IMAGE_ASPECT_PLANE_1_BIT, or (for 3-plane formats only) VK_IMAGE_ASPECT_PLANE_2_BIT. Querying each of these image subresource layouts will return a different `offset` and `size` representing the region of memory used for that plane. If the image is *disjoint*, then the `offset` is relative to the base address of the plane. If the image is *non-disjoint*, then the `offset` is relative to the base address of the image.

If the image's tiling is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then the `aspectMask` member of `VkImageSubresource` **must** be one of `VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT`, where the maximum allowed plane index *i* is defined by the VkDrmFormatModifierPropertiesEXT ::`drmFormatModifierPlaneCount` associated with the image's VkImageCreateInfo::`format` and modifier. The memory range used by the subresource is described by `offset` and `size`. If the image is *disjoint*, then the `offset` is relative to the base address of the *memory plane*. If the image is *non-disjoint*, then the `offset` is relative to the base address of the image. If the image is non-linear, then `rowPitch`, `arrayPitch`, and `depthPitch` have an implementation-dependent meaning.

To query the memory layout of an image subresource, call:

```
// Provided by VK_VERSION_1_4
void vkGetImageSubresourceLayout2(
    VkDevice                                    device,
    VkImage                                     image,
    const VkImageSubresource2*                  pSubresource,
    VkSubresourceLayout2*                       pLayout);
```

```
// Provided by VK_KHR_maintenance5
// Equivalent to vkGetImageSubresourceLayout2
void vkGetImageSubresourceLayout2KHR(
    VkDevice                                    device,
    VkImage                                     image,
    const VkImageSubresource2*                  pSubresource,
    VkSubresourceLayout2*                       pLayout);
```

```
// Provided by VK_EXT_host_image_copy, VK_EXT_image_compression_control
// Equivalent to vkGetImageSubresourceLayout2
void vkGetImageSubresourceLayout2EXT(
    VkDevice                                    device,
    VkImage                                     image,
    const VkImageSubresource2*                  pSubresource,
    VkSubresourceLayout2*                       pLayout);
```

- `device` is the logical device that owns the image.
- `image` is the image whose layout is being queried.
- `pSubresource` is a pointer to a VkImageSubresource2 structure selecting a specific image for the image subresource.
- `pLayout` is a pointer to a VkSubresourceLayout2 structure in which the layout is returned.

`vkGetImageSubresourceLayout2` behaves similarly to vkGetImageSubresourceLayout, with the ability to specify extended inputs via chained input structures, and to return extended information via chained output structures.

It is legal to call `vkGetImageSubresourceLayout2` with an `image` created with `tiling` equal to VK_IMAGE_TILING_OPTIMAL, but the members of VkSubresourceLayout2::`subresourceLayout` will have undefined values in this case.

NOTE Structures chained from VkImageSubresource2::`pNext` will also be updated when `tiling` is equal to VK_IMAGE_TILING_OPTIMAL.

Valid Usage

- VUID-vkGetImageSubresourceLayout2-aspectMask-00997
  The `aspectMask` member of `pSubresource` **must** only have a single bit set
- VUID-vkGetImageSubresourceLayout2-mipLevel-01716
  The `mipLevel` member of `pSubresource` **must** be less than the `mipLevels` specified in `image`
- VUID-vkGetImageSubresourceLayout2-arrayLayer-01717
  The `arrayLayer` member of `pSubresource` **must** be less than the `arrayLayers` specified in `image`
- VUID-vkGetImageSubresourceLayout2-format-08886
  If `format` of the `image` is a color format that is not a multi-planar format, and `tiling` of the `image` is VK_IMAGE_TILING_LINEAR or VK_IMAGE_TILING_OPTIMAL, the `aspectMask` member of `pSubresource` **must** be VK_IMAGE_ASPECT_COLOR_BIT
- VUID-vkGetImageSubresourceLayout2-format-04462
  If `format` of the `image` has a depth component, the `aspectMask` member of `pSubresource` **must** contain VK_IMAGE_ASPECT_DEPTH_BIT
- VUID-vkGetImageSubresourceLayout2-format-04463
  If `format` of the `image` has a stencil component, the `aspectMask` member of `pSubresource` **must** contain VK_IMAGE_ASPECT_STENCIL_BIT

- VUID-vkGetImageSubresourceLayout2-format-04464
  If `format` of the `image` does not contain a stencil or depth component, the `aspectMask` member of `pSubresource` **must** not contain VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkGetImageSubresourceLayout2-tiling-08717
  If the `tiling` of the `image` is VK_IMAGE_TILING_LINEAR and has a multi-planar format, then the `aspectMask` member of `pSubresource` **must** be a single valid multi-planar aspect mask bit
- VUID-vkGetImageSubresourceLayout2-tiling-09435
  If the `tiling` of the `image` is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then the `aspectMask` member of `pSubresource` **must** be `VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT` and the index *i* **must** be less than the VkDrmFormatModifierPropertiesEXT::`drmFormatModifierPlaneCount` associated with the image's `format` and VkImageDrmFormatModifierPropertiesEXT::`drmFormatModifier`

Valid Usage (Implicit)

- VUID-vkGetImageSubresourceLayout2-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetImageSubresourceLayout2-image-parameter
  `image` **must** be a valid VkImage handle
- VUID-vkGetImageSubresourceLayout2-pSubresource-parameter
  `pSubresource` **must** be a valid pointer to a valid VkImageSubresource2 structure
- VUID-vkGetImageSubresourceLayout2-pLayout-parameter
  `pLayout` **must** be a valid pointer to a VkSubresourceLayout2 structure
- VUID-vkGetImageSubresourceLayout2-image-parent
  `image` **must** have been created, allocated, or retrieved from `device`

The `VkImageSubresource2` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkImageSubresource2 {
    VkStructureType       sType;
    void*                 pNext;
    VkImageSubresource    imageSubresource;
} VkImageSubresource2;
```

```
// Provided by VK_KHR_maintenance5
// Equivalent to VkImageSubresource2
typedef VkImageSubresource2 VkImageSubresource2KHR;
```

```
// Provided by VK_EXT_host_image_copy, VK_EXT_image_compression_control
// Equivalent to VkImageSubresource2
typedef VkImageSubresource2 VkImageSubresource2EXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `imageSubresource` is a VkImageSubresource structure.

Valid Usage (Implicit)

- VUID-VkImageSubresource2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_SUBRESOURCE_2
- VUID-VkImageSubresource2-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkImageSubresource2-imageSubresource-parameter
  `imageSubresource` **must** be a valid VkImageSubresource structure

Information about the layout of the image subresource is returned in a `VkSubresourceLayout2` structure:

```
// Provided by VK_VERSION_1_4
typedef struct VkSubresourceLayout2 {
    VkStructureType        sType;
    void*                  pNext;
    VkSubresourceLayout    subresourceLayout;
} VkSubresourceLayout2;
```

```
// Provided by VK_KHR_maintenance5
// Equivalent to VkSubresourceLayout2
typedef VkSubresourceLayout2 VkSubresourceLayout2KHR;
```

```
// Provided by VK_EXT_host_image_copy, VK_EXT_image_compression_control
// Equivalent to VkSubresourceLayout2
typedef VkSubresourceLayout2 VkSubresourceLayout2EXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `subresourceLayout` is a VkSubresourceLayout structure.

Valid Usage (Implicit)

- VUID-VkSubresourceLayout2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SUBRESOURCE_LAYOUT_2
- VUID-VkSubresourceLayout2-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkImageCompressionPropertiesEXT or VkSubresourceHostMemcpySize
- VUID-VkSubresourceLayout2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

To query the memory size needed to copy to or from an image using vkCopyMemoryToImage or vkCopyImageToMemory when the VK_HOST_IMAGE_COPY_MEMCPY_BIT flag is specified, add a VkSubresourceHostMemcpySize structure to the `pNext` chain of the VkSubresourceLayout2 structure in a call to vkGetImageSubresourceLayout2.

The `VkSubresourceHostMemcpySize` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkSubresourceHostMemcpySize {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceSize       size;
} VkSubresourceHostMemcpySize;
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to VkSubresourceHostMemcpySize
typedef VkSubresourceHostMemcpySize VkSubresourceHostMemcpySizeEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `size` is the size in bytes of the image subresource.

Valid Usage (Implicit)

- VUID-VkSubresourceHostMemcpySize-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SUBRESOURCE_HOST_MEMCPY_SIZE

Structure Chaining

Extends the structure

- VkSubresourceLayout2

To query the memory layout of an image subresource, without an image object, call:

```
// Provided by VK_VERSION_1_4
void vkGetDeviceImageSubresourceLayout(
    VkDevice                                    device,
    const VkDeviceImageSubresourceInfo*         pInfo,
    VkSubresourceLayout2*                       pLayout);
```

```
// Provided by VK_KHR_maintenance5
// Equivalent to vkGetDeviceImageSubresourceLayout
void vkGetDeviceImageSubresourceLayoutKHR(
    VkDevice                                    device,
    const VkDeviceImageSubresourceInfo*         pInfo,
    VkSubresourceLayout2*                       pLayout);
```

- `device` is the logical device that owns the image.
- `pInfo` is a pointer to a VkDeviceImageSubresourceInfo structure containing parameters required for the subresource layout query.
- `pLayout` is a pointer to a VkSubresourceLayout2 structure in which the layout is returned.

`vkGetDeviceImageSubresourceLayout` behaves similarly to vkGetImageSubresourceLayout2, but uses a VkImageCreateInfo structure to specify the image rather than a VkImage object.

Valid Usage (Implicit)

- VUID-vkGetDeviceImageSubresourceLayout-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDeviceImageSubresourceLayout-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkDeviceImageSubresourceInfo structure
- VUID-vkGetDeviceImageSubresourceLayout-pLayout-parameter
  `pLayout` **must** be a valid pointer to a VkSubresourceLayout2 structure

The `VkDeviceImageSubresourceInfo` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkDeviceImageSubresourceInfo {
    VkStructureType               sType;
    const void*                   pNext;
    const VkImageCreateInfo*      pCreateInfo;
    const VkImageSubresource2*    pSubresource;
} VkDeviceImageSubresourceInfo;
```

```
// Provided by VK_KHR_maintenance5
// Equivalent to VkDeviceImageSubresourceInfo
typedef VkDeviceImageSubresourceInfo VkDeviceImageSubresourceInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pCreateInfo` is a pointer to a VkImageCreateInfo structure containing parameters affecting creation of the image to query.
- `pSubresource` is a pointer to a VkImageSubresource2 structure selecting a specific image subresource for the query.

Valid Usage

- VUID-VkDeviceImageSubresourceInfo-aspectMask-00997
  The `aspectMask` member of `pSubresource` **must** only have a single bit set
- VUID-VkDeviceImageSubresourceInfo-mipLevel-01716
  The `mipLevel` member of `pSubresource` **must** be less than the `mipLevels` specified in `pCreateInfo`
- VUID-VkDeviceImageSubresourceInfo-arrayLayer-01717
  The `arrayLayer` member of `pSubresource` **must** be less than the `arrayLayers` specified in `pCreateInfo`
- VUID-VkDeviceImageSubresourceInfo-format-08886
  If `format` of the `pCreateInfo` is a color format that is not a multi-planar format, and `tiling` of the `pCreateInfo` is VK_IMAGE_TILING_LINEAR or VK_IMAGE_TILING_OPTIMAL, the `aspectMask` member of `pSubresource` **must** be VK_IMAGE_ASPECT_COLOR_BIT
- VUID-VkDeviceImageSubresourceInfo-format-04462
  If `format` of the `pCreateInfo` has a depth component, the `aspectMask` member of `pSubresource` **must** contain VK_IMAGE_ASPECT_DEPTH_BIT
- VUID-VkDeviceImageSubresourceInfo-format-04463
  If `format` of the `pCreateInfo` has a stencil component, the `aspectMask` member of `pSubresource` **must** contain VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkDeviceImageSubresourceInfo-format-04464
  If `format` of the `pCreateInfo` does not contain a stencil or depth component, the `aspectMask` member of `pSubresource` **must** not contain VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkDeviceImageSubresourceInfo-tiling-08717
  If the `tiling` of the `pCreateInfo` is VK_IMAGE_TILING_LINEAR and has a multi-planar format, then the `aspectMask` member of `pSubresource` **must** be a single valid multi-planar aspect mask bit

Valid Usage (Implicit)

- VUID-VkDeviceImageSubresourceInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_IMAGE_SUBRESOURCE_INFO
- VUID-VkDeviceImageSubresourceInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDeviceImageSubresourceInfo-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkImageCreateInfo structure
- VUID-VkDeviceImageSubresourceInfo-pSubresource-parameter
  `pSubresource` **must** be a valid pointer to a valid VkImageSubresource2 structure

If an image was created with VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then the image has a Linux DRM format modifier. To query the *modifier*, call:

```
// Provided by VK_EXT_image_drm_format_modifier
VkResult vkGetImageDrmFormatModifierPropertiesEXT(
    VkDevice                                    device,
    VkImage                                     image,
    VkImageDrmFormatModifierPropertiesEXT*      pProperties);
```

- `device` is the logical device that owns the image.
- `image` is the queried image.
- `pProperties` is a pointer to a VkImageDrmFormatModifierPropertiesEXT structure in which properties of the image's *DRM format modifier* are returned.

Valid Usage

- VUID-vkGetImageDrmFormatModifierPropertiesEXT-image-02272
  `image` **must** have been created with `tiling` equal to VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT

Valid Usage (Implicit)

- VUID-vkGetImageDrmFormatModifierPropertiesEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetImageDrmFormatModifierPropertiesEXT-image-parameter
  `image` **must** be a valid VkImage handle
- VUID-vkGetImageDrmFormatModifierPropertiesEXT-pProperties-parameter
  `pProperties` **must** be a valid pointer to a VkImageDrmFormatModifierPropertiesEXT structure
- VUID-vkGetImageDrmFormatModifierPropertiesEXT-image-parent

`image` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkImageDrmFormatModifierPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier
typedef struct VkImageDrmFormatModifierPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint64_t           drmFormatModifier;
} VkImageDrmFormatModifierPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `drmFormatModifier` returns the image's Linux DRM format modifier.

If the `image` was created with VkImageDrmFormatModifierListCreateInfoEXT, then the returned `drmFormatModifier` **must** belong to the list of modifiers provided at time of image creation in VkImageDrmFormatModifierListCreateInfoEXT::`pDrmFormatModifiers`. If the `image` was created with VkImageDrmFormatModifierExplicitCreateInfoEXT, then the returned `drmFormatModifier` **must** be the modifier provided at time of image creation in VkImageDrmFormatModifierExplicitCreateInfoEXT::`drmFormatModifier`.

Valid Usage (Implicit)

- VUID-VkImageDrmFormatModifierPropertiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_PROPERTIES_EXT
- VUID-VkImageDrmFormatModifierPropertiesEXT-pNext-pNext
  `pNext` **must** be `NULL`

To destroy an image, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyImage(
    VkDevice                                    device,
    VkImage                                     image,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the image.
- `image` is the image to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyImage-image-01000
  All submitted commands that refer to `image`, either directly or via a `VkImageView`, **must** have completed execution
- VUID-vkDestroyImage-image-01001
  If `VkAllocationCallbacks` were provided when `image` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyImage-image-01002
  If no `VkAllocationCallbacks` were provided when `image` was created, `pAllocator` **must** be `NULL`
- VUID-vkDestroyImage-image-04882
  `image` **must** not have been acquired from vkGetSwapchainImagesKHR

Valid Usage (Implicit)

- VUID-vkDestroyImage-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyImage-image-parameter
  If `image` is not VK_NULL_HANDLE, `image` **must** be a valid VkImage handle
- VUID-vkDestroyImage-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyImage-image-parent
  If `image` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `image` **must** be externally synchronized

12.4.1. Image Format Features

Valid uses of a VkImage **may** depend on the image's *format features*, defined below. Such constraints are documented in the affected valid usage statement.

- If the image was created with VK_IMAGE_TILING_LINEAR, then its set of *format features* is the value of VkFormatProperties::`linearTilingFeatures` found by calling vkGetPhysicalDeviceFormatProperties on the same `format` as VkImageCreateInfo::`format`.
- If the image was created with VK_IMAGE_TILING_OPTIMAL, then its set of *format features* is the value of VkFormatProperties::`optimalTilingFeatures` found by calling vkGetPhysicalDeviceFormatProperties on the same `format` as VkImageCreateInfo::`format`.
- If the image was created with VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then:
  - The image's DRM format modifier is the value of VkImageDrmFormatModifierPropertiesEXT::`drmFormatModifier` found by calling vkGetImageDrmFormatModifierPropertiesEXT.
  - Let VkDrmFormatModifierPropertiesListEXT::`pDrmFormatModifierProperties` be the array found by calling vkGetPhysicalDeviceFormatProperties2 on the same `format` as VkImageCreateInfo::`format`.
  - Let `VkDrmFormatModifierPropertiesEXT prop` be the array element whose `drmFormatModifier` member is the value of the image's DRM format modifier.
  - Then the image's set of *format features* is the value of `prop`::`drmFormatModifierTilingFeatures`.

12.4.2. Image Mip Level Sizing

A *complete mipmap chain* is the full set of mip levels, from the largest mip level provided, down to the *minimum mip level size*.

Conventional Images

For conventional images, the dimensions of each successive mip level, n+1, are:

$$width_{n+1} = \max(\lfloor width_n/2 \rfloor, 1)$$

$$height_{n+1} = \max(\lfloor height_n/2 \rfloor, 1)$$

$$depth_{n+1} = \max(\lfloor depth_n/2 \rfloor, 1)$$

where $width_n$, $height_n$, and $depth_n$ are the dimensions of the next larger mip level, n.

The minimum mip level size is:

- 1 for one-dimensional images,
- 1x1 for two-dimensional images, and

- 1x1x1 for three-dimensional images.

The number of levels in a complete mipmap chain is:

$$\lfloor \log_2(\max(\texttt{width}_0, \texttt{height}_0, \texttt{depth}_0)) \rfloor + 1$$

where $\texttt{width}_0$, $\texttt{height}_0$, and $\texttt{depth}_0$ are the dimensions of the largest (most detailed) mip level, `0`.

12.4.3. Image Capture Replay for Descriptor Heaps

To allow for a replay tool to capture and replay image descriptors used with descriptor heaps, opaque data **must** be captured during the initial recording and provided when recreating the image during replay, using the following functions:

To get the opaque capture descriptor data for images, call:

```
// Provided by VK_EXT_descriptor_heap
VkResult vkGetImageOpaqueCaptureDataEXT(
    VkDevice                                    device,
    uint32_t                                    imageCount,
    const VkImage*                              pImages,
    VkHostAddressRangeEXT*                      pDatas);
```

- `device` is the logical device that gets the data.
- `imageCount` is the number of images to retrieve data from.
- `pImages` is a pointer to an array of VkImage objects to retrieve the opaque capture data from.
- `pDatas` is a pointer to an array of VkHostAddressRangeEXT structures defining the host address ranges where each image's opaque capture data will be written.

Valid Usage

- VUID-vkGetImageOpaqueCaptureDataEXT-descriptorHeapCaptureReplay-11282
  The `descriptorHeapCaptureReplay` feature **must** be enabled
- VUID-vkGetImageOpaqueCaptureDataEXT-size-11283
  The `size` member of each element of `pDatas` **must** be equal to `imageCaptureReplayOpaqueDataSize`
- VUID-vkGetImageOpaqueCaptureDataEXT-device-11284
  If `device` was created with multiple physical devices, then the `bufferDeviceAddressMultiDevice` feature **must** be enabled
- VUID-vkGetImageOpaqueCaptureDataEXT-pImages-11285
  Each element of `pImages` **must** have been created with VK_IMAGE_CREATE_DESCRIPTOR_HEAP_CAPTURE_REPLAY_BIT_EXT set in VkImageCreateInfo::`flags`

Valid Usage (Implicit)

- VUID-vkGetImageOpaqueCaptureDataEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetImageOpaqueCaptureDataEXT-pImages-parameter
  `pImages` **must** be a valid pointer to an array of `imageCount` valid VkImage handles
- VUID-vkGetImageOpaqueCaptureDataEXT-pDatas-parameter
  `pDatas` **must** be a valid pointer to an array of `imageCount` VkHostAddressRangeEXT structures
- VUID-vkGetImageOpaqueCaptureDataEXT-imageCount-arraylength
  `imageCount` **must** be greater than `0`
- VUID-vkGetImageOpaqueCaptureDataEXT-pImages-parent
  Each element of `pImages` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkOpaqueCaptureDataCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkOpaqueCaptureDataCreateInfoEXT {
    VkStructureType                     sType;
    const void*                         pNext;
    const VkHostAddressRangeConstEXT*   pData;
} VkOpaqueCaptureDataCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pData` is a pointer to the range of host memory containing opaque data previously captured via vkGetImageOpaqueCaptureDataEXT.

When an image is created with VK_IMAGE_CREATE_DESCRIPTOR_HEAP_CAPTURE_REPLAY_BIT_EXT set in VkImageCreateInfo ::`flags`, if the `pNext` chain of VkImageCreateInfo includes this structure, and `pData` is not `NULL`, the

implementation will attempt to recreate the image such that descriptors written with vkWriteResourceDescriptorsEXT will be reproduced with the same bit pattern as during capture if possible. If the implementation is unable to recreate the image based on this data, image creation will fail and return VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS.

If this structure is not present, it is equivalent to setting `pData` to `NULL`.

Valid Usage (Implicit)

- VUID-VkOpaqueCaptureDataCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_OPAQUE_CAPTURE_DATA_CREATE_INFO_EXT
- VUID-VkOpaqueCaptureDataCreateInfoEXT-pData-parameter
  If `pData` is not `NULL`, `pData` **must** be a valid pointer to a valid VkHostAddressRangeConstEXT structure

Structure Chaining

Extends the structure

- VkImageCreateInfo

12.5. Image Layouts

Images are stored in implementation-dependent opaque layouts in memory. Each layout has limitations on what kinds of operations are supported for image subresources using the layout. At any given time, the data representing an image subresource in memory exists in a particular layout which is determined by the most recent layout transition that was performed on that image subresource. Applications have control over which layout each image subresource uses, and **can** transition an image subresource from one layout to another. Transitions **can** happen with an image memory barrier, included as part of a vkCmdPipelineBarrier or a vkCmdWaitEvents command buffer command (see Image Memory Barriers), or as part of a subpass dependency within a render pass (see `VkSubpassDependency`).

Image layout is per-image subresource. Separate image subresources of the same image **can** be in different layouts at the same time, with the exception that depth and stencil aspects of a given image subresource **can** only be in different layouts if the `separateDepthStencilLayouts` feature is enabled. When a `VkImageView` descriptor is accessed on the device, all image subresources must be in a valid image layout.

NOTE

Each layout **may** offer optimal performance for a specific usage of image memory. For example, an image with a layout of VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL **may** provide optimal performance for use as a color attachment, but be unsupported for use in transfer commands. Applications **can** transition an image subresource from one layout to another in order to achieve optimal performance when the image subresource is used for multiple kinds of operations. After initialization, applications need not use

> any layout other than the general layout, though this **may** produce suboptimal performance on some implementations.

Upon creation, all image subresources of an image are initially in the same layout, where that layout is selected by the `VkImageCreateInfo::initialLayout` member. The `initialLayout` **must** be either VK_IMAGE_LAYOUT_UNDEFINED, VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT, or VK_IMAGE_LAYOUT_PREINITIALIZED. If it is VK_IMAGE_LAYOUT_PREINITIALIZED and the image has linear tiling, then the image data **can** be preinitialized by the host while using this layout, and the transition away from this layout will preserve that data. If it is VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT, then the image data is interpreted to be zeroed. If the memory bound to this image is not fully zeroed when the image is transitioned to a different layout, the results are undefined. If it is VK_IMAGE_LAYOUT_UNDEFINED, then the contents of the data are considered to be undefined, and the transition away from this layout is not guaranteed to preserve that data. For either of these initial layouts, any image subresources **must** be transitioned to another layout before they are accessed by the device.

Host access to image memory is only well-defined for linear images and for image subresources of those images which are currently in any of the following layouts:

- VK_IMAGE_LAYOUT_PREINITIALIZED
- VK_IMAGE_LAYOUT_GENERAL
- VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT

Calling vkGetImageSubresourceLayout for a linear image returns a subresource layout mapping that is valid for either of those image layouts.

The set of image layouts consists of:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageLayout {
    VK_IMAGE_LAYOUT_UNDEFINED = 0,
    VK_IMAGE_LAYOUT_GENERAL = 1,
    VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL = 2,
    VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL = 3,
    VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL = 4,
    VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL = 5,
    VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL = 6,
    VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL = 7,
    VK_IMAGE_LAYOUT_PREINITIALIZED = 8,
  // Provided by VK_VERSION_1_1
    VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL = 1000117000,
  // Provided by VK_VERSION_1_1
    VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL = 1000117001,
  // Provided by VK_VERSION_1_2
    VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL = 1000241000,
  // Provided by VK_VERSION_1_2
    VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL = 1000241001,
  // Provided by VK_VERSION_1_2
    VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL = 1000241002,
```

```
  // Provided by VK_VERSION_1_2
    VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL = 1000241003,
  // Provided by VK_VERSION_1_3
    VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL = 1000314000,
  // Provided by VK_VERSION_1_3
    VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL = 1000314001,
  // Provided by VK_VERSION_1_4
    VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ = 1000232000,
  // Provided by VK_KHR_swapchain
    VK_IMAGE_LAYOUT_PRESENT_SRC_KHR = 1000001002,
  // Provided by VK_KHR_video_decode_queue
    VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR = 1000024000,
  // Provided by VK_KHR_video_decode_queue
    VK_IMAGE_LAYOUT_VIDEO_DECODE_SRC_KHR = 1000024001,
  // Provided by VK_KHR_video_decode_queue
    VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR = 1000024002,
  // Provided by VK_KHR_shared_presentable_image
    VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR = 1000111000,
  // Provided by VK_EXT_fragment_density_map
    VK_IMAGE_LAYOUT_FRAGMENT_DENSITY_MAP_OPTIMAL_EXT = 1000218000,
  // Provided by VK_KHR_fragment_shading_rate
    VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR = 1000164003,
  // Provided by VK_KHR_video_encode_queue
    VK_IMAGE_LAYOUT_VIDEO_ENCODE_DST_KHR = 1000299000,
  // Provided by VK_KHR_video_encode_queue
    VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR = 1000299001,
  // Provided by VK_KHR_video_encode_queue
    VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR = 1000299002,
  // Provided by VK_EXT_attachment_feedback_loop_layout
    VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT = 1000339000,
  // Provided by VK_KHR_video_encode_quantization_map
    VK_IMAGE_LAYOUT_VIDEO_ENCODE_QUANTIZATION_MAP_KHR = 1000553000,
  // Provided by VK_EXT_zero_initialize_device_memory
    VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT = 1000620000,
  // Provided by VK_KHR_maintenance2
    VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL_KHR =
VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL,
  // Provided by VK_KHR_maintenance2
    VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL_KHR =
VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL,
  // Provided by VK_KHR_dynamic_rendering_local_read
    VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR = VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ,
  // Provided by VK_KHR_separate_depth_stencil_layouts
    VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL_KHR =
VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL,
  // Provided by VK_KHR_separate_depth_stencil_layouts
    VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL_KHR =
VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL,
  // Provided by VK_KHR_separate_depth_stencil_layouts
    VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL_KHR =
VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL,
```

```
  // Provided by VK_KHR_separate_depth_stencil_layouts
    VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL_KHR =
VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL,
  // Provided by VK_KHR_synchronization2
    VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR = VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL,
  // Provided by VK_KHR_synchronization2
    VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR = VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL,
} VkImageLayout;
```

The type(s) of device access supported by each layout are:

- VK_IMAGE_LAYOUT_UNDEFINED specifies that the layout is unknown. Image memory **cannot** be transitioned into this layout. This layout **can** be used as the `initialLayout` member of VkImageCreateInfo. This layout **can** be used in place of the current image layout in a layout transition, but doing so will cause the contents of the image's memory to be undefined.
- VK_IMAGE_LAYOUT_PREINITIALIZED specifies that a linear image's memory is in a defined layout and **can** be populated by data, but that it has not yet been initialized by the driver. Image memory **cannot** be transitioned into this layout. This layout **can** be used as the `initialLayout` member of VkImageCreateInfo when `tiling` is VK_IMAGE_TILING_LINEAR. This layout is intended to be used as the initial layout for an image whose contents are written by the host, and hence the data **can** be written to memory immediately, without first executing a layout transition.
- VK_IMAGE_LAYOUT_GENERAL supports all types of device access, unless specified otherwise.
- VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT specifies that an image's memory is in a defined layout and is zeroed, but that it has not yet been initialized by the driver. Image memory **cannot** be transitioned into this layout. This layout **can** be used as the `initialLayout` member of VkImageCreateInfo. This layout is intended to be used as the initial layout for an image whose contents are already zeroed, either from being explicitly set to zero by an application or from being allocated with VK_MEMORY_ALLOCATE_ZERO_INITIALIZE_BIT_EXT.
- VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL specifies a layout that **must** only be used with attachment accesses in the graphics pipeline.
- VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL specifies a layout allowing read only access as an attachment, or in shaders as a sampled image, combined image/sampler, or input attachment.
- VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL **must** only be used as a color or resolve attachment in a `VkFramebuffer`. This layout is valid only for image subresources of images created with the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT usage flag set.
- VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL specifies a layout for both the depth and stencil aspects of a depth/stencil format image allowing read and write access as a depth/stencil attachment. It is equivalent to VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL and VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL.
- VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL specifies a layout for both the depth and stencil aspects of a depth/stencil format image allowing read only access as a depth/stencil attachment or in shaders as a sampled image, combined image/sampler, or input attachment. It is equivalent to VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL and

VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL.

- VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL specifies a layout for depth/stencil format images allowing read and write access to the stencil aspect as a stencil attachment, and read only access to the depth aspect as a depth attachment or in shaders as a sampled image, combined image/sampler, or input attachment. It is equivalent to VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL and VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL.
- VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL specifies a layout for depth/stencil format images allowing read and write access to the depth aspect as a depth attachment, and read only access to the stencil aspect as a stencil attachment or in shaders as a sampled image, combined image/sampler, or input attachment. It is equivalent to VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL and VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL.
- VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL specifies a layout for the depth aspect of a depth/stencil format image allowing read and write access as a depth attachment.
- VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL specifies a layout for the depth aspect of a depth/stencil format image allowing read-only access as a depth attachment or in shaders as a sampled image, combined image/sampler, or input attachment.
- VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL specifies a layout for the stencil aspect of a depth/stencil format image allowing read and write access as a stencil attachment.
- VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL specifies a layout for the stencil aspect of a depth/stencil format image allowing read-only access as a stencil attachment or in shaders as a sampled image, combined image/sampler, or input attachment.
- VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL specifies a layout allowing read-only access in a shader as a sampled image, combined image/sampler, or input attachment. This layout is valid only for image subresources of images created with the VK_IMAGE_USAGE_SAMPLED_BIT or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage bits enabled.
- VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL **must** only be used as a source image of a transfer command (see the definition of VK_PIPELINE_STAGE_TRANSFER_BIT). This layout is valid only for image subresources of images created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set.
- VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL **must** only be used as a destination image of a transfer command. This layout is valid only for image subresources of images created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set.
- VK_IMAGE_LAYOUT_PRESENT_SRC_KHR **must** only be used for presenting a presentable image for display.
- VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR is valid only for shared presentable images, and **must** be used for any usage the image supports.
- VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR **must** only be used as a fragment shading rate attachment or This layout is valid only for image subresources of images created with the VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR usage flag set.

- VK_IMAGE_LAYOUT_FRAGMENT_DENSITY_MAP_OPTIMAL_EXT **must** only be used as a fragment density map attachment in a VkRenderPass. This layout is valid only for image subresources of images created with the VK_IMAGE_USAGE_FRAGMENT_DENSITY_MAP_BIT_EXT usage flag set.
- VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR **must** only be used as a decode output picture in a video decode operation. This layout is valid only for image subresources of images created with the VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR usage flag set.
- VK_IMAGE_LAYOUT_VIDEO_DECODE_SRC_KHR is reserved for future use.
- VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR **must** only be used as an output reconstructed picture or an input reference picture in a video decode operation. This layout is valid only for image subresources of images created with the VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR usage flag set.
- VK_IMAGE_LAYOUT_VIDEO_ENCODE_DST_KHR is reserved for future use.
- VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR **must** only be used as an encode input picture in a video encode operation. This layout is valid only for image subresources of images created with the VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR usage flag set.
- VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR **must** only be used as an output reconstructed picture or an input reference picture in a video encode operation. This layout is valid only for image subresources of images created with the VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR usage flag set.
- VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT **must** only be used as either a color attachment or depth/stencil attachment and/or read-only access in a shader as a sampled image, combined image/sampler, or input attachment. This layout is valid only for image subresources of images created with the VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT usage flag set, and either the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flags set, and either the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT or VK_IMAGE_USAGE_SAMPLED_BIT usage flags set.
- VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ **must** only be used as either a storage image, or a color or depth/stencil attachment and an input attachment. This layout is valid only for image subresources of images created with either the VK_IMAGE_USAGE_STORAGE_BIT usage flag set, or both the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT and either of the VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT usage flags set.
- VK_IMAGE_LAYOUT_VIDEO_ENCODE_QUANTIZATION_MAP_KHR **must** only be used as a quantization map in a video encode operation. This layout is valid only for image subresources of images created with the VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR usage flags set.

The layout of each image subresource is not a state of the image subresource itself, but is rather a property of how the data in memory is organized, and thus for each mechanism of accessing an image in the API the application **must** specify a parameter or structure member that indicates

which image layout the image subresource(s) are considered to be in when the image will be accessed. For transfer commands, this is a parameter to the command (see Clear Commands and Copy Commands). For use as a framebuffer attachment, this is a member in the substructures of the VkRenderPassCreateInfo (see Render Pass). For use in a descriptor set, this is a member in the `VkDescriptorImageInfo` structure (see Descriptor Set Updates).

If the `unifiedImageLayouts` feature is enabled, the VK_IMAGE_LAYOUT_GENERAL image layout **may** be used in place of the other layouts where allowed with no loss of performance.

NOTE

VK_IMAGE_LAYOUT_GENERAL can be a useful catch-all image layout, but there are situations where a dedicated image layout must be used instead. Some examples include:

- VK_IMAGE_LAYOUT_PRESENT_SRC_KHR
- VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR
- VK_IMAGE_LAYOUT_VIDEO_DECODE_SRC_KHR, VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR, and VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR without the `unifiedImageLayoutsVideo` feature
- VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR, VK_IMAGE_LAYOUT_VIDEO_ENCODE_DST_KHR, and VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR without the `unifiedImageLayoutsVideo` feature
- VK_IMAGE_LAYOUT_VIDEO_ENCODE_QUANTIZATION_MAP_KHR without the `unifiedImageLayoutsVideo` feature

NOTE

While VK_IMAGE_LAYOUT_GENERAL suggests that all types of device access are possible, it does not mean that all patterns of memory accesses are safe in all situations. Common Render Pass Data Races outlines some situations where data races are unavoidable. For example, when a subresource is used as both an attachment and a sampled image (i.e., not an input attachment), enabling feedback loop adds extra guarantees which VK_IMAGE_LAYOUT_GENERAL alone does not.

12.5.1. Image Layout Matching Rules

At the time that any command buffer command accessing an image executes on any queue, the layouts of the image subresources that are accessed **must** all match exactly the layout specified via the API controlling those accesses, except in case of accesses to an image with a depth/stencil format performed through descriptors referring to only a single aspect of the image, where the following relaxed matching rules apply:

- Descriptors referring just to the depth aspect of a depth/stencil image only need to match in the image layout of the depth aspect, thus VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL and VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL are considered to match.

- Descriptors referring just to the stencil aspect of a depth/stencil image only need to match in the image layout of the stencil aspect, thus VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL and VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL are considered to match.

When performing a layout transition on an image subresource, the old layout value **must** either equal the current layout of the image subresource (at the time the transition executes), or else be VK_IMAGE_LAYOUT_UNDEFINED (implying that the contents of the image subresource need not be preserved). The new layout used in a transition **must** not be any of:

- VK_IMAGE_LAYOUT_UNDEFINED
- VK_IMAGE_LAYOUT_PREINITIALIZED
- VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT

The image layout of each image subresource of a depth/stencil image created with VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT is dependent on the last sample locations used to render to the image subresource as a depth/stencil attachment, thus applications **must** provide the same sample locations that were last used to render to the given image subresource whenever a layout transition of the image subresource happens, otherwise the contents of the depth aspect of the image subresource become undefined.

In addition, depth reads from a depth/stencil attachment referring to an image subresource range of a depth/stencil image created with VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT using different sample locations than what have been last used to perform depth writes to the image subresources of the same image subresource range return undefined values.

Similarly, depth writes to a depth/stencil attachment referring to an image subresource range of a depth/stencil image created with VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT using different sample locations than what have been last used to perform depth writes to the image subresources of the same image subresource range make the contents of the depth aspect of those image subresources undefined.

12.5.2. Image Layout Rules with External Memory

When images created with external memory handle types listed in the external image implied layouts table are accessed outside of Vulkan, they **must** be in the layout specified in the table, and **can** be assumed to be in that layout when they are accessed in Vulkan after external access.

Table 13. External Image Implied Layouts

External Memory Handle Type	Implied Layout When Accessed Externally
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT	VK_IMAGE_LAYOUT_GENERAL
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT	VK_IMAGE_LAYOUT_GENERAL

12.6. Image Views

Image objects are not directly accessed by pipeline shaders for reading or writing image data. Instead, *image views* representing contiguous ranges of the image subresources and containing additional metadata are used for that purpose. Views **must** be created on images of compatible types, and **must** represent a valid subset of image subresources.

Image views are represented by VkImageView handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkImageView)
```

VK_REMAINING_ARRAY_LAYERS is a special constant value used for image views to indicate that all remaining array layers in an image after the base layer should be included in the view.

```
#define VK_REMAINING_ARRAY_LAYERS         (~0U)
```

VK_REMAINING_MIP_LEVELS is a special constant value used for image views to indicate that all remaining mipmap levels in an image after the base level should be included in the view.

```
#define VK_REMAINING_MIP_LEVELS           (~0U)
```

The types of image views that **can** be created are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageViewType {
    VK_IMAGE_VIEW_TYPE_1D = 0,
    VK_IMAGE_VIEW_TYPE_2D = 1,
    VK_IMAGE_VIEW_TYPE_3D = 2,
    VK_IMAGE_VIEW_TYPE_CUBE = 3,
    VK_IMAGE_VIEW_TYPE_1D_ARRAY = 4,
    VK_IMAGE_VIEW_TYPE_2D_ARRAY = 5,
    VK_IMAGE_VIEW_TYPE_CUBE_ARRAY = 6,
} VkImageViewType;
```

To create an image view, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateImageView(
    VkDevice                                    device,
    const VkImageViewCreateInfo*                pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkImageView*                                pView);
```

- `device` is the logical device that creates the image view.
- `pCreateInfo` is a pointer to a VkImageViewCreateInfo structure containing parameters to be used to create the image view.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pView` is a pointer to a VkImageView handle in which the resulting image view object is returned.

Valid Usage

- VUID-vkCreateImageView-device-09667
  `device` **must** support at least one queue family with one of the VK_QUEUE_VIDEO_ENCODE_BIT_KHR, VK_QUEUE_VIDEO_DECODE_BIT_KHR, VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT capabilities
- VUID-vkCreateImageView-image-09179
  VkImageViewCreateInfo::`image` **must** have been created from `device`

Valid Usage (Implicit)

- VUID-vkCreateImageView-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateImageView-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkImageViewCreateInfo structure
- VUID-vkCreateImageView-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateImageView-pView-parameter
  `pView` **must** be a valid pointer to a VkImageView handle
- VUID-vkCreateImageView-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkImageViewCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageViewCreateInfo {
    VkStructureType            sType;
    const void*                pNext;
    VkImageViewCreateFlags     flags;
    VkImage                    image;
    VkImageViewType            viewType;
    VkFormat                   format;
    VkComponentMapping         components;
    VkImageSubresourceRange    subresourceRange;
} VkImageViewCreateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask of VkImageViewCreateFlagBits specifying additional parameters of the image view.
- image is a VkImage on which the view will be created.
- viewType is a VkImageViewType value specifying the type of the image view.
- format is a VkFormat specifying the format and type used to interpret texel blocks of the image.
- components is a VkComponentMapping structure specifying a remapping of color components (or of depth or stencil components after they have been converted into color components).
- subresourceRange is a VkImageSubresourceRange structure selecting the set of mipmap levels and array layers to be accessible to the view.

Some of the image creation parameters are inherited by the view. In particular, image view creation inherits the implicit parameter usage specifying the allowed usages of the image view that, by default, takes the value of the corresponding usage parameter specified in VkImageCreateInfo at image creation time. The implicit usage **can** be overridden by adding a VkImageViewUsageCreateInfo or VkImageViewUsage2CreateInfoKHR structure to the pNext chain, but the view usage **must** be a subset of the image usage. If image has a depth-stencil format and was created with a VkImageStencilUsageCreateInfo structure included in the pNext chain of VkImageCreateInfo, the usage is calculated based on the subresourceRange.aspectMask provided:

- If aspectMask includes only VK_IMAGE_ASPECT_STENCIL_BIT, the implicit usage is equal to VkImageStencilUsageCreateInfo::stencilUsage.
- If aspectMask includes only VK_IMAGE_ASPECT_DEPTH_BIT, the implicit usage is equal to VkImageCreateInfo::usage.
- If both aspects are included in aspectMask, the implicit usage is equal to the intersection of VkImageCreateInfo::usage and VkImageStencilUsageCreateInfo::stencilUsage.

If image is a 3D image, its Z range **can** be restricted to a subset by adding a VkImageViewSlicedCreateInfoEXT to the pNext chain.

If `image` was not created with VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT, `format` **must** equal the `format` used to create `image`.

If `image` was created with VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT, `format` **can** be set to a different format as follows:

- If `image` was created with the VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT flag and `format` is an uncompressed format, it **must** be *size-compatible* with the `format` used to create `image`.
- If creating a single-plane view, `format` **must** equal one of the *plane view formats*.
- If creating a multi-plane view, `format` **must** equal the `format` used to create `image`.
- Otherwise, `format` **must** be compatible with the `format` of `image`.

Views of compatible formats will have the same mapping between texel coordinates and memory locations irrespective of the `format`, with only the interpretation of the bit pattern changing.

NOTE

Values intended to be used with one view format **may** not be exactly preserved when written or read through a different format. For example, an integer value that happens to have the bit pattern of a floating-point denorm or NaN **may** be flushed or canonicalized when written or read through a view with a floating-point format. Similarly, a value written through a signed normalized format that has a bit pattern exactly equal to -2^b **may** be changed to $-2^b + 1$ as described in Conversion from Normalized Fixed-Point to Floating-Point.

The texel dimensions of the view are derived as follows:

- If `image` was created with the VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT flag and `format` is an uncompressed format, the dimensions equal the dimensions of the selected mip level divided by the compressed texel block size and rounded up.
- If creating a single-plane view, the dimensions equal the *plane view dimensions*.
- Otherwise, the dimensions are equal to the dimensions of the selected mip level as defined in Image Mip Level Sizing.

If `subresourceRange.aspectMask` is one of the multi-planar aspect mask bits, a single-plane image view will be created. The `image` of a single-plane view **must** have been created with a multi-planar format and the VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT create flag.

The set of *plane view formats* for the plane is the compatible plane format and the formats compatible with it.

The *plane view dimensions* are derived from the dimensions of the `image` in the manner listed for plane compatibility for the plane.

The plane view's aspect mask is considered to be equivalent to VK_IMAGE_ASPECT_COLOR_BIT when used as a framebuffer attachment.

If the `format` of the `image` is a multi-planar format and `subresourceRange.aspectMask` is equal to VK_IMAGE_ASPECT_COLOR_BIT, a multi-plane image view will be created.

The VkComponentMapping `components` member describes a remapping from components of the image to components of the vector returned by shader image instructions. This remapping **must** be the identity swizzle for any `VkImageView` used with a combined image sampler that enables sampler $Y'C_BC_R$ conversion, input attachment descriptors, framebuffer attachments, and storage image descriptors.

Image views with formats that require YC_BC_R conversion and `usage` containing VK_IMAGE_USAGE_SAMPLED_BIT **must** be created with a VkSamplerYcbcrConversionInfo structure in the `pNext` chain to be used with sampling. Any image view created with a VkSamplerYcbcrConversionInfo structure in the `pNext` chain **must** only be sampled using a sampler with an identically defined VkSamplerYcbcrConversion. When used in a video coding operation, the specified sampler $Y'C_BC_R$ conversion has no effect.

Table 14. Image Type and Image View Type Compatibility Requirements

Image View Type	Compatible Image Types
VK_IMAGE_VIEW_TYPE_1D	VK_IMAGE_TYPE_1D
VK_IMAGE_VIEW_TYPE_1D_ARRAY	VK_IMAGE_TYPE_1D
VK_IMAGE_VIEW_TYPE_2D	VK_IMAGE_TYPE_2D , VK_IMAGE_TYPE_3D
VK_IMAGE_VIEW_TYPE_2D_ARRAY	VK_IMAGE_TYPE_2D , VK_IMAGE_TYPE_3D
VK_IMAGE_VIEW_TYPE_CUBE	VK_IMAGE_TYPE_2D
VK_IMAGE_VIEW_TYPE_CUBE_ARRAY	VK_IMAGE_TYPE_2D
VK_IMAGE_VIEW_TYPE_3D	VK_IMAGE_TYPE_3D

Valid Usage

- VUID-VkImageViewCreateInfo-image-01003
  If `image` was not created with VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT then `viewType` **must** not be VK_IMAGE_VIEW_TYPE_CUBE or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-VkImageViewCreateInfo-viewType-01004
  If the `imageCubeArray` feature is not enabled, `viewType` **must** not be VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-VkImageViewCreateInfo-image-06723
  If `image` was created with VK_IMAGE_TYPE_3D but without VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set then `viewType` **must** not be VK_IMAGE_VIEW_TYPE_2D_ARRAY
- VUID-VkImageViewCreateInfo-image-06728
  If `image` was created with VK_IMAGE_TYPE_3D but without VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT or VK_IMAGE_CREATE_2D_VIEW_COMPATIBLE_BIT_EXT set, then `viewType` **must** not be VK_IMAGE_VIEW_TYPE_2D
- VUID-VkImageViewCreateInfo-image-04970
  If `image` was created with VK_IMAGE_TYPE_3D and `viewType` is VK_IMAGE_VIEW_TYPE_2D or VK_IMAGE_VIEW_TYPE_2D_ARRAY then `subresourceRange.levelCount` **must** be 1

- VUID-VkImageViewCreateInfo-image-04972
  If `image` was created with a `samples` value not equal to VK_SAMPLE_COUNT_1_BIT then `viewType` **must** be either VK_IMAGE_VIEW_TYPE_2D or VK_IMAGE_VIEW_TYPE_2D_ARRAY
- VUID-VkImageViewCreateInfo-image-04441
  `image` **must** have been created with a `usage` value containing at least one of the following:
  - VK_IMAGE_USAGE_SAMPLED_BIT
  - VK_IMAGE_USAGE_STORAGE_BIT
  - VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT
  - VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT
  - VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT
  - VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT
  - VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
  - VK_IMAGE_USAGE_FRAGMENT_DENSITY_MAP_BIT_EXT
  - VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR
  - VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR
  - VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR
  - VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR
  - VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR
  - VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR
- VUID-VkImageViewCreateInfo-None-02273
  The format features of the resultant image view **must** contain at least one bit
- VUID-VkImageViewCreateInfo-usage-02274
  If `usage` contains VK_IMAGE_USAGE_SAMPLED_BIT, then the format features of the resultant image view **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT
- VUID-VkImageViewCreateInfo-usage-02275
  If `usage` contains VK_IMAGE_USAGE_STORAGE_BIT, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT
- VUID-VkImageViewCreateInfo-usage-02276
  If `usage` contains VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT, then the image view's format features **must** contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT
- VUID-VkImageViewCreateInfo-usage-02277
  If `usage` contains VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, then the image view's format features **must** contain VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkImageViewCreateInfo-image-08333
  If `usage` contains VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR, then the image view's format features **must** contain VK_FORMAT_FEATURE_VIDEO_DECODE_OUTPUT_BIT_KHR
- VUID-VkImageViewCreateInfo-image-08334
  If `usage` contains VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR, then the image

view's format features **must** contain VK_FORMAT_FEATURE_VIDEO_DECODE_DPB_BIT_KHR

- VUID-VkImageViewCreateInfo-image-08335
  `usage` **must** not include VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR
- VUID-VkImageViewCreateInfo-image-08336
  If `usage` contains VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR, then the image view's format features **must** contain VK_FORMAT_FEATURE_VIDEO_ENCODE_INPUT_BIT_KHR
- VUID-VkImageViewCreateInfo-image-08337
  If `usage` contains VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR, then the image view's format features **must** contain VK_FORMAT_FEATURE_VIDEO_ENCODE_DPB_BIT_KHR
- VUID-VkImageViewCreateInfo-image-08338
  `usage` **must** not include VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR
- VUID-VkImageViewCreateInfo-usage-10259
  If `usage` contains VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR, then the image view's format features **must** contain VK_FORMAT_FEATURE_2_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR
- VUID-VkImageViewCreateInfo-usage-10260
  If `usage` contains VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR, then the image view's format features **must** contain VK_FORMAT_FEATURE_2_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR
- VUID-VkImageViewCreateInfo-usage-08932
  If `usage` contains VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT,

  then the image view's format features **must** contain at least one of VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT or VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkImageViewCreateInfo-subresourceRange-01478
  `subresourceRange.baseMipLevel` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `image` was created
- VUID-VkImageViewCreateInfo-subresourceRange-01718
  If `subresourceRange.levelCount` is not VK_REMAINING_MIP_LEVELS, `subresourceRange.baseMipLevel` + `subresourceRange.levelCount` **must** be less than or equal to the `mipLevels` specified in VkImageCreateInfo when `image` was created
- VUID-VkImageViewCreateInfo-usage-12488
  If `usage` contains VK_IMAGE_USAGE_FRAGMENT_DENSITY_MAP_BIT_EXT, then the image view's format features **must** contain VK_FORMAT_FEATURE_2_FRAGMENT_DENSITY_MAP_BIT_EXT
- VUID-VkImageViewCreateInfo-image-02571
  If `image` was created with the VK_IMAGE_USAGE_FRAGMENT_DENSITY_MAP_BIT_EXT usage flag set, `subresourceRange.levelCount` **must** be `1`
- VUID-VkImageViewCreateInfo-image-06724

If image is not a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT or VK_IMAGE_CREATE_2D_VIEW_COMPATIBLE_BIT_EXT set, or viewType is not VK_IMAGE_VIEW_TYPE_2D or VK_IMAGE_VIEW_TYPE_2D_ARRAY, subresourceRange.baseArrayLayer **must** be less than the arrayLayers specified in VkImageCreateInfo when image was created

- VUID-VkImageViewCreateInfo-subresourceRange-06725
  If subresourceRange.layerCount is not VK_REMAINING_ARRAY_LAYERS, image is not a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT or VK_IMAGE_CREATE_2D_VIEW_COMPATIBLE_BIT_EXT set, or viewType is not VK_IMAGE_VIEW_TYPE_2D or VK_IMAGE_VIEW_TYPE_2D_ARRAY, subresourceRange.layerCount **must** be non-zero and subresourceRange.baseArrayLayer + subresourceRange.layerCount **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when image was created
- VUID-VkImageViewCreateInfo-image-02724
  If image is a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set, and viewType is VK_IMAGE_VIEW_TYPE_2D or VK_IMAGE_VIEW_TYPE_2D_ARRAY, subresourceRange.baseArrayLayer **must** be less than the depth computed from baseMipLevel and extent.depth specified in VkImageCreateInfo when image was created, according to the formula defined in Image Mip Level Sizing
- VUID-VkImageViewCreateInfo-subresourceRange-02725
  If subresourceRange.layerCount is not VK_REMAINING_ARRAY_LAYERS, image is a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT set, and viewType is VK_IMAGE_VIEW_TYPE_2D or VK_IMAGE_VIEW_TYPE_2D_ARRAY, subresourceRange.layerCount **must** be non-zero and subresourceRange.baseArrayLayer + subresourceRange.layerCount **must** be less than or equal to the depth computed from baseMipLevel and extent.depth specified in VkImageCreateInfo when image was created, according to the formula defined in Image Mip Level Sizing
- VUID-VkImageViewCreateInfo-image-01761
  If image was created with the VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT flag, but without the VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT flag, and a single-plane view and a multi-plane view is not being created, format **must** be compatible with the format used to create image, as defined in Format Compatibility Classes
- VUID-VkImageViewCreateInfo-image-01583
  If image was created with the VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT flag, format **must** be compatible with, or **must** be an uncompressed format that is size-compatible with, the format used to create image
- VUID-VkImageViewCreateInfo-image-07072
  If image was created with the VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT flag and format is a non-compressed format, the levelCount member of subresourceRange **must** be 1
- VUID-VkImageViewCreateInfo-image-09487
  If image was created with the VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT flag, the VkPhysicalDeviceMaintenance6Properties::blockTexelViewCompatibleMultipleLayers property is not VK_TRUE, and format is a non-compressed format, then the layerCount member of subresourceRange **must** be 1

- VUID-VkImageViewCreateInfo-pNext-01585
  If a VkImageFormatListCreateInfo structure was included in the pNext chain of the VkImageCreateInfo structure used when creating image and VkImageFormatListCreateInfo::viewFormatCount is not zero then format **must** be one of the formats in VkImageFormatListCreateInfo::pViewFormats
- VUID-VkImageViewCreateInfo-image-12397
  If image was not created with the VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT flag, format **must** be identical to the format used to create image
- VUID-VkImageViewCreateInfo-image-01586
  If creating a single-plane view, format **must** be compatible with the VkFormat for the plane of the image format indicated by subresourceRange.aspectMask, as defined in Compatible Formats of Planes of Multi-Planar Formats
- VUID-VkImageViewCreateInfo-subresourceRange-07818
  subresourceRange.aspectMask **must** only have at most 1 valid multi-planar aspect mask bit
- VUID-VkImageViewCreateInfo-format-12398
  If creating a multi-plane view, format **must** be identical to the format used to create image
- VUID-VkImageViewCreateInfo-format-06415
  If the image view's format requires a sampler $Y'C_BC_R$ conversion and usage contains VK_IMAGE_USAGE_SAMPLED_BIT, then the pNext chain **must** include a VkSamplerYcbcrConversionInfo structure with a conversion value other than VK_NULL_HANDLE
- VUID-VkImageViewCreateInfo-format-04714
  If format has a _422 or _420 suffix then image **must** have been created with a width that is a multiple of 2
- VUID-VkImageViewCreateInfo-format-04715
  If format has a _420 suffix then image **must** have been created with a height that is a multiple of 2
- VUID-VkImageViewCreateInfo-pNext-01970
  If the pNext chain includes a VkSamplerYcbcrConversionInfo structure with a conversion value other than VK_NULL_HANDLE, all members of components **must** have the identity swizzle
- VUID-VkImageViewCreateInfo-pNext-06658
  If the pNext chain includes a VkSamplerYcbcrConversionInfo structure with a conversion value other than VK_NULL_HANDLE, format **must** be the same used in VkSamplerYcbcrConversionCreateInfo::format
- VUID-VkImageViewCreateInfo-image-01020
  If image is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-VkImageViewCreateInfo-subResourceRange-01021
  viewType **must** be compatible with the type of image as shown in the view type compatibility table
- VUID-VkImageViewCreateInfo-image-02086
  If image was created with the

VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR usage flag set, viewType **must** be VK_IMAGE_VIEW_TYPE_2D or VK_IMAGE_VIEW_TYPE_2D_ARRAY

- VUID-VkImageViewCreateInfo-attachmentFragmentShadingRate-12386
  If the attachmentFragmentShadingRate feature is not enabled, the usage for the image view **must** not contain VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkImageViewCreateInfo-usage-04550
  If the usage for the image view includes VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR, then the image view's format features **must** contain VK_FORMAT_FEATURE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-VkImageViewCreateInfo-usage-04551
  If the usage for the image view includes VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR, and layeredShadingRateAttachments is VK_FALSE, subresourceRange.layerCount **must** be 1
- VUID-VkImageViewCreateInfo-flags-02572
  If the fragmentDensityMapDynamic feature is not enabled, flags **must** not contain VK_IMAGE_VIEW_CREATE_FRAGMENT_DENSITY_MAP_DYNAMIC_BIT_EXT
- VUID-VkImageViewCreateInfo-flags-03567
  If the fragmentDensityMapDeferred feature is not enabled, flags **must** not contain VK_IMAGE_VIEW_CREATE_FRAGMENT_DENSITY_MAP_DEFERRED_BIT_EXT
- VUID-VkImageViewCreateInfo-flags-03568
  If flags contains VK_IMAGE_VIEW_CREATE_FRAGMENT_DENSITY_MAP_DEFERRED_BIT_EXT, flags **must** not contain VK_IMAGE_VIEW_CREATE_FRAGMENT_DENSITY_MAP_DYNAMIC_BIT_EXT
- VUID-VkImageViewCreateInfo-image-03569
  If image was created with flags containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT and the VK_IMAGE_USAGE_SAMPLED_BIT usage flag set, subresourceRange.layerCount **must** be less than or equal to VkPhysicalDeviceFragmentDensityMap2PropertiesEXT::maxSubsampledArrayLayers
- VUID-VkImageViewCreateInfo-flags-04116
  If flags does not contain VK_IMAGE_VIEW_CREATE_FRAGMENT_DENSITY_MAP_DYNAMIC_BIT_EXT, and image was created with the VK_IMAGE_USAGE_FRAGMENT_DENSITY_MAP_BIT_EXT usage flag set, its flags **must** not contain any of VK_IMAGE_CREATE_PROTECTED_BIT, VK_IMAGE_CREATE_SPARSE_BINDING_BIT, VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT, or VK_IMAGE_CREATE_SPARSE_ALIASED_BIT
- VUID-VkImageViewCreateInfo-pNext-02662
  If the pNext chain includes a VkImageViewUsageCreateInfo structure, and image was not created with a VkImageStencilUsageCreateInfo structure included in the pNext chain of VkImageCreateInfo, its usage member **must** not include any bits that were not set in the usage member of the VkImageCreateInfo structure used to create image
- VUID-VkImageViewCreateInfo-pNext-02663
  If the pNext chain includes a VkImageViewUsageCreateInfo structure, image was created

with a VkImageStencilUsageCreateInfo structure included in the pNext chain of VkImageCreateInfo, and subresourceRange.aspectMask includes VK_IMAGE_ASPECT_STENCIL_BIT, the usage member of the VkImageViewUsageCreateInfo structure **must** not include any bits that were not set in the usage member of the VkImageStencilUsageCreateInfo structure used to create image

- VUID-VkImageViewCreateInfo-pNext-02664
  If the pNext chain includes a VkImageViewUsageCreateInfo structure, image was created with a VkImageStencilUsageCreateInfo structure included in the pNext chain of VkImageCreateInfo, and subresourceRange.aspectMask includes bits other than VK_IMAGE_ASPECT_STENCIL_BIT, the usage member of the VkImageViewUsageCreateInfo structure **must** not include any bits that were not set in the usage member of the VkImageCreateInfo structure used to create image
- VUID-VkImageViewCreateInfo-imageViewType-04973
  If viewType is VK_IMAGE_VIEW_TYPE_1D, VK_IMAGE_VIEW_TYPE_2D, or VK_IMAGE_VIEW_TYPE_3D; and subresourceRange.layerCount is not VK_REMAINING_ARRAY_LAYERS, then subresourceRange.layerCount **must** be 1
- VUID-VkImageViewCreateInfo-imageViewType-04974
  If viewType is VK_IMAGE_VIEW_TYPE_1D, VK_IMAGE_VIEW_TYPE_2D, or VK_IMAGE_VIEW_TYPE_3D; and subresourceRange.layerCount is VK_REMAINING_ARRAY_LAYERS, then the remaining number of layers **must** be 1
- VUID-VkImageViewCreateInfo-viewType-02960
  If viewType is VK_IMAGE_VIEW_TYPE_CUBE and subresourceRange.layerCount is not VK_REMAINING_ARRAY_LAYERS, subresourceRange.layerCount **must** be 6
- VUID-VkImageViewCreateInfo-viewType-02961
  If viewType is VK_IMAGE_VIEW_TYPE_CUBE_ARRAY and subresourceRange.layerCount is not VK_REMAINING_ARRAY_LAYERS, subresourceRange.layerCount **must** be a multiple of 6
- VUID-VkImageViewCreateInfo-viewType-02962
  If viewType is VK_IMAGE_VIEW_TYPE_CUBE and subresourceRange.layerCount is VK_REMAINING_ARRAY_LAYERS, the remaining number of layers **must** be 6
- VUID-VkImageViewCreateInfo-viewType-02963
  If viewType is VK_IMAGE_VIEW_TYPE_CUBE_ARRAY and subresourceRange.layerCount is VK_REMAINING_ARRAY_LAYERS, the remaining number of layers **must** be a multiple of 6
- VUID-VkImageViewCreateInfo-imageViewFormatSwizzle-04465
  If the VK_KHR_portability_subset extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::imageViewFormatSwizzle is VK_FALSE, all elements of components **must** have the identity swizzle
- VUID-VkImageViewCreateInfo-imageViewFormatReinterpretation-04466
  If the VK_KHR_portability_subset extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::imageViewFormatReinterpretation is VK_FALSE, the VkFormat in format **must** not contain a different number of components, or a different number of bits in each component, than the format of the VkImage in image
- VUID-VkImageViewCreateInfo-image-04817
  If image was created with the VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR usage flag set, VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR, or

VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR, then the viewType **must** be VK_IMAGE_VIEW_TYPE_2D or VK_IMAGE_VIEW_TYPE_2D_ARRAY

- VUID-VkImageViewCreateInfo-image-04818
  If image was created with the VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR usage flag set, VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR, or VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR, then the viewType **must** be VK_IMAGE_VIEW_TYPE_2D or VK_IMAGE_VIEW_TYPE_2D_ARRAY
- VUID-VkImageViewCreateInfo-image-10261
  If image was created with the VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR usage flags set, then viewType **must** be VK_IMAGE_VIEW_TYPE_2D or VK_IMAGE_VIEW_TYPE_2D_ARRAY
- VUID-VkImageViewCreateInfo-flags-08106
  If flags includes VK_IMAGE_VIEW_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT, the descriptorBufferCaptureReplay feature **must** be enabled
- VUID-VkImageViewCreateInfo-pNext-08107
  If the pNext chain includes a VkOpaqueCaptureDescriptorDataCreateInfoEXT structure, flags **must** contain VK_IMAGE_VIEW_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT
- VUID-VkImageViewCreateInfo-pNext-06787
  If the pNext chain includes a VkExportMetalObjectCreateInfoEXT structure, its exportObjectType member **must** be VK_EXPORT_METAL_OBJECT_TYPE_METAL_TEXTURE_BIT_EXT
- VUID-VkImageViewCreateInfo-subresourceRange-09594
  subresourceRange.aspectMask **must** be valid for the format the image was created with
- VUID-VkImageViewCreateInfo-image-13357
  If image was created with VK_IMAGE_CREATE_ALIAS_SINGLE_LAYER_DESCRIPTOR_BIT_KHR, format **must** not be a multi-planar format
- VUID-VkImageViewCreateInfo-pNext-12508
  The pNext chain **must** contain at most one of the VkImageViewUsageCreateInfo and VkImageViewUsage2CreateInfoKHR structures

Valid Usage (Implicit)

- VUID-VkImageViewCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_IMAGE_VIEW_CREATE_INFO
- VUID-VkImageViewCreateInfo-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkExportMetalObjectCreateInfoEXT, VkImageViewASTCDecodeModeEXT, VkImageViewMinLodCreateInfoEXT, VkImageViewSlicedCreateInfoEXT, VkImageViewUsage2CreateInfoKHR,

VkImageViewUsageCreateInfo, VkOpaqueCaptureDescriptorDataCreateInfoEXT, or VkSamplerYcbcrConversionInfo

- VUID-VkImageViewCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique, with the exception of structures of type VkExportMetalObjectCreateInfoEXT
- VUID-VkImageViewCreateInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkImageViewCreateFlagBits values
- VUID-VkImageViewCreateInfo-image-parameter
  `image` **must** be a valid VkImage handle
- VUID-VkImageViewCreateInfo-viewType-parameter
  `viewType` **must** be a valid VkImageViewType value
- VUID-VkImageViewCreateInfo-format-parameter
  `format` **must** be a valid VkFormat value
- VUID-VkImageViewCreateInfo-components-parameter
  `components` **must** be a valid VkComponentMapping structure
- VUID-VkImageViewCreateInfo-subresourceRange-parameter
  `subresourceRange` **must** be a valid VkImageSubresourceRange structure

Bits which **can** be set in VkImageViewCreateInfo::`flags`, specifying additional parameters of an image view, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageViewCreateFlagBits {
  // Provided by VK_EXT_fragment_density_map
    VK_IMAGE_VIEW_CREATE_FRAGMENT_DENSITY_MAP_DYNAMIC_BIT_EXT = 0x00000001,
  // Provided by VK_EXT_descriptor_buffer
    VK_IMAGE_VIEW_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT = 0x00000004,
  // Provided by VK_EXT_fragment_density_map2
    VK_IMAGE_VIEW_CREATE_FRAGMENT_DENSITY_MAP_DEFERRED_BIT_EXT = 0x00000002,
} VkImageViewCreateFlagBits;
```

- VK_IMAGE_VIEW_CREATE_FRAGMENT_DENSITY_MAP_DYNAMIC_BIT_EXT specifies that the fragment density map will be read by the device during VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT.
- VK_IMAGE_VIEW_CREATE_FRAGMENT_DENSITY_MAP_DEFERRED_BIT_EXT specifies that the fragment density map will be read by the host during vkEndCommandBuffer for the primary command buffer that the render pass is recorded into.
- VK_IMAGE_VIEW_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT specifies that the image view **can** be used with descriptor buffers when capturing and replaying (e.g. for trace capture and replay), see VkOpaqueCaptureDescriptorDataCreateInfoEXT for more detail.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkImageViewCreateFlags;
```

VkImageViewCreateFlags is a bitmask type for setting a mask of zero or more VkImageViewCreateFlagBits.

The set of usages for the created image view **can** be restricted compared to the parent image's usage flags by adding a VkImageViewUsageCreateInfo structure to the pNext chain of VkImageViewCreateInfo.

The VkImageViewUsageCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkImageViewUsageCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkImageUsageFlags    usage;
} VkImageViewUsageCreateInfo;
```

```
// Provided by VK_KHR_maintenance2
// Equivalent to VkImageViewUsageCreateInfo
typedef VkImageViewUsageCreateInfo VkImageViewUsageCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- usage is a bitmask of VkImageUsageFlagBits specifying allowed usages of the image view.

When this structure is chained to VkImageViewCreateInfo the usage field overrides the implicit usage parameter inherited from image creation time and its value is used instead for the purposes of determining the valid usage conditions of VkImageViewCreateInfo.

Valid Usage (Implicit)

- VUID-VkImageViewUsageCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_IMAGE_VIEW_USAGE_CREATE_INFO
- VUID-VkImageViewUsageCreateInfo-usage-parameter
  usage **must** be a valid combination of VkImageUsageFlagBits values

Structure Chaining

Extends the structure

- VkImageViewCreateInfo

The set of usages for the created image view **can** be restricted compared to the parent image's `usage` flags by adding a `VkImageViewUsage2CreateInfoKHR` structure to the `pNext` chain of VkImageViewCreateInfo.

The `VkImageViewUsage2CreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_extended_flags
typedef struct VkImageViewUsage2CreateInfoKHR {
    VkStructureType          sType;
    void*                    pNext;
    VkImageUsageFlags2KHR    usage;
} VkImageViewUsage2CreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `usage` is a bitmask of VkImageUsageFlagBits2KHR specifying allowed usages of the image view.

When this structure is chained to VkImageViewCreateInfo the `usage` field overrides the implicit `usage` parameter inherited from image creation time and its value is used instead for the purposes of determining the valid usage conditions of VkImageViewCreateInfo.

Valid Usage (Implicit)

- VUID-VkImageViewUsage2CreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_VIEW_USAGE_2_CREATE_INFO_KHR
- VUID-VkImageViewUsage2CreateInfoKHR-usage-parameter
  `usage` **must** be a valid combination of VkImageUsageFlagBits2KHR values

Structure Chaining

Extends the structure

- VkImageViewCreateInfo

The range of 3D slices for the created image view **can** be restricted to a subset of the parent image's Z range by adding a `VkImageViewSlicedCreateInfoEXT` structure to the `pNext` chain of VkImageViewCreateInfo.

The `VkImageViewSlicedCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_image_sliced_view_of_3d
typedef struct VkImageViewSlicedCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           sliceOffset;
    uint32_t           sliceCount;
} VkImageViewSlicedCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `sliceOffset` is the Z-offset for the first 3D slice accessible to the image view.
- `sliceCount` is the number of 3D slices accessible to the image view.

When this structure is chained to VkImageViewCreateInfo the `sliceOffset` field is treated as a Z-offset for the sliced view and `sliceCount` specifies the range. Shader accesses using a Z coordinate of 0 will access the depth slice corresponding to `sliceOffset` in the image, and in a shader, the maximum in-bounds Z coordinate for the view is `sliceCount` - 1.

A sliced 3D view **must** only be used with a single mip level. The slice coordinates are integer coordinates within the `subresourceRange.baseMipLevel` used to create the image view.

The effective view depth is equal to `extent.depth` used to create the `image` for this view adjusted by `subresourceRange.baseMipLevel` as specified in Image Mip Level Sizing.

Shader access to this image view is only affected by `VkImageViewSlicedCreateInfoEXT` if it uses a descriptor of type VK_DESCRIPTOR_TYPE_STORAGE_IMAGE. For access using any other descriptor type, the contents of `VkImageViewSlicedCreateInfoEXT` are ignored; instead, `sliceOffset` is treated as being equal to 0, and `sliceCount` is treated as being equal to VK_REMAINING_3D_SLICES_EXT.

Valid Usage

- VUID-VkImageViewSlicedCreateInfoEXT-sliceOffset-07867
  `sliceOffset` **must** be less than the effective view depth as specified in Image Mip Level Sizing
- VUID-VkImageViewSlicedCreateInfoEXT-sliceCount-07868
  If `sliceCount` is not VK_REMAINING_3D_SLICES_EXT, it **must** be non-zero and `sliceOffset` + `sliceCount` **must** be less than or equal to the effective view depth as specified in Image Mip Level Sizing
- VUID-VkImageViewSlicedCreateInfoEXT-image-07869
  `image` **must** have been created with `imageType` equal to VK_IMAGE_TYPE_3D
- VUID-VkImageViewSlicedCreateInfoEXT-viewType-07909
  `viewType` **must** be VK_IMAGE_VIEW_TYPE_3D
- VUID-VkImageViewSlicedCreateInfoEXT-None-07870
  The image view **must** reference exactly 1 mip level

- VUID-VkImageViewSlicedCreateInfoEXT-None-07871
  The imageSlicedViewOf3D feature **must** be enabled

Valid Usage (Implicit)

- VUID-VkImageViewSlicedCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_VIEW_SLICED_CREATE_INFO_EXT

Structure Chaining

Extends the structure

- VkImageViewCreateInfo

VK_REMAINING_3D_SLICES_EXT is a special constant value used for VkImageViewSlicedCreateInfoEXT::`sliceCount` to indicate that all remaining 3D slices in an image after the first slice offset specified should be included in the view.

```
#define VK_REMAINING_3D_SLICES_EXT        (~0U)
```

The `VkImageSubresourceRange` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageSubresourceRange {
    VkImageAspectFlags    aspectMask;
    uint32_t              baseMipLevel;
    uint32_t              levelCount;
    uint32_t              baseArrayLayer;
    uint32_t              layerCount;
} VkImageSubresourceRange;
```

- `aspectMask` is a bitmask of VkImageAspectFlagBits specifying which aspect(s) of the image are included in the view.
- `baseMipLevel` is the first mipmap level accessible to the view.
- `levelCount` is the number of mipmap levels (starting from `baseMipLevel`) accessible to the view.
- `baseArrayLayer` is the first array layer accessible to the view.
- `layerCount` is the number of array layers (starting from `baseArrayLayer`) accessible to the view.

The number of mipmap levels and array layers **must** be a subset of the image subresources in the image. If an application wants to use all mip levels or layers in an image after the `baseMipLevel` or `baseArrayLayer`, it **can** set `levelCount` and `layerCount` to the special values VK_REMAINING_MIP_LEVELS and VK_REMAINING_ARRAY_LAYERS without knowing the exact number of mip levels or layers.

For cube and cube array image views, the layers of the image view starting at `baseArrayLayer` correspond to faces in the order +X, -X, +Y, -Y, +Z, -Z. For cube arrays, each set of six sequential layers is a single cube, so the number of cube maps in a cube map array view is `layerCount` / *6*, and image array layer (`baseArrayLayer` + i) is face index (i mod 6) of cube *i* / *6*. If the number of layers in the view, whether set explicitly in `layerCount` or implied by VK_REMAINING_ARRAY_LAYERS, is not a multiple of 6, the last cube map in the array **must** not be accessed.

`aspectMask` **must** be only VK_IMAGE_ASPECT_COLOR_BIT, VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT if `format` is a color, depth-only or stencil-only format, respectively.

When creating a VkImageView, if `image` was created with VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT, and the image `format` is a multi-planar format, then `aspectMask` **can** be set to one of the multi-planar aspect bits.

If using a depth/stencil format with both depth and stencil components, `aspectMask` **must** include at least one of VK_IMAGE_ASPECT_DEPTH_BIT and VK_IMAGE_ASPECT_STENCIL_BIT, and **can** include both.

When creating an image view of a depth/stencil image to be used as a descriptor (e.g. for sampling in the shader, or for use as an input attachment), the `aspectMask` **must** only include one bit, which selects whether the image view is used for depth reads (i.e. using a floating-point sampler or input attachment in the shader) or stencil reads (i.e. using an unsigned integer sampler or input attachment in the shader). When an image view of a depth/stencil image is used as a depth/stencil framebuffer attachment, the `aspectMask` is ignored and both depth and stencil image subresources are used.

When the `VkImageSubresourceRange` structure is used to select a subset of the slices of a 3D image's mip level in order to create a 2D or 2D array image view of a 3D image created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT, `baseArrayLayer` and `layerCount` specify the first slice index and the number of slices to include in the created image view. Such an image view **can** be used as a framebuffer attachment that refers only to the specified range of slices of the selected mip level. If the `maintenance9` feature is not enabled, any layout transitions performed on such an attachment view during a render pass instance still apply to the entire subresource referenced which includes all the slices of the selected mip level.

Valid Usage

- VUID-VkImageSubresourceRange-levelCount-01720
  If `levelCount` is not VK_REMAINING_MIP_LEVELS, it **must** be greater than `0`
- VUID-VkImageSubresourceRange-layerCount-01721
  If `layerCount` is not VK_REMAINING_ARRAY_LAYERS, it **must** be greater than `0`
- VUID-VkImageSubresourceRange-aspectMask-01670
  If `aspectMask` includes VK_IMAGE_ASPECT_COLOR_BIT, then it **must** not include any of the multi-planar aspect mask bits
- VUID-VkImageSubresourceRange-aspectMask-02278
  `aspectMask` **must** not include `VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT` for any index *i*

Valid Usage (Implicit)

- VUID-VkImageSubresourceRange-aspectMask-parameter
  `aspectMask` **must** be a valid combination of VkImageAspectFlagBits values

Bits which **can** be set in an aspect mask to specify aspects of an image for purposes such as identifying a subresource, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageAspectFlagBits {
    VK_IMAGE_ASPECT_COLOR_BIT = 0x00000001,
    VK_IMAGE_ASPECT_DEPTH_BIT = 0x00000002,
    VK_IMAGE_ASPECT_STENCIL_BIT = 0x00000004,
    VK_IMAGE_ASPECT_METADATA_BIT = 0x00000008,
  // Provided by VK_VERSION_1_1
    VK_IMAGE_ASPECT_PLANE_0_BIT = 0x00000010,
  // Provided by VK_VERSION_1_1
    VK_IMAGE_ASPECT_PLANE_1_BIT = 0x00000020,
  // Provided by VK_VERSION_1_1
    VK_IMAGE_ASPECT_PLANE_2_BIT = 0x00000040,
  // Provided by VK_VERSION_1_3
    VK_IMAGE_ASPECT_NONE = 0,
  // Provided by VK_EXT_image_drm_format_modifier
    VK_IMAGE_ASPECT_MEMORY_PLANE_0_BIT_EXT = 0x00000080,
  // Provided by VK_EXT_image_drm_format_modifier
    VK_IMAGE_ASPECT_MEMORY_PLANE_1_BIT_EXT = 0x00000100,
  // Provided by VK_EXT_image_drm_format_modifier
    VK_IMAGE_ASPECT_MEMORY_PLANE_2_BIT_EXT = 0x00000200,
  // Provided by VK_EXT_image_drm_format_modifier
    VK_IMAGE_ASPECT_MEMORY_PLANE_3_BIT_EXT = 0x00000400,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_IMAGE_ASPECT_PLANE_0_BIT_KHR = VK_IMAGE_ASPECT_PLANE_0_BIT,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_IMAGE_ASPECT_PLANE_1_BIT_KHR = VK_IMAGE_ASPECT_PLANE_1_BIT,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_IMAGE_ASPECT_PLANE_2_BIT_KHR = VK_IMAGE_ASPECT_PLANE_2_BIT,
  // Provided by VK_KHR_maintenance4
    VK_IMAGE_ASPECT_NONE_KHR = VK_IMAGE_ASPECT_NONE,
} VkImageAspectFlagBits;
```

- VK_IMAGE_ASPECT_NONE specifies no image aspect, or the image aspect is not applicable.
- VK_IMAGE_ASPECT_COLOR_BIT specifies the color aspect.
- VK_IMAGE_ASPECT_DEPTH_BIT specifies the depth aspect.
- VK_IMAGE_ASPECT_STENCIL_BIT specifies the stencil aspect.
- VK_IMAGE_ASPECT_METADATA_BIT specifies the metadata aspect used for sparse resource operations.

- VK_IMAGE_ASPECT_PLANE_0_BIT specifies plane 0 of a *multi-planar* image format.
- VK_IMAGE_ASPECT_PLANE_1_BIT specifies plane 1 of a *multi-planar* image format.
- VK_IMAGE_ASPECT_PLANE_2_BIT specifies plane 2 of a *multi-planar* image format.
- VK_IMAGE_ASPECT_MEMORY_PLANE_0_BIT_EXT specifies *memory plane* 0.
- VK_IMAGE_ASPECT_MEMORY_PLANE_1_BIT_EXT specifies *memory plane* 1.
- VK_IMAGE_ASPECT_MEMORY_PLANE_2_BIT_EXT specifies *memory plane* 2.
- VK_IMAGE_ASPECT_MEMORY_PLANE_3_BIT_EXT specifies *memory plane* 3.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkImageAspectFlags;
```

`VkImageAspectFlags` is a bitmask type for setting a mask of zero or more VkImageAspectFlagBits.

The `VkComponentMapping` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkComponentMapping {
    VkComponentSwizzle    r;
    VkComponentSwizzle    g;
    VkComponentSwizzle    b;
    VkComponentSwizzle    a;
} VkComponentMapping;
```

- `r` is a VkComponentSwizzle specifying the component value placed in the R component of the output vector.
- `g` is a VkComponentSwizzle specifying the component value placed in the G component of the output vector.
- `b` is a VkComponentSwizzle specifying the component value placed in the B component of the output vector.
- `a` is a VkComponentSwizzle specifying the component value placed in the A component of the output vector.

Valid Usage (Implicit)

- VUID-VkComponentMapping-r-parameter
  `r` **must** be a valid VkComponentSwizzle value
- VUID-VkComponentMapping-g-parameter
  `g` **must** be a valid VkComponentSwizzle value
- VUID-VkComponentMapping-b-parameter
  `b` **must** be a valid VkComponentSwizzle value
- VUID-VkComponentMapping-a-parameter
  `a` **must** be a valid VkComponentSwizzle value

Possible values of the members of VkComponentMapping, specifying the component values placed in each component of the output vector, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkComponentSwizzle {
    VK_COMPONENT_SWIZZLE_IDENTITY = 0,
    VK_COMPONENT_SWIZZLE_ZERO = 1,
    VK_COMPONENT_SWIZZLE_ONE = 2,
    VK_COMPONENT_SWIZZLE_R = 3,
    VK_COMPONENT_SWIZZLE_G = 4,
    VK_COMPONENT_SWIZZLE_B = 5,
    VK_COMPONENT_SWIZZLE_A = 6,
} VkComponentSwizzle;
```

- VK_COMPONENT_SWIZZLE_IDENTITY specifies that the component is set to the identity swizzle.
- VK_COMPONENT_SWIZZLE_ZERO specifies that the component is set to zero.
- VK_COMPONENT_SWIZZLE_ONE specifies that the component is set to either 1 or 1.0, depending on whether the type of the image view format is integer or floating-point respectively, as determined by the Format Definition section for each VkFormat.
- VK_COMPONENT_SWIZZLE_R specifies that the component is set to the value of the R component of the image.
- VK_COMPONENT_SWIZZLE_G specifies that the component is set to the value of the G component of the image.
- VK_COMPONENT_SWIZZLE_B specifies that the component is set to the value of the B component of the image.
- VK_COMPONENT_SWIZZLE_A specifies that the component is set to the value of the A component of the image.

Setting the identity swizzle on a component is equivalent to setting the identity mapping on that component. That is:

Table 15. Component Mappings Equivalent To VK_COMPONENT_SWIZZLE_IDENTITY

Component	Identity Mapping
components.r	VK_COMPONENT_SWIZZLE_R
components.g	VK_COMPONENT_SWIZZLE_G
components.b	VK_COMPONENT_SWIZZLE_B
components.a	VK_COMPONENT_SWIZZLE_A

If the pNext chain includes a VkImageViewASTCDecodeModeEXT structure, then that structure includes a parameter specifying the decode mode for image views using ASTC compressed formats.

The VkImageViewASTCDecodeModeEXT structure is defined as:

```
// Provided by VK_EXT_astc_decode_mode
typedef struct VkImageViewASTCDecodeModeEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkFormat           decodeMode;
} VkImageViewASTCDecodeModeEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `decodeMode` is the intermediate format used to decode ASTC compressed formats.

Valid Usage

- VUID-VkImageViewASTCDecodeModeEXT-decodeMode-02230
  `decodeMode` **must** be one of VK_FORMAT_R16G16B16A16_SFLOAT, VK_FORMAT_R8G8B8A8_UNORM, or VK_FORMAT_E5B9G9R9_UFLOAT_PACK32
- VUID-VkImageViewASTCDecodeModeEXT-decodeMode-02231
  If the `decodeModeSharedExponent` feature is not enabled, `decodeMode` **must** not be VK_FORMAT_E5B9G9R9_UFLOAT_PACK32
- VUID-VkImageViewASTCDecodeModeEXT-decodeMode-02232
  If `decodeMode` is VK_FORMAT_R8G8B8A8_UNORM, the image view **must** not include blocks using any of the ASTC HDR modes
- VUID-VkImageViewASTCDecodeModeEXT-format-04084
  `format` of the image view **must** be one of the ASTC Compressed Image Formats

If `format` uses sRGB encoding then the `decodeMode` has no effect.

Valid Usage (Implicit)

- VUID-VkImageViewASTCDecodeModeEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_VIEW_ASTC_DECODE_MODE_EXT
- VUID-VkImageViewASTCDecodeModeEXT-decodeMode-parameter
  `decodeMode` **must** be a valid VkFormat value

Structure Chaining

Extends the structure

- VkImageViewCreateInfo

To destroy an image view, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyImageView(
    VkDevice                                    device,
    VkImageView                                 imageView,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the image view.
- `imageView` is the image view to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyImageView-imageView-01026
  All submitted commands that refer to `imageView` **must** have completed execution
- VUID-vkDestroyImageView-imageView-01027
  If `VkAllocationCallbacks` were provided when `imageView` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyImageView-imageView-01028
  If no `VkAllocationCallbacks` were provided when `imageView` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyImageView-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyImageView-imageView-parameter
  If `imageView` is not VK_NULL_HANDLE, `imageView` **must** be a valid VkImageView handle
- VUID-vkDestroyImageView-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyImageView-imageView-parent
  If `imageView` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `imageView` **must** be externally synchronized

12.6.1. Image View Format Features

Valid uses of a VkImageView **may** depend on the image view's *format features*, defined below. Such

constraints are documented in the affected valid usage statement.

- If Vulkan 1.3 is supported or the VK_KHR_format_feature_flags2 extension is supported, and VkImageViewCreateInfo::image was created with VK_IMAGE_TILING_LINEAR, then the image view's set of *format features* is the value of VkFormatProperties3::linearTilingFeatures found by calling vkGetPhysicalDeviceFormatProperties2 on the same format as VkImageViewCreateInfo::format.
- If Vulkan 1.3 is not supported and the VK_KHR_format_feature_flags2 extension is not supported, and VkImageViewCreateInfo::image was created with VK_IMAGE_TILING_LINEAR, then the image view's set of *format features* is the union of the value of VkFormatProperties::linearTilingFeatures found by calling vkGetPhysicalDeviceFormatProperties on the same format as VkImageViewCreateInfo::format, with:
  - VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT if the format is a depth/stencil format and the image view features also contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT.
  - VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT if the format is one of the extended storage formats and the shaderStorageImageReadWithoutFormat feature is enabled.
  - VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT if the format is one of the extended storage formats and the shaderStorageImageWriteWithoutFormat feature is enabled.
- If Vulkan 1.3 is supported or the VK_KHR_format_feature_flags2 extension is supported, and VkImageViewCreateInfo::image was created with VK_IMAGE_TILING_OPTIMAL, then the image view's set of *format features* is the value of VkFormatProperties::optimalTilingFeatures or VkFormatProperties3::optimalTilingFeatures found by calling vkGetPhysicalDeviceFormatProperties or vkGetPhysicalDeviceFormatProperties2 on the same format as VkImageViewCreateInfo::format.
- If Vulkan 1.3 is not supported and the VK_KHR_format_feature_flags2 extension is not supported, and VkImageViewCreateInfo::image was created with VK_IMAGE_TILING_OPTIMAL, then the image view's set of *format features* is the union of the value of VkFormatProperties::optimalTilingFeatures found by calling vkGetPhysicalDeviceFormatProperties on the same format as VkImageViewCreateInfo::format, with:
  - VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT if the format is a depth/stencil format and the image view features also contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT.
  - VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT if the format is one of the extended storage formats and the shaderStorageImageReadWithoutFormat feature is enabled.
  - VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT if the format is one of the extended storage formats and the shaderStorageImageWriteWithoutFormat feature is enabled.
- If VkImageViewCreateInfo::image was created with VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then:
  - The image's DRM format modifier is the value of

VkImageDrmFormatModifierPropertiesEXT::drmFormatModifier found by calling vkGetImageDrmFormatModifierPropertiesEXT.

- Let VkDrmFormatModifierPropertiesListEXT::pDrmFormatModifierProperties be the array found by calling vkGetPhysicalDeviceFormatProperties2 on the same format as VkImageViewCreateInfo::format.
- Let VkDrmFormatModifierPropertiesEXT prop be the array element whose drmFormatModifier member is the value of the image's DRM format modifier.
- Then the image view's set of *format features* is prop::drmFormatModifierTilingFeatures.

The VkImageViewMinLodCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_image_view_min_lod
typedef struct VkImageViewMinLodCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    float              minLod;
} VkImageViewMinLodCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- minLod is the value to clamp the minimum LOD accessible by this VkImageView.

If the pNext chain includes a VkImageViewMinLodCreateInfoEXT structure, then that structure includes a parameter specifying a value to clamp the minimum LOD value during Image Level(s) Selection, Texel Gathering and Integer Texel Coordinate Operations.

If the image view contains VkImageViewMinLodCreateInfoEXT and it is used as part of a sampling operation:

$\text{minLodFloat}_{\text{imageView}}$ = minLod

otherwise:

$\text{minLodFloat}_{\text{imageView}}$ = 0.0

An integer variant of this parameter is also defined for sampling operations which access integer mipmap levels:

$\text{minLodInteger}_{\text{imageView}} = \square\text{minLodFloat}_{\text{imageView}}\square$

Valid Usage

- VUID-VkImageViewMinLodCreateInfoEXT-minLod-06455
  If the minLod feature is not enabled, minLod **must** be 0.0
- VUID-VkImageViewMinLodCreateInfoEXT-minLod-06456
  minLod **must** be less than or equal to the index of the last mipmap level accessible to the

view

Valid Usage (Implicit)

- VUID-VkImageViewMinLodCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_VIEW_MIN_LOD_CREATE_INFO_EXT

Structure Chaining

Extends the structure

- VkImageViewCreateInfo

12.7. Acceleration Structures

Acceleration structures are opaque data structures that are built by the implementation to more efficiently perform spatial queries on the provided geometric data. For these extensions, an acceleration structure is either a top-level acceleration structure containing a set of bottom-level acceleration structures or a bottom-level acceleration structure containing either a set of axis-aligned bounding boxes for custom geometry or a set of triangles.

Each instance in the top-level acceleration structure contains a reference to a bottom-level acceleration structure as well as an instance transform plus information required to index into the shader bindings. The top-level acceleration structure is what is bound to the acceleration descriptor, for example to trace inside the shader in the ray tracing pipeline.

Acceleration structures for the `VK_KHR_acceleration_structure` extension and micromaps for the `VK_KHR_opacity_micromap` extension are represented by `VkAccelerationStructureKHR` handles:

```
// Provided by VK_KHR_acceleration_structure
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkAccelerationStructureKHR)
```

To create an acceleration structure using a device address, call:

```
// Provided by VK_KHR_acceleration_structure with VK_KHR_device_address_commands
VkResult vkCreateAccelerationStructure2KHR(
    VkDevice                                    device,
    const VkAccelerationStructureCreateInfo2KHR* pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkAccelerationStructureKHR*                 pAccelerationStructure);
```

- `device` is the logical device that creates the acceleration structure object.
- `pCreateInfo` is a pointer to a VkAccelerationStructureCreateInfo2KHR structure containing parameters affecting creation of the acceleration structure.

- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pAccelerationStructure` is a pointer to a `VkAccelerationStructureKHR` handle in which the resulting acceleration structure object is returned.

Similar to other objects in Vulkan, the acceleration structure creation merely creates an object with a specific "shape". The type and quantity of geometry that can be built into an acceleration structure is determined by the parameters of VkAccelerationStructureCreateInfo2KHR.

The acceleration structure data is stored in the object referred to by `VkAccelerationStructureCreateInfo2KHR`::`addressRange.address`.

Once the `VkAccelerationStructureKHR` object has been created, it **must** be populated by acceleration structure build or acceleration structure copy commands such as vkCmdBuildAccelerationStructuresKHR and vkCmdCopyAccelerationStructureKHR. Acceleration structures created with this command **must** not be used by host commands.

NOTE The expected usage for a trace capture/replay tool is that it will serialize and later deserialize the acceleration structure data using acceleration structure copy commands. During capture the tool will use vkCmdCopyAccelerationStructureToMemoryKHR with a `mode` of VK_COPY_ACCELERATION_STRUCTURE_MODE_SERIALIZE_KHR, and vkCmdCopyMemoryToAccelerationStructureKHR with a `mode` of VK_COPY_ACCELERATION_STRUCTURE_MODE_DESERIALIZE_KHR during replay.

Valid Usage

- VUID-vkCreateAccelerationStructure2KHR-accelerationStructure-03611
  The `VkPhysicalDeviceAccelerationStructureFeaturesKHR`::`accelerationStructure` feature **must** be enabled
- VUID-vkCreateAccelerationStructure2KHR-deviceAddressCommands-13086
  The `VkPhysicalDeviceDeviceAddressCommandsFeaturesKHR`::`deviceAddressCommands` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCreateAccelerationStructure2KHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateAccelerationStructure2KHR-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkAccelerationStructureCreateInfo2KHR structure
- VUID-vkCreateAccelerationStructure2KHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateAccelerationStructure2KHR-pAccelerationStructure-parameter
  `pAccelerationStructure` **must** be a valid pointer to a VkAccelerationStructureKHR handle

- VUID-vkCreateAccelerationStructure2KHR-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkAccelerationStructureCreateInfo2KHR structure is defined as:

```
// Provided by VK_KHR_acceleration_structure with VK_KHR_device_address_commands
typedef struct VkAccelerationStructureCreateInfo2KHR {
    VkStructureType                          sType;
    const void*                              pNext;
    VkAccelerationStructureCreateFlagsKHR    createFlags;
    VkDeviceAddressRangeKHR                  addressRange;
    VkAddressCommandFlagsKHR                 addressFlags;
    VkAccelerationStructureTypeKHR           type;
} VkAccelerationStructureCreateInfo2KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- createFlags is a bitmask of VkAccelerationStructureCreateFlagBitsKHR specifying additional creation parameters of the acceleration structure.
- addressRange is a VkDeviceAddressRangeKHR structure defining the size required for the acceleration structure and its starting address.
- addressFlags is a VkAddressCommandFlagsKHR value defining the flags for the address range.
- type is a VkAccelerationStructureTypeKHR value specifying the type of acceleration structure that will be created.

Applications **should** create an acceleration structure with a specific VkAccelerationStructureTypeKHR other than VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR.

NOTE VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR is intended to be used by API translation layers. This can be used at acceleration structure creation time in cases where the actual acceleration structure type (top or bottom) is not yet known. The

actual acceleration structure type must be specified as VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR, VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR or VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR when the build is performed.

During replay, if `createFlags` includes VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR, `addressRange.address` **must** be an address used to create an identical acceleration structure on the same implementation. The address **must** be in the range of an identically created VkBuffer at the same offset.

Applications **should** avoid creating acceleration structures with application-provided addresses and implementation-provided addresses in the same process, to reduce the likelihood of VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR errors.

NOTE

The expected usage for this is that a trace capture/replay tool will add the VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT flag and VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT to all buffers used as storage for an acceleration structure with VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR included in `createFlags`. This also means that the tool will need to add VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT to memory allocations to allow the VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT flag to be set where the application may not have otherwise required it. During capture the tool will save the queried opaque device addresses in the trace. During replay, the buffers will be created specifying the original address so any address values stored in the trace data will remain valid.

Implementations are expected to separate such buffers in the GPU address space so normal allocations will avoid using these addresses. Applications and tools should avoid mixing application-provided and implementation-provided addresses for buffers created with VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT, to avoid address space allocation conflicts.

If the acceleration structure will be the target of a build operation, the required size for an acceleration structure **can** be queried with vkGetAccelerationStructureBuildSizesKHR. If the acceleration structure is going to be the target of a copy, vkCmdWriteAccelerationStructuresPropertiesKHR **can** be used to obtain the size required depending on the type of copy.

Valid Usage

- VUID-VkAccelerationStructureCreateInfo2KHR-addressRange-13097
  If the range specified by `addressRange` is not bound completely to memory when accessed, `addressFlags` **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR

- VUID-VkAccelerationStructureCreateInfo2KHR-addressRange-13098
  If the buffer from which the range specified by `addressRange` was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `addressFlags` **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkAccelerationStructureCreateInfo2KHR-addressRange-13099
  If the buffer from which the range specified by `addressRange` was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `addressFlags` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkAccelerationStructureCreateInfo2KHR-addressFlags-13100
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkAccelerationStructureCreateInfo2KHR-addressRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkAccelerationStructureCreateInfo2KHR-addressRange-13123
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkAccelerationStructureCreateInfo2KHR-addressFlags-13101
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkAccelerationStructureCreateInfo2KHR-addressRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `addressFlags` **must** include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkAccelerationStructureCreateInfo2KHR-addressRange-13125
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created without VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `addressFlags` **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkAccelerationStructureCreateInfo2KHR-addressRange-11602
  `addressRange.address` **must** be a valid VkDeviceAddress
- VUID-VkAccelerationStructureCreateInfo2KHR-addressRange-11603
  `addressRange` **must** specify a range within a valid address retrieved from `buffer` that was created with a `usage` value containing

VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR

- VUID-VkAccelerationStructureCreateInfo2KHR-addressRange-11604
  `addressRange` **must** specify a range within a valid address retrieved from `buffer` that was not created with a `flags` value containing VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT
- VUID-VkAccelerationStructureCreateInfo2KHR-addressRange-11605
  `addressRange.address` **must** be a multiple of `256` bytes
- VUID-VkAccelerationStructureCreateInfo2KHR-createFlags-11606
  If the `address` member of `addressRange` was retrieved from a VkBuffer created with VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT, `createFlags` **must** include VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR
- VUID-VkAccelerationStructureCreateInfo2KHR-createFlags-11607
  If the `address` member of `addressRange` was retrieved from a VkBuffer not created with VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT, `createFlags` **must** not include VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR
- VUID-VkAccelerationStructureCreateInfo2KHR-addressRange-11608
  `addressRange.size` **must** be greater than zero
- VUID-VkAccelerationStructureCreateInfo2KHR-micromap-11609
  If `type` is VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR, the `VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap` feature **must** be enabled
- VUID-VkAccelerationStructureCreateInfo2KHR-createFlags-11610
  If `type` is VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR, `createFlags` **must** not include VK_ACCELERATION_STRUCTURE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT
- VUID-VkAccelerationStructureCreateInfo2KHR-createFlags-03613
  If `createFlags` includes VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR, VkPhysicalDeviceAccelerationStructureFeaturesKHR::`accelerationStructureCaptureReplay` **must** be VK_TRUE
- VUID-VkAccelerationStructureCreateInfo2KHR-createFlags-11673
  If `createFlags` includes VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR and `device` was created with multiple physical devices, the `bufferDeviceAddressMultiDevice` feature **must** be enabled
- VUID-VkAccelerationStructureCreateInfo2KHR-createFlags-08108
  If `createFlags` includes VK_ACCELERATION_STRUCTURE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT, the `descriptorBufferCaptureReplay` feature **must** be enabled
- VUID-VkAccelerationStructureCreateInfo2KHR-pNext-08109
  If the `pNext` chain includes a VkOpaqueCaptureDescriptorDataCreateInfoEXT structure, `createFlags` **must** contain VK_ACCELERATION_STRUCTURE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_

EXT

Valid Usage (Implicit)

- VUID-VkAccelerationStructureCreateInfo2KHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CREATE_INFO_2_KHR
- VUID-VkAccelerationStructureCreateInfo2KHR-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkOpaqueCaptureDescriptorDataCreateInfoEXT
- VUID-VkAccelerationStructureCreateInfo2KHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkAccelerationStructureCreateInfo2KHR-createFlags-parameter
  If `createFlags` is not `0`, `createFlags` **must** be a valid combination of VkAccelerationStructureCreateFlagBitsKHR values
- VUID-VkAccelerationStructureCreateInfo2KHR-addressFlags-parameter
  If `addressFlags` is not `0`, `addressFlags` **must** be a valid combination of VkAddressCommandFlagBitsKHR values
- VUID-VkAccelerationStructureCreateInfo2KHR-type-parameter
  `type` **must** be a valid VkAccelerationStructureTypeKHR value

To create an acceleration structure using a buffer, call:

WARNING This functionality is superseded by vkCreateAccelerationStructure2KHR. See Legacy Functionality for more information.

```
// Provided by VK_KHR_acceleration_structure
VkResult vkCreateAccelerationStructureKHR(
    VkDevice                                    device,
    const VkAccelerationStructureCreateInfoKHR* pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkAccelerationStructureKHR*                 pAccelerationStructure);
```

- `device` is the logical device that creates the acceleration structure object.
- `pCreateInfo` is a pointer to a VkAccelerationStructureCreateInfoKHR structure containing parameters affecting creation of the acceleration structure.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pAccelerationStructure` is a pointer to a `VkAccelerationStructureKHR` handle in which the resulting acceleration structure object is returned.

Similar to other objects in Vulkan, the acceleration structure creation merely creates an object with a specific “shape”. The type and quantity of geometry that can be built into an acceleration structure is determined by the parameters of VkAccelerationStructureCreateInfoKHR.

The acceleration structure data is stored in the object referred to by VkAccelerationStructureCreateInfoKHR::buffer. Once memory has been bound to that buffer, it **must** be populated by acceleration structure build or acceleration structure copy commands such as vkCmdBuildAccelerationStructuresKHR, vkBuildAccelerationStructuresKHR, vkCmdCopyAccelerationStructureKHR, and vkCopyAccelerationStructureKHR.

NOTE The expected usage for a trace capture/replay tool is that it will serialize and later deserialize the acceleration structure data using acceleration structure copy commands. During capture the tool will use vkCopyAccelerationStructureToMemoryKHR or vkCmdCopyAccelerationStructureToMemoryKHR with a mode of VK_COPY_ACCELERATION_STRUCTURE_MODE_SERIALIZE_KHR, and vkCopyMemoryToAccelerationStructureKHR or vkCmdCopyMemoryToAccelerationStructureKHR with a mode of VK_COPY_ACCELERATION_STRUCTURE_MODE_DESERIALIZE_KHR during replay.

NOTE Memory does not need to be bound to the underlying buffer when vkCreateAccelerationStructureKHR is called.

Valid Usage

- VUID-vkCreateAccelerationStructureKHR-accelerationStructure-03611
  The VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure feature **must** be enabled
- VUID-vkCreateAccelerationStructureKHR-deviceAddress-03488
  If VkAccelerationStructureCreateInfoKHR::deviceAddress is not zero, the accelerationStructureCaptureReplay feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCreateAccelerationStructureKHR-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkCreateAccelerationStructureKHR-pCreateInfo-parameter
  pCreateInfo **must** be a valid pointer to a valid VkAccelerationStructureCreateInfoKHR structure
- VUID-vkCreateAccelerationStructureKHR-pAllocator-parameter
  If pAllocator is not NULL, pAllocator **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateAccelerationStructureKHR-pAccelerationStructure-parameter
  pAccelerationStructure **must** be a valid pointer to a VkAccelerationStructureKHR handle
- VUID-vkCreateAccelerationStructureKHR-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkAccelerationStructureCreateInfoKHR structure is defined as:

WARNING	This functionality is superseded by VkAccelerationStructureCreateInfo2KHR. See Legacy Functionality for more information.

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureCreateInfoKHR {
    VkStructureType                           sType;
    const void*                               pNext;
    VkAccelerationStructureCreateFlagsKHR     createFlags;
    VkBuffer                                  buffer;
    VkDeviceSize                              offset;
    VkDeviceSize                              size;
    VkAccelerationStructureTypeKHR            type;
    VkDeviceAddress                           deviceAddress;
} VkAccelerationStructureCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- createFlags is a bitmask of VkAccelerationStructureCreateFlagBitsKHR specifying additional creation parameters of the acceleration structure.
- buffer is the buffer on which the acceleration structure will be stored.
- offset is an offset in bytes from the base address of the buffer at which the acceleration structure will be stored, and **must** be a multiple of 256.
- size is the size required for the acceleration structure.
- type is a VkAccelerationStructureTypeKHR value specifying the type of acceleration structure that will be created.
- deviceAddress is the device address requested for the acceleration structure, obtained from vkGetAccelerationStructureDeviceAddressKHR, if the accelerationStructureCaptureReplay feature is being used. If deviceAddress is zero, no specific address is requested.

Applications **should** avoid creating acceleration structures with application-provided addresses

and implementation-provided addresses in the same process, to reduce the likelihood of VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR errors.

NOTE

The expected usage for this is that a trace capture/replay tool will add the VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT flag to all buffers that use VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT, and will add VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT to all buffers used as storage for an acceleration structure where `deviceAddress` is not zero. This also means that the tool will need to add VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT to memory allocations to allow the flag to be set where the application may not have otherwise required it. During capture the tool will save the queried opaque device addresses in the trace. During replay, the buffers will be created specifying the original address so any address values stored in the trace data will remain valid.

Implementations are expected to separate such buffers in the GPU address space so normal allocations will avoid using these addresses. Applications and tools should avoid mixing application-provided and implementation-provided addresses for buffers created with VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT, to avoid address space allocation conflicts.

Applications **should** create an acceleration structure with a specific VkAccelerationStructureTypeKHR other than VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR.

NOTE

VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR is intended to be used by API translation layers. This can be used at acceleration structure creation time in cases where the actual acceleration structure type (top or bottom) is not yet known. The actual acceleration structure type is determined when the acceleration structure is constructed as VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR or VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR. For acceleration structures constructed by build commands, this type is specified by the build parameters. For acceleration structures constructed by acceleration structure copy commands, this type is determined by the source acceleration structure or serialized data, depending on the copy command.

If the acceleration structure will be the target of a build operation, the required size for an acceleration structure **can** be queried with vkGetAccelerationStructureBuildSizesKHR. If the acceleration structure is going to be the target of a compacting copy, vkCmdWriteAccelerationStructuresPropertiesKHR or vkWriteAccelerationStructuresPropertiesKHR **can** be used to obtain the compacted size required.

Valid Usage

- VUID-VkAccelerationStructureCreateInfoKHR-deviceAddress-03612
  If `deviceAddress` is not zero, `createFlags` **must** include VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR

- VUID-VkAccelerationStructureCreateInfoKHR-deviceAddress-09488
  If `deviceAddress` is not zero, it **must** have been retrieved from an identically created acceleration structure, except for `buffer` and `deviceAddress`
- VUID-VkAccelerationStructureCreateInfoKHR-deviceAddress-09489
  If `deviceAddress` is not zero, `buffer` **must** have been created identically to the `buffer` used to create the acceleration structure from which `deviceAddress` was retrieved, except for VkBufferOpaqueCaptureAddressCreateInfo::`opaqueCaptureAddress`
- VUID-VkAccelerationStructureCreateInfoKHR-deviceAddress-09490
  If `deviceAddress` is not zero, `buffer` **must** have been created with a VkBufferOpaqueCaptureAddressCreateInfo::`opaqueCaptureAddress` that was retrieved from vkGetBufferOpaqueCaptureAddress for the `buffer` that was used to create the acceleration structure from which `deviceAddress` was retrieved
- VUID-VkAccelerationStructureCreateInfoKHR-buffer-03614
  `buffer` **must** have been created with the VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR usage flag set
- VUID-VkAccelerationStructureCreateInfoKHR-buffer-03615
  `buffer` **must** not have been created with VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT
- VUID-VkAccelerationStructureCreateInfoKHR-offset-03616
  The sum of `offset` and `size` **must** be less than or equal to the size of `buffer`
- VUID-VkAccelerationStructureCreateInfoKHR-offset-03734
  `offset` **must** be a multiple of `256` bytes
- VUID-VkAccelerationStructureCreateInfoKHR-type-11600
  `type` **must** not equal VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR
- VUID-VkAccelerationStructureCreateInfoKHR-createFlags-03613
  If `createFlags` includes VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR, VkPhysicalDeviceAccelerationStructureFeaturesKHR::`accelerationStructureCaptureReplay` **must** be VK_TRUE
- VUID-VkAccelerationStructureCreateInfoKHR-createFlags-11673
  If `createFlags` includes VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR and `device` was created with multiple physical devices, the `bufferDeviceAddressMultiDevice` feature **must** be enabled
- VUID-VkAccelerationStructureCreateInfoKHR-createFlags-08108
  If `createFlags` includes VK_ACCELERATION_STRUCTURE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT, the `descriptorBufferCaptureReplay` feature **must** be enabled
- VUID-VkAccelerationStructureCreateInfoKHR-pNext-08109
  If the `pNext` chain includes a VkOpaqueCaptureDescriptorDataCreateInfoEXT structure, `createFlags` **must** contain VK_ACCELERATION_STRUCTURE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT

Valid Usage (Implicit)

- VUID-VkAccelerationStructureCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CREATE_INFO_KHR
- VUID-VkAccelerationStructureCreateInfoKHR-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkOpaqueCaptureDescriptorDataCreateInfoEXT
- VUID-VkAccelerationStructureCreateInfoKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkAccelerationStructureCreateInfoKHR-createFlags-parameter
  If `createFlags` is not `0`, `createFlags` **must** be a valid combination of VkAccelerationStructureCreateFlagBitsKHR values
- VUID-VkAccelerationStructureCreateInfoKHR-buffer-parameter
  `buffer` **must** be a valid VkBuffer handle
- VUID-VkAccelerationStructureCreateInfoKHR-type-parameter
  `type` **must** be a valid VkAccelerationStructureTypeKHR value
- VUID-VkAccelerationStructureCreateInfoKHR-deviceAddress-parameter
  If `deviceAddress` is not `0`, `deviceAddress` **must** be a valid VkDeviceAddress value

To get the build sizes for an acceleration structure, call:

```
// Provided by VK_KHR_acceleration_structure
void vkGetAccelerationStructureBuildSizesKHR(
    VkDevice                                    device,
    VkAccelerationStructureBuildTypeKHR         buildType,
    const VkAccelerationStructureBuildGeometryInfoKHR* pBuildInfo,
    const uint32_t*                             pMaxPrimitiveCounts,
    VkAccelerationStructureBuildSizesInfoKHR*   pSizeInfo);
```

- `device` is the logical device that will be used for creating the acceleration structure.
- `buildType` defines whether host or device operations (or both) are being queried for.
- `pBuildInfo` is a pointer to a VkAccelerationStructureBuildGeometryInfoKHR structure describing parameters of a build operation.
- `pMaxPrimitiveCounts` is a pointer to an array of `pBuildInfo->geometryCount` `uint32_t` values defining the number of primitives built into each geometry.
- `pSizeInfo` is a pointer to a VkAccelerationStructureBuildSizesInfoKHR structure which returns the size required for an acceleration structure and the sizes required for the scratch buffers, given the build parameters. The size requirements for a scratch buffer **may** be zero.

The `srcAccelerationStructure`, `dstAccelerationStructure`, and `mode` members of `pBuildInfo` are ignored. Any VkDeviceOrHostAddressKHR, VkDeviceOrHostAddressConstKHR, or VkDeviceAddress members of `pBuildInfo` are ignored by this command, except that the `hostAddress` member of

VkAccelerationStructureGeometryTrianglesDataKHR::transformData will be examined to check if it is NULL.

An acceleration structure created with the accelerationStructureSize returned by this command supports any build or update with a VkAccelerationStructureBuildGeometryInfoKHR structure and array of VkAccelerationStructureBuildRangeInfoKHR structures subject to the following properties:

- The build command is a host build command, and buildType is VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_KHR or VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_OR_DEVICE_KHR
- The build command is a device build command, and buildType is VK_ACCELERATION_STRUCTURE_BUILD_TYPE_DEVICE_KHR or VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_OR_DEVICE_KHR
- For VkAccelerationStructureBuildGeometryInfoKHR:
  - Its type and flags members are equal to pBuildInfo->type and pBuildInfo->flags, respectively.
  - geometryCount is less than or equal to pBuildInfo->geometryCount.
  - For each element of either pGeometries or ppGeometries at a given index, its geometryType member is equal to pBuildInfo->geometryType.
  - For each element of either pGeometries or ppGeometries at a given index, its flags member is equal to the corresponding member of the same element in pBuildInfo.
  - For each element of either pGeometries or ppGeometries at a given index, with a geometryType member equal to VK_GEOMETRY_TYPE_TRIANGLES_KHR, the vertexFormat and indexType members of geometry.triangles are equal to the corresponding members of the same element in pBuildInfo.
  - For each element of either pGeometries or ppGeometries at a given index, with a geometryType member equal to VK_GEOMETRY_TYPE_TRIANGLES_KHR, the maxVertex member of geometry.triangles is less than or equal to the corresponding member of the same element in pBuildInfo.
  - For each element of either pGeometries or ppGeometries at a given index, with a geometryType member equal to VK_GEOMETRY_TYPE_TRIANGLES_KHR, if the applicable address in the transformData member of geometry.triangles is not NULL, the corresponding transformData.hostAddress parameter in pBuildInfo is not NULL.
  - For each element of either pGeometries or ppGeometries at a given index, with a geometryType member equal to VK_GEOMETRY_TYPE_MICROMAP_KHR, the sum of usage information in either the pUsageCounts or ppUsageCounts members of the VkAccelerationStructureGeometryMicromapDataKHR structure in its pNext chain is equal to the sum of usage information in the corresponding members of pBuildInfo.
  - For each element of either pGeometries or ppGeometries at a given index, with a geometryType member equal to VK_GEOMETRY_TYPE_TRIANGLES_KHR, if the pNext chain contains VkAccelerationStructureTrianglesOpacityMicromapKHR the corresponding member of pBuildInfo also contains VkAccelerationStructureTrianglesOpacityMicromapKHR and with an equivalent micromap.

- For each VkAccelerationStructureBuildRangeInfoKHR corresponding to the VkAccelerationStructureBuildGeometryInfoKHR:
  - Its `primitiveCount` member is less than or equal to the corresponding element of `pMaxPrimitiveCounts`.
  - For each element of either `pGeometries` or `ppGeometries` at a given index, with a `geometryType` member equal to VK_GEOMETRY_TYPE_TRIANGLES_KHR, if the `pNext` chain contains VkAccelerationStructureTrianglesOpacityMicromapEXT the corresponding member of `pBuildInfo` also contains VkAccelerationStructureTrianglesOpacityMicromapEXT and with an equivalent `micromap`.
- For each VkAccelerationStructureBuildRangeInfoKHR corresponding to the VkAccelerationStructureBuildGeometryInfoKHR:
  - Its `primitiveCount` member is less than or equal to the corresponding element of `pMaxPrimitiveCounts`.

Similarly, the `updateScratchSize` value will support any build command specifying the VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR `mode` under the above conditions, and the `buildScratchSize` value will support any build command specifying the VK_BUILD_ACCELERATION_STRUCTURE_MODE_BUILD_KHR `mode` under the above conditions.

Valid Usage

- VUID-vkGetAccelerationStructureBuildSizesKHR-accelerationStructure-08933
  The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled
- VUID-vkGetAccelerationStructureBuildSizesKHR-device-03618
  If `device` was created with multiple physical devices, then the `bufferDeviceAddressMultiDevice` feature **must** be enabled
- VUID-vkGetAccelerationStructureBuildSizesKHR-pMaxPrimitiveCounts-11612
  If `pBuildInfo->geometryCount` is not `0` and `pBuildInfo->type` is not VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR , `pMaxPrimitiveCounts` **must** be a valid pointer to an array of `pBuildInfo->geometryCount` `uint32_t` values
- VUID-vkGetAccelerationStructureBuildSizesKHR-pMaxPrimitiveCounts-11613
  If `pBuildInfo->type` is VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR, `pMaxPrimitiveCounts` **must** equal `NULL`
- VUID-vkGetAccelerationStructureBuildSizesKHR-buildType-11614
  If `pBuildInfo->type` is VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR, `buildType` **must** equal VK_ACCELERATION_STRUCTURE_BUILD_TYPE_DEVICE_KHR
- VUID-vkGetAccelerationStructureBuildSizesKHR-pBuildInfo-03785
  If `pBuildInfo->pGeometries` or `pBuildInfo->ppGeometries` has a `geometryType` of VK_GEOMETRY_TYPE_INSTANCES_KHR, each `pMaxPrimitiveCounts`[i] **must** be less than or equal to VkPhysicalDeviceAccelerationStructurePropertiesKHR::`maxInstanceCount`

Valid Usage (Implicit)

- VUID-vkGetAccelerationStructureBuildSizesKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetAccelerationStructureBuildSizesKHR-buildType-parameter
  `buildType` **must** be a valid VkAccelerationStructureBuildTypeKHR value
- VUID-vkGetAccelerationStructureBuildSizesKHR-pBuildInfo-parameter
  `pBuildInfo` **must** be a valid pointer to a valid VkAccelerationStructureBuildGeometryInfoKHR structure
- VUID-vkGetAccelerationStructureBuildSizesKHR-pMaxPrimitiveCounts-parameter
  If `pMaxPrimitiveCounts` is not `NULL`, `pMaxPrimitiveCounts` **must** be a valid pointer to an array of `pBuildInfo->geometryCount` `uint32_t` values
- VUID-vkGetAccelerationStructureBuildSizesKHR-pSizeInfo-parameter
  `pSizeInfo` **must** be a valid pointer to a VkAccelerationStructureBuildSizesInfoKHR structure

The `VkAccelerationStructureBuildSizesInfoKHR` structure describes the required build sizes for an acceleration structure and scratch buffers and is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureBuildSizesInfoKHR {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceSize       accelerationStructureSize;
    VkDeviceSize       updateScratchSize;
    VkDeviceSize       buildScratchSize;
} VkAccelerationStructureBuildSizesInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `accelerationStructureSize` is the size in bytes required in a VkAccelerationStructureKHR for a build or update operation.
- `updateScratchSize` is the size in bytes required in a scratch buffer for an update operation.
- `buildScratchSize` is the size in bytes required in a scratch buffer for a build operation.

Acceleration structure objects that represent micromaps **cannot** be the target for update commands. Implementations **must** set `updateScratchSize` equal to `0` if the `pBuildInfo->type` is VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR.

Valid Usage (Implicit)

- VUID-VkAccelerationStructureBuildSizesInfoKHR-sType-sType
  `sType` **must** be

VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_BUILD_SIZES_INFO_KHR

- VUID-VkAccelerationStructureBuildSizesInfoKHR-pNext-pNext
  pNext **must** be NULL

Values which **can** be set in VkAccelerationStructureCreateInfoKHR::type specifying the type of acceleration structure, are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkAccelerationStructureTypeKHR {
    VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR = 0,
    VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR = 1,
    VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR = 2,
  // Provided by VK_KHR_opacity_micromap
    VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR = 1000623000,
} VkAccelerationStructureTypeKHR;
```

- VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR is a top-level acceleration structure containing instance data referring to bottom-level acceleration structures.
- VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR is a bottom-level acceleration structure containing the AABBs or geometry to be intersected.
- VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR is an acceleration structure whose type is determined at build time used for special circumstances. In these cases, the acceleration structure type is not known at creation time, but **must** be specified at build time as micromap, top, or bottom.
- VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR is a micromap containing opacity information, which **can** be referenced by geometry in bottom-level acceleration structures.

Bits which **can** be set in VkAccelerationStructureCreateInfoKHR::createFlags, specifying additional creation parameters for acceleration structures, are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkAccelerationStructureCreateFlagBitsKHR {
    VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR =
0x00000001,
  // Provided by VK_EXT_descriptor_buffer
    VK_ACCELERATION_STRUCTURE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT =
0x00000008,
} VkAccelerationStructureCreateFlagBitsKHR;
```

- VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR specifies that the acceleration structure's address **can** be saved and reused on a subsequent run.
- VK_ACCELERATION_STRUCTURE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT specifies that the acceleration structure **can** be used with descriptor buffers when capturing and replaying (e.g. for trace capture and replay), see

VkOpaqueCaptureDescriptorDataCreateInfoEXT for more detail.

```
// Provided by VK_KHR_acceleration_structure
typedef VkFlags VkAccelerationStructureCreateFlagsKHR;
```

VkAccelerationStructureCreateFlagsKHR is a bitmask type for setting a mask of zero or more VkAccelerationStructureCreateFlagBitsKHR.

Bits which **can** be set in VkAccelerationStructureBuildGeometryInfoKHR::`flags` specifying additional parameters for acceleration structure builds, are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkBuildAccelerationStructureFlagBitsKHR {
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR = 0x00000001,
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_COMPACTION_BIT_KHR = 0x00000002,
    VK_BUILD_ACCELERATION_STRUCTURE_PREFER_FAST_TRACE_BIT_KHR = 0x00000004,
    VK_BUILD_ACCELERATION_STRUCTURE_PREFER_FAST_BUILD_BIT_KHR = 0x00000008,
    VK_BUILD_ACCELERATION_STRUCTURE_LOW_MEMORY_BIT_KHR = 0x00000010,
  // Provided by VK_EXT_opacity_micromap
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_BIT_EXT =
0x00000100,
  // Provided by VK_KHR_ray_tracing_position_fetch
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_BIT_KHR = 0x00000800,
  // Provided by VK_KHR_opacity_micromap
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_KHR =
0x00000040,
  // Provided by VK_KHR_opacity_micromap
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_BIT_KHR =
0x00000080,
  // Provided by VK_KHR_opacity_micromap
    VK_BUILD_ACCELERATION_STRUCTURE_MICROMAP_LOSSY_BIT_KHR = 0x00000400,
  // Provided by VK_EXT_opacity_micromap
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_EXT =
VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_KHR,
  // Provided by VK_EXT_opacity_micromap
  // VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_EXT is a legacy
alias
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_EXT =
VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_EXT,
  // Provided by VK_EXT_opacity_micromap
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_BIT_EXT =
VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_BIT_KHR,
  // Provided by VK_EXT_opacity_micromap
  // VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_EXT is a legacy
alias
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_EXT =
VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_BIT_EXT,
  // Provided by VK_EXT_opacity_micromap
  // VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_EXT is a
legacy alias
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_EXT =
VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_BIT_EXT,
  // Provided by VK_KHR_ray_tracing_position_fetch
  // VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_KHR is a legacy alias
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_KHR =
VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_BIT_KHR,
} VkBuildAccelerationStructureFlagBitsKHR;
```

- VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR specifies that the specified acceleration structure **can** be updated with a mode of

VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR in VkAccelerationStructureBuildGeometryInfoKHR .

- VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_COMPACTION_BIT_KHR specifies that the specified acceleration structure **can** act as the source for a copy acceleration structure command with `mode` of VK_COPY_ACCELERATION_STRUCTURE_MODE_COMPACT_KHR to produce a compacted acceleration structure.
- VK_BUILD_ACCELERATION_STRUCTURE_PREFER_FAST_TRACE_BIT_KHR specifies that the given acceleration structure build **should** prioritize trace performance over build time.
- VK_BUILD_ACCELERATION_STRUCTURE_PREFER_FAST_BUILD_BIT_KHR specifies that the given acceleration structure build **should** prioritize build time over trace performance.
- VK_BUILD_ACCELERATION_STRUCTURE_LOW_MEMORY_BIT_KHR specifies that this acceleration structure **should** minimize the size of the scratch memory and the final result acceleration structure, potentially at the expense of build time or trace performance.
- VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_KHR specifies that the opacity micromaps associated with the specified acceleration structure **may** change with an acceleration structure update.
- VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_BIT_EXT specifies that the data of the opacity micromaps associated with the specified acceleration structure **may** change with an acceleration structure update.
- VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_BIT_KHR specifies that the specified acceleration structure **may** be referenced in an instance with VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_BIT_KHR set.
- VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_BIT_KHR specifies that the specified acceleration structure **can** be used when fetching the vertex positions of a hit triangle.
- VK_BUILD_ACCELERATION_STRUCTURE_MICROMAP_LOSSY_BIT_KHR specifies that micromaps **can** be compressed with a lossy encoding as described in Ray Opacity Micromap, which **may** support additional subdivision levels.

NOTE VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR and VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_COMPACTION_BIT_KHR **may** take more time and memory than a normal build, and so **should** only be used when those features are needed.

NOTE VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR and VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_COMPACTION_BIT_KHR are allowed to be used together. In that case, the result of the compaction copy is used as the source of a build with `mode` of VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR to perform the compacted update.

```
// Provided by VK_KHR_acceleration_structure
typedef VkFlags VkBuildAccelerationStructureFlagsKHR;
```

VkBuildAccelerationStructureFlagsKHR is a bitmask type for setting a mask of zero or more VkBuildAccelerationStructureFlagBitsKHR.

Geometry types are specified by VkGeometryTypeKHR, which takes values:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkGeometryTypeKHR {
    VK_GEOMETRY_TYPE_TRIANGLES_KHR = 0,
    VK_GEOMETRY_TYPE_AABBS_KHR = 1,
    VK_GEOMETRY_TYPE_INSTANCES_KHR = 2,
  // Provided by VK_KHR_opacity_micromap
    VK_GEOMETRY_TYPE_MICROMAP_KHR = 1000623000,
} VkGeometryTypeKHR;
```

- VK_GEOMETRY_TYPE_TRIANGLES_KHR specifies a geometry type consisting of triangles.
- VK_GEOMETRY_TYPE_AABBS_KHR specifies a geometry type consisting of axis-aligned bounding boxes.
- VK_GEOMETRY_TYPE_INSTANCES_KHR specifies a geometry type consisting of acceleration structure instances.
- VK_GEOMETRY_TYPE_MICROMAP_KHR specifies a geometry type consisting of a micromap array.

Bits specifying additional parameters for geometries in acceleration structure builds, are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkGeometryFlagBitsKHR {
    VK_GEOMETRY_OPAQUE_BIT_KHR = 0x00000001,
    VK_GEOMETRY_NO_DUPLICATE_ANY_HIT_INVOCATION_BIT_KHR = 0x00000002,
} VkGeometryFlagBitsKHR;
```

- VK_GEOMETRY_OPAQUE_BIT_KHR specifies that this geometry does not invoke the any-hit shaders even if present in a hit group.
- VK_GEOMETRY_NO_DUPLICATE_ANY_HIT_INVOCATION_BIT_KHR specifies that the implementation **must** only call the any-hit shader a single time for each primitive in this geometry. If this bit is absent an implementation **may** invoke the any-hit shader more than once for this geometry.

```
// Provided by VK_KHR_acceleration_structure
typedef VkFlags VkGeometryFlagsKHR;
```

VkGeometryFlagsKHR is a bitmask type for setting a mask of zero or more VkGeometryFlagBitsKHR.

To destroy an acceleration structure, call:

```
// Provided by VK_KHR_acceleration_structure
void vkDestroyAccelerationStructureKHR(
    VkDevice                                    device,
    VkAccelerationStructureKHR                  accelerationStructure,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the acceleration structure.
- `accelerationStructure` is the acceleration structure to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyAccelerationStructureKHR-accelerationStructure-08934
  The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled
- VUID-vkDestroyAccelerationStructureKHR-accelerationStructure-02442
  All submitted commands that refer to `accelerationStructure` **must** have completed execution
- VUID-vkDestroyAccelerationStructureKHR-accelerationStructure-02443
  If `VkAllocationCallbacks` were provided when `accelerationStructure` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyAccelerationStructureKHR-accelerationStructure-02444
  If no `VkAllocationCallbacks` were provided when `accelerationStructure` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyAccelerationStructureKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyAccelerationStructureKHR-accelerationStructure-parameter
  If `accelerationStructure` is not VK_NULL_HANDLE, `accelerationStructure` **must** be a valid VkAccelerationStructureKHR handle
- VUID-vkDestroyAccelerationStructureKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyAccelerationStructureKHR-accelerationStructure-parent
  If `accelerationStructure` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `accelerationStructure` **must** be externally synchronized

Possible values of `buildType` in vkGetAccelerationStructureBuildSizesKHR are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkAccelerationStructureBuildTypeKHR {
    VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_KHR = 0,
    VK_ACCELERATION_STRUCTURE_BUILD_TYPE_DEVICE_KHR = 1,
    VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_OR_DEVICE_KHR = 2,
} VkAccelerationStructureBuildTypeKHR;
```

- VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_KHR requests the memory requirement for operations performed by the host.
- VK_ACCELERATION_STRUCTURE_BUILD_TYPE_DEVICE_KHR requests the memory requirement for operations performed by the device.
- VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_OR_DEVICE_KHR requests the memory requirement for operations performed by either the host, or the device.

To query the 64-bit device address for an acceleration structure, call:

```
// Provided by VK_KHR_acceleration_structure
VkDeviceAddress vkGetAccelerationStructureDeviceAddressKHR(
    VkDevice                                    device,
    const VkAccelerationStructureDeviceAddressInfoKHR* pInfo);
```

- `device` is the logical device that the acceleration structure was created on.
- `pInfo` is a pointer to a VkAccelerationStructureDeviceAddressInfoKHR structure specifying the acceleration structure to retrieve an address for.

The 64-bit return value is an address of the acceleration structure, which can be used for device and shader operations that involve acceleration structures, such as ray traversal and acceleration structure building.

If the acceleration structure was created with vkCreateAccelerationStructure2KHR, the return value will be the same address as `addressRange.address`.

If the acceleration structure was created with vkCreateAccelerationStructureKHR with a non-zero value of VkAccelerationStructureCreateInfoKHR::`deviceAddress`, the return value will be the same address.

Otherwise, if the acceleration structure was created with a `type` of VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR, the returned address **must** be consistent with the relative offset to other acceleration structures with `type`

VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR allocated with the same VkBuffer. That is, the difference in returned addresses between the two **must** be the same as the difference in offsets provided at acceleration structure creation.

The returned address **must** be aligned to 256 bytes.

NOTE For acceleration structures created with vkCreateAccelerationStructureKHR, their device address **may** be different from the buffer device address corresponding to the acceleration structure's start offset in its storage buffer for acceleration structure types other than VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR.

Valid Usage

- VUID-vkGetAccelerationStructureDeviceAddressKHR-accelerationStructure-08935
  The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled
- VUID-vkGetAccelerationStructureDeviceAddressKHR-device-03504
  If `device` was created with multiple physical devices, then the `bufferDeviceAddressMultiDevice` feature **must** be enabled
- VUID-vkGetAccelerationStructureDeviceAddressKHR-pInfo-09541
  If the buffer on which `pInfo->accelerationStructure` was placed is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkGetAccelerationStructureDeviceAddressKHR-pInfo-09542
  The buffer on which `pInfo->accelerationStructure` was placed **must** have been created with the VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT usage flag set

Valid Usage (Implicit)

- VUID-vkGetAccelerationStructureDeviceAddressKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetAccelerationStructureDeviceAddressKHR-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkAccelerationStructureDeviceAddressInfoKHR structure

The `VkAccelerationStructureDeviceAddressInfoKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureDeviceAddressInfoKHR {
    VkStructureType              sType;
    const void*                  pNext;
    VkAccelerationStructureKHR   accelerationStructure;
} VkAccelerationStructureDeviceAddressInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `accelerationStructure` specifies the acceleration structure whose address is being queried.

Valid Usage (Implicit)

- VUID-VkAccelerationStructureDeviceAddressInfoKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_DEVICE_ADDRESS_INFO_KHR`
- VUID-VkAccelerationStructureDeviceAddressInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkAccelerationStructureDeviceAddressInfoKHR-accelerationStructure-parameter
  `accelerationStructure` **must** be a valid `VkAccelerationStructureKHR` handle

12.8. Micromaps

Micromaps are opaque data structures that are built by the implementation to encode sub-triangle data to be included in an acceleration structure.

Micromaps are represented by `VkAccelerationStructureKHR` handles for use with `VK_KHR_opacity_micromap` commands.

Micromaps are represented by `VkMicromapEXT` handles for use with `VK_EXT_opacity_micromap` commands:

```
// Provided by VK_EXT_opacity_micromap
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkMicromapEXT)
```

To create a micromap, call:

```
// Provided by VK_EXT_opacity_micromap
VkResult vkCreateMicromapEXT(
    VkDevice                                    device,
    const VkMicromapCreateInfoEXT*              pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkMicromapEXT*                              pMicromap);
```

- `device` is the logical device that creates the micromap structure object.
- `pCreateInfo` is a pointer to a `VkMicromapCreateInfoEXT` structure containing parameters affecting creation of the micromap.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pMicromap` is a pointer to a `VkMicromapEXT` handle in which the resulting micromap object is returned.

Similar to other objects in Vulkan, the micromap creation merely creates an object with a specific “shape”. The type and quantity of geometry that can be built into a micromap is determined by the parameters of VkMicromapCreateInfoEXT.

The micromap data is stored in the object referred to by VkMicromapCreateInfoEXT::buffer. Once memory has been bound to that buffer, it must be populated by micromap build or micromap copy commands such as vkCmdBuildMicromapsEXT, vkBuildMicromapsEXT, vkCmdCopyMicromapEXT, and vkCopyMicromapEXT.

NOTE

The expected usage for a trace capture/replay tool is that it will serialize and later deserialize the micromap data using micromap copy commands. During capture the tool will set VkCopyMicromapToMemoryInfoEXT::mode to VK_COPY_MICROMAP_MODE_SERIALIZE_EXT, and during replay the tool will set VkCopyMemoryToMicromapInfoEXT::mode to VK_COPY_MICROMAP_MODE_DESERIALIZE_EXT.

Valid Usage

- VUID-vkCreateMicromapEXT-micromap-11615
  The VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCreateMicromapEXT-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkCreateMicromapEXT-pCreateInfo-parameter
  pCreateInfo **must** be a valid pointer to a valid VkMicromapCreateInfoEXT structure
- VUID-vkCreateMicromapEXT-pAllocator-parameter
  If pAllocator is not NULL, pAllocator **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateMicromapEXT-pMicromap-parameter
  pMicromap **must** be a valid pointer to a VkMicromapEXT handle
- VUID-vkCreateMicromapEXT-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR
- VK_ERROR_OUT_OF_HOST_MEMORY

- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkMicromapCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkMicromapCreateInfoEXT {
    VkStructureType             sType;
    const void*                 pNext;
    VkMicromapCreateFlagsEXT    createFlags;
    VkBuffer                    buffer;
    VkDeviceSize                offset;
    VkDeviceSize                size;
    VkMicromapTypeEXT           type;
    VkDeviceAddress             deviceAddress;
} VkMicromapCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- createFlags is a bitmask of VkMicromapCreateFlagBitsEXT specifying additional creation parameters of the micromap.
- buffer is the buffer on which the micromap will be stored.
- offset is an offset in bytes from the base address of the buffer at which the micromap will be stored, and **must** be a multiple of 256.
- size is the size required for the micromap.
- type is a VkMicromapTypeEXT value specifying the type of micromap that will be created.
- deviceAddress is the device address requested for the micromap if the VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromapCaptureReplay feature is being used.

If deviceAddress is zero, no specific address is requested.

If deviceAddress is not zero, deviceAddress **must** be an address retrieved from an identically created micromap on the same implementation. The micromap **must** also be placed on an identically created buffer and at the same offset.

Applications **should** avoid creating micromaps with application-provided addresses and implementation-provided addresses in the same process, to reduce the likelihood of VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR errors.

NOTE

The expected usage for this is that a trace capture/replay tool will add the VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT flag and VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT to all buffers used as storage for a micromap with VK_MICROMAP_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_EXT included in createFlags. This also means that the tool will need to add

> VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT to memory allocations to allow the VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT flag to be set where the application may not have otherwise required it. During capture the tool will save the queried opaque device addresses in the trace. During replay, the buffers will be created specifying the original address so any address values stored in the trace data will remain valid.
>
> Implementations are expected to separate such buffers in the GPU address space so normal allocations will avoid using these addresses. Applications and tools should avoid mixing application-provided and implementation-provided addresses for buffers created with VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT, to avoid address space allocation conflicts.

If the micromap will be the target of a build operation, the required size for a micromap **can** be queried with vkGetMicromapBuildSizesEXT.

Valid Usage

- VUID-VkMicromapCreateInfoEXT-deviceAddress-07433
  If `deviceAddress` is not zero, `createFlags` **must** include VK_MICROMAP_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_EXT
- VUID-VkMicromapCreateInfoEXT-micromapCaptureReplay-11616
  If `createFlags` includes VK_MICROMAP_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_EXT, `VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromapCaptureReplay` **must** be VK_TRUE
- VUID-VkMicromapCreateInfoEXT-bufferDeviceAddressMultiDevice-11617
  If `createFlags` includes VK_MICROMAP_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_EXT and `device` was created with multiple physical devices, the `bufferDeviceAddressMultiDevice` feature **must** be enabled
- VUID-VkMicromapCreateInfoEXT-buffer-07435
  `buffer` **must** have been created with the VK_BUFFER_USAGE_MICROMAP_STORAGE_BIT_EXT usage flag set
- VUID-VkMicromapCreateInfoEXT-buffer-07436
  `buffer` **must** not have been created with VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT
- VUID-VkMicromapCreateInfoEXT-offset-07437
  The sum of `offset` and `size` **must** be less than or equal to the size of `buffer`
- VUID-VkMicromapCreateInfoEXT-offset-07438
  `offset` **must** be a multiple of `256` bytes

Valid Usage (Implicit)

- VUID-VkMicromapCreateInfoEXT-sType-sType

sType **must** be VK_STRUCTURE_TYPE_MICROMAP_CREATE_INFO_EXT

- VUID-VkMicromapCreateInfoEXT-pNext-pNext
  pNext **must** be NULL
- VUID-VkMicromapCreateInfoEXT-createFlags-parameter
  If createFlags is not 0, createFlags **must** be a valid combination of VkMicromapCreateFlagBitsEXT values
- VUID-VkMicromapCreateInfoEXT-buffer-parameter
  buffer **must** be a valid VkBuffer handle
- VUID-VkMicromapCreateInfoEXT-type-parameter
  type **must** be a valid VkMicromapTypeEXT value
- VUID-VkMicromapCreateInfoEXT-deviceAddress-parameter
  If deviceAddress is not 0, deviceAddress **must** be a valid VkDeviceAddress value

To get the build sizes for a micromap, call:

```
// Provided by VK_EXT_opacity_micromap
void vkGetMicromapBuildSizesEXT(
    VkDevice                                    device,
    VkAccelerationStructureBuildTypeKHR         buildType,
    const VkMicromapBuildInfoEXT*               pBuildInfo,
    VkMicromapBuildSizesInfoEXT*                pSizeInfo);
```

- device is the logical device that will be used for creating the micromap.
- buildType defines whether host or device operations (or both) are being queried for.
- pBuildInfo is a pointer to a VkMicromapBuildInfoEXT structure describing parameters of a build operation.
- pSizeInfo is a pointer to a VkMicromapBuildSizesInfoEXT structure which returns the size required for a micromap and the sizes required for the scratch buffers, given the build parameters. The size requirements for a scratch buffer **may** be zero.

The dstMicromap, mode, data, scratchData, and triangleArray members of pBuildInfo are ignored by this command.

A micromap created with the micromapSize returned by this command supports any build with a VkMicromapBuildInfoEXT structure subject to the following properties:

- The build command is a host build command, and buildType is VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_KHR or VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_OR_DEVICE_KHR
- The build command is a device build command, and buildType is VK_ACCELERATION_STRUCTURE_BUILD_TYPE_DEVICE_KHR or VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_OR_DEVICE_KHR
- For VkMicromapBuildInfoEXT:

- Its `type` and `flags` members are equal to `pBuildInfo->type` and `pBuildInfo->flags`, respectively.
- The sum of usage information in either `pUsageCounts` or `ppUsageCounts` is equal to the sum of usage information in either `pBuildInfo->pUsageCounts` or `pBuildInfo->ppUsageCounts`.

Similarly, the `buildScratchSize` value will support any build command specifying the VK_BUILD_MICROMAP_MODE_BUILD_EXT `mode` under the above conditions.

Valid Usage

- VUID-vkGetMicromapBuildSizesEXT-micromap-11618
  The `VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap` feature **must** be enabled
- VUID-vkGetMicromapBuildSizesEXT-device-07440
  If `device` was created with multiple physical devices, the `bufferDeviceAddressMultiDevice` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetMicromapBuildSizesEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetMicromapBuildSizesEXT-buildType-parameter
  `buildType` **must** be a valid VkAccelerationStructureBuildTypeKHR value
- VUID-vkGetMicromapBuildSizesEXT-pBuildInfo-parameter
  `pBuildInfo` **must** be a valid pointer to a valid VkMicromapBuildInfoEXT structure
- VUID-vkGetMicromapBuildSizesEXT-pSizeInfo-parameter
  `pSizeInfo` **must** be a valid pointer to a VkMicromapBuildSizesInfoEXT structure

The `VkMicromapBuildSizesInfoEXT` structure describes the required build sizes for a micromap and scratch buffers and is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkMicromapBuildSizesInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceSize       micromapSize;
    VkDeviceSize       buildScratchSize;
    VkBool32           discardable;
} VkMicromapBuildSizesInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `micromapSize` is the size in bytes required in a VkMicromapEXT for a build or update operation.

- `buildScratchSize` is the size in bytes required in a scratch buffer for a build operation.
- `discardable` indicates whether or not the micromap object may be destroyed after an acceleration structure build or update. A false value means that acceleration structures built with this micromap **may** contain references to the data contained therein, and the application **must** not destroy the micromap until ray traversal has concluded. A true value means that the information in the micromap will be copied by value into the acceleration structure, and the micromap **can** be destroyed after the acceleration structure build concludes.

Valid Usage (Implicit)

- VUID-VkMicromapBuildSizesInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MICROMAP_BUILD_SIZES_INFO_EXT
- VUID-VkMicromapBuildSizesInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`

Values which **can** be set in VkMicromapCreateInfoEXT::`type` specifying the type of micromap, are:

```
// Provided by VK_EXT_opacity_micromap
typedef enum VkMicromapTypeEXT {
    VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT = 0,
} VkMicromapTypeEXT;
```

- VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT is a micromap containing data to control the opacity of a triangle.

Bits which **can** be set in VkMicromapCreateInfoEXT::`createFlags`, specifying additional creation parameters for micromaps, are:

```
// Provided by VK_EXT_opacity_micromap
typedef enum VkMicromapCreateFlagBitsEXT {
    VK_MICROMAP_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_EXT = 0x00000001,
} VkMicromapCreateFlagBitsEXT;
```

- VK_MICROMAP_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_EXT specifies that the micromap's address **can** be saved and reused on a subsequent run.

```
// Provided by VK_EXT_opacity_micromap
typedef VkFlags VkMicromapCreateFlagsEXT;
```

`VkMicromapCreateFlagsEXT` is a bitmask type for setting a mask of zero or more VkMicromapCreateFlagBitsEXT.

To destroy a micromap, call:

```
// Provided by VK_EXT_opacity_micromap
void vkDestroyMicromapEXT(
    VkDevice                                    device,
    VkMicromapEXT                               micromap,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the micromap.
- `micromap` is the micromap to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyMicromapEXT-micromap-11619
  The `VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap` feature **must** be enabled
- VUID-vkDestroyMicromapEXT-micromap-07441
  All submitted commands that refer to `micromap` **must** have completed execution
- VUID-vkDestroyMicromapEXT-micromap-07442
  If `VkAllocationCallbacks` were provided when `micromap` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyMicromapEXT-micromap-07443
  If no `VkAllocationCallbacks` were provided when `micromap` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyMicromapEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyMicromapEXT-micromap-parameter
  If `micromap` is not VK_NULL_HANDLE, `micromap` **must** be a valid VkMicromapEXT handle
- VUID-vkDestroyMicromapEXT-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyMicromapEXT-micromap-parent
  If `micromap` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `micromap` **must** be externally synchronized

12.9. Resource Memory Association

Resources are initially created as *virtual allocations* with no backing memory. Device memory is allocated separately (see Device Memory) and then associated with the resource. This association is done differently for sparse and non-sparse resources.

Resources created with any of the sparse creation flags are considered sparse resources. Resources created without these flags are non-sparse. The details on resource memory association for sparse resources are described in Sparse Resources.

Non-sparse resources **must** be bound completely and contiguously to a single `VkDeviceMemory` object before the resource is passed as a parameter to any of the following operations:

- creating image or buffer views
- updating descriptor sets
- recording commands in a command buffer

Once bound, the memory binding is immutable for the lifetime of the resource.

In a logical device representing more than one physical device, buffer and image resources exist on all physical devices but **can** be bound to memory differently on each. Each such replicated resource is an *instance* of the resource. For sparse resources, each instance **can** be bound to memory arbitrarily differently. For non-sparse resources, each instance **can** either be bound to the local or a peer instance of the memory, or for images **can** be bound to rectangular regions from the local and/or peer instances. When a resource is used in a descriptor set, each physical device interprets the descriptor according to its own instance's binding to memory.

NOTE	There are no new copy commands to transfer data between physical devices. Instead, an application **can** create a resource with a peer mapping and use it as the source or destination of a transfer command executed by a single physical device to copy the data from one physical device to another.

To determine the memory requirements for a buffer resource, call:

```
// Provided by VK_VERSION_1_0
void vkGetBufferMemoryRequirements(
    VkDevice                                    device,
    VkBuffer                                    buffer,
    VkMemoryRequirements*                       pMemoryRequirements);
```

- `device` is the logical device that owns the buffer.
- `buffer` is the buffer to query.
- `pMemoryRequirements` is a pointer to a VkMemoryRequirements structure in which the memory requirements of the buffer object are returned.

Valid Usage (Implicit)

- VUID-vkGetBufferMemoryRequirements-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetBufferMemoryRequirements-buffer-parameter
  `buffer` **must** be a valid VkBuffer handle
- VUID-vkGetBufferMemoryRequirements-pMemoryRequirements-parameter
  `pMemoryRequirements` **must** be a valid pointer to a VkMemoryRequirements structure
- VUID-vkGetBufferMemoryRequirements-buffer-parent
  `buffer` **must** have been created, allocated, or retrieved from `device`

To determine the memory requirements for an image resource which is not created with the VK_IMAGE_CREATE_DISJOINT_BIT flag set, call:

```
// Provided by VK_VERSION_1_0
void vkGetImageMemoryRequirements(
    VkDevice                                    device,
    VkImage                                     image,
    VkMemoryRequirements*                       pMemoryRequirements);
```

- `device` is the logical device that owns the image.
- `image` is the image to query.
- `pMemoryRequirements` is a pointer to a VkMemoryRequirements structure in which the memory requirements of the image object are returned.

Valid Usage

- VUID-vkGetImageMemoryRequirements-image-01588
  `image` **must** not have been created with the VK_IMAGE_CREATE_DISJOINT_BIT flag set

Valid Usage (Implicit)

- VUID-vkGetImageMemoryRequirements-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetImageMemoryRequirements-image-parameter
  `image` **must** be a valid VkImage handle
- VUID-vkGetImageMemoryRequirements-pMemoryRequirements-parameter
  `pMemoryRequirements` **must** be a valid pointer to a VkMemoryRequirements structure
- VUID-vkGetImageMemoryRequirements-image-parent
  `image` **must** have been created, allocated, or retrieved from `device`

The VkMemoryRequirements structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMemoryRequirements {
    VkDeviceSize    size;
    VkDeviceSize    alignment;
    uint32_t        memoryTypeBits;
} VkMemoryRequirements;
```

- size is the size, in bytes, of the memory allocation **required** for the resource.
- alignment is the alignment, in bytes, of the offset within the allocation **required** for the resource.
- memoryTypeBits is a bitmask and contains one bit set for every supported memory type for the resource. Bit i is set if and only if the memory type i in the VkPhysicalDeviceMemoryProperties structure for the physical device is supported for the resource.

The value of size has no meaning and **should** be ignored if the resource being queried was created with any of the following external memory handle types:

- VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLTEXTURE_BIT_EXT

The implementation guarantees certain properties about the memory requirements returned by vkGetBufferMemoryRequirements2, vkGetImageMemoryRequirements2, vkGetDeviceBufferMemoryRequirements, vkGetDeviceImageMemoryRequirements, vkGetBufferMemoryRequirements and vkGetImageMemoryRequirements:

- The memoryTypeBits member always contains at least one bit set.
- If buffer is a VkBuffer not created with the VK_BUFFER_CREATE_SPARSE_BINDING_BIT or VK_BUFFER_CREATE_PROTECTED_BIT bits set, or if image is a linear image that was not created with the VK_IMAGE_CREATE_PROTECTED_BIT bit set, then the memoryTypeBits member always contains at least one bit set corresponding to a VkMemoryType with a propertyFlags that has both the VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT bit and the VK_MEMORY_PROPERTY_HOST_COHERENT_BIT bit set. In other words, mappable coherent memory **can** always be attached to these objects.
- If buffer was created with VkExternalMemoryBufferCreateInfo::handleTypes set to 0 or image was created with VkExternalMemoryImageCreateInfo::handleTypes set to 0, the memoryTypeBits member always contains at least one bit set corresponding to a VkMemoryType with a propertyFlags that has the VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT bit set.
- The memoryTypeBits member is identical for all VkBuffer objects created with the same value for the flags and usage members in the VkBufferCreateInfo structure and the handleTypes member of the VkExternalMemoryBufferCreateInfo structure passed to vkCreateBuffer. Further, if usage1 and usage2 of type VkBufferUsageFlags are such that the bits set in usage2 are a subset of

the bits set in usage1, and they have the same flags and VkExternalMemoryBufferCreateInfo ::handleTypes, then the bits set in memoryTypeBits returned for usage1 **must** be a subset of the bits set in memoryTypeBits returned for usage2, for all values of flags.

- The alignment member is a power of two.
- The alignment member is identical for all VkBuffer objects created with the same combination of values for the usage and flags members in the VkBufferCreateInfo structure passed to vkCreateBuffer.
- If the maintenance4 feature is enabled, then the alignment member is identical for all VkImage objects created with the same combination of values for the flags, imageType, format, extent, mipLevels, arrayLayers, samples, tiling and usage members in the VkImageCreateInfo structure , and the tilingControl member of the VkImageTilingControlCreateInfoEXT structure if in the pNext chain, passed to vkCreateImage.
- The alignment member satisfies the buffer descriptor offset alignment requirements associated with the VkBuffer's usage:
  - If usage included VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT or VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT, alignment **must** be an integer multiple of VkPhysicalDeviceLimits::minTexelBufferOffsetAlignment.
  - If usage included VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT, alignment **must** be an integer multiple of VkPhysicalDeviceLimits::minUniformBufferOffsetAlignment.
  - If usage included VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, alignment **must** be an integer multiple of VkPhysicalDeviceLimits::minStorageBufferOffsetAlignment.
- For images created with a color format, the memoryTypeBits member is identical for all VkImage objects created with the same combination of values for the tiling member, the VK_IMAGE_CREATE_SPARSE_BINDING_BIT bit and VK_IMAGE_CREATE_PROTECTED_BIT bit of the flags member, the VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT bit of the flags member, the VK_IMAGE_USAGE_HOST_TRANSFER_BIT bit of the usage member if the VkPhysicalDeviceHostImageCopyProperties::identicalMemoryTypeRequirements property is VK_FALSE, handleTypes member of VkExternalMemoryImageCreateInfo, and the VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT of the usage member in the VkImageCreateInfo structure passed to vkCreateImage.
- For images created with a depth/stencil format, the memoryTypeBits member is identical for all VkImage objects created with the same combination of values for the format member, the tiling member, the VK_IMAGE_CREATE_SPARSE_BINDING_BIT bit and VK_IMAGE_CREATE_PROTECTED_BIT bit of the flags member, the VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT bit of the flags member, the VK_IMAGE_USAGE_HOST_TRANSFER_BIT bit of the usage member if the VkPhysicalDeviceHostImageCopyProperties::identicalMemoryTypeRequirements property is VK_FALSE, handleTypes member of VkExternalMemoryImageCreateInfo, and the VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT of the usage member in the VkImageCreateInfo structure passed to vkCreateImage.
- If the memory requirements are for a VkImage, the memoryTypeBits member **must** not refer to a VkMemoryType with a propertyFlags that has the VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT bit set if the image did not have VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT bit set in the usage member of the

VkImageCreateInfo structure passed to vkCreateImage.

- If the memory requirements are for a VkBuffer, the memoryTypeBits member **must** not refer to a VkMemoryType with a propertyFlags that has the VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT bit set.

> **NOTE** The implication of this requirement is that lazily allocated memory is disallowed for buffers in all cases.

- The size member is identical for all VkBuffer objects created with the same combination of creation parameters specified in VkBufferCreateInfo and its pNext chain.
- The size member is identical for all VkImage objects created with the same combination of creation parameters specified in VkImageCreateInfo and its pNext chain.

> **NOTE** This, however, does not imply that they interpret the contents of the bound memory identically with each other. That additional guarantee, however, **can** be explicitly requested using VK_IMAGE_CREATE_ALIAS_BIT.

- If the maintenance4 feature is enabled, these additional guarantees apply:
  - For a VkBuffer, the size memory requirement is never greater than that of another VkBuffer created with a greater or equal size specified in VkBufferCreateInfo, all other creation parameters being identical.
  - For a VkBuffer, the size memory requirement is never greater than the result of aligning VkBufferCreateInfo::size with the alignment memory requirement.
  - For a VkImage, the size memory requirement is never greater than that of another VkImage created with a greater or equal value in each of extent.width, extent.height, and extent.depth; all other creation parameters being identical.
  - The memory requirements returned by vkGetDeviceBufferMemoryRequirements are identical to those that would be returned by vkGetBufferMemoryRequirements2 if it were called with a VkBuffer created with the same VkBufferCreateInfo values.
  - The memory requirements returned by vkGetDeviceImageMemoryRequirements are identical to those that would be returned by vkGetImageMemoryRequirements2 if it were called with a VkImage created with the same VkImageCreateInfo values.

To determine the memory requirements for a buffer resource, call:

```
// Provided by VK_VERSION_1_1
void vkGetBufferMemoryRequirements2(
    VkDevice                                    device,
    const VkBufferMemoryRequirementsInfo2*      pInfo,
    VkMemoryRequirements2*                      pMemoryRequirements);
```

```
// Provided by VK_KHR_get_memory_requirements2
// Equivalent to vkGetBufferMemoryRequirements2
void vkGetBufferMemoryRequirements2KHR(
    VkDevice                                    device,
    const VkBufferMemoryRequirementsInfo2*      pInfo,
    VkMemoryRequirements2*                      pMemoryRequirements);
```

- `device` is the logical device that owns the buffer.
- `pInfo` is a pointer to a VkBufferMemoryRequirementsInfo2 structure containing parameters required for the memory requirements query.
- `pMemoryRequirements` is a pointer to a VkMemoryRequirements2 structure in which the memory requirements of the buffer object are returned.

Valid Usage (Implicit)

- VUID-vkGetBufferMemoryRequirements2-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetBufferMemoryRequirements2-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkBufferMemoryRequirementsInfo2 structure
- VUID-vkGetBufferMemoryRequirements2-pMemoryRequirements-parameter
  `pMemoryRequirements` **must** be a valid pointer to a VkMemoryRequirements2 structure

To determine the memory requirements for a buffer resource without creating an object, call:

```
// Provided by VK_VERSION_1_3
void vkGetDeviceBufferMemoryRequirements(
    VkDevice                                    device,
    const VkDeviceBufferMemoryRequirements*     pInfo,
    VkMemoryRequirements2*                      pMemoryRequirements);
```

```
// Provided by VK_KHR_maintenance4
// Equivalent to vkGetDeviceBufferMemoryRequirements
void vkGetDeviceBufferMemoryRequirementsKHR(
    VkDevice                                    device,
    const VkDeviceBufferMemoryRequirements*     pInfo,
    VkMemoryRequirements2*                      pMemoryRequirements);
```

- `device` is the logical device intended to own the buffer.
- `pInfo` is a pointer to a VkDeviceBufferMemoryRequirements structure containing parameters required for the memory requirements query.
- `pMemoryRequirements` is a pointer to a VkMemoryRequirements2 structure in which the memory requirements of the buffer object are returned.

Valid Usage (Implicit)

- VUID-vkGetDeviceBufferMemoryRequirements-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDeviceBufferMemoryRequirements-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkDeviceBufferMemoryRequirements structure
- VUID-vkGetDeviceBufferMemoryRequirements-pMemoryRequirements-parameter
  `pMemoryRequirements` **must** be a valid pointer to a VkMemoryRequirements2 structure

The `VkBufferMemoryRequirementsInfo2` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkBufferMemoryRequirementsInfo2 {
    VkStructureType    sType;
    const void*        pNext;
    VkBuffer           buffer;
} VkBufferMemoryRequirementsInfo2;
```

```
// Provided by VK_KHR_get_memory_requirements2
// Equivalent to VkBufferMemoryRequirementsInfo2
typedef VkBufferMemoryRequirementsInfo2 VkBufferMemoryRequirementsInfo2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `buffer` is the buffer to query.

Valid Usage (Implicit)

- VUID-VkBufferMemoryRequirementsInfo2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BUFFER_MEMORY_REQUIREMENTS_INFO_2
- VUID-VkBufferMemoryRequirementsInfo2-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkBufferMemoryRequirementsInfo2-buffer-parameter
  `buffer` **must** be a valid VkBuffer handle

The `VkDeviceBufferMemoryRequirements` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkDeviceBufferMemoryRequirements {
    VkStructureType              sType;
    const void*                  pNext;
    const VkBufferCreateInfo*    pCreateInfo;
} VkDeviceBufferMemoryRequirements;
```

```
// Provided by VK_KHR_maintenance4
// Equivalent to VkDeviceBufferMemoryRequirements
typedef VkDeviceBufferMemoryRequirements VkDeviceBufferMemoryRequirementsKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pCreateInfo` is a pointer to a VkBufferCreateInfo structure containing parameters affecting creation of the buffer to query.

Valid Usage (Implicit)

- VUID-VkDeviceBufferMemoryRequirements-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_BUFFER_MEMORY_REQUIREMENTS
- VUID-VkDeviceBufferMemoryRequirements-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDeviceBufferMemoryRequirements-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkBufferCreateInfo structure

To determine the memory requirements for an image resource, call:

```
// Provided by VK_VERSION_1_1
void vkGetImageMemoryRequirements2(
    VkDevice                                    device,
    const VkImageMemoryRequirementsInfo2*       pInfo,
    VkMemoryRequirements2*                      pMemoryRequirements);
```

```
// Provided by VK_KHR_get_memory_requirements2
// Equivalent to vkGetImageMemoryRequirements2
void vkGetImageMemoryRequirements2KHR(
    VkDevice                                    device,
    const VkImageMemoryRequirementsInfo2*       pInfo,
    VkMemoryRequirements2*                      pMemoryRequirements);
```

- `device` is the logical device that owns the image.
- `pInfo` is a pointer to a VkImageMemoryRequirementsInfo2 structure containing parameters

required for the memory requirements query.

- `pMemoryRequirements` is a pointer to a VkMemoryRequirements2 structure in which the memory requirements of the image object are returned.

Valid Usage (Implicit)

- VUID-vkGetImageMemoryRequirements2-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetImageMemoryRequirements2-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkImageMemoryRequirementsInfo2 structure
- VUID-vkGetImageMemoryRequirements2-pMemoryRequirements-parameter
  `pMemoryRequirements` **must** be a valid pointer to a VkMemoryRequirements2 structure

To determine the memory requirements for an image resource without creating an object, call:

```
// Provided by VK_VERSION_1_3
void vkGetDeviceImageMemoryRequirements(
    VkDevice                                    device,
    const VkDeviceImageMemoryRequirements*      pInfo,
    VkMemoryRequirements2*                      pMemoryRequirements);
```

```
// Provided by VK_KHR_maintenance4
// Equivalent to vkGetDeviceImageMemoryRequirements
void vkGetDeviceImageMemoryRequirementsKHR(
    VkDevice                                    device,
    const VkDeviceImageMemoryRequirements*      pInfo,
    VkMemoryRequirements2*                      pMemoryRequirements);
```

- `device` is the logical device intended to own the image.
- `pInfo` is a pointer to a VkDeviceImageMemoryRequirements structure containing parameters required for the memory requirements query.
- `pMemoryRequirements` is a pointer to a VkMemoryRequirements2 structure in which the memory requirements of the image object are returned.

Valid Usage (Implicit)

- VUID-vkGetDeviceImageMemoryRequirements-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDeviceImageMemoryRequirements-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkDeviceImageMemoryRequirements structure
- VUID-vkGetDeviceImageMemoryRequirements-pMemoryRequirements-parameter
  `pMemoryRequirements` **must** be a valid pointer to a VkMemoryRequirements2 structure

The VkImageMemoryRequirementsInfo2 structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkImageMemoryRequirementsInfo2 {
    VkStructureType    sType;
    const void*        pNext;
    VkImage            image;
} VkImageMemoryRequirementsInfo2;
```

```
// Provided by VK_KHR_get_memory_requirements2
// Equivalent to VkImageMemoryRequirementsInfo2
typedef VkImageMemoryRequirementsInfo2 VkImageMemoryRequirementsInfo2KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- image is the image to query.

Valid Usage

- VUID-VkImageMemoryRequirementsInfo2-image-01589
  If image was created with a *multi-planar* format and the VK_IMAGE_CREATE_DISJOINT_BIT flag, there **must** be a VkImagePlaneMemoryRequirementsInfo included in the pNext chain of the VkImageMemoryRequirementsInfo2 structure
- VUID-VkImageMemoryRequirementsInfo2-image-02279
  If image was created with VK_IMAGE_CREATE_DISJOINT_BIT and with VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then there **must** be a VkImagePlaneMemoryRequirementsInfo included in the pNext chain of the VkImageMemoryRequirementsInfo2 structure
- VUID-VkImageMemoryRequirementsInfo2-image-01590
  If image was not created with the VK_IMAGE_CREATE_DISJOINT_BIT flag, there **must** not be a VkImagePlaneMemoryRequirementsInfo included in the pNext chain of the VkImageMemoryRequirementsInfo2 structure
- VUID-VkImageMemoryRequirementsInfo2-image-02280
  If image was created with a single-plane format and with any tiling other than VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then there **must** not be a VkImagePlaneMemoryRequirementsInfo included in the pNext chain of the VkImageMemoryRequirementsInfo2 structure

Valid Usage (Implicit)

- VUID-VkImageMemoryRequirementsInfo2-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_IMAGE_MEMORY_REQUIREMENTS_INFO_2

- VUID-VkImageMemoryRequirementsInfo2-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkImagePlaneMemoryRequirementsInfo
- VUID-VkImageMemoryRequirementsInfo2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkImageMemoryRequirementsInfo2-image-parameter
  `image` **must** be a valid VkImage handle

The `VkDeviceImageMemoryRequirements` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkDeviceImageMemoryRequirements {
    VkStructureType             sType;
    const void*                 pNext;
    const VkImageCreateInfo*    pCreateInfo;
    VkImageAspectFlagBits       planeAspect;
} VkDeviceImageMemoryRequirements;
```

```
// Provided by VK_KHR_maintenance4
// Equivalent to VkDeviceImageMemoryRequirements
typedef VkDeviceImageMemoryRequirements VkDeviceImageMemoryRequirementsKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pCreateInfo` is a pointer to a VkImageCreateInfo structure containing parameters affecting creation of the image to query.
- `planeAspect` is a VkImageAspectFlagBits value specifying the aspect corresponding to the image plane to query. This parameter is ignored unless `pCreateInfo->tiling` is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, or `pCreateInfo->flags` has VK_IMAGE_CREATE_DISJOINT_BIT set.

Valid Usage

- VUID-VkDeviceImageMemoryRequirements-pCreateInfo-06416
  The `pCreateInfo->pNext` chain **must** not contain a VkImageSwapchainCreateInfoKHR structure
- VUID-VkDeviceImageMemoryRequirements-pCreateInfo-12489
  `pCreateInfo->tiling` **must** not be VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT
- VUID-VkDeviceImageMemoryRequirements-pCreateInfo-06417
  If `pCreateInfo->format` specifies a *multi-planar* format and `pCreateInfo->flags` has VK_IMAGE_CREATE_DISJOINT_BIT set then `planeAspect` **must** not be VK_IMAGE_ASPECT_NONE_KHR

- VUID-VkDeviceImageMemoryRequirements-pCreateInfo-06419
  If `pCreateInfo->flags` has VK_IMAGE_CREATE_DISJOINT_BIT set and if the `pCreateInfo->tiling` is VK_IMAGE_TILING_LINEAR or VK_IMAGE_TILING_OPTIMAL, then `planeAspect` **must** be a single valid multi-planar aspect mask bit
- VUID-VkDeviceImageMemoryRequirements-planeAspect-12399
  If `planeAspect` is not 0 and `pCreateInfo->flags` does not have VK_IMAGE_CREATE_DISJOINT_BIT set, then `planeAspect` must be a valid VkImageAspectFlagBits value

Valid Usage (Implicit)

- VUID-VkDeviceImageMemoryRequirements-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_IMAGE_MEMORY_REQUIREMENTS
- VUID-VkDeviceImageMemoryRequirements-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDeviceImageMemoryRequirements-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkImageCreateInfo structure

To determine the memory requirements for a plane of a disjoint image, add a `VkImagePlaneMemoryRequirementsInfo` structure to the `pNext` chain of the `VkImageMemoryRequirementsInfo2` structure.

The `VkImagePlaneMemoryRequirementsInfo` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkImagePlaneMemoryRequirementsInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkImageAspectFlagBits    planeAspect;
} VkImagePlaneMemoryRequirementsInfo;
```

```
// Provided by VK_KHR_sampler_ycbcr_conversion
// Equivalent to VkImagePlaneMemoryRequirementsInfo
typedef VkImagePlaneMemoryRequirementsInfo VkImagePlaneMemoryRequirementsInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `planeAspect` is a VkImageAspectFlagBits value specifying the aspect corresponding to the image plane to query.

Valid Usage

- VUID-VkImagePlaneMemoryRequirementsInfo-planeAspect-02281
  If the image's `tiling` is VK_IMAGE_TILING_LINEAR or VK_IMAGE_TILING_OPTIMAL, then `planeAspect` **must** be a single valid multi-planar aspect mask bit
- VUID-VkImagePlaneMemoryRequirementsInfo-planeAspect-02282
  If the image's `tiling` is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then `planeAspect` **must** be a single valid *memory plane* for the image (that is, `planeAspect` **must** specify a plane index that is less than the VkDrmFormatModifierPropertiesEXT::`drmFormatModifierPlaneCount` associated with the image's `format` and VkImageDrmFormatModifierPropertiesEXT::`drmFormatModifier`)

Valid Usage (Implicit)

- VUID-VkImagePlaneMemoryRequirementsInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_PLANE_MEMORY_REQUIREMENTS_INFO
- VUID-VkImagePlaneMemoryRequirementsInfo-planeAspect-parameter
  `planeAspect` **must** be a valid VkImageAspectFlagBits value

Structure Chaining

Extends the structure

- VkImageMemoryRequirementsInfo2

The `VkMemoryRequirements2` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkMemoryRequirements2 {
    VkStructureType         sType;
    void*                   pNext;
    VkMemoryRequirements    memoryRequirements;
} VkMemoryRequirements2;
```

```
// Provided by VK_KHR_get_memory_requirements2
// Equivalent to VkMemoryRequirements2
typedef VkMemoryRequirements2 VkMemoryRequirements2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryRequirements` is a VkMemoryRequirements structure describing the memory requirements of the resource.

Valid Usage (Implicit)

- VUID-VkMemoryRequirements2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_REQUIREMENTS_2
- VUID-VkMemoryRequirements2-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkMemoryDedicatedRequirements
- VUID-VkMemoryRequirements2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

The `VkMemoryDedicatedRequirements` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkMemoryDedicatedRequirements {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           prefersDedicatedAllocation;
    VkBool32           requiresDedicatedAllocation;
} VkMemoryDedicatedRequirements;
```

```
// Provided by VK_KHR_dedicated_allocation
// Equivalent to VkMemoryDedicatedRequirements
typedef VkMemoryDedicatedRequirements VkMemoryDedicatedRequirementsKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `prefersDedicatedAllocation` specifies that the implementation would prefer a dedicated allocation for this resource. The application is still free to suballocate the resource but it **may** get better performance if a dedicated allocation is used.
- `requiresDedicatedAllocation` specifies that a dedicated allocation is required for this resource.

To determine the dedicated allocation requirements of a buffer or image resource, add a VkMemoryDedicatedRequirements structure to the `pNext` chain of the VkMemoryRequirements2 structure passed as the `pMemoryRequirements` parameter of vkGetBufferMemoryRequirements2, vkGetDeviceBufferMemoryRequirements, vkGetDeviceImageMemoryRequirements, or vkGetImageMemoryRequirements2 respectively.

Constraints on the values returned for buffer resources are:

- `requiresDedicatedAllocation` **may** be VK_TRUE if the `pNext` chain of VkBufferCreateInfo for the call to `vkCreateBuffer` used to create the buffer being queried included a VkExternalMemoryBufferCreateInfo structure, and any of the handle types specified in VkExternalMemoryBufferCreateInfo::`handleTypes` requires dedicated allocation, as reported by vkGetPhysicalDeviceExternalBufferProperties in `VkExternalBufferProperties` ::`externalMemoryProperties.externalMemoryFeatures`. Otherwise, `requiresDedicatedAllocation` will

be VK_FALSE.

- When the implementation sets `requiresDedicatedAllocation` to VK_TRUE, it **must** also set `prefersDedicatedAllocation` to VK_TRUE.
- If VK_BUFFER_CREATE_SPARSE_BINDING_BIT was set in VkBufferCreateInfo::`flags` when `buffer` was created, then both `prefersDedicatedAllocation` and `requiresDedicatedAllocation` will be VK_FALSE.

Constraints on the values returned for image resources are:

- `requiresDedicatedAllocation` **may** be VK_TRUE if the `pNext` chain of VkImageCreateInfo for the call to vkCreateImage used to create the image being queried included a VkExternalMemoryImageCreateInfo structure, and any of the handle types specified in VkExternalMemoryImageCreateInfo::`handleTypes` requires dedicated allocation, as reported by vkGetPhysicalDeviceImageFormatProperties2 in `VkExternalImageFormatProperties` ::`externalMemoryProperties.externalMemoryFeatures`.
- `requiresDedicatedAllocation` **may** be VK_TRUE if the image's tiling is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT.
- `requiresDedicatedAllocation` will otherwise be VK_FALSE.
- If VK_IMAGE_CREATE_SPARSE_BINDING_BIT was set in VkImageCreateInfo::`flags` when `image` was created, then both `prefersDedicatedAllocation` and `requiresDedicatedAllocation` will be VK_FALSE.

Valid Usage (Implicit)

- VUID-VkMemoryDedicatedRequirements-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_DEDICATED_REQUIREMENTS

Structure Chaining

Extends the structure

- VkMemoryRequirements2

To attach memory to a buffer object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkBindBufferMemory(
    VkDevice                                    device,
    VkBuffer                                    buffer,
    VkDeviceMemory                              memory,
    VkDeviceSize                                memoryOffset);
```

- `device` is the logical device that owns the buffer and memory.
- `buffer` is the buffer to be attached to memory.

- `memory` is a VkDeviceMemory object describing the device memory to attach.
- `memoryOffset` is the start offset of the region of `memory` which is to be bound to the buffer. The number of bytes returned in the `VkMemoryRequirements`::`size` member in `memory`, starting from `memoryOffset` bytes, will be bound to the specified buffer.

`vkBindBufferMemory` is equivalent to passing the same parameters through VkBindBufferMemoryInfo to vkBindBufferMemory2.

Valid Usage

- VUID-vkBindBufferMemory-buffer-07459
  `buffer` **must** not have been bound to a memory object
- VUID-vkBindBufferMemory-buffer-01030
  `buffer` **must** not have been created with any sparse memory binding flags
- VUID-vkBindBufferMemory-memoryOffset-01031
  `memoryOffset` **must** be less than the size of `memory`
- VUID-vkBindBufferMemory-memory-01035
  `memory` **must** have been allocated using one of the memory types allowed in the `memoryTypeBits` member of the `VkMemoryRequirements` structure returned from a call to `vkGetBufferMemoryRequirements` with `buffer`
- VUID-vkBindBufferMemory-None-10739
  `memoryOffset` **must** be an integer multiple of the `alignment` member of the `VkMemoryRequirements` structure returned from a call to `vkGetBufferMemoryRequirements` with `buffer`
- VUID-vkBindBufferMemory-None-10741
  the `size` member of the `VkMemoryRequirements` structure returned from a call to `vkGetBufferMemoryRequirements` with `buffer` **must** be less than or equal to the size of `memory` minus `memoryOffset`
- VUID-vkBindBufferMemory-buffer-01444
  If `buffer` requires a dedicated allocation (as reported by vkGetBufferMemoryRequirements2 in VkMemoryDedicatedRequirements::`requiresDedicatedAllocation` for `buffer`), `memory` **must** have been allocated with VkMemoryDedicatedAllocateInfo::`buffer` equal to `buffer`
- VUID-vkBindBufferMemory-memory-01508
  If the `VkMemoryAllocateInfo` provided when `memory` was allocated included a VkMemoryDedicatedAllocateInfo structure in its `pNext` chain, and VkMemoryDedicatedAllocateInfo::`buffer` was not VK_NULL_HANDLE, then `buffer` **must** equal VkMemoryDedicatedAllocateInfo::`buffer`, and `memoryOffset` **must** be zero
- VUID-vkBindBufferMemory-memory-10925
  If the `VkMemoryAllocateInfo` provided when `memory` was allocated included a VkMemoryDedicatedAllocateInfo structure in its `pNext` chain, VkMemoryDedicatedAllocateInfo::`image` **must** have been VK_NULL_HANDLE
- VUID-vkBindBufferMemory-None-01898
  If `buffer` was created with the VK_BUFFER_CREATE_PROTECTED_BIT bit set, the buffer

must be bound to a memory object allocated with a memory type that reports VK_MEMORY_PROPERTY_PROTECTED_BIT

- VUID-vkBindBufferMemory-None-01899
  If `buffer` was created with the VK_BUFFER_CREATE_PROTECTED_BIT bit not set, the buffer **must** not be bound to a memory object allocated with a memory type that reports VK_MEMORY_PROPERTY_PROTECTED_BIT
- VUID-vkBindBufferMemory-memory-02726
  If the value of VkExportMemoryAllocateInfo::`handleTypes` used to allocate `memory` is not `0`, it **must** include at least one of the handles set in VkExternalMemoryBufferCreateInfo ::`handleTypes` when `buffer` was created
- VUID-vkBindBufferMemory-memory-02985
  If `memory` was allocated by a memory import operation, the external handle type of the imported memory **must** also have been set in VkExternalMemoryBufferCreateInfo ::`handleTypes` when `buffer` was created
- VUID-vkBindBufferMemory-bufferDeviceAddress-03339
  If the VkPhysicalDeviceBufferDeviceAddressFeatures::`bufferDeviceAddress` feature is enabled and `buffer` was created with the VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT usage flag set, `memory` **must** have been allocated with the VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT bit set
- VUID-vkBindBufferMemory-bufferDeviceAddressCaptureReplay-09200
  If the VkPhysicalDeviceBufferDeviceAddressFeatures::`bufferDeviceAddressCaptureReplay` feature is enabled and `buffer` was created with the VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT bit set, `memory` **must** have been allocated with the VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT bit set
- VUID-vkBindBufferMemory-descriptorBufferCaptureReplay-08112
  If the `buffer` was created with the VK_BUFFER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT bit set, `memory` **must** have been allocated with the VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT bit set
- VUID-vkBindBufferMemory-buffer-09201
  If the `buffer` was created with the VK_BUFFER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT bit set, `memory` **must** have been allocated with the VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT bit set
- VUID-vkBindBufferMemory-buffer-11408
  If the `buffer` was created with the VK_BUFFER_USAGE_DESCRIPTOR_HEAP_BIT_EXT or VK_BUFFER_USAGE_2_DESCRIPTOR_HEAP_BIT_EXT bit set, `memory` **must** have been allocated with the VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT bit set

Valid Usage (Implicit)

- VUID-vkBindBufferMemory-device-parameter

`device` **must** be a valid VkDevice handle

- VUID-vkBindBufferMemory-buffer-parameter
  `buffer` **must** be a valid VkBuffer handle
- VUID-vkBindBufferMemory-memory-parameter
  `memory` **must** be a valid VkDeviceMemory handle
- VUID-vkBindBufferMemory-buffer-parent
  `buffer` **must** have been created, allocated, or retrieved from `device`
- VUID-vkBindBufferMemory-memory-parent
  `memory` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `buffer` **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To attach memory to buffer objects for one or more buffers at a time, call:

```
// Provided by VK_VERSION_1_1
VkResult vkBindBufferMemory2(
    VkDevice                                    device,
    uint32_t                                    bindInfoCount,
    const VkBindBufferMemoryInfo*               pBindInfos);
```

```
// Provided by VK_KHR_bind_memory2
// Equivalent to vkBindBufferMemory2
VkResult vkBindBufferMemory2KHR(
    VkDevice                                    device,
    uint32_t                                    bindInfoCount,
    const VkBindBufferMemoryInfo*               pBindInfos);
```

- `device` is the logical device that owns the buffers and memory.
- `bindInfoCount` is the number of elements in `pBindInfos`.
- `pBindInfos` is a pointer to an array of `bindInfoCount` VkBindBufferMemoryInfo structures describing buffers and memory to bind.

On some implementations, it **may** be more efficient to batch memory bindings into a single command.

If the `maintenance6` feature is enabled, this command **must** attempt to perform all of the memory binding operations described by `pBindInfos`, and **must** not early exit on the first failure.

If any of the memory binding operations described by `pBindInfos` fail, the VkResult returned by this command **must** be the return value of any one of the memory binding operations which did not return VK_SUCCESS.

NOTE

If the `vkBindBufferMemory2` command failed, VkBindMemoryStatus structures were not included in the `pNext` chains of each element of `pBindInfos`, and `bindInfoCount` was greater than one, then the buffers referenced by `pBindInfos` will be in an indeterminate state, and must not be used.

Applications should destroy these buffers.

Valid Usage (Implicit)

- VUID-vkBindBufferMemory2-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkBindBufferMemory2-pBindInfos-parameter
  `pBindInfos` **must** be a valid pointer to an array of `bindInfoCount` valid VkBindBufferMemoryInfo structures
- VUID-vkBindBufferMemory2-bindInfoCount-arraylength
  `bindInfoCount` **must** be greater than `0`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

VkBindBufferMemoryInfo contains members corresponding to the parameters of vkBindBufferMemory.

The VkBindBufferMemoryInfo structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkBindBufferMemoryInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkBuffer           buffer;
    VkDeviceMemory     memory;
    VkDeviceSize       memoryOffset;
} VkBindBufferMemoryInfo;
```

```
// Provided by VK_KHR_bind_memory2
// Equivalent to VkBindBufferMemoryInfo
typedef VkBindBufferMemoryInfo VkBindBufferMemoryInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- buffer is the buffer to be attached to memory.
- memory is a VkDeviceMemory object describing the device memory to attach.
- memoryOffset is the start offset of the region of memory which is to be bound to the buffer. The number of bytes returned in the VkMemoryRequirements::size member in memory, starting from memoryOffset bytes, will be bound to the specified buffer.

Valid Usage

- VUID-VkBindBufferMemoryInfo-buffer-07459
  buffer **must** not have been bound to a memory object
- VUID-VkBindBufferMemoryInfo-buffer-01030
  buffer **must** not have been created with any sparse memory binding flags
- VUID-VkBindBufferMemoryInfo-memoryOffset-01031
  memoryOffset **must** be less than the size of memory
- VUID-VkBindBufferMemoryInfo-memory-01035
  memory **must** have been allocated using one of the memory types allowed in the memoryTypeBits member of the VkMemoryRequirements structure returned from a call to vkGetBufferMemoryRequirements with buffer
- VUID-VkBindBufferMemoryInfo-None-10739
  memoryOffset **must** be an integer multiple of the alignment member of the VkMemoryRequirements structure returned from a call to vkGetBufferMemoryRequirements with buffer
- VUID-VkBindBufferMemoryInfo-None-10741

the size member of the VkMemoryRequirements structure returned from a call to vkGetBufferMemoryRequirements with buffer **must** be less than or equal to the size of memory minus memoryOffset

- VUID-VkBindBufferMemoryInfo-buffer-01444
  If buffer requires a dedicated allocation (as reported by vkGetBufferMemoryRequirements2 in VkMemoryDedicatedRequirements::requiresDedicatedAllocation for buffer), memory **must** have been allocated with VkMemoryDedicatedAllocateInfo::buffer equal to buffer
- VUID-VkBindBufferMemoryInfo-memory-01508
  If the VkMemoryAllocateInfo provided when memory was allocated included a VkMemoryDedicatedAllocateInfo structure in its pNext chain, and VkMemoryDedicatedAllocateInfo::buffer was not VK_NULL_HANDLE, then buffer **must** equal VkMemoryDedicatedAllocateInfo::buffer, and memoryOffset **must** be zero
- VUID-VkBindBufferMemoryInfo-memory-10925
  If the VkMemoryAllocateInfo provided when memory was allocated included a VkMemoryDedicatedAllocateInfo structure in its pNext chain, VkMemoryDedicatedAllocateInfo::image **must** have been VK_NULL_HANDLE
- VUID-VkBindBufferMemoryInfo-None-01898
  If buffer was created with the VK_BUFFER_CREATE_PROTECTED_BIT bit set, the buffer **must** be bound to a memory object allocated with a memory type that reports VK_MEMORY_PROPERTY_PROTECTED_BIT
- VUID-VkBindBufferMemoryInfo-None-01899
  If buffer was created with the VK_BUFFER_CREATE_PROTECTED_BIT bit not set, the buffer **must** not be bound to a memory object allocated with a memory type that reports VK_MEMORY_PROPERTY_PROTECTED_BIT
- VUID-VkBindBufferMemoryInfo-memory-02726
  If the value of VkExportMemoryAllocateInfo::handleTypes used to allocate memory is not 0, it **must** include at least one of the handles set in VkExternalMemoryBufferCreateInfo::handleTypes when buffer was created
- VUID-VkBindBufferMemoryInfo-memory-02985
  If memory was allocated by a memory import operation, the external handle type of the imported memory **must** also have been set in VkExternalMemoryBufferCreateInfo::handleTypes when buffer was created
- VUID-VkBindBufferMemoryInfo-bufferDeviceAddress-03339
  If the VkPhysicalDeviceBufferDeviceAddressFeatures::bufferDeviceAddress feature is enabled and buffer was created with the VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT usage flag set, memory **must** have been allocated with the VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT bit set
- VUID-VkBindBufferMemoryInfo-bufferDeviceAddressCaptureReplay-09200
  If the VkPhysicalDeviceBufferDeviceAddressFeatures::bufferDeviceAddressCaptureReplay feature is enabled and buffer was created with the VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT bit set, memory **must** have been allocated with the VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT bit set

- VUID-VkBindBufferMemoryInfo-descriptorBufferCaptureReplay-08112
  If the `buffer` was created with the VK_BUFFER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT bit set, `memory` **must** have been allocated with the VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT bit set
- VUID-VkBindBufferMemoryInfo-buffer-09201
  If the `buffer` was created with the VK_BUFFER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT bit set, `memory` **must** have been allocated with the VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT bit set
- VUID-VkBindBufferMemoryInfo-buffer-11408
  If the `buffer` was created with the VK_BUFFER_USAGE_DESCRIPTOR_HEAP_BIT_EXT or VK_BUFFER_USAGE_2_DESCRIPTOR_HEAP_BIT_EXT bit set, `memory` **must** have been allocated with the VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT bit set
- VUID-VkBindBufferMemoryInfo-pNext-01605
  If the `pNext` chain includes a VkBindBufferMemoryDeviceGroupInfo structure, all instances of `memory` specified by VkBindBufferMemoryDeviceGroupInfo::`pDeviceIndices` **must** have been allocated

Valid Usage (Implicit)

- VUID-VkBindBufferMemoryInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_INFO
- VUID-VkBindBufferMemoryInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkBindBufferMemoryDeviceGroupInfo or VkBindMemoryStatus
- VUID-VkBindBufferMemoryInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkBindBufferMemoryInfo-buffer-parameter
  `buffer` **must** be a valid VkBuffer handle
- VUID-VkBindBufferMemoryInfo-memory-parameter
  `memory` **must** be a valid VkDeviceMemory handle
- VUID-VkBindBufferMemoryInfo-commonparent
  Both of `buffer`, and `memory` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `buffer` **must** be externally synchronized

The `VkBindBufferMemoryDeviceGroupInfo` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkBindBufferMemoryDeviceGroupInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           deviceIndexCount;
    const uint32_t*    pDeviceIndices;
} VkBindBufferMemoryDeviceGroupInfo;
```

```
// Provided by VK_KHR_bind_memory2 with VK_KHR_device_group
// Equivalent to VkBindBufferMemoryDeviceGroupInfo
typedef VkBindBufferMemoryDeviceGroupInfo VkBindBufferMemoryDeviceGroupInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `deviceIndexCount` is the number of elements in `pDeviceIndices`.
- `pDeviceIndices` is a pointer to an array of device indices.

If the `pNext` chain of VkBindBufferMemoryInfo includes a `VkBindBufferMemoryDeviceGroupInfo` structure, then that structure determines how memory is bound to buffers across multiple devices in a device group.

If `deviceIndexCount` is greater than zero, then on device index i the buffer is attached to the instance of `memory` on the physical device with device index `pDeviceIndices`[i].

If `deviceIndexCount` is zero and `memory` comes from a memory heap with the VK_MEMORY_HEAP_MULTI_INSTANCE_BIT bit set, then it is as if `pDeviceIndices` contains consecutive indices from zero to the number of physical devices in the logical device, minus one. In other words, by default each physical device attaches to its own instance of `memory`.

If `deviceIndexCount` is zero and `memory` comes from a memory heap without the VK_MEMORY_HEAP_MULTI_INSTANCE_BIT bit set, then it is as if `pDeviceIndices` contains an array of zeros. In other words, by default each physical device attaches to instance zero.

Valid Usage

- VUID-VkBindBufferMemoryDeviceGroupInfo-deviceIndexCount-01606
  `deviceIndexCount` **must** either be zero or equal to the number of physical devices in the logical device
- VUID-VkBindBufferMemoryDeviceGroupInfo-pDeviceIndices-01607
  All elements of `pDeviceIndices` **must** be valid device indices

Valid Usage (Implicit)

- VUID-VkBindBufferMemoryDeviceGroupInfo-sType-sType

sType **must** be VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_DEVICE_GROUP_INFO

- VUID-VkBindBufferMemoryDeviceGroupInfo-pDeviceIndices-parameter
  If deviceIndexCount is not 0, pDeviceIndices **must** be a valid pointer to an array of deviceIndexCount uint32_t values

Structure Chaining

Extends the structure

- VkBindBufferMemoryInfo

The VkBindMemoryStatus structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkBindMemoryStatus {
    VkStructureType    sType;
    const void*        pNext;
    VkResult*          pResult;
} VkBindMemoryStatus;
```

```
// Provided by VK_KHR_maintenance6
// Equivalent to VkBindMemoryStatus
typedef VkBindMemoryStatus VkBindMemoryStatusKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pResult is a pointer to a VkResult value.

If the pNext chain of VkBindBufferMemoryInfo or VkBindImageMemoryInfo includes a VkBindMemoryStatus structure, then the VkBindMemoryStatus::pResult will be populated with a value describing the result of the corresponding memory binding operation.

Valid Usage (Implicit)

- VUID-VkBindMemoryStatus-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_BIND_MEMORY_STATUS
- VUID-VkBindMemoryStatus-pResult-parameter
  pResult **must** be a valid pointer to a VkResult value

Structure Chaining

Extends the structures

- VkBindBufferMemoryInfo
- VkBindImageMemoryInfo

To attach memory to a VkImage object created without the VK_IMAGE_CREATE_DISJOINT_BIT set, call:

```
// Provided by VK_VERSION_1_0
VkResult vkBindImageMemory(
    VkDevice                                    device,
    VkImage                                     image,
    VkDeviceMemory                              memory,
    VkDeviceSize                                memoryOffset);
```

- device is the logical device that owns the image and memory.
- image is the image.
- memory is the VkDeviceMemory object describing the device memory to attach.
- memoryOffset is the start offset of the region of memory which is to be bound to the image. The number of bytes returned in the VkMemoryRequirements::size member in memory, starting from memoryOffset bytes, will be bound to the specified image.

vkBindImageMemory is equivalent to passing the same parameters through VkBindImageMemoryInfo to vkBindImageMemory2.

Valid Usage

- VUID-vkBindImageMemory-image-07460
  image **must** not have been bound to a memory object
- VUID-vkBindImageMemory-image-01045
  image **must** not have been created with any sparse memory binding flags
- VUID-vkBindImageMemory-memoryOffset-01046
  memoryOffset **must** be less than the size of memory
- VUID-vkBindImageMemory-image-01445
  If image requires a dedicated allocation (as reported by vkGetImageMemoryRequirements2 in VkMemoryDedicatedRequirements::requiresDedicatedAllocation for image), memory **must** have been created with VkMemoryDedicatedAllocateInfo::image equal to image
- VUID-vkBindImageMemory-memory-02628
  If the VkMemoryAllocateInfo provided when memory was allocated included a VkMemoryDedicatedAllocateInfo structure in its pNext chain, and VkMemoryDedicatedAllocateInfo::image was not VK_NULL_HANDLE, then image **must**

equal VkMemoryDedicatedAllocateInfo::image and memoryOffset **must** be zero

- VUID-vkBindImageMemory-memory-10926
  If the VkMemoryAllocateInfo provided when memory was allocated included a VkMemoryDedicatedAllocateInfo structure in its pNext chain, VkMemoryDedicatedAllocateInfo::buffer **must** have been VK_NULL_HANDLE
- VUID-vkBindImageMemory-None-01901
  If image was created with the VK_IMAGE_CREATE_PROTECTED_BIT bit set, the image **must** be bound to a memory object allocated with a memory type that reports VK_MEMORY_PROPERTY_PROTECTED_BIT
- VUID-vkBindImageMemory-None-01902
  If image was created with the VK_IMAGE_CREATE_PROTECTED_BIT bit not set, the image **must** not be bound to a memory object created with a memory type that reports VK_MEMORY_PROPERTY_PROTECTED_BIT
- VUID-vkBindImageMemory-memory-02728
  If the value of VkExportMemoryAllocateInfo::handleTypes used to allocate memory is not 0, it **must** include at least one of the handles set in VkExternalMemoryImageCreateInfo ::handleTypes when image was created
- VUID-vkBindImageMemory-memory-02989
  If memory was created by a memory import operation, the external handle type of the imported memory **must** also have been set in VkExternalMemoryImageCreateInfo ::handleTypes when image was created
- VUID-vkBindImageMemory-image-08113
  If the image was created with the VK_IMAGE_CREATE_DESCRIPTOR_HEAP_CAPTURE_REPLAY_BIT_EXT bit set, memory **must** have been allocated with the VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT bit set
- VUID-vkBindImageMemory-image-09202
  If the image was created with the VK_IMAGE_CREATE_DESCRIPTOR_HEAP_CAPTURE_REPLAY_BIT_EXT bit set, memory **must** have been allocated with the VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT bit set
- VUID-vkBindImageMemory-image-01608
  image **must** not have been created with the VK_IMAGE_CREATE_DISJOINT_BIT set
- VUID-vkBindImageMemory-memory-01047
  memory **must** have been allocated using one of the memory types allowed in the memoryTypeBits member of the VkMemoryRequirements structure returned from a call to vkGetImageMemoryRequirements with image
- VUID-vkBindImageMemory-None-10735
  memoryOffset **must** be an integer multiple of the alignment member of the VkMemoryRequirements structure returned from a call to vkGetImageMemoryRequirements with image
- VUID-vkBindImageMemory-None-10737
  size member of the VkMemoryRequirements structure returned from a call to vkGetImageMemoryRequirements with image **must** be less than or equal to the size of memory

minus `memoryOffset`

Valid Usage (Implicit)

- VUID-vkBindImageMemory-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkBindImageMemory-image-parameter
  `image` **must** be a valid VkImage handle
- VUID-vkBindImageMemory-memory-parameter
  `memory` **must** be a valid VkDeviceMemory handle
- VUID-vkBindImageMemory-image-parent
  `image` **must** have been created, allocated, or retrieved from `device`
- VUID-vkBindImageMemory-memory-parent
  `memory` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `image` **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To attach memory to image objects for one or more images at a time, call:

```
// Provided by VK_VERSION_1_1
VkResult vkBindImageMemory2(
    VkDevice                                    device,
    uint32_t                                    bindInfoCount,
    const VkBindImageMemoryInfo*                pBindInfos);
```

```
// Provided by VK_KHR_bind_memory2
// Equivalent to vkBindImageMemory2
VkResult vkBindImageMemory2KHR(
    VkDevice                                    device,
    uint32_t                                    bindInfoCount,
    const VkBindImageMemoryInfo*                pBindInfos);
```

- `device` is the logical device that owns the images and memory.
- `bindInfoCount` is the number of elements in `pBindInfos`.
- `pBindInfos` is a pointer to an array of VkBindImageMemoryInfo structures, describing images and memory to bind.

On some implementations, it **may** be more efficient to batch memory bindings into a single command.

If the `maintenance6` feature is enabled, this command **must** attempt to perform all of the memory binding operations described by `pBindInfos`, and **must** not early exit on the first failure.

If any of the memory binding operations described by `pBindInfos` fail, the VkResult returned by this command **must** be the return value of any one of the memory binding operations which did not return VK_SUCCESS.

NOTE

If the `vkBindImageMemory2` command failed, VkBindMemoryStatus structures were not included in the `pNext` chains of each element of `pBindInfos`, and `bindInfoCount` was greater than one, then the images referenced by `pBindInfos` will be in an indeterminate state, and must not be used.

Applications should destroy these images.

Valid Usage

- VUID-vkBindImageMemory2-pBindInfos-02858
  If any VkBindImageMemoryInfo::`image` was created with VK_IMAGE_CREATE_DISJOINT_BIT then all planes of VkBindImageMemoryInfo::`image` **must** be bound individually in separate `pBindInfos`
- VUID-vkBindImageMemory2-pBindInfos-04006
  `pBindInfos` **must** not refer to the same image subresource more than once

Valid Usage (Implicit)

- VUID-vkBindImageMemory2-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkBindImageMemory2-pBindInfos-parameter
  `pBindInfos` **must** be a valid pointer to an array of `bindInfoCount` valid VkBindImageMemoryInfo structures

- VUID-vkBindImageMemory2-bindInfoCount-arraylength
  `bindInfoCount` **must** be greater than `0`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

`VkBindImageMemoryInfo` contains members corresponding to the parameters of vkBindImageMemory.

The `VkBindImageMemoryInfo` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkBindImageMemoryInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkImage            image;
    VkDeviceMemory     memory;
    VkDeviceSize       memoryOffset;
} VkBindImageMemoryInfo;
```

```
// Provided by VK_KHR_bind_memory2
// Equivalent to VkBindImageMemoryInfo
typedef VkBindImageMemoryInfo VkBindImageMemoryInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `image` is the image to be attached to memory.
- `memory` is a VkDeviceMemory object describing the device memory to attach.
- `memoryOffset` is the start offset of the region of `memory` which is to be bound to the image. The number of bytes returned in the `VkMemoryRequirements`::`size` member in `memory`, starting from `memoryOffset` bytes, will be bound to the specified image.

Valid Usage

- VUID-VkBindImageMemoryInfo-image-07460
  `image` **must** not have been bound to a memory object
- VUID-VkBindImageMemoryInfo-image-01045
  `image` **must** not have been created with any sparse memory binding flags
- VUID-VkBindImageMemoryInfo-memoryOffset-01046
  `memoryOffset` **must** be less than the size of `memory`
- VUID-VkBindImageMemoryInfo-image-01445
  If `image` requires a dedicated allocation (as reported by vkGetImageMemoryRequirements2 in VkMemoryDedicatedRequirements::`requiresDedicatedAllocation` for `image`), `memory` **must** have been created with VkMemoryDedicatedAllocateInfo::`image` equal to `image`
- VUID-VkBindImageMemoryInfo-memory-02628
  If the `VkMemoryAllocateInfo` provided when `memory` was allocated included a VkMemoryDedicatedAllocateInfo structure in its `pNext` chain, and VkMemoryDedicatedAllocateInfo::`image` was not VK_NULL_HANDLE, then `image` **must** equal VkMemoryDedicatedAllocateInfo::`image` and `memoryOffset` **must** be zero
- VUID-VkBindImageMemoryInfo-memory-10926
  If the `VkMemoryAllocateInfo` provided when `memory` was allocated included a VkMemoryDedicatedAllocateInfo structure in its `pNext` chain, VkMemoryDedicatedAllocateInfo::`buffer` **must** have been VK_NULL_HANDLE
- VUID-VkBindImageMemoryInfo-None-01901
  If image was created with the VK_IMAGE_CREATE_PROTECTED_BIT bit set, the image **must** be bound to a memory object allocated with a memory type that reports VK_MEMORY_PROPERTY_PROTECTED_BIT
- VUID-VkBindImageMemoryInfo-None-01902
  If image was created with the VK_IMAGE_CREATE_PROTECTED_BIT bit not set, the image **must** not be bound to a memory object created with a memory type that reports VK_MEMORY_PROPERTY_PROTECTED_BIT
- VUID-VkBindImageMemoryInfo-memory-02728
  If the value of VkExportMemoryAllocateInfo::`handleTypes` used to allocate `memory` is not `0`, it **must** include at least one of the handles set in VkExternalMemoryImageCreateInfo::`handleTypes` when `image` was created
- VUID-VkBindImageMemoryInfo-memory-02989
  If `memory` was created by a memory import operation, the external handle type of the imported memory **must** also have been set in VkExternalMemoryImageCreateInfo::`handleTypes` when `image` was created
- VUID-VkBindImageMemoryInfo-image-08113
  If the `image` was created with the VK_IMAGE_CREATE_DESCRIPTOR_HEAP_CAPTURE_REPLAY_BIT_EXT bit set, `memory` **must** have been allocated with the VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT bit set
- VUID-VkBindImageMemoryInfo-image-09202

If the image was created with the VK_IMAGE_CREATE_DESCRIPTOR_HEAP_CAPTURE_REPLAY_BIT_EXT bit set, memory **must** have been allocated with the VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT bit set

- VUID-VkBindImageMemoryInfo-pNext-01615
  If the pNext chain does not include a VkBindImagePlaneMemoryInfo structure, then memory **must** have been allocated using one of the memory types allowed in the memoryTypeBits member of the VkMemoryRequirements structure returned from a call to vkGetImageMemoryRequirements2 with image
- VUID-VkBindImageMemoryInfo-pNext-01616
  If the pNext chain does not include a VkBindImagePlaneMemoryInfo structure, then memoryOffset **must** be an integer multiple of the alignment member of the VkMemoryRequirements structure returned from a call to vkGetImageMemoryRequirements2 with image
- VUID-VkBindImageMemoryInfo-pNext-01617
  If the pNext chain does not include a VkBindImagePlaneMemoryInfo structure, then the difference of the size of memory and memoryOffset **must** be greater than or equal to the size member of the VkMemoryRequirements structure returned from a call to vkGetImageMemoryRequirements2 with the same image
- VUID-VkBindImageMemoryInfo-pNext-01618
  If the pNext chain includes a VkBindImagePlaneMemoryInfo structure, image **must** have been created with the VK_IMAGE_CREATE_DISJOINT_BIT bit set
- VUID-VkBindImageMemoryInfo-image-07736
  If image was created with the VK_IMAGE_CREATE_DISJOINT_BIT bit set, then the pNext chain **must** include a VkBindImagePlaneMemoryInfo structure
- VUID-VkBindImageMemoryInfo-pNext-01619
  If the pNext chain includes a VkBindImagePlaneMemoryInfo structure, memory **must** have been allocated using one of the memory types allowed in the memoryTypeBits member of the VkMemoryRequirements structure returned from a call to vkGetImageMemoryRequirements2 with image and where VkBindImagePlaneMemoryInfo::planeAspect corresponds to the VkImagePlaneMemoryRequirementsInfo::planeAspect in the VkImageMemoryRequirementsInfo2 structure's pNext chain
- VUID-VkBindImageMemoryInfo-pNext-01620
  If the pNext chain includes a VkBindImagePlaneMemoryInfo structure, memoryOffset **must** be an integer multiple of the alignment member of the VkMemoryRequirements structure returned from a call to vkGetImageMemoryRequirements2 with image and where VkBindImagePlaneMemoryInfo::planeAspect corresponds to the VkImagePlaneMemoryRequirementsInfo::planeAspect in the VkImageMemoryRequirementsInfo2 structure's pNext chain
- VUID-VkBindImageMemoryInfo-pNext-01621
  If the pNext chain includes a VkBindImagePlaneMemoryInfo structure, the difference of the size of memory and memoryOffset **must** be greater than or equal to the size member of the VkMemoryRequirements structure returned from a call to vkGetImageMemoryRequirements2 with the same image and where

VkBindImagePlaneMemoryInfo::planeAspect corresponds to the VkImagePlaneMemoryRequirementsInfo::planeAspect in the VkImageMemoryRequirementsInfo2 structure's pNext chain

- VUID-VkBindImageMemoryInfo-pNext-01626
  If the pNext chain includes a VkBindImageMemoryDeviceGroupInfo structure, all instances of memory specified by VkBindImageMemoryDeviceGroupInfo::pDeviceIndices **must** have been allocated
- VUID-VkBindImageMemoryInfo-pNext-01627
  If the pNext chain includes a VkBindImageMemoryDeviceGroupInfo structure, and VkBindImageMemoryDeviceGroupInfo::splitInstanceBindRegionCount is not zero, then image **must** have been created with the VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT bit set
- VUID-VkBindImageMemoryInfo-pNext-01628
  If the pNext chain includes a VkBindImageMemoryDeviceGroupInfo structure, all elements of VkBindImageMemoryDeviceGroupInfo::pSplitInstanceBindRegions **must** be valid rectangles contained within the dimensions of image
- VUID-VkBindImageMemoryInfo-pNext-01629
  If the pNext chain includes a VkBindImageMemoryDeviceGroupInfo structure, the union of the areas of all elements of VkBindImageMemoryDeviceGroupInfo ::pSplitInstanceBindRegions that correspond to the same instance of image **must** cover the entire image
- VUID-VkBindImageMemoryInfo-image-01630
  If image was created with a valid swapchain handle in VkImageSwapchainCreateInfoKHR::swapchain, then the pNext chain **must** include a VkBindImageMemorySwapchainInfoKHR structure containing the same swapchain handle
- VUID-VkBindImageMemoryInfo-pNext-01631
  If the pNext chain includes a VkBindImageMemorySwapchainInfoKHR structure, memory **must** be VK_NULL_HANDLE
- VUID-VkBindImageMemoryInfo-pNext-01632
  If the pNext chain does not include a VkBindImageMemorySwapchainInfoKHR structure, memory **must** be a valid VkDeviceMemory handle

Valid Usage (Implicit)

- VUID-VkBindImageMemoryInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_INFO
- VUID-VkBindImageMemoryInfo-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkBindImageMemoryDeviceGroupInfo, VkBindImageMemorySwapchainInfoKHR, VkBindImagePlaneMemoryInfo, or VkBindMemoryStatus
- VUID-VkBindImageMemoryInfo-sType-unique

The sType value of each structure in the pNext chain **must** be unique

- VUID-VkBindImageMemoryInfo-image-parameter
  image **must** be a valid VkImage handle
- VUID-VkBindImageMemoryInfo-commonparent
  Both of image, and memory that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to image **must** be externally synchronized

The VkBindImageMemoryDeviceGroupInfo structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkBindImageMemoryDeviceGroupInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           deviceIndexCount;
    const uint32_t*    pDeviceIndices;
    uint32_t           splitInstanceBindRegionCount;
    const VkRect2D*    pSplitInstanceBindRegions;
} VkBindImageMemoryDeviceGroupInfo;
```

```
// Provided by VK_KHR_bind_memory2 with VK_KHR_device_group
// Equivalent to VkBindImageMemoryDeviceGroupInfo
typedef VkBindImageMemoryDeviceGroupInfo VkBindImageMemoryDeviceGroupInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- deviceIndexCount is the number of elements in pDeviceIndices.
- pDeviceIndices is a pointer to an array of device indices.
- splitInstanceBindRegionCount is the number of elements in pSplitInstanceBindRegions.
- pSplitInstanceBindRegions is a pointer to an array of VkRect2D structures describing which regions of the image are attached to each instance of memory.

If the pNext chain of VkBindImageMemoryInfo includes a VkBindImageMemoryDeviceGroupInfo structure, then that structure determines how memory is bound to images across multiple devices in a device group.

If deviceIndexCount is greater than zero, then on device index i image is attached to the instance of the memory on the physical device with device index pDeviceIndices[i].

Let N be the number of physical devices in the logical device. If splitInstanceBindRegionCount is

greater than zero, then pSplitInstanceBindRegions is a pointer to an array of N^2 rectangles, where the image region specified by the rectangle at element i*N+j in resource instance i is bound to the memory instance j. The blocks of the memory that are bound to each sparse image block region use an offset in memory, relative to memoryOffset, computed as if the whole image was being bound to a contiguous range of memory. In other words, horizontally adjacent image blocks use consecutive blocks of memory, vertically adjacent image blocks are separated by the number of bytes per block multiplied by the width in blocks of image, and the block at (0,0) corresponds to memory starting at memoryOffset.

If splitInstanceBindRegionCount and deviceIndexCount are zero and the memory comes from a memory heap with the VK_MEMORY_HEAP_MULTI_INSTANCE_BIT bit set, then it is as if pDeviceIndices contains consecutive indices from zero to the number of physical devices in the logical device, minus one. In other words, by default each physical device attaches to its own instance of the memory.

If splitInstanceBindRegionCount and deviceIndexCount are zero and the memory comes from a memory heap without the VK_MEMORY_HEAP_MULTI_INSTANCE_BIT bit set, then it is as if pDeviceIndices contains an array of zeros. In other words, by default each physical device attaches to instance zero.

Valid Usage

- VUID-VkBindImageMemoryDeviceGroupInfo-deviceIndexCount-01633
  At least one of deviceIndexCount and splitInstanceBindRegionCount **must** be zero
- VUID-VkBindImageMemoryDeviceGroupInfo-deviceIndexCount-01634
  deviceIndexCount **must** either be zero or equal to the number of physical devices in the logical device
- VUID-VkBindImageMemoryDeviceGroupInfo-pDeviceIndices-01635
  All elements of pDeviceIndices **must** be valid device indices
- VUID-VkBindImageMemoryDeviceGroupInfo-splitInstanceBindRegionCount-01636
  splitInstanceBindRegionCount **must** either be zero or equal to the number of physical devices in the logical device squared
- VUID-VkBindImageMemoryDeviceGroupInfo-pSplitInstanceBindRegions-01637
  Elements of pSplitInstanceBindRegions that correspond to the same instance of an image **must** not overlap
- VUID-VkBindImageMemoryDeviceGroupInfo-offset-01638
  The offset.x member of any element of pSplitInstanceBindRegions **must** be a multiple of the sparse image block width (VkSparseImageFormatProperties::imageGranularity.width) of all non-metadata aspects of the image
- VUID-VkBindImageMemoryDeviceGroupInfo-offset-01639
  The offset.y member of any element of pSplitInstanceBindRegions **must** be a multiple of the sparse image block height (VkSparseImageFormatProperties::imageGranularity.height) of all non-metadata aspects of the image
- VUID-VkBindImageMemoryDeviceGroupInfo-extent-01640
  The extent.width member of any element of pSplitInstanceBindRegions **must** either be a

multiple of the sparse image block width of all non-metadata aspects of the image, or else `extent.width` + `offset.x` **must** equal the width of the image subresource

- VUID-VkBindImageMemoryDeviceGroupInfo-extent-01641
  The `extent.height` member of any element of `pSplitInstanceBindRegions` **must** either be a multiple of the sparse image block height of all non-metadata aspects of the image, or else `extent.height` + `offset.y` **must** equal the height of the image subresource

Valid Usage (Implicit)

- VUID-VkBindImageMemoryDeviceGroupInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_DEVICE_GROUP_INFO
- VUID-VkBindImageMemoryDeviceGroupInfo-pDeviceIndices-parameter
  If `deviceIndexCount` is not `0`, `pDeviceIndices` **must** be a valid pointer to an array of `deviceIndexCount` `uint32_t` values
- VUID-VkBindImageMemoryDeviceGroupInfo-pSplitInstanceBindRegions-parameter
  If `splitInstanceBindRegionCount` is not `0`, `pSplitInstanceBindRegions` **must** be a valid pointer to an array of `splitInstanceBindRegionCount` VkRect2D structures

Structure Chaining

Extends the structure

- VkBindImageMemoryInfo

If the `pNext` chain of VkBindImageMemoryInfo includes a `VkBindImageMemorySwapchainInfoKHR` structure, then that structure includes a swapchain handle and image index indicating that the image will be bound to memory from that swapchain.

The `VkBindImageMemorySwapchainInfoKHR` structure is defined as:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
typedef struct VkBindImageMemorySwapchainInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkSwapchainKHR     swapchain;
    uint32_t           imageIndex;
} VkBindImageMemorySwapchainInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `swapchain` is VK_NULL_HANDLE or a swapchain handle.
- `imageIndex` is an image index within `swapchain`.

If swapchain is not NULL, the swapchain and imageIndex are used to determine the memory that the image is bound to, instead of memory and memoryOffset.

Memory **can** be bound to a swapchain and use the pDeviceIndices or pSplitInstanceBindRegions members of VkBindImageMemoryDeviceGroupInfo.

Valid Usage

- VUID-VkBindImageMemorySwapchainInfoKHR-imageIndex-01644
  imageIndex **must** be less than the number of images in swapchain
- VUID-VkBindImageMemorySwapchainInfoKHR-swapchain-07756
  If the swapchain has been created with VK_SWAPCHAIN_CREATE_DEFERRED_MEMORY_ALLOCATION_BIT_KHR, imageIndex **must** be one that has previously been returned by vkAcquireNextImageKHR or vkAcquireNextImage2KHR

Valid Usage (Implicit)

- VUID-VkBindImageMemorySwapchainInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_SWAPCHAIN_INFO_KHR
- VUID-VkBindImageMemorySwapchainInfoKHR-swapchain-parameter
  swapchain **must** be a valid VkSwapchainKHR handle

Host Synchronization

- Host access to swapchain **must** be externally synchronized

Structure Chaining

Extends the structure

- VkBindImageMemoryInfo

In order to bind *planes* of a *disjoint image*, add a VkBindImagePlaneMemoryInfo structure to the pNext chain of VkBindImageMemoryInfo.

The VkBindImagePlaneMemoryInfo structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkBindImagePlaneMemoryInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkImageAspectFlagBits    planeAspect;
} VkBindImagePlaneMemoryInfo;
```

```
// Provided by VK_KHR_sampler_ycbcr_conversion
// Equivalent to VkBindImagePlaneMemoryInfo
typedef VkBindImagePlaneMemoryInfo VkBindImagePlaneMemoryInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `planeAspect` is a VkImageAspectFlagBits value specifying the aspect of the disjoint image plane to bind.

Valid Usage

- VUID-VkBindImagePlaneMemoryInfo-planeAspect-02283
  If the image's `tiling` is VK_IMAGE_TILING_LINEAR or VK_IMAGE_TILING_OPTIMAL, then `planeAspect` **must** be a single valid multi-planar aspect mask bit
- VUID-VkBindImagePlaneMemoryInfo-planeAspect-02284
  If the image's `tiling` is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then `planeAspect` **must** be a single valid *memory plane* for the image (that is, `planeAspect` **must** specify a plane index that is less than the VkDrmFormatModifierPropertiesEXT::`drmFormatModifierPlaneCount` associated with the image's `format` and VkImageDrmFormatModifierPropertiesEXT::`drmFormatModifier`)

Valid Usage (Implicit)

- VUID-VkBindImagePlaneMemoryInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BIND_IMAGE_PLANE_MEMORY_INFO
- VUID-VkBindImagePlaneMemoryInfo-planeAspect-parameter
  `planeAspect` **must** be a valid VkImageAspectFlagBits value

Structure Chaining

Extends the structure

- VkBindImageMemoryInfo

12.9.1. Buffer-Image Granularity

The implementation-dependent limit `bufferImageGranularity` specifies a page-like granularity at which linear and non-linear resources **must** be placed in adjacent memory locations to avoid aliasing. Two resources which do not satisfy this granularity requirement are said to alias. `bufferImageGranularity` is specified in bytes, and **must** be a power of two. Implementations which do not impose a granularity restriction **may** report a `bufferImageGranularity` value of one.

> **NOTE** Despite its name, `bufferImageGranularity` is really a granularity between "linear" and "non-linear" resources.

Given resourceA at the lower memory offset and resourceB at the higher memory offset in the same `VkDeviceMemory` object, where one resource is linear and the other is non-linear (as defined in the Glossary), and the following:

```
resourceA.end       = resourceA.memoryOffset + resourceA.size - 1
resourceA.endPage   = resourceA.end & ~(bufferImageGranularity-1)
resourceB.start     = resourceB.memoryOffset
resourceB.startPage = resourceB.start & ~(bufferImageGranularity-1)
```

The following property **must** hold:

```
resourceA.endPage < resourceB.startPage
```

That is, the end of the first resource (A) and the beginning of the second resource (B) **must** be on separate "pages" of size `bufferImageGranularity`. `bufferImageGranularity` **may** be different than the physical page size of the memory heap. This restriction is only needed when a linear resource and a non-linear resource are adjacent in memory and will be used simultaneously. The memory ranges of adjacent resources **can** be closer than `bufferImageGranularity`, provided they meet the `alignment` requirement for the objects in question.

Sparse block size in bytes and sparse image and buffer memory alignments **must** all be multiples of the `bufferImageGranularity`. Therefore, memory bound to sparse resources naturally satisfies the `bufferImageGranularity`.

12.10. Resource Sharing Mode

Buffer and image objects are created with a *sharing mode* controlling how they **can** be accessed from queues. The supported sharing modes are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSharingMode {
    VK_SHARING_MODE_EXCLUSIVE = 0,
    VK_SHARING_MODE_CONCURRENT = 1,
} VkSharingMode;
```

- VK_SHARING_MODE_EXCLUSIVE specifies that access to any range or image subresource of the object will be exclusive to a single queue family at a time.
- VK_SHARING_MODE_CONCURRENT specifies that concurrent access to any range or image subresource of the object from multiple queue families is supported.

NOTE	VK_SHARING_MODE_CONCURRENT **may** result in lower performance access to the buffer or image than VK_SHARING_MODE_EXCLUSIVE.

Ranges of buffers and image subresources of image objects created using VK_SHARING_MODE_EXCLUSIVE **must** only be accessed by queues in the queue family that has *ownership* of the resource. Upon creation, such resources are not owned by any queue family; ownership is implicitly acquired upon first use within a queue. Once a resource using VK_SHARING_MODE_EXCLUSIVE is owned by some queue family, unless the `maintenance9` feature is enabled, the application **must** perform a queue family ownership transfer if it wishes to make the memory contents of a range or image subresource accessible to a different queue family. VK_SHARING_MODE_EXCLUSIVE resources that are already owned by a queue family **may** be acquired by a different queue family without a queue family ownership transfer, but unless the `maintenance9` feature is enabled, their contents become undefined.

If the `maintenance9` feature is enabled, the contents of buffer resources, and of linear image resources (i.e., those created with `tiling` set to VK_IMAGE_TILING_LINEAR) are always preserved when they are implicitly acquired by a different queue family on the same logical device (i.e., neither queue family is VK_QUEUE_FAMILY_FOREIGN_EXT or VK_QUEUE_FAMILY_EXTERNAL). This means that whenever the `maintenance9` feature is enabled, explicit queue family ownership transfers of buffer and linear image resources between different queue families on the same logical device are **optional**.

Additionally, if the `maintenance9` feature is enabled, the contents of some optimal image resources (i.e., those created with VK_IMAGE_TILING_OPTIMAL) are always preserved when they are implicitly acquired by a different queue family on the same logical device (i.e., neither queue family is VK_QUEUE_FAMILY_FOREIGN_EXT or VK_QUEUE_FAMILY_EXTERNAL). This applies only to optimal images that are being implicitly acquired by a queue family whose index bit is set in the current queue family's VkQueueFamilyOwnershipTransferPropertiesKHR ::`optimalImageTransferToQueueFamilies`, and that were created without any of the following bits set in `usage`:

- VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT
- VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT
- VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT
- VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT
- VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT
- VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR

This means that whenever the `maintenance9` feature is enabled, explicit queue family ownership transfers of such image resources between such combinations of queue families are **optional**. For all other optimal images and/or combinations of queue families, the application **must** still perform

an explicit queue family ownership transfer if it wishes to make the memory contents of an optimal image subresource already owned by a queue family accessible to a different queue family.

NOTE

Applications are allowed to perform explicit queue family ownership transfers in circumstances where they are not required, but there is no functional nor performance advantage in doing so. Performing explicit transfers in such cases remains supported for backward compatibility and is not recommended for new applications.

NOTE

Before being used on the first queue, images still require a layout transition from these layouts:

- VK_IMAGE_LAYOUT_UNDEFINED
- VK_IMAGE_LAYOUT_PREINITIALIZED
- VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT

A queue family **can** take ownership of an image subresource or buffer range of a resource created with VK_SHARING_MODE_EXCLUSIVE, without an ownership transfer, in the same way as for a resource that was just created; however, taking ownership in this way has the effect that the contents of the image subresource or buffer range are undefined.

Ranges of buffers and image subresources of image objects created using VK_SHARING_MODE_CONCURRENT **must** only be accessed by queues from the queue families specified through the `queueFamilyIndexCount` and `pQueueFamilyIndices` members of the corresponding create info structures.

12.10.1. External Resource Sharing

Resources **should** only be accessed in the Vulkan instance that has exclusive ownership of their underlying memory. Only one Vulkan instance has exclusive ownership of a resource's underlying memory at a given time, regardless of whether the resource was created using VK_SHARING_MODE_EXCLUSIVE or VK_SHARING_MODE_CONCURRENT. Applications can transfer ownership of a resource's underlying memory only if the memory has been imported from or exported to another instance or external API using external memory handles. The semantics for transferring ownership outside of the instance are similar to those used for transferring ownership of VK_SHARING_MODE_EXCLUSIVE resources between queues, and is also accomplished using VkBufferMemoryBarrier or VkImageMemoryBarrier operations. To make the contents of the underlying memory accessible in the destination instance or API, applications **must**

1. Release exclusive ownership from the source instance or API.
2. Ensure the release operation has completed using semaphores or fences.
3. Acquire exclusive ownership in the destination instance or API.

Unlike queue family ownership transfers, the destination instance or API is not specified explicitly when releasing ownership, nor is the source instance or API specified when acquiring ownership. Instead, the image or memory barrier's `dstQueueFamilyIndex` or `srcQueueFamilyIndex` parameters are set to the reserved queue family index VK_QUEUE_FAMILY_EXTERNAL or

VK_QUEUE_FAMILY_FOREIGN_EXT to represent the external destination or source respectively.

Binding a resource to a memory object shared between multiple Vulkan instances or other APIs does not change the ownership of the underlying memory. The first entity to access the resource implicitly acquires ownership. An entity **can** also implicitly take ownership from another entity in the same way without an explicit ownership transfer. However, taking ownership in this way has the effect that the contents of the underlying memory are undefined.

Accessing a resource backed by memory that is owned by a particular instance or API has the same semantics as accessing a VK_SHARING_MODE_EXCLUSIVE resource, with one exception: Implementations **must** ensure layout transitions performed on one member of a set of identical subresources of identical images that alias the same range of an underlying memory object affect the layout of all the subresources in the set.

As a corollary, writes to any image subresources in such a set **must** not make the contents of memory used by other subresources in the set undefined. An application **can** define the content of a subresource of one image by performing device writes to an identical subresource of another image provided both images are bound to the same region of external memory. Applications **may** also add resources to such a set after the content of the existing set members has been defined without making the content undefined by creating a new image with the initial layout VK_IMAGE_LAYOUT_UNDEFINED and binding it to the same region of external memory as the existing images.

NOTE

Because layout transitions apply to all identical images aliasing the same region of external memory, the actual layout of the memory backing a new image as well as an existing image with defined content will not be undefined. Such an image is not usable until it acquires ownership of its memory from the existing owner. Therefore, the layout specified as part of this transition will be the true initial layout of the image. The undefined layout specified when creating it is a placeholder to simplify valid usage requirements.

When sharing resources with other APIs, it is generally left to those APIs to define how they interact with the Vulkan concepts of image layouts and fine-grained synchronization. However, some external memory handle types require use of specific image layouts when transitioning to or from external usage, as defined in the external image implied layouts table.

12.11. Memory Aliasing

A range of a `VkDeviceMemory` allocation is *aliased* if it is bound to multiple resources simultaneously, as described below, via vkBindImageMemory, vkBindBufferMemory, via sparse memory bindings, or by binding the memory to resources in multiple Vulkan instances or external APIs using external memory handle export and import mechanisms.

Consider two resources, resource_A and resource_B, bound respectively to memory range_A and range_B. Let paddedRange_A and paddedRange_B be, respectively, range_A and range_B aligned to `bufferImageGranularity`. If the resources are both linear or both non-linear (as defined in the Glossary), then the resources *alias* the memory in the intersection of range_A and range_B. If one resource is linear and the other is non-linear, then the resources *alias* the memory in the

intersection of paddedRange_A and paddedRange_B.

Applications **can** alias memory, but use of multiple aliases is subject to several constraints.

NOTE	Memory aliasing **can** be useful to reduce the total device memory footprint of an application, if some large resources are used for disjoint periods of time.

When a non-linear, non-VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT image is bound to an aliased range, all image subresources of the image *overlap* the range. When a linear image is bound to an aliased range, the image subresources that (according to the image's advertised layout) include bytes from the aliased range overlap the range. When a VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT image has sparse image blocks bound to an aliased range, only image subresources including those sparse image blocks overlap the range, and when the memory bound to the image's mip tail overlaps an aliased range all image subresources in the mip tail overlap the range.

Buffers, and linear image subresources are considered *host-accessible subresources* when they are in any of these layouts:

- VK_IMAGE_LAYOUT_PREINITIALIZED
- VK_IMAGE_LAYOUT_GENERAL
- VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT

That is, the host has a well-defined addressing scheme to interpret the contents, and thus the layout of the data in memory **can** be consistently interpreted across aliases if each of those aliases is a host-accessible subresource. Non-linear images, and linear image subresources in other layouts, are not host-accessible.

If two aliases are both host-accessible, then they interpret the contents of the memory in consistent ways, and data written to one alias **can** be read by the other alias.

For an acceleration structure `AS_2` that is an alias of another acceleration structure `AS_1`, `AS_2` **can** be used in place of `AS_1` for operations acting on acceleration structures if the following conditions are met:

- The buffer referred to by the `buffer` member of the VkAccelerationStructureCreateInfoKHR that `AS_2` was created with **must** be bound to the same VkDeviceMemory as the buffer referred to by the `buffer` member of the VkAccelerationStructureCreateInfoKHR that `AS_1` was created with.
- The start of the memory range occupied by `AS_2`, defined by the sum of the `offset` member of the VkAccelerationStructureCreateInfoKHR that `AS_2` was created with and the offset at which the buffer associated with `AS_2` is bound to a VkDeviceMemory, **must** match the start of the memory range occupied by `AS_1`.
- If the `deviceAddress` member of the VkAccelerationStructureCreateInfoKHR that `AS_2` was created with is non-zero, it **must** match the `deviceAddress` member of the VkAccelerationStructureCreateInfoKHR that `AS_1` was created with.
- The `createFlags` member of the VkAccelerationStructureCreateInfoKHR that `AS_2` was created with **must** match the `createFlags` member of the VkAccelerationStructureCreateInfoKHR that

`AS_1` was created with.

- The `type` member of the VkAccelerationStructureCreateInfoKHR that `AS_2` was created with **must** match the `type` member of the VkAccelerationStructureCreateInfoKHR that `AS_1` was created with.
- The `size` member of the VkAccelerationStructureCreateInfoKHR that `AS_2` was created with **must** be greater than or equal to the `size` returned by vkGetAccelerationStructureBuildSizesKHR for the build parameters `AS_1` was built with.

After an acceleration structure object is destroyed, aliased acceleration structures **may** continue being used to refer to that acceleration structure for operations acting on acceleration structures.

If two aliases are both images that were created with identical creation parameters, both were created with the VK_IMAGE_CREATE_ALIAS_BIT flag set, and both are bound identically to memory except for VkBindImageMemoryDeviceGroupInfo::`pDeviceIndices` and VkBindImageMemoryDeviceGroupInfo::`pSplitInstanceBindRegions`, then they interpret the contents of the memory in consistent ways, and data written to one alias **can** be read by the other alias.

Additionally, if an individual plane of a multi-planar image and a single-plane image alias the same memory, then they also interpret the contents of the memory in consistent ways under the same conditions, but with the following modifications:

- Both **must** have been created with the VK_IMAGE_CREATE_DISJOINT_BIT flag.
- The single-plane image **must** have a VkFormat that is equivalent to that of the multi-planar image's individual plane.
- The single-plane image and the individual plane of the multi-planar image **must** be bound identically to memory except for VkBindImageMemoryDeviceGroupInfo::`pDeviceIndices` and VkBindImageMemoryDeviceGroupInfo::`pSplitInstanceBindRegions`.
- The `width` and `height` of the single-plane image are derived from the multi-planar image's dimensions in the manner listed for plane compatibility for the aliased plane.
- If either image's `tiling` is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then both images **must** be linear.
- All other creation parameters **must** be identical.

Aliases created by binding the same memory to resources in multiple Vulkan instances or external APIs using external memory handle export and import mechanisms interpret the contents of the memory in consistent ways, and data written to one alias **can** be read by the other alias.

Otherwise, the aliases interpret the contents of the memory differently, and writes via one alias make the contents of memory partially or completely undefined to the other alias. If the first alias is a host-accessible subresource, then the bytes affected are those written by the memory operations according to its addressing scheme. If the first alias is not host-accessible, then the bytes affected are those overlapped by the image subresources that were written. If the second alias is a host-accessible subresource, the affected bytes become undefined. If the second alias is not host-accessible, all sparse image blocks (for sparse partially-resident images) or all image subresources (for non-sparse image and fully resident sparse images) that overlap the affected bytes become undefined.

If any image subresources are made undefined due to writes to an alias, then each of those image subresources **must** have its layout transitioned from VK_IMAGE_LAYOUT_UNDEFINED to a valid layout before it is used, from VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT if it is zeroed, or from VK_IMAGE_LAYOUT_PREINITIALIZED if the memory has been written by the host. If any sparse blocks of a sparse image have been made undefined, then only the image subresources containing them **must** be transitioned.

Use of an overlapping range by two aliases **must** be separated by a memory dependency using the appropriate access types if at least one of those uses performs writes, whether the aliases interpret memory consistently or not. If buffer or image memory barriers are used, the scope of the barrier **must** contain the entire range and/or set of image subresources that overlap.

If two aliasing image views are used in the same framebuffer, then the render pass **must** declare the attachments using the VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT, and follow the other rules listed in that section.

NOTE	Memory recycled via an application suballocator (i.e. without freeing and reallocating the memory objects) is not substantially different from memory aliasing. However, a suballocator usually waits on a fence before recycling a region of memory, and signaling a fence involves sufficient implicit dependencies to satisfy all the above requirements.

12.11.1. Resource Memory Overlap

Applications **can** safely access resources concurrently via separate device and host operations as long as the accessed memory locations are guaranteed to not overlap, as defined in Memory Location, and the operation, resource, and access are otherwise independently valid.

Some operations have alignment requirements or access ambiguous memory locations, so the semantics of a particular operation **should** be considered when determining the overlap. Such requirements will be described alongside the operation. Operations between host and device when using non-coherent memory are aligned to `nonCoherentAtomSize`, as defined by vkFlushMappedMemoryRanges and vkInvalidateMappedMemoryRanges.

NOTE	The intent is that buffers (or linear images) can be accessed concurrently, even when they share cache lines, but otherwise do not access the same memory range. The concept of a device cache line size is not exposed in the memory model.

Chapter 13. Images

Images are specialized resources that have multi-dimensional access rather than the typical linear access to memory. These resources allow implementations to optimize the memory layout for common access patterns, by mapping multi-dimensional coordinates to an implementation-dependent offset in the underlying memory.

Additionally, images are homogeneous, with every discrete coordinate associated with data that is in the same format as the data associated with any other coordinate. Each set of data associated with one of these coordinates is referred to in this specification as a *texel*. A texel can consist of up to 4 separate components, labeled as (R,G,B,A) in this chapter. Each texel block in an image maps to a distinct set of memory locations within the memory they are bound to. Unless otherwise noted by the specification, the mapping from texels to memory is implementation-defined.

NOTE

This terminology is historical; texel is a combination of the words "**tex**ture" and "**el**ement", and (R,G,B,A) are abbreviations for **R**ed, **G**reen, **B**lue, and **A**lpha. Early interactive computer graphics only supported operations that allowed image data to be used to add color "texture" to rendered objects, which is just a small subset of what the image operations on images described here enable. These terms are deeply embedded in the industry, and so are still used here by convention, despite images being used in far more varied ways than they were in the past.

Some image formats identify components as D for **d**epth, S for **s**tencil, or X for padding elements. Image format components identified as D are treated as R, and S components are treated as G for the purpose of image accesses. X components are ignored when reading, and **may** be modified in implementation-dependent ways when writing to that texel.

The coordinates used to identify a texel are six-dimensional, made up of the following integer indices:

- x - The first spatial index
- y - The second spatial index
- z - The third spatial index
- layer - The array index for arrayed images
- sample - The sample index
- level - The detail level

Each image is constructed with a number of texels in each dimension, with the integer size in each dimension for that image identified as:

- width - The number of x indices
- height - The number of y indices
- depth - The number of z indices
- layers - The number of layers
- samples - The number of samples

- levels - The number of detail levels

Of these sizes, the number of levels is somewhat unique - each further level reduces the number of indices in each of the x, y, and z dimensions by half, according to these formulae:

- $width_{level}$ = max(1, ⌊width / 2^{level}⌋)
- $height_{level}$ = max(1, ⌊height / 2^{level}⌋)
- $depth_{level}$ = max(1, ⌊depth / 2^{level}⌋)

An image will have no more than one level where all of the above dimensions are set to 1. See Image Mip Level Sizing for more details.

The Vulkan specification allows the creation of resources with fewer dimensions than this (e.g. texel buffers); these can be considered equivalent to an image with all dimensions specified, with the missing dimensions having a size of 1, and an implicit coordinate value of 0.

These parameters are set for an image during image creation, and are adjusted when accessing an image through an image view.

13.1. Image Coordinate Validation

When accessing an image, a set of (x,y,z,layer,sample,level) coordinates are used to indicate which texel is accessed. These coordinates are first checked to see if they refer to texels within the image dimensions, according to the following equations:

- x < $width_{level}$
- y < $height_{level}$
- z < $depth_{level}$
- layer < layers
- sample < samples
- level < levels

If any of these equations evaluates to false, the coordinate is considered out of bounds, otherwise they are *in bounds*.

13.2. Image Reads

Image reads use a set of (x,y,z,layer,sample,level) coordinates, validated as per Image Coordinate Validation, and return a converted value for the texel at that coordinate. If the coordinates are out of bounds, behavior of the read is as described in Shader Out-of-Bounds Memory Access. If the coordinates are in bounds, but the texel is not backed by memory, behavior of the read is as described in Accessing Unbound Regions. Otherwise, the read proceeds as follows.

13.2.1. Texel Decode

The formatted value of the texel at the (x,y,z,layer,sample,level) coordinate is read and decoded

according to the procedures outlined in the Khronos Data Format Specification.

For sRGB formats, the (R,G,B,A) components are first converted as if they are UNORM formats, and then sRGB to linear conversion is performed on the converted (R,G,B) components, as described in the “sRGB EOTF” section of the Khronos Data Format Specification.

13.2.2. Component Substitution

If after conversion, less than four of the (R,G,B,A) components are present, missing components are substituted by the components of (0,0,0,1) for missing (R,G,B,A) components, respectively.

13.2.3. Numeric Encoding

The values are encoded according to the bit width and numeric format of each component:

- Components with a fixed-point numeric format or with a floating-point numeric format and a bit width less than or equal to 32 are encoded into the IEEE-754 binary32 format.
- Components with a floating-point numeric format and a bit width greater than 32 are encoded into the IEEE-754 binary64 format.
- Components with an integer numeric format and a bit width less than or equal to 32 are directly encoded as 32-bit integer values with the same signedness.
- Components with an integer numeric format and a bit width greater than 32 are directly encoded as 64-bit integer values with the same signedness.

These values are then returned as the result of the image read.

13.3. Image Writes

Image writes also use a set of (x,y,z,layer,sample,level) coordinates, validated as per Image Coordinate Validation, and with a value to be written. If the coordinates are out of bounds, the write is discarded. If the coordinates are in bounds, but the texel is not backed by memory, the write is silently discarded. Otherwise, an image write is performed as follows:

13.3.1. Texel Encode

If the image format is sRGB, a linear to sRGB conversion is applied to the (R,G,B) components of value as described in the “sRGB EOTF” section of the Khronos Data Format Specification.

The converted (R,G,B) and original A values are then encoded to the image format according to the procedures outlined in the Khronos Data Format Specification. Components not present in the image format are discarded.

The final value is then written to the texel at (x,y,z,layer,sample,level).

13.4. SPIR-V Image Access

Images are represented in SPIR-V by variables with the OpTypeImage type, and operated on by

image instructions.

Images are mapped to a shader using image views, which define access to a subset of the base image resource, specified by VkImageSubresourceRange.

13.4.1. SPIR-V Image Read Instructions

Texels **can** be read from an image using OpImageRead or OpImageSparseRead. The elements of the *Coordinate* operand and the optional `Sample` image operand map to image coordinates, modified by the parameters of the image view used, as:

- x = `u`
- y = `v`
- z = `w`
- layer = `array_layer`
- sample = `Sample`
- level = 0

Any index missing in the instruction is implicitly set to 0.

After the coordinates are validated, some indices are offset before performing the access as follows:

- layer += VkImageSubresourceRange::`baseArrayLayer`
- level += VkImageSubresourceRange::`baseMipLevel`
- z += VkImageViewSlicedCreateInfoEXT::`sliceOffset`

13.4.2. SPIR-V Image Write Instructions

Texels **can** be written to an image using OpImageWrite. The elements of the *Coordinate* operand and the optional `Sample` image operand map to image coordinates, modified by the parameters of the image view used, in the same ways as for image reads.

13.4.3. SPIR-V Image Atomic Instructions

SPIR-V atomic instructions operate on pointers rather than texels, but **can** be used with texels via OpImageTexelPointer. OpImageTexelPointer **can** be used to obtain a valid pointer to a specific texel in an image. The elements of the *Coordinate* operand and the optional `Sample` image operand map to image coordinates, modified by the parameters of the image view used, in the same ways as for image reads. The resulting pointer **can** be used to access the identified texel, with atomic operations potentially reading and writing to that texel.

13.4.4. SPIR-V Image Query Instructions

Query operations return properties of an image, allowing the accessible size of each dimension of an image to be queried from within a shader. Query operations do not access any texels in the image.

- `OpImageQuerySizeLod` returns the values of $width_{level}$, $height_{level}$, $depth_{level}$, and layers for the image at a specified level, where level is equal to the sum of the `Level of Detail` operand and VkImageSubresourceRange::`baseMipLevel`. If the image is not a sampled image and VkImageViewSlicedCreateInfoEXT::`sliceCount` is not set to VK_REMAINING_3D_SLICES_EXT, `sliceCount` will be returned in place of $depth_{level}$. If the image is not a sampled image and VkImageViewSlicedCreateInfoEXT::`sliceCount` is set to VK_REMAINING_3D_SLICES_EXT, the value returned will be equal to $depth_{level}$:minus: VkImageViewSlicedCreateInfoEXT::`sliceOffset`. If `Level of Detail` is greater than or equal to VkImageSubresourceRange::`levelCount`, the result is undefined.
- `OpImageQuerySize` is equivalent to `OpImageQuerySizeLod` with level set to VkImageSubresourceRange::`baseMipLevel`.
- `OpImageQueryLevels` returns the value of VkImageSubresourceRange::`levelCount` specified for the image view used.
- `OpImageQuerySamples` returns the value of samples for the image.

> **NOTE** `OpImageQueryLod` is performed as a sampling operation, so is not included here.

If the descriptor bound to the image is a null descriptor, all values returned by the above queries are replaced with 0.

13.4.5. SPIR-V Image Sampling Instructions

Sampling operations **can** also be performed on images, typically in conjunction with a sampler, as defined in the Sampling chapter.

Chapter 14. Samplers

VkSampler objects represent the state of an image sampler which is used by the implementation to read image data and apply filtering and other transformations for the shader.

Samplers are represented by VkSampler handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkSampler)
```

To create a sampler object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateSampler(
    VkDevice                                    device,
    const VkSamplerCreateInfo*                  pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkSampler*                                  pSampler);
```

- device is the logical device that creates the sampler.
- pCreateInfo is a pointer to a VkSamplerCreateInfo structure specifying the state of the sampler object.
- pAllocator controls host memory allocation as described in the Memory Allocation chapter.
- pSampler is a pointer to a VkSampler handle in which the resulting sampler object is returned.

Valid Usage

- VUID-vkCreateSampler-device-09668
  device **must** support at least one queue family with one of the VK_QUEUE_COMPUTE_BIT or VK_QUEUE_GRAPHICS_BIT capabilities
- VUID-vkCreateSampler-maxSamplerAllocationCount-04110
  There **must** be less than VkPhysicalDeviceLimits::maxSamplerAllocationCount VkSampler objects currently created on the device
- VUID-vkCreateSampler-maxSamplerAllocationCount-11412
  If there are any pipelines or shaders with embedded samplers currently created on the device, there **must** be less than (maxSamplerAllocationCount - (minSamplerHeapReservedRangeWithEmbedded / samplerDescriptorSize)) VkSampler objects currently created on the device

Valid Usage (Implicit)

- VUID-vkCreateSampler-device-parameter
  device **must** be a valid VkDevice handle

- VUID-vkCreateSampler-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkSamplerCreateInfo structure
- VUID-vkCreateSampler-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateSampler-pSampler-parameter
  `pSampler` **must** be a valid pointer to a VkSampler handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkSamplerCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSamplerCreateInfo {
    VkStructureType         sType;
    const void*             pNext;
    VkSamplerCreateFlags    flags;
    VkFilter                magFilter;
    VkFilter                minFilter;
    VkSamplerMipmapMode     mipmapMode;
    VkSamplerAddressMode    addressModeU;
    VkSamplerAddressMode    addressModeV;
    VkSamplerAddressMode    addressModeW;
    float                   mipLodBias;
    VkBool32                anisotropyEnable;
    float                   maxAnisotropy;
    VkBool32                compareEnable;
    VkCompareOp             compareOp;
    float                   minLod;
    float                   maxLod;
    VkBorderColor           borderColor;
    VkBool32                unnormalizedCoordinates;
} VkSamplerCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkSamplerCreateFlagBits describing additional parameters of the sampler.
- `magFilter` is a VkFilter value specifying the magnification filter to apply to lookups.
- `minFilter` is a VkFilter value specifying the minification filter to apply to lookups.
- `mipmapMode` is a VkSamplerMipmapMode value specifying the mipmap filter to apply to lookups.
- `addressModeU` is a VkSamplerAddressMode value specifying the wrapping operation used when the i coordinate used to sample the image would be out of bounds.
- `addressModeV` is a VkSamplerAddressMode value specifying the wrapping operation used when the j coordinate used to sample the image would be out of bounds.
- `addressModeW` is a VkSamplerAddressMode value specifying the wrapping operation used when the k coordinate used to sample the image would be out of bounds. If `unnormalizedCoordinates` is VK_TRUE, `addressModeW` is ignored.
- `mipLodBias` is the bias to be added to mipmap LOD calculation and bias provided by image sampling functions in SPIR-V, as described in the LOD Operation section.
- `anisotropyEnable` is VK_TRUE to enable anisotropic filtering, as described in the Texel Anisotropic Filtering section, or VK_FALSE otherwise.
- `maxAnisotropy` is the anisotropy value clamp used by the sampler when `anisotropyEnable` is VK_TRUE. If `anisotropyEnable` is VK_FALSE, `maxAnisotropy` is ignored.
- `compareEnable` is VK_TRUE to enable comparison against a reference value during lookups, or VK_FALSE otherwise.
  - Note: Some implementations will default to shader state if this member does not match.
- `compareOp` is a VkCompareOp value specifying the comparison operator to apply to fetched data before filtering as described in the Depth Compare Operation section.
- `minLod` is used to clamp the minimum of the computed LOD value.
- `maxLod` is used to clamp the maximum of the computed LOD value. To avoid clamping the maximum value, set `maxLod` to the constant VK_LOD_CLAMP_NONE.
- `borderColor` is a VkBorderColor value specifying the predefined border color to use.
- `unnormalizedCoordinates` controls whether to use unnormalized or normalized texel coordinates to address texels of the image. When `unnormalizedCoordinates` is VK_TRUE, the range of the image coordinates used to lookup the texel is in the range of zero to the image size in each dimension. When `unnormalizedCoordinates` is VK_FALSE, the range of image coordinates is zero to one.

  When `unnormalizedCoordinates` is VK_TRUE, images the sampler is used with in the shader have the following requirements:

  - The `viewType` **must** be either VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D.
  - The image view **must** have a single layer and a single mip level.

    When `unnormalizedCoordinates` is VK_TRUE, image built-in functions in the shader that use

the sampler have the following requirements:

- The functions **must** not use projection.
- The functions **must** not use offsets.

Mapping of OpenGL to Vulkan Filter Modes

`magFilter` values of VK_FILTER_NEAREST and VK_FILTER_LINEAR directly correspond to `GL_NEAREST` and `GL_LINEAR` magnification filters. `minFilter` and `mipmapMode` combine to correspond to the similarly named OpenGL minification filter of `GL_minFilter_MIPMAP_mipmapMode` (e.g. `minFilter` of VK_FILTER_LINEAR and `mipmapMode` of VK_SAMPLER_MIPMAP_MODE_NEAREST correspond to `GL_LINEAR_MIPMAP_NEAREST`).

NOTE

There are no Vulkan filter modes that directly correspond to OpenGL minification filters of `GL_LINEAR` or `GL_NEAREST`, but they **can** be emulated using VK_SAMPLER_MIPMAP_MODE_NEAREST, `minLod` = 0, and `maxLod` = 0.25, and using `minFilter` = VK_FILTER_LINEAR or `minFilter` = VK_FILTER_NEAREST, respectively.

Note that using a `maxLod` of zero would cause magnification to always be performed, and the `magFilter` to always be used. This is valid, just not an exact match for OpenGL behavior. Clamping the maximum LOD to 0.25 allows the λ value to be non-zero and minification to be performed, while still always rounding down to the base level. If the `minFilter` and `magFilter` are equal, then using a `maxLod` of zero also works.

The maximum number of sampler objects which **can** be simultaneously created on a device is implementation-dependent and specified by the `maxSamplerAllocationCount` member of the VkPhysicalDeviceLimits structure.

NOTE

For historical reasons, if `maxSamplerAllocationCount` is exceeded, some implementations may return VK_ERROR_TOO_MANY_OBJECTS. Exceeding this limit will result in undefined behavior, and an application should not rely on the use of the returned error code in order to identify when the limit is reached.

Since VkSampler is a non-dispatchable handle type, implementations **may** return the same handle for sampler state vectors that are identical. In such cases, all such objects would only count once against the `maxSamplerAllocationCount` limit.

When this structure is used to write a descriptor via vkWriteSamplerDescriptorsEXT, applications **can** give the descriptor a debug name in a similar way to naming an object, via the VkDebugUtilsObjectNameInfoEXT structure. However, as there is no actual object, VkDebugUtilsObjectNameInfoEXT **must** be passed via the `pNext` chain of this structure, with a `objectType` of VK_OBJECT_TYPE_UNKNOWN and a `objectHandle` of VK_NULL_HANDLE. The name is attached to the unique set of descriptor bits written by the implementation, and writing the same bits again with new debug info **may** rename the original descriptor.

NOTE

Implementations are not prevented from returning the same bits for different descriptors. This can result in multiple different samplers mapping to the same

name.

VkDebugUtilsObjectNameInfoEXT **can** also be chained in the same way when defining an embedded sampler via VkDescriptorSetAndBindingMappingEXT, naming the embedded sampler.

Valid Usage

- VUID-VkSamplerCreateInfo-mipLodBias-01069
  The absolute value of `mipLodBias` **must** be less than or equal to `VkPhysicalDeviceLimits::maxSamplerLodBias`
- VUID-VkSamplerCreateInfo-samplerMipLodBias-04467
  If the `VK_KHR_portability_subset` extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::`samplerMipLodBias` is VK_FALSE, `mipLodBias` **must** be zero
- VUID-VkSamplerCreateInfo-maxLod-01973
  `maxLod` **must** be greater than or equal to `minLod`
- VUID-VkSamplerCreateInfo-anisotropyEnable-01070
  If the `samplerAnisotropy` feature is not enabled, `anisotropyEnable` **must** be VK_FALSE
- VUID-VkSamplerCreateInfo-anisotropyEnable-01071
  If `anisotropyEnable` is VK_TRUE, `maxAnisotropy` **must** be between `1.0` and `VkPhysicalDeviceLimits::maxSamplerAnisotropy`, inclusive
- VUID-VkSamplerCreateInfo-minFilter-01645
  If sampler $Y'C_BC_R$ conversion is enabled and the potential format features of the sampler $Y'C_BC_R$ conversion do not support VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT, `minFilter` and `magFilter` **must** be equal to the sampler $Y'C_BC_R$ conversion's `chromaFilter`
- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01072
  If `unnormalizedCoordinates` is VK_TRUE, `minFilter` and `magFilter` **must** be equal
- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01073
  If `unnormalizedCoordinates` is VK_TRUE, `mipmapMode` **must** be VK_SAMPLER_MIPMAP_MODE_NEAREST
- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01074
  If `unnormalizedCoordinates` is VK_TRUE, `minLod` and `maxLod` **must** be zero
- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01075
  If `unnormalizedCoordinates` is VK_TRUE, `addressModeU` and `addressModeV` **must** each be either VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE or VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_BORDER
- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01076
  If `unnormalizedCoordinates` is VK_TRUE, `anisotropyEnable` **must** be VK_FALSE
- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01077
  If `unnormalizedCoordinates` is VK_TRUE, `compareEnable` **must** be VK_FALSE
- VUID-VkSamplerCreateInfo-addressModeU-01078

If any of addressModeU, addressModeV or addressModeW are VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_BORDER, borderColor **must** be a valid VkBorderColor value

- VUID-VkSamplerCreateInfo-addressModeU-01646
  If sampler $Y'C_BC_R$ conversion is enabled, addressModeU, addressModeV, and addressModeW **must** be VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE, anisotropyEnable **must** be VK_FALSE, and unnormalizedCoordinates **must** be VK_FALSE
- VUID-VkSamplerCreateInfo-None-01647
  If sampler $Y'C_BC_R$ conversion is enabled and the pNext chain includes a VkSamplerReductionModeCreateInfo structure, then the sampler reduction mode **must** be VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE
- VUID-VkSamplerCreateInfo-pNext-06726
  If the samplerFilterMinmax feature is not enabled and the pNext chain includes a VkSamplerReductionModeCreateInfo structure, then the sampler reduction mode **must** be VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE
- VUID-VkSamplerCreateInfo-addressModeU-01079
  If the samplerMirrorClampToEdge feature is not enabled, and if the VK_KHR_sampler_mirror_clamp_to_edge extension is not enabled, addressModeU, addressModeV and addressModeW **must** not be VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE
- VUID-VkSamplerCreateInfo-compareEnable-01080
  If compareEnable is VK_TRUE, compareOp **must** be a valid VkCompareOp value
- VUID-VkSamplerCreateInfo-magFilter-01081
  If either magFilter or minFilter is VK_FILTER_CUBIC_EXT, anisotropyEnable **must** be VK_FALSE
- VUID-VkSamplerCreateInfo-compareEnable-01423
  If compareEnable is VK_TRUE, the reductionMode member of VkSamplerReductionModeCreateInfo **must** be VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE
- VUID-VkSamplerCreateInfo-flags-02574
  If flags includes VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT, then minFilter and magFilter **must** be equal
- VUID-VkSamplerCreateInfo-flags-02575
  If flags includes VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT, then mipmapMode **must** be VK_SAMPLER_MIPMAP_MODE_NEAREST
- VUID-VkSamplerCreateInfo-flags-02576
  If flags includes VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT, then minLod and maxLod **must** be zero
- VUID-VkSamplerCreateInfo-flags-02577
  If flags includes VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT, then addressModeU and addressModeV **must** each be either VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE or VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_BORDER
- VUID-VkSamplerCreateInfo-flags-02578
  If flags includes VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT, then anisotropyEnable

must be VK_FALSE

- VUID-VkSamplerCreateInfo-flags-02579
  If `flags` includes VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT, then `compareEnable` **must** be VK_FALSE
- VUID-VkSamplerCreateInfo-flags-02580
  If `flags` includes VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT, then `unnormalizedCoordinates` **must** be VK_FALSE
- VUID-VkSamplerCreateInfo-nonSeamlessCubeMap-06788
  If the `nonSeamlessCubeMap` feature is not enabled, `flags` **must** not include VK_SAMPLER_CREATE_NON_SEAMLESS_CUBE_MAP_BIT_EXT
- VUID-VkSamplerCreateInfo-borderColor-04011
  If `borderColor` is one of VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT, then a VkSamplerCustomBorderColorCreateInfoEXT **must** be included in the `pNext` chain
- VUID-VkSamplerCreateInfo-customBorderColors-04085
  If the `customBorderColors` feature is not enabled, `borderColor` **must** not be VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT
- VUID-VkSamplerCreateInfo-borderColor-04442
  If `borderColor` is one of VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT, and VkSamplerCustomBorderColorCreateInfoEXT::`format` is not VK_FORMAT_UNDEFINED, VkSamplerCustomBorderColorCreateInfoEXT::`customBorderColor` **must** be within the range of values representable in `format`
- VUID-VkSamplerCreateInfo-None-04012
  The maximum number of samplers with custom border colors which **can** be simultaneously created on a device is implementation-dependent and specified by the `maxCustomBorderColorSamplers` member of the VkPhysicalDeviceCustomBorderColorPropertiesEXT structure
- VUID-VkSamplerCreateInfo-flags-08110
  If `flags` includes VK_SAMPLER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT, the `descriptorBufferCaptureReplay` feature **must** be enabled
- VUID-VkSamplerCreateInfo-pNext-08111
  If the `pNext` chain includes a VkOpaqueCaptureDescriptorDataCreateInfoEXT structure, `flags` **must** contain VK_SAMPLER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT

Valid Usage (Implicit)

- VUID-VkSamplerCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SAMPLER_CREATE_INFO
- VUID-VkSamplerCreateInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either

`NULL` or a pointer to a valid instance of VkDebugUtilsObjectNameInfoEXT, VkOpaqueCaptureDescriptorDataCreateInfoEXT, VkSamplerBorderColorComponentMappingCreateInfoEXT, VkSamplerCustomBorderColorCreateInfoEXT, VkSamplerCustomBorderColorIndexCreateInfoEXT, VkSamplerReductionModeCreateInfo, or VkSamplerYcbcrConversionInfo

- VUID-VkSamplerCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkSamplerCreateInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkSamplerCreateFlagBits values
- VUID-VkSamplerCreateInfo-magFilter-parameter
  `magFilter` **must** be a valid VkFilter value
- VUID-VkSamplerCreateInfo-minFilter-parameter
  `minFilter` **must** be a valid VkFilter value
- VUID-VkSamplerCreateInfo-mipmapMode-parameter
  `mipmapMode` **must** be a valid VkSamplerMipmapMode value
- VUID-VkSamplerCreateInfo-addressModeU-parameter
  `addressModeU` **must** be a valid VkSamplerAddressMode value
- VUID-VkSamplerCreateInfo-addressModeV-parameter
  `addressModeV` **must** be a valid VkSamplerAddressMode value
- VUID-VkSamplerCreateInfo-addressModeW-parameter
  `addressModeW` **must** be a valid VkSamplerAddressMode value

VK_LOD_CLAMP_NONE is a special constant value used for VkSamplerCreateInfo::`maxLod` to indicate that maximum LOD clamping should not be performed.

```
#define VK_LOD_CLAMP_NONE                 1000.0F
```

Bits which **can** be set in VkSamplerCreateInfo::`flags`, specifying additional parameters of a sampler, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSamplerCreateFlagBits {
  // Provided by VK_EXT_fragment_density_map
    VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT = 0x00000001,
  // Provided by VK_EXT_fragment_density_map
    VK_SAMPLER_CREATE_SUBSAMPLED_COARSE_RECONSTRUCTION_BIT_EXT = 0x00000002,
  // Provided by VK_EXT_descriptor_buffer
    VK_SAMPLER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT = 0x00000008,
  // Provided by VK_EXT_non_seamless_cube_map
    VK_SAMPLER_CREATE_NON_SEAMLESS_CUBE_MAP_BIT_EXT = 0x00000004,
} VkSamplerCreateFlagBits;
```

- VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT specifies that the sampler will read from an image created with flags containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT.
- VK_SAMPLER_CREATE_SUBSAMPLED_COARSE_RECONSTRUCTION_BIT_EXT specifies that the implementation **may** use approximations when reconstructing a full color value for texture access from a subsampled image.
- VK_SAMPLER_CREATE_NON_SEAMLESS_CUBE_MAP_BIT_EXT specifies that cube map edge handling is not performed.

NOTE	The approximations used when VK_SAMPLER_CREATE_SUBSAMPLED_COARSE_RECONSTRUCTION_BIT_EXT is specified are implementation defined. Some implementations **may** interpolate between fragment density levels in a subsampled image. In that case, this bit **may** be used to decide whether the interpolation factors are calculated per fragment or at a coarser granularity.

- VK_SAMPLER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT specifies that the sampler **can** be used with descriptor buffers when capturing and replaying (e.g. for trace capture and replay), see VkOpaqueCaptureDescriptorDataCreateInfoEXT for more detail.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSamplerCreateFlags;
```

VkSamplerCreateFlags is a bitmask type for setting a mask of zero or more VkSamplerCreateFlagBits.

The VkSamplerReductionModeCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkSamplerReductionModeCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkSamplerReductionMode    reductionMode;
} VkSamplerReductionModeCreateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- reductionMode is a VkSamplerReductionMode value controlling how texture filtering combines texel values.

If the pNext chain of VkSamplerCreateInfo includes a VkSamplerReductionModeCreateInfo structure, then that structure includes a mode controlling how texture filtering combines texel values.

If this structure is not present, reductionMode is considered to be VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE.

Valid Usage (Implicit)

- VUID-VkSamplerReductionModeCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SAMPLER_REDUCTION_MODE_CREATE_INFO
- VUID-VkSamplerReductionModeCreateInfo-reductionMode-parameter
  `reductionMode` **must** be a valid VkSamplerReductionMode value

Structure Chaining

Extends the structure

- VkSamplerCreateInfo

Reduction modes are specified by VkSamplerReductionMode, which takes values:

```
// Provided by VK_VERSION_1_2
typedef enum VkSamplerReductionMode {
    VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE = 0,
    VK_SAMPLER_REDUCTION_MODE_MIN = 1,
    VK_SAMPLER_REDUCTION_MODE_MAX = 2,
} VkSamplerReductionMode;
```

- VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE specifies that texel values are combined by computing a weighted average of values in the footprint, using weights as specified in the image operations chapter.
- VK_SAMPLER_REDUCTION_MODE_MIN specifies that texel values are combined by taking the component-wise minimum of values in the footprint with non-zero weights.
- VK_SAMPLER_REDUCTION_MODE_MAX specifies that texel values are combined by taking the component-wise maximum of values in the footprint with non-zero weights.

Possible values of the VkSamplerCreateInfo::`magFilter` and `minFilter` parameters, specifying filters used for texture lookups, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkFilter {
    VK_FILTER_NEAREST = 0,
    VK_FILTER_LINEAR = 1,
  // Provided by VK_EXT_filter_cubic
    VK_FILTER_CUBIC_EXT = 1000015000,
} VkFilter;
```

- VK_FILTER_NEAREST specifies nearest filtering.
- VK_FILTER_LINEAR specifies linear filtering.
- VK_FILTER_CUBIC_EXT specifies cubic filtering.

These filters are described in detail in Texel Filtering.

Possible values of the VkSamplerCreateInfo::mipmapMode, specifying the mipmap mode used for texture lookups, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSamplerMipmapMode {
    VK_SAMPLER_MIPMAP_MODE_NEAREST = 0,
    VK_SAMPLER_MIPMAP_MODE_LINEAR = 1,
} VkSamplerMipmapMode;
```

- VK_SAMPLER_MIPMAP_MODE_NEAREST specifies nearest filtering.
- VK_SAMPLER_MIPMAP_MODE_LINEAR specifies linear filtering.

These modes are described in detail in Texel Filtering.

Possible values of the VkSamplerCreateInfo::addressMode* parameters, corresponding to different wrapping operations used during sampling, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSamplerAddressMode {
    VK_SAMPLER_ADDRESS_MODE_REPEAT = 0,
    VK_SAMPLER_ADDRESS_MODE_MIRRORED_REPEAT = 1,
    VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE = 2,
    VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_BORDER = 3,
  // Provided by VK_VERSION_1_2, VK_KHR_sampler_mirror_clamp_to_edge
    VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE = 4,
  // Provided by VK_KHR_sampler_mirror_clamp_to_edge
  // VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE_KHR is a legacy alias
    VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE_KHR =
VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE,
} VkSamplerAddressMode;
```

- VK_SAMPLER_ADDRESS_MODE_REPEAT specifies that the repeat wrap mode will be used.
- VK_SAMPLER_ADDRESS_MODE_MIRRORED_REPEAT specifies that the mirrored repeat wrap mode will be used.
- VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE specifies that the clamp to edge wrap mode will be used.
- VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_BORDER specifies that the clamp to border wrap mode will be used.
- VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE specifies that the mirror clamp to edge wrap mode will be used. This is only valid if the samplerMirrorClampToEdge feature is enabled, or if the VK_KHR_sampler_mirror_clamp_to_edge extension is enabled.

Comparison operators compare a *reference* and a *test* value, and return a true (“passed”) or false (“failed”) value depending on the comparison operator chosen. The supported operators are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCompareOp {
    VK_COMPARE_OP_NEVER = 0,
    VK_COMPARE_OP_LESS = 1,
    VK_COMPARE_OP_EQUAL = 2,
    VK_COMPARE_OP_LESS_OR_EQUAL = 3,
    VK_COMPARE_OP_GREATER = 4,
    VK_COMPARE_OP_NOT_EQUAL = 5,
    VK_COMPARE_OP_GREATER_OR_EQUAL = 6,
    VK_COMPARE_OP_ALWAYS = 7,
} VkCompareOp;
```

- VK_COMPARE_OP_NEVER specifies that the comparison always evaluates false.
- VK_COMPARE_OP_LESS specifies that the comparison evaluates *reference* < *test*.
- VK_COMPARE_OP_EQUAL specifies that the comparison evaluates *reference* = *test*.
- VK_COMPARE_OP_LESS_OR_EQUAL specifies that the comparison evaluates *reference* ≤ *test*.
- VK_COMPARE_OP_GREATER specifies that the comparison evaluates *reference* > *test*.
- VK_COMPARE_OP_NOT_EQUAL specifies that the comparison evaluates *reference* ≠ *test*.
- VK_COMPARE_OP_GREATER_OR_EQUAL specifies that the comparison evaluates *reference* ≥ *test*.
- VK_COMPARE_OP_ALWAYS specifies that the comparison always evaluates true.

Comparison operators are used for:

- The Depth Compare Operation operator for a sampler, specified by VkSamplerCreateInfo ::`compareOp`.
- The stencil comparison operator for the stencil test, specified by vkCmdSetStencilOp::`compareOp` or VkStencilOpState::`compareOp`.
- The Depth Comparison operator for the depth test, specified by vkCmdSetDepthCompareOp ::`depthCompareOp` or VkPipelineDepthStencilStateCreateInfo::`depthCompareOp`.

Each such use describes how the *reference* and *test* values for that comparison are determined.

Possible values of VkSamplerCreateInfo::`borderColor`, specifying the border color used for texture lookups, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkBorderColor {
    VK_BORDER_COLOR_FLOAT_TRANSPARENT_BLACK = 0,
    VK_BORDER_COLOR_INT_TRANSPARENT_BLACK = 1,
    VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK = 2,
    VK_BORDER_COLOR_INT_OPAQUE_BLACK = 3,
    VK_BORDER_COLOR_FLOAT_OPAQUE_WHITE = 4,
    VK_BORDER_COLOR_INT_OPAQUE_WHITE = 5,
  // Provided by VK_EXT_custom_border_color
    VK_BORDER_COLOR_FLOAT_CUSTOM_EXT = 1000287003,
  // Provided by VK_EXT_custom_border_color
    VK_BORDER_COLOR_INT_CUSTOM_EXT = 1000287004,
} VkBorderColor;
```

- VK_BORDER_COLOR_FLOAT_TRANSPARENT_BLACK specifies a transparent, floating-point format, black color.
- VK_BORDER_COLOR_INT_TRANSPARENT_BLACK specifies a transparent, integer format, black color.
- VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK specifies an opaque, floating-point format, black color.
- VK_BORDER_COLOR_INT_OPAQUE_BLACK specifies an opaque, integer format, black color.
- VK_BORDER_COLOR_FLOAT_OPAQUE_WHITE specifies an opaque, floating-point format, white color.
- VK_BORDER_COLOR_INT_OPAQUE_WHITE specifies an opaque, integer format, white color.
- VK_BORDER_COLOR_FLOAT_CUSTOM_EXT specifies that a VkSamplerCustomBorderColorCreateInfoEXT structure is included in the VkSamplerCreateInfo::pNext chain containing the color data in floating-point format.
- VK_BORDER_COLOR_INT_CUSTOM_EXT specifies that a VkSamplerCustomBorderColorCreateInfoEXT structure is included in the VkSamplerCreateInfo::pNext chain containing the color data in integer format.

These colors are described in detail in Border Replacement.

To destroy a sampler, call:

```
// Provided by VK_VERSION_1_0
void vkDestroySampler(
    VkDevice                                    device,
    VkSampler                                   sampler,
    const VkAllocationCallbacks*                pAllocator);
```

- device is the logical device that destroys the sampler.
- sampler is the sampler to destroy.
- pAllocator controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroySampler-sampler-01082
  All submitted commands that refer to `sampler` **must** have completed execution
- VUID-vkDestroySampler-sampler-01083
  If `VkAllocationCallbacks` were provided when `sampler` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroySampler-sampler-01084
  If no `VkAllocationCallbacks` were provided when `sampler` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroySampler-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroySampler-sampler-parameter
  If `sampler` is not VK_NULL_HANDLE, `sampler` **must** be a valid VkSampler handle
- VUID-vkDestroySampler-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroySampler-sampler-parent
  If `sampler` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `sampler` **must** be externally synchronized

14.1. Sampler $Y'C_BC_R$ Conversion

To create a sampler with $Y'C_BC_R$ conversion enabled, add a VkSamplerYcbcrConversionInfo structure to the `pNext` chain of the VkSamplerCreateInfo structure. To create a sampler $Y'C_BC_R$ conversion, the `samplerYcbcrConversion` feature **must** be enabled. Conversion **must** be fixed at pipeline creation time, through use of a combined embedded sampler and image mapping if using descriptor heaps, or a combined image sampler with an immutable sampler in `VkDescriptorSetLayoutBinding`.

A VkSamplerYcbcrConversionInfo **must** be provided for samplers to be used with image views that access VK_IMAGE_ASPECT_COLOR_BIT if the format is one of the formats that require a sampler $Y'C_BC_R$ conversion .

The `VkSamplerYcbcrConversionInfo` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkSamplerYcbcrConversionInfo {
    VkStructureType             sType;
    const void*                 pNext;
    VkSamplerYcbcrConversion    conversion;
} VkSamplerYcbcrConversionInfo;
```

```
// Provided by VK_KHR_sampler_ycbcr_conversion
// Equivalent to VkSamplerYcbcrConversionInfo
typedef VkSamplerYcbcrConversionInfo VkSamplerYcbcrConversionInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `conversion` is a VkSamplerYcbcrConversion handle created with vkCreateSamplerYcbcrConversion.

Valid Usage (Implicit)

- VUID-VkSamplerYcbcrConversionInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_INFO
- VUID-VkSamplerYcbcrConversionInfo-conversion-parameter
  `conversion` **must** be a valid VkSamplerYcbcrConversion handle

Structure Chaining

Extends the structures

- VkImageViewCreateInfo
- VkSamplerCreateInfo

A sampler $Y'C_BC_R$ conversion is an opaque representation of a device-specific sampler $Y'C_BC_R$ conversion description, represented as a `VkSamplerYcbcrConversion` handle:

```
// Provided by VK_VERSION_1_1
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkSamplerYcbcrConversion)
```

```
// Provided by VK_KHR_sampler_ycbcr_conversion
// Equivalent to VkSamplerYcbcrConversion
typedef VkSamplerYcbcrConversion VkSamplerYcbcrConversionKHR;
```

To create a VkSamplerYcbcrConversion, call:

```
// Provided by VK_VERSION_1_1
VkResult vkCreateSamplerYcbcrConversion(
    VkDevice                                    device,
    const VkSamplerYcbcrConversionCreateInfo*   pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkSamplerYcbcrConversion*                   pYcbcrConversion);
```

```
// Provided by VK_KHR_sampler_ycbcr_conversion
// Equivalent to vkCreateSamplerYcbcrConversion
VkResult vkCreateSamplerYcbcrConversionKHR(
    VkDevice                                    device,
    const VkSamplerYcbcrConversionCreateInfo*   pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkSamplerYcbcrConversion*                   pYcbcrConversion);
```

- `device` is the logical device that creates the sampler $Y'C_BC_R$ conversion.
- `pCreateInfo` is a pointer to a VkSamplerYcbcrConversionCreateInfo structure specifying the requested sampler $Y'C_BC_R$ conversion.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pYcbcrConversion` is a pointer to a VkSamplerYcbcrConversion handle in which the resulting sampler $Y'C_BC_R$ conversion is returned.

The interpretation of the configured sampler $Y'C_BC_R$ conversion is described in more detail in the description of sampler $Y'C_BC_R$ conversion in the Image Operations chapter.

Valid Usage

- VUID-vkCreateSamplerYcbcrConversion-None-01648
  The `samplerYcbcrConversion` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCreateSamplerYcbcrConversion-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateSamplerYcbcrConversion-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkSamplerYcbcrConversionCreateInfo structure
- VUID-vkCreateSamplerYcbcrConversion-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateSamplerYcbcrConversion-pYcbcrConversion-parameter
  `pYcbcrConversion` **must** be a valid pointer to a VkSamplerYcbcrConversion handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkSamplerYcbcrConversionCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkSamplerYcbcrConversionCreateInfo {
    VkStructureType                 sType;
    const void*                     pNext;
    VkFormat                        format;
    VkSamplerYcbcrModelConversion   ycbcrModel;
    VkSamplerYcbcrRange             ycbcrRange;
    VkComponentMapping              components;
    VkChromaLocation                xChromaOffset;
    VkChromaLocation                yChromaOffset;
    VkFilter                        chromaFilter;
    VkBool32                        forceExplicitReconstruction;
} VkSamplerYcbcrConversionCreateInfo;
```

```
// Provided by VK_KHR_sampler_ycbcr_conversion
// Equivalent to VkSamplerYcbcrConversionCreateInfo
typedef VkSamplerYcbcrConversionCreateInfo VkSamplerYcbcrConversionCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- format is the format of the image from which color information will be retrieved.
- ycbcrModel describes the color matrix for conversion between color models.
- ycbcrRange describes whether the encoded values have headroom and foot room, or whether the encoding uses the full numerical range.
- components applies a *swizzle* based on VkComponentSwizzle enums prior to range expansion and color model conversion.
- xChromaOffset describes the sample location associated with downsampled chroma components in the x dimension. xChromaOffset has no effect for formats in which chroma components are not downsampled horizontally.

- `yChromaOffset` describes the sample location associated with downsampled chroma components in the y dimension. `yChromaOffset` has no effect for formats in which the chroma components are not downsampled vertically.
- `chromaFilter` is the filter for chroma reconstruction.
- `forceExplicitReconstruction` **can** be used to ensure that reconstruction is done explicitly, if supported.

NOTE

Setting `forceExplicitReconstruction` to VK_TRUE **may** have a performance penalty on implementations where explicit reconstruction is not the default mode of operation.

If `format` supports VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT the `forceExplicitReconstruction` value behaves as if it were VK_TRUE.

Sampler $Y'C_BC_R$ conversion objects do not support *external format conversion* without additional extensions defining *external formats*.

Valid Usage

- VUID-VkSamplerYcbcrConversionCreateInfo-format-04061
  `format` **must** represent unsigned normalized values (i.e. the format **must** be a `UNORM` format)
- VUID-VkSamplerYcbcrConversionCreateInfo-format-01650
  The potential format features of the sampler $Y'C_BC_R$ conversion **must** support VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT or VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT
- VUID-VkSamplerYcbcrConversionCreateInfo-xChromaOffset-01651
  If the potential format features of the sampler $Y'C_BC_R$ conversion do not support VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT, `xChromaOffset` and `yChromaOffset` **must** not be VK_CHROMA_LOCATION_COSITED_EVEN if the corresponding components are downsampled
- VUID-VkSamplerYcbcrConversionCreateInfo-xChromaOffset-01652
  If the potential format features of the sampler $Y'C_BC_R$ conversion do not support VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT, `xChromaOffset` and `yChromaOffset` **must** not be VK_CHROMA_LOCATION_MIDPOINT if the corresponding components are downsampled
- VUID-VkSamplerYcbcrConversionCreateInfo-components-02581
  If the format has a `_422` or `_420` suffix, then `components.g` **must** be the identity swizzle
- VUID-VkSamplerYcbcrConversionCreateInfo-components-02582
  If the format has a `_422` or `_420` suffix, then `components.a` **must** be the identity swizzle, VK_COMPONENT_SWIZZLE_ONE, or VK_COMPONENT_SWIZZLE_ZERO
- VUID-VkSamplerYcbcrConversionCreateInfo-components-02583

If the format has a _422 or _420 suffix, then components.r **must** be the identity swizzle or VK_COMPONENT_SWIZZLE_B

- VUID-VkSamplerYcbcrConversionCreateInfo-components-02584
  If the format has a _422 or _420 suffix, then components.b **must** be the identity swizzle or VK_COMPONENT_SWIZZLE_R
- VUID-VkSamplerYcbcrConversionCreateInfo-components-02585
  If the format has a _422 or _420 suffix, and if either components.r or components.b is the identity swizzle, both values **must** be the identity swizzle
- VUID-VkSamplerYcbcrConversionCreateInfo-ycbcrModel-01655
  If ycbcrModel is not VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY, then components.r, components.g, and components.b **must** correspond to components of the format; that is, components.r, components.g, and components.b **must** not be VK_COMPONENT_SWIZZLE_ZERO or VK_COMPONENT_SWIZZLE_ONE, and **must** not correspond to a component containing zero or one as a consequence of component substitution
- VUID-VkSamplerYcbcrConversionCreateInfo-ycbcrRange-02748
  If ycbcrRange is VK_SAMPLER_YCBCR_RANGE_ITU_NARROW, then the R, G, and B components obtained by applying the component swizzle to format **must** each have a bit-depth greater than or equal to 8
- VUID-VkSamplerYcbcrConversionCreateInfo-forceExplicitReconstruction-01656
  If the potential format features of the sampler $Y'C_BC_R$ conversion do not support VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT forceExplicitReconstruction **must** be VK_FALSE
- VUID-VkSamplerYcbcrConversionCreateInfo-chromaFilter-01657
  If the potential format features of the sampler $Y'C_BC_R$ conversion do not support VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT, chromaFilter **must** not be VK_FILTER_LINEAR

Valid Usage (Implicit)

- VUID-VkSamplerYcbcrConversionCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_CREATE_INFO
- VUID-VkSamplerYcbcrConversionCreateInfo-pNext-pNext
  pNext **must** be NULL
- VUID-VkSamplerYcbcrConversionCreateInfo-format-parameter
  format **must** be a valid VkFormat value
- VUID-VkSamplerYcbcrConversionCreateInfo-ycbcrModel-parameter
  ycbcrModel **must** be a valid VkSamplerYcbcrModelConversion value
- VUID-VkSamplerYcbcrConversionCreateInfo-ycbcrRange-parameter
  ycbcrRange **must** be a valid VkSamplerYcbcrRange value
- VUID-VkSamplerYcbcrConversionCreateInfo-components-parameter
  components **must** be a valid VkComponentMapping structure

- VUID-VkSamplerYcbcrConversionCreateInfo-xChromaOffset-parameter
  `xChromaOffset` **must** be a valid VkChromaLocation value
- VUID-VkSamplerYcbcrConversionCreateInfo-yChromaOffset-parameter
  `yChromaOffset` **must** be a valid VkChromaLocation value
- VUID-VkSamplerYcbcrConversionCreateInfo-chromaFilter-parameter
  `chromaFilter` **must** be a valid VkFilter value

If `chromaFilter` is VK_FILTER_NEAREST, chroma samples are reconstructed to luma component resolution using nearest-neighbor sampling. Otherwise, chroma samples are reconstructed using interpolation. More details can be found in the description of sampler $Y'C_BC_R$ conversion in the Image Operations chapter.

VkSamplerYcbcrModelConversion defines the conversion from the source color model to the shader color model. Possible values are:

```
// Provided by VK_VERSION_1_1
typedef enum VkSamplerYcbcrModelConversion {
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY = 0,
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_IDENTITY = 1,
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_709 = 2,
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_601 = 3,
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_2020 = 4,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY_KHR =
VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_IDENTITY_KHR =
VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_IDENTITY,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_709_KHR =
VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_709,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_601_KHR =
VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_601,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_2020_KHR =
VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_2020,
} VkSamplerYcbcrModelConversion;
```

```
// Provided by VK_KHR_sampler_ycbcr_conversion
// Equivalent to VkSamplerYcbcrModelConversion
typedef VkSamplerYcbcrModelConversion VkSamplerYcbcrModelConversionKHR;
```

- VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY specifies that the input values to the conversion are unmodified.
- VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_IDENTITY specifies no model conversion

but the inputs are range expanded as for $Y'C_BC_R$.

- VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_709 specifies the color model conversion from $Y'C_BC_R$ to R'G'B' defined in BT.709 and described in the "BT.709 $Y'C_BC_R$ conversion" section of the Khronos Data Format Specification.
- VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_601 specifies the color model conversion from $Y'C_BC_R$ to R'G'B' defined in BT.601 and described in the "BT.601 $Y'C_BC_R$ conversion" section of the Khronos Data Format Specification.
- VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_2020 specifies the color model conversion from $Y'C_BC_R$ to R'G'B' defined in BT.2020 and described in the "BT.2020 $Y'C_BC_R$ conversion" section of the Khronos Data Format Specification.

In the `VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_*` color models, for the input to the sampler $Y'C_BC_R$ range expansion and model conversion:

- the Y (Y' luma) component corresponds to the G component of an RGB image.
- the CB (C_B or "U" blue color difference) component corresponds to the B component of an RGB image.
- the CR (C_R or "V" red color difference) component corresponds to the R component of an RGB image.
- the alpha component, if present, is not modified by color model conversion.

These rules reflect the mapping of components after the component swizzle operation (controlled by VkSamplerYcbcrConversionCreateInfo::`components`).

NOTE

For example, a "YUVA" 32-bit format comprising four 8-bit components can be implemented as VK_FORMAT_R8G8B8A8_UNORM with a component mapping:

- `components.a` = VK_COMPONENT_SWIZZLE_IDENTITY
- `components.r` = VK_COMPONENT_SWIZZLE_B
- `components.g` = VK_COMPONENT_SWIZZLE_R
- `components.b` = VK_COMPONENT_SWIZZLE_G

The VkSamplerYcbcrRange enum describes whether color components are encoded using the full range of numerical values or whether values are reserved for headroom and foot room. VkSamplerYcbcrRange is defined as:

```
// Provided by VK_VERSION_1_1
typedef enum VkSamplerYcbcrRange {
    VK_SAMPLER_YCBCR_RANGE_ITU_FULL = 0,
    VK_SAMPLER_YCBCR_RANGE_ITU_NARROW = 1,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_SAMPLER_YCBCR_RANGE_ITU_FULL_KHR = VK_SAMPLER_YCBCR_RANGE_ITU_FULL,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_SAMPLER_YCBCR_RANGE_ITU_NARROW_KHR = VK_SAMPLER_YCBCR_RANGE_ITU_NARROW,
} VkSamplerYcbcrRange;
```

```
// Provided by VK_KHR_sampler_ycbcr_conversion
// Equivalent to VkSamplerYcbcrRange
typedef VkSamplerYcbcrRange VkSamplerYcbcrRangeKHR;
```

- VK_SAMPLER_YCBCR_RANGE_ITU_FULL specifies that the full range of the encoded values are valid and interpreted according to the ITU “full range” quantization rules.
- VK_SAMPLER_YCBCR_RANGE_ITU_NARROW specifies that headroom and foot room are reserved in the numerical range of encoded values, and the remaining values are expanded according to the ITU “narrow range” quantization rules.

The formulae for these conversions are described in the Sampler $Y'C_BC_R$ Range Expansion section of the Image Operations chapter.

No range modification takes place if `ycbcrModel` is VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY; the `ycbcrRange` field of VkSamplerYcbcrConversionCreateInfo is ignored in this case.

The VkChromaLocation enum defines the location of downsampled chroma component samples relative to the luma samples, and is defined as:

```
// Provided by VK_VERSION_1_1
typedef enum VkChromaLocation {
    VK_CHROMA_LOCATION_COSITED_EVEN = 0,
    VK_CHROMA_LOCATION_MIDPOINT = 1,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_CHROMA_LOCATION_COSITED_EVEN_KHR = VK_CHROMA_LOCATION_COSITED_EVEN,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_CHROMA_LOCATION_MIDPOINT_KHR = VK_CHROMA_LOCATION_MIDPOINT,
} VkChromaLocation;
```

```
// Provided by VK_KHR_sampler_ycbcr_conversion
// Equivalent to VkChromaLocation
typedef VkChromaLocation VkChromaLocationKHR;
```

- VK_CHROMA_LOCATION_COSITED_EVEN specifies that downsampled chroma samples are

aligned with luma samples with even coordinates.

- VK_CHROMA_LOCATION_MIDPOINT specifies that downsampled chroma samples are located half way between each even luma sample and the nearest higher odd luma sample.

To destroy a sampler $Y'C_BC_R$ conversion, call:

```
// Provided by VK_VERSION_1_1
void vkDestroySamplerYcbcrConversion(
    VkDevice                                    device,
    VkSamplerYcbcrConversion                    ycbcrConversion,
    const VkAllocationCallbacks*                pAllocator);
```

```
// Provided by VK_KHR_sampler_ycbcr_conversion
// Equivalent to vkDestroySamplerYcbcrConversion
void vkDestroySamplerYcbcrConversionKHR(
    VkDevice                                    device,
    VkSamplerYcbcrConversion                    ycbcrConversion,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the $Y'C_BC_R$ conversion.
- `ycbcrConversion` is the conversion to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage (Implicit)

- VUID-vkDestroySamplerYcbcrConversion-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroySamplerYcbcrConversion-ycbcrConversion-parameter
  If `ycbcrConversion` is not VK_NULL_HANDLE, `ycbcrConversion` **must** be a valid VkSamplerYcbcrConversion handle
- VUID-vkDestroySamplerYcbcrConversion-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroySamplerYcbcrConversion-ycbcrConversion-parent
  If `ycbcrConversion` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `ycbcrConversion` **must** be externally synchronized

14.2. Custom Border Colors

In addition to the predefined border color values, applications **can** provide a custom border color value by including the VkSamplerCustomBorderColorCreateInfoEXT structure in the VkSamplerCreateInfo::pNext chain.

The VkSamplerCustomBorderColorCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_custom_border_color
typedef struct VkSamplerCustomBorderColorCreateInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkClearColorValue    customBorderColor;
    VkFormat             format;
} VkSamplerCustomBorderColorCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- customBorderColor is a VkClearColorValue representing the desired custom sampler border color.
- format is a VkFormat representing the format of the sampled image view(s). This field may be VK_FORMAT_UNDEFINED if the customBorderColorWithoutFormat feature is enabled.

NOTE

If format is a depth/stencil format, the aspect is determined by the value of VkSamplerCreateInfo::borderColor. If VkSamplerCreateInfo::borderColor is VK_BORDER_COLOR_FLOAT_CUSTOM_EXT, the depth aspect is considered. If VkSamplerCreateInfo::borderColor is VK_BORDER_COLOR_INT_CUSTOM_EXT, the stencil aspect is considered.

If format is VK_FORMAT_UNDEFINED, the VkSamplerCreateInfo::borderColor is VK_BORDER_COLOR_INT_CUSTOM_EXT, and the sampler is used with an image with a stencil format, then the implementation **must** source the custom border color from either the first or second components of VkSamplerCreateInfo ::borderColor and **should** source it from the first component.

In some cases, implementations **may** return undefined values instead of the specified custom border color when used on VK_QUEUE_COMPUTE_BIT queues that do not also support VK_QUEUE_GRAPHICS_BIT. Undefined: values **may** also be returned in task shaders.

NOTE

Tests added in 2026 revealed issues around custom border colors on AMD in compute-only queues and with task shaders, that were not fixable in software. While this will be fixed in future hardware, there is no reasonable workaround for existing hardware. Use of standard border colors is unaffected.

Valid Usage

- VUID-VkSamplerCustomBorderColorCreateInfoEXT-format-07605
  If `format` is not VK_FORMAT_UNDEFINED and `format` is not a depth/stencil format then the VkSamplerCreateInfo::`borderColor` type **must** match the sampled type of the provided `format`, as shown in the *SPIR-V Type* column of the Interpretation of Numeric Format table
- VUID-VkSamplerCustomBorderColorCreateInfoEXT-format-04014
  If the `customBorderColorWithoutFormat` feature is not enabled then `format` **must** not be VK_FORMAT_UNDEFINED
- VUID-VkSamplerCustomBorderColorCreateInfoEXT-format-04015
  If the sampler is used to sample an image view of VK_FORMAT_B4G4R4A4_UNORM_PACK16, VK_FORMAT_B5G6R5_UNORM_PACK16, VK_FORMAT_A1B5G5R5_UNORM_PACK16, or VK_FORMAT_B5G5R5A1_UNORM_PACK16 format then `format` **must** not be VK_FORMAT_UNDEFINED

Valid Usage (Implicit)

- VUID-VkSamplerCustomBorderColorCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SAMPLER_CUSTOM_BORDER_COLOR_CREATE_INFO_EXT
- VUID-VkSamplerCustomBorderColorCreateInfoEXT-format-parameter
  `format` **must** be a valid VkFormat value

Structure Chaining

Extends the structure

- VkSamplerCreateInfo

If the sampler is created with VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK, VK_BORDER_COLOR_INT_OPAQUE_BLACK, VK_BORDER_COLOR_FLOAT_CUSTOM_EXT, or VK_BORDER_COLOR_INT_CUSTOM_EXT `borderColor`, and that sampler will be combined with an image view that does not have an identity swizzle, and VkPhysicalDeviceBorderColorSwizzleFeaturesEXT::`borderColorSwizzleFromImage` is not enabled, then it is necessary to specify the component mapping of the border color, by including the `VkSamplerBorderColorComponentMappingCreateInfoEXT` structure in the VkSamplerCreateInfo::`pNext` chain, to get defined results.

The `VkSamplerBorderColorComponentMappingCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_border_color_swizzle
typedef struct VkSamplerBorderColorComponentMappingCreateInfoEXT {
    VkStructureType       sType;
    const void*           pNext;
    VkComponentMapping    components;
    VkBool32              srgb;
} VkSamplerBorderColorComponentMappingCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `components` is a VkComponentMapping structure specifying a remapping of the border color components.
- `srgb` indicates that the sampler will be combined with an image view that has an image format which is sRGB encoded.

The VkComponentMapping `components` member describes a remapping from components of the border color to components of the vector returned by shader image instructions when the border color is used.

Valid Usage

- VUID-VkSamplerBorderColorComponentMappingCreateInfoEXT-borderColorSwizzle-06437
  The `borderColorSwizzle` feature **must** be enabled

Valid Usage (Implicit)

- VUID-VkSamplerBorderColorComponentMappingCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SAMPLER_BORDER_COLOR_COMPONENT_MAPPING_CREATE_INFO_EXT
- VUID-VkSamplerBorderColorComponentMappingCreateInfoEXT-components-parameter
  `components` **must** be a valid VkComponentMapping structure

Structure Chaining

Extends the structure

- VkSamplerCreateInfo

14.2.1. Custom Border Color Registration

When using descriptor heaps, it is possible to create a sampler descriptor without creating a

sampler object. Without this object, implementations using a fixed palette of border colors have no way to know which custom border colors are active or not.

In order to use custom border colors with descriptor heaps, border colors **must** instead be registered with the implementation and unregistered when they are no longer in use, and the registered index provided when writing the sampler descriptor.

To register a custom border color, call:

```
// Provided by VK_EXT_custom_border_color with VK_EXT_descriptor_heap
VkResult vkRegisterCustomBorderColorEXT(
    VkDevice                                    device,
    const VkSamplerCustomBorderColorCreateInfoEXT* pBorderColor,
    VkBool32                                    requestIndex,
    uint32_t*                                   pIndex);
```

- `device` is the logical device where the border color is registered.
- `pBorderColor` is a pointer to a VkSamplerCustomBorderColorCreateInfoEXT structure specifying the custom border color value to register.
- `requestIndex` is a Boolean value indicating if a specific index is requested or not.
- `pIndex` is a pointer to a `uint32_t` index value that will be written by the command upon success.

If `requestIndex` is VK_TRUE, the value present in `pIndex` when passed to the command is a requested index, and rather than returning a new index, the implementation will attempt to register that index, leaving the value intact. If the implementation is unable to register a requested index, VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS will be returned. If an index has not been registered (either explicitly or implicitly by creating a sampler object), or if it has been subsequently unregistered, the implementation **must** register that index successfully.

If `requestIndex` is VK_FALSE, the value stored in `pIndex` is ignored, and a new index will be returned if the implementation is able to register a new index. If the implementation is unable to register a new index, VK_ERROR_TOO_MANY_OBJECTS will be returned.

If an index is successfully registered, it **can** be used when writing a sampler descriptor or creating a sampler object to use with the custom border color, via VkSamplerCustomBorderColorIndexCreateInfoEXT.

NOTE The type of border color is not specified by this command (VK_BORDER_COLOR_FLOAT_CUSTOM_EXT vs. VK_BORDER_COLOR_INT_CUSTOM_EXT); the data will be interpreted at the point the border color is sampled with an actual sampler. Implementations are expected to store the data as raw bytes if they do not need the format to be specified.

Valid Usage

- VUID-vkRegisterCustomBorderColorEXT-requestIndex-11287
  If `requestIndex` is VK_TRUE, the value stored in `pIndex` **must** be less than

maxCustomBorderColorSamplers

Valid Usage (Implicit)

- VUID-vkRegisterCustomBorderColorEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkRegisterCustomBorderColorEXT-pBorderColor-parameter
  `pBorderColor` **must** be a valid pointer to a valid VkSamplerCustomBorderColorCreateInfoEXT structure
- VUID-vkRegisterCustomBorderColorEXT-pIndex-parameter
  `pIndex` **must** be a valid pointer to a `uint32_t` value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_TOO_MANY_OBJECTS
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To unregister a custom border color, call:

```
// Provided by VK_EXT_custom_border_color with VK_EXT_descriptor_heap
void vkUnregisterCustomBorderColorEXT(
    VkDevice                                    device,
    uint32_t                                    index);
```

- `device` is the logical device where the border color is registered.
- `index` is the `uint32_t` index value to unregister.

Valid Usage

- VUID-vkUnregisterCustomBorderColorEXT-index-11288
  `index` **must** be less than `maxCustomBorderColorSamplers`

Valid Usage (Implicit)

- VUID-vkUnregisterCustomBorderColorEXT-device-parameter
  `device` **must** be a valid VkDevice handle

The `VkSamplerCustomBorderColorIndexCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_custom_border_color with VK_EXT_descriptor_heap
typedef struct VkSamplerCustomBorderColorIndexCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           index;
} VkSamplerCustomBorderColorIndexCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `index` is the `uint32_t` index value to use with the sampler.

If this structure is included in the `pNext` chain of VkSamplerCreateInfo, the value of `index` will be used for the custom border color registration. `index` does not need to be registered at the point that a sampler object is created or a sampler descriptor is written; as long as it is registered when any use of the sampler is recorded to a command, and remains registered while the sampler is in use. The color registered with the index and the color specified in the sampler **must** be identically defined.

If this structure is not provided when creating a sampler object with a custom border color, it is equivalent to registering a new custom border color by calling vkRegisterCustomBorderColorEXT with that custom border color value, and using that value as `index` in this structure. This implicit registration will be implicitly unregistered when the sampler is destroyed.

If this structure is not provided when creating a sampler object without a custom border color, it is equivalent to setting `index` to 0.

Valid Usage

- VUID-VkSamplerCustomBorderColorIndexCreateInfoEXT-index-11289
  `index` **must** be less than `maxCustomBorderColorSamplers`

Valid Usage (Implicit)

- VUID-VkSamplerCustomBorderColorIndexCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SAMPLER_CUSTOM_BORDER_COLOR_INDEX_CREATE_INFO_EXT

Structure Chaining

Extends the structure

- VkSamplerCreateInfo

Chapter 15. Resource Descriptors

A *descriptor* is an opaque data structure used to access shader resources such as buffers, images, or samplers. Rather than existing as distinct objects, descriptors are handled as opaque data, which can be accessed by a shader through descriptor heaps, descriptor buffers, or descriptor sets.

Shaders access descriptors via the `ResourceHeapEXT` and `SamplerHeapEXT` built-ins, or through variables decorated with `DescriptorSet` and `Binding` values linking them to the API. Details of the shader interface mapping are described in the Shader Resource Interface section.

NOTE Shaders can also access buffers without going through descriptors by using Physical Storage Buffer Access to access them through 64-bit addresses.

15.1. Descriptor Types

There are a number of different types of descriptor supported by Vulkan, corresponding to different resources or usage. The following sections describe the API definitions of each descriptor type. The mapping of each type to SPIR-V is listed in the Shader Resource and Descriptor Type Correspondence and Shader Resource and Storage Class Correspondence tables in the Shader Interfaces chapter.

Possible descriptor types are:

```
// Provided by VK_VERSION_1_0
typedef enum VkDescriptorType {
    VK_DESCRIPTOR_TYPE_SAMPLER = 0,
    VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER = 1,
    VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE = 2,
    VK_DESCRIPTOR_TYPE_STORAGE_IMAGE = 3,
    VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER = 4,
    VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER = 5,
    VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER = 6,
    VK_DESCRIPTOR_TYPE_STORAGE_BUFFER = 7,
    VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC = 8,
    VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC = 9,
    VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT = 10,
  // Provided by VK_VERSION_1_3
    VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK = 1000138000,
  // Provided by VK_KHR_acceleration_structure
    VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR = 1000150000,
  // Provided by VK_EXT_mutable_descriptor_type
    VK_DESCRIPTOR_TYPE_MUTABLE_EXT = 1000351000,
} VkDescriptorType;
```

- VK_DESCRIPTOR_TYPE_SAMPLER specifies a sampler descriptor.
- VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER specifies a combined image sampler descriptor.

- VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE specifies a sampled image descriptor.
- VK_DESCRIPTOR_TYPE_STORAGE_IMAGE specifies a storage image descriptor.
- VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER specifies a uniform texel buffer descriptor.
- VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER specifies a storage texel buffer descriptor.
- VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER specifies a uniform buffer descriptor.
- VK_DESCRIPTOR_TYPE_STORAGE_BUFFER specifies a storage buffer descriptor.
- VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC specifies a dynamic uniform buffer descriptor.
- VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC specifies a dynamic storage buffer descriptor.
- VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT specifies an input attachment descriptor.
- VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK specifies an inline uniform block.
- VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR specifies an acceleration structure descriptor.
- VK_DESCRIPTOR_TYPE_MUTABLE_EXT specifies a descriptor of mutable type.

15.1.1. Storage Image

A *storage image* (VK_DESCRIPTOR_TYPE_STORAGE_IMAGE) is a descriptor type associated with an image resource via an image view that load, store, and atomic operations **can** be performed on.

Storage image loads are supported in all shader stages for image views whose format features contain VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT.

Stores to storage images are supported in task, mesh and compute shaders for image views whose format features contain VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT.

Atomic operations on storage images are supported in task, mesh and compute shaders for image views whose format features contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT.

When the `fragmentStoresAndAtomics` feature is enabled, stores and atomic operations are also supported for storage images in fragment shaders with the same set of image formats as supported in compute shaders. When the `vertexPipelineStoresAndAtomics` feature is enabled, stores and atomic operations are also supported in vertex, tessellation, and geometry shaders with the same set of image formats as supported in compute shaders.

The image subresources for a storage image **must** be in the VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR or VK_IMAGE_LAYOUT_GENERAL layout in order to access its data in a shader.

15.1.2. Sampler

A *sampler descriptor* (VK_DESCRIPTOR_TYPE_SAMPLER) is a descriptor type associated with a sampler object, used to control the behavior of sampling operations performed on a sampled image.

15.1.3. Sampled Image

A *sampled image* (VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE) is a descriptor type associated with an image resource via an image view that sampling operations **can** be performed on.

Shaders combine a sampled image variable and a sampler variable to perform sampling operations.

Sampled images are supported in all shader stages for image views whose format features contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT.

The image subresources for a sampled image **must** be in one of the following layouts:

- VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL
- VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL
- VK_IMAGE_LAYOUT_GENERAL
- VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR
- VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL
- VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL
- VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL
- VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT

15.1.4. Combined Image Sampler

A *combined image sampler* (VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER) is a single descriptor type associated with both a sampler and an image resource, combining both a sampler and sampled image descriptor into a single descriptor.

If the descriptor refers to a sampler that performs $Y'C_BC_R$ conversion or samples a subsampled image, the sampler **must** only be used to sample the image in the same descriptor. Otherwise, the sampler and image in this type of descriptor **can** be used freely with any other samplers and images.

The image subresources for a combined image sampler **must** be in one of the following layouts:

- VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL
- VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL
- VK_IMAGE_LAYOUT_GENERAL
- VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR
- VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL
- VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL
- VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL

- VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT

NOTE	On some implementations, it **may** be more efficient to sample from an image using a combination of sampler and sampled image that are stored together in the descriptor set in a combined descriptor.

15.1.5. Uniform Texel Buffer

A *uniform texel buffer* (VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER) is a descriptor type associated with a buffer resource via a buffer view that image sampling operations **can** be performed on.

Uniform texel buffers define a tightly-packed 1-dimensional linear array of texels, with texels going through format conversion when read in a shader in the same way as they are for an image.

Load operations from uniform texel buffers are supported in all shader stages for buffer view formats which report format features support for VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT.

15.1.6. Storage Texel Buffer

A *storage texel buffer* (VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER) is a descriptor type associated with a buffer resource via a buffer view that image load, store, and atomic operations **can** be performed on.

Storage texel buffers define a tightly-packed 1-dimensional linear array of texels, with texels going through format conversion when read in a shader in the same way as they are for an image. Unlike uniform texel buffers, these buffers can also be written to in the same way as for storage images.

Storage texel buffer loads are supported in all shader stages for texel buffer view formats which report format features support for VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT.

Stores to storage texel buffers are supported in task, mesh and compute shaders for texel buffer formats which report format features support for VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT.

Atomic operations on storage texel buffers are supported in task, mesh and compute shaders for texel buffer formats which report format features support for VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT.

When the `fragmentStoresAndAtomics` feature is enabled, stores and atomic operations are also supported for storage texel buffers in fragment shaders with the same set of texel buffer formats as supported in compute shaders. When the `vertexPipelineStoresAndAtomics` feature is enabled, stores and atomic operations are also supported in vertex, tessellation, and geometry shaders with the same set of texel buffer formats as supported in compute shaders.

15.1.7. Storage Buffer

A *storage buffer* (VK_DESCRIPTOR_TYPE_STORAGE_BUFFER) is a descriptor type associated with a buffer resource directly, described in a shader as a structure with various members that load, store, and atomic operations **can** be performed on.

NOTE	Atomic operations **can** only be performed on members of certain types as defined in the SPIR-V environment appendix.

15.1.8. Uniform Buffer

A *uniform buffer* (VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER) is a descriptor type associated with a buffer resource directly, described in a shader as a structure with various members that load operations **can** be performed on.

15.1.9. Dynamic Uniform Buffer

A *dynamic uniform buffer* (VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC) is almost identical to a uniform buffer, and differs only in how the offset into the buffer is specified. This descriptor type is only valid when using descriptor sets. The base offset calculated by VkDescriptorBufferInfo when initially updating a descriptor set is added to a dynamic offset when binding a descriptor set.

15.1.10. Dynamic Storage Buffer

A *dynamic storage buffer* (VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC) is almost identical to a storage buffer, and differs only in how the offset into the buffer is specified. This descriptor type is only valid when using descriptor sets. The base offset calculated by VkDescriptorBufferInfo when initially updating a descriptor set is added to a dynamic offset when binding a descriptor set.

15.1.11. Inline Uniform Block

An *inline uniform block* (VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK) is almost identical to a uniform buffer in how it is accessed in the shader. Where it differs is that its storage is taken directly from a containing descriptor set or descriptor buffer, instead of being backed by a separate buffer object. This descriptor type is not valid when using descriptor heaps; applications **can** directly access the heap pointer in a shader or use the VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT mapping.

Inline uniform blocks are typically used to access a small set of constant data that does not require the additional flexibility provided by the indirection enabled when using a uniform buffer where the descriptor and the referenced buffer memory are decoupled. Compared to push constants, they allow reusing the same set of constant data across multiple disjoint sets of drawing and dispatching commands.

Inline uniform block descriptors **cannot** be aggregated into arrays. Instead, the array size specified for an inline uniform block descriptor binding specifies the binding's capacity in bytes.

15.1.12. Input Attachment

An *input attachment* (VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT) is a descriptor type associated with an image resource via an image view that **can** be used for framebuffer local load operations in fragment shaders.

All image formats that are supported for color attachments (VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT) or depth/stencil attachments (VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT) for a given image tiling mode are also supported for input attachments.

An image view used as an input attachment **must** be in one of the following layouts:

- VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL
- VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL
- VK_IMAGE_LAYOUT_GENERAL
- VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR
- VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL
- VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL
- VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL
- VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT
- VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ

15.1.13. Acceleration Structure

An *acceleration structure* (VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR) is a descriptor type that is used to retrieve scene geometry from within shaders that are used for ray traversal. Shaders have read-only access to the memory.

15.1.14. Mutable

A descriptor of *mutable* (VK_DESCRIPTOR_TYPE_MUTABLE_EXT) type indicates that this descriptor **can** mutate to a number of different types. This descriptor type is not valid when using descriptor heaps, as the elements of a descriptor heap **can** be manually adjusted to hold different descriptor types already. When specified in a descriptor set layout, any of the descriptor types given in the VkMutableDescriptorTypeListEXT::`pDescriptorTypes` list of descriptor types in the `pNext` chain of VkDescriptorSetLayoutCreateInfo for this binding. At any point, each individual descriptor of mutable type has an active descriptor type. The active descriptor type **can** be any one of the declared types in `pDescriptorTypes`. Additionally, a mutable descriptor's active descriptor type **can** be of the VK_DESCRIPTOR_TYPE_MUTABLE_EXT type, which is the initial active descriptor type. The active descriptor type **can** change when the descriptor is updated. When a descriptor is consumed by binding a descriptor buffer or set, the active descriptor type is considered, not VK_DESCRIPTOR_TYPE_MUTABLE_EXT.

An active descriptor type of VK_DESCRIPTOR_TYPE_MUTABLE_EXT is considered an undefined descriptor. If a descriptor is consumed where the active descriptor type does not match what the shader expects, the descriptor is considered an undefined descriptor.

NOTE

To find which descriptor types are supported as VK_DESCRIPTOR_TYPE_MUTABLE_EXT, the application **can** use vkGetDescriptorSetLayoutSupport with a VK_DESCRIPTOR_TYPE_MUTABLE_EXT binding, with the list of descriptor types to query in the VkMutableDescriptorTypeCreateInfoEXT::`pMutableDescriptorTypeLists` array for that binding.

NOTE

The intention of a mutable descriptor type is that implementations allocate N bytes per descriptor, where N is determined by the maximum descriptor size for a given descriptor binding. Implementations are not expected to keep track of the active descriptor type, and it should be considered a C-like union type.

A mutable descriptor type is not considered as efficient in terms of runtime performance as using a non-mutable descriptor type, and applications are not encouraged to use them outside API layering efforts. Mutable descriptor types can be more efficient if the alternative is using many different descriptors to emulate mutable descriptor types.

15.2. Physical Storage Buffer Access

Buffer device addresses **can** also be used to access buffer memory in a shader, using the `SPV_KHR_physical_storage_buffer` extension and the `PhysicalStorageBuffer` storage class. For example, this value **can** be stored in a uniform buffer, and the shader **can** read the value from the uniform buffer and use it to do a dependent read/write to this buffer. All loads, stores, and atomics in a shader through `PhysicalStorageBuffer` pointers **must** access addresses in the address range of some buffer.

Chapter 16. Descriptor Heaps

When the descriptorHeap feature is enabled, applications can use descriptor heaps to specify the descriptors that they will be accessing from shaders. Descriptor heaps are not objects, rather they are state that is bound to a command buffer indicating the device address of descriptors that can be accessed from a shader. There are two heaps available to the application - the sampler heap and the resource heap. The sampler heap contains sampler descriptors, the resource heap contains image, acceleration structure, and buffer descriptors.

16.1. Writing Descriptors

Descriptors can be obtained for the different heaps with commands that write descriptors to a host address, with the application responsible for transferring those descriptors into device memory for access via heaps.

To write sampler descriptors to memory, call:

```
// Provided by VK_EXT_descriptor_heap
VkResult vkWriteSamplerDescriptorsEXT(
    VkDevice                                    device,
    uint32_t                                    samplerCount,
    const VkSamplerCreateInfo*                  pSamplers,
    const VkHostAddressRangeEXT*                pDescriptors);
```

- device is the logical device that the descriptors are for.
- samplerCount is the number of elements in pSamplers and pDescriptors.
- pSamplers is a pointer to an array of VkSamplerCreateInfo structures defining properties of the sampler descriptors that will be written.
- pDescriptors is a pointer to an array of VkHostAddressRangeEXT structures defining the host address ranges that will be written to for each descriptor.

Each descriptor will be written to pDescriptors[i].address where i is the index of its create info in pSamplers.

Descriptors written using a fully identical VkSamplerCreateInfo structure on the same VkDevice will always return the same bit pattern. If the descriptorHeapCaptureReplay feature is enabled, descriptors written using a fully identical VkSamplerCreateInfo structure on a VkDevice created from the same VkPhysicalDevice with identical parameters will always return the same bit pattern.

NOTE YC_BC_R samplers must be embedded in a shader by using VkShaderDescriptorSetAndBindingMappingInfoEXT, they cannot be specified here.

Valid Usage

- VUID-vkWriteSamplerDescriptorsEXT-descriptorHeap-11202

The descriptorHeap feature **must** be enabled

- VUID-vkWriteSamplerDescriptorsEXT-size-11203
  The size member of each element of pDescriptors **must** be greater than or equal to the value returned by vkGetPhysicalDeviceDescriptorSizeEXT with a descriptorType equal to VK_DESCRIPTOR_TYPE_SAMPLER
- VUID-vkWriteSamplerDescriptorsEXT-pSamplers-11204
  Elements of pSamplers **must** not include VkSamplerYcbcrConversionInfo structures in their pNext chains
- VUID-vkWriteSamplerDescriptorsEXT-borderColor-11444
  If the borderColor of any element of pSamplers is VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT, VkSamplerCustomBorderColorIndexCreateInfoEXT **must** be included in the pNext chain of that element
- VUID-vkWriteSamplerDescriptorsEXT-borderColor-11205
  If the borderColor of any element of pSamplers is VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT, VkSamplerCustomBorderColorIndexCreateInfoEXT::index **must** be a value less than maxCustomBorderColorSamplers
- VUID-vkWriteSamplerDescriptorsEXT-pNext-11400
  If there is a VkDebugUtilsObjectNameInfoEXT structure in the pNext chain of any element of pSamplers, its objectType **must** be VK_OBJECT_TYPE_UNKNOWN
- VUID-vkWriteSamplerDescriptorsEXT-pDescriptors-12531
  Each element of pDescriptors **must** specify a unique, non-overlapping address range

Valid Usage (Implicit)

- VUID-vkWriteSamplerDescriptorsEXT-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkWriteSamplerDescriptorsEXT-pSamplers-parameter
  pSamplers **must** be a valid pointer to an array of samplerCount valid VkSamplerCreateInfo structures
- VUID-vkWriteSamplerDescriptorsEXT-pDescriptors-parameter
  pDescriptors **must** be a valid pointer to an array of samplerCount valid VkHostAddressRangeEXT structures
- VUID-vkWriteSamplerDescriptorsEXT-samplerCount-arraylength
  samplerCount **must** be greater than 0

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To write resource descriptors to memory, call:

```
// Provided by VK_EXT_descriptor_heap
VkResult vkWriteResourceDescriptorsEXT(
    VkDevice                                    device,
    uint32_t                                    resourceCount,
    const VkResourceDescriptorInfoEXT*          pResources,
    const VkHostAddressRangeEXT*                pDescriptors);
```

- `device` is the logical device that the descriptors are for.
- `resourceCount` is the number of elements in `pResources` and `pDescriptors`.
- `pResources` is a pointer to an array of VkResourceDescriptorInfoEXT structures defining properties of the resource descriptors that will be written.
- `pDescriptors` is a pointer to an array of VkHostAddressRangeEXT structures defining the host address ranges that will be written to for each descriptor.

Each descriptor will be written to `pDescriptors`[i].address where i is the index of its create info in `pResources`.

If any image descriptor written by this command includes a VkSamplerYcbcrConversion, multiple descriptors will be written adjacent to each other for that descriptor, equal to VkSamplerYcbcrConversionImageFormatProperties::`combinedImageSamplerDescriptorCount` for the image .

If any image descriptor written by this command is for an image created with `flags` containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT, multiple descriptors will be written adjacent to each other for that descriptor, equal to VkSubsampledImageFormatPropertiesEXT ::`subsampledImageDescriptorCount` for the image.

Descriptors using the same `type` and written using a fully identical VkTexelBufferDescriptorInfoEXT or VkDeviceAddressRangeKHR structure on the same VkDevice will always return the same bit pattern. If the descriptorHeapCaptureReplay feature is enabled, this applies to any VkDevice created with identical parameters from the same VkPhysicalDevice.

NOTE Recreating the same buffer descriptor during replay of a prior capture requires that the device address is the same, which requires additional data to be captured and provided during replay when creating a buffer and allocating memory for it.

Image descriptors using the same `type` and written using a fully identical

VkImageDescriptorInfoEXT other than VkImageDescriptorInfoEXT::pView->image, where image was successfully created with VK_IMAGE_CREATE_DESCRIPTOR_HEAP_CAPTURE_REPLAY_BIT_EXT and a VkOpaqueCaptureDataCreateInfoEXT with data captured via vkGetImageOpaqueCaptureDataEXT from an image used previously, will write a descriptor with the same bit pattern if possible; if the same bit pattern cannot be generated, VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS will be returned instead.

NOTE

Image creation is sufficiently complex that it may not be possible to recreate all possible descriptors from an image during replay, even if the image itself was successfully recreated. The conditions for this happening will be largely the same as those which could cause allocating a buffer with the same device address during replay to fail. Replay tools are advised to recreate captured descriptors for an image immediately after recreating the image itself wherever possible.

Valid Usage

- VUID-vkWriteResourceDescriptorsEXT-descriptorHeap-11206
  The descriptorHeap feature **must** be enabled
- VUID-vkWriteResourceDescriptorsEXT-size-11207
  The size member of each element of pDescriptors **must** be greater than or equal to the value returned by vkGetPhysicalDeviceDescriptorSizeEXT with a descriptorType equal to type
- VUID-vkWriteResourceDescriptorsEXT-pResources-11208
  If any element of pResources specifies a VkImageViewCreateInfo structure with a VkSamplerYcbcrConversionInfo structure in its pNext chain, the corresponding element of pDescriptors **must** have a size member that is greater than or equal to the product of the value returned by vkGetPhysicalDeviceDescriptorSizeEXT with a descriptorType equal to type and VkSamplerYcbcrConversionImageFormatProperties::combinedImageSamplerDescriptorCount, as queried from VkPhysicalDeviceImageFormatInfo2 with image format info equivalent to the image view the descriptor is being created for
- VUID-vkWriteResourceDescriptorsEXT-pResources-11209
  If any element of pResources specifies a VkImageViewCreateInfo structure with an image created with flags containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT, the corresponding element of pDescriptors **must** have a size member that is greater than or equal to the product of the value returned by vkGetPhysicalDeviceDescriptorSizeEXT with a descriptorType equal to type and VkSubsampledImageFormatPropertiesEXT::subsampledImageDescriptorCount, as queried from VkPhysicalDeviceImageFormatInfo2 with image format info equivalent to the image view the descriptor is being created for
- VUID-vkWriteResourceDescriptorsEXT-pResources-12494
  If any element of pResources specifies a VkImageViewCreateInfo structure, device **must** support at least one queue family with one of the VK_QUEUE_VIDEO_ENCODE_BIT_KHR, VK_QUEUE_VIDEO_DECODE_BIT_KHR, VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT capabilities
- VUID-vkWriteResourceDescriptorsEXT-pResources-12495

If any element of pResources specifies a VkImageViewCreateInfo structure, VkImageViewCreateInfo::image **must** have been created from device

- VUID-vkWriteResourceDescriptorsEXT-pDescriptors-12532
  Each element of pDescriptors **must** specify a unique, non-overlapping address range

Valid Usage (Implicit)

- VUID-vkWriteResourceDescriptorsEXT-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkWriteResourceDescriptorsEXT-pResources-parameter
  pResources **must** be a valid pointer to an array of resourceCount valid VkResourceDescriptorInfoEXT structures
- VUID-vkWriteResourceDescriptorsEXT-pDescriptors-parameter
  pDescriptors **must** be a valid pointer to an array of resourceCount valid VkHostAddressRangeEXT structures
- VUID-vkWriteResourceDescriptorsEXT-resourceCount-arraylength
  resourceCount **must** be greater than 0

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

VkResourceDescriptorInfoEXT is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkResourceDescriptorInfoEXT {
    VkStructureType                 sType;
    const void*                     pNext;
    VkDescriptorType                type;
    VkResourceDescriptorDataEXT     data;
} VkResourceDescriptorInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.

- `type` is the type of descriptor to get.
- `data` is a VkResourceDescriptorDataEXT union defining the properties of a resource descriptor according to `type`.

If `type` is VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, `data->pTexelBuffer` is used to construct the descriptor.

If `type` is VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR, VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, `data->pAddressRange` is used to construct the descriptor. For acceleration structures, the size of the range is not used by the descriptor, and **can** be set to 0. If a non-zero size is provided though, it **must** be a valid range.

> **NOTE** Applications may wish to provide a valid range as a way to check their own assumptions about the range they are binding; but it has no bearing on anything except validation. Implementations cannot make any assumptions based on the size of the provided range.

If `type` is VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, `data->pImage` is used to construct the descriptor.

If the `nullDescriptor` feature is enabled, the corresponding element of `data` **may** be `NULL` to generate a null descriptor.

Applications **can** give resource descriptors a debug name in a similar way to naming an object, via the VkDebugUtilsObjectNameInfoEXT structure. However, as there is no actual object, this structure **must** be passed via the `pNext` chain of this structure, with a `objectType` of VK_OBJECT_TYPE_UNKNOWN and a `objectHandle` of VK_NULL_HANDLE. The name is attached to the unique set of descriptor bits written by the implementation, and writing the same bits again with new debug info **may** rename the original descriptor.

> **NOTE** Implementations are not prevented from returning the same bits for different descriptors. This can result in multiple different resources mapping to the same name. A common case for this might be something like a uniform buffer and storage buffer with the same device address range.

If a descriptor becomes invalid due to the underlying resource becoming invalid, implementations **may** remove the name association.

Valid Usage

- VUID-VkResourceDescriptorInfoEXT-type-11210
  `type` **must** be one of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER, VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER,

VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT

- VUID-VkResourceDescriptorInfoEXT-None-11211
  If the nullDescriptor feature is not enabled, and type is VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE or VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, data->pImage **must** not be NULL
- VUID-VkResourceDescriptorInfoEXT-type-11469
  If type is VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, data->pImage **must** not be NULL
- VUID-VkResourceDescriptorInfoEXT-None-11212
  If the nullDescriptor feature is not enabled, and type is VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, data->pTexelBuffer **must** not be NULL
- VUID-VkResourceDescriptorInfoEXT-None-11213
  If the nullDescriptor feature is not enabled, and type is VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR, VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, data->pAddressRange **must** not be NULL
- VUID-VkResourceDescriptorInfoEXT-type-12349
  If type is VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, data->pTexelBuffer->addressRange.address **must** be a multiple of the effective alignment requirement of data->pTexelBuffer->format as determined by minTexelBufferOffsetAlignment
- VUID-VkResourceDescriptorInfoEXT-type-12350
  If type is VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, data->pAddressRange->address **must** be a multiple of minUniformBufferOffsetAlignment
- VUID-VkResourceDescriptorInfoEXT-type-12351
  If type is VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, data->pAddressRange->address **must** be a multiple of minStorageBufferOffsetAlignment
- VUID-VkResourceDescriptorInfoEXT-type-11454
  If type is one of
  - VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR

    data->pAddressRange->address **must** be a multiple of 256
- VUID-VkResourceDescriptorInfoEXT-type-12496
  If type is VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, data->pAddressRange->size **must** be less than or equal to maxUniformBufferRange
- VUID-VkResourceDescriptorInfoEXT-type-12497
  If type is VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, and the shader64BitIndexing feature is not enabled, data->pAddressRange->size **must** be less than or equal to maxStorageBufferRange
- VUID-VkResourceDescriptorInfoEXT-pNext-11401
  If there is a VkDebugUtilsObjectNameInfoEXT structure in the pNext chain, its objectType **must** be VK_OBJECT_TYPE_UNKNOWN

- VUID-VkResourceDescriptorInfoEXT-type-11422
  If `type` is VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, `data.pImage->pView->image` **must** not have been created with an `imageType` of VK_IMAGE_TYPE_3D
- VUID-VkResourceDescriptorInfoEXT-type-11424
  If `type` is VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, `data.pImage->pView->viewType` is VK_IMAGE_VIEW_TYPE_2D, and the `image2DViewOf3D` feature is not enabled, `data.pImage->pView->image` **must** not have been created with an `imageType` of VK_IMAGE_TYPE_3D
- VUID-VkResourceDescriptorInfoEXT-type-11425
  If `type` is VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, `data.pImage->pView->viewType` is VK_IMAGE_VIEW_TYPE_2D, and the `sampler2DViewOf3D` feature is not enabled, `data.pImage->pView->image` **must** not have been created with an `imageType` of VK_IMAGE_TYPE_3D
- VUID-VkResourceDescriptorInfoEXT-type-11433
  If `type` is VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, `data.pAddressRange->size` **must** not be 0
- VUID-VkResourceDescriptorInfoEXT-type-11458
  If `type` is VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE and `data.pImage` is not `NULL`, the effective `usage` of the VkImageViewCreateInfo **must** include the VK_IMAGE_USAGE_SAMPLED_BIT usage flag
- VUID-VkResourceDescriptorInfoEXT-type-11459
  If `type` is VK_DESCRIPTOR_TYPE_STORAGE_IMAGE and `data.pImage` is not `NULL`, the effective `usage` of the VkImageViewCreateInfo **must** include the VK_IMAGE_USAGE_STORAGE_BIT usage flag
- VUID-VkResourceDescriptorInfoEXT-type-11460
  If `type` is VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT and `data.pImage` is not `NULL`, the effective `usage` of the VkImageViewCreateInfo **must** include the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flag
- VUID-VkResourceDescriptorInfoEXT-type-11461
  If `type` is VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER and `data.pAddressRange` is not `NULL`, `data.pAddressRange` **must** be a device address range allocated to the application from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-VkResourceDescriptorInfoEXT-type-11462
  If `type` is VK_DESCRIPTOR_TYPE_STORAGE_BUFFER and `data.pAddressRange` is not `NULL`, `data.pAddressRange` **must** be a device address range allocated to the application from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-VkResourceDescriptorInfoEXT-type-11463
  If `type` is VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER and `data.pTexelBuffer` is not `NULL`, `data.pTexelBuffer->addressRange` **must** be a device address range allocated to the application from a buffer created with the VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT usage flag set
- VUID-VkResourceDescriptorInfoEXT-type-11464
  If `type` is VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER and `data.pTexelBuffer` is not `NULL`, `data.pTexelBuffer->addressRange` **must** be a device address range allocated to the application from a buffer created with the

VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT usage flag set

- VUID-VkResourceDescriptorInfoEXT-type-11483
  If `type` is VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR, and `data.pAddressRange` is not `NULL`, `data.pAddressRange->address` **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-VkResourceDescriptorInfoEXT-type-11484
  If `type` is VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR, `data.pAddressRange` is not `NULL`, and `data.pAddressRange->size` is not 0, `data.pAddressRange` **must** be a device address range allocated to the application from the buffer used to create the acceleration structure that `data.pAddressRange->address` was retrieved from, and within the buffer range bound to that acceleration structure

Valid Usage (Implicit)

- VUID-VkResourceDescriptorInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RESOURCE_DESCRIPTOR_INFO_EXT
- VUID-VkResourceDescriptorInfoEXT-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkDebugUtilsObjectNameInfoEXT
- VUID-VkResourceDescriptorInfoEXT-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkResourceDescriptorInfoEXT-type-parameter
  `type` **must** be a valid VkDescriptorType value
- VUID-VkResourceDescriptorInfoEXT-pImage-parameter
  If `type` is VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, VK_DESCRIPTOR_TYPE_BLOCK_MATCH_IMAGE_QCOM, or VK_DESCRIPTOR_TYPE_SAMPLE_WEIGHT_IMAGE_QCOM, and if `pImage` is not `NULL`, the `pImage` member of `data` **must** be a valid pointer to a valid VkImageDescriptorInfoEXT structure
- VUID-VkResourceDescriptorInfoEXT-pTexelBuffer-parameter
  If `type` is VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, and if `pTexelBuffer` is not `NULL`, the `pTexelBuffer` member of `data` **must** be a valid pointer to a valid VkTexelBufferDescriptorInfoEXT structure
- VUID-VkResourceDescriptorInfoEXT-pAddressRange-parameter
  If `type` is VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR, VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, and if `pAddressRange` is not `NULL`, the `pAddressRange` member of `data` **must** be a valid pointer to a valid VkDeviceAddressRangeEXT structure
- VUID-VkResourceDescriptorInfoEXT-pTensorARM-parameter
  If `type` is VK_DESCRIPTOR_TYPE_TENSOR_ARM, and if `pTensorARM` is not `NULL`, the `pTensorARM` member of `data` **must** be a valid pointer to a valid

VkTensorViewCreateInfoARM structure

The VkResourceDescriptorDataEXT union is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef union VkResourceDescriptorDataEXT {
    const VkImageDescriptorInfoEXT*          pImage;
    const VkTexelBufferDescriptorInfoEXT*    pTexelBuffer;
    const VkDeviceAddressRangeEXT*           pAddressRange;
    const VkTensorViewCreateInfoARM*         pTensorARM;
} VkResourceDescriptorDataEXT;
```

- pImage is a pointer to a VkImageDescriptorInfoEXT structure.
- pTexelBuffer is a pointer to a VkTexelBufferDescriptorInfoEXT structure.
- pAddressRange is a pointer to a VkDeviceAddressRangeKHR structure.

VkTexelBufferDescriptorInfoEXT is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkTexelBufferDescriptorInfoEXT {
    VkStructureType            sType;
    const void*                pNext;
    VkFormat                   format;
    VkDeviceAddressRangeEXT    addressRange;
} VkTexelBufferDescriptorInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- format is the VkFormat of the descriptor.
- addressRange is a VkDeviceAddressRangeKHR defining the range of data backing the descriptor.

Valid Usage (Implicit)

- VUID-VkTexelBufferDescriptorInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_TEXEL_BUFFER_DESCRIPTOR_INFO_EXT
- VUID-VkTexelBufferDescriptorInfoEXT-pNext-pNext
  pNext **must** be NULL
- VUID-VkTexelBufferDescriptorInfoEXT-format-parameter
  format **must** be a valid VkFormat value

VkImageDescriptorInfoEXT is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkImageDescriptorInfoEXT {
    VkStructureType                 sType;
    const void*                     pNext;
    const VkImageViewCreateInfo*    pView;
    VkImageLayout                   layout;
} VkImageDescriptorInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pView` is an VkImageViewCreateInfo describing the descriptor.
- `layout` is the VkImageLayout that the image view will be in when accessed as a descriptor.

Valid Usage

- VUID-VkImageDescriptorInfoEXT-pView-11426
  `pView->viewType` **must** not be VK_IMAGE_VIEW_TYPE_2D_ARRAY if `pView->image` was created with an `imageType` of VK_IMAGE_TYPE_3D
- VUID-VkImageDescriptorInfoEXT-pView-11427
  If `pView->viewType` is VK_IMAGE_VIEW_TYPE_2D and `pView->image` was created with an `imageType` of VK_IMAGE_TYPE_3D, `pView->image` **must** have been created with VK_IMAGE_CREATE_2D_VIEW_COMPATIBLE_BIT_EXT set
- VUID-VkImageDescriptorInfoEXT-layout-11219
  `layout` **must** be VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR, VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT, or VK_IMAGE_LAYOUT_GENERAL
- VUID-VkImageDescriptorInfoEXT-layout-11221
  If `layout` is VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL, or VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL, then `pView->subresourceRange.aspectMask` **must** not include VK_IMAGE_ASPECT_COLOR_BIT
- VUID-VkImageDescriptorInfoEXT-pView-11430
  If `pView->image` is a depth/stencil image, `pView->subresourceRange.aspectMask` **must** include either VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT but not both

Valid Usage (Implicit)

- VUID-VkImageDescriptorInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_DESCRIPTOR_INFO_EXT
- VUID-VkImageDescriptorInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkImageDescriptorInfoEXT-pView-parameter
  `pView` **must** be a valid pointer to a valid VkImageViewCreateInfo structure
- VUID-VkImageDescriptorInfoEXT-layout-parameter
  `layout` **must** be a valid VkImageLayout value

16.2. Using Heaps

One descriptor heap of each type can be bound to a command buffer for use with shaders. When using descriptor heaps, pipelines **must** be created with VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT and shaders created with VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT in order to make use of the heaps.

When any heap state command is recorded to a command buffer, it immediately invalidates all descriptor set and descriptor buffer state set by vkCmdBindDescriptorSets2, vkCmdPushDescriptorSet2, vkCmdPushDescriptorSetWithTemplate2, vkCmdSetDescriptorBufferOffsets2EXT, vkCmdBindDescriptorBufferEmbeddedSamplers2EXT, vkCmdPushConstants2, vkCmdBindDescriptorSets, vkCmdPushDescriptorSet, vkCmdPushDescriptorSetWithTemplate, vkCmdBindDescriptorBuffersEXT, vkCmdSetDescriptorBufferOffsetsEXT, vkCmdBindDescriptorBufferEmbeddedSamplersEXT, or vkCmdPushConstants. Similarly, recording any of these commands immediately invalidates all state set by commands in this chapter.

Implementations **may** require storage in descriptor heaps for their own internal descriptors. Storage for these extra descriptors **must** be allocated by the application as part of each descriptor heap address range, and **must** not be accessed by the application while it is bound as a reserved range, in any command buffer, until all such command buffers are freed or reset. The amount of storage required by an implementation is advertised by `minResourceHeapReservedRange` for the resource heap, and `minSamplerHeapReservedRange` or `minSamplerHeapReservedRangeWithEmbedded` for the sampler heap (the latter being required when using embedded samplers).

Applications **can** set different heaps to use the same address ranges, but **must** take care to ensure that the reserved ranges for each heap do not overlap with each other or with user ranges.

To bind a sampler heap to a command buffer, call:

```
// Provided by VK_EXT_descriptor_heap
void vkCmdBindSamplerHeapEXT(
    VkCommandBuffer                             commandBuffer,
    const VkBindHeapInfoEXT*                    pBindInfo);
```

- `commandBuffer` is the command buffer that the sampler heap will be bound to.
- `pBindInfo` is a VkBindHeapInfoEXT specifying the device address range used for the heap and any implementation reservations.

Addresses in the range defined by `pBindInfo->heapRange` are bound as the sampler heap. The application **can** access samplers and data through this heap anywhere except for the reserved range specified by `pBindInfo->reservedRangeOffset`. Addresses in the range [`pBindInfo->reservedRangeOffset`, `pBindInfo->reservedRangeOffset` + `pBindInfo->reservedRangeSize`) are reserved for the implementation and **must** not be accessed by the application at any time from when this command is recorded until there are no command buffers with that range bound.

The reserved range is bound to the command buffer until it has been reset.

NOTE	Implementations may require a larger sampler heap reservation to store embedded sampler descriptors when used in a mapping, as advertised by `minSamplerHeapReservedRangeWithEmbedded`.

Shaders executed by commands recorded after this command **can** use the specified sampler heap to access resources. `pBindInfo->heapRange.address` will be available to shaders to access samplers and data through the `SamplerHeapEXT` `BuiltIn` or via shader bindings.

When `vkCmdBindSamplerHeapEXT` is recorded, it immediately invalidates all non-heap descriptor state. Similarly, recording any non-heap descriptor state commands immediately invalidates state set by this command.

Valid Usage

- VUID-vkCmdBindSamplerHeapEXT-pBindInfo-11223
  The sum of `pBindInfo->reservedRangeOffset` and `pBindInfo->reservedRangeSize` **must** be less than or equal to `pBindInfo->heapRange.size`
- VUID-vkCmdBindSamplerHeapEXT-pBindInfo-11224
  `pBindInfo->reservedRangeSize` **must** be greater than or equal to `minSamplerHeapReservedRange`
- VUID-vkCmdBindSamplerHeapEXT-pBindInfo-11225
  `pBindInfo->heapRange.size` **must** be less than or equal to `maxSamplerHeapSize`
- VUID-vkCmdBindSamplerHeapEXT-pBindInfo-11226
  `pBindInfo->heapRange.address` **must** be a multiple of `samplerHeapAlignment`
- VUID-vkCmdBindSamplerHeapEXT-pBindInfo-11434
  `pBindInfo->reservedRangeOffset` **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdBindSamplerHeapEXT-pBindInfo-11228
  Memory bound to addresses in the range [`pBindInfo->heapRange.address` + `pBindInfo->reservedRangeOffset`, `pBindInfo->heapRange.address` + `pBindInfo->reservedRangeOffset` + `pBindInfo->reservedRangeSize`) **must** not be bound to any other command buffer as a reserved range for any heap unless the reserved range matches exactly and it is the same heap type

- VUID-vkCmdBindSamplerHeapEXT-heapRange-11230
  `heapRange` **must** be a device address range allocated to the application from a buffer created with the VK_BUFFER_USAGE_DESCRIPTOR_HEAP_BIT_EXT usage flag set
- VUID-vkCmdBindSamplerHeapEXT-commandBuffer-11231
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pSamplerHeapBindInfo` equal to `NULL`

Valid Usage (Implicit)

- VUID-vkCmdBindSamplerHeapEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindSamplerHeapEXT-pBindInfo-parameter
  `pBindInfo` **must** be a valid pointer to a valid VkBindHeapInfoEXT structure
- VUID-vkCmdBindSamplerHeapEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBindSamplerHeapEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindSamplerHeapEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindSamplerHeapEXT is not affected by conditional rendering

To bind a resource heap to a command buffer, call:

```
// Provided by VK_EXT_descriptor_heap
void vkCmdBindResourceHeapEXT(
    VkCommandBuffer                             commandBuffer,
    const VkBindHeapInfoEXT*                    pBindInfo);
```

- `commandBuffer` is the command buffer that the resource heap will be bound to.
- `pBindInfo` is a VkBindHeapInfoEXT specifying the device address range used for the heap and any implementation reservations.

Addresses in the range defined by `pBindInfo->heapRange` are bound as the resource heap. The application **can** access resources and data through this heap anywhere except for the reserved range specified by `pBindInfo->reservedRangeOffset`. Addresses in the range [`pBindInfo->reservedRangeOffset`, `pBindInfo->reservedRangeOffset` + `pBindInfo->reservedRangeSize`) are reserved for the implementation and **must** not be accessed by the application at any time from when this command is recorded until there are no command buffers with that range bound.

The reserved range is bound to the command buffer until it has been reset.

Shaders executed by commands recorded after this command **can** use the specified resource heap to access resources. `pBindInfo->heapRange.address` will be available to shaders to access resources through the `ResourceHeapEXT BuiltIn` or via shader bindings.

When `vkCmdBindResourceHeapEXT` is recorded, it immediately invalidates all non-heap descriptor state. Similarly, recording any non-heap descriptor state commands immediately invalidates state set by this command.

Valid Usage

- VUID-vkCmdBindResourceHeapEXT-pBindInfo-11232
  The sum of `pBindInfo->reservedRangeOffset` and `pBindInfo->reservedRangeSize` **must** be less than or equal to `pBindInfo->heapRange.size`
- VUID-vkCmdBindResourceHeapEXT-pBindInfo-11233
  `pBindInfo->reservedRangeSize` **must** be greater than or equal to `minResourceHeapReservedRange`
- VUID-vkCmdBindResourceHeapEXT-pBindInfo-11234
  `pBindInfo->heapRange.size` **must** be less than or equal to `maxResourceHeapSize`
- VUID-vkCmdBindResourceHeapEXT-pBindInfo-11235
  `pBindInfo->heapRange.address` **must** be a multiple of `resourceHeapAlignment`
- VUID-vkCmdBindResourceHeapEXT-pBindInfo-11435

`pBindInfo->reservedRangeOffset` **must** be a multiple of `bufferDescriptorAlignment`

- VUID-vkCmdBindResourceHeapEXT-pBindInfo-11436
  `pBindInfo->reservedRangeOffset` **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdBindResourceHeapEXT-pBindInfo-11236
  Memory bound to addresses in the range [`pBindInfo->heapRange.address` + `pBindInfo->reservedRangeOffset`, `pBindInfo->heapRange.address` + `pBindInfo->reservedRangeOffset` + `pBindInfo->reservedRangeSize`) **must** not be bound to any other command buffer as a reserved range for any heap unless the reserved range matches exactly and it is the same heap type
- VUID-vkCmdBindResourceHeapEXT-heapRange-11237
  `heapRange` **must** be a device address range allocated to the application from a buffer created with the VK_BUFFER_USAGE_DESCRIPTOR_HEAP_BIT_EXT usage flag set
- VUID-vkCmdBindResourceHeapEXT-commandBuffer-11238
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pResourceHeapBindInfo` equal to `NULL`

Valid Usage (Implicit)

- VUID-vkCmdBindResourceHeapEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindResourceHeapEXT-pBindInfo-parameter
  `pBindInfo` **must** be a valid pointer to a valid VkBindHeapInfoEXT structure
- VUID-vkCmdBindResourceHeapEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBindResourceHeapEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindResourceHeapEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindResourceHeapEXT is not affected by conditional rendering

`VkBindHeapInfoEXT` is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkBindHeapInfoEXT {
    VkStructureType            sType;
    const void*                pNext;
    VkDeviceAddressRangeEXT    heapRange;
    VkDeviceSize               reservedRangeOffset;
    VkDeviceSize               reservedRangeSize;
} VkBindHeapInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `heapRange` is a VkDeviceAddressRangeKHR defining the device address range used for the heap, inclusive of the implementation reserved range.
- `reservedRangeOffset` is the offset within `heapRange` to the start of the reserved range for the implementation.
- `reservedRangeSize` is the size of the reserved range for the implementation within `heapRange`.

Valid Usage (Implicit)

- VUID-VkBindHeapInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BIND_HEAP_INFO_EXT
- VUID-VkBindHeapInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`

`SamplerHeapEXT`

Decorating a variable with the `SamplerHeapEXT` built-in decoration will back it with the contents of

the sampler heap bound by vkCmdBindSamplerHeapEXT.

Valid Usage

- VUID-SamplerHeapEXT-SamplerHeapEXT-11239
  The variable decorated with `SamplerHeapEXT` **must** be declared using the `UniformConstant` Storage Class

`ResourceHeapEXT`

Decorating a variable with the `ResourceHeapEXT` built-in decoration will back it with the contents of the resource heap bound by vkCmdBindResourceHeapEXT.

Valid Usage

- VUID-ResourceHeapEXT-ResourceHeapEXT-11241
  The variable decorated with `ResourceHeapEXT` **must** be declared using the `UniformConstant` Storage Class

16.2.1. Push Data

Push constants specified by vkCmdPushConstants or vkCmdPushConstants2KHR rely on descriptor set layout state, and are not compatible with descriptor heaps. A new push interface is provided for use with descriptor heaps:

To update push data when using descriptor heaps, call:

```
// Provided by VK_EXT_descriptor_heap
void vkCmdPushDataEXT(
    VkCommandBuffer                             commandBuffer,
    const VkPushDataInfoEXT*                    pPushDataInfo);
```

- `commandBuffer` is the command buffer in which the push data update will be recorded.
- `pPushDataInfo` is a pointer to a VkPushDataInfoEXT structure.

When `vkCmdPushDataEXT` is recorded, it immediately invalidates all non-heap descriptor state. Similarly, recording any non-heap descriptor state commands immediately invalidates state set by this command.

All push data is available to all shaders using the existing `PushConstant` Storage Class.

NOTE Device addresses in push data are intended as the fast path for shader-constant data that does not fit into push data directly. These addresses can also be mapped to an existing buffer declaration in the shader using set and binding decorations, which will be the preferred path for some implementations initially, though such implementations are expected to lean less on this mechanism over time.

Valid Usage (Implicit)

- VUID-vkCmdPushDataEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdPushDataEXT-pPushDataInfo-parameter
  `pPushDataInfo` **must** be a valid pointer to a valid VkPushDataInfoEXT structure
- VUID-vkCmdPushDataEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdPushDataEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdPushDataEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdPushDataEXT is not affected by conditional rendering

The `VkPushDataInfoEXT` structure is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkPushDataInfoEXT {
    VkStructureType               sType;
    const void*                   pNext;
    uint32_t                      offset;
    VkHostAddressRangeConstEXT    data;
} VkPushDataInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `offset` is the start offset of the push data range to update, in units of bytes.
- `data` is the host address range containing the push data to update.

Valid Usage

- VUID-VkPushDataInfoEXT-offset-11243
  The sum of `offset` and `data.size` **must** be less than or equal to `maxPushDataSize`
- VUID-VkPushDataInfoEXT-offset-11418
  `offset` **must** be a multiple of 4
- VUID-VkPushDataInfoEXT-data-11419
  `data.size` **must** be a multiple of 4

Valid Usage (Implicit)

- VUID-VkPushDataInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PUSH_DATA_INFO_EXT
- VUID-VkPushDataInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkPushDataInfoEXT-data-parameter
  `data` **must** be a valid VkHostAddressRangeConstEXT structure

16.3. Shader Bindings

While descriptor heaps can be accessed directly through the `SamplerHeapEXT` and `ResourceHeapEXT` built-ins, shaders using the existing `DescriptorSet` and `Binding` decorations **can** map these to heap offsets. In place of descriptor set layouts and pipeline layouts, information **can** be provided at pipeline or shader creation time to indicate how these bindings are mapped, through a combination of constants, push data, and indirections through device addresses. This interface provides significantly more flexibility than descriptor set layouts, enabling applications to specify precisely where they expect each descriptor to be.

The `VkShaderDescriptorSetAndBindingMappingInfoEXT` structure is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkShaderDescriptorSetAndBindingMappingInfoEXT {
    VkStructureType                                 sType;
    const void*                                     pNext;
    uint32_t                                        mappingCount;
    const VkDescriptorSetAndBindingMappingEXT*      pMappings;
} VkShaderDescriptorSetAndBindingMappingInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `mappingCount` is the number of elements in `pMappings`.
- `pMappings` is a pointer to an array of VkDescriptorSetAndBindingMappingEXT structures specifying mappings for a set of descriptors.

Including this structure in the `pNext` chain of VkPipelineShaderStageCreateInfo will set mappings for the shader defined by that structure. Similarly, including this structure in the `pNext` chain of a VkShaderCreateInfoEXT with a `codeType` of VK_SHADER_CODE_TYPE_SPIRV_EXT will set mappings for that shader.

If this structure is not present, it is equivalent to setting `mappingCount` to 0.

Valid Usage

- VUID-VkShaderDescriptorSetAndBindingMappingInfoEXT-pMappings-11244
  Any two elements of `pMappings` **must** not have the same value of `descriptorSet`, an overlapping range specified by `firstBinding` and `bindingCount`, and any overlapping bits in `resourceMask`

Valid Usage (Implicit)

- VUID-VkShaderDescriptorSetAndBindingMappingInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SHADER_DESCRIPTOR_SET_AND_BINDING_MAPPING_INFO_EXT
- VUID-VkShaderDescriptorSetAndBindingMappingInfoEXT-pMappings-parameter
  If `mappingCount` is not `0`, `pMappings` **must** be a valid pointer to an array of `mappingCount` valid VkDescriptorSetAndBindingMappingEXT structures

Structure Chaining

Extends the structures

- VkPipelineShaderStageCreateInfo
- VkShaderCreateInfoEXT

The VkDescriptorSetAndBindingMappingEXT structure is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkDescriptorSetAndBindingMappingEXT {
    VkStructureType                     sType;
    const void*                         pNext;
    uint32_t                            descriptorSet;
    uint32_t                            firstBinding;
    uint32_t                            bindingCount;
    VkSpirvResourceTypeFlagsEXT         resourceMask;
    VkDescriptorMappingSourceEXT        source;
    VkDescriptorMappingSourceDataEXT    sourceData;
} VkDescriptorSetAndBindingMappingEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- descriptorSet is the value of DescriptorSet for resources that this mapping affects.
- firstBinding is the first value of Binding of resources that this mapping affects.
- bindingCount is the number of consecutive Binding values of resources that this mapping affects.
- resourceMask is a mask of VkSpirvResourceTypeFlagBitsEXT values indicating which resource types are specified by this mapping.
- source is a VkDescriptorMappingSourceEXT value specifying the method of mapping specified for the affected resources.
- sourceData is a VkDescriptorMappingSourceDataEXT that provides the details of how each mapping is specified according to source.

Resources specified in a shader with a DescriptorSet decoration set to descriptorSet, a Binding decoration greater than or equal to firstBinding and less than the sum of firstBinding and bindingCount, and a resource type matching one of the bits in resourceMask will be mapped according to source and sourceData.

NOTE

Applications are free to overspecify bindings that are not present; allowing reuse of the same mapping structures with multiple shaders, even when those shaders only partially reuse those mappings. This includes things like setting binding counts higher than the number used in the shader, specifying bindings that are not present in the shader, and setting resourceMask to all possible resources that may be encountered.

If source selects an element of sourceData defined by a structure, the description of that structure defines how resources are mapped. Source mappings using a single base type are defined here.

If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_CONSTANT_OFFSET_EXT, the resource will be backed by heap data as specified by constantOffset.

If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, the resource will

be backed by heap data as specified by pushIndex.

If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, the resource will be backed by heap data as specified by indirectIndex.

If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the resource will be backed by heap data as specified by indirectIndexArray.

If source is VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, the resource will be backed by heap data as specified by heapData.

If source is VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, the resource will be backed by push data at a range from pushDataOffset to the size of the resource, allowing a uniform buffer to be backed by push data access push data. Accessing data in the uniform buffer at an offset of shaderOffset in the shader will access push data at an offset equal to

offset = shaderOffset + pushDataOffset.

If source is VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, the resource will be backed by data pointed to by a device address in push data at an offset of pushAddressOffset. Accessing data via the mapped resource in the shader will access data backing the address specified in push data:

address = ((VkDeviceAddress*)pPushData)[pushAddressOffset/8]

where pPushData is the total set of push data specified by vkCmdPushDataEXT. If the shader resource is an acceleration structure, the address **must** be a valid acceleration structure address.

If source is VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the resource will be backed by heap data as specified by indirectAddress.

NOTE Accesses to resources using mappings to anything that is not a descriptor in a heap are not subject to robustness guarantees; resources for such mappings must not be accessed out of bounds.

If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, the resource will be backed by heap data as specified by shaderRecordIndex.

If source is VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, the resource will be backed by shader record data at a range from shaderRecordDataOffset to the size of the resource, allowing a uniform buffer to be used as a way to access shader record data. Accessing data in the uniform buffer at an offset shaderOffset in the shader will access shader record data at an offset equal to

offset = shaderOffset + shaderRecordDataOffset.

If source is VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, the resource will be backed by data pointed to by a device address in the shader record at

`shaderRecordAddressOffset`. Accessing data via the mapped resource in the shader will access data backing the address specified in shader record data:

```
address = ((VkDeviceAddress*)pShaderRecordData)[shaderRecordAddressOffset/8]
```

where pShaderRecordData is the memory associated with a given shader as its shader record. If the shader resource is an acceleration structure, the address **must** be a valid acceleration structure address.

NOTE	Accesses to resources using VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT mappings are not subject to robustness guarantees; data must not be accessed outside of the allocated memory range.

Mappings must be declared for all variables with a `DescriptorSet` and `Binding` in the shader resource interface.

Valid Usage

- VUID-VkDescriptorSetAndBindingMappingEXT-source-11245
  If `source` is VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, `bindingCount` **must** be 1
- VUID-VkDescriptorSetAndBindingMappingEXT-source-11246
  If `source` is VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, `sourceData.pushDataOffset` **must** be a multiple of 4
- VUID-VkDescriptorSetAndBindingMappingEXT-source-12452
  If `source` is VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, `sourceData.pushDataOffset` **must** be less than or equal to `maxPushDataSize` - 4
- VUID-VkDescriptorSetAndBindingMappingEXT-source-11247
  If `source` is VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, `sourceData.pushAddressOffset` **must** be a multiple of 8
- VUID-VkDescriptorSetAndBindingMappingEXT-source-12453
  If `source` is VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, `sourceData.pushAddressOffset` **must** be less than or equal to `maxPushDataSize` - 8
- VUID-VkDescriptorSetAndBindingMappingEXT-source-11248
  If `source` is VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, `bindingCount` **must** be 1
- VUID-VkDescriptorSetAndBindingMappingEXT-source-11249
  If `source` is VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, `sourceData.shaderRecordDataOffset` **must** be a multiple of 4

- VUID-VkDescriptorSetAndBindingMappingEXT-source-11250
  If `source` is `VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT`, `sourceData.shaderRecordAddressOffset` **must** be a multiple of 8
- VUID-VkDescriptorSetAndBindingMappingEXT-source-11251
  If `source` is `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_CONSTANT_OFFSET_EXT`, `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT`, `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT`, `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT`, or `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT`, and `descriptorSet`, `firstBinding`, and `bindingCount` identify any `OpTypeImage` variables, any `heapOffset`, and `heapArrayStride` members of the corresponding member of `sourceData` **must** be 0 or a multiple of `imageDescriptorAlignment`
- VUID-VkDescriptorSetAndBindingMappingEXT-source-11252
  If `source` is `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_CONSTANT_OFFSET_EXT`, `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT`, `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT`, `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT`, or `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT`, and `descriptorSet`, `firstBinding`, and `bindingCount` identify any `OpTypeStruct` variables, any `heapOffset`, and `heapArrayStride` members of the corresponding member of `sourceData` **must** be 0 or a multiple of `bufferDescriptorAlignment`
- VUID-VkDescriptorSetAndBindingMappingEXT-source-11253
  If `source` is `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_CONSTANT_OFFSET_EXT`, `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT`, `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT`, `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT`, or `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT`, and `descriptorSet`, `firstBinding`, and `bindingCount` identify any `OpTypeSampler` variables, any `heapOffset` and `heapArrayStride` members of the corresponding member of `sourceData` **must** be 0 or a multiple of `samplerDescriptorAlignment`
- VUID-VkDescriptorSetAndBindingMappingEXT-source-11254
  If `source` is `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_CONSTANT_OFFSET_EXT`, `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT`, `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT`, `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT`, or `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT`, and `descriptorSet`, `firstBinding`, and `bindingCount` identify any `OpTypeSampledImage` variables, and `pEmbeddedSampler` is `NULL`, then any `samplerHeapOffset` and `samplerHeapArrayStride` members of the corresponding member of `sourceData` **must** be 0 or a multiple of `samplerDescriptorAlignment`
- VUID-VkDescriptorSetAndBindingMappingEXT-source-12406
  If `source` is `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_CONSTANT_OFFSET_EXT`, `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT`, `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT`, `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT`, or `VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT`, and

descriptorSet, firstBinding, and bindingCount identify any OpTypeSampledImage variables, any heapOffset and heapArrayStride members of the corresponding member of sourceData **must** be 0 or a multiple of imageDescriptorAlignment

- VUID-VkDescriptorSetAndBindingMappingEXT-source-12456
  If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, and descriptorSet, firstBinding, and bindingCount identify any OpTypeSampledImage variables, and pEmbeddedSampler is NULL, samplerPushOffset **must** be a multiple of 4
- VUID-VkDescriptorSetAndBindingMappingEXT-source-12457
  If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, and descriptorSet, firstBinding, and bindingCount identify any OpTypeSampledImage variables, and pEmbeddedSampler is NULL, samplerPushOffset **must** be less than or equal to maxPushDataSize - 4
- VUID-VkDescriptorSetAndBindingMappingEXT-source-12458
  If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, and descriptorSet, firstBinding, and bindingCount identify any OpTypeSampledImage variables, and pEmbeddedSampler is NULL, samplerPushOffset **must** be a multiple of 8
- VUID-VkDescriptorSetAndBindingMappingEXT-source-12459
  If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, and descriptorSet, firstBinding, and bindingCount identify any OpTypeSampledImage variables, and pEmbeddedSampler is NULL, samplerPushOffset **must** be less than or equal to maxPushDataSize - 8
- VUID-VkDescriptorSetAndBindingMappingEXT-source-12460
  If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, and descriptorSet, firstBinding, and bindingCount identify any OpTypeSampledImage variables, and pEmbeddedSampler is NULL, samplerAddressOffset **must** be a multiple of 4
- VUID-VkDescriptorSetAndBindingMappingEXT-source-11356
  If source is VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, resourceMask **must** include VK_SPIRV_RESOURCE_TYPE_UNIFORM_BUFFER_BIT_EXT
- VUID-VkDescriptorSetAndBindingMappingEXT-source-11357
  If source is VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, resourceMask **must** include at least one of VK_SPIRV_RESOURCE_TYPE_UNIFORM_BUFFER_BIT_EXT, VK_SPIRV_RESOURCE_TYPE_READ_ONLY_STORAGE_BUFFER_BIT_EXT, VK_SPIRV_RESOURCE_TYPE_READ_WRITE_STORAGE_BUFFER_BIT_EXT, or VK_SPIRV_RESOURCE_TYPE_ACCELERATION_STRUCTURE_BIT_EXT
- VUID-VkDescriptorSetAndBindingMappingEXT-source-11358
  If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or

VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, and the mapping sets useCombinedImageSamplerIndex to VK_TRUE, resourceMask **must** include at least one of VK_SPIRV_RESOURCE_TYPE_COMBINED_SAMPLED_IMAGE_BIT_EXT, VK_SPIRV_RESOURCE_TYPE_SAMPLED_IMAGE_BIT_EXT, or VK_SPIRV_RESOURCE_TYPE_SAMPLER_BIT_EXT

- VUID-VkDescriptorSetAndBindingMappingEXT-source-11389
  If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_CONSTANT_OFFSET_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, and bindingCount is not 1, the pEmbeddedSampler member of the corresponding mapping structure **must** be NULL

Valid Usage (Implicit)

- VUID-VkDescriptorSetAndBindingMappingEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DESCRIPTOR_SET_AND_BINDING_MAPPING_EXT
- VUID-VkDescriptorSetAndBindingMappingEXT-pNext-pNext
  pNext **must** be NULL
- VUID-VkDescriptorSetAndBindingMappingEXT-resourceMask-parameter
  resourceMask **must** be a valid combination of VkSpirvResourceTypeFlagBitsEXT values
- VUID-VkDescriptorSetAndBindingMappingEXT-source-parameter
  source **must** be a valid VkDescriptorMappingSourceEXT value
- VUID-VkDescriptorSetAndBindingMappingEXT-constantOffset-parameter
  If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_CONSTANT_OFFSET_EXT, the constantOffset member of sourceData **must** be a valid VkDescriptorMappingSourceConstantOffsetEXT structure
- VUID-VkDescriptorSetAndBindingMappingEXT-pushIndex-parameter
  If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, the pushIndex member of sourceData **must** be a valid VkDescriptorMappingSourcePushIndexEXT structure
- VUID-VkDescriptorSetAndBindingMappingEXT-indirectIndex-parameter
  If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, the indirectIndex member of sourceData **must** be a valid VkDescriptorMappingSourceIndirectIndexEXT structure
- VUID-VkDescriptorSetAndBindingMappingEXT-indirectIndexArray-parameter
  If source is VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the indirectIndexArray member of sourceData **must** be a valid VkDescriptorMappingSourceIndirectIndexArrayEXT structure
- VUID-VkDescriptorSetAndBindingMappingEXT-shaderRecordIndex-parameter
  If source is

VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, the shaderRecordIndex member of sourceData **must** be a valid VkDescriptorMappingSourceShaderRecordIndexEXT structure

Bits which **can** be set in VkDescriptorSetAndBindingMappingEXT::resourceMask, are:

```
// Provided by VK_EXT_descriptor_heap
typedef enum VkSpirvResourceTypeFlagBitsEXT {
    VK_SPIRV_RESOURCE_TYPE_ALL_EXT = 0x7FFFFFFF,
    VK_SPIRV_RESOURCE_TYPE_SAMPLER_BIT_EXT = 0x00000001,
    VK_SPIRV_RESOURCE_TYPE_SAMPLED_IMAGE_BIT_EXT = 0x00000002,
    VK_SPIRV_RESOURCE_TYPE_READ_ONLY_IMAGE_BIT_EXT = 0x00000004,
    VK_SPIRV_RESOURCE_TYPE_READ_WRITE_IMAGE_BIT_EXT = 0x00000008,
    VK_SPIRV_RESOURCE_TYPE_COMBINED_SAMPLED_IMAGE_BIT_EXT = 0x00000010,
    VK_SPIRV_RESOURCE_TYPE_UNIFORM_BUFFER_BIT_EXT = 0x00000020,
    VK_SPIRV_RESOURCE_TYPE_READ_ONLY_STORAGE_BUFFER_BIT_EXT = 0x00000040,
    VK_SPIRV_RESOURCE_TYPE_READ_WRITE_STORAGE_BUFFER_BIT_EXT = 0x00000080,
  // Provided by VK_EXT_descriptor_heap with VK_KHR_ray_tracing_pipeline or
VK_NV_ray_tracing
    VK_SPIRV_RESOURCE_TYPE_ACCELERATION_STRUCTURE_BIT_EXT = 0x00000100,
} VkSpirvResourceTypeFlagBitsEXT;
```

- VK_SPIRV_RESOURCE_TYPE_ALL_EXT specifies that all resource declarations are included.
- VK_SPIRV_RESOURCE_TYPE_SAMPLER_BIT_EXT specifies OpTypeSampler variables.
- VK_SPIRV_RESOURCE_TYPE_SAMPLED_IMAGE_BIT_EXT specifies OpTypeImage variables with a Sampled parameter of 1.
- VK_SPIRV_RESOURCE_TYPE_READ_ONLY_IMAGE_BIT_EXT specifies OpTypeImage variables with a Sampled parameter of 2 and decorated with NonWritable.
- VK_SPIRV_RESOURCE_TYPE_READ_WRITE_IMAGE_BIT_EXT specifies OpTypeImage variables with a Sampled parameter of 2 and not decorated with NonWritable.
- VK_SPIRV_RESOURCE_TYPE_COMBINED_SAMPLED_IMAGE_BIT_EXT specifies OpTypeSampledImage variables.
- VK_SPIRV_RESOURCE_TYPE_UNIFORM_BUFFER_BIT_EXT specifies OpTypeStruct variables in the Uniform storage class decorated with Block.
- VK_SPIRV_RESOURCE_TYPE_READ_ONLY_STORAGE_BUFFER_BIT_EXT specifies OpTypeStruct variables either in the StorageBuffer storage class decorated with Block or in the Uniform storage class decorated with BufferBlock, and decorated with NonWritable.
- VK_SPIRV_RESOURCE_TYPE_READ_WRITE_STORAGE_BUFFER_BIT_EXT specifies OpTypeStruct variables either in the StorageBuffer storage class decorated with Block or in the Uniform storage class decorated with BufferBlock, but not decorated with NonWritable.
- VK_SPIRV_RESOURCE_TYPE_ACCELERATION_STRUCTURE_BIT_EXT specifies OpTypeAccelerationStructureKHR variables.

```
// Provided by VK_EXT_descriptor_heap
typedef VkFlags VkSpirvResourceTypeFlagsEXT;
```

VkSpirvResourceTypeFlagsEXT is a bitmask type for setting a mask of zero or more VkSpirvResourceTypeFlagBitsEXT.

The possible mapping sources for a shader binding are:

```
// Provided by VK_EXT_descriptor_heap
typedef enum VkDescriptorMappingSourceEXT {
    VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_CONSTANT_OFFSET_EXT = 0,
    VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT = 1,
    VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT = 2,
    VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT = 3,
    VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT = 4,
    VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT = 5,
    VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT = 6,
    VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT = 7,
  // Provided by VK_EXT_descriptor_heap with VK_KHR_ray_tracing_pipeline or
VK_NV_ray_tracing
    VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT = 8,
  // Provided by VK_EXT_descriptor_heap with VK_KHR_ray_tracing_pipeline or
VK_NV_ray_tracing
    VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT = 9,
  // Provided by VK_EXT_descriptor_heap with VK_KHR_ray_tracing_pipeline or
VK_NV_ray_tracing
    VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT = 10,
} VkDescriptorMappingSourceEXT;
```

- VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_CONSTANT_OFFSET_EXT specifies that the resource will be backed by a descriptor from the heap at a constant index.
- VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT specifies that the resource will be backed by a descriptor from the heap at an index sourced from push data, added to a constant index.
- VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT specifies that the resource will be backed by a descriptor from the heap at an index sourced from an address in push data, added to a constant index. If the mapping is an array, the array will be mapped to a base offset in indirect memory, and subsequent elements are mapped as offsets to that base.
- VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT specifies that the resource will be backed by a descriptor from the heap at an index sourced from an address in push data, added to a constant index. If the mapping is an array, each array element will be mapped to a separate index in indirect memory.
- VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT specifies that the resource will be backed by heap data directly.
- VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT specifies that the resource will be backed

by push data directly.

- VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT specifies that the resource will be backed by an address in push data.
- VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT specifies that the resource will be backed by an address sourced via another address in push data.
- VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT specifies that the resource will be backed by a descriptor from the heap at an index sourced from shader record data, added to a constant index.
- VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT specifies that the resource will be backed by shader record data directly.
- VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT specifies that the resource will be backed by an address in shader record data.

The VkDescriptorMappingSourceDataEXT union is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef union VkDescriptorMappingSourceDataEXT {
    VkDescriptorMappingSourceConstantOffsetEXT        constantOffset;
    VkDescriptorMappingSourcePushIndexEXT             pushIndex;
    VkDescriptorMappingSourceIndirectIndexEXT         indirectIndex;
    VkDescriptorMappingSourceIndirectIndexArrayEXT    indirectIndexArray;
    VkDescriptorMappingSourceHeapDataEXT              heapData;
    uint32_t                                          pushDataOffset;
    uint32_t                                          pushAddressOffset;
    VkDescriptorMappingSourceIndirectAddressEXT       indirectAddress;
    VkDescriptorMappingSourceShaderRecordIndexEXT     shaderRecordIndex;
    uint32_t                                          shaderRecordDataOffset;
    uint32_t                                          shaderRecordAddressOffset;
} VkDescriptorMappingSourceDataEXT;
```

- constantOffset is a VkDescriptorMappingSourceConstantOffsetEXT structure specifying the mapping for resources at a constant byte offset into a heap.
- pushIndex is a VkDescriptorMappingSourcePushIndexEXT structure specifying the mapping for resources at an index into a heap sourced from push data.
- indirectIndex is a VkDescriptorMappingSourceIndirectIndexEXT structure specifying the mapping for resources at an index into a heap sourced from an address in push data.
- indirectIndexArray is a VkDescriptorMappingSourceIndirectIndexArrayEXT structure specifying the mapping for resources to an array of indices into a heap sourced from an address in push data.
- heapData is a VkDescriptorMappingSourceHeapDataEXT structure specifying an offset into heap data for a uniform buffer to map to.
- pushDataOffset is an offset into push data for a uniform buffer to map to.
- pushAddressOffset is an offset into push data storing an address for a resource to map to.

- `indirectAddress` is a VkDescriptorMappingSourceIndirectAddressEXT structure specifying an address in push data containing another address for a resource to map to.
- `shaderRecordIndex` is a VkDescriptorMappingSourceShaderRecordIndexEXT structure specifying the mapping for resources at an index into a heap sourced from shader record data.
- `shaderRecordDataOffset` is an offset into shader record data for a uniform buffer to map to.
- `shaderRecordAddressOffset` is an offset into shader record data storing an address for a resource to map to.

The `VkDescriptorMappingSourceConstantOffsetEXT` structure is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkDescriptorMappingSourceConstantOffsetEXT {
    uint32_t                    heapOffset;
    uint32_t                    heapArrayStride;
    const VkSamplerCreateInfo*  pEmbeddedSampler;
    uint32_t                    samplerHeapOffset;
    uint32_t                    samplerHeapArrayStride;
} VkDescriptorMappingSourceConstantOffsetEXT;
```

- `heapOffset` is a constant byte offset added to the heap address for the mapped resource or sampler.
- `heapArrayStride` is a constant byte stride that multiplies the shader binding and array index.
- `pEmbeddedSampler` is an optional VkSamplerCreateInfo structure specifying a sampler to embed into the shader, in place of looking the sampler up in a heap.
- `samplerHeapOffset` is used only when mapping a combined image sampler, used in place of `heapOffset` to retrieve the sampler.
- `samplerHeapArrayStride` is used only when mapping a combined image sampler, used in place of `heapArrayStride` to retrieve the sampler.

Resources using this mapping will be backed by a descriptor in the heap, at an offset calculated as

shaderIndex = (Binding - `firstBinding`) + arrayIndex

offset = heapOffset + (shaderIndex * heapArrayStride)

where Binding is the binding value in the shader, arrayIndex is the index into the array if the shader binding is declared as an array.

If the mapped resource is a `OpTypeSampledImage`, offset is instead calculated for the sampler as

offset = samplerHeapOffset + (shaderIndex * samplerHeapArrayStride)

If the mapped resource is a `OpTypeSampler` or `OpTypeSampledImage`, and `pEmbeddedSampler` is not `NULL`,

the specified embedded sampler will be used rather than accessing the sampler heap.

Valid Usage

- VUID-VkDescriptorMappingSourceConstantOffsetEXT-pEmbeddedSampler-11445
  If `pEmbeddedSampler` is a valid pointer to a VkSamplerCreateInfo, its `borderColor` **must** not be VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT
- VUID-VkDescriptorMappingSourceConstantOffsetEXT-pEmbeddedSampler-11415
  If `pEmbeddedSampler` is a valid pointer to a VkSamplerCreateInfo, and there is a VkDebugUtilsObjectNameInfoEXT structure in its `pNext` chain, its `objectType` **must** be VK_OBJECT_TYPE_UNKNOWN
- VUID-VkDescriptorMappingSourceConstantOffsetEXT-pEmbeddedSampler-12432
  If `pEmbeddedSampler` is a valid pointer to a VkSamplerCreateInfo, the VkSamplerCreateInfo ::`pNext` chain **must** not contain VkOpaqueCaptureDescriptorDataCreateInfoEXT

Valid Usage (Implicit)

- VUID-VkDescriptorMappingSourceConstantOffsetEXT-pEmbeddedSampler-parameter
  If `pEmbeddedSampler` is not `NULL`, `pEmbeddedSampler` **must** be a valid pointer to a valid VkSamplerCreateInfo structure

The `VkDescriptorMappingSourcePushIndexEXT` structure is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkDescriptorMappingSourcePushIndexEXT {
    uint32_t                    heapOffset;
    uint32_t                    pushOffset;
    uint32_t                    heapIndexStride;
    uint32_t                    heapArrayStride;
    const VkSamplerCreateInfo*  pEmbeddedSampler;
    VkBool32                    useCombinedImageSamplerIndex;
    uint32_t                    samplerHeapOffset;
    uint32_t                    samplerPushOffset;
    uint32_t                    samplerHeapIndexStride;
    uint32_t                    samplerHeapArrayStride;
} VkDescriptorMappingSourcePushIndexEXT;
```

- `heapOffset` is a constant byte offset added to the heap address for the mapped resource or sampler.
- `pushOffset` is an index into push data where an index into the heap for the mapped resource will be retrieved.
- `heapIndexStride` is a constant byte stride that multiplies the index in push data.
- `heapArrayStride` is a constant byte stride that multiplies the shader binding and array index.

- `pEmbeddedSampler` is an optional VkSamplerCreateInfo structure specifying a sampler to embed into the shader, in place of looking the sampler up in a heap.
- `useCombinedImageSamplerIndex` specifies whether the generated index value will be decoded as two packed indices if the mapped resource is an `OpTypeSampledImage`.
- `samplerHeapOffset` is used only when mapping a combined image sampler, used in place of `heapOffset` to retrieve the sampler.
- `samplerPushOffset` is used only when mapping a combined image sampler, used in place of `pushOffset` to retrieve the sampler.
- `samplerHeapIndexStride` is used only when mapping a combined image sampler, used in place of `heapIndexStride` to retrieve the sampler.
- `samplerHeapArrayStride` is used only when mapping a combined image sampler, used in place of `heapArrayStride` to retrieve the sampler.

Resources using this mapping will be backed by a descriptor in the heap, at an offset calculated as

pushIndex = ((uint32_t*)pPushData)[pushOffset/4]

shaderIndex = (Binding - firstBinding) + arrayIndex

offset = heapOffset + (pushIndex × heapIndexStride) + (shaderIndex × heapArrayStride)

where Binding is the binding value in the shader, arrayIndex is the index into the array if the shader binding is declared as an array, and pPushData is the total set of push data specified by vkCmdPushDataEXT.

If the mapped resource is a `OpTypeSampledImage`, offset is instead calculated for the sampler as

samplerPushIndex = ((uint32_t*)pPushData)[samplerPushOffset/4]

offset = samplerHeapOffset + (samplerPushIndex × samplerHeapIndexStride) + (shaderIndex × samplerHeapArrayStride)

If `useCombinedImageSamplerIndex` is VK_TRUE, and the mapped resource is a `OpTypeSampledImage`, pushIndex and samplerPushIndex in the above equations are instead calculated as

pushIndex = ((uint32_t*)pPushData)[pushOffset/4] & 0xFFFFF

samplerPushIndex = (((uint32_t*)pPushData)[pushOffset/4] >> 20) & 0xFFF

If the mapped resource is a `OpTypeSampler` or `OpTypeSampledImage`, and `pEmbeddedSampler` is not `NULL`,

the specified embedded sampler will be used rather than accessing the sampler heap.

Valid Usage

- VUID-VkDescriptorMappingSourcePushIndexEXT-pushOffset-11258
  `pushOffset` **must** be a multiple of 4
- VUID-VkDescriptorMappingSourcePushIndexEXT-pushOffset-11259
  `pushOffset` **must** be less than or equal to `maxPushDataSize` - 4
- VUID-VkDescriptorMappingSourcePushIndexEXT-pEmbeddedSampler-11445
  If `pEmbeddedSampler` is a valid pointer to a VkSamplerCreateInfo, its `borderColor` **must** not be VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT
- VUID-VkDescriptorMappingSourcePushIndexEXT-pEmbeddedSampler-11415
  If `pEmbeddedSampler` is a valid pointer to a VkSamplerCreateInfo, and there is a VkDebugUtilsObjectNameInfoEXT structure in its `pNext` chain, its `objectType` **must** be VK_OBJECT_TYPE_UNKNOWN
- VUID-VkDescriptorMappingSourcePushIndexEXT-pEmbeddedSampler-12432
  If `pEmbeddedSampler` is a valid pointer to a VkSamplerCreateInfo, the VkSamplerCreateInfo ::`pNext` chain **must** not contain VkOpaqueCaptureDescriptorDataCreateInfoEXT

Valid Usage (Implicit)

- VUID-VkDescriptorMappingSourcePushIndexEXT-pEmbeddedSampler-parameter
  If `pEmbeddedSampler` is not `NULL`, `pEmbeddedSampler` **must** be a valid pointer to a valid VkSamplerCreateInfo structure

The `VkDescriptorMappingSourceIndirectIndexEXT` structure is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkDescriptorMappingSourceIndirectIndexEXT {
    uint32_t                        heapOffset;
    uint32_t                        pushOffset;
    uint32_t                        addressOffset;
    uint32_t                        heapIndexStride;
    uint32_t                        heapArrayStride;
    const VkSamplerCreateInfo*      pEmbeddedSampler;
    VkBool32                        useCombinedImageSamplerIndex;
    uint32_t                        samplerHeapOffset;
    uint32_t                        samplerPushOffset;
    uint32_t                        samplerAddressOffset;
    uint32_t                        samplerHeapIndexStride;
    uint32_t                        samplerHeapArrayStride;
} VkDescriptorMappingSourceIndirectIndexEXT;
```

- `heapOffset` is a constant byte offset added to the heap address for the mapped resource or

sampler.

- `pushOffset` is an offset into push data where the indirect address will be.
- `addressOffset` is an index into the address in push data where an index into the heap for the mapped resource will be retrieved.
- `heapIndexStride` is a constant byte stride that multiplies the index in indirect data.
- `heapArrayStride` is a constant byte stride that multiplies the shader binding and array index.
- `pEmbeddedSampler` is an optional VkSamplerCreateInfo structure specifying a sampler to embed into the shader, in place of looking the sampler up in a heap.
- `useCombinedImageSamplerIndex` specifies whether the generated index value will be decoded as two packed indices if the mapped resource is an `OpTypeSampledImage`.
- `samplerHeapOffset` is used only when mapping a combined image sampler, used in place of `heapOffset` to retrieve the sampler.
- `samplerPushOffset` is used only when mapping a combined image sampler, used in place of `pushOffset` to retrieve the sampler.
- `samplerAddressOffset` is used only when mapping a combined image sampler, used in place of `addressOffset` to retrieve the sampler.
- `samplerHeapIndexStride` is used only when mapping a combined image sampler, used in place of `heapIndexStride` to retrieve the sampler.
- `samplerHeapArrayStride` is used only when mapping a combined image sampler, used in place of `heapArrayStride` to retrieve the sampler.

Resources using this mapping will be backed by a descriptor in the heap, at an offset calculated as

```
uint32_t *indirectAddress = ((VkDeviceAddress*)pPushData)[pushOffset/8]

indirectIndex = indirectAddress[(addressOffset / 4)]

shaderIndex = (Binding - firstBinding) + arrayIndex

offset = heapOffset + (indirectIndex × heapIndexStride) + (shaderIndex × heapArrayStride)
```

where Binding is the binding value in the shader, arrayIndex is the index into the array if the shader binding is declared as an array, and pPushData is the total set of push data specified by vkCmdPushDataEXT. The value of the address in push data **must** be a multiple of 4. Index reads through indirectAddress are performed as non-volatile uniform buffer reads, and can be synchronized using VK_ACCESS_2_UNIFORM_READ_BIT. The value in memory **must** remain static while any shader invocation using this mapping is in flight to avoid a data race.

If the mapped resource is a `OpTypeSampledImage`, offset is instead calculated for the sampler as

uint32_t *samplerIndirectAddress = ((VkDeviceAddress*)pPushData)[samplerPushOffset/8]

samplerIndirectIndex = samplerIndirectAddress[(samplerAddressOffset / 4)]

offset = samplerHeapOffset + (samplerIndirectIndex × samplerHeapIndexStride) + (shaderIndex × samplerHeapArrayStride)

If `useCombinedImageSamplerIndex` is VK_TRUE, and the mapped resource is a `OpTypeSampledImage`, indirectIndex and samplerIndirectIndex in the above equations are instead calculated as

indirectIndex = indirectAddress[addressOffset/4] & 0xFFFFF

samplerIndirectIndex = indirectAddress[addressOffset/4] >> 20) & 0xFFF

If the mapped resource is a `OpTypeSampler` or `OpTypeSampledImage`, and `pEmbeddedSampler` is not `NULL`, the specified embedded sampler will be used rather than accessing the sampler heap.

Valid Usage

- VUID-VkDescriptorMappingSourceIndirectIndexEXT-pushOffset-11260
  `pushOffset` **must** be a multiple of 8
- VUID-VkDescriptorMappingSourceIndirectIndexEXT-pushOffset-11261
  `pushOffset` **must** be less than or equal to `maxPushDataSize` - 8
- VUID-VkDescriptorMappingSourceIndirectIndexEXT-addressOffset-11262
  `addressOffset` **must** be a multiple of 4
- VUID-VkDescriptorMappingSourceIndirectIndexEXT-pEmbeddedSampler-11445
  If `pEmbeddedSampler` is a valid pointer to a VkSamplerCreateInfo, its `borderColor` **must** not be VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT
- VUID-VkDescriptorMappingSourceIndirectIndexEXT-pEmbeddedSampler-11415
  If `pEmbeddedSampler` is a valid pointer to a VkSamplerCreateInfo, and there is a VkDebugUtilsObjectNameInfoEXT structure in its `pNext` chain, its `objectType` **must** be VK_OBJECT_TYPE_UNKNOWN
- VUID-VkDescriptorMappingSourceIndirectIndexEXT-pEmbeddedSampler-12432
  If `pEmbeddedSampler` is a valid pointer to a VkSamplerCreateInfo, the VkSamplerCreateInfo ::`pNext` chain **must** not contain VkOpaqueCaptureDescriptorDataCreateInfoEXT

Valid Usage (Implicit)

- VUID-VkDescriptorMappingSourceIndirectIndexEXT-pEmbeddedSampler-parameter

If `pEmbeddedSampler` is not `NULL`, `pEmbeddedSampler` **must** be a valid pointer to a valid VkSamplerCreateInfo structure

The `VkDescriptorMappingSourceIndirectIndexArrayEXT` structure is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkDescriptorMappingSourceIndirectIndexArrayEXT {
    uint32_t                    heapOffset;
    uint32_t                    pushOffset;
    uint32_t                    addressOffset;
    uint32_t                    heapIndexStride;
    const VkSamplerCreateInfo*  pEmbeddedSampler;
    VkBool32                    useCombinedImageSamplerIndex;
    uint32_t                    samplerHeapOffset;
    uint32_t                    samplerPushOffset;
    uint32_t                    samplerAddressOffset;
    uint32_t                    samplerHeapIndexStride;
} VkDescriptorMappingSourceIndirectIndexArrayEXT;
```

- `heapOffset` is a constant byte offset added to the heap address for the mapped resource or sampler.
- `pushOffset` is an offset into push data where the indirect address will be.
- `addressOffset` is an index into the address in push data where an index into the heap for the mapped resource will be retrieved.
- `heapIndexStride` is a constant byte stride that multiplies the index in indirect data.
- `pEmbeddedSampler` is an optional VkSamplerCreateInfo structure specifying a sampler to embed into the shader, in place of looking the sampler up in a heap.
- `useCombinedImageSamplerIndex` specifies whether the generated index value will be decoded as two packed indices if the mapped resource is an `OpTypeSampledImage`.
- `samplerHeapOffset` is used only when mapping a combined image sampler, used in place of `heapOffset` to retrieve the sampler.
- `samplerPushOffset` is used only when mapping a combined image sampler, used in place of `pushOffset` to retrieve the sampler.
- `samplerAddressOffset` is used only when mapping a combined image sampler, used in place of `addressOffset` to retrieve the sampler.
- `samplerHeapIndexStride` is used only when mapping a combined image sampler, used in place of `heapIndexStride` to retrieve the sampler.

Resources using this mapping will be backed by a descriptor in the heap, at an offset calculated as

```
uint32_t *indirectAddress = ((VkDeviceAddress*)pPushData)[pushOffset/8]
```

shaderIndex = (Binding - firstBinding) + arrayIndex

indirectIndex = indirectAddress[(addressOffset / 4) + shaderIndex]

offset = heapOffset + (indirectIndex × heapIndexStride)

where Binding is the binding value in the shader, arrayIndex is the index into the array if the shader binding is declared as an array, and pPushData is the total set of push data specified by vkCmdPushDataEXT. The value of the address in push data **must** be a multiple of 4. Index reads through indirectAddress are performed as non-volatile uniform buffer reads, and can be synchronized using VK_ACCESS_2_UNIFORM_READ_BIT. The value in memory **must** remain static while any shader invocation using this mapping is in flight to avoid a data race.

If the mapped resource is a `OpTypeSampledImage`, offset is instead calculated for the sampler as

uint32_t *samplerIndirectAddress = ((VkDeviceAddress*)pPushData)[samplerPushOffset/8]

samplerIndirectIndex = samplerAddr[(samplerAddressOffset / 4) + shaderIndex]

offset = samplerHeapOffset + (samplerIndirectIndex × samplerHeapIndexStride)

If `useCombinedImageSamplerIndex` is VK_TRUE, and the mapped resource is a `OpTypeSampledImage`, indirectIndex and samplerIndirectIndex in the above equations are instead calculated as

indirectIndex = indirectAddress[addressOffset/4 + shaderIndex] & 0xFFFFF

samplerIndirectIndex = indirectAddress[addressOffset/4 + shaderIndex] >> 20) & 0xFFF

If the mapped resource is a `OpTypeSampler` or `OpTypeSampledImage`, and `pEmbeddedSampler` is not `NULL`, the specified embedded sampler will be used rather than accessing the sampler heap.

Valid Usage

- VUID-VkDescriptorMappingSourceIndirectIndexArrayEXT-pushOffset-11359
  `pushOffset` **must** be a multiple of 8
- VUID-VkDescriptorMappingSourceIndirectIndexArrayEXT-pushOffset-11360
  `pushOffset` **must** be less than or equal to `maxPushDataSize` - 8
- VUID-VkDescriptorMappingSourceIndirectIndexArrayEXT-addressOffset-11361
  `addressOffset` **must** be a multiple of 4

- VUID-VkDescriptorMappingSourceIndirectIndexArrayEXT-pEmbeddedSampler-11445
  If `pEmbeddedSampler` is a valid pointer to a VkSamplerCreateInfo, its `borderColor` **must** not be VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT
- VUID-VkDescriptorMappingSourceIndirectIndexArrayEXT-pEmbeddedSampler-11415
  If `pEmbeddedSampler` is a valid pointer to a VkSamplerCreateInfo, and there is a VkDebugUtilsObjectNameInfoEXT structure in its `pNext` chain, its `objectType` **must** be VK_OBJECT_TYPE_UNKNOWN
- VUID-VkDescriptorMappingSourceIndirectIndexArrayEXT-pEmbeddedSampler-12432
  If `pEmbeddedSampler` is a valid pointer to a VkSamplerCreateInfo, the VkSamplerCreateInfo ::`pNext` chain **must** not contain VkOpaqueCaptureDescriptorDataCreateInfoEXT

Valid Usage (Implicit)

- VUID-VkDescriptorMappingSourceIndirectIndexArrayEXT-pEmbeddedSampler-parameter
  If `pEmbeddedSampler` is not `NULL`, `pEmbeddedSampler` **must** be a valid pointer to a valid VkSamplerCreateInfo structure

The `VkDescriptorMappingSourceHeapDataEXT` structure is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkDescriptorMappingSourceHeapDataEXT {
    uint32_t    heapOffset;
    uint32_t    pushOffset;
} VkDescriptorMappingSourceHeapDataEXT;
```

- `heapOffset` is a constant byte offset added to the heap address for the mapped buffer.
- `pushOffset` is an index into push data where an additional offset into the heap for the mapped resource will be retrieved.

Uniform buffers using this mapping will be backed directly by data in the heap. Accessing data in the uniform buffer at an offset of shaderOffset in the shader will access heap data at an offset equal to

offset = shaderOffset + heapOffset + ((uint32_t*)pPushData)[pushOffset/4]

where pPushData is the total set of push data specified by vkCmdPushDataEXT. Shader reads through the heap mapped in this way are performed according to the mapped resource.

Valid Usage

- VUID-VkDescriptorMappingSourceHeapDataEXT-heapOffset-11263
  `heapOffset` **must** be a multiple of `minUniformBufferOffsetAlignment`

- VUID-VkDescriptorMappingSourceHeapDataEXT-pushOffset-11264
  `pushOffset` **must** be a multiple of 4
- VUID-VkDescriptorMappingSourceHeapDataEXT-pushOffset-11265
  `pushOffset` **must** be less than or equal to `maxPushDataSize` - 4

The `VkDescriptorMappingSourceIndirectAddressEXT` structure is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkDescriptorMappingSourceIndirectAddressEXT {
    uint32_t    pushOffset;
    uint32_t    addressOffset;
} VkDescriptorMappingSourceIndirectAddressEXT;
```

- `pushOffset` is a byte offset into push data where an indirect address containing the address for the mapped resource will be retrieved.
- `addressOffset` is a byte offset into the indirect address where the address for the mapped resource will be retrieved.

Accessing data via the mapped resource in the shader will access data backing the address specified in the indirect address at the supplied offset:

indirectAddress = ((VkDeviceAddress*)pPushData)[pushOffset/8]

resourceAddress = ((VkDeviceAddress*)indirectAddress)[addressOffset/8]

where pPushData is the total set of push data specified by vkCmdPushDataEXT. Reads through indirectAddress are performed as non-volatile uniform buffer reads, and can be synchronized using VK_ACCESS_2_UNIFORM_READ_BIT. Shader reads through resourceAddress are performed according to the mapped resource. If the shader resource is an acceleration structure, the address **must** be a valid acceleration structure address.

Valid Usage

- VUID-VkDescriptorMappingSourceIndirectAddressEXT-pushOffset-11266
  `pushOffset` **must** be a multiple of 8
- VUID-VkDescriptorMappingSourceIndirectAddressEXT-pushOffset-11267
  `pushOffset` **must** be less than or equal to `maxPushDataSize` - 8
- VUID-VkDescriptorMappingSourceIndirectAddressEXT-addressOffset-11268
  `addressOffset` **must** be a multiple of 8

The `VkDescriptorMappingSourceShaderRecordIndexEXT` structure is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkDescriptorMappingSourceShaderRecordIndexEXT {
    uint32_t                        heapOffset;
    uint32_t                        shaderRecordOffset;
    uint32_t                        heapIndexStride;
    uint32_t                        heapArrayStride;
    const VkSamplerCreateInfo*      pEmbeddedSampler;
    VkBool32                        useCombinedImageSamplerIndex;
    uint32_t                        samplerHeapOffset;
    uint32_t                        samplerShaderRecordOffset;
    uint32_t                        samplerHeapIndexStride;
    uint32_t                        samplerHeapArrayStride;
} VkDescriptorMappingSourceShaderRecordIndexEXT;
```

- `heapOffset` is a constant byte offset added to the heap address for the mapped resource or sampler.
- `shaderRecordOffset` is an index into shader record data where an index into the heap for the mapped resource will be retrieved.
- `heapIndexStride` is a constant byte stride that multiplies the index in shader record data.
- `heapArrayStride` is a constant byte stride that multiplies the shader binding and array index.
- `pEmbeddedSampler` is an optional VkSamplerCreateInfo structure specifying a sampler to embed into the shader, in place of looking the sampler up in a heap.
- `useCombinedImageSamplerIndex` specifies whether the generated index value will be decoded as two packed indices if the mapped resource is an `OpTypeSampledImage`.
- `samplerHeapOffset` is used only when mapping a combined image sampler, used in place of `heapOffset` to retrieve the sampler.
- `samplerShaderRecordOffset` is used only when mapping a combined image sampler, used in place of `shaderRecordOffset` to retrieve the sampler.
- `samplerHeapIndexStride` is used only when mapping a combined image sampler, used in place of `heapIndexStride` to retrieve the sampler.
- `samplerHeapArrayStride` is used only when mapping a combined image sampler, used in place of `heapArrayStride` to retrieve the sampler.

Resources using this mapping will be backed by a descriptor in the heap, at an offset calculated as

shaderRecordIndex = ((uint32_t*)pShaderRecordData)[shaderRecordOffset/4]

shaderIndex = (Binding - firstBinding) + arrayIndex

offset = heapOffset + (shaderRecordIndex × heapIndexStride) + (shaderIndex × heapArrayStride)

where Binding is the binding value in the shader, arrayIndex is the index into the array if the shader binding is declared as an array, and pShaderRecordData is the set of shader record data accessible to the shader.

If the mapped resource is a `OpTypeSampledImage`, offset is instead calculated for the sampler as

samplerShaderRecordIndex = ((uint32_t*)pShaderRecordData)[samplerShaderRecordOffset/4]

offset = samplerHeapOffset + (samplerShaderRecordIndex × samplerHeapIndexStride) + (shaderIndex × samplerHeapArrayStride)

If `useCombinedImageSamplerIndex` is VK_TRUE, and the mapped resource is a `OpTypeSampledImage`, shaderRecordIndex and samplerShaderRecordIndex in the above equations are instead calculated as

shaderRecordIndex = ((uint32_t*)pShaderRecordData)[shaderRecordOffset/4] & 0xFFFFF

samplerShaderRecordIndex = (((uint32_t*)pShaderRecordData)[shaderRecordOffset/4] >> 20) & 0xFFF

If the mapped resource is a `OpTypeSampler` or `OpTypeSampledImage`, and `pEmbeddedSampler` is not `NULL`, the specified embedded sampler will be used rather than accessing the sampler heap.

Valid Usage

- VUID-VkDescriptorMappingSourceShaderRecordIndexEXT-shaderRecordOffset-11269
  `shaderRecordOffset` **must** be a multiple of 4
- VUID-VkDescriptorMappingSourceShaderRecordIndexEXT-shaderRecordOffset-11270
  `shaderRecordOffset` **must** be less than or equal to `maxShaderGroupStride` - 4
- VUID-VkDescriptorMappingSourceShaderRecordIndexEXT-pEmbeddedSampler-11445
  If `pEmbeddedSampler` is a valid pointer to a VkSamplerCreateInfo, its `borderColor` **must** not be VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT
- VUID-VkDescriptorMappingSourceShaderRecordIndexEXT-pEmbeddedSampler-11415
  If `pEmbeddedSampler` is a valid pointer to a VkSamplerCreateInfo, and there is a VkDebugUtilsObjectNameInfoEXT structure in its `pNext` chain, its `objectType` **must** be VK_OBJECT_TYPE_UNKNOWN
- VUID-VkDescriptorMappingSourceShaderRecordIndexEXT-pEmbeddedSampler-12432
  If `pEmbeddedSampler` is a valid pointer to a VkSamplerCreateInfo, the VkSamplerCreateInfo ::`pNext` chain **must** not contain VkOpaqueCaptureDescriptorDataCreateInfoEXT

Valid Usage (Implicit)

- VUID-VkDescriptorMappingSourceShaderRecordIndexEXT-pEmbeddedSampler-parameter
  If `pEmbeddedSampler` is not `NULL`, `pEmbeddedSampler` **must** be a valid pointer to a valid VkSamplerCreateInfo structure

16.4. Packing descriptors more tightly

For simplicity, descriptor sizes are advertised and treated by default as equal to the advertised descriptor size limits:

- `samplerDescriptorSize`
- `bufferDescriptorSize`
- `imageDescriptorSize`

However, for some specific use cases it is useful to be able to pack specific descriptors into memory more tightly than this when the implementation allows for this.

To query the size of a heap descriptor for a specific VkDescriptorType, call:

```
// Provided by VK_EXT_descriptor_heap
VkDeviceSize vkGetPhysicalDeviceDescriptorSizeEXT(
    VkPhysicalDevice                            physicalDevice,
    VkDescriptorType                            descriptorType);
```

- `physicalDevice` is the physical device from which to query the descriptor sizes.
- `descriptorType` is a VkDescriptorType specifying the type of heap descriptor to query the size for.

The return value of this function will be a VkDeviceSize indicating the size in bytes (N) of a heap descriptor with a type equal to `descriptorType`. When a descriptor of this type is written by vkWriteResourceDescriptorsEXT or vkWriteSamplerDescriptorsEXT, only the first N bytes are written; the rest will not be accessed and **can** be safely discarded when copying descriptors around. Additionally, those first N bytes are the only bytes that will be accessed when the descriptor is accessed in the shader. N will never be larger than the applicable limits in VkPhysicalDeviceDescriptorHeapPropertiesEXT.

NOTE	Values returned by this function may be non power-of-two.
NOTE	This command is not intended for general use, and is for tools that already take advantage of tighter packing with other similar features (e.g. `VK_EXT_descriptor_buffer`) to optimize accesses in some cases. Applications can safely ignore this function and are advised to do so, to avoid depending on non-portable packing.

Valid Usage

- VUID-vkGetPhysicalDeviceDescriptorSizeEXT-type-11362 `descriptorType` **must** be one of VK_DESCRIPTOR_TYPE_SAMPLER, VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER, VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceDescriptorSizeEXT-physicalDevice-parameter `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceDescriptorSizeEXT-descriptorType-parameter `descriptorType` **must** be a valid VkDescriptorType value

Chapter 17. Descriptor Sets

A descriptor set object is an opaque object containing storage for a set of descriptors, where the types and number of descriptors is defined by a descriptor set layout. The layout object **may** be used to define the association of each descriptor binding with memory or other implementation resources. The layout is used both for determining the resources that need to be associated with the descriptor set, and determining the interface between shader stages and shader resources.

17.1. Descriptor Set Layout

A descriptor set layout object is defined by an array of zero or more descriptor bindings. Each individual descriptor binding is specified by a descriptor type, a count (array size) of the number of descriptors in the binding, a set of shader stages that **can** access the binding, and (if using immutable samplers) an array of sampler descriptors.

Descriptor set layout objects are represented by `VkDescriptorSetLayout` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDescriptorSetLayout)
```

To create descriptor set layout objects, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateDescriptorSetLayout(
    VkDevice                                    device,
    const VkDescriptorSetLayoutCreateInfo*      pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkDescriptorSetLayout*                      pSetLayout);
```

- `device` is the logical device that creates the descriptor set layout.
- `pCreateInfo` is a pointer to a VkDescriptorSetLayoutCreateInfo structure specifying the state of the descriptor set layout object.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pSetLayout` is a pointer to a VkDescriptorSetLayout handle in which the resulting descriptor set layout object is returned.

Valid Usage

- VUID-vkCreateDescriptorSetLayout-support-09582
  If the descriptor layout exceeds the limits reported through the physical device limits, then vkGetDescriptorSetLayoutSupport **must** have returned VkDescriptorSetLayoutSupport with `supported` equal to VK_TRUE for `pCreateInfo`

Valid Usage (Implicit)

- VUID-vkCreateDescriptorSetLayout-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateDescriptorSetLayout-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkDescriptorSetLayoutCreateInfo structure
- VUID-vkCreateDescriptorSetLayout-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateDescriptorSetLayout-pSetLayout-parameter
  `pSetLayout` **must** be a valid pointer to a VkDescriptorSetLayout handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Information about the descriptor set layout is passed in a `VkDescriptorSetLayoutCreateInfo` structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorSetLayoutCreateInfo {
    VkStructureType                        sType;
    const void*                            pNext;
    VkDescriptorSetLayoutCreateFlags       flags;
    uint32_t                               bindingCount;
    const VkDescriptorSetLayoutBinding*    pBindings;
} VkDescriptorSetLayoutCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkDescriptorSetLayoutCreateFlagBits specifying options for descriptor set layout creation.
- `bindingCount` is the number of elements in `pBindings`.
- `pBindings` is a pointer to an array of VkDescriptorSetLayoutBinding structures.

Valid Usage

- VUID-VkDescriptorSetLayoutCreateInfo-binding-00279
  The VkDescriptorSetLayoutBinding::binding members of the elements of the pBindings array **must** each have different values
- VUID-VkDescriptorSetLayoutCreateInfo-flags-10354
  If flags contains VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT, and the VK_KHR_push_descriptor extension is not enabled, pushDescriptor **must** be enabled
- VUID-VkDescriptorSetLayoutCreateInfo-flags-00280
  If flags contains VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT, then all elements of pBindings **must** not have a descriptorType of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC
- VUID-VkDescriptorSetLayoutCreateInfo-flags-02208
  If flags contains VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT, then all elements of pBindings **must** not have a descriptorType of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK
- VUID-VkDescriptorSetLayoutCreateInfo-flags-00281
  If flags contains VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT, then the total number of elements of all bindings **must** be less than or equal to VkPhysicalDevicePushDescriptorProperties::maxPushDescriptors
- VUID-VkDescriptorSetLayoutCreateInfo-flags-04590
  If flags contains VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT, flags **must** not contain VK_DESCRIPTOR_SET_LAYOUT_CREATE_HOST_ONLY_POOL_BIT_EXT
- VUID-VkDescriptorSetLayoutCreateInfo-flags-04591
  If flags contains VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT, pBindings **must** not have a descriptorType of VK_DESCRIPTOR_TYPE_MUTABLE_EXT
- VUID-VkDescriptorSetLayoutCreateInfo-flags-03000
  If any binding has the VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT bit set, flags **must** include VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT
- VUID-VkDescriptorSetLayoutCreateInfo-descriptorType-03001
  If any binding has the VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT bit set, then all bindings **must** not have descriptorType of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC
- VUID-VkDescriptorSetLayoutCreateInfo-flags-04592
  If flags contains VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT, flags **must** not contain VK_DESCRIPTOR_SET_LAYOUT_CREATE_HOST_ONLY_POOL_BIT_EXT
- VUID-VkDescriptorSetLayoutCreateInfo-pBindings-07303
  If any element pBindings[i] has a descriptorType of VK_DESCRIPTOR_TYPE_MUTABLE_EXT, then the pNext chain **must** include a VkMutableDescriptorTypeCreateInfoEXT structure, and mutableDescriptorTypeListCount **must** be greater than i

- VUID-VkDescriptorSetLayoutCreateInfo-descriptorType-04594
  If a binding has a `descriptorType` value of VK_DESCRIPTOR_TYPE_MUTABLE_EXT, then `pImmutableSamplers` **must** be `NULL`
- VUID-VkDescriptorSetLayoutCreateInfo-mutableDescriptorType-04595
  If VkPhysicalDeviceMutableDescriptorTypeFeaturesEXT::`mutableDescriptorType` is not enabled, `pBindings` **must** not contain a `descriptorType` of VK_DESCRIPTOR_TYPE_MUTABLE_EXT
- VUID-VkDescriptorSetLayoutCreateInfo-flags-04596
  If `flags` contains VK_DESCRIPTOR_SET_LAYOUT_CREATE_HOST_ONLY_POOL_BIT_EXT, VkPhysicalDeviceMutableDescriptorTypeFeaturesEXT::`mutableDescriptorType` **must** be enabled
- VUID-VkDescriptorSetLayoutCreateInfo-flags-08000
  If `flags` contains VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, then all elements of `pBindings` **must** not have a `descriptorType` of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC
- VUID-VkDescriptorSetLayoutCreateInfo-flags-08001
  If `flags` contains VK_DESCRIPTOR_SET_LAYOUT_CREATE_EMBEDDED_IMMUTABLE_SAMPLERS_BIT_EXT, `flags` **must** also contain VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-VkDescriptorSetLayoutCreateInfo-flags-08002
  If `flags` contains VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, then `flags` **must** not contain VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT
- VUID-VkDescriptorSetLayoutCreateInfo-flags-08003
  If `flags` contains VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, then `flags` **must** not contain VK_DESCRIPTOR_SET_LAYOUT_CREATE_HOST_ONLY_POOL_BIT_EXT

Valid Usage (Implicit)

- VUID-VkDescriptorSetLayoutCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_CREATE_INFO
- VUID-VkDescriptorSetLayoutCreateInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkDescriptorSetLayoutBindingFlagsCreateInfo or VkMutableDescriptorTypeCreateInfoEXT
- VUID-VkDescriptorSetLayoutCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkDescriptorSetLayoutCreateInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkDescriptorSetLayoutCreateFlagBits values

- VUID-VkDescriptorSetLayoutCreateInfo-pBindings-parameter
  If `bindingCount` is not `0`, `pBindings` **must** be a valid pointer to an array of `bindingCount` valid VkDescriptorSetLayoutBinding structures

If the `pNext` chain of a VkDescriptorSetLayoutCreateInfo or VkDescriptorPoolCreateInfo structure includes a VkMutableDescriptorTypeCreateInfoEXT structure, then that structure specifies information about the possible descriptor types for mutable descriptor types.

The `VkMutableDescriptorTypeCreateInfoEXT` structure is defined as:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_mutable_descriptor_type
typedef struct VkMutableDescriptorTypeCreateInfoEXT {
    VkStructureType                          sType;
    const void*                              pNext;
    uint32_t                                 mutableDescriptorTypeListCount;
    const VkMutableDescriptorTypeListEXT*    pMutableDescriptorTypeLists;
} VkMutableDescriptorTypeCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `mutableDescriptorTypeListCount` is the number of elements in `pMutableDescriptorTypeLists`.
- `pMutableDescriptorTypeLists` is a pointer to an array of `VkMutableDescriptorTypeListEXT` structures.

If `mutableDescriptorTypeListCount` is zero or if this structure is not included in the `pNext` chain, the VkMutableDescriptorTypeListEXT for each element is considered to be zero or `NULL` for each member. Otherwise, the descriptor set layout binding at VkDescriptorSetLayoutCreateInfo ::`pBindings`[i] uses the descriptor type lists in VkMutableDescriptorTypeCreateInfoEXT ::`pMutableDescriptorTypeLists`[i].

Valid Usage (Implicit)

- VUID-VkMutableDescriptorTypeCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MUTABLE_DESCRIPTOR_TYPE_CREATE_INFO_EXT
- VUID-VkMutableDescriptorTypeCreateInfoEXT-pMutableDescriptorTypeLists-parameter
  If `mutableDescriptorTypeListCount` is not `0`, `pMutableDescriptorTypeLists` **must** be a valid pointer to an array of `mutableDescriptorTypeListCount` valid VkMutableDescriptorTypeListEXT structures

Structure Chaining

Extends the structures

- VkDescriptorPoolCreateInfo
- VkDescriptorSetLayoutCreateInfo

The list of potential descriptor types a given mutable descriptor **can** mutate to is passed in a `VkMutableDescriptorTypeListEXT` structure.

The `VkMutableDescriptorTypeListEXT` structure is defined as:

WARNING	This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_mutable_descriptor_type
typedef struct VkMutableDescriptorTypeListEXT {
    uint32_t                  descriptorTypeCount;
    const VkDescriptorType*   pDescriptorTypes;
} VkMutableDescriptorTypeListEXT;
```

- `descriptorTypeCount` is the number of elements in `pDescriptorTypes`.
- `pDescriptorTypes` is `NULL` or a pointer to an array of `descriptorTypeCount` VkDescriptorType values defining which descriptor types a given binding may mutate to.

Valid Usage

- VUID-VkMutableDescriptorTypeListEXT-descriptorTypeCount-04597
  `descriptorTypeCount` **must** not be `0` if the corresponding binding is of VK_DESCRIPTOR_TYPE_MUTABLE_EXT
- VUID-VkMutableDescriptorTypeListEXT-pDescriptorTypes-04598
  `pDescriptorTypes` **must** be a valid pointer to an array of `descriptorTypeCount` valid, unique VkDescriptorType values if the given binding is of VK_DESCRIPTOR_TYPE_MUTABLE_EXT type
- VUID-VkMutableDescriptorTypeListEXT-descriptorTypeCount-04599
  `descriptorTypeCount` **must** be `0` if the corresponding binding is not of VK_DESCRIPTOR_TYPE_MUTABLE_EXT
- VUID-VkMutableDescriptorTypeListEXT-pDescriptorTypes-04600
  `pDescriptorTypes` **must** not contain VK_DESCRIPTOR_TYPE_MUTABLE_EXT
- VUID-VkMutableDescriptorTypeListEXT-pDescriptorTypes-04601
  `pDescriptorTypes` **must** not contain VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC
- VUID-VkMutableDescriptorTypeListEXT-pDescriptorTypes-04602
  `pDescriptorTypes` **must** not contain VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC

- VUID-VkMutableDescriptorTypeListEXT-pDescriptorTypes-04603
  `pDescriptorTypes` **must** not contain VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK

Valid Usage (Implicit)

- VUID-VkMutableDescriptorTypeListEXT-pDescriptorTypes-parameter
  If `descriptorTypeCount` is not `0`, `pDescriptorTypes` **must** be a valid pointer to an array of `descriptorTypeCount` valid VkDescriptorType values

Bits which **can** be set in VkDescriptorSetLayoutCreateInfo::`flags`, specifying options for descriptor set layout, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkDescriptorSetLayoutCreateFlagBits {
  // Provided by VK_VERSION_1_2
    VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT = 0x00000002,
  // Provided by VK_VERSION_1_4
    VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT = 0x00000001,
  // Provided by VK_EXT_descriptor_buffer
    VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT = 0x00000010,
  // Provided by VK_EXT_descriptor_buffer
    VK_DESCRIPTOR_SET_LAYOUT_CREATE_EMBEDDED_IMMUTABLE_SAMPLERS_BIT_EXT = 0x00000020,
  // Provided by VK_EXT_mutable_descriptor_type
    VK_DESCRIPTOR_SET_LAYOUT_CREATE_HOST_ONLY_POOL_BIT_EXT = 0x00000004,
  // Provided by VK_KHR_push_descriptor
    VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR =
VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT,
  // Provided by VK_EXT_descriptor_indexing
    VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT_EXT =
VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT,
} VkDescriptorSetLayoutCreateFlagBits;
```

- VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT specifies that descriptor sets **must** not be allocated using this layout, and descriptors are instead pushed by vkCmdPushDescriptorSet.
- VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT specifies that descriptor sets using this layout **must** be allocated from a descriptor pool created with the VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT bit set. Descriptor set layouts created with this bit set have alternate limits for the maximum number of descriptors per-stage and per-pipeline layout. The non-UpdateAfterBind limits only count descriptors in sets created without this flag. The UpdateAfterBind limits count all descriptors, but the limits **may** be higher than the non-UpdateAfterBind limits.
- VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT specifies that this layout **must** only be used with descriptor buffers.
- VK_DESCRIPTOR_SET_LAYOUT_CREATE_EMBEDDED_IMMUTABLE_SAMPLERS_BIT_EXT

specifies that this is a layout only containing immutable samplers that **can** be bound by vkCmdBindDescriptorBufferEmbeddedSamplersEXT. Unlike normal immutable samplers, embedded immutable samplers do not require the application to provide them in a descriptor buffer.

- VK_DESCRIPTOR_SET_LAYOUT_CREATE_HOST_ONLY_POOL_BIT_EXT specifies that descriptor sets using this layout **must** be allocated from a descriptor pool created with the VK_DESCRIPTOR_POOL_CREATE_HOST_ONLY_BIT_EXT bit set. Descriptor set layouts created with this bit have no expressible limit for maximum number of descriptors per-stage. Host descriptor sets are limited only by available host memory, but **may** be limited for implementation specific reasons. Implementations **may** limit the number of supported descriptors to UpdateAfterBind limits or non-UpdateAfterBind limits, whichever is larger.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDescriptorSetLayoutCreateFlags;
```

VkDescriptorSetLayoutCreateFlags is a bitmask type for setting a mask of zero or more VkDescriptorSetLayoutCreateFlagBits.

The VkDescriptorSetLayoutBinding structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorSetLayoutBinding {
    uint32_t              binding;
    VkDescriptorType      descriptorType;
    uint32_t              descriptorCount;
    VkShaderStageFlags    stageFlags;
    const VkSampler*      pImmutableSamplers;
} VkDescriptorSetLayoutBinding;
```

- binding is the binding number of this entry and corresponds to a resource of the same binding number in the shader stages.
- descriptorType is a VkDescriptorType specifying which type of resource descriptors are used for this binding.
- descriptorCount is the number of descriptors contained in the binding, accessed in a shader as an array, except if descriptorType is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK in which case descriptorCount is the size in bytes of the inline uniform block. If descriptorCount is zero this binding entry is reserved and the resource **must** not be accessed from any stage via this binding within any pipeline using the set layout.
- stageFlags member is a bitmask of VkShaderStageFlagBits specifying which pipeline shader stages **can** access a resource for this binding. VK_SHADER_STAGE_ALL is a shorthand specifying that all defined shader stages, including any additional stages defined by extensions, **can** access the resource.

  If a shader stage is not included in stageFlags, then a resource **must** not be accessed from that stage via this binding within any pipeline using the set layout. Other than input attachments

which are limited to the fragment shader, there are no limitations on what combinations of stages **can** use a descriptor binding, and in particular a binding **can** be used by both graphics stages and the compute stage.

- `pImmutableSamplers` affects initialization of samplers. If `descriptorType` specifies a VK_DESCRIPTOR_TYPE_SAMPLER or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER type descriptor, then `pImmutableSamplers` **can** be used to initialize a set of *immutable samplers*. Immutable samplers are permanently bound into the set layout and **must** not be changed; updating a VK_DESCRIPTOR_TYPE_SAMPLER descriptor with immutable samplers is not allowed and updates to a VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER descriptor with immutable samplers do not modify the samplers (the image views are updated, but the sampler updates are ignored). If `pImmutableSamplers` is not `NULL`, then it is a pointer to an array of sampler handles that will be copied into the set layout and used for the corresponding binding. Only the sampler handles are copied; the sampler objects **must** not be destroyed before the final use of the set layout and any descriptor pools and sets created using it. If `pImmutableSamplers` is `NULL`, then the sampler slots are dynamic and sampler handles **must** be bound into descriptor sets using this layout. If `descriptorType` is not one of these descriptor types, then `pImmutableSamplers` is ignored.

The above layout definition allows the descriptor bindings to be specified sparsely such that not all binding numbers between 0 and the maximum binding number need to be specified in the `pBindings` array. Bindings that are not specified have a `descriptorCount` and `stageFlags` of zero, and the value of `descriptorType` is undefined. However, all binding numbers between 0 and the maximum binding number in the VkDescriptorSetLayoutCreateInfo::`pBindings` array **may** consume memory in the descriptor set layout even if not all descriptor bindings are used, though it **should** not consume additional memory from the descriptor pool.

NOTE	The maximum binding number specified **should** be as compact as possible to avoid wasted memory.

Valid Usage

- VUID-VkDescriptorSetLayoutBinding-descriptorType-00282
  If `descriptorType` is VK_DESCRIPTOR_TYPE_SAMPLER or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, and `descriptorCount` is not `0` and `pImmutableSamplers` is not `NULL`, `pImmutableSamplers` **must** be a valid pointer to an array of `descriptorCount` valid `VkSampler` handles
- VUID-VkDescriptorSetLayoutBinding-descriptorType-12200
  If `descriptorType` is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, and `descriptorCount` is not `0` and `pImmutableSamplers` is not `NULL`, either each element of `pImmutableSamplers` **must** be a `VkSampler` that enables sampler $Y'C_BC_R$ conversion or none of them enable sampler $Y'C_BC_R$ conversion
- VUID-VkDescriptorSetLayoutBinding-descriptorType-12215
  If `descriptorType` is VK_DESCRIPTOR_TYPE_SAMPLER, each element of `pImmutableSamplers` **must** not be a `VkSampler` object that enables sampler $Y'C_BC_R$ conversion
- VUID-VkDescriptorSetLayoutBinding-descriptorType-04604

If the inlineUniformBlock feature is not enabled, descriptorType **must** not be VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK

- VUID-VkDescriptorSetLayoutBinding-descriptorType-02209
  If descriptorType is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK then descriptorCount **must** be a multiple of 4
- VUID-VkDescriptorSetLayoutBinding-descriptorType-08004
  If descriptorType is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK and VkDescriptorSetLayoutCreateInfo::flags does not contain VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT then descriptorCount **must** be less than or equal to VkPhysicalDeviceInlineUniformBlockProperties::maxInlineUniformBlockSize
- VUID-VkDescriptorSetLayoutBinding-flags-08005
  If VkDescriptorSetLayoutCreateInfo::flags contains VK_DESCRIPTOR_SET_LAYOUT_CREATE_EMBEDDED_IMMUTABLE_SAMPLERS_BIT_EXT, descriptorType **must** be VK_DESCRIPTOR_TYPE_SAMPLER
- VUID-VkDescriptorSetLayoutBinding-flags-08006
  If VkDescriptorSetLayoutCreateInfo::flags contains VK_DESCRIPTOR_SET_LAYOUT_CREATE_EMBEDDED_IMMUTABLE_SAMPLERS_BIT_EXT, descriptorCount **must** be less than or equal to 1
- VUID-VkDescriptorSetLayoutBinding-flags-08007
  If VkDescriptorSetLayoutCreateInfo::flags contains VK_DESCRIPTOR_SET_LAYOUT_CREATE_EMBEDDED_IMMUTABLE_SAMPLERS_BIT_EXT, and descriptorCount is equal to 1, pImmutableSamplers **must** not be NULL
- VUID-VkDescriptorSetLayoutBinding-descriptorCount-09465
  If descriptorCount is not 0, stageFlags **must** be VK_SHADER_STAGE_ALL or a valid combination of other VkShaderStageFlagBits values
- VUID-VkDescriptorSetLayoutBinding-descriptorType-01510
  If descriptorType is VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT and descriptorCount is not 0, then stageFlags **must** be 0 or VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-VkDescriptorSetLayoutBinding-pImmutableSamplers-04009
  The sampler objects indicated by pImmutableSamplers **must** not have a borderColor with one of the values VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT
- VUID-VkDescriptorSetLayoutBinding-descriptorType-04605
  If descriptorType is VK_DESCRIPTOR_TYPE_MUTABLE_EXT, then pImmutableSamplers **must** be NULL

Valid Usage (Implicit)

- VUID-VkDescriptorSetLayoutBinding-descriptorType-parameter
  descriptorType **must** be a valid VkDescriptorType value

If the pNext chain of a VkDescriptorSetLayoutCreateInfo structure includes a

VkDescriptorSetLayoutBindingFlagsCreateInfo structure, then that structure includes an array of flags, one for each descriptor set layout binding.

The VkDescriptorSetLayoutBindingFlagsCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkDescriptorSetLayoutBindingFlagsCreateInfo {
    VkStructureType                    sType;
    const void*                        pNext;
    uint32_t                           bindingCount;
    const VkDescriptorBindingFlags*    pBindingFlags;
} VkDescriptorSetLayoutBindingFlagsCreateInfo;
```

```
// Provided by VK_EXT_descriptor_indexing
// Equivalent to VkDescriptorSetLayoutBindingFlagsCreateInfo
typedef VkDescriptorSetLayoutBindingFlagsCreateInfo
VkDescriptorSetLayoutBindingFlagsCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- bindingCount is zero or the number of elements in pBindingFlags.
- pBindingFlags is a pointer to an array of VkDescriptorBindingFlags bitfields, one for each descriptor set layout binding.

If bindingCount is zero or if this structure is not included in the pNext chain, the VkDescriptorBindingFlags for each descriptor set layout binding is considered to be zero. Otherwise, the descriptor set layout binding at VkDescriptorSetLayoutCreateInfo::pBindings[i] uses the flags in pBindingFlags[i].

Valid Usage

- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-bindingCount-03002
  If bindingCount is not zero, bindingCount **must** equal VkDescriptorSetLayoutCreateInfo ::bindingCount
- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-flags-03003
  If VkDescriptorSetLayoutCreateInfo::flags includes VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT, then all elements of pBindingFlags **must** not include VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT, VK_DESCRIPTOR_BINDING_UPDATE_UNUSED_WHILE_PENDING_BIT, or VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT
- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-pBindingFlags-03004
  If an element of pBindingFlags includes VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT, then it **must** be the element with the highest binding number

- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-descriptorBindingUniformBufferUpdateAfterBind-03005
  If VkPhysicalDeviceDescriptorIndexingFeatures::descriptorBindingUniformBufferUpdateAfterBind is not enabled, all bindings with descriptor type VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER **must** not use VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT
- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-descriptorBindingSampledImageUpdateAfterBind-03006
  If VkPhysicalDeviceDescriptorIndexingFeatures::descriptorBindingSampledImageUpdateAfterBind is not enabled, all bindings with descriptor type VK_DESCRIPTOR_TYPE_SAMPLER, VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, or VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE **must** not use VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT
- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-descriptorBindingStorageImageUpdateAfterBind-03007
  If VkPhysicalDeviceDescriptorIndexingFeatures::descriptorBindingStorageImageUpdateAfterBind is not enabled, all bindings with descriptor type VK_DESCRIPTOR_TYPE_STORAGE_IMAGE **must** not use VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT
- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-descriptorBindingStorageBufferUpdateAfterBind-03008
  If VkPhysicalDeviceDescriptorIndexingFeatures::descriptorBindingStorageBufferUpdateAfterBind is not enabled, all bindings with descriptor type VK_DESCRIPTOR_TYPE_STORAGE_BUFFER **must** not use VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT
- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-descriptorBindingUniformTexelBufferUpdateAfterBind-03009
  If VkPhysicalDeviceDescriptorIndexingFeatures::descriptorBindingUniformTexelBufferUpdateAfterBind is not enabled, all bindings with descriptor type VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER **must** not use VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT
- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-descriptorBindingStorageTexelBufferUpdateAfterBind-03010
  If VkPhysicalDeviceDescriptorIndexingFeatures::descriptorBindingStorageTexelBufferUpdateAfterBind is not enabled, all bindings with descriptor type VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER **must** not use VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT
- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-descriptorBindingInlineUniformBlockUpdateAfterBind-02211
  If VkPhysicalDeviceInlineUniformBlockFeatures::descriptorBindingInlineUniformBlockUpdateAfterBind is not enabled, all bindings with descriptor type VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK **must** not use VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT
- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-

descriptorBindingAccelerationStructureUpdateAfterBind-03570
If VkPhysicalDeviceAccelerationStructureFeaturesKHR ::descriptorBindingAccelerationStructureUpdateAfterBind is not enabled, all bindings with descriptor type VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR or VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_NV **must** not use VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT

- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-None-03011
  All bindings with descriptor type VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC, or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC **must** not use VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT
- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-descriptorBindingUpdateUnusedWhilePending-03012
  If VkPhysicalDeviceDescriptorIndexingFeatures ::descriptorBindingUpdateUnusedWhilePending is not enabled, all elements of pBindingFlags **must** not include VK_DESCRIPTOR_BINDING_UPDATE_UNUSED_WHILE_PENDING_BIT
- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-descriptorBindingPartiallyBound-03013
  If VkPhysicalDeviceDescriptorIndexingFeatures::descriptorBindingPartiallyBound is not enabled, all elements of pBindingFlags **must** not include VK_DESCRIPTOR_BINDING_PARTIALLY_BOUND_BIT
- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-descriptorBindingVariableDescriptorCount-03014
  If VkPhysicalDeviceDescriptorIndexingFeatures ::descriptorBindingVariableDescriptorCount is not enabled, all elements of pBindingFlags **must** not include VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT
- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-pBindingFlags-03015
  If an element of pBindingFlags includes VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT, that element's descriptorType **must** not be VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC

Valid Usage (Implicit)

- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_BINDING_FLAGS_CREATE_INFO
- VUID-VkDescriptorSetLayoutBindingFlagsCreateInfo-pBindingFlags-parameter
  If bindingCount is not 0, pBindingFlags **must** be a valid pointer to an array of bindingCount valid combinations of VkDescriptorBindingFlagBits values

Structure Chaining

Extends the structure

- VkDescriptorSetLayoutCreateInfo

Bits which **can** be set in each element of VkDescriptorSetLayoutBindingFlagsCreateInfo ::pBindingFlags, specifying options for the corresponding descriptor set layout binding, are:

```
// Provided by VK_VERSION_1_2
typedef enum VkDescriptorBindingFlagBits {
    VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT = 0x00000001,
    VK_DESCRIPTOR_BINDING_UPDATE_UNUSED_WHILE_PENDING_BIT = 0x00000002,
    VK_DESCRIPTOR_BINDING_PARTIALLY_BOUND_BIT = 0x00000004,
    VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT = 0x00000008,
  // Provided by VK_EXT_descriptor_indexing
    VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT_EXT =
VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT,
  // Provided by VK_EXT_descriptor_indexing
    VK_DESCRIPTOR_BINDING_UPDATE_UNUSED_WHILE_PENDING_BIT_EXT =
VK_DESCRIPTOR_BINDING_UPDATE_UNUSED_WHILE_PENDING_BIT,
  // Provided by VK_EXT_descriptor_indexing
    VK_DESCRIPTOR_BINDING_PARTIALLY_BOUND_BIT_EXT =
VK_DESCRIPTOR_BINDING_PARTIALLY_BOUND_BIT,
  // Provided by VK_EXT_descriptor_indexing
    VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT_EXT =
VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT,
} VkDescriptorBindingFlagBits;
```

```
// Provided by VK_EXT_descriptor_indexing
// Equivalent to VkDescriptorBindingFlagBits
typedef VkDescriptorBindingFlagBits VkDescriptorBindingFlagBitsEXT;
```

- VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT specifies that if descriptors in this binding are updated between when the descriptor set is bound in a command buffer and when that command buffer is submitted to a queue, then the submission will use the most recently set descriptors for this binding and the updates do not invalidate the command buffer. Descriptor bindings created with this flag are also partially exempt from the external synchronization requirement in vkUpdateDescriptorSetWithTemplateKHR and vkUpdateDescriptorSets. Multiple descriptors with this flag set **can** be updated concurrently in different threads, though the same descriptor **must** not be updated concurrently by two threads. Descriptors with this flag set **can** be updated concurrently with the set being bound to a command buffer in another thread, but not concurrently with the set being reset or freed.
- VK_DESCRIPTOR_BINDING_PARTIALLY_BOUND_BIT specifies that descriptors in this binding that are not *dynamically used* need not contain valid descriptors at the time the descriptors are consumed. A descriptor is dynamically used if any shader invocation executes an instruction

that performs any memory access using the descriptor. If a descriptor is not dynamically used, any resource referenced by the descriptor is not considered to be referenced during command execution.

- VK_DESCRIPTOR_BINDING_UPDATE_UNUSED_WHILE_PENDING_BIT specifies that descriptors in this binding **can** be updated after a command buffer has bound this descriptor set, or while a command buffer that uses this descriptor set is pending execution, as long as the descriptors that are updated are not used by those command buffers. Descriptor bindings created with this flag are also partially exempt from the external synchronization requirement in vkUpdateDescriptorSetWithTemplateKHR and vkUpdateDescriptorSets in the same way as for VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT. If VK_DESCRIPTOR_BINDING_PARTIALLY_BOUND_BIT is also set, then descriptors **can** be updated as long as they are not dynamically used by any shader invocations. If VK_DESCRIPTOR_BINDING_PARTIALLY_BOUND_BIT is not set, then descriptors **can** be updated as long as they are not statically used by any shader invocations.
- VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT specifies that this is a *variable-sized descriptor binding* whose size will be specified when a descriptor set is allocated using this layout. The value of `descriptorCount` is treated as an upper bound on the size of the binding. This **must** only be used for the last binding in the descriptor set layout (i.e. the binding with the largest value of `binding`). For the purposes of counting against limits such as `maxDescriptorSet*` and `maxPerStageDescriptor*`, the full value of `descriptorCount` is counted, except for descriptor bindings with a descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK, when VkDescriptorSetLayoutCreateInfo::`flags` does not contain VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT. In this case, `descriptorCount` specifies the upper bound on the byte size of the binding; thus it counts against the `maxInlineUniformTotalSize` limit instead.

NOTE Note that while VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT and VK_DESCRIPTOR_BINDING_UPDATE_UNUSED_WHILE_PENDING_BIT both involve updates to descriptor sets after they are bound, VK_DESCRIPTOR_BINDING_UPDATE_UNUSED_WHILE_PENDING_BIT is a weaker requirement since it is only about descriptors that are not used, whereas VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT requires the implementation to observe updates to descriptors that are used.

```
// Provided by VK_VERSION_1_2
typedef VkFlags VkDescriptorBindingFlags;
```

```
// Provided by VK_EXT_descriptor_indexing
// Equivalent to VkDescriptorBindingFlags
typedef VkDescriptorBindingFlags VkDescriptorBindingFlagsEXT;
```

`VkDescriptorBindingFlags` is a bitmask type for setting a mask of zero or more VkDescriptorBindingFlagBits.

To query information about whether a descriptor set layout **can** be created, call:

```
// Provided by VK_VERSION_1_1
void vkGetDescriptorSetLayoutSupport(
    VkDevice                                    device,
    const VkDescriptorSetLayoutCreateInfo*      pCreateInfo,
    VkDescriptorSetLayoutSupport*               pSupport);
```

```
// Provided by VK_KHR_maintenance3
// Equivalent to vkGetDescriptorSetLayoutSupport
void vkGetDescriptorSetLayoutSupportKHR(
    VkDevice                                    device,
    const VkDescriptorSetLayoutCreateInfo*      pCreateInfo,
    VkDescriptorSetLayoutSupport*               pSupport);
```

- `device` is the logical device that would create the descriptor set layout.
- `pCreateInfo` is a pointer to a VkDescriptorSetLayoutCreateInfo structure specifying the state of the descriptor set layout object.
- `pSupport` is a pointer to a VkDescriptorSetLayoutSupport structure, in which information about support for the descriptor set layout object is returned.

Some implementations have limitations on what fits in a descriptor set which are not easily expressible in terms of existing limits like `maxDescriptorSet*`, for example if all descriptor types share a limited space in memory but each descriptor is a different size or alignment. This command returns information about whether a descriptor set satisfies this limit. If the descriptor set layout satisfies the VkPhysicalDeviceMaintenance3Properties::`maxPerSetDescriptors` limit, this command is guaranteed to return VK_TRUE in VkDescriptorSetLayoutSupport::`supported`. If the descriptor set layout exceeds the VkPhysicalDeviceMaintenance3Properties::`maxPerSetDescriptors` limit, whether the descriptor set layout is supported is implementation-dependent and **may** depend on whether the descriptor sizes and alignments cause the layout to exceed an internal limit.

This command does not consider other limits such as `maxPerStageDescriptor*`, and so a descriptor set layout that is supported according to this command **must** still satisfy the pipeline layout limits such as `maxPerStageDescriptor*` in order to be used in a pipeline layout.

NOTE This is a `VkDevice` query rather than `VkPhysicalDevice` because the answer **may** depend on enabled features.

Valid Usage (Implicit)

- VUID-vkGetDescriptorSetLayoutSupport-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDescriptorSetLayoutSupport-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkDescriptorSetLayoutCreateInfo structure
- VUID-vkGetDescriptorSetLayoutSupport-pSupport-parameter
  `pSupport` **must** be a valid pointer to a VkDescriptorSetLayoutSupport structure

Information about support for the descriptor set layout is returned in a VkDescriptorSetLayoutSupport structure:

```
// Provided by VK_VERSION_1_1
typedef struct VkDescriptorSetLayoutSupport {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           supported;
} VkDescriptorSetLayoutSupport;
```

```
// Provided by VK_KHR_maintenance3
// Equivalent to VkDescriptorSetLayoutSupport
typedef VkDescriptorSetLayoutSupport VkDescriptorSetLayoutSupportKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- supported specifies whether the descriptor set layout **can** be created.

supported will be VK_TRUE if the descriptor set **can** be created, or else VK_FALSE.

Valid Usage (Implicit)

- VUID-VkDescriptorSetLayoutSupport-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_SUPPORT
- VUID-VkDescriptorSetLayoutSupport-pNext-pNext
  pNext **must** be NULL or a pointer to a valid instance of VkDescriptorSetVariableDescriptorCountLayoutSupport
- VUID-VkDescriptorSetLayoutSupport-sType-unique
  The sType value of each structure in the pNext chain **must** be unique

If the pNext chain of a VkDescriptorSetLayoutSupport structure includes a VkDescriptorSetVariableDescriptorCountLayoutSupport structure, then that structure returns additional information about whether the descriptor set layout is supported.

```
// Provided by VK_VERSION_1_2
typedef struct VkDescriptorSetVariableDescriptorCountLayoutSupport {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxVariableDescriptorCount;
} VkDescriptorSetVariableDescriptorCountLayoutSupport;
```

```
// Provided by VK_EXT_descriptor_indexing
// Equivalent to VkDescriptorSetVariableDescriptorCountLayoutSupport
typedef VkDescriptorSetVariableDescriptorCountLayoutSupport
VkDescriptorSetVariableDescriptorCountLayoutSupportEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxVariableDescriptorCount` indicates the maximum number of descriptors supported in the highest numbered binding of the layout, if that binding is variable-sized. If the highest numbered binding of the layout has a descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK then `maxVariableDescriptorCount` indicates the maximum byte size supported for the binding, if that binding is variable-sized.

If the VkDescriptorSetLayoutCreateInfo structure specified in vkGetDescriptorSetLayoutSupport ::`pCreateInfo` includes a variable-sized descriptor, then `supported` is determined assuming the requested size of the variable-sized descriptor, and `maxVariableDescriptorCount` is the maximum size of that descriptor that **can** be successfully created (which is greater than or equal to the requested size passed in). If the VkDescriptorSetLayoutCreateInfo structure does not include a variable-sized descriptor, or if the VkPhysicalDeviceDescriptorIndexingFeatures ::`descriptorBindingVariableDescriptorCount` feature is not enabled, then `maxVariableDescriptorCount` is zero. For the purposes of this command, a variable-sized descriptor binding with a `descriptorCount` of zero is treated as having a `descriptorCount` of four if `descriptorType` is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK, or one otherwise, and thus the binding is not ignored and the maximum descriptor count will be returned. If the layout is not supported, then the value written to `maxVariableDescriptorCount` is undefined.

Valid Usage (Implicit)

- VUID-VkDescriptorSetVariableDescriptorCountLayoutSupport-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DESCRIPTOR_SET_VARIABLE_DESCRIPTOR_COUNT_LAYOUT_SUPPORT

Structure Chaining

Extends the structure

- VkDescriptorSetLayoutSupport

The following examples show a shader snippet using two descriptor sets, and application code that creates corresponding descriptor set layouts.

GLSL Example

```
//
// binding to a single sampled image descriptor in set 0
```

```
//
layout (set=0, binding=0) uniform texture2D mySampledImage;

//
// binding to an array of sampled image descriptors in set 0
//
layout (set=0, binding=1) uniform texture2D myArrayOfSampledImages[12];

//
// binding to a single uniform buffer descriptor in set 1
//
layout (set=1, binding=0) uniform myUniformBuffer
{
    vec4 myElement[32];
};
```

SPIR-V Example

```
          ...
     %1 = OpExtInstImport "GLSL.std.450"
          ...
          OpName %9 "mySampledImage"
          OpName %14 "myArrayOfSampledImages"
          OpName %18 "myUniformBuffer"
          OpMemberName %18 0 "myElement"
          OpName %20 ""
          OpDecorate %9 DescriptorSet 0
          OpDecorate %9 Binding 0
          OpDecorate %14 DescriptorSet 0
          OpDecorate %14 Binding 1
          OpDecorate %17 ArrayStride 16
          OpMemberDecorate %18 0 Offset 0
          OpDecorate %18 Block
          OpDecorate %20 DescriptorSet 1
          OpDecorate %20 Binding 0
     %2 = OpTypeVoid
     %3 = OpTypeFunction %2
     %6 = OpTypeFloat 32
     %7 = OpTypeImage %6 2D 0 0 0 1 Unknown
     %8 = OpTypePointer UniformConstant %7
     %9 = OpVariable %8 UniformConstant
    %10 = OpTypeInt 32 0
    %11 = OpConstant %10 12
    %12 = OpTypeArray %7 %11
    %13 = OpTypePointer UniformConstant %12
    %14 = OpVariable %13 UniformConstant
    %15 = OpTypeVector %6 4
    %16 = OpConstant %10 32
    %17 = OpTypeArray %15 %16
    %18 = OpTypeStruct %17
```

```
        %19 = OpTypePointer Uniform %18
        %20 = OpVariable %19 Uniform
              ...
```

API Example

```
VkResult myResult;

const VkDescriptorSetLayoutBinding myDescriptorSetLayoutBinding[] =
{
    // binding to a single image descriptor
    {
        .dstBinding = 0,
        .descriptorType = VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE,
        .descriptorCount = 1,
        .stageFlags = VK_SHADER_STAGE_FRAGMENT_BIT,
        .pImmutableSamplers = NULL
    },

    // binding to an array of image descriptors
    {
        .dstBinding = 1,
        .descriptorType = VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE,
        .descriptorCount = 12,
        .stageFlags = VK_SHADER_STAGE_FRAGMENT_BIT,
        .pImmutableSamplers = NULL
    },

    // binding to a single uniform buffer descriptor
    {
        .dstBinding = 0,
        .descriptorType = VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER,
        .descriptorCount = 1,
        .stageFlags = VK_SHADER_STAGE_FRAGMENT_BIT,
        .pImmutableSamplers = NULL
    }
};

const VkDescriptorSetLayoutCreateInfo myDescriptorSetLayoutCreateInfo[] =
{
    // Information for first descriptor set with two descriptor bindings
    {
        .sType = VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_CREATE_INFO,
        .pNext = NULL,
        .flags = 0,
        .bindingCount = 2,
        .pBindings = &myDescriptorSetLayoutBinding[0]
    },

    // Information for second descriptor set with one descriptor binding
```

```
    {
        .sType = VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_CREATE_INFO,
        .pNext = NULL,
        .flags = 0,
        .bindingCount = 1,
        .pBindings = &myDescriptorSetLayoutBinding[2]
    }
};

VkDescriptorSetLayout myDescriptorSetLayout[2];

//
// Create first descriptor set layout
//
myResult = vkCreateDescriptorSetLayout(
    myDevice,
    &myDescriptorSetLayoutCreateInfo[0],
    NULL,
    &myDescriptorSetLayout[0]);

//
// Create second descriptor set layout
//
myResult = vkCreateDescriptorSetLayout(
    myDevice,
    &myDescriptorSetLayoutCreateInfo[1],
    NULL,
    &myDescriptorSetLayout[1]);
```

To destroy a descriptor set layout, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyDescriptorSetLayout(
    VkDevice                                    device,
    VkDescriptorSetLayout                       descriptorSetLayout,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the descriptor set layout.
- `descriptorSetLayout` is the descriptor set layout to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyDescriptorSetLayout-descriptorSetLayout-00284
  If `VkAllocationCallbacks` were provided when `descriptorSetLayout` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyDescriptorSetLayout-descriptorSetLayout-00285

If no VkAllocationCallbacks were provided when descriptorSetLayout was created, pAllocator **must** be NULL

Valid Usage (Implicit)

- VUID-vkDestroyDescriptorSetLayout-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkDestroyDescriptorSetLayout-descriptorSetLayout-parameter
  If descriptorSetLayout is not VK_NULL_HANDLE, descriptorSetLayout **must** be a valid VkDescriptorSetLayout handle
- VUID-vkDestroyDescriptorSetLayout-pAllocator-parameter
  If pAllocator is not NULL, pAllocator **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyDescriptorSetLayout-descriptorSetLayout-parent
  If descriptorSetLayout is a valid handle, it **must** have been created, allocated, or retrieved from device

Host Synchronization

- Host access to descriptorSetLayout **must** be externally synchronized

17.2. Pipeline Layouts

Access to descriptor sets from a pipeline is accomplished through a *pipeline layout*. Zero or more descriptor set layouts and zero or more push constant ranges are combined to form a pipeline layout object describing the complete set of resources that **can** be accessed by a pipeline. The pipeline layout represents a sequence of descriptor sets with each having a specific layout. This sequence of layouts is used to determine the interface between shader stages and shader resources. Each pipeline is created using a pipeline layout.

Pipeline layout objects are represented by VkPipelineLayout handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkPipelineLayout)
```

To create a pipeline layout, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreatePipelineLayout(
    VkDevice                                    device,
    const VkPipelineLayoutCreateInfo*           pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkPipelineLayout*                           pPipelineLayout);
```

- `device` is the logical device that creates the pipeline layout.
- `pCreateInfo` is a pointer to a VkPipelineLayoutCreateInfo structure specifying the state of the pipeline layout object.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pPipelineLayout` is a pointer to a VkPipelineLayout handle in which the resulting pipeline layout object is returned.

Valid Usage (Implicit)

- VUID-vkCreatePipelineLayout-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreatePipelineLayout-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkPipelineLayoutCreateInfo structure
- VUID-vkCreatePipelineLayout-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreatePipelineLayout-pPipelineLayout-parameter
  `pPipelineLayout` **must** be a valid pointer to a VkPipelineLayout handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkPipelineLayoutCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineLayoutCreateInfo {
    VkStructureType                 sType;
    const void*                     pNext;
    VkPipelineLayoutCreateFlags     flags;
    uint32_t                        setLayoutCount;
    const VkDescriptorSetLayout*    pSetLayouts;
    uint32_t                        pushConstantRangeCount;
    const VkPushConstantRange*      pPushConstantRanges;
} VkPipelineLayoutCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkPipelineLayoutCreateFlagBits specifying options for pipeline layout creation.
- `setLayoutCount` is the number of descriptor sets included in the pipeline layout.
- `pSetLayouts` is a pointer to an array of `VkDescriptorSetLayout` objects. The implementation **must** not access these objects outside of the duration of the command this structure is passed to.
- `pushConstantRangeCount` is the number of push constant ranges included in the pipeline layout.
- `pPushConstantRanges` is a pointer to an array of VkPushConstantRange structures defining a set of push constant ranges for use in a single pipeline layout. In addition to descriptor set layouts, a pipeline layout also describes how many push constants **can** be accessed by each stage of the pipeline.

NOTE	Push constants represent a high speed path to modify constant data in pipelines that is expected to outperform memory-backed resource updates.

Valid Usage

- VUID-VkPipelineLayoutCreateInfo-setLayoutCount-00286
  `setLayoutCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxBoundDescriptorSets`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03016
  The total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a `descriptorType` of VK_DESCRIPTOR_TYPE_SAMPLER and VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER accessible to any given shader stage across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageDescriptorSamplers`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03017
  The total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a `descriptorType` of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER and

VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC accessible to any given shader stage across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceLimits::maxPerStageDescriptorUniformBuffers

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03018
  The total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_STORAGE_BUFFER and VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC accessible to any given shader stage across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceLimits::maxPerStageDescriptorStorageBuffers
- VUID-VkPipelineLayoutCreateInfo-descriptorType-06939
  The total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, and VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER, accessible to any given shader stage across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceLimits::maxPerStageDescriptorSampledImages
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03020
  The total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, and VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER accessible to any given shader stage across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceLimits::maxPerStageDescriptorStorageImages
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03021
  The total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT accessible to any given shader stage across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceLimits::maxPerStageDescriptorInputAttachments
- VUID-VkPipelineLayoutCreateInfo-descriptorType-02214
  The total number of bindings in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set and with a descriptorType of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK accessible to any given shader stage across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceInlineUniformBlockProperties::maxPerStageDescriptorInlineUniformBlocks
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03022
  The total number of descriptors with a descriptorType of VK_DESCRIPTOR_TYPE_SAMPLER and VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER accessible to any given shader stage across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceLimits::maxPerStageDescriptorSamplers and VkPhysicalDeviceDescriptorIndexingProperties::maxPerStageDescriptorUpdateAfterBindSamplers

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03023
  The total number of descriptors with a descriptorType of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER and VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC accessible to any given shader stage across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceLimits::maxPerStageDescriptorUniformBuffers and VkPhysicalDeviceDescriptorIndexingProperties::maxPerStageDescriptorUpdateAfterBindUniformBuffers
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03024
  The total number of descriptors with a descriptorType of VK_DESCRIPTOR_TYPE_STORAGE_BUFFER and VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC accessible to any given shader stage across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceLimits::maxPerStageDescriptorStorageBuffers and VkPhysicalDeviceDescriptorIndexingProperties::maxPerStageDescriptorUpdateAfterBindStorageBuffers
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03025
  The total number of descriptors with a descriptorType of VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, and VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER accessible to any given shader stage across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceLimits::maxPerStageDescriptorSampledImages and VkPhysicalDeviceDescriptorIndexingProperties::maxPerStageDescriptorUpdateAfterBindSampledImages
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03026
  The total number of descriptors with a descriptorType of VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, and VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER accessible to any given shader stage across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceLimits::maxPerStageDescriptorStorageImages and VkPhysicalDeviceDescriptorIndexingProperties::maxPerStageDescriptorUpdateAfterBindStorageImages
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03027
  The total number of descriptors with a descriptorType of VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT accessible to any given shader stage across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceLimits::maxPerStageDescriptorInputAttachments and VkPhysicalDeviceDescriptorIndexingProperties::maxPerStageDescriptorUpdateAfterBindInputAttachments
- VUID-VkPipelineLayoutCreateInfo-descriptorType-02215
  The total number of bindings with a descriptorType of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK accessible to any given shader stage across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceInlineUniformBlockProperties::maxPerStageDescriptorInlineUniformBlocks and VkPhysicalDeviceInlineUniformBlockProperties

::maxPerStageDescriptorUpdateAfterBindInlineUniformBlocks

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03028
  The total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_SAMPLER and VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceLimits::maxDescriptorSetSamplers
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03029
  The total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceLimits::maxDescriptorSetUniformBuffers
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03030
  If the maintenance7 feature is not enabled, the total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceLimits::maxDescriptorSetUniformBuffersDynamic
- VUID-VkPipelineLayoutCreateInfo-maintenance7-10003
  If the maintenance7 feature is enabled, the total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceMaintenance7PropertiesKHR::maxDescriptorSetTotalUniformBuffersDynamic
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03031
  The total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_STORAGE_BUFFER accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceLimits::maxDescriptorSetStorageBuffers
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03032
  If the maintenance7 feature is not enabled, the total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceLimits::maxDescriptorSetStorageBuffersDynamic
- VUID-VkPipelineLayoutCreateInfo-maintenance7-10004
  If the maintenance7 feature is enabled, the total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a

descriptorType of VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceMaintenance7PropertiesKHR::maxDescriptorSetTotalStorageBuffersDynamic

- VUID-VkPipelineLayoutCreateInfo-None-10005
  The total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceMaintenance7PropertiesKHR::maxDescriptorSetTotalBuffersDynamic
- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-10006
  The total number of descriptors of the type VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceMaintenance7PropertiesKHR::maxDescriptorSetTotalBuffersDynamic and VkPhysicalDeviceMaintenance7PropertiesKHR::maxDescriptorSetUpdateAfterBindTotalBuffersDynamic
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03033
  The total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, and VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceLimits::maxDescriptorSetSampledImages
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03034
  The total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, and VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceLimits::maxDescriptorSetStorageImages
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03035
  The total number of descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceLimits::maxDescriptorSetInputAttachments
- VUID-VkPipelineLayoutCreateInfo-descriptorType-02216
  The total number of bindings in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceInlineUniformBlockProperties::maxDescriptorSetInlineUniformBlocks

- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-03036
  The total number of descriptors of the type VK_DESCRIPTOR_TYPE_SAMPLER and VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceLimits::maxDescriptorSetSamplers and VkPhysicalDeviceDescriptorIndexingProperties::maxDescriptorSetUpdateAfterBindSamplers
- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-03037
  The total number of descriptors of the type VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceLimits::maxDescriptorSetUniformBuffers and VkPhysicalDeviceDescriptorIndexingProperties::maxDescriptorSetUpdateAfterBindUniformBuffers
- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-03038
  If the maintenance7 feature is not enabled, the total number of descriptors of the type VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceLimits::maxDescriptorSetUniformBuffersDynamic and VkPhysicalDeviceDescriptorIndexingProperties::maxDescriptorSetUpdateAfterBindUniformBuffersDynamic
- VUID-VkPipelineLayoutCreateInfo-maintenance7-10007
  If the maintenance7 feature is enabled, the total number of descriptors of the type VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceMaintenance7PropertiesKHR::maxDescriptorSetTotalUniformBuffersDynamic and VkPhysicalDeviceMaintenance7PropertiesKHR::maxDescriptorSetUpdateAfterBindTotalUniformBuffersDynamic
- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-03039
  The total number of descriptors of the type VK_DESCRIPTOR_TYPE_STORAGE_BUFFER accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceLimits::maxDescriptorSetStorageBuffers and VkPhysicalDeviceDescriptorIndexingProperties::maxDescriptorSetUpdateAfterBindStorageBuffers
- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-03040
  If the maintenance7 feature is not enabled, the total number of descriptors of the type VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceLimits::maxDescriptorSetStorageBuffersDynamic and VkPhysicalDeviceDescriptorIndexingProperties::maxDescriptorSetUpdateAfterBindStorageBuffersDynamic
- VUID-VkPipelineLayoutCreateInfo-maintenance7-10008
  If the maintenance7 feature is enabled, the total number of descriptors of the type VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceMaintenance7PropertiesKHR::maxDescriptorSetTotalStorageBuffersDynamic and VkPhysicalDeviceMaintenance7PropertiesKHR

::maxDescriptorSetUpdateAfterBindTotalStorageBuffersDynamic

- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-03041
  The total number of descriptors of the type VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, and VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceLimits::maxDescriptorSetSampledImages and VkPhysicalDeviceDescriptorIndexingProperties::maxDescriptorSetUpdateAfterBindSampledImages
- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-03042
  The total number of descriptors of the type VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, and VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceLimits::maxDescriptorSetStorageImages and VkPhysicalDeviceDescriptorIndexingProperties::maxDescriptorSetUpdateAfterBindStorageImages
- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-03043
  The total number of descriptors of the type VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceLimits::maxDescriptorSetInputAttachments and VkPhysicalDeviceDescriptorIndexingProperties::maxDescriptorSetUpdateAfterBindInputAttachments
- VUID-VkPipelineLayoutCreateInfo-descriptorType-02217
  The total number of bindings with a descriptorType of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceInlineUniformBlockProperties::maxDescriptorSetInlineUniformBlocks and VkPhysicalDeviceInlineUniformBlockProperties::maxDescriptorSetUpdateAfterBindInlineUniformBlocks
- VUID-VkPipelineLayoutCreateInfo-descriptorType-06531
  The total number of descriptors with a descriptorType of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceVulkan13Properties::maxInlineUniformTotalSize
- VUID-VkPipelineLayoutCreateInfo-pPushConstantRanges-00292
  Any two elements of pPushConstantRanges **must** not include the same stage in stageFlags
- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-00293
  pSetLayouts **must** not contain more than one descriptor set layout that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT set
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03571
  The total number of bindings in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR accessible to any given shader stage across all elements of pSetLayouts **must** be less than or equal to

VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxPerStageDescriptorAccelerationStructures

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03572
  The total number of bindings with a descriptorType of VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR accessible to any given shader stage across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxPerStageDescriptorAccelerationStructures and VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxPerStageDescriptorUpdateAfterBindAccelerationStructures
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03573
  The total number of bindings in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set with a descriptorType of VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxDescriptorSetAccelerationStructures
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03574
  The total number of bindings with a descriptorType of VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR accessible across all shader stages and across all elements of pSetLayouts **must** be less than or equal to the maximum of VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxDescriptorSetAccelerationStructures and VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxDescriptorSetUpdateAfterBindAccelerationStructures
- VUID-VkPipelineLayoutCreateInfo-pImmutableSamplers-03566
  The total number of pImmutableSamplers created with flags containing VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT or VK_SAMPLER_CREATE_SUBSAMPLED_COARSE_RECONSTRUCTION_BIT_EXT across all shader stages and across all elements of pSetLayouts **must** be less than or equal to VkPhysicalDeviceFragmentDensityMap2PropertiesEXT::maxDescriptorSetSubsampledSamplers
- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-04606
  Any element of pSetLayouts **must** not have been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_HOST_ONLY_POOL_BIT_EXT bit set
- VUID-VkPipelineLayoutCreateInfo-graphicsPipelineLibrary-06753
  If the graphicsPipelineLibrary feature is not enabled, elements of pSetLayouts **must** be valid VkDescriptorSetLayout objects
- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-08008
  If any element of pSetLayouts was created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT bit set, all elements of pSetLayouts **must** have been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT bit set
- VUID-VkPipelineLayoutCreateInfo-flags-13352
  If flags has VK_PIPELINE_LAYOUT_CREATE_NO_TASK_SHADER_BIT_KHR set, then VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT **must** also be set

Valid Usage (Implicit)

- VUID-VkPipelineLayoutCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_LAYOUT_CREATE_INFO
- VUID-VkPipelineLayoutCreateInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkPipelineLayoutCreateFlagBits values
- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-parameter
  If `setLayoutCount` is not `0`, `pSetLayouts` **must** be a valid pointer to an array of `setLayoutCount` valid or VK_NULL_HANDLE VkDescriptorSetLayout handles
- VUID-VkPipelineLayoutCreateInfo-pPushConstantRanges-parameter
  If `pushConstantRangeCount` is not `0`, `pPushConstantRanges` **must** be a valid pointer to an array of `pushConstantRangeCount` valid VkPushConstantRange structures

Structure Chaining

Extends the structures

- VkBindDescriptorBufferEmbeddedSamplersInfoEXT
- VkBindDescriptorSetsInfo
- VkIndirectCommandsLayoutCreateInfoEXT
- VkPushConstantsInfo
- VkPushDescriptorSetInfo
- VkPushDescriptorSetWithTemplateInfo
- VkSetDescriptorBufferOffsetsInfoEXT

```
// Provided by VK_VERSION_1_0, VK_EXT_graphics_pipeline_library
typedef enum VkPipelineLayoutCreateFlagBits {
  // Provided by VK_KHR_maintenance11 with VK_EXT_shader_object,
VK_EXT_graphics_pipeline_library
    VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT = 0x00000002,
  // Provided by VK_KHR_maintenance11 with VK_EXT_shader_object and
(VK_EXT_mesh_shader or VK_NV_mesh_shader)
    VK_PIPELINE_LAYOUT_CREATE_NO_TASK_SHADER_BIT_KHR = 0x00000004,
} VkPipelineLayoutCreateFlagBits;
```

- VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT specifies that implementations **must** ensure that the properties and/or absence of a particular descriptor set do not influence any other properties of the pipeline layout. This allows pipeline libraries linked without VK_PIPELINE_CREATE_LINK_TIME_OPTIMIZATION_BIT_EXT to be created with a subset of the total descriptor sets.
- VK_PIPELINE_LAYOUT_CREATE_NO_TASK_SHADER_BIT_KHR, when used in combination with

VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT, specifies that this pipeline layout will only be used to draw with shader objects created with VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineLayoutCreateFlags;
```

VkPipelineLayoutCreateFlags is a bitmask type for setting a mask of VkPipelineLayoutCreateFlagBits.

The VkPushConstantRange structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPushConstantRange {
    VkShaderStageFlags    stageFlags;
    uint32_t              offset;
    uint32_t              size;
} VkPushConstantRange;
```

- stageFlags is a set of stage flags describing the shader stages that will access a range of push constants. If a particular stage is not included in the range, then accessing members of that range of push constants from the corresponding shader stage will return undefined values.
- offset and size are the start offset and size, respectively, consumed by the range. Both offset and size are in units of bytes and **must** be a multiple of 4. The layout of the push constant variables is specified in the shader.

Valid Usage

- VUID-VkPushConstantRange-offset-00294
  offset **must** be less than VkPhysicalDeviceLimits::maxPushConstantsSize
- VUID-VkPushConstantRange-offset-00295
  offset **must** be a multiple of 4
- VUID-VkPushConstantRange-size-00296
  size **must** be greater than 0
- VUID-VkPushConstantRange-size-00297
  size **must** be a multiple of 4
- VUID-VkPushConstantRange-size-00298
  size **must** be less than or equal to VkPhysicalDeviceLimits::maxPushConstantsSize minus offset

Valid Usage (Implicit)

- VUID-VkPushConstantRange-stageFlags-parameter
  stageFlags **must** be a valid combination of VkShaderStageFlagBits values

Once created, pipeline layouts **can** be used as part of pipeline creation (see Pipelines), as part of binding descriptor sets (see Descriptor Set Binding), and as part of setting push constants (see Push Constant Updates). Pipeline creation accepts a pipeline layout as input, and the layout **may** be used to map (set, binding, arrayElement) tuples to implementation resources or memory locations within a descriptor set. The assignment of implementation resources depends only on the bindings defined in the descriptor sets that comprise the pipeline layout, and not on any shader source.

All resource variables statically used in all shaders in a pipeline **must** be declared with a (set, binding, arrayElement) that exists in the corresponding descriptor set layout and is of an appropriate descriptor type and includes the set of shader stages it is used by in `stageFlags`. The pipeline layout **can** include entries that are not used by a particular pipeline. The pipeline layout allows the application to provide a consistent set of bindings across multiple pipeline compiles, which enables those pipelines to be compiled in a way that the implementation **may** cheaply switch pipelines without reprogramming the bindings.

Similarly, the push constant block declared in each shader (if present) **must** only place variables at offsets that are each included in a push constant range with `stageFlags` including the bit corresponding to the shader stage that uses it. The pipeline layout **can** include ranges or portions of ranges that are not used by a particular pipeline.

There is a limit on the total number of resources of each type that **can** be included in bindings in all descriptor set layouts in a pipeline layout as shown in Pipeline Layout Resource Limits. The “Total Resources Available” column gives the limit on the number of each type of resource that **can** be included in bindings in all descriptor sets in the pipeline layout. Some resource types count against multiple limits. Additionally, there are limits on the total number of each type of resource that **can** be used in any pipeline stage as described in Shader Resource Limits.

Table 16. Pipeline Layout Resource Limits

Total Resources Available	Resource Types
`maxDescriptorSetSamplers` or `maxDescriptorSetUpdateAfterBindSamplers`	sampler
	combined image sampler
`maxDescriptorSetSampledImages` or `maxDescriptorSetUpdateAfterBindSampledImages`	sampled image
	combined image sampler
	uniform texel buffer
`maxDescriptorSetStorageImages` or `maxDescriptorSetUpdateAfterBindStorageImages`	storage image
	storage texel buffer
`maxDescriptorSetUniformBuffers` or `maxDescriptorSetUpdateAfterBindUniformBuffers`	uniform buffer
	uniform buffer dynamic
`maxDescriptorSetUniformBuffersDynamic` or `maxDescriptorSetUpdateAfterBindUniformBuffersDynamic` or `maxDescriptorSetUpdateAfterBindTotalUniformBuffersDynamic`	uniform buffer dynamic

Total Resources Available	Resource Types
maxDescriptorSetStorageBuffers or maxDescriptorSetUpdateAfterBindStorageBuffers	storage buffer
	storage buffer dynamic
maxDescriptorSetStorageBuffersDynamic or maxDescriptorSetUpdateAfterBindStorageBuffersDynamic or maxDescriptorSetUpdateAfterBindTotalStorageBuffersDynamic	storage buffer dynamic
maxDescriptorSetInputAttachments or maxDescriptorSetUpdateAfterBindInputAttachments	input attachment
maxDescriptorSetInlineUniformBlocks or maxDescriptorSetUpdateAfterBindInlineUniformBlocks	inline uniform block
maxDescriptorSetAccelerationStructures or maxDescriptorSetUpdateAfterBindAccelerationStructures	acceleration structure

To destroy a pipeline layout, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyPipelineLayout(
    VkDevice                                    device,
    VkPipelineLayout                            pipelineLayout,
    const VkAllocationCallbacks*                pAllocator);
```

- device is the logical device that destroys the pipeline layout.
- pipelineLayout is the pipeline layout to destroy.
- pAllocator controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyPipelineLayout-pipelineLayout-00299
  If VkAllocationCallbacks were provided when pipelineLayout was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyPipelineLayout-pipelineLayout-00300
  If no VkAllocationCallbacks were provided when pipelineLayout was created, pAllocator **must** be NULL

Valid Usage (Implicit)

- VUID-vkDestroyPipelineLayout-device-parameter
  device **must** be a valid VkDevice handle

- VUID-vkDestroyPipelineLayout-pipelineLayout-parameter
  If `pipelineLayout` is not VK_NULL_HANDLE, `pipelineLayout` **must** be a valid VkPipelineLayout handle
- VUID-vkDestroyPipelineLayout-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyPipelineLayout-pipelineLayout-parent
  If `pipelineLayout` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `pipelineLayout` **must** be externally synchronized

17.2.1. Pipeline Layout Compatibility

Two pipeline layouts are defined to be "compatible for push constants" if they were created with identical push constant ranges. Two pipeline layouts are defined to be "compatible for set N" if they were created with identically defined descriptor set layouts for sets zero through N, if both of them either were or were not created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT, and if they were created with identical push constant ranges.

When binding a descriptor set (see Descriptor Set Binding) to set number N, a previously bound descriptor set bound with lower index M than N is disturbed if the pipeline layouts for set M and N are not compatible for set M. Otherwise, the bound descriptor set in M is not disturbed.

If, additionally, the previously bound descriptor set for set N was bound using a pipeline layout not compatible for set N, then all bindings in sets numbered greater than N are disturbed.

When binding a pipeline, the pipeline **can** correctly access any previously bound descriptor set N if it was bound with a compatible pipeline layout for set N, and it was not disturbed.

Layout compatibility means that descriptor sets **can** be bound to a command buffer for use by any pipeline created with a compatible pipeline layout, and without having bound a particular pipeline first. It also means that descriptor sets **can** remain valid across a pipeline change, and the same resources will be accessible to the newly bound pipeline.

When a descriptor set is disturbed by binding descriptor sets, the disturbed set is considered to contain undefined descriptors bound with the same pipeline layout as the disturbing descriptor set.

Implementor's Note

A consequence of layout compatibility is that when the implementation compiles a pipeline layout and maps pipeline resources to implementation resources, the mechanism for set N

should only be a function of sets [0..N].

NOTE	Place the least frequently changing descriptor sets near the start of the pipeline layout, and place the descriptor sets representing the most frequently changing resources near the end. When pipelines are switched, only the descriptor set bindings that have been invalidated will need to be updated and the remainder of the descriptor set bindings will remain in place.

The maximum number of descriptor sets that **can** be bound to a pipeline layout is queried from physical device properties (see `maxBoundDescriptorSets` in Limits).

API Example

```
const VkDescriptorSetLayout layouts[] = { layout1, layout2 };

const VkPushConstantRange ranges[] =
{
    {
        .stageFlags = VK_SHADER_STAGE_VERTEX_BIT,
        .offset = 0,
        .size = 4
    },
    {
        .stageFlags = VK_SHADER_STAGE_FRAGMENT_BIT,
        .offset = 4,
        .size = 4
    },
};

const VkPipelineLayoutCreateInfo createInfo =
{
    .sType = VK_STRUCTURE_TYPE_PIPELINE_LAYOUT_CREATE_INFO,
    .pNext = NULL,
    .flags = 0,
    .setLayoutCount = 2,
    .pSetLayouts = layouts,
    .pushConstantRangeCount = 2,
    .pPushConstantRanges = ranges
};

VkPipelineLayout myPipelineLayout;
myResult = vkCreatePipelineLayout(
    myDevice,
    &createInfo,
    NULL,
    &myPipelineLayout);
```

17.3. Allocation of Descriptor Sets

A *descriptor pool* maintains a pool of descriptors, from which descriptor sets are allocated. Descriptor pools are externally synchronized, meaning that the application **must** not allocate and/or free descriptor sets from the same pool in multiple threads simultaneously.

Descriptor pools are represented by `VkDescriptorPool` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDescriptorPool)
```

To create a descriptor pool object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateDescriptorPool(
    VkDevice                                    device,
    const VkDescriptorPoolCreateInfo*           pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkDescriptorPool*                           pDescriptorPool);
```

- `device` is the logical device that creates the descriptor pool.
- `pCreateInfo` is a pointer to a VkDescriptorPoolCreateInfo structure specifying the state of the descriptor pool object.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pDescriptorPool` is a pointer to a VkDescriptorPool handle in which the resulting descriptor pool object is returned.

The created descriptor pool is returned in `pDescriptorPool`.

Valid Usage (Implicit)

- VUID-vkCreateDescriptorPool-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateDescriptorPool-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkDescriptorPoolCreateInfo structure
- VUID-vkCreateDescriptorPool-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateDescriptorPool-pDescriptorPool-parameter
  `pDescriptorPool` **must** be a valid pointer to a VkDescriptorPool handle
- VUID-vkCreateDescriptorPool-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_FRAGMENTATION_EXT
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Additional information about the pool is passed in a `VkDescriptorPoolCreateInfo` structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorPoolCreateInfo {
    VkStructureType                sType;
    const void*                    pNext;
    VkDescriptorPoolCreateFlags    flags;
    uint32_t                       maxSets;
    uint32_t                       poolSizeCount;
    const VkDescriptorPoolSize*    pPoolSizes;
} VkDescriptorPoolCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkDescriptorPoolCreateFlagBits specifying certain supported operations on the pool.
- `maxSets` is the maximum number of descriptor sets that **can** be allocated from the pool.
- `poolSizeCount` is the number of elements in `pPoolSizes`.
- `pPoolSizes` is a pointer to an array of VkDescriptorPoolSize structures, each containing a descriptor type and number of descriptors of that type to be allocated in the pool.

If multiple `VkDescriptorPoolSize` structures containing the same descriptor type appear in the `pPoolSizes` array then the pool will be created with enough storage for the total number of descriptors of each type.

Fragmentation of a descriptor pool is possible and **may** lead to descriptor set allocation failures. A failure due to fragmentation is defined as failing a descriptor set allocation despite the sum of all outstanding descriptor set allocations from the pool plus the requested allocation requiring no more than the total number of descriptors requested at pool creation. Implementations provide certain guarantees of when fragmentation **must** not cause allocation failure, as described below.

If a descriptor pool has not had any descriptor sets freed since it was created or most recently reset

then fragmentation **must** not cause an allocation failure (note that this is always the case for a pool created without the VK_DESCRIPTOR_POOL_CREATE_FREE_DESCRIPTOR_SET_BIT bit set). Additionally, if all sets allocated from the pool since it was created or most recently reset use the same number of descriptors (of each type) and the requested allocation also uses that same number of descriptors (of each type), then fragmentation **must** not cause an allocation failure.

If an allocation failure occurs due to fragmentation, an application **can** create an additional descriptor pool to perform further descriptor set allocations.

If `flags` has the VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT bit set, descriptor pool creation **may** fail with the error VK_ERROR_FRAGMENTATION if the total number of descriptors across all pools (including this one) created with this bit set exceeds `maxUpdateAfterBindDescriptorsInAllPools`, or if fragmentation of the underlying hardware resources occurs.

If a `pPoolSizes`[i]::`type` is VK_DESCRIPTOR_TYPE_MUTABLE_EXT, a VkMutableDescriptorTypeCreateInfoEXT structure in the `pNext` chain **can** be used to specify which mutable descriptor types **can** be allocated from the pool. If included in the `pNext` chain, VkMutableDescriptorTypeCreateInfoEXT::`pMutableDescriptorTypeLists`[i] specifies which kind of VK_DESCRIPTOR_TYPE_MUTABLE_EXT descriptors **can** be allocated from this pool entry. If VkMutableDescriptorTypeCreateInfoEXT does not exist in the `pNext` chain, or VkMutableDescriptorTypeCreateInfoEXT::`pMutableDescriptorTypeLists`[i] is out of range, the descriptor pool allocates enough memory to be able to allocate a VK_DESCRIPTOR_TYPE_MUTABLE_EXT descriptor with any supported VkDescriptorType as a mutable descriptor. A mutable descriptor **can** be allocated from a pool entry if the type list in VkDescriptorSetLayoutCreateInfo is a subset of the type list declared in the descriptor pool, or if the pool entry is created without a descriptor type list. Multiple `pPoolSizes` entries with VK_DESCRIPTOR_TYPE_MUTABLE_EXT **can** be declared. When multiple such pool entries are present in `pPoolSizes`, they specify sets of supported descriptor types which either fully overlap, partially overlap, or are disjoint. Two sets fully overlap if the sets of supported descriptor types are equal. If the sets are not disjoint they partially overlap. A pool entry without a `VkMutableDescriptorTypeListEXT` assigned to it is considered to partially overlap any other pool entry which has a `VkMutableDescriptorTypeListEXT` assigned to it. The application **must** ensure that partial overlap does not exist in `pPoolSizes`.

NOTE	The requirement of no partial overlap is intended to resolve ambiguity for validation as there is no confusion which `pPoolSizes` entries will be allocated from. An implementation is not expected to depend on this requirement.

Valid Usage

- VUID-VkDescriptorPoolCreateInfo-descriptorPoolOverallocation-09227
  `maxSets` **must** be greater than `0`
- VUID-VkDescriptorPoolCreateInfo-flags-04607
  If `flags` has the VK_DESCRIPTOR_POOL_CREATE_HOST_ONLY_BIT_EXT bit set, then the VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT bit **must** not be set
- VUID-VkDescriptorPoolCreateInfo-mutableDescriptorType-04608

If VkPhysicalDeviceMutableDescriptorTypeFeaturesEXT::mutableDescriptorType is not enabled, pPoolSizes **must** not contain a descriptorType of VK_DESCRIPTOR_TYPE_MUTABLE_EXT

- VUID-VkDescriptorPoolCreateInfo-flags-04609
  If flags has the VK_DESCRIPTOR_POOL_CREATE_HOST_ONLY_BIT_EXT bit set, VkPhysicalDeviceMutableDescriptorTypeFeaturesEXT::mutableDescriptorType **must** be enabled
- VUID-VkDescriptorPoolCreateInfo-pPoolSizes-04787
  If pPoolSizes contains a descriptorType of VK_DESCRIPTOR_TYPE_MUTABLE_EXT, any other VK_DESCRIPTOR_TYPE_MUTABLE_EXT element in pPoolSizes **must** not have sets of supported descriptor types which partially overlap
- VUID-VkDescriptorPoolCreateInfo-pPoolSizes-09424
  If pPoolSizes contains a descriptorType of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK, the pNext chain **must** include a VkDescriptorPoolInlineUniformBlockCreateInfo structure whose maxInlineUniformBlockBindings member is not zero

Valid Usage (Implicit)

- VUID-VkDescriptorPoolCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DESCRIPTOR_POOL_CREATE_INFO
- VUID-VkDescriptorPoolCreateInfo-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkDescriptorPoolInlineUniformBlockCreateInfo or VkMutableDescriptorTypeCreateInfoEXT
- VUID-VkDescriptorPoolCreateInfo-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkDescriptorPoolCreateInfo-flags-parameter
  If flags is not 0, flags **must** be a valid combination of VkDescriptorPoolCreateFlagBits values
- VUID-VkDescriptorPoolCreateInfo-pPoolSizes-parameter
  If poolSizeCount is not 0, pPoolSizes **must** be a valid pointer to an array of poolSizeCount valid VkDescriptorPoolSize structures

In order to be able to allocate descriptor sets having inline uniform block bindings the descriptor pool **must** be created with specifying the inline uniform block binding capacity of the descriptor pool, in addition to the total inline uniform data capacity in bytes which is specified through a VkDescriptorPoolSize structure with a descriptorType value of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK. This **can** be done by adding a VkDescriptorPoolInlineUniformBlockCreateInfo structure to the pNext chain of VkDescriptorPoolCreateInfo.

The VkDescriptorPoolInlineUniformBlockCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkDescriptorPoolInlineUniformBlockCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           maxInlineUniformBlockBindings;
} VkDescriptorPoolInlineUniformBlockCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxInlineUniformBlockBindings` is the number of inline uniform block bindings to allocate.

Valid Usage (Implicit)

- VUID-VkDescriptorPoolInlineUniformBlockCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DESCRIPTOR_POOL_INLINE_UNIFORM_BLOCK_CREATE_INFO

Structure Chaining

Extends the structure

- VkDescriptorPoolCreateInfo

Bits which **can** be set in VkDescriptorPoolCreateInfo::`flags`, enabling operations on a descriptor pool, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkDescriptorPoolCreateFlagBits {
    VK_DESCRIPTOR_POOL_CREATE_FREE_DESCRIPTOR_SET_BIT = 0x00000001,
  // Provided by VK_VERSION_1_2
    VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT = 0x00000002,
  // Provided by VK_EXT_mutable_descriptor_type
    VK_DESCRIPTOR_POOL_CREATE_HOST_ONLY_BIT_EXT = 0x00000004,
  // Provided by VK_EXT_descriptor_indexing
    VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT_EXT =
VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT,
} VkDescriptorPoolCreateFlagBits;
```

- VK_DESCRIPTOR_POOL_CREATE_FREE_DESCRIPTOR_SET_BIT specifies that descriptor sets **can** return their individual allocations to the pool, i.e. all of vkAllocateDescriptorSets, vkFreeDescriptorSets, and vkResetDescriptorPool are allowed. Otherwise, descriptor sets allocated from the pool **must** not be individually freed back to the pool, i.e. only vkAllocateDescriptorSets and vkResetDescriptorPool are allowed.
- VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT specifies that descriptor sets

allocated from this pool **can** include bindings with the VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT bit set. It is valid to allocate descriptor sets that have bindings that do not set the VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT bit from a pool that has VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT set.

- VK_DESCRIPTOR_POOL_CREATE_HOST_ONLY_BIT_EXT specifies that this descriptor pool and the descriptor sets allocated from it reside entirely in host memory and cannot be bound. Similar to descriptor sets allocated without this flag, applications **can** copy-from and copy-to descriptor sets allocated from this descriptor pool. Descriptor sets allocated from this pool are partially exempt from the external synchronization requirement in vkUpdateDescriptorSetWithTemplateKHR and vkUpdateDescriptorSets. Descriptor sets and their descriptors can be updated concurrently in different threads, though the same descriptor **must** not be updated concurrently by two threads.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDescriptorPoolCreateFlags;
```

VkDescriptorPoolCreateFlags is a bitmask type for setting a mask of zero or more VkDescriptorPoolCreateFlagBits.

The VkDescriptorPoolSize structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorPoolSize {
    VkDescriptorType    type;
    uint32_t            descriptorCount;
} VkDescriptorPoolSize;
```

- type is the type of descriptor.
- descriptorCount is the number of descriptors of that type to allocate. If type is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK then descriptorCount is the number of bytes to allocate for descriptors of this type.

NOTE

When creating a descriptor pool that will contain descriptors for combined image samplers with $Y'C_BC_R$ conversion, an application needs to account for non-trivial descriptor consumption when choosing the descriptorCount value, as indicated by VkSamplerYcbcrConversionImageFormatProperties::combinedImageSamplerDescriptorCount.

For simplicity, the application **can** use the VkPhysicalDeviceMaintenance6Properties::maxCombinedImageSamplerDescriptorCount property to conservatively size the descriptorCount value instead of querying combinedImageSamplerDescriptorCount for each format.

Valid Usage

- VUID-VkDescriptorPoolSize-descriptorCount-00302
  `descriptorCount` **must** be greater than `0`
- VUID-VkDescriptorPoolSize-type-02218
  If `type` is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK then `descriptorCount` **must** be a multiple of `4`

Valid Usage (Implicit)

- VUID-VkDescriptorPoolSize-type-parameter
  `type` **must** be a valid VkDescriptorType value

To destroy a descriptor pool, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyDescriptorPool(
    VkDevice                                    device,
    VkDescriptorPool                            descriptorPool,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the descriptor pool.
- `descriptorPool` is the descriptor pool to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

When a pool is destroyed, all descriptor sets allocated from the pool are implicitly freed and become invalid. Descriptor sets allocated from a given pool do not need to be freed before destroying that descriptor pool.

Valid Usage

- VUID-vkDestroyDescriptorPool-descriptorPool-00303
  All submitted commands that refer to `descriptorPool` (via any allocated descriptor sets) **must** have completed execution
- VUID-vkDestroyDescriptorPool-descriptorPool-00304
  If `VkAllocationCallbacks` were provided when `descriptorPool` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyDescriptorPool-descriptorPool-00305
  If no `VkAllocationCallbacks` were provided when `descriptorPool` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyDescriptorPool-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyDescriptorPool-descriptorPool-parameter
  If `descriptorPool` is not VK_NULL_HANDLE, `descriptorPool` **must** be a valid VkDescriptorPool handle
- VUID-vkDestroyDescriptorPool-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyDescriptorPool-descriptorPool-parent
  If `descriptorPool` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `descriptorPool` **must** be externally synchronized

Descriptor sets are allocated from descriptor pool objects, and are represented by `VkDescriptorSet` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDescriptorSet)
```

To allocate descriptor sets from a descriptor pool, call:

```
// Provided by VK_VERSION_1_0
VkResult vkAllocateDescriptorSets(
    VkDevice                                    device,
    const VkDescriptorSetAllocateInfo*          pAllocateInfo,
    VkDescriptorSet*                            pDescriptorSets);
```

- `device` is the logical device that owns the descriptor pool.
- `pAllocateInfo` is a pointer to a VkDescriptorSetAllocateInfo structure describing parameters of the allocation.
- `pDescriptorSets` is a pointer to an array of VkDescriptorSet handles in which the resulting descriptor set objects are returned.

The allocated descriptor sets are returned in `pDescriptorSets`.

When a descriptor set is allocated, the initial state is largely uninitialized and all descriptors are undefined, with the exception that samplers with a non-null `pImmutableSamplers` are initialized on allocation. Descriptors also become undefined if the underlying resource or view object is

destroyed. Descriptor sets containing undefined descriptors **can** still be bound and used, subject to the following conditions:

- For descriptor set bindings created with the VK_DESCRIPTOR_BINDING_PARTIALLY_BOUND_BIT bit set, all descriptors in that binding that are dynamically used **must** have been populated before the descriptor set is consumed.
- For descriptor set bindings created without the VK_DESCRIPTOR_BINDING_PARTIALLY_BOUND_BIT bit set, all descriptors in that binding that are statically used **must** have been populated before the descriptor set is consumed.
- Descriptor bindings with descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK **can** be undefined when the descriptor set is consumed; though values in that block will be undefined.
- Entries that are not used by a pipeline **can** have undefined descriptors.

If a call to `vkAllocateDescriptorSets` would cause the total number of descriptor sets allocated from the pool to exceed the value of VkDescriptorPoolCreateInfo::`maxSets` used to create `pAllocateInfo->descriptorPool`, then the allocation **may** fail due to lack of space in the descriptor pool. Similarly, the allocation **may** fail due to lack of space if the call to `vkAllocateDescriptorSets` would cause the number of any given descriptor type to exceed the sum of all the `descriptorCount` members of each element of VkDescriptorPoolCreateInfo::`pPoolSizes` with a `type` equal to that type.

Additionally, the allocation **may** also fail if a call to `vkAllocateDescriptorSets` would cause the total number of inline uniform block bindings allocated from the pool to exceed the value of VkDescriptorPoolInlineUniformBlockCreateInfo::`maxInlineUniformBlockBindings` used to create the descriptor pool.

If the allocation fails due to no more space in the descriptor pool, and not because of system or device memory exhaustion, then VK_ERROR_OUT_OF_POOL_MEMORY **must** be returned.

`vkAllocateDescriptorSets` **can** be used to create multiple descriptor sets. If the creation of any of those descriptor sets fails, then the implementation **must** destroy all successfully created descriptor set objects from this command, set all entries of the `pDescriptorSets` array to VK_NULL_HANDLE and return the error.

Valid Usage (Implicit)

- VUID-vkAllocateDescriptorSets-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkAllocateDescriptorSets-pAllocateInfo-parameter
  `pAllocateInfo` **must** be a valid pointer to a valid VkDescriptorSetAllocateInfo structure
- VUID-vkAllocateDescriptorSets-pDescriptorSets-parameter
  `pDescriptorSets` **must** be a valid pointer to an array of `pAllocateInfo->descriptorSetCount` VkDescriptorSet handles
- VUID-vkAllocateDescriptorSets-device-queuecount
  The device **must** have been created with at least 1 queue
- VUID-vkAllocateDescriptorSets-pAllocateInfo::descriptorSetCount-arraylength

pAllocateInfo->descriptorSetCount **must** be greater than 0

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_FRAGMENTED_POOL
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_OUT_OF_POOL_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkDescriptorSetAllocateInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorSetAllocateInfo {
    VkStructureType                 sType;
    const void*                     pNext;
    VkDescriptorPool                descriptorPool;
    uint32_t                        descriptorSetCount;
    const VkDescriptorSetLayout*    pSetLayouts;
} VkDescriptorSetAllocateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- descriptorPool is the pool which the sets will be allocated from.
- descriptorSetCount determines the number of descriptor sets to be allocated from the pool.
- pSetLayouts is a pointer to an array of descriptor set layouts, with each member specifying how the corresponding descriptor set is allocated.

Valid Usage

- VUID-VkDescriptorSetAllocateInfo-apiVersion-07895
  If the VK_KHR_maintenance1 extension is not enabled and VkPhysicalDeviceProperties ::apiVersion is less than Vulkan 1.1, descriptorSetCount **must** not be greater than the number of sets that are currently available for allocation in descriptorPool
- VUID-VkDescriptorSetAllocateInfo-apiVersion-07896
  If the VK_KHR_maintenance1 extension is not enabled and VkPhysicalDeviceProperties ::apiVersion is less than Vulkan 1.1, descriptorPool **must** have enough free descriptor

capacity remaining to allocate the descriptor sets of the specified layouts

- VUID-VkDescriptorSetAllocateInfo-pSetLayouts-00308
  Each element of `pSetLayouts` **must** not have been created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT set
- VUID-VkDescriptorSetAllocateInfo-pSetLayouts-03044
  If any element of `pSetLayouts` was created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set, `descriptorPool` **must** have been created with the VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT flag set
- VUID-VkDescriptorSetAllocateInfo-pSetLayouts-09380
  If `pSetLayouts`[i] was created with an element of `pBindingFlags` that includes VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT, and VkDescriptorSetVariableDescriptorCountAllocateInfo is included in the `pNext` chain, and `VkDescriptorSetVariableDescriptorCountAllocateInfo`::`descriptorSetCount` is not zero, then VkDescriptorSetVariableDescriptorCountAllocateInfo::`pDescriptorCounts`[i] **must** be less than or equal to VkDescriptorSetLayoutBinding::`descriptorCount` for the corresponding binding used to create `pSetLayouts`[i]
- VUID-VkDescriptorSetAllocateInfo-pSetLayouts-04610
  If any element of `pSetLayouts` was created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_HOST_ONLY_POOL_BIT_EXT bit set, `descriptorPool` **must** have been created with the VK_DESCRIPTOR_POOL_CREATE_HOST_ONLY_BIT_EXT flag set
- VUID-VkDescriptorSetAllocateInfo-pSetLayouts-08009
  Each element of `pSetLayouts` **must** not have been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT bit set

Valid Usage (Implicit)

- VUID-VkDescriptorSetAllocateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DESCRIPTOR_SET_ALLOCATE_INFO
- VUID-VkDescriptorSetAllocateInfo-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkDescriptorSetVariableDescriptorCountAllocateInfo
- VUID-VkDescriptorSetAllocateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkDescriptorSetAllocateInfo-descriptorPool-parameter
  `descriptorPool` **must** be a valid VkDescriptorPool handle
- VUID-VkDescriptorSetAllocateInfo-pSetLayouts-parameter
  `pSetLayouts` **must** be a valid pointer to an array of `descriptorSetCount` valid VkDescriptorSetLayout handles
- VUID-VkDescriptorSetAllocateInfo-descriptorSetCount-arraylength
  `descriptorSetCount` **must** be greater than `0`

- VUID-VkDescriptorSetAllocateInfo-commonparent
  Both of `descriptorPool`, and the elements of `pSetLayouts` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `descriptorPool` **must** be externally synchronized

If the `pNext` chain of a VkDescriptorSetAllocateInfo structure includes a `VkDescriptorSetVariableDescriptorCountAllocateInfo` structure, then that structure includes an array of descriptor counts for variable-sized descriptor bindings, one for each descriptor set being allocated.

The `VkDescriptorSetVariableDescriptorCountAllocateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkDescriptorSetVariableDescriptorCountAllocateInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           descriptorSetCount;
    const uint32_t*    pDescriptorCounts;
} VkDescriptorSetVariableDescriptorCountAllocateInfo;
```

```
// Provided by VK_EXT_descriptor_indexing
// Equivalent to VkDescriptorSetVariableDescriptorCountAllocateInfo
typedef VkDescriptorSetVariableDescriptorCountAllocateInfo
VkDescriptorSetVariableDescriptorCountAllocateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `descriptorSetCount` is zero or the number of elements in `pDescriptorCounts`.
- `pDescriptorCounts` is a pointer to an array of descriptor counts, with each member specifying the number of descriptors in a variable-sized descriptor binding in the corresponding descriptor set being allocated.

If `descriptorSetCount` is zero or this structure is not included in the `pNext` chain, then the variable lengths are considered to be zero. Otherwise, `pDescriptorCounts`[i] is the number of descriptors in the variable-sized descriptor binding in the corresponding descriptor set layout. If the variable-sized descriptor binding in the corresponding descriptor set layout has a descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK then `pDescriptorCounts`[i] specifies the binding's capacity in bytes. If VkDescriptorSetAllocateInfo::`pSetLayouts`[i] does not include a variable-sized descriptor binding, then `pDescriptorCounts`[i] is ignored.

Valid Usage

- VUID-VkDescriptorSetVariableDescriptorCountAllocateInfo-descriptorSetCount-03045
  If `descriptorSetCount` is not zero, `descriptorSetCount` **must** equal VkDescriptorSetAllocateInfo::`descriptorSetCount`

Valid Usage (Implicit)

- VUID-VkDescriptorSetVariableDescriptorCountAllocateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DESCRIPTOR_SET_VARIABLE_DESCRIPTOR_COUNT_ALLOCATE_INFO
- VUID-VkDescriptorSetVariableDescriptorCountAllocateInfo-pDescriptorCounts-parameter
  If `descriptorSetCount` is not `0`, `pDescriptorCounts` **must** be a valid pointer to an array of `descriptorSetCount` `uint32_t` values

Structure Chaining

Extends the structure

- VkDescriptorSetAllocateInfo

To free allocated descriptor sets, call:

```
// Provided by VK_VERSION_1_0
VkResult vkFreeDescriptorSets(
    VkDevice                                    device,
    VkDescriptorPool                            descriptorPool,
    uint32_t                                    descriptorSetCount,
    const VkDescriptorSet*                      pDescriptorSets);
```

- `device` is the logical device that owns the descriptor pool.
- `descriptorPool` is the descriptor pool from which the descriptor sets were allocated.
- `descriptorSetCount` is the number of elements in the `pDescriptorSets` array.
- `pDescriptorSets` is a pointer to an array of handles to VkDescriptorSet objects.

After calling `vkFreeDescriptorSets`, all descriptor sets in `pDescriptorSets` are invalid.

Valid Usage

- VUID-vkFreeDescriptorSets-pDescriptorSets-00309
  All submitted commands that refer to any element of `pDescriptorSets` **must** have completed execution

- VUID-vkFreeDescriptorSets-pDescriptorSets-00310
  `pDescriptorSets` **must** be a valid pointer to an array of `descriptorSetCount` `VkDescriptorSet` handles, each element of which **must** either be a valid handle or VK_NULL_HANDLE
- VUID-vkFreeDescriptorSets-descriptorPool-00312
  `descriptorPool` **must** have been created with the VK_DESCRIPTOR_POOL_CREATE_FREE_DESCRIPTOR_SET_BIT flag

Valid Usage (Implicit)

- VUID-vkFreeDescriptorSets-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkFreeDescriptorSets-descriptorPool-parameter
  `descriptorPool` **must** be a valid VkDescriptorPool handle
- VUID-vkFreeDescriptorSets-descriptorSetCount-arraylength
  `descriptorSetCount` **must** be greater than `0`
- VUID-vkFreeDescriptorSets-descriptorPool-parent
  `descriptorPool` **must** have been created, allocated, or retrieved from `device`
- VUID-vkFreeDescriptorSets-pDescriptorSets-parent
  Each element of `pDescriptorSets` that is a valid handle **must** have been created, allocated, or retrieved from `descriptorPool`

Host Synchronization

- Host access to `descriptorPool` **must** be externally synchronized
- Host access to each member of `pDescriptorSets` **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To return all descriptor sets allocated from a given pool to the pool, rather than freeing individual descriptor sets, call:

```
// Provided by VK_VERSION_1_0
VkResult vkResetDescriptorPool(
    VkDevice                                    device,
    VkDescriptorPool                            descriptorPool,
    VkDescriptorPoolResetFlags                  flags);
```

- `device` is the logical device that owns the descriptor pool.
- `descriptorPool` is the descriptor pool to be reset.
- `flags` is reserved for future use.

Resetting a descriptor pool recycles all of the resources from all of the descriptor sets allocated from the descriptor pool back to the descriptor pool, and the descriptor sets are implicitly freed.

Valid Usage

- VUID-vkResetDescriptorPool-descriptorPool-00313
  All uses of `descriptorPool` (via any allocated descriptor sets) **must** have completed execution

Valid Usage (Implicit)

- VUID-vkResetDescriptorPool-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkResetDescriptorPool-descriptorPool-parameter
  `descriptorPool` **must** be a valid VkDescriptorPool handle
- VUID-vkResetDescriptorPool-flags-zerobitmask
  `flags` **must** be `0`
- VUID-vkResetDescriptorPool-descriptorPool-parent
  `descriptorPool` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `descriptorPool` **must** be externally synchronized
- Host access to any `VkDescriptorSet` objects allocated from `descriptorPool` **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDescriptorPoolResetFlags;
```

VkDescriptorPoolResetFlags is a bitmask type for setting a mask, but is currently reserved for future use.

17.4. Descriptor Set Updates

Once allocated, descriptor sets **can** be updated with a combination of write and copy operations. To update descriptor sets, call:

```
// Provided by VK_VERSION_1_0
void vkUpdateDescriptorSets(
    VkDevice                                    device,
    uint32_t                                    descriptorWriteCount,
    const VkWriteDescriptorSet*                 pDescriptorWrites,
    uint32_t                                    descriptorCopyCount,
    const VkCopyDescriptorSet*                  pDescriptorCopies);
```

- device is the logical device that updates the descriptor sets.
- descriptorWriteCount is the number of elements in the pDescriptorWrites array.
- pDescriptorWrites is a pointer to an array of VkWriteDescriptorSet structures describing the descriptor sets to write to.
- descriptorCopyCount is the number of elements in the pDescriptorCopies array.
- pDescriptorCopies is a pointer to an array of VkCopyDescriptorSet structures describing the descriptor sets to copy between.

The operations described by pDescriptorWrites are performed first, followed by the operations described by pDescriptorCopies. Within each array, the operations are performed in the order they appear in the array.

Each element in the pDescriptorWrites array describes an operation updating the descriptor set using descriptors for resources specified in the structure.

Each element in the pDescriptorCopies array is a VkCopyDescriptorSet structure describing an operation copying descriptors between sets.

If the dstSet member of any element of pDescriptorWrites or pDescriptorCopies is bound, accessed, or modified by any command that was recorded to a command buffer which is currently in the recording or executable state, and any of the descriptor bindings that are updated were not created

with the VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT or VK_DESCRIPTOR_BINDING_UPDATE_UNUSED_WHILE_PENDING_BIT bits set, that command buffer becomes invalid.

Copying a descriptor from a descriptor set does not constitute a use of the referenced resource or view, as it is the reference itself that is copied. Applications **can** copy a descriptor referencing a destroyed resource, and it **can** copy an undefined descriptor. The destination descriptor becomes undefined in both cases.

Valid Usage

- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06236
  For each element i where pDescriptorWrites[i].descriptorType is VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, elements of the pTexelBufferView member of pDescriptorWrites[i] **must** have been created on device
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06237
  For each element i where pDescriptorWrites[i].descriptorType is VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC, or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC, the buffer member of any element of the pBufferInfo member of pDescriptorWrites[i] **must** have been created on device
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06238
  For each element i where pDescriptorWrites[i].descriptorType is VK_DESCRIPTOR_TYPE_SAMPLER or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, and dstSet was not allocated with a layout that included immutable samplers for dstBinding with descriptorType, the sampler member of any element of the pImageInfo member of pDescriptorWrites[i] **must** have been created on device
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06239
  For each element i where pDescriptorWrites[i].descriptorType is VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER the imageView member of any element of the pImageInfo member of pDescriptorWrites[i] **must** have been created on device
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06240
  For each element i where pDescriptorWrites[i].descriptorType is VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR, elements of the pAccelerationStructures member of a VkWriteDescriptorSetAccelerationStructureKHR structure in the pNext chain of pDescriptorWrites[i] **must** have been created on device
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06493
  For each element i where pDescriptorWrites[i].descriptorType is VK_DESCRIPTOR_TYPE_SAMPLER, VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, pDescriptorWrites[i].pImageInfo **must** be a

valid pointer to an array of pDescriptorWrites[i].descriptorCount valid VkDescriptorImageInfo structures

- VUID-vkUpdateDescriptorSets-None-03047
  The dstSet member of each element of pDescriptorWrites or pDescriptorCopies for bindings which were created without the VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT or VK_DESCRIPTOR_BINDING_UPDATE_UNUSED_WHILE_PENDING_BIT bits set **must** not be used by any command that was recorded to a command buffer which is in the pending state
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06993
  Host access to pDescriptorWrites[i].dstSet and pDescriptorCopies[i].dstSet **must** be externally synchronized unless explicitly denoted otherwise for specific flags

Valid Usage (Implicit)

- VUID-vkUpdateDescriptorSets-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-parameter
  If descriptorWriteCount is not 0, pDescriptorWrites **must** be a valid pointer to an array of descriptorWriteCount valid VkWriteDescriptorSet structures
- VUID-vkUpdateDescriptorSets-pDescriptorCopies-parameter
  If descriptorCopyCount is not 0, pDescriptorCopies **must** be a valid pointer to an array of descriptorCopyCount valid VkCopyDescriptorSet structures

The VkWriteDescriptorSet structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkWriteDescriptorSet {
    VkStructureType                  sType;
    const void*                      pNext;
    VkDescriptorSet                  dstSet;
    uint32_t                         dstBinding;
    uint32_t                         dstArrayElement;
    uint32_t                         descriptorCount;
    VkDescriptorType                 descriptorType;
    const VkDescriptorImageInfo*     pImageInfo;
    const VkDescriptorBufferInfo*    pBufferInfo;
    const VkBufferView*              pTexelBufferView;
} VkWriteDescriptorSet;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- dstSet is the destination descriptor set to update.

- dstBinding is the descriptor binding within that set.
- dstArrayElement is the starting element in that array. If the descriptor binding identified by dstSet and dstBinding has a descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK then dstArrayElement specifies the starting byte offset within the binding.
- descriptorCount is the number of descriptors to update. If the descriptor binding identified by dstSet and dstBinding has a descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK, then descriptorCount specifies the number of bytes to update. Otherwise, descriptorCount is one of
  - the number of elements in pImageInfo
  - the number of elements in pBufferInfo
  - the number of elements in pTexelBufferView
  - a value matching the dataSize member of a VkWriteDescriptorSetInlineUniformBlock structure in the pNext chain
  - a value matching the accelerationStructureCount of a VkWriteDescriptorSetAccelerationStructureKHR structure in the pNext chain
- descriptorType is a VkDescriptorType specifying the type of each descriptor in pImageInfo, pBufferInfo, or pTexelBufferView, as described below. If VkDescriptorSetLayoutBinding for dstSet at dstBinding is not equal to VK_DESCRIPTOR_TYPE_MUTABLE_EXT, descriptorType **must** be the same type as the descriptorType specified in VkDescriptorSetLayoutBinding for dstSet at dstBinding. The type of the descriptor also controls which array the descriptors are taken from.
- pImageInfo is a pointer to an array of VkDescriptorImageInfo structures or is ignored, as described below.
- pBufferInfo is a pointer to an array of VkDescriptorBufferInfo structures or is ignored, as described below.
- pTexelBufferView is a pointer to an array of VkBufferView handles as described in the Buffer Views section or is ignored, as described below.

Members of pImageInfo, pBufferInfo and pTexelBufferView are only accessed by the implementation when they correspond to a descriptor type being defined - otherwise they are ignored. The members accessed are as follows for each descriptor type:

- For VK_DESCRIPTOR_TYPE_SAMPLER, only the sampler member of each element of VkWriteDescriptorSet::pImageInfo is accessed.
- For VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, only the imageView and imageLayout members of each element of VkWriteDescriptorSet::pImageInfo are accessed.
- For VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, all members of each element of VkWriteDescriptorSet::pImageInfo are accessed.
- For VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC, or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC, all members of each element of VkWriteDescriptorSet::pBufferInfo are accessed.

- For VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, each element of VkWriteDescriptorSet::pTexelBufferView is accessed.

When updating descriptor sets with a descriptorType of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK, none of the pImageInfo, pBufferInfo, or pTexelBufferView members are accessed, instead the source data of the descriptor update operation is taken from the VkWriteDescriptorSetInlineUniformBlock structure in the pNext chain of VkWriteDescriptorSet. When updating descriptor sets with a descriptorType of VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR, none of the pImageInfo, pBufferInfo, or pTexelBufferView members are accessed, instead the source data of the descriptor update operation is taken from the VkWriteDescriptorSetAccelerationStructureKHR structure in the pNext chain of VkWriteDescriptorSet.

If the nullDescriptor feature is enabled, the buffer, acceleration structure, imageView, or bufferView **can** be VK_NULL_HANDLE. Loads from a null descriptor return zero values and stores and atomics to a null descriptor are discarded. A null acceleration structure descriptor results in the miss shader being invoked.

If the destination descriptor is a mutable descriptor, the active descriptor type for the destination descriptor becomes descriptorType.

Consecutive Binding Updates

If the dstBinding has fewer than descriptorCount array elements remaining starting from dstArrayElement, then the remainder will be used to update the subsequent binding - dstBinding+1 starting at array element zero. If a binding has a descriptorCount of zero, it is skipped. This behavior applies recursively, with the update affecting consecutive bindings as needed to update all descriptorCount descriptors. Consecutive bindings **must** have identical VkDescriptorType, VkShaderStageFlags, VkDescriptorBindingFlagBits, and immutable samplers references. In addition, if the VkDescriptorType is VK_DESCRIPTOR_TYPE_MUTABLE_EXT, the supported descriptor types in VkMutableDescriptorTypeCreateInfoEXT **must** be equally defined.

NOTE The same behavior applies to bindings with a descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK where descriptorCount specifies the number of bytes to update while dstArrayElement specifies the starting byte offset, thus in this case if the dstBinding has a smaller byte size than the sum of dstArrayElement and descriptorCount, then the remainder will be used to update the subsequent binding - dstBinding+1 starting at offset zero. This falls out as a special case of the above rule.

Valid Usage

- VUID-VkWriteDescriptorSet-dstBinding-00315
  dstBinding **must** be less than or equal to the maximum value of binding of all VkDescriptorSetLayoutBinding structures specified when dstSet's descriptor set layout was created

- VUID-VkWriteDescriptorSet-dstBinding-00316
  `dstBinding` **must** be a binding with a non-zero `descriptorCount`
- VUID-VkWriteDescriptorSet-dstBinding-10009
  `dstBinding` **must** be a binding with a non-zero VkDescriptorSetLayoutCreateInfo ::`bindingCount`
- VUID-VkWriteDescriptorSet-descriptorCount-00317
  All consecutive bindings updated via a single `VkWriteDescriptorSet` structure, except those with a `descriptorCount` of zero, **must** have identical `descriptorType`
- VUID-VkWriteDescriptorSet-descriptorCount-10776
  All consecutive bindings updated via a single `VkWriteDescriptorSet` structure, except those with a `descriptorCount` of zero, **must** have identical `stageFlags`
- VUID-VkWriteDescriptorSet-descriptorCount-00318
  All consecutive bindings updated via a single `VkWriteDescriptorSet` structure, except those with a `descriptorCount` of zero, **must** all either use immutable samplers or **must** all not use immutable samplers
- VUID-VkWriteDescriptorSet-descriptorCount-10777
  All consecutive bindings updated via a single `VkWriteDescriptorSet` structure, except those with a `descriptorCount` of zero, **must** have identical VkDescriptorBindingFlagBits
- VUID-VkWriteDescriptorSet-descriptorType-00319
  `descriptorType` **must** match the type of `dstBinding` within `dstSet`
- VUID-VkWriteDescriptorSet-dstSet-00320
  `dstSet` **must** be a valid VkDescriptorSet handle
- VUID-VkWriteDescriptorSet-dstArrayElement-00321
  The sum of `dstArrayElement` and `descriptorCount` **must** be less than or equal to the number of array elements in the descriptor set binding specified by `dstBinding`, and all applicable consecutive bindings
- VUID-VkWriteDescriptorSet-descriptorType-02219
  If `descriptorType` is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK, `dstArrayElement` **must** be an integer multiple of `4`
- VUID-VkWriteDescriptorSet-descriptorType-02220
  If `descriptorType` is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK, `descriptorCount` **must** be an integer multiple of `4`
- VUID-VkWriteDescriptorSet-descriptorType-02994
  If `descriptorType` is VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, each element of `pTexelBufferView` **must** be either a valid `VkBufferView` handle or VK_NULL_HANDLE
- VUID-VkWriteDescriptorSet-descriptorType-02995
  If `descriptorType` is VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER and the `nullDescriptor` feature is not enabled, each element of `pTexelBufferView` **must** not be VK_NULL_HANDLE
- VUID-VkWriteDescriptorSet-descriptorType-00324
  If `descriptorType` is VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, VK_DESCRIPTOR_TYPE_STORAGE_BUFFER,

VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC, or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC, pBufferInfo **must** be a valid pointer to an array of descriptorCount valid VkDescriptorBufferInfo structures

- VUID-VkWriteDescriptorSet-descriptorType-00325
  If descriptorType is VK_DESCRIPTOR_TYPE_SAMPLER or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, and dstSet was not allocated with a layout that included immutable samplers for dstBinding with descriptorType, the sampler member of each element of pImageInfo **must** be a valid VkSampler object
- VUID-VkWriteDescriptorSet-descriptorType-02996
  If descriptorType is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, or VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, the imageView member of each element of pImageInfo **must** be either a valid VkImageView handle or VK_NULL_HANDLE
- VUID-VkWriteDescriptorSet-descriptorType-02997
  If descriptorType is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, or VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, and the nullDescriptor feature is not enabled, the imageView member of each element of pImageInfo **must** not be VK_NULL_HANDLE
- VUID-VkWriteDescriptorSet-descriptorType-07683
  If descriptorType is VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, then the imageView member of each element of pImageInfo **must** not be VK_NULL_HANDLE
- VUID-VkWriteDescriptorSet-descriptorType-02221
  If descriptorType is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK, the pNext chain **must** include a VkWriteDescriptorSetInlineUniformBlock structure whose dataSize member equals descriptorCount
- VUID-VkWriteDescriptorSet-descriptorType-02382
  If descriptorType is VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR, the pNext chain **must** include a VkWriteDescriptorSetAccelerationStructureKHR structure whose accelerationStructureCount member equals descriptorCount
- VUID-VkWriteDescriptorSet-descriptorType-01946
  If descriptorType is VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, then the imageView member of each pImageInfo element **must** have been created without a VkSamplerYcbcrConversionInfo structure in its pNext chain
- VUID-VkWriteDescriptorSet-descriptorType-02738
  If descriptorType is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, and if any element of pImageInfo has an imageView member that was created with a VkSamplerYcbcrConversionInfo structure in its pNext chain, then dstSet **must** have been allocated with a layout that included immutable samplers for dstBinding, and the corresponding immutable sampler **must** have been created with an identically defined VkSamplerYcbcrConversionInfo object
- VUID-VkWriteDescriptorSet-descriptorType-01948
  If descriptorType is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, and dstSet was allocated with a layout that included immutable samplers for dstBinding, then the imageView member of each element of pImageInfo which corresponds to an immutable

sampler that enables sampler $Y'C_BC_R$ conversion **must** have been created with a VkSamplerYcbcrConversionInfo structure in its pNext chain with an identically defined VkSamplerYcbcrConversionInfo to the corresponding immutable sampler

- VUID-VkWriteDescriptorSet-descriptorType-09506
  If descriptorType is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, dstSet was allocated with a layout that included immutable samplers for dstBinding, and those samplers enable sampler $Y'C_BC_R$ conversion, then imageView **must** not be VK_NULL_HANDLE
- VUID-VkWriteDescriptorSet-descriptorType-00327
  If descriptorType is VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC, the offset member of each element of pBufferInfo **must** be a multiple of VkPhysicalDeviceLimits::minUniformBufferOffsetAlignment
- VUID-VkWriteDescriptorSet-descriptorType-00328
  If descriptorType is VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC, the offset member of each element of pBufferInfo **must** be a multiple of VkPhysicalDeviceLimits::minStorageBufferOffsetAlignment
- VUID-VkWriteDescriptorSet-descriptorType-00329
  If descriptorType is VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC, VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC, and the buffer member of any element of pBufferInfo is the handle of a non-sparse buffer, then that buffer **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-VkWriteDescriptorSet-descriptorType-00330
  If descriptorType is VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC, the buffer member of each element of pBufferInfo **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-VkWriteDescriptorSet-descriptorType-00331
  If descriptorType is VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC, the buffer member of each element of pBufferInfo **must** have been created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-VkWriteDescriptorSet-descriptorType-00332
  If descriptorType is VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC, the range member of each element of pBufferInfo, or the effective range if range is VK_WHOLE_SIZE, **must** be less than or equal to VkPhysicalDeviceLimits::maxUniformBufferRange
- VUID-VkWriteDescriptorSet-descriptorType-00333
  If descriptorType is VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC, and the shader64BitIndexing feature is not enabled, the range member of each element of pBufferInfo, or the effective range if range is VK_WHOLE_SIZE, **must** be less than or equal to VkPhysicalDeviceLimits

::maxStorageBufferRange

- VUID-VkWriteDescriptorSet-descriptorType-08765
  If descriptorType is VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER, the pTexelBufferView buffer view usage **must** include VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT
- VUID-VkWriteDescriptorSet-descriptorType-08766
  If descriptorType is VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, the pTexelBufferView buffer view usage **must** include VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT
- VUID-VkWriteDescriptorSet-descriptorType-00336
  If descriptorType is VK_DESCRIPTOR_TYPE_STORAGE_IMAGE or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, the imageView member of each element of pImageInfo **must** have been created with the identity swizzle
- VUID-VkWriteDescriptorSet-descriptorType-00337
  If descriptorType is VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, the imageView member of each element of pImageInfo **must** have been created with the VK_IMAGE_USAGE_SAMPLED_BIT usage flag set
- VUID-VkWriteDescriptorSet-descriptorType-04149
  If descriptorType is VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE the imageLayout member of each element of pImageInfo **must** be a member of the list given in Sampled Image
- VUID-VkWriteDescriptorSet-descriptorType-04150
  If descriptorType is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER the imageLayout member of each element of pImageInfo **must** be a member of the list given in Combined Image Sampler
- VUID-VkWriteDescriptorSet-descriptorType-04151
  If descriptorType is VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT the imageLayout member of each element of pImageInfo **must** be a member of the list given in Input Attachment
- VUID-VkWriteDescriptorSet-descriptorType-04152
  If descriptorType is VK_DESCRIPTOR_TYPE_STORAGE_IMAGE the imageLayout member of each element of pImageInfo **must** be a member of the list given in Storage Image
- VUID-VkWriteDescriptorSet-descriptorType-00338
  If descriptorType is VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, the imageView member of each element of pImageInfo **must** have been created with the VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT usage flag set
- VUID-VkWriteDescriptorSet-descriptorType-00339
  If descriptorType is VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, the imageView member of each element of pImageInfo **must** have been created with the VK_IMAGE_USAGE_STORAGE_BIT usage flag set
- VUID-VkWriteDescriptorSet-descriptorType-02752
  If descriptorType is VK_DESCRIPTOR_TYPE_SAMPLER, then dstSet **must** not have been allocated with a layout that included immutable samplers for dstBinding
- VUID-VkWriteDescriptorSet-dstSet-04611

If the VkDescriptorSetLayoutBinding for dstSet at dstBinding is VK_DESCRIPTOR_TYPE_MUTABLE_EXT, the new active descriptor type descriptorType **must** exist in the corresponding pMutableDescriptorTypeLists list for dstBinding

- VUID-VkWriteDescriptorSet-descriptorType-06450
  If descriptorType is VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, the imageView member of each element of pImageInfo **must** have either been created without a VkImageViewMinLodCreateInfoEXT included in the pNext chain or with a VkImageViewMinLodCreateInfoEXT::minLod of 0.0

Valid Usage (Implicit)

- VUID-VkWriteDescriptorSet-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_WRITE_DESCRIPTOR_SET
- VUID-VkWriteDescriptorSet-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkWriteDescriptorSetAccelerationStructureKHR or VkWriteDescriptorSetInlineUniformBlock
- VUID-VkWriteDescriptorSet-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkWriteDescriptorSet-descriptorType-parameter
  descriptorType **must** be a valid VkDescriptorType value
- VUID-VkWriteDescriptorSet-descriptorCount-arraylength
  descriptorCount **must** be greater than 0
- VUID-VkWriteDescriptorSet-commonparent
  Both of dstSet, and the elements of pTexelBufferView that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

The VkDescriptorBufferInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorBufferInfo {
    VkBuffer        buffer;
    VkDeviceSize    offset;
    VkDeviceSize    range;
} VkDescriptorBufferInfo;
```

- buffer is VK_NULL_HANDLE or the buffer resource.
- offset is the offset in bytes from the start of buffer. Access to buffer memory via this descriptor uses addressing that is relative to this starting offset.
- range is the size in bytes that is used for this descriptor update, or VK_WHOLE_SIZE to use the range from offset to the end of the buffer.

NOTE

When setting `range` to VK_WHOLE_SIZE, the effective range **must** not be larger than the maximum range for the descriptor type (`maxUniformBufferRange` or `maxStorageBufferRange`). This means that VK_WHOLE_SIZE is not typically useful in the common case where uniform buffer descriptors are suballocated from a buffer that is much larger than `maxUniformBufferRange`.

For VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC and VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC descriptor types, `offset` is the base offset from which the dynamic offset is applied and `range` is the static size used for all dynamic offsets.

When `range` is VK_WHOLE_SIZE the effective range is calculated at vkUpdateDescriptorSets by taking the size of `buffer` minus the `offset`.

Valid Usage

- VUID-VkDescriptorBufferInfo-offset-00340
  `offset` **must** be less than the size of `buffer`
- VUID-VkDescriptorBufferInfo-range-00341
  If `range` is not equal to VK_WHOLE_SIZE, `range` **must** be greater than `0`
- VUID-VkDescriptorBufferInfo-range-00342
  If `range` is not equal to VK_WHOLE_SIZE, `range` **must** be less than or equal to the size of `buffer` minus `offset`
- VUID-VkDescriptorBufferInfo-buffer-02998
  If the `nullDescriptor` feature is not enabled, `buffer` **must** not be VK_NULL_HANDLE
- VUID-VkDescriptorBufferInfo-buffer-02999
  If `buffer` is VK_NULL_HANDLE, `offset` **must** be zero and `range` **must** be VK_WHOLE_SIZE

Valid Usage (Implicit)

- VUID-VkDescriptorBufferInfo-buffer-parameter
  If `buffer` is not VK_NULL_HANDLE, `buffer` **must** be a valid VkBuffer handle

The `VkDescriptorImageInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorImageInfo {
    VkSampler        sampler;
    VkImageView      imageView;
    VkImageLayout    imageLayout;
} VkDescriptorImageInfo;
```

- `sampler` is a sampler handle, and is used in descriptor updates for types VK_DESCRIPTOR_TYPE_SAMPLER and VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER if the binding being updated does not use immutable samplers.

- `imageView` is VK_NULL_HANDLE or an image view handle, and is used in descriptor updates for types VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, and VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT.
- `imageLayout` is the layout that the image subresources accessible from `imageView` will be in at the time this descriptor is accessed. `imageLayout` is used in descriptor updates for types VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, and VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT.

Members of `VkDescriptorImageInfo` that are not used in an update (as described above) are ignored.

Valid Usage

- VUID-VkDescriptorImageInfo-imageView-06712
  `imageView` **must** not be a 2D array image view created from a 3D image
- VUID-VkDescriptorImageInfo-imageView-07795
  If `imageView` is a 2D view created from a 3D image, then `descriptorType` **must** be VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER
- VUID-VkDescriptorImageInfo-imageView-07796
  If `imageView` is a 2D view created from a 3D image, then the image **must** have been created with VK_IMAGE_CREATE_2D_VIEW_COMPATIBLE_BIT_EXT set
- VUID-VkDescriptorImageInfo-descriptorType-06713
  If the `image2DViewOf3D` feature is not enabled or `descriptorType` is not VK_DESCRIPTOR_TYPE_STORAGE_IMAGE then `imageView` **must** not be a 2D view created from a 3D image
- VUID-VkDescriptorImageInfo-descriptorType-06714
  If the `sampler2DViewOf3D` feature is not enabled or `descriptorType` is not VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER then `imageView` **must** not be a 2D view created from a 3D image
- VUID-VkDescriptorImageInfo-imageView-01976
  If `imageView` is created from a depth/stencil image, the `aspectMask` used to create the `imageView` **must** include either VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT but not both
- VUID-VkDescriptorImageInfo-imageLayout-09425
  If `imageLayout` is VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL, then the `aspectMask` used to create `imageView` **must** not include either VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkDescriptorImageInfo-imageLayout-09426
  If `imageLayout` is VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL,

VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL, then the `aspectMask` used to create `imageView` **must** not include VK_IMAGE_ASPECT_COLOR_BIT

- VUID-VkDescriptorImageInfo-sampler-01564
  If `sampler` is used and the VkFormat of the image is a multi-planar format, the image **must** have been created with VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT, and the `aspectMask` of the `imageView` **must** be a valid multi-planar aspect mask bit
- VUID-VkDescriptorImageInfo-mutableComparisonSamplers-04450
  If the `VK_KHR_portability_subset` extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::`mutableComparisonSamplers` is VK_FALSE, then `sampler` **must** have been created with VkSamplerCreateInfo::`compareEnable` set to VK_FALSE

Valid Usage (Implicit)

- VUID-VkDescriptorImageInfo-commonparent
  Both of `imageView`, and `sampler` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

If the `descriptorType` member of VkWriteDescriptorSet is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK then the data to write to the descriptor set is specified through a `VkWriteDescriptorSetInlineUniformBlock` structure included in the `pNext` chain of `VkWriteDescriptorSet`.

The `VkWriteDescriptorSetInlineUniformBlock` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkWriteDescriptorSetInlineUniformBlock {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           dataSize;
    const void*        pData;
} VkWriteDescriptorSetInlineUniformBlock;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `dataSize` is the number of bytes of inline uniform block data pointed to by `pData`.
- `pData` is a pointer to `dataSize` number of bytes of data to write to the inline uniform block.

Valid Usage

- VUID-VkWriteDescriptorSetInlineUniformBlock-dataSize-02222
  `dataSize` **must** be an integer multiple of `4`

Valid Usage (Implicit)

- VUID-VkWriteDescriptorSetInlineUniformBlock-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_WRITE_DESCRIPTOR_SET_INLINE_UNIFORM_BLOCK
- VUID-VkWriteDescriptorSetInlineUniformBlock-pData-parameter
  `pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-VkWriteDescriptorSetInlineUniformBlock-dataSize-arraylength
  `dataSize` **must** be greater than `0`

Structure Chaining

Extends the structure

- VkWriteDescriptorSet

The `VkWriteDescriptorSetAccelerationStructureKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkWriteDescriptorSetAccelerationStructureKHR {
    VkStructureType                      sType;
    const void*                          pNext;
    uint32_t                             accelerationStructureCount;
    const VkAccelerationStructureKHR*    pAccelerationStructures;
} VkWriteDescriptorSetAccelerationStructureKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `accelerationStructureCount` is the number of elements in `pAccelerationStructures`.
- `pAccelerationStructures` is a pointer to an array of VkAccelerationStructureKHR structures specifying the acceleration structures to update.

Valid Usage

- VUID-VkWriteDescriptorSetAccelerationStructureKHR-pAccelerationStructures-03579
  Each acceleration structure in `pAccelerationStructures` **must** have been created with a `type` of VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR or

VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR

- VUID-VkWriteDescriptorSetAccelerationStructureKHR-pAccelerationStructures-03580
  If the `nullDescriptor` feature is not enabled, each element of `pAccelerationStructures` **must** not be VK_NULL_HANDLE

Valid Usage (Implicit)

- VUID-VkWriteDescriptorSetAccelerationStructureKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_WRITE_DESCRIPTOR_SET_ACCELERATION_STRUCTURE_KHR
- VUID-VkWriteDescriptorSetAccelerationStructureKHR-pAccelerationStructures-parameter
  `pAccelerationStructures` **must** be a valid pointer to an array of `accelerationStructureCount` valid or VK_NULL_HANDLE VkAccelerationStructureKHR handles
- VUID-VkWriteDescriptorSetAccelerationStructureKHR-accelerationStructureCount-arraylength
  `accelerationStructureCount` **must** be greater than `0`

Structure Chaining

Extends the structure

- VkWriteDescriptorSet

The `VkCopyDescriptorSet` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkCopyDescriptorSet {
    VkStructureType    sType;
    const void*        pNext;
    VkDescriptorSet    srcSet;
    uint32_t           srcBinding;
    uint32_t           srcArrayElement;
    VkDescriptorSet    dstSet;
    uint32_t           dstBinding;
    uint32_t           dstArrayElement;
    uint32_t           descriptorCount;
} VkCopyDescriptorSet;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcSet`, `srcBinding`, and `srcArrayElement` are the source set, binding, and array element, respectively. If the descriptor binding identified by `srcSet` and `srcBinding` has a descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK then `srcArrayElement` specifies the starting

byte offset within the binding to copy from.

- `dstSet`, `dstBinding`, and `dstArrayElement` are the destination set, binding, and array element, respectively. If the descriptor binding identified by `dstSet` and `dstBinding` has a descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK then `dstArrayElement` specifies the starting byte offset within the binding to copy to.
- `descriptorCount` is the number of descriptors to copy from the source to destination. If `descriptorCount` is greater than the number of remaining array elements in the source or destination binding, those affect consecutive bindings in a manner similar to VkWriteDescriptorSet above. If the descriptor binding identified by `srcSet` and `srcBinding` has a descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK then `descriptorCount` specifies the number of bytes to copy and the remaining array elements in the source or destination binding refer to the remaining number of bytes in those.

If the `VkDescriptorSetLayoutBinding` for `dstBinding` is VK_DESCRIPTOR_TYPE_MUTABLE_EXT and `srcBinding` is not VK_DESCRIPTOR_TYPE_MUTABLE_EXT, the new active descriptor type becomes the descriptor type of `srcBinding`. If both `VkDescriptorSetLayoutBinding` for `srcBinding` and `dstBinding` are VK_DESCRIPTOR_TYPE_MUTABLE_EXT, the active descriptor type in each source descriptor is copied into the corresponding destination descriptor. The active descriptor type **can** be different for each source descriptor.

NOTE The intention is that copies to and from mutable descriptors are simple memcpy operations. Copies between non-mutable and mutable descriptors are expected to require one memcpy per descriptor to handle the difference in size, but this use case with more than one `descriptorCount` is considered rare.

Valid Usage

- VUID-VkCopyDescriptorSet-srcBinding-00345
  `srcBinding` **must** be a valid binding within `srcSet`
- VUID-VkCopyDescriptorSet-srcArrayElement-00346
  The sum of `srcArrayElement` and `descriptorCount` **must** be less than or equal to the number of array elements in the descriptor set binding specified by `srcBinding`, and all applicable consecutive bindings
- VUID-VkCopyDescriptorSet-dstBinding-00347
  `dstBinding` **must** be a valid binding within `dstSet`
- VUID-VkCopyDescriptorSet-dstArrayElement-00348
  The sum of `dstArrayElement` and `descriptorCount` **must** be less than or equal to the number of array elements in the descriptor set binding specified by `dstBinding`, and all applicable consecutive bindings
- VUID-VkCopyDescriptorSet-dstBinding-02632
  The type of `dstBinding` within `dstSet` **must** be equal to the type of `srcBinding` within `srcSet`
- VUID-VkCopyDescriptorSet-srcSet-00349
  If `srcSet` is equal to `dstSet`, then the source and destination ranges of descriptors **must** not overlap, where the ranges **may** include array elements from consecutive bindings

- VUID-VkCopyDescriptorSet-srcBinding-02223
  If the descriptor type of the descriptor set binding specified by `srcBinding` is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK, `srcArrayElement` **must** be an integer multiple of `4`
- VUID-VkCopyDescriptorSet-dstBinding-02224
  If the descriptor type of the descriptor set binding specified by `dstBinding` is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK, `dstArrayElement` **must** be an integer multiple of `4`
- VUID-VkCopyDescriptorSet-srcBinding-02225
  If the descriptor type of the descriptor set binding specified by either `srcBinding` or `dstBinding` is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK, `descriptorCount` **must** be an integer multiple of `4`
- VUID-VkCopyDescriptorSet-srcSet-01918
  If `srcSet`'s layout was created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT flag set, then `dstSet`'s layout **must** also have been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT flag set
- VUID-VkCopyDescriptorSet-srcSet-04885
  If `srcSet`'s layout was created without either the VK_DESCRIPTOR_SET_LAYOUT_CREATE_HOST_ONLY_POOL_BIT_EXT flag or the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT flag set, then `dstSet`'s layout **must** have been created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT flag set
- VUID-VkCopyDescriptorSet-srcSet-01920
  If the descriptor pool from which `srcSet` was allocated was created with the VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT flag set, then the descriptor pool from which `dstSet` was allocated **must** also have been created with the VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT flag set
- VUID-VkCopyDescriptorSet-srcSet-04887
  If the descriptor pool from which `srcSet` was allocated was created without either the VK_DESCRIPTOR_POOL_CREATE_HOST_ONLY_BIT_EXT flag or the VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT flag set, then the descriptor pool from which `dstSet` was allocated **must** have been created without the VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT flag set
- VUID-VkCopyDescriptorSet-dstBinding-02753
  If the descriptor type of the descriptor set binding specified by `dstBinding` is VK_DESCRIPTOR_TYPE_SAMPLER, then `dstSet` **must** not have been allocated with a layout that included immutable samplers for `dstBinding`
- VUID-VkCopyDescriptorSet-dstSet-04612
  If `VkDescriptorSetLayoutBinding` for `dstSet` at `dstBinding` is VK_DESCRIPTOR_TYPE_MUTABLE_EXT, the new active descriptor type **must** exist in the corresponding `pMutableDescriptorTypeLists` list for `dstBinding` if the new active descriptor type is not VK_DESCRIPTOR_TYPE_MUTABLE_EXT
- VUID-VkCopyDescriptorSet-srcSet-04613

If VkDescriptorSetLayoutBinding for srcSet at srcBinding is VK_DESCRIPTOR_TYPE_MUTABLE_EXT and the VkDescriptorSetLayoutBinding for dstSet at dstBinding is not VK_DESCRIPTOR_TYPE_MUTABLE_EXT, the active descriptor type for the source descriptor **must** match the descriptor type of dstBinding

- VUID-VkCopyDescriptorSet-dstSet-04614
  If VkDescriptorSetLayoutBinding for dstSet at dstBinding is VK_DESCRIPTOR_TYPE_MUTABLE_EXT, and the new active descriptor type is VK_DESCRIPTOR_TYPE_MUTABLE_EXT, the pMutableDescriptorTypeLists for srcBinding and dstBinding **must** match exactly

Valid Usage (Implicit)

- VUID-VkCopyDescriptorSet-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_COPY_DESCRIPTOR_SET
- VUID-VkCopyDescriptorSet-pNext-pNext
  pNext **must** be NULL
- VUID-VkCopyDescriptorSet-srcSet-parameter
  srcSet **must** be a valid VkDescriptorSet handle
- VUID-VkCopyDescriptorSet-dstSet-parameter
  dstSet **must** be a valid VkDescriptorSet handle
- VUID-VkCopyDescriptorSet-commonparent
  Both of dstSet, and srcSet **must** have been created, allocated, or retrieved from the same VkDevice

17.5. Descriptor Update Templates

A descriptor update template specifies a mapping from descriptor update information in host memory to descriptors in a descriptor set. It is designed to avoid passing redundant information to the driver when frequently updating the same set of descriptors in descriptor sets.

Descriptor update template objects are represented by VkDescriptorUpdateTemplate handles:

```
// Provided by VK_VERSION_1_1
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDescriptorUpdateTemplate)
```

```
// Provided by VK_KHR_descriptor_update_template
// Equivalent to VkDescriptorUpdateTemplate
typedef VkDescriptorUpdateTemplate VkDescriptorUpdateTemplateKHR;
```

17.6. Descriptor Set Updates With Templates

Updating a large VkDescriptorSet array **can** be an expensive operation since an application **must** specify one VkWriteDescriptorSet structure for each descriptor or descriptor array to update, each of which re-specifies the same state when updating the same descriptor in multiple descriptor sets. For cases when an application wishes to update the same set of descriptors in multiple descriptor sets allocated using the same VkDescriptorSetLayout, vkUpdateDescriptorSetWithTemplate **can** be used as a replacement for vkUpdateDescriptorSets.

VkDescriptorUpdateTemplate allows implementations to convert a set of descriptor update operations on a single descriptor set to an internal format. In conjunction with vkCmdPushDescriptorSetWithTemplate or vkUpdateDescriptorSetWithTemplate, this **can** be more efficient compared to calling vkCmdPushDescriptorSet or vkUpdateDescriptorSets. The descriptors themselves are not specified in the VkDescriptorUpdateTemplate, rather, offsets into an application provided pointer to host memory are specified, which are combined with a pointer passed to vkCmdPushDescriptorSetWithTemplate or vkUpdateDescriptorSetWithTemplate. This allows large batches of updates to be executed without having to convert application data structures into a strictly-defined Vulkan data structure.

To create a descriptor update template, call:

```
// Provided by VK_VERSION_1_1
VkResult vkCreateDescriptorUpdateTemplate(
    VkDevice                                    device,
    const VkDescriptorUpdateTemplateCreateInfo* pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkDescriptorUpdateTemplate*                 pDescriptorUpdateTemplate);
```

```
// Provided by VK_KHR_descriptor_update_template
// Equivalent to vkCreateDescriptorUpdateTemplate
VkResult vkCreateDescriptorUpdateTemplateKHR(
    VkDevice                                    device,
    const VkDescriptorUpdateTemplateCreateInfo* pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkDescriptorUpdateTemplate*                 pDescriptorUpdateTemplate);
```

- device is the logical device that creates the descriptor update template.
- pCreateInfo is a pointer to a VkDescriptorUpdateTemplateCreateInfo structure specifying the set of descriptors to update with a single call to vkCmdPushDescriptorSetWithTemplate or vkUpdateDescriptorSetWithTemplate.
- pAllocator controls host memory allocation as described in the Memory Allocation chapter.
- pDescriptorUpdateTemplate is a pointer to a VkDescriptorUpdateTemplate handle in which the resulting descriptor update template object is returned.

Valid Usage (Implicit)

- VUID-vkCreateDescriptorUpdateTemplate-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateDescriptorUpdateTemplate-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkDescriptorUpdateTemplateCreateInfo structure
- VUID-vkCreateDescriptorUpdateTemplate-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateDescriptorUpdateTemplate-pDescriptorUpdateTemplate-parameter
  `pDescriptorUpdateTemplate` **must** be a valid pointer to a VkDescriptorUpdateTemplate handle
- VUID-vkCreateDescriptorUpdateTemplate-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkDescriptorUpdateTemplateCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkDescriptorUpdateTemplateCreateInfo {
    VkStructureType                           sType;
    const void*                               pNext;
    VkDescriptorUpdateTemplateCreateFlags     flags;
    uint32_t                                  descriptorUpdateEntryCount;
    const VkDescriptorUpdateTemplateEntry*    pDescriptorUpdateEntries;
    VkDescriptorUpdateTemplateType            templateType;
    VkDescriptorSetLayout                     descriptorSetLayout;
    VkPipelineBindPoint                       pipelineBindPoint;
    VkPipelineLayout                          pipelineLayout;
    uint32_t                                  set;
} VkDescriptorUpdateTemplateCreateInfo;
```

```
// Provided by VK_KHR_descriptor_update_template
// Equivalent to VkDescriptorUpdateTemplateCreateInfo
typedef VkDescriptorUpdateTemplateCreateInfo VkDescriptorUpdateTemplateCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `descriptorUpdateEntryCount` is the number of elements in the `pDescriptorUpdateEntries` array.
- `pDescriptorUpdateEntries` is a pointer to an array of VkDescriptorUpdateTemplateEntry structures describing the descriptors to be updated by the descriptor update template.
- `templateType` specifies the type of the descriptor update template. If set to VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET it **can** only be used to update descriptor sets with a fixed `descriptorSetLayout`. If set to VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS it **can** only be used to push descriptor sets using the provided `pipelineBindPoint`, `pipelineLayout`, and `set` number.
- `descriptorSetLayout` is the descriptor set layout used to build the descriptor update template. All descriptor sets which are going to be updated through the newly created descriptor update template **must** be created with a layout that matches (is the same as, or defined identically to) this layout. This parameter is ignored if `templateType` is not VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET. The implementation **must** not access this object outside of the duration of the command this structure is passed to.
- `pipelineBindPoint` is a VkPipelineBindPoint indicating the type of the pipeline that will use the descriptors. This parameter is ignored if `templateType` is not VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS
- `pipelineLayout` is a VkPipelineLayout object used to program the bindings. This parameter is ignored if `templateType` is not VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS
- `set` is the set number of the descriptor set in the pipeline layout that will be updated. This parameter is ignored if `templateType` is not VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS

Valid Usage

- VUID-VkDescriptorUpdateTemplateCreateInfo-templateType-00350
  If `templateType` is VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET, `descriptorSetLayout` **must** be a valid `VkDescriptorSetLayout` handle
- VUID-VkDescriptorUpdateTemplateCreateInfo-templateType-10355
  If `templateType` is VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS, and the VK_KHR_push_descriptor extension is not enabled, `pushDescriptor` **must** be enabled
- VUID-VkDescriptorUpdateTemplateCreateInfo-templateType-00351
  If `templateType` is VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS, `pipelineBindPoint` **must** be a valid VkPipelineBindPoint value
- VUID-VkDescriptorUpdateTemplateCreateInfo-templateType-00352

If `templateType` is VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS, `pipelineLayout` **must** be a valid `VkPipelineLayout` handle

- VUID-VkDescriptorUpdateTemplateCreateInfo-templateType-00353
  If `templateType` is VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS, `set` **must** be the unique set number in the pipeline layout that uses a descriptor set layout that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT
- VUID-VkDescriptorUpdateTemplateCreateInfo-templateType-04615
  If `templateType` is VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET, `descriptorSetLayout` **must** not contain a binding with type VK_DESCRIPTOR_TYPE_MUTABLE_EXT

Valid Usage (Implicit)

- VUID-VkDescriptorUpdateTemplateCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_CREATE_INFO
- VUID-VkDescriptorUpdateTemplateCreateInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDescriptorUpdateTemplateCreateInfo-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkDescriptorUpdateTemplateCreateInfo-pDescriptorUpdateEntries-parameter
  `pDescriptorUpdateEntries` **must** be a valid pointer to an array of `descriptorUpdateEntryCount` valid VkDescriptorUpdateTemplateEntry structures
- VUID-VkDescriptorUpdateTemplateCreateInfo-templateType-parameter
  `templateType` **must** be a valid VkDescriptorUpdateTemplateType value
- VUID-VkDescriptorUpdateTemplateCreateInfo-descriptorUpdateEntryCount-arraylength
  `descriptorUpdateEntryCount` **must** be greater than `0`
- VUID-VkDescriptorUpdateTemplateCreateInfo-commonparent
  Both of `descriptorSetLayout`, and `pipelineLayout` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

```
// Provided by VK_VERSION_1_1
typedef VkFlags VkDescriptorUpdateTemplateCreateFlags;
```

```
// Provided by VK_KHR_descriptor_update_template
// Equivalent to VkDescriptorUpdateTemplateCreateFlags
typedef VkDescriptorUpdateTemplateCreateFlags
VkDescriptorUpdateTemplateCreateFlagsKHR;
```

`VkDescriptorUpdateTemplateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

The descriptor update template type is determined by the VkDescriptorUpdateTemplateCreateInfo ::templateType property, which takes the following values:

```
// Provided by VK_VERSION_1_1
typedef enum VkDescriptorUpdateTemplateType {
    VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET = 0,
  // Provided by VK_VERSION_1_4
    VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS = 1,
  // Provided by VK_KHR_descriptor_update_template with VK_KHR_push_descriptor,
VK_KHR_push_descriptor with VK_VERSION_1_1 or VK_KHR_descriptor_update_template
    VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS_KHR =
VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS,
  // Provided by VK_KHR_descriptor_update_template
    VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET_KHR =
VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET,
} VkDescriptorUpdateTemplateType;
```

```
// Provided by VK_KHR_descriptor_update_template
// Equivalent to VkDescriptorUpdateTemplateType
typedef VkDescriptorUpdateTemplateType VkDescriptorUpdateTemplateTypeKHR;
```

- VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET specifies that the descriptor update template will be used for descriptor set updates only.
- VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS specifies that the descriptor update template will be used for push descriptor updates only.

The VkDescriptorUpdateTemplateEntry structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkDescriptorUpdateTemplateEntry {
    uint32_t            dstBinding;
    uint32_t            dstArrayElement;
    uint32_t            descriptorCount;
    VkDescriptorType    descriptorType;
    size_t              offset;
    size_t              stride;
} VkDescriptorUpdateTemplateEntry;
```

```
// Provided by VK_KHR_descriptor_update_template
// Equivalent to VkDescriptorUpdateTemplateEntry
typedef VkDescriptorUpdateTemplateEntry VkDescriptorUpdateTemplateEntryKHR;
```

- dstBinding is the descriptor binding to update when using this descriptor update template.
- dstArrayElement is the starting element in the array belonging to dstBinding. If the descriptor binding identified by dstBinding has a descriptor type of

VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK then dstArrayElement specifies the starting byte offset to update.

- descriptorCount is the number of descriptors to update. If descriptorCount is greater than the number of remaining array elements in the destination binding, those affect consecutive bindings in a manner similar to VkWriteDescriptorSet above. If the descriptor binding identified by dstBinding has a descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK then descriptorCount specifies the number of bytes to update and the remaining array elements in the destination binding refer to the remaining number of bytes in it.
- descriptorType is a VkDescriptorType specifying the type of the descriptor.
- offset is the offset in bytes of the first binding in the raw data structure.
- stride is the stride in bytes between two consecutive array elements of the descriptor update information in the raw data structure. The actual pointer ptr for each array element j of update entry i is computed using the following formula:

```
    const char *ptr = (const char *)pData + pDescriptorUpdateEntries[i].offset + j
* pDescriptorUpdateEntries[i].stride
```

The stride is useful in case the bindings are stored in structs along with other data. If descriptorType is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK then the value of stride is ignored and the stride is assumed to be 1, i.e. the descriptor update information for them is always specified as a contiguous range.

Valid Usage

- VUID-VkDescriptorUpdateTemplateEntry-dstBinding-00354
  dstBinding **must** be a valid binding in the descriptor set layout implicitly specified when using a descriptor update template to update descriptors
- VUID-VkDescriptorUpdateTemplateEntry-dstArrayElement-00355
  dstArrayElement and descriptorCount **must** be less than or equal to the number of array elements in the descriptor set binding implicitly specified when using a descriptor update template to update descriptors, and all applicable consecutive bindings
- VUID-VkDescriptorUpdateTemplateEntry-descriptor-02226
  If descriptorType is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK, dstArrayElement **must** be an integer multiple of 4
- VUID-VkDescriptorUpdateTemplateEntry-descriptor-02227
  If descriptorType is VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK, descriptorCount **must** be an integer multiple of 4

Valid Usage (Implicit)

- VUID-VkDescriptorUpdateTemplateEntry-descriptorType-parameter
  descriptorType **must** be a valid VkDescriptorType value

To destroy a descriptor update template, call:

```
// Provided by VK_VERSION_1_1
void vkDestroyDescriptorUpdateTemplate(
    VkDevice                                    device,
    VkDescriptorUpdateTemplate                  descriptorUpdateTemplate,
    const VkAllocationCallbacks*                pAllocator);
```

```
// Provided by VK_KHR_descriptor_update_template
// Equivalent to vkDestroyDescriptorUpdateTemplate
void vkDestroyDescriptorUpdateTemplateKHR(
    VkDevice                                    device,
    VkDescriptorUpdateTemplate                  descriptorUpdateTemplate,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that has been used to create the descriptor update template.
- `descriptorUpdateTemplate` is the descriptor update template to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyDescriptorUpdateTemplate-descriptorSetLayout-00356
  If `VkAllocationCallbacks` were provided when `descriptorUpdateTemplate` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyDescriptorUpdateTemplate-descriptorSetLayout-00357
  If no `VkAllocationCallbacks` were provided when `descriptorUpdateTemplate` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyDescriptorUpdateTemplate-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyDescriptorUpdateTemplate-descriptorUpdateTemplate-parameter
  If `descriptorUpdateTemplate` is not VK_NULL_HANDLE, `descriptorUpdateTemplate` **must** be a valid VkDescriptorUpdateTemplate handle
- VUID-vkDestroyDescriptorUpdateTemplate-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyDescriptorUpdateTemplate-descriptorUpdateTemplate-parent
  If `descriptorUpdateTemplate` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `descriptorUpdateTemplate` **must** be externally synchronized

Once a `VkDescriptorUpdateTemplate` has been created, descriptor sets **can** be updated by calling:

```
// Provided by VK_VERSION_1_1
void vkUpdateDescriptorSetWithTemplate(
    VkDevice                                    device,
    VkDescriptorSet                             descriptorSet,
    VkDescriptorUpdateTemplate                  descriptorUpdateTemplate,
    const void*                                 pData);
```

```
// Provided by VK_KHR_descriptor_update_template
// Equivalent to vkUpdateDescriptorSetWithTemplate
void vkUpdateDescriptorSetWithTemplateKHR(
    VkDevice                                    device,
    VkDescriptorSet                             descriptorSet,
    VkDescriptorUpdateTemplate                  descriptorUpdateTemplate,
    const void*                                 pData);
```

- `device` is the logical device that updates the descriptor set.
- `descriptorSet` is the descriptor set to update.
- `descriptorUpdateTemplate` is a VkDescriptorUpdateTemplate object specifying the update mapping between `pData` and the descriptor set to update.
- `pData` is a pointer to memory containing one or more VkDescriptorImageInfo, VkDescriptorBufferInfo, or VkBufferView structures or VkAccelerationStructureKHR handles used to write the descriptors.

Valid Usage

- VUID-vkUpdateDescriptorSetWithTemplate-pData-01685
  `pData` **must** be a valid pointer to a memory containing one or more valid instances of VkAccelerationStructureKHR handles, VkDescriptorImageInfo, VkDescriptorBufferInfo, or VkBufferView in a layout defined by `descriptorUpdateTemplate` when it was created with vkCreateDescriptorUpdateTemplate
- VUID-vkUpdateDescriptorSetWithTemplate-descriptorSet-06995
  Host access to `descriptorSet` **must** be externally synchronized unless explicitly denoted otherwise for specific flags

Valid Usage (Implicit)

- VUID-vkUpdateDescriptorSetWithTemplate-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkUpdateDescriptorSetWithTemplate-descriptorSet-parameter
  `descriptorSet` **must** be a valid VkDescriptorSet handle
- VUID-vkUpdateDescriptorSetWithTemplate-descriptorUpdateTemplate-parameter
  `descriptorUpdateTemplate` **must** be a valid VkDescriptorUpdateTemplate handle
- VUID-vkUpdateDescriptorSetWithTemplate-descriptorSet-parent
  `descriptorSet` **must** have been created, allocated, or retrieved from `device`
- VUID-vkUpdateDescriptorSetWithTemplate-descriptorUpdateTemplate-parent
  `descriptorUpdateTemplate` **must** have been created, allocated, or retrieved from `device`

API Example

```
struct AppBufferView {
    VkBufferView bufferView;
    uint32_t     applicationRelatedInformation;
};

struct AppDataStructure
{
    VkDescriptorImageInfo  imageInfo;          // a single image info
    VkDescriptorBufferInfo bufferInfoArray[3]; // 3 buffer infos in an array
    AppBufferView          bufferView[2];      // An application-defined structure
containing a bufferView
    // ... some more application-related data
};

const VkDescriptorUpdateTemplateEntry descriptorUpdateTemplateEntries[] =
{
    // binding to a single image descriptor
    {
        .dstBinding = 0,
        .dstArrayElement = 0,
        .descriptorCount = 1,
        .descriptorType = VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER,
        .offset = offsetof(AppDataStructure, imageInfo),
        .stride = 0         // stride not required if descriptorCount is 1
    },

    // binding to an array of buffer descriptors
    {
        .dstBinding = 1,
        .dstArrayElement = 0,
        .descriptorCount = 3,
        .descriptorType = VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER,
```

```
        .offset = offsetof(AppDataStructure, bufferInfoArray),
        .stride = sizeof(VkDescriptorBufferInfo)    // descriptor buffer infos are
compact
    },

    // binding to an array of buffer views
    {
        .dstBinding = 2,
        .dstArrayElement = 0,
        .descriptorCount = 2,
        .descriptorType = VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER,
        .offset = offsetof(AppDataStructure, bufferView) +
                  offsetof(AppBufferView, bufferView),
        .stride = sizeof(AppBufferView)             // bufferViews do not have to be
compact
    },
};

// create a descriptor update template for descriptor set updates
const VkDescriptorUpdateTemplateCreateInfo createInfo =
{
    .sType = VK_STRUCTURE_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_CREATE_INFO,
    .pNext = NULL,
    .flags = 0,
    .descriptorUpdateEntryCount = 3,
    .pDescriptorUpdateEntries = descriptorUpdateTemplateEntries,
    .templateType = VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET,
    .descriptorSetLayout = myLayout,
    .pipelineBindPoint = 0,     // ignored by given templateType
    .pipelineLayout = 0,        // ignored by given templateType
    .set = 0,                   // ignored by given templateType
};

VkDescriptorUpdateTemplate myDescriptorUpdateTemplate;
myResult = vkCreateDescriptorUpdateTemplate(
    myDevice,
    &createInfo,
    NULL,
    &myDescriptorUpdateTemplate);

AppDataStructure appData;

// fill appData here or cache it in your engine
vkUpdateDescriptorSetWithTemplate(myDevice, myDescriptorSet,
myDescriptorUpdateTemplate, &appData);
```

17.7. Descriptor Set Binding

To bind one or more descriptor sets to a command buffer, call:

```
// Provided by VK_VERSION_1_0
void vkCmdBindDescriptorSets(
    VkCommandBuffer                             commandBuffer,
    VkPipelineBindPoint                         pipelineBindPoint,
    VkPipelineLayout                            layout,
    uint32_t                                    firstSet,
    uint32_t                                    descriptorSetCount,
    const VkDescriptorSet*                      pDescriptorSets,
    uint32_t                                    dynamicOffsetCount,
    const uint32_t*                             pDynamicOffsets);
```

- `commandBuffer` is the command buffer that the descriptor sets will be bound to.
- `pipelineBindPoint` is a VkPipelineBindPoint indicating the type of the pipeline that will use the descriptors. There is a separate set of bind points for each pipeline type, so binding one does not disturb the others.
- `layout` is a VkPipelineLayout object used to program the bindings.
- `firstSet` is the set number of the first descriptor set to be bound.
- `descriptorSetCount` is the number of elements in the `pDescriptorSets` array.
- `pDescriptorSets` is a pointer to an array of handles to VkDescriptorSet objects describing the descriptor sets to bind to.
- `dynamicOffsetCount` is the number of dynamic offsets in the `pDynamicOffsets` array.
- `pDynamicOffsets` is a pointer to an array of `uint32_t` values specifying dynamic offsets.

`vkCmdBindDescriptorSets` binds descriptor sets `pDescriptorSets`[0..`descriptorSetCount`-1] to set numbers [`firstSet`..`firstSet`+`descriptorSetCount`-1] for subsequent bound pipeline commands set by `pipelineBindPoint`. Any bindings that were previously applied via these sets , or calls to vkCmdSetDescriptorBufferOffsetsEXT or vkCmdBindDescriptorBufferEmbeddedSamplersEXT, are no longer valid.

Once bound, a descriptor set affects rendering of subsequent commands that interact with the given pipeline type in the command buffer until either a different set is bound to the same set number, or the set is disturbed as described in Pipeline Layout Compatibility.

A compatible descriptor set **must** be bound for all set numbers that any shaders in a pipeline access, at the time that a drawing or dispatching command is recorded to execute using that pipeline. However, if none of the shaders in a pipeline statically use any bindings with a particular set number, then no descriptor set need be bound for that set number, even if the pipeline layout includes a non-trivial descriptor set layout for that set number.

When consuming a descriptor, a descriptor is considered valid if the descriptor is not undefined as described by descriptor set allocation. If the `nullDescriptor` feature is enabled, a null descriptor is also considered valid. A descriptor that was disturbed by Pipeline Layout Compatibility, or was never bound by `vkCmdBindDescriptorSets` is not considered valid. For any given descriptor, VkDescriptorBindingFlagBits and VkDescriptorSetLayoutCreateFlagBits determine if validity is defined in terms of the descriptor being statically accessed, or dynamically accessed. If the

descriptor is determined to be accessed by the appropriate definition, the consuming descriptor type in the pipeline **must** match the VkDescriptorType in VkDescriptorSetLayoutCreateInfo for the descriptor to be considered valid. If a descriptor is a mutable descriptor, the consuming descriptor type in the pipeline **must** match the active descriptor type for the descriptor to be considered valid.

NOTE	Further validation may be carried out beyond validation for descriptor types, e.g. Texel Input Validation.

If any of the sets being bound include dynamic uniform or storage buffers, then `pDynamicOffsets` includes one element for each array element in each dynamic descriptor type binding in each set. Values are taken from `pDynamicOffsets` in an order such that all entries for set N come before set N+1; within a set, entries are ordered by the binding numbers in the descriptor set layouts; and within a binding array, elements are in order. `dynamicOffsetCount` **must** equal the total number of dynamic descriptors in the sets being bound.

The effective offset used for dynamic uniform and storage buffer bindings is the sum of the relative offset taken from `pDynamicOffsets`, and the base address of the buffer plus base offset in the descriptor set. The range of the dynamic uniform and storage buffer bindings is the buffer range as specified in the descriptor set.

Each of the `pDescriptorSets` **must** be compatible with the pipeline layout specified by `layout`. The layout used to program the bindings **must** also be compatible with the pipeline used in subsequent bound pipeline commands with that pipeline type, as defined in the Pipeline Layout Compatibility section.

The descriptor set contents bound by a call to `vkCmdBindDescriptorSets` **may** be consumed at the following times:

- For descriptor bindings created with the VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT bit set, the contents **may** be consumed when the command buffer is submitted to a queue, or during shader execution of the resulting draws and dispatches, or any time in between. Otherwise,
- during host execution of the command, or during shader execution of the resulting draws and dispatches, or any time in between.

Thus, the contents of a descriptor set binding **must** not be altered (overwritten by an update command, or freed) between the first point in time that it **may** be consumed, and when the command completes executing on the queue.

The contents of `pDynamicOffsets` are consumed immediately during execution of `vkCmdBindDescriptorSets`. Once all pending uses have completed, it is legal to update and reuse a descriptor set.

Valid Usage

- VUID-vkCmdBindDescriptorSets-commandBuffer-11295
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pSamplerHeapBindInfo` equal to

NULL

- VUID-vkCmdBindDescriptorSets-commandBuffer-11296
  If commandBuffer is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::pResourceHeapBindInfo equal to NULL

- VUID-vkCmdBindDescriptorSets-pDescriptorSets-00358
  Each element of pDescriptorSets that is not VK_NULL_HANDLE **must** have been allocated with a VkDescriptorSetLayout that matches (is the same as, or identically defined as) the VkDescriptorSetLayout at set *n* in layout, where *n* is the sum of firstSet and the index into pDescriptorSets
- VUID-vkCmdBindDescriptorSets-dynamicOffsetCount-00359
  dynamicOffsetCount **must** be equal to the total number of dynamic descriptors in pDescriptorSets
- VUID-vkCmdBindDescriptorSets-firstSet-00360
  The sum of firstSet and descriptorSetCount **must** be less than or equal to VkPipelineLayoutCreateInfo::setLayoutCount provided when layout was created
- VUID-vkCmdBindDescriptorSets-pDynamicOffsets-01971
  Each element of pDynamicOffsets which corresponds to a descriptor binding with type VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC **must** be a multiple of VkPhysicalDeviceLimits::minUniformBufferOffsetAlignment
- VUID-vkCmdBindDescriptorSets-pDynamicOffsets-01972
  Each element of pDynamicOffsets which corresponds to a descriptor binding with type VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC **must** be a multiple of VkPhysicalDeviceLimits::minStorageBufferOffsetAlignment
- VUID-vkCmdBindDescriptorSets-pDescriptorSets-01979
  For each dynamic uniform or storage buffer binding in pDescriptorSets, the sum of the effective offset and the range of the binding **must** be less than or equal to the size of the buffer
- VUID-vkCmdBindDescriptorSets-pDescriptorSets-06715
  For each dynamic uniform or storage buffer binding in pDescriptorSets, if the range was set with VK_WHOLE_SIZE then pDynamicOffsets which corresponds to the descriptor binding **must** be 0
- VUID-vkCmdBindDescriptorSets-pDescriptorSets-04616
  Each element of pDescriptorSets **must** not have been allocated from a VkDescriptorPool with the VK_DESCRIPTOR_POOL_CREATE_HOST_ONLY_BIT_EXT flag set
- VUID-vkCmdBindDescriptorSets-pDescriptorSets-06563
  If the graphicsPipelineLibrary feature is not enabled, each element of pDescriptorSets **must** be a valid VkDescriptorSet
- VUID-vkCmdBindDescriptorSets-pDescriptorSets-08010
  Each element of pDescriptorSets **must** have been allocated with a VkDescriptorSetLayout which was not created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdBindDescriptorSets-pipelineBindPoint-00361

`pipelineBindPoint` **must** be supported by the `commandBuffer`'s parent `VkCommandPool`'s queue family

Valid Usage (Implicit)

- VUID-vkCmdBindDescriptorSets-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindDescriptorSets-pipelineBindPoint-parameter
  `pipelineBindPoint` **must** be a valid VkPipelineBindPoint value
- VUID-vkCmdBindDescriptorSets-layout-parameter
  `layout` **must** be a valid VkPipelineLayout handle
- VUID-vkCmdBindDescriptorSets-pDescriptorSets-parameter
  `pDescriptorSets` **must** be a valid pointer to an array of `descriptorSetCount` valid or VK_NULL_HANDLE VkDescriptorSet handles
- VUID-vkCmdBindDescriptorSets-pDynamicOffsets-parameter
  If `dynamicOffsetCount` is not `0`, `pDynamicOffsets` **must** be a valid pointer to an array of `dynamicOffsetCount` `uint32_t` values
- VUID-vkCmdBindDescriptorSets-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBindDescriptorSets-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindDescriptorSets-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindDescriptorSets-descriptorSetCount-arraylength
  `descriptorSetCount` **must** be greater than `0`
- VUID-vkCmdBindDescriptorSets-commonparent
  Each of `commandBuffer`, `layout`, and the elements of `pDescriptorSets` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindDescriptorSets is not affected by conditional rendering

To bind one or more descriptor sets to a command buffer, call:

```
// Provided by VK_VERSION_1_4
void vkCmdBindDescriptorSets2(
    VkCommandBuffer                             commandBuffer,
    const VkBindDescriptorSetsInfo*             pBindDescriptorSetsInfo);
```

```
// Provided by VK_KHR_maintenance6
// Equivalent to vkCmdBindDescriptorSets2
void vkCmdBindDescriptorSets2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkBindDescriptorSetsInfo*             pBindDescriptorSetsInfo);
```

- `commandBuffer` is the command buffer that the descriptor sets will be bound to.
- `pBindDescriptorSetsInfo` is a pointer to a `VkBindDescriptorSetsInfo` structure.

Valid Usage

- VUID-vkCmdBindDescriptorSets2-commandBuffer-11295
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pSamplerHeapBindInfo` equal to `NULL`
- VUID-vkCmdBindDescriptorSets2-commandBuffer-11296
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pResourceHeapBindInfo` equal to `NULL`
- VUID-vkCmdBindDescriptorSets2-pBindDescriptorSetsInfo-09467
  Each bit in `pBindDescriptorSetsInfo->stageFlags` **must** be a stage supported by the

commandBuffer's parent VkCommandPool's queue family

Valid Usage (Implicit)

- VUID-vkCmdBindDescriptorSets2-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindDescriptorSets2-pBindDescriptorSetsInfo-parameter
  pBindDescriptorSetsInfo **must** be a valid pointer to a valid VkBindDescriptorSetsInfo structure
- VUID-vkCmdBindDescriptorSets2-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdBindDescriptorSets2-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindDescriptorSets2-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindDescriptorSets2 is not affected by conditional rendering

The VkBindDescriptorSetsInfo structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkBindDescriptorSetsInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkShaderStageFlags        stageFlags;
    VkPipelineLayout          layout;
    uint32_t                  firstSet;
    uint32_t                  descriptorSetCount;
    const VkDescriptorSet*    pDescriptorSets;
    uint32_t                  dynamicOffsetCount;
    const uint32_t*           pDynamicOffsets;
} VkBindDescriptorSetsInfo;
```

```
// Provided by VK_KHR_maintenance6
// Equivalent to VkBindDescriptorSetsInfo
typedef VkBindDescriptorSetsInfo VkBindDescriptorSetsInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stageFlags` is a bitmask of VkShaderStageFlagBits specifying the shader stages the descriptor sets will be bound to.
- `layout` is a VkPipelineLayout object used to program the bindings.
- `firstSet` is the set number of the first descriptor set to be bound.
- `descriptorSetCount` is the number of elements in the `pDescriptorSets` array.
- `pDescriptorSets` is a pointer to an array of handles to VkDescriptorSet objects describing the descriptor sets to bind to.
- `dynamicOffsetCount` is the number of dynamic offsets in the `pDynamicOffsets` array.
- `pDynamicOffsets` is a pointer to an array of `uint32_t` values specifying dynamic offsets.

If `stageFlags` specifies a subset of all stages corresponding to one or more pipeline bind points, the binding operation still affects all stages corresponding to the given pipeline bind point(s) as if the equivalent original version of this command had been called with the same parameters. For example, specifying a `stageFlags` value of VK_SHADER_STAGE_VERTEX_BIT | VK_SHADER_STAGE_FRAGMENT_BIT | VK_SHADER_STAGE_COMPUTE_BIT is equivalent to calling the original version of this command once with VK_PIPELINE_BIND_POINT_GRAPHICS and once with VK_PIPELINE_BIND_POINT_COMPUTE.

Valid Usage

- VUID-VkBindDescriptorSetsInfo-pDescriptorSets-00358
  Each element of `pDescriptorSets` that is not VK_NULL_HANDLE **must** have been allocated with a `VkDescriptorSetLayout` that matches (is the same as, or identically defined as) the `VkDescriptorSetLayout` at set *n* in `layout`, where *n* is the sum of `firstSet` and the index into

pDescriptorSets

- VUID-VkBindDescriptorSetsInfo-dynamicOffsetCount-00359
  dynamicOffsetCount **must** be equal to the total number of dynamic descriptors in pDescriptorSets
- VUID-VkBindDescriptorSetsInfo-firstSet-00360
  The sum of firstSet and descriptorSetCount **must** be less than or equal to VkPipelineLayoutCreateInfo::setLayoutCount provided when layout was created
- VUID-VkBindDescriptorSetsInfo-pDynamicOffsets-01971
  Each element of pDynamicOffsets which corresponds to a descriptor binding with type VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC **must** be a multiple of VkPhysicalDeviceLimits::minUniformBufferOffsetAlignment
- VUID-VkBindDescriptorSetsInfo-pDynamicOffsets-01972
  Each element of pDynamicOffsets which corresponds to a descriptor binding with type VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC **must** be a multiple of VkPhysicalDeviceLimits::minStorageBufferOffsetAlignment
- VUID-VkBindDescriptorSetsInfo-pDescriptorSets-01979
  For each dynamic uniform or storage buffer binding in pDescriptorSets, the sum of the effective offset and the range of the binding **must** be less than or equal to the size of the buffer
- VUID-VkBindDescriptorSetsInfo-pDescriptorSets-06715
  For each dynamic uniform or storage buffer binding in pDescriptorSets, if the range was set with VK_WHOLE_SIZE then pDynamicOffsets which corresponds to the descriptor binding **must** be 0
- VUID-VkBindDescriptorSetsInfo-pDescriptorSets-04616
  Each element of pDescriptorSets **must** not have been allocated from a VkDescriptorPool with the VK_DESCRIPTOR_POOL_CREATE_HOST_ONLY_BIT_EXT flag set
- VUID-VkBindDescriptorSetsInfo-pDescriptorSets-06563
  If the graphicsPipelineLibrary feature is not enabled, each element of pDescriptorSets **must** be a valid VkDescriptorSet
- VUID-VkBindDescriptorSetsInfo-pDescriptorSets-08010
  Each element of pDescriptorSets **must** have been allocated with a VkDescriptorSetLayout which was not created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-VkBindDescriptorSetsInfo-None-09495
  layout **must** be a valid VkPipelineLayout handle

Valid Usage (Implicit)

- VUID-VkBindDescriptorSetsInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_BIND_DESCRIPTOR_SETS_INFO
- VUID-VkBindDescriptorSetsInfo-pNext-pNext
  pNext **must** be NULL or a pointer to a valid instance of VkPipelineLayoutCreateInfo

- VUID-VkBindDescriptorSetsInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkBindDescriptorSetsInfo-stageFlags-parameter
  `stageFlags` **must** be a valid combination of VkShaderStageFlagBits values
- VUID-VkBindDescriptorSetsInfo-layout-parameter
  If `layout` is not VK_NULL_HANDLE, `layout` **must** be a valid VkPipelineLayout handle
- VUID-VkBindDescriptorSetsInfo-pDescriptorSets-parameter
  `pDescriptorSets` **must** be a valid pointer to an array of `descriptorSetCount` valid VkDescriptorSet handles
- VUID-VkBindDescriptorSetsInfo-pDynamicOffsets-parameter
  If `dynamicOffsetCount` is not `0`, `pDynamicOffsets` **must** be a valid pointer to an array of `dynamicOffsetCount` `uint32_t` values
- VUID-VkBindDescriptorSetsInfo-descriptorSetCount-arraylength
  `descriptorSetCount` **must** be greater than `0`
- VUID-VkBindDescriptorSetsInfo-commonparent
  Both of `layout`, and the elements of `pDescriptorSets` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

17.8. Push Descriptor Updates

In addition to allocating descriptor sets and binding them to a command buffer, an application **can** record descriptor updates into the command buffer.

To push descriptor updates into a command buffer, call:

```
// Provided by VK_VERSION_1_4
void vkCmdPushDescriptorSet(
    VkCommandBuffer                             commandBuffer,
    VkPipelineBindPoint                         pipelineBindPoint,
    VkPipelineLayout                            layout,
    uint32_t                                    set,
    uint32_t                                    descriptorWriteCount,
    const VkWriteDescriptorSet*                 pDescriptorWrites);
```

```
// Provided by VK_KHR_push_descriptor
// Equivalent to vkCmdPushDescriptorSet
void vkCmdPushDescriptorSetKHR(
    VkCommandBuffer                             commandBuffer,
    VkPipelineBindPoint                         pipelineBindPoint,
    VkPipelineLayout                            layout,
    uint32_t                                    set,
    uint32_t                                    descriptorWriteCount,
    const VkWriteDescriptorSet*                 pDescriptorWrites);
```

- `commandBuffer` is the command buffer that the descriptors will be recorded in.
- `pipelineBindPoint` is a VkPipelineBindPoint indicating the type of the pipeline that will use the descriptors. There is a separate set of push descriptor bindings for each pipeline type, so binding one does not disturb the others.
- `layout` is a VkPipelineLayout object used to program the bindings.
- `set` is the set number of the descriptor set in the pipeline layout that will be updated.
- `descriptorWriteCount` is the number of elements in the `pDescriptorWrites` array.
- `pDescriptorWrites` is a pointer to an array of VkWriteDescriptorSet structures describing the descriptors to be updated.

Push descriptors are a small bank of descriptors whose storage is internally managed by the command buffer rather than being written into a descriptor set and later bound to a command buffer. Push descriptors allow for incremental updates of descriptors without managing the lifetime of descriptor sets.

When a command buffer begins recording, all push descriptors are undefined. Push descriptors **can** be updated incrementally and cause shaders to use the updated descriptors for subsequent bound pipeline commands with the pipeline type set by `pipelineBindPoint` until the descriptor is overwritten, or else until the set is disturbed as described in Pipeline Layout Compatibility. When the set is disturbed or push descriptors with a different descriptor set layout are set, all push descriptors are undefined.

Push descriptors that are statically used by a pipeline **must** not be undefined at the time that a drawing or dispatching command is recorded to execute using that pipeline. This includes immutable sampler descriptors, which **must** be pushed before they are accessed by a pipeline (the immutable samplers are pushed, rather than the samplers in `pDescriptorWrites`). Push descriptors that are not statically used **can** remain undefined.

Push descriptors do not use dynamic offsets. Instead, the corresponding non-dynamic descriptor types **can** be used and the `offset` member of VkDescriptorBufferInfo **can** be changed each time the descriptor is written.

Each element of `pDescriptorWrites` is interpreted as in VkWriteDescriptorSet, except the `dstSet` member is ignored.

To push an immutable sampler, use a VkWriteDescriptorSet with `dstBinding` and `dstArrayElement` selecting the immutable sampler's binding. If the descriptor type is VK_DESCRIPTOR_TYPE_SAMPLER, the `pImageInfo` parameter is ignored and the immutable sampler is taken from the push descriptor set layout in the pipeline layout. If the descriptor type is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, the `sampler` member of the `pImageInfo` parameter is ignored and the immutable sampler is taken from the push descriptor set layout in the pipeline layout.

Valid Usage

- VUID-vkCmdPushDescriptorSet-commandBuffer-11295
  If `commandBuffer` is a secondary command buffer, it **must** have begun with

VkCommandBufferInheritanceDescriptorHeapInfoEXT::pSamplerHeapBindInfo equal to NULL

- VUID-vkCmdPushDescriptorSet-commandBuffer-11296
  If commandBuffer is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::pResourceHeapBindInfo equal to NULL
- VUID-vkCmdPushDescriptorSet-set-00364
  set **must** be less than VkPipelineLayoutCreateInfo::setLayoutCount provided when layout was created
- VUID-vkCmdPushDescriptorSet-set-00365
  set **must** be the unique set number in the pipeline layout that uses a descriptor set layout that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT
- VUID-vkCmdPushDescriptorSet-pDescriptorWrites-06494
  For each element i where pDescriptorWrites[i].descriptorType is VK_DESCRIPTOR_TYPE_SAMPLER, VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, pDescriptorWrites[i].pImageInfo **must** be a valid pointer to an array of pDescriptorWrites[i].descriptorCount valid VkDescriptorImageInfo structures
- VUID-vkCmdPushDescriptorSet-pipelineBindPoint-00363
  pipelineBindPoint **must** be supported by the commandBuffer's parent VkCommandPool's queue family
- VUID-vkCmdPushDescriptorSet-None-10356
  If the VK_KHR_push_descriptor extension is not enabled, pushDescriptor **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdPushDescriptorSet-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdPushDescriptorSet-pipelineBindPoint-parameter
  pipelineBindPoint **must** be a valid VkPipelineBindPoint value
- VUID-vkCmdPushDescriptorSet-layout-parameter
  layout **must** be a valid VkPipelineLayout handle
- VUID-vkCmdPushDescriptorSet-pDescriptorWrites-parameter
  pDescriptorWrites **must** be a valid pointer to an array of descriptorWriteCount valid VkWriteDescriptorSet structures
- VUID-vkCmdPushDescriptorSet-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdPushDescriptorSet-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdPushDescriptorSet-videocoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdPushDescriptorSet-descriptorWriteCount-arraylength
  `descriptorWriteCount` **must** be greater than `0`
- VUID-vkCmdPushDescriptorSet-commonparent
  Both of `commandBuffer`, and `layout` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdPushDescriptorSet is not affected by conditional rendering

To push descriptor updates into a command buffer, call:

```
// Provided by VK_VERSION_1_4
void vkCmdPushDescriptorSet2(
    VkCommandBuffer                             commandBuffer,
    const VkPushDescriptorSetInfo*              pPushDescriptorSetInfo);
```

```
// Provided by VK_KHR_maintenance6 with VK_KHR_push_descriptor
// Equivalent to vkCmdPushDescriptorSet2
void vkCmdPushDescriptorSet2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkPushDescriptorSetInfo*              pPushDescriptorSetInfo);
```

- `commandBuffer` is the command buffer that the descriptors will be recorded in.

- pPushDescriptorSetInfo is a pointer to a VkPushDescriptorSetInfo structure.

Valid Usage

- VUID-vkCmdPushDescriptorSet2-commandBuffer-11295
  If commandBuffer is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::pSamplerHeapBindInfo equal to NULL
- VUID-vkCmdPushDescriptorSet2-commandBuffer-11296
  If commandBuffer is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::pResourceHeapBindInfo equal to NULL
- VUID-vkCmdPushDescriptorSet2-pPushDescriptorSetInfo-09468
  Each bit in pPushDescriptorSetInfo->stageFlags **must** be a stage supported by the commandBuffer's parent VkCommandPool's queue family
- VUID-vkCmdPushDescriptorSet2-None-10357
  If the VK_KHR_push_descriptor extension is not enabled, pushDescriptor **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdPushDescriptorSet2-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdPushDescriptorSet2-pPushDescriptorSetInfo-parameter
  pPushDescriptorSetInfo **must** be a valid pointer to a valid VkPushDescriptorSetInfo structure
- VUID-vkCmdPushDescriptorSet2-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdPushDescriptorSet2-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdPushDescriptorSet2-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdPushDescriptorSet2 is not affected by conditional rendering

The VkPushDescriptorSetInfo structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPushDescriptorSetInfo {
    VkStructureType                sType;
    const void*                    pNext;
    VkShaderStageFlags             stageFlags;
    VkPipelineLayout               layout;
    uint32_t                       set;
    uint32_t                       descriptorWriteCount;
    const VkWriteDescriptorSet*    pDescriptorWrites;
} VkPushDescriptorSetInfo;
```

```
// Provided by VK_KHR_maintenance6 with VK_KHR_push_descriptor
// Equivalent to VkPushDescriptorSetInfo
typedef VkPushDescriptorSetInfo VkPushDescriptorSetInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- stageFlags is a bitmask of VkShaderStageFlagBits specifying the shader stages that will use the descriptors.
- layout is a VkPipelineLayout object used to program the bindings.
- set is the set number of the descriptor set in the pipeline layout that will be updated.
- descriptorWriteCount is the number of elements in the pDescriptorWrites array.
- pDescriptorWrites is a pointer to an array of VkWriteDescriptorSet structures describing the descriptors to be updated.

If stageFlags specifies a subset of all stages corresponding to one or more pipeline bind points, the

binding operation still affects all stages corresponding to the given pipeline bind point(s) as if the equivalent original version of this command had been called with the same parameters. For example, specifying a `stageFlags` value of VK_SHADER_STAGE_VERTEX_BIT | VK_SHADER_STAGE_FRAGMENT_BIT | VK_SHADER_STAGE_COMPUTE_BIT is equivalent to calling the original version of this command once with VK_PIPELINE_BIND_POINT_GRAPHICS and once with VK_PIPELINE_BIND_POINT_COMPUTE.

Valid Usage

- VUID-VkPushDescriptorSetInfo-set-00364
  `set` **must** be less than VkPipelineLayoutCreateInfo::`setLayoutCount` provided when `layout` was created
- VUID-VkPushDescriptorSetInfo-set-00365
  `set` **must** be the unique set number in the pipeline layout that uses a descriptor set layout that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT
- VUID-VkPushDescriptorSetInfo-pDescriptorWrites-06494
  For each element i where `pDescriptorWrites`[i].`descriptorType` is VK_DESCRIPTOR_TYPE_SAMPLER, VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, `pDescriptorWrites`[i].`pImageInfo` **must** be a valid pointer to an array of `pDescriptorWrites`[i].`descriptorCount` valid `VkDescriptorImageInfo` structures
- VUID-VkPushDescriptorSetInfo-None-09495
  `layout` **must** be a valid VkPipelineLayout handle

Valid Usage (Implicit)

- VUID-VkPushDescriptorSetInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_INFO
- VUID-VkPushDescriptorSetInfo-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkPipelineLayoutCreateInfo
- VUID-VkPushDescriptorSetInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkPushDescriptorSetInfo-stageFlags-parameter
  `stageFlags` **must** be a valid combination of VkShaderStageFlagBits values
- VUID-VkPushDescriptorSetInfo-layout-parameter
  If `layout` is not VK_NULL_HANDLE, `layout` **must** be a valid VkPipelineLayout handle
- VUID-VkPushDescriptorSetInfo-pDescriptorWrites-parameter
  `pDescriptorWrites` **must** be a valid pointer to an array of `descriptorWriteCount` valid VkWriteDescriptorSet structures
- VUID-VkPushDescriptorSetInfo-descriptorWriteCount-arraylength
  `descriptorWriteCount` **must** be greater than `0`

17.9. Push Descriptor Updates With Descriptor Update Templates

To use a descriptor update template to specify the push descriptors to update in a command buffer, call:

```
// Provided by VK_VERSION_1_4
void vkCmdPushDescriptorSetWithTemplate(
    VkCommandBuffer                             commandBuffer,
    VkDescriptorUpdateTemplate                  descriptorUpdateTemplate,
    VkPipelineLayout                            layout,
    uint32_t                                    set,
    const void*                                 pData);
```

```
// Provided by VK_KHR_descriptor_update_template with VK_KHR_push_descriptor,
VK_KHR_push_descriptor with VK_VERSION_1_1 or VK_KHR_descriptor_update_template
// Equivalent to vkCmdPushDescriptorSetWithTemplate
void vkCmdPushDescriptorSetWithTemplateKHR(
    VkCommandBuffer                             commandBuffer,
    VkDescriptorUpdateTemplate                  descriptorUpdateTemplate,
    VkPipelineLayout                            layout,
    uint32_t                                    set,
    const void*                                 pData);
```

- `commandBuffer` is the command buffer that the descriptors will be recorded in.
- `descriptorUpdateTemplate` is a descriptor update template defining how to interpret the descriptor information in `pData`.
- `layout` is a VkPipelineLayout object used to program the bindings. It **must** be compatible with the layout used to create the `descriptorUpdateTemplate` handle.
- `set` is the set number of the descriptor set in the pipeline layout that will be updated. This **must** be the same number used to create the `descriptorUpdateTemplate` handle.
- `pData` is a pointer to memory containing descriptors for the templated update.

Valid Usage

- VUID-vkCmdPushDescriptorSetWithTemplate-commandBuffer-11295
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pSamplerHeapBindInfo` equal to `NULL`
- VUID-vkCmdPushDescriptorSetWithTemplate-commandBuffer-11296
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pResourceHeapBindInfo` equal to `NULL`

- VUID-vkCmdPushDescriptorSetWithTemplate-commandBuffer-00366
  The `pipelineBindPoint` specified during the creation of the descriptor update template **must** be supported by the `commandBuffer`'s parent `VkCommandPool`'s queue family
- VUID-vkCmdPushDescriptorSetWithTemplate-pData-01686
  `pData` **must** be a valid pointer to memory containing one or more valid instances of VkDescriptorImageInfo, VkDescriptorBufferInfo, or VkBufferView in a layout defined by `descriptorUpdateTemplate` when it was created with vkCreateDescriptorUpdateTemplate
- VUID-vkCmdPushDescriptorSetWithTemplate-layout-07993
  `layout` **must** be compatible with the layout used to create `descriptorUpdateTemplate`
- VUID-vkCmdPushDescriptorSetWithTemplate-descriptorUpdateTemplate-07994
  `descriptorUpdateTemplate` **must** have been created with a `templateType` of VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS
- VUID-vkCmdPushDescriptorSetWithTemplate-set-07995
  `set` **must** be the same value used to create `descriptorUpdateTemplate`
- VUID-vkCmdPushDescriptorSetWithTemplate-set-07304
  `set` **must** be less than VkPipelineLayoutCreateInfo::`setLayoutCount` provided when `layout` was created
- VUID-vkCmdPushDescriptorSetWithTemplate-set-11854
  `set` **must** reference a valid VkDescriptorSetLayout handle in `layout`
- VUID-vkCmdPushDescriptorSetWithTemplate-set-07305
  `set` **must** be the unique set number in the pipeline layout that uses a descriptor set layout that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT
- VUID-vkCmdPushDescriptorSetWithTemplate-None-10358
  If the VK_KHR_push_descriptor extension is not enabled, `pushDescriptor` **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdPushDescriptorSetWithTemplate-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdPushDescriptorSetWithTemplate-descriptorUpdateTemplate-parameter
  `descriptorUpdateTemplate` **must** be a valid VkDescriptorUpdateTemplate handle
- VUID-vkCmdPushDescriptorSetWithTemplate-layout-parameter
  `layout` **must** be a valid VkPipelineLayout handle
- VUID-vkCmdPushDescriptorSetWithTemplate-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdPushDescriptorSetWithTemplate-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdPushDescriptorSetWithTemplate-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdPushDescriptorSetWithTemplate-commonparent

Each of `commandBuffer`, `descriptorUpdateTemplate`, and `layout` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdPushDescriptorSetWithTemplate is not affected by conditional rendering

API Example

```
struct AppDataStructure
{
    VkDescriptorImageInfo  imageInfo;          // a single image info
    // ... some more application-related data
};

const VkDescriptorUpdateTemplateEntry descriptorUpdateTemplateEntries[] =
{
    // binding to a single image descriptor
    {
        .dstBinding = 0,
        .dstArrayElement = 0,
        .descriptorCount = 1,
        .descriptorType = VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER,
        .offset = offsetof(AppDataStructure, imageInfo),
        .stride = 0     // not required if descriptorCount is 1
    }
};

// create a descriptor update template for push descriptor set updates
```

```
const VkDescriptorUpdateTemplateCreateInfo createInfo =
{
    .sType = VK_STRUCTURE_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_CREATE_INFO,
    .pNext = NULL,
    .flags = 0,
    .descriptorUpdateEntryCount = 1,
    .pDescriptorUpdateEntries = descriptorUpdateTemplateEntries,
    .templateType = VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS,
    .descriptorSetLayout = 0,   // ignored by given templateType
    .pipelineBindPoint = VK_PIPELINE_BIND_POINT_GRAPHICS,
    .pipelineLayout = myPipelineLayout,
    .set = 0,
};

VkDescriptorUpdateTemplate myDescriptorUpdateTemplate;
myResult = vkCreateDescriptorUpdateTemplate(
    myDevice,
    &createInfo,
    NULL,
    &myDescriptorUpdateTemplate);

AppDataStructure appData;
// fill appData here or cache it in your engine
vkCmdPushDescriptorSetWithTemplate(myCmdBuffer, myDescriptorUpdateTemplate,
myPipelineLayout, 0,&appData);
```

To use a descriptor update template to specify the push descriptors to update in a command buffer, call:

```
// Provided by VK_VERSION_1_4
void vkCmdPushDescriptorSetWithTemplate2(
    VkCommandBuffer                             commandBuffer,
    const VkPushDescriptorSetWithTemplateInfo*  pPushDescriptorSetWithTemplateInfo);
```

```
// Provided by VK_KHR_maintenance6 with VK_KHR_push_descriptor
// Equivalent to vkCmdPushDescriptorSetWithTemplate2
void vkCmdPushDescriptorSetWithTemplate2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkPushDescriptorSetWithTemplateInfo*  pPushDescriptorSetWithTemplateInfo);
```

- `commandBuffer` is the command buffer that the descriptors will be recorded in.
- `pPushDescriptorSetWithTemplateInfo` is a pointer to a `VkPushDescriptorSetWithTemplateInfo` structure.

Valid Usage

- VUID-vkCmdPushDescriptorSetWithTemplate2-commandBuffer-11295

If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pSamplerHeapBindInfo` equal to `NULL`

- VUID-vkCmdPushDescriptorSetWithTemplate2-commandBuffer-11296
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pResourceHeapBindInfo` equal to `NULL`

Valid Usage (Implicit)

- VUID-vkCmdPushDescriptorSetWithTemplate2-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdPushDescriptorSetWithTemplate2-pPushDescriptorSetWithTemplateInfo-parameter
  `pPushDescriptorSetWithTemplateInfo` **must** be a valid pointer to a valid VkPushDescriptorSetWithTemplateInfo structure
- VUID-vkCmdPushDescriptorSetWithTemplate2-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdPushDescriptorSetWithTemplate2-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdPushDescriptorSetWithTemplate2-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdPushDescriptorSetWithTemplate2 is not affected by conditional rendering

The VkPushDescriptorSetWithTemplateInfo structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPushDescriptorSetWithTemplateInfo {
    VkStructureType               sType;
    const void*                   pNext;
    VkDescriptorUpdateTemplate    descriptorUpdateTemplate;
    VkPipelineLayout              layout;
    uint32_t                      set;
    const void*                   pData;
} VkPushDescriptorSetWithTemplateInfo;
```

```
// Provided by VK_KHR_maintenance6 with VK_KHR_push_descriptor
// Equivalent to VkPushDescriptorSetWithTemplateInfo
typedef VkPushDescriptorSetWithTemplateInfo VkPushDescriptorSetWithTemplateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- descriptorUpdateTemplate is a descriptor update template defining how to interpret the descriptor information in pData.
- layout is a VkPipelineLayout object used to program the bindings. It **must** be compatible with the layout used to create the descriptorUpdateTemplate handle.
- set is the set number of the descriptor set in the pipeline layout that will be updated. This **must** be the same number used to create the descriptorUpdateTemplate handle.
- pData is a pointer to memory containing descriptors for the templated update.

Valid Usage

- VUID-VkPushDescriptorSetWithTemplateInfo-commandBuffer-00366
  The pipelineBindPoint specified during the creation of the descriptor update template **must** be supported by the commandBuffer's parent VkCommandPool's queue family
- VUID-VkPushDescriptorSetWithTemplateInfo-pData-01686
  pData **must** be a valid pointer to memory containing one or more valid instances of VkDescriptorImageInfo, VkDescriptorBufferInfo, or VkBufferView in a layout defined by descriptorUpdateTemplate when it was created with vkCreateDescriptorUpdateTemplate
- VUID-VkPushDescriptorSetWithTemplateInfo-layout-07993
  layout **must** be compatible with the layout used to create descriptorUpdateTemplate
- VUID-VkPushDescriptorSetWithTemplateInfo-descriptorUpdateTemplate-07994

descriptorUpdateTemplate **must** have been created with a templateType of VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS

- VUID-VkPushDescriptorSetWithTemplateInfo-set-07995
set **must** be the same value used to create descriptorUpdateTemplate
- VUID-VkPushDescriptorSetWithTemplateInfo-set-07304
set **must** be less than VkPipelineLayoutCreateInfo::setLayoutCount provided when layout was created
- VUID-VkPushDescriptorSetWithTemplateInfo-set-11854
set **must** reference a valid VkDescriptorSetLayout handle in layout
- VUID-VkPushDescriptorSetWithTemplateInfo-set-07305
set **must** be the unique set number in the pipeline layout that uses a descriptor set layout that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT
- VUID-VkPushDescriptorSetWithTemplateInfo-None-09495
layout **must** be a valid VkPipelineLayout handle
- VUID-VkPushDescriptorSetWithTemplateInfo-None-10359
If the VK_KHR_push_descriptor extension is not enabled, pushDescriptor **must** be enabled

Valid Usage (Implicit)

- VUID-VkPushDescriptorSetWithTemplateInfo-sType-sType
sType **must** be VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_WITH_TEMPLATE_INFO
- VUID-VkPushDescriptorSetWithTemplateInfo-pNext-pNext
pNext **must** be NULL or a pointer to a valid instance of VkPipelineLayoutCreateInfo
- VUID-VkPushDescriptorSetWithTemplateInfo-sType-unique
The sType value of each structure in the pNext chain **must** be unique
- VUID-VkPushDescriptorSetWithTemplateInfo-descriptorUpdateTemplate-parameter
descriptorUpdateTemplate **must** be a valid VkDescriptorUpdateTemplate handle
- VUID-VkPushDescriptorSetWithTemplateInfo-layout-parameter
If layout is not VK_NULL_HANDLE, layout **must** be a valid VkPipelineLayout handle
- VUID-VkPushDescriptorSetWithTemplateInfo-pData-parameter
pData **must** be a pointer value
- VUID-VkPushDescriptorSetWithTemplateInfo-commonparent
Both of descriptorUpdateTemplate, and layout that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

17.10. Push Constant Updates

As described above in section Pipeline Layouts, the pipeline layout defines shader push constants which are updated via Vulkan commands rather than via writes to memory or copy commands.

NOTE Push constants represent a high speed path to modify constant data in pipelines that is expected to outperform memory-backed resource updates.

To update push constants, call:

```
// Provided by VK_VERSION_1_0
void vkCmdPushConstants(
    VkCommandBuffer                             commandBuffer,
    VkPipelineLayout                            layout,
    VkShaderStageFlags                          stageFlags,
    uint32_t                                    offset,
    uint32_t                                    size,
    const void*                                 pValues);
```

- `commandBuffer` is the command buffer in which the push constant update will be recorded.
- `layout` is the pipeline layout used to program the push constant updates.
- `stageFlags` is a bitmask of VkShaderStageFlagBits specifying the shader stages that will use the push constants in the updated range.
- `offset` is the start offset of the push constant range to update, in units of bytes.
- `size` is the size of the push constant range to update, in units of bytes.
- `pValues` is a pointer to an array of `size` bytes containing the new push constant values.

When a command buffer begins recording, all push constant values are undefined. Reads of undefined push constant values by the executing shader return undefined values.

Push constant values **can** be updated incrementally, causing shader stages in `stageFlags` to read the new data from `pValues` for push constants modified by this command, while still reading the previous data for push constants not modified by this command. When a bound pipeline command is issued, the bound pipeline's layout **must** be compatible with the layouts used to set the values of all push constants in the pipeline layout's push constant ranges, as described in Pipeline Layout Compatibility. Binding a pipeline with a layout that is not compatible with the push constant layout does not disturb the push constant values.

NOTE As `stageFlags` needs to include all flags the relevant push constant ranges were created with, any flags that are not supported by the queue family that the VkCommandPool used to allocate `commandBuffer` was created on are ignored.

Valid Usage

- VUID-vkCmdPushConstants-commandBuffer-11295
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pSamplerHeapBindInfo` equal to `NULL`
- VUID-vkCmdPushConstants-commandBuffer-11296
  If `commandBuffer` is a secondary command buffer, it **must** have begun with

VkCommandBufferInheritanceDescriptorHeapInfoEXT::pResourceHeapBindInfo equal to NULL

- VUID-vkCmdPushConstants-offset-01795
  For each byte in the range specified by offset and size and for each shader stage in stageFlags, there **must** be a push constant range in layout that includes that byte and that stage
- VUID-vkCmdPushConstants-offset-01796
  For each byte in the range specified by offset and size and for each push constant range that overlaps that byte, stageFlags **must** include all stages in that push constant range's VkPushConstantRange::stageFlags
- VUID-vkCmdPushConstants-offset-00368
  offset **must** be a multiple of 4
- VUID-vkCmdPushConstants-size-00369
  size **must** be a multiple of 4
- VUID-vkCmdPushConstants-offset-00370
  offset **must** be less than VkPhysicalDeviceLimits::maxPushConstantsSize
- VUID-vkCmdPushConstants-size-00371
  size **must** be less than or equal to VkPhysicalDeviceLimits::maxPushConstantsSize minus offset

Valid Usage (Implicit)

- VUID-vkCmdPushConstants-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdPushConstants-layout-parameter
  layout **must** be a valid VkPipelineLayout handle
- VUID-vkCmdPushConstants-stageFlags-parameter
  stageFlags **must** be a valid combination of VkShaderStageFlagBits values
- VUID-vkCmdPushConstants-pValues-parameter
  pValues **must** be a valid pointer to an array of size bytes
- VUID-vkCmdPushConstants-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdPushConstants-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdPushConstants-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdPushConstants-size-arraylength
  size **must** be greater than 0
- VUID-vkCmdPushConstants-commonparent
  Both of commandBuffer, and layout **must** have been created, allocated, or retrieved from

the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdPushConstants is not affected by conditional rendering

To update push constants, call:

```
// Provided by VK_VERSION_1_4
void vkCmdPushConstants2(
    VkCommandBuffer                             commandBuffer,
    const VkPushConstantsInfo*                  pPushConstantsInfo);
```

```
// Provided by VK_KHR_maintenance6
// Equivalent to vkCmdPushConstants2
void vkCmdPushConstants2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkPushConstantsInfo*                  pPushConstantsInfo);
```

- `commandBuffer` is the command buffer in which the push constant update will be recorded.
- `pPushConstantsInfo` is a pointer to a VkPushConstantsInfo structure.

Valid Usage

- VUID-vkCmdPushConstants2-commandBuffer-11295

If commandBuffer is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::pSamplerHeapBindInfo equal to NULL

- VUID-vkCmdPushConstants2-commandBuffer-11296
  If commandBuffer is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::pResourceHeapBindInfo equal to NULL

Valid Usage (Implicit)

- VUID-vkCmdPushConstants2-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdPushConstants2-pPushConstantsInfo-parameter
  pPushConstantsInfo **must** be a valid pointer to a valid VkPushConstantsInfo structure
- VUID-vkCmdPushConstants2-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdPushConstants2-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdPushConstants2-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdPushConstants2 is not affected by conditional rendering

The VkPushConstantsInfo structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPushConstantsInfo {
    VkStructureType       sType;
    const void*           pNext;
    VkPipelineLayout      layout;
    VkShaderStageFlags    stageFlags;
    uint32_t              offset;
    uint32_t              size;
    const void*           pValues;
} VkPushConstantsInfo;
```

```
// Provided by VK_KHR_maintenance6
// Equivalent to VkPushConstantsInfo
typedef VkPushConstantsInfo VkPushConstantsInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- layout is the pipeline layout used to program the push constant updates.
- stageFlags is a bitmask of VkShaderStageFlagBits specifying the shader stages that will use the push constants in the updated range.
- offset is the start offset of the push constant range to update, in units of bytes.
- size is the size of the push constant range to update, in units of bytes.
- pValues is a pointer to an array of size bytes containing the new push constant values.

Valid Usage

- VUID-VkPushConstantsInfo-offset-01795
  For each byte in the range specified by offset and size and for each shader stage in stageFlags, there **must** be a push constant range in layout that includes that byte and that stage
- VUID-VkPushConstantsInfo-offset-01796
  For each byte in the range specified by offset and size and for each push constant range that overlaps that byte, stageFlags **must** include all stages in that push constant range's VkPushConstantRange::stageFlags
- VUID-VkPushConstantsInfo-offset-00368
  offset **must** be a multiple of 4

- VUID-VkPushConstantsInfo-size-00369
  `size` **must** be a multiple of `4`
- VUID-VkPushConstantsInfo-offset-00370
  `offset` **must** be less than `VkPhysicalDeviceLimits`::`maxPushConstantsSize`
- VUID-VkPushConstantsInfo-size-00371
  `size` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxPushConstantsSize` minus `offset`
- VUID-VkPushConstantsInfo-None-09495
  `layout` **must** be a valid VkPipelineLayout handle

Valid Usage (Implicit)

- VUID-VkPushConstantsInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PUSH_CONSTANTS_INFO
- VUID-VkPushConstantsInfo-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkPipelineLayoutCreateInfo
- VUID-VkPushConstantsInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkPushConstantsInfo-layout-parameter
  If `layout` is not VK_NULL_HANDLE, `layout` **must** be a valid VkPipelineLayout handle
- VUID-VkPushConstantsInfo-stageFlags-parameter
  `stageFlags` **must** be a valid combination of VkShaderStageFlagBits values
- VUID-VkPushConstantsInfo-pValues-parameter
  `pValues` **must** be a valid pointer to an array of `size` bytes
- VUID-VkPushConstantsInfo-size-arraylength
  `size` **must** be greater than `0`

Chapter 18. Descriptor Buffers

If the descriptorBuffer feature is enabled, an alternative way to specify descriptor sets is via buffers, rather than descriptor set objects.

18.1. Putting Descriptors in Memory

Commands are provided to retrieve descriptor data, and also to locate where in memory that data **must** be written to match the given descriptor set layout.

To determine the amount of memory needed to store all descriptors with a given layout, call:

WARNING	This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
void vkGetDescriptorSetLayoutSizeEXT(
    VkDevice                                    device,
    VkDescriptorSetLayout                       layout,
    VkDeviceSize*                               pLayoutSizeInBytes);
```

- device is the logical device that gets the size.
- layout is the descriptor set layout being queried.
- pLayoutSizeInBytes is a pointer to VkDeviceSize where the size in bytes will be written.

The size of a descriptor set layout will be at least as large as the sum total of the size of all descriptors in the layout, and **may** be larger. This size represents the amount of memory that will be required to store all of the descriptors for this layout in memory, when placed according to the layout's offsets as obtained by vkGetDescriptorSetLayoutBindingOffsetEXT.

If any binding in layout is VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT, the returned size includes space for the maximum descriptorCount descriptors as declared for that binding. To compute the required size of a descriptor set with a VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT:

$$size = offset + descriptorSize \times variableDescriptorCount$$

where offset is obtained by vkGetDescriptorSetLayoutBindingOffsetEXT and descriptorSize is the size of the relevant descriptor as obtained from VkPhysicalDeviceDescriptorBufferPropertiesEXT, and variableDescriptorCount is the equivalent of VkDescriptorSetVariableDescriptorCountAllocateInfo::pDescriptorCounts. For VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK, variableDescriptorCount is the size in bytes for the inline uniform block, and descriptorSize is 1.

If VkPhysicalDeviceDescriptorBufferPropertiesEXT::combinedImageSamplerDescriptorSingleArray is

VK_FALSE and the variable descriptor type is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, variableDescriptorCount is always considered to be the upper bound.

Valid Usage

- VUID-vkGetDescriptorSetLayoutSizeEXT-None-08011
  The `descriptorBuffer` feature **must** be enabled
- VUID-vkGetDescriptorSetLayoutSizeEXT-layout-08012
  `layout` **must** have been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT flag set
- VUID-vkGetDescriptorSetLayoutSizeEXT-layout-11811
  `layout` **must** have not been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT flag set
- VUID-vkGetDescriptorSetLayoutSizeEXT-layout-11812
  `layout` **must** have not been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_EMBEDDED_IMMUTABLE_SAMPLERS_BIT_EXT flag set

Valid Usage (Implicit)

- VUID-vkGetDescriptorSetLayoutSizeEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDescriptorSetLayoutSizeEXT-layout-parameter
  `layout` **must** be a valid VkDescriptorSetLayout handle
- VUID-vkGetDescriptorSetLayoutSizeEXT-pLayoutSizeInBytes-parameter
  `pLayoutSizeInBytes` **must** be a valid pointer to a VkDeviceSize value
- VUID-vkGetDescriptorSetLayoutSizeEXT-layout-parent
  `layout` **must** have been created, allocated, or retrieved from `device`

To get the offset of a binding within a descriptor set layout in memory, call:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
void vkGetDescriptorSetLayoutBindingOffsetEXT(
    VkDevice                                    device,
    VkDescriptorSetLayout                       layout,
    uint32_t                                    binding,
    VkDeviceSize*                               pOffset);
```

- `device` is the logical device that gets the offset.

- `layout` is the descriptor set layout being queried.
- `binding` is the binding number being queried.
- `pOffset` is a pointer to VkDeviceSize where the byte offset of the binding will be written.

Each binding in a descriptor set layout is assigned an offset in memory by the implementation. When a shader accesses a resource with that binding, it will access the bound descriptor buffer from that offset to look for its descriptor. This command provides an application with that offset, so that descriptors can be placed in the correct locations. The precise location accessed by a shader for a given descriptor is as follows:

$$location = bufferAddress + setOffset + descriptorOffset + (arrayElement \times descriptorSize)$$

where bufferAddress and setOffset are the base address and offset for the identified descriptor set as specified by vkCmdBindDescriptorBuffersEXT and vkCmdSetDescriptorBufferOffsetsEXT, descriptorOffset is the offset for the binding returned by this command, arrayElement is the index into the array specified in the shader, and descriptorSize is the size of the relevant descriptor as obtained from VkPhysicalDeviceDescriptorBufferPropertiesEXT. Applications are responsible for placing valid descriptors at the expected location in order for a shader to access it. The overall offset added to bufferAddress to calculate location **must** be less than VkPhysicalDeviceDescriptorBufferPropertiesEXT::`maxSamplerDescriptorBufferRange` for samplers and VkPhysicalDeviceDescriptorBufferPropertiesEXT::`maxResourceDescriptorBufferRange` for resources.

If any `binding` in `layout` is VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT, that `binding` **must** have the largest offset of any `binding`.

A descriptor `binding` with type VK_DESCRIPTOR_TYPE_MUTABLE_EXT **can** be used. Any potential types in VkMutableDescriptorTypeCreateInfoEXT::`pDescriptorTypes` for `binding` share the same offset. If the size of the mutable descriptor is larger than the size of a concrete descriptor type being accessed, the padding area is ignored by the implementation.

Valid Usage

- VUID-vkGetDescriptorSetLayoutBindingOffsetEXT-None-08013
  The `descriptorBuffer` feature **must** be enabled
- VUID-vkGetDescriptorSetLayoutBindingOffsetEXT-layout-08014
  `layout` **must** have been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT flag set
- VUID-vkGetDescriptorSetLayoutBindingOffsetEXT-layout-11813
  `layout` **must** have not been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT flag set
- VUID-vkGetDescriptorSetLayoutBindingOffsetEXT-layout-11814
  `layout` **must** have not been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_EMBEDDED_IMMUTABLE_SAMPLERS_BIT_EXT flag set

Valid Usage (Implicit)

- VUID-vkGetDescriptorSetLayoutBindingOffsetEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDescriptorSetLayoutBindingOffsetEXT-layout-parameter
  `layout` **must** be a valid VkDescriptorSetLayout handle
- VUID-vkGetDescriptorSetLayoutBindingOffsetEXT-pOffset-parameter
  `pOffset` **must** be a valid pointer to a VkDeviceSize value
- VUID-vkGetDescriptorSetLayoutBindingOffsetEXT-layout-parent
  `layout` **must** have been created, allocated, or retrieved from `device`

To get descriptor data to place in a buffer, call:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
void vkGetDescriptorEXT(
    VkDevice                                    device,
    const VkDescriptorGetInfoEXT*               pDescriptorInfo,
    size_t                                      dataSize,
    void*                                       pDescriptor);
```

- `device` is the logical device that gets the descriptor.
- `pDescriptorInfo` is a pointer to a VkDescriptorGetInfoEXT structure specifying the parameters of the descriptor to get.
- `dataSize` is the amount of the descriptor data to get in bytes.
- `pDescriptor` is a pointer to an application-allocated buffer where the descriptor will be written.

The size of the data for each descriptor type is determined by the value in VkPhysicalDeviceDescriptorBufferPropertiesEXT. This value also defines the stride in bytes for arrays of that descriptor type.

If the VkPhysicalDeviceDescriptorBufferPropertiesEXT::`combinedImageSamplerDescriptorSingleArray` property is VK_FALSE the implementation requires an array of VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER descriptors to be written into a descriptor buffer as an array of image descriptors, immediately followed by an array of sampler descriptors. Applications **must** write the first VkPhysicalDeviceDescriptorBufferPropertiesEXT::`sampledImageDescriptorSize` bytes of the data returned through `pDescriptor` to the first array, and the remaining VkPhysicalDeviceDescriptorBufferPropertiesEXT::`samplerDescriptorSize` bytes of the data to the second array. For variable-sized descriptor bindings of VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER descriptors, the two arrays each have a size equal to the upper bound `descriptorCount` of that binding.

A descriptor obtained by this command references the underlying VkImageView or VkSampler, and these objects **must** not be destroyed before the last time a descriptor is dynamically accessed. For descriptor types which consume an address instead of an object, the underlying VkBuffer is referenced instead.

Valid Usage

- VUID-vkGetDescriptorEXT-None-08015
  The `descriptorBuffer` feature **must** be enabled
- VUID-vkGetDescriptorEXT-dataSize-08125
  If `pDescriptorInfo->type` is not VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER or `pDescriptorInfo->data.pCombinedImageSampler` has an `imageView` member that was not created with a `VkSamplerYcbcrConversionInfo` structure in its `pNext` chain, `dataSize` **must** equal the size of a descriptor of type VkDescriptorGetInfoEXT::`type` determined by the value in VkPhysicalDeviceDescriptorBufferPropertiesEXT , or determined by VkPhysicalDeviceDescriptorBufferDensityMapPropertiesEXT::`combinedImageSamplerDensityMapDescriptorSize` if `pDescriptorInfo` specifies a VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER whose VkSampler was created with VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT set
- VUID-vkGetDescriptorEXT-descriptorType-09469
  If `pDescriptorInfo->type` is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER and `pDescriptorInfo->data.pCombinedImageSampler` has an `imageView` member that was created with a `VkSamplerYcbcrConversionInfo` structure in its `pNext` chain, `dataSize` **must** equal the size of VkPhysicalDeviceDescriptorBufferPropertiesEXT::`combinedImageSamplerDescriptorSize` times VkSamplerYcbcrConversionImageFormatProperties::`combinedImageSamplerDescriptorCount`
- VUID-vkGetDescriptorEXT-pDescriptorInfo-09507
  If `pDescriptorInfo->type` is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER and it has a `imageView` that is VK_NULL_HANDLE then `dataSize` **must** be equal to the size of VkPhysicalDeviceDescriptorBufferPropertiesEXT::`combinedImageSamplerDescriptorSize`

Valid Usage (Implicit)

- VUID-vkGetDescriptorEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDescriptorEXT-pDescriptorInfo-parameter
  `pDescriptorInfo` **must** be a valid pointer to a valid VkDescriptorGetInfoEXT structure
- VUID-vkGetDescriptorEXT-pDescriptor-parameter
  `pDescriptor` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-vkGetDescriptorEXT-dataSize-arraylength
  `dataSize` **must** be greater than `0`

Information about the descriptor to get is passed in a `VkDescriptorGetInfoEXT` structure:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
typedef struct VkDescriptorGetInfoEXT {
    VkStructureType        sType;
    const void*            pNext;
    VkDescriptorType       type;
    VkDescriptorDataEXT    data;
} VkDescriptorGetInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `type` is the type of descriptor to get.
- `data` is a VkDescriptorDataEXT union containing the information needed to get the descriptor.

Valid Usage

- VUID-VkDescriptorGetInfoEXT-type-08018
  `type` **must** not be VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC, VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC or VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK
- VUID-VkDescriptorGetInfoEXT-type-08019
  If `type` is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, the `pCombinedImageSampler->sampler` member of `data` **must** be a VkSampler created on `device`
- VUID-VkDescriptorGetInfoEXT-type-08020
  If `type` is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, the `pCombinedImageSampler->imageView` member of `data` **must** be a VkImageView created on `device`, or VK_NULL_HANDLE
- VUID-VkDescriptorGetInfoEXT-type-08021
  If `type` is VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, the `pInputAttachmentImage->imageView` member of `data` **must** be a VkImageView created on `device`
- VUID-VkDescriptorGetInfoEXT-type-08022
  If `type` is VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, and if `pSampledImage` is not `NULL`, the `pSampledImage->imageView` member of `data` **must** be a VkImageView created on `device`, or VK_NULL_HANDLE
- VUID-VkDescriptorGetInfoEXT-type-08023
  If `type` is VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, and if `pStorageImage` is not `NULL`, the `pStorageImage->imageView` member of `data` **must** be a VkImageView created on `device`, or VK_NULL_HANDLE
- VUID-VkDescriptorGetInfoEXT-type-09427
  If `type` is VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER, `pUniformBuffer` is not `NULL`, the number of texel buffer elements given by (⌊`pUniformBuffer->range` / (texel block size)⌋ ×

(texels per block)) where texel block size and texels per block are as defined in the Compatible Formats table for pUniformBuffer->format, **must** be less than or equal to VkPhysicalDeviceLimits::maxTexelBufferElements

- VUID-VkDescriptorGetInfoEXT-type-09428
  If type is VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, pStorageBuffer is not NULL, the number of texel buffer elements given by (⌊pStorageBuffer->range / (texel block size)⌋ × (texels per block)) where texel block size and texels per block are as defined in the Compatible Formats table for pStorageBuffer->format, **must** be less than or equal to VkPhysicalDeviceLimits::maxTexelBufferElements
- VUID-VkDescriptorGetInfoEXT-type-08028
  If type is VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR and accelerationStructure is not 0, accelerationStructure **must** contain the address of a VkAccelerationStructureKHR created on device
- VUID-VkDescriptorGetInfoEXT-type-12216
  If type is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER and pCombinedImageSampler->imageView is not VK_NULL_HANDLE, the pCombinedImageSampler->imageView member of data **must** have been created with VK_IMAGE_USAGE_SAMPLED_BIT set
- VUID-VkDescriptorGetInfoEXT-type-12217
  If type is VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, pSampledImage is not NULL, and pSampledImage->imageView is not VK_NULL_HANDLE, the pSampledImage->imageView member of data **must** have been created with VK_IMAGE_USAGE_SAMPLED_BIT set
- VUID-VkDescriptorGetInfoEXT-type-12218
  If type is VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, pStorageImage is not NULL, and pStorageImage->imageView is not VK_NULL_HANDLE, the pStorageImage->imageView member of data **must** have been created with VK_IMAGE_USAGE_STORAGE_BIT set
- VUID-VkDescriptorGetInfoEXT-type-12219
  If type is VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, the pInputAttachmentImage->imageView member of data **must** have been created with VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT set
- VUID-VkDescriptorGetInfoEXT-type-12220
  If type is VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, pUniformBuffer is not NULL and pUniformBuffer->address is not zero, pUniformBuffer->address **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-VkDescriptorGetInfoEXT-type-12221
  If type is VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, pStorageBuffer is not NULL and pStorageBuffer->address is not zero, pStorageBuffer->address **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-VkDescriptorGetInfoEXT-type-12222
  If type is VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER, pUniformTexelBuffer is not NULL and pUniformTexelBuffer->address is not zero, pUniformTexelBuffer->address **must** be a device address allocated to the application from a buffer created with the

VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT usage flag set

- VUID-VkDescriptorGetInfoEXT-type-12223
  If `type` is VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, `pStorageTexelBuffer` is not `NULL` and `pStorageTexelBuffer->address` is not zero, `pStorageTexelBuffer->address` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT usage flag set
- VUID-VkDescriptorGetInfoEXT-type-12265
  If `type` is VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER and `pUniformBuffer` is not `NULL`, `pUniformBuffer->address` **must** be aligned to `VkPhysicalDeviceLimits::minUniformBufferOffsetAlignment`
- VUID-VkDescriptorGetInfoEXT-type-12266
  If `type` is VK_DESCRIPTOR_TYPE_STORAGE_BUFFER and `pStorageBuffer` is not `NULL`, `pStorageBuffer->address` **must** be aligned to `VkPhysicalDeviceLimits::minStorageBufferOffsetAlignment`
- VUID-VkDescriptorGetInfoEXT-type-12269
  If `type` is VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER, and `pUniformTexelBuffer` is not `NULL`, `pUniformTexelBuffer->address` **must** be a multiple of the effective alignment requirement as determined by `minTexelBufferOffsetAlignment`
- VUID-VkDescriptorGetInfoEXT-type-12270
  If `type` is VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, and `pStorageTexelBuffer` is not `NULL`, `pStorageTexelBuffer->address` **must** be a multiple of the effective alignment requirement as determined by `minTexelBufferOffsetAlignment`
- VUID-VkDescriptorGetInfoEXT-type-12492
  If `type` is VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER and `pUniformBuffer->address` is not 0, `pUniformBuffer->range` **must** be less than or equal to `maxUniformBufferRange`
- VUID-VkDescriptorGetInfoEXT-type-12493
  If `type` is VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, the `shader64BitIndexing` feature is not enabled, and `pStorageBuffer->address` is not 0, `pStorageBuffer->range` **must** be less than or equal to `maxStorageBufferRange`

Valid Usage (Implicit)

- VUID-VkDescriptorGetInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DESCRIPTOR_GET_INFO_EXT
- VUID-VkDescriptorGetInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDescriptorGetInfoEXT-type-parameter
  `type` **must** be a valid VkDescriptorType value
- VUID-VkDescriptorGetInfoEXT-pSampler-parameter
  If `type` is VK_DESCRIPTOR_TYPE_SAMPLER, the `pSampler` member of `data` **must** be a valid pointer to a valid VkSampler handle
- VUID-VkDescriptorGetInfoEXT-pCombinedImageSampler-parameter

If `type` is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, the `pCombinedImageSampler` member of `data` **must** be a valid pointer to a valid VkDescriptorImageInfo structure

- VUID-VkDescriptorGetInfoEXT-pInputAttachmentImage-parameter
  If `type` is VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, the `pInputAttachmentImage` member of `data` **must** be a valid pointer to a valid VkDescriptorImageInfo structure
- VUID-VkDescriptorGetInfoEXT-pSampledImage-parameter
  If `type` is VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, and if `pSampledImage` is not `NULL`, the `pSampledImage` member of `data` **must** be a valid pointer to a valid VkDescriptorImageInfo structure
- VUID-VkDescriptorGetInfoEXT-pStorageImage-parameter
  If `type` is VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, and if `pStorageImage` is not `NULL`, the `pStorageImage` member of `data` **must** be a valid pointer to a valid VkDescriptorImageInfo structure
- VUID-VkDescriptorGetInfoEXT-pUniformTexelBuffer-parameter
  If `type` is VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER, and if `pUniformTexelBuffer` is not `NULL`, the `pUniformTexelBuffer` member of `data` **must** be a valid pointer to a valid VkDescriptorAddressInfoEXT structure
- VUID-VkDescriptorGetInfoEXT-pStorageTexelBuffer-parameter
  If `type` is VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, and if `pStorageTexelBuffer` is not `NULL`, the `pStorageTexelBuffer` member of `data` **must** be a valid pointer to a valid VkDescriptorAddressInfoEXT structure
- VUID-VkDescriptorGetInfoEXT-pUniformBuffer-parameter
  If `type` is VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, and if `pUniformBuffer` is not `NULL`, the `pUniformBuffer` member of `data` **must** be a valid pointer to a valid VkDescriptorAddressInfoEXT structure
- VUID-VkDescriptorGetInfoEXT-pStorageBuffer-parameter
  If `type` is VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, and if `pStorageBuffer` is not `NULL`, the `pStorageBuffer` member of `data` **must** be a valid pointer to a valid VkDescriptorAddressInfoEXT structure
- VUID-VkDescriptorGetInfoEXT-accelerationStructure-parameter
  If `type` is VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR or VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_NV, the `accelerationStructure` member of `data` **must** be a valid VkDeviceAddress value

Data describing the descriptor is passed in a `VkDescriptorDataEXT` structure:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
typedef union VkDescriptorDataEXT {
    const VkSampler*                    pSampler;
    const VkDescriptorImageInfo*        pCombinedImageSampler;
    const VkDescriptorImageInfo*        pInputAttachmentImage;
    const VkDescriptorImageInfo*        pSampledImage;
    const VkDescriptorImageInfo*        pStorageImage;
    const VkDescriptorAddressInfoEXT*   pUniformTexelBuffer;
    const VkDescriptorAddressInfoEXT*   pStorageTexelBuffer;
    const VkDescriptorAddressInfoEXT*   pUniformBuffer;
    const VkDescriptorAddressInfoEXT*   pStorageBuffer;
    VkDeviceAddress                     accelerationStructure;
} VkDescriptorDataEXT;
```

- `pSampler` is a pointer to a VkSampler handle specifying the parameters of a VK_DESCRIPTOR_TYPE_SAMPLER descriptor.
- `pCombinedImageSampler` is a pointer to a VkDescriptorImageInfo structure specifying the parameters of a VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER descriptor.
- `pInputAttachmentImage` is a pointer to a VkDescriptorImageInfo structure specifying the parameters of a VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT descriptor.
- `pSampledImage` is a pointer to a VkDescriptorImageInfo structure specifying the parameters of a VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE descriptor.
- `pStorageImage` is a pointer to a VkDescriptorImageInfo structure specifying the parameters of a VK_DESCRIPTOR_TYPE_STORAGE_IMAGE descriptor.
- `pUniformTexelBuffer` is a pointer to a VkDescriptorAddressInfoEXT structure specifying the parameters of a VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER descriptor.
- `pStorageTexelBuffer` is a pointer to a VkDescriptorAddressInfoEXT structure specifying the parameters of a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor.
- `pUniformBuffer` is a pointer to a VkDescriptorAddressInfoEXT structure specifying the parameters of a VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER descriptor.
- `pStorageBuffer` is a pointer to a VkDescriptorAddressInfoEXT structure specifying the parameters of a VK_DESCRIPTOR_TYPE_STORAGE_BUFFER descriptor.
- `accelerationStructure` is reserved for future use and is ignored.

If the `nullDescriptor` feature is enabled, `pSampledImage`, `pStorageImage`, `pUniformTexelBuffer`, `pStorageTexelBuffer`, `pUniformBuffer`, and `pStorageBuffer` **can** each be `NULL`. Loads from a null descriptor return zero values and stores and atomics to a null descriptor are discarded.

If the `nullDescriptor` feature is enabled, `accelerationStructure` **can** be `0`. A null acceleration structure descriptor results in the miss shader being invoked.

Valid Usage

- VUID-VkDescriptorDataEXT-type-08034

If VkDescriptorGetInfoEXT::type is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, and the nullDescriptor feature is not enabled, pCombinedImageSampler->imageView **must** not be VK_NULL_HANDLE

- VUID-VkDescriptorDataEXT-type-08035
  If VkDescriptorGetInfoEXT::type is VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, and the nullDescriptor feature is not enabled, pSampledImage **must** not be NULL and pSampledImage->imageView **must** not be VK_NULL_HANDLE
- VUID-VkDescriptorDataEXT-type-08036
  If VkDescriptorGetInfoEXT::type is VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, and the nullDescriptor feature is not enabled, pStorageImage **must** not be NULL and pStorageImage->imageView **must** not be VK_NULL_HANDLE
- VUID-VkDescriptorDataEXT-type-08037
  If VkDescriptorGetInfoEXT::type is VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER, and the nullDescriptor feature is not enabled, pUniformTexelBuffer **must** not be NULL
- VUID-VkDescriptorDataEXT-type-08038
  If VkDescriptorGetInfoEXT::type is VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, and the nullDescriptor feature is not enabled, pStorageTexelBuffer **must** not be NULL
- VUID-VkDescriptorDataEXT-type-08039
  If VkDescriptorGetInfoEXT::type is VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, and the nullDescriptor feature is not enabled, pUniformBuffer **must** not be NULL
- VUID-VkDescriptorDataEXT-type-08040
  If VkDescriptorGetInfoEXT::type is VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, and the nullDescriptor feature is not enabled, pStorageBuffer **must** not be NULL
- VUID-VkDescriptorDataEXT-type-08041
  If VkDescriptorGetInfoEXT::type is VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR, and the nullDescriptor feature is not enabled, accelerationStructure **must** not be 0

Data describing a VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER, or VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is passed in a VkDescriptorAddressInfoEXT structure:

WARNING	This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
typedef struct VkDescriptorAddressInfoEXT {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceAddress    address;
    VkDeviceSize       range;
    VkFormat           format;
} VkDescriptorAddressInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `address` is either `0` or a device address at an offset in a buffer, where the base address can be queried from vkGetBufferDeviceAddress.
- `range` is the size in bytes of the buffer or buffer view used by the descriptor.
- `format` is the format of the data elements in the buffer view and is ignored for buffers.

Valid Usage

- VUID-VkDescriptorAddressInfoEXT-None-09508
  If `address` is not zero, and the descriptor is of type VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, then `format` **must** not be VK_FORMAT_UNDEFINED
- VUID-VkDescriptorAddressInfoEXT-address-08043
  If the `nullDescriptor` feature is not enabled, `address` **must** not be zero
- VUID-VkDescriptorAddressInfoEXT-nullDescriptor-08938
  If `address` is zero, `range` **must** be VK_WHOLE_SIZE
- VUID-VkDescriptorAddressInfoEXT-nullDescriptor-08939
  If `address` is not zero, `range` **must** not be VK_WHOLE_SIZE
- VUID-VkDescriptorAddressInfoEXT-range-08045
  If `address` is not zero, then `range` **must** be less than or equal to the size of the buffer containing `address` minus the offset of `address` from the base address of the buffer
- VUID-VkDescriptorAddressInfoEXT-range-08940
  `range` **must** not be zero

Valid Usage (Implicit)

- VUID-VkDescriptorAddressInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DESCRIPTOR_ADDRESS_INFO_EXT
- VUID-VkDescriptorAddressInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`

- VUID-VkDescriptorAddressInfoEXT-address-parameter
  If `address` is not `0`, `address` **must** be a valid VkDeviceAddress value
- VUID-VkDescriptorAddressInfoEXT-format-parameter
  `format` **must** be a valid VkFormat value

If the `nullDescriptor` feature is enabled, `address` **can** be zero. Loads from a null descriptor return zero values and stores and atomics to a null descriptor are discarded.

Immutable samplers specified in a descriptor set layout through `pImmutableSamplers` **must** be provided by applications when obtaining descriptor data. Immutable samplers written in a descriptor buffer **must** have identical parameters to the immutable samplers in the descriptor set layout that consumes the sampler.

NOTE If the descriptor set layout was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_EMBEDDED_IMMUTABLE_SAMPLERS_BIT_EXT, there is no buffer backing for the immutable sampler, so this requirement does not exist. The implementation handles allocation of these descriptors internally.

NOTE As descriptors are now in regular memory, drivers cannot hide copies of immutable samplers that end up in descriptor sets from the application. As such, applications are required to provide these samplers as if they were not provided immutably.

18.2. Binding Descriptor Buffers

Descriptor buffers have their own separate binding point on the command buffer, with buffers bound using vkCmdBindDescriptorBuffersEXT. vkCmdSetDescriptorBufferOffsetsEXT assigns pairs of buffer binding indices and buffer offsets to the same binding point on the command buffer as vkCmdBindDescriptorSets, allowing subsequent bound pipeline commands to use the specified descriptor buffers. Bindings applied via vkCmdBindDescriptorSets **cannot** exist simultaneously with those applied via calls to vkCmdSetDescriptorBufferOffsetsEXT or vkCmdBindDescriptorBufferEmbeddedSamplersEXT, as calls to vkCmdSetDescriptorBufferOffsetsEXT or vkCmdBindDescriptorBufferEmbeddedSamplersEXT invalidate any bindings by previous calls to vkCmdBindDescriptorSets and vice-versa.

To bind descriptor buffers to a command buffer, call:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
void vkCmdBindDescriptorBuffersEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    bufferCount,
    const VkDescriptorBufferBindingInfoEXT*     pBindingInfos);
```

- `commandBuffer` is the command buffer that the descriptor buffers will be bound to.
- `bufferCount` is the number of elements in the `pBindingInfos` array.
- `pBindingInfos` is a pointer to an array of VkDescriptorBufferBindingInfoEXT structures.

`vkCmdBindDescriptorBuffersEXT` causes any offsets previously set by vkCmdSetDescriptorBufferOffsetsEXT that use the bindings numbered [`0`.. `bufferCount`-1] to be no longer valid for subsequent bound pipeline commands. Any previously bound buffers at binding points greater than or equal to `bufferCount` are unbound.

Valid Usage

- VUID-vkCmdBindDescriptorBuffersEXT-commandBuffer-11295
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pSamplerHeapBindInfo` equal to `NULL`
- VUID-vkCmdBindDescriptorBuffersEXT-commandBuffer-11296
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pResourceHeapBindInfo` equal to `NULL`
- VUID-vkCmdBindDescriptorBuffersEXT-None-08047
  The `descriptorBuffer` feature **must** be enabled
- VUID-vkCmdBindDescriptorBuffersEXT-maxSamplerDescriptorBufferBindings-08048
  There **must** be no more than VkPhysicalDeviceDescriptorBufferPropertiesEXT::`maxSamplerDescriptorBufferBindings` elements in `pBindingInfos` with VkDescriptorBufferBindingInfoEXT::`usage` containing VK_BUFFER_USAGE_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdBindDescriptorBuffersEXT-maxResourceDescriptorBufferBindings-08049
  There **must** be no more than VkPhysicalDeviceDescriptorBufferPropertiesEXT::`maxResourceDescriptorBufferBindings` elements in `pBindingInfos` with VkDescriptorBufferBindingInfoEXT::`usage` containing VK_BUFFER_USAGE_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdBindDescriptorBuffersEXT-None-08050
  There **must** be no more than `1` element in `pBindingInfos` with VkDescriptorBufferBindingInfoEXT::`usage` containing VK_BUFFER_USAGE_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdBindDescriptorBuffersEXT-bufferCount-08051
  `bufferCount` **must** be less than or equal to VkPhysicalDeviceDescriptorBufferPropertiesEXT::`maxDescriptorBufferBindings`
- VUID-vkCmdBindDescriptorBuffersEXT-pBindingInfos-08053
  For each element of `pBindingInfos`, the buffer from which `address` was queried **must** have been created with the VK_BUFFER_USAGE_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT usage flag set if it contains sampler descriptor data
- VUID-vkCmdBindDescriptorBuffersEXT-pBindingInfos-08054

For each element of pBindingInfos, the buffer from which address was queried **must** have been created with the VK_BUFFER_USAGE_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT usage flag set if it contains resource descriptor data

- VUID-vkCmdBindDescriptorBuffersEXT-pBindingInfos-08055
  For each element of pBindingInfos, at least one buffer from which address was queried must contain usage

Valid Usage (Implicit)

- VUID-vkCmdBindDescriptorBuffersEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindDescriptorBuffersEXT-pBindingInfos-parameter
  pBindingInfos **must** be a valid pointer to an array of bufferCount valid VkDescriptorBufferBindingInfoEXT structures
- VUID-vkCmdBindDescriptorBuffersEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdBindDescriptorBuffersEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindDescriptorBuffersEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindDescriptorBuffersEXT-bufferCount-arraylength
  bufferCount **must** be greater than 0

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindDescriptorBuffersEXT is not affected by conditional rendering

Data describing a descriptor buffer binding is passed in a VkDescriptorBufferBindingInfoEXT structure:

WARNING	This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
typedef struct VkDescriptorBufferBindingInfoEXT {
    VkStructureType       sType;
    const void*           pNext;
    VkDeviceAddress       address;
    VkBufferUsageFlags    usage;
} VkDescriptorBufferBindingInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- address is a VkDeviceAddress specifying the device address defining the descriptor buffer to be bound.
- usage is a bitmask of VkBufferUsageFlagBits specifying the VkBufferCreateInfo::usage for the buffer from which address was queried. Usage flags other than VK_BUFFER_USAGE_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT, VK_BUFFER_USAGE_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT, and VK_BUFFER_USAGE_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT are ignored.

If the pNext chain includes a VkBufferUsageFlags2CreateInfo structure, VkBufferUsageFlags2CreateInfo::usage from that structure is used instead of usage from this structure.

Valid Usage

- VUID-VkDescriptorBufferBindingInfoEXT-None-09499
  If the pNext chain does not include a VkBufferUsageFlags2CreateInfo structure, usage **must** be a valid combination of VkBufferUsageFlagBits values
- VUID-VkDescriptorBufferBindingInfoEXT-None-09500
  If the pNext chain does not include a VkBufferUsageFlags2CreateInfo structure, usage **must** not be 0
- VUID-VkDescriptorBufferBindingInfoEXT-usage-10998
  The usage must include at least one of VK_BUFFER_USAGE_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT, VK_BUFFER_USAGE_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT, or

VK_BUFFER_USAGE_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT

- VUID-VkDescriptorBufferBindingInfoEXT-bufferlessPushDescriptors-08056
  If VkPhysicalDeviceDescriptorBufferPropertiesEXT::bufferlessPushDescriptors is VK_FALSE, and usage contains VK_BUFFER_USAGE_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT, then the pNext chain **must** include a VkDescriptorBufferBindingPushDescriptorBufferHandleEXT structure
- VUID-VkDescriptorBufferBindingInfoEXT-address-08057
  address **must** be aligned to VkPhysicalDeviceDescriptorBufferPropertiesEXT::descriptorBufferOffsetAlignment
- VUID-VkDescriptorBufferBindingInfoEXT-usage-08122
  If usage includes VK_BUFFER_USAGE_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT, address **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT usage flag set
- VUID-VkDescriptorBufferBindingInfoEXT-usage-08123
  If usage includes VK_BUFFER_USAGE_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT, address **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT usage flag set
- VUID-VkDescriptorBufferBindingInfoEXT-usage-08124
  If usage includes VK_BUFFER_USAGE_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT, address **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT usage flag set

Valid Usage (Implicit)

- VUID-VkDescriptorBufferBindingInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DESCRIPTOR_BUFFER_BINDING_INFO_EXT
- VUID-VkDescriptorBufferBindingInfoEXT-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkBufferUsageFlags2CreateInfo or VkDescriptorBufferBindingPushDescriptorBufferHandleEXT
- VUID-VkDescriptorBufferBindingInfoEXT-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkDescriptorBufferBindingInfoEXT-address-parameter
  address **must** be a valid VkDeviceAddress value

When the VkPhysicalDeviceDescriptorBufferPropertiesEXT::bufferlessPushDescriptors property is VK_FALSE, the VkBuffer handle of the buffer for push descriptors is passed in a VkDescriptorBufferBindingPushDescriptorBufferHandleEXT structure:

WARNING	This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
typedef struct VkDescriptorBufferBindingPushDescriptorBufferHandleEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkBuffer           buffer;
} VkDescriptorBufferBindingPushDescriptorBufferHandleEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `buffer` is the `VkBuffer` handle of the buffer for push descriptors.

Valid Usage

- VUID-VkDescriptorBufferBindingPushDescriptorBufferHandleEXT-bufferlessPushDescriptors-08059 `VkPhysicalDeviceDescriptorBufferPropertiesEXT::bufferlessPushDescriptors` **must** be VK_FALSE

Valid Usage (Implicit)

- VUID-VkDescriptorBufferBindingPushDescriptorBufferHandleEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_DESCRIPTOR_BUFFER_BINDING_PUSH_DESCRIPTOR_BUFFER_HANDLE_EXT
- VUID-VkDescriptorBufferBindingPushDescriptorBufferHandleEXT-buffer-parameter `buffer` **must** be a valid VkBuffer handle

Structure Chaining

Extends the structure

- VkDescriptorBufferBindingInfoEXT

To set descriptor buffer offsets in a command buffer, call:

WARNING	This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
void vkCmdSetDescriptorBufferOffsetsEXT(
    VkCommandBuffer                             commandBuffer,
    VkPipelineBindPoint                         pipelineBindPoint,
    VkPipelineLayout                            layout,
    uint32_t                                    firstSet,
    uint32_t                                    setCount,
    const uint32_t*                             pBufferIndices,
    const VkDeviceSize*                         pOffsets);
```

- `commandBuffer` is the command buffer in which the descriptor buffer offsets will be set.
- `pipelineBindPoint` is a VkPipelineBindPoint indicating the type of the pipeline that will use the descriptors.
- `layout` is a VkPipelineLayout object used to program the bindings.
- `firstSet` is the number of the first set to be bound.
- `setCount` is the number of elements in the `pBufferIndices` and `pOffsets` arrays.
- `pBufferIndices` is a pointer to an array of indices into the descriptor buffer binding points set by vkCmdBindDescriptorBuffersEXT.
- `pOffsets` is a pointer to an array of VkDeviceSize offsets to apply to the bound descriptor buffers.

`vkCmdSetDescriptorBufferOffsetsEXT` binds `setCount` pairs of descriptor buffers, specified by indices into the binding points bound using vkCmdBindDescriptorBuffersEXT, and buffer offsets to set numbers [`firstSet`..`firstSet`+`setCount`-1] for subsequent bound pipeline commands set by `pipelineBindPoint`. Set [`firstSet` + i] is bound to the descriptor buffer at binding `pBufferIndices`[i] at an offset of `pOffsets`[i]. Any bindings that were previously applied via these sets, or calls to vkCmdBindDescriptorSets, are no longer valid. Other sets will also be invalidated upon calling this command if `layout` differs from the pipeline layout used to bind those other sets, as described in Pipeline Layout Compatibility.

After binding descriptors, applications **can** modify descriptor memory either by performing writes on the host or with device commands. When descriptor memory is updated with device commands, visibility for the shader stage accessing a descriptor is ensured with the VK_ACCESS_2_DESCRIPTOR_BUFFER_READ_BIT_EXT access flag. Implementations **must** not access resources referenced by these descriptors unless they are dynamically accessed by shaders. Descriptors bound with this call **can** be undefined if they are not dynamically accessed by shaders.

Implementations **may** read descriptor data for any statically accessed descriptor if the `binding` in `layout` is not declared with the VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT flag. If the `binding` in `layout` is declared with VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT, implementations **must** not read descriptor data that is not dynamically accessed.

Applications **must** ensure that any descriptor which the implementation **may** read **must** be in-bounds of the underlying descriptor buffer binding.

NOTE Applications can freely decide how large a variable descriptor buffer binding is, so it may not be safe to read such descriptor payloads statically. The intention of these rules is to allow implementations to speculatively prefetch descriptor payloads where feasible.

Dynamically accessing a resource through descriptor data from an unbound region of a sparse partially-resident buffer will result in invalid descriptor data being read, and therefore undefined behavior.

NOTE For descriptors written by the host, visibility is implied through the automatic visibility operation on queue submit, and there is no need to consider `VK_ACCESS_2_DESCRIPTOR_BUFFER_READ_BIT`. Explicit synchronization for descriptors is only required when descriptors are updated on the device.

NOTE The requirements above imply that all descriptor bindings have been defined with the equivalent of VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT, VK_DESCRIPTOR_BINDING_UPDATE_UNUSED_WHILE_PENDING_BIT and VK_DESCRIPTOR_BINDING_PARTIALLY_BOUND_BIT, but enabling those features is not required to get this behavior.

Valid Usage

- VUID-vkCmdSetDescriptorBufferOffsetsEXT-commandBuffer-11295
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pSamplerHeapBindInfo` equal to `NULL`
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-commandBuffer-11296
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pResourceHeapBindInfo` equal to `NULL`
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-pOffsets-08061
  The offsets in `pOffsets` **must** be aligned to VkPhysicalDeviceDescriptorBufferPropertiesEXT::`descriptorBufferOffsetAlignment`
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-pOffsets-08063
  The offsets in `pOffsets` **must** be small enough such that any descriptor binding referenced by `layout` without the VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT flag computes a valid address inside the underlying VkBuffer
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-pOffsets-08126
  The offsets in `pOffsets` **must** be small enough such that any location accessed by a shader as a sampler descriptor **must** be within VkPhysicalDeviceDescriptorBufferPropertiesEXT::`maxSamplerDescriptorBufferRange` of the sampler descriptor buffer binding
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-pOffsets-08127
  The offsets in `pOffsets` **must** be small enough such that any location accessed by a shader as a resource descriptor **must** be within

VkPhysicalDeviceDescriptorBufferPropertiesEXT::maxResourceDescriptorBufferRange of the resource descriptor buffer binding

- VUID-vkCmdSetDescriptorBufferOffsetsEXT-pBufferIndices-08064
  Each element of pBufferIndices **must** be less than VkPhysicalDeviceDescriptorBufferPropertiesEXT::maxDescriptorBufferBindings
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-pBufferIndices-08065
  Each element of pBufferIndices **must** reference a valid descriptor buffer binding set by a previous call to vkCmdBindDescriptorBuffersEXT in commandBuffer
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-firstSet-08066
  The sum of firstSet and setCount **must** be less than or equal to VkPipelineLayoutCreateInfo::setLayoutCount provided when layout was created
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-firstSet-09006
  The VkDescriptorSetLayout for each set from firstSet to firstSet + setCount when layout was created **must** have been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT bit set
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-firstSet-11803
  The VkDescriptorSetLayout for each set from firstSet to firstSet + setCount when layout was created **must** not have been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR bit set
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-firstSet-11804
  The VkDescriptorSetLayout for each set from firstSet to firstSet + setCount when layout was created **must** not have been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_EMBEDDED_IMMUTABLE_SAMPLERS_BIT_EXT bit set
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-None-08060
  The descriptorBuffer feature **must** be enabled
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-pipelineBindPoint-08067
  pipelineBindPoint **must** be supported by the commandBuffer's parent VkCommandPool's queue family

Valid Usage (Implicit)

- VUID-vkCmdSetDescriptorBufferOffsetsEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-pipelineBindPoint-parameter
  pipelineBindPoint **must** be a valid VkPipelineBindPoint value
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-layout-parameter
  layout **must** be a valid VkPipelineLayout handle
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-pBufferIndices-parameter
  pBufferIndices **must** be a valid pointer to an array of setCount uint32_t values
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-pOffsets-parameter
  pOffsets **must** be a valid pointer to an array of setCount VkDeviceSize values

- VUID-vkCmdSetDescriptorBufferOffsetsEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-setCount-arraylength
  `setCount` **must** be greater than `0`
- VUID-vkCmdSetDescriptorBufferOffsetsEXT-commonparent
  Both of `commandBuffer`, and `layout` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDescriptorBufferOffsetsEXT is not affected by conditional rendering

To set descriptor buffer offsets in a command buffer, call:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_KHR_maintenance6 with VK_EXT_descriptor_buffer
void vkCmdSetDescriptorBufferOffsets2EXT(
    VkCommandBuffer                             commandBuffer,
    const VkSetDescriptorBufferOffsetsInfoEXT*  pSetDescriptorBufferOffsetsInfo);
```

- `commandBuffer` is the command buffer in which the descriptor buffer offsets will be set.
- `pSetDescriptorBufferOffsetsInfo` is a pointer to a `VkSetDescriptorBufferOffsetsInfoEXT` structure.

Valid Usage

- VUID-vkCmdSetDescriptorBufferOffsets2EXT-commandBuffer-11295
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pSamplerHeapBindInfo` equal to `NULL`
- VUID-vkCmdSetDescriptorBufferOffsets2EXT-commandBuffer-11296
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pResourceHeapBindInfo` equal to `NULL`
- VUID-vkCmdSetDescriptorBufferOffsets2EXT-descriptorBuffer-09470
  The `descriptorBuffer` feature **must** be enabled
- VUID-vkCmdSetDescriptorBufferOffsets2EXT-pSetDescriptorBufferOffsetsInfo-09471
  Each bit in `pSetDescriptorBufferOffsetsInfo->stageFlags` **must** be a stage supported by the `commandBuffer`'s parent `VkCommandPool`'s queue family

Valid Usage (Implicit)

- VUID-vkCmdSetDescriptorBufferOffsets2EXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDescriptorBufferOffsets2EXT-pSetDescriptorBufferOffsetsInfo-parameter
  `pSetDescriptorBufferOffsetsInfo` **must** be a valid pointer to a valid VkSetDescriptorBufferOffsetsInfoEXT structure
- VUID-vkCmdSetDescriptorBufferOffsets2EXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetDescriptorBufferOffsets2EXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDescriptorBufferOffsets2EXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDescriptorBufferOffsets2EXT is not affected by conditional rendering

The `VkSetDescriptorBufferOffsetsInfoEXT` structure is defined as:

```
// Provided by VK_KHR_maintenance6 with VK_EXT_descriptor_buffer
typedef struct VkSetDescriptorBufferOffsetsInfoEXT {
    VkStructureType        sType;
    const void*            pNext;
    VkShaderStageFlags     stageFlags;
    VkPipelineLayout       layout;
    uint32_t               firstSet;
    uint32_t               setCount;
    const uint32_t*        pBufferIndices;
    const VkDeviceSize*    pOffsets;
} VkSetDescriptorBufferOffsetsInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stageFlags` is a bitmask of VkShaderStageFlagBits specifying the shader stages the descriptor sets will be bound to.
- `layout` is a VkPipelineLayout object used to program the bindings.
- `firstSet` is the number of the first set to be bound.
- `setCount` is the number of elements in the `pBufferIndices` and `pOffsets` arrays.
- `pBufferIndices` is a pointer to an array of indices into the descriptor buffer binding points set by

vkCmdBindDescriptorBuffersEXT.

- `pOffsets` is a pointer to an array of VkDeviceSize offsets to apply to the bound descriptor buffers.

If `stageFlags` specifies a subset of all stages corresponding to one or more pipeline bind points, the binding operation still affects all stages corresponding to the given pipeline bind point(s) as if the equivalent original version of this command had been called with the same parameters. For example, specifying a `stageFlags` value of VK_SHADER_STAGE_VERTEX_BIT | VK_SHADER_STAGE_FRAGMENT_BIT | VK_SHADER_STAGE_COMPUTE_BIT is equivalent to calling the original version of this command once with VK_PIPELINE_BIND_POINT_GRAPHICS and once with VK_PIPELINE_BIND_POINT_COMPUTE.

Valid Usage

- VUID-VkSetDescriptorBufferOffsetsInfoEXT-pOffsets-08061
  The offsets in `pOffsets` **must** be aligned to VkPhysicalDeviceDescriptorBufferPropertiesEXT::`descriptorBufferOffsetAlignment`
- VUID-VkSetDescriptorBufferOffsetsInfoEXT-pOffsets-08063
  The offsets in `pOffsets` **must** be small enough such that any descriptor binding referenced by `layout` without the VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT flag computes a valid address inside the underlying VkBuffer
- VUID-VkSetDescriptorBufferOffsetsInfoEXT-pOffsets-08126
  The offsets in `pOffsets` **must** be small enough such that any location accessed by a shader as a sampler descriptor **must** be within VkPhysicalDeviceDescriptorBufferPropertiesEXT::`maxSamplerDescriptorBufferRange` of the sampler descriptor buffer binding
- VUID-VkSetDescriptorBufferOffsetsInfoEXT-pOffsets-08127
  The offsets in `pOffsets` **must** be small enough such that any location accessed by a shader as a resource descriptor **must** be within VkPhysicalDeviceDescriptorBufferPropertiesEXT::`maxResourceDescriptorBufferRange` of the resource descriptor buffer binding
- VUID-VkSetDescriptorBufferOffsetsInfoEXT-pBufferIndices-08064
  Each element of `pBufferIndices` **must** be less than VkPhysicalDeviceDescriptorBufferPropertiesEXT::`maxDescriptorBufferBindings`
- VUID-VkSetDescriptorBufferOffsetsInfoEXT-pBufferIndices-08065
  Each element of `pBufferIndices` **must** reference a valid descriptor buffer binding set by a previous call to vkCmdBindDescriptorBuffersEXT in `commandBuffer`
- VUID-VkSetDescriptorBufferOffsetsInfoEXT-firstSet-08066
  The sum of `firstSet` and `setCount` **must** be less than or equal to VkPipelineLayoutCreateInfo::`setLayoutCount` provided when `layout` was created
- VUID-VkSetDescriptorBufferOffsetsInfoEXT-firstSet-09006
  The VkDescriptorSetLayout for each set from `firstSet` to `firstSet` + `setCount` when `layout` was created **must** have been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT bit set
- VUID-VkSetDescriptorBufferOffsetsInfoEXT-firstSet-11803

The VkDescriptorSetLayout for each set from `firstSet` to `firstSet` + `setCount` when `layout` was created **must** not have been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR bit set

- VUID-VkSetDescriptorBufferOffsetsInfoEXT-firstSet-11804
  The VkDescriptorSetLayout for each set from `firstSet` to `firstSet` + `setCount` when `layout` was created **must** not have been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_EMBEDDED_IMMUTABLE_SAMPLERS_BIT_EXT bit set

- VUID-VkSetDescriptorBufferOffsetsInfoEXT-None-09495
  `layout` **must** be a valid VkPipelineLayout handle

Valid Usage (Implicit)

- VUID-VkSetDescriptorBufferOffsetsInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SET_DESCRIPTOR_BUFFER_OFFSETS_INFO_EXT
- VUID-VkSetDescriptorBufferOffsetsInfoEXT-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkPipelineLayoutCreateInfo
- VUID-VkSetDescriptorBufferOffsetsInfoEXT-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkSetDescriptorBufferOffsetsInfoEXT-stageFlags-parameter
  `stageFlags` **must** be a valid combination of VkShaderStageFlagBits values
- VUID-VkSetDescriptorBufferOffsetsInfoEXT-layout-parameter
  If `layout` is not VK_NULL_HANDLE, `layout` **must** be a valid VkPipelineLayout handle
- VUID-VkSetDescriptorBufferOffsetsInfoEXT-pBufferIndices-parameter
  `pBufferIndices` **must** be a valid pointer to an array of `setCount` `uint32_t` values
- VUID-VkSetDescriptorBufferOffsetsInfoEXT-pOffsets-parameter
  `pOffsets` **must** be a valid pointer to an array of `setCount` VkDeviceSize values
- VUID-VkSetDescriptorBufferOffsetsInfoEXT-setCount-arraylength
  `setCount` **must** be greater than `0`

To bind an embedded immutable sampler set to a command buffer, call:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
void vkCmdBindDescriptorBufferEmbeddedSamplersEXT(
    VkCommandBuffer                             commandBuffer,
    VkPipelineBindPoint                         pipelineBindPoint,
    VkPipelineLayout                            layout,
    uint32_t                                    set);
```

- `commandBuffer` is the command buffer that the embedded immutable samplers will be bound to.
- `pipelineBindPoint` is a VkPipelineBindPoint indicating the type of the pipeline that will use the embedded immutable samplers.
- `layout` is a VkPipelineLayout object used to program the bindings.
- `set` is the number of the set to be bound.

`vkCmdBindDescriptorBufferEmbeddedSamplersEXT` binds the embedded immutable samplers in `set` of `layout` to `set` for the command buffer for subsequent bound pipeline commands set by `pipelineBindPoint`. Any previous binding to this set by vkCmdSetDescriptorBufferOffsetsEXT or this command is overwritten. Any sets that were last bound by a call to vkCmdBindDescriptorSets are invalidated upon calling this command. Other sets will also be invalidated upon calling this command if `layout` differs from the pipeline layout used to bind those other sets, as described in Pipeline Layout Compatibility.

Valid Usage

- VUID-vkCmdBindDescriptorBufferEmbeddedSamplersEXT-commandBuffer-11295
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pSamplerHeapBindInfo` equal to `NULL`
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplersEXT-commandBuffer-11296
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pResourceHeapBindInfo` equal to `NULL`
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplersEXT-set-08070
  The VkDescriptorSetLayout at index `set` when `layout` was created **must** have been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_EMBEDDED_IMMUTABLE_SAMPLERS_BIT_EXT bit set
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplersEXT-set-08071
  `set` **must** be less than VkPipelineLayoutCreateInfo::`setLayoutCount` provided when `layout` was created
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplersEXT-None-08068
  The `descriptorBuffer` feature **must** be enabled
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplersEXT-pipelineBindPoint-08069
  `pipelineBindPoint` **must** be supported by the `commandBuffer`'s parent `VkCommandPool`'s queue family

Valid Usage (Implicit)

- VUID-vkCmdBindDescriptorBufferEmbeddedSamplersEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplersEXT-pipelineBindPoint-parameter

`pipelineBindPoint` **must** be a valid VkPipelineBindPoint value

- VUID-vkCmdBindDescriptorBufferEmbeddedSamplersEXT-layout-parameter
  `layout` **must** be a valid VkPipelineLayout handle
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplersEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplersEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplersEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplersEXT-commonparent
  Both of `commandBuffer`, and `layout` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindDescriptorBufferEmbeddedSamplersEXT is not affected by conditional rendering

To bind an embedded immutable sampler set to a command buffer, call:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_KHR_maintenance6 with VK_EXT_descriptor_buffer
void vkCmdBindDescriptorBufferEmbeddedSamplers2EXT(
    VkCommandBuffer                             commandBuffer,
    const VkBindDescriptorBufferEmbeddedSamplersInfoEXT*
pBindDescriptorBufferEmbeddedSamplersInfo);
```

- `commandBuffer` is the command buffer that the embedded immutable samplers will be bound to.
- `pBindDescriptorBufferEmbeddedSamplersInfo` is a pointer to a `VkBindDescriptorBufferEmbeddedSamplersInfoEXT` structure.

Valid Usage

- VUID-vkCmdBindDescriptorBufferEmbeddedSamplers2EXT-commandBuffer-11295
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pSamplerHeapBindInfo` equal to `NULL`
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplers2EXT-commandBuffer-11296
  If `commandBuffer` is a secondary command buffer, it **must** have begun with VkCommandBufferInheritanceDescriptorHeapInfoEXT::`pResourceHeapBindInfo` equal to `NULL`
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplers2EXT-descriptorBuffer-09472
  The `descriptorBuffer` feature **must** be enabled
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplers2EXT-pBindDescriptorBufferEmbeddedSamplersInfo-09473
  Each bit in `pBindDescriptorBufferEmbeddedSamplersInfo->stageFlags` **must** be a stage supported by the `commandBuffer`'s parent `VkCommandPool`'s queue family

Valid Usage (Implicit)

- VUID-vkCmdBindDescriptorBufferEmbeddedSamplers2EXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplers2EXT-pBindDescriptorBufferEmbeddedSamplersInfo-parameter
  `pBindDescriptorBufferEmbeddedSamplersInfo` **must** be a valid pointer to a valid VkBindDescriptorBufferEmbeddedSamplersInfoEXT structure
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplers2EXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplers2EXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindDescriptorBufferEmbeddedSamplers2EXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindDescriptorBufferEmbeddedSamplers2EXT is not affected by conditional rendering

The `VkBindDescriptorBufferEmbeddedSamplersInfoEXT` structure is defined as:

```
// Provided by VK_KHR_maintenance6 with VK_EXT_descriptor_buffer
typedef struct VkBindDescriptorBufferEmbeddedSamplersInfoEXT {
    VkStructureType       sType;
    const void*           pNext;
    VkShaderStageFlags    stageFlags;
    VkPipelineLayout      layout;
    uint32_t              set;
} VkBindDescriptorBufferEmbeddedSamplersInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stageFlags` is a bitmask of VkShaderStageFlagBits specifying the shader stages that will use the embedded immutable samplers.
- `layout` is a VkPipelineLayout object used to program the bindings.
- `set` is the number of the set to be bound.

If `stageFlags` specifies a subset of all stages corresponding to one or more pipeline bind points, the binding operation still affects all stages corresponding to the given pipeline bind point(s) as if the equivalent original version of this command had been called with the same parameters. For example, specifying a `stageFlags` value of VK_SHADER_STAGE_VERTEX_BIT |

VK_SHADER_STAGE_FRAGMENT_BIT | VK_SHADER_STAGE_COMPUTE_BIT is equivalent to calling the original version of this command once with VK_PIPELINE_BIND_POINT_GRAPHICS and once with VK_PIPELINE_BIND_POINT_COMPUTE.

Valid Usage

- VUID-VkBindDescriptorBufferEmbeddedSamplersInfoEXT-set-08070
  The VkDescriptorSetLayout at index `set` when `layout` was created **must** have been created with the VK_DESCRIPTOR_SET_LAYOUT_CREATE_EMBEDDED_IMMUTABLE_SAMPLERS_BIT_EXT bit set
- VUID-VkBindDescriptorBufferEmbeddedSamplersInfoEXT-set-08071
  `set` **must** be less than VkPipelineLayoutCreateInfo::`setLayoutCount` provided when `layout` was created
- VUID-VkBindDescriptorBufferEmbeddedSamplersInfoEXT-None-09495
  `layout` **must** be a valid VkPipelineLayout handle

Valid Usage (Implicit)

- VUID-VkBindDescriptorBufferEmbeddedSamplersInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BIND_DESCRIPTOR_BUFFER_EMBEDDED_SAMPLERS_INFO_EXT
- VUID-VkBindDescriptorBufferEmbeddedSamplersInfoEXT-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkPipelineLayoutCreateInfo
- VUID-VkBindDescriptorBufferEmbeddedSamplersInfoEXT-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkBindDescriptorBufferEmbeddedSamplersInfoEXT-stageFlags-parameter
  `stageFlags` **must** be a valid combination of VkShaderStageFlagBits values
- VUID-VkBindDescriptorBufferEmbeddedSamplersInfoEXT-layout-parameter
  If `layout` is not VK_NULL_HANDLE, `layout` **must** be a valid VkPipelineLayout handle

18.3. Updating Descriptor Buffers

Updates to descriptor data in buffers **can** be performed by any operation on either the host or device that **can** access memory.

Descriptor buffer reads **can** be synchronized using VK_ACCESS_2_DESCRIPTOR_BUFFER_READ_BIT_EXT in the relevant shader stage.

18.4. Push Descriptors With Descriptor Buffers

If the `descriptorBufferPushDescriptors` feature is enabled, push descriptors **can** be used with

descriptor buffers in the same way as with descriptor sets.

The VkPhysicalDeviceDescriptorBufferPropertiesEXT::bufferlessPushDescriptors property indicates whether the implementation requires a buffer to back push descriptors. If the property is VK_FALSE then before recording any push descriptors, the application **must** bind exactly 1 descriptor buffer that was created with the VK_BUFFER_USAGE_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT usage flag set. When this buffer is bound, any previously recorded push descriptors that are required for a subsequent command **must** be recorded again.

18.5. Capture and Replay

In a similar way to bufferDeviceAddressCaptureReplay, the descriptorBufferCaptureReplay feature allows the creation of opaque handles for objects at capture time that **can** be passed into object creation calls in a future replay, causing descriptors to be created with the same data. The opaque memory address for any memory used by these resources **must** have been captured using vkGetDeviceMemoryOpaqueCaptureAddress and be replayed using VkMemoryOpaqueCaptureAddressAllocateInfo.

To get the opaque descriptor data for a buffer, call:

WARNING	This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
VkResult vkGetBufferOpaqueCaptureDescriptorDataEXT(
    VkDevice                                    device,
    const VkBufferCaptureDescriptorDataInfoEXT* pInfo,
    void*                                       pData);
```

- device is the logical device that gets the data.
- pInfo is a pointer to a VkBufferCaptureDescriptorDataInfoEXT structure specifying the buffer.
- pData is a pointer to an application-allocated buffer where the data will be written.

Valid Usage

- VUID-vkGetBufferOpaqueCaptureDescriptorDataEXT-None-08072
  The descriptorBufferCaptureReplay feature **must** be enabled
- VUID-vkGetBufferOpaqueCaptureDescriptorDataEXT-pData-08073
  pData **must** point to a buffer that is at least VkPhysicalDeviceDescriptorBufferPropertiesEXT::bufferCaptureReplayDescriptorDataSize bytes in size
- VUID-vkGetBufferOpaqueCaptureDescriptorDataEXT-device-08074
  If device was created with multiple physical devices, then the bufferDeviceAddressMultiDevice feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetBufferOpaqueCaptureDescriptorDataEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetBufferOpaqueCaptureDescriptorDataEXT-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkBufferCaptureDescriptorDataInfoEXT structure
- VUID-vkGetBufferOpaqueCaptureDescriptorDataEXT-pData-parameter
  `pData` **must** be a pointer value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Information about the buffer to get descriptor buffer capture data for is passed in a `VkBufferCaptureDescriptorDataInfoEXT` structure:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
typedef struct VkBufferCaptureDescriptorDataInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkBuffer           buffer;
} VkBufferCaptureDescriptorDataInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `buffer` is the `VkBuffer` handle of the buffer to get opaque capture data for.

Valid Usage

- VUID-VkBufferCaptureDescriptorDataInfoEXT-buffer-08075
  `buffer` **must** have been created with VK_BUFFER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT set in

VkBufferCreateInfo::flags

Valid Usage (Implicit)

- VUID-VkBufferCaptureDescriptorDataInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_BUFFER_CAPTURE_DESCRIPTOR_DATA_INFO_EXT
- VUID-VkBufferCaptureDescriptorDataInfoEXT-pNext-pNext
  pNext **must** be NULL
- VUID-VkBufferCaptureDescriptorDataInfoEXT-buffer-parameter
  buffer **must** be a valid VkBuffer handle

To get the opaque capture descriptor data for an image, call:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
VkResult vkGetImageOpaqueCaptureDescriptorDataEXT(
    VkDevice                                    device,
    const VkImageCaptureDescriptorDataInfoEXT*  pInfo,
    void*                                       pData);
```

- device is the logical device that gets the data.
- pInfo is a pointer to a VkImageCaptureDescriptorDataInfoEXT structure specifying the image.
- pData is a pointer to an application-allocated buffer where the data will be written.

Valid Usage

- VUID-vkGetImageOpaqueCaptureDescriptorDataEXT-None-08076
  The descriptorBufferCaptureReplay feature **must** be enabled
- VUID-vkGetImageOpaqueCaptureDescriptorDataEXT-pData-08077
  pData **must** point to a buffer that is at least VkPhysicalDeviceDescriptorBufferPropertiesEXT::imageCaptureReplayDescriptorDataSize bytes in size
- VUID-vkGetImageOpaqueCaptureDescriptorDataEXT-device-08078
  If device was created with multiple physical devices, then the bufferDeviceAddressMultiDevice feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetImageOpaqueCaptureDescriptorDataEXT-device-parameter

`device` **must** be a valid VkDevice handle

- VUID-vkGetImageOpaqueCaptureDescriptorDataEXT-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkImageCaptureDescriptorDataInfoEXT structure
- VUID-vkGetImageOpaqueCaptureDescriptorDataEXT-pData-parameter
  `pData` **must** be a pointer value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Information about the image to get descriptor buffer capture data for is passed in a `VkImageCaptureDescriptorDataInfoEXT` structure:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
typedef struct VkImageCaptureDescriptorDataInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkImage            image;
} VkImageCaptureDescriptorDataInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `image` is the `VkImage` handle of the image to get opaque capture data for.

Valid Usage

- VUID-VkImageCaptureDescriptorDataInfoEXT-image-08079
  `image` **must** have been created with VK_IMAGE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT set in VkImageCreateInfo::`flags`

Valid Usage (Implicit)

- VUID-VkImageCaptureDescriptorDataInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_CAPTURE_DESCRIPTOR_DATA_INFO_EXT
- VUID-VkImageCaptureDescriptorDataInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkImageCaptureDescriptorDataInfoEXT-image-parameter
  `image` **must** be a valid VkImage handle

To get the opaque capture descriptor data for an image view, call:

WARNING	This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
VkResult vkGetImageViewOpaqueCaptureDescriptorDataEXT(
    VkDevice                                    device,
    const VkImageViewCaptureDescriptorDataInfoEXT* pInfo,
    void*                                       pData);
```

- `device` is the logical device that gets the data.
- `pInfo` is a pointer to a VkImageViewCaptureDescriptorDataInfoEXT structure specifying the image view.
- `pData` is a pointer to an application-allocated buffer where the data will be written.

Valid Usage

- VUID-vkGetImageViewOpaqueCaptureDescriptorDataEXT-None-08080
  The `descriptorBufferCaptureReplay` feature **must** be enabled
- VUID-vkGetImageViewOpaqueCaptureDescriptorDataEXT-pData-08081
  `pData` **must** point to a buffer that is at least VkPhysicalDeviceDescriptorBufferPropertiesEXT::`imageViewCaptureReplayDescriptorDataSize` bytes in size
- VUID-vkGetImageViewOpaqueCaptureDescriptorDataEXT-device-08082
  If `device` was created with multiple physical devices, then the `bufferDeviceAddressMultiDevice` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetImageViewOpaqueCaptureDescriptorDataEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetImageViewOpaqueCaptureDescriptorDataEXT-pInfo-parameter

pInfo **must** be a valid pointer to a valid VkImageViewCaptureDescriptorDataInfoEXT structure

- VUID-vkGetImageViewOpaqueCaptureDescriptorDataEXT-pData-parameter
  pData **must** be a pointer value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Information about the image view to get descriptor buffer capture data for is passed in a VkImageViewCaptureDescriptorDataInfoEXT structure:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
typedef struct VkImageViewCaptureDescriptorDataInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkImageView        imageView;
} VkImageViewCaptureDescriptorDataInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- imageView is the VkImageView handle of the image view to get opaque capture data for.

Valid Usage

- VUID-VkImageViewCaptureDescriptorDataInfoEXT-imageView-08083
  imageView **must** have been created with VK_IMAGE_VIEW_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT set in VkImageViewCreateInfo::flags

Valid Usage (Implicit)

- VUID-VkImageViewCaptureDescriptorDataInfoEXT-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_VIEW_CAPTURE_DESCRIPTOR_DATA_INFO_EXT`
- VUID-VkImageViewCaptureDescriptorDataInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkImageViewCaptureDescriptorDataInfoEXT-imageView-parameter
  `imageView` **must** be a valid VkImageView handle

To get the opaque capture descriptor data for a sampler, call:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
VkResult vkGetSamplerOpaqueCaptureDescriptorDataEXT(
    VkDevice                                    device,
    const VkSamplerCaptureDescriptorDataInfoEXT* pInfo,
    void*                                       pData);
```

- `device` is the logical device that gets the data.
- `pInfo` is a pointer to a VkSamplerCaptureDescriptorDataInfoEXT structure specifying the sampler.
- `pData` is a pointer to an application-allocated buffer where the data will be written.

Valid Usage

- VUID-vkGetSamplerOpaqueCaptureDescriptorDataEXT-None-08084
  The `descriptorBufferCaptureReplay` feature **must** be enabled
- VUID-vkGetSamplerOpaqueCaptureDescriptorDataEXT-pData-08085
  `pData` **must** point to a buffer that is at least VkPhysicalDeviceDescriptorBufferPropertiesEXT::`samplerCaptureReplayDescriptorDataSize` bytes in size
- VUID-vkGetSamplerOpaqueCaptureDescriptorDataEXT-device-08086
  If `device` was created with multiple physical devices, then the `bufferDeviceAddressMultiDevice` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetSamplerOpaqueCaptureDescriptorDataEXT-device-parameter
  `device` **must** be a valid VkDevice handle

- VUID-vkGetSamplerOpaqueCaptureDescriptorDataEXT-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkSamplerCaptureDescriptorDataInfoEXT structure
- VUID-vkGetSamplerOpaqueCaptureDescriptorDataEXT-pData-parameter
  `pData` **must** be a pointer value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Information about the sampler to get descriptor buffer capture data for is passed in a `VkSamplerCaptureDescriptorDataInfoEXT` structure:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
typedef struct VkSamplerCaptureDescriptorDataInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkSampler          sampler;
} VkSamplerCaptureDescriptorDataInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `sampler` is the `VkSampler` handle of the sampler to get opaque capture data for.

Valid Usage

- VUID-VkSamplerCaptureDescriptorDataInfoEXT-sampler-08087
  `sampler` **must** have been created with VK_SAMPLER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT set in VkSamplerCreateInfo::`flags`

Valid Usage (Implicit)

- VUID-VkSamplerCaptureDescriptorDataInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SAMPLER_CAPTURE_DESCRIPTOR_DATA_INFO_EXT
- VUID-VkSamplerCaptureDescriptorDataInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkSamplerCaptureDescriptorDataInfoEXT-sampler-parameter
  `sampler` **must** be a valid VkSampler handle

To get the opaque capture descriptor data for an acceleration structure, call:

WARNING	This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer with VK_KHR_acceleration_structure or
VK_NV_ray_tracing
VkResult vkGetAccelerationStructureOpaqueCaptureDescriptorDataEXT(
    VkDevice                                    device,
    const VkAccelerationStructureCaptureDescriptorDataInfoEXT* pInfo,
    void*                                       pData);
```

- `device` is the logical device that gets the data.
- `pInfo` is a pointer to a VkAccelerationStructureCaptureDescriptorDataInfoEXT structure specifying the acceleration structure.
- `pData` is a pointer to an application-allocated buffer where the data will be written.

Valid Usage

- VUID-vkGetAccelerationStructureOpaqueCaptureDescriptorDataEXT-None-08088
  The `descriptorBufferCaptureReplay` feature **must** be enabled
- VUID-vkGetAccelerationStructureOpaqueCaptureDescriptorDataEXT-pData-08089
  `pData` **must** point to a buffer that is at least VkPhysicalDeviceDescriptorBufferPropertiesEXT::`accelerationStructureCaptureReplayDescriptorDataSize` bytes in size
- VUID-vkGetAccelerationStructureOpaqueCaptureDescriptorDataEXT-device-08090
  If `device` was created with multiple physical devices, then the `bufferDeviceAddressMultiDevice` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetAccelerationStructureOpaqueCaptureDescriptorDataEXT-device-parameter
  `device` **must** be a valid VkDevice handle

- VUID-vkGetAccelerationStructureOpaqueCaptureDescriptorDataEXT-pInfo-parameter `pInfo` **must** be a valid pointer to a valid VkAccelerationStructureCaptureDescriptorDataInfoEXT structure
- VUID-vkGetAccelerationStructureOpaqueCaptureDescriptorDataEXT-pData-parameter `pData` **must** be a pointer value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Information about the acceleration structure to get descriptor buffer capture data for is passed in a `VkAccelerationStructureCaptureDescriptorDataInfoEXT` structure:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer with VK_KHR_acceleration_structure or
VK_NV_ray_tracing
typedef struct VkAccelerationStructureCaptureDescriptorDataInfoEXT {
    VkStructureType               sType;
    const void*                   pNext;
    VkAccelerationStructureKHR    accelerationStructure;
    VkAccelerationStructureNV     accelerationStructureNV;
} VkAccelerationStructureCaptureDescriptorDataInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `accelerationStructure` is the `VkAccelerationStructureKHR` handle of the acceleration structure to get opaque capture data for.
- `accelerationStructureNV` is the `VkAccelerationStructureNV` handle of the acceleration structure to get opaque capture data for.

Valid Usage

- VUID-VkAccelerationStructureCaptureDescriptorDataInfoEXT-accelerationStructure-08091

If accelerationStructure is not VK_NULL_HANDLE then accelerationStructure **must** have been created with VK_ACCELERATION_STRUCTURE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT set in VkAccelerationStructureCreateInfoKHR::createFlags

- VUID-VkAccelerationStructureCaptureDescriptorDataInfoEXT-accelerationStructure-08093
  If accelerationStructure is not VK_NULL_HANDLE then accelerationStructureNV **must** be VK_NULL_HANDLE
- VUID-VkAccelerationStructureCaptureDescriptorDataInfoEXT-accelerationStructureNV-08094
  If accelerationStructureNV is not VK_NULL_HANDLE then accelerationStructure **must** be VK_NULL_HANDLE
- VUID-VkAccelerationStructureCaptureDescriptorDataInfoEXT-None-12430
  If the accelerationStructure feature is not enabled, accelerationStructure **must** be VK_NULL_HANDLE
- VUID-VkAccelerationStructureCaptureDescriptorDataInfoEXT-accelerationStructureNV-12431
  If the VK_NV_ray_tracing extension is not enabled, then accelerationStructureNV **must** be VK_NULL_HANDLE

Valid Usage (Implicit)

- VUID-VkAccelerationStructureCaptureDescriptorDataInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CAPTURE_DESCRIPTOR_DATA_INFO_EXT
- VUID-VkAccelerationStructureCaptureDescriptorDataInfoEXT-pNext-pNext
  pNext **must** be NULL
- VUID-VkAccelerationStructureCaptureDescriptorDataInfoEXT-accelerationStructure-parameter
  If accelerationStructure is not VK_NULL_HANDLE, accelerationStructure **must** be a valid VkAccelerationStructureKHR handle
- VUID-VkAccelerationStructureCaptureDescriptorDataInfoEXT-accelerationStructureNV-parameter
  If accelerationStructureNV is not VK_NULL_HANDLE, accelerationStructureNV **must** be a valid VkAccelerationStructureNV handle
- VUID-VkAccelerationStructureCaptureDescriptorDataInfoEXT-commonparent
  Both of accelerationStructure, and accelerationStructureNV that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

The VkOpaqueCaptureDescriptorDataCreateInfoEXT structure is defined as:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
typedef struct VkOpaqueCaptureDescriptorDataCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    const void*        opaqueCaptureDescriptorData;
} VkOpaqueCaptureDescriptorDataCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `opaqueCaptureDescriptorData` is a pointer to an application-allocated buffer containing opaque capture data retrieved using vkGetBufferOpaqueCaptureDescriptorDataEXT, vkGetImageOpaqueCaptureDescriptorDataEXT, vkGetImageViewOpaqueCaptureDescriptorDataEXT, vkGetSamplerOpaqueCaptureDescriptorDataEXT, or vkGetAccelerationStructureOpaqueCaptureDescriptorDataEXT.

During replay, opaque descriptor capture data **can** be specified by adding a `VkOpaqueCaptureDescriptorDataCreateInfoEXT` structure to the relevant `pNext` chain of a VkAccelerationStructureCreateInfoKHR, VkBufferCreateInfo, VkImageCreateInfo, VkImageViewCreateInfo, or VkSamplerCreateInfo structure.

When providing opaque capture data for an image, if the `pNext` chain of VkImageCreateInfo contains an instance of both this structure and VkOpaqueCaptureDataCreateInfoEXT, they **should** both specify data from the same original resource. If they have capture data from different original resources, resource creation is much more likely to fail.

Valid Usage (Implicit)

- VUID-VkOpaqueCaptureDescriptorDataCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_OPAQUE_CAPTURE_DESCRIPTOR_DATA_CREATE_INFO_EXT
- VUID-VkOpaqueCaptureDescriptorDataCreateInfoEXT-opaqueCaptureDescriptorData-parameter
  `opaqueCaptureDescriptorData` **must** be a pointer value

Structure Chaining

Extends the structures

- VkAccelerationStructureCreateInfo2KHR
- VkAccelerationStructureCreateInfoKHR
- VkBufferCreateInfo

- VkImageCreateInfo
- VkImageViewCreateInfo
- VkSamplerCreateInfo
- VkTensorViewCreateInfoARM

Chapter 19. Shader Interfaces

When a pipeline is created, the set of shaders specified in the corresponding `VkPipelineCreateInfo` structure is implicitly linked at a number of different interfaces.

- Shader Input and Output Interface
- Vertex Input Interface
- Fragment Output Interface
- Fragment Tile Image Interface
- Fragment Input Attachment Interface
- Ray Tracing Pipeline Interface
- Shader Resource Interface

This chapter describes valid uses for a set of SPIR-V decorations. Any other use of one of these decorations is invalid, with the exception that, when using SPIR-V versions 1.4 and earlier: `Block`, `BufferBlock`, `Offset`, `ArrayStride`, and `MatrixStride` can also decorate types and type members used by variables in the `Private` and `Function` storage classes.

NOTE In this chapter, there are references to SPIR-V terms such as the `MeshNV` execution model. These terms will appear even in a build of the specification which does not support any extensions. This is as intended, since these terms appear in the unified SPIR-V specification without such qualifiers.

19.1. Shader Input and Output Interfaces

When multiple stages are present in a pipeline, the outputs of one stage form an interface with the inputs of the next stage. When such an interface involves a shader, shader outputs are matched against the inputs of the next stage, and shader inputs are matched against the outputs of the previous stage.

All the variables forming the shader input and output *interfaces* are listed as operands to the `OpEntryPoint` instruction and are declared with the `Input` or `Output` storage classes, respectively, in the SPIR-V module. These generally form the interfaces between consecutive shader stages, regardless of any non-shader stages between the consecutive shader stages.

There are two classes of variables that **can** be matched between shader stages, built-in variables and user-defined variables. Each class has a different set of matching criteria.

For compute shaders, the input interface is formed by the built-in interface. The output interface is empty.

19.1.1. Built-In Interface Block

Shader built-in variables meeting the following requirements define the *built-in interface block*. They **must**

- be explicitly declared (there are no implicit built-ins),
- be identified with a `BuiltIn` decoration,
- form object types as described in the Built-in Variables section, and
- be declared in a block whose top-level members are the built-ins.

There **must** be no more than one built-in interface block per shader per interface , except for the mesh output interface where there **must** be at most one built-in interface block decorated with the `PerPrimitiveEXT` decoration and at most one built-in interface block without this decoration .

Built-ins **must** not have any `Location` or `Component` decorations.

19.1.2. User-Defined Variable Interface

The non-built-in variables listed by `OpEntryPoint` with the `Input` or `Output` storage class form the *user-defined variable interface*. These **must** have numeric type or, recursively, composite types of such types. If an implementation supports `storageInputOutput16`, components **can** have a width of 16 bits. These variables **must** be identified with a `Location` decoration and **can** also be identified with a `Component` decoration.

19.1.3. Interface Matching

An output variable, block, or structure member in a given shader stage has an interface match with an input variable, block, or structure member in a subsequent shader stage if they both adhere to the following conditions:

- They have equivalent decorations, other than:
  - `XfbBuffer`, `XfbStride`, `Offset`, and `Stream`
  - one is not decorated with `Component` and the other is declared with a `Component` of `0`
  - Interpolation decorations
  - `RelaxedPrecision` if one is an input variable and the other an output variable
- Their types match as follows:
  - if the input is declared in a tessellation control or geometry shader as an `OpTypeArray` with an `Element Type` equivalent to the `OpType*` declaration of the output, and neither is a structure member; or
  - if the `maintenance4` feature is enabled, they are declared as `OpTypeVector` variables, and the output has a `Component Count` value higher than that of the input but the same `Component Type`; or
  - if the output is declared in a mesh shader as an `OpTypeArray` with an `Element Type` equivalent to the `OpType*` declaration of the input, and neither is a structure member; or
  - if the input is decorated with `PerVertexKHR`, and is declared in a fragment shader as an `OpTypeArray` with an `Element Type` equivalent to the `OpType*` declaration of the output, and neither the input nor the output is a structure member; or
  - if in any other case they are declared with an equivalent `OpType*` declaration.

- If both are structures and every member has an interface match.

NOTE	The word “structure” above refers to both variables that have an `OpTypeStruct` type and interface blocks (which are also declared as `OpTypeStruct`).

If the pipeline is compiled as separate graphics pipeline libraries and the `graphicsPipelineLibraryIndependentInterpolationDecoration` limit is not supported, matches are not found if the interpolation decorations differ between the last pre-rasterization shader stage and the fragment shader stage.

All input variables and blocks **must** have an interface match in the preceding shader stage, except for built-in variables in fragment shaders. Shaders **can** declare and write to output variables that are not declared or read by the subsequent stage.

The value of an input variable is undefined if the preceding stage does not write to a matching output variable, as described above.

19.1.4. Location and Component Assignment

User-defined variables in interfaces between shader stages in the graphics pipeline consume a unique set of `Location` and `Component` values.

Available space for user-defined interface variables is partitioned into a number of 32-bit four-component vectors, each identified by a `Location` value. Each individual 32-bit component of a vector is then further identified by a `Component` value.

16-bit scalar or vector values consume one `Component` slot per 16-bit component and **must** be specified within a single `Location`. 32-bit scalar or vector values consume one `Component` slot per 32-bit component and **must** be specified within a single `Location`. 64-bit scalar or vector values consume two consecutive `Component` slots per 64-bit component from up to two consecutive `Location` slots.

For any shader interface variable where one level of the array is disregarded for type matching, the outer array level is also disregarded when assigning `Location` slots.

An array of size n with elements consuming l `Location` slots each will consume l × n `Location` slots. Each element of the array will consume `Component` slots in each `Location` slot identically to a declaration using the element type.

NOTE	Applications running on Vulkan implementations advertising a VkPhysicalDeviceDriverProperties::`conformanceVersion` less than 1.4.6.0 should be aware that arrays may consume all four `Component` slots in each `Location` slot.

Matrices of size n × m are assigned locations identically to arrays of size n of vectors of length m with an identical element type.

When a variable with a structure type is decorated with a `Location`, the members in the structure type **must** not be decorated with a `Location`. The variable’s members are assigned consecutive locations in declaration order, starting from the first member, which is assigned the location

decoration from the variable. The `Location` slots consumed by structure members are determined by applying the rules above in a depth-first traversal of the instantiated members as though the structure or block member were declared as an input or output variable of the same type.

A variable with a structure type that is not decorated with `Block` **must** be decorated with a `Location`.

When a variable with a structure type decorated with `Block` is declared without a `Location` decoration, each member in the structure **must** be decorated with a `Location`. Types nested deeper than the top-level members **must** not have `Location` decorations.

Multiple variable declarations in the same storage class **must** not have overlapping `Component` slots within the same `Location`.

The number of input and output locations available for a shader input or output interface depends on the shader stage as described in Shader Input and Output Locations. All variables in both the built-in interface block and the user-defined variable interface count against these limits. Each effective `Location` **must** have a value less than the number of `Location` slots available for the given interface, as specified in the "Locations Available" column in Shader Input and Output Locations.

Table 17. Shader Input and Output Locations

Shader Interface	Locations Available
vertex input	`maxVertexInputAttributes`
vertex output	`maxVertexOutputComponents` / 4
tessellation control input	`maxTessellationControlPerVertexInputComponents` / 4
tessellation control output	`maxTessellationControlPerVertexOutputComponents` / 4
tessellation evaluation input	`maxTessellationEvaluationInputComponents` / 4
tessellation evaluation output	`maxTessellationEvaluationOutputComponents` / 4
geometry input	`maxGeometryInputComponents` / 4
geometry output	`maxGeometryOutputComponents` / 4
fragment input	`maxFragmentInputComponents` / 4
fragment output	`maxFragmentOutputAttachments`
mesh output	`maxMeshOutputComponents` / 4

19.2. Vertex Input Interface

When the vertex stage is present in a pipeline, the vertex shader input variables form an interface with the vertex input attributes. The vertex shader input variables are matched by the `Location` and `Component` decorations to the vertex input attributes specified in the `pVertexInputState` member of the VkGraphicsPipelineCreateInfo structure.

The vertex shader input variables listed by `OpEntryPoint` with the `Input` storage class form the *vertex input interface*. These variables **must** be identified with a `Location` decoration and **can** also be

identified with a Component decoration.

For the purposes of interface matching: variables declared without a Component decoration are considered to have a Component decoration of zero. The number of available vertex input Location slots is given by the maxVertexInputAttributes member of the VkPhysicalDeviceLimits structure.

See Attribute Location and Component Assignment for details.

All vertex shader inputs declared as above **must** have a corresponding attribute and binding in the pipeline.

Components and locations are consumed as defined for Location and Component Assignment. Multiple user-defined input variable declarations **must** not have overlapping Component slots within the same Location.

19.3. Fragment Output Interface

When the fragment stage is present in a pipeline, the fragment shader outputs form an interface with the output attachments defined by a render pass instance. The fragment shader output variables are matched by the Location and Component decorations to specified color attachments.

The fragment shader output variables listed by OpEntryPoint with the Output storage class form the *fragment output interface*. These variables **must** be identified with a Location decoration. They **can** also be identified with a Component decoration and/or an Index decoration. For the purposes of interface matching: variables declared without a Component decoration are considered to have a Component decoration of zero, and variables declared without an Index decoration are considered to have an Index decoration of zero.

A fragment shader output variable identified with a Location decoration of *i* is associated with the color attachment indicated by VkRenderingInfo::pColorAttachments[*i*]. When using render pass objects, it is associated with the color attachment indicated by VkSubpassDescription ::pColorAttachments[*i*]. Values are written to those attachments after passing through the blending unit as described in Blending, if enabled. The number of available fragment output Location slots is given by the maxFragmentOutputAttachments member of the VkPhysicalDeviceLimits structure.

If the dynamicRenderingLocalRead feature is supported, fragment output locations **can** be remapped when using dynamic rendering.

To set the fragment output location mappings during rendering, call:

```
// Provided by VK_VERSION_1_4
void vkCmdSetRenderingAttachmentLocations(
    VkCommandBuffer                             commandBuffer,
    const VkRenderingAttachmentLocationInfo*    pLocationInfo);
```

```
// Provided by VK_KHR_dynamic_rendering_local_read
// Equivalent to vkCmdSetRenderingAttachmentLocations
void vkCmdSetRenderingAttachmentLocationsKHR(
    VkCommandBuffer                             commandBuffer,
    const VkRenderingAttachmentLocationInfo*    pLocationInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pLocationInfo` is a VkRenderingAttachmentLocationInfo structure indicating the new mappings.

This command sets the attachment location mappings for subsequent drawing commands, and **must** match the mappings provided to the bound pipeline, if one is bound, which **can** be set by chaining VkRenderingAttachmentLocationInfo to VkGraphicsPipelineCreateInfo.

Until this command is called, mappings in the command buffer state are treated as each color attachment specified in vkCmdBeginRendering having a location equal to its index in VkRenderingInfo::`pColorAttachments`. This state is reset whenever vkCmdBeginRendering is called.

Valid Usage

- VUID-vkCmdSetRenderingAttachmentLocations-dynamicRenderingLocalRead-09509
  `dynamicRenderingLocalRead` **must** be enabled
- VUID-vkCmdSetRenderingAttachmentLocations-pLocationInfo-09510
  `pLocationInfo->colorAttachmentCount` **must** be equal to the value of VkRenderingInfo ::`colorAttachmentCount` used to begin the current render pass instance
- VUID-vkCmdSetRenderingAttachmentLocations-commandBuffer-09511
  The current render pass instance **must** have been started or resumed by vkCmdBeginRendering in this `commandBuffer`

Valid Usage (Implicit)

- VUID-vkCmdSetRenderingAttachmentLocations-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetRenderingAttachmentLocations-pLocationInfo-parameter
  `pLocationInfo` **must** be a valid pointer to a valid VkRenderingAttachmentLocationInfo structure
- VUID-vkCmdSetRenderingAttachmentLocations-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetRenderingAttachmentLocations-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetRenderingAttachmentLocations-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdSetRenderingAttachmentLocations-videocoding

This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetRenderingAttachmentLocations is not affected by conditional rendering

The `VkRenderingAttachmentLocationInfo` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkRenderingAttachmentLocationInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           colorAttachmentCount;
    const uint32_t*    pColorAttachmentLocations;
} VkRenderingAttachmentLocationInfo;
```

```
// Provided by VK_KHR_dynamic_rendering_local_read
// Equivalent to VkRenderingAttachmentLocationInfo
typedef VkRenderingAttachmentLocationInfo VkRenderingAttachmentLocationInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `colorAttachmentCount` is the number of elements in `pColorAttachmentLocations`.
- `pColorAttachmentLocations` is a pointer to an array of `colorAttachmentCount uint32_t` values defining remapped locations for color attachments.

This structure allows applications to remap the locations of color attachments to different fragment

shader output locations.

Each element of pColorAttachmentLocations set to VK_ATTACHMENT_UNUSED will be inaccessible to this pipeline as a color attachment; no location will map to it. Each element of pColorAttachmentLocations set to any other value will map the specified location value to the color attachment specified in the render pass at the corresponding index in the pColorAttachmentLocations array. Any writes to a fragment output location that is not mapped to an attachment **must** be discarded.

If pColorAttachmentLocations is NULL, it is equivalent to setting each element to its index within the array.

This structure **can** be included in the pNext chain of a VkGraphicsPipelineCreateInfo structure to set this state for a pipeline. If this structure is not included in the pNext chain of VkGraphicsPipelineCreateInfo, it is equivalent to specifying this structure with the following properties:

- colorAttachmentCount set to VkPipelineRenderingCreateInfo::colorAttachmentCount.
- pColorAttachmentLocations set to NULL.

This structure **can** be included in the pNext chain of a VkCommandBufferInheritanceInfo structure to specify inherited state from the primary command buffer. If VkCommandBufferInheritanceInfo ::renderPass is not VK_NULL_HANDLE, or VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT is not specified in VkCommandBufferBeginInfo::flags, members of this structure are ignored. If this structure is not included in the pNext chain of VkCommandBufferInheritanceInfo, it is equivalent to specifying this structure with the following properties:

- colorAttachmentCount set to VkCommandBufferInheritanceRenderingInfo::colorAttachmentCount.
- pColorAttachmentLocations set to NULL.

Valid Usage

- VUID-VkRenderingAttachmentLocationInfo-dynamicRenderingLocalRead-09512
  If the dynamicRenderingLocalRead feature is not enabled, and pColorAttachmentLocations is not NULL, each element **must** be the value of its index within the array
- VUID-VkRenderingAttachmentLocationInfo-pColorAttachmentLocations-09513
  Elements of pColorAttachmentLocations that are not VK_ATTACHMENT_UNUSED **must** each be unique
- VUID-VkRenderingAttachmentLocationInfo-colorAttachmentCount-09514
  colorAttachmentCount **must** be less than or equal to maxColorAttachments
- VUID-VkRenderingAttachmentLocationInfo-pColorAttachmentLocations-09515
  Each element of pColorAttachmentLocations **must** be less than maxColorAttachments

Valid Usage (Implicit)

- VUID-VkRenderingAttachmentLocationInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_LOCATION_INFO

Structure Chaining

Extends the structures

- VkCommandBufferInheritanceInfo
- VkGraphicsPipelineCreateInfo

When an active fragment shader invocation finishes, the values of all fragment shader outputs are copied out and used as blend inputs or color attachment writes. If an output variable is declared, but there is no color attachment at that `Location`, any values written to it are discarded. If a color attachment at a given `Location` does not have an output variable declared, the values **must** not be modified.

Output `Component` words identified as 0, 1, 2, and 3 will be directed to the R, G, B, and A inputs to the blending unit, respectively, or to the output attachment if blending is disabled. If two variables are placed within the same `Location`, they **must** have the same numeric type. `Component` words which do not correspond to any fragment shader output will also result in undefined values for blending or color attachment writes.

Fragment outputs identified with an `Index` of zero are directed to the first input of the blending unit associated with the corresponding `Location`. Outputs identified with an `Index` of one are directed to the second input of the corresponding blending unit.

Components and locations are consumed as defined for Location and Component Assignment. Output variable declarations **must** not consume any of the same `Component` slots within the same `Location` and with the same `Index` value as any other output variable declaration.

Output values written by a fragment shader **must** be declared with either `OpTypeFloat` or `OpTypeInt`, and a `Width` of 32. If `storageInputOutput16` is supported, output values written by a fragment shader **can** be also declared with either `OpTypeFloat` or `OpTypeInt` and a `Width` of 16. Composites of these types are also permitted. If the color attachment has a signed or unsigned normalized fixed-point format, color values are assumed to be floating-point and are converted to fixed-point as described in Conversion From Floating-Point to Normalized Fixed-Point; if the color attachment has an integer format, color values are assumed to be integers and converted to the bit-depth of the target. Any value that cannot be represented in the attachment's format is undefined. For any other attachment format no conversion is performed. If the type of the values written by the fragment shader does not match the format of the corresponding color attachment, the resulting values are undefined for those components.

19.4. Legacy Dithering

The application **can** enable dithering to be applied to the color output of a subpass, by using the VK_SUBPASS_DESCRIPTION_ENABLE_LEGACY_DITHERING_BIT_EXT flag. For use in a dynamic render pass, the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag **must** be used. In that case, the pipelines used **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT.

When dithering is enabled, the implementation **may** modify the output color value c by one ULP. This modification **must** only depend on the framebuffer coordinates (x_f,y_f) of the sample, as well as on the value of c.

The exact details of the dithering algorithm are unspecified, including the algorithm itself, the formats dithering is applied to, and the stage in which it is applied.

NOTE	This extension is intended only for use by OpenGL emulation layers, and as such the dithering algorithm applied to the subpass **should** be equivalent to the vendor's OpenGL implementation, if any.

19.5. Fragment Tile Image Interface

When a fragment stage is present in a pipeline, the fragment shader tile image variables decorated with `Location` form an interface with the color attachments defined by the render pass instance. The fragment shader tile image variables are matched by `Location` decorations to the color attachments specified in the `pColorAttachments` array of the VkRenderingInfoKHR structure describing the render pass instance the fragment shader is executed in.

The fragment shader variables listed by `OpEntryPoint` with the `TileImageEXT` storage class and a decoration of `Location` form the *fragment tile image interface*. These variables **must** be declared with a type of `OpTypeImage`, and a `Dim` operand of `TileImageDataEXT`. The `Component` decoration is not supported for these variables.

Reading from a tile image variable with a `Location` decoration of *i* reads from the color attachment identified by the element of VkRenderingInfoKHR::`pColorAttachments` with a `location` equal to *i*. If the tile image variable is declared as an array of size N, it consumes N consecutive tile image locations, starting with the index specified. There **must** not be more than one tile image variable with the same `Location` whether explicitly declared or implied by an array declaration. The number of available tile image locations is the same as the number of available fragment output locations as given by the `maxFragmentOutputAttachments` member of the `VkPhysicalDeviceLimits` structure.

The basic data type (floating-point, integer, unsigned integer) of the tile image variable **must** match the basic format of the corresponding color attachment, or the values read from the tile image variables are poison.

19.6. Fragment Input Attachment Interface

When a fragment stage is present in a pipeline, the fragment shader subpass inputs form an interface with the input attachments of the current subpass. The fragment shader subpass input

variables are matched by InputAttachmentIndex decorations to the input attachments specified in the pInputAttachments array of the VkSubpassDescription structure describing the subpass that the fragment shader is executed in.

The fragment shader subpass input variables with the UniformConstant storage class and a decoration of InputAttachmentIndex that are statically used by OpEntryPoint form the *fragment input attachment interface*. These variables **must** be declared with a type of OpTypeImage, a Dim operand of SubpassData, an Arrayed operand of 0, and a Sampled operand of 2. The MS operand of the OpTypeImage **must** be 0 if the samples field of the corresponding VkAttachmentDescription is VK_SAMPLE_COUNT_1_BIT and multisampled-render-to-single-sampled is not enabled, and 1 otherwise.

A subpass input variable identified with an InputAttachmentIndex decoration of *i* reads from the input attachment indicated by pInputAttachments[*i*] member of VkSubpassDescription. If the subpass input variable is declared as an array of size N, or a runtime-sized array, it consumes consecutive input attachments, starting with the index specified. For runtime-sized arrays, the number of input attachment indices consumed is equal to VkDescriptorSetLayoutBinding::descriptorCount. There **must** not be more than one input variable with the same InputAttachmentIndex whether explicitly declared or implied by an array declaration per image aspect. A multi-aspect image (e.g. a depth/stencil format) **can** use the same input variable. The number of available input attachment indices is given by the maxPerStageDescriptorInputAttachments member of the VkPhysicalDeviceLimits structure.

When using dynamic rendering with the dynamicRenderingLocalRead feature enabled, a subpass input variable with a InputAttachmentIndex decoration of *i* can be mapped to a color, depth, or stencil attachment.

To set the input attachment index mappings during dynamic rendering, call:

```
// Provided by VK_VERSION_1_4
void vkCmdSetRenderingInputAttachmentIndices(
    VkCommandBuffer                             commandBuffer,
    const VkRenderingInputAttachmentIndexInfo*  pInputAttachmentIndexInfo);
```

```
// Provided by VK_KHR_dynamic_rendering_local_read
// Equivalent to vkCmdSetRenderingInputAttachmentIndices
void vkCmdSetRenderingInputAttachmentIndicesKHR(
    VkCommandBuffer                             commandBuffer,
    const VkRenderingInputAttachmentIndexInfo*  pInputAttachmentIndexInfo);
```

- commandBuffer is the command buffer into which the command will be recorded.
- pInputAttachmentIndexInfo is a VkRenderingInputAttachmentIndexInfo structure indicating the new mappings.

This command sets the input attachment index mappings for subsequent drawing commands, and **must** match the mappings provided to the bound pipeline, if one is bound, which **can** be set by chaining VkRenderingInputAttachmentIndexInfo to VkGraphicsPipelineCreateInfo.

Until this command is called, mappings in the command buffer state are treated as each color attachment specified in vkCmdBeginRendering mapping to subpass inputs with a `InputAttachmentIndex` equal to its index in VkRenderingInfo::`pColorAttachments`, and depth/stencil attachments mapping to input attachments without these decorations. This state is reset whenever vkCmdBeginRendering is called.

Valid Usage

- VUID-vkCmdSetRenderingInputAttachmentIndices-dynamicRenderingLocalRead-09516
  `dynamicRenderingLocalRead` **must** be enabled
- VUID-vkCmdSetRenderingInputAttachmentIndices-pInputAttachmentIndexInfo-09517
  `pInputAttachmentIndexInfo->colorAttachmentCount` **must** be equal to the value of VkRenderingInfo::`colorAttachmentCount` used to begin the current render pass instance
- VUID-vkCmdSetRenderingInputAttachmentIndices-commandBuffer-09518
  The current render pass instance **must** have been started or resumed by vkCmdBeginRendering in this `commandBuffer`

Valid Usage (Implicit)

- VUID-vkCmdSetRenderingInputAttachmentIndices-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetRenderingInputAttachmentIndices-pInputAttachmentIndexInfo-parameter
  `pInputAttachmentIndexInfo` **must** be a valid pointer to a valid VkRenderingInputAttachmentIndexInfo structure
- VUID-vkCmdSetRenderingInputAttachmentIndices-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetRenderingInputAttachmentIndices-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetRenderingInputAttachmentIndices-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdSetRenderingInputAttachmentIndices-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetRenderingInputAttachmentIndices is not affected by conditional rendering

The VkRenderingInputAttachmentIndexInfo structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkRenderingInputAttachmentIndexInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           colorAttachmentCount;
    const uint32_t*    pColorAttachmentInputIndices;
    const uint32_t*    pDepthInputAttachmentIndex;
    const uint32_t*    pStencilInputAttachmentIndex;
} VkRenderingInputAttachmentIndexInfo;
```

```
// Provided by VK_KHR_dynamic_rendering_local_read
// Equivalent to VkRenderingInputAttachmentIndexInfo
typedef VkRenderingInputAttachmentIndexInfo VkRenderingInputAttachmentIndexInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- colorAttachmentCount is the number of elements in pColorAttachmentInputIndices.
- pColorAttachmentInputIndices is a pointer to an array of colorAttachmentCount uint32_t values defining indices for color attachments to be used as input attachments.
- pDepthInputAttachmentIndex is either NULL, or a pointer to a uint32_t value defining the index for the depth attachment to be used as an input attachment.
- pStencilInputAttachmentIndex is either NULL, or a pointer to a uint32_t value defining the index for the stencil attachment to be used as an input attachment.

This structure allows applications to remap attachments to different input attachment indices.

Each element of pColorAttachmentInputIndices set to a value of VK_ATTACHMENT_UNUSED indicates that the corresponding attachment will not be used as an input attachment in this pipeline. Any other value in each of those elements will map the corresponding attachment to a

InputAttachmentIndex value defined in shader code.

If pColorAttachmentInputIndices is NULL, it is equivalent to setting each element to its index within the array.

If pDepthInputAttachmentIndex or pStencilInputAttachmentIndex are set to NULL, they map to input attachments without a InputAttachmentIndex decoration. If they point to a value of VK_ATTACHMENT_UNUSED, it indicates that the corresponding attachment will not be used as an input attachment in this pipeline. If they point to any other value it maps the corresponding attachment to a InputAttachmentIndex value defined in shader code.

This structure **can** be included in the pNext chain of a VkGraphicsPipelineCreateInfo structure to set this state for a pipeline. If this structure is not included in the pNext chain of VkGraphicsPipelineCreateInfo, it is equivalent to specifying this structure with the following properties:

- colorAttachmentCount set to VkPipelineRenderingCreateInfo::colorAttachmentCount.
- pColorAttachmentInputIndices set to NULL.
- pDepthInputAttachmentIndex set to NULL.
- pStencilInputAttachmentIndex set to NULL.

This structure **can** be included in the pNext chain of a VkCommandBufferInheritanceInfo structure to specify inherited state from the primary command buffer. If this structure is not included in the pNext chain of VkCommandBufferInheritanceInfo, it is equivalent to specifying this structure with the following properties:

- colorAttachmentCount set to VkCommandBufferInheritanceRenderingInfo::colorAttachmentCount.
- pColorAttachmentInputIndices set to NULL.
- pDepthInputAttachmentIndex set to NULL.
- pStencilInputAttachmentIndex set to NULL.

Valid Usage

- VUID-VkRenderingInputAttachmentIndexInfo-dynamicRenderingLocalRead-09519
  If the dynamicRenderingLocalRead feature is not enabled, and pColorAttachmentInputIndices is not NULL, each element **must** be VK_ATTACHMENT_UNUSED
- VUID-VkRenderingInputAttachmentIndexInfo-dynamicRenderingLocalRead-09520
  If the dynamicRenderingLocalRead feature is not enabled, pDepthInputAttachmentIndex **must** be a valid pointer to a value of VK_ATTACHMENT_UNUSED
- VUID-VkRenderingInputAttachmentIndexInfo-dynamicRenderingLocalRead-09521
  If the dynamicRenderingLocalRead feature is not enabled, pStencilInputAttachmentIndex **must** be a valid pointer to a value of VK_ATTACHMENT_UNUSED
- VUID-VkRenderingInputAttachmentIndexInfo-pColorAttachmentInputIndices-09522
  Elements of pColorAttachmentInputIndices that are not VK_ATTACHMENT_UNUSED **must** each be unique

- VUID-VkRenderingInputAttachmentIndexInfo-pColorAttachmentInputIndices-09523 Elements of `pColorAttachmentInputIndices` that are not VK_ATTACHMENT_UNUSED **must** not take the same value as the content of `pDepthInputAttachmentIndex`
- VUID-VkRenderingInputAttachmentIndexInfo-pColorAttachmentInputIndices-09524 Elements of `pColorAttachmentInputIndices` that are not VK_ATTACHMENT_UNUSED **must** not take the same value as the content of `pStencilInputAttachmentIndex`
- VUID-VkRenderingInputAttachmentIndexInfo-colorAttachmentCount-09525 `colorAttachmentCount` **must** be less than or equal to `maxColorAttachments`
- VUID-VkRenderingInputAttachmentIndexInfo-pDepthInputAttachmentIndex-12274 Elements of `pDepthInputAttachmentIndex`, `pStencilInputAttachmentIndex`, and `pColorAttachmentInputIndices` that are not VK_ATTACHMENT_UNUSED **must** be less than `maxPerStageDescriptorInputAttachments`

Valid Usage (Implicit)

- VUID-VkRenderingInputAttachmentIndexInfo-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_RENDERING_INPUT_ATTACHMENT_INDEX_INFO
- VUID-VkRenderingInputAttachmentIndexInfo-pColorAttachmentInputIndices-parameter If `colorAttachmentCount` is not `0`, and `pColorAttachmentInputIndices` is not `NULL`, `pColorAttachmentInputIndices` **must** be a valid pointer to an array of `colorAttachmentCount` `uint32_t` values
- VUID-VkRenderingInputAttachmentIndexInfo-pDepthInputAttachmentIndex-parameter If `pDepthInputAttachmentIndex` is not `NULL`, `pDepthInputAttachmentIndex` **must** be a valid pointer to a valid `uint32_t` value
- VUID-VkRenderingInputAttachmentIndexInfo-pStencilInputAttachmentIndex-parameter If `pStencilInputAttachmentIndex` is not `NULL`, `pStencilInputAttachmentIndex` **must** be a valid pointer to a valid `uint32_t` value

Structure Chaining

Extends the structures

- VkCommandBufferInheritanceInfo
- VkGraphicsPipelineCreateInfo

Variables identified with the `InputAttachmentIndex` **must** only be used by a fragment stage. The numeric format of the subpass input **must** match the format of the corresponding input attachment, or the values of subpass loads from these variables are poison. If the framebuffer attachment contains both depth and stencil aspects, the numeric format of the subpass input determines if depth or stencil aspect is accessed by the shader.

See Input Attachment for more details.

19.6.1. Fragment Input Attachment Compatibility

An input attachment that is statically accessed by a fragment shader **must** be backed by a descriptor that is equivalent to the VkImageView in the VkFramebuffer, except for `subresourceRange.aspectMask`. The `aspectMask` **must** be equal to the aspect accessed by the shader.

19.7. Ray Tracing Pipeline Interface

Ray tracing pipelines **may** have more stages than other pipelines with multiple instances of each stage and more dynamic interactions between the stages, but still have interface structures that obey the same general rules as interfaces between shader stages in other pipelines. The three types of inter-stage interface variables for ray tracing pipelines are:

- Ray payloads containing data preserved for the entire lifetime of the ray.
- Hit attributes containing data about a specific hit for the duration of its processing.
- Callable data for passing data into and out of a callable shader.

Ray payloads and callable data are used in explicit shader call instructions, so they have an incoming variant to distinguish the parameter passed to the invocation from any other payloads or data being used by subsequent shader call instructions.

An interface structure between stages **must** match between the stages using it. Specifically:

- If an intersection shader is present, the hit attribute structure read in an any-hit or closest hit shader **must** be the same structure as the hit attribute structure written in the corresponding intersection shader in the same hit group.
- If an intersection shader is not present, the hit attribute structure read in an any-hit or closest hit shader **must** be a vector of 2 32-bit floating-point values that accepts the barycentric coordinates for triangle hits.
- The incoming callable data for a callable shader **must** be the same structure as the callable data referenced by the execute callable instruction in the calling shader.
- The ray payload for a shader invoked by a ray tracing command **must** be the same structure for all shader stages using the payload for that ray, and **must** be declared in the shader even if it is not referenced.

Any shader with an incoming ray payload, incoming callable data, or hit attribute **must** only declare one variable of that type.

Table 18. Ray Pipeline Shader Interface

Shader Stage	Ray Payload	Incoming Ray Payload	Hit Attribute	Callable Data	Incoming Callable Data
Ray Generation	r/w			r/w	
Intersection			r/w		

Shader Stage	Ray Payload	Incoming Ray Payload	Hit Attribute	Callable Data	Incoming Callable Data
Any-Hit		r/w	r		
Closest Hit	r/w	r/w	r	r/w	
Miss	r/w	r/w		r/w	
Callable				r/w	r/w

19.8. Shader Resource Interface

Shaders **can** access resources and samplers through Resource Descriptors created in the API, via the Descriptor Heap Interface or the Descriptor Set Interface.

The set of shader variables that form the *shader resource interface* for a stage are the variables statically used by that stage's `OpEntryPoint` with a storage class of `Uniform`, `UniformConstant`, `StorageBuffer`, or `PushConstant`. For the fragment shader, this includes the fragment input attachment interface.

The shader resource interface is made up of the sub-interfaces defined in the following sections.

19.8.1. Push Constant Interface

The shader variables defined with a storage class of `PushConstant` that are statically used by the shader entry points for the pipeline define the *push constant interface*. They **must** be:

- typed as `OpTypeStruct`,
- identified with a `Block` decoration, and
- laid out explicitly using the `Offset`, `ArrayStride`, and `MatrixStride` decorations as specified in Offset and Stride Assignment.

There **must** be no more than one push constant block statically used per shader entry point.

When using descriptor heaps, each statically used member of a push constant block **must** be placed at an `Offset` such that the entire member is entirely contained within the range specified by vkCmdPushDataEXT in the command buffer. The `Offset` decoration for any member of a push constant block **must** not cause the space required for that member to extend outside the range [0, `maxPushDataSize`).

When using descriptor buffers or descriptor sets, each statically used member of a push constant block **must** be placed at an `Offset` such that the entire member is entirely contained within the VkPushConstantRange for each `OpEntryPoint` that uses it, and the `stageFlags` for that range **must** specify the appropriate VkShaderStageFlagBits for that stage. The `Offset` decoration for any member of a push constant block **must** not cause the space required for that member to extend outside the range [0, `maxPushConstantsSize`).

Any member of a push constant block that is declared as an array **must** only be accessed with

dynamically uniform indices.

19.8.2. Descriptor Heap Interface

The *descriptor heap interface* provides access to resources through pointers to heaps of different types, without many of the restrictions that apply to the descriptor set interface.

Two built-in pointers are available to shaders:

- `SamplerHeapEXT` for samplers
- `ResourceHeapEXT` for resources

These built-ins **must** be declared as pointers in the `UniformConstant Storage Class`. These built-ins **must** not be used to access data outside of the heap bound to them.

These built-ins **can** be accessed non-uniformly, with no further decoration required, and with no dependency on other features or properties. The `UniformId` decoration **can** be applied to the result of accesses to indicate that the data will be accessed uniformly to a given scope, as a hint to improve performance on some implementations.

Resources retrieved from each heap **must** have been created with descriptors that match the variable being declared, as follows:

Table 19. Heap Resource Type Correspondence

Descriptor Type	Heap	SPIR-V Type
VK_DESCRIPTOR_TYPE_SAMPLER	`SamplerHeapEXT`	`OpTypeSampler`
VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE	`ResourceHeapEXT`	`OpTypeImage`
VK_DESCRIPTOR_TYPE_STORAGE_IMAGE	`ResourceHeapEXT`	`OpTypeImage`
VK_DESCRIPTOR_TYPE_SAMPLE_WEIGHT_IMAGE_QCOM	`ResourceHeapEXT`	`OpTypeImage`
VK_DESCRIPTOR_TYPE_BLOCK_MATCH_IMAGE_QCOM	`ResourceHeapEXT`	`OpTypeImage`
VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT	`ResourceHeapEXT`	`OpTypeImage`
VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER	`ResourceHeapEXT`	`OpTypeImage`
VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER	`ResourceHeapEXT`	`OpTypeImage`
VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER	`ResourceHeapEXT`	`OpTypeBufferEXT` with the `Uniform Storage Class`
VK_DESCRIPTOR_TYPE_STORAGE_BUFFER	`ResourceHeapEXT`	`OpTypeBufferEXT` with the `StorageBuffer Storage Class`

Descriptor Type	Heap	SPIR-V Type
VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR	ResourceHeapEXT	OpTypeAccelerationStructureKHR

While the built-in heap pointers **can** be declared and dereferenced as pointing to any type, applications **must** not access data types valid for one heap from any other heap.

When one of the types above is read from a heap in the shader, it will read a number of bytes equal to the value advertised for the VkDescriptorType as returned by vkGetPhysicalDeviceDescriptorSizeEXT.

For image types, there are further restrictions on the operands used for the type, according to the descriptor type:

Table 20. Heap Image Operand Correspondence

Descriptor Type	Dim	Arrayed	MS	Sampled
VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE	*	*	0	1
VK_DESCRIPTOR_TYPE_STORAGE_IMAGE	*	*	0 or 1	2
VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT	SubpassData	0	0 or 1	2
VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER	Buffer	0	0	2
VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER	Buffer	0	0	2
VK_DESCRIPTOR_TYPE_SAMPLE_WEIGHT_IMAGE_QCOM	2D	1	0	1
VK_DESCRIPTOR_TYPE_BLOCK_MATCH_IMAGE_QCOM	2D	0	0	1

For storage images and input attachments, MS is 0 if the image has one sample per pixel, or 1 otherwise. For sampled and storage images, the Dim and Arrayed qualifiers depend on the VkImageViewType specified when writing the descriptor:

Table 21. Heap Image View Type Dimensionality

Image View Type	Dim	Arrayed
VK_IMAGE_VIEW_TYPE_1D	1D	0
VK_IMAGE_VIEW_TYPE_2D	2D	0
VK_IMAGE_VIEW_TYPE_3D	3D	0
VK_IMAGE_VIEW_TYPE_CUBE	Cube	0

Image View Type	Dim	Arrayed
VK_IMAGE_VIEW_TYPE_1D_ARRAY	1D	1
VK_IMAGE_VIEW_TYPE_2D_ARRAY	2D	1
VK_IMAGE_VIEW_TYPE_CUBE_ARRAY	Cube	1

The type and format of the image resource **must** also match between the API and SPIR-V.

Descriptors accessed via the ResourceHeapEXT built-in **must** be explicitly laid out.

There is no further limit to the number of resources that **can** be accessed by a shader through a heap pointer beyond the size of the bound range for each heap.

19.8.3. Descriptor Set Interface

The *descriptor set interface* is comprised of the shader variables with the storage class of StorageBuffer, Uniform, or UniformConstant (including the variables in the fragment input attachment interface) that are statically used by the shader entry points for the pipeline.

When using descriptor heaps, this interface is not used directly to access heaps, but **may** be accessed by specifying Shader Bindings to map to the descriptor heaps.

These variables **must** have DescriptorSet and Binding decorations specified, which are assigned and matched with the VkDescriptorSetLayout objects in the pipeline layout as described in DescriptorSet and Binding Assignment.

The Image Format of an OpTypeImage declaration **must** not be **Unknown**, for variables which are used for OpImageRead, OpImageSparseRead, or OpImageWrite operations, except under the following conditions:

- For OpImageWrite, if the image format is listed in the storage without format list and if the shaderStorageImageWriteWithoutFormat feature is enabled and the shader module declares the StorageImageWriteWithoutFormat capability.
- For OpImageWrite, if the image format supports VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT and the shader module declares the StorageImageWriteWithoutFormat capability.
- For OpImageRead or OpImageSparseRead, if the image format is listed in the storage without format list and if the shaderStorageImageReadWithoutFormat feature is enabled and the shader module declares the StorageImageReadWithoutFormat capability.
- For OpImageRead or OpImageSparseRead, if the image format supports VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT and the shader module declares the StorageImageReadWithoutFormat capability.
- For OpImageRead, if Dim is SubpassData (indicating a read from an input attachment).

The Image Format of an OpTypeImage declaration **must** not be **Unknown**, for variables which are used

for OpAtomic* operations.

Variables identified with the Uniform storage class are used to access transparent buffer backed resources. Such variables **must** be:

- typed as OpTypeStruct, or an array of this type,
- identified with a Block or BufferBlock decoration, and
- laid out explicitly using the Offset, ArrayStride, and MatrixStride decorations as specified in Offset and Stride Assignment.

Variables identified with the StorageBuffer storage class are used to access transparent buffer backed resources. Such variables **must** be:

- typed as OpTypeStruct, or an array of this type,
- identified with a Block decoration, and
- laid out explicitly using the Offset, ArrayStride, and MatrixStride decorations as specified in Offset and Stride Assignment.

The Offset decoration for any member of a Block-decorated variable in the Uniform storage class **must** not cause the space required for that variable to extend outside the range [0, maxUniformBufferRange). The Offset decoration for any member of a Block-decorated variable in the StorageBuffer storage class **must** not cause the space required for that variable to extend outside the range [0, maxStorageBufferRange).

Variables identified with the Uniform storage class **can** also be used to access transparent descriptor set backed resources when the variable is assigned to a descriptor set layout binding with a descriptorType of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK. In this case the variable **must** be typed as OpTypeStruct and **cannot** be aggregated into arrays of that type. Further, the Offset decoration for any member of such a variable **must** not cause the space required for that variable to extend outside the range [0,maxInlineUniformBlockSize).

Variables identified with a storage class of UniformConstant and a decoration of InputAttachmentIndex **must** be declared as described in Fragment Input Attachment Interface.

SPIR-V variables decorated with a descriptor set and binding that identify a combined image sampler descriptor **can** have a type of OpTypeImage, OpTypeSampler (Sampled=1), or OpTypeSampledImage.

NOTE	When using descriptor heaps, there is no combined image sampler descriptor type, but shader variables of OpTypeSampledImage can be mapped to a separate image and sampler for compatibility.

When accessing a resource through such a variable, the resource **must** be selected via compile time constant expressions unless features are enabled to allow dynamically uniform or non-uniform expressions, as described below:

- Storage images (except storage texel buffers and input attachments):
  - Dynamically uniform: shaderStorageImageArrayDynamicIndexing and StorageImageArrayDynamicIndexing

    - Non-uniform: `shaderStorageImageArrayNonUniformIndexing` and `StorageImageArrayNonUniformIndexing`
- Storage texel buffers:
    - Dynamically uniform: `shaderStorageTexelBufferArrayDynamicIndexing` and `StorageTexelBufferArrayDynamicIndexing`
    - Non-uniform: `shaderStorageTexelBufferArrayNonUniformIndexing` and `StorageTexelBufferArrayNonUniformIndexing`
- Input attachments:
    - Dynamically uniform: `shaderInputAttachmentArrayDynamicIndexing` and `InputAttachmentArrayDynamicIndexing`
    - Non-uniform: `shaderInputAttachmentArrayNonUniformIndexing` and `InputAttachmentArrayNonUniformIndexing`
- Sampled images (except uniform texel buffers), samplers and combined image samplers:
    - Dynamically uniform: `shaderSampledImageArrayDynamicIndexing` and `SampledImageArrayDynamicIndexing`
    - Non-uniform: `shaderSampledImageArrayNonUniformIndexing` and `SampledImageArrayNonUniformIndexing`
- Uniform texel buffers:
    - Dynamically uniform: `shaderUniformTexelBufferArrayDynamicIndexing` and `UniformTexelBufferArrayDynamicIndexing`
    - Non-uniform: `shaderUniformTexelBufferArrayNonUniformIndexing` and `UniformTexelBufferArrayNonUniformIndexing`
- Uniform buffers:
    - Dynamically uniform: `shaderUniformBufferArrayDynamicIndexing` and `UniformBufferArrayDynamicIndexing`
    - Non-uniform: `shaderUniformBufferArrayNonUniformIndexing` and `UniformBufferArrayNonUniformIndexing`
- Storage buffers:
    - Dynamically uniform: `shaderStorageBufferArrayDynamicIndexing` and `StorageBufferArrayDynamicIndexing`
    - Non-uniform: `shaderStorageBufferArrayNonUniformIndexing` and `StorageBufferArrayNonUniformIndexing`
- Acceleration structures:
    - Dynamically uniform: Always supported.
    - Non-uniform: Always supported.

NOTE

Implementations must take care when implementing this if subgroups are not necessarily a subset of the invocation group for their hardware (e.g. if multiple draw calls are packed together). If such an implementation needs uniformity across the subgroup for any resource access, the implementation is responsible for

> ensuring that indices that are only dynamically uniform across the invocation group still function as expected if they advertise the respective feature.

A combined image sampler in an array that enables sampler $Y'C_BC_R$ conversion or samples a subsampled image **must** only be indexed by constant integral expressions.

Table 22. Shader Resource and Descriptor Type Correspondence

Resource type	Descriptor Type
sampler	VK_DESCRIPTOR_TYPE_SAMPLER or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER
sampled image	VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER
storage image	VK_DESCRIPTOR_TYPE_STORAGE_IMAGE
combined image sampler	VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER or both VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE and VK_DESCRIPTOR_TYPE_SAMPLER when using heaps
uniform texel buffer	VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER
storage texel buffer	VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER
uniform buffer	VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC
storage buffer	VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC
input attachment	VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT
inline uniform block	VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK
acceleration structure	VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR

Table 23. Shader Resource and Storage Class Correspondence

Resource type	Storage Class	Type[1]	Decoration(s)[2]
sampler	UniformConstant	OpTypeSampler	
sampled image	UniformConstant	OpTypeImage (Sampled=1)	
storage image	UniformConstant	OpTypeImage (Sampled=2)	
combined image sampler	UniformConstant	OpTypeSampledImage OpTypeImage (Sampled=1) OpTypeSampler	
uniform texel buffer	UniformConstant	OpTypeImage (Dim=Buffer, Sampled=1)	
storage texel buffer	UniformConstant	OpTypeImage (Dim=Buffer, Sampled=2)	
uniform buffer	Uniform	OpTypeStruct	Block, Offset, (ArrayStride), (MatrixStride)

Resource type	Storage Class	Type[1]	Decoration(s)[2]
storage buffer	Uniform	OpTypeStruct	BufferBlock, Offset, (ArrayStride), (MatrixStride)
	StorageBuffer		Block, Offset, (ArrayStride), (MatrixStride)
input attachment	UniformConstant	OpTypeImage (Dim =SubpassData, Sampled=2)	InputAttachmentIndex
inline uniform block	Uniform	OpTypeStruct	Block, Offset, (ArrayStride), (MatrixStride)
acceleration structure	UniformConstant	OpTypeAccelerationStructureKHR	

1

Where OpTypeImage is referenced, the Dim values Buffer and SubpassData are only accepted where they are specifically referenced. They do not correspond to resource types where a generic OpTypeImage is specified.

2

In addition to DescriptorSet and Binding.

19.8.4. DescriptorSet and Binding Assignment

A variable decorated with a DescriptorSet decoration of s and a Binding decoration of b indicates that this variable is associated with the VkDescriptorSetLayoutBinding that has a binding equal to b in pSetLayouts[s] that was specified in VkPipelineLayoutCreateInfo. If using descriptor heaps, such a variable will instead be associated with a shader binding.

If not using descriptor heaps, DescriptorSet decoration values **must** be between zero and maxBoundDescriptorSets minus one, inclusive. If a pipeline is created with VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, or a shader is created with VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT, DescriptorSet decorations **can** be any 32-bit unsigned integer value. Binding decoration values **can** be any 32-bit unsigned integer value. Each descriptor set has its own binding name space.

If the Binding decoration is used with an array, the entire array is assigned that binding value. The decorated array **must** have an Element Type corresponding to a descriptor type, and the size of the array **must** be no larger than the number of descriptors in the binding. If the array is runtime-sized, then array elements greater than or equal to the size of that binding in the bound descriptor set **must** not be used. If the array is runtime-sized, the runtimeDescriptorArray feature **must** be enabled and the RuntimeDescriptorArray capability **must** be declared. The index of each element of the array is referred to as the *arrayElement*. For the purposes of interface matching and descriptor set operations, if a resource variable is not an array, it is treated as if it has an arrayElement of zero.

NOTE Even though an array is assigned to a single Binding, when using descriptor heap mappings, each element will be assigned a consecutive offset according to the stride for that type. Applications should take care to set subsequent bindings with this in

mind to avoid unintentional overlap.

There is a limit on the number of resources of each type that **can** be accessed by a pipeline stage as shown in Shader Resource Limits. The “Resources Per Stage” column gives the limit on the number of each type of resource that **can** be statically used for an entry point in any given stage in a pipeline. The “Resource Types” column lists which resource types are counted against the limit. Some resource types count against multiple limits. The VK_DESCRIPTOR_TYPE_MUTABLE_EXT descriptor type counts as one individual resource and one for every unique resource limit per descriptor set type that is present in the associated binding’s VkMutableDescriptorTypeListEXT. If multiple descriptor types in VkMutableDescriptorTypeListEXT map to the same resource limit, only one descriptor is consumed for purposes of computing resource limits. These limits only apply to resources accessed with `DescriptorSet` and `Binding` values.

A pipeline layout **may** include descriptor sets and bindings which are not referenced by any variables statically used by the entry points for the shader stages in the binding’s `stageFlags`. Similarly, descriptor heap bindings **may** include mappings that are unused by the shader.

However, if a variable assigned to a given `DescriptorSet` and `Binding` is statically used by the entry point for a shader stage, the heap bindings **must** specify a mapping for it when using heaps, or the pipeline layout **must** contain a descriptor set layout binding in that descriptor set layout and for that binding number, and that binding’s `stageFlags` **must** include the appropriate VkShaderStageFlagBits for that stage. The variable **must** be of a valid resource type determined by its SPIR-V type and storage class, as defined in Shader Resource and Storage Class Correspondence. The descriptor set layout binding **must** be of a corresponding descriptor type, as defined in Shader Resource and Descriptor Type Correspondence.

NOTE

There are no limits on the number of shader variables that can have overlapping set and binding values in a shader; but which resources are statically used has an impact. If any shader variable identifying a resource is statically used in a shader, then the underlying descriptor bound at the declared set and binding **must** support the declared type in the shader when the shader executes.

If multiple shader variables are declared with the same set and binding values, and with the same underlying descriptor type, they can all be statically used within the same shader. However, accesses are not automatically synchronized, and `Aliased` decorations should be used to avoid data hazards (see section 2.18.2 Aliasing in the SPIR-V specification).

If multiple shader variables with the same set and binding values are declared in a single shader, but with different declared types, where any of those are not supported by the relevant bound descriptor, that shader can only be executed if the variables with the unsupported type are not statically used.

A noteworthy example of using multiple statically-used shader variables sharing the same descriptor set and binding values is a descriptor of type VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER that has multiple corresponding shader variables in the `UniformConstant` storage class, where some could be `OpTypeImage` (`Sampled`=1), some could be `OpTypeSampler`, and some could be

`OpTypeSampledImage`.

Table 24. Shader Resource Limits

Resources per Stage	Resource Types
`maxPerStageDescriptorSamplers` or `maxPerStageDescriptorUpdateAfterBindSamplers`	sampler
	combined image sampler
`maxPerStageDescriptorSampledImages` or `maxPerStageDescriptorUpdateAfterBindSampledImages`	sampled image
	combined image sampler
	uniform texel buffer
`maxPerStageDescriptorStorageImages` or `maxPerStageDescriptorUpdateAfterBindStorageImages`	storage image
	storage texel buffer
`maxPerStageDescriptorUniformBuffers` or `maxPerStageDescriptorUpdateAfterBindUniformBuffers`	uniform buffer
	uniform buffer dynamic
`maxPerStageDescriptorStorageBuffers` or `maxPerStageDescriptorUpdateAfterBindStorageBuffers`	storage buffer
	storage buffer dynamic
`maxPerStageDescriptorInputAttachments` or `maxPerStageDescriptorUpdateAfterBindInputAttachments`	input attachment[1]
`maxPerStageDescriptorInlineUniformBlocks` or `maxPerStageDescriptorUpdateAfterBindInlineUniformBlocks`	inline uniform block
`maxPerStageDescriptorAccelerationStructures` or `maxPerStageDescriptorUpdateAfterBindAccelerationStructures`	acceleration structure

NOTE Resources accessed directly through the descriptor heap interface do not count towards these limits.

1

Input attachments **can** only be used in the fragment shader stage.

19.8.5. Offset and Stride Assignment

When a SPIR-V object is declared using an explicit layout, it **must** be laid out according to the following additional requirements.

NOTE The numeric order of `Offset` decorations does not need to follow member declaration order.

Alignment Requirements

There are different alignment requirements depending on the specific resources and on the features enabled.

Matrix types are defined in terms of arrays as follows:

- A column-major matrix with C columns and R rows is equivalent to a C element array of vectors with R components.
- A row-major matrix with C columns and R rows is equivalent to an R element array of vectors with C components.

The *scalar alignment* of the type of an `OpTypeStruct` member is defined recursively as follows:

- A scalar of size N has a scalar alignment of N.
- A vector type has a scalar alignment equal to that of its component type.
- An array type has a scalar alignment equal to that of its element type.
- A structure has a scalar alignment equal to the largest scalar alignment of any of its members.
- A matrix type inherits *scalar alignment* from the equivalent array declaration.
- `OpTypeImage` has a scalar alignment equal to the value of `imageDescriptorAlignment`
- `OpTypeBufferEXT` has a scalar alignment equal to the value of `bufferDescriptorAlignment`
- `OpTypeSampler` has a scalar alignment equal to the value of `samplerDescriptorAlignment`

The aligned size of a `OpTypeImage`, `OpTypeBufferEXT`, or `OpTypeSampler` can be queried from within SPIR-V using `OpConstantSizeOfEXT`, which can be used with the `OffsetIdEXT` or `ArrayStrideIdEXT` decorations to lay out types in a descriptor heap. `OpConstantSizeOfEXT` returns the following values for each type:

Table 25. Shader Descriptor Sizes

Type	Size
`OpTypeSampler`	`samplerDescriptorSize` aligned to `samplerDescriptorAlignment`
`OpTypeImage`	`imageDescriptorSize` aligned to `imageDescriptorAlignment`
`OpTypeBufferEXT`	`bufferDescriptorSize` aligned to `bufferDescriptorAlignment`
`OpTypeAccelerationStructureKHR`	`bufferDescriptorSize` aligned to `bufferDescriptorAlignment`

The *base alignment* of the type of an `OpTypeStruct` member is defined recursively as follows:

- A scalar has a base alignment equal to its scalar alignment.
- A two-component vector has a base alignment equal to twice its scalar alignment.
- A three- or four-component vector has a base alignment equal to four times its scalar alignment.
- An array has a base alignment equal to the base alignment of its element type.
- A structure has a base alignment equal to the largest base alignment of any of its members. An empty structure has a base alignment equal to the size of the smallest scalar type permitted by the capabilities declared in the SPIR-V module. For example, for a 1 byte aligned empty

structure in the StorageBuffer storage class, StorageBuffer8BitAccess or UniformAndStorageBuffer8BitAccess **must** be declared in the SPIR-V module.

- A matrix type inherits *base alignment* from the equivalent array declaration.

The *extended alignment* of the type of an OpTypeStruct member is similarly defined as follows:

- A scalar or vector type has an extended alignment equal to its base alignment.
- An array or structure type has an extended alignment equal to the largest extended alignment of any of its members, rounded up to a multiple of 16.
- A matrix type inherits extended alignment from the equivalent array declaration.

A member is defined to *improperly straddle* if either of the following are true:

- It is a vector with total size less than or equal to 16 bytes, and has Offset decorations placing its first byte at F and its last byte at L, where floor(F / 16) != floor(L / 16).
- It is a vector with total size greater than 16 bytes and has its Offset decorations placing its first byte at a non-integer multiple of 16.

Standard Buffer Layout

Every member of an OpTypeStruct that is required to be explicitly laid out **must** be aligned according to the first matching rule as follows. If the structure is contained in pointer types of multiple storage classes, it **must** satisfy the requirements for every storage class used to reference it.

1. If the scalarBlockLayout feature is enabled and the storage class is Uniform, StorageBuffer, PhysicalStorageBuffer, ShaderRecordBufferKHR, or PushConstant, or the storage class is UniformConstant and the type is decorated with either SamplerHeapEXT or ResourceHeapEXT, then every member **must** be aligned according to its scalar alignment.
2. If the workgroupMemoryExplicitLayoutScalarBlockLayout feature is enabled and the storage class is Workgroup then every member **must** be aligned according to its scalar alignment.
3. All vectors **must** be aligned according to their scalar alignment.
4. If the uniformBufferStandardLayout feature is not enabled, then any member of an OpTypeStruct with a storage class of Uniform and a decoration of Block **must** be aligned according to its extended alignment.
5. Every other member **must** be aligned according to its base alignment.

NOTE	Even if scalar alignment is supported, it is generally more performant to use the *base alignment*.

The memory layout **must** obey the following rules:

- The Offset or OffsetIdEXT decoration of any member **must** be a multiple of its alignment.
- Any ArrayStride, or ArrayStrideIdEXT, or MatrixStride decoration **must** be a multiple of the alignment of the array or matrix as defined above.

If one of the conditions below applies

- The storage class is `Uniform`, `StorageBuffer`, `PhysicalStorageBuffer`, `ShaderRecordBufferKHR`, or `PushConstant`, and the `scalarBlockLayout` feature is not enabled.
- The storage class is `Workgroup`, and either the structure member is not part of a `Block` or the `workgroupMemoryExplicitLayoutScalarBlockLayout` feature is not enabled.
- The storage class is any other storage class.

the memory layout **must** also obey the following rules:

- Vectors **must** not improperly straddle, as defined above.
- The `Offset` decoration of a member **must** not place it between the end of a structure, an array, or a matrix and the next multiple of the alignment of that structure, array, or matrix.

NOTE	The **std430 layout** in GLSL satisfies these rules for types using the base alignment. The **std140 layout** satisfies the rules for types using the extended alignment.

NOTE	An alternate explanation of the shader memory requirements, discussing the underlying extensions and core versions which affect them, is found in the Vulkan Guide.

19.9. Built-In Variables

Built-in variables are accessed in shaders by declaring a variable decorated with a `BuiltIn` SPIR-V decoration. The meaning of each `BuiltIn` decoration is as follows. In the remainder of this section, the name of a built-in is used interchangeably with a term equivalent to a variable decorated with that particular built-in. Built-ins that represent integer values **can** be declared as either signed or unsigned 32-bit integers.

As mentioned above, some inputs and outputs have an additional level of arrayness relative to other shader inputs and outputs. This level of arrayness is not included in the type descriptions below, but **must** be included when declaring the built-in.

Any two variables declared in the `Input` storage class listed as operands on the same `OpEntryPoint` **must** not have the same `BuiltIn` decoration. Any two variables declared in the `Output` storage class listed as operands on the same `OpEntryPoint` **must** not have the same `BuiltIn` decoration.

Built-in values for descriptor heaps are listed in the descriptor heap chapter:

- [SamplerHeapEXT]
- [ResourceHeapEXT]

Types used to access these built-ins **must** be laid out explicitly using the `Offset`, `OffsetIdEXT`, `ArrayStride`, `ArrayStrideIdEXT`, and `MatrixStride` decorations as specified in Offset and Stride Assignment.

`BaryCoordKHR`

The `BaryCoordKHR` decoration **can** be used to decorate a fragment shader input variable. This variable will contain a three-component floating-point vector with barycentric weights that indicate the location of the fragment relative to the screen-space locations of vertices of its primitive, obtained using perspective interpolation.

Valid Usage

- VUID-BaryCoordKHR-BaryCoordKHR-04154
  The `BaryCoordKHR` decoration **must** be used only within the `Fragment Execution Model`
- VUID-BaryCoordKHR-BaryCoordKHR-04155
  The variable decorated with `BaryCoordKHR` **must** be declared using the `Input Storage Class`
- VUID-BaryCoordKHR-BaryCoordKHR-04156
  The variable decorated with `BaryCoordKHR` **must** be declared as a three-component vector of 32-bit floating-point values

`BaryCoordNoPerspKHR`

The `BaryCoordNoPerspKHR` decoration **can** be used to decorate a fragment shader input variable. This variable will contain a three-component floating-point vector with barycentric weights that indicate the location of the fragment relative to the screen-space locations of vertices of its primitive, obtained using linear interpolation.

Valid Usage

- VUID-BaryCoordNoPerspKHR-BaryCoordNoPerspKHR-04160
  The `BaryCoordNoPerspKHR` decoration **must** be used only within the `Fragment Execution Model`
- VUID-BaryCoordNoPerspKHR-BaryCoordNoPerspKHR-04161
  The variable decorated with `BaryCoordNoPerspKHR` **must** be declared using the `Input Storage Class`
- VUID-BaryCoordNoPerspKHR-BaryCoordNoPerspKHR-04162
  The variable decorated with `BaryCoordNoPerspKHR` **must** be declared as a three-component vector of 32-bit floating-point values

`BaseInstance`

Decorating a variable with the `BaseInstance` built-in will make that variable contain the integer value corresponding to the first instance that was passed to the command that invoked the current vertex shader invocation. `BaseInstance` is the `firstInstance` parameter to a *direct drawing command* or the `firstInstance` member of a structure consumed by an *indirect drawing command*.

Valid Usage

- VUID-BaseInstance-BaseInstance-04181

The BaseInstance decoration **must** be used only within the Vertex Execution Model

- VUID-BaseInstance-BaseInstance-04182
  The variable decorated with BaseInstance **must** be declared using the Input Storage Class
- VUID-BaseInstance-BaseInstance-04183
  The variable decorated with BaseInstance **must** be declared as a scalar 32-bit integer value

BaseVertex

Decorating a variable with the BaseVertex built-in will make that variable contain the integer value corresponding to the first vertex or vertex offset that was passed to the command that invoked the current vertex shader invocation. For *non-indexed drawing commands*, this variable is the firstVertex parameter to a *direct drawing command* or the firstVertex member of the structure consumed by an *indirect drawing command*. For *indexed drawing commands*, this variable is the vertexOffset parameter to a *direct drawing command* or the vertexOffset member of the structure consumed by an *indirect drawing command*.

Valid Usage

- VUID-BaseVertex-BaseVertex-04184
  The BaseVertex decoration **must** be used only within the Vertex Execution Model
- VUID-BaseVertex-BaseVertex-04185
  The variable decorated with BaseVertex **must** be declared using the Input Storage Class
- VUID-BaseVertex-BaseVertex-04186
  The variable decorated with BaseVertex **must** be declared as a scalar 32-bit integer value

ClipDistance

Decorating a variable with the ClipDistance built-in decoration will make that variable contain the mechanism for controlling user clipping. ClipDistance is an array such that the i^{th} element of the array specifies the clip distance for plane i. A clip distance of 0 means the vertex is on the plane, a positive distance means the vertex is inside the clip half-space, and a negative distance means the vertex is outside the clip half-space.

NOTE The array variable decorated with ClipDistance is explicitly sized by the shader.

NOTE In the last pre-rasterization shader stage, these values will be linearly interpolated across the primitive and the portion of the primitive with interpolated distances less than 0 will be considered outside the clip volume. If ClipDistance is then used by a fragment shader, ClipDistance contains these linearly interpolated values.

Valid Usage

- VUID-ClipDistance-ClipDistance-04187
  The ClipDistance decoration **must** be used only within the MeshEXT, MeshNV, Vertex,

Fragment, TessellationControl, TessellationEvaluation, or Geometry Execution Model

- VUID-ClipDistance-ClipDistance-04188
  The variable decorated with ClipDistance within the MeshEXT, MeshNV, or Vertex Execution Model **must** be declared using the Output Storage Class
- VUID-ClipDistance-ClipDistance-04189
  The variable decorated with ClipDistance within the Fragment Execution Model **must** be declared using the Input Storage Class
- VUID-ClipDistance-ClipDistance-04190
  The variable decorated with ClipDistance within the TessellationControl, TessellationEvaluation, or Geometry Execution Model **must** not be declared in a Storage Class other than Input or Output
- VUID-ClipDistance-ClipDistance-04191
  The variable decorated with ClipDistance **must** be declared as an array of 32-bit floating-point values

CullDistance

Decorating a variable with the CullDistance built-in decoration will make that variable contain the mechanism for controlling user culling. If any member of this array is assigned a negative value for all vertices belonging to a primitive, then the primitive is discarded before rasterization.

NOTE In fragment shaders, the values of the CullDistance array are linearly interpolated across each primitive.

NOTE If CullDistance decorates an input variable, that variable will contain the corresponding value from the CullDistance decorated output variable from the previous shader stage.

Valid Usage

- VUID-CullDistance-CullDistance-04196
  The CullDistance decoration **must** be used only within the MeshEXT, MeshNV, Vertex, Fragment, TessellationControl, TessellationEvaluation, or Geometry Execution Model
- VUID-CullDistance-CullDistance-04197
  The variable decorated with CullDistance within the MeshEXT, MeshNV or Vertex Execution Model **must** be declared using the Output Storage Class
- VUID-CullDistance-CullDistance-04198
  The variable decorated with CullDistance within the Fragment Execution Model **must** be declared using the Input Storage Class
- VUID-CullDistance-CullDistance-04199
  The variable decorated with CullDistance within the TessellationControl, TessellationEvaluation, or Geometry Execution Model **must** not be declared using a Storage Class other than Input or Output

- VUID-CullDistance-CullDistance-04200
  The variable decorated with `CullDistance` **must** be declared as an array of 32-bit floating-point values

`CullPrimitiveEXT`

Decorating a variable with the `CullPrimitiveEXT` built-in decoration will make that variable contain the culling state of output primitives. If the per-primitive boolean value is `true`, the primitive will be culled, if it is `false` it will not be culled.

Valid Usage

- VUID-CullPrimitiveEXT-CullPrimitiveEXT-07034
  The `CullPrimitiveEXT` decoration **must** be used only within the `MeshEXT Execution Model`
- VUID-CullPrimitiveEXT-CullPrimitiveEXT-07035
  The variable decorated with `CullPrimitiveEXT` **must** be declared using the `Output Storage Class`
- VUID-CullPrimitiveEXT-CullPrimitiveEXT-07036
  `CullPrimitiveEXT` **must** decorate a scalar boolean member of a structure decorated as `Block`, or decorate a variable of type `OpTypeArray` of boolean values
- VUID-CullPrimitiveEXT-CullPrimitiveEXT-10589
  If `CullPrimitiveEXT` is declared as an array of boolean values, the size of the array **must** match the value specified by `OutputPrimitivesEXT`
- VUID-CullPrimitiveEXT-CullPrimitiveEXT-10590
  If `CullPrimitiveEXT` decorates a member of a structure, the variable declaration of the containing `Block` type **must** have an array size that matches the value specified by `OutputPrimitivesEXT`
- VUID-CullPrimitiveEXT-CullPrimitiveEXT-10591
  There must be only one declaration of the `CullPrimitiveEXT` associated with an entry point's interface
- VUID-CullPrimitiveEXT-CullPrimitiveEXT-07038
  The variable decorated with `CullPrimitiveEXT` within the `MeshEXT Execution Model` **must** also be decorated with the `PerPrimitiveEXT` decoration

`CullMaskKHR`

A variable decorated with the `CullMaskKHR` decoration will specify the cull mask of the ray being processed. The value is given by the `Cull Mask` parameter passed into one of the `OpTrace*` instructions.

Valid Usage

- VUID-CullMaskKHR-CullMaskKHR-06735
  The `CullMaskKHR` decoration **must** be used only within the `IntersectionKHR`, `AnyHitKHR`, `ClosestHitKHR`, or `MissKHR Execution Model`

- VUID-CullMaskKHR-CullMaskKHR-06736
  The variable decorated with `CullMaskKHR` **must** be declared using the `Input Storage Class`
- VUID-CullMaskKHR-CullMaskKHR-06737
  The variable decorated with `CullMaskKHR` **must** be declared as a scalar 32-bit integer value

`DeviceIndex`

The `DeviceIndex` decoration **can** be applied to a shader input which will be filled with the device index of the physical device that is executing the current shader invocation. This value will be in the range $[0, max(1, physicalDeviceCount))$, where physicalDeviceCount is the `physicalDeviceCount` member of VkDeviceGroupDeviceCreateInfo.

Valid Usage

- VUID-DeviceIndex-DeviceIndex-04205
  The variable decorated with `DeviceIndex` **must** be declared using the `Input Storage Class`
- VUID-DeviceIndex-DeviceIndex-04206
  The variable decorated with `DeviceIndex` **must** be declared as a scalar 32-bit integer value

`DrawIndex`

Decorating a variable with the `DrawIndex` built-in will make that variable contain the integer value corresponding to the zero-based index of the draw that invoked the current task, mesh, or vertex shader invocation. For *indirect drawing commands*, `DrawIndex` begins at zero and increments by one for each draw executed. The number of draws is given by the `drawCount` parameter. For *direct drawing commands*, if vkCmdDrawMultiEXT or vkCmdDrawMultiIndexedEXT is used, this variable contains the integer value corresponding to the zero-based index of the draw. Otherwise `DrawIndex` is always zero. `DrawIndex` is dynamically uniform.

When task or mesh shaders are used, only the first active stage will have proper access to the variable. The value read by other stages is poison.

Valid Usage

- VUID-DrawIndex-DrawIndex-04207
  The `DrawIndex` decoration **must** be used only within the `Vertex`, `MeshEXT`, `TaskEXT`, `MeshNV`, or `TaskNV Execution Model`
- VUID-DrawIndex-DrawIndex-04208
  The variable decorated with `DrawIndex` **must** be declared using the `Input Storage Class`
- VUID-DrawIndex-DrawIndex-04209
  The variable decorated with `DrawIndex` **must** be declared as a scalar 32-bit integer value

`FragCoord`

Decorating a variable with the `FragCoord` built-in decoration will make that variable contain the

coordinates (x,y,z,1/w) of the fragment being processed.

The (x,y) values are the framebuffer coordinates (x_f,y_f) of the fragment.

When Sample Shading is enabled, the x and y components of `FragCoord` reflect the location of one of the samples corresponding to the shader invocation.

Otherwise, the x and y components of `FragCoord` reflect the location of the center of the fragment.

The z component of `FragCoord` is the interpolated depth value of the primitive.

The w component is the interpolated $\frac{1}{w}$.

The `Centroid` interpolation decoration is ignored, but allowed, on `FragCoord`.

Valid Usage

- VUID-FragCoord-FragCoord-04210
  The `FragCoord` decoration **must** be used only within the `Fragment Execution Model`
- VUID-FragCoord-FragCoord-04211
  The variable decorated with `FragCoord` **must** be declared using the `Input Storage Class`
- VUID-FragCoord-FragCoord-04212
  The variable decorated with `FragCoord` **must** be declared as a four-component vector of 32-bit floating-point values

`FragDepth`

To have a shader supply a fragment-depth value, the shader **must** declare the `DepthReplacing` execution mode. Such a shader's fragment-depth value will come from the variable decorated with the `FragDepth` built-in decoration.

This value will be used for any subsequent depth testing performed by the implementation or writes to the depth attachment. See fragment shader depth replacement for details.

Valid Usage

- VUID-FragDepth-FragDepth-04213
  The `FragDepth` decoration **must** be used only within the `Fragment Execution Model`
- VUID-FragDepth-FragDepth-04214
  The variable decorated with `FragDepth` **must** be declared using the `Output Storage Class`
- VUID-FragDepth-FragDepth-04215
  The variable decorated with `FragDepth` **must** be declared as a scalar 32-bit floating-point value
- VUID-FragDepth-FragDepth-04216
  If the shader dynamically writes to the variable decorated with `FragDepth`, the `DepthReplacing Execution Mode` **must** be declared

`FragInvocationCountEXT`

Decorating a variable with the `FragInvocationCountEXT` built-in decoration will make that variable contain the maximum number of fragment shader invocations for the fragment, as determined by `minSampleShading`.

If Sample Shading is not enabled, `FragInvocationCountEXT` will be filled with a value of 1.

Valid Usage

- VUID-FragInvocationCountEXT-FragInvocationCountEXT-04217
  The `FragInvocationCountEXT` decoration **must** be used only within the `Fragment Execution Model`
- VUID-FragInvocationCountEXT-FragInvocationCountEXT-04218
  The variable decorated with `FragInvocationCountEXT` **must** be declared using the `Input Storage Class`
- VUID-FragInvocationCountEXT-FragInvocationCountEXT-04219
  The variable decorated with `FragInvocationCountEXT` **must** be declared as a scalar 32-bit integer value

`FragSizeEXT`

Decorating a variable with the `FragSizeEXT` built-in decoration will make that variable contain the dimensions in pixels of the area that the fragment covers for that invocation.

> **NOTE** When used in a custom resolve operation and the `customResolve` feature is enabled, the dimensions in pixels returned **may** be (1,1) if the fragment area was reduced.

If fragment density map is not enabled, `FragSizeEXT` will be filled with a value of (1,1).

Valid Usage

- VUID-FragSizeEXT-FragSizeEXT-04220
  The `FragSizeEXT` decoration **must** be used only within the `Fragment Execution Model`
- VUID-FragSizeEXT-FragSizeEXT-04221
  The variable decorated with `FragSizeEXT` **must** be declared using the `Input Storage Class`
- VUID-FragSizeEXT-FragSizeEXT-04222
  The variable decorated with `FragSizeEXT` **must** be declared as a two-component vector of 32-bit integer values

`FragStencilRefEXT`

Decorating a variable with the `FragStencilRefEXT` built-in decoration will make that variable contain the new stencil reference value for all samples covered by the fragment. This value will be used as the stencil reference value used in stencil testing.

To write to FragStencilRefEXT, a shader **must** declare the StencilRefReplacingEXT execution mode. If a shader declares the StencilRefReplacingEXT execution mode and there is an execution path through the shader that does not set FragStencilRefEXT, then the fragment's stencil reference value is undefined for executions of the shader that take that path.

Only the least significant **s** bits of the integer value of the variable decorated with FragStencilRefEXT are considered for stencil testing, where **s** is the number of bits in the stencil framebuffer attachment, and higher order bits are discarded.

See fragment shader stencil reference replacement for more details.

Valid Usage

- VUID-FragStencilRefEXT-FragStencilRefEXT-04223
  The FragStencilRefEXT decoration **must** be used only within the Fragment Execution Model
- VUID-FragStencilRefEXT-FragStencilRefEXT-04224
  The variable decorated with FragStencilRefEXT **must** be declared using the Output Storage Class
- VUID-FragStencilRefEXT-FragStencilRefEXT-04225
  The variable decorated with FragStencilRefEXT **must** be declared as a scalar integer value

FrontFacing

Decorating a variable with the FrontFacing built-in decoration will make that variable contain whether the fragment is front or back facing. This variable is non-zero if the current fragment is considered to be part of a front-facing polygon primitive or of a non-polygon primitive and is zero if the fragment is considered to be part of a back-facing polygon primitive.

Valid Usage

- VUID-FrontFacing-FrontFacing-04229
  The FrontFacing decoration **must** be used only within the Fragment Execution Model
- VUID-FrontFacing-FrontFacing-04230
  The variable decorated with FrontFacing **must** be declared using the Input Storage Class
- VUID-FrontFacing-FrontFacing-04231
  The variable decorated with FrontFacing **must** be declared as a boolean value

FullyCoveredEXT

Decorating a variable with the FullyCoveredEXT built-in decoration will make that variable indicate whether the fragment area is fully covered by the generating primitive. This variable is non-zero if conservative rasterization is enabled and the current fragment area is fully covered by the generating primitive, and is zero if the fragment is not covered or partially covered, or conservative rasterization is disabled.

Valid Usage

- VUID-FullyCoveredEXT-FullyCoveredEXT-04232
  The FullyCoveredEXT decoration **must** be used only within the Fragment Execution Model
- VUID-FullyCoveredEXT-FullyCoveredEXT-04233
  The variable decorated with FullyCoveredEXT **must** be declared using the Input Storage Class
- VUID-FullyCoveredEXT-FullyCoveredEXT-04234
  The variable decorated with FullyCoveredEXT **must** be declared as a boolean value
- VUID-FullyCoveredEXT-conservativeRasterizationPostDepthCoverage-04235
  If VkPhysicalDeviceConservativeRasterizationPropertiesEXT ::conservativeRasterizationPostDepthCoverage is not supported, the PostDepthCoverage Execution Mode **must** not be declared, when a variable with the FullyCoveredEXT decoration is declared

GlobalInvocationId

Decorating a variable with the GlobalInvocationId built-in decoration will make that variable contain the location of the current invocation within the global workgroup. Each component is equal to the index of the local workgroup multiplied by the size of the local workgroup plus LocalInvocationId.

Valid Usage

- VUID-GlobalInvocationId-GlobalInvocationId-04236
  The GlobalInvocationId decoration **must** be used only within the GLCompute, MeshEXT, TaskEXT, MeshNV, or TaskNV Execution Model
- VUID-GlobalInvocationId-GlobalInvocationId-04237
  The variable decorated with GlobalInvocationId **must** be declared using the Input Storage Class
- VUID-GlobalInvocationId-GlobalInvocationId-04238
  The variable decorated with GlobalInvocationId **must** be declared as a three-component vector of 32-bit integer values

HelperInvocation

Decorating a variable with the HelperInvocation built-in decoration will make that variable contain whether the current invocation is a helper invocation. This variable is non-zero if the current fragment being shaded is a helper invocation and zero otherwise. A helper invocation is an invocation of the shader that is produced to satisfy internal requirements such as the generation of derivatives.

NOTE It is very likely that a helper invocation will have a value of SampleMask fragment shader input value that is zero.

Valid Usage

- VUID-HelperInvocation-HelperInvocation-04239
  The `HelperInvocation` decoration **must** be used only within the `Fragment` `Execution Model`
- VUID-HelperInvocation-HelperInvocation-04240
  The variable decorated with `HelperInvocation` **must** be declared using the `Input` `Storage Class`
- VUID-HelperInvocation-HelperInvocation-04241
  The variable decorated with `HelperInvocation` **must** be declared as a boolean value

`HitKindKHR`

A variable decorated with the `HitKindKHR` decoration will describe the intersection that triggered the execution of the current shader. The values are determined by the intersection shader. For user-defined intersection shaders this is the value that was passed to the "Hit Kind" operand of `OpReportIntersectionKHR`. For triangle intersection candidates, this will be one of `HitKindFrontFacingTriangleKHR` or `HitKindBackFacingTriangleKHR`.

Valid Usage

- VUID-HitKindKHR-HitKindKHR-04242
  The `HitKindKHR` decoration **must** be used only within the `AnyHitKHR` or `ClosestHitKHR` `Execution Model`
- VUID-HitKindKHR-HitKindKHR-04243
  The variable decorated with `HitKindKHR` **must** be declared using the `Input` `Storage Class`
- VUID-HitKindKHR-HitKindKHR-04244
  The variable decorated with `HitKindKHR` **must** be declared as a scalar 32-bit integer value

`HitTriangleVertexPositionsKHR`

A variable decorated with the `HitTriangleVertexPositionsKHR` decoration will specify the object space vertices of the triangle at the current intersection in application-provided order. The positions returned are transformed by the geometry transform, which is performed at standard floating-point precision, but without a specifically defined order of floating-point operations to perform the matrix multiplication.

Valid Usage

- VUID-HitTriangleVertexPositionsKHR-HitTriangleVertexPositionsKHR-08747
  The `HitTriangleVertexPositionsKHR` decoration **must** be used only within the `AnyHitKHR` or `ClosestHitKHR` `Execution Model`
- VUID-HitTriangleVertexPositionsKHR-HitTriangleVertexPositionsKHR-08748
  The variable decorated with `HitTriangleVertexPositionsKHR` **must** be declared using the `Input` `Storage Class`
- VUID-HitTriangleVertexPositionsKHR-HitTriangleVertexPositionsKHR-08749

The variable decorated with HitTriangleVertexPositionsKHR **must** be declared as an array of three vectors of three 32-bit float values

- VUID-HitTriangleVertexPositionsKHR-HitTriangleVertexPositionsKHR-08750
  The variable decorated with HitTriangleVertexPositionsKHR **must** be used only if the value of HitKindKHR is HitKindFrontFacingTriangleKHR or HitKindBackFacingTriangleKHR
- VUID-HitTriangleVertexPositionsKHR-None-08751
  The acceleration structure corresponding to the current intersection **must** have been built with VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_BIT_KHR

IncomingRayFlagsKHR

A variable with the IncomingRayFlagsKHR decoration will contain the ray flags passed in to the trace call that invoked this particular shader. Setting pipeline flags on the ray tracing pipeline **must** not cause any corresponding flags to be set in variables with this decoration.

Valid Usage

- VUID-IncomingRayFlagsKHR-IncomingRayFlagsKHR-04248
  The IncomingRayFlagsKHR decoration **must** be used only within the IntersectionKHR, AnyHitKHR, ClosestHitKHR, or MissKHR Execution Model
- VUID-IncomingRayFlagsKHR-IncomingRayFlagsKHR-04249
  The variable decorated with IncomingRayFlagsKHR **must** be declared using the Input Storage Class
- VUID-IncomingRayFlagsKHR-IncomingRayFlagsKHR-04250
  The variable decorated with IncomingRayFlagsKHR **must** be declared as a scalar 32-bit integer value

InstanceCustomIndexKHR

A variable decorated with the InstanceCustomIndexKHR decoration will contain the application-defined value of the instance that intersects the current ray. This variable contains the value that was specified in VkAccelerationStructureInstanceKHR::instanceCustomIndex for the current acceleration structure instance in the lower 24 bits and the upper 8 bits will be zero.

Valid Usage

- VUID-InstanceCustomIndexKHR-InstanceCustomIndexKHR-04251
  The InstanceCustomIndexKHR decoration **must** be used only within the IntersectionKHR, AnyHitKHR, or ClosestHitKHR Execution Model
- VUID-InstanceCustomIndexKHR-InstanceCustomIndexKHR-04252
  The variable decorated with InstanceCustomIndexKHR **must** be declared using the Input Storage Class
- VUID-InstanceCustomIndexKHR-InstanceCustomIndexKHR-04253
  The variable decorated with InstanceCustomIndexKHR **must** be declared as a scalar 32-bit integer value

`InstanceId`

Decorating a variable in an intersection, any-hit, or closest hit shader with the `InstanceId` decoration will make that variable contain the index of the instance that intersects the current ray.

Valid Usage

- VUID-InstanceId-InstanceId-04254
  The `InstanceId` decoration **must** be used only within the `IntersectionKHR`, `AnyHitKHR`, or `ClosestHitKHR` `Execution Model`
- VUID-InstanceId-InstanceId-04255
  The variable decorated with `InstanceId` **must** be declared using the `Input` `Storage Class`
- VUID-InstanceId-InstanceId-04256
  The variable decorated with `InstanceId` **must** be declared as a scalar 32-bit integer value

`InvocationId`

Decorating a variable with the `InvocationId` built-in decoration will make that variable contain the index of the current shader invocation in a geometry shader, or the index of the output patch vertex in a tessellation control shader.

In a geometry shader, the index of the current shader invocation ranges from zero to the number of instances declared in the shader minus one. If the instance count of the geometry shader is one or is not specified, then `InvocationId` will be zero.

Valid Usage

- VUID-InvocationId-InvocationId-04257
  The `InvocationId` decoration **must** be used only within the `TessellationControl` or `Geometry` `Execution Model`
- VUID-InvocationId-InvocationId-04258
  The variable decorated with `InvocationId` **must** be declared using the `Input` `Storage Class`
- VUID-InvocationId-InvocationId-04259
  The variable decorated with `InvocationId` **must** be declared as a scalar 32-bit integer value

`InstanceIndex`

Decorating a variable in a vertex shader with the `InstanceIndex` built-in decoration will make that variable contain the index of the instance that is being processed by the current vertex shader invocation. `InstanceIndex` begins at the `firstInstance` parameter to vkCmdDraw or vkCmdDrawIndexed or at the `firstInstance` member of a structure consumed by vkCmdDrawIndirect or vkCmdDrawIndexedIndirect.

Valid Usage

- VUID-InstanceIndex-InstanceIndex-04263
  The `InstanceIndex` decoration **must** be used only within the `Vertex` Execution Model
- VUID-InstanceIndex-InstanceIndex-04264
  The variable decorated with `InstanceIndex` **must** be declared using the `Input` Storage Class
- VUID-InstanceIndex-InstanceIndex-04265
  The variable decorated with `InstanceIndex` **must** be declared as a scalar 32-bit integer value

`LaunchIdKHR`

A variable decorated with the `LaunchIdKHR` decoration will specify the index of the work item being processed. One work item is generated for each of the `width` × `height` × `depth` items dispatched by a vkCmdTraceRaysKHR command. All shader invocations inherit the same value for variables decorated with `LaunchIdKHR`.

Valid Usage

- VUID-LaunchIdKHR-LaunchIdKHR-04266
  The `LaunchIdKHR` decoration **must** be used only within the `RayGenerationKHR`, `IntersectionKHR`, `AnyHitKHR`, `ClosestHitKHR`, `MissKHR`, or `CallableKHR` Execution Model
- VUID-LaunchIdKHR-LaunchIdKHR-04267
  The variable decorated with `LaunchIdKHR` **must** be declared using the `Input` Storage Class
- VUID-LaunchIdKHR-LaunchIdKHR-04268
  The variable decorated with `LaunchIdKHR` **must** be declared as a three-component vector of 32-bit integer values

`LaunchSizeKHR`

A variable decorated with the `LaunchSizeKHR` decoration will contain the `width`, `height`, and `depth` dimensions passed to the vkCmdTraceRaysKHR command that initiated this shader execution. The `width` is in the first component, the `height` is in the second component, and the `depth` is in the third component.

Valid Usage

- VUID-LaunchSizeKHR-LaunchSizeKHR-04269
  The `LaunchSizeKHR` decoration **must** be used only within the `RayGenerationKHR`, `IntersectionKHR`, `AnyHitKHR`, `ClosestHitKHR`, `MissKHR`, or `CallableKHR` Execution Model
- VUID-LaunchSizeKHR-LaunchSizeKHR-04270
  The variable decorated with `LaunchSizeKHR` **must** be declared using the `Input` Storage Class
- VUID-LaunchSizeKHR-LaunchSizeKHR-04271
  The variable decorated with `LaunchSizeKHR` **must** be declared as a three-component vector of 32-bit integer values

`Layer`

Decorating a variable with the `Layer` built-in decoration will make that variable contain the selected layer of a multi-layer framebuffer attachment.

In a mesh, vertex, tessellation evaluation, or geometry shader, any variable decorated with `Layer` can be written with the framebuffer layer index to which the primitive produced by that shader will be directed.

The last active pre-rasterization shader stage (in pipeline order) controls the `Layer` that is used. Outputs in previous shader stages are not used, even if the last stage fails to write the `Layer`.

If the last active pre-rasterization shader stage shader entry point's interface does not include a variable decorated with `Layer`, then the first layer is used. If a pre-rasterization shader stage shader entry point's interface includes a variable decorated with `Layer`, it **must** write the same value to `Layer` for all output vertices of a given primitive. If the `Layer` value is less than 0 or greater than or equal to the number of layers in the framebuffer, then primitives **may** still be rasterized, fragment shaders **may** be executed, and the framebuffer values for all layers are undefined. In a mesh shader this also applies when the `Layer` value is greater than or equal to the `maxMeshOutputLayers` limit.

In a fragment shader, a variable decorated with `Layer` contains the layer index of the primitive that the fragment invocation belongs to.

Valid Usage

- VUID-Layer-Layer-04272
  The `Layer` decoration **must** be used only within the `MeshEXT`, `MeshNV`, `Vertex`, `TessellationEvaluation`, `Geometry`, or `Fragment` `Execution Model`
- VUID-Layer-Layer-04273
  If the `shaderOutputLayer` feature is not enabled then the `Layer` decoration **must** be used only within the `Geometry` or `Fragment` `Execution Model`
- VUID-Layer-Layer-04274
  The variable decorated with `Layer` within the `MeshEXT`, `MeshNV`, `Vertex`, `TessellationEvaluation`, or `Geometry` `Execution Model` **must** be declared using the `Output` `Storage Class`
- VUID-Layer-Layer-04275
  The variable decorated with `Layer` within the `Fragment` `Execution Model` **must** be declared using the `Input` `Storage Class`
- VUID-Layer-Layer-04276
  The variable decorated with `Layer` **must** be declared as a scalar 32-bit integer value for all supported execution models except `MeshEXT`
- VUID-Layer-Layer-07039
  The variable decorated with `Layer` within the `MeshEXT` `Execution Model` **must** also be decorated with the `PerPrimitiveEXT` decoration
- VUID-Layer-Layer-10592
  `Layer` within the `MeshEXT` `Execution Model` **must** decorate a scalar 32-bit integer member of

a structure decorated as Block, or decorate a variable of type OpTypeArray of scalar 32-bit integer values

- VUID-Layer-Layer-10593
  If Layer is declared as an array of 32-bit integer values, within the MeshEXT Execution Model, the size of the array **must** match the value specified by OutputPrimitivesEXT
- VUID-Layer-Layer-10594
  If Layer decorates a member of a structure, the variable declaration of the containing Block type **must** have an array size that matches the value specified by OutputPrimitivesEXT

LocalInvocationId

Decorating a variable with the LocalInvocationId built-in decoration will make that variable contain the location of the current task, mesh, or compute shader invocation within the local workgroup. Each component ranges from zero through to the size of the workgroup in that dimension minus one.

NOTE If the size of the workgroup in a particular dimension is one, then the LocalInvocationId in that dimension will be zero. If the workgroup is effectively two-dimensional, then LocalInvocationId.z will be zero. If the workgroup is effectively one-dimensional, then both LocalInvocationId.y and LocalInvocationId.z will be zero.

Valid Usage

- VUID-LocalInvocationId-LocalInvocationId-04281
  The LocalInvocationId decoration **must** be used only within the GLCompute, MeshEXT, TaskEXT, MeshNV, or TaskNV Execution Model
- VUID-LocalInvocationId-LocalInvocationId-04282
  The variable decorated with LocalInvocationId **must** be declared using the Input Storage Class
- VUID-LocalInvocationId-LocalInvocationId-04283
  The variable decorated with LocalInvocationId **must** be declared as a three-component vector of 32-bit integer values

LocalInvocationIndex

Decorating a variable with the LocalInvocationIndex built-in decoration will make that variable contain a one-dimensional representation of LocalInvocationId. This is computed as:

```
LocalInvocationIndex =
    LocalInvocationId.z * WorkgroupSize.x * WorkgroupSize.y +
    LocalInvocationId.y * WorkgroupSize.x +
    LocalInvocationId.x;
```

Valid Usage

- VUID-LocalInvocationIndex-LocalInvocationIndex-04284
  The `LocalInvocationIndex` decoration **must** be used only within the `GLCompute`, `MeshEXT`, `TaskEXT`, `MeshNV`, or `TaskNV` `Execution Model`
- VUID-LocalInvocationIndex-LocalInvocationIndex-04285
  The variable decorated with `LocalInvocationIndex` **must** be declared using the `Input` `Storage Class`
- VUID-LocalInvocationIndex-LocalInvocationIndex-04286
  The variable decorated with `LocalInvocationIndex` **must** be declared as a scalar 32-bit integer value

`NumSubgroups`

Decorating a variable with the `NumSubgroups` built-in decoration will make that variable contain the number of subgroups in the local workgroup.

Valid Usage

- VUID-NumSubgroups-NumSubgroups-04293
  The `NumSubgroups` decoration **must** be used only within the `GLCompute`, `MeshEXT`, `TaskEXT`, `MeshNV`, or `TaskNV` `Execution Model`
- VUID-NumSubgroups-NumSubgroups-04294
  The variable decorated with `NumSubgroups` **must** be declared using the `Input` `Storage Class`
- VUID-NumSubgroups-NumSubgroups-04295
  The variable decorated with `NumSubgroups` **must** be declared as a scalar 32-bit integer value

`NumWorkgroups`

Decorating a variable with the `NumWorkgroups` built-in decoration will make that variable contain the number of local workgroups that are part of the dispatch that the invocation belongs to. Each component is equal to the values of the workgroup count parameters passed into the dispatching commands.

Valid Usage

- VUID-NumWorkgroups-NumWorkgroups-04296
  The `NumWorkgroups` decoration **must** be used only within the `GLCompute`, `MeshEXT`, or `TaskEXT` `Execution Model`
- VUID-NumWorkgroups-NumWorkgroups-04297
  The variable decorated with `NumWorkgroups` **must** be declared using the `Input` `Storage Class`
- VUID-NumWorkgroups-NumWorkgroups-04298
  The variable decorated with `NumWorkgroups` **must** be declared as a three-component vector of 32-bit integer values

`ObjectRayDirectionKHR`

A variable decorated with the `ObjectRayDirectionKHR` decoration will specify the direction of the ray being processed, in object space.

Valid Usage

- VUID-ObjectRayDirectionKHR-ObjectRayDirectionKHR-04299
  The `ObjectRayDirectionKHR` decoration **must** be used only within the `IntersectionKHR`, `AnyHitKHR`, or `ClosestHitKHR` `Execution Model`
- VUID-ObjectRayDirectionKHR-ObjectRayDirectionKHR-04300
  The variable decorated with `ObjectRayDirectionKHR` **must** be declared using the `Input` `Storage Class`
- VUID-ObjectRayDirectionKHR-ObjectRayDirectionKHR-04301
  The variable decorated with `ObjectRayDirectionKHR` **must** be declared as a three-component vector of 32-bit floating-point values

`ObjectRayOriginKHR`

A variable decorated with the `ObjectRayOriginKHR` decoration will specify the origin of the ray being processed, in object space.

Valid Usage

- VUID-ObjectRayOriginKHR-ObjectRayOriginKHR-04302
  The `ObjectRayOriginKHR` decoration **must** be used only within the `IntersectionKHR`, `AnyHitKHR`, or `ClosestHitKHR` `Execution Model`
- VUID-ObjectRayOriginKHR-ObjectRayOriginKHR-04303
  The variable decorated with `ObjectRayOriginKHR` **must** be declared using the `Input` `Storage Class`
- VUID-ObjectRayOriginKHR-ObjectRayOriginKHR-04304
  The variable decorated with `ObjectRayOriginKHR` **must** be declared as a three-component vector of 32-bit floating-point values

`ObjectToWorldKHR`

A variable decorated with the `ObjectToWorldKHR` decoration will contain the current object-to-world transformation matrix, which is determined by the instance of the current intersection.

Valid Usage

- VUID-ObjectToWorldKHR-ObjectToWorldKHR-04305
  The `ObjectToWorldKHR` decoration **must** be used only within the `IntersectionKHR`, `AnyHitKHR`, or `ClosestHitKHR` `Execution Model`
- VUID-ObjectToWorldKHR-ObjectToWorldKHR-04306
  The variable decorated with `ObjectToWorldKHR` **must** be declared using the `Input` `Storage`

Class

- VUID-ObjectToWorldKHR-ObjectToWorldKHR-04307
  The variable decorated with ObjectToWorldKHR **must** be declared as a matrix with four columns of three-component vectors of 32-bit floating-point values

PatchVertices

Decorating a variable with the PatchVertices built-in decoration will make that variable contain the number of vertices in the input patch being processed by the shader. In a Tessellation Control Shader, this is the same as the patchControlPoints member of VkPipelineTessellationStateCreateInfo. In a Tessellation Evaluation Shader, PatchVertices is equal to the tessellation control output patch size. When the same shader is used in different pipelines where the patch sizes are configured differently, the value of the PatchVertices variable will also differ.

Valid Usage

- VUID-PatchVertices-PatchVertices-04308
  The PatchVertices decoration **must** be used only within the TessellationControl or TessellationEvaluation Execution Model
- VUID-PatchVertices-PatchVertices-04309
  The variable decorated with PatchVertices **must** be declared using the Input Storage Class
- VUID-PatchVertices-PatchVertices-04310
  The variable decorated with PatchVertices **must** be declared as a scalar 32-bit integer value

PointCoord

Decorating a variable with the PointCoord built-in decoration will make that variable contain the coordinate of the current fragment within the point being rasterized, normalized to the size of the point with origin in the upper left corner of the point, as described in Basic Point Rasterization. If the primitive the fragment shader invocation belongs to is not a point, then the variable decorated with PointCoord contains poison.

NOTE Depending on how the point is rasterized, PointCoord **may** never reach (0,0) or (1,1).

Valid Usage

- VUID-PointCoord-PointCoord-04311
  The PointCoord decoration **must** be used only within the Fragment Execution Model
- VUID-PointCoord-PointCoord-04312
  The variable decorated with PointCoord **must** be declared using the Input Storage Class
- VUID-PointCoord-PointCoord-04313
  The variable decorated with PointCoord **must** be declared as a two-component vector of 32-bit floating-point values

PointSize

Decorating a variable with the PointSize built-in decoration will make that variable contain the size of point primitives or the final rasterization of polygons if polygon mode is VK_POLYGON_MODE_POINT when VkPhysicalDeviceMaintenance5Properties ::polygonModePointSize is set to VK_TRUE . The value written to the variable decorated with PointSize by the last pre-rasterization shader stage in the pipeline is used as the framebuffer-space size of points produced by rasterization. If the maintenance5 feature is enabled and a value is not written to a variable decorated with PointSize, a value of 1.0 is used as the size of points.

NOTE When PointSize decorates a variable in the Input Storage Class, it contains the data written to the output variable decorated with PointSize from the previous shader stage.

Valid Usage

- VUID-PointSize-PointSize-04314
  The PointSize decoration **must** be used only within the MeshEXT, MeshNV, Vertex, TessellationControl, TessellationEvaluation, or Geometry Execution Model
- VUID-PointSize-PointSize-04315
  The variable decorated with PointSize within the MeshEXT, MeshNV, or Vertex Execution Model **must** be declared using the Output Storage Class
- VUID-PointSize-PointSize-04316
  The variable decorated with PointSize within the TessellationControl, TessellationEvaluation, or Geometry Execution Model **must** not be declared using a Storage Class other than Input or Output
- VUID-PointSize-PointSize-04317
  The variable decorated with PointSize **must** be declared as a scalar 32-bit floating-point value

Position

Decorating a variable with the Position built-in decoration will make that variable contain the position of the current vertex. In the last pre-rasterization shader stage, the value of the variable decorated with Position is used in subsequent primitive assembly, clipping, and rasterization operations.

NOTE When Position decorates a variable in the Input Storage Class, it contains the data written to the output variable decorated with Position from the previous shader stage.

Valid Usage

- VUID-Position-Position-04318
  The Position decoration **must** be used only within the MeshEXT, MeshNV, Vertex, TessellationControl, TessellationEvaluation, or Geometry Execution Model

- VUID-Position-Position-04319
  The variable decorated with `Position` within the `MeshEXT`, `MeshNV`, or `Vertex Execution Model` **must** be declared using the `Output Storage Class`
- VUID-Position-Position-04320
  The variable decorated with `Position` within the `TessellationControl`, `TessellationEvaluation`, or `Geometry Execution Model` **must** not be declared using a `Storage Class` other than `Input` or `Output`
- VUID-Position-Position-04321
  The variable decorated with `Position` **must** be declared as a four-component vector of 32-bit floating-point values

`PrimitiveId`

Decorating a variable with the `PrimitiveId` built-in decoration will make that variable contain the index of the current primitive.

The index of the first primitive generated by a drawing command is zero, and the index is incremented after every individual point, line, or triangle primitive is processed.

For triangles drawn as points or line segments (see Polygon Mode), the primitive index is incremented only once, even if multiple points or lines are eventually drawn.

Variables decorated with `PrimitiveId` are reset to zero between each instance drawn.

Restarting a primitive topology using primitive restart has no effect on the value of variables decorated with `PrimitiveId`.

In tessellation control and tessellation evaluation shaders, it will contain the index of the patch within the current set of rendering primitives that corresponds to the shader invocation.

In a geometry shader, it will contain the number of primitives presented as input to the shader since the current set of rendering primitives was started. If the previous stage is a tessellation evaluation shader, it will contain the same value as the tessellation shaders would read.

In a fragment shader, it will contain the primitive index written by the mesh shader if a mesh shader is present, or the primitive index written by the geometry shader if a geometry shader is present, or with the value that would have been presented as input to the geometry shader had it been present.

In an intersection, any-hit, or closest hit shader, it will contain the index within the geometry of the triangle or bounding box being processed.

NOTE

When the `PrimitiveId` decoration is applied to an output variable in the mesh shader or geometry shader, the resulting value is seen through the `PrimitiveId` decorated input variable in the fragment shader.

The fragment shader using `PrimitiveId` will need to declare either the `MeshShadingEXT`, `Geometry` or `Tessellation` capability to satisfy the requirement SPIR-V has to use `PrimitiveId`.

Valid Usage

- VUID-PrimitiveId-PrimitiveId-04330
  The PrimitiveId decoration **must** be used only within the MeshEXT, MeshNV, IntersectionKHR, AnyHitKHR, ClosestHitKHR, TessellationControl, TessellationEvaluation, Geometry, or Fragment Execution Model
- VUID-PrimitiveId-Fragment-04331
  If the pipeline contains both the Fragment and Geometry Execution Model and a variable decorated with PrimitiveId is read from Fragment shader, then the Geometry shader **must** write to the output variables decorated with PrimitiveId in all execution paths
- VUID-PrimitiveId-Fragment-04332
  If the pipeline contains both the Fragment and MeshEXT or MeshNV Execution Model and a variable decorated with PrimitiveId is read from Fragment shader, then the MeshEXT or MeshNV shader **must** write to the output variables decorated with PrimitiveId in all execution paths
- VUID-PrimitiveId-Fragment-04333
  If Fragment Execution Model contains a variable decorated with PrimitiveId, then either the MeshShadingEXT, MeshShadingNV, Geometry or Tessellation capability **must** also be declared
- VUID-PrimitiveId-PrimitiveId-04334
  The variable decorated with PrimitiveId within the TessellationControl, TessellationEvaluation, Fragment, IntersectionKHR, AnyHitKHR, or ClosestHitKHR Execution Model **must** be declared using the Input Storage Class
- VUID-PrimitiveId-PrimitiveId-04335
  The variable decorated with PrimitiveId within the Geometry Execution Model **must** be declared using the Input or Output Storage Class
- VUID-PrimitiveId-PrimitiveId-04336
  The variable decorated with PrimitiveId within the MeshEXT or MeshNV Execution Model **must** be declared using the Output Storage Class
- VUID-PrimitiveId-PrimitiveId-04337
  The variable decorated with PrimitiveId **must** be declared as a scalar 32-bit integer value for all supported execution models except MeshEXT
- VUID-PrimitiveId-PrimitiveId-07040
  The variable decorated with PrimitiveId within the MeshEXT Execution Model **must** also be decorated with the PerPrimitiveEXT decoration
- VUID-PrimitiveId-PrimitiveId-10595
  PrimitiveId within the MeshEXT Execution Model **must** decorate a scalar 32-bit integer member of a structure decorated as Block, or decorate a variable of type OpTypeArray of 32-bit integer values
- VUID-PrimitiveId-PrimitiveId-10596
  If PrimitiveId is declared as an array of 32-bit integer values, within the MeshEXT Execution Model, the size of the array **must** match the value specified by OutputPrimitivesEXT
- VUID-PrimitiveId-PrimitiveId-10597
  If PrimitiveId decorates a member of a structure, the variable declaration of the

containing `Block` type **must** have an array size that matches the value specified by `OutputPrimitivesEXT`

`PrimitivePointIndicesEXT`

Decorating a variable with the `PrimitivePointIndicesEXT` decoration will make that variable contain the output array of vertex index values for point primitives.

Valid Usage

- VUID-PrimitivePointIndicesEXT-PrimitivePointIndicesEXT-07041
  The `PrimitivePointIndicesEXT` decoration **must** be used only within the `MeshEXT Execution Model`
- VUID-PrimitivePointIndicesEXT-PrimitivePointIndicesEXT-07042
  The `PrimitivePointIndicesEXT` decoration **must** be used with the `OutputPoints Execution Mode`
- VUID-PrimitivePointIndicesEXT-PrimitivePointIndicesEXT-07043
  The variable decorated with `PrimitivePointIndicesEXT` **must** be declared using the `Output Storage Class`
- VUID-PrimitivePointIndicesEXT-PrimitivePointIndicesEXT-07044
  The variable decorated with `PrimitivePointIndicesEXT` **must** be declared as an array of scalar 32-bit integer values
- VUID-PrimitivePointIndicesEXT-PrimitivePointIndicesEXT-07045
  The index to access the array decorated with `PrimitivePointIndicesEXT` **must** be in the range [0, N-1], where N is the value specified by the “Primitive Count” operand of `OpSetMeshOutputsEXT`
- VUID-PrimitivePointIndicesEXT-PrimitivePointIndicesEXT-12335
  All index values written to the array decorated with `PrimitivePointIndicesEXT` **must** be in the range [0, N-1], where N is the value specified by the “Vertex Count” operand of `OpSetMeshOutputsEXT`
- VUID-PrimitivePointIndicesEXT-PrimitivePointIndicesEXT-07046
  The size of the array decorated with `PrimitivePointIndicesEXT` **must** match the value specified by `OutputPrimitivesEXT`

`PrimitiveLineIndicesEXT`

Decorating a variable with the `PrimitiveLineIndicesEXT` decoration will make that variable contain the output array of vertex index values for line primitives.

Valid Usage

- VUID-PrimitiveLineIndicesEXT-PrimitiveLineIndicesEXT-07047
  The `PrimitiveLineIndicesEXT` decoration **must** be used only within the `MeshEXT Execution Model`
- VUID-PrimitiveLineIndicesEXT-PrimitiveLineIndicesEXT-07048

The PrimitiveLineIndicesEXT decoration **must** be used with the OutputLinesEXT Execution Mode

- VUID-PrimitiveLineIndicesEXT-PrimitiveLineIndicesEXT-07049
  The variable decorated with PrimitiveLineIndicesEXT **must** be declared using the Output Storage Class
- VUID-PrimitiveLineIndicesEXT-PrimitiveLineIndicesEXT-07050
  The variable decorated with PrimitiveLineIndicesEXT **must** be declared as an array of two component vector 32-bit integer values
- VUID-PrimitiveLineIndicesEXT-PrimitiveLineIndicesEXT-07051
  The index to access the array decorated with PrimitiveLineIndicesEXT **must** be in the range [0, N-1], where N is the value specified by the “Primitive Count” operand of OpSetMeshOutputsEXT
- VUID-PrimitiveLineIndicesEXT-PrimitiveLineIndicesEXT-12336
  All index values written to the array decorated with PrimitiveLineIndicesEXT **must** be in the range [0, N-1], where N is the value specified by the “Vertex Count” operand of OpSetMeshOutputsEXT
- VUID-PrimitiveLineIndicesEXT-PrimitiveLineIndicesEXT-07052
  The size of the array decorated with PrimitiveLineIndicesEXT **must** match the value specified by OutputPrimitivesEXT

PrimitiveTriangleIndicesEXT

Decorating a variable with the PrimitiveTriangleIndicesEXT decoration will make that variable contain the output array of vertex index values for triangle primitives.

Valid Usage

- VUID-PrimitiveTriangleIndicesEXT-PrimitiveTriangleIndicesEXT-07053
  The PrimitiveTriangleIndicesEXT decoration **must** be used only within the MeshEXT Execution Model
- VUID-PrimitiveTriangleIndicesEXT-PrimitiveTriangleIndicesEXT-07054
  The PrimitiveTriangleIndicesEXT decoration **must** be used with the OutputTrianglesEXT Execution Mode
- VUID-PrimitiveTriangleIndicesEXT-PrimitiveTriangleIndicesEXT-07055
  The variable decorated with PrimitiveTriangleIndicesEXT **must** be declared using the Output Storage Class
- VUID-PrimitiveTriangleIndicesEXT-PrimitiveTriangleIndicesEXT-07056
  The variable decorated with PrimitiveTriangleIndicesEXT **must** be declared as an array of three component vector 32-bit integer values
- VUID-PrimitiveTriangleIndicesEXT-PrimitiveTriangleIndicesEXT-07057
  The index to access the array decorated with PrimitiveTriangleIndicesEXT **must** be in the range [0, N-1], where N is the value specified by the “Primitive Count” operand of OpSetMeshOutputsEXT
- VUID-PrimitiveTriangleIndicesEXT-PrimitiveTriangleIndicesEXT-12337

All index values written to the array decorated with `PrimitiveTriangleIndicesEXT` **must** be in the range [0, N-1], where N is the value specified by the "Vertex Count" operand of `OpSetMeshOutputsEXT`

- VUID-PrimitiveTriangleIndicesEXT-PrimitiveTriangleIndicesEXT-07058
  The size of the array decorated with `PrimitiveTriangleIndicesEXT` **must** match the value specified by `OutputPrimitivesEXT`

`PrimitiveShadingRateKHR`

Decorating a variable with the `PrimitiveShadingRateKHR` built-in decoration will make that variable contain the primitive fragment shading rate.

The value written to the variable decorated with `PrimitiveShadingRateKHR` by the last pre-rasterization shader stage in the pipeline is used as the primitive fragment shading rate. Outputs in previous shader stages are ignored.

If the last active pre-rasterization shader stage shader entry point's interface does not include a variable decorated with `PrimitiveShadingRateKHR`, then it is as if the shader specified a fragment shading rate value of 0, indicating a horizontal and vertical rate of 1 pixel.

Valid Usage

- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-04484
  The `PrimitiveShadingRateKHR` decoration **must** be used only within the `MeshEXT`, `MeshNV`, `Vertex`, or `Geometry` `Execution Model`
- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-04485
  The variable decorated with `PrimitiveShadingRateKHR` **must** be declared using the `Output` `Storage Class`
- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-04486
  The variable decorated with `PrimitiveShadingRateKHR` **must** be declared as a scalar 32-bit integer value for all supported execution models except `MeshEXT`
- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-04487
  The value written to `PrimitiveShadingRateKHR` **must** include no more than one of `Vertical2Pixels` and `Vertical4Pixels`
- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-04488
  The value written to `PrimitiveShadingRateKHR` **must** include no more than one of `Horizontal2Pixels` and `Horizontal4Pixels`
- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-04489
  The value written to `PrimitiveShadingRateKHR` **must** not have any bits set other than those defined by **Fragment Shading Rate Flags** enumerants in the SPIR-V specification
- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-07059
  The variable decorated with `PrimitiveShadingRateKHR` within the `MeshEXT` `Execution Model` **must** also be decorated with the `PerPrimitiveEXT` decoration
- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-10598
  `PrimitiveShadingRateKHR` within the `MeshEXT` `Execution Model` **must** decorate a scalar 32-bit

integer member of a structure decorated as Block, or decorate a variable of type OpTypeArray of 32-bit integer values

- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-10599
  If PrimitiveShadingRateKHR is declared as an array of 32-bit integer values, within the MeshEXT Execution Model, the size of the array **must** match the value specified by OutputPrimitivesEXT
- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-10600
  If PrimitiveShadingRateKHR decorates a member of a structure, the variable declaration of the containing Block type **must** have an array size that matches the value specified by OutputPrimitivesEXT
- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-12275
  If PrimitiveShadingRateKHR is declared in the MeshEXT Execution Model, the primitiveFragmentShadingRateMeshShader feature **must** be enabled

RayGeometryIndexKHR

A variable decorated with the RayGeometryIndexKHR decoration will contain the geometry index for the acceleration structure geometry currently being shaded.

Valid Usage

- VUID-RayGeometryIndexKHR-RayGeometryIndexKHR-04345
  The RayGeometryIndexKHR decoration **must** be used only within the IntersectionKHR, AnyHitKHR, or ClosestHitKHR Execution Model
- VUID-RayGeometryIndexKHR-RayGeometryIndexKHR-04346
  The variable decorated with RayGeometryIndexKHR **must** be declared using the Input Storage Class
- VUID-RayGeometryIndexKHR-RayGeometryIndexKHR-04347
  The variable decorated with RayGeometryIndexKHR **must** be declared as a scalar 32-bit integer value

RayTmaxKHR

A variable decorated with the RayTmaxKHR decoration will contain the parametric t_{max} value of the ray being processed. The value is independent of the space in which the ray origin and direction exist. The value is initialized to the parameter passed into the pipeline trace ray instruction.

The t_{max} value changes throughout the lifetime of the ray that produced the intersection. In the closest hit shader, the value reflects the closest distance to the intersected primitive. In the any-hit shader, it reflects the distance to the primitive currently being intersected. In the intersection shader, it reflects the distance to the closest primitive intersected so far or the initial value. The value can change in the intersection shader after calling OpReportIntersectionKHR if the corresponding any-hit shader does not ignore the intersection. In a miss shader, the value is identical to the parameter passed into the pipeline trace ray instruction.

Valid Usage

- VUID-RayTmaxKHR-RayTmaxKHR-04348
  The RayTmaxKHR decoration **must** be used only within the IntersectionKHR, AnyHitKHR, ClosestHitKHR, or MissKHR Execution Model
- VUID-RayTmaxKHR-RayTmaxKHR-04349
  The variable decorated with RayTmaxKHR **must** be declared using the Input Storage Class
- VUID-RayTmaxKHR-RayTmaxKHR-04350
  The variable decorated with RayTmaxKHR **must** be declared as a scalar 32-bit floating-point value

RayTminKHR

A variable decorated with the RayTminKHR decoration will contain the parametric t_{min} value of the ray being processed. The value is independent of the space in which the ray origin and direction exist. The value is the parameter passed into the pipeline trace ray instruction.

The t_{min} value remains constant for the duration of the ray query.

Valid Usage

- VUID-RayTminKHR-RayTminKHR-04351
  The RayTminKHR decoration **must** be used only within the IntersectionKHR, AnyHitKHR, ClosestHitKHR, or MissKHR Execution Model
- VUID-RayTminKHR-RayTminKHR-04352
  The variable decorated with RayTminKHR **must** be declared using the Input Storage Class
- VUID-RayTminKHR-RayTminKHR-04353
  The variable decorated with RayTminKHR **must** be declared as a scalar 32-bit floating-point value

SampleId

Decorating a variable with the SampleId built-in decoration will make that variable contain the coverage index for the current fragment shader invocation. SampleId ranges from zero to the number of samples in the framebuffer minus one. If a fragment shader entry point's interface includes an input variable decorated with SampleId, Sample Shading is considered enabled with a minSampleShading value of 1.0.

Valid Usage

- VUID-SampleId-SampleId-04354
  The SampleId decoration **must** be used only within the Fragment Execution Model
- VUID-SampleId-SampleId-04355
  The variable decorated with SampleId **must** be declared using the Input Storage Class
- VUID-SampleId-SampleId-04356

The variable decorated with `SampleId` **must** be declared as a scalar 32-bit integer value

`SampleMask`

Decorating a variable with the `SampleMask` built-in decoration will make that variable contain the sample mask for the current fragment shader invocation.

A variable in the `Input` storage class decorated with `SampleMask` will contain a bitmask of the set of samples covered by the primitive generating the fragment during rasterization. It has a sample bit set if and only if the sample is considered covered for this fragment shader invocation. `SampleMask`[] is an array of integers. Bits are mapped to samples in a manner where bit B of mask M (`SampleMask[M]`) corresponds to sample 32 × M + B.

A variable in the `Output` storage class decorated with `SampleMask` is an array of integers forming a bit array in a manner similar to an input variable decorated with `SampleMask`, but where each bit represents coverage as computed by the shader. This computed `SampleMask` is combined with the generated coverage mask in the multisample coverage operation.

Variables decorated with `SampleMask` **must** be either an unsized array, or explicitly sized to be no larger than the implementation-dependent maximum sample-mask (as an array of 32-bit elements), determined by the maximum number of samples.

If a fragment shader entry point's interface does not include an output variable decorated with `SampleMask`, the sample mask has no effect on the processing of a fragment.

Valid Usage

- VUID-SampleMask-SampleMask-04357
  The `SampleMask` decoration **must** be used only within the `Fragment` `Execution` `Model`
- VUID-SampleMask-SampleMask-04358
  The variable decorated with `SampleMask` **must** be declared using the `Input` or `Output` `Storage` `Class`
- VUID-SampleMask-SampleMask-04359
  The variable decorated with `SampleMask` **must** be declared as an array of 32-bit integer values

`SamplePosition`

Decorating a variable with the `SamplePosition` built-in decoration will make that variable contain the sub-pixel position of the sample being shaded. The top left of the pixel is considered to be at coordinate (0,0) and the bottom right of the pixel is considered to be at coordinate (1,1).

If the render pass has a fragment density map attachment, the variable will instead contain the sub-fragment position of the sample being shaded. The top left of the fragment is considered to be at coordinate (0,0) and the bottom right of the fragment is considered to be at coordinate (1,1) for any fragment area.

If a fragment shader entry point's interface includes an input variable decorated with

SamplePosition, Sample Shading is considered enabled with a minSampleShading value of 1.0.

If the current pipeline uses custom sample locations the value of any variable decorated with the SamplePosition built-in decoration is poison.

Valid Usage

- VUID-SamplePosition-SamplePosition-04360
  The SamplePosition decoration **must** be used only within the Fragment Execution Model
- VUID-SamplePosition-SamplePosition-04361
  The variable decorated with SamplePosition **must** be declared using the Input Storage Class
- VUID-SamplePosition-SamplePosition-04362
  The variable decorated with SamplePosition **must** be declared as a two-component vector of 32-bit floating-point values

ShadingRateKHR

Decorating a variable with the ShadingRateKHR built-in decoration will make that variable contain the fragment shading rate for the current fragment invocation.

Valid Usage

- VUID-ShadingRateKHR-ShadingRateKHR-04490
  The ShadingRateKHR decoration **must** be used only within the Fragment Execution Model
- VUID-ShadingRateKHR-ShadingRateKHR-04491
  The variable decorated with ShadingRateKHR **must** be declared using the Input Storage Class
- VUID-ShadingRateKHR-ShadingRateKHR-04492
  The variable decorated with ShadingRateKHR **must** be declared as a scalar 32-bit integer value

SubgroupId

Decorating a variable with the SubgroupId built-in decoration will make that variable contain the index of the subgroup within the local workgroup. This variable is in range [0, NumSubgroups-1].

Valid Usage

- VUID-SubgroupId-SubgroupId-04367
  The SubgroupId decoration **must** be used only within the GLCompute, MeshEXT, TaskEXT, MeshNV, or TaskNV Execution Model
- VUID-SubgroupId-SubgroupId-04368
  The variable decorated with SubgroupId **must** be declared using the Input Storage Class
- VUID-SubgroupId-SubgroupId-04369

The variable decorated with SubgroupId **must** be declared as a scalar 32-bit integer value

SubgroupEqMask

Decorating a variable with the SubgroupEqMask builtin decoration will make that variable contain the *subgroup mask* of the current subgroup invocation. The bit corresponding to the SubgroupLocalInvocationId is set in the variable decorated with SubgroupEqMask. All other bits are set to zero.

SubgroupEqMaskKHR is an alias of SubgroupEqMask.

Valid Usage

- VUID-SubgroupEqMask-SubgroupEqMask-04370
  The variable decorated with SubgroupEqMask **must** be declared using the Input Storage Class
- VUID-SubgroupEqMask-SubgroupEqMask-04371
  The variable decorated with SubgroupEqMask **must** be declared as a four-component vector of 32-bit integer values

SubgroupGeMask

Decorating a variable with the SubgroupGeMask builtin decoration will make that variable contain the *subgroup mask* of the current subgroup invocation. The bits corresponding to the invocations greater than or equal to SubgroupLocalInvocationId through SubgroupSize-1 are set in the variable decorated with SubgroupGeMask. All other bits are set to zero.

SubgroupGeMaskKHR is an alias of SubgroupGeMask.

Valid Usage

- VUID-SubgroupGeMask-SubgroupGeMask-04372
  The variable decorated with SubgroupGeMask **must** be declared using the Input Storage Class
- VUID-SubgroupGeMask-SubgroupGeMask-04373
  The variable decorated with SubgroupGeMask **must** be declared as a four-component vector of 32-bit integer values

SubgroupGtMask

Decorating a variable with the SubgroupGtMask builtin decoration will make that variable contain the *subgroup mask* of the current subgroup invocation. The bits corresponding to the invocations greater than SubgroupLocalInvocationId through SubgroupSize-1 are set in the variable decorated with SubgroupGtMask. All other bits are set to zero.

SubgroupGtMaskKHR is an alias of SubgroupGtMask.

Valid Usage

- VUID-SubgroupGtMask-SubgroupGtMask-04374
  The variable decorated with `SubgroupGtMask` **must** be declared using the `Input` `Storage Class`
- VUID-SubgroupGtMask-SubgroupGtMask-04375
  The variable decorated with `SubgroupGtMask` **must** be declared as a four-component vector of 32-bit integer values

`SubgroupLeMask`

Decorating a variable with the `SubgroupLeMask` builtin decoration will make that variable contain the *subgroup mask* of the current subgroup invocation. The bits corresponding to the invocations less than or equal to `SubgroupLocalInvocationId` are set in the variable decorated with `SubgroupLeMask`. All other bits are set to zero.

`SubgroupLeMaskKHR` is an alias of `SubgroupLeMask`.

Valid Usage

- VUID-SubgroupLeMask-SubgroupLeMask-04376
  The variable decorated with `SubgroupLeMask` **must** be declared using the `Input` `Storage Class`
- VUID-SubgroupLeMask-SubgroupLeMask-04377
  The variable decorated with `SubgroupLeMask` **must** be declared as a four-component vector of 32-bit integer values

`SubgroupLtMask`

Decorating a variable with the `SubgroupLtMask` builtin decoration will make that variable contain the *subgroup mask* of the current subgroup invocation. The bits corresponding to the invocations less than `SubgroupLocalInvocationId` are set in the variable decorated with `SubgroupLtMask`. All other bits are set to zero.

`SubgroupLtMaskKHR` is an alias of `SubgroupLtMask`.

Valid Usage

- VUID-SubgroupLtMask-SubgroupLtMask-04378
  The variable decorated with `SubgroupLtMask` **must** be declared using the `Input` `Storage Class`
- VUID-SubgroupLtMask-SubgroupLtMask-04379
  The variable decorated with `SubgroupLtMask` **must** be declared as a four-component vector of 32-bit integer values

SubgroupLocalInvocationId

Decorating a variable with the SubgroupLocalInvocationId builtin decoration will make that variable contain the index of the invocation within the subgroup. This variable is in range [0,SubgroupSize-1].

NOTE

There is no direct relationship between SubgroupLocalInvocationId and LocalInvocationId or LocalInvocationIndex. If the shader was created with full subgroups, applications can compute their own local invocation index to serve the same purpose:

index = SubgroupLocalInvocationId + SubgroupId × SubgroupSize

If full subgroups are not enabled, some subgroups may be dispatched with inactive invocations that do not correspond to a local workgroup invocation, making the value of index unreliable.

Valid Usage

- VUID-SubgroupLocalInvocationId-SubgroupLocalInvocationId-04380
  The variable decorated with SubgroupLocalInvocationId **must** be declared using the Input Storage Class
- VUID-SubgroupLocalInvocationId-SubgroupLocalInvocationId-04381
  The variable decorated with SubgroupLocalInvocationId **must** be declared as a scalar 32-bit integer value

SubgroupSize

Decorating a variable with the SubgroupSize builtin decoration will make that variable contain the implementation-dependent number of invocations in a subgroup. This value **must** be a power-of-two integer.

If the shader was created with varying subgroup size allowed, the SubgroupSize decorated variable will contain the subgroup size for each subgroup that gets dispatched. This value **must** be between minSubgroupSize and maxSubgroupSize and **must** be uniform with subgroup scope.

If the shader was created with a required subgroup size, the SubgroupSize decorated variable will match that value.

If the shader does not allow varying subgroup sizes, the variable decorated with SubgroupSize will match subgroupSize.

The maximum number of invocations that an implementation can support per subgroup is 128.

Valid Usage

- VUID-SubgroupSize-SubgroupSize-04382
  The variable decorated with SubgroupSize **must** be declared using the Input Storage Class

- VUID-SubgroupSize-SubgroupSize-04383
  The variable decorated with `SubgroupSize` **must** be declared as a scalar 32-bit integer value

`TessCoord`

Decorating a variable with the `TessCoord` built-in decoration will make that variable contain the three-dimensional (u,v,w) barycentric coordinate of the tessellated vertex within the patch. u, v, and w are in the range [0,1] and vary linearly across the primitive being subdivided. For the tessellation modes of `Quads` or `IsoLines`, the third component is always zero.

Valid Usage

- VUID-TessCoord-TessCoord-04387
  The `TessCoord` decoration **must** be used only within the `TessellationEvaluation` `Execution Model`
- VUID-TessCoord-TessCoord-04388
  The variable decorated with `TessCoord` **must** be declared using the `Input` `Storage Class`
- VUID-TessCoord-TessCoord-04389
  The variable decorated with `TessCoord` **must** be declared as a three-component vector of 32-bit floating-point values

`TessLevelOuter`

Decorating a variable with the `TessLevelOuter` built-in decoration will make that variable contain the outer tessellation levels for the current patch.

In tessellation control shaders, the variable decorated with `TessLevelOuter` **can** be written to, controlling the tessellation factors for the resulting patch. These values are used by the tessellator to control primitive tessellation and **can** be read by tessellation evaluation shaders.

In tessellation evaluation shaders, the variable decorated with `TessLevelOuter` **can** read the values written by the tessellation control shader.

Valid Usage

- VUID-TessLevelOuter-TessLevelOuter-04390
  The `TessLevelOuter` decoration **must** be used only within the `TessellationControl` or `TessellationEvaluation` `Execution Model`
- VUID-TessLevelOuter-TessLevelOuter-04391
  The variable decorated with `TessLevelOuter` within the `TessellationControl` `Execution Model` **must** be declared using the `Output` `Storage Class`
- VUID-TessLevelOuter-TessLevelOuter-04392
  The variable decorated with `TessLevelOuter` within the `TessellationEvaluation` `Execution Model` **must** be declared using the `Input` `Storage Class`
- VUID-TessLevelOuter-TessLevelOuter-04393

The variable decorated with `TessLevelOuter` **must** be declared as an array of size four, containing 32-bit floating-point values

`TessLevelInner`

Decorating a variable with the `TessLevelInner` built-in decoration will make that variable contain the inner tessellation levels for the current patch.

In tessellation control shaders, the variable decorated with `TessLevelInner` **can** be written to, controlling the tessellation factors for the resulting patch. These values are used by the tessellator to control primitive tessellation and **can** be read by tessellation evaluation shaders.

In tessellation evaluation shaders, the variable decorated with `TessLevelInner` **can** read the values written by the tessellation control shader.

Valid Usage

- VUID-TessLevelInner-TessLevelInner-04394
  The `TessLevelInner` decoration **must** be used only within the `TessellationControl` or `TessellationEvaluation` `Execution Model`
- VUID-TessLevelInner-TessLevelInner-04395
  The variable decorated with `TessLevelInner` within the `TessellationControl` `Execution Model` **must** be declared using the `Output` `Storage Class`
- VUID-TessLevelInner-TessLevelInner-04396
  The variable decorated with `TessLevelInner` within the `TessellationEvaluation` `Execution Model` **must** be declared using the `Input` `Storage Class`
- VUID-TessLevelInner-TessLevelInner-04397
  The variable decorated with `TessLevelInner` **must** be declared as an array of size two, containing 32-bit floating-point values

`VertexIndex`

Decorating a variable with the `VertexIndex` built-in decoration will make that variable contain the index of the vertex that is being processed by the current vertex shader invocation. For non-indexed draws, this variable begins at the `firstVertex` parameter to vkCmdDraw or the `firstVertex` member of a structure consumed by vkCmdDrawIndirect and increments by one for each vertex in the draw. For indexed draws, its value is the content of the index buffer for the vertex plus the `vertexOffset` parameter to vkCmdDrawIndexed or the `vertexOffset` member of the structure consumed by vkCmdDrawIndexedIndirect.

NOTE `VertexIndex` starts at the same starting value for each instance.

Valid Usage

- VUID-VertexIndex-VertexIndex-04398
  The `VertexIndex` decoration **must** be used only within the `Vertex` `Execution Model`

- VUID-VertexIndex-VertexIndex-04399
  The variable decorated with `VertexIndex` **must** be declared using the `Input` `Storage Class`
- VUID-VertexIndex-VertexIndex-04400
  The variable decorated with `VertexIndex` **must** be declared as a scalar 32-bit integer value

`ViewIndex`

The `ViewIndex` decoration **can** be applied to a shader input which will be filled with the index of the view that is being processed by the current shader invocation.

If multiview is enabled in the render pass, this value will be the index of one of the bits set in the view mask of the subpass the pipeline is compiled against. If multiview is not enabled in the render pass, this value will be zero.

Valid Usage

- VUID-ViewIndex-ViewIndex-04401
  The `ViewIndex` decoration **must** be used only within the `MeshEXT`, `Vertex`, `Geometry`, `TessellationControl`, `TessellationEvaluation` or `Fragment` `Execution Model`
- VUID-ViewIndex-ViewIndex-04402
  The variable decorated with `ViewIndex` **must** be declared using the `Input` `Storage Class`
- VUID-ViewIndex-ViewIndex-04403
  The variable decorated with `ViewIndex` **must** be declared as a scalar 32-bit integer value

`ViewportIndex`

Decorating a variable with the `ViewportIndex` built-in decoration will make that variable contain the index of the viewport.

In a mesh, vertex, tessellation evaluation, or geometry shader, the variable decorated with `ViewportIndex` can be written to with the viewport index to which the primitive produced by that shader will be directed.

The selected viewport index is used to select the viewport transform and scissor rectangle.

The last active *pre-rasterization shader stage* (in pipeline order) controls the `ViewportIndex` that is used. Outputs in previous shader stages are not used, even if the last stage fails to write the `ViewportIndex`.

If the last active pre-rasterization shader stage shader entry point's interface does not include a variable decorated with `ViewportIndex` then the first viewport is used. If a pre-rasterization shader stage shader entry point's interface includes a variable decorated with `ViewportIndex`, it **must** write the same value to `ViewportIndex` for all output vertices of a given primitive.

In a fragment shader, the variable decorated with `ViewportIndex` contains the viewport index of the primitive that the fragment invocation belongs to.

Valid Usage

- VUID-ViewportIndex-ViewportIndex-04404
  The `ViewportIndex` decoration **must** be used only within the `MeshEXT`, `MeshNV`, `Vertex`, `TessellationEvaluation`, `Geometry`, or `Fragment` `Execution Model`
- VUID-ViewportIndex-ViewportIndex-04405
  If the `shaderOutputViewportIndex` feature is not enabled then the `ViewportIndex` decoration **must** be used only within the `Geometry` or `Fragment` `Execution Model`
- VUID-ViewportIndex-ViewportIndex-04406
  The variable decorated with `ViewportIndex` within the `MeshEXT`, `MeshNV`, `Vertex`, `TessellationEvaluation`, or `Geometry` `Execution Model` **must** be declared using the `Output` `Storage Class`
- VUID-ViewportIndex-ViewportIndex-04407
  The variable decorated with `ViewportIndex` within the `Fragment` `Execution Model` **must** be declared using the `Input` `Storage Class`
- VUID-ViewportIndex-ViewportIndex-04408
  The variable decorated with `ViewportIndex` **must** be declared as a scalar 32-bit integer value for all supported execution models except `MeshEXT`
- VUID-ViewportIndex-ViewportIndex-07060
  The variable decorated with `ViewportIndex` within the `MeshEXT` `Execution Model` **must** also be decorated with the `PerPrimitiveEXT` decoration
- VUID-ViewportIndex-ViewportIndex-10601
  `ViewportIndex` within the `MeshEXT` `Execution Model` **must** decorate a scalar 32-bit integer member of a structure decorated as `Block`, or decorate a variable of type `OpTypeArray` of scalar 32-bit integer values
- VUID-ViewportIndex-ViewportIndex-10602
  If `ViewportIndex` is declared as an array of 32-bit integer values, within the `MeshEXT` `Execution Model`, the size of the array **must** match the value specified by `OutputPrimitivesEXT`
- VUID-ViewportIndex-ViewportIndex-10603
  If `ViewportIndex` decorates a member of a structure, the variable declaration of the containing `Block` type **must** have an array size that matches the value specified by `OutputPrimitivesEXT`

`WorkgroupId`

Decorating a variable with the `WorkgroupId` built-in decoration will make that variable contain the global coordinate of the local workgroup that the current invocation is a member of. Each component is in the range [base,base + count), where base and count are based on the parameters passed into the dispatching or drawing commands in each dimension.

Valid Usage

- VUID-WorkgroupId-WorkgroupId-04422

The WorkgroupId decoration **must** be used only within the GLCompute, MeshEXT, TaskEXT, MeshNV, or TaskNV Execution Model

- VUID-WorkgroupId-WorkgroupId-04423
  The variable decorated with WorkgroupId **must** be declared using the Input Storage Class
- VUID-WorkgroupId-WorkgroupId-04424
  The variable decorated with WorkgroupId **must** be declared as a three-component vector of 32-bit integer values

WorkgroupSize

NOTE	SPIR-V 1.6 deprecated WorkgroupSize in favor of using the LocalSizeId Execution Mode instead. Support for LocalSizeId was added with VK_KHR_maintenance4 and promoted to core in Version 1.3.

Decorating an object with the WorkgroupSize built-in decoration will make that object contain the dimensions of a local workgroup. If an object is decorated with the WorkgroupSize decoration, this takes precedence over any LocalSize or LocalSizeId execution mode.

Valid Usage

- VUID-WorkgroupSize-WorkgroupSize-04425
  The WorkgroupSize decoration **must** be used only within the GLCompute, MeshEXT, TaskEXT, MeshNV, or TaskNV Execution Model
- VUID-WorkgroupSize-WorkgroupSize-04426
  The variable decorated with WorkgroupSize **must** be a specialization constant or a constant
- VUID-WorkgroupSize-WorkgroupSize-04427
  The variable decorated with WorkgroupSize **must** be declared as a three-component vector of 32-bit integer values

WorldRayDirectionKHR

A variable decorated with the WorldRayDirectionKHR decoration will specify the direction of the ray being processed, in world space. The value is the parameter passed into the pipeline trace ray instruction.

Valid Usage

- VUID-WorldRayDirectionKHR-WorldRayDirectionKHR-04428
  The WorldRayDirectionKHR decoration **must** be used only within the IntersectionKHR, AnyHitKHR, ClosestHitKHR, or MissKHR Execution Model
- VUID-WorldRayDirectionKHR-WorldRayDirectionKHR-04429
  The variable decorated with WorldRayDirectionKHR **must** be declared using the Input Storage Class
- VUID-WorldRayDirectionKHR-WorldRayDirectionKHR-04430

The variable decorated with `WorldRayDirectionKHR` **must** be declared as a three-component vector of 32-bit floating-point values

`WorldRayOriginKHR`

A variable decorated with the `WorldRayOriginKHR` decoration will specify the origin of the ray being processed, in world space. The value is the parameter passed into the pipeline trace ray instruction.

Valid Usage

- VUID-WorldRayOriginKHR-WorldRayOriginKHR-04431
  The `WorldRayOriginKHR` decoration **must** be used only within the `IntersectionKHR`, `AnyHitKHR`, `ClosestHitKHR`, or `MissKHR` `Execution Model`
- VUID-WorldRayOriginKHR-WorldRayOriginKHR-04432
  The variable decorated with `WorldRayOriginKHR` **must** be declared using the `Input` `Storage Class`
- VUID-WorldRayOriginKHR-WorldRayOriginKHR-04433
  The variable decorated with `WorldRayOriginKHR` **must** be declared as a three-component vector of 32-bit floating-point values

`WorldToObjectKHR`

A variable decorated with the `WorldToObjectKHR` decoration will contain the current world-to-object transformation matrix, which is determined by the instance of the current intersection.

Valid Usage

- VUID-WorldToObjectKHR-WorldToObjectKHR-04434
  The `WorldToObjectKHR` decoration **must** be used only within the `IntersectionKHR`, `AnyHitKHR`, or `ClosestHitKHR` `Execution Model`
- VUID-WorldToObjectKHR-WorldToObjectKHR-04435
  The variable decorated with `WorldToObjectKHR` **must** be declared using the `Input` `Storage Class`
- VUID-WorldToObjectKHR-WorldToObjectKHR-04436
  The variable decorated with `WorldToObjectKHR` **must** be declared as a matrix with four columns of three-component vectors of 32-bit floating-point values

`HitIsSphereNV`

A variable decorated with the `HitIsSphereNV` decoration will contain a non-zero value if the current ray hit a sphere primitive or zero otherwise.

Valid Usage

- VUID-HitIsSphereNV-HitIsSphereNV-10513

The HitIsSphereNV decoration **must** be used only within the AnyHitKHR, or ClosestHitKHR Execution Model

- VUID-HitIsSphereNV-HitIsSphereNV-10514
  The variable decorated with HitIsSphereNV **must** be declared using the Input Storage Class
- VUID-HitIsSphereNV-HitIsSphereNV-10515
  The variable decorated with HitIsSphereNV **must** be declared as a boolean value

HitIsLSSNV

A variable decorated with the HitIsLSSNV decoration will contain a non-zero value if the current ray hit an LSS primitive or zero otherwise.

Valid Usage

- VUID-HitIsLSSNV-HitIsLSSNV-10516
  The HitIsLSSNV decoration **must** be used only within the AnyHitKHR, or ClosestHitKHR Execution Model
- VUID-HitIsLSSNV-HitIsLSSNV-10517
  The variable decorated with HitIsLSSNV **must** be declared using the Input Storage Class
- VUID-HitIsLSSNV-HitIsLSSNV-10518
  The variable decorated with HitIsLSSNV **must** be declared as a boolean value

HitSpherePositionNV

A variable decorated with the HitSpherePositionNV decoration will contain the position of the sphere primitive intersected by the current ray.

Valid Usage

- VUID-HitSpherePositionNV-HitSpherePositionNV-10519
  The HitSpherePositionNV decoration **must** be used only within the AnyHitKHR, or ClosestHitKHR Execution Model
- VUID-HitSpherePositionNV-HitSpherePositionNV-10520
  The variable decorated with HitSpherePositionNV **must** be declared using the Input Storage Class
- VUID-HitSpherePositionNV-HitSpherePositionNV-10521
  The variable decorated with HitSpherePositionNV **must** be declared as a three-component vector of 32-bit floating-point values

HitSphereRadiusNV

A variable decorated with the HitSphereRadiusNV decoration will contain the radius of the sphere primitive intersected by the current ray.

Valid Usage

- VUID-HitSphereRadiusNV-HitSphereRadiusNV-10522
  The `HitSphereRadiusNV` decoration **must** be used only within the `AnyHitKHR`, or `ClosestHitKHR` `Execution Model`
- VUID-HitSphereRadiusNV-HitSphereRadiusNV-10523
  The variable decorated with `HitSphereRadiusNV` **must** be declared using the `Input` `Storage Class`
- VUID-HitSphereRadiusNV-HitSphereRadiusNV-10524
  The variable decorated with `HitSphereRadiusNV` **must** be declared as a scalar 32-bit floating-point value

`HitLSSPositionsNV`

A variable decorated with the `HitLSSPositionsNV` decoration will contain the position of the LSS primitive intersected by the current ray.

Valid Usage

- VUID-HitLSSPositionsNV-HitLSSPositionsNV-10525
  The `HitLSSPositionsNV` decoration **must** be used only within the `AnyHitKHR`, or `ClosestHitKHR` `Execution Model`
- VUID-HitLSSPositionsNV-HitLSSPositionsNV-10526
  The variable decorated with `HitLSSPositionsNV` **must** be declared using the `Input` `Storage Class`
- VUID-HitLSSPositionsNV-HitLSSPositionsNV-10527
  The variable decorated with `HitLSSPositionsNV` **must** be declared as an array of size two, containing three-component vectors of 32-bit floating-point values

`HitLSSRadiiNV`

A variable decorated with the `HitLSSRadiiNV` decoration will contain the radii of the LSS primitive intersected by the current ray.

Valid Usage

- VUID-HitLSSRadiiNV-HitLSSRadiiNV-10528
  The `HitLSSRadiiNV` decoration **must** be used only within the `AnyHitKHR`, or `ClosestHitKHR` `Execution Model`
- VUID-HitLSSRadiiNV-HitLSSRadiiNV-10529
  The variable decorated with `HitLSSRadiiNV` **must** be declared using the `Input` `Storage Class`
- VUID-HitLSSRadiiNV-HitLSSRadiiNV-10530
  The variable decorated with `HitLSSRadiiNV` **must** be declared as an array of size two, containing 32-bit floating-point values

`ClusterIDNV`

A variable decorated with the `ClusterIDNV` decoration will contain the triangle cluster ID of a hit triangle in a cluster acceleration structure if the current ray hit a triangle primitive or `-1` otherwise.

Valid Usage

- VUID-ClusterIDNV-ClusterIDNV-10531
  The `ClusterIDNV` decoration **must** be used only within the `AnyHitKHR`, or `ClosestHitKHR` `Execution Model`
- VUID-ClusterIDNV-ClusterIDNV-10532
  The variable decorated with `ClusterIDNV` **must** be declared using the `Input` `Storage Class`
- VUID-ClusterIDNV-ClusterIDNV-10533
  The variable decorated with `ClusterIDNV` **must** be declared as a scalar 32-bit integer value

Chapter 20. Sampling

Sampling is performed on images, typically in conjunction with a sampler, to return a value based on a neighborhood of texels in an image.

Sampling instructions include the functionality of the following SPIR-V Image Instructions:

- `OpImageSample*` and `OpImageSparseSample*` read one or more neighboring texels of the image, and filter the texel values based on the state of the sampler.
  - Instructions with `ImplicitLod` in the name determine the LOD used in the sampling operation based on the coordinates used in neighboring fragments.
  - Instructions with `ExplicitLod` in the name determine the LOD used in the sampling operation based on additional coordinates.
  - Instructions with `Proj` in the name apply homogeneous projection to the coordinates.
- `OpImageFetch` and `OpImageSparseFetch` return a single texel of the image. No sampler is used.
- `OpImage*Gather`, and `OpImageSparse*Gather` read neighboring texels and return a single component of each.
- `OpImage*Dref*` instructions apply depth comparison on the texel values.
- `OpImageSparse*` instructions additionally return a sparse residency code.
- `OpImageQueryLod` returns the LOD parameters that would be used in a sample operation. The actual operation is not performed.

20.1. Sampling Coordinate Systems

There are three sampling coordinate systems used in this chapter:

- Normalized Sampling Coordinates
- Unnormalized Sampling Coordinates
- Integer Sampling Coordinates

SPIR-V `OpImageFetch`, and `OpImageSparseFetch` instructions use integer sampling coordinates.

Other image instructions **can** use either normalized or unnormalized sampling coordinates (selected by the `unnormalizedCoordinates` state of the sampler used in the instruction), but there are limitations on what operations, image state, and sampler state are supported. Normalized coordinates are logically converted to unnormalized as part of image operations, and certain steps are only performed on normalized coordinates. The array layer coordinate is always treated as unnormalized even when other coordinates are normalized.

20.1.1. Normalized Sampling Coordinates

Normalized sampling coordinates are referred to as (s,t,r,q,a), with the coordinates having the following meanings:

- s: Coordinate in the first dimension of an image.
- t: Coordinate in the second dimension of an image.
- r: Coordinate in the third dimension of an image.
  - (s,t,r) are interpreted as a direction vector for Cube images.
- q: Fourth coordinate, for homogeneous (projective) coordinates.
- a: Coordinate for array layer.

The values s,t,r,q,a are all floating-point values. Values of s,t,r in the normalized range [0.0,1.0] and a in the range [0.0, layers - 1] correspond to locations in the image being sampled, where layers is the dimension of the image with the same name. The value q is used as a divisor for each of s,t,r, rather than directly corresponding to a location in an image.

The coordinates are extracted from the SPIR-V operand based on the dimensionality of the image variable and type of instruction. For `Proj` instructions, the components are in order (s, [t,] [r,] q), with t and r being conditionally present based on the `Dim` of the image. For non-`Proj` instructions, the coordinates are (s [,t] [,r] [,a]), with t and r being conditionally present based on the `Dim` of the image and a being conditionally present based on the `Arrayed` property of the image. Projective image instructions are not supported on `Arrayed` images.

20.1.2. Unnormalized Sampling Coordinates

Unnormalized sampling coordinates are referred to as (u,v,w,a), with the coordinates having the following meanings:

- u: Coordinate in the first dimension of an image.
- v: Coordinate in the second dimension of an image.
- w: Coordinate in the third dimension of an image.
- a: Coordinate for array layer.

The values u,v,w,a are all floating-point values, with values in the ranges [0.0,width), [0.0, height), [0.0, depth), and [0,layers-1] corresponding to locations in the image being sampled. The values width, height, depth, and layers are the dimensions of the accessed image.

Only the u and v coordinates are directly extracted from the SPIR-V operand, because only 1D and 2D (non-`Arrayed`) dimensionalities support unnormalized coordinates. The components are in order (u [,v]), with v being conditionally present when the dimensionality is 2D. When normalized coordinates are converted to unnormalized coordinates, all four coordinates are used.

20.1.3. Integer Sampling Coordinates

Integer sampling coordinates are referred to as (i,j,k,l,n), with the coordinates having the following meanings:

- i: Coordinate in the first dimension of an image.
- j: Coordinate in the second dimension of an image.

- k: Coordinate in the third dimension of an image.
- l: Coordinate for array layer.
- n: Index of the sample within the texel.

The values i,j,k,l,n are all integer values, with values in the ranges [0,width), [0, height), [0, depth), [0,layers), and [0,samples) corresponding to locations in the image being sampled. The values width, height, depth, layers, and samples are the dimensions of the accessed image.

They are extracted from the SPIR-V operand in order (i [,j] [,k] [,l] [,n]), with j and k conditionally present based on the `Dim` of the image, and l conditionally present based on the `Arrayed` property of the image. n is conditionally present and is taken from the `Sample` image operand.

Integer coordinates are used as image coordinates to perform an image read after sampling calculations, directly translating each coordinate as follows:

- i → x
- j → y
- k → z
- l → layer
- n → sample

level is calculated separately via the `Lod` image operand if present, or is set to 0 otherwise.

Integer sampling coordinates are used as image coordinates to perform an image read after sampling calculations, directly translating each coordinate as follows:

- x = i
- y = j
- z = k
- layer = l + VkImageSubresourceRange::`baseArrayLayer`
- sample = n

20.1.4. Sampling Coordinate Diagrams

The following diagrams illustrate the different sampling coordinate systems for 2D images.

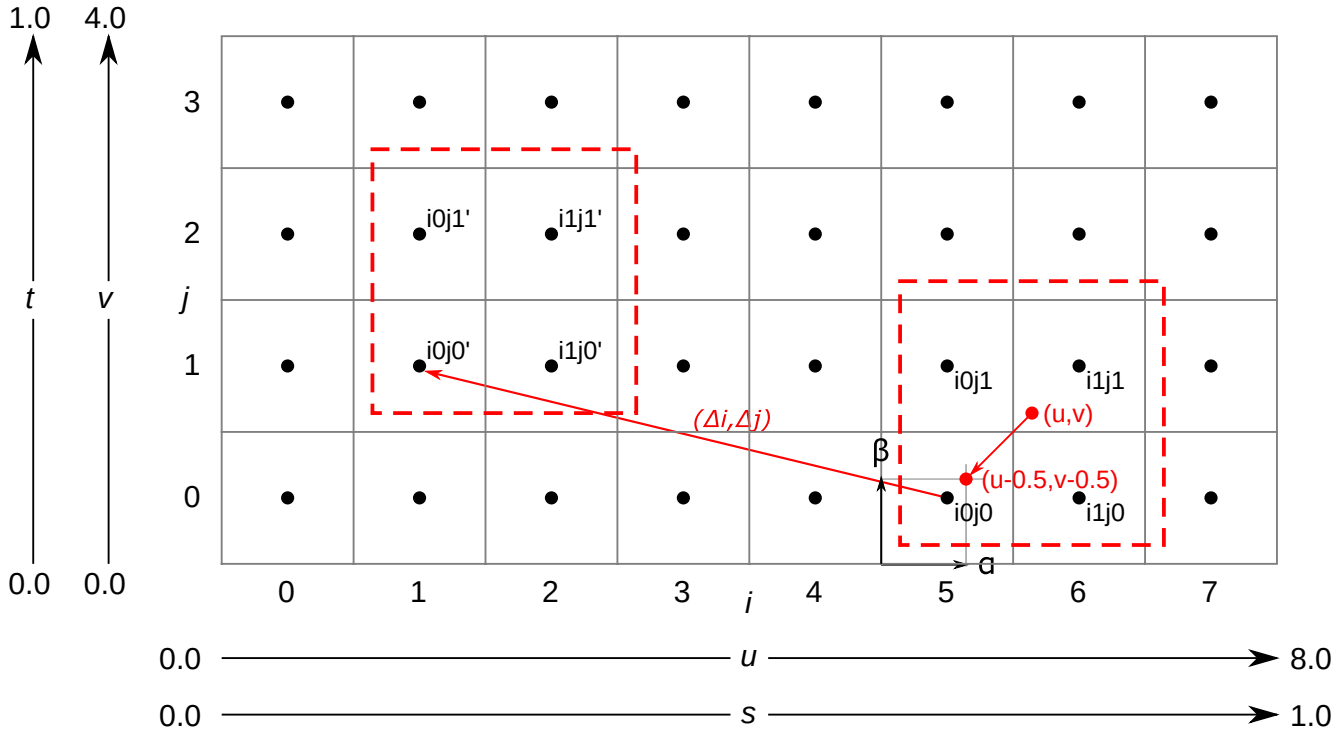


Figure 3. Texel Coordinate Systems, Linear Filtering

The Texel Coordinate Systems - For the example shown of an 8×4 texel two dimensional image.

- Normalized texel coordinates:
  - The s coordinate goes from 0.0 to 1.0.
  - The t coordinate goes from 0.0 to 1.0.
- Unnormalized texel coordinates:
  - The u coordinate within the range 0.0 to 8.0 is within the image, otherwise it is outside the image.
  - The v coordinate within the range 0.0 to 4.0 is within the image, otherwise it is outside the image.
- Integer texel coordinates:
  - The i coordinate within the range 0 to 7 addresses texels within the image, otherwise it is outside the image.
  - The j coordinate within the range 0 to 3 addresses texels within the image, otherwise it is outside the image.
- Also shown for linear filtering:
  - Given the unnormalized coordinates (u,v), the four texels selected are i_0j_0, i_1j_0, i_0j_1, and i_1j_1.
  - The fractions α and β.
  - Given the offset Δ_i and Δ_j, the four texels selected by the offset are $i_0j'_0$, $i_1j'_0$, $i_0j'_1$, and $i_1j'_1$.

NOTE For formats with reduced-resolution components, Δ_i and Δ_j are relative to the resolution of the highest-resolution component, and therefore may be divided by two relative to the unnormalized coordinate space of the lower-resolution components.

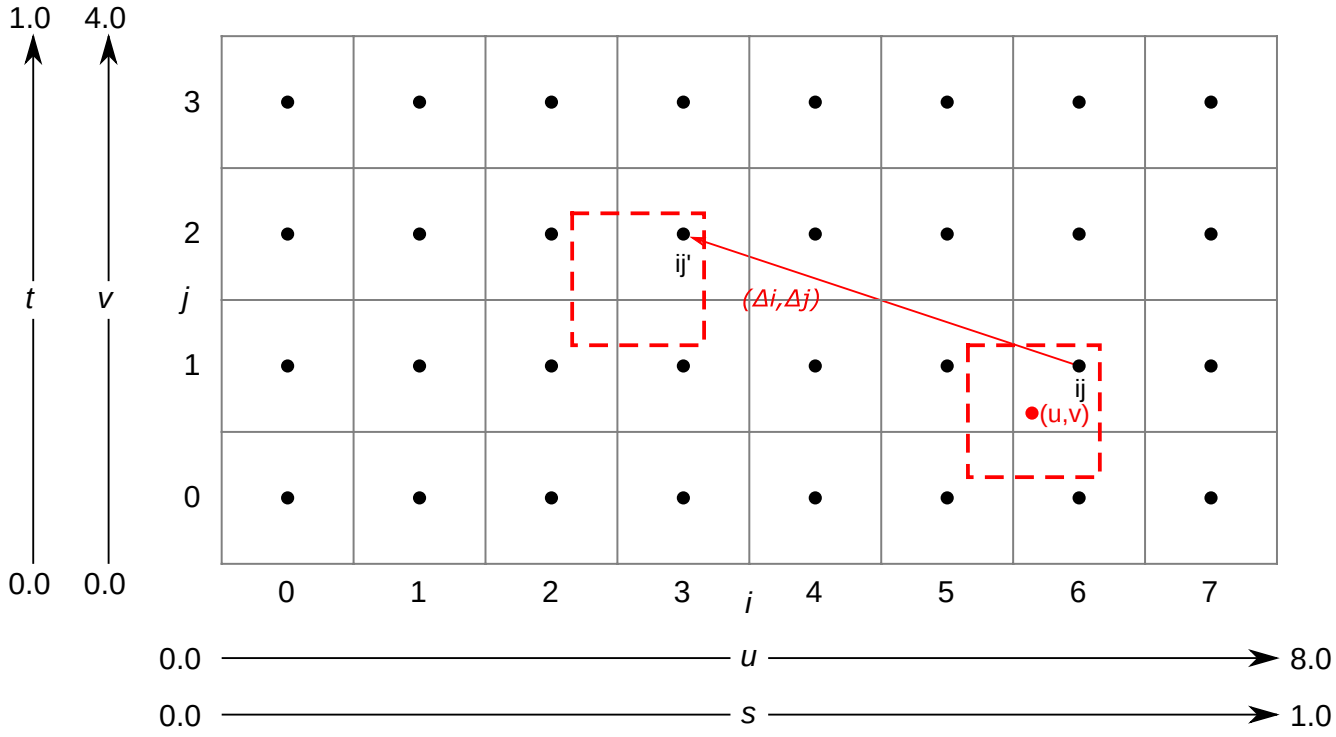


Figure 4. Texel Coordinate Systems, Nearest Filtering

The Texel Coordinate Systems - For the example shown of an 8×4 texel two dimensional image.

- Texel coordinates as above. Also shown for nearest filtering:
  - Given the unnormalized coordinates (u,v), the texel selected is ij.
  - Given the offset Δ_i and Δ_j, the texel selected by the offset is ij'.

20.2. Sampling Operations

Sampling instructions are SPIR-V image instructions that read from an image with a sampler. *Sampling operations* are a set of steps that are performed on state, coordinates, and texel values while processing a sampling instruction, and which are common to some or all sampling instructions. They include the following steps, which are performed in the listed order:

- Validation operations
  - Instruction/Format/Sampler/Image validation
  - Coordinate validation
  - Layout validation
  - Cube Map Edge Handling
- Border Replacement
- Texel Reads
- Depth comparison
- Component swizzle
- Chroma reconstruction
- Y'C_BC_R conversion

For sampling instructions involving multiple texels (for sampling or gathering), these steps are applied for each texel that is used in the instruction. Depending on the type of sampling instruction, other steps are conditionally performed between these steps or involving multiple coordinate or texel values.

If Chroma Reconstruction is implicit, Texel Filtering instead takes place during chroma reconstruction, before sampler $Y'C_BC_R$ conversion occurs.

20.2.1. Texel Input Validation Operations

Texel input validation operations inspect instruction/image/sampler state or coordinates, and in certain circumstances cause the texel value to be replaced or become undefined. There are a series of validations that the texel undergoes.

Instruction/Format/Sampler/Image View Validation

There are a number of cases where a SPIR-V instruction **can** mismatch with the sampler, the image view, or both, and a number of further cases where the sampler **can** mismatch with the image view. In such cases the value of the texel returned is poison.

These cases include:

- The sampler `borderColor` is an integer type and the image view `format` is not one of the VkFormat integer types or a stencil component of a depth/stencil format.
- The sampler `borderColor` is a float type and the image view `format` is not one of the VkFormat float types or a depth component of a depth/stencil format.
- The sampler `borderColor` is one of the opaque black colors (VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK or VK_BORDER_COLOR_INT_OPAQUE_BLACK) and the image view VkComponentSwizzle for any of the VkComponentMapping components is not the identity swizzle, and VkPhysicalDeviceBorderColorSwizzleFeaturesEXT::`borderColorSwizzleFromImage` feature is not enabled, and VkSamplerBorderColorComponentMappingCreateInfoEXT is not specified.
- VkSamplerBorderColorComponentMappingCreateInfoEXT::`components`, if specified, has a component swizzle that does not match the component swizzle of the image view, and either component swizzle is not a form of identity swizzle.
- VkSamplerBorderColorComponentMappingCreateInfoEXT::`srgb`, if specified, does not match the sRGB encoding of the image view.
- The sampler `borderColor` is a custom color (VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT) and the supplied VkSamplerCustomBorderColorCreateInfoEXT::`customBorderColor` is outside the bounds of the values representable in the image view's `format`.
- The sampler `borderColor` is a custom color (VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT) and the image view VkComponentSwizzle for any of the VkComponentMapping components is not the identity swizzle, and VkPhysicalDeviceBorderColorSwizzleFeaturesEXT::`borderColorSwizzleFromImage` feature is not enabled, and VkSamplerBorderColorComponentMappingCreateInfoEXT is not specified.

- The VkImageLayout of any subresource in the image view does not match the VkDescriptorImageInfo::imageLayout used to write the image descriptor.
- The SPIR-V Image Format is not compatible with the image view's or buffer view's format.
- The sampler unnormalizedCoordinates is VK_TRUE and any of the limitations of unnormalized coordinates are violated.
- The sampler was created with flags containing VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT and the image was not created with flags containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT.
- The sampler was not created with flags containing VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT and the image was created with flags containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT.
- The sampler was created with flags containing VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT and is used with a function that is not OpImageSampleImplicitLod or OpImageSampleExplicitLod, or is used with operands Offset or ConstOffsets.
- The SPIR-V instruction is one of the OpImage*Dref* instructions and the sampler compareEnable is VK_FALSE.
- The SPIR-V instruction is not one of the OpImage*Dref* instructions and the sampler compareEnable is VK_TRUE.
- The SPIR-V instruction is one of the OpImage*Dref* instructions, the image view format is one of the depth/stencil formats, and the image view aspect is not VK_IMAGE_ASPECT_DEPTH_BIT.
- The SPIR-V instruction's image variable's properties are not compatible with the image view:
  - If the image view's viewType is VK_IMAGE_VIEW_TYPE_1D_ARRAY or VK_IMAGE_VIEW_TYPE_2D_ARRAY, and the image was not created with VK_IMAGE_CREATE_ALIAS_SINGLE_LAYER_DESCRIPTOR_BIT_KHR, then the instruction **must** have Arrayed = 1.
  - If the image view's viewType is VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D, and the image was not created with VK_IMAGE_CREATE_ALIAS_SINGLE_LAYER_DESCRIPTOR_BIT_KHR, the instruction **must** have Arrayed = 0.
  - If the image view's viewType is VK_IMAGE_VIEW_TYPE_1D_ARRAY or VK_IMAGE_VIEW_TYPE_2D_ARRAY and Arrayed = 0, the image view **must** have been created with a single layer.
  - If the image view's viewType is VK_IMAGE_VIEW_TYPE_CUBE_ARRAY then the instruction **must** have Arrayed = 1.
  - If the image view's viewType is VK_IMAGE_VIEW_TYPE_CUBE then the instruction **must** have Arrayed = 0.
  - If the image was created with VkImageCreateInfo::samples equal to VK_SAMPLE_COUNT_1_BIT, the instruction **must** have MS = 0.
  - If the image was created with VkImageCreateInfo::samples not equal to VK_SAMPLE_COUNT_1_BIT, the instruction **must** have MS = 1.
  - If the Sampled Type of the OpTypeImage does not match the SPIR-V Type.

    - If the signedness of any read or sample operation does not match the signedness of the image's format.
- The sampler was created with a specified VkSamplerCustomBorderColorCreateInfoEXT::`format` which does not match the VkFormat of the image view(s) it is sampling.
- The sampler is sampling an image view of VK_FORMAT_B4G4R4A4_UNORM_PACK16, VK_FORMAT_B5G6R5_UNORM_PACK16, or VK_FORMAT_B5G5R5A1_UNORM_PACK16 format without a specified VkSamplerCustomBorderColorCreateInfoEXT::`format`.

If the underlying `VkImage` format has an X component in its format description, undefined values are read from those bits.

NOTE

If the `VkImage` format and `VkImageView` format are the same, these bits will be unused by format conversion and this will have no effect. However, if the `VkImageView` format is different, then some bits of the result may be undefined. For example, when a VK_FORMAT_R10X6_UNORM_PACK16 `VkImage` is sampled via a VK_FORMAT_R16_UNORM `VkImageView`, the low 6 bits of the value before format conversion are undefined and format conversion may return a range of different values.

NOTE

When accessing storage images or texel buffers, there are many valid usage statements that prevent the `OpTypeImage` and VkFormat from being different with respect to things such as numeric type, number of components, etc. Texel output validation covers cases such as using `R32ui` with VK_FORMAT_R8_UINT that for historical reasons are not considered invalid usage, but still result in undefined values. Validation layers will provide a warning in these undefined value cases.

NOTE

Some implementations will return undefined values in the case where a sampler uses a VkSamplerAddressMode of VK_SAMPLER_ADDRESS_MODE_MIRRORED_REPEAT, the sampler is used with operands `Offset`, `ConstOffset`, or `ConstOffsets`, and the value of the offset is greater than or equal to the corresponding width, height, or depth of any accessed image level.

This behavior was not tested prior to Vulkan conformance test suite version 1.3.8.0. Affected implementations will have a conformance test waiver for this issue.

Layout Validation

If all planes of a *disjoint multi-planar* image are not in the same image layout, the image **must** not be sampled with sampler $Y'C_BC_R$ conversion enabled.

Coordinate Validation

Once the normalized or unnormalized coordinates have been converted to integer image coordinates, the integer coordinates are validated as image coordinates, as outlined in Image Coordinate Validation, converted as follows:

- x = i
- y = j
- z = k
- layer = l
- sample = n
- level = d

Cube Map Edge Handling

When sampling a cube map, if the image coordinates are out of bounds of the selected cube map face, the following steps are performed.

> **NOTE** This does not occur when using VK_FILTER_NEAREST filtering within a mip level, since VK_FILTER_NEAREST is treated as using VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE.

- Cube Map Edge Texel
  - If the texel lies beyond the selected cube map face in either only x or only y, then the coordinates (x,y,layer) are transformed to select the adjacent texel from the appropriate neighboring face.
- Cube Map Corner Texel
  - If the texel lies beyond the selected cube map face in both x and y, then there is no unique neighboring face from which to read that texel. The texel **should** be replaced by the average of the three values of the adjacent texels in each incident face. However, implementations **may** replace the cube map corner texel by other methods. The methods are subject to the constraint that for linear filtering if the three available texels have the same value, the resulting filtered texel **must** have that value, and for cubic filtering if the twelve available samples have the same value, the resulting filtered texel **must** have that value.

20.2.2. Border Replacement

If the sampler includes a border, out of bounds texels are replaced with a value based on the image format and the `borderColor` of the sampler. The border color is:

Table 26. Border Color B, Custom Border Color VkSamplerCustomBorderColorCreateInfoEXT ::customBorderColor U

Sampler `borderColor`	**Corresponding Border Color**
VK_BORDER_COLOR_FLOAT_TRANSPARENT_BLACK	$[B_r, B_g, B_b, B_a] = [0.0, 0.0, 0.0, 0.0]$
VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK	$[B_r, B_g, B_b, B_a] = [0.0, 0.0, 0.0, 1.0]$
VK_BORDER_COLOR_FLOAT_OPAQUE_WHITE	$[B_r, B_g, B_b, B_a] = [1.0, 1.0, 1.0, 1.0]$
VK_BORDER_COLOR_INT_TRANSPARENT_BLACK	$[B_r, B_g, B_b, B_a] = [0, 0, 0, 0]$
VK_BORDER_COLOR_INT_OPAQUE_BLACK	$[B_r, B_g, B_b, B_a] = [0, 0, 0, 1]$

Sampler `borderColor`	Corresponding Border Color
VK_BORDER_COLOR_INT_OPAQUE_WHITE	$[B_r, B_g, B_b, B_a] = [1, 1, 1, 1]$
VK_BORDER_COLOR_FLOAT_CUSTOM_EXT	$[B_r, B_g, B_b, B_a] = [U_r, U_g, U_b, U_a]$
VK_BORDER_COLOR_INT_CUSTOM_EXT	$[B_r, B_g, B_b, B_a] = [U_r, U_g, U_b, U_a]$

The custom border color (U) **may** be rounded by implementations prior to texel replacement, but the error introduced by such a rounding **must** not exceed one ULP of the image's `format`.

NOTE The names `VK_BORDER_COLOR_*_TRANSPARENT_BLACK`, `VK_BORDER_COLOR_*_OPAQUE_BLACK`, and `VK_BORDER_COLOR_*_OPAQUE_WHITE` are meant to describe which components are zeros and ones in the vocabulary of compositing, and are not meant to imply that the numerical value of VK_BORDER_COLOR_INT_OPAQUE_WHITE is a saturating value for integers.

This is substituted for the texel value by replacing the number of components in the image format.

Table 27. Border Texel Components After Replacement

Texel Aspect or Format	Component Assignment
Depth aspect	$D = B_r$
Stencil aspect	$S = B_r$†
One component color format	$Color_r = B_r$
Two component color format	$[Color_r,Color_g] = [B_r,B_g]$
Three component color format	$[Color_r,Color_g,Color_b] = [B_r,B_g,B_b]$
Four component color format	$[Color_r,Color_g,Color_b,Color_a] = [B_r,B_g,B_b,B_a]$
Single component alpha format	$[Color_r,Color_g,Color_b, Color_a] = [0,0,0,B_a]$

† $S = B_g$ **may** be substituted as the replacement method by the implementation when VkSamplerCreateInfo::`borderColor` is VK_BORDER_COLOR_INT_CUSTOM_EXT and VkSamplerCustomBorderColorCreateInfoEXT::`format` is VK_FORMAT_UNDEFINED. Implementations **should** use $S = B_r$ as the replacement method.

If `rgba4OpaqueBlackSwizzled` is VK_FALSE, the implementation **may** swap the blue and alpha channels when sampling non-custom border colors with the VK_FORMAT_B4G4R4A4_UNORM_PACK16 format, or the red and alpha channels with the VK_FORMAT_R4G4B4A4_UNORM_PACK16 format.

NOTE As VK_FORMAT_B4G4R4A4_UNORM_PACK16 is required by Vulkan, support must be advertised for this format. Some Vulkan implementations on Apple hardware implement these formats through a hardware format with a different channel order, swizzled to match Vulkan's expectations. Unfortunately the swizzle cannot be readily applied to the fixed border colors - resulting in the apparent channel swap. For most standard border colors this does not result in a modification to the sampled output. However, `VK_BORDER_COLOR_*_OPAQUE_BLACK` will instead be sampled as transparent red or blue. If the `customBorderColorWithoutFormat` feature is

> supported and enabled, this functionality is expected to work without issue, but this feature may come with a performance cost.

When border color replacement occurs, texel reads are skipped, and the replaced color is used for ongoing operations instead.

20.2.3. Texel Reads

A texel is read from an image, performed as outlined in Image Reads, using the converted image coordinates.

The returned components of each texel are then processed by further input operations.

20.2.4. Depth Compare Operation

If the image view has a depth/stencil format, the depth component is selected by the `aspectMask`, and the operation is an `OpImage*Dref*` instruction, a depth comparison is performed. The result is 1.0 if the comparison evaluates to true, and 0.0 otherwise. This value replaces the depth component D.

The compare operation is selected by the VkCompareOp value set by VkSamplerCreateInfo ::`compareOp`. The reference value from the SPIR-V operand D_{ref} and the texel depth value D_{tex} are used as the *reference* and *test* values, respectively, in that operation.

If the image being sampled has an unsigned normalized fixed-point format, then D_{ref} is clamped to [0,1] before the compare operation.

If the value of `magFilter` is VK_FILTER_LINEAR, or the value of `minFilter` is VK_FILTER_LINEAR, then D may be computed in an implementation-dependent manner which differs from the normal rules of linear filtering. The resulting value **must** be in the range [0,1] and should be proportional to, or a weighted average of, the number of comparison passes or failures.

20.2.5. Component Swizzle

All texel input instructions apply a *swizzle* based on:

- the VkComponentSwizzle enums in the `components` member of the VkImageViewCreateInfo structure for the image being read if sampler $Y'C_BC_R$ conversion is not enabled, and
- the VkComponentSwizzle enums in the `components` member of the VkSamplerYcbcrConversionCreateInfo structure for the sampler $Y'C_BC_R$ conversion if sampler $Y'C_BC_R$ conversion is enabled.

The swizzle **can** rearrange the components of the texel, or substitute zero or one for any components. It is defined as follows for each color component:

$$Color'_{component} = \begin{cases} Color_r & \text{for RED swizzle} \\ Color_g & \text{for GREEN swizzle} \\ Color_b & \text{for BLUE swizzle} \\ Color_a & \text{for ALPHA swizzle} \\ 0 & \text{for ZERO swizzle} \\ one & \text{for ONE swizzle} \\ identity & \text{for IDENTITY swizzle} \end{cases}$$

where:

$$one = \begin{cases} 1.0\text{f} & \text{for floating-point components} \\ 1 & \text{for integer components} \end{cases}$$

$$identity = \begin{cases} Color_r & \text{for } component = r \\ Color_g & \text{for } component = g \\ Color_b & \text{for } component = b \\ Color_a & \text{for } component = a \end{cases}$$

If the border color is one of the VK_BORDER_COLOR_*_OPAQUE_BLACK enums and the VkComponentSwizzle is not the identity swizzle for all components, the value of the texel after swizzle is undefined.

If the image view has a depth/stencil format and the VkComponentSwizzle is VK_COMPONENT_SWIZZLE_ONE, and VkPhysicalDeviceMaintenance5Properties ::depthStencilSwizzleOneSupport is not VK_TRUE, the value of the texel after swizzle is undefined.

20.2.6. Sparse Residency

OpImageSparse* instructions return a structure which includes a *residency code* indicating whether any texels accessed by the instruction are sparse unbound texels. This code **can** be interpreted by the OpImageSparseTexelsResident instruction which converts the residency code to a boolean value.

20.2.7. Chroma Reconstruction

In some color models, the color representation is defined in terms of monochromatic light intensity (often called “luma”) and color differences relative to this intensity, often called “chroma”. It is common for color models other than RGB to represent the chroma components at lower spatial resolution than the luma component. This approach is used to take advantage of the eye’s lower spatial sensitivity to color compared with its sensitivity to brightness. Less commonly, the same approach is used with additive color, since the green component dominates the eye’s sensitivity to light intensity and the spatial sensitivity to color introduced by red and blue is lower.

Lower-resolution components are “downsampled” by resizing them to a lower spatial resolution than the component representing luminance. This process is also commonly known as “chroma subsampling”. There is one luminance sample in each texture texel, but each chrominance sample may be shared among several texels in one or both texture dimensions.

- “_444” formats do not spatially downsample chroma values compared with luma: there are

unique chroma samples for each texel.

- "_422" formats have downsampling in the x dimension (corresponding to u or s coordinates): they are sampled at half the resolution of luma in that dimension.
- "_420" formats have downsampling in the x dimension (corresponding to u or s coordinates) and the y dimension (corresponding to v or t coordinates): they are sampled at half the resolution of luma in both dimensions.

The process of reconstructing a full color value for texture access involves accessing both chroma and luma values at the same location. To generate the color accurately, the values of the lower-resolution components at the location of the luma samples are reconstructed from the lower-resolution sample locations, an operation known here as "chroma reconstruction" irrespective of the actual color model.

The location of the chroma samples relative to the luma coordinates is determined by the xChromaOffset and yChromaOffset members of the VkSamplerYcbcrConversionCreateInfo structure used to create the sampler $Y'C_BC_R$ conversion.

The following diagrams show the relationship between unnormalized (u,v) coordinates and (i,j) integer texel positions in the luma component (shown in black, with circles showing integer sample positions) and the texel coordinates of reduced-resolution chroma components, shown as crosses in red.

NOTE

If the chroma values are reconstructed at the locations of the luma samples by means of interpolation, chroma samples from outside the image bounds are needed; these are determined according to Wrapping Operation. These diagrams represent this by showing the bounds of the "chroma texel" extending beyond the image bounds, and including additional chroma sample positions where required for interpolation. The limits of a sample for NEAREST sampling are shown as a grid.

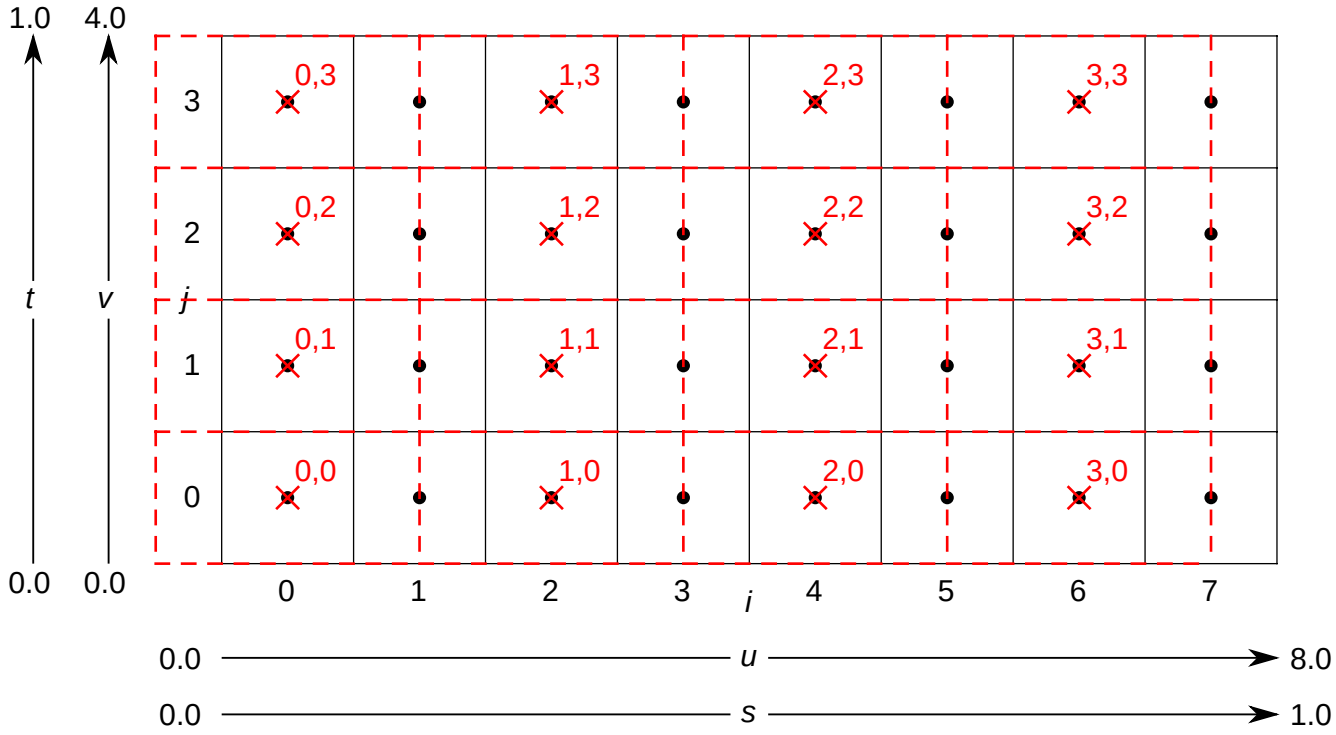


Figure 5. 422 downsampling, xChromaOffset=COSITED_EVEN

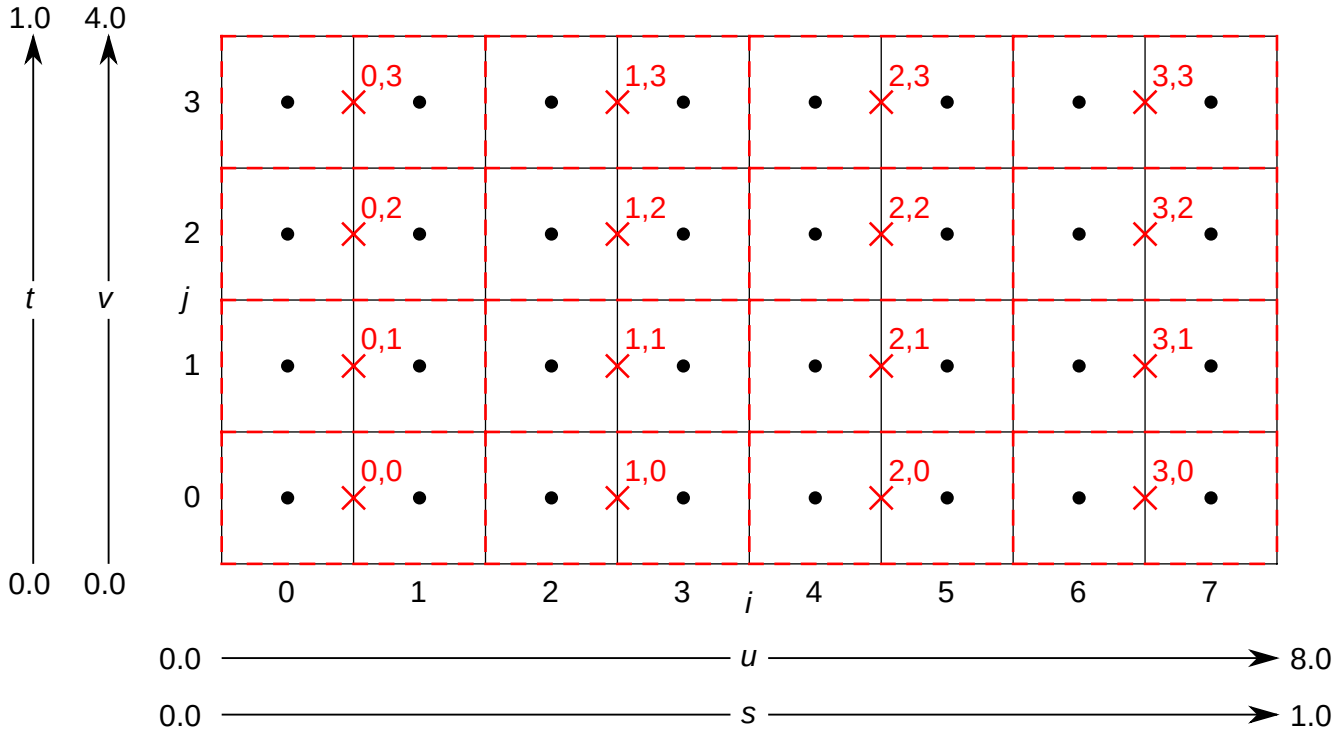


Figure 6. 422 downsampling, xChromaOffset=MIDPOINT

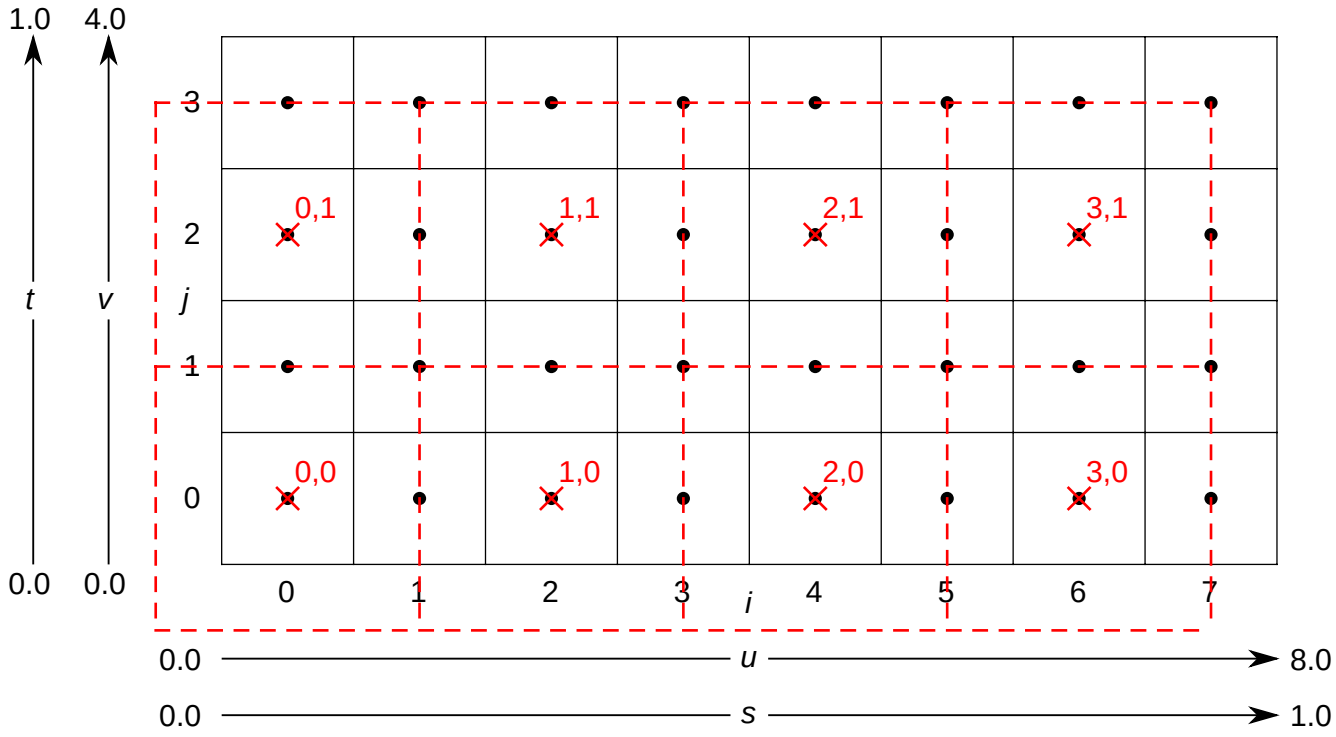


Figure 7. 420 downsampling, xChromaOffset=COSITED_EVEN, yChromaOffset=COSITED_EVEN

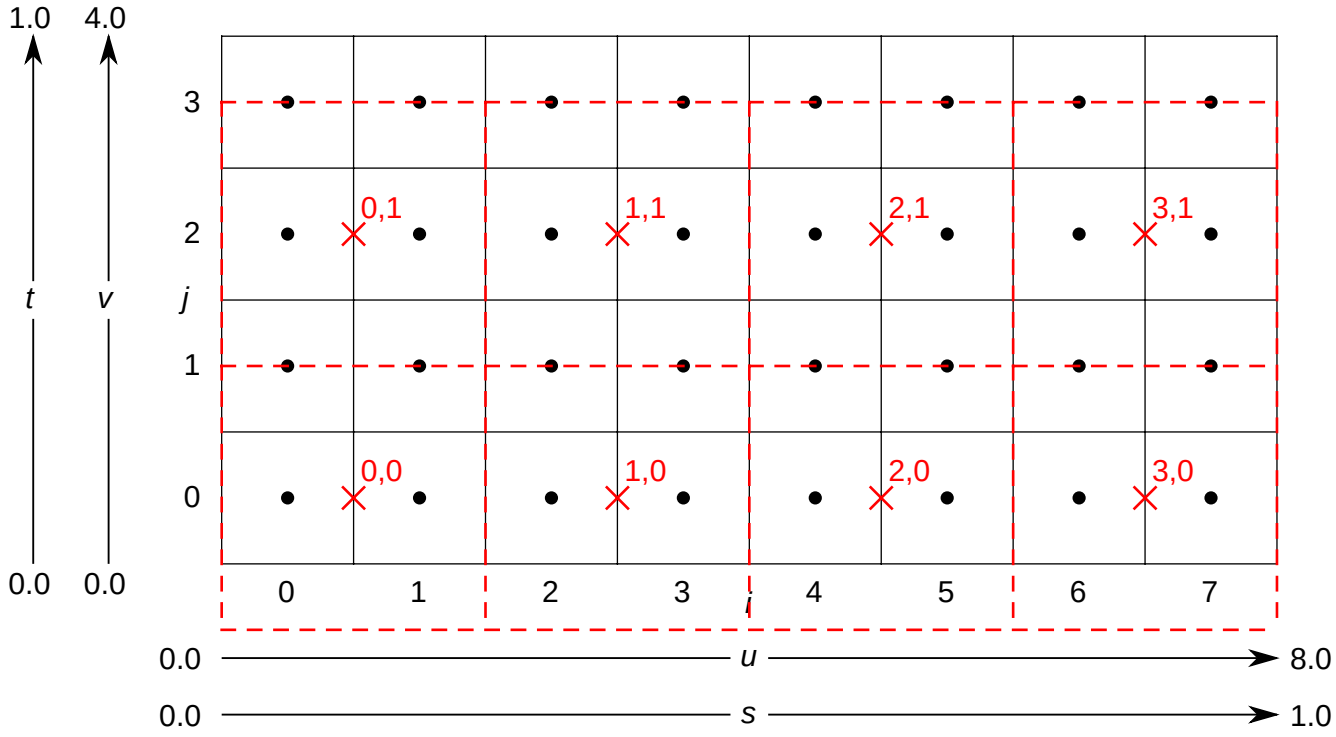


Figure 8. 420 downsampling, xChromaOffset=MIDPOINT, yChromaOffset=COSITED_EVEN

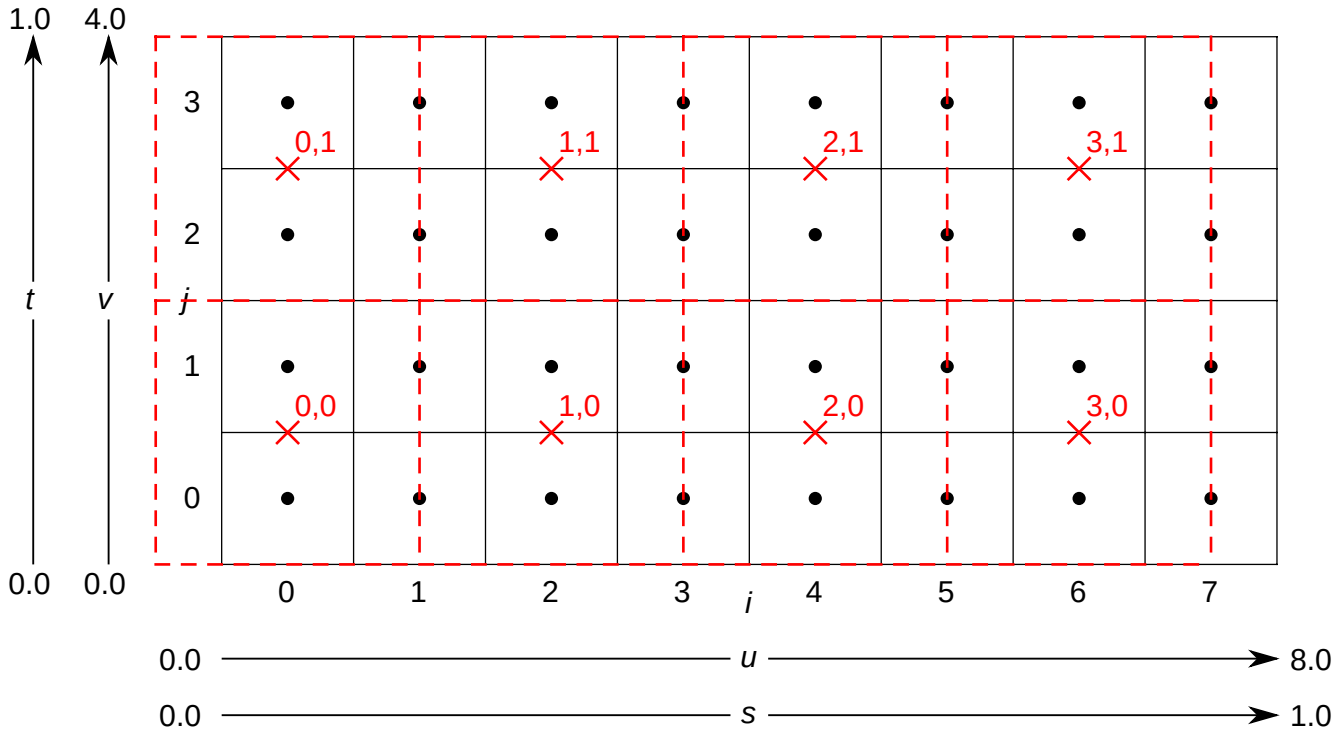


Figure 9. 420 downsampling, xChromaOffset=COSITED_EVEN, yChromaOffset=MIDPOINT

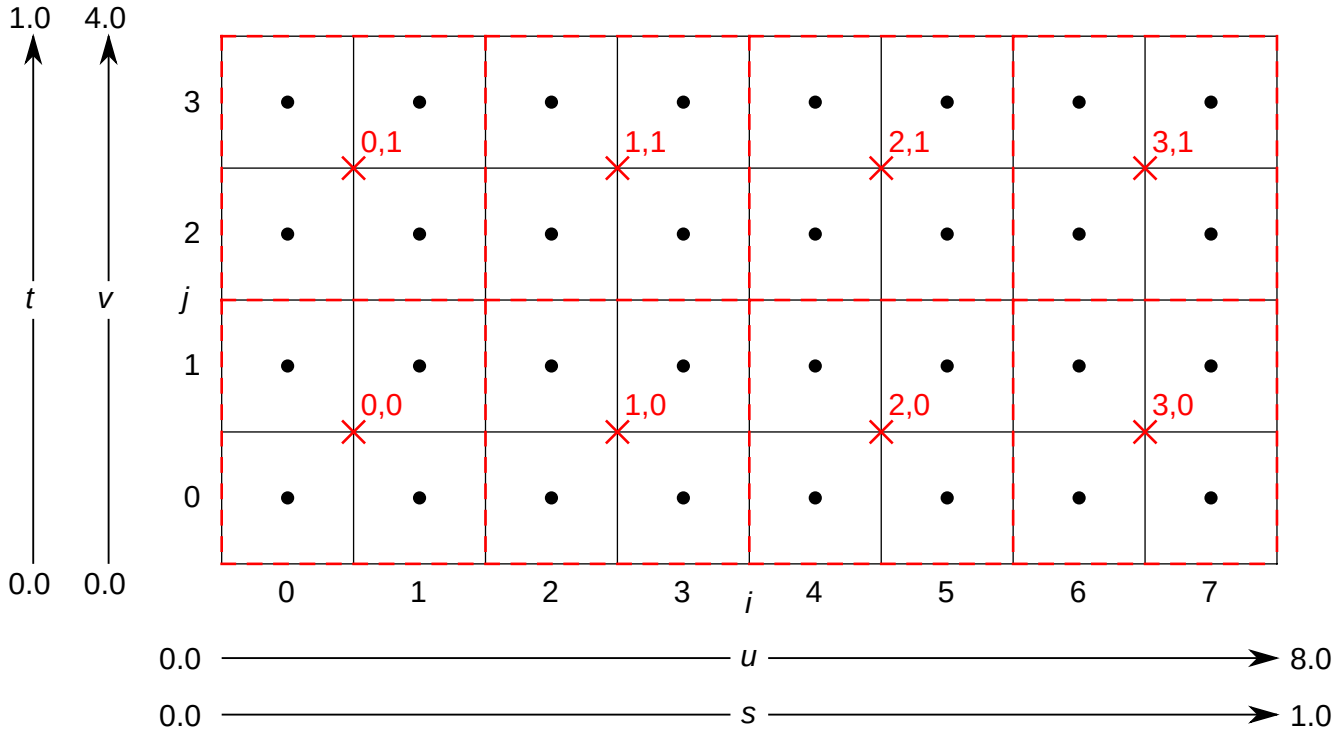


Figure 10. 420 downsampling, xChromaOffset=MIDPOINT, yChromaOffset=MIDPOINT

Reconstruction is implemented in one of two ways:

If the format of the image that is to be sampled sets VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT, or the VkSamplerYcbcrConversionCreateInfo's `forceExplicitReconstruction` is VK_TRUE, reconstruction is performed as an explicit step independent of filtering, described in the Explicit Reconstruction section.

If the format of the image that is to be sampled does not set VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT and if the VkSamplerYcbcrConversionCreateInfo's `forceExplicitReconstruction` is VK_FALSE, reconstruction is performed as an implicit part of filtering prior to color model conversion, with no separate post-conversion texel filtering step, as described in the Implicit Reconstruction section.

Explicit Reconstruction

- If the `chromaFilter` member of the VkSamplerYcbcrConversionCreateInfo structure is VK_FILTER_NEAREST:
  - If the format's R and B components are reduced in resolution in just width by a factor of two relative to the G component (i.e. this is a "`_422`" format), the $\tau_{ijk}[level]$ values accessed by texel filtering are reconstructed as follows:

$$\tau_R{}'(i,\ j) = \tau_R(\lfloor i \times 0.5 \rfloor,\ j)[level]$$
$$\tau_B{}'(i,\ j) = \tau_B(\lfloor i \times 0.5 \rfloor,\ j)[level]$$

  - If the format's R and B components are reduced in resolution in width and height by a factor of two relative to the G component (i.e. this is a "`_420`" format), the $\tau_{ijk}[level]$ values accessed by texel filtering are reconstructed as follows:

$$\tau_R{}'(i,\ j) = \tau_R(\lfloor i \times 0.5 \rfloor,\ \lfloor j \times 0.5 \rfloor)[level]$$
$$\tau_B{}'(i,\ j) = \tau_B(\lfloor i \times 0.5 \rfloor,\ \lfloor j \times 0.5 \rfloor)[level]$$

NOTE `xChromaOffset` and `yChromaOffset` have no effect if `chromaFilter` is VK_FILTER_NEAREST for explicit reconstruction.

- If the `chromaFilter` member of the VkSamplerYcbcrConversionCreateInfo structure is VK_FILTER_LINEAR:
  - If the format's R and B components are reduced in resolution in just width by a factor of two relative to the G component (i.e. this is a "`_422`" format):
    - If `xChromaOffset` is VK_CHROMA_LOCATION_COSITED_EVEN:

$$\tau_{RB}{}'(i,\ j) = \begin{cases} \tau_{RB}(\lfloor i \times 0.5 \rfloor,\ j)[level], & 0.5 \times i = \lfloor 0.5 \times i \rfloor \\ 0.5 \times \tau_{RB}(\lfloor i \times 0.5 \rfloor,\ j)[level] + \\ 0.5 \times \tau_{RB}(\lfloor i \times 0.5 \rfloor + 1,\ j)[level], & 0.5 \times i \neq \lfloor 0.5 \times i \rfloor \end{cases}$$

    - If `xChromaOffset` is VK_CHROMA_LOCATION_MIDPOINT:

$$\tau_{RB}{}'(i,\ j) = \begin{cases} 0.25 \times \tau_{RB}(\lfloor i \times 0.5 \rfloor - 1,\ j)[level] + \\ 0.75 \times \tau_{RB}(\lfloor i \times 0.5 \rfloor,\ j)[level], & 0.5 \times i = \lfloor 0.5 \times i \rfloor \\ 0.75 \times \tau_{RB}(\lfloor i \times 0.5 \rfloor,\ j)[level] + \\ 0.25 \times \tau_{RB}(\lfloor i \times 0.5 \rfloor + 1,\ j)[level], & 0.5 \times i \neq \lfloor 0.5 \times i \rfloor \end{cases}$$

  - If the format's R and B components are reduced in resolution in width and height by a factor of two relative to the G component (i.e. this is a "`_420`" format), a similar relationship applies. Due to the number of options, these formulae are expressed more concisely as follows:

$$i_{RB} = \begin{cases} 0.5 \times (i) & \text{xChromaOffset=COSITED_EVEN} \\ 0.5 \times (i - 0.5) & \text{xChromaOffset=MIDPOINT} \end{cases}$$
$$j_{RB} = \begin{cases} 0.5 \times (j) & \text{yChromaOffset=COSITED_EVEN} \\ 0.5 \times (j - 0.5) & \text{yChromaOffset=MIDPOINT} \end{cases}$$
$$i_{floor} = \lfloor i_{RB} \rfloor$$
$$j_{floor} = \lfloor j_{RB} \rfloor$$
$$i_{frac} = i_{RB} - i_{floor}$$
$$j_{frac} = j_{RB} - j_{floor}$$

$$\begin{aligned} \tau_{RB}{}'(i,\ j) = {} & \tau_{RB}(i_{floor},\ j_{floor})[level] && \times (1 - i_{frac}) && \times && (1 - j_{frac}) + \\ & \tau_{RB}(1 + i_{floor},\ j_{floor})[level] && \times (i_{frac}) && \times && (1 - j_{frac}) + \\ & \tau_{RB}(i_{floor},\ 1 + j_{floor})[level] && \times (1 - i_{frac}) && \times && (j_{frac}) + \\ & \tau_{RB}(1 + i_{floor},\ 1 + j_{floor})[level] && \times (i_{frac}) && \times && (j_{frac}) \end{aligned}$$

NOTE In the case where the texture itself is bilinearly interpolated as described in Texel Filtering, thus requiring four full-color samples for the filtering operation, and where the reconstruction of these samples uses bilinear interpolation in the chroma

components due to chromaFilter=VK_FILTER_LINEAR, up to nine chroma samples may be required, depending on the sample location.

Implicit Reconstruction

Implicit reconstruction takes place by the samples being interpolated, as required by the filter settings of the sampler, except that chromaFilter takes precedence for the chroma samples.

If chromaFilter is VK_FILTER_NEAREST, an implementation **may** behave as if xChromaOffset and yChromaOffset were both VK_CHROMA_LOCATION_MIDPOINT, irrespective of the values set.

NOTE This will not have any visible effect if the locations of the luma samples coincide with the location of the samples used for rasterization.

The sample coordinates are adjusted by the downsample factor of the component (such that, for example, the sample coordinates are divided by two if the component has a downsample factor of two relative to the luma component):

$$u_{RB}'(422/420) = \begin{cases} 0.5 \times (u + 0.5), & \text{xChromaOffset=COSITED_EVEN} \\ 0.5 \times u, & \text{xChromaOffset=MIDPOINT} \end{cases}$$

$$v_{RB}'(420) = \begin{cases} 0.5 \times (v + 0.5), & \text{yChromaOffset=COSITED_EVEN} \\ 0.5 \times v, & \text{yChromaOffset=MIDPOINT} \end{cases}$$

20.2.8. Sampler $Y'C_BC_R$ Conversion

Sampler $Y'C_BC_R$ conversion performs the following operations on sampled data, in order:

1. Sampler $Y'C_BC_R$ Component Swizzle
2. Sampler $Y'C_BC_R$ Range Expansion
3. Sampler $Y'C_BC_R$ Model Conversion

An implementation **may** combine the range expansion and model conversion into a single mathematical operation.

These operations are applied to color component values after sampling operations which are not specific to sampler $Y'C_BC_R$ conversion have completed.

Sampler $Y'C_BC_R$ Component Swizzle

Rather than the component swizzle performed as part of sampling, which is banned for $Y'C_BC_R$ image views used with sampler $Y'C_BC_R$ conversion, the component swizzle specified by VkSamplerYcbcrConversionCreateInfo::components is applied to the sampled data instead. This is applied in the same way as the component swizzle usually performed during sampling.

Sampler $Y'C_BC_R$ Range Expansion

Sampler $Y'C_BC_R$ range expansion is not applied if ycbcrModel is VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY.

For other values of ycbcrModel, range expansion applies independently to each component of the sampled data. For the purposes of range expansion and $Y'C_BC_R$ model conversion, the R and B components contain color difference (chroma) values and the G component contains luma. The A component is not modified by sampler $Y'C_BC_R$ range expansion.

The range expansion to be applied is defined by the ycbcrRange member of the VkSamplerYcbcrConversionCreateInfo structure:

- If ycbcrRange is VK_SAMPLER_YCBCR_RANGE_ITU_FULL, the following transformations are applied:

$$
\begin{aligned}
Y' &= C'_{rgba}[G] \\
C_B &= C'_{rgba}[B] - \frac{2^{(n-1)}}{(2^n)-1} \\
C_R &= C'_{rgba}[R] - \frac{2^{(n-1)}}{(2^n)-1}
\end{aligned}
$$

> **NOTE**
>
> These formulae correspond to the “full range” encoding in the “Quantization schemes” chapter of the Khronos Data Format Specification.
>
> Should any future amendments be made to the ITU specifications from which these equations are derived, the formulae used by Vulkan **may** also be updated to maintain parity.

- If ycbcrRange is VK_SAMPLER_YCBCR_RANGE_ITU_NARROW, the following transformations are applied:

$$
\begin{aligned}
Y' &= \frac{C'_{rgba}[G] \times (2^n - 1) - 16 \times 2^{n-8}}{219 \times 2^{n-8}} \\
C_B &= \frac{C'_{rgba}[B] \times (2^n - 1) - 128 \times 2^{n-8}}{224 \times 2^{n-8}} \\
C_R &= \frac{C'_{rgba}[R] \times (2^n - 1) - 128 \times 2^{n-8}}{224 \times 2^{n-8}}
\end{aligned}
$$

> **NOTE**
>
> These formulae correspond to the “narrow range” encoding in the “Quantization schemes” chapter of the Khronos Data Format Specification.

- n is the bit-depth of the components in the format.

The precision of the operations performed during range expansion **must** be at least that of the source format.

An implementation **may** clamp the results of these range expansion operations such that Y' falls in the range [0,1], and/or such that C_B and C_R fall in the range [-0.5,0.5].

Sampler $Y'C_BC_R$ Model Conversion

The range-expanded values are converted between color models, according to the color model

conversion specified in the `ycbcrModel` member:

VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY

The color components are not modified by the color model conversion since they are assumed already to represent the desired color model in which the shader is operating; $Y'C_BC_R$ range expansion is also ignored.

VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_IDENTITY

The color components are not modified by the color model conversion and are assumed to be treated as though in $Y'C_BC_R$ form both in memory and in the shader; $Y'C_BC_R$ range expansion is applied to the components as for other $Y'C_BC_R$ models, with the vector (C_R,Y',C_B,A) provided to the shader.

VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_709

The color components are transformed from a $Y'C_BC_R$ representation to an R'G'B' representation as described in the "BT.709 $Y'C_BC_R$ conversion" section of the Khronos Data Format Specification.

VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_601

The color components are transformed from a $Y'C_BC_R$ representation to an R'G'B' representation as described in the "BT.601 $Y'C_BC_R$ conversion" section of the Khronos Data Format Specification.

VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_2020

The color components are transformed from a $Y'C_BC_R$ representation to an R'G'B' representation as described in the "BT.2020 $Y'C_BC_R$ conversion" section of the Khronos Data Format Specification.

In this operation, each output component is dependent on each input component.

An implementation **may** clamp the R'G'B' results of these conversions to the range [0,1].

The precision of the operations performed during model conversion **must** be at least that of the source format.

The alpha component is not modified by these model conversions.

NOTE

Sampling operations in a non-linear color space can introduce color and intensity shifts at sharp transition boundaries. To avoid this issue, the technically precise color correction sequence described in the "Introduction to Color Conversions" chapter of the Khronos Data Format Specification may be performed as follows:

- Calculate the unnormalized texel coordinates corresponding to the desired sample position.
- For a `minFilter` or `magFilter` of VK_FILTER_NEAREST:
  1. Calculate (*i*,*j*) for the sample location as described under the "nearest filtering" formulae in (u,v,w,a) to (i,j,k,l,n) Transformation and Array Layer Selection
  2. Calculate the normalized texel coordinates corresponding to these integer coordinates.

3. Sample using sampler $Y'C_BC_R$ conversion at this location.

- For a `minFilter` or `magFilter` of VK_FILTER_LINEAR:

  1. Calculate $(i_{[0,1]},j_{[0,1]})$ for the sample location as described under the "linear filtering" formulae in (u,v,w,a) to (i,j,k,l,n) Transformation and Array Layer Selection
  2. Calculate the normalized texel coordinates corresponding to these integer coordinates.
  3. Sample using sampler $Y'C_BC_R$ conversion at each of these locations.
  4. Convert the non-linear A'R'G'B' outputs of the $Y'C_BC_R$ conversions to linear ARGB values as described in the "Transfer Functions" chapter of the Khronos Data Format Specification.
  5. Interpolate the linear ARGB values using the α and β values described in the "linear filtering" section of (u,v,w,a) to (i,j,k,l,n) Transformation and Array Layer Selection and the equations in Texel Filtering.

The additional calculations and, especially, additional number of sampling operations in the VK_FILTER_LINEAR case can be expected to have a performance impact compared with using the outputs directly. Since the variations from "correct" results are subtle for most content, the application author should determine whether a more costly implementation is strictly necessary.

If `chromaFilter`, and `minFilter` or `magFilter` are both VK_FILTER_NEAREST, these operations are redundant and sampling using sampler $Y'C_BC_R$ conversion at the desired sample coordinates will produce the "correct" results without further processing.

20.3. Normalized Texel Coordinate Operations

If the image sampler instruction provides normalized texel coordinates, some of the following operations are performed.

20.3.1. Projection Operation

For `Proj` image operations, the normalized texel coordinates (s,t,r,q,a) and (if present) the D_{ref} coordinate are transformed as follows:

$$s = \frac{s}{q}, \qquad \text{for 1D, 2D, or 3D image}$$

$$t = \frac{t}{q}, \qquad \text{for 2D or 3D image}$$

$$r = \frac{r}{q}, \qquad \text{for 3D image}$$

$$D_{ref} = \frac{D_{ref}}{q}, \qquad \text{if provided}$$

20.3.2. Derivative Image Operations

Derivatives are used for LOD selection. These derivatives are either implicit (in an `ImplicitLod` image instruction in a mesh, task, compute, or fragment shader) or explicit (provided explicitly by shader to the image instruction in any shader).

For implicit derivatives image instructions, the derivatives of texel coordinates are calculated in the same manner as derivative operations. That is:

$$\begin{aligned}
\partial s/\partial x &= dPdx(s), & \partial s/\partial y &= dPdy(s), && \text{for 1D, 2D, Cube, or 3D image} \\
\partial t/\partial x &= dPdx(t), & \partial t/\partial y &= dPdy(t), && \text{for 2D, Cube, or 3D image} \\
\partial r/\partial x &= dPdx(r), & \partial r/\partial y &= dPdy(r), && \text{for Cube or 3D image}
\end{aligned}$$

Partial derivatives not defined above for certain image dimensionalities are set to zero.

For explicit LOD image instructions, if the **optional** SPIR-V operand `Grad` is provided, then the operand values are used for the derivatives. The number of components present in each derivative for a given image dimensionality matches the number of partial derivatives computed above.

If the **optional** SPIR-V operand `Lod` is provided, then derivatives are set to zero, the cube map derivative transformation is skipped, and the scale factor operation is skipped. Instead, the floating-point scalar coordinate is directly assigned to λ_{base} as described in LOD Operation.

If the image or sampler object used by an implicit derivative image instruction is not uniform across the quad and `quadDivergentImplicitLod` is not supported, then the derivative and LOD values are undefined. Implicit derivatives are well-defined when the image and sampler and control flow are uniform across the quad, even if they diverge between different quads.

If `quadDivergentImplicitLod` is supported, then derivatives and implicit LOD values are well-defined even if the image or sampler object are not uniform within a quad. The derivatives are computed as specified above, and the implicit LOD calculation proceeds for each shader invocation using its respective image and sampler object.

20.3.3. Cube Map Face Selection and Transformations

For cube map image instructions, the (s,t,r) coordinates are treated as a direction vector (r_x,r_y,r_z). The direction vector is used to select a cube map face. The direction vector is transformed to a per-face texel coordinate system (s_{face},t_{face}). The direction vector is also used to transform the derivatives to per-face derivatives.

20.3.4. Cube Map Face Selection

The direction vector selects one of the cube map's faces based on the largest magnitude coordinate direction (the major axis direction). Since two or more coordinates **can** have identical magnitude, the implementation **must** have rules to disambiguate this situation.

The rules **should** have as the first rule that r_z wins over r_y and r_x, and the second rule that r_y wins over r_x. An implementation **may** choose other rules, but the rules **must** be deterministic and depend only on (r_x,r_y,r_z).

The layer number (corresponding to a cube map face), the coordinate selections for s_c, t_c, r_c, and the selection of derivatives, are determined by the major axis direction as specified in the following two tables.

Table 28. Cube Map Face and Coordinate Selection

Major Axis Direction	Layer Number	Cube Map Face	s_c	t_c	r_c
$+r_x$	0	Positive X	$-r_z$	$-r_y$	r_x
$-r_x$	1	Negative X	$+r_z$	$-r_y$	r_x
$+r_y$	2	Positive Y	$+r_x$	$+r_z$	r_y
$-r_y$	3	Negative Y	$+r_x$	$-r_z$	r_y
$+r_z$	4	Positive Z	$+r_x$	$-r_y$	r_z
$-r_z$	5	Negative Z	$-r_x$	$-r_y$	r_z

Table 29. Cube Map Derivative Selection

Major Axis Direction	$\partial s_c / \partial x$	$\partial s_c / \partial y$	$\partial t_c / \partial x$	$\partial t_c / \partial y$	$\partial r_c / \partial x$	$\partial r_c / \partial y$
$+r_x$	$-\partial r_z / \partial x$	$-\partial r_z / \partial y$	$-\partial r_y / \partial x$	$-\partial r_y / \partial y$	$+\partial r_x / \partial x$	$+\partial r_x / \partial y$
$-r_x$	$+\partial r_z / \partial x$	$+\partial r_z / \partial y$	$-\partial r_y / \partial x$	$-\partial r_y / \partial y$	$-\partial r_x / \partial x$	$-\partial r_x / \partial y$
$+r_y$	$+\partial r_x / \partial x$	$+\partial r_x / \partial y$	$+\partial r_z / \partial x$	$+\partial r_z / \partial y$	$+\partial r_y / \partial x$	$+\partial r_y / \partial y$
$-r_y$	$+\partial r_x / \partial x$	$+\partial r_x / \partial y$	$-\partial r_z / \partial x$	$-\partial r_z / \partial y$	$-\partial r_y / \partial x$	$-\partial r_y / \partial y$
$+r_z$	$+\partial r_x / \partial x$	$+\partial r_x / \partial y$	$-\partial r_y / \partial x$	$-\partial r_y / \partial y$	$+\partial r_z / \partial x$	$+\partial r_z / \partial y$
$-r_z$	$-\partial r_x / \partial x$	$-\partial r_x / \partial y$	$-\partial r_y / \partial x$	$-\partial r_y / \partial y$	$-\partial r_z / \partial x$	$-\partial r_z / \partial y$

20.3.5. Cube Map Coordinate Transformation

$$s_{face} = \frac{1}{2} \times \frac{s_c}{|r_c|} + \frac{1}{2}$$
$$t_{face} = \frac{1}{2} \times \frac{t_c}{|r_c|} + \frac{1}{2}$$

20.3.6. Cube Map Derivative Transformation

The partial derivatives of the Cube Map Coordinate Transformations can be computed as:

$$\begin{aligned}\frac{\partial s_{face}}{\partial x} &= \frac{\partial}{\partial x}\left(\frac{1}{2} \times \frac{s_c}{|r_c|} + \frac{1}{2}\right) \\ &= \frac{1}{2} \times \frac{\partial}{\partial x}\left(\frac{s_c}{|r_c|}\right) \\ &= \frac{1}{2} \times \left(\frac{|r_c| \times \partial s_c / \partial x - s_c \times \partial r_c / \partial x}{(r_c)^2}\right)\end{aligned}$$

The other derivatives are simplified similarly, resulting in

$$\begin{aligned}\frac{\partial s_{face}}{\partial y} &= \frac{1}{2} \times \left(\frac{|r_c| \times \partial s_c / \partial y - s_c \times \partial r_c / \partial y}{(r_c)^2}\right) \\ \frac{\partial t_{face}}{\partial x} &= \frac{1}{2} \times \left(\frac{|r_c| \times \partial t_c / \partial x - t_c \times \partial r_c / \partial x}{(r_c)^2}\right) \\ \frac{\partial t_{face}}{\partial y} &= \frac{1}{2} \times \left(\frac{|r_c| \times \partial t_c / \partial y - t_c \times \partial r_c / \partial y}{(r_c)^2}\right)\end{aligned}$$

20.3.7. Scale Factor Operation, LOD Operation and Image Level(s) Selection

LOD selection **can** be either explicit (provided explicitly by the image instruction) or implicit (determined from a scale factor calculated from the derivatives). The LOD **must** be computed with `mipmapPrecisionBits` of accuracy.

Scale Factor Operation

The magnitude of the derivatives are calculated by:

$m_{ux} = |\partial s/\partial x| \times w_{base}$

$m_{vx} = |\partial t/\partial x| \times h_{base}$

$m_{wx} = |\partial r/\partial x| \times d_{base}$

$m_{uy} = |\partial s/\partial y| \times w_{base}$

$m_{vy} = |\partial t/\partial y| \times h_{base}$

$m_{wy} = |\partial r/\partial y| \times d_{base}$

where:

$\partial t/\partial x = \partial t/\partial y = 0$ (for 1D images)

$\partial r/\partial x = \partial r/\partial y = 0$ (for 1D, 2D, or Cube images)

and:

w_{base} = image.w

h_{base} = image.h

d_{base} = image.d

(for the `baseMipLevel`, from the image descriptor).

A point sampled in screen space has an elliptical footprint in texture space. The minimum and maximum scale factors (ρ_{min}, ρ_{max}) **should** be the minor and major axes of this ellipse.

The *scale factors* ρ_x and ρ_y, calculated from the magnitude of the derivatives in x and y, are used to compute the minimum and maximum scale factors.

ρ_x and ρ_y **may** be approximated with functions f_x and f_y, subject to the following constraints:

$$f_x \text{is continuous and monotonically increasing in each of} m_{ux},\ m_{vx},\ \text{and} m_{wx}$$
$$f_y \text{is continuous and monotonically increasing in each of} m_{uy},\ m_{vy},\ \text{and} m_{wy}$$

$$\max(|m_{ux}|,\ |m_{vx}|,\ |m_{wx}|) \le f_x \le \sqrt{2}(|m_{ux}| + |m_{vx}| + |m_{wx}|)$$
$$\max(|m_{uy}|,\ |m_{vy}|,\ |m_{wy}|) \le f_y \le \sqrt{2}(|m_{uy}| + |m_{vy}| + |m_{wy}|)$$

The minimum and maximum scale factors (ρ_{min},ρ_{max}) are determined by:

$\rho_{max} = \max(\rho_x, \rho_y)$

$\rho_{min} = \min(\rho_x, \rho_y)$

The ratio of anisotropy is determined by:

$\eta = \min(\rho_{max}/\rho_{min}, \max_{Aniso})$

where:

$sampler.max_{Aniso}$ = `maxAnisotropy` (from sampler descriptor)

$limits.max_{Aniso}$ = `maxSamplerAnisotropy` (from physical device limits)

$max_{Aniso} = \min(sampler.max_{Aniso}, limits.max_{Aniso})$

If $\rho_{max} = \rho_{min} = 0$, then all the partial derivatives are zero, the fragment's footprint in texel space is a point, and η **should** be treated as 1. If $\rho_{max} \neq 0$ and $\rho_{min} = 0$ then all partial derivatives along one axis are zero, the fragment's footprint in texel space is a line segment, and η **should** be treated as max_{Aniso}. However, anytime the footprint is small in texel space the implementation **may** use a smaller value of η, even when ρ_{min} is zero or close to zero. If either VkPhysicalDeviceFeatures ::`samplerAnisotropy` or VkSamplerCreateInfo::`anisotropyEnable` are VK_FALSE, max_{Aniso} is set to 1.

If $\eta = 1$, sampling is isotropic. If $\eta > 1$, sampling is anisotropic.

The sampling rate (N) is derived as:

$N = \lceil \eta \rceil$

An implementation **may** round N up to the nearest supported sampling rate. An implementation **may** use the value of N as an approximation of η.

LOD Operation

The LOD parameter λ is computed as follows:

$$\lambda_{base}(x, y) = \begin{cases} shaderOp.Lod & \text{(from optional SPIR-V operand)} \\ \log_2\left(\frac{\rho_{max}}{\eta}\right) & \text{otherwise} \end{cases}$$

$$\lambda'(x, y) = \lambda_{base} + \text{clamp}\left(sampler.bias + shaderOp.bias, -maxSamplerLodBias, maxSamplerLodBias\right)$$

$$\lambda = \begin{cases} lod_{max}, & \lambda' > lod_{max} \\ \lambda', & lod_{min} \leq \lambda' \leq lod_{max} \\ lod_{min}, & \lambda' < lod_{min} \\ undefined, & lod_{min} > lod_{max} \end{cases}$$

where:

$$sampler.bias = mipLodBias \quad \text{(from sampler descriptor)}$$

$$shaderOp.bias = \begin{cases} Bias & \text{(from optional SPIR-V operand)} \\ 0 & \text{otherwise} \end{cases}$$

$$sampler.lod_{min} = minLod \quad \text{(from sampler descriptor)}$$

$$shaderOp.lod_{min} = \begin{cases} MinLod & \text{(from optional SPIR-V operand)} \\ 0 & \text{otherwise} \end{cases}$$

$$lod_{min} = \max(sampler.lod_{min}, shaderOp.lod_{min})$$

$$lod_{max} = maxLod \quad \text{(from sampler descriptor)}$$

and maxSamplerLodBias is the value of the VkPhysicalDeviceLimits feature `maxSamplerLodBias`.

Image Level(s) Selection

The image level(s) d, d_{hi}, and d_{lo} which texels are read from are determined by an image-level parameter d_l, which is computed based on the LOD parameter, as follows:

$$d_l = \begin{cases} nearest(d'), & \text{mipmapMode is VK_SAMPLER_MIPMAP_MODE_NEAREST} \\ d', & \text{otherwise} \end{cases}$$

where:

$$d' = max(level_{base} + \text{clamp}(\lambda, 0, q), minLod_{imageView})$$

$$nearest(d') = \begin{cases} \lceil d' + 0.5 \rceil - 1, & \text{preferred} \\ \lfloor d' + 0.5 \rfloor, & \text{alternative} \end{cases}$$

and:

$$minLod_{imageView} = \begin{cases} minLodFloat_{imageView}, & \text{preferred} \\ minLodInteger_{imageView}, & \text{alternative} \end{cases}$$
$$level_{base} = baseMipLevel$$
$$q = levelCount - 1$$

`baseMipLevel` and `levelCount` are taken from the `subresourceRange` of the image view.

$minLod_{imageView}$ **must** be less than or equal to $level_{base}$
q.

If the sampler's `mipmapMode` is VK_SAMPLER_MIPMAP_MODE_NEAREST, then the level selected is d = d_l.

If the sampler's `mipmapMode` is VK_SAMPLER_MIPMAP_MODE_LINEAR, two neighboring levels are selected:

$$d_{hi} = \lfloor d_l \rfloor$$
$$d_{lo} = min(d_{hi} + 1, level_{base} + q)$$
$$\delta = d_l - d_{hi}$$

δ is the fractional value, quantized to the number of mipmap precision bits, used for linear filtering between levels.

20.3.8. (s,t,r,q,a) to (u,v,w,a) Transformation

The normalized texel coordinates are scaled by the image level dimensions and the array layer is selected.

This transformation is performed once for each level used in filtering (either d, or d_{hi} and d_{lo}).

$$u(x,\ y) = s(x,\ y) \times width_{scale} + \Delta_i$$

$$v(x,\ y) = \begin{cases} 0 & \text{for 1D images} \\ t(x,\ y) \times height_{scale} + \Delta_j & \text{otherwise} \end{cases}$$

$$w(x,\ y) = \begin{cases} 0 & \text{for 2D or Cube images} \\ r(x,\ y) \times depth_{scale} + \Delta_k & \text{otherwise} \end{cases}$$

$$a(x,\ y) = \begin{cases} a(x,\ y) & \text{for array images} \\ 0 & \text{otherwise} \end{cases}$$

where:

$width_{scale} = width_{level}$

$height_{scale} = height_{level}$

$depth_{scale} = depth_{level}$

and where (Δ_i, Δ_j, Δ_k) are taken from the image instruction if it includes a `ConstOffset` or `Offset` operand, otherwise they are taken to be zero.

Operations then proceed to Unnormalized Texel Coordinate Operations.

20.4. Unnormalized Texel Coordinate Operations

20.4.1. (u,v,w,a) to (i,j,k,l,n) Transformation and Array Layer Selection

The unnormalized texel coordinates are transformed to integer texel coordinates relative to the selected mipmap level.

The layer index l is computed as:

l = clamp(RNE(a), 0, `layerCount` - 1) + `baseArrayLayer`

where `layerCount` is the number of layers in the image subresource range of the image view, `baseArrayLayer` is the first layer from the subresource range, and where:

$$\text{RNE}\,(a) = \begin{cases} \text{roundTiesToEven}\,(a) & \text{preferred, from IEEE Std 754-2008 Floating-Point Arithmetic} \\ \lfloor a + 0.5 \rfloor & \text{alternative} \end{cases}$$

The sample index n is assigned the value 0.

Nearest filtering (VK_FILTER_NEAREST) computes the integer texel coordinates that the unnormalized coordinates lie within:

$$
\begin{aligned}
i &= \lfloor u + shift \rfloor \\
j &= \lfloor v + shift \rfloor \\
k &= \lfloor w + shift \rfloor
\end{aligned}
$$

where:

shift = 0.0

Linear filtering (VK_FILTER_LINEAR) computes a set of neighboring coordinates which bound the unnormalized coordinates. The integer texel coordinates are combinations of i_0 or i_1, j_0 or j_1, k_0 or k_1, as well as weights α, β, and γ.

$$
\begin{aligned}
i_0 &= \lfloor u - shift \rfloor \\
i_1 &= i_0 + 1 \\
j_0 &= \lfloor v - shift \rfloor \\
j_1 &= j_0 + 1 \\
k_0 &= \lfloor w - shift \rfloor \\
k_1 &= k_0 + 1
\end{aligned}
$$

$$
\begin{aligned}
\alpha &= \text{frac}\,(u - shift) \\
\beta &= \text{frac}\,(v - shift) \\
\gamma &= \text{frac}\,(w - shift)
\end{aligned}
$$

where:

shift = 0.5

and where:

$$
\text{frac}\,(x) = x - \lfloor x \rfloor
$$

where the number of fraction bits retained is specified by VkPhysicalDeviceLimits ::subTexelPrecisionBits.

Cubic filtering (VK_FILTER_CUBIC_EXT) computes a set of neighboring coordinates which bound the unnormalized coordinates. The integer texel coordinates are combinations of i_0, i_1, i_2 or i_3, j_0, j_1, j_2 or j_3, k_0, k_1, k_2 or k_3, as well as weights α, β, and γ.

$$
\begin{aligned}
i_0 &= \lfloor u - \frac{3}{2} \rfloor & i_1 &= i_0 + 1 & i_2 &= i_1 + 1 & i_3 &= i_2 + 1 \\
j_0 &= \lfloor v - \frac{3}{2} \rfloor & j_1 &= j_0 + 1 & j_2 &= j_1 + 1 & j_3 &= j_2 + 1 \\
k_0 &= \lfloor w - \frac{3}{2} \rfloor & k_1 &= k_0 + 1 & k_2 &= k_1 + 1 & k_3 &= k_2 + 1
\end{aligned}
$$

$$\alpha = \text{frac}\left(u - \frac{1}{2}\right)$$

$$\beta = \text{frac}\left(v - \frac{1}{2}\right)$$

$$\gamma = \text{frac}\left(w - \frac{1}{2}\right)$$

where:

$$\text{frac}(x) = x - \lfloor x \rfloor$$

where the number of fraction bits retained is specified by `VkPhysicalDeviceLimits::subTexelPrecisionBits`.

20.5. Integer Texel Coordinate Operations

The `OpImageFetch` and `OpImageSparseFetch` SPIR-V instructions **may** supply a LOD from which texels are to be fetched using the optional SPIR-V operand `Lod`. Other integer-coordinate operations **must** not. If the `Lod` is provided then it **must** be an integer.

The image level selected is:

$$d = level_{base} + \begin{cases} Lod & \text{(from optional SPIR-V operand)} \\ 0 & \text{otherwise} \end{cases}$$

If d does not lie in the range [`baseMipLevel`, `baseMipLevel` + `levelCount`) or d is less than $\text{minLodInteger}_{\text{imageView}}$, then any values fetched are zero if the `robustImageAccess2` feature is enabled, otherwise are undefined, and any writes (if supported) are discarded.

20.6. Image Sample Operations

20.6.1. Wrapping Operation

If the used sampler was created without VK_SAMPLER_CREATE_NON_SEAMLESS_CUBE_MAP_BIT_EXT, `Cube` images ignore the wrap modes specified in the sampler. Instead, if VK_FILTER_NEAREST is used within a mip level then VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE is used, and if VK_FILTER_LINEAR is used within a mip level then sampling at the edges is performed as described earlier in the Cube map edge handling section.

The first integer texel coordinate i is transformed based on the `addressModeU` parameter of the sampler.

$$i = \begin{cases} i \bmod size & \text{for repeat} \\ (size-1) - \text{mirror}\left((i \bmod (2 \times size)) - size\right) & \text{for mirrored repeat} \\ \text{clamp}\left(i,\ 0,\ size-1\right) & \text{for clamp to edge} \\ \text{clamp}\left(i,\ -1,\ size\right) & \text{for clamp to border} \\ \text{clamp}\left(\text{mirror}\left(i\right),\ 0,\ size-1\right) & \text{for mirror clamp to edge} \end{cases}$$

where:

$$\text{mirror}\left(n\right) = \begin{cases} n & \text{for } n \geq 0 \\ -(1+n) & \text{otherwise} \end{cases}$$

j (for 2D and Cube image) and k (for 3D image) are similarly transformed based on the `addressModeV` and `addressModeW` parameters of the sampler, respectively.

20.6.2. Texel Gathering

SPIR-V instructions with `Gather` in the name return a vector derived from 4 texels in the base level of the image view.

The rules for the VK_FILTER_LINEAR minification filter are applied to identify the four selected texels for `OpImage*Gather` .

If an `OpImage*Gather` operation does not use the `ConstOffsets` image operand then the four texels form the 2 × 2 rectangle used for texture filtering:

$$\begin{aligned} \tau[R] &= \tau_{i0\,j1}[level_{base}][comp] \\ \tau[G] &= \tau_{i1\,j1}[level_{base}][comp] \\ \tau[B] &= \tau_{i1\,j0}[level_{base}][comp] \\ \tau[A] &= \tau_{i0\,j0}[level_{base}][comp] \end{aligned}$$

If an `OpImage*Gather` operation uses the `ConstOffsets` image operand then the offsets allow a custom filter to be defined:

$$\begin{aligned} \tau[R] &= \tau_{i0\,j0\,+\,\Delta_0}[level_{base}][comp] \\ \tau[G] &= \tau_{i0\,j0\,+\,\Delta_1}[level_{base}][comp] \\ \tau[B] &= \tau_{i0\,j0\,+\,\Delta_2}[level_{base}][comp] \\ \tau[A] &= \tau_{i0\,j0\,+\,\Delta_3}[level_{base}][comp] \end{aligned}$$

where:

$$\tau[level_{base}][comp] = \begin{cases} \tau[level_{base}][R], & \text{for } comp = 0 \\ \tau[level_{base}][G], & \text{for } comp = 1 \\ \tau[level_{base}][B], & \text{for } comp = 2 \\ \tau[level_{base}][A], & \text{for } comp = 3 \end{cases}$$

$comp$ from SPIR-V operand Component

Each texel is then converted to an RGBA value according to component substitution and then swizzled.

`OpImage*Gather` **must** not be used on a sampled image with sampler $Y'C_BC_R$ conversion enabled.

If $level_{base}$ < $minLodInteger_{imageView}$, then any values fetched are zero if the `robustImageAccess2` feature is enabled. Otherwise values are undefined.

20.6.3. Texel Filtering

Texel filtering is first performed for each level (either d or d_{hi} and d_{lo}).

If λ is less than or equal to zero, the texture is said to be *magnified*, and the filter mode within a mip level is selected by the `magFilter` in the sampler. If λ is greater than zero, the texture is said to be *minified*, and the filter mode within a mip level is selected by the `minFilter` in the sampler.

Texel Nearest Filtering

Within a mip level, VK_FILTER_NEAREST filtering selects a single value using the (i, j, k) texel coordinates, with all texels taken from layer l.

$$\tau[level] = \begin{cases} \tau_{ijk}[level], & \text{for 3D image} \\ \tau_{ij}[level], & \text{for 2D or Cube image} \\ \tau_{i}[level], & \text{for 1D image} \end{cases}$$

Texel Linear Filtering

Within a mip level, VK_FILTER_LINEAR filtering combines 8 (for 3D), 4 (for 2D or Cube), or 2 (for 1D) texel values, together with their linear weights. The linear weights are derived from the fractions computed earlier:

$$\begin{aligned} w_{i_0} &= (1-\alpha) \\ w_{i_1} &= (\alpha) \\ w_{j_0} &= (1-\beta) \\ w_{j_1} &= (\beta) \\ w_{k_0} &= (1-\gamma) \\ w_{k_1} &= (\gamma) \end{aligned}$$

The values of multiple texels, together with their weights, are combined to produce a filtered value.

The VkSamplerReductionModeCreateInfo::`reductionMode` **can** control the process by which multiple texels, together with their weights, are combined to produce a filtered texture value.

When the `reductionMode` is set (explicitly or implicitly) to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, a weighted average is computed:

$$\tau_{3D} = \sum_{k=k_0}^{k_1} \sum_{j=j_0}^{j_1} \sum_{i=i_0}^{i_1} (w_i)(w_j)(w_k)\tau_{ijk}$$

$$\tau_{2D} = \sum_{j=j_0}^{j_1} \sum_{i=i_0}^{i_1} (w_i)(w_j)\tau_{ij}$$

$$\tau_{1D} = \sum_{i=i_0}^{i_1} (w_i)\tau_i$$

However, if the reduction mode is VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX, the process operates on the above set of multiple texels, together with their weights, computing a component-wise minimum or maximum, respectively, of the components of the set of texels with non-zero weights.

Texel Cubic Filtering

Within a mip level, VK_FILTER_CUBIC_EXT filtering computes a weighted average of 64 (for 3D), 16 (for 2D), or 4 (for 1D) texel values, together with their Catmull-Rom weights.

Catmull-Rom weights are derived from the fractions computed earlier.

$$\begin{bmatrix} w_{i_0} & w_{i_1} & w_{i_2} & w_{i_3} \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & \alpha & \alpha^2 & \alpha^3 \end{bmatrix}\begin{bmatrix} 0 & 2 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 2 & -5 & 4 & -1 \\ -1 & 3 & -3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} w_{j_0} & w_{j_1} & w_{j_2} & w_{j_3} \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & \beta & \beta^2 & \beta^3 \end{bmatrix}\begin{bmatrix} 0 & 2 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 2 & -5 & 4 & -1 \\ -1 & 3 & -3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} w_{k_0} & w_{k_1} & w_{k_2} & w_{k_3} \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & \gamma & \gamma^2 & \gamma^3 \end{bmatrix}\begin{bmatrix} 0 & 2 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 2 & -5 & 4 & -1 \\ -1 & 3 & -3 & 1 \end{bmatrix}$$

The values of multiple texels, together with their weights, are combined to produce a filtered value.

The VkSamplerReductionModeCreateInfo::`reductionMode` **can** control the process by which multiple texels, together with their weights, are combined to produce a filtered texture value.

When the `reductionMode` is set (explicitly or implicitly) to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE , a weighted average is computed:

$$\tau_{3D} = \sum_{k=j_0}^{k_3} \sum_{j=j_0}^{j_3} \sum_{i=i_0}^{i_3} (w_i)(w_j)(w_k)\tau_{ijk}$$
$$\tau_{2D} = \sum_{j=j_0}^{j_3} \sum_{i=i_0}^{i_3} (w_i)(w_j)\tau_{ij}$$
$$\tau_{1D} = \sum_{i=i_0}^{i_3} (w_i)\tau_i$$

However, if the reduction mode is VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX, the process operates on the above set of multiple texels, together with their weights, computing a component-wise minimum or maximum, respectively, of the components of the set of texels with non-zero weights.

Texel Mipmap Filtering

VK_SAMPLER_MIPMAP_MODE_NEAREST filtering returns the value of a single mipmap level,

$\tau = \tau[d]$.

VK_SAMPLER_MIPMAP_MODE_LINEAR filtering combines the values of multiple mipmap levels (τ[hi] and τ[lo]), together with their linear weights.

The linear weights are derived from the fraction computed earlier:

$$w_{hi} = (1 - \delta)$$
$$w_{lo} = (\delta)$$

The values of multiple mipmap levels, together with their weights, are combined to produce a final filtered value.

The VkSamplerReductionModeCreateInfo::`reductionMode` **can** control the process by which multiple texels, together with their weights, are combined to produce a filtered texture value.

When the `reductionMode` is set (explicitly or implicitly) to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, a weighted average is computed:

$$\tau = (w_{hi})\tau[hi] + (w_{lo})\tau[lo]$$

However, if the reduction mode is VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX, the process operates on the above values, together with their weights, computing a component-wise minimum or maximum, respectively, of the components of the values with non-zero weights.

Texel Anisotropic Filtering

Anisotropic filtering is enabled by the `anisotropyEnable` in the sampler. When enabled, the image filtering scheme accounts for a degree of anisotropy.

The particular scheme for anisotropic texture filtering is implementation-dependent. Implementations **should** consider the `magFilter`, `minFilter` and `mipmapMode` of the sampler to control the specifics of the anisotropic filtering scheme used. In addition, implementations **should** consider `minLod` and `maxLod` of the sampler.

For historical reasons, vendor implementations of anisotropic filtering interpret these sampler parameters in different ways, particularly in corner cases such as `magFilter`, `minFilter` of VK_FILTER_NEAREST or `maxAnisotropy` equal to 1.0. Applications should not expect consistent behavior in such cases, and should use anisotropic filtering only with parameters which are expected to give a quality improvement relative to `LINEAR` filtering.

The following describes one particular approach to implementing anisotropic filtering for the 2D Image case; implementations **may** choose other methods:

Given a `magFilter`, `minFilter` of VK_FILTER_LINEAR and a `mipmapMode` of VK_SAMPLER_MIPMAP_MODE_NEAREST:

NOTE

Instead of a single isotropic sample, N isotropic samples are sampled within the image footprint of the image level d to approximate an anisotropic filter. The sum $\tau_{2Daniso}$ is defined using the single isotropic $\tau_{2D}(u,v)$ at level d.

$$\tau_{2Daniso} = \frac{1}{N}\sum_{i=1}^{N} \tau_{2D}\left(u\left(x - \frac{1}{2} + \frac{i}{N+1}, y\right), v\left(x - \frac{1}{2} + \frac{i}{N+1}, y\right)\right), \quad \text{when } \rho_x > \rho_y$$

$$\tau_{2Daniso} = \frac{1}{N}\sum_{i=1}^{N} \tau_{2D}\left(u\left(x, y - \frac{1}{2} + \frac{i}{N+1}\right), v\left(x, y - \frac{1}{2} + \frac{i}{N+1}\right)\right), \quad \text{when } \rho_y \geq \rho_x$$

When VkSamplerReductionModeCreateInfo::`reductionMode` is VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, the above summation is used. However, if the reduction mode is VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX, the process operates on the above values, together with their weights, computing a component-wise minimum or maximum, respectively, of the components of the values with non-zero weights.

20.7. Image Operation Steps

Each step described in this chapter is performed by a subset of the image instructions:

- Texel Input Validation Operations, Format Conversion, Texel Replacement, Conversion to RGBA, and Component Swizzle: Performed by all sampling instructions.
- Depth Comparison: Performed by `OpImage*Dref` instructions.
- Projection: Performed by all `OpImage*Proj` instructions.
- Derivative Image Operations, Cube Map Operations, Scale Factor Operation, LOD Operation and Image Level(s) Selection, and Texel Anisotropic Filtering: Performed by all `OpImageSample*` and `OpImageSparseSample*` instructions.
- (s,t,r,q,a) to (u,v,w,a) Transformation, Wrapping, and (u,v,w,a) to (i,j,k,l,n) Transformation And

Array Layer Selection: Performed by all `OpImageSample`, `OpImageSparseSample`, and `OpImage*Gather` instructions.

- Texel Gathering: Performed by `OpImage*Gather` instructions.
- Texel Filtering: Performed by all `OpImageSample*` and `OpImageSparseSample*` instructions.
- Sparse Residency: Performed by all `OpImageSparse*` instructions.

20.8. LOD Query

`OpImageQueryLod` returns the LOD parameters that would be used in a sampling instruction with the given image and coordinates. If the descriptor that would be accessed is a null descriptor then (0,0) is returned. Otherwise, the steps described in this chapter are performed as if for `OpImageSampleImplicitLod`, up to Scale Factor Operation, LOD Operation and Image Level(s) Selection. The return value is the vector (d_l - $level_{base}$, λ'). These values **may** be subject to implementation-specific maxima and minima for very large, out-of-range values.

Chapter 21. Fragment Density Map Operations

21.1. Fragment Density Map Operations Overview

When a fragment is generated in a render pass that has a fragment density map attachment, its area is determined by the properties of the local framebuffer region that the fragment occupies. The framebuffer is divided into a uniform grid of these local regions, and their fragment area property is derived from the density map with the following operations:

- Fetch density value
  - Component swizzle
  - Component mapping
- Fragment area conversion
  - Fragment area filter
  - Fragment area clamp

21.2. Fetch Density Value

Each local framebuffer region at center coordinate (x,y) reads a texel from the fragment density map.

First, the local framebuffer region center coordinate (x,y) is offset by the value specified in VkRenderPassFragmentDensityMapOffsetEndInfoEXT. If no offset is specified, then the default offset (0,0) is used. The offset coordinate (x',y') is computed as follows:

$$x' = x - pFragmentDensityOffsets[layer]_x$$
$$y' = y - pFragmentDensityOffsets[layer]_y$$

NOTE

The offset is relative to the fragment density map, so it is subtracted from the framebuffer coordinates to make the adjustment relative to the framebuffer.

In other words, applying a positive offset in the x component will shift the fragment density map to the right relative to the framebuffer. This means the framebuffer coordinates need to undergo a shift to the left.

The offset fragment coordinate (x',y') reads a texel from the fragment density map at image coordinates (x,y,0,layer,0,0), where (x,y) are calculated as:

$$x = \text{clamp}\left(\left\lfloor \frac{x'}{fragmentDensityTexelSize_{width}} \right\rfloor, 0, fragmentDensityMap_{width} - 1\right)$$

$$y = \text{clamp}\left(\left\lfloor \frac{y'}{fragmentDensityTexelSize_{height}} \right\rfloor, 0, fragmentDensityMap_{height} - 1\right)$$

Where the size of each region in the framebuffer is:

$$fragmentDensityTexelSize'_{width} = 2^{\lceil \log_2(\lfloor \frac{framebuffer_{width}}{fragmentDensityMap_{width}} \rfloor) \rceil}$$

$$fragmentDensityTexelSize'_{height} = 2^{\lceil \log_2(\lfloor \frac{framebuffer_{height}}{fragmentDensityMap_{height}} \rfloor) \rceil}$$

NOTE

The original spec did not include the floor in the equation above. Implementations have shipped with and without the floor. Given that the earliest implementations shipped with the floor, the spec now reflects that as most content was written with this behavior.

All implementations exposing support for version 3 or greater of the extension **must** implement with the floor included.

If using vkCmdBeginRendering, then $framebuffer_{width}$ and $framebuffer_{height}$ are defined as:

$$framebuffer_{width} = renderAreaOffset_{x} plus renderAreaExtent_{width}$$

$$framebuffer_{height} = renderAreaOffset_{y} plus renderAreaExtent_{height}$$

using VkRenderingInfo::`renderArea`.

This region is subject to the limits in `VkPhysicalDeviceFragmentDensityMapPropertiesEXT` and therefore the final region size is clamped:

$$fragmentDensityTexelSize_{width} = \text{clamp}(fragmentDensityTexelSize'_{width}, minFragmentDensityTexelSize_{width}, maxFragmentDensityTexelSize_{width})$$

$$fragmentDensityTexelSize_{height} = \text{clamp}(fragmentDensityTexelSize'_{height}, minFragmentDensityTexelSize_{height}, maxFragmentDensityTexelSize_{height})$$

When multiview is enabled for the render pass and the fragment density map attachment view was created with `layerCount` greater than 1, the layer used for offsets and for fetching from the fragment density map is:

$$layer = baseArrayLayer + ViewIndex$$

Otherwise:

$$layer = baseArrayLayer$$

The texel fetched from the density map at (x,y,0,layer,0,0) is next converted to density with the

following operations.

21.2.1. Component Swizzle

The `components` member of VkImageViewCreateInfo is applied to the fetched texel as defined in Image component swizzle.

21.2.2. Component Mapping

The swizzled texel's components are mapped to a density value:

$$densityValue_{xy} = (C'_r, C'_g)$$

21.3. Fragment Area Conversion

Fragment area for the framebuffer region is undefined if the density fetched is not a normalized floating-point value greater than `0.0`. Otherwise, the fetched fragment area for that region is derived as:

$$fragmentArea_{wh} = \frac{1.0}{densityValue_{xy}}$$

21.3.1. Fragment Area Filter

Optionally, the implementation **may** fetch additional density map texels in an implementation defined window around (i,j). The texels follow the standard conversion steps up to and including fragment area conversion.

A single fetched fragment area for the framebuffer region is chosen by the implementation and **must** have an area between the *min* and *max* areas of the fetched set.

21.3.2. Fragment Area Clamp

The implementation **may** clamp the fetched fragment area to one that it supports. The clamped fragment area **must** have a size less than or equal to the original fetched value. Implementations **may** vary the supported set of fragment areas per framebuffer region. Fragment area (1,1) **must** always be in the supported set.

NOTE

For example, if the fetched fragment area is (1,4) but the implementation only supports areas of {(1,1),(2,2)}, it could choose to clamp the area to (2,2) since it has the same size as (1,4). While this would produce fragments that have lower quality strictly in the x-axis, the overall density is maintained.

The clamped fragment area is assigned to the corresponding framebuffer region.

Chapter 22. Queries

Queries provide a mechanism to return information about the processing of a sequence of Vulkan commands. Query operations are asynchronous, and as such, their results are not returned immediately. Instead, their results, and their availability status are stored in a Query Pool. The state of these queries **can** be read back on the host, or copied to a buffer object on the device.

The supported query types are:

- Occlusion Queries
- Pipeline Statistics Queries
- Timestamp Queries
- Result Status Queries
- Video Encode Feedback Queries
- Performance Queries
- Transform Feedback Queries
- Mesh Shader Queries

Several additional queries with specific purposes associated with ray tracing are available if the corresponding extensions are supported, as described for VkQueryType.

22.1. Query Pools

Queries are managed using *query pool* objects. Each query pool is a collection of a specific number of queries of a particular type.

Query pools are represented by `VkQueryPool` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkQueryPool)
```

To create a query pool, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateQueryPool(
    VkDevice                                    device,
    const VkQueryPoolCreateInfo*                pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkQueryPool*                                pQueryPool);
```

- `device` is the logical device that creates the query pool.
- `pCreateInfo` is a pointer to a VkQueryPoolCreateInfo structure containing the number and type of queries to be managed by the pool.

- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pQueryPool` is a pointer to a VkQueryPool handle in which the resulting query pool object is returned.

Valid Usage

- VUID-vkCreateQueryPool-device-09663
  `device` **must** support at least one queue family with one of the VK_QUEUE_VIDEO_ENCODE_BIT_KHR, VK_QUEUE_VIDEO_DECODE_BIT_KHR, VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT capabilities

Valid Usage (Implicit)

- VUID-vkCreateQueryPool-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateQueryPool-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkQueryPoolCreateInfo structure
- VUID-vkCreateQueryPool-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateQueryPool-pQueryPool-parameter
  `pQueryPool` **must** be a valid pointer to a VkQueryPool handle
- VUID-vkCreateQueryPool-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkQueryPoolCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkQueryPoolCreateInfo {
    VkStructureType                 sType;
    const void*                     pNext;
    VkQueryPoolCreateFlags          flags;
    VkQueryType                     queryType;
    uint32_t                        queryCount;
    VkQueryPipelineStatisticFlags   pipelineStatistics;
} VkQueryPoolCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkQueryPoolCreateFlagBits.
- `queryType` is a VkQueryType value specifying the type of queries managed by the pool.
- `queryCount` is the number of queries managed by the pool.
- `pipelineStatistics` is a bitmask of VkQueryPipelineStatisticFlagBits specifying which counters will be returned in queries on the new pool, as described below in Pipeline Statistics Queries. `pipelineStatistics` is ignored if `queryType` is not VK_QUERY_TYPE_PIPELINE_STATISTICS.

Valid Usage

- VUID-VkQueryPoolCreateInfo-queryType-00791
  If the `pipelineStatisticsQuery` feature is not enabled, `queryType` **must** not be VK_QUERY_TYPE_PIPELINE_STATISTICS
- VUID-VkQueryPoolCreateInfo-meshShaderQueries-07068
  If the `meshShaderQueries` feature is not enabled, `queryType` **must** not be VK_QUERY_TYPE_MESH_PRIMITIVES_GENERATED_EXT
- VUID-VkQueryPoolCreateInfo-meshShaderQueries-07069
  If the `meshShaderQueries` feature is not enabled, and `queryType` is VK_QUERY_TYPE_PIPELINE_STATISTICS, `pipelineStatistics` **must** not contain VK_QUERY_PIPELINE_STATISTIC_TASK_SHADER_INVOCATIONS_BIT_EXT or VK_QUERY_PIPELINE_STATISTIC_MESH_SHADER_INVOCATIONS_BIT_EXT
- VUID-VkQueryPoolCreateInfo-queryType-00792
  If `queryType` is VK_QUERY_TYPE_PIPELINE_STATISTICS, `pipelineStatistics` **must** be a valid combination of VkQueryPipelineStatisticFlagBits values
- VUID-VkQueryPoolCreateInfo-queryType-09534
  If `queryType` is VK_QUERY_TYPE_PIPELINE_STATISTICS, `pipelineStatistics` **must** not be zero
- VUID-VkQueryPoolCreateInfo-queryType-03222
  If `queryType` is VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, the `pNext` chain **must** include a VkQueryPoolPerformanceCreateInfoKHR structure
- VUID-VkQueryPoolCreateInfo-queryCount-02763

queryCount **must** be greater than 0

- VUID-VkQueryPoolCreateInfo-queryType-11839
  If queryType is VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR, then at least one of the queue families of the device **must** support result status queries, as indicated by VkQueueFamilyQueryResultStatusPropertiesKHR::queryResultStatusSupport
- VUID-VkQueryPoolCreateInfo-pNext-10779
  If the pNext chain includes a VkVideoProfileInfoKHR structure and its videoCodecOperation member is VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR, then the videoDecodeVP9 feature **must** be enabled
- VUID-VkQueryPoolCreateInfo-queryType-07133
  If queryType is VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR, then the pNext chain **must** include a VkVideoProfileInfoKHR structure with videoCodecOperation specifying an encode operation
- VUID-VkQueryPoolCreateInfo-queryType-07906
  If queryType is VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR, then the pNext chain **must** include a VkQueryPoolVideoEncodeFeedbackCreateInfoKHR structure
- VUID-VkQueryPoolCreateInfo-queryType-07907
  If queryType is VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR, and the pNext chain includes a VkVideoProfileInfoKHR structure and a VkQueryPoolVideoEncodeFeedbackCreateInfoKHR structure, then VkQueryPoolVideoEncodeFeedbackCreateInfoKHR::encodeFeedbackFlags **must** not contain any bits that are not set in VkVideoEncodeCapabilitiesKHR::supportedEncodeFeedbackFlags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile described by VkVideoProfileInfoKHR and its pNext chain
- VUID-VkQueryPoolCreateInfo-queryType-12437
  If queryType is VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR and the pNext chain includes a VkQueryPoolVideoEncodeFeedbackCreateInfoKHR structure with its encodeFeedbackFlags member containing VK_VIDEO_ENCODE_FEEDBACK_AVERAGE_QUANTIZATION_BIT_KHR, VK_VIDEO_ENCODE_FEEDBACK_MIN_QUANTIZATION_BIT_KHR, VK_VIDEO_ENCODE_FEEDBACK_MAX_QUANTIZATION_BIT_KHR, VK_VIDEO_ENCODE_FEEDBACK_INTRA_PIXELS_BIT_KHR, VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR, VK_VIDEO_ENCODE_FEEDBACK_SKIPPED_PIXELS_BIT_KHR, or VK_VIDEO_ENCODE_FEEDBACK_PICTURE_PARTITION_COUNT_BIT_KHR, then videoEncodeFeedback2 **must** be enabled
- VUID-VkQueryPoolCreateInfo-queryType-12438
  If queryType is VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR and the pNext chain includes a VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR structure with a non-zero maxPerPartitionFeedbackEntries member, then videoEncodeFeedback2 **must** be enabled
- VUID-VkQueryPoolCreateInfo-queryType-12439
  If queryType is VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR and the pNext chain includes a VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR structure with a non-zero maxPerPartitionFeedbackEntries member, then

VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR::perPartitionEncodeFeedbackFlags **must** not be zero

- VUID-VkQueryPoolCreateInfo-queryType-12440
  If queryType is VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR and the pNext chain includes a VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR structure, then its maxPerPartitionFeedbackEntries **must** be less than or equal to VkVideoEncodeFeedback2CapabilitiesKHR::maxPerPartitionFeedbackEntries, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile described by VkVideoProfileInfoKHR and its pNext chain
- VUID-VkQueryPoolCreateInfo-queryType-12441
  If queryType is VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR and the pNext chain includes a VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR structure, then its perPartitionEncodeFeedbackFlags **must** not contain any bits that are not set in VkVideoEncodeFeedback2CapabilitiesKHR::supportedPerPartitionEncodeFeedbackFlags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile described by VkVideoProfileInfoKHR and its pNext chain
- VUID-VkQueryPoolCreateInfo-pNext-10248
  If the pNext chain includes a VkVideoProfileInfoKHR structure and its videoCodecOperation member is VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then the videoEncodeAV1 feature **must** be enabled

Valid Usage (Implicit)

- VUID-VkQueryPoolCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_QUERY_POOL_CREATE_INFO
- VUID-VkQueryPoolCreateInfo-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkQueryPoolPerformanceCreateInfoKHR, VkQueryPoolVideoEncodeFeedbackCreateInfoKHR, VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR, VkVideoDecodeAV1ProfileInfoKHR, VkVideoDecodeH264ProfileInfoKHR, VkVideoDecodeH265ProfileInfoKHR, VkVideoDecodeUsageInfoKHR, VkVideoDecodeVP9ProfileInfoKHR, VkVideoEncodeAV1ProfileInfoKHR, VkVideoEncodeH264ProfileInfoKHR, VkVideoEncodeH265ProfileInfoKHR, VkVideoEncodeUsageInfoKHR, or VkVideoProfileInfoKHR
- VUID-VkQueryPoolCreateInfo-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkQueryPoolCreateInfo-flags-parameter
  If flags is not 0, flags **must** be a valid combination of VkQueryPoolCreateFlagBits values
- VUID-VkQueryPoolCreateInfo-queryType-parameter
  queryType **must** be a valid VkQueryType value

Bits which **can** be set in VkQueryPoolCreateInfo::flags, specifying options for query pools, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueryPoolCreateFlagBits {
  // Provided by VK_KHR_maintenance9
    VK_QUERY_POOL_CREATE_RESET_BIT_KHR = 0x00000001,
} VkQueryPoolCreateFlagBits;
```

- VK_QUERY_POOL_CREATE_RESET_BIT_KHR specifies that queries in the query pool are initialized on creation and do not need to be reset before first use.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkQueryPoolCreateFlags;
```

VkQueryPoolCreateFlags is a bitmask type for setting a mask of zero or more VkQueryPoolCreateFlagBits.

The VkQueryPoolPerformanceCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_performance_query
typedef struct VkQueryPoolPerformanceCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           queueFamilyIndex;
    uint32_t           counterIndexCount;
    const uint32_t*    pCounterIndices;
} VkQueryPoolPerformanceCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- queueFamilyIndex is the queue family index to create this performance query pool for.
- counterIndexCount is the length of the pCounterIndices array.
- pCounterIndices is a pointer to an array of indices into the vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR::pCounters to enable in this performance query pool.

Valid Usage

- VUID-VkQueryPoolPerformanceCreateInfoKHR-queueFamilyIndex-03236
  queueFamilyIndex **must** be a valid queue family index of the device
- VUID-VkQueryPoolPerformanceCreateInfoKHR-performanceCounterQueryPools-03237
  The performanceCounterQueryPools feature **must** be enabled
- VUID-VkQueryPoolPerformanceCreateInfoKHR-pCounterIndices-03321
  Each element of pCounterIndices **must** be in the range of counters reported by vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR for the queue family

specified in queueFamilyIndex

Valid Usage (Implicit)

- VUID-VkQueryPoolPerformanceCreateInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_QUERY_POOL_PERFORMANCE_CREATE_INFO_KHR
- VUID-VkQueryPoolPerformanceCreateInfoKHR-pCounterIndices-parameter
  pCounterIndices **must** be a valid pointer to an array of counterIndexCount uint32_t values
- VUID-VkQueryPoolPerformanceCreateInfoKHR-counterIndexCount-arraylength
  counterIndexCount **must** be greater than 0

Structure Chaining

Extends the structure

- VkQueryPoolCreateInfo

To query the number of passes required to query a performance query pool on a physical device, call:

```
// Provided by VK_KHR_performance_query
void vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR(
    VkPhysicalDevice                            physicalDevice,
    const VkQueryPoolPerformanceCreateInfoKHR*  pPerformanceQueryCreateInfo,
    uint32_t*                                   pNumPasses);
```

- physicalDevice is the handle to the physical device whose queue family performance query counter properties will be queried.
- pPerformanceQueryCreateInfo is a pointer to a VkQueryPoolPerformanceCreateInfoKHR of the performance query that is to be created.
- pNumPasses is a pointer to an integer related to the number of passes required to query the performance query pool, as described below.

The pPerformanceQueryCreateInfo member VkQueryPoolPerformanceCreateInfoKHR::queueFamilyIndex **must** be a queue family of physicalDevice. The number of passes required to capture the counters specified in the pPerformanceQueryCreateInfo member VkQueryPoolPerformanceCreateInfoKHR ::pCounterIndices is returned in pNumPasses.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR-physicalDevice-parameter
  physicalDevice **must** be a valid VkPhysicalDevice handle

- VUID-vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR-pPerformanceQueryCreateInfo-parameter
  `pPerformanceQueryCreateInfo` **must** be a valid pointer to a valid VkQueryPoolPerformanceCreateInfoKHR structure
- VUID-vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR-pNumPasses-parameter
  `pNumPasses` **must** be a valid pointer to a `uint32_t` value

To destroy a query pool, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyQueryPool(
    VkDevice                                    device,
    VkQueryPool                                 queryPool,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the query pool.
- `queryPool` is the query pool to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyQueryPool-queryPool-00793
  All submitted commands that refer to `queryPool` **must** have completed execution
- VUID-vkDestroyQueryPool-queryPool-00794
  If `VkAllocationCallbacks` were provided when `queryPool` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyQueryPool-queryPool-00795
  If no `VkAllocationCallbacks` were provided when `queryPool` was created, `pAllocator` **must** be `NULL`

NOTE

Applications **can** verify that `queryPool` **can** be destroyed by checking that `vkGetQueryPoolResults`() without the VK_QUERY_RESULT_PARTIAL_BIT flag returns VK_SUCCESS for all queries that are used in command buffers submitted for execution.

Valid Usage (Implicit)

- VUID-vkDestroyQueryPool-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyQueryPool-queryPool-parameter
  If `queryPool` is not VK_NULL_HANDLE, `queryPool` **must** be a valid VkQueryPool handle

- VUID-vkDestroyQueryPool-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyQueryPool-queryPool-parent
  If `queryPool` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `queryPool` **must** be externally synchronized

Possible values of VkQueryPoolCreateInfo::`queryType`, specifying the type of queries managed by the pool, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueryType {
    VK_QUERY_TYPE_OCCLUSION = 0,
    VK_QUERY_TYPE_PIPELINE_STATISTICS = 1,
    VK_QUERY_TYPE_TIMESTAMP = 2,
  // Provided by VK_KHR_video_queue
    VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR = 1000023000,
  // Provided by VK_EXT_transform_feedback
    VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT = 1000028004,
  // Provided by VK_KHR_performance_query
    VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR = 1000116000,
  // Provided by VK_KHR_acceleration_structure
    VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR = 1000150000,
  // Provided by VK_KHR_acceleration_structure
    VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR = 1000150001,
  // Provided by VK_KHR_video_encode_queue
    VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR = 1000299000,
  // Provided by VK_EXT_mesh_shader
    VK_QUERY_TYPE_MESH_PRIMITIVES_GENERATED_EXT = 1000328000,
  // Provided by VK_EXT_primitives_generated_query
    VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT = 1000382000,
  // Provided by VK_KHR_ray_tracing_maintenance1
    VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR =
1000386000,
  // Provided by VK_KHR_ray_tracing_maintenance1
    VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR = 1000386001,
  // Provided by VK_EXT_opacity_micromap
    VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT = 1000396000,
  // Provided by VK_EXT_opacity_micromap
    VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT = 1000396001,
} VkQueryType;
```

- VK_QUERY_TYPE_OCCLUSION specifies an occlusion query.

- VK_QUERY_TYPE_PIPELINE_STATISTICS specifies a pipeline statistics query.
- VK_QUERY_TYPE_TIMESTAMP specifies a timestamp query.
- VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR specifies a performance query.
- VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT specifies a transform feedback query.
- VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT specifies a primitives generated query.
- VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR specifies an acceleration structure size query for use with vkCmdWriteAccelerationStructuresPropertiesKHR or vkWriteAccelerationStructuresPropertiesKHR.
- VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR specifies a serialization acceleration structure size query.
- VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR specifies an acceleration structure size query for use with vkCmdWriteAccelerationStructuresPropertiesKHR or vkWriteAccelerationStructuresPropertiesKHR.
- VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR specifies a serialization acceleration structure pointer count query.
- VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR specifies a result status query.
- VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR specifies a video encode feedback query.
- VK_QUERY_TYPE_MESH_PRIMITIVES_GENERATED_EXT specifies a generated mesh primitives query.
- VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT specifies a micromap query to get the size required by a serialized micromap.
- VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT specifies a micromap query to get the size required by a compacted micromap.

22.2. Query Operation

The operation of queries is controlled by the commands vkCmdBeginQuery, vkCmdEndQuery, vkCmdBeginQueryIndexedEXT, vkCmdEndQueryIndexedEXT, vkCmdResetQueryPool, vkCmdCopyQueryPoolResults, vkCmdWriteAccelerationStructuresPropertiesKHR, vkCmdWriteMicromapsPropertiesEXT, vkCmdCopyQueryPoolResultsToMemoryKHR, vkCmdWriteTimestamp2, and vkCmdWriteTimestamp.

In order for a `VkCommandBuffer` to record query management commands, the queue family for which its `VkCommandPool` was created **must** support the appropriate type of operations (graphics, compute) suitable for the query type of a given query pool.

Each query in a query pool has a status that is either *unavailable* or *available*, and also has state to store the numerical results of a query operation of the type requested when the query pool was created. Resetting a query via vkCmdResetQueryPool or vkResetQueryPool sets the status to unavailable and makes the numerical results undefined. A query is made available by the operation of vkCmdEndQuery, vkCmdEndQueryIndexedEXT, vkCmdWriteAccelerationStructuresPropertiesKHR, vkCmdWriteMicromapsPropertiesEXT,

vkCmdWriteTimestamp2, or vkCmdWriteTimestamp. Both the availability status and numerical results **can** be retrieved by calling either vkGetQueryPoolResults or vkCmdCopyQueryPoolResults.

After query pool creation, where VK_QUERY_POOL_CREATE_RESET_BIT_KHR was not set in VkQueryPoolCreateInfo::`flags` each query is in an uninitialized state and **must** be reset before it is used. Queries **must** also be reset between uses.

If a logical device includes multiple physical devices, then each command that writes a query **must** execute on a single physical device, and any call to vkCmdBeginQuery **must** execute the corresponding vkCmdEndQuery command on the same physical device.

To reset a range of queries in a query pool on a queue, call:

```
// Provided by VK_VERSION_1_0
void vkCmdResetQueryPool(
    VkCommandBuffer                             commandBuffer,
    VkQueryPool                                 queryPool,
    uint32_t                                    firstQuery,
    uint32_t                                    queryCount);
```

- `commandBuffer` is the command buffer into which this command will be recorded.
- `queryPool` is the handle of the query pool managing the queries being reset.
- `firstQuery` is the initial query index to reset.
- `queryCount` is the number of queries to reset.

When executed on a queue, this command sets the status of query indices [`firstQuery`, `firstQuery` + `queryCount` - 1] to unavailable.

This command defines an execution dependency between other query commands that reference the same query.

The first synchronization scope includes all commands which reference the queries in `queryPool` indicated by `firstQuery` and `queryCount` that occur earlier in submission order.

The second synchronization scope includes all commands which reference the queries in `queryPool` indicated by `firstQuery` and `queryCount` that occur later in submission order.

The operation of this command happens after the first scope and happens before the second scope.

If the `queryType` used to create `queryPool` was VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, this command sets the status of query indices [`firstQuery`, `firstQuery` + `queryCount` - 1] to unavailable for each pass of `queryPool`, as indicated by a call to vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR.

NOTE

Because `vkCmdResetQueryPool` resets all the passes of the indicated queries, applications must not record a `vkCmdResetQueryPool` command for a `queryPool` created with VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR in a command buffer that needs to be submitted multiple times as indicated by a call to

vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR. Otherwise applications will never be able to complete the recorded queries.

Valid Usage

- VUID-vkCmdResetQueryPool-firstQuery-09436
  `firstQuery` **must** be less than the number of queries in `queryPool`
- VUID-vkCmdResetQueryPool-firstQuery-09437
  The sum of `firstQuery` and `queryCount` **must** be less than or equal to the number of queries in `queryPool`
- VUID-vkCmdResetQueryPool-None-02841
  All queries used by the command **must** not be active
- VUID-vkCmdResetQueryPool-firstQuery-02862
  If `queryPool` was created with VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, this command **must** not be recorded in a command buffer that, either directly or through secondary command buffers, also contains begin commands for a query from the set of queries [`firstQuery`, `firstQuery` + `queryCount` - 1]

Valid Usage (Implicit)

- VUID-vkCmdResetQueryPool-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdResetQueryPool-queryPool-parameter
  `queryPool` **must** be a valid VkQueryPool handle
- VUID-vkCmdResetQueryPool-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdResetQueryPool-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdResetQueryPool-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdResetQueryPool-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdResetQueryPool-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdResetQueryPool-commonparent
  Both of `commandBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Action

Conditional Rendering

vkCmdResetQueryPool is not affected by conditional rendering

To reset a range of queries in a query pool on the host, call:

```
// Provided by VK_VERSION_1_2
void vkResetQueryPool(
    VkDevice                                    device,
    VkQueryPool                                 queryPool,
    uint32_t                                    firstQuery,
    uint32_t                                    queryCount);
```

- `device` is the logical device that owns the query pool.
- `queryPool` is the handle of the query pool managing the queries being reset.
- `firstQuery` is the initial query index to reset.
- `queryCount` is the number of queries to reset.

This command sets the status of query indices [`firstQuery`, `firstQuery` + `queryCount` - 1] to unavailable.

If queryPool is VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR this command sets the status of query indices [firstQuery, firstQuery + queryCount - 1] to unavailable for each pass.

Valid Usage

- VUID-vkResetQueryPool-firstQuery-09436
  firstQuery **must** be less than the number of queries in queryPool
- VUID-vkResetQueryPool-firstQuery-09437
  The sum of firstQuery and queryCount **must** be less than or equal to the number of queries in queryPool
- VUID-vkResetQueryPool-None-02665
  The hostQueryReset feature **must** be enabled
- VUID-vkResetQueryPool-firstQuery-02741
  Submitted commands that refer to the range specified by firstQuery and queryCount in queryPool **must** have completed execution
- VUID-vkResetQueryPool-firstQuery-02742
  The range of queries specified by firstQuery and queryCount in queryPool **must** not be in use by calls to vkGetQueryPoolResults or vkResetQueryPool in other threads

Valid Usage (Implicit)

- VUID-vkResetQueryPool-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkResetQueryPool-queryPool-parameter
  queryPool **must** be a valid VkQueryPool handle
- VUID-vkResetQueryPool-queryPool-parent
  queryPool **must** have been created, allocated, or retrieved from device

Once queries are reset and ready for use, query commands **can** be issued to a command buffer. Pipeline statistics queries count pipeline stage invocations, occlusion queries count drawn samples, resulting from commands that are recorded between a vkCmdBeginQuery command and a vkCmdEndQuery command within a specified command buffer, effectively scoping a set of dispatching and/or drawing commands. Timestamp queries write timestamps to a query pool. Performance queries record performance counters to a query pool.

A query **must** begin and end in the same command buffer, although if it is a primary command buffer, and the inheritedQueries feature is enabled, it **can** execute secondary command buffers during the query operation. For a secondary command buffer to be executed while an occlusion query or pipeline statistics query is active, it **must** set the occlusionQueryEnable, queryFlags, and/or pipelineStatistics members of VkCommandBufferInheritanceInfo to conservative values, as described in the Command Buffer Recording section. A query **must** either begin and end inside the same subpass of a render pass instance, or **must** both begin and end outside of a render pass instance (i.e. contain entire render pass instances).

If queries are used while executing a render pass instance that has multiview enabled, the query uses N consecutive query indices in the query pool (starting at `query`) where N is the number of bits set in the view mask in the subpass the query is used in. How the numerical results of the query are distributed among the queries is implementation-dependent. For example, some implementations **may** write each view's results to a distinct query, while other implementations **may** write the total result to the first query and write zero to the other queries. However, the sum of the results in all the queries **must** accurately reflect the total result of the query summed over all views. Applications **can** sum the results from all the queries to compute the total result.

Queries used with multiview rendering **must** not span subpasses, i.e. they **must** begin and end in the same subpass.

A query **must** either begin and end inside the same video coding scope, or **must** both begin and end outside of a video coding scope and **must** not contain entire video coding scopes.

To begin a query, call:

```
// Provided by VK_VERSION_1_0
void vkCmdBeginQuery(
    VkCommandBuffer                             commandBuffer,
    VkQueryPool                                 queryPool,
    uint32_t                                    query,
    VkQueryControlFlags                         flags);
```

- `commandBuffer` is the command buffer into which this command will be recorded.
- `queryPool` is the query pool that will manage the results of the query.
- `query` is the query index within the query pool that will contain the results.
- `flags` is a bitmask of VkQueryControlFlagBits specifying constraints on the types of queries that **can** be performed.

If the `queryType` of the pool is VK_QUERY_TYPE_OCCLUSION and `flags` contains VK_QUERY_CONTROL_PRECISE_BIT, an implementation **must** return a result that matches the actual number of samples passed. This is described in more detail in Occlusion Queries.

Calling `vkCmdBeginQuery` is equivalent to calling vkCmdBeginQueryIndexedEXT with the `index` parameter set to zero.

After beginning a query, that query is considered *active* within the command buffer it was called in until that same query is ended. Queries active in a primary command buffer when secondary command buffers are executed are considered active for those secondary command buffers.

Furthermore, if the query is started within a video coding scope, the following command buffer states are initialized for the query type:

- The *active query index* is set to the value specified by `query`.
- The *last activatable query index* is also set to the value specified by `query`.

Each video coding operation stores a result to the query corresponding to the current active query

index, followed by incrementing the active query index. If the active query index gets incremented past the last activatable query index, issuing any further video coding operations results in undefined behavior.

NOTE	In practice, this means that currently no more than a single video coding operation **must** be issued between a begin and end query pair.

This command defines an execution dependency between other query commands that reference the same query.

The first synchronization scope includes all commands which reference the queries in `queryPool` indicated by `query` that occur earlier in submission order.

The second synchronization scope includes all commands which reference the queries in `queryPool` indicated by `query` that occur later in submission order.

The operation of this command happens after the first scope and happens before the second scope.

Valid Usage

- VUID-vkCmdBeginQuery-None-00807
  All queries used by the command **must** be *unavailable*
- VUID-vkCmdBeginQuery-queryType-02804
  The `queryType` used to create `queryPool` **must** not be VK_QUERY_TYPE_TIMESTAMP
- VUID-vkCmdBeginQuery-queryType-04728
  The `queryType` used to create `queryPool` **must** not be VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR or VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR
- VUID-vkCmdBeginQuery-queryType-06741
  The `queryType` used to create `queryPool` **must** not be VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR or VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR
- VUID-vkCmdBeginQuery-queryType-08972
  The `queryType` used to create `queryPool` **must** not be VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT or VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT
- VUID-vkCmdBeginQuery-queryType-00800
  If the `occlusionQueryPrecise` feature is not enabled, or the `queryType` used to create `queryPool` was not VK_QUERY_TYPE_OCCLUSION, `flags` **must** not contain VK_QUERY_CONTROL_PRECISE_BIT
- VUID-vkCmdBeginQuery-query-00802
  `query` **must** be less than the number of queries in `queryPool`
- VUID-vkCmdBeginQuery-queryType-00803
  If the `queryType` used to create `queryPool` was VK_QUERY_TYPE_OCCLUSION, the

VkCommandPool that commandBuffer was allocated from **must** support graphics operations

- VUID-vkCmdBeginQuery-queryType-00804
  If the queryType used to create queryPool was VK_QUERY_TYPE_PIPELINE_STATISTICS and any of the pipelineStatistics indicate graphics operations, the VkCommandPool that commandBuffer was allocated from **must** support graphics operations
- VUID-vkCmdBeginQuery-queryType-00805
  If the queryType used to create queryPool was VK_QUERY_TYPE_PIPELINE_STATISTICS and any of the pipelineStatistics indicate compute operations, the VkCommandPool that commandBuffer was allocated from **must** support compute operations
- VUID-vkCmdBeginQuery-commandBuffer-01885
  commandBuffer **must** not be a protected command buffer
- VUID-vkCmdBeginQuery-query-00808
  If called within a render pass instance, the sum of query and the number of bits set in the current subpass's view mask **must** be less than or equal to the number of queries in queryPool
- VUID-vkCmdBeginQuery-queryType-07126
  If the queryType used to create queryPool was VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR, then the VkCommandPool that commandBuffer was allocated from **must** have been created with a queue family index that supports result status queries, as indicated by VkQueueFamilyQueryResultStatusPropertiesKHR::queryResultStatusSupport
- VUID-vkCmdBeginQuery-None-07127
  If there is a bound video session, then there **must** be no active queries
- VUID-vkCmdBeginQuery-None-08370
  If there is a bound video session, then it **must** not have been created with VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR
- VUID-vkCmdBeginQuery-queryType-07128
  If the queryType used to create queryPool was VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR and there is a bound video session, then queryPool **must** have been created with a VkVideoProfileInfoKHR structure included in the pNext chain of VkQueryPoolCreateInfo identical to the one specified in VkVideoSessionCreateInfoKHR::pVideoProfile the bound video session was created with
- VUID-vkCmdBeginQuery-queryType-07129
  If the queryType used to create queryPool was VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR, then there **must** be a bound video session
- VUID-vkCmdBeginQuery-queryType-07130
  If the queryType used to create queryPool was VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR and there is a bound video session, then queryPool **must** have been created with a VkVideoProfileInfoKHR structure included in the pNext chain of VkQueryPoolCreateInfo identical to the one specified in VkVideoSessionCreateInfoKHR::pVideoProfile the bound video session was created with
- VUID-vkCmdBeginQuery-queryType-07131

If the queryType used to create queryPool was not VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR or VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR, then there **must** be no bound video session

- VUID-vkCmdBeginQuery-queryPool-01922
  queryPool **must** have been created with a queryType that differs from that of any queries that are active within commandBuffer
- VUID-vkCmdBeginQuery-queryType-07070
  If the queryType used to create queryPool was VK_QUERY_TYPE_MESH_PRIMITIVES_GENERATED_EXT the VkCommandPool that commandBuffer was allocated from **must** support graphics operations
- VUID-vkCmdBeginQuery-queryType-02327
  If the queryType used to create queryPool was VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT the VkCommandPool that commandBuffer was allocated from **must** support graphics operations
- VUID-vkCmdBeginQuery-queryType-02328
  If the queryType used to create queryPool was VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT then VkPhysicalDeviceTransformFeedbackPropertiesEXT::transformFeedbackQueries **must** be supported
- VUID-vkCmdBeginQuery-queryType-06687
  If the queryType used to create queryPool was VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT the VkCommandPool that commandBuffer was allocated from **must** support graphics operations
- VUID-vkCmdBeginQuery-queryType-06688
  If the queryType used to create queryPool was VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT then primitivesGeneratedQuery **must** be enabled
- VUID-vkCmdBeginQuery-queryPool-07289
  If queryPool was created with a queryType of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, then the VkQueryPoolPerformanceCreateInfoKHR::queueFamilyIndex queryPool was created with **must** equal the queue family index of the VkCommandPool that commandBuffer was allocated from
- VUID-vkCmdBeginQuery-queryPool-03223
  If queryPool was created with a queryType of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, the profiling lock **must** have been held before vkBeginCommandBuffer was called on commandBuffer
- VUID-vkCmdBeginQuery-queryPool-03224
  If queryPool was created with a queryType of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR and one of the counters used to create queryPool was VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_BUFFER_KHR, the query begin **must** be the first recorded command in commandBuffer

- VUID-vkCmdBeginQuery-queryPool-03225
  If `queryPool` was created with a `queryType` of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR and one of the counters used to create `queryPool` was VK_PERFORMANCE_COUNTER_SCOPE_RENDER_PASS_KHR, the begin command **must** not be recorded within a render pass instance
- VUID-vkCmdBeginQuery-queryPool-03226
  If `queryPool` was created with a `queryType` of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR and another query pool with a `queryType` VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR has been used within `commandBuffer`, its parent primary command buffer or secondary command buffer recorded within the same parent primary command buffer as `commandBuffer`, the `performanceCounterMultipleQueryPools` feature **must** be enabled
- VUID-vkCmdBeginQuery-None-02863
  If `queryPool` was created with a `queryType` of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, this command **must** not be recorded in a command buffer that, either directly or through secondary command buffers, also contains a `vkCmdResetQueryPool` command affecting the same query

Valid Usage (Implicit)

- VUID-vkCmdBeginQuery-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBeginQuery-queryPool-parameter
  `queryPool` **must** be a valid VkQueryPool handle
- VUID-vkCmdBeginQuery-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkQueryControlFlagBits values
- VUID-vkCmdBeginQuery-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBeginQuery-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdBeginQuery-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdBeginQuery-commonparent
  Both of `commandBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized

- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Action State

Conditional Rendering

vkCmdBeginQuery is not affected by conditional rendering

To begin an indexed query, call:

```
// Provided by VK_EXT_transform_feedback
void vkCmdBeginQueryIndexedEXT(
    VkCommandBuffer                             commandBuffer,
    VkQueryPool                                 queryPool,
    uint32_t                                    query,
    VkQueryControlFlags                         flags,
    uint32_t                                    index);
```

- commandBuffer is the command buffer into which this command will be recorded.
- queryPool is the query pool that will manage the results of the query.
- query is the query index within the query pool that will contain the results.
- flags is a bitmask of VkQueryControlFlagBits specifying constraints on the types of queries that **can** be performed.
- index is the query type specific index. When the query type is VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT or VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT, the index represents the vertex stream.

The vkCmdBeginQueryIndexedEXT command operates the same as the vkCmdBeginQuery command,

except that it also accepts a query type specific `index` parameter.

This command defines an execution dependency between other query commands that reference the same query index.

The first synchronization scope includes all commands which reference the queries in `queryPool` indicated by `query` and `index` that occur earlier in submission order.

The second synchronization scope includes all commands which reference the queries in `queryPool` indicated by `query` and `index` that occur later in submission order.

The operation of this command happens after the first scope and happens before the second scope.

Valid Usage

- VUID-vkCmdBeginQueryIndexedEXT-None-00807
  All queries used by the command **must** be *unavailable*
- VUID-vkCmdBeginQueryIndexedEXT-queryType-02804
  The `queryType` used to create `queryPool` **must** not be VK_QUERY_TYPE_TIMESTAMP
- VUID-vkCmdBeginQueryIndexedEXT-queryType-04728
  The `queryType` used to create `queryPool` **must** not be VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR or VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR
- VUID-vkCmdBeginQueryIndexedEXT-queryType-06741
  The `queryType` used to create `queryPool` **must** not be VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR or VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR
- VUID-vkCmdBeginQueryIndexedEXT-queryType-08972
  The `queryType` used to create `queryPool` **must** not be VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT or VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT
- VUID-vkCmdBeginQueryIndexedEXT-queryType-00800
  If the `occlusionQueryPrecise` feature is not enabled, or the `queryType` used to create `queryPool` was not VK_QUERY_TYPE_OCCLUSION, `flags` **must** not contain VK_QUERY_CONTROL_PRECISE_BIT
- VUID-vkCmdBeginQueryIndexedEXT-query-00802
  `query` **must** be less than the number of queries in `queryPool`
- VUID-vkCmdBeginQueryIndexedEXT-queryType-00803
  If the `queryType` used to create `queryPool` was VK_QUERY_TYPE_OCCLUSION, the `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdBeginQueryIndexedEXT-queryType-00804
  If the `queryType` used to create `queryPool` was VK_QUERY_TYPE_PIPELINE_STATISTICS and any of the `pipelineStatistics` indicate graphics operations, the `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

- VUID-vkCmdBeginQueryIndexedEXT-queryType-00805
  If the queryType used to create queryPool was VK_QUERY_TYPE_PIPELINE_STATISTICS and any of the pipelineStatistics indicate compute operations, the VkCommandPool that commandBuffer was allocated from **must** support compute operations
- VUID-vkCmdBeginQueryIndexedEXT-commandBuffer-01885
  commandBuffer **must** not be a protected command buffer
- VUID-vkCmdBeginQueryIndexedEXT-query-00808
  If called within a render pass instance, the sum of query and the number of bits set in the current subpass's view mask **must** be less than or equal to the number of queries in queryPool
- VUID-vkCmdBeginQueryIndexedEXT-queryType-07126
  If the queryType used to create queryPool was VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR, then the VkCommandPool that commandBuffer was allocated from **must** have been created with a queue family index that supports result status queries, as indicated by VkQueueFamilyQueryResultStatusPropertiesKHR::queryResultStatusSupport
- VUID-vkCmdBeginQueryIndexedEXT-None-07127
  If there is a bound video session, then there **must** be no active queries
- VUID-vkCmdBeginQueryIndexedEXT-None-08370
  If there is a bound video session, then it **must** not have been created with VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR
- VUID-vkCmdBeginQueryIndexedEXT-queryType-07128
  If the queryType used to create queryPool was VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR and there is a bound video session, then queryPool **must** have been created with a VkVideoProfileInfoKHR structure included in the pNext chain of VkQueryPoolCreateInfo identical to the one specified in VkVideoSessionCreateInfoKHR::pVideoProfile the bound video session was created with
- VUID-vkCmdBeginQueryIndexedEXT-queryType-07129
  If the queryType used to create queryPool was VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR, then there **must** be a bound video session
- VUID-vkCmdBeginQueryIndexedEXT-queryType-07130
  If the queryType used to create queryPool was VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR and there is a bound video session, then queryPool **must** have been created with a VkVideoProfileInfoKHR structure included in the pNext chain of VkQueryPoolCreateInfo identical to the one specified in VkVideoSessionCreateInfoKHR::pVideoProfile the bound video session was created with
- VUID-vkCmdBeginQueryIndexedEXT-queryType-07131
  If the queryType used to create queryPool was not VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR or VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR, then there **must** be no bound video session
- VUID-vkCmdBeginQueryIndexedEXT-queryPool-04753
  If the queryPool was created with the same queryType as that of another active query

within commandBuffer, then index **must** not match the index used for the active query

- VUID-vkCmdBeginQueryIndexedEXT-queryType-02338
  If the queryType used to create queryPool was VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT the VkCommandPool that commandBuffer was allocated from **must** support graphics operations
- VUID-vkCmdBeginQueryIndexedEXT-queryType-02339
  If the queryType used to create queryPool was VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT the index parameter **must** be less than VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams
- VUID-vkCmdBeginQueryIndexedEXT-queryType-06692
  If the queryType used to create queryPool was not VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT and not VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT, the index **must** be zero
- VUID-vkCmdBeginQueryIndexedEXT-queryType-06689
  If the queryType used to create queryPool was VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT the VkCommandPool that commandBuffer was allocated from **must** support graphics operations
- VUID-vkCmdBeginQueryIndexedEXT-queryType-06690
  If the queryType used to create queryPool was VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT the index parameter **must** be less than VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams
- VUID-vkCmdBeginQueryIndexedEXT-queryType-06691
  If the queryType used to create queryPool was VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT and the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled, the index parameter **must** be zero
- VUID-vkCmdBeginQueryIndexedEXT-queryType-06693
  If the queryType used to create queryPool was VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT then primitivesGeneratedQuery **must** be enabled
- VUID-vkCmdBeginQueryIndexedEXT-queryType-02341
  If the queryType used to create queryPool was VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT then VkPhysicalDeviceTransformFeedbackPropertiesEXT::transformFeedbackQueries **must** be supported
- VUID-vkCmdBeginQueryIndexedEXT-queryType-07071
  The queryType used to create queryPool **must** not be VK_QUERY_TYPE_MESH_PRIMITIVES_GENERATED_EXT
- VUID-vkCmdBeginQueryIndexedEXT-queryPool-07289
  If queryPool was created with a queryType of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, then the VkQueryPoolPerformanceCreateInfoKHR::queueFamilyIndex queryPool was created with **must** equal the queue family index of the VkCommandPool that commandBuffer was allocated

from

- VUID-vkCmdBeginQueryIndexedEXT-queryPool-03223
  If `queryPool` was created with a `queryType` of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, the profiling lock **must** have been held before vkBeginCommandBuffer was called on `commandBuffer`
- VUID-vkCmdBeginQueryIndexedEXT-queryPool-03224
  If `queryPool` was created with a `queryType` of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR and one of the counters used to create `queryPool` was VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_BUFFER_KHR, the query begin **must** be the first recorded command in `commandBuffer`
- VUID-vkCmdBeginQueryIndexedEXT-queryPool-03225
  If `queryPool` was created with a `queryType` of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR and one of the counters used to create `queryPool` was VK_PERFORMANCE_COUNTER_SCOPE_RENDER_PASS_KHR, the begin command **must** not be recorded within a render pass instance
- VUID-vkCmdBeginQueryIndexedEXT-queryPool-03226
  If `queryPool` was created with a `queryType` of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR and another query pool with a `queryType` VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR has been used within `commandBuffer`, its parent primary command buffer or secondary command buffer recorded within the same parent primary command buffer as `commandBuffer`, the `performanceCounterMultipleQueryPools` feature **must** be enabled
- VUID-vkCmdBeginQueryIndexedEXT-None-02863
  If `queryPool` was created with a `queryType` of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, this command **must** not be recorded in a command buffer that, either directly or through secondary command buffers, also contains a `vkCmdResetQueryPool` command affecting the same query

Valid Usage (Implicit)

- VUID-vkCmdBeginQueryIndexedEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBeginQueryIndexedEXT-queryPool-parameter
  `queryPool` **must** be a valid VkQueryPool handle
- VUID-vkCmdBeginQueryIndexedEXT-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkQueryControlFlagBits values
- VUID-vkCmdBeginQueryIndexedEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBeginQueryIndexedEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations

- VUID-vkCmdBeginQueryIndexedEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdBeginQueryIndexedEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBeginQueryIndexedEXT-commonparent
  Both of `commandBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Action State

Conditional Rendering

vkCmdBeginQueryIndexedEXT is not affected by conditional rendering

Bits which **can** be set in vkCmdBeginQuery::`flags`, specifying constraints on the types of queries that **can** be performed, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueryControlFlagBits {
    VK_QUERY_CONTROL_PRECISE_BIT = 0x00000001,
} VkQueryControlFlagBits;
```

- VK_QUERY_CONTROL_PRECISE_BIT specifies the precision of occlusion queries.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkQueryControlFlags;
```

VkQueryControlFlags is a bitmask type for setting a mask of zero or more VkQueryControlFlagBits.

To end a query after the set of desired commands is recorded, call:

```
// Provided by VK_VERSION_1_0
void vkCmdEndQuery(
    VkCommandBuffer                             commandBuffer,
    VkQueryPool                                 queryPool,
    uint32_t                                    query);
```

- commandBuffer is the command buffer into which this command will be recorded.
- queryPool is the query pool that is managing the results of the query.
- query is the query index within the query pool where the result is stored.

The command completes the query in queryPool identified by query, and marks it as available.

This command defines an execution dependency between other query commands that reference the same query.

The first synchronization scope includes all commands which reference the queries in queryPool indicated by query that occur earlier in submission order.

The second synchronization scope includes only the operation of this command.

Calling vkCmdEndQuery is equivalent to calling vkCmdEndQueryIndexedEXT with the index parameter set to zero.

Valid Usage

- VUID-vkCmdEndQuery-None-01923
  All queries used by the command **must** be active
- VUID-vkCmdEndQuery-query-00810
  query **must** be less than the number of queries in queryPool
- VUID-vkCmdEndQuery-commandBuffer-01886
  commandBuffer **must** not be a protected command buffer
- VUID-vkCmdEndQuery-query-00812
  If vkCmdEndQuery is called within a render pass instance, the sum of query and the number of bits set in the current subpass's view mask **must** be less than or equal to the number of queries in queryPool
- VUID-vkCmdEndQuery-queryPool-03227

If queryPool was created with a queryType of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR and one or more of the counters used to create queryPool was VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_BUFFER_KHR, the vkCmdEndQuery **must** be the last recorded command in commandBuffer

- VUID-vkCmdEndQuery-queryPool-03228
  If queryPool was created with a queryType of VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR and one or more of the counters used to create queryPool was VK_PERFORMANCE_COUNTER_SCOPE_RENDER_PASS_KHR, the vkCmdEndQuery **must** not be recorded within a render pass instance

- VUID-vkCmdEndQuery-None-07007
  If called within a subpass of a render pass instance, the corresponding vkCmdBeginQuery* command **must** have been called previously within the same subpass

Valid Usage (Implicit)

- VUID-vkCmdEndQuery-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdEndQuery-queryPool-parameter
  queryPool **must** be a valid VkQueryPool handle
- VUID-vkCmdEndQuery-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdEndQuery-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdEndQuery-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdEndQuery-commonparent
  Both of commandBuffer, and queryPool **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Action State

Conditional Rendering

vkCmdEndQuery is not affected by conditional rendering

To end an indexed query after the set of desired drawing or dispatching commands is recorded, call:

```
// Provided by VK_EXT_transform_feedback
void vkCmdEndQueryIndexedEXT(
    VkCommandBuffer                             commandBuffer,
    VkQueryPool                                 queryPool,
    uint32_t                                    query,
    uint32_t                                    index);
```

- `commandBuffer` is the command buffer into which this command will be recorded.
- `queryPool` is the query pool that is managing the results of the query.
- `query` is the query index within the query pool where the result is stored.
- `index` is the query type specific index.

The command completes the query in `queryPool` identified by `query` and `index`, and marks it as available.

The `vkCmdEndQueryIndexedEXT` command operates the same as the vkCmdEndQuery command, except that it also accepts a query type specific `index` parameter.

This command defines an execution dependency between other query commands that reference the same query index.

The first synchronization scope includes all commands which reference the queries in `queryPool`

indicated by query that occur earlier in submission order.

The second synchronization scope includes only the operation of this command.

Valid Usage

- VUID-vkCmdEndQueryIndexedEXT-None-02342
  All queries used by the command **must** be active
- VUID-vkCmdEndQueryIndexedEXT-query-02343
  query **must** be less than the number of queries in queryPool
- VUID-vkCmdEndQueryIndexedEXT-commandBuffer-02344
  commandBuffer **must** not be a protected command buffer
- VUID-vkCmdEndQueryIndexedEXT-query-02345
  If vkCmdEndQueryIndexedEXT is called within a render pass instance, the sum of query and the number of bits set in the current subpass's view mask **must** be less than or equal to the number of queries in queryPool
- VUID-vkCmdEndQueryIndexedEXT-queryType-06694
  If the queryType used to create queryPool was VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT or VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT, the index parameter **must** be less than VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams
- VUID-vkCmdEndQueryIndexedEXT-queryType-06695
  If the queryType used to create queryPool was not VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT and not VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT, the index **must** be zero
- VUID-vkCmdEndQueryIndexedEXT-queryType-06696
  If the queryType used to create queryPool was VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT or VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT, index **must** equal the index used to begin the query
- VUID-vkCmdEndQueryIndexedEXT-None-07007
  If called within a subpass of a render pass instance, the corresponding vkCmdBeginQuery* command **must** have been called previously within the same subpass

Valid Usage (Implicit)

- VUID-vkCmdEndQueryIndexedEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdEndQueryIndexedEXT-queryPool-parameter
  queryPool **must** be a valid VkQueryPool handle
- VUID-vkCmdEndQueryIndexedEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state

- VUID-vkCmdEndQueryIndexedEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdEndQueryIndexedEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdEndQueryIndexedEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdEndQueryIndexedEXT-commonparent
  Both of `commandBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Action State

Conditional Rendering

vkCmdEndQueryIndexedEXT is not affected by conditional rendering

An application **can** retrieve results either by requesting they be written into application-provided memory, or by requesting they be copied into a `VkBuffer`. In either case, the layout in memory is defined as follows:

- The first query's result is written starting at the first byte requested by the command, and each subsequent query's result begins `stride` bytes later.
- Each query writes results in a tightly packed array of result values as follows:
  - Occlusion queries write one unsigned integer value - the number of samples passed.
  - Pipeline statistics queries write one unsigned integer value for each bit that is enabled in the `pipelineStatistics` when the pool is created, and the statistics values are written in bit order starting from the least significant bit.
  - Timestamp queries write one unsigned integer value.
  - Performance queries write one VkPerformanceCounterResultKHR value for each VkPerformanceCounterKHR in the query.
  - Transform feedback queries write two unsigned integer values; the first integer is the number of primitives successfully written to the corresponding transform feedback buffer and the second is the number of primitives output to the vertex stream, regardless of whether they were successfully captured or not. In other words, if the transform feedback buffer was sized too small for the number of primitives output by the vertex stream, the first integer represents the number of primitives actually written and the second is the number that would have been written if all the transform feedback buffers associated with that vertex stream were large enough.
  - Primitives generated queries write one unsigned integer value: the number of primitives output to the vertex stream, regardless of whether transform feedback is active or not, or whether they were successfully captured by transform feedback or not. This is identical to the second integer result value produced by transform feedback queries if transform feedback is active.
  - Mesh shader queries write a single unsigned integer.
  - Video encode feedback queries write one or more signed or unsigned integer values for each bit that is enabled in VkQueryPoolVideoEncodeFeedbackCreateInfoKHR ::`encodeFeedbackFlags` when the pool is created, and the feedback values are written in bit order starting from the least significant bit, as described here.
- If VK_QUERY_RESULT_WITH_AVAILABILITY_BIT is specified, an additional unsigned integer result value is written indicating whether the query's result is available, with any non-zero value indicating that it is available.
- If VK_QUERY_RESULT_WITH_STATUS_BIT_KHR is specified, an additional signed integer result value is written indicating the status of the query result. Positive values indicate success, negative values indicate failure, and zero indicates that the result is not yet available. Specific error codes are as defined in the VkQueryResultStatusKHR enumeration.
- Depending on the query type, additional results are written for each query in a tightly packed array of result values as follows:
  - If per picture partition feedback is requested, video encode feedback queries write one or more integer values after the primary query results for each encoded picture partition for each bit that is enabled in VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR ::`perPartitionEncodeFeedbackFlags` when the pool is created. The feedback values are written in picture-partition-major order and, for each encoded picture partition but at most for VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR::`maxPerPartitionFeedbackEntr`

ies number of picture partitions, in bit order starting from the least significant bit.

NOTE Per picture partition feedback values follow all primary query results, including the availability or result status written as the final element of the primary query results in response to VK_QUERY_RESULT_WITH_AVAILABILITY_BIT or VK_QUERY_RESULT_WITH_STATUS_BIT_KHR, respectively. Implementations will never write more per picture partition results than the maxPerPartitionFeedbackEntries requested when the query pool was created.

- The bit width of all integer result values written by all query types is either 32- or 64-bits, as requested by the command. The only exceptions are the VkPerformanceCounterResultKHR values written by performance queries whose type is determined by the unit member of the corresponding VkPerformanceCounterKHR.
- If more than one query is retrieved and stride is not at least as large as the size of the array of values corresponding to a single query, the values written to memory are undefined.

To retrieve status and results for a set of queries, call:

```
// Provided by VK_VERSION_1_0
VkResult vkGetQueryPoolResults(
    VkDevice                                    device,
    VkQueryPool                                 queryPool,
    uint32_t                                    firstQuery,
    uint32_t                                    queryCount,
    size_t                                      dataSize,
    void*                                       pData,
    VkDeviceSize                                stride,
    VkQueryResultFlags                          flags);
```

- device is the logical device that owns the query pool.
- queryPool is the query pool managing the queries containing the desired results.
- firstQuery is the initial query index.
- queryCount is the number of queries to read.
- dataSize is the size in bytes of the buffer pointed to by pData.
- pData is a pointer to an application-allocated buffer where the results will be written.
- stride is the stride in bytes between results for individual queries within pData.
- flags is a bitmask of VkQueryResultFlagBits specifying how and when results are returned.

Any results written for a query are written according to a layout dependent on the query type.

If no bits are set in flags, and all requested queries are in the available state, results are written as an array of 32-bit unsigned integer values. Behavior when not all queries are available is described below.

If VK_QUERY_RESULT_WITH_AVAILABILITY_BIT is set, results for all queries in `queryPool` identified by `firstQuery` and `queryCount` are copied to `pData`, along with an extra availability or status value written directly after the results of each query and interpreted as an unsigned integer. A value of zero indicates that the results are not yet available, otherwise the query is complete and results are available. The size of the availability or status values is 64 bits if VK_QUERY_RESULT_64_BIT is set in `flags`. Otherwise, it is 32 bits.

If VK_QUERY_RESULT_WITH_STATUS_BIT_KHR is set, results for all queries in `queryPool` identified by `firstQuery` and `queryCount` are copied to `pData`, along with an extra status value written directly after the results of each query and interpreted as a signed integer. A value of zero indicates that the results are not yet available. Positive values indicate that the operations within the query completed successfully, and the query results are valid. Negative values indicate that the operations within the query completed unsuccessfully.

VkQueryResultStatusKHR defines specific meaning for values returned here, though implementations are free to return other values.

If the status value written is negative, indicating that the operations within the query completed unsuccessfully, then all other results written by this command are undefined unless otherwise specified for any of the results of the used query type.

NOTE

If VK_QUERY_RESULT_WITH_AVAILABILITY_BIT or VK_QUERY_RESULT_WITH_STATUS_BIT_KHR is set, the layout of data in the buffer is a *(result,availability)* or *(result,status)* pair for each query returned, and `stride` is the stride between each pair.

Results for any available query written by this command are final and represent the final result of the query. If VK_QUERY_RESULT_PARTIAL_BIT is set, then for any query that is unavailable, an intermediate result between zero and the final result value is written for that query. Otherwise, any result written by this command is undefined.

If VK_QUERY_RESULT_64_BIT is set, results and, if returned, availability or status values for all queries are written as an array of 64-bit values. If the `queryPool` was created with VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, results for each query are written as an array of the type indicated by VkPerformanceCounterKHR::`storage` for the counter being queried. Otherwise, results and availability or status values are written as an array of 32-bit values. If an unsigned integer query's value overflows the result type, the value **may** either wrap or saturate. If the `maintenance7` feature is enabled, for an unsigned integer query, the 32-bit result value **must** be equal to the 32 least significant bits of the equivalent 64-bit result value. If a signed integer query's value overflows the result type, the value is undefined. If a floating-point query's value is not representable as the result type, the value is undefined.

If VK_QUERY_RESULT_WAIT_BIT is set, this command defines an execution dependency with any earlier commands that write one of the identified queries. The first synchronization scope includes all instances of vkCmdEndQuery, vkCmdEndQueryIndexedEXT, vkCmdWriteAccelerationStructuresPropertiesKHR, vkCmdWriteMicromapsPropertiesEXT, vkCmdWriteTimestamp2, and vkCmdWriteTimestamp that reference any query in `queryPool` indicated by `firstQuery` and `queryCount`. The second synchronization scope includes the host operations of this command.

If VK_QUERY_RESULT_WAIT_BIT is not set, vkGetQueryPoolResults **may** return VK_NOT_READY if there are queries in the unavailable state.

NOTE

Applications **must** take care to ensure that use of the VK_QUERY_RESULT_WAIT_BIT bit has the desired effect.

For example, if a query has been used previously and a command buffer records the commands vkCmdResetQueryPool, vkCmdBeginQuery, and vkCmdEndQuery for that query, then the query will remain in the available state until vkResetQueryPool is called or the vkCmdResetQueryPool command executes on a queue. Applications **can** use fences or events to ensure that a query has already been reset before checking for its results or availability status. Otherwise, a stale value could be returned from a previous use of the query.

The above also applies when VK_QUERY_RESULT_WAIT_BIT is used in combination with VK_QUERY_RESULT_WITH_AVAILABILITY_BIT. In this case, the returned availability status **may** reflect the result of a previous use of the query unless vkResetQueryPool is called or the vkCmdResetQueryPool command has been executed since the last use of the query.

A similar situation can arise with the VK_QUERY_RESULT_WITH_STATUS_BIT_KHR flag.

NOTE

Applications **can** double-buffer query pool usage, with a pool per frame, and reset queries at the end of the frame in which they are read.

Valid Usage

- VUID-vkGetQueryPoolResults-firstQuery-09436
  `firstQuery` **must** be less than the number of queries in `queryPool`
- VUID-vkGetQueryPoolResults-firstQuery-09437
  The sum of `firstQuery` and `queryCount` **must** be less than or equal to the number of queries in `queryPool`
- VUID-vkGetQueryPoolResults-queryCount-09438
  If `queryCount` is greater than 1, `stride` **must** not be zero
- VUID-vkGetQueryPoolResults-queryType-09439
  If the `queryType` used to create `queryPool` was VK_QUERY_TYPE_TIMESTAMP, `flags` **must** not contain VK_QUERY_RESULT_PARTIAL_BIT
- VUID-vkGetQueryPoolResults-queryType-09440
  If the `queryType` used to create `queryPool` was VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, `flags` **must** not contain VK_QUERY_RESULT_WITH_AVAILABILITY_BIT, VK_QUERY_RESULT_WITH_STATUS_BIT_KHR, VK_QUERY_RESULT_PARTIAL_BIT, or VK_QUERY_RESULT_64_BIT
- VUID-vkGetQueryPoolResults-queryType-09441

If the queryType used to create queryPool was VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, the queryPool **must** have been recorded once for each pass as retrieved via a call to vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR

- VUID-vkGetQueryPoolResults-queryType-11874
  If the queryType used to create queryPool was not VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR or VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR, then flags **must** not include VK_QUERY_RESULT_WITH_STATUS_BIT_KHR
- VUID-vkGetQueryPoolResults-queryType-09442
  If the queryType used to create queryPool was VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR, then flags **must** include VK_QUERY_RESULT_WITH_STATUS_BIT_KHR
- VUID-vkGetQueryPoolResults-flags-09443
  If flags includes VK_QUERY_RESULT_WITH_STATUS_BIT_KHR, then it **must** not include VK_QUERY_RESULT_WITH_AVAILABILITY_BIT
- VUID-vkGetQueryPoolResults-None-09401
  All queries used by the command **must** not be uninitialized
- VUID-vkGetQueryPoolResults-flags-02828
  If VK_QUERY_RESULT_64_BIT is not set in flags and the queryType used to create queryPool was not VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, then pData **must** be aligned to a multiple of 4
- VUID-vkGetQueryPoolResults-queryCount-12251
  If queryCount is greater than 1, VK_QUERY_RESULT_64_BIT is not set in flags and the queryType used to create queryPool was not VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, then stride **must** be a multiple of 4
- VUID-vkGetQueryPoolResults-flags-00815
  If VK_QUERY_RESULT_64_BIT is set in flags then pData **must** be aligned to a multiple of 8
- VUID-vkGetQueryPoolResults-queryCount-12252
  If queryCount is greater than 1 and VK_QUERY_RESULT_64_BIT is set in flags, then stride **must** be a multiple of 8
- VUID-vkGetQueryPoolResults-stride-08993
  If queryCount is greater than 1 and VK_QUERY_RESULT_WITH_AVAILABILITY_BIT is set, stride **must** be large enough to contain the unsigned integer representing availability or status in addition to the query result
- VUID-vkGetQueryPoolResults-queryType-03229
  If the queryType used to create queryPool was VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, then pData **must** be aligned to a multiple of the size of VkPerformanceCounterResultKHR
- VUID-vkGetQueryPoolResults-queryCount-12253
  If queryCount is greater than 1 and the queryType used to create queryPool was VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, then stride **must** be a multiple of the size of VkPerformanceCounterResultKHR

- VUID-vkGetQueryPoolResults-queryType-04519
  If `queryCount` is greater than 1 and the `queryType` used to create `queryPool` was VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, then `stride` **must** be large enough to contain the VkQueryPoolPerformanceCreateInfoKHR::`counterIndexCount` used to create `queryPool` times the size of VkPerformanceCounterResultKHR
- VUID-vkGetQueryPoolResults-dataSize-00817
  `dataSize` **must** be large enough to contain the result of each query, as described here

Valid Usage (Implicit)

- VUID-vkGetQueryPoolResults-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetQueryPoolResults-queryPool-parameter
  `queryPool` **must** be a valid VkQueryPool handle
- VUID-vkGetQueryPoolResults-pData-parameter
  `pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-vkGetQueryPoolResults-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkQueryResultFlagBits values
- VUID-vkGetQueryPoolResults-dataSize-arraylength
  `dataSize` **must** be greater than `0`
- VUID-vkGetQueryPoolResults-queryPool-parent
  `queryPool` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_NOT_READY
- VK_SUCCESS

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Bits which **can** be set in vkGetQueryPoolResults::`flags` and vkCmdCopyQueryPoolResults::`flags`, specifying how and when results are returned, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueryResultFlagBits {
    VK_QUERY_RESULT_64_BIT = 0x00000001,
    VK_QUERY_RESULT_WAIT_BIT = 0x00000002,
    VK_QUERY_RESULT_WITH_AVAILABILITY_BIT = 0x00000004,
    VK_QUERY_RESULT_PARTIAL_BIT = 0x00000008,
  // Provided by VK_KHR_video_queue
    VK_QUERY_RESULT_WITH_STATUS_BIT_KHR = 0x00000010,
} VkQueryResultFlagBits;
```

- VK_QUERY_RESULT_64_BIT specifies the results will be written as an array of 64-bit unsigned integer values. If this bit is not set, the results will be written as an array of 32-bit unsigned integer values.
- VK_QUERY_RESULT_WAIT_BIT specifies that Vulkan will wait for each query's status to become available before retrieving its results.
- VK_QUERY_RESULT_WITH_AVAILABILITY_BIT specifies that the availability status accompanies the results.
- VK_QUERY_RESULT_PARTIAL_BIT specifies that returning partial results is acceptable.
- VK_QUERY_RESULT_WITH_STATUS_BIT_KHR specifies that the last value returned in the query is a VkQueryResultStatusKHR value. See result status query for information on how an application can determine whether the use of this flag bit is supported.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkQueryResultFlags;
```

VkQueryResultFlags is a bitmask type for setting a mask of zero or more VkQueryResultFlagBits.

Specific status codes that **can** be returned from a query are:

```
// Provided by VK_KHR_video_queue
typedef enum VkQueryResultStatusKHR {
    VK_QUERY_RESULT_STATUS_ERROR_KHR = -1,
    VK_QUERY_RESULT_STATUS_NOT_READY_KHR = 0,
    VK_QUERY_RESULT_STATUS_COMPLETE_KHR = 1,
  // Provided by VK_KHR_video_encode_queue
    VK_QUERY_RESULT_STATUS_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_KHR = -1000299000,
} VkQueryResultStatusKHR;
```

- VK_QUERY_RESULT_STATUS_NOT_READY_KHR specifies that the query result is not yet available.
- VK_QUERY_RESULT_STATUS_ERROR_KHR specifies that operations did not complete successfully.
- VK_QUERY_RESULT_STATUS_COMPLETE_KHR specifies that operations completed successfully and the query result is available.

- VK_QUERY_RESULT_STATUS_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_KHR specifies that a video encode operation did not complete successfully due to the destination video bitstream buffer range not being sufficiently large to fit the encoded bitstream data.

To copy query statuses and numerical results directly to buffer memory, call:

```
// Provided by VK_KHR_device_address_commands
void vkCmdCopyQueryPoolResultsToMemoryKHR(
    VkCommandBuffer                             commandBuffer,
    VkQueryPool                                 queryPool,
    uint32_t                                    firstQuery,
    uint32_t                                    queryCount,
    const VkStridedDeviceAddressRangeKHR*       pDstRange,
    VkAddressCommandFlagsKHR                    dstFlags,
    VkQueryResultFlags                          queryResultFlags);
```

- commandBuffer is the command buffer into which this command will be recorded.
- queryPool is the query pool managing the queries containing the desired results.
- firstQuery is the initial query index.
- queryCount is the number of queries. firstQuery and queryCount together define a range of queries.
- pDstRange is a pointer to a VkStridedDeviceAddressRangeKHR describing a range of memory addressed with a stride that will receive the results of the copy command.
- dstFlags is a VkAddressCommandFlagsKHR value defining the flags for the destination address range.
- queryResultFlags is a bitmask of VkQueryResultFlagBits specifying how and when results are returned.

Any results written for a query are written according to a layout dependent on the query type.

Results for any query in queryPool identified by firstQuery and queryCount that is available are copied to pDstRange.

If VK_QUERY_RESULT_WITH_AVAILABILITY_BIT is set, results for all queries in queryPool identified by firstQuery and queryCount are copied to pDstRange, along with an extra availability value written directly after the results of each query and interpreted as an unsigned integer. A value of zero indicates that the results are not yet available, otherwise the query is complete and results are available.

If VK_QUERY_RESULT_WITH_STATUS_BIT_KHR is set, results for all queries in queryPool identified by firstQuery and queryCount are copied to pDstRange, along with an extra status value written directly after the results of each query and interpreted as a signed integer. A value of zero indicates that the results are not yet available. Positive values indicate that the operations within the query completed successfully, and the query results are valid. Negative values indicate that the operations within the query completed unsuccessfully.

VkQueryResultStatusKHR defines specific meaning for values returned here, though implementations are free to return other values.

If the status value written is negative, indicating that the operations within the query completed unsuccessfully, then all other results written by this command are undefined unless otherwise specified for any of the results of the used query type.

Results for any available query written by this command are final and represent the final result of the query. If VK_QUERY_RESULT_PARTIAL_BIT is set, then for any query that is unavailable, an intermediate result between zero and the final result value is written for that query. Otherwise, any result written by this command is undefined.

If VK_QUERY_RESULT_64_BIT is set, results and availability or status values for all queries are written as an array of 64-bit values. If the `queryPool` was created with VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, results for each query are written as an array of the type indicated by VkPerformanceCounterKHR::`storage` for the counter being queried. Otherwise, results and availability or status values are written as an array of 32-bit values. If an unsigned integer query's value overflows the result type, the value **may** either wrap or saturate. If the `maintenance7` feature is enabled, for an unsigned integer query, the 32-bit result value **must** be equal to the 32 least significant bits of the equivalent 64-bit result value. If a signed integer query's value overflows the result type, the value is undefined. If a floating-point query's value is not representable as the result type, the value is undefined.

This command defines an execution dependency between other query commands that reference the same query.

The first synchronization scope includes all commands which reference the queries in `queryPool` indicated by `firstQuery` and `queryCount` that occur earlier in submission order. If `queryResultFlags` does not include VK_QUERY_RESULT_WAIT_BIT, vkCmdEndQueryIndexedEXT, vkCmdWriteTimestamp2, vkCmdEndQuery, and vkCmdWriteTimestamp are excluded from this scope.

The second synchronization scope includes all commands which reference the queries in `queryPool` indicated by `firstQuery` and `queryCount` that occur later in submission order.

The operation of this command happens after the first scope and happens before the second scope.

`vkCmdCopyQueryPoolResultsToMemoryKHR` is considered to be a transfer operation, and its writes to buffer memory **must** be synchronized using VK_PIPELINE_STAGE_TRANSFER_BIT and VK_ACCESS_TRANSFER_WRITE_BIT before using the results.

Valid Usage

- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-pDstRange-13097
  If the range specified by `pDstRange` is not bound completely to memory when accessed, `dstFlags` **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-pDstRange-13098
  If the buffer from which the range specified by `pDstRange` was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `dstFlags`

must include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR

- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-pDstRange-13099
  If the buffer from which the range specified by `pDstRange` was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `dstFlags` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-dstFlags-13100
  `dstFlags` **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-pDstRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `pDstRange`, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `dstFlags` **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-pDstRange-13123
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `pDstRange`, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `dstFlags` **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-dstFlags-13101
  `dstFlags` **must** not include both VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-pDstRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `pDstRange`, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `dstFlags` **must** include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-pDstRange-13125
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `pDstRange`, was created without VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `dstFlags` **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-firstQuery-09436
  `firstQuery` **must** be less than the number of queries in `queryPool`
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-firstQuery-09437
  The sum of `firstQuery` and `queryCount` **must** be less than or equal to the number of queries in `queryPool`
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-queryCount-09438
  If `queryCount` is greater than 1, `stride` **must** not be zero
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-queryType-09439

If the queryType used to create queryPool was VK_QUERY_TYPE_TIMESTAMP, queryResultFlags **must** not contain VK_QUERY_RESULT_PARTIAL_BIT

- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-queryType-09440
  If the queryType used to create queryPool was VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, queryResultFlags **must** not contain VK_QUERY_RESULT_WITH_AVAILABILITY_BIT, VK_QUERY_RESULT_WITH_STATUS_BIT_KHR, VK_QUERY_RESULT_PARTIAL_BIT, or VK_QUERY_RESULT_64_BIT
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-queryType-09441
  If the queryType used to create queryPool was VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, the queryPool **must** have been recorded once for each pass as retrieved via a call to vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-queryType-11874
  If the queryType used to create queryPool was not VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR or VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR, then queryResultFlags **must** not include VK_QUERY_RESULT_WITH_STATUS_BIT_KHR
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-queryType-09442
  If the queryType used to create queryPool was VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR, then queryResultFlags **must** include VK_QUERY_RESULT_WITH_STATUS_BIT_KHR
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-flags-09443
  If queryResultFlags includes VK_QUERY_RESULT_WITH_STATUS_BIT_KHR, then it **must** not include VK_QUERY_RESULT_WITH_AVAILABILITY_BIT
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-None-13076
  All queries used by the command **must** not be uninitialized when the command is executed
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-flags-13077
  If VK_QUERY_RESULT_64_BIT is not set in queryResultFlags then pDstRange->address and pDstRange->stride **must** be multiples of 4
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-flags-13078
  If VK_QUERY_RESULT_64_BIT is set in queryResultFlags then pDstRange->address and pDstRange->stride **must** be multiples of 8
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-pDstRange-13079
  pDstRange->size **must** be large enough to contain the result of each query, as described here
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-pDstRange-13080
  The buffer from which the range defined by pDstRange was queried **must** have been created with VK_BUFFER_USAGE_TRANSFER_DST_BIT usage flag
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-queryType-13081
  If the queryType used to create queryPool was VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR,

VkPhysicalDevicePerformanceQueryPropertiesKHR::allowCommandBufferQueryCopies **must** be VK_TRUE

- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-None-13083
  All queries used by the command **must** not be active
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-None-13084
  All queries used by the command **must** have been made *available* by prior executed commands
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-dstFlags-13085
  dstFlags **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR

Valid Usage (Implicit)

- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-queryPool-parameter
  queryPool **must** be a valid VkQueryPool handle
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-pDstRange-parameter
  pDstRange **must** be a valid pointer to a valid VkStridedDeviceAddressRangeKHR structure
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-dstFlags-parameter
  If dstFlags is not 0, dstFlags **must** be a valid combination of VkAddressCommandFlagBitsKHR values
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-queryResultFlags-parameter
  If queryResultFlags is not 0, queryResultFlags **must** be a valid combination of VkQueryResultFlagBits values
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_TRANSFER_BIT operations
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdCopyQueryPoolResultsToMemoryKHR-commonparent
  Both of commandBuffer, and queryPool **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdCopyQueryPoolResultsToMemoryKHR is not affected by conditional rendering

To copy query statuses and numerical results directly to buffer memory, call:

```
// Provided by VK_VERSION_1_0
void vkCmdCopyQueryPoolResults(
    VkCommandBuffer                             commandBuffer,
    VkQueryPool                                 queryPool,
    uint32_t                                    firstQuery,
    uint32_t                                    queryCount,
    VkBuffer                                    dstBuffer,
    VkDeviceSize                                dstOffset,
    VkDeviceSize                                stride,
    VkQueryResultFlags                          flags);
```

- `commandBuffer` is the command buffer into which this command will be recorded.
- `queryPool` is the query pool managing the queries containing the desired results.
- `firstQuery` is the initial query index.
- `queryCount` is the number of queries. `firstQuery` and `queryCount` together define a range of queries.
- `dstBuffer` is a VkBuffer object that will receive the results of the copy command.
- `dstOffset` is an offset into `dstBuffer`.
- `stride` is the stride in bytes between results for individual queries within `dstBuffer`. The required size of the backing memory for `dstBuffer` is determined as described above for vkGetQueryPoolResults.

- `flags` is a bitmask of VkQueryResultFlagBits specifying how and when results are returned.

Any results written for a query are written according to a layout dependent on the query type.

Results for any query in `queryPool` identified by `firstQuery` and `queryCount` that is available are copied to `dstBuffer`.

If VK_QUERY_RESULT_WITH_AVAILABILITY_BIT is set, results for all queries in `queryPool` identified by `firstQuery` and `queryCount` are copied to `dstBuffer`, along with an extra availability value written directly after the results of each query and interpreted as an unsigned integer. A value of zero indicates that the results are not yet available, otherwise the query is complete and results are available.

If VK_QUERY_RESULT_WITH_STATUS_BIT_KHR is set, results for all queries in `queryPool` identified by `firstQuery` and `queryCount` are copied to `dstBuffer`, along with an extra status value written directly after the results of each query and interpreted as a signed integer. A value of zero indicates that the results are not yet available. Positive values indicate that the operations within the query completed successfully, and the query results are valid. Negative values indicate that the operations within the query completed unsuccessfully.

VkQueryResultStatusKHR defines specific meaning for values returned here, though implementations are free to return other values.

If the status value written is negative, indicating that the operations within the query completed unsuccessfully, then all other results written by this command are undefined unless otherwise specified for any of the results of the used query type.

Results for any available query written by this command are final and represent the final result of the query. If VK_QUERY_RESULT_PARTIAL_BIT is set, then for any query that is unavailable, an intermediate result between zero and the final result value is written for that query. Otherwise, any result written by this command is undefined.

If VK_QUERY_RESULT_64_BIT is set, results and availability or status values for all queries are written as an array of 64-bit values. If the `queryPool` was created with VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, results for each query are written as an array of the type indicated by VkPerformanceCounterKHR::`storage` for the counter being queried. Otherwise, results and availability or status values are written as an array of 32-bit values. If an unsigned integer query's value overflows the result type, the value **may** either wrap or saturate. If the `maintenance7` feature is enabled, for an unsigned integer query, the 32-bit result value **must** be equal to the 32 least significant bits of the equivalent 64-bit result value. If a signed integer query's value overflows the result type, the value is undefined. If a floating-point query's value is not representable as the result type, the value is undefined.

This command defines an execution dependency between other query commands that reference the same query.

The first synchronization scope includes all commands which reference the queries in `queryPool` indicated by `firstQuery` and `queryCount` that occur earlier in submission order. If `flags` does not include VK_QUERY_RESULT_WAIT_BIT, vkCmdEndQuery, vkCmdEndQueryIndexedEXT, vkCmdWriteAccelerationStructuresPropertiesKHR, vkCmdWriteMicromapsPropertiesEXT,

vkCmdWriteTimestamp2, and vkCmdWriteTimestamp are excluded from this scope.

The second synchronization scope includes all commands which reference the queries in queryPool indicated by firstQuery and queryCount that occur later in submission order.

The operation of this command happens after the first scope and happens before the second scope.

vkCmdCopyQueryPoolResults is considered to be a transfer operation, and its writes to buffer memory **must** be synchronized using VK_PIPELINE_STAGE_TRANSFER_BIT and VK_ACCESS_TRANSFER_WRITE_BIT before using the results.

Valid Usage

- VUID-vkCmdCopyQueryPoolResults-firstQuery-09436
  firstQuery **must** be less than the number of queries in queryPool
- VUID-vkCmdCopyQueryPoolResults-firstQuery-09437
  The sum of firstQuery and queryCount **must** be less than or equal to the number of queries in queryPool
- VUID-vkCmdCopyQueryPoolResults-queryCount-09438
  If queryCount is greater than 1, stride **must** not be zero
- VUID-vkCmdCopyQueryPoolResults-queryType-09439
  If the queryType used to create queryPool was VK_QUERY_TYPE_TIMESTAMP, flags **must** not contain VK_QUERY_RESULT_PARTIAL_BIT
- VUID-vkCmdCopyQueryPoolResults-queryType-09440
  If the queryType used to create queryPool was VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, flags **must** not contain VK_QUERY_RESULT_WITH_AVAILABILITY_BIT, VK_QUERY_RESULT_WITH_STATUS_BIT_KHR, VK_QUERY_RESULT_PARTIAL_BIT, or VK_QUERY_RESULT_64_BIT
- VUID-vkCmdCopyQueryPoolResults-queryType-09441
  If the queryType used to create queryPool was VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, the queryPool **must** have been recorded once for each pass as retrieved via a call to vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR
- VUID-vkCmdCopyQueryPoolResults-queryType-11874
  If the queryType used to create queryPool was not VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR or VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR, then flags **must** not include VK_QUERY_RESULT_WITH_STATUS_BIT_KHR
- VUID-vkCmdCopyQueryPoolResults-queryType-09442
  If the queryType used to create queryPool was VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR, then flags **must** include VK_QUERY_RESULT_WITH_STATUS_BIT_KHR
- VUID-vkCmdCopyQueryPoolResults-flags-09443
  If flags includes VK_QUERY_RESULT_WITH_STATUS_BIT_KHR, then it **must** not include

VK_QUERY_RESULT_WITH_AVAILABILITY_BIT

- VUID-vkCmdCopyQueryPoolResults-None-09402
  All queries used by the command **must** not be uninitialized when the command is executed
- VUID-vkCmdCopyQueryPoolResults-dstOffset-00819
  `dstOffset` **must** be less than the size of `dstBuffer`
- VUID-vkCmdCopyQueryPoolResults-flags-00822
  If VK_QUERY_RESULT_64_BIT is not set in `flags` then `dstOffset` **must** be a multiple of `4`
- VUID-vkCmdCopyQueryPoolResults-queryCount-12254
  If `queryCount` is greater than 1 and VK_QUERY_RESULT_64_BIT is not set in `flags` then `stride` **must** be a multiple of `4`
- VUID-vkCmdCopyQueryPoolResults-flags-00823
  If VK_QUERY_RESULT_64_BIT is set in `flags` then `dstOffset` **must** be a multiple of `8`
- VUID-vkCmdCopyQueryPoolResults-queryCount-12255
  If `queryCount` is greater than 1 and VK_QUERY_RESULT_64_BIT is set in `flags` then `stride` **must** be a multiple of `8`
- VUID-vkCmdCopyQueryPoolResults-dstBuffer-00824
  `dstBuffer` **must** have enough storage, from `dstOffset`, to contain the result of each query, as described here
- VUID-vkCmdCopyQueryPoolResults-dstBuffer-00825
  `dstBuffer` **must** have been created with the VK_BUFFER_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-vkCmdCopyQueryPoolResults-dstBuffer-00826
  If `dstBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdCopyQueryPoolResults-queryType-03232
  If the `queryType` used to create `queryPool` was VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR, VkPhysicalDevicePerformanceQueryPropertiesKHR::`allowCommandBufferQueryCopies` **must** be VK_TRUE
- VUID-vkCmdCopyQueryPoolResults-None-07429
  All queries used by the command **must** not be active
- VUID-vkCmdCopyQueryPoolResults-None-08752
  All queries used by the command **must** have been made *available* by prior executed commands

Valid Usage (Implicit)

- VUID-vkCmdCopyQueryPoolResults-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyQueryPoolResults-queryPool-parameter

`queryPool` **must** be a valid VkQueryPool handle

- VUID-vkCmdCopyQueryPoolResults-dstBuffer-parameter
  `dstBuffer` **must** be a valid VkBuffer handle
- VUID-vkCmdCopyQueryPoolResults-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkQueryResultFlagBits values
- VUID-vkCmdCopyQueryPoolResults-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdCopyQueryPoolResults-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdCopyQueryPoolResults-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyQueryPoolResults-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyQueryPoolResults-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdCopyQueryPoolResults-commonparent
  Each of `commandBuffer`, `dstBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdCopyQueryPoolResults is not affected by conditional rendering

Rendering operations such as clears, MSAA resolves, attachment load/store operations, and blits

may count towards the results of queries. This behavior is implementation-dependent and **may** vary depending on the path used within an implementation. For example, some implementations have several types of clears, some of which **may** include vertices and some not.

22.3. Occlusion Queries

Occlusion queries track the number of samples that pass the per-fragment tests for a set of drawing commands. As such, occlusion queries are only available on queue families supporting graphics operations. The application **can** then use these results to inform future rendering decisions. An occlusion query is begun and ended by calling `vkCmdBeginQuery` and `vkCmdEndQuery`, respectively. When an occlusion query begins, the count of passing samples always starts at zero. For each drawing command, the count is incremented as described in Sample Counting. If `flags` does not contain VK_QUERY_CONTROL_PRECISE_BIT an implementation **may** generate any non-zero result value for the query if the count of passing samples is non-zero.

NOTE Not setting VK_QUERY_CONTROL_PRECISE_BIT mode **may** be more efficient on some implementations, and **should** be used where it is sufficient to know a boolean result on whether any samples passed the per-fragment tests. In this case, some implementations **may** only return zero or one, regardless of the actual number of samples passing the per-fragment tests.

Setting VK_QUERY_CONTROL_PRECISE_BIT does not guarantee that different implementations return the same number of samples in an occlusion query. Some implementations **may** kill fragments in the pre-rasterization shader stage, and these killed fragments do not contribute to the final result of the query. It is possible that some implementations generate a zero result value for the query, while others generate a non-zero value.

Some implementations **may** track the number of samples that pass the per-fragment tests multiplied for each color output. Such implementations **may** produce inconsistent values if more than one color output is written in the fragment shader.

When an occlusion query finishes, the result for that query is marked as available. The application **can** then either copy the result to a buffer (via `vkCmdCopyQueryPoolResults`) or request it be put into host memory (via `vkGetQueryPoolResults`).

NOTE If occluding geometry is not drawn first, samples **can** pass the depth test, but still not be visible in a final image.

22.4. Pipeline Statistics Queries

Pipeline statistics queries allow the application to sample a specified set of `VkPipeline` counters. These counters are accumulated by Vulkan for a set of either drawing or dispatching commands while a pipeline statistics query is active. As such, pipeline statistics queries are available on queue families supporting either graphics or compute operations. The availability of pipeline statistics queries is indicated by the `pipelineStatisticsQuery` member of the `VkPhysicalDeviceFeatures` object (see `vkGetPhysicalDeviceFeatures` and `vkCreateDevice` for detecting and requesting this query type on a `VkDevice`).

A pipeline statistics query is begun and ended by calling vkCmdBeginQuery and vkCmdEndQuery, respectively. When a pipeline statistics query begins, all statistics counters are set to zero. While the query is active, the pipeline type determines which set of statistics are available, but these **must** be configured on the query pool when it is created. If a statistic counter is issued on a command buffer that does not support the corresponding operation, or the counter corresponds to a shading stage which is missing from any of the pipelines used while the query is active, the value of that counter is undefined after the query has been made available. At least one statistic counter relevant to the operations supported on the recording command buffer **must** be enabled.

Bits which **can** be set in VkQueryPoolCreateInfo::pipelineStatistics for query pools and in VkCommandBufferInheritanceInfo::pipelineStatistics for secondary command buffers, individually enabling pipeline statistics counters, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueryPipelineStatisticFlagBits {
    VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_VERTICES_BIT = 0x00000001,
    VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_PRIMITIVES_BIT = 0x00000002,
    VK_QUERY_PIPELINE_STATISTIC_VERTEX_SHADER_INVOCATIONS_BIT = 0x00000004,
    VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_INVOCATIONS_BIT = 0x00000008,
    VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_PRIMITIVES_BIT = 0x00000010,
    VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT = 0x00000020,
    VK_QUERY_PIPELINE_STATISTIC_CLIPPING_PRIMITIVES_BIT = 0x00000040,
    VK_QUERY_PIPELINE_STATISTIC_FRAGMENT_SHADER_INVOCATIONS_BIT = 0x00000080,
    VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_CONTROL_SHADER_PATCHES_BIT = 0x00000100,
    VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_EVALUATION_SHADER_INVOCATIONS_BIT =
0x00000200,
    VK_QUERY_PIPELINE_STATISTIC_COMPUTE_SHADER_INVOCATIONS_BIT = 0x00000400,
  // Provided by VK_EXT_mesh_shader
    VK_QUERY_PIPELINE_STATISTIC_TASK_SHADER_INVOCATIONS_BIT_EXT = 0x00000800,
  // Provided by VK_EXT_mesh_shader
    VK_QUERY_PIPELINE_STATISTIC_MESH_SHADER_INVOCATIONS_BIT_EXT = 0x00001000,
} VkQueryPipelineStatisticFlagBits;
```

- VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_VERTICES_BIT specifies that queries managed by the pool will count the number of vertices processed by the input assembly stage. Vertices corresponding to incomplete primitives **may** contribute to the count.
- VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_PRIMITIVES_BIT specifies that queries managed by the pool will count the number of primitives processed by the input assembly stage. If primitive restart is enabled, restarting the primitive topology has no effect on the count. Incomplete primitives **may** be counted.
- VK_QUERY_PIPELINE_STATISTIC_VERTEX_SHADER_INVOCATIONS_BIT specifies that queries managed by the pool will count the number of vertex shader invocations. This counter's value is incremented each time a vertex shader is invoked.
- VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_INVOCATIONS_BIT specifies that queries managed by the pool will count the number of geometry shader invocations. This counter's value is incremented each time a geometry shader is invoked. In the case of instanced geometry shaders, the geometry shader invocations count is incremented for each separate instanced

invocation.

- VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_PRIMITIVES_BIT specifies that queries managed by the pool will count the number of primitives generated by geometry shader invocations. The counter's value is incremented each time the geometry shader emits a primitive. Restarting primitive topology using the SPIR-V instructions `OpEndPrimitive` or `OpEndStreamPrimitive` has no effect on the geometry shader output primitives count.
- VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT specifies that queries managed by the pool will count the number of primitives processed by the Primitive Clipping stage of the pipeline. The counter's value is incremented each time a primitive reaches the primitive clipping stage.
- VK_QUERY_PIPELINE_STATISTIC_CLIPPING_PRIMITIVES_BIT specifies that queries managed by the pool will count the number of primitives output by the Primitive Clipping stage of the pipeline. The counter's value is incremented each time a primitive passes the primitive clipping stage. The actual number of primitives output by the primitive clipping stage for a particular input primitive is implementation-dependent but **must** satisfy the following conditions:
  - If at least one vertex of the input primitive lies inside the clipping volume, and either the primitive is not rasterized according to the Basic Polygon Rasterization rules, or at least one sample point for a fragment lies within the rasterized triangle according to those rules, the counter is incremented by one or more.
  - Otherwise, the counter is incremented by zero or more.
- VK_QUERY_PIPELINE_STATISTIC_FRAGMENT_SHADER_INVOCATIONS_BIT specifies that queries managed by the pool will count the number of fragment shader invocations. The counter's value is incremented each time the fragment shader is invoked.
- VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_CONTROL_SHADER_PATCHES_BIT specifies that queries managed by the pool will count the number of patches processed by the tessellation control shader. The counter's value is incremented once for each patch for which a tessellation control shader is invoked.
- VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_EVALUATION_SHADER_INVOCATIONS_BIT specifies that queries managed by the pool will count the number of invocations of the tessellation evaluation shader. The counter's value is incremented each time the tessellation evaluation shader is invoked.
- VK_QUERY_PIPELINE_STATISTIC_COMPUTE_SHADER_INVOCATIONS_BIT specifies that queries managed by the pool will count the number of compute shader invocations. The counter's value is incremented every time the compute shader is invoked. Implementations **may** skip the execution of certain compute shader invocations or execute additional compute shader invocations for implementation-dependent reasons as long as the results of rendering otherwise remain unchanged.
- VK_QUERY_PIPELINE_STATISTIC_TASK_SHADER_INVOCATIONS_BIT_EXT specifies that queries managed by the pool will count the number of task shader invocations. The counter's value is incremented every time the task shader is invoked.
- VK_QUERY_PIPELINE_STATISTIC_MESH_SHADER_INVOCATIONS_BIT_EXT specifies that queries managed by the pool will count the number of mesh shader invocations. The counter's value is incremented every time the mesh shader is invoked.

These values are intended to measure relative statistics on one implementation. Various device architectures will count these values differently. Any or all counters **may** be affected by the issues described in Query Operation.

This counting difference is especially true if the pipeline contains mesh or task shaders, which may affect several of the counters in unexpected ways.

NOTE	For example, tile-based rendering devices **may** need to replay the scene multiple times, affecting some of the counts.

If a pipeline has `rasterizerDiscardEnable` enabled, implementations **may** discard primitives after the final pre-rasterization shader stage. As a result, if `rasterizerDiscardEnable` is enabled, the clipping input and output primitives counters **may** not be incremented.

When a pipeline statistics query finishes, the result for that query is marked as available. The application **can** copy the result to a buffer (via `vkCmdCopyQueryPoolResults`), or request it be put into host memory (via `vkGetQueryPoolResults`).

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkQueryPipelineStatisticFlags;
```

`VkQueryPipelineStatisticFlags` is a bitmask type for setting a mask of zero or more VkQueryPipelineStatisticFlagBits.

22.5. Timestamp Queries

Timestamps provide applications with a mechanism for monotonically tracking the execution of commands. A timestamp is an integer value generated by the `VkPhysicalDevice`. Unlike other queries, timestamps do not operate over a range, and so do not use vkCmdBeginQuery or vkCmdEndQuery. The mechanism is built around a set of commands that allow the application to tell the `VkPhysicalDevice` to write timestamp values to a query pool and then either read timestamp values on the host (using vkGetQueryPoolResults) or copy timestamp values to a `VkBuffer` (using vkCmdCopyQueryPoolResults).

The number of valid bits in a timestamp value is determined by the `VkQueueFamilyProperties` ::`timestampValidBits` property of the queue on which the timestamp is written. Timestamps are supported on any queue which reports a non-zero value for `timestampValidBits` via vkGetPhysicalDeviceQueueFamilyProperties. If the `timestampComputeAndGraphics` limit is VK_TRUE, timestamps are supported by every queue family that supports either graphics or compute operations (see VkQueueFamilyProperties).

The number of nanoseconds it takes for a timestamp value to be incremented by 1 **can** be obtained from `VkPhysicalDeviceLimits`::`timestampPeriod` after a call to `vkGetPhysicalDeviceProperties`.

To request a timestamp and write the value to memory, call:

```
// Provided by VK_VERSION_1_3
void vkCmdWriteTimestamp2(
    VkCommandBuffer                             commandBuffer,
    VkPipelineStageFlags2                       stage,
    VkQueryPool                                 queryPool,
    uint32_t                                    query);
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to vkCmdWriteTimestamp2
void vkCmdWriteTimestamp2KHR(
    VkCommandBuffer                             commandBuffer,
    VkPipelineStageFlags2                       stage,
    VkQueryPool                                 queryPool,
    uint32_t                                    query);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `stage` specifies a stage of the pipeline.
- `queryPool` is the query pool that will manage the timestamp.
- `query` is the query within the query pool that will contain the timestamp.

When `vkCmdWriteTimestamp2` is submitted to a queue, it defines an execution dependency on commands that were submitted before it, and writes a timestamp to a query pool.

The first synchronization scope includes all commands that occur earlier in submission order. The synchronization scope is limited to operations on the pipeline stage specified by `stage`.

The second synchronization scope includes only the timestamp write operation.

> **NOTE** Implementations may write the timestamp at any stage that is logically later than `stage`.

Any timestamp write that happens-after another timestamp write in the same submission **must** not have a lower value unless its value overflows the maximum supported integer bit width of the query. If `VK_KHR_calibrated_timestamps` is enabled, this extends to timestamp writes across all submissions on the same logical device: any timestamp write that happens-after another **must** not have a lower value unless its value overflows the maximum supported integer bit width of the query. Timestamps written by this command **must** be in the VK_TIME_DOMAIN_DEVICE_KHR time domain. If an overflow occurs, the timestamp value **must** wrap back to zero.

If `vkCmdWriteTimestamp2` is called while executing a render pass instance that has multiview enabled, the timestamp uses N consecutive query indices in the query pool (starting at `query`) where N is the number of bits set in the view mask of the subpass or dynamic render pass the command is executed in. The resulting query values are determined by an implementation-dependent choice of one of the following behaviors:

- The first query is a timestamp value and (if more than one bit is set in the view mask) zero is

written to the remaining queries.

- All N queries are timestamp values.

Either way, if two timestamps are written in the same subpass or dynamic render pass with multiview enabled, each of the N consecutive queries written for a timestamp **must** not have a lower value than the queries with corresponding indices written by the timestamp that happens-before unless the value overflows the maximum supported integer bit width of the query.

Valid Usage

- VUID-vkCmdWriteTimestamp2-stage-03929
  If the `geometryShader` feature is not enabled, `stage` **must** not contain VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT
- VUID-vkCmdWriteTimestamp2-stage-03930
  If the `tessellationShader` feature is not enabled, `stage` **must** not contain VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-vkCmdWriteTimestamp2-stage-03931
  If the `conditionalRendering` feature is not enabled, `stage` **must** not contain VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT
- VUID-vkCmdWriteTimestamp2-stage-03932
  If the `fragmentDensityMap` feature is not enabled, `stage` **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-vkCmdWriteTimestamp2-stage-03933
  If the `transformFeedback` feature is not enabled, `stage` **must** not contain VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-vkCmdWriteTimestamp2-stage-03934
  If the `meshShader` feature is not enabled, `stage` **must** not contain VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT
- VUID-vkCmdWriteTimestamp2-stage-03935
  If the `taskShader` feature is not enabled, `stage` **must** not contain VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT
- VUID-vkCmdWriteTimestamp2-stage-07317
  If the `attachmentFragmentShadingRate` feature is not enabled, `stage` **must** not contain VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdWriteTimestamp2-stage-07947
  If the `rayTracingPipeline` feature is not enabled, `stage` **must** not contain VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR
- VUID-vkCmdWriteTimestamp2-stage-10751
  If the `accelerationStructure` feature is not enabled, `stage` **must** not contain VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- VUID-vkCmdWriteTimestamp2-stage-10752
  If the `rayTracingMaintenance1` feature is not enabled, `stage` **must** not contain

VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR

- VUID-vkCmdWriteTimestamp2-stage-11541
  If the VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap feature is not enabled, stage **must** not contain VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- VUID-vkCmdWriteTimestamp2-synchronization2-03858
  The synchronization2 feature **must** be enabled
- VUID-vkCmdWriteTimestamp2-stage-03859
  stage **must** only include a single pipeline stage
- VUID-vkCmdWriteTimestamp2-stage-03860
  stage **must** only include stages valid for the queue family that was used to create the command pool that commandBuffer was allocated from
- VUID-vkCmdWriteTimestamp2-queryPool-03861
  queryPool **must** have been created with a queryType of VK_QUERY_TYPE_TIMESTAMP
- VUID-vkCmdWriteTimestamp2-timestampValidBits-03863
  The command pool's queue family **must** support a non-zero timestampValidBits
- VUID-vkCmdWriteTimestamp2-query-04903
  query **must** be less than the number of queries in queryPool
- VUID-vkCmdWriteTimestamp2-None-03864
  All queries used by the command **must** be *unavailable*
- VUID-vkCmdWriteTimestamp2-query-03865
  If vkCmdWriteTimestamp2 is called within a render pass instance, the sum of query and the number of bits set in the current subpass's view mask **must** be less than or equal to the number of queries in queryPool

Valid Usage (Implicit)

- VUID-vkCmdWriteTimestamp2-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdWriteTimestamp2-stage-parameter
  If stage is not 0, stage **must** be a valid combination of VkPipelineStageFlagBits2 values
- VUID-vkCmdWriteTimestamp2-queryPool-parameter
  queryPool **must** be a valid VkQueryPool handle
- VUID-vkCmdWriteTimestamp2-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdWriteTimestamp2-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_TRANSFER_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdWriteTimestamp2-suspended
  This command **must** not be called between suspended render pass instances

- VUID-vkCmdWriteTimestamp2-commonparent
  Both of `commandBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Action

Conditional Rendering

vkCmdWriteTimestamp2 is not affected by conditional rendering

To request a timestamp and write the value to memory, call:

WARNING This functionality is superseded by vkCmdWriteTimestamp2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkCmdWriteTimestamp(
    VkCommandBuffer                             commandBuffer,
    VkPipelineStageFlagBits                     pipelineStage,
    VkQueryPool                                 queryPool,
    uint32_t                                    query);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pipelineStage` is a VkPipelineStageFlagBits value, specifying a stage of the pipeline.
- `queryPool` is the query pool that will manage the timestamp.
- `query` is the query within the query pool that will contain the timestamp.

When `vkCmdWriteTimestamp` is submitted to a queue, it defines an execution dependency on commands that were submitted before it, and writes a timestamp to a query pool.

The first synchronization scope includes all commands that occur earlier in submission order. The synchronization scope is limited to operations on the pipeline stage specified by `pipelineStage`.

The second synchronization scope includes only the timestamp write operation.

> **NOTE** Implementations may write the timestamp at any stage that is logically later than `pipelineStage`.

Any timestamp write that happens-after another timestamp write in the same submission **must** not have a lower value unless its value overflows the maximum supported integer bit width of the query. If `VK_KHR_calibrated_timestamps` is enabled, this extends to timestamp writes across all submissions on the same logical device: any timestamp write that happens-after another **must** not have a lower value unless its value overflows the maximum supported integer bit width of the query. Timestamps written by this command **must** be in the VK_TIME_DOMAIN_DEVICE_KHR time domain. If an overflow occurs, the timestamp value **must** wrap back to zero.

If `vkCmdWriteTimestamp` is called while executing a render pass instance that has multiview enabled, the timestamp uses N consecutive query indices in the query pool (starting at `query`) where N is the number of bits set in the view mask of the subpass or dynamic render pass the command is executed in. The resulting query values are determined by an implementation-dependent choice of one of the following behaviors:

- The first query is a timestamp value and (if more than one bit is set in the view mask) zero is written to the remaining queries.
- All N queries are timestamp values.

Either way, if two timestamps are written in the same subpass or dynamic render pass with multiview enabled, each of the N consecutive queries written for a timestamp **must** not have a lower value than the queries with corresponding indices written by the timestamp that happens-before unless the value overflows the maximum supported integer bit width of the query.

Valid Usage

- VUID-vkCmdWriteTimestamp-pipelineStage-04074
  `pipelineStage` **must** be a valid stage for the queue family that was used to create the command pool that `commandBuffer` was allocated from
- VUID-vkCmdWriteTimestamp-pipelineStage-04075
  If the `geometryShader` feature is not enabled, `pipelineStage` **must** not be VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT

- VUID-vkCmdWriteTimestamp-pipelineStage-04076
  If the `tessellationShader` feature is not enabled, `pipelineStage` **must** not be VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-vkCmdWriteTimestamp-pipelineStage-04077
  If the `conditionalRendering` feature is not enabled, `pipelineStage` **must** not be VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT
- VUID-vkCmdWriteTimestamp-pipelineStage-04078
  If the `fragmentDensityMap` feature is not enabled, `pipelineStage` **must** not be VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- VUID-vkCmdWriteTimestamp-pipelineStage-04079
  If the `transformFeedback` feature is not enabled, `pipelineStage` **must** not be VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-vkCmdWriteTimestamp-pipelineStage-04080
  If the `meshShader` feature is not enabled, `pipelineStage` **must** not be VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT
- VUID-vkCmdWriteTimestamp-pipelineStage-07077
  If the `taskShader` feature is not enabled, `pipelineStage` **must** not be VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT
- VUID-vkCmdWriteTimestamp-fragmentShadingRate-07315
  If the `attachmentFragmentShadingRate` feature is not enabled, `pipelineStage` **must** not be VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdWriteTimestamp-synchronization2-06489
  If the `synchronization2` feature is not enabled, `pipelineStage` **must** not be VK_PIPELINE_STAGE_NONE
- VUID-vkCmdWriteTimestamp-rayTracingPipeline-07944
  If the `rayTracingPipeline` feature is not enabled, `pipelineStage` **must** not be VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR
- VUID-vkCmdWriteTimestamp-queryPool-01416
  `queryPool` **must** have been created with a `queryType` of VK_QUERY_TYPE_TIMESTAMP
- VUID-vkCmdWriteTimestamp-timestampValidBits-00829
  The command pool's queue family **must** support a non-zero `timestampValidBits`
- VUID-vkCmdWriteTimestamp-query-04904
  `query` **must** be less than the number of queries in `queryPool`
- VUID-vkCmdWriteTimestamp-None-00830
  All queries used by the command **must** be *unavailable*
- VUID-vkCmdWriteTimestamp-query-00831
  If `vkCmdWriteTimestamp` is called within a render pass instance, the sum of `query` and the number of bits set in the current subpass's view mask **must** be less than or equal to the number of queries in `queryPool`

Valid Usage (Implicit)

- VUID-vkCmdWriteTimestamp-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdWriteTimestamp-pipelineStage-parameter
  `pipelineStage` **must** be a valid VkPipelineStageFlagBits value
- VUID-vkCmdWriteTimestamp-queryPool-parameter
  `queryPool` **must** be a valid VkQueryPool handle
- VUID-vkCmdWriteTimestamp-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdWriteTimestamp-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_TRANSFER_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdWriteTimestamp-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdWriteTimestamp-commonparent
  Both of `commandBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Action

Conditional Rendering

vkCmdWriteTimestamp is not affected by conditional rendering

22.6. Performance Queries

Performance queries provide applications with a mechanism for getting performance counter information about the execution of command buffers, render passes, and commands.

Each queue family advertises the performance counters that **can** be queried on a queue of that family via a call to vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR. Implementations **may** limit access to performance counters based on platform requirements or only to specialized drivers for development purposes.

NOTE This may include no performance counters being enumerated, or a reduced set. Please refer to platform-specific documentation for guidance on any such restrictions.

Performance queries use the existing vkCmdBeginQuery and vkCmdEndQuery to control what command buffers, render passes, or commands to get performance information for.

Implementations **may** require multiple passes where the command buffer, render passes, or commands being recorded are the same and are executed on the same queue to record performance counter data. This is achieved by submitting the same batch and providing a VkPerformanceQuerySubmitInfoKHR structure containing a counter pass index. The number of passes required for a given performance query pool **can** be queried via a call to vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR.

NOTE	Command buffers recorded with VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT **must** not be re-submitted. Changing command buffer usage bits **may** affect performance. To avoid this, the application **should** re-record any command buffers with the VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT when multiple counter passes are required.

Performance counter results from a performance query pool **can** be obtained with the command vkGetQueryPoolResults.

The VkPerformanceCounterResultKHR union is defined as:

```
// Provided by VK_KHR_performance_query
typedef union VkPerformanceCounterResultKHR {
    int32_t     int32;
    int64_t     int64;
    uint32_t    uint32;
    uint64_t    uint64;
    float       float32;
    double      float64;
} VkPerformanceCounterResultKHR;
```

- int32 is a 32-bit signed integer value.
- int64 is a 64-bit signed integer value.
- uint32 is a 32-bit unsigned integer value.
- uint64 is a 64-bit unsigned integer value.
- float32 is a 32-bit floating-point value.
- float64 is a 64-bit floating-point value.

Performance query results are returned in an array of VkPerformanceCounterResultKHR unions containing the data associated with each counter in the query, stored in the same order as the counters supplied in pCounterIndices when creating the performance query. VkPerformanceCounterKHR::storage specifies how to parse the counter data.

22.6.1. Profiling Lock

To record and submit a command buffer containing a performance query pool the profiling lock **must** be held. The profiling lock **must** be acquired prior to any call to vkBeginCommandBuffer that will be using a performance query pool. The profiling lock **must** be held while any command buffer containing a performance query pool is in the *recording*, *executable*, or *pending state*. To acquire the profiling lock, call:

```
// Provided by VK_KHR_performance_query
VkResult vkAcquireProfilingLockKHR(
    VkDevice                                    device,
    const VkAcquireProfilingLockInfoKHR*        pInfo);
```

- `device` is the logical device to profile.
- `pInfo` is a pointer to a VkAcquireProfilingLockInfoKHR structure containing information about how the profiling is to be acquired.

Implementations **may** allow multiple actors to hold the profiling lock concurrently.

Valid Usage (Implicit)

- VUID-vkAcquireProfilingLockKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkAcquireProfilingLockKHR-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkAcquireProfilingLockInfoKHR structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED
- VK_TIMEOUT

The `VkAcquireProfilingLockInfoKHR` structure is defined as:

```
// Provided by VK_KHR_performance_query
typedef struct VkAcquireProfilingLockInfoKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    VkAcquireProfilingLockFlagsKHR  flags;
    uint64_t                        timeout;
} VkAcquireProfilingLockInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.

- `timeout` indicates how long the function waits, in nanoseconds, if the profiling lock is not available.

Valid Usage (Implicit)

- VUID-VkAcquireProfilingLockInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_ACQUIRE_PROFILING_LOCK_INFO_KHR
- VUID-VkAcquireProfilingLockInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkAcquireProfilingLockInfoKHR-flags-zerobitmask
  `flags` **must** be `0`

If `timeout` is 0, `vkAcquireProfilingLockKHR` will not block while attempting to acquire the profiling lock. If `timeout` is `UINT64_MAX`, the function will not return until the profiling lock was acquired.

```
// Provided by VK_KHR_performance_query
typedef enum VkAcquireProfilingLockFlagBitsKHR {
} VkAcquireProfilingLockFlagBitsKHR;
```

```
// Provided by VK_KHR_performance_query
typedef VkFlags VkAcquireProfilingLockFlagsKHR;
```

VkAcquireProfilingLockFlagsKHR is a bitmask type for setting a mask, but is currently reserved for future use.

To release the profiling lock, call:

```
// Provided by VK_KHR_performance_query
void vkReleaseProfilingLockKHR(
    VkDevice                                    device);
```

- `device` is the logical device to cease profiling on.

Valid Usage

- VUID-vkReleaseProfilingLockKHR-device-03235
  The profiling lock of `device` **must** have been held via a previous successful call to vkAcquireProfilingLockKHR

Valid Usage (Implicit)

- VUID-vkReleaseProfilingLockKHR-device-parameter

`device` **must** be a valid VkDevice handle

22.7. Transform Feedback Queries

Transform feedback queries track the number of primitives attempted to be written and actually written, by the vertex stream being captured, to a transform feedback buffer. This query is updated during drawing commands while transform feedback is active. The number of primitives actually written will be less than the number attempted to be written if the bound transform feedback buffer size was too small for the number of primitives actually drawn. Primitives are not written beyond the bound range of the transform feedback buffer. A transform feedback query is begun and ended by calling `vkCmdBeginQuery` and `vkCmdEndQuery`, respectively to query for vertex stream zero. `vkCmdBeginQueryIndexedEXT` and `vkCmdEndQueryIndexedEXT` **can** be used to begin and end transform feedback queries for any supported vertex stream. When a transform feedback query begins, the count of primitives written and primitives needed starts from zero. For each drawing command, the count is incremented as vertex attribute outputs are captured to the transform feedback buffers while transform feedback is active.

When a transform feedback query finishes, the result for that query is marked as available. The application **can** then either copy the result to a buffer (via `vkCmdCopyQueryPoolResults`) or request it be put into host memory (via `vkGetQueryPoolResults`).

22.8. Primitives Generated Queries

When a generated primitive query for a vertex stream is active, the primitives-generated count is incremented every time a primitive emitted to that stream reaches the transform feedback stage, whether or not transform feedback is active. A primitives generated query is begun and ended by calling `vkCmdBeginQuery` and `vkCmdEndQuery`, respectively to query for vertex stream zero. `vkCmdBeginQueryIndexedEXT` and `vkCmdEndQueryIndexedEXT` **can** be used to begin and end primitives generated queries for any supported vertex stream. When a primitives generated query begins, the count of primitives generated starts from zero.

When a primitives generated query finishes, the result for that query is marked as available. The application **can** then either copy the result to a buffer (via `vkCmdCopyQueryPoolResults`) or request it be put into host memory (via `vkGetQueryPoolResults`).

NOTE The result of this query is typically identical to VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT, but the primitives generated query is deterministic, i.e. it must be identical to the number of primitives processed. VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT may vary for implementation-dependent reasons, e.g. the same primitive may be processed multiple times for purposes of clipping.

22.9. Mesh Shader Queries

When a generated mesh primitives query is active, the mesh-primitives-generated count is

incremented every time a primitive emitted from the mesh shader stage reaches the fragment shader stage. When a generated mesh primitives query begins, the mesh-primitives-generated count starts from zero.

Mesh and task shader pipeline statistics queries function the same way that invocation queries work for other shader stages, counting the number of times the respective shader stage has been run. When the statistics query begins, the invocation counters start from zero.

22.10. Result Status Queries

Result status queries serve a single purpose: allowing the application to determine whether a set of operations have completed successfully or not, as indicated by the VkQueryResultStatusKHR value written when retrieving the result of a query using the VK_QUERY_RESULT_WITH_STATUS_BIT_KHR flag.

Unlike other query types, result status queries do not track or maintain any other data beyond the completion status, thus no other data is written when retrieving their results.

Support for result status queries is indicated by VkQueueFamilyQueryResultStatusPropertiesKHR ::`queryResultStatusSupport` , as returned by vkGetPhysicalDeviceQueueFamilyProperties2 for the queue family in question.

22.11. Video Encode Feedback Queries

Video encode feedback queries allow the application to capture feedback values generated by video encode operations. As such, video encode feedback queries are available on queue families supporting video encode operations. The availability of individual video encode feedback values is indicated by the bits of VkVideoEncodeCapabilitiesKHR::`supportedEncodeFeedbackFlags`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the queries are intended to be used with.

The set of enabled video encode feedback values **must** be configured on the query pool when it is created using the `encodeFeedbackFlags` member of the VkQueryPoolVideoEncodeFeedbackCreateInfoKHR structure included in the `pNext` chain of VkQueryPoolCreateInfo.

The `VkQueryPoolVideoEncodeFeedbackCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkQueryPoolVideoEncodeFeedbackCreateInfoKHR {
    VkStructureType                  sType;
    const void*                      pNext;
    VkVideoEncodeFeedbackFlagsKHR    encodeFeedbackFlags;
} VkQueryPoolVideoEncodeFeedbackCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `encodeFeedbackFlags` is a bitmask of VkVideoEncodeFeedbackFlagBitsKHR values specifying the set of enabled video encode feedback values captured by queries of the new pool.

Valid Usage (Implicit)

- VUID-VkQueryPoolVideoEncodeFeedbackCreateInfoKHR-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_QUERY_POOL_VIDEO_ENCODE_FEEDBACK_CREATE_INFO_KHR
- VUID-VkQueryPoolVideoEncodeFeedbackCreateInfoKHR-encodeFeedbackFlags-parameter `encodeFeedbackFlags` **must** be a valid combination of VkVideoEncodeFeedbackFlagBitsKHR values

Structure Chaining

Extends the structure

- VkQueryPoolCreateInfo

Bits which **can** be set in VkQueryPoolVideoEncodeFeedbackCreateInfoKHR::`encodeFeedbackFlags` for video encode feedback query pools are:

```
// Provided by VK_KHR_video_encode_queue
typedef enum VkVideoEncodeFeedbackFlagBitsKHR {
    VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR = 0x00000004,
  // Provided by VK_KHR_video_encode_feedback2
    VK_VIDEO_ENCODE_FEEDBACK_AVERAGE_QUANTIZATION_BIT_KHR = 0x00000008,
  // Provided by VK_KHR_video_encode_feedback2
    VK_VIDEO_ENCODE_FEEDBACK_MIN_QUANTIZATION_BIT_KHR = 0x00000010,
  // Provided by VK_KHR_video_encode_feedback2
    VK_VIDEO_ENCODE_FEEDBACK_MAX_QUANTIZATION_BIT_KHR = 0x00000020,
  // Provided by VK_KHR_video_encode_feedback2
    VK_VIDEO_ENCODE_FEEDBACK_INTRA_PIXELS_BIT_KHR = 0x00000040,
  // Provided by VK_KHR_video_encode_feedback2
    VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR = 0x00000080,
  // Provided by VK_KHR_video_encode_feedback2
    VK_VIDEO_ENCODE_FEEDBACK_SKIPPED_PIXELS_BIT_KHR = 0x00000100,
  // Provided by VK_KHR_video_encode_feedback2
    VK_VIDEO_ENCODE_FEEDBACK_PICTURE_PARTITION_COUNT_BIT_KHR = 0x00000200,
} VkVideoEncodeFeedbackFlagBitsKHR;
```

- VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR specifies that queries managed by the pool will capture the byte offset of the bitstream data written by the video encode operation to the bitstream buffer specified in VkVideoEncodeInfoKHR::`dstBuffer` relative to the offset specified in VkVideoEncodeInfoKHR::`dstBufferOffset`. For the first video

encode operation issued by any video encode command, this value will always be zero, meaning that bitstream data is always written to the buffer specified in VkVideoEncodeInfoKHR::`dstBuffer` starting from the offset specified in VkVideoEncodeInfoKHR::`dstBufferOffset`.

- VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR specifies that queries managed by the pool will capture the number of bytes written by the video encode operation to the bitstream buffer specified in VkVideoEncodeInfoKHR::`dstBuffer`.
- VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR specifies that queries managed by the pool will capture a boolean value indicating that the data written to the bitstream buffer specified in VkVideoEncodeInfoKHR::`dstBuffer` contains overridden parameters.
- VK_VIDEO_ENCODE_FEEDBACK_AVERAGE_QUANTIZATION_BIT_KHR specifies that queries managed by the pool will capture the average quantization used across the encoded coding blocks.
- VK_VIDEO_ENCODE_FEEDBACK_MIN_QUANTIZATION_BIT_KHR specifies that queries managed by the pool will capture the minimum quantization used across the encoded coding blocks.
- VK_VIDEO_ENCODE_FEEDBACK_MAX_QUANTIZATION_BIT_KHR specifies that queries managed by the pool will capture the maximum quantization used across the encoded coding blocks.
- VK_VIDEO_ENCODE_FEEDBACK_INTRA_PIXELS_BIT_KHR specifies that queries managed by the pool will capture the number of pixels encoded as intra blocks, including or excluding any skipped blocks, as described below.
- VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR specifies that queries managed by the pool will capture the number of pixels encoded as inter blocks, including or excluding any skipped blocks, as described below.
- VK_VIDEO_ENCODE_FEEDBACK_SKIPPED_PIXELS_BIT_KHR specifies that queries managed by the pool will capture the number of pixels encoded as skipped blocks.
- VK_VIDEO_ENCODE_FEEDBACK_PICTURE_PARTITION_COUNT_BIT_KHR specifies that queries managed by the pool will capture the number of encoded picture partitions.

The average, minimum, and maximum quantization values captured when VK_VIDEO_ENCODE_FEEDBACK_AVERAGE_QUANTIZATION_BIT_KHR, VK_VIDEO_ENCODE_FEEDBACK_MIN_QUANTIZATION_BIT_KHR, and/or VK_VIDEO_ENCODE_FEEDBACK_MAX_QUANTIZATION_BIT_KHR are specified, respectively, depend on the used video profile:

- In case of H.264 encode profiles, this value is a signed integer indicating the average, minimum, or maximum QP used across the encoded coding blocks, respectively.
- In case of H.265 encode profiles this value is a signed integer indicating the average, minimum, or maximum QP used across the encoded coding blocks, respectively.
- In case of AV1 encode profiles this value is an unsigned integer indicating the average, minimum, or maximum quantizer index used across the encoded coding blocks, respectively.

The rounding mode used to calculate the average value captured when VK_VIDEO_ENCODE_FEEDBACK_AVERAGE_QUANTIZATION_BIT_KHR is specified is undefined.

The number of pixels captured when VK_VIDEO_ENCODE_FEEDBACK_INTRA_PIXELS_BIT_KHR, VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR, and/or VK_VIDEO_ENCODE_FEEDBACK_SKIPPED_PIXELS_BIT_KHR are specified includes padding pixels beyond the requested coded extent that are part of the complete coding blocks encoded for the picture.

If VK_VIDEO_ENCODE_FEEDBACK_SKIPPED_PIXELS_BIT_KHR is not supported for a video encode profile, then the results captured for VK_VIDEO_ENCODE_FEEDBACK_INTRA_PIXELS_BIT_KHR and VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR will always include pixels in any intra or inter blocks, respectively, that are encoded as skipped blocks. Otherwise, the behavior depends on the used video profile:

- In case of H.264 encode profiles, skipped blocks are always inter blocks and the results captured for VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR will always include pixels encoded as skipped blocks.

  NOTE Implementations that would otherwise not count skipped pixels as inter pixels can simply sum the captured non-skipped inter pixel count and skipped pixel counts in the reported value for VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR.

- In case of H.265 encode profiles, skipped blocks are always inter blocks and the results captured for VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR will always include pixels encoded as skipped blocks.

  NOTE Implementations that would otherwise not count skipped pixels as inter pixels can simply sum the captured non-skipped inter pixel count and skipped pixel counts in the reported value for VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR.

- In case of AV1 encode profiles, both inter and intra blocks **can** be skipped blocks. Implementations **should** include any pixels encoded as skipped intra or inter blocks in the results captured for VK_VIDEO_ENCODE_FEEDBACK_INTRA_PIXELS_BIT_KHR and VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR, respectively, if possible. However, some implementations **may** not be able to do so if they do not differentiate between skipped intra and inter blocks. Therefore, in case of AV1 encode profiles that do support VK_VIDEO_ENCODE_FEEDBACK_SKIPPED_PIXELS_BIT_KHR implementations **must** either include skipped pixels part of intra and inter blocks in both or neither of the results captured by VK_VIDEO_ENCODE_FEEDBACK_INTRA_PIXELS_BIT_KHR and VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR, respectively.

  NOTE Applications can detect whether the reported intra and inter pixel counts include the skipped pixels or not, by checking whether the sum of the intra and inter pixels adds up to the coded extent padded to entire coding blocks.

When retrieving the results of video encode feedback queries, the values corresponding to each enabled video encode feedback are written in the order of the bits defined above, followed by an optional value indicating availability or result status if

VK_QUERY_RESULT_WITH_AVAILABILITY_BIT or VK_QUERY_RESULT_WITH_STATUS_BIT_KHR is specified, respectively. If per picture partition feedback is enabled, this is followed by the values corresponding to each enabled per picture partition video encode feedback for each encoded picture partition but at most for VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR ::maxPerPartitionFeedbackEntries number of picture partitions, in picture-partition-major order.

If the result status of a video encode feedback query is negative, then the results of all enabled video encode feedback values will be undefined.

NOTE

Applications should always specify VK_QUERY_RESULT_WITH_STATUS_BIT_KHR when retrieving the results of video encode feedback queries and ignore such undefined video encode feedback values for any unsuccessfully completed video encode operations.

```
// Provided by VK_KHR_video_encode_queue
typedef VkFlags VkVideoEncodeFeedbackFlagsKHR;
```

VkVideoEncodeFeedbackFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeFeedbackFlagBitsKHR.

22.11.1. Per Picture Partition Feedback

If the videoEncodeFeedback2 feature is enabled, video encode feedback queries also allow the application to capture additional feedback values for each picture partition generated by video encode operations. The availability of individual per picture partition video encode feedback values is indicated by the bits of VkVideoEncodeFeedback2CapabilitiesKHR ::supportedPerPartitionEncodeFeedbackFlags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the queries are intended to be used with.

The set of enabled per picture partition video encode feedback values **must** be configured on the query pool at the time of its creation by using the perPartitionEncodeFeedbackFlags member of the VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR structure included in the pNext chain of VkQueryPoolCreateInfo.

The VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_feedback2
typedef struct VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR {
    VkStructureType                                 sType;
    const void*                                     pNext;
    uint32_t                                        maxPerPartitionFeedbackEntries;
    VkVideoEncodePerPartitionFeedbackFlagsKHR       perPartitionEncodeFeedbackFlags;
} VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.

- `maxPerPartitionFeedbackEntries` specifies the maximum number of picture partitions for which per picture partition video encode feedback values **can** be captured by queries of the new pool.
- `perPartitionEncodeFeedbackFlags` is a bitmask of VkVideoEncodePerPartitionFeedbackFlagBitsKHR values specifying the set of enabled per picture partition video encode feedback values captured by queries of the new pool.

Valid Usage (Implicit)

- VUID-VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_QUERY_POOL_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_CREATE_INFO_KHR
- VUID-VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR-perPartitionEncodeFeedbackFlags-parameter
  If `perPartitionEncodeFeedbackFlags` is not `0`, `perPartitionEncodeFeedbackFlags` **must** be a valid combination of VkVideoEncodePerPartitionFeedbackFlagBitsKHR values

Structure Chaining

Extends the structure

- VkQueryPoolCreateInfo

Bits which **can** be set in VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR::`perPartitionEncodeFeedbackFlags` for video encode feedback query pools are:

```
// Provided by VK_KHR_video_encode_feedback2
typedef enum VkVideoEncodePerPartitionFeedbackFlagBitsKHR {
    VK_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_STATUS_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR =
0x00000002,
    VK_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR =
0x00000004,
} VkVideoEncodePerPartitionFeedbackFlagBitsKHR;
```

- VK_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_STATUS_BIT_KHR specifies that queries managed by the pool will capture the result status of each encoded picture partition. These result status values are reported as VkQueryResultStatusKHR values, similar to result status queries but indicate the result status of the encoding of individual picture partitions instead of that of the whole video encode operation.
- VK_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR specifies that queries managed by the pool will capture the byte offset of the bitstream data of the corresponding picture partition written to the bitstream buffer specified in VkVideoEncodeInfoKHR::`dstBuffer` relative to the offset specified in VkVideoEncodeInfoKHR::`dstBufferOffset`.

- VK_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR specifies that queries managed by the pool will capture the number of bytes written by the video encode operation for the corresponding picture partition to the bitstream buffer specified in VkVideoEncodeInfoKHR::dstBuffer.

The bitstream offsets reported for subsequent picture partitions are always monotonically increasing and the bitstream buffer ranges corresponding to individual picture partitions do not overlap.

NOTE

The bitstream range corresponding to the entire frame, as reported through the VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR and VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR feedback values may contain additional data such as frame headers. In addition, certain implementations or video compression standards may have additional payloads in between subsequent picture partitions. For example, AV1 bitstreams contain the tile_size_minus_1 syntax element in between subsequent tiles that are not part of the actual tile data. Therefore the specification only guarantees monotonically increasing offsets and non-overlapping ranges for picture partitions but does not guarantee that the picture partition ranges are tightly packed or that the union of the picture partition ranges covers the entire range of the bitstream output.

When retrieving the results of video encode feedback queries with per picture partition video encode feedback values, the values corresponding to each enabled per picture partition video encode feedback are written for each encoded picture partition in the order of the bits defined above, in picture-partition-major order.

If the video encode operation results in more encoded picture partitions than the maxPerPartitionFeedbackEntries specified in the VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR structure the query pool was created with, then the per picture partition feedback values corresponding to additional picture partitions are not available for retrieval.

If the per picture partition feedback status is negative, then all other per partition feedback values corresponding to that picture partition will be undefined. Specifically, implementations **may** report a per picture partition feedback status of VK_QUERY_RESULT_STATUS_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_KHR for the picture partition(s) whose encoded bitstream data no longer fit into the destination video bitstream buffer range.

```
// Provided by VK_KHR_video_encode_feedback2
typedef VkFlags VkVideoEncodePerPartitionFeedbackFlagsKHR;
```

VkVideoEncodePerPartitionFeedbackFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodePerPartitionFeedbackFlagBitsKHR.

Chapter 23. Clear Commands

23.1. Clearing Images Outside a Render Pass Instance

Color and depth/stencil images **can** be cleared outside a render pass instance using vkCmdClearColorImage or vkCmdClearDepthStencilImage, respectively. These commands are only allowed outside of a render pass instance.

To clear one or more subranges of a color image, call:

```
// Provided by VK_VERSION_1_0
void vkCmdClearColorImage(
    VkCommandBuffer                             commandBuffer,
    VkImage                                     image,
    VkImageLayout                               imageLayout,
    const VkClearColorValue*                    pColor,
    uint32_t                                    rangeCount,
    const VkImageSubresourceRange*              pRanges);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `image` is the image to be cleared.
- `imageLayout` specifies the current layout of the image subresource ranges to be cleared, and **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_GENERAL or VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL.
- `pColor` is a pointer to a VkClearColorValue structure containing the values that the image subresource ranges will be cleared to (see Clear Values below).
- `rangeCount` is the number of image subresource range structures in `pRanges`.
- `pRanges` is a pointer to an array of VkImageSubresourceRange structures describing a range of mipmap levels, array layers, and aspects to be cleared, as described in Image Views.

Each specified range in `pRanges` is cleared to the value specified by `pColor`.

Valid Usage

- VUID-vkCmdClearColorImage-image-01993
  The format features of `image` **must** contain VK_FORMAT_FEATURE_TRANSFER_DST_BIT
- VUID-vkCmdClearColorImage-image-00002
  `image` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-vkCmdClearColorImage-image-01545
  `image` **must** not use any of the formats that require a sampler $Y'C_BC_R$ conversion
- VUID-vkCmdClearColorImage-image-00003
  If `image` is non-sparse then it **must** be bound completely and contiguously to a single

VkDeviceMemory object

- VUID-vkCmdClearColorImage-imageLayout-00004
  imageLayout **must** specify the layout of the image subresource ranges of image specified in pRanges at the time this command is executed on a VkDevice
- VUID-vkCmdClearColorImage-imageLayout-01394
  imageLayout **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL or VK_IMAGE_LAYOUT_GENERAL
- VUID-vkCmdClearColorImage-aspectMask-02498
  The VkImageSubresourceRange::aspectMask members of the elements of the pRanges array **must** each only include VK_IMAGE_ASPECT_COLOR_BIT
- VUID-vkCmdClearColorImage-baseMipLevel-01470
  The VkImageSubresourceRange::baseMipLevel members of the elements of the pRanges array **must** each be less than the mipLevels specified in VkImageCreateInfo when image was created
- VUID-vkCmdClearColorImage-pRanges-01692
  For each VkImageSubresourceRange element of pRanges, if the levelCount member is not VK_REMAINING_MIP_LEVELS, then baseMipLevel + levelCount **must** be less than or equal to the mipLevels specified in VkImageCreateInfo when image was created
- VUID-vkCmdClearColorImage-baseArrayLayer-01472
  The VkImageSubresourceRange::baseArrayLayer members of the elements of the pRanges array **must** each be less than the arrayLayers specified in VkImageCreateInfo when image was created
- VUID-vkCmdClearColorImage-pRanges-01693
  For each VkImageSubresourceRange element of pRanges, if the layerCount member is not VK_REMAINING_ARRAY_LAYERS, then baseArrayLayer + layerCount **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when image was created
- VUID-vkCmdClearColorImage-image-00007
  image **must** not have a compressed or depth/stencil format
- VUID-vkCmdClearColorImage-pColor-04961
  pColor **must** be a valid pointer to a VkClearColorValue union
- VUID-vkCmdClearColorImage-commandBuffer-01805
  If commandBuffer is an unprotected command buffer and protectedNoFault is not supported, image **must** not be a protected image
- VUID-vkCmdClearColorImage-commandBuffer-01806
  If commandBuffer is a protected command buffer and protectedNoFault is not supported, image **must** not be an unprotected image
- VUID-vkCmdClearColorImage-image-09678
  If image's format has components other than R and G, it **must** not have a 64-bit component width

Valid Usage (Implicit)

- VUID-vkCmdClearColorImage-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdClearColorImage-image-parameter
  `image` **must** be a valid VkImage handle
- VUID-vkCmdClearColorImage-imageLayout-parameter
  `imageLayout` **must** be a valid VkImageLayout value
- VUID-vkCmdClearColorImage-pRanges-parameter
  `pRanges` **must** be a valid pointer to an array of `rangeCount` valid VkImageSubresourceRange structures
- VUID-vkCmdClearColorImage-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdClearColorImage-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdClearColorImage-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdClearColorImage-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdClearColorImage-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdClearColorImage-rangeCount-arraylength
  `rangeCount` **must** be greater than `0`
- VUID-vkCmdClearColorImage-commonparent
  Both of `commandBuffer`, and `image` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdClearColorImage is not affected by conditional rendering

To clear one or more subranges of a depth/stencil image, call:

```
// Provided by VK_VERSION_1_0
void vkCmdClearDepthStencilImage(
    VkCommandBuffer                             commandBuffer,
    VkImage                                     image,
    VkImageLayout                               imageLayout,
    const VkClearDepthStencilValue*             pDepthStencil,
    uint32_t                                    rangeCount,
    const VkImageSubresourceRange*              pRanges);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `image` is the image to be cleared.
- `imageLayout` specifies the current layout of the image subresource ranges to be cleared, and **must** be VK_IMAGE_LAYOUT_GENERAL or VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL.
- `pDepthStencil` is a pointer to a VkClearDepthStencilValue structure containing the values that the depth and stencil image subresource ranges will be cleared to (see Clear Values below).
- `rangeCount` is the number of image subresource range structures in `pRanges`.
- `pRanges` is a pointer to an array of VkImageSubresourceRange structures describing a range of mipmap levels, array layers, and aspects to be cleared, as described in Image Views.

Valid Usage

- VUID-vkCmdClearDepthStencilImage-image-01994
  The format features of `image` **must** contain VK_FORMAT_FEATURE_TRANSFER_DST_BIT
- VUID-vkCmdClearDepthStencilImage-pRanges-02658
  If the `aspectMask` member of any element of `pRanges` includes VK_IMAGE_ASPECT_STENCIL_BIT, and `image` was created with separate stencil usage, it must have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set

- VUID-vkCmdClearDepthStencilImage-pRanges-02659
  If the aspectMask member of any element of pRanges includes VK_IMAGE_ASPECT_STENCIL_BIT, and image was not created with separate stencil usage, the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag **must** have been set in the VkImageCreateInfo::usage used to create image
- VUID-vkCmdClearDepthStencilImage-pRanges-02660
  If the aspectMask member of any element of pRanges includes VK_IMAGE_ASPECT_DEPTH_BIT, image **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-vkCmdClearDepthStencilImage-image-00010
  If image is non-sparse then it **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-vkCmdClearDepthStencilImage-imageLayout-00011
  imageLayout **must** specify the layout of the image subresource ranges of image specified in pRanges at the time this command is executed on a VkDevice
- VUID-vkCmdClearDepthStencilImage-imageLayout-00012
  imageLayout **must** be either of VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL or VK_IMAGE_LAYOUT_GENERAL
- VUID-vkCmdClearDepthStencilImage-aspectMask-02824
  The VkImageSubresourceRange::aspectMask member of each element of the pRanges array **must** not include bits other than VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkCmdClearDepthStencilImage-image-02825
  If the image's format does not have a stencil component, then the VkImageSubresourceRange::aspectMask member of each element of the pRanges array **must** not include the VK_IMAGE_ASPECT_STENCIL_BIT bit
- VUID-vkCmdClearDepthStencilImage-image-02826
  If the image's format does not have a depth component, then the VkImageSubresourceRange::aspectMask member of each element of the pRanges array **must** not include the VK_IMAGE_ASPECT_DEPTH_BIT bit
- VUID-vkCmdClearDepthStencilImage-baseMipLevel-01474
  The VkImageSubresourceRange::baseMipLevel members of the elements of the pRanges array **must** each be less than the mipLevels specified in VkImageCreateInfo when image was created
- VUID-vkCmdClearDepthStencilImage-pRanges-01694
  For each VkImageSubresourceRange element of pRanges, if the levelCount member is not VK_REMAINING_MIP_LEVELS, then baseMipLevel + levelCount **must** be less than or equal to the mipLevels specified in VkImageCreateInfo when image was created
- VUID-vkCmdClearDepthStencilImage-baseArrayLayer-01476
  The VkImageSubresourceRange::baseArrayLayer members of the elements of the pRanges array **must** each be less than the arrayLayers specified in VkImageCreateInfo when image was created
- VUID-vkCmdClearDepthStencilImage-pRanges-01695

For each VkImageSubresourceRange element of pRanges, if the layerCount member is not VK_REMAINING_ARRAY_LAYERS, then baseArrayLayer + layerCount **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when image was created

- VUID-vkCmdClearDepthStencilImage-image-00014
  image **must** have a depth/stencil format
- VUID-vkCmdClearDepthStencilImage-commandBuffer-01807
  If commandBuffer is an unprotected command buffer and protectedNoFault is not supported, image **must** not be a protected image
- VUID-vkCmdClearDepthStencilImage-commandBuffer-01808
  If commandBuffer is a protected command buffer and protectedNoFault is not supported, image **must** not be an unprotected image

Valid Usage (Implicit)

- VUID-vkCmdClearDepthStencilImage-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdClearDepthStencilImage-image-parameter
  image **must** be a valid VkImage handle
- VUID-vkCmdClearDepthStencilImage-imageLayout-parameter
  imageLayout **must** be a valid VkImageLayout value
- VUID-vkCmdClearDepthStencilImage-pDepthStencil-parameter
  pDepthStencil **must** be a valid pointer to a valid VkClearDepthStencilValue structure
- VUID-vkCmdClearDepthStencilImage-pRanges-parameter
  pRanges **must** be a valid pointer to an array of rangeCount valid VkImageSubresourceRange structures
- VUID-vkCmdClearDepthStencilImage-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdClearDepthStencilImage-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdClearDepthStencilImage-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdClearDepthStencilImage-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdClearDepthStencilImage-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdClearDepthStencilImage-rangeCount-arraylength
  rangeCount **must** be greater than 0
- VUID-vkCmdClearDepthStencilImage-commonparent
  Both of commandBuffer, and image **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdClearDepthStencilImage is not affected by conditional rendering

Clears outside render pass instances are treated as transfer operations for the purposes of memory barriers.

23.2. Clearing Images Inside a Render Pass Instance

To clear one or more regions of color and depth/stencil attachments inside a render pass instance, call:

```
// Provided by VK_VERSION_1_0
void vkCmdClearAttachments(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    attachmentCount,
    const VkClearAttachment*                    pAttachments,
    uint32_t                                    rectCount,
    const VkClearRect*                          pRects);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `attachmentCount` is the number of entries in the `pAttachments` array.
- `pAttachments` is a pointer to an array of VkClearAttachment structures defining the attachments to clear and the clear values to use.
- `rectCount` is the number of entries in the `pRects` array.
- `pRects` is a pointer to an array of VkClearRect structures defining regions to clear for every attachment in `pAttachments`.

If the render pass has a fragment density map attachment, clears follow the operations of fragment density maps as if each clear region was a primitive which generates fragments. The clear color is applied to all pixels inside each fragment's area regardless if the pixels lie outside of the clear region. Clears **may** have a different set of supported fragment areas than draws.

Unlike other clear commands, vkCmdClearAttachments is not a transfer command. It performs its operations in rasterization order. For color attachments, the operations are executed as color attachment writes, by the VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT stage. For depth/stencil attachments, the operations are executed as depth writes and stencil writes by the VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT and VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT stages.

`vkCmdClearAttachments` is not affected by the bound pipeline state.

NOTE	It is generally preferable to clear attachments by using the VK_ATTACHMENT_LOAD_OP_CLEAR load operation at the start of rendering, as it is more efficient on some implementations.

If any attachment's `aspectMask` to be cleared is not backed by an image view, the clear has no effect on that aspect.

If an attachment being cleared refers to an image view created with an `aspectMask` equal to one of VK_IMAGE_ASPECT_PLANE_0_BIT, VK_IMAGE_ASPECT_PLANE_1_BIT or VK_IMAGE_ASPECT_PLANE_2_BIT, it is considered to be VK_IMAGE_ASPECT_COLOR_BIT for purposes of this command, and **must** be cleared with the VK_IMAGE_ASPECT_COLOR_BIT aspect as specified by image view creation.

Valid Usage

- VUID-vkCmdClearAttachments-aspectMask-07884
  If the current render pass instance does not use dynamic rendering, and the `aspectMask` member of any element of `pAttachments` contains VK_IMAGE_ASPECT_DEPTH_BIT, the current subpass instance's depth-stencil attachment **must** be either VK_ATTACHMENT_UNUSED or the attachment `format` **must** contain a depth component
- VUID-vkCmdClearAttachments-aspectMask-07885
  If the current render pass instance does not use dynamic rendering, and the `aspectMask` member of any element of `pAttachments` contains VK_IMAGE_ASPECT_STENCIL_BIT, the current subpass instance's depth-stencil attachment **must** be either VK_ATTACHMENT_UNUSED or the attachment `format` **must** contain a stencil component
- VUID-vkCmdClearAttachments-aspectMask-07271
  If the `aspectMask` member of any element of `pAttachments` contains VK_IMAGE_ASPECT_COLOR_BIT, the `colorAttachment` **must** be a valid color attachment index in the current render pass instance
- VUID-vkCmdClearAttachments-rect-02682
  The `rect` member of each element of `pRects` **must** have an `extent.width` greater than `0`
- VUID-vkCmdClearAttachments-rect-02683

The `rect` member of each element of `pRects` **must** have an `extent.height` greater than `0`

- VUID-vkCmdClearAttachments-pRects-00016
  The rectangular region specified by each element of `pRects` **must** be contained within the render area of the current render pass instance
- VUID-vkCmdClearAttachments-pRects-06937
  The layers specified by each element of `pRects` **must** be contained within every attachment that `pAttachments` refers to, i.e. for each element of `pRects`, VkClearRect ::`baseArrayLayer` + VkClearRect::`layerCount` **must** be less than or equal to the number of layers rendered to in the current render pass instance
- VUID-vkCmdClearAttachments-layerCount-01934
  The `layerCount` member of each element of `pRects` **must** not be `0`
- VUID-vkCmdClearAttachments-commandBuffer-02504
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, each attachment to be cleared **must** not be a protected image
- VUID-vkCmdClearAttachments-commandBuffer-02505
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, each attachment to be cleared **must** not be an unprotected image
- VUID-vkCmdClearAttachments-baseArrayLayer-00018
  If the render pass instance this is recorded in uses multiview, then `baseArrayLayer` **must** be zero and `layerCount` **must** be one
- VUID-vkCmdClearAttachments-colorAttachment-09503
  The `colorAttachment` member of each element of `pAttachments` **must** not identify a color attachment that is currently mapped to VK_ATTACHMENT_UNUSED in `commandBuffer` via VkRenderingAttachmentLocationInfo
- VUID-vkCmdClearAttachments-None-09679
  If the attachment format has components other than R and G, it **must** not have a 64-bit component width

Valid Usage (Implicit)

- VUID-vkCmdClearAttachments-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdClearAttachments-pAttachments-parameter
  `pAttachments` **must** be a valid pointer to an array of `attachmentCount` valid VkClearAttachment structures
- VUID-vkCmdClearAttachments-pRects-parameter
  `pRects` **must** be a valid pointer to an array of `rectCount` VkClearRect structures
- VUID-vkCmdClearAttachments-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdClearAttachments-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations

- VUID-vkCmdClearAttachments-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdClearAttachments-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdClearAttachments-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdClearAttachments-attachmentCount-arraylength
  `attachmentCount` **must** be greater than `0`
- VUID-vkCmdClearAttachments-rectCount-arraylength
  `rectCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdClearAttachments is affected by conditional rendering

The `VkClearRect` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkClearRect {
    VkRect2D    rect;
    uint32_t    baseArrayLayer;
    uint32_t    layerCount;
} VkClearRect;
```

- `rect` is the two-dimensional region to be cleared.
- `baseArrayLayer` is the first layer to be cleared.
- `layerCount` is the number of layers to clear.

The layers [baseArrayLayer, baseArrayLayer + layerCount) counting from the base layer of the attachment image view are cleared.

The VkClearAttachment structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkClearAttachment {
    VkImageAspectFlags    aspectMask;
    uint32_t              colorAttachment;
    VkClearValue          clearValue;
} VkClearAttachment;
```

- aspectMask is a mask selecting the color, depth and/or stencil aspects of the attachment to be cleared.
- colorAttachment is only meaningful if VK_IMAGE_ASPECT_COLOR_BIT is set in aspectMask, in which case it is an index into the bound color attachments.
- clearValue is the color or depth/stencil value to clear the attachment to, as described in Clear Values below.

Valid Usage

- VUID-VkClearAttachment-aspectMask-00019
  If aspectMask includes VK_IMAGE_ASPECT_COLOR_BIT, it **must** not include VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkClearAttachment-aspectMask-00020
  aspectMask **must** not include VK_IMAGE_ASPECT_METADATA_BIT
- VUID-VkClearAttachment-aspectMask-02246
  aspectMask **must** not include VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT for any index *i*

Valid Usage (Implicit)

- VUID-VkClearAttachment-aspectMask-parameter
  aspectMask **must** be a valid combination of VkImageAspectFlagBits values

23.3. Clear Values

The VkClearColorValue structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef union VkClearColorValue {
    float       float32[4];
    int32_t     int32[4];
    uint32_t    uint32[4];
} VkClearColorValue;
```

- `float32` are the color clear values when the format of the image or attachment is one of the numeric formats with a numeric type that is floating-point. Floating-point values are automatically converted to the format of the image, with the clear value being treated as linear if the image is sRGB.
- `int32` are the color clear values when the format of the image or attachment has a numeric type that is signed integer (`SINT`). Signed integer values are converted to the format of the image by casting to the smaller type (with negative 32-bit values mapping to negative values in the smaller type). If the integer clear value is not representable in the target type (e.g. would overflow in conversion to that type), the clear value is undefined.
- `uint32` are the color clear values when the format of the image or attachment has a numeric type that is unsigned integer (`UINT`). Unsigned integer values are converted to the format of the image by casting to the integer type with fewer bits.

The four array elements of the clear color map to R, G, B, and A components of image formats, in order.

If the image has more than one sample, the same value is written to all samples for any pixels being cleared.

If the image or attachment format has a 64-bit component width, the first 2 array elements of each of the arrays above are reinterpreted as a single 64-bit element for the R component. The next 2 array elements are used in the same way for the G component. In other words, the union behaves as if it had the following additional members:

```
double float64[2];
int64_t int64[2];
uint64_t uint64[2];
```

The `VkClearDepthStencilValue` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkClearDepthStencilValue {
    float       depth;
    uint32_t    stencil;
} VkClearDepthStencilValue;
```

- `depth` is the clear value for the depth aspect of the depth/stencil attachment. It is a floating-point value which is automatically converted to the attachment's format.

- `stencil` is the clear value for the stencil aspect of the depth/stencil attachment. It is a 32-bit integer value which is converted to the attachment's format by taking the appropriate number of LSBs.

Valid Usage

- VUID-VkClearDepthStencilValue-depth-00022
  Unless the `VK_EXT_depth_range_unrestricted` extension is enabled `depth` **must** be between `0.0` and `1.0`, inclusive

The `VkClearValue` union is defined as:

```
// Provided by VK_VERSION_1_0
typedef union VkClearValue {
    VkClearColorValue           color;
    VkClearDepthStencilValue    depthStencil;
} VkClearValue;
```

- `color` specifies the color image clear values to use when clearing a color image or attachment.
- `depthStencil` specifies the depth and stencil clear values to use when clearing a depth/stencil image or attachment.

This union is used where part of the API requires either color or depth/stencil clear values, depending on the attachment, and defines the initial clear values in the VkRenderPassBeginInfo structure.

23.4. Filling Memory

To fill a memory range with a fixed 4-byte bit pattern, call:

```
// Provided by VK_KHR_device_address_commands
void vkCmdFillMemoryKHR(
    VkCommandBuffer                             commandBuffer,
    const VkDeviceAddressRangeKHR*              pDstRange,
    VkAddressCommandFlagsKHR                    dstFlags,
    uint32_t                                    data);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pDstRange` is a pointer to the VkDeviceAddressRangeKHR selecting the memory range to be filled.
- `dstFlags` is a VkAddressCommandFlagsKHR value defining the copy flags for the destination address range.
- `data` is the 4-byte word written repeatedly to the destination range to fill `pDstRange->size` bytes of data.

NOTE The bit pattern of `data` is determined by the host endianness. For example, an unsigned integer value of `1` will result in a different bit pattern on a little endian machine compared to a big endian machine.

Valid Usage

- VUID-vkCmdFillMemoryKHR-pDstRange-13097
  If the range specified by `pDstRange` is not bound completely to memory when accessed, `dstFlags` **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR
- VUID-vkCmdFillMemoryKHR-pDstRange-13098
  If the buffer from which the range specified by `pDstRange` was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `dstFlags` **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-vkCmdFillMemoryKHR-pDstRange-13099
  If the buffer from which the range specified by `pDstRange` was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `dstFlags` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-vkCmdFillMemoryKHR-dstFlags-13100
  `dstFlags` **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdFillMemoryKHR-pDstRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `pDstRange`, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `dstFlags` **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdFillMemoryKHR-pDstRange-13123
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `pDstRange`, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `dstFlags` **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdFillMemoryKHR-dstFlags-13101
  `dstFlags` **must** not include both VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdFillMemoryKHR-pDstRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `pDstRange`, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `dstFlags` **must** include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdFillMemoryKHR-pDstRange-13125
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range

backing pDstRange, was created without VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, dstFlags **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR

- VUID-vkCmdFillMemoryKHR-dstRange-13000
  The buffer from which pDstRange was queried **must** have been created with VK_BUFFER_USAGE_TRANSFER_DST_BIT usage flag
- VUID-vkCmdFillMemoryKHR-pDstRange-13001
  pDstRange->address **must** be a multiple of 4
- VUID-vkCmdFillMemoryKHR-pDstRange-13002
  pDstRange->size **must** be a multiple of 4
- VUID-vkCmdFillMemoryKHR-commandBuffer-13003
  If commandBuffer is an unprotected command buffer and protectedNoFault is not supported, dstFlags **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-vkCmdFillMemoryKHR-commandBuffer-13004
  If commandBuffer is a protected command buffer and protectedNoFault is not supported, dstFlags **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR

Valid Usage (Implicit)

- VUID-vkCmdFillMemoryKHR-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdFillMemoryKHR-pDstRange-parameter
  pDstRange **must** be a valid pointer to a valid VkDeviceAddressRangeKHR structure
- VUID-vkCmdFillMemoryKHR-dstFlags-parameter
  If dstFlags is not 0, dstFlags **must** be a valid combination of VkAddressCommandFlagBitsKHR values
- VUID-vkCmdFillMemoryKHR-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdFillMemoryKHR-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_TRANSFER_BIT operations
- VUID-vkCmdFillMemoryKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdFillMemoryKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdFillMemoryKHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized

- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdFillMemoryKHR is not affected by conditional rendering

23.5. Updating Memory

To update device memory inline in a command buffer, call:

```
// Provided by VK_KHR_device_address_commands
void vkCmdUpdateMemoryKHR(
    VkCommandBuffer                             commandBuffer,
    const VkDeviceAddressRangeKHR*              pDstRange,
    VkAddressCommandFlagsKHR                    dstFlags,
    VkDeviceSize                                dataSize,
    const void*                                 pData);
```

- commandBuffer is the command buffer into which the command will be recorded.
- pDstRange is a pointer to the VkDeviceAddressRangeKHR selecting the memory range to be updated.
- dstFlags is a VkAddressCommandFlagsKHR value defining the copy flags for the destination address range.
- dataSize is the number of bytes to update, and **must** be a multiple of 4.
- pData is a pointer to the source data for the buffer update, and **must** be at least dataSize bytes in size.

The source data is copied from pData to the command buffer when the command is called, and then copied to dstRange when the command is executed on a device.

NOTE Due to the data being copied into command buffers, memory range copies are recommended for larger data transfers, to avoid excessive memory consumption.

Valid Usage

- VUID-vkCmdUpdateMemoryKHR-pDstRange-13097
  If the range specified by `pDstRange` is not bound completely to memory when accessed, `dstFlags` **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR
- VUID-vkCmdUpdateMemoryKHR-pDstRange-13098
  If the buffer from which the range specified by `pDstRange` was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `dstFlags` **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-vkCmdUpdateMemoryKHR-pDstRange-13099
  If the buffer from which the range specified by `pDstRange` was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `dstFlags` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-vkCmdUpdateMemoryKHR-dstFlags-13100
  `dstFlags` **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdUpdateMemoryKHR-pDstRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `pDstRange`, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `dstFlags` **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdUpdateMemoryKHR-pDstRange-13123
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `pDstRange`, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `dstFlags` **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdUpdateMemoryKHR-dstFlags-13101
  `dstFlags` **must** not include both VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdUpdateMemoryKHR-pDstRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `pDstRange`, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `dstFlags` **must** include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdUpdateMemoryKHR-pDstRange-13125
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `pDstRange`, was created without VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `dstFlags` **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-vkCmdUpdateMemoryKHR-dstRange-13005

The buffer from which `pDstRange` was queried **must** have been created with VK_BUFFER_USAGE_TRANSFER_DST_BIT usage flag

- VUID-vkCmdUpdateMemoryKHR-pDstRange-13006
  `pDstRange->address` **must** be a multiple of 4
- VUID-vkCmdUpdateMemoryKHR-pDstRange-13007
  `pDstRange->size` **must** be less than or equal to 65536
- VUID-vkCmdUpdateMemoryKHR-dataSize-13008
  `dataSize` **must** be less than or equal to `pDstRange->size`
- VUID-vkCmdUpdateMemoryKHR-dataSize-13009
  `dataSize` **must** be a multiple of 4
- VUID-vkCmdUpdateMemoryKHR-commandBuffer-13010
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `dstFlags` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-vkCmdUpdateMemoryKHR-commandBuffer-13011
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, `dstFlags` **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR

Valid Usage (Implicit)

- VUID-vkCmdUpdateMemoryKHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdUpdateMemoryKHR-pDstRange-parameter
  `pDstRange` **must** be a valid pointer to a valid VkDeviceAddressRangeKHR structure
- VUID-vkCmdUpdateMemoryKHR-dstFlags-parameter
  If `dstFlags` is not `0`, `dstFlags` **must** be a valid combination of VkAddressCommandFlagBitsKHR values
- VUID-vkCmdUpdateMemoryKHR-pData-parameter
  `pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-vkCmdUpdateMemoryKHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdUpdateMemoryKHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_TRANSFER_BIT operations
- VUID-vkCmdUpdateMemoryKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdUpdateMemoryKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdUpdateMemoryKHR-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdUpdateMemoryKHR-dataSize-arraylength
  `dataSize` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdUpdateMemoryKHR is not affected by conditional rendering

23.6. Filling Buffers

To clear buffer data, call:

```
// Provided by VK_VERSION_1_0
void vkCmdFillBuffer(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    dstBuffer,
    VkDeviceSize                                dstOffset,
    VkDeviceSize                                size,
    uint32_t                                    data);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `dstBuffer` is the buffer to be filled.
- `dstOffset` is the byte offset into the buffer at which to start filling, and **must** be a multiple of 4.
- `size` is the number of bytes to fill, and **must** be either a multiple of 4, or VK_WHOLE_SIZE to fill the range from `dstOffset` to the end of the buffer. If VK_WHOLE_SIZE is used and the remaining size of the buffer is not a multiple of 4, then the nearest smaller multiple is used.
- `data` is the 4-byte word written repeatedly to the buffer to fill `size` bytes of data. The data word is written to memory according to the host endianness.

`vkCmdFillBuffer` is treated as a “transfer” operation for the purposes of synchronization barriers. It is only compatible with buffers created with the VK_BUFFER_USAGE_TRANSFER_DST_BIT usage flag set.

Valid Usage

- VUID-vkCmdFillBuffer-dstOffset-00024
  `dstOffset` **must** be less than the size of `dstBuffer`
- VUID-vkCmdFillBuffer-dstOffset-00025
  `dstOffset` **must** be a multiple of `4`
- VUID-vkCmdFillBuffer-size-00026
  If `size` is not equal to VK_WHOLE_SIZE, `size` **must** be greater than `0`
- VUID-vkCmdFillBuffer-size-00027
  If `size` is not equal to VK_WHOLE_SIZE, `size` **must** be less than or equal to the size of `dstBuffer` minus `dstOffset`
- VUID-vkCmdFillBuffer-size-00028
  If `size` is not equal to VK_WHOLE_SIZE, `size` **must** be a multiple of `4`
- VUID-vkCmdFillBuffer-dstBuffer-00029
  `dstBuffer` **must** have been created with the VK_BUFFER_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-vkCmdFillBuffer-apiVersion-07894
  If the VK_KHR_maintenance1 extension is not enabled and VkPhysicalDeviceProperties ::`apiVersion` is less than Vulkan 1.1, the VkCommandPool that `commandBuffer` was allocated from **must** support graphics or compute operations
- VUID-vkCmdFillBuffer-dstBuffer-00031
  If `dstBuffer` is non-sparse then it **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-vkCmdFillBuffer-commandBuffer-01811
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `dstBuffer` **must** not be a protected buffer
- VUID-vkCmdFillBuffer-commandBuffer-01812
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, `dstBuffer` **must** not be an unprotected buffer

Valid Usage (Implicit)

- VUID-vkCmdFillBuffer-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdFillBuffer-dstBuffer-parameter
  `dstBuffer` **must** be a valid VkBuffer handle
- VUID-vkCmdFillBuffer-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdFillBuffer-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, or VK_QUEUE_TRANSFER_BIT

operations

- VUID-vkCmdFillBuffer-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdFillBuffer-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdFillBuffer-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdFillBuffer-commonparent
  Both of `commandBuffer`, and `dstBuffer` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdFillBuffer is not affected by conditional rendering

23.7. Updating Buffers

To update buffer data inline in a command buffer, call:

```
// Provided by VK_VERSION_1_0
void vkCmdUpdateBuffer(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    dstBuffer,
    VkDeviceSize                                dstOffset,
    VkDeviceSize                                dataSize,
    const void*                                 pData);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `dstBuffer` is a handle to the buffer to be updated.
- `dstOffset` is the byte offset into the buffer to start updating, and **must** be a multiple of 4.
- `dataSize` is the number of bytes to update, and **must** be a multiple of 4.
- `pData` is a pointer to the source data for the buffer update, and **must** be at least `dataSize` bytes in size.

`dataSize` **must** be less than or equal to 65536 bytes. For larger updates, applications **can** use buffer to buffer copies.

NOTE

Buffer updates performed with `vkCmdUpdateBuffer` first copy the data into command buffer memory when the command is recorded (which requires additional storage and may incur an additional allocation), and then copy the data from the command buffer into `dstBuffer` when the command is executed on a device.

The additional cost of this functionality compared to buffer to buffer copies means it should only be used for very small amounts of data, and is why it is limited to at most 65536 bytes. Applications **can** work around this restriction by issuing multiple `vkCmdUpdateBuffer` commands to different ranges of the same buffer, but doing so is not recommended.

The source data is copied from `pData` to the command buffer when the command is called.

`vkCmdUpdateBuffer` is only allowed outside of a render pass. This command is treated as a "transfer" operation for the purposes of synchronization barriers. The VK_BUFFER_USAGE_TRANSFER_DST_BIT **must** be specified in `usage` of VkBufferCreateInfo in order for the buffer to be compatible with `vkCmdUpdateBuffer`.

Valid Usage

- VUID-vkCmdUpdateBuffer-dstOffset-00032
  `dstOffset` **must** be less than the size of `dstBuffer`
- VUID-vkCmdUpdateBuffer-dataSize-00033
  `dataSize` **must** be less than or equal to the size of `dstBuffer` minus `dstOffset`
- VUID-vkCmdUpdateBuffer-dstBuffer-00034
  `dstBuffer` **must** have been created with the VK_BUFFER_USAGE_TRANSFER_DST_BIT usage flag set

- VUID-vkCmdUpdateBuffer-dstBuffer-00035
  If `dstBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdUpdateBuffer-dstOffset-00036
  `dstOffset` **must** be a multiple of `4`
- VUID-vkCmdUpdateBuffer-dataSize-00037
  `dataSize` **must** be less than or equal to `65536`
- VUID-vkCmdUpdateBuffer-dataSize-00038
  `dataSize` **must** be a multiple of `4`
- VUID-vkCmdUpdateBuffer-commandBuffer-01813
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `dstBuffer` **must** not be a protected buffer
- VUID-vkCmdUpdateBuffer-commandBuffer-01814
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, `dstBuffer` **must** not be an unprotected buffer

Valid Usage (Implicit)

- VUID-vkCmdUpdateBuffer-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdUpdateBuffer-dstBuffer-parameter
  `dstBuffer` **must** be a valid VkBuffer handle
- VUID-vkCmdUpdateBuffer-pData-parameter
  `pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-vkCmdUpdateBuffer-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdUpdateBuffer-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, or VK_QUEUE_TRANSFER_BIT operations
- VUID-vkCmdUpdateBuffer-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdUpdateBuffer-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdUpdateBuffer-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdUpdateBuffer-dataSize-arraylength
  `dataSize` **must** be greater than `0`
- VUID-vkCmdUpdateBuffer-commonparent
  Both of `commandBuffer`, and `dstBuffer` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdUpdateBuffer is not affected by conditional rendering

NOTE The `pData` parameter was of type `uint32_t*` instead of `void*` prior to version 1.0.19 of the Specification and VK_HEADER_VERSION 19 of the Vulkan Header Files. This was a historical anomaly, as the source data may be of other types.

Chapter 24. Copy Commands

An application **can** copy buffer and image data using several methods described in this chapter, depending on the type of data transfer.

All copy commands are treated as “transfer” operations for the purposes of synchronization barriers.

All copy commands that have a source format with an X component in its format description read undefined values from those bits.

All copy commands that have a destination format with an X component in its format description write undefined values to those bits.

24.1. Copying Data Between Memory Ranges

To copy data between memory ranges, call:

```
// Provided by VK_KHR_device_address_commands
void vkCmdCopyMemoryKHR(
    VkCommandBuffer                             commandBuffer,
    const VkCopyDeviceMemoryInfoKHR*            pCopyMemoryInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pCopyMemoryInfo` is a pointer to a VkCopyDeviceMemoryInfoKHR structure describing the copies to perform.

Valid Usage

- VUID-vkCmdCopyMemoryKHR-commandBuffer-13012
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, the `srcFlags` member of all elements of `pCopyMemoryInfo->pRegions` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-vkCmdCopyMemoryKHR-commandBuffer-13013
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, the `dstFlags` member of all elements of `pCopyMemoryInfo->pRegions` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-vkCmdCopyMemoryKHR-commandBuffer-13014
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, the `dstFlags` member of all elements of `pCopyMemoryInfo->pRegions` **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR

Valid Usage (Implicit)

- VUID-vkCmdCopyMemoryKHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyMemoryKHR-pCopyMemoryInfo-parameter
  If `pCopyMemoryInfo` is not `NULL`, `pCopyMemoryInfo` **must** be a valid pointer to a valid VkCopyDeviceMemoryInfoKHR structure
- VUID-vkCmdCopyMemoryKHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdCopyMemoryKHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_TRANSFER_BIT operations
- VUID-vkCmdCopyMemoryKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyMemoryKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyMemoryKHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdCopyMemoryKHR is not affected by conditional rendering

The `VkCopyDeviceMemoryInfoKHR` structure is defined as:

```
// Provided by VK_KHR_device_address_commands
typedef struct VkCopyDeviceMemoryInfoKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    uint32_t                        regionCount;
    const VkDeviceMemoryCopyKHR*    pRegions;
} VkCopyDeviceMemoryInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `regionCount` is the number of copies to be performed.
- `pRegions` is a pointer to an array of VkDeviceMemoryCopyKHR structures describing individual copy operations between two memory ranges.

Valid Usage

- VUID-VkCopyDeviceMemoryInfoKHR-srcRange-13015
  The range of memory backing the address range defined by the `srcRange` member of any element of `pRegions` **must** not overlap the memory backing the address range defined by the `srcRange` or `dstRange` of any element of `pRegions`

Valid Usage (Implicit)

- VUID-VkCopyDeviceMemoryInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COPY_DEVICE_MEMORY_INFO_KHR
- VUID-VkCopyDeviceMemoryInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkCopyDeviceMemoryInfoKHR-pRegions-parameter
  `pRegions` **must** be a valid pointer to an array of `regionCount` valid VkDeviceMemoryCopyKHR structures
- VUID-VkCopyDeviceMemoryInfoKHR-regionCount-arraylength
  `regionCount` **must** be greater than `0`

The `VkDeviceMemoryCopyKHR` structure is defined as:

```
// Provided by VK_KHR_device_address_commands
typedef struct VkDeviceMemoryCopyKHR {
    VkStructureType             sType;
    const void*                 pNext;
    VkDeviceAddressRangeKHR     srcRange;
    VkAddressCommandFlagsKHR    srcFlags;
    VkDeviceAddressRangeKHR     dstRange;
    VkAddressCommandFlagsKHR    dstFlags;
} VkDeviceMemoryCopyKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcRange` is a VkDeviceAddressRangeKHR structure defining the source memory to copy from.
- `srcFlags` is a VkAddressCommandFlagsKHR value defining the copy flags for the source address range.
- `dstRange` is a VkDeviceAddressRangeKHR structure defining the destination memory to copy to.
- `dstFlags` is a VkAddressCommandFlagsKHR value defining the copy flags for the destination address range.

This structure defines a copy operation where `srcRange.size` bytes will be copied from `srcRange.address` to `dstRange.address`.

Valid Usage

- VUID-VkDeviceMemoryCopyKHR-srcRange-13097
  If the range specified by `srcRange` is not bound completely to memory when accessed, `srcFlags` **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-srcRange-13098
  If the buffer from which the range specified by `srcRange` was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `srcFlags` **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-srcRange-13099
  If the buffer from which the range specified by `srcRange` was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `srcFlags` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-srcFlags-13100
  `srcFlags` **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-srcRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `srcRange`, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `srcFlags` **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR or

VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR

- VUID-VkDeviceMemoryCopyKHR-srcRange-13123
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `srcRange`, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `srcFlags` **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-srcFlags-13101
  `srcFlags` **must** not include both VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-srcRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `srcRange`, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `srcFlags` **must** include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-srcRange-13125
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `srcRange`, was created without VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `srcFlags` **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-dstRange-13097
  If the range specified by `dstRange` is not bound completely to memory when accessed, `dstFlags` **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-dstRange-13098
  If the buffer from which the range specified by `dstRange` was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `dstFlags` **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-dstRange-13099
  If the buffer from which the range specified by `dstRange` was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `dstFlags` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-dstFlags-13100
  `dstFlags` **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-dstRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `dstRange`, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `dstFlags` **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-dstRange-13123

If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing dstRange, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, dstFlags **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR

- VUID-VkDeviceMemoryCopyKHR-dstFlags-13101
  dstFlags **must** not include both VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-dstRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing dstRange, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, dstFlags **must** include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-dstRange-13125
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing dstRange, was created without VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, dstFlags **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDeviceMemoryCopyKHR-size-13016
  The size member of dstRange **must** be greater than or equal to the size member of srcRange
- VUID-VkDeviceMemoryCopyKHR-srcRange-13017
  The buffer from which srcRange was queried **must** have been created with VK_BUFFER_USAGE_TRANSFER_SRC_BIT
- VUID-VkDeviceMemoryCopyKHR-dstRange-13018
  The buffer from which dstRange was queried **must** have been created with VK_BUFFER_USAGE_TRANSFER_DST_BIT

Valid Usage (Implicit)

- VUID-VkDeviceMemoryCopyKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DEVICE_MEMORY_COPY_KHR
- VUID-VkDeviceMemoryCopyKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkDeviceMemoryCopyKHR-srcFlags-parameter
  If srcFlags is not 0, srcFlags **must** be a valid combination of VkAddressCommandFlagBitsKHR values
- VUID-VkDeviceMemoryCopyKHR-dstFlags-parameter
  If dstFlags is not 0, dstFlags **must** be a valid combination of VkAddressCommandFlagBitsKHR values

24.2. Copying Data Between Buffers

To copy data between buffer objects, call:

```
// Provided by VK_VERSION_1_0
void vkCmdCopyBuffer(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    srcBuffer,
    VkBuffer                                    dstBuffer,
    uint32_t                                    regionCount,
    const VkBufferCopy*                         pRegions);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `srcBuffer` is the source buffer.
- `dstBuffer` is the destination buffer.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of VkBufferCopy structures specifying the regions to copy.

Each source region specified by `pRegions` is copied from the source buffer to the destination region of the destination buffer.

Valid Usage

- VUID-vkCmdCopyBuffer-commandBuffer-01822
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `srcBuffer` **must** not be a protected buffer
- VUID-vkCmdCopyBuffer-commandBuffer-01823
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `dstBuffer` **must** not be a protected buffer
- VUID-vkCmdCopyBuffer-commandBuffer-01824
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, `dstBuffer` **must** not be an unprotected buffer
- VUID-vkCmdCopyBuffer-srcOffset-00113
  The `srcOffset` member of each element of `pRegions` **must** be less than the size of `srcBuffer`
- VUID-vkCmdCopyBuffer-dstOffset-00114
  The `dstOffset` member of each element of `pRegions` **must** be less than the size of `dstBuffer`
- VUID-vkCmdCopyBuffer-size-00115
  The `size` member of each element of `pRegions` **must** be less than or equal to the size of `srcBuffer` minus `srcOffset`
- VUID-vkCmdCopyBuffer-size-00116
  The `size` member of each element of `pRegions` **must** be less than or equal to the size of `dstBuffer` minus `dstOffset`

- VUID-vkCmdCopyBuffer-pRegions-00117
  Each memory location accessed by this command **must** not be accessed by both a source region and a destination region
- VUID-vkCmdCopyBuffer-pRegions-12482
  Each memory location accessed by this command **must** not be accessed by more than one destination region
- VUID-vkCmdCopyBuffer-srcBuffer-00118
  `srcBuffer` **must** have been created with the VK_BUFFER_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-vkCmdCopyBuffer-srcBuffer-00119
  If `srcBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdCopyBuffer-dstBuffer-00120
  `dstBuffer` **must** have been created with the VK_BUFFER_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-vkCmdCopyBuffer-dstBuffer-00121
  If `dstBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

Valid Usage (Implicit)

- VUID-vkCmdCopyBuffer-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyBuffer-srcBuffer-parameter
  `srcBuffer` **must** be a valid VkBuffer handle
- VUID-vkCmdCopyBuffer-dstBuffer-parameter
  `dstBuffer` **must** be a valid VkBuffer handle
- VUID-vkCmdCopyBuffer-pRegions-parameter
  `pRegions` **must** be a valid pointer to an array of `regionCount` valid VkBufferCopy structures
- VUID-vkCmdCopyBuffer-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdCopyBuffer-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, or VK_QUEUE_TRANSFER_BIT operations
- VUID-vkCmdCopyBuffer-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyBuffer-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyBuffer-videocoding
  This command **must** only be called outside of a video coding scope

- VUID-vkCmdCopyBuffer-regionCount-arraylength
  `regionCount` **must** be greater than `0`
- VUID-vkCmdCopyBuffer-commonparent
  Each of `commandBuffer`, `dstBuffer`, and `srcBuffer` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdCopyBuffer is not affected by conditional rendering

The `VkBufferCopy` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBufferCopy {
    VkDeviceSize    srcOffset;
    VkDeviceSize    dstOffset;
    VkDeviceSize    size;
} VkBufferCopy;
```

- `srcOffset` is the starting offset in bytes from the start of `srcBuffer`.
- `dstOffset` is the starting offset in bytes from the start of `dstBuffer`.
- `size` is the number of bytes to copy.

Valid Usage

- VUID-VkBufferCopy-size-01988
  The `size` **must** be greater than `0`

A more extensible version of the copy buffer command is defined below.

To copy data between buffer objects, call:

```
// Provided by VK_VERSION_1_3
void vkCmdCopyBuffer2(
    VkCommandBuffer                             commandBuffer,
    const VkCopyBufferInfo2*                    pCopyBufferInfo);
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to vkCmdCopyBuffer2
void vkCmdCopyBuffer2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkCopyBufferInfo2*                    pCopyBufferInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pCopyBufferInfo` is a pointer to a VkCopyBufferInfo2 structure describing the copy parameters.

Each source region specified by `pCopyBufferInfo->pRegions` is copied from the source buffer to the destination region of the destination buffer.

Valid Usage

- VUID-vkCmdCopyBuffer2-commandBuffer-01822
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `srcBuffer` **must** not be a protected buffer
- VUID-vkCmdCopyBuffer2-commandBuffer-01823
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `dstBuffer` **must** not be a protected buffer
- VUID-vkCmdCopyBuffer2-commandBuffer-01824
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, `dstBuffer` **must** not be an unprotected buffer

Valid Usage (Implicit)

- VUID-vkCmdCopyBuffer2-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyBuffer2-pCopyBufferInfo-parameter

pCopyBufferInfo **must** be a valid pointer to a valid VkCopyBufferInfo2 structure

- VUID-vkCmdCopyBuffer2-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdCopyBuffer2-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, or VK_QUEUE_TRANSFER_BIT operations
- VUID-vkCmdCopyBuffer2-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyBuffer2-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyBuffer2-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdCopyBuffer2 is not affected by conditional rendering

The VkCopyBufferInfo2 structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkCopyBufferInfo2 {
    VkStructureType         sType;
    const void*             pNext;
    VkBuffer                srcBuffer;
    VkBuffer                dstBuffer;
    uint32_t                regionCount;
    const VkBufferCopy2*    pRegions;
} VkCopyBufferInfo2;
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to VkCopyBufferInfo2
typedef VkCopyBufferInfo2 VkCopyBufferInfo2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcBuffer` is the source buffer.
- `dstBuffer` is the destination buffer.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of VkBufferCopy2 structures specifying the regions to copy.

Valid Usage

- VUID-VkCopyBufferInfo2-srcOffset-00113
  The `srcOffset` member of each element of `pRegions` **must** be less than the size of `srcBuffer`
- VUID-VkCopyBufferInfo2-dstOffset-00114
  The `dstOffset` member of each element of `pRegions` **must** be less than the size of `dstBuffer`
- VUID-VkCopyBufferInfo2-size-00115
  The `size` member of each element of `pRegions` **must** be less than or equal to the size of `srcBuffer` minus `srcOffset`
- VUID-VkCopyBufferInfo2-size-00116
  The `size` member of each element of `pRegions` **must** be less than or equal to the size of `dstBuffer` minus `dstOffset`
- VUID-VkCopyBufferInfo2-pRegions-00117
  Each memory location accessed by this command **must** not be accessed by both a source region and a destination region
- VUID-VkCopyBufferInfo2-pRegions-12482
  Each memory location accessed by this command **must** not be accessed by more than one destination region
- VUID-VkCopyBufferInfo2-srcBuffer-00118
  `srcBuffer` **must** have been created with the VK_BUFFER_USAGE_TRANSFER_SRC_BIT usage flag set

- VUID-VkCopyBufferInfo2-srcBuffer-00119
  If `srcBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkCopyBufferInfo2-dstBuffer-00120
  `dstBuffer` **must** have been created with the VK_BUFFER_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-VkCopyBufferInfo2-dstBuffer-00121
  If `dstBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

Valid Usage (Implicit)

- VUID-VkCopyBufferInfo2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COPY_BUFFER_INFO_2
- VUID-VkCopyBufferInfo2-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkCopyBufferInfo2-srcBuffer-parameter
  `srcBuffer` **must** be a valid VkBuffer handle
- VUID-VkCopyBufferInfo2-dstBuffer-parameter
  `dstBuffer` **must** be a valid VkBuffer handle
- VUID-VkCopyBufferInfo2-pRegions-parameter
  `pRegions` **must** be a valid pointer to an array of `regionCount` valid VkBufferCopy2 structures
- VUID-VkCopyBufferInfo2-regionCount-arraylength
  `regionCount` **must** be greater than `0`
- VUID-VkCopyBufferInfo2-commonparent
  Both of `dstBuffer`, and `srcBuffer` **must** have been created, allocated, or retrieved from the same VkDevice

The `VkBufferCopy2` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkBufferCopy2 {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceSize       srcOffset;
    VkDeviceSize       dstOffset;
    VkDeviceSize       size;
} VkBufferCopy2;
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to VkBufferCopy2
typedef VkBufferCopy2 VkBufferCopy2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcOffset` is the starting offset in bytes from the start of `srcBuffer`.
- `dstOffset` is the starting offset in bytes from the start of `dstBuffer`.
- `size` is the number of bytes to copy.

Valid Usage

- VUID-VkBufferCopy2-size-01988
  The `size` **must** be greater than `0`

Valid Usage (Implicit)

- VUID-VkBufferCopy2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BUFFER_COPY_2
- VUID-VkBufferCopy2-pNext-pNext
  `pNext` **must** be `NULL`

24.3. Copying Data Between Images

To copy data between image objects, call:

```
// Provided by VK_VERSION_1_0
void vkCmdCopyImage(
    VkCommandBuffer                             commandBuffer,
    VkImage                                     srcImage,
    VkImageLayout                               srcImageLayout,
    VkImage                                     dstImage,
    VkImageLayout                               dstImageLayout,
    uint32_t                                    regionCount,
    const VkImageCopy*                          pRegions);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `srcImage` is the source image.
- `srcImageLayout` is the current layout of the source image subresource.
- `dstImage` is the destination image.

- `dstImageLayout` is the current layout of the destination image subresource.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of VkImageCopy structures specifying the regions to copy.

Each source region specified by `pRegions` is copied from the source image to the destination region of the destination image.

If any region accesses a depth aspect in `dstImage` and the `VK_EXT_depth_range_unrestricted` extension is not enabled, values copied from `srcImage` outside of the range [0,1] will be written as undefined values to the destination image.

Copy regions for each image **must** be aligned to a multiple of the texel block extent in each dimension, except at the edges of the image, where region extents **must** match the edge of the image.

The texels in the images are accessed as follows.

Texels at each coordinate (x,y,z,layer) in an image subresource are accessed, where:

x is in the range [offset.x, offset.x + extent.width),

y is in the range [offset.y, offset.y + extent.height),

z is in the range [offset.z, offset.z + extent.depth),

layer is in the range [subresource.baseArrayLayer, subresource.baseArrayLayer + subresource.layerCount)

where for `srcImage`, *offset*, *extent*, and *subresource* are `srcOffset`, `extent`, and `srcSubresource` for each region, respectively.

For `dstImage`, *offset*, *extent*, and *subresource* are `dstOffset`, `extent`, and `dstSubresource` for each region, respectively, where `dstOffset` and `extent` are first size-adjusted if the format of the destination image has a different block extent than the format of the source image (e.g. one is a compressed format).

Multi-planar images **can** only be copied on a per-plane basis, and the subresources used in each region when copying to or from such images **must** specify only one plane, though different regions **can** specify different planes. When copying planes of multi-planar images, the format considered is the compatible format for that plane, rather than the format of the multi-planar image.

Image data **can** be copied between images with different image types. If one image is VK_IMAGE_TYPE_3D and the other image is VK_IMAGE_TYPE_2D with multiple layers, then each slice is copied to or from a different layer; `depth` slices in the 3D image correspond to `layerCount` layers in the 2D image, with an effective `depth` of `1` used for the 2D image. If the `maintenance5` feature

is enabled, all other combinations are allowed and function as if 1D images are 2D images with a height of 1. Otherwise, other combinations of image types are disallowed.

The memory locations accessed by the source and destination images overlap if they alias each other, both `srcSubresource` and `dstSubresource` overlap the aliased range, and any of the following are true:

- Their contents are not able to be interpreted in consistent ways
- If the aliased regions are *host-accessible*, their accessed regions defined above overlap, where the memory locations accessed by the texels of a linear image are according to its VkSubresourceLayout
- Otherwise, their accessed texel ranges share any of the same texel coordinates

The memory locations of an image overlap if those two ranges share any of the same texel coordinates in the same image aspects.

Valid Usage

- VUID-vkCmdCopyImage-commandBuffer-01825
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `srcImage` **must** not be a protected image
- VUID-vkCmdCopyImage-commandBuffer-01826
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `dstImage` **must** not be a protected image
- VUID-vkCmdCopyImage-commandBuffer-01827
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, `dstImage` **must** not be an unprotected image
- VUID-vkCmdCopyImage-commandBuffer-10217
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT, and the `maintenance10` feature is not enabled, for each element of `pRegions`, where the `aspectMask` member of `srcSubresource` is VK_IMAGE_ASPECT_COLOR_BIT, the `aspectMask` of `dstSubresource` **must** not be VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkCmdCopyImage-commandBuffer-11782
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support VK_QUEUE_COMPUTE_BIT, and in any element of `pRegions` the `aspectMask` member of `srcSubresource` is VK_IMAGE_ASPECT_COLOR_BIT and the `aspectMask` of `dstSubresource` is VK_IMAGE_ASPECT_DEPTH_BIT, then the format features of `dstImage` **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_COMPUTE_QUEUE_BIT_KHR
- VUID-vkCmdCopyImage-commandBuffer-11783
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of `pRegions` the `aspectMask` member of `srcSubresource` is VK_IMAGE_ASPECT_COLOR_BIT and the `aspectMask` of

dstSubresource is VK_IMAGE_ASPECT_DEPTH_BIT, then the format features of dstImage **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_TRANSFER_QUEUE_BIT_KHR

- VUID-vkCmdCopyImage-commandBuffer-11784
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support VK_QUEUE_COMPUTE_BIT, and in any element of pRegions the aspectMask member of srcSubresource is VK_IMAGE_ASPECT_COLOR_BIT and the aspectMask of dstSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of dstImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_COMPUTE_QUEUE_BIT_KHR
- VUID-vkCmdCopyImage-commandBuffer-11785
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of pRegions the aspectMask member of srcSubresource is VK_IMAGE_ASPECT_COLOR_BIT and the aspectMask of dstSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of dstImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_TRANSFER_QUEUE_BIT_KHR
- VUID-vkCmdCopyImage-commandBuffer-12449
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT and srcSubresource.aspectMask or dstSubresource.aspectMask of any element of pRegions contain VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT, then both srcImage and dstImage **must** have a sample count equal to VK_SAMPLE_COUNT_1_BIT
- VUID-vkCmdCopyImage-commandBuffer-10218
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT, and the maintenance10 feature is not enabled, for each element of pRegions, where the aspectMask member of dstSubresource is VK_IMAGE_ASPECT_COLOR_BIT then the aspectMask of srcSubresource **must** not be VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkCmdCopyImage-commandBuffer-11786
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support VK_QUEUE_COMPUTE_BIT, and in any element of pRegions the aspectMask member of dstSubresource is VK_IMAGE_ASPECT_COLOR_BIT and the aspectMask of srcSubresource is VK_IMAGE_ASPECT_DEPTH_BIT, then the format features of srcImage **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_COMPUTE_QUEUE_BIT_KHR
- VUID-vkCmdCopyImage-commandBuffer-11787
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of pRegions the aspectMask member of dstSubresource is VK_IMAGE_ASPECT_COLOR_BIT and the aspectMask of srcSubresource is VK_IMAGE_ASPECT_DEPTH_BIT, then the format features of srcImage **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_TRANSFER_QUEUE_BIT_KHR
- VUID-vkCmdCopyImage-commandBuffer-11788
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support

VK_QUEUE_COMPUTE_BIT, and in any element of pRegions the aspectMask member of dstSubresource is VK_IMAGE_ASPECT_COLOR_BIT and the aspectMask of srcSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of srcImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_COMPUTE_QUEUE_BIT_KHR

- VUID-vkCmdCopyImage-commandBuffer-11789
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of pRegions the aspectMask member of dstSubresource is VK_IMAGE_ASPECT_COLOR_BIT and the aspectMask of srcSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of srcImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_TRANSFER_QUEUE_BIT_KHR
- VUID-vkCmdCopyImage-pRegions-00124
  Each memory location accessed by this command **must** not be accessed by both a source region and a destination region, as specified by pRegions and accessed according to Copy Image Addressing
- VUID-vkCmdCopyImage-pRegions-12484
  Each memory location accessed by this command **must** not be accessed by more than one destination region, as specified by pRegions and accessed according to Copy Image Addressing
- VUID-vkCmdCopyImage-srcImage-01995
  The format features of srcImage **must** contain VK_FORMAT_FEATURE_TRANSFER_SRC_BIT
- VUID-vkCmdCopyImage-srcImageLayout-00128
  srcImageLayout **must** specify the layout of the image subresources of srcImage specified in pRegions at the time this command is executed on a VkDevice
- VUID-vkCmdCopyImage-srcImageLayout-01917
  srcImageLayout **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL, or VK_IMAGE_LAYOUT_GENERAL
- VUID-vkCmdCopyImage-srcImage-09460
  If srcImage and dstImage are the same, and any element of pRegions contains the srcSubresource and dstSubresource with matching mipLevel and overlapping array layers, then the srcImageLayout and dstImageLayout **must** be VK_IMAGE_LAYOUT_GENERAL or VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR
- VUID-vkCmdCopyImage-dstImage-01996
  The format features of dstImage **must** contain VK_FORMAT_FEATURE_TRANSFER_DST_BIT
- VUID-vkCmdCopyImage-dstImageLayout-00133
  dstImageLayout **must** specify the layout of the image subresources of dstImage specified in pRegions at the time this command is executed on a VkDevice
- VUID-vkCmdCopyImage-dstImageLayout-01395
  dstImageLayout **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL, or VK_IMAGE_LAYOUT_GENERAL
- VUID-vkCmdCopyImage-srcImage-01548
  If the VkFormat of each of srcImage and dstImage is not a multi-planar format, the VkFormat of each of srcImage and dstImage **must** be size-compatible

- VUID-vkCmdCopyImage-None-01549
  In a copy to or from a plane of a multi-planar image, the VkFormat of the image and plane **must** be compatible according to the description of compatible planes for the plane being copied
- VUID-vkCmdCopyImage-srcImage-09247
  If the VkFormat of each of `srcImage` and `dstImage` is a compressed image format, the formats **must** have the same texel block extent
- VUID-vkCmdCopyImage-srcImage-00136
  The sample count of `srcImage` and `dstImage` **must** match
- VUID-vkCmdCopyImage-srcOffset-01783
  The `srcOffset` and `extent` members of each element of `pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in VkQueueFamilyProperties
- VUID-vkCmdCopyImage-dstOffset-01784
  The `dstOffset` and `extent` members of each element of `pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in VkQueueFamilyProperties
- VUID-vkCmdCopyImage-srcImage-01551
  If neither `srcImage` nor `dstImage` has a multi-planar format and the `maintenance8` feature is not enabled then for each element of `pRegions`, `srcSubresource.aspectMask` and `dstSubresource.aspectMask` **must** match
- VUID-vkCmdCopyImage-pRegions-12201
  For each element of `pRegions` where `srcSubresource.aspectMask` and `dstSubresource.aspectMask` each contain at least one of VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT, `srcSubresource.aspectMask` and `dstSubresource.aspectMask` **must** match
- VUID-vkCmdCopyImage-srcSubresource-10214
  If `srcSubresource.aspectMask` is VK_IMAGE_ASPECT_COLOR_BIT, then `dstSubresource.aspectMask` **must** not contain both VK_IMAGE_ASPECT_DEPTH_BIT and VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkCmdCopyImage-dstSubresource-10215
  If `dstSubresource.aspectMask` is VK_IMAGE_ASPECT_COLOR_BIT, then `srcSubresource.aspectMask` **must** not contain both VK_IMAGE_ASPECT_DEPTH_BIT and VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkCmdCopyImage-srcImage-08713
  If `srcImage` has a multi-planar format, then for each element of `pRegions`, `srcSubresource.aspectMask` **must** be a single valid multi-planar aspect mask bit
- VUID-vkCmdCopyImage-dstImage-08714
  If `dstImage` has a multi-planar format, then for each element of `pRegions`, `dstSubresource.aspectMask` **must** be a single valid multi-planar aspect mask bit
- VUID-vkCmdCopyImage-srcImage-01556
  If `srcImage` has a multi-planar format and the `dstImage` does not have a multi-planar image format, then for each element of `pRegions`, `dstSubresource.aspectMask` **must** be

VK_IMAGE_ASPECT_COLOR_BIT

- VUID-vkCmdCopyImage-dstImage-01557
  If dstImage has a multi-planar format and the srcImage does not have a multi-planar image format, then for each element of pRegions, srcSubresource.aspectMask **must** be VK_IMAGE_ASPECT_COLOR_BIT
- VUID-vkCmdCopyImage-srcSubresource-10211
  If srcSubresource.aspectMask is VK_IMAGE_ASPECT_COLOR_BIT and dstSubresource.aspectMask is VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT, then the VkFormat values of srcImage and dstImage **must** be compatible according to the list of compatible depth-stencil and color formats
- VUID-vkCmdCopyImage-srcSubresource-10212
  If srcSubresource.aspectMask is VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT and dstSubresource.aspectMask is VK_IMAGE_ASPECT_COLOR_BIT, then the VkFormat values of srcImage and dstImage **must** be compatible according to the list of compatible depth-stencil and color formats
- VUID-vkCmdCopyImage-apiVersion-07932
  If the VK_KHR_maintenance1 extension is not enabled, or VkPhysicalDeviceProperties ::apiVersion is less than Vulkan 1.1, and either srcImage or dstImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, srcSubresource.baseArrayLayer and dstSubresource.baseArrayLayer **must** both be 0, and srcSubresource.layerCount and dstSubresource.layerCount **must** both be 1
- VUID-vkCmdCopyImage-srcImage-04443
  If srcImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, srcSubresource.baseArrayLayer **must** be 0 and srcSubresource.layerCount **must** be 1
- VUID-vkCmdCopyImage-dstImage-04444
  If dstImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, dstSubresource.baseArrayLayer **must** be 0 and dstSubresource.layerCount **must** be 1
- VUID-vkCmdCopyImage-aspectMask-00142
  For each element of pRegions, srcSubresource.aspectMask **must** specify aspects present in srcImage
- VUID-vkCmdCopyImage-aspectMask-00143
  For each element of pRegions, dstSubresource.aspectMask **must** specify aspects present in dstImage
- VUID-vkCmdCopyImage-srcOffset-00144
  For each element of pRegions, srcOffset.x and (extent.width + srcOffset.x) **must** both be greater than or equal to 0 and less than or equal to the width of the specified srcSubresource of srcImage
- VUID-vkCmdCopyImage-srcOffset-00145
  For each element of pRegions, srcOffset.y and (extent.height + srcOffset.y) **must** both be greater than or equal to 0 and less than or equal to the height of the specified srcSubresource of srcImage
- VUID-vkCmdCopyImage-srcImage-00146
  If srcImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, srcOffset.y

must be `0` and `extent.height` **must** be `1`

- VUID-vkCmdCopyImage-srcOffset-00147
  If `srcImage` is of type VK_IMAGE_TYPE_3D, then for each element of `pRegions`, `srcOffset.z` and (`extent.depth` + `srcOffset.z`) **must** both be greater than or equal to `0` and less than or equal to the depth of the specified `srcSubresource` of `srcImage`
- VUID-vkCmdCopyImage-srcImage-01785
  If `srcImage` is of type VK_IMAGE_TYPE_1D, then for each element of `pRegions`, `srcOffset.z` **must** be `0` and `extent.depth` **must** be `1`
- VUID-vkCmdCopyImage-dstImage-01786
  If `dstImage` is of type VK_IMAGE_TYPE_1D, then for each element of `pRegions`, `dstOffset.z` **must** be `0`
- VUID-vkCmdCopyImage-srcImage-10907
  If either the VkFormat of each of `srcImage` and `dstImage` is not a compressed image format, and `dstImage` is of type VK_IMAGE_TYPE_1D, then for each element of `pRegions`, `extent.depth` **must** be `1`
- VUID-vkCmdCopyImage-srcImage-01787
  If `srcImage` is of type VK_IMAGE_TYPE_2D, then for each element of `pRegions`, `srcOffset.z` **must** be `0`
- VUID-vkCmdCopyImage-dstImage-01788
  If `dstImage` is of type VK_IMAGE_TYPE_2D, then for each element of `pRegions`, `dstOffset.z` **must** be `0`
- VUID-vkCmdCopyImage-apiVersion-07933
  If the VK_KHR_maintenance1 extension is not enabled, and VkPhysicalDeviceProperties ::`apiVersion` is less than Vulkan 1.1, `srcImage` and `dstImage` **must** have the same VkImageType
- VUID-vkCmdCopyImage-apiVersion-08969
  If the VK_KHR_maintenance1 extension is not enabled, and VkPhysicalDeviceProperties ::`apiVersion` is less than Vulkan 1.1, `srcImage` or `dstImage` is of type VK_IMAGE_TYPE_2D, then for each element of `pRegions`, `extent.depth` **must** be `1`
- VUID-vkCmdCopyImage-srcImage-07743
  If `srcImage` and `dstImage` have a different VkImageType, and the `maintenance5` feature is not enabled, one **must** be VK_IMAGE_TYPE_3D and the other **must** be VK_IMAGE_TYPE_2D
- VUID-vkCmdCopyImage-srcImage-08793
  If `srcImage` and `dstImage` have the same VkImageType, for each element of `pRegions`, if neither of the `layerCount` members of `srcSubresource` or `dstSubresource` are VK_REMAINING_ARRAY_LAYERS, the `layerCount` members of `srcSubresource` or `dstSubresource` **must** match
- VUID-vkCmdCopyImage-srcImage-08794
  If `srcImage` and `dstImage` have the same VkImageType, and one of the `layerCount` members of `srcSubresource` or `dstSubresource` is VK_REMAINING_ARRAY_LAYERS, the other member **must** be either VK_REMAINING_ARRAY_LAYERS or equal to the `arrayLayers` member of the VkImageCreateInfo used to create the image minus `baseArrayLayer`
- VUID-vkCmdCopyImage-srcImage-01790

If srcImage and dstImage are both of type VK_IMAGE_TYPE_2D, then for each element of pRegions, extent.depth **must** be 1

- VUID-vkCmdCopyImage-srcImage-01791
  If srcImage is of type VK_IMAGE_TYPE_2D, and dstImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, extent.depth **must** equal srcSubresource.layerCount
- VUID-vkCmdCopyImage-dstImage-01792
  If dstImage is of type VK_IMAGE_TYPE_2D, and srcImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, extent.depth **must** equal dstSubresource.layerCount
- VUID-vkCmdCopyImage-dstOffset-00150
  For each element of pRegions, dstOffset.x and (extent.width + dstOffset.x), where extent is adjusted for size-compatibility, **must** both be greater than or equal to 0 and less than or equal to the width of the specified dstSubresource of dstImage
- VUID-vkCmdCopyImage-dstOffset-00151
  For each element of pRegions, dstOffset.y and (extent.height + dstOffset.y), where extent is adjusted for size-compatibility, **must** both be greater than or equal to 0 and less than or equal to the height of the specified dstSubresource of dstImage
- VUID-vkCmdCopyImage-dstImage-00152
  If dstImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, dstOffset.y **must** be 0
- VUID-vkCmdCopyImage-srcImage-10908
  If either the VkFormat of each of srcImage and dstImage is not a compressed image format, and dstImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, extent.height **must** be 1, where extent is adjusted for size-compatibility
- VUID-vkCmdCopyImage-dstOffset-00153
  If dstImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, dstOffset.z and (extent.depth + dstOffset.z), where extent is adjusted for size-compatibility, **must** both be greater than or equal to 0 and less than or equal to the depth of the specified dstSubresource of dstImage
- VUID-vkCmdCopyImage-pRegions-07278
  For each element of pRegions, srcOffset.x **must** be a multiple of the texel block extent width of the VkFormat of srcImage
- VUID-vkCmdCopyImage-pRegions-07279
  For each element of pRegions, srcOffset.y **must** be a multiple of the texel block extent height of the VkFormat of srcImage
- VUID-vkCmdCopyImage-pRegions-07280
  For each element of pRegions, srcOffset.z **must** be a multiple of the texel block extent depth of the VkFormat of srcImage
- VUID-vkCmdCopyImage-pRegions-07281
  For each element of pRegions, dstOffset.x **must** be a multiple of the texel block extent width of the VkFormat of dstImage
- VUID-vkCmdCopyImage-pRegions-07282
  For each element of pRegions, dstOffset.y **must** be a multiple of the texel block extent height of the VkFormat of dstImage

- VUID-vkCmdCopyImage-pRegions-07283
  For each element of pRegions, dstOffset.z **must** be a multiple of the texel block extent depth of the VkFormat of dstImage
- VUID-vkCmdCopyImage-srcImage-01728
  For each element of pRegions, if the sum of srcOffset.x and extent.width does not equal the width of the subresource specified by srcSubresource, extent.width **must** be a multiple of the texel block extent width of the VkFormat of srcImage
- VUID-vkCmdCopyImage-srcImage-01729
  For each element of pRegions, if the sum of srcOffset.y and extent.height does not equal the height of the subresource specified by srcSubresource, extent.height **must** be a multiple of the texel block extent height of the VkFormat of srcImage
- VUID-vkCmdCopyImage-srcImage-01730
  For each element of pRegions, if the sum of srcOffset.z and extent.depth does not equal the depth of the subresource specified by srcSubresource, extent.depth **must** be a multiple of the texel block extent depth of the VkFormat of srcImage
- VUID-vkCmdCopyImage-aspect-06662
  If the aspect member of any element of pRegions includes any flag other than VK_IMAGE_ASPECT_STENCIL_BIT or srcImage was not created with separate stencil usage, srcImage **must** have been created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-vkCmdCopyImage-aspect-06663
  If the aspect member of any element of pRegions includes any flag other than VK_IMAGE_ASPECT_STENCIL_BIT or dstImage was not created with separate stencil usage, dstImage **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-vkCmdCopyImage-aspect-06664
  If the aspect member of any element of pRegions includes VK_IMAGE_ASPECT_STENCIL_BIT, and srcImage was created with separate stencil usage, srcImage **must** have been created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-vkCmdCopyImage-aspect-06665
  If the aspect member of any element of pRegions includes VK_IMAGE_ASPECT_STENCIL_BIT, and dstImage was created with separate stencil usage, srcImage **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-vkCmdCopyImage-srcImage-07966
  If srcImage is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-vkCmdCopyImage-srcSubresource-07967
  The srcSubresource.mipLevel member of each element of pRegions **must** be less than the mipLevels specified in VkImageCreateInfo when srcImage was created
- VUID-vkCmdCopyImage-srcSubresource-07968
  If srcSubresource.layerCount is not VK_REMAINING_ARRAY_LAYERS,

`srcSubresource.baseArrayLayer` + `srcSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in VkImageCreateInfo when `srcImage` was created

- VUID-vkCmdCopyImage-srcImage-07969
  `srcImage` **must** not have been created with `flags` containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-vkCmdCopyImage-dstImage-07966
  If `dstImage` is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdCopyImage-dstSubresource-07967
  The `dstSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `dstImage` was created
- VUID-vkCmdCopyImage-dstSubresource-07968
  If `dstSubresource.layerCount` is not VK_REMAINING_ARRAY_LAYERS, `dstSubresource.baseArrayLayer` + `dstSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in VkImageCreateInfo when `dstImage` was created
- VUID-vkCmdCopyImage-dstImage-07969
  `dstImage` **must** not have been created with `flags` containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT

Valid Usage (Implicit)

- VUID-vkCmdCopyImage-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyImage-srcImage-parameter
  `srcImage` **must** be a valid VkImage handle
- VUID-vkCmdCopyImage-srcImageLayout-parameter
  `srcImageLayout` **must** be a valid VkImageLayout value
- VUID-vkCmdCopyImage-dstImage-parameter
  `dstImage` **must** be a valid VkImage handle
- VUID-vkCmdCopyImage-dstImageLayout-parameter
  `dstImageLayout` **must** be a valid VkImageLayout value
- VUID-vkCmdCopyImage-pRegions-parameter
  `pRegions` **must** be a valid pointer to an array of `regionCount` valid VkImageCopy structures
- VUID-vkCmdCopyImage-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdCopyImage-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, or VK_QUEUE_TRANSFER_BIT operations

- VUID-vkCmdCopyImage-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyImage-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyImage-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdCopyImage-regionCount-arraylength
  `regionCount` **must** be greater than `0`
- VUID-vkCmdCopyImage-commonparent
  Each of `commandBuffer`, `dstImage`, and `srcImage` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdCopyImage is not affected by conditional rendering

The `VkImageCopy` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageCopy {
    VkImageSubresourceLayers    srcSubresource;
    VkOffset3D                  srcOffset;
    VkImageSubresourceLayers    dstSubresource;
    VkOffset3D                  dstOffset;
    VkExtent3D                  extent;
} VkImageCopy;
```

- `srcSubresource` and `dstSubresource` are VkImageSubresourceLayers structures specifying the image subresources of the images used for the source and destination image data, respectively.
- `srcOffset` and `dstOffset` select the initial `x`, `y`, and `z` offsets in texels of the sub-regions of the source and destination image data.
- `extent` is the size in texels of the image to copy in `width`, `height` and `depth`.

Valid Usage

- VUID-VkImageCopy-apiVersion-07940
  If the VK_KHR_sampler_ycbcr_conversion extension is not enabled, and VkPhysicalDeviceProperties::`apiVersion` is less than Vulkan 1.1, the `aspectMask` member of `srcSubresource` and `dstSubresource` **must** match
- VUID-VkImageCopy-apiVersion-07941
  If the VK_KHR_maintenance1 extension is not enabled, and VkPhysicalDeviceProperties ::`apiVersion` is less than Vulkan 1.1, the `layerCount` member of `srcSubresource` and `dstSubresource` **must** match
- VUID-VkImageCopy-extent-06668
  `extent.width` **must** not be 0
- VUID-VkImageCopy-extent-06669
  `extent.height` **must** not be 0
- VUID-VkImageCopy-extent-06670
  `extent.depth` **must** not be 0

Valid Usage (Implicit)

- VUID-VkImageCopy-srcSubresource-parameter
  `srcSubresource` **must** be a valid VkImageSubresourceLayers structure
- VUID-VkImageCopy-dstSubresource-parameter
  `dstSubresource` **must** be a valid VkImageSubresourceLayers structure

The `VkImageSubresourceLayers` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageSubresourceLayers {
    VkImageAspectFlags    aspectMask;
    uint32_t              mipLevel;
    uint32_t              baseArrayLayer;
    uint32_t              layerCount;
} VkImageSubresourceLayers;
```

- `aspectMask` is a combination of VkImageAspectFlagBits, selecting the color, depth and/or stencil aspects to be copied.
- `mipLevel` is the mipmap level to copy.
- `baseArrayLayer` and `layerCount` are the starting layer and number of layers to copy.

Valid Usage

- VUID-VkImageSubresourceLayers-aspectMask-00167
  If `aspectMask` contains VK_IMAGE_ASPECT_COLOR_BIT, it **must** not contain either of VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkImageSubresourceLayers-aspectMask-00168
  `aspectMask` **must** not contain VK_IMAGE_ASPECT_METADATA_BIT
- VUID-VkImageSubresourceLayers-aspectMask-02247
  `aspectMask` **must** not include `VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT` for any index *i*
- VUID-VkImageSubresourceLayers-layerCount-09243
  If the `maintenance5` feature is not enabled, `layerCount` **must** not be VK_REMAINING_ARRAY_LAYERS
- VUID-VkImageSubresourceLayers-layerCount-01700
  If `layerCount` is not VK_REMAINING_ARRAY_LAYERS, it **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkImageSubresourceLayers-aspectMask-parameter
  `aspectMask` **must** be a valid combination of VkImageAspectFlagBits values

A more extensible version of the copy image command is defined below.

To copy data between image objects, call:

```
// Provided by VK_VERSION_1_3
void vkCmdCopyImage2(
    VkCommandBuffer                             commandBuffer,
    const VkCopyImageInfo2*                     pCopyImageInfo);
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to vkCmdCopyImage2
void vkCmdCopyImage2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkCopyImageInfo2*                     pCopyImageInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pCopyImageInfo` is a pointer to a VkCopyImageInfo2 structure describing the copy parameters.

This command is functionally identical to vkCmdCopyImage, but includes extensible sub-structures that include `sType` and `pNext` parameters, allowing them to be more easily extended.

Valid Usage

- VUID-vkCmdCopyImage2-commandBuffer-01825
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `srcImage` **must** not be a protected image
- VUID-vkCmdCopyImage2-commandBuffer-01826
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `dstImage` **must** not be a protected image
- VUID-vkCmdCopyImage2-commandBuffer-01827
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, `dstImage` **must** not be an unprotected image
- VUID-vkCmdCopyImage2-commandBuffer-10217
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT, and the `maintenance10` feature is not enabled, for each element of `pCopyImageInfo->pRegions`, where the `aspectMask` member of `srcSubresource` is VK_IMAGE_ASPECT_COLOR_BIT, the `aspectMask` of `dstSubresource` **must** not be VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkCmdCopyImage2-commandBuffer-11782
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support VK_QUEUE_COMPUTE_BIT, and in any element of `pCopyImageInfo->pRegions` the `aspectMask` member of `srcSubresource` is VK_IMAGE_ASPECT_COLOR_BIT and the `aspectMask` of `dstSubresource` is VK_IMAGE_ASPECT_DEPTH_BIT, then the format features of `dstImage` **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_COMPUTE_QUEUE_BIT_KHR
- VUID-vkCmdCopyImage2-commandBuffer-11783
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of `pCopyImageInfo->pRegions` the `aspectMask` member of `srcSubresource` is VK_IMAGE_ASPECT_COLOR_BIT and the `aspectMask` of `dstSubresource` is VK_IMAGE_ASPECT_DEPTH_BIT, then the format features of `dstImage` **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_TRANSFER_QUEUE_BIT_KHR

- VUID-vkCmdCopyImage2-commandBuffer-11784
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support VK_QUEUE_COMPUTE_BIT, and in any element of pCopyImageInfo->pRegions the aspectMask member of srcSubresource is VK_IMAGE_ASPECT_COLOR_BIT and the aspectMask of dstSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of dstImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_COMPUTE_QUEUE_BIT_KHR
- VUID-vkCmdCopyImage2-commandBuffer-11785
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of pCopyImageInfo->pRegions the aspectMask member of srcSubresource is VK_IMAGE_ASPECT_COLOR_BIT and the aspectMask of dstSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of dstImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_TRANSFER_QUEUE_BIT_KHR
- VUID-vkCmdCopyImage2-commandBuffer-12449
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT and srcSubresource.aspectMask or dstSubresource.aspectMask of any element of pCopyImageInfo->pRegions contain VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT, then both srcImage and dstImage **must** have a sample count equal to VK_SAMPLE_COUNT_1_BIT
- VUID-vkCmdCopyImage2-commandBuffer-10218
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT, and the maintenance10 feature is not enabled, for each element of pCopyImageInfo->pRegions, where the aspectMask member of dstSubresource is VK_IMAGE_ASPECT_COLOR_BIT then the aspectMask of srcSubresource **must** not be VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkCmdCopyImage2-commandBuffer-11786
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support VK_QUEUE_COMPUTE_BIT, and in any element of pCopyImageInfo->pRegions the aspectMask member of dstSubresource is VK_IMAGE_ASPECT_COLOR_BIT and the aspectMask of srcSubresource is VK_IMAGE_ASPECT_DEPTH_BIT, then the format features of srcImage **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_COMPUTE_QUEUE_BIT_KHR
- VUID-vkCmdCopyImage2-commandBuffer-11787
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of pCopyImageInfo->pRegions the aspectMask member of dstSubresource is VK_IMAGE_ASPECT_COLOR_BIT and the aspectMask of srcSubresource is VK_IMAGE_ASPECT_DEPTH_BIT, then the format features of srcImage **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_TRANSFER_QUEUE_BIT_KHR
- VUID-vkCmdCopyImage2-commandBuffer-11788
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support

VK_QUEUE_COMPUTE_BIT, and in any element of pCopyImageInfo->pRegions the aspectMask member of dstSubresource is VK_IMAGE_ASPECT_COLOR_BIT and the aspectMask of srcSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of srcImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_COMPUTE_QUEUE_BIT_KHR

- VUID-vkCmdCopyImage2-commandBuffer-11789
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of pCopyImageInfo->pRegions the aspectMask member of dstSubresource is VK_IMAGE_ASPECT_COLOR_BIT and the aspectMask of srcSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of srcImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_TRANSFER_QUEUE_BIT_KHR

Valid Usage (Implicit)

- VUID-vkCmdCopyImage2-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyImage2-pCopyImageInfo-parameter
  pCopyImageInfo **must** be a valid pointer to a valid VkCopyImageInfo2 structure
- VUID-vkCmdCopyImage2-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdCopyImage2-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, or VK_QUEUE_TRANSFER_BIT operations
- VUID-vkCmdCopyImage2-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyImage2-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyImage2-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdCopyImage2 is not affected by conditional rendering

The VkCopyImageInfo2 structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkCopyImageInfo2 {
    VkStructureType        sType;
    const void*            pNext;
    VkImage                srcImage;
    VkImageLayout          srcImageLayout;
    VkImage                dstImage;
    VkImageLayout          dstImageLayout;
    uint32_t               regionCount;
    const VkImageCopy2*    pRegions;
} VkCopyImageInfo2;
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to VkCopyImageInfo2
typedef VkCopyImageInfo2 VkCopyImageInfo2KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- srcImage is the source image.
- srcImageLayout is the current layout of the source image subresource.
- dstImage is the destination image.
- dstImageLayout is the current layout of the destination image subresource.
- regionCount is the number of regions to copy.
- pRegions is a pointer to an array of VkImageCopy2 structures specifying the regions to copy.

Valid Usage

- VUID-VkCopyImageInfo2-pRegions-00124
  Each memory location accessed by this command **must** not be accessed by both a source region and a destination region, as specified by `pRegions` and accessed according to Copy Image Addressing
- VUID-VkCopyImageInfo2-pRegions-12484
  Each memory location accessed by this command **must** not be accessed by more than one destination region, as specified by `pRegions` and accessed according to Copy Image Addressing
- VUID-VkCopyImageInfo2-srcImage-01995
  The format features of `srcImage` **must** contain VK_FORMAT_FEATURE_TRANSFER_SRC_BIT
- VUID-VkCopyImageInfo2-srcImageLayout-00128
  `srcImageLayout` **must** specify the layout of the image subresources of `srcImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-VkCopyImageInfo2-srcImageLayout-01917
  `srcImageLayout` **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL, or VK_IMAGE_LAYOUT_GENERAL
- VUID-VkCopyImageInfo2-srcImage-09460
  If `srcImage` and `dstImage` are the same, and any element of `pRegions` contains the `srcSubresource` and `dstSubresource` with matching `mipLevel` and overlapping array layers, then the `srcImageLayout` and `dstImageLayout` **must** be VK_IMAGE_LAYOUT_GENERAL or VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR
- VUID-VkCopyImageInfo2-dstImage-01996
  The format features of `dstImage` **must** contain VK_FORMAT_FEATURE_TRANSFER_DST_BIT
- VUID-VkCopyImageInfo2-dstImageLayout-00133
  `dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-VkCopyImageInfo2-dstImageLayout-01395
  `dstImageLayout` **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL, or VK_IMAGE_LAYOUT_GENERAL
- VUID-VkCopyImageInfo2-srcImage-01548
  If the VkFormat of each of `srcImage` and `dstImage` is not a multi-planar format, the VkFormat of each of `srcImage` and `dstImage` **must** be size-compatible
- VUID-VkCopyImageInfo2-None-01549
  In a copy to or from a plane of a multi-planar image, the VkFormat of the image and plane **must** be compatible according to the description of compatible planes for the plane being copied
- VUID-VkCopyImageInfo2-srcImage-09247
  If the VkFormat of each of `srcImage` and `dstImage` is a compressed image format, the formats **must** have the same texel block extent
- VUID-VkCopyImageInfo2-srcImage-00136

The sample count of `srcImage` and `dstImage` **must** match

- VUID-VkCopyImageInfo2-srcOffset-01783
  The `srcOffset` and `extent` members of each element of `pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in VkQueueFamilyProperties
- VUID-VkCopyImageInfo2-dstOffset-01784
  The `dstOffset` and `extent` members of each element of `pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in VkQueueFamilyProperties
- VUID-VkCopyImageInfo2-srcImage-01551
  If neither `srcImage` nor `dstImage` has a multi-planar format and the `maintenance8` feature is not enabled then for each element of `pRegions`, `srcSubresource.aspectMask` and `dstSubresource.aspectMask` **must** match
- VUID-VkCopyImageInfo2-pRegions-12201
  For each element of `pRegions` where `srcSubresource.aspectMask` and `dstSubresource.aspectMask` each contain at least one of VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT, `srcSubresource.aspectMask` and `dstSubresource.aspectMask` **must** match
- VUID-VkCopyImageInfo2-srcSubresource-10214
  If `srcSubresource.aspectMask` is VK_IMAGE_ASPECT_COLOR_BIT, then `dstSubresource.aspectMask` **must** not contain both VK_IMAGE_ASPECT_DEPTH_BIT and VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkCopyImageInfo2-dstSubresource-10215
  If `dstSubresource.aspectMask` is VK_IMAGE_ASPECT_COLOR_BIT, then `srcSubresource.aspectMask` **must** not contain both VK_IMAGE_ASPECT_DEPTH_BIT and VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkCopyImageInfo2-srcImage-08713
  If `srcImage` has a multi-planar format, then for each element of `pRegions`, `srcSubresource.aspectMask` **must** be a single valid multi-planar aspect mask bit
- VUID-VkCopyImageInfo2-dstImage-08714
  If `dstImage` has a multi-planar format, then for each element of `pRegions`, `dstSubresource.aspectMask` **must** be a single valid multi-planar aspect mask bit
- VUID-VkCopyImageInfo2-srcImage-01556
  If `srcImage` has a multi-planar format and the `dstImage` does not have a multi-planar image format, then for each element of `pRegions`, `dstSubresource.aspectMask` **must** be VK_IMAGE_ASPECT_COLOR_BIT
- VUID-VkCopyImageInfo2-dstImage-01557
  If `dstImage` has a multi-planar format and the `srcImage` does not have a multi-planar image format, then for each element of `pRegions`, `srcSubresource.aspectMask` **must** be VK_IMAGE_ASPECT_COLOR_BIT
- VUID-VkCopyImageInfo2-srcSubresource-10211
  If `srcSubresource.aspectMask` is VK_IMAGE_ASPECT_COLOR_BIT and `dstSubresource.aspectMask` is VK_IMAGE_ASPECT_DEPTH_BIT or

VK_IMAGE_ASPECT_STENCIL_BIT, then the VkFormat values of srcImage and dstImage **must** be compatible according to the list of compatible depth-stencil and color formats

- VUID-VkCopyImageInfo2-srcSubresource-10212
  If srcSubresource.aspectMask is VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT and dstSubresource.aspectMask is VK_IMAGE_ASPECT_COLOR_BIT, then the VkFormat values of srcImage and dstImage **must** be compatible according to the list of compatible depth-stencil and color formats
- VUID-VkCopyImageInfo2-apiVersion-07932
  If the VK_KHR_maintenance1 extension is not enabled, or VkPhysicalDeviceProperties ::apiVersion is less than Vulkan 1.1, and either srcImage or dstImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, srcSubresource.baseArrayLayer and dstSubresource.baseArrayLayer **must** both be 0, and srcSubresource.layerCount and dstSubresource.layerCount **must** both be 1
- VUID-VkCopyImageInfo2-srcImage-04443
  If srcImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, srcSubresource.baseArrayLayer **must** be 0 and srcSubresource.layerCount **must** be 1
- VUID-VkCopyImageInfo2-dstImage-04444
  If dstImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, dstSubresource.baseArrayLayer **must** be 0 and dstSubresource.layerCount **must** be 1
- VUID-VkCopyImageInfo2-aspectMask-00142
  For each element of pRegions, srcSubresource.aspectMask **must** specify aspects present in srcImage
- VUID-VkCopyImageInfo2-aspectMask-00143
  For each element of pRegions, dstSubresource.aspectMask **must** specify aspects present in dstImage
- VUID-VkCopyImageInfo2-srcOffset-00144
  For each element of pRegions, srcOffset.x and (extent.width + srcOffset.x) **must** both be greater than or equal to 0 and less than or equal to the width of the specified srcSubresource of srcImage
- VUID-VkCopyImageInfo2-srcOffset-00145
  For each element of pRegions, srcOffset.y and (extent.height + srcOffset.y) **must** both be greater than or equal to 0 and less than or equal to the height of the specified srcSubresource of srcImage
- VUID-VkCopyImageInfo2-srcImage-00146
  If srcImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, srcOffset.y **must** be 0 and extent.height **must** be 1
- VUID-VkCopyImageInfo2-srcOffset-00147
  If srcImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, srcOffset.z and (extent.depth + srcOffset.z) **must** both be greater than or equal to 0 and less than or equal to the depth of the specified srcSubresource of srcImage
- VUID-VkCopyImageInfo2-srcImage-01785
  If srcImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, srcOffset.z **must** be 0 and extent.depth **must** be 1

- VUID-VkCopyImageInfo2-dstImage-01786
  If `dstImage` is of type VK_IMAGE_TYPE_1D, then for each element of `pRegions`, `dstOffset.z` **must** be `0`
- VUID-VkCopyImageInfo2-srcImage-10907
  If either the VkFormat of each of `srcImage` and `dstImage` is not a compressed image format, and `dstImage` is of type VK_IMAGE_TYPE_1D, then for each element of `pRegions`, `extent.depth` **must** be `1`
- VUID-VkCopyImageInfo2-srcImage-01787
  If `srcImage` is of type VK_IMAGE_TYPE_2D, then for each element of `pRegions`, `srcOffset.z` **must** be `0`
- VUID-VkCopyImageInfo2-dstImage-01788
  If `dstImage` is of type VK_IMAGE_TYPE_2D, then for each element of `pRegions`, `dstOffset.z` **must** be `0`
- VUID-VkCopyImageInfo2-apiVersion-07933
  If the VK_KHR_maintenance1 extension is not enabled, and VkPhysicalDeviceProperties ::`apiVersion` is less than Vulkan 1.1, `srcImage` and `dstImage` **must** have the same VkImageType
- VUID-VkCopyImageInfo2-apiVersion-08969
  If the VK_KHR_maintenance1 extension is not enabled, and VkPhysicalDeviceProperties ::`apiVersion` is less than Vulkan 1.1, `srcImage` or `dstImage` is of type VK_IMAGE_TYPE_2D, then for each element of `pRegions`, `extent.depth` **must** be `1`
- VUID-VkCopyImageInfo2-srcImage-07743
  If `srcImage` and `dstImage` have a different VkImageType, and the `maintenance5` feature is not enabled, one **must** be VK_IMAGE_TYPE_3D and the other **must** be VK_IMAGE_TYPE_2D
- VUID-VkCopyImageInfo2-srcImage-08793
  If `srcImage` and `dstImage` have the same VkImageType, for each element of `pRegions`, if neither of the `layerCount` members of `srcSubresource` or `dstSubresource` are VK_REMAINING_ARRAY_LAYERS, the `layerCount` members of `srcSubresource` or `dstSubresource` **must** match
- VUID-VkCopyImageInfo2-srcImage-08794
  If `srcImage` and `dstImage` have the same VkImageType, and one of the `layerCount` members of `srcSubresource` or `dstSubresource` is VK_REMAINING_ARRAY_LAYERS, the other member **must** be either VK_REMAINING_ARRAY_LAYERS or equal to the `arrayLayers` member of the VkImageCreateInfo used to create the image minus `baseArrayLayer`
- VUID-VkCopyImageInfo2-srcImage-01790
  If `srcImage` and `dstImage` are both of type VK_IMAGE_TYPE_2D, then for each element of `pRegions`, `extent.depth` **must** be `1`
- VUID-VkCopyImageInfo2-srcImage-01791
  If `srcImage` is of type VK_IMAGE_TYPE_2D, and `dstImage` is of type VK_IMAGE_TYPE_3D, then for each element of `pRegions`, `extent.depth` **must** equal `srcSubresource.layerCount`
- VUID-VkCopyImageInfo2-dstImage-01792
  If `dstImage` is of type VK_IMAGE_TYPE_2D, and `srcImage` is of type VK_IMAGE_TYPE_3D, then for each element of `pRegions`, `extent.depth` **must** equal `dstSubresource.layerCount`

- VUID-VkCopyImageInfo2-dstOffset-00150
  For each element of pRegions, dstOffset.x and (extent.width + dstOffset.x), where extent is adjusted for size-compatibility, **must** both be greater than or equal to 0 and less than or equal to the width of the specified dstSubresource of dstImage
- VUID-VkCopyImageInfo2-dstOffset-00151
  For each element of pRegions, dstOffset.y and (extent.height + dstOffset.y), where extent is adjusted for size-compatibility, **must** both be greater than or equal to 0 and less than or equal to the height of the specified dstSubresource of dstImage
- VUID-VkCopyImageInfo2-dstImage-00152
  If dstImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, dstOffset.y **must** be 0
- VUID-VkCopyImageInfo2-srcImage-10908
  If either the VkFormat of each of srcImage and dstImage is not a compressed image format, and dstImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, extent.height **must** be 1, where extent is adjusted for size-compatibility
- VUID-VkCopyImageInfo2-dstOffset-00153
  If dstImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, dstOffset.z and (extent.depth + dstOffset.z), where extent is adjusted for size-compatibility, **must** both be greater than or equal to 0 and less than or equal to the depth of the specified dstSubresource of dstImage
- VUID-VkCopyImageInfo2-pRegions-07278
  For each element of pRegions, srcOffset.x **must** be a multiple of the texel block extent width of the VkFormat of srcImage
- VUID-VkCopyImageInfo2-pRegions-07279
  For each element of pRegions, srcOffset.y **must** be a multiple of the texel block extent height of the VkFormat of srcImage
- VUID-VkCopyImageInfo2-pRegions-07280
  For each element of pRegions, srcOffset.z **must** be a multiple of the texel block extent depth of the VkFormat of srcImage
- VUID-VkCopyImageInfo2-pRegions-07281
  For each element of pRegions, dstOffset.x **must** be a multiple of the texel block extent width of the VkFormat of dstImage
- VUID-VkCopyImageInfo2-pRegions-07282
  For each element of pRegions, dstOffset.y **must** be a multiple of the texel block extent height of the VkFormat of dstImage
- VUID-VkCopyImageInfo2-pRegions-07283
  For each element of pRegions, dstOffset.z **must** be a multiple of the texel block extent depth of the VkFormat of dstImage
- VUID-VkCopyImageInfo2-srcImage-01728
  For each element of pRegions, if the sum of srcOffset.x and extent.width does not equal the width of the subresource specified by srcSubresource, extent.width **must** be a multiple of the texel block extent width of the VkFormat of srcImage
- VUID-VkCopyImageInfo2-srcImage-01729

For each element of pRegions, if the sum of srcOffset.y and extent.height does not equal the height of the subresource specified by srcSubresource, extent.height **must** be a multiple of the texel block extent height of the VkFormat of srcImage

- VUID-VkCopyImageInfo2-srcImage-01730
  For each element of pRegions, if the sum of srcOffset.z and extent.depth does not equal the depth of the subresource specified by srcSubresource, extent.depth **must** be a multiple of the texel block extent depth of the VkFormat of srcImage
- VUID-VkCopyImageInfo2-aspect-06662
  If the aspect member of any element of pRegions includes any flag other than VK_IMAGE_ASPECT_STENCIL_BIT or srcImage was not created with separate stencil usage, srcImage **must** have been created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-VkCopyImageInfo2-aspect-06663
  If the aspect member of any element of pRegions includes any flag other than VK_IMAGE_ASPECT_STENCIL_BIT or dstImage was not created with separate stencil usage, dstImage **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-VkCopyImageInfo2-aspect-06664
  If the aspect member of any element of pRegions includes VK_IMAGE_ASPECT_STENCIL_BIT, and srcImage was created with separate stencil usage, srcImage **must** have been created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-VkCopyImageInfo2-aspect-06665
  If the aspect member of any element of pRegions includes VK_IMAGE_ASPECT_STENCIL_BIT, and dstImage was created with separate stencil usage, srcImage **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-VkCopyImageInfo2-srcImage-07966
  If srcImage is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-VkCopyImageInfo2-srcSubresource-07967
  The srcSubresource.mipLevel member of each element of pRegions **must** be less than the mipLevels specified in VkImageCreateInfo when srcImage was created
- VUID-VkCopyImageInfo2-srcSubresource-07968
  If srcSubresource.layerCount is not VK_REMAINING_ARRAY_LAYERS, srcSubresource.baseArrayLayer + srcSubresource.layerCount of each element of pRegions **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when srcImage was created
- VUID-VkCopyImageInfo2-srcImage-07969
  srcImage **must** not have been created with flags containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-VkCopyImageInfo2-dstImage-07966
  If dstImage is non-sparse then the image or each specified *disjoint* plane **must** be bound

completely and contiguously to a single VkDeviceMemory object

- VUID-VkCopyImageInfo2-dstSubresource-07967
  The dstSubresource.mipLevel member of each element of pRegions **must** be less than the mipLevels specified in VkImageCreateInfo when dstImage was created
- VUID-VkCopyImageInfo2-dstSubresource-07968
  If dstSubresource.layerCount is not VK_REMAINING_ARRAY_LAYERS, dstSubresource.baseArrayLayer + dstSubresource.layerCount of each element of pRegions **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when dstImage was created
- VUID-VkCopyImageInfo2-dstImage-07969
  dstImage **must** not have been created with flags containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT

Valid Usage (Implicit)

- VUID-VkCopyImageInfo2-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_COPY_IMAGE_INFO_2
- VUID-VkCopyImageInfo2-pNext-pNext
  pNext **must** be NULL
- VUID-VkCopyImageInfo2-srcImage-parameter
  srcImage **must** be a valid VkImage handle
- VUID-VkCopyImageInfo2-srcImageLayout-parameter
  srcImageLayout **must** be a valid VkImageLayout value
- VUID-VkCopyImageInfo2-dstImage-parameter
  dstImage **must** be a valid VkImage handle
- VUID-VkCopyImageInfo2-dstImageLayout-parameter
  dstImageLayout **must** be a valid VkImageLayout value
- VUID-VkCopyImageInfo2-pRegions-parameter
  pRegions **must** be a valid pointer to an array of regionCount valid VkImageCopy2 structures
- VUID-VkCopyImageInfo2-regionCount-arraylength
  regionCount **must** be greater than 0
- VUID-VkCopyImageInfo2-commonparent
  Both of dstImage, and srcImage **must** have been created, allocated, or retrieved from the same VkDevice

The VkImageCopy2 structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkImageCopy2 {
    VkStructureType             sType;
    const void*                 pNext;
    VkImageSubresourceLayers    srcSubresource;
    VkOffset3D                  srcOffset;
    VkImageSubresourceLayers    dstSubresource;
    VkOffset3D                  dstOffset;
    VkExtent3D                  extent;
} VkImageCopy2;
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to VkImageCopy2
typedef VkImageCopy2 VkImageCopy2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcSubresource` and `dstSubresource` are VkImageSubresourceLayers structures specifying the image subresources of the images used for the source and destination image data, respectively.
- `srcOffset` and `dstOffset` select the initial `x`, `y`, and `z` offsets in texels of the sub-regions of the source and destination image data.
- `extent` is the size in texels of the image to copy in `width`, `height` and `depth`.

Valid Usage

- VUID-VkImageCopy2-apiVersion-07940
  If the VK_KHR_sampler_ycbcr_conversion extension is not enabled, and VkPhysicalDeviceProperties::`apiVersion` is less than Vulkan 1.1, the `aspectMask` member of `srcSubresource` and `dstSubresource` **must** match
- VUID-VkImageCopy2-apiVersion-07941
  If the VK_KHR_maintenance1 extension is not enabled, and VkPhysicalDeviceProperties ::`apiVersion` is less than Vulkan 1.1, the `layerCount` member of `srcSubresource` and `dstSubresource` **must** match
- VUID-VkImageCopy2-extent-06668
  `extent.width` **must** not be 0
- VUID-VkImageCopy2-extent-06669
  `extent.height` **must** not be 0
- VUID-VkImageCopy2-extent-06670
  `extent.depth` **must** not be 0

Valid Usage (Implicit)

- VUID-VkImageCopy2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_COPY_2
- VUID-VkImageCopy2-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkImageCopy2-srcSubresource-parameter
  `srcSubresource` **must** be a valid VkImageSubresourceLayers structure
- VUID-VkImageCopy2-dstSubresource-parameter
  `dstSubresource` **must** be a valid VkImageSubresourceLayers structure

24.4. Copying Data Between Memory Ranges and Images

To copy data from an image to memory ranges, call:

```
// Provided by VK_KHR_device_address_commands
void vkCmdCopyMemoryToImageKHR(
    VkCommandBuffer                             commandBuffer,
    const VkCopyDeviceMemoryImageInfoKHR*       pCopyMemoryInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pCopyMemoryInfo` is a pointer to a VkCopyDeviceMemoryImageInfoKHR structure describing the copies to perform.

Valid Usage

- VUID-vkCmdCopyMemoryToImageKHR-commandBuffer-13102
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, the `addressCopyFlags` member of all elements of `pCopyMemoryInfo->pRegions` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-vkCmdCopyMemoryToImageKHR-commandBuffer-13103
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `pCopyMemoryInfo->image` **must** not have been created with VK_IMAGE_CREATE_PROTECTED_BIT
- VUID-vkCmdCopyMemoryToImageKHR-commandBuffer-13104
  If the `maintenance11` feature is not enabled, and the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT, the `addressRange.pname`:address member of any element of `pCopyMemoryInfo->pRegions` **must** be a multiple of 4
- VUID-vkCmdCopyMemoryToImageKHR-imageOffset-13105
  The `imageOffset` and `imageExtent` members of each element of `pCopyMemoryInfo->pRegions`

must respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in VkQueueFamilyProperties

- VUID-vkCmdCopyMemoryToImageKHR-commandBuffer-13106
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT, for each element of `pCopyMemoryInfo->pRegions`, the `aspectMask` member of `imageSubresource` **must** not be VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkCmdCopyMemoryToImageKHR-imageLayout-13019
  The `imageLayout` member of each element of `pCopyMemoryInfo->pRegions` **must** be VK_IMAGE_LAYOUT_GENERAL, VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, or VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL
- VUID-vkCmdCopyMemoryToImageKHR-pCopyMemoryInfo-13020
  `pCopyMemoryInfo->image` **must** have been created with VK_IMAGE_USAGE_TRANSFER_DST_BIT
- VUID-vkCmdCopyMemoryToImageKHR-commandBuffer-13021
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, `pCopyMemoryInfo->image` **must** have been created with VK_IMAGE_CREATE_PROTECTED_BIT
- VUID-vkCmdCopyMemoryToImageKHR-None-13022
  If VK_EXT_depth_range_unrestricted is not enabled, for each element of `pCopyMemoryInfo->pRegions` whose `imageSubresource` contains a depth aspect, each data element in its `addressRange` **must** be in the range [0,1]
- VUID-vkCmdCopyMemoryToImageKHR-addressRange-13128
  At least one buffer containing the total address range specified by the `addressRange` member of each element of `pCopyMemoryInfo->pRegions` **must** have been created with VK_BUFFER_USAGE_TRANSFER_SRC_BIT

Valid Usage (Implicit)

- VUID-vkCmdCopyMemoryToImageKHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyMemoryToImageKHR-pCopyMemoryInfo-parameter
  If `pCopyMemoryInfo` is not `NULL`, `pCopyMemoryInfo` **must** be a valid pointer to a valid VkCopyDeviceMemoryImageInfoKHR structure
- VUID-vkCmdCopyMemoryToImageKHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdCopyMemoryToImageKHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_TRANSFER_BIT operations
- VUID-vkCmdCopyMemoryToImageKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyMemoryToImageKHR-suspended

This command **must** not be called between suspended render pass instances

- VUID-vkCmdCopyMemoryToImageKHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdCopyMemoryToImageKHR is not affected by conditional rendering

To copy data from an image to memory ranges, call:

```
// Provided by VK_KHR_device_address_commands
void vkCmdCopyImageToMemoryKHR(
    VkCommandBuffer                             commandBuffer,
    const VkCopyDeviceMemoryImageInfoKHR*       pCopyMemoryInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pCopyMemoryInfo` is a pointer to a VkCopyDeviceMemoryImageInfoKHR structure describing the copies to perform.

Valid Usage

- VUID-vkCmdCopyImageToMemoryKHR-commandBuffer-13102
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, the `addressCopyFlags` member of all elements of `pCopyMemoryInfo->pRegions` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-vkCmdCopyImageToMemoryKHR-commandBuffer-13103
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `pCopyMemoryInfo->image` **must** not have been created with

VK_IMAGE_CREATE_PROTECTED_BIT

- VUID-vkCmdCopyImageToMemoryKHR-commandBuffer-13104
  If the maintenance11 feature is not enabled, and the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT, the addressRange.pname:address member of any element of pCopyMemoryInfo->pRegions **must** be a multiple of 4
- VUID-vkCmdCopyImageToMemoryKHR-imageOffset-13105
  The imageOffset and imageExtent members of each element of pCopyMemoryInfo->pRegions **must** respect the image transfer granularity requirements of commandBuffer's command pool's queue family, as described in VkQueueFamilyProperties
- VUID-vkCmdCopyImageToMemoryKHR-commandBuffer-13106
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT, for each element of pCopyMemoryInfo->pRegions, the aspectMask member of imageSubresource **must** not be VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkCmdCopyImageToMemoryKHR-imageLayout-13023
  The imageLayout member of each element of pCopyMemoryInfo->pRegions **must** be VK_IMAGE_LAYOUT_GENERAL, VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, or VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL
- VUID-vkCmdCopyImageToMemoryKHR-pCopyMemoryInfo-13024
  pCopyMemoryInfo->image **must** have been created with VK_IMAGE_USAGE_TRANSFER_SRC_BIT
- VUID-vkCmdCopyImageToMemoryKHR-commandBuffer-13025
  If commandBuffer is a protected command buffer and protectedNoFault is not supported, the addressFlags member of all elements of pCopyMemoryInfo->pRegions **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-vkCmdCopyImageToMemoryKHR-addressRange-13129
  At least one buffer containing the total address range specified by the addressRange member of each element of pCopyMemoryInfo->pRegions **must** have been created with VK_BUFFER_USAGE_TRANSFER_DST_BIT

Valid Usage (Implicit)

- VUID-vkCmdCopyImageToMemoryKHR-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyImageToMemoryKHR-pCopyMemoryInfo-parameter
  If pCopyMemoryInfo is not NULL, pCopyMemoryInfo **must** be a valid pointer to a valid VkCopyDeviceMemoryImageInfoKHR structure
- VUID-vkCmdCopyImageToMemoryKHR-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdCopyImageToMemoryKHR-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support

VK_QUEUE_TRANSFER_BIT operations

- VUID-vkCmdCopyImageToMemoryKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyImageToMemoryKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyImageToMemoryKHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdCopyImageToMemoryKHR is not affected by conditional rendering

The `VkCopyDeviceMemoryImageInfoKHR` structure is defined as:

```
// Provided by VK_KHR_device_address_commands
typedef struct VkCopyDeviceMemoryImageInfoKHR {
    VkStructureType                       sType;
    const void*                           pNext;
    VkImage                               image;
    uint32_t                              regionCount;
    const VkDeviceMemoryImageCopyKHR*     pRegions;
} VkCopyDeviceMemoryImageInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `image` is the base image accessed by the copy command.
- `regionCount` is the number of copies to be performed.

- `pRegions` is a pointer to an array of VkDeviceMemoryImageCopyKHR structures describing individual copy operations between an image and a memory range.

Valid Usage

- VUID-VkCopyDeviceMemoryImageInfoKHR-addressRange-13027
  Each memory location accessed by this command **must** not be accessed by both a source region and a destination region, as specified in `pRegions` and accessed according to Buffer and Image Addressing
- VUID-VkCopyDeviceMemoryImageInfoKHR-pRegions-12473
  Each memory location accessed by this command **must** not be accessed by more than one destination region, as specified by `pRegions` and accessed according to Buffer and Image Addressing
- VUID-VkCopyDeviceMemoryImageInfoKHR-imageLayout-13028
  The `imageLayout` member of each element of `pRegions` **must** specify the layout of the image subresource specified for that element of `pRegions` at the time this command is executed on a VkDevice
- VUID-VkCopyDeviceMemoryImageInfoKHR-image-07966
  If `image` is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkCopyDeviceMemoryImageInfoKHR-imageSubresource-07967
  The `imageSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `image` was created
- VUID-VkCopyDeviceMemoryImageInfoKHR-imageSubresource-07968
  If `imageSubresource.layerCount` is not VK_REMAINING_ARRAY_LAYERS, `imageSubresource.baseArrayLayer` + `imageSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in VkImageCreateInfo when `image` was created
- VUID-VkCopyDeviceMemoryImageInfoKHR-image-07969
  `image` **must** not have been created with `flags` containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-VkCopyDeviceMemoryImageInfoKHR-image-07973
  `image` **must** have a sample count equal to VK_SAMPLE_COUNT_1_BIT
- VUID-VkCopyDeviceMemoryImageInfoKHR-image-07979
  If `image` is of type VK_IMAGE_TYPE_1D, then for each element of `pRegions`, `imageOffset.y` **must** be `0` and `imageExtent.height` **must** be `1`
- VUID-VkCopyDeviceMemoryImageInfoKHR-imageOffset-09104
  For each element of `pRegions`, `imageOffset.z` and (`imageExtent.depth` + `imageOffset.z`) **must** both be greater than or equal to `0` and less than or equal to the depth of the specified `imageSubresource` of `image`
- VUID-VkCopyDeviceMemoryImageInfoKHR-image-07980
  If `image` is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of

pRegions, imageOffset.z **must** be 0 and imageExtent.depth **must** be 1

- VUID-VkCopyDeviceMemoryImageInfoKHR-image-07274
  For each element of pRegions, imageOffset.x **must** be a multiple of the texel block extent width of the VkFormat of image
- VUID-VkCopyDeviceMemoryImageInfoKHR-image-07275
  For each element of pRegions, imageOffset.y **must** be a multiple of the texel block extent height of the VkFormat of image
- VUID-VkCopyDeviceMemoryImageInfoKHR-image-07276
  For each element of pRegions, imageOffset.z **must** be a multiple of the texel block extent depth of the VkFormat of image
- VUID-VkCopyDeviceMemoryImageInfoKHR-image-00207
  For each element of pRegions, if the sum of imageOffset.x and extent.width does not equal the width of the subresource specified by imageSubresource, extent.width **must** be a multiple of the texel block extent width of the VkFormat of image
- VUID-VkCopyDeviceMemoryImageInfoKHR-image-00208
  For each element of pRegions, if the sum of imageOffset.y and extent.height does not equal the height of the subresource specified by imageSubresource, extent.height **must** be a multiple of the texel block extent height of the VkFormat of image
- VUID-VkCopyDeviceMemoryImageInfoKHR-image-00209
  For each element of pRegions, if the sum of imageOffset.z and extent.depth does not equal the depth of the subresource specified by srcSubresource, extent.depth **must** be a multiple of the texel block extent depth of the VkFormat of image
- VUID-VkCopyDeviceMemoryImageInfoKHR-imageSubresource-09105
  For each element of pRegions, imageSubresource.aspectMask **must** specify aspects present in image
- VUID-VkCopyDeviceMemoryImageInfoKHR-image-07981
  If image has a multi-planar format, then for each element of pRegions, imageSubresource.aspectMask **must** be a single valid multi-planar aspect mask bit
- VUID-VkCopyDeviceMemoryImageInfoKHR-image-07983
  If image is of type VK_IMAGE_TYPE_3D, for each element of pRegions, imageSubresource.baseArrayLayer **must** be 0 and imageSubresource.layerCount **must** be 1
- VUID-VkCopyDeviceMemoryImageInfoKHR-addressRowLength-09106
  For each element of pRegions, addressRowLength **must** be a multiple of the texel block extent width of the VkFormat of image
- VUID-VkCopyDeviceMemoryImageInfoKHR-addressImageHeight-09107
  For each element of pRegions, addressImageHeight **must** be a multiple of the texel block extent height of the VkFormat of image
- VUID-VkCopyDeviceMemoryImageInfoKHR-addressRowLength-09108
  For each element of pRegions, addressRowLength divided by the texel block extent width and then multiplied by the texel block size of image **must** be less than or equal to 2^{31}-1
- VUID-VkCopyDeviceMemoryImageInfoKHR-image-13029
  If image does not have either a depth/stencil format or a multi-planar format, then for

each element of pRegions, addressRange.address **must** be a multiple of the texel block size

- VUID-VkCopyDeviceMemoryImageInfoKHR-image-13030
  If image has a multi-planar format, then for each element of pRegions, addressRange.address **must** be a multiple of the element size of the compatible format for the format and the aspectMask of the imageSubresource as defined in Compatible Formats of Planes of Multi-Planar Formats
- VUID-VkCopyDeviceMemoryImageInfoKHR-image-13031
  If image has a depth/stencil format, the addressRange.address member of any element of pRegions **must** be a multiple of 4
- VUID-VkCopyDeviceMemoryImageInfoKHR-pRegions-13032
  The image region specified by each element of pRegions **must** be contained within the specified imageSubresource of image

Valid Usage (Implicit)

- VUID-VkCopyDeviceMemoryImageInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_COPY_DEVICE_MEMORY_IMAGE_INFO_KHR
- VUID-VkCopyDeviceMemoryImageInfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkCopyDeviceMemoryImageInfoKHR-image-parameter
  image **must** be a valid VkImage handle
- VUID-VkCopyDeviceMemoryImageInfoKHR-pRegions-parameter
  pRegions **must** be a valid pointer to an array of regionCount valid VkDeviceMemoryImageCopyKHR structures
- VUID-VkCopyDeviceMemoryImageInfoKHR-regionCount-arraylength
  regionCount **must** be greater than 0

The VkDeviceMemoryImageCopyKHR structure is defined as:

```
// Provided by VK_KHR_device_address_commands
typedef struct VkDeviceMemoryImageCopyKHR {
    VkStructureType             sType;
    const void*                 pNext;
    VkDeviceAddressRangeKHR     addressRange;
    VkAddressCommandFlagsKHR    addressFlags;
    uint32_t                    addressRowLength;
    uint32_t                    addressImageHeight;
    VkImageSubresourceLayers    imageSubresource;
    VkImageLayout               imageLayout;
    VkOffset3D                  imageOffset;
    VkExtent3D                  imageExtent;
} VkDeviceMemoryImageCopyKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `addressRange` is a VkDeviceAddressRangeKHR structure defining the address range accessed by the copy operation.
- `addressFlags` is a VkAddressCommandFlagsKHR value defining the copy flags for the source address range.
- `addressRowLength` specifies the total number of texels in the address range representing a row of texels in the x dimension. Each new row of pixels in the copy will be read at an offset increment by this number of texels.
- `addressImageHeight` specifies the total number of rows of texels in the address range representing a 2D slice of texels in the x and y dimensions. Each new 2D slice of pixels in the copy will be read at an offset increment by this number of rows of texels.
- `imageSubresource` is a VkImageSubresourceLayers used to specify the specific image subresources of the image used for the source or destination image data.
- `imageLayout` is the VkImageLayout of the image subresource accessed by this copy.
- `imageOffset` selects the initial `x`, `y`, `z` offsets in texels of the sub-region of the source or destination image data.
- `imageExtent` is the size in texels of the image to copy in `width`, `height` and `depth`.

This structure defines a copy operation where `srcRange.size` bytes will be copied from `srcRange.address` to `dstRange.address`.

Valid Usage

- VUID-VkDeviceMemoryImageCopyKHR-addressRange-13097
  If the range specified by `addressRange` is not bound completely to memory when accessed, `addressFlags` **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR
- VUID-VkDeviceMemoryImageCopyKHR-addressRange-13098
  If the buffer from which the range specified by `addressRange` was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `addressFlags` **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkDeviceMemoryImageCopyKHR-addressRange-13099
  If the buffer from which the range specified by `addressRange` was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `addressFlags` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkDeviceMemoryImageCopyKHR-addressFlags-13100
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDeviceMemoryImageCopyKHR-addressRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR

or VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR

- VUID-VkDeviceMemoryImageCopyKHR-addressRange-13123
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDeviceMemoryImageCopyKHR-addressFlags-13101
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDeviceMemoryImageCopyKHR-addressRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `addressFlags` **must** include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDeviceMemoryImageCopyKHR-addressRange-13125
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created without VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `addressFlags` **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDeviceMemoryImageCopyKHR-size-13037
  The `size` member of `addressRange` **must** be large enough to contain all address locations that are accessed by this copy according to Buffer and Image Addressing
- VUID-VkDeviceMemoryImageCopyKHR-addressRowLength-09101
  `addressRowLength` **must** be `0`, or greater than or equal to the `width` member of `imageExtent`
- VUID-VkDeviceMemoryImageCopyKHR-addressImageHeight-09102
  `addressImageHeight` **must** be `0`, or greater than or equal to the `height` member of `imageExtent`
- VUID-VkDeviceMemoryImageCopyKHR-aspectMask-09103
  The `aspectMask` member of `imageSubresource` **must** only have a single bit set
- VUID-VkDeviceMemoryImageCopyKHR-imageExtent-06659
  `imageExtent.width` **must** not be 0
- VUID-VkDeviceMemoryImageCopyKHR-imageExtent-06660
  `imageExtent.height` **must** not be 0
- VUID-VkDeviceMemoryImageCopyKHR-imageExtent-06661
  `imageExtent.depth` **must** not be 0

Valid Usage (Implicit)

- VUID-VkDeviceMemoryImageCopyKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_MEMORY_IMAGE_COPY_KHR
- VUID-VkDeviceMemoryImageCopyKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDeviceMemoryImageCopyKHR-addressFlags-parameter
  If `addressFlags` is not `0`, `addressFlags` **must** be a valid combination of VkAddressCommandFlagBitsKHR values
- VUID-VkDeviceMemoryImageCopyKHR-imageSubresource-parameter
  `imageSubresource` **must** be a valid VkImageSubresourceLayers structure
- VUID-VkDeviceMemoryImageCopyKHR-imageLayout-parameter
  `imageLayout` **must** be a valid VkImageLayout value

24.5. Copying Data Between Buffers and Images

Data **can** be copied between buffers and images, enabling applications to load and store data between images and application-defined offsets in buffer memory.

When copying between an address or a buffer and an image, texels in the image and bytes in memory are accessed as follows.

Texels at each coordinate (x,y,z,layer) in an image subresource are accessed, where:

x is in the range [`imageOffset.x`, `imageOffset.x` + `imageExtent.width`),

y is in the range [`imageOffset.y`, `imageOffset.y` + `imageExtent.height`),

z is in the range [`imageOffset.z`, `imageOffset.z` + `imageExtent.depth`),

layer is in the range [`imageSubresource.baseArrayLayer`, `imageSubresource.baseArrayLayer` + `imageSubresource.layerCount`)

For each (x,y,z,layer) coordinate in the image, bytes in memory are accessed at offsets in the range [texelOffset, texelOffset + blockSize), where:

texelOffset = baseAddress + (⌊(x - imageOffset.x) / blockWidth⌋ × blockSize) + (⌊(y - imageOffset.y) / blockHeight⌋ × rowExtent) + (⌊(z - imageOffset.z) / blockDepth⌋ × sliceExtent) + ((layer - `imageSubresource.baseArrayLayer`) × layerExtent)

rowExtent = ⌈ max(`bufferRowLength`, `imageExtent.width`) / blockWidth ⌉ × blockSize

sliceExtent = ⌈ max(`bufferImageHeight`, `imageExtent.height`) / blockHeight ⌉ × rowExtent

layerExtent = ⌈ `imageExtent.depth` / blockDepth ⌉ × sliceExtent

and where blockSize, blockWidth, blockHeight, and blockDepth are the texel block size and extents of the image's format, and baseAddress is either the supplied address for the copy, or the address of the buffer plus any supplied `bufferOffset` value.

When copying between a buffer and the depth or stencil aspect of an image, data in the buffer is assumed to be laid out as separate planes rather than interleaved. Addressing calculations are thus performed for a different format than the base image, according to the aspect, as described in the following table:

Table 30. Depth/Stencil Aspect Copy

Base Format	Depth Aspect Format	Stencil Aspect Format
VK_FORMAT_D16_UNORM	VK_FORMAT_D16_UNORM	-
VK_FORMAT_X8_D24_UNORM_PACK32	VK_FORMAT_X8_D24_UNORM_PACK32	-
VK_FORMAT_D32_SFLOAT	VK_FORMAT_D32_SFLOAT	-
VK_FORMAT_S8_UINT	-	VK_FORMAT_S8_UINT
VK_FORMAT_D16_UNORM_S8_UINT	VK_FORMAT_D16_UNORM	VK_FORMAT_S8_UINT
VK_FORMAT_D24_UNORM_S8_UINT	VK_FORMAT_X8_D24_UNORM_PACK32	VK_FORMAT_S8_UINT
VK_FORMAT_D32_SFLOAT_S8_UINT	VK_FORMAT_D32_SFLOAT	VK_FORMAT_S8_UINT

When copying between a buffer and any plane of a multi-planar image, addressing calculations are performed using the compatible format for that plane, rather than the format of the multi-planar image.

Each texel block is copied from one resource to the other according to the above addressing equations.

The memory locations accessed by a buffer and image overlap if they alias each other, the `imageSubresource` overlaps the aliased range, and either: * their contents are not interpreted in consistent ways, or * their accessed regions defined above overlap, where the memory locations accessed by the texels of a linear image are according to its VkSubresourceLayout

The memory locations of an image overlap if those two ranges share any of the same texel coordinates in the same image aspects.

To copy data from a buffer object to an image object, call:

```
// Provided by VK_VERSION_1_0
void vkCmdCopyBufferToImage(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    srcBuffer,
    VkImage                                     dstImage,
    VkImageLayout                               dstImageLayout,
    uint32_t                                    regionCount,
    const VkBufferImageCopy*                    pRegions);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `srcBuffer` is the source buffer.
- `dstImage` is the destination image.
- `dstImageLayout` is the layout of the destination image subresources for the copy.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of VkBufferImageCopy structures specifying the regions to copy.

Each source region specified by `pRegions` is copied from the source buffer to the destination region of the destination image according to the addressing calculations for each resource. If any region accesses a depth aspect in `dstImage` and the `VK_EXT_depth_range_unrestricted` extension is not enabled, values copied from `srcBuffer` outside of the range [0,1] will be written as undefined values to the destination image.

Copy regions for the image **must** be aligned to a multiple of the texel block extent in each dimension, except at the edges of the image, where region extents **must** match the edge of the image.

Valid Usage

- VUID-vkCmdCopyBufferToImage-dstImage-07966
  If `dstImage` is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdCopyBufferToImage-imageSubresource-07967
  The `imageSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `dstImage` was created
- VUID-vkCmdCopyBufferToImage-imageSubresource-07968
  If `imageSubresource.layerCount` is not VK_REMAINING_ARRAY_LAYERS, `imageSubresource.baseArrayLayer + imageSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in VkImageCreateInfo when `dstImage` was created
- VUID-vkCmdCopyBufferToImage-dstImage-07969
  `dstImage` **must** not have been created with `flags` containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT

- VUID-vkCmdCopyBufferToImage-imageSubresource-07971
  For each element of pRegions, imageOffset.x and (imageExtent.width + imageOffset.x) **must** both be greater than or equal to 0 and less than or equal to the width of the specified imageSubresource of dstImage
- VUID-vkCmdCopyBufferToImage-imageSubresource-07972
  For each element of pRegions, imageOffset.y and (imageExtent.height + imageOffset.y) **must** both be greater than or equal to 0 and less than or equal to the height of the specified imageSubresource of dstImage
- VUID-vkCmdCopyBufferToImage-dstImage-07973
  dstImage **must** have a sample count equal to VK_SAMPLE_COUNT_1_BIT
- VUID-vkCmdCopyBufferToImage-commandBuffer-01828
  If commandBuffer is an unprotected command buffer and protectedNoFault is not supported, srcBuffer **must** not be a protected buffer
- VUID-vkCmdCopyBufferToImage-commandBuffer-01829
  If commandBuffer is an unprotected command buffer and protectedNoFault is not supported, dstImage **must** not be a protected image
- VUID-vkCmdCopyBufferToImage-commandBuffer-01830
  If commandBuffer is a protected command buffer and protectedNoFault is not supported, dstImage **must** not be an unprotected image
- VUID-vkCmdCopyBufferToImage-commandBuffer-07737
  If the maintenance11 feature is not enabled, and the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT, the bufferOffset member of any element of pRegions **must** be a multiple of 4
- VUID-vkCmdCopyBufferToImage-imageOffset-07738
  The imageOffset and imageExtent members of each element of pRegions **must** respect the image transfer granularity requirements of commandBuffer's command pool's queue family, as described in VkQueueFamilyProperties
- VUID-vkCmdCopyBufferToImage-commandBuffer-07739
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT, and the maintenance10 feature is not enabled, for each element of pRegions, the aspectMask member of imageSubresource **must** not be VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkCmdCopyBufferToImage-commandBuffer-11778
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support VK_QUEUE_COMPUTE_BIT, and in any element of pRegions the aspectMask member of imageSubresource is VK_IMAGE_ASPECT_DEPTH_BIT, then the format features of dstImage **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_COMPUTE_QUEUE_BIT_KHR
- VUID-vkCmdCopyBufferToImage-commandBuffer-11779
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of pRegions the aspectMask

member of imageSubresource is VK_IMAGE_ASPECT_DEPTH_BIT, then the format features of dstImage **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_TRANSFER_QUEUE_BIT_KHR

- VUID-vkCmdCopyBufferToImage-commandBuffer-11780
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support VK_QUEUE_COMPUTE_BIT, and in any element of pRegions the aspectMask member of imageSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of dstImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_COMPUTE_QUEUE_BIT_KHR
- VUID-vkCmdCopyBufferToImage-commandBuffer-11781
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of pRegions the aspectMask member of imageSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of dstImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_TRANSFER_QUEUE_BIT_KHR
- VUID-vkCmdCopyBufferToImage-pRegions-00171
  srcBuffer **must** be large enough to contain all buffer locations that are accessed according to Buffer and Image Addressing, for each element of pRegions
- VUID-vkCmdCopyBufferToImage-pRegions-00173
  Each memory location accessed by this command **must** not be accessed by both a source region and a destination region, as specified in pRegions and accessed according to Buffer and Image Addressing
- VUID-vkCmdCopyBufferToImage-pRegions-12483
  Each memory location accessed by this command **must** not be accessed by more than one destination region, as specified by pRegions and accessed according to Buffer and Image Addressing
- VUID-vkCmdCopyBufferToImage-srcBuffer-00174
  srcBuffer **must** have been created with the VK_BUFFER_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-vkCmdCopyBufferToImage-dstImage-01997
  The format features of dstImage **must** contain VK_FORMAT_FEATURE_TRANSFER_DST_BIT
- VUID-vkCmdCopyBufferToImage-srcBuffer-00176
  If srcBuffer is non-sparse then it **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-vkCmdCopyBufferToImage-dstImage-00177
  dstImage **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-vkCmdCopyBufferToImage-dstImageLayout-00180
  dstImageLayout **must** specify the layout of the image subresources of dstImage specified in pRegions at the time this command is executed on a VkDevice
- VUID-vkCmdCopyBufferToImage-dstImageLayout-01396
  dstImageLayout **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR,

VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL, or VK_IMAGE_LAYOUT_GENERAL

- VUID-vkCmdCopyBufferToImage-pRegions-07931
  If VK_EXT_depth_range_unrestricted is not enabled, for each element of pRegions whose imageSubresource contains a depth aspect, the data in srcBuffer **must** be in the range [0,1]
- VUID-vkCmdCopyBufferToImage-dstImage-07979
  If dstImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, imageOffset.y **must** be 0 and imageExtent.height **must** be 1
- VUID-vkCmdCopyBufferToImage-imageOffset-09104
  For each element of pRegions, imageOffset.z and (imageExtent.depth + imageOffset.z) **must** both be greater than or equal to 0 and less than or equal to the depth of the specified imageSubresource of dstImage
- VUID-vkCmdCopyBufferToImage-dstImage-07980
  If dstImage is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of pRegions, imageOffset.z **must** be 0 and imageExtent.depth **must** be 1
- VUID-vkCmdCopyBufferToImage-dstImage-07274
  For each element of pRegions, imageOffset.x **must** be a multiple of the texel block extent width of the VkFormat of dstImage
- VUID-vkCmdCopyBufferToImage-dstImage-07275
  For each element of pRegions, imageOffset.y **must** be a multiple of the texel block extent height of the VkFormat of dstImage
- VUID-vkCmdCopyBufferToImage-dstImage-07276
  For each element of pRegions, imageOffset.z **must** be a multiple of the texel block extent depth of the VkFormat of dstImage
- VUID-vkCmdCopyBufferToImage-dstImage-00207
  For each element of pRegions, if the sum of imageOffset.x and extent.width does not equal the width of the subresource specified by imageSubresource, extent.width **must** be a multiple of the texel block extent width of the VkFormat of dstImage
- VUID-vkCmdCopyBufferToImage-dstImage-00208
  For each element of pRegions, if the sum of imageOffset.y and extent.height does not equal the height of the subresource specified by imageSubresource, extent.height **must** be a multiple of the texel block extent height of the VkFormat of dstImage
- VUID-vkCmdCopyBufferToImage-dstImage-00209
  For each element of pRegions, if the sum of imageOffset.z and extent.depth does not equal the depth of the subresource specified by srcSubresource, extent.depth **must** be a multiple of the texel block extent depth of the VkFormat of dstImage
- VUID-vkCmdCopyBufferToImage-imageSubresource-09105
  For each element of pRegions, imageSubresource.aspectMask **must** specify aspects present in dstImage
- VUID-vkCmdCopyBufferToImage-dstImage-07981
  If dstImage has a multi-planar format, then for each element of pRegions, imageSubresource.aspectMask **must** be a single valid multi-planar aspect mask bit
- VUID-vkCmdCopyBufferToImage-dstImage-07983

If `dstImage` is of type VK_IMAGE_TYPE_3D, for each element of `pRegions`, `imageSubresource.baseArrayLayer` **must** be `0` and `imageSubresource.layerCount` **must** be `1`

- VUID-vkCmdCopyBufferToImage-bufferRowLength-09106
  For each element of `pRegions`, `bufferRowLength` **must** be a multiple of the texel block extent width of the VkFormat of `dstImage`
- VUID-vkCmdCopyBufferToImage-bufferImageHeight-09107
  For each element of `pRegions`, `bufferImageHeight` **must** be a multiple of the texel block extent height of the VkFormat of `dstImage`
- VUID-vkCmdCopyBufferToImage-bufferRowLength-09108
  For each element of `pRegions`, `bufferRowLength` divided by the texel block extent width and then multiplied by the texel block size of `dstImage` **must** be less than or equal to 2^{31}-1
- VUID-vkCmdCopyBufferToImage-dstImage-07975
  If `dstImage` does not have either a depth/stencil format or a multi-planar format, then for each element of `pRegions`, `bufferOffset` **must** be a multiple of the texel block size
- VUID-vkCmdCopyBufferToImage-dstImage-07976
  If `dstImage` has a multi-planar format, then for each element of `pRegions`, `bufferOffset` **must** be a multiple of the element size of the compatible format for the format and the `aspectMask` of the `imageSubresource` as defined in Compatible Formats of Planes of Multi-Planar Formats
- VUID-vkCmdCopyBufferToImage-dstImage-07978
  If `dstImage` has a depth/stencil format, the `bufferOffset` member of any element of `pRegions` **must** be a multiple of `4`

Valid Usage (Implicit)

- VUID-vkCmdCopyBufferToImage-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyBufferToImage-srcBuffer-parameter
  `srcBuffer` **must** be a valid VkBuffer handle
- VUID-vkCmdCopyBufferToImage-dstImage-parameter
  `dstImage` **must** be a valid VkImage handle
- VUID-vkCmdCopyBufferToImage-dstImageLayout-parameter
  `dstImageLayout` **must** be a valid VkImageLayout value
- VUID-vkCmdCopyBufferToImage-pRegions-parameter
  `pRegions` **must** be a valid pointer to an array of `regionCount` valid VkBufferImageCopy structures
- VUID-vkCmdCopyBufferToImage-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdCopyBufferToImage-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, or VK_QUEUE_TRANSFER_BIT

operations

- VUID-vkCmdCopyBufferToImage-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyBufferToImage-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyBufferToImage-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdCopyBufferToImage-regionCount-arraylength
  `regionCount` **must** be greater than `0`
- VUID-vkCmdCopyBufferToImage-commonparent
  Each of `commandBuffer`, `dstImage`, and `srcBuffer` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdCopyBufferToImage is not affected by conditional rendering

To copy data from an image object to a buffer object, call:

```
// Provided by VK_VERSION_1_0
void vkCmdCopyImageToBuffer(
    VkCommandBuffer                             commandBuffer,
    VkImage                                     srcImage,
    VkImageLayout                               srcImageLayout,
    VkBuffer                                    dstBuffer,
    uint32_t                                    regionCount,
    const VkBufferImageCopy*                    pRegions);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `srcImage` is the source image.
- `srcImageLayout` is the layout of the source image subresources for the copy.
- `dstBuffer` is the destination buffer.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of VkBufferImageCopy structures specifying the regions to copy.

Each source region specified by `pRegions` is copied from the source image to the destination region of the destination buffer according to the addressing calculations for each resource.

Copy regions for the image **must** be aligned to a multiple of the texel block extent in each dimension, except at the edges of the image, where region extents **must** match the edge of the image.

Valid Usage

- VUID-vkCmdCopyImageToBuffer-srcImage-07966
  If `srcImage` is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdCopyImageToBuffer-imageSubresource-07967
  The `imageSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `srcImage` was created
- VUID-vkCmdCopyImageToBuffer-imageSubresource-07968
  If `imageSubresource.layerCount` is not VK_REMAINING_ARRAY_LAYERS, `imageSubresource.baseArrayLayer` + `imageSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in VkImageCreateInfo when `srcImage` was created
- VUID-vkCmdCopyImageToBuffer-srcImage-07969
  `srcImage` **must** not have been created with `flags` containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-vkCmdCopyImageToBuffer-imageSubresource-07971
  For each element of `pRegions`, `imageOffset.x` and (`imageExtent.width` + `imageOffset.x`) **must** both be greater than or equal to `0` and less than or equal to the width of the specified `imageSubresource` of `srcImage`

- VUID-vkCmdCopyImageToBuffer-imageSubresource-07972
  For each element of `pRegions`, `imageOffset.y` and (`imageExtent.height` + `imageOffset.y`) **must** both be greater than or equal to `0` and less than or equal to the height of the specified `imageSubresource` of `srcImage`

- VUID-vkCmdCopyImageToBuffer-srcImage-07973
  `srcImage` **must** have a sample count equal to VK_SAMPLE_COUNT_1_BIT

- VUID-vkCmdCopyImageToBuffer-commandBuffer-01831
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `srcImage` **must** not be a protected image

- VUID-vkCmdCopyImageToBuffer-commandBuffer-01832
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `dstBuffer` **must** not be a protected buffer

- VUID-vkCmdCopyImageToBuffer-commandBuffer-01833
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, `dstBuffer` **must** not be an unprotected buffer

- VUID-vkCmdCopyImageToBuffer-commandBuffer-07746
  If the `maintenance11` feature is not enabled, and the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT, the `bufferOffset` member of any element of `pRegions` **must** be a multiple of `4`

- VUID-vkCmdCopyImageToBuffer-imageOffset-07747
  The `imageOffset` and `imageExtent` members of each element of `pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in VkQueueFamilyProperties

- VUID-vkCmdCopyImageToBuffer-commandBuffer-10216
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT, and the `maintenance10` feature is not enabled, for each element of `pRegions`, the `aspectMask` member of `imageSubresource` **must** not be VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT

- VUID-vkCmdCopyImageToBuffer-commandBuffer-11790
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support VK_QUEUE_COMPUTE_BIT, and in any element of `pRegions` the `aspectMask` member of `imageSubresource` is VK_IMAGE_ASPECT_DEPTH_BIT, then the format features of `srcImage` **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_COMPUTE_QUEUE_BIT_KHR

- VUID-vkCmdCopyImageToBuffer-commandBuffer-11791
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of `pRegions` the `aspectMask` member of `imageSubresource` is VK_IMAGE_ASPECT_DEPTH_BIT, then the format features of `srcImage` **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_TRANSFER_QUEUE_BIT_KHR

- VUID-vkCmdCopyImageToBuffer-commandBuffer-11792

If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support VK_QUEUE_COMPUTE_BIT, and in any element of pRegions the aspectMask member of imageSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of srcImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_COMPUTE_QUEUE_BIT_KHR

- VUID-vkCmdCopyImageToBuffer-commandBuffer-11793
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of pRegions the aspectMask member of imageSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of srcImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_TRANSFER_QUEUE_BIT_KHR
- VUID-vkCmdCopyImageToBuffer-pRegions-00183
  dstBuffer **must** be large enough to contain all buffer locations that are accessed according to Buffer and Image Addressing, for each element of pRegions
- VUID-vkCmdCopyImageToBuffer-pRegions-00184
  Each memory location accessed by this command **must** not be accessed by both a source region and a destination region, as specified in pRegions and accessed according to Buffer and Image Addressing
- VUID-vkCmdCopyImageToBuffer-pRegions-12485
  Each memory location accessed by this command **must** not be accessed by more than one destination region, as specified by pRegions and accessed according to Buffer and Image Addressing
- VUID-vkCmdCopyImageToBuffer-srcImage-00186
  srcImage **must** have been created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-vkCmdCopyImageToBuffer-srcImage-01998
  The format features of srcImage **must** contain VK_FORMAT_FEATURE_TRANSFER_SRC_BIT
- VUID-vkCmdCopyImageToBuffer-dstBuffer-00191
  dstBuffer **must** have been created with the VK_BUFFER_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-vkCmdCopyImageToBuffer-dstBuffer-00192
  If dstBuffer is non-sparse then it **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-vkCmdCopyImageToBuffer-srcImageLayout-00189
  srcImageLayout **must** specify the layout of the image subresources of srcImage specified in pRegions at the time this command is executed on a VkDevice
- VUID-vkCmdCopyImageToBuffer-srcImageLayout-01397
  srcImageLayout **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL, or VK_IMAGE_LAYOUT_GENERAL
- VUID-vkCmdCopyImageToBuffer-srcImage-07979
  If srcImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, imageOffset.y **must** be 0 and imageExtent.height **must** be 1

- VUID-vkCmdCopyImageToBuffer-imageOffset-09104
  For each element of pRegions, imageOffset.z and (imageExtent.depth + imageOffset.z) **must** both be greater than or equal to 0 and less than or equal to the depth of the specified imageSubresource of srcImage
- VUID-vkCmdCopyImageToBuffer-srcImage-07980
  If srcImage is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of pRegions, imageOffset.z **must** be 0 and imageExtent.depth **must** be 1
- VUID-vkCmdCopyImageToBuffer-srcImage-07274
  For each element of pRegions, imageOffset.x **must** be a multiple of the texel block extent width of the VkFormat of srcImage
- VUID-vkCmdCopyImageToBuffer-srcImage-07275
  For each element of pRegions, imageOffset.y **must** be a multiple of the texel block extent height of the VkFormat of srcImage
- VUID-vkCmdCopyImageToBuffer-srcImage-07276
  For each element of pRegions, imageOffset.z **must** be a multiple of the texel block extent depth of the VkFormat of srcImage
- VUID-vkCmdCopyImageToBuffer-srcImage-00207
  For each element of pRegions, if the sum of imageOffset.x and extent.width does not equal the width of the subresource specified by imageSubresource, extent.width **must** be a multiple of the texel block extent width of the VkFormat of srcImage
- VUID-vkCmdCopyImageToBuffer-srcImage-00208
  For each element of pRegions, if the sum of imageOffset.y and extent.height does not equal the height of the subresource specified by imageSubresource, extent.height **must** be a multiple of the texel block extent height of the VkFormat of srcImage
- VUID-vkCmdCopyImageToBuffer-srcImage-00209
  For each element of pRegions, if the sum of imageOffset.z and extent.depth does not equal the depth of the subresource specified by srcSubresource, extent.depth **must** be a multiple of the texel block extent depth of the VkFormat of srcImage
- VUID-vkCmdCopyImageToBuffer-imageSubresource-09105
  For each element of pRegions, imageSubresource.aspectMask **must** specify aspects present in srcImage
- VUID-vkCmdCopyImageToBuffer-srcImage-07981
  If srcImage has a multi-planar format, then for each element of pRegions, imageSubresource.aspectMask **must** be a single valid multi-planar aspect mask bit
- VUID-vkCmdCopyImageToBuffer-srcImage-07983
  If srcImage is of type VK_IMAGE_TYPE_3D, for each element of pRegions, imageSubresource.baseArrayLayer **must** be 0 and imageSubresource.layerCount **must** be 1
- VUID-vkCmdCopyImageToBuffer-bufferRowLength-09106
  For each element of pRegions, bufferRowLength **must** be a multiple of the texel block extent width of the VkFormat of srcImage
- VUID-vkCmdCopyImageToBuffer-bufferImageHeight-09107
  For each element of pRegions, bufferImageHeight **must** be a multiple of the texel block extent height of the VkFormat of srcImage

- VUID-vkCmdCopyImageToBuffer-bufferRowLength-09108
  For each element of `pRegions`, `bufferRowLength` divided by the texel block extent width and then multiplied by the texel block size of `srcImage` **must** be less than or equal to 2^{31}-1
- VUID-vkCmdCopyImageToBuffer-srcImage-07975
  If `srcImage` does not have either a depth/stencil format or a multi-planar format, then for each element of `pRegions`, `bufferOffset` **must** be a multiple of the texel block size
- VUID-vkCmdCopyImageToBuffer-srcImage-07976
  If `srcImage` has a multi-planar format, then for each element of `pRegions`, `bufferOffset` **must** be a multiple of the element size of the compatible format for the format and the `aspectMask` of the `imageSubresource` as defined in Compatible Formats of Planes of Multi-Planar Formats
- VUID-vkCmdCopyImageToBuffer-srcImage-07978
  If `srcImage` has a depth/stencil format, the `bufferOffset` member of any element of `pRegions` **must** be a multiple of `4`

Valid Usage (Implicit)

- VUID-vkCmdCopyImageToBuffer-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyImageToBuffer-srcImage-parameter
  `srcImage` **must** be a valid VkImage handle
- VUID-vkCmdCopyImageToBuffer-srcImageLayout-parameter
  `srcImageLayout` **must** be a valid VkImageLayout value
- VUID-vkCmdCopyImageToBuffer-dstBuffer-parameter
  `dstBuffer` **must** be a valid VkBuffer handle
- VUID-vkCmdCopyImageToBuffer-pRegions-parameter
  `pRegions` **must** be a valid pointer to an array of `regionCount` valid VkBufferImageCopy structures
- VUID-vkCmdCopyImageToBuffer-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdCopyImageToBuffer-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, or VK_QUEUE_TRANSFER_BIT operations
- VUID-vkCmdCopyImageToBuffer-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyImageToBuffer-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyImageToBuffer-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdCopyImageToBuffer-regionCount-arraylength

`regionCount` **must** be greater than `0`

- VUID-vkCmdCopyImageToBuffer-commonparent
  Each of `commandBuffer`, `dstBuffer`, and `srcImage` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdCopyImageToBuffer is not affected by conditional rendering

For both vkCmdCopyBufferToImage and vkCmdCopyImageToBuffer, each element of `pRegions` is a structure defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBufferImageCopy {
    VkDeviceSize                bufferOffset;
    uint32_t                    bufferRowLength;
    uint32_t                    bufferImageHeight;
    VkImageSubresourceLayers    imageSubresource;
    VkOffset3D                  imageOffset;
    VkExtent3D                  imageExtent;
} VkBufferImageCopy;
```

- `bufferOffset` is the offset in bytes from the start of the buffer object where the image data is copied from or to.
- `bufferRowLength` and `bufferImageHeight` specify in texels a subregion of a larger two- or three-

dimensional image in buffer memory, and control the addressing calculations. If either of these values is zero, that aspect of the buffer memory is considered to be tightly packed according to the `imageExtent`.

- `imageSubresource` is a VkImageSubresourceLayers used to specify the specific image subresources of the image used for the source or destination image data.
- `imageOffset` selects the initial `x`, `y`, `z` offsets in texels of the sub-region of the source or destination image data.
- `imageExtent` is the size in texels of the image to copy in `width`, `height` and `depth`.

Valid Usage

- VUID-VkBufferImageCopy-bufferRowLength-09101
  `bufferRowLength` **must** be `0`, or greater than or equal to the `width` member of `imageExtent`
- VUID-VkBufferImageCopy-bufferImageHeight-09102
  `bufferImageHeight` **must** be `0`, or greater than or equal to the `height` member of `imageExtent`
- VUID-VkBufferImageCopy-aspectMask-09103
  The `aspectMask` member of `imageSubresource` **must** only have a single bit set
- VUID-VkBufferImageCopy-imageExtent-06659
  `imageExtent.width` **must** not be 0
- VUID-VkBufferImageCopy-imageExtent-06660
  `imageExtent.height` **must** not be 0
- VUID-VkBufferImageCopy-imageExtent-06661
  `imageExtent.depth` **must** not be 0

Valid Usage (Implicit)

- VUID-VkBufferImageCopy-imageSubresource-parameter
  `imageSubresource` **must** be a valid VkImageSubresourceLayers structure

More extensible versions of the commands to copy between buffers and images are defined below.

To copy data from a buffer object to an image object, call:

```
// Provided by VK_VERSION_1_3
void vkCmdCopyBufferToImage2(
    VkCommandBuffer                             commandBuffer,
    const VkCopyBufferToImageInfo2*             pCopyBufferToImageInfo);
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to vkCmdCopyBufferToImage2
void vkCmdCopyBufferToImage2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkCopyBufferToImageInfo2*             pCopyBufferToImageInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pCopyBufferToImageInfo` is a pointer to a VkCopyBufferToImageInfo2 structure describing the copy parameters.

This command is functionally identical to vkCmdCopyBufferToImage, but includes extensible sub-structures that include `sType` and `pNext` parameters, allowing them to be more easily extended.

Valid Usage

- VUID-vkCmdCopyBufferToImage2-commandBuffer-01828
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `srcBuffer` **must** not be a protected buffer
- VUID-vkCmdCopyBufferToImage2-commandBuffer-01829
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `dstImage` **must** not be a protected image
- VUID-vkCmdCopyBufferToImage2-commandBuffer-01830
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, `dstImage` **must** not be an unprotected image
- VUID-vkCmdCopyBufferToImage2-commandBuffer-07737
  If the `maintenance11` feature is not enabled, and the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT, the `bufferOffset` member of any element of `pCopyBufferToImageInfo->pRegions` **must** be a multiple of `4`
- VUID-vkCmdCopyBufferToImage2-imageOffset-07738
  The `imageOffset` and `imageExtent` members of each element of `pCopyBufferToImageInfo->pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in VkQueueFamilyProperties
- VUID-vkCmdCopyBufferToImage2-commandBuffer-07739
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT, and the `maintenance10` feature is not enabled, for each element of `pCopyBufferToImageInfo->pRegions`, the `aspectMask` member of `imageSubresource` **must** not be VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkCmdCopyBufferToImage2-commandBuffer-11778
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support VK_QUEUE_COMPUTE_BIT, and in any element of `pCopyBufferToImageInfo->pRegions` the `aspectMask` member of `imageSubresource` is VK_IMAGE_ASPECT_DEPTH_BIT, then the

format features of dstImage **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_COMPUTE_QUEUE_BIT_KHR

- VUID-vkCmdCopyBufferToImage2-commandBuffer-11779
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of pCopyBufferToImageInfo->pRegions the aspectMask member of imageSubresource is VK_IMAGE_ASPECT_DEPTH_BIT, then the format features of dstImage **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_TRANSFER_QUEUE_BIT_KHR
- VUID-vkCmdCopyBufferToImage2-commandBuffer-11780
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support VK_QUEUE_COMPUTE_BIT, and in any element of pCopyBufferToImageInfo->pRegions the aspectMask member of imageSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of dstImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_COMPUTE_QUEUE_BIT_KHR
- VUID-vkCmdCopyBufferToImage2-commandBuffer-11781
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of pCopyBufferToImageInfo->pRegions the aspectMask member of imageSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of dstImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_TRANSFER_QUEUE_BIT_KHR

Valid Usage (Implicit)

- VUID-vkCmdCopyBufferToImage2-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyBufferToImage2-pCopyBufferToImageInfo-parameter
  pCopyBufferToImageInfo **must** be a valid pointer to a valid VkCopyBufferToImageInfo2 structure
- VUID-vkCmdCopyBufferToImage2-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdCopyBufferToImage2-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, or VK_QUEUE_TRANSFER_BIT operations
- VUID-vkCmdCopyBufferToImage2-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyBufferToImage2-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyBufferToImage2-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdCopyBufferToImage2 is not affected by conditional rendering

The `VkCopyBufferToImageInfo2` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkCopyBufferToImageInfo2 {
    VkStructureType             sType;
    const void*                 pNext;
    VkBuffer                    srcBuffer;
    VkImage                     dstImage;
    VkImageLayout               dstImageLayout;
    uint32_t                    regionCount;
    const VkBufferImageCopy2*   pRegions;
} VkCopyBufferToImageInfo2;
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to VkCopyBufferToImageInfo2
typedef VkCopyBufferToImageInfo2 VkCopyBufferToImageInfo2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcBuffer` is the source buffer.

- `dstImage` is the destination image.
- `dstImageLayout` is the layout of the destination image subresources for the copy.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of VkBufferImageCopy2 structures specifying the regions to copy.

Valid Usage

- VUID-VkCopyBufferToImageInfo2-pRegions-04565
  The image region specified by each element of `pRegions` **must** be contained within the specified `imageSubresource` of `dstImage`
- VUID-VkCopyBufferToImageInfo2-pRegions-00171
  `srcBuffer` **must** be large enough to contain all buffer locations that are accessed according to Buffer and Image Addressing, for each element of `pRegions`
- VUID-VkCopyBufferToImageInfo2-pRegions-00173
  Each memory location accessed by this command **must** not be accessed by both a source region and a destination region, as specified in `pRegions` and accessed according to Buffer and Image Addressing
- VUID-VkCopyBufferToImageInfo2-pRegions-12483
  Each memory location accessed by this command **must** not be accessed by more than one destination region, as specified by `pRegions` and accessed according to Buffer and Image Addressing
- VUID-VkCopyBufferToImageInfo2-srcBuffer-00174
  `srcBuffer` **must** have been created with the VK_BUFFER_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-VkCopyBufferToImageInfo2-dstImage-01997
  The format features of `dstImage` **must** contain VK_FORMAT_FEATURE_TRANSFER_DST_BIT
- VUID-VkCopyBufferToImageInfo2-srcBuffer-00176
  If `srcBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkCopyBufferToImageInfo2-dstImage-00177
  `dstImage` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-VkCopyBufferToImageInfo2-dstImageLayout-00180
  `dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-VkCopyBufferToImageInfo2-dstImageLayout-01396
  `dstImageLayout` **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL, or VK_IMAGE_LAYOUT_GENERAL
- VUID-VkCopyBufferToImageInfo2-pRegions-07931
  If VK_EXT_depth_range_unrestricted is not enabled, for each element of `pRegions` whose `imageSubresource` contains a depth aspect, the data in `srcBuffer` **must** be in the range [0,1]

- VUID-VkCopyBufferToImageInfo2-dstImage-07966
  If `dstImage` is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkCopyBufferToImageInfo2-imageSubresource-07967
  The `imageSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `dstImage` was created
- VUID-VkCopyBufferToImageInfo2-imageSubresource-07968
  If `imageSubresource.layerCount` is not VK_REMAINING_ARRAY_LAYERS, `imageSubresource.baseArrayLayer` + `imageSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in VkImageCreateInfo when `dstImage` was created
- VUID-VkCopyBufferToImageInfo2-dstImage-07969
  `dstImage` **must** not have been created with `flags` containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-VkCopyBufferToImageInfo2-dstImage-07973
  `dstImage` **must** have a sample count equal to VK_SAMPLE_COUNT_1_BIT
- VUID-VkCopyBufferToImageInfo2-dstImage-07979
  If `dstImage` is of type VK_IMAGE_TYPE_1D, then for each element of `pRegions`, `imageOffset.y` **must** be `0` and `imageExtent.height` **must** be `1`
- VUID-VkCopyBufferToImageInfo2-imageOffset-09104
  For each element of `pRegions`, `imageOffset.z` and (`imageExtent.depth` + `imageOffset.z`) **must** both be greater than or equal to `0` and less than or equal to the depth of the specified `imageSubresource` of `dstImage`
- VUID-VkCopyBufferToImageInfo2-dstImage-07980
  If `dstImage` is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of `pRegions`, `imageOffset.z` **must** be `0` and `imageExtent.depth` **must** be `1`
- VUID-VkCopyBufferToImageInfo2-dstImage-07274
  For each element of `pRegions`, `imageOffset.x` **must** be a multiple of the texel block extent width of the VkFormat of `dstImage`
- VUID-VkCopyBufferToImageInfo2-dstImage-07275
  For each element of `pRegions`, `imageOffset.y` **must** be a multiple of the texel block extent height of the VkFormat of `dstImage`
- VUID-VkCopyBufferToImageInfo2-dstImage-07276
  For each element of `pRegions`, `imageOffset.z` **must** be a multiple of the texel block extent depth of the VkFormat of `dstImage`
- VUID-VkCopyBufferToImageInfo2-dstImage-00207
  For each element of `pRegions`, if the sum of `imageOffset.x` and `extent.width` does not equal the width of the subresource specified by `imageSubresource`, `extent.width` **must** be a multiple of the texel block extent width of the VkFormat of `dstImage`
- VUID-VkCopyBufferToImageInfo2-dstImage-00208
  For each element of `pRegions`, if the sum of `imageOffset.y` and `extent.height` does not equal the height of the subresource specified by `imageSubresource`, `extent.height` **must** be a

multiple of the texel block extent height of the VkFormat of dstImage

- VUID-VkCopyBufferToImageInfo2-dstImage-00209
  For each element of pRegions, if the sum of imageOffset.z and extent.depth does not equal the depth of the subresource specified by srcSubresource, extent.depth **must** be a multiple of the texel block extent depth of the VkFormat of dstImage
- VUID-VkCopyBufferToImageInfo2-imageSubresource-09105
  For each element of pRegions, imageSubresource.aspectMask **must** specify aspects present in dstImage
- VUID-VkCopyBufferToImageInfo2-dstImage-07981
  If dstImage has a multi-planar format, then for each element of pRegions, imageSubresource.aspectMask **must** be a single valid multi-planar aspect mask bit
- VUID-VkCopyBufferToImageInfo2-dstImage-07983
  If dstImage is of type VK_IMAGE_TYPE_3D, for each element of pRegions, imageSubresource.baseArrayLayer **must** be 0 and imageSubresource.layerCount **must** be 1
- VUID-VkCopyBufferToImageInfo2-bufferRowLength-09106
  For each element of pRegions, bufferRowLength **must** be a multiple of the texel block extent width of the VkFormat of dstImage
- VUID-VkCopyBufferToImageInfo2-bufferImageHeight-09107
  For each element of pRegions, bufferImageHeight **must** be a multiple of the texel block extent height of the VkFormat of dstImage
- VUID-VkCopyBufferToImageInfo2-bufferRowLength-09108
  For each element of pRegions, bufferRowLength divided by the texel block extent width and then multiplied by the texel block size of dstImage **must** be less than or equal to 2^{31}-1
- VUID-VkCopyBufferToImageInfo2-dstImage-07975
  If dstImage does not have either a depth/stencil format or a multi-planar format, then for each element of pRegions, bufferOffset **must** be a multiple of the texel block size
- VUID-VkCopyBufferToImageInfo2-dstImage-07976
  If dstImage has a multi-planar format, then for each element of pRegions, bufferOffset **must** be a multiple of the element size of the compatible format for the format and the aspectMask of the imageSubresource as defined in Compatible Formats of Planes of Multi-Planar Formats
- VUID-VkCopyBufferToImageInfo2-dstImage-07978
  If dstImage has a depth/stencil format, the bufferOffset member of any element of pRegions **must** be a multiple of 4
- VUID-VkCopyBufferToImageInfo2-pRegions-06223
  For each element of pRegions not containing VkCopyCommandTransformInfoQCOM in its pNext chain, imageOffset.x and (imageExtent.width + imageOffset.x) **must** both be greater than or equal to 0 and less than or equal to the width of the specified imageSubresource of dstImage
- VUID-VkCopyBufferToImageInfo2-pRegions-06224
  For each element of pRegions not containing VkCopyCommandTransformInfoQCOM in its pNext chain, imageOffset.y and (imageExtent.height + imageOffset.y) **must** both be greater than or equal to 0 and less than or equal to the height of the specified imageSubresource of

dstImage

Valid Usage (Implicit)

- VUID-VkCopyBufferToImageInfo2-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_COPY_BUFFER_TO_IMAGE_INFO_2
- VUID-VkCopyBufferToImageInfo2-pNext-pNext
  pNext **must** be NULL
- VUID-VkCopyBufferToImageInfo2-srcBuffer-parameter
  srcBuffer **must** be a valid VkBuffer handle
- VUID-VkCopyBufferToImageInfo2-dstImage-parameter
  dstImage **must** be a valid VkImage handle
- VUID-VkCopyBufferToImageInfo2-dstImageLayout-parameter
  dstImageLayout **must** be a valid VkImageLayout value
- VUID-VkCopyBufferToImageInfo2-pRegions-parameter
  pRegions **must** be a valid pointer to an array of regionCount valid VkBufferImageCopy2 structures
- VUID-VkCopyBufferToImageInfo2-regionCount-arraylength
  regionCount **must** be greater than 0
- VUID-VkCopyBufferToImageInfo2-commonparent
  Both of dstImage, and srcBuffer **must** have been created, allocated, or retrieved from the same VkDevice

To copy data from an image object to a buffer object, call:

```
// Provided by VK_VERSION_1_3
void vkCmdCopyImageToBuffer2(
    VkCommandBuffer                             commandBuffer,
    const VkCopyImageToBufferInfo2*             pCopyImageToBufferInfo);
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to vkCmdCopyImageToBuffer2
void vkCmdCopyImageToBuffer2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkCopyImageToBufferInfo2*             pCopyImageToBufferInfo);
```

- commandBuffer is the command buffer into which the command will be recorded.
- pCopyImageToBufferInfo is a pointer to a VkCopyImageToBufferInfo2 structure describing the copy parameters.

This command is functionally identical to vkCmdCopyImageToBuffer, but includes extensible sub-structures that include sType and pNext parameters, allowing them to be more easily extended.

Valid Usage

- VUID-vkCmdCopyImageToBuffer2-commandBuffer-01831
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `srcImage` **must** not be a protected image
- VUID-vkCmdCopyImageToBuffer2-commandBuffer-01832
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `dstBuffer` **must** not be a protected buffer
- VUID-vkCmdCopyImageToBuffer2-commandBuffer-01833
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, `dstBuffer` **must** not be an unprotected buffer
- VUID-vkCmdCopyImageToBuffer2-commandBuffer-07746
  If the `maintenance11` feature is not enabled, and the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT, the `bufferOffset` member of any element of `pCopyImageToBufferInfo->pRegions` **must** be a multiple of `4`
- VUID-vkCmdCopyImageToBuffer2-imageOffset-07747
  The `imageOffset` and `imageExtent` members of each element of `pCopyImageToBufferInfo->pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in VkQueueFamilyProperties
- VUID-vkCmdCopyImageToBuffer2-commandBuffer-10216
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT, and the `maintenance10` feature is not enabled, for each element of `pCopyImageToBufferInfo->pRegions`, the `aspectMask` member of `imageSubresource` **must** not be VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-vkCmdCopyImageToBuffer2-commandBuffer-11790
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support VK_QUEUE_COMPUTE_BIT, and in any element of `pCopyImageToBufferInfo->pRegions` the `aspectMask` member of `imageSubresource` is VK_IMAGE_ASPECT_DEPTH_BIT, then the format features of `srcImage` **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_COMPUTE_QUEUE_BIT_KHR
- VUID-vkCmdCopyImageToBuffer2-commandBuffer-11791
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of `pCopyImageToBufferInfo->pRegions` the `aspectMask` member of `imageSubresource` is VK_IMAGE_ASPECT_DEPTH_BIT, then the format features of `srcImage` **must** contain VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_TRANSFER_QUEUE_BIT_KHR
- VUID-vkCmdCopyImageToBuffer2-commandBuffer-11792
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT but does support VK_QUEUE_COMPUTE_BIT, and in any element of `pCopyImageToBufferInfo->pRegions` the

aspectMask member of imageSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of srcImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_COMPUTE_QUEUE_BIT_KHR

- VUID-vkCmdCopyImageToBuffer2-commandBuffer-11793
  If the queue family used to create the VkCommandPool which commandBuffer was allocated from does not support VK_QUEUE_GRAPHICS_BIT and VK_QUEUE_COMPUTE_BIT, but does support VK_QUEUE_TRANSFER_BIT, and in any element of pCopyImageToBufferInfo->pRegions the aspectMask member of imageSubresource is VK_IMAGE_ASPECT_STENCIL_BIT, then the format features of srcImage **must** contain VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_TRANSFER_QUEUE_BIT_KHR

Valid Usage (Implicit)

- VUID-vkCmdCopyImageToBuffer2-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyImageToBuffer2-pCopyImageToBufferInfo-parameter
  pCopyImageToBufferInfo **must** be a valid pointer to a valid VkCopyImageToBufferInfo2 structure
- VUID-vkCmdCopyImageToBuffer2-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdCopyImageToBuffer2-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, or VK_QUEUE_TRANSFER_BIT operations
- VUID-vkCmdCopyImageToBuffer2-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyImageToBuffer2-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyImageToBuffer2-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdCopyImageToBuffer2 is not affected by conditional rendering

The VkCopyImageToBufferInfo2 structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkCopyImageToBufferInfo2 {
    VkStructureType              sType;
    const void*                  pNext;
    VkImage                      srcImage;
    VkImageLayout                srcImageLayout;
    VkBuffer                     dstBuffer;
    uint32_t                     regionCount;
    const VkBufferImageCopy2*    pRegions;
} VkCopyImageToBufferInfo2;
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to VkCopyImageToBufferInfo2
typedef VkCopyImageToBufferInfo2 VkCopyImageToBufferInfo2KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- srcImage is the source image.
- srcImageLayout is the layout of the source image subresources for the copy.
- dstBuffer is the destination buffer.
- regionCount is the number of regions to copy.
- pRegions is a pointer to an array of VkBufferImageCopy2 structures specifying the regions to copy.

Valid Usage

- VUID-VkCopyImageToBufferInfo2-pRegions-04566
  The image region specified by each element of `pRegions` **must** be contained within the specified `imageSubresource` of `srcImage`
- VUID-VkCopyImageToBufferInfo2-pRegions-00183
  `dstBuffer` **must** be large enough to contain all buffer locations that are accessed according to Buffer and Image Addressing, for each element of `pRegions`
- VUID-VkCopyImageToBufferInfo2-pRegions-00184
  Each memory location accessed by this command **must** not be accessed by both a source region and a destination region, as specified in `pRegions` and accessed according to Buffer and Image Addressing
- VUID-VkCopyImageToBufferInfo2-pRegions-12485
  Each memory location accessed by this command **must** not be accessed by more than one destination region, as specified by `pRegions` and accessed according to Buffer and Image Addressing
- VUID-VkCopyImageToBufferInfo2-srcImage-00186
  `srcImage` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-VkCopyImageToBufferInfo2-srcImage-01998
  The format features of `srcImage` **must** contain VK_FORMAT_FEATURE_TRANSFER_SRC_BIT
- VUID-VkCopyImageToBufferInfo2-dstBuffer-00191
  `dstBuffer` **must** have been created with the VK_BUFFER_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-VkCopyImageToBufferInfo2-dstBuffer-00192
  If `dstBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkCopyImageToBufferInfo2-srcImageLayout-00189
  `srcImageLayout` **must** specify the layout of the image subresources of `srcImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-VkCopyImageToBufferInfo2-srcImageLayout-01397
  `srcImageLayout` **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL, or VK_IMAGE_LAYOUT_GENERAL
- VUID-VkCopyImageToBufferInfo2-srcImage-07966
  If `srcImage` is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkCopyImageToBufferInfo2-imageSubresource-07967
  The `imageSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `srcImage` was created
- VUID-VkCopyImageToBufferInfo2-imageSubresource-07968
  If `imageSubresource.layerCount` is not VK_REMAINING_ARRAY_LAYERS, `imageSubresource.baseArrayLayer` + `imageSubresource.layerCount` of each element of

pRegions **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when srcImage was created

- VUID-VkCopyImageToBufferInfo2-srcImage-07969
  srcImage **must** not have been created with flags containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-VkCopyImageToBufferInfo2-srcImage-07973
  srcImage **must** have a sample count equal to VK_SAMPLE_COUNT_1_BIT
- VUID-VkCopyImageToBufferInfo2-srcImage-07979
  If srcImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, imageOffset.y **must** be 0 and imageExtent.height **must** be 1
- VUID-VkCopyImageToBufferInfo2-imageOffset-09104
  For each element of pRegions, imageOffset.z and (imageExtent.depth + imageOffset.z) **must** both be greater than or equal to 0 and less than or equal to the depth of the specified imageSubresource of srcImage
- VUID-VkCopyImageToBufferInfo2-srcImage-07980
  If srcImage is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of pRegions, imageOffset.z **must** be 0 and imageExtent.depth **must** be 1
- VUID-VkCopyImageToBufferInfo2-srcImage-07274
  For each element of pRegions, imageOffset.x **must** be a multiple of the texel block extent width of the VkFormat of srcImage
- VUID-VkCopyImageToBufferInfo2-srcImage-07275
  For each element of pRegions, imageOffset.y **must** be a multiple of the texel block extent height of the VkFormat of srcImage
- VUID-VkCopyImageToBufferInfo2-srcImage-07276
  For each element of pRegions, imageOffset.z **must** be a multiple of the texel block extent depth of the VkFormat of srcImage
- VUID-VkCopyImageToBufferInfo2-srcImage-00207
  For each element of pRegions, if the sum of imageOffset.x and extent.width does not equal the width of the subresource specified by imageSubresource, extent.width **must** be a multiple of the texel block extent width of the VkFormat of srcImage
- VUID-VkCopyImageToBufferInfo2-srcImage-00208
  For each element of pRegions, if the sum of imageOffset.y and extent.height does not equal the height of the subresource specified by imageSubresource, extent.height **must** be a multiple of the texel block extent height of the VkFormat of srcImage
- VUID-VkCopyImageToBufferInfo2-srcImage-00209
  For each element of pRegions, if the sum of imageOffset.z and extent.depth does not equal the depth of the subresource specified by srcSubresource, extent.depth **must** be a multiple of the texel block extent depth of the VkFormat of srcImage
- VUID-VkCopyImageToBufferInfo2-imageSubresource-09105
  For each element of pRegions, imageSubresource.aspectMask **must** specify aspects present in srcImage
- VUID-VkCopyImageToBufferInfo2-srcImage-07981

If srcImage has a multi-planar format, then for each element of pRegions, imageSubresource.aspectMask **must** be a single valid multi-planar aspect mask bit

- VUID-VkCopyImageToBufferInfo2-srcImage-07983
  If srcImage is of type VK_IMAGE_TYPE_3D, for each element of pRegions, imageSubresource.baseArrayLayer **must** be 0 and imageSubresource.layerCount **must** be 1
- VUID-VkCopyImageToBufferInfo2-bufferRowLength-09106
  For each element of pRegions, bufferRowLength **must** be a multiple of the texel block extent width of the VkFormat of srcImage
- VUID-VkCopyImageToBufferInfo2-bufferImageHeight-09107
  For each element of pRegions, bufferImageHeight **must** be a multiple of the texel block extent height of the VkFormat of srcImage
- VUID-VkCopyImageToBufferInfo2-bufferRowLength-09108
  For each element of pRegions, bufferRowLength divided by the texel block extent width and then multiplied by the texel block size of srcImage **must** be less than or equal to 2^{31}-1
- VUID-VkCopyImageToBufferInfo2-srcImage-07975
  If srcImage does not have either a depth/stencil format or a multi-planar format, then for each element of pRegions, bufferOffset **must** be a multiple of the texel block size
- VUID-VkCopyImageToBufferInfo2-srcImage-07976
  If srcImage has a multi-planar format, then for each element of pRegions, bufferOffset **must** be a multiple of the element size of the compatible format for the format and the aspectMask of the imageSubresource as defined in Compatible Formats of Planes of Multi-Planar Formats
- VUID-VkCopyImageToBufferInfo2-srcImage-07978
  If srcImage has a depth/stencil format, the bufferOffset member of any element of pRegions **must** be a multiple of 4
- VUID-VkCopyImageToBufferInfo2-imageOffset-00197
  For each element of pRegions not containing VkCopyCommandTransformInfoQCOM in its pNext chain, imageOffset.x and (imageExtent.width + imageOffset.x) **must** both be greater than or equal to 0 and less than or equal to the width of the specified imageSubresource of srcImage
- VUID-VkCopyImageToBufferInfo2-imageOffset-00198
  For each element of pRegions not containing VkCopyCommandTransformInfoQCOM in its pNext chain, imageOffset.y and (imageExtent.height + imageOffset.y) **must** both be greater than or equal to 0 and less than or equal to the height of the specified imageSubresource of srcImage

Valid Usage (Implicit)

- VUID-VkCopyImageToBufferInfo2-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_COPY_IMAGE_TO_BUFFER_INFO_2
- VUID-VkCopyImageToBufferInfo2-pNext-pNext
  pNext **must** be NULL
- VUID-VkCopyImageToBufferInfo2-srcImage-parameter

`srcImage` **must** be a valid VkImage handle

- VUID-VkCopyImageToBufferInfo2-srcImageLayout-parameter
  `srcImageLayout` **must** be a valid VkImageLayout value
- VUID-VkCopyImageToBufferInfo2-dstBuffer-parameter
  `dstBuffer` **must** be a valid VkBuffer handle
- VUID-VkCopyImageToBufferInfo2-pRegions-parameter
  `pRegions` **must** be a valid pointer to an array of `regionCount` valid VkBufferImageCopy2 structures
- VUID-VkCopyImageToBufferInfo2-regionCount-arraylength
  `regionCount` **must** be greater than `0`
- VUID-VkCopyImageToBufferInfo2-commonparent
  Both of `dstBuffer`, and `srcImage` **must** have been created, allocated, or retrieved from the same VkDevice

For both vkCmdCopyBufferToImage2 and vkCmdCopyImageToBuffer2, each element of `pRegions` is a structure defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkBufferImageCopy2 {
    VkStructureType             sType;
    const void*                 pNext;
    VkDeviceSize                bufferOffset;
    uint32_t                    bufferRowLength;
    uint32_t                    bufferImageHeight;
    VkImageSubresourceLayers    imageSubresource;
    VkOffset3D                  imageOffset;
    VkExtent3D                  imageExtent;
} VkBufferImageCopy2;
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to VkBufferImageCopy2
typedef VkBufferImageCopy2 VkBufferImageCopy2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `bufferOffset` is the offset in bytes from the start of the buffer object where the image data is copied from or to.
- `bufferRowLength` and `bufferImageHeight` specify in texels a subregion of a larger two- or three-dimensional image in buffer memory, and control the addressing calculations. If either of these values is zero, that aspect of the buffer memory is considered to be tightly packed according to the `imageExtent`.
- `imageSubresource` is a VkImageSubresourceLayers used to specify the specific image

subresources of the image used for the source or destination image data.

- `imageOffset` selects the initial `x`, `y`, `z` offsets in texels of the sub-region of the source or destination image data.
- `imageExtent` is the size in texels of the image to copy in `width`, `height` and `depth`.

This structure is functionally identical to VkBufferImageCopy, but adds `sType` and `pNext` parameters, allowing it to be more easily extended.

Valid Usage

- VUID-VkBufferImageCopy2-bufferRowLength-09101
  `bufferRowLength` **must** be `0`, or greater than or equal to the `width` member of `imageExtent`
- VUID-VkBufferImageCopy2-bufferImageHeight-09102
  `bufferImageHeight` **must** be `0`, or greater than or equal to the `height` member of `imageExtent`
- VUID-VkBufferImageCopy2-aspectMask-09103
  The `aspectMask` member of `imageSubresource` **must** only have a single bit set
- VUID-VkBufferImageCopy2-imageExtent-06659
  `imageExtent.width` **must** not be 0
- VUID-VkBufferImageCopy2-imageExtent-06660
  `imageExtent.height` **must** not be 0
- VUID-VkBufferImageCopy2-imageExtent-06661
  `imageExtent.depth` **must** not be 0

Valid Usage (Implicit)

- VUID-VkBufferImageCopy2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BUFFER_IMAGE_COPY_2
- VUID-VkBufferImageCopy2-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkBufferImageCopy2-imageSubresource-parameter
  `imageSubresource` **must** be a valid VkImageSubresourceLayers structure

The following commands can be used to copy between host memory and images. Bytes in host memory and texels in images are accessed as specified in Copying Data Between Buffers and Images, with buffers replaced with host memory.

To copy data from host memory to an image object, call:

```
// Provided by VK_VERSION_1_4
VkResult vkCopyMemoryToImage(
    VkDevice                                    device,
    const VkCopyMemoryToImageInfo*              pCopyMemoryToImageInfo);
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to vkCopyMemoryToImage
VkResult vkCopyMemoryToImageEXT(
    VkDevice                                    device,
    const VkCopyMemoryToImageInfo*              pCopyMemoryToImageInfo);
```

- `device` is the device which owns `pCopyMemoryToImageInfo->dstImage`.
- `pCopyMemoryToImageInfo` is a pointer to a VkCopyMemoryToImageInfo structure describing the copy parameters.

This command is functionally similar to vkCmdCopyBufferToImage2, except it is executed on the host and reads from host memory instead of a buffer. The memory of `pCopyMemoryToImageInfo->dstImage` is accessed by the host as if coherent.

NOTE	Because queue submissions automatically make host memory visible to the device, there would not be a need for a memory barrier before using the results of this copy operation on the device.

Valid Usage (Implicit)

- VUID-vkCopyMemoryToImage-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCopyMemoryToImage-pCopyMemoryToImageInfo-parameter
  `pCopyMemoryToImageInfo` **must** be a valid pointer to a valid VkCopyMemoryToImageInfo structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_MEMORY_MAP_FAILED
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkCopyMemoryToImageInfo` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkCopyMemoryToImageInfo {
    VkStructureType               sType;
    const void*                   pNext;
    VkHostImageCopyFlags          flags;
    VkImage                       dstImage;
    VkImageLayout                 dstImageLayout;
    uint32_t                      regionCount;
    const VkMemoryToImageCopy*    pRegions;
} VkCopyMemoryToImageInfo;
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to VkCopyMemoryToImageInfo
typedef VkCopyMemoryToImageInfo VkCopyMemoryToImageInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkHostImageCopyFlagBits values describing additional copy parameters.
- `dstImage` is the destination image.
- `dstImageLayout` is the layout of the destination image subresources for the copy.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of VkMemoryToImageCopy structures specifying the regions to copy.

`vkCopyMemoryToImage` does not check whether the device memory associated with `dstImage` is currently in use before performing the copy. The application **must** guarantee that any previously submitted command that reads from or writes to the copy regions has completed before the host performs the copy.

Copy regions for the image **must** be aligned to a multiple of the texel block extent in each dimension, except at the edges of the image, where region extents **must** match the edge of the image.

Valid Usage

- VUID-VkCopyMemoryToImageInfo-dstImage-09109
  If `dstImage` is sparse then all memory ranges accessed by the copy command **must** be bound as described in Binding Resource Memory
- VUID-VkCopyMemoryToImageInfo-dstImage-09111
  If the stencil aspect of `dstImage` is accessed, and `dstImage` was not created with separate stencil usage, `dstImage` **must** have been created with the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag set
- VUID-VkCopyMemoryToImageInfo-dstImage-09112

If the stencil aspect of dstImage is accessed, and dstImage was created with separate stencil usage, dstImage **must** have been created with the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag set

- VUID-VkCopyMemoryToImageInfo-dstImage-09113
  If non-stencil aspects of dstImage are accessed, dstImage **must** have been created with the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag set
- VUID-VkCopyMemoryToImageInfo-imageOffset-09114
  If flags contains VK_HOST_IMAGE_COPY_MEMCPY_BIT, the x, y, and z members of the imageOffset member of each element of pRegions **must** be 0
- VUID-VkCopyMemoryToImageInfo-dstImage-09115
  If flags contains VK_HOST_IMAGE_COPY_MEMCPY_BIT, the imageExtent member of each element of pRegions **must** equal the extents of dstImage identified by imageSubresource
- VUID-VkCopyMemoryToImageInfo-dstImage-07966
  If dstImage is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-VkCopyMemoryToImageInfo-imageSubresource-07967
  The imageSubresource.mipLevel member of each element of pRegions **must** be less than the mipLevels specified in VkImageCreateInfo when dstImage was created
- VUID-VkCopyMemoryToImageInfo-imageSubresource-07968
  If imageSubresource.layerCount is not VK_REMAINING_ARRAY_LAYERS, imageSubresource.baseArrayLayer + imageSubresource.layerCount of each element of pRegions **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when dstImage was created
- VUID-VkCopyMemoryToImageInfo-dstImage-07969
  dstImage **must** not have been created with flags containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-VkCopyMemoryToImageInfo-imageSubresource-07971
  For each element of pRegions, imageOffset.x and (imageExtent.width + imageOffset.x) **must** both be greater than or equal to 0 and less than or equal to the width of the specified imageSubresource of dstImage
- VUID-VkCopyMemoryToImageInfo-imageSubresource-07972
  For each element of pRegions, imageOffset.y and (imageExtent.height + imageOffset.y) **must** both be greater than or equal to 0 and less than or equal to the height of the specified imageSubresource of dstImage
- VUID-VkCopyMemoryToImageInfo-dstImage-07973
  dstImage **must** have a sample count equal to VK_SAMPLE_COUNT_1_BIT
- VUID-VkCopyMemoryToImageInfo-dstImage-07979
  If dstImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, imageOffset.y **must** be 0 and imageExtent.height **must** be 1
- VUID-VkCopyMemoryToImageInfo-imageOffset-09104
  For each element of pRegions, imageOffset.z and (imageExtent.depth + imageOffset.z) **must**

both be greater than or equal to `0` and less than or equal to the depth of the specified `imageSubresource` of `dstImage`

- VUID-VkCopyMemoryToImageInfo-dstImage-07980
  If `dstImage` is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of `pRegions`, `imageOffset.z` **must** be `0` and `imageExtent.depth` **must** be `1`
- VUID-VkCopyMemoryToImageInfo-dstImage-07274
  For each element of `pRegions`, `imageOffset.x` **must** be a multiple of the texel block extent width of the VkFormat of `dstImage`
- VUID-VkCopyMemoryToImageInfo-dstImage-07275
  For each element of `pRegions`, `imageOffset.y` **must** be a multiple of the texel block extent height of the VkFormat of `dstImage`
- VUID-VkCopyMemoryToImageInfo-dstImage-07276
  For each element of `pRegions`, `imageOffset.z` **must** be a multiple of the texel block extent depth of the VkFormat of `dstImage`
- VUID-VkCopyMemoryToImageInfo-dstImage-00207
  For each element of `pRegions`, if the sum of `imageOffset.x` and `extent.width` does not equal the width of the subresource specified by `imageSubresource`, `extent.width` **must** be a multiple of the texel block extent width of the VkFormat of `dstImage`
- VUID-VkCopyMemoryToImageInfo-dstImage-00208
  For each element of `pRegions`, if the sum of `imageOffset.y` and `extent.height` does not equal the height of the subresource specified by `imageSubresource`, `extent.height` **must** be a multiple of the texel block extent height of the VkFormat of `dstImage`
- VUID-VkCopyMemoryToImageInfo-dstImage-00209
  For each element of `pRegions`, if the sum of `imageOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `srcSubresource`, `extent.depth` **must** be a multiple of the texel block extent depth of the VkFormat of `dstImage`
- VUID-VkCopyMemoryToImageInfo-imageSubresource-09105
  For each element of `pRegions`, `imageSubresource.aspectMask` **must** specify aspects present in `dstImage`
- VUID-VkCopyMemoryToImageInfo-dstImage-07981
  If `dstImage` has a multi-planar format, then for each element of `pRegions`, `imageSubresource.aspectMask` **must** be a single valid multi-planar aspect mask bit
- VUID-VkCopyMemoryToImageInfo-dstImage-07983
  If `dstImage` is of type VK_IMAGE_TYPE_3D, for each element of `pRegions`, `imageSubresource.baseArrayLayer` **must** be `0` and `imageSubresource.layerCount` **must** be `1`
- VUID-VkCopyMemoryToImageInfo-memoryRowLength-09106
  For each element of `pRegions`, `memoryRowLength` **must** be a multiple of the texel block extent width of the VkFormat of `dstImage`
- VUID-VkCopyMemoryToImageInfo-memoryImageHeight-09107
  For each element of `pRegions`, `memoryImageHeight` **must** be a multiple of the texel block extent height of the VkFormat of `dstImage`
- VUID-VkCopyMemoryToImageInfo-memoryRowLength-09108

For each element of pRegions, memoryRowLength divided by the texel block extent width and then multiplied by the texel block size of dstImage **must** be less than or equal to 2^{31}-1

- VUID-VkCopyMemoryToImageInfo-pRegions-09062
  Each memory location accessed by this command **must** not be accessed in both a source region and a destination region, as specified in pRegions and accessed according to Buffer and Image Addressing
- VUID-VkCopyMemoryToImageInfo-pRegions-12474
  Each memory location accessed by this command **must** not be accessed in more than one destination region, as specified by pRegions and accessed according to Buffer and Image Addressing
- VUID-VkCopyMemoryToImageInfo-dstImageLayout-09059
  dstImageLayout **must** specify the current layout of the image subresources of dstImage specified in pRegions
- VUID-VkCopyMemoryToImageInfo-dstImageLayout-09060
  dstImageLayout **must** be one of the image layouts returned in VkPhysicalDeviceHostImageCopyProperties::pCopyDstLayouts
- VUID-VkCopyMemoryToImageInfo-flags-09393
  If flags includes VK_HOST_IMAGE_COPY_MEMCPY_BIT, for each region in pRegions, memoryRowLength and memoryImageHeight **must** both be 0

Valid Usage (Implicit)

- VUID-VkCopyMemoryToImageInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_COPY_MEMORY_TO_IMAGE_INFO
- VUID-VkCopyMemoryToImageInfo-pNext-pNext
  pNext **must** be NULL
- VUID-VkCopyMemoryToImageInfo-flags-parameter
  If flags is not 0, flags **must** be a valid combination of VkHostImageCopyFlagBits values
- VUID-VkCopyMemoryToImageInfo-dstImage-parameter
  dstImage **must** be a valid VkImage handle
- VUID-VkCopyMemoryToImageInfo-dstImageLayout-parameter
  dstImageLayout **must** be a valid VkImageLayout value
- VUID-VkCopyMemoryToImageInfo-pRegions-parameter
  pRegions **must** be a valid pointer to an array of regionCount valid VkMemoryToImageCopy structures
- VUID-VkCopyMemoryToImageInfo-regionCount-arraylength
  regionCount **must** be greater than 0

Each element of VkCopyMemoryToImageInfo::pRegions is a structure defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkMemoryToImageCopy {
    VkStructureType             sType;
    const void*                 pNext;
    const void*                 pHostPointer;
    uint32_t                    memoryRowLength;
    uint32_t                    memoryImageHeight;
    VkImageSubresourceLayers    imageSubresource;
    VkOffset3D                  imageOffset;
    VkExtent3D                  imageExtent;
} VkMemoryToImageCopy;
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to VkMemoryToImageCopy
typedef VkMemoryToImageCopy VkMemoryToImageCopyEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pHostPointer` is the host memory address which is the source of the copy.
- `memoryRowLength` and `memoryImageHeight` specify in texels a subregion of a larger two- or three-dimensional image in host memory, and control the addressing calculations. If either of these values is zero, that aspect of the host memory is considered to be tightly packed according to the `imageExtent`.
- `imageSubresource` is a VkImageSubresourceLayers used to specify the specific image subresources of the image used for the source or destination image data.
- `imageOffset` selects the initial `x`, `y`, `z` offsets in texels of the sub-region of the destination image data.
- `imageExtent` is the size in texels of the image to copy in `width`, `height` and `depth`.

This structure is functionally similar to VkBufferImageCopy2, except it defines host memory as the source of copy instead of a buffer. In particular, the same data packing rules and restrictions as that structure apply here as well.

Valid Usage

- VUID-VkMemoryToImageCopy-pHostPointer-09061
  `pHostPointer` **must** point to memory that is large enough to contain all memory locations that are accessed according to Buffer and Image Addressing
- VUID-VkMemoryToImageCopy-memoryRowLength-09101
  `memoryRowLength` **must** be `0`, or greater than or equal to the `width` member of `imageExtent`
- VUID-VkMemoryToImageCopy-memoryImageHeight-09102
  `memoryImageHeight` **must** be `0`, or greater than or equal to the `height` member of `imageExtent`

- VUID-VkMemoryToImageCopy-aspectMask-09103
  The `aspectMask` member of `imageSubresource` **must** only have a single bit set
- VUID-VkMemoryToImageCopy-imageExtent-06659
  `imageExtent.width` **must** not be 0
- VUID-VkMemoryToImageCopy-imageExtent-06660
  `imageExtent.height` **must** not be 0
- VUID-VkMemoryToImageCopy-imageExtent-06661
  `imageExtent.depth` **must** not be 0

Valid Usage (Implicit)

- VUID-VkMemoryToImageCopy-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MEMORY_TO_IMAGE_COPY
- VUID-VkMemoryToImageCopy-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkMemoryToImageCopy-pHostPointer-parameter
  `pHostPointer` **must** be a pointer value
- VUID-VkMemoryToImageCopy-imageSubresource-parameter
  `imageSubresource` **must** be a valid VkImageSubresourceLayers structure

To copy data from an image object to host memory, call:

```
// Provided by VK_VERSION_1_4
VkResult vkCopyImageToMemory(
    VkDevice                                    device,
    const VkCopyImageToMemoryInfo*              pCopyImageToMemoryInfo);
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to vkCopyImageToMemory
VkResult vkCopyImageToMemoryEXT(
    VkDevice                                    device,
    const VkCopyImageToMemoryInfo*              pCopyImageToMemoryInfo);
```

- `device` is the device which owns `pCopyImageToMemoryInfo->srcImage`.
- `pCopyImageToMemoryInfo` is a pointer to a VkCopyImageToMemoryInfo structure describing the copy parameters.

This command is functionally similar to vkCmdCopyImageToBuffer2, except it is executed on the host and writes to host memory instead of a buffer. The memory of `pCopyImageToMemoryInfo->srcImage` is accessed by the host as if coherent.

NOTE If the device has written to the image memory, it is not automatically made

> available to the host. Before this copy command can be called, a memory barrier for this image **must** have been issued on the device with the second synchronization scope including VK_PIPELINE_STAGE_HOST_BIT and VK_ACCESS_HOST_READ_BIT.

Valid Usage (Implicit)

- VUID-vkCopyImageToMemory-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCopyImageToMemory-pCopyImageToMemoryInfo-parameter
  `pCopyImageToMemoryInfo` **must** be a valid pointer to a valid VkCopyImageToMemoryInfo structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_MEMORY_MAP_FAILED
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkCopyImageToMemoryInfo` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkCopyImageToMemoryInfo {
    VkStructureType               sType;
    const void*                   pNext;
    VkHostImageCopyFlags          flags;
    VkImage                       srcImage;
    VkImageLayout                 srcImageLayout;
    uint32_t                      regionCount;
    const VkImageToMemoryCopy*    pRegions;
} VkCopyImageToMemoryInfo;
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to VkCopyImageToMemoryInfo
typedef VkCopyImageToMemoryInfo VkCopyImageToMemoryInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkHostImageCopyFlagBits values describing additional copy parameters.
- `srcImage` is the source image.
- `srcImageLayout` is the layout of the source image subresources for the copy.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of VkImageToMemoryCopy structures specifying the regions to copy.

`vkCopyImageToMemory` does not check whether the device memory associated with `srcImage` is currently in use before performing the copy. The application **must** guarantee that any previously submitted command that writes to the copy regions has completed before the host performs the copy.

Copy regions for the image **must** be aligned to a multiple of the texel block extent in each dimension, except at the edges of the image, where region extents **must** match the edge of the image.

Valid Usage

- VUID-VkCopyImageToMemoryInfo-srcImage-09109
  If `srcImage` is sparse then all memory ranges accessed by the copy command **must** be bound as described in Binding Resource Memory
- VUID-VkCopyImageToMemoryInfo-srcImage-09111
  If the stencil aspect of `srcImage` is accessed, and `srcImage` was not created with separate stencil usage, `srcImage` **must** have been created with the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag set
- VUID-VkCopyImageToMemoryInfo-srcImage-09112
  If the stencil aspect of `srcImage` is accessed, and `srcImage` was created with separate stencil usage, `srcImage` **must** have been created with the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag set
- VUID-VkCopyImageToMemoryInfo-srcImage-09113
  If non-stencil aspects of `srcImage` are accessed, `srcImage` **must** have been created with the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag set
- VUID-VkCopyImageToMemoryInfo-imageOffset-09114
  If `flags` contains VK_HOST_IMAGE_COPY_MEMCPY_BIT, the `x`, `y`, and `z` members of the `imageOffset` member of each element of `pRegions` **must** be `0`
- VUID-VkCopyImageToMemoryInfo-srcImage-09115
  If `flags` contains VK_HOST_IMAGE_COPY_MEMCPY_BIT, the `imageExtent` member of each element of `pRegions` **must** equal the extents of `srcImage` identified by `imageSubresource`
- VUID-VkCopyImageToMemoryInfo-srcImage-07966
  If `srcImage` is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-VkCopyImageToMemoryInfo-imageSubresource-07967
  The `imageSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `srcImage` was created
- VUID-VkCopyImageToMemoryInfo-imageSubresource-07968
  If `imageSubresource.layerCount` is not VK_REMAINING_ARRAY_LAYERS, `imageSubresource.baseArrayLayer` + `imageSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in VkImageCreateInfo when `srcImage` was created
- VUID-VkCopyImageToMemoryInfo-srcImage-07969
  `srcImage` **must** not have been created with `flags` containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-VkCopyImageToMemoryInfo-imageSubresource-07971
  For each element of `pRegions`, `imageOffset.x` and (`imageExtent.width` + `imageOffset.x`) **must** both be greater than or equal to `0` and less than or equal to the width of the specified `imageSubresource` of `srcImage`
- VUID-VkCopyImageToMemoryInfo-imageSubresource-07972
  For each element of `pRegions`, `imageOffset.y` and (`imageExtent.height` + `imageOffset.y`) **must** both be greater than or equal to `0` and less than or equal to the height of the specified `imageSubresource` of `srcImage`
- VUID-VkCopyImageToMemoryInfo-srcImage-07973
  `srcImage` **must** have a sample count equal to VK_SAMPLE_COUNT_1_BIT
- VUID-VkCopyImageToMemoryInfo-srcImage-07979
  If `srcImage` is of type VK_IMAGE_TYPE_1D, then for each element of `pRegions`, `imageOffset.y` **must** be `0` and `imageExtent.height` **must** be `1`
- VUID-VkCopyImageToMemoryInfo-imageOffset-09104
  For each element of `pRegions`, `imageOffset.z` and (`imageExtent.depth` + `imageOffset.z`) **must** both be greater than or equal to `0` and less than or equal to the depth of the specified `imageSubresource` of `srcImage`
- VUID-VkCopyImageToMemoryInfo-srcImage-07980
  If `srcImage` is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of `pRegions`, `imageOffset.z` **must** be `0` and `imageExtent.depth` **must** be `1`
- VUID-VkCopyImageToMemoryInfo-srcImage-07274
  For each element of `pRegions`, `imageOffset.x` **must** be a multiple of the texel block extent width of the VkFormat of `srcImage`
- VUID-VkCopyImageToMemoryInfo-srcImage-07275
  For each element of `pRegions`, `imageOffset.y` **must** be a multiple of the texel block extent height of the VkFormat of `srcImage`
- VUID-VkCopyImageToMemoryInfo-srcImage-07276
  For each element of `pRegions`, `imageOffset.z` **must** be a multiple of the texel block extent depth of the VkFormat of `srcImage`
- VUID-VkCopyImageToMemoryInfo-srcImage-00207
  For each element of `pRegions`, if the sum of `imageOffset.x` and `extent.width` does not equal

the width of the subresource specified by `imageSubresource`, `extent.width` **must** be a multiple of the texel block extent width of the VkFormat of `srcImage`

- VUID-VkCopyImageToMemoryInfo-srcImage-00208
  For each element of `pRegions`, if the sum of `imageOffset.y` and `extent.height` does not equal the height of the subresource specified by `imageSubresource`, `extent.height` **must** be a multiple of the texel block extent height of the VkFormat of `srcImage`
- VUID-VkCopyImageToMemoryInfo-srcImage-00209
  For each element of `pRegions`, if the sum of `imageOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `srcSubresource`, `extent.depth` **must** be a multiple of the texel block extent depth of the VkFormat of `srcImage`
- VUID-VkCopyImageToMemoryInfo-imageSubresource-09105
  For each element of `pRegions`, `imageSubresource.aspectMask` **must** specify aspects present in `srcImage`
- VUID-VkCopyImageToMemoryInfo-srcImage-07981
  If `srcImage` has a multi-planar format, then for each element of `pRegions`, `imageSubresource.aspectMask` **must** be a single valid multi-planar aspect mask bit
- VUID-VkCopyImageToMemoryInfo-srcImage-07983
  If `srcImage` is of type VK_IMAGE_TYPE_3D, for each element of `pRegions`, `imageSubresource.baseArrayLayer` **must** be `0` and `imageSubresource.layerCount` **must** be `1`
- VUID-VkCopyImageToMemoryInfo-memoryRowLength-09106
  For each element of `pRegions`, `memoryRowLength` **must** be a multiple of the texel block extent width of the VkFormat of `srcImage`
- VUID-VkCopyImageToMemoryInfo-memoryImageHeight-09107
  For each element of `pRegions`, `memoryImageHeight` **must** be a multiple of the texel block extent height of the VkFormat of `srcImage`
- VUID-VkCopyImageToMemoryInfo-memoryRowLength-09108
  For each element of `pRegions`, `memoryRowLength` divided by the texel block extent width and then multiplied by the texel block size of `srcImage` **must** be less than or equal to 2^{31}-1
- VUID-VkImageToMemoryCopy-pRegions-09067
  Each memory location accessed by this command **must** not be accessed in both a source region and a destination region, as specified in `pRegions` and accessed according to Buffer and Image Addressing
- VUID-VkCopyImageToMemoryInfo-pRegions-12475
  Each memory location accessed by this command **must** not be accessed in more than one destination region, as specified by `pRegions` and accessed according to Buffer and Image Addressing
- VUID-VkCopyImageToMemoryInfo-srcImageLayout-09064
  `srcImageLayout` **must** specify the current layout of the image subresources of `srcImage` specified in `pRegions`
- VUID-VkCopyImageToMemoryInfo-srcImageLayout-09065
  `srcImageLayout` **must** be one of the image layouts returned in VkPhysicalDeviceHostImageCopyProperties::`pCopySrcLayouts`

- VUID-VkCopyImageToMemoryInfo-flags-09394
  If `flags` includes VK_HOST_IMAGE_COPY_MEMCPY_BIT, for each region in `pRegions`, `memoryRowLength` and `memoryImageHeight` **must** both be 0

Valid Usage (Implicit)

- VUID-VkCopyImageToMemoryInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COPY_IMAGE_TO_MEMORY_INFO
- VUID-VkCopyImageToMemoryInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkCopyImageToMemoryInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkHostImageCopyFlagBits values
- VUID-VkCopyImageToMemoryInfo-srcImage-parameter
  `srcImage` **must** be a valid VkImage handle
- VUID-VkCopyImageToMemoryInfo-srcImageLayout-parameter
  `srcImageLayout` **must** be a valid VkImageLayout value
- VUID-VkCopyImageToMemoryInfo-pRegions-parameter
  `pRegions` **must** be a valid pointer to an array of `regionCount` valid VkImageToMemoryCopy structures
- VUID-VkCopyImageToMemoryInfo-regionCount-arraylength
  `regionCount` **must** be greater than `0`

Each element of VkCopyImageToMemoryInfo::`pRegions` is a structure defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkImageToMemoryCopy {
    VkStructureType             sType;
    const void*                 pNext;
    void*                       pHostPointer;
    uint32_t                    memoryRowLength;
    uint32_t                    memoryImageHeight;
    VkImageSubresourceLayers    imageSubresource;
    VkOffset3D                  imageOffset;
    VkExtent3D                  imageExtent;
} VkImageToMemoryCopy;
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to VkImageToMemoryCopy
typedef VkImageToMemoryCopy VkImageToMemoryCopyEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `pHostPointer` is the host memory address which is the destination of the copy.
- `memoryRowLength` and `memoryImageHeight` specify in texels a subregion of a larger two- or three-dimensional image in host memory, and control the addressing calculations. If either of these values is zero, that aspect of the host memory is considered to be tightly packed according to the `imageExtent`.
- `imageSubresource` is a VkImageSubresourceLayers used to specify the specific image subresources of the image used for the source or destination image data.
- `imageOffset` selects the initial `x`, `y`, `z` offsets in texels of the sub-region of the source image data.
- `imageExtent` is the size in texels of the image to copy in `width`, `height` and `depth`.

This structure is functionally similar to VkBufferImageCopy2, except it defines host memory as the target of copy instead of a buffer. In particular, the same data packing rules and restrictions as that structure apply here as well.

Valid Usage

- VUID-VkImageToMemoryCopy-pHostPointer-09066
  `pHostPointer` **must** point to memory that is large enough to contain all memory locations that are accessed according to Buffer and Image Addressing
- VUID-VkImageToMemoryCopy-memoryRowLength-09101
  `memoryRowLength` **must** be `0`, or greater than or equal to the `width` member of `imageExtent`
- VUID-VkImageToMemoryCopy-memoryImageHeight-09102
  `memoryImageHeight` **must** be `0`, or greater than or equal to the `height` member of `imageExtent`
- VUID-VkImageToMemoryCopy-aspectMask-09103
  The `aspectMask` member of `imageSubresource` **must** only have a single bit set
- VUID-VkImageToMemoryCopy-imageExtent-06659
  `imageExtent.width` **must** not be 0
- VUID-VkImageToMemoryCopy-imageExtent-06660
  `imageExtent.height` **must** not be 0
- VUID-VkImageToMemoryCopy-imageExtent-06661
  `imageExtent.depth` **must** not be 0

Valid Usage (Implicit)

- VUID-VkImageToMemoryCopy-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_TO_MEMORY_COPY
- VUID-VkImageToMemoryCopy-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkImageToMemoryCopy-pHostPointer-parameter
  `pHostPointer` **must** be a pointer value
- VUID-VkImageToMemoryCopy-imageSubresource-parameter

`imageSubresource` **must** be a valid VkImageSubresourceLayers structure

Bits which **can** be set in VkCopyMemoryToImageInfo::`flags`, VkCopyImageToMemoryInfo::`flags`, and VkCopyImageToImageInfo::`flags`, specifying additional copy parameters are:

```
// Provided by VK_VERSION_1_4
typedef enum VkHostImageCopyFlagBits {
    VK_HOST_IMAGE_COPY_MEMCPY_BIT = 0x00000001,
  // VK_HOST_IMAGE_COPY_MEMCPY is a legacy alias
    VK_HOST_IMAGE_COPY_MEMCPY = VK_HOST_IMAGE_COPY_MEMCPY_BIT,
  // Provided by VK_EXT_host_image_copy
    VK_HOST_IMAGE_COPY_MEMCPY_BIT_EXT = VK_HOST_IMAGE_COPY_MEMCPY_BIT,
  // Provided by VK_EXT_host_image_copy
  // VK_HOST_IMAGE_COPY_MEMCPY_EXT is a legacy alias
    VK_HOST_IMAGE_COPY_MEMCPY_EXT = VK_HOST_IMAGE_COPY_MEMCPY_BIT,
} VkHostImageCopyFlagBits;
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to VkHostImageCopyFlagBits
typedef VkHostImageCopyFlagBits VkHostImageCopyFlagBitsEXT;
```

- VK_HOST_IMAGE_COPY_MEMCPY_BIT specifies that no memory layout swizzling is to be applied during data copy. For copies between memory and images, this flag indicates that image data in host memory is swizzled in exactly the same way as the image data on the device. Using this flag indicates that the implementations **may** use a simple memory copy to transfer the data between the host memory and the device memory. The format of the swizzled data in host memory is platform dependent and is not defined in this specification.

```
// Provided by VK_VERSION_1_4
typedef VkFlags VkHostImageCopyFlags;
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to VkHostImageCopyFlags
typedef VkHostImageCopyFlags VkHostImageCopyFlagsEXT;
```

`VkHostImageCopyFlags` is a bitmask type for setting a mask of zero or more VkHostImageCopyFlagBits.

To copy data from an image object to another image object using the host, call:

```
// Provided by VK_VERSION_1_4
VkResult vkCopyImageToImage(
    VkDevice                                    device,
    const VkCopyImageToImageInfo*               pCopyImageToImageInfo);
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to vkCopyImageToImage
VkResult vkCopyImageToImageEXT(
    VkDevice                                    device,
    const VkCopyImageToImageInfo*               pCopyImageToImageInfo);
```

- `device` is the device which owns `pCopyImageToImageInfo->srcImage` and `pCopyImageToImageInfo->dstImage`.
- `pCopyImageToImageInfo` is a pointer to a VkCopyImageToImageInfo structure describing the copy parameters.

This command is functionally similar to vkCmdCopyImage2, except it is executed on the host. The memory of `pCopyImageToImageInfo->srcImage` and `pCopyImageToImageInfo->dstImage` is accessed by the host as if coherent.

NOTE

If the device has written to the memory of `pCopyImageToImageInfo->srcImage`, it is not automatically made available to the host. Before this copy command can be called, a memory barrier for this image **must** have been issued on the device with the second synchronization scope including VK_PIPELINE_STAGE_HOST_BIT and VK_ACCESS_HOST_READ_BIT.

Because queue submissions automatically make host memory visible to the device, there would not be a need for a memory barrier before using the results of this copy operation in `pCopyImageToImageInfo->dstImage` on the device.

Valid Usage (Implicit)

- VUID-vkCopyImageToImage-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCopyImageToImage-pCopyImageToImageInfo-parameter
  `pCopyImageToImageInfo` **must** be a valid pointer to a valid VkCopyImageToImageInfo structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_MEMORY_MAP_FAILED
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkCopyImageToImageInfo` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkCopyImageToImageInfo {
    VkStructureType         sType;
    const void*             pNext;
    VkHostImageCopyFlags    flags;
    VkImage                 srcImage;
    VkImageLayout           srcImageLayout;
    VkImage                 dstImage;
    VkImageLayout           dstImageLayout;
    uint32_t                regionCount;
    const VkImageCopy2*     pRegions;
} VkCopyImageToImageInfo;
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to VkCopyImageToImageInfo
typedef VkCopyImageToImageInfo VkCopyImageToImageInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkHostImageCopyFlagBits values describing additional copy parameters.
- `srcImage` is the source image.
- `srcImageLayout` is the layout of the source image subresources for the copy.
- `dstImage` is the destination image.
- `dstImageLayout` is the layout of the destination image subresources for the copy.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of VkImageCopy2 structures specifying the regions to copy.

`vkCopyImageToImage` does not check whether the device memory associated with `srcImage` or `dstImage` is currently in use before performing the copy. The application **must** guarantee that any previously submitted command that writes to the copy regions has completed before the host performs the copy.

Valid Usage

- VUID-VkCopyImageToImageInfo-srcImage-09069
  `srcImage` and `dstImage` **must** have been created with identical image creation parameters
- VUID-VkCopyImageToImageInfo-pRegions-12476
  Each memory location accessed by this command **must** not be accessed by both a source region and a destination region, as specified by `pRegions` and accessed according to Copy Image Addressing
- VUID-VkCopyImageToImageInfo-pRegions-12477
  Each memory location accessed by this command **must** not be accessed by more than one destination region, as specified by `pRegions` and accessed according to Copy Image Addressing
- VUID-VkCopyImageToImageInfo-srcImage-09109
  If `srcImage` is sparse then all memory ranges accessed by the copy command **must** be bound as described in Binding Resource Memory
- VUID-VkCopyImageToImageInfo-srcImage-09111
  If the stencil aspect of `srcImage` is accessed, and `srcImage` was not created with separate stencil usage, `srcImage` **must** have been created with the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag set
- VUID-VkCopyImageToImageInfo-srcImage-09112
  If the stencil aspect of `srcImage` is accessed, and `srcImage` was created with separate stencil usage, `srcImage` **must** have been created with the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag set
- VUID-VkCopyImageToImageInfo-srcImage-09113
  If non-stencil aspects of `srcImage` are accessed, `srcImage` **must** have been created with the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag set
- VUID-VkCopyImageToImageInfo-srcOffset-09114
  If `flags` contains VK_HOST_IMAGE_COPY_MEMCPY_BIT, the `x`, `y`, and `z` members of the `srcOffset` member of each element of `pRegions` **must** be `0`
- VUID-VkCopyImageToImageInfo-srcImage-09115
  If `flags` contains VK_HOST_IMAGE_COPY_MEMCPY_BIT, the `extent` member of each element of `pRegions` **must** equal the extents of `srcImage` identified by `srcSubresource`
- VUID-VkCopyImageToImageInfo-srcImage-07966
  If `srcImage` is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkCopyImageToImageInfo-srcSubresource-07967
  The `srcSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `srcImage` was created
- VUID-VkCopyImageToImageInfo-srcSubresource-07968
  If `srcSubresource.layerCount` is not VK_REMAINING_ARRAY_LAYERS, `srcSubresource.baseArrayLayer` + `srcSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in VkImageCreateInfo when

srcImage was created

- VUID-VkCopyImageToImageInfo-srcImage-07969
  srcImage **must** not have been created with flags containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-VkCopyImageToImageInfo-srcSubresource-07971
  For each element of pRegions, srcOffset.x and (extent.width + srcOffset.x) **must** both be greater than or equal to 0 and less than or equal to the width of the specified srcSubresource of srcImage
- VUID-VkCopyImageToImageInfo-srcSubresource-07972
  For each element of pRegions, srcOffset.y and (extent.height + srcOffset.y) **must** both be greater than or equal to 0 and less than or equal to the height of the specified srcSubresource of srcImage
- VUID-VkCopyImageToImageInfo-srcImage-07979
  If srcImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, srcOffset.y **must** be 0 and extent.height **must** be 1
- VUID-VkCopyImageToImageInfo-srcOffset-09104
  For each element of pRegions, srcOffset.z and (extent.depth + srcOffset.z) **must** both be greater than or equal to 0 and less than or equal to the depth of the specified srcSubresource of srcImage
- VUID-VkCopyImageToImageInfo-srcImage-07980
  If srcImage is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of pRegions, srcOffset.z **must** be 0 and extent.depth **must** be 1
- VUID-VkCopyImageToImageInfo-srcImage-07274
  For each element of pRegions, srcOffset.x **must** be a multiple of the texel block extent width of the VkFormat of srcImage
- VUID-VkCopyImageToImageInfo-srcImage-07275
  For each element of pRegions, srcOffset.y **must** be a multiple of the texel block extent height of the VkFormat of srcImage
- VUID-VkCopyImageToImageInfo-srcImage-07276
  For each element of pRegions, srcOffset.z **must** be a multiple of the texel block extent depth of the VkFormat of srcImage
- VUID-VkCopyImageToImageInfo-srcImage-00207
  For each element of pRegions, if the sum of srcOffset.x and extent.width does not equal the width of the subresource specified by srcSubresource, extent.width **must** be a multiple of the texel block extent width of the VkFormat of srcImage
- VUID-VkCopyImageToImageInfo-srcImage-00208
  For each element of pRegions, if the sum of srcOffset.y and extent.height does not equal the height of the subresource specified by srcSubresource, extent.height **must** be a multiple of the texel block extent height of the VkFormat of srcImage
- VUID-VkCopyImageToImageInfo-srcImage-00209
  For each element of pRegions, if the sum of srcOffset.z and extent.depth does not equal the depth of the subresource specified by srcSubresource, extent.depth **must** be a multiple

of the texel block extent depth of the VkFormat of srcImage

- VUID-VkCopyImageToImageInfo-srcSubresource-09105
  For each element of pRegions, srcSubresource.aspectMask **must** specify aspects present in srcImage
- VUID-VkCopyImageToImageInfo-srcImage-07981
  If srcImage has a multi-planar format, then for each element of pRegions, srcSubresource.aspectMask **must** be a single valid multi-planar aspect mask bit
- VUID-VkCopyImageToImageInfo-srcImage-07983
  If srcImage is of type VK_IMAGE_TYPE_3D, for each element of pRegions, srcSubresource.baseArrayLayer **must** be 0 and srcSubresource.layerCount **must** be 1
- VUID-VkCopyImageToImageInfo-dstImage-09109
  If dstImage is sparse then all memory ranges accessed by the copy command **must** be bound as described in Binding Resource Memory
- VUID-VkCopyImageToImageInfo-dstImage-09111
  If the stencil aspect of dstImage is accessed, and dstImage was not created with separate stencil usage, dstImage **must** have been created with the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag set
- VUID-VkCopyImageToImageInfo-dstImage-09112
  If the stencil aspect of dstImage is accessed, and dstImage was created with separate stencil usage, dstImage **must** have been created with the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag set
- VUID-VkCopyImageToImageInfo-dstImage-09113
  If non-stencil aspects of dstImage are accessed, dstImage **must** have been created with the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag set
- VUID-VkCopyImageToImageInfo-dstOffset-09114
  If flags contains VK_HOST_IMAGE_COPY_MEMCPY_BIT, the x, y, and z members of the dstOffset member of each element of pRegions **must** be 0
- VUID-VkCopyImageToImageInfo-dstImage-09115
  If flags contains VK_HOST_IMAGE_COPY_MEMCPY_BIT, the extent member of each element of pRegions **must** equal the extents of dstImage identified by dstSubresource
- VUID-VkCopyImageToImageInfo-dstImage-07966
  If dstImage is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-VkCopyImageToImageInfo-dstSubresource-07967
  The dstSubresource.mipLevel member of each element of pRegions **must** be less than the mipLevels specified in VkImageCreateInfo when dstImage was created
- VUID-VkCopyImageToImageInfo-dstSubresource-07968
  If dstSubresource.layerCount is not VK_REMAINING_ARRAY_LAYERS, dstSubresource.baseArrayLayer + dstSubresource.layerCount of each element of pRegions **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when dstImage was created
- VUID-VkCopyImageToImageInfo-dstImage-07969

dstImage **must** not have been created with flags containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT

- VUID-VkCopyImageToImageInfo-dstSubresource-07971
  For each element of pRegions, dstOffset.x and (extent.width + dstOffset.x) **must** both be greater than or equal to 0 and less than or equal to the width of the specified dstSubresource of dstImage
- VUID-VkCopyImageToImageInfo-dstSubresource-07972
  For each element of pRegions, dstOffset.y and (extent.height + dstOffset.y) **must** both be greater than or equal to 0 and less than or equal to the height of the specified dstSubresource of dstImage
- VUID-VkCopyImageToImageInfo-dstImage-07979
  If dstImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, dstOffset.y **must** be 0 and extent.height **must** be 1
- VUID-VkCopyImageToImageInfo-dstOffset-09104
  For each element of pRegions, dstOffset.z and (extent.depth + dstOffset.z) **must** both be greater than or equal to 0 and less than or equal to the depth of the specified dstSubresource of dstImage
- VUID-VkCopyImageToImageInfo-dstImage-07980
  If dstImage is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of pRegions, dstOffset.z **must** be 0 and extent.depth **must** be 1
- VUID-VkCopyImageToImageInfo-dstImage-07274
  For each element of pRegions, dstOffset.x **must** be a multiple of the texel block extent width of the VkFormat of dstImage
- VUID-VkCopyImageToImageInfo-dstImage-07275
  For each element of pRegions, dstOffset.y **must** be a multiple of the texel block extent height of the VkFormat of dstImage
- VUID-VkCopyImageToImageInfo-dstImage-07276
  For each element of pRegions, dstOffset.z **must** be a multiple of the texel block extent depth of the VkFormat of dstImage
- VUID-VkCopyImageToImageInfo-dstImage-00207
  For each element of pRegions, if the sum of dstOffset.x and extent.width does not equal the width of the subresource specified by dstSubresource, extent.width **must** be a multiple of the texel block extent width of the VkFormat of dstImage
- VUID-VkCopyImageToImageInfo-dstImage-00208
  For each element of pRegions, if the sum of dstOffset.y and extent.height does not equal the height of the subresource specified by dstSubresource, extent.height **must** be a multiple of the texel block extent height of the VkFormat of dstImage
- VUID-VkCopyImageToImageInfo-dstImage-00209
  For each element of pRegions, if the sum of dstOffset.z and extent.depth does not equal the depth of the subresource specified by srcSubresource, extent.depth **must** be a multiple of the texel block extent depth of the VkFormat of dstImage
- VUID-VkCopyImageToImageInfo-dstSubresource-09105

For each element of pRegions, dstSubresource.aspectMask **must** specify aspects present in dstImage

- VUID-VkCopyImageToImageInfo-dstImage-07981
  If dstImage has a multi-planar format, then for each element of pRegions, dstSubresource.aspectMask **must** be a single valid multi-planar aspect mask bit
- VUID-VkCopyImageToImageInfo-dstImage-07983
  If dstImage is of type VK_IMAGE_TYPE_3D, for each element of pRegions, dstSubresource.baseArrayLayer **must** be 0 and dstSubresource.layerCount **must** be 1
- VUID-VkCopyImageToImageInfo-srcImageLayout-09070
  srcImageLayout **must** specify the current layout of the image subresources of srcImage specified in pRegions
- VUID-VkCopyImageToImageInfo-dstImageLayout-09071
  dstImageLayout **must** specify the current layout of the image subresources of dstImage specified in pRegions
- VUID-VkCopyImageToImageInfo-srcImageLayout-09072
  srcImageLayout **must** be one of the image layouts returned in VkPhysicalDeviceHostImageCopyProperties::pCopySrcLayouts
- VUID-VkCopyImageToImageInfo-dstImageLayout-09073
  dstImageLayout **must** be one of the image layouts returned in VkPhysicalDeviceHostImageCopyProperties::pCopyDstLayouts

Valid Usage (Implicit)

- VUID-VkCopyImageToImageInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_COPY_IMAGE_TO_IMAGE_INFO
- VUID-VkCopyImageToImageInfo-pNext-pNext
  pNext **must** be NULL
- VUID-VkCopyImageToImageInfo-flags-parameter
  If flags is not 0, flags **must** be a valid combination of VkHostImageCopyFlagBits values
- VUID-VkCopyImageToImageInfo-srcImage-parameter
  srcImage **must** be a valid VkImage handle
- VUID-VkCopyImageToImageInfo-srcImageLayout-parameter
  srcImageLayout **must** be a valid VkImageLayout value
- VUID-VkCopyImageToImageInfo-dstImage-parameter
  dstImage **must** be a valid VkImage handle
- VUID-VkCopyImageToImageInfo-dstImageLayout-parameter
  dstImageLayout **must** be a valid VkImageLayout value
- VUID-VkCopyImageToImageInfo-pRegions-parameter
  pRegions **must** be a valid pointer to an array of regionCount valid VkImageCopy2 structures
- VUID-VkCopyImageToImageInfo-regionCount-arraylength

regionCount **must** be greater than 0

- VUID-VkCopyImageToImageInfo-commonparent
  Both of dstImage, and srcImage **must** have been created, allocated, or retrieved from the same VkDevice

24.6. Indirect Copies

An application can use indirect copies when the copy parameters are not known during the command buffer creation time.

To copy data between two memory regions by specifying copy parameters indirectly in memory, call:

```
// Provided by VK_KHR_copy_memory_indirect
void vkCmdCopyMemoryIndirectKHR(
    VkCommandBuffer                             commandBuffer,
    const VkCopyMemoryIndirectInfoKHR*          pCopyMemoryIndirectInfo);
```

- commandBuffer is the command buffer into which the command will be recorded.
- pCopyMemoryIndirectInfo is a pointer to a VkCopyMemoryIndirectInfoKHR structure containing the copy parameters, including the number of copies to execute and a strided array of VkCopyMemoryIndirectCommandKHR structures.

Each region specified in the memory referenced by pCopyMemoryIndirectInfo->copyAddressRange is copied from the source region to the specified destination region.

Valid Usage

- VUID-vkCmdCopyMemoryIndirectKHR-indirectMemoryCopy-10935
  The indirectMemoryCopy feature **must** be enabled
- VUID-vkCmdCopyMemoryIndirectKHR-commandBuffer-10936
  The VkCommandPool that commandBuffer was allocated from **must** support at least one of the queue types specified in VkPhysicalDeviceCopyMemoryIndirectPropertiesKHR ::supportedQueues
- VUID-vkCmdCopyMemoryIndirectKHR-commandBuffer-10937
  commandBuffer must not be a protected command buffer

Valid Usage (Implicit)

- VUID-vkCmdCopyMemoryIndirectKHR-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyMemoryIndirectKHR-pCopyMemoryIndirectInfo-parameter
  pCopyMemoryIndirectInfo **must** be a valid pointer to a valid

VkCopyMemoryIndirectInfoKHR structure

- VUID-vkCmdCopyMemoryIndirectKHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdCopyMemoryIndirectKHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, or VK_QUEUE_TRANSFER_BIT operations
- VUID-vkCmdCopyMemoryIndirectKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyMemoryIndirectKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyMemoryIndirectKHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdCopyMemoryIndirectKHR is not affected by conditional rendering

The VkCopyMemoryIndirectInfoKHR structure is defined as:

```
// Provided by VK_KHR_copy_memory_indirect
typedef struct VkCopyMemoryIndirectInfoKHR {
    VkStructureType                   sType;
    const void*                       pNext;
    VkAddressCopyFlagsKHR             srcCopyFlags;
    VkAddressCopyFlagsKHR             dstCopyFlags;
    uint32_t                          copyCount;
    VkStridedDeviceAddressRangeKHR    copyAddressRange;
} VkCopyMemoryIndirectInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcCopyFlags` is a VkAddressCopyFlagsKHR value defining the copy flags for the source address range.
- `dstCopyFlags` is a VkAddressCopyFlagsKHR value defining the copy flags for the destination address range.
- `copyCount` is the number of copies to execute, and **can** be zero.
- `copyAddressRange` is a memory region specifying the copy parameters. It is laid out as an array of VkCopyMemoryIndirectCommandKHR structures.

Valid Usage

- VUID-VkCopyMemoryIndirectInfoKHR-srcCopyFlags-10938
  If `srcCopyFlags` contains VK_ADDRESS_COPY_SPARSE_BIT_KHR, the source memory regions accessed **must** be bound to memory
- VUID-VkCopyMemoryIndirectInfoKHR-dstCopyFlags-10939
  If `dstCopyFlags` contains VK_ADDRESS_COPY_SPARSE_BIT_KHR, the destination memory regions accessed **must** be bound to memory
- VUID-VkCopyMemoryIndirectInfoKHR-srcCopyFlags-10940
  `srcCopyFlags` **must** not contain VK_ADDRESS_COPY_PROTECTED_BIT_KHR
- VUID-VkCopyMemoryIndirectInfoKHR-dstCopyFlags-10941
  `dstCopyFlags` **must** not contain VK_ADDRESS_COPY_PROTECTED_BIT_KHR
- VUID-VkCopyMemoryIndirectInfoKHR-copyAddressRange-10942
  `copyAddressRange.address` **must** be 4 byte aligned
- VUID-VkCopyMemoryIndirectInfoKHR-copyAddressRange-10943
  `copyAddressRange.stride` **must** be a multiple of `4` and **must** be greater than or equal to sizeof(VkCopyMemoryIndirectCommandKHR)
- VUID-VkCopyMemoryIndirectInfoKHR-copyCount-10944
  `copyCount` **must** be less than or equal to `copyAddressRange.size` / `copyAddressRange.stride`
- VUID-VkCopyMemoryIndirectInfoKHR-copyAddressRange-10945
  Each memory location accessed by this command **must** not be accessed by both a source region and a destination region, and **must** not be accessed by more than one destination

region, as specified in `copyAddressRange`

- VUID-VkCopyMemoryIndirectInfoKHR-copyAddressRange-12210
  `copyAddressRange` **must** be a device address range allocated to the application from a buffer created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set

Valid Usage (Implicit)

- VUID-VkCopyMemoryIndirectInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COPY_MEMORY_INDIRECT_INFO_KHR
- VUID-VkCopyMemoryIndirectInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkCopyMemoryIndirectInfoKHR-srcCopyFlags-parameter
  If `srcCopyFlags` is not `0`, `srcCopyFlags` **must** be a valid combination of VkAddressCopyFlagBitsKHR values
- VUID-VkCopyMemoryIndirectInfoKHR-dstCopyFlags-parameter
  If `dstCopyFlags` is not `0`, `dstCopyFlags` **must** be a valid combination of VkAddressCopyFlagBitsKHR values

The structure describing source and destination memory regions, `VkCopyMemoryIndirectCommandKHR` is defined as:

```
// Provided by VK_KHR_copy_memory_indirect
typedef struct VkCopyMemoryIndirectCommandKHR {
    VkDeviceAddress    srcAddress;
    VkDeviceAddress    dstAddress;
    VkDeviceSize       size;
} VkCopyMemoryIndirectCommandKHR;
```

- `srcAddress` is the starting address of the source device memory to copy from.
- `dstAddress` is the starting address of the destination device memory to copy to.
- `size` is the size of the copy in bytes.

Valid Usage

- VUID-VkCopyMemoryIndirectCommandKHR-srcAddress-10958
  The `srcAddress` **must** be 4 byte aligned
- VUID-VkCopyMemoryIndirectCommandKHR-dstAddress-10959
  The `dstAddress` **must** be 4 byte aligned
- VUID-VkCopyMemoryIndirectCommandKHR-size-10960
  The `size` **must** be 4 byte aligned
- VUID-VkCopyMemoryIndirectCommandKHR-srcAddress-10961
  The memory in range [`srcAddress`, `srcAddress` + `size` - 1] **must** be within the bounds of the

memory allocation backing `srcAddress`

- VUID-VkCopyMemoryIndirectCommandKHR-dstAddress-10962
  The memory in range [`dstAddress`, `dstAddress` + `size` - 1] **must** be within the bounds of the memory allocation backing `dstAddress`
- VUID-VkCopyMemoryIndirectCommandKHR-srcAddress-12211
  The range of memory defined by `srcAddress` and `size` **must** be a device address range allocated to the application from a buffer created with the VK_BUFFER_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-VkCopyMemoryIndirectCommandKHR-dstAddress-12212
  The range of memory defined by `dstAddress` and `size` **must** be a device address range allocated to the application from a buffer created with the VK_BUFFER_USAGE_TRANSFER_DST_BIT usage flag set

Valid Usage (Implicit)

- VUID-VkCopyMemoryIndirectCommandKHR-srcAddress-parameter
  `srcAddress` **must** be a valid VkDeviceAddress value
- VUID-VkCopyMemoryIndirectCommandKHR-dstAddress-parameter
  `dstAddress` **must** be a valid VkDeviceAddress value

Bits which **can** be set in a VkAddressCopyFlagsKHR mask are:

```
// Provided by VK_KHR_copy_memory_indirect
typedef enum VkAddressCopyFlagBitsKHR {
    VK_ADDRESS_COPY_DEVICE_LOCAL_BIT_KHR = 0x00000001,
    VK_ADDRESS_COPY_SPARSE_BIT_KHR = 0x00000002,
    VK_ADDRESS_COPY_PROTECTED_BIT_KHR = 0x00000004,
} VkAddressCopyFlagBitsKHR;
```

- VK_ADDRESS_COPY_DEVICE_LOCAL_BIT_KHR specifies that the address range is expected to be resident in device local memory. Specifying this value is optional, but **may** lead to improved performance if set accurately.
- VK_ADDRESS_COPY_PROTECTED_BIT_KHR specifies that the address range is allocated from protected memory.
- VK_ADDRESS_COPY_SPARSE_BIT_KHR specifies that the address range may not be fully bound to physical memory when accessed.

```
// Provided by VK_KHR_copy_memory_indirect
typedef VkFlags VkAddressCopyFlagsKHR;
```

`VkAddressCopyFlagsKHR` is a bitmask type for setting a mask of zero or more VkAddressCopyFlagBitsKHR.

To copy data from a memory region to an image object by specifying copy parameters in memory, call:

```
// Provided by VK_KHR_copy_memory_indirect
void vkCmdCopyMemoryToImageIndirectKHR(
    VkCommandBuffer                             commandBuffer,
    const VkCopyMemoryToImageIndirectInfoKHR*   pCopyMemoryToImageIndirectInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pCopyMemoryToImageIndirectInfo` is a pointer to a VkCopyMemoryToImageIndirectInfoKHR structure which contains the copy parameters, including the number of copies to execute and a strided array of VkCopyMemoryToImageIndirectCommandKHR structures.

Each region specified in the memory referenced by `pCopyMemoryToImageIndirectInfo->copyAddressRange` is copied from the source region to an image region in the destination image. If the destination image is of type VK_IMAGE_TYPE_3D, the starting slice and number of slices to copy are specified in `pImageSubresources->baseArrayLayer` and `pImageSubresources->layerCount` respectively as `imageOffset` and `imageExtent` from VkCopyMemoryToImageIndirectCommandKHR are only available at device execution time.

Valid Usage

- VUID-vkCmdCopyMemoryToImageIndirectKHR-indirectMemoryToImageCopy-10947
  The `indirectMemoryToImageCopy` feature **must** be enabled
- VUID-vkCmdCopyMemoryToImageIndirectKHR-commandBuffer-10948
  The VkCommandPool that `commandBuffer` was allocated from **must** support at least one of the queue types specified in VkPhysicalDeviceCopyMemoryIndirectPropertiesKHR ::`supportedQueues`
- VUID-vkCmdCopyMemoryToImageIndirectKHR-commandBuffer-10949
  `commandBuffer` must not be a protected command buffer

Valid Usage (Implicit)

- VUID-vkCmdCopyMemoryToImageIndirectKHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyMemoryToImageIndirectKHR-pCopyMemoryToImageIndirectInfo-parameter
  `pCopyMemoryToImageIndirectInfo` **must** be a valid pointer to a valid VkCopyMemoryToImageIndirectInfoKHR structure
- VUID-vkCmdCopyMemoryToImageIndirectKHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdCopyMemoryToImageIndirectKHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support

VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, or VK_QUEUE_TRANSFER_BIT operations

- VUID-vkCmdCopyMemoryToImageIndirectKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyMemoryToImageIndirectKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyMemoryToImageIndirectKHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT	Action

Conditional Rendering

vkCmdCopyMemoryToImageIndirectKHR is not affected by conditional rendering

The VkCopyMemoryToImageIndirectInfoKHR structure is defined as:

```
// Provided by VK_KHR_copy_memory_indirect
typedef struct VkCopyMemoryToImageIndirectInfoKHR {
    VkStructureType                     sType;
    const void*                         pNext;
    VkAddressCopyFlagsKHR               srcCopyFlags;
    uint32_t                            copyCount;
    VkStridedDeviceAddressRangeKHR      copyAddressRange;
    VkImage                             dstImage;
    VkImageLayout                       dstImageLayout;
    const VkImageSubresourceLayers*     pImageSubresources;
} VkCopyMemoryToImageIndirectInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcCopyFlags` is a VkAddressCopyFlagsKHR value defining the copy flags for the source address range.
- `copyCount` is the number of copies to execute, and **can** be zero.
- `copyAddressRange` is a memory region specifying the copy parameters. It is laid out as an array of VkCopyMemoryToImageIndirectCommandKHR structures.
- `dstImage` is the destination image.
- `dstImageLayout` is the layout of the destination image subresources for the copy.
- `pImageSubresources` is a pointer to an array of `copyCount` VkImageSubresourceLayers structures, specifying the image subresources of the destination image data for the copy operation.

Valid Usage

- VUID-VkCopyMemoryToImageIndirectInfoKHR-srcCopyFlags-10950
  If `srcCopyFlags` contains VK_ADDRESS_COPY_SPARSE_BIT_KHR, the source memory regions accessed **must** be bound to memory
- VUID-VkCopyMemoryToImageIndirectInfoKHR-copyCount-10951
  `copyCount` **must** be less than or equal to `copyAddressRange.size` / `copyAddressRange.stride`
- VUID-VkCopyMemoryToImageIndirectInfoKHR-copyAddressRange-10952
  `copyAddressRange.address` **must** be 4 byte aligned
- VUID-VkCopyMemoryToImageIndirectInfoKHR-copyAddressRange-10953
  `copyAddressRange.stride` **must** be a multiple of `4` and **must** be greater than or equal to sizeof(VkCopyMemoryToImageIndirectCommandKHR)
- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImage-10955
  The format features of `dstImage` **must** contain VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR
- VUID-VkCopyMemoryToImageIndirectInfoKHR-copyAddressRange-12213
  `copyAddressRange` **must** be a device address range allocated to the application from a buffer created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set

- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImage-07661
  `dstImage` **must** not be a protected image
- VUID-VkCopyMemoryToImageIndirectInfoKHR-aspectMask-07662
  The `aspectMask` member for every subresource in `pImageSubresources` **must** only have a single bit set
- VUID-VkCopyMemoryToImageIndirectInfoKHR-aspectMask-12287
  The `aspectMask` member for every subresource in `pImageSubresources` **must** specify an aspect present in `dstImage`
- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImage-07664
  `dstImage` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImage-07665
  If `dstImage` is non-sparse then the image or each specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImage-07973
  `dstImage` **must** have a sample count equal to VK_SAMPLE_COUNT_1_BIT
- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImageLayout-07667
  `dstImageLayout` **must** specify the layout of the image subresources of `dstImage` at the time this command is executed on a `VkDevice`
- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImageLayout-07669
  `dstImageLayout` **must** be VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL, VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, or VK_IMAGE_LAYOUT_GENERAL
- VUID-VkCopyMemoryToImageIndirectInfoKHR-mipLevel-07670
  The specified `mipLevel` of each region in `pImageSubresources` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `dstImage` was created
- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImage-12288
  If `dstImage` is not of type VK_IMAGE_TYPE_3D, and the specified `layerCount` of each region in `pImageSubresources` is not VK_REMAINING_ARRAY_LAYERS, the specified `baseArrayLayer` + `layerCount` of each region in `pImageSubresources` **must** be less than or equal to the `arrayLayers` specified in VkImageCreateInfo when `dstImage` was created
- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImage-12289
  If `dstImage` is of type VK_IMAGE_TYPE_3D, and the specified `layerCount` of each region in `pImageSubresources` is not VK_REMAINING_ARRAY_LAYERS, for each destination region, (`imageSubresource.baseArrayLayer` + `imageSubresource.layerCount`) **must** be less than or equal to the depth of the specified subresource
- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImage-12290
  If `dstImage` is of type VK_IMAGE_TYPE_3D, and the specified `layerCount` of each region in `pImageSubresources` is not VK_REMAINING_ARRAY_LAYERS, for each destination region, if (`imageSubresource.baseArrayLayer` + `imageSubresource.layerCount`) does not equal the depth of the specified subresource, `imageSubresource.layerCount` **must** be a multiple of the texel block extent depth of the VkFormat of `dstImage`
- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImage-12291
  If `dstImage` is of type VK_IMAGE_TYPE_3D, for each destination region,

`imageSubresource.baseArrayLayer` **must** be a multiple of the texel block extent depth of the VkFormat of `dstImage`

- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImage-12292
  If `dstImage` is of type VK_IMAGE_TYPE_3D, for each destination region, `imageSubresource.baseArrayLayer` **must** be less than or equal to the depth of the specified subresource
- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImage-07673
  `dstImage` **must** not have been created with `flags` containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-VkCopyMemoryToImageIndirectInfoKHR-commandBuffer-07674
  If the queue family used to create the VkCommandPool which `commandBuffer` was allocated from does not support VK_QUEUE_GRAPHICS_BIT, for each region, the `aspectMask` member of `pImageSubresources` **must** not be VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImage-10974
  The format features of `dstImage` **must** contain VK_FORMAT_FEATURE_TRANSFER_DST_BIT
- VUID-VkCopyMemoryToImageIndirectInfoKHR-copyAddressRange-10975
  Each memory location accessed by this command **must** not be accessed by both a source region and a destination region, and **must** not be accessed by more than one destination region, as specified in `copyAddressRange` and accessed according to Buffer and Image Addressing

Valid Usage (Implicit)

- VUID-VkCopyMemoryToImageIndirectInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COPY_MEMORY_TO_IMAGE_INDIRECT_INFO_KHR
- VUID-VkCopyMemoryToImageIndirectInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkCopyMemoryToImageIndirectInfoKHR-srcCopyFlags-parameter
  If `srcCopyFlags` is not `0`, `srcCopyFlags` **must** be a valid combination of VkAddressCopyFlagBitsKHR values
- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImage-parameter
  `dstImage` **must** be a valid VkImage handle
- VUID-VkCopyMemoryToImageIndirectInfoKHR-dstImageLayout-parameter
  `dstImageLayout` **must** be a valid VkImageLayout value
- VUID-VkCopyMemoryToImageIndirectInfoKHR-pImageSubresources-parameter
  `pImageSubresources` **must** be a valid pointer to an array of `copyCount` valid VkImageSubresourceLayers structures
- VUID-VkCopyMemoryToImageIndirectInfoKHR-copyCount-arraylength
  `copyCount` **must** be greater than `0`

The structure describing source and destination memory regions,

VkCopyMemoryToImageIndirectCommandKHR is defined as:

```
// Provided by VK_KHR_copy_memory_indirect
typedef struct VkCopyMemoryToImageIndirectCommandKHR {
    VkDeviceAddress             srcAddress;
    uint32_t                    bufferRowLength;
    uint32_t                    bufferImageHeight;
    VkImageSubresourceLayers    imageSubresource;
    VkOffset3D                  imageOffset;
    VkExtent3D                  imageExtent;
} VkCopyMemoryToImageIndirectCommandKHR;
```

- srcAddress is the starting address of the source device memory to copy from.
- bufferRowLength and bufferImageHeight specify in texels a subregion of a larger two- or three-dimensional image in buffer memory, and control the addressing calculations. If either of these values is zero, that aspect of the buffer memory is considered to be tightly packed according to the imageExtent.
- imageSubresource is a VkImageSubresourceLayers structure used to specify the specific image subresources of the image used for the destination image data, which **must** match the value specified in the corresponding index of the pCopyMemoryToImageIndirectInfo->pImageSubresources array of vkCmdCopyMemoryToImageIndirectKHR during command recording.
- imageOffset selects the initial x, y, z offsets in texels of the sub-region of the destination image data.
- imageExtent is the size in texels of the destination image in width, height and depth.

Valid Usage

- VUID-VkCopyMemoryToImageIndirectCommandKHR-srcAddress-10963
  The srcAddress **must** be 4 byte aligned
- VUID-VkCopyMemoryToImageIndirectCommandKHR-dstImage-12282
  If dstImage does not have either a depth/stencil format or a multi-planar format, srcAddress **must** be a multiple of the texel block size
- VUID-VkCopyMemoryToImageIndirectCommandKHR-dstImage-12283
  If dstImage has a multi-planar format, srcAddress **must** be a multiple of the element size of the compatible format for the format and the aspectMask of the imageSubresource as defined in Compatible Formats of Planes of Multi-Planar Formats
- VUID-VkCopyMemoryToImageIndirectCommandKHR-bufferRowLength-10964
  bufferRowLength **must** be 0, or greater than or equal to the width member of imageExtent
- VUID-VkCopyMemoryToImageIndirectCommandKHR-bufferImageHeight-10965
  bufferImageHeight **must** be 0, or greater than or equal to the height member of imageExtent
- VUID-VkCopyMemoryToImageIndirectCommandKHR-imageOffset-10966
  imageOffset **must** specify a valid offset in the destination image
- VUID-VkCopyMemoryToImageIndirectCommandKHR-imageExtent-10967

`imageExtent` **must** specify a valid region in the destination image and **can** be `0`

- VUID-VkCopyMemoryToImageIndirectCommandKHR-srcAddress-10968
  The memory allocation backing `srcAddress` **must** be large enough to contain all locations that are accessed according to Buffer and Image Addressing
- VUID-VkCopyMemoryToImageIndirectCommandKHR-imageOffset-10969
  The `imageOffset` and `imageExtent` region **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in VkQueueFamilyProperties
- VUID-VkCopyMemoryToImageIndirectCommandKHR-imageOffset-10970
  For each destination region, `imageOffset.x` and (`imageExtent.width` + `imageOffset.x`) **must** both be greater than or equal to `0` and less than or equal to the width of the specified subresource
- VUID-VkCopyMemoryToImageIndirectCommandKHR-imageOffset-10971
  For each destination region, `imageOffset.y` and (`imageExtent.height` + `imageOffset.y`) **must** both be greater than or equal to `0` and less than or equal to the height of the specified subresource
- VUID-VkCopyMemoryToImageIndirectCommandKHR-imageSubresource-12284
  The members of `imageSubresource` **must** be identical to the members of the VkImageSubresourceLayers structure specified in the corresponding index of the `pCopyMemoryToImageIndirectInfo->pImageSubresources` array of vkCmdCopyMemoryToImageIndirectKHR during command recording
- VUID-VkCopyMemoryToImageIndirectCommandKHR-dstImage-12285
  If `dstImage` is of type VK_IMAGE_TYPE_1D, `imageOffset.y` **must** be `0` and `imageExtent.height` **must** be `1`
- VUID-VkCopyMemoryToImageIndirectCommandKHR-dstImage-12286
  If `dstImage` is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, `imageOffset.z` **must** be `0` and `imageExtent.depth` **must** be `1`
- VUID-VkCopyMemoryToImageIndirectCommandKHR-srcAddress-12214
  `srcAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_TRANSFER_SRC_BIT usage flag set

Valid Usage (Implicit)

- VUID-VkCopyMemoryToImageIndirectCommandKHR-srcAddress-parameter
  `srcAddress` **must** be a valid VkDeviceAddress value
- VUID-VkCopyMemoryToImageIndirectCommandKHR-imageSubresource-parameter
  `imageSubresource` **must** be a valid VkImageSubresourceLayers structure

24.7. Image Copies With Scaling

To copy regions of a source image into a destination image, potentially performing format conversion, arbitrary scaling, and filtering, call:

```
// Provided by VK_VERSION_1_0
void vkCmdBlitImage(
    VkCommandBuffer                             commandBuffer,
    VkImage                                     srcImage,
    VkImageLayout                               srcImageLayout,
    VkImage                                     dstImage,
    VkImageLayout                               dstImageLayout,
    uint32_t                                    regionCount,
    const VkImageBlit*                          pRegions,
    VkFilter                                    filter);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `srcImage` is the source image.
- `srcImageLayout` is the layout of the source image subresources for the blit.
- `dstImage` is the destination image.
- `dstImageLayout` is the layout of the destination image subresources for the blit.
- `regionCount` is the number of regions to blit.
- `pRegions` is a pointer to an array of VkImageBlit structures specifying the regions to blit.
- `filter` is a VkFilter specifying the filter to apply if the blits require scaling.

`vkCmdBlitImage` **must** not be used for multisampled source or destination images. Use vkCmdResolveImage for this purpose.

As the sizes of the source and destination extents **can** differ in any dimension, texels in the source extent are scaled and filtered to the destination extent. Scaling occurs via the following operations:

- For each destination texel, the integer coordinate of that texel is converted to an unnormalized texture coordinate, using the effective inverse of the equations described in unnormalized to integer conversion:

$$u_{base} = i + \frac{1}{2}$$

$$v_{base} = j + \frac{1}{2}$$

$$w_{base} = k + \frac{1}{2}$$

- These base coordinates are then offset by the first destination offset:

$$u_{offset} = u_{base} - x_{dst0}$$

$$v_{offset} = v_{base} - y_{dst0}$$

$$w_{offset} = w_{base} - z_{dst0}$$

$$a_{offset} = a - \texttt{baseArrayLayer}_{dst}$$

- The scale is determined from the source and destination regions, and applied to the offset coordinates:

$$scale_u = (x_{src1} - x_{src0}) / (x_{dst1} - x_{dst0})$$

$$scale_v = (y_{src1} - y_{src0}) / (y_{dst1} - y_{dst0})$$

$$scale_w = (z_{src1} - z_{src0}) / (z_{dst1} - z_{dst0})$$

$$u_{scaled} = u_{offset} \times scale_u$$

$$v_{scaled} = v_{offset} \times scale_v$$

$$w_{scaled} = w_{offset} \times scale_w$$

- Finally the source offset is added to the scaled coordinates, to determine the final unnormalized coordinates used to sample from `srcImage`:

$$u = u_{scaled} + x_{src0}$$

$$v = v_{scaled} + y_{src0}$$

$$w = w_{scaled} + z_{src0}$$

$$q = \texttt{mipLevel}$$

$$a = a_{offset} + \texttt{baseArrayLayer}_{src}$$

These coordinates are used to sample from the source image, as described in Image Operations, with the filter mode equal to that of `filter`, a mipmap mode of VK_SAMPLER_MIPMAP_MODE_NEAREST and an address mode of VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE. Implementations **must** clamp at the edge of the

source image, and **may** additionally clamp to the edge of the source region.

The destination offsets form a box which contains all texels which are written. A component in `dstOffsets`[1] **can** be smaller than the corresponding component in `dstOffsets`[0] and forms the same box as swapped `dstOffsets`. A destination texel exists if the half-texel coordinates u_{base}, v_{base} and w_{base} are all contained inside the region.

NOTE

Due to allowable rounding errors in the generation of the source texture coordinates, it is not always possible to guarantee exactly which source texels will be sampled for a given blit. As rounding errors are implementation-dependent, the exact results of a blitting operation are also implementation-dependent.

While blitting with equal destination coordinates is not permitted, blitting with source coordinates being equal is. However, the resulting coordinate is integer and will approximately land in-between texels when sampled, which may produce unexpected results. Especially for nearest filtering, allowed imprecision in generation of the source coordinate may end up sampling an unexpected texel due to rounding errors if the coordinate is off by one ULP. Implementations are not required to implement the formula above exactly as written. E.g., varying interpolation in a fragment shader may be used.

Blits are done layer by layer starting with the `baseArrayLayer` member of `srcSubresource` for the source and `dstSubresource` for the destination. `layerCount` layers are blitted to the destination image.

When blitting 3D textures, slices in the destination region bounded by `dstOffsets`[0].z and `dstOffsets`[1].z are sampled from slices in the source region bounded by `srcOffsets`[0].z and `srcOffsets`[1].z. If the `filter` parameter is VK_FILTER_LINEAR then the value sampled from the source image is taken by doing linear filtering using the interpolated **z** coordinate represented by **w** in the previous equations. If the `filter` parameter is VK_FILTER_NEAREST then the value sampled from the source image is taken from the single nearest slice, with an implementation-dependent arithmetic rounding mode.

The following filtering and conversion rules apply:

- Integer formats **can** only be converted to other integer formats with the same signedness.
- No format conversion is supported between depth/stencil images. The formats **must** match.
- Format conversions on unorm, snorm, scaled and packed float formats of the copied aspect of the image are performed by first converting the pixels to float values.
- For sRGB source formats, nonlinear RGB values are converted to linear representation prior to filtering.
- After filtering, the float values are first clamped and then cast to the destination image format. In case of sRGB destination format, linear RGB values are converted to nonlinear representation before writing the pixel to the image.

Signed and unsigned integers are converted by first clamping to the representable range of the destination format, then casting the value.

Valid Usage

- VUID-vkCmdBlitImage-commandBuffer-01834
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `srcImage` **must** not be a protected image
- VUID-vkCmdBlitImage-commandBuffer-01835
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `dstImage` **must** not be a protected image
- VUID-vkCmdBlitImage-commandBuffer-01836
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, `dstImage` **must** not be an unprotected image
- VUID-vkCmdBlitImage-pRegions-00217
  The union of all destination regions, specified by the elements of `pRegions`, **must** not overlap in memory with any texel that **may** be sampled during the blit operation
- VUID-vkCmdBlitImage-srcImage-01999
  The format features of `srcImage` **must** contain VK_FORMAT_FEATURE_BLIT_SRC_BIT
- VUID-vkCmdBlitImage-srcImage-06421
  `srcImage` **must** not use a format that requires a sampler $Y'C_BC_R$ conversion
- VUID-vkCmdBlitImage-srcImage-00219
  `srcImage` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-vkCmdBlitImage-srcImage-00220
  If `srcImage` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBlitImage-srcImageLayout-00221
  `srcImageLayout` **must** specify the layout of the image subresources of `srcImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-vkCmdBlitImage-srcImageLayout-01398
  `srcImageLayout` **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL or VK_IMAGE_LAYOUT_GENERAL
- VUID-vkCmdBlitImage-srcImage-09459
  If `srcImage` and `dstImage` are the same, and an element of `pRegions` contains the `srcSubresource` and `dstSubresource` with matching `mipLevel` and overlapping array layers, then the `srcImageLayout` and `dstImageLayout` **must** be VK_IMAGE_LAYOUT_GENERAL or VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR
- VUID-vkCmdBlitImage-dstImage-02000
  The format features of `dstImage` **must** contain VK_FORMAT_FEATURE_BLIT_DST_BIT
- VUID-vkCmdBlitImage-dstImage-06422
  `dstImage` **must** not use a format that requires a sampler $Y'C_BC_R$ conversion
- VUID-vkCmdBlitImage-dstImage-00224
  `dstImage` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set

- VUID-vkCmdBlitImage-dstImage-00225
  If `dstImage` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBlitImage-dstImageLayout-00226
  `dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-vkCmdBlitImage-dstImageLayout-01399
  `dstImageLayout` **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL or VK_IMAGE_LAYOUT_GENERAL
- VUID-vkCmdBlitImage-srcImage-00229
  If either of `srcImage` or `dstImage` was created with a signed integer VkFormat, the other **must** also have been created with a signed integer VkFormat
- VUID-vkCmdBlitImage-srcImage-00230
  If either of `srcImage` or `dstImage` was created with an unsigned integer color VkFormat, the other **must** also have been created with an unsigned integer color VkFormat
- VUID-vkCmdBlitImage-srcImage-00231
  If either of `srcImage` or `dstImage` was created with a depth/stencil format, the other **must** have exactly the same format
- VUID-vkCmdBlitImage-srcImage-00232
  If `srcImage` was created with a depth/stencil format, `filter` **must** be VK_FILTER_NEAREST
- VUID-vkCmdBlitImage-srcImage-00233
  `srcImage` **must** have been created with a `samples` value of VK_SAMPLE_COUNT_1_BIT
- VUID-vkCmdBlitImage-dstImage-00234
  `dstImage` **must** have been created with a `samples` value of VK_SAMPLE_COUNT_1_BIT
- VUID-vkCmdBlitImage-filter-02001
  If `filter` is VK_FILTER_LINEAR, then the format features of `srcImage` **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdBlitImage-filter-02002
  If `filter` is VK_FILTER_CUBIC_EXT, then the format features of `srcImage` **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdBlitImage-filter-00237
  If `filter` is VK_FILTER_CUBIC_EXT, `srcImage` **must** be of type VK_IMAGE_TYPE_2D
- VUID-vkCmdBlitImage-srcSubresource-01705
  The `srcSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `srcImage` was created
- VUID-vkCmdBlitImage-dstSubresource-01706
  The `dstSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `dstImage` was created
- VUID-vkCmdBlitImage-srcSubresource-01707
  If `srcSubresource.layerCount` is not VK_REMAINING_ARRAY_LAYERS, `srcSubresource.baseArrayLayer` + `srcSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in VkImageCreateInfo when

srcImage was created

- VUID-vkCmdBlitImage-dstSubresource-01708
  If dstSubresource.layerCount is not VK_REMAINING_ARRAY_LAYERS, dstSubresource.baseArrayLayer + dstSubresource.layerCount of each element of pRegions **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when dstImage was created
- VUID-vkCmdBlitImage-dstImage-02545
  dstImage and srcImage **must** not have been created with flags containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-vkCmdBlitImage-maintenance8-10207
  If the maintenance8 feature is enabled and srcImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, srcSubresource.baseArrayLayer **must** be 0, and srcSubresource.layerCount and dstSubresource.layerCount **must** each be 1
- VUID-vkCmdBlitImage-maintenance8-10208
  If the maintenance8 feature is enabled and dstImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, dstSubresource.baseArrayLayer **must** be 0, and srcSubresource.layerCount and dstSubresource.layerCount **must** each be 1
- VUID-vkCmdBlitImage-maintenance8-10579
  If the maintenance8 feature is enabled, dstImage is VK_IMAGE_TYPE_3D, and srcImage is not of type VK_IMAGE_TYPE_3D, then for each element of pRegions, the absolute difference of the z member of each member of dstOffsets **must** equal srcSubresource.layerCount
- VUID-vkCmdBlitImage-maintenance8-10580
  If the maintenance8 feature is enabled, srcImage is VK_IMAGE_TYPE_3D, and dstImage is not of type VK_IMAGE_TYPE_3D, then for each element of pRegions, the absolute difference of the z member of each member of srcOffsets **must** equal dstSubresource.layerCount
- VUID-vkCmdBlitImage-srcImage-00240
  If the maintenance8 feature is not enabled and either srcImage or dstImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, srcSubresource.baseArrayLayer and dstSubresource.baseArrayLayer **must** each be 0, and srcSubresource.layerCount and dstSubresource.layerCount **must** each be 1
- VUID-vkCmdBlitImage-aspectMask-00241
  For each element of pRegions, srcSubresource.aspectMask **must** specify aspects present in srcImage
- VUID-vkCmdBlitImage-aspectMask-00242
  For each element of pRegions, dstSubresource.aspectMask **must** specify aspects present in dstImage
- VUID-vkCmdBlitImage-srcOffset-00243
  For each element of pRegions, srcOffsets[0].x and srcOffsets[1].x **must** both be greater than or equal to 0 and less than or equal to the width of the specified srcSubresource of srcImage
- VUID-vkCmdBlitImage-srcOffset-00244
  For each element of pRegions, srcOffsets[0].y and srcOffsets[1].y **must** both be greater than or equal to 0 and less than or equal to the height of the specified srcSubresource of

srcImage

- VUID-vkCmdBlitImage-srcImage-00245
  If srcImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, srcOffsets[0].y **must** be 0 and srcOffsets[1].y **must** be 1
- VUID-vkCmdBlitImage-srcOffset-00246
  For each element of pRegions, srcOffsets[0].z and srcOffsets[1].z **must** both be greater than or equal to 0 and less than or equal to the depth of the specified srcSubresource of srcImage
- VUID-vkCmdBlitImage-srcImage-00247
  If srcImage is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of pRegions, srcOffsets[0].z **must** be 0 and srcOffsets[1].z **must** be 1
- VUID-vkCmdBlitImage-dstOffset-00248
  For each element of pRegions, dstOffsets[0].x and dstOffsets[1].x **must** both be greater than or equal to 0 and less than or equal to the width of the specified dstSubresource of dstImage
- VUID-vkCmdBlitImage-dstOffset-00249
  For each element of pRegions, dstOffsets[0].y and dstOffsets[1].y **must** both be greater than or equal to 0 and less than or equal to the height of the specified dstSubresource of dstImage
- VUID-vkCmdBlitImage-dstImage-00250
  If dstImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, dstOffsets[0].y **must** be 0 and dstOffsets[1].y **must** be 1
- VUID-vkCmdBlitImage-dstOffset-00251
  For each element of pRegions, dstOffsets[0].z and dstOffsets[1].z **must** both be greater than or equal to 0 and less than or equal to the depth of the specified dstSubresource of dstImage
- VUID-vkCmdBlitImage-dstImage-00252
  If dstImage is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of pRegions, dstOffsets[0].z **must** be 0 and dstOffsets[1].z **must** be 1
- VUID-vkCmdBlitImage-dstOffsets-12518
  dstOffsets[0].x **must** not be equal to dstOffsets[1].x
- VUID-vkCmdBlitImage-dstOffsets-12519
  dstOffsets[0].y **must** not be equal to dstOffsets[1].y
- VUID-vkCmdBlitImage-dstOffsets-12520
  dstOffsets[0].z **must** not be equal to dstOffsets[1].z

Valid Usage (Implicit)

- VUID-vkCmdBlitImage-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBlitImage-srcImage-parameter
  srcImage **must** be a valid VkImage handle

- VUID-vkCmdBlitImage-srcImageLayout-parameter
  `srcImageLayout` **must** be a valid VkImageLayout value
- VUID-vkCmdBlitImage-dstImage-parameter
  `dstImage` **must** be a valid VkImage handle
- VUID-vkCmdBlitImage-dstImageLayout-parameter
  `dstImageLayout` **must** be a valid VkImageLayout value
- VUID-vkCmdBlitImage-pRegions-parameter
  `pRegions` **must** be a valid pointer to an array of `regionCount` valid VkImageBlit structures
- VUID-vkCmdBlitImage-filter-parameter
  `filter` **must** be a valid VkFilter value
- VUID-vkCmdBlitImage-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBlitImage-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBlitImage-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdBlitImage-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdBlitImage-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBlitImage-regionCount-arraylength
  `regionCount` **must** be greater than `0`
- VUID-vkCmdBlitImage-commonparent
  Each of `commandBuffer`, `dstImage`, and `srcImage` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdBlitImage is not affected by conditional rendering

The VkImageBlit structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageBlit {
    VkImageSubresourceLayers    srcSubresource;
    VkOffset3D                  srcOffsets[2];
    VkImageSubresourceLayers    dstSubresource;
    VkOffset3D                  dstOffsets[2];
} VkImageBlit;
```

- srcSubresource is the subresource to blit from.
- srcOffsets is a pointer to an array of two VkOffset3D structures specifying the bounds of the source region within srcSubresource.
- dstSubresource is the subresource to blit into.
- dstOffsets is a pointer to an array of two VkOffset3D structures specifying the bounds of the destination region within dstSubresource.

For each element of the pRegions array, a blit operation is performed for the specified source and destination regions.

Valid Usage

- VUID-VkImageBlit-aspectMask-00238
  The aspectMask member of srcSubresource and dstSubresource **must** match
- VUID-VkImageBlit-layerCount-08800
  If neither of the layerCount members of srcSubresource or dstSubresource are VK_REMAINING_ARRAY_LAYERS, the layerCount members of srcSubresource and dstSubresource **must** match
- VUID-VkImageBlit-layerCount-08801
  If one of the layerCount members of srcSubresource or dstSubresource is VK_REMAINING_ARRAY_LAYERS, the other member **must** be either VK_REMAINING_ARRAY_LAYERS or equal to the arrayLayers member of the VkImageCreateInfo used to create the image minus baseArrayLayer

Valid Usage (Implicit)

- VUID-VkImageBlit-srcSubresource-parameter
  srcSubresource **must** be a valid VkImageSubresourceLayers structure
- VUID-VkImageBlit-dstSubresource-parameter

`dstSubresource` **must** be a valid VkImageSubresourceLayers structure

A more extensible version of the blit image command is defined below.

To copy regions of a source image into a destination image, potentially performing format conversion, arbitrary scaling, and filtering, call:

```
// Provided by VK_VERSION_1_3
void vkCmdBlitImage2(
    VkCommandBuffer                             commandBuffer,
    const VkBlitImageInfo2*                     pBlitImageInfo);
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to vkCmdBlitImage2
void vkCmdBlitImage2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkBlitImageInfo2*                     pBlitImageInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pBlitImageInfo` is a pointer to a VkBlitImageInfo2 structure describing the blit parameters.

This command is functionally identical to vkCmdBlitImage, but includes extensible sub-structures that include `sType` and `pNext` parameters, allowing them to be more easily extended.

Valid Usage

- VUID-vkCmdBlitImage2-commandBuffer-01834
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `srcImage` **must** not be a protected image
- VUID-vkCmdBlitImage2-commandBuffer-01835
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `dstImage` **must** not be a protected image
- VUID-vkCmdBlitImage2-commandBuffer-01836
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, `dstImage` **must** not be an unprotected image

Valid Usage (Implicit)

- VUID-vkCmdBlitImage2-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBlitImage2-pBlitImageInfo-parameter
  `pBlitImageInfo` **must** be a valid pointer to a valid VkBlitImageInfo2 structure
- VUID-vkCmdBlitImage2-commandBuffer-recording

commandBuffer **must** be in the recording state

- VUID-vkCmdBlitImage2-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBlitImage2-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdBlitImage2-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdBlitImage2-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdBlitImage2 is not affected by conditional rendering

The VkBlitImageInfo2 structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkBlitImageInfo2 {
    VkStructureType        sType;
    const void*            pNext;
    VkImage                srcImage;
    VkImageLayout          srcImageLayout;
    VkImage                dstImage;
    VkImageLayout          dstImageLayout;
    uint32_t               regionCount;
    const VkImageBlit2*    pRegions;
    VkFilter               filter;
} VkBlitImageInfo2;
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to VkBlitImageInfo2
typedef VkBlitImageInfo2 VkBlitImageInfo2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcImage` is the source image.
- `srcImageLayout` is the layout of the source image subresources for the blit.
- `dstImage` is the destination image.
- `dstImageLayout` is the layout of the destination image subresources for the blit.
- `regionCount` is the number of regions to blit.
- `pRegions` is a pointer to an array of VkImageBlit2 structures specifying the regions to blit.
- `filter` is a VkFilter specifying the filter to apply if the blits require scaling.

Valid Usage

- VUID-VkBlitImageInfo2-pRegions-00217
  The union of all destination regions, specified by the elements of `pRegions`, **must** not overlap in memory with any texel that **may** be sampled during the blit operation
- VUID-VkBlitImageInfo2-srcImage-01999
  The format features of `srcImage` **must** contain VK_FORMAT_FEATURE_BLIT_SRC_BIT
- VUID-VkBlitImageInfo2-srcImage-06421
  `srcImage` **must** not use a format that requires a sampler $Y'C_BC_R$ conversion
- VUID-VkBlitImageInfo2-srcImage-00219
  `srcImage` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-VkBlitImageInfo2-srcImage-00220
  If `srcImage` is non-sparse then it **must** be bound completely and contiguously to a single

VkDeviceMemory object

- VUID-VkBlitImageInfo2-srcImageLayout-00221
  srcImageLayout **must** specify the layout of the image subresources of srcImage specified in pRegions at the time this command is executed on a VkDevice
- VUID-VkBlitImageInfo2-srcImageLayout-01398
  srcImageLayout **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL or VK_IMAGE_LAYOUT_GENERAL
- VUID-VkBlitImageInfo2-srcImage-09459
  If srcImage and dstImage are the same, and an element of pRegions contains the srcSubresource and dstSubresource with matching mipLevel and overlapping array layers, then the srcImageLayout and dstImageLayout **must** be VK_IMAGE_LAYOUT_GENERAL or VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR
- VUID-VkBlitImageInfo2-dstImage-02000
  The format features of dstImage **must** contain VK_FORMAT_FEATURE_BLIT_DST_BIT
- VUID-VkBlitImageInfo2-dstImage-06422
  dstImage **must** not use a format that requires a sampler $Y'C_BC_R$ conversion
- VUID-VkBlitImageInfo2-dstImage-00224
  dstImage **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-VkBlitImageInfo2-dstImage-00225
  If dstImage is non-sparse then it **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-VkBlitImageInfo2-dstImageLayout-00226
  dstImageLayout **must** specify the layout of the image subresources of dstImage specified in pRegions at the time this command is executed on a VkDevice
- VUID-VkBlitImageInfo2-dstImageLayout-01399
  dstImageLayout **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL or VK_IMAGE_LAYOUT_GENERAL
- VUID-VkBlitImageInfo2-srcImage-00229
  If either of srcImage or dstImage was created with a signed integer VkFormat, the other **must** also have been created with a signed integer VkFormat
- VUID-VkBlitImageInfo2-srcImage-00230
  If either of srcImage or dstImage was created with an unsigned integer color VkFormat, the other **must** also have been created with an unsigned integer color VkFormat
- VUID-VkBlitImageInfo2-srcImage-00231
  If either of srcImage or dstImage was created with a depth/stencil format, the other **must** have exactly the same format
- VUID-VkBlitImageInfo2-srcImage-00232
  If srcImage was created with a depth/stencil format, filter **must** be VK_FILTER_NEAREST
- VUID-VkBlitImageInfo2-srcImage-00233
  srcImage **must** have been created with a samples value of VK_SAMPLE_COUNT_1_BIT
- VUID-VkBlitImageInfo2-dstImage-00234

dstImage **must** have been created with a samples value of VK_SAMPLE_COUNT_1_BIT

- VUID-VkBlitImageInfo2-filter-02001
  If filter is VK_FILTER_LINEAR, then the format features of srcImage **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-VkBlitImageInfo2-filter-02002
  If filter is VK_FILTER_CUBIC_EXT, then the format features of srcImage **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-VkBlitImageInfo2-filter-00237
  If filter is VK_FILTER_CUBIC_EXT, srcImage **must** be of type VK_IMAGE_TYPE_2D
- VUID-VkBlitImageInfo2-srcSubresource-01705
  The srcSubresource.mipLevel member of each element of pRegions **must** be less than the mipLevels specified in VkImageCreateInfo when srcImage was created
- VUID-VkBlitImageInfo2-dstSubresource-01706
  The dstSubresource.mipLevel member of each element of pRegions **must** be less than the mipLevels specified in VkImageCreateInfo when dstImage was created
- VUID-VkBlitImageInfo2-srcSubresource-01707
  If srcSubresource.layerCount is not VK_REMAINING_ARRAY_LAYERS, srcSubresource.baseArrayLayer + srcSubresource.layerCount of each element of pRegions **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when srcImage was created
- VUID-VkBlitImageInfo2-dstSubresource-01708
  If dstSubresource.layerCount is not VK_REMAINING_ARRAY_LAYERS, dstSubresource.baseArrayLayer + dstSubresource.layerCount of each element of pRegions **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when dstImage was created
- VUID-VkBlitImageInfo2-dstImage-02545
  dstImage and srcImage **must** not have been created with flags containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-VkBlitImageInfo2-maintenance8-10207
  If the maintenance8 feature is enabled and srcImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, srcSubresource.baseArrayLayer **must** be 0, and srcSubresource.layerCount and dstSubresource.layerCount **must** each be 1
- VUID-VkBlitImageInfo2-maintenance8-10208
  If the maintenance8 feature is enabled and dstImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, dstSubresource.baseArrayLayer **must** be 0, and srcSubresource.layerCount and dstSubresource.layerCount **must** each be 1
- VUID-VkBlitImageInfo2-maintenance8-10579
  If the maintenance8 feature is enabled, dstImage is VK_IMAGE_TYPE_3D, and srcImage is not of type VK_IMAGE_TYPE_3D, then for each element of pRegions, the absolute difference of the z member of each member of dstOffsets **must** equal srcSubresource.layerCount
- VUID-VkBlitImageInfo2-maintenance8-10580
  If the maintenance8 feature is enabled, srcImage is VK_IMAGE_TYPE_3D, and dstImage is not of type VK_IMAGE_TYPE_3D, then for each element of pRegions, the absolute difference of

the z member of each member of srcOffsets **must** equal dstSubresource.layerCount

- VUID-VkBlitImageInfo2-srcImage-00240
  If the maintenance8 feature is not enabled and either srcImage or dstImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, srcSubresource.baseArrayLayer and dstSubresource.baseArrayLayer **must** each be 0, and srcSubresource.layerCount and dstSubresource.layerCount **must** each be 1
- VUID-VkBlitImageInfo2-aspectMask-00241
  For each element of pRegions, srcSubresource.aspectMask **must** specify aspects present in srcImage
- VUID-VkBlitImageInfo2-aspectMask-00242
  For each element of pRegions, dstSubresource.aspectMask **must** specify aspects present in dstImage
- VUID-VkBlitImageInfo2-srcOffset-00243
  For each element of pRegions, srcOffsets[0].x and srcOffsets[1].x **must** both be greater than or equal to 0 and less than or equal to the width of the specified srcSubresource of srcImage
- VUID-VkBlitImageInfo2-srcOffset-00244
  For each element of pRegions, srcOffsets[0].y and srcOffsets[1].y **must** both be greater than or equal to 0 and less than or equal to the height of the specified srcSubresource of srcImage
- VUID-VkBlitImageInfo2-srcImage-00245
  If srcImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, srcOffsets[0].y **must** be 0 and srcOffsets[1].y **must** be 1
- VUID-VkBlitImageInfo2-srcOffset-00246
  For each element of pRegions, srcOffsets[0].z and srcOffsets[1].z **must** both be greater than or equal to 0 and less than or equal to the depth of the specified srcSubresource of srcImage
- VUID-VkBlitImageInfo2-srcImage-00247
  If srcImage is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of pRegions, srcOffsets[0].z **must** be 0 and srcOffsets[1].z **must** be 1
- VUID-VkBlitImageInfo2-dstOffset-00248
  For each element of pRegions, dstOffsets[0].x and dstOffsets[1].x **must** both be greater than or equal to 0 and less than or equal to the width of the specified dstSubresource of dstImage
- VUID-VkBlitImageInfo2-dstOffset-00249
  For each element of pRegions, dstOffsets[0].y and dstOffsets[1].y **must** both be greater than or equal to 0 and less than or equal to the height of the specified dstSubresource of dstImage
- VUID-VkBlitImageInfo2-dstImage-00250
  If dstImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, dstOffsets[0].y **must** be 0 and dstOffsets[1].y **must** be 1
- VUID-VkBlitImageInfo2-dstOffset-00251
  For each element of pRegions, dstOffsets[0].z and dstOffsets[1].z **must** both be greater

than or equal to `0` and less than or equal to the depth of the specified `dstSubresource` of `dstImage`

- VUID-VkBlitImageInfo2-dstImage-00252
  If `dstImage` is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of `pRegions`, `dstOffsets`[0].z **must** be `0` and `dstOffsets`[1].z **must** be `1`
- VUID-VkBlitImageInfo2-dstOffsets-12518
  `dstOffsets`[0].x **must** not be equal to `dstOffsets`[1].x
- VUID-VkBlitImageInfo2-dstOffsets-12519
  `dstOffsets`[0].y **must** not be equal to `dstOffsets`[1].y
- VUID-VkBlitImageInfo2-dstOffsets-12520
  `dstOffsets`[0].z **must** not be equal to `dstOffsets`[1].z

Valid Usage (Implicit)

- VUID-VkBlitImageInfo2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BLIT_IMAGE_INFO_2
- VUID-VkBlitImageInfo2-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkBlitImageInfo2-srcImage-parameter
  `srcImage` **must** be a valid VkImage handle
- VUID-VkBlitImageInfo2-srcImageLayout-parameter
  `srcImageLayout` **must** be a valid VkImageLayout value
- VUID-VkBlitImageInfo2-dstImage-parameter
  `dstImage` **must** be a valid VkImage handle
- VUID-VkBlitImageInfo2-dstImageLayout-parameter
  `dstImageLayout` **must** be a valid VkImageLayout value
- VUID-VkBlitImageInfo2-pRegions-parameter
  `pRegions` **must** be a valid pointer to an array of `regionCount` valid VkImageBlit2 structures
- VUID-VkBlitImageInfo2-filter-parameter
  `filter` **must** be a valid VkFilter value
- VUID-VkBlitImageInfo2-regionCount-arraylength
  `regionCount` **must** be greater than `0`
- VUID-VkBlitImageInfo2-commonparent
  Both of `dstImage`, and `srcImage` **must** have been created, allocated, or retrieved from the same VkDevice

The `VkImageBlit2` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkImageBlit2 {
    VkStructureType             sType;
    const void*                 pNext;
    VkImageSubresourceLayers    srcSubresource;
    VkOffset3D                  srcOffsets[2];
    VkImageSubresourceLayers    dstSubresource;
    VkOffset3D                  dstOffsets[2];
} VkImageBlit2;
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to VkImageBlit2
typedef VkImageBlit2 VkImageBlit2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcSubresource` is the subresource to blit from.
- `srcOffsets` is a pointer to an array of two VkOffset3D structures specifying the bounds of the source region within `srcSubresource`.
- `dstSubresource` is the subresource to blit into.
- `dstOffsets` is a pointer to an array of two VkOffset3D structures specifying the bounds of the destination region within `dstSubresource`.

For each element of the `pRegions` array, a blit operation is performed for the specified source and destination regions.

Valid Usage

- VUID-VkImageBlit2-aspectMask-00238
  The `aspectMask` member of `srcSubresource` and `dstSubresource` **must** match
- VUID-VkImageBlit2-layerCount-08800
  If neither of the `layerCount` members of `srcSubresource` or `dstSubresource` are VK_REMAINING_ARRAY_LAYERS, the `layerCount` members of `srcSubresource` and `dstSubresource` **must** match
- VUID-VkImageBlit2-layerCount-08801
  If one of the `layerCount` members of `srcSubresource` or `dstSubresource` is VK_REMAINING_ARRAY_LAYERS, the other member **must** be either VK_REMAINING_ARRAY_LAYERS or equal to the `arrayLayers` member of the VkImageCreateInfo used to create the image minus `baseArrayLayer`

Valid Usage (Implicit)

- VUID-VkImageBlit2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_BLIT_2
- VUID-VkImageBlit2-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkImageBlit2-srcSubresource-parameter
  `srcSubresource` **must** be a valid VkImageSubresourceLayers structure
- VUID-VkImageBlit2-dstSubresource-parameter
  `dstSubresource` **must** be a valid VkImageSubresourceLayers structure

24.8. Resolving Multisample Images

To resolve a multisample color image to a non-multisample color image, call:

```
// Provided by VK_VERSION_1_0
void vkCmdResolveImage(
    VkCommandBuffer                             commandBuffer,
    VkImage                                     srcImage,
    VkImageLayout                               srcImageLayout,
    VkImage                                     dstImage,
    VkImageLayout                               dstImageLayout,
    uint32_t                                    regionCount,
    const VkImageResolve*                       pRegions);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `srcImage` is the source image.
- `srcImageLayout` is the layout of the source image subresources for the resolve.
- `dstImage` is the destination image.
- `dstImageLayout` is the layout of the destination image subresources for the resolve.
- `regionCount` is the number of regions to resolve.
- `pRegions` is a pointer to an array of VkImageResolve structures specifying the regions to resolve.

During the resolve the samples corresponding to each pixel location in the source are converted to a single sample before being written to the destination.

If the source format is a floating-point or normalized type, the resolve mode is chosen as implementation-dependent behavior. If the resolve mode requires to calculate the result from multiple samples, such as by computing an average or weighted average of the samples, the values for each pixel are resolved with implementation-defined numerical precision.

If the numeric format of `srcImage` uses sRGB encoding and the resolve mode requires the implementation to convert the samples to floating-point to perform the calculations, the

implementation **should** convert samples from nonlinear to linear before resolving the samples as described in the "sRGB EOTF" section of the Khronos Data Format Specification. In this case, the implementation **must** convert the linear averaged value to nonlinear before writing the resolved result to `dstImage`. If the `maintenance10` feature is enabled, whether a nonlinear to linear conversion happens for sRGB encoded resolve is controlled by `resolveSrgbFormatAppliesTransferFunction`.

If the source format is an integer type, a single sample's value is selected for each pixel.

`srcOffset` and `dstOffset` select the initial `x`, `y`, and `z` offsets in texels of the sub-regions of the source and destination image data. `extent` is the size in texels of the source image to resolve in `width`, `height` and `depth`. Each element of `pRegions` **must** be a region that is contained within its corresponding image.

Resolves are done layer by layer starting with the `baseArrayLayer` member of `srcSubresource` for the source and `dstSubresource` for the destination. `layerCount` layers are resolved to the destination image.

Valid Usage

- VUID-vkCmdResolveImage-commandBuffer-01837
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `srcImage` **must** not be a protected image
- VUID-vkCmdResolveImage-commandBuffer-01838
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `dstImage` **must** not be a protected image
- VUID-vkCmdResolveImage-commandBuffer-01839
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, `dstImage` **must** not be an unprotected image
- VUID-vkCmdResolveImage-pRegions-00255
  The union of all source regions, and the union of all destination regions, specified by the elements of `pRegions`, **must** not overlap in memory
- VUID-vkCmdResolveImage-srcImage-00256
  If `srcImage` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdResolveImage-srcImage-00257
  `srcImage` **must** have a sample count equal to any valid sample count value other than VK_SAMPLE_COUNT_1_BIT
- VUID-vkCmdResolveImage-dstImage-00258
  If `dstImage` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdResolveImage-dstImage-00259
  `dstImage` **must** have a sample count equal to VK_SAMPLE_COUNT_1_BIT
- VUID-vkCmdResolveImage-srcImageLayout-00260
  `srcImageLayout` **must** specify the layout of the image subresources of `srcImage` specified in `pRegions` at the time this command is executed on a `VkDevice`

- VUID-vkCmdResolveImage-srcImageLayout-01400
  `srcImageLayout` **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL or VK_IMAGE_LAYOUT_GENERAL
- VUID-vkCmdResolveImage-dstImageLayout-00262
  `dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-vkCmdResolveImage-dstImageLayout-01401
  `dstImageLayout` **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL or VK_IMAGE_LAYOUT_GENERAL
- VUID-vkCmdResolveImage-maintenance10-11799
  If the `maintenance10` feature is enabled, the format features of `dstImage` **must** contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT or VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-vkCmdResolveImage-dstImage-02003
  The format features of `dstImage` **must** contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT if the `maintenance10` feature is not enabled
- VUID-vkCmdResolveImage-srcImage-01386
  `srcImage` and `dstImage` **must** have been created with the same image format
- VUID-vkCmdResolveImage-srcSubresource-01709
  The `srcSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `srcImage` was created
- VUID-vkCmdResolveImage-dstSubresource-01710
  The `dstSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `dstImage` was created
- VUID-vkCmdResolveImage-srcSubresource-01711
  If `srcSubresource.layerCount` is not VK_REMAINING_ARRAY_LAYERS, `srcSubresource.baseArrayLayer` + `srcSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in VkImageCreateInfo when `srcImage` was created
- VUID-vkCmdResolveImage-dstSubresource-01712
  If `dstSubresource.layerCount` is not VK_REMAINING_ARRAY_LAYERS, `dstSubresource.baseArrayLayer` + `dstSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in VkImageCreateInfo when `dstImage` was created
- VUID-vkCmdResolveImage-dstImage-02546
  `dstImage` and `srcImage` **must** not have been created with `flags` containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-vkCmdResolveImage-srcImage-04446
  If `dstImage` is of type VK_IMAGE_TYPE_3D, then for each element of `pRegions`, `srcSubresource.layerCount` **must** be `1`
- VUID-vkCmdResolveImage-srcImage-04447
  If `dstImage` is of type VK_IMAGE_TYPE_3D, then for each element of `pRegions`,

dstSubresource.baseArrayLayer **must** be 0 and dstSubresource.layerCount **must** be 1

- VUID-vkCmdResolveImage-srcOffset-00269
  For each element of pRegions, srcOffset.x and (extent.width + srcOffset.x) **must** both be greater than or equal to 0 and less than or equal to the width of the specified srcSubresource of srcImage
- VUID-vkCmdResolveImage-srcOffset-00270
  For each element of pRegions, srcOffset.y and (extent.height + srcOffset.y) **must** both be greater than or equal to 0 and less than or equal to the height of the specified srcSubresource of srcImage
- VUID-vkCmdResolveImage-srcImage-00271
  If srcImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, srcOffset.y **must** be 0 and extent.height **must** be 1
- VUID-vkCmdResolveImage-srcOffset-00272
  For each element of pRegions, srcOffset.z and (extent.depth + srcOffset.z) **must** both be greater than or equal to 0 and less than or equal to the depth of the specified srcSubresource of srcImage
- VUID-vkCmdResolveImage-srcImage-00273
  If srcImage is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of pRegions, srcOffset.z **must** be 0 and extent.depth **must** be 1
- VUID-vkCmdResolveImage-dstOffset-00274
  For each element of pRegions, dstOffset.x and (extent.width + dstOffset.x) **must** both be greater than or equal to 0 and less than or equal to the width of the specified dstSubresource of dstImage
- VUID-vkCmdResolveImage-dstOffset-00275
  For each element of pRegions, dstOffset.y and (extent.height + dstOffset.y) **must** both be greater than or equal to 0 and less than or equal to the height of the specified dstSubresource of dstImage
- VUID-vkCmdResolveImage-dstImage-00276
  If dstImage is of type VK_IMAGE_TYPE_1D, then for each element of pRegions, dstOffset.y **must** be 0 and extent.height **must** be 1
- VUID-vkCmdResolveImage-dstOffset-00277
  For each element of pRegions, dstOffset.z and (extent.depth + dstOffset.z) **must** both be greater than or equal to 0 and less than or equal to the depth of the specified dstSubresource of dstImage
- VUID-vkCmdResolveImage-dstImage-00278
  If dstImage is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of pRegions, dstOffset.z **must** be 0 and extent.depth **must** be 1
- VUID-vkCmdResolveImage-srcImage-06762
  srcImage **must** have been created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-vkCmdResolveImage-srcImage-06763
  The format features of srcImage **must** contain VK_FORMAT_FEATURE_TRANSFER_SRC_BIT
- VUID-vkCmdResolveImage-dstImage-06764

`dstImage` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set

- VUID-vkCmdResolveImage-dstImage-06765
  The format features of `dstImage` **must** contain VK_FORMAT_FEATURE_TRANSFER_DST_BIT
- VUID-vkCmdResolveImage-srcSubresource-11800
  `srcSubresource.aspectMask` for each element in `pRegions` **must** not specify an aspect which is not part of the image format of `srcImage`
- VUID-vkCmdResolveImage-dstSubresource-11801
  `dstSubresource.aspectMask` for each element in `pRegions` **must** not specify an aspect which is not part of the image format of `dstImage`
- VUID-vkCmdResolveImage-srcSubresource-11802
  `srcSubresource.aspectMask` **must** equal `dstSubresource.aspectMask` for each element in `pRegions`

Valid Usage (Implicit)

- VUID-vkCmdResolveImage-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdResolveImage-srcImage-parameter
  `srcImage` **must** be a valid VkImage handle
- VUID-vkCmdResolveImage-srcImageLayout-parameter
  `srcImageLayout` **must** be a valid VkImageLayout value
- VUID-vkCmdResolveImage-dstImage-parameter
  `dstImage` **must** be a valid VkImage handle
- VUID-vkCmdResolveImage-dstImageLayout-parameter
  `dstImageLayout` **must** be a valid VkImageLayout value
- VUID-vkCmdResolveImage-pRegions-parameter
  `pRegions` **must** be a valid pointer to an array of `regionCount` valid VkImageResolve structures
- VUID-vkCmdResolveImage-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdResolveImage-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdResolveImage-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdResolveImage-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdResolveImage-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdResolveImage-regionCount-arraylength

`regionCount` **must** be greater than `0`

- VUID-vkCmdResolveImage-commonparent
  Each of `commandBuffer`, `dstImage`, and `srcImage` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdResolveImage is not affected by conditional rendering

The `VkImageResolve` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageResolve {
    VkImageSubresourceLayers    srcSubresource;
    VkOffset3D                  srcOffset;
    VkImageSubresourceLayers    dstSubresource;
    VkOffset3D                  dstOffset;
    VkExtent3D                  extent;
} VkImageResolve;
```

- `srcSubresource` and `dstSubresource` are VkImageSubresourceLayers structures specifying the image subresources of the images used for the source and destination image data, respectively. Resolve of depth/stencil images is not supported. VkImageResolve2 **can** support resolve of depth/stencil images with `maintenance10` instead.
- `srcOffset` and `dstOffset` select the initial `x`, `y`, and `z` offsets in texels of the sub-regions of the source and destination image data.
- `extent` is the size in texels of the source image to resolve in `width`, `height` and `depth`.

Valid Usage

- VUID-VkImageResolve-aspectMask-10981
  The `aspectMask` member of `srcSubresource` and `dstSubresource` **must** only contain VK_IMAGE_ASPECT_COLOR_BIT
- VUID-VkImageResolve-layerCount-08803
  If neither of the `layerCount` members of `srcSubresource` or `dstSubresource` are VK_REMAINING_ARRAY_LAYERS, the `layerCount` member of `srcSubresource` and `dstSubresource` **must** match
- VUID-VkImageResolve-layerCount-08804
  If one of the `layerCount` members of `srcSubresource` or `dstSubresource` is VK_REMAINING_ARRAY_LAYERS, the other member **must** be either VK_REMAINING_ARRAY_LAYERS or equal to the `arrayLayers` member of the VkImageCreateInfo used to create the image minus `baseArrayLayer`

Valid Usage (Implicit)

- VUID-VkImageResolve-srcSubresource-parameter
  `srcSubresource` **must** be a valid VkImageSubresourceLayers structure
- VUID-VkImageResolve-dstSubresource-parameter
  `dstSubresource` **must** be a valid VkImageSubresourceLayers structure

A more extensible version of the resolve image command is defined below.

To resolve a multisample image to a non-multisample image, call:

```
// Provided by VK_VERSION_1_3
void vkCmdResolveImage2(
    VkCommandBuffer                             commandBuffer,
    const VkResolveImageInfo2*                  pResolveImageInfo);
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to vkCmdResolveImage2
void vkCmdResolveImage2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkResolveImageInfo2*                  pResolveImageInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pResolveImageInfo` is a pointer to a VkResolveImageInfo2 structure describing the resolve parameters.

This command is functionally identical to vkCmdResolveImage, but includes extensible sub-

structures that include `sType` and `pNext` parameters, allowing them to be more easily extended.

Valid Usage

- VUID-vkCmdResolveImage2-commandBuffer-01837
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `srcImage` **must** not be a protected image
- VUID-vkCmdResolveImage2-commandBuffer-01838
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, `dstImage` **must** not be a protected image
- VUID-vkCmdResolveImage2-commandBuffer-01839
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, `dstImage` **must** not be an unprotected image

Valid Usage (Implicit)

- VUID-vkCmdResolveImage2-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdResolveImage2-pResolveImageInfo-parameter
  `pResolveImageInfo` **must** be a valid pointer to a valid VkResolveImageInfo2 structure
- VUID-vkCmdResolveImage2-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdResolveImage2-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdResolveImage2-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdResolveImage2-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdResolveImage2-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdResolveImage2 is not affected by conditional rendering

The VkResolveImageInfo2 structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkResolveImageInfo2 {
    VkStructureType           sType;
    const void*               pNext;
    VkImage                   srcImage;
    VkImageLayout             srcImageLayout;
    VkImage                   dstImage;
    VkImageLayout             dstImageLayout;
    uint32_t                  regionCount;
    const VkImageResolve2*    pRegions;
} VkResolveImageInfo2;
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to VkResolveImageInfo2
typedef VkResolveImageInfo2 VkResolveImageInfo2KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- srcImage is the source image.
- srcImageLayout is the layout of the source image subresources for the resolve.
- dstImage is the destination image.
- dstImageLayout is the layout of the destination image subresources for the resolve.
- regionCount is the number of regions to resolve.
- pRegions is a pointer to an array of VkImageResolve2 structures specifying the regions to resolve.

If the source format is a floating-point or normalized type, the resolve mode is chosen as implementation-dependent behavior, unless VkResolveImageModeInfoKHR is included in the pNext

chain, in which case it is defined by VkResolveImageModeInfoKHR::resolveMode. If the resolve mode requires to calculate the result from multiple samples, such as by computing an average or weighted average of the samples, the values for each pixel are resolved with implementation-defined numerical precision.

If the numeric format of srcImage uses sRGB encoding and the resolve mode requires the implementation to convert the samples to floating-point to perform the calculations, the implementation **should** convert samples from nonlinear to linear before resolving the samples as described in the "sRGB EOTF" section of the Khronos Data Format Specification. In this case, the implementation **must** convert the linear averaged value to nonlinear before writing the resolved result to dstImage. If the maintenance10 feature is enabled, whether a nonlinear to linear conversion happens for sRGB encoded resolve is controlled by resolveSrgbFormatAppliesTransferFunction. If VkResolveImageModeInfoKHR is included in the pNext chain, this default behavior **can** be overridden with VK_RESOLVE_IMAGE_SKIP_TRANSFER_FUNCTION_BIT_KHR or VK_RESOLVE_IMAGE_ENABLE_TRANSFER_FUNCTION_BIT_KHR flags.

If the source format is an integer type, a single sample's value is selected for each pixel, unless VkResolveImageModeInfoKHR is included in the pNext chain, in which case it is defined by VkResolveImageModeInfoKHR::resolveMode or VkResolveImageModeInfoKHR::stencilResolveMode depending on which aspect is being resolved.

Valid Usage

- VUID-VkResolveImageInfo2-pRegions-00255
  The union of all source regions, and the union of all destination regions, specified by the elements of pRegions, **must** not overlap in memory
- VUID-VkResolveImageInfo2-srcImage-00256
  If srcImage is non-sparse then it **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-VkResolveImageInfo2-srcImage-00257
  srcImage **must** have a sample count equal to any valid sample count value other than VK_SAMPLE_COUNT_1_BIT
- VUID-VkResolveImageInfo2-dstImage-00258
  If dstImage is non-sparse then it **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-VkResolveImageInfo2-dstImage-00259
  dstImage **must** have a sample count equal to VK_SAMPLE_COUNT_1_BIT
- VUID-VkResolveImageInfo2-srcImageLayout-00260
  srcImageLayout **must** specify the layout of the image subresources of srcImage specified in pRegions at the time this command is executed on a VkDevice
- VUID-VkResolveImageInfo2-srcImageLayout-01400
  srcImageLayout **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL or VK_IMAGE_LAYOUT_GENERAL
- VUID-VkResolveImageInfo2-dstImageLayout-00262
  dstImageLayout **must** specify the layout of the image subresources of dstImage specified in

pRegions at the time this command is executed on a VkDevice

- VUID-VkResolveImageInfo2-dstImageLayout-01401
  dstImageLayout **must** be VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR, VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL or VK_IMAGE_LAYOUT_GENERAL
- VUID-VkResolveImageInfo2-maintenance10-11799
  If the maintenance10 feature is enabled, the format features of dstImage **must** contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT or VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkResolveImageInfo2-dstImage-02003
  The format features of dstImage **must** contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT if the maintenance10 feature is not enabled
- VUID-VkResolveImageInfo2-srcImage-01386
  srcImage and dstImage **must** have been created with the same image format
- VUID-VkResolveImageInfo2-srcSubresource-01709
  The srcSubresource.mipLevel member of each element of pRegions **must** be less than the mipLevels specified in VkImageCreateInfo when srcImage was created
- VUID-VkResolveImageInfo2-dstSubresource-01710
  The dstSubresource.mipLevel member of each element of pRegions **must** be less than the mipLevels specified in VkImageCreateInfo when dstImage was created
- VUID-VkResolveImageInfo2-srcSubresource-01711
  If srcSubresource.layerCount is not VK_REMAINING_ARRAY_LAYERS, srcSubresource.baseArrayLayer + srcSubresource.layerCount of each element of pRegions **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when srcImage was created
- VUID-VkResolveImageInfo2-dstSubresource-01712
  If dstSubresource.layerCount is not VK_REMAINING_ARRAY_LAYERS, dstSubresource.baseArrayLayer + dstSubresource.layerCount of each element of pRegions **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when dstImage was created
- VUID-VkResolveImageInfo2-dstImage-02546
  dstImage and srcImage **must** not have been created with flags containing VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- VUID-VkResolveImageInfo2-srcImage-04446
  If dstImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, srcSubresource.layerCount **must** be 1
- VUID-VkResolveImageInfo2-srcImage-04447
  If dstImage is of type VK_IMAGE_TYPE_3D, then for each element of pRegions, dstSubresource.baseArrayLayer **must** be 0 and dstSubresource.layerCount **must** be 1
- VUID-VkResolveImageInfo2-srcOffset-00269
  For each element of pRegions, srcOffset.x and (extent.width + srcOffset.x) **must** both be greater than or equal to 0 and less than or equal to the width of the specified srcSubresource of srcImage

- VUID-VkResolveImageInfo2-srcOffset-00270
  For each element of `pRegions`, `srcOffset.y` and (`extent.height` + `srcOffset.y`) **must** both be greater than or equal to `0` and less than or equal to the height of the specified `srcSubresource` of `srcImage`
- VUID-VkResolveImageInfo2-srcImage-00271
  If `srcImage` is of type VK_IMAGE_TYPE_1D, then for each element of `pRegions`, `srcOffset.y` **must** be `0` and `extent.height` **must** be `1`
- VUID-VkResolveImageInfo2-srcOffset-00272
  For each element of `pRegions`, `srcOffset.z` and (`extent.depth` + `srcOffset.z`) **must** both be greater than or equal to `0` and less than or equal to the depth of the specified `srcSubresource` of `srcImage`
- VUID-VkResolveImageInfo2-srcImage-00273
  If `srcImage` is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of `pRegions`, `srcOffset.z` **must** be `0` and `extent.depth` **must** be `1`
- VUID-VkResolveImageInfo2-dstOffset-00274
  For each element of `pRegions`, `dstOffset.x` and (`extent.width` + `dstOffset.x`) **must** both be greater than or equal to `0` and less than or equal to the width of the specified `dstSubresource` of `dstImage`
- VUID-VkResolveImageInfo2-dstOffset-00275
  For each element of `pRegions`, `dstOffset.y` and (`extent.height` + `dstOffset.y`) **must** both be greater than or equal to `0` and less than or equal to the height of the specified `dstSubresource` of `dstImage`
- VUID-VkResolveImageInfo2-dstImage-00276
  If `dstImage` is of type VK_IMAGE_TYPE_1D, then for each element of `pRegions`, `dstOffset.y` **must** be `0` and `extent.height` **must** be `1`
- VUID-VkResolveImageInfo2-dstOffset-00277
  For each element of `pRegions`, `dstOffset.z` and (`extent.depth` + `dstOffset.z`) **must** both be greater than or equal to `0` and less than or equal to the depth of the specified `dstSubresource` of `dstImage`
- VUID-VkResolveImageInfo2-dstImage-00278
  If `dstImage` is of type VK_IMAGE_TYPE_1D or VK_IMAGE_TYPE_2D, then for each element of `pRegions`, `dstOffset.z` **must** be `0` and `extent.depth` **must** be `1`
- VUID-VkResolveImageInfo2-srcImage-06762
  `srcImage` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_SRC_BIT usage flag set
- VUID-VkResolveImageInfo2-srcImage-06763
  The format features of `srcImage` **must** contain VK_FORMAT_FEATURE_TRANSFER_SRC_BIT
- VUID-VkResolveImageInfo2-dstImage-06764
  `dstImage` **must** have been created with the VK_IMAGE_USAGE_TRANSFER_DST_BIT usage flag set
- VUID-VkResolveImageInfo2-dstImage-06765
  The format features of `dstImage` **must** contain VK_FORMAT_FEATURE_TRANSFER_DST_BIT
- VUID-VkResolveImageInfo2-srcSubresource-11800

srcSubresource.aspectMask for each element in pRegions **must** not specify an aspect which is not part of the image format of srcImage

- VUID-VkResolveImageInfo2-dstSubresource-11801
  dstSubresource.aspectMask for each element in pRegions **must** not specify an aspect which is not part of the image format of dstImage
- VUID-VkResolveImageInfo2-srcSubresource-11802
  srcSubresource.aspectMask **must** equal dstSubresource.aspectMask for each element in pRegions
- VUID-VkResolveImageInfo2-pNext-10982
  If VkResolveImageModeInfoKHR is included in the pNext chain, flags includes VK_RESOLVE_IMAGE_SKIP_TRANSFER_FUNCTION_BIT_KHR or VK_RESOLVE_IMAGE_ENABLE_TRANSFER_FUNCTION_BIT_KHR, then the format of srcImage and dstImage **must** use sRGB encoding
- VUID-VkResolveImageInfo2-srcImage-10983
  If srcImage has a color format and VkResolveImageModeInfoKHR is included in the pNext chain, its resolveMode **must** not be VK_RESOLVE_MODE_NONE
- VUID-VkResolveImageInfo2-srcImage-10984
  If srcImage has a non-integer color format, and VkResolveImageModeInfoKHR is included in the pNext chain, its resolveMode **must** be VK_RESOLVE_MODE_AVERAGE_BIT
- VUID-VkResolveImageInfo2-srcImage-10985
  If srcImage has an integer color format, and VkResolveImageModeInfoKHR is included in the pNext chain, its resolveMode **must** be VK_RESOLVE_MODE_SAMPLE_ZERO_BIT
- VUID-VkResolveImageInfo2-srcImage-10986
  If srcImage has a depth-stencil format, VkResolveImageModeInfoKHR **must** be included in the pNext chain
- VUID-VkResolveImageInfo2-srcImage-10987
  If srcImage has a depth-stencil format, and a depth aspect is referenced by pRegions, VkResolveImageModeInfoKHR::resolveMode **must** not be VK_RESOLVE_MODE_NONE
- VUID-VkResolveImageInfo2-srcImage-10988
  If srcImage has a depth-stencil format, and a stencil aspect is referenced by pRegions, VkResolveImageModeInfoKHR::stencilResolveMode **must** not be VK_RESOLVE_MODE_NONE
- VUID-VkResolveImageInfo2-srcImage-10989
  If srcImage has a depth-stencil format, and a depth aspect is referenced by pRegions, VkResolveImageModeInfoKHR::resolveMode **must** be one of the bits set in VkPhysicalDeviceDepthStencilResolveProperties::supportedDepthResolveModes
- VUID-VkResolveImageInfo2-srcImage-10990
  If srcImage has a depth-stencil format, and a stencil aspect is referenced by pRegions, VkResolveImageModeInfoKHR::stencilResolveMode **must** be one of the bits set in VkPhysicalDeviceDepthStencilResolveProperties::supportedStencilResolveModes
- VUID-VkResolveImageInfo2-srcImage-10991
  If srcImage has a depth-stencil format, and both a depth aspect and stencil aspect are referenced by pRegions, and VkPhysicalDeviceDepthStencilResolveProperties

::independentResolve is VK_FALSE, VkResolveImageModeInfoKHR::resolveMode **must** be equal to VkResolveImageModeInfoKHR::stencilResolveMode

- VUID-VkResolveImageInfo2-srcImage-10992
  If srcImage has a depth-stencil format containing both a depth aspect and stencil aspect, and VkPhysicalDeviceDepthStencilResolveProperties::independentResolveNone is VK_FALSE, every element of pRegions **must** contain both depth and stencil aspects

Valid Usage (Implicit)

- VUID-VkResolveImageInfo2-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_RESOLVE_IMAGE_INFO_2
- VUID-VkResolveImageInfo2-pNext-pNext
  pNext **must** be NULL or a pointer to a valid instance of VkResolveImageModeInfoKHR
- VUID-VkResolveImageInfo2-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkResolveImageInfo2-srcImage-parameter
  srcImage **must** be a valid VkImage handle
- VUID-VkResolveImageInfo2-srcImageLayout-parameter
  srcImageLayout **must** be a valid VkImageLayout value
- VUID-VkResolveImageInfo2-dstImage-parameter
  dstImage **must** be a valid VkImage handle
- VUID-VkResolveImageInfo2-dstImageLayout-parameter
  dstImageLayout **must** be a valid VkImageLayout value
- VUID-VkResolveImageInfo2-pRegions-parameter
  pRegions **must** be a valid pointer to an array of regionCount valid VkImageResolve2 structures
- VUID-VkResolveImageInfo2-regionCount-arraylength
  regionCount **must** be greater than 0
- VUID-VkResolveImageInfo2-commonparent
  Both of dstImage, and srcImage **must** have been created, allocated, or retrieved from the same VkDevice

The VkImageResolve2 structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkImageResolve2 {
    VkStructureType            sType;
    const void*                pNext;
    VkImageSubresourceLayers   srcSubresource;
    VkOffset3D                 srcOffset;
    VkImageSubresourceLayers   dstSubresource;
    VkOffset3D                 dstOffset;
    VkExtent3D                 extent;
} VkImageResolve2;
```

```
// Provided by VK_KHR_copy_commands2
// Equivalent to VkImageResolve2
typedef VkImageResolve2 VkImageResolve2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcSubresource` and `dstSubresource` are VkImageSubresourceLayers structures specifying the image subresources of the images used for the source and destination image data, respectively.
- `srcOffset` and `dstOffset` select the initial `x`, `y`, and `z` offsets in texels of the sub-regions of the source and destination image data.
- `extent` is the size in texels of the source image to resolve in `width`, `height` and `depth`.

Valid Usage

- VUID-VkImageResolve2-aspectMask-10993
  The `aspectMask` member of `srcSubresource` and `dstSubresource` **must** only contain VK_IMAGE_ASPECT_DEPTH_BIT, VK_IMAGE_ASPECT_STENCIL_BIT, or VK_IMAGE_ASPECT_COLOR_BIT
- VUID-VkImageResolve2-maintenance10-10994
  If the `maintenance10` feature is not enabled, `srcSubresource` and `dstSubresource` **must** not contain VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT
- VUID-VkImageResolve2-layerCount-08803
  If neither of the `layerCount` members of `srcSubresource` or `dstSubresource` are VK_REMAINING_ARRAY_LAYERS, the `layerCount` member of `srcSubresource` and `dstSubresource` **must** match
- VUID-VkImageResolve2-layerCount-08804
  If one of the `layerCount` members of `srcSubresource` or `dstSubresource` is VK_REMAINING_ARRAY_LAYERS, the other member **must** be either VK_REMAINING_ARRAY_LAYERS or equal to the `arrayLayers` member of the VkImageCreateInfo used to create the image minus `baseArrayLayer`

Valid Usage (Implicit)

- VUID-VkImageResolve2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_RESOLVE_2
- VUID-VkImageResolve2-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkImageResolve2-srcSubresource-parameter
  `srcSubresource` **must** be a valid VkImageSubresourceLayers structure
- VUID-VkImageResolve2-dstSubresource-parameter
  `dstSubresource` **must** be a valid VkImageSubresourceLayers structure

The `VkResolveImageModeInfoKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance10
typedef struct VkResolveImageModeInfoKHR {
    VkStructureType           sType;
    const void*               pNext;
    VkResolveImageFlagsKHR    flags;
    VkResolveModeFlagBits     resolveMode;
    VkResolveModeFlagBits     stencilResolveMode;
} VkResolveImageModeInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkResolveImageFlagBitsKHR.
- `resolveMode` is a VkResolveModeFlagBits value defining how `srcImage` will be resolved into `dstImage` when resolving non-stencil values.
- `stencilResolveMode` is a VkResolveModeFlagBits value defining how `srcImage` will be resolved into `dstImage` when resolving stencil values.

Valid Usage

- VUID-VkResolveImageModeInfoKHR-flags-10995
  If `flags` includes VK_RESOLVE_IMAGE_SKIP_TRANSFER_FUNCTION_BIT_KHR, `flags` **must** not include VK_RESOLVE_IMAGE_ENABLE_TRANSFER_FUNCTION_BIT_KHR
- VUID-VkResolveImageModeInfoKHR-flags-10996
  If `flags` includes VK_RESOLVE_IMAGE_SKIP_TRANSFER_FUNCTION_BIT_KHR or VK_RESOLVE_IMAGE_ENABLE_TRANSFER_FUNCTION_BIT_KHR, `resolveSrgbFormatSupportsTransferFunctionControl` **must** be VK_TRUE
- VUID-VkResolveImageModeInfoKHR-flags-10997
  If `flags` includes VK_RESOLVE_IMAGE_SKIP_TRANSFER_FUNCTION_BIT_KHR or VK_RESOLVE_IMAGE_ENABLE_TRANSFER_FUNCTION_BIT_KHR, `resolveMode` **must** be

equal to VK_RESOLVE_MODE_AVERAGE_BIT

Valid Usage (Implicit)

- VUID-VkResolveImageModeInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_RESOLVE_IMAGE_MODE_INFO_KHR
- VUID-VkResolveImageModeInfoKHR-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkResolveImageFlagBitsKHR values
- VUID-VkResolveImageModeInfoKHR-resolveMode-parameter
  If `resolveMode` is not `0`, `resolveMode` **must** be a valid VkResolveModeFlagBits value
- VUID-VkResolveImageModeInfoKHR-stencilResolveMode-parameter
  If `stencilResolveMode` is not `0`, `stencilResolveMode` **must** be a valid VkResolveModeFlagBits value

Structure Chaining

Extends the structure

- VkResolveImageInfo2

Bits which **can** be set in VkResolveImageModeInfoKHR::`flags`, describing additional properties of a resolve operation, are:

```
// Provided by VK_KHR_maintenance10
typedef enum VkResolveImageFlagBitsKHR {
  // Provided by VK_KHR_maintenance10 with VK_VERSION_1_3 or VK_KHR_copy_commands2
    VK_RESOLVE_IMAGE_SKIP_TRANSFER_FUNCTION_BIT_KHR = 0x00000001,
  // Provided by VK_KHR_maintenance10 with VK_VERSION_1_3 or VK_KHR_copy_commands2
    VK_RESOLVE_IMAGE_ENABLE_TRANSFER_FUNCTION_BIT_KHR = 0x00000002,
} VkResolveImageFlagBitsKHR;
```

- VK_RESOLVE_IMAGE_SKIP_TRANSFER_FUNCTION_BIT_KHR specifies that resolve operations happening to an sRGB encoded image **must** not convert samples from nonlinear to linear before averaging.
- VK_RESOLVE_IMAGE_ENABLE_TRANSFER_FUNCTION_BIT_KHR specifies that resolve operations happening to an sRGB encoded image **must** convert samples from nonlinear to linear before averaging.

```
// Provided by VK_KHR_maintenance10
typedef VkFlags VkResolveImageFlagsKHR;
```

`VkResolveImageFlagsKHR` is a bitmask type for setting a mask of zero or more VkResolveImageFlagBitsKHR.

Chapter 25. Drawing Commands

Drawing commands (commands with `Draw` in the name) provoke work in a graphics pipeline. Drawing commands are recorded into a command buffer and when executed by a queue, will produce work which executes according to the bound graphics pipeline, or if the `shaderObject` feature is enabled, any shader objects bound to graphics stages. A graphics pipeline or a combination of one or more graphics shader objects **must** be bound to a command buffer before any drawing commands are recorded in that command buffer.

Drawing can be achieved in two modes:

- Programmable Mesh Shading, the mesh shader assembles primitives, or
- Programmable Primitive Shading, the input primitives are assembled as follows.

Each draw is made up of zero or more vertices and zero or more instances, which are processed by the device and result in the assembly of primitives. Primitives are assembled according to the `pInputAssemblyState` member of the VkGraphicsPipelineCreateInfo structure, which is of type `VkPipelineInputAssemblyStateCreateInfo`:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineInputAssemblyStateCreateInfo {
    VkStructureType                            sType;
    const void*                                pNext;
    VkPipelineInputAssemblyStateCreateFlags    flags;
    VkPrimitiveTopology                        topology;
    VkBool32                                   primitiveRestartEnable;
} VkPipelineInputAssemblyStateCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `topology` is a VkPrimitiveTopology defining the primitive topology, as described below.
- `primitiveRestartEnable` controls whether a special vertex index value is treated as restarting the assembly of primitives. This enable only applies to indexed draws (vkCmdDrawIndexed, vkCmdDrawMultiIndexedEXT, and vkCmdDrawIndexedIndirect), and the special index value is either 0xFFFFFFFF when the `indexType` parameter of Vulkan 1.4 or `vkCmdBindIndexBuffer2` or `vkCmdBindIndexBuffer` is equal to VK_INDEX_TYPE_UINT32; 0xFF when `indexType` is equal to VK_INDEX_TYPE_UINT8; or 0xFFFF when `indexType` is equal to VK_INDEX_TYPE_UINT16. Primitive restart is not allowed for “list” topologies, unless one of the features `primitiveTopologyPatchListRestart` (for VK_PRIMITIVE_TOPOLOGY_PATCH_LIST) or `primitiveTopologyListRestart` (for all other list topologies) is enabled. If the `primitiveRestartIndex` feature is enabled, vkCmdSetPrimitiveRestartIndexEXT **can** be used to set a custom index for primitive restart.

Restarting the assembly of primitives discards the most recent index values if those elements formed an incomplete primitive, and restarts the primitive assembly using the subsequent indices,

but only assembling the immediately following element through the end of the originally specified elements. The primitive restart index value comparison is performed before adding the `vertexOffset` value to the index value.

Valid Usage

- VUID-VkPipelineInputAssemblyStateCreateInfo-topology-06252
  If the `primitiveTopologyListRestart` feature is not enabled, and `topology` is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, `primitiveRestartEnable` **must** be VK_FALSE
- VUID-VkPipelineInputAssemblyStateCreateInfo-topology-06253
  If the `primitiveTopologyPatchListRestart` feature is not enabled, and `topology` is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, `primitiveRestartEnable` **must** be VK_FALSE
- VUID-VkPipelineInputAssemblyStateCreateInfo-topology-00429
  If the `geometryShader` feature is not enabled, `topology` **must** not be any of VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY
- VUID-VkPipelineInputAssemblyStateCreateInfo-topology-00430
  If the `tessellationShader` feature is not enabled, `topology` **must** not be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST
- VUID-VkPipelineInputAssemblyStateCreateInfo-triangleFans-04452
  If the `VK_KHR_portability_subset` extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::`triangleFans` is VK_FALSE, `topology` **must** not be VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN

Valid Usage (Implicit)

- VUID-VkPipelineInputAssemblyStateCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_INPUT_ASSEMBLY_STATE_CREATE_INFO
- VUID-VkPipelineInputAssemblyStateCreateInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkPipelineInputAssemblyStateCreateInfo-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkPipelineInputAssemblyStateCreateInfo-topology-parameter
  `topology` **must** be a valid VkPrimitiveTopology value

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineInputAssemblyStateCreateFlags;
```

VkPipelineInputAssemblyStateCreateFlags is a bitmask type for setting a mask, but is currently reserved for future use.

To dynamically control whether a special vertex index value is treated as restarting the assembly of primitives, call:

```
// Provided by VK_VERSION_1_3
void vkCmdSetPrimitiveRestartEnable(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    primitiveRestartEnable);
```

```
// Provided by VK_EXT_extended_dynamic_state2, VK_EXT_shader_object
// Equivalent to vkCmdSetPrimitiveRestartEnable
void vkCmdSetPrimitiveRestartEnableEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    primitiveRestartEnable);
```

- commandBuffer is the command buffer into which the command will be recorded.
- primitiveRestartEnable controls whether a special vertex index value is treated as restarting the assembly of primitives. It behaves in the same way as VkPipelineInputAssemblyStateCreateInfo ::primitiveRestartEnable.

This command sets the primitive restart enable for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE set in VkPipelineDynamicStateCreateInfo ::pDynamicStates. Otherwise, this state is specified by the VkPipelineInputAssemblyStateCreateInfo ::primitiveRestartEnable value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetPrimitiveRestartEnable-None-08970
  At least one of the following **must** be true:
  - the extendedDynamicState2 feature is enabled
  - the shaderObject feature is enabled
  - the value of VkApplicationInfo::apiVersion used to create the VkInstance parent of commandBuffer is greater than or equal to Version 1.3

Valid Usage (Implicit)

- VUID-vkCmdSetPrimitiveRestartEnable-commandBuffer-parameter

`commandBuffer` **must** be a valid VkCommandBuffer handle

- VUID-vkCmdSetPrimitiveRestartEnable-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetPrimitiveRestartEnable-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetPrimitiveRestartEnable-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetPrimitiveRestartEnable is not affected by conditional rendering

To dynamically control which special vertex index value is treated as restarting the assembly of primitives, call:

```
// Provided by VK_EXT_primitive_restart_index
void vkCmdSetPrimitiveRestartIndexEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    primitiveRestartIndex);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `primitiveRestartIndex` controls which special vertex index value is treated as restarting the assembly of primitives. This overrides the default restart index value as defined as part of the description of the VkPipelineInputAssemblyStateCreateInfo::`primitiveRestartEnable` state. Default values are applied when binding an index buffer.

This command sets a custom primitive restart index for subsequent drawing commands. Binding

an index buffer invalidates the custom index value.

Valid Usage

- VUID-vkCmdSetPrimitiveRestartIndexEXT-primitiveRestartIndex-12395
  The `primitiveRestartIndex` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdSetPrimitiveRestartIndexEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetPrimitiveRestartIndexEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetPrimitiveRestartIndexEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetPrimitiveRestartIndexEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetPrimitiveRestartIndexEXT is not affected by conditional rendering

25.1. Primitive Topologies

Primitive topology determines how consecutive vertices are organized into primitives, and determines the type of primitive that is used at the beginning of the graphics pipeline. The effective

topology for later stages of the pipeline is altered by tessellation or geometry shading (if either is in use) and depends on the execution modes of those shaders. In the case of mesh shading the only effective topology is defined by the execution mode of the mesh shader.

The primitive topologies defined by VkPrimitiveTopology are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPrimitiveTopology {
    VK_PRIMITIVE_TOPOLOGY_POINT_LIST = 0,
    VK_PRIMITIVE_TOPOLOGY_LINE_LIST = 1,
    VK_PRIMITIVE_TOPOLOGY_LINE_STRIP = 2,
    VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST = 3,
    VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP = 4,
    VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN = 5,
    VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY = 6,
    VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY = 7,
    VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY = 8,
    VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY = 9,
    VK_PRIMITIVE_TOPOLOGY_PATCH_LIST = 10,
} VkPrimitiveTopology;
```

- VK_PRIMITIVE_TOPOLOGY_POINT_LIST specifies a series of separate point primitives.
- VK_PRIMITIVE_TOPOLOGY_LINE_LIST specifies a series of separate line primitives.
- VK_PRIMITIVE_TOPOLOGY_LINE_STRIP specifies a series of connected line primitives with consecutive lines sharing a vertex.
- VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST specifies a series of separate triangle primitives.
- VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP specifies a series of connected triangle primitives with consecutive triangles sharing an edge.
- VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN specifies a series of connected triangle primitives with all triangles sharing a common vertex. If the VK_KHR_portability_subset extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::triangleFans is VK_FALSE, then triangle fans are not supported by the implementation, and VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN **must** not be used.
- VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY specifies a series of separate line primitives with adjacency.
- VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY specifies a series of connected line primitives with adjacency, with consecutive primitives sharing three vertices.
- VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY specifies a series of separate triangle primitives with adjacency.
- VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY specifies connected triangle primitives with adjacency, with consecutive triangles sharing an edge.
- VK_PRIMITIVE_TOPOLOGY_PATCH_LIST specifies separate patch primitives.

Each primitive topology, and its construction from a list of vertices, is described in detail below

with a supporting diagram, according to the following key:

•	Vertex	A point in 3-dimensional space. Positions chosen within the diagrams are arbitrary and for illustration only.
5	Vertex Number	Sequence position of a vertex within the provided vertex data.
	Provoking Vertex	Provoking vertex within the main primitive. The tail is angled towards the relevant primitive. Used in flat shading.
	Primitive Edge	An edge connecting the points of a main primitive.
	Adjacency Edge	Points connected by these lines do not contribute to a main primitive, and are only accessible in a geometry shader.
	Winding Order	The relative order in which vertices are defined within a primitive, used in the facing determination. This ordering has no specific start or end point.

The diagrams are supported with mathematical definitions where the vertices (v) and primitives (p) are numbered starting from 0; v_0 is the first vertex in the provided data and p_0 is the first primitive in the set of primitives defined by the vertices and topology.

To dynamically set primitive topology, call:

```
// Provided by VK_VERSION_1_3
void vkCmdSetPrimitiveTopology(
    VkCommandBuffer                             commandBuffer,
    VkPrimitiveTopology                         primitiveTopology);
```

```
// Provided by VK_EXT_extended_dynamic_state, VK_EXT_shader_object
// Equivalent to vkCmdSetPrimitiveTopology
void vkCmdSetPrimitiveTopologyEXT(
    VkCommandBuffer                             commandBuffer,
    VkPrimitiveTopology                         primitiveTopology);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `primitiveTopology` specifies the primitive topology to use for drawing.

This command sets the primitive topology for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineInputAssemblyStateCreateInfo ::`topology` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetPrimitiveTopology-None-08971
  At least one of the following **must** be true:

- the `extendedDynamicState` feature is enabled
- the `shaderObject` feature is enabled
- the value of VkApplicationInfo::`apiVersion` used to create the VkInstance parent of `commandBuffer` is greater than or equal to Version 1.3

Valid Usage (Implicit)

- VUID-vkCmdSetPrimitiveTopology-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetPrimitiveTopology-primitiveTopology-parameter
  `primitiveTopology` **must** be a valid VkPrimitiveTopology value
- VUID-vkCmdSetPrimitiveTopology-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetPrimitiveTopology-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetPrimitiveTopology-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetPrimitiveTopology is not affected by conditional rendering

25.1.1. Topology Class

The primitive topologies are grouped into the following topology classes:

Table 31. Topology Classes

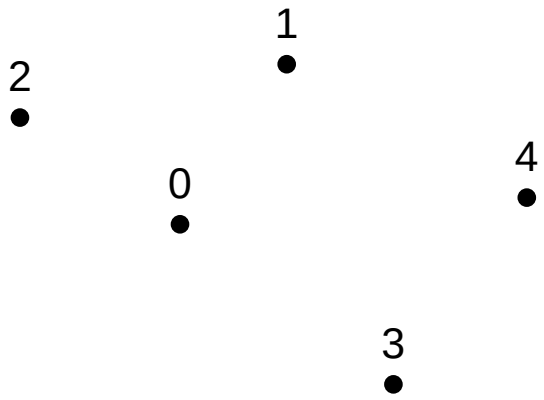
Topology Class	Primitive Topology
Point	VK_PRIMITIVE_TOPOLOGY_POINT_LIST
Line	VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_STRIP, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY
Triangle	VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY
Patch	VK_PRIMITIVE_TOPOLOGY_PATCH_LIST

25.1.2. Point Lists

When the topology is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, each consecutive vertex defines a single point primitive, according to the equation:

$$p_i = \{v_i\}$$

As there is only one vertex, that vertex is the provoking vertex. The number of primitives generated is equal to `vertexCount`.



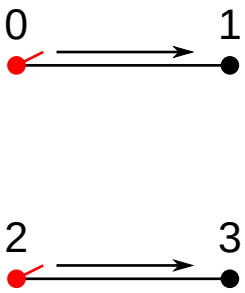
25.1.3. Line Lists

When the primitive topology is VK_PRIMITIVE_TOPOLOGY_LINE_LIST, each consecutive pair of vertices defines a single line primitive, according to the equation:

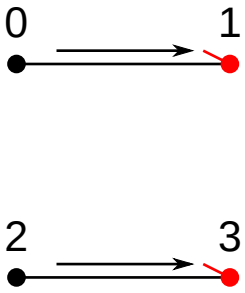
$$p_i = \{v_{2i}, v_{2i+1}\}$$

The number of primitives generated is equal to ⌊vertexCount/2⌋.

When the provokingVertexMode is VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT, the provoking vertex for p_i is v_{2i}.



When the provokingVertexMode is VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT, the provoking vertex for p_i is v_{2i+1}.



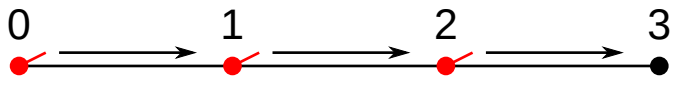
25.1.4. Line Strips

When the primitive topology is VK_PRIMITIVE_TOPOLOGY_LINE_STRIP, one line primitive is defined by each vertex and the following vertex, according to the equation:

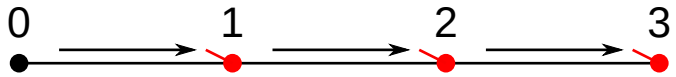
$$p_i = \{v_i, v_{i+1}\}$$

The number of primitives generated is equal to max(0,vertexCount-1).

When the provokingVertexMode is VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT, the provoking vertex for p_i is v_i.



When the provokingVertexMode is VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT, the provoking vertex for p_i is v_{i+1}.



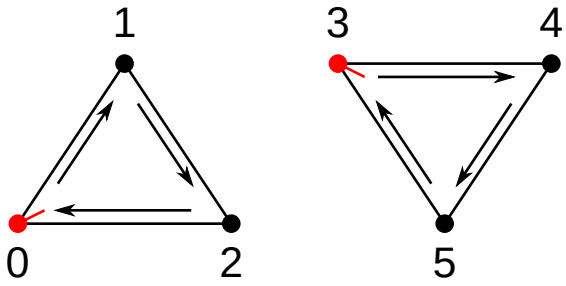
25.1.5. Triangle Lists

When the primitive topology is VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, each consecutive set of three vertices defines a single triangle primitive, according to the equation:

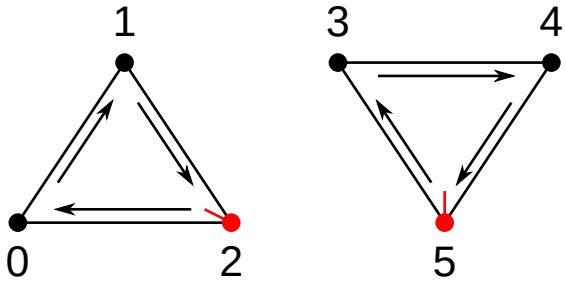
$p_i = \{v_{3i}, v_{3i+1}, v_{3i+2}\}$

The number of primitives generated is equal to ⌊vertexCount/3⌋.

When the provokingVertexMode is VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT, the provoking vertex for p_i is v_{3i}.



When the provokingVertexMode is VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT, the provoking vertex for p_i is v_{3i+2}.



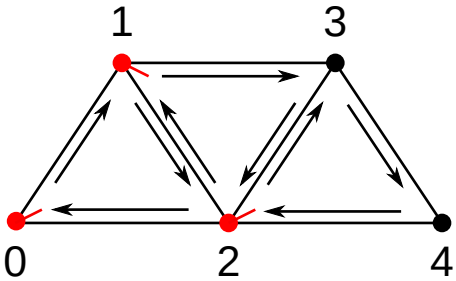
25.1.6. Triangle Strips

When the primitive topology is VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP, one triangle primitive is defined by each vertex and the two vertices that follow it, according to the equation:

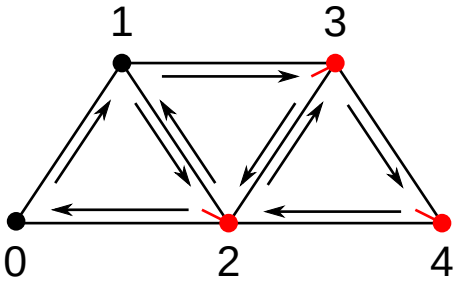
$p_i = \{v_i, v_{i+(1+i\%2)}, v_{i+(2-i\%2)}\}$

The number of primitives generated is equal to max(0,vertexCount-2).

When the provokingVertexMode is VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT, the provoking vertex for p_i is v_i.



When the provokingVertexMode is VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT, the provoking vertex for p_i is v_{i+2}.



NOTE The ordering of the vertices in each successive triangle is reversed, so that the winding order is consistent throughout the strip.

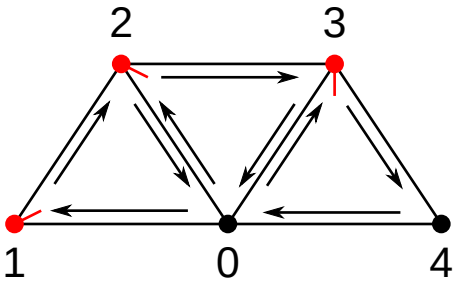
25.1.7. Triangle Fans

When the primitive topology is VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN, triangle primitives are defined around a shared common vertex, according to the equation:

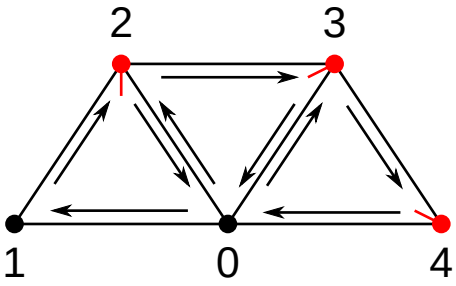
$$p_i = \{v_{i+1}, v_{i+2}, v_0\}$$

The number of primitives generated is equal to max(0,`vertexCount`-2).

When the `provokingVertexMode` is VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT, the provoking vertex for p_i is v_{i+1}.



When the `provokingVertexMode` is VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT, the provoking vertex for p_i is v_{i+2}.



NOTE If the VK_KHR_portability_subset extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::`triangleFans` is VK_FALSE, then triangle fans are not supported by the implementation, and VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN **must** not be used.

25.1.8. Line Lists With Adjacency

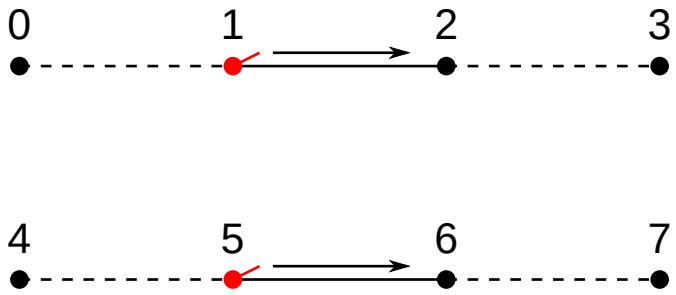
When the primitive topology is VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, each consecutive set of four vertices defines a single line primitive with adjacency, according to the equation:

$$p_i = \{v_{4i}, v_{4i+1}, v_{4i+2}, v_{4i+3}\}$$

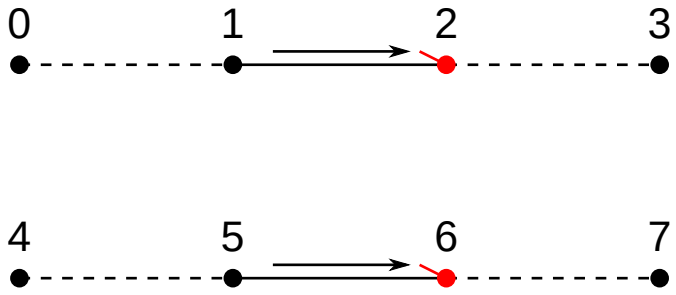
A line primitive is described by the second and third vertices of the total primitive, with the remaining two vertices only accessible in a geometry shader.

The number of primitives generated is equal to ⌊vertexCount/4⌋.

When the provokingVertexMode is VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT, the provoking vertex for p_i is v_{4i+1}.



When the provokingVertexMode is VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT, the provoking vertex for p_i is v_{4i+2}.



25.1.9. Line Strips With Adjacency

When the primitive topology is VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY, one line primitive with adjacency is defined by each vertex and the following vertex, according to the equation:

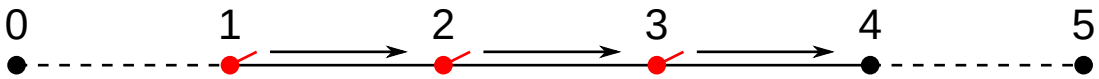
$$p_i = \{v_i, v_{i+1}, v_{i+2}, v_{i+3}\}$$

A line primitive is described by the second and third vertices of the total primitive, with the remaining two vertices only accessible in a geometry shader.

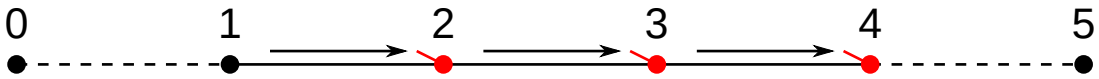
The number of primitives generated is equal to max(0,vertexCount-3).

When the provokingVertexMode is VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT, the

provoking vertex for p_i is v_{i+1}.



When the `provokingVertexMode` is VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT, the provoking vertex for p_i is v_{i+2}.



25.1.10. Triangle Lists With Adjacency

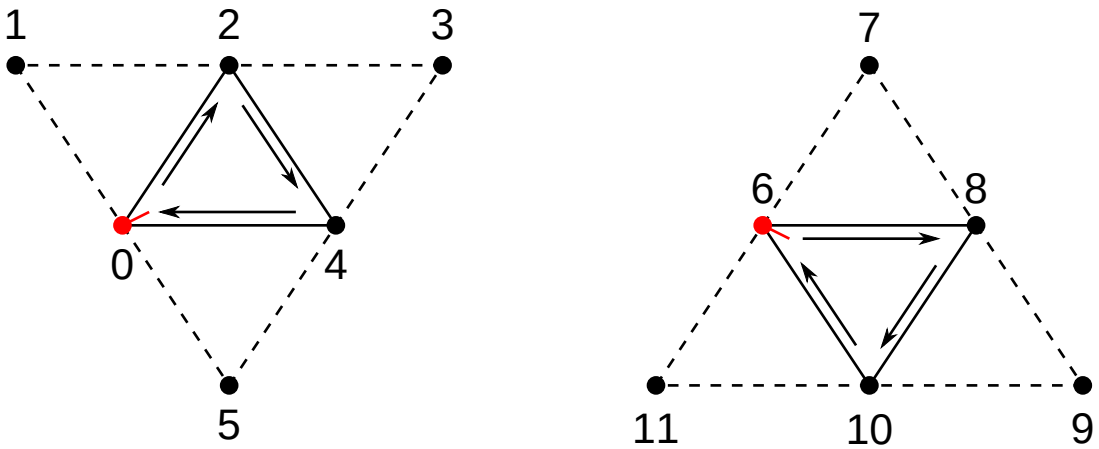
When the primitive topology is VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, each consecutive set of six vertices defines a single triangle primitive with adjacency, according to the equations:

$$p_i = \{v_{6i}, v_{6i+1}, v_{6i+2}, v_{6i+3}, v_{6i+4}, v_{6i+5}\}$$

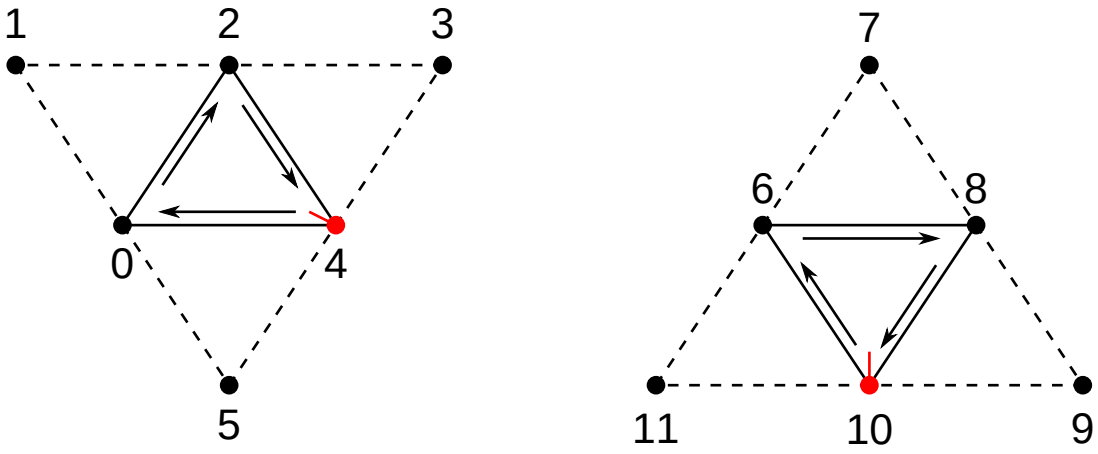
A triangle primitive is described by the first, third, and fifth vertices of the total primitive, with the remaining three vertices only accessible in a geometry shader.

The number of primitives generated is equal to ⌊`vertexCount`/6⌋.

When the `provokingVertexMode` is VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT, the provoking vertex for p_i is v_{6i}.



When the `provokingVertexMode` is VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT, the provoking vertex for p_i is v_{6i+4}.



25.1.11. Triangle Strips With Adjacency

When the primitive topology is VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY, one triangle primitive with adjacency is defined by each vertex and the following 5 vertices.

The number of primitives generated, n, is equal to ⌊max(0, `vertexCount` - 4)/2⌋.

If n=1, the primitive is defined as:

$$p = \{v_0, v_1, v_2, v_5, v_4, v_3\}$$

If n>1, the total primitive consists of different vertices according to where it is in the strip:

$$p_i = \{v_{2i}, v_{2i+1}, v_{2i+2}, v_{2i+6}, v_{2i+4}, v_{2i+3}\} \text{ when } i=0$$

$$p_i = \{v_{2i}, v_{2i+3}, v_{2i+4}, v_{2i+6}, v_{2i+2}, v_{2i-2}\} \text{ when } i>0,\ i<n-1, \text{ and } i\%2=1$$

$$p_i = \{v_{2i}, v_{2i-2}, v_{2i+2}, v_{2i+6}, v_{2i+4}, v_{2i+3}\} \text{ when } i>0,\ i<n-1, \text{ and } i\%2=0$$

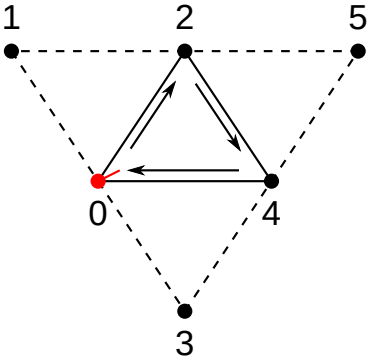
$$p_i = \{v_{2i}, v_{2i+3}, v_{2i+4}, v_{2i+5}, v_{2i+2}, v_{2i-2}\} \text{ when } i=n-1 \text{ and } i\%2=1$$

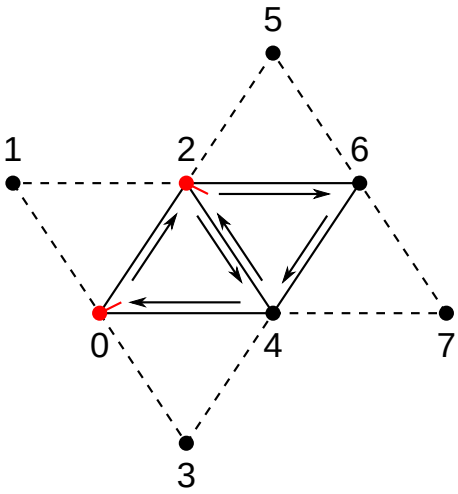
$$p_i = \{v_{2i}, v_{2i-2}, v_{2i+2}, v_{2i+5}, v_{2i+4}, v_{2i+3}\} \text{ when } i=n-1 \text{ and } i\%2=0$$

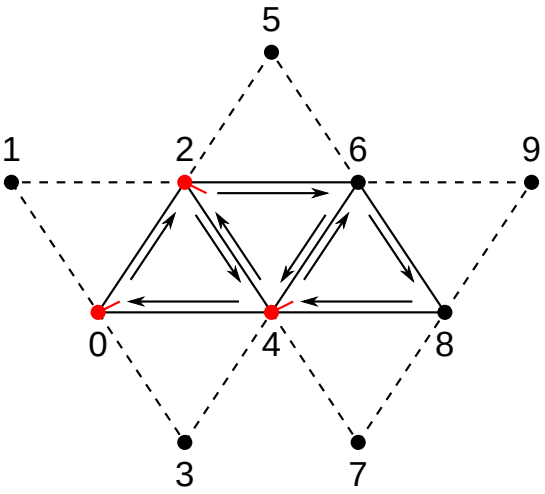
A triangle primitive is described by the first, third, and fifth vertices of the total primitive in all cases, with the remaining three vertices only accessible in a geometry shader.

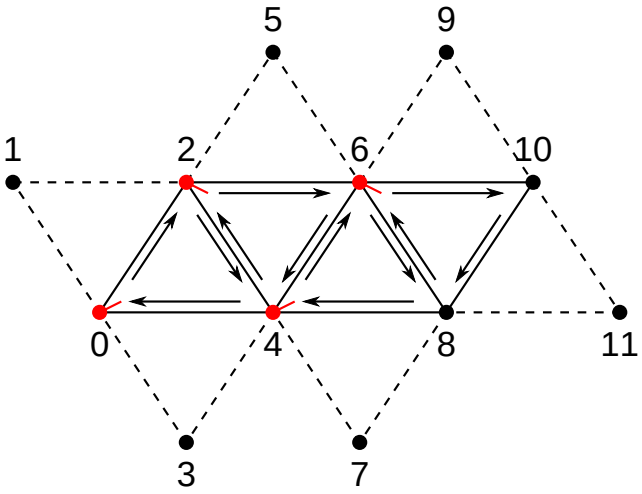
NOTE	The ordering of the vertices in each successive triangle is altered so that the winding order is consistent throughout the strip.

When the `provokingVertexMode` is VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT, the provoking vertex for p_i is always v_{2i}.

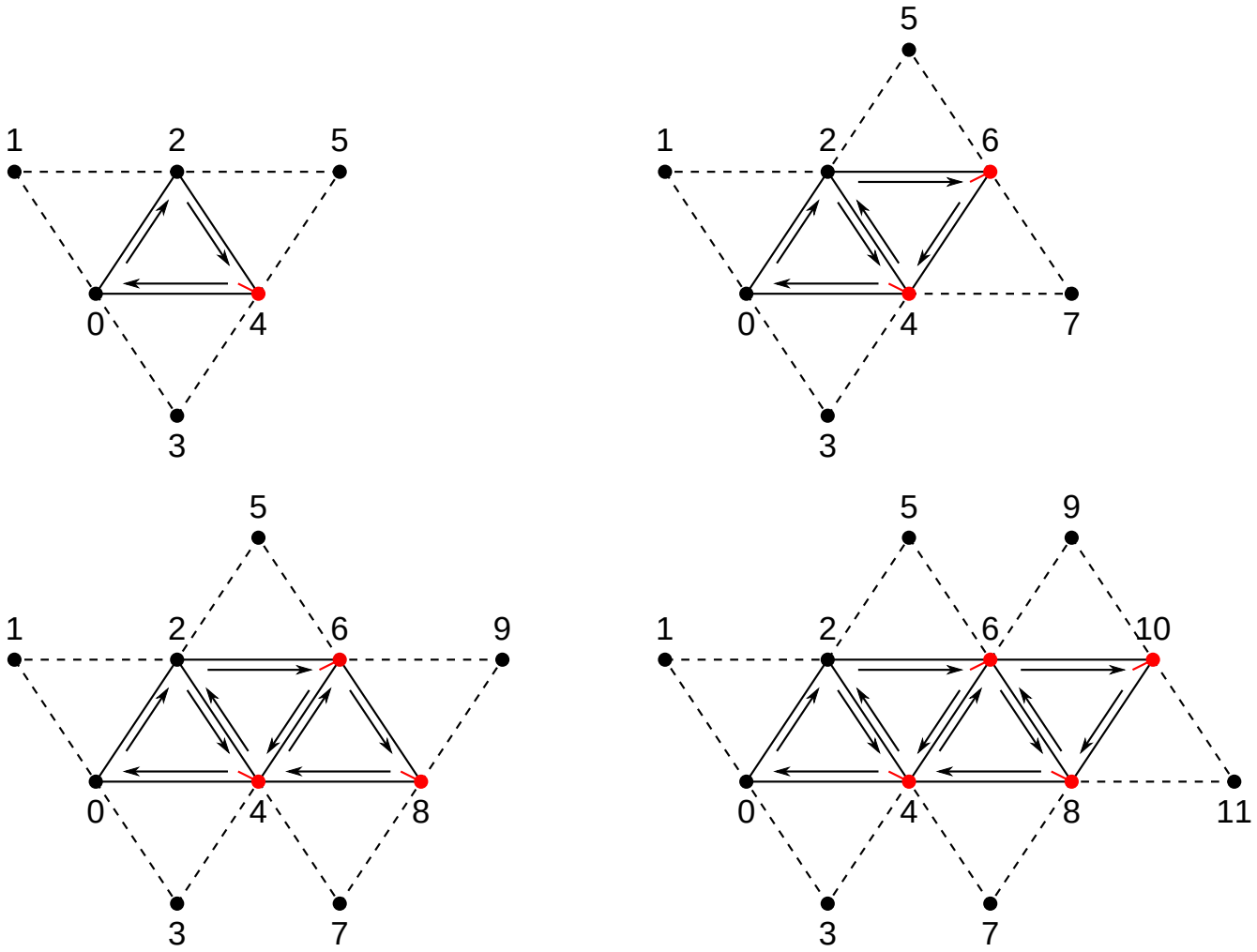








When the `provokingVertexMode` is VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT, the provoking vertex for p_i is always v_{2i+4}.



25.1.12. Patch Lists

When the primitive topology is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, each consecutive set of m vertices defines a single patch primitive, according to the equation:

$$p_i = \{v_{mi}, v_{mi+1}, \ldots, v_{mi+(m-2)}, v_{mi+(m-1)}\}$$

where m is equal to VkPipelineTessellationStateCreateInfo::`patchControlPoints`.

Patch lists are never passed to vertex post-processing, and as such no provoking vertex is defined for patch primitives. The number of primitives generated is equal to ⌊`vertexCount`/m⌋.

The vertices comprising a patch have no implied geometry, and are used as inputs to tessellation shaders and the fixed-function tessellator to generate new point, line, or triangle primitives.

25.2. Effective Primitive Topology

The effective primitive topology is the primitive topology that is consumed or produced by a given stage of the graphics pipeline. The interpretation of this topology depends on the active shader stages and other state set in the graphics pipeline.

25.2.1. Vertex Input Assembler Topology

This is the topology set when the VK_SHADER_STAGE_VERTEX_BIT stage is used and Programmable Primitive Shading occurs. It **can** be set with VkPipelineInputAssemblyStateCreateInfo::`topology` or vkCmdSetPrimitiveTopology.

25.2.2. Clip Space Topology

This is the output of the last pre-rasterization shader stage involved.

- If VK_SHADER_STAGE_VERTEX_BIT is the last pre-rasterization stage, the topology is not changed from what was set in the input assembly.
- If VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT is the last pre-rasterization stage, the topology depends on the `OpExecutionMode` found in either tessellation stage.
  - The `Isolines` `Execution Mode` is in line topology class.
  - The `Triangles` and `Quads` `Execution Mode` is in triangle topology class.
  - The `PointMode` `Execution Mode` is in point topology class and will take precedence over the other tessellation stage `Execution Mode`.
- If VK_SHADER_STAGE_GEOMETRY_BIT is the last pre-rasterization stage, the topology depends on the `OpExecutionMode` found in this shader stage.
  - The `OutputPoints` `Execution Mode` is in point topology class.
  - The `OutputLineStrip` `Execution Mode` is in line topology class.
  - The `OutputTriangleStrip` `Execution Mode` is in triangle topology class.
- If VK_SHADER_STAGE_MESH_BIT_EXT is the last pre-rasterization stage, the topology depends on the `OpExecutionMode` found in this shader stage.
  - The `OutputPoints` `Execution Mode` is in point topology class.
  - The `OutputLinesEXT` `Execution Mode` is in line topology class.
  - The `OutputTrianglesEXT` `Execution Mode` is in triangle topology class.

25.2.3. Rasterization Input Topology

After clipping and viewport transformations, the Polygon Mode **can** set a topology as an input for rasterization. This **can** be different from what was used in Clip Space Topology.

If VkPipelineRasterizationStateCreateInfo::`rasterizerDiscardEnable` or vkCmdSetRasterizerDiscardEnable::`rasterizerDiscardEnable` is set to VK_TRUE, then this Rasterization Input Topology is not observed.

If the clip space topology is in triangle topology class then VK_POLYGON_MODE_LINE will turn it into line topology and VK_POLYGON_MODE_POINT will turn it into point topology.

If the clip space topology is in line topology class then VK_POLYGON_MODE_POINT will turn it into point topology.

25.3. Primitive Order

Primitives generated by drawing commands progress through the stages of the graphics pipeline in *primitive order*. Primitive order is initially determined in the following way:

1. Submission order determines the initial ordering.
2. For indirect drawing commands, the order in which accessed instances of the VkDrawIndirectCommand are stored in `buffer`, from lower indirect buffer addresses to higher addresses.
3. If a drawing command includes multiple instances, the order in which instances are executed, from lower numbered instances to higher.
4. The order in which primitives are specified by a drawing command:
    - For non-indexed draws, from vertices with a lower numbered `vertexIndex` to a higher numbered `vertexIndex`.
    - For indexed draws, vertices sourced from lower index buffer addresses to higher addresses.
    - For draws using mesh shaders, the order is provided by mesh shading.

Within this order implementations further sort primitives:

5. If tessellation shading is active, by an implementation-dependent order of new primitives generated by tessellation.
6. If geometry shading is active, by the order new primitives are generated by geometry shading.
7. If the polygon mode is not VK_POLYGON_MODE_FILL, by an implementation-dependent ordering of the new primitives generated within the original primitive.

Primitive order is later used to define rasterization order, which determines the order in which fragments output results to a framebuffer.

25.4. Programmable Primitive Shading

Once primitives are assembled, they proceed to the vertex shading stage of the pipeline. If the draw includes multiple instances, then the set of primitives is sent to the vertex shading stage multiple times, once for each instance.

It is implementation-dependent whether vertex shading occurs on vertices that are discarded as part of incomplete primitives, but if it does occur then it operates as if they were vertices in complete primitives and such invocations **can** have side effects.

Vertex shading receives two per-vertex inputs from the primitive assembly stage - the `vertexIndex` and the `instanceIndex`. How these values are generated is defined below, with each command.

Drawing commands fall roughly into two categories:

- Non-indexed drawing commands present a sequential `vertexIndex` to the vertex shader. The sequential index is generated automatically by the device (see Fixed-Function Vertex Processing for details on both specifying the vertex attributes indexed by `vertexIndex`, as well as binding

vertex buffers containing those attributes to a command buffer). These commands are:

  - vkCmdDraw
  - vkCmdDrawIndirect
  - vkCmdDrawIndirectCount
  - vkCmdDrawIndirectCountKHR
  - vkCmdDrawMultiEXT

- Indexed drawing commands read index values from an *index buffer* and use this to compute the `vertexIndex` value for the vertex shader. These commands are:
  - vkCmdDrawIndexed
  - vkCmdDrawIndexedIndirect
  - vkCmdDrawIndexedIndirectCount
  - vkCmdDrawIndexedIndirectCountKHR
  - vkCmdDrawMultiIndexedEXT

For indexed drawing commands, indices are sourced from a range of memory bound to the command buffer, referred to as the *bound index buffer range*, and specified by one of the following commands.

To bind an address range as an index buffer to a command buffer, call:

```
// Provided by VK_KHR_device_address_commands
void vkCmdBindIndexBuffer3KHR(
    VkCommandBuffer                             commandBuffer,
    const VkBindIndexBuffer3InfoKHR*            pInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `pInfo` is a pointer to a VkBindIndexBuffer3InfoKHR structure defining parameters of this command.

The bound index buffer range is set to the range of memory indicated by `pInfo->addressRange`.

If the `nullDescriptor` feature is enabled, `pInfo->addressRange.size` and `pInfo->addressRange.address` **can** be 0. When the size and address are 0, every index read from this binding will return a value of zero.

Valid Usage (Implicit)

- VUID-vkCmdBindIndexBuffer3KHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindIndexBuffer3KHR-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkBindIndexBuffer3InfoKHR structure
- VUID-vkCmdBindIndexBuffer3KHR-commandBuffer-recording

commandBuffer **must** be in the recording state

- VUID-vkCmdBindIndexBuffer3KHR-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindIndexBuffer3KHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindIndexBuffer3KHR is not affected by conditional rendering

VkBindIndexBuffer3InfoKHR is defined as:

```
// Provided by VK_KHR_device_address_commands
typedef struct VkBindIndexBuffer3InfoKHR {
    VkStructureType             sType;
    const void*                 pNext;
    VkDeviceAddressRangeKHR     addressRange;
    VkAddressCommandFlagsKHR    addressFlags;
    VkIndexType                 indexType;
} VkBindIndexBuffer3InfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- addressRange is the VkDeviceAddressRangeKHR of the address range to bind.
- addressFlags is a VkAddressCommandFlagsKHR value defining the flags for the address range.
- indexType is a VkIndexType value specifying the size of the indices.

Valid Usage

- VUID-VkBindIndexBuffer3InfoKHR-addressRange-13097
  If the range specified by `addressRange` is not bound completely to memory when accessed, `addressFlags` **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR
- VUID-VkBindIndexBuffer3InfoKHR-addressRange-13098
  If the buffer from which the range specified by `addressRange` was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `addressFlags` **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkBindIndexBuffer3InfoKHR-addressRange-13099
  If the buffer from which the range specified by `addressRange` was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `addressFlags` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkBindIndexBuffer3InfoKHR-addressFlags-13100
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkBindIndexBuffer3InfoKHR-addressRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkBindIndexBuffer3InfoKHR-addressRange-13123
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkBindIndexBuffer3InfoKHR-addressFlags-13101
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkBindIndexBuffer3InfoKHR-addressRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `addressFlags` **must** include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkBindIndexBuffer3InfoKHR-addressRange-13125
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created without VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `addressFlags` **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR

- VUID-VkBindIndexBuffer3InfoKHR-addressRange-13051
  If `addressRange.address` is not 0, the buffer from which `addressRange` was queried **must** have been created with VK_BUFFER_USAGE_INDEX_BUFFER_BIT
- VUID-VkBindIndexBuffer3InfoKHR-addressRange-13052
  `addressRange.address` **must** be a multiple of the size of the type indicated by `indexType`
- VUID-VkBindIndexBuffer3InfoKHR-indexType-13053
  `indexType` **must** not be VK_INDEX_TYPE_NONE_KHR
- VUID-VkBindIndexBuffer3InfoKHR-indexType-13054
  If `indexType` is VK_INDEX_TYPE_UINT8_KHR, the `indexTypeUint8` feature **must** be enabled
- VUID-VkBindIndexBuffer3InfoKHR-None-13055
  If the `nullDescriptor` feature is not enabled, `addressRange.size` **must** be greater than 0
- VUID-VkBindIndexBuffer3InfoKHR-addressRange-13056
  If `addressRange.size` is 0, `addressRange.address` **must** be 0

Valid Usage (Implicit)

- VUID-VkBindIndexBuffer3InfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BIND_INDEX_BUFFER_3_INFO_KHR
- VUID-VkBindIndexBuffer3InfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkBindIndexBuffer3InfoKHR-addressFlags-parameter
  If `addressFlags` is not `0`, `addressFlags` **must** be a valid combination of VkAddressCommandFlagBitsKHR values
- VUID-VkBindIndexBuffer3InfoKHR-indexType-parameter
  `indexType` **must** be a valid VkIndexType value

To bind an index buffer, along with its size, to a command buffer, call:

```
// Provided by VK_VERSION_1_4
void vkCmdBindIndexBuffer2(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    buffer,
    VkDeviceSize                                offset,
    VkDeviceSize                                size,
    VkIndexType                                 indexType);
```

```
// Provided by VK_KHR_maintenance5
// Equivalent to vkCmdBindIndexBuffer2
void vkCmdBindIndexBuffer2KHR(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    buffer,
    VkDeviceSize                                offset,
    VkDeviceSize                                size,
    VkIndexType                                 indexType);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `buffer` is the buffer being bound.
- `offset` is the starting offset in bytes within `buffer` used in index buffer address calculations.
- `size` is the size in bytes of index data bound from `buffer`.
- `indexType` is a VkIndexType value specifying the size of the indices.

`buffer`, `offset`, and `size` specify the bound index buffer range, with a range of memory bound from [`base` + `offset`, `base` + `offset` + `size`), where `base` is the start of `buffer`. If `size` is VK_WHOLE_SIZE then the range is from `offset` to the end of the `buffer`.

If the `maintenance6` feature is enabled, `buffer` **can** be VK_NULL_HANDLE. If `buffer` is VK_NULL_HANDLE and the `nullDescriptor` feature is enabled, every index fetched results in a value of zero.

Valid Usage

- VUID-vkCmdBindIndexBuffer2-offset-08782
  `offset` **must** be less than the size of `buffer`
- VUID-vkCmdBindIndexBuffer2-offset-08783
  The sum of `offset` and the base address of the range of `VkDeviceMemory` object that is backing `buffer` **must** be a multiple of the size of the type indicated by `indexType`
- VUID-vkCmdBindIndexBuffer2-buffer-08784
  `buffer` **must** have been created with the VK_BUFFER_USAGE_INDEX_BUFFER_BIT usage flag set
- VUID-vkCmdBindIndexBuffer2-buffer-08785
  If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBindIndexBuffer2-indexType-08786
  `indexType` **must** not be VK_INDEX_TYPE_NONE_KHR
- VUID-vkCmdBindIndexBuffer2-indexType-08787
  If `indexType` is VK_INDEX_TYPE_UINT8, the `indexTypeUint8` feature **must** be enabled
- VUID-vkCmdBindIndexBuffer2-None-09493
  If the `maintenance6` feature is not enabled, `buffer` **must** not be VK_NULL_HANDLE
- VUID-vkCmdBindIndexBuffer2-buffer-09494

If `buffer` is VK_NULL_HANDLE, offset **must** be zero

- VUID-vkCmdBindIndexBuffer2-size-08767
  If `size` is not VK_WHOLE_SIZE, `size` **must** be a multiple of the size of the type indicated by `indexType`
- VUID-vkCmdBindIndexBuffer2-size-08768
  If `size` is not VK_WHOLE_SIZE, the sum of `offset` and `size` **must** be less than or equal to the size of `buffer`

Valid Usage (Implicit)

- VUID-vkCmdBindIndexBuffer2-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindIndexBuffer2-buffer-parameter
  If `buffer` is not VK_NULL_HANDLE, `buffer` **must** be a valid VkBuffer handle
- VUID-vkCmdBindIndexBuffer2-indexType-parameter
  `indexType` **must** be a valid VkIndexType value
- VUID-vkCmdBindIndexBuffer2-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBindIndexBuffer2-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindIndexBuffer2-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindIndexBuffer2-commonparent
  Both of `buffer`, and `commandBuffer` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindIndexBuffer2 is not affected by conditional rendering

To bind an index buffer to a command buffer, call:

```
// Provided by VK_VERSION_1_0
void vkCmdBindIndexBuffer(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    buffer,
    VkDeviceSize                                offset,
    VkIndexType                                 indexType);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `buffer` is the buffer being bound.
- `offset` is the starting offset in bytes within `buffer` used in index buffer address calculations.
- `indexType` is a VkIndexType value specifying the size of the indices.

`buffer` and `offset` specify the bound index buffer range, with a range of memory bound from [`base` + `offset`, `base` + `offset` + `size`), where `size` is from `offset` to the end of the `buffer`.

If the `maintenance6` feature is enabled, `buffer` **can** be VK_NULL_HANDLE. If `buffer` is VK_NULL_HANDLE and the `nullDescriptor` feature is enabled, every index fetched results in a value of zero.

Valid Usage

- VUID-vkCmdBindIndexBuffer-offset-08782
  `offset` **must** be less than the size of `buffer`
- VUID-vkCmdBindIndexBuffer-offset-08783
  The sum of `offset` and the base address of the range of `VkDeviceMemory` object that is backing `buffer` **must** be a multiple of the size of the type indicated by `indexType`
- VUID-vkCmdBindIndexBuffer-buffer-08784
  `buffer` **must** have been created with the VK_BUFFER_USAGE_INDEX_BUFFER_BIT usage flag set
- VUID-vkCmdBindIndexBuffer-buffer-08785
  If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBindIndexBuffer-indexType-08786
  `indexType` **must** not be VK_INDEX_TYPE_NONE_KHR
- VUID-vkCmdBindIndexBuffer-indexType-08787
  If `indexType` is VK_INDEX_TYPE_UINT8, the `indexTypeUint8` feature **must** be enabled
- VUID-vkCmdBindIndexBuffer-None-09493

If the maintenance6 feature is not enabled, buffer **must** not be VK_NULL_HANDLE

- VUID-vkCmdBindIndexBuffer-buffer-09494
  If buffer is VK_NULL_HANDLE, offset **must** be zero

Valid Usage (Implicit)

- VUID-vkCmdBindIndexBuffer-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindIndexBuffer-buffer-parameter
  If buffer is not VK_NULL_HANDLE, buffer **must** be a valid VkBuffer handle
- VUID-vkCmdBindIndexBuffer-indexType-parameter
  indexType **must** be a valid VkIndexType value
- VUID-vkCmdBindIndexBuffer-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdBindIndexBuffer-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindIndexBuffer-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindIndexBuffer-commonparent
  Both of buffer, and commandBuffer that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindIndexBuffer is not affected by conditional rendering

Possible values of vkCmdBindIndexBuffer2::indexType and vkCmdBindIndexBuffer::indexType, specifying the size of indices, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkIndexType {
    VK_INDEX_TYPE_UINT16 = 0,
    VK_INDEX_TYPE_UINT32 = 1,
  // Provided by VK_VERSION_1_4
    VK_INDEX_TYPE_UINT8 = 1000265000,
  // Provided by VK_KHR_acceleration_structure
    VK_INDEX_TYPE_NONE_KHR = 1000165000,
  // Provided by VK_KHR_index_type_uint8
    VK_INDEX_TYPE_UINT8_KHR = VK_INDEX_TYPE_UINT8,
} VkIndexType;
```

- VK_INDEX_TYPE_UINT16 specifies that indices are 16-bit unsigned integer values.
- VK_INDEX_TYPE_UINT32 specifies that indices are 32-bit unsigned integer values.
- VK_INDEX_TYPE_NONE_KHR specifies that no indices are provided.
- VK_INDEX_TYPE_UINT8 specifies that indices are 8-bit unsigned integer values.

The parameters for each drawing command are specified directly in the command or read from buffer memory, depending on the command. Drawing commands that source their parameters from buffer memory are known as *indirect* drawing commands.

All drawing commands interact with the robustBufferAccess feature.

To record a non-indexed draw, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDraw(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    vertexCount,
    uint32_t                                    instanceCount,
    uint32_t                                    firstVertex,
    uint32_t                                    firstInstance);
```

- commandBuffer is the command buffer into which the command is recorded.
- vertexCount is the number of vertices to draw.
- instanceCount is the number of instances to draw.
- firstVertex is the index of the first vertex to draw.
- firstInstance is the instance ID of the first instance to draw.

When the command is executed, primitives are assembled using the current primitive topology and vertexCount consecutive vertex indices with the first vertexIndex value equal to firstVertex. The primitives are drawn instanceCount times with instanceIndex starting with firstInstance and

increasing sequentially for each instance. The assembled primitives execute the bound graphics pipeline.

Valid Usage

- VUID-vkCmdDraw-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDraw-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDraw-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDraw-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDraw-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDraw-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDraw-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDraw-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDraw-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this

command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT

- VUID-vkCmdDraw-magFilter-09598
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDraw-mipmapMode-04770
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDraw-mipmapMode-09599
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDraw-unnormalizedCoordinates-09635
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDraw-None-08609
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `viewType` **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDraw-None-08610
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions with `ImplicitLod`, `Dref` or `Proj` in their name
- VUID-vkCmdDraw-None-08611
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions that include a LOD bias or any offset values
- VUID-vkCmdDraw-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDraw-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain

VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT

- VUID-vkCmdDraw-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDraw-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDraw-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDraw-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDraw-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDraw-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDraw-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDraw-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDraw-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDraw-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges

and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader

- VUID-vkCmdDraw-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDraw-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDraw-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDraw-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDraw-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDraw-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDraw-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDraw-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between

SPIR-V Image Dimensions and Vulkan ImageView Types

- VUID-vkCmdDraw-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match
- VUID-vkCmdDraw-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDraw-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have four components
- VUID-vkCmdDraw-OpImageWrite-04469
  If a VkBufferView is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDraw-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdDraw-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdDraw-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdDraw-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdDraw-sparseImageInt64Atomics-04474
  If the `sparseImageInt64Atomics` feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdDraw-sparseImageInt64Atomics-04475
  If the `sparseImageInt64Atomics` feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command

- VUID-vkCmdDraw-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDraw-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDraw-micromap-11636
  If an acceleration structure is accessed as a result of this command through an `OpRayQueryProceedKHR` operation , the `VK_EXT_opacity_micromap` extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its `pNext` chain, the shader **must** have specified a `OpacityMicromapIdKHR` execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDraw-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDraw-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with `flags` that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDraw-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with `flags` that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDraw-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDraw-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or

VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a OpTypeStruct decorated with Block or BufferBlock using that mapping, the calculated offset for the resource heap **must** be a multiple of bufferDescriptorAlignment

- VUID-vkCmdDraw-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeImage or OpTypeSampledImage using that mapping, the calculated offset for the resource heap **must** be a multiple of imageDescriptorAlignment
- VUID-vkCmdDraw-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeSampler or OpTypeSampledImage using that mapping, the calculated offset for the sampler heap **must** be a multiple of samplerDescriptorAlignment
- VUID-vkCmdDraw-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDraw-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDraw-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDraw-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping

using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8

- VUID-vkCmdDraw-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDraw-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDraw-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDraw-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDraw-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDraw-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDraw-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not

enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdDraw-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDraw-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDraw-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDraw-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDraw-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDraw-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT,

VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`

- VUID-vkCmdDraw-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDraw-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDraw-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDraw-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDraw-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDraw-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDraw-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDraw-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDraw-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDraw-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDraw-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by VkRenderingInputAttachmentIndexInfo, or no `InputAttachmentIndex` if VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` or VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are `NULL`, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDraw-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` and VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are non-`NULL`
- VUID-vkCmdDraw-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDraw-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command

- VUID-vkCmdDraw-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDraw-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDraw-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDraw-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDraw-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDraw-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDraw-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDraw-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDraw-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDraw-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must**

have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::`maxMultiviewInstanceIndex`
- VUID-vkCmdDraw-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDraw-None-07634
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-06666
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `sampleLocationsEnable` is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07846
  If the `depthBounds` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** match the `scissorCount` parameter of `vkCmdSetScissorWithCount`
- VUID-vkCmdDraw-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be `1`
- VUID-vkCmdDraw-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, and any shader object bound to a graphics stage writes to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of

vkCmdSetViewportWithCount **must** be 1

- VUID-vkCmdDraw-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of blendEnable **must** be VK_FALSE
- VUID-vkCmdDraw-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and none of the following is enabled:
  - the multisampledRenderToSingleSampled feature

  then the current value of rasterizationSamples **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDraw-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDraw-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::viewMask equal to VkRenderingInfo::viewMask
- VUID-vkCmdDraw-colorAttachmentCount-06179
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08910
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08912
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDraw-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDraw-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT ::discardRectangleCount
- VUID-vkCmdDraw-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT ::maxDiscardRectangles
- VUID-vkCmdDraw-None-07880

If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound

graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->imageView

- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo ::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDraw-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDraw-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDraw-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDraw-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDraw-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDraw-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and

vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView

- VUID-vkCmdDraw-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDraw-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDraw-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDraw-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDraw-pDepthAttachment-08964
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDraw-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the

VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDraw-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDraw-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDraw-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDraw-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDraw-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDraw-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and `alphaToCoverageEnable` was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDraw-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set `alphaToCoverageEnable` to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDraw-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07625
  If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07626
  If the `logicOp` feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current

value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07630
  If the geometryStreams feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07631
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07632
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of conservativeRasterizationMode is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-conservativePointAndLineRasterization-07499
  If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-vkCmdDraw-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated

in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-08669
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current lineRasterizationMode is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current alphaToCoverageEnable, alphaToOneEnable and sampleShadingEnable states **must** all be VK_FALSE
- VUID-vkCmdDraw-None-07639
  If the depthClipControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the

VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-09650
  If the depthClampControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of depthClampEnable is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDraw-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::rasterizationSamples parameter used to create the bound graphics pipeline
- VUID-vkCmdDraw-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the rasterizationSamples parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDraw-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the rasterizationSamples member of that structure
- VUID-vkCmdDraw-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDraw-firstAttachment-07478

If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments

- VUID-vkCmdDraw-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE, then the number of active color attachments **must** not exceed `advancedBlendMaxColorAttachments`

- VUID-vkCmdDraw-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE

- VUID-vkCmdDraw-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE

- VUID-vkCmdDraw-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE

- VUID-vkCmdDraw-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the `rasterizationStream` to

zero

- VUID-vkCmdDraw-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDraw-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDraw-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDraw-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT ::`maxSampleLocationGridSize.width` as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a `samples` parameter equaling `rasterizationSamples`
- VUID-vkCmdDraw-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT ::`maxSampleLocationGridSize.height` as returned by

vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdDraw-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDraw-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDraw-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDraw-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDraw-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled
- VUID-vkCmdDraw-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if

the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled

- VUID-vkCmdDraw-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDraw-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDraw-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDraw-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a nextStage including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDraw-None-08684
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDraw-None-08685
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDraw-None-08686
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled,

vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT

- VUID-vkCmdDraw-None-08687
  If there is no bound graphics pipeline, and the geometryShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDraw-None-08688
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDraw-None-08689
  If there is no bound graphics pipeline, and the taskShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDraw-None-08690
  If there is no bound graphics pipeline, and the meshShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDraw-None-08693
  If there is no bound graphics pipeline, and at least one of the taskShader and meshShader features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid VkShaderEXT bound, and the other **must** have no VkShaderEXT bound
- VUID-vkCmdDraw-None-08696
  If there is no bound graphics pipeline, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no VkShaderEXT bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDraw-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDraw-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any VkShaderEXT bound
- VUID-vkCmdDraw-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDraw-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically

defined arrays of descriptor set layouts

- VUID-vkCmdDraw-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDraw-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to vkCmdSetStencilWriteMask **must** be `0`
- VUID-vkCmdDraw-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the `pColorWriteMasks` parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDraw-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of `Location` for any output attachment statically used in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`
- VUID-vkCmdDraw-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::`pColorAttachmentLocations` in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance
- VUID-vkCmdDraw-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::`pColorAttachmentInputIndices` in the bound pipeline, if either `pColorAttachmentInputIndices`[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an `InputAttachmentIndex` decoration, the two values **must** match
- VUID-vkCmdDraw-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` in the bound pipeline or the corresponding value set currently in the current render pass instance is

statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match

- VUID-vkCmdDraw-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDraw-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDraw-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDraw-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDraw-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::flags which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDraw-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDraw-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDraw-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT ::customResolve as VK_TRUE
- VUID-vkCmdDraw-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render

pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE

- VUID-vkCmdDraw-None-11861
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount

- VUID-vkCmdDraw-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline

- VUID-vkCmdDraw-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDraw-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDraw-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo

::pDepthAttachment->resolveImageView

- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDraw-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDraw-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView
- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDraw-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDraw-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDraw-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering,

vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView

- VUID-vkCmdDraw-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDraw-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDraw-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDraw-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDraw-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDraw-flags-13363
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDraw-commandBuffer-02712
  If commandBuffer is a protected command buffer and protectedNoFault is not supported, any resource written to by the VkPipeline object bound to the pipeline bind point used by this command **must** not be an unprotected resource

- VUID-vkCmdDraw-commandBuffer-02713
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, pipeline stages other than the framebuffer-space and compute stages in the `VkPipeline` object bound to the pipeline bind point used by this command **must** not write to any resource
- VUID-vkCmdDraw-commandBuffer-04617
  If any of the shader stages of the `VkPipeline` bound to the pipeline bind point used by this command uses the `RayQueryKHR` capability, then `commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdDraw-None-04007
  All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or VK_NULL_HANDLE buffers bound
- VUID-vkCmdDraw-None-04008
  If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be VK_NULL_HANDLE
- VUID-vkCmdDraw-None-02721
  If the `robustBufferAccess` feature is not enabled, and that pipeline was created without enabling VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS for `vertexInputs`, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in Vertex Input Description
- VUID-vkCmdDraw-format-10389
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is a packed format, and the `legacyVertexAttributes` feature is not enabled, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the size of the `format`
- VUID-vkCmdDraw-format-10390
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is not a packed format, and either the `legacyVertexAttributes` feature is not enabled or `format` has 64-bit components, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the component size of the `format`
- VUID-vkCmdDraw-None-07842
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled then vkCmdSetPrimitiveTopology **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-dynamicPrimitiveTopologyUnrestricted-07500
  If the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled and the dynamicPrimitiveTopologyUnrestricted is VK_FALSE, then the primitiveTopology parameter of vkCmdSetPrimitiveTopology **must** be of the same topology class as the pipeline VkPipelineInputAssemblyStateCreateInfo::topology state

- VUID-vkCmdDraw-primitiveTopology-10286
  If a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage is bound, then the current value of primitiveTopology **must** be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command
- VUID-vkCmdDraw-primitiveTopology-10747
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command, then a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage **must** be bound
- VUID-vkCmdDraw-primitiveTopology-10748
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_POINT_LIST prior to this drawing command, the maintenance5 feature is not enabled, both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT and VK_SHADER_STAGE_GEOMETRY_BIT stage are not bound, then the Vertex Execution Model **must** have a PointSize decorated variable that is statically written to
- VUID-vkCmdDraw-pStrides-04913
  If the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, but without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, then vkCmdBindVertexBuffers2 with a non-NULL pStrides parameter or vkCmdBindVertexBuffers3KHR with setStride set to VK_TRUE **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDraw-None-13118
  If the bound graphics pipeline was created without the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, and vkCmdBindVertexBuffers3KHR was called and not subsequently invalidated in the current command buffer prior to this draw command, the value of setStride in each of its pBindInfos elements must have been VK_FALSE
- VUID-vkCmdDraw-None-04914
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDraw-Input-07939
  If the maintenance9 feature is not enabled, and there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then all variables with the Input storage class decorated with Location in the Vertex Execution Model OpEntryPoint **must** contain a location in VkVertexInputAttributeDescription2EXT::location

- VUID-vkCmdDraw-Input-08734
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and either the `legacyVertexAttributes` feature is not enabled or the SPIR-V Type associated with a given `Input` variable of the corresponding `Location` in the `Vertex` `Execution` `Model` `OpEntryPoint` is 64-bit, then the numeric type associated with all `Input` variables of the corresponding `Location` in the `Vertex` `Execution` `Model` `OpEntryPoint` **must** be the same as VkVertexInputAttributeDescription2EXT::`format`
- VUID-vkCmdDraw-format-08936
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::`format` has a 64-bit component, then the scalar width associated with all `Input` variables of the corresponding `Location` in the `Vertex` `Execution` `Model` `OpEntryPoint` **must** be 64-bit
- VUID-vkCmdDraw-format-08937
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and the scalar width associated with a `Location` decorated `Input` variable in the `Vertex` `Execution` `Model` `OpEntryPoint` is 64-bit, then the corresponding VkVertexInputAttributeDescription2EXT ::`format` **must** have a 64-bit component
- VUID-vkCmdDraw-None-09203
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::`format` has a 64-bit component, then all `Input` variables at the corresponding `Location` in the `Vertex` `Execution` `Model` `OpEntryPoint` **must** not use components that are not present in the format
- VUID-vkCmdDraw-None-04875
  If there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with both a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage and the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state enabled, and the current value of `primitiveTopology` is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, then vkCmdSetPatchControlPointsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-04879
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-09637

If the primitiveTopologyListRestart feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled, then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE

- VUID-vkCmdDraw-None-10909
  If the primitiveTopologyPatchListRestart feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDraw-stage-06481
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::stage member of any element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDraw-None-08885
  There **must** be no shader object bound to either of the VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT stages
- VUID-vkCmdDraw-None-07619
  If a shader object is bound to the VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage and the VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT dynamic state enabled, then vkCmdSetTessellationDomainOriginEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-OpExecutionMode-12239
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the type of subdivision, they **must** be the same
- VUID-vkCmdDraw-OpExecutionMode-12240
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the orientation of triangles, they **must** be the same
- VUID-vkCmdDraw-OpExecutionMode-12241
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the segment spacing, they **must** be the same
- VUID-vkCmdDraw-OpExecutionMode-12242
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain

an OpExecutionMode instruction specifying the output patch size, they **must** be the same

- VUID-vkCmdDraw-pNext-09461
  If the bound graphics pipeline state was created with VkPipelineVertexInputDivisorStateCreateInfo in the pNext chain of VkGraphicsPipelineCreateInfo::pVertexInputState, any member of VkPipelineVertexInputDivisorStateCreateInfo::pVertexBindingDivisors has a value other than 1 in divisor, and VkPhysicalDeviceVertexAttributeDivisorProperties ::supportsNonZeroFirstInstance is VK_FALSE, then firstInstance **must** be 0
- VUID-vkCmdDraw-None-09462
  If shader objects are used for drawing or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, any member of the pVertexBindingDescriptions parameter to the vkCmdSetVertexInputEXT call that sets this dynamic state has a value other than 1 in divisor, and VkPhysicalDeviceVertexAttributeDivisorProperties::supportsNonZeroFirstInstance is VK_FALSE, then firstInstance **must** be 0

Valid Usage (Implicit)

- VUID-vkCmdDraw-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDraw-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdDraw-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDraw-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDraw-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDraw-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDraw is affected by conditional rendering

To record an indexed draw, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDrawIndexed(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    indexCount,
    uint32_t                                    instanceCount,
    uint32_t                                    firstIndex,
    int32_t                                     vertexOffset,
    uint32_t                                    firstInstance);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `indexCount` is the number of vertices to draw.
- `instanceCount` is the number of instances to draw.
- `firstIndex` is the base index within the index buffer.
- `vertexOffset` is the value added to the vertex index before indexing into the vertex buffer.
- `firstInstance` is the instance ID of the first instance to draw.

When the command is executed, primitives are assembled using the current primitive topology and `indexCount` vertices whose indices are retrieved from the index buffer. The index buffer is treated as an array of tightly packed unsigned integers of size defined by the vkCmdBindIndexBuffer2::`indexType` or the vkCmdBindIndexBuffer::`indexType` parameter with which the buffer was bound.

The first vertex index is read from `indexAddress` + (`firstIndex` × `indexSize`), where `indexAddress` is the effective address specified for the index buffer by vkCmdBindIndexBuffer3KHR, vkCmdBindIndexBuffer2, or vkCmdBindIndexBuffer, and `indexSize` is the byte size of the type specified by `indexType`. Subsequent index values are retrieved from consecutive locations in the index buffer. Indices are first compared to the primitive restart value, then zero extended to 32 bits (if the `indexType` is VK_INDEX_TYPE_UINT8 or VK_INDEX_TYPE_UINT16) and have `vertexOffset` added to them, before being supplied as the `vertexIndex` value.

The primitives are drawn `instanceCount` times with `instanceIndex` starting with `firstInstance` and

increasing sequentially for each instance. The assembled primitives execute the bound graphics pipeline.

Valid Usage

- VUID-vkCmdDrawIndexed-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawIndexed-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawIndexed-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawIndexed-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexed-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexed-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexed-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexed-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawIndexed-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this

command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT

- VUID-vkCmdDrawIndexed-magFilter-09598
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndexed-mipmapMode-04770
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndexed-mipmapMode-09599
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndexed-unnormalizedCoordinates-09635
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDrawIndexed-None-08609
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `viewType` **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawIndexed-None-08610
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions with `ImplicitLod`, `Dref` or `Proj` in their name
- VUID-vkCmdDrawIndexed-None-08611
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions that include a LOD bias or any offset values
- VUID-vkCmdDrawIndexed-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawIndexed-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain

VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT

- VUID-vkCmdDrawIndexed-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDrawIndexed-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawIndexed-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawIndexed-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndexed-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndexed-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexed-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexed-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexed-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexed-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges

and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader

- VUID-vkCmdDrawIndexed-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawIndexed-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDrawIndexed-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawIndexed-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexed-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexed-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexed-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexed-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between

SPIR-V Image Dimensions and Vulkan ImageView Types

- VUID-vkCmdDrawIndexed-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match
- VUID-vkCmdDrawIndexed-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawIndexed-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have four components
- VUID-vkCmdDrawIndexed-OpImageWrite-04469
  If a VkBufferView is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawIndexed-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdDrawIndexed-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdDrawIndexed-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdDrawIndexed-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdDrawIndexed-sparseImageInt64Atomics-04474
  If the `sparseImageInt64Atomics` feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdDrawIndexed-sparseImageInt64Atomics-04475
  If the `sparseImageInt64Atomics` feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command

- VUID-vkCmdDrawIndexed-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawIndexed-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawIndexed-micromap-11636
  If an acceleration structure is accessed as a result of this command through an `OpRayQueryProceedKHR` operation , the `VK_EXT_opacity_micromap` extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its `pNext` chain, the shader **must** have specified a `OpacityMicromapIdKHR` execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawIndexed-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawIndexed-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with `flags` that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawIndexed-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with `flags` that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDrawIndexed-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawIndexed-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or

VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a OpTypeStruct decorated with Block or BufferBlock using that mapping, the calculated offset for the resource heap **must** be a multiple of bufferDescriptorAlignment

- VUID-vkCmdDrawIndexed-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeImage or OpTypeSampledImage using that mapping, the calculated offset for the resource heap **must** be a multiple of imageDescriptorAlignment
- VUID-vkCmdDrawIndexed-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeSampler or OpTypeSampledImage using that mapping, the calculated offset for the sampler heap **must** be a multiple of samplerDescriptorAlignment
- VUID-vkCmdDrawIndexed-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDrawIndexed-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndexed-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndexed-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping

using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8

- VUID-vkCmdDrawIndexed-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndexed-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndexed-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawIndexed-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawIndexed-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndexed-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndexed-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not

enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdDrawIndexed-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`

- VUID-vkCmdDrawIndexed-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT

- VUID-vkCmdDrawIndexed-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds

- VUID-vkCmdDrawIndexed-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set

- VUID-vkCmdDrawIndexed-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set

- VUID-vkCmdDrawIndexed-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT,

VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`

- VUID-vkCmdDrawIndexed-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndexed-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawIndexed-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDrawIndexed-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDrawIndexed-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawIndexed-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawIndexed-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndexed-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndexed-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawIndexed-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDrawIndexed-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by VkRenderingInputAttachmentIndexInfo, or no `InputAttachmentIndex` if VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` or VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are `NULL`, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawIndexed-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` and VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are non-`NULL`
- VUID-vkCmdDrawIndexed-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawIndexed-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexed-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexed-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexed-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexed-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawIndexed-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawIndexed-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawIndexed-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawIndexed-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDrawIndexed-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawIndexed-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must**

have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::`maxMultiviewInstanceIndex`
- VUID-vkCmdDrawIndexed-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndexed-None-07634
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-06666
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `sampleLocationsEnable` is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07846
  If the `depthBounds` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** match the `scissorCount` parameter of `vkCmdSetScissorWithCount`
- VUID-vkCmdDrawIndexed-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be `1`
- VUID-vkCmdDrawIndexed-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, and any shader object bound to a graphics stage writes to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of

vkCmdSetViewportWithCount **must** be 1

- VUID-vkCmdDrawIndexed-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of blendEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndexed-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and none of the following is enabled:
  - the multisampledRenderToSingleSampled feature

  then the current value of rasterizationSamples **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDrawIndexed-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDrawIndexed-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::viewMask equal to VkRenderingInfo::viewMask
- VUID-vkCmdDrawIndexed-colorAttachmentCount-06179
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08910
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08912
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawIndexed-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT ::discardRectangleCount
- VUID-vkCmdDrawIndexed-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT ::maxDiscardRectangles
- VUID-vkCmdDrawIndexed-None-07880

If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound

graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->imageView

- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo ::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawIndexed-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawIndexed-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawIndexed-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawIndexed-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndexed-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and

vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView

- VUID-vkCmdDrawIndexed-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDrawIndexed-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawIndexed-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-pDepthAttachment-08964
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the

VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexed-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawIndexed-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawIndexed-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDrawIndexed-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and `alphaToCoverageEnable` was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawIndexed-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set `alphaToCoverageEnable` to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawIndexed-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07625
  If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07626
  If the `logicOp` feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current

value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07630
  If the geometryStreams feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07631
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07632
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of conservativeRasterizationMode is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-conservativePointAndLineRasterization-07499
  If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-vkCmdDrawIndexed-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated

in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-08669
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current lineRasterizationMode is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current alphaToCoverageEnable, alphaToOneEnable and sampleShadingEnable states **must** all be VK_FALSE
- VUID-vkCmdDrawIndexed-None-07639
  If the depthClipControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the

VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-09650
  If the depthClampControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of depthClampEnable is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawIndexed-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::rasterizationSamples parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawIndexed-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the rasterizationSamples parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexed-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the rasterizationSamples member of that structure
- VUID-vkCmdDrawIndexed-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndexed-firstAttachment-07478

If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments

- VUID-vkCmdDrawIndexed-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE, then the number of active color attachments **must** not exceed `advancedBlendMaxColorAttachments`

- VUID-vkCmdDrawIndexed-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE

- VUID-vkCmdDrawIndexed-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE

- VUID-vkCmdDrawIndexed-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE

- VUID-vkCmdDrawIndexed-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the `rasterizationStream` to

zero

- VUID-vkCmdDrawIndexed-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawIndexed-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexed-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndexed-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT ::`maxSampleLocationGridSize.width` as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a `samples` parameter equaling `rasterizationSamples`
- VUID-vkCmdDrawIndexed-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT ::`maxSampleLocationGridSize.height` as returned by

vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdDrawIndexed-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDrawIndexed-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexed-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexed-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexed-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled
- VUID-vkCmdDrawIndexed-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if

the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled

- VUID-vkCmdDrawIndexed-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawIndexed-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawIndexed-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawIndexed-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a nextStage including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDrawIndexed-None-08684
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDrawIndexed-None-08685
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawIndexed-None-08686
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled,

vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT

- VUID-vkCmdDrawIndexed-None-08687
  If there is no bound graphics pipeline, and the geometryShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawIndexed-None-08688
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawIndexed-None-08689
  If there is no bound graphics pipeline, and the taskShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawIndexed-None-08690
  If there is no bound graphics pipeline, and the meshShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndexed-None-08693
  If there is no bound graphics pipeline, and at least one of the taskShader and meshShader features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid VkShaderEXT bound, and the other **must** have no VkShaderEXT bound
- VUID-vkCmdDrawIndexed-None-08696
  If there is no bound graphics pipeline, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no VkShaderEXT bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawIndexed-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDrawIndexed-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any VkShaderEXT bound
- VUID-vkCmdDrawIndexed-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawIndexed-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically

defined arrays of descriptor set layouts

- VUID-vkCmdDrawIndexed-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndexed-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to vkCmdSetStencilWriteMask **must** be `0`
- VUID-vkCmdDrawIndexed-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the `pColorWriteMasks` parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawIndexed-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of `Location` for any output attachment statically used in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`
- VUID-vkCmdDrawIndexed-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::`pColorAttachmentLocations` in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance
- VUID-vkCmdDrawIndexed-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::`pColorAttachmentInputIndices` in the bound pipeline, if either `pColorAttachmentInputIndices`[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an `InputAttachmentIndex` decoration, the two values **must** match
- VUID-vkCmdDrawIndexed-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` in the bound pipeline or the corresponding value set currently in the current render pass instance is

statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match

- VUID-vkCmdDrawIndexed-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndexed-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawIndexed-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDrawIndexed-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawIndexed-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::flags which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawIndexed-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndexed-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndexed-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT ::customResolve as VK_TRUE
- VUID-vkCmdDrawIndexed-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render

pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE

- VUID-vkCmdDrawIndexed-None-11861
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndexed-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawIndexed-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo

::pDepthAttachment->resolveImageView

- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexed-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawIndexed-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndexed-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering,

vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView

- VUID-vkCmdDrawIndexed-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndexed-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndexed-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawIndexed-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndexed-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndexed-flags-13363
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndexed-commandBuffer-02712
  If commandBuffer is a protected command buffer and protectedNoFault is not supported, any resource written to by the VkPipeline object bound to the pipeline bind point used by this command **must** not be an unprotected resource

- VUID-vkCmdDrawIndexed-commandBuffer-02713
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, pipeline stages other than the framebuffer-space and compute stages in the `VkPipeline` object bound to the pipeline bind point used by this command **must** not write to any resource
- VUID-vkCmdDrawIndexed-commandBuffer-04617
  If any of the shader stages of the `VkPipeline` bound to the pipeline bind point used by this command uses the `RayQueryKHR` capability, then `commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdDrawIndexed-None-04007
  All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or VK_NULL_HANDLE buffers bound
- VUID-vkCmdDrawIndexed-None-04008
  If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be VK_NULL_HANDLE
- VUID-vkCmdDrawIndexed-None-02721
  If the `robustBufferAccess` feature is not enabled, and that pipeline was created without enabling VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS for `vertexInputs`, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in Vertex Input Description
- VUID-vkCmdDrawIndexed-format-10389
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is a packed format, and the `legacyVertexAttributes` feature is not enabled, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the size of the `format`
- VUID-vkCmdDrawIndexed-format-10390
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is not a packed format, and either the `legacyVertexAttributes` feature is not enabled or `format` has 64-bit components, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the component size of the `format`
- VUID-vkCmdDrawIndexed-None-07842
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled then vkCmdSetPrimitiveTopology **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-dynamicPrimitiveTopologyUnrestricted-07500
  If the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled and the dynamicPrimitiveTopologyUnrestricted is VK_FALSE, then the primitiveTopology parameter of vkCmdSetPrimitiveTopology **must** be of the same topology class as the pipeline VkPipelineInputAssemblyStateCreateInfo::topology state

- VUID-vkCmdDrawIndexed-primitiveTopology-10286
  If a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage is bound, then the current value of primitiveTopology **must** be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command
- VUID-vkCmdDrawIndexed-primitiveTopology-10747
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command, then a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage **must** be bound
- VUID-vkCmdDrawIndexed-primitiveTopology-10748
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_POINT_LIST prior to this drawing command, the maintenance5 feature is not enabled, both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT and VK_SHADER_STAGE_GEOMETRY_BIT stage are not bound, then the Vertex Execution Model **must** have a PointSize decorated variable that is statically written to
- VUID-vkCmdDrawIndexed-pStrides-04913
  If the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, but without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, then vkCmdBindVertexBuffers2 with a non-NULL pStrides parameter or vkCmdBindVertexBuffers3KHR with setStride set to VK_TRUE **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndexed-None-13118
  If the bound graphics pipeline was created without the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, and vkCmdBindVertexBuffers3KHR was called and not subsequently invalidated in the current command buffer prior to this draw command, the value of setStride in each of its pBindInfos elements must have been VK_FALSE
- VUID-vkCmdDrawIndexed-None-04914
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndexed-Input-07939
  If the maintenance9 feature is not enabled, and there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then all variables with the Input storage class decorated with Location in the Vertex Execution Model OpEntryPoint **must** contain a location in VkVertexInputAttributeDescription2EXT ::location

- VUID-vkCmdDrawIndexed-Input-08734
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and either the `legacyVertexAttributes` feature is not enabled or the SPIR-V Type associated with a given `Input` variable of the corresponding `Location` in the `Vertex` `Execution` `Model` `OpEntryPoint` is 64-bit, then the numeric type associated with all `Input` variables of the corresponding `Location` in the `Vertex` `Execution` `Model` `OpEntryPoint` **must** be the same as VkVertexInputAttributeDescription2EXT::`format`
- VUID-vkCmdDrawIndexed-format-08936
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::`format` has a 64-bit component, then the scalar width associated with all `Input` variables of the corresponding `Location` in the `Vertex` `Execution` `Model` `OpEntryPoint` **must** be 64-bit
- VUID-vkCmdDrawIndexed-format-08937
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and the scalar width associated with a `Location` decorated `Input` variable in the `Vertex` `Execution` `Model` `OpEntryPoint` is 64-bit, then the corresponding VkVertexInputAttributeDescription2EXT::`format` **must** have a 64-bit component
- VUID-vkCmdDrawIndexed-None-09203
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::`format` has a 64-bit component, then all `Input` variables at the corresponding `Location` in the `Vertex` `Execution` `Model` `OpEntryPoint` **must** not use components that are not present in the format
- VUID-vkCmdDrawIndexed-None-04875
  If there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with both a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage and the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state enabled, and the current value of `primitiveTopology` is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, then vkCmdSetPatchControlPointsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-04879
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-09637

If the primitiveTopologyListRestart feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled, then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE

- VUID-vkCmdDrawIndexed-None-10909
  If the primitiveTopologyPatchListRestart feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndexed-stage-06481
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::stage member of any element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndexed-None-08885
  There **must** be no shader object bound to either of the VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT stages
- VUID-vkCmdDrawIndexed-None-07619
  If a shader object is bound to the VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage and the VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT dynamic state enabled, then vkCmdSetTessellationDomainOriginEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-OpExecutionMode-12239
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the type of subdivision, they **must** be the same
- VUID-vkCmdDrawIndexed-OpExecutionMode-12240
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the orientation of triangles, they **must** be the same
- VUID-vkCmdDrawIndexed-OpExecutionMode-12241
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the segment spacing, they **must** be the same
- VUID-vkCmdDrawIndexed-OpExecutionMode-12242
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain

an OpExecutionMode instruction specifying the output patch size, they **must** be the same

- VUID-vkCmdDrawIndexed-None-07312
  If the maintenance6 feature is not enabled, a valid index buffer **must** be bound
- VUID-vkCmdDrawIndexed-primitiveRestartIndex-12401
  If the primitive restart index for this draw operation was set with vkCmdSetPrimitiveRestartIndexEXT, then primitiveRestartIndex **must** not be greater than the maximum representable value for the bound VkIndexType
- VUID-vkCmdDrawIndexed-pNext-09461
  If the bound graphics pipeline state was created with VkPipelineVertexInputDivisorStateCreateInfo in the pNext chain of VkGraphicsPipelineCreateInfo::pVertexInputState, any member of VkPipelineVertexInputDivisorStateCreateInfo::pVertexBindingDivisors has a value other than 1 in divisor, and VkPhysicalDeviceVertexAttributeDivisorProperties ::supportsNonZeroFirstInstance is VK_FALSE, then firstInstance **must** be 0
- VUID-vkCmdDrawIndexed-None-09462
  If shader objects are used for drawing or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, any member of the pVertexBindingDescriptions parameter to the vkCmdSetVertexInputEXT call that sets this dynamic state has a value other than 1 in divisor, and VkPhysicalDeviceVertexAttributeDivisorProperties::supportsNonZeroFirstInstance is VK_FALSE, then firstInstance **must** be 0
- VUID-vkCmdDrawIndexed-robustBufferAccess2-08798
  If the robustBufferAccess2 feature is not enabled, (indexSize × (firstIndex + indexCount)) **must** be less than or equal to the size of the bound index buffer range, with indexSize being based on the type specified by indexType, and the other parameters sourced from this command

Valid Usage (Implicit)

- VUID-vkCmdDrawIndexed-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDrawIndexed-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdDrawIndexed-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDrawIndexed-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndexed-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawIndexed-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawIndexed is affected by conditional rendering

To record an ordered sequence of draws which have no state changes between them, call:

```
// Provided by VK_EXT_multi_draw
void vkCmdDrawMultiEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    drawCount,
    const VkMultiDrawInfoEXT*                   pVertexInfo,
    uint32_t                                    instanceCount,
    uint32_t                                    firstInstance,
    uint32_t                                    stride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `drawCount` is the number of draws to execute, and **can** be zero.
- `pVertexInfo` is a pointer to an array of VkMultiDrawInfoEXT with vertex information to be drawn.
- `instanceCount` is the number of instances per draw.
- `firstInstance` is the instance ID of the first instance in each draw.
- `stride` is the byte stride between consecutive elements of `pVertexInfo`.

The number of draws recorded is `drawCount`, with each draw reading, sequentially, a `firstVertex` and a `vertexCount` from `pVertexInfo`. For each recorded draw, primitives are assembled as for vkCmdDraw, and drawn `instanceCount` times with `instanceIndex` starting with `firstInstance` and sequentially for each instance.

Valid Usage

- VUID-vkCmdDrawMultiEXT-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawMultiEXT-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawMultiEXT-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawMultiEXT-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMultiEXT-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMultiEXT-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMultiEXT-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMultiEXT-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawMultiEXT-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawMultiEXT-magFilter-09598

If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT

- VUID-vkCmdDrawMultiEXT-mipmapMode-04770
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawMultiEXT-mipmapMode-09599
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawMultiEXT-unnormalizedCoordinates-09635
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1
- VUID-vkCmdDrawMultiEXT-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawMultiEXT-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdDrawMultiEXT-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias or any offset values
- VUID-vkCmdDrawMultiEXT-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawMultiEXT-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawMultiEXT-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic

operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT

- VUID-vkCmdDrawMultiEXT-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawMultiEXT-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawMultiEXT-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawMultiEXT-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawMultiEXT-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMultiEXT-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMultiEXT-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMultiEXT-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMultiEXT-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader

- VUID-vkCmdDrawMultiEXT-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawMultiEXT-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDrawMultiEXT-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawMultiEXT-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMultiEXT-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMultiEXT-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMultiEXT-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMultiEXT-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawMultiEXT-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the

numeric type of the view's format and the Sampled Type operand of the OpTypeImage **must** match

- VUID-vkCmdDrawMultiEXT-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawMultiEXT-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components
- VUID-vkCmdDrawMultiEXT-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawMultiEXT-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawMultiEXT-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawMultiEXT-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawMultiEXT-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawMultiEXT-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawMultiEXT-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawMultiEXT-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawMultiEXT-None-09600

If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdDrawMultiEXT-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawMultiEXT-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawMultiEXT-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with flags that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawMultiEXT-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with flags that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDrawMultiEXT-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawMultiEXT-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a OpTypeStruct decorated with Block or BufferBlock using that mapping, the calculated offset for the resource heap **must** be a multiple of bufferDescriptorAlignment

- VUID-vkCmdDrawMultiEXT-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdDrawMultiEXT-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdDrawMultiEXT-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDrawMultiEXT-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawMultiEXT-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawMultiEXT-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8

- VUID-vkCmdDrawMultiEXT-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawMultiEXT-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawMultiEXT-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawMultiEXT-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawMultiEXT-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMultiEXT-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMultiEXT-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer

as specified by VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdDrawMultiEXT-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDrawMultiEXT-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDrawMultiEXT-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDrawMultiEXT-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMultiEXT-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMultiEXT-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to

must be aligned to `minUniformBufferOffsetAlignment`

- VUID-vkCmdDrawMultiEXT-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMultiEXT-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawMultiEXT-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDrawMultiEXT-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDrawMultiEXT-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawMultiEXT-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by

this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawMultiEXT-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawMultiEXT-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawMultiEXT-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawMultiEXT-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' `pInputAttachments`[`InputAttachmentIndex`] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDrawMultiEXT-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by VkRenderingInputAttachmentIndexInfo, or no `InputAttachmentIndex` if VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` or VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are `NULL`, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawMultiEXT-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` and VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are non-`NULL`
- VUID-vkCmdDrawMultiEXT-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawMultiEXT-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawMultiEXT-None-10795

If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawMultiEXT-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMultiEXT-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMultiEXT-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawMultiEXT-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawMultiEXT-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawMultiEXT-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawMultiEXT-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDrawMultiEXT-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawMultiEXT-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior

to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07838

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::`maxMultiviewInstanceIndex`
- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawMultiEXT-None-07634
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-06666
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `sampleLocationsEnable` is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the

current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07846
  If the depthBounds feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state

enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** match the `scissorCount` parameter of `vkCmdSetScissorWithCount`
- VUID-vkCmdDrawMultiEXT-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be 1
- VUID-vkCmdDrawMultiEXT-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, and any shader object bound to a graphics stage writes to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be 1

- VUID-vkCmdDrawMultiEXT-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of `blendEnable` **must** be VK_FALSE
- VUID-vkCmdDrawMultiEXT-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of `rasterizerDiscardEnable` is VK_FALSE, and none of the following is enabled:

  ◦ the `multisampledRenderToSingleSampled` feature

  then the current value of `rasterizationSamples` **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDrawMultiEXT-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDrawMultiEXT-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::`viewMask` equal to VkRenderingInfo::`viewMask`
- VUID-vkCmdDrawMultiEXT-colorAttachmentCount-06179
  If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::`colorAttachmentCount` equal to VkRenderingInfo::`colorAttachmentCount`
- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08910
  If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `imageView` not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::`pColorAttachmentFormats` used to create the bound graphics pipeline
- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08912
  If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `imageView` equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::`pColorAttachmentFormats` used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08911

If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMultiEXT-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawMultiEXT-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT ::discardRectangleCount
- VUID-vkCmdDrawMultiEXT-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT ::maxDiscardRectangles
- VUID-vkCmdDrawMultiEXT-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any

graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView

- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo

::pStencilAttachment->imageView

- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiEXT-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawMultiEXT-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawMultiEXT-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawMultiEXT-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawMultiEXT-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawMultiEXT-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance,

and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView

- VUID-vkCmdDrawMultiEXT-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDrawMultiEXT-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawMultiEXT-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiEXT-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiEXT-pDepthAttachment-08964
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiEXT-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must**

not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMultiEXT-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiEXT-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiEXT-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawMultiEXT-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawMultiEXT-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDrawMultiEXT-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state

enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and alphaToCoverageEnable was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-vkCmdDrawMultiEXT-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set alphaToCoverageEnable to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-vkCmdDrawMultiEXT-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07625
  If the alphaToOne feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07626
  If the logicOp feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound,

then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07630
  If the `geometryStreams` feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07631
  If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07632
  If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `conservativeRasterizationMode` is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-conservativePointAndLineRasterization-07499
  If the `VK_EXT_conservative_rasterization` extension is enabled, `conservativePointAndLineRasterization` is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of `conservativeRasterizationMode` **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-vkCmdDrawMultiEXT-None-07633
  If the `depthClipEnable` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-08669
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current lineRasterizationMode is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current alphaToCoverageEnable, alphaToOneEnable and sampleShadingEnable states **must** all be VK_FALSE
- VUID-vkCmdDrawMultiEXT-None-07639
  If the depthClipControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not

subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-09650
  If the depthClampControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of depthClampEnable is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawMultiEXT-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::rasterizationSamples parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawMultiEXT-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the rasterizationSamples parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMultiEXT-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the rasterizationSamples member of that structure
- VUID-vkCmdDrawMultiEXT-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawMultiEXT-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the

VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments

- VUID-vkCmdDrawMultiEXT-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE, then the number of active color attachments **must** not exceed `advancedBlendMaxColorAttachments`

- VUID-vkCmdDrawMultiEXT-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE

- VUID-vkCmdDrawMultiEXT-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE

- VUID-vkCmdDrawMultiEXT-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE

- VUID-vkCmdDrawMultiEXT-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the `rasterizationStream` to zero

- VUID-vkCmdDrawMultiEXT-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawMultiEXT-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT ::`maxSampleLocationGridSize.width` as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a `samples` parameter equaling `rasterizationSamples`
- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT ::`maxSampleLocationGridSize.height` as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a `samples` parameter equaling

rasterizationSamples

- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMultiEXT-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled
- VUID-vkCmdDrawMultiEXT-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode

state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled

- VUID-vkCmdDrawMultiEXT-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawMultiEXT-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawMultiEXT-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawMultiEXT-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a nextStage including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDrawMultiEXT-None-08684
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDrawMultiEXT-None-08685
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawMultiEXT-None-08686
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages

with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT

- VUID-vkCmdDrawMultiEXT-None-08687
  If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawMultiEXT-None-08688
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawMultiEXT-None-08689
  If there is no bound graphics pipeline, and the `taskShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawMultiEXT-None-08690
  If there is no bound graphics pipeline, and the `meshShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawMultiEXT-None-08693
  If there is no bound graphics pipeline, and at least one of the `taskShader` and `meshShader` features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid `VkShaderEXT` bound, and the other **must** have no `VkShaderEXT` bound
- VUID-vkCmdDrawMultiEXT-None-08696
  If there is no bound graphics pipeline, and a valid `VkShaderEXT` is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no `VkShaderEXT` bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawMultiEXT-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDrawMultiEXT-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any `VkShaderEXT` bound
- VUID-vkCmdDrawMultiEXT-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawMultiEXT-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts

- VUID-vkCmdDrawMultiEXT-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpDepthAttachmentReadEXT, the depthWriteEnable parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawMultiEXT-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpStencilAttachmentReadEXT, the writeMask parameter in the last call to vkCmdSetStencilWriteMask **must** be 0
- VUID-vkCmdDrawMultiEXT-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the pColorWriteMasks parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawMultiEXT-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of Location for any output attachment statically used in the Fragment Execution Model executed by this command **must** be less than maxFragmentDualSrcAttachments
- VUID-vkCmdDrawMultiEXT-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::pColorAttachmentLocations in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance
- VUID-vkCmdDrawMultiEXT-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::pColorAttachmentInputIndices in the bound pipeline, if either pColorAttachmentInputIndices[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawMultiEXT-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two

values **must** match

- VUID-vkCmdDrawMultiEXT-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawMultiEXT-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawMultiEXT-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDrawMultiEXT-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawMultiEXT-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::flags which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawMultiEXT-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawMultiEXT-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawMultiEXT-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT ::customResolve as VK_TRUE
- VUID-vkCmdDrawMultiEXT-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with

VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE

- VUID-vkCmdDrawMultiEXT-None-11861
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawMultiEXT-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawMultiEXT-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiEXT-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiEXT-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->resolveImageView

- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiEXT-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiEXT-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiEXT-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawMultiEXT-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawMultiEXT-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and

VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView

- VUID-vkCmdDrawMultiEXT-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawMultiEXT-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawMultiEXT-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawMultiEXT-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawMultiEXT-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawMultiEXT-flags-13363
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawMultiEXT-commandBuffer-02712
  If commandBuffer is a protected command buffer and protectedNoFault is not supported, any resource written to by the VkPipeline object bound to the pipeline bind point used by this command **must** not be an unprotected resource

- VUID-vkCmdDrawMultiEXT-commandBuffer-02713
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, pipeline stages other than the framebuffer-space and compute stages in the `VkPipeline` object bound to the pipeline bind point used by this command **must** not write to any resource
- VUID-vkCmdDrawMultiEXT-commandBuffer-04617
  If any of the shader stages of the `VkPipeline` bound to the pipeline bind point used by this command uses the `RayQueryKHR` capability, then `commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdDrawMultiEXT-None-04007
  All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or VK_NULL_HANDLE buffers bound
- VUID-vkCmdDrawMultiEXT-None-04008
  If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be VK_NULL_HANDLE
- VUID-vkCmdDrawMultiEXT-None-02721
  If the `robustBufferAccess` feature is not enabled, and that pipeline was created without enabling VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS for `vertexInputs`, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in Vertex Input Description
- VUID-vkCmdDrawMultiEXT-format-10389
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is a packed format, and the `legacyVertexAttributes` feature is not enabled, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the size of the `format`
- VUID-vkCmdDrawMultiEXT-format-10390
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is not a packed format, and either the `legacyVertexAttributes` feature is not enabled or `format` has 64-bit components, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the component size of the `format`
- VUID-vkCmdDrawMultiEXT-None-07842
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled then vkCmdSetPrimitiveTopology **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-dynamicPrimitiveTopologyUnrestricted-07500
  If the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled and the dynamicPrimitiveTopologyUnrestricted is VK_FALSE, then the primitiveTopology parameter of vkCmdSetPrimitiveTopology **must** be of the same topology class as the pipeline VkPipelineInputAssemblyStateCreateInfo::topology state

- VUID-vkCmdDrawMultiEXT-primitiveTopology-10286
  If a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage is bound, then the current value of primitiveTopology **must** be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command
- VUID-vkCmdDrawMultiEXT-primitiveTopology-10747
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command, then a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage **must** be bound
- VUID-vkCmdDrawMultiEXT-primitiveTopology-10748
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_POINT_LIST prior to this drawing command, the maintenance5 feature is not enabled, both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT and VK_SHADER_STAGE_GEOMETRY_BIT stage are not bound, then the Vertex Execution Model **must** have a PointSize decorated variable that is statically written to
- VUID-vkCmdDrawMultiEXT-pStrides-04913
  If the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, but without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, then vkCmdBindVertexBuffers2 with a non-NULL pStrides parameter or vkCmdBindVertexBuffers3KHR with setStride set to VK_TRUE **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawMultiEXT-None-13118
  If the bound graphics pipeline was created without the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, and vkCmdBindVertexBuffers3KHR was called and not subsequently invalidated in the current command buffer prior to this draw command, the value of setStride in each of its pBindInfos elements must have been VK_FALSE
- VUID-vkCmdDrawMultiEXT-None-04914
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawMultiEXT-Input-07939
  If the maintenance9 feature is not enabled, and there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then all variables with the Input storage class decorated with Location in the Vertex Execution Model OpEntryPoint **must** contain a location in VkVertexInputAttributeDescription2EXT ::location

- VUID-vkCmdDrawMultiEXT-Input-08734
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and either the `legacyVertexAttributes` feature is not enabled or the SPIR-V Type associated with a given `Input` variable of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the numeric type associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be the same as VkVertexInputAttributeDescription2EXT::`format`
- VUID-vkCmdDrawMultiEXT-format-08936
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::`format` has a 64-bit component, then the scalar width associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be 64-bit
- VUID-vkCmdDrawMultiEXT-format-08937
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and the scalar width associated with a `Location` decorated `Input` variable in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the corresponding VkVertexInputAttributeDescription2EXT::`format` **must** have a 64-bit component
- VUID-vkCmdDrawMultiEXT-None-09203
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::`format` has a 64-bit component, then all `Input` variables at the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** not use components that are not present in the format
- VUID-vkCmdDrawMultiEXT-None-04875
  If there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with both a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage and the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state enabled, and the current value of `primitiveTopology` is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, then vkCmdSetPatchControlPointsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-04879
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-09637

If the primitiveTopologyListRestart feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled, then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE

- VUID-vkCmdDrawMultiEXT-None-10909
  If the primitiveTopologyPatchListRestart feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawMultiEXT-stage-06481
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::stage member of any element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawMultiEXT-None-08885
  There **must** be no shader object bound to either of the VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT stages
- VUID-vkCmdDrawMultiEXT-None-07619
  If a shader object is bound to the VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage and the VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT dynamic state enabled, then vkCmdSetTessellationDomainOriginEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-OpExecutionMode-12239
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the type of subdivision, they **must** be the same
- VUID-vkCmdDrawMultiEXT-OpExecutionMode-12240
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the orientation of triangles, they **must** be the same
- VUID-vkCmdDrawMultiEXT-OpExecutionMode-12241
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the segment spacing, they **must** be the same
- VUID-vkCmdDrawMultiEXT-OpExecutionMode-12242
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain

an OpExecutionMode instruction specifying the output patch size, they **must** be the same

- VUID-vkCmdDrawMultiEXT-pNext-09461
  If the bound graphics pipeline state was created with VkPipelineVertexInputDivisorStateCreateInfo in the pNext chain of VkGraphicsPipelineCreateInfo::pVertexInputState, any member of VkPipelineVertexInputDivisorStateCreateInfo::pVertexBindingDivisors has a value other than 1 in divisor, and VkPhysicalDeviceVertexAttributeDivisorProperties ::supportsNonZeroFirstInstance is VK_FALSE, then firstInstance **must** be 0
- VUID-vkCmdDrawMultiEXT-None-09462
  If shader objects are used for drawing or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, any member of the pVertexBindingDescriptions parameter to the vkCmdSetVertexInputEXT call that sets this dynamic state has a value other than 1 in divisor, and VkPhysicalDeviceVertexAttributeDivisorProperties::supportsNonZeroFirstInstance is VK_FALSE, then firstInstance **must** be 0
- VUID-vkCmdDrawMultiEXT-None-04933
  The multiDraw feature **must** be enabled
- VUID-vkCmdDrawMultiEXT-drawCount-04934
  drawCount **must** be less than VkPhysicalDeviceMultiDrawPropertiesEXT::maxMultiDrawCount
- VUID-vkCmdDrawMultiEXT-drawCount-04935
  If drawCount is greater than zero, pVertexInfo **must** be a valid pointer to memory containing one or more valid instances of VkMultiDrawInfoEXT structures
- VUID-vkCmdDrawMultiEXT-drawCount-09628
  If drawCount is greater than 1, stride **must** be a multiple of 4 and **must** be greater than or equal to sizeof(VkMultiDrawInfoEXT)

Valid Usage (Implicit)

- VUID-vkCmdDrawMultiEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDrawMultiEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdDrawMultiEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDrawMultiEXT-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawMultiEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawMultiEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawMultiEXT is affected by conditional rendering

To record an ordered sequence of indexed draws which have no state changes between them, call:

```
// Provided by VK_EXT_multi_draw
void vkCmdDrawMultiIndexedEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    drawCount,
    const VkMultiDrawIndexedInfoEXT*            pIndexInfo,
    uint32_t                                    instanceCount,
    uint32_t                                    firstInstance,
    uint32_t                                    stride,
    const int32_t*                              pVertexOffset);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `drawCount` is the number of draws to execute, and **can** be zero.
- `pIndexInfo` is a pointer to an array of VkMultiDrawIndexedInfoEXT with index information to be drawn.
- `instanceCount` is the number of instances per draw.
- `firstInstance` is the instance ID of the first instance in each draw.
- `stride` is the byte stride between consecutive elements of `pIndexInfo`.
- `pVertexOffset` is `NULL` or a pointer to the value added to the vertex index before indexing into the vertex buffer. When specified, `VkMultiDrawIndexedInfoEXT::offset` is ignored.

The number of draws recorded is `drawCount`, with each draw reading, sequentially, a `firstIndex` and an `indexCount` from `pIndexInfo`. For each recorded draw, primitives are assembled as for

vkCmdDrawIndexed, and drawn instanceCount times with instanceIndex starting with firstInstance and sequentially for each instance. If pVertexOffset is NULL, a vertexOffset is also read from pIndexInfo, otherwise the value from dereferencing pVertexOffset is used.

Valid Usage

- VUID-vkCmdDrawMultiIndexedEXT-None-08606
  If the shaderObject feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawMultiIndexedEXT-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawMultiIndexedEXT-commandBuffer-02707
  If commandBuffer is an unprotected command buffer and protectedNoFault is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawMultiIndexedEXT-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMultiIndexedEXT-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMultiIndexedEXT-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMultiIndexedEXT-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMultiIndexedEXT-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a VkDescriptorSetLayout that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawMultiIndexedEXT-magFilter-04553
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and

compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT

- VUID-vkCmdDrawMultiIndexedEXT-magFilter-09598
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawMultiIndexedEXT-mipmapMode-04770
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawMultiIndexedEXT-mipmapMode-09599
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawMultiIndexedEXT-unnormalizedCoordinates-09635
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1
- VUID-vkCmdDrawMultiIndexedEXT-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawMultiIndexedEXT-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdDrawMultiIndexedEXT-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias or any offset values
- VUID-vkCmdDrawMultiIndexedEXT-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawMultiIndexedEXT-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then

the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT

- VUID-vkCmdDrawMultiIndexedEXT-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDrawMultiIndexedEXT-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawMultiIndexedEXT-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawMultiIndexedEXT-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawMultiIndexedEXT-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawMultiIndexedEXT-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMultiIndexedEXT-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMultiIndexedEXT-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMultiIndexedEXT-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMultiIndexedEXT-None-08601

If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader

- VUID-vkCmdDrawMultiIndexedEXT-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawMultiIndexedEXT-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDrawMultiIndexedEXT-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawMultiIndexedEXT-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMultiIndexedEXT-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMultiIndexedEXT-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMultiIndexedEXT-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMultiIndexedEXT-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType`

must match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types

- VUID-vkCmdDrawMultiIndexedEXT-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match
- VUID-vkCmdDrawMultiIndexedEXT-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawMultiIndexedEXT-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have four components
- VUID-vkCmdDrawMultiIndexedEXT-OpImageWrite-04469
  If a VkBufferView is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawMultiIndexedEXT-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdDrawMultiIndexedEXT-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdDrawMultiIndexedEXT-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdDrawMultiIndexedEXT-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdDrawMultiIndexedEXT-sparseImageInt64Atomics-04474
  If the `sparseImageInt64Atomics` feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdDrawMultiIndexedEXT-sparseImageInt64Atomics-04475
  If the `sparseImageInt64Atomics` feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command

- VUID-vkCmdDrawMultiIndexedEXT-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawMultiIndexedEXT-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawMultiIndexedEXT-micromap-11636
  If an acceleration structure is accessed as a result of this command through an `OpRayQueryProceedKHR` operation , the `VK_EXT_opacity_micromap` extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its `pNext` chain, the shader **must** have specified a `OpacityMicromapIdKHR` execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawMultiIndexedEXT-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawMultiIndexedEXT-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with `flags` that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawMultiIndexedEXT-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with `flags` that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDrawMultiIndexedEXT-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawMultiIndexedEXT-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or

VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a OpTypeStruct decorated with Block or BufferBlock using that mapping, the calculated offset for the resource heap **must** be a multiple of bufferDescriptorAlignment

- VUID-vkCmdDrawMultiIndexedEXT-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeImage or OpTypeSampledImage using that mapping, the calculated offset for the resource heap **must** be a multiple of imageDescriptorAlignment
- VUID-vkCmdDrawMultiIndexedEXT-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeSampler or OpTypeSampledImage using that mapping, the calculated offset for the sampler heap **must** be a multiple of samplerDescriptorAlignment
- VUID-vkCmdDrawMultiIndexedEXT-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDrawMultiIndexedEXT-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawMultiIndexedEXT-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawMultiIndexedEXT-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping

using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8

- VUID-vkCmdDrawMultiIndexedEXT-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawMultiIndexedEXT-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawMultiIndexedEXT-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawMultiIndexedEXT-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawMultiIndexedEXT-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMultiIndexedEXT-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMultiIndexedEXT-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not

enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdDrawMultiIndexedEXT-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDrawMultiIndexedEXT-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDrawMultiIndexedEXT-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDrawMultiIndexedEXT-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMultiIndexedEXT-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMultiIndexedEXT-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT,

VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`

- VUID-vkCmdDrawMultiIndexedEXT-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMultiIndexedEXT-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawMultiIndexedEXT-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDrawMultiIndexedEXT-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDrawMultiIndexedEXT-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawMultiIndexedEXT-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawMultiIndexedEXT-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawMultiIndexedEXT-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawMultiIndexedEXT-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawMultiIndexedEXT-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDrawMultiIndexedEXT-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by VkRenderingInputAttachmentIndexInfo, or no `InputAttachmentIndex` if VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` or VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are `NULL`, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawMultiIndexedEXT-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` and VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are non-`NULL`
- VUID-vkCmdDrawMultiIndexedEXT-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawMultiIndexedEXT-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command

- VUID-vkCmdDrawMultiIndexedEXT-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMultiIndexedEXT-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMultiIndexedEXT-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMultiIndexedEXT-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawMultiIndexedEXT-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawMultiIndexedEXT-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawMultiIndexedEXT-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawMultiIndexedEXT-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDrawMultiIndexedEXT-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawMultiIndexedEXT-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must**

have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::`maxMultiviewInstanceIndex`
- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawMultiIndexedEXT-None-07634
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-06666
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `sampleLocationsEnable` is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07846
  If the `depthBounds` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** match the `scissorCount` parameter of `vkCmdSetScissorWithCount`
- VUID-vkCmdDrawMultiIndexedEXT-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be `1`
- VUID-vkCmdDrawMultiIndexedEXT-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, and any shader object bound to a graphics stage writes to the `PrimitiveShadingRateKHR` built-in,

then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1

- VUID-vkCmdDrawMultiIndexedEXT-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of blendEnable **must** be VK_FALSE
- VUID-vkCmdDrawMultiIndexedEXT-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and none of the following is enabled:
  - the multisampledRenderToSingleSampled feature

  then the current value of rasterizationSamples **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDrawMultiIndexedEXT-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDrawMultiIndexedEXT-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::viewMask equal to VkRenderingInfo::viewMask
- VUID-vkCmdDrawMultiIndexedEXT-colorAttachmentCount-06179
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08910
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08912
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the

corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawMultiIndexedEXT-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount
- VUID-vkCmdDrawMultiIndexedEXT-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT

::maxDiscardRectangles

- VUID-vkCmdDrawMultiIndexedEXT-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo

::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView

- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawMultiIndexedEXT-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawMultiIndexedEXT-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawMultiIndexedEXT-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView

- VUID-vkCmdDrawMultiIndexedEXT-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawMultiIndexedEXT-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDrawMultiIndexedEXT-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawMultiIndexedEXT-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-pDepthAttachment-08964
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo ::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a

graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMultiIndexedEXT-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawMultiIndexedEXT-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawMultiIndexedEXT-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDrawMultiIndexedEXT-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then

vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and `alphaToCoverageEnable` was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawMultiIndexedEXT-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set `alphaToCoverageEnable` to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawMultiIndexedEXT-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07625
  If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07626
  If the `logicOp` feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07630
  If the geometryStreams feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07631
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07632
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of conservativeRasterizationMode is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-conservativePointAndLineRasterization-07499
  If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-vkCmdDrawMultiIndexedEXT-None-07633

If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-08669
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current lineRasterizationMode is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current alphaToCoverageEnable, alphaToOneEnable and sampleShadingEnable states **must** all be VK_FALSE

- VUID-vkCmdDrawMultiIndexedEXT-None-07639
  If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-09650
  If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of `depthClampEnable` is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawMultiIndexedEXT-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::`rasterizationSamples` parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawMultiIndexedEXT-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the `rasterizationSamples` parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMultiIndexedEXT-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with `multisampledRenderToSingleSampledEnable` equal to VK_TRUE, then the `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the `rasterizationSamples` member of that structure
- VUID-vkCmdDrawMultiIndexedEXT-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to

vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments

- VUID-vkCmdDrawMultiIndexedEXT-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawMultiIndexedEXT-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE, then the number of active color attachments **must** not exceed `advancedBlendMaxColorAttachments`
- VUID-vkCmdDrawMultiIndexedEXT-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawMultiIndexedEXT-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawMultiIndexedEXT-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawMultiIndexedEXT-primitivesGeneratedQueryWithNonZeroStreams-07481

If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the rasterizationStream to zero

- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.width in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT ::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples
- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if

sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.height in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMultiIndexedEXT-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled

- VUID-vkCmdDrawMultiIndexedEXT-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled
- VUID-vkCmdDrawMultiIndexedEXT-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawMultiIndexedEXT-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawMultiIndexedEXT-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawMultiIndexedEXT-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a nextStage including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDrawMultiIndexedEXT-None-08684
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDrawMultiIndexedEXT-None-08685
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled,

vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT

- VUID-vkCmdDrawMultiIndexedEXT-None-08686
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-vkCmdDrawMultiIndexedEXT-None-08687
  If there is no bound graphics pipeline, and the geometryShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawMultiIndexedEXT-None-08688
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawMultiIndexedEXT-None-08689
  If there is no bound graphics pipeline, and the taskShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawMultiIndexedEXT-None-08690
  If there is no bound graphics pipeline, and the meshShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawMultiIndexedEXT-None-08693
  If there is no bound graphics pipeline, and at least one of the taskShader and meshShader features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid VkShaderEXT bound, and the other **must** have no VkShaderEXT bound
- VUID-vkCmdDrawMultiIndexedEXT-None-08696
  If there is no bound graphics pipeline, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no VkShaderEXT bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawMultiIndexedEXT-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDrawMultiIndexedEXT-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any VkShaderEXT bound
- VUID-vkCmdDrawMultiIndexedEXT-None-08878
  All bound graphics shader objects **must** have been created with identical or identically

defined push constant ranges

- VUID-vkCmdDrawMultiIndexedEXT-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts
- VUID-vkCmdDrawMultiIndexedEXT-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawMultiIndexedEXT-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to vkCmdSetStencilWriteMask **must** be `0`
- VUID-vkCmdDrawMultiIndexedEXT-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the `pColorWriteMasks` parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawMultiIndexedEXT-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of `Location` for any output attachment statically used in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`
- VUID-vkCmdDrawMultiIndexedEXT-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::`pColorAttachmentLocations` in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance
- VUID-vkCmdDrawMultiIndexedEXT-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::`pColorAttachmentInputIndices` in the bound pipeline, if either `pColorAttachmentInputIndices`[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an `InputAttachmentIndex` decoration, the two values **must** match
- VUID-vkCmdDrawMultiIndexedEXT-None-10927

If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match

- VUID-vkCmdDrawMultiIndexedEXT-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawMultiIndexedEXT-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawMultiIndexedEXT-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDrawMultiIndexedEXT-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawMultiIndexedEXT-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::flags which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawMultiIndexedEXT-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawMultiIndexedEXT-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawMultiIndexedEXT-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT ::customResolve as VK_TRUE

- VUID-vkCmdDrawMultiIndexedEXT-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::`customResolve` as VK_FALSE
- VUID-vkCmdDrawMultiIndexedEXT-None-11861
  If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::`colorAttachmentCount` equal to VkRenderingInfo::`colorAttachmentCount`
- VUID-vkCmdDrawMultiIndexedEXT-None-11862
  If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `resolveImageView` not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::`pColorAttachmentFormats` used to create the bound graphics pipeline
- VUID-vkCmdDrawMultiIndexedEXT-None-11863
  If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `resolveImageView` equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::`pColorAttachmentFormats` used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-11864
  If the `dynamicRenderingUnusedAttachments` feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `resolveImageView` not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::`pColorAttachmentFormats` used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::`pColorAttachmentFormats`, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the `dynamicRenderingUnusedAttachments` feature is not enabled, and VkRenderingInfo::`pDepthAttachment->resolveImageView` was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::`depthAttachmentFormat` used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the `dynamicRenderingUnusedAttachments` feature is not enabled, and

VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView

- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMultiIndexedEXT-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawMultiIndexedEXT-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the

sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView

- VUID-vkCmdDrawMultiIndexedEXT-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawMultiIndexedEXT-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawMultiIndexedEXT-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawMultiIndexedEXT-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawMultiIndexedEXT-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawMultiIndexedEXT-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawMultiIndexedEXT-flags-13363
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT

- VUID-vkCmdDrawMultiIndexedEXT-commandBuffer-02712
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, any resource written to by the `VkPipeline` object bound to the pipeline bind point used by this command **must** not be an unprotected resource
- VUID-vkCmdDrawMultiIndexedEXT-commandBuffer-02713
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, pipeline stages other than the framebuffer-space and compute stages in the `VkPipeline` object bound to the pipeline bind point used by this command **must** not write to any resource
- VUID-vkCmdDrawMultiIndexedEXT-commandBuffer-04617
  If any of the shader stages of the `VkPipeline` bound to the pipeline bind point used by this command uses the `RayQueryKHR` capability, then `commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdDrawMultiIndexedEXT-None-04007
  All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or VK_NULL_HANDLE buffers bound
- VUID-vkCmdDrawMultiIndexedEXT-None-04008
  If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be VK_NULL_HANDLE
- VUID-vkCmdDrawMultiIndexedEXT-None-02721
  If the `robustBufferAccess` feature is not enabled, and that pipeline was created without enabling VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS for `vertexInputs`, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in Vertex Input Description
- VUID-vkCmdDrawMultiIndexedEXT-format-10389
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is a packed format, and the `legacyVertexAttributes` feature is not enabled, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the size of the `format`
- VUID-vkCmdDrawMultiIndexedEXT-format-10390
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is not a packed format, and either the `legacyVertexAttributes` feature is not enabled or `format` has 64-bit components, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the component size of the `format`
- VUID-vkCmdDrawMultiIndexedEXT-None-07842
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled then

vkCmdSetPrimitiveTopology **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-dynamicPrimitiveTopologyUnrestricted-07500
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled and the dynamicPrimitiveTopologyUnrestricted is VK_FALSE, then the primitiveTopology parameter of vkCmdSetPrimitiveTopology **must** be of the same topology class as the pipeline VkPipelineInputAssemblyStateCreateInfo::topology state
- VUID-vkCmdDrawMultiIndexedEXT-primitiveTopology-10286
  If a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage is bound, then the current value of primitiveTopology **must** be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-primitiveTopology-10747
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command, then a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage **must** be bound
- VUID-vkCmdDrawMultiIndexedEXT-primitiveTopology-10748
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_POINT_LIST prior to this drawing command, the maintenance5 feature is not enabled, both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT and VK_SHADER_STAGE_GEOMETRY_BIT stage are not bound, then the Vertex Execution Model **must** have a PointSize decorated variable that is statically written to
- VUID-vkCmdDrawMultiIndexedEXT-pStrides-04913
  If the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, but without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, then vkCmdBindVertexBuffers2 with a non-NULL pStrides parameter or vkCmdBindVertexBuffers3KHR with setStride set to VK_TRUE **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawMultiIndexedEXT-None-13118
  If the bound graphics pipeline was created without the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, and vkCmdBindVertexBuffers3KHR was called and not subsequently invalidated in the current command buffer prior to this draw command, the value of setStride in each of its pBindInfos elements must have been VK_FALSE
- VUID-vkCmdDrawMultiIndexedEXT-None-04914
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawMultiIndexedEXT-Input-07939
  If the maintenance9 feature is not enabled, and there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created

with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then all variables with the Input storage class decorated with Location in the Vertex Execution Model OpEntryPoint **must** contain a location in VkVertexInputAttributeDescription2EXT::location

- VUID-vkCmdDrawMultiIndexedEXT-Input-08734
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and either the legacyVertexAttributes feature is not enabled or the SPIR-V Type associated with a given Input variable of the corresponding Location in the Vertex Execution Model OpEntryPoint is 64-bit, then the numeric type associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be the same as VkVertexInputAttributeDescription2EXT::format
- VUID-vkCmdDrawMultiIndexedEXT-format-08936
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then the scalar width associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be 64-bit
- VUID-vkCmdDrawMultiIndexedEXT-format-08937
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and the scalar width associated with a Location decorated Input variable in the Vertex Execution Model OpEntryPoint is 64-bit, then the corresponding VkVertexInputAttributeDescription2EXT::format **must** have a 64-bit component
- VUID-vkCmdDrawMultiIndexedEXT-None-09203
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then all Input variables at the corresponding Location in the Vertex Execution Model OpEntryPoint **must** not use components that are not present in the format
- VUID-vkCmdDrawMultiIndexedEXT-None-04875
  If there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with both a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage and the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state enabled, and the current value of primitiveTopology is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, then vkCmdSetPatchControlPointsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-04879
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-09637
  If the `primitiveTopologyListRestart` feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled, then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawMultiIndexedEXT-None-10909
  If the `primitiveTopologyPatchListRestart` feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawMultiIndexedEXT-stage-06481
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::`stage` member of any element of VkGraphicsPipelineCreateInfo::`pStages` set to VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawMultiIndexedEXT-None-08885
  There **must** be no shader object bound to either of the VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT stages
- VUID-vkCmdDrawMultiIndexedEXT-None-07619
  If a shader object is bound to the VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage and the VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT dynamic state enabled, then vkCmdSetTessellationDomainOriginEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-OpExecutionMode-12239
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the type of subdivision, they **must** be the same
- VUID-vkCmdDrawMultiIndexedEXT-OpExecutionMode-12240
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the orientation of triangles, they **must** be the same
- VUID-vkCmdDrawMultiIndexedEXT-OpExecutionMode-12241
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain

an OpExecutionMode instruction specifying the segment spacing, they **must** be the same

- VUID-vkCmdDrawMultiIndexedEXT-OpExecutionMode-12242
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the output patch size, they **must** be the same

- VUID-vkCmdDrawMultiIndexedEXT-None-07312
  If the maintenance6 feature is not enabled, a valid index buffer **must** be bound

- VUID-vkCmdDrawMultiIndexedEXT-primitiveRestartIndex-12401
  If the primitive restart index for this draw operation was set with vkCmdSetPrimitiveRestartIndexEXT, then primitiveRestartIndex **must** not be greater than the maximum representable value for the bound VkIndexType

- VUID-vkCmdDrawMultiIndexedEXT-pNext-09461
  If the bound graphics pipeline state was created with VkPipelineVertexInputDivisorStateCreateInfo in the pNext chain of VkGraphicsPipelineCreateInfo::pVertexInputState, any member of VkPipelineVertexInputDivisorStateCreateInfo::pVertexBindingDivisors has a value other than 1 in divisor, and VkPhysicalDeviceVertexAttributeDivisorProperties ::supportsNonZeroFirstInstance is VK_FALSE, then firstInstance **must** be 0

- VUID-vkCmdDrawMultiIndexedEXT-None-09462
  If shader objects are used for drawing or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, any member of the pVertexBindingDescriptions parameter to the vkCmdSetVertexInputEXT call that sets this dynamic state has a value other than 1 in divisor, and VkPhysicalDeviceVertexAttributeDivisorProperties::supportsNonZeroFirstInstance is VK_FALSE, then firstInstance **must** be 0

- VUID-vkCmdDrawMultiIndexedEXT-robustBufferAccess2-08798
  If the robustBufferAccess2 feature is not enabled, (indexSize × (firstIndex + indexCount)) **must** be less than or equal to the size of the bound index buffer range, with indexSize being based on the type specified by indexType, and the other parameters sourced from this command

- VUID-vkCmdDrawMultiIndexedEXT-None-04937
  The multiDraw feature **must** be enabled

- VUID-vkCmdDrawMultiIndexedEXT-drawCount-04939
  drawCount **must** be less than VkPhysicalDeviceMultiDrawPropertiesEXT::maxMultiDrawCount

- VUID-vkCmdDrawMultiIndexedEXT-drawCount-04940
  If drawCount is greater than zero, pIndexInfo **must** be a valid pointer to memory containing one or more valid instances of VkMultiDrawIndexedInfoEXT structures

- VUID-vkCmdDrawMultiIndexedEXT-drawCount-09629
  If drawCount is greater than 1, stride **must** be a multiple of 4 and **must** be greater than or equal to sizeof(VkMultiDrawIndexedInfoEXT)

Valid Usage (Implicit)

- VUID-vkCmdDrawMultiIndexedEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDrawMultiIndexedEXT-pVertexOffset-parameter
  If `pVertexOffset` is not `NULL`, `pVertexOffset` **must** be a valid pointer to a valid `int32_t` value
- VUID-vkCmdDrawMultiIndexedEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdDrawMultiIndexedEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDrawMultiIndexedEXT-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawMultiIndexedEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawMultiIndexedEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawMultiIndexedEXT is affected by conditional rendering

The `VkMultiDrawInfoEXT` structure is defined as:

```
// Provided by VK_EXT_multi_draw
typedef struct VkMultiDrawInfoEXT {
    uint32_t    firstVertex;
    uint32_t    vertexCount;
} VkMultiDrawInfoEXT;
```

- `firstVertex` is the first vertex to draw.
- `vertexCount` is the number of vertices to draw.

The members of `VkMultiDrawInfoEXT` have the same meaning as the `firstVertex` and `vertexCount` parameters in vkCmdDraw.

The `VkMultiDrawIndexedInfoEXT` structure is defined as:

```
// Provided by VK_EXT_multi_draw
typedef struct VkMultiDrawIndexedInfoEXT {
    uint32_t    firstIndex;
    uint32_t    indexCount;
    int32_t     vertexOffset;
} VkMultiDrawIndexedInfoEXT;
```

- `firstIndex` is the first index to draw.
- `indexCount` is the number of vertices to draw.
- `vertexOffset` is the value added to the vertex index before indexing into the vertex buffer for indexed multidraws.

The `firstIndex`, `indexCount`, and `vertexOffset` members of `VkMultiDrawIndexedInfoEXT` have the same meaning as the `firstIndex`, `indexCount`, and `vertexOffset` parameters, respectively, of vkCmdDrawIndexed.

`VkDrawIndirect2InfoKHR` is defined as:

```
// Provided by VK_KHR_device_address_commands
typedef struct VkDrawIndirect2InfoKHR {
    VkStructureType                   sType;
    const void*                       pNext;
    VkStridedDeviceAddressRangeKHR    addressRange;
    VkAddressCommandFlagsKHR          addressFlags;
    uint32_t                          drawCount;
} VkDrawIndirect2InfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `addressRange` is the VkStridedDeviceAddressRangeKHR containing draw parameters.
- `addressFlags` is a VkAddressCommandFlagsKHR value defining the flags for the address range.

- `drawCount` is the number of draws to execute, and **can** be zero.

Valid Usage

- VUID-VkDrawIndirect2InfoKHR-addressRange-13097
  If the range specified by `addressRange` is not bound completely to memory when accessed, `addressFlags` **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR
- VUID-VkDrawIndirect2InfoKHR-addressRange-13098
  If the buffer from which the range specified by `addressRange` was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `addressFlags` **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkDrawIndirect2InfoKHR-addressRange-13099
  If the buffer from which the range specified by `addressRange` was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `addressFlags` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkDrawIndirect2InfoKHR-addressFlags-13100
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDrawIndirect2InfoKHR-addressRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDrawIndirect2InfoKHR-addressRange-13123
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDrawIndirect2InfoKHR-addressFlags-13101
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDrawIndirect2InfoKHR-addressRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `addressFlags` **must** include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDrawIndirect2InfoKHR-addressRange-13125
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created without

VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, addressFlags **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR

- VUID-VkDrawIndirect2InfoKHR-addressRange-13107
  The buffer from which addressRange was queried **must** have been created with VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag
- VUID-VkDrawIndirect2InfoKHR-protectedNoFault-13108
  If protectedNoFault is not supported, the buffer from which addressRange was queried **must** not have been created with VK_BUFFER_CREATE_PROTECTED_BIT
- VUID-VkDrawIndirect2InfoKHR-addressRange-13109
  addressRange.address **must** be a multiple of 4
- VUID-VkDrawIndirect2InfoKHR-drawCount-02718
  If the multiDrawIndirect feature is not enabled, drawCount **must** be 0 or 1
- VUID-VkDrawIndirect2InfoKHR-drawCount-02719
  drawCount **must** be less than or equal to VkPhysicalDeviceLimits::maxDrawIndirectCount

Valid Usage (Implicit)

- VUID-VkDrawIndirect2InfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DRAW_INDIRECT_2_INFO_KHR
- VUID-VkDrawIndirect2InfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkDrawIndirect2InfoKHR-addressFlags-parameter
  If addressFlags is not 0, addressFlags **must** be a valid combination of VkAddressCommandFlagBitsKHR values

To record a non-indexed indirect drawing command, call:

```
// Provided by VK_KHR_device_address_commands
void vkCmdDrawIndirect2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkDrawIndirect2InfoKHR*               pInfo);
```

- commandBuffer is the command buffer into which the command is recorded.
- pInfo is a pointer to a VkDrawIndirect2InfoKHR structure defining parameters of this command.

vkCmdDrawIndirect2KHR behaves similarly to vkCmdDraw except that the parameters are read by the device from an address range during execution. pInfo->drawCount draws are executed by the command, with parameters taken from pInfo->addressRange starting at addressRange.address and increasing by addressRange.stride bytes for each successive draw. The parameters of each draw are encoded in an array of VkDrawIndirectCommand structures.

Valid Usage

- VUID-vkCmdDrawIndirect2KHR-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawIndirect2KHR-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawIndirect2KHR-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawIndirect2KHR-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirect2KHR-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirect2KHR-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirect2KHR-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirect2KHR-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawIndirect2KHR-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndirect2KHR-magFilter-09598

If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT

- VUID-vkCmdDrawIndirect2KHR-mipmapMode-04770
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndirect2KHR-mipmapMode-09599
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndirect2KHR-unnormalizedCoordinates-09635
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1
- VUID-vkCmdDrawIndirect2KHR-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawIndirect2KHR-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdDrawIndirect2KHR-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias or any offset values
- VUID-vkCmdDrawIndirect2KHR-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawIndirect2KHR-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawIndirect2KHR-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic

operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT

- VUID-vkCmdDrawIndirect2KHR-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawIndirect2KHR-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawIndirect2KHR-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndirect2KHR-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndirect2KHR-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirect2KHR-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirect2KHR-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirect2KHR-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirect2KHR-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader

- VUID-vkCmdDrawIndirect2KHR-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawIndirect2KHR-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDrawIndirect2KHR-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawIndirect2KHR-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirect2KHR-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirect2KHR-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirect2KHR-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirect2KHR-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawIndirect2KHR-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the

numeric type of the view's format and the Sampled Type operand of the OpTypeImage **must** match

- VUID-vkCmdDrawIndirect2KHR-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawIndirect2KHR-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components
- VUID-vkCmdDrawIndirect2KHR-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawIndirect2KHR-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawIndirect2KHR-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawIndirect2KHR-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawIndirect2KHR-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawIndirect2KHR-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawIndirect2KHR-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawIndirect2KHR-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawIndirect2KHR-None-09600

If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdDrawIndirect2KHR-micromap-11636
  If an acceleration structure is accessed as a result of this command through an `OpRayQueryProceedKHR` operation , the `VK_EXT_opacity_micromap` extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its `pNext` chain, the shader **must** have specified a `OpacityMicromapIdKHR` execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawIndirect2KHR-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawIndirect2KHR-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with `flags` that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawIndirect2KHR-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with `flags` that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDrawIndirect2KHR-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawIndirect2KHR-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`

- VUID-vkCmdDrawIndirect2KHR-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdDrawIndirect2KHR-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdDrawIndirect2KHR-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDrawIndirect2KHR-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndirect2KHR-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndirect2KHR-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8

- VUID-vkCmdDrawIndirect2KHR-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndirect2KHR-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndirect2KHR-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawIndirect2KHR-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawIndirect2KHR-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndirect2KHR-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndirect2KHR-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer

as specified by VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdDrawIndirect2KHR-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDrawIndirect2KHR-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDrawIndirect2KHR-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDrawIndirect2KHR-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirect2KHR-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirect2KHR-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to

must be aligned to `minUniformBufferOffsetAlignment`

- VUID-vkCmdDrawIndirect2KHR-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirect2KHR-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawIndirect2KHR-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDrawIndirect2KHR-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDrawIndirect2KHR-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawIndirect2KHR-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by

this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawIndirect2KHR-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndirect2KHR-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndirect2KHR-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawIndirect2KHR-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDrawIndirect2KHR-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by VkRenderingInputAttachmentIndexInfo, or no `InputAttachmentIndex` if VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` or VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are `NULL`, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawIndirect2KHR-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` and VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are non-`NULL`
- VUID-vkCmdDrawIndirect2KHR-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawIndirect2KHR-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirect2KHR-None-10795

If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirect2KHR-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirect2KHR-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirect2KHR-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirect2KHR-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirect2KHR-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirect2KHR-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawIndirect2KHR-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDrawIndirect2KHR-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawIndirect2KHR-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior

to this drawing command

- VUID-vkCmdDrawIndirect2KHR-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07838

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect2KHR-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::`maxMultiviewInstanceIndex`
- VUID-vkCmdDrawIndirect2KHR-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndirect2KHR-None-07634
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-06666
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `sampleLocationsEnable` is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the

current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect2KHR-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07846
  If the depthBounds feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state

enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect2KHR-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** match the `scissorCount` parameter of `vkCmdSetScissorWithCount`
- VUID-vkCmdDrawIndirect2KHR-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be `1`
- VUID-vkCmdDrawIndirect2KHR-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, and any shader object bound to a graphics stage writes to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command

buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1

- VUID-vkCmdDrawIndirect2KHR-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of blendEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirect2KHR-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and none of the following is enabled:
  - the multisampledRenderToSingleSampled feature

  then the current value of rasterizationSamples **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDrawIndirect2KHR-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDrawIndirect2KHR-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::viewMask equal to VkRenderingInfo::viewMask
- VUID-vkCmdDrawIndirect2KHR-colorAttachmentCount-06179
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndirect2KHR-dynamicRenderingUnusedAttachments-08910
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawIndirect2KHR-dynamicRenderingUnusedAttachments-08912
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used

to create the bound pipeline equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirect2KHR-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawIndirect2KHR-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT ::discardRectangleCount
- VUID-vkCmdDrawIndirect2KHR-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT ::maxDiscardRectangles

- VUID-vkCmdDrawIndirect2KHR-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndirect2KHR-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of

VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView

- VUID-vkCmdDrawIndirect2KHR-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawIndirect2KHR-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawIndirect2KHR-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawIndirect2KHR-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawIndirect2KHR-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndirect2KHR-multisampledRenderToSingleSampled-07287

If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView

- VUID-vkCmdDrawIndirect2KHR-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDrawIndirect2KHR-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawIndirect2KHR-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-pDepthAttachment-08964
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo

::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirect2KHR-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo ::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo ::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawIndirect2KHR-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawIndirect2KHR-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDrawIndirect2KHR-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07622

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect2KHR-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and alphaToCoverageEnable was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-vkCmdDrawIndirect2KHR-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set alphaToCoverageEnable to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-vkCmdDrawIndirect2KHR-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07625
  If the alphaToOne feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07626
  If the logicOp feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with

VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect2KHR-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07630
  If the geometryStreams feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07631
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07632
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of conservativeRasterizationMode is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-conservativePointAndLineRasterization-07499
  If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-vkCmdDrawIndirect2KHR-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then

vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect2KHR-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-08669
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current lineRasterizationMode is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current alphaToCoverageEnable, alphaToOneEnable and sampleShadingEnable states **must** all be VK_FALSE
- VUID-vkCmdDrawIndirect2KHR-None-07639
  If the depthClipControl feature is enabled, and a shader object is bound to any graphics

stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect2KHR-None-09650
  If the depthClampControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of depthClampEnable is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawIndirect2KHR-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::rasterizationSamples parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawIndirect2KHR-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the rasterizationSamples parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirect2KHR-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the rasterizationSamples member of that structure
- VUID-vkCmdDrawIndirect2KHR-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments

- VUID-vkCmdDrawIndirect2KHR-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndirect2KHR-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE, then the number of active color attachments **must** not exceed `advancedBlendMaxColorAttachments`
- VUID-vkCmdDrawIndirect2KHR-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawIndirect2KHR-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawIndirect2KHR-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawIndirect2KHR-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the

last call to vkCmdSetRasterizationStreamEXT **must** have set the `rasterizationStream` to zero

- VUID-vkCmdDrawIndirect2KHR-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawIndirect2KHR-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirect2KHR-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndirect2KHR-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::`maxSampleLocationGridSize.width` as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a `samples` parameter equaling `rasterizationSamples`
- VUID-vkCmdDrawIndirect2KHR-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT

::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdDrawIndirect2KHR-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDrawIndirect2KHR-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirect2KHR-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirect2KHR-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirect2KHR-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled
- VUID-vkCmdDrawIndirect2KHR-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or

VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled

- VUID-vkCmdDrawIndirect2KHR-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawIndirect2KHR-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawIndirect2KHR-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawIndirect2KHR-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a nextStage including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDrawIndirect2KHR-None-08684
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDrawIndirect2KHR-None-08685
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawIndirect2KHR-None-08686

If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT

- VUID-vkCmdDrawIndirect2KHR-None-08687
  If there is no bound graphics pipeline, and the geometryShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawIndirect2KHR-None-08688
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawIndirect2KHR-None-08689
  If there is no bound graphics pipeline, and the taskShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawIndirect2KHR-None-08690
  If there is no bound graphics pipeline, and the meshShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndirect2KHR-None-08693
  If there is no bound graphics pipeline, and at least one of the taskShader and meshShader features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid VkShaderEXT bound, and the other **must** have no VkShaderEXT bound
- VUID-vkCmdDrawIndirect2KHR-None-08696
  If there is no bound graphics pipeline, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no VkShaderEXT bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawIndirect2KHR-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDrawIndirect2KHR-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any VkShaderEXT bound
- VUID-vkCmdDrawIndirect2KHR-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawIndirect2KHR-None-08879
  All bound graphics shader objects **must** have either been created with the

VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts

- VUID-vkCmdDrawIndirect2KHR-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirect2KHR-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to vkCmdSetStencilWriteMask **must** be `0`
- VUID-vkCmdDrawIndirect2KHR-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the `pColorWriteMasks` parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawIndirect2KHR-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of `Location` for any output attachment statically used in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`
- VUID-vkCmdDrawIndirect2KHR-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::`pColorAttachmentLocations` in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance
- VUID-vkCmdDrawIndirect2KHR-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::`pColorAttachmentInputIndices` in the bound pipeline, if either `pColorAttachmentInputIndices`[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an `InputAttachmentIndex` decoration, the two values **must** match
- VUID-vkCmdDrawIndirect2KHR-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` in the bound

pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an `InputAttachmentIndex` decoration, the two values **must** match

- VUID-vkCmdDrawIndirect2KHR-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an `InputAttachmentIndex` decoration, the two values **must** match
- VUID-vkCmdDrawIndirect2KHR-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawIndirect2KHR-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDrawIndirect2KHR-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawIndirect2KHR-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::`flags` which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawIndirect2KHR-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the `dynamicRenderingUnusedAttachments` feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndirect2KHR-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the `dynamicRenderingUnusedAttachments` feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndirect2KHR-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT ::`customResolve` as VK_TRUE
- VUID-vkCmdDrawIndirect2KHR-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a

custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE

- VUID-vkCmdDrawIndirect2KHR-None-11861
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndirect2KHR-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawIndirect2KHR-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound

graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView

- VUID-vkCmdDrawIndirect2KHR-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndirect2KHR-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect2KHR-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawIndirect2KHR-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndirect2KHR-pStencilAttachment-11873

If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView

- VUID-vkCmdDrawIndirect2KHR-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndirect2KHR-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndirect2KHR-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawIndirect2KHR-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndirect2KHR-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndirect2KHR-flags-13363
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndirect2KHR-None-04007
  All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or VK_NULL_HANDLE buffers

bound

- VUID-vkCmdDrawIndirect2KHR-None-04008
  If the nullDescriptor feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be VK_NULL_HANDLE
- VUID-vkCmdDrawIndirect2KHR-None-02721
  If the robustBufferAccess feature is not enabled, and that pipeline was created without enabling VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS for vertexInputs, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in Vertex Input Description
- VUID-vkCmdDrawIndirect2KHR-format-10389
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::format or VkVertexInputAttributeDescription2EXT ::format is a packed format, and the legacyVertexAttributes feature is not enabled, the value of attribAddress, calculated as described in Vertex Input Calculation, **must** be a multiple of the size of the format
- VUID-vkCmdDrawIndirect2KHR-format-10390
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::format or VkVertexInputAttributeDescription2EXT ::format is not a packed format, and either the legacyVertexAttributes feature is not enabled or format has 64-bit components, the value of attribAddress, calculated as described in Vertex Input Calculation, **must** be a multiple of the component size of the format
- VUID-vkCmdDrawIndirect2KHR-None-07842
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled then vkCmdSetPrimitiveTopology **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-dynamicPrimitiveTopologyUnrestricted-07500
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled and the dynamicPrimitiveTopologyUnrestricted is VK_FALSE, then the primitiveTopology parameter of vkCmdSetPrimitiveTopology **must** be of the same topology class as the pipeline VkPipelineInputAssemblyStateCreateInfo::topology state
- VUID-vkCmdDrawIndirect2KHR-primitiveTopology-10286
  If a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage is bound, then the current value of primitiveTopology **must** be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-primitiveTopology-10747
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command, then a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage **must** be bound

- VUID-vkCmdDrawIndirect2KHR-primitiveTopology-10748
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_POINT_LIST prior to this drawing command, the maintenance5 feature is not enabled, both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT and VK_SHADER_STAGE_GEOMETRY_BIT stage are not bound, then the Vertex Execution Model **must** have a PointSize decorated variable that is statically written to
- VUID-vkCmdDrawIndirect2KHR-pStrides-04913
  If the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, but without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, then vkCmdBindVertexBuffers2 with a non-NULL pStrides parameter or vkCmdBindVertexBuffers3KHR with setStride set to VK_TRUE **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndirect2KHR-None-13118
  If the bound graphics pipeline was created without the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, and vkCmdBindVertexBuffers3KHR was called and not subsequently invalidated in the current command buffer prior to this draw command, the value of setStride in each of its pBindInfos elements must have been VK_FALSE
- VUID-vkCmdDrawIndirect2KHR-None-04914
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndirect2KHR-Input-07939
  If the maintenance9 feature is not enabled, and there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then all variables with the Input storage class decorated with Location in the Vertex Execution Model OpEntryPoint **must** contain a location in VkVertexInputAttributeDescription2EXT::location
- VUID-vkCmdDrawIndirect2KHR-Input-08734
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and either the legacyVertexAttributes feature is not enabled or the SPIR-V Type associated with a given Input variable of the corresponding Location in the Vertex Execution Model OpEntryPoint is 64-bit, then the numeric type associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be the same as VkVertexInputAttributeDescription2EXT::format
- VUID-vkCmdDrawIndirect2KHR-format-08936
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then the scalar width associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be 64-bit

- VUID-vkCmdDrawIndirect2KHR-format-08937
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and the scalar width associated with a Location decorated Input variable in the Vertex Execution Model OpEntryPoint is 64-bit, then the corresponding VkVertexInputAttributeDescription2EXT ::format **must** have a 64-bit component
- VUID-vkCmdDrawIndirect2KHR-None-09203
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then all Input variables at the corresponding Location in the Vertex Execution Model OpEntryPoint **must** not use components that are not present in the format
- VUID-vkCmdDrawIndirect2KHR-None-04875
  If there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with both a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage and the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state enabled, and the current value of primitiveTopology is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, then vkCmdSetPatchControlPointsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-04879
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-None-09637
  If the primitiveTopologyListRestart feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled, then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirect2KHR-None-10909
  If the primitiveTopologyPatchListRestart feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline

state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE

- VUID-vkCmdDrawIndirect2KHR-stage-06481
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::`stage` member of any element of VkGraphicsPipelineCreateInfo::`pStages` set to VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndirect2KHR-None-08885
  There **must** be no shader object bound to either of the VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT stages
- VUID-vkCmdDrawIndirect2KHR-None-07619
  If a shader object is bound to the VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage and the VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT dynamic state enabled, then vkCmdSetTessellationDomainOriginEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect2KHR-OpExecutionMode-12239
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the type of subdivision, they **must** be the same
- VUID-vkCmdDrawIndirect2KHR-OpExecutionMode-12240
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the orientation of triangles, they **must** be the same
- VUID-vkCmdDrawIndirect2KHR-OpExecutionMode-12241
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the segment spacing, they **must** be the same
- VUID-vkCmdDrawIndirect2KHR-OpExecutionMode-12242
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the output patch size, they **must** be the same
- VUID-vkCmdDrawIndirect2KHR-pInfo-13110
  If `pInfo->drawCount` is greater than `0`, `pInfo->addressRange.size` **must** be greater than or equal to (`pInfo->drawCount` - 1) × `pInfo->addressRange.stride` + `sizeof`(VkDrawIndirectCommand)
- VUID-vkCmdDrawIndirect2KHR-pInfo-13111
  If `pInfo->drawCount` is greater than 1, `pInfo->addressRange.stride` **must** not be 0
- VUID-vkCmdDrawIndirect2KHR-pInfo-13112
  If `pInfo->addressRange.stride` is not 0, `pInfo->addressRange.stride` **must** be greater than or equal to `sizeof`(VkDrawIndirectCommand)

- VUID-vkCmdDrawIndirect2KHR-pInfo-13113
  If `pInfo->addressRange.stride` is not 0, `pInfo->addressRange.stride` **must** be a multiple of `4`
- VUID-vkCmdDrawIndirect2KHR-commandBuffer-13057
  `commandBuffer` **must** not be a protected command buffer

Valid Usage (Implicit)

- VUID-vkCmdDrawIndirect2KHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDrawIndirect2KHR-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkDrawIndirect2InfoKHR structure
- VUID-vkCmdDrawIndirect2KHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdDrawIndirect2KHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDrawIndirect2KHR-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndirect2KHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawIndirect2KHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawIndirect2KHR is affected by conditional rendering

To record a non-indexed indirect drawing command, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDrawIndirect(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    buffer,
    VkDeviceSize                                offset,
    uint32_t                                    drawCount,
    uint32_t                                    stride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `buffer` is the buffer containing draw parameters.
- `offset` is the byte offset into `buffer` where parameters begin.
- `drawCount` is the number of draws to execute, and **can** be zero.
- `stride` is the byte stride between successive sets of draw parameters.

`vkCmdDrawIndirect` behaves similarly to vkCmdDraw except that the parameters are read by the device from a buffer during execution. `drawCount` draws are executed by the command, with parameters taken from `buffer` starting at `offset` and increasing by `stride` bytes for each successive draw. The parameters of each draw are encoded in an array of VkDrawIndirectCommand structures. If `drawCount` is less than or equal to one, `stride` is ignored.

Valid Usage

- VUID-vkCmdDrawIndirect-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawIndirect-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawIndirect-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawIndirect-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirect-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT

- VUID-vkCmdDrawIndirect-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirect-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirect-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawIndirect-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndirect-magFilter-09598
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndirect-mipmapMode-04770
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndirect-mipmapMode-09599
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndirect-unnormalizedCoordinates-09635
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDrawIndirect-None-08609
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample

a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D

- VUID-vkCmdDrawIndirect-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdDrawIndirect-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias or any offset values
- VUID-vkCmdDrawIndirect-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawIndirect-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawIndirect-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDrawIndirect-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawIndirect-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawIndirect-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndirect-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax returned by vkGetPhysicalDeviceImageFormatProperties2

- VUID-vkCmdDrawIndirect-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the OpTypeImage is Unknown, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirect-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the OpTypeImage is Unknown, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirect-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the OpTypeImage is Unknown, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirect-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the OpTypeImage is Unknown, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirect-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDrawIndirect-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawIndirect-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDrawIndirect-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawIndirect-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for uniformBuffers, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirect-None-08612
  If the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage

corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirect-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirect-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirect-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawIndirect-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match
- VUID-vkCmdDrawIndirect-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawIndirect-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have four components
- VUID-vkCmdDrawIndirect-OpImageWrite-04469
  If a VkBufferView is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawIndirect-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdDrawIndirect-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32

- VUID-vkCmdDrawIndirect-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawIndirect-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawIndirect-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawIndirect-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawIndirect-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawIndirect-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawIndirect-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawIndirect-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawIndirect-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not

have been created with flags that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR

- VUID-vkCmdDrawIndirect-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with flags that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDrawIndirect-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawIndirect-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a OpTypeStruct decorated with Block or BufferBlock using that mapping, the calculated offset for the resource heap **must** be a multiple of bufferDescriptorAlignment
- VUID-vkCmdDrawIndirect-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeImage or OpTypeSampledImage using that mapping, the calculated offset for the resource heap **must** be a multiple of imageDescriptorAlignment
- VUID-vkCmdDrawIndirect-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeSampler or OpTypeSampledImage using that mapping, the calculated offset for the sampler heap **must** be a multiple of samplerDescriptorAlignment
- VUID-vkCmdDrawIndirect-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location

in push data **must** be a multiple of 4

- VUID-vkCmdDrawIndirect-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndirect-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndirect-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdDrawIndirect-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndirect-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndirect-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawIndirect-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawIndirect-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this

command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdDrawIndirect-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndirect-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndirect-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDrawIndirect-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDrawIndirect-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDrawIndirect-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or

VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set

- VUID-vkCmdDrawIndirect-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirect-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdDrawIndirect-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirect-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawIndirect-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is

mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR

- VUID-vkCmdDrawIndirect-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color

- VUID-vkCmdDrawIndirect-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawIndirect-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawIndirect-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS

- VUID-vkCmdDrawIndirect-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS

- VUID-vkCmdDrawIndirect-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass

- VUID-vkCmdDrawIndirect-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' `pInputAttachments`[`InputAttachmentIndex`] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility

- VUID-vkCmdDrawIndirect-pDepthInputAttachmentIndex-09595

Input attachment views accessed in a dynamic render pass with a InputAttachmentIndex referenced by VkRenderingInputAttachmentIndexInfo, or no InputAttachmentIndex if VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex or VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex are NULL, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo

- VUID-vkCmdDrawIndirect-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an InputAttachmentIndex if both VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex and VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex are non-NULL
- VUID-vkCmdDrawIndirect-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an InputAttachmentIndex, the InputAttachmentIndex **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawIndirect-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirect-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirect-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirect-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirect-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirect-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirect-None-12340

If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command

- VUID-vkCmdDrawIndirect-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawIndirect-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDrawIndirect-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawIndirect-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR,

VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::`maxMultiviewInstanceIndex`
- VUID-vkCmdDrawIndirect-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndirect-None-07634
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the

current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-06666
  If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of sampleLocationsEnable is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07846
  If the depthBounds feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the

VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** match the `scissorCount` parameter of `vkCmdSetScissorWithCount`
- VUID-vkCmdDrawIndirect-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics

pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be `1`
- VUID-vkCmdDrawIndirect-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, and any shader object bound to a graphics stage writes to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be `1`
- VUID-vkCmdDrawIndirect-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of `blendEnable` **must** be VK_FALSE
- VUID-vkCmdDrawIndirect-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of `rasterizerDiscardEnable` is VK_FALSE, and none of the following is enabled:

  ◦ the `multisampledRenderToSingleSampled` feature

  then the current value of `rasterizationSamples` **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDrawIndirect-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDrawIndirect-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo ::`viewMask` equal to VkRenderingInfo::`viewMask`
- VUID-vkCmdDrawIndirect-colorAttachmentCount-06179
  If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render

pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount

- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08910
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08912
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawIndirect-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state

enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT ::discardRectangleCount

- VUID-vkCmdDrawIndirect-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT ::maxDiscardRectangles
- VUID-vkCmdDrawIndirect-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo ::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo ::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->imageView

- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawIndirect-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawIndirect-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not

VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT

- VUID-vkCmdDrawIndirect-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawIndirect-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndirect-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawIndirect-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDrawIndirect-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawIndirect-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a

graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirect-pDepthAttachment-08964
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawIndirect-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawIndirect-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the

VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream

- VUID-vkCmdDrawIndirect-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and alphaToCoverageEnable was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-vkCmdDrawIndirect-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set alphaToCoverageEnable to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-vkCmdDrawIndirect-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07625
  If the alphaToOne feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07626
  If the logicOp feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07630
  If the geometryStreams feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07631
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07632
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the

VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of conservativeRasterizationMode is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-conservativePointAndLineRasterization-07499
  If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-vkCmdDrawIndirect-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-08669
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-07849

If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current lineRasterizationMode is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current alphaToCoverageEnable, alphaToOneEnable and sampleShadingEnable states **must** all be VK_FALSE
- VUID-vkCmdDrawIndirect-None-07639
  If the depthClipControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-09650
  If the depthClampControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of depthClampEnable is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawIndirect-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::rasterizationSamples parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawIndirect-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the samples

parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the `rasterizationSamples` parameter in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawIndirect-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with `multisampledRenderToSingleSampledEnable` equal to VK_TRUE, then the `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the `rasterizationSamples` member of that structure
- VUID-vkCmdDrawIndirect-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndirect-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndirect-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE, then the number of active color attachments **must** not exceed `advancedBlendMaxColorAttachments`
- VUID-vkCmdDrawIndirect-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawIndirect-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then

vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE

- VUID-vkCmdDrawIndirect-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawIndirect-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the `rasterizationStream` to zero
- VUID-vkCmdDrawIndirect-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawIndirect-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirect-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndirect-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound

graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.width in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdDrawIndirect-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.height in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples
- VUID-vkCmdDrawIndirect-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDrawIndirect-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirect-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the

value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawIndirect-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirect-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled
- VUID-vkCmdDrawIndirect-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled
- VUID-vkCmdDrawIndirect-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawIndirect-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawIndirect-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawIndirect-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the

VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a nextStage including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDrawIndirect-None-08684
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDrawIndirect-None-08685
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawIndirect-None-08686
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-vkCmdDrawIndirect-None-08687
  If there is no bound graphics pipeline, and the geometryShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawIndirect-None-08688
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawIndirect-None-08689
  If there is no bound graphics pipeline, and the taskShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawIndirect-None-08690
  If there is no bound graphics pipeline, and the meshShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndirect-None-08693
  If there is no bound graphics pipeline, and at least one of the taskShader and meshShader features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid VkShaderEXT bound, and the other **must** have no VkShaderEXT bound
- VUID-vkCmdDrawIndirect-None-08696
  If there is no bound graphics pipeline, and a valid VkShaderEXT is bound to the

VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no VkShaderEXT bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage

- VUID-vkCmdDrawIndirect-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDrawIndirect-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any VkShaderEXT bound
- VUID-vkCmdDrawIndirect-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawIndirect-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts
- VUID-vkCmdDrawIndirect-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpDepthAttachmentReadEXT, the depthWriteEnable parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirect-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpStencilAttachmentReadEXT, the writeMask parameter in the last call to vkCmdSetStencilWriteMask **must** be 0
- VUID-vkCmdDrawIndirect-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the pColorWriteMasks parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawIndirect-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of Location for any output attachment statically used in the Fragment Execution Model executed by this

command **must** be less than `maxFragmentDualSrcAttachments`

- VUID-vkCmdDrawIndirect-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::`pColorAttachmentLocations` in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance

- VUID-vkCmdDrawIndirect-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::`pColorAttachmentInputIndices` in the bound pipeline, if either `pColorAttachmentInputIndices`[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an `InputAttachmentIndex` decoration, the two values **must** match

- VUID-vkCmdDrawIndirect-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an `InputAttachmentIndex` decoration, the two values **must** match

- VUID-vkCmdDrawIndirect-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an `InputAttachmentIndex` decoration, the two values **must** match

- VUID-vkCmdDrawIndirect-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT

- VUID-vkCmdDrawIndirect-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag

- VUID-vkCmdDrawIndirect-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass

- VUID-vkCmdDrawIndirect-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::`flags` which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for

sample shading **must** equal 0.0

- VUID-vkCmdDrawIndirect-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the `dynamicRenderingUnusedAttachments` feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndirect-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the `dynamicRenderingUnusedAttachments` feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndirect-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::`customResolve` as VK_TRUE
- VUID-vkCmdDrawIndirect-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::`customResolve` as VK_FALSE
- VUID-vkCmdDrawIndirect-None-11861
  If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::`colorAttachmentCount` equal to VkRenderingInfo::`colorAttachmentCount`
- VUID-vkCmdDrawIndirect-None-11862
  If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `resolveImageView` not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::`pColorAttachmentFormats` used to create the bound graphics pipeline
- VUID-vkCmdDrawIndirect-None-11863
  If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `resolveImageView` equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::`pColorAttachmentFormats` used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-11864
  If the `dynamicRenderingUnusedAttachments` feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the

VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirect-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirect-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the

bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirect-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawIndirect-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndirect-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndirect-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndirect-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndirect-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawIndirect-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndirect-flags-12491
  If there are any shader objects bound that were created with flags that includes the

VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set

- VUID-vkCmdDrawIndirect-flags-13363
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT

- VUID-vkCmdDrawIndirect-None-04007
  All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or VK_NULL_HANDLE buffers bound

- VUID-vkCmdDrawIndirect-None-04008
  If the nullDescriptor feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be VK_NULL_HANDLE

- VUID-vkCmdDrawIndirect-None-02721
  If the robustBufferAccess feature is not enabled, and that pipeline was created without enabling VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS for vertexInputs, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in Vertex Input Description

- VUID-vkCmdDrawIndirect-format-10389
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::format or VkVertexInputAttributeDescription2EXT ::format is a packed format, and the legacyVertexAttributes feature is not enabled, the value of attribAddress, calculated as described in Vertex Input Calculation, **must** be a multiple of the size of the format

- VUID-vkCmdDrawIndirect-format-10390
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::format or VkVertexInputAttributeDescription2EXT ::format is not a packed format, and either the legacyVertexAttributes feature is not enabled or format has 64-bit components, the value of attribAddress, calculated as described in Vertex Input Calculation, **must** be a multiple of the component size of the format

- VUID-vkCmdDrawIndirect-None-07842
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled then

vkCmdSetPrimitiveTopology **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-dynamicPrimitiveTopologyUnrestricted-07500
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled and the dynamicPrimitiveTopologyUnrestricted is VK_FALSE, then the primitiveTopology parameter of vkCmdSetPrimitiveTopology **must** be of the same topology class as the pipeline VkPipelineInputAssemblyStateCreateInfo::topology state
- VUID-vkCmdDrawIndirect-primitiveTopology-10286
  If a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage is bound, then the current value of primitiveTopology **must** be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command
- VUID-vkCmdDrawIndirect-primitiveTopology-10747
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command, then a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage **must** be bound
- VUID-vkCmdDrawIndirect-primitiveTopology-10748
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_POINT_LIST prior to this drawing command, the maintenance5 feature is not enabled, both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT and VK_SHADER_STAGE_GEOMETRY_BIT stage are not bound, then the Vertex Execution Model **must** have a PointSize decorated variable that is statically written to
- VUID-vkCmdDrawIndirect-pStrides-04913
  If the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, but without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, then vkCmdBindVertexBuffers2 with a non-NULL pStrides parameter or vkCmdBindVertexBuffers3KHR with setStride set to VK_TRUE **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndirect-None-13118
  If the bound graphics pipeline was created without the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, and vkCmdBindVertexBuffers3KHR was called and not subsequently invalidated in the current command buffer prior to this draw command, the value of setStride in each of its pBindInfos elements must have been VK_FALSE
- VUID-vkCmdDrawIndirect-None-04914
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndirect-Input-07939
  If the maintenance9 feature is not enabled, and there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created

with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then all variables with the Input storage class decorated with Location in the Vertex Execution Model OpEntryPoint **must** contain a location in VkVertexInputAttributeDescription2EXT::location

- VUID-vkCmdDrawIndirect-Input-08734
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and either the legacyVertexAttributes feature is not enabled or the SPIR-V Type associated with a given Input variable of the corresponding Location in the Vertex Execution Model OpEntryPoint is 64-bit, then the numeric type associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be the same as VkVertexInputAttributeDescription2EXT::format
- VUID-vkCmdDrawIndirect-format-08936
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then the scalar width associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be 64-bit
- VUID-vkCmdDrawIndirect-format-08937
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and the scalar width associated with a Location decorated Input variable in the Vertex Execution Model OpEntryPoint is 64-bit, then the corresponding VkVertexInputAttributeDescription2EXT::format **must** have a 64-bit component
- VUID-vkCmdDrawIndirect-None-09203
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then all Input variables at the corresponding Location in the Vertex Execution Model OpEntryPoint **must** not use components that are not present in the format
- VUID-vkCmdDrawIndirect-None-04875
  If there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with both a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage and the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state enabled, and the current value of primitiveTopology is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, then vkCmdSetPatchControlPointsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-04879
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-09637
  If the `primitiveTopologyListRestart` feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled, then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirect-None-10909
  If the `primitiveTopologyPatchListRestart` feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirect-stage-06481
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::`stage` member of any element of VkGraphicsPipelineCreateInfo::`pStages` set to VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndirect-None-08885
  There **must** be no shader object bound to either of the VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT stages
- VUID-vkCmdDrawIndirect-None-07619
  If a shader object is bound to the VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage and the VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT dynamic state enabled, then vkCmdSetTessellationDomainOriginEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-OpExecutionMode-12239
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the type of subdivision, they **must** be the same
- VUID-vkCmdDrawIndirect-OpExecutionMode-12240
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the orientation of triangles, they **must** be the same
- VUID-vkCmdDrawIndirect-OpExecutionMode-12241
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain

an OpExecutionMode instruction specifying the segment spacing, they **must** be the same

- VUID-vkCmdDrawIndirect-OpExecutionMode-12242
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the output patch size, they **must** be the same
- VUID-vkCmdDrawIndirect-buffer-02708
  If buffer is non-sparse then it **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-vkCmdDrawIndirect-buffer-02709
  buffer **must** have been created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirect-offset-02710
  offset **must** be a multiple of 4
- VUID-vkCmdDrawIndirect-commandBuffer-02711
  commandBuffer **must** not be a protected command buffer
- VUID-vkCmdDrawIndirect-drawCount-02718
  If the multiDrawIndirect feature is not enabled, drawCount **must** be 0 or 1
- VUID-vkCmdDrawIndirect-drawCount-02719
  drawCount **must** be less than or equal to VkPhysicalDeviceLimits::maxDrawIndirectCount
- VUID-vkCmdDrawIndirect-drawCount-00476
  If drawCount is greater than 1, stride **must** be a multiple of 4 and **must** be greater than or equal to sizeof(VkDrawIndirectCommand)
- VUID-vkCmdDrawIndirect-drawCount-00487
  If drawCount is equal to 1, (offset + sizeof(VkDrawIndirectCommand)) **must** be less than or equal to the size of buffer
- VUID-vkCmdDrawIndirect-drawCount-00488
  If drawCount is greater than 1, (stride × (drawCount - 1) + offset + sizeof (VkDrawIndirectCommand)) **must** be less than or equal to the size of buffer

Valid Usage (Implicit)

- VUID-vkCmdDrawIndirect-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDrawIndirect-buffer-parameter
  buffer **must** be a valid VkBuffer handle
- VUID-vkCmdDrawIndirect-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdDrawIndirect-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations

- VUID-vkCmdDrawIndirect-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndirect-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawIndirect-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdDrawIndirect-commonparent
  Both of `buffer`, and `commandBuffer` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawIndirect is affected by conditional rendering

The `VkDrawIndirectCommand` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDrawIndirectCommand {
    uint32_t    vertexCount;
    uint32_t    instanceCount;
    uint32_t    firstVertex;
    uint32_t    firstInstance;
} VkDrawIndirectCommand;
```

- `vertexCount` is the number of vertices to draw.
- `instanceCount` is the number of instances to draw.
- `firstVertex` is the index of the first vertex to draw.

- `firstInstance` is the instance ID of the first instance to draw.

The members of `VkDrawIndirectCommand` have the same meaning as the similarly named parameters of vkCmdDraw.

Valid Usage

- VUID-VkDrawIndirectCommand-pNext-09461
  If the bound graphics pipeline state was created with VkPipelineVertexInputDivisorStateCreateInfo in the `pNext` chain of VkGraphicsPipelineCreateInfo::`pVertexInputState`, any member of VkPipelineVertexInputDivisorStateCreateInfo::`pVertexBindingDivisors` has a value other than `1` in `divisor`, and VkPhysicalDeviceVertexAttributeDivisorProperties::`supportsNonZeroFirstInstance` is VK_FALSE, then `firstInstance` **must** be `0`
- VUID-VkDrawIndirectCommand-None-09462
  If shader objects are used for drawing or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, any member of the `pVertexBindingDescriptions` parameter to the vkCmdSetVertexInputEXT call that sets this dynamic state has a value other than `1` in `divisor`, and VkPhysicalDeviceVertexAttributeDivisorProperties::`supportsNonZeroFirstInstance` is VK_FALSE, then `firstInstance` **must** be `0`
- VUID-VkDrawIndirectCommand-firstInstance-00501
  If the `drawIndirectFirstInstance` feature is not enabled, `firstInstance` **must** be `0`

`VkDrawIndirectCount2InfoKHR` is defined as:

```
// Provided by VK_KHR_device_address_commands
typedef struct VkDrawIndirectCount2InfoKHR {
    VkStructureType                  sType;
    const void*                      pNext;
    VkStridedDeviceAddressRangeKHR   addressRange;
    VkAddressCommandFlagsKHR         addressFlags;
    VkDeviceAddressRangeKHR          countAddressRange;
    VkAddressCommandFlagsKHR         countAddressFlags;
    uint32_t                         maxDrawCount;
} VkDrawIndirectCount2InfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `addressRange` is the VkStridedDeviceAddressRangeKHR containing draw parameters.
- `addressFlags` is a VkAddressCommandFlagsKHR value defining the flags for the address range.
- `countAddressRange` is the VkDeviceAddressRangeKHR containing the draw count.
- `countAddressFlags` is a VkAddressCommandFlagsKHR value defining the flags for the count address range.

- `maxDrawCount` specifies the maximum number of draws that will be executed. The actual number of executed draw calls is the minimum of the count specified in `countAddressRange` and `maxDrawCount`.

Valid Usage

- VUID-VkDrawIndirectCount2InfoKHR-addressRange-13097
  If the range specified by `addressRange` is not bound completely to memory when accessed, `addressFlags` **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-addressRange-13098
  If the buffer from which the range specified by `addressRange` was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `addressFlags` **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-addressRange-13099
  If the buffer from which the range specified by `addressRange` was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `addressFlags` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-addressFlags-13100
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-addressRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-addressRange-13123
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-addressFlags-13101
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-addressRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `addressFlags` **must** include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-addressRange-13125

If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing addressRange, was created without VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, addressFlags **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR

- VUID-VkDrawIndirectCount2InfoKHR-addressRange-13107
  The buffer from which addressRange was queried **must** have been created with VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag
- VUID-VkDrawIndirectCount2InfoKHR-protectedNoFault-13108
  If protectedNoFault is not supported, the buffer from which addressRange was queried **must** not have been created with VK_BUFFER_CREATE_PROTECTED_BIT
- VUID-VkDrawIndirectCount2InfoKHR-addressRange-13109
  addressRange.address **must** be a multiple of 4
- VUID-VkDrawIndirectCount2InfoKHR-countAddressRange-13097
  If the range specified by countAddressRange is not bound completely to memory when accessed, countAddressFlags **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-countAddressRange-13098
  If the buffer from which the range specified by countAddressRange was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and protectedNoFault is not supported, countAddressFlags **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-countAddressRange-13099
  If the buffer from which the range specified by countAddressRange was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and protectedNoFault is not supported, countAddressFlags **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-countAddressFlags-13100
  countAddressFlags **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-countAddressRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing countAddressRange, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, countAddressFlags **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-countAddressRange-13123
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing countAddressRange, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, countAddressFlags **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-countAddressFlags-13101
  countAddressFlags **must** not include both

VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR

- VUID-VkDrawIndirectCount2InfoKHR-countAddressRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `countAddressRange`, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `countAddressFlags` **must** include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-countAddressRange-13125
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `countAddressRange`, was created without VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `countAddressFlags` **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDrawIndirectCount2InfoKHR-countAddressRange-13114
  The buffer from which `countAddressRange` was queried **must** have been created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag
- VUID-VkDrawIndirectCount2InfoKHR-countAddressRange-13115
  `countAddressRange.address` **must** be a multiple of `4`
- VUID-VkDrawIndirectCount2InfoKHR-countAddressRange-13116
  The count stored in `countAddressRange` **must** be less than or equal to VkPhysicalDeviceLimits::`maxDrawIndirectCount`
- VUID-VkDrawIndirectCount2InfoKHR-countAddressRange-13117
  `countAddressRange.size` **must** be greater than or equal to 4

Valid Usage (Implicit)

- VUID-VkDrawIndirectCount2InfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DRAW_INDIRECT_COUNT_2_INFO_KHR
- VUID-VkDrawIndirectCount2InfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDrawIndirectCount2InfoKHR-addressFlags-parameter
  If `addressFlags` is not `0`, `addressFlags` **must** be a valid combination of VkAddressCommandFlagBitsKHR values
- VUID-VkDrawIndirectCount2InfoKHR-countAddressFlags-parameter
  If `countAddressFlags` is not `0`, `countAddressFlags` **must** be a valid combination of VkAddressCommandFlagBitsKHR values

To record a non-indexed indirect drawing command, call:

```
// Provided by VK_KHR_device_address_commands with VK_KHR_draw_indirect_count or
VK_VERSION_1_2
void vkCmdDrawIndirectCount2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkDrawIndirectCount2InfoKHR*          pInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `pInfo` is a pointer to a VkDrawIndirectCount2InfoKHR structure defining parameters of this command.

`vkCmdDrawIndirectCount2KHR` behaves similarly to vkCmdDrawIndirect2KHR except that the draw count is read by the device from an address range during execution. The command will read an unsigned 32-bit integer from `pInfo->countAddressRange` and use this as the draw count.

Valid Usage

- VUID-vkCmdDrawIndirectCount2KHR-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawIndirectCount2KHR-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawIndirectCount2KHR-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawIndirectCount2KHR-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirectCount2KHR-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirectCount2KHR-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirectCount2KHR-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created

with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT

- VUID-vkCmdDrawIndirectCount2KHR-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a VkDescriptorSetLayout that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawIndirectCount2KHR-magFilter-04553
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndirectCount2KHR-magFilter-09598
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndirectCount2KHR-mipmapMode-04770
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndirectCount2KHR-mipmapMode-09599
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndirectCount2KHR-unnormalizedCoordinates-09635
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1
- VUID-vkCmdDrawIndirectCount2KHR-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawIndirectCount2KHR-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name

- VUID-vkCmdDrawIndirectCount2KHR-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias or any offset values
- VUID-vkCmdDrawIndirectCount2KHR-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawIndirectCount2KHR-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawIndirectCount2KHR-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDrawIndirectCount2KHR-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawIndirectCount2KHR-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawIndirectCount2KHR-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndirectCount2KHR-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndirectCount2KHR-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the OpTypeImage is Unknown, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirectCount2KHR-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the OpTypeImage is Unknown, the view's format features **must** contain

VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT

- VUID-vkCmdDrawIndirectCount2KHR-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirectCount2KHR-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirectCount2KHR-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDrawIndirectCount2KHR-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawIndirectCount2KHR-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDrawIndirectCount2KHR-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawIndirectCount2KHR-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirectCount2KHR-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirectCount2KHR-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or

VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for storageBuffers, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirectCount2KHR-None-08613
  If the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirectCount2KHR-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's viewType **must** match the Dim operand of the OpTypeImage as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawIndirectCount2KHR-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's format and the Sampled Type operand of the OpTypeImage **must** match
- VUID-vkCmdDrawIndirectCount2KHR-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawIndirectCount2KHR-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components
- VUID-vkCmdDrawIndirectCount2KHR-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawIndirectCount2KHR-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawIndirectCount2KHR-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawIndirectCount2KHR-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawIndirectCount2KHR-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that

instruction **must** have a `Width` of 32

- VUID-vkCmdDrawIndirectCount2KHR-sparseImageInt64Atomics-04474
  If the `sparseImageInt64Atomics` feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdDrawIndirectCount2KHR-sparseImageInt64Atomics-04475
  If the `sparseImageInt64Atomics` feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdDrawIndirectCount2KHR-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawIndirectCount2KHR-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawIndirectCount2KHR-micromap-11636
  If an acceleration structure is accessed as a result of this command through an `OpRayQueryProceedKHR` operation , the `VK_EXT_opacity_micromap` extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its `pNext` chain, the shader **must** have specified a `OpacityMicromapIdKHR` execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawIndirectCount2KHR-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawIndirectCount2KHR-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with `flags` that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawIndirectCount2KHR-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap

array, any VkShaderEXT that accessed the descriptor **must** not have been created with `flags` that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT

- VUID-vkCmdDrawIndirectCount2KHR-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawIndirectCount2KHR-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`
- VUID-vkCmdDrawIndirectCount2KHR-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdDrawIndirectCount2KHR-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdDrawIndirectCount2KHR-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDrawIndirectCount2KHR-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location

in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping

- VUID-vkCmdDrawIndirectCount2KHR-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndirectCount2KHR-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdDrawIndirectCount2KHR-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndirectCount2KHR-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndirectCount2KHR-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawIndirectCount2KHR-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawIndirectCount2KHR-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdDrawIndirectCount2KHR-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndirectCount2KHR-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndirectCount2KHR-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDrawIndirectCount2KHR-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDrawIndirectCount2KHR-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDrawIndirectCount2KHR-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirectCount2KHR-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping

using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set

- VUID-vkCmdDrawIndirectCount2KHR-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdDrawIndirectCount2KHR-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirectCount2KHR-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawIndirectCount2KHR-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDrawIndirectCount2KHR-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered

before this command was recorded, and still be registered during the sampling operation, with an identically defined color

- VUID-vkCmdDrawIndirectCount2KHR-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawIndirectCount2KHR-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawIndirectCount2KHR-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndirectCount2KHR-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndirectCount2KHR-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawIndirectCount2KHR-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDrawIndirectCount2KHR-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by VkRenderingInputAttachmentIndexInfo, or no `InputAttachmentIndex` if VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` or VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are `NULL`, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawIndirectCount2KHR-pDepthInputAttachmentIndex-09596

Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` and VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are non-`NULL`

- VUID-vkCmdDrawIndirectCount2KHR-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawIndirectCount2KHR-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectCount2KHR-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectCount2KHR-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectCount2KHR-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectCount2KHR-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirectCount2KHR-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirectCount2KHR-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirectCount2KHR-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an

attachment, storage image, or sampled image by this command

- VUID-vkCmdDrawIndirectCount2KHR-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDrawIndirectCount2KHR-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawIndirectCount2KHR-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the

current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthBoundsTestEnable is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount2KHR-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::maxMultiviewInstanceIndex
- VUID-vkCmdDrawIndirectCount2KHR-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndirectCount2KHR-None-07634
  If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-06666
  If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value

of rasterizerDiscardEnable is VK_FALSE, and the current value of sampleLocationsEnable is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount2KHR-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07846
  If the depthBounds feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state

enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount2KHR-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the viewportCount parameter of vkCmdSetViewportWithCount **must** match the scissorCount parameter of vkCmdSetScissorWithCount
- VUID-vkCmdDrawIndirectCount2KHR-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of logicOpEnable is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-primitiveFragmentShadingRateWithMultipleViewports-04552

If the primitiveFragmentShadingRateWithMultipleViewports limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1

- VUID-vkCmdDrawIndirectCount2KHR-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the primitiveFragmentShadingRateWithMultipleViewports limit is not supported, and any shader object bound to a graphics stage writes to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1
- VUID-vkCmdDrawIndirectCount2KHR-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of blendEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirectCount2KHR-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and none of the following is enabled:
  - the multisampledRenderToSingleSampled feature

  then the current value of rasterizationSamples **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDrawIndirectCount2KHR-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDrawIndirectCount2KHR-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::viewMask equal to VkRenderingInfo::viewMask
- VUID-vkCmdDrawIndirectCount2KHR-colorAttachmentCount-06179
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndirectCount2KHR-dynamicRenderingUnusedAttachments-08910
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render

pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline

- VUID-vkCmdDrawIndirectCount2KHR-dynamicRenderingUnusedAttachments-08912
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount2KHR-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount2KHR-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawIndirectCount2KHR-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT

::discardRectangleCount

- VUID-vkCmdDrawIndirectCount2KHR-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT ::maxDiscardRectangles
- VUID-vkCmdDrawIndirectCount2KHR-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo ::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount2KHR-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo ::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->imageView
- VUID-vkCmdDrawIndirectCount2KHR-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo ::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics

pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirectCount2KHR-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo ::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount2KHR-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo ::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->imageView
- VUID-vkCmdDrawIndirectCount2KHR-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo ::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount2KHR-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawIndirectCount2KHR-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawIndirectCount2KHR-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawIndirectCount2KHR-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance,

and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline

- VUID-vkCmdDrawIndirectCount2KHR-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndirectCount2KHR-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawIndirectCount2KHR-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDrawIndirectCount2KHR-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawIndirectCount2KHR-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount2KHR-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline

must not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirectCount2KHR-pDepthAttachment-08964
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount2KHR-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount2KHR-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount2KHR-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount2KHR-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawIndirectCount2KHR-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawIndirectCount2KHR-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream

- VUID-vkCmdDrawIndirectCount2KHR-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and alphaToCoverageEnable was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-vkCmdDrawIndirectCount2KHR-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set alphaToCoverageEnable to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-vkCmdDrawIndirectCount2KHR-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07625
  If the alphaToOne feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the

VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount2KHR-None-07626
  If the logicOp feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07630
  If the geometryStreams feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07631
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07632
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of conservativeRasterizationMode is

VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount2KHR-conservativePointAndLineRasterization-07499
  If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-vkCmdDrawIndirectCount2KHR-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-08669
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic

state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount2KHR-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current lineRasterizationMode is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current alphaToCoverageEnable, alphaToOneEnable and sampleShadingEnable states **must** all be VK_FALSE
- VUID-vkCmdDrawIndirectCount2KHR-None-07639
  If the depthClipControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-09650
  If the depthClampControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of depthClampEnable is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawIndirectCount2KHR-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::rasterizationSamples parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawIndirectCount2KHR-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the rasterizationSamples parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectCount2KHR-None-09211

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with `multisampledRenderToSingleSampledEnable` equal to VK_TRUE, then the `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the `rasterizationSamples` member of that structure

- VUID-vkCmdDrawIndirectCount2KHR-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndirectCount2KHR-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndirectCount2KHR-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE, then the number of active color attachments **must** not exceed `advancedBlendMaxColorAttachments`
- VUID-vkCmdDrawIndirectCount2KHR-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawIndirectCount2KHR-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE

- VUID-vkCmdDrawIndirectCount2KHR-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawIndirectCount2KHR-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the `rasterizationStream` to zero
- VUID-vkCmdDrawIndirectCount2KHR-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawIndirectCount2KHR-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectCount2KHR-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndirectCount2KHR-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if

sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.width in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdDrawIndirectCount2KHR-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.height in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples
- VUID-vkCmdDrawIndirectCount2KHR-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDrawIndirectCount2KHR-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectCount2KHR-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectCount2KHR-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawIndirectCount2KHR-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled
- VUID-vkCmdDrawIndirectCount2KHR-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled
- VUID-vkCmdDrawIndirectCount2KHR-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawIndirectCount2KHR-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawIndirectCount2KHR-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawIndirectCount2KHR-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not

subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount2KHR-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a `nextStage` including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDrawIndirectCount2KHR-None-08684
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDrawIndirectCount2KHR-None-08685
  If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawIndirectCount2KHR-None-08686
  If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-vkCmdDrawIndirectCount2KHR-None-08687
  If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawIndirectCount2KHR-None-08688
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawIndirectCount2KHR-None-08689
  If there is no bound graphics pipeline, and the `taskShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawIndirectCount2KHR-None-08690
  If there is no bound graphics pipeline, and the `meshShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndirectCount2KHR-None-08693
  If there is no bound graphics pipeline, and at least one of the `taskShader` and `meshShader` features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid `VkShaderEXT` bound, and the other **must** have no `VkShaderEXT` bound
- VUID-vkCmdDrawIndirectCount2KHR-None-08696
  If there is no bound graphics pipeline, and a valid `VkShaderEXT` is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no `VkShaderEXT` bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage

- VUID-vkCmdDrawIndirectCount2KHR-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDrawIndirectCount2KHR-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any `VkShaderEXT` bound
- VUID-vkCmdDrawIndirectCount2KHR-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawIndirectCount2KHR-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts
- VUID-vkCmdDrawIndirectCount2KHR-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirectCount2KHR-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to vkCmdSetStencilWriteMask **must** be `0`
- VUID-vkCmdDrawIndirectCount2KHR-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the `pColorWriteMasks` parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawIndirectCount2KHR-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of `Location` for any output attachment statically used in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`
- VUID-vkCmdDrawIndirectCount2KHR-None-09548

If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::pColorAttachmentLocations in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance

- VUID-vkCmdDrawIndirectCount2KHR-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::pColorAttachmentInputIndices in the bound pipeline, if either pColorAttachmentInputIndices[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndirectCount2KHR-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndirectCount2KHR-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndirectCount2KHR-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawIndirectCount2KHR-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDrawIndirectCount2KHR-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawIndirectCount2KHR-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::flags which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawIndirectCount2KHR-None-11522

If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT

- VUID-vkCmdDrawIndirectCount2KHR-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndirectCount2KHR-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndirectCount2KHR-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndirectCount2KHR-None-11861
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndirectCount2KHR-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawIndirectCount2KHR-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount2KHR-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding

element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirectCount2KHR-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount2KHR-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndirectCount2KHR-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount2KHR-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount2KHR-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndirectCount2KHR-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView, the value of the format **must** be

VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirectCount2KHR-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawIndirectCount2KHR-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndirectCount2KHR-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndirectCount2KHR-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndirectCount2KHR-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndirectCount2KHR-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawIndirectCount2KHR-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndirectCount2KHR-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of

the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set

- VUID-vkCmdDrawIndirectCount2KHR-flags-13363
  If there are any shader objects bound that were created with `flags` that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::`stageFlags` set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::`stageFlags` set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT

- VUID-vkCmdDrawIndirectCount2KHR-None-04007
  All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or VK_NULL_HANDLE buffers bound

- VUID-vkCmdDrawIndirectCount2KHR-None-04008
  If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be VK_NULL_HANDLE

- VUID-vkCmdDrawIndirectCount2KHR-None-02721
  If the `robustBufferAccess` feature is not enabled, and that pipeline was created without enabling VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS for `vertexInputs`, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in Vertex Input Description

- VUID-vkCmdDrawIndirectCount2KHR-format-10389
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is a packed format, and the `legacyVertexAttributes` feature is not enabled, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the size of the `format`

- VUID-vkCmdDrawIndirectCount2KHR-format-10390
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is not a packed format, and either the `legacyVertexAttributes` feature is not enabled or `format` has 64-bit components, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the component size of the `format`

- VUID-vkCmdDrawIndirectCount2KHR-None-07842
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled then vkCmdSetPrimitiveTopology **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount2KHR-dynamicPrimitiveTopologyUnrestricted-07500
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled and the dynamicPrimitiveTopologyUnrestricted is VK_FALSE, then the primitiveTopology parameter of vkCmdSetPrimitiveTopology **must** be of the same topology class as the pipeline VkPipelineInputAssemblyStateCreateInfo::topology state
- VUID-vkCmdDrawIndirectCount2KHR-primitiveTopology-10286
  If a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage is bound, then the current value of primitiveTopology **must** be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-primitiveTopology-10747
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command, then a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage **must** be bound
- VUID-vkCmdDrawIndirectCount2KHR-primitiveTopology-10748
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_POINT_LIST prior to this drawing command, the maintenance5 feature is not enabled, both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT and VK_SHADER_STAGE_GEOMETRY_BIT stage are not bound, then the Vertex Execution Model **must** have a PointSize decorated variable that is statically written to
- VUID-vkCmdDrawIndirectCount2KHR-pStrides-04913
  If the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, but without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, then vkCmdBindVertexBuffers2 with a non-NULL pStrides parameter or vkCmdBindVertexBuffers3KHR with setStride set to VK_TRUE **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndirectCount2KHR-None-13118
  If the bound graphics pipeline was created without the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, and vkCmdBindVertexBuffers3KHR was called and not subsequently invalidated in the current command buffer prior to this draw command, the value of setStride in each of its pBindInfos elements must have been VK_FALSE
- VUID-vkCmdDrawIndirectCount2KHR-None-04914
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndirectCount2KHR-Input-07939
  If the maintenance9 feature is not enabled, and there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then all variables with the Input storage class decorated with Location in the Vertex Execution

Model OpEntryPoint **must** contain a location in VkVertexInputAttributeDescription2EXT::location

- VUID-vkCmdDrawIndirectCount2KHR-Input-08734
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and either the legacyVertexAttributes feature is not enabled or the SPIR-V Type associated with a given Input variable of the corresponding Location in the Vertex Execution Model OpEntryPoint is 64-bit, then the numeric type associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be the same as VkVertexInputAttributeDescription2EXT::format
- VUID-vkCmdDrawIndirectCount2KHR-format-08936
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then the scalar width associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be 64-bit
- VUID-vkCmdDrawIndirectCount2KHR-format-08937
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and the scalar width associated with a Location decorated Input variable in the Vertex Execution Model OpEntryPoint is 64-bit, then the corresponding VkVertexInputAttributeDescription2EXT::format **must** have a 64-bit component
- VUID-vkCmdDrawIndirectCount2KHR-None-09203
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then all Input variables at the corresponding Location in the Vertex Execution Model OpEntryPoint **must** not use components that are not present in the format
- VUID-vkCmdDrawIndirectCount2KHR-None-04875
  If there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with both a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage and the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state enabled, and the current value of primitiveTopology is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, then vkCmdSetPatchControlPointsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-None-04879
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** have been called and not subsequently

invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount2KHR-None-09637
  If the primitiveTopologyListRestart feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled, then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirectCount2KHR-None-10909
  If the primitiveTopologyPatchListRestart feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirectCount2KHR-stage-06481
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::stage member of any element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndirectCount2KHR-None-08885
  There **must** be no shader object bound to either of the VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT stages
- VUID-vkCmdDrawIndirectCount2KHR-None-07619
  If a shader object is bound to the VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage and the VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT dynamic state enabled, then vkCmdSetTessellationDomainOriginEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount2KHR-OpExecutionMode-12239
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the type of subdivision, they **must** be the same
- VUID-vkCmdDrawIndirectCount2KHR-OpExecutionMode-12240
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the orientation of triangles, they **must** be the same
- VUID-vkCmdDrawIndirectCount2KHR-OpExecutionMode-12241
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the segment spacing, they **must** be the same

- VUID-vkCmdDrawIndirectCount2KHR-OpExecutionMode-12242
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the output patch size, they **must** be the same
- VUID-vkCmdDrawIndirectCount2KHR-pInfo-13110
  If pInfo->maxDrawCount is greater than 0, pInfo->addressRange.size **must** be greater than or equal to (pInfo->maxDrawCount - 1) × pInfo->addressRange.stride + sizeof (VkDrawIndirectCommand)
- VUID-vkCmdDrawIndirectCount2KHR-pInfo-13111
  If pInfo->maxDrawCount is greater than 1, pInfo->addressRange.stride **must** not be 0
- VUID-vkCmdDrawIndirectCount2KHR-pInfo-13112
  If pInfo->addressRange.stride is not 0, pInfo->addressRange.stride **must** be greater than or equal to sizeof(VkDrawIndirectCommand)
- VUID-vkCmdDrawIndirectCount2KHR-pInfo-13113
  If pInfo->addressRange.stride is not 0, pInfo->addressRange.stride **must** be a multiple of 4
- VUID-vkCmdDrawIndirectCount2KHR-commandBuffer-13058
  commandBuffer **must** not be a protected command buffer
- VUID-vkCmdDrawIndirectCount2KHR-drawIndirectCount-13119
  If the drawIndirectCount feature is not enabled this function **must** not be used

Valid Usage (Implicit)

- VUID-vkCmdDrawIndirectCount2KHR-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDrawIndirectCount2KHR-pInfo-parameter
  pInfo **must** be a valid pointer to a valid VkDrawIndirectCount2InfoKHR structure
- VUID-vkCmdDrawIndirectCount2KHR-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdDrawIndirectCount2KHR-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDrawIndirectCount2KHR-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndirectCount2KHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawIndirectCount2KHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawIndirectCount2KHR is affected by conditional rendering

To record a non-indexed draw call with a draw call count sourced from a buffer, call:

```
// Provided by VK_VERSION_1_2
void vkCmdDrawIndirectCount(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    buffer,
    VkDeviceSize                                offset,
    VkBuffer                                    countBuffer,
    VkDeviceSize                                countBufferOffset,
    uint32_t                                    maxDrawCount,
    uint32_t                                    stride);
```

```
// Provided by VK_KHR_draw_indirect_count
// Equivalent to vkCmdDrawIndirectCount
void vkCmdDrawIndirectCountKHR(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    buffer,
    VkDeviceSize                                offset,
    VkBuffer                                    countBuffer,
    VkDeviceSize                                countBufferOffset,
    uint32_t                                    maxDrawCount,
    uint32_t                                    stride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `buffer` is the buffer containing draw parameters.

- `offset` is the byte offset into `buffer` where parameters begin.
- `countBuffer` is the buffer containing the draw count.
- `countBufferOffset` is the byte offset into `countBuffer` where the draw count begins.
- `maxDrawCount` specifies the maximum number of draws that will be executed. The actual number of executed draw calls is the minimum of the count specified in `countBuffer` and `maxDrawCount`.
- `stride` is the byte stride between successive sets of draw parameters.

`vkCmdDrawIndirectCount` behaves similarly to vkCmdDrawIndirect except that the draw count is read by the device from a buffer during execution. The command will read an unsigned 32-bit integer from `countBuffer` located at `countBufferOffset` and use this as the draw count.

Valid Usage

- VUID-vkCmdDrawIndirectCount-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawIndirectCount-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawIndirectCount-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawIndirectCount-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirectCount-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirectCount-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirectCount-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirectCount-None-08119

If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a VkDescriptorSetLayout that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident

- VUID-vkCmdDrawIndirectCount-magFilter-04553
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndirectCount-magFilter-09598
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndirectCount-mipmapMode-04770
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndirectCount-mipmapMode-09599
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndirectCount-unnormalizedCoordinates-09635
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1
- VUID-vkCmdDrawIndirectCount-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawIndirectCount-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdDrawIndirectCount-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any

of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias or any offset values

- VUID-vkCmdDrawIndirectCount-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawIndirectCount-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawIndirectCount-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDrawIndirectCount-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawIndirectCount-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawIndirectCount-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndirectCount-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndirectCount-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the OpTypeImage is Unknown, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirectCount-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the OpTypeImage is Unknown, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirectCount-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format

field of the OpTypeImage is Unknown, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT

- VUID-vkCmdDrawIndirectCount-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the OpTypeImage is Unknown, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirectCount-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDrawIndirectCount-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawIndirectCount-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDrawIndirectCount-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawIndirectCount-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for uniformBuffers, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirectCount-None-08612
  If the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirectCount-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for storageBuffers, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirectCount-None-08613
  If the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirectCount-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's viewType **must** match the Dim operand of the OpTypeImage as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawIndirectCount-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's format and the Sampled Type operand of the OpTypeImage **must** match
- VUID-vkCmdDrawIndirectCount-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawIndirectCount-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components
- VUID-vkCmdDrawIndirectCount-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawIndirectCount-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawIndirectCount-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawIndirectCount-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawIndirectCount-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawIndirectCount-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic

instructions through an OpTypeImage with a SampledType with a Width of 64 by this command

- VUID-vkCmdDrawIndirectCount-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawIndirectCount-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawIndirectCount-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawIndirectCount-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawIndirectCount-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawIndirectCount-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with flags that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawIndirectCount-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with flags that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDrawIndirectCount-None-12472

Descriptors dynamically accessed in bound descriptor heaps **must** be valid

- VUID-vkCmdDrawIndirectCount-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`
- VUID-vkCmdDrawIndirectCount-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdDrawIndirectCount-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdDrawIndirectCount-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDrawIndirectCount-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndirectCount-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound

to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress

- VUID-vkCmdDrawIndirectCount-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdDrawIndirectCount-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndirectCount-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndirectCount-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawIndirectCount-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawIndirectCount-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndirectCount-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either

VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdDrawIndirectCount-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndirectCount-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDrawIndirectCount-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDrawIndirectCount-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDrawIndirectCount-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirectCount-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to

must have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set

- VUID-vkCmdDrawIndirectCount-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdDrawIndirectCount-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirectCount-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawIndirectCount-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDrawIndirectCount-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDrawIndirectCount-protectedNoFault-11455

If protectedNoFault is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawIndirectCount-protectedNoFault-11456
  If protectedNoFault is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawIndirectCount-renderPass-02684
  The current render pass **must** be compatible with the renderPass member of the VkGraphicsPipelineCreateInfo structure specified when creating the VkPipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndirectCount-subpass-02685
  The subpass index of the current render pass **must** be equal to the subpass member of the VkGraphicsPipelineCreateInfo structure specified when creating the VkPipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndirectCount-OpTypeImage-07468
  If any shader executed by this pipeline accesses an OpTypeImage variable with a Dim operand of SubpassData, it **must** be decorated with an InputAttachmentIndex that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawIndirectCount-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' pInputAttachments [InputAttachmentIndex] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDrawIndirectCount-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a InputAttachmentIndex referenced by VkRenderingInputAttachmentIndexInfo, or no InputAttachmentIndex if VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex or VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex are NULL, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawIndirectCount-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an InputAttachmentIndex if both VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex and VkRenderingInputAttachmentIndexInfo

::pStencilInputAttachmentIndex are non-NULL

- VUID-vkCmdDrawIndirectCount-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an InputAttachmentIndex, the InputAttachmentIndex **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawIndirectCount-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectCount-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectCount-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectCount-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectCount-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirectCount-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirectCount-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirectCount-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawIndirectCount-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only

layout for the depth aspect, depth writes **must** be disabled

- VUID-vkCmdDrawIndirectCount-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawIndirectCount-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing

command

- VUID-vkCmdDrawIndirectCount-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::`maxMultiviewInstanceIndex`
- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndirectCount-None-07634
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-06666
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `sampleLocationsEnable` is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07846
  If the `depthBounds` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** match the `scissorCount` parameter of `vkCmdSetScissorWithCount`
- VUID-vkCmdDrawIndirectCount-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the

shader stages of the bound graphics pipeline write to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1

- VUID-vkCmdDrawIndirectCount-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the primitiveFragmentShadingRateWithMultipleViewports limit is not supported, and any shader object bound to a graphics stage writes to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1
- VUID-vkCmdDrawIndirectCount-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of blendEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirectCount-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and none of the following is enabled:
  - the multisampledRenderToSingleSampled feature

  then the current value of rasterizationSamples **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDrawIndirectCount-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDrawIndirectCount-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::viewMask equal to VkRenderingInfo::viewMask
- VUID-vkCmdDrawIndirectCount-colorAttachmentCount-06179
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08910
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have

been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline

- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08912
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawIndirectCount-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount
- VUID-vkCmdDrawIndirectCount-rasterizerDiscardEnable-09236

If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT ::maxDiscardRectangles

- VUID-vkCmdDrawIndirectCount-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo ::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo ::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->imageView
- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo ::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawIndirectCount-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawIndirectCount-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawIndirectCount-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of

the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline

- VUID-vkCmdDrawIndirectCount-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndirectCount-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawIndirectCount-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDrawIndirectCount-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawIndirectCount-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount-pDepthAttachment-08964

If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirectCount-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawIndirectCount-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawIndirectCount-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDrawIndirectCount-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the

current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and alphaToCoverageEnable was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-vkCmdDrawIndirectCount-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set alphaToCoverageEnable to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-vkCmdDrawIndirectCount-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07625
  If the alphaToOne feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07626
  If the `logicOp` feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07630
  If the `geometryStreams` feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07631
  If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07632
  If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `conservativeRasterizationMode` is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-conservativePointAndLineRasterization-07499

If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT

- VUID-vkCmdDrawIndirectCount-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-08669
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current `alphaToCoverageEnable`, `alphaToOneEnable` and `sampleShadingEnable` states **must** all be VK_FALSE
- VUID-vkCmdDrawIndirectCount-None-07639
  If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-09650
  If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of `depthClampEnable` is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawIndirectCount-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::`rasterizationSamples` parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawIndirectCount-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the `rasterizationSamples` parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectCount-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with

multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the rasterizationSamples member of that structure

- VUID-vkCmdDrawIndirectCount-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndirectCount-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndirectCount-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and an active color attachment current value of blendEnable is VK_TRUE, then the number of active color attachments **must** not exceed advancedBlendMaxColorAttachments
- VUID-vkCmdDrawIndirectCount-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE
- VUID-vkCmdDrawIndirectCount-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE
- VUID-vkCmdDrawIndirectCount-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and

VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE

- VUID-vkCmdDrawIndirectCount-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the rasterizationStream to zero
- VUID-vkCmdDrawIndirectCount-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawIndirectCount-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.width in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT ::maxSampleLocationGridSize.width as returned by

vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.height in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples
- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawIndirectCount-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled
- VUID-vkCmdDrawIndirectCount-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled
- VUID-vkCmdDrawIndirectCount-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawIndirectCount-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawIndirectCount-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawIndirectCount-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a nextStage including the corresponding bit to the shader object bound to the following graphics stage

in the logical pipeline

- VUID-vkCmdDrawIndirectCount-None-08684
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDrawIndirectCount-None-08685
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawIndirectCount-None-08686
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-vkCmdDrawIndirectCount-None-08687
  If there is no bound graphics pipeline, and the geometryShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawIndirectCount-None-08688
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawIndirectCount-None-08689
  If there is no bound graphics pipeline, and the taskShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawIndirectCount-None-08690
  If there is no bound graphics pipeline, and the meshShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndirectCount-None-08693
  If there is no bound graphics pipeline, and at least one of the taskShader and meshShader features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid VkShaderEXT bound, and the other **must** have no VkShaderEXT bound
- VUID-vkCmdDrawIndirectCount-None-08696
  If there is no bound graphics pipeline, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no VkShaderEXT bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawIndirectCount-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound

- VUID-vkCmdDrawIndirectCount-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any VkShaderEXT bound
- VUID-vkCmdDrawIndirectCount-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawIndirectCount-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts
- VUID-vkCmdDrawIndirectCount-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpDepthAttachmentReadEXT, the depthWriteEnable parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirectCount-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpStencilAttachmentReadEXT, the writeMask parameter in the last call to vkCmdSetStencilWriteMask **must** be 0
- VUID-vkCmdDrawIndirectCount-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the pColorWriteMasks parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawIndirectCount-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of Location for any output attachment statically used in the Fragment Execution Model executed by this command **must** be less than maxFragmentDualSrcAttachments
- VUID-vkCmdDrawIndirectCount-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::pColorAttachmentLocations in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance

- VUID-vkCmdDrawIndirectCount-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::pColorAttachmentInputIndices in the bound pipeline, if either pColorAttachmentInputIndices[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndirectCount-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndirectCount-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndirectCount-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawIndirectCount-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDrawIndirectCount-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawIndirectCount-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::flags which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawIndirectCount-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndirectCount-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not

contain a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT

- VUID-vkCmdDrawIndirectCount-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndirectCount-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndirectCount-None-11861
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndirectCount-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawIndirectCount-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount-None-11865

If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirectCount-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectCount-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for

each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline

- VUID-vkCmdDrawIndirectCount-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndirectCount-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndirectCount-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndirectCount-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndirectCount-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawIndirectCount-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndirectCount-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndirectCount-flags-13363
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set

layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT

- VUID-vkCmdDrawIndirectCount-None-04007
  All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or VK_NULL_HANDLE buffers bound
- VUID-vkCmdDrawIndirectCount-None-04008
  If the nullDescriptor feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be VK_NULL_HANDLE
- VUID-vkCmdDrawIndirectCount-None-02721
  If the robustBufferAccess feature is not enabled, and that pipeline was created without enabling VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS for vertexInputs, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in Vertex Input Description
- VUID-vkCmdDrawIndirectCount-format-10389
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::format or VkVertexInputAttributeDescription2EXT ::format is a packed format, and the legacyVertexAttributes feature is not enabled, the value of attribAddress, calculated as described in Vertex Input Calculation, **must** be a multiple of the size of the format
- VUID-vkCmdDrawIndirectCount-format-10390
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::format or VkVertexInputAttributeDescription2EXT ::format is not a packed format, and either the legacyVertexAttributes feature is not enabled or format has 64-bit components, the value of attribAddress, calculated as described in Vertex Input Calculation, **must** be a multiple of the component size of the format
- VUID-vkCmdDrawIndirectCount-None-07842
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled then vkCmdSetPrimitiveTopology **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-dynamicPrimitiveTopologyUnrestricted-07500
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled and the dynamicPrimitiveTopologyUnrestricted is VK_FALSE, then the primitiveTopology parameter of vkCmdSetPrimitiveTopology **must** be of the same topology class as the pipeline

VkPipelineInputAssemblyStateCreateInfo::topology state

- VUID-vkCmdDrawIndirectCount-primitiveTopology-10286
  If a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage is bound, then the current value of primitiveTopology **must** be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command
- VUID-vkCmdDrawIndirectCount-primitiveTopology-10747
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command, then a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage **must** be bound
- VUID-vkCmdDrawIndirectCount-primitiveTopology-10748
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_POINT_LIST prior to this drawing command, the maintenance5 feature is not enabled, both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT and VK_SHADER_STAGE_GEOMETRY_BIT stage are not bound, then the Vertex Execution Model **must** have a PointSize decorated variable that is statically written to
- VUID-vkCmdDrawIndirectCount-pStrides-04913
  If the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, but without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, then vkCmdBindVertexBuffers2 with a non-NULL pStrides parameter or vkCmdBindVertexBuffers3KHR with setStride set to VK_TRUE **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndirectCount-None-13118
  If the bound graphics pipeline was created without the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, and vkCmdBindVertexBuffers3KHR was called and not subsequently invalidated in the current command buffer prior to this draw command, the value of setStride in each of its pBindInfos elements must have been VK_FALSE
- VUID-vkCmdDrawIndirectCount-None-04914
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndirectCount-Input-07939
  If the maintenance9 feature is not enabled, and there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then all variables with the Input storage class decorated with Location in the Vertex Execution Model OpEntryPoint **must** contain a location in VkVertexInputAttributeDescription2EXT::location
- VUID-vkCmdDrawIndirectCount-Input-08734
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and either the legacyVertexAttributes feature is not enabled or the SPIR-V Type associated with a given Input variable of the corresponding Location in the Vertex Execution Model OpEntryPoint is 64-bit, then the numeric type associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be the same as VkVertexInputAttributeDescription2EXT::format

- VUID-vkCmdDrawIndirectCount-format-08936
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then the scalar width associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be 64-bit
- VUID-vkCmdDrawIndirectCount-format-08937
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and the scalar width associated with a Location decorated Input variable in the Vertex Execution Model OpEntryPoint is 64-bit, then the corresponding VkVertexInputAttributeDescription2EXT ::format **must** have a 64-bit component
- VUID-vkCmdDrawIndirectCount-None-09203
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then all Input variables at the corresponding Location in the Vertex Execution Model OpEntryPoint **must** not use components that are not present in the format
- VUID-vkCmdDrawIndirectCount-None-04875
  If there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with both a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage and the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state enabled, and the current value of primitiveTopology is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, then vkCmdSetPatchControlPointsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-04879
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-None-09637
  If the primitiveTopologyListRestart feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST,

VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled, then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE

- VUID-vkCmdDrawIndirectCount-None-10909
  If the `primitiveTopologyPatchListRestart` feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirectCount-stage-06481
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::`stage` member of any element of VkGraphicsPipelineCreateInfo::`pStages` set to VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndirectCount-None-08885
  There **must** be no shader object bound to either of the VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT stages
- VUID-vkCmdDrawIndirectCount-None-07619
  If a shader object is bound to the VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage and the VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT dynamic state enabled, then vkCmdSetTessellationDomainOriginEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectCount-OpExecutionMode-12239
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the type of subdivision, they **must** be the same
- VUID-vkCmdDrawIndirectCount-OpExecutionMode-12240
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the orientation of triangles, they **must** be the same
- VUID-vkCmdDrawIndirectCount-OpExecutionMode-12241
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the segment spacing, they **must** be the same
- VUID-vkCmdDrawIndirectCount-OpExecutionMode-12242
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the output patch size, they **must** be the same
- VUID-vkCmdDrawIndirectCount-buffer-02708

If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdDrawIndirectCount-buffer-02709
  `buffer` **must** have been created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirectCount-offset-02710
  `offset` **must** be a multiple of `4`
- VUID-vkCmdDrawIndirectCount-commandBuffer-02711
  `commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdDrawIndirectCount-countBuffer-02714
  If `countBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdDrawIndirectCount-countBuffer-02715
  `countBuffer` **must** have been created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirectCount-countBufferOffset-02716
  `countBufferOffset` **must** be a multiple of `4`
- VUID-vkCmdDrawIndirectCount-countBuffer-02717
  The count stored in `countBuffer` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDrawIndirectCount`
- VUID-vkCmdDrawIndirectCount-countBufferOffset-04129
  (`countBufferOffset` + `sizeof`(uint32_t)) **must** be less than or equal to the size of `countBuffer`
- VUID-vkCmdDrawIndirectCount-None-04445
  If the `drawIndirectCount` feature is not enabled this function **must** not be used
- VUID-vkCmdDrawIndirectCount-stride-03110
  `stride` **must** be a multiple of `4` and **must** be greater than or equal to sizeof(`VkDrawIndirectCommand`)
- VUID-vkCmdDrawIndirectCount-maxDrawCount-03111
  If `maxDrawCount` is greater than or equal to `1`, (`stride` × (`maxDrawCount` - 1) + `offset` + sizeof(`VkDrawIndirectCommand`)) **must** be less than or equal to the size of `buffer`

Valid Usage (Implicit)

- VUID-vkCmdDrawIndirectCount-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDrawIndirectCount-buffer-parameter
  `buffer` **must** be a valid VkBuffer handle
- VUID-vkCmdDrawIndirectCount-countBuffer-parameter
  `countBuffer` **must** be a valid VkBuffer handle
- VUID-vkCmdDrawIndirectCount-commandBuffer-recording
  `commandBuffer` **must** be in the recording state

- VUID-vkCmdDrawIndirectCount-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDrawIndirectCount-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndirectCount-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawIndirectCount-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdDrawIndirectCount-commonparent
  Each of `buffer`, `commandBuffer`, and `countBuffer` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawIndirectCount is affected by conditional rendering

To record an indexed indirect drawing command, call:

```
// Provided by VK_KHR_device_address_commands
void vkCmdDrawIndexedIndirect2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkDrawIndirect2InfoKHR*               pInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `pInfo` is a pointer to a VkDrawIndirect2InfoKHR structure defining parameters of this command.

vkCmdDrawIndexedIndirect2KHR behaves similarly to vkCmdDrawIndexed except that the parameters are read by the device from an address range during execution. pInfo->drawCount draws are executed by the command, with parameters taken from pInfo->addressRange starting at addressRange.address and increasing by addressRange.stride bytes for each successive draw. The parameters of each draw are encoded in an array of VkDrawIndexedIndirectCommand structures.

Valid Usage

- VUID-vkCmdDrawIndexedIndirect2KHR-None-08606
  If the shaderObject feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawIndexedIndirect2KHR-commandBuffer-02707
  If commandBuffer is an unprotected command buffer and protectedNoFault is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a VkDescriptorSetLayout that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawIndexedIndirect2KHR-magFilter-04553

If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT

- VUID-vkCmdDrawIndexedIndirect2KHR-magFilter-09598
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndexedIndirect2KHR-mipmapMode-04770
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndexedIndirect2KHR-mipmapMode-09599
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndexedIndirect2KHR-unnormalizedCoordinates-09635
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias or any offset values
- VUID-vkCmdDrawIndexedIndirect2KHR-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT

- VUID-vkCmdDrawIndexedIndirect2KHR-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawIndexedIndirect2KHR-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndexedIndirect2KHR-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndexedIndirect2KHR-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexedIndirect2KHR-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexedIndirect2KHR-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexedIndirect2KHR-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain

VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT

- VUID-vkCmdDrawIndexedIndirect2KHR-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDrawIndexedIndirect2KHR-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawIndexedIndirect2KHR-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawIndexedIndirect2KHR-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexedIndirect2KHR-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexedIndirect2KHR-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawIndexedIndirect2KHR-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match
- VUID-vkCmdDrawIndexedIndirect2KHR-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawIndexedIndirect2KHR-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have four components
- VUID-vkCmdDrawIndexedIndirect2KHR-OpImageWrite-04469
  If a VkBufferView is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawIndexedIndirect2KHR-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdDrawIndexedIndirect2KHR-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdDrawIndexedIndirect2KHR-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdDrawIndexedIndirect2KHR-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdDrawIndexedIndirect2KHR-sparseImageInt64Atomics-04474
  If the `sparseImageInt64Atomics` feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdDrawIndexedIndirect2KHR-sparseImageInt64Atomics-04475
  If the `sparseImageInt64Atomics` feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic

instructions through an OpTypeImage with a SampledType with a Width of 64 by this command

- VUID-vkCmdDrawIndexedIndirect2KHR-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawIndexedIndirect2KHR-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawIndexedIndirect2KHR-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawIndexedIndirect2KHR-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawIndexedIndirect2KHR-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with flags that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawIndexedIndirect2KHR-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with flags that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT,

VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`

- VUID-vkCmdDrawIndexedIndirect2KHR-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11304

If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8

- VUID-vkCmdDrawIndexedIndirect2KHR-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdDrawIndexedIndirect2KHR-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndexedIndirect2KHR-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping

using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`

- VUID-vkCmdDrawIndexedIndirect2KHR-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDrawIndexedIndirect2KHR-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDrawIndexedIndirect2KHR-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the

resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawIndexedIndirect2KHR-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawIndexedIndirect2KHR-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndexedIndirect2KHR-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndexedIndirect2KHR-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDrawIndexedIndirect2KHR-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by VkRenderingInputAttachmentIndexInfo, or no `InputAttachmentIndex` if VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` or VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are `NULL`, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawIndexedIndirect2KHR-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` and VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are non-`NULL`
- VUID-vkCmdDrawIndexedIndirect2KHR-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in VkRenderingInputAttachmentIndexInfo

- VUID-vkCmdDrawIndexedIndirect2KHR-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDrawIndexedIndirect2KHR-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP

- VUID-vkCmdDrawIndexedIndirect2KHR-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value

of stencilTestEnable is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect2KHR-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::maxMultiviewInstanceIndex
- VUID-vkCmdDrawIndexedIndirect2KHR-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07634
  If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-06666
  If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of sampleLocationsEnable is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior

to this drawing command

- VUID-vkCmdDrawIndexedIndirect2KHR-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07846
  If the `depthBounds` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not

subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect2KHR-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** match the `scissorCount` parameter of `vkCmdSetScissorWithCount`
- VUID-vkCmdDrawIndexedIndirect2KHR-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be 1
- VUID-vkCmdDrawIndexedIndirect2KHR-

primitiveFragmentShadingRateWithMultipleViewports-08642
If the primitiveFragmentShadingRateWithMultipleViewports limit is not supported, and any shader object bound to a graphics stage writes to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1

- VUID-vkCmdDrawIndexedIndirect2KHR-blendEnable-04727
If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of blendEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08644
If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and none of the following is enabled:
  - the multisampledRenderToSingleSampled feature

  then the current value of rasterizationSamples **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08876
If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDrawIndexedIndirect2KHR-viewMask-06178
If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::viewMask equal to VkRenderingInfo::viewMask
- VUID-vkCmdDrawIndexedIndirect2KHR-colorAttachmentCount-06179
If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndexedIndirect2KHR-dynamicRenderingUnusedAttachments-08910
If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirect2KHR-dynamicRenderingUnusedAttachments-08912
If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render

pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexedIndirect2KHR-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT ::discardRectangleCount
- VUID-vkCmdDrawIndexedIndirect2KHR-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is

VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT::maxDiscardRectangles

- VUID-vkCmdDrawIndexedIndirect2KHR-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect2KHR-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndexedIndirect2KHR-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect2KHR-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexedIndirect2KHR-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawIndexedIndirect2KHR-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect2KHR-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawIndexedIndirect2KHR-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawIndexedIndirect2KHR-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect2KHR-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirect2KHR-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance,

and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView

- VUID-vkCmdDrawIndexedIndirect2KHR-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawIndexedIndirect2KHR-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDrawIndexedIndirect2KHR-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawIndexedIndirect2KHR-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect2KHR-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect2KHR-pDepthAttachment-08964
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexedIndirect2KHR-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect2KHR-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect2KHR-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect2KHR-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawIndexedIndirect2KHR-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawIndexedIndirect2KHR-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07621

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect2KHR-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and `alphaToCoverageEnable` was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawIndexedIndirect2KHR-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set `alphaToCoverageEnable` to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07625
  If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07626
  If the `logicOp` feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled,

and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect2KHR-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07630
  If the geometryStreams feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07631
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07632
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of conservativeRasterizationMode is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-conservativePointAndLineRasterization-07499
  If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in

the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT

- VUID-vkCmdDrawIndexedIndirect2KHR-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08669
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and

the current lineRasterizationMode is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current alphaToCoverageEnable, alphaToOneEnable and sampleShadingEnable states **must** all be VK_FALSE

- VUID-vkCmdDrawIndexedIndirect2KHR-None-07639
  If the depthClipControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-09650
  If the depthClampControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of depthClampEnable is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawIndexedIndirect2KHR-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::rasterizationSamples parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirect2KHR-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the rasterizationSamples parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the rasterizationSamples member of that structure
- VUID-vkCmdDrawIndexedIndirect2KHR-firstAttachment-07476

If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments

- VUID-vkCmdDrawIndexedIndirect2KHR-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndexedIndirect2KHR-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE, then the number of active color attachments **must** not exceed `advancedBlendMaxColorAttachments`
- VUID-vkCmdDrawIndexedIndirect2KHR-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawIndexedIndirect2KHR-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawIndexedIndirect2KHR-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of

VK_TRUE

- VUID-vkCmdDrawIndexedIndirect2KHR-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the rasterizationStream to zero
- VUID-vkCmdDrawIndexedIndirect2KHR-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawIndexedIndirect2KHR-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirect2KHR-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndexedIndirect2KHR-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.width in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples
- VUID-vkCmdDrawIndexedIndirect2KHR-sampleLocationsEnable-07486

If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.height in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdDrawIndexedIndirect2KHR-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDrawIndexedIndirect2KHR-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirect2KHR-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirect2KHR-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirect2KHR-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or

VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled

- VUID-vkCmdDrawIndexedIndirect2KHR-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled
- VUID-vkCmdDrawIndexedIndirect2KHR-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawIndexedIndirect2KHR-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawIndexedIndirect2KHR-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a nextStage including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08684
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the

current command buffer with pStages with an element of VK_SHADER_STAGE_VERTEX_BIT

- VUID-vkCmdDrawIndexedIndirect2KHR-None-08685
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08686
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08687
  If there is no bound graphics pipeline, and the geometryShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08688
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08689
  If there is no bound graphics pipeline, and the taskShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08690
  If there is no bound graphics pipeline, and the meshShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08693
  If there is no bound graphics pipeline, and at least one of the taskShader and meshShader features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid VkShaderEXT bound, and the other **must** have no VkShaderEXT bound
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08696
  If there is no bound graphics pipeline, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no VkShaderEXT bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the

VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any VkShaderEXT bound

- VUID-vkCmdDrawIndexedIndirect2KHR-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts
- VUID-vkCmdDrawIndexedIndirect2KHR-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpDepthAttachmentReadEXT, the depthWriteEnable parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndexedIndirect2KHR-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpStencilAttachmentReadEXT, the writeMask parameter in the last call to vkCmdSetStencilWriteMask **must** be 0
- VUID-vkCmdDrawIndexedIndirect2KHR-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the pColorWriteMasks parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawIndexedIndirect2KHR-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of Location for any output attachment statically used in the Fragment Execution Model executed by this command **must** be less than maxFragmentDualSrcAttachments
- VUID-vkCmdDrawIndexedIndirect2KHR-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::pColorAttachmentLocations in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance
- VUID-vkCmdDrawIndexedIndirect2KHR-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::pColorAttachmentInputIndices in the bound

pipeline, if either pColorAttachmentInputIndices[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match

- VUID-vkCmdDrawIndexedIndirect2KHR-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndexedIndirect2KHR-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndexedIndirect2KHR-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDrawIndexedIndirect2KHR-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawIndexedIndirect2KHR-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::flags which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT

- VUID-vkCmdDrawIndexedIndirect2KHR-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndexedIndirect2KHR-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11861
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect2KHR-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value

of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexedIndirect2KHR-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndexedIndirect2KHR-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect2KHR-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndexedIndirect2KHR-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect2KHR-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline

- VUID-vkCmdDrawIndexedIndirect2KHR-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndexedIndirect2KHR-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndexedIndirect2KHR-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndexedIndirect2KHR-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndexedIndirect2KHR-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawIndexedIndirect2KHR-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndexedIndirect2KHR-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndexedIndirect2KHR-flags-13363
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT,

VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::`stageFlags` set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT

- VUID-vkCmdDrawIndexedIndirect2KHR-None-04007
  All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or VK_NULL_HANDLE buffers bound
- VUID-vkCmdDrawIndexedIndirect2KHR-None-04008
  If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be VK_NULL_HANDLE
- VUID-vkCmdDrawIndexedIndirect2KHR-None-02721
  If the `robustBufferAccess` feature is not enabled, and that pipeline was created without enabling VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS for `vertexInputs`, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in Vertex Input Description
- VUID-vkCmdDrawIndexedIndirect2KHR-format-10389
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is a packed format, and the `legacyVertexAttributes` feature is not enabled, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the size of the `format`
- VUID-vkCmdDrawIndexedIndirect2KHR-format-10390
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is not a packed format, and either the `legacyVertexAttributes` feature is not enabled or `format` has 64-bit components, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the component size of the `format`
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07842
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled then vkCmdSetPrimitiveTopology **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-dynamicPrimitiveTopologyUnrestricted-07500
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled and the `dynamicPrimitiveTopologyUnrestricted` is VK_FALSE, then the `primitiveTopology` parameter of `vkCmdSetPrimitiveTopology` **must** be of the same topology class as the pipeline VkPipelineInputAssemblyStateCreateInfo::`topology` state
- VUID-vkCmdDrawIndexedIndirect2KHR-primitiveTopology-10286

If a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage is bound, then the current value of `primitiveTopology` **must** be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect2KHR-primitiveTopology-10747
  If vkCmdSetPrimitiveTopology set `primitiveTopology` to VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command, then a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage **must** be bound
- VUID-vkCmdDrawIndexedIndirect2KHR-primitiveTopology-10748
  If vkCmdSetPrimitiveTopology set `primitiveTopology` to VK_PRIMITIVE_TOPOLOGY_POINT_LIST prior to this drawing command, the `maintenance5` feature is not enabled, both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT and VK_SHADER_STAGE_GEOMETRY_BIT stage are not bound, then the `Vertex Execution Model` **must** have a `PointSize` decorated variable that is statically written to
- VUID-vkCmdDrawIndexedIndirect2KHR-pStrides-04913
  If the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, but without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, then vkCmdBindVertexBuffers2 with a non-`NULL` `pStrides` parameter or vkCmdBindVertexBuffers3KHR with `setStride` set to VK_TRUE **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-13118
  If the bound graphics pipeline was created without the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, and vkCmdBindVertexBuffers3KHR was called and not subsequently invalidated in the current command buffer prior to this draw command, the value of `setStride` in each of its `pBindInfos` elements must have been VK_FALSE
- VUID-vkCmdDrawIndexedIndirect2KHR-None-04914
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndexedIndirect2KHR-Input-07939
  If the `maintenance9` feature is not enabled, and there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then all variables with the `Input` storage class decorated with `Location` in the `Vertex Execution Model OpEntryPoint` **must** contain a location in VkVertexInputAttributeDescription2EXT::`location`
- VUID-vkCmdDrawIndexedIndirect2KHR-Input-08734
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and either the `legacyVertexAttributes` feature is not enabled or the SPIR-V Type associated with a given

Input variable of the corresponding Location in the Vertex Execution Model OpEntryPoint is 64-bit, then the numeric type associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be the same as VkVertexInputAttributeDescription2EXT::format

- VUID-vkCmdDrawIndexedIndirect2KHR-format-08936
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then the scalar width associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be 64-bit
- VUID-vkCmdDrawIndexedIndirect2KHR-format-08937
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and the scalar width associated with a Location decorated Input variable in the Vertex Execution Model OpEntryPoint is 64-bit, then the corresponding VkVertexInputAttributeDescription2EXT ::format **must** have a 64-bit component
- VUID-vkCmdDrawIndexedIndirect2KHR-None-09203
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then all Input variables at the corresponding Location in the Vertex Execution Model OpEntryPoint **must** not use components that are not present in the format
- VUID-vkCmdDrawIndexedIndirect2KHR-None-04875
  If there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with both a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage and the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state enabled, and the current value of primitiveTopology is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, then vkCmdSetPatchControlPointsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-04879
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-None-09637
  If the primitiveTopologyListRestart feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, there is a shader object

bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled, then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE

- VUID-vkCmdDrawIndexedIndirect2KHR-None-10909
  If the `primitiveTopologyPatchListRestart` feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndexedIndirect2KHR-stage-06481
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::`stage` member of any element of VkGraphicsPipelineCreateInfo::`pStages` set to VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect2KHR-None-08885
  There **must** be no shader object bound to either of the VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT stages
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07619
  If a shader object is bound to the VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage and the VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT dynamic state enabled, then vkCmdSetTessellationDomainOriginEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect2KHR-OpExecutionMode-12239
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the type of subdivision, they **must** be the same
- VUID-vkCmdDrawIndexedIndirect2KHR-OpExecutionMode-12240
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the orientation of triangles, they **must** be the same
- VUID-vkCmdDrawIndexedIndirect2KHR-OpExecutionMode-12241
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the segment spacing, they **must** be the same
- VUID-vkCmdDrawIndexedIndirect2KHR-OpExecutionMode-12242
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the output patch size, they **must** be the same
- VUID-vkCmdDrawIndexedIndirect2KHR-None-07312
  If the `maintenance6` feature is not enabled, a valid index buffer **must** be bound

- VUID-vkCmdDrawIndexedIndirect2KHR-primitiveRestartIndex-12401
  If the primitive restart index for this draw operation was set with vkCmdSetPrimitiveRestartIndexEXT, then `primitiveRestartIndex` **must** not be greater than the maximum representable value for the bound VkIndexType
- VUID-vkCmdDrawIndexedIndirect2KHR-pInfo-13110
  If `pInfo->drawCount` is greater than `0`, `pInfo->addressRange.size` **must** be greater than or equal to (`pInfo->drawCount` - 1) × `pInfo->addressRange.stride` + `sizeof`(VkDrawIndexedIndirectCommand)
- VUID-vkCmdDrawIndexedIndirect2KHR-pInfo-13111
  If `pInfo->drawCount` is greater than 1, `pInfo->addressRange.stride` **must** not be 0
- VUID-vkCmdDrawIndexedIndirect2KHR-pInfo-13112
  If `pInfo->addressRange.stride` is not 0, `pInfo->addressRange.stride` **must** be greater than or equal to `sizeof`(VkDrawIndexedIndirectCommand)
- VUID-vkCmdDrawIndexedIndirect2KHR-pInfo-13113
  If `pInfo->addressRange.stride` is not 0, `pInfo->addressRange.stride` **must** be a multiple of `4`
- VUID-vkCmdDrawIndexedIndirect2KHR-commandBuffer-13059
  `commandBuffer` **must** not be a protected command buffer

Valid Usage (Implicit)

- VUID-vkCmdDrawIndexedIndirect2KHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDrawIndexedIndirect2KHR-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkDrawIndirect2InfoKHR structure
- VUID-vkCmdDrawIndexedIndirect2KHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdDrawIndexedIndirect2KHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDrawIndexedIndirect2KHR-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndexedIndirect2KHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawIndexedIndirect2KHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally

synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawIndexedIndirect2KHR is affected by conditional rendering

To record an indexed indirect drawing command, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDrawIndexedIndirect(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    buffer,
    VkDeviceSize                                offset,
    uint32_t                                    drawCount,
    uint32_t                                    stride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `buffer` is the buffer containing draw parameters.
- `offset` is the byte offset into `buffer` where parameters begin.
- `drawCount` is the number of draws to execute, and **can** be zero.
- `stride` is the byte stride between successive sets of draw parameters.

`vkCmdDrawIndexedIndirect` behaves similarly to vkCmdDrawIndexed except that the parameters are read by the device from a buffer during execution. `drawCount` draws are executed by the command, with parameters taken from `buffer` starting at `offset` and increasing by `stride` bytes for each successive draw. The parameters of each draw are encoded in an array of VkDrawIndexedIndirectCommand structures. If `drawCount` is less than or equal to one, `stride` is ignored.

Valid Usage

- VUID-vkCmdDrawIndexedIndirect-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawIndexedIndirect-None-08600

If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility

- VUID-vkCmdDrawIndexedIndirect-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawIndexedIndirect-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawIndexedIndirect-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndexedIndirect-magFilter-09598
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndexedIndirect-mipmapMode-04770

If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT

- VUID-vkCmdDrawIndexedIndirect-mipmapMode-09599
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndexedIndirect-unnormalizedCoordinates-09635
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDrawIndexedIndirect-None-08609
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `viewType` **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawIndexedIndirect-None-08610
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions with `ImplicitLod`, `Dref` or `Proj` in their name
- VUID-vkCmdDrawIndexedIndirect-None-08611
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions that include a LOD bias or any offset values
- VUID-vkCmdDrawIndexedIndirect-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawIndexedIndirect-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawIndexedIndirect-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDrawIndexedIndirect-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT

- VUID-vkCmdDrawIndexedIndirect-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawIndexedIndirect-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndexedIndirect-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndexedIndirect-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexedIndirect-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexedIndirect-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexedIndirect-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexedIndirect-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDrawIndexedIndirect-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawIndexedIndirect-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the

subresource accessible from the VkImageView as defined by the image layout matching rules

- VUID-vkCmdDrawIndexedIndirect-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawIndexedIndirect-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexedIndirect-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexedIndirect-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexedIndirect-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexedIndirect-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawIndexedIndirect-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match
- VUID-vkCmdDrawIndexedIndirect-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format

- VUID-vkCmdDrawIndexedIndirect-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components
- VUID-vkCmdDrawIndexedIndirect-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawIndexedIndirect-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawIndexedIndirect-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawIndexedIndirect-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawIndexedIndirect-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawIndexedIndirect-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawIndexedIndirect-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawIndexedIndirect-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawIndexedIndirect-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawIndexedIndirect-micromap-11636
  If an acceleration structure is accessed as a result of this command through an

OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE

- VUID-vkCmdDrawIndexedIndirect-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawIndexedIndirect-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with flags that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawIndexedIndirect-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with flags that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawIndexedIndirect-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a OpTypeStruct decorated with Block or BufferBlock using that mapping, the calculated offset for the resource heap **must** be a multiple of bufferDescriptorAlignment
- VUID-vkCmdDrawIndexedIndirect-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader

accesses an OpTypeImage or OpTypeSampledImage using that mapping, the calculated offset for the resource heap **must** be a multiple of imageDescriptorAlignment

- VUID-vkCmdDrawIndexedIndirect-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeSampler or OpTypeSampledImage using that mapping, the calculated offset for the sampler heap **must** be a multiple of samplerDescriptorAlignment
- VUID-vkCmdDrawIndexedIndirect-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDrawIndexedIndirect-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndexedIndirect-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndexedIndirect-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdDrawIndexedIndirect-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping

- VUID-vkCmdDrawIndexedIndirect-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndexedIndirect-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawIndexedIndirect-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawIndexedIndirect-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndexedIndirect-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndexedIndirect-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndexedIndirect-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDrawIndexedIndirect-None-11376
  If a bound shader was created as a VkShaderEXT with the

VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT

- VUID-vkCmdDrawIndexedIndirect-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDrawIndexedIndirect-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndexedIndirect-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndexedIndirect-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdDrawIndexedIndirect-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must**

have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set

- VUID-vkCmdDrawIndexedIndirect-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawIndexedIndirect-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDrawIndexedIndirect-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDrawIndexedIndirect-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawIndexedIndirect-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawIndexedIndirect-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the

VkGraphicsPipelineCreateInfo structure specified when creating the VkPipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS

- VUID-vkCmdDrawIndexedIndirect-subpass-02685
  The subpass index of the current render pass **must** be equal to the subpass member of the VkGraphicsPipelineCreateInfo structure specified when creating the VkPipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndexedIndirect-OpTypeImage-07468
  If any shader executed by this pipeline accesses an OpTypeImage variable with a Dim operand of SubpassData, it **must** be decorated with an InputAttachmentIndex that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawIndexedIndirect-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' pInputAttachments [InputAttachmentIndex] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDrawIndexedIndirect-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a InputAttachmentIndex referenced by VkRenderingInputAttachmentIndexInfo, or no InputAttachmentIndex if VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex or VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex are NULL, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawIndexedIndirect-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an InputAttachmentIndex if both VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex and VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex are non-NULL
- VUID-vkCmdDrawIndexedIndirect-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an InputAttachmentIndex, the InputAttachmentIndex **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawIndexedIndirect-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirect-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirect-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way

other than as an attachment by this command

- VUID-vkCmdDrawIndexedIndirect-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirect-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawIndexedIndirect-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawIndexedIndirect-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawIndexedIndirect-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawIndexedIndirect-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDrawIndexedIndirect-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawIndexedIndirect-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the

current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthBiasEnable is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and an active color attachment current value of blendEnable is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthBoundsTestEnable is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::`maxMultiviewInstanceIndex`
- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndexedIndirect-None-07634
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-06666
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `sampleLocationsEnable` is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in

the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07846
  If the `depthBounds` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-viewportCount-03419

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the viewportCount parameter of vkCmdSetViewportWithCount **must** match the scissorCount parameter of vkCmdSetScissorWithCount

- VUID-vkCmdDrawIndexedIndirect-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of logicOpEnable is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the primitiveFragmentShadingRateWithMultipleViewports limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1
- VUID-vkCmdDrawIndexedIndirect-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the primitiveFragmentShadingRateWithMultipleViewports limit is not supported, and any shader object bound to a graphics stage writes to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1
- VUID-vkCmdDrawIndexedIndirect-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain

VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of `blendEnable` **must** be VK_FALSE

- VUID-vkCmdDrawIndexedIndirect-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of `rasterizerDiscardEnable` is VK_FALSE, and none of the following is enabled:

  - the `multisampledRenderToSingleSampled` feature

  then the current value of `rasterizationSamples` **must** be the same as the current color and/or depth/stencil attachments

- VUID-vkCmdDrawIndexedIndirect-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDrawIndexedIndirect-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::`viewMask` equal to VkRenderingInfo::`viewMask`
- VUID-vkCmdDrawIndexedIndirect-colorAttachmentCount-06179
  If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::`colorAttachmentCount` equal to VkRenderingInfo::`colorAttachmentCount`
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08910
  If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `imageView` not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::`pColorAttachmentFormats` used to create the bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08912
  If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `imageView` equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::`pColorAttachmentFormats` used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08911
  If the `dynamicRenderingUnusedAttachments` feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `imageView` not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::`pColorAttachmentFormats` used to create the bound

graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexedIndirect-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawIndexedIndirect-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT ::discardRectangleCount
- VUID-vkCmdDrawIndexedIndirect-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT ::maxDiscardRectangles
- VUID-vkCmdDrawIndexedIndirect-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of

VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexedIndirect-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawIndexedIndirect-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirect-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndexedIndirect-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawIndexedIndirect-pNext-07935
  If this command is called inside a render pass instance started with

vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples

- VUID-vkCmdDrawIndexedIndirect-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawIndexedIndirect-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect-pDepthAttachment-08964
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo ::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo ::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo

::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexedIndirect-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo ::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawIndexedIndirect-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawIndexedIndirect-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDrawIndexedIndirect-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and `alphaToCoverageEnable` was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawIndexedIndirect-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set `alphaToCoverageEnable` to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawIndexedIndirect-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07625
  If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07626
  If the `logicOp` feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07629

If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07630
  If the `geometryStreams` feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07631
  If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07632
  If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `conservativeRasterizationMode` is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-conservativePointAndLineRasterization-07499
  If the `VK_EXT_conservative_rasterization` extension is enabled, `conservativePointAndLineRasterization` is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of `conservativeRasterizationMode` **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-vkCmdDrawIndexedIndirect-None-07633
  If the `depthClipEnable` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07636
  If the `VK_EXT_provoking_vertex` extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created

with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-08666
  If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-08669
  If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07849
  If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stippledLineEnable` is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current `alphaToCoverageEnable`, `alphaToOneEnable` and `sampleShadingEnable` states **must** all be VK_FALSE
- VUID-vkCmdDrawIndexedIndirect-None-07639
  If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-09650

If the depthClampControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of depthClampEnable is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawIndexedIndirect-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::rasterizationSamples parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirect-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the rasterizationSamples parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirect-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the rasterizationSamples member of that structure
- VUID-vkCmdDrawIndexedIndirect-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndexedIndirect-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to

vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments

- VUID-vkCmdDrawIndexedIndirect-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE, then the number of active color attachments **must** not exceed `advancedBlendMaxColorAttachments`
- VUID-vkCmdDrawIndexedIndirect-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawIndexedIndirect-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawIndexedIndirect-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawIndexedIndirect-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the `rasterizationStream` to zero
- VUID-vkCmdDrawIndexedIndirect-sampleLocationsPerPixel-07482

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with

- VUID-vkCmdDrawIndexedIndirect-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.width in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT ::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples
- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.height in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT ::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirect-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled
- VUID-vkCmdDrawIndexedIndirect-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled

- VUID-vkCmdDrawIndexedIndirect-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawIndexedIndirect-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawIndexedIndirect-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawIndexedIndirect-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a nextStage including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDrawIndexedIndirect-None-08684
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDrawIndexedIndirect-None-08685
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawIndexedIndirect-None-08686
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-vkCmdDrawIndexedIndirect-None-08687

If there is no bound graphics pipeline, and the geometryShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_GEOMETRY_BIT

- VUID-vkCmdDrawIndexedIndirect-None-08688
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawIndexedIndirect-None-08689
  If there is no bound graphics pipeline, and the taskShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect-None-08690
  If there is no bound graphics pipeline, and the meshShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect-None-08693
  If there is no bound graphics pipeline, and at least one of the taskShader and meshShader features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid VkShaderEXT bound, and the other **must** have no VkShaderEXT bound
- VUID-vkCmdDrawIndexedIndirect-None-08696
  If there is no bound graphics pipeline, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no VkShaderEXT bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawIndexedIndirect-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDrawIndexedIndirect-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any VkShaderEXT bound
- VUID-vkCmdDrawIndexedIndirect-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawIndexedIndirect-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts
- VUID-vkCmdDrawIndexedIndirect-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics

pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpDepthAttachmentReadEXT, the depthWriteEnable parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE

- VUID-vkCmdDrawIndexedIndirect-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpStencilAttachmentReadEXT, the writeMask parameter in the last call to vkCmdSetStencilWriteMask **must** be 0
- VUID-vkCmdDrawIndexedIndirect-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the pColorWriteMasks parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawIndexedIndirect-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of Location for any output attachment statically used in the Fragment Execution Model executed by this command **must** be less than maxFragmentDualSrcAttachments
- VUID-vkCmdDrawIndexedIndirect-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::pColorAttachmentLocations in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance
- VUID-vkCmdDrawIndexedIndirect-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::pColorAttachmentInputIndices in the bound pipeline, if either pColorAttachmentInputIndices[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndexedIndirect-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndexedIndirect-None-10928

If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match

- VUID-vkCmdDrawIndexedIndirect-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDrawIndexedIndirect-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawIndexedIndirect-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::flags which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawIndexedIndirect-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndexedIndirect-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndexedIndirect-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT ::customResolve as VK_TRUE
- VUID-vkCmdDrawIndexedIndirect-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndexedIndirect-None-11861

If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount

- VUID-vkCmdDrawIndexedIndirect-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirect-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a

custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexedIndirect-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirect-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirect-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndexedIndirect-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the

sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView

- VUID-vkCmdDrawIndexedIndirect-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndexedIndirect-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndexedIndirect-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawIndexedIndirect-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndexedIndirect-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndexedIndirect-flags-13363
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndexedIndirect-None-04007
  All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or VK_NULL_HANDLE buffers bound
- VUID-vkCmdDrawIndexedIndirect-None-04008
  If the nullDescriptor feature is not enabled, all vertex input bindings accessed via vertex

input variables declared in the vertex shader entry point's interface **must** not be VK_NULL_HANDLE

- VUID-vkCmdDrawIndexedIndirect-None-02721
  If the `robustBufferAccess` feature is not enabled, and that pipeline was created without enabling VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS for `vertexInputs`, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in Vertex Input Description
- VUID-vkCmdDrawIndexedIndirect-format-10389
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is a packed format, and the `legacyVertexAttributes` feature is not enabled, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the size of the `format`
- VUID-vkCmdDrawIndexedIndirect-format-10390
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is not a packed format, and either the `legacyVertexAttributes` feature is not enabled or `format` has 64-bit components, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the component size of the `format`
- VUID-vkCmdDrawIndexedIndirect-None-07842
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled then vkCmdSetPrimitiveTopology **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-dynamicPrimitiveTopologyUnrestricted-07500
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled and the `dynamicPrimitiveTopologyUnrestricted` is VK_FALSE, then the `primitiveTopology` parameter of `vkCmdSetPrimitiveTopology` **must** be of the same topology class as the pipeline VkPipelineInputAssemblyStateCreateInfo::`topology` state
- VUID-vkCmdDrawIndexedIndirect-primitiveTopology-10286
  If a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage is bound, then the current value of `primitiveTopology` **must** be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-primitiveTopology-10747
  If vkCmdSetPrimitiveTopology set `primitiveTopology` to VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command, then a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage **must** be bound
- VUID-vkCmdDrawIndexedIndirect-primitiveTopology-10748
  If vkCmdSetPrimitiveTopology set `primitiveTopology` to VK_PRIMITIVE_TOPOLOGY_POINT_LIST prior to this drawing command, the `maintenance5` feature is not enabled, both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT and

VK_SHADER_STAGE_GEOMETRY_BIT stage are not bound, then the Vertex Execution Model **must** have a PointSize decorated variable that is statically written to

- VUID-vkCmdDrawIndexedIndirect-pStrides-04913
  If the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, but without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, then vkCmdBindVertexBuffers2 with a non-NULL pStrides parameter or vkCmdBindVertexBuffers3KHR with setStride set to VK_TRUE **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndexedIndirect-None-13118
  If the bound graphics pipeline was created without the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, and vkCmdBindVertexBuffers3KHR was called and not subsequently invalidated in the current command buffer prior to this draw command, the value of setStride in each of its pBindInfos elements must have been VK_FALSE
- VUID-vkCmdDrawIndexedIndirect-None-04914
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndexedIndirect-Input-07939
  If the maintenance9 feature is not enabled, and there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then all variables with the Input storage class decorated with Location in the Vertex Execution Model OpEntryPoint **must** contain a location in VkVertexInputAttributeDescription2EXT::location
- VUID-vkCmdDrawIndexedIndirect-Input-08734
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and either the legacyVertexAttributes feature is not enabled or the SPIR-V Type associated with a given Input variable of the corresponding Location in the Vertex Execution Model OpEntryPoint is 64-bit, then the numeric type associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be the same as VkVertexInputAttributeDescription2EXT::format
- VUID-vkCmdDrawIndexedIndirect-format-08936
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then the scalar width associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be 64-bit

- VUID-vkCmdDrawIndexedIndirect-format-08937
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and the scalar width associated with a `Location` decorated `Input` variable in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the corresponding VkVertexInputAttributeDescription2EXT ::`format` **must** have a 64-bit component
- VUID-vkCmdDrawIndexedIndirect-None-09203
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::`format` has a 64-bit component, then all `Input` variables at the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** not use components that are not present in the format
- VUID-vkCmdDrawIndexedIndirect-None-04875
  If there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with both a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage and the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state enabled, and the current value of `primitiveTopology` is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, then vkCmdSetPatchControlPointsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-04879
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-09637
  If the `primitiveTopologyListRestart` feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled, then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndexedIndirect-None-10909
  If the `primitiveTopologyPatchListRestart` feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndexedIndirect-stage-06481
  The bound graphics pipeline **must** not have been created with the

VkPipelineShaderStageCreateInfo::stage member of any element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT

- VUID-vkCmdDrawIndexedIndirect-None-08885
  There **must** be no shader object bound to either of the VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT stages
- VUID-vkCmdDrawIndexedIndirect-None-07619
  If a shader object is bound to the VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage and the VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT dynamic state enabled, then vkCmdSetTessellationDomainOriginEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-OpExecutionMode-12239
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the type of subdivision, they **must** be the same
- VUID-vkCmdDrawIndexedIndirect-OpExecutionMode-12240
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the orientation of triangles, they **must** be the same
- VUID-vkCmdDrawIndexedIndirect-OpExecutionMode-12241
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the segment spacing, they **must** be the same
- VUID-vkCmdDrawIndexedIndirect-OpExecutionMode-12242
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the output patch size, they **must** be the same
- VUID-vkCmdDrawIndexedIndirect-buffer-02708
  If buffer is non-sparse then it **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-vkCmdDrawIndexedIndirect-buffer-02709
  buffer **must** have been created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndexedIndirect-offset-02710
  offset **must** be a multiple of 4
- VUID-vkCmdDrawIndexedIndirect-commandBuffer-02711
  commandBuffer **must** not be a protected command buffer
- VUID-vkCmdDrawIndexedIndirect-drawCount-02718
  If the multiDrawIndirect feature is not enabled, drawCount **must** be 0 or 1

- VUID-vkCmdDrawIndexedIndirect-drawCount-02719
  `drawCount` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxDrawIndirectCount`
- VUID-vkCmdDrawIndexedIndirect-None-07312
  If the `maintenance6` feature is not enabled, a valid index buffer **must** be bound
- VUID-vkCmdDrawIndexedIndirect-primitiveRestartIndex-12401
  If the primitive restart index for this draw operation was set with vkCmdSetPrimitiveRestartIndexEXT, then `primitiveRestartIndex` **must** not be greater than the maximum representable value for the bound VkIndexType
- VUID-vkCmdDrawIndexedIndirect-drawCount-00528
  If `drawCount` is greater than `1`, `stride` **must** be a multiple of `4` and **must** be greater than or equal to `sizeof`(`VkDrawIndexedIndirectCommand`)
- VUID-vkCmdDrawIndexedIndirect-drawCount-00539
  If `drawCount` is equal to `1`, (`offset` + `sizeof`(`VkDrawIndexedIndirectCommand`)) **must** be less than or equal to the size of `buffer`
- VUID-vkCmdDrawIndexedIndirect-drawCount-00540
  If `drawCount` is greater than `1`, (`stride` × (`drawCount` - 1) + `offset` + `sizeof`(`VkDrawIndexedIndirectCommand`)) **must** be less than or equal to the size of `buffer`

Valid Usage (Implicit)

- VUID-vkCmdDrawIndexedIndirect-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDrawIndexedIndirect-buffer-parameter
  `buffer` **must** be a valid VkBuffer handle
- VUID-vkCmdDrawIndexedIndirect-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdDrawIndexedIndirect-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDrawIndexedIndirect-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndexedIndirect-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawIndexedIndirect-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdDrawIndexedIndirect-commonparent
  Both of `buffer`, and `commandBuffer` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawIndexedIndirect is affected by conditional rendering

The `VkDrawIndexedIndirectCommand` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDrawIndexedIndirectCommand {
    uint32_t    indexCount;
    uint32_t    instanceCount;
    uint32_t    firstIndex;
    int32_t     vertexOffset;
    uint32_t    firstInstance;
} VkDrawIndexedIndirectCommand;
```

- `indexCount` is the number of vertices to draw.
- `instanceCount` is the number of instances to draw.
- `firstIndex` is the base index within the index buffer.
- `vertexOffset` is the value added to the vertex index before indexing into the vertex buffer.
- `firstInstance` is the instance ID of the first instance to draw.

The members of `VkDrawIndexedIndirectCommand` have the same meaning as the similarly named parameters of vkCmdDrawIndexed.

Valid Usage

- VUID-VkDrawIndexedIndirectCommand-pNext-09461
  If the bound graphics pipeline state was created with

VkPipelineVertexInputDivisorStateCreateInfo in the pNext chain of VkGraphicsPipelineCreateInfo::pVertexInputState, any member of VkPipelineVertexInputDivisorStateCreateInfo::pVertexBindingDivisors has a value other than 1 in divisor, and VkPhysicalDeviceVertexAttributeDivisorProperties::supportsNonZeroFirstInstance is VK_FALSE, then firstInstance **must** be 0

- VUID-VkDrawIndexedIndirectCommand-None-09462
  If shader objects are used for drawing or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, any member of the pVertexBindingDescriptions parameter to the vkCmdSetVertexInputEXT call that sets this dynamic state has a value other than 1 in divisor, and VkPhysicalDeviceVertexAttributeDivisorProperties::supportsNonZeroFirstInstance is VK_FALSE, then firstInstance **must** be 0
- VUID-VkDrawIndexedIndirectCommand-robustBufferAccess2-08798
  If the robustBufferAccess2 feature is not enabled, (indexSize × (firstIndex + indexCount)) **must** be less than or equal to the size of the bound index buffer range, with indexSize being based on the type specified by indexType, and the other parameters sourced from this command
- VUID-VkDrawIndexedIndirectCommand-firstInstance-00554
  If the drawIndirectFirstInstance feature is not enabled, firstInstance **must** be 0

To record a indexed indirect drawing command, call:

```
// Provided by VK_KHR_device_address_commands with VK_KHR_draw_indirect_count or
VK_VERSION_1_2
void vkCmdDrawIndexedIndirectCount2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkDrawIndirectCount2InfoKHR*          pInfo);
```

- commandBuffer is the command buffer into which the command is recorded.
- pInfo is a pointer to a VkDrawIndirectCount2InfoKHR structure defining parameters of this command.

vkCmdDrawIndexedIndirectCount2KHR behaves similarly to vkCmdDrawIndexedIndirect2KHR except that the draw count is read by the device from an address range during execution. The command will read an unsigned 32-bit integer from pInfo->countAddressRange and use this as the draw count.

Valid Usage

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08606
  If the shaderObject feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts

compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility

- VUID-vkCmdDrawIndexedIndirectCount2KHR-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawIndexedIndirectCount2KHR-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-magFilter-09598
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-mipmapMode-04770
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and

compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT

- VUID-vkCmdDrawIndexedIndirectCount2KHR-mipmapMode-09599
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-unnormalizedCoordinates-09635
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias or any offset values
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with

VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY

- VUID-vkCmdDrawIndexedIndirectCount2KHR-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndexedIndirectCount2KHR-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndexedIndirectCount2KHR-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawIndexedIndirectCount2KHR-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexedIndirectCount2KHR-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexedIndirectCount2KHR-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawIndexedIndirectCount2KHR-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match
- VUID-vkCmdDrawIndexedIndirectCount2KHR-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawIndexedIndirectCount2KHR-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using

OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components

- VUID-vkCmdDrawIndexedIndirectCount2KHR-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawIndexedIndirectCount2KHR-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawIndexedIndirectCount2KHR-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawIndexedIndirectCount2KHR-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawIndexedIndirectCount2KHR-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawIndexedIndirectCount2KHR-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawIndexedIndirectCount2KHR-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a

VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE

- VUID-vkCmdDrawIndexedIndirectCount2KHR-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawIndexedIndirectCount2KHR-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with flags that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawIndexedIndirectCount2KHR-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with flags that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a OpTypeStruct decorated with Block or BufferBlock using that mapping, the calculated offset for the resource heap **must** be a multiple of bufferDescriptorAlignment
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeImage or OpTypeSampledImage using that mapping, the calculated offset for the resource heap **must** be a multiple of imageDescriptorAlignment

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound

to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses

a push constant value, that value **must** have been set by vkCmdPushDataEXT

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDrawIndexedIndirectCount2KHR-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDrawIndexedIndirectCount2KHR-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawIndexedIndirectCount2KHR-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawIndexedIndirectCount2KHR-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS

- VUID-vkCmdDrawIndexedIndirectCount2KHR-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndexedIndirectCount2KHR-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by VkRenderingInputAttachmentIndexInfo, or no `InputAttachmentIndex` if VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` or VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are `NULL`, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` and VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are non-`NULL`
- VUID-vkCmdDrawIndexedIndirectCount2KHR-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-10797

If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been

called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of

stencilTestEnable is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount2KHR-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::maxMultiviewInstanceIndex
- VUID-vkCmdDrawIndexedIndirectCount2KHR-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07634
  If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-06666
  If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of sampleLocationsEnable is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07844

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07846
  If the `depthBounds` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and

VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the viewportCount parameter of vkCmdSetViewportWithCount **must** match the scissorCount parameter of vkCmdSetScissorWithCount

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of logicOpEnable is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the primitiveFragmentShadingRateWithMultipleViewports limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1
- VUID-vkCmdDrawIndexedIndirectCount2KHR-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the primitiveFragmentShadingRateWithMultipleViewports limit is not supported, and any shader object bound to a graphics stage writes to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1
- VUID-vkCmdDrawIndexedIndirectCount2KHR-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of blendEnable **must** be VK_FALSE

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of `rasterizerDiscardEnable` is VK_FALSE, and none of the following is enabled:

  - the `multisampledRenderToSingleSampled` feature

  then the current value of `rasterizationSamples` **must** be the same as the current color and/or depth/stencil attachments

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering

- VUID-vkCmdDrawIndexedIndirectCount2KHR-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::`viewMask` equal to VkRenderingInfo::`viewMask`

- VUID-vkCmdDrawIndexedIndirectCount2KHR-colorAttachmentCount-06179
  If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::`colorAttachmentCount` equal to VkRenderingInfo::`colorAttachmentCount`

- VUID-vkCmdDrawIndexedIndirectCount2KHR-dynamicRenderingUnusedAttachments-08910
  If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `imageView` not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::`pColorAttachmentFormats` used to create the bound graphics pipeline

- VUID-vkCmdDrawIndexedIndirectCount2KHR-dynamicRenderingUnusedAttachments-08912
  If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `imageView` equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::`pColorAttachmentFormats` used to create the bound pipeline equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexedIndirectCount2KHR-dynamicRenderingUnusedAttachments-08911
  If the `dynamicRenderingUnusedAttachments` feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `imageView` not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::`pColorAttachmentFormats` used to create the bound

graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT ::discardRectangleCount
- VUID-vkCmdDrawIndexedIndirectCount2KHR-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT ::maxDiscardRectangles
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount2KHR-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndexedIndirectCount2KHR-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount2KHR-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount2KHR-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView

- VUID-vkCmdDrawIndexedIndirectCount2KHR-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount2KHR-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawIndexedIndirectCount2KHR-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawIndexedIndirectCount2KHR-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirectCount2KHR-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndexedIndirectCount2KHR-multisampledRenderToSingleSampled-

07287
If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView

- VUID-vkCmdDrawIndexedIndirectCount2KHR-pNext-07935
If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDrawIndexedIndirectCount2KHR-renderPass-06198
If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pColorAttachments-08963
If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pColorAttachments-11539
If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pDepthAttachment-08964
If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo ::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pDepthAttachment-11540
If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a

graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexedIndirectCount2KHR-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount2KHR-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawIndexedIndirectCount2KHR-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawIndexedIndirectCount2KHR-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then

vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and `alphaToCoverageEnable` was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawIndexedIndirectCount2KHR-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set `alphaToCoverageEnable` to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07625
  If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07626
  If the `logicOp` feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07630
  If the `geometryStreams` feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07631
  If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07632
  If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `conservativeRasterizationMode` is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-conservativePointAndLineRasterization-07499
  If the `VK_EXT_conservative_rasterization` extension is enabled, `conservativePointAndLineRasterization` is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of `conservativeRasterizationMode` **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08669
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current lineRasterizationMode is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current alphaToCoverageEnable, alphaToOneEnable and sampleShadingEnable states **must** all be

VK_FALSE

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07639
  If the depthClipControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-09650
  If the depthClampControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of depthClampEnable is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawIndexedIndirectCount2KHR-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::rasterizationSamples parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirectCount2KHR-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the rasterizationSamples parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the rasterizationSamples member of that structure
- VUID-vkCmdDrawIndexedIndirectCount2KHR-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the

current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments

- VUID-vkCmdDrawIndexedIndirectCount2KHR-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndexedIndirectCount2KHR-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and an active color attachment current value of blendEnable is VK_TRUE, then the number of active color attachments **must** not exceed advancedBlendMaxColorAttachments
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE
- VUID-vkCmdDrawIndexedIndirectCount2KHR-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE

- VUID-vkCmdDrawIndexedIndirectCount2KHR-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the `rasterizationStream` to zero
- VUID-vkCmdDrawIndexedIndirectCount2KHR-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawIndexedIndirectCount2KHR-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndexedIndirectCount2KHR-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::`maxSampleLocationGridSize.width` as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a `samples` parameter equaling `rasterizationSamples`
- VUID-vkCmdDrawIndexedIndirectCount2KHR-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.height in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdDrawIndexedIndirectCount2KHR-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample

- VUID-vkCmdDrawIndexedIndirectCount2KHR-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawIndexedIndirectCount2KHR-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawIndexedIndirectCount2KHR-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawIndexedIndirectCount2KHR-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode

state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled

- VUID-vkCmdDrawIndexedIndirectCount2KHR-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled
- VUID-vkCmdDrawIndexedIndirectCount2KHR-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawIndexedIndirectCount2KHR-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawIndexedIndirectCount2KHR-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a nextStage including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08684
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_VERTEX_BIT

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08685
  If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08686
  If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08687
  If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08688
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08689
  If there is no bound graphics pipeline, and the `taskShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08690
  If there is no bound graphics pipeline, and the `meshShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08693
  If there is no bound graphics pipeline, and at least one of the `taskShader` and `meshShader` features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid `VkShaderEXT` bound, and the other **must** have no `VkShaderEXT` bound
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08696
  If there is no bound graphics pipeline, and a valid `VkShaderEXT` is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no `VkShaderEXT` bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any `VkShaderEXT` bound

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to vkCmdSetStencilWriteMask **must** be `0`
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the `pColorWriteMasks` parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawIndexedIndirectCount2KHR-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of `Location` for any output attachment statically used in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::`pColorAttachmentLocations` in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::`pColorAttachmentInputIndices` in the bound pipeline, if either `pColorAttachmentInputIndices`[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound

pipeline in an InputAttachmentIndex decoration, the two values **must** match

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawIndexedIndirectCount2KHR-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::flags which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and

vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE

- VUID-vkCmdDrawIndexedIndirectCount2KHR-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11861
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount2KHR-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound

graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the `dynamicRenderingUnusedAttachments` feature is not enabled, and VkRenderingInfo::`pDepthAttachment->resolveImageView` was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::`depthAttachmentFormat` used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::`pDepthAttachment->resolveImageView`
- VUID-vkCmdDrawIndexedIndirectCount2KHR-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the `dynamicRenderingUnusedAttachments` feature is enabled, VkRenderingInfo::`pDepthAttachment->resolveImageView` was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::`depthAttachmentFormat` used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::`pDepthAttachment->resolveImageView`, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the `dynamicRenderingUnusedAttachments` feature is not enabled, and VkRenderingInfo::`pStencilAttachment->resolveImageView` was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::`stencilAttachmentFormat` used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the `dynamicRenderingUnusedAttachments` feature is not enabled, and VkRenderingInfo::`pStencilAttachment->resolveImageView` was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::`stencilAttachmentFormat` used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::`pStencilAttachment->resolveImageView`
- VUID-vkCmdDrawIndexedIndirectCount2KHR-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the `dynamicRenderingUnusedAttachments` feature is enabled, VkRenderingInfo::`pStencilAttachment->resolveImageView` was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::`stencilAttachmentFormat` used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::`pStencilAttachment->resolveImageView`, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount2KHR-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::`colorAttachmentCount` parameter greater than `0` and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::`pColorAttachments` array with a `resolveImageView` not equal to VK_NULL_HANDLE, the `resolveImageView` **must** have been created with a

sample count equal to the value of rasterizationSamples for the bound graphics pipeline

- VUID-vkCmdDrawIndexedIndirectCount2KHR-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndexedIndirectCount2KHR-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndexedIndirectCount2KHR-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndexedIndirectCount2KHR-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawIndexedIndirectCount2KHR-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndexedIndirectCount2KHR-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndexedIndirectCount2KHR-flags-13363
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::stageFlags set with

VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::`stageFlags` set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-04007
  All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or VK_NULL_HANDLE buffers bound
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-04008
  If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be VK_NULL_HANDLE
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-02721
  If the `robustBufferAccess` feature is not enabled, and that pipeline was created without enabling VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS for `vertexInputs`, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in Vertex Input Description
- VUID-vkCmdDrawIndexedIndirectCount2KHR-format-10389
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is a packed format, and the `legacyVertexAttributes` feature is not enabled, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the size of the `format`
- VUID-vkCmdDrawIndexedIndirectCount2KHR-format-10390
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is not a packed format, and either the `legacyVertexAttributes` feature is not enabled or `format` has 64-bit components, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the component size of the `format`
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07842
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled then vkCmdSetPrimitiveTopology **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-dynamicPrimitiveTopologyUnrestricted-07500
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled and the `dynamicPrimitiveTopologyUnrestricted` is VK_FALSE, then the `primitiveTopology` parameter of `vkCmdSetPrimitiveTopology` **must** be of the same topology class as the pipeline VkPipelineInputAssemblyStateCreateInfo::`topology` state

- VUID-vkCmdDrawIndexedIndirectCount2KHR-primitiveTopology-10286
  If a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage is bound, then the current value of `primitiveTopology` **must** be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-primitiveTopology-10747
  If vkCmdSetPrimitiveTopology set `primitiveTopology` to VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command, then a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage **must** be bound
- VUID-vkCmdDrawIndexedIndirectCount2KHR-primitiveTopology-10748
  If vkCmdSetPrimitiveTopology set `primitiveTopology` to VK_PRIMITIVE_TOPOLOGY_POINT_LIST prior to this drawing command, the `maintenance5` feature is not enabled, both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT and VK_SHADER_STAGE_GEOMETRY_BIT stage are not bound, then the `Vertex Execution Model` **must** have a `PointSize` decorated variable that is statically written to
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pStrides-04913
  If the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, but without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, then vkCmdBindVertexBuffers2 with a non-`NULL` `pStrides` parameter or vkCmdBindVertexBuffers3KHR with `setStride` set to VK_TRUE **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-13118
  If the bound graphics pipeline was created without the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, and vkCmdBindVertexBuffers3KHR was called and not subsequently invalidated in the current command buffer prior to this draw command, the value of `setStride` in each of its `pBindInfos` elements must have been VK_FALSE
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-04914
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-Input-07939
  If the `maintenance9` feature is not enabled, and there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then all variables with the `Input` storage class decorated with `Location` in the `Vertex Execution Model OpEntryPoint` **must** contain a location in VkVertexInputAttributeDescription2EXT ::`location`
- VUID-vkCmdDrawIndexedIndirectCount2KHR-Input-08734
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and either the

legacyVertexAttributes feature is not enabled or the SPIR-V Type associated with a given Input variable of the corresponding Location in the Vertex Execution Model OpEntryPoint is 64-bit, then the numeric type associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be the same as VkVertexInputAttributeDescription2EXT::format

- VUID-vkCmdDrawIndexedIndirectCount2KHR-format-08936
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then the scalar width associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be 64-bit
- VUID-vkCmdDrawIndexedIndirectCount2KHR-format-08937
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and the scalar width associated with a Location decorated Input variable in the Vertex Execution Model OpEntryPoint is 64-bit, then the corresponding VkVertexInputAttributeDescription2EXT::format **must** have a 64-bit component
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-09203
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then all Input variables at the corresponding Location in the Vertex Execution Model OpEntryPoint **must** not use components that are not present in the format
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-04875
  If there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with both a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage and the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state enabled, and the current value of primitiveTopology is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, then vkCmdSetPatchControlPointsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-04879
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-09637
  If the primitiveTopologyListRestart feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, or

VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled, then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE

- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-10909
  If the `primitiveTopologyPatchListRestart` feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndexedIndirectCount2KHR-stage-06481
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::`stage` member of any element of VkGraphicsPipelineCreateInfo::`pStages` set to VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-08885
  There **must** be no shader object bound to either of the VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT stages
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07619
  If a shader object is bound to the VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage and the VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT dynamic state enabled, then vkCmdSetTessellationDomainOriginEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount2KHR-OpExecutionMode-12239
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the type of subdivision, they **must** be the same
- VUID-vkCmdDrawIndexedIndirectCount2KHR-OpExecutionMode-12240
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the orientation of triangles, they **must** be the same
- VUID-vkCmdDrawIndexedIndirectCount2KHR-OpExecutionMode-12241
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the segment spacing, they **must** be the same
- VUID-vkCmdDrawIndexedIndirectCount2KHR-OpExecutionMode-12242
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the output patch size, they **must** be the same
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pInfo-13110
  If `pInfo->maxDrawCount` is greater than `0`, `pInfo->addressRange.size` **must** be greater than or

equal to (pInfo->maxDrawCount - 1) × pInfo->addressRange.stride + sizeof (VkDrawIndexedIndirectCommand)

- VUID-vkCmdDrawIndexedIndirectCount2KHR-pInfo-13111
  If pInfo->maxDrawCount is greater than 1, pInfo->addressRange.stride **must** not be 0
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pInfo-13112
  If pInfo->addressRange.stride is not 0, pInfo->addressRange.stride **must** be greater than or equal to sizeof(VkDrawIndexedIndirectCommand)
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pInfo-13113
  If pInfo->addressRange.stride is not 0, pInfo->addressRange.stride **must** be a multiple of 4
- VUID-vkCmdDrawIndexedIndirectCount2KHR-None-07312
  If the maintenance6 feature is not enabled, a valid index buffer **must** be bound
- VUID-vkCmdDrawIndexedIndirectCount2KHR-primitiveRestartIndex-12401
  If the primitive restart index for this draw operation was set with vkCmdSetPrimitiveRestartIndexEXT, then primitiveRestartIndex **must** not be greater than the maximum representable value for the bound VkIndexType
- VUID-vkCmdDrawIndexedIndirectCount2KHR-commandBuffer-13060
  commandBuffer **must** not be a protected command buffer
- VUID-vkCmdDrawIndexedIndirectCount2KHR-drawIndirectCount-13120
  If the drawIndirectCount feature is not enabled this function **must** not be used

Valid Usage (Implicit)

- VUID-vkCmdDrawIndexedIndirectCount2KHR-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDrawIndexedIndirectCount2KHR-pInfo-parameter
  pInfo **must** be a valid pointer to a valid VkDrawIndirectCount2InfoKHR structure
- VUID-vkCmdDrawIndexedIndirectCount2KHR-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdDrawIndexedIndirectCount2KHR-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDrawIndexedIndirectCount2KHR-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndexedIndirectCount2KHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawIndexedIndirectCount2KHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawIndexedIndirectCount2KHR is affected by conditional rendering

To record an indexed draw call with a draw call count sourced from a buffer, call:

```
// Provided by VK_VERSION_1_2
void vkCmdDrawIndexedIndirectCount(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    buffer,
    VkDeviceSize                                offset,
    VkBuffer                                    countBuffer,
    VkDeviceSize                                countBufferOffset,
    uint32_t                                    maxDrawCount,
    uint32_t                                    stride);
```

```
// Provided by VK_KHR_draw_indirect_count
// Equivalent to vkCmdDrawIndexedIndirectCount
void vkCmdDrawIndexedIndirectCountKHR(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    buffer,
    VkDeviceSize                                offset,
    VkBuffer                                    countBuffer,
    VkDeviceSize                                countBufferOffset,
    uint32_t                                    maxDrawCount,
    uint32_t                                    stride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `buffer` is the buffer containing draw parameters.

- `offset` is the byte offset into `buffer` where parameters begin.
- `countBuffer` is the buffer containing the draw count.
- `countBufferOffset` is the byte offset into `countBuffer` where the draw count begins.
- `maxDrawCount` specifies the maximum number of draws that will be executed. The actual number of executed draw calls is the minimum of the count specified in `countBuffer` and `maxDrawCount`.
- `stride` is the byte stride between successive sets of draw parameters.

`vkCmdDrawIndexedIndirectCount` behaves similarly to vkCmdDrawIndexedIndirect except that the draw count is read by the device from a buffer during execution. The command will read an unsigned 32-bit integer from `countBuffer` located at `countBufferOffset` and use this as the draw count.

Valid Usage

- VUID-vkCmdDrawIndexedIndirectCount-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawIndexedIndirectCount-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawIndexedIndirectCount-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawIndexedIndirectCount-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT

- VUID-vkCmdDrawIndexedIndirectCount-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a VkDescriptorSetLayout that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawIndexedIndirectCount-magFilter-04553
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndexedIndirectCount-magFilter-09598
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndexedIndirectCount-mipmapMode-04770
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndexedIndirectCount-mipmapMode-09599
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndexedIndirectCount-unnormalizedCoordinates-09635
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1
- VUID-vkCmdDrawIndexedIndirectCount-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawIndexedIndirectCount-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdDrawIndexedIndirectCount-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample

a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions that include a LOD bias or any offset values

- VUID-vkCmdDrawIndexedIndirectCount-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawIndexedIndirectCount-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawIndexedIndirectCount-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDrawIndexedIndirectCount-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawIndexedIndirectCount-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndexedIndirectCount-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndexedIndirectCount-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexedIndirectCount-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexedIndirectCount-OpTypeImage-07029

For any VkBufferView being written as a storage texel buffer where the image format field of the OpTypeImage is Unknown, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT

- VUID-vkCmdDrawIndexedIndirectCount-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the OpTypeImage is Unknown, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndexedIndirectCount-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDrawIndexedIndirectCount-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawIndexedIndirectCount-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDrawIndexedIndirectCount-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawIndexedIndirectCount-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for uniformBuffers, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexedIndirectCount-None-08612
  If the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexedIndirectCount-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for storageBuffers, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to

the same pipeline bind point

- VUID-vkCmdDrawIndexedIndirectCount-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndexedIndirectCount-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawIndexedIndirectCount-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match
- VUID-vkCmdDrawIndexedIndirectCount-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawIndexedIndirectCount-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have four components
- VUID-vkCmdDrawIndexedIndirectCount-OpImageWrite-04469
  If a VkBufferView is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawIndexedIndirectCount-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdDrawIndexedIndirectCount-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdDrawIndexedIndirectCount-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdDrawIndexedIndirectCount-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdDrawIndexedIndirectCount-sparseImageInt64Atomics-04474

If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command

- VUID-vkCmdDrawIndexedIndirectCount-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawIndexedIndirectCount-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawIndexedIndirectCount-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawIndexedIndirectCount-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawIndexedIndirectCount-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawIndexedIndirectCount-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with flags that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawIndexedIndirectCount-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with flags that includes

VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT

- VUID-vkCmdDrawIndexedIndirectCount-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawIndexedIndirectCount-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`
- VUID-vkCmdDrawIndexedIndirectCount-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdDrawIndexedIndirectCount-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdDrawIndexedIndirectCount-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDrawIndexedIndirectCount-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping

- VUID-vkCmdDrawIndexedIndirectCount-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndexedIndirectCount-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdDrawIndexedIndirectCount-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndexedIndirectCount-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndexedIndirectCount-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawIndexedIndirectCount-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawIndexedIndirectCount-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndexedIndirectCount-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this

command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdDrawIndexedIndirectCount-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndexedIndirectCount-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDrawIndexedIndirectCount-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDrawIndexedIndirectCount-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDrawIndexedIndirectCount-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndexedIndirectCount-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or

VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set

- VUID-vkCmdDrawIndexedIndirectCount-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdDrawIndexedIndirectCount-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndexedIndirectCount-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawIndexedIndirectCount-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDrawIndexedIndirectCount-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color

- VUID-vkCmdDrawIndexedIndirectCount-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawIndexedIndirectCount-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawIndexedIndirectCount-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndexedIndirectCount-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndexedIndirectCount-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawIndexedIndirectCount-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDrawIndexedIndirectCount-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by VkRenderingInputAttachmentIndexInfo, or no `InputAttachmentIndex` if VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` or VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are `NULL`, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawIndexedIndirectCount-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both VkRenderingInputAttachmentIndexInfo

::pDepthInputAttachmentIndex and VkRenderingInputAttachmentIndexInfo ::pStencilInputAttachmentIndex are non-NULL

- VUID-vkCmdDrawIndexedIndirectCount-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an InputAttachmentIndex, the InputAttachmentIndex **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawIndexedIndirectCount-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirectCount-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirectCount-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirectCount-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirectCount-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawIndexedIndirectCount-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawIndexedIndirectCount-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawIndexedIndirectCount-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawIndexedIndirectCount-None-06886

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled

- VUID-vkCmdDrawIndexedIndirectCount-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawIndexedIndirectCount-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called

and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::`maxMultiviewInstanceIndex`
- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndexedIndirectCount-None-07634
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-06666
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `sampleLocationsEnable` is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not

subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07846
  If the `depthBounds` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in

the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** match the `scissorCount` parameter of `vkCmdSetScissorWithCount`
- VUID-vkCmdDrawIndexedIndirectCount-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the

VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1

- VUID-vkCmdDrawIndexedIndirectCount-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the primitiveFragmentShadingRateWithMultipleViewports limit is not supported, and any shader object bound to a graphics stage writes to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1
- VUID-vkCmdDrawIndexedIndirectCount-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of blendEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndexedIndirectCount-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and none of the following is enabled:
  - the multisampledRenderToSingleSampled feature

  then the current value of rasterizationSamples **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDrawIndexedIndirectCount-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDrawIndexedIndirectCount-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::viewMask equal to VkRenderingInfo::viewMask
- VUID-vkCmdDrawIndexedIndirectCount-colorAttachmentCount-06179
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08910
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo

::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline

- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08912
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawIndexedIndirectCount-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount

- VUID-vkCmdDrawIndexedIndirectCount-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT ::maxDiscardRectangles
- VUID-vkCmdDrawIndexedIndirectCount-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo ::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo ::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->imageView
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo ::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawIndexedIndirectCount-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of

the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline

- VUID-vkCmdDrawIndexedIndirectCount-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndexedIndirectCount-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawIndexedIndirectCount-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDrawIndexedIndirectCount-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawIndexedIndirectCount-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount-pDepthAttachment-08964

If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexedIndirectCount-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawIndexedIndirectCount-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawIndexedIndirectCount-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDrawIndexedIndirectCount-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a

graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and `alphaToCoverageEnable` was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawIndexedIndirectCount-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set `alphaToCoverageEnable` to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawIndexedIndirectCount-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07625
  If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then

vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07626
  If the `logicOp` feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07630
  If the `geometryStreams` feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07631
  If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07632
  If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `conservativeRasterizationMode` is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not

subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-conservativePointAndLineRasterization-07499
  If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-vkCmdDrawIndexedIndirectCount-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-08669
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been

called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current `alphaToCoverageEnable`, `alphaToOneEnable` and `sampleShadingEnable` states **must** all be VK_FALSE
- VUID-vkCmdDrawIndexedIndirectCount-None-07639
  If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-09650
  If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of `depthClampEnable` is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawIndexedIndirectCount-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::`rasterizationSamples` parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirectCount-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the `rasterizationSamples` parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirectCount-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is

bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the rasterizationSamples member of that structure

- VUID-vkCmdDrawIndexedIndirectCount-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndexedIndirectCount-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndexedIndirectCount-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and an active color attachment current value of blendEnable is VK_TRUE, then the number of active color attachments **must** not exceed advancedBlendMaxColorAttachments
- VUID-vkCmdDrawIndexedIndirectCount-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE
- VUID-vkCmdDrawIndexedIndirectCount-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE
- VUID-vkCmdDrawIndexedIndirectCount-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics

pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE

- VUID-vkCmdDrawIndexedIndirectCount-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the `rasterizationStream` to zero
- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the

sampleLocationsInfo.sampleLocationGridSize.width in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.height in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples
- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current

value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawIndexedIndirectCount-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled
- VUID-vkCmdDrawIndexedIndirectCount-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled
- VUID-vkCmdDrawIndexedIndirectCount-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawIndexedIndirectCount-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawIndexedIndirectCount-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawIndexedIndirectCount-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-nextStage-10745

For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a `nextStage` including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline

- VUID-vkCmdDrawIndexedIndirectCount-None-08684
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDrawIndexedIndirectCount-None-08685
  If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawIndexedIndirectCount-None-08686
  If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-vkCmdDrawIndexedIndirectCount-None-08687
  If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawIndexedIndirectCount-None-08688
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawIndexedIndirectCount-None-08689
  If there is no bound graphics pipeline, and the `taskShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount-None-08690
  If there is no bound graphics pipeline, and the `meshShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount-None-08693
  If there is no bound graphics pipeline, and at least one of the `taskShader` and `meshShader` features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid `VkShaderEXT` bound, and the other **must** have no `VkShaderEXT` bound
- VUID-vkCmdDrawIndexedIndirectCount-None-08696
  If there is no bound graphics pipeline, and a valid `VkShaderEXT` is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no `VkShaderEXT` bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawIndexedIndirectCount-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the

VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound

- VUID-vkCmdDrawIndexedIndirectCount-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any `VkShaderEXT` bound
- VUID-vkCmdDrawIndexedIndirectCount-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawIndexedIndirectCount-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts
- VUID-vkCmdDrawIndexedIndirectCount-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndexedIndirectCount-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to vkCmdSetStencilWriteMask **must** be `0`
- VUID-vkCmdDrawIndexedIndirectCount-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the `pColorWriteMasks` parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawIndexedIndirectCount-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of `Location` for any output attachment statically used in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`
- VUID-vkCmdDrawIndexedIndirectCount-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::`pColorAttachmentLocations` in the bound pipeline

must match the value for the corresponding locations set currently in the current render pass instance

- VUID-vkCmdDrawIndexedIndirectCount-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::pColorAttachmentInputIndices in the bound pipeline, if either pColorAttachmentInputIndices[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndexedIndirectCount-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndexedIndirectCount-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndexedIndirectCount-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDrawIndexedIndirectCount-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawIndexedIndirectCount-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::flags which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawIndexedIndirectCount-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT

- VUID-vkCmdDrawIndexedIndirectCount-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndexedIndirectCount-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndexedIndirectCount-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndexedIndirectCount-None-11861
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndexedIndirectCount-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirectCount-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be

VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndexedIndirectCount-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndexedIndirectCount-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a

VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline

- VUID-vkCmdDrawIndexedIndirectCount-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndexedIndirectCount-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndexedIndirectCount-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndexedIndirectCount-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndexedIndirectCount-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawIndexedIndirectCount-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndexedIndirectCount-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndexedIndirectCount-flags-13363

If there are any shader objects bound that were created with `flags` that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::`stageFlags` set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::`stageFlags` set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT

- VUID-vkCmdDrawIndexedIndirectCount-None-04007
  All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or VK_NULL_HANDLE buffers bound
- VUID-vkCmdDrawIndexedIndirectCount-None-04008
  If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be VK_NULL_HANDLE
- VUID-vkCmdDrawIndexedIndirectCount-None-02721
  If the `robustBufferAccess` feature is not enabled, and that pipeline was created without enabling VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS for `vertexInputs`, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in Vertex Input Description
- VUID-vkCmdDrawIndexedIndirectCount-format-10389
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is a packed format, and the `legacyVertexAttributes` feature is not enabled, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the size of the `format`
- VUID-vkCmdDrawIndexedIndirectCount-format-10390
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is not a packed format, and either the `legacyVertexAttributes` feature is not enabled or `format` has 64-bit components, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the component size of the `format`
- VUID-vkCmdDrawIndexedIndirectCount-None-07842
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled then vkCmdSetPrimitiveTopology **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-dynamicPrimitiveTopologyUnrestricted-07500
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled and the

dynamicPrimitiveTopologyUnrestricted is VK_FALSE, then the primitiveTopology parameter of vkCmdSetPrimitiveTopology **must** be of the same topology class as the pipeline VkPipelineInputAssemblyStateCreateInfo::topology state

- VUID-vkCmdDrawIndexedIndirectCount-primitiveTopology-10286
  If a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage is bound, then the current value of primitiveTopology **must** be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-primitiveTopology-10747
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command, then a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage **must** be bound
- VUID-vkCmdDrawIndexedIndirectCount-primitiveTopology-10748
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_POINT_LIST prior to this drawing command, the maintenance5 feature is not enabled, both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT and VK_SHADER_STAGE_GEOMETRY_BIT stage are not bound, then the Vertex Execution Model **must** have a PointSize decorated variable that is statically written to
- VUID-vkCmdDrawIndexedIndirectCount-pStrides-04913
  If the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, but without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, then vkCmdBindVertexBuffers2 with a non-NULL pStrides parameter or vkCmdBindVertexBuffers3KHR with setStride set to VK_TRUE **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndexedIndirectCount-None-13118
  If the bound graphics pipeline was created without the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, and vkCmdBindVertexBuffers3KHR was called and not subsequently invalidated in the current command buffer prior to this draw command, the value of setStride in each of its pBindInfos elements must have been VK_FALSE
- VUID-vkCmdDrawIndexedIndirectCount-None-04914
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndexedIndirectCount-Input-07939
  If the maintenance9 feature is not enabled, and there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then all variables with the Input storage class decorated with Location in the Vertex Execution Model OpEntryPoint **must** contain a location in VkVertexInputAttributeDescription2EXT ::location
- VUID-vkCmdDrawIndexedIndirectCount-Input-08734

If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and either the legacyVertexAttributes feature is not enabled or the SPIR-V Type associated with a given Input variable of the corresponding Location in the Vertex Execution Model OpEntryPoint is 64-bit, then the numeric type associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be the same as VkVertexInputAttributeDescription2EXT::format

- VUID-vkCmdDrawIndexedIndirectCount-format-08936
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then the scalar width associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be 64-bit
- VUID-vkCmdDrawIndexedIndirectCount-format-08937
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and the scalar width associated with a Location decorated Input variable in the Vertex Execution Model OpEntryPoint is 64-bit, then the corresponding VkVertexInputAttributeDescription2EXT ::format **must** have a 64-bit component
- VUID-vkCmdDrawIndexedIndirectCount-None-09203
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then all Input variables at the corresponding Location in the Vertex Execution Model OpEntryPoint **must** not use components that are not present in the format
- VUID-vkCmdDrawIndexedIndirectCount-None-04875
  If there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with both a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage and the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state enabled, and the current value of primitiveTopology is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, then vkCmdSetPatchControlPointsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-04879
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-09637
  If the primitiveTopologyListRestart feature is not enabled, the input assembly is

VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled, then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE

- VUID-vkCmdDrawIndexedIndirectCount-None-10909
  If the `primitiveTopologyPatchListRestart` feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndexedIndirectCount-stage-06481
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::`stage` member of any element of VkGraphicsPipelineCreateInfo::`pStages` set to VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndexedIndirectCount-None-08885
  There **must** be no shader object bound to either of the VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT stages
- VUID-vkCmdDrawIndexedIndirectCount-None-07619
  If a shader object is bound to the VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage and the VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT dynamic state enabled, then vkCmdSetTessellationDomainOriginEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-OpExecutionMode-12239
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the type of subdivision, they **must** be the same
- VUID-vkCmdDrawIndexedIndirectCount-OpExecutionMode-12240
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the orientation of triangles, they **must** be the same
- VUID-vkCmdDrawIndexedIndirectCount-OpExecutionMode-12241
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the segment spacing, they **must** be the same
- VUID-vkCmdDrawIndexedIndirectCount-OpExecutionMode-12242
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the output patch size, they **must** be the same

- VUID-vkCmdDrawIndexedIndirectCount-buffer-02708
  If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdDrawIndexedIndirectCount-buffer-02709
  `buffer` **must** have been created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndexedIndirectCount-offset-02710
  `offset` **must** be a multiple of `4`
- VUID-vkCmdDrawIndexedIndirectCount-commandBuffer-02711
  `commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdDrawIndexedIndirectCount-countBuffer-02714
  If `countBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdDrawIndexedIndirectCount-countBuffer-02715
  `countBuffer` **must** have been created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndexedIndirectCount-countBufferOffset-02716
  `countBufferOffset` **must** be a multiple of `4`
- VUID-vkCmdDrawIndexedIndirectCount-countBuffer-02717
  The count stored in `countBuffer` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDrawIndirectCount`
- VUID-vkCmdDrawIndexedIndirectCount-countBufferOffset-04129
  (`countBufferOffset` + `sizeof`(uint32_t)) **must** be less than or equal to the size of `countBuffer`
- VUID-vkCmdDrawIndexedIndirectCount-None-04445
  If the `drawIndirectCount` feature is not enabled this function **must** not be used
- VUID-vkCmdDrawIndexedIndirectCount-None-07312
  If the `maintenance6` feature is not enabled, a valid index buffer **must** be bound
- VUID-vkCmdDrawIndexedIndirectCount-primitiveRestartIndex-12401
  If the primitive restart index for this draw operation was set with vkCmdSetPrimitiveRestartIndexEXT, then `primitiveRestartIndex` **must** not be greater than the maximum representable value for the bound VkIndexType
- VUID-vkCmdDrawIndexedIndirectCount-stride-03142
  `stride` **must** be a multiple of `4` and **must** be greater than or equal to sizeof(`VkDrawIndexedIndirectCommand`)
- VUID-vkCmdDrawIndexedIndirectCount-maxDrawCount-03143
  If `maxDrawCount` is greater than or equal to `1`, (`stride` × (`maxDrawCount` - 1) + `offset` + sizeof(`VkDrawIndexedIndirectCommand`)) **must** be less than or equal to the size of `buffer`

Valid Usage (Implicit)

- VUID-vkCmdDrawIndexedIndirectCount-commandBuffer-parameter

`commandBuffer` **must** be a valid VkCommandBuffer handle

- VUID-vkCmdDrawIndexedIndirectCount-buffer-parameter
  `buffer` **must** be a valid VkBuffer handle
- VUID-vkCmdDrawIndexedIndirectCount-countBuffer-parameter
  `countBuffer` **must** be a valid VkBuffer handle
- VUID-vkCmdDrawIndexedIndirectCount-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdDrawIndexedIndirectCount-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDrawIndexedIndirectCount-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndexedIndirectCount-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawIndexedIndirectCount-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdDrawIndexedIndirectCount-commonparent
  Each of `buffer`, `commandBuffer`, and `countBuffer` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawIndexedIndirectCount is affected by conditional rendering

25.4.1. Drawing Transform Feedback

It is possible to draw vertex data that was previously captured during active transform feedback by binding one or more of the transform feedback buffers as vertex buffers. A pipeline barrier is required between using the buffers as transform feedback buffers and vertex buffers to ensure all writes to the transform feedback buffers are visible when the data is read as vertex attributes. The source access is VK_ACCESS_TRANSFORM_FEEDBACK_WRITE_BIT_EXT and the destination access is VK_ACCESS_VERTEX_ATTRIBUTE_READ_BIT for the pipeline stages VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT and VK_PIPELINE_STAGE_VERTEX_INPUT_BIT respectively. The value written to the counter buffer by vkCmdEndTransformFeedbackEXT **can** be used to determine the vertex count for the draw. A pipeline barrier is required between using the counter buffer for `vkCmdEndTransformFeedbackEXT` and `vkCmdDrawIndirectByteCountEXT` where the source access is VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT and the destination access is VK_ACCESS_INDIRECT_COMMAND_READ_BIT for the pipeline stages VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT and VK_PIPELINE_STAGE_DRAW_INDIRECT_BIT respectively.

To record a non-indexed draw call, where the vertex count is based on a byte count read from a memory range and the passed in vertex stride parameter, call:

```
// Provided by VK_KHR_device_address_commands with VK_EXT_transform_feedback
void vkCmdDrawIndirectByteCount2EXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    instanceCount,
    uint32_t                                    firstInstance,
    const VkBindTransformFeedbackBuffer2InfoEXT* pCounterInfo,
    uint32_t                                    counterOffset,
    uint32_t                                    vertexStride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `instanceCount` is the number of instances to draw.
- `firstInstance` is the instance ID of the first instance to draw.
- `pCounterInfo` is a pointer to a VkBindTransformFeedbackBuffer2InfoEXT defining the range of memory that the byte count is read from.
- `counterOffset` is subtracted from the byte count.
- `vertexStride` is the stride in bytes between each element of the vertex data that is used to calculate the vertex count from the counter value. This value is typically the same value that was used in the graphics pipeline state when the transform feedback was captured as the `XfbStride`.

When the command is executed, primitives are assembled in the same way as done with vkCmdDraw except the `vertexCount` is calculated based on the byte count read from `counterBuffer` at offset `counterBufferOffset`. The assembled primitives execute the bound graphics pipeline.

The effective `vertexCount` is calculated as follows:

```
const uint32_t * counterBufferPtr = (const uint8_t *)counterRange.address;
vertexCount = floor(max(0, (*counterBufferPtr - counterOffset)) / vertexStride);
```

The effective `firstVertex` is zero.

Valid Usage

- VUID-vkCmdDrawIndirectByteCount2EXT-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawIndirectByteCount2EXT-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawIndirectByteCount2EXT-magFilter-04553

If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT

- VUID-vkCmdDrawIndirectByteCount2EXT-magFilter-09598
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndirectByteCount2EXT-mipmapMode-04770
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndirectByteCount2EXT-mipmapMode-09599
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndirectByteCount2EXT-unnormalizedCoordinates-09635
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias or any offset values
- VUID-vkCmdDrawIndirectByteCount2EXT-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT

- VUID-vkCmdDrawIndirectByteCount2EXT-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawIndirectByteCount2EXT-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndirectByteCount2EXT-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndirectByteCount2EXT-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirectByteCount2EXT-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirectByteCount2EXT-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirectByteCount2EXT-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain

VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT

- VUID-vkCmdDrawIndirectByteCount2EXT-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDrawIndirectByteCount2EXT-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawIndirectByteCount2EXT-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawIndirectByteCount2EXT-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirectByteCount2EXT-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirectByteCount2EXT-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawIndirectByteCount2EXT-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match
- VUID-vkCmdDrawIndirectByteCount2EXT-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawIndirectByteCount2EXT-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have four components
- VUID-vkCmdDrawIndirectByteCount2EXT-OpImageWrite-04469
  If a VkBufferView is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawIndirectByteCount2EXT-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdDrawIndirectByteCount2EXT-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdDrawIndirectByteCount2EXT-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdDrawIndirectByteCount2EXT-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdDrawIndirectByteCount2EXT-sparseImageInt64Atomics-04474
  If the `sparseImageInt64Atomics` feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdDrawIndirectByteCount2EXT-sparseImageInt64Atomics-04475
  If the `sparseImageInt64Atomics` feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic

instructions through an OpTypeImage with a SampledType with a Width of 64 by this command

- VUID-vkCmdDrawIndirectByteCount2EXT-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawIndirectByteCount2EXT-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawIndirectByteCount2EXT-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawIndirectByteCount2EXT-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawIndirectByteCount2EXT-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with flags that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawIndirectByteCount2EXT-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with flags that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT,

VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`

- VUID-vkCmdDrawIndirectByteCount2EXT-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11304

If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8

- VUID-vkCmdDrawIndirectByteCount2EXT-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdDrawIndirectByteCount2EXT-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndirectByteCount2EXT-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping

using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`

- VUID-vkCmdDrawIndirectByteCount2EXT-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDrawIndirectByteCount2EXT-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDrawIndirectByteCount2EXT-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the

resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawIndirectByteCount2EXT-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawIndirectByteCount2EXT-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndirectByteCount2EXT-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawIndirectByteCount2EXT-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDrawIndirectByteCount2EXT-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by VkRenderingInputAttachmentIndexInfo, or no `InputAttachmentIndex` if VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` or VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are `NULL`, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawIndirectByteCount2EXT-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` and VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are non-`NULL`
- VUID-vkCmdDrawIndirectByteCount2EXT-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in VkRenderingInputAttachmentIndexInfo

- VUID-vkCmdDrawIndirectByteCount2EXT-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDrawIndirectByteCount2EXT-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP

- VUID-vkCmdDrawIndirectByteCount2EXT-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value

of stencilTestEnable is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCount2EXT-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::maxMultiviewInstanceIndex
- VUID-vkCmdDrawIndirectByteCount2EXT-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07634
  If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-06666
  If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of sampleLocationsEnable is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior

to this drawing command

- VUID-vkCmdDrawIndirectByteCount2EXT-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07846
  If the `depthBounds` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not

subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCount2EXT-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** match the `scissorCount` parameter of `vkCmdSetScissorWithCount`
- VUID-vkCmdDrawIndirectByteCount2EXT-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be 1
- VUID-vkCmdDrawIndirectByteCount2EXT-

primitiveFragmentShadingRateWithMultipleViewports-08642
If the primitiveFragmentShadingRateWithMultipleViewports limit is not supported, and any shader object bound to a graphics stage writes to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1

- VUID-vkCmdDrawIndirectByteCount2EXT-blendEnable-04727
If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of blendEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08644
If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and none of the following is enabled:
  - the multisampledRenderToSingleSampled feature

  then the current value of rasterizationSamples **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08876
If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDrawIndirectByteCount2EXT-viewMask-06178
If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::viewMask equal to VkRenderingInfo::viewMask
- VUID-vkCmdDrawIndirectByteCount2EXT-colorAttachmentCount-06179
If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndirectByteCount2EXT-dynamicRenderingUnusedAttachments-08910
If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawIndirectByteCount2EXT-dynamicRenderingUnusedAttachments-08912
If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render

pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirectByteCount2EXT-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT ::discardRectangleCount
- VUID-vkCmdDrawIndirectByteCount2EXT-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is

VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT::maxDiscardRectangles

- VUID-vkCmdDrawIndirectByteCount2EXT-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCount2EXT-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndirectByteCount2EXT-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCount2EXT-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirectByteCount2EXT-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawIndirectByteCount2EXT-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCount2EXT-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawIndirectByteCount2EXT-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawIndirectByteCount2EXT-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawIndirectByteCount2EXT-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawIndirectByteCount2EXT-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance,

and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView

- VUID-vkCmdDrawIndirectByteCount2EXT-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawIndirectByteCount2EXT-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDrawIndirectByteCount2EXT-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawIndirectByteCount2EXT-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCount2EXT-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCount2EXT-pDepthAttachment-08964
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo ::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirectByteCount2EXT-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCount2EXT-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCount2EXT-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCount2EXT-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawIndirectByteCount2EXT-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawIndirectByteCount2EXT-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07621

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCount2EXT-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and `alphaToCoverageEnable` was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawIndirectByteCount2EXT-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set `alphaToCoverageEnable` to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07625
  If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07626
  If the `logicOp` feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled,

and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCount2EXT-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07630
  If the geometryStreams feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07631
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07632
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of conservativeRasterizationMode is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-conservativePointAndLineRasterization-07499
  If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in

the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT

- VUID-vkCmdDrawIndirectByteCount2EXT-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08669
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and

the current lineRasterizationMode is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current alphaToCoverageEnable, alphaToOneEnable and sampleShadingEnable states **must** all be VK_FALSE

- VUID-vkCmdDrawIndirectByteCount2EXT-None-07639
  If the depthClipControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-09650
  If the depthClampControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of depthClampEnable is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawIndirectByteCount2EXT-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::rasterizationSamples parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawIndirectByteCount2EXT-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the rasterizationSamples parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the rasterizationSamples member of that structure
- VUID-vkCmdDrawIndirectByteCount2EXT-firstAttachment-07476

If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments

- VUID-vkCmdDrawIndirectByteCount2EXT-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndirectByteCount2EXT-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE, then the number of active color attachments **must** not exceed `advancedBlendMaxColorAttachments`
- VUID-vkCmdDrawIndirectByteCount2EXT-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawIndirectByteCount2EXT-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawIndirectByteCount2EXT-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of

VK_TRUE

- VUID-vkCmdDrawIndirectByteCount2EXT-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the rasterizationStream to zero
- VUID-vkCmdDrawIndirectByteCount2EXT-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawIndirectByteCount2EXT-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectByteCount2EXT-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndirectByteCount2EXT-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.width in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples
- VUID-vkCmdDrawIndirectByteCount2EXT-sampleLocationsEnable-07486

If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.height in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdDrawIndirectByteCount2EXT-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDrawIndirectByteCount2EXT-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectByteCount2EXT-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectByteCount2EXT-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectByteCount2EXT-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or

VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled

- VUID-vkCmdDrawIndirectByteCount2EXT-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled
- VUID-vkCmdDrawIndirectByteCount2EXT-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawIndirectByteCount2EXT-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawIndirectByteCount2EXT-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a nextStage including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08684
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the

current command buffer with pStages with an element of VK_SHADER_STAGE_VERTEX_BIT

- VUID-vkCmdDrawIndirectByteCount2EXT-None-08685
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08686
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08687
  If there is no bound graphics pipeline, and the geometryShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08688
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08689
  If there is no bound graphics pipeline, and the taskShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08690
  If there is no bound graphics pipeline, and the meshShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08693
  If there is no bound graphics pipeline, and at least one of the taskShader and meshShader features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid VkShaderEXT bound, and the other **must** have no VkShaderEXT bound
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08696
  If there is no bound graphics pipeline, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no VkShaderEXT bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the

VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any VkShaderEXT bound

- VUID-vkCmdDrawIndirectByteCount2EXT-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts
- VUID-vkCmdDrawIndirectByteCount2EXT-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpDepthAttachmentReadEXT, the depthWriteEnable parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirectByteCount2EXT-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpStencilAttachmentReadEXT, the writeMask parameter in the last call to vkCmdSetStencilWriteMask **must** be 0
- VUID-vkCmdDrawIndirectByteCount2EXT-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the pColorWriteMasks parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawIndirectByteCount2EXT-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of Location for any output attachment statically used in the Fragment Execution Model executed by this command **must** be less than maxFragmentDualSrcAttachments
- VUID-vkCmdDrawIndirectByteCount2EXT-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::pColorAttachmentLocations in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance
- VUID-vkCmdDrawIndirectByteCount2EXT-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::pColorAttachmentInputIndices in the bound

pipeline, if either pColorAttachmentInputIndices[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match

- VUID-vkCmdDrawIndirectByteCount2EXT-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndirectByteCount2EXT-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndirectByteCount2EXT-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDrawIndirectByteCount2EXT-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawIndirectByteCount2EXT-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::flags which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT

- VUID-vkCmdDrawIndirectByteCount2EXT-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndirectByteCount2EXT-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11861
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCount2EXT-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value

of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirectByteCount2EXT-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndirectByteCount2EXT-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCount2EXT-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndirectByteCount2EXT-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCount2EXT-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline

- VUID-vkCmdDrawIndirectByteCount2EXT-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndirectByteCount2EXT-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndirectByteCount2EXT-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndirectByteCount2EXT-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawIndirectByteCount2EXT-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawIndirectByteCount2EXT-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndirectByteCount2EXT-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndirectByteCount2EXT-flags-13363
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT,

VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::`stageFlags` set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT

- VUID-vkCmdDrawIndirectByteCount2EXT-None-04007
  All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or VK_NULL_HANDLE buffers bound
- VUID-vkCmdDrawIndirectByteCount2EXT-None-04008
  If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be VK_NULL_HANDLE
- VUID-vkCmdDrawIndirectByteCount2EXT-None-02721
  If the `robustBufferAccess` feature is not enabled, and that pipeline was created without enabling VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS for `vertexInputs`, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in Vertex Input Description
- VUID-vkCmdDrawIndirectByteCount2EXT-format-10389
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is a packed format, and the `legacyVertexAttributes` feature is not enabled, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the size of the `format`
- VUID-vkCmdDrawIndirectByteCount2EXT-format-10390
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is not a packed format, and either the `legacyVertexAttributes` feature is not enabled or `format` has 64-bit components, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a multiple of the component size of the `format`
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07842
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled then vkCmdSetPrimitiveTopology **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-dynamicPrimitiveTopologyUnrestricted-07500
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled and the `dynamicPrimitiveTopologyUnrestricted` is VK_FALSE, then the `primitiveTopology` parameter of `vkCmdSetPrimitiveTopology` **must** be of the same topology class as the pipeline VkPipelineInputAssemblyStateCreateInfo::`topology` state
- VUID-vkCmdDrawIndirectByteCount2EXT-primitiveTopology-10286

If a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage is bound, then the current value of `primitiveTopology` **must** be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command

- VUID-vkCmdDrawIndirectByteCount2EXT-primitiveTopology-10747
  If vkCmdSetPrimitiveTopology set `primitiveTopology` to VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command, then a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage **must** be bound
- VUID-vkCmdDrawIndirectByteCount2EXT-primitiveTopology-10748
  If vkCmdSetPrimitiveTopology set `primitiveTopology` to VK_PRIMITIVE_TOPOLOGY_POINT_LIST prior to this drawing command, the `maintenance5` feature is not enabled, both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT and VK_SHADER_STAGE_GEOMETRY_BIT stage are not bound, then the `Vertex Execution Model` **must** have a `PointSize` decorated variable that is statically written to
- VUID-vkCmdDrawIndirectByteCount2EXT-pStrides-04913
  If the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, but without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, then vkCmdBindVertexBuffers2 with a non-`NULL` `pStrides` parameter or vkCmdBindVertexBuffers3KHR with `setStride` set to VK_TRUE **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-13118
  If the bound graphics pipeline was created without the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, and vkCmdBindVertexBuffers3KHR was called and not subsequently invalidated in the current command buffer prior to this draw command, the value of `setStride` in each of its `pBindInfos` elements must have been VK_FALSE
- VUID-vkCmdDrawIndirectByteCount2EXT-None-04914
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndirectByteCount2EXT-Input-07939
  If the `maintenance9` feature is not enabled, and there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then all variables with the `Input` storage class decorated with `Location` in the `Vertex Execution Model OpEntryPoint` **must** contain a location in VkVertexInputAttributeDescription2EXT ::`location`
- VUID-vkCmdDrawIndirectByteCount2EXT-Input-08734
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and either the `legacyVertexAttributes` feature is not enabled or the SPIR-V Type associated with a given

Input variable of the corresponding Location in the Vertex Execution Model OpEntryPoint is 64-bit, then the numeric type associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be the same as VkVertexInputAttributeDescription2EXT::format

- VUID-vkCmdDrawIndirectByteCount2EXT-format-08936
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then the scalar width associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be 64-bit
- VUID-vkCmdDrawIndirectByteCount2EXT-format-08937
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and the scalar width associated with a Location decorated Input variable in the Vertex Execution Model OpEntryPoint is 64-bit, then the corresponding VkVertexInputAttributeDescription2EXT ::format **must** have a 64-bit component
- VUID-vkCmdDrawIndirectByteCount2EXT-None-09203
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then all Input variables at the corresponding Location in the Vertex Execution Model OpEntryPoint **must** not use components that are not present in the format
- VUID-vkCmdDrawIndirectByteCount2EXT-None-04875
  If there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with both a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage and the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state enabled, and the current value of primitiveTopology is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, then vkCmdSetPatchControlPointsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-04879
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-None-09637
  If the primitiveTopologyListRestart feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, there is a shader object

bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled, then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE

- VUID-vkCmdDrawIndirectByteCount2EXT-None-10909
  If the `primitiveTopologyPatchListRestart` feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirectByteCount2EXT-stage-06481
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::`stage` member of any element of VkGraphicsPipelineCreateInfo::`pStages` set to VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndirectByteCount2EXT-None-08885
  There **must** be no shader object bound to either of the VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT stages
- VUID-vkCmdDrawIndirectByteCount2EXT-None-07619
  If a shader object is bound to the VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage and the VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT dynamic state enabled, then vkCmdSetTessellationDomainOriginEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCount2EXT-OpExecutionMode-12239
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the type of subdivision, they **must** be the same
- VUID-vkCmdDrawIndirectByteCount2EXT-OpExecutionMode-12240
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the orientation of triangles, they **must** be the same
- VUID-vkCmdDrawIndirectByteCount2EXT-OpExecutionMode-12241
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the segment spacing, they **must** be the same
- VUID-vkCmdDrawIndirectByteCount2EXT-OpExecutionMode-12242
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an `OpExecutionMode` instruction specifying the output patch size, they **must** be the same
- VUID-vkCmdDrawIndirectByteCount2EXT-pNext-09461
  If the bound graphics pipeline state was created with VkPipelineVertexInputDivisorStateCreateInfo in the `pNext` chain of

VkGraphicsPipelineCreateInfo::pVertexInputState, any member of VkPipelineVertexInputDivisorStateCreateInfo::pVertexBindingDivisors has a value other than 1 in divisor, and VkPhysicalDeviceVertexAttributeDivisorProperties ::supportsNonZeroFirstInstance is VK_FALSE, then firstInstance **must** be 0

- VUID-vkCmdDrawIndirectByteCount2EXT-None-09462
  If shader objects are used for drawing or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, any member of the pVertexBindingDescriptions parameter to the vkCmdSetVertexInputEXT call that sets this dynamic state has a value other than 1 in divisor, and VkPhysicalDeviceVertexAttributeDivisorProperties::supportsNonZeroFirstInstance is VK_FALSE, then firstInstance **must** be 0
- VUID-vkCmdDrawIndirectByteCount2EXT-counterOffset-09474
  counterOffset **must** be a multiple of 4
- VUID-vkCmdDrawIndirectByteCount2EXT-transformFeedback-02287
  VkPhysicalDeviceTransformFeedbackFeaturesEXT::transformFeedback **must** be enabled
- VUID-vkCmdDrawIndirectByteCount2EXT-transformFeedbackDraw-02288
  The implementation **must** support VkPhysicalDeviceTransformFeedbackPropertiesEXT ::transformFeedbackDraw
- VUID-vkCmdDrawIndirectByteCount2EXT-vertexStride-02289
  vertexStride **must** be greater than 0 and less than or equal to VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBufferDataStride
- VUID-vkCmdDrawIndirectByteCount2EXT-vertexStride-09475
  vertexStride **must** be a multiple of 4
- VUID-vkCmdDrawIndirectByteCount2EXT-commandBuffer-02646
  commandBuffer **must** not be a protected command buffer
- VUID-vkCmdDrawIndirectByteCount2EXT-pInfo-13061
  The VkBuffer that pInfo->counterRange was queried from **must** have been created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT bit set
- VUID-vkCmdDrawIndirectByteCount2EXT-pInfo-13062
  pInfo->counterRange.address **must** be a multiple of 4

Valid Usage (Implicit)

- VUID-vkCmdDrawIndirectByteCount2EXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDrawIndirectByteCount2EXT-pCounterInfo-parameter
  pCounterInfo **must** be a valid pointer to a valid VkBindTransformFeedbackBuffer2InfoEXT structure
- VUID-vkCmdDrawIndirectByteCount2EXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdDrawIndirectByteCount2EXT-commandBuffer-cmdpool

The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations

- VUID-vkCmdDrawIndirectByteCount2EXT-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndirectByteCount2EXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawIndirectByteCount2EXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawIndirectByteCount2EXT is affected by conditional rendering

To record a non-indexed draw call, where the vertex count is based on a byte count read from a buffer and the passed in vertex stride parameter, call:

WARNING This functionality is superseded by vkCmdDrawIndirectByteCount2EXT. See Legacy Functionality for more information.

```
// Provided by VK_EXT_transform_feedback
void vkCmdDrawIndirectByteCountEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    instanceCount,
    uint32_t                                    firstInstance,
    VkBuffer                                    counterBuffer,
    VkDeviceSize                                counterBufferOffset,
    uint32_t                                    counterOffset,
    uint32_t                                    vertexStride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `instanceCount` is the number of instances to draw.
- `firstInstance` is the instance ID of the first instance to draw.
- `counterBuffer` is the buffer handle from where the byte count is read.
- `counterBufferOffset` is the offset into the buffer used to read the byte count, which is used to calculate the vertex count for this draw call.
- `counterOffset` is subtracted from the byte count read from the `counterBuffer` at the `counterBufferOffset`.
- `vertexStride` is the stride in bytes between each element of the vertex data that is used to calculate the vertex count from the counter value. This value is typically the same value that was used in the graphics pipeline state when the transform feedback was captured as the `XfbStride`.

When the command is executed, primitives are assembled in the same way as done with vkCmdDraw except the `vertexCount` is calculated based on the byte count read from `counterBuffer` at offset `counterBufferOffset`. The assembled primitives execute the bound graphics pipeline.

The effective `vertexCount` is calculated as follows:

```
const uint32_t * counterBufferPtr = (const uint8_t *)counterBuffer.address +
counterBufferOffset;
vertexCount = floor(max(0, (*counterBufferPtr - counterOffset)) / vertexStride);
```

The effective `firstVertex` is zero.

Valid Usage

- VUID-vkCmdDrawIndirectByteCountEXT-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawIndirectByteCountEXT-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawIndirectByteCountEXT-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawIndirectByteCountEXT-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT

- VUID-vkCmdDrawIndirectByteCountEXT-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirectByteCountEXT-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirectByteCountEXT-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawIndirectByteCountEXT-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawIndirectByteCountEXT-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-magFilter-09598
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-mipmapMode-04770
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-mipmapMode-09599
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-unnormalizedCoordinates-09635
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample

a VkImageView as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1

- VUID-vkCmdDrawIndirectByteCountEXT-None-08609
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `viewType` **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawIndirectByteCountEXT-None-08610
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions with `ImplicitLod`, `Dref` or `Proj` in their name
- VUID-vkCmdDrawIndirectByteCountEXT-None-08611
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions that include a LOD bias or any offset values
- VUID-vkCmdDrawIndirectByteCountEXT-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawIndirectByteCountEXT-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawIndirectByteCountEXT-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawIndirectByteCountEXT-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX

as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2

- VUID-vkCmdDrawIndirectByteCountEXT-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDrawIndirectByteCountEXT-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawIndirectByteCountEXT-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDrawIndirectByteCountEXT-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawIndirectByteCountEXT-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not

access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirectByteCountEXT-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirectByteCountEXT-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirectByteCountEXT-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirectByteCountEXT-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawIndirectByteCountEXT-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match
- VUID-vkCmdDrawIndirectByteCountEXT-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawIndirectByteCountEXT-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have four components
- VUID-vkCmdDrawIndirectByteCountEXT-OpImageWrite-04469
  If a VkBufferView is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawIndirectByteCountEXT-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64

- VUID-vkCmdDrawIndirectByteCountEXT-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawIndirectByteCountEXT-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawIndirectByteCountEXT-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawIndirectByteCountEXT-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawIndirectByteCountEXT-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawIndirectByteCountEXT-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawIndirectByteCountEXT-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawIndirectByteCountEXT-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device

- VUID-vkCmdDrawIndirectByteCountEXT-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with `flags` that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawIndirectByteCountEXT-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with `flags` that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDrawIndirectByteCountEXT-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawIndirectByteCountEXT-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`
- VUID-vkCmdDrawIndirectByteCountEXT-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdDrawIndirectByteCountEXT-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdDrawIndirectByteCountEXT-None-11300

If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4

- VUID-vkCmdDrawIndirectByteCountEXT-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndirectByteCountEXT-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndirectByteCountEXT-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdDrawIndirectByteCountEXT-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawIndirectByteCountEXT-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawIndirectByteCountEXT-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawIndirectByteCountEXT-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the

VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by vkCmdBind*HeapEXT commands

- VUID-vkCmdDrawIndirectByteCountEXT-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for uniformBuffers, the robustBufferAccess2 feature is not enabled, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndirectByteCountEXT-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for storageBuffers, the robustBufferAccess2 feature is not enabled, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndirectByteCountEXT-None-11374
  If the robustBufferAccess2 feature is not enabled, the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawIndirectByteCountEXT-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of pBindInfo->reservedRangeSize set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to minSamplerHeapReservedRangeWithEmbedded
- VUID-vkCmdDrawIndirectByteCountEXT-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDrawIndirectByteCountEXT-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds

- VUID-vkCmdDrawIndirectByteCountEXT-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirectByteCountEXT-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirectByteCountEXT-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdDrawIndirectByteCountEXT-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirectByteCountEXT-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawIndirectByteCountEXT-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound

to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR

- VUID-vkCmdDrawIndirectByteCountEXT-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color

- VUID-vkCmdDrawIndirectByteCountEXT-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawIndirectByteCountEXT-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawIndirectByteCountEXT-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS

- VUID-vkCmdDrawIndirectByteCountEXT-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS

- VUID-vkCmdDrawIndirectByteCountEXT-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass

- VUID-vkCmdDrawIndirectByteCountEXT-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat

as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' pInputAttachments [InputAttachmentIndex] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility

- VUID-vkCmdDrawIndirectByteCountEXT-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a InputAttachmentIndex referenced by VkRenderingInputAttachmentIndexInfo, or no InputAttachmentIndex if VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex or VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex are NULL, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawIndirectByteCountEXT-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an InputAttachmentIndex if both VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex and VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex are non-NULL
- VUID-vkCmdDrawIndirectByteCountEXT-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an InputAttachmentIndex, the InputAttachmentIndex **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawIndirectByteCountEXT-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectByteCountEXT-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectByteCountEXT-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectByteCountEXT-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndirectByteCountEXT-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirectByteCountEXT-None-12339

If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command

- VUID-vkCmdDrawIndirectByteCountEXT-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawIndirectByteCountEXT-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawIndirectByteCountEXT-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDrawIndirectByteCountEXT-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawIndirectByteCountEXT-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07835

If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::`maxMultiviewInstanceIndex`
- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` set to VK_TRUE,

then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set

- VUID-vkCmdDrawIndirectByteCountEXT-None-07634
  If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-06666
  If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of sampleLocationsEnable is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of

depthTestEnable is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07846
  If the depthBounds feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the viewportCount parameter of vkCmdSetViewportWithCount **must** match the scissorCount parameter of vkCmdSetScissorWithCount
- VUID-vkCmdDrawIndirectByteCountEXT-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be `1`
- VUID-vkCmdDrawIndirectByteCountEXT-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, and any shader object bound to a graphics stage writes to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be `1`
- VUID-vkCmdDrawIndirectByteCountEXT-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of `blendEnable` **must** be VK_FALSE
- VUID-vkCmdDrawIndirectByteCountEXT-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of `rasterizerDiscardEnable` is VK_FALSE, and none of the following is enabled:
  - the `multisampledRenderToSingleSampled` feature

  then the current value of `rasterizationSamples` **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDrawIndirectByteCountEXT-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must**

have been begun with vkCmdBeginRendering

- VUID-vkCmdDrawIndirectByteCountEXT-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo ::viewMask equal to VkRenderingInfo::viewMask
- VUID-vkCmdDrawIndirectByteCountEXT-colorAttachmentCount-06179
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08910
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08912
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCountEXT-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state

enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments

- VUID-vkCmdDrawIndirectByteCountEXT-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT ::discardRectangleCount
- VUID-vkCmdDrawIndirectByteCountEXT-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT ::maxDiscardRectangles
- VUID-vkCmdDrawIndirectByteCountEXT-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo ::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCountEXT-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is

specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR

- VUID-vkCmdDrawIndirectByteCountEXT-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawIndirectByteCountEXT-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawIndirectByteCountEXT-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawIndirectByteCountEXT-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawIndirectByteCountEXT-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDrawIndirectByteCountEXT-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawIndirectByteCountEXT-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the

corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirectByteCountEXT-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCountEXT-pDepthAttachment-08964
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCountEXT-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCountEXT-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCountEXT-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCountEXT-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless

VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set

- VUID-vkCmdDrawIndirectByteCountEXT-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawIndirectByteCountEXT-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDrawIndirectByteCountEXT-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and alphaToCoverageEnable was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-vkCmdDrawIndirectByteCountEXT-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to

vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set `alphaToCoverageEnable` to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0

- VUID-vkCmdDrawIndirectByteCountEXT-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07625
  If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07626
  If the `logicOp` feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07630
  If the `geometryStreams` feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07631

If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07632
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of conservativeRasterizationMode is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-conservativePointAndLineRasterization-07499
  If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-vkCmdDrawIndirectByteCountEXT-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-08669

If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current lineRasterizationMode is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current alphaToCoverageEnable, alphaToOneEnable and sampleShadingEnable states **must** all be VK_FALSE
- VUID-vkCmdDrawIndirectByteCountEXT-None-07639
  If the depthClipControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-09650
  If the depthClampControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of depthClampEnable is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawIndirectByteCountEXT-samples-07472
  If the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::`rasterizationSamples` parameter used to create the bound graphics pipeline

- VUID-vkCmdDrawIndirectByteCountEXT-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the `rasterizationSamples` parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectByteCountEXT-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with `multisampledRenderToSingleSampledEnable` equal to VK_TRUE, then the `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the `rasterizationSamples` member of that structure
- VUID-vkCmdDrawIndirectByteCountEXT-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndirectByteCountEXT-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawIndirectByteCountEXT-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE, then the number of active color attachments **must** not exceed `advancedBlendMaxColorAttachments`
- VUID-vkCmdDrawIndirectByteCountEXT-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the

current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE

- VUID-vkCmdDrawIndirectByteCountEXT-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE
- VUID-vkCmdDrawIndirectByteCountEXT-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE
- VUID-vkCmdDrawIndirectByteCountEXT-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the rasterizationStream to zero
- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples parameter of the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.width in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples
- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.height in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples
- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by

vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawIndirectByteCountEXT-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled
- VUID-vkCmdDrawIndirectByteCountEXT-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled
- VUID-vkCmdDrawIndirectByteCountEXT-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawIndirectByteCountEXT-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines

feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE

- VUID-vkCmdDrawIndirectByteCountEXT-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawIndirectByteCountEXT-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a nextStage including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDrawIndirectByteCountEXT-None-08684
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-None-08685
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-None-08686
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-None-08687
  If there is no bound graphics pipeline, and the geometryShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-None-08688
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawIndirectByteCountEXT-None-08689
  If there is no bound graphics pipeline, and the taskShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawIndirectByteCountEXT-None-08690

If there is no bound graphics pipeline, and the meshShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_MESH_BIT_EXT

- VUID-vkCmdDrawIndirectByteCountEXT-None-08693
  If there is no bound graphics pipeline, and at least one of the taskShader and meshShader features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid VkShaderEXT bound, and the other **must** have no VkShaderEXT bound
- VUID-vkCmdDrawIndirectByteCountEXT-None-08696
  If there is no bound graphics pipeline, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no VkShaderEXT bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawIndirectByteCountEXT-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDrawIndirectByteCountEXT-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any VkShaderEXT bound
- VUID-vkCmdDrawIndirectByteCountEXT-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawIndirectByteCountEXT-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts
- VUID-vkCmdDrawIndirectByteCountEXT-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpDepthAttachmentReadEXT, the depthWriteEnable parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirectByteCountEXT-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpStencilAttachmentReadEXT, the writeMask parameter in the last call to vkCmdSetStencilWriteMask **must** be 0

- VUID-vkCmdDrawIndirectByteCountEXT-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the pColorWriteMasks parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawIndirectByteCountEXT-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of Location for any output attachment statically used in the Fragment Execution Model executed by this command **must** be less than maxFragmentDualSrcAttachments
- VUID-vkCmdDrawIndirectByteCountEXT-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::pColorAttachmentLocations in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance
- VUID-vkCmdDrawIndirectByteCountEXT-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::pColorAttachmentInputIndices in the bound pipeline, if either pColorAttachmentInputIndices[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndirectByteCountEXT-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndirectByteCountEXT-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawIndirectByteCountEXT-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawIndirectByteCountEXT-None-09643

If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag

- VUID-vkCmdDrawIndirectByteCountEXT-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawIndirectByteCountEXT-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::`flags` which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawIndirectByteCountEXT-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the `dynamicRenderingUnusedAttachments` feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndirectByteCountEXT-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the `dynamicRenderingUnusedAttachments` feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawIndirectByteCountEXT-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::`customResolve` as VK_TRUE
- VUID-vkCmdDrawIndirectByteCountEXT-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::`customResolve` as VK_FALSE
- VUID-vkCmdDrawIndirectByteCountEXT-None-11861
  If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::`colorAttachmentCount` equal to VkRenderingInfo::`colorAttachmentCount`
- VUID-vkCmdDrawIndirectByteCountEXT-None-11862
  If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `resolveImageView` not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::`pColorAttachmentFormats` used to create the bound graphics pipeline

- VUID-vkCmdDrawIndirectByteCountEXT-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCountEXT-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCountEXT-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCountEXT-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawIndirectByteCountEXT-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawIndirectByteCountEXT-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawIndirectByteCountEXT-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawIndirectByteCountEXT-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawIndirectByteCountEXT-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawIndirectByteCountEXT-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE

- VUID-vkCmdDrawIndirectByteCountEXT-flags-13361
  If there are any shader objects bound that were created with `flags` that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawIndirectByteCountEXT-flags-13362
  If there are any shader objects bound that were created with `flags` that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a `setLayoutCount` not equal to 0, and with `pSetLayouts` not equal to `NULL`, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndirectByteCountEXT-flags-12491
  If there are any shader objects bound that were created with `flags` that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a `pushConstantRangeCount` not equal to 0, and with `pPushConstantRanges` not equal to `NULL`, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawIndirectByteCountEXT-flags-13363
  If there are any shader objects bound that were created with `flags` that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::`stageFlags` set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::`stageFlags` set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndirectByteCountEXT-None-04007
  All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or VK_NULL_HANDLE buffers bound
- VUID-vkCmdDrawIndirectByteCountEXT-None-04008
  If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be VK_NULL_HANDLE
- VUID-vkCmdDrawIndirectByteCountEXT-None-02721
  If the `robustBufferAccess` feature is not enabled, and that pipeline was created without enabling VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS for `vertexInputs`, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in Vertex Input Description
- VUID-vkCmdDrawIndirectByteCountEXT-format-10389
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::`format` or VkVertexInputAttributeDescription2EXT::`format` is a packed format, and the `legacyVertexAttributes` feature is not enabled, the value of `attribAddress`, calculated as described in Vertex Input Calculation, **must** be a

multiple of the size of the format

- VUID-vkCmdDrawIndirectByteCountEXT-format-10390
  For each vertex attribute accessed by this command, if its VkVertexInputAttributeDescription::format or VkVertexInputAttributeDescription2EXT::format is not a packed format, and either the legacyVertexAttributes feature is not enabled or format has 64-bit components, the value of attribAddress, calculated as described in Vertex Input Calculation, **must** be a multiple of the component size of the format
- VUID-vkCmdDrawIndirectByteCountEXT-None-07842
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled then vkCmdSetPrimitiveTopology **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-dynamicPrimitiveTopologyUnrestricted-07500
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY dynamic state enabled and the dynamicPrimitiveTopologyUnrestricted is VK_FALSE, then the primitiveTopology parameter of vkCmdSetPrimitiveTopology **must** be of the same topology class as the pipeline VkPipelineInputAssemblyStateCreateInfo::topology state
- VUID-vkCmdDrawIndirectByteCountEXT-primitiveTopology-10286
  If a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage is bound, then the current value of primitiveTopology **must** be VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-primitiveTopology-10747
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_PATCH_LIST prior to this drawing command, then a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage **must** be bound
- VUID-vkCmdDrawIndirectByteCountEXT-primitiveTopology-10748
  If vkCmdSetPrimitiveTopology set primitiveTopology to VK_PRIMITIVE_TOPOLOGY_POINT_LIST prior to this drawing command, the maintenance5 feature is not enabled, both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT and VK_SHADER_STAGE_GEOMETRY_BIT stage are not bound, then the Vertex Execution Model **must** have a PointSize decorated variable that is statically written to
- VUID-vkCmdDrawIndirectByteCountEXT-pStrides-04913
  If the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, but without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, then vkCmdBindVertexBuffers2 with a non-NULL pStrides parameter or vkCmdBindVertexBuffers3KHR with setStride set to VK_TRUE **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command
- VUID-vkCmdDrawIndirectByteCountEXT-None-13118
  If the bound graphics pipeline was created without the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, without the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, and

vkCmdBindVertexBuffers3KHR was called and not subsequently invalidated in the current command buffer prior to this draw command, the value of setStride in each of its pBindInfos elements must have been VK_FALSE

- VUID-vkCmdDrawIndirectByteCountEXT-None-04914
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this draw command

- VUID-vkCmdDrawIndirectByteCountEXT-Input-07939
  If the maintenance9 feature is not enabled, and there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then all variables with the Input storage class decorated with Location in the Vertex Execution Model OpEntryPoint **must** contain a location in VkVertexInputAttributeDescription2EXT::location

- VUID-vkCmdDrawIndirectByteCountEXT-Input-08734
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and either the legacyVertexAttributes feature is not enabled or the SPIR-V Type associated with a given Input variable of the corresponding Location in the Vertex Execution Model OpEntryPoint is 64-bit, then the numeric type associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be the same as VkVertexInputAttributeDescription2EXT::format

- VUID-vkCmdDrawIndirectByteCountEXT-format-08936
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then the scalar width associated with all Input variables of the corresponding Location in the Vertex Execution Model OpEntryPoint **must** be 64-bit

- VUID-vkCmdDrawIndirectByteCountEXT-format-08937
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and the scalar width associated with a Location decorated Input variable in the Vertex Execution Model OpEntryPoint is 64-bit, then the corresponding VkVertexInputAttributeDescription2EXT::format **must** have a 64-bit component

- VUID-vkCmdDrawIndirectByteCountEXT-None-09203
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled and VkVertexInputAttributeDescription2EXT::format has a 64-bit component, then all Input variables at the corresponding Location in the Vertex Execution Model OpEntryPoint **must**

not use components that are not present in the format

- VUID-vkCmdDrawIndirectByteCountEXT-None-04875
  If there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with both a VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage and the VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT dynamic state enabled, and the current value of `primitiveTopology` is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, then vkCmdSetPatchControlPointsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-04879
  If there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-09637
  If the `primitiveTopologyListRestart` feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, there is a shader object bound to the VK_SHADER_STAGE_VERTEX_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled, then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirectByteCountEXT-None-10909
  If the `primitiveTopologyPatchListRestart` feature is not enabled, the input assembly is VK_PRIMITIVE_TOPOLOGY_PATCH_LIST, there is a shader object bound to the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE dynamic state enabled then vkCmdSetPrimitiveRestartEnable **must** be VK_FALSE
- VUID-vkCmdDrawIndirectByteCountEXT-stage-06481
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::`stage` member of any element of VkGraphicsPipelineCreateInfo::`pStages` set to VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawIndirectByteCountEXT-None-08885
  There **must** be no shader object bound to either of the VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT stages
- VUID-vkCmdDrawIndirectByteCountEXT-None-07619
  If a shader object is bound to the VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage and the VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT dynamic state enabled, then vkCmdSetTessellationDomainOriginEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-OpExecutionMode-12239
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the type of subdivision, they **must** be the same
- VUID-vkCmdDrawIndirectByteCountEXT-OpExecutionMode-12240
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the orientation of triangles, they **must** be the same
- VUID-vkCmdDrawIndirectByteCountEXT-OpExecutionMode-12241
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the segment spacing, they **must** be the same
- VUID-vkCmdDrawIndirectByteCountEXT-OpExecutionMode-12242
  If a shader is bound to both the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT and VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stages, and if both stages contain an OpExecutionMode instruction specifying the output patch size, they **must** be the same
- VUID-vkCmdDrawIndirectByteCountEXT-pNext-09461
  If the bound graphics pipeline state was created with VkPipelineVertexInputDivisorStateCreateInfo in the pNext chain of VkGraphicsPipelineCreateInfo::pVertexInputState, any member of VkPipelineVertexInputDivisorStateCreateInfo::pVertexBindingDivisors has a value other than 1 in divisor, and VkPhysicalDeviceVertexAttributeDivisorProperties::supportsNonZeroFirstInstance is VK_FALSE, then firstInstance **must** be 0
- VUID-vkCmdDrawIndirectByteCountEXT-None-09462
  If shader objects are used for drawing or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, any member of the pVertexBindingDescriptions parameter to the vkCmdSetVertexInputEXT call that sets this dynamic state has a value other than 1 in divisor, and VkPhysicalDeviceVertexAttributeDivisorProperties::supportsNonZeroFirstInstance is VK_FALSE, then firstInstance **must** be 0
- VUID-vkCmdDrawIndirectByteCountEXT-counterOffset-09474
  counterOffset **must** be a multiple of 4
- VUID-vkCmdDrawIndirectByteCountEXT-transformFeedback-02287
  VkPhysicalDeviceTransformFeedbackFeaturesEXT::transformFeedback **must** be enabled
- VUID-vkCmdDrawIndirectByteCountEXT-transformFeedbackDraw-02288
  The implementation **must** support VkPhysicalDeviceTransformFeedbackPropertiesEXT::transformFeedbackDraw
- VUID-vkCmdDrawIndirectByteCountEXT-vertexStride-02289
  vertexStride **must** be greater than 0 and less than or equal to VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBufferDataStride
- VUID-vkCmdDrawIndirectByteCountEXT-vertexStride-09475
  vertexStride **must** be a multiple of 4

- VUID-vkCmdDrawIndirectByteCountEXT-commandBuffer-02646
  `commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdDrawIndirectByteCountEXT-counterBuffer-04567
  If `counterBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdDrawIndirectByteCountEXT-counterBuffer-02290
  `counterBuffer` **must** have been created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set
- VUID-vkCmdDrawIndirectByteCountEXT-counterBufferOffset-04568
  `counterBufferOffset` **must** be a multiple of `4`

Valid Usage (Implicit)

- VUID-vkCmdDrawIndirectByteCountEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDrawIndirectByteCountEXT-counterBuffer-parameter
  `counterBuffer` **must** be a valid VkBuffer handle
- VUID-vkCmdDrawIndirectByteCountEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdDrawIndirectByteCountEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDrawIndirectByteCountEXT-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndirectByteCountEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawIndirectByteCountEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdDrawIndirectByteCountEXT-commonparent
  Both of `commandBuffer`, and `counterBuffer` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawIndirectByteCountEXT is affected by conditional rendering

25.5. Conditional Rendering

Certain rendering commands **can** be executed conditionally based on a value in buffer memory. These rendering commands are limited to drawing commands, dispatching commands, and clearing attachments with vkCmdClearAttachments within a conditional rendering block which is defined by commands vkCmdBeginConditionalRenderingEXT and vkCmdEndConditionalRenderingEXT. Other rendering commands remain unaffected by conditional rendering.

The complete list of commands affected by conditional rendering is:

Commands Affected by Conditional Rendering

- vkCmdBeginCustomResolveEXT
- vkCmdClearAttachments
- vkCmdDispatch
- vkCmdDispatchBase
- vkCmdDispatchBaseKHR
- vkCmdDispatchIndirect
- vkCmdDispatchIndirect2KHR
- vkCmdDraw
- vkCmdDrawIndexed
- vkCmdDrawIndexedIndirect
- vkCmdDrawIndexedIndirect2KHR
- vkCmdDrawIndexedIndirectCount
- vkCmdDrawIndexedIndirectCount2KHR
- vkCmdDrawIndexedIndirectCountKHR
- vkCmdDrawIndirect

- vkCmdDrawIndirect2KHR
- vkCmdDrawIndirectByteCount2EXT
- vkCmdDrawIndirectByteCountEXT
- vkCmdDrawIndirectCount
- vkCmdDrawIndirectCount2KHR
- vkCmdDrawIndirectCountKHR
- vkCmdDrawMeshTasksEXT
- vkCmdDrawMeshTasksIndirect2EXT
- vkCmdDrawMeshTasksIndirectCount2EXT
- vkCmdDrawMeshTasksIndirectCountEXT
- vkCmdDrawMeshTasksIndirectEXT
- vkCmdDrawMultiEXT
- vkCmdDrawMultiIndexedEXT
- vkCmdExecuteGeneratedCommandsEXT

After beginning conditional rendering, it is considered *active* within the command buffer it was called until it is ended with vkCmdEndConditionalRenderingEXT.

Conditional rendering **must** begin and end in the same command buffer. When conditional rendering is active, a primary command buffer **can** execute secondary command buffers if the `inheritedConditionalRendering` feature is enabled. For a secondary command buffer to be executed while conditional rendering is active in the primary command buffer, it **must** set the `conditionalRenderingEnable` flag of VkCommandBufferInheritanceConditionalRenderingInfoEXT, as described in the Command Buffer Recording section.

Conditional rendering **must** also either begin and end inside the same subpass of a render pass instance, or **must** both begin and end outside of a render pass instance (i.e. contain entire render pass instances).

To begin conditional rendering, call:

WARNING	This functionality is superseded by vkCmdBeginConditionalRendering2EXT. See Legacy Functionality for more information.

```
// Provided by VK_EXT_conditional_rendering
void vkCmdBeginConditionalRenderingEXT(
    VkCommandBuffer                             commandBuffer,
    const VkConditionalRenderingBeginInfoEXT*   pConditionalRenderingBegin);
```

- `commandBuffer` is the command buffer into which this command will be recorded.
- `pConditionalRenderingBegin` is a pointer to a VkConditionalRenderingBeginInfoEXT structure specifying parameters of conditional rendering.

Valid Usage

- VUID-vkCmdBeginConditionalRenderingEXT-None-01980
  Conditional rendering **must** not already be active

Valid Usage (Implicit)

- VUID-vkCmdBeginConditionalRenderingEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBeginConditionalRenderingEXT-pConditionalRenderingBegin-parameter
  `pConditionalRenderingBegin` **must** be a valid pointer to a valid VkConditionalRenderingBeginInfoEXT structure
- VUID-vkCmdBeginConditionalRenderingEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBeginConditionalRenderingEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBeginConditionalRenderingEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdBeginConditionalRenderingEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	Action State

Conditional Rendering

vkCmdBeginConditionalRenderingEXT is not affected by conditional rendering

The VkConditionalRenderingBeginInfoEXT structure is defined as:

WARNING	This functionality is superseded by VkConditionalRenderingBeginInfo2EXT. See Legacy Functionality for more information.

```
// Provided by VK_EXT_conditional_rendering
typedef struct VkConditionalRenderingBeginInfoEXT {
    VkStructureType                 sType;
    const void*                     pNext;
    VkBuffer                        buffer;
    VkDeviceSize                    offset;
    VkConditionalRenderingFlagsEXT  flags;
} VkConditionalRenderingBeginInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- buffer is a buffer containing the predicate for conditional rendering.
- offset is the byte offset into buffer where the predicate is located.
- flags is a bitmask of VkConditionalRenderingFlagsEXT specifying the behavior of conditional rendering.

If the 32-bit value at offset in buffer memory is zero, then the rendering commands are discarded, otherwise they are executed as normal. If the value of the predicate in buffer memory changes while conditional rendering is active, the rendering commands **may** be discarded in an implementation-dependent way. Some implementations may latch the value of the predicate upon beginning conditional rendering while others may read it before every rendering command.

Valid Usage

- VUID-VkConditionalRenderingBeginInfoEXT-buffer-01981
  If buffer is non-sparse then it **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-VkConditionalRenderingBeginInfoEXT-buffer-01982
  buffer **must** have been created with the VK_BUFFER_USAGE_CONDITIONAL_RENDERING_BIT_EXT usage flag set
- VUID-VkConditionalRenderingBeginInfoEXT-offset-01983
  offset **must** be less than the size of buffer by at least 32 bits
- VUID-VkConditionalRenderingBeginInfoEXT-offset-01984
  offset **must** be a multiple of 4

Valid Usage (Implicit)

- VUID-VkConditionalRenderingBeginInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_CONDITIONAL_RENDERING_BEGIN_INFO_EXT
- VUID-VkConditionalRenderingBeginInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkConditionalRenderingBeginInfoEXT-buffer-parameter
  `buffer` **must** be a valid VkBuffer handle
- VUID-VkConditionalRenderingBeginInfoEXT-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkConditionalRenderingFlagBitsEXT values

To begin conditional rendering, call:

```
// Provided by VK_KHR_device_address_commands with VK_EXT_conditional_rendering
void vkCmdBeginConditionalRendering2EXT(
    VkCommandBuffer                             commandBuffer,
    const VkConditionalRenderingBeginInfo2EXT*  pConditionalRenderingBegin);
```

- `commandBuffer` is the command buffer into which this command will be recorded.
- `pConditionalRenderingBegin` is a pointer to a VkConditionalRenderingBeginInfo2EXT structure specifying parameters of conditional rendering.

Valid Usage

- VUID-vkCmdBeginConditionalRendering2EXT-None-13063
  Conditional rendering **must** not already be active

Valid Usage (Implicit)

- VUID-vkCmdBeginConditionalRendering2EXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBeginConditionalRendering2EXT-pConditionalRenderingBegin-parameter
  `pConditionalRenderingBegin` **must** be a valid pointer to a valid VkConditionalRenderingBeginInfo2EXT structure
- VUID-vkCmdBeginConditionalRendering2EXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBeginConditionalRendering2EXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBeginConditionalRendering2EXT-suspended

This command **must** not be called between suspended render pass instances

- VUID-vkCmdBeginConditionalRendering2EXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	Action State

Conditional Rendering

vkCmdBeginConditionalRendering2EXT is not affected by conditional rendering

The `VkConditionalRenderingBeginInfo2EXT` structure is defined as:

```
// Provided by VK_KHR_device_address_commands with VK_EXT_conditional_rendering
typedef struct VkConditionalRenderingBeginInfo2EXT {
    VkStructureType                   sType;
    const void*                       pNext;
    VkDeviceAddressRangeKHR           addressRange;
    VkAddressCommandFlagsKHR          addressFlags;
    VkConditionalRenderingFlagsEXT    flags;
} VkConditionalRenderingBeginInfo2EXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `addressRange` is the VkDeviceAddressRangeKHR containing the predicate for conditional rendering.
- `addressFlags` is a VkAddressCommandFlagsKHR value defining the flags for the address range.
- `flags` is a bitmask of VkConditionalRenderingFlagsEXT specifying the behavior of conditional

rendering.

If the 32-bit value at addressRange.address is zero, then the rendering commands are discarded, otherwise they are executed as normal. If the value of the predicate in memory changes while conditional rendering is active, the rendering commands **may** be discarded in an implementation-dependent way. Some implementations **may** latch the value of the predicate upon beginning conditional rendering while others **may** read it before every rendering command.

Valid Usage

- VUID-VkConditionalRenderingBeginInfo2EXT-addressRange-13097
  If the range specified by addressRange is not bound completely to memory when accessed, addressFlags **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR
- VUID-VkConditionalRenderingBeginInfo2EXT-addressRange-13098
  If the buffer from which the range specified by addressRange was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and protectedNoFault is not supported, addressFlags **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkConditionalRenderingBeginInfo2EXT-addressRange-13099
  If the buffer from which the range specified by addressRange was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and protectedNoFault is not supported, addressFlags **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkConditionalRenderingBeginInfo2EXT-addressFlags-13100
  addressFlags **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkConditionalRenderingBeginInfo2EXT-addressRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing addressRange, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, addressFlags **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkConditionalRenderingBeginInfo2EXT-addressRange-13123
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing addressRange, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, addressFlags **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkConditionalRenderingBeginInfo2EXT-addressFlags-13101
  addressFlags **must** not include both VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkConditionalRenderingBeginInfo2EXT-addressRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing addressRange, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, addressFlags **must** include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or

VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR

- VUID-VkConditionalRenderingBeginInfo2EXT-addressRange-13125
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created without VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `addressFlags` **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkConditionalRenderingBeginInfo2EXT-addressRange-13064
  The VkBuffer that `addressRange` was queried from **must** have been created with VK_BUFFER_USAGE_CONDITIONAL_RENDERING_BIT_EXT
- VUID-VkConditionalRenderingBeginInfo2EXT-addressRange-13065
  `addressRange.address` **must** be a multiple of 4
- VUID-VkConditionalRenderingBeginInfo2EXT-addressRange-13066
  `addressRange.size` **must** be greater than or equal to 4

Valid Usage (Implicit)

- VUID-VkConditionalRenderingBeginInfo2EXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_CONDITIONAL_RENDERING_BEGIN_INFO_2_EXT
- VUID-VkConditionalRenderingBeginInfo2EXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkConditionalRenderingBeginInfo2EXT-addressFlags-parameter
  If `addressFlags` is not `0`, `addressFlags` **must** be a valid combination of VkAddressCommandFlagBitsKHR values
- VUID-VkConditionalRenderingBeginInfo2EXT-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkConditionalRenderingFlagBitsEXT values

Bits which **can** be set in vkCmdBeginConditionalRenderingEXT::`flags`, specifying the behavior of conditional rendering, are:

```
// Provided by VK_EXT_conditional_rendering
typedef enum VkConditionalRenderingFlagBitsEXT {
    VK_CONDITIONAL_RENDERING_INVERTED_BIT_EXT = 0x00000001,
} VkConditionalRenderingFlagBitsEXT;
```

- VK_CONDITIONAL_RENDERING_INVERTED_BIT_EXT specifies the condition used to determine whether to discard rendering commands or not. That is, if the 32-bit predicate read from `buffer` memory at `offset` is zero, the rendering commands are not discarded, and if non zero, then they are discarded.

```
// Provided by VK_EXT_conditional_rendering
typedef VkFlags VkConditionalRenderingFlagsEXT;
```

VkConditionalRenderingFlagsEXT is a bitmask type for setting a mask of zero or more VkConditionalRenderingFlagBitsEXT.

To end conditional rendering, call:

```
// Provided by VK_EXT_conditional_rendering
void vkCmdEndConditionalRenderingEXT(
    VkCommandBuffer                             commandBuffer);
```

- commandBuffer is the command buffer into which this command will be recorded.

Once ended, conditional rendering becomes inactive.

Valid Usage

- VUID-vkCmdEndConditionalRenderingEXT-None-01985
  Conditional rendering **must** be active
- VUID-vkCmdEndConditionalRenderingEXT-None-01986
  If conditional rendering was made active outside of a render pass instance, it **must** not be ended inside a render pass instance
- VUID-vkCmdEndConditionalRenderingEXT-None-01987
  If conditional rendering was made active within a subpass it **must** be ended in the same subpass

Valid Usage (Implicit)

- VUID-vkCmdEndConditionalRenderingEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdEndConditionalRenderingEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdEndConditionalRenderingEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdEndConditionalRenderingEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdEndConditionalRenderingEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	Action State

Conditional Rendering

vkCmdEndConditionalRenderingEXT is not affected by conditional rendering

25.6. Programmable Mesh Shading

In this drawing approach, primitives are assembled by the mesh shader stage. Mesh shading operates similarly to dispatching compute as the shaders make use of workgroups.

To record a mesh tasks drawing command, call:

```
// Provided by VK_EXT_mesh_shader
void vkCmdDrawMeshTasksEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    groupCountX,
    uint32_t                                    groupCountY,
    uint32_t                                    groupCountZ);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `groupCountX` is the number of local workgroups to dispatch in the X dimension.
- `groupCountY` is the number of local workgroups to dispatch in the Y dimension.
- `groupCountZ` is the number of local workgroups to dispatch in the Z dimension.

When the command is executed, a global workgroup consisting of `groupCountX` × `groupCountY` × `groupCountZ` local workgroups is assembled.

Valid Usage

- VUID-vkCmdDrawMeshTasksEXT-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawMeshTasksEXT-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawMeshTasksEXT-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawMeshTasksEXT-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksEXT-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksEXT-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksEXT-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksEXT-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawMeshTasksEXT-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawMeshTasksEXT-magFilter-09598

If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT

- VUID-vkCmdDrawMeshTasksEXT-mipmapMode-04770
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawMeshTasksEXT-mipmapMode-09599
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawMeshTasksEXT-unnormalizedCoordinates-09635
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1
- VUID-vkCmdDrawMeshTasksEXT-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawMeshTasksEXT-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdDrawMeshTasksEXT-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias or any offset values
- VUID-vkCmdDrawMeshTasksEXT-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawMeshTasksEXT-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawMeshTasksEXT-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic

operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT

- VUID-vkCmdDrawMeshTasksEXT-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawMeshTasksEXT-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawMeshTasksEXT-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawMeshTasksEXT-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawMeshTasksEXT-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksEXT-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksEXT-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksEXT-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksEXT-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader

- VUID-vkCmdDrawMeshTasksEXT-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawMeshTasksEXT-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDrawMeshTasksEXT-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawMeshTasksEXT-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksEXT-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksEXT-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksEXT-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksEXT-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawMeshTasksEXT-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the

numeric type of the view's format and the Sampled Type operand of the OpTypeImage **must** match

- VUID-vkCmdDrawMeshTasksEXT-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawMeshTasksEXT-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components
- VUID-vkCmdDrawMeshTasksEXT-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawMeshTasksEXT-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawMeshTasksEXT-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawMeshTasksEXT-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawMeshTasksEXT-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawMeshTasksEXT-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawMeshTasksEXT-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawMeshTasksEXT-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawMeshTasksEXT-None-09600

If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdDrawMeshTasksEXT-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawMeshTasksEXT-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawMeshTasksEXT-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with flags that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawMeshTasksEXT-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with flags that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDrawMeshTasksEXT-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawMeshTasksEXT-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a OpTypeStruct decorated with Block or BufferBlock using that mapping, the calculated offset for the resource heap **must** be a multiple of bufferDescriptorAlignment

- VUID-vkCmdDrawMeshTasksEXT-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeImage or OpTypeSampledImage using that mapping, the calculated offset for the resource heap **must** be a multiple of imageDescriptorAlignment
- VUID-vkCmdDrawMeshTasksEXT-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeSampler or OpTypeSampledImage using that mapping, the calculated offset for the sampler heap **must** be a multiple of samplerDescriptorAlignment
- VUID-vkCmdDrawMeshTasksEXT-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDrawMeshTasksEXT-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawMeshTasksEXT-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawMeshTasksEXT-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8

- VUID-vkCmdDrawMeshTasksEXT-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawMeshTasksEXT-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawMeshTasksEXT-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawMeshTasksEXT-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawMeshTasksEXT-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMeshTasksEXT-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMeshTasksEXT-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer

as specified by VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdDrawMeshTasksEXT-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDrawMeshTasksEXT-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDrawMeshTasksEXT-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDrawMeshTasksEXT-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMeshTasksEXT-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMeshTasksEXT-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to

must be aligned to `minUniformBufferOffsetAlignment`

- VUID-vkCmdDrawMeshTasksEXT-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMeshTasksEXT-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawMeshTasksEXT-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDrawMeshTasksEXT-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDrawMeshTasksEXT-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawMeshTasksEXT-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by

this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawMeshTasksEXT-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawMeshTasksEXT-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawMeshTasksEXT-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawMeshTasksEXT-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDrawMeshTasksEXT-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by VkRenderingInputAttachmentIndexInfo, or no `InputAttachmentIndex` if VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` or VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are `NULL`, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawMeshTasksEXT-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` and VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are non-`NULL`
- VUID-vkCmdDrawMeshTasksEXT-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawMeshTasksEXT-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksEXT-None-10795

If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawMeshTasksEXT-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksEXT-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksEXT-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawMeshTasksEXT-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawMeshTasksEXT-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawMeshTasksEXT-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawMeshTasksEXT-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDrawMeshTasksEXT-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawMeshTasksEXT-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior

to this drawing command

- VUID-vkCmdDrawMeshTasksEXT-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07838

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksEXT-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::`maxMultiviewInstanceIndex`
- VUID-vkCmdDrawMeshTasksEXT-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawMeshTasksEXT-None-07634
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-06666
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `sampleLocationsEnable` is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the

current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksEXT-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07846
  If the depthBounds feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state

enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksEXT-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** match the `scissorCount` parameter of `vkCmdSetScissorWithCount`
- VUID-vkCmdDrawMeshTasksEXT-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be `1`
- VUID-vkCmdDrawMeshTasksEXT-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, and any shader object bound to a graphics stage writes to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command

buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1

- VUID-vkCmdDrawMeshTasksEXT-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of blendEnable **must** be VK_FALSE
- VUID-vkCmdDrawMeshTasksEXT-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and none of the following is enabled:
    - the multisampledRenderToSingleSampled feature

  then the current value of rasterizationSamples **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDrawMeshTasksEXT-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDrawMeshTasksEXT-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::viewMask equal to VkRenderingInfo::viewMask
- VUID-vkCmdDrawMeshTasksEXT-colorAttachmentCount-06179
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawMeshTasksEXT-dynamicRenderingUnusedAttachments-08910
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksEXT-dynamicRenderingUnusedAttachments-08912
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used

to create the bound pipeline equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMeshTasksEXT-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawMeshTasksEXT-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT ::discardRectangleCount
- VUID-vkCmdDrawMeshTasksEXT-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT ::maxDiscardRectangles

- VUID-vkCmdDrawMeshTasksEXT-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawMeshTasksEXT-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of

VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView

- VUID-vkCmdDrawMeshTasksEXT-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawMeshTasksEXT-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawMeshTasksEXT-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawMeshTasksEXT-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksEXT-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawMeshTasksEXT-multisampledRenderToSingleSampled-07287

If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView

- VUID-vkCmdDrawMeshTasksEXT-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDrawMeshTasksEXT-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawMeshTasksEXT-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-pDepthAttachment-08964
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo

::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMeshTasksEXT-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo ::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo ::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawMeshTasksEXT-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawMeshTasksEXT-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDrawMeshTasksEXT-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07622

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksEXT-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and alphaToCoverageEnable was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-vkCmdDrawMeshTasksEXT-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set alphaToCoverageEnable to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-vkCmdDrawMeshTasksEXT-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07625
  If the alphaToOne feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07626
  If the logicOp feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with

VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksEXT-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07630
  If the geometryStreams feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07631
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07632
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of conservativeRasterizationMode is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-conservativePointAndLineRasterization-07499
  If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-vkCmdDrawMeshTasksEXT-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then

vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksEXT-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-08669
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current lineRasterizationMode is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current alphaToCoverageEnable, alphaToOneEnable and sampleShadingEnable states **must** all be VK_FALSE
- VUID-vkCmdDrawMeshTasksEXT-None-07639
  If the depthClipControl feature is enabled, and a shader object is bound to any graphics

stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksEXT-None-09650
  If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of `depthClampEnable` is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawMeshTasksEXT-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::`rasterizationSamples` parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksEXT-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the `rasterizationSamples` parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksEXT-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with `multisampledRenderToSingleSampledEnable` equal to VK_TRUE, then the `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the `rasterizationSamples` member of that structure
- VUID-vkCmdDrawMeshTasksEXT-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments

- VUID-vkCmdDrawMeshTasksEXT-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawMeshTasksEXT-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE, then the number of active color attachments **must** not exceed `advancedBlendMaxColorAttachments`
- VUID-vkCmdDrawMeshTasksEXT-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawMeshTasksEXT-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawMeshTasksEXT-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawMeshTasksEXT-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the

last call to vkCmdSetRasterizationStreamEXT **must** have set the `rasterizationStream` to zero

- VUID-vkCmdDrawMeshTasksEXT-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawMeshTasksEXT-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksEXT-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawMeshTasksEXT-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::`maxSampleLocationGridSize.width` as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a `samples` parameter equaling `rasterizationSamples`
- VUID-vkCmdDrawMeshTasksEXT-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT

::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdDrawMeshTasksEXT-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDrawMeshTasksEXT-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksEXT-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksEXT-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksEXT-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled
- VUID-vkCmdDrawMeshTasksEXT-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or

VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled

- VUID-vkCmdDrawMeshTasksEXT-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawMeshTasksEXT-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawMeshTasksEXT-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawMeshTasksEXT-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksEXT-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a nextStage including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDrawMeshTasksEXT-None-08684
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDrawMeshTasksEXT-None-08685
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawMeshTasksEXT-None-08686

If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT

- VUID-vkCmdDrawMeshTasksEXT-None-08687
  If there is no bound graphics pipeline, and the geometryShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawMeshTasksEXT-None-08688
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawMeshTasksEXT-None-08689
  If there is no bound graphics pipeline, and the taskShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawMeshTasksEXT-None-08690
  If there is no bound graphics pipeline, and the meshShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawMeshTasksEXT-None-08693
  If there is no bound graphics pipeline, and at least one of the taskShader and meshShader features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid VkShaderEXT bound, and the other **must** have no VkShaderEXT bound
- VUID-vkCmdDrawMeshTasksEXT-None-08696
  If there is no bound graphics pipeline, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no VkShaderEXT bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksEXT-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDrawMeshTasksEXT-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any VkShaderEXT bound
- VUID-vkCmdDrawMeshTasksEXT-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawMeshTasksEXT-None-08879
  All bound graphics shader objects **must** have either been created with the

VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts

- VUID-vkCmdDrawMeshTasksEXT-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawMeshTasksEXT-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to vkCmdSetStencilWriteMask **must** be `0`
- VUID-vkCmdDrawMeshTasksEXT-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the `pColorWriteMasks` parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawMeshTasksEXT-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of `Location` for any output attachment statically used in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`
- VUID-vkCmdDrawMeshTasksEXT-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::`pColorAttachmentLocations` in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance
- VUID-vkCmdDrawMeshTasksEXT-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::`pColorAttachmentInputIndices` in the bound pipeline, if either `pColorAttachmentInputIndices`[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an `InputAttachmentIndex` decoration, the two values **must** match
- VUID-vkCmdDrawMeshTasksEXT-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` in the bound

pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an `InputAttachmentIndex` decoration, the two values **must** match

- VUID-vkCmdDrawMeshTasksEXT-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an `InputAttachmentIndex` decoration, the two values **must** match
- VUID-vkCmdDrawMeshTasksEXT-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawMeshTasksEXT-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDrawMeshTasksEXT-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawMeshTasksEXT-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::`flags` which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawMeshTasksEXT-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the `dynamicRenderingUnusedAttachments` feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawMeshTasksEXT-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the `dynamicRenderingUnusedAttachments` feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawMeshTasksEXT-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT ::`customResolve` as VK_TRUE
- VUID-vkCmdDrawMeshTasksEXT-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a

custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE

- VUID-vkCmdDrawMeshTasksEXT-None-11861
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawMeshTasksEXT-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksEXT-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound

graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView

- VUID-vkCmdDrawMeshTasksEXT-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawMeshTasksEXT-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksEXT-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksEXT-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawMeshTasksEXT-pStencilAttachment-11873

If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView

- VUID-vkCmdDrawMeshTasksEXT-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawMeshTasksEXT-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawMeshTasksEXT-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawMeshTasksEXT-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawMeshTasksEXT-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawMeshTasksEXT-flags-13363
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawMeshTasksEXT-stage-06480
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::stage member of any element of

VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT

- VUID-vkCmdDrawMeshTasksEXT-pStages-10680
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** not have last bound the pStages element of VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT with a valid shader object other than VK_NULL_HANDLE
- VUID-vkCmdDrawMeshTasksEXT-None-07074
  Transform Feedback Queries **must** not be active
- VUID-vkCmdDrawMeshTasksEXT-None-07075
  Primitives Generated Queries **must** not be active
- VUID-vkCmdDrawMeshTasksEXT-pipelineStatistics-07076
  The pipelineStatistics member used to create any active Pipeline Statistics Query **must** not contain VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_VERTICES_BIT, VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_PRIMITIVES_BIT, VK_QUERY_PIPELINE_STATISTIC_VERTEX_SHADER_INVOCATIONS_BIT, VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_INVOCATIONS_BIT, VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_PRIMITIVES_BIT, VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT, VK_QUERY_PIPELINE_STATISTIC_CLIPPING_PRIMITIVES_BIT, VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_CONTROL_SHADER_PATCHES_BIT, or VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_EVALUATION_SHADER_INVOCATIONS_BIT
- VUID-vkCmdDrawMeshTasksEXT-None-08694
  If there is no bound graphics pipeline, and the meshShader feature is enabled, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_MESH_BIT_EXT stage, and that VkShaderEXT was created without the VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT flag, a valid VkShaderEXT **must** be bound to the VK_SHADER_STAGE_TASK_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksEXT-None-08695
  If there is no bound graphics pipeline, and the meshShader feature is enabled, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_MESH_BIT_EXT stage, and that VkShaderEXT was created with the VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT flag, there **must** be no VkShaderEXT bound to the VK_SHADER_STAGE_TASK_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksEXT-flags-13364
  If there are any shader objects bound that were created with flags that includes both the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR and VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT bits, the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_NO_TASK_SHADER_BIT_KHR set
- VUID-vkCmdDrawMeshTasksEXT-flags-13365
  If there are any shader objects bound that were created with flags that includes both the

VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR and VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT bits, there **must** be no shader object bound to the VK_SHADER_STAGE_TASK_BIT_EXT stage

- VUID-vkCmdDrawMeshTasksEXT-TaskEXT-07322
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS contains a shader using the `TaskEXT Execution Model`, `groupCountX` **must** be less than or equal to `VkPhysicalDeviceMeshShaderPropertiesEXT`::`maxTaskWorkGroupCount`[0]
- VUID-vkCmdDrawMeshTasksEXT-TaskEXT-07323
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS contains a shader using the `TaskEXT Execution Model`, `groupCountY` **must** be less than or equal to `VkPhysicalDeviceMeshShaderPropertiesEXT`::`maxTaskWorkGroupCount`[1]
- VUID-vkCmdDrawMeshTasksEXT-TaskEXT-07324
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS contains a shader using the `TaskEXT Execution Model`, `groupCountZ` **must** be less than or equal to `VkPhysicalDeviceMeshShaderPropertiesEXT`::`maxTaskWorkGroupCount`[2]
- VUID-vkCmdDrawMeshTasksEXT-TaskEXT-07325
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS contains a shader using the `TaskEXT Execution Model`, the product of `groupCountX`, `groupCountY` and `groupCountZ` **must** be less than or equal to `VkPhysicalDeviceMeshShaderPropertiesEXT`::`maxTaskWorkGroupTotalCount`
- VUID-vkCmdDrawMeshTasksEXT-TaskEXT-07326
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS does not contain a shader using the `TaskEXT Execution Model`, `groupCountX` **must** be less than or equal to `VkPhysicalDeviceMeshShaderPropertiesEXT`::`maxMeshWorkGroupCount`[0]
- VUID-vkCmdDrawMeshTasksEXT-TaskEXT-07327
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS does not contain a shader using the `TaskEXT Execution Model`, `groupCountY` **must** be less than or equal to `VkPhysicalDeviceMeshShaderPropertiesEXT`::`maxMeshWorkGroupCount`[1]
- VUID-vkCmdDrawMeshTasksEXT-TaskEXT-07328
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS does not contain a shader using the `TaskEXT Execution Model`, `groupCountZ` **must** be less than or equal to `VkPhysicalDeviceMeshShaderPropertiesEXT`::`maxMeshWorkGroupCount`[2]
- VUID-vkCmdDrawMeshTasksEXT-TaskEXT-07329
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS does not contain a shader using the `TaskEXT Execution Model`, the product of `groupCountX`, `groupCountY` and `groupCountZ` **must** be less than or equal to `VkPhysicalDeviceMeshShaderPropertiesEXT`::`maxMeshWorkGroupTotalCount`

- VUID-vkCmdDrawMeshTasksEXT-MeshEXT-07087
  The current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS **must** contain a shader stage using the `MeshEXT Execution Model`

Valid Usage (Implicit)

- VUID-vkCmdDrawMeshTasksEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDrawMeshTasksEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdDrawMeshTasksEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDrawMeshTasksEXT-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawMeshTasksEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawMeshTasksEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawMeshTasksEXT is affected by conditional rendering

To record an indirect mesh tasks drawing command, call:

```
// Provided by VK_KHR_device_address_commands with VK_EXT_mesh_shader
void vkCmdDrawMeshTasksIndirect2EXT(
    VkCommandBuffer                             commandBuffer,
    const VkDrawIndirect2InfoKHR*               pInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `pInfo` is a pointer to a VkDrawIndirect2InfoKHR structure defining parameters of this command.

`vkCmdDrawMeshTasksIndirect2EXT` behaves similarly to vkCmdDrawMeshTasksEXT except that the parameters are read by the device from an address range during execution. `pInfo->drawCount` draws are executed by the command, with parameters taken from `pInfo->addressRange` starting at `addressRange.address` and increasing by `addressRange.stride` bytes for each successive draw. The parameters of each draw are encoded in an array of VkDrawMeshTasksIndirectCommandEXT structures.

Valid Usage

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawMeshTasksIndirect2EXT-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08119
  If a descriptor is dynamically used with a VkPipeline created with

VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a VkDescriptorSetLayout that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident

- VUID-vkCmdDrawMeshTasksIndirect2EXT-magFilter-04553
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-magFilter-09598
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-mipmapMode-04770
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-mipmapMode-09599
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-unnormalizedCoordinates-09635
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias

or any offset values

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawMeshTasksIndirect2EXT-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawMeshTasksIndirect2EXT-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawMeshTasksIndirect2EXT-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain

VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT

- VUID-vkCmdDrawMeshTasksIndirect2EXT-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawMeshTasksIndirect2EXT-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksIndirect2EXT-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08613
  If the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksIndirect2EXT-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's viewType **must** match the Dim operand of the OpTypeImage as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawMeshTasksIndirect2EXT-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's format and the Sampled Type operand of the OpTypeImage **must** match
- VUID-vkCmdDrawMeshTasksIndirect2EXT-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawMeshTasksIndirect2EXT-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components
- VUID-vkCmdDrawMeshTasksIndirect2EXT-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawMeshTasksIndirect2EXT-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawMeshTasksIndirect2EXT-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawMeshTasksIndirect2EXT-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawMeshTasksIndirect2EXT-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawMeshTasksIndirect2EXT-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic

instructions through an OpTypeImage with a SampledType with a Width of 64 by this command

- VUID-vkCmdDrawMeshTasksIndirect2EXT-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawMeshTasksIndirect2EXT-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirect2EXT-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawMeshTasksIndirect2EXT-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with flags that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawMeshTasksIndirect2EXT-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with flags that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-12472

Descriptors dynamically accessed in bound descriptor heaps **must** be valid

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound

to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either

VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for storageBuffers, the robustBufferAccess2 feature is not enabled, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11374
  If the robustBufferAccess2 feature is not enabled, the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMeshTasksIndirect2EXT-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of pBindInfo->reservedRangeSize set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to minSamplerHeapReservedRangeWithEmbedded
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to

must have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDrawMeshTasksIndirect2EXT-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDrawMeshTasksIndirect2EXT-protectedNoFault-11455

If protectedNoFault is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawMeshTasksIndirect2EXT-protectedNoFault-11456
  If protectedNoFault is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawMeshTasksIndirect2EXT-renderPass-02684
  The current render pass **must** be compatible with the renderPass member of the VkGraphicsPipelineCreateInfo structure specified when creating the VkPipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawMeshTasksIndirect2EXT-subpass-02685
  The subpass index of the current render pass **must** be equal to the subpass member of the VkGraphicsPipelineCreateInfo structure specified when creating the VkPipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawMeshTasksIndirect2EXT-OpTypeImage-07468
  If any shader executed by this pipeline accesses an OpTypeImage variable with a Dim operand of SubpassData, it **must** be decorated with an InputAttachmentIndex that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' pInputAttachments [InputAttachmentIndex] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDrawMeshTasksIndirect2EXT-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a InputAttachmentIndex referenced by VkRenderingInputAttachmentIndexInfo, or no InputAttachmentIndex if VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex or VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex are NULL, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawMeshTasksIndirect2EXT-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an InputAttachmentIndex if both VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex and VkRenderingInputAttachmentIndexInfo

::pStencilInputAttachmentIndex are non-NULL

- VUID-vkCmdDrawMeshTasksIndirect2EXT-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an InputAttachmentIndex, the InputAttachmentIndex **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only

layout for the depth aspect, depth writes **must** be disabled

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing

command

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::`maxMultiviewInstanceIndex`
- VUID-vkCmdDrawMeshTasksIndirect2EXT-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07634
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-06666
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `sampleLocationsEnable` is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07846
  If the `depthBounds` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** match the `scissorCount` parameter of `vkCmdSetScissorWithCount`
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the

shader stages of the bound graphics pipeline write to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1

- VUID-vkCmdDrawMeshTasksIndirect2EXT-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the primitiveFragmentShadingRateWithMultipleViewports limit is not supported, and any shader object bound to a graphics stage writes to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1
- VUID-vkCmdDrawMeshTasksIndirect2EXT-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of blendEnable **must** be VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and none of the following is enabled:
  - the multisampledRenderToSingleSampled feature

  then the current value of rasterizationSamples **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDrawMeshTasksIndirect2EXT-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::viewMask equal to VkRenderingInfo::viewMask
- VUID-vkCmdDrawMeshTasksIndirect2EXT-colorAttachmentCount-06179
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawMeshTasksIndirect2EXT-dynamicRenderingUnusedAttachments-08910
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have

been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline

- VUID-vkCmdDrawMeshTasksIndirect2EXT-dynamicRenderingUnusedAttachments-08912
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirect2EXT-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount
- VUID-vkCmdDrawMeshTasksIndirect2EXT-rasterizerDiscardEnable-09236

If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT ::maxDiscardRectangles

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo ::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirect2EXT-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo ::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->imageView
- VUID-vkCmdDrawMeshTasksIndirect2EXT-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo ::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMeshTasksIndirect2EXT-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirect2EXT-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawMeshTasksIndirect2EXT-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirect2EXT-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawMeshTasksIndirect2EXT-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawMeshTasksIndirect2EXT-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of

the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline

- VUID-vkCmdDrawMeshTasksIndirect2EXT-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawMeshTasksIndirect2EXT-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawMeshTasksIndirect2EXT-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDrawMeshTasksIndirect2EXT-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawMeshTasksIndirect2EXT-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirect2EXT-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirect2EXT-pDepthAttachment-08964

If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMeshTasksIndirect2EXT-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirect2EXT-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirect2EXT-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirect2EXT-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawMeshTasksIndirect2EXT-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawMeshTasksIndirect2EXT-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a

graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and `alphaToCoverageEnable` was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawMeshTasksIndirect2EXT-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set `alphaToCoverageEnable` to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07625
  If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then

vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07626
  If the `logicOp` feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07630
  If the `geometryStreams` feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07631
  If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07632
  If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `conservativeRasterizationMode` is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not

subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirect2EXT-conservativePointAndLineRasterization-07499
  If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08669
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been

called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current `alphaToCoverageEnable`, `alphaToOneEnable` and `sampleShadingEnable` states **must** all be VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07639
  If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-09650
  If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of `depthClampEnable` is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawMeshTasksIndirect2EXT-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::`rasterizationSamples` parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksIndirect2EXT-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the `rasterizationSamples` parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is

bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with `multisampledRenderToSingleSampledEnable` equal to VK_TRUE, then the `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the `rasterizationSamples` member of that structure

- VUID-vkCmdDrawMeshTasksIndirect2EXT-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawMeshTasksIndirect2EXT-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawMeshTasksIndirect2EXT-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE, then the number of active color attachments **must** not exceed `advancedBlendMaxColorAttachments`
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirect2EXT-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics

pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE

- VUID-vkCmdDrawMeshTasksIndirect2EXT-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the `rasterizationStream` to zero
- VUID-vkCmdDrawMeshTasksIndirect2EXT-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawMeshTasksIndirect2EXT-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawMeshTasksIndirect2EXT-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the

sampleLocationsInfo.sampleLocationGridSize.width in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdDrawMeshTasksIndirect2EXT-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.height in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples
- VUID-vkCmdDrawMeshTasksIndirect2EXT-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDrawMeshTasksIndirect2EXT-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current

value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawMeshTasksIndirect2EXT-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled
- VUID-vkCmdDrawMeshTasksIndirect2EXT-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled
- VUID-vkCmdDrawMeshTasksIndirect2EXT-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawMeshTasksIndirect2EXT-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirect2EXT-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirect2EXT-nextStage-10745

For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a `nextStage` including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08684
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08685
  If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08686
  If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08687
  If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08688
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08689
  If there is no bound graphics pipeline, and the `taskShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08690
  If there is no bound graphics pipeline, and the `meshShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08693
  If there is no bound graphics pipeline, and at least one of the `taskShader` and `meshShader` features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid `VkShaderEXT` bound, and the other **must** have no `VkShaderEXT` bound
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08696
  If there is no bound graphics pipeline, and a valid `VkShaderEXT` is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no `VkShaderEXT` bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the

VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any `VkShaderEXT` bound
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts
- VUID-vkCmdDrawMeshTasksIndirect2EXT-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirect2EXT-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to vkCmdSetStencilWriteMask **must** be `0`
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the `pColorWriteMasks` parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawMeshTasksIndirect2EXT-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of `Location` for any output attachment statically used in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::`pColorAttachmentLocations` in the bound pipeline

must match the value for the corresponding locations set currently in the current render pass instance

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::pColorAttachmentInputIndices in the bound pipeline, if either pColorAttachmentInputIndices[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawMeshTasksIndirect2EXT-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::flags which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirect2EXT-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11861
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirect2EXT-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be

VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawMeshTasksIndirect2EXT-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawMeshTasksIndirect2EXT-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirect2EXT-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a

VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline

- VUID-vkCmdDrawMeshTasksIndirect2EXT-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawMeshTasksIndirect2EXT-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawMeshTasksIndirect2EXT-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirect2EXT-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirect2EXT-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawMeshTasksIndirect2EXT-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawMeshTasksIndirect2EXT-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawMeshTasksIndirect2EXT-flags-13363

If there are any shader objects bound that were created with `flags` that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::`stageFlags` set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::`stageFlags` set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT

- VUID-vkCmdDrawMeshTasksIndirect2EXT-stage-06480
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::`stage` member of any element of VkGraphicsPipelineCreateInfo::`pStages` set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-pStages-10680
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** not have last bound the `pStages` element of VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT with a valid shader object other than VK_NULL_HANDLE
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07074
  Transform Feedback Queries **must** not be active
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-07075
  Primitives Generated Queries **must** not be active
- VUID-vkCmdDrawMeshTasksIndirect2EXT-pipelineStatistics-07076
  The `pipelineStatistics` member used to create any active Pipeline Statistics Query **must** not contain VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_VERTICES_BIT, VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_PRIMITIVES_BIT, VK_QUERY_PIPELINE_STATISTIC_VERTEX_SHADER_INVOCATIONS_BIT, VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_INVOCATIONS_BIT, VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_PRIMITIVES_BIT, VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT, VK_QUERY_PIPELINE_STATISTIC_CLIPPING_PRIMITIVES_BIT, VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_CONTROL_SHADER_PATCHES_BIT, or VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_EVALUATION_SHADER_INVOCATIONS_BIT
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08694
  If there is no bound graphics pipeline, and the `meshShader` feature is enabled, and a valid `VkShaderEXT` is bound to the VK_SHADER_STAGE_MESH_BIT_EXT stage, and that `VkShaderEXT` was created without the VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT flag, a valid `VkShaderEXT` **must** be bound to the VK_SHADER_STAGE_TASK_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksIndirect2EXT-None-08695

If there is no bound graphics pipeline, and the meshShader feature is enabled, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_MESH_BIT_EXT stage, and that VkShaderEXT was created with the VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT flag, there **must** be no VkShaderEXT bound to the VK_SHADER_STAGE_TASK_BIT_EXT stage

- VUID-vkCmdDrawMeshTasksIndirect2EXT-flags-13364
  If there are any shader objects bound that were created with flags that includes both the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR and VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT bits, the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_NO_TASK_SHADER_BIT_KHR set
- VUID-vkCmdDrawMeshTasksIndirect2EXT-flags-13365
  If there are any shader objects bound that were created with flags that includes both the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR and VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT bits, there **must** be no shader object bound to the VK_SHADER_STAGE_TASK_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksIndirect2EXT-commandBuffer-13067
  commandBuffer **must** not be a protected command buffer

Valid Usage (Implicit)

- VUID-vkCmdDrawMeshTasksIndirect2EXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDrawMeshTasksIndirect2EXT-pInfo-parameter
  pInfo **must** be a valid pointer to a valid VkDrawIndirect2InfoKHR structure
- VUID-vkCmdDrawMeshTasksIndirect2EXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdDrawMeshTasksIndirect2EXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDrawMeshTasksIndirect2EXT-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawMeshTasksIndirect2EXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawMeshTasksIndirect2EXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawMeshTasksIndirect2EXT is affected by conditional rendering

To record an indirect mesh tasks drawing command, call:

WARNING This functionality is superseded by vkCmdDrawMeshTasksIndirect2EXT. See Legacy Functionality for more information.

```
// Provided by VK_EXT_mesh_shader
void vkCmdDrawMeshTasksIndirectEXT(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    buffer,
    VkDeviceSize                                offset,
    uint32_t                                    drawCount,
    uint32_t                                    stride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `buffer` is the buffer containing draw parameters.
- `offset` is the byte offset into `buffer` where parameters begin.
- `drawCount` is the number of draws to execute, and **can** be zero.
- `stride` is the byte stride between successive sets of draw parameters.

`vkCmdDrawMeshTasksIndirectEXT` behaves similarly to vkCmdDrawMeshTasksEXT except that the parameters are read by the device from a buffer during execution. `drawCount` draws are executed by the command, with parameters taken from `buffer` starting at `offset` and increasing by `stride` bytes for each successive draw. The parameters of each draw are encoded in an array of VkDrawMeshTasksIndirectCommandEXT structures. If `drawCount` is less than or equal to one, `stride` is ignored.

Valid Usage

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawMeshTasksIndirectEXT-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawMeshTasksIndirectEXT-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawMeshTasksIndirectEXT-magFilter-09598
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT

- VUID-vkCmdDrawMeshTasksIndirectEXT-mipmapMode-04770
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawMeshTasksIndirectEXT-mipmapMode-09599
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawMeshTasksIndirectEXT-unnormalizedCoordinates-09635
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias or any offset values
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain

VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawMeshTasksIndirectEXT-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawMeshTasksIndirectEXT-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawMeshTasksIndirectEXT-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksIndirectEXT-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksIndirectEXT-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksIndirectEXT-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawMeshTasksIndirectEXT-imageLayout-00344

If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawMeshTasksIndirectEXT-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksIndirectEXT-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksIndirectEXT-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawMeshTasksIndirectEXT-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match
- VUID-vkCmdDrawMeshTasksIndirectEXT-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that

instruction **must** have at least as many components as the image view's format

- VUID-vkCmdDrawMeshTasksIndirectEXT-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have four components
- VUID-vkCmdDrawMeshTasksIndirectEXT-OpImageWrite-04469
  If a VkBufferView is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawMeshTasksIndirectEXT-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdDrawMeshTasksIndirectEXT-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdDrawMeshTasksIndirectEXT-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdDrawMeshTasksIndirectEXT-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdDrawMeshTasksIndirectEXT-sparseImageInt64Atomics-04474
  If the `sparseImageInt64Atomics` feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdDrawMeshTasksIndirectEXT-sparseImageInt64Atomics-04475
  If the `sparseImageInt64Atomics` feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdDrawMeshTasksIndirectEXT-micromap-11636
  If an acceleration structure is accessed as a result of this command through an `OpRayQueryProceedKHR` operation , the `VK_EXT_opacity_micromap` extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its `pNext` chain, the shader **must** have specified a `OpacityMicromapIdKHR` execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectEXT-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawMeshTasksIndirectEXT-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with `flags` that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawMeshTasksIndirectEXT-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with `flags` that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT,

VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeImage or OpTypeSampledImage using that mapping, the calculated offset for the resource heap **must** be a multiple of imageDescriptorAlignment

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeSampler or OpTypeSampledImage using that mapping, the calculated offset for the sampler heap **must** be a multiple of samplerDescriptorAlignment
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location

in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMeshTasksIndirectEXT-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or

VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDrawMeshTasksIndirectEXT-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDrawMeshTasksIndirectEXT-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawMeshTasksIndirectEXT-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawMeshTasksIndirectEXT-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawMeshTasksIndirectEXT-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawMeshTasksIndirectEXT-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDrawMeshTasksIndirectEXT-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by VkRenderingInputAttachmentIndexInfo, or no `InputAttachmentIndex` if VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` or VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are `NULL`, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawMeshTasksIndirectEXT-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` and VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are non-`NULL`
- VUID-vkCmdDrawMeshTasksIndirectEXT-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load,

store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08617

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::`maxMultiviewInstanceIndex`
- VUID-vkCmdDrawMeshTasksIndirectEXT-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07634
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-06666
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `sampleLocationsEnable` is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled,

and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07846
  If the depthBounds feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently

invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectEXT-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** match the `scissorCount` parameter of `vkCmdSetScissorWithCount`
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be 1
- VUID-vkCmdDrawMeshTasksIndirectEXT-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, and any shader object bound to a graphics stage writes to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be 1
- VUID-vkCmdDrawMeshTasksIndirectEXT-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the

current value of rasterizerDiscardEnable is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of blendEnable **must** be VK_FALSE

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and none of the following is enabled:

  - the multisampledRenderToSingleSampled feature

  then the current value of rasterizationSamples **must** be the same as the current color and/or depth/stencil attachments

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDrawMeshTasksIndirectEXT-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::viewMask equal to VkRenderingInfo::viewMask
- VUID-vkCmdDrawMeshTasksIndirectEXT-colorAttachmentCount-06179
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawMeshTasksIndirectEXT-dynamicRenderingUnusedAttachments-08910
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksIndirectEXT-dynamicRenderingUnusedAttachments-08912
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectEXT-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have

been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount
- VUID-vkCmdDrawMeshTasksIndirectEXT-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT::maxDiscardRectangles
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently

invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectEXT-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawMeshTasksIndirectEXT-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectEXT-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectEXT-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawMeshTasksIndirectEXT-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo

::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMeshTasksIndirectEXT-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawMeshTasksIndirectEXT-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawMeshTasksIndirectEXT-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksIndirectEXT-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawMeshTasksIndirectEXT-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawMeshTasksIndirectEXT-pNext-07935

If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples

- VUID-vkCmdDrawMeshTasksIndirectEXT-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawMeshTasksIndirectEXT-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectEXT-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectEXT-pDepthAttachment-08964
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo ::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectEXT-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo ::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectEXT-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance,

there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo ::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMeshTasksIndirectEXT-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo ::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectEXT-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawMeshTasksIndirectEXT-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawMeshTasksIndirectEXT-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently

invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and `alphaToCoverageEnable` was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawMeshTasksIndirectEXT-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set `alphaToCoverageEnable` to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07625
  If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07626
  If the `logicOp` feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07630
  If the geometryStreams feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07631
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07632
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of conservativeRasterizationMode is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-conservativePointAndLineRasterization-07499
  If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the

VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08666
  If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08669
  If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07849
  If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stippledLineEnable` is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current `alphaToCoverageEnable`, `alphaToOneEnable` and `sampleShadingEnable` states **must** all be VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07639
  If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-09650
  If the depthClampControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of depthClampEnable is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawMeshTasksIndirectEXT-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::rasterizationSamples parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksIndirectEXT-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the rasterizationSamples parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the rasterizationSamples member of that structure
- VUID-vkCmdDrawMeshTasksIndirectEXT-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawMeshTasksIndirectEXT-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current

value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments

- VUID-vkCmdDrawMeshTasksIndirectEXT-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and an active color attachment current value of blendEnable is VK_TRUE, then the number of active color attachments **must** not exceed advancedBlendMaxColorAttachments
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectEXT-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectEXT-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the rasterizationStream to zero

- VUID-vkCmdDrawMeshTasksIndirectEXT-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawMeshTasksIndirectEXT-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawMeshTasksIndirectEXT-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT ::`maxSampleLocationGridSize.width` as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a `samples` parameter equaling `rasterizationSamples`
- VUID-vkCmdDrawMeshTasksIndirectEXT-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT ::`maxSampleLocationGridSize.height` as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a `samples` parameter equaling

rasterizationSamples

- VUID-vkCmdDrawMeshTasksIndirectEXT-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdDrawMeshTasksIndirectEXT-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled
- VUID-vkCmdDrawMeshTasksIndirectEXT-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode

state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled

- VUID-vkCmdDrawMeshTasksIndirectEXT-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawMeshTasksIndirectEXT-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectEXT-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectEXT-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a nextStage including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08684
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08685
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08686
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages

with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08687
  If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08688
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08689
  If there is no bound graphics pipeline, and the `taskShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08690
  If there is no bound graphics pipeline, and the `meshShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08693
  If there is no bound graphics pipeline, and at least one of the `taskShader` and `meshShader` features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid `VkShaderEXT` bound, and the other **must** have no `VkShaderEXT` bound
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08696
  If there is no bound graphics pipeline, and a valid `VkShaderEXT` is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no `VkShaderEXT` bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any `VkShaderEXT` bound
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts

- VUID-vkCmdDrawMeshTasksIndirectEXT-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpDepthAttachmentReadEXT, the depthWriteEnable parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirectEXT-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpStencilAttachmentReadEXT, the writeMask parameter in the last call to vkCmdSetStencilWriteMask **must** be 0
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the pColorWriteMasks parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawMeshTasksIndirectEXT-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of Location for any output attachment statically used in the Fragment Execution Model executed by this command **must** be less than maxFragmentDualSrcAttachments
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::pColorAttachmentLocations in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::pColorAttachmentInputIndices in the bound pipeline, if either pColorAttachmentInputIndices[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two

values **must** match

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an `InputAttachmentIndex` decoration, the two values **must** match
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawMeshTasksIndirectEXT-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::`flags` which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the `dynamicRenderingUnusedAttachments` feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the `dynamicRenderingUnusedAttachments` feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT ::`customResolve` as VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectEXT-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with

VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE

- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11861
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectEXT-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->resolveImageView

- VUID-vkCmdDrawMeshTasksIndirectEXT-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawMeshTasksIndirectEXT-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectEXT-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksIndirectEXT-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawMeshTasksIndirectEXT-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and

VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView

- VUID-vkCmdDrawMeshTasksIndirectEXT-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectEXT-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirectEXT-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawMeshTasksIndirectEXT-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawMeshTasksIndirectEXT-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawMeshTasksIndirectEXT-flags-13363
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectEXT-stage-06480
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::stage member of any element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT,

VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT

- VUID-vkCmdDrawMeshTasksIndirectEXT-pStages-10680
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** not have last bound the pStages element of VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT with a valid shader object other than VK_NULL_HANDLE
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07074
  Transform Feedback Queries **must** not be active
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-07075
  Primitives Generated Queries **must** not be active
- VUID-vkCmdDrawMeshTasksIndirectEXT-pipelineStatistics-07076
  The pipelineStatistics member used to create any active Pipeline Statistics Query **must** not contain VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_VERTICES_BIT, VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_PRIMITIVES_BIT, VK_QUERY_PIPELINE_STATISTIC_VERTEX_SHADER_INVOCATIONS_BIT, VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_INVOCATIONS_BIT, VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_PRIMITIVES_BIT, VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT, VK_QUERY_PIPELINE_STATISTIC_CLIPPING_PRIMITIVES_BIT, VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_CONTROL_SHADER_PATCHES_BIT, or VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_EVALUATION_SHADER_INVOCATIONS_BIT
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08694
  If there is no bound graphics pipeline, and the meshShader feature is enabled, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_MESH_BIT_EXT stage, and that VkShaderEXT was created without the VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT flag, a valid VkShaderEXT **must** be bound to the VK_SHADER_STAGE_TASK_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksIndirectEXT-None-08695
  If there is no bound graphics pipeline, and the meshShader feature is enabled, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_MESH_BIT_EXT stage, and that VkShaderEXT was created with the VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT flag, there **must** be no VkShaderEXT bound to the VK_SHADER_STAGE_TASK_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksIndirectEXT-flags-13364
  If there are any shader objects bound that were created with flags that includes both the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR and VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT bits, the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_NO_TASK_SHADER_BIT_KHR set
- VUID-vkCmdDrawMeshTasksIndirectEXT-flags-13365
  If there are any shader objects bound that were created with flags that includes both the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR and VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT bits, there **must** be no shader object

bound to the VK_SHADER_STAGE_TASK_BIT_EXT stage

- VUID-vkCmdDrawMeshTasksIndirectEXT-buffer-02708
  If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdDrawMeshTasksIndirectEXT-buffer-02709
  `buffer` **must** have been created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMeshTasksIndirectEXT-offset-02710
  `offset` **must** be a multiple of `4`
- VUID-vkCmdDrawMeshTasksIndirectEXT-commandBuffer-02711
  `commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdDrawMeshTasksIndirectEXT-drawCount-02718
  If the `multiDrawIndirect` feature is not enabled, `drawCount` **must** be `0` or `1`
- VUID-vkCmdDrawMeshTasksIndirectEXT-drawCount-02719
  `drawCount` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxDrawIndirectCount`
- VUID-vkCmdDrawMeshTasksIndirectEXT-drawCount-07088
  If `drawCount` is greater than `1`, `stride` **must** be a multiple of `4` and **must** be greater than or equal to `sizeof`(`VkDrawMeshTasksIndirectCommandEXT`)
- VUID-vkCmdDrawMeshTasksIndirectEXT-drawCount-07089
  If `drawCount` is equal to `1`, (`offset` + `sizeof`(VkDrawMeshTasksIndirectCommandEXT)) **must** be less than or equal to the size of `buffer`
- VUID-vkCmdDrawMeshTasksIndirectEXT-drawCount-07090
  If `drawCount` is greater than `1`, (`stride` × (`drawCount` - 1) + `offset` + `sizeof`(VkDrawMeshTasksIndirectCommandEXT)) **must** be less than or equal to the size of `buffer`
- VUID-vkCmdDrawMeshTasksIndirectEXT-MeshEXT-07091
  The current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS **must** contain a shader stage using the `MeshEXT` `Execution` `Model`

Valid Usage (Implicit)

- VUID-vkCmdDrawMeshTasksIndirectEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDrawMeshTasksIndirectEXT-buffer-parameter
  `buffer` **must** be a valid VkBuffer handle
- VUID-vkCmdDrawMeshTasksIndirectEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdDrawMeshTasksIndirectEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations

- VUID-vkCmdDrawMeshTasksIndirectEXT-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawMeshTasksIndirectEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawMeshTasksIndirectEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdDrawMeshTasksIndirectEXT-commonparent
  Both of `buffer`, and `commandBuffer` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawMeshTasksIndirectEXT is affected by conditional rendering

The `VkDrawMeshTasksIndirectCommandEXT` structure is defined as:

```
// Provided by VK_EXT_mesh_shader
typedef struct VkDrawMeshTasksIndirectCommandEXT {
    uint32_t    groupCountX;
    uint32_t    groupCountY;
    uint32_t    groupCountZ;
} VkDrawMeshTasksIndirectCommandEXT;
```

- `groupCountX` is the number of local workgroups to dispatch in the X dimension.
- `groupCountY` is the number of local workgroups to dispatch in the Y dimension.
- `groupCountZ` is the number of local workgroups to dispatch in the Z dimension.

The members of VkDrawMeshTasksIndirectCommandEXT have the same meaning as the similarly named parameters of vkCmdDrawMeshTasksEXT.

Valid Usage

- VUID-VkDrawMeshTasksIndirectCommandEXT-TaskEXT-07322
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS contains a shader using the TaskEXT Execution Model, groupCountX **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskWorkGroupCount[0]
- VUID-VkDrawMeshTasksIndirectCommandEXT-TaskEXT-07323
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS contains a shader using the TaskEXT Execution Model, groupCountY **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskWorkGroupCount[1]
- VUID-VkDrawMeshTasksIndirectCommandEXT-TaskEXT-07324
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS contains a shader using the TaskEXT Execution Model, groupCountZ **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskWorkGroupCount[2]
- VUID-VkDrawMeshTasksIndirectCommandEXT-TaskEXT-07325
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS contains a shader using the TaskEXT Execution Model, the product of groupCountX, groupCountY and groupCountZ **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskWorkGroupTotalCount
- VUID-VkDrawMeshTasksIndirectCommandEXT-TaskEXT-07326
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS does not contain a shader using the TaskEXT Execution Model, groupCountX **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupCount[0]
- VUID-VkDrawMeshTasksIndirectCommandEXT-TaskEXT-07327
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS does not contain a shader using the TaskEXT Execution Model, groupCountY **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupCount[1]
- VUID-VkDrawMeshTasksIndirectCommandEXT-TaskEXT-07328
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS does not contain a shader using the TaskEXT Execution Model, groupCountZ **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupCount[2]
- VUID-VkDrawMeshTasksIndirectCommandEXT-TaskEXT-07329
  If the current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS does not contain a shader using the TaskEXT Execution Model, the product of groupCountX, groupCountY and groupCountZ **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupTotalCount

To record an indirect mesh tasks drawing command with the draw count sourced from an address range, call:

```
// Provided by VK_KHR_device_address_commands with (VK_KHR_draw_indirect_count or
VK_VERSION_1_2) and VK_EXT_mesh_shader
void vkCmdDrawMeshTasksIndirectCount2EXT(
    VkCommandBuffer                             commandBuffer,
    const VkDrawIndirectCount2InfoKHR*          pInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `pInfo` is a pointer to a VkDrawIndirectCount2InfoKHR structure defining parameters of this command.

`vkCmdDrawMeshTasksIndirectCount2EXT` behaves similarly to vkCmdDrawMeshTasksIndirect2EXT except that the draw count is read by the device from an address range during execution. The command will read an unsigned 32-bit integer from `pInfo->countAddressRange` and use this as the draw count.

Valid Usage

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified

via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-magFilter-09598
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-mipmapMode-04770
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-mipmapMode-09599
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-unnormalizedCoordinates-09635
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08609
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `viewType` **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08610
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions with `ImplicitLod`, `Dref` or `Proj` in their name

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias or any offset values
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the OpTypeImage is Unknown, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the OpTypeImage is Unknown, the view's format features **must** contain

VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or

VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for storageBuffers, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08613
  If the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's viewType **must** match the Dim operand of the OpTypeImage as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's format and the Sampled Type operand of the OpTypeImage **must** match
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that

instruction **must** have a `Width` of 32

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-sparseImageInt64Atomics-04474
  If the `sparseImageInt64Atomics` feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-sparseImageInt64Atomics-04475
  If the `sparseImageInt64Atomics` feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-micromap-11636
  If an acceleration structure is accessed as a result of this command through an `OpRayQueryProceedKHR` operation , the `VK_EXT_opacity_micromap` extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its `pNext` chain, the shader **must** have specified a `OpacityMicromapIdKHR` execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with `flags` that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap

array, any VkShaderEXT that accessed the descriptor **must** not have been created with `flags` that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location

in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping

using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered

before this command was recorded, and still be registered during the sampling operation, with an identically defined color

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by VkRenderingInputAttachmentIndexInfo, or no `InputAttachmentIndex` if VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` or VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are `NULL`, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pDepthInputAttachmentIndex-09596

Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` and VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are non-`NULL`

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an

attachment, storage image, or sampled image by this command

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the

current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthBoundsTestEnable is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::maxMultiviewInstanceIndex
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07634
  If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-06666
  If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value

of rasterizerDiscardEnable is VK_FALSE, and the current value of sampleLocationsEnable is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07846
  If the depthBounds feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state

enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the viewportCount parameter of vkCmdSetViewportWithCount **must** match the scissorCount parameter of vkCmdSetScissorWithCount
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of logicOpEnable is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-primitiveFragmentShadingRateWithMultipleViewports-04552

If the primitiveFragmentShadingRateWithMultipleViewports limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the primitiveFragmentShadingRateWithMultipleViewports limit is not supported, and any shader object bound to a graphics stage writes to the PrimitiveShadingRateKHR built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the viewportCount parameter of vkCmdSetViewportWithCount **must** be 1
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of blendEnable **must** be VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and none of the following is enabled:
  - the multisampledRenderToSingleSampled feature

  then the current value of rasterizationSamples **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::viewMask equal to VkRenderingInfo::viewMask
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-colorAttachmentCount-06179
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-dynamicRenderingUnusedAttachments-08910

If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-dynamicRenderingUnusedAttachments-08912
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-attachmentCount-07750
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is

VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT::maxDiscardRectangles
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not

VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pDepthAttachment-08964
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the `primitivesGeneratedQueryWithRasterizerDiscard` feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in `VkPipelineRasterizationStateStreamCreateInfoEXT`::`rasterizationStream`
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07620
  If the `depthClamp` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and `alphaToCoverageEnable` was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set

alphaToCoverageEnable to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07625
  If the alphaToOne feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07626
  If the logicOp feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07630
  If the geometryStreams feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07631
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to

any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07632
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of conservativeRasterizationMode is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-conservativePointAndLineRasterization-07499
  If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08669

If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current lineRasterizationMode is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current alphaToCoverageEnable, alphaToOneEnable and sampleShadingEnable states **must** all be VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07639
  If the depthClipControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-09650
  If the depthClampControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of depthClampEnable is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-samples-07472
  If the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::`rasterizationSamples` parameter used to create the bound graphics pipeline

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the `rasterizationSamples` parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with `multisampledRenderToSingleSampledEnable` equal to VK_TRUE, then the `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the `rasterizationSamples` member of that structure
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE, then the number of active color attachments **must** not exceed `advancedBlendMaxColorAttachments`
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the

VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the rasterizationStream to zero
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples parameter of the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::`maxSampleLocationGridSize.width` as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a `samples` parameter equaling `rasterizationSamples`
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::`maxSampleLocationGridSize.height` as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a `samples` parameter equaling `rasterizationSamples`
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction `InterpolateAtSample`
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then `sampleLocationsInfo.sampleLocationGridSize.width` **must** evenly divide VkMultisamplePropertiesEXT::`maxSampleLocationGridSize.width` as returned by

vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines

feature **must** be enabled and VkPhysicalDeviceLimits::`strictLines` **must** be VK_TRUE

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::`stage` member of an element of VkGraphicsPipelineCreateInfo::`pStages` set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a `nextStage` including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08684
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08685
  If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08686
  If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08687
  If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08688
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08689
  If there is no bound graphics pipeline, and the `taskShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08690

If there is no bound graphics pipeline, and the meshShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_MESH_BIT_EXT

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08693
  If there is no bound graphics pipeline, and at least one of the taskShader and meshShader features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid VkShaderEXT bound, and the other **must** have no VkShaderEXT bound
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08696
  If there is no bound graphics pipeline, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no VkShaderEXT bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any VkShaderEXT bound
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpDepthAttachmentReadEXT, the depthWriteEnable parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpStencilAttachmentReadEXT, the writeMask parameter in the last call to vkCmdSetStencilWriteMask **must** be 0

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the pColorWriteMasks parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of Location for any output attachment statically used in the Fragment Execution Model executed by this command **must** be less than maxFragmentDualSrcAttachments
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::pColorAttachmentLocations in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::pColorAttachmentInputIndices in the bound pipeline, if either pColorAttachmentInputIndices[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-09643

If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::`flags` which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the `dynamicRenderingUnusedAttachments` feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the `dynamicRenderingUnusedAttachments` feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::`customResolve` as VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::`customResolve` as VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11861
  If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::`colorAttachmentCount` equal to VkRenderingInfo::`colorAttachmentCount`
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11862
  If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `resolveImageView` not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::`pColorAttachmentFormats` used to create the bound graphics pipeline

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the

bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the `dynamicRenderingUnusedAttachments` feature is not enabled, and VkRenderingInfo::`pStencilAttachment->resolveImageView` was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::`stencilAttachmentFormat` used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo ::`pStencilAttachment->resolveImageView`
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the `dynamicRenderingUnusedAttachments` feature is enabled, VkRenderingInfo::`pStencilAttachment->resolveImageView` was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::`stencilAttachmentFormat` used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo ::`pStencilAttachment->resolveImageView`, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::`colorAttachmentCount` parameter greater than `0` and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::`pColorAttachments` array with a `resolveImageView` not equal to VK_NULL_HANDLE, the `resolveImageView` **must** have been created with a sample count equal to the value of `rasterizationSamples` for the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::`pDepthAttachment->resolveImageView` was not VK_NULL_HANDLE, the value of `rasterizationSamples` for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::`pDepthAttachment->resolveImageView`
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::`pStencilAttachment->resolveImageView` was not VK_NULL_HANDLE, the value of `rasterizationSamples` for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::`pStencilAttachment->resolveImageView`
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::`customResolve` as VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called,

then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::`customResolve` as VK_FALSE

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-flags-13361
  If there are any shader objects bound that were created with `flags` that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-flags-13362
  If there are any shader objects bound that were created with `flags` that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a `setLayoutCount` not equal to 0, and with `pSetLayouts` not equal to `NULL`, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-flags-12491
  If there are any shader objects bound that were created with `flags` that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a `pushConstantRangeCount` not equal to 0, and with `pPushConstantRanges` not equal to `NULL`, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-flags-13363
  If there are any shader objects bound that were created with `flags` that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::`stageFlags` set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::`stageFlags` set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-stage-06480
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::`stage` member of any element of VkGraphicsPipelineCreateInfo::`pStages` set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pStages-10680
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** not have last bound the `pStages` element of VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT with a valid shader object other than VK_NULL_HANDLE
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07074
  Transform Feedback Queries **must** not be active

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-07075
  Primitives Generated Queries **must** not be active
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pipelineStatistics-07076
  The `pipelineStatistics` member used to create any active Pipeline Statistics Query **must** not contain VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_VERTICES_BIT, VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_PRIMITIVES_BIT, VK_QUERY_PIPELINE_STATISTIC_VERTEX_SHADER_INVOCATIONS_BIT, VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_INVOCATIONS_BIT, VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_PRIMITIVES_BIT, VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT, VK_QUERY_PIPELINE_STATISTIC_CLIPPING_PRIMITIVES_BIT, VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_CONTROL_SHADER_PATCHES_BIT, or VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_EVALUATION_SHADER_INVOCATIONS_BIT
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08694
  If there is no bound graphics pipeline, and the `meshShader` feature is enabled, and a valid `VkShaderEXT` is bound to the VK_SHADER_STAGE_MESH_BIT_EXT stage, and that `VkShaderEXT` was created without the VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT flag, a valid `VkShaderEXT` **must** be bound to the VK_SHADER_STAGE_TASK_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-None-08695
  If there is no bound graphics pipeline, and the `meshShader` feature is enabled, and a valid `VkShaderEXT` is bound to the VK_SHADER_STAGE_MESH_BIT_EXT stage, and that `VkShaderEXT` was created with the VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT flag, there **must** be no `VkShaderEXT` bound to the VK_SHADER_STAGE_TASK_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-flags-13364
  If there are any shader objects bound that were created with `flags` that includes both the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR and VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT bits, the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_NO_TASK_SHADER_BIT_KHR set
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-flags-13365
  If there are any shader objects bound that were created with `flags` that includes both the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR and VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT bits, there **must** be no shader object bound to the VK_SHADER_STAGE_TASK_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-commandBuffer-13068
  `commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-drawIndirectCount-13069
  If the `drawIndirectCount` feature is not enabled this function **must** not be used

Valid Usage (Implicit)

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle

- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkDrawIndirectCount2InfoKHR structure
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawMeshTasksIndirectCount2EXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawMeshTasksIndirectCount2EXT is affected by conditional rendering

To record an indirect mesh tasks drawing command with the draw count sourced from a buffer, call:

WARNING This functionality is superseded by vkCmdDrawMeshTasksIndirectCount2EXT. See Legacy Functionality for more information.

```
// Provided by VK_EXT_mesh_shader with VK_VERSION_1_2 or VK_KHR_draw_indirect_count or
VK_AMD_draw_indirect_count
void vkCmdDrawMeshTasksIndirectCountEXT(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    buffer,
    VkDeviceSize                                offset,
    VkBuffer                                    countBuffer,
    VkDeviceSize                                countBufferOffset,
    uint32_t                                    maxDrawCount,
    uint32_t                                    stride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `buffer` is the buffer containing draw parameters.
- `offset` is the byte offset into `buffer` where parameters begin.
- `countBuffer` is the buffer containing the draw count.
- `countBufferOffset` is the byte offset into `countBuffer` where the draw count begins.
- `maxDrawCount` specifies the maximum number of draws that will be executed. The actual number of executed draw calls is the minimum of the count specified in `countBuffer` and `maxDrawCount`.
- `stride` is the byte stride between successive sets of draw parameters.

`vkCmdDrawMeshTasksIndirectCountEXT` behaves similarly to vkCmdDrawMeshTasksIndirectEXT except that the draw count is read by the device from a buffer during execution. The command will read an unsigned 32-bit integer from `countBuffer` located at `countBufferOffset` and use this as the draw count.

Valid Usage

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-magFilter-09598
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-mipmapMode-04770
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-mipmapMode-09599
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-unnormalizedCoordinates-09635
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample

a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias or any offset values
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX

as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not

access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08612
  If the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for storageBuffers, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08613
  If the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's viewType **must** match the Dim operand of the OpTypeImage as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's format and the Sampled Type operand of the OpTypeImage **must** match
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with `flags` that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with `flags` that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11300

If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the

VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by vkCmdBind*HeapEXT commands

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for uniformBuffers, the robustBufferAccess2 feature is not enabled, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for storageBuffers, the robustBufferAccess2 feature is not enabled, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11374
  If the robustBufferAccess2 feature is not enabled, the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of pBindInfo->reservedRangeSize set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to minSamplerHeapReservedRangeWithEmbedded
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound

to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat

as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' pInputAttachments [InputAttachmentIndex] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a InputAttachmentIndex referenced by VkRenderingInputAttachmentIndexInfo, or no InputAttachmentIndex if VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex or VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex are NULL, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an InputAttachmentIndex if both VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex and VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex are non-NULL
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an InputAttachmentIndex, the InputAttachmentIndex **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-12339

If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07835

If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::`maxMultiviewInstanceIndex`
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` set to VK_TRUE,

then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07634
  If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-06666
  If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of sampleLocationsEnable is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07843
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of depthTestEnable is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of

depthTestEnable is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07846
  If the depthBounds feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stencilTestEnable is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the viewportCount parameter of vkCmdSetViewportWithCount **must** match the scissorCount parameter of vkCmdSetScissorWithCount
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be `1`
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, and any shader object bound to a graphics stage writes to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be `1`
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of `blendEnable` **must** be VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of `rasterizerDiscardEnable` is VK_FALSE, and none of the following is enabled:
  - the `multisampledRenderToSingleSampled` feature

  then the current value of `rasterizationSamples` **must** be the same as the current color and/or depth/stencil attachments
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must**

have been begun with vkCmdBeginRendering

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo ::viewMask equal to VkRenderingInfo::viewMask
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-colorAttachmentCount-06179
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-dynamicRenderingUnusedAttachments-08910
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-dynamicRenderingUnusedAttachments-08912
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-dynamicRenderingUnusedAttachments-08911
  If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07749
  If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-attachmentCount-07750

If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07751
  If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT ::discardRectangleCount
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-rasterizerDiscardEnable-09236
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT ::maxDiscardRectangles
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07880
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the

dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-dynamicRenderingUnusedAttachments-08917
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-imageView-06183

If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-multisampledRenderToSingleSampled-07286
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-multisampledRenderToSingleSampled-07287
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pNext-07935
  If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with

multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-renderPass-06198
  If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pColorAttachments-08963
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pColorAttachments-11539
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pDepthAttachment-08964
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must**

not be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo ::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and `alphaToCoverageEnable` was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set `alphaToCoverageEnable` to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07625
  If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07626
  If the `logicOp` feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with

VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07630
  If the geometryStreams feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07631
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07632
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of conservativeRasterizationMode is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-conservativePointAndLineRasterization-07499
  If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07633
  If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07636
  If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled,

and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08666
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08669
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07849
  If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current lineRasterizationMode is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current alphaToCoverageEnable, alphaToOneEnable and sampleShadingEnable states **must** all be VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07639
  If the depthClipControl feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-09650
  If the depthClampControl feature is enabled, and a shader object is bound to any graphics

stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of depthClampEnable is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::rasterizationSamples parameter used to create the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the samples parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the rasterizationSamples parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the rasterizationSamples member of that structure
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing

command **must** have set a value for all active color attachments

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE, then the number of active color attachments **must** not exceed `advancedBlendMaxColorAttachments`
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the `rasterizationStream` to zero
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-sampleLocationsPerPixel-07482

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then the sampleLocationsPerPixel member of pSampleLocationsInfo in the last call to vkCmdSetSampleLocationsEXT **must** equal the rasterizationSamples parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.width in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT ::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.height in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT ::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction `InterpolateAtSample`
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then `sampleLocationsInfo.sampleLocationGridSize.width` **must** evenly divide VkMultisamplePropertiesEXT::`maxSampleLocationGridSize.width` as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then `sampleLocationsInfo.sampleLocationGridSize.height` **must** evenly divide VkMultisamplePropertiesEXT::`maxSampleLocationGridSize.height` as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then `sampleLocationsInfo.sampleLocationsPerPixel` **must** equal `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current `stippledLineEnable` state is VK_TRUE and the current `lineRasterizationMode` state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the `stippledRectangularLines` feature **must** be enabled
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current `stippledLineEnable` state is VK_TRUE and the current `lineRasterizationMode` state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the `stippledBresenhamLines` feature **must** be enabled

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to the last graphics stage in the logical pipeline, it **must** have been created with a nextStage including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08684
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08685
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08686
  If there is no bound graphics pipeline, and the tessellationShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08687

If there is no bound graphics pipeline, and the geometryShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_GEOMETRY_BIT

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08688
  If there is no bound graphics pipeline, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08689
  If there is no bound graphics pipeline, and the taskShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08690
  If there is no bound graphics pipeline, and the meshShader feature is enabled, vkCmdBindShadersEXT **must** have been called in the current command buffer with pStages with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08693
  If there is no bound graphics pipeline, and at least one of the taskShader and meshShader features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid VkShaderEXT bound, and the other **must** have no VkShaderEXT bound
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08696
  If there is no bound graphics pipeline, and a valid VkShaderEXT is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no VkShaderEXT bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call **must** also be bound
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any VkShaderEXT bound
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics

pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpDepthAttachmentReadEXT, the depthWriteEnable parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpStencilAttachmentReadEXT, the writeMask parameter in the last call to vkCmdSetStencilWriteMask **must** be 0
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the pColorWriteMasks parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of Location for any output attachment statically used in the Fragment Execution Model executed by this command **must** be less than maxFragmentDualSrcAttachments
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::pColorAttachmentLocations in the bound pipeline **must** match the value for the corresponding locations set currently in the current render pass instance
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::pColorAttachmentInputIndices in the bound pipeline, if either pColorAttachmentInputIndices[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-10928

If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::flags which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11861

If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-dynamicRenderingUnusedAttachments-

11867
If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering,

vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-flags-13363
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::stageFlags set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-stage-06480
  The bound graphics pipeline **must** not have been created with the VkPipelineShaderStageCreateInfo::stage member of any element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT,

VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pStages-10680
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** not have last bound the `pStages` element of VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT with a valid shader object other than VK_NULL_HANDLE
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07074
  Transform Feedback Queries **must** not be active
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-07075
  Primitives Generated Queries **must** not be active
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-pipelineStatistics-07076
  The `pipelineStatistics` member used to create any active Pipeline Statistics Query **must** not contain VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_VERTICES_BIT, VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_PRIMITIVES_BIT, VK_QUERY_PIPELINE_STATISTIC_VERTEX_SHADER_INVOCATIONS_BIT, VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_INVOCATIONS_BIT, VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_PRIMITIVES_BIT, VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT, VK_QUERY_PIPELINE_STATISTIC_CLIPPING_PRIMITIVES_BIT, VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_CONTROL_SHADER_PATCHES_BIT, or VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_EVALUATION_SHADER_INVOCATIONS_BIT
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08694
  If there is no bound graphics pipeline, and the `meshShader` feature is enabled, and a valid `VkShaderEXT` is bound to the VK_SHADER_STAGE_MESH_BIT_EXT stage, and that `VkShaderEXT` was created without the VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT flag, a valid `VkShaderEXT` **must** be bound to the VK_SHADER_STAGE_TASK_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-08695
  If there is no bound graphics pipeline, and the `meshShader` feature is enabled, and a valid `VkShaderEXT` is bound to the VK_SHADER_STAGE_MESH_BIT_EXT stage, and that `VkShaderEXT` was created with the VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT flag, there **must** be no `VkShaderEXT` bound to the VK_SHADER_STAGE_TASK_BIT_EXT stage
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-flags-13364
  If there are any shader objects bound that were created with `flags` that includes both the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR and VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT bits, the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_NO_TASK_SHADER_BIT_KHR set
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-flags-13365
  If there are any shader objects bound that were created with `flags` that includes both the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR and

VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT bits, there **must** be no shader object bound to the VK_SHADER_STAGE_TASK_BIT_EXT stage

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-buffer-02708
  If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-buffer-02709
  `buffer` **must** have been created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-offset-02710
  `offset` **must** be a multiple of `4`
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-commandBuffer-02711
  `commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-countBuffer-02714
  If `countBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-countBuffer-02715
  `countBuffer` **must** have been created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-countBufferOffset-02716
  `countBufferOffset` **must** be a multiple of `4`
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-countBuffer-02717
  The count stored in `countBuffer` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDrawIndirectCount`
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-countBufferOffset-04129
  (`countBufferOffset` + `sizeof`(uint32_t)) **must** be less than or equal to the size of `countBuffer`
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-None-04445
  If the `drawIndirectCount` feature is not enabled this function **must** not be used
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-stride-07096
  `stride` **must** be a multiple of `4` and **must** be greater than or equal to `sizeof`(`VkDrawMeshTasksIndirectCommandEXT`)
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-maxDrawCount-07097
  If `maxDrawCount` is greater than or equal to `1`, (`stride` × (`maxDrawCount` - 1) + `offset` + `sizeof`(`VkDrawMeshTasksIndirectCommandEXT`)) **must** be less than or equal to the size of `buffer`
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-MeshEXT-07100
  The current pipeline bound to VK_PIPELINE_BIND_POINT_GRAPHICS **must** contain a shader stage using the `MeshEXT` `Execution` `Model`

Valid Usage (Implicit)

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle

- VUID-vkCmdDrawMeshTasksIndirectCountEXT-buffer-parameter
  `buffer` **must** be a valid VkBuffer handle
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-countBuffer-parameter
  `countBuffer` **must** be a valid VkBuffer handle
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdDrawMeshTasksIndirectCountEXT-commonparent
  Each of `buffer`, `commandBuffer`, and `countBuffer` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDrawMeshTasksIndirectCountEXT is affected by conditional rendering

Chapter 26. Fixed-Function Vertex Processing

Vertex fetching is controlled via configurable state, as a logically distinct graphics pipeline stage.

26.1. Vertex Attributes

Vertex shaders **can** define input variables, which receive *vertex attribute* data transferred from one or more VkBuffer(s) by drawing commands. Vertex shader input variables are bound to buffers via an indirect binding where the vertex shader associates a *vertex input attribute* number with each variable, vertex input attributes are associated to *vertex input bindings* on a per-pipeline basis, and vertex input bindings are associated with specific buffers on a per-draw basis via the vkCmdBindVertexBuffers command. Vertex input attribute and vertex input binding descriptions also contain format information controlling how data is extracted from buffer memory and converted to the format expected by the vertex shader.

There are VkPhysicalDeviceLimits::maxVertexInputAttributes number of vertex input attributes and VkPhysicalDeviceLimits::maxVertexInputBindings number of vertex input bindings (each referred to by zero-based indices), where there are at least as many vertex input attributes as there are vertex input bindings. Applications **can** store multiple vertex input attributes interleaved in a single buffer, and use a single vertex input binding to access those attributes.

In GLSL, vertex shaders associate input variables with a vertex input attribute number using the location layout qualifier. The Component layout qualifier associates components of a vertex shader input variable with components of a vertex input attribute.

GLSL Example

```
// Assign location M to variableName
layout (location=M, component=2) in vec2 variableName;

// Assign locations [N,N+L) to the array elements of variableNameArray
layout (location=N) in vec4 variableNameArray[L];
```

In SPIR-V, vertex shaders associate input variables with a vertex input attribute number using the Location decoration. The Component decoration associates components of a vertex shader input variable with components of a vertex input attribute. The Location and Component decorations are specified via the OpDecorate instruction.

SPIR-V Example

```
          ...
     %1 = OpExtInstImport "GLSL.std.450"
          ...
          OpName %9 "variableName"
          OpName %15 "variableNameArray"
          OpDecorate %18 BuiltIn VertexIndex
```

```
        OpDecorate %19 BuiltIn InstanceIndex
        OpDecorate %9 Location M
        OpDecorate %9 Component 2
        OpDecorate %15 Location N
        ...
 %2 = OpTypeVoid
 %3 = OpTypeFunction %2
 %6 = OpTypeFloat 32
 %7 = OpTypeVector %6 2
 %8 = OpTypePointer Input %7
 %9 = OpVariable %8 Input
%10 = OpTypeVector %6 4
%11 = OpTypeInt 32 0
%12 = OpConstant %11 L
%13 = OpTypeArray %10 %12
%14 = OpTypePointer Input %13
%15 = OpVariable %14 Input
        ...
```

26.1.1. Attribute Location and Component Assignment

The `Location` decoration specifies which vertex input attribute is used to read and interpret the data that a variable will consume.

When a vertex shader input variable declared using a 16- or 32-bit scalar or vector data type is assigned a `Location`, its value(s) are taken from the components of the input attribute specified with the corresponding `VkVertexInputAttributeDescription`::`location`. The components used depend on the type of variable and the `Component` decoration specified in the variable declaration, as identified in Input Attribute Components Accessed By 16-Bit and 32-Bit Input Variables. Any 16-bit or 32-bit scalar or vector input will consume a single `Location`. For 16-bit and 32-bit data types, missing components are filled in with default values as described below.

If an implementation supports `storageInputOutput16`, vertex shader input variables **can** have a width of 16 bits.

In all the following component assignment specifications, if or the `maintenance9` feature is enabled, and there is no `VkVertexInputAttributeDescription`::`location` specified for the shader vertex attribute `Location` being read, the value (0,0,0,0) or (0,0,0,1) is used for each of the equivalent (x,y,z,w) components consumed entries as specified below.

Table 32. Input Attribute Components Accessed By 16-Bit and 32-Bit Input Variables

16-bit or 32-bit data type	`Component` **decoration**	**Components consumed**
scalar	0 or unspecified	(x, 0, 0, 0)
scalar	1	(0, y, 0, 0)
scalar	2	(0, 0, z, 0)

16-bit or 32-bit data type	`Component` decoration	Components consumed
scalar	3	(o, o, o, w)
two-component vector	0 or unspecified	(x, y, o, o)
two-component vector	1	(o, y, z, o)
two-component vector	2	(o, o, z, w)
three-component vector	0 or unspecified	(x, y, z, o)
three-component vector	1	(o, y, z, w)
four-component vector	0 or unspecified	(x, y, z, w)

Components indicated by “o” are available for use by other input variables which are sourced from the same attribute, and if used, are either filled with the corresponding component from the input format (if present), or the default value.

When a vertex shader input variable declared using a 32-bit floating-point matrix type is assigned a `Location` *i*, its values are taken from consecutive input attributes starting with the corresponding `VkVertexInputAttributeDescription`::`location`. Such matrices are treated as an array of column vectors with values taken from the input attributes identified in Input Attributes Accessed by 32-Bit Input Matrix Variables. The `VkVertexInputAttributeDescription`::`format` **must** be specified with a VkFormat that corresponds to the appropriate type of column vector. The `Component` decoration **must** not be used with matrix types.

Table 33. Input Attributes Accessed by 32-Bit Input Matrix Variables

Data type	Column vector type	Locations consumed	Components consumed
mat2	two-component vector	i, i+1	(x, y, o, o), (x, y, o, o)
mat2x3	three-component vector	i, i+1	(x, y, z, o), (x, y, z, o)
mat2x4	four-component vector	i, i+1	(x, y, z, w), (x, y, z, w)
mat3x2	two-component vector	i, i+1, i+2	(x, y, o, o), (x, y, o, o), (x, y, o, o)
mat3	three-component vector	i, i+1, i+2	(x, y, z, o), (x, y, z, o), (x, y, z, o)
mat3x4	four-component vector	i, i+1, i+2	(x, y, z, w), (x, y, z, w), (x, y, z, w)
mat4x2	two-component vector	i, i+1, i+2, i+3	(x, y, o, o), (x, y, o, o), (x, y, o, o), (x, y, o, o)
mat4x3	three-component vector	i, i+1, i+2, i+3	(x, y, z, o), (x, y, z, o), (x, y, z, o), (x, y, z, o)
mat4	four-component vector	i, i+1, i+2, i+3	(x, y, z, w), (x, y, z, w), (x, y, z, w), (x, y, z, w)

Components indicated by “o” are available for use by other input variables which are sourced from the same attribute, and if used, are either filled with the corresponding component from the input (if present), or the default value.

When a vertex shader input variable declared using a scalar or vector 64-bit data type is assigned a `Location` *i*, its values are taken from consecutive input attributes starting with the corresponding `VkVertexInputAttributeDescription`::`location`. The `Location` slots and `Component` words used depend on the type of variable and the `Component` decoration specified in the variable declaration, as identified in Input Attribute Locations and Components Accessed by 64-Bit Input Variables. For 64-bit data types, no default attribute values are provided. Input variables **must** not use more components than provided by the attribute.

Table 34. Input Attribute Locations and Components Accessed by 64-Bit Input Variables

Input format	Locations consumed	64-bit data type	`Location` decoration	`Component` decoration	32-bit components consumed
R64	i	scalar	i	0 or unspecified	(x, y, -, -)
R64G64	i	scalar	i	0 or unspecified	(x, y, o, o)
		scalar	i	2	(o, o, z, w)
		two-component vector	i	0 or unspecified	(x, y, z, w)
R64G64B64	i, i+1	scalar	i	0 or unspecified	(x, y, o, o), (o, o, -, -)
		scalar	i	2	(o, o, z, w), (o, o, -, -)
		scalar	i+1	0 or unspecified	(o, o, o, o), (x, y, -, -)
		two-component vector	i	0 or unspecified	(x, y, z, w), (o, o, -, -)
		three-component vector	i	unspecified	(x, y, z, w), (x, y, -, -)

Input format	Locations consumed	64-bit data type	`Location` decoration	`Component` decoration	32-bit components consumed
R64G64B64A64	i, i+1	scalar	i	0 or unspecified	(x, y, o, o), (o, o, o, o)
		scalar	i	2	(o, o, z, w), (o, o, o, o)
		scalar	i+1	0 or unspecified	(o, o, o, o), (x, y, o, o)
		scalar	i+1	2	(o, o, o, o), (o, o, z, w)
		two-component vector	i	0 or unspecified	(x, y, z, w), (o, o, o, o)
		two-component vector	i+1	0 or unspecified	(o, o, o, o), (x, y, z, w)
		three-component vector	i	unspecified	(x, y, z, w), (x, y, o, o)
		four-component vector	i	unspecified	(x, y, z, w), (x, y, z, w)

Components indicated by “o” are available for use by other input variables which are sourced from the same attribute. Components indicated by “-” are not available for input variables as there are no default values provided for 64-bit data types, and there is no data provided by the input format.

When a vertex shader input variable declared using a 64-bit floating-point matrix type is assigned a `Location` *i*, its values are taken from consecutive input attribute locations. Such matrices are treated as an array of column vectors with values taken from the input attributes as shown in Input Attribute Locations and Components Accessed by 64-Bit Input Variables. Each column vector starts at the `Location` immediately following the last `Location` of the previous column vector. The number of attributes and components assigned to each matrix is determined by the matrix dimensions and ranges from two to eight locations.

When a vertex shader input variable declared using an array type is assigned a location, its values are taken from consecutive input attributes starting with the corresponding `VkVertexInputAttributeDescription`::`location`. The number of attributes and components assigned to each element are determined according to the data type of the array elements and `Component` decoration (if any) specified in the declaration of the array, as described above. Each element of the array, in order, is assigned to consecutive locations, but all at the same specified component within each location.

Only input variables declared with the data types and component decorations as specified above are supported. Two variables are allowed to share the same `Location` slot only if their `Component` words do not overlap. If multiple variables share the same `Location` slot, they **must** all have the same SPIR-V floating-point component type or all have the same width scalar type components.

26.2. Vertex Input Description

Applications specify vertex input attribute and vertex input binding descriptions as part of graphics pipeline creation by setting the VkGraphicsPipelineCreateInfo::pVertexInputState pointer to a VkPipelineVertexInputStateCreateInfo structure. Alternatively, if the graphics pipeline is created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled, then the vertex input attribute and vertex input binding descriptions are specified dynamically with vkCmdSetVertexInputEXT, and the VkGraphicsPipelineCreateInfo::pVertexInputState pointer is ignored.

The VkPipelineVertexInputStateCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineVertexInputStateCreateInfo {
    VkStructureType                             sType;
    const void*                                 pNext;
    VkPipelineVertexInputStateCreateFlags       flags;
    uint32_t                                    vertexBindingDescriptionCount;
    const VkVertexInputBindingDescription*      pVertexBindingDescriptions;
    uint32_t                                    vertexAttributeDescriptionCount;
    const VkVertexInputAttributeDescription*    pVertexAttributeDescriptions;
} VkPipelineVertexInputStateCreateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is reserved for future use.
- vertexBindingDescriptionCount is the number of vertex binding descriptions provided in pVertexBindingDescriptions.
- pVertexBindingDescriptions is a pointer to an array of VkVertexInputBindingDescription structures.
- vertexAttributeDescriptionCount is the number of vertex attribute descriptions provided in pVertexAttributeDescriptions.
- pVertexAttributeDescriptions is a pointer to an array of VkVertexInputAttributeDescription structures.

Valid Usage

- VUID-VkPipelineVertexInputStateCreateInfo-vertexBindingDescriptionCount-00613 vertexBindingDescriptionCount **must** be less than or equal to VkPhysicalDeviceLimits::maxVertexInputBindings
- VUID-VkPipelineVertexInputStateCreateInfo-vertexAttributeDescriptionCount-00614 vertexAttributeDescriptionCount **must** be less than or equal to VkPhysicalDeviceLimits::maxVertexInputAttributes
- VUID-VkPipelineVertexInputStateCreateInfo-binding-00615

For every `binding` specified by each element of `pVertexAttributeDescriptions`, a VkVertexInputBindingDescription **must** exist in `pVertexBindingDescriptions` with the same value of `binding`

- VUID-VkPipelineVertexInputStateCreateInfo-pVertexBindingDescriptions-00616
  All elements of `pVertexBindingDescriptions` **must** describe distinct binding numbers
- VUID-VkPipelineVertexInputStateCreateInfo-pVertexAttributeDescriptions-00617
  All elements of `pVertexAttributeDescriptions` **must** describe distinct attribute locations

Valid Usage (Implicit)

- VUID-VkPipelineVertexInputStateCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_VERTEX_INPUT_STATE_CREATE_INFO
- VUID-VkPipelineVertexInputStateCreateInfo-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkPipelineVertexInputDivisorStateCreateInfo
- VUID-VkPipelineVertexInputStateCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkPipelineVertexInputStateCreateInfo-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkPipelineVertexInputStateCreateInfo-pVertexBindingDescriptions-parameter
  If `vertexBindingDescriptionCount` is not `0`, `pVertexBindingDescriptions` **must** be a valid pointer to an array of `vertexBindingDescriptionCount` valid VkVertexInputBindingDescription structures
- VUID-VkPipelineVertexInputStateCreateInfo-pVertexAttributeDescriptions-parameter
  If `vertexAttributeDescriptionCount` is not `0`, `pVertexAttributeDescriptions` **must** be a valid pointer to an array of `vertexAttributeDescriptionCount` valid VkVertexInputAttributeDescription structures

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineVertexInputStateCreateFlags;
```

`VkPipelineVertexInputStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

Each vertex input binding is specified by the `VkVertexInputBindingDescription` structure, defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkVertexInputBindingDescription {
    uint32_t             binding;
    uint32_t             stride;
    VkVertexInputRate    inputRate;
} VkVertexInputBindingDescription;
```

- `binding` is the binding number that this structure describes.
- `stride` is the byte stride between consecutive elements within the buffer.
- `inputRate` is a VkVertexInputRate value specifying whether vertex attribute addressing is a function of the vertex index or of the instance index.

Valid Usage

- VUID-VkVertexInputBindingDescription-binding-00618
  `binding` **must** be less than `VkPhysicalDeviceLimits`::`maxVertexInputBindings`
- VUID-VkVertexInputBindingDescription-stride-00619
  `stride` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxVertexInputBindingStride`
- VUID-VkVertexInputBindingDescription-stride-04456
  If the `VK_KHR_portability_subset` extension is enabled, `stride` **must** be a multiple of, and at least as large as, VkPhysicalDevicePortabilitySubsetPropertiesKHR::`minVertexInputBindingStrideAlignment`

Valid Usage (Implicit)

- VUID-VkVertexInputBindingDescription-inputRate-parameter
  `inputRate` **must** be a valid VkVertexInputRate value

Possible values of VkVertexInputBindingDescription::`inputRate`, specifying the rate at which vertex attributes are pulled from buffers, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkVertexInputRate {
    VK_VERTEX_INPUT_RATE_VERTEX = 0,
    VK_VERTEX_INPUT_RATE_INSTANCE = 1,
} VkVertexInputRate;
```

- VK_VERTEX_INPUT_RATE_VERTEX specifies that vertex attribute addressing is a function of the vertex index.
- VK_VERTEX_INPUT_RATE_INSTANCE specifies that vertex attribute addressing is a function of the instance index.

Each vertex input attribute is specified by the `VkVertexInputAttributeDescription` structure, defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkVertexInputAttributeDescription {
    uint32_t    location;
    uint32_t    binding;
    VkFormat    format;
    uint32_t    offset;
} VkVertexInputAttributeDescription;
```

- `location` is the shader input location number for this attribute.
- `binding` is the binding number which this attribute takes its data from.
- `format` is the size and type of the vertex attribute data.
- `offset` is a byte offset of this attribute relative to the start of an element in the vertex input binding.

Valid Usage

- VUID-VkVertexInputAttributeDescription-location-00620
  `location` **must** be less than `VkPhysicalDeviceLimits`::`maxVertexInputAttributes`
- VUID-VkVertexInputAttributeDescription-binding-00621
  `binding` **must** be less than `VkPhysicalDeviceLimits`::`maxVertexInputBindings`
- VUID-VkVertexInputAttributeDescription-offset-00622
  `offset` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxVertexInputAttributeOffset`
- VUID-VkVertexInputAttributeDescription-format-00623
  The format features of `format` **must** contain VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT
- VUID-VkVertexInputAttributeDescription-vertexAttributeAccessBeyondStride-04457
  If the `VK_KHR_portability_subset` extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::`vertexAttributeAccessBeyondStride` is VK_FALSE, the sum of `offset` plus the size of the vertex attribute data described by `format` **must** not be greater than `stride` in the VkVertexInputBindingDescription referenced in `binding`

Valid Usage (Implicit)

- VUID-VkVertexInputAttributeDescription-format-parameter
  `format` **must** be a valid VkFormat value

To dynamically set the vertex input attribute and vertex input binding descriptions, call:

```
// Provided by VK_EXT_shader_object, VK_EXT_vertex_input_dynamic_state
void vkCmdSetVertexInputEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    vertexBindingDescriptionCount,
    const VkVertexInputBindingDescription2EXT*  pVertexBindingDescriptions,
    uint32_t                                    vertexAttributeDescriptionCount,
    const VkVertexInputAttributeDescription2EXT* pVertexAttributeDescriptions);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `vertexBindingDescriptionCount` is the number of vertex binding descriptions provided in `pVertexBindingDescriptions`.
- `pVertexBindingDescriptions` is a pointer to an array of VkVertexInputBindingDescription2EXT structures.
- `vertexAttributeDescriptionCount` is the number of vertex attribute descriptions provided in `pVertexAttributeDescriptions`.
- `pVertexAttributeDescriptions` is a pointer to an array of VkVertexInputAttributeDescription2EXT structures.

This command sets the vertex input attribute and vertex input binding descriptions state for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_VERTEX_INPUT_EXT set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the VkGraphicsPipelineCreateInfo::`pVertexInputState` values used to create the currently active pipeline.

If drawing using shader objects, or if the bound pipeline state object was also created with the VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE dynamic state enabled, then vkCmdBindVertexBuffers2 can be used instead of `vkCmdSetVertexInputEXT` to dynamically set the stride.

The vertex attribute description for any location in the range [0,`VkPhysicalDeviceLimits`::`maxVertexInputAttributes`) not specified in the `pVertexAttributeDescriptions` array becomes undefined.

Valid Usage

- VUID-vkCmdSetVertexInputEXT-None-08546
  Either the `vertexInputDynamicState` feature or the `shaderObject` feature or both **must** be enabled
- VUID-vkCmdSetVertexInputEXT-vertexBindingDescriptionCount-04791
  `vertexBindingDescriptionCount` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxVertexInputBindings`
- VUID-vkCmdSetVertexInputEXT-vertexAttributeDescriptionCount-04792
  `vertexAttributeDescriptionCount` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxVertexInputAttributes`

- VUID-vkCmdSetVertexInputEXT-binding-04793
  For every `binding` specified by each element of `pVertexAttributeDescriptions`, a VkVertexInputBindingDescription2EXT **must** exist in `pVertexBindingDescriptions` with the same value of `binding`
- VUID-vkCmdSetVertexInputEXT-pVertexBindingDescriptions-04794
  All elements of `pVertexBindingDescriptions` **must** describe distinct binding numbers
- VUID-vkCmdSetVertexInputEXT-pVertexAttributeDescriptions-04795
  All elements of `pVertexAttributeDescriptions` **must** describe distinct attribute locations

Valid Usage (Implicit)

- VUID-vkCmdSetVertexInputEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetVertexInputEXT-pVertexBindingDescriptions-parameter
  If `vertexBindingDescriptionCount` is not `0`, `pVertexBindingDescriptions` **must** be a valid pointer to an array of `vertexBindingDescriptionCount` valid VkVertexInputBindingDescription2EXT structures
- VUID-vkCmdSetVertexInputEXT-pVertexAttributeDescriptions-parameter
  If `vertexAttributeDescriptionCount` is not `0`, `pVertexAttributeDescriptions` **must** be a valid pointer to an array of `vertexAttributeDescriptionCount` valid VkVertexInputAttributeDescription2EXT structures
- VUID-vkCmdSetVertexInputEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetVertexInputEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetVertexInputEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetVertexInputEXT is not affected by conditional rendering

The VkVertexInputBindingDescription2EXT structure is defined as:

```
// Provided by VK_EXT_shader_object, VK_EXT_vertex_input_dynamic_state
typedef struct VkVertexInputBindingDescription2EXT {
    VkStructureType      sType;
    void*                pNext;
    uint32_t             binding;
    uint32_t             stride;
    VkVertexInputRate    inputRate;
    uint32_t             divisor;
} VkVertexInputBindingDescription2EXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- binding is the binding number that this structure describes.
- stride is the byte stride between consecutive elements within the buffer.
- inputRate is a VkVertexInputRate value specifying whether vertex attribute addressing is a function of the vertex index or of the instance index.
- divisor is the number of successive instances that will use the same value of the vertex attribute when instanced rendering is enabled. This member **can** be a value other than 1 if the vertexAttributeInstanceRateDivisor feature is enabled. For example, if the divisor is N, the same vertex attribute will be applied to N successive instances before moving on to the next vertex attribute. The maximum value of divisor is implementation-dependent and can be queried using VkPhysicalDeviceVertexAttributeDivisorPropertiesEXT::maxVertexAttribDivisor. A value of 0 **can** be used for the divisor if the vertexAttributeInstanceRateZeroDivisor feature is enabled. In this case, the same vertex attribute will be applied to all instances.

Valid Usage

- VUID-VkVertexInputBindingDescription2EXT-binding-04796
  binding **must** be less than VkPhysicalDeviceLimits::maxVertexInputBindings

- VUID-VkVertexInputBindingDescription2EXT-stride-04797
  `stride` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxVertexInputBindingStride`
- VUID-VkVertexInputBindingDescription2EXT-divisor-04798
  If the `vertexAttributeInstanceRateZeroDivisor` feature is not enabled, `divisor` **must** not be `0`
- VUID-VkVertexInputBindingDescription2EXT-divisor-04799
  If the `vertexAttributeInstanceRateDivisor` feature is not enabled, `divisor` **must** be `1`
- VUID-VkVertexInputBindingDescription2EXT-divisor-06226
  `divisor` **must** be a value between `0` and `VkPhysicalDeviceVertexAttributeDivisorPropertiesEXT`::`maxVertexAttribDivisor`, inclusive
- VUID-VkVertexInputBindingDescription2EXT-divisor-06227
  If `divisor` is not `1` then `inputRate` **must** be of type VK_VERTEX_INPUT_RATE_INSTANCE

Valid Usage (Implicit)

- VUID-VkVertexInputBindingDescription2EXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VERTEX_INPUT_BINDING_DESCRIPTION_2_EXT
- VUID-VkVertexInputBindingDescription2EXT-inputRate-parameter
  `inputRate` **must** be a valid VkVertexInputRate value

The `VkVertexInputAttributeDescription2EXT` structure is defined as:

```
// Provided by VK_EXT_shader_object, VK_EXT_vertex_input_dynamic_state
typedef struct VkVertexInputAttributeDescription2EXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           location;
    uint32_t           binding;
    VkFormat           format;
    uint32_t           offset;
} VkVertexInputAttributeDescription2EXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `location` is the shader input location number for this attribute.
- `binding` is the binding number which this attribute takes its data from.
- `format` is the size and type of the vertex attribute data.
- `offset` is a byte offset of this attribute relative to the start of an element in the vertex input binding.

Valid Usage

- VUID-VkVertexInputAttributeDescription2EXT-location-06228
  `location` **must** be less than `VkPhysicalDeviceLimits`::`maxVertexInputAttributes`
- VUID-VkVertexInputAttributeDescription2EXT-binding-06229
  `binding` **must** be less than `VkPhysicalDeviceLimits`::`maxVertexInputBindings`
- VUID-VkVertexInputAttributeDescription2EXT-offset-06230
  `offset` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxVertexInputAttributeOffset`
- VUID-VkVertexInputAttributeDescription2EXT-format-04805
  The format features of `format` **must** contain VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT
- VUID-VkVertexInputAttributeDescription2EXT-vertexAttributeAccessBeyondStride-04806
  If the `VK_KHR_portability_subset` extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::`vertexAttributeAccessBeyondStride` is VK_FALSE, the sum of `offset` plus the size of the vertex attribute data described by `format` **must** not be greater than `stride` in the VkVertexInputBindingDescription2EXT referenced in `binding`

Valid Usage (Implicit)

- VUID-VkVertexInputAttributeDescription2EXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VERTEX_INPUT_ATTRIBUTE_DESCRIPTION_2_EXT
- VUID-VkVertexInputAttributeDescription2EXT-format-parameter
  `format` **must** be a valid VkFormat value

To bind strided address ranges as vertex buffers to a command buffer for use in subsequent drawing commands, call:

```
// Provided by VK_KHR_device_address_commands
void vkCmdBindVertexBuffers3KHR(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstBinding,
    uint32_t                                    bindingCount,
    const VkBindVertexBuffer3InfoKHR*           pBindingInfos);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `firstBinding` is the index of the first vertex input binding whose state is updated by the command.
- `bindingCount` is the number of vertex input bindings whose state is updated by the command.
- `pInfos` is a pointer to an array of VkBindVertexBuffer3InfoKHR structures defining address ranges to bind as vertex buffers.

The values taken from the address ranges defined by elements i of pInfos replace the current state for the vertex input binding firstBinding + i, for i in [0, bindingCount). The vertex input binding is updated to the range of memory indicated by pInfos[i]. All vertex input attributes that use each of these bindings will use these updated addresses in their address calculations for subsequent drawing commands. If the nullDescriptor feature is enabled, elements of pInfos **can** have a addressRange.size and addressRange.address of 0. If a vertex input attribute is bound to a vertex input binding with a size and address of 0, the values taken from memory are considered to be zero, and missing G, B, or A components are filled with (0,0,1).

This command also dynamically sets the byte strides between consecutive elements within the strided range defined by pInfos[i] to the value of pInfos[i].addressRange.stride when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE set in VkPipelineDynamicStateCreateInfo ::pDynamicStates. Otherwise, strides are specified by the VkVertexInputBindingDescription::stride values used to create the currently active pipeline.

If drawing using shader objects or if the bound pipeline state object was also created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **can** be used instead of vkCmdBindVertexBuffers3KHR to set the stride.

NOTE Unlike the static state to set the same, stride for each address range must be between 0 and the maximum extent of the attributes in the binding. vkCmdSetVertexInputEXT does not have this restriction so can be used if other stride values are desired.

Valid Usage

- VUID-vkCmdBindVertexBuffers3KHR-firstBinding-13070
  firstBinding **must** be less than VkPhysicalDeviceLimits::maxVertexInputBindings
- VUID-vkCmdBindVertexBuffers3KHR-firstBinding-13071
  The sum of firstBinding and bindingCount **must** be less than or equal to VkPhysicalDeviceLimits::maxVertexInputBindings
- VUID-vkCmdBindVertexBuffers3KHR-addressRange-13073
  The addressRange.stride of all elements of pInfos **must** be either 0 or greater than or equal to the maximum extent of all vertex input attributes fetched from the corresponding binding, where the extent is calculated as the VkVertexInputAttributeDescription::offset plus VkVertexInputAttributeDescription ::format size

Valid Usage (Implicit)

- VUID-vkCmdBindVertexBuffers3KHR-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindVertexBuffers3KHR-pBindingInfos-parameter
  pBindingInfos **must** be a valid pointer to an array of bindingCount valid

VkBindVertexBuffer3InfoKHR structures

- VUID-vkCmdBindVertexBuffers3KHR-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdBindVertexBuffers3KHR-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindVertexBuffers3KHR-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindVertexBuffers3KHR-bindingCount-arraylength
  bindingCount **must** be greater than 0

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindVertexBuffers3KHR is not affected by conditional rendering

VkBindVertexBuffer3InfoKHR is defined as:

```
// Provided by VK_KHR_device_address_commands
typedef struct VkBindVertexBuffer3InfoKHR {
    VkStructureType                  sType;
    const void*                      pNext;
    VkBool32                         setStride;
    VkStridedDeviceAddressRangeKHR   addressRange;
    VkAddressCommandFlagsKHR         addressFlags;
} VkBindVertexBuffer3InfoKHR;
```

- sType is a VkStructureType value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `setStride` is a VkBool32 value indicating whether `addressRange.stride` sets the stride for the buffer. If `setStride` is VK_TRUE, the dynamic stride is set for all attributes from this buffer. If `setStride` is VK_FALSE, the stride value is not set.
- `addressRange` is the VkStridedDeviceAddressRangeKHR of the address range to bind.
- `addressFlags` is a VkAddressCommandFlagsKHR value defining the flags for the address range.

Valid Usage

- VUID-VkBindVertexBuffer3InfoKHR-addressRange-13097
  If the range specified by `addressRange` is not bound completely to memory when accessed, `addressFlags` **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR
- VUID-VkBindVertexBuffer3InfoKHR-addressRange-13098
  If the buffer from which the range specified by `addressRange` was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `addressFlags` **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkBindVertexBuffer3InfoKHR-addressRange-13099
  If the buffer from which the range specified by `addressRange` was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and `protectedNoFault` is not supported, `addressFlags` **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkBindVertexBuffer3InfoKHR-addressFlags-13100
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkBindVertexBuffer3InfoKHR-addressRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkBindVertexBuffer3InfoKHR-addressRange-13123
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkBindVertexBuffer3InfoKHR-addressFlags-13101
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkBindVertexBuffer3InfoKHR-addressRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `addressFlags` **must**

include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR

- VUID-VkBindVertexBuffer3InfoKHR-addressRange-13125
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing addressRange, was created without VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, addressFlags **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkBindVertexBuffer3InfoKHR-addressRange-13074
  If addressRange.address is not 0, the buffer from which addressRange was queried **must** have been created with the VK_BUFFER_USAGE_VERTEX_BUFFER_BIT usage flag set
- VUID-VkBindVertexBuffer3InfoKHR-addressRange-13075
  If addressRange.size is 0, addressRange.address **must** be 0
- VUID-VkBindVertexBuffer3InfoKHR-size-13072
  If the nullDescriptor feature is not enabled, addressRange.size **must** not be 0
- VUID-VkBindVertexBuffer3InfoKHR-setStride-13126
  If setStride is VK_TRUE, addressRange.stride **must** be less than or equal to VkPhysicalDeviceLimits::maxVertexInputBindingStride
- VUID-VkBindVertexBuffer3InfoKHR-setStride-13127
  If setStride is VK_FALSE, addressRange.stride **must** be 0

Valid Usage (Implicit)

- VUID-VkBindVertexBuffer3InfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_BIND_VERTEX_BUFFER_3_INFO_KHR
- VUID-VkBindVertexBuffer3InfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkBindVertexBuffer3InfoKHR-addressFlags-parameter
  If addressFlags is not 0, addressFlags **must** be a valid combination of VkAddressCommandFlagBitsKHR values

To bind vertex buffers, along with their sizes and strides, to a command buffer for use in subsequent drawing commands, call:

```
// Provided by VK_VERSION_1_3
void vkCmdBindVertexBuffers2(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstBinding,
    uint32_t                                    bindingCount,
    const VkBuffer*                             pBuffers,
    const VkDeviceSize*                         pOffsets,
    const VkDeviceSize*                         pSizes,
    const VkDeviceSize*                         pStrides);
```

```
// Provided by VK_EXT_extended_dynamic_state, VK_EXT_shader_object
// Equivalent to vkCmdBindVertexBuffers2
void vkCmdBindVertexBuffers2EXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstBinding,
    uint32_t                                    bindingCount,
    const VkBuffer*                             pBuffers,
    const VkDeviceSize*                         pOffsets,
    const VkDeviceSize*                         pSizes,
    const VkDeviceSize*                         pStrides);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `firstBinding` is the index of the first vertex input binding whose state is updated by the command.
- `bindingCount` is the number of vertex input bindings whose state is updated by the command.
- `pBuffers` is a pointer to an array of buffer handles.
- `pOffsets` is a pointer to an array of buffer offsets.
- `pSizes` is `NULL` or a pointer to an array of the size in bytes of vertex data bound from `pBuffers`.
- `pStrides` is `NULL` or a pointer to an array of buffer strides.

The values taken from elements i of `pBuffers` and `pOffsets` replace the current state for the vertex input binding `firstBinding` + i, for i in [0, `bindingCount`). The vertex input binding is updated to start at the offset indicated by `pOffsets`[i] from the start of the buffer `pBuffers`[i]. If `pSizes` is not `NULL` then `pSizes`[i] specifies the bound size of the vertex buffer starting from the corresponding elements of `pBuffers`[i] plus `pOffsets`[i]. If `pSizes`[i] is VK_WHOLE_SIZE then the bound size is from `pBuffers`[i] plus `pOffsets`[i] to the end of the buffer `pBuffers`[i]. All vertex input attributes that use each of these bindings will use these updated addresses in their address calculations for subsequent drawing commands. If the `nullDescriptor` feature is enabled, elements of `pBuffers` **can** be VK_NULL_HANDLE, and **can** be used by the vertex shader. If a vertex input attribute is bound to a vertex input binding that is VK_NULL_HANDLE, the values taken from memory are considered to be zero, and missing G, B, or A components are filled with (0,0,1).

This command also dynamically sets the byte strides between consecutive elements within buffer `pBuffers`[i] to the corresponding `pStrides`[i] value when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE set in

VkPipelineDynamicStateCreateInfo::pDynamicStates. Otherwise, strides are specified by the VkVertexInputBindingDescription::stride values used to create the currently active pipeline.

If drawing using shader objects or if the bound pipeline state object was also created with the VK_DYNAMIC_STATE_VERTEX_INPUT_EXT dynamic state enabled then vkCmdSetVertexInputEXT **can** be used instead of vkCmdBindVertexBuffers2 to set the stride.

NOTE	Unlike the static state to set the same, pStrides must be between 0 and the maximum extent of the attributes in the binding. vkCmdSetVertexInputEXT does not have this restriction so can be used if other stride values are desired.

Valid Usage

- VUID-vkCmdBindVertexBuffers2-None-08971
  At least one of the following **must** be true:
  - the extendedDynamicState feature is enabled
  - the shaderObject feature is enabled
  - the value of VkApplicationInfo::apiVersion used to create the VkInstance parent of commandBuffer is greater than or equal to Version 1.3
- VUID-vkCmdBindVertexBuffers2-firstBinding-03355
  firstBinding **must** be less than VkPhysicalDeviceLimits::maxVertexInputBindings
- VUID-vkCmdBindVertexBuffers2-firstBinding-03356
  The sum of firstBinding and bindingCount **must** be less than or equal to VkPhysicalDeviceLimits::maxVertexInputBindings
- VUID-vkCmdBindVertexBuffers2-pOffsets-03357
  If pSizes is not NULL, all elements of pOffsets **must** be less than the size of the corresponding element in pBuffers
- VUID-vkCmdBindVertexBuffers2-pSizes-03358
  If pSizes is not NULL, all elements of pOffsets plus pSizes , where pSizes is not VK_WHOLE_SIZE, **must** be less than or equal to the size of the corresponding element in pBuffers
- VUID-vkCmdBindVertexBuffers2-pBuffers-03359
  All elements of pBuffers **must** have been created with the VK_BUFFER_USAGE_VERTEX_BUFFER_BIT usage flag set
- VUID-vkCmdBindVertexBuffers2-pBuffers-03360
  Each element of pBuffers that is non-sparse **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-vkCmdBindVertexBuffers2-pBuffers-04111
  If the nullDescriptor feature is not enabled, all elements of pBuffers **must** not be VK_NULL_HANDLE
- VUID-vkCmdBindVertexBuffers2-pBuffers-04112
  If an element of pBuffers is VK_NULL_HANDLE, then the corresponding element of pOffsets **must** be zero

- VUID-vkCmdBindVertexBuffers2-pStrides-03362
  If `pStrides` is not `NULL` each element of `pStrides` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxVertexInputBindingStride`
- VUID-vkCmdBindVertexBuffers2-pStrides-06209
  If `pStrides` is not `NULL` each element of `pStrides` **must** be either 0 or greater than or equal to the maximum extent of all vertex input attributes fetched from the corresponding binding, where the extent is calculated as the VkVertexInputAttributeDescription::`offset` plus VkVertexInputAttributeDescription::`format` size

Valid Usage (Implicit)

- VUID-vkCmdBindVertexBuffers2-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindVertexBuffers2-pBuffers-parameter
  `pBuffers` **must** be a valid pointer to an array of `bindingCount` valid or VK_NULL_HANDLE VkBuffer handles
- VUID-vkCmdBindVertexBuffers2-pOffsets-parameter
  `pOffsets` **must** be a valid pointer to an array of `bindingCount` VkDeviceSize values
- VUID-vkCmdBindVertexBuffers2-pSizes-parameter
  If `pSizes` is not `NULL`, `pSizes` **must** be a valid pointer to an array of `bindingCount` VkDeviceSize values
- VUID-vkCmdBindVertexBuffers2-pStrides-parameter
  If `pStrides` is not `NULL`, `pStrides` **must** be a valid pointer to an array of `bindingCount` VkDeviceSize values
- VUID-vkCmdBindVertexBuffers2-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBindVertexBuffers2-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindVertexBuffers2-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindVertexBuffers2-bindingCount-arraylength
  If any of `pSizes`, or `pStrides` are not `NULL`, `bindingCount` **must** be greater than `0`
- VUID-vkCmdBindVertexBuffers2-commonparent
  Both of `commandBuffer`, and the elements of `pBuffers` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally

synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindVertexBuffers2 is not affected by conditional rendering

To bind vertex buffers to a command buffer for use in subsequent drawing commands, call:

```
// Provided by VK_VERSION_1_0
void vkCmdBindVertexBuffers(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstBinding,
    uint32_t                                    bindingCount,
    const VkBuffer*                             pBuffers,
    const VkDeviceSize*                         pOffsets);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `firstBinding` is the index of the first vertex input binding whose state is updated by the command.
- `bindingCount` is the number of vertex input bindings whose state is updated by the command.
- `pBuffers` is a pointer to an array of buffer handles.
- `pOffsets` is a pointer to an array of buffer offsets.

The values taken from elements i of `pBuffers` and `pOffsets` replace the current state for the vertex input binding `firstBinding` + i, for i in [0, `bindingCount`). The vertex input binding is updated to start at the offset indicated by `pOffsets`[i] from the start of the buffer `pBuffers`[i]. All vertex input attributes that use each of these bindings will use these updated addresses in their address calculations for subsequent drawing commands. If the `nullDescriptor` feature is enabled, elements of `pBuffers` **can** be VK_NULL_HANDLE, and **can** be used by the vertex shader. If a vertex input attribute is bound to a vertex input binding that is VK_NULL_HANDLE, the values taken from memory are considered to be zero, and missing G, B, or A components are filled with (0,0,1).

Valid Usage

- VUID-vkCmdBindVertexBuffers-firstBinding-00624

`firstBinding` **must** be less than `VkPhysicalDeviceLimits`::`maxVertexInputBindings`

- VUID-vkCmdBindVertexBuffers-firstBinding-00625
  The sum of `firstBinding` and `bindingCount` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxVertexInputBindings`
- VUID-vkCmdBindVertexBuffers-pOffsets-00626
  All elements of `pOffsets` **must** be less than the size of the corresponding element in `pBuffers`
- VUID-vkCmdBindVertexBuffers-pBuffers-00627
  All elements of `pBuffers` **must** have been created with the VK_BUFFER_USAGE_VERTEX_BUFFER_BIT flag
- VUID-vkCmdBindVertexBuffers-pBuffers-00628
  Each element of `pBuffers` that is non-sparse **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBindVertexBuffers-pBuffers-04001
  If the `nullDescriptor` feature is not enabled, all elements of `pBuffers` **must** not be VK_NULL_HANDLE
- VUID-vkCmdBindVertexBuffers-pBuffers-04002
  If an element of `pBuffers` is VK_NULL_HANDLE, then the corresponding element of `pOffsets` **must** be zero

Valid Usage (Implicit)

- VUID-vkCmdBindVertexBuffers-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindVertexBuffers-pBuffers-parameter
  `pBuffers` **must** be a valid pointer to an array of `bindingCount` valid or VK_NULL_HANDLE VkBuffer handles
- VUID-vkCmdBindVertexBuffers-pOffsets-parameter
  `pOffsets` **must** be a valid pointer to an array of `bindingCount` VkDeviceSize values
- VUID-vkCmdBindVertexBuffers-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBindVertexBuffers-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindVertexBuffers-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindVertexBuffers-bindingCount-arraylength
  `bindingCount` **must** be greater than `0`
- VUID-vkCmdBindVertexBuffers-commonparent
  Both of `commandBuffer`, and the elements of `pBuffers` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindVertexBuffers is not affected by conditional rendering

26.3. Vertex Attribute Divisor in Instanced Rendering

If the `vertexAttributeInstanceRateDivisor` feature is enabled and the `pNext` chain of VkPipelineVertexInputStateCreateInfo includes a `VkPipelineVertexInputDivisorStateCreateInfo` structure, then that structure controls how vertex attributes are assigned to an instance when instanced rendering is enabled.

The `VkPipelineVertexInputDivisorStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPipelineVertexInputDivisorStateCreateInfo {
    VkStructureType                                 sType;
    const void*                                     pNext;
    uint32_t                                        vertexBindingDivisorCount;
    const VkVertexInputBindingDivisorDescription*   pVertexBindingDivisors;
} VkPipelineVertexInputDivisorStateCreateInfo;
```

```
// Provided by VK_KHR_vertex_attribute_divisor
// Equivalent to VkPipelineVertexInputDivisorStateCreateInfo
typedef VkPipelineVertexInputDivisorStateCreateInfo
VkPipelineVertexInputDivisorStateCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `vertexBindingDivisorCount` is the number of elements in the `pVertexBindingDivisors` array.
- `pVertexBindingDivisors` is a pointer to an array of VkVertexInputBindingDivisorDescription structures specifying the divisor value for each binding.

Valid Usage (Implicit)

- VUID-VkPipelineVertexInputDivisorStateCreateInfo-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_VERTEX_INPUT_DIVISOR_STATE_CREATE_INFO`
- VUID-VkPipelineVertexInputDivisorStateCreateInfo-pVertexBindingDivisors-parameter
  `pVertexBindingDivisors` **must** be a valid pointer to an array of `vertexBindingDivisorCount` VkVertexInputBindingDivisorDescription structures
- VUID-VkPipelineVertexInputDivisorStateCreateInfo-vertexBindingDivisorCount-arraylength
  `vertexBindingDivisorCount` **must** be greater than `0`

Structure Chaining

Extends the structure

- VkPipelineVertexInputStateCreateInfo

The individual divisor values per binding are specified using the `VkVertexInputBindingDivisorDescription` structure which is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkVertexInputBindingDivisorDescription {
    uint32_t    binding;
    uint32_t    divisor;
} VkVertexInputBindingDivisorDescription;
```

```
// Provided by VK_KHR_vertex_attribute_divisor
// Equivalent to VkVertexInputBindingDivisorDescription
typedef VkVertexInputBindingDivisorDescription
VkVertexInputBindingDivisorDescriptionKHR;
```

- `binding` is the binding number for which the divisor is specified.
- `divisor` is the number of successive instances that will use the same value of the vertex attribute when instanced rendering is enabled. For example, if the divisor is N, the same vertex attribute will be applied to N successive instances before moving on to the next vertex attribute. The maximum value of `divisor` is implementation-dependent and can be queried using VkPhysicalDeviceVertexAttributeDivisorProperties::`maxVertexAttribDivisor`. A value of `0` **can** be used for the divisor if the `vertexAttributeInstanceRateZeroDivisor` feature is enabled. In this

case, the same vertex attribute will be applied to all instances.

If this structure is not used to define a divisor value for an attribute, then the divisor has a logical default value of 1.

Valid Usage

- VUID-VkVertexInputBindingDivisorDescription-binding-01869
  `binding` **must** be less than VkPhysicalDeviceLimits::`maxVertexInputBindings`
- VUID-VkVertexInputBindingDivisorDescription-vertexAttributeInstanceRateZeroDivisor-02228
  If the `vertexAttributeInstanceRateZeroDivisor` feature is not enabled, `divisor` **must** not be `0`
- VUID-VkVertexInputBindingDivisorDescription-vertexAttributeInstanceRateDivisor-02229
  If the `vertexAttributeInstanceRateDivisor` feature is not enabled, `divisor` **must** be `1`
- VUID-VkVertexInputBindingDivisorDescription-divisor-01870
  `divisor` **must** be a value between `0` and VkPhysicalDeviceVertexAttributeDivisorProperties::`maxVertexAttribDivisor`, inclusive
- VUID-VkVertexInputBindingDivisorDescription-inputRate-01871
  VkVertexInputBindingDescription::`inputRate` **must** be of type VK_VERTEX_INPUT_RATE_INSTANCE for this `binding`

26.4. Vertex Input Address Calculation

The address of each attribute for each `vertexIndex` and `instanceIndex` is calculated as follows:

- Let `attribDesc` be the member of VkPipelineVertexInputStateCreateInfo::`pVertexAttributeDescriptions` with VkVertexInputAttributeDescription::`location` equal to the vertex input attribute number.
- Let `bindingDesc` be the member of VkPipelineVertexInputStateCreateInfo::`pVertexBindingDescriptions` with VkVertexInputAttributeDescription::`binding` equal to `attribDesc.binding`.
- Let `vertexIndex` be the index of the vertex within the draw (a value between `firstVertex` and `firstVertex`+`vertexCount` for `vkCmdDraw`, or a value taken from the index buffer plus `vertexOffset` for `vkCmdDrawIndexed`), and let `instanceIndex` be the instance number of the draw (a value between `firstInstance` and `firstInstance`+`instanceCount`).
- Let `offset` be an array of offsets into the bound vertex buffers specified during `vkCmdBindVertexBuffers` or `vkCmdBindVertexBuffers2` with `pOffsets`.
- Let `divisor` be the member of VkPipelineVertexInputDivisorStateCreateInfo::`pVertexBindingDivisors` with VkVertexInputBindingDivisorDescription::`binding` equal to `attribDesc.binding`. If the vertex binding state is dynamically set, instead let `divisor` be the member of the `pVertexBindingDescriptions` parameter to the vkCmdSetVertexInputEXT call with VkVertexInputBindingDescription2EXT::`binding` equal to `attribDesc.binding`.

- Let `stride` be the member of VkPipelineVertexInputStateCreateInfo ::`pVertexBindingDescriptions->stride` unless there is dynamic state causing the value to be ignored. In this case the value is set from the last value from one of the following
  - vkCmdSetVertexInputEXT::`pVertexBindingDescriptions->stride`
  - vkCmdBindVertexBuffers2::`pStrides`, if not `NULL`

```
bufferBindingAddress = buffer[binding].baseAddress + offset[binding];

if (bindingDesc.inputRate == VK_VERTEX_INPUT_RATE_VERTEX)
    effectiveVertexOffset = vertexIndex * stride;
else
    if (divisor == 0)
        effectiveVertexOffset = firstInstance * stride;
    else
        effectiveVertexOffset = (firstInstance + ((instanceIndex - firstInstance) /
divisor)) * stride;

attribAddress = bufferBindingAddress + effectiveVertexOffset + attribDesc.offset;
```

26.4.1. Vertex Input Extraction

For each attribute, raw data is extracted starting at `attribAddress` and is converted from the VkVertexInputAttributeDescription's `format` in the same manner as described for image reads as if a texel were read from that address. The numeric type of the attribute's `format` **must** match the numeric type of the input variable in the shader. The input variable in the shader **must** be declared as a 64-bit data type if and only if `format` is a 64-bit data type. If the sum of `attribAddress` and the data extracted is outside of the bound vertex buffer, behavior is as described by Shader Out-of-Bounds Memory Access.

If either `format` is a 64-bit format or the `legacyVertexAttributes` feature is not enabled, and `format` is a packed format, `attribAddress` **must** be a multiple of the size in bytes of the size of the format as described in Packed Formats. Otherwise, if either `format` is a 64-bit format or the `legacyVertexAttributes` feature is not enabled, `attribAddress` **must** be a multiple of the size in bytes of the component type indicated by `format` (see Formats).

The number of components in the vertex shader input variable need not exactly match the number of components in the format. If the vertex shader has fewer components, the extra components are discarded. If the numeric format of `format` uses sRGB encoding, and the `maintenance10` feature is enabled, the implementation **must** convert values from nonlinear to linear as described in the "sRGB EOTF" section of the Khronos Data Format Specification. If the `maintenance10` feature is not enabled, the implementation **should** convert values from nonlinear to linear. Implementations which do not support `maintenance10`, and which do not convert nonlinear to linear for sRGB formats **should** not expose VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT for such formats.

Chapter 27. Tessellation

Tessellation involves three pipeline stages. First, a tessellation control shader transforms control points of a patch and **can** produce per-patch data. Second, a fixed-function tessellator generates multiple primitives corresponding to a tessellation of the patch in (u,v) or (u,v,w) parameter space. Third, a tessellation evaluation shader transforms the vertices of the tessellated patch, for example to compute their positions and attributes as part of the tessellated surface. The tessellator is enabled when the pipeline contains both a tessellation control shader and a tessellation evaluation shader.

27.1. Tessellator

If a pipeline includes both tessellation shaders (control and evaluation), the tessellator consumes each input patch (after vertex shading) and produces a new set of independent primitives (points, lines, or triangles). These primitives are logically produced by subdividing a geometric primitive (rectangle or triangle) according to the per-patch outer and inner tessellation levels written by the tessellation control shader. These levels are specified using the built-in variables `TessLevelOuter` and `TessLevelInner`, respectively. This subdivision is performed in an implementation-dependent manner. If no tessellation shaders are present in the pipeline, the tessellator is disabled and incoming primitives are passed through without modification.

The type of subdivision performed by the tessellator is specified by an `OpExecutionMode` instruction using one of the `Triangles`, `Quads`, or `IsoLines` execution modes. When using shader objects, this instruction **must** be specified in the tessellation evaluation shader, and **may** also be specified in the tessellation control shader. When using pipelines, this instruction **may** be specified in either the tessellation evaluation or tessellation control shader. When using shader objects, `OutputVertices` **must** be specified in the tessellation control shader, and **may** also be specified in the tessellation evaluation shader. Other tessellation-related execution modes **can** also be specified in either the tessellation control or tessellation evaluation shaders.

Any tessellation-related modes specified in both the tessellation control and tessellation evaluation shaders **must** be the same.

Tessellation execution modes include:

- `Triangles`, `Quads`, and `IsoLines`. These control the type of subdivision and topology of the output primitives. When using shader objects, one mode **must** be set in at least the tessellation evaluation stage. When using pipelines, one mode **must** be set in at least one of the tessellation shader stages. If the `VK_KHR_portability_subset` extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::`tessellationIsolines` is VK_FALSE, then isoline tessellation is not supported by the implementation, and `IsoLines` **must** not be used in either tessellation shader stage.
- `VertexOrderCw` and `VertexOrderCcw`. These control the orientation of triangles generated by the tessellator. One mode **must** be set in at least one of the tessellation shader stages.
- `PointMode`. Controls generation of points rather than triangles or lines. This functionality defaults to disabled, and is enabled if either shader stage includes the execution mode. If the `VK_KHR_portability_subset` extension is enabled, and

VkPhysicalDevicePortabilitySubsetFeaturesKHR::tessellationPointMode is VK_FALSE, then point mode tessellation is not supported by the implementation, and PointMode **must** not be used in either tessellation shader stage.

- SpacingEqual, SpacingFractionalEven, and SpacingFractionalOdd. Controls the spacing of segments on the edges of tessellated primitives. One mode **must** be set in at least one of the tessellation shader stages.
- OutputVertices. Controls the size of the output patch of the tessellation control shader. When using shader objects, one value **must** be set in at least the tessellation control stage. When using pipelines, one value **must** be set in at least one of the tessellation shader stages.

For triangles, the tessellator subdivides a triangle primitive into smaller triangles. For quads, the tessellator subdivides a rectangle primitive into smaller triangles. For isolines, the tessellator subdivides a rectangle primitive into a collection of line segments arranged in strips stretching across the rectangle in the u dimension (i.e. the coordinates in TessCoord are of the form (0,x) through (1,x) for all tessellation evaluation shader invocations that share a line).

Each vertex produced by the tessellator has an associated (u,v,w) or (u,v) position in a normalized parameter space, with parameter values in the range [0,1], as illustrated in figures Domain parameterization for tessellation primitive modes (upper-left origin) and Domain parameterization for tessellation primitive modes (lower-left origin). The domain space **can** have either an upper-left or lower-left origin, selected by the domainOrigin member of VkPipelineTessellationDomainOriginStateCreateInfo.

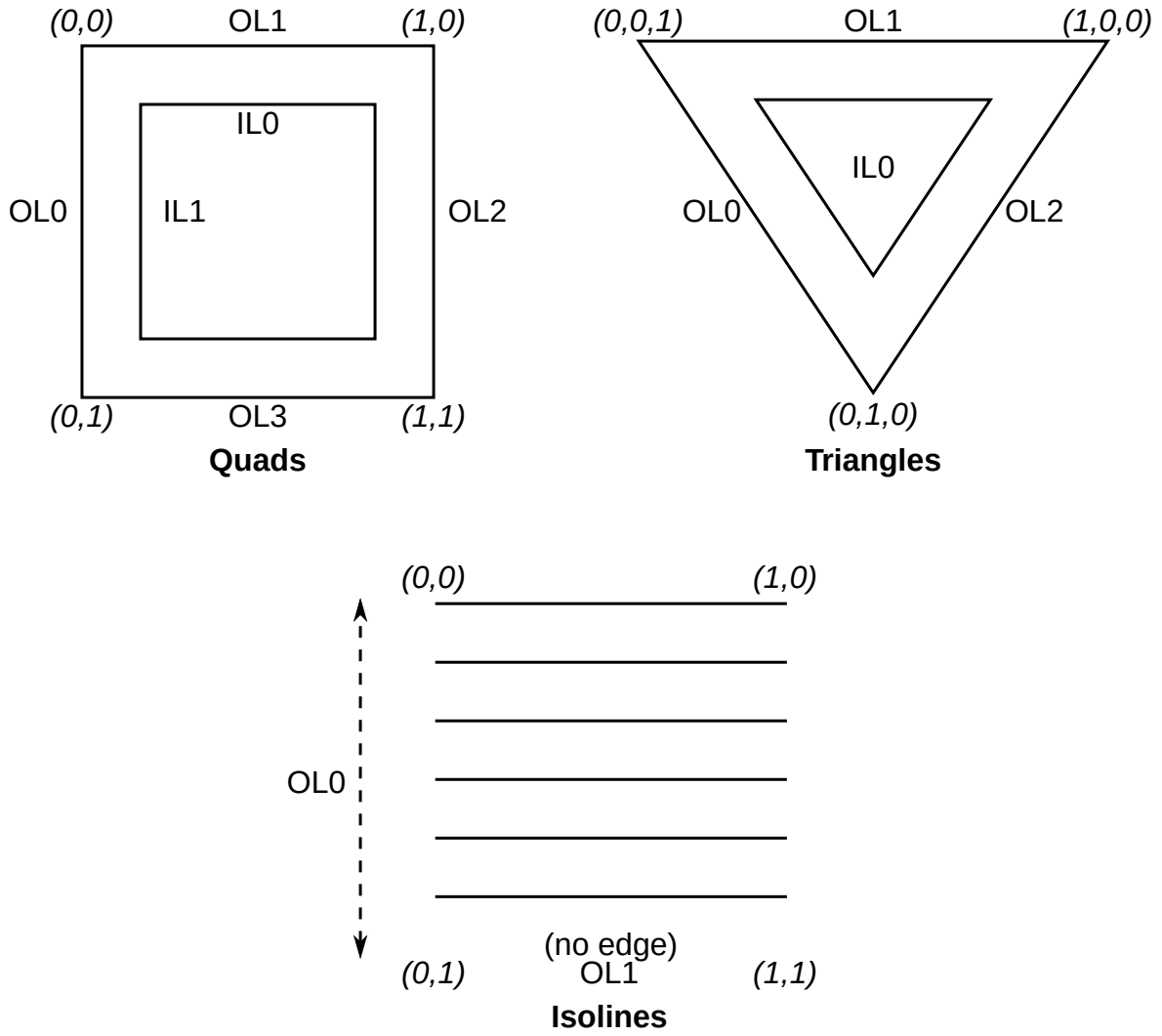


Figure 11. Domain parameterization for tessellation primitive modes (upper-left origin)

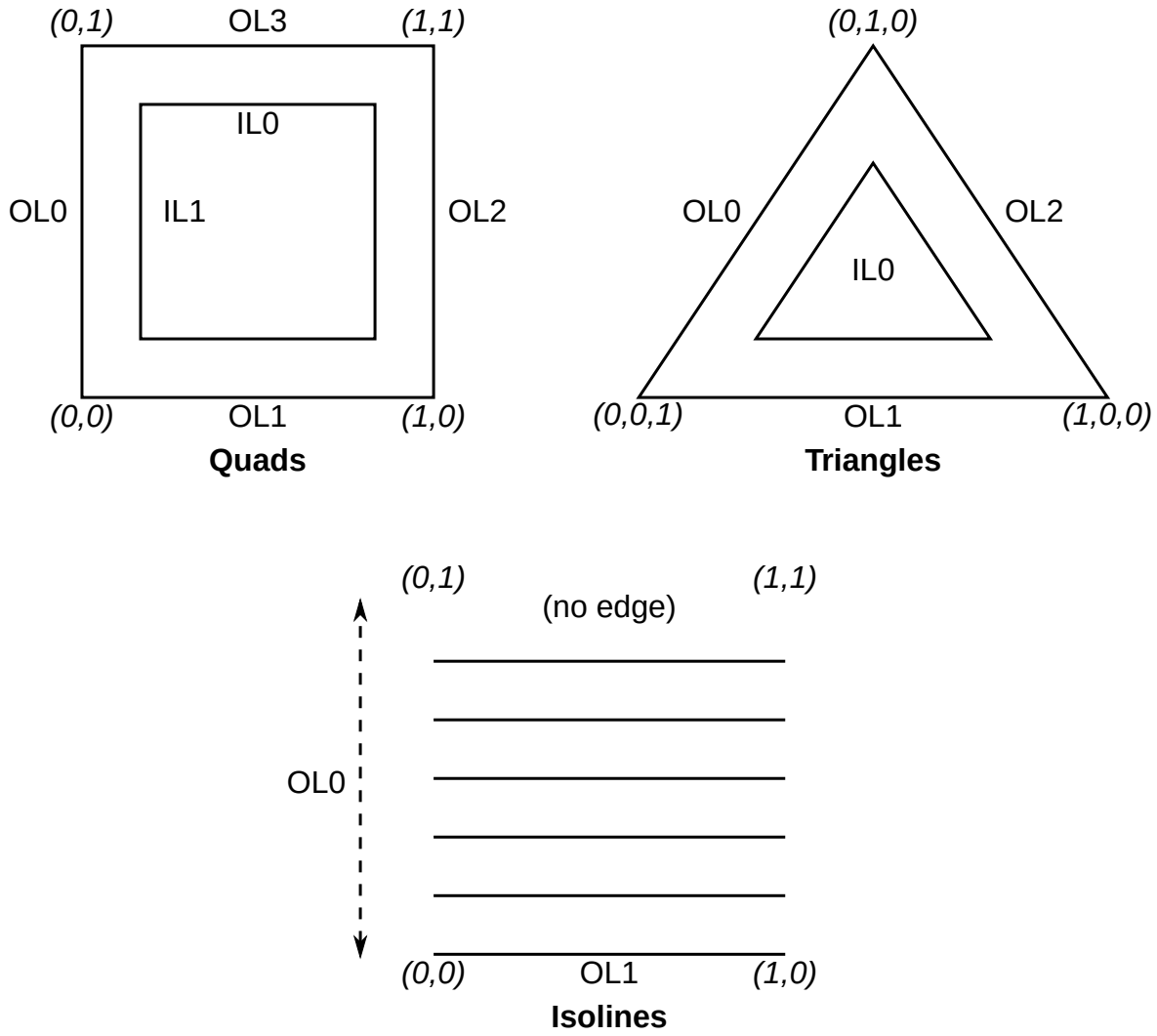


Figure 12. Domain parameterization for tessellation primitive modes (lower-left origin)

Caption

In the domain parameterization diagrams, the coordinates illustrate the value of `TessCoord` at the corners of the domain. The labels on the edges indicate the inner (IL0 and IL1) and outer (OL0 through OL3) tessellation level values used to control the number of subdivisions along each edge of the domain.

For triangles, the vertex's position is a barycentric coordinate (u,v,w), where u + v + w = 1.0, and indicates the relative influence of the three vertices of the triangle on the position of the vertex. For quads and isolines, the position is a (u,v) coordinate indicating the relative horizontal and vertical position of the vertex relative to the subdivided rectangle. The subdivision process is explained in more detail in subsequent sections.

27.2. Tessellator Patch Discard

A patch is discarded by the tessellator if any relevant outer tessellation level is less than or equal to zero.

Patches will also be discarded if any relevant outer tessellation level corresponds to a floating-point

NaN (not a number) in implementations supporting NaN.

No new primitives are generated and the tessellation evaluation shader is not executed for patches that are discarded. For `Quads`, all four outer levels are relevant. For `Triangles` and `IsoLines`, only the first three or two outer levels, respectively, are relevant. Negative inner levels will not cause a patch to be discarded; they will be clamped as described below.

27.3. Tessellator Spacing

Each of the tessellation levels is used to determine the number and spacing of segments used to subdivide a corresponding edge. The method used to derive the number and spacing of segments is specified by an `OpExecutionMode` in the tessellation control or tessellation evaluation shader using one of the identifiers `SpacingEqual`, `SpacingFractionalEven`, or `SpacingFractionalOdd`.

If `SpacingEqual` is used, the floating-point tessellation level is first clamped to [1, `maxLevel`], where `maxLevel` is the implementation-dependent maximum tessellation level (`VkPhysicalDeviceLimits ::maxTessellationGenerationLevel`). The result is rounded up to the nearest integer n, and the corresponding edge is divided into n segments of equal length in (u,v) space.

If `SpacingFractionalEven` is used, the tessellation level is first clamped to [2, `maxLevel`] and then rounded up to the nearest even integer n. If `SpacingFractionalOdd` is used, the tessellation level is clamped to [1, `maxLevel` - 1] and then rounded up to the nearest odd integer n. If n is one, the edge will not be subdivided. Otherwise, the corresponding edge will be divided into n - 2 segments of equal length, and two additional segments of equal length that are typically shorter than the other segments. The length of the two additional segments relative to the others will decrease monotonically with n - f, where f is the clamped floating-point tessellation level. When n - f is zero, the additional segments will have equal length to the other segments. As n - f approaches 2.0, the relative length of the additional segments approaches zero. The two additional segments **must** be placed symmetrically on opposite sides of the subdivided edge. The relative location of these two segments is implementation-dependent, but **must** be identical for any pair of subdivided edges with identical values of f.

When tessellating triangles or quads using point mode with fractional odd spacing, the tessellator **may** produce *interior vertices* that are positioned on the edge of the patch if an inner tessellation level is less than or equal to one. Such vertices are considered distinct from vertices produced by subdividing the outer edge of the patch, even if there are pairs of vertices with identical coordinates.

27.4. Tessellation Primitive Ordering

Few guarantees are provided for the relative ordering of primitives produced by tessellation, as they pertain to primitive order.

- The output primitives generated from each input primitive are passed to subsequent pipeline stages in an implementation-dependent order.
- All output primitives generated from a given input primitive are passed to subsequent pipeline stages before any output primitives generated from subsequent input primitives.

27.5. Tessellator Vertex Winding Order

When the tessellator produces triangles (in the `Triangles` or `Quads` modes), the orientation of all triangles is specified with an `OpExecutionMode` of `VertexOrderCw` or `VertexOrderCcw` in the tessellation control or tessellation evaluation shaders. If the order is `VertexOrderCw`, the vertices of all generated triangles will have clockwise ordering in (u,v) or (u,v,w) space. If the order is `VertexOrderCcw`, the vertices will have counter-clockwise ordering in that space.

If the tessellation domain has an upper-left origin, the vertices of a triangle have counter-clockwise ordering if

$$a = u_0 v_1 - u_1 v_0 + u_1 v_2 - u_2 v_1 + u_2 v_0 - u_0 v_2$$

is negative, and clockwise ordering if a is positive. u_i and v_i are the u and v coordinates in normalized parameter space of the ith vertex of the triangle. If the tessellation domain has a lower-left origin, the vertices of a triangle have counter-clockwise ordering if a is positive, and clockwise ordering if a is negative.

NOTE

The value a is proportional (with a positive factor) to the signed area of the triangle.

In `Triangles` mode, even though the vertex coordinates have a w value, it does not participate directly in the computation of a, being an affine combination of u and v.

27.6. Triangle Tessellation

If the tessellation primitive mode is `Triangles`, an equilateral triangle is subdivided into a collection of triangles covering the area of the original triangle. First, the original triangle is subdivided into a collection of concentric equilateral triangles. The edges of each of these triangles are subdivided, and the area between each triangle pair is filled by triangles produced by joining the vertices on the subdivided edges. The number of concentric triangles and the number of subdivisions along each triangle except the outermost is derived from the first inner tessellation level. The edges of the outermost triangle are subdivided independently, using the first, second, and third outer tessellation levels to control the number of subdivisions of the u = 0 (left), v = 0 (bottom), and w = 0 (right) edges, respectively. The second inner tessellation level and the fourth outer tessellation level have no effect in this mode.

If the first inner tessellation level and all three outer tessellation levels are exactly one after clamping and rounding, only a single triangle with (u,v,w) coordinates of (0,0,1), (1,0,0), and (0,1,0) is generated. If the inner tessellation level is one and any of the outer tessellation levels is greater than one, the inner tessellation level is treated as though it were originally specified as 1 + ε and will result in a two- or three-segment subdivision depending on the tessellation spacing. When used with fractional odd spacing, the three-segment subdivision **may** produce *inner vertices* positioned on the edge of the triangle.

If any tessellation level is greater than one, tessellation begins by producing a set of concentric inner triangles and subdividing their edges. First, the three outer edges are temporarily subdivided using the clamped and rounded first inner tessellation level and the specified tessellation spacing,

generating n segments. For the outermost inner triangle, the inner triangle is degenerate — a single point at the center of the triangle — if n is two. Otherwise, for each corner of the outer triangle, an inner triangle corner is produced at the intersection of two lines extended perpendicular to the corner's two adjacent edges running through the vertex of the subdivided outer edge nearest that corner. If n is three, the edges of the inner triangle are not subdivided and it is the final triangle in the set of concentric triangles. Otherwise, each edge of the inner triangle is divided into n - 2 segments, with the n - 1 vertices of this subdivision produced by intersecting the inner edge with lines perpendicular to the edge running through the n - 1 innermost vertices of the subdivision of the outer edge. Once the outermost inner triangle is subdivided, the previous subdivision process repeats itself, using the generated triangle as an outer triangle. This subdivision process is illustrated in Inner Triangle Tessellation.

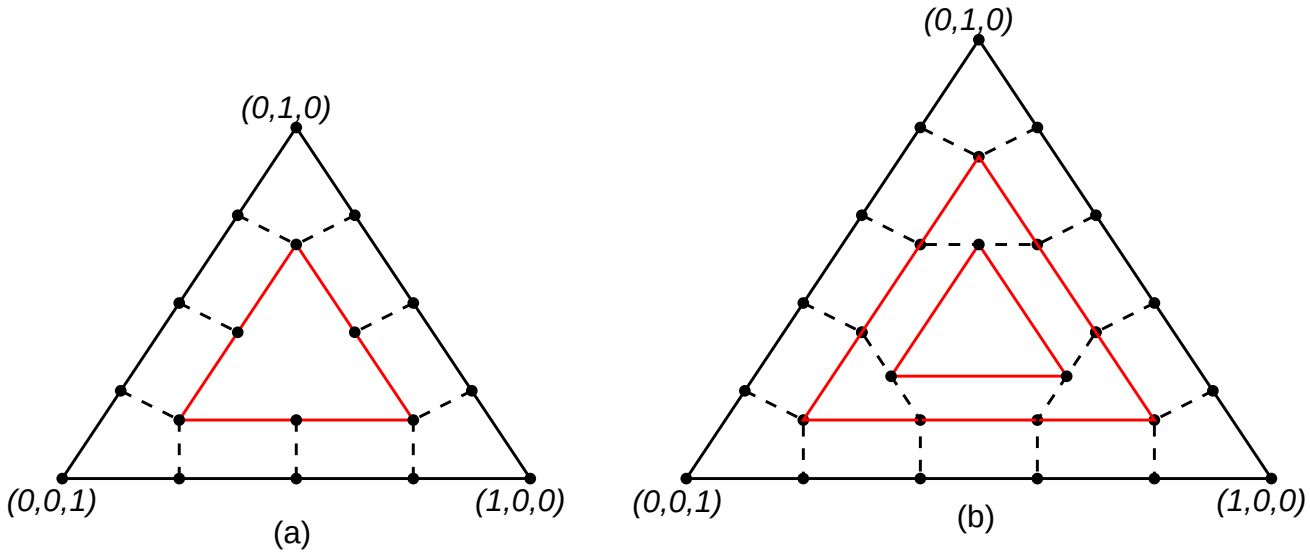


Figure 13. Inner Triangle Tessellation

Caption

In the Inner Triangle Tessellation diagram, inner tessellation levels of (a) four and (b) five are shown (not to scale). Solid black circles depict vertices along the edges of the concentric triangles. The edges of inner triangles are subdivided by intersecting the edge with segments perpendicular to the edge passing through each inner vertex of the subdivided outer edge. Dotted lines depict edges connecting corresponding vertices on the inner and outer triangle edges.

Once all the concentric triangles are produced and their edges are subdivided, the area between each pair of adjacent inner triangles is filled completely with a set of non-overlapping triangles. In this subdivision, two of the three vertices of each triangle are taken from adjacent vertices on a subdivided edge of one triangle; the third is one of the vertices on the corresponding edge of the other triangle. If the innermost triangle is degenerate (i.e., a point), the triangle containing it is subdivided into six triangles by connecting each of the six vertices on that triangle with the center point. If the innermost triangle is not degenerate, that triangle is added to the set of generated triangles as-is.

After the area corresponding to any inner triangles is filled, the tessellator generates triangles to cover the area between the outermost triangle and the outermost inner triangle. To do this, the

temporary subdivision of the outer triangle edge above is discarded. Instead, the u = 0, v = 0, and w = 0 edges are subdivided according to the first, second, and third outer tessellation levels, respectively, and the tessellation spacing. The original subdivision of the first inner triangle is retained. The area between the outer and first inner triangles is completely filled by non-overlapping triangles as described above. If the first (and only) inner triangle is degenerate, a set of triangles is produced by connecting each vertex on the outer triangle edges with the center point.

After all triangles are generated, each vertex in the subdivided triangle is assigned a barycentric (u,v,w) coordinate based on its location relative to the three vertices of the outer triangle.

The algorithm used to subdivide the triangular domain in (u,v,w) space into individual triangles is implementation-dependent. However, the set of triangles produced will completely cover the domain, and no portion of the domain will be covered by multiple triangles.

Output triangles are generated with a topology similar to triangle lists, except that the order in which each triangle is generated, and the order in which the vertices are generated for each triangle, are implementation-dependent. However, the order of vertices in each triangle is consistent across the domain as described in Tessellator Vertex Winding Order.

27.7. Quad Tessellation

If the tessellation primitive mode is `Quads`, a rectangle is subdivided into a collection of triangles covering the area of the original rectangle. First, the original rectangle is subdivided into a regular mesh of rectangles, where the number of rectangles along the u = 0 and u = 1 (vertical) and v = 0 and v = 1 (horizontal) edges are derived from the first and second inner tessellation levels, respectively. All rectangles, except those adjacent to one of the outer rectangle edges, are decomposed into triangle pairs. The outermost rectangle edges are subdivided independently, using the first, second, third, and fourth outer tessellation levels to control the number of subdivisions of the u = 0 (left), v = 0 (bottom), u = 1 (right), and v = 1 (top) edges, respectively. The area between the inner rectangles of the mesh and the outer rectangle edges is filled by triangles produced by joining the vertices on the subdivided outer edges to the vertices on the edge of the inner rectangle mesh.

If both clamped inner tessellation levels and all four clamped outer tessellation levels are exactly one, only a single triangle pair covering the outer rectangle is generated. Otherwise, if either clamped inner tessellation level is one, that tessellation level is treated as though it was originally specified as 1 + ε and will result in a two- or three-segment subdivision depending on the tessellation spacing. When used with fractional odd spacing, the three-segment subdivision **may** produce *inner vertices* positioned on the edge of the rectangle.

If any tessellation level is greater than one, tessellation begins by subdividing the u = 0 and u = 1 edges of the outer rectangle into m segments using the clamped and rounded first inner tessellation level and the tessellation spacing. The v = 0 and v = 1 edges are subdivided into n segments using the second inner tessellation level. Each vertex on the u = 0 and v = 0 edges is joined with the corresponding vertex on the u = 1 and v = 1 edges to produce a set of vertical and horizontal lines that divide the rectangle into a grid of smaller rectangles. The primitive generator emits a pair of non-overlapping triangles covering each such rectangle not adjacent to an edge of the outer rectangle. The boundary of the region covered by these triangles forms an inner rectangle, the edges of which are subdivided by the grid vertices that lie on the edge. If either m or n is two, the

inner rectangle is degenerate, and one or both of the rectangle's *edges* consist of a single point. This subdivision is illustrated in Figure Inner Quad Tessellation.

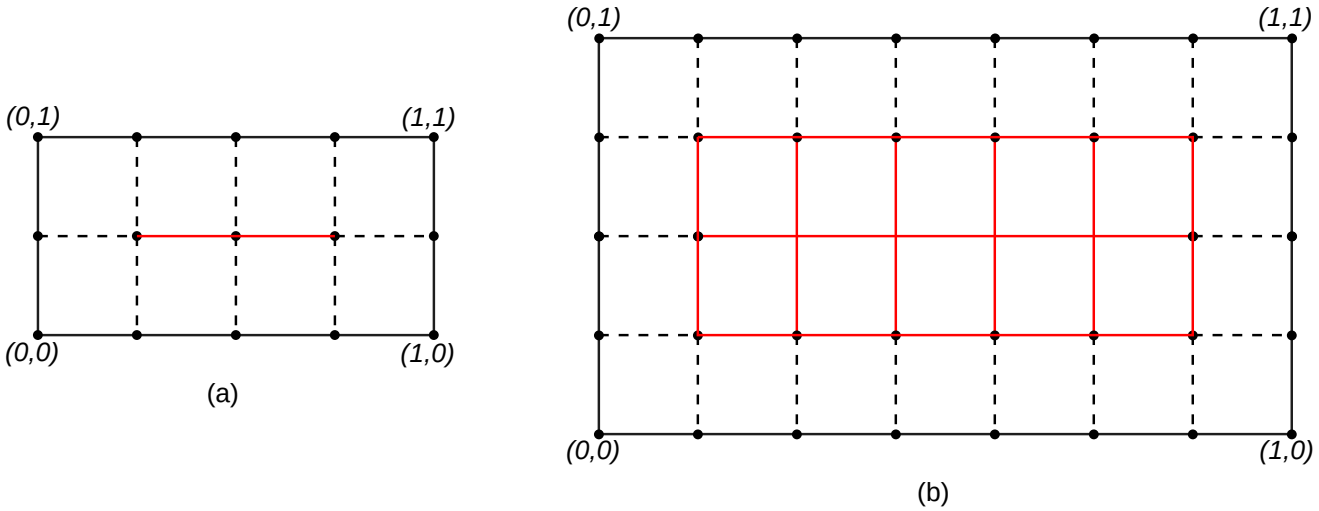


Figure 14. Inner Quad Tessellation

Caption

In the Inner Quad Tessellation diagram, inner quad tessellation levels of (a) (4,2) and (b) (7,4) are shown. The regions highlighted in red in figure (b) depict the 10 inner rectangles, each of which will be subdivided into two triangles. Solid black circles depict vertices on the boundary of the outer and inner rectangles, where the inner rectangle of figure (a) is degenerate (a single line segment). Dotted lines depict the horizontal and vertical edges connecting corresponding vertices on the inner and outer rectangle edges.

After the area corresponding to the inner rectangle is filled, the tessellator **must** produce triangles to cover the area between the inner and outer rectangles. To do this, the subdivision of the outer rectangle edge above is discarded. Instead, the u = 0, v = 0, u = 1, and v = 1 edges are subdivided according to the first, second, third, and fourth outer tessellation levels, respectively, and the tessellation spacing. The original subdivision of the inner rectangle is retained. The area between the outer and inner rectangles is completely filled by non-overlapping triangles. Two of the three vertices of each triangle are adjacent vertices on a subdivided edge of one rectangle; the third is one of the vertices on the corresponding edge of the other rectangle. If either edge of the innermost rectangle is degenerate, the area near the corresponding outer edges is filled by connecting each vertex on the outer edge with the single vertex making up the *inner edge*.

The algorithm used to subdivide the rectangular domain in (u,v) space into individual triangles is implementation-dependent. However, the set of triangles produced will completely cover the domain, and no portion of the domain will be covered by multiple triangles.

Output triangles are generated with a topology similar to triangle lists, except that the order in which each triangle is generated, and the order in which the vertices are generated for each triangle, are implementation-dependent. However, the order of vertices in each triangle is consistent across the domain as described in Tessellator Vertex Winding Order.

27.8. Isoline Tessellation

If the tessellation primitive mode is `IsoLines`, a set of independent horizontal line segments is drawn. The segments are arranged into connected strips called *isolines*, where the vertices of each isoline have a constant v coordinate and u coordinates covering the full range [0,1]. The number of isolines generated is derived from the first outer tessellation level; the number of segments in each isoline is derived from the second outer tessellation level. Both inner tessellation levels and the third and fourth outer tessellation levels have no effect in this mode.

As with quad tessellation above, isoline tessellation begins with a rectangle. The u = 0 and u = 1 edges of the rectangle are subdivided according to the first outer tessellation level. For the purposes of this subdivision, the tessellation spacing mode is ignored and treated as `SpacingEqual`. An isoline is drawn connecting each vertex on the u = 0 rectangle edge to the corresponding vertex on the u = 1 rectangle edge, except that no line is drawn between (0,1) and (1,1). If the number of isolines on the subdivided u = 0 and u = 1 edges is n, this process will result in n equally spaced lines with constant v coordinates of 0, $\frac{1}{n}$, $\frac{2}{n}$, ..., $\frac{n-1}{n}$.

Each of the n isolines is then subdivided according to the second outer tessellation level and the tessellation spacing, resulting in m line segments. Each segment of each line is emitted by the tessellator. These line segments are generated with a topology similar to line lists, except that the order in which each line is generated, and the order in which the vertices are generated for each line segment, are implementation-dependent.

NOTE If the `VK_KHR_portability_subset` extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::`tessellationIsolines` is VK_FALSE, then isoline tessellation is not supported by the implementation.

27.9. Tessellation Point Mode

For all primitive modes, the tessellator is capable of generating points instead of lines or triangles. If the tessellation control or tessellation evaluation shader specifies the `OpExecutionMode PointMode`, the primitive generator will generate one point for each distinct vertex produced by tessellation, rather than emitting triangles or lines. Otherwise, the tessellator will produce a collection of line segments or triangles according to the primitive mode. These points are generated with a topology similar to point lists, except the order in which the points are generated for each input primitive is undefined.

NOTE If the `VK_KHR_portability_subset` extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::`tessellationPointMode` is VK_FALSE, then tessellation point mode is not supported by the implementation.

27.10. Tessellation Pipeline State

The `pTessellationState` member of VkGraphicsPipelineCreateInfo is a pointer to a `VkPipelineTessellationStateCreateInfo` structure.

The `VkPipelineTessellationStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineTessellationStateCreateInfo {
    VkStructureType                            sType;
    const void*                                pNext;
    VkPipelineTessellationStateCreateFlags     flags;
    uint32_t                                   patchControlPoints;
} VkPipelineTessellationStateCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `patchControlPoints` is the number of control points per patch.

Valid Usage

- VUID-VkPipelineTessellationStateCreateInfo-patchControlPoints-01214
  `patchControlPoints` **must** be greater than zero and less than or equal to `VkPhysicalDeviceLimits`::`maxTessellationPatchSize`

Valid Usage (Implicit)

- VUID-VkPipelineTessellationStateCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_STATE_CREATE_INFO
- VUID-VkPipelineTessellationStateCreateInfo-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkPipelineTessellationDomainOriginStateCreateInfo
- VUID-VkPipelineTessellationStateCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkPipelineTessellationStateCreateInfo-flags-zerobitmask
  `flags` **must** be `0`

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineTessellationStateCreateFlags;
```

`VkPipelineTessellationStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

The `VkPipelineTessellationDomainOriginStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPipelineTessellationDomainOriginStateCreateInfo {
    VkStructureType               sType;
    const void*                   pNext;
    VkTessellationDomainOrigin    domainOrigin;
} VkPipelineTessellationDomainOriginStateCreateInfo;
```

```
// Provided by VK_KHR_maintenance2
// Equivalent to VkPipelineTessellationDomainOriginStateCreateInfo
typedef VkPipelineTessellationDomainOriginStateCreateInfo
VkPipelineTessellationDomainOriginStateCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `domainOrigin` is a VkTessellationDomainOrigin value controlling the origin of the tessellation domain space.

If the `VkPipelineTessellationDomainOriginStateCreateInfo` structure is included in the `pNext` chain of VkPipelineTessellationStateCreateInfo, it controls the origin of the tessellation domain. If this structure is not present, it is as if `domainOrigin` was VK_TESSELLATION_DOMAIN_ORIGIN_UPPER_LEFT.

Valid Usage (Implicit)

- VUID-VkPipelineTessellationDomainOriginStateCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_DOMAIN_ORIGIN_STATE_CREATE_INFO
- VUID-VkPipelineTessellationDomainOriginStateCreateInfo-domainOrigin-parameter
  `domainOrigin` **must** be a valid VkTessellationDomainOrigin value

Structure Chaining

Extends the structure

- VkPipelineTessellationStateCreateInfo

The possible tessellation domain origins are specified by the VkTessellationDomainOrigin enumeration:

```
// Provided by VK_VERSION_1_1
typedef enum VkTessellationDomainOrigin {
    VK_TESSELLATION_DOMAIN_ORIGIN_UPPER_LEFT = 0,
    VK_TESSELLATION_DOMAIN_ORIGIN_LOWER_LEFT = 1,
  // Provided by VK_KHR_maintenance2
    VK_TESSELLATION_DOMAIN_ORIGIN_UPPER_LEFT_KHR =
VK_TESSELLATION_DOMAIN_ORIGIN_UPPER_LEFT,
  // Provided by VK_KHR_maintenance2
    VK_TESSELLATION_DOMAIN_ORIGIN_LOWER_LEFT_KHR =
VK_TESSELLATION_DOMAIN_ORIGIN_LOWER_LEFT,
} VkTessellationDomainOrigin;
```

```
// Provided by VK_KHR_maintenance2
// Equivalent to VkTessellationDomainOrigin
typedef VkTessellationDomainOrigin VkTessellationDomainOriginKHR;
```

- VK_TESSELLATION_DOMAIN_ORIGIN_UPPER_LEFT specifies that the origin of the domain space is in the upper left corner, as shown in figure Domain parameterization for tessellation primitive modes (upper-left origin).
- VK_TESSELLATION_DOMAIN_ORIGIN_LOWER_LEFT specifies that the origin of the domain space is in the lower left corner, as shown in figure Domain parameterization for tessellation primitive modes (lower-left origin).

This enum affects how the VertexOrderCw and VertexOrderCcw tessellation execution modes are interpreted, since the winding is defined relative to the orientation of the domain.

To dynamically set the origin of the tessellation domain space, call:

```
// Provided by VK_EXT_extended_dynamic_state3 with VK_KHR_maintenance2 or
VK_VERSION_1_1, VK_EXT_shader_object
void vkCmdSetTessellationDomainOriginEXT(
    VkCommandBuffer                             commandBuffer,
    VkTessellationDomainOrigin                  domainOrigin);
```

- commandBuffer is the command buffer into which the command will be recorded.
- domainOrigin specifies the origin of the tessellation domain space.

This command sets the origin of the tessellation domain space for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT set in VkPipelineDynamicStateCreateInfo::pDynamicStates. Otherwise, this state is specified by the VkPipelineTessellationDomainOriginStateCreateInfo::domainOrigin value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetTessellationDomainOriginEXT-None-09423
  At least one of the following **must** be true:
  - The `extendedDynamicState3TessellationDomainOrigin` feature is enabled
  - The `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetTessellationDomainOriginEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetTessellationDomainOriginEXT-domainOrigin-parameter
  `domainOrigin` **must** be a valid VkTessellationDomainOrigin value
- VUID-vkCmdSetTessellationDomainOriginEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetTessellationDomainOriginEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetTessellationDomainOriginEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetTessellationDomainOriginEXT is not affected by conditional rendering

Chapter 28. Geometry Shading

The geometry shader operates on a group of vertices and their associated data assembled from a single input primitive, and emits zero or more output primitives and the group of vertices and their associated data required for each output primitive. Geometry shading is enabled when a geometry shader is included in the pipeline.

28.1. Geometry Shader Input Primitives

Each geometry shader invocation has access to all vertices in the primitive (and their associated data), which are presented to the shader as an array of inputs.

The input primitive type expected by the geometry shader is specified with an `OpExecutionMode` instruction in the geometry shader, and **must** match the incoming primitive type specified by either the pipeline's primitive topology if tessellation is inactive, or the tessellation mode if tessellation is active, as follows:

- An input primitive type of `InputPoints` **must** only be used with a pipeline topology of VK_PRIMITIVE_TOPOLOGY_POINT_LIST, or with a tessellation shader specifying `PointMode`. The input arrays always contain one element, as described by the point list topology or tessellation in point mode.
- An input primitive type of `InputLines` **must** only be used with a pipeline topology of VK_PRIMITIVE_TOPOLOGY_LINE_LIST or VK_PRIMITIVE_TOPOLOGY_LINE_STRIP, or with a tessellation shader specifying `IsoLines` that does not specify `PointMode`. The input arrays always contain two elements, as described by the line list topology or line strip topology, or by isoline tessellation.
- An input primitive type of `InputLinesAdjacency` **must** only be used when tessellation is inactive, with a pipeline topology of VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY or VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY. The input arrays always contain four elements, as described by the line list with adjacency topology or line strip with adjacency topology.
- An input primitive type of `Triangles` **must** only be used with a pipeline topology of VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP, or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN; or with a tessellation shader specifying `Quads` or `Triangles` that does not specify `PointMode`. The input arrays always contain three elements, as described by the triangle list topology, triangle strip topology, or triangle fan topology, or by triangle or quad tessellation. Vertices **may** be in a different absolute order than specified by the topology, but **must** adhere to the specified winding order.
- An input primitive type of `InputTrianglesAdjacency` **must** only be used when tessellation is inactive, with a pipeline topology of VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY or VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY. The input arrays always contain six elements, as described by the triangle list with adjacency topology or triangle strip with adjacency topology. Vertices **may** be in a different absolute order than specified by the topology, but **must** adhere to the specified winding order, and the vertices making up the main primitive **must** still occur at the first, third, and fifth index.

28.2. Geometry Shader Output Primitives

A geometry shader generates primitives in one of three output modes: points, line strips, or triangle strips. The primitive mode is specified in the shader using an `OpExecutionMode` instruction with the `OutputPoints`, `OutputLineStrip` or `OutputTriangleStrip` modes, respectively. Each geometry shader **must** include exactly one output primitive mode.

The vertices output by the geometry shader are assembled into points, lines, or triangles based on the output primitive type and the resulting primitives are then further processed as described in Rasterization. If the number of vertices emitted by the geometry shader is not sufficient to produce a single primitive, vertices corresponding to incomplete primitives are not processed by subsequent pipeline stages. The number of vertices output by the geometry shader is limited to a maximum count specified in the shader.

The maximum output vertex count is specified in the shader using an `OpExecutionMode` instruction with the mode set to `OutputVertices` and the maximum number of vertices that will be produced by the geometry shader specified as a literal. Each geometry shader **must** specify a maximum output vertex count.

28.3. Multiple Invocations of Geometry Shaders

Geometry shaders **can** be invoked more than one time for each input primitive. This is known as *geometry shader instancing* and is requested by including an `OpExecutionMode` instruction with `mode` specified as `Invocations` and the number of invocations specified as an integer literal.

In this mode, the geometry shader will execute at least n times for each input primitive, where n is the number of invocations specified in the `OpExecutionMode` instruction. The instance number is available to each invocation as a built-in input using `InvocationId`.

28.4. Geometry Shader Primitive Ordering

Limited guarantees are provided for the relative ordering of primitives produced by a geometry shader, as they pertain to primitive order.

- For instanced geometry shaders, the output primitives generated from each input primitive are passed to subsequent pipeline stages using the invocation number to order the primitives, from least to greatest.
- All output primitives generated from a given input primitive are passed to subsequent pipeline stages before any output primitives generated from subsequent input primitives.

Chapter 29. Mesh Shading

Task and mesh shaders operate in workgroups to produce a collection of primitives that will be processed by subsequent stages of the graphics pipeline.

Work on the mesh pipeline is initiated by the application drawing a set of mesh tasks organized in global workgroups. If the optional task shader is active, each workgroup triggers the execution of task shader invocations that will create a new set of mesh workgroups upon completion. Each of these created workgroups, or each of the original workgroups if no task shader is present, triggers the execution of mesh shader invocations.

Each mesh shader workgroup emits zero or more output primitives along with the group of vertices and their associated data required for each output primitive.

29.1. Task Shader Input

For every workgroup issued via the drawing commands a group of task shader invocations is executed. There are no inputs other than the builtin workgroup identifiers.

29.2. Task Shader Output

The task shader can emit zero or more mesh workgroups to be generated. Shaders using the `TaskEXT` `Execution Model` can do so using the `OpEmitMeshTasksEXT` instruction. The `groupCountX`, `groupCountY` and `groupCountZ` arguments passed to this instruction **must** be less than or equal to the respective dimension within `VkPhysicalDeviceMeshShaderPropertiesEXT`::`maxMeshWorkGroupCount`. The product of these arguments **must** be less than or equal to `VkPhysicalDeviceMeshShaderPropertiesEXT` ::`maxMeshWorkGroupTotalCount`.

The task shader can also pass user-defined data to all mesh shader invocations that it creates. Shaders using the `TaskEXT Execution Model` can do so by writing to a payload variable with `TaskPayloadWorkgroupEXT` storage class that is passed to the `OpEmitMeshTasksEXT` instruction.

29.3. Mesh Generation

If a task shader exists, the mesh assembler creates a variable amount of mesh workgroups depending on each task's output. If there is no task shader, the drawing commands emit the mesh shader invocations directly.

29.4. Mesh Shader Input

The only inputs available to the mesh shader are variables identifying the specific workgroup and invocation and, if applicable, the payload variable passed to the `OpEmitMeshTasksEXT` instruction by the task shader that spawned the mesh shader's workgroup. The mesh shader can operate without a task shader as well.

29.5. Mesh Shader Output

A mesh shader generates primitives in one of three output modes: points, lines, or triangles. For shaders using the `MeshEXT Execution Model` the primitive mode is specified in the shader using an `OpExecutionMode` instruction with the `OutputPoints`, `OutputLinesEXT`, or `OutputTrianglesEXT` modes, respectively. Each mesh shader **must** include exactly one output primitive mode.

For shaders using the `MeshEXT Execution Model` the maximum output vertex count is specified as a literal in the shader using an `OpExecutionMode` instruction with the mode set to `OutputVertices` and **must** be less than or equal to `VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshOutputVertices`.

For shaders using the `MeshEXT Execution Model` the maximum output primitive count is specified as a literal in the shader using an `OpExecutionMode` instruction with the mode set to `OutputPrimitivesEXT` and **must** be less than or equal to `VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshOutputPrimitives`.

For shaders using the `MeshEXT Execution Model` the number of vertices and primitives output by the mesh shader is provided via calling the `OpSetMeshOutputsEXT` instruction. The `vertexCount` argument **must** be less than or equal to the maximum output vertex count specified in the shader. The `primitiveCount` argument **must** be less than or equal to the maximum output primitive count specified in the shader.

Depending on the output primitive mode an appropriately-decorated variable is the output array of local index values into the vertex output arrays from which primitives are assembled according to the output primitive type:

- `OutputPoints` uses the `PrimitivePointIndicesEXT` decoration.
- `OutputLinesEXT` uses the `PrimitiveLineIndicesEXT` decoration.
- `OutputTrianglesEXT` uses the `PrimitiveTriangleIndicesEXT` decoration.

These resulting primitives are then further processed as described in Rasterization.

With the exception of primitive indices, all output built-ins and custom attributes count towards the total storage size occupied by output variables in mesh shaders. This size can be calculated as follows, taking into account the fact that the number of effective scalar attributes is 4 times the number of effective locations used according to the location assignment rules. Let v be the number of views, n_v be the number of effective scalar per-vertex attributes not dependent on `ViewIndex`, n_{vpv} be the number of effective scalar per-vertex attributes dependent on `ViewIndex`, m_v be the maximum number of vertices specified by the `OutputVertices Execution Mode`, g_v be `meshOutputPerVertexGranularity`, n_p be the number of effective scalar per-primitive attributes not dependent on `ViewIndex`, n_{ppv} be the number of effective scalar per-primitive attributes dependent on `ViewIndex`, m_p be the maximum number of primitives specified by the `OutputPrimitivesEXT Execution Mode` and g_p be `meshOutputPerPrimitiveGranularity`:

$$(n_v + (n_{vpv} \times v)) \times 4 \times \text{align}(m_v, g_v) + (n_p + (n_{ppv} \times v)) \times 4 \times \text{align}(m_p, g_p)$$

29.6. Mesh Shader Primitive Ordering

The following guarantees are provided for the relative ordering of primitives produced by a mesh shader, as they pertain to primitive order.

- When a task shader is used, mesh workgroups spawned from lower tasks will be ordered prior to those workgroups from subsequent tasks.
- All output primitives generated from a given mesh workgroup are passed to subsequent pipeline stages before any output primitives generated from subsequent input workgroups.
- All output primitives within a mesh workgroup will be generated in the ordering provided by the builtin primitive indexbuffer (from low address to high address).

Chapter 30. Fixed-Function Vertex Post-Processing

After pre-rasterization shader stages, the following fixed-function operations are applied to vertices of the resulting primitives:

- Transform feedback (see Transform Feedback)
- Flat shading (see Flat Shading).
- Primitive clipping, including application-defined half-spaces (see Primitive Clipping).
- Shader output attribute clipping (see Clipping Shader Outputs).
- Perspective division on clip coordinates (see Coordinate Transformations).
- Viewport mapping, including depth range scaling (see Controlling the Viewport).
- Front face determination for polygon primitives (see Basic Polygon Rasterization).

Next, rasterization is performed on primitives as described in chapter Rasterization.

30.1. Transform Feedback

Before any other fixed-function vertex post-processing, vertex outputs from the last shader in the pre-rasterization shader stage **can** be written out to one or more transform feedback buffers bound to the command buffer. To capture vertex outputs the last pre-rasterization shader stage shader **must** be declared with the `Xfb` execution mode. Outputs decorated with `XfbBuffer` will be written out to the corresponding transform feedback buffers bound to the command buffer when transform feedback is active. Transform feedback buffers are bound to the command buffer by using vkCmdBindTransformFeedbackBuffersEXT. Transform feedback is made active by calling vkCmdBeginTransformFeedbackEXT and made inactive by calling vkCmdEndTransformFeedbackEXT. After vertex data is written it is possible to use vkCmdDrawIndirectByteCountEXT to start a new draw where the `vertexCount` is derived from the number of bytes written by a previous transform feedback.

When an individual point, line, or triangle primitive reaches the transform feedback stage while transform feedback is active, the values of the specified output variables are assembled into primitives and appended to the bound transform feedback buffers. Any primitive for which any pair of vertices have the same position **should** be assembled for transform feedback normally but it **may** be discarded prior to primitive assembly instead. After activating transform feedback, the values of the first assembled primitive are written at the starting offsets of the bound transform feedback buffers, and subsequent primitives are appended to the buffer. If the optional `pCounterBuffers` and `pCounterBufferOffsets` parameters are specified, the starting points within the transform feedback buffers are adjusted so data is appended to the previously written values indicated by the value stored by the implementation in the counter buffer.

For multi-vertex primitives, all values for a given vertex are written before writing values for any other vertex. When the `transformFeedbackPreservesProvokingVertex` feature is not enabled, implementations **may** write out any vertex within the primitive first, but all subsequent vertices for

that primitive **must** be written out in a consistent winding order defined as follows:

- If neither geometry nor tessellation shading is active, vertices within a primitive are appended according to the winding order described by the primitive topology defined by the VkPipelineInputAssemblyStateCreateInfo::`topology` used to execute the drawing command.
- If geometry shading is active, vertices within a primitive are appended according to the winding order described by the primitive topology defined by the `OutputPoints`, `OutputLineStrip`, or `OutputTriangleStrip` execution mode.
- If tessellation shading is active but geometry shading is not, vertices within a primitive are appended according to the winding order defined by triangle tessellation, quad tessellation, and isoline tessellation.

When the `transformFeedbackPreservesProvokingVertex` feature is enabled, then in addition to writing vertices with a consistent winding order, the vertex order **must** preserve the provoking vertex of each primitive:

- When the pipeline's provoking vertex mode is VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT, the primitive's provoking vertex **must** be the first vertex written.
- When the pipeline's provoking vertex mode is VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT, the primitive's provoking vertex **must** be the last vertex written.

If `transformFeedbackPreservesTriangleFanProvokingVertex` is VK_FALSE, neither geometry nor tessellation shading is active, and the primitive topology is VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN, then the first vertex written from each primitive is implementation-defined even when the `transformFeedbackPreservesProvokingVertex` feature is enabled.

When capturing vertices, the stride associated with each transform feedback buffer, as indicated by the `XfbStride` decoration, indicates the number of bytes of storage reserved for each vertex in the transform feedback buffer. For every vertex captured, each output attribute with a `Offset` decoration will be written to the storage reserved for the vertex at the associated transform feedback buffer. When writing output variables that are arrays or structures, individual array elements or structure members are written tightly packed in order. For vector types, individual components are written in order. For matrix types, outputs are written as an array of column vectors.

If any component of an output with an assigned transform feedback offset was not written to by its shader, the value recorded for that component is undefined, and the primitives needed and primitives written counts in the corresponding VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT query will increment as normal. All components of an output variable **must** be written at an offset aligned to the size of the component. The size of each component of an output variable **must** be at least 32-bits. When capturing a vertex, any portion of the reserved storage not associated with an output variable with an assigned transform feedback offset will be unmodified.

When transform feedback is inactive, no vertices are recorded. If there is a valid counter buffer

handle and counter buffer offset in the pCounterBuffers and pCounterBufferOffsets arrays, writes to the corresponding transform feedback buffer will start at the byte offset represented by the value stored in the counter buffer location.

Individual lines or triangles of a strip or fan primitive will be extracted and recorded separately. Incomplete primitives are not recorded.

When using a geometry shader that emits vertices to multiple vertex streams, a primitive will be assembled and output for each stream when there are enough vertices emitted for the output primitive type. All outputs assigned to a given transform feedback buffer are required to come from a single vertex stream.

The sizes of the transform feedback buffers are defined by the vkCmdBindTransformFeedbackBuffersEXT pSizes parameter for each of the bound buffers, or the size of the bound buffer, whichever is the lesser. If there is less space remaining in any of the transform feedback buffers than the size of all of the vertex data for that primitive based on the XfbStride for that XfbBuffer then no vertex data of that primitive is recorded in any transform feedback buffer, and the value for the number of primitives written in the corresponding VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT query for all transform feedback buffers is no longer incremented.

Any outputs made to a XfbBuffer that is not bound to a transform feedback buffer are ignored.

To bind transform feedback memory ranges to a command buffer for use in subsequent drawing commands, call:

```
// Provided by VK_KHR_device_address_commands with VK_EXT_transform_feedback
void vkCmdBindTransformFeedbackBuffers2EXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstBinding,
    uint32_t                                    bindingCount,
    const VkBindTransformFeedbackBuffer2InfoEXT* pBindingInfos);
```

- commandBuffer is the command buffer into which the command is recorded.
- firstBinding is the index of the first transform feedback binding whose state is updated by the command.
- bindingCount is the number of transform feedback bindings whose state is updated by the command.
- pBindingInfos is a pointer to an array of VkBindTransformFeedbackBuffer2InfoEXT structures specifying the ranges of memory to be used to capture transform feedback data.

Element i of pBindingInfos replaces the current state for the transform feedback binding firstBinding + i, for i in [0,bindingCount).

Valid Usage

- VUID-vkCmdBindTransformFeedbackBuffers2EXT-transformFeedback-02355

VkPhysicalDeviceTransformFeedbackFeaturesEXT::transformFeedback **must** be enabled

- VUID-vkCmdBindTransformFeedbackBuffers2EXT-firstBinding-02356
  firstBinding **must** be less than VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBuffers
- VUID-vkCmdBindTransformFeedbackBuffers2EXT-firstBinding-02357
  The sum of firstBinding and bindingCount **must** be less than or equal to VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBuffers
- VUID-vkCmdBindTransformFeedbackBuffers2EXT-None-02365
  Transform feedback **must** not be active when the vkCmdBindTransformFeedbackBuffers2EXT command is recorded
- VUID-vkCmdBindTransformFeedbackBuffers2EXT-addressRange-13090
  The addressRange.address member of all elements of pBindingInfos **must** be a multiple of 4
- VUID-vkCmdBindTransformFeedbackBuffers2EXT-addressRange-13091
  The buffer that the addressRange of each element of pBindingInfos was queried from **must** have been created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT
- VUID-vkCmdBindTransformFeedbackBuffers2EXT-addressRange-13092
  The addressRange.size member of all elements of pBindingInfos **must** be less than or equal to VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBufferSize

Valid Usage (Implicit)

- VUID-vkCmdBindTransformFeedbackBuffers2EXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindTransformFeedbackBuffers2EXT-pBindingInfos-parameter
  If pBindingInfos is not NULL, pBindingInfos **must** be a valid pointer to an array of bindingCount valid VkBindTransformFeedbackBuffer2InfoEXT structures
- VUID-vkCmdBindTransformFeedbackBuffers2EXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdBindTransformFeedbackBuffers2EXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindTransformFeedbackBuffers2EXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindTransformFeedbackBuffers2EXT-bindingCount-arraylength
  If pBindingInfos is not NULL, bindingCount **must** be greater than 0

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally

synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindTransformFeedbackBuffers2EXT is not affected by conditional rendering

VkBindTransformFeedbackBuffer2InfoEXT is defined as:

```
// Provided by VK_KHR_device_address_commands with VK_EXT_transform_feedback
typedef struct VkBindTransformFeedbackBuffer2InfoEXT {
    VkStructureType             sType;
    const void*                 pNext;
    VkDeviceAddressRangeKHR     addressRange;
    VkAddressCommandFlagsKHR    addressFlags;
} VkBindTransformFeedbackBuffer2InfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- addressRange is the VkDeviceAddressRangeKHR of the address range to use.
- addressFlags is a VkAddressCommandFlagsKHR value defining the flags for the address range.

Valid Usage

- VUID-VkBindTransformFeedbackBuffer2InfoEXT-addressRange-13097
  If the range specified by addressRange is not bound completely to memory when accessed, addressFlags **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR
- VUID-VkBindTransformFeedbackBuffer2InfoEXT-addressRange-13098
  If the buffer from which the range specified by addressRange was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and protectedNoFault is not supported, addressFlags **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkBindTransformFeedbackBuffer2InfoEXT-addressRange-13099
  If the buffer from which the range specified by addressRange was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and protectedNoFault is not supported, addressFlags **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR

- VUID-VkBindTransformFeedbackBuffer2InfoEXT-addressFlags-13100
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkBindTransformFeedbackBuffer2InfoEXT-addressRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkBindTransformFeedbackBuffer2InfoEXT-addressRange-13123
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, `addressFlags` **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkBindTransformFeedbackBuffer2InfoEXT-addressFlags-13101
  `addressFlags` **must** not include both VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkBindTransformFeedbackBuffer2InfoEXT-addressRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `addressFlags` **must** include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkBindTransformFeedbackBuffer2InfoEXT-addressRange-13125
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing `addressRange`, was created without VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, `addressFlags` **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR

Valid Usage (Implicit)

- VUID-VkBindTransformFeedbackBuffer2InfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_BIND_TRANSFORM_FEEDBACK_BUFFER_2_INFO_EXT
- VUID-VkBindTransformFeedbackBuffer2InfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkBindTransformFeedbackBuffer2InfoEXT-addressFlags-parameter
  If `addressFlags` is not `0`, `addressFlags` **must** be a valid combination of VkAddressCommandFlagBitsKHR values

To bind transform feedback buffers to a command buffer for use in subsequent drawing commands, call:

WARNING	This functionality is superseded by vkCmdBindTransformFeedbackBuffers2EXT. See Legacy Functionality for more information.

```
// Provided by VK_EXT_transform_feedback
void vkCmdBindTransformFeedbackBuffersEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstBinding,
    uint32_t                                    bindingCount,
    const VkBuffer*                             pBuffers,
    const VkDeviceSize*                         pOffsets,
    const VkDeviceSize*                         pSizes);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `firstBinding` is the index of the first transform feedback binding whose state is updated by the command.
- `bindingCount` is the number of transform feedback bindings whose state is updated by the command.
- `pBuffers` is a pointer to an array of buffer handles.
- `pOffsets` is a pointer to an array of buffer offsets.
- `pSizes` is `NULL` or a pointer to an array of VkDeviceSize buffer sizes, specifying the maximum number of bytes to capture to the corresponding transform feedback buffer. If `pSizes` is `NULL`, it is equivalent to setting a `pSizes` array where every element is VK_WHOLE_SIZE.

The values taken from elements i of `pBuffers`, `pOffsets` and `pSizes` replace the current state for the transform feedback binding `firstBinding` + i, for i in [0, `bindingCount`). The transform feedback binding is updated to start at the offset indicated by `pOffsets`[i] from the start of the buffer `pBuffers`[i].

When an element of `pSizes`[i] is VK_WHOLE_SIZE, or `pSizes` is `NULL`, the effective range is calculated by taking the size of `pBuffers`[i] minus `pOffsets`[i]. Otherwise, the effective range is equal to the element in `pSizes`[i].

Valid Usage

- VUID-vkCmdBindTransformFeedbackBuffersEXT-transformFeedback-02355
  `VkPhysicalDeviceTransformFeedbackFeaturesEXT`::`transformFeedback` **must** be enabled
- VUID-vkCmdBindTransformFeedbackBuffersEXT-firstBinding-02356
  `firstBinding` **must** be less than `VkPhysicalDeviceTransformFeedbackPropertiesEXT`::`maxTransformFeedbackBuffers`
- VUID-vkCmdBindTransformFeedbackBuffersEXT-firstBinding-02357

The sum of `firstBinding` and `bindingCount` **must** be less than or equal to `VkPhysicalDeviceTransformFeedbackPropertiesEXT`::`maxTransformFeedbackBuffers`

- VUID-vkCmdBindTransformFeedbackBuffersEXT-None-02365
  Transform feedback **must** not be active when the `vkCmdBindTransformFeedbackBuffersEXT` command is recorded
- VUID-vkCmdBindTransformFeedbackBuffersEXT-pOffsets-02358
  All elements of `pOffsets` **must** be less than the size of the corresponding element in `pBuffers`
- VUID-vkCmdBindTransformFeedbackBuffersEXT-pOffsets-02359
  All elements of `pOffsets` **must** be a multiple of 4
- VUID-vkCmdBindTransformFeedbackBuffersEXT-pBuffers-02360
  All elements of `pBuffers` **must** have been created with the VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT usage flag set
- VUID-vkCmdBindTransformFeedbackBuffersEXT-pOffsets-02363
  All elements of `pOffsets` plus the effective size of the element, **must** be less than or equal to the size of the corresponding buffer in `pBuffers`
- VUID-vkCmdBindTransformFeedbackBuffersEXT-pBuffers-02364
  Each element of `pBuffers` that is non-sparse **must** be bound completely and contiguously to a single `VkDeviceMemory` object

Valid Usage (Implicit)

- VUID-vkCmdBindTransformFeedbackBuffersEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBindTransformFeedbackBuffersEXT-pBuffers-parameter
  `pBuffers` **must** be a valid pointer to an array of `bindingCount` valid VkBuffer handles
- VUID-vkCmdBindTransformFeedbackBuffersEXT-pOffsets-parameter
  `pOffsets` **must** be a valid pointer to an array of `bindingCount` VkDeviceSize values
- VUID-vkCmdBindTransformFeedbackBuffersEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBindTransformFeedbackBuffersEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBindTransformFeedbackBuffersEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindTransformFeedbackBuffersEXT-bindingCount-arraylength
  `bindingCount` **must** be greater than `0`
- VUID-vkCmdBindTransformFeedbackBuffersEXT-commonparent
  Both of `commandBuffer`, and the elements of `pBuffers` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBindTransformFeedbackBuffersEXT is not affected by conditional rendering

Transform feedback for specific transform feedback buffers is made active by calling:

```
// Provided by VK_KHR_device_address_commands with VK_EXT_transform_feedback
void vkCmdBeginTransformFeedback2EXT(
    VkCommandBuffer                                 commandBuffer,
    uint32_t                                        firstCounterRange,
    uint32_t                                        counterRangeCount,
    const VkBindTransformFeedbackBuffer2InfoEXT* pCounterInfos);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `firstCounterRange` is the index of the first transform feedback buffer corresponding to `pCounterInfos`[0].
- `counterRangeCount` is the size of the `pCounterInfos` array.
- `pCounterInfos` is `NULL` or a pointer to an array of VkBindTransformFeedbackBuffer2InfoEXT structures defining memory ranges containing counters used to resume transform feedback from a previous location.

If `pCounterInfos` is `NULL`, it is equivalent to an array of VkBindTransformFeedbackBuffer2InfoEXT structures with each element having a `addressRange.size` of 0.

If the `addressRange.size` of an element of `pCounterInfos` at index i is 0, transform feedback data written to `XfbBuffer` with a `XFB Buffer Number` equal to the sum of i and `firstCounterRange` will be written starting at an offset of 0 in the bound transform feedback buffer. If the `addressRange.size` of an element of `pCounterInfos` at index i is not 0, it will instead be written starting at an offset equal to the 32-bit value found at `pCounterInfos`[i].addressRange.address.

The active transform feedback buffers will capture primitives emitted from the corresponding `XfbBuffer` in the bound graphics pipeline. Any `XfbBuffer` emitted that does not output to an active transform feedback buffer will not be captured.

Valid Usage

- VUID-vkCmdBeginTransformFeedback2EXT-transformFeedback-02366
  `VkPhysicalDeviceTransformFeedbackFeaturesEXT`::`transformFeedback` **must** be enabled
- VUID-vkCmdBeginTransformFeedback2EXT-None-02367
  Transform feedback **must** not be active
- VUID-vkCmdBeginTransformFeedback2EXT-firstCounter-02368
  `firstCounterRange` **must** be less than `VkPhysicalDeviceTransformFeedbackPropertiesEXT`::`maxTransformFeedbackBuffers`
- VUID-vkCmdBeginTransformFeedback2EXT-firstCounter-02369
  The sum of `firstCounterRange` and `counterRangeCount` **must** be less than or equal to `VkPhysicalDeviceTransformFeedbackPropertiesEXT`::`maxTransformFeedbackBuffers`
- VUID-vkCmdBeginTransformFeedback2EXT-firstCounter-09630
  The sum of `firstCounterRange` and `counterRangeCount` **must** be less than or equal to the number of transform feedback buffers bound by vkCmdBindTransformFeedbackBuffers2EXT
- VUID-vkCmdBeginTransformFeedback2EXT-None-06233
  If the `shaderObject` feature is not enabled, a valid graphics pipeline **must** be bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdBeginTransformFeedback2EXT-None-04128
  The last pre-rasterization shader stage of the bound graphics pipeline **must** have been declared with the `Xfb` execution mode
- VUID-vkCmdBeginTransformFeedback2EXT-None-02373
  Transform feedback **must** not be made active in a render pass instance with multiview enabled
- VUID-vkCmdBeginTransformFeedback2EXT-pCounterInfos-13093
  Each element of `pCounterInfos` **must** have a `addressRange.size` that is either 0 or greater than or equal to 4
- VUID-vkCmdBeginTransformFeedback2EXT-pCounterInfos-13094
  For each element of `pCounterInfos` with a non-zero address, the buffer the address was queried from **must** have been created with the VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT usage flag set

Valid Usage (Implicit)

- VUID-vkCmdBeginTransformFeedback2EXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBeginTransformFeedback2EXT-pCounterInfos-parameter

If `counterRangeCount` is not `0`, and `pCounterInfos` is not `NULL`, `pCounterInfos` **must** be a valid pointer to an array of `counterRangeCount` valid VkBindTransformFeedbackBuffer2InfoEXT structures

- VUID-vkCmdBeginTransformFeedback2EXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBeginTransformFeedback2EXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBeginTransformFeedback2EXT-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdBeginTransformFeedback2EXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBeginTransformFeedback2EXT is not affected by conditional rendering

Transform feedback for specific transform feedback buffers is made inactive by calling:

```
// Provided by VK_KHR_device_address_commands with VK_EXT_transform_feedback
void vkCmdEndTransformFeedback2EXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstCounterRange,
    uint32_t                                    counterRangeCount,
    const VkBindTransformFeedbackBuffer2InfoEXT* pCounterInfos);
```

- `commandBuffer` is the command buffer into which the command is recorded.

- `firstCounterRange` is the index of the first transform feedback buffer corresponding to `pCounterInfos`[0].
- `counterRangeCount` is the size of the `pCounterInfos` array.
- `pCounterInfos` is `NULL` or a pointer to an array of VkBindTransformFeedbackBuffer2InfoEXT structures defining memory ranges used to write counters used to later resume transform feedback.

If `pCounterInfos` is `NULL`, it is equivalent to an array of VkBindTransformFeedbackBuffer2InfoEXT structures with each element having a `addressRange.size` of 0.

If the `addressRange.size` of an element of `pCounterInfos` at index i is 0, no data is written to the address range. If the `addressRange.size` of an element of `pCounterInfos` at index i is not 0, the byte offset, where the next vertex data would be written to in the transform feedback buffer at a binding equal to the sum of i and `firstCounterRange`, is written to that range's `address`.

Valid Usage

- VUID-vkCmdEndTransformFeedback2EXT-transformFeedback-02374
  `VkPhysicalDeviceTransformFeedbackFeaturesEXT`::`transformFeedback` **must** be enabled
- VUID-vkCmdEndTransformFeedback2EXT-None-02375
  Transform feedback **must** be active
- VUID-vkCmdEndTransformFeedback2EXT-firstCounterBuffer-02376
  `firstCounterBuffer` **must** be less than `VkPhysicalDeviceTransformFeedbackPropertiesEXT`::`maxTransformFeedbackBuffers`
- VUID-vkCmdEndTransformFeedback2EXT-firstCounterBuffer-02377
  The sum of `firstCounterBuffer` and `counterRangeCount` **must** be less than or equal to `VkPhysicalDeviceTransformFeedbackPropertiesEXT`::`maxTransformFeedbackBuffers`
- VUID-vkCmdEndTransformFeedback2EXT-pCounterInfos-13095
  Each element of `pCounterInfos` **must** have a `addressRange.size` that is either 0 or greater than or equal to 4
- VUID-vkCmdEndTransformFeedback2EXT-pCounterInfos-13096
  For each element of `pCounterInfos` with a non-zero address, the buffer the address was queried from **must** have been created with the VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT usage flag set

Valid Usage (Implicit)

- VUID-vkCmdEndTransformFeedback2EXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdEndTransformFeedback2EXT-pCounterInfos-parameter
  If `counterRangeCount` is not `0`, and `pCounterInfos` is not `NULL`, `pCounterInfos` **must** be a valid pointer to an array of `counterRangeCount` valid VkBindTransformFeedbackBuffer2InfoEXT structures

- VUID-vkCmdEndTransformFeedback2EXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdEndTransformFeedback2EXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdEndTransformFeedback2EXT-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdEndTransformFeedback2EXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdEndTransformFeedback2EXT is not affected by conditional rendering

Transform feedback for specific transform feedback buffers is made active by calling:

WARNING This functionality is superseded by vkCmdBeginTransformFeedback2EXT. See Legacy Functionality for more information.

```
// Provided by VK_EXT_transform_feedback
void vkCmdBeginTransformFeedbackEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstCounterBuffer,
    uint32_t                                    counterBufferCount,
    const VkBuffer*                             pCounterBuffers,
    const VkDeviceSize*                         pCounterBufferOffsets);
```

- `commandBuffer` is the command buffer into which the command is recorded.

- `firstCounterBuffer` is the index of the first transform feedback buffer corresponding to `pCounterBuffers`[0] and `pCounterBufferOffsets`[0].
- `counterBufferCount` is the size of the `pCounterBuffers` and `pCounterBufferOffsets` arrays.
- `pCounterBuffers` is `NULL` or a pointer to an array of VkBuffer handles to counter buffers. Each buffer contains a 4 byte integer value representing the byte offset from the start of the corresponding transform feedback buffer from where to start capturing vertex data. If the byte offset stored to the counter buffer location was done using vkCmdEndTransformFeedbackEXT it can be used to resume transform feedback from the previous location. In that case, a pipeline barrier is required between the calls to `vkCmdEndTransformFeedbackEXT` and `vkCmdBeginTransformFeedbackEXT`, with VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT as the source and destination stages, VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT as the source access and VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT as the destination access. If `pCounterBuffers` is `NULL`, then transform feedback will start capturing vertex data to byte offset zero in all bound transform feedback buffers. For each element of `pCounterBuffers` that is VK_NULL_HANDLE, transform feedback will start capturing vertex data to byte zero in the corresponding bound transform feedback buffer.
- `pCounterBufferOffsets` is `NULL` or a pointer to an array of VkDeviceSize values specifying offsets within each of the `pCounterBuffers` where the counter values were previously written. The location in each counter buffer at these offsets **must** be large enough to contain 4 bytes of data. This data is the number of bytes captured by the previous transform feedback to this buffer. If `pCounterBufferOffsets` is `NULL`, then it is assumed the offsets are zero.

The active transform feedback buffers will capture primitives emitted from the corresponding `XfbBuffer` in the bound graphics pipeline. Any `XfbBuffer` emitted that does not output to an active transform feedback buffer will not be captured.

Valid Usage

- VUID-vkCmdBeginTransformFeedbackEXT-transformFeedback-02366
  `VkPhysicalDeviceTransformFeedbackFeaturesEXT`::`transformFeedback` **must** be enabled
- VUID-vkCmdBeginTransformFeedbackEXT-None-02367
  Transform feedback **must** not be active
- VUID-vkCmdBeginTransformFeedbackEXT-firstCounter-02368
  `firstCounterBuffer` **must** be less than `VkPhysicalDeviceTransformFeedbackPropertiesEXT`::`maxTransformFeedbackBuffers`
- VUID-vkCmdBeginTransformFeedbackEXT-firstCounter-02369
  The sum of `firstCounterBuffer` and `counterBufferCount` **must** be less than or equal to `VkPhysicalDeviceTransformFeedbackPropertiesEXT`::`maxTransformFeedbackBuffers`
- VUID-vkCmdBeginTransformFeedbackEXT-firstCounter-09630
  The sum of `firstCounterBuffer` and `counterBufferCount` **must** be less than or equal to the number of transform feedback buffers bound by vkCmdBindTransformFeedbackBuffersEXT
- VUID-vkCmdBeginTransformFeedbackEXT-None-06233

If the shaderObject feature is not enabled, a valid graphics pipeline **must** be bound to VK_PIPELINE_BIND_POINT_GRAPHICS

- VUID-vkCmdBeginTransformFeedbackEXT-None-04128
  The last pre-rasterization shader stage of the bound graphics pipeline **must** have been declared with the Xfb execution mode
- VUID-vkCmdBeginTransformFeedbackEXT-None-02373
  Transform feedback **must** not be made active in a render pass instance with multiview enabled
- VUID-vkCmdBeginTransformFeedbackEXT-counterBufferCount-02607
  If counterBufferCount is not 0, and pCounterBuffers is not NULL, pCounterBuffers **must** be a valid pointer to an array of counterBufferCount VkBuffer handles that are either valid or VK_NULL_HANDLE
- VUID-vkCmdBeginTransformFeedbackEXT-pCounterBufferOffsets-02370
  For each buffer handle in the array, if it is not VK_NULL_HANDLE it **must** reference a buffer large enough to hold 4 bytes at the corresponding offset from the pCounterBufferOffsets array
- VUID-vkCmdBeginTransformFeedbackEXT-pCounterBuffer-02371
  If pCounterBuffers is NULL, then pCounterBufferOffsets **must** also be NULL
- VUID-vkCmdBeginTransformFeedbackEXT-pCounterBuffers-02372
  For each buffer handle in the pCounterBuffers array that is not VK_NULL_HANDLE it **must** have been created with the VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT usage flag set

Valid Usage (Implicit)

- VUID-vkCmdBeginTransformFeedbackEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBeginTransformFeedbackEXT-pCounterBufferOffsets-parameter
  If counterBufferCount is not 0, and pCounterBufferOffsets is not NULL, pCounterBufferOffsets **must** be a valid pointer to an array of counterBufferCount VkDeviceSize values
- VUID-vkCmdBeginTransformFeedbackEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdBeginTransformFeedbackEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdBeginTransformFeedbackEXT-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdBeginTransformFeedbackEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBeginTransformFeedbackEXT-commonparent
  Both of commandBuffer, and the elements of pCounterBuffers that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same

VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdBeginTransformFeedbackEXT is not affected by conditional rendering

Transform feedback for specific transform feedback buffers is made inactive by calling:

WARNING This functionality is superseded by vkCmdEndTransformFeedback2EXT. See Legacy Functionality for more information.

```
// Provided by VK_EXT_transform_feedback
void vkCmdEndTransformFeedbackEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstCounterBuffer,
    uint32_t                                    counterBufferCount,
    const VkBuffer*                             pCounterBuffers,
    const VkDeviceSize*                         pCounterBufferOffsets);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `firstCounterBuffer` is the index of the first transform feedback buffer corresponding to `pCounterBuffers`[0] and `pCounterBufferOffsets`[0].
- `counterBufferCount` is the size of the `pCounterBuffers` and `pCounterBufferOffsets` arrays.
- `pCounterBuffers` is `NULL` or a pointer to an array of VkBuffer handles to counter buffers. The counter buffers are used to record the current byte positions of each transform feedback buffer where the next vertex output data would be captured. This **can** be used by a subsequent vkCmdBeginTransformFeedbackEXT call to resume transform feedback capture from this position. It can also be used by vkCmdDrawIndirectByteCountEXT to determine the vertex count

of the draw call.

- pCounterBufferOffsets is NULL or a pointer to an array of VkDeviceSize values specifying offsets within each of the pCounterBuffers where the counter values can be written. The location in each counter buffer at these offsets **must** be large enough to contain 4 bytes of data. The data stored at this location is the byte offset from the start of the transform feedback buffer binding where the next vertex data would be written. If pCounterBufferOffsets is NULL, then it is assumed the offsets are zero.

Valid Usage

- VUID-vkCmdEndTransformFeedbackEXT-transformFeedback-02374
  VkPhysicalDeviceTransformFeedbackFeaturesEXT::transformFeedback **must** be enabled
- VUID-vkCmdEndTransformFeedbackEXT-None-02375
  Transform feedback **must** be active
- VUID-vkCmdEndTransformFeedbackEXT-firstCounterBuffer-02376
  firstCounterBuffer **must** be less than VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBuffers
- VUID-vkCmdEndTransformFeedbackEXT-firstCounterBuffer-02377
  The sum of firstCounterBuffer and counterBufferCount **must** be less than or equal to VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBuffers
- VUID-vkCmdEndTransformFeedbackEXT-counterBufferCount-02608
  If counterBufferCount is not 0, and pCounterBuffers is not NULL, pCounterBuffers **must** be a valid pointer to an array of counterBufferCount VkBuffer handles that are either valid or VK_NULL_HANDLE
- VUID-vkCmdEndTransformFeedbackEXT-pCounterBufferOffsets-02378
  For each buffer handle in the array, if it is not VK_NULL_HANDLE it **must** reference a buffer large enough to hold 4 bytes at the corresponding offset from the pCounterBufferOffsets array
- VUID-vkCmdEndTransformFeedbackEXT-pCounterBuffer-02379
  If pCounterBuffers is NULL, then pCounterBufferOffsets **must** also be NULL
- VUID-vkCmdEndTransformFeedbackEXT-pCounterBuffers-02380
  For each buffer handle in the pCounterBuffers array that is not VK_NULL_HANDLE it **must** have been created with the VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT usage flag set

Valid Usage (Implicit)

- VUID-vkCmdEndTransformFeedbackEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdEndTransformFeedbackEXT-pCounterBufferOffsets-parameter
  If counterBufferCount is not 0, and pCounterBufferOffsets is not NULL, pCounterBufferOffsets **must** be a valid pointer to an array of counterBufferCount VkDeviceSize values

- VUID-vkCmdEndTransformFeedbackEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdEndTransformFeedbackEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdEndTransformFeedbackEXT-renderpass
  This command **must** only be called inside of a render pass instance
- VUID-vkCmdEndTransformFeedbackEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdEndTransformFeedbackEXT-commonparent
  Both of `commandBuffer`, and the elements of `pCounterBuffers` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdEndTransformFeedbackEXT is not affected by conditional rendering

30.2. Flat Shading

Flat shading a vertex output attribute means to assign all vertices of the primitive the same value for that output. The output values assigned are those of the *provoking vertex* of the primitive. Flat shading is applied to those vertex attributes that match fragment input attributes which are decorated as `Flat`.

If neither mesh, geometry nor tessellation shading is active, the provoking vertex is determined by the primitive topology defined by VkPipelineInputAssemblyStateCreateInfo::`topology` used to execute the drawing command.

If a shader using MeshEXT Execution Model is active, the provoking vertex is determined by the primitive topology defined by the OutputPoints, OutputLinesEXT, or OutputTrianglesEXT execution mode.

If geometry shading is active, the provoking vertex is determined by the primitive topology defined by the OutputPoints, OutputLineStrip, or OutputTriangleStrip execution mode.

If tessellation shading is active but geometry shading is not, the provoking vertex **may** be any of the vertices in each primitive.

For a given primitive topology, the pipeline's provoking vertex mode determines which vertex is the provoking vertex. To specify the provoking vertex mode, include a VkPipelineRasterizationProvokingVertexStateCreateInfoEXT structure in the VkPipelineRasterizationStateCreateInfo::pNext chain when creating the pipeline.

The VkPipelineRasterizationProvokingVertexStateCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_provoking_vertex
typedef struct VkPipelineRasterizationProvokingVertexStateCreateInfoEXT {
    VkStructureType             sType;
    const void*                 pNext;
    VkProvokingVertexModeEXT    provokingVertexMode;
} VkPipelineRasterizationProvokingVertexStateCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- provokingVertexMode is a VkProvokingVertexModeEXT value selecting the provoking vertex mode.

If this structure is not provided when creating the pipeline, the pipeline will use the VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT mode.

If the provokingVertexModePerPipeline limit is VK_FALSE, then all pipelines bound within a render pass instance **must** have the same provokingVertexMode.

Valid Usage

- VUID-VkPipelineRasterizationProvokingVertexStateCreateInfoEXT-provokingVertexMode-04883
  If provokingVertexMode is VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT, then the provokingVertexLast feature **must** be enabled

Valid Usage (Implicit)

- VUID-VkPipelineRasterizationProvokingVertexStateCreateInfoEXT-sType-sType
  sType **must** be

VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_PROVOKING_VERTEX_STATE_CREATE_INFO_EXT

- VUID-VkPipelineRasterizationProvokingVertexStateCreateInfoEXT-provokingVertexMode-parameter
  `provokingVertexMode` **must** be a valid VkProvokingVertexModeEXT value

Structure Chaining

Extends the structure

- VkPipelineRasterizationStateCreateInfo

Possible values of VkPipelineRasterizationProvokingVertexStateCreateInfoEXT::`provokingVertexMode` are:

```
// Provided by VK_EXT_provoking_vertex
typedef enum VkProvokingVertexModeEXT {
    VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT = 0,
    VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT = 1,
} VkProvokingVertexModeEXT;
```

- VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT specifies that the provoking vertex is the first non-adjacency vertex in the list of vertices used by a primitive.
- VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT specifies that the provoking vertex is the last non-adjacency vertex in the list of vertices used by a primitive.

These modes are described more precisely in Primitive Topologies.

To dynamically set the `provokingVertexMode` state, call:

```
// Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_provoking_vertex,
VK_EXT_provoking_vertex with VK_EXT_shader_object
void vkCmdSetProvokingVertexModeEXT(
    VkCommandBuffer                             commandBuffer,
    VkProvokingVertexModeEXT                    provokingVertexMode);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `provokingVertexMode` specifies the `provokingVertexMode` state.

This command sets the `provokingVertexMode` state for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineRasterizationProvokingVertexStateCreateInfoEXT::`provokingVertexMode` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetProvokingVertexModeEXT-None-09423
  At least one of the following **must** be true:
  - The `extendedDynamicState3ProvokingVertexMode` feature is enabled
  - The `shaderObject` feature is enabled
- VUID-vkCmdSetProvokingVertexModeEXT-provokingVertexMode-07447
  If `provokingVertexMode` is VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT, then the `provokingVertexLast` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdSetProvokingVertexModeEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetProvokingVertexModeEXT-provokingVertexMode-parameter
  `provokingVertexMode` **must** be a valid VkProvokingVertexModeEXT value
- VUID-vkCmdSetProvokingVertexModeEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetProvokingVertexModeEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetProvokingVertexModeEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetProvokingVertexModeEXT is not affected by conditional rendering

30.3. Primitive Clipping

Primitives are culled against the *cull volume* and then clipped to the *clip volume*. In clip coordinates, the *view volume* is defined by:

$$-w_c \leq x_c \leq w_c$$
$$-w_c \leq y_c \leq w_c$$
$$z_m \leq z_c \leq w_c$$

where if VkPipelineViewportDepthClipControlCreateInfoEXT::`negativeOneToOne` is VK_TRUE z_m is equal to $-w_c$ otherwise z_m is equal to zero.

This view volume **can** be further restricted by as many as `VkPhysicalDeviceLimits`::`maxClipDistances` application-defined half-spaces.

The cull volume is the intersection of up to `VkPhysicalDeviceLimits`::`maxCullDistances` application-defined half-spaces (if no application-defined cull half-spaces are enabled, culling against the cull volume is skipped).

A shader **must** write a single cull distance for each enabled cull half-space to elements of the `CullDistance` array. If the cull distance for any enabled cull half-space is negative for all of the vertices of the primitive under consideration, the primitive is discarded. Otherwise the primitive is clipped against the clip volume as defined below.

The clip volume is the intersection of up to `VkPhysicalDeviceLimits`::`maxClipDistances` application-defined half-spaces with the view volume (if no application-defined clip half-spaces are enabled, the clip volume is the view volume).

A shader **must** write a single clip distance for each enabled clip half-space to elements of the `ClipDistance` array. Clip half-space i is then given by the set of points satisfying the inequality

$$c_i(\mathbf{P}) \geq 0$$

where $c_i(\mathbf{P})$ is the clip distance i at point **P**. For point primitives, $c_i(\mathbf{P})$ is simply the clip distance for the vertex in question. For line and triangle primitives, per-vertex clip distances are interpolated using a weighted mean, with weights derived according to the algorithms described in sections Basic Line Segment Rasterization and Basic Polygon Rasterization, using the perspective interpolation equations.

The number of application-defined clip and cull half-spaces that are enabled is determined by the explicit size of the built-in arrays `ClipDistance` and `CullDistance`, respectively, declared as an output in the interface of the entry point of the final shader stage before clipping.

If VkPipelineRasterizationDepthClipStateCreateInfoEXT is present in the graphics pipeline state

then depth clipping is disabled if VkPipelineRasterizationDepthClipStateCreateInfoEXT ::depthClipEnable is VK_FALSE. Otherwise, if VkPipelineRasterizationDepthClipStateCreateInfoEXT is not present, depth clipping is disabled when VkPipelineRasterizationStateCreateInfo ::depthClampEnable is VK_TRUE.

To dynamically set enable or disable depth clamping, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetDepthClampEnableEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    depthClampEnable);
```

- commandBuffer is the command buffer into which the command will be recorded.
- depthClampEnable specifies whether depth clamping is enabled.

This command sets whether depth clamping is enabled or disabled for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT set in VkPipelineDynamicStateCreateInfo ::pDynamicStates. Otherwise, this state is specified by the VkPipelineRasterizationStateCreateInfo ::depthClampEnable value used to create the currently active pipeline.

If the depth clamping state is changed dynamically, and the pipeline was not created with VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT enabled, then depth clipping is enabled when depth clamping is disabled and vice versa.

If the maintenance11 feature is enabled, the above statement is further clarified to state that the depth clipping state is only affected by the current value of depthClampEnable if the pipeline was not created with the VkPipelineRasterizationDepthClipStateCreateInfoEXT structure. If VkPipelineRasterizationDepthClipStateCreateInfoEXT was included in the pipeline creation then the depth clipping state is set explicitly by VkPipelineRasterizationDepthClipStateCreateInfoEXT ::depthClipEnable and not changed when the depth clamping state is modified.

Valid Usage

- VUID-vkCmdSetDepthClampEnableEXT-None-09423
  At least one of the following **must** be true:
  - The extendedDynamicState3DepthClampEnable feature is enabled
  - The shaderObject feature is enabled
- VUID-vkCmdSetDepthClampEnableEXT-depthClamp-07449
  If the depthClamp feature is not enabled, depthClampEnable **must** be VK_FALSE

Valid Usage (Implicit)

- VUID-vkCmdSetDepthClampEnableEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle

- VUID-vkCmdSetDepthClampEnableEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetDepthClampEnableEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDepthClampEnableEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDepthClampEnableEXT is not affected by conditional rendering

To dynamically set enable or disable depth clipping, call:

```
// Provided by VK_EXT_depth_clip_enable with VK_EXT_extended_dynamic_state3,
VK_EXT_depth_clip_enable with VK_EXT_shader_object
void vkCmdSetDepthClipEnableEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    depthClipEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthClipEnable` specifies whether depth clipping is enabled.

This command sets whether depth clipping is enabled or disabled for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineRasterizationDepthClipStateCreateInfoEXT::`depthClipEnable` value used to create the

currently active pipeline. If VkPipelineRasterizationDepthClipStateCreateInfoEXT was not specified, depth clipping is set to the inverse of the current value of `depthClampEnable`.

Valid Usage

- VUID-vkCmdSetDepthClipEnableEXT-None-09423
  At least one of the following **must** be true:
  - The `extendedDynamicState3DepthClipEnable` feature is enabled
  - The `shaderObject` feature is enabled
- VUID-vkCmdSetDepthClipEnableEXT-depthClipEnable-07451
  The `depthClipEnable` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdSetDepthClipEnableEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDepthClipEnableEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetDepthClipEnableEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDepthClipEnableEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDepthClipEnableEXT is not affected by conditional rendering

When depth clipping is disabled, the plane equation

$$z_m \leq z_c \leq w_c$$

(see the clip volume definition above) is ignored by view volume clipping (effectively, there is no near or far plane clipping).

If the primitive under consideration is a point or line segment, then clipping passes it unchanged if its vertices lie entirely within the clip volume.

Possible values of VkPhysicalDevicePointClippingProperties::pointClippingBehavior, specifying clipping behavior of a point primitive whose vertex lies outside the clip volume, are:

```
// Provided by VK_VERSION_1_1
typedef enum VkPointClippingBehavior {
    VK_POINT_CLIPPING_BEHAVIOR_ALL_CLIP_PLANES = 0,
    VK_POINT_CLIPPING_BEHAVIOR_USER_CLIP_PLANES_ONLY = 1,
  // Provided by VK_KHR_maintenance2
    VK_POINT_CLIPPING_BEHAVIOR_ALL_CLIP_PLANES_KHR =
VK_POINT_CLIPPING_BEHAVIOR_ALL_CLIP_PLANES,
  // Provided by VK_KHR_maintenance2
    VK_POINT_CLIPPING_BEHAVIOR_USER_CLIP_PLANES_ONLY_KHR =
VK_POINT_CLIPPING_BEHAVIOR_USER_CLIP_PLANES_ONLY,
} VkPointClippingBehavior;
```

```
// Provided by VK_KHR_maintenance2
// Equivalent to VkPointClippingBehavior
typedef VkPointClippingBehavior VkPointClippingBehaviorKHR;
```

- VK_POINT_CLIPPING_BEHAVIOR_ALL_CLIP_PLANES specifies that the primitive is discarded if the vertex lies outside any clip plane, including the planes bounding the view volume.
- VK_POINT_CLIPPING_BEHAVIOR_USER_CLIP_PLANES_ONLY specifies that the primitive is discarded only if the vertex lies outside any user clip plane.

If either of a line segment's vertices lies outside of the clip volume, the line segment **may** be clipped, with new vertex coordinates computed for each vertex that lies outside the clip volume. A clipped line segment endpoint lies on both the original line segment and the boundary of the clip volume.

This clipping produces a value, $0 \leq t \leq 1$, for each clipped vertex. If the coordinates of a clipped vertex are $\mathbf{P}$ and the unclipped line segment's vertex coordinates are $\mathbf{P}_1$ and $\mathbf{P}_2$, then t satisfies the following equation

$\mathbf{P} = t\,\mathbf{P}_1 + (1-t)\,\mathbf{P}_2$.

t is used to clip vertex output attributes as described in Clipping Shader Outputs.

If the primitive is a polygon, it passes unchanged if every one of its edges lies entirely inside the clip volume, and is either clipped or discarded otherwise. If the edges of the polygon intersect the boundary of the clip volume, the intersecting edges are reconnected by new edges that lie along the boundary of the clip volume - in some cases requiring the introduction of new vertices into a polygon.

If a polygon intersects an edge of the clip volume's boundary, the clipped polygon **must** include a point on this boundary edge.

Primitives rendered with application-defined half-spaces **must** satisfy a complementarity criterion. Suppose a series of primitives is drawn where each vertex i has a single specified clip distance d_i (or a number of similarly specified clip distances, if multiple half-spaces are enabled). Next, suppose that the same series of primitives are drawn again with each such clip distance replaced by $-d_i$ (and the graphics pipeline is otherwise the same). In this case, primitives **must** not be missing any pixels, and pixels **must** not be drawn twice in regions where those primitives are cut by the clip planes.

The `VkPipelineViewportDepthClipControlCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_depth_clip_control
typedef struct VkPipelineViewportDepthClipControlCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           negativeOneToOne;
} VkPipelineViewportDepthClipControlCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `negativeOneToOne` sets the z_m in the *view volume* to $-w_c$.

Valid Usage

- VUID-VkPipelineViewportDepthClipControlCreateInfoEXT-negativeOneToOne-06470
  If the `depthClipControl` feature is not enabled, `negativeOneToOne` **must** be VK_FALSE

Valid Usage (Implicit)

- VUID-VkPipelineViewportDepthClipControlCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_DEPTH_CLIP_CONTROL_CREATE_INFO_EXT

Structure Chaining

Extends the structure

- VkPipelineViewportStateCreateInfo

To dynamically set `negativeOneToOne`, call:

```
// Provided by VK_EXT_depth_clip_control with VK_EXT_extended_dynamic_state3,
VK_EXT_depth_clip_control with VK_EXT_shader_object
void vkCmdSetDepthClipNegativeOneToOneEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    negativeOneToOne);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `negativeOneToOne` specifies the `negativeOneToOne` state.

This command sets the `negativeOneToOne` state for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineViewportDepthClipControlCreateInfoEXT::`negativeOneToOne` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthClipNegativeOneToOneEXT-None-09423
  At least one of the following **must** be true:
  - The `extendedDynamicState3DepthClipNegativeOneToOne` feature is enabled
  - The `shaderObject` feature is enabled
- VUID-vkCmdSetDepthClipNegativeOneToOneEXT-depthClipControl-07453
  The `depthClipControl` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdSetDepthClipNegativeOneToOneEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDepthClipNegativeOneToOneEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetDepthClipNegativeOneToOneEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDepthClipNegativeOneToOneEXT-videocoding

This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDepthClipNegativeOneToOneEXT is not affected by conditional rendering

30.4. Clipping Shader Outputs

Next, vertex output attributes are clipped. The output values associated with a vertex that lies within the clip volume are unaffected by clipping. If a primitive is clipped, however, the output values assigned to vertices produced by clipping are clipped.

Let the output values assigned to the two vertices $\mathbf{P}_1$ and $\mathbf{P}_2$ of an unclipped edge be $\mathbf{c}_1$ and $\mathbf{c}_2$. The value of t (see Primitive Clipping) for a clipped point $\mathbf{P}$ is used to obtain the output value associated with $\mathbf{P}$ as

$$\mathbf{c} = t\,\mathbf{c}_1 + (1-t)\,\mathbf{c}_2.$$

(Multiplying an output value by a scalar means multiplying each of *x*, *y*, *z*, and *w* by the scalar.)

Since this computation is performed in clip space before division by w_c, clipped output values are perspective-correct.

Polygon clipping creates a clipped vertex along an edge of the clip volume's boundary. This situation is handled by noting that polygon clipping proceeds by clipping against one half-space at a time. Output value clipping is done in the same way, so that clipped points always occur at the intersection of polygon edges (possibly already clipped) with the clip volume's boundary.

For vertex output attributes whose matching fragment input attributes are decorated with

`NoPerspective`, the value of t used to obtain the output value associated with **P** will be adjusted to produce results that vary linearly in framebuffer space.

Output attributes of integer or unsigned integer type **must** always be flat shaded. Flat shaded attributes are constant over the primitive being rasterized (see Basic Line Segment Rasterization and Basic Polygon Rasterization), and no interpolation is performed. The output value **c** is taken from either $\mathbf{c}_1$ or $\mathbf{c}_2$, since flat shading has already occurred and the two values are identical.

30.5. Coordinate Transformations

Clip coordinates for a vertex result from shader execution, which yields a vertex coordinate `Position`.

Perspective division on clip coordinates yields *normalized device coordinates*, followed by a *viewport* transformation (see Controlling the Viewport) to convert these coordinates into *framebuffer coordinates*.

If a vertex in clip coordinates has a position given by

$$\begin{pmatrix} x_c \\ y_c \\ z_c \\ w_c \end{pmatrix}$$

then the vertex's normalized device coordinates are

$$\begin{pmatrix} x_d \\ y_d \\ z_d \end{pmatrix} = \begin{pmatrix} \frac{x_c}{w_c} \\ \frac{y_c}{w_c} \\ \frac{z_c}{w_c} \end{pmatrix}$$

30.6. Controlling the Viewport

The viewport transformation is determined by the selected viewport's width and height in pixels, p_x and p_y, respectively, and its center (o_x, o_y) (also in pixels), as well as its depth range min and max determining a depth range scale value p_z and a depth range bias value o_z (defined below). The vertex's framebuffer coordinates (x_f, y_f) and depth z_f are given by

$$x_f = (p_x / 2)\, x_d + o_x$$

$$y_f = (p_y / 2)\, y_d + o_y$$

$$z_f = p_z \times z_d + o_z$$

Multiple viewports are available, numbered zero up to VkPhysicalDeviceLimits::maxViewports minus one. The number of viewports used by a pipeline is controlled by the viewportCount member of the VkPipelineViewportStateCreateInfo structure used in pipeline creation.

x_f and y_f have limited precision, where the number of fractional bits retained is specified by VkPhysicalDeviceLimits::subPixelPrecisionBits. When rasterizing line segments, the number of fractional bits is specified by VkPhysicalDeviceLineRasterizationProperties::lineSubPixelPrecisionBits.

The VkPipelineViewportStateCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineViewportStateCreateInfo {
    VkStructureType                       sType;
    const void*                           pNext;
    VkPipelineViewportStateCreateFlags    flags;
    uint32_t                              viewportCount;
    const VkViewport*                     pViewports;
    uint32_t                              scissorCount;
    const VkRect2D*                       pScissors;
} VkPipelineViewportStateCreateInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is reserved for future use.
- viewportCount is the number of viewports used by the pipeline.
- pViewports is a pointer to an array of VkViewport structures, defining the viewport transforms. If the viewport state is dynamic, this member is ignored.
- scissorCount is the number of scissors and **must** match the number of viewports.
- pScissors is a pointer to an array of VkRect2D structures defining the rectangular bounds of the scissor for the corresponding viewport. If the scissor state is dynamic, this member is ignored.

Valid Usage

- VUID-VkPipelineViewportStateCreateInfo-viewportCount-01216
  If the multiViewport feature is not enabled, viewportCount **must** not be greater than 1
- VUID-VkPipelineViewportStateCreateInfo-scissorCount-01217
  If the multiViewport feature is not enabled, scissorCount **must** not be greater than 1
- VUID-VkPipelineViewportStateCreateInfo-viewportCount-01218
  viewportCount **must** be less than or equal to VkPhysicalDeviceLimits::maxViewports
- VUID-VkPipelineViewportStateCreateInfo-scissorCount-01219
  scissorCount **must** be less than or equal to VkPhysicalDeviceLimits::maxViewports
- VUID-VkPipelineViewportStateCreateInfo-x-02821
  The x and y members of offset member of any element of pScissors **must** be greater than

or equal to `0`

- VUID-VkPipelineViewportStateCreateInfo-offset-02822
  Evaluation of (`offset.x` + `extent.width`) **must** not cause a signed integer addition overflow for any element of `pScissors`
- VUID-VkPipelineViewportStateCreateInfo-offset-02823
  Evaluation of (`offset.y` + `extent.height`) **must** not cause a signed integer addition overflow for any element of `pScissors`
- VUID-VkPipelineViewportStateCreateInfo-scissorCount-04134
  If `scissorCount` and `viewportCount` are both not dynamic, then `scissorCount` and `viewportCount` **must** be identical
- VUID-VkPipelineViewportStateCreateInfo-viewportCount-04135
  If the graphics pipeline is being created with VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT set then `viewportCount` **must** be `0`, otherwise `viewportCount` **must** be greater than `0`
- VUID-VkPipelineViewportStateCreateInfo-scissorCount-04136
  If the graphics pipeline is being created with VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT set then `scissorCount` **must** be `0`, otherwise `scissorCount` **must** be greater than `0`

Valid Usage (Implicit)

- VUID-VkPipelineViewportStateCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_STATE_CREATE_INFO
- VUID-VkPipelineViewportStateCreateInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkPipelineViewportDepthClampControlCreateInfoEXT or VkPipelineViewportDepthClipControlCreateInfoEXT
- VUID-VkPipelineViewportStateCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkPipelineViewportStateCreateInfo-flags-zerobitmask
  `flags` **must** be `0`

To dynamically set the viewport count and viewports, call:

```
// Provided by VK_VERSION_1_3
void vkCmdSetViewportWithCount(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    viewportCount,
    const VkViewport*                           pViewports);
```

```
// Provided by VK_EXT_extended_dynamic_state, VK_EXT_shader_object
// Equivalent to vkCmdSetViewportWithCount
void vkCmdSetViewportWithCountEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    viewportCount,
    const VkViewport*                           pViewports);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `viewportCount` specifies the viewport count.
- `pViewports` specifies the viewports to use for drawing.

This command sets the viewport count and viewports state for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. Otherwise, this state is specified by the corresponding VkPipelineViewportStateCreateInfo::`viewportCount` and `pViewports` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetViewportWithCount-None-08971
  At least one of the following **must** be true:
  - the `extendedDynamicState` feature is enabled
  - the `shaderObject` feature is enabled
  - the value of VkApplicationInfo::`apiVersion` used to create the VkInstance parent of `commandBuffer` is greater than or equal to Version 1.3
- VUID-vkCmdSetViewportWithCount-viewportCount-03394
  `viewportCount` **must** be between 1 and `VkPhysicalDeviceLimits`::`maxViewports`, inclusive
- VUID-vkCmdSetViewportWithCount-viewportCount-03395
  If the `multiViewport` feature is not enabled, `viewportCount` **must** be 1

Valid Usage (Implicit)

- VUID-vkCmdSetViewportWithCount-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetViewportWithCount-pViewports-parameter
  `pViewports` **must** be a valid pointer to an array of `viewportCount` valid VkViewport structures
- VUID-vkCmdSetViewportWithCount-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetViewportWithCount-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support

VK_QUEUE_GRAPHICS_BIT operations

- VUID-vkCmdSetViewportWithCount-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetViewportWithCount-viewportCount-arraylength
  `viewportCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetViewportWithCount is not affected by conditional rendering

To dynamically set the scissor count and scissor rectangular bounds, call:

```
// Provided by VK_VERSION_1_3
void vkCmdSetScissorWithCount(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    scissorCount,
    const VkRect2D*                             pScissors);
```

```
// Provided by VK_EXT_extended_dynamic_state, VK_EXT_shader_object
// Equivalent to vkCmdSetScissorWithCount
void vkCmdSetScissorWithCountEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    scissorCount,
    const VkRect2D*                             pScissors);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `scissorCount` specifies the scissor count.

- `pScissors` specifies the scissors to use for drawing.

This command sets the scissor count and scissor rectangular bounds state for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the corresponding VkPipelineViewportStateCreateInfo::`scissorCount` and `pScissors` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetScissorWithCount-None-08971
  At least one of the following **must** be true:
  - the `extendedDynamicState` feature is enabled
  - the `shaderObject` feature is enabled
  - the value of VkApplicationInfo::`apiVersion` used to create the VkInstance parent of `commandBuffer` is greater than or equal to Version 1.3
- VUID-vkCmdSetScissorWithCount-scissorCount-03397
  `scissorCount` **must** be between `1` and `VkPhysicalDeviceLimits`::`maxViewports`, inclusive
- VUID-vkCmdSetScissorWithCount-scissorCount-03398
  If the `multiViewport` feature is not enabled, `scissorCount` **must** be `1`
- VUID-vkCmdSetScissorWithCount-x-03399
  The `x` and `y` members of `offset` member of any element of `pScissors` **must** be greater than or equal to `0`
- VUID-vkCmdSetScissorWithCount-offset-03400
  Evaluation of (`offset.x` + `extent.width`) **must** not cause a signed integer addition overflow for any element of `pScissors`
- VUID-vkCmdSetScissorWithCount-offset-03401
  Evaluation of (`offset.y` + `extent.height`) **must** not cause a signed integer addition overflow for any element of `pScissors`

Valid Usage (Implicit)

- VUID-vkCmdSetScissorWithCount-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetScissorWithCount-pScissors-parameter
  `pScissors` **must** be a valid pointer to an array of `scissorCount` VkRect2D structures
- VUID-vkCmdSetScissorWithCount-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetScissorWithCount-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations

- VUID-vkCmdSetScissorWithCount-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetScissorWithCount-scissorCount-arraylength
  `scissorCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetScissorWithCount is not affected by conditional rendering

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineViewportStateCreateFlags;
```

`VkPipelineViewportStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

A *pre-rasterization shader stage* **can** direct each primitive to one of several viewports. The destination viewport for a primitive is selected by the last active pre-rasterization shader stage that has an output variable decorated with `ViewportIndex`. The viewport transform uses the viewport corresponding to the value assigned to `ViewportIndex`, and taken from an implementation-dependent vertex of each primitive. If `ViewportIndex` is outside the range zero to `viewportCount` minus one for a primitive, or if the last active pre-rasterization shader stage did not assign a value to `ViewportIndex` for all vertices of a primitive due to flow control, the values resulting from the viewport transformation of the vertices of such primitives are undefined. If the last pre-rasterization shader stage does not have an output decorated with `ViewportIndex`, the viewport numbered zero is used by the viewport transformation.

A single vertex **can** be used in more than one individual primitive, in primitives such as VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP. In this case, the viewport transformation is applied separately for each primitive.

To dynamically set the viewport transformation parameters, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetViewport(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstViewport,
    uint32_t                                    viewportCount,
    const VkViewport*                           pViewports);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `firstViewport` is the index of the first viewport whose parameters are updated by the command.
- `viewportCount` is the number of viewports whose parameters are updated by the command.
- `pViewports` is a pointer to an array of VkViewport structures specifying viewport parameters.

This command sets the viewport transformation parameters state for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_VIEWPORT set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the `VkPipelineViewportStateCreateInfo`::`pViewports` values used to create the currently active pipeline.

The viewport parameters taken from element i of `pViewports` replace the current state for the viewport index `firstViewport` + i, for i in [0, `viewportCount`).

Valid Usage

- VUID-vkCmdSetViewport-firstViewport-01223
  The sum of `firstViewport` and `viewportCount` **must** be between `1` and `VkPhysicalDeviceLimits`::`maxViewports`, inclusive
- VUID-vkCmdSetViewport-firstViewport-01224
  If the `multiViewport` feature is not enabled, `firstViewport` **must** be `0`
- VUID-vkCmdSetViewport-viewportCount-01225
  If the `multiViewport` feature is not enabled, `viewportCount` **must** be `1`

Valid Usage (Implicit)

- VUID-vkCmdSetViewport-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetViewport-pViewports-parameter
  `pViewports` **must** be a valid pointer to an array of `viewportCount` valid VkViewport structures
- VUID-vkCmdSetViewport-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetViewport-commandBuffer-cmdpool

The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations

- VUID-vkCmdSetViewport-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetViewport-viewportCount-arraylength
  viewportCount **must** be greater than 0

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetViewport is not affected by conditional rendering

Both VkPipelineViewportStateCreateInfo and vkCmdSetViewport use VkViewport to set the viewport transformation parameters.

The VkViewport structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkViewport {
    float    x;
    float    y;
    float    width;
    float    height;
    float    minDepth;
    float    maxDepth;
} VkViewport;
```

- x and y are the viewport's upper left corner (x,y).
- width and height are the viewport's width and height, respectively.

- `minDepth` and `maxDepth` are the depth range for the viewport.

NOTE	Despite their names, `minDepth` **can** be less than, equal to, or greater than `maxDepth`.

The framebuffer depth coordinate z_f **may** be represented using either a fixed-point or floating-point representation. However, a floating-point representation **must** be used if the depth/stencil attachment has a floating-point depth component. If an m-bit fixed-point representation is used, we assume that it represents each value $\frac{k}{2^m - 1}$, where k ∈ { 0, 1, ..., 2^m-1 }, as k (e.g. 1.0 is represented in binary as a string of all ones).

The viewport parameters shown in the above equations are found from these values as

o_x = `x` + `width` / 2

o_y = `y` + `height` / 2

o_z = `minDepth` (or (`maxDepth` + `minDepth`) / 2 if VkPipelineViewportDepthClipControlCreateInfoEXT::`negativeOneToOne` is VK_TRUE)

p_x = `width`

p_y = `height`

p_z = `maxDepth` - `minDepth` (or (`maxDepth` - `minDepth`) / 2 if VkPipelineViewportDepthClipControlCreateInfoEXT::`negativeOneToOne` is VK_TRUE)

The application **can** specify a negative term for `height`, which has the effect of negating the y coordinate in clip space before performing the transform. When using a negative `height`, the application **should** also adjust the `y` value to point to the lower left corner of the viewport instead of the upper left corner. Using the negative `height` allows the application to avoid having to negate the y component of the `Position` output from the last pre-rasterization shader stage.

The width and height of the implementation-dependent maximum viewport dimensions **must** be greater than or equal to the width and height of the largest image which **can** be created and attached to a framebuffer.

The floating-point viewport bounds are represented with an implementation-dependent precision.

Valid Usage

- VUID-VkViewport-width-01770
  `width` **must** be greater than `0.0`

- VUID-VkViewport-width-01771
  `width` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxViewportDimensions`[0]
- VUID-VkViewport-apiVersion-07917
  If the VK_KHR_maintenance1 extension is not enabled, the `VK_AMD_negative_viewport_height` extension is not enabled, and VkPhysicalDeviceProperties::`apiVersion` is less than Vulkan 1.1, `height` **must** be greater than `0.0`
- VUID-VkViewport-height-01773
  The absolute value of `height` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxViewportDimensions`[1]
- VUID-VkViewport-x-01774
  `x` **must** be greater than or equal to `viewportBoundsRange`[0]
- VUID-VkViewport-x-01232
  (`x` + `width`) **must** be less than or equal to `viewportBoundsRange`[1]
- VUID-VkViewport-y-01775
  `y` **must** be greater than or equal to `viewportBoundsRange`[0]
- VUID-VkViewport-y-01776
  `y` **must** be less than or equal to `viewportBoundsRange`[1]
- VUID-VkViewport-y-01777
  (`y` + `height`) **must** be greater than or equal to `viewportBoundsRange`[0]
- VUID-VkViewport-y-01233
  (`y` + `height`) **must** be less than or equal to `viewportBoundsRange`[1]
- VUID-VkViewport-minDepth-01234
  If the `VK_EXT_depth_range_unrestricted` extension is not enabled, `minDepth` **must** be between `0.0` and `1.0`, inclusive
- VUID-VkViewport-maxDepth-01235
  If the `VK_EXT_depth_range_unrestricted` extension is not enabled, `maxDepth` **must** be between `0.0` and `1.0`, inclusive

Chapter 31. Rasterization

Rasterization is the process by which a primitive is converted to a two-dimensional image. Each discrete location of this image contains associated data such as depth, color, or other attributes.

Rasterizing a primitive begins by determining which squares of an integer grid in framebuffer coordinates are occupied by the primitive, and assigning one or more depth values to each such square. This process is described below for points, lines, and polygons.

A grid square, including its (x,y) framebuffer coordinates, z (depth), and associated data added by fragment shaders, is called a fragment. A fragment is located by its upper left corner, which lies on integer grid coordinates.

Rasterization operations also refer to a fragment's sample locations, which are offset by fractional values from its upper left corner. The rasterization rules for points, lines, and triangles involve testing whether each sample location is inside the primitive. Fragments need not actually be square, and rasterization rules are not affected by the aspect ratio of fragments. Display of non-square grids, however, will cause rasterized points and line segments to appear fatter in one direction than the other.

We assume that fragments are square, since it simplifies antialiasing and texturing. After rasterization, fragments are processed by fragment operations.

Several factors affect rasterization, including the members of VkPipelineRasterizationStateCreateInfo and VkPipelineMultisampleStateCreateInfo.

The `VkPipelineRasterizationStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineRasterizationStateCreateInfo {
    VkStructureType                            sType;
    const void*                                pNext;
    VkPipelineRasterizationStateCreateFlags    flags;
    VkBool32                                   depthClampEnable;
    VkBool32                                   rasterizerDiscardEnable;
    VkPolygonMode                              polygonMode;
    VkCullModeFlags                            cullMode;
    VkFrontFace                                frontFace;
    VkBool32                                   depthBiasEnable;
    float                                      depthBiasConstantFactor;
    float                                      depthBiasClamp;
    float                                      depthBiasSlopeFactor;
    float                                      lineWidth;
} VkPipelineRasterizationStateCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.

- `depthClampEnable` controls whether to clamp the fragment's depth values as described in Depth Test. If the pipeline is not created with VkPipelineRasterizationDepthClipStateCreateInfoEXT present then enabling depth clamp will also disable clipping primitives to the z planes of the frustum as described in Primitive Clipping. Otherwise depth clipping is controlled by the state set in VkPipelineRasterizationDepthClipStateCreateInfoEXT.
- `rasterizerDiscardEnable` controls whether primitives are discarded immediately before the rasterization stage.
- `polygonMode` is the triangle rendering mode. See VkPolygonMode.
- `cullMode` is the triangle facing direction used for primitive culling. See VkCullModeFlagBits.
- `frontFace` is a VkFrontFace value specifying the front-facing triangle orientation to be used for culling.
- `depthBiasEnable` controls whether to bias fragment depth values.
- `depthBiasConstantFactor` is a scalar factor controlling the constant depth value added to each fragment.
- `depthBiasClamp` is the maximum (or minimum) depth bias of a fragment.
- `depthBiasSlopeFactor` is a scalar factor applied to a fragment's slope in depth bias calculations.
- `lineWidth` is the width of rasterized line segments.

Valid Usage

- VUID-VkPipelineRasterizationStateCreateInfo-depthClampEnable-00782
  If the `depthClamp` feature is not enabled, `depthClampEnable` **must** be VK_FALSE
- VUID-VkPipelineRasterizationStateCreateInfo-polygonMode-01507
  If the `fillModeNonSolid` feature is not enabled, `polygonMode` **must** be VK_POLYGON_MODE_FILL
- VUID-VkPipelineRasterizationStateCreateInfo-pointPolygons-04458
  If the `VK_KHR_portability_subset` extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::`pointPolygons` is VK_FALSE, and `rasterizerDiscardEnable` is VK_FALSE, `polygonMode` **must** not be VK_POLYGON_MODE_POINT

Valid Usage (Implicit)

- VUID-VkPipelineRasterizationStateCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_STATE_CREATE_INFO
- VUID-VkPipelineRasterizationStateCreateInfo-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkDepthBiasRepresentationInfoEXT, VkPipelineRasterizationConservativeStateCreateInfoEXT, VkPipelineRasterizationDepthClipStateCreateInfoEXT, VkPipelineRasterizationLineStateCreateInfo, VkPipelineRasterizationProvokingVertexStateCreateInfoEXT, or

VkPipelineRasterizationStateStreamCreateInfoEXT

- VUID-VkPipelineRasterizationStateCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkPipelineRasterizationStateCreateInfo-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkPipelineRasterizationStateCreateInfo-polygonMode-parameter
  `polygonMode` **must** be a valid VkPolygonMode value
- VUID-VkPipelineRasterizationStateCreateInfo-cullMode-parameter
  If `cullMode` is not `0`, `cullMode` **must** be a valid combination of VkCullModeFlagBits values
- VUID-VkPipelineRasterizationStateCreateInfo-frontFace-parameter
  `frontFace` **must** be a valid VkFrontFace value

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineRasterizationStateCreateFlags;
```

`VkPipelineRasterizationStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

If the `pNext` chain of VkPipelineRasterizationStateCreateInfo includes a `VkPipelineRasterizationDepthClipStateCreateInfoEXT` structure, then that structure controls whether depth clipping is enabled or disabled.

The `VkPipelineRasterizationDepthClipStateCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_depth_clip_enable
typedef struct VkPipelineRasterizationDepthClipStateCreateInfoEXT {
    VkStructureType                                    sType;
    const void*                                        pNext;
    VkPipelineRasterizationDepthClipStateCreateFlagsEXT    flags;
    VkBool32                                           depthClipEnable;
} VkPipelineRasterizationDepthClipStateCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `depthClipEnable` controls whether depth clipping is enabled as described in Primitive Clipping.

Valid Usage (Implicit)

- VUID-VkPipelineRasterizationDepthClipStateCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_DEPTH_CLIP_STATE_CREATE_INFO_EXT

- VUID-VkPipelineRasterizationDepthClipStateCreateInfoEXT-flags-zerobitmask
  `flags` **must** be `0`

Structure Chaining

Extends the structure

- VkPipelineRasterizationStateCreateInfo

```
// Provided by VK_EXT_depth_clip_enable
typedef VkFlags VkPipelineRasterizationDepthClipStateCreateFlagsEXT;
```

`VkPipelineRasterizationDepthClipStateCreateFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

The `VkPipelineMultisampleStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineMultisampleStateCreateInfo {
    VkStructureType                         sType;
    const void*                             pNext;
    VkPipelineMultisampleStateCreateFlags   flags;
    VkSampleCountFlagBits                   rasterizationSamples;
    VkBool32                                sampleShadingEnable;
    float                                   minSampleShading;
    const VkSampleMask*                     pSampleMask;
    VkBool32                                alphaToCoverageEnable;
    VkBool32                                alphaToOneEnable;
} VkPipelineMultisampleStateCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `rasterizationSamples` is a VkSampleCountFlagBits value specifying the number of samples used in rasterization. This value is ignored for the purposes of setting the number of samples used in rasterization if the pipeline is created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state set, but if VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state is not set, it is still used to define the size of the `pSampleMask` array as described below.
- `sampleShadingEnable` **can** be used to enable Sample Shading.
- `minSampleShading` specifies a minimum fraction of sample shading if `sampleShadingEnable` is VK_TRUE.
- `pSampleMask` is a pointer to an array of VkSampleMask values used in the sample mask test.

- `alphaToCoverageEnable` controls whether a temporary coverage value is generated based on the alpha component of the fragment's first color output as specified in the Multisample Coverage section.
- `alphaToOneEnable` controls whether the alpha component of the fragment's first color output is replaced with one as described in Multisample Coverage.

Each bit in the sample mask is associated with a unique sample index as defined for the coverage mask. Each bit b for mask word w in the sample mask corresponds to sample index i, where i = 32 × w + b. `pSampleMask` has a length equal to ⌈ `rasterizationSamples` / 32 ⌉ words.

If `pSampleMask` is `NULL`, it is treated as if the mask has all bits set to `1`.

Valid Usage

- VUID-VkPipelineMultisampleStateCreateInfo-sampleShadingEnable-00784
  If the `sampleRateShading` feature is not enabled, `sampleShadingEnable` **must** be VK_FALSE
- VUID-VkPipelineMultisampleStateCreateInfo-alphaToOneEnable-00785
  If the `alphaToOne` feature is not enabled, `alphaToOneEnable` **must** be VK_FALSE
- VUID-VkPipelineMultisampleStateCreateInfo-minSampleShading-00786
  `minSampleShading` **must** be in the range [0,1]

Valid Usage (Implicit)

- VUID-VkPipelineMultisampleStateCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_MULTISAMPLE_STATE_CREATE_INFO
- VUID-VkPipelineMultisampleStateCreateInfo-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkPipelineSampleLocationsStateCreateInfoEXT
- VUID-VkPipelineMultisampleStateCreateInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkPipelineMultisampleStateCreateInfo-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkPipelineMultisampleStateCreateInfo-rasterizationSamples-parameter
  `rasterizationSamples` **must** be a valid VkSampleCountFlagBits value

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineMultisampleStateCreateFlags;
```

`VkPipelineMultisampleStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

The elements of the sample mask array are of type VkSampleMask, each representing 32 bits of coverage information:

```
// Provided by VK_VERSION_1_0
typedef uint32_t VkSampleMask;
```

Rasterization only generates fragments which cover one or more pixels inside the framebuffer. Pixels outside the framebuffer are never considered covered in the fragment. Fragments which would be produced by application of any of the primitive rasterization rules described below but which lie outside the framebuffer are not produced, nor are they processed by any later stage of the pipeline, including any of the fragment operations.

Surviving fragments are processed by fragment shaders. Fragment shaders determine associated data for fragments, and **can** also modify or replace their assigned depth values.

31.1. Discarding Primitives Before Rasterization

Primitives are discarded before rasterization if the `rasterizerDiscardEnable` member of VkPipelineRasterizationStateCreateInfo is enabled. When enabled, primitives are discarded after they are processed by the last active shader stage in the pipeline before rasterization.

To dynamically enable whether primitives are discarded before the rasterization stage, call:

```
// Provided by VK_VERSION_1_3
void vkCmdSetRasterizerDiscardEnable(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    rasterizerDiscardEnable);
```

```
// Provided by VK_EXT_extended_dynamic_state2, VK_EXT_shader_object
// Equivalent to vkCmdSetRasterizerDiscardEnable
void vkCmdSetRasterizerDiscardEnableEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    rasterizerDiscardEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `rasterizerDiscardEnable` controls whether primitives are discarded immediately before the rasterization stage.

This command sets the discard enable for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineRasterizationStateCreateInfo ::`rasterizerDiscardEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetRasterizerDiscardEnable-None-08970
  At least one of the following **must** be true:

- the `extendedDynamicState2` feature is enabled
- the `shaderObject` feature is enabled
- the value of VkApplicationInfo::`apiVersion` used to create the VkInstance parent of `commandBuffer` is greater than or equal to Version 1.3

Valid Usage (Implicit)

- VUID-vkCmdSetRasterizerDiscardEnable-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetRasterizerDiscardEnable-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetRasterizerDiscardEnable-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetRasterizerDiscardEnable-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetRasterizerDiscardEnable is not affected by conditional rendering

31.2. Controlling the Vertex Stream Used for Rasterization

By default vertex data output from the last pre-rasterization shader stage are directed to vertex stream zero. Geometry shaders **can** emit primitives to multiple independent vertex streams. Each

vertex emitted by the geometry shader is directed at one of the vertex streams. As vertices are received on each vertex stream, they are arranged into primitives of the type specified by the geometry shader output primitive type. The shading language instructions `OpEndPrimitive` and `OpEndStreamPrimitive` **can** be used to end the primitive being assembled on a given vertex stream and start a new empty primitive of the same type. An implementation supports up to `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams` streams, which is at least 1. The individual streams are numbered 0 through `maxTransformFeedbackStreams` minus 1. There is no requirement on the order of the streams to which vertices are emitted, and the number of vertices emitted to each vertex stream **can** be completely independent, subject only to the `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreamDataSize` and `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBufferDataSize` limits. The primitives output from all vertex streams are passed to the transform feedback stage to be captured to transform feedback buffers in the manner specified by the last pre-rasterization shader stage shader's `XfbBuffer`, `XfbStride`, and `Offsets` decorations on the output interface variables in the graphics pipeline. To use a vertex stream other than zero, or to use multiple streams, the `GeometryStreams` capability **must** be specified.

By default, the primitives output from vertex stream zero are rasterized. If the implementation supports the VkPhysicalDeviceTransformFeedbackPropertiesEXT::`transformFeedbackRasterizationStreamSelect` property it is possible to rasterize a vertex stream other than zero.

By default, geometry shaders that emit vertices to multiple vertex streams are limited to using only the `OutputPoints` output primitive type. If the implementation supports the VkPhysicalDeviceTransformFeedbackPropertiesEXT::`transformFeedbackStreamsLinesTriangles` property it is possible to emit `OutputLineStrip` or `OutputTriangleStrip` in addition to `OutputPoints`.

The vertex stream used for rasterization is specified by adding a `VkPipelineRasterizationStateStreamCreateInfoEXT` structure to the `pNext` chain of a VkPipelineRasterizationStateCreateInfo structure.

The `VkPipelineRasterizationStateStreamCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_transform_feedback
typedef struct VkPipelineRasterizationStateStreamCreateInfoEXT {
    VkStructureType                                    sType;
    const void*                                        pNext;
    VkPipelineRasterizationStateStreamCreateFlagsEXT   flags;
    uint32_t                                           rasterizationStream;
} VkPipelineRasterizationStateStreamCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `rasterizationStream` is the vertex stream selected for rasterization.

If this structure is not present, `rasterizationStream` is assumed to be zero.

Valid Usage

- VUID-VkPipelineRasterizationStateStreamCreateInfoEXT-geometryStreams-02324 `VkPhysicalDeviceTransformFeedbackFeaturesEXT::geometryStreams` **must** be enabled
- VUID-VkPipelineRasterizationStateStreamCreateInfoEXT-rasterizationStream-02325 `rasterizationStream` **must** be less than `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams`
- VUID-VkPipelineRasterizationStateStreamCreateInfoEXT-rasterizationStream-02326 `rasterizationStream` **must** be zero if `VkPhysicalDeviceTransformFeedbackPropertiesEXT::transformFeedbackRasterizationStreamSelect` is `VK_FALSE`

Valid Usage (Implicit)

- VUID-VkPipelineRasterizationStateStreamCreateInfoEXT-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_STATE_STREAM_CREATE_INFO_EXT`
- VUID-VkPipelineRasterizationStateStreamCreateInfoEXT-flags-zerobitmask `flags` **must** be `0`

Structure Chaining

Extends the structure

- `VkPipelineRasterizationStateCreateInfo`

```
// Provided by VK_EXT_transform_feedback
typedef VkFlags VkPipelineRasterizationStateStreamCreateFlagsEXT;
```

`VkPipelineRasterizationStateStreamCreateFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

To dynamically set the `rasterizationStream` state, call:

```
// Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_transform_feedback,
VK_EXT_shader_object with VK_EXT_transform_feedback
void vkCmdSetRasterizationStreamEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    rasterizationStream);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `rasterizationStream` specifies the `rasterizationStream` state.

This command sets the rasterizationStream state for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT set in VkPipelineDynamicStateCreateInfo::pDynamicStates. Otherwise, this state is specified by the VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetRasterizationStreamEXT-None-09423
  At least one of the following **must** be true:
  - The extendedDynamicState3RasterizationStream feature is enabled
  - The shaderObject feature is enabled
- VUID-vkCmdSetRasterizationStreamEXT-transformFeedback-07411
  The transformFeedback feature **must** be enabled
- VUID-vkCmdSetRasterizationStreamEXT-rasterizationStream-07412
  rasterizationStream **must** be less than VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams
- VUID-vkCmdSetRasterizationStreamEXT-rasterizationStream-07413
  rasterizationStream **must** be zero if VkPhysicalDeviceTransformFeedbackPropertiesEXT::transformFeedbackRasterizationStreamSelect is VK_FALSE

Valid Usage (Implicit)

- VUID-vkCmdSetRasterizationStreamEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetRasterizationStreamEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdSetRasterizationStreamEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetRasterizationStreamEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetRasterizationStreamEXT is not affected by conditional rendering

31.3. Rasterization Order

Within a subpass of a render pass instance, for a given (x,y,layer,sample) sample location, the following operations are guaranteed to execute in *rasterization order*, for each separate primitive that includes that sample location:

1. Fragment operations, in the order defined
2. Blending, logic operations, and color writes

Execution of these operations for each primitive in a subpass occurs in primitive order.

31.4. Multisampling

Multisampling is a mechanism to antialias all Vulkan primitives: points, lines, and polygons. The technique is to sample all primitives multiple times at each pixel. Each sample in each framebuffer attachment has storage for a color, depth, and/or stencil value, such that per-fragment operations apply to each sample independently. The color sample values **can** be later *resolved* to a single color (see Resolving Multisample Images and the Render Pass chapter for more details on how to resolve multisample images to non-multisample images).

Vulkan defines rasterization rules for single-sample modes in a way that is equivalent to a multisample mode with a single sample in the center of each fragment.

Each fragment includes a coverage mask with a single bit for each sample in the fragment, and a number of depth values and associated data for each sample.

It is understood that each pixel has `rasterizationSamples` locations associated with it. These locations are exact positions, rather than regions or areas, and each is referred to as a sample point. The sample points associated with a pixel **must** be located inside or on the boundary of the unit square that is considered to bound the pixel. Furthermore, the relative locations of sample points **may** be identical for each pixel in the framebuffer, or they **may** differ.

If the render pass has a fragment density map attachment, each fragment only has `rasterizationSamples` locations associated with it regardless of how many pixels are covered in the

fragment area. Fragment sample locations are defined as if the fragment had an area of (1,1) and its sample points **must** be located within these bounds. Their actual location in the framebuffer is calculated by scaling the sample location by the fragment area. Attachments with storage for multiple samples per pixel are located at the pixel sample locations. Otherwise, the fragment's sample locations are generally used for evaluation of associated data and fragment operations.

If the current pipeline includes a fragment shader with one or more variables in its interface decorated with `Sample` and `Input`, the data associated with those variables will be assigned independently for each sample. The values for each sample **must** be evaluated at the location of the sample. The data associated with any other variables not decorated with `Sample` and `Input` need not be evaluated independently for each sample.

A *coverage mask* is generated for each fragment, based on which samples within that fragment are determined to be within the area of the primitive that generated the fragment.

Single pixel fragments and multi-pixel fragments defined by a fragment density map have one set of samples. Multi-pixel fragments defined by setting the fragment shading rate have one set of samples per pixel. Each set of samples has a number of samples determined by VkPipelineMultisampleStateCreateInfo::`rasterizationSamples`. Each sample in a set is assigned a unique *sample index* i in the range [0, `rasterizationSamples`).

To dynamically set the `rasterizationSamples`, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetRasterizationSamplesEXT(
    VkCommandBuffer                             commandBuffer,
    VkSampleCountFlagBits                       rasterizationSamples);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `rasterizationSamples` specifies `rasterizationSamples`.

This command sets the `rasterizationSamples` for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineMultisampleStateCreateInfo::`rasterizationSamples` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetRasterizationSamplesEXT-None-09423
  At least one of the following **must** be true:
  - The `extendedDynamicState3RasterizationSamples` feature is enabled
  - The `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetRasterizationSamplesEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetRasterizationSamplesEXT-rasterizationSamples-parameter
  `rasterizationSamples` **must** be a valid VkSampleCountFlagBits value
- VUID-vkCmdSetRasterizationSamplesEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetRasterizationSamplesEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetRasterizationSamplesEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetRasterizationSamplesEXT is not affected by conditional rendering

Each sample in a fragment is also assigned a unique *coverage index* j in the range [0, n × `rasterizationSamples`), where n is the number of sets in the fragment. If the fragment contains a single set of samples, the *coverage index* is always equal to the *sample index*.

If the fragment shading rate is set, the coverage index j is determined as a function of the *pixel index* p, the *sample index* i, and the number of rasterization samples r as:

$$j = i + r \times ((f_w \times f_h) - 1 - p)$$

where the pixel index p is determined as a function of the pixel's framebuffer location (x,y) and the fragment size (f_w,f_h):

$$p_x = x \% f_w$$

$$p_y = y \% f_h$$

$$p = p_x + (p_y \times f_w)$$

The tables below illustrate the pixel index for multi-pixel fragments:

Table 35. Pixel Indices - 1 Wide

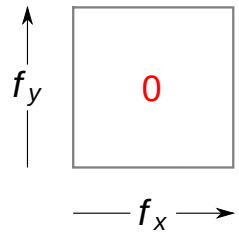
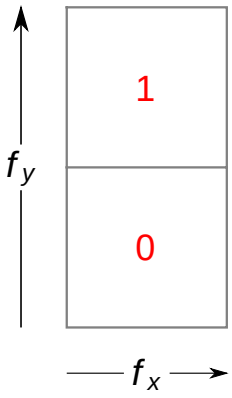
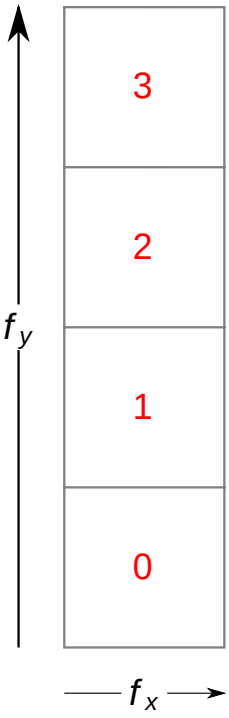
1x1	1x2	1x4
		

Table 36. Pixel Indices - 2 Wide

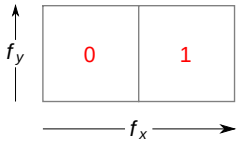
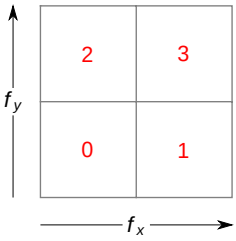
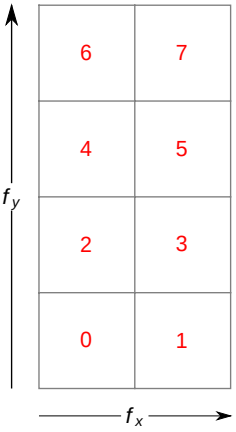
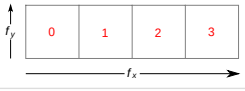
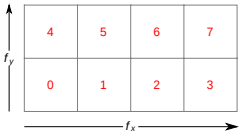
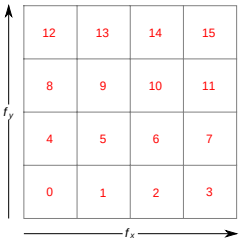
2x1	2x2	2x4
		

Table 37. Pixel Indices - 4 Wide

4x1	4x2	4x4
		

The coverage mask includes B bits packed into W words, defined as:

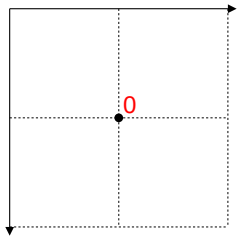
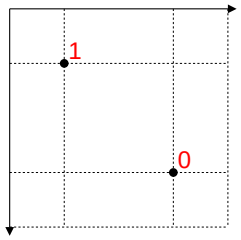
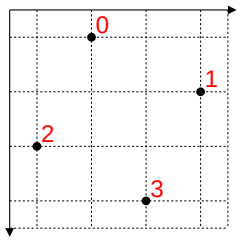
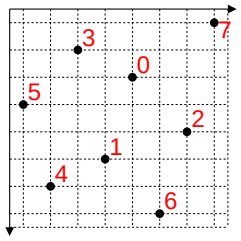
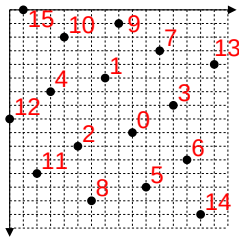
$$B = n \times \texttt{rasterizationSamples}$$

$$W = \lceil B/32 \rceil$$

Bit b in coverage mask word w is `1` if the sample with coverage index j = 32×w + b is covered, and `0` otherwise.

If the `standardSampleLocations` member of VkPhysicalDeviceLimits is VK_TRUE, then the sample counts VK_SAMPLE_COUNT_1_BIT, VK_SAMPLE_COUNT_2_BIT, VK_SAMPLE_COUNT_4_BIT, VK_SAMPLE_COUNT_8_BIT, and VK_SAMPLE_COUNT_16_BIT have sample locations as listed in the following table, with the ith entry in the table corresponding to sample index i. VK_SAMPLE_COUNT_32_BIT and VK_SAMPLE_COUNT_64_BIT do not have standard sample locations. Locations are defined relative to an origin in the upper left corner of the fragment.

Table 38. Standard Sample Locations

Sample count	Sample Locations	
VK_SAMPLE_COUNT_1_BIT	(0.5,0.5)	
VK_SAMPLE_COUNT_2_BIT	(0.75,0.75) (0.25,0.25)	
VK_SAMPLE_COUNT_4_BIT	(0.375, 0.125) (0.875, 0.375) (0.125, 0.625) (0.625, 0.875)	
VK_SAMPLE_COUNT_8_BIT	(0.5625, 0.3125) (0.4375, 0.6875) (0.8125, 0.5625) (0.3125, 0.1875) (0.1875, 0.8125) (0.0625, 0.4375) (0.6875, 0.9375) (0.9375, 0.0625)	
VK_SAMPLE_COUNT_16_BIT	(0.5625, 0.5625) (0.4375, 0.3125) (0.3125, 0.625) (0.75, 0.4375) (0.1875, 0.375) (0.625, 0.8125) (0.8125, 0.6875) (0.6875, 0.1875) (0.375, 0.875) (0.5, 0.0625) (0.25, 0.125) (0.125, 0.75) (0.0, 0.5) (0.9375, 0.25) (0.875, 0.9375) (0.0625, 0.0)	

31.5. Custom Sample Locations

Applications **can** also control the sample locations used for rasterization.

If the pNext chain of the VkPipelineMultisampleStateCreateInfo structure specified at pipeline creation time includes a VkPipelineSampleLocationsStateCreateInfoEXT structure, then that structure controls the sample locations used when rasterizing primitives with the pipeline.

The VkPipelineSampleLocationsStateCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_sample_locations
typedef struct VkPipelineSampleLocationsStateCreateInfoEXT {
    VkStructureType             sType;
    const void*                 pNext;
    VkBool32                    sampleLocationsEnable;
    VkSampleLocationsInfoEXT    sampleLocationsInfo;
} VkPipelineSampleLocationsStateCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- sampleLocationsEnable controls whether custom sample locations are used. If sampleLocationsEnable is VK_FALSE, the default sample locations are used and the values specified in sampleLocationsInfo are ignored.
- sampleLocationsInfo is the sample locations to use during rasterization if sampleLocationsEnable is VK_TRUE and the graphics pipeline is not created with VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT.

Valid Usage (Implicit)

- VUID-VkPipelineSampleLocationsStateCreateInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PIPELINE_SAMPLE_LOCATIONS_STATE_CREATE_INFO_EXT
- VUID-VkPipelineSampleLocationsStateCreateInfoEXT-sampleLocationsInfo-parameter
  sampleLocationsInfo **must** be a valid VkSampleLocationsInfoEXT structure

Structure Chaining

Extends the structure

- VkPipelineMultisampleStateCreateInfo

The VkSampleLocationsInfoEXT structure is defined as:

```
// Provided by VK_EXT_sample_locations
typedef struct VkSampleLocationsInfoEXT {
    VkStructureType                sType;
    const void*                    pNext;
    VkSampleCountFlagBits          sampleLocationsPerPixel;
    VkExtent2D                     sampleLocationGridSize;
    uint32_t                       sampleLocationsCount;
    const VkSampleLocationEXT*     pSampleLocations;
} VkSampleLocationsInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `sampleLocationsPerPixel` is a VkSampleCountFlagBits value specifying the number of sample locations per pixel.
- `sampleLocationGridSize` is the size of the sample location grid to select custom sample locations for.
- `sampleLocationsCount` is the number of sample locations in `pSampleLocations`.
- `pSampleLocations` is a pointer to an array of `sampleLocationsCount` VkSampleLocationEXT structures.

This structure **can** be used either to specify the sample locations to be used for rendering or to specify the set of sample locations an image subresource has been last rendered with for the purposes of layout transitions of depth/stencil images created with VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT.

The sample locations in `pSampleLocations` specify `sampleLocationsPerPixel` number of sample locations for each pixel in the grid of the size specified in `sampleLocationGridSize`. The sample location for sample i at the pixel grid location (x,y) is taken from `pSampleLocations`[(x + y × `sampleLocationGridSize.width`) × `sampleLocationsPerPixel` + i].

If the render pass has a fragment density map, the implementation will choose the sample locations for the fragment and the contents of `pSampleLocations` **may** be ignored.

Valid Usage

- VUID-VkSampleLocationsInfoEXT-sampleLocationsPerPixel-01526
  `sampleLocationsPerPixel` **must** be a valid VkSampleCountFlagBits value that is set in VkPhysicalDeviceSampleLocationsPropertiesEXT::`sampleLocationSampleCounts`
- VUID-VkSampleLocationsInfoEXT-sampleLocationsCount-01527
  `sampleLocationsCount` **must** equal `sampleLocationsPerPixel` × `sampleLocationGridSize.width` × `sampleLocationGridSize.height`

Valid Usage (Implicit)

- VUID-VkSampleLocationsInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SAMPLE_LOCATIONS_INFO_EXT
- VUID-VkSampleLocationsInfoEXT-pSampleLocations-parameter
  If `sampleLocationsCount` is not `0`, `pSampleLocations` **must** be a valid pointer to an array of `sampleLocationsCount` VkSampleLocationEXT structures

Structure Chaining

Extends the structures

- VkImageMemoryBarrier
- VkImageMemoryBarrier2

The `VkSampleLocationEXT` structure is defined as:

```
// Provided by VK_EXT_sample_locations
typedef struct VkSampleLocationEXT {
    float    x;
    float    y;
} VkSampleLocationEXT;
```

- `x` is the horizontal coordinate of the sample's location.
- `y` is the vertical coordinate of the sample's location.

The domain space of the sample location coordinates has an upper-left origin within the pixel in framebuffer space.

The values specified in a `VkSampleLocationEXT` structure are always clamped to the implementation-dependent sample location coordinate range [`sampleLocationCoordinateRange`[0],`sampleLocationCoordinateRange`[1]] that **can** be queried using VkPhysicalDeviceSampleLocationsPropertiesEXT.

To dynamically set the `sampleLocationsEnable` state, call:

```
// Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_sample_locations,
VK_EXT_sample_locations with VK_EXT_shader_object
void vkCmdSetSampleLocationsEnableEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    sampleLocationsEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `sampleLocationsEnable` specifies the `sampleLocationsEnable` state.

This command sets the sampleLocationsEnable state for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT set in VkPipelineDynamicStateCreateInfo::pDynamicStates. Otherwise, this state is specified by the VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetSampleLocationsEnableEXT-None-09423
  At least one of the following **must** be true:
  - The extendedDynamicState3SampleLocationsEnable feature is enabled
  - The shaderObject feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetSampleLocationsEnableEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetSampleLocationsEnableEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdSetSampleLocationsEnableEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetSampleLocationsEnableEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetSampleLocationsEnableEXT is not affected by conditional rendering

To dynamically set the sample locations used for rasterization, call:

```
// Provided by VK_EXT_sample_locations
void vkCmdSetSampleLocationsEXT(
    VkCommandBuffer                             commandBuffer,
    const VkSampleLocationsInfoEXT*             pSampleLocationsInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pSampleLocationsInfo` is the sample locations state to set.

This command sets the custom sample locations for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`, and when the VkPipelineSampleLocationsStateCreateInfoEXT ::`sampleLocationsEnable` property of the bound graphics pipeline is VK_TRUE. Otherwise, this state is specified by the VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsInfo` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetSampleLocationsEXT-variableSampleLocations-01530
  If VkPhysicalDeviceSampleLocationsPropertiesEXT::`variableSampleLocations` is VK_FALSE then the current render pass **must** have been begun by specifying a VkRenderPassSampleLocationsBeginInfoEXT structure whose `pPostSubpassSampleLocations` member contains an element with a `subpassIndex` matching the current subpass index and the `sampleLocationsInfo` member of that element **must** match the sample locations state pointed to by `pSampleLocationsInfo`

Valid Usage (Implicit)

- VUID-vkCmdSetSampleLocationsEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetSampleLocationsEXT-pSampleLocationsInfo-parameter
  `pSampleLocationsInfo` **must** be a valid pointer to a valid VkSampleLocationsInfoEXT structure
- VUID-vkCmdSetSampleLocationsEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetSampleLocationsEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support

VK_QUEUE_GRAPHICS_BIT operations

- VUID-vkCmdSetSampleLocationsEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetSampleLocationsEXT is not affected by conditional rendering

31.6. Fragment Shading Rates

The features advertised by VkPhysicalDeviceFragmentShadingRateFeaturesKHR allow an application to control the shading rate of a given fragment shader invocation.

The fragment shading rate strongly interacts with Multisampling, and the set of available rates for an implementation **may** be restricted by sample rate.

To query available shading rates, call:

```
// Provided by VK_KHR_fragment_shading_rate
VkResult vkGetPhysicalDeviceFragmentShadingRatesKHR(
    VkPhysicalDevice                            physicalDevice,
    uint32_t*                                   pFragmentShadingRateCount,
    VkPhysicalDeviceFragmentShadingRateKHR*     pFragmentShadingRates);
```

- `physicalDevice` is the handle to the physical device whose properties will be queried.
- `pFragmentShadingRateCount` is a pointer to an integer related to the number of fragment shading rates available or queried, as described below.
- `pFragmentShadingRates` is either `NULL` or a pointer to an array of

VkPhysicalDeviceFragmentShadingRateKHR structures.

If pFragmentShadingRates is NULL, then the number of fragment shading rates available is returned in pFragmentShadingRateCount. Otherwise, pFragmentShadingRateCount **must** point to a variable set by the application to the number of elements in the pFragmentShadingRates array, and on return the variable is overwritten with the number of structures actually written to pFragmentShadingRates. If pFragmentShadingRateCount is less than the number of fragment shading rates available, at most pFragmentShadingRateCount structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available fragment shading rates were returned.

The returned array of fragment shading rates **must** be ordered from largest fragmentSize.width value to smallest, and each set of fragment shading rates with the same fragmentSize.width value **must** be ordered from largest fragmentSize.height to smallest. Any two entries in the array **must** not have the same fragmentSize values.

For any entry in the array, the following rules also apply:

- The value of fragmentSize.width **must** be less than or equal to maxFragmentSize.width.
- The value of fragmentSize.width **must** be greater than or equal to 1.
- The value of fragmentSize.width **must** be a power-of-two.
- The value of fragmentSize.height **must** be less than or equal to maxFragmentSize.height.
- The value of fragmentSize.height **must** be greater than or equal to 1.
- The value of fragmentSize.height **must** be a power-of-two.
- The highest sample count in sampleCounts **must** be less than or equal to maxFragmentShadingRateRasterizationSamples.
- The product of fragmentSize.width, fragmentSize.height, and the highest sample count in sampleCounts **must** be less than or equal to maxFragmentShadingRateCoverageSamples.

Implementations **must** support at least the following shading rates:

sampleCounts	fragmentSize
VK_SAMPLE_COUNT_1_BIT | VK_SAMPLE_COUNT_4_BIT	{2,2}
VK_SAMPLE_COUNT_1_BIT | VK_SAMPLE_COUNT_4_BIT	{2,1}
~0	{1,1}

If framebufferColorSampleCounts includes VK_SAMPLE_COUNT_2_BIT, the required rates **must** also include VK_SAMPLE_COUNT_2_BIT.

NOTE Including the {1,1} fragment size is done for completeness; it has no actual effect on the support of rendering without setting the fragment size. All sample counts are supported for this rate.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceFragmentShadingRatesKHR-physicalDevice-parameter

physicalDevice **must** be a valid VkPhysicalDevice handle

- VUID-vkGetPhysicalDeviceFragmentShadingRatesKHR-pFragmentShadingRateCount-parameter
  pFragmentShadingRateCount **must** be a valid pointer to a uint32_t value
- VUID-vkGetPhysicalDeviceFragmentShadingRatesKHR-pFragmentShadingRates-parameter
  If the value referenced by pFragmentShadingRateCount is not 0, and pFragmentShadingRates is not NULL, pFragmentShadingRates **must** be a valid pointer to an array of pFragmentShadingRateCount VkPhysicalDeviceFragmentShadingRateKHR structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkPhysicalDeviceFragmentShadingRateKHR structure is defined as:

```
// Provided by VK_KHR_fragment_shading_rate
typedef struct VkPhysicalDeviceFragmentShadingRateKHR {
    VkStructureType       sType;
    void*                 pNext;
    VkSampleCountFlags    sampleCounts;
    VkExtent2D            fragmentSize;
} VkPhysicalDeviceFragmentShadingRateKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- sampleCounts is a bitmask of sample counts for which the shading rate described by fragmentSize is supported.
- fragmentSize is a VkExtent2D describing the width and height of a supported shading rate.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentShadingRateKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_KHR

- VUID-VkPhysicalDeviceFragmentShadingRateKHR-pNext-pNext
  `pNext` **must** be `NULL`

Fragment shading rates **can** be set at three points, with the three rates combined to determine the final shading rate.

31.6.1. Pipeline Fragment Shading Rate

The *pipeline fragment shading rate* **can** be set on a per-draw basis by either setting the rate in a graphics pipeline, or dynamically via vkCmdSetFragmentShadingRateKHR.

The `VkPipelineFragmentShadingRateStateCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_fragment_shading_rate
typedef struct VkPipelineFragmentShadingRateStateCreateInfoKHR {
    VkStructureType                         sType;
    const void*                             pNext;
    VkExtent2D                              fragmentSize;
    VkFragmentShadingRateCombinerOpKHR      combinerOps[2];
} VkPipelineFragmentShadingRateStateCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `fragmentSize` specifies a VkExtent2D structure containing the fragment size used to define the pipeline fragment shading rate for drawing commands using this pipeline.
- `combinerOps` specifies a VkFragmentShadingRateCombinerOpKHR value determining how the pipeline, primitive, and attachment shading rates are combined for fragments generated by drawing commands using the created pipeline.

If the `pNext` chain of VkGraphicsPipelineCreateInfo includes a `VkPipelineFragmentShadingRateStateCreateInfoKHR` structure, then that structure includes parameters controlling the pipeline fragment shading rate.

If this structure is not present, `fragmentSize` is considered to be equal to (1,1), and both elements of `combinerOps` are considered to be equal to VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR.

Valid Usage (Implicit)

- VUID-VkPipelineFragmentShadingRateStateCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_FRAGMENT_SHADING_RATE_STATE_CREATE_INFO_KHR

Structure Chaining

Extends the structure

- VkGraphicsPipelineCreateInfo

To dynamically set the pipeline fragment shading rate and combiner operation, call:

```
// Provided by VK_KHR_fragment_shading_rate
void vkCmdSetFragmentShadingRateKHR(
    VkCommandBuffer                             commandBuffer,
    const VkExtent2D*                           pFragmentSize,
    const VkFragmentShadingRateCombinerOpKHR    combinerOps[2]);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pFragmentSize` specifies the pipeline fragment shading rate for subsequent drawing commands.
- `combinerOps` specifies a VkFragmentShadingRateCombinerOpKHR determining how the pipeline, primitive, and attachment shading rates are combined for fragments generated by subsequent drawing commands.

This command sets the pipeline fragment shading rate and combiner operation for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineFragmentShadingRateStateCreateInfoKHR values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetFragmentShadingRateKHR-pipelineFragmentShadingRate-04507
  If the `pipelineFragmentShadingRate` feature is not enabled, `pFragmentSize->width` **must** be `1`
- VUID-vkCmdSetFragmentShadingRateKHR-pipelineFragmentShadingRate-04508
  If the `pipelineFragmentShadingRate` feature is not enabled, `pFragmentSize->height` **must** be `1`
- VUID-vkCmdSetFragmentShadingRateKHR-pipelineFragmentShadingRate-04509
  One of the `pipelineFragmentShadingRate`, `primitiveFragmentShadingRate`, or `attachmentFragmentShadingRate` features **must** be enabled
- VUID-vkCmdSetFragmentShadingRateKHR-primitiveFragmentShadingRate-04510
  If the `primitiveFragmentShadingRate` feature is not enabled, `combinerOps`[0] **must** be VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR
- VUID-vkCmdSetFragmentShadingRateKHR-attachmentFragmentShadingRate-04511
  If the `attachmentFragmentShadingRate` feature is not enabled, `combinerOps`[1] **must** be VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR
- VUID-vkCmdSetFragmentShadingRateKHR-fragmentSizeNonTrivialCombinerOps-04512
  If the `fragmentShadingRateNonTrivialCombinerOps` limit is not supported, elements of

`combinerOps` **must** be either VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR or VK_FRAGMENT_SHADING_RATE_COMBINER_OP_REPLACE_KHR

- VUID-vkCmdSetFragmentShadingRateKHR-pFragmentSize-04513
  `pFragmentSize->width` **must** be greater than or equal to `1`
- VUID-vkCmdSetFragmentShadingRateKHR-pFragmentSize-04514
  `pFragmentSize->height` **must** be greater than or equal to `1`
- VUID-vkCmdSetFragmentShadingRateKHR-pFragmentSize-04515
  `pFragmentSize->width` **must** be a power-of-two value
- VUID-vkCmdSetFragmentShadingRateKHR-pFragmentSize-04516
  `pFragmentSize->height` **must** be a power-of-two value
- VUID-vkCmdSetFragmentShadingRateKHR-pFragmentSize-04517
  `pFragmentSize->width` **must** be less than or equal to `4`
- VUID-vkCmdSetFragmentShadingRateKHR-pFragmentSize-04518
  `pFragmentSize->height` **must** be less than or equal to `4`

Valid Usage (Implicit)

- VUID-vkCmdSetFragmentShadingRateKHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetFragmentShadingRateKHR-pFragmentSize-parameter
  `pFragmentSize` **must** be a valid pointer to a valid VkExtent2D structure
- VUID-vkCmdSetFragmentShadingRateKHR-combinerOps-parameter
  Each element of `combinerOps` **must** be a valid VkFragmentShadingRateCombinerOpKHR value
- VUID-vkCmdSetFragmentShadingRateKHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetFragmentShadingRateKHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetFragmentShadingRateKHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetFragmentShadingRateKHR is not affected by conditional rendering

31.6.2. Primitive Fragment Shading Rate

The *primitive fragment shading rate* **can** be set via the `PrimitiveShadingRateKHR` built-in in the last active pre-rasterization shader stage. If the last pre-rasterization shader stage is using the `MeshEXT Execution Model`, the rate associated with a given primitive is sourced from the value written to the per-primitive `PrimitiveShadingRateKHR`. Otherwise the rate associated with a given primitive is sourced from the value written to `PrimitiveShadingRateKHR` by that primitive's provoking vertex.

31.6.3. Attachment Fragment Shading Rate

The *attachment shading rate* **can** be set by including VkFragmentShadingRateAttachmentInfoKHR in a subpass to define a *fragment shading rate attachment*. Each pixel in the framebuffer is assigned an attachment fragment shading rate by the corresponding texel in the fragment shading rate attachment, according to:

$$x' = \text{floor}(x / \text{region}_x)$$

$$y' = \text{floor}(y / \text{region}_y)$$

where x' and y' are the coordinates of a texel in the fragment shading rate attachment, x and y are the coordinates of the pixel in the framebuffer, and region_x and region_y are the size of the region each texel corresponds to, as defined by the `shadingRateAttachmentTexelSize` member of VkFragmentShadingRateAttachmentInfoKHR.

If multiview is enabled and the shading rate attachment has multiple layers, the shading rate attachment texel is selected using layer = `ViewIndex`. If multiview is disabled, and both the shading rate attachment and the framebuffer have multiple layers, the shading rate attachment texel is selected using layer = `Layer`. Otherwise, layer = 0.

The texel is read from the fragment shading rate attachment image using image coordinates equal to (x',y',0,layer,0,0). The fragment size is encoded into the first component of the result of that operation as follows:

$$size_w = 2^{((texel/4)\&3)}$$

$$size_h = 2^{(texel\&3)}$$

where texel is the value in the first component of the returned value, and $size_w$ and $size_h$ are the width and height of the fragment size, decoded from the texel.

If no fragment shading rate attachment is specified, this size is calculated as $size_w = size_h = 1$. Applications **must** not specify a width or height greater than 4 by this method.

The *Fragment Shading Rate* enumeration in SPIR-V adheres to the above encoding.

31.6.4. Combining the Fragment Shading Rates

The final rate (C_{xy}') used for fragment shading **must** be one of the rates returned by vkGetPhysicalDeviceFragmentShadingRatesKHR for the sample count used by rasterization.

If any of the following conditions are met, C_{xy}' is set to {1,1} by the implementation:

- If Sample Shading is enabled.
- The `fragmentShadingRateWithSampleMask` limit is not supported, and VkPipelineMultisampleStateCreateInfo::`pSampleMask` contains a zero value in any bit used by fragment operations.
- The `fragmentShadingRateWithShaderSampleMask` is not supported, and the fragment shader has `SampleMask` in the input or output interface.
- The `fragmentShadingRateWithShaderDepthStencilWrites` limit is not supported, and the fragment shader declares the `FragDepth` or `FragStencilRefEXT` built-in.
- The `fragmentShadingRateWithConservativeRasterization` limit is not supported, and VkPipelineRasterizationConservativeStateCreateInfoEXT::`conservativeRasterizationMode` is not VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT.
- The `fragmentShadingRateWithFragmentShaderInterlock` limit is not supported, and the fragment shader declares any of the fragment shader interlock execution modes.
- The `fragmentShadingRateWithCustomSampleLocations` limit is not supported, and VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` is VK_TRUE.
- The fragment shader declares any of the `TileImageColorReadAccessEXT`, `TileImageDepthReadAccessEXT`, or `TileImageStencilReadAccessEXT` capabilities.

Otherwise, each of the specified shading rates are combined and then used to derive the value of C_{xy}'. As there are three ways to specify shading rates, two combiner operations are specified - between the pipeline and primitive shading rates, and between the result of that and the attachment shading rate.

The equation used for each combiner operation is defined by VkFragmentShadingRateCombinerOpKHR:

```
// Provided by VK_KHR_fragment_shading_rate
typedef enum VkFragmentShadingRateCombinerOpKHR {
    VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR = 0,
    VK_FRAGMENT_SHADING_RATE_COMBINER_OP_REPLACE_KHR = 1,
    VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MIN_KHR = 2,
    VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MAX_KHR = 3,
    VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MUL_KHR = 4,
} VkFragmentShadingRateCombinerOpKHR;
```

- VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR specifies a combiner operation of combine(A_{xy},B_{xy}) = A_{xy}.
- VK_FRAGMENT_SHADING_RATE_COMBINER_OP_REPLACE_KHR specifies a combiner operation of combine(A_{xy},B_{xy}) = B_{xy}.
- VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MIN_KHR specifies a combiner operation of combine(A_{xy},B_{xy}) = min(A_{xy},B_{xy}).
- VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MAX_KHR specifies a combiner operation of combine(A_{xy},B_{xy}) = max(A_{xy},B_{xy}).
- VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MUL_KHR specifies a combiner operation of combine(A_{xy},B_{xy}) = A_{xy}*B_{xy}.

where combine(A_{xy},B_{xy}) is the combine operation, and A_{xy} and B_{xy} are the inputs to the operation.

If fragmentShadingRateStrictMultiplyCombiner is VK_FALSE, using VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MUL_KHR with values of 1 for both A and B in the same dimension results in the value 2 being produced for that dimension. See the definition of fragmentShadingRateStrictMultiplyCombiner for more information.

These operations are performed in a component-wise fashion.

This is used to generate a combined fragment area using the equation:

$$C_{xy} = \text{combine}(A_{xy},B_{xy})$$

where C_{xy} is the combined fragment area result, and A_{xy} and B_{xy} are the fragment areas of the fragment shading rates being combined.

Two combine operations are performed, first with A_{xy} equal to the pipeline fragment shading rate and B_{xy} equal to the primitive fragment shading rate, with the combine() operation selected by combinerOps[0]. A second combination is then performed, with A_{xy} equal to the result of the first combination and B_{xy} equal to the attachment fragment shading rate, with the combine() operation selected by combinerOps[1]. The result of the second combination is used as the final fragment shading rate, reported via the ShadingRateKHR built-in.

Implementations **should** clamp the inputs to the combiner operations A_{xy} and B_{xy}, and **must** do so if VkPhysicalDeviceMaintenance6Properties::fragmentShadingRateClampCombinerInputs is VK_TRUE. All implementations **must** clamp the result of the second combiner operation.

A fragment shading rate R_{xy} representing any of A_{xy}, B_{xy} or C_{xy} is clamped as follows. If R_{xy} is one of the rates returned by vkGetPhysicalDeviceFragmentShadingRatesKHR for the sample count used by rasterization, the clamped shading rate R_{xy}' is R_{xy}. Otherwise, the clamped shading rate is selected from the rates returned by vkGetPhysicalDeviceFragmentShadingRatesKHR for the sample count used by rasterization. From this list of supported rates, the following steps are applied in order, to select a single value:

1. Keep only rates where $R_x' \leq R_x$ and $R_y' \leq R_y$.
   - Implementations **may** also keep rates where $R_x' \leq R_y$ and $R_y' \leq R_x$.
2. Keep only rates with the highest area ($R_x' \times R_y'$).
3. Keep only rates with the lowest aspect ratio ($R_x' + R_y'$).
4. In cases where a wide (e.g. 4x1) and tall (e.g. 1x4) rate remain, the implementation **may** choose either rate. However, it **must** choose this rate consistently for the same shading rates, and combiner operations for the lifetime of the VkDevice.

31.7. Sample Shading

Sample shading **can** be used to specify a minimum number of unique samples to process for each fragment. If sample shading is enabled, an implementation **must** invoke the fragment shader at least max(⌈ VkPipelineMultisampleStateCreateInfo::`minSampleShading` × VkPipelineMultisampleStateCreateInfo::`rasterizationSamples` ⌉, 1) times per fragment. If VkPipelineMultisampleStateCreateInfo::`sampleShadingEnable` is VK_TRUE, sample shading is enabled.

If a fragment shader entry point statically uses an input variable decorated with a `BuiltIn` of `SampleId` or `SamplePosition`, sample shading is enabled and a value of `1.0` is used instead of `minSampleShading`. If a fragment shader entry point statically uses an input variable decorated with `Sample`, sample shading **may** be enabled and a value of `1.0` will be used instead of `minSampleShading` if it is.

NOTE

If a shader decorates an input variable with `Sample` and that value meaningfully impacts the output of a shader, sample shading will be enabled to ensure that the input is in fact interpolated per-sample. This is inherent to the specification and not spelled out here - if an application simply declares such a variable it is implementation-defined whether sample shading is enabled or not. It is possible to see the effects of this by using atomics in the shader or using a pipeline statistics query to query the number of fragment invocations, even if the shader itself does not use any per-sample variables.

If there are fewer fragment invocations than covered samples, implementations **may** include those samples in fragment shader invocations in any manner as long as covered samples are all shaded at least once, and each invocation that is not a helper invocation covers at least one sample.

31.8. Barycentric Interpolation

When the `fragmentShaderBarycentric` feature is enabled, the `PerVertexKHR` interpolation decoration

can be used with fragment shader inputs to indicate that the decorated inputs do not have associated data in the fragment. Such inputs **can** only be accessed in a fragment shader using an array index whose value (0, 1, or 2) identifies one of the vertices of the primitive that produced the fragment. Reads of per-vertex values for missing vertices, such as the third vertex of a line primitive, will return values from the valid vertex with the highest index. This means that the per-vertex values of indices 1 and 2 for point primitives will be equal to those of index 0, and the per-vertex values of index 2 for line primitives will be equal to those of index 1.

When tessellation, geometry shading, and mesh shading are not active, fragment shader inputs decorated with `PerVertexKHR` will take values from one of the vertices of the primitive that produced the fragment, identified by the extra index provided in SPIR-V code accessing the input. If the *n* vertices passed to a draw call are numbered 0 through *n*-1, and the point, line, and triangle primitives produced by the draw call are numbered with consecutive integers beginning with zero, the following table indicates the original vertex numbers used when the provoking vertex mode is VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT for index values of 0, 1, and 2. If an input decorated with `PerVertexKHR` is accessed with any other vertex index value, or is accessed while rasterizing a polygon when the VkPipelineRasterizationStateCreateInfo::`polygonMode` property of the currently active pipeline is not VK_POLYGON_MODE_FILL, an undefined value is returned.

Primitive Topology	Vertex 0	Vertex 1	Vertex 2
VK_PRIMITIVE_TOPOLOGY_POINT_LIST	i	i	i
VK_PRIMITIVE_TOPOLOGY_LINE_LIST	2i	2i+1	2i+1
VK_PRIMITIVE_TOPOLOGY_LINE_STRIP	i	i+1	i+1
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST	3i	3i+1	3i+2
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP (even)	i	i+1	i+2
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP (odd)	i	i+2	i+1
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN	i+1	i+2	0
VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY	4i+1	4i+2	4i+2
VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY	i+1	i+2	i+2

Primitive Topology	Vertex 0	Vertex 1	Vertex 2
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY	6i	6i+2	6i+4
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY (even)	2i	2i+2	2i+4
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY (odd)	2i	2i+4	2i+2

When the provoking vertex mode is VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT, the original vertex numbers used are the same as above except as indicated in the table below.

Primitive Topology	Vertex 0	Vertex 1	Vertex 2
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP (odd, and triStripVertexOrderIndependentOfProvokingVertex of VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR is VK_FALSE)	i+1	i	i+2
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN	0	i+1	i+2
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY (odd)	2i+2	2i	2i+4

When geometry or mesh shading is active, primitives processed by fragment shaders are assembled from the vertices emitted by the geometry or mesh shader. In this case, the vertices used for fragment shader inputs decorated with PerVertexKHR are derived by treating the primitives produced by the shader as though they were specified by a draw call and consulting the table above.

When using tessellation without geometry shading, the tessellator produces primitives in an implementation-dependent manner. While there is no defined vertex ordering for inputs decorated with PerVertexKHR, the vertex ordering used in this case will be consistent with the ordering used to derive the values of inputs decorated with BaryCoordKHR or BaryCoordNoPerspKHR.

Fragment shader inputs decorated with BaryCoordKHR or BaryCoordNoPerspKHR hold three-component

vectors with barycentric weights that indicate the location of the fragment relative to the screen-space locations of vertices of its primitive. For point primitives, such variables are always assigned the value (1,0,0). For line primitives, the built-ins are obtained by interpolating an attribute whose values for the vertices numbered 0 and 1 are (1,0,0) and (0,1,0), respectively. For polygon primitives, the built-ins are obtained by interpolating an attribute whose values for the vertices numbered 0, 1, and 2 are (1,0,0), (0,1,0), and (0,0,1), respectively. For `BaryCoordKHR`, the values are obtained using perspective interpolation. For `BaryCoordNoPerspKHR`, the values are obtained using linear interpolation. The values of `BaryCoordKHR` and `BaryCoordNoPerspKHR` are undefined while rasterizing a polygon when the VkPipelineRasterizationStateCreateInfo::`polygonMode` property of the currently active pipeline is not VK_POLYGON_MODE_FILL.

31.9. Points

A point is drawn by generating a set of fragments in the shape of a square centered around the vertex of the point. Each vertex has an associated point size controlling the width/height of that square. The point size is taken from the (potentially clipped) shader built-in `PointSize` written by:

- the geometry shader, if active;
- the tessellation evaluation shader, if active and no geometry shader is active;
- the vertex shader, otherwise

and clamped to the implementation-dependent point size range [`pointSizeRange`[0], `pointSizeRange`[1]]. The value written to `PointSize` **must** be greater than zero. If the `maintenance5` feature is enabled, and a value is not written to `PointSize`, the point size takes a default value of 1.0.

The range of supported sizes and the size of evenly-spaced gradations within that range are implementation-dependent. A size of 1.0 will always be supported - support for additional sizes is advertised by the `largePoints` feature and the `pointSizeRange` and `pointSizeGranularity` limits indicate the available size range and gradations. If, for instance, the size range is from 0.1 to 2.0 and the gradation size is 0.1, then the sizes 0.1, 0.2, ..., 1.9, 2.0 are supported. Additional point sizes **may** also be supported. There is no requirement that these sizes be equally spaced. If an unsupported size is requested, the nearest supported size is used instead.

Further, if the render pass has a fragment density map attachment, point size **may** be rounded by the implementation to a multiple of the fragment's width or height.

31.9.1. Basic Point Rasterization

Point rasterization produces a fragment for each fragment area group of framebuffer pixels with one or more sample points that intersect a region centered at the point's (x_f,y_f). This region is a square with side equal to the current point size. Coverage bits that correspond to sample points that intersect the region are 1, other coverage bits are 0. All fragments produced in rasterizing a point are assigned the same associated data, which are those of the vertex corresponding to the point. However, the fragment shader built-in `PointCoord` contains point sprite texture coordinates. The s and t point sprite texture coordinates vary from zero to one across the point horizontally left-to-right and vertically top-to-bottom, respectively. The following formulas are used to evaluate s and t:

$$s = \frac{1}{2} + \frac{(x_p - x_f)}{\text{size}}$$

$$t = \frac{1}{2} + \frac{(y_p - y_f)}{\text{size}}$$

where size is the point's size; (x_p,y_p) is the location at which the point sprite coordinates are evaluated - this **may** be the framebuffer coordinates of the fragment center, or the location of a sample; and (x_f,y_f) is the exact, unrounded framebuffer coordinate of the vertex for the point.

31.10. Line Segments

Line segment rasterization options are controlled by the VkPipelineRasterizationLineStateCreateInfo structure.

The VkPipelineRasterizationLineStateCreateInfo structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPipelineRasterizationLineStateCreateInfo {
    VkStructureType            sType;
    const void*                pNext;
    VkLineRasterizationMode    lineRasterizationMode;
    VkBool32                   stippledLineEnable;
    uint32_t                   lineStippleFactor;
    uint16_t                   lineStipplePattern;
} VkPipelineRasterizationLineStateCreateInfo;
```

```
// Provided by VK_KHR_line_rasterization
// Equivalent to VkPipelineRasterizationLineStateCreateInfo
typedef VkPipelineRasterizationLineStateCreateInfo
VkPipelineRasterizationLineStateCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- lineRasterizationMode is a VkLineRasterizationMode value selecting the style of line rasterization.
- stippledLineEnable enables stippled line rasterization.
- lineStippleFactor is the repeat factor used in stippled line rasterization.
- lineStipplePattern is the bit pattern used in stippled line rasterization.

If stippledLineEnable is VK_FALSE, the values of lineStippleFactor and lineStipplePattern are ignored.

Valid Usage

- VUID-VkPipelineRasterizationLineStateCreateInfo-lineRasterizationMode-02768
  If `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the `rectangularLines` feature **must** be enabled
- VUID-VkPipelineRasterizationLineStateCreateInfo-lineRasterizationMode-02769
  If `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the `bresenhamLines` feature **must** be enabled
- VUID-VkPipelineRasterizationLineStateCreateInfo-lineRasterizationMode-02770
  If `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the `smoothLines` feature **must** be enabled
- VUID-VkPipelineRasterizationLineStateCreateInfo-stippledLineEnable-02771
  If `stippledLineEnable` is VK_TRUE and `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the `stippledRectangularLines` feature **must** be enabled
- VUID-VkPipelineRasterizationLineStateCreateInfo-stippledLineEnable-02772
  If `stippledLineEnable` is VK_TRUE and `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the `stippledBresenhamLines` feature **must** be enabled
- VUID-VkPipelineRasterizationLineStateCreateInfo-stippledLineEnable-02773
  If `stippledLineEnable` is VK_TRUE and `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the `stippledSmoothLines` feature **must** be enabled
- VUID-VkPipelineRasterizationLineStateCreateInfo-stippledLineEnable-02774
  If `stippledLineEnable` is VK_TRUE and `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the `stippledRectangularLines` feature **must** be enabled and VkPhysicalDeviceLimits::`strictLines` **must** be VK_TRUE

Valid Usage (Implicit)

- VUID-VkPipelineRasterizationLineStateCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_LINE_STATE_CREATE_INFO
- VUID-VkPipelineRasterizationLineStateCreateInfo-lineRasterizationMode-parameter
  `lineRasterizationMode` **must** be a valid VkLineRasterizationMode value

Structure Chaining

Extends the structure

- VkPipelineRasterizationStateCreateInfo

Possible values of VkPipelineRasterizationLineStateCreateInfo::lineRasterizationMode, specifying the line rasterization mode, are:

```
// Provided by VK_VERSION_1_4
typedef enum VkLineRasterizationMode {
    VK_LINE_RASTERIZATION_MODE_DEFAULT = 0,
    VK_LINE_RASTERIZATION_MODE_RECTANGULAR = 1,
    VK_LINE_RASTERIZATION_MODE_BRESENHAM = 2,
    VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH = 3,
  // Provided by VK_KHR_line_rasterization
    VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR = VK_LINE_RASTERIZATION_MODE_DEFAULT,
  // Provided by VK_KHR_line_rasterization
    VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR =
VK_LINE_RASTERIZATION_MODE_RECTANGULAR,
  // Provided by VK_KHR_line_rasterization
    VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR = VK_LINE_RASTERIZATION_MODE_BRESENHAM,
  // Provided by VK_KHR_line_rasterization
    VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR =
VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH,
} VkLineRasterizationMode;
```

```
// Provided by VK_KHR_line_rasterization
// Equivalent to VkLineRasterizationMode
typedef VkLineRasterizationMode VkLineRasterizationModeKHR;
```

- VK_LINE_RASTERIZATION_MODE_DEFAULT is equivalent to VK_LINE_RASTERIZATION_MODE_RECTANGULAR if VkPhysicalDeviceLimits::strictLines is VK_TRUE, otherwise lines are drawn as non-strictLines parallelograms. Both of these modes are defined in Basic Line Segment Rasterization.
- VK_LINE_RASTERIZATION_MODE_RECTANGULAR specifies lines drawn as if they were rectangles extruded from the line.
- VK_LINE_RASTERIZATION_MODE_BRESENHAM specifies lines drawn by determining which pixel diamonds the line intersects and exits, as defined in Bresenham Line Segment Rasterization.
- VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH specifies lines drawn as if they were rectangles extruded from the line, with alpha falloff, as defined in Smooth Lines.

To dynamically set the line width, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetLineWidth(
    VkCommandBuffer                             commandBuffer,
    float                                       lineWidth);
```

- commandBuffer is the command buffer into which the command will be recorded.

- `lineWidth` is the width of rasterized line segments.

This command sets the line width for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_LINE_WIDTH set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineRasterizationStateCreateInfo::`lineWidth` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetLineWidth-lineWidth-00788
  If the `wideLines` feature is not enabled, `lineWidth` **must** be `1.0`

Valid Usage (Implicit)

- VUID-vkCmdSetLineWidth-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetLineWidth-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetLineWidth-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetLineWidth-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetLineWidth is not affected by conditional rendering

Not all line widths need be supported for line segment rasterization, but width 1.0 antialiased segments **must** be provided. The range and gradations are obtained from the `lineWidthRange` and `lineWidthGranularity` members of VkPhysicalDeviceLimits. If, for instance, the size range is from 0.1 to 2.0 and the gradation size is 0.1, then the sizes 0.1, 0.2, ..., 1.9, 2.0 are supported. Additional line widths **may** also be supported. There is no requirement that these widths be equally spaced. If an unsupported width is requested, the nearest supported width is used instead.

Further, if the render pass has a fragment density map attachment, line width **may** be rounded by the implementation to a multiple of the fragment's width or height.

31.10.1. Basic Line Segment Rasterization

If the `lineRasterizationMode` member of VkPipelineRasterizationLineStateCreateInfo is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, rasterized line segments produce fragments which intersect a rectangle centered on the line segment. Two of the edges are parallel to the specified line segment; each is at a distance of one-half the current width from that segment in directions perpendicular to the direction of the line. The other two edges pass through the line endpoints and are perpendicular to the direction of the specified line segment. Coverage bits that correspond to sample points that intersect the rectangle are 1, other coverage bits are 0.

Next we specify how the data associated with each rasterized fragment are obtained. Let $\mathbf{p}_r = (x_d, y_d)$ be the framebuffer coordinates at which associated data are evaluated. This **may** be the center of a fragment or the location of a sample within the fragment. When `rasterizationSamples` is VK_SAMPLE_COUNT_1_BIT, the fragment center **must** be used. Let $\mathbf{p}_a = (x_a, y_a)$ and $\mathbf{p}_b = (x_b, y_b)$ be initial and final endpoints of the line segment, respectively. Set

$$t = \frac{(\mathbf{p}_r - \mathbf{p}_a) \cdot (\mathbf{p}_b - \mathbf{p}_a)}{\| \mathbf{p}_b - \mathbf{p}_a \|^2}$$

(Note that t = 0 at $\mathbf{p}_a$ and t = 1 at $\mathbf{p}_b$. Also note that this calculation projects the vector from $\mathbf{p}_a$ to $\mathbf{p}_r$ onto the line, and thus computes the normalized distance of the fragment along the line.)

If `strictLines` is VK_TRUE, line segments are rasterized using perspective or linear interpolation.

Perspective interpolation for a line segment interpolates two values in a manner that is correct when taking the perspective of the viewport into consideration, by way of the line segment's clip coordinates. An interpolated value f can be determined by

$$f = \frac{(1-t) f_a / w_a + t f_b / w_b}{(1-t) / w_a + t / w_b}$$

where f_a and f_b are the data associated with the starting and ending endpoints of the segment, respectively; w_a and w_b are the clip w coordinates of the starting and ending endpoints of the segment, respectively.

Linear interpolation for a line segment directly interpolates two values, and an interpolated value f can be determined by

$$f = (1 - t) f_a + t f_b$$

where f_a and f_b are the data associated with the starting and ending endpoints of the segment, respectively.

The clip coordinate w for a sample is determined using perspective interpolation. The depth value z for a sample is determined using linear interpolation. Interpolation of fragment shader input values is determined by Interpolation decorations.

The above description documents the preferred method of line rasterization, and **must** be used when `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_RECTANGULAR.

By default, when `strictLines` is VK_FALSE, and when the `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_DEFAULT, the edges of the lines are generated as a parallelogram surrounding the original line. The major axis is chosen by noting the axis in which there is the greatest distance between the line start and end points. If the difference is equal in both directions then the X axis is chosen as the major axis. Edges 2 and 3 are aligned to the minor axis and are centered on the endpoints of the line as in Non strict lines, and each is `lineWidth` long. Edges 0 and 1 are parallel to the line and connect the endpoints of edges 2 and 3. Coverage bits that correspond to sample points that intersect the parallelogram are 1, other coverage bits are 0.

Samples that fall exactly on the edge of the parallelogram follow the polygon rasterization rules.

Interpolation occurs as if the parallelogram was decomposed into two triangles where each pair of vertices at each end of the line has identical attributes.

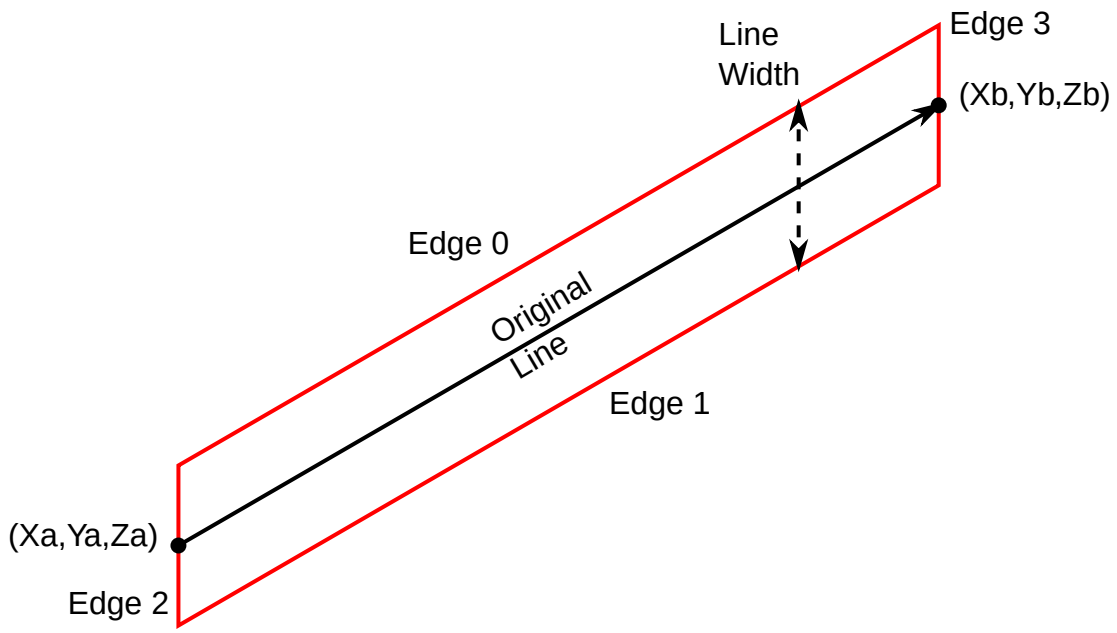


Figure 15. Non strict lines

Only when `strictLines` is VK_FALSE implementations **may** deviate from the non-strict line algorithm described above in the following ways:

- Implementations **may** instead interpolate each fragment according to the formula in Basic Line Segment Rasterization using the original line segment endpoints.
- Rasterization of non-antialiased non-strict line segments **may** be performed using the rules defined in Bresenham Line Segment Rasterization.

If `VkPhysicalDeviceMaintenance5Properties`::`nonStrictSinglePixelWideLinesUseParallelogram` is VK_TRUE, and `strictLines` is VK_FALSE, non-strict lines of width 1.0 are rasterized as parallelograms, otherwise they are rasterized using Bresenham's algorithm.

If `VkPhysicalDeviceMaintenance5Properties`::`nonStrictWideLinesUseParallelogram` is VK_TRUE, and `strictLines` is VK_FALSE, non-strict lines of width greater than 1.0 are rasterized as parallelograms, otherwise they are rasterized using Bresenham's algorithm.

31.10.2. Bresenham Line Segment Rasterization

If `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the following rules replace the line rasterization rules defined in Basic Line Segment Rasterization.

Non-strict lines **may** also follow these rasterization rules for non-antialiased lines.

Line segment rasterization begins by characterizing the segment as either *x-major* or *y-major*. x-major line segments have slope in the closed interval [-1,1]; all other line segments are y-major (slope is determined by the segment's endpoints). We specify rasterization only for x-major segments except in cases where the modifications for y-major segments are not self-evident.

Ideally, Vulkan uses a *diamond-exit* rule to determine those fragments that are produced by rasterizing a line segment. For each fragment f with center at framebuffer coordinates x_f and y_f, define a diamond-shaped region that is the intersection of four half planes:

$$R_f = \{(x, y) \mid |x - x_f| + |y - y_f| < \frac{1}{2}\}$$

Essentially, a line segment starting at p_a and ending at p_b produces those fragments f for which the segment intersects R_f, except if p_b is contained in R_f.

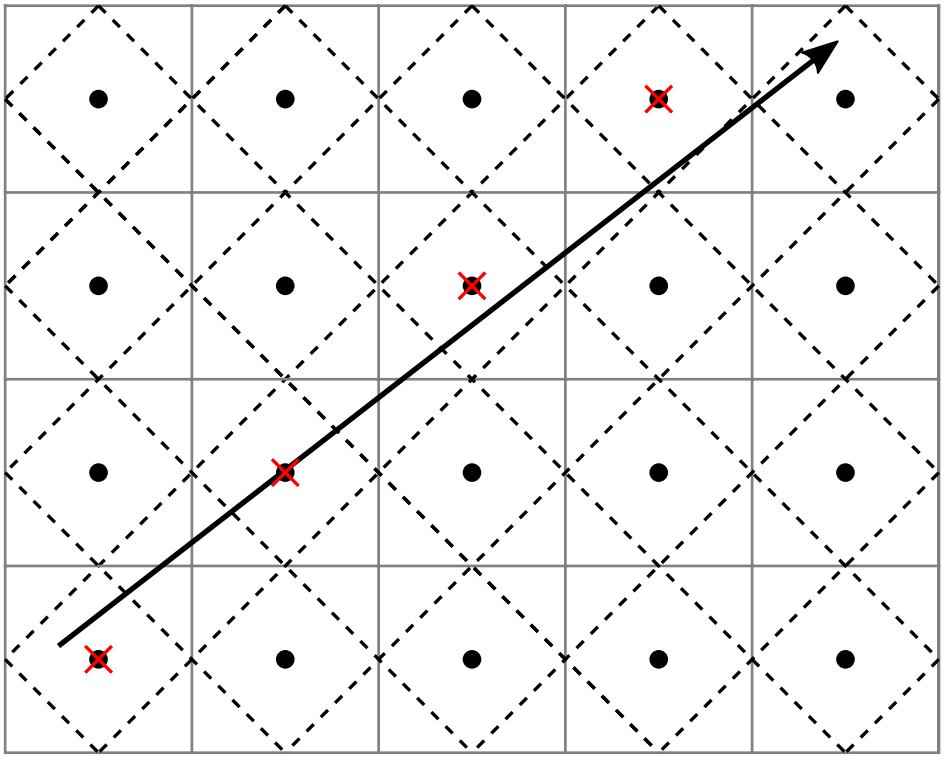


Figure 16. Visualization of Bresenham's algorithm

To avoid difficulties when an endpoint lies on a boundary of R_f we (in principle) perturb the supplied endpoints by a tiny amount. Let p_a and p_b have framebuffer coordinates (x_a, y_a) and (x_b, y_b), respectively. Obtain the perturbed endpoints p_a' given by $(x_a, y_a) - (\varepsilon, \varepsilon^2)$ and p_b' given by $(x_b, y_b) - (\varepsilon, \varepsilon^2)$. Rasterizing the line segment starting at p_a and ending at p_b produces those fragments f for which the segment starting at p_a' and ending on p_b' intersects R_f, except if p_b' is contained in R_f. ε is chosen to be so small that rasterizing the line segment produces the same fragments when δ is substituted for ε for any $0 < \delta \leq \varepsilon$.

When p_a and p_b lie on fragment centers, this characterization of fragments reduces to Bresenham's algorithm with one modification: lines produced in this description are "half-open", meaning that the final fragment (corresponding to p_b) is not drawn. This means that when rasterizing a series of connected line segments, shared endpoints will be produced only once rather than twice (as would occur with Bresenham's algorithm).

Implementations **may** use other line segment rasterization algorithms, subject to the following rules:

- The coordinates of a fragment produced by the algorithm **must** not deviate by more than one unit in either x or y framebuffer coordinates from a corresponding fragment produced by the diamond-exit rule.
- The total number of fragments produced by the algorithm **must** not differ from that produced by the diamond-exit rule by more than one.
- For an x-major line, two fragments that lie in the same framebuffer-coordinate column **must** not be produced (for a y-major line, two fragments that lie in the same framebuffer-coordinate row **must** not be produced).

- If two line segments share a common endpoint, and both segments are either x-major (both left-to-right or both right-to-left) or y-major (both bottom-to-top or both top-to-bottom), then rasterizing both segments **must** not produce duplicate fragments. Fragments also **must** not be omitted so as to interrupt continuity of the connected segments.

The actual width w of Bresenham lines is determined by rounding the line width to the nearest integer, clamping it to the implementation-dependent `lineWidthRange` (with both values rounded to the nearest integer), then clamping it to be no less than 1.

Bresenham line segments of width other than one are rasterized by offsetting them in the minor direction (for an x-major line, the minor direction is y, and for a y-major line, the minor direction is x) and producing a row or column of fragments in the minor direction. If the line segment has endpoints given by (x_0, y_0) and (x_1, y_1) in framebuffer coordinates, the segment with endpoints $(x_0, y_0 - \frac{w-1}{2})$ and $(x_1, y_1 - \frac{w-1}{2})$ is rasterized, but instead of a single fragment, a column of fragments of height w (a row of fragments of length w for a y-major segment) is produced at each x (y for y-major) location. The lowest fragment of this column is the fragment that would be produced by rasterizing the segment of width 1 with the modified coordinates.

The preferred method of attribute interpolation for a wide line is to generate the same attribute values for all fragments in the row or column described above, as if the adjusted line was used for interpolation and those values replicated to the other fragments, except for `FragCoord` which is interpolated as usual. Implementations **may** instead interpolate each fragment according to the formula in Basic Line Segment Rasterization, using the original line segment endpoints.

When Bresenham lines are being rasterized, sample locations **may** all be treated as being at the pixel center (this **may** affect attribute and depth interpolation).

NOTE

The sample locations described above are **not** used for determining coverage, they are only used for things like attribute interpolation. The rasterization rules that determine coverage are defined in terms of whether the line intersects **pixels**, as opposed to the point sampling rules used for other primitive types. So these rules are independent of the sample locations. One consequence of this is that Bresenham lines cover the same pixels regardless of the number of rasterization samples, and cover all samples in those pixels (unless masked out or killed).

31.10.3. Line Stipple

If the `stippledLineEnable` member of VkPipelineRasterizationLineStateCreateInfo is VK_TRUE, then lines are rasterized with a *line stipple* determined by `lineStippleFactor` and `lineStipplePattern`. `lineStipplePattern` is an unsigned 16-bit integer that determines which fragments are to be drawn or discarded when the line is rasterized. `lineStippleFactor` is a count that is used to modify the effective line stipple by causing each bit in `lineStipplePattern` to be used `lineStippleFactor` times.

Line stippling discards certain fragments that are produced by rasterization. The masking is achieved using three parameters: the 16-bit line stipple pattern *p*, the line stipple factor *r*, and an integer stipple counter *s*. Let

$$b = \lfloor \frac{s}{r} \rfloor \bmod 16$$

Then a fragment is produced if the *b*'th bit of *p* is 1, and discarded otherwise. The bits of *p* are numbered with 0 being the least significant and 15 being the most significant.

The initial value of *s* is zero. For VK_LINE_RASTERIZATION_MODE_BRESENHAM lines, *s* is incremented after production of each fragment of a line segment (fragments are produced in order, beginning at the starting point and working towards the ending point). For VK_LINE_RASTERIZATION_MODE_RECTANGULAR and VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH lines, the rectangular region is subdivided into adjacent unit-length rectangles, and s is incremented once for each rectangle. Rectangles with a value of *s* such that the *b*'th bit of *p* is zero are discarded. If the last rectangle in a line segment is shorter than unit-length, then the remainder **may** carry over to the next line segment in the line strip using the same value of *s* (this is the preferred behavior, for the stipple pattern to appear more consistent through the strip).

s is reset to 0 at the start of each strip (for line strips), and before every line segment in a group of independent segments.

If the line segment has been clipped, then the value of *s* at the beginning of the line segment is implementation-dependent.

To dynamically set the line stipple state, call:

```
// Provided by VK_VERSION_1_4
void vkCmdSetLineStipple(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    lineStippleFactor,
    uint16_t                                    lineStipplePattern);
```

```
// Provided by VK_KHR_line_rasterization
// Equivalent to vkCmdSetLineStipple
void vkCmdSetLineStippleKHR(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    lineStippleFactor,
    uint16_t                                    lineStipplePattern);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `lineStippleFactor` is the repeat factor used in stippled line rasterization.
- `lineStipplePattern` is the bit pattern used in stippled line rasterization.

This command sets the line stipple state for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_LINE_STIPPLE set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineRasterizationLineStateCreateInfo::`lineStippleFactor` and VkPipelineRasterizationLineStateCreateInfo::`lineStipplePattern` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetLineStipple-lineStippleFactor-02776
  `lineStippleFactor` **must** be in the range [1,256]

Valid Usage (Implicit)

- VUID-vkCmdSetLineStipple-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetLineStipple-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetLineStipple-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetLineStipple-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetLineStipple is not affected by conditional rendering

31.10.4. Smooth Lines

If the `lineRasterizationMode` member of VkPipelineRasterizationLineStateCreateInfo is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then lines are considered to be rectangles using the same geometry as for VK_LINE_RASTERIZATION_MODE_RECTANGULAR lines. The rules for determining which pixels are covered are implementation-dependent, and **may**

include nearby pixels where no sample locations are covered or where the rectangle does not intersect the pixel at all. For each pixel that is considered covered, the fragment computes a coverage value that approximates the area of the intersection of the rectangle with the pixel square, and this coverage value is multiplied into the color location 0's alpha value after fragment shading, as described in Multisample Coverage.

NOTE	The details of the rasterization rules and area calculation are left intentionally vague, to allow implementations to generate coverage and values that are aesthetically pleasing.

31.11. Polygons

A polygon results from the decomposition of a triangle strip, triangle fan or a series of independent triangles. Like points and line segments, polygon rasterization is controlled by several variables in the VkPipelineRasterizationStateCreateInfo structure.

31.11.1. Basic Polygon Rasterization

The first step of polygon rasterization is to determine whether the triangle is *back-facing* or *front-facing*. This determination is made based on the sign of the (clipped or unclipped) polygon's area computed in framebuffer coordinates. One way to compute this area is:

$$a = -\frac{1}{2}\sum_{i=0}^{n-1} x_f^i y_f^{i \oplus 1} - x_f^{i \oplus 1} y_f^i$$

where x_f^i and y_f^i are the x and y framebuffer coordinates of the ith vertex of the n-vertex polygon (vertices are numbered starting at zero for the purposes of this computation) and i □ 1 is (i + 1) mod n.

The interpretation of the sign of a is determined by the VkPipelineRasterizationStateCreateInfo ::`frontFace` property of the currently active pipeline. Possible values are:

```
// Provided by VK_VERSION_1_0
typedef enum VkFrontFace {
    VK_FRONT_FACE_COUNTER_CLOCKWISE = 0,
    VK_FRONT_FACE_CLOCKWISE = 1,
} VkFrontFace;
```

- VK_FRONT_FACE_COUNTER_CLOCKWISE specifies that a triangle with positive area is considered front-facing.
- VK_FRONT_FACE_CLOCKWISE specifies that a triangle with negative area is considered front-facing.

Any triangle which is not front-facing is back-facing, including zero-area triangles.

To dynamically set the front face orientation, call:

```
// Provided by VK_VERSION_1_3
void vkCmdSetFrontFace(
    VkCommandBuffer                             commandBuffer,
    VkFrontFace                                 frontFace);
```

```
// Provided by VK_EXT_extended_dynamic_state, VK_EXT_shader_object
// Equivalent to vkCmdSetFrontFace
void vkCmdSetFrontFaceEXT(
    VkCommandBuffer                             commandBuffer,
    VkFrontFace                                 frontFace);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `frontFace` is a VkFrontFace value specifying the front-facing triangle orientation to be used for culling.

This command sets the front face orientation for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_FRONT_FACE set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineRasterizationStateCreateInfo::`frontFace` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetFrontFace-None-08971
  At least one of the following **must** be true:
  - the `extendedDynamicState` feature is enabled
  - the `shaderObject` feature is enabled
  - the value of VkApplicationInfo::`apiVersion` used to create the VkInstance parent of `commandBuffer` is greater than or equal to Version 1.3

Valid Usage (Implicit)

- VUID-vkCmdSetFrontFace-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetFrontFace-frontFace-parameter
  `frontFace` **must** be a valid VkFrontFace value
- VUID-vkCmdSetFrontFace-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetFrontFace-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetFrontFace-videocoding

This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetFrontFace is not affected by conditional rendering

Once the orientation of triangles is determined, they are culled according to the VkPipelineRasterizationStateCreateInfo::`cullMode` property of the currently active pipeline. Possible values are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCullModeFlagBits {
    VK_CULL_MODE_NONE = 0,
    VK_CULL_MODE_FRONT_BIT = 0x00000001,
    VK_CULL_MODE_BACK_BIT = 0x00000002,
    VK_CULL_MODE_FRONT_AND_BACK = 0x00000003,
} VkCullModeFlagBits;
```

- VK_CULL_MODE_NONE specifies that no triangles are discarded.
- VK_CULL_MODE_FRONT_BIT specifies that front-facing triangles are discarded.
- VK_CULL_MODE_BACK_BIT specifies that back-facing triangles are discarded.
- VK_CULL_MODE_FRONT_AND_BACK specifies that all triangles are discarded.

Following culling, fragments are produced for any triangles which have not been discarded.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkCullModeFlags;
```

VkCullModeFlags is a bitmask type for setting a mask of zero or more VkCullModeFlagBits.

To dynamically set the cull mode, call:

```
// Provided by VK_VERSION_1_3
void vkCmdSetCullMode(
    VkCommandBuffer                             commandBuffer,
    VkCullModeFlags                             cullMode);
```

```
// Provided by VK_EXT_extended_dynamic_state, VK_EXT_shader_object
// Equivalent to vkCmdSetCullMode
void vkCmdSetCullModeEXT(
    VkCommandBuffer                             commandBuffer,
    VkCullModeFlags                             cullMode);
```

- commandBuffer is the command buffer into which the command will be recorded.
- cullMode specifies the cull mode property to use for drawing.

This command sets the cull mode for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_CULL_MODE set in VkPipelineDynamicStateCreateInfo::pDynamicStates. Otherwise, this state is specified by the VkPipelineRasterizationStateCreateInfo::cullMode value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetCullMode-None-08971
  At least one of the following **must** be true:
  - the extendedDynamicState feature is enabled
  - the shaderObject feature is enabled
  - the value of VkApplicationInfo::apiVersion used to create the VkInstance parent of commandBuffer is greater than or equal to Version 1.3

Valid Usage (Implicit)

- VUID-vkCmdSetCullMode-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetCullMode-cullMode-parameter
  If cullMode is not 0, cullMode **must** be a valid combination of VkCullModeFlagBits values
- VUID-vkCmdSetCullMode-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdSetCullMode-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support

VK_QUEUE_GRAPHICS_BIT operations

- VUID-vkCmdSetCullMode-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetCullMode is not affected by conditional rendering

The rule for determining which fragments are produced by polygon rasterization is called *point sampling*. The two-dimensional projection obtained by taking the x and y framebuffer coordinates of the polygon's vertices is formed. Fragments are produced for any fragment area groups of pixels for which any sample points lie inside of this polygon. Coverage bits that correspond to sample points that satisfy the point sampling criteria are 1, other coverage bits are 0. Special treatment is given to a sample whose sample location lies on a polygon edge. In such a case, if two polygons lie on either side of a common edge (with identical endpoints) on which a sample point lies, then exactly one of the polygons **must** result in a covered sample for that fragment during rasterization. As for the data associated with each fragment produced by rasterizing a polygon, we begin by specifying how these values are produced for fragments in a triangle.

Barycentric coordinates are a set of three numbers, a, b, and c, each in the range [0,1], with a + b + c = 1. These coordinates uniquely specify any point p within the triangle or on the triangle's boundary as

$$p = a\,p_a + b\,p_b + c\,p_c$$

where p_a, p_b, and p_c are the vertices of the triangle. a, b, and c are determined by:

$$a = \frac{\mathrm{A}(pp_bp_c)}{\mathrm{A}(p_ap_bp_c)}, \quad b = \frac{\mathrm{A}(pp_ap_c)}{\mathrm{A}(p_ap_bp_c)}, \quad c = \frac{\mathrm{A}(pp_ap_b)}{\mathrm{A}(p_ap_bp_c)},$$

where A(lmn) denotes the area in framebuffer coordinates of the triangle with vertices l, m, and n.

Denote an associated datum at p_a, p_b, or p_c as f_a, f_b, or f_c, respectively.

Perspective interpolation for a triangle interpolates three values in a manner that is correct when taking the perspective of the viewport into consideration, by way of the triangle's clip coordinates. An interpolated value f can be determined by

$$f = \frac{af_a / w_a + bf_b / w_b + cf_c / w_c}{a / w_a + b / w_b + c / w_c}$$

where w_a, w_b, and w_c are the clip w coordinates of p_a, p_b, and p_c, respectively. a, b, and c are the barycentric coordinates of the location at which the data are produced.

Linear interpolation for a triangle directly interpolates three values, and an interpolated value f can be determined by

$$f = a f_a + b f_b + c f_c$$

where f_a, f_b, and f_c are the data associated with p_a, p_b, and p_c, respectively.

The clip coordinate w for a sample is determined using perspective interpolation. The depth value z for a sample is determined using linear interpolation. Interpolation of fragment shader input values is determined by Interpolation decorations.

For a polygon with more than three edges, such as are produced by clipping a triangle, a convex combination of the values of the datum at the polygon's vertices **must** be used to obtain the value assigned to each fragment produced by the rasterization algorithm. That is, it **must** be the case that at every fragment

$$f = \sum_{i=1}^{n} a_i f_i$$

where n is the number of vertices in the polygon and f_i is the value of f at vertex i. For each i, $0 \le a_i \le 1$ and $\sum_{i=1}^{n} a_i = 1$. The values of a_i **may** differ from fragment to fragment, but at vertex i, $a_i = 1$ and $a_j = 0$ for $j \neq i$.

NOTE

One algorithm that achieves the required behavior is to triangulate a polygon (without adding any vertices) and then treat each triangle individually as already discussed. A scan-line rasterizer that linearly interpolates data along each edge and then linearly interpolates data across each horizontal span from edge to edge also satisfies the restrictions (in this case the numerator and denominator of perspective interpolation are iterated independently, and a division is performed for each fragment).

31.11.2. Polygon Mode

Possible values of the VkPipelineRasterizationStateCreateInfo::`polygonMode` property of the currently active pipeline, specifying the method of rasterization for polygons, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPolygonMode {
    VK_POLYGON_MODE_FILL = 0,
    VK_POLYGON_MODE_LINE = 1,
    VK_POLYGON_MODE_POINT = 2,
} VkPolygonMode;
```

- VK_POLYGON_MODE_POINT specifies that polygon vertices are drawn as points.
- VK_POLYGON_MODE_LINE specifies that polygon edges are drawn as line segments.
- VK_POLYGON_MODE_FILL specifies that polygons are rendered using the polygon rasterization rules in this section.

These modes affect only the final rasterization of polygons: in particular, a polygon's vertices are shaded and the polygon is clipped and possibly culled before these modes are applied.

If VkPhysicalDeviceMaintenance5Properties::polygonModePointSize is VK_TRUE, the point size of the final rasterization of polygons is taken from PointSize when polygon mode is VK_POLYGON_MODE_POINT.

Otherwise, if VkPhysicalDeviceMaintenance5Properties::polygonModePointSize is VK_FALSE, the point size of the final rasterization of polygons is 1.0 when polygon mode is VK_POLYGON_MODE_POINT.

To dynamically set the polygon mode, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetPolygonModeEXT(
    VkCommandBuffer                             commandBuffer,
    VkPolygonMode                               polygonMode);
```

- commandBuffer is the command buffer into which the command will be recorded.
- polygonMode specifies polygon mode.

This command sets the polygon mode for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_POLYGON_MODE_EXT set in VkPipelineDynamicStateCreateInfo::pDynamicStates. Otherwise, this state is specified by the VkPipelineRasterizationStateCreateInfo::polygonMode value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetPolygonModeEXT-None-09423
  At least one of the following **must** be true:
  - The extendedDynamicState3PolygonMode feature is enabled
  - The shaderObject feature is enabled
- VUID-vkCmdSetPolygonModeEXT-fillModeNonSolid-07424

If the `fillModeNonSolid` feature is not enabled, `polygonMode` **must** be VK_POLYGON_MODE_FILL

Valid Usage (Implicit)

- VUID-vkCmdSetPolygonModeEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetPolygonModeEXT-polygonMode-parameter
  `polygonMode` **must** be a valid VkPolygonMode value
- VUID-vkCmdSetPolygonModeEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetPolygonModeEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetPolygonModeEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetPolygonModeEXT is not affected by conditional rendering

31.11.3. Depth Bias

The depth values of all fragments generated by the rasterization of a polygon **can** be biased (offset) by a single depth bias value o that is computed for that polygon.

Depth Bias Enable

The depth bias computation is enabled by the `depthBiasEnable` set with vkCmdSetDepthBiasEnable and `vkCmdSetDepthBiasEnableEXT`, or the corresponding VkPipelineRasterizationStateCreateInfo ::`depthBiasEnable` value used to create the currently active pipeline. If the depth bias enable is VK_FALSE, no bias is applied and the fragment's depth values are unchanged.

To dynamically enable whether to bias fragment depth values, call:

```
// Provided by VK_VERSION_1_3
void vkCmdSetDepthBiasEnable(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    depthBiasEnable);
```

```
// Provided by VK_EXT_extended_dynamic_state2, VK_EXT_shader_object
// Equivalent to vkCmdSetDepthBiasEnable
void vkCmdSetDepthBiasEnableEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    depthBiasEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthBiasEnable` controls whether to bias fragment depth values.

This command sets the depth bias enable for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineRasterizationStateCreateInfo ::`depthBiasEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthBiasEnable-None-08970
  At least one of the following **must** be true:
  - the `extendedDynamicState2` feature is enabled
  - the `shaderObject` feature is enabled
  - the value of VkApplicationInfo::`apiVersion` used to create the VkInstance parent of `commandBuffer` is greater than or equal to Version 1.3

Valid Usage (Implicit)

- VUID-vkCmdSetDepthBiasEnable-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDepthBiasEnable-commandBuffer-recording
  `commandBuffer` **must** be in the recording state

- VUID-vkCmdSetDepthBiasEnable-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDepthBiasEnable-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDepthBiasEnable is not affected by conditional rendering

Depth Bias Computation

The depth bias depends on three parameters:

- `depthBiasSlopeFactor` scales the maximum depth slope m of the polygon
- `depthBiasConstantFactor` scales the parameter r of the depth attachment
- the scaled terms are summed to produce a value which is then clamped to a minimum or maximum value specified by `depthBiasClamp`

`depthBiasSlopeFactor`, `depthBiasConstantFactor`, and `depthBiasClamp` **can** each be positive, negative, or zero. These parameters are set as described for vkCmdSetDepthBias and vkCmdSetDepthBias2EXT below.

The maximum depth slope m of a triangle is

$$m = \sqrt{\left(\frac{\partial z_f}{\partial x_f}\right)^2 + \left(\frac{\partial z_f}{\partial y_f}\right)^2}$$

where (x_f, y_f, z_f) is a point on the triangle. m **may** be approximated as

$$m = \max\left(\left|\frac{\partial z_f}{\partial x_f}\right|, \left|\frac{\partial z_f}{\partial y_f}\right|\right).$$

In a pipeline with a depth bias representation of VK_DEPTH_BIAS_REPRESENTATION_FLOAT_EXT, r, for the given primitive is defined as

r = 1

Otherwise r is the minimum resolvable difference that depends on the depth attachment representation. If VkDepthBiasRepresentationInfoEXT::depthBiasExact is VK_FALSE it is the smallest difference in a sample's depth z_f values that is guaranteed to remain distinct throughout polygon rasterization and in the depth attachment. All pairs of fragments generated by the rasterization of two polygons with otherwise identical vertices, but z_f values that differ by r, will have distinct depth values.

For fixed-point depth attachment representations, or in a pipeline with a depth bias representation of VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORCE_UNORM_EXT, r is constant throughout the range of the entire depth attachment. If VkDepthBiasRepresentationInfoEXT::depthBiasExact is VK_TRUE, then its value **must** be

$r = 2^{-n}$

Otherwise its value is implementation-dependent but **must** be at most

$r = 2 \times 2^{-n}$

where n is the number of bits used for the depth aspect when using a fixed-point attachment, or the number of mantissa bits plus one when using a floating-point attachment.

Otherwise for floating-point depth attachment, there is no single minimum resolvable difference. In this case, the minimum resolvable difference for a given polygon is dependent on the maximum exponent, e, in the range of z values spanned by the primitive. If n is the number of bits in the floating-point mantissa, the minimum resolvable difference, r, for the given primitive is defined as

$r = 2^{e-n}$

If no depth attachment is present, r is undefined.

The bias value o for a polygon is

$$o = \text{dbclamp}(m \times depthBiasSlopeFactor + r \times depthBiasConstantFactor)$$

$$\text{where} \quad \text{dbclamp}(x) = \begin{cases} x & depthBiasClamp = 0 \text{ or } NaN \\ \min(x, depthBiasClamp) & depthBiasClamp > 0 \\ \max(x, depthBiasClamp) & depthBiasClamp < 0 \end{cases}$$

m is computed as described above. If the depth attachment uses a fixed-point representation, m is a function of depth values in the range [0,1], and o is applied to depth values in the same range.

Depth bias is applied to triangle topology primitives received by the rasterizer regardless of polygon mode. Depth bias **may** also be applied to line and point topology primitives received by the rasterizer.

To dynamically set the depth bias parameters, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetDepthBias(
    VkCommandBuffer                             commandBuffer,
    float                                       depthBiasConstantFactor,
    float                                       depthBiasClamp,
    float                                       depthBiasSlopeFactor);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthBiasConstantFactor` is a scalar factor controlling the constant depth value added to each fragment.
- `depthBiasClamp` is the maximum (or minimum) depth bias of a fragment.
- `depthBiasSlopeFactor` is a scalar factor applied to a fragment's slope in depth bias calculations.

This command sets the depth bias parameters for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_DEPTH_BIAS set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the corresponding VkPipelineRasterizationStateCreateInfo ::`depthBiasConstantFactor`, `depthBiasClamp`, and `depthBiasSlopeFactor` values used to create the currently active pipeline.

Calling this function is equivalent to calling `vkCmdSetDepthBias2EXT` without a `VkDepthBiasRepresentationInfoEXT` structure in the `pNext` chain of `VkDepthBiasInfoEXT`.

Valid Usage

- VUID-vkCmdSetDepthBias-depthBiasClamp-00790
  If the `depthBiasClamp` feature is not enabled, `depthBiasClamp` **must** be `0.0`

Valid Usage (Implicit)

- VUID-vkCmdSetDepthBias-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDepthBias-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetDepthBias-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support

VK_QUEUE_GRAPHICS_BIT operations

- VUID-vkCmdSetDepthBias-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDepthBias is not affected by conditional rendering

The `VkDepthBiasRepresentationInfoEXT` structure is defined as:

```
// Provided by VK_EXT_depth_bias_control
typedef struct VkDepthBiasRepresentationInfoEXT {
    VkStructureType                 sType;
    const void*                     pNext;
    VkDepthBiasRepresentationEXT    depthBiasRepresentation;
    VkBool32                        depthBiasExact;
} VkDepthBiasRepresentationInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `depthBiasRepresentation` is a VkDepthBiasRepresentationEXT value specifying the depth bias representation.
- `depthBiasExact` specifies that the implementation is not allowed to scale the depth bias value to ensure a minimum resolvable distance.

Valid Usage

- VUID-VkDepthBiasRepresentationInfoEXT-leastRepresentableValueForceUnormRepresentation-08947
  If the `leastRepresentableValueForceUnormRepresentation` feature is not enabled, `depthBiasRepresentation` **must** not be VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORCE_UNORM_EXT
- VUID-VkDepthBiasRepresentationInfoEXT-floatRepresentation-08948
  If the `floatRepresentation` feature is not enabled, `depthBiasRepresentation` **must** not be VK_DEPTH_BIAS_REPRESENTATION_FLOAT_EXT
- VUID-VkDepthBiasRepresentationInfoEXT-depthBiasExact-08949
  If the `depthBiasExact` feature is not enabled, `depthBiasExact` **must** be VK_FALSE

Valid Usage (Implicit)

- VUID-VkDepthBiasRepresentationInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEPTH_BIAS_REPRESENTATION_INFO_EXT
- VUID-VkDepthBiasRepresentationInfoEXT-depthBiasRepresentation-parameter
  `depthBiasRepresentation` **must** be a valid VkDepthBiasRepresentationEXT value

Structure Chaining

Extends the structures

- VkDepthBiasInfoEXT
- VkPipelineRasterizationStateCreateInfo

Possible values of VkDepthBiasRepresentationInfoEXT::`depthBiasRepresentation`, specifying the depth bias representation are:

```
// Provided by VK_EXT_depth_bias_control
typedef enum VkDepthBiasRepresentationEXT {
    VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORMAT_EXT = 0,
    VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORCE_UNORM_EXT = 1,
    VK_DEPTH_BIAS_REPRESENTATION_FLOAT_EXT = 2,
} VkDepthBiasRepresentationEXT;
```

- VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORMAT_EXT specifies that the depth bias representation is a factor of the format's r as described in Depth Bias Computation.
- VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORCE_UNORM_EXT specifies that the depth bias representation is a factor of a constant r defined by the bit-size or

mantissa of the format as described in Depth Bias Computation.

- VK_DEPTH_BIAS_REPRESENTATION_FLOAT_EXT specifies that the depth bias representation is a factor of constant r equal to 1.

The VkDepthBiasInfoEXT structure is defined as:

```
// Provided by VK_EXT_depth_bias_control
typedef struct VkDepthBiasInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    float              depthBiasConstantFactor;
    float              depthBiasClamp;
    float              depthBiasSlopeFactor;
} VkDepthBiasInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- depthBiasConstantFactor is a scalar factor controlling the constant depth value added to each fragment.
- depthBiasClamp is the maximum (or minimum) depth bias of a fragment.
- depthBiasSlopeFactor is a scalar factor applied to a fragment's slope in depth bias calculations.

If pNext does not contain a VkDepthBiasRepresentationInfoEXT structure, then this command is equivalent to including a VkDepthBiasRepresentationInfoEXT with depthBiasExact set to VK_FALSE and depthBiasRepresentation set to VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORMAT_EXT.

Valid Usage

- VUID-VkDepthBiasInfoEXT-depthBiasClamp-08950
  If the depthBiasClamp feature is not enabled, depthBiasClamp **must** be 0.0

Valid Usage (Implicit)

- VUID-VkDepthBiasInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DEPTH_BIAS_INFO_EXT
- VUID-VkDepthBiasInfoEXT-pNext-pNext
  pNext **must** be NULL or a pointer to a valid instance of VkDepthBiasRepresentationInfoEXT
- VUID-VkDepthBiasInfoEXT-sType-unique
  The sType value of each structure in the pNext chain **must** be unique

To dynamically set the depth bias parameters, call:

```
// Provided by VK_EXT_depth_bias_control
void vkCmdSetDepthBias2EXT(
    VkCommandBuffer                             commandBuffer,
    const VkDepthBiasInfoEXT*                   pDepthBiasInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pDepthBiasInfo` is a pointer to a VkDepthBiasInfoEXT structure specifying depth bias parameters.

This command is functionally identical to vkCmdSetDepthBias, but includes extensible sub-structures that include `sType` and `pNext` parameters, allowing them to be more easily extended.

Valid Usage (Implicit)

- VUID-vkCmdSetDepthBias2EXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDepthBias2EXT-pDepthBiasInfo-parameter
  `pDepthBiasInfo` **must** be a valid pointer to a valid VkDepthBiasInfoEXT structure
- VUID-vkCmdSetDepthBias2EXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetDepthBias2EXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDepthBias2EXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDepthBias2EXT is not affected by conditional rendering

31.11.4. Conservative Rasterization

If the pNext chain of VkPipelineRasterizationStateCreateInfo includes a VkPipelineRasterizationConservativeStateCreateInfoEXT structure, then that structure includes parameters controlling conservative rasterization.

VkPipelineRasterizationConservativeStateCreateInfoEXT is defined as:

```
// Provided by VK_EXT_conservative_rasterization
typedef struct VkPipelineRasterizationConservativeStateCreateInfoEXT {
    VkStructureType                                         sType;
    const void*                                             pNext;
    VkPipelineRasterizationConservativeStateCreateFlagsEXT    flags;
    VkConservativeRasterizationModeEXT
conservativeRasterizationMode;
    float
extraPrimitiveOverestimationSize;
} VkPipelineRasterizationConservativeStateCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is reserved for future use.
- conservativeRasterizationMode is the conservative rasterization mode to use.
- extraPrimitiveOverestimationSize is the extra size in pixels to increase the generating primitive during conservative rasterization at each of its edges in X and Y equally in screen space beyond the base overestimation specified in VkPhysicalDeviceConservativeRasterizationPropertiesEXT ::primitiveOverestimationSize. If conservativeRasterizationMode is not VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, this value is ignored.

If this structure is not included in the pNext chain, conservativeRasterizationMode is considered to be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT, and conservative rasterization is disabled.

Polygon rasterization **can** be made conservative by setting conservativeRasterizationMode to VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT or VK_CONSERVATIVE_RASTERIZATION_MODE_UNDERESTIMATE_EXT in VkPipelineRasterizationConservativeStateCreateInfoEXT.

NOTE If conservativePointAndLineRasterization is supported, conservative rasterization can be applied to line and point primitives, otherwise it must be disabled.

Valid Usage

- VUID-VkPipelineRasterizationConservativeStateCreateInfoEXT-extraPrimitiveOverestimationSize-01769
  `extraPrimitiveOverestimationSize` **must** be in the range of `0.0` to `VkPhysicalDeviceConservativeRasterizationPropertiesEXT`::`maxExtraPrimitiveOverestimationSize` inclusive

Valid Usage (Implicit)

- VUID-VkPipelineRasterizationConservativeStateCreateInfoEXT-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_CONSERVATIVE_STATE_CREATE_INFO_EXT`
- VUID-VkPipelineRasterizationConservativeStateCreateInfoEXT-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkPipelineRasterizationConservativeStateCreateInfoEXT-conservativeRasterizationMode-parameter
  `conservativeRasterizationMode` **must** be a valid `VkConservativeRasterizationModeEXT` value

Structure Chaining

Extends the structure

- `VkPipelineRasterizationStateCreateInfo`

```
// Provided by VK_EXT_conservative_rasterization
typedef VkFlags VkPipelineRasterizationConservativeStateCreateFlagsEXT;
```

`VkPipelineRasterizationConservativeStateCreateFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

Possible values of `VkPipelineRasterizationConservativeStateCreateInfoEXT`::`conservativeRasterizationMode`, specifying the conservative rasterization mode are:

```
// Provided by VK_EXT_conservative_rasterization
typedef enum VkConservativeRasterizationModeEXT {
    VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT = 0,
    VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT = 1,
    VK_CONSERVATIVE_RASTERIZATION_MODE_UNDERESTIMATE_EXT = 2,
} VkConservativeRasterizationModeEXT;
```

- VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT specifies that conservative rasterization is disabled and rasterization proceeds as normal.
- VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT specifies that conservative rasterization is enabled in overestimation mode.
- VK_CONSERVATIVE_RASTERIZATION_MODE_UNDERESTIMATE_EXT specifies that conservative rasterization is enabled in underestimation mode.

To dynamically set the conservativeRasterizationMode, call:

```
// Provided by VK_EXT_conservative_rasterization with VK_EXT_extended_dynamic_state3,
VK_EXT_conservative_rasterization with VK_EXT_shader_object
void vkCmdSetConservativeRasterizationModeEXT(
    VkCommandBuffer                             commandBuffer,
    VkConservativeRasterizationModeEXT          conservativeRasterizationMode);
```

- commandBuffer is the command buffer into which the command will be recorded.
- conservativeRasterizationMode specifies the conservativeRasterizationMode state.

This command sets the conservativeRasterizationMode state for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT set in VkPipelineDynamicStateCreateInfo::pDynamicStates. Otherwise, this state is specified by the VkPipelineRasterizationConservativeStateCreateInfoEXT::conservativeRasterizationMode value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetConservativeRasterizationModeEXT-None-09423
  At least one of the following **must** be true:
  - The extendedDynamicState3ConservativeRasterizationMode feature is enabled
  - The shaderObject feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetConservativeRasterizationModeEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetConservativeRasterizationModeEXT-conservativeRasterizationMode-parameter
  conservativeRasterizationMode **must** be a valid VkConservativeRasterizationModeEXT value
- VUID-vkCmdSetConservativeRasterizationModeEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdSetConservativeRasterizationModeEXT-commandBuffer-cmdpool

The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations

- VUID-vkCmdSetConservativeRasterizationModeEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetConservativeRasterizationModeEXT is not affected by conditional rendering

To dynamically set the extraPrimitiveOverestimationSize, call:

```
// Provided by VK_EXT_conservative_rasterization with VK_EXT_extended_dynamic_state3,
VK_EXT_conservative_rasterization with VK_EXT_shader_object
void vkCmdSetExtraPrimitiveOverestimationSizeEXT(
    VkCommandBuffer                             commandBuffer,
    float                                       extraPrimitiveOverestimationSize);
```

- commandBuffer is the command buffer into which the command will be recorded.
- extraPrimitiveOverestimationSize specifies the extraPrimitiveOverestimationSize.

This command sets the extraPrimitiveOverestimationSize for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT set in VkPipelineDynamicStateCreateInfo::pDynamicStates. Otherwise, this state is specified by the VkPipelineRasterizationConservativeStateCreateInfoEXT::extraPrimitiveOverestimationSize value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetExtraPrimitiveOverestimationSizeEXT-None-09423
  At least one of the following **must** be true:
  - The `extendedDynamicState3ExtraPrimitiveOverestimationSize` feature is enabled
  - The `shaderObject` feature is enabled
- VUID-vkCmdSetExtraPrimitiveOverestimationSizeEXT-extraPrimitiveOverestimationSize-07428
  `extraPrimitiveOverestimationSize` **must** be in the range of `0.0` to `VkPhysicalDeviceConservativeRasterizationPropertiesEXT`::`maxExtraPrimitiveOverestimationSize` inclusive

Valid Usage (Implicit)

- VUID-vkCmdSetExtraPrimitiveOverestimationSizeEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetExtraPrimitiveOverestimationSizeEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetExtraPrimitiveOverestimationSizeEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetExtraPrimitiveOverestimationSizeEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetExtraPrimitiveOverestimationSizeEXT is not affected by conditional rendering

When overestimate conservative rasterization is enabled, rather than evaluating coverage at individual sample locations, a determination is made whether any portion of the pixel (including its edges and corners) is covered by the primitive. If any portion of the pixel is covered, then all bits of the coverage mask for the fragment corresponding to that pixel are enabled. If the render pass has a fragment density map attachment and any bit of the coverage mask for the fragment is enabled, then all bits of the coverage mask for the fragment are enabled.

For the purposes of evaluating which pixels are covered by the primitive, implementations **can** increase the size of the primitive by up to VkPhysicalDeviceConservativeRasterizationPropertiesEXT::primitiveOverestimationSize pixels at each of the primitive edges. This **may** increase the number of fragments generated by this primitive and represents an overestimation of the pixel coverage.

This overestimation size can be increased further by setting the extraPrimitiveOverestimationSize value above 0.0 in steps of VkPhysicalDeviceConservativeRasterizationPropertiesEXT::extraPrimitiveOverestimationSizeGranularity up to and including VkPhysicalDeviceConservativeRasterizationPropertiesEXT::maxExtraPrimitiveOverestimationSize. This **may** further increase the number of fragments generated by this primitive.

The actual precision of the overestimation size used for conservative rasterization **may** vary between implementations and produce results that only approximate the primitiveOverestimationSize and extraPrimitiveOverestimationSizeGranularity properties. Implementations **may** especially vary these approximations when the render pass has a fragment density map and the fragment area covers multiple pixels.

For triangles if VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT is enabled, fragments will be generated if the primitive area covers any portion of any pixel inside the fragment area, including their edges or corners. The tie-breaking rule described in Basic Polygon Rasterization does not apply during conservative rasterization and coverage is set for all fragments generated from shared edges of polygons. Degenerate triangles that evaluate to zero area after rasterization, even for pixels containing a vertex or edge of the zero-area polygon, will be culled if VkPhysicalDeviceConservativeRasterizationPropertiesEXT::degenerateTrianglesRasterized is VK_FALSE or will generate fragments if degenerateTrianglesRasterized is VK_TRUE. The fragment input values for these degenerate triangles take their attribute and depth values from the provoking vertex. Degenerate triangles are considered backfacing and the application **can** enable backface culling if desired. Triangles that are zero area before rasterization **may** be culled regardless.

For lines if VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT is enabled, and the implementation sets VkPhysicalDeviceConservativeRasterizationPropertiesEXT::conservativePointAndLineRasterization to VK_TRUE, fragments will be generated if the line covers any portion of any pixel inside the fragment area, including their edges or corners. Degenerate lines that evaluate to zero length after rasterization will be culled if VkPhysicalDeviceConservativeRasterizationPropertiesEXT::degenerateLinesRasterized is VK_FALSE or will generate fragments if degenerateLinesRasterized is VK_TRUE. The fragment input values for

these degenerate lines take their attribute and depth values from the provoking vertex. Lines that are zero length before rasterization **may** be culled regardless.

For points if VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT is enabled, and the implementation sets VkPhysicalDeviceConservativeRasterizationPropertiesEXT ::conservativePointAndLineRasterization to VK_TRUE, fragments will be generated if the point square covers any portion of any pixel inside the fragment area, including their edges or corners.

When underestimate conservative rasterization is enabled, rather than evaluating coverage at individual sample locations, a determination is made whether all of the pixel (including its edges and corners) is covered by the primitive. If the entire pixel is covered, then a fragment is generated with all bits of its coverage mask corresponding to the pixel enabled, otherwise the pixel is not considered covered even if some portion of the pixel is covered. The fragment is discarded if no pixels inside the fragment area are considered covered. If the render pass has a fragment density map attachment and any pixel inside the fragment area is not considered covered, then the fragment is discarded even if some pixels are considered covered.

For triangles, if VK_CONSERVATIVE_RASTERIZATION_MODE_UNDERESTIMATE_EXT is enabled, fragments will only be generated if any pixel inside the fragment area is fully covered by the generating primitive, including its edges and corners.

For lines, if VK_CONSERVATIVE_RASTERIZATION_MODE_UNDERESTIMATE_EXT is enabled, fragments will be generated if any pixel inside the fragment area, including its edges and corners, is entirely covered by the line.

For points, if VK_CONSERVATIVE_RASTERIZATION_MODE_UNDERESTIMATE_EXT is enabled, fragments will only be generated if the point square covers the entirety of any pixel square inside the fragment area, including its edges or corners.

If the render pass has a fragment density map and VK_CONSERVATIVE_RASTERIZATION_MODE_UNDERESTIMATE_EXT is enabled, fragments will only be generated if the entirety of all pixels inside the fragment area are covered by the generating primitive, line, or point.

For both overestimate and underestimate conservative rasterization modes a fragment that has all of its pixel squares fully covered by the generating primitive **must** set FullyCoveredEXT to VK_TRUE if the implementation enables the VkPhysicalDeviceConservativeRasterizationPropertiesEXT ::fullyCoveredFragmentShaderInputVariable feature.

When setting the fragment shading rate results in fragments covering multiple pixels, coverage for conservative rasterization is still evaluated on a per-pixel basis and may result in fragments with partial coverage. For fragment shader inputs decorated with FullyCoveredEXT, a fragment is considered fully covered if and only if all pixels in the fragment are fully covered by the generating primitive.

Chapter 32. Fragment Operations

Fragments produced by rasterization go through a number of operations to determine whether or how values produced by fragment shading are written to the framebuffer.

The following fragment operations adhere to rasterization order, and are typically performed in this order:

1. Discard rectangles test
2. Scissor test
3. Sample mask test
4. Certain Fragment shading operations:
    - Sample Mask Accesses
    - Tile Image Reads
    - Depth Replacement
    - Stencil Reference Replacement
    - Interlocked Operations
5. Multisample coverage
6. Depth bounds test
7. Stencil test
8. Depth test
9. Sample counting
10. Coverage reduction

The coverage mask generated by rasterization describes the initial coverage of each sample covered by the fragment. Fragment operations will update the coverage mask to add or subtract coverage where appropriate. If a fragment operation results in all bits of the coverage mask being `0`, the fragment is discarded, and no further operations are performed.

When one of the fragment operations in this chapter is described as “replacing” a fragment shader output, that output is replaced unconditionally, even if no fragment shader previously wrote to that output.

If there is a fragment shader and it declares the `PostDepthCoverage` execution mode, the sample mask test is instead performed after the depth test.

If `VkPhysicalDeviceMaintenance5Properties`::`earlyFragmentMultisampleCoverageAfterSampleCounting` is VK_TRUE and there is a fragment shader which declares the `EarlyFragmentTests` execution mode, fragment shading and multisample coverage operations **must** be performed after sample counting.

Otherwise, if `VkPhysicalDeviceMaintenance5Properties`::`earlyFragmentMultisampleCoverageAfterSampleCounting` is VK_FALSE and there is a fragment shader which declares the `EarlyFragmentTests` execution mode, fragment shading and multisample

coverage operations **should** instead be performed after sample counting, but **may** be performed before sample counting.

If VkPhysicalDeviceMaintenance5Properties::earlyFragmentSampleMaskTestBeforeSampleCounting is VK_TRUE and there is a fragment shader which declares the EarlyFragmentTests execution mode, sample mask test operations **must** follow the order of fragment operations from above.

Otherwise, if VkPhysicalDeviceMaintenance5Properties::earlyFragmentSampleMaskTestBeforeSampleCounting is VK_FALSE and there is a fragment shader which declares the EarlyFragmentTests execution mode, sample mask test operations **should** follow the order of fragment operations from above but **may** instead be performed after sample counting.

For a pipeline with the following properties:

- a fragment shader is specified
- the fragment shader does not write to storage resources;
- the fragment shader specifies the DepthReplacing execution mode; and
- either
  - the fragment shader specifies the DepthUnchanged execution mode;
  - the fragment shader specifies the DepthLess execution mode and the pipeline uses a VkPipelineDepthStencilStateCreateInfo::depthCompareOp of VK_COMPARE_OP_GREATER or VK_COMPARE_OP_GREATER_OR_EQUAL; or
  - the fragment shader specifies the DepthGreater execution mode and the pipeline uses a VkPipelineDepthStencilStateCreateInfo::depthCompareOp of VK_COMPARE_OP_LESS or VK_COMPARE_OP_LESS_OR_EQUAL

the implementation **may** perform depth bounds test before fragment shading and perform an additional depth test immediately after that using the interpolated depth value generated by rasterization.

Once all fragment operations have completed, fragment shader outputs for covered color attachment samples pass through framebuffer operations.

32.1. Discard Rectangles Test

The discard rectangle test compares the framebuffer coordinates (x_f,y_f) of each sample covered by a fragment against a set of *discard rectangles*.

Each discard rectangle is defined by a VkRect2D. These values are either set by the VkPipelineDiscardRectangleStateCreateInfoEXT structure during pipeline creation, or dynamically by the vkCmdSetDiscardRectangleEXT command.

A given sample is considered inside a discard rectangle if the x_f is in the range [VkRect2D::offset.x, VkRect2D::offset.x + VkRect2D::extent.width), and y_f is in the range [VkRect2D::offset.y, VkRect2D::offset.y + VkRect2D::extent.height). If the test is set to be inclusive, samples that are not inside any of the discard rectangles will have their coverage set to 0. If the test is set to be exclusive, samples that are inside any of the discard rectangles will have their coverage set to 0.

If no discard rectangles are specified, the coverage mask is unmodified by this operation.

The VkPipelineDiscardRectangleStateCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_discard_rectangles
typedef struct VkPipelineDiscardRectangleStateCreateInfoEXT {
    VkStructureType                                    sType;
    const void*                                        pNext;
    VkPipelineDiscardRectangleStateCreateFlagsEXT      flags;
    VkDiscardRectangleModeEXT                          discardRectangleMode;
    uint32_t                                           discardRectangleCount;
    const VkRect2D*                                    pDiscardRectangles;
} VkPipelineDiscardRectangleStateCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is reserved for future use.
- discardRectangleMode is a VkDiscardRectangleModeEXT value determining whether the discard rectangle test is inclusive or exclusive.
- discardRectangleCount is the number of discard rectangles to use.
- pDiscardRectangles is a pointer to an array of VkRect2D structures defining discard rectangles.

If the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state is enabled for a pipeline, the pDiscardRectangles member is ignored. If the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state is not enabled for the pipeline the presence of this structure in the VkGraphicsPipelineCreateInfo chain, and a discardRectangleCount greater than zero, implicitly enables discard rectangles in the pipeline, otherwise discard rectangles **must** be enabled or disabled by vkCmdSetDiscardRectangleEnableEXT. If the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state is enabled for the pipeline, the discardRectangleMode member is ignored, and the discard rectangle mode **must** be set by vkCmdSetDiscardRectangleModeEXT.

When this structure is included in the pNext chain of VkGraphicsPipelineCreateInfo, it defines parameters of the discard rectangle test. If the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state is not enabled, and this structure is not included in the pNext chain, it is equivalent to specifying this structure with a discardRectangleCount of 0. If all VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT, VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT, and VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic states are enabled, the application **can** omit this structure from the pNext chain of VkGraphicsPipelineCreateInfo and still use discard rectangles by setting all state dynamically. In this case vkCmdSetDiscardRectangleEXT **must** be called to set the discard rectangle for all indices [0, maxDiscardRectangles) before drawing with discard rectangles enabled. Individual discard rectangles **can** be made ineffective by setting their VkRect2D::extent.width and VkRect2D::extent.height to zero.

Valid Usage

- VUID-VkPipelineDiscardRectangleStateCreateInfoEXT-discardRectangleCount-00582 `discardRectangleCount` **must** be less than or equal to `VkPhysicalDeviceDiscardRectanglePropertiesEXT`::`maxDiscardRectangles`

Valid Usage (Implicit)

- VUID-VkPipelineDiscardRectangleStateCreateInfoEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_DISCARD_RECTANGLE_STATE_CREATE_INFO_EXT
- VUID-VkPipelineDiscardRectangleStateCreateInfoEXT-flags-zerobitmask `flags` **must** be `0`
- VUID-VkPipelineDiscardRectangleStateCreateInfoEXT-discardRectangleMode-parameter `discardRectangleMode` **must** be a valid VkDiscardRectangleModeEXT value

Structure Chaining

Extends the structure

- VkGraphicsPipelineCreateInfo

```
// Provided by VK_EXT_discard_rectangles
typedef VkFlags VkPipelineDiscardRectangleStateCreateFlagsEXT;
```

`VkPipelineDiscardRectangleStateCreateFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

VkDiscardRectangleModeEXT values are:

```
// Provided by VK_EXT_discard_rectangles
typedef enum VkDiscardRectangleModeEXT {
    VK_DISCARD_RECTANGLE_MODE_INCLUSIVE_EXT = 0,
    VK_DISCARD_RECTANGLE_MODE_EXCLUSIVE_EXT = 1,
} VkDiscardRectangleModeEXT;
```

- VK_DISCARD_RECTANGLE_MODE_INCLUSIVE_EXT specifies that the discard rectangle test is inclusive.
- VK_DISCARD_RECTANGLE_MODE_EXCLUSIVE_EXT specifies that the discard rectangle test is exclusive.

To dynamically set the discard rectangles, call:

```
// Provided by VK_EXT_discard_rectangles
void vkCmdSetDiscardRectangleEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstDiscardRectangle,
    uint32_t                                    discardRectangleCount,
    const VkRect2D*                             pDiscardRectangles);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `firstDiscardRectangle` is the index of the first discard rectangle whose state is updated by the command.
- `discardRectangleCount` is the number of discard rectangles whose state is updated by the command.
- `pDiscardRectangles` is a pointer to an array of VkRect2D structures specifying discard rectangles.

The discard rectangle taken from element i of `pDiscardRectangles` replaces the current state for the discard rectangle at index `firstDiscardRectangle` + i, for i in [0, `discardRectangleCount`).

This command sets the discard rectangles for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineDiscardRectangleStateCreateInfoEXT::`pDiscardRectangles` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDiscardRectangleEXT-firstDiscardRectangle-00585
  The sum of `firstDiscardRectangle` and `discardRectangleCount` **must** be less than or equal to VkPhysicalDeviceDiscardRectanglePropertiesEXT::`maxDiscardRectangles`
- VUID-vkCmdSetDiscardRectangleEXT-x-00587
  The `x` and `y` members of `offset` in each VkRect2D element of `pDiscardRectangles` **must** be greater than or equal to `0`
- VUID-vkCmdSetDiscardRectangleEXT-offset-00588
  Evaluation of (`offset.x` + `extent.width`) in each VkRect2D element of `pDiscardRectangles` **must** not cause a signed integer addition overflow
- VUID-vkCmdSetDiscardRectangleEXT-offset-00589
  Evaluation of (`offset.y` + `extent.height`) in each VkRect2D element of `pDiscardRectangles` **must** not cause a signed integer addition overflow

Valid Usage (Implicit)

- VUID-vkCmdSetDiscardRectangleEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle

- VUID-vkCmdSetDiscardRectangleEXT-pDiscardRectangles-parameter
  `pDiscardRectangles` **must** be a valid pointer to an array of `discardRectangleCount` VkRect2D structures
- VUID-vkCmdSetDiscardRectangleEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetDiscardRectangleEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDiscardRectangleEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetDiscardRectangleEXT-discardRectangleCount-arraylength
  `discardRectangleCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDiscardRectangleEXT is not affected by conditional rendering

To dynamically set whether discard rectangles are enabled, call:

```
// Provided by VK_EXT_discard_rectangles
void vkCmdSetDiscardRectangleEnableEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    discardRectangleEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `discardRectangleEnable` specifies whether discard rectangles are enabled or not.

This command sets the discard rectangle enable for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT set in VkPipelineDynamicStateCreateInfo::pDynamicStates. Otherwise, this state is implied by the VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount value used to create the currently active pipeline, where a non-zero discardRectangleCount implicitly enables discard rectangles, otherwise they are disabled.

Valid Usage

- VUID-vkCmdSetDiscardRectangleEnableEXT-specVersion-07851
  The VK_EXT_discard_rectangles extension **must** be enabled, and the implementation **must** support at least specVersion 2 of this extension

Valid Usage (Implicit)

- VUID-vkCmdSetDiscardRectangleEnableEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDiscardRectangleEnableEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdSetDiscardRectangleEnableEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDiscardRectangleEnableEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDiscardRectangleEnableEXT is not affected by conditional rendering

To dynamically set the discard rectangle mode, call:

```
// Provided by VK_EXT_discard_rectangles
void vkCmdSetDiscardRectangleModeEXT(
    VkCommandBuffer                             commandBuffer,
    VkDiscardRectangleModeEXT                   discardRectangleMode);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `discardRectangleMode` specifies the discard rectangle mode for all discard rectangles, either inclusive or exclusive.

This command sets the discard rectangle mode for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineDiscardRectangleStateCreateInfoEXT::`discardRectangleMode` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDiscardRectangleModeEXT-specVersion-07852
  The `VK_EXT_discard_rectangles` extension **must** be enabled, and the implementation **must** support at least `specVersion` `2` of this extension

Valid Usage (Implicit)

- VUID-vkCmdSetDiscardRectangleModeEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDiscardRectangleModeEXT-discardRectangleMode-parameter
  `discardRectangleMode` **must** be a valid VkDiscardRectangleModeEXT value
- VUID-vkCmdSetDiscardRectangleModeEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetDiscardRectangleModeEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDiscardRectangleModeEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDiscardRectangleModeEXT is not affected by conditional rendering

32.2. Scissor Test

The scissor test compares the framebuffer coordinates (x_f,y_f) of each sample covered by a fragment against a *scissor rectangle* at the index equal to the fragment's `ViewportIndex`.

Each scissor rectangle is defined by a VkRect2D. These values are either set by the VkPipelineViewportStateCreateInfo structure during pipeline creation, or dynamically by the vkCmdSetScissor command.

A given sample is considered inside a scissor rectangle if x_f is in the range [VkRect2D::`offset.x`, VkRect2D::`offset.x` + VkRect2D::`extent.width`), and y_f is in the range [VkRect2D::`offset.y`, VkRect2D::`offset.y` + VkRect2D::`extent.height`). Samples with coordinates outside the scissor rectangle at the corresponding `ViewportIndex` will have their coverage set to `0`.

To dynamically set the scissor rectangles, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetScissor(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstScissor,
    uint32_t                                    scissorCount,
    const VkRect2D*                             pScissors);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `firstScissor` is the index of the first scissor whose state is updated by the command.

- `scissorCount` is the number of scissors whose rectangles are updated by the command.
- `pScissors` is a pointer to an array of VkRect2D structures defining scissor rectangles.

The scissor rectangles taken from element i of `pScissors` replace the current state for the scissor index `firstScissor` + i, for i in [0, `scissorCount`).

This command sets the scissor rectangles for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_SCISSOR set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineViewportStateCreateInfo::`pScissors` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetScissor-firstScissor-00592
  The sum of `firstScissor` and `scissorCount` **must** be between `1` and `VkPhysicalDeviceLimits`::`maxViewports`, inclusive
- VUID-vkCmdSetScissor-firstScissor-00593
  If the `multiViewport` feature is not enabled, `firstScissor` **must** be `0`
- VUID-vkCmdSetScissor-scissorCount-00594
  If the `multiViewport` feature is not enabled, `scissorCount` **must** be `1`
- VUID-vkCmdSetScissor-x-00595
  The `x` and `y` members of `offset` member of any element of `pScissors` **must** be greater than or equal to `0`
- VUID-vkCmdSetScissor-offset-00596
  Evaluation of (`offset.x` + `extent.width`) **must** not cause a signed integer addition overflow for any element of `pScissors`
- VUID-vkCmdSetScissor-offset-00597
  Evaluation of (`offset.y` + `extent.height`) **must** not cause a signed integer addition overflow for any element of `pScissors`

Valid Usage (Implicit)

- VUID-vkCmdSetScissor-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetScissor-pScissors-parameter
  `pScissors` **must** be a valid pointer to an array of `scissorCount` VkRect2D structures
- VUID-vkCmdSetScissor-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetScissor-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetScissor-videocoding
  This command **must** only be called outside of a video coding scope

- VUID-vkCmdSetScissor-scissorCount-arraylength
  `scissorCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetScissor is not affected by conditional rendering

32.3. Sample Mask Test

The sample mask test compares the coverage mask for a fragment with the *sample mask* defined by VkPipelineMultisampleStateCreateInfo::`pSampleMask`.

To dynamically set the sample mask, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetSampleMaskEXT(
    VkCommandBuffer                             commandBuffer,
    VkSampleCountFlagBits                       samples,
    const VkSampleMask*                         pSampleMask);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `samples` specifies the number of sample bits in the `pSampleMask`.
- `pSampleMask` is a pointer to an array of VkSampleMask values, where the array size is based on the `samples` parameter. If the `maintenance10` feature is enabled, and this parameter is set to `NULL`, it is treated as if the mask has all bits set to `1`.

This command sets the sample mask for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with

VK_DYNAMIC_STATE_SAMPLE_MASK_EXT set in VkPipelineDynamicStateCreateInfo ::pDynamicStates. Otherwise, this state is specified by the VkPipelineMultisampleStateCreateInfo ::pSampleMask value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetSampleMaskEXT-None-09423
  At least one of the following **must** be true:
  - The extendedDynamicState3SampleMask feature is enabled
  - The shaderObject feature is enabled
- VUID-vkCmdSetSampleMaskEXT-pSampleMask-10999
  pSampleMask **must** not be NULL if the maintenance10 feature is not enabled

Valid Usage (Implicit)

- VUID-vkCmdSetSampleMaskEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetSampleMaskEXT-samples-parameter
  samples **must** be a valid VkSampleCountFlagBits value
- VUID-vkCmdSetSampleMaskEXT-pSampleMask-parameter
  If pSampleMask is not NULL, pSampleMask **must** be a valid pointer to an array of $\lceil \frac{samples}{32} \rceil$ VkSampleMask values
- VUID-vkCmdSetSampleMaskEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdSetSampleMaskEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetSampleMaskEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetSampleMaskEXT is not affected by conditional rendering

Each bit of the coverage mask is associated with a sample index as described in the rasterization chapter. If the bit in VkPipelineMultisampleStateCreateInfo::`pSampleMask` which is associated with that same sample index is set to `0`, the coverage mask bit is set to `0`.

32.4. Fragment Shading

Fragment shaders are invoked for each fragment, or as helper invocations.

Most operations in the fragment shader are not performed in rasterization order, with exceptions called out in the following sections.

For fragment shaders invoked by fragments, the following rules apply:

- A fragment shader **must** not be executed if a fragment operation that executes before fragment shading discards the fragment.
- A fragment shader **may** not be executed if:
  - An implementation determines that another fragment shader, invoked by a subsequent primitive in primitive order, overwrites all results computed by the shader (including writes to storage resources).
  - Any other fragment operation discards the fragment, and the shader does not write to any storage resources.
  - If a fragment shader statically computes the same values for different framebuffer locations, and does not write to any storage resources, multiple fragments **may** be shaded by one fragment shader invocation. This **may** affect VK_QUERY_PIPELINE_STATISTIC_FRAGMENT_SHADER_INVOCATIONS_BIT results, but **must** otherwise not be visible behavior to applications.
- Otherwise, at least one fragment shader **must** be executed.
  - If sample shading is enabled and multiple invocations per fragment are **required**, additional invocations **must** be executed as specified.
  - Each covered sample **must** be included in at least one fragment shader invocation.

If no fragment shader is included in the pipeline, no fragment shader is executed, and undefined

values **may** be written to all color attachment outputs during this fragment operation.

NOTE

Multiple fragment shader invocations may be executed for the same fragment for any number of implementation-dependent reasons. When there is more than one fragment shader invocation per fragment, the association of samples to invocations is implementation-dependent. Stores and atomics performed by these additional invocations have the normal effect.

For example, if the subpass includes multiple views in its view mask, a fragment shader may be invoked separately for each view.

Similarly, if the render pass has a fragment density map attachment, more than one fragment shader invocation may be invoked for each covered sample. Such additional invocations are only produced if `VkPhysicalDeviceFragmentDensityMapPropertiesEXT::fragmentDensityInvocations` is `VK_TRUE`. Implementations may generate these additional fragment shader invocations in order to make transitions between fragment areas with different fragment densities more smooth.

32.4.1. Sample Mask

Reading from the `SampleMask` built-in in the `Input` storage class will return the coverage mask for the current fragment as calculated by fragment operations that executed prior to fragment shading.

If sample shading is enabled, fragment shaders will only see values of `1` for samples being shaded - other bits will be `0`.

Each bit of the coverage mask is associated with a sample index as described in the rasterization chapter. If the bit in `SampleMask` which is associated with that same sample index is set to `0`, that coverage mask bit is set to `0`.

Values written to the `SampleMask` built-in in the `Output` storage class will be used by the multisample coverage operation, with the same encoding as the input built-in.

32.4.2. Shader Termination and Demotion

If a fragment shader is terminated early or demoted to a helper invocation by using `OpKill`, `OpTerminateInvocation`, or `OpDemoteToHelperInvocation`, samples covered by that shader invocation have their coverage mask bits set to `0`.

32.4.3. Fragment Shader Tile Image Reads

If the `VK_EXT_shader_tile_image` extension is enabled, implementations divide the framebuffer into a grid of tiles. A tile image is a view of a framebuffer attachment tile for fragments with locations within the tile.

Within a render pass instance initiated by vkCmdBeginRenderingKHR, fragment shader invocations **can** read the framebuffer color, depth, and stencil values at the fragment location via tile images.

NOTE

Even though fragment shader invocations can only read from the corresponding fragment location, the abstraction of a tile image is introduced for the following reasons:

- Tile dimensions will be exposed in a future extension
- Future functionality such as executing compute dispatches within render passes via tile shaders can leverage tile images.

Enabling shaderTileImageColorReadAccess, shaderTileImageDepthReadAccess, shaderTileImageStencilReadAccess enables fragment shader invocations to read from color, depth, and stencil, respectively.

Color values are read from tile image variables with `OpColorAttachmentReadEXT`. Tile image variables are linked to specific color attachments using `Location` decoration. See Fragment Tile Image Interface for more details.

Depth values are read with `OpDepthAttachmentReadEXT`.

Stencil values are read with `OpStencilAttachmentReadEXT`.

The sample to read is specified by a sample index value specified as the `Sample` operand to `OpColorAttachmentReadEXT`, `OpDepthAttachmentReadEXT`, or `OpStencilAttachmentReadEXT`.

If sample shading is disabled, a fragment invocation **can** read from all sample locations associated with the fragment regardless of the fragment's coverage. This functionality is supported for VkPipelineMultisampleStateCreateInfo::`rasterizationSamples` > 1 when VkPhysicalDeviceShaderTileImagePropertiesEXT::`shaderTileImageReadSampleFromPixelRateInvocation` is VK_TRUE.

If sample shading is enabled, and `minSampleShading` is 1.0, a fragment invocation **must** only read from the coverage index sample. Tile image access **must** not be used if the value of `minSampleShading` is not 1.0.

If the fragment shader declares the `EarlyFragmentTests` execution mode, depth reads are allowed only if depth writes are disabled and stencil reads are allowed only if stencil writes are disabled.

If VkPhysicalDeviceShaderTileImagePropertiesEXT::`shaderTileImageReadFromHelperInvocation` is VK_FALSE, values read from helper invocations are poison, otherwise the values read are subject to the coherency guarantees described below.

`OpDepthAttachmentReadEXT` returns a poison value if no depth attachment is present. `OpStencilAttachmentReadEXT` returns a poison value if no stencil attachment is present.

Tile image reads from color, depth, and stencil attachments are said to be coherent when the accesses happen in raster order and without data race with respect to accesses to the attachments from framebuffer-space pipeline stages. The samples which qualify for coherent access and the enabling conditions are described below.

- Let Rc be the set of components being read from an attachment A in a draw call

- Let Wc be the set of components being written to A by the draw call

The samples which qualify for coherent tile image reads from an attachment A are:

- All samples in a pixel when Rc is disjoint with Wc.
- The samples with coverage in a fragment when Rc is not disjoint with Wc. The samples with coverage are determined by the coverage mask for the fragment as calculated by fragment operations that executed prior to fragment shading, including early fragment tests if enabled for the draw call.

A fragment shader **can** declare `NonCoherentColorAttachmentReadEXT`, `NonCoherentDepthAttachmentReadEXT`, or `NonCoherentStencilAttachmentReadEXT` execution modes to enable non-coherent tile image reads which require an explicit vkCmdPipelineBarrier2 call for the writes to an attachment to be made visible via tile image reads.

When VkPhysicalDeviceShaderTileImagePropertiesEXT::`shaderTileImageCoherentReadAccelerated` is VK_TRUE, the implementation prefers that coherent tile image reads are used, otherwise the implementation prefers that non-coherent tile image reads are used.

NOTE

In practice, the most common tile image reads usage patterns fall under one of the following:

- Programmable blending - each fragment reads from a single sample (SampleID) at its location. Per-sample shading is typically enabled when multisampled rendertargets are used.
- G-buffer generation and shading in one render pass - in the shading phase a fragment reads from a single sample at its location.
- Programmable resolve - a fragment reads from all samples at its location (per-sample shading is disabled). This requires the use of a "full-screen triangle" instead of a rectangle composed of two triangles in order to avoid data races along the shared edge of the triangles.
- 1:1 texturing with LOD - in use cases such as deferred screen space decals a fragment reads a single sample (SampleID) from depth buffer, but requires being able to read from helper threads to derive the texture LOD. This use case is supported as long as the attachment components being read are not overwritten by color, depth, or stencil attachment writes.

All of the above use cases are supported by coherent tile image reads, but only the latter three are supported when non-coherent reads are used as there is no mechanism to synchronize non-coherent reads with writes within a draw call.

32.4.4. Depth Replacement

Writing to the `FragDepth` built-in will replace the fragment's calculated depth values for each sample in the input `SampleMask`. Depth testing performed after the fragment shader for this fragment will use this new value as z_f.

32.4.5. Stencil Reference Replacement

Writing to the `FragStencilRefEXT` built-in will replace the fragment's stencil reference value for each sample in the input `SampleMask`. Stencil testing performed after the fragment shader for this fragment will use this new value as s_r.

32.4.6. Interlocked Operations

`OpBeginInvocationInterlockEXT` and `OpEndInvocationInterlockEXT` define a section of a fragment shader which imposes additional ordering constraints on operations performed within them. These operations are defined as *interlocked operations*. How interlocked operations are ordered against other fragment shader invocations depends on the specified execution modes.

If the `ShadingRateInterlockOrderedEXT` execution mode is specified, any interlocked operations in a fragment shader **must** happen before interlocked operations in fragment shader invocations that execute later in rasterization order and cover at least one sample in the same fragment area, and **must** happen after interlocked operations in a fragment shader that executes earlier in rasterization order and cover at least one sample in the same fragment area.

If the `ShadingRateInterlockUnorderedEXT` execution mode is specified, any interlocked operations in a fragment shader **must** happen before or after interlocked operations in fragment shader invocations that execute earlier or later in rasterization order and cover at least one sample in the same fragment area.

If the `PixelInterlockOrderedEXT` execution mode is specified, any interlocked operations in a fragment shader **must** happen before interlocked operations in fragment shader invocations that execute later in rasterization order and cover at least one sample in the same pixel, and **must** happen after interlocked operations in a fragment shader that executes earlier in rasterization order and cover at least one sample in the same pixel.

If the `PixelInterlockUnorderedEXT` execution mode is specified, any interlocked operations in a fragment shader **must** happen before or after interlocked operations in fragment shader invocations that execute earlier or later in rasterization order and cover at least one sample in the same pixel.

If the `SampleInterlockOrderedEXT` execution mode is specified, any interlocked operations in a fragment shader **must** happen before interlocked operations in fragment shader invocations that execute later in rasterization order and cover at least one of the same samples, and **must** happen after interlocked operations in a fragment shader that executes earlier in rasterization order and cover at least one of the same samples.

If the `SampleInterlockUnorderedEXT` execution mode is specified, any interlocked operations in a fragment shader **must** happen before or after interlocked operations in fragment shader invocations that execute earlier or later in rasterization order and cover at least one of the same samples.

32.5. Multisample Coverage

If a fragment shader is active and its entry point's interface includes a built-in output variable

decorated with SampleMask, the coverage mask is ANDed with the bits of the SampleMask built-in to generate a new coverage mask. If sample shading is enabled, bits written to SampleMask corresponding to samples that are not being shaded by the fragment shader invocation are ignored. If no fragment shader is active, or if the active fragment shader does not include SampleMask in its interface, the coverage mask is not modified.

Next, the fragment alpha value and coverage mask are modified based on the line coverage factor if the lineRasterizationMode member of the VkPipelineRasterizationStateCreateInfo structure is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, and the alphaToCoverageEnable and alphaToOneEnable members of the VkPipelineMultisampleStateCreateInfo structure.

To dynamically set the alphaToCoverageEnable state, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetAlphaToCoverageEnableEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    alphaToCoverageEnable);
```

- commandBuffer is the command buffer into which the command will be recorded.
- alphaToCoverageEnable specifies the alphaToCoverageEnable state.

This command sets the alphaToCoverageEnable state for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT set in VkPipelineDynamicStateCreateInfo::pDynamicStates. Otherwise, this state is specified by the VkPipelineMultisampleStateCreateInfo::alphaToCoverageEnable value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetAlphaToCoverageEnableEXT-None-09423
  At least one of the following **must** be true:
  - The extendedDynamicState3AlphaToCoverageEnable feature is enabled
  - The shaderObject feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetAlphaToCoverageEnableEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetAlphaToCoverageEnableEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdSetAlphaToCoverageEnableEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations

- VUID-vkCmdSetAlphaToCoverageEnableEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetAlphaToCoverageEnableEXT is not affected by conditional rendering

To dynamically set the `alphaToOneEnable` state, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetAlphaToOneEnableEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    alphaToOneEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `alphaToOneEnable` specifies the `alphaToOneEnable` state.

This command sets the `alphaToOneEnable` state for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineMultisampleStateCreateInfo ::`alphaToOneEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetAlphaToOneEnableEXT-None-09423
  At least one of the following **must** be true:
  - The `extendedDynamicState3AlphaToOneEnable` feature is enabled

- The `shaderObject` feature is enabled

- VUID-vkCmdSetAlphaToOneEnableEXT-alphaToOne-07607
  If the `alphaToOne` feature is not enabled, `alphaToOneEnable` **must** be VK_FALSE

Valid Usage (Implicit)

- VUID-vkCmdSetAlphaToOneEnableEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetAlphaToOneEnableEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetAlphaToOneEnableEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetAlphaToOneEnableEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetAlphaToOneEnableEXT is not affected by conditional rendering

All alpha values in this section refer only to the alpha component of the fragment shader output that has a `Location` and `Index` decoration of zero (see the Fragment Output Interface section). If that shader output has an integer or unsigned integer type, then these operations are skipped.

If the `lineRasterizationMode` member of the VkPipelineRasterizationStateCreateInfo structure is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH and the fragment came from a line segment, then the alpha value is replaced by multiplying it by the coverage factor for the fragment

computed during smooth line rasterization.

If `alphaToCoverageEnable` is enabled, a temporary coverage mask is generated where each bit is determined by the fragment's alpha value, which is ANDed with the fragment coverage mask.

No specific algorithm is specified for converting the alpha value to a temporary coverage mask. It is intended that the number of 1's in this value be proportional to the alpha value (clamped to [0,1]), with all 1's corresponding to a value of 1.0 and all 0's corresponding to 0.0. The algorithm **may** be different at different framebuffer coordinates.

NOTE	Using different algorithms at different framebuffer coordinates **may** help to avoid artifacts caused by regular coverage sample locations.

Finally, if `alphaToOneEnable` is enabled, each alpha value is replaced by the maximum representable alpha value for fixed-point color attachments, or by 1.0 for floating-point attachments. Otherwise, the alpha values are not changed.

32.6. Depth and Stencil Operations

Pipeline state controlling the depth bounds tests, stencil test, and depth test is specified through the members of the `VkPipelineDepthStencilStateCreateInfo` structure.

The `VkPipelineDepthStencilStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineDepthStencilStateCreateInfo {
    VkStructureType                           sType;
    const void*                               pNext;
    VkPipelineDepthStencilStateCreateFlags    flags;
    VkBool32                                  depthTestEnable;
    VkBool32                                  depthWriteEnable;
    VkCompareOp                               depthCompareOp;
    VkBool32                                  depthBoundsTestEnable;
    VkBool32                                  stencilTestEnable;
    VkStencilOpState                          front;
    VkStencilOpState                          back;
    float                                     minDepthBounds;
    float                                     maxDepthBounds;
} VkPipelineDepthStencilStateCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkPipelineDepthStencilStateCreateFlagBits specifying additional depth/stencil state information.
- `depthTestEnable` controls whether depth testing is enabled.
- `depthWriteEnable` controls whether depth writes are enabled when `depthTestEnable` is VK_TRUE. Depth writes are always disabled when `depthTestEnable` is VK_FALSE.

- `depthCompareOp` is a VkCompareOp value specifying the comparison operator to use in the Depth Comparison step of the depth test.
- `depthBoundsTestEnable` controls whether depth bounds testing is enabled.
- `stencilTestEnable` controls whether stencil testing is enabled.
- `front` and `back` are VkStencilOpState values controlling the corresponding parameters of the stencil test.
- `minDepthBounds` is the minimum depth bound used in the depth bounds test.
- `maxDepthBounds` is the maximum depth bound used in the depth bounds test.

Valid Usage

- VUID-VkPipelineDepthStencilStateCreateInfo-depthBoundsTestEnable-00598
  If the `depthBounds` feature is not enabled, `depthBoundsTestEnable` **must** be VK_FALSE
- VUID-VkPipelineDepthStencilStateCreateInfo-separateStencilMaskRef-04453
  If the `VK_KHR_portability_subset` extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::`separateStencilMaskRef` is VK_FALSE, and the value of VkPipelineDepthStencilStateCreateInfo::`stencilTestEnable` is VK_TRUE, and the value of VkPipelineRasterizationStateCreateInfo::`cullMode` is VK_CULL_MODE_NONE, the value of `reference` in each of the VkStencilOpState structs in `front` and `back` **must** be the same
- VUID-VkPipelineDepthStencilStateCreateInfo-rasterizationOrderDepthAttachmentAccess-06463
  If the `rasterizationOrderDepthAttachmentAccess` feature is not enabled, `flags` **must** not include VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT_EXT
- VUID-VkPipelineDepthStencilStateCreateInfo-rasterizationOrderStencilAttachmentAccess-06464
  If the `rasterizationOrderStencilAttachmentAccess` feature is not enabled, `flags` **must** not include VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_BIT_EXT

Valid Usage (Implicit)

- VUID-VkPipelineDepthStencilStateCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_DEPTH_STENCIL_STATE_CREATE_INFO
- VUID-VkPipelineDepthStencilStateCreateInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkPipelineDepthStencilStateCreateInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkPipelineDepthStencilStateCreateFlagBits values

- VUID-VkPipelineDepthStencilStateCreateInfo-depthCompareOp-parameter
  `depthCompareOp` **must** be a valid VkCompareOp value
- VUID-VkPipelineDepthStencilStateCreateInfo-front-parameter
  `front` **must** be a valid VkStencilOpState structure
- VUID-VkPipelineDepthStencilStateCreateInfo-back-parameter
  `back` **must** be a valid VkStencilOpState structure

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineDepthStencilStateCreateFlags;
```

`VkPipelineDepthStencilStateCreateFlags` is a bitmask type for setting a mask of zero or more VkPipelineDepthStencilStateCreateFlagBits.

Bits which **can** be set in the VkPipelineDepthStencilStateCreateInfo::`flags` parameter are:

```
// Provided by VK_VERSION_1_0, VK_EXT_rasterization_order_attachment_access
typedef enum VkPipelineDepthStencilStateCreateFlagBits {
  // Provided by VK_EXT_rasterization_order_attachment_access

VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT
_EXT = 0x00000001,
  // Provided by VK_EXT_rasterization_order_attachment_access

VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_B
IT_EXT = 0x00000002,
} VkPipelineDepthStencilStateCreateFlagBits;
```

- VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT_EXT specifies that access to the depth aspects of depth/stencil and input attachments will have implicit framebuffer-local memory dependencies.
- VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_BIT_EXT specifies that access to the stencil aspects of depth/stencil and input attachments will have implicit framebuffer-local memory dependencies.

When VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT_EXT is included in a pipeline, it forms a framebuffer-local memory dependency for each fragment generated by draw commands for that pipeline with the following scopes:

- The first synchronization scope includes VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT and VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT pipeline stages executed by all previous fragments (as defined by primitive order) in the corresponding framebuffer regions including those generated by the same draw command.
- The second synchronization scope includes VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT stage executed by the generated fragment.

- The first access scope includes all writes to the depth aspect of depth/stencil attachments.
- The second access scope includes all reads from the depth aspect of input attachments.

When VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_BIT_EXT is included in a pipeline, it forms a framebuffer-local memory dependency for each fragment generated by draw commands for that pipeline with the following scopes:

- The first synchronization scope includes VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT and VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT pipeline stages executed by all previous fragments (as defined by primitive order) in the corresponding framebuffer regions including those generated by the same draw command.
- The second synchronization scope includes VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT and VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT pipeline stages executed by the generated fragment.
- The first access scope includes all writes to the stencil aspect of depth/stencil attachments.
- The second access scope includes all reads from the stencil aspect of input attachments.

32.7. Depth Bounds Test

The depth bounds test compares the depth value z_a in the depth/stencil attachment at each sample's framebuffer coordinates (x_f,y_f) and sample index i against a set of *depth bounds*.

The depth bounds are determined by two floating-point values defining a minimum (`minDepthBounds`) and maximum (`maxDepthBounds`) depth value. These values are either set by the VkPipelineDepthStencilStateCreateInfo structure during pipeline creation, or dynamically by vkCmdSetDepthBoundsTestEnable and vkCmdSetDepthBounds.

A given sample is considered within the depth bounds if z_a is in the range [`minDepthBounds`,`maxDepthBounds`]. Samples with depth attachment values outside of the depth bounds will have their coverage set to `0`.

If the depth bounds test is disabled, or if there is no depth attachment, the coverage mask is unmodified by this operation.

To dynamically enable or disable the depth bounds test, call:

```
// Provided by VK_VERSION_1_3
void vkCmdSetDepthBoundsTestEnable(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    depthBoundsTestEnable);
```

```
// Provided by VK_EXT_extended_dynamic_state, VK_EXT_shader_object
// Equivalent to vkCmdSetDepthBoundsTestEnable
void vkCmdSetDepthBoundsTestEnableEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    depthBoundsTestEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthBoundsTestEnable` specifies if the depth bounds test is enabled.

This command sets the depth bounds enable for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineDepthStencilStateCreateInfo::`depthBoundsTestEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthBoundsTestEnable-None-08971
  At least one of the following **must** be true:
  - the `extendedDynamicState` feature is enabled
  - the `shaderObject` feature is enabled
  - the value of VkApplicationInfo::`apiVersion` used to create the VkInstance parent of `commandBuffer` is greater than or equal to Version 1.3
- VUID-vkCmdSetDepthBoundsTestEnable-depthBounds-10010
  If the `depthBounds` feature is not enabled, `depthBoundsTestEnable` **must** be VK_FALSE

Valid Usage (Implicit)

- VUID-vkCmdSetDepthBoundsTestEnable-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDepthBoundsTestEnable-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetDepthBoundsTestEnable-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDepthBoundsTestEnable-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized

- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDepthBoundsTestEnable is not affected by conditional rendering

To dynamically set the depth bounds range, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetDepthBounds(
    VkCommandBuffer                             commandBuffer,
    float                                       minDepthBounds,
    float                                       maxDepthBounds);
```

- commandBuffer is the command buffer into which the command will be recorded.
- minDepthBounds is the minimum depth bound.
- maxDepthBounds is the maximum depth bound.

This command sets the depth bounds range for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_DEPTH_BOUNDS set in VkPipelineDynamicStateCreateInfo::pDynamicStates. Otherwise, this state is specified by the VkPipelineDepthStencilStateCreateInfo::minDepthBounds and VkPipelineDepthStencilStateCreateInfo::maxDepthBounds values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthBounds-minDepthBounds-00600
  If the VK_EXT_depth_range_unrestricted extension is not enabled minDepthBounds **must** be between 0.0 and 1.0, inclusive
- VUID-vkCmdSetDepthBounds-maxDepthBounds-00601
  If the VK_EXT_depth_range_unrestricted extension is not enabled maxDepthBounds **must** be between 0.0 and 1.0, inclusive
- VUID-vkCmdSetDepthBounds-minDepthBounds-10912

`minDepthBounds` **must** be less than or equal to `maxDepthBounds`

Valid Usage (Implicit)

- VUID-vkCmdSetDepthBounds-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDepthBounds-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetDepthBounds-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDepthBounds-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDepthBounds is not affected by conditional rendering

32.8. Stencil Test

The stencil test compares the stencil attachment value s_a in the depth/stencil attachment at each sample's framebuffer coordinates (x_f,y_f) and sample index i against a *stencil reference value*.

If the render pass has a fragment density map attachment and the fragment covers multiple pixels, there is an implementation-dependent association of coverage samples to stencil attachment samples within the fragment. However, if all samples in the fragment are covered, and the stencil attachment value is updated as a result of this test, all stencil attachment samples will be updated.

If the stencil test is not enabled, as specified by vkCmdSetStencilTestEnable or VkPipelineDepthStencilStateCreateInfo::`stencilTestEnable`, or if there is no stencil attachment, the coverage mask is unmodified by this operation.

The stencil test is controlled by one of two sets of stencil-related state, the front stencil state and the back stencil state. Stencil tests and writes use the back stencil state when processing fragments generated by back-facing polygons, and the front stencil state when processing fragments generated by front-facing polygons or any other primitives.

The comparison operation performed is determined by the VkCompareOp value set by vkCmdSetStencilOp::`compareOp`, or by VkStencilOpState::`compareOp` during pipeline creation.

The compare mask s_c and stencil reference value s_r of the front or the back stencil state set determine arguments of the comparison operation. s_c is set by the VkPipelineDepthStencilStateCreateInfo structure during pipeline creation, or by the vkCmdSetStencilCompareMask command. s_r is set by VkPipelineDepthStencilStateCreateInfo or by vkCmdSetStencilReference.

s_r and s_a are each independently combined with s_c using a bitwise `AND` operation to create masked reference and attachment values s'_r and s'_a. s'_r and s'_a are used as the *reference* and *test* values, respectively, in the operation specified by the VkCompareOp.

If the comparison evaluates to false, the coverage for the sample is set to `0`.

A new stencil value s_g is generated according to a stencil operation defined by VkStencilOp parameters set by vkCmdSetStencilOp or VkPipelineDepthStencilStateCreateInfo. If the stencil test fails, `failOp` defines the stencil operation used. If the stencil test passes however, the stencil op used is based on the depth test - if it passes, VkStencilOpState::`passOp` is used, otherwise VkStencilOpState::`depthFailOp` is used.

The stencil attachment value s_a is then updated with the generated stencil value s_g according to the write mask s_w defined by `writeMask` in VkPipelineDepthStencilStateCreateInfo::`front` and VkPipelineDepthStencilStateCreateInfo::`back` as:

$$s_a = (s_a \mathbin{\square} \neg s_w) \mathbin{\square} (s_g \mathbin{\square} s_w)$$

To dynamically enable or disable the stencil test, call:

```
// Provided by VK_VERSION_1_3
void vkCmdSetStencilTestEnable(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    stencilTestEnable);
```

```
// Provided by VK_EXT_extended_dynamic_state, VK_EXT_shader_object
// Equivalent to vkCmdSetStencilTestEnable
void vkCmdSetStencilTestEnableEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    stencilTestEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `stencilTestEnable` specifies if the stencil test is enabled.

This command sets the stencil test enable for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineDepthStencilStateCreateInfo ::`stencilTestEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetStencilTestEnable-None-08971
  At least one of the following **must** be true:
  - the `extendedDynamicState` feature is enabled
  - the `shaderObject` feature is enabled
  - the value of VkApplicationInfo::`apiVersion` used to create the VkInstance parent of `commandBuffer` is greater than or equal to Version 1.3

Valid Usage (Implicit)

- VUID-vkCmdSetStencilTestEnable-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetStencilTestEnable-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetStencilTestEnable-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetStencilTestEnable-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetStencilTestEnable is not affected by conditional rendering

To dynamically set the stencil operation, call:

```
// Provided by VK_VERSION_1_3
void vkCmdSetStencilOp(
    VkCommandBuffer                             commandBuffer,
    VkStencilFaceFlags                          faceMask,
    VkStencilOp                                 failOp,
    VkStencilOp                                 passOp,
    VkStencilOp                                 depthFailOp,
    VkCompareOp                                 compareOp);
```

```
// Provided by VK_EXT_extended_dynamic_state, VK_EXT_shader_object
// Equivalent to vkCmdSetStencilOp
void vkCmdSetStencilOpEXT(
    VkCommandBuffer                             commandBuffer,
    VkStencilFaceFlags                          faceMask,
    VkStencilOp                                 failOp,
    VkStencilOp                                 passOp,
    VkStencilOp                                 depthFailOp,
    VkCompareOp                                 compareOp);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `faceMask` is a bitmask of VkStencilFaceFlagBits specifying the set of stencil state for which to update the stencil operation.
- `failOp` is a VkStencilOp value specifying the action performed on samples that fail the stencil test.
- `passOp` is a VkStencilOp value specifying the action performed on samples that pass both the depth and stencil tests.
- `depthFailOp` is a VkStencilOp value specifying the action performed on samples that pass the stencil test and fail the depth test.
- `compareOp` is a VkCompareOp value specifying the comparison operator used in the stencil test.

This command sets the stencil operation for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_STENCIL_OP set in VkPipelineDynamicStateCreateInfo::pDynamicStates. Otherwise, this state is specified by the corresponding VkPipelineDepthStencilStateCreateInfo::failOp, passOp, depthFailOp, and compareOp values used to create the currently active pipeline, for both front and back faces.

Valid Usage

- VUID-vkCmdSetStencilOp-None-08971
  At least one of the following **must** be true:
  - the extendedDynamicState feature is enabled
  - the shaderObject feature is enabled
  - the value of VkApplicationInfo::apiVersion used to create the VkInstance parent of commandBuffer is greater than or equal to Version 1.3

Valid Usage (Implicit)

- VUID-vkCmdSetStencilOp-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetStencilOp-faceMask-parameter
  faceMask **must** be a valid combination of VkStencilFaceFlagBits values
- VUID-vkCmdSetStencilOp-failOp-parameter
  failOp **must** be a valid VkStencilOp value
- VUID-vkCmdSetStencilOp-passOp-parameter
  passOp **must** be a valid VkStencilOp value
- VUID-vkCmdSetStencilOp-depthFailOp-parameter
  depthFailOp **must** be a valid VkStencilOp value
- VUID-vkCmdSetStencilOp-compareOp-parameter
  compareOp **must** be a valid VkCompareOp value
- VUID-vkCmdSetStencilOp-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdSetStencilOp-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetStencilOp-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally

synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetStencilOp is not affected by conditional rendering

The VkStencilOpState structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkStencilOpState {
    VkStencilOp    failOp;
    VkStencilOp    passOp;
    VkStencilOp    depthFailOp;
    VkCompareOp    compareOp;
    uint32_t       compareMask;
    uint32_t       writeMask;
    uint32_t       reference;
} VkStencilOpState;
```

- failOp is a VkStencilOp value specifying the action performed on samples that fail the stencil test.
- passOp is a VkStencilOp value specifying the action performed on samples that pass both the depth and stencil tests.
- depthFailOp is a VkStencilOp value specifying the action performed on samples that pass the stencil test and fail the depth test.
- compareOp is a VkCompareOp value specifying the comparison operator used in the stencil test.
- compareMask selects the bits of the unsigned integer stencil values participating in the stencil test.
- writeMask selects the bits of the unsigned integer stencil values updated by the stencil test in the stencil framebuffer attachment.
- reference is an integer stencil reference value that is used in the unsigned stencil comparison.

Valid Usage (Implicit)

- VUID-VkStencilOpState-failOp-parameter

failOp **must** be a valid VkStencilOp value

- VUID-VkStencilOpState-passOp-parameter
  passOp **must** be a valid VkStencilOp value
- VUID-VkStencilOpState-depthFailOp-parameter
  depthFailOp **must** be a valid VkStencilOp value
- VUID-VkStencilOpState-compareOp-parameter
  compareOp **must** be a valid VkCompareOp value

To dynamically set the stencil compare mask, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetStencilCompareMask(
    VkCommandBuffer                             commandBuffer,
    VkStencilFaceFlags                          faceMask,
    uint32_t                                    compareMask);
```

- commandBuffer is the command buffer into which the command will be recorded.
- faceMask is a bitmask of VkStencilFaceFlagBits specifying the set of stencil state for which to update the compare mask.
- compareMask is the new value to use as the stencil compare mask.

This command sets the stencil compare mask for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK set in VkPipelineDynamicStateCreateInfo ::pDynamicStates. Otherwise, this state is specified by the VkStencilOpState::compareMask value used to create the currently active pipeline, for both front and back faces.

Valid Usage (Implicit)

- VUID-vkCmdSetStencilCompareMask-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetStencilCompareMask-faceMask-parameter
  faceMask **must** be a valid combination of VkStencilFaceFlagBits values
- VUID-vkCmdSetStencilCompareMask-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdSetStencilCompareMask-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetStencilCompareMask-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetStencilCompareMask is not affected by conditional rendering

VkStencilFaceFlagBits values are:

```
// Provided by VK_VERSION_1_0
typedef enum VkStencilFaceFlagBits {
    VK_STENCIL_FACE_FRONT_BIT = 0x00000001,
    VK_STENCIL_FACE_BACK_BIT = 0x00000002,
    VK_STENCIL_FACE_FRONT_AND_BACK = 0x00000003,
  // VK_STENCIL_FRONT_AND_BACK is a legacy alias
    VK_STENCIL_FRONT_AND_BACK = VK_STENCIL_FACE_FRONT_AND_BACK,
} VkStencilFaceFlagBits;
```

- VK_STENCIL_FACE_FRONT_BIT specifies that only the front set of stencil state is updated.
- VK_STENCIL_FACE_BACK_BIT specifies that only the back set of stencil state is updated.
- VK_STENCIL_FACE_FRONT_AND_BACK is the combination of VK_STENCIL_FACE_FRONT_BIT and VK_STENCIL_FACE_BACK_BIT, and specifies that both sets of stencil state are updated.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkStencilFaceFlags;
```

`VkStencilFaceFlags` is a bitmask type for setting a mask of zero or more VkStencilFaceFlagBits.

To dynamically set the stencil write mask, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetStencilWriteMask(
    VkCommandBuffer                             commandBuffer,
    VkStencilFaceFlags                          faceMask,
    uint32_t                                    writeMask);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `faceMask` is a bitmask of VkStencilFaceFlagBits specifying the set of stencil state for which to update the write mask, as described above for vkCmdSetStencilCompareMask.
- `writeMask` is the new value to use as the stencil write mask.

This command sets the stencil write mask for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_STENCIL_WRITE_MASK set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. Otherwise, this state is specified by the `writeMask` value used to create the currently active pipeline, for both VkPipelineDepthStencilStateCreateInfo::`front` and VkPipelineDepthStencilStateCreateInfo::`back` faces.

Valid Usage (Implicit)

- VUID-vkCmdSetStencilWriteMask-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetStencilWriteMask-faceMask-parameter
  `faceMask` **must** be a valid combination of VkStencilFaceFlagBits values
- VUID-vkCmdSetStencilWriteMask-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetStencilWriteMask-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetStencilWriteMask-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetStencilWriteMask is not affected by conditional rendering

To dynamically set the stencil reference value, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetStencilReference(
    VkCommandBuffer                             commandBuffer,
    VkStencilFaceFlags                          faceMask,
    uint32_t                                    reference);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `faceMask` is a bitmask of VkStencilFaceFlagBits specifying the set of stencil state for which to update the reference value, as described above for vkCmdSetStencilCompareMask.
- `reference` is the new value to use as the stencil reference value.

This command sets the stencil reference value for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_STENCIL_REFERENCE set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. Otherwise, this state is specified by the VkStencilOpState::`reference` value used to create the currently active pipeline, for both front and back faces.

Valid Usage (Implicit)

- VUID-vkCmdSetStencilReference-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetStencilReference-faceMask-parameter
  `faceMask` **must** be a valid combination of VkStencilFaceFlagBits values
- VUID-vkCmdSetStencilReference-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetStencilReference-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetStencilReference-videocoding

This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetStencilReference is not affected by conditional rendering

Possible values of the `failOp`, `passOp`, and `depthFailOp` members of VkStencilOpState, specifying what happens to the stored stencil value if this or certain subsequent tests fail or pass, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkStencilOp {
    VK_STENCIL_OP_KEEP = 0,
    VK_STENCIL_OP_ZERO = 1,
    VK_STENCIL_OP_REPLACE = 2,
    VK_STENCIL_OP_INCREMENT_AND_CLAMP = 3,
    VK_STENCIL_OP_DECREMENT_AND_CLAMP = 4,
    VK_STENCIL_OP_INVERT = 5,
    VK_STENCIL_OP_INCREMENT_AND_WRAP = 6,
    VK_STENCIL_OP_DECREMENT_AND_WRAP = 7,
} VkStencilOp;
```

- VK_STENCIL_OP_KEEP keeps the current value.
- VK_STENCIL_OP_ZERO sets the value to 0.
- VK_STENCIL_OP_REPLACE sets the value to `reference`.
- VK_STENCIL_OP_INCREMENT_AND_CLAMP increments the current value and clamps to the maximum representable unsigned value.
- VK_STENCIL_OP_DECREMENT_AND_CLAMP decrements the current value and clamps to 0.
- VK_STENCIL_OP_INVERT bitwise-inverts the current value.

- VK_STENCIL_OP_INCREMENT_AND_WRAP increments the current value and wraps to 0 when the maximum value would have been exceeded.
- VK_STENCIL_OP_DECREMENT_AND_WRAP decrements the current value and wraps to the maximum possible value when the value would go below 0.

For purposes of increment and decrement, the stencil bits are considered as an unsigned integer.

32.9. Depth Test

The depth test compares the depth value z_a in the depth/stencil attachment at each sample's framebuffer coordinates (x_f,y_f) and sample index i against the sample's depth value z_f. If there is no depth attachment then the depth test is skipped.

If the render pass has a fragment density map attachment and the fragment covers multiple pixels, there is an implementation-dependent association of rasterization samples to depth attachment samples within the fragment. However, if all samples in the fragment are covered, and the depth attachment value is updated as a result of this test, all depth attachment samples will be updated.

The depth test occurs in three stages, as detailed in the following sections.

32.9.1. Depth Clamping and Range Adjustment

If VkPipelineRasterizationStateCreateInfo::`depthClampEnable` is enabled, z_f is clamped to [z_{min}, z_{max}], where z_{min} = min(n,f), z_{max} = max(n,f), and n and f are the `minDepth` and `maxDepth` depth range values of the viewport used by this fragment, respectively. If the `depthClampControl` feature is enabled and VkPipelineViewportDepthClampControlCreateInfoEXT::`depthClampMode` is VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT, then z_{min} and z_{max} are equal to the `minDepthClamp` and `maxDepthClamp` values of the VkDepthClampRangeEXT structure pointed to by `pDepthClampRange`.

If VkPhysicalDeviceDepthClampZeroOneFeaturesKHR::`depthClampZeroOne` is enabled:

- If the depth attachment has a floating-point format and VK_EXT_depth_range_unrestricted is enabled then z_f is unchanged.
- Otherwise, z_f is clamped to the range [0, 1].

Otherwise:

- If z_f is not in the range [z_{min}, z_{max}], then z_f is undefined following this step.
- If the depth attachment has a fixed-point format and z_f is not in the range [0, 1], then z_f is undefined following this step.

The `VkPipelineViewportDepthClampControlCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_depth_clamp_control
typedef struct VkPipelineViewportDepthClampControlCreateInfoEXT {
    VkStructureType                sType;
    const void*                    pNext;
    VkDepthClampModeEXT            depthClampMode;
    const VkDepthClampRangeEXT*    pDepthClampRange;
} VkPipelineViewportDepthClampControlCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `depthClampMode` determines how the clamp range is determined for each viewport.
- `pDepthClampRange` sets the depth clamp range for all viewports if `depthClampMode` is VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT.

This structure extends `VkPipelineViewportStateCreateInfo` and specifies the depth clamp range used in the pipeline. If this structure is not provided in the `pNext` chain then `depthClampMode` defaults to VK_DEPTH_CLAMP_MODE_VIEWPORT_RANGE_EXT.

Valid Usage

- VUID-VkPipelineViewportDepthClampControlCreateInfoEXT-pDepthClampRange-09646
  If `depthClampMode` is VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT, and the pipeline is not created with VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT, then `pDepthClampRange` **must** be a valid pointer to a valid `VkDepthClampRangeEXT` structure

Valid Usage (Implicit)

- VUID-VkPipelineViewportDepthClampControlCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_DEPTH_CLAMP_CONTROL_CREATE_INFO_EXT
- VUID-VkPipelineViewportDepthClampControlCreateInfoEXT-depthClampMode-parameter
  `depthClampMode` **must** be a valid VkDepthClampModeEXT value
- VUID-VkPipelineViewportDepthClampControlCreateInfoEXT-pDepthClampRange-parameter
  If `pDepthClampRange` is not `NULL`, `pDepthClampRange` **must** be a valid pointer to a valid VkDepthClampRangeEXT structure

Structure Chaining

Extends the structure

- VkPipelineViewportStateCreateInfo

Possible values of VkPipelineViewportDepthClampControlCreateInfoEXT::depthClampMode, specifying which range should be used for depth clamping, are:

```
// Provided by VK_EXT_depth_clamp_control
typedef enum VkDepthClampModeEXT {
    VK_DEPTH_CLAMP_MODE_VIEWPORT_RANGE_EXT = 0,
    VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT = 1,
} VkDepthClampModeEXT;
```

- VK_DEPTH_CLAMP_MODE_VIEWPORT_RANGE_EXT specifies that the depth clamp range follows the viewport depth range. The depth clamp range of each viewport will implicitly be set to z_{min} = min(n,f) and z_{max} = max(n,f), where n and f are the minDepth and maxDepth depth range values of the viewport.
- VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT specifies that a single user-defined depth clamp range will be used for all viewports. The user-defined depth clamp range is defined by the minDepthClamp and maxDepthClamp members of VkDepthClampRangeEXT.

To dynamically set the viewport depth clamp range parameters, call:

```
// Provided by VK_EXT_depth_clamp_control, VK_EXT_depth_clamp_control with
VK_EXT_shader_object
void vkCmdSetDepthClampRangeEXT(
    VkCommandBuffer                             commandBuffer,
    VkDepthClampModeEXT                         depthClampMode,
    const VkDepthClampRangeEXT*                 pDepthClampRange);
```

- commandBuffer is the command buffer into which the command will be recorded.
- depthClampMode determines how the clamp range is determined for each viewport.
- pDepthClampRange sets the depth clamp range for all viewports if depthClampMode is VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT.

This command sets the viewport depth clamp range for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT set in VkPipelineDynamicStateCreateInfo::pDynamicStates. Otherwise, this state is specified by the VkPipelineViewportDepthClampControlCreateInfoEXT::depthClampMode value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthClampRangeEXT-pDepthClampRange-09647
  If depthClampMode is VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT, then pDepthClampRange **must** be a valid pointer to a valid VkDepthClampRangeEXT structure

Valid Usage (Implicit)

- VUID-vkCmdSetDepthClampRangeEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDepthClampRangeEXT-depthClampMode-parameter
  `depthClampMode` **must** be a valid VkDepthClampModeEXT value
- VUID-vkCmdSetDepthClampRangeEXT-pDepthClampRange-parameter
  If `pDepthClampRange` is not `NULL`, `pDepthClampRange` **must** be a valid pointer to a valid VkDepthClampRangeEXT structure
- VUID-vkCmdSetDepthClampRangeEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetDepthClampRangeEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDepthClampRangeEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDepthClampRangeEXT is not affected by conditional rendering

Both VkPipelineViewportDepthClampControlCreateInfoEXT and vkCmdSetDepthClampRangeEXT use `VkDepthClampRangeEXT` to set the viewport depth clamp range parameters.

The `VkDepthClampRangeEXT` structure is defined as:

```
// Provided by VK_EXT_depth_clamp_control
typedef struct VkDepthClampRangeEXT {
    float    minDepthClamp;
    float    maxDepthClamp;
} VkDepthClampRangeEXT;
```

- `minDepthClamp` sets z_{min} in the depth clamp range of the viewport.
- `maxDepthClamp` sets z_{max} in the depth clamp range of the viewport.

Valid Usage

- VUID-VkDepthClampRangeEXT-pDepthClampRange-00999
  `minDepthClamp` **must** be less than or equal to `maxDepthClamp`
- VUID-VkDepthClampRangeEXT-pDepthClampRange-09648
  If the `VK_EXT_depth_range_unrestricted` extension is not enabled, `minDepthClamp` **must** be greater than or equal to `0.0`
- VUID-VkDepthClampRangeEXT-pDepthClampRange-09649
  If the `VK_EXT_depth_range_unrestricted` extension is not enabled, `maxDepthClamp` **must** be less than or equal to `1.0`

32.9.2. Depth Comparison

If the depth test is not enabled, as specified by vkCmdSetDepthTestEnable or VkPipelineDepthStencilStateCreateInfo::`depthTestEnable`, then this step is skipped.

The comparison operation performed is determined by the VkCompareOp value set by vkCmdSetDepthCompareOp, or by VkPipelineDepthStencilStateCreateInfo::`depthCompareOp` during pipeline creation. z_f and z_a are used as the *reference* and *test* values, respectively, in the operation specified by the VkCompareOp.

If the comparison evaluates to false, the coverage for the sample is set to `0`.

32.9.3. Depth Attachment Writes

If depth writes are enabled, as specified by vkCmdSetDepthWriteEnable or VkPipelineDepthStencilStateCreateInfo::`depthWriteEnable`, and the comparison evaluated to true, the depth attachment value z_a is set to the sample's depth value z_f. If there is no depth attachment, no value is written.

To dynamically enable or disable the depth test, call:

```
// Provided by VK_VERSION_1_3
void vkCmdSetDepthTestEnable(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    depthTestEnable);
```

```
// Provided by VK_EXT_extended_dynamic_state, VK_EXT_shader_object
// Equivalent to vkCmdSetDepthTestEnable
void vkCmdSetDepthTestEnableEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    depthTestEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthTestEnable` specifies if the depth test is enabled.

This command sets the depth test enable for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineDepthStencilStateCreateInfo ::`depthTestEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthTestEnable-None-08971
  At least one of the following **must** be true:
  - the `extendedDynamicState` feature is enabled
  - the `shaderObject` feature is enabled
  - the value of VkApplicationInfo::`apiVersion` used to create the VkInstance parent of `commandBuffer` is greater than or equal to Version 1.3

Valid Usage (Implicit)

- VUID-vkCmdSetDepthTestEnable-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDepthTestEnable-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetDepthTestEnable-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDepthTestEnable-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDepthTestEnable is not affected by conditional rendering

To dynamically set the depth compare operator, call:

```
// Provided by VK_VERSION_1_3
void vkCmdSetDepthCompareOp(
    VkCommandBuffer                             commandBuffer,
    VkCompareOp                                 depthCompareOp);
```

```
// Provided by VK_EXT_extended_dynamic_state, VK_EXT_shader_object
// Equivalent to vkCmdSetDepthCompareOp
void vkCmdSetDepthCompareOpEXT(
    VkCommandBuffer                             commandBuffer,
    VkCompareOp                                 depthCompareOp);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthCompareOp` is a VkCompareOp value specifying the comparison operator used for the Depth Comparison step of the depth test.

This command sets the depth comparison operator for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_DEPTH_COMPARE_OP set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineDepthStencilStateCreateInfo::`depthCompareOp` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthCompareOp-None-08971
  At least one of the following **must** be true:
  - the `extendedDynamicState` feature is enabled
  - the `shaderObject` feature is enabled
  - the value of VkApplicationInfo::`apiVersion` used to create the VkInstance parent of `commandBuffer` is greater than or equal to Version 1.3

Valid Usage (Implicit)

- VUID-vkCmdSetDepthCompareOp-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDepthCompareOp-depthCompareOp-parameter
  `depthCompareOp` **must** be a valid VkCompareOp value
- VUID-vkCmdSetDepthCompareOp-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetDepthCompareOp-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDepthCompareOp-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDepthCompareOp is not affected by conditional rendering

To dynamically set the depth write enable, call:

```
// Provided by VK_VERSION_1_3
void vkCmdSetDepthWriteEnable(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    depthWriteEnable);
```

```
// Provided by VK_EXT_extended_dynamic_state, VK_EXT_shader_object
// Equivalent to vkCmdSetDepthWriteEnable
void vkCmdSetDepthWriteEnableEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    depthWriteEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthWriteEnable` specifies if depth writes are enabled.

This command sets the depth write enable for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineDepthStencilStateCreateInfo::`depthWriteEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthWriteEnable-None-08971
  At least one of the following **must** be true:
  - the `extendedDynamicState` feature is enabled
  - the `shaderObject` feature is enabled
  - the value of VkApplicationInfo::`apiVersion` used to create the VkInstance parent of `commandBuffer` is greater than or equal to Version 1.3

Valid Usage (Implicit)

- VUID-vkCmdSetDepthWriteEnable-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetDepthWriteEnable-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetDepthWriteEnable-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetDepthWriteEnable-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetDepthWriteEnable is not affected by conditional rendering

32.10. Sample Counting

Occlusion queries use query pool entries to track the number of samples that pass all the per-fragment tests. The mechanism of collecting an occlusion query value is described in Occlusion Queries.

The occlusion query sample counter increments by one for each sample with a coverage value of 1 in each fragment that survives all the per-fragment tests, including scissor, sample mask, alpha to coverage, stencil, and depth tests.

32.11. Coverage Reduction

Coverage reduction takes the coverage information for a fragment and converts that to a boolean coverage value for each color sample in each pixel covered by the fragment.

32.11.1. Pixel Coverage

Coverage for each pixel is first extracted from the total fragment coverage mask. This consists of `rasterizationSamples` unique coverage samples for each pixel in the fragment area, each with a unique sample index. If the fragment only contains a single pixel, coverage for the pixel is equivalent to the fragment coverage.

If the render pass has a fragment density map attachment and the fragment covers multiple pixels, pixel coverage is generated in an implementation-dependent manner. If all samples in the fragment are covered, all samples will be covered in each pixel coverage.

If the fragment shading rate is set, and the fragment covers multiple pixels, each pixel's coverage consists of the coverage samples with a pixel index matching that pixel, and each sample retains its unique sample index i.

32.11.2. Color Sample Coverage

Once pixel coverage is determined, coverage for each individual color sample corresponding to that pixel is determined.

The number of rasterizationSamples is identical to the number of samples in the color attachments. A color sample is covered if the pixel coverage sample with the same sample index i is covered.

If the pNext chain of VkSubpassDescription2 or VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with the multisampledRenderToSingleSampledEnable field equal to VK_TRUE, sample coverage is calculated as if the attachment has VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples samples.

Chapter 33. The Framebuffer

33.1. Blending

Blending combines the incoming *source* fragment's R, G, B, and A values with the *destination* R, G, B, and A values of each sample stored in the framebuffer at the fragment's (x_f,y_f) location. If any components are missing in the framebuffer attachment, they are replaced as described in Component Substitution. Blending is performed for each color sample covered by the fragment, rather than just once for each fragment.

Source and destination values are combined according to the blend operation, quadruplets of source and destination weighting factors determined by the blend factors, and a blend constant, to obtain a new set of R, G, B, and A values, as described below.

Blending is computed and applied separately to each color attachment used by the subpass, with separate controls for each attachment.

Prior to performing the blend operation, signed and unsigned normalized fixed-point color components undergo an implied conversion to floating-point as specified by Conversion from Normalized Fixed-Point to Floating-Point. Blending computations are treated as if carried out in floating-point, and basic blend operations are performed with a precision and dynamic range no lower than that used to represent destination components. Advanced blending operations are performed with a precision and dynamic range no lower than the smaller of that used to represent destination components or that used to represent 16-bit floating-point values.

NOTE	Blending is only defined for floating-point, UNORM, SNORM, and sRGB formats. Within those formats, the implementation may only support blending on some subset of them. Which formats support blending is indicated by VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT.

The pipeline blend state is included in the `VkPipelineColorBlendStateCreateInfo` structure during graphics pipeline creation:

The `VkPipelineColorBlendStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineColorBlendStateCreateInfo {
    VkStructureType                               sType;
    const void*                                   pNext;
    VkPipelineColorBlendStateCreateFlags          flags;
    VkBool32                                      logicOpEnable;
    VkLogicOp                                     logicOp;
    uint32_t                                      attachmentCount;
    const VkPipelineColorBlendAttachmentState*    pAttachments;
    float                                         blendConstants[4];
} VkPipelineColorBlendStateCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkPipelineColorBlendStateCreateFlagBits specifying additional color blending information.
- `logicOpEnable` controls whether to apply Logical Operations.
- `logicOp` selects which logical operation to apply.
- `attachmentCount` is the number of VkPipelineColorBlendAttachmentState elements in `pAttachments`. It is ignored if the pipeline is created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT, VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT, and VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic states set, and either VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT set or the advancedBlendCoherentOperations feature is not enabled.
- `pAttachments` is a pointer to an array of VkPipelineColorBlendAttachmentState structures defining blend state for each color attachment. It is ignored if the pipeline is created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT, VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT, and VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic states set, and either VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT set or the advancedBlendCoherentOperations feature is not enabled.
- `blendConstants` is a pointer to an array of four values used as the R, G, B, and A components of the blend constant that are used in blending, depending on the blend factor.

Valid Usage

- VUID-VkPipelineColorBlendStateCreateInfo-pAttachments-00605
  If the `independentBlend` feature is not enabled, all elements of `pAttachments` **must** be identical
- VUID-VkPipelineColorBlendStateCreateInfo-logicOpEnable-00606
  If the `logicOp` feature is not enabled, `logicOpEnable` **must** be VK_FALSE
- VUID-VkPipelineColorBlendStateCreateInfo-logicOpEnable-00607
  If `logicOpEnable` is VK_TRUE, `logicOp` **must** be a valid VkLogicOp value
- VUID-VkPipelineColorBlendStateCreateInfo-rasterizationOrderColorAttachmentAccess-06465
  If the `rasterizationOrderColorAttachmentAccess` feature is not enabled, `flags` **must** not include VK_PIPELINE_COLOR_BLEND_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_ACCESS_BIT_EXT
- VUID-VkPipelineColorBlendStateCreateInfo-pAttachments-07353
  If `attachmentCount` is not `0`, and any of VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT, VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT, or

VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT are not set, pAttachments **must** be a valid pointer to an array of attachmentCount valid VkPipelineColorBlendAttachmentState structures

Valid Usage (Implicit)

- VUID-VkPipelineColorBlendStateCreateInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PIPELINE_COLOR_BLEND_STATE_CREATE_INFO
- VUID-VkPipelineColorBlendStateCreateInfo-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkPipelineColorBlendAdvancedStateCreateInfoEXT or VkPipelineColorWriteCreateInfoEXT
- VUID-VkPipelineColorBlendStateCreateInfo-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkPipelineColorBlendStateCreateInfo-flags-parameter
  If flags is not 0, flags **must** be a valid combination of VkPipelineColorBlendStateCreateFlagBits values
- VUID-VkPipelineColorBlendStateCreateInfo-pAttachments-parameter
  If attachmentCount is not 0, and pAttachments is not NULL, pAttachments **must** be a valid pointer to an array of attachmentCount valid VkPipelineColorBlendAttachmentState structures

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineColorBlendStateCreateFlags;
```

VkPipelineColorBlendStateCreateFlags is a bitmask type for setting a mask of zero or more VkPipelineColorBlendStateCreateFlagBits.

Bits which **can** be set in the VkPipelineColorBlendStateCreateInfo::flags parameter are:

```
// Provided by VK_VERSION_1_0, VK_EXT_rasterization_order_attachment_access
typedef enum VkPipelineColorBlendStateCreateFlagBits {
  // Provided by VK_EXT_rasterization_order_attachment_access
    VK_PIPELINE_COLOR_BLEND_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_ACCESS_BIT_EXT
= 0x00000001,
} VkPipelineColorBlendStateCreateFlagBits;
```

- VK_PIPELINE_COLOR_BLEND_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_ACCESS_BIT_EXT specifies that access to color and input attachments will have implicit framebuffer-local memory dependencies, allowing applications to express custom blending operations in a fragment shader.

When

VK_PIPELINE_COLOR_BLEND_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_ACCESS_BIT_EXT is included in a pipeline, it forms a framebuffer-local memory dependency for each fragment generated by draw commands for that pipeline with the following scopes:

- The first synchronization scope includes the VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT pipeline stage executed by all previous fragments (as defined by primitive order) in the corresponding framebuffer regions including those generated by the same draw command.
- The second synchronization scope includes the VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT pipeline stage executed by the generated fragment.
- The first access scope includes all writes to color attachments.
- The second access scope includes all reads from input attachments.

The VkPipelineColorBlendAttachmentState structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineColorBlendAttachmentState {
    VkBool32                 blendEnable;
    VkBlendFactor            srcColorBlendFactor;
    VkBlendFactor            dstColorBlendFactor;
    VkBlendOp                colorBlendOp;
    VkBlendFactor            srcAlphaBlendFactor;
    VkBlendFactor            dstAlphaBlendFactor;
    VkBlendOp                alphaBlendOp;
    VkColorComponentFlags    colorWriteMask;
} VkPipelineColorBlendAttachmentState;
```

- blendEnable controls whether blending is enabled for the corresponding color attachment. If blending is not enabled, the source fragment's color for that attachment is passed through unmodified.
- srcColorBlendFactor selects which blend factor is used to determine the source factors (S_r,S_g,S_b).
- dstColorBlendFactor selects which blend factor is used to determine the destination factors (D_r,D_g,D_b).
- colorBlendOp selects which blend operation is used to calculate the RGB values to write to the color attachment.
- srcAlphaBlendFactor selects which blend factor is used to determine the source factor S_a.
- dstAlphaBlendFactor selects which blend factor is used to determine the destination factor D_a.
- alphaBlendOp selects which blend operation is used to calculate the alpha values to write to the color attachment.
- colorWriteMask is a bitmask of VkColorComponentFlagBits specifying which of the R, G, B, and/or A components are enabled for writing, as described for the Color Write Mask.

Valid Usage

- VUID-VkPipelineColorBlendAttachmentState-srcColorBlendFactor-00608
  If the dualSrcBlend feature is not enabled, srcColorBlendFactor **must** not be VK_BLEND_FACTOR_SRC1_COLOR, VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR, VK_BLEND_FACTOR_SRC1_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA
- VUID-VkPipelineColorBlendAttachmentState-dstColorBlendFactor-00609
  If the dualSrcBlend feature is not enabled, dstColorBlendFactor **must** not be VK_BLEND_FACTOR_SRC1_COLOR, VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR, VK_BLEND_FACTOR_SRC1_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA
- VUID-VkPipelineColorBlendAttachmentState-srcAlphaBlendFactor-00610
  If the dualSrcBlend feature is not enabled, srcAlphaBlendFactor **must** not be VK_BLEND_FACTOR_SRC1_COLOR, VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR, VK_BLEND_FACTOR_SRC1_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA
- VUID-VkPipelineColorBlendAttachmentState-dstAlphaBlendFactor-00611
  If the dualSrcBlend feature is not enabled, dstAlphaBlendFactor **must** not be VK_BLEND_FACTOR_SRC1_COLOR, VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR, VK_BLEND_FACTOR_SRC1_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA
- VUID-VkPipelineColorBlendAttachmentState-colorBlendOp-01406
  If either of colorBlendOp or alphaBlendOp is an advanced blend operation, then colorBlendOp **must** equal alphaBlendOp
- VUID-VkPipelineColorBlendAttachmentState-advancedBlendIndependentBlend-01407
  If VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT::advancedBlendIndependentBlend is VK_FALSE and colorBlendOp is an advanced blend operation, then colorBlendOp **must** be the same for all attachments
- VUID-VkPipelineColorBlendAttachmentState-advancedBlendIndependentBlend-01408
  If VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT::advancedBlendIndependentBlend is VK_FALSE and alphaBlendOp is an advanced blend operation, then alphaBlendOp **must** be the same for all attachments
- VUID-VkPipelineColorBlendAttachmentState-advancedBlendAllOperations-01409
  If VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT::advancedBlendAllOperations is VK_FALSE, then colorBlendOp **must** not be VK_BLEND_OP_ZERO_EXT, VK_BLEND_OP_SRC_EXT, VK_BLEND_OP_DST_EXT, VK_BLEND_OP_SRC_OVER_EXT, VK_BLEND_OP_DST_OVER_EXT, VK_BLEND_OP_SRC_IN_EXT, VK_BLEND_OP_DST_IN_EXT, VK_BLEND_OP_SRC_OUT_EXT, VK_BLEND_OP_DST_OUT_EXT, VK_BLEND_OP_SRC_ATOP_EXT, VK_BLEND_OP_DST_ATOP_EXT, VK_BLEND_OP_XOR_EXT, VK_BLEND_OP_INVERT_EXT, VK_BLEND_OP_INVERT_RGB_EXT, VK_BLEND_OP_LINEARDODGE_EXT, VK_BLEND_OP_LINEARBURN_EXT, VK_BLEND_OP_VIVIDLIGHT_EXT, VK_BLEND_OP_LINEARLIGHT_EXT, VK_BLEND_OP_PINLIGHT_EXT, VK_BLEND_OP_HARDMIX_EXT, VK_BLEND_OP_PLUS_EXT, VK_BLEND_OP_PLUS_CLAMPED_EXT, VK_BLEND_OP_PLUS_CLAMPED_ALPHA_EXT, VK_BLEND_OP_PLUS_DARKER_EXT, VK_BLEND_OP_MINUS_EXT, VK_BLEND_OP_MINUS_CLAMPED_EXT, VK_BLEND_OP_CONTRAST_EXT, VK_BLEND_OP_INVERT_OVG_EXT,

VK_BLEND_OP_RED_EXT, VK_BLEND_OP_GREEN_EXT, or VK_BLEND_OP_BLUE_EXT

- VUID-VkPipelineColorBlendAttachmentState-colorBlendOp-01410
  If `colorBlendOp` or `alphaBlendOp` is an advanced blend operation, then `colorAttachmentCount` of the subpass this pipeline is compiled against **must** be less than or equal to VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT::`advancedBlendMaxColorAttachments`
- VUID-VkPipelineColorBlendAttachmentState-constantAlphaColorBlendFactors-04454
  If the `VK_KHR_portability_subset` extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::`constantAlphaColorBlendFactors` is VK_FALSE, `srcColorBlendFactor` **must** not be VK_BLEND_FACTOR_CONSTANT_ALPHA or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA
- VUID-VkPipelineColorBlendAttachmentState-constantAlphaColorBlendFactors-04455
  If the `VK_KHR_portability_subset` extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::`constantAlphaColorBlendFactors` is VK_FALSE, `dstColorBlendFactor` **must** not be VK_BLEND_FACTOR_CONSTANT_ALPHA or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA

Valid Usage (Implicit)

- VUID-VkPipelineColorBlendAttachmentState-srcColorBlendFactor-parameter
  `srcColorBlendFactor` **must** be a valid VkBlendFactor value
- VUID-VkPipelineColorBlendAttachmentState-dstColorBlendFactor-parameter
  `dstColorBlendFactor` **must** be a valid VkBlendFactor value
- VUID-VkPipelineColorBlendAttachmentState-colorBlendOp-parameter
  `colorBlendOp` **must** be a valid VkBlendOp value
- VUID-VkPipelineColorBlendAttachmentState-srcAlphaBlendFactor-parameter
  `srcAlphaBlendFactor` **must** be a valid VkBlendFactor value
- VUID-VkPipelineColorBlendAttachmentState-dstAlphaBlendFactor-parameter
  `dstAlphaBlendFactor` **must** be a valid VkBlendFactor value
- VUID-VkPipelineColorBlendAttachmentState-alphaBlendOp-parameter
  `alphaBlendOp` **must** be a valid VkBlendOp value
- VUID-VkPipelineColorBlendAttachmentState-colorWriteMask-parameter
  If `colorWriteMask` is not `0`, `colorWriteMask` **must** be a valid combination of VkColorComponentFlagBits values

To dynamically set `blendEnable`, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetColorBlendEnableEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstAttachment,
    uint32_t                                    attachmentCount,
    const VkBool32*                             pColorBlendEnables);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `firstAttachment` the first color attachment the color blending enable applies to.
- `attachmentCount` the number of color blending enables in the `pColorBlendEnables` array.
- `pColorBlendEnables` an array of booleans to indicate whether color blending is enabled for the corresponding attachment.

This command sets the color blending enable of the specified color attachments for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineColorBlendAttachmentState ::`blendEnable` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetColorBlendEnableEXT-None-09423
  At least one of the following **must** be true:
  - The `extendedDynamicState3ColorBlendEnable` feature is enabled
  - The `shaderObject` feature is enabled
- VUID-vkCmdSetColorBlendEnableEXT-firstAttachment-12525
  The sum of `firstAttachment` and `attachmentCount` **must** be less than or equal to `maxColorAttachments`

Valid Usage (Implicit)

- VUID-vkCmdSetColorBlendEnableEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetColorBlendEnableEXT-pColorBlendEnables-parameter
  `pColorBlendEnables` **must** be a valid pointer to an array of `attachmentCount` VkBool32 values
- VUID-vkCmdSetColorBlendEnableEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetColorBlendEnableEXT-commandBuffer-cmdpool
  The VkCommandPool that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetColorBlendEnableEXT-videocoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdSetColorBlendEnableEXT-attachmentCount-arraylength
  `attachmentCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetColorBlendEnableEXT is not affected by conditional rendering

To dynamically set color blend factors and operations, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetColorBlendEquationEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstAttachment,
    uint32_t                                    attachmentCount,
    const VkColorBlendEquationEXT*              pColorBlendEquations);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `firstAttachment` the first color attachment the color blend factors and operations apply to.
- `attachmentCount` the number of VkColorBlendEquationEXT elements in the `pColorBlendEquations` array.
- `pColorBlendEquations` an array of VkColorBlendEquationEXT structs that specify the color blend factors and operations for the corresponding attachments.

This command sets the color blending factors and operations of the specified attachments for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the

VkPipelineColorBlendAttachmentState::srcColorBlendFactor, VkPipelineColorBlendAttachmentState::dstColorBlendFactor, VkPipelineColorBlendAttachmentState::colorBlendOp, VkPipelineColorBlendAttachmentState::srcAlphaBlendFactor, VkPipelineColorBlendAttachmentState::dstAlphaBlendFactor, and VkPipelineColorBlendAttachmentState::alphaBlendOp values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetColorBlendEquationEXT-None-09423
  At least one of the following **must** be true:
  - The extendedDynamicState3ColorBlendEquation feature is enabled
  - The shaderObject feature is enabled
- VUID-vkCmdSetColorBlendEquationEXT-firstAttachment-12526
  The sum of firstAttachment and attachmentCount **must** be less than or equal to maxColorAttachments

Valid Usage (Implicit)

- VUID-vkCmdSetColorBlendEquationEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetColorBlendEquationEXT-pColorBlendEquations-parameter
  pColorBlendEquations **must** be a valid pointer to an array of attachmentCount valid VkColorBlendEquationEXT structures
- VUID-vkCmdSetColorBlendEquationEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdSetColorBlendEquationEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetColorBlendEquationEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetColorBlendEquationEXT-attachmentCount-arraylength
  attachmentCount **must** be greater than 0

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetColorBlendEquationEXT is not affected by conditional rendering

The VkColorBlendEquationEXT structure is defined as:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
typedef struct VkColorBlendEquationEXT {
    VkBlendFactor    srcColorBlendFactor;
    VkBlendFactor    dstColorBlendFactor;
    VkBlendOp        colorBlendOp;
    VkBlendFactor    srcAlphaBlendFactor;
    VkBlendFactor    dstAlphaBlendFactor;
    VkBlendOp        alphaBlendOp;
} VkColorBlendEquationEXT;
```

- srcColorBlendFactor selects which blend factor is used to determine the source factors (S_r,S_g,S_b).
- dstColorBlendFactor selects which blend factor is used to determine the destination factors (D_r,D_g,D_b).
- colorBlendOp selects which blend operation is used to calculate the RGB values to write to the color attachment.
- srcAlphaBlendFactor selects which blend factor is used to determine the source factor S_a.
- dstAlphaBlendFactor selects which blend factor is used to determine the destination factor D_a.
- alphaBlendOp selects which blend operation is used to calculate the alpha values to write to the color attachment.

Valid Usage

- VUID-VkColorBlendEquationEXT-dualSrcBlend-07357
  If the dualSrcBlend feature is not enabled, srcColorBlendFactor **must** not be VK_BLEND_FACTOR_SRC1_COLOR, VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR, VK_BLEND_FACTOR_SRC1_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA
- VUID-VkColorBlendEquationEXT-dualSrcBlend-07358
  If the dualSrcBlend feature is not enabled, dstColorBlendFactor **must** not be VK_BLEND_FACTOR_SRC1_COLOR, VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR,

VK_BLEND_FACTOR_SRC1_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA

- VUID-VkColorBlendEquationEXT-dualSrcBlend-07359
  If the dualSrcBlend feature is not enabled, srcAlphaBlendFactor **must** not be VK_BLEND_FACTOR_SRC1_COLOR, VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR, VK_BLEND_FACTOR_SRC1_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA
- VUID-VkColorBlendEquationEXT-dualSrcBlend-07360
  If the dualSrcBlend feature is not enabled, dstAlphaBlendFactor **must** not be VK_BLEND_FACTOR_SRC1_COLOR, VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR, VK_BLEND_FACTOR_SRC1_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA
- VUID-VkColorBlendEquationEXT-colorBlendOp-07361
  colorBlendOp and alphaBlendOp **must** not be VK_BLEND_OP_ZERO_EXT, VK_BLEND_OP_SRC_EXT, VK_BLEND_OP_DST_EXT, VK_BLEND_OP_SRC_OVER_EXT, VK_BLEND_OP_DST_OVER_EXT, VK_BLEND_OP_SRC_IN_EXT, VK_BLEND_OP_DST_IN_EXT, VK_BLEND_OP_SRC_OUT_EXT, VK_BLEND_OP_DST_OUT_EXT, VK_BLEND_OP_SRC_ATOP_EXT, VK_BLEND_OP_DST_ATOP_EXT, VK_BLEND_OP_XOR_EXT, VK_BLEND_OP_MULTIPLY_EXT, VK_BLEND_OP_SCREEN_EXT, VK_BLEND_OP_OVERLAY_EXT, VK_BLEND_OP_DARKEN_EXT, VK_BLEND_OP_LIGHTEN_EXT, VK_BLEND_OP_COLORDODGE_EXT, VK_BLEND_OP_COLORBURN_EXT, VK_BLEND_OP_HARDLIGHT_EXT, VK_BLEND_OP_SOFTLIGHT_EXT, VK_BLEND_OP_DIFFERENCE_EXT, VK_BLEND_OP_EXCLUSION_EXT, VK_BLEND_OP_INVERT_EXT, VK_BLEND_OP_INVERT_RGB_EXT, VK_BLEND_OP_LINEARDODGE_EXT, VK_BLEND_OP_LINEARBURN_EXT, VK_BLEND_OP_VIVIDLIGHT_EXT, VK_BLEND_OP_LINEARLIGHT_EXT, VK_BLEND_OP_PINLIGHT_EXT, VK_BLEND_OP_HARDMIX_EXT, VK_BLEND_OP_HSL_HUE_EXT, VK_BLEND_OP_HSL_SATURATION_EXT, VK_BLEND_OP_HSL_COLOR_EXT, VK_BLEND_OP_HSL_LUMINOSITY_EXT, VK_BLEND_OP_PLUS_EXT, VK_BLEND_OP_PLUS_CLAMPED_EXT, VK_BLEND_OP_PLUS_CLAMPED_ALPHA_EXT, VK_BLEND_OP_PLUS_DARKER_EXT, VK_BLEND_OP_MINUS_EXT, VK_BLEND_OP_MINUS_CLAMPED_EXT, VK_BLEND_OP_CONTRAST_EXT, VK_BLEND_OP_INVERT_OVG_EXT, VK_BLEND_OP_RED_EXT, VK_BLEND_OP_GREEN_EXT, or VK_BLEND_OP_BLUE_EXT
- VUID-VkColorBlendEquationEXT-constantAlphaColorBlendFactors-07362
  If the VK_KHR_portability_subset extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::constantAlphaColorBlendFactors is VK_FALSE, srcColorBlendFactor **must** not be VK_BLEND_FACTOR_CONSTANT_ALPHA or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA
- VUID-VkColorBlendEquationEXT-constantAlphaColorBlendFactors-07363
  If the VK_KHR_portability_subset extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::constantAlphaColorBlendFactors is VK_FALSE, dstColorBlendFactor **must** not be VK_BLEND_FACTOR_CONSTANT_ALPHA or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA

Valid Usage (Implicit)

- VUID-VkColorBlendEquationEXT-srcColorBlendFactor-parameter
  `srcColorBlendFactor` **must** be a valid VkBlendFactor value
- VUID-VkColorBlendEquationEXT-dstColorBlendFactor-parameter
  `dstColorBlendFactor` **must** be a valid VkBlendFactor value
- VUID-VkColorBlendEquationEXT-colorBlendOp-parameter
  `colorBlendOp` **must** be a valid VkBlendOp value
- VUID-VkColorBlendEquationEXT-srcAlphaBlendFactor-parameter
  `srcAlphaBlendFactor` **must** be a valid VkBlendFactor value
- VUID-VkColorBlendEquationEXT-dstAlphaBlendFactor-parameter
  `dstAlphaBlendFactor` **must** be a valid VkBlendFactor value
- VUID-VkColorBlendEquationEXT-alphaBlendOp-parameter
  `alphaBlendOp` **must** be a valid VkBlendOp value

To dynamically set the color write masks, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetColorWriteMaskEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstAttachment,
    uint32_t                                    attachmentCount,
    const VkColorComponentFlags*                pColorWriteMasks);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `firstAttachment` the first color attachment the color write masks apply to.
- `attachmentCount` the number of VkColorComponentFlags values in the `pColorWriteMasks` array.
- `pColorWriteMasks` an array of VkColorComponentFlags values that specify the color write masks of the corresponding attachments.

This command sets the color write masks of the specified attachments for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineColorBlendAttachmentState ::`colorWriteMask` values used to create the currently active pipeline.

NOTE

Formats with bits that are shared between components specified by VkColorComponentFlagBits, such as VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, cannot have their channels individually masked by this functionality; either all components that share bits have to be enabled, or none of them.

Valid Usage

- VUID-vkCmdSetColorWriteMaskEXT-None-09423
  At least one of the following **must** be true:
  - The `extendedDynamicState3ColorWriteMask` feature is enabled
  - The `shaderObject` feature is enabled
- VUID-vkCmdSetColorWriteMaskEXT-firstAttachment-12527
  The sum of `firstAttachment` and `attachmentCount` **must** be less than or equal to `maxColorAttachments`

Valid Usage (Implicit)

- VUID-vkCmdSetColorWriteMaskEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetColorWriteMaskEXT-pColorWriteMasks-parameter
  `pColorWriteMasks` **must** be a valid pointer to an array of `attachmentCount` valid combinations of VkColorComponentFlagBits values
- VUID-vkCmdSetColorWriteMaskEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetColorWriteMaskEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetColorWriteMaskEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetColorWriteMaskEXT-attachmentCount-arraylength
  `attachmentCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetColorWriteMaskEXT is not affected by conditional rendering

33.1.1. Blend Factors

The source and destination color and alpha blending factors are selected from the enum:

```
// Provided by VK_VERSION_1_0
typedef enum VkBlendFactor {
    VK_BLEND_FACTOR_ZERO = 0,
    VK_BLEND_FACTOR_ONE = 1,
    VK_BLEND_FACTOR_SRC_COLOR = 2,
    VK_BLEND_FACTOR_ONE_MINUS_SRC_COLOR = 3,
    VK_BLEND_FACTOR_DST_COLOR = 4,
    VK_BLEND_FACTOR_ONE_MINUS_DST_COLOR = 5,
    VK_BLEND_FACTOR_SRC_ALPHA = 6,
    VK_BLEND_FACTOR_ONE_MINUS_SRC_ALPHA = 7,
    VK_BLEND_FACTOR_DST_ALPHA = 8,
    VK_BLEND_FACTOR_ONE_MINUS_DST_ALPHA = 9,
    VK_BLEND_FACTOR_CONSTANT_COLOR = 10,
    VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR = 11,
    VK_BLEND_FACTOR_CONSTANT_ALPHA = 12,
    VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA = 13,
    VK_BLEND_FACTOR_SRC_ALPHA_SATURATE = 14,
    VK_BLEND_FACTOR_SRC1_COLOR = 15,
    VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR = 16,
    VK_BLEND_FACTOR_SRC1_ALPHA = 17,
    VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA = 18,
} VkBlendFactor;
```

The semantics of the enum values are described in the table below:

Table 39. Blend Factors

VkBlendFactor	RGB Blend Factors (S_r,S_g,S_b) or (D_r,D_g,D_b)	Alpha Blend Factor (S_a or D_a)
VK_BLEND_FACTOR_ZERO	(0,0,0)	0
VK_BLEND_FACTOR_ONE	(1,1,1)	1
VK_BLEND_FACTOR_SRC_COLOR	(R_{s0},G_{s0},B_{s0})	A_{s0}
VK_BLEND_FACTOR_ONE_MINUS_SRC_COLOR	(1-R_{s0},1-G_{s0},1-B_{s0})	1-A_{s0}
VK_BLEND_FACTOR_DST_COLOR	(R_d,G_d,B_d)	A_d
VK_BLEND_FACTOR_ONE_MINUS_DST_COLOR	(1-R_d,1-G_d,1-B_d)	1-A_d

VkBlendFactor	RGB Blend Factors (S_r,S_g,S_b) or (D_r,D_g,D_b)	Alpha Blend Factor (S_a or D_a)
VK_BLEND_FACTOR_SRC_ALPHA	(A_{s0},A_{s0},A_{s0})	A_{s0}
VK_BLEND_FACTOR_ONE_MINUS_SRC_ALPHA	($1-A_{s0}$,$1-A_{s0}$,$1-A_{s0}$)	$1-A_{s0}$
VK_BLEND_FACTOR_DST_ALPHA	(A_d,A_d,A_d)	A_d
VK_BLEND_FACTOR_ONE_MINUS_DST_ALPHA	($1-A_d$,$1-A_d$,$1-A_d$)	$1-A_d$
VK_BLEND_FACTOR_CONSTANT_COLOR	(R_c,G_c,B_c)	A_c
VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR	($1-R_c$,$1-G_c$,$1-B_c$)	$1-A_c$
VK_BLEND_FACTOR_CONSTANT_ALPHA	(A_c,A_c,A_c)	A_c
VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA	($1-A_c$,$1-A_c$,$1-A_c$)	$1-A_c$
VK_BLEND_FACTOR_SRC_ALPHA_SATURATE	(f,f,f); f = $\min(A_{s0},1-A_d)$	1
VK_BLEND_FACTOR_SRC1_COLOR	(R_{s1},G_{s1},B_{s1})	A_{s1}
VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR	($1-R_{s1}$,$1-G_{s1}$,$1-B_{s1}$)	$1-A_{s1}$
VK_BLEND_FACTOR_SRC1_ALPHA	(A_{s1},A_{s1},A_{s1})	A_{s1}
VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA	($1-A_{s1}$,$1-A_{s1}$,$1-A_{s1}$)	$1-A_{s1}$

In this table, the following conventions are used:

- R_{s0},G_{s0},B_{s0} and A_{s0} represent the first source color R, G, B, and A components, respectively, for the fragment output location corresponding to the color attachment being blended.
- R_{s1},G_{s1},B_{s1} and A_{s1} represent the second source color R, G, B, and A components, respectively, used in dual source blending modes, for the fragment output location corresponding to the color attachment being blended.
- R_d,G_d,B_d and A_d represent the R, G, B, and A components of the destination color. That is, the color currently in the corresponding color attachment for this fragment/sample.
- R_c,G_c,B_c and A_c represent the blend constant R, G, B, and A components, respectively.

To dynamically set and change the blend constants, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetBlendConstants(
    VkCommandBuffer                             commandBuffer,
    const float                                 blendConstants[4]);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `blendConstants` is a pointer to an array of four values specifying the R_c, G_c, B_c, and A_c components of the blend constant color used in blending, depending on the blend factor.

This command sets blend constants for subsequent drawing commands when drawing using

shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_BLEND_CONSTANTS set in VkPipelineDynamicStateCreateInfo::pDynamicStates. Otherwise, this state is specified by the VkPipelineColorBlendStateCreateInfo::blendConstants values used to create the currently active pipeline.

Valid Usage (Implicit)

- VUID-vkCmdSetBlendConstants-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetBlendConstants-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdSetBlendConstants-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetBlendConstants-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetBlendConstants is not affected by conditional rendering

33.1.2. Dual-Source Blending

Blend factors that use the secondary color input (R_{s1},G_{s1},B_{s1},A_{s1}) (VK_BLEND_FACTOR_SRC1_COLOR, VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR, VK_BLEND_FACTOR_SRC1_ALPHA, and VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA) **may** consume implementation resources that could otherwise be used for rendering to multiple color attachments. Therefore, the number of color attachments that **can** be used in a framebuffer **may** be lower when using dual-source

blending.

Dual-source blending is only supported if the `dualSrcBlend` feature is enabled.

The maximum number of color attachments that **can** be used in a subpass when using dual-source blending functions is implementation-dependent and is reported as the `maxFragmentDualSrcAttachments` member of `VkPhysicalDeviceLimits`.

Color outputs **can** be bound to the first and second inputs of the blender using the `Index` decoration, as described in Fragment Output Interface. If the second color input to the blender is not written in the shader, or if no output is bound to the second input of a blender, the value of the second input is undefined.

33.1.3. Blend Operations

Once the source and destination blend factors have been selected, they along with the source and destination components are passed to the blending operations. RGB and alpha components **can** use different operations. Possible values of VkBlendOp, specifying the operations, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkBlendOp {
    VK_BLEND_OP_ADD = 0,
    VK_BLEND_OP_SUBTRACT = 1,
    VK_BLEND_OP_REVERSE_SUBTRACT = 2,
    VK_BLEND_OP_MIN = 3,
    VK_BLEND_OP_MAX = 4,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_ZERO_EXT = 1000148000,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_SRC_EXT = 1000148001,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_DST_EXT = 1000148002,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_SRC_OVER_EXT = 1000148003,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_DST_OVER_EXT = 1000148004,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_SRC_IN_EXT = 1000148005,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_DST_IN_EXT = 1000148006,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_SRC_OUT_EXT = 1000148007,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_DST_OUT_EXT = 1000148008,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_SRC_ATOP_EXT = 1000148009,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_DST_ATOP_EXT = 1000148010,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_XOR_EXT = 1000148011,
```

```
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_MULTIPLY_EXT = 1000148012,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_SCREEN_EXT = 1000148013,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_OVERLAY_EXT = 1000148014,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_DARKEN_EXT = 1000148015,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_LIGHTEN_EXT = 1000148016,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_COLORDODGE_EXT = 1000148017,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_COLORBURN_EXT = 1000148018,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_HARDLIGHT_EXT = 1000148019,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_SOFTLIGHT_EXT = 1000148020,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_DIFFERENCE_EXT = 1000148021,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_EXCLUSION_EXT = 1000148022,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_INVERT_EXT = 1000148023,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_INVERT_RGB_EXT = 1000148024,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_LINEARDODGE_EXT = 1000148025,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_LINEARBURN_EXT = 1000148026,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_VIVIDLIGHT_EXT = 1000148027,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_LINEARLIGHT_EXT = 1000148028,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_PINLIGHT_EXT = 1000148029,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_HARDMIX_EXT = 1000148030,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_HSL_HUE_EXT = 1000148031,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_HSL_SATURATION_EXT = 1000148032,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_HSL_COLOR_EXT = 1000148033,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_HSL_LUMINOSITY_EXT = 1000148034,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_PLUS_EXT = 1000148035,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_PLUS_CLAMPED_EXT = 1000148036,
  // Provided by VK_EXT_blend_operation_advanced
```

```
    VK_BLEND_OP_PLUS_CLAMPED_ALPHA_EXT = 1000148037,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_PLUS_DARKER_EXT = 1000148038,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_MINUS_EXT = 1000148039,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_MINUS_CLAMPED_EXT = 1000148040,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_CONTRAST_EXT = 1000148041,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_INVERT_OVG_EXT = 1000148042,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_RED_EXT = 1000148043,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_GREEN_EXT = 1000148044,
  // Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_BLUE_EXT = 1000148045,
} VkBlendOp;
```

The semantics of the basic blend operations are described in the table below:

Table 40. Basic Blend Operations

VkBlendOp	RGB Components	Alpha Component
VK_BLEND_OP_ADD	$R = R_{s0} \times S_r + R_d \times D_r$ $G = G_{s0} \times S_g + G_d \times D_g$ $B = B_{s0} \times S_b + B_d \times D_b$	$A = A_{s0} \times S_a + A_d \times D_a$
VK_BLEND_OP_SUBTRACT	$R = R_{s0} \times S_r - R_d \times D_r$ $G = G_{s0} \times S_g - G_d \times D_g$ $B = B_{s0} \times S_b - B_d \times D_b$	$A = A_{s0} \times S_a - A_d \times D_a$
VK_BLEND_OP_REVERSE_SUBTRACT	$R = R_d \times D_r - R_{s0} \times S_r$ $G = G_d \times D_g - G_{s0} \times S_g$ $B = B_d \times D_b - B_{s0} \times S_b$	$A = A_d \times D_a - A_{s0} \times S_a$
VK_BLEND_OP_MIN	$R = \min(R_{s0},R_d)$ $G = \min(G_{s0},G_d)$ $B = \min(B_{s0},B_d)$	$A = \min(A_{s0},A_d)$
VK_BLEND_OP_MAX	$R = \max(R_{s0},R_d)$ $G = \max(G_{s0},G_d)$ $B = \max(B_{s0},B_d)$	$A = \max(A_{s0},A_d)$

In this table, the following conventions are used:

- R_{s0}, G_{s0}, B_{s0} and A_{s0} represent the first source color R, G, B, and A components, respectively.
- R_d, G_d, B_d and A_d represent the R, G, B, and A components of the destination color. That is, the color currently in the corresponding color attachment for this fragment/sample.
- S_r, S_g, S_b and S_a represent the source blend factor R, G, B, and A components, respectively.
- D_r, D_g, D_b and D_a represent the destination blend factor R, G, B, and A components, respectively.

The blending operation produces a new set of values R, G, B and A, which are written to the framebuffer attachment. If blending is not enabled for this attachment, then R, G, B and A are assigned R_{s0}, G_{s0}, B_{s0} and A_{s0}, respectively.

If the color attachment is fixed-point, the components of the source and destination values and blend factors are each clamped to [0,1] or [-1,1] respectively for an unsigned normalized or signed normalized color attachment prior to evaluating the blend operations. If the color attachment is floating-point, no clamping occurs.

If the numeric format of a framebuffer attachment uses sRGB encoding, the R, G, and B destination color values (after conversion from fixed-point to floating-point) are considered to be encoded for the sRGB color space and hence are linearized prior to their use in blending. Each R, G, and B component is converted from nonlinear to linear as described in the “sRGB EOTF” section of the Khronos Data Format Specification. If the format is not sRGB, no linearization is performed.

If the numeric format of a framebuffer attachment uses sRGB encoding, then the final R, G, and B values are converted into the nonlinear sRGB representation before being written to the framebuffer attachment as described in the “sRGB $EOTF^{-1}$” section of the Khronos Data Format

Specification.

If the numeric format of a framebuffer color attachment is not sRGB encoded, then the resulting c_s values for R, G, and B are unmodified.

The value of A is never sRGB encoded. That is, the alpha component is always stored in memory as linear.

If the framebuffer color attachment is VK_ATTACHMENT_UNUSED, no writes are performed through that attachment. Writes are not performed to framebuffer color attachments greater than or equal to the VkSubpassDescription::colorAttachmentCount or VkSubpassDescription2::colorAttachmentCount value.

33.1.4. Advanced Blend Operations

The *advanced blend operations* are those listed in tables f/X/Y/Z Advanced Blend Operations, Hue-Saturation-Luminosity Advanced Blend Operations, and Additional RGB Blend Operations.

If the pNext chain of VkPipelineColorBlendStateCreateInfo includes a VkPipelineColorBlendAdvancedStateCreateInfoEXT structure, then that structure includes parameters that affect advanced blend operations.

The VkPipelineColorBlendAdvancedStateCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_blend_operation_advanced
typedef struct VkPipelineColorBlendAdvancedStateCreateInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkBool32             srcPremultiplied;
    VkBool32             dstPremultiplied;
    VkBlendOverlapEXT    blendOverlap;
} VkPipelineColorBlendAdvancedStateCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- srcPremultiplied specifies whether the source color of the blend operation is treated as premultiplied.
- dstPremultiplied specifies whether the destination color of the blend operation is treated as premultiplied.
- blendOverlap is a VkBlendOverlapEXT value specifying how the source and destination sample's coverage is correlated.

If this structure is not present, srcPremultiplied and dstPremultiplied are both considered to be VK_TRUE, and blendOverlap is considered to be VK_BLEND_OVERLAP_UNCORRELATED_EXT.

Valid Usage

- VUID-VkPipelineColorBlendAdvancedStateCreateInfoEXT-srcPremultiplied-01424
  If the non-premultiplied source color property is not supported, `srcPremultiplied` **must** be VK_TRUE
- VUID-VkPipelineColorBlendAdvancedStateCreateInfoEXT-dstPremultiplied-01425
  If the non-premultiplied destination color property is not supported, `dstPremultiplied` **must** be VK_TRUE
- VUID-VkPipelineColorBlendAdvancedStateCreateInfoEXT-blendOverlap-01426
  If the correlated overlap property is not supported, `blendOverlap` **must** be VK_BLEND_OVERLAP_UNCORRELATED_EXT

Valid Usage (Implicit)

- VUID-VkPipelineColorBlendAdvancedStateCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_COLOR_BLEND_ADVANCED_STATE_CREATE_INFO_EXT
- VUID-VkPipelineColorBlendAdvancedStateCreateInfoEXT-blendOverlap-parameter
  `blendOverlap` **must** be a valid VkBlendOverlapEXT value

Structure Chaining

Extends the structure

- VkPipelineColorBlendStateCreateInfo

To dynamically set the advanced blend state, call:

```
// Provided by VK_EXT_blend_operation_advanced with VK_EXT_extended_dynamic_state3,
VK_EXT_blend_operation_advanced with VK_EXT_shader_object
void vkCmdSetColorBlendAdvancedEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    firstAttachment,
    uint32_t                                    attachmentCount,
    const VkColorBlendAdvancedEXT*              pColorBlendAdvanced);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `firstAttachment` the first color attachment the advanced blend parameters apply to.
- `attachmentCount` the number of VkColorBlendAdvancedEXT elements in the `pColorBlendAdvanced` array.
- `pColorBlendAdvanced` an array of VkColorBlendAdvancedEXT structs that specify the advanced color blend parameters for the corresponding attachments.

This command sets the advanced blend operation parameters of the specified attachments for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT set in VkPipelineDynamicStateCreateInfo::pDynamicStates. Otherwise, this state is specified by the VkPipelineColorBlendAdvancedStateCreateInfoEXT::srcPremultiplied, VkPipelineColorBlendAdvancedStateCreateInfoEXT::dstPremultiplied, and VkPipelineColorBlendAdvancedStateCreateInfoEXT::blendOverlap values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetColorBlendAdvancedEXT-None-09423
  At least one of the following **must** be true:
  - The extendedDynamicState3ColorBlendAdvanced feature is enabled
  - The shaderObject feature is enabled
- VUID-vkCmdSetColorBlendAdvancedEXT-firstAttachment-12528
  The sum of firstAttachment and attachmentCount **must** be less than or equal to maxColorAttachments

Valid Usage (Implicit)

- VUID-vkCmdSetColorBlendAdvancedEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetColorBlendAdvancedEXT-pColorBlendAdvanced-parameter
  pColorBlendAdvanced **must** be a valid pointer to an array of attachmentCount valid VkColorBlendAdvancedEXT structures
- VUID-vkCmdSetColorBlendAdvancedEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdSetColorBlendAdvancedEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetColorBlendAdvancedEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetColorBlendAdvancedEXT-attachmentCount-arraylength
  attachmentCount **must** be greater than 0

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetColorBlendAdvancedEXT is not affected by conditional rendering

The VkColorBlendAdvancedEXT structure is defined as:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
typedef struct VkColorBlendAdvancedEXT {
    VkBlendOp            advancedBlendOp;
    VkBool32             srcPremultiplied;
    VkBool32             dstPremultiplied;
    VkBlendOverlapEXT    blendOverlap;
    VkBool32             clampResults;
} VkColorBlendAdvancedEXT;
```

- advancedBlendOp selects which blend operation is used to calculate the RGB values to write to the color attachment.
- srcPremultiplied specifies whether the source color of the blend operation is treated as premultiplied.
- dstPremultiplied specifies whether the destination color of the blend operation is treated as premultiplied.
- blendOverlap is a VkBlendOverlapEXT value specifying how the source and destination sample's coverage is correlated.
- clampResults specifies that results **must** be clamped to the [0,1] range before writing to the attachment, which is useful when the attachment format is not normalized fixed-point.

Valid Usage

- VUID-VkColorBlendAdvancedEXT-srcPremultiplied-07505
  If the non-premultiplied source color property is not supported, srcPremultiplied **must** be VK_TRUE
- VUID-VkColorBlendAdvancedEXT-dstPremultiplied-07506
  If the non-premultiplied destination color property is not supported, dstPremultiplied **must** be VK_TRUE
- VUID-VkColorBlendAdvancedEXT-blendOverlap-07507

If the correlated overlap property is not supported, `blendOverlap` **must** be VK_BLEND_OVERLAP_UNCORRELATED_EXT

Valid Usage (Implicit)

- VUID-VkColorBlendAdvancedEXT-advancedBlendOp-parameter `advancedBlendOp` **must** be a valid VkBlendOp value
- VUID-VkColorBlendAdvancedEXT-blendOverlap-parameter `blendOverlap` **must** be a valid VkBlendOverlapEXT value

When using one of the operations in table f/X/Y/Z Advanced Blend Operations or Hue-Saturation-Luminosity Advanced Blend Operations, blending is performed according to the following equations:

$$\begin{aligned} R = & \quad f(R_s', R_d') * p_0(A_s, A_d) + & Y * R_s' * p_1(A_s, A_d) + & \quad Z * R_d' * p_2(A_s, A_d) \\ G = & \quad f(G_s', G_d') * p_0(A_s, A_d) + & Y * G_s' * p_1(A_s, A_d) + & \quad Z * G_d' * p_2(A_s, A_d) \\ B = & \quad f(B_s', B_d') * p_0(A_s, A_d) + & Y * B_s' * p_1(A_s, A_d) + & \quad Z * B_d' * p_2(A_s, A_d) \\ A = & \quad X * p_0(A_s, A_d) + & Y * p_1(A_s, A_d) + & \quad Z * p_2(A_s, A_d) \end{aligned}$$

where the function f and terms X, Y, and Z are specified in the table. The R, G, and B components of the source color used for blending are derived according to `srcPremultiplied`. If `srcPremultiplied` is VK_TRUE, the fragment color components are considered to have been premultiplied by the A component prior to blending. The base source color (R_s',G_s',B_s') is obtained by dividing through by the A component:

$$(R_s', G_s', B_s') = \begin{cases} (0, 0, 0) & A_s = 0 \\ (\frac{R_s}{A_s}, \frac{G_s}{A_s}, \frac{B_s}{A_s}) & \text{otherwise} \end{cases}$$

If `srcPremultiplied` is VK_FALSE, the fragment color components are used as the base color:

$$(R_s', G_s', B_s') = (R_s, G_s, B_s)$$

The R, G, and B components of the destination color used for blending are derived according to `dstPremultiplied`. If `dstPremultiplied` is VK_TRUE, the destination components are considered to have been premultiplied by the A component prior to blending. The base destination color (R_d',G_d',B_d') is obtained by dividing through by the A component:

$$(R_d', G_d', B_d') = \begin{cases} (0, 0, 0) & A_d = 0 \\ (\frac{R_d}{A_d}, \frac{G_d}{A_d}, \frac{B_d}{A_d}) & \text{otherwise} \end{cases}$$

If `dstPremultiplied` is VK_FALSE, the destination color components are used as the base color:

$$(R_d', G_d', B_d') = (R_d, G_d, B_d)$$

When blending using advanced blend operations, we expect that the R, G, and B components of premultiplied source and destination color inputs be stored as the product of non-premultiplied R, G, and B component values and the A component of the color. If any R, G, or B component of a premultiplied input color is non-zero and the A component is zero, the color is considered ill-formed, and the corresponding component of the blend result is undefined.

All of the advanced blend operation formulas in this chapter compute the result as a premultiplied color. If `dstPremultiplied` is VK_FALSE, that result color's R, G, and B components are divided by the A component before being written to the framebuffer. If any R, G, or B component of the color is non-zero and the A component is zero, the result is considered ill-formed, and the corresponding component of the blend result is undefined. If all components are zero, that value is unchanged.

If the A component of any input or result color is less than zero, the color is considered ill-formed, and all components of the blend result are undefined.

The weighting functions p_0, p_1, and p_2 are defined in table Advanced Blend Overlap Modes. In these functions, the A components of the source and destination colors are taken to indicate the portion of the pixel covered by the fragment (source) and the fragments previously accumulated in the pixel (destination). The functions p_0, p_1, and p_2 approximate the relative portion of the pixel covered by the intersection of the source and destination, covered only by the source, and covered only by the destination, respectively.

Possible values of VkPipelineColorBlendAdvancedStateCreateInfoEXT::`blendOverlap`, specifying the blend overlap functions, are:

```
// Provided by VK_EXT_blend_operation_advanced
typedef enum VkBlendOverlapEXT {
    VK_BLEND_OVERLAP_UNCORRELATED_EXT = 0,
    VK_BLEND_OVERLAP_DISJOINT_EXT = 1,
    VK_BLEND_OVERLAP_CONJOINT_EXT = 2,
} VkBlendOverlapEXT;
```

- VK_BLEND_OVERLAP_UNCORRELATED_EXT specifies that there is no correlation between the source and destination coverage.
- VK_BLEND_OVERLAP_CONJOINT_EXT specifies that the source and destination coverage are considered to have maximal overlap.
- VK_BLEND_OVERLAP_DISJOINT_EXT specifies that the source and destination coverage are considered to have minimal overlap.

Table 41. Advanced Blend Overlap Modes

Overlap Mode	Weighting Equations
VK_BLEND_OVERLAP_UNCORRELATED_EXT	$p_0(A_s, A_d) = A_s A_d$ $p_1(A_s, A_d) = A_s(1 - A_d)$ $p_2(A_s, A_d) = A_d(1 - A_s)$

Overlap Mode	Weighting Equations
VK_BLEND_OVERLAP_CONJOINT_EXT	$p_0(A_s, A_d) = min(A_s, A_d)$ $p_1(A_s, A_d) = max(A_s - A_d, 0)$ $p_2(A_s, A_d) = max(A_d - A_s, 0)$
VK_BLEND_OVERLAP_DISJOINT_EXT	$p_0(A_s, A_d) = max(A_s + A_d - 1, 0)$ $p_1(A_s, A_d) = min(A_s, 1 - A_d)$ $p_2(A_s, A_d) = min(A_d, 1 - A_s)$

Table 42. f/X/Y/Z Advanced Blend Operations

Mode	Blend Coefficients
VK_BLEND_OP_ZERO_EXT	$(X, Y, Z) = (0, 0, 0)$ $f(C_s, C_d) = 0$
VK_BLEND_OP_SRC_EXT	$(X, Y, Z) = (1, 1, 0)$ $f(C_s, C_d) = C_s$
VK_BLEND_OP_DST_EXT	$(X, Y, Z) = (1, 0, 1)$ $f(C_s, C_d) = C_d$
VK_BLEND_OP_SRC_OVER_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = C_s$
VK_BLEND_OP_DST_OVER_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = C_d$
VK_BLEND_OP_SRC_IN_EXT	$(X, Y, Z) = (1, 0, 0)$ $f(C_s, C_d) = C_s$
VK_BLEND_OP_DST_IN_EXT	$(X, Y, Z) = (1, 0, 0)$ $f(C_s, C_d) = C_d$
VK_BLEND_OP_SRC_OUT_EXT	$(X, Y, Z) = (0, 1, 0)$ $f(C_s, C_d) = 0$
VK_BLEND_OP_DST_OUT_EXT	$(X, Y, Z) = (0, 0, 1)$ $f(C_s, C_d) = 0$
VK_BLEND_OP_SRC_ATOP_EXT	$(X, Y, Z) = (1, 0, 1)$ $f(C_s, C_d) = C_s$
VK_BLEND_OP_DST_ATOP_EXT	$(X, Y, Z) = (1, 1, 0)$ $f(C_s, C_d) = C_d$
VK_BLEND_OP_XOR_EXT	$(X, Y, Z) = (0, 1, 1)$ $f(C_s, C_d) = 0$
VK_BLEND_OP_MULTIPLY_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = C_s C_d$

Mode	Blend Coefficients
VK_BLEND_OP_SCREEN_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = C_s + C_d - C_s C_d$
VK_BLEND_OP_OVERLAY_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 2C_s C_d & C_d \le 0.5 \\ 1 - 2(1 - C_s)(1 - C_d) & \text{otherwise} \end{cases}$
VK_BLEND_OP_DARKEN_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = min(C_s, C_d)$
VK_BLEND_OP_LIGHTEN_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = max(C_s, C_d)$
VK_BLEND_OP_COLORDODGE_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 0 & C_d \le 0 \\ min(1, \frac{C_d}{1 - C_s}) & C_d > 0 \text{and} C_s < 1 \\ 1 & C_d > 0 \text{and} C_s \ge 1 \end{cases}$
VK_BLEND_OP_COLORBURN_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 1 & C_d \ge 1 \\ 1 - min(1, \frac{1 - C_d}{C_s}) & C_d < 1 \text{and} C_s > 0 \\ 0 & C_d < 1 \text{and} C_s \le 0 \end{cases}$
VK_BLEND_OP_HARDLIGHT_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 2C_s C_d & C_s \le 0.5 \\ 1 - 2(1 - C_s)(1 - C_d) & \text{otherwise} \end{cases}$
VK_BLEND_OP_SOFTLIGHT_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} C_d - (1 - 2C_s) C_d (1 - C_d) & C_s \le 0.5 \\ C_d + (2C_s - 1) C_d ((16C_d - 12) C_d + 3) & C_s > 0.5 \text{and} C_d \le 0.25 \\ C_d + (2C_s - 1)(\sqrt{C_d} - C_d) & C_s > 0.5 \text{and} C_d > 0.25 \end{cases}$
VK_BLEND_OP_DIFFERENCE_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \lvert C_d - C_s \rvert$
VK_BLEND_OP_EXCLUSION_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = C_s + C_d - 2C_s C_d$
VK_BLEND_OP_INVERT_EXT	$(X, Y, Z) = (1, 0, 1)$ $f(C_s, C_d) = 1 - C_d$
VK_BLEND_OP_INVERT_RGB_EXT	$(X, Y, Z) = (1, 0, 1)$ $f(C_s, C_d) = C_s(1 - C_d)$
VK_BLEND_OP_LINEARDODGE_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} C_s + C_d & C_s + C_d \le 1 \\ 1 & \text{otherwise} \end{cases}$
VK_BLEND_OP_LINEARBURN_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} C_s + C_d - 1 & C_s + C_d > 1 \\ 0 & \text{otherwise} \end{cases}$

Mode	Blend Coefficients
VK_BLEND_OP_VIVIDLIGHT_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 1 - min(1, \frac{1-C_d}{2C_s}) & 0 < C_s < 0.5 \\ 0 & C_s \le 0 \\ min(1, \frac{C_d}{2(1-C_s)}) & 0.5 \le C_s < 1 \\ 1 & C_s \ge 1 \end{cases}$
VK_BLEND_OP_LINEARLIGHT_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 1 & 2C_s + C_d > 2 \\ 2C_s + C_d - 1 & 1 < 2C_s + C_d \le 2 \\ 0 & 2C_s + C_d \le 1 \end{cases}$
VK_BLEND_OP_PINLIGHT_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 0 & 2C_s - 1 > C_d \text{and} C_s < 0.5 \\ 2C_s - 1 & 2C_s - 1 > C_d \text{and} C_s \ge 0.5 \\ 2C_s & 2C_s - 1 \le C_d \text{and} C_s < 0.5C_d \\ C_d & 2C_s - 1 \le C_d \text{and} C_s \ge 0.5C_d \end{cases}$
VK_BLEND_OP_HARDMIX_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 0 & C_s + C_d < 1 \\ 1 & \text{otherwise} \end{cases}$

When using one of the HSL blend operations in table Hue-Saturation-Luminosity Advanced Blend Operations as the blend operation, the RGB color components produced by the function f are effectively obtained by converting both the non-premultiplied source and destination colors to the HSL (hue, saturation, luminosity) color space, generating a new HSL color by selecting H, S, and L components from the source or destination according to the blend operation, and then converting the result back to RGB. In the equations below, a blended RGB color is produced according to the following pseudocode:

```
float minv3(vec3 c) {
  return min(min(c.r, c.g), c.b);
}
float maxv3(vec3 c) {
  return max(max(c.r, c.g), c.b);
}
float lumv3(vec3 c) {
  return dot(c, vec3(0.30, 0.59, 0.11));
}
float satv3(vec3 c) {
  return maxv3(c) - minv3(c);
}

// If any color components are outside [0,1], adjust the color to
// get the components in range.
vec3 ClipColor(vec3 color) {
  float lum = lumv3(color);
  float mincol = minv3(color);
  float maxcol = maxv3(color);
  if (mincol < 0.0) {
    color = lum + ((color-lum)*lum) / (lum-mincol);
```

```
    }
    if (maxcol > 1.0) {
      color = lum + ((color-lum)*(1-lum)) / (maxcol-lum);
    }
    return color;
  }

  // Take the base RGB color <cbase> and override its luminosity
  // with that of the RGB color <clum>.
  vec3 SetLum(vec3 cbase, vec3 clum) {
    float lbase = lumv3(cbase);
    float llum = lumv3(clum);
    float ldiff = llum - lbase;
    vec3 color = cbase + vec3(ldiff);
    return ClipColor(color);
  }

  // Take the base RGB color <cbase> and override its saturation with
  // that of the RGB color <csat>.  Then override the luminosity of the
  // result with that of the RGB color <clum>.
  vec3 SetLumSat(vec3 cbase, vec3 csat, vec3 clum)
  {
    float minbase = minv3(cbase);
    float sbase = satv3(cbase);
    float ssat = satv3(csat);
    vec3 color;
    if (sbase > 0) {
      // Equivalent (modulo rounding errors) to setting the
      // smallest (R,G,B) component to 0, the largest to <ssat>,
      // and interpolating the "middle" component based on its
      // original value relative to the smallest/largest.
      color = (cbase - minbase) * ssat / sbase;
    } else {
      color = vec3(0.0);
    }
    return SetLum(color, clum);
  }
```

Table 43. Hue-Saturation-Luminosity Advanced Blend Operations

Mode	Result
VK_BLEND_OP_HSL_HUE_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = SetLumSat(C_s, C_d, C_d)$
VK_BLEND_OP_HSL_SATURATION_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = SetLumSat(C_d, C_s, C_d)$
VK_BLEND_OP_HSL_COLOR_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = SetLum(C_s, C_d)$

Mode	Result
VK_BLEND_OP_HSL_LUMINOSITY_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = SetLum(C_d, C_s)$

When using one of the operations in table Additional RGB Blend Operations as the blend operation, the source and destination colors used by these blending operations are interpreted according to `srcPremultiplied` and `dstPremultiplied`. The blending operations below are evaluated where the RGB source and destination color components are both considered to have been premultiplied by the corresponding A component.

$$(R_s{}', G_s{}', B_s{}') = \begin{cases} (R_s, G_s, B_s) & \text{if srcPremultiplied is VK_TRUE} \\ (R_s A_s, G_s A_s, B_s A_s) & \text{if srcPremultiplied is VK_FALSE} \end{cases}$$

$$(R_d{}', G_d{}', B_d{}') = \begin{cases} (R_d, G_d, B_d) & \text{if dstPremultiplied is VK_TRUE} \\ (R_d A_d, G_d A_d, B_d A_d) & \text{if dstPremultiplied is VK_FALSE} \end{cases}$$

Table 44. Additional RGB Blend Operations

Mode	Result
VK_BLEND_OP_PLUS_EXT	$(R, G, B, A) = (R_s{}' + R_d{}', G_s{}' + G_d{}', B_s{}' + B_d{}', A_s + A_d)$
VK_BLEND_OP_PLUS_CLAMPED_EXT	$(R, G, B, A) = (min(1, R_s{}' + R_d{}'), min(1, G_s{}' + G_d{}'), min(1, B_s{}' + B_d{}'), min(1, A_s + A_d))$
VK_BLEND_OP_PLUS_CLAMPED_ALPHA_EXT	$(R, G, B, A) = (min(min(1, A_s + A_d), R_s{}' + R_d{}'), min(min(1, A_s + A_d), G_s{}' + G_d{}'), min(min(1, A_s + A_d), B_s{}' + B_d{}'), min(1, A_s + A_d))$
VK_BLEND_OP_PLUS_DARKER_EXT	$(R, G, B, A) = (max(0, min(1, A_s + A_d) - ((A_s - R_s{}') + (A_d - R_d{}'))), max(0, min(1, A_s + A_d) - ((A_s - G_s{}') + (A_d - G_d{}'))), max(0, min(1, A_s + A_d) - ((A_s - B_s{}') + (A_d - B_d{}'))), min(1, A_s + A_d))$
VK_BLEND_OP_MINUS_EXT	$(R, G, B, A) = (R_d{}' - R_s{}', G_d{}' - G_s{}', B_d{}' - B_s{}', A_d - A_s)$
VK_BLEND_OP_MINUS_CLAMPED_EXT	$(R, G, B, A) = (max(0, R_d{}' - R_s{}'), max(0, G_d{}' - G_s{}'), max(0, B_d{}' - B_s{}'), max(0, A_d - A_s))$

Mode	Result
VK_BLEND_OP_CONTRAST_EXT	$(R, G, B, A) = (\frac{A_d}{2} + 2(R_d{}' - \frac{A_d}{2})(R_s{}' - \frac{A_s}{2}), \frac{A_d}{2} + 2(G_d{}' - \frac{A_d}{2})(G_s{}' - \frac{A_s}{2}), \frac{A_d}{2} + 2(B_d{}' - \frac{A_d}{2})(B_s{}' - \frac{A_s}{2}), A_d)$
VK_BLEND_OP_INVERT_OVG_EXT	$(R, G, B, A) = (A_s(1 - R_d{}') + (1 - A_s)R_d{}', A_s(1 - G_d{}') + (1 - A_s)G_d{}', A_s(1 - B_d{}') + (1 - A_s)B_d{}', A_s + A_d - A_sA_d)$
VK_BLEND_OP_RED_EXT	$(R, G, B, A) = (R_s{}', G_d{}', B_d{}', A_d)$
VK_BLEND_OP_GREEN_EXT	$(R, G, B, A) = (R_d{}', G_s{}', B_d{}', A_d)$
VK_BLEND_OP_BLUE_EXT	$(R, G, B, A) = (R_d{}', G_d{}', B_s{}', A_d)$

33.2. Logical Operations

The application **can** enable a *logical operation* between the fragment's color values and the existing value in the framebuffer attachment. This logical operation is applied prior to updating the framebuffer attachment. Logical operations are applied only for signed and unsigned integer and normalized integer framebuffers. Logical operations are not applied to floating-point or sRGB format color attachments.

Logical operations are controlled by the `logicOpEnable` and `logicOp` members of VkPipelineColorBlendStateCreateInfo. The `logicOpEnable` state can also be controlled by vkCmdSetLogicOpEnableEXT if the graphics pipeline is created with VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. The `logicOp` state can also be controlled by vkCmdSetLogicOpEXT if the graphics pipeline is created with VK_DYNAMIC_STATE_LOGIC_OP_EXT set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. If `logicOpEnable` is VK_TRUE, then a logical operation selected by `logicOp` is applied between each color attachment and the fragment's corresponding output value, and blending of all attachments is treated as if it were disabled. Any attachments using color formats for which logical operations are not supported simply pass through the color values unmodified. The logical operation is applied independently for each of the red, green, blue, and alpha components. The `logicOp` is selected from the following operations:

```
// Provided by VK_VERSION_1_0
typedef enum VkLogicOp {
    VK_LOGIC_OP_CLEAR = 0,
    VK_LOGIC_OP_AND = 1,
    VK_LOGIC_OP_AND_REVERSE = 2,
    VK_LOGIC_OP_COPY = 3,
    VK_LOGIC_OP_AND_INVERTED = 4,
    VK_LOGIC_OP_NO_OP = 5,
    VK_LOGIC_OP_XOR = 6,
    VK_LOGIC_OP_OR = 7,
    VK_LOGIC_OP_NOR = 8,
    VK_LOGIC_OP_EQUIVALENT = 9,
    VK_LOGIC_OP_INVERT = 10,
    VK_LOGIC_OP_OR_REVERSE = 11,
    VK_LOGIC_OP_COPY_INVERTED = 12,
    VK_LOGIC_OP_OR_INVERTED = 13,
    VK_LOGIC_OP_NAND = 14,
    VK_LOGIC_OP_SET = 15,
} VkLogicOp;
```

The logical operations supported by Vulkan are summarized in the following table in which

- ¬ is bitwise invert,
- ∧ is bitwise and,
- ∨ is bitwise or,
- ⊕ is bitwise exclusive or,
- s is the fragment's R_{s0}, G_{s0}, B_{s0} or A_{s0} component value for the fragment output corresponding to the color attachment being updated, and
- d is the color attachment's R, G, B or A component value:

Table 45. Logical Operations

Mode	Operation
VK_LOGIC_OP_CLEAR	0
VK_LOGIC_OP_AND	s ∧ d
VK_LOGIC_OP_AND_REVERSE	s ∧ ¬ d
VK_LOGIC_OP_COPY	s
VK_LOGIC_OP_AND_INVERTED	¬ s ∧ d
VK_LOGIC_OP_NO_OP	d
VK_LOGIC_OP_XOR	s ⊕ d
VK_LOGIC_OP_OR	s ∨ d
VK_LOGIC_OP_NOR	¬ (s ∨ d)
VK_LOGIC_OP_EQUIVALENT	¬ (s ⊕ d)
VK_LOGIC_OP_INVERT	¬ d
VK_LOGIC_OP_OR_REVERSE	s ∨ ¬ d
VK_LOGIC_OP_COPY_INVERTED	¬ s
VK_LOGIC_OP_OR_INVERTED	¬ s ∨ d
VK_LOGIC_OP_NAND	¬ (s ∧ d)
VK_LOGIC_OP_SET	all 1s

The result of the logical operation is then written to the color attachment as controlled by the component write mask, described in Color Write Mask.

To dynamically set whether logical operations are enabled, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetLogicOpEnableEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    logicOpEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `logicOpEnable` specifies whether logical operations are enabled.

This command sets whether logical operations are enabled for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineColorBlendStateCreateInfo ::`logicOpEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetLogicOpEnableEXT-None-09423
  At least one of the following **must** be true:
  - The `extendedDynamicState3LogicOpEnable` feature is enabled
  - The `shaderObject` feature is enabled
- VUID-vkCmdSetLogicOpEnableEXT-logicOp-07366
  If the `logicOp` feature is not enabled, `logicOpEnable` **must** be VK_FALSE

Valid Usage (Implicit)

- VUID-vkCmdSetLogicOpEnableEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetLogicOpEnableEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetLogicOpEnableEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetLogicOpEnableEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetLogicOpEnableEXT is not affected by conditional rendering

To dynamically set the logical operation to apply for blend state, call:

```
// Provided by VK_EXT_extended_dynamic_state2, VK_EXT_shader_object
void vkCmdSetLogicOpEXT(
    VkCommandBuffer                             commandBuffer,
    VkLogicOp                                   logicOp);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `logicOp` specifies the logical operation to apply for blend state.

This command sets the logical operation for blend state for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_LOGIC_OP_EXT set in VkPipelineDynamicStateCreateInfo::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineColorBlendStateCreateInfo::`logicOp` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetLogicOpEXT-None-09422
  At least one of the following **must** be true:
  - The `extendedDynamicState2LogicOp` feature is enabled
  - The `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetLogicOpEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetLogicOpEXT-logicOp-parameter
  `logicOp` **must** be a valid VkLogicOp value
- VUID-vkCmdSetLogicOpEXT-commandBuffer-recording

commandBuffer **must** be in the recording state

- VUID-vkCmdSetLogicOpEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetLogicOpEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetLogicOpEXT is not affected by conditional rendering

33.3. Color Write Mask

Bits which **can** be set in VkPipelineColorBlendAttachmentState::colorWriteMask, determining whether the final color values R, G, B and A are written to the framebuffer attachment, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkColorComponentFlagBits {
    VK_COLOR_COMPONENT_R_BIT = 0x00000001,
    VK_COLOR_COMPONENT_G_BIT = 0x00000002,
    VK_COLOR_COMPONENT_B_BIT = 0x00000004,
    VK_COLOR_COMPONENT_A_BIT = 0x00000008,
} VkColorComponentFlagBits;
```

- VK_COLOR_COMPONENT_R_BIT specifies that the R value is written to the color attachment for the appropriate sample. Otherwise, the value in memory is unmodified.
- VK_COLOR_COMPONENT_G_BIT specifies that the G value is written to the color attachment for the appropriate sample. Otherwise, the value in memory is unmodified.

- VK_COLOR_COMPONENT_B_BIT specifies that the B value is written to the color attachment for the appropriate sample. Otherwise, the value in memory is unmodified.
- VK_COLOR_COMPONENT_A_BIT specifies that the A value is written to the color attachment for the appropriate sample. Otherwise, the value in memory is unmodified.

The color write mask operation is applied regardless of whether blending is enabled.

The color write mask operation is applied only if Color Write Enable is enabled for the respective attachment. Otherwise the color write mask is ignored and writes to all components of the attachment are disabled.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkColorComponentFlags;
```

VkColorComponentFlags is a bitmask type for setting a mask of zero or more VkColorComponentFlagBits.

33.4. Color Write Enable

The VkPipelineColorWriteCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_color_write_enable
typedef struct VkPipelineColorWriteCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           attachmentCount;
    const VkBool32*    pColorWriteEnables;
} VkPipelineColorWriteCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- attachmentCount is the number of VkBool32 elements in pColorWriteEnables.
- pColorWriteEnables is a pointer to an array of per target attachment boolean values specifying whether color writes are enabled for the given attachment.

When this structure is included in the pNext chain of VkPipelineColorBlendStateCreateInfo, it defines per-attachment color write state. If this structure is not included in the pNext chain, it is equivalent to specifying this structure with attachmentCount equal to the attachmentCount member of VkPipelineColorBlendStateCreateInfo, and pColorWriteEnables pointing to an array of as many VK_TRUE values.

If the colorWriteEnable feature is not enabled, all VkBool32 elements in the pColorWriteEnables array **must** be VK_TRUE.

Color Write Enable interacts with the Color Write Mask as follows:

- If `colorWriteEnable` is VK_TRUE, writes to the attachment are determined by the `colorWriteMask`.
- If `colorWriteEnable` is VK_FALSE, the `colorWriteMask` is ignored and writes to all components of the attachment are disabled. This is equivalent to specifying a `colorWriteMask` of 0.

Valid Usage

- VUID-VkPipelineColorWriteCreateInfoEXT-pAttachments-04801
  If the `colorWriteEnable` feature is not enabled, all elements of `pColorWriteEnables` **must** be VK_TRUE
- VUID-VkPipelineColorWriteCreateInfoEXT-attachmentCount-07608
  If the pipeline is being created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT, VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT, or VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic states not set, `attachmentCount` **must** be equal to the `attachmentCount` member of the `VkPipelineColorBlendStateCreateInfo` structure specified during pipeline creation
- VUID-VkPipelineColorWriteCreateInfoEXT-attachmentCount-06655
  `attachmentCount` **must** be less than or equal to `maxColorAttachments`

Valid Usage (Implicit)

- VUID-VkPipelineColorWriteCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PIPELINE_COLOR_WRITE_CREATE_INFO_EXT
- VUID-VkPipelineColorWriteCreateInfoEXT-pColorWriteEnables-parameter
  If `attachmentCount` is not `0`, `pColorWriteEnables` **must** be a valid pointer to an array of `attachmentCount` VkBool32 values

Structure Chaining

Extends the structure

- VkPipelineColorBlendStateCreateInfo

To dynamically enable or disable writes to a color attachment, call:

```
// Provided by VK_EXT_color_write_enable
void vkCmdSetColorWriteEnableEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    attachmentCount,
    const VkBool32*                             pColorWriteEnables);
```

- `commandBuffer` is the command buffer into which the command will be recorded.

- `attachmentCount` is the number of VkBool32 elements in `pColorWriteEnables`.
- `pColorWriteEnables` is a pointer to an array of per target attachment boolean values specifying whether color writes are enabled for the given attachment.

This command sets the color write enables for subsequent drawing commands when drawing using shader objects, or when the graphics pipeline is created with VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT set in VkPipelineDynamicStateCreateInfo ::`pDynamicStates`. Otherwise, this state is specified by the VkPipelineColorWriteCreateInfoEXT ::`pColorWriteEnables` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetColorWriteEnableEXT-None-04803
  The `colorWriteEnable` feature **must** be enabled
- VUID-vkCmdSetColorWriteEnableEXT-attachmentCount-06656
  `attachmentCount` **must** be less than or equal to `maxColorAttachments`

Valid Usage (Implicit)

- VUID-vkCmdSetColorWriteEnableEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetColorWriteEnableEXT-pColorWriteEnables-parameter
  `pColorWriteEnables` **must** be a valid pointer to an array of `attachmentCount` VkBool32 values
- VUID-vkCmdSetColorWriteEnableEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdSetColorWriteEnableEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdSetColorWriteEnableEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetColorWriteEnableEXT-attachmentCount-arraylength
  `attachmentCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_GRAPHICS_BIT	State

Conditional Rendering

vkCmdSetColorWriteEnableEXT is not affected by conditional rendering

Chapter 34. Dispatching Commands

The *dispatching commands* described in this chapter provoke work in a compute pipeline. Dispatching commands are recorded into a command buffer and when executed by a queue, will produce work which executes according to the bound compute pipeline. A compute pipeline **must** be bound to a command buffer before any dispatching commands are recorded in that command buffer.

To record a dispatch, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDispatch(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    groupCountX,
    uint32_t                                    groupCountY,
    uint32_t                                    groupCountZ);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `groupCountX` is the number of local workgroups to dispatch in the X dimension.
- `groupCountY` is the number of local workgroups to dispatch in the Y dimension.
- `groupCountZ` is the number of local workgroups to dispatch in the Z dimension.

When the command is executed, a global workgroup consisting of `groupCountX` × `groupCountY` × `groupCountZ` local workgroups is assembled.

Valid Usage

- VUID-vkCmdDispatch-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDispatch-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDispatch-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDispatch-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDispatch-None-08115

If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT

- VUID-vkCmdDispatch-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDispatch-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDispatch-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDispatch-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDispatch-magFilter-09598
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDispatch-mipmapMode-04770
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDispatch-mipmapMode-09599
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDispatch-unnormalizedCoordinates-09635
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `levelCount` and

`layerCount` **must** be 1

- VUID-vkCmdDispatch-None-08609
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `viewType` **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDispatch-None-08610
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions with `ImplicitLod`, `Dref` or `Proj` in their name
- VUID-vkCmdDispatch-None-08611
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions that include a LOD bias or any offset values
- VUID-vkCmdDispatch-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDispatch-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDispatch-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDispatch-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDispatch-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDispatch-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDispatch-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports

cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2

- VUID-vkCmdDispatch-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDispatch-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDispatch-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDispatch-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDispatch-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDispatch-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDispatch-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDispatch-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDispatch-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to

the same pipeline bind point

- VUID-vkCmdDispatch-None-08612
  If the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatch-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for storageBuffers, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatch-None-08613
  If the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatch-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's viewType **must** match the Dim operand of the OpTypeImage as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDispatch-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's format and the Sampled Type operand of the OpTypeImage **must** match
- VUID-vkCmdDispatch-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDispatch-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components
- VUID-vkCmdDispatch-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDispatch-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDispatch-SampledType-04471

If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32

- VUID-vkCmdDispatch-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDispatch-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDispatch-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDispatch-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDispatch-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDispatch-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDispatch-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDispatch-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDispatch-micromap-11638

If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with flags that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR

- VUID-vkCmdDispatch-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with flags that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDispatch-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDispatch-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a OpTypeStruct decorated with Block or BufferBlock using that mapping, the calculated offset for the resource heap **must** be a multiple of bufferDescriptorAlignment
- VUID-vkCmdDispatch-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeImage or OpTypeSampledImage using that mapping, the calculated offset for the resource heap **must** be a multiple of imageDescriptorAlignment
- VUID-vkCmdDispatch-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeSampler or OpTypeSampledImage using that mapping, the calculated offset for the sampler heap **must** be a multiple of samplerDescriptorAlignment
- VUID-vkCmdDispatch-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound

to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4

- VUID-vkCmdDispatch-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDispatch-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDispatch-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdDispatch-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDispatch-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDispatch-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDispatch-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command

must not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands

- VUID-vkCmdDispatch-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDispatch-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDispatch-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDispatch-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDispatch-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDispatch-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDispatch-None-11437

If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set

- VUID-vkCmdDispatch-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDispatch-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdDispatch-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDispatch-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDispatch-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping

using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR

- VUID-vkCmdDispatch-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDispatch-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDispatch-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDispatch-None-10672
  this command **must** be called outside of a render pass instance
- VUID-vkCmdDispatch-None-10743
  If there is no bound compute pipeline, a valid `VkShaderEXT` **must** be bound to the VK_SHADER_STAGE_COMPUTE_BIT stage
- VUID-vkCmdDispatch-commandBuffer-02712
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, any resource written to by the `VkPipeline` object bound to the pipeline bind point used by this command **must** not be an unprotected resource
- VUID-vkCmdDispatch-commandBuffer-02713
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, pipeline stages other than the framebuffer-space and compute stages in the `VkPipeline` object bound to the pipeline bind point used by this command **must** not write to any resource

- VUID-vkCmdDispatch-commandBuffer-04617
  If any of the shader stages of the VkPipeline bound to the pipeline bind point used by this command uses the RayQueryKHR capability, then commandBuffer **must** not be a protected command buffer
- VUID-vkCmdDispatch-groupCountX-00386
  groupCountX **must** be less than or equal to VkPhysicalDeviceLimits::maxComputeWorkGroupCount[0]
- VUID-vkCmdDispatch-groupCountY-00387
  groupCountY **must** be less than or equal to VkPhysicalDeviceLimits::maxComputeWorkGroupCount[1]
- VUID-vkCmdDispatch-groupCountZ-00388
  groupCountZ **must** be less than or equal to VkPhysicalDeviceLimits::maxComputeWorkGroupCount[2]

Valid Usage (Implicit)

- VUID-vkCmdDispatch-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDispatch-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdDispatch-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdDispatch-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDispatch-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdDispatch is affected by conditional rendering

VkDispatchIndirect2InfoKHR is defined as:

```
// Provided by VK_KHR_device_address_commands
typedef struct VkDispatchIndirect2InfoKHR {
    VkStructureType             sType;
    const void*                 pNext;
    VkDeviceAddressRangeKHR     addressRange;
    VkAddressCommandFlagsKHR    addressFlags;
} VkDispatchIndirect2InfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- addressRange is the VkDeviceAddressRangeKHR containing dispatch parameters.
- addressFlags is a VkAddressCommandFlagsKHR value defining the flags for the address range.

Valid Usage

- VUID-VkDispatchIndirect2InfoKHR-addressRange-13097
  If the range specified by addressRange is not bound completely to memory when accessed, addressFlags **must** not include VK_ADDRESS_COMMAND_FULLY_BOUND_BIT_KHR
- VUID-VkDispatchIndirect2InfoKHR-addressRange-13098
  If the buffer from which the range specified by addressRange was queried was created with VK_BUFFER_CREATE_PROTECTED_BIT, and protectedNoFault is not supported, addressFlags **must** include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkDispatchIndirect2InfoKHR-addressRange-13099
  If the buffer from which the range specified by addressRange was queried was created without VK_BUFFER_CREATE_PROTECTED_BIT, and protectedNoFault is not supported, addressFlags **must** not include VK_ADDRESS_COMMAND_PROTECTED_BIT_KHR
- VUID-VkDispatchIndirect2InfoKHR-addressFlags-13100
  addressFlags **must** not include both VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDispatchIndirect2InfoKHR-addressRange-13122
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing addressRange, was created with VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, addressFlags **must** include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_STORAGE_BUFFER_USAGE_BIT_KHR
- VUID-VkDispatchIndirect2InfoKHR-addressRange-13123
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range

backing addressRange, was created without VK_BUFFER_USAGE_STORAGE_BUFFER_BIT, addressFlags **must** not include VK_ADDRESS_COMMAND_STORAGE_BUFFER_USAGE_BIT_KHR

- VUID-VkDispatchIndirect2InfoKHR-addressFlags-13101
  addressFlags **must** not include both VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR and VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDispatchIndirect2InfoKHR-addressRange-13124
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing addressRange, was created with VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, addressFlags **must** include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR or VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDispatchIndirect2InfoKHR-addressRange-13125
  If any buffer, which is bound to a range of VkDeviceMemory that overlaps the range backing addressRange, was created without VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT, addressFlags **must** not include VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- VUID-VkDispatchIndirect2InfoKHR-addressRange-13107
  The buffer from which addressRange was queried **must** have been created with VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag
- VUID-VkDispatchIndirect2InfoKHR-protectedNoFault-13108
  If protectedNoFault is not supported, the buffer from which addressRange was queried **must** not have been created with VK_BUFFER_CREATE_PROTECTED_BIT
- VUID-VkDispatchIndirect2InfoKHR-addressRange-13109
  addressRange.address **must** be a multiple of 4

Valid Usage (Implicit)

- VUID-VkDispatchIndirect2InfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DISPATCH_INDIRECT_2_INFO_KHR
- VUID-VkDispatchIndirect2InfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkDispatchIndirect2InfoKHR-addressFlags-parameter
  If addressFlags is not 0, addressFlags **must** be a valid combination of VkAddressCommandFlagBitsKHR values

To record an indirect dispatching command, call:

```
// Provided by VK_KHR_device_address_commands
void vkCmdDispatchIndirect2KHR(
    VkCommandBuffer                             commandBuffer,
    const VkDispatchIndirect2InfoKHR*           pInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `pInfo` is a pointer to a VkDispatchIndirect2InfoKHR structure defining parameters of this command.

`vkCmdDispatchIndirect2KHR` behaves similarly to vkCmdDispatchIndirect except that the parameters, encoded in a VkDispatchIndirectCommand structure, are read by the device from an address range during execution.

Valid Usage

- VUID-vkCmdDispatchIndirect2KHR-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDispatchIndirect2KHR-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdDispatchIndirect2KHR-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDispatchIndirect2KHR-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDispatchIndirect2KHR-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDispatchIndirect2KHR-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDispatchIndirect2KHR-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT

- VUID-vkCmdDispatchIndirect2KHR-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a VkDescriptorSetLayout that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDispatchIndirect2KHR-magFilter-04553
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDispatchIndirect2KHR-magFilter-09598
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDispatchIndirect2KHR-mipmapMode-04770
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDispatchIndirect2KHR-mipmapMode-09599
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDispatchIndirect2KHR-unnormalizedCoordinates-09635
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1
- VUID-vkCmdDispatchIndirect2KHR-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDispatchIndirect2KHR-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdDispatchIndirect2KHR-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample

a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions that include a LOD bias or any offset values

- VUID-vkCmdDispatchIndirect2KHR-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDispatchIndirect2KHR-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDispatchIndirect2KHR-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDispatchIndirect2KHR-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDispatchIndirect2KHR-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdDispatchIndirect2KHR-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDispatchIndirect2KHR-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDispatchIndirect2KHR-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDispatchIndirect2KHR-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDispatchIndirect2KHR-OpTypeImage-07029

For any VkBufferView being written as a storage texel buffer where the image format field of the OpTypeImage is Unknown, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT

- VUID-vkCmdDispatchIndirect2KHR-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the OpTypeImage is Unknown, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDispatchIndirect2KHR-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDispatchIndirect2KHR-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDispatchIndirect2KHR-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDispatchIndirect2KHR-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDispatchIndirect2KHR-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for uniformBuffers, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchIndirect2KHR-None-08612
  If the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchIndirect2KHR-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for storageBuffers, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to

the same pipeline bind point

- VUID-vkCmdDispatchIndirect2KHR-None-08613
  If the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchIndirect2KHR-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's viewType **must** match the Dim operand of the OpTypeImage as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDispatchIndirect2KHR-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's format and the Sampled Type operand of the OpTypeImage **must** match
- VUID-vkCmdDispatchIndirect2KHR-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDispatchIndirect2KHR-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components
- VUID-vkCmdDispatchIndirect2KHR-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDispatchIndirect2KHR-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDispatchIndirect2KHR-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDispatchIndirect2KHR-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDispatchIndirect2KHR-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDispatchIndirect2KHR-sparseImageInt64Atomics-04474

If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command

- VUID-vkCmdDispatchIndirect2KHR-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDispatchIndirect2KHR-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDispatchIndirect2KHR-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDispatchIndirect2KHR-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdDispatchIndirect2KHR-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDispatchIndirect2KHR-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with flags that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDispatchIndirect2KHR-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with flags that includes

VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT

- VUID-vkCmdDispatchIndirect2KHR-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDispatchIndirect2KHR-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`
- VUID-vkCmdDispatchIndirect2KHR-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdDispatchIndirect2KHR-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdDispatchIndirect2KHR-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDispatchIndirect2KHR-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping

- VUID-vkCmdDispatchIndirect2KHR-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDispatchIndirect2KHR-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdDispatchIndirect2KHR-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDispatchIndirect2KHR-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDispatchIndirect2KHR-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDispatchIndirect2KHR-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDispatchIndirect2KHR-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDispatchIndirect2KHR-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this

command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdDispatchIndirect2KHR-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDispatchIndirect2KHR-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDispatchIndirect2KHR-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDispatchIndirect2KHR-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDispatchIndirect2KHR-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDispatchIndirect2KHR-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or

VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set

- VUID-vkCmdDispatchIndirect2KHR-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdDispatchIndirect2KHR-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDispatchIndirect2KHR-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDispatchIndirect2KHR-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDispatchIndirect2KHR-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color

- VUID-vkCmdDispatchIndirect2KHR-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDispatchIndirect2KHR-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDispatchIndirect2KHR-commandBuffer-13049
  `commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdDispatchIndirect2KHR-pInfo-13050
  `pInfo->addressRange.size` **must** be greater than or equal to `sizeof`(VkDispatchIndirectCommand)

Valid Usage (Implicit)

- VUID-vkCmdDispatchIndirect2KHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDispatchIndirect2KHR-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkDispatchIndirect2InfoKHR structure
- VUID-vkCmdDispatchIndirect2KHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdDispatchIndirect2KHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdDispatchIndirect2KHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdDispatchIndirect2KHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDispatchIndirect2KHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdDispatchIndirect2KHR is affected by conditional rendering

To record an indirect dispatching command, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDispatchIndirect(
    VkCommandBuffer                             commandBuffer,
    VkBuffer                                    buffer,
    VkDeviceSize                                offset);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `buffer` is the buffer containing dispatch parameters.
- `offset` is the byte offset into `buffer` where parameters begin.

`vkCmdDispatchIndirect` behaves similarly to vkCmdDispatch except that the parameters are read by the device from a buffer during execution. The parameters of the dispatch are encoded in a VkDispatchIndirectCommand structure taken from `buffer` starting at `offset`.

Valid Usage

- VUID-vkCmdDispatchIndirect-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDispatchIndirect-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts

compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility

- VUID-vkCmdDispatchIndirect-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDispatchIndirect-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDispatchIndirect-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDispatchIndirect-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDispatchIndirect-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDispatchIndirect-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDispatchIndirect-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDispatchIndirect-magFilter-09598
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDispatchIndirect-mipmapMode-04770
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and

compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT

- VUID-vkCmdDispatchIndirect-mipmapMode-09599
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDispatchIndirect-unnormalizedCoordinates-09635
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1
- VUID-vkCmdDispatchIndirect-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDispatchIndirect-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdDispatchIndirect-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias or any offset values
- VUID-vkCmdDispatchIndirect-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDispatchIndirect-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDispatchIndirect-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDispatchIndirect-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDispatchIndirect-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with

VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY

- VUID-vkCmdDispatchIndirect-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDispatchIndirect-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDispatchIndirect-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDispatchIndirect-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDispatchIndirect-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDispatchIndirect-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDispatchIndirect-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDispatchIndirect-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDispatchIndirect-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules

- VUID-vkCmdDispatchIndirect-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDispatchIndirect-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchIndirect-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchIndirect-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchIndirect-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchIndirect-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDispatchIndirect-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match
- VUID-vkCmdDispatchIndirect-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDispatchIndirect-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using

OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components

- VUID-vkCmdDispatchIndirect-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDispatchIndirect-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDispatchIndirect-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDispatchIndirect-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDispatchIndirect-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDispatchIndirect-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDispatchIndirect-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDispatchIndirect-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDispatchIndirect-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDispatchIndirect-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a

VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE

- VUID-vkCmdDispatchIndirect-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDispatchIndirect-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with flags that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDispatchIndirect-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with flags that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDispatchIndirect-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDispatchIndirect-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a OpTypeStruct decorated with Block or BufferBlock using that mapping, the calculated offset for the resource heap **must** be a multiple of bufferDescriptorAlignment
- VUID-vkCmdDispatchIndirect-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeImage or OpTypeSampledImage using that mapping, the calculated offset for the resource heap **must** be a multiple of imageDescriptorAlignment

- VUID-vkCmdDispatchIndirect-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdDispatchIndirect-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDispatchIndirect-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDispatchIndirect-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDispatchIndirect-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdDispatchIndirect-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDispatchIndirect-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound

to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress

- VUID-vkCmdDispatchIndirect-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDispatchIndirect-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDispatchIndirect-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDispatchIndirect-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDispatchIndirect-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDispatchIndirect-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDispatchIndirect-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses

a push constant value, that value **must** have been set by vkCmdPushDataEXT

- VUID-vkCmdDispatchIndirect-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdDispatchIndirect-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDispatchIndirect-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDispatchIndirect-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdDispatchIndirect-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set

- VUID-vkCmdDispatchIndirect-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdDispatchIndirect-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDispatchIndirect-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDispatchIndirect-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDispatchIndirect-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDispatchIndirect-None-10672
  this command **must** be called outside of a render pass instance
- VUID-vkCmdDispatchIndirect-None-10743
  If there is no bound compute pipeline, a valid `VkShaderEXT` **must** be bound to the

VK_SHADER_STAGE_COMPUTE_BIT stage

- VUID-vkCmdDispatchIndirect-buffer-02708
  If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdDispatchIndirect-buffer-02709
  `buffer` **must** have been created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set
- VUID-vkCmdDispatchIndirect-offset-02710
  `offset` **must** be a multiple of `4`
- VUID-vkCmdDispatchIndirect-commandBuffer-02711
  `commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdDispatchIndirect-offset-00407
  The sum of `offset` and the size of `VkDispatchIndirectCommand` **must** be less than or equal to the size of `buffer`

Valid Usage (Implicit)

- VUID-vkCmdDispatchIndirect-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDispatchIndirect-buffer-parameter
  `buffer` **must** be a valid VkBuffer handle
- VUID-vkCmdDispatchIndirect-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdDispatchIndirect-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdDispatchIndirect-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDispatchIndirect-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdDispatchIndirect-commonparent
  Both of `buffer`, and `commandBuffer` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdDispatchIndirect is affected by conditional rendering

The VkDispatchIndirectCommand structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDispatchIndirectCommand {
    uint32_t    x;
    uint32_t    y;
    uint32_t    z;
} VkDispatchIndirectCommand;
```

- x is the number of local workgroups to dispatch in the X dimension.
- y is the number of local workgroups to dispatch in the Y dimension.
- z is the number of local workgroups to dispatch in the Z dimension.

The members of VkDispatchIndirectCommand have the same meaning as the corresponding parameters of vkCmdDispatch.

Valid Usage

- VUID-VkDispatchIndirectCommand-x-00417
  x **must** be less than or equal to VkPhysicalDeviceLimits::maxComputeWorkGroupCount[0]
- VUID-VkDispatchIndirectCommand-y-00418
  y **must** be less than or equal to VkPhysicalDeviceLimits::maxComputeWorkGroupCount[1]
- VUID-VkDispatchIndirectCommand-z-00419
  z **must** be less than or equal to VkPhysicalDeviceLimits::maxComputeWorkGroupCount[2]

To record a dispatch using non-zero base values for the components of WorkgroupId, call:

```
// Provided by VK_VERSION_1_1
void vkCmdDispatchBase(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    baseGroupX,
    uint32_t                                    baseGroupY,
    uint32_t                                    baseGroupZ,
    uint32_t                                    groupCountX,
    uint32_t                                    groupCountY,
    uint32_t                                    groupCountZ);
```

```
// Provided by VK_KHR_device_group
// Equivalent to vkCmdDispatchBase
void vkCmdDispatchBaseKHR(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    baseGroupX,
    uint32_t                                    baseGroupY,
    uint32_t                                    baseGroupZ,
    uint32_t                                    groupCountX,
    uint32_t                                    groupCountY,
    uint32_t                                    groupCountZ);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `baseGroupX` is the start value for the X component of `WorkgroupId`.
- `baseGroupY` is the start value for the Y component of `WorkgroupId`.
- `baseGroupZ` is the start value for the Z component of `WorkgroupId`.
- `groupCountX` is the number of local workgroups to dispatch in the X dimension.
- `groupCountY` is the number of local workgroups to dispatch in the Y dimension.
- `groupCountZ` is the number of local workgroups to dispatch in the Z dimension.

When the command is executed, a global workgroup consisting of `groupCountX` × `groupCountY` × `groupCountZ` local workgroups is assembled, with `WorkgroupId` values ranging from [`baseGroup*`, `baseGroup*` + `groupCount*`) in each component. vkCmdDispatch is equivalent to `vkCmdDispatchBase(0,0,0,groupCountX,groupCountY,groupCountZ)`.

Valid Usage

- VUID-vkCmdDispatchBase-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDispatchBase-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility

- VUID-vkCmdDispatchBase-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdDispatchBase-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDispatchBase-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDispatchBase-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDispatchBase-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdDispatchBase-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdDispatchBase-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdDispatchBase-magFilter-09598
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDispatchBase-mipmapMode-04770
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT

- VUID-vkCmdDispatchBase-mipmapMode-09599
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdDispatchBase-unnormalizedCoordinates-09635
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDispatchBase-None-08609
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `viewType` **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdDispatchBase-None-08610
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions with `ImplicitLod`, `Dref` or `Proj` in their name
- VUID-vkCmdDispatchBase-None-08611
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions that include a LOD bias or any offset values
- VUID-vkCmdDispatchBase-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdDispatchBase-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdDispatchBase-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdDispatchBase-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdDispatchBase-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY

- VUID-vkCmdDispatchBase-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDispatchBase-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdDispatchBase-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDispatchBase-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDispatchBase-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdDispatchBase-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdDispatchBase-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdDispatchBase-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDispatchBase-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdDispatchBase-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in

the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdDispatchBase-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchBase-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchBase-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchBase-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchBase-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdDispatchBase-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match
- VUID-vkCmdDispatchBase-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDispatchBase-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have four components

- VUID-vkCmdDispatchBase-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDispatchBase-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDispatchBase-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDispatchBase-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdDispatchBase-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdDispatchBase-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDispatchBase-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdDispatchBase-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdDispatchBase-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDispatchBase-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its

specialization constant operand equal to VK_TRUE

- VUID-vkCmdDispatchBase-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdDispatchBase-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with `flags` that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdDispatchBase-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with `flags` that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdDispatchBase-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdDispatchBase-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`
- VUID-vkCmdDispatchBase-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdDispatchBase-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound

to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`

- VUID-vkCmdDispatchBase-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdDispatchBase-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDispatchBase-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdDispatchBase-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdDispatchBase-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdDispatchBase-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader

accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress

- VUID-vkCmdDispatchBase-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdDispatchBase-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdDispatchBase-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDispatchBase-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDispatchBase-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdDispatchBase-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdDispatchBase-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdDispatchBase-None-11398

If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds

- VUID-vkCmdDispatchBase-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDispatchBase-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdDispatchBase-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdDispatchBase-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdDispatchBase-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound

to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`

- VUID-vkCmdDispatchBase-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdDispatchBase-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdDispatchBase-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDispatchBase-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdDispatchBase-None-10672
  this command **must** be called outside of a render pass instance
- VUID-vkCmdDispatchBase-None-10743
  If there is no bound compute pipeline, a valid `VkShaderEXT` **must** be bound to the VK_SHADER_STAGE_COMPUTE_BIT stage
- VUID-vkCmdDispatchBase-commandBuffer-02712

If commandBuffer is a protected command buffer and protectedNoFault is not supported, any resource written to by the VkPipeline object bound to the pipeline bind point used by this command **must** not be an unprotected resource

- VUID-vkCmdDispatchBase-commandBuffer-02713
  If commandBuffer is a protected command buffer and protectedNoFault is not supported, pipeline stages other than the framebuffer-space and compute stages in the VkPipeline object bound to the pipeline bind point used by this command **must** not write to any resource
- VUID-vkCmdDispatchBase-commandBuffer-04617
  If any of the shader stages of the VkPipeline bound to the pipeline bind point used by this command uses the RayQueryKHR capability, then commandBuffer **must** not be a protected command buffer
- VUID-vkCmdDispatchBase-baseGroupX-00421
  baseGroupX **must** be less than VkPhysicalDeviceLimits::maxComputeWorkGroupCount[0]
- VUID-vkCmdDispatchBase-baseGroupX-00422
  baseGroupY **must** be less than VkPhysicalDeviceLimits::maxComputeWorkGroupCount[1]
- VUID-vkCmdDispatchBase-baseGroupZ-00423
  baseGroupZ **must** be less than VkPhysicalDeviceLimits::maxComputeWorkGroupCount[2]
- VUID-vkCmdDispatchBase-groupCountX-00424
  groupCountX **must** be less than or equal to VkPhysicalDeviceLimits::maxComputeWorkGroupCount[0] minus baseGroupX
- VUID-vkCmdDispatchBase-groupCountY-00425
  groupCountY **must** be less than or equal to VkPhysicalDeviceLimits::maxComputeWorkGroupCount[1] minus baseGroupY
- VUID-vkCmdDispatchBase-groupCountZ-00426
  groupCountZ **must** be less than or equal to VkPhysicalDeviceLimits::maxComputeWorkGroupCount[2] minus baseGroupZ
- VUID-vkCmdDispatchBase-baseGroupX-00427
  If any of baseGroupX, baseGroupY, or baseGroupZ are not zero, then the bound compute pipeline **must** have been created with the VK_PIPELINE_CREATE_DISPATCH_BASE_BIT flag or the bound compute shader object **must** have been created with the VK_SHADER_CREATE_DISPATCH_BASE_BIT_EXT flag

Valid Usage (Implicit)

- VUID-vkCmdDispatchBase-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDispatchBase-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdDispatchBase-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations

- VUID-vkCmdDispatchBase-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDispatchBase-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdDispatchBase is affected by conditional rendering

Chapter 35. Device-Generated Commands

This chapter discusses the generation of command buffer content on the device, for which these principal steps are to be taken:

- Define a layout describing the sequence of commands which should be generated.
- Optionally set up device-bindable shaders.
- Retrieve device addresses for setting buffers on the device.
- Fill one or more `VkBuffer` with the appropriate content that gets interpreted by the command layout.
- Create a `preprocess` `VkBuffer` using the device-queried allocation information.
- Optionally preprocess the input data in a separate action.
- Generate and execute the actual commands.

The preprocessing step executes in a separate logical pipeline from either graphics or compute. When preprocessing commands in a separate step they **must** be explicitly synchronized against the command execution. When not preprocessing in a separate step, the preprocessing is automatically synchronized against the command execution.

35.1. Indirect Commands Layout

The device-side command generation happens through an iterative processing of an atomic sequence comprised of command tokens, which are represented by:

```
// Provided by VK_EXT_device_generated_commands
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkIndirectCommandsLayoutEXT)
```

Each indirect command layout **must** have exactly one action command token and it **must** be the last token in the sequence.

NOTE	If the indirect commands layout contains only 1 token, it will be an action command token, and the contents of the indirect buffer will be a sequence of indirect command structures, similar to the ones used for indirect draws and dispatches. On some implementations, using indirect draws and dispatches for these cases will result in increased performance compared to using device-generated commands, due to the overhead that results from using the latter.

35.1.1. Creation and Deletion

Indirect command layouts for `VK_EXT_device_generated_commands` are created by:

```
// Provided by VK_EXT_device_generated_commands
VkResult vkCreateIndirectCommandsLayoutEXT(
    VkDevice                                    device,
    const VkIndirectCommandsLayoutCreateInfoEXT* pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkIndirectCommandsLayoutEXT*                pIndirectCommandsLayout);
```

- `device` is the logical device that creates the indirect command layout.
- `pCreateInfo` is a pointer to a VkIndirectCommandsLayoutCreateInfoEXT structure containing parameters affecting creation of the indirect command layout.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pIndirectCommandsLayout` is a pointer to a `VkIndirectCommandsLayoutEXT` handle in which the resulting indirect command layout is returned.

Valid Usage

- VUID-vkCreateIndirectCommandsLayoutEXT-deviceGeneratedCommands-11089
  The `VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT::deviceGeneratedCommands` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCreateIndirectCommandsLayoutEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateIndirectCommandsLayoutEXT-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkIndirectCommandsLayoutCreateInfoEXT structure
- VUID-vkCreateIndirectCommandsLayoutEXT-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateIndirectCommandsLayoutEXT-pIndirectCommandsLayout-parameter
  `pIndirectCommandsLayout` **must** be a valid pointer to a VkIndirectCommandsLayoutEXT handle
- VUID-vkCreateIndirectCommandsLayoutEXT-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkIndirectCommandsLayoutCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectCommandsLayoutCreateInfoEXT {
    VkStructureType                                 sType;
    const void*                                     pNext;
    VkIndirectCommandsLayoutUsageFlagsEXT           flags;
    VkShaderStageFlags                              shaderStages;
    uint32_t                                        indirectStride;
    VkPipelineLayout                                pipelineLayout;
    uint32_t                                        tokenCount;
    const VkIndirectCommandsLayoutTokenEXT*         pTokens;
} VkIndirectCommandsLayoutCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask of VkIndirectCommandsLayoutUsageFlagBitsEXT specifying usage rules for this layout.
- shaderStages is the VkShaderStageFlags that this layout supports.
- indirectStride is the distance in bytes between sequences in the indirect buffer.
- pipelineLayout is the optional VkPipelineLayout that tokens in this layout use. If the dynamicGeneratedPipelineLayout feature is enabled, pipelineLayout **can** be VK_NULL_HANDLE and the layout **must** be specified by chaining the VkPipelineLayoutCreateInfo structure off the pNext. If the descriptorHeap feature is enabled, pipelineLayout **can** be VK_NULL_HANDLE without providing a VkPipelineLayoutCreateInfo structure.
- tokenCount is the length of the individual command sequence.
- pTokens is a pointer to an array of VkIndirectCommandsLayoutTokenEXT describing each command token in detail.

The following code illustrates some of the flags:

```
void cmdProcessAllSequences(cmd, indirectExecutionSet, indirectCommandsLayout,
indirectAddress, sequencesCount)
{
  for (s = 0; s < sequencesCount; s++)
  {
    sUsed = s;
```

```
    if (indirectCommandsLayout.flags &
VK_INDIRECT_COMMANDS_LAYOUT_USAGE_UNORDERED_SEQUENCES_BIT_EXT) {
      sUsed = incoherent_implementation_dependent_permutation[ sUsed ];
    }

    cmdProcessSequence( cmd, indirectExecutionSet, indirectCommandsLayout,
indirectAddress, sUsed );
  }
}
```

When tokens are consumed, an offset is computed based on token offset and stream stride. The resulting offset is required to be aligned. The alignment for a specific token is equal to the scalar alignment of the data type as defined in Alignment Requirements, or 4, whichever is lower.

Valid Usage

- VUID-VkIndirectCommandsLayoutCreateInfoEXT-indirectStride-11090
  `indirectStride` **must** be less than or equal to VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::`maxIndirectCommandsIndirectStride`
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-shaderStages-11091
  `shaderStages` **must** only contain stages supported by VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::`supportedIndirectCommandsShaderStages`
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-tokenCount-11092
  `tokenCount` **must** be less than or equal to VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::`maxIndirectCommandsTokenCount`
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11093
  The number of tokens in the `pTokens` array with `type` equal to VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT **must** be less than or equal to 1
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11145
  The number of tokens in the `pTokens` array with `type` equal to VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT **must** be less than or equal to 1
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11094
  The number of tokens in the `pTokens` array with `type` equal to VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT or VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_SEQUENCE_INDEX_EXT **must** be less than or equal to 1
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11095
  If the action command token in the `pTokens` array is not an indexed draw token, then `pTokens` **must** not contain a member with `type` set to

VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT

- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11096
  If the action command token in the pTokens array is not a non-mesh draw token, then pTokens **must** not contain a member with type set to VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11097
  If the pTokens array contains multiple tokens with type equal to VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT, then there **must** be no duplicate VkIndirectCommandsVertexBufferTokenEXT::vertexBindingUnit values
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11099
  For all VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_SEQUENCE_INDEX_EXT, and VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT type tokens in pTokens, there **must** be no overlapping ranges between any specified push constant ranges
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11100
  The action command token **must** be the last token in the pTokens array
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11139
  If the pTokens array contains a VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, then this token **must** be the first token in the array
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11101
  For any element of pTokens, if type is VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT or VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT and the dynamicGeneratedPipelineLayout feature is not enabled, then the pipelineLayout **must** not be VK_NULL_HANDLE
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11102
  For any element of pTokens, if type is either VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT or VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT and pipelineLayout is VK_NULL_HANDLE, then the pNext chain **must** include a VkPipelineLayoutCreateInfo struct
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11103
  For any element of pTokens, the offset **must** be greater than or equal to the offset member of the previous tokens
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11104
  For any element of pTokens, if type is VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_COUNT_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_COUNT_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_EXT,

VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_COUNT_EXT, or VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_EXT, then `shaderStages` **must** contain graphics stages

- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11105
  For any element of `pTokens`, if `type` is VK_INDIRECT_COMMANDS_TOKEN_TYPE_DISPATCH_EXT, then `shaderStages` **must** be VK_SHADER_STAGE_COMPUTE_BIT
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11106
  For any element of `pTokens`, if `type` is VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_EXT or VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_COUNT_EXT, then `shaderStages` **must** contain VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11108
  For any element of `pTokens`, if `type` is VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT, then `shaderStages` **must** contain ray tracing stages
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-shaderStages-11109
  If `shaderStages` contains graphics stages then the state tokens in `pTokens` **must** not include VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_DISPATCH_EXT
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-shaderStages-11110
  If `shaderStages` is VK_SHADER_STAGE_COMPUTE_BIT then the state tokens in `pTokens` **must** only include VK_INDIRECT_COMMANDS_TOKEN_TYPE_DISPATCH_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_SEQUENCE_INDEX_EXT, or VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-shaderStages-11111
  If `shaderStages` contains ray tracing stages then the state tokens in `pTokens` **must** only include VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_SEQUENCE_INDEX_EXT, or VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-shaderStages-11112
  The `shaderStages` **must** only contain stages from one of the following:
  - VK_SHADER_STAGE_ALL (if the `descriptorHeap` feature is enabled)
  - graphics stages
  - VK_SHADER_STAGE_COMPUTE_BIT
  - mesh stages and VK_SHADER_STAGE_FRAGMENT_BIT
  - ray tracing stages

- VUID-VkIndirectCommandsLayoutCreateInfoEXT-shaderStages-11113
  If `shaderStages` contains VK_SHADER_STAGE_FRAGMENT_BIT, then `shaderStages` **must** also contain VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT

Valid Usage (Implicit)

- VUID-VkIndirectCommandsLayoutCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_INDIRECT_COMMANDS_LAYOUT_CREATE_INFO_EXT
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkPipelineLayoutCreateInfo
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkIndirectCommandsLayoutUsageFlagBitsEXT values
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-shaderStages-parameter
  `shaderStages` **must** be a valid combination of VkShaderStageFlagBits values
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pipelineLayout-parameter
  If `pipelineLayout` is not VK_NULL_HANDLE, `pipelineLayout` **must** be a valid VkPipelineLayout handle
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-parameter
  `pTokens` **must** be a valid pointer to an array of `tokenCount` valid VkIndirectCommandsLayoutTokenEXT structures
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-tokenCount-arraylength
  `tokenCount` **must** be greater than `0`

Bits which **can** be set in VkIndirectCommandsLayoutCreateInfoEXT::`flags`, specifying usage hints of an indirect command layout, are:

```
// Provided by VK_EXT_device_generated_commands
typedef enum VkIndirectCommandsLayoutUsageFlagBitsEXT {
    VK_INDIRECT_COMMANDS_LAYOUT_USAGE_EXPLICIT_PREPROCESS_BIT_EXT = 0x00000001,
    VK_INDIRECT_COMMANDS_LAYOUT_USAGE_UNORDERED_SEQUENCES_BIT_EXT = 0x00000002,
} VkIndirectCommandsLayoutUsageFlagBitsEXT;
```

- VK_INDIRECT_COMMANDS_LAYOUT_USAGE_EXPLICIT_PREPROCESS_BIT_EXT specifies that the layout is always used with the manual preprocessing step through calling vkCmdPreprocessGeneratedCommandsEXT and executed by vkCmdExecuteGeneratedCommandsEXT with `isPreprocessed` set to VK_TRUE.
- VK_INDIRECT_COMMANDS_LAYOUT_USAGE_UNORDERED_SEQUENCES_BIT_EXT specifies that the processing of sequences will happen at an implementation-dependent order, which is not guaranteed to be deterministic using the same input data. This flag is ignored when the

`shaderStages` is VK_SHADER_STAGE_COMPUTE_BIT as it is implied that the dispatch sequence is always unordered.

```
// Provided by VK_EXT_device_generated_commands
typedef VkFlags VkIndirectCommandsLayoutUsageFlagsEXT;
```

`VkIndirectCommandsLayoutUsageFlagsEXT` is a bitmask type for setting a mask of zero or more VkIndirectCommandsLayoutUsageFlagBitsEXT.

Indirect command layouts for `VK_EXT_device_generated_commands` are destroyed by:

```
// Provided by VK_EXT_device_generated_commands
void vkDestroyIndirectCommandsLayoutEXT(
    VkDevice                                    device,
    VkIndirectCommandsLayoutEXT                 indirectCommandsLayout,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the layout.
- `indirectCommandsLayout` is the layout to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyIndirectCommandsLayoutEXT-indirectCommandsLayout-11114
  All submitted commands that refer to `indirectCommandsLayout` **must** have completed execution
- VUID-vkDestroyIndirectCommandsLayoutEXT-indirectCommandsLayout-11115
  If `VkAllocationCallbacks` were provided when `indirectCommandsLayout` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyIndirectCommandsLayoutEXT-indirectCommandsLayout-11116
  If no `VkAllocationCallbacks` were provided when `indirectCommandsLayout` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyIndirectCommandsLayoutEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyIndirectCommandsLayoutEXT-indirectCommandsLayout-parameter
  If `indirectCommandsLayout` is not VK_NULL_HANDLE, `indirectCommandsLayout` **must** be a valid VkIndirectCommandsLayoutEXT handle
- VUID-vkDestroyIndirectCommandsLayoutEXT-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure

- VUID-vkDestroyIndirectCommandsLayoutEXT-indirectCommandsLayout-parent
  If `indirectCommandsLayout` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `indirectCommandsLayout` **must** be externally synchronized

35.1.2. Token Input Streams

For `VK_EXT_device_generated_commands`, the input streams **can** contain raw `uint32_t` values, existing indirect commands such as:

- VkDrawIndirectCommand
- VkDrawIndexedIndirectCommand
- VkDispatchIndirectCommand
- VkDrawMeshTasksIndirectCommandEXT
- VkTraceRaysIndirectCommandKHR
- VkTraceRaysIndirectCommand2KHR

or additional commands as listed below. How the data is used is described in the next section.

The `VkBindIndexBufferIndirectCommandEXT` structure specifies the input data for the VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT token.

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkBindIndexBufferIndirectCommandEXT {
    VkDeviceAddress    bufferAddress;
    uint32_t           size;
    VkIndexType        indexType;
} VkBindIndexBufferIndirectCommandEXT;
```

- `bufferAddress` specifies a physical address of the VkBuffer used as index buffer.
- `size` is the byte size range which is available for this operation from the provided address.
- `indexType` is a VkIndexType value specifying how indices are treated.

Valid Usage

- VUID-VkBindIndexBufferIndirectCommandEXT-None-11117
  The buffer's usage flags from which the address was acquired **must** have the VK_BUFFER_USAGE_INDEX_BUFFER_BIT bit set
- VUID-VkBindIndexBufferIndirectCommandEXT-bufferAddress-11118
  The `bufferAddress` **must** be aligned to the VkIndexType of the `indexType` used

Valid Usage (Implicit)

- VUID-VkBindIndexBufferIndirectCommandEXT-bufferAddress-parameter
  `bufferAddress` **must** be a valid VkDeviceAddress value
- VUID-VkBindIndexBufferIndirectCommandEXT-indexType-parameter
  `indexType` **must** be a valid VkIndexType value

The `VkBindVertexBufferIndirectCommandEXT` structure specifies the input data for the VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT token.

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkBindVertexBufferIndirectCommandEXT {
    VkDeviceAddress    bufferAddress;
    uint32_t           size;
    uint32_t           stride;
} VkBindVertexBufferIndirectCommandEXT;
```

- `bufferAddress` specifies a physical address of the VkBuffer used as vertex input binding.
- `size` is the byte size range which is available for this operation from the provided address.
- `stride` is the byte size stride for this vertex input binding as in `VkVertexInputBindingDescription` ::`stride`.

Valid Usage

- VUID-VkBindVertexBufferIndirectCommandEXT-None-11120
  The buffer's usage flag from which the address was acquired **must** have the VK_BUFFER_USAGE_VERTEX_BUFFER_BIT bit set

Valid Usage (Implicit)

- VUID-VkBindVertexBufferIndirectCommandEXT-bufferAddress-parameter
  `bufferAddress` **must** be a valid VkDeviceAddress value

The `VkDrawIndirectCountIndirectCommandEXT` structure specifies the input data for all draw-type tokens.

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkDrawIndirectCountIndirectCommandEXT {
    VkDeviceAddress    bufferAddress;
    uint32_t           stride;
    uint32_t           commandCount;
} VkDrawIndirectCountIndirectCommandEXT;
```

- `bufferAddress` specifies a physical address of the VkBuffer used for draw commands.
- `stride` is the byte size stride for the command arguments.
- `commandCount` is the number of commands to execute.

The corresponding indirect draw structure data will be read from the buffer address.

Valid Usage

- VUID-VkDrawIndirectCountIndirectCommandEXT-None-11122
  The buffer's usage flag from which the address was acquired **must** have the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT bit set

Valid Usage (Implicit)

- VUID-VkDrawIndirectCountIndirectCommandEXT-bufferAddress-parameter
  `bufferAddress` **must** be a valid VkDeviceAddress value

35.1.3. Tokenized Command Processing

The processing for `VK_EXT_device_generated_commands` is in principle illustrated below:

```
void cmdProcessSequence(cmd, indirectExecutionSet, indirectCommandsLayout,
indirectAddress, s)
{
  for (t = 0; t < indirectCommandsLayout.tokenCount; t++)
  {
    uint32_t offset  = indirectCommandsLayout.pTokens[t].offset;
    uint32_t stride  = indirectCommandsLayout.indirectStride;
    VkDeviceAddress streamData = indirectAddress;
    const void* input = streamData + stride * s + offset;

    // further details later
    indirectCommandsLayout.pTokens[t].command (cmd, indirectExecutionSet, input, s);
  }
}

void cmdProcessAllSequences(cmd, indirectExecutionSet, indirectCommandsLayout,
indirectAddress, sequencesCount)
{
  for (s = 0; s < sequencesCount; s++)
  {
    sUsed = s;

    if (indirectCommandsLayout.flags &
VK_INDIRECT_COMMANDS_LAYOUT_USAGE_UNORDERED_SEQUENCES_BIT_EXT) {
      sUsed = incoherent_implementation_dependent_permutation[ sUsed ];
```

```
    }

    cmdProcessSequence( cmd, indirectExecutionSet, indirectCommandsLayout,
indirectAddress, sUsed );
  }
}
```

The processing of each sequence is considered stateless, therefore all state changes **must** occur prior to action commands within the sequence. A single sequence is strictly targeting the VkShaderStageFlags it was created with.

The primary input data for each token is provided through VkBuffer content at preprocessing using vkCmdPreprocessGeneratedCommandsEXT or execution time using vkCmdExecuteGeneratedCommandsEXT, however some functional arguments, for example push constant layouts, are specified at layout creation time. The input size is different for each token.

Possible values of those elements of the VkIndirectCommandsLayoutCreateInfoEXT::pTokens array specifying command tokens (other elements of the array specify command parameters) are:

```
// Provided by VK_EXT_device_generated_commands
typedef enum VkIndirectCommandsTokenTypeEXT {
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT = 0,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT = 1,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT = 2,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT = 3,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT = 4,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_EXT = 5,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_EXT = 6,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_COUNT_EXT = 7,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_COUNT_EXT = 8,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_DISPATCH_EXT = 9,
  // Provided by VK_EXT_descriptor_heap with VK_EXT_device_generated_commands
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_EXT = 1000135000,
  // Provided by VK_EXT_descriptor_heap with VK_EXT_device_generated_commands
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_SEQUENCE_INDEX_EXT = 1000135001,
  // Provided by VK_EXT_device_generated_commands with VK_EXT_mesh_shader
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_EXT = 1000328000,
  // Provided by VK_EXT_device_generated_commands with VK_EXT_mesh_shader
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_COUNT_EXT = 1000328001,
  // Provided by VK_KHR_ray_tracing_maintenance1 with VK_EXT_device_generated_commands
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT = 1000386004,
} VkIndirectCommandsTokenTypeEXT;
```

Table 46. Supported Indirect Command Tokens

Common Tokens	Command Data
VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT	u32[] array of indices into the indirect execution set

Common Tokens	Command Data
VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT	`u32[]` raw data
VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_EXT	`u8[]` raw data
VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT	`u32` placeholder data (not accessed by shader)
VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_SEQUENCE_INDEX_EXT	`u32` placeholder data (not accessed by shader)
Compute Tokens	
VK_INDIRECT_COMMANDS_TOKEN_TYPE_DISPATCH_EXT	VkDispatchIndirectCommand
Ray Tracing Tokens	
VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT	VkTraceRaysIndirectCommand2KHR
Graphics State Tokens	
VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT	VkBindIndexBufferIndirectCommandEXT
VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT	VkBindVertexBufferIndirectCommandEXT
Graphics Draw Tokens	
VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_EXT	VkDrawIndexedIndirectCommand
VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_EXT	VkDrawIndirectCommand
VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_EXT	VkDrawMeshTasksIndirectCommandEXT
Graphics Draw Count Tokens	
VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_COUNT_EXT	VkDrawIndirectCountIndirectCommandEXT with VkDrawIndexedIndirectCommand
VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_COUNT_EXT	VkDrawIndirectCountIndirectCommandEXT with VkDrawIndirectCommand
VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_COUNT_EXT	VkDrawIndirectCountIndirectCommandEXT with VkDrawMeshTasksIndirectCommandEXT

The VkIndirectCommandsLayoutTokenEXT structure specifies details to the function arguments that need to be known at layout creation time:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectCommandsLayoutTokenEXT {
    VkStructureType                 sType;
    const void*                     pNext;
    VkIndirectCommandsTokenTypeEXT  type;
    VkIndirectCommandsTokenDataEXT  data;
    uint32_t                        offset;
} VkIndirectCommandsLayoutTokenEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- type specifies the VkIndirectCommandsTokenTypeEXT for data.
- data specifies a VkIndirectCommandsTokenDataEXT containing token-specific details for command execution. It is ignored if type does not match any member of the VkIndirectCommandsTokenDataEXT union.
- offset is the relative byte offset for the token within one sequence of the indirect buffer. The data stored at that offset is the command data for the token, e.g. VkDispatchIndirectCommand.

Valid Usage

- VUID-VkIndirectCommandsLayoutTokenEXT-offset-11124
  offset **must** be less than or equal to VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::maxIndirectCommandsTokenOffset
- VUID-VkIndirectCommandsLayoutTokenEXT-offset-11125
  offset **must** be aligned to 4
- VUID-VkIndirectCommandsLayoutTokenEXT-meshShader-11126
  If meshShader or taskShader are not enabled, type **must** not be VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_COUNT_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_NV_EXT or VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_COUNT_NV_EXT
- VUID-VkIndirectCommandsLayoutTokenEXT-rayTracingMaintenance1-11128
  If the rayTracingMaintenance1 feature is not enabled, type **must** not be VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT
- VUID-VkIndirectCommandsLayoutTokenEXT-deviceGeneratedCommandsMultiDrawIndirectCount-11129
  If VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::deviceGeneratedCommandsMultiDrawIndirectCount is not supported, type **must** not be VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_COUNT_EXT or VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_COUNT_EXT

- VUID-VkIndirectCommandsLayoutTokenEXT-deviceGeneratedCommandsMultiDrawIndirectCount-11130
  If VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::deviceGeneratedCommandsMultiDrawIndirectCount is not supported, type **must** not be VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_COUNT_EXT
- VUID-VkIndirectCommandsLayoutTokenEXT-descriptorHeap-11332
  If the descriptorHeap feature is not enabled, type **must** not be VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_EXT or VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_SEQUENCE_INDEX_EXT
- VUID-VkIndirectCommandsLayoutTokenEXT-type-11333
  If type is VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_EXT or VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_SEQUENCE_INDEX_EXT VkIndirectCommandsPushConstantTokenEXT::updateRange.shaderStages **must** be VK_SHADER_STAGE_ALL

Valid Usage (Implicit)

- VUID-VkIndirectCommandsLayoutTokenEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_INDIRECT_COMMANDS_LAYOUT_TOKEN_EXT
- VUID-VkIndirectCommandsLayoutTokenEXT-type-parameter
  type **must** be a valid VkIndirectCommandsTokenTypeEXT value
- VUID-VkIndirectCommandsLayoutTokenEXT-pPushConstant-parameter
  If type is VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_EXT, or VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_SEQUENCE_INDEX_EXT, the pPushConstant member of data **must** be a valid pointer to a valid VkIndirectCommandsPushConstantTokenEXT structure
- VUID-VkIndirectCommandsLayoutTokenEXT-pVertexBuffer-parameter
  If type is VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT, the pVertexBuffer member of data **must** be a valid pointer to a valid VkIndirectCommandsVertexBufferTokenEXT structure
- VUID-VkIndirectCommandsLayoutTokenEXT-pIndexBuffer-parameter
  If type is VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT, the pIndexBuffer member of data **must** be a valid pointer to a valid VkIndirectCommandsIndexBufferTokenEXT structure
- VUID-VkIndirectCommandsLayoutTokenEXT-pExecutionSet-parameter
  If type is VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT, the pExecutionSet member of data **must** be a valid pointer to a valid VkIndirectCommandsExecutionSetTokenEXT structure

The VkIndirectCommandsTokenDataEXT structure provides token-specific details used to generate the indirect execution layout.

```
// Provided by VK_EXT_device_generated_commands
typedef union VkIndirectCommandsTokenDataEXT {
    const VkIndirectCommandsPushConstantTokenEXT*    pPushConstant;
    const VkIndirectCommandsVertexBufferTokenEXT*    pVertexBuffer;
    const VkIndirectCommandsIndexBufferTokenEXT*     pIndexBuffer;
    const VkIndirectCommandsExecutionSetTokenEXT*    pExecutionSet;
} VkIndirectCommandsTokenDataEXT;
```

- pPushConstant is a pointer to a VkIndirectCommandsPushConstantTokenEXT structure needed for VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_SEQUENCE_INDEX_EXT, and VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT tokens
- pVertexBuffer is a pointer to a VkIndirectCommandsVertexBufferTokenEXT structure needed for VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT tokens
- pIndexBuffer is a pointer to a VkIndirectCommandsIndexBufferTokenEXT structure needed for VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT tokens
- pExecutionSet is a pointer to a VkIndirectCommandsExecutionSetTokenEXT structure needed for VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT tokens

The appropriate member of the union **must** be set for each token.

The following code provides detailed information on how an individual sequence is processed. For valid usage, all restrictions from the regular commands apply.

```
void cmdProcessSequence(cmd, indirectExecutionSet, indirectCommandsLayout,
indirectAddress, s)
{
  for (uint32_t t = 0; t < indirectCommandsLayout.tokenCount; t++) {
    VkIndirectCommandsLayoutTokenEXT *token = &indirectCommandsLayout.pTokens[t];

    uint32_t offset  = token->offset;
    uint32_t stride  = indirectCommandsLayout.indirectStride;
    VkDeviceAddress streamData = indirectAddress;
    const void* input = streamData + stride * s + offset;

    switch (token->tokenType) {
    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT:
      uint32_t *bind = input;
      VkIndirectCommandsExecutionSetTokenEXT *info = token->data.pExecutionSet;

      if (info->type == VK_INDIRECT_EXECUTION_SET_INFO_TYPE_PIPELINES_EXT) {
        vkCmdBindPipeline(cmd, indirectExecutionSet.pipelineBindPoint,
indirectExecutionSet.pipelines[*bind]);
      } else {
        VkShaderStageFlagBits stages[];
        VkShaderEXT shaders[];
```

```
        uint32_t i = 0;
        IterateBitmaskLSBToMSB(iter, info->shaderStages) {
            stages[i] = iter;
            shaders[i] = indirectExecutionSet.shaders[bind[i]].shaderObject;
            i++;
        }
        vkCmdBindShadersEXT(cmd, i, stages, shaders);
      }
      break;

    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT:
      uint32_t* data = input;
      VkPushConstantsInfoKHR info = {
        VK_STRUCTURE_TYPE_PUSH_CONSTANTS_INFO_KHR,
        // this can also use `dynamicGeneratedPipelineLayout' to pass a
VkPipelineLayoutCreateInfo from pNext
        indirectCommandsLayout.pipelineLayout,
        token->token.pushConstant.updateRange.shaderStages,
        token->token.pushConstant.updateRange.offset,
        token->token.pushConstant.updateRange.size,
        data
      };

      vkCmdPushConstants2KHR(cmd, &info);
      break;

    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_EXT:

      VkHostAddressRangeConstEXT addressRange = {
        token->token.pushConstant.updateRange.data.size,
        input
      };

      VkPushDataInfoEXT info = {
        VK_STRUCTURE_TYPE_PUSH_DATA_INFO_EXT,
        NULL,
        token->token.pushConstant.updateRange.offset,
        addressRange
      };

      vkCmdPushDataEXT(cmd, &info);
      break;

    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT:
      VkPushConstantsInfoKHR info = {
        VK_STRUCTURE_TYPE_PUSH_CONSTANTS_INFO_KHR,
        // this can also use `dynamicGeneratedPipelineLayout' to pass a
VkPipelineLayoutCreateInfo from pNext
        indirectCommandsLayout.pipelineLayout,
        token->token.pushConstant.updateRange.shaderStages,
        token->token.pushConstant.updateRange.offset,
```

```
        // this must be 4
        token->token.pushConstant.updateRange.size,
        // this just updates the sequence index
        &s
      };

      vkCmdPushConstants2KHR(cmd, &info);
      break;

    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_SEQUENCE_INDEX_EXT:

      VkHostAddressRangeConstEXT addressRange = {
        // this must be 4
        token->token.pushConstant.updateRange.data.size,
        // this just updates the sequence index
        &s
      };

      VkPushDataInfoEXT info = {
        VK_STRUCTURE_TYPE_PUSH_DATA_INFO_EXT,
        NULL,
        token->token.pushConstant.updateRange.offset,
        addressRange
      };

      vkCmdPushDataEXT(cmd, &info);
      break;

    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT:
      VkBindIndexBufferIndirectCommandEXT* data = input;

      vkCmdBindIndexBuffer(cmd, deriveBuffer(data->bufferAddress), deriveOffset(data-
>bufferAddress), data->indexType);
      break;

    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT:
      VkBindVertexBufferIndirectCommandEXT* data = input;

      vkCmdBindVertexBuffers2(cmd, token->token.vertexBuffer->vertexBindingUnit, 1,
&deriveBuffer(data->bufferAddress),
                              &deriveOffset(data->bufferAddress), data->size, data-
>stride);
      break;

    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_EXT:
      VkDrawIndexedIndirectCommand *data = input;

      vkCmdDrawIndexed(cmd, data->indexCount, data->instanceCount, data->firstIndex,
data->vertexOffset, data->firstInstance);
      break;
    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_COUNT_EXT:
```

```
      VkDrawIndirectCountIndirectCommandEXT* data = input;

      vkCmdDrawIndexedIndirect(cmd, deriveBuffer(data->bufferAddress),  deriveoffset
(data->bufferAddress), min(data->commandCount, indirectCommandsLayout.maxDrawCount),
data->stride);
      break;

    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_EXT:
      VkDrawIndirectCommand* data = input;

      vkCmdDraw(cmd, data->vertex_count, data->instanceCount, data->firstVertex, data
->firstIndex);
      break;

    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_COUNT_EXT:
      VkDrawIndirectCountIndirectCommandEXT* data = input;

      vkCmdDrawIndirect(cmd, deriveBuffer(data->bufferAddress), deriveoffset(data-
>bufferAddress), min(data->commandCount, indirectCommandsLayout.maxDrawCount), data-
>stride);
      break;

    // only available if VK_NV_mesh_shader is enabled
    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_NV_EXT:
      VkDrawMeshTasksIndirectCommandNV *data = input;

      vkCmdDrawMeshTasksNV(cmd, data->taskCount, data->firstTask);
     break;

    // only available if VK_NV_mesh_shader is enabled
    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_COUNT_NV_EXT:
      VkDrawIndirectCountIndirectCommandEXT* data = input;

      vkCmdDrawMeshTasksIndirectCountNV(cmd, deriveBuffer(data->bufferAddress),
deriveoffset(data->bufferAddress), min(data->commandCount, indirectCommandsLayout
.maxDrawCount), data->stride);
      break;

    // only available if VK_EXT_mesh_shader is enabled
    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_EXT:
      VkDrawMeshTasksIndirectCommandEXT *data = input;

      vkCmdDrawMeshTasksEXT(cmd, data->groupCountX, data->groupCountY, data-
>groupCountZ);
     break;

    // only available if VK_EXT_mesh_shader is enabled
    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_COUNT_EXT:
      VkDrawIndirectCountIndirectCommandEXT* data = input;

      vkCmdDrawMeshTasksIndirectCountEXT(cmd, deriveBuffer(data->bufferAddress),
```

```
deriveoffset(data->bufferAddress), min(data->commandCount, indirectCommandsLayout
.maxDrawCount), data->stride);
      break;

    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DISPATCH_EXT:
      VkDispatchIndirectCommand *data = input;

      vkCmdDispatch(cmd, data->x, data->y, data->z);
      break;

    // only available if VK_KHR_ray_tracing_maintenance1 is enabled
    case VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT:
      vkCmdTraceRaysIndirect2KHR(cmd, deriveBuffer(input));
      break;
    }
  }
}
```

The VkIndirectCommandsPushConstantTokenEXT structure specifies the layout token info for VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_SEQUENCE_INDEX_EXT, and VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT tokens.

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectCommandsPushConstantTokenEXT {
    VkPushConstantRange    updateRange;
} VkIndirectCommandsPushConstantTokenEXT;
```

- updateRange is the push constant range that will be updated by the token.

The stageFlags member of updateRange is ignored.

Valid Usage

- VUID-VkIndirectCommandsPushConstantTokenEXT-updateRange-11132
  If the token type is VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT or VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT, updateRange **must** be contained within the push constant info used by VkIndirectCommandsLayoutCreateInfoEXT
- VUID-VkIndirectCommandsPushConstantTokenEXT-size-11133
  If the token type is VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_SEQUENCE_INDEX_EXT or VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT, the size member of updateRange **must** be 4

Valid Usage (Implicit)

- VUID-VkIndirectCommandsPushConstantTokenEXT-updateRange-parameter
  `updateRange` **must** be a valid VkPushConstantRange structure

The `VkIndirectCommandsVertexBufferTokenEXT` structure specifies the layout token info for the VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT token.

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectCommandsVertexBufferTokenEXT {
    uint32_t    vertexBindingUnit;
} VkIndirectCommandsVertexBufferTokenEXT;
```

- `vertexBindingUnit` is the vertex input binding number to be bound.

Valid Usage

- VUID-VkIndirectCommandsVertexBufferTokenEXT-vertexBindingUnit-11134
  `vertexBindingUnit` **must** be less than the total number of vertex input bindings in use by the current graphics state

The `VkIndirectCommandsIndexBufferTokenEXT` structure specifies the layout token info for the VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT token.

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectCommandsIndexBufferTokenEXT {
    VkIndirectCommandsInputModeFlagBitsEXT    mode;
} VkIndirectCommandsIndexBufferTokenEXT;
```

- `mode` specifies the mode to use with this token.

This allows for easy layering of Vulkan atop other APIs. When VK_INDIRECT_COMMANDS_INPUT_MODE_DXGI_INDEX_BUFFER_EXT is specified, the indirect buffer can contain a `D3D12_INDEX_BUFFER_VIEW` instead of VkBindIndexBufferIndirectCommandEXT as D3D's DXGI format value is mapped to the VkIndexType. It works as both structs are otherwise binary compatible.

Valid Usage

- VUID-VkIndirectCommandsIndexBufferTokenEXT-mode-11135
  `mode` **must** be non-zero
- VUID-VkIndirectCommandsIndexBufferTokenEXT-mode-11136
  `mode` **must** be one of the bits supported in VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::`supportedIndirectCommandsIn`

putModes

Valid Usage (Implicit)

- VUID-VkIndirectCommandsIndexBufferTokenEXT-mode-parameter
  mode **must** be a valid VkIndirectCommandsInputModeFlagBitsEXT value

Bits which are set in VkIndirectCommandsIndexBufferTokenEXT::mode, specifying how an index buffer is used, are:

```
// Provided by VK_EXT_device_generated_commands
typedef enum VkIndirectCommandsInputModeFlagBitsEXT {
    VK_INDIRECT_COMMANDS_INPUT_MODE_VULKAN_INDEX_BUFFER_EXT = 0x00000001,
    VK_INDIRECT_COMMANDS_INPUT_MODE_DXGI_INDEX_BUFFER_EXT = 0x00000002,
} VkIndirectCommandsInputModeFlagBitsEXT;
```

- VK_INDIRECT_COMMANDS_INPUT_MODE_VULKAN_INDEX_BUFFER_EXT specifies that the indirect buffer contains VkBindIndexBufferIndirectCommandEXT.
- VK_INDIRECT_COMMANDS_INPUT_MODE_DXGI_INDEX_BUFFER_EXT specifies that the indirect buffer contains D3D12_INDEX_BUFFER_VIEW.

```
// Provided by VK_EXT_device_generated_commands
typedef VkFlags VkIndirectCommandsInputModeFlagsEXT;
```

VkIndirectCommandsInputModeFlagsEXT is a bitmask type for setting a mask of zero or more VkIndirectCommandsInputModeFlagBitsEXT.

The VkIndirectCommandsExecutionSetTokenEXT structure specifies the input data for the VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token.

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectCommandsExecutionSetTokenEXT {
    VkIndirectExecutionSetInfoTypeEXT    type;
    VkShaderStageFlags                   shaderStages;
} VkIndirectCommandsExecutionSetTokenEXT;
```

- type describes the type of indirect execution set in use.
- shaderStages specifies the shaders that will be changed by this token.

Valid Usage

- VUID-VkIndirectCommandsExecutionSetTokenEXT-shaderStages-11137
  Each bit in shaderStages **must** be supported by

VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStagesPipelineBinding or VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStagesShaderBinding

Valid Usage (Implicit)

- VUID-VkIndirectCommandsExecutionSetTokenEXT-type-parameter
  `type` **must** be a valid VkIndirectExecutionSetInfoTypeEXT value
- VUID-VkIndirectCommandsExecutionSetTokenEXT-shaderStages-parameter
  `shaderStages` **must** be a valid combination of VkShaderStageFlagBits values

35.2. Indirect Commands Generation and Execution

The generation of commands on the device requires a `preprocess` buffer.

With `VK_EXT_device_generated_commands`, to retrieve the memory size and alignment requirements of a particular execution state call:

```
// Provided by VK_EXT_device_generated_commands
void vkGetGeneratedCommandsMemoryRequirementsEXT(
    VkDevice                                    device,
    const VkGeneratedCommandsMemoryRequirementsInfoEXT* pInfo,
    VkMemoryRequirements2*                      pMemoryRequirements);
```

- `device` is the logical device that owns the buffer.
- `pInfo` is a pointer to a VkGeneratedCommandsMemoryRequirementsInfoEXT structure containing parameters required for the memory requirements query.
- `pMemoryRequirements` is a pointer to a VkMemoryRequirements2 structure in which the memory requirements of the buffer object are returned.

If the size returned is zero, the preprocessing step can be skipped for this layout.

Valid Usage (Implicit)

- VUID-vkGetGeneratedCommandsMemoryRequirementsEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetGeneratedCommandsMemoryRequirementsEXT-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkGeneratedCommandsMemoryRequirementsInfoEXT structure
- VUID-vkGetGeneratedCommandsMemoryRequirementsEXT-pMemoryRequirements-parameter
  `pMemoryRequirements` **must** be a valid pointer to a VkMemoryRequirements2 structure

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkGeneratedCommandsMemoryRequirementsInfoEXT {
    VkStructureType                sType;
    const void*                    pNext;
    VkIndirectExecutionSetEXT      indirectExecutionSet;
    VkIndirectCommandsLayoutEXT    indirectCommandsLayout;
    uint32_t                       maxSequenceCount;
    uint32_t                       maxDrawCount;
} VkGeneratedCommandsMemoryRequirementsInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `indirectExecutionSet` is the indirect execution set to be used for binding shaders.
- `indirectCommandsLayout` is the VkIndirectCommandsLayoutEXT that this buffer memory is intended to be used with.
- `maxSequenceCount` is the maximum number of sequences that this buffer memory can be used with.
- `maxDrawCount` is the maximum number of indirect draws that can be executed by any COUNT-type multi-draw indirect tokens. The draw count in the indirect buffer is clamped to this value for these token types.

If the action command token for the layout is not a COUNT-type multi-draw indirect token, `maxDrawCount` is ignored.

Valid Usage

- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-maxSequencesCount-11009
  `maxSequencesCount` **must** be less than or equal to VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::`maxIndirectSequenceCount`
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-indirectCommandsLayout-11010
  If `indirectCommandsLayout` was created with a token sequence that contained the VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, `indirectExecutionSet` **must** not be VK_NULL_HANDLE
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-indirectCommandsLayout-11151
  If `indirectCommandsLayout` was created with a token sequence that contained the VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, the shader stages used to create the initial shader state of `indirectExecutionSet` **must** equal the VkIndirectCommandsExecutionSetTokenEXT::`shaderStages` used to create `indirectCommandsLayout`
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-indirectCommandsLayout-11011
  If `indirectCommandsLayout` was not created with a token sequence that contained the

VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, indirectExecutionSet **must** be VK_NULL_HANDLE

- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-maxDrawCount-11146
  When not ignored, maxDrawCount × maxSequenceCount **must** be less than 2^24
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-indirectExecutionSet-11012
  If indirectExecutionSet is VK_NULL_HANDLE, either a VkGeneratedCommandsPipelineInfoEXT or a VkGeneratedCommandsShaderInfoEXT **must** be included in the pNext chain

Valid Usage (Implicit)

- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_GENERATED_COMMANDS_MEMORY_REQUIREMENTS_INFO_EXT
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkGeneratedCommandsPipelineInfoEXT or VkGeneratedCommandsShaderInfoEXT
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-indirectExecutionSet-parameter
  If indirectExecutionSet is not VK_NULL_HANDLE, indirectExecutionSet **must** be a valid VkIndirectExecutionSetEXT handle
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-indirectCommandsLayout-parameter
  indirectCommandsLayout **must** be a valid VkIndirectCommandsLayoutEXT handle
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-commonparent
  Both of indirectCommandsLayout, and indirectExecutionSet that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkGeneratedCommandsPipelineInfoEXT {
    VkStructureType    sType;
    void*              pNext;
    VkPipeline         pipeline;
} VkGeneratedCommandsPipelineInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pipeline is a valid pipeline object.

Valid Usage (Implicit)

- VUID-VkGeneratedCommandsPipelineInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_GENERATED_COMMANDS_PIPELINE_INFO_EXT
- VUID-VkGeneratedCommandsPipelineInfoEXT-pipeline-parameter
  `pipeline` **must** be a valid VkPipeline handle

Structure Chaining

Extends the structures

- VkGeneratedCommandsInfoEXT
- VkGeneratedCommandsMemoryRequirementsInfoEXT

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkGeneratedCommandsShaderInfoEXT {
    VkStructureType       sType;
    void*                 pNext;
    uint32_t              shaderCount;
    const VkShaderEXT*    pShaders;
} VkGeneratedCommandsShaderInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderCount` is the size of the `pShaders` array.
- `pShaders` is a pointer to an array of shader objects.

Valid Usage

- VUID-VkGeneratedCommandsShaderInfoEXT-pShaders-11127
  `pShaders` **must** not contain more than one shader object for a given VkShaderStageFlagBits stage

Valid Usage (Implicit)

- VUID-VkGeneratedCommandsShaderInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_GENERATED_COMMANDS_SHADER_INFO_EXT
- VUID-VkGeneratedCommandsShaderInfoEXT-pShaders-parameter
  `pShaders` **must** be a valid pointer to an array of `shaderCount` valid VkShaderEXT handles
- VUID-VkGeneratedCommandsShaderInfoEXT-shaderCount-arraylength
  `shaderCount` **must** be greater than `0`

Structure Chaining

Extends the structures

- VkGeneratedCommandsInfoEXT
- VkGeneratedCommandsMemoryRequirementsInfoEXT

35.2.1. Indirect Execution Sets

Indirect Execution Sets contain sets of pipelines or shader objects which can be bound individually.

```
// Provided by VK_EXT_device_generated_commands
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkIndirectExecutionSetEXT)
```

Indirect Execution Sets allow the device to bind different shaders and pipeline states using Device-Generated Commands.

Indirect Execution Sets are created by calling:

```
// Provided by VK_EXT_device_generated_commands
VkResult vkCreateIndirectExecutionSetEXT(
    VkDevice                                    device,
    const VkIndirectExecutionSetCreateInfoEXT*  pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkIndirectExecutionSetEXT*                  pIndirectExecutionSet);
```

- `device` is the logical device that creates the indirect execution set.
- `pCreateInfo` is a pointer to a VkIndirectExecutionSetCreateInfoEXT structure containing parameters affecting creation of the indirect execution set.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pIndirectExecutionSet` is a pointer to a VkIndirectExecutionSetEXT handle in which the resulting indirect execution set is returned.

Valid Usage

- VUID-vkCreateIndirectExecutionSetEXT-deviceGeneratedCommands-11013
  The `VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT::deviceGeneratedCommands` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCreateIndirectExecutionSetEXT-device-parameter
  `device` **must** be a valid VkDevice handle

- VUID-vkCreateIndirectExecutionSetEXT-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkIndirectExecutionSetCreateInfoEXT structure
- VUID-vkCreateIndirectExecutionSetEXT-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateIndirectExecutionSetEXT-pIndirectExecutionSet-parameter
  `pIndirectExecutionSet` **must** be a valid pointer to a VkIndirectExecutionSetEXT handle
- VUID-vkCreateIndirectExecutionSetEXT-device-queuecount
  The device **must** have been created with at least `1` queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkIndirectExecutionSetCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectExecutionSetCreateInfoEXT {
    VkStructureType                       sType;
    const void*                           pNext;
    VkIndirectExecutionSetInfoTypeEXT     type;
    VkIndirectExecutionSetInfoEXT         info;
} VkIndirectExecutionSetCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `type` is a VkIndirectExecutionSetInfoTypeEXT describing the type of set being created and determining which field of the `info` union will be used.
- `info` is a VkIndirectExecutionSetInfoEXT union containing layout information for the set.

Valid Usage

- VUID-VkIndirectExecutionSetCreateInfoEXT-maxIndirectShaderObjectCount-11014
  If `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT`::`maxIndirectShaderObjectCount` is

zero or the shaderObject feature is not enabled type **must** not be VK_INDIRECT_EXECUTION_SET_INFO_TYPE_SHADER_OBJECTS_EXT

Valid Usage (Implicit)

- VUID-VkIndirectExecutionSetCreateInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_CREATE_INFO_EXT
- VUID-VkIndirectExecutionSetCreateInfoEXT-type-parameter
  type **must** be a valid VkIndirectExecutionSetInfoTypeEXT value
- VUID-VkIndirectExecutionSetCreateInfoEXT-pPipelineInfo-parameter
  If type is VK_INDIRECT_EXECUTION_SET_INFO_TYPE_PIPELINES_EXT, the pPipelineInfo member of info **must** be a valid pointer to a valid VkIndirectExecutionSetPipelineInfoEXT structure
- VUID-VkIndirectExecutionSetCreateInfoEXT-pShaderInfo-parameter
  If type is VK_INDIRECT_EXECUTION_SET_INFO_TYPE_SHADER_OBJECTS_EXT, the pShaderInfo member of info **must** be a valid pointer to a valid VkIndirectExecutionSetShaderInfoEXT structure

Values which **can** be set in VkIndirectExecutionSetCreateInfoEXT::type, specifying contents of an indirect execution set, are:

```
// Provided by VK_EXT_device_generated_commands
typedef enum VkIndirectExecutionSetInfoTypeEXT {
    VK_INDIRECT_EXECUTION_SET_INFO_TYPE_PIPELINES_EXT = 0,
    VK_INDIRECT_EXECUTION_SET_INFO_TYPE_SHADER_OBJECTS_EXT = 1,
} VkIndirectExecutionSetInfoTypeEXT;
```

- VK_INDIRECT_EXECUTION_SET_INFO_TYPE_PIPELINES_EXT specifies that the indirect execution set contains VkPipeline objects.
- VK_INDIRECT_EXECUTION_SET_INFO_TYPE_SHADER_OBJECTS_EXT specifies that the indirect execution set contains VkShaderEXT objects.

The VkIndirectExecutionSetInfoEXT union is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef union VkIndirectExecutionSetInfoEXT {
    const VkIndirectExecutionSetPipelineInfoEXT*    pPipelineInfo;
    const VkIndirectExecutionSetShaderInfoEXT*      pShaderInfo;
} VkIndirectExecutionSetInfoEXT;
```

- pPipelineInfo is a pointer to a VkIndirectExecutionSetPipelineInfoEXT structure containing pipeline layout information for the set.
- pShaderInfo is a pointer to a VkIndirectExecutionSetShaderInfoEXT structure containing shader

object layout information for the set.

The VkIndirectExecutionSetPipelineInfoEXT structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectExecutionSetPipelineInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkPipeline         initialPipeline;
    uint32_t           maxPipelineCount;
} VkIndirectExecutionSetPipelineInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- initialPipeline is the initial pipeline for the set. This pipeline will be automatically added to the set at index 0.
- maxPipelineCount is the maximum number of pipelines stored in the set.

The characteristics of initialPipeline will be used to validate all pipelines added to the set even if they are removed from the set or destroyed.

When an Indirect Execution Set created with pipelines is used, initialPipeline constitutes the initial shader state.

Valid Usage

- VUID-VkIndirectExecutionSetPipelineInfoEXT-supportedIndirectCommandsShaderStagesPipelineBinding-11015
  If VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStagesPipelineBinding does not contain VK_SHADER_STAGE_COMPUTE_BIT, the VkPipelineBindPoint of initialPipeline **must** not be VK_PIPELINE_BIND_POINT_COMPUTE
- VUID-VkIndirectExecutionSetPipelineInfoEXT-supportedIndirectCommandsShaderStagesPipelineBinding-11016
  If VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStagesPipelineBinding does not contain VK_SHADER_STAGE_FRAGMENT_BIT, the VkPipelineBindPoint of initialPipeline **must** not be VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-VkIndirectExecutionSetPipelineInfoEXT-supportedIndirectCommandsShaderStagesPipelineBinding-11017
  If VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStagesPipelineBinding does not contain ray tracing stages, the VkPipelineBindPoint of initialPipeline **must** not be VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR
- VUID-VkIndirectExecutionSetPipelineInfoEXT-maxPipelineCount-11018

maxPipelineCount must be between 1 and VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::maxIndirectPipelineCount

- VUID-VkIndirectExecutionSetPipelineInfoEXT-initialPipeline-11019
  initialPipeline must not use descriptors of type VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC
- VUID-VkIndirectExecutionSetPipelineInfoEXT-initialPipeline-11153
  initialPipeline must have been created with VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT

Valid Usage (Implicit)

- VUID-VkIndirectExecutionSetPipelineInfoEXT-sType-sType
  sType must be VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_PIPELINE_INFO_EXT
- VUID-VkIndirectExecutionSetPipelineInfoEXT-initialPipeline-parameter
  initialPipeline must be a valid VkPipeline handle

The VkIndirectExecutionSetShaderInfoEXT structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectExecutionSetShaderInfoEXT {
    VkStructureType                                 sType;
    const void*                                     pNext;
    uint32_t                                        shaderCount;
    const VkShaderEXT*                              pInitialShaders;
    const VkIndirectExecutionSetShaderLayoutInfoEXT* pSetLayoutInfos;
    uint32_t                                        maxShaderCount;
    uint32_t                                        pushConstantRangeCount;
    const VkPushConstantRange*                      pPushConstantRanges;
} VkIndirectExecutionSetShaderInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderCount is the number of members in the pInitialShaders and pSetLayoutInfos arrays.
- pInitialShaders is a pointer to an array containing a VkShaderEXT object for each shader stage that will be used in the set. These shaders will be automatically added to the set beginning at index 0.
- pSetLayoutInfos is NULL or a pointer to an array containing a VkIndirectExecutionSetShaderLayoutInfoEXT used by each corresponding pInitialShaders shader stage in the set.
- maxShaderCount is the maximum number of shader objects stored in the set.
- pushConstantRangeCount is the number of members in the pPushConstantRanges array.

- `pPushConstantRanges` is a pointer to the array of VkPushConstantRange ranges used by all shaders in the set.

The characteristics of `pInitialShaders` will be used to validate all shaders added to the set even if they are removed from the set or destroyed.

When an Indirect Execution Set created with shader objects is used, `pInitialShaders` constitutes the initial shader state.

If `pSetLayoutInfos` is `NULL`, the descriptor layout parameters are inherited from the shader object.

Valid Usage

- VUID-VkIndirectExecutionSetShaderInfoEXT-pInitialShaders-11020
  All members of `pInitialShaders` **must** have a `stage` supported by VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::`supportedIndirectCommandsShaderStagesShaderBinding`
- VUID-VkIndirectExecutionSetShaderInfoEXT-maxShaderCount-11021
  `maxShaderCount` **must** not be zero
- VUID-VkIndirectExecutionSetShaderInfoEXT-maxShaderCount-11022
  `maxShaderCount` **must** be less than or equal to `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT`::`maxIndirectShaderObjectCount`
- VUID-VkIndirectExecutionSetShaderInfoEXT-maxShaderCount-11036
  `maxShaderCount` **must** be greater than or equal to `shaderCount`
- VUID-VkIndirectExecutionSetShaderInfoEXT-stage-11023
  The `stage` of each element in the `pInitialShaders` array **must** be unique
- VUID-VkIndirectExecutionSetShaderInfoEXT-pInitialShaders-11154
  Each member of `pInitialShaders` **must** have been created with VK_SHADER_CREATE_INDIRECT_BINDABLE_BIT_EXT
- VUID-VkIndirectExecutionSetShaderInfoEXT-pSetLayoutInfos-10929
  If `pSetLayoutInfos` is not `NULL`, the descriptor layout values specified **must** be compatible with the descriptor set layouts defined at the creation of the shader object
- VUID-VkIndirectExecutionSetShaderInfoEXT-pInitialShaders-11321
  If any element of `pInitialShaders` was created with VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT, all elements of `pInitialShaders` **must** have been created with VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT
- VUID-VkIndirectExecutionSetShaderInfoEXT-pInitialShaders-11322
  If any element of `pInitialShaders` was created without VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT, all elements of `pInitialShaders` **must** have been created without VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT
- VUID-VkIndirectExecutionSetShaderInfoEXT-pInitialShaders-11323
  If elements of `pInitialShaders` were created with VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT, `pSetLayoutInfos` **must** be `NULL`

Valid Usage (Implicit)

- VUID-VkIndirectExecutionSetShaderInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_SHADER_INFO_EXT
- VUID-VkIndirectExecutionSetShaderInfoEXT-pInitialShaders-parameter
  `pInitialShaders` **must** be a valid pointer to an array of `shaderCount` valid VkShaderEXT handles
- VUID-VkIndirectExecutionSetShaderInfoEXT-pSetLayoutInfos-parameter
  If `pSetLayoutInfos` is not `NULL`, `pSetLayoutInfos` **must** be a valid pointer to an array of `shaderCount` valid VkIndirectExecutionSetShaderLayoutInfoEXT structures
- VUID-VkIndirectExecutionSetShaderInfoEXT-pPushConstantRanges-parameter
  If `pushConstantRangeCount` is not `0`, `pPushConstantRanges` **must** be a valid pointer to an array of `pushConstantRangeCount` valid VkPushConstantRange structures
- VUID-VkIndirectExecutionSetShaderInfoEXT-shaderCount-arraylength
  `shaderCount` **must** be greater than `0`

The `VkIndirectExecutionSetShaderLayoutInfoEXT` structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectExecutionSetShaderLayoutInfoEXT {
    VkStructureType                 sType;
    const void*                     pNext;
    uint32_t                        setLayoutCount;
    const VkDescriptorSetLayout*    pSetLayouts;
} VkIndirectExecutionSetShaderLayoutInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `setLayoutCount` is the number of members in the `pSetLayouts` array.
- `pSetLayouts` is a pointer to an array containing VkDescriptorSetLayout objects used by the shader stage. The implementation **must** not access these objects outside of the duration of the command this structure is passed to.

Valid Usage

- VUID-VkIndirectExecutionSetShaderLayoutInfoEXT-pSetLayouts-11024
  All members of `pSetLayouts` **must** not contain descriptors of type VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC

Valid Usage (Implicit)

- VUID-VkIndirectExecutionSetShaderLayoutInfoEXT-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_SHADER_LAYOUT_INFO_EXT`
- VUID-VkIndirectExecutionSetShaderLayoutInfoEXT-pSetLayouts-parameter
  If `setLayoutCount` is not `0`, `pSetLayouts` **must** be a valid pointer to an array of `setLayoutCount` valid or VK_NULL_HANDLE VkDescriptorSetLayout handles

Destroy an Indirect Execution Set by calling:

```
// Provided by VK_EXT_device_generated_commands
void vkDestroyIndirectExecutionSetEXT(
    VkDevice                                    device,
    VkIndirectExecutionSetEXT                   indirectExecutionSet,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that owns the indirect execution set.
- `indirectExecutionSet` is the indirect execution set to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyIndirectExecutionSetEXT-indirectExecutionSet-11025
  All submitted commands that refer to `indirectExecutionSet` **must** have completed execution

Valid Usage (Implicit)

- VUID-vkDestroyIndirectExecutionSetEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyIndirectExecutionSetEXT-indirectExecutionSet-parameter
  If `indirectExecutionSet` is not VK_NULL_HANDLE, `indirectExecutionSet` **must** be a valid VkIndirectExecutionSetEXT handle
- VUID-vkDestroyIndirectExecutionSetEXT-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyIndirectExecutionSetEXT-indirectExecutionSet-parent
  If `indirectExecutionSet` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `indirectExecutionSet` **must** be externally synchronized

The `VkWriteIndirectExecutionSetPipelineEXT` structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkWriteIndirectExecutionSetPipelineEXT {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           index;
    VkPipeline         pipeline;
} VkWriteIndirectExecutionSetPipelineEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `index` is the element of the set to update.
- `pipeline` is the pipeline to store in the indirect execution set.

Valid Usage

- VUID-VkWriteIndirectExecutionSetPipelineEXT-index-11026
  `index` **must** be less than the value of `VkIndirectExecutionSetPipelineInfoEXT::maxPipelineCount` used to create the set
- VUID-VkWriteIndirectExecutionSetPipelineEXT-pipeline-11027
  `pipeline` **must** have been created with VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT
- VUID-VkWriteIndirectExecutionSetPipelineEXT-index-11029
  `index` **must** not be referenced by submitted command buffers
- VUID-VkWriteIndirectExecutionSetPipelineEXT-pipeline-11030
  The shader stages contained in `pipeline` **must** be supported by `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStagesPipelineBinding`

Valid Usage (Implicit)

- VUID-VkWriteIndirectExecutionSetPipelineEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_WRITE_INDIRECT_EXECUTION_SET_PIPELINE_EXT
- VUID-VkWriteIndirectExecutionSetPipelineEXT-pipeline-parameter
  `pipeline` **must** be a valid VkPipeline handle

The VkWriteIndirectExecutionSetShaderEXT structure is defined as:

```
// Provided by VK_EXT_device_generated_commands with VK_EXT_shader_object
typedef struct VkWriteIndirectExecutionSetShaderEXT {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           index;
    VkShaderEXT        shader;
} VkWriteIndirectExecutionSetShaderEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- index is the element of the set to update.
- shader is the shader to store in the indirect execution set.

Shaders need not be stored in the Indirect Execution Set according to their stage. The only restriction for shader indices within a set is that the value of the index **must** be less than the maximum number of shaders in the set.

Valid Usage

- VUID-VkWriteIndirectExecutionSetShaderEXT-index-11031
  index **must** be less than VkIndirectExecutionSetShaderInfoEXT::maxShaderCount
- VUID-VkWriteIndirectExecutionSetShaderEXT-shader-11032
  shader **must** have been created with VK_SHADER_CREATE_INDIRECT_BINDABLE_BIT_EXT
- VUID-VkWriteIndirectExecutionSetShaderEXT-pInitialShaders-11033
  A shader created with the same VkShaderStageFlagBits **must** have been passed in the VkIndirectExecutionSetShaderInfoEXT::pInitialShaders array
- VUID-VkWriteIndirectExecutionSetShaderEXT-index-11034
  index **must** not be in use by submitted command buffers

Valid Usage (Implicit)

- VUID-VkWriteIndirectExecutionSetShaderEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_WRITE_INDIRECT_EXECUTION_SET_SHADER_EXT
- VUID-VkWriteIndirectExecutionSetShaderEXT-shader-parameter
  shader **must** be a valid VkShaderEXT handle

Pipeline elements in an Indirect Execution Set can be updated by calling:

```
// Provided by VK_EXT_device_generated_commands
void vkUpdateIndirectExecutionSetPipelineEXT(
    VkDevice                                    device,
    VkIndirectExecutionSetEXT                   indirectExecutionSet,
    uint32_t                                    executionSetWriteCount,
    const VkWriteIndirectExecutionSetPipelineEXT* pExecutionSetWrites);
```

- `device` is the logical device that owns the indirect execution set.
- `indirectExecutionSet` is the indirect execution set being updated.
- `executionSetWriteCount` is the number of elements in the `pExecutionSetWrites` array.
- `pExecutionSetWrites` is a pointer to an array of VkWriteIndirectExecutionSetPipelineEXT structures describing the elements to update.

Valid Usage

- VUID-vkUpdateIndirectExecutionSetPipelineEXT-indirectExecutionSet-11035
  `indirectExecutionSet` **must** have been created with type VK_INDIRECT_EXECUTION_SET_INFO_TYPE_PIPELINES_EXT
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-executionSetWriteCount-11037
  `executionSetWriteCount` **must** be less than or equal to `VkIndirectExecutionSetPipelineInfoEXT`::`maxPipelineCount`
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-pExecutionSetWrites-11042
  Each element in the `pExecutionSetWrites` array **must** have a unique `VkWriteIndirectExecutionSetPipelineEXT`::`index`
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-None-11038
  Each member of the Indirect Execution Set referenced by the update command **must** not be in use by the device
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-initialPipeline-11324
  If `initialPipeline` was created without VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, each pipeline in `pExecutionSetWrites` **must** also have been created without VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-initialPipeline-11325
  If `initialPipeline` was created with VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, each pipeline in `pExecutionSetWrites` **must** also have been created with VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-None-11039
  If `initialPipeline` was created without VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT, the layout of each pipeline in `pExecutionSetWrites` **must** be compatible with the `initialPipeline` used to create the Indirect Execution Set
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-None-11040

Each pipeline in the Indirect Execution Set **must** have identically defined static and dynamic state values to the `initialPipeline` used to create the Indirect Execution Set

- VUID-vkUpdateIndirectExecutionSetPipelineEXT-initialPipeline-11147
  Each pipeline in the Indirect Execution Set **must** have identically defined fragment outputs interface to the `initialPipeline` used to create the Indirect Execution Set
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-initialPipeline-11152
  Each pipeline in the Indirect Execution Set **must** match the `initialPipeline` used to create the Indirect Execution Set in its included shader stages
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-initialPipeline-11098
  Each pipeline in the Indirect Execution Set **must** match the `initialPipeline` used to create the Indirect Execution Set in its use of `FragDepth`
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-initialPipeline-11086
  Each pipeline in the Indirect Execution Set **must** match the `initialPipeline` used to create the Indirect Execution Set in its use of `SampleMask`
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-initialPipeline-11085
  Each pipeline in the Indirect Execution Set **must** match the `initialPipeline` used to create the Indirect Execution Set in its use of `StencilExportEXT`

Valid Usage (Implicit)

- VUID-vkUpdateIndirectExecutionSetPipelineEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-indirectExecutionSet-parameter
  `indirectExecutionSet` **must** be a valid VkIndirectExecutionSetEXT handle
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-pExecutionSetWrites-parameter
  `pExecutionSetWrites` **must** be a valid pointer to an array of `executionSetWriteCount` valid VkWriteIndirectExecutionSetPipelineEXT structures
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-executionSetWriteCount-arraylength
  `executionSetWriteCount` **must** be greater than `0`
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-indirectExecutionSet-parent
  `indirectExecutionSet` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `indirectExecutionSet` **must** be externally synchronized

Shader object elements in an Indirect Execution Set can be updated by calling:

```
// Provided by VK_EXT_device_generated_commands
void vkUpdateIndirectExecutionSetShaderEXT(
    VkDevice                                    device,
    VkIndirectExecutionSetEXT                   indirectExecutionSet,
    uint32_t                                    executionSetWriteCount,
    const VkWriteIndirectExecutionSetShaderEXT* pExecutionSetWrites);
```

- `device` is the logical device that owns the indirect execution set.
- `indirectExecutionSet` is the indirect execution set being updated.
- `executionSetWriteCount` is the number of elements in the `pExecutionSetWrites` array.
- `pExecutionSetWrites` is a pointer to an array of VkWriteIndirectExecutionSetShaderEXT structures describing the elements to update.

Valid Usage

- VUID-vkUpdateIndirectExecutionSetShaderEXT-indirectExecutionSet-11041
  `indirectExecutionSet` **must** have been created with type VK_INDIRECT_EXECUTION_SET_INFO_TYPE_SHADER_OBJECTS_EXT
- VUID-vkUpdateIndirectExecutionSetShaderEXT-pExecutionSetWrites-11043
  Each element in the `pExecutionSetWrites` array **must** have a unique `VkWriteIndirectExecutionSetShaderEXT::index`
- VUID-vkUpdateIndirectExecutionSetShaderEXT-None-11044
  Each member of the Indirect Execution Set referenced by the update command **must** not be in use by the device
- VUID-vkUpdateIndirectExecutionSetShaderEXT-pInitialShaders-11326
  If the shaders in `pInitialShaders` were created without VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT, each shader in `pExecutionSetWrites` **must** also have been created without VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT
- VUID-vkUpdateIndirectExecutionSetShaderEXT-pInitialShaders-11327
  If the shaders in `pInitialShaders` were created with VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT, each shader in `pExecutionSetWrites` **must** also have been created with VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT
- VUID-vkUpdateIndirectExecutionSetShaderEXT-pExecutionSetWrites-11140
  If the shaders in `pInitialShaders` were created without VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT, the descriptor layout of each shader in `pExecutionSetWrites` **must** be compatible with the initial layout info used to create the Indirect Execution Set
- VUID-vkUpdateIndirectExecutionSetShaderEXT-None-11148
  Each fragment shader element in the Indirect Execution Set **must** have identically defined fragment outputs interface to the initial shader state used to create the Indirect Execution Set
- VUID-vkUpdateIndirectExecutionSetShaderEXT-FragDepth-11054
  Each fragment shader element in the Indirect Execution Set **must** match the initial shader

state used to create the Indirect Execution Set in its use of FragDepth

- VUID-vkUpdateIndirectExecutionSetShaderEXT-SampleMask-11050
  Each fragment shader element in the Indirect Execution Set **must** match the initial shader state used to create the Indirect Execution Set in its use of SampleMask
- VUID-vkUpdateIndirectExecutionSetShaderEXT-StencilExportEXT-11003
  Each fragment shader element in the Indirect Execution Set **must** match the initial shader state used to create the Indirect Execution Set in its use of StencilExportEXT

Valid Usage (Implicit)

- VUID-vkUpdateIndirectExecutionSetShaderEXT-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkUpdateIndirectExecutionSetShaderEXT-indirectExecutionSet-parameter
  indirectExecutionSet **must** be a valid VkIndirectExecutionSetEXT handle
- VUID-vkUpdateIndirectExecutionSetShaderEXT-pExecutionSetWrites-parameter
  pExecutionSetWrites **must** be a valid pointer to an array of executionSetWriteCount valid VkWriteIndirectExecutionSetShaderEXT structures
- VUID-vkUpdateIndirectExecutionSetShaderEXT-executionSetWriteCount-arraylength
  executionSetWriteCount **must** be greater than 0
- VUID-vkUpdateIndirectExecutionSetShaderEXT-indirectExecutionSet-parent
  indirectExecutionSet **must** have been created, allocated, or retrieved from device

Host Synchronization

- Host access to indirectExecutionSet **must** be externally synchronized

It is legal to update an Indirect Execution Set that is in flight as long as the element indices in pExecutionSetWrites are not in use. Any change to an indirect execution set requires recalculating memory requirements by calling vkGetGeneratedCommandsMemoryRequirementsEXT for commands that use that modified state. Commands that are in flight or those not using updated elements require no changes.

The lifetimes of pipelines and shader objects contained in a set **must** match or exceed the lifetime of the set.

With VK_EXT_device_generated_commands, the actual generation of commands as well as their execution on the device is handled as a single action with:

```
// Provided by VK_EXT_device_generated_commands
void vkCmdExecuteGeneratedCommandsEXT(
    VkCommandBuffer                             commandBuffer,
    VkBool32                                    isPreprocessed,
    const VkGeneratedCommandsInfoEXT*           pGeneratedCommandsInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `isPreprocessed` represents whether the input data has already been preprocessed on the device. If it is VK_FALSE this command will implicitly trigger the preprocessing step, otherwise not.
- `pGeneratedCommandsInfo` is a pointer to a VkGeneratedCommandsInfoEXT structure containing parameters affecting the generation of commands.

If the VK_INDIRECT_COMMANDS_LAYOUT_USAGE_UNORDERED_SEQUENCES_BIT_EXT flag was used to create the VkGeneratedCommandsInfoEXT::`indirectCommandsLayout` then the execution of sequences through this command **may** use implementation-defined ordering which is not guaranteed to be coherent using the same input data. It does not affect the order of token processing within a sequence. This is the implied ordering with VK_INDIRECT_COMMANDS_TOKEN_TYPE_DISPATCH_EXT.

After a call to `vkCmdExecuteGeneratedCommandsEXT`, command buffer state will become undefined according to the tokens executed. This table specifies the relationship between tokens used and state invalidation.

Table 47. Indirect Execution State Invalidation

Common Tokens	States Invalidated
VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT	Bound shaders and pipelines
VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT	Push constant data
VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT	Push constant data
VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_EXT	Push data
VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_SEQUENCE_INDEX_EXT	Push data
VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT	Index buffer
VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT	Vertex buffer

Valid Usage

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08606

If the shaderObject feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdExecuteGeneratedCommandsEXT-commandBuffer-02707
  If commandBuffer is an unprotected command buffer and protectedNoFault is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a VkDescriptorSetLayout that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdExecuteGeneratedCommandsEXT-magFilter-04553
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-magFilter-09598
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of

this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT

- VUID-vkCmdExecuteGeneratedCommandsEXT-mipmapMode-04770
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-mipmapMode-09599
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-unnormalizedCoordinates-09635
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08609
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `viewType` **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08610
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions with `ImplicitLod`, `Dref` or `Proj` in their name
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08611
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions that include a LOD bias or any offset values
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-02692

If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdExecuteGeneratedCommandsEXT-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdExecuteGeneratedCommandsEXT-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdExecuteGeneratedCommandsEXT-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in

the descriptor sets used by this command

- VUID-vkCmdExecuteGeneratedCommandsEXT-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdExecuteGeneratedCommandsEXT-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdExecuteGeneratedCommandsEXT-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdExecuteGeneratedCommandsEXT-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdExecuteGeneratedCommandsEXT-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match

- VUID-vkCmdExecuteGeneratedCommandsEXT-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdExecuteGeneratedCommandsEXT-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components
- VUID-vkCmdExecuteGeneratedCommandsEXT-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdExecuteGeneratedCommandsEXT-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdExecuteGeneratedCommandsEXT-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdExecuteGeneratedCommandsEXT-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdExecuteGeneratedCommandsEXT-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdExecuteGeneratedCommandsEXT-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdExecuteGeneratedCommandsEXT-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all

image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdExecuteGeneratedCommandsEXT-micromap-11636
  If an acceleration structure is accessed as a result of this command through an `OpRayQueryProceedKHR` operation , the `VK_EXT_opacity_micromap` extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its `pNext` chain, the shader **must** have specified a `OpacityMicromapIdKHR` execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdExecuteGeneratedCommandsEXT-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdExecuteGeneratedCommandsEXT-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with `flags` that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdExecuteGeneratedCommandsEXT-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with `flags` that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping

using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeImage or OpTypeSampledImage using that mapping, the calculated offset for the resource heap **must** be a multiple of imageDescriptorAlignment

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an OpTypeSampler or OpTypeSampledImage using that mapping, the calculated offset for the sampler heap **must** be a multiple of samplerDescriptorAlignment
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping

using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdExecuteGeneratedCommandsEXT-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of

`pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound

to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdExecuteGeneratedCommandsEXT-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdExecuteGeneratedCommandsEXT-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdExecuteGeneratedCommandsEXT-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the

address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdExecuteGeneratedCommandsEXT-renderPass-02684
  The current render pass **must** be compatible with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdExecuteGeneratedCommandsEXT-subpass-02685
  The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to VK_PIPELINE_BIND_POINT_GRAPHICS
- VUID-vkCmdExecuteGeneratedCommandsEXT-OpTypeImage-07468
  If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07469
  Input attachment views accessed in a subpass **must** be created with the same VkFormat as the corresponding subpass definition, and be created with a VkImageView that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the bound VkFramebuffer as specified by Fragment Input Attachment Compatibility
- VUID-vkCmdExecuteGeneratedCommandsEXT-pDepthInputAttachmentIndex-09595
  Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by VkRenderingInputAttachmentIndexInfo, or no `InputAttachmentIndex` if VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` or VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are `NULL`, **must** be created with a VkImageView that is compatible with the corresponding color, depth, or stencil attachment in VkRenderingInfo
- VUID-vkCmdExecuteGeneratedCommandsEXT-pDepthInputAttachmentIndex-09596
  Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both VkRenderingInputAttachmentIndexInfo::`pDepthInputAttachmentIndex` and VkRenderingInputAttachmentIndexInfo::`pStencilInputAttachmentIndex` are non-`NULL`
- VUID-vkCmdExecuteGeneratedCommandsEXT-InputAttachmentIndex-09597
  If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in VkRenderingInputAttachmentIndexInfo
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-06537
  Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-10795
  If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-10796
  If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-10797
  If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-12338
  If a color attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_COLOR_BIT on that attachment, the color attachment **must** not be written to by this command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-12339
  If a depth attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_DEPTH_BIT on that attachment, the depth attachment **must** not be written to by this command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-12340
  If a stencil attachment is read in this command in any way other than as an attachment, or has been read by any prior command in this subpass as a non-attachment, and feedback loop is not enabled for VK_IMAGE_ASPECT_STENCIL_BIT on that attachment, the stencil attachment **must** not be written to by this command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09003
  If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-06886
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, depth writes **must** be disabled
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-06887
  If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, all stencil ops **must** be VK_STENCIL_OP_KEEP
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07831
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07832
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not

subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08617
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineWidth **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07834
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBiasEnable` is VK_TRUE, then vkCmdSetDepthBias or vkCmdSetDepthBias2EXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07835
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_BLEND_CONSTANTS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and an active color attachment current value of `blendEnable` is VK_TRUE with a blend equation where any VkBlendFactor member is VK_BLEND_FACTOR_CONSTANT_COLOR, VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR, VK_BLEND_FACTOR_CONSTANT_ALPHA, or VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA, then vkCmdSetBlendConstants **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07836
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthBoundsTestEnable` is VK_TRUE, then vkCmdSetDepthBounds **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07837
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilCompareMask **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07838
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilWriteMask **must** have been called

and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07839
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_REFERENCE dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilReference **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-maxMultiviewInstanceIndex-02688
  If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to VkPhysicalDeviceMultiviewProperties::`maxMultiviewInstanceIndex`
- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-02689
  If the bound graphics pipeline was created with VkPipelineSampleLocationsStateCreateInfoEXT::`sampleLocationsEnable` set to VK_TRUE, then the active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07634
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-06666
  If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `sampleLocationsEnable` is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07840
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CULL_MODE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetCullMode **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07841
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_FRONT_FACE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetFrontFace **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07843

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07844
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthWriteEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07845
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_COMPARE_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `depthTestEnable` is VK_TRUE, then vkCmdSetDepthCompareOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07846
  If the `depthBounds` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBoundsTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07847
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetStencilTestEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07848
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_OP dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `stencilTestEnable` is VK_TRUE, then vkCmdSetStencilOp **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-viewportCount-03417
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, then vkCmdSetViewportWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-scissorCount-03418
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT dynamic state enabled, then vkCmdSetScissorWithCount **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-viewportCount-03419
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with both the VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT and VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic states enabled, then the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** match the `scissorCount` parameter of `vkCmdSetScissorWithCount`
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-04876
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE dynamic state enabled, then vkCmdSetRasterizerDiscardEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-04877
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then vkCmdSetDepthBiasEnable **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-logicOp-04878
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_EXT dynamic state enabled, the current value of `rasterizerDiscardEnable` is VK_FALSE, and the current value of `logicOpEnable` is VK_TRUE, then vkCmdSetLogicOpEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-primitiveFragmentShadingRateWithMultipleViewports-04552
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, the bound graphics pipeline was created with the VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT dynamic state enabled, and any of the shader stages of the bound graphics pipeline write to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be `1`
- VUID-vkCmdExecuteGeneratedCommandsEXT-primitiveFragmentShadingRateWithMultipleViewports-08642
  If the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, and any shader object bound to a graphics stage writes to the `PrimitiveShadingRateKHR` built-in, then vkCmdSetViewportWithCount **must** have been called in the current command buffer prior to this drawing command, and the `viewportCount` parameter of `vkCmdSetViewportWithCount` **must** be `1`
- VUID-vkCmdExecuteGeneratedCommandsEXT-blendEnable-04727
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics

pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then for each color attachment, if the corresponding image view's format features do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding current value of `blendEnable` **must** be VK_FALSE

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08644
  If a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of `rasterizerDiscardEnable` is VK_FALSE, and none of the following is enabled:

  - the `multisampledRenderToSingleSampled` feature

  then the current value of `rasterizationSamples` **must** be the same as the current color and/or depth/stencil attachments

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08876
  If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with vkCmdBeginRendering
- VUID-vkCmdExecuteGeneratedCommandsEXT-viewMask-06178
  If the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::`viewMask` equal to VkRenderingInfo::`viewMask`
- VUID-vkCmdExecuteGeneratedCommandsEXT-colorAttachmentCount-06179
  If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering, the bound graphics pipeline **must** have been created with a VkPipelineRenderingCreateInfo::`colorAttachmentCount` equal to VkRenderingInfo::`colorAttachmentCount`
- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08910
  If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `imageView` not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::`pColorAttachmentFormats` used to create the bound graphics pipeline
- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08912
  If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo::`colorAttachmentCount` greater than `0`, then each element of the VkRenderingInfo::`pColorAttachments` array with an `imageView` equal to VK_NULL_HANDLE **must** have the corresponding element of VkPipelineRenderingCreateInfo::`pColorAttachmentFormats` used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-

08911
If the dynamicRenderingUnusedAttachments feature is enabled, and the current render pass instance was begun with vkCmdBeginRendering and VkRenderingInfo ::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo ::pColorAttachments array with an imageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkPipelineRenderingCreateInfo::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07749
If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorWriteEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-attachmentCount-07750
If the colorWriteEnable feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the attachmentCount parameter of the most recent call to vkCmdSetColorWriteEnableEXT in the current command buffer **must** be greater than or equal to the number of active color attachments
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07751
If the VK_EXT_discard_rectangles extension is enabled, a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo included a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPipelineDiscardRectangleStateCreateInfoEXT ::discardRectangleCount
- VUID-vkCmdExecuteGeneratedCommandsEXT-rasterizerDiscardEnable-09236
If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT dynamic state enabled and the pNext chain of VkGraphicsPipelineCreateInfo did not include a VkPipelineDiscardRectangleStateCreateInfoEXT structure, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for each discard rectangle in VkPhysicalDeviceDiscardRectanglePropertiesEXT ::maxDiscardRectangles
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07880

If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDiscardRectangleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07881
  If the VK_EXT_discard_rectangles extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of discardRectangleEnable is VK_TRUE, then vkCmdSetDiscardRectangleModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08913
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08914
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView
- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08915
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08916
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-

08917
If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView

- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08918
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, and the value of VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->imageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteGeneratedCommandsEXT-imageView-06183
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingLocalRead-11797
  If the current render pass instance was begun with vkCmdBeginRendering, the dynamicRenderingLocalRead feature is enabled, the VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR flag is specified, and an attachment is being used as a feedback loop as specified by VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR, VkRenderingAttachmentFlagsInfoKHR::flags for that attachment **must** include VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- VUID-vkCmdExecuteGeneratedCommandsEXT-imageView-06184
  If the current render pass instance was begun with vkCmdBeginRendering and VkRenderingFragmentDensityMapAttachmentInfoEXT::imageView was not VK_NULL_HANDLE, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-multisampledRenderToSingleSampled-07285
  If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with a imageView not equal to VK_NULL_HANDLE **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdExecuteGeneratedCommandsEXT-multisampledRenderToSingleSampled-

07286
If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->imageView

- VUID-vkCmdExecuteGeneratedCommandsEXT-multisampledRenderToSingleSampled-07287
If the multisampledRenderToSingleSampled feature is not enabled, and vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->imageView
- VUID-vkCmdExecuteGeneratedCommandsEXT-pNext-07935
If this command is called inside a render pass instance started with vkCmdBeginRendering, and the pNext chain of VkRenderingInfo includes a VkMultisampledRenderToSingleSampledInfoEXT structure with multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the value of rasterizationSamples for the bound graphics pipeline **must** be equal to VkMultisampledRenderToSingleSampledInfoEXT::rasterizationSamples
- VUID-vkCmdExecuteGeneratedCommandsEXT-renderPass-06198
If the current render pass instance was begun with vkCmdBeginRendering, the bound pipeline **must** have been created with a VkGraphicsPipelineCreateInfo::renderPass equal to VK_NULL_HANDLE
- VUID-vkCmdExecuteGeneratedCommandsEXT-pColorAttachments-08963
If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->imageView was not VK_NULL_HANDLE, then the corresponding element of VkPipelineRenderingCreateInfo ::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteGeneratedCommandsEXT-pColorAttachments-11539
If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the VkRenderingInfo::pColorAttachments->resolveImageView was not VK_NULL_HANDLE, then the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteGeneratedCommandsEXT-pDepthAttachment-08964
If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance,

there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED

- VUID-vkCmdExecuteGeneratedCommandsEXT-pDepthAttachment-11540
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, depth test is enabled, and the VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteGeneratedCommandsEXT-pStencilAttachment-08965
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has not yet been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->imageView was not VK_NULL_HANDLE, then the VkPipelineRenderingCreateInfo::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteGeneratedCommandsEXT-pStencilAttachment-11860
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, there is a graphics pipeline bound, stencil test is enabled and the VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, then the VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the pipeline **must** not be VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteGeneratedCommandsEXT-flags-10582
  If the current render pass instance was begun with a vkCmdBeginRendering call in commandBuffer, its VkRenderingInfo::flags parameter **must** not have VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT set unless VK_RENDERING_CONTENTS_INLINE_BIT_KHR is also set
- VUID-vkCmdExecuteGeneratedCommandsEXT-primitivesGeneratedQueryWithRasterizerDiscard-06708
  If the primitivesGeneratedQueryWithRasterizerDiscard feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, rasterization discard **must** not be enabled
- VUID-vkCmdExecuteGeneratedCommandsEXT-primitivesGeneratedQueryWithNonZeroStreams-06709
  If the primitivesGeneratedQueryWithNonZeroStreams feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, the bound graphics pipeline **must** not have been created with a non-zero value in VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07620
  If the depthClamp feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the

current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetDepthClampEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07621
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetPolygonModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07622
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetRasterizationSamplesEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07623
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-alphaToCoverageEnable-08919
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and alphaToCoverageEnable was VK_TRUE in the last call to vkCmdSetAlphaToCoverageEnableEXT, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-vkCmdExecuteGeneratedCommandsEXT-alphaToCoverageEnable-08920
  If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetAlphaToCoverageEnableEXT in the current command buffer set alphaToCoverageEnable to VK_TRUE, then the Fragment Output Interface **must** contain a variable for the alpha Component word in Location 0 at Index 0
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07624
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToCoverageEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07625
  If the alphaToOne feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAlphaToOneEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07626
  If the logicOp feature is enabled, a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetLogicOpEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07627
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorBlendEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07629
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and there are color attachments bound, then vkCmdSetColorWriteMaskEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07630
  If the geometryStreams feature is enabled, and a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with both a VK_SHADER_STAGE_GEOMETRY_BIT stage and the VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT dynamic state enabled, then vkCmdSetRasterizationStreamEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07631
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetConservativeRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07632
  If the VK_EXT_conservative_rasterization extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of conservativeRasterizationMode is VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT, then vkCmdSetExtraPrimitiveOverestimationSizeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-conservativePointAndLineRasterization-

07499
If the VK_EXT_conservative_rasterization extension is enabled, conservativePointAndLineRasterization is not supported, a shader object is bound to any graphics stage or a graphics pipeline is bound, the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line or point topology class, then the current value of conservativeRasterizationMode **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07633
If the depthClipEnable feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT dynamic state enabled, then vkCmdSetDepthClipEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07636
If the VK_EXT_provoking_vertex extension is enabled, a shader object is bound to the VK_SHADER_STAGE_VERTEX_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetProvokingVertexModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08666
If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineRasterizationModeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08669
If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, and the effective rasterization input topology is in the line topology class, then vkCmdSetLineStippleEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07849
If any of the stippledRectangularLines, stippledBresenhamLines or stippledSmoothLines features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE dynamic state enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and the current value of stippledLineEnable is VK_TRUE, then vkCmdSetLineStipple **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-10608
  If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic state enabled, the effective rasterization input topology is in the line topology class, and the current `lineRasterizationMode` is VK_LINE_RASTERIZATION_MODE_BRESENHAM or VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the current `alphaToCoverageEnable`, `alphaToOneEnable` and `sampleShadingEnable` states **must** all be VK_FALSE
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07639
  If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT dynamic state enabled, then vkCmdSetDepthClipNegativeOneToOneEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09650
  If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT dynamic state enabled, and the current value of `depthClampEnable` is VK_TRUE, then vkCmdSetDepthClampRangeEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-rasterizationSamples-07471
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` in the last call to vkCmdSetRasterizationSamplesEXT **must** follow the rules for a zero-attachment subpass
- VUID-vkCmdExecuteGeneratedCommandsEXT-samples-07472
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the VkPipelineMultisampleStateCreateInfo::`rasterizationSamples` parameter used to create the bound graphics pipeline
- VUID-vkCmdExecuteGeneratedCommandsEXT-samples-07473
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT state and VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT states enabled, then the `samples` parameter in the last call to vkCmdSetSampleMaskEXT **must** be greater than or equal to the `rasterizationSamples` parameter in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09211
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, or a shader object is bound to any graphics stage, and the current render pass instance includes a VkMultisampledRenderToSingleSampledInfoEXT structure with

multisampledRenderToSingleSampledEnable equal to VK_TRUE, then the rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT **must** be the same as the rasterizationSamples member of that structure

- VUID-vkCmdExecuteGeneratedCommandsEXT-firstAttachment-07476
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorBlendEnableEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdExecuteGeneratedCommandsEXT-firstAttachment-07478
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then the last call to vkCmdSetColorWriteMaskEXT in the current command buffer prior to this drawing command **must** have set a value for all active color attachments
- VUID-vkCmdExecuteGeneratedCommandsEXT-advancedBlendMaxColorAttachments-07480
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT dynamic states enabled, the current value of rasterizerDiscardEnable is VK_FALSE, and an active color attachment current value of blendEnable is VK_TRUE, then the number of active color attachments **must** not exceed advancedBlendMaxColorAttachments
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-10862
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT , but not the VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE
- VUID-vkCmdExecuteGeneratedCommandsEXT-rasterizerDiscardEnable-10863
  If a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT, but not the VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetColorBlendAdvancedEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the blendEnable current value of VK_TRUE
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-10864
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with

VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT and VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT dynamic state enabled, and the current value of `rasterizerDiscardEnable` is VK_FALSE, then either vkCmdSetColorBlendAdvancedEXT or vkCmdSetColorBlendEquationEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command for all active color attachments with the `blendEnable` current value of VK_TRUE

- VUID-vkCmdExecuteGeneratedCommandsEXT-primitivesGeneratedQueryWithNonZeroStreams-07481
  If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query is active, a shader object is bound to the VK_SHADER_STAGE_GEOMETRY_BIT stage or a graphics pipeline is bound which was created with VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT state enabled, the last call to vkCmdSetRasterizationStreamEXT **must** have set the `rasterizationStream` to zero
- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsPerPixel-07482
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state disabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` member of the VkPipelineMultisampleStateCreateInfo structure the bound graphics pipeline has been created with
- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsPerPixel-07483
  If a shader object is bound to any graphics stage or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of `sampleLocationsEnable` is VK_TRUE, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to vkCmdSetSampleLocationsEXT **must** equal the `rasterizationSamples` parameter of the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-07484
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the current active depth attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set
- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-07485
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if `sampleLocationsEnable` was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to

vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples

- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-07486
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state enabled and the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, then the sampleLocationsInfo.sampleLocationGridSize.height in the last call to vkCmdSetSampleLocationsEXT **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling rasterizationSamples
- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-07487
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, or the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT state enabled, and if sampleLocationsEnable was VK_TRUE in the last call to vkCmdSetSampleLocationsEnableEXT, the fragment shader code **must** not statically use the extended instruction InterpolateAtSample
- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-07936
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.width **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.width as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-07937
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::maxSampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-07938
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then

sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdExecuteGeneratedCommandsEXT-stippledLineEnable-07495
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR, then the stippledRectangularLines feature **must** be enabled
- VUID-vkCmdExecuteGeneratedCommandsEXT-stippledLineEnable-07496
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM, then the stippledBresenhamLines feature **must** be enabled
- VUID-vkCmdExecuteGeneratedCommandsEXT-stippledLineEnable-07497
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH, then the stippledSmoothLines feature **must** be enabled
- VUID-vkCmdExecuteGeneratedCommandsEXT-stippledLineEnable-07498
  If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE
- VUID-vkCmdExecuteGeneratedCommandsEXT-stage-07073
  If the bound pipeline was created with the VkPipelineShaderStageCreateInfo::stage member of an element of VkGraphicsPipelineCreateInfo::pStages set to VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then Mesh Shader Queries **must** not be active
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08877
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetAttachmentFeedbackLoopEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command
- VUID-vkCmdExecuteGeneratedCommandsEXT-nextStage-10745
  For each shader object bound to a graphics stage, except for the shader object bound to

the last graphics stage in the logical pipeline, it **must** have been created with a `nextStage` including the corresponding bit to the shader object bound to the following graphics stage in the logical pipeline

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08684
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_VERTEX_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08685
  If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08686
  If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08687
  If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_GEOMETRY_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08688
  If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_FRAGMENT_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08689
  If there is no bound graphics pipeline, and the `taskShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_TASK_BIT_EXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08690
  If there is no bound graphics pipeline, and the `meshShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of VK_SHADER_STAGE_MESH_BIT_EXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08693
  If there is no bound graphics pipeline, and at least one of the `taskShader` and `meshShader` features is enabled, one of the VK_SHADER_STAGE_VERTEX_BIT or VK_SHADER_STAGE_MESH_BIT_EXT stages **must** have a valid `VkShaderEXT` bound, and the other **must** have no `VkShaderEXT` bound
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08696
  If there is no bound graphics pipeline, and a valid `VkShaderEXT` is bound to the VK_SHADER_STAGE_VERTEX_BIT stage, there **must** be no `VkShaderEXT` bound to either the VK_SHADER_STAGE_TASK_BIT_EXT stage or the VK_SHADER_STAGE_MESH_BIT_EXT stage
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08698
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, then all shaders created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag in the same vkCreateShadersEXT call

must also be bound

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08699
  If any graphics shader is bound which was created with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag, any stages in between stages whose shaders which did not create a shader with the VK_SHADER_CREATE_LINK_STAGE_BIT_EXT flag as part of the same vkCreateShadersEXT call **must** not have any VkShaderEXT bound
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08878
  All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08879
  All bound graphics shader objects **must** have either been created with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag set, or with identical or identically defined arrays of descriptor set layouts
- VUID-vkCmdExecuteGeneratedCommandsEXT-pDynamicStates-08715
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpDepthAttachmentReadEXT, the depthWriteEnable parameter in the last call to vkCmdSetDepthWriteEnable **must** be VK_FALSE
- VUID-vkCmdExecuteGeneratedCommandsEXT-pDynamicStates-08716
  If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_STENCIL_WRITE_MASK dynamic state enabled, and the fragment shader declares the EarlyFragmentTests execution mode and uses OpStencilAttachmentReadEXT, the writeMask parameter in the last call to vkCmdSetStencilWriteMask **must** be 0
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09116
  If a shader object is bound to any graphics stage or the bound graphics pipeline was created with VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT, and the format of any color attachment is VK_FORMAT_E5B9G9R9_UFLOAT_PACK32, the corresponding element of the pColorWriteMasks parameter of vkCmdSetColorWriteMaskEXT **must** either include all of VK_COLOR_COMPONENT_R_BIT, VK_COLOR_COMPONENT_G_BIT, VK_COLOR_COMPONENT_B_BIT, and VK_COLOR_COMPONENT_A_BIT, or none of them
- VUID-vkCmdExecuteGeneratedCommandsEXT-maxFragmentDualSrcAttachments-09239
  If blending is enabled for any attachment where either the source or destination blend factors for that attachment use the secondary color input, the maximum value of Location for any output attachment statically used in the Fragment Execution Model executed by this command **must** be less than maxFragmentDualSrcAttachments
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09548
  If the current render pass was begun with vkCmdBeginRendering, there is no shader object bound to any graphics stage, the value of each element of VkRenderingAttachmentLocationInfo::pColorAttachmentLocations in the bound pipeline **must** match the value for the corresponding locations set currently in the current render

pass instance

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09549
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, for each element of VkRenderingInputAttachmentIndexInfo::pColorAttachmentInputIndices in the bound pipeline, if either pColorAttachmentInputIndices[i] or the value for the corresponding index set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-10927
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pDepthInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-10928
  If the current render pass was begun with vkCmdBeginRendering, and there is no shader object bound to any graphics stage, if either the value of VkRenderingInputAttachmentIndexInfo::pStencilInputAttachmentIndex in the bound pipeline or the corresponding value set currently in the current render pass instance is statically used in the bound pipeline in an InputAttachmentIndex decoration, the two values **must** match
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09642
  If the current render pass was begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag, the bound graphics pipeline **must** have been created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09643
  If the bound graphics pipeline was created with VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT, the current render pass **must** have begun with vkCmdBeginRendering with the VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT flag
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-10772
  If a shader object is bound to any graphics stage, *multiview* functionality **must** not be enabled in the current render pass
- VUID-vkCmdExecuteGeneratedCommandsEXT-flags-11521
  If the current render pass instance was begun with vkCmdBeginRendering with VkRenderingInfo::flags which includes VK_RENDERING_FRAGMENT_REGION_BIT_EXT, and if sample shading is enabled (explicitly or implicitly), then the minimum fraction for sample shading **must** equal 0.0
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11522
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** have been created with a VkCustomResolveCreateInfoEXT

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11523
  If the current render pass instance was begun with vkCmdBeginRendering and does not contain a custom resolve, and the dynamicRenderingUnusedAttachments feature is not enabled, the graphics pipeline bound **must** not have been created with a VkCustomResolveCreateInfoEXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-customResolve-11524
  If the current render pass instance was begun with vkCmdBeginRendering and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdExecuteGeneratedCommandsEXT-customResolve-11525
  If the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, and vkCmdBeginCustomResolveEXT has not been recorded in the render pass instance, the graphics pipeline bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11861
  If the dynamicRenderingUnusedAttachments feature is not enabled and the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, the bound graphics pipeline **must** have been created with a VkCustomResolveCreateInfoEXT::colorAttachmentCount equal to VkRenderingInfo::colorAttachmentCount
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11862
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11863
  If the dynamicRenderingUnusedAttachments feature is not enabled, and the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView equal to VK_NULL_HANDLE **must** have the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound pipeline equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-11864
  If the dynamicRenderingUnusedAttachments feature is enabled, the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, and VkRenderingInfo::colorAttachmentCount greater than 0, then each element of the VkRenderingInfo::pColorAttachments array with an resolveImageView not equal to VK_NULL_HANDLE **must** have been created with a VkFormat equal to the corresponding element of VkCustomResolveCreateInfoEXT::pColorAttachmentFormats used to create the bound graphics pipeline, or the corresponding element of

VkCustomResolveCreateInfoEXT::pColorAttachmentFormats, if it exists, **must** be VK_FORMAT_UNDEFINED

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11865
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11866
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-11867
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::depthAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pDepthAttachment->resolveImageView, the value of the format **must** be VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11868
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to VK_FORMAT_UNDEFINED
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11869
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is not enabled, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline **must** be equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-11870
  If the current render pass instance was begun with vkCmdBeginRendering, it contains a custom resolve, the dynamicRenderingUnusedAttachments feature is enabled, VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, and the value of VkCustomResolveCreateInfoEXT::stencilAttachmentFormat used to create the bound graphics pipeline was not equal to the VkFormat used to create VkRenderingInfo::pStencilAttachment->resolveImageView, the value of the format **must** be

VK_FORMAT_UNDEFINED

- VUID-vkCmdExecuteGeneratedCommandsEXT-colorAttachmentCount-11871
  If the current render pass instance was begun with vkCmdBeginRendering with a VkRenderingInfo::colorAttachmentCount parameter greater than 0 and vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, then for each element of the VkRenderingInfo::pColorAttachments array with a resolveImageView not equal to VK_NULL_HANDLE, the resolveImageView **must** have been created with a sample count equal to the value of rasterizationSamples for the bound graphics pipeline
- VUID-vkCmdExecuteGeneratedCommandsEXT-pDepthAttachment-11872
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pDepthAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pDepthAttachment->resolveImageView
- VUID-vkCmdExecuteGeneratedCommandsEXT-pStencilAttachment-11873
  If the current render pass instance was begun with vkCmdBeginRendering, vkCmdBeginCustomResolveEXT has been recorded in the render pass instance, and VkRenderingInfo::pStencilAttachment->resolveImageView was not VK_NULL_HANDLE, the value of rasterizationSamples for the bound graphics pipeline **must** be equal to the sample count used to create VkRenderingInfo::pStencilAttachment->resolveImageView
- VUID-vkCmdExecuteGeneratedCommandsEXT-customResolve-11529
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_TRUE
- VUID-vkCmdExecuteGeneratedCommandsEXT-customResolve-11530
  If a shader object is bound to the fragment stage, the current render pass instance was begun with vkCmdBeginRendering and contains a custom resolve, a fragment density map attachment is active, and vkCmdBeginCustomResolveEXT has not yet been called, then the fragment shader object bound **must** have been created with VkCustomResolveCreateInfoEXT::customResolve as VK_FALSE
- VUID-vkCmdExecuteGeneratedCommandsEXT-flags-13361
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, then all bound shader objects **must** have been created with the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit
- VUID-vkCmdExecuteGeneratedCommandsEXT-flags-13362
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a setLayoutCount not equal to 0, and with pSetLayouts not equal to NULL, then the pipeline layout of the bound descriptor sets **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set
- VUID-vkCmdExecuteGeneratedCommandsEXT-flags-12491
  If there are any shader objects bound that were created with flags that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, with a pushConstantRangeCount not equal to 0, and with pPushConstantRanges not equal to NULL, then the pipeline layout of

the bound push constants **must** have been created with VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT set

- VUID-vkCmdExecuteGeneratedCommandsEXT-flags-13363
  If there are any shader objects bound that were created with `flags` that includes the VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR bit, and any of the descriptor set layouts used to create the pipeline layout of the bound descriptor sets were created with any binding's VkDescriptorSetLayoutBinding::`stageFlags` set with VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT, then there **must** be no binding with VkDescriptorSetLayoutBinding::`stageFlags` set with VK_SHADER_STAGE_TASK_BIT_EXT or VK_SHADER_STAGE_MESH_BIT_EXT

- VUID-vkCmdExecuteGeneratedCommandsEXT-commandBuffer-11045
  `commandBuffer` **must** not be a protected command buffer

- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11046
  If `isPreprocessed` is VK_TRUE and vkGetGeneratedCommandsMemoryRequirementsEXT did not return a required size of zero then vkCmdPreprocessGeneratedCommandsEXT **must** have already been executed on the device before this command executes, and the preprocessing command **must** have used the same `pGeneratedCommandsInfo` content as well as the content of the input buffers it references (all except VkGeneratedCommandsInfoEXT::`preprocessAddress`)

- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11047
  If `isPreprocessed` is VK_TRUE then the `indirectCommandsLayout` member of `pGeneratedCommandsInfo` **must** have been created with the VK_INDIRECT_COMMANDS_LAYOUT_USAGE_EXPLICIT_PREPROCESS_BIT_EXT bit set

- VUID-vkCmdExecuteGeneratedCommandsEXT-indirectCommandsLayout-11141
  If the `indirectCommandsLayout` member of `pGeneratedCommandsInfo` was created with the VK_INDIRECT_COMMANDS_LAYOUT_USAGE_EXPLICIT_PREPROCESS_BIT_EXT bit set, then `isPreprocessed` **must** be VK_TRUE

- VUID-vkCmdExecuteGeneratedCommandsEXT-preprocessAddress-11142
  The contents of the `preprocessAddress` member of `pGeneratedCommandsInfo` **must** not have been previously used to record another vkCmdExecuteGeneratedCommandsEXT

- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11048
  If `isPreprocessed` is VK_TRUE then the bound descriptor sets and push constants **must** match identically with those bound during recording of the corresponding call to vkCmdPreprocessGeneratedCommandsEXT

- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-10198
  If `isPreprocessed` is VK_TRUE then the conditional render state and its predicate value **must** match identically with the state and value set during execution of the corresponding call to vkCmdPreprocessGeneratedCommandsEXT

- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11049
  If `isPreprocessed` is VK_TRUE and the `indirectCommandsLayout` member of `pGeneratedCommandsInfo` contains a draw token, then the graphics state bound on `commandBuffer` **must** match identically with the graphics state bound on the

stateCommandBuffer passed to vkCmdPreprocessGeneratedCommandsEXT

- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11149
  If isPreprocessed is VK_TRUE, then the queue family index of commandBuffer **must** be the same as the queue family index used to allocate the stateCommandBuffer passed to vkCmdPreprocessGeneratedCommandsEXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11051
  If isPreprocessed is VK_TRUE and the indirectCommandsLayout member of pGeneratedCommandsInfo contains a dispatch token, then the compute state bound on commandBuffer **must** match identically with the compute state bound on the stateCommandBuffer passed to vkCmdPreprocessGeneratedCommandsEXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11052
  If isPreprocessed is VK_TRUE and the indirectCommandsLayout member of pGeneratedCommandsInfo contains a ray tracing token, then the ray tracing state bound on commandBuffer **must** match identically with the ray tracing state bound on the stateCommandBuffer passed to vkCmdPreprocessGeneratedCommandsEXT
- VUID-vkCmdExecuteGeneratedCommandsEXT-indirectCommandsLayout-11053
  If the token sequence of the passed VkGeneratedCommandsInfoEXT::indirectCommandsLayout contains a VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, the initial shader state of VkGeneratedCommandsInfoEXT::indirectExecutionSet **must** be bound on commandBuffer
- VUID-vkCmdExecuteGeneratedCommandsEXT-indirectCommandsLayout-11004
  If indirectCommandsLayout was created with a token sequence that contained the VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token and indirectExecutionSet was created using VK_INDIRECT_EXECUTION_SET_INFO_TYPE_SHADER_OBJECTS_EXT, every executed VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token **must** bind all the shader stages set in the VkIndirectCommandsExecutionSetTokenEXT::shaderStages used to create indirectCommandsLayout
- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11055
  If isPreprocessed is VK_TRUE and the token sequence of the passed VkGeneratedCommandsInfoEXT::indirectCommandsLayout contains a VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, the members of VkGeneratedCommandsInfoEXT::indirectExecutionSet accessed by this command **must** not have been modified since the preprocess buffer was generated
- VUID-vkCmdExecuteGeneratedCommandsEXT-indirectCommandsLayout-11056
  If the indirectCommandsLayout member of pGeneratedCommandsInfo contains a draw token, then the active render pass **must** not have a specified fragment density map
- VUID-vkCmdExecuteGeneratedCommandsEXT-deviceGeneratedCommandsTransformFeedback-11057
  If deviceGeneratedCommandsTransformFeedback is not supported on device, transform feedback **must** not be active
- VUID-vkCmdExecuteGeneratedCommandsEXT-indirectExecutionSet-11058
  If transform feedback is active, VkGeneratedCommandsInfoEXT::indirectExecutionSet

must be VK_NULL_HANDLE

- VUID-vkCmdExecuteGeneratedCommandsEXT-deviceGeneratedCommands-11059
  The VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT::deviceGeneratedCommands feature **must** be enabled
- VUID-vkCmdExecuteGeneratedCommandsEXT-supportedIndirectCommandsShaderStages-11060
  The bound shader stages **must** be supported by VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStages
- VUID-vkCmdExecuteGeneratedCommandsEXT-supportedIndirectCommandsShaderStages-11061
  Only stages specified in VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStages **can** be set in pGeneratedCommandsInfo->shaderStages
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11062
  If a rendering pass is currently active, either the view mask **must** be 0 or indirectExecutionSet **must** be VK_NULL_HANDLE
- VUID-vkCmdExecuteGeneratedCommandsEXT-commandBuffer-11143
  commandBuffer **must** not have been recorded with VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT
- VUID-vkCmdExecuteGeneratedCommandsEXT-indirectCommandsLayout-10769
  If the indirectCommandsLayout member of pGeneratedCommandsInfo contains a draw token, this command **must** not be called outside a render pass instance
- VUID-vkCmdExecuteGeneratedCommandsEXT-indirectCommandsLayout-12202
  If the indirectCommandsLayout member of pGeneratedCommandsInfo does not contain a draw token, this command **must** not be called inside a render pass instance

Valid Usage (Implicit)

- VUID-vkCmdExecuteGeneratedCommandsEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdExecuteGeneratedCommandsEXT-pGeneratedCommandsInfo-parameter
  pGeneratedCommandsInfo **must** be a valid pointer to a valid VkGeneratedCommandsInfoEXT structure
- VUID-vkCmdExecuteGeneratedCommandsEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdExecuteGeneratedCommandsEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdExecuteGeneratedCommandsEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdExecuteGeneratedCommandsEXT-videocoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdExecuteGeneratedCommandsEXT-bufferlevel
  `commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Both	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	Action Indirection

Conditional Rendering

vkCmdExecuteGeneratedCommandsEXT is affected by conditional rendering

The `VkGeneratedCommandsInfoEXT` structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkGeneratedCommandsInfoEXT {
    VkStructureType                 sType;
    const void*                     pNext;
    VkShaderStageFlags              shaderStages;
    VkIndirectExecutionSetEXT       indirectExecutionSet;
    VkIndirectCommandsLayoutEXT     indirectCommandsLayout;
    VkDeviceAddress                 indirectAddress;
    VkDeviceSize                    indirectAddressSize;
    VkDeviceAddress                 preprocessAddress;
    VkDeviceSize                    preprocessSize;
    uint32_t                        maxSequenceCount;
    VkDeviceAddress                 sequenceCountAddress;
    uint32_t                        maxDrawCount;
} VkGeneratedCommandsInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderStages` is the mask of shader stages used by the commands.
- `indirectExecutionSet` is the indirect execution set to be used for binding shaders.
- `indirectCommandsLayout` is the VkIndirectCommandsLayoutEXT that specifies the command sequence data.
- `indirectAddress` is an address that holds the indirect buffer data.
- `indirectAddressSize` is the size in bytes of indirect buffer data starting at `indirectAddress`.
- `preprocessAddress` specifies a physical address of the `VkBuffer` used for preprocessing the input data for execution. If this structure is used with vkCmdExecuteGeneratedCommandsEXT with its `isPreprocessed` set to VK_TRUE, then the preprocessing step is skipped but data in this address **may** still be modified. The contents and the layout of this address are opaque to applications and **must** not be modified outside functions related to device-generated commands or copied to another buffer for reuse.
- `preprocessSize` is the maximum byte size within `preprocessAddress` that is available for preprocessing.
- `maxSequenceCount` is used to determine the number of sequences to execute.
- `sequenceCountAddress` specifies an optional physical address of a single `uint32_t` value containing the requested number of sequences to execute.
- `maxDrawCount` is the maximum number of indirect draws that can be executed by any COUNT-type multi-draw indirect tokens. The draw count in the indirect buffer is clamped to this value for these token types.

If `sequenceCountAddress` is not `NULL`, then `maxSequenceCount` is the maximum number of sequences that can be executed. The actual number is `min(maxSequenceCount, *sequenceCountAddress)`. If `sequenceCountAddress` is `NULL`, then `maxSequenceCount` is the exact number of sequences to execute.

If the action command token for the layout is not a COUNT-type multi-draw indirect token, `maxDrawCount` is ignored.

Valid Usage

- VUID-VkGeneratedCommandsInfoEXT-preprocessAddress-11063
  If vkGetGeneratedCommandsMemoryRequirementsEXT returns a non-zero size, `preprocessAddress` **must** not be `0`
- VUID-VkGeneratedCommandsInfoEXT-preprocessAddress-11064
  `VkDeviceMemory` objects bound to the underlying buffer for `preprocessAddress` **must** have been allocated using one of the memory types allowed in the `memoryTypeBits` member of the VkMemoryRequirements structure returned by vkGetGeneratedCommandsMemoryRequirementsEXT
- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11065
  If the `indirectCommandsLayout` uses a token of VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT, then the `indirectExecutionSet`'s push constant layout **must** contain the `updateRange` specified in

VkIndirectCommandsPushConstantTokenEXT

- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11066
  If the `indirectCommandsLayout` uses a token of VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT, then the `indirectExecutionSet`'s push constant layout **must** contain the `updateRange` specified in VkIndirectCommandsPushConstantTokenEXT
- VUID-VkGeneratedCommandsInfoEXT-maxSequenceCount-11067
  `maxSequenceCount` **must** be less than or equal to VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::`maxIndirectSequenceCount` and VkGeneratedCommandsMemoryRequirementsInfoEXT::`maxSequenceCount` that was used to determine the `preprocessSize`
- VUID-VkGeneratedCommandsInfoEXT-sequenceCountAddress-11068
  If `sequenceCountAddress` is not `NULL`, the value contained in the address **must** be less than or equal to VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::`maxIndirectSequenceCount` and VkGeneratedCommandsMemoryRequirementsInfoEXT::`maxSequenceCount` that was used to determine the `preprocessSize`
- VUID-VkGeneratedCommandsInfoEXT-maxSequenceCount-10246
  `maxSequenceCount` **must** not be zero
- VUID-VkGeneratedCommandsInfoEXT-preprocessAddress-11069
  `preprocessAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_2_PREPROCESS_BUFFER_BIT_EXT usage flag set
- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11144
  If the `indirectCommandsLayout` contains a VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, and there is a descriptor and push constant layout info provided either by `pipelineLayout` or through a VkPipelineLayoutCreateInfo in `pNext` of the VkIndirectCommandsLayoutCreateInfoEXT used to create `indirectCommandsLayout`, the pipeline layout **must** be compatible with the descriptor and push constant layout info used by `indirectExecutionSet`
- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11328
  If the `indirectCommandsLayout` contains a VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, and there was no descriptor and push constant layout info provided either by `pipelineLayout` or through a VkPipelineLayoutCreateInfo in `pNext` of the VkIndirectCommandsLayoutCreateInfoEXT used to create `indirectCommandsLayout`, pipelines in `indirectExecutionSet` **must** have been created with VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT
- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11329
  If the `indirectCommandsLayout` contains a VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, and there was a descriptor and push constant layout info provided either by `pipelineLayout` or through a VkPipelineLayoutCreateInfo in `pNext` of the VkIndirectCommandsLayoutCreateInfoEXT used to create `indirectCommandsLayout`, pipelines in `indirectExecutionSet` **must** have been created without VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT
- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11330
  If the `indirectCommandsLayout` contains a

VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, and there was no descriptor and push constant layout info provided either by pipelineLayout or through a VkPipelineLayoutCreateInfo in pNext of the VkIndirectCommandsLayoutCreateInfoEXT used to create indirectCommandsLayout, shaders in indirectExecutionSet **must** have been created with VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT

- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11331
  If the indirectCommandsLayout contains a VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, and there was a descriptor and push constant layout info provided either by pipelineLayout or through a VkPipelineLayoutCreateInfo in pNext of the VkIndirectCommandsLayoutCreateInfoEXT used to create indirectCommandsLayout, shaders in indirectExecutionSet **must** have been created without VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT
- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11002
  If indirectCommandsLayout was created with a token sequence that contained the VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, the shader stages used to create the initial shader state of indirectExecutionSet **must** equal the VkIndirectCommandsExecutionSetTokenEXT::shaderStages used to create indirectCommandsLayout
- VUID-VkGeneratedCommandsInfoEXT-preprocessSize-11071
  preprocessSize **must** be greater than or equal to the memory requirement's size returned by vkGetGeneratedCommandsMemoryRequirementsEXT using the matching inputs (indirectCommandsLayout, ...) as within this structure
- VUID-VkGeneratedCommandsInfoEXT-sequenceCountAddress-11072
  The underlying buffer for sequenceCountAddress **must** have the VK_BUFFER_USAGE_2_INDIRECT_BUFFER_BIT_KHR bit set in its usage flag
- VUID-VkGeneratedCommandsInfoEXT-sequenceCountAddress-11073
  If sequenceCountAddress is not NULL, sequenceCountAddress **must** be aligned to 4
- VUID-VkGeneratedCommandsInfoEXT-indirectAddress-12407
  The underlying buffer for indirectAddress **must** have the VK_BUFFER_USAGE_2_INDIRECT_BUFFER_BIT_KHR bit set in its usage flag
- VUID-VkGeneratedCommandsInfoEXT-indirectAddress-11074
  indirectAddress **must** be aligned to 4
- VUID-VkGeneratedCommandsInfoEXT-indirectAddressSize-11077
  indirectAddressSize **must** be greater than zero
- VUID-VkGeneratedCommandsInfoEXT-maxDrawCount-11078
  When not ignored, maxDrawCount × maxSequenceCount **must** be less than 2^24
- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11079
  If indirectCommandsLayout was created using a VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT token and shader objects are not bound then the bound graphics pipeline **must** have been created with VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE in pDynamicStates
- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11083
  If the token sequence of the passed indirectCommandsLayout contains a

VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, the indirectExecutionSet **must** not be VK_NULL_HANDLE

- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-10241
  If the token sequence of the passed indirectCommandsLayout does not contain a VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, the indirectExecutionSet **must** be VK_NULL_HANDLE
- VUID-VkGeneratedCommandsInfoEXT-indirectExecutionSet-11080
  If indirectExecutionSet is VK_NULL_HANDLE, a VkGeneratedCommandsPipelineInfoEXT or VkGeneratedCommandsShaderInfoEXT **must** be included in the pNext chain

Valid Usage (Implicit)

- VUID-VkGeneratedCommandsInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_GENERATED_COMMANDS_INFO_EXT
- VUID-VkGeneratedCommandsInfoEXT-shaderStages-parameter
  shaderStages **must** be a valid combination of VkShaderStageFlagBits values
- VUID-VkGeneratedCommandsInfoEXT-indirectExecutionSet-parameter
  If indirectExecutionSet is not VK_NULL_HANDLE, indirectExecutionSet **must** be a valid VkIndirectExecutionSetEXT handle
- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-parameter
  indirectCommandsLayout **must** be a valid VkIndirectCommandsLayoutEXT handle
- VUID-VkGeneratedCommandsInfoEXT-indirectAddress-parameter
  indirectAddress **must** be a valid VkDeviceAddress value
- VUID-VkGeneratedCommandsInfoEXT-preprocessAddress-parameter
  If preprocessAddress is not 0, preprocessAddress **must** be a valid VkDeviceAddress value
- VUID-VkGeneratedCommandsInfoEXT-sequenceCountAddress-parameter
  If sequenceCountAddress is not 0, sequenceCountAddress **must** be a valid VkDeviceAddress value
- VUID-VkGeneratedCommandsInfoEXT-commonparent
  Both of indirectCommandsLayout, and indirectExecutionSet that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Referencing the functions defined in Indirect Commands Layout, vkCmdExecuteGeneratedCommandsEXT behaves as:

```
uint32_t sequencesCount = sequenceCountAddress ?
      min(maxSequenceCount, sequenceCountAddress.load_uint32()) :
      maxSequenceCount;


cmdProcessAllSequences(commandBuffer, indirectExecutionSet,
                       indirectCommandsLayout, indirectAddress,
```

```
                        sequencesCount);

// The stateful commands within indirectCommandsLayout will not
// affect the state of subsequent commands in the target
// command buffer (cmd)
```

NOTE It is important to note that the affected values of all state related to the `shaderStages` used are undefined after this command. This means that e.g., if this command indirectly alters push constants, the push constant state becomes undefined.

Commands **can** be preprocessed prior to execution using the following command:

```
// Provided by VK_EXT_device_generated_commands
void vkCmdPreprocessGeneratedCommandsEXT(
    VkCommandBuffer                             commandBuffer,
    const VkGeneratedCommandsInfoEXT*           pGeneratedCommandsInfo,
    VkCommandBuffer                             stateCommandBuffer);
```

- `commandBuffer` is the command buffer which does the preprocessing.
- `pGeneratedCommandsInfo` is a pointer to a VkGeneratedCommandsInfoEXT structure containing parameters affecting the preprocessing step.
- `stateCommandBuffer` is a command buffer from which to snapshot current states affecting the preprocessing step. When a graphics action command token is used, graphics state is snapshotted. When a compute action command token is used, compute state is snapshotted. When a ray tracing action command token is used, ray tracing state is snapshotted. It can be deleted at any time after this command has been recorded.

NOTE `stateCommandBuffer` access is not synchronized by the driver, meaning that this command buffer **must** not be modified between threads in an unsafe manner.

Valid Usage

- VUID-vkCmdPreprocessGeneratedCommandsEXT-commandBuffer-11081
  `commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdPreprocessGeneratedCommandsEXT-pGeneratedCommandsInfo-11082
  `pGeneratedCommandsInfo`'s `indirectCommandsLayout` **must** have been created with the VK_INDIRECT_COMMANDS_LAYOUT_USAGE_EXPLICIT_PREPROCESS_BIT_EXT bit set
- VUID-vkCmdPreprocessGeneratedCommandsEXT-indirectCommandsLayout-11084
  If the token sequence of the passed VkGeneratedCommandsInfoEXT ::`indirectCommandsLayout` contains a VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, the initial shader state of VkGeneratedCommandsInfoEXT::`indirectExecutionSet` **must** be bound on `stateCommandBuffer`
- VUID-vkCmdPreprocessGeneratedCommandsEXT-stateCommandBuffer-11138

stateCommandBuffer **must** be in the recording state

- VUID-vkCmdPreprocessGeneratedCommandsEXT-deviceGeneratedCommands-11087
  The VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT::deviceGeneratedCommands feature **must** be enabled
- VUID-vkCmdPreprocessGeneratedCommandsEXT-supportedIndirectCommandsShaderStages-11088
  Only stages specified in VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStages **can** be set in pGeneratedCommandsInfo->shaderStages

Valid Usage (Implicit)

- VUID-vkCmdPreprocessGeneratedCommandsEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdPreprocessGeneratedCommandsEXT-pGeneratedCommandsInfo-parameter
  pGeneratedCommandsInfo **must** be a valid pointer to a valid VkGeneratedCommandsInfoEXT structure
- VUID-vkCmdPreprocessGeneratedCommandsEXT-stateCommandBuffer-parameter
  stateCommandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdPreprocessGeneratedCommandsEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdPreprocessGeneratedCommandsEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdPreprocessGeneratedCommandsEXT-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdPreprocessGeneratedCommandsEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdPreprocessGeneratedCommandsEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdPreprocessGeneratedCommandsEXT-bufferlevel
  commandBuffer **must** be a primary VkCommandBuffer
- VUID-vkCmdPreprocessGeneratedCommandsEXT-commonparent
  Both of commandBuffer, and stateCommandBuffer **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to stateCommandBuffer **must** be externally synchronized

- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdPreprocessGeneratedCommandsEXT is not affected by conditional rendering

The bound descriptor sets and push constants that will be used with indirect command generation **must** already be specified on stateCommandBuffer at the time of preprocessing commands with vkCmdPreprocessGeneratedCommandsEXT. They **must** match the bound descriptor sets and push constants used in the execution of indirect commands with vkCmdExecuteGeneratedCommandsEXT.

If push constants for shader stages are also specified in the VkGeneratedCommandsInfoEXT ::indirectCommandsLayout with a VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT or VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT token, then those values override the push constants that were previously pushed.

All State-setting commands that are bound on stateCommandBuffer will be used. Any parameters set by those commands **must** be set identically in a command buffer that vkCmdExecuteGeneratedCommandsEXT is recorded into, at the point vkCmdExecuteGeneratedCommandsEXT is recorded. If conditional rendering is used, the predicate value at preprocessing time **must** match the one at execution time. The queue family index stateCommandBuffer was allocated from **must** be the same as the queue family index of the command buffer used in vkCmdExecuteGeneratedCommandsEXT.

On some implementations, preprocessing **may** have no effect on performance.

vkCmdExecuteGeneratedCommandsEXT **may** write to the preprocess buffer, no matter the isPreprocess parameter. In this case, the implementation **must** insert appropriate synchronization automatically, which corresponds to the following pseudocode:

- Barrier
  - srcStageMask = DRAW_INDIRECT
  - srcAccessMask = 0

    - dstStageMask = COMMAND_PREPROCESS_BIT
    - dstAccessMask = COMMAND_PREPROCESS_WRITE_BIT | COMMAND_PREPROCESS_READ_BIT
- Do internal writes
- Barrier
    - srcStageMask = COMMAND_PREPROCESS_BIT
    - srcAccessMask = COMMAND_PREPROCESS_WRITE_BIT
    - dstStageMask = DRAW_INDIRECT
    - dstAccessMask = INDIRECT_COMMAND_READ_BIT
- Execute

Chapter 36. Sparse Resources

As documented in Resource Memory Association, `VkBuffer` and `VkImage` resources in Vulkan **must** be bound completely and contiguously to a single `VkDeviceMemory` object. This binding **must** be done before the resource is used, and the binding is immutable for the lifetime of the resource.

Sparse resources relax these restrictions and provide these additional features:

- Sparse resources **can** be bound non-contiguously to one or more `VkDeviceMemory` allocations.
- Sparse resources **can** be re-bound to different memory allocations over the lifetime of the resource.
- Sparse resources **can** have descriptors generated and used orthogonally with memory binding commands.

36.1. Sparse Resource Features

Sparse resources have several features that **must** be enabled explicitly at resource creation time. The features are enabled by including bits in the `flags` parameter of VkImageCreateInfo or VkBufferCreateInfo. Each feature also has one or more corresponding feature enables specified in VkPhysicalDeviceFeatures.

- The `sparseBinding` feature is the base, and provides the following capabilities:
  - Resources **can** be bound at some defined (sparse block) granularity.
  - The entire resource **must** be bound to memory before use regardless of regions actually accessed.
  - No specific mapping of image region to memory offset is defined, i.e. the location that each texel corresponds to in memory is implementation-dependent.
  - Sparse buffers have a well-defined mapping of buffer range to memory range, where an offset into a range of the buffer that is bound to a single contiguous range of memory corresponds to an identical offset within that range of memory.
  - Requested via the VK_IMAGE_CREATE_SPARSE_BINDING_BIT and VK_BUFFER_CREATE_SPARSE_BINDING_BIT bits.
  - A sparse image created using VK_IMAGE_CREATE_SPARSE_BINDING_BIT (but not VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT) supports all formats that non-sparse usage supports, and supports both VK_IMAGE_TILING_OPTIMAL and VK_IMAGE_TILING_LINEAR tiling.
- *Sparse Residency* builds on (and requires) the `sparseBinding` feature. It includes the following capabilities:
  - Resources do not have to be completely bound to memory before use on the device.
  - Images have a prescribed sparse image block layout, allowing specific rectangular regions of the image to be bound to specific offsets in memory allocations.
  - Consistency of access to unbound regions of the resource is defined by the absence or presence of `VkPhysicalDeviceSparseProperties`::`residencyNonResidentStrict`. If this property is

present, accesses to unbound regions of the resource are well defined and behave as if the data bound is populated with all zeros; writes are discarded. When this property is absent, accesses are considered safe, but reads will return undefined values.

  - Requested via the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT and VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT bits.
  - Sparse residency support is advertised on a finer grain via the following features:
    - The `sparseResidencyBuffer` feature provides support for creating `VkBuffer` objects with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT.
    - The `sparseResidencyImage2D` feature provides support for creating 2D single-sampled `VkImage` objects with VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT.
    - The `sparseResidencyImage3D` feature provides support for creating 3D `VkImage` objects with VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT.
    - The `sparseResidency2Samples` feature provides support for creating 2D `VkImage` objects with 2 samples and VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT.
    - The `sparseResidency4Samples` feature provides support for creating 2D `VkImage` objects with 4 samples and VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT.
    - The `sparseResidency8Samples` feature provides support for creating 2D `VkImage` objects with 8 samples and VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT.
    - The `sparseResidency16Samples` feature provides support for creating 2D `VkImage` objects with 16 samples and VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT.

    Implementations supporting `sparseResidencyImage2D` are only **required** to support sparse 2D, single-sampled images. Support for sparse 3D and MSAA images is **optional** and **can** be enabled via `sparseResidencyImage3D`, `sparseResidency2Samples`, `sparseResidency4Samples`, `sparseResidency8Samples`, and `sparseResidency16Samples`.
  - A sparse image created using VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT supports all non-compressed color formats with power-of-two element size that non-sparse usage supports. Additional formats **may** also be supported and **can** be queried via vkGetPhysicalDeviceSparseImageFormatProperties. VK_IMAGE_TILING_LINEAR tiling is not supported.
- The `sparseResidencyAliased` feature provides the following capability that **can** be enabled per resource:

  Allows physical memory ranges to be shared between multiple locations in the same sparse resource or between multiple sparse resources, with each binding of a memory location observing a consistent interpretation of the memory contents.

  See Sparse Memory Aliasing for more information.

36.2. Sparse Buffers and Fully-Resident Images

Both `VkBuffer` and `VkImage` objects created with the VK_IMAGE_CREATE_SPARSE_BINDING_BIT or VK_BUFFER_CREATE_SPARSE_BINDING_BIT bits **can** be thought of as a linear region of address

space. In the VkImage case if VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT is not used, this linear region is entirely opaque, meaning that there is no application-visible mapping between texel location and memory offset.

Unless VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT or VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT are also used, the entire resource **must** be bound to one or more VkDeviceMemory objects before use.

36.2.1. Sparse Buffer and Fully-Resident Image Block Size

The sparse block size in bytes for sparse buffers and fully-resident images is reported as VkMemoryRequirements::alignment. alignment represents both the memory alignment requirement and the binding granularity (in bytes) for sparse resources.

36.3. Sparse Partially-Resident Buffers

VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT bit allow the buffer to be made only partially resident. Partially resident VkBuffer objects are allocated and bound identically to VkBuffer objects using only the VK_BUFFER_CREATE_SPARSE_BINDING_BIT feature. The only difference is the ability for some regions of the buffer to be unbound during device use. After creating a VkBuffer with VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT, the entire buffer region is considered unbound.

36.4. Sparse Partially-Resident Images

VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT bit allow specific rectangular regions of the image called sparse image blocks to be bound to specific ranges of memory. This allows the application to manage residency at either image subresource or sparse image block granularity. Each image subresource (outside of the mip tail) starts on a sparse block boundary and has dimensions that are integer multiples of the corresponding dimensions of the sparse image block. After creating a VkImage with VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT, all image subresources are considered unbound.

NOTE

Applications **can** use these types of images to control LOD based on total memory consumption. If memory pressure becomes an issue the application **can** unbind and disable specific mipmap levels of images without having to recreate resources or modify texel data of unaffected levels.

The application **can** also use this functionality to access subregions of the image in a “megatexture” fashion. The application **can** create a large image and only populate the region of the image that is currently being used in the scene.

36.4.1. Accessing Unbound Regions

The following member of VkPhysicalDeviceSparseProperties affects how data in unbound regions of sparse resources are handled by the implementation:

- residencyNonResidentStrict

If this property is not present, reads of unbound regions of the image will return undefined values. Both reads and writes are still considered *safe* and will not affect other resources or populated regions of the image.

If this property is present, all reads of unbound regions of the image will behave as if the region was bound to memory populated with all zeros; writes will be discarded.

Image operations performed on unbound memory **may** still alter some component values in the natural way for those accesses, e.g. substituting a value of one for alpha in formats that do not have an alpha component.

> Example: Reading the alpha component of an unbacked VK_FORMAT_R8_UNORM image will return a value of 1.0f.

If a value was previously written to the same unbound sparse memory location in the same shader invocation, that value **may** be returned instead; using the `VolatileTexel` image operand, the `Volatile` memory semantic, or the `Volatile` decoration to load the value will prevent prior stored values from being returned.

NOTE	Getting the value of the previous store is possible as implementations are free to optimize multiple accesses in the general case. There are other ways this **can** be prevented, but using volatile loads is by far the simplest.

See Physical Device Enumeration for instructions for retrieving physical device properties.

> **Implementor's Note**
>
> For implementations that **cannot** natively handle access to unbound regions of a resource, the implementation **may** allocate and bind memory to the unbound regions. Reads and writes to unbound regions will access the implementation-managed memory instead.
>
> Given that the values resulting from reads of unbound regions are undefined in this scenario, implementations **may** use the same physical memory for all unbound regions of multiple resources within the same process.

36.4.2. Mip Tail Regions

Sparse images created using VK_IMAGE_CREATE_SPARSE_BINDING_BIT (without also using VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT) have no specific mapping of image region or image subresource to memory offset defined, so the entire image **can** be thought of as a linear opaque address region. However, images created with VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT do have a prescribed sparse image block layout, and hence each image subresource **must** start on a sparse block boundary. Within each array layer, the set of mip levels that have a smaller size than the sparse block size in bytes are grouped together into a *mip tail region*.

If the VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT flag is present in the `flags` member of

VkSparseImageFormatProperties, for the image's format, then any mip level which has dimensions that are not integer multiples of the corresponding dimensions of the sparse image block, and all subsequent mip levels, are also included in the mip tail region.

The following member of VkPhysicalDeviceSparseProperties **may** affect how the implementation places mip levels in the mip tail region:

- residencyAlignedMipSize

Each mip tail region is bound to memory as an opaque region (i.e. **must** be bound using a VkSparseImageOpaqueMemoryBindInfo structure) and **may** be of a size greater than or equal to the sparse block size in bytes. This size is guaranteed to be an integer multiple of the sparse block size in bytes.

An implementation **may** choose to allow each array-layer's mip tail region to be bound to memory independently or require that all array-layer's mip tail regions be treated as one. This is dictated by VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT in VkSparseImageMemoryRequirements::flags.

The following diagrams depict how VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT and VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT alter memory usage and requirements.

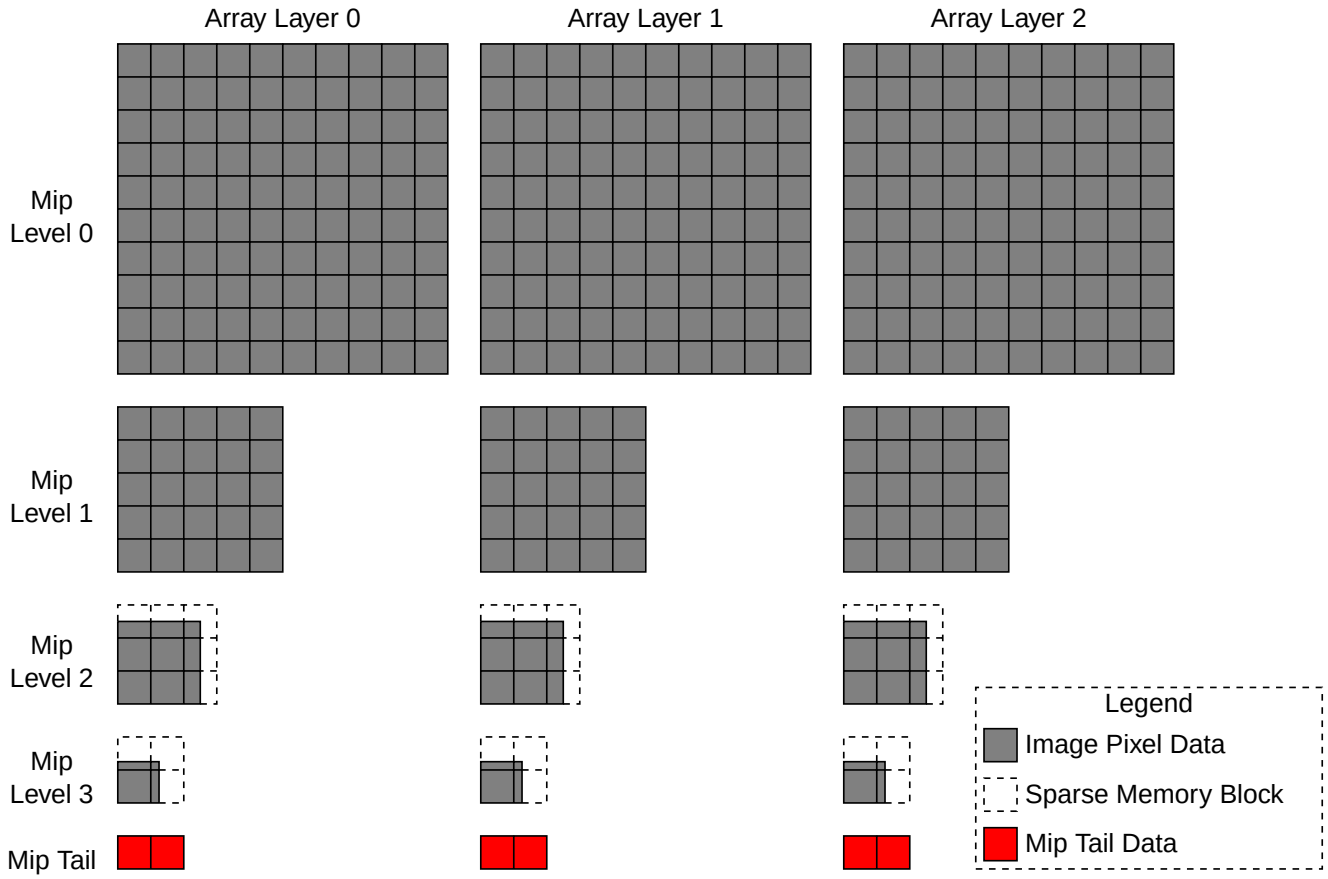


Figure 17. Sparse Image

In the absence of VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT and VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT, each array layer contains a mip tail region containing texel data for all mip levels smaller than the sparse image block in any dimension.

Mip levels that are as large as or larger than a sparse image block in all dimensions **can** be bound

individually. Right-edges and bottom-edges of each level are allowed to have partially used sparse blocks. Any bound partially-used-sparse-blocks **must** still have their full sparse block size in bytes allocated in memory.

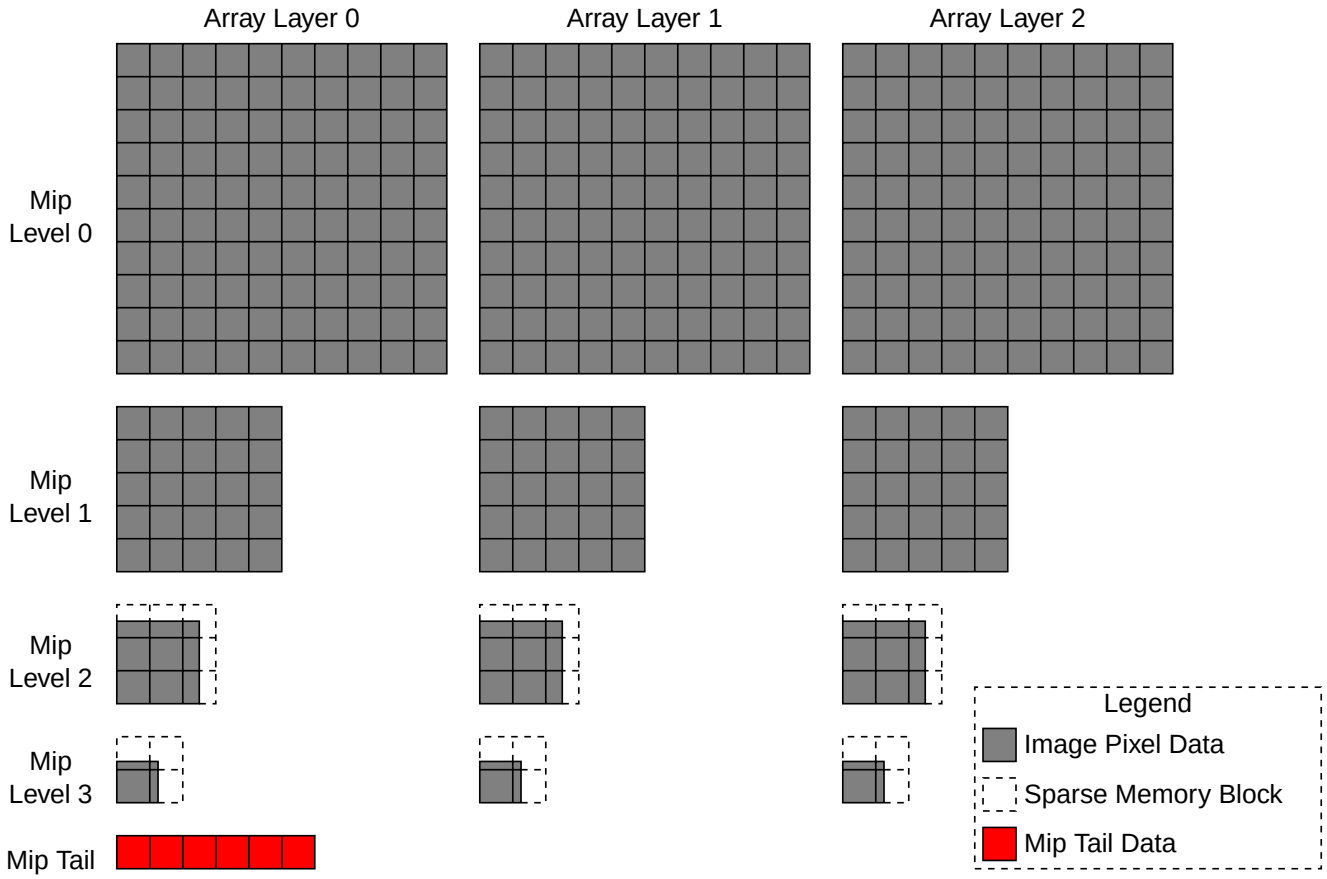


Figure 18. Sparse Image with Single Mip Tail

When VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT is present all array layers will share a single mip tail region.

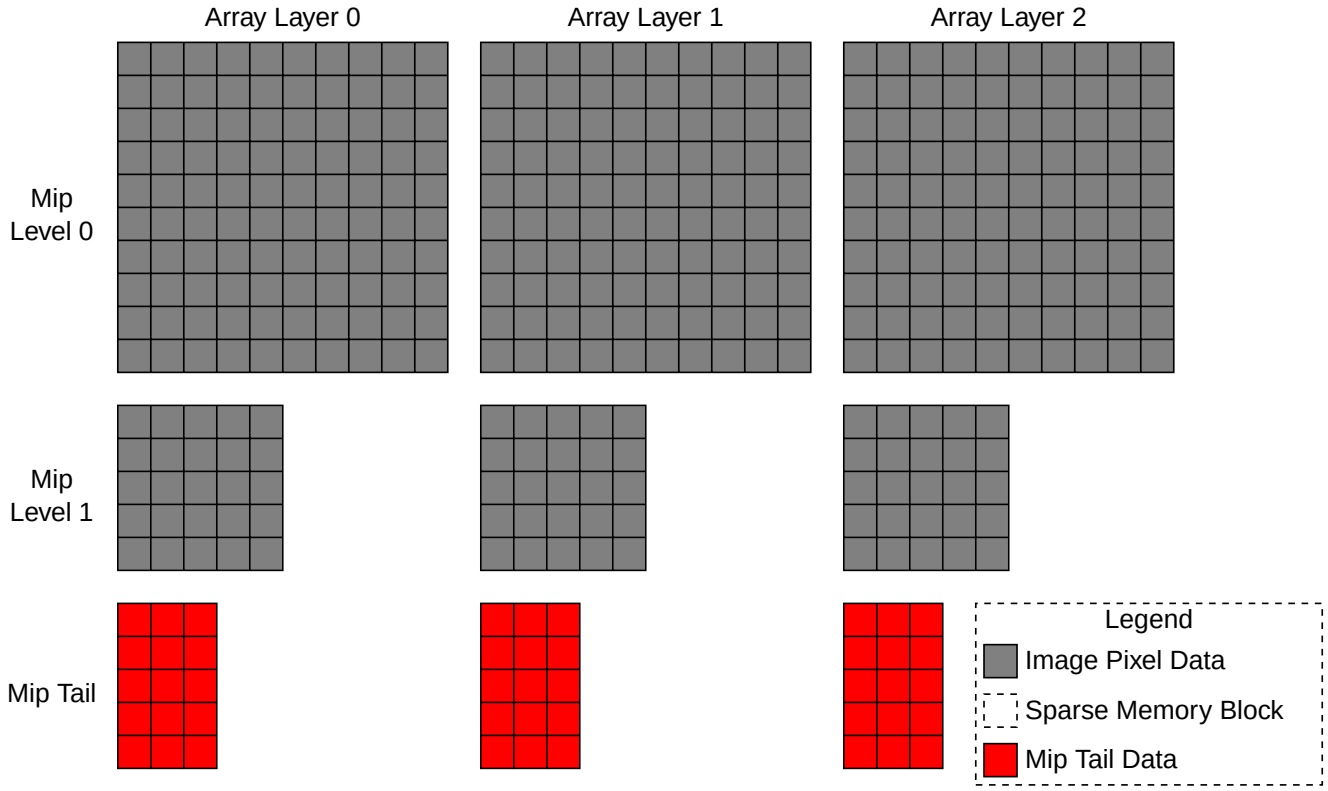


Figure 19. Sparse Image with Aligned Mip Size

NOTE	The mip tail regions are presented here in 2D arrays simply for figure size reasons. Each mip tail is logically a single array of sparse blocks with an implementation-dependent mapping of texels or compressed texel blocks to sparse blocks.

When VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT is present the first mip level that would contain partially used sparse blocks begins the mip tail region. This level and all subsequent levels are placed in the mip tail. Only the first N mip levels whose dimensions are an exact multiple of the sparse image block dimensions **can** be bound and unbound on a sparse block basis.

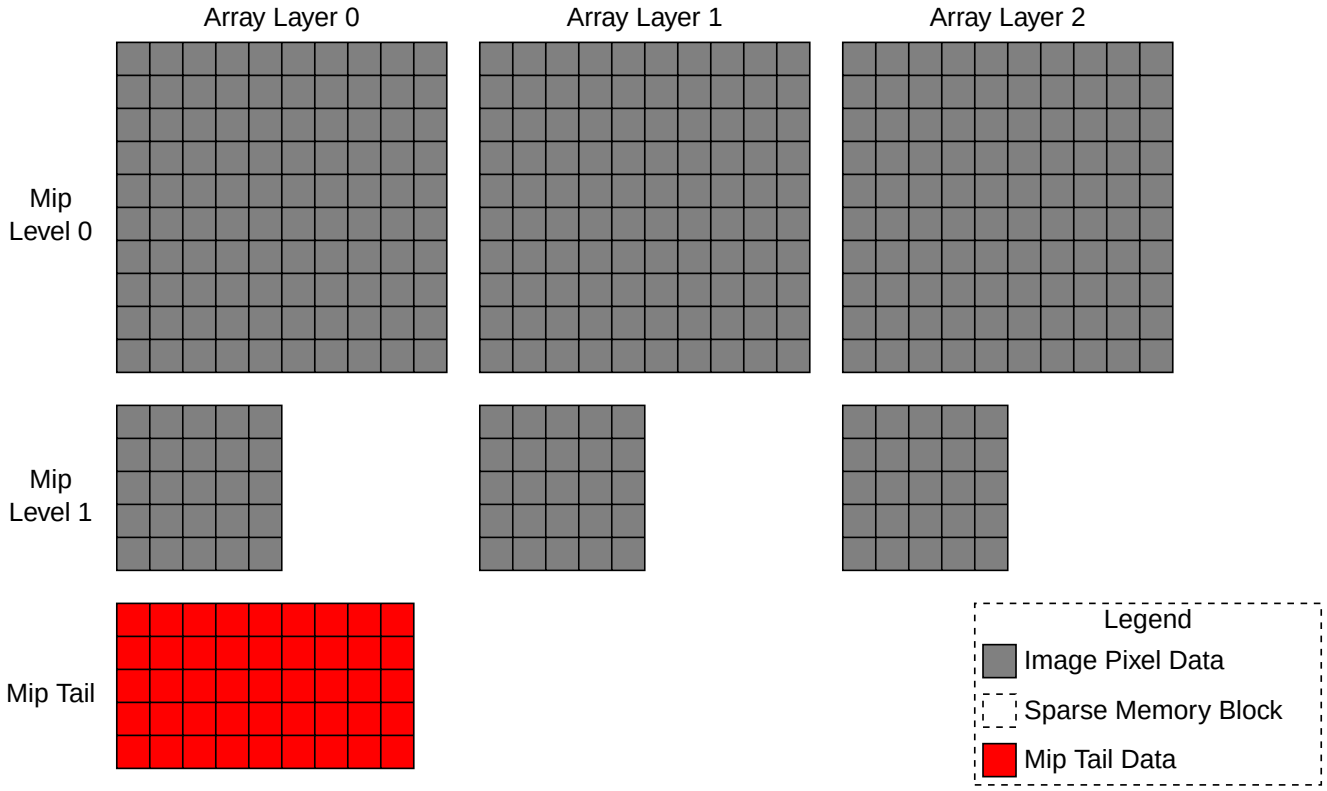


Figure 20. Sparse Image with Aligned Mip Size and Single Mip Tail

NOTE	The mip tail region is presented here in a 2D array simply for figure size reasons. It is logically a single array of sparse blocks with an implementation-dependent mapping of texels or compressed texel blocks to sparse blocks.

When both VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT and VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT are present the constraints from each of these flags are in effect.

36.4.3. Standard Sparse Image Block Shapes

Standard sparse image block shapes define a standard set of dimensions for sparse image blocks that depend on the format of the image. Layout of texels or compressed texel blocks within a sparse image block is implementation-dependent. All currently defined standard sparse image block shapes are 64 KB in size.

For block-compressed formats (e.g. VK_FORMAT_BC5_UNORM_BLOCK), the texel size is the size of the compressed texel block (e.g. 128-bit for `BC5`) thus the dimensions of the standard sparse image block shapes apply in terms of compressed texel blocks.

NOTE	For block-compressed formats, the dimensions of a sparse image block in terms of texels **can** be calculated by multiplying the sparse image block dimensions by the compressed texel block dimensions.

Table 48. Standard Sparse Image Block Shapes (Single Sample)

TEXEL SIZE (bits)	Block Shape (2D)	Block Shape (3D)
8-Bit	256 × 256 × 1	64 × 32 × 32
16-Bit	256 × 128 × 1	32 × 32 × 32
32-Bit	128 × 128 × 1	32 × 32 × 16
64-Bit	128 × 64 × 1	32 × 16 × 16
128-Bit	64 × 64 × 1	16 × 16 × 16

Table 49. Standard Sparse Image Block Shapes (MSAA)

TEXEL SIZE (bits)	Block Shape (2X)	Block Shape (4X)	Block Shape (8X)	Block Shape (16X)
8-Bit	128 × 256 × 1	128 × 128 × 1	64 × 128 × 1	64 × 64 × 1
16-Bit	128 × 128 × 1	128 × 64 × 1	64 × 64 × 1	64 × 32 × 1
32-Bit	64 × 128 × 1	64 × 64 × 1	32 × 64 × 1	32 × 32 × 1
64-Bit	64 × 64 × 1	64 × 32 × 1	32 × 32 × 1	32 × 16 × 1
128-Bit	32 × 64 × 1	32 × 32 × 1	16 × 32 × 1	16 × 16 × 1

Implementations that support the standard sparse image block shape for all formats listed in the Standard Sparse Image Block Shapes (Single Sample) and Standard Sparse Image Block Shapes (MSAA) tables **may** advertise the following `VkPhysicalDeviceSparseProperties`:

- `residencyStandard2DBlockShape`
- `residencyStandard2DMultisampleBlockShape`
- `residencyStandard3DBlockShape`

Reporting each of these features does *not* imply that all possible image types are supported as sparse. Instead, this indicates that no supported sparse image of the corresponding type will use custom sparse image block dimensions for any formats that have a corresponding standard sparse image block shape.

36.4.4. Custom Sparse Image Block Shapes

An implementation that does not support a standard image block shape for a particular sparse partially-resident image **may** choose to support a custom sparse image block shape for it instead. The dimensions of such a custom sparse image block shape are reported in `VkSparseImageFormatProperties`::`imageGranularity`. As with standard sparse image block shapes, the size in bytes of the custom sparse image block shape will be reported in `VkMemoryRequirements`::`alignment`.

Custom sparse image block dimensions are reported through `vkGetPhysicalDeviceSparseImageFormatProperties` and `vkGetImageSparseMemoryRequirements`.

An implementation **must** not support both the standard sparse image block shape and a custom

sparse image block shape for the same image. The standard sparse image block shape **must** be used if it is supported.

36.4.5. Multiple Aspects

Partially resident images are allowed to report separate sparse properties for different aspects of the image. One example is for depth/stencil images where the implementation separates the depth and stencil data into separate planes. Another reason for multiple aspects is to allow the application to manage memory allocation for implementation-private *metadata* associated with the image. See the figure below:

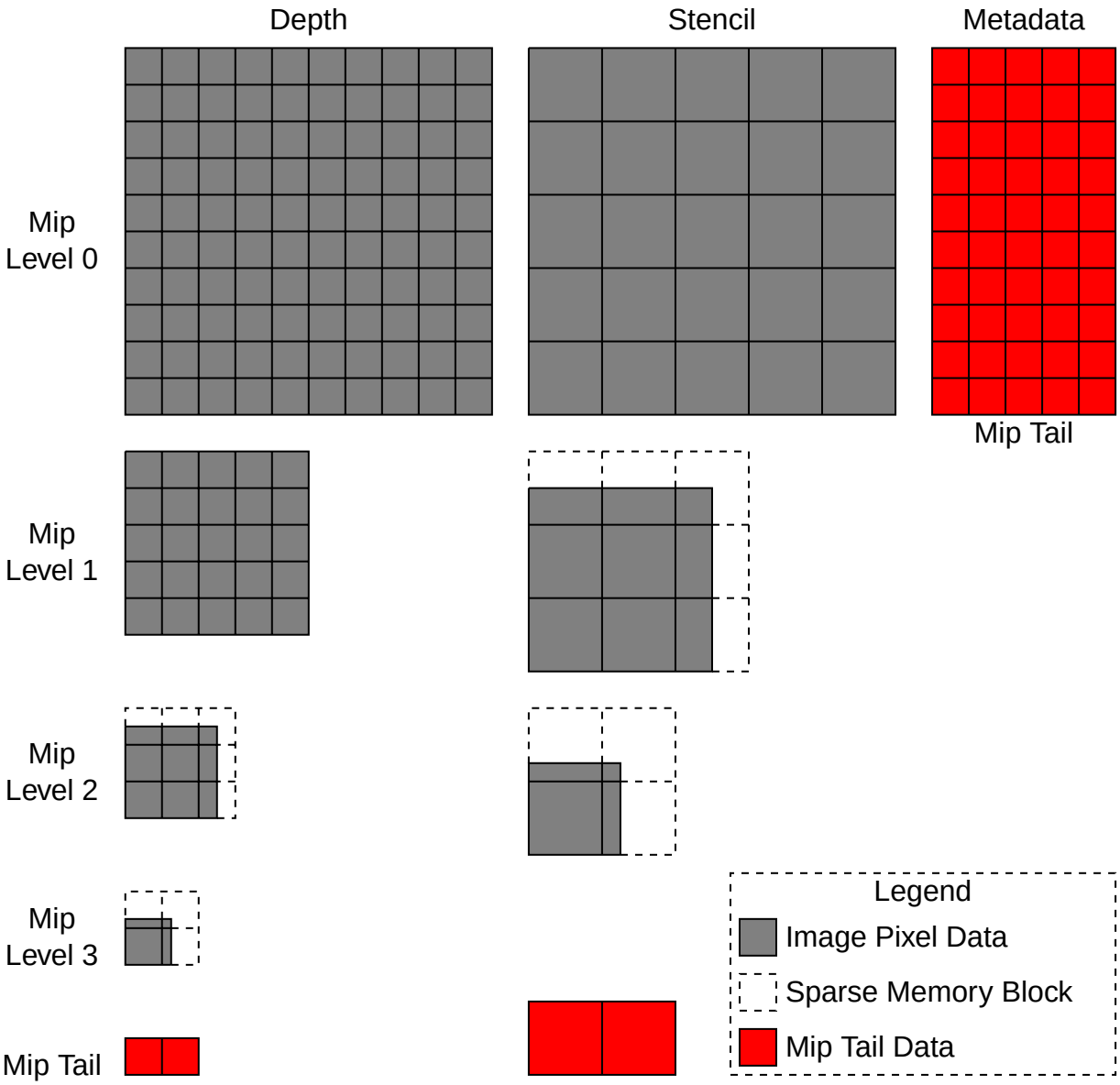


Figure 21. Multiple Aspect Sparse Image

NOTE The mip tail regions are presented here in 2D arrays simply for figure size reasons. Each mip tail is logically a single array of sparse blocks with an implementation-dependent mapping of texels or compressed texel blocks to sparse blocks.

In the figure above the depth, stencil, and metadata aspects all have unique sparse properties. The per-texel stencil data is ¼ the size of the depth data, hence the stencil sparse blocks include 4 × the number of texels. The sparse block size in bytes for all of the aspects is identical and defined by

`VkMemoryRequirements::alignment`.

Metadata

The metadata aspect of an image has the following constraints:

- All metadata is reported in the mip tail region of the metadata aspect.
- All metadata **must** be bound prior to device use of the sparse image.

36.5. Sparse Memory Aliasing

By default sparse resources have the same aliasing rules as non-sparse resources. See Memory Aliasing for more information.

`VkDevice` objects that have the `sparseResidencyAliased` feature enabled are able to use the VK_BUFFER_CREATE_SPARSE_ALIASED_BIT and VK_IMAGE_CREATE_SPARSE_ALIASED_BIT flags for resource creation. These flags allow resources to access physical memory bound into multiple locations within one or more sparse resources in a *data consistent* fashion. This means that reading physical memory from multiple aliased locations will return the same value.

Care **must** be taken when performing a write operation to aliased physical memory. Memory dependencies **must** be used to separate writes to one alias from reads or writes to another alias. Writes to aliased memory that are not properly guarded against accesses to different aliases will have undefined results for all accesses to the aliased memory.

Applications that wish to make use of data consistent sparse memory aliasing **must** abide by the following guidelines:

- All sparse resources that are bound to aliased physical memory **must** be created with the VK_BUFFER_CREATE_SPARSE_ALIASED_BIT / VK_IMAGE_CREATE_SPARSE_ALIASED_BIT flag.
- All resources that access aliased physical memory **must** interpret the memory in the same way. This implies the following:
  - Buffers and images **cannot** alias the same physical memory in a data consistent fashion. The physical memory ranges **must** be used exclusively by buffers or used exclusively by images for data consistency to be guaranteed.
  - Memory in sparse image mip tail regions **cannot** access aliased memory in a data consistent fashion.
  - Sparse images that alias the same physical memory **must** have compatible formats and be using the same sparse image block shape in order to access aliased memory in a data consistent fashion.

Failure to follow any of the above guidelines will require the application to abide by the normal, non-sparse resource aliasing rules. In this case memory **cannot** be accessed in a data consistent fashion.

NOTE	Enabling sparse resource memory aliasing **can** be a way to lower physical memory use, but it **may** reduce performance on some implementations. An application

> developer **can** test on their target HW and balance the memory / performance trade-offs measured.

36.6. Sparse Resource Implementation Guidelines (Informative)

This section is Informative. It is included to aid in implementors' understanding of sparse resources.

Device Virtual Address

The basic `sparseBinding` feature allows the resource to reserve its own device virtual address range at resource creation time rather than relying on a bind operation to set this. Without any other creation flags, no other constraints are relaxed compared to normal resources. All pages **must** be bound to physical memory before the device accesses the resource.

The `sparseResidency` features allow sparse resources to be used even when not all pages are bound to memory. Implementations that support access to unbound pages without causing a fault **may** support `residencyNonResidentStrict`.

Not faulting on access to unbound pages is not enough to support `residencyNonResidentStrict`. An implementation **must** also guarantee that reads after writes to unbound regions of the resource always return data for the read as if the memory contains zeros. Depending on any caching hierarchy of the implementation this **may** not always be possible.

Any implementation that does not fault, but does not guarantee correct read values **must** not support `residencyNonResidentStrict`.

Any implementation that **cannot** access unbound pages without causing a fault will require the implementation to bind the entire device virtual address range to physical memory. Any pages that the application does not bind to memory **may** be bound to one (or more) "placeholder" physical page(s) allocated by the implementation. Given the following properties:

- A process **must** not access memory from another process
- Reads return undefined values

It is sufficient for each host process to allocate these placeholder pages and use them for all resources in that process. Implementations **may** allocate more often (per instance, per device, or per resource).

Binding Memory

The byte size reported in `VkMemoryRequirements`::`size` **must** be greater than or equal to the amount of physical memory **required** to fully populate the resource. Some implementations require "holes" in the device virtual address range that are never accessed. These holes **may** be included in the `size` reported for the resource.

Including or not including the device virtual address holes in the resource size will alter how the implementation provides support for VkSparseImageOpaqueMemoryBindInfo. This operation **must** be supported for all sparse images, even ones created with VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT.

- If the holes are included in the size, this bind function becomes very easy. In most cases the resourceOffset is simply a device virtual address offset and the implementation can easily determine what device virtual address to bind. The cost is that the application **may** allocate more physical memory for the resource than it needs.
- If the holes are not included in the size, the application **can** allocate less physical memory than otherwise for the resource. However, in this case the implementation **must** account for the holes when mapping resourceOffset to the actual device virtual address intended to be mapped.

NOTE

If the application always uses VkSparseImageMemoryBindInfo to bind memory for the non-tail mip levels, any holes that are present in the resource size **may** never be bound.

Since VkSparseImageMemoryBindInfo uses texel locations to determine which device virtual addresses to bind, it is impossible to bind device virtual address holes with this operation.

Binding Metadata Memory

All metadata for sparse images have their own sparse properties and are embedded in the mip tail region for said properties. See the Multiaspect section for details.

Given that metadata is in a mip tail region, and the mip tail region **must** be reported as contiguous (either globally or per-array-layer), some implementations will have to resort to complicated offset → device virtual address mapping for handling VkSparseImageOpaqueMemoryBindInfo.

To make this easier on the implementation, the VK_SPARSE_MEMORY_BIND_METADATA_BIT explicitly specifies when metadata is bound with VkSparseImageOpaqueMemoryBindInfo. When this flag is not present, the resourceOffset **may** be treated as a strict device virtual address offset.

When VK_SPARSE_MEMORY_BIND_METADATA_BIT is present, the resourceOffset **must** have been derived explicitly from the imageMipTailOffset in the sparse resource properties returned for the metadata aspect. By manipulating the value returned for imageMipTailOffset, the resourceOffset does not have to correlate directly to a device virtual address offset, and **may** instead be whatever value makes it easiest for the implementation to derive the correct device virtual address.

36.7. Sparse Resource API

The APIs related to sparse resources are grouped into the following categories:

- Physical Device Features
- Physical Device Sparse Properties
- Sparse Image Format Properties
- Sparse Resource Creation
- Sparse Resource Memory Requirements
- Binding Resource Memory

36.7.1. Physical Device Features

Some sparse-resource related features are reported and enabled in `VkPhysicalDeviceFeatures`. These features **must** be supported and enabled on the `VkDevice` object before applications **can** use them. See Physical Device Features for information on how to get and set enabled device features, and for more detailed explanations of these features.

Sparse Physical Device Features

- `sparseBinding`: Support for creating VkBuffer and `VkImage` objects with the VK_BUFFER_CREATE_SPARSE_BINDING_BIT and VK_IMAGE_CREATE_SPARSE_BINDING_BIT flags, respectively.
- `sparseResidencyBuffer`: Support for creating VkBuffer objects with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag.
- `sparseResidencyImage2D`: Support for creating 2D single-sampled `VkImage` objects with VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT.
- `sparseResidencyImage3D`: Support for creating 3D VkImage objects with VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT.
- `sparseResidency2Samples`: Support for creating 2D VkImage objects with 2 samples and VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT.
- `sparseResidency4Samples`: Support for creating 2D VkImage objects with 4 samples and VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT.
- `sparseResidency8Samples`: Support for creating 2D VkImage objects with 8 samples and VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT.
- `sparseResidency16Samples`: Support for creating 2D VkImage objects with 16 samples and VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT.
- `sparseResidencyAliased`: Support for creating VkBuffer and `VkImage` objects with the VK_BUFFER_CREATE_SPARSE_ALIASED_BIT and VK_IMAGE_CREATE_SPARSE_ALIASED_BIT flags, respectively.

36.7.2. Physical Device Sparse Properties

Some features of the implementation are not possible to disable, and are reported to allow applications to alter their sparse resource usage accordingly. These read-only capabilities are reported in the VkPhysicalDeviceProperties::`sparseProperties` member, which is a `VkPhysicalDeviceSparseProperties` structure.

The VkPhysicalDeviceSparseProperties structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPhysicalDeviceSparseProperties {
    VkBool32    residencyStandard2DBlockShape;
    VkBool32    residencyStandard2DMultisampleBlockShape;
    VkBool32    residencyStandard3DBlockShape;
    VkBool32    residencyAlignedMipSize;
    VkBool32    residencyNonResidentStrict;
} VkPhysicalDeviceSparseProperties;
```

- residencyStandard2DBlockShape is VK_TRUE if the physical device will access all single-sample 2D sparse resources using the standard sparse image block shapes (based on image format), as described in the Standard Sparse Image Block Shapes (Single Sample) table. If this property is not supported the value returned in the imageGranularity member of the VkSparseImageFormatProperties structure for single-sample 2D images is not **required** to match the standard sparse image block dimensions listed in the table.
- residencyStandard2DMultisampleBlockShape is VK_TRUE if the physical device will access all multisample 2D sparse resources using the standard sparse image block shapes (based on image format), as described in the Standard Sparse Image Block Shapes (MSAA) table. If this property is not supported, the value returned in the imageGranularity member of the VkSparseImageFormatProperties structure for multisample 2D images is not **required** to match the standard sparse image block dimensions listed in the table.
- residencyStandard3DBlockShape is VK_TRUE if the physical device will access all 3D sparse resources using the standard sparse image block shapes (based on image format), as described in the Standard Sparse Image Block Shapes (Single Sample) table. If this property is not supported, the value returned in the imageGranularity member of the VkSparseImageFormatProperties structure for 3D images is not **required** to match the standard sparse image block dimensions listed in the table.
- residencyAlignedMipSize is VK_TRUE if images with mip level dimensions that are not integer multiples of the corresponding dimensions of the sparse image block **may** be placed in the mip tail. If this property is not reported, only mip levels with dimensions smaller than the imageGranularity member of the VkSparseImageFormatProperties structure will be placed in the mip tail. If this property is reported the implementation is allowed to return VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT in the flags member of VkSparseImageFormatProperties, indicating that mip level dimensions that are not integer multiples of the corresponding dimensions of the sparse image block will be placed in the mip tail.
- residencyNonResidentStrict specifies whether the physical device **can** consistently access non-resident regions of a resource. If this property is VK_TRUE, access to non-resident regions of resources will be guaranteed to return values as if the resource was populated with 0; writes to non-resident regions will be discarded.

36.7.3. Sparse Image Format Properties

Given that certain aspects of sparse image support, including the sparse image block dimensions,

may be implementation-dependent, vkGetPhysicalDeviceSparseImageFormatProperties **can** be used to query for sparse image format properties prior to resource creation. This command is used to check whether a given set of sparse image parameters is supported and what the sparse image block shape will be.

Sparse Image Format Properties API

The VkSparseImageFormatProperties structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseImageFormatProperties {
    VkImageAspectFlags          aspectMask;
    VkExtent3D                  imageGranularity;
    VkSparseImageFormatFlags    flags;
} VkSparseImageFormatProperties;
```

- aspectMask is a bitmask of VkImageAspectFlagBits specifying which aspects of the image the properties apply to.
- imageGranularity is the width, height, and depth of the sparse image block in texels.
- flags is a bitmask of VkSparseImageFormatFlagBits specifying additional information about the sparse resource.

Bits which **may** be set in VkSparseImageFormatProperties::flags, specifying additional information about the sparse resource, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSparseImageFormatFlagBits {
    VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT = 0x00000001,
    VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT = 0x00000002,
    VK_SPARSE_IMAGE_FORMAT_NONSTANDARD_BLOCK_SIZE_BIT = 0x00000004,
} VkSparseImageFormatFlagBits;
```

- VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT specifies that the image uses a single mip tail region for all array layers.
- VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT specifies that the first mip level whose dimensions are not integer multiples of the corresponding dimensions of the sparse image block begins the mip tail region.
- VK_SPARSE_IMAGE_FORMAT_NONSTANDARD_BLOCK_SIZE_BIT specifies that the image uses non-standard sparse image block dimensions, and the imageGranularity values do not match the standard sparse image block dimensions for the given format.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSparseImageFormatFlags;
```

VkSparseImageFormatFlags is a bitmask type for setting a mask of zero or more

VkSparseImageFormatFlagBits.

vkGetPhysicalDeviceSparseImageFormatProperties returns an array of VkSparseImageFormatProperties. Each element describes properties for one set of image aspects that are bound simultaneously for a VkImage created with the provided image creation parameters. This is usually one element for each aspect in the image, but for interleaved depth/stencil images there is only one element describing the combined aspects.

WARNING	This functionality is superseded by vkGetPhysicalDeviceSparseImageFormatProperties2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceSparseImageFormatProperties(
    VkPhysicalDevice                            physicalDevice,
    VkFormat                                    format,
    VkImageType                                 type,
    VkSampleCountFlagBits                       samples,
    VkImageUsageFlags                           usage,
    VkImageTiling                               tiling,
    uint32_t*                                   pPropertyCount,
    VkSparseImageFormatProperties*              pProperties);
```

- physicalDevice is the physical device from which to query the sparse image format properties.
- format is the image format.
- type is the dimensionality of the image.
- samples is a VkSampleCountFlagBits value specifying the number of samples per texel.
- usage is a bitmask describing the intended usage of the image.
- tiling is the tiling arrangement of the texel blocks in memory.
- pPropertyCount is a pointer to an integer related to the number of sparse format properties available or queried, as described below.
- pProperties is either NULL or a pointer to an array of VkSparseImageFormatProperties structures.

If pProperties is NULL, then the number of sparse format properties available is returned in pPropertyCount. Otherwise, pPropertyCount **must** point to a variable set by the application to the number of elements in the pProperties array, and on return the variable is overwritten with the number of structures actually written to pProperties. If pPropertyCount is less than the number of sparse format properties available, at most pPropertyCount structures will be written.

If VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT is not supported for the given arguments, pPropertyCount will be zero upon return, and no data will be written to pProperties.

Multiple aspects are returned for depth/stencil images that are implemented as separate planes by the implementation. The depth and stencil data planes each have unique

VkSparseImageFormatProperties data.

Depth/stencil images with depth and stencil data interleaved into a single plane will return a single VkSparseImageFormatProperties structure with the aspectMask set to VK_IMAGE_ASPECT_DEPTH_BIT | VK_IMAGE_ASPECT_STENCIL_BIT.

Valid Usage

- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-samples-01094
  samples **must** be a valid VkSampleCountFlagBits value that is set in VkImageFormatProperties::sampleCounts returned by vkGetPhysicalDeviceImageFormatProperties or vkGetPhysicalDeviceImageFormatProperties2 with format, type, tiling, and usage equal to those in this command

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-physicalDevice-parameter
  physicalDevice **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-format-parameter
  format **must** be a valid VkFormat value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-type-parameter
  type **must** be a valid VkImageType value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-samples-parameter
  samples **must** be a valid VkSampleCountFlagBits value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-usage-parameter
  usage **must** be a valid combination of VkImageUsageFlagBits values
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-tiling-parameter
  tiling **must** be a valid VkImageTiling value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-pPropertyCount-parameter
  pPropertyCount **must** be a valid pointer to a uint32_t value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-pProperties-parameter
  If the value referenced by pPropertyCount is not 0, and pProperties is not NULL, pProperties **must** be a valid pointer to an array of pPropertyCount VkSparseImageFormatProperties structures

vkGetPhysicalDeviceSparseImageFormatProperties2 returns an array of VkSparseImageFormatProperties2. Each element describes properties for one set of image aspects that are bound simultaneously for a VkImage created with the provided image creation parameters. This is usually one element for each aspect in the image, but for interleaved depth/stencil images there is only one element describing the combined aspects.

```
// Provided by VK_VERSION_1_1
void vkGetPhysicalDeviceSparseImageFormatProperties2(
    VkPhysicalDevice                            physicalDevice,
    const VkPhysicalDeviceSparseImageFormatInfo2* pFormatInfo,
    uint32_t*                                   pPropertyCount,
    VkSparseImageFormatProperties2*             pProperties);
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to vkGetPhysicalDeviceSparseImageFormatProperties2
void vkGetPhysicalDeviceSparseImageFormatProperties2KHR(
    VkPhysicalDevice                            physicalDevice,
    const VkPhysicalDeviceSparseImageFormatInfo2* pFormatInfo,
    uint32_t*                                   pPropertyCount,
    VkSparseImageFormatProperties2*             pProperties);
```

- `physicalDevice` is the physical device from which to query the sparse image format properties.
- `pFormatInfo` is a pointer to a VkPhysicalDeviceSparseImageFormatInfo2 structure containing input parameters to the command.
- `pPropertyCount` is a pointer to an integer related to the number of sparse format properties available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of VkSparseImageFormatProperties2 structures.

`vkGetPhysicalDeviceSparseImageFormatProperties2` behaves identically to vkGetPhysicalDeviceSparseImageFormatProperties, with the ability to return extended information by adding extending structures to the `pNext` chain of its `pProperties` parameter.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSparseImageFormatProperties2-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties2-pFormatInfo-parameter
  `pFormatInfo` **must** be a valid pointer to a valid VkPhysicalDeviceSparseImageFormatInfo2 structure
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties2-pPropertyCount-parameter
  `pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties2-pProperties-parameter
  If the value referenced by `pPropertyCount` is not `0`, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` VkSparseImageFormatProperties2 structures

The VkPhysicalDeviceSparseImageFormatInfo2 structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceSparseImageFormatInfo2 {
    VkStructureType          sType;
    const void*              pNext;
    VkFormat                 format;
    VkImageType              type;
    VkSampleCountFlagBits    samples;
    VkImageUsageFlags        usage;
    VkImageTiling            tiling;
} VkPhysicalDeviceSparseImageFormatInfo2;
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to VkPhysicalDeviceSparseImageFormatInfo2
typedef VkPhysicalDeviceSparseImageFormatInfo2
VkPhysicalDeviceSparseImageFormatInfo2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `format` is the image format.
- `type` is the dimensionality of the image.
- `samples` is a VkSampleCountFlagBits value specifying the number of samples per texel.
- `usage` is a bitmask of VkImageUsageFlagBits describing the intended usage of the image.
- `tiling` is the tiling arrangement of the texel blocks in memory.

Only usage flags representable in VkImageUsageFlagBits are returned in `usage`. If the `pNext` chain includes a VkImageUsageFlags2CreateInfoKHR structure, all usage flags are returned in VkImageUsageFlags2CreateInfoKHR::`usage`.

Valid Usage

- VUID-VkPhysicalDeviceSparseImageFormatInfo2-samples-01095
  `samples` **must** be a valid VkSampleCountFlagBits value that is set in `VkImageFormatProperties`::`sampleCounts` returned by `vkGetPhysicalDeviceImageFormatProperties` or vkGetPhysicalDeviceImageFormatProperties2 with `format`, `type`, `tiling`, and `usage` equal to those in this command

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSparseImageFormatInfo2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SPARSE_IMAGE_FORMAT_INFO_2

- VUID-VkPhysicalDeviceSparseImageFormatInfo2-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkImageUsageFlags2CreateInfoKHR
- VUID-VkPhysicalDeviceSparseImageFormatInfo2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkPhysicalDeviceSparseImageFormatInfo2-format-parameter
  `format` **must** be a valid VkFormat value
- VUID-VkPhysicalDeviceSparseImageFormatInfo2-type-parameter
  `type` **must** be a valid VkImageType value
- VUID-VkPhysicalDeviceSparseImageFormatInfo2-samples-parameter
  `samples` **must** be a valid VkSampleCountFlagBits value
- VUID-VkPhysicalDeviceSparseImageFormatInfo2-usage-parameter
  `usage` **must** be a valid combination of VkImageUsageFlagBits values
- VUID-VkPhysicalDeviceSparseImageFormatInfo2-tiling-parameter
  `tiling` **must** be a valid VkImageTiling value

The `VkSparseImageFormatProperties2` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkSparseImageFormatProperties2 {
    VkStructureType                 sType;
    void*                           pNext;
    VkSparseImageFormatProperties   properties;
} VkSparseImageFormatProperties2;
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to VkSparseImageFormatProperties2
typedef VkSparseImageFormatProperties2 VkSparseImageFormatProperties2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `properties` is a VkSparseImageFormatProperties structure which is populated with the same values as in vkGetPhysicalDeviceSparseImageFormatProperties.

Valid Usage (Implicit)

- VUID-VkSparseImageFormatProperties2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SPARSE_IMAGE_FORMAT_PROPERTIES_2
- VUID-VkSparseImageFormatProperties2-pNext-pNext
  `pNext` **must** be `NULL`

36.7.4. Sparse Resource Creation

Sparse resources require that one or more sparse feature flags be specified (as part of the VkPhysicalDeviceFeatures structure described previously in the Physical Device Features section) when calling vkCreateDevice. When the appropriate device features are enabled, the VK_BUFFER_CREATE_SPARSE_* and VK_IMAGE_CREATE_SPARSE_* flags **can** be used. See vkCreateBuffer and vkCreateImage for details of the resource creation APIs.

NOTE	Specifying VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT or VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT requires specifying VK_BUFFER_CREATE_SPARSE_BINDING_BIT or VK_IMAGE_CREATE_SPARSE_BINDING_BIT, respectively, as well. This means that resources **must** be created with the appropriate *_SPARSE_BINDING_BIT to be used with the sparse binding command (vkQueueBindSparse).

36.7.5. Sparse Resource Memory Requirements

Sparse resources have specific memory requirements related to binding sparse memory. These memory requirements are reported differently for VkBuffer objects and VkImage objects.

Buffer and Fully-Resident Images

Buffers (both fully and partially resident) and fully-resident images **can** be bound to memory using only the data from VkMemoryRequirements. For all sparse resources the VkMemoryRequirements ::alignment member specifies both the binding granularity in bytes and the **required** alignment of VkDeviceMemory.

Partially Resident Images

Partially resident images have a different method for binding memory. As with buffers and fully resident images, the VkMemoryRequirements::alignment field specifies the binding granularity in bytes for the image.

Requesting sparse memory requirements for VkImage objects using vkGetImageSparseMemoryRequirements will return an array of one or more VkSparseImageMemoryRequirements structures. Each structure describes the sparse memory requirements for a group of aspects of the image.

The sparse image **must** have been created using the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag to retrieve valid sparse image memory requirements.

Sparse Image Memory Requirements

The VkSparseImageMemoryRequirements structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseImageMemoryRequirements {
    VkSparseImageFormatProperties    formatProperties;
    uint32_t                         imageMipTailFirstLod;
    VkDeviceSize                     imageMipTailSize;
    VkDeviceSize                     imageMipTailOffset;
    VkDeviceSize                     imageMipTailStride;
} VkSparseImageMemoryRequirements;
```

- `formatProperties` is a VkSparseImageFormatProperties structure specifying properties of the image format.
- `imageMipTailFirstLod` is the first mip level at which image subresources are included in the mip tail region.
- `imageMipTailSize` is the memory size (in bytes) of the mip tail region. If `formatProperties.flags` contains VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT, this is the size of the whole mip tail, otherwise this is the size of the mip tail of a single array layer. This value is guaranteed to be a multiple of the sparse block size in bytes.
- `imageMipTailOffset` is the opaque memory offset used with VkSparseImageOpaqueMemoryBindInfo to bind the mip tail region(s).
- `imageMipTailStride` is the offset stride between each array-layer's mip tail, if `formatProperties.flags` does not contain VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT (otherwise the value is undefined).

To query sparse memory requirements for an image, call:

```
// Provided by VK_VERSION_1_0
void vkGetImageSparseMemoryRequirements(
    VkDevice                                    device,
    VkImage                                     image,
    uint32_t*                                   pSparseMemoryRequirementCount,
    VkSparseImageMemoryRequirements*            pSparseMemoryRequirements);
```

- `device` is the logical device that owns the image.
- `image` is the VkImage object to get the memory requirements for.
- `pSparseMemoryRequirementCount` is a pointer to an integer related to the number of sparse memory requirements available or queried, as described below.
- `pSparseMemoryRequirements` is either `NULL` or a pointer to an array of `VkSparseImageMemoryRequirements` structures.

If `pSparseMemoryRequirements` is `NULL`, then the number of sparse memory requirements available is returned in `pSparseMemoryRequirementCount`. Otherwise, `pSparseMemoryRequirementCount` **must** point to a variable set by the application to the number of elements in the `pSparseMemoryRequirements` array, and on return the variable is overwritten with the number of structures actually written to `pSparseMemoryRequirements`. If `pSparseMemoryRequirementCount` is less than the number of sparse

memory requirements available, at most pSparseMemoryRequirementCount structures will be written.

If the image was not created with VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT then pSparseMemoryRequirementCount will be zero and pSparseMemoryRequirements will not be written to.

NOTE

It is legal for an implementation to report a larger value in VkMemoryRequirements::size than would be obtained by adding together memory sizes for all VkSparseImageMemoryRequirements returned by vkGetImageSparseMemoryRequirements. This **may** occur when the implementation requires unused padding in the address range describing the resource.

Valid Usage (Implicit)

- VUID-vkGetImageSparseMemoryRequirements-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkGetImageSparseMemoryRequirements-image-parameter
  image **must** be a valid VkImage handle
- VUID-vkGetImageSparseMemoryRequirements-pSparseMemoryRequirementCount-parameter
  pSparseMemoryRequirementCount **must** be a valid pointer to a uint32_t value
- VUID-vkGetImageSparseMemoryRequirements-pSparseMemoryRequirements-parameter
  If the value referenced by pSparseMemoryRequirementCount is not 0, and pSparseMemoryRequirements is not NULL, pSparseMemoryRequirements **must** be a valid pointer to an array of pSparseMemoryRequirementCount VkSparseImageMemoryRequirements structures
- VUID-vkGetImageSparseMemoryRequirements-image-parent
  image **must** have been created, allocated, or retrieved from device

To query sparse memory requirements for an image, call:

```
// Provided by VK_VERSION_1_1
void vkGetImageSparseMemoryRequirements2(
    VkDevice                                    device,
    const VkImageSparseMemoryRequirementsInfo2* pInfo,
    uint32_t*                                   pSparseMemoryRequirementCount,
    VkSparseImageMemoryRequirements2*           pSparseMemoryRequirements);
```

```
// Provided by VK_KHR_get_memory_requirements2
// Equivalent to vkGetImageSparseMemoryRequirements2
void vkGetImageSparseMemoryRequirements2KHR(
    VkDevice                                    device,
    const VkImageSparseMemoryRequirementsInfo2* pInfo,
    uint32_t*                                   pSparseMemoryRequirementCount,
    VkSparseImageMemoryRequirements2*           pSparseMemoryRequirements);
```

- `device` is the logical device that owns the image.
- `pInfo` is a pointer to a VkImageSparseMemoryRequirementsInfo2 structure containing parameters required for the memory requirements query.
- `pSparseMemoryRequirementCount` is a pointer to an integer related to the number of sparse memory requirements available or queried, as described below.
- `pSparseMemoryRequirements` is either `NULL` or a pointer to an array of `VkSparseImageMemoryRequirements2` structures.

Valid Usage (Implicit)

- VUID-vkGetImageSparseMemoryRequirements2-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetImageSparseMemoryRequirements2-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkImageSparseMemoryRequirementsInfo2 structure
- VUID-vkGetImageSparseMemoryRequirements2-pSparseMemoryRequirementCount-parameter
  `pSparseMemoryRequirementCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetImageSparseMemoryRequirements2-pSparseMemoryRequirements-parameter
  If the value referenced by `pSparseMemoryRequirementCount` is not `0`, and `pSparseMemoryRequirements` is not `NULL`, `pSparseMemoryRequirements` **must** be a valid pointer to an array of `pSparseMemoryRequirementCount` VkSparseImageMemoryRequirements2 structures

To determine the sparse memory requirements for an image resource without creating an object, call:

```
// Provided by VK_VERSION_1_3
void vkGetDeviceImageSparseMemoryRequirements(
    VkDevice                                    device,
    const VkDeviceImageMemoryRequirements*      pInfo,
    uint32_t*                                   pSparseMemoryRequirementCount,
    VkSparseImageMemoryRequirements2*           pSparseMemoryRequirements);
```

```
// Provided by VK_KHR_maintenance4
// Equivalent to vkGetDeviceImageSparseMemoryRequirements
void vkGetDeviceImageSparseMemoryRequirementsKHR(
    VkDevice                                    device,
    const VkDeviceImageMemoryRequirements*      pInfo,
    uint32_t*                                   pSparseMemoryRequirementCount,
    VkSparseImageMemoryRequirements2*           pSparseMemoryRequirements);
```

- `device` is the logical device intended to own the image.
- `pInfo` is a pointer to a VkDeviceImageMemoryRequirements structure containing parameters required for the memory requirements query.
- `pSparseMemoryRequirementCount` is a pointer to an integer related to the number of sparse memory requirements available or queried, as described below.
- `pSparseMemoryRequirements` is either `NULL` or a pointer to an array of `VkSparseImageMemoryRequirements2` structures.

Valid Usage (Implicit)

- VUID-vkGetDeviceImageSparseMemoryRequirements-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDeviceImageSparseMemoryRequirements-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkDeviceImageMemoryRequirements structure
- VUID-vkGetDeviceImageSparseMemoryRequirements-pSparseMemoryRequirementCount-parameter
  `pSparseMemoryRequirementCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetDeviceImageSparseMemoryRequirements-pSparseMemoryRequirements-parameter
  If the value referenced by `pSparseMemoryRequirementCount` is not `0`, and `pSparseMemoryRequirements` is not `NULL`, `pSparseMemoryRequirements` **must** be a valid pointer to an array of `pSparseMemoryRequirementCount` VkSparseImageMemoryRequirements2 structures

The `VkImageSparseMemoryRequirementsInfo2` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkImageSparseMemoryRequirementsInfo2 {
    VkStructureType    sType;
    const void*        pNext;
    VkImage            image;
} VkImageSparseMemoryRequirementsInfo2;
```

```
// Provided by VK_KHR_get_memory_requirements2
// Equivalent to VkImageSparseMemoryRequirementsInfo2
typedef VkImageSparseMemoryRequirementsInfo2 VkImageSparseMemoryRequirementsInfo2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `image` is the image to query.

Valid Usage (Implicit)

- VUID-VkImageSparseMemoryRequirementsInfo2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_SPARSE_MEMORY_REQUIREMENTS_INFO_2
- VUID-VkImageSparseMemoryRequirementsInfo2-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkImageSparseMemoryRequirementsInfo2-image-parameter
  `image` **must** be a valid VkImage handle

The `VkSparseImageMemoryRequirements2` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkSparseImageMemoryRequirements2 {
    VkStructureType                     sType;
    void*                               pNext;
    VkSparseImageMemoryRequirements     memoryRequirements;
} VkSparseImageMemoryRequirements2;
```

```
// Provided by VK_KHR_get_memory_requirements2
// Equivalent to VkSparseImageMemoryRequirements2
typedef VkSparseImageMemoryRequirements2 VkSparseImageMemoryRequirements2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryRequirements` is a VkSparseImageMemoryRequirements structure describing the memory requirements of the sparse image.

Valid Usage (Implicit)

- VUID-VkSparseImageMemoryRequirements2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SPARSE_IMAGE_MEMORY_REQUIREMENTS_2
- VUID-VkSparseImageMemoryRequirements2-pNext-pNext
  `pNext` **must** be `NULL`

36.7.6. Binding Resource Memory

Non-sparse resources are backed by a single physical allocation prior to device use (via `vkBindImageMemory` or `vkBindBufferMemory`), and their backing **must** not be changed. On the other hand, sparse resources **can** be bound to memory non-contiguously and these bindings **can** be altered during the lifetime of the resource.

NOTE It is important to note that freeing a `VkDeviceMemory` object with `vkFreeMemory` will not

cause resources (or resource regions) bound to the memory object to become unbound. Applications **must** not access resources bound to memory that has been freed.

Sparse memory bindings execute on a queue that includes the VK_QUEUE_SPARSE_BINDING_BIT bit. Applications **must** use synchronization primitives to guarantee that other queues do not access ranges of memory concurrently with a binding change. Applications **can** access other ranges of the same resource while a bind operation is executing.

NOTE

Implementations **must** provide a guarantee that simultaneously binding sparse blocks while another queue accesses those same sparse blocks via a sparse resource **must** not access memory owned by another process or otherwise corrupt the system.

While some implementations **may** include VK_QUEUE_SPARSE_BINDING_BIT support in queue families that also include graphics and compute support, other implementations **may** only expose a VK_QUEUE_SPARSE_BINDING_BIT-only queue family. In either case, applications **must** use synchronization primitives to explicitly request any ordering dependencies between sparse memory binding operations and other graphics/compute/transfer operations, as sparse binding operations are not automatically ordered against command buffer execution, even within a single queue.

When binding memory explicitly for the VK_IMAGE_ASPECT_METADATA_BIT the application **must** use the VK_SPARSE_MEMORY_BIND_METADATA_BIT in the `VkSparseMemoryBind::flags` field when binding memory. Binding memory for metadata is done the same way as binding memory for the mip tail, with the addition of the VK_SPARSE_MEMORY_BIND_METADATA_BIT flag.

Binding the mip tail for any aspect **must** only be performed using VkSparseImageOpaqueMemoryBindInfo. If `formatProperties.flags` contains VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT, then it **can** be bound with a single VkSparseMemoryBind structure, with `resourceOffset` = `imageMipTailOffset` and `size` = `imageMipTailSize`.

If `formatProperties.flags` does not contain VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT then the offset for the mip tail in each array layer is given as:

```
arrayMipTailOffset = imageMipTailOffset + arrayLayer * imageMipTailStride;
```

and the mip tail **can** be bound with `layerCount` VkSparseMemoryBind structures, each using `size` = `imageMipTailSize` and `resourceOffset` = `arrayMipTailOffset` as defined above.

Sparse memory binding is handled by the following APIs and related data structures.

Sparse Memory Binding Functions

The `VkSparseMemoryBind` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseMemoryBind {
    VkDeviceSize               resourceOffset;
    VkDeviceSize               size;
    VkDeviceMemory             memory;
    VkDeviceSize               memoryOffset;
    VkSparseMemoryBindFlags    flags;
} VkSparseMemoryBind;
```

- `resourceOffset` is the offset into the resource.
- `size` is the size of the memory region to be bound.
- `memory` is the VkDeviceMemory object that the range of the resource is bound to. If `memory` is VK_NULL_HANDLE, the range is unbound.
- `memoryOffset` is the offset into the VkDeviceMemory object to bind the resource range to. If `memory` is VK_NULL_HANDLE, this value is ignored.
- `flags` is a bitmask of VkSparseMemoryBindFlagBits specifying usage of the binding operation.

The *binding range* [`resourceOffset`, `resourceOffset` + `size`) has different constraints based on `flags`. If `flags` contains VK_SPARSE_MEMORY_BIND_METADATA_BIT, the binding range **must** be within the mip tail region of the metadata aspect. This metadata region is defined by:

metadataRegion = [base, base + `imageMipTailSize`)

base = `imageMipTailOffset` + `imageMipTailStride` × n

and `imageMipTailOffset`, `imageMipTailSize`, and `imageMipTailStride` values are from the VkSparseImageMemoryRequirements corresponding to the metadata aspect of the image, and n is a valid array layer index for the image.

`imageMipTailStride` is considered to be zero for aspects where `VkSparseImageMemoryRequirements::formatProperties.flags` contains VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT.

If `flags` does not contain VK_SPARSE_MEMORY_BIND_METADATA_BIT, the binding range **must** be within the range [0,VkMemoryRequirements::`size`).

Valid Usage

- VUID-VkSparseMemoryBind-memory-01096
  If `memory` is not VK_NULL_HANDLE, `memory` and `memoryOffset` **must** match the memory requirements of the resource, as described in section Resource Memory Association
- VUID-VkSparseMemoryBind-resourceOffset-09491
  If the resource being bound is a `VkBuffer`, `resourceOffset`, `memoryOffset` and `size` **must** be an integer multiple of the `alignment` of the VkMemoryRequirements structure returned from a call to vkGetBufferMemoryRequirements with the buffer resource

- VUID-VkSparseMemoryBind-resourceOffset-09492
  If the resource being bound is a `VkImage`, `resourceOffset` and `memoryOffset` **must** be an integer multiple of the `alignment` of the VkMemoryRequirements structure returned from a call to vkGetImageMemoryRequirements with the image resource
- VUID-VkSparseMemoryBind-memory-01097
  If `memory` is not VK_NULL_HANDLE, `memory` **must** not have been created with a memory type that reports VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT bit set
- VUID-VkSparseMemoryBind-size-01098
  `size` **must** be greater than `0`
- VUID-VkSparseMemoryBind-resourceOffset-01099
  `resourceOffset` **must** be less than the size of the resource
- VUID-VkSparseMemoryBind-size-01100
  `size` **must** be less than or equal to the size of the resource minus `resourceOffset`
- VUID-VkSparseMemoryBind-memoryOffset-01101
  `memoryOffset` **must** be less than the size of `memory`
- VUID-VkSparseMemoryBind-size-01102
  `size` **must** be less than or equal to the size of `memory` minus `memoryOffset`
- VUID-VkSparseMemoryBind-memory-02730
  If `memory` was created with VkExportMemoryAllocateInfo::`handleTypes` not equal to `0`, at least one handle type it contained **must** also have been set in VkExternalMemoryBufferCreateInfo::`handleTypes` or VkExternalMemoryImageCreateInfo::`handleTypes` when the resource was created
- VUID-VkSparseMemoryBind-memory-02731
  If `memory` was created by a memory import operation, the external handle type of the imported memory **must** also have been set in VkExternalMemoryBufferCreateInfo::`handleTypes` or VkExternalMemoryImageCreateInfo::`handleTypes` when the resource was created

Valid Usage (Implicit)

- VUID-VkSparseMemoryBind-memory-parameter
  If `memory` is not VK_NULL_HANDLE, `memory` **must** be a valid VkDeviceMemory handle
- VUID-VkSparseMemoryBind-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkSparseMemoryBindFlagBits values

Bits which **can** be set in VkSparseMemoryBind::`flags`, specifying usage of a sparse memory binding operation, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSparseMemoryBindFlagBits {
    VK_SPARSE_MEMORY_BIND_METADATA_BIT = 0x00000001,
} VkSparseMemoryBindFlagBits;
```

- VK_SPARSE_MEMORY_BIND_METADATA_BIT specifies that the memory being bound is only for the metadata aspect.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSparseMemoryBindFlags;
```

VkSparseMemoryBindFlags is a bitmask type for setting a mask of zero or more VkSparseMemoryBindFlagBits.

Memory is bound to VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_BINDING_BIT flag using the following structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseBufferMemoryBindInfo {
    VkBuffer                     buffer;
    uint32_t                     bindCount;
    const VkSparseMemoryBind*    pBinds;
} VkSparseBufferMemoryBindInfo;
```

- buffer is the VkBuffer object to be bound.
- bindCount is the number of VkSparseMemoryBind structures in the pBinds array.
- pBinds is a pointer to an array of VkSparseMemoryBind structures.

Valid Usage (Implicit)

- VUID-VkSparseBufferMemoryBindInfo-buffer-parameter
  buffer **must** be a valid VkBuffer handle
- VUID-VkSparseBufferMemoryBindInfo-pBinds-parameter
  pBinds **must** be a valid pointer to an array of bindCount valid VkSparseMemoryBind structures
- VUID-VkSparseBufferMemoryBindInfo-bindCount-arraylength
  bindCount **must** be greater than 0

Memory is bound to opaque regions of VkImage objects created with the VK_IMAGE_CREATE_SPARSE_BINDING_BIT flag using the following structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseImageOpaqueMemoryBindInfo {
    VkImage                      image;
    uint32_t                     bindCount;
    const VkSparseMemoryBind*    pBinds;
} VkSparseImageOpaqueMemoryBindInfo;
```

- `image` is the VkImage object to be bound.
- `bindCount` is the number of VkSparseMemoryBind structures in the `pBinds` array.
- `pBinds` is a pointer to an array of VkSparseMemoryBind structures.

NOTE

This structure is normally used to bind memory to fully-resident sparse images or for mip tail regions of partially resident images. However, it **can** also be used to bind memory for the entire binding range of partially resident images.

If the `pBinds`[i].flags of an element *i* of `pBinds` does not contain VK_SPARSE_MEMORY_BIND_METADATA_BIT, the `resourceOffset` is in the range [0, VkMemoryRequirements::`size`). This range includes data from all aspects of the image, including metadata. For most implementations this will probably mean that the `resourceOffset` is a simple device address offset within the resource. It is possible for an application to bind a range of memory that includes both resource data and metadata. However, the application would not know what part of the image the memory is used for, or if any range is being used for metadata.

If the `pBinds`[i].flags of an element *i* of `pBinds` contains VK_SPARSE_MEMORY_BIND_METADATA_BIT, the binding range specified **must** be within the mip tail region of the metadata aspect. In this case the `resourceOffset` is not **required** to be a simple device address offset within the resource. However, it *is* defined to be within [`imageMipTailOffset`, `imageMipTailOffset` + `imageMipTailSize`) for the metadata aspect. See VkSparseMemoryBind for the full constraints on binding region with this flag present.

Valid Usage

- VUID-VkSparseImageOpaqueMemoryBindInfo-pBinds-01103
  If the `flags` member of any element of `pBinds` contains VK_SPARSE_MEMORY_BIND_METADATA_BIT, the binding range defined **must** be within the mip tail region of the metadata aspect of `image`

Valid Usage (Implicit)

- VUID-VkSparseImageOpaqueMemoryBindInfo-image-parameter
  `image` **must** be a valid VkImage handle
- VUID-VkSparseImageOpaqueMemoryBindInfo-pBinds-parameter

`pBinds` **must** be a valid pointer to an array of `bindCount` valid VkSparseMemoryBind structures

- VUID-VkSparseImageOpaqueMemoryBindInfo-bindCount-arraylength
  `bindCount` **must** be greater than `0`

Memory **can** be bound to sparse image blocks of `VkImage` objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag using the following structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseImageMemoryBindInfo {
    VkImage                          image;
    uint32_t                         bindCount;
    const VkSparseImageMemoryBind*   pBinds;
} VkSparseImageMemoryBindInfo;
```

- `image` is the VkImage object to be bound.
- `bindCount` is the number of VkSparseImageMemoryBind structures in the `pBinds` array.
- `pBinds` is a pointer to an array of VkSparseImageMemoryBind structures.

Valid Usage

- VUID-VkSparseImageMemoryBindInfo-subresource-01722
  The `subresource.mipLevel` member of each element of `pBinds` **must** be less than the `mipLevels` specified in VkImageCreateInfo when `image` was created
- VUID-VkSparseImageMemoryBindInfo-subresource-01723
  The `subresource.arrayLayer` member of each element of `pBinds` **must** be less than the `arrayLayers` specified in VkImageCreateInfo when `image` was created
- VUID-VkSparseImageMemoryBindInfo-subresource-01106
  The `subresource.aspectMask` member of each element of `pBinds` **must** be valid for the `format` specified in VkImageCreateInfo when `image` was created
- VUID-VkSparseImageMemoryBindInfo-image-02901
  `image` **must** have been created with VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT set

Valid Usage (Implicit)

- VUID-VkSparseImageMemoryBindInfo-image-parameter
  `image` **must** be a valid VkImage handle
- VUID-VkSparseImageMemoryBindInfo-pBinds-parameter
  `pBinds` **must** be a valid pointer to an array of `bindCount` valid VkSparseImageMemoryBind structures
- VUID-VkSparseImageMemoryBindInfo-bindCount-arraylength
  `bindCount` **must** be greater than `0`

The VkSparseImageMemoryBind structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseImageMemoryBind {
    VkImageSubresource         subresource;
    VkOffset3D                 offset;
    VkExtent3D                 extent;
    VkDeviceMemory             memory;
    VkDeviceSize               memoryOffset;
    VkSparseMemoryBindFlags    flags;
} VkSparseImageMemoryBind;
```

- subresource is the image *aspect* and region of interest in the image.
- offset are the coordinates of the first texel within the image subresource to bind.
- extent is the size in texels of the region within the image subresource to bind. The extent **must** be a multiple of the sparse image block dimensions, except when binding sparse image blocks along the edge of an image subresource it **can** instead be such that any coordinate of offset + extent equals the corresponding dimensions of the image subresource.
- memory is the VkDeviceMemory object that the sparse image blocks of the image are bound to. If memory is VK_NULL_HANDLE, the sparse image blocks are unbound.
- memoryOffset is an offset into the VkDeviceMemory object. If memory is VK_NULL_HANDLE, this value is ignored.
- flags are sparse memory binding flags.

Valid Usage

- VUID-VkSparseImageMemoryBind-memory-01104
  If the sparseResidencyAliased feature is not enabled, and if any other resources are bound to ranges of memory, the range of memory being bound **must** not overlap with those bound ranges
- VUID-VkSparseImageMemoryBind-memory-01105
  memory and memoryOffset **must** match the memory requirements of the calling command's image, as described in section Resource Memory Association
- VUID-VkSparseImageMemoryBind-offset-01107
  offset.x **must** be a multiple of the sparse image block width (VkSparseImageFormatProperties::imageGranularity.width) of the image
- VUID-VkSparseImageMemoryBind-extent-09388
  extent.width **must** be greater than 0
- VUID-VkSparseImageMemoryBind-extent-01108
  extent.width **must** either be a multiple of the sparse image block width of the image, or else (extent.width + offset.x) **must** equal the width of the image subresource
- VUID-VkSparseImageMemoryBind-offset-01109
  offset.y **must** be a multiple of the sparse image block height

(VkSparseImageFormatProperties::imageGranularity.height) of the image

- VUID-VkSparseImageMemoryBind-extent-09389
  extent.height **must** be greater than 0
- VUID-VkSparseImageMemoryBind-extent-01110
  extent.height **must** either be a multiple of the sparse image block height of the image, or else (extent.height + offset.y) **must** equal the height of the image subresource
- VUID-VkSparseImageMemoryBind-offset-01111
  offset.z **must** be a multiple of the sparse image block depth (VkSparseImageFormatProperties::imageGranularity.depth) of the image
- VUID-VkSparseImageMemoryBind-extent-09390
  extent.depth **must** be greater than 0
- VUID-VkSparseImageMemoryBind-extent-01112
  extent.depth **must** either be a multiple of the sparse image block depth of the image, or else (extent.depth + offset.z) **must** equal the depth of the image subresource
- VUID-VkSparseImageMemoryBind-memory-02732
  If memory was created with VkExportMemoryAllocateInfo::handleTypes not equal to 0, at least one handle type it contained **must** also have been set in VkExternalMemoryImageCreateInfo::handleTypes when the image was created
- VUID-VkSparseImageMemoryBind-memory-02733
  If memory was created by a memory import operation, the external handle type of the imported memory **must** also have been set in VkExternalMemoryImageCreateInfo ::handleTypes when image was created

Valid Usage (Implicit)

- VUID-VkSparseImageMemoryBind-subresource-parameter
  subresource **must** be a valid VkImageSubresource structure
- VUID-VkSparseImageMemoryBind-memory-parameter
  If memory is not VK_NULL_HANDLE, memory **must** be a valid VkDeviceMemory handle
- VUID-VkSparseImageMemoryBind-flags-parameter
  If flags is not 0, flags **must** be a valid combination of VkSparseMemoryBindFlagBits values

To submit sparse binding operations to a queue, call:

```
// Provided by VK_VERSION_1_0
VkResult vkQueueBindSparse(
    VkQueue                                     queue,
    uint32_t                                    bindInfoCount,
    const VkBindSparseInfo*                     pBindInfo,
    VkFence                                     fence);
```

- `queue` is the queue that the sparse binding operations will be submitted to.
- `bindInfoCount` is the number of elements in the `pBindInfo` array.
- `pBindInfo` is a pointer to an array of VkBindSparseInfo structures, each specifying a sparse binding submission batch.
- `fence` is an **optional** handle to a fence to be signaled. If `fence` is not VK_NULL_HANDLE, it defines a fence signal operation.

`vkQueueBindSparse` is a queue submission command, with each batch defined by an element of `pBindInfo` as a VkBindSparseInfo structure. Batches begin execution in the order they appear in `pBindInfo`, but **may** complete out of order.

Within a batch, a given range of a resource **must** not be bound more than once. Across batches, if a range is to be bound to one allocation and offset and then to another allocation and offset, then the application **must** guarantee (usually using semaphores) that the binding operations are executed in the correct order, as well as to order binding operations against the execution of command buffer submissions.

As no operation to vkQueueBindSparse causes any pipeline stage to access memory, synchronization primitives used in this command effectively only define execution dependencies.

Additional information about fence and semaphore operation is described in the synchronization chapter.

Valid Usage

- VUID-vkQueueBindSparse-fence-01113
  If `fence` is not VK_NULL_HANDLE, `fence` **must** be unsignaled
- VUID-vkQueueBindSparse-fence-01114
  If `fence` is not VK_NULL_HANDLE, `fence` **must** not be associated with any other queue command that has not yet completed execution on that queue
- VUID-vkQueueBindSparse-pSignalSemaphores-01115
  Each element of the `pSignalSemaphores` member of each element of `pBindInfo` **must** be unsignaled when the semaphore signal operation it defines is executed on the device
- VUID-vkQueueBindSparse-pWaitSemaphores-01116
  When a semaphore wait operation referring to a binary semaphore defined by any element of the `pWaitSemaphores` member of any element of `pBindInfo` executes on `queue`, there **must** be no other queues waiting on the same semaphore
- VUID-vkQueueBindSparse-pWaitSemaphores-03245
  All elements of the `pWaitSemaphores` member of all elements of `pBindInfo` referring to a semaphore created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_BINARY **must** reference a semaphore signal operation that has been submitted for execution and any semaphore signal operations on which it depends **must** have also been submitted for execution

Valid Usage (Implicit)

- VUID-vkQueueBindSparse-queue-parameter
  `queue` **must** be a valid VkQueue handle
- VUID-vkQueueBindSparse-pBindInfo-parameter
  If `bindInfoCount` is not `0`, `pBindInfo` **must** be a valid pointer to an array of `bindInfoCount` valid VkBindSparseInfo structures
- VUID-vkQueueBindSparse-fence-parameter
  If `fence` is not VK_NULL_HANDLE, `fence` **must** be a valid VkFence handle
- VUID-vkQueueBindSparse-queuetype
  The `queue` **must** support VK_QUEUE_SPARSE_BINDING_BIT operations
- VUID-vkQueueBindSparse-commonparent
  Both of `fence`, and `queue` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `queue` **must** be externally synchronized if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR
- Host access to `fence` **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	VK_QUEUE_SPARSE_BINDING_BIT	-

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkBindSparseInfo structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBindSparseInfo {
    VkStructureType                              sType;
    const void*                                  pNext;
    uint32_t                                     waitSemaphoreCount;
    const VkSemaphore*                           pWaitSemaphores;
    uint32_t                                     bufferBindCount;
    const VkSparseBufferMemoryBindInfo*          pBufferBinds;
    uint32_t                                     imageOpaqueBindCount;
    const VkSparseImageOpaqueMemoryBindInfo*     pImageOpaqueBinds;
    uint32_t                                     imageBindCount;
    const VkSparseImageMemoryBindInfo*           pImageBinds;
    uint32_t                                     signalSemaphoreCount;
    const VkSemaphore*                           pSignalSemaphores;
} VkBindSparseInfo;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- waitSemaphoreCount is the number of semaphores upon which to wait before executing the sparse binding operations for the batch.
- pWaitSemaphores is a pointer to an array of semaphores upon which to wait before the sparse binding operations for this batch begin execution. If semaphores to wait on are provided, they define a semaphore wait operation.
- bufferBindCount is the number of sparse buffer bindings to perform in the batch.
- pBufferBinds is a pointer to an array of VkSparseBufferMemoryBindInfo structures.
- imageOpaqueBindCount is the number of opaque sparse image bindings to perform.
- pImageOpaqueBinds is a pointer to an array of VkSparseImageOpaqueMemoryBindInfo structures, indicating opaque sparse image bindings to perform.
- imageBindCount is the number of sparse image bindings to perform.
- pImageBinds is a pointer to an array of VkSparseImageMemoryBindInfo structures, indicating sparse image bindings to perform.
- signalSemaphoreCount is the number of semaphores to be signaled once the sparse binding operations specified by the structure have completed execution.
- pSignalSemaphores is a pointer to an array of semaphores which will be signaled when the sparse binding operations for this batch have completed execution. If semaphores to be signaled are provided, they define a semaphore signal operation.

The first synchronization scope of each semaphore signal operation defined by this structure includes all sparse binding operations defined by this structure.

The second synchronization scope of each semaphore wait operation defined by this structure includes all sparse binding operations defined by this structure.

Valid Usage

- VUID-VkBindSparseInfo-pWaitSemaphores-03246
  If any element of pWaitSemaphores or pSignalSemaphores was created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE then the pNext chain **must** include a VkTimelineSemaphoreSubmitInfo structure
- VUID-VkBindSparseInfo-pNext-03247
  If the pNext chain of this structure includes a VkTimelineSemaphoreSubmitInfo structure and any element of pWaitSemaphores was created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE then its waitSemaphoreValueCount member **must** equal waitSemaphoreCount
- VUID-VkBindSparseInfo-pNext-03248
  If the pNext chain of this structure includes a VkTimelineSemaphoreSubmitInfo structure and any element of pSignalSemaphores was created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE then its signalSemaphoreValueCount member **must** equal signalSemaphoreCount
- VUID-VkBindSparseInfo-pSignalSemaphores-03249
  For each element of pSignalSemaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE the corresponding element of VkTimelineSemaphoreSubmitInfo::pSignalSemaphoreValues **must** have a value greater than the current value of the semaphore when the semaphore signal operation is executed
- VUID-VkBindSparseInfo-pWaitSemaphores-03250
  For each element of pWaitSemaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE the corresponding element of VkTimelineSemaphoreSubmitInfo::pWaitSemaphoreValues **must** have a value which does not differ from the current value of the semaphore or from the value of any outstanding semaphore wait or signal operation on that semaphore by more than maxTimelineSemaphoreValueDifference
- VUID-VkBindSparseInfo-pSignalSemaphores-03251
  For each element of pSignalSemaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE the corresponding element of VkTimelineSemaphoreSubmitInfo::pSignalSemaphoreValues **must** have a value which does not differ from the current value of the semaphore or from the value of any outstanding semaphore wait or signal operation on that semaphore by more than maxTimelineSemaphoreValueDifference

Valid Usage (Implicit)

- VUID-VkBindSparseInfo-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_BIND_SPARSE_INFO
- VUID-VkBindSparseInfo-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkDeviceGroupBindSparseInfo,

VkFrameBoundaryEXT, or VkTimelineSemaphoreSubmitInfo

- VUID-VkBindSparseInfo-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkBindSparseInfo-pWaitSemaphores-parameter
  If waitSemaphoreCount is not 0, pWaitSemaphores **must** be a valid pointer to an array of waitSemaphoreCount valid VkSemaphore handles
- VUID-VkBindSparseInfo-pBufferBinds-parameter
  If bufferBindCount is not 0, pBufferBinds **must** be a valid pointer to an array of bufferBindCount valid VkSparseBufferMemoryBindInfo structures
- VUID-VkBindSparseInfo-pImageOpaqueBinds-parameter
  If imageOpaqueBindCount is not 0, pImageOpaqueBinds **must** be a valid pointer to an array of imageOpaqueBindCount valid VkSparseImageOpaqueMemoryBindInfo structures
- VUID-VkBindSparseInfo-pImageBinds-parameter
  If imageBindCount is not 0, pImageBinds **must** be a valid pointer to an array of imageBindCount valid VkSparseImageMemoryBindInfo structures
- VUID-VkBindSparseInfo-pSignalSemaphores-parameter
  If signalSemaphoreCount is not 0, pSignalSemaphores **must** be a valid pointer to an array of signalSemaphoreCount valid VkSemaphore handles
- VUID-VkBindSparseInfo-commonparent
  Both of the elements of pSignalSemaphores, and the elements of pWaitSemaphores that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

To specify the values to use when waiting for and signaling semaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE, add a VkTimelineSemaphoreSubmitInfo structure to the pNext chain of the VkBindSparseInfo structure.

If the pNext chain of VkBindSparseInfo includes a VkDeviceGroupBindSparseInfo structure, then that structure includes device indices specifying which instance of the resources and memory are bound.

The VkDeviceGroupBindSparseInfo structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkDeviceGroupBindSparseInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           resourceDeviceIndex;
    uint32_t           memoryDeviceIndex;
} VkDeviceGroupBindSparseInfo;
```

```
// Provided by VK_KHR_device_group
// Equivalent to VkDeviceGroupBindSparseInfo
typedef VkDeviceGroupBindSparseInfo VkDeviceGroupBindSparseInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `resourceDeviceIndex` is a device index indicating which instance of the resource is bound.
- `memoryDeviceIndex` is a device index indicating which instance of the memory the resource instance is bound to.

These device indices apply to all buffer and image memory binds included in the batch pointing to this structure. The semaphore waits and signals for the batch are executed only by the physical device specified by the `resourceDeviceIndex`.

If this structure is not present, `resourceDeviceIndex` and `memoryDeviceIndex` are assumed to be zero.

Valid Usage

- VUID-VkDeviceGroupBindSparseInfo-resourceDeviceIndex-01118
  `resourceDeviceIndex` and `memoryDeviceIndex` **must** both be valid device indices
- VUID-VkDeviceGroupBindSparseInfo-memoryDeviceIndex-01119
  Each memory allocation bound in this batch **must** have allocated an instance for `memoryDeviceIndex`

Valid Usage (Implicit)

- VUID-VkDeviceGroupBindSparseInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_GROUP_BIND_SPARSE_INFO

Structure Chaining

Extends the structure

- VkBindSparseInfo

Chapter 37. Window System Integration (WSI)

This chapter discusses the window system integration (WSI) between the Vulkan API and the various forms of displaying the results of rendering to a user. Since the Vulkan API **can** be used without displaying results, WSI is provided through the use of optional Vulkan extensions. This chapter provides an overview of WSI. See the appendix for additional details of each WSI extension, including which extensions **must** be enabled in order to use each of the functions described in this chapter.

37.1. WSI Platform

A platform is an abstraction for a window system, OS, etc. Some examples include MS Windows, Android, and Wayland. The Vulkan API **may** be integrated in a unique manner for each platform.

The Vulkan API does not define any type of platform object. Platform-specific WSI extensions are defined, each containing platform-specific functions for using WSI. Use of these extensions is guarded by preprocessor symbols as defined in the Window System-Specific Header Control appendix.

In order for an application to be compiled to use WSI with a given platform, it **must** either:

- `#define` the appropriate preprocessor symbol prior to including the `vulkan.h` header file, or
- include `vulkan_core.h` and any native platform headers, followed by the appropriate platform-specific header.

The preprocessor symbols and platform-specific headers are defined in the Window System Extensions and Headers table.

Each platform-specific extension is an instance extension. The application **must** enable instance extensions with `vkCreateInstance` before using them.

37.2. WSI Surface

Native platform surface or window objects are abstracted by surface objects, which are represented by `VkSurfaceKHR` handles:

```
// Provided by VK_KHR_surface
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkSurfaceKHR)
```

The `VK_KHR_surface` extension declares the `VkSurfaceKHR` object, and provides a function for destroying `VkSurfaceKHR` objects. Separate platform-specific extensions each provide a function for creating a `VkSurfaceKHR` object for the respective platform. From the application's perspective this is an opaque handle, just like the handles of other Vulkan objects.

NOTE On certain platforms, the Vulkan loader and ICDs **may** have conventions that treat

the handle as a pointer to a structure containing the platform-specific information about the surface. This will be described in the documentation for the loader-ICD interface, and in the `vk_icd.h` header file of the LoaderAndTools source-code repository. This does not affect the loader-layer interface; layers **may** wrap `VkSurfaceKHR` objects.

37.2.1. Android Platform

To create a `VkSurfaceKHR` object for an Android native window, call:

```
// Provided by VK_KHR_android_surface
VkResult vkCreateAndroidSurfaceKHR(
    VkInstance                                  instance,
    const VkAndroidSurfaceCreateInfoKHR*        pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkSurfaceKHR*                               pSurface);
```

- `instance` is the instance to associate the surface with.
- `pCreateInfo` is a pointer to a VkAndroidSurfaceCreateInfoKHR structure containing parameters affecting the creation of the surface object.
- `pAllocator` is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see Memory Allocation).
- `pSurface` is a pointer to a VkSurfaceKHR handle in which the created surface object is returned.

During the lifetime of a surface created using a particular ANativeWindow handle any attempts to create another surface for the same ANativeWindow and any attempts to connect to the same ANativeWindow through other platform mechanisms will fail.

NOTE In particular, only one `VkSurfaceKHR` **can** exist at a time for a given window. Similarly, a native window **cannot** be used by both a `VkSurfaceKHR` and `EGLSurface` simultaneously.

If successful, `vkCreateAndroidSurfaceKHR` increments the ANativeWindow's reference count, and `vkDestroySurfaceKHR` will decrement it.

On Android, when a swapchain's `imageExtent` does not match the surface's `currentExtent`, the presentable images will be scaled to the surface's dimensions during presentation. `minImageExtent` is (1,1), and `maxImageExtent` is the maximum image size supported by the consumer. For the system compositor, `currentExtent` is the window size (i.e. the consumer's preferred size).

Valid Usage (Implicit)

- VUID-vkCreateAndroidSurfaceKHR-instance-parameter
  `instance` **must** be a valid VkInstance handle
- VUID-vkCreateAndroidSurfaceKHR-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkAndroidSurfaceCreateInfoKHR structure

- VUID-vkCreateAndroidSurfaceKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateAndroidSurfaceKHR-pSurface-parameter
  `pSurface` **must** be a valid pointer to a VkSurfaceKHR handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_NATIVE_WINDOW_IN_USE_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkAndroidSurfaceCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_android_surface
typedef struct VkAndroidSurfaceCreateInfoKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    VkAndroidSurfaceCreateFlagsKHR  flags;
    struct ANativeWindow*           window;
} VkAndroidSurfaceCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `window` is a pointer to the ANativeWindow to associate the surface with.

Valid Usage

- VUID-VkAndroidSurfaceCreateInfoKHR-window-01248
  `window` **must** point to a valid Android ANativeWindow

Valid Usage (Implicit)

- VUID-VkAndroidSurfaceCreateInfoKHR-sType-sType

sType **must** be VK_STRUCTURE_TYPE_ANDROID_SURFACE_CREATE_INFO_KHR

- VUID-VkAndroidSurfaceCreateInfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkAndroidSurfaceCreateInfoKHR-flags-zerobitmask
  flags **must** be 0

To remove an unnecessary compile time dependency, an incomplete type definition of ANativeWindow is provided in the Vulkan headers:

```
// Provided by VK_KHR_android_surface
struct ANativeWindow;
```

The actual ANativeWindow type is defined in Android NDK headers.

```
// Provided by VK_KHR_android_surface
typedef VkFlags VkAndroidSurfaceCreateFlagsKHR;
```

VkAndroidSurfaceCreateFlagsKHR is a bitmask type for setting a mask, but is currently reserved for future use.

37.2.2. Wayland Platform

To create a VkSurfaceKHR object for a Wayland surface, call:

```
// Provided by VK_KHR_wayland_surface
VkResult vkCreateWaylandSurfaceKHR(
    VkInstance                                  instance,
    const VkWaylandSurfaceCreateInfoKHR*        pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkSurfaceKHR*                               pSurface);
```

- instance is the instance to associate the surface with.
- pCreateInfo is a pointer to a VkWaylandSurfaceCreateInfoKHR structure containing parameters affecting the creation of the surface object.
- pAllocator is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see Memory Allocation).
- pSurface is a pointer to a VkSurfaceKHR handle in which the created surface object is returned.

Valid Usage (Implicit)

- VUID-vkCreateWaylandSurfaceKHR-instance-parameter
  instance **must** be a valid VkInstance handle

- VUID-vkCreateWaylandSurfaceKHR-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkWaylandSurfaceCreateInfoKHR structure
- VUID-vkCreateWaylandSurfaceKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateWaylandSurfaceKHR-pSurface-parameter
  `pSurface` **must** be a valid pointer to a VkSurfaceKHR handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkWaylandSurfaceCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_wayland_surface
typedef struct VkWaylandSurfaceCreateInfoKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    VkWaylandSurfaceCreateFlagsKHR  flags;
    struct wl_display*              display;
    struct wl_surface*              surface;
} VkWaylandSurfaceCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `display` and `surface` are pointers to the Wayland `wl_display` and `wl_surface` to associate the surface with.

Valid Usage

- VUID-VkWaylandSurfaceCreateInfoKHR-display-01304
  `display` **must** point to a valid Wayland `wl_display`
- VUID-VkWaylandSurfaceCreateInfoKHR-surface-01305
  `surface` **must** point to a valid Wayland `wl_surface`

Valid Usage (Implicit)

- VUID-VkWaylandSurfaceCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_WAYLAND_SURFACE_CREATE_INFO_KHR
- VUID-VkWaylandSurfaceCreateInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkWaylandSurfaceCreateInfoKHR-flags-zerobitmask
  `flags` **must** be `0`

On Wayland, `currentExtent` is the special value (0xFFFFFFFF, 0xFFFFFFFF), indicating that the surface size will be determined by the extent of a swapchain targeting the surface. Whatever the application sets a swapchain's `imageExtent` to will be the size of the window, after the first image is presented. `minImageExtent` is (1,1), and `maxImageExtent` is the maximum supported surface size. Any calls to vkGetPhysicalDeviceSurfacePresentModesKHR on a surface created with `vkCreateWaylandSurfaceKHR` are **required** to return VK_PRESENT_MODE_MAILBOX_KHR as one of the valid present modes.

Some Vulkan functions **may** send protocol over the specified `wl_display` connection when using a swapchain or presentable images created from a `VkSurfaceKHR` referring to a `wl_surface`. Applications **must** therefore ensure that both the `wl_display` and the `wl_surface` remain valid for the lifetime of any `VkSwapchainKHR` objects created from a particular `wl_display` and `wl_surface`. Also, calling vkQueuePresentKHR will result in Vulkan sending `wl_surface.commit` requests to the underlying `wl_surface` of each The `wl_surface.attach`, `wl_surface.damage`, and `wl_surface.commit` requests **must** be issued by the implementation during the call to vkQueuePresentKHR and **must** not be issued by the implementation outside of vkQueuePresentKHR. This ensures that any Wayland requests sent by the client after the call to vkQueuePresentKHR returns will be received by the compositor after the `wl_surface.commit`. Regardless of the mode of swapchain creation, a new `wl_event_queue` **must** be created for each successful vkCreateWaylandSurfaceKHR call, and every Wayland object created by the implementation **must** be assigned to this event queue. If the platform provides Wayland 1.11 or greater, this **must** be implemented by the use of Wayland proxy object wrappers, to avoid race conditions.

If the application wishes to synchronize any window changes with a particular frame, such requests **must** be sent to the Wayland display server prior to calling vkQueuePresentKHR.

The implementation **must** ensure that no `wp_color_management_surface_v1` object exists for a surface, which has no swapchains, or only swapchains with a `imageColorSpace` of VK_COLOR_SPACE_PASS_THROUGH_EXT associated with it.

```
// Provided by VK_KHR_wayland_surface
typedef VkFlags VkWaylandSurfaceCreateFlagsKHR;
```

`VkWaylandSurfaceCreateFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

37.2.3. Win32 Platform

To create a VkSurfaceKHR object for a Win32 window, call:

```
// Provided by VK_KHR_win32_surface
VkResult vkCreateWin32SurfaceKHR(
    VkInstance                                  instance,
    const VkWin32SurfaceCreateInfoKHR*          pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkSurfaceKHR*                               pSurface);
```

- instance is the instance to associate the surface with.
- pCreateInfo is a pointer to a VkWin32SurfaceCreateInfoKHR structure containing parameters affecting the creation of the surface object.
- pAllocator is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see Memory Allocation).
- pSurface is a pointer to a VkSurfaceKHR handle in which the created surface object is returned.

Valid Usage (Implicit)

- VUID-vkCreateWin32SurfaceKHR-instance-parameter
  instance **must** be a valid VkInstance handle
- VUID-vkCreateWin32SurfaceKHR-pCreateInfo-parameter
  pCreateInfo **must** be a valid pointer to a valid VkWin32SurfaceCreateInfoKHR structure
- VUID-vkCreateWin32SurfaceKHR-pAllocator-parameter
  If pAllocator is not NULL, pAllocator **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateWin32SurfaceKHR-pSurface-parameter
  pSurface **must** be a valid pointer to a VkSurfaceKHR handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Some Vulkan functions **may** call the SendMessage system API when interacting with a VkSurfaceKHR

through a VkSwapchainKHR. In a multithreaded environment, calling SendMessage from a thread that is not the thread associated with pCreateInfo->hwnd will block until the application has processed the window message. Thus, applications **should** either call these Vulkan functions on the message pump thread, or make sure their message pump is actively running. Failing to do so **may** result in deadlocks.

The functions subject to this requirement are:

- vkCreateSwapchainKHR
- vkDestroySwapchainKHR
- vkAcquireNextImageKHR and vkAcquireNextImage2KHR
- vkQueuePresentKHR
- vkReleaseSwapchainImagesKHR
- vkAcquireFullScreenExclusiveModeEXT
- vkReleaseFullScreenExclusiveModeEXT
- vkSetHdrMetadataEXT

The VkWin32SurfaceCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_win32_surface
typedef struct VkWin32SurfaceCreateInfoKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    VkWin32SurfaceCreateFlagsKHR    flags;
    HINSTANCE                       hinstance;
    HWND                            hwnd;
} VkWin32SurfaceCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is reserved for future use.
- hinstance is the Win32 HINSTANCE for the window to associate the surface with.
- hwnd is the Win32 HWND for the window to associate the surface with.

Valid Usage

- VUID-VkWin32SurfaceCreateInfoKHR-hinstance-01307
  hinstance **must** be a valid Win32 HINSTANCE
- VUID-VkWin32SurfaceCreateInfoKHR-hwnd-01308
  hwnd **must** be a valid Win32 HWND

Valid Usage (Implicit)

- VUID-VkWin32SurfaceCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_WIN32_SURFACE_CREATE_INFO_KHR
- VUID-VkWin32SurfaceCreateInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkWin32SurfaceCreateInfoKHR-flags-zerobitmask
  `flags` **must** be `0`

With Win32, `minImageExtent`, `maxImageExtent`, and `currentExtent` **must** always equal the window size.

The `currentExtent` of a Win32 surface **must** have both `width` and `height` greater than 0, or both of them 0.

NOTE

Due to the above restrictions, unless VkSwapchainPresentScalingCreateInfoKHR is used to specify handling of disparities between surface and swapchain dimensions, it is only possible to create a new swapchain on this platform with `imageExtent` being equal to the current size of the window, as reported in VkSurfaceCapabilitiesKHR::`currentExtent`.

The window size **may** become (0, 0) on this platform (e.g. when the window is minimized), and so a swapchain **cannot** be created until the size changes.

```
// Provided by VK_KHR_win32_surface
typedef VkFlags VkWin32SurfaceCreateFlagsKHR;
```

`VkWin32SurfaceCreateFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

37.2.4. XCB Platform

To create a `VkSurfaceKHR` object for an X11 window, using the XCB client-side library, call:

```
// Provided by VK_KHR_xcb_surface
VkResult vkCreateXcbSurfaceKHR(
    VkInstance                                  instance,
    const VkXcbSurfaceCreateInfoKHR*            pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkSurfaceKHR*                               pSurface);
```

- `instance` is the instance to associate the surface with.
- `pCreateInfo` is a pointer to a VkXcbSurfaceCreateInfoKHR structure containing parameters affecting the creation of the surface object.
- `pAllocator` is the allocator used for host memory allocated for the surface object when there is

no more specific allocator available (see Memory Allocation).

- `pSurface` is a pointer to a VkSurfaceKHR handle in which the created surface object is returned.

Valid Usage (Implicit)

- VUID-vkCreateXcbSurfaceKHR-instance-parameter
  `instance` **must** be a valid VkInstance handle
- VUID-vkCreateXcbSurfaceKHR-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkXcbSurfaceCreateInfoKHR structure
- VUID-vkCreateXcbSurfaceKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateXcbSurfaceKHR-pSurface-parameter
  `pSurface` **must** be a valid pointer to a VkSurfaceKHR handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkXcbSurfaceCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_xcb_surface
typedef struct VkXcbSurfaceCreateInfoKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    VkXcbSurfaceCreateFlagsKHR      flags;
    xcb_connection_t*               connection;
    xcb_window_t                    window;
} VkXcbSurfaceCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `connection` is a pointer to an `xcb_connection_t` to the X server.

- `window` is the `xcb_window_t` for the X11 window to associate the surface with.

Valid Usage

- VUID-VkXcbSurfaceCreateInfoKHR-connection-01310
  `connection` **must** point to a valid X11 `xcb_connection_t`
- VUID-VkXcbSurfaceCreateInfoKHR-window-01311
  `window` **must** be a valid X11 `xcb_window_t`

Valid Usage (Implicit)

- VUID-VkXcbSurfaceCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_XCB_SURFACE_CREATE_INFO_KHR
- VUID-VkXcbSurfaceCreateInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkXcbSurfaceCreateInfoKHR-flags-zerobitmask
  `flags` **must** be `0`

With Xcb, `minImageExtent`, `maxImageExtent`, and `currentExtent` **must** always equal the window size.

The `currentExtent` of an Xcb surface **must** have both `width` and `height` greater than 0, or both of them 0.

NOTE

Due to the above restrictions, unless VkSwapchainPresentScalingCreateInfoKHR is used to specify handling of disparities between surface and swapchain dimensions, it is only possible to create a new swapchain on this platform with `imageExtent` being equal to the current size of the window, as reported in VkSurfaceCapabilitiesKHR::`currentExtent`.

The window size **may** become (0, 0) on this platform (e.g. when the window is minimized), and so a swapchain **cannot** be created until the size changes.

Some Vulkan functions **may** send protocol over the specified xcb connection when using a swapchain or presentable images created from a VkSurfaceKHR referring to an xcb window. Applications **must** therefore ensure the xcb connection is available to Vulkan for the duration of any functions that manipulate such swapchains or their presentable images, and any functions that build or queue command buffers that operate on such presentable images. Specifically, applications using Vulkan with xcb-based swapchains **must**

- Avoid holding a server grab on an xcb connection while waiting for Vulkan operations to complete using a swapchain derived from a different xcb connection referring to the same X server instance. Failing to do so **may** result in deadlock.

```
// Provided by VK_KHR_xcb_surface
typedef VkFlags VkXcbSurfaceCreateFlagsKHR;
```

VkXcbSurfaceCreateFlagsKHR is a bitmask type for setting a mask, but is currently reserved for future use.

37.2.5. Xlib Platform

To create a VkSurfaceKHR object for an X11 window, using the Xlib client-side library, call:

```
// Provided by VK_KHR_xlib_surface
VkResult vkCreateXlibSurfaceKHR(
    VkInstance                                  instance,
    const VkXlibSurfaceCreateInfoKHR*           pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkSurfaceKHR*                               pSurface);
```

- instance is the instance to associate the surface with.
- pCreateInfo is a pointer to a VkXlibSurfaceCreateInfoKHR structure containing the parameters affecting the creation of the surface object.
- pAllocator is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see Memory Allocation).
- pSurface is a pointer to a VkSurfaceKHR handle in which the created surface object is returned.

Valid Usage (Implicit)

- VUID-vkCreateXlibSurfaceKHR-instance-parameter
  instance **must** be a valid VkInstance handle
- VUID-vkCreateXlibSurfaceKHR-pCreateInfo-parameter
  pCreateInfo **must** be a valid pointer to a valid VkXlibSurfaceCreateInfoKHR structure
- VUID-vkCreateXlibSurfaceKHR-pAllocator-parameter
  If pAllocator is not NULL, pAllocator **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateXlibSurfaceKHR-pSurface-parameter
  pSurface **must** be a valid pointer to a VkSurfaceKHR handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkXlibSurfaceCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_xlib_surface
typedef struct VkXlibSurfaceCreateInfoKHR {
    VkStructureType                sType;
    const void*                    pNext;
    VkXlibSurfaceCreateFlagsKHR    flags;
    Display*                       dpy;
    Window                         window;
} VkXlibSurfaceCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is reserved for future use.
- dpy is a pointer to an Xlib Display connection to the X server.
- window is an Xlib Window to associate the surface with.

Valid Usage

- VUID-VkXlibSurfaceCreateInfoKHR-dpy-01313
  dpy **must** point to a valid Xlib Display
- VUID-VkXlibSurfaceCreateInfoKHR-window-01314
  window **must** be a valid Xlib Window

Valid Usage (Implicit)

- VUID-VkXlibSurfaceCreateInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_XLIB_SURFACE_CREATE_INFO_KHR
- VUID-VkXlibSurfaceCreateInfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkXlibSurfaceCreateInfoKHR-flags-zerobitmask
  flags **must** be 0

With Xlib, minImageExtent, maxImageExtent, and currentExtent **must** always equal the window size.

The currentExtent of an Xlib surface **must** have both width and height greater than 0, or both of them 0.

NOTE

Due to the above restrictions, unless VkSwapchainPresentScalingCreateInfoKHR is used to specify handling of disparities between surface and swapchain dimensions, it is only possible to create a new swapchain on this platform with `imageExtent` being equal to the current size of the window, as reported in VkSurfaceCapabilitiesKHR::`currentExtent`.

The window size **may** become (0, 0) on this platform (e.g. when the window is minimized), and so a swapchain **cannot** be created until the size changes.

Some Vulkan functions **may** send protocol over the specified Xlib `Display` connection when using a swapchain or presentable images created from a VkSurfaceKHR referring to an Xlib window. Applications **must** therefore ensure the display connection is available to Vulkan for the duration of any functions that manipulate such swapchains or their presentable images, and any functions that build or queue command buffers that operate on such presentable images. Specifically, applications using Vulkan with Xlib-based swapchains **must**

- Avoid holding a server grab on a display connection while waiting for Vulkan operations to complete using a swapchain derived from a different display connection referring to the same X server instance. Failing to do so **may** result in deadlock.

Some implementations may require threads to implement some presentation modes so applications **must** call `XInitThreads`() before calling any other Xlib functions.

```
// Provided by VK_KHR_xlib_surface
typedef VkFlags VkXlibSurfaceCreateFlagsKHR;
```

`VkXlibSurfaceCreateFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

37.2.6. DirectFB Platform

To create a `VkSurfaceKHR` object for a DirectFB surface, call:

```
// Provided by VK_EXT_directfb_surface
VkResult vkCreateDirectFBSurfaceEXT(
    VkInstance                                  instance,
    const VkDirectFBSurfaceCreateInfoEXT*       pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkSurfaceKHR*                               pSurface);
```

- `instance` is the instance to associate the surface with.
- `pCreateInfo` is a pointer to a VkDirectFBSurfaceCreateInfoEXT structure containing parameters affecting the creation of the surface object.
- `pAllocator` is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see Memory Allocation).
- `pSurface` is a pointer to a VkSurfaceKHR handle in which the created surface object is returned.

Valid Usage (Implicit)

- VUID-vkCreateDirectFBSurfaceEXT-instance-parameter
  `instance` **must** be a valid VkInstance handle
- VUID-vkCreateDirectFBSurfaceEXT-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkDirectFBSurfaceCreateInfoEXT structure
- VUID-vkCreateDirectFBSurfaceEXT-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateDirectFBSurfaceEXT-pSurface-parameter
  `pSurface` **must** be a valid pointer to a VkSurfaceKHR handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkDirectFBSurfaceCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_directfb_surface
typedef struct VkDirectFBSurfaceCreateInfoEXT {
    VkStructureType                  sType;
    const void*                      pNext;
    VkDirectFBSurfaceCreateFlagsEXT  flags;
    IDirectFB*                       dfb;
    IDirectFBSurface*                surface;
} VkDirectFBSurfaceCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `dfb` is a pointer to the `IDirectFB` main interface of DirectFB.
- `surface` is a pointer to a `IDirectFBSurface` surface interface.

Valid Usage

- VUID-VkDirectFBSurfaceCreateInfoEXT-dfb-04117
  `dfb` **must** point to a valid DirectFB `IDirectFB`
- VUID-VkDirectFBSurfaceCreateInfoEXT-surface-04118
  `surface` **must** point to a valid DirectFB `IDirectFBSurface`

Valid Usage (Implicit)

- VUID-VkDirectFBSurfaceCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DIRECTFB_SURFACE_CREATE_INFO_EXT
- VUID-VkDirectFBSurfaceCreateInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDirectFBSurfaceCreateInfoEXT-flags-zerobitmask
  `flags` **must** be `0`

With DirectFB, `minImageExtent`, `maxImageExtent`, and `currentExtent` **must** always equal the surface size.

```
// Provided by VK_EXT_directfb_surface
typedef VkFlags VkDirectFBSurfaceCreateFlagsEXT;
```

`VkDirectFBSurfaceCreateFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

37.2.7. Metal Platform

To create a `VkSurfaceKHR` object for a CAMetalLayer, call:

```
// Provided by VK_EXT_metal_surface
VkResult vkCreateMetalSurfaceEXT(
    VkInstance                                  instance,
    const VkMetalSurfaceCreateInfoEXT*          pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkSurfaceKHR*                               pSurface);
```

- `instance` is the instance with which to associate the surface.
- `pCreateInfo` is a pointer to a `VkMetalSurfaceCreateInfoEXT` structure specifying parameters affecting the creation of the surface object.
- `pAllocator` is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see Memory Allocation).
- `pSurface` is a pointer to a `VkSurfaceKHR` handle in which the created surface object is returned.

Valid Usage (Implicit)

- VUID-vkCreateMetalSurfaceEXT-instance-parameter
  `instance` **must** be a valid VkInstance handle
- VUID-vkCreateMetalSurfaceEXT-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkMetalSurfaceCreateInfoEXT structure
- VUID-vkCreateMetalSurfaceEXT-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateMetalSurfaceEXT-pSurface-parameter
  `pSurface` **must** be a valid pointer to a VkSurfaceKHR handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_NATIVE_WINDOW_IN_USE_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkMetalSurfaceCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_metal_surface
typedef struct VkMetalSurfaceCreateInfoEXT {
    VkStructureType                 sType;
    const void*                     pNext;
    VkMetalSurfaceCreateFlagsEXT    flags;
    const CAMetalLayer*             pLayer;
} VkMetalSurfaceCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `pLayer` is a reference to a CAMetalLayer object representing a renderable surface.

Valid Usage (Implicit)

- VUID-VkMetalSurfaceCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_METAL_SURFACE_CREATE_INFO_EXT
- VUID-VkMetalSurfaceCreateInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkMetalSurfaceCreateInfoEXT-flags-zerobitmask
  `flags` **must** be `0`

To remove an unnecessary compile time dependency, an incomplete type definition of CAMetalLayer is provided in the Vulkan headers:

```
// Provided by VK_EXT_metal_surface
#ifdef __OBJC__
@class CAMetalLayer;
#else
typedef void CAMetalLayer;
#endif
```

The actual CAMetalLayer type is defined in the QuartzCore framework.

```
// Provided by VK_EXT_metal_surface
typedef VkFlags VkMetalSurfaceCreateFlagsEXT;
```

`VkMetalSurfaceCreateFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

37.2.8. Platform-Independent Information

Once created, `VkSurfaceKHR` objects **can** be used in this and other extensions, in particular the `VK_KHR_swapchain` extension.

Several WSI functions return VK_ERROR_SURFACE_LOST_KHR if the surface becomes no longer available. After such an error, the surface (and any child swapchain, if one exists) **should** be destroyed, as there is no way to restore them to a not-lost state. Applications **may** attempt to create a new `VkSurfaceKHR` using the same native platform window object, but whether such re-creation will succeed is platform-dependent and **may** depend on the reason the surface became unavailable. A lost surface does not otherwise cause devices to be lost.

To destroy a `VkSurfaceKHR` object, call:

```
// Provided by VK_KHR_surface
void vkDestroySurfaceKHR(
    VkInstance                                  instance,
    VkSurfaceKHR                                surface,
    const VkAllocationCallbacks*                pAllocator);
```

- `instance` is the instance used to create the surface.
- `surface` is the surface to destroy.
- `pAllocator` is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see Memory Allocation).

Destroying a `VkSurfaceKHR` merely severs the connection between Vulkan and the native surface, and does not imply destroying the native surface, closing a window, or similar behavior.

Valid Usage

- VUID-vkDestroySurfaceKHR-surface-01266
  All `VkSwapchainKHR` objects created for `surface` **must** have been destroyed prior to destroying `surface`
- VUID-vkDestroySurfaceKHR-surface-01267
  If `VkAllocationCallbacks` were provided when `surface` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroySurfaceKHR-surface-01268
  If no `VkAllocationCallbacks` were provided when `surface` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroySurfaceKHR-instance-parameter
  `instance` **must** be a valid VkInstance handle
- VUID-vkDestroySurfaceKHR-surface-parameter
  If `surface` is not VK_NULL_HANDLE, `surface` **must** be a valid VkSurfaceKHR handle
- VUID-vkDestroySurfaceKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroySurfaceKHR-surface-parent
  If `surface` is a valid handle, it **must** have been created, allocated, or retrieved from `instance`

Host Synchronization

- Host access to `surface` **must** be externally synchronized

37.3. Presenting Directly to Display Devices

In some environments applications **can** also present Vulkan rendering directly to display devices without using an intermediate windowing system. This **can** be useful for embedded applications, or implementing the rendering/presentation backend of a windowing system using Vulkan. The VK_KHR_display extension provides the functionality necessary to enumerate display devices and create VkSurfaceKHR objects that target displays.

37.3.1. Display Enumeration

Displays are represented by VkDisplayKHR handles:

```
// Provided by VK_KHR_display
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDisplayKHR)
```

Various functions are provided for enumerating the available display devices present on a Vulkan physical device. To query information about the available displays, call:

```
// Provided by VK_KHR_display
VkResult vkGetPhysicalDeviceDisplayPropertiesKHR(
    VkPhysicalDevice                            physicalDevice,
    uint32_t*                                   pPropertyCount,
    VkDisplayPropertiesKHR*                     pProperties);
```

- physicalDevice is a physical device.
- pPropertyCount is a pointer to an integer related to the number of display devices available or queried, as described below.
- pProperties is either NULL or a pointer to an array of VkDisplayPropertiesKHR structures.

If pProperties is NULL, then the number of display devices available for physicalDevice is returned in pPropertyCount. Otherwise, pPropertyCount **must** point to a variable set by the application to the number of elements in the pProperties array, and on return the variable is overwritten with the number of structures actually written to pProperties. If the value of pPropertyCount is less than the number of display devices for physicalDevice, at most pPropertyCount structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available properties were returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceDisplayPropertiesKHR-physicalDevice-parameter
  physicalDevice **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceDisplayPropertiesKHR-pPropertyCount-parameter
  pPropertyCount **must** be a valid pointer to a uint32_t value
- VUID-vkGetPhysicalDeviceDisplayPropertiesKHR-pProperties-parameter

If the value referenced by pPropertyCount is not 0, and pProperties is not NULL, pProperties **must** be a valid pointer to an array of pPropertyCount VkDisplayPropertiesKHR structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkDisplayPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_display
typedef struct VkDisplayPropertiesKHR {
    VkDisplayKHR                  display;
    const char*                   displayName;
    VkExtent2D                    physicalDimensions;
    VkExtent2D                    physicalResolution;
    VkSurfaceTransformFlagsKHR    supportedTransforms;
    VkBool32                      planeReorderPossible;
    VkBool32                      persistentContent;
} VkDisplayPropertiesKHR;
```

- display is a handle that is used to refer to the display described here. This handle will be valid for the lifetime of the Vulkan instance.
- displayName is NULL or a pointer to a null-terminated UTF-8 string containing the name of the display. Generally, this will be the name provided by the display's EDID. If NULL, no suitable name is available. If not NULL, the string pointed to **must** remain accessible and unmodified as long as display is valid.
- physicalDimensions describes the physical width and height of the visible portion of the display, in millimeters.
- physicalResolution describes the physical, native, or preferred resolution of the display.

NOTE For devices which have no natural value to return here, implementations **should** return the maximum resolution supported.

- supportedTransforms is a bitmask of VkSurfaceTransformFlagBitsKHR describing which transforms are supported by this display.

- `planeReorderPossible` tells whether the planes on this display **can** have their z order changed. If this is VK_TRUE, the application **can** re-arrange the planes on this display in any order relative to each other.
- `persistentContent` tells whether the display supports self-refresh/internal buffering. If this is true, the application **can** submit persistent present operations on swapchains created against this display.

NOTE	Persistent presents **may** have higher latency, and **may** use less power when the screen content is updated infrequently, or when only a portion of the screen needs to be updated in most frames.

To query information about the available displays, call:

```
// Provided by VK_KHR_get_display_properties2
VkResult vkGetPhysicalDeviceDisplayProperties2KHR(
    VkPhysicalDevice                            physicalDevice,
    uint32_t*                                   pPropertyCount,
    VkDisplayProperties2KHR*                    pProperties);
```

- `physicalDevice` is a physical device.
- `pPropertyCount` is a pointer to an integer related to the number of display devices available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of `VkDisplayProperties2KHR` structures.

`vkGetPhysicalDeviceDisplayProperties2KHR` behaves similarly to vkGetPhysicalDeviceDisplayPropertiesKHR, with the ability to return extended information via chained output structures.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceDisplayProperties2KHR-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceDisplayProperties2KHR-pPropertyCount-parameter
  `pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceDisplayProperties2KHR-pProperties-parameter
  If the value referenced by `pPropertyCount` is not `0`, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` VkDisplayProperties2KHR structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkDisplayProperties2KHR structure is defined as:

```
// Provided by VK_KHR_get_display_properties2
typedef struct VkDisplayProperties2KHR {
    VkStructureType           sType;
    void*                     pNext;
    VkDisplayPropertiesKHR    displayProperties;
} VkDisplayProperties2KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- displayProperties is a VkDisplayPropertiesKHR structure.

Valid Usage (Implicit)

- VUID-VkDisplayProperties2KHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DISPLAY_PROPERTIES_2_KHR
- VUID-VkDisplayProperties2KHR-pNext-pNext
  pNext **must** be NULL

Acquiring and Releasing Displays

On some platforms, access to displays is limited to a single process or native driver instance. On such platforms, some or all of the displays may not be available to Vulkan if they are already in use by a native windowing system or other application.

To acquire permission to directly access a display in Vulkan from an X11 server, call:

```
// Provided by VK_EXT_acquire_xlib_display
VkResult vkAcquireXlibDisplayEXT(
    VkPhysicalDevice                            physicalDevice,
    Display*                                    dpy,
    VkDisplayKHR                                display);
```

- physicalDevice is the physical device the display is on.
- dpy is a connection to the X11 server that currently owns display.

- `display` is the display the caller wishes to control in Vulkan.

All permissions necessary to control the display are granted to the Vulkan instance associated with `physicalDevice` until the display is released or the X11 connection specified by `dpy` is terminated. Permission to access the display **may** be temporarily revoked during periods when the X11 server from which control was acquired itself loses access to `display`. During such periods, operations which require access to the display **must** fail with an appropriate error code. If the X11 server associated with `dpy` does not own `display`, or if permission to access it has already been acquired by another entity, the call **must** return the error code VK_ERROR_INITIALIZATION_FAILED.

NOTE	One example of when an X11 server loses access to a display is when it loses ownership of its virtual terminal.

Valid Usage (Implicit)

- VUID-vkAcquireXlibDisplayEXT-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkAcquireXlibDisplayEXT-dpy-parameter
  `dpy` **must** be a valid pointer to a `Display` value
- VUID-vkAcquireXlibDisplayEXT-display-parameter
  `display` **must** be a valid VkDisplayKHR handle
- VUID-vkAcquireXlibDisplayEXT-display-parent
  `display` **must** have been created, allocated, or retrieved from `physicalDevice`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

When acquiring displays from an X11 server, an application may also wish to enumerate and identify them using a native handle rather than a `VkDisplayKHR` handle. To determine the `VkDisplayKHR` handle corresponding to an X11 RandR Output, call:

```
// Provided by VK_EXT_acquire_xlib_display
VkResult vkGetRandROutputDisplayEXT(
    VkPhysicalDevice                            physicalDevice,
    Display*                                    dpy,
    RROutput                                    rrOutput,
    VkDisplayKHR*                               pDisplay);
```

- `physicalDevice` is the physical device to query the display handle on.
- `dpy` is a connection to the X11 server from which `rrOutput` was queried.
- `rrOutput` is an X11 RandR output ID.
- `pDisplay` is a pointer to a VkDisplayKHR handle where the display is returned.

If there is no VkDisplayKHR corresponding to `rrOutput` on `physicalDevice`, VK_NULL_HANDLE will be returned in `pDisplay`.

Valid Usage (Implicit)

- VUID-vkGetRandROutputDisplayEXT-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetRandROutputDisplayEXT-dpy-parameter
  `dpy` **must** be a valid pointer to a `Display` value
- VUID-vkGetRandROutputDisplayEXT-pDisplay-parameter
  `pDisplay` **must** be a valid pointer to a VkDisplayKHR handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To acquire permission to directly control a display in Vulkan from the Direct Rendering Manager (DRM) interface, call:

```
// Provided by VK_EXT_acquire_drm_display
VkResult vkAcquireDrmDisplayEXT(
    VkPhysicalDevice                            physicalDevice,
    int32_t                                     drmFd,
    VkDisplayKHR                                display);
```

- `physicalDevice` is the physical device the display is on.
- `drmFd` is the DRM primary file descriptor.
- `display` is the display the caller wishes Vulkan to control.

All permissions necessary to control the display are granted to the Vulkan instance associated with the provided `physicalDevice` until the display is either released or the connector is unplugged. The provided `drmFd` **must** correspond to the one owned by the `physicalDevice`. If not, the error code VK_ERROR_UNKNOWN **must** be returned. The DRM FD must have DRM master permissions. If any error is encountered during the acquisition of the display, the call **must** return the error code VK_ERROR_INITIALIZATION_FAILED.

The provided DRM fd should not be closed before the display is released, attempting to do it may result in undefined behavior.

Valid Usage (Implicit)

- VUID-vkAcquireDrmDisplayEXT-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkAcquireDrmDisplayEXT-display-parameter
  `display` **must** be a valid VkDisplayKHR handle
- VUID-vkAcquireDrmDisplayEXT-display-parent
  `display` **must** have been created, allocated, or retrieved from `physicalDevice`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Before acquiring a display from the DRM interface, the caller may want to select a specific `VkDisplayKHR` handle by identifying it using a `connectorId`. To do so, call:

```
// Provided by VK_EXT_acquire_drm_display
VkResult vkGetDrmDisplayEXT(
    VkPhysicalDevice                            physicalDevice,
    int32_t                                     drmFd,
    uint32_t                                    connectorId,
    VkDisplayKHR*                               display);
```

- `physicalDevice` is the physical device to query the display from.

- drmFd is the DRM primary file descriptor.
- connectorId is the identifier of the specified DRM connector.
- pDisplay is a pointer to a VkDisplayKHR handle where the display is returned.

If there is no VkDisplayKHR corresponding to the connectorId on the physicalDevice, the returned display will be VK_NULL_HANDLE. The provided drmFd **must** correspond to the one owned by the physicalDevice. If not, the error code VK_ERROR_UNKNOWN will be returned. Master permissions are not required, because the file descriptor is just used for information gathering purposes. The given connectorId **must** be a resource owned by the provided drmFd. If not, the error code VK_ERROR_UNKNOWN will be returned. If any error is encountered during the identification of the display, the call **must** return the error code VK_ERROR_INITIALIZATION_FAILED.

Valid Usage (Implicit)

- VUID-vkGetDrmDisplayEXT-physicalDevice-parameter
  physicalDevice **must** be a valid VkPhysicalDevice handle
- VUID-vkGetDrmDisplayEXT-display-parameter
  display **must** be a valid pointer to a VkDisplayKHR handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To release a previously acquired display, call:

```
// Provided by VK_EXT_direct_mode_display
VkResult vkReleaseDisplayEXT(
    VkPhysicalDevice                            physicalDevice,
    VkDisplayKHR                                display);
```

- physicalDevice is the physical device the display is on.
- display is the display to release control of.

Valid Usage (Implicit)

- VUID-vkReleaseDisplayEXT-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkReleaseDisplayEXT-display-parameter
  `display` **must** be a valid VkDisplayKHR handle
- VUID-vkReleaseDisplayEXT-display-parent
  `display` **must** have been created, allocated, or retrieved from `physicalDevice`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Display Planes

Images are presented to individual planes on a display. Devices **must** support at least one plane on each display. Planes **can** be stacked and blended to composite multiple images on one display. Devices **may** support only a fixed stacking order and fixed mapping between planes and displays, or they **may** allow arbitrary application-specified stacking orders and mappings between planes and displays. To query the properties of device display planes, call:

```
// Provided by VK_KHR_display
VkResult vkGetPhysicalDeviceDisplayPlanePropertiesKHR(
    VkPhysicalDevice                            physicalDevice,
    uint32_t*                                   pPropertyCount,
    VkDisplayPlanePropertiesKHR*                pProperties);
```

- `physicalDevice` is a physical device.
- `pPropertyCount` is a pointer to an integer related to the number of display planes available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of `VkDisplayPlanePropertiesKHR` structures.

If `pProperties` is `NULL`, then the number of display planes available for `physicalDevice` is returned in `pPropertyCount`. Otherwise, `pPropertyCount` **must** point to a variable set by the application to the number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If the value of `pPropertyCount` is less than the number of display planes for `physicalDevice`, at most `pPropertyCount` structures will be written.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceDisplayPlanePropertiesKHR-physicalDevice-parameter

physicalDevice **must** be a valid VkPhysicalDevice handle

- VUID-vkGetPhysicalDeviceDisplayPlanePropertiesKHR-pPropertyCount-parameter
  pPropertyCount **must** be a valid pointer to a uint32_t value
- VUID-vkGetPhysicalDeviceDisplayPlanePropertiesKHR-pProperties-parameter
  If the value referenced by pPropertyCount is not 0, and pProperties is not NULL, pProperties **must** be a valid pointer to an array of pPropertyCount VkDisplayPlanePropertiesKHR structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkDisplayPlanePropertiesKHR structure is defined as:

```
// Provided by VK_KHR_display
typedef struct VkDisplayPlanePropertiesKHR {
    VkDisplayKHR    currentDisplay;
    uint32_t        currentStackIndex;
} VkDisplayPlanePropertiesKHR;
```

- currentDisplay is the handle of the display the plane is currently associated with. If the plane is not currently attached to any displays, this will be VK_NULL_HANDLE.
- currentStackIndex is the current z-order of the plane. This will be between 0 and the value returned by vkGetPhysicalDeviceDisplayPlanePropertiesKHR in pPropertyCount.

To query the properties of a device's display planes, call:

```
// Provided by VK_KHR_get_display_properties2
VkResult vkGetPhysicalDeviceDisplayPlaneProperties2KHR(
    VkPhysicalDevice                            physicalDevice,
    uint32_t*                                   pPropertyCount,
    VkDisplayPlaneProperties2KHR*               pProperties);
```

- physicalDevice is a physical device.

- pPropertyCount is a pointer to an integer related to the number of display planes available or queried, as described below.
- pProperties is either NULL or a pointer to an array of VkDisplayPlaneProperties2KHR structures.

vkGetPhysicalDeviceDisplayPlaneProperties2KHR behaves similarly to vkGetPhysicalDeviceDisplayPlanePropertiesKHR, with the ability to return extended information via chained output structures.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceDisplayPlaneProperties2KHR-physicalDevice-parameter
  physicalDevice **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceDisplayPlaneProperties2KHR-pPropertyCount-parameter
  pPropertyCount **must** be a valid pointer to a uint32_t value
- VUID-vkGetPhysicalDeviceDisplayPlaneProperties2KHR-pProperties-parameter
  If the value referenced by pPropertyCount is not 0, and pProperties is not NULL, pProperties **must** be a valid pointer to an array of pPropertyCount VkDisplayPlaneProperties2KHR structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkDisplayPlaneProperties2KHR structure is defined as:

```
// Provided by VK_KHR_get_display_properties2
typedef struct VkDisplayPlaneProperties2KHR {
    VkStructureType                sType;
    void*                          pNext;
    VkDisplayPlanePropertiesKHR    displayPlaneProperties;
} VkDisplayPlaneProperties2KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.

- `displayPlaneProperties` is a VkDisplayPlanePropertiesKHR structure.

Valid Usage (Implicit)

- VUID-VkDisplayPlaneProperties2KHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DISPLAY_PLANE_PROPERTIES_2_KHR
- VUID-VkDisplayPlaneProperties2KHR-pNext-pNext
  `pNext` **must** be `NULL`

To determine which displays a plane is usable with, call:

```
// Provided by VK_KHR_display
VkResult vkGetDisplayPlaneSupportedDisplaysKHR(
    VkPhysicalDevice                            physicalDevice,
    uint32_t                                    planeIndex,
    uint32_t*                                   pDisplayCount,
    VkDisplayKHR*                               pDisplays);
```

- `physicalDevice` is a physical device.
- `planeIndex` is the plane which the application wishes to use, and **must** be in the range [0, physical device plane count - 1].
- `pDisplayCount` is a pointer to an integer related to the number of displays available or queried, as described below.
- `pDisplays` is either `NULL` or a pointer to an array of `VkDisplayKHR` handles.

If `pDisplays` is `NULL`, then the number of displays usable with the specified `planeIndex` for `physicalDevice` is returned in `pDisplayCount`. Otherwise, `pDisplayCount` **must** point to a variable set by the application to the number of elements in the `pDisplays` array, and on return the variable is overwritten with the number of handles actually written to `pDisplays`. If the value of `pDisplayCount` is less than the number of usable display-plane pairs for `physicalDevice`, at most `pDisplayCount` handles will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available pairs were returned.

Valid Usage

- VUID-vkGetDisplayPlaneSupportedDisplaysKHR-planeIndex-01249
  `planeIndex` **must** be less than the number of display planes supported by the device as determined by calling `vkGetPhysicalDeviceDisplayPlanePropertiesKHR`

Valid Usage (Implicit)

- VUID-vkGetDisplayPlaneSupportedDisplaysKHR-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle

- VUID-vkGetDisplayPlaneSupportedDisplaysKHR-pDisplayCount-parameter
  `pDisplayCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetDisplayPlaneSupportedDisplaysKHR-pDisplays-parameter
  If the value referenced by `pDisplayCount` is not `0`, and `pDisplays` is not `NULL`, `pDisplays` **must** be a valid pointer to an array of `pDisplayCount` VkDisplayKHR handles

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Additional properties of displays are queried using specialized query functions.

Display Modes

Display modes are represented by `VkDisplayModeKHR` handles:

```
// Provided by VK_KHR_display
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDisplayModeKHR)
```

Each display has one or more supported modes associated with it by default. These built-in modes are queried by calling:

```
// Provided by VK_KHR_display
VkResult vkGetDisplayModePropertiesKHR(
    VkPhysicalDevice                            physicalDevice,
    VkDisplayKHR                                display,
    uint32_t*                                   pPropertyCount,
    VkDisplayModePropertiesKHR*                 pProperties);
```

- `physicalDevice` is the physical device associated with `display`.
- `display` is the display to query.
- `pPropertyCount` is a pointer to an integer related to the number of display modes available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of `VkDisplayModePropertiesKHR` structures.

If pProperties is NULL, then the number of display modes available on the specified display for physicalDevice is returned in pPropertyCount. Otherwise, pPropertyCount **must** point to a variable set by the application to the number of elements in the pProperties array, and on return the variable is overwritten with the number of structures actually written to pProperties. If the value of pPropertyCount is less than the number of display modes for physicalDevice, at most pPropertyCount structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available display modes were returned.

Valid Usage (Implicit)

- VUID-vkGetDisplayModePropertiesKHR-physicalDevice-parameter
  physicalDevice **must** be a valid VkPhysicalDevice handle
- VUID-vkGetDisplayModePropertiesKHR-display-parameter
  display **must** be a valid VkDisplayKHR handle
- VUID-vkGetDisplayModePropertiesKHR-pPropertyCount-parameter
  pPropertyCount **must** be a valid pointer to a uint32_t value
- VUID-vkGetDisplayModePropertiesKHR-pProperties-parameter
  If the value referenced by pPropertyCount is not 0, and pProperties is not NULL, pProperties **must** be a valid pointer to an array of pPropertyCount VkDisplayModePropertiesKHR structures
- VUID-vkGetDisplayModePropertiesKHR-display-parent
  display **must** have been created, allocated, or retrieved from physicalDevice

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkDisplayModePropertiesKHR structure is defined as:

```
// Provided by VK_KHR_display
typedef struct VkDisplayModePropertiesKHR {
    VkDisplayModeKHR              displayMode;
    VkDisplayModeParametersKHR    parameters;
} VkDisplayModePropertiesKHR;
```

- `displayMode` is a handle to the display mode described in this structure. This handle will be valid for the lifetime of the Vulkan instance.
- `parameters` is a VkDisplayModeParametersKHR structure describing the display parameters associated with `displayMode`.

```
// Provided by VK_KHR_display
typedef VkFlags VkDisplayModeCreateFlagsKHR;
```

`VkDisplayModeCreateFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

To query the properties of a device's built-in display modes, call:

```
// Provided by VK_KHR_get_display_properties2
VkResult vkGetDisplayModeProperties2KHR(
    VkPhysicalDevice                            physicalDevice,
    VkDisplayKHR                                display,
    uint32_t*                                   pPropertyCount,
    VkDisplayModeProperties2KHR*                pProperties);
```

- `physicalDevice` is the physical device associated with `display`.
- `display` is the display to query.
- `pPropertyCount` is a pointer to an integer related to the number of display modes available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of `VkDisplayModeProperties2KHR` structures.

`vkGetDisplayModeProperties2KHR` behaves similarly to vkGetDisplayModePropertiesKHR, with the ability to return extended information via chained output structures.

Valid Usage (Implicit)

- VUID-vkGetDisplayModeProperties2KHR-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetDisplayModeProperties2KHR-display-parameter
  `display` **must** be a valid VkDisplayKHR handle
- VUID-vkGetDisplayModeProperties2KHR-pPropertyCount-parameter
  `pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetDisplayModeProperties2KHR-pProperties-parameter
  If the value referenced by `pPropertyCount` is not `0`, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` VkDisplayModeProperties2KHR structures
- VUID-vkGetDisplayModeProperties2KHR-display-parent
  `display` **must** have been created, allocated, or retrieved from `physicalDevice`

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkDisplayModeProperties2KHR structure is defined as:

```
// Provided by VK_KHR_get_display_properties2
typedef struct VkDisplayModeProperties2KHR {
    VkStructureType               sType;
    void*                         pNext;
    VkDisplayModePropertiesKHR    displayModeProperties;
} VkDisplayModeProperties2KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- displayModeProperties is a VkDisplayModePropertiesKHR structure.

Valid Usage (Implicit)

- VUID-VkDisplayModeProperties2KHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DISPLAY_MODE_PROPERTIES_2_KHR
- VUID-VkDisplayModeProperties2KHR-pNext-pNext
  pNext **must** be NULL

The VkDisplayModeParametersKHR structure is defined as:

```
// Provided by VK_KHR_display
typedef struct VkDisplayModeParametersKHR {
    VkExtent2D    visibleRegion;
    uint32_t      refreshRate;
} VkDisplayModeParametersKHR;
```

- visibleRegion is the 2D extents of the visible region.
- refreshRate is a uint32_t that is the number of times the display is refreshed each second

multiplied by 1000.

NOTE For example, a 60Hz display mode would report a `refreshRate` of 60,000.

Valid Usage

- VUID-VkDisplayModeParametersKHR-width-01990
  The `width` member of `visibleRegion` **must** be greater than `0`
- VUID-VkDisplayModeParametersKHR-height-01991
  The `height` member of `visibleRegion` **must** be greater than `0`
- VUID-VkDisplayModeParametersKHR-refreshRate-01992
  `refreshRate` **must** be greater than `0`

Additional modes **may** also be created by calling:

```
// Provided by VK_KHR_display
VkResult vkCreateDisplayModeKHR(
    VkPhysicalDevice                            physicalDevice,
    VkDisplayKHR                                display,
    const VkDisplayModeCreateInfoKHR*           pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkDisplayModeKHR*                           pMode);
```

- `physicalDevice` is the physical device associated with `display`.
- `display` is the display to create an additional mode for.
- `pCreateInfo` is a pointer to a VkDisplayModeCreateInfoKHR structure describing the new mode to create.
- `pAllocator` is the allocator used for host memory allocated for the display mode object when there is no more specific allocator available (see Memory Allocation).
- `pMode` is a pointer to a VkDisplayModeKHR handle in which the mode created is returned.

Valid Usage (Implicit)

- VUID-vkCreateDisplayModeKHR-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkCreateDisplayModeKHR-display-parameter
  `display` **must** be a valid VkDisplayKHR handle
- VUID-vkCreateDisplayModeKHR-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkDisplayModeCreateInfoKHR structure
- VUID-vkCreateDisplayModeKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure

- VUID-vkCreateDisplayModeKHR-pMode-parameter
  `pMode` **must** be a valid pointer to a VkDisplayModeKHR handle
- VUID-vkCreateDisplayModeKHR-display-parent
  `display` **must** have been created, allocated, or retrieved from `physicalDevice`

Host Synchronization

- Host access to `display` **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkDisplayModeCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_display
typedef struct VkDisplayModeCreateInfoKHR {
    VkStructureType                sType;
    const void*                    pNext;
    VkDisplayModeCreateFlagsKHR    flags;
    VkDisplayModeParametersKHR     parameters;
} VkDisplayModeCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use, and **must** be zero.
- `parameters` is a VkDisplayModeParametersKHR structure describing the display parameters to use in creating the new mode. If the parameters are not compatible with the specified display, the implementation **must** return VK_ERROR_INITIALIZATION_FAILED.

Valid Usage (Implicit)

- VUID-VkDisplayModeCreateInfoKHR-sType-sType

`sType` **must** be VK_STRUCTURE_TYPE_DISPLAY_MODE_CREATE_INFO_KHR

- VUID-VkDisplayModeCreateInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDisplayModeCreateInfoKHR-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkDisplayModeCreateInfoKHR-parameters-parameter
  `parameters` **must** be a valid VkDisplayModeParametersKHR structure

Applications that wish to present directly to a display **must** select which layer, or "plane" of the display they wish to target, and a mode to use with the display. Each display supports at least one plane. The capabilities of a given mode and plane combination are determined by calling:

```
// Provided by VK_KHR_display
VkResult vkGetDisplayPlaneCapabilitiesKHR(
    VkPhysicalDevice                            physicalDevice,
    VkDisplayModeKHR                            mode,
    uint32_t                                    planeIndex,
    VkDisplayPlaneCapabilitiesKHR*              pCapabilities);
```

- `physicalDevice` is the physical device associated with the display specified by `mode`.
- `mode` is the display mode the application intends to program when using the specified plane. Note this parameter also implicitly specifies a display.
- `planeIndex` is the plane which the application intends to use with the display, and is less than the number of display planes supported by the device.
- `pCapabilities` is a pointer to a VkDisplayPlaneCapabilitiesKHR structure in which the capabilities are returned.

Valid Usage (Implicit)

- VUID-vkGetDisplayPlaneCapabilitiesKHR-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetDisplayPlaneCapabilitiesKHR-mode-parameter
  `mode` **must** be a valid VkDisplayModeKHR handle
- VUID-vkGetDisplayPlaneCapabilitiesKHR-pCapabilities-parameter
  `pCapabilities` **must** be a valid pointer to a VkDisplayPlaneCapabilitiesKHR structure
- VUID-vkGetDisplayPlaneCapabilitiesKHR-mode-parent
  `mode` **must** have been created, allocated, or retrieved from `physicalDevice`

Host Synchronization

- Host access to `mode` **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkDisplayPlaneCapabilitiesKHR` structure is defined as:

```
// Provided by VK_KHR_display
typedef struct VkDisplayPlaneCapabilitiesKHR {
    VkDisplayPlaneAlphaFlagsKHR    supportedAlpha;
    VkOffset2D                     minSrcPosition;
    VkOffset2D                     maxSrcPosition;
    VkExtent2D                     minSrcExtent;
    VkExtent2D                     maxSrcExtent;
    VkOffset2D                     minDstPosition;
    VkOffset2D                     maxDstPosition;
    VkExtent2D                     minDstExtent;
    VkExtent2D                     maxDstExtent;
} VkDisplayPlaneCapabilitiesKHR;
```

- `supportedAlpha` is a bitmask of VkDisplayPlaneAlphaFlagBitsKHR describing the supported alpha blending modes.
- `minSrcPosition` is the minimum source rectangle offset supported by this plane using the specified mode.
- `maxSrcPosition` is the maximum source rectangle offset supported by this plane using the specified mode. The `x` and `y` components of `maxSrcPosition` **must** each be greater than or equal to the `x` and `y` components of `minSrcPosition`, respectively.
- `minSrcExtent` is the minimum source rectangle size supported by this plane using the specified mode.
- `maxSrcExtent` is the maximum source rectangle size supported by this plane using the specified mode.
- `minDstPosition`, `maxDstPosition`, `minDstExtent`, `maxDstExtent` all have similar semantics to their corresponding `*Src*` equivalents, but apply to the output region within the mode rather than the input region within the source image. Unlike the `*Src*` offsets, `minDstPosition` and `maxDstPosition` **may** contain negative values.

The minimum and maximum position and extent fields describe the implementation limits, if any, as they apply to the specified display mode and plane. Vendors **may** support displaying a subset of a

swapchain's presentable images on the specified display plane. This is expressed by returning `minSrcPosition`, `maxSrcPosition`, `minSrcExtent`, and `maxSrcExtent` values that indicate a range of possible positions and sizes which **may** be used to specify the region within the presentable images that source pixels will be read from when creating a swapchain on the specified display mode and plane.

Vendors **may** also support mapping the presentable images' content to a subset or superset of the visible region in the specified display mode. This is expressed by returning `minDstPosition`, `maxDstPosition`, `minDstExtent` and `maxDstExtent` values that indicate a range of possible positions and sizes which **may** be used to describe the region within the display mode that the source pixels will be mapped to.

Other vendors **may** support only a 1-1 mapping between pixels in the presentable images and the display mode. This **may** be indicated by returning (0,0) for `minSrcPosition`, `maxSrcPosition`, `minDstPosition`, and `maxDstPosition`, and (display mode width, display mode height) for `minSrcExtent`, `maxSrcExtent`, `minDstExtent`, and `maxDstExtent`.

The value `supportedAlpha` **must** contain at least one valid VkDisplayPlaneAlphaFlagBitsKHR bit.

These values indicate the limits of the implementation's individual fields. Not all combinations of values within the offset and extent ranges returned in `VkDisplayPlaneCapabilitiesKHR` are guaranteed to be supported. Presentation requests specifying unsupported combinations **may** fail.

To query the capabilities of a given mode and plane combination, call:

```
// Provided by VK_KHR_get_display_properties2
VkResult vkGetDisplayPlaneCapabilities2KHR(
    VkPhysicalDevice                            physicalDevice,
    const VkDisplayPlaneInfo2KHR*               pDisplayPlaneInfo,
    VkDisplayPlaneCapabilities2KHR*             pCapabilities);
```

- `physicalDevice` is the physical device associated with `pDisplayPlaneInfo`.
- `pDisplayPlaneInfo` is a pointer to a VkDisplayPlaneInfo2KHR structure describing the plane and mode.
- `pCapabilities` is a pointer to a VkDisplayPlaneCapabilities2KHR structure in which the capabilities are returned.

`vkGetDisplayPlaneCapabilities2KHR` behaves similarly to vkGetDisplayPlaneCapabilitiesKHR, with the ability to specify extended inputs via chained input structures, and to return extended information via chained output structures.

Valid Usage (Implicit)

- VUID-vkGetDisplayPlaneCapabilities2KHR-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetDisplayPlaneCapabilities2KHR-pDisplayPlaneInfo-parameter
  `pDisplayPlaneInfo` **must** be a valid pointer to a valid VkDisplayPlaneInfo2KHR structure

- VUID-vkGetDisplayPlaneCapabilities2KHR-pCapabilities-parameter
  `pCapabilities` **must** be a valid pointer to a VkDisplayPlaneCapabilities2KHR structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkDisplayPlaneInfo2KHR` structure is defined as:

```
// Provided by VK_KHR_get_display_properties2
typedef struct VkDisplayPlaneInfo2KHR {
    VkStructureType     sType;
    const void*         pNext;
    VkDisplayModeKHR    mode;
    uint32_t            planeIndex;
} VkDisplayPlaneInfo2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `mode` is the display mode the application intends to program when using the specified plane.

> **NOTE** This parameter also implicitly specifies a display.

- `planeIndex` is the plane which the application intends to use with the display.

The members of `VkDisplayPlaneInfo2KHR` correspond to the arguments to vkGetDisplayPlaneCapabilitiesKHR, with `sType` and `pNext` added for extensibility.

Valid Usage (Implicit)

- VUID-VkDisplayPlaneInfo2KHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DISPLAY_PLANE_INFO_2_KHR
- VUID-VkDisplayPlaneInfo2KHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDisplayPlaneInfo2KHR-mode-parameter
  `mode` **must** be a valid VkDisplayModeKHR handle

Host Synchronization

- Host access to `mode` **must** be externally synchronized

The `VkDisplayPlaneCapabilities2KHR` structure is defined as:

```
// Provided by VK_KHR_get_display_properties2
typedef struct VkDisplayPlaneCapabilities2KHR {
    VkStructureType                 sType;
    void*                           pNext;
    VkDisplayPlaneCapabilitiesKHR   capabilities;
} VkDisplayPlaneCapabilities2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `capabilities` is a VkDisplayPlaneCapabilitiesKHR structure.

Valid Usage (Implicit)

- VUID-VkDisplayPlaneCapabilities2KHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DISPLAY_PLANE_CAPABILITIES_2_KHR
- VUID-VkDisplayPlaneCapabilities2KHR-pNext-pNext
  `pNext` **must** be `NULL`

37.3.2. Display Control

To set the power state of a display, call:

```
// Provided by VK_EXT_display_control
VkResult vkDisplayPowerControlEXT(
    VkDevice                                    device,
    VkDisplayKHR                                display,
    const VkDisplayPowerInfoEXT*                pDisplayPowerInfo);
```

- `device` is a logical device associated with `display`.
- `display` is the display whose power state is modified.
- `pDisplayPowerInfo` is a pointer to a VkDisplayPowerInfoEXT structure specifying the new power state of `display`.

Valid Usage (Implicit)

- VUID-vkDisplayPowerControlEXT-device-parameter

`device` **must** be a valid VkDevice handle

- VUID-vkDisplayPowerControlEXT-display-parameter
  `display` **must** be a valid VkDisplayKHR handle
- VUID-vkDisplayPowerControlEXT-pDisplayPowerInfo-parameter
  `pDisplayPowerInfo` **must** be a valid pointer to a valid VkDisplayPowerInfoEXT structure
- VUID-vkDisplayPowerControlEXT-commonparent
  Both of `device`, and `display` **must** have been created, allocated, or retrieved from the same VkPhysicalDevice

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkDisplayPowerInfoEXT` structure is defined as:

```
// Provided by VK_EXT_display_control
typedef struct VkDisplayPowerInfoEXT {
    VkStructureType           sType;
    const void*               pNext;
    VkDisplayPowerStateEXT    powerState;
} VkDisplayPowerInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `powerState` is a VkDisplayPowerStateEXT value specifying the new power state of the display.

Valid Usage (Implicit)

- VUID-VkDisplayPowerInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DISPLAY_POWER_INFO_EXT
- VUID-VkDisplayPowerInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDisplayPowerInfoEXT-powerState-parameter
  `powerState` **must** be a valid VkDisplayPowerStateEXT value

Possible values of VkDisplayPowerInfoEXT::powerState, specifying the new power state of a display, are:

```
// Provided by VK_EXT_display_control
typedef enum VkDisplayPowerStateEXT {
    VK_DISPLAY_POWER_STATE_OFF_EXT = 0,
    VK_DISPLAY_POWER_STATE_SUSPEND_EXT = 1,
    VK_DISPLAY_POWER_STATE_ON_EXT = 2,
} VkDisplayPowerStateEXT;
```

- VK_DISPLAY_POWER_STATE_OFF_EXT specifies that the display is powered down.
- VK_DISPLAY_POWER_STATE_SUSPEND_EXT specifies that the display is put into a low power mode, from which it **may** be able to transition back to VK_DISPLAY_POWER_STATE_ON_EXT more quickly than if it were in VK_DISPLAY_POWER_STATE_OFF_EXT. This state **may** be the same as VK_DISPLAY_POWER_STATE_OFF_EXT.
- VK_DISPLAY_POWER_STATE_ON_EXT specifies that the display is powered on.

37.3.3. Display Surfaces

A complete display configuration includes a mode, one or more display planes and any parameters describing their behavior, and parameters describing some aspects of the images associated with those planes. Display surfaces describe the configuration of a single plane within a complete display configuration. To create a VkSurfaceKHR object for a display plane, call:

```
// Provided by VK_KHR_display
VkResult vkCreateDisplayPlaneSurfaceKHR(
    VkInstance                                  instance,
    const VkDisplaySurfaceCreateInfoKHR*        pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkSurfaceKHR*                               pSurface);
```

- instance is the instance corresponding to the physical device the targeted display is on.
- pCreateInfo is a pointer to a VkDisplaySurfaceCreateInfoKHR structure specifying which mode, plane, and other parameters to use, as described below.
- pAllocator is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see Memory Allocation).
- pSurface is a pointer to a VkSurfaceKHR handle in which the created surface is returned.

Valid Usage (Implicit)

- VUID-vkCreateDisplayPlaneSurfaceKHR-instance-parameter
  instance **must** be a valid VkInstance handle
- VUID-vkCreateDisplayPlaneSurfaceKHR-pCreateInfo-parameter
  pCreateInfo **must** be a valid pointer to a valid VkDisplaySurfaceCreateInfoKHR structure

- VUID-vkCreateDisplayPlaneSurfaceKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateDisplayPlaneSurfaceKHR-pSurface-parameter
  `pSurface` **must** be a valid pointer to a VkSurfaceKHR handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkDisplaySurfaceCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_display
typedef struct VkDisplaySurfaceCreateInfoKHR {
    VkStructureType                    sType;
    const void*                        pNext;
    VkDisplaySurfaceCreateFlagsKHR     flags;
    VkDisplayModeKHR                   displayMode;
    uint32_t                           planeIndex;
    uint32_t                           planeStackIndex;
    VkSurfaceTransformFlagBitsKHR      transform;
    float                              globalAlpha;
    VkDisplayPlaneAlphaFlagBitsKHR     alphaMode;
    VkExtent2D                         imageExtent;
} VkDisplaySurfaceCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use, and **must** be zero.
- `displayMode` is a VkDisplayModeKHR handle specifying the mode to use when displaying this surface.
- `planeIndex` is the plane on which this surface appears.
- `planeStackIndex` is the z-order of the plane.
- `transform` is a VkSurfaceTransformFlagBitsKHR value specifying the transformation to apply to images as part of the scanout operation.

- `globalAlpha` is the global alpha value. This value is ignored if `alphaMode` is not VK_DISPLAY_PLANE_ALPHA_GLOBAL_BIT_KHR.
- `alphaMode` is a VkDisplayPlaneAlphaFlagBitsKHR value specifying the type of alpha blending to use.
- `imageExtent` is the size of the presentable images to use with the surface.

NOTE	Creating a display surface **must** not modify the state of the displays, planes, or other resources it names. For example, it **must** not apply the specified mode to be set on the associated display. Application of display configuration occurs as a side effect of presenting to a display surface.

Valid Usage

- VUID-VkDisplaySurfaceCreateInfoKHR-planeIndex-01252
  `planeIndex` **must** be less than the number of display planes supported by the device as determined by calling `vkGetPhysicalDeviceDisplayPlanePropertiesKHR`
- VUID-VkDisplaySurfaceCreateInfoKHR-planeReorderPossible-01253
  If the `planeReorderPossible` member of the `VkDisplayPropertiesKHR` structure returned by `vkGetPhysicalDeviceDisplayPropertiesKHR` for the display corresponding to `displayMode` is VK_TRUE then `planeStackIndex` **must** be less than the number of display planes supported by the device as determined by calling `vkGetPhysicalDeviceDisplayPlanePropertiesKHR`; otherwise `planeStackIndex` **must** equal the `currentStackIndex` member of `VkDisplayPlanePropertiesKHR` returned by `vkGetPhysicalDeviceDisplayPlanePropertiesKHR` for the display plane corresponding to `displayMode`
- VUID-VkDisplaySurfaceCreateInfoKHR-alphaMode-01254
  If `alphaMode` is VK_DISPLAY_PLANE_ALPHA_GLOBAL_BIT_KHR then `globalAlpha` **must** be between `0` and `1`, inclusive
- VUID-VkDisplaySurfaceCreateInfoKHR-alphaMode-01255
  `alphaMode` **must** be one of the bits present in the `supportedAlpha` member of `VkDisplayPlaneCapabilitiesKHR` for the display plane corresponding to `displayMode`
- VUID-VkDisplaySurfaceCreateInfoKHR-transform-06740
  `transform` **must** be one of the bits present in the `supportedTransforms` member of `VkDisplayPropertiesKHR` for the display corresponding to `displayMode`
- VUID-VkDisplaySurfaceCreateInfoKHR-width-01256
  The `width` and `height` members of `imageExtent` **must** be less than or equal to VkPhysicalDeviceLimits::`maxImageDimension2D`

Valid Usage (Implicit)

- VUID-VkDisplaySurfaceCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DISPLAY_SURFACE_CREATE_INFO_KHR
- VUID-VkDisplaySurfaceCreateInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`

- VUID-VkDisplaySurfaceCreateInfoKHR-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkDisplaySurfaceCreateInfoKHR-displayMode-parameter
  `displayMode` **must** be a valid VkDisplayModeKHR handle
- VUID-VkDisplaySurfaceCreateInfoKHR-transform-parameter
  `transform` **must** be a valid VkSurfaceTransformFlagBitsKHR value
- VUID-VkDisplaySurfaceCreateInfoKHR-alphaMode-parameter
  `alphaMode` **must** be a valid VkDisplayPlaneAlphaFlagBitsKHR value

```
// Provided by VK_KHR_display
typedef VkFlags VkDisplaySurfaceCreateFlagsKHR;
```

`VkDisplaySurfaceCreateFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

Bits which **can** be set in VkDisplaySurfaceCreateInfoKHR::`alphaMode`, specifying the type of alpha blending to use on a display, are:

```
// Provided by VK_KHR_display
typedef enum VkDisplayPlaneAlphaFlagBitsKHR {
    VK_DISPLAY_PLANE_ALPHA_OPAQUE_BIT_KHR = 0x00000001,
    VK_DISPLAY_PLANE_ALPHA_GLOBAL_BIT_KHR = 0x00000002,
    VK_DISPLAY_PLANE_ALPHA_PER_PIXEL_BIT_KHR = 0x00000004,
    VK_DISPLAY_PLANE_ALPHA_PER_PIXEL_PREMULTIPLIED_BIT_KHR = 0x00000008,
} VkDisplayPlaneAlphaFlagBitsKHR;
```

- VK_DISPLAY_PLANE_ALPHA_OPAQUE_BIT_KHR specifies that the source image will be treated as opaque.
- VK_DISPLAY_PLANE_ALPHA_GLOBAL_BIT_KHR specifies that a global alpha value **must** be specified that will be applied to all pixels in the source image.
- VK_DISPLAY_PLANE_ALPHA_PER_PIXEL_BIT_KHR specifies that the alpha value will be determined by the alpha component of the source image's pixels. If the source format contains no alpha values, no blending will be applied. The source alpha values are not premultiplied into the source image's other color components.
- VK_DISPLAY_PLANE_ALPHA_PER_PIXEL_PREMULTIPLIED_BIT_KHR is equivalent to VK_DISPLAY_PLANE_ALPHA_PER_PIXEL_BIT_KHR, except the source alpha values are assumed to be premultiplied into the source image's other color components.

```
// Provided by VK_KHR_display
typedef VkFlags VkDisplayPlaneAlphaFlagsKHR;
```

`VkDisplayPlaneAlphaFlagsKHR` is a bitmask type for setting a mask of zero or more VkDisplayPlaneAlphaFlagBitsKHR.

37.3.4. Presenting to Headless Surfaces

Vulkan rendering can be presented to a headless surface, where the presentation operation is a no-op producing no externally-visible result.

NOTE

Because there is no real presentation target, the headless presentation engine may be extended to impose an arbitrary or customizable set of restrictions and features. This makes it a useful portable test target for applications targeting a wide range of presentation engines where the actual target presentation engines might be scarce, unavailable or otherwise undesirable or inconvenient to use for general Vulkan application development.

The usual surface query mechanisms must be used to determine the actual restrictions and features of the implementation.

To create a headless `VkSurfaceKHR` object, call:

```
// Provided by VK_EXT_headless_surface
VkResult vkCreateHeadlessSurfaceEXT(
    VkInstance                                  instance,
    const VkHeadlessSurfaceCreateInfoEXT*       pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkSurfaceKHR*                               pSurface);
```

- `instance` is the instance to associate the surface with.
- `pCreateInfo` is a pointer to a VkHeadlessSurfaceCreateInfoEXT structure containing parameters affecting the creation of the surface object.
- `pAllocator` is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see Memory Allocation).
- `pSurface` is a pointer to a VkSurfaceKHR handle in which the created surface object is returned.

Valid Usage (Implicit)

- VUID-vkCreateHeadlessSurfaceEXT-instance-parameter
  `instance` **must** be a valid VkInstance handle
- VUID-vkCreateHeadlessSurfaceEXT-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkHeadlessSurfaceCreateInfoEXT structure
- VUID-vkCreateHeadlessSurfaceEXT-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateHeadlessSurfaceEXT-pSurface-parameter
  `pSurface` **must** be a valid pointer to a VkSurfaceKHR handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkHeadlessSurfaceCreateInfoEXT structure is defined as:

```
// Provided by VK_EXT_headless_surface
typedef struct VkHeadlessSurfaceCreateInfoEXT {
    VkStructureType                    sType;
    const void*                        pNext;
    VkHeadlessSurfaceCreateFlagsEXT    flags;
} VkHeadlessSurfaceCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is reserved for future use.

Valid Usage (Implicit)

- VUID-VkHeadlessSurfaceCreateInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_HEADLESS_SURFACE_CREATE_INFO_EXT
- VUID-VkHeadlessSurfaceCreateInfoEXT-pNext-pNext
  pNext **must** be NULL
- VUID-VkHeadlessSurfaceCreateInfoEXT-flags-zerobitmask
  flags **must** be 0

For headless surfaces, currentExtent is the reserved value (0xFFFFFFFF, 0xFFFFFFFF). Whatever the application sets a swapchain's imageExtent to will be the size of the surface, after the first image is presented.

```
// Provided by VK_EXT_headless_surface
typedef VkFlags VkHeadlessSurfaceCreateFlagsEXT;
```

VkHeadlessSurfaceCreateFlagsEXT is a bitmask type for setting a mask, but is currently reserved for future use.

37.4. Querying for WSI Support

Not all physical devices will include WSI support. Within a physical device, not all queue families will support presentation. WSI support and compatibility **can** be determined in a platform-neutral manner (which determines support for presentation to a particular surface object) and additionally **may** be determined in platform-specific manners (which determine support for presentation on the specified physical device but do not guarantee support for presentation to a particular surface object).

To determine whether a queue family of a physical device supports presentation to a given surface, call:

```
// Provided by VK_KHR_surface
VkResult vkGetPhysicalDeviceSurfaceSupportKHR(
    VkPhysicalDevice                            physicalDevice,
    uint32_t                                    queueFamilyIndex,
    VkSurfaceKHR                                surface,
    VkBool32*                                   pSupported);
```

- `physicalDevice` is the physical device.
- `queueFamilyIndex` is the queue family.
- `surface` is the surface.
- `pSupported` is a pointer to a VkBool32. VK_TRUE indicates support, and VK_FALSE indicates no support.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfaceSupportKHR-queueFamilyIndex-01269
  `queueFamilyIndex` **must** be less than `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties` for the given `physicalDevice`

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfaceSupportKHR-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceSurfaceSupportKHR-surface-parameter
  `surface` **must** be a valid VkSurfaceKHR handle
- VUID-vkGetPhysicalDeviceSurfaceSupportKHR-pSupported-parameter
  `pSupported` **must** be a valid pointer to a VkBool32 value
- VUID-vkGetPhysicalDeviceSurfaceSupportKHR-commonparent
  Both of `physicalDevice`, and `surface` **must** have been created, allocated, or retrieved from the same VkInstance

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

37.4.1. Android Platform

On Android, all physical devices and queue families **must** be capable of presentation with any native window. As a result there is no Android-specific query for these capabilities.

37.4.2. Wayland Platform

To determine whether a queue family of a physical device supports presentation to a Wayland compositor, call:

```
// Provided by VK_KHR_wayland_surface
VkBool32 vkGetPhysicalDeviceWaylandPresentationSupportKHR(
    VkPhysicalDevice                            physicalDevice,
    uint32_t                                    queueFamilyIndex,
    struct wl_display*                          display);
```

- `physicalDevice` is the physical device.
- `queueFamilyIndex` is the queue family index.
- `display` is a pointer to the `wl_display` associated with a Wayland compositor.

This platform-specific function **can** be called prior to creating a surface.

Valid Usage

- VUID-vkGetPhysicalDeviceWaylandPresentationSupportKHR-queueFamilyIndex-01306 `queueFamilyIndex` **must** be less than `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties` for the given `physicalDevice`

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceWaylandPresentationSupportKHR-physicalDevice-parameter

physicalDevice **must** be a valid VkPhysicalDevice handle

- VUID-vkGetPhysicalDeviceWaylandPresentationSupportKHR-display-parameter
  display **must** be a valid pointer to a wl_display value

37.4.3. Win32 Platform

To determine whether a queue family of a physical device supports presentation to the Microsoft Windows desktop, call:

```
// Provided by VK_KHR_win32_surface
VkBool32 vkGetPhysicalDeviceWin32PresentationSupportKHR(
    VkPhysicalDevice                            physicalDevice,
    uint32_t                                    queueFamilyIndex);
```

- physicalDevice is the physical device.
- queueFamilyIndex is the queue family index.

This platform-specific function **can** be called prior to creating a surface.

Valid Usage

- VUID-vkGetPhysicalDeviceWin32PresentationSupportKHR-queueFamilyIndex-01309
  queueFamilyIndex **must** be less than pQueueFamilyPropertyCount returned by vkGetPhysicalDeviceQueueFamilyProperties for the given physicalDevice

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceWin32PresentationSupportKHR-physicalDevice-parameter
  physicalDevice **must** be a valid VkPhysicalDevice handle

37.4.4. XCB Platform

To determine whether a queue family of a physical device supports presentation to an X11 server, using the XCB client-side library, call:

```
// Provided by VK_KHR_xcb_surface
VkBool32 vkGetPhysicalDeviceXcbPresentationSupportKHR(
    VkPhysicalDevice                            physicalDevice,
    uint32_t                                    queueFamilyIndex,
    xcb_connection_t*                           connection,
    xcb_visualid_t                              visual_id);
```

- physicalDevice is the physical device.

- `queueFamilyIndex` is the queue family index.
- `connection` is a pointer to an `xcb_connection_t` to the X server.
- `visual_id` is an X11 visual (`xcb_visualid_t`).

This platform-specific function **can** be called prior to creating a surface.

Valid Usage

- VUID-vkGetPhysicalDeviceXcbPresentationSupportKHR-queueFamilyIndex-01312 `queueFamilyIndex` **must** be less than `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties` for the given `physicalDevice`

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceXcbPresentationSupportKHR-physicalDevice-parameter `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceXcbPresentationSupportKHR-connection-parameter `connection` **must** be a valid pointer to an `xcb_connection_t` value

37.4.5. Xlib Platform

To determine whether a queue family of a physical device supports presentation to an X11 server, using the Xlib client-side library, call:

```
// Provided by VK_KHR_xlib_surface
VkBool32 vkGetPhysicalDeviceXlibPresentationSupportKHR(
    VkPhysicalDevice                            physicalDevice,
    uint32_t                                    queueFamilyIndex,
    Display*                                    dpy,
    VisualID                                    visualID);
```

- `physicalDevice` is the physical device.
- `queueFamilyIndex` is the queue family index.
- `dpy` is a pointer to an Xlib `Display` connection to the server.
- `visualID` is an X11 visual (`VisualID`).

This platform-specific function **can** be called prior to creating a surface.

Valid Usage

- VUID-vkGetPhysicalDeviceXlibPresentationSupportKHR-queueFamilyIndex-01315 `queueFamilyIndex` **must** be less than `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties` for the given `physicalDevice`

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceXlibPresentationSupportKHR-physicalDevice-parameter `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceXlibPresentationSupportKHR-dpy-parameter `dpy` **must** be a valid pointer to a `Display` value

37.4.6. DirectFB Platform

To determine whether a queue family of a physical device supports presentation with DirectFB library, call:

```
// Provided by VK_EXT_directfb_surface
VkBool32 vkGetPhysicalDeviceDirectFBPresentationSupportEXT(
    VkPhysicalDevice                            physicalDevice,
    uint32_t                                    queueFamilyIndex,
    IDirectFB*                                  dfb);
```

- `physicalDevice` is the physical device.
- `queueFamilyIndex` is the queue family index.
- `dfb` is a pointer to the `IDirectFB` main interface of DirectFB.

This platform-specific function **can** be called prior to creating a surface.

Valid Usage

- VUID-vkGetPhysicalDeviceDirectFBPresentationSupportEXT-queueFamilyIndex-04119 `queueFamilyIndex` **must** be less than `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties` for the given `physicalDevice`

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceDirectFBPresentationSupportEXT-physicalDevice-parameter `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceDirectFBPresentationSupportEXT-dfb-parameter `dfb` **must** be a valid pointer to an `IDirectFB` value

37.4.7. Metal Platform

On Apple platforms with Metal support, all physical devices and queue families **must** be capable of presentation with any layer. As a result there is no Apple-specific query for these capabilities.

37.5. Surface Queries

The capabilities of a swapchain targeting a surface are the intersection of the capabilities of the WSI platform, the native window or display, and the physical device. The resulting capabilities **can** be obtained with the queries listed below in this section.

NOTE	In addition to the surface capabilities as obtained by surface queries below, swapchain images are also subject to ordinary image creation limits as reported by vkGetPhysicalDeviceImageFormatProperties. As an application is instructed by the appropriate Valid Usage sections, both the surface capabilities and the image creation limits have to be satisfied whenever swapchain images are created.

37.5.1. Surface Capabilities

To query the basic capabilities of a surface, needed in order to create a swapchain, call:

WARNING	This functionality is superseded by vkGetPhysicalDeviceSurfaceCapabilities2KHR. See Legacy Functionality for more information.

```
// Provided by VK_KHR_surface
VkResult vkGetPhysicalDeviceSurfaceCapabilitiesKHR(
    VkPhysicalDevice                            physicalDevice,
    VkSurfaceKHR                                surface,
    VkSurfaceCapabilitiesKHR*                   pSurfaceCapabilities);
```

- `physicalDevice` is the physical device that will be associated with the swapchain to be created. See vkCreateSwapchainKHR for details of swapchain creation.
- `surface` is the surface that will be associated with the swapchain.
- `pSurfaceCapabilities` is a pointer to a VkSurfaceCapabilitiesKHR structure in which the capabilities are returned.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfaceCapabilitiesKHR-surface-06211
  `surface` **must** be supported by `physicalDevice`, as reported by vkGetPhysicalDeviceSurfaceSupportKHR or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfaceCapabilitiesKHR-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceSurfaceCapabilitiesKHR-surface-parameter
  `surface` **must** be a valid VkSurfaceKHR handle

- VUID-vkGetPhysicalDeviceSurfaceCapabilitiesKHR-pSurfaceCapabilities-parameter
  `pSurfaceCapabilities` **must** be a valid pointer to a VkSurfaceCapabilitiesKHR structure
- VUID-vkGetPhysicalDeviceSurfaceCapabilitiesKHR-commonparent
  Both of `physicalDevice`, and `surface` **must** have been created, allocated, or retrieved from the same VkInstance

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkSurfaceCapabilitiesKHR` structure is defined as:

WARNING This functionality is superseded by VK_KHR_get_surface_capabilities2. See Legacy Functionality for more information.

```
// Provided by VK_KHR_surface
typedef struct VkSurfaceCapabilitiesKHR {
    uint32_t                         minImageCount;
    uint32_t                         maxImageCount;
    VkExtent2D                       currentExtent;
    VkExtent2D                       minImageExtent;
    VkExtent2D                       maxImageExtent;
    uint32_t                         maxImageArrayLayers;
    VkSurfaceTransformFlagsKHR       supportedTransforms;
    VkSurfaceTransformFlagBitsKHR    currentTransform;
    VkCompositeAlphaFlagsKHR         supportedCompositeAlpha;
    VkImageUsageFlags                supportedUsageFlags;
} VkSurfaceCapabilitiesKHR;
```

- `minImageCount` is the minimum number of images the specified device supports for a swapchain created for the surface, and will be at least one.
- `maxImageCount` is the maximum number of images the specified device supports for a swapchain created for the surface, and will be either 0, or greater than or equal to `minImageCount`. A value of 0 means that there is no limit on the number of images, though there **may** be limits related to the total amount of memory used by presentable images.

- `currentExtent` is the current width and height of the surface, or the special value (0xFFFFFFFF, 0xFFFFFFFF) indicating that the surface size will be determined by the extent of a swapchain targeting the surface.
- `minImageExtent` contains the smallest valid swapchain extent for the surface on the specified device. The `width` and `height` of the extent will each be less than or equal to the corresponding `width` and `height` of `currentExtent`, unless `currentExtent` has the special value described above.
- `maxImageExtent` contains the largest valid swapchain extent for the surface on the specified device. The `width` and `height` of the extent will each be greater than or equal to the corresponding `width` and `height` of `minImageExtent`. The `width` and `height` of the extent will each be greater than or equal to the corresponding `width` and `height` of `currentExtent`, unless `currentExtent` has the special value described above.
- `maxImageArrayLayers` is the maximum number of layers presentable images **can** have for a swapchain created for this device and surface, and will be at least one.
- `supportedTransforms` is a bitmask of VkSurfaceTransformFlagBitsKHR indicating the presentation transforms supported for the surface on the specified device. At least one bit will be set.
- `currentTransform` is a VkSurfaceTransformFlagBitsKHR value indicating the surface's current transform relative to the presentation engine's natural orientation.
- `supportedCompositeAlpha` is a bitmask of VkCompositeAlphaFlagBitsKHR, representing the alpha compositing modes supported by the presentation engine for the surface on the specified device, and at least one bit will be set. Opaque composition **can** be achieved in any alpha compositing mode by either using an image format that has no alpha component, or by ensuring that all pixels in the presentable images have an alpha value of 1.0.
- `supportedUsageFlags` is a bitmask of VkImageUsageFlagBits representing the ways the application **can** use the presentable images of a swapchain created with VkPresentModeKHR set to VK_PRESENT_MODE_FIFO_LATEST_READY_KHR, VK_PRESENT_MODE_IMMEDIATE_KHR, VK_PRESENT_MODE_MAILBOX_KHR, VK_PRESENT_MODE_FIFO_KHR or VK_PRESENT_MODE_FIFO_RELAXED_KHR for the surface on the specified device. VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT **must** be included in the set. Implementations **may** support additional usages.

NOTE	Supported usage flags of a presentable image when using VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR or VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR presentation mode are provided by VkSharedPresentSurfaceCapabilities2KHR or VkSharedPresentSurfaceCapabilitiesKHR.

NOTE	Formulas such as min(N, `maxImageCount`) are not correct, since `maxImageCount` **may** be zero.

To query the basic capabilities of a surface defined by the core or extensions, call:

```
// Provided by VK_KHR_get_surface_capabilities2
VkResult vkGetPhysicalDeviceSurfaceCapabilities2KHR(
    VkPhysicalDevice                            physicalDevice,
    const VkPhysicalDeviceSurfaceInfo2KHR*      pSurfaceInfo,
    VkSurfaceCapabilities2KHR*                  pSurfaceCapabilities);
```

- `physicalDevice` is the physical device that will be associated with the swapchain to be created, as described for vkCreateSwapchainKHR.
- `pSurfaceInfo` is a pointer to a VkPhysicalDeviceSurfaceInfo2KHR structure describing the surface and other fixed parameters that would be consumed by vkCreateSwapchainKHR.
- `pSurfaceCapabilities` is a pointer to a VkSurfaceCapabilities2KHR structure in which the capabilities are returned.

`vkGetPhysicalDeviceSurfaceCapabilities2KHR` behaves similarly to vkGetPhysicalDeviceSurfaceCapabilitiesKHR, with the ability to specify extended inputs via chained input structures, and to return extended information via chained output structures.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfaceCapabilities2KHR-pSurfaceInfo-06521
  `pSurfaceInfo->surface` **must** be a valid VkSurfaceKHR handle
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2KHR-pSurfaceInfo-06522
  `pSurfaceInfo->surface` **must** be supported by `physicalDevice`, as reported by vkGetPhysicalDeviceSurfaceSupportKHR or an equivalent platform-specific mechanism
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2KHR-pNext-02671
  If a VkSurfaceCapabilitiesFullScreenExclusiveEXT structure is included in the `pNext` chain of `pSurfaceCapabilities`, a VkSurfaceFullScreenExclusiveWin32InfoEXT structure **must** be included in the `pNext` chain of `pSurfaceInfo`
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2KHR-pNext-07776
  If a VkSurfacePresentModeCompatibilityKHR structure is included in the `pNext` chain of `pSurfaceCapabilities`, a VkSurfacePresentModeKHR structure **must** be included in the `pNext` chain of `pSurfaceInfo`
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2KHR-pNext-07777
  If a VkSurfacePresentScalingCapabilitiesKHR structure is included in the `pNext` chain of `pSurfaceCapabilities`, a VkSurfacePresentModeKHR structure **must** be included in the `pNext` chain of `pSurfaceInfo`

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfaceCapabilities2KHR-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2KHR-pSurfaceInfo-parameter
  `pSurfaceInfo` **must** be a valid pointer to a valid VkPhysicalDeviceSurfaceInfo2KHR

structure

- VUID-vkGetPhysicalDeviceSurfaceCapabilities2KHR-pSurfaceCapabilities-parameter `pSurfaceCapabilities` **must** be a valid pointer to a VkSurfaceCapabilities2KHR structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkPhysicalDeviceSurfaceInfo2KHR` structure is defined as:

```
// Provided by VK_KHR_get_surface_capabilities2
typedef struct VkPhysicalDeviceSurfaceInfo2KHR {
    VkStructureType    sType;
    const void*        pNext;
    VkSurfaceKHR       surface;
} VkPhysicalDeviceSurfaceInfo2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `surface` is the surface that will be associated with the swapchain.

The members of `VkPhysicalDeviceSurfaceInfo2KHR` correspond to the arguments to vkGetPhysicalDeviceSurfaceCapabilitiesKHR, with `sType` and `pNext` added for extensibility.

Additional capabilities of a surface **may** be available to swapchains created with different full-screen exclusive settings - particularly if exclusive full-screen access is application controlled. These additional capabilities **can** be queried by adding a VkSurfaceFullScreenExclusiveInfoEXT structure to the `pNext` chain of this structure when used to query surface properties. Additionally, for Win32 surfaces with application controlled exclusive full-screen access, chaining a VkSurfaceFullScreenExclusiveWin32InfoEXT structure **may** also report additional surface capabilities. These additional capabilities only apply to swapchains created with the same parameters included in the `pNext` chain of VkSwapchainCreateInfoKHR.

Valid Usage

- VUID-VkPhysicalDeviceSurfaceInfo2KHR-pNext-02672
  If the `pNext` chain includes a VkSurfaceFullScreenExclusiveInfoEXT structure with its `fullScreenExclusive` member set to VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT, and `surface` was created using vkCreateWin32SurfaceKHR, a VkSurfaceFullScreenExclusiveWin32InfoEXT structure **must** be included in the `pNext` chain
- VUID-VkPhysicalDeviceSurfaceInfo2KHR-surface-07919
  If surface is not VK_NULL_HANDLE, `surface` **must** be a valid VkSurfaceKHR handle

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSurfaceInfo2KHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SURFACE_INFO_2_KHR
- VUID-VkPhysicalDeviceSurfaceInfo2KHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkSurfaceFullScreenExclusiveInfoEXT, VkSurfaceFullScreenExclusiveWin32InfoEXT, or VkSurfacePresentModeKHR
- VUID-VkPhysicalDeviceSurfaceInfo2KHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

If the `pNext` chain of VkSwapchainCreateInfoKHR includes a `VkSurfaceFullScreenExclusiveInfoEXT` structure, then that structure specifies the application's preferred full-screen transition behavior.

The `VkSurfaceFullScreenExclusiveInfoEXT` structure is defined as:

```
// Provided by VK_EXT_full_screen_exclusive
typedef struct VkSurfaceFullScreenExclusiveInfoEXT {
    VkStructureType             sType;
    void*                       pNext;
    VkFullScreenExclusiveEXT    fullScreenExclusive;
} VkSurfaceFullScreenExclusiveInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `fullScreenExclusive` is a VkFullScreenExclusiveEXT value specifying the preferred full-screen transition behavior.

If this structure is not present, `fullScreenExclusive` is considered to be VK_FULL_SCREEN_EXCLUSIVE_DEFAULT_EXT.

Valid Usage (Implicit)

- VUID-VkSurfaceFullScreenExclusiveInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SURFACE_FULL_SCREEN_EXCLUSIVE_INFO_EXT
- VUID-VkSurfaceFullScreenExclusiveInfoEXT-fullScreenExclusive-parameter
  `fullScreenExclusive` **must** be a valid VkFullScreenExclusiveEXT value

Structure Chaining

Extends the structures

- VkPhysicalDeviceSurfaceInfo2KHR
- VkSwapchainCreateInfoKHR

Possible values of `VkSurfaceFullScreenExclusiveInfoEXT`::`fullScreenExclusive` are:

```
// Provided by VK_EXT_full_screen_exclusive
typedef enum VkFullScreenExclusiveEXT {
    VK_FULL_SCREEN_EXCLUSIVE_DEFAULT_EXT = 0,
    VK_FULL_SCREEN_EXCLUSIVE_ALLOWED_EXT = 1,
    VK_FULL_SCREEN_EXCLUSIVE_DISALLOWED_EXT = 2,
    VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT = 3,
} VkFullScreenExclusiveEXT;
```

- VK_FULL_SCREEN_EXCLUSIVE_DEFAULT_EXT specifies that the implementation **should** determine the appropriate full-screen method by whatever means it deems appropriate.
- VK_FULL_SCREEN_EXCLUSIVE_ALLOWED_EXT specifies that the implementation **may** use full-screen exclusive mechanisms when available. Such mechanisms **may** result in better performance and/or the availability of different presentation capabilities, but **may** require a more disruptive transition during swapchain initialization, first presentation and/or destruction.
- VK_FULL_SCREEN_EXCLUSIVE_DISALLOWED_EXT specifies that the implementation **should** avoid using full-screen mechanisms which rely on disruptive transitions.
- VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT specifies that the application will manage full-screen exclusive mode by using the vkAcquireFullScreenExclusiveModeEXT and vkReleaseFullScreenExclusiveModeEXT commands.

The `VkSurfaceFullScreenExclusiveWin32InfoEXT` structure is defined as:

```
// Provided by VK_KHR_win32_surface with VK_EXT_full_screen_exclusive
typedef struct VkSurfaceFullScreenExclusiveWin32InfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    HMONITOR           hmonitor;
} VkSurfaceFullScreenExclusiveWin32InfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `hmonitor` is the Win32 `HMONITOR` handle identifying the display to create the surface with.

NOTE If `hmonitor` is invalidated (e.g. the monitor is unplugged) during the lifetime of a swapchain created with this structure, operations on that swapchain will return VK_ERROR_OUT_OF_DATE_KHR.

NOTE It is the responsibility of the application to change the display settings of the targeted Win32 display using the appropriate platform APIs. Such changes **may** alter the surface capabilities reported for the created surface.

Valid Usage

- VUID-VkSurfaceFullScreenExclusiveWin32InfoEXT-hmonitor-02673
  `hmonitor` **must** be a valid `HMONITOR`

Valid Usage (Implicit)

- VUID-VkSurfaceFullScreenExclusiveWin32InfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SURFACE_FULL_SCREEN_EXCLUSIVE_WIN32_INFO_EXT

Structure Chaining

Extends the structures

- VkPhysicalDeviceSurfaceInfo2KHR
- VkSwapchainCreateInfoKHR

The `VkSurfaceCapabilities2KHR` structure is defined as:

```
// Provided by VK_KHR_get_surface_capabilities2
typedef struct VkSurfaceCapabilities2KHR {
    VkStructureType             sType;
    void*                       pNext;
    VkSurfaceCapabilitiesKHR    surfaceCapabilities;
} VkSurfaceCapabilities2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `surfaceCapabilities` is a VkSurfaceCapabilitiesKHR structure describing the capabilities of the specified surface.

Only usage flags representable in VkImageUsageFlagBits are returned in `surfaceCapabilities.supportedUsageFlags`. If the `pNext` chain includes a VkImageUsageFlags2CreateInfoKHR structure, all usage flags are returned in VkImageUsageFlags2CreateInfoKHR::`usage`.

Valid Usage (Implicit)

- VUID-VkSurfaceCapabilities2KHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_2_KHR
- VUID-VkSurfaceCapabilities2KHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkImageUsageFlags2CreateInfoKHR, VkPresentTimingSurfaceCapabilitiesEXT, VkSharedPresentSurfaceCapabilities2KHR, VkSharedPresentSurfaceCapabilitiesKHR, VkSurfaceCapabilitiesFullScreenExclusiveEXT, VkSurfaceCapabilitiesPresentId2KHR, VkSurfaceCapabilitiesPresentWait2KHR, VkSurfacePresentModeCompatibilityKHR, VkSurfacePresentScalingCapabilitiesKHR, VkSurfaceProtectedCapabilitiesKHR, or VkSwapchainFlagsSurfaceCapabilitiesEXT
- VUID-VkSurfaceCapabilities2KHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

An application queries if a protected VkSurfaceKHR is displayable on a specific windowing system using `VkSurfaceProtectedCapabilitiesKHR`, which **can** be passed in `pNext` parameter of `VkSurfaceCapabilities2KHR`.

The `VkSurfaceProtectedCapabilitiesKHR` structure is defined as:

```
// Provided by VK_KHR_surface_protected_capabilities
typedef struct VkSurfaceProtectedCapabilitiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           supportsProtected;
} VkSurfaceProtectedCapabilitiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `supportsProtected` specifies whether a protected swapchain created from VkPhysicalDeviceSurfaceInfo2KHR::`surface` for a particular windowing system **can** be displayed on screen or not. If `supportsProtected` is VK_TRUE, then creation of swapchains with the VK_SWAPCHAIN_CREATE_PROTECTED_BIT_KHR flag set **must** be supported for `surface`.

Valid Usage (Implicit)

- VUID-VkSurfaceProtectedCapabilitiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SURFACE_PROTECTED_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkSurfaceCapabilities2KHR

The `VkSurfacePresentScalingCapabilitiesKHR` structure is defined as:

```
// Provided by VK_KHR_surface_maintenance1
typedef struct VkSurfacePresentScalingCapabilitiesKHR {
    VkStructureType             sType;
    void*                       pNext;
    VkPresentScalingFlagsKHR    supportedPresentScaling;
    VkPresentGravityFlagsKHR    supportedPresentGravityX;
    VkPresentGravityFlagsKHR    supportedPresentGravityY;
    VkExtent2D                  minScaledImageExtent;
    VkExtent2D                  maxScaledImageExtent;
} VkSurfacePresentScalingCapabilitiesKHR;
```

```
// Provided by VK_EXT_surface_maintenance1
// Equivalent to VkSurfacePresentScalingCapabilitiesKHR
typedef VkSurfacePresentScalingCapabilitiesKHR VkSurfacePresentScalingCapabilitiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `supportedPresentScaling` is a bitmask of VkPresentScalingFlagBitsKHR representing the scaling methods supported by the surface, or `0` if application-defined scaling is not supported.
- `supportedPresentGravityX` is a bitmask of VkPresentGravityFlagBitsKHR representing the X-axis pixel gravity supported by the surface, or `0` if Vulkan-defined pixel gravity is not supported for the X axis.
- `supportedPresentGravityY` is a bitmask of VkPresentGravityFlagBitsKHR representing the Y-axis

pixel gravity supported by the surface, or `0` if Vulkan-defined pixel gravity is not supported for the Y axis.

- `minScaledImageExtent` contains the smallest valid swapchain extent for the surface on the specified device when one of the scaling methods specified in `supportedPresentScaling` is used, or the special value (0xFFFFFFFF, 0xFFFFFFFF) indicating that the surface size will be determined by the extent of a swapchain targeting the surface. The `width` and `height` of the extent will each be less than or equal to the corresponding `width` and `height` of VkSurfaceCapabilitiesKHR::`minImageExtent`.
- `maxScaledImageExtent` contains the largest valid swapchain extent for the surface on the specified device when one of the scaling methods specified in `supportedPresentScaling` is used, or the special value described above for `minScaledImageExtent`. The `width` and `height` of the extent will each be greater than or equal to the corresponding `width` and `height` of VkSurfaceCapabilitiesKHR::`maxImageExtent`.

To query the set of supported scaling modes for a given present mode, add a VkSurfacePresentModeKHR structure in the `pNext` chain of VkPhysicalDeviceSurfaceInfo2KHR when calling vkGetPhysicalDeviceSurfaceCapabilities2KHR. The implementation **must** return the same values in `VkSurfacePresentScalingCapabilitiesKHR` for any of the compatible present modes as obtained through VkSurfacePresentModeCompatibilityKHR.

The application **can** specify the scaling mode when creating a swapchain through the use of VkSwapchainPresentScalingCreateInfoKHR.

Valid Usage (Implicit)

- VUID-VkSurfacePresentScalingCapabilitiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SURFACE_PRESENT_SCALING_CAPABILITIES_KHR
- VUID-VkSurfacePresentScalingCapabilitiesKHR-supportedPresentScaling-parameter
  If `supportedPresentScaling` is not `0`, `supportedPresentScaling` **must** be a valid combination of VkPresentScalingFlagBitsKHR values
- VUID-VkSurfacePresentScalingCapabilitiesKHR-supportedPresentGravityX-parameter
  If `supportedPresentGravityX` is not `0`, `supportedPresentGravityX` **must** be a valid combination of VkPresentGravityFlagBitsKHR values
- VUID-VkSurfacePresentScalingCapabilitiesKHR-supportedPresentGravityY-parameter
  If `supportedPresentGravityY` is not `0`, `supportedPresentGravityY` **must** be a valid combination of VkPresentGravityFlagBitsKHR values

Structure Chaining

Extends the structure

- VkSurfaceCapabilities2KHR

Bits which **may** be set in VkSurfacePresentScalingCapabilitiesKHR::`supportedPresentScaling`, specifying scaling modes supported by the surface, are:

```
// Provided by VK_KHR_surface_maintenance1
typedef enum VkPresentScalingFlagBitsKHR {
    VK_PRESENT_SCALING_ONE_TO_ONE_BIT_KHR = 0x00000001,
    VK_PRESENT_SCALING_ASPECT_RATIO_STRETCH_BIT_KHR = 0x00000002,
    VK_PRESENT_SCALING_STRETCH_BIT_KHR = 0x00000004,
    VK_PRESENT_SCALING_ONE_TO_ONE_BIT_EXT = VK_PRESENT_SCALING_ONE_TO_ONE_BIT_KHR,
    VK_PRESENT_SCALING_ASPECT_RATIO_STRETCH_BIT_EXT =
VK_PRESENT_SCALING_ASPECT_RATIO_STRETCH_BIT_KHR,
    VK_PRESENT_SCALING_STRETCH_BIT_EXT = VK_PRESENT_SCALING_STRETCH_BIT_KHR,
} VkPresentScalingFlagBitsKHR;
```

```
// Provided by VK_EXT_surface_maintenance1
// Equivalent to VkPresentScalingFlagBitsKHR
typedef VkPresentScalingFlagBitsKHR VkPresentScalingFlagBitsEXT;
```

- VK_PRESENT_SCALING_ONE_TO_ONE_BIT_KHR specifies that no scaling occurs, and pixels in the swapchain image are mapped to one and only one pixel in the surface. The mapping between pixels is defined by the chosen presentation gravity.
- VK_PRESENT_SCALING_ASPECT_RATIO_STRETCH_BIT_KHR specifies that the swapchain image will be minified or magnified such that at least one of the resulting width or height is equal to the corresponding surface dimension, and the other resulting dimension is less than or equal to the corresponding surface dimension, with the aspect ratio of the resulting image being identical to that of the original swapchain image.
- VK_PRESENT_SCALING_STRETCH_BIT_KHR specifies that the swapchain image will be minified or magnified such that the resulting image dimensions are equal to those of the surface.

```
// Provided by VK_KHR_surface_maintenance1
typedef VkFlags VkPresentScalingFlagsKHR;
```

```
// Provided by VK_EXT_surface_maintenance1
// Equivalent to VkPresentScalingFlagsKHR
typedef VkPresentScalingFlagsKHR VkPresentScalingFlagsEXT;
```

VkPresentScalingFlagsKHR is a bitmask type for setting a mask of zero or more VkPresentScalingFlagBitsKHR.

Bits which **may** be set in the VkSurfacePresentScalingCapabilitiesKHR::supportedPresentGravityX or supportedPresentGravityY fields, specifying the gravity of presented pixels supported by the surface, are:

```
// Provided by VK_KHR_surface_maintenance1
typedef enum VkPresentGravityFlagBitsKHR {
    VK_PRESENT_GRAVITY_MIN_BIT_KHR = 0x00000001,
    VK_PRESENT_GRAVITY_MAX_BIT_KHR = 0x00000002,
    VK_PRESENT_GRAVITY_CENTERED_BIT_KHR = 0x00000004,
    VK_PRESENT_GRAVITY_MIN_BIT_EXT = VK_PRESENT_GRAVITY_MIN_BIT_KHR,
    VK_PRESENT_GRAVITY_MAX_BIT_EXT = VK_PRESENT_GRAVITY_MAX_BIT_KHR,
    VK_PRESENT_GRAVITY_CENTERED_BIT_EXT = VK_PRESENT_GRAVITY_CENTERED_BIT_KHR,
} VkPresentGravityFlagBitsKHR;
```

```
// Provided by VK_EXT_surface_maintenance1
// Equivalent to VkPresentGravityFlagBitsKHR
typedef VkPresentGravityFlagBitsKHR VkPresentGravityFlagBitsEXT;
```

- VK_PRESENT_GRAVITY_MIN_BIT_KHR means that the pixels will gravitate towards the top or left side of the surface.
- VK_PRESENT_GRAVITY_MAX_BIT_KHR means that the pixels will gravitate towards the bottom or right side of the surface.
- VK_PRESENT_GRAVITY_CENTERED_BIT_KHR means that the pixels will be centered in the surface.

If the value in VkSurfaceCapabilitiesKHR::currentTransform is not VK_SURFACE_TRANSFORM_IDENTITY_BIT_KHR, it is implementation-defined whether the gravity configuration applies to the presented image before or after transformation.

```
// Provided by VK_KHR_surface_maintenance1
typedef VkFlags VkPresentGravityFlagsKHR;
```

```
// Provided by VK_EXT_surface_maintenance1
// Equivalent to VkPresentGravityFlagsKHR
typedef VkPresentGravityFlagsKHR VkPresentGravityFlagsEXT;
```

VkPresentGravityFlagsKHR is a bitmask type for setting a mask of zero or more VkPresentGravityFlagBitsKHR.

The VkSurfacePresentModeKHR structure is defined as:

```
// Provided by VK_KHR_surface_maintenance1
typedef struct VkSurfacePresentModeKHR {
    VkStructureType     sType;
    void*               pNext;
    VkPresentModeKHR    presentMode;
} VkSurfacePresentModeKHR;
```

```
// Provided by VK_EXT_surface_maintenance1
// Equivalent to VkSurfacePresentModeKHR
typedef VkSurfacePresentModeKHR VkSurfacePresentModeEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `presentMode` is the presentation mode the swapchain will use.

If the `VkSurfacePresentModeKHR` structure is included in the `pNext` chain of VkPhysicalDeviceSurfaceInfo2KHR, the values returned in VkSurfaceCapabilitiesKHR::`minImageCount`, VkSurfaceCapabilitiesKHR::`maxImageCount`, VkSurfacePresentScalingCapabilitiesKHR::`minScaledImageExtent`, and VkSurfacePresentScalingCapabilitiesKHR::`maxScaledImageExtent` are valid only for the specified `presentMode`. If `presentMode` is VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR or VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR, the per-present mode image counts **must** both be one. The per-present mode image counts **may** be less-than or greater-than the image counts returned when `VkSurfacePresentModeKHR` is not provided.

NOTE

If VkSwapchainPresentModesCreateInfoKHR is provided to swapchain creation, the requirements for forward progress may be less strict. For example, a FIFO swapchain might only require 2 images to guarantee forward progress, but a MAILBOX one might require 4. Without the per-present image counts, such an implementation would have to return 4 in VkSurfaceCapabilitiesKHR::`minImageCount`, which pessimizes FIFO. Conversely, an implementation may return a low number for minImageCount, but internally bump the image count when the application queries vkGetSwapchainImagesKHR, which can surprise applications, and is not discoverable until swapchain creation. Using `VkSurfacePresentModeKHR` and VkSwapchainPresentModesCreateInfoKHR together effectively removes this problem.

VkSwapchainPresentModesCreateInfoKHR is required for the specification to be backwards compatible with applications that do not know about, or make use of this feature.

Valid Usage

- VUID-VkSurfacePresentModeKHR-presentMode-07780
  `presentMode` **must** be a value reported by vkGetPhysicalDeviceSurfacePresentModesKHR for the specified surface

Valid Usage (Implicit)

- VUID-VkSurfacePresentModeKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SURFACE_PRESENT_MODE_KHR

- VUID-VkSurfacePresentModeKHR-presentMode-parameter
  `presentMode` **must** be a valid VkPresentModeKHR value

Structure Chaining

Extends the structure

- VkPhysicalDeviceSurfaceInfo2KHR

The `VkSurfacePresentModeCompatibilityKHR` structure is defined as:

```
// Provided by VK_KHR_surface_maintenance1
typedef struct VkSurfacePresentModeCompatibilityKHR {
    VkStructureType      sType;
    void*                pNext;
    uint32_t             presentModeCount;
    VkPresentModeKHR*    pPresentModes;
} VkSurfacePresentModeCompatibilityKHR;
```

```
// Provided by VK_EXT_surface_maintenance1
// Equivalent to VkSurfacePresentModeCompatibilityKHR
typedef VkSurfacePresentModeCompatibilityKHR VkSurfacePresentModeCompatibilityEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `presentModeCount` is an integer related to the number of present modes available or queried, as described below.
- `pPresentModes` is a pointer to an array of VkPresentModeKHR in which present modes compatible with a given present mode are returned.

If `pPresentModes` is `NULL`, then the number of present modes that are compatible with the one specified in VkSurfacePresentModeKHR is returned in `presentModeCount`. Otherwise, `presentModeCount` **must** be set by the application to the number of elements in the `pPresentModes` array, and on return is overwritten with the number of values actually written to `pPresentModes`. If the value of `presentModeCount` is less than the number of compatible present modes that are supported, at most `presentModeCount` values will be written to `pPresentModes`. The implementation **must** include the present mode passed to VkSurfacePresentModeKHR in `pPresentModes`, unless `presentModeCount` is zero.

To query the set of present modes compatible with a given initial present mode, add a VkSurfacePresentModeKHR structure in the `pNext` chain of VkPhysicalDeviceSurfaceInfo2KHR when calling vkGetPhysicalDeviceSurfaceCapabilities2KHR.

The application **can** create a swapchain whose present mode **can** be modified through the use of VkSwapchainPresentModesCreateInfoKHR.

Valid Usage (Implicit)

- VUID-VkSurfacePresentModeCompatibilityKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SURFACE_PRESENT_MODE_COMPATIBILITY_KHR
- VUID-VkSurfacePresentModeCompatibilityKHR-pPresentModes-parameter
  If `presentModeCount` is not `0`, and `pPresentModes` is not `NULL`, `pPresentModes` **must** be a valid pointer to an array of `presentModeCount` VkPresentModeKHR values

Structure Chaining

Extends the structure

- VkSurfaceCapabilities2KHR

The `VkSharedPresentSurfaceCapabilitiesKHR` structure is defined as:

```
// Provided by VK_KHR_shared_presentable_image
typedef struct VkSharedPresentSurfaceCapabilitiesKHR {
    VkStructureType      sType;
    void*                pNext;
    VkImageUsageFlags    sharedPresentSupportedUsageFlags;
} VkSharedPresentSurfaceCapabilitiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `sharedPresentSupportedUsageFlags` is a bitmask of VkImageUsageFlagBits representing the ways the application **can** use the shared presentable image from a swapchain created with VkPresentModeKHR set to VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR or VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR for the surface on the specified device. VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT **must** be included in the set but implementations **may** support additional usages.

This structure **can** be included in the `pNext` chain of VkSurfaceCapabilities2KHR to determine supported usage when using VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR or VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR presentation mode.

Valid Usage (Implicit)

- VUID-VkSharedPresentSurfaceCapabilitiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SHARED_PRESENT_SURFACE_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkSurfaceCapabilities2KHR

The VkSharedPresentSurfaceCapabilities2KHR structure is defined as:

```
// Provided by VK_KHR_extended_flags with VK_KHR_shared_presentable_image
typedef struct VkSharedPresentSurfaceCapabilities2KHR {
    VkStructureType          sType;
    void*                    pNext;
    VkImageUsageFlags2KHR    sharedPresentSupportedUsageFlags;
} VkSharedPresentSurfaceCapabilities2KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- sharedPresentSupportedUsageFlags is a bitmask of VkImageUsageFlagBits2KHR representing the ways the application **can** use the shared presentable image from a swapchain created with VkPresentModeKHR set to VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR or VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR for the surface on the specified device. VK_IMAGE_USAGE_2_COLOR_ATTACHMENT_BIT_KHR **must** be included in the set but implementations **may** support additional usages.

This structure **can** be included in the pNext chain of VkSurfaceCapabilities2KHR to determine supported usage when using VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR or VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR presentation mode.

Valid Usage (Implicit)

- VUID-VkSharedPresentSurfaceCapabilities2KHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_SHARED_PRESENT_SURFACE_CAPABILITIES_2_KHR

Structure Chaining

Extends the structure

- VkSurfaceCapabilities2KHR

The VkSwapchainFlagsSurfaceCapabilitiesEXT structure is defined as:

```
// Provided by VK_EXT_multisampled_render_to_swapchain
typedef struct VkSwapchainFlagsSurfaceCapabilitiesEXT {
    VkStructureType              sType;
    void*                        pNext;
    VkSwapchainCreateFlagsKHR    swapchainSupportedFlags;
} VkSwapchainFlagsSurfaceCapabilitiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `swapchainSupportedFlags` is a bitmask of VkSwapchainCreateFlagsKHR representing the supported flags in VkSwapchainCreateInfoKHR when creating a swapchain for the surface on the specified device. If a VkSurfacePresentModeKHR structure is included in the `pNext` chain of `pSurfaceInfo` in vkGetPhysicalDeviceSurfaceCapabilities2KHR, this value represents the supported flags for the specified present mode and those compatible with it as returned in VkSurfacePresentModeCompatibilityKHR.

Valid Usage (Implicit)

- VUID-VkSwapchainFlagsSurfaceCapabilitiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SWAPCHAIN_FLAGS_SURFACE_CAPABILITIES_EXT

Structure Chaining

Extends the structure

- VkSurfaceCapabilities2KHR

The `VkSurfaceCapabilitiesFullScreenExclusiveEXT` structure is defined as:

```
// Provided by VK_EXT_full_screen_exclusive
typedef struct VkSurfaceCapabilitiesFullScreenExclusiveEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           fullScreenExclusiveSupported;
} VkSurfaceCapabilitiesFullScreenExclusiveEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `fullScreenExclusiveSupported` is a boolean describing whether the surface is able to make use of exclusive full-screen access.

This structure **can** be included in the `pNext` chain of VkSurfaceCapabilities2KHR to determine support for exclusive full-screen access. If `fullScreenExclusiveSupported` is VK_FALSE, it indicates that exclusive full-screen access is not obtainable for this surface.

Applications **must** not attempt to create swapchains with VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT set if `fullScreenExclusiveSupported` is VK_FALSE.

Valid Usage (Implicit)

- VUID-VkSurfaceCapabilitiesFullScreenExclusiveEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_FULL_SCREEN_EXCLUSIVE_EXT

Structure Chaining

Extends the structure

- VkSurfaceCapabilities2KHR

The `VkSurfaceCapabilitiesPresentId2KHR` structure is defined as:

```
// Provided by VK_KHR_present_id2
typedef struct VkSurfaceCapabilitiesPresentId2KHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           presentId2Supported;
} VkSurfaceCapabilitiesPresentId2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `presentId2Supported` is a boolean describing whether the surface is able to support the present-ID extension.

This structure **can** be included in the `pNext` chain of VkSurfaceCapabilities2KHR to determine support for present-wait. If `presentId2Supported` is VK_FALSE, it indicates that attaching an ID to presentation requests is not possible for this surface.

Applications **must** not attempt to include VkPresentId2KHR in the `pNext` chain of a VkPresentInfoKHR if `presentId2Supported` is VK_FALSE.

Valid Usage (Implicit)

- VUID-VkSurfaceCapabilitiesPresentId2KHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_PRESENT_ID_2_KHR

Structure Chaining

Extends the structure

- VkSurfaceCapabilities2KHR

The VkSurfaceCapabilitiesPresentWait2KHR structure is defined as:

```
// Provided by VK_KHR_present_wait2
typedef struct VkSurfaceCapabilitiesPresentWait2KHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           presentWait2Supported;
} VkSurfaceCapabilitiesPresentWait2KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- presentWait2Supported is a boolean describing whether the surface is able to support the present-wait extension.

This structure **can** be included in the pNext chain of VkSurfaceCapabilities2KHR to determine support for present-wait. If presentWait2Supported is VK_FALSE, it indicates that waiting for presentation is not possible for this surface.

Applications **must** not attempt to call vkWaitForPresent2KHR on a swapchain if presentWait2Supported is VK_FALSE.

Valid Usage (Implicit)

- VUID-VkSurfaceCapabilitiesPresentWait2KHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_PRESENT_WAIT_2_KHR

Structure Chaining

Extends the structure

- VkSurfaceCapabilities2KHR

The VkPresentTimingSurfaceCapabilitiesEXT structure is defined as:

```
// Provided by VK_EXT_present_timing
typedef struct VkPresentTimingSurfaceCapabilitiesEXT {
    VkStructureType           sType;
    void*                     pNext;
    VkBool32                  presentTimingSupported;
    VkBool32                  presentAtAbsoluteTimeSupported;
    VkBool32                  presentAtRelativeTimeSupported;
    VkPresentStageFlagsEXT    presentStageQueries;
} VkPresentTimingSurfaceCapabilitiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `presentTimingSupported` indicates whether querying presentation timestamps is supported for a swapchain created from VkPhysicalDeviceSurfaceInfo2KHR::`surface`.
- `presentAtAbsoluteTimeSupported` indicates whether a swapchain created from VkPhysicalDeviceSurfaceInfo2KHR::`surface` supports presenting images with absolute times.
- `presentAtRelativeTimeSupported` indicates whether a swapchain created from VkPhysicalDeviceSurfaceInfo2KHR::`surface` supports presenting images with relative times.
- `presentStageQueries` is a bitmask of VkPresentStageFlagBitsEXT indicating which present stages a swapchain created from VkPhysicalDeviceSurfaceInfo2KHR::`surface` is able to provide timing information for.

If `presentTimingSupported` is VK_TRUE, the implementation **must** also advertise support for at least the VK_PRESENT_STAGE_QUEUE_OPERATIONS_END_BIT_EXT present stage in `presentStageQueries`.

Valid Usage (Implicit)

- VUID-VkPresentTimingSurfaceCapabilitiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PRESENT_TIMING_SURFACE_CAPABILITIES_EXT

Structure Chaining

Extends the structure

- VkSurfaceCapabilities2KHR

To query the basic capabilities of a surface, needed in order to create a swapchain, call:

```
// Provided by VK_EXT_display_surface_counter
VkResult vkGetPhysicalDeviceSurfaceCapabilities2EXT(
    VkPhysicalDevice                            physicalDevice,
    VkSurfaceKHR                                surface,
    VkSurfaceCapabilities2EXT*                  pSurfaceCapabilities);
```

- `physicalDevice` is the physical device that will be associated with the swapchain to be created, as described for vkCreateSwapchainKHR.
- `surface` is the surface that will be associated with the swapchain.
- `pSurfaceCapabilities` is a pointer to a VkSurfaceCapabilities2EXT structure in which the capabilities are returned.

`vkGetPhysicalDeviceSurfaceCapabilities2EXT` behaves similarly to vkGetPhysicalDeviceSurfaceCapabilitiesKHR, with the ability to return extended information by adding extending structures to the `pNext` chain of its `pSurfaceCapabilities` parameter.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfaceCapabilities2EXT-surface-06211
  `surface` **must** be supported by `physicalDevice`, as reported by vkGetPhysicalDeviceSurfaceSupportKHR or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfaceCapabilities2EXT-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2EXT-surface-parameter
  `surface` **must** be a valid VkSurfaceKHR handle
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2EXT-pSurfaceCapabilities-parameter
  `pSurfaceCapabilities` **must** be a valid pointer to a VkSurfaceCapabilities2EXT structure
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2EXT-commonparent
  Both of `physicalDevice`, and `surface` **must** have been created, allocated, or retrieved from the same VkInstance

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkSurfaceCapabilities2EXT` structure is defined as:

```
// Provided by VK_EXT_display_surface_counter
typedef struct VkSurfaceCapabilities2EXT {
    VkStructureType                  sType;
    void*                            pNext;
    uint32_t                         minImageCount;
    uint32_t                         maxImageCount;
    VkExtent2D                       currentExtent;
    VkExtent2D                       minImageExtent;
    VkExtent2D                       maxImageExtent;
    uint32_t                         maxImageArrayLayers;
    VkSurfaceTransformFlagsKHR       supportedTransforms;
    VkSurfaceTransformFlagBitsKHR    currentTransform;
    VkCompositeAlphaFlagsKHR         supportedCompositeAlpha;
    VkImageUsageFlags                supportedUsageFlags;
    VkSurfaceCounterFlagsEXT         supportedSurfaceCounters;
} VkSurfaceCapabilities2EXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `minImageCount` is the minimum number of images the specified device supports for a swapchain created for the surface, and will be at least one.
- `maxImageCount` is the maximum number of images the specified device supports for a swapchain created for the surface, and will be either 0, or greater than or equal to `minImageCount`. A value of 0 means that there is no limit on the number of images, though there **may** be limits related to the total amount of memory used by presentable images.
- `currentExtent` is the current width and height of the surface, or the special value (0xFFFFFFFF, 0xFFFFFFFF) indicating that the surface size will be determined by the extent of a swapchain targeting the surface.
- `minImageExtent` contains the smallest valid swapchain extent for the surface on the specified device. The `width` and `height` of the extent will each be less than or equal to the corresponding `width` and `height` of `currentExtent`, unless `currentExtent` has the special value described above.
- `maxImageExtent` contains the largest valid swapchain extent for the surface on the specified device. The `width` and `height` of the extent will each be greater than or equal to the corresponding `width` and `height` of `minImageExtent`. The `width` and `height` of the extent will each be greater than or equal to the corresponding `width` and `height` of `currentExtent`, unless `currentExtent` has the special value described above.
- `maxImageArrayLayers` is the maximum number of layers presentable images **can** have for a swapchain created for this device and surface, and will be at least one.
- `supportedTransforms` is a bitmask of VkSurfaceTransformFlagBitsKHR indicating the presentation transforms supported for the surface on the specified device. At least one bit will be set.
- `currentTransform` is a VkSurfaceTransformFlagBitsKHR value indicating the surface's current transform relative to the presentation engine's natural orientation.
- `supportedCompositeAlpha` is a bitmask of VkCompositeAlphaFlagBitsKHR, representing the alpha

compositing modes supported by the presentation engine for the surface on the specified device, and at least one bit will be set. Opaque composition **can** be achieved in any alpha compositing mode by either using an image format that has no alpha component, or by ensuring that all pixels in the presentable images have an alpha value of 1.0.

- `supportedUsageFlags` is a bitmask of VkImageUsageFlagBits representing the ways the application **can** use the presentable images of a swapchain created with VkPresentModeKHR set to VK_PRESENT_MODE_FIFO_LATEST_READY_KHR, VK_PRESENT_MODE_IMMEDIATE_KHR, VK_PRESENT_MODE_MAILBOX_KHR, VK_PRESENT_MODE_FIFO_KHR or VK_PRESENT_MODE_FIFO_RELAXED_KHR for the surface on the specified device. VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT **must** be included in the set. Implementations **may** support additional usages.
- `supportedSurfaceCounters` is a bitmask of VkSurfaceCounterFlagBitsEXT indicating the supported surface counter types. It **must** not include VK_SURFACE_COUNTER_VBLANK_BIT_EXT unless the surface queried is a display surface.

Valid Usage (Implicit)

- VUID-VkSurfaceCapabilities2EXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_2_EXT
- VUID-VkSurfaceCapabilities2EXT-pNext-pNext
  `pNext` **must** be `NULL`

Bits which **can** be set in VkSurfaceCapabilities2EXT::`supportedSurfaceCounters`, indicating supported surface counter types, are:

```
// Provided by VK_EXT_display_surface_counter
typedef enum VkSurfaceCounterFlagBitsEXT {
    VK_SURFACE_COUNTER_VBLANK_BIT_EXT = 0x00000001,
  // VK_SURFACE_COUNTER_VBLANK_EXT is a legacy alias
    VK_SURFACE_COUNTER_VBLANK_EXT = VK_SURFACE_COUNTER_VBLANK_BIT_EXT,
} VkSurfaceCounterFlagBitsEXT;
```

- VK_SURFACE_COUNTER_VBLANK_BIT_EXT specifies a counter incrementing once every time a vertical blanking period occurs on the display associated with the surface.

```
// Provided by VK_EXT_display_surface_counter
typedef VkFlags VkSurfaceCounterFlagsEXT;
```

`VkSurfaceCounterFlagsEXT` is a bitmask type for setting a mask of zero or more VkSurfaceCounterFlagBitsEXT.

Bits which **may** be set in VkSurfaceCapabilitiesKHR::`supportedTransforms` indicating the presentation transforms supported for the surface on the specified device, and possible values of VkSurfaceCapabilitiesKHR::`currentTransform` indicating the surface's current transform relative to the presentation engine's natural orientation, are:

```
// Provided by VK_KHR_surface
typedef enum VkSurfaceTransformFlagBitsKHR {
    VK_SURFACE_TRANSFORM_IDENTITY_BIT_KHR = 0x00000001,
    VK_SURFACE_TRANSFORM_ROTATE_90_BIT_KHR = 0x00000002,
    VK_SURFACE_TRANSFORM_ROTATE_180_BIT_KHR = 0x00000004,
    VK_SURFACE_TRANSFORM_ROTATE_270_BIT_KHR = 0x00000008,
    VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_BIT_KHR = 0x00000010,
    VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_ROTATE_90_BIT_KHR = 0x00000020,
    VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_ROTATE_180_BIT_KHR = 0x00000040,
    VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_ROTATE_270_BIT_KHR = 0x00000080,
    VK_SURFACE_TRANSFORM_INHERIT_BIT_KHR = 0x00000100,
} VkSurfaceTransformFlagBitsKHR;
```

- VK_SURFACE_TRANSFORM_IDENTITY_BIT_KHR specifies that image content is presented without being transformed.
- VK_SURFACE_TRANSFORM_ROTATE_90_BIT_KHR specifies that image content is rotated 90 degrees clockwise.
- VK_SURFACE_TRANSFORM_ROTATE_180_BIT_KHR specifies that image content is rotated 180 degrees clockwise.
- VK_SURFACE_TRANSFORM_ROTATE_270_BIT_KHR specifies that image content is rotated 270 degrees clockwise.
- VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_BIT_KHR specifies that image content is mirrored horizontally.
- VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_ROTATE_90_BIT_KHR specifies that image content is mirrored horizontally, then rotated 90 degrees clockwise.
- VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_ROTATE_180_BIT_KHR specifies that image content is mirrored horizontally, then rotated 180 degrees clockwise.
- VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_ROTATE_270_BIT_KHR specifies that image content is mirrored horizontally, then rotated 270 degrees clockwise.
- VK_SURFACE_TRANSFORM_INHERIT_BIT_KHR specifies that the presentation transform is not specified, and is instead determined by platform-specific considerations and mechanisms outside Vulkan.

```
// Provided by VK_KHR_display, VK_KHR_surface
typedef VkFlags VkSurfaceTransformFlagsKHR;
```

VkSurfaceTransformFlagsKHR is a bitmask type for setting a mask of zero or more VkSurfaceTransformFlagBitsKHR.

The supportedCompositeAlpha member is of type VkCompositeAlphaFlagBitsKHR, containing the following values:

```
// Provided by VK_KHR_surface
typedef enum VkCompositeAlphaFlagBitsKHR {
    VK_COMPOSITE_ALPHA_OPAQUE_BIT_KHR = 0x00000001,
    VK_COMPOSITE_ALPHA_PRE_MULTIPLIED_BIT_KHR = 0x00000002,
    VK_COMPOSITE_ALPHA_POST_MULTIPLIED_BIT_KHR = 0x00000004,
    VK_COMPOSITE_ALPHA_INHERIT_BIT_KHR = 0x00000008,
} VkCompositeAlphaFlagBitsKHR;
```

These values are described as follows:

- VK_COMPOSITE_ALPHA_OPAQUE_BIT_KHR: The alpha component, if it exists, of the images is ignored in the compositing process. Instead, the image is treated as if it has a constant alpha of 1.0.
- VK_COMPOSITE_ALPHA_PRE_MULTIPLIED_BIT_KHR: The alpha component, if it exists, of the images is respected in the compositing process. The non-alpha components of the image are expected to already be multiplied by the alpha component by the application.
- VK_COMPOSITE_ALPHA_POST_MULTIPLIED_BIT_KHR: The alpha component, if it exists, of the images is respected in the compositing process. The non-alpha components of the image are not expected to already be multiplied by the alpha component by the application; instead, the compositor will multiply the non-alpha components of the image by the alpha component during compositing.
- VK_COMPOSITE_ALPHA_INHERIT_BIT_KHR: The way in which the presentation engine treats the alpha component in the images is unknown to the Vulkan API. Instead, the application is responsible for setting the composite alpha blending mode using native window system commands. If the application does not set the blending mode using native window system commands, then a platform-specific default will be used.

```
// Provided by VK_KHR_surface
typedef VkFlags VkCompositeAlphaFlagsKHR;
```

VkCompositeAlphaFlagsKHR is a bitmask type for setting a mask of zero or more VkCompositeAlphaFlagBitsKHR.

37.5.2. Surface Format Support

To query the supported swapchain format-color space pairs for a surface, call:

WARNING	This functionality is superseded by vkGetPhysicalDeviceSurfaceFormats2KHR. See Legacy Functionality for more information.

```
// Provided by VK_KHR_surface
VkResult vkGetPhysicalDeviceSurfaceFormatsKHR(
    VkPhysicalDevice                            physicalDevice,
    VkSurfaceKHR                                surface,
    uint32_t*                                   pSurfaceFormatCount,
    VkSurfaceFormatKHR*                         pSurfaceFormats);
```

- `physicalDevice` is the physical device that will be associated with the swapchain to be created. See vkCreateSwapchainKHR for details of swapchain creation.
- `surface` is the surface that will be associated with the swapchain.
- `pSurfaceFormatCount` is a pointer to an integer related to the number of format pairs available or queried, as described below.
- `pSurfaceFormats` is either `NULL` or a pointer to an array of `VkSurfaceFormatKHR` structures.

If `pSurfaceFormats` is `NULL`, then the number of format pairs supported for the given `surface` is returned in `pSurfaceFormatCount`. Otherwise, `pSurfaceFormatCount` **must** point to a variable set by the application to the number of elements in the `pSurfaceFormats` array, and on return the variable is overwritten with the number of structures actually written to `pSurfaceFormats`. If the value of `pSurfaceFormatCount` is less than the number of format pairs supported, at most `pSurfaceFormatCount` structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available format pairs were returned.

The number of format pairs supported **must** be greater than or equal to 1. `pSurfaceFormats` **must** not contain an entry whose value for `format` is VK_FORMAT_UNDEFINED.

If `pSurfaceFormats` includes an entry whose value for `colorSpace` is VK_COLOR_SPACE_SRGB_NONLINEAR_KHR and whose value for `format` is a UNORM (or SRGB) format and the corresponding SRGB (or UNORM) format is a color renderable format for VK_IMAGE_TILING_OPTIMAL, then `pSurfaceFormats` **must** also contain an entry with the same value for `colorSpace` and `format` equal to the corresponding SRGB (or UNORM) format.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfaceFormatsKHR-surface-06524
  `surface` **must** be a valid VkSurfaceKHR handle
- VUID-vkGetPhysicalDeviceSurfaceFormatsKHR-surface-06525
  `surface` **must** be supported by `physicalDevice`, as reported by vkGetPhysicalDeviceSurfaceSupportKHR or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfaceFormatsKHR-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceSurfaceFormatsKHR-surface-parameter

If `surface` is not VK_NULL_HANDLE, `surface` **must** be a valid VkSurfaceKHR handle

- VUID-vkGetPhysicalDeviceSurfaceFormatsKHR-pSurfaceFormatCount-parameter
  `pSurfaceFormatCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceSurfaceFormatsKHR-pSurfaceFormats-parameter
  If the value referenced by `pSurfaceFormatCount` is not `0`, and `pSurfaceFormats` is not `NULL`, `pSurfaceFormats` **must** be a valid pointer to an array of `pSurfaceFormatCount` VkSurfaceFormatKHR structures
- VUID-vkGetPhysicalDeviceSurfaceFormatsKHR-commonparent
  Both of `physicalDevice`, and `surface` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkInstance

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkSurfaceFormatKHR` structure is defined as:

WARNING This functionality is superseded by VK_KHR_get_surface_capabilities2. See Legacy Functionality for more information.

```
// Provided by VK_KHR_surface
typedef struct VkSurfaceFormatKHR {
    VkFormat           format;
    VkColorSpaceKHR    colorSpace;
} VkSurfaceFormatKHR;
```

- `format` is a VkFormat that is compatible with the specified surface.
- `colorSpace` is a presentation VkColorSpaceKHR that is compatible with the surface.

To query the supported swapchain format tuples for a surface, call:

```
// Provided by VK_KHR_get_surface_capabilities2
VkResult vkGetPhysicalDeviceSurfaceFormats2KHR(
    VkPhysicalDevice                            physicalDevice,
    const VkPhysicalDeviceSurfaceInfo2KHR*      pSurfaceInfo,
    uint32_t*                                   pSurfaceFormatCount,
    VkSurfaceFormat2KHR*                        pSurfaceFormats);
```

- `physicalDevice` is the physical device that will be associated with the swapchain to be created, as described for vkCreateSwapchainKHR.
- `pSurfaceInfo` is a pointer to a VkPhysicalDeviceSurfaceInfo2KHR structure describing the surface and other fixed parameters that would be consumed by vkCreateSwapchainKHR.
- `pSurfaceFormatCount` is a pointer to an integer related to the number of format tuples available or queried, as described below.
- `pSurfaceFormats` is either `NULL` or a pointer to an array of VkSurfaceFormat2KHR structures.

vkGetPhysicalDeviceSurfaceFormats2KHR behaves similarly to vkGetPhysicalDeviceSurfaceFormatsKHR, with the ability to be extended via `pNext` chains.

If `pSurfaceFormats` is `NULL`, then the number of format tuples supported for the given `surface` is returned in `pSurfaceFormatCount`. Otherwise, `pSurfaceFormatCount` **must** point to a variable set by the application to the number of elements in the `pSurfaceFormats` array, and on return the variable is overwritten with the number of structures actually written to `pSurfaceFormats`. If the value of `pSurfaceFormatCount` is less than the number of format tuples supported, at most `pSurfaceFormatCount` structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available values were returned.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfaceFormats2KHR-pSurfaceInfo-06521
  `pSurfaceInfo->surface` **must** be a valid VkSurfaceKHR handle
- VUID-vkGetPhysicalDeviceSurfaceFormats2KHR-pSurfaceInfo-06522
  `pSurfaceInfo->surface` **must** be supported by `physicalDevice`, as reported by vkGetPhysicalDeviceSurfaceSupportKHR or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfaceFormats2KHR-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceSurfaceFormats2KHR-pSurfaceInfo-parameter
  `pSurfaceInfo` **must** be a valid pointer to a valid VkPhysicalDeviceSurfaceInfo2KHR structure
- VUID-vkGetPhysicalDeviceSurfaceFormats2KHR-pSurfaceFormatCount-parameter
  `pSurfaceFormatCount` **must** be a valid pointer to a `uint32_t` value

- VUID-vkGetPhysicalDeviceSurfaceFormats2KHR-pSurfaceFormats-parameter
  If the value referenced by `pSurfaceFormatCount` is not `0`, and `pSurfaceFormats` is not `NULL`, `pSurfaceFormats` **must** be a valid pointer to an array of `pSurfaceFormatCount` VkSurfaceFormat2KHR structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkSurfaceFormat2KHR` structure is defined as:

```
// Provided by VK_KHR_get_surface_capabilities2
typedef struct VkSurfaceFormat2KHR {
    VkStructureType       sType;
    void*                 pNext;
    VkSurfaceFormatKHR    surfaceFormat;
} VkSurfaceFormat2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `surfaceFormat` is a VkSurfaceFormatKHR structure describing a format-color space pair that is compatible with the specified surface.

If the `imageCompressionControlSwapchain` feature is supported and a VkImageCompressionPropertiesEXT structure is included in the `pNext` chain of this structure, then it will be filled with the compression properties that are supported for the `surfaceFormat`.

Valid Usage

- VUID-VkSurfaceFormat2KHR-pNext-06750
  If the `VK_EXT_image_compression_control_swapchain` extension is not supported, the `pNext` chain **must** not include an VkImageCompressionPropertiesEXT structure

Valid Usage (Implicit)

- VUID-VkSurfaceFormat2KHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SURFACE_FORMAT_2_KHR
- VUID-VkSurfaceFormat2KHR-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkImageCompressionPropertiesEXT
- VUID-VkSurfaceFormat2KHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

While the `format` of a presentable image refers to the encoding of each pixel, the `colorSpace` determines how the presentation engine interprets the pixel values. A color space in this document refers to a specific color space (defined by the chromaticities of its primaries and a white point in CIE Lab), and transfer functions indicating the mapping between the image data and the colorimetry with respect to the given color space.

Possible values of VkSurfaceFormatKHR::`colorSpace`, specifying the color spaces that a presentation engine can accept, are:

```
// Provided by VK_KHR_surface
typedef enum VkColorSpaceKHR {
    VK_COLOR_SPACE_SRGB_NONLINEAR_KHR = 0,
  // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_DISPLAY_P3_NONLINEAR_EXT = 1000104001,
  // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_EXTENDED_SRGB_LINEAR_EXT = 1000104002,
  // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_DISPLAY_P3_LINEAR_EXT = 1000104003,
  // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_DCI_P3_NONLINEAR_EXT = 1000104004,
  // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_BT709_LINEAR_EXT = 1000104005,
  // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_BT709_NONLINEAR_EXT = 1000104006,
  // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_BT2020_LINEAR_EXT = 1000104007,
  // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_HDR10_ST2084_EXT = 1000104008,
  // Provided by VK_EXT_swapchain_colorspace
  // VK_COLOR_SPACE_DOLBYVISION_EXT is legacy, but no reason was given in the API XML
    VK_COLOR_SPACE_DOLBYVISION_EXT = 1000104009,
  // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_HDR10_HLG_EXT = 1000104010,
  // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_ADOBERGB_LINEAR_EXT = 1000104011,
  // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_ADOBERGB_NONLINEAR_EXT = 1000104012,
  // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_PASS_THROUGH_EXT = 1000104013,
  // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_EXTENDED_SRGB_NONLINEAR_EXT = 1000104014,
  // VK_COLORSPACE_SRGB_NONLINEAR_KHR is a legacy alias
    VK_COLORSPACE_SRGB_NONLINEAR_KHR = VK_COLOR_SPACE_SRGB_NONLINEAR_KHR,
  // Provided by VK_EXT_swapchain_colorspace
  // VK_COLOR_SPACE_DCI_P3_LINEAR_EXT is a legacy alias
    VK_COLOR_SPACE_DCI_P3_LINEAR_EXT = VK_COLOR_SPACE_DISPLAY_P3_LINEAR_EXT,
} VkColorSpaceKHR;
```

- VK_COLOR_SPACE_SRGB_NONLINEAR_KHR specifies support for the images in sRGB color space, encoded according to the sRGB specification.
- VK_COLOR_SPACE_DISPLAY_P3_NONLINEAR_EXT specifies support for the images in Display-P3 color space, encoded using a Display-P3 transfer function.
- VK_COLOR_SPACE_EXTENDED_SRGB_LINEAR_EXT specifies support for the images in extended sRGB color space, encoded using a linear transfer function.
- VK_COLOR_SPACE_EXTENDED_SRGB_NONLINEAR_EXT specifies support for the images in extended sRGB color space, encoded according to the scRGB specification.

- VK_COLOR_SPACE_DISPLAY_P3_LINEAR_EXT specifies support for the images in Display-P3 color space, encoded using a linear transfer function.
- VK_COLOR_SPACE_DCI_P3_NONLINEAR_EXT specifies support for the images in DCI-P3 color space, encoded according to the DCI-P3 specification. Note that values in such an image are interpreted as XYZ encoded color data by the presentation engine.
- VK_COLOR_SPACE_BT709_LINEAR_EXT specifies support for the images in BT709 color space, encoded using a linear transfer function.
- VK_COLOR_SPACE_BT709_NONLINEAR_EXT specifies support for the images in BT709 color space, encoded according to the BT709 specification.
- VK_COLOR_SPACE_BT2020_LINEAR_EXT specifies support for the images in BT2020 color space, encoded using a linear transfer function.
- VK_COLOR_SPACE_HDR10_ST2084_EXT specifies support for the images in HDR10 (BT2020) color space, encoded according to SMPTE ST2084 Perceptual Quantizer (PQ) specification.
- VK_COLOR_SPACE_HDR10_HLG_EXT specifies support for the images in HDR10 (BT2020) color space, encoded according to the Hybrid Log Gamma (HLG) specification.
- VK_COLOR_SPACE_ADOBERGB_LINEAR_EXT specifies support for images in Adobe RGB color space, encoded using a linear transfer function.
- VK_COLOR_SPACE_ADOBERGB_NONLINEAR_EXT specifies support for the images in Adobe RGB color space, encoded according to the Adobe RGB specification (approximately Gamma 2.2).
- VK_COLOR_SPACE_PASS_THROUGH_EXT specifies that color components are used “as is”. This is intended to allow applications to supply data for color spaces not described here.

NOTE

In the initial release of the VK_KHR_surface and VK_KHR_swapchain extensions, the token VK_COLORSPACE_SRGB_NONLINEAR_KHR was used. Starting in the 2016-05-13 updates to the extension branches, matching release 1.0.13 of the core API specification, VK_COLOR_SPACE_SRGB_NONLINEAR_KHR is used instead for consistency with Vulkan naming rules. The older enum is still available for backwards compatibility.

NOTE

In older versions of this extension VK_COLOR_SPACE_DISPLAY_P3_LINEAR_EXT was misnamed VK_COLOR_SPACE_DCI_P3_LINEAR_EXT. This has been updated to indicate that it uses RGB color encoding, not XYZ. The old name is legacy but is maintained for backwards compatibility.

NOTE

In older versions of the VK_EXT_swapchain_colorspace extension, VK_COLOR_SPACE_DOLBYVISION_EXT was exposed. The intent was to indicate the presentation engine shall decode an image using the SMPTE ST 2084 Perceptual Quantizer (PQ) EOTF, and then apply a proprietary OOTF to process the image. However, Dolby Vision profile 8.4 describes an encoding using the Hybrid Log Gamma (HLG) OETF, and there is no swapchain extension for signaling Dolby Vision metadata to be used by a proprietary OOTF. This enum is legacy but is maintained for backwards compatibility.

NOTE For a traditional “Linear” or non-gamma transfer function color space use VK_COLOR_SPACE_PASS_THROUGH_EXT.

NOTE On Wayland, VK_COLOR_SPACE_PASS_THROUGH_EXT can be used to disable color management by the WSI on a surface, which makes it possible for the application to create a `wp_color_management_surface_v1` object without triggering a `surface_exists` protocol error.

See vkCreateWaylandSurfaceKHR.

The presentation engine interprets the pixel values of the R, G, and B components as having been encoded using an appropriate transfer function. Applications **should** ensure that the appropriate transfer function has been applied. Texel encode requires that all implementations implicitly apply the sRGB $EOTF^{-1}$ on R, G, and B components when shaders write to an sRGB pixel format image, which is useful for sRGB color spaces. For sRGB color spaces with other pixel formats, or other non-linear color spaces, applications **can** apply the transfer function explicitly in a shader. The A channel is always interpreted as linearly encoded.

This extension defines enums for VkColorSpaceKHR that correspond to the following color spaces:

Table 50. Color Spaces and Attributes

Name	Red Primary	Green Primary	Blue Primary	White-point	Transfer function
DCI-P3	1.000, 0.000	0.000, 1.000	0.000, 0.000	0.3333, 0.3333	DCI P3
Display-P3	0.680, 0.320	0.265, 0.690	0.150, 0.060	0.3127, 0.3290 (D65)	Display-P3
BT709	0.640, 0.330	0.300, 0.600	0.150, 0.060	0.3127, 0.3290 (D65)	BT709
sRGB	0.640, 0.330	0.300, 0.600	0.150, 0.060	0.3127, 0.3290 (D65)	sRGB
extended sRGB	0.640, 0.330	0.300, 0.600	0.150, 0.060	0.3127, 0.3290 (D65)	scRGB
HDR10_ST2084	0.708, 0.292	0.170, 0.797	0.131, 0.046	0.3127, 0.3290 (D65)	ST2084 PQ
HDR10_HLG	0.708, 0.292	0.170, 0.797	0.131, 0.046	0.3127, 0.3290 (D65)	HLG
Adobe RGB	0.640, 0.330	0.210, 0.710	0.150, 0.060	0.3127, 0.3290 (D65)	Adobe RGB

The transfer functions are described in the “Transfer Functions” chapter of the Khronos Data Format Specification.

Except Display-P3 OETF, which is:

$$E = \begin{cases} 1.055 \times L^{\frac{1}{2.4}} - 0.055 & \text{for } 0.0030186 \leq L \leq 1 \\ 12.92 \times L & \text{for } 0 \leq L < 0.0030186 \end{cases}$$

where L is the linear value of a color component and E is the encoded value (as stored in the image in memory).

NOTE For most uses, the sRGB OETF is equivalent.

37.5.3. Surface Presentation Mode Support

To query the supported presentation modes for a surface, call:

```
// Provided by VK_KHR_surface
VkResult vkGetPhysicalDeviceSurfacePresentModesKHR(
    VkPhysicalDevice                            physicalDevice,
    VkSurfaceKHR                                surface,
    uint32_t*                                   pPresentModeCount,
    VkPresentModeKHR*                           pPresentModes);
```

- `physicalDevice` is the physical device that will be associated with the swapchain to be created. See vkCreateSwapchainKHR for details of swapchain creation.
- `surface` is the surface that will be associated with the swapchain.
- `pPresentModeCount` is a pointer to an integer related to the number of presentation modes available or queried, as described below.
- `pPresentModes` is either `NULL` or a pointer to an array of VkPresentModeKHR values, indicating the supported presentation modes.

If `pPresentModes` is `NULL`, then the number of presentation modes supported for the given `surface` is returned in `pPresentModeCount`. Otherwise, `pPresentModeCount` **must** point to a variable set by the application to the number of elements in the `pPresentModes` array, and on return the variable is overwritten with the number of values actually written to `pPresentModes`. If the value of `pPresentModeCount` is less than the number of presentation modes supported, at most `pPresentModeCount` values will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available modes were returned.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfacePresentModesKHR-surface-06524
  `surface` **must** be a valid VkSurfaceKHR handle
- VUID-vkGetPhysicalDeviceSurfacePresentModesKHR-surface-06525
  `surface` **must** be supported by `physicalDevice`, as reported by vkGetPhysicalDeviceSurfaceSupportKHR or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfacePresentModesKHR-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceSurfacePresentModesKHR-surface-parameter
  If `surface` is not VK_NULL_HANDLE, `surface` **must** be a valid VkSurfaceKHR handle
- VUID-vkGetPhysicalDeviceSurfacePresentModesKHR-pPresentModeCount-parameter
  `pPresentModeCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceSurfacePresentModesKHR-pPresentModes-parameter
  If the value referenced by `pPresentModeCount` is not `0`, and `pPresentModes` is not `NULL`, `pPresentModes` **must** be a valid pointer to an array of `pPresentModeCount` VkPresentModeKHR values
- VUID-vkGetPhysicalDeviceSurfacePresentModesKHR-commonparent
  Both of `physicalDevice`, and `surface` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkInstance

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To query the supported presentation modes for a surface combined with select other fixed swapchain creation parameters, call:

```
// Provided by VK_EXT_full_screen_exclusive
VkResult vkGetPhysicalDeviceSurfacePresentModes2EXT(
    VkPhysicalDevice                            physicalDevice,
    const VkPhysicalDeviceSurfaceInfo2KHR*      pSurfaceInfo,
    uint32_t*                                   pPresentModeCount,
    VkPresentModeKHR*                           pPresentModes);
```

- `physicalDevice` is the physical device that will be associated with the swapchain to be created, as described for vkCreateSwapchainKHR.

- `pSurfaceInfo` is a pointer to a VkPhysicalDeviceSurfaceInfo2KHR structure describing the surface and other fixed parameters that would be consumed by vkCreateSwapchainKHR.
- `pPresentModeCount` is a pointer to an integer related to the number of presentation modes available or queried, as described below.
- `pPresentModes` is either `NULL` or a pointer to an array of VkPresentModeKHR values, indicating the supported presentation modes.

`vkGetPhysicalDeviceSurfacePresentModes2EXT` behaves similarly to vkGetPhysicalDeviceSurfacePresentModesKHR, with the ability to specify extended inputs via chained input structures.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfacePresentModes2EXT-pSurfaceInfo-06521
  `pSurfaceInfo->surface` **must** be a valid VkSurfaceKHR handle
- VUID-vkGetPhysicalDeviceSurfacePresentModes2EXT-pSurfaceInfo-06522
  `pSurfaceInfo->surface` **must** be supported by `physicalDevice`, as reported by vkGetPhysicalDeviceSurfaceSupportKHR or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfacePresentModes2EXT-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceSurfacePresentModes2EXT-pSurfaceInfo-parameter
  `pSurfaceInfo` **must** be a valid pointer to a valid VkPhysicalDeviceSurfaceInfo2KHR structure
- VUID-vkGetPhysicalDeviceSurfacePresentModes2EXT-pPresentModeCount-parameter
  `pPresentModeCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceSurfacePresentModes2EXT-pPresentModes-parameter
  If the value referenced by `pPresentModeCount` is not `0`, and `pPresentModes` is not `NULL`, `pPresentModes` **must** be a valid pointer to an array of `pPresentModeCount` VkPresentModeKHR values

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY

- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Possible values of elements of the vkGetPhysicalDeviceSurfacePresentModesKHR::pPresentModes array, indicating the supported presentation modes for a surface, are:

```
// Provided by VK_KHR_surface
typedef enum VkPresentModeKHR {
    VK_PRESENT_MODE_IMMEDIATE_KHR = 0,
    VK_PRESENT_MODE_MAILBOX_KHR = 1,
    VK_PRESENT_MODE_FIFO_KHR = 2,
    VK_PRESENT_MODE_FIFO_RELAXED_KHR = 3,
  // Provided by VK_KHR_shared_presentable_image
    VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR = 1000111000,
  // Provided by VK_KHR_shared_presentable_image
    VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR = 1000111001,
  // Provided by VK_KHR_present_mode_fifo_latest_ready
    VK_PRESENT_MODE_FIFO_LATEST_READY_KHR = 1000361000,
  // Provided by VK_EXT_present_mode_fifo_latest_ready
    VK_PRESENT_MODE_FIFO_LATEST_READY_EXT = VK_PRESENT_MODE_FIFO_LATEST_READY_KHR,
} VkPresentModeKHR;
```

- VK_PRESENT_MODE_IMMEDIATE_KHR specifies that the presentation engine does not wait for a vertical blanking period to update the current image, meaning this mode **may** result in visible tearing. No internal queuing of presentation requests is needed, as the requests are applied immediately.
- VK_PRESENT_MODE_MAILBOX_KHR specifies that the presentation engine waits for the next vertical blanking period to update the current image. Tearing **cannot** be observed. An internal single-entry queue is used to hold pending presentation requests. If the queue is full when a new presentation request is received, the new request replaces the existing entry, and any images associated with the prior entry become available for reuse by the application. One request is removed from the queue and processed during each vertical blanking period in which the queue is non-empty.
- VK_PRESENT_MODE_FIFO_KHR specifies that the presentation engine waits for the next vertical blanking period to update the current image. Tearing **cannot** be observed. An internal queue is used to hold pending presentation requests. New requests are appended to the end of the queue, and one request is removed from the beginning of the queue and processed during each vertical blanking period in which the queue is non-empty. This is the only value of presentMode that is **required** to be supported.
- VK_PRESENT_MODE_FIFO_RELAXED_KHR specifies that the presentation engine generally waits for the next vertical blanking period to update the current image. If a vertical blanking period has already passed since the last update of the current image then the presentation engine does not wait for another vertical blanking period for the update, meaning this mode **may** result in visible tearing in this case. This mode is useful for reducing visual stutter with an application

that will mostly present a new image before the next vertical blanking period, but may occasionally be late, and present a new image just after the next vertical blanking period. An internal queue is used to hold pending presentation requests. New requests are appended to the end of the queue, and one request is removed from the beginning of the queue and processed during or after each vertical blanking period in which the queue is non-empty.

- VK_PRESENT_MODE_FIFO_LATEST_READY_KHR specifies that the presentation engine waits for the next vertical blanking period to update the current image. Tearing **cannot** be observed. An internal queue is used to hold pending presentation requests. New requests are appended to the end of the queue. At each vertical blanking period, the presentation engine dequeues all successive requests that are ready to be presented from the beginning of the queue. If using the `presentAtAbsoluteTime` feature to provide a target present time, the presentation engine checks the specified time for each image. If the target present time is less-than or equal-to the current time, the presentation engine dequeues the image and checks the next one. The image of the last dequeued request is presented. The other dequeued requests are dropped.
- VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR specifies that the presentation engine and application have concurrent access to a single image, which is referred to as a *shared presentable image*. The presentation engine is only required to update the current image after a new presentation request is received. Therefore the application **must** make a presentation request whenever an update is required. However, the presentation engine **may** update the current image at any point, meaning this mode **may** result in visible tearing.
- VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR specifies that the presentation engine and application have concurrent access to a single image, which is referred to as a *shared presentable image*. The presentation engine periodically updates the current image on its regular refresh cycle. The application is only required to make one initial presentation request, after which the presentation engine **must** update the current image without any need for further presentation requests. The application **can** indicate the image contents have been updated by making a presentation request, but this does not guarantee the timing of when it will be updated. This mode **may** result in visible tearing if rendering to the image is not timed correctly.

The supported VkImageUsageFlagBits of the presentable images of a swapchain created for a surface **may** differ depending on the presentation mode, and can be determined as per the table below:

Table 51. Presentable Image Usage Queries

Presentation mode	Image usage flags
VK_PRESENT_MODE_IMMEDIATE_KHR	VkSurfaceCapabilitiesKHR::`supportedUsageFlags`
VK_PRESENT_MODE_MAILBOX_KHR	VkSurfaceCapabilitiesKHR::`supportedUsageFlags`
VK_PRESENT_MODE_FIFO_KHR	VkSurfaceCapabilitiesKHR::`supportedUsageFlags`
VK_PRESENT_MODE_FIFO_RELAXED_KHR	VkSurfaceCapabilitiesKHR::`supportedUsageFlags`
VK_PRESENT_MODE_FIFO_LATEST_READY_KHR	VkSurfaceCapabilitiesKHR::`supportedUsageFlags`
VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR	VkSharedPresentSurfaceCapabilitiesKHR::`sharedPresentSupportedUsageFlags`

Presentation mode	Image usage flags
VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR	VkSharedPresentSurfaceCapabilitiesKHR::sharedPresentSupportedUsageFlags

NOTE For reference, the mode indicated by VK_PRESENT_MODE_FIFO_KHR is equivalent to the behavior of {wgl|glX|egl}SwapBuffers with a swap interval of 1, while the mode indicated by VK_PRESENT_MODE_FIFO_RELAXED_KHR is equivalent to the behavior of {wgl|glX}SwapBuffers with a swap interval of -1 (from the {WGL|GLX}_EXT_swap_control_tear extensions).

37.6. Full Screen Exclusive Control

Swapchains created with `fullScreenExclusive` set to VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT **must** acquire and release exclusive full-screen access explicitly, using the following commands.

To acquire exclusive full-screen access for a swapchain, call:

```
// Provided by VK_EXT_full_screen_exclusive
VkResult vkAcquireFullScreenExclusiveModeEXT(
    VkDevice                                    device,
    VkSwapchainKHR                              swapchain);
```

- `device` is the device associated with `swapchain`.
- `swapchain` is the swapchain to acquire exclusive full-screen access for.

Valid Usage

- VUID-vkAcquireFullScreenExclusiveModeEXT-swapchain-02674
  `swapchain` **must** not be in the retired state
- VUID-vkAcquireFullScreenExclusiveModeEXT-swapchain-02675
  `swapchain` **must** be a swapchain created with a VkSurfaceFullScreenExclusiveInfoEXT structure, with `fullScreenExclusive` set to VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT
- VUID-vkAcquireFullScreenExclusiveModeEXT-swapchain-02676
  `swapchain` **must** not currently have exclusive full-screen access

A return value of VK_SUCCESS indicates that the `swapchain` successfully acquired exclusive full-screen access. The swapchain will retain this exclusivity until either the application releases exclusive full-screen access with vkReleaseFullScreenExclusiveModeEXT, destroys the swapchain, or if any of the swapchain commands return VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT indicating that the mode was lost because of platform-specific changes.

If the swapchain was unable to acquire exclusive full-screen access to the display then VK_ERROR_INITIALIZATION_FAILED is returned. An application **can** attempt to acquire exclusive full-screen access again for the same swapchain even if this command fails, or if VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT has been returned by a swapchain command.

Valid Usage (Implicit)

- VUID-vkAcquireFullScreenExclusiveModeEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkAcquireFullScreenExclusiveModeEXT-swapchain-parameter
  `swapchain` **must** be a valid VkSwapchainKHR handle
- VUID-vkAcquireFullScreenExclusiveModeEXT-swapchain-parent
  `swapchain` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To release exclusive full-screen access from a swapchain, call:

```
// Provided by VK_EXT_full_screen_exclusive
VkResult vkReleaseFullScreenExclusiveModeEXT(
    VkDevice                                    device,
    VkSwapchainKHR                              swapchain);
```

- `device` is the device associated with `swapchain`.
- `swapchain` is the swapchain to release exclusive full-screen access from.

NOTE Applications will not be able to present to `swapchain` after this call until exclusive full-screen access is reacquired. This is usually useful to handle when an application is minimized or otherwise intends to stop presenting for a time.

Valid Usage

- VUID-vkReleaseFullScreenExclusiveModeEXT-swapchain-02677
  `swapchain` **must** not be in the retired state
- VUID-vkReleaseFullScreenExclusiveModeEXT-swapchain-02678
  `swapchain` **must** be a swapchain created with a VkSurfaceFullScreenExclusiveInfoEXT structure, with `fullScreenExclusive` set to VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT

Valid Usage (Implicit)

- VUID-vkReleaseFullScreenExclusiveModeEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkReleaseFullScreenExclusiveModeEXT-swapchain-parameter
  `swapchain` **must** be a valid VkSwapchainKHR handle
- VUID-vkReleaseFullScreenExclusiveModeEXT-swapchain-parent
  `swapchain` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

37.7. Device Group Queries

A logical device that represents multiple physical devices **may** support presenting from images on more than one physical device, or combining images from multiple physical devices.

To query these capabilities, call:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_surface
VkResult vkGetDeviceGroupPresentCapabilitiesKHR(
    VkDevice                                    device,
    VkDeviceGroupPresentCapabilitiesKHR*        pDeviceGroupPresentCapabilities);
```

- `device` is the logical device.
- `pDeviceGroupPresentCapabilities` is a pointer to a VkDeviceGroupPresentCapabilitiesKHR structure in which the device's capabilities are returned.

Valid Usage (Implicit)

- VUID-vkGetDeviceGroupPresentCapabilitiesKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDeviceGroupPresentCapabilitiesKHR-pDeviceGroupPresentCapabilities-parameter
  `pDeviceGroupPresentCapabilities` **must** be a valid pointer to a VkDeviceGroupPresentCapabilitiesKHR structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkDeviceGroupPresentCapabilitiesKHR` structure is defined as:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_surface
typedef struct VkDeviceGroupPresentCapabilitiesKHR {
    VkStructureType                      sType;
    void*                                pNext;
    uint32_t                             presentMask[VK_MAX_DEVICE_GROUP_SIZE];
    VkDeviceGroupPresentModeFlagsKHR     modes;
} VkDeviceGroupPresentCapabilitiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `presentMask` is an array of VK_MAX_DEVICE_GROUP_SIZE `uint32_t` masks, where the mask at element i is non-zero if physical device i has a presentation engine, and where bit j is set in element i if physical device i **can** present swapchain images from physical device j. If element i is non-zero, then bit i **must** be set.
- `modes` is a bitmask of VkDeviceGroupPresentModeFlagBitsKHR indicating which device group presentation modes are supported.

`modes` always has VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR set.

The present mode flags are also used when presenting an image, in VkDeviceGroupPresentInfoKHR::`mode`.

If a device group only includes a single physical device, then `modes` **must** equal VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR.

Valid Usage (Implicit)

- VUID-VkDeviceGroupPresentCapabilitiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_CAPABILITIES_KHR
- VUID-VkDeviceGroupPresentCapabilitiesKHR-pNext-pNext
  `pNext` **must** be `NULL`

Bits which **may** be set in VkDeviceGroupPresentCapabilitiesKHR::`modes`, indicating which device group presentation modes are supported, are:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_surface
typedef enum VkDeviceGroupPresentModeFlagBitsKHR {
    VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR = 0x00000001,
    VK_DEVICE_GROUP_PRESENT_MODE_REMOTE_BIT_KHR = 0x00000002,
    VK_DEVICE_GROUP_PRESENT_MODE_SUM_BIT_KHR = 0x00000004,
    VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_MULTI_DEVICE_BIT_KHR = 0x00000008,
} VkDeviceGroupPresentModeFlagBitsKHR;
```

- VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR specifies that any physical device with a presentation engine **can** present its own swapchain images.
- VK_DEVICE_GROUP_PRESENT_MODE_REMOTE_BIT_KHR specifies that any physical device with a presentation engine **can** present swapchain images from any physical device in its `presentMask`.
- VK_DEVICE_GROUP_PRESENT_MODE_SUM_BIT_KHR specifies that any physical device with a presentation engine **can** present the sum of swapchain images from any physical devices in its `presentMask`.
- VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_MULTI_DEVICE_BIT_KHR specifies that multiple physical devices with a presentation engine **can** each present their own swapchain images.

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_surface
typedef VkFlags VkDeviceGroupPresentModeFlagsKHR;
```

VkDeviceGroupPresentModeFlagsKHR is a bitmask type for setting a mask of zero or more VkDeviceGroupPresentModeFlagBitsKHR.

Some surfaces **may** not be capable of using all the device group present modes.

To query the supported device group present modes for a particular surface, call:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_surface
VkResult vkGetDeviceGroupSurfacePresentModesKHR(
    VkDevice                                    device,
    VkSurfaceKHR                                surface,
    VkDeviceGroupPresentModeFlagsKHR*           pModes);
```

- device is the logical device.
- surface is the surface.
- pModes is a pointer to a VkDeviceGroupPresentModeFlagsKHR in which the supported device group present modes for the surface are returned.

The modes returned by this command are not invariant, and **may** change in response to the surface being moved, resized, or occluded. These modes **must** be a subset of the modes returned by vkGetDeviceGroupPresentCapabilitiesKHR.

Valid Usage

- VUID-vkGetDeviceGroupSurfacePresentModesKHR-surface-06212
  surface **must** be supported by all physical devices associated with device, as reported by vkGetPhysicalDeviceSurfaceSupportKHR or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetDeviceGroupSurfacePresentModesKHR-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkGetDeviceGroupSurfacePresentModesKHR-surface-parameter
  surface **must** be a valid VkSurfaceKHR handle
- VUID-vkGetDeviceGroupSurfacePresentModesKHR-pModes-parameter
  pModes **must** be a valid pointer to a VkDeviceGroupPresentModeFlagsKHR value
- VUID-vkGetDeviceGroupSurfacePresentModesKHR-commonparent
  Both of device, and surface **must** have been created, allocated, or retrieved from the same

VkInstance

Host Synchronization

- Host access to `surface` **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To query the supported device group presentation modes for a surface combined with select other fixed swapchain creation parameters, call:

```
// Provided by VK_EXT_full_screen_exclusive with VK_KHR_device_group or VK_VERSION_1_1
VkResult vkGetDeviceGroupSurfacePresentModes2EXT(
    VkDevice                                    device,
    const VkPhysicalDeviceSurfaceInfo2KHR*      pSurfaceInfo,
    VkDeviceGroupPresentModeFlagsKHR*           pModes);
```

- `device` is the logical device.
- `pSurfaceInfo` is a pointer to a VkPhysicalDeviceSurfaceInfo2KHR structure describing the surface and other fixed parameters that would be consumed by vkCreateSwapchainKHR.
- `pModes` is a pointer to a VkDeviceGroupPresentModeFlagsKHR in which the supported device group present modes for the surface are returned.

`vkGetDeviceGroupSurfacePresentModes2EXT` behaves similarly to vkGetDeviceGroupSurfacePresentModesKHR, with the ability to specify extended inputs via chained input structures.

Valid Usage

- VUID-vkGetDeviceGroupSurfacePresentModes2EXT-pSurfaceInfo-06213
  `pSurfaceInfo->surface` **must** be supported by all physical devices associated with `device`, as reported by vkGetPhysicalDeviceSurfaceSupportKHR or an equivalent platform-

specific mechanism

Valid Usage (Implicit)

- VUID-vkGetDeviceGroupSurfacePresentModes2EXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDeviceGroupSurfacePresentModes2EXT-pSurfaceInfo-parameter
  `pSurfaceInfo` **must** be a valid pointer to a valid VkPhysicalDeviceSurfaceInfo2KHR structure
- VUID-vkGetDeviceGroupSurfacePresentModes2EXT-pModes-parameter
  `pModes` **must** be a valid pointer to a VkDeviceGroupPresentModeFlagsKHR value

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

When using VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_MULTI_DEVICE_BIT_KHR, the application **may** need to know which regions of the surface are used when presenting locally on each physical device. Presentation of swapchain images to this surface need only have valid contents in the regions returned by this command.

To query a set of rectangles used in presentation on the physical device, call:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_surface
VkResult vkGetPhysicalDevicePresentRectanglesKHR(
    VkPhysicalDevice                            physicalDevice,
    VkSurfaceKHR                                surface,
    uint32_t*                                   pRectCount,
    VkRect2D*                                   pRects);
```

- `physicalDevice` is the physical device.
- `surface` is the surface.
- `pRectCount` is a pointer to an integer related to the number of rectangles available or queried, as

described below.

- pRects is either NULL or a pointer to an array of VkRect2D structures.

If pRects is NULL, then the number of rectangles used when presenting the given surface is returned in pRectCount. Otherwise, pRectCount **must** point to a variable set by the application to the number of elements in the pRects array, and on return the variable is overwritten with the number of structures actually written to pRects. If the value of pRectCount is less than the number of rectangles, at most pRectCount structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available rectangles were returned.

The values returned by this command are not invariant, and **may** change in response to the surface being moved, resized, or occluded.

The rectangles returned by this command **must** not overlap.

Valid Usage

- VUID-vkGetPhysicalDevicePresentRectanglesKHR-surface-06211
  surface **must** be supported by physicalDevice, as reported by vkGetPhysicalDeviceSurfaceSupportKHR or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetPhysicalDevicePresentRectanglesKHR-physicalDevice-parameter
  physicalDevice **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDevicePresentRectanglesKHR-surface-parameter
  surface **must** be a valid VkSurfaceKHR handle
- VUID-vkGetPhysicalDevicePresentRectanglesKHR-pRectCount-parameter
  pRectCount **must** be a valid pointer to a uint32_t value
- VUID-vkGetPhysicalDevicePresentRectanglesKHR-pRects-parameter
  If the value referenced by pRectCount is not 0, and pRects is not NULL, pRects **must** be a valid pointer to an array of pRectCount VkRect2D structures
- VUID-vkGetPhysicalDevicePresentRectanglesKHR-commonparent
  Both of physicalDevice, and surface **must** have been created, allocated, or retrieved from the same VkInstance

Host Synchronization

- Host access to surface **must** be externally synchronized

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

37.8. Present Timing Queries

Traditional game and real-time-animation applications frequently use VK_PRESENT_MODE_FIFO_KHR so that presentable images are updated during the vertical blanking period of a given refresh cycle (RC) of the presentation engine's display. On fixed refresh rate displays, this avoids the visual anomaly known as tearing.

However, synchronizing the presentation of images with the RC does not prevent all forms of visual anomalies. Stuttering occurs when the geometry for each presentable image is not accurately positioned for when that image will be displayed. The geometry may appear to move too little some RCs, and too much for others. Sometimes the animation appears to freeze, when the same image is used for more RCs than other images.

In order to minimize stuttering, an application needs to: 1) render and present images at a consistent rate that is, on fixed refresh rate displays, a multiple of the presentation engine's refresh rate; 2) correctly position its geometry for when the presentable image will be displayed to the user. The `VK_EXT_present_timing` extension allows an application to satisfy these needs.

The presentation engine's display typically refreshes the pixels that are displayed to the user on a periodic basis. This period **may** be fixed (Fixed Refresh Rate, FRR) or variable (Variable Refresh Rate, VRR).

In order to collect timing information about presentation, a swapchain needs an internal queue to store asynchronously updated results until applications collect them.

To allocate the swapchain's internal timing results queue, call:

```
// Provided by VK_EXT_present_timing
VkResult vkSetSwapchainPresentTimingQueueSizeEXT(
    VkDevice                                    device,
    VkSwapchainKHR                              swapchain,
    uint32_t                                    size);
```

- `device` is the device associated with `swapchain`.
- `swapchain` is the swapchain to allocate a results queue for.
- `size` is the requested number of slots in the internal results queue.

If this function is called multiple times, the internal queue is reallocated to fit the new `size`. If the new `size` is less than the current number of outstanding results, VK_NOT_READY is returned and no allocation is performed.

Valid Usage

- VUID-vkSetSwapchainPresentTimingQueueSizeEXT-swapchain-12229
  `swapchain` **must** have been created with `VkSwapchainCreateInfoKHR`::`flags` containing VK_SWAPCHAIN_CREATE_PRESENT_TIMING_BIT_EXT

Valid Usage (Implicit)

- VUID-vkSetSwapchainPresentTimingQueueSizeEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkSetSwapchainPresentTimingQueueSizeEXT-swapchain-parameter
  `swapchain` **must** be a valid VkSwapchainKHR handle
- VUID-vkSetSwapchainPresentTimingQueueSizeEXT-swapchain-parent
  `swapchain` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `swapchain` **must** be externally synchronized

Return Codes

Success

- VK_NOT_READY
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The implementation maintains an internal monotonically increasing counter which updates when the presentation engine's timing properties are modified.

To query the presentation engine's current timing properties for a given swapchain, call:

```
// Provided by VK_EXT_present_timing
VkResult vkGetSwapchainTimingPropertiesEXT(
    VkDevice                                    device,
    VkSwapchainKHR                              swapchain,
    VkSwapchainTimingPropertiesEXT*             pSwapchainTimingProperties,
    uint64_t*                                   pSwapchainTimingPropertiesCounter);
```

- `device` is the device associated with `swapchain`.
- `swapchain` is the swapchain to obtain timing properties for.
- `pSwapchainTimingProperties` is a pointer to an instance of the VkSwapchainTimingPropertiesEXT structure.
- `pSwapchainTimingPropertiesCounter` is `NULL` or a pointer to a 64-bit unsigned integer set by the implementation to the current value of the swapchain's internal timing properties counter.

If `vkGetSwapchainTimingPropertiesEXT` returns VK_NOT_READY, the implementation was not able to determine the current refresh cycle duration. Some platforms **may** not provide timing properties until after at least one image has been presented to the `swapchain`. If timing properties change for the `swapchain`, these platforms **may** not provide updated results until after at least one additional image has been presented to the `swapchain`.

Valid Usage (Implicit)

- VUID-vkGetSwapchainTimingPropertiesEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetSwapchainTimingPropertiesEXT-swapchain-parameter
  `swapchain` **must** be a valid VkSwapchainKHR handle
- VUID-vkGetSwapchainTimingPropertiesEXT-pSwapchainTimingProperties-parameter
  `pSwapchainTimingProperties` **must** be a valid pointer to a VkSwapchainTimingPropertiesEXT structure
- VUID-vkGetSwapchainTimingPropertiesEXT-pSwapchainTimingPropertiesCounter-parameter
  If `pSwapchainTimingPropertiesCounter` is not `NULL`, `pSwapchainTimingPropertiesCounter` **must** be a valid pointer to a `uint64_t` value
- VUID-vkGetSwapchainTimingPropertiesEXT-swapchain-parent
  `swapchain` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `swapchain` **must** be externally synchronized

Return Codes

Success

- VK_NOT_READY
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkSwapchainTimingPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_present_timing
typedef struct VkSwapchainTimingPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint64_t           refreshDuration;
    uint64_t           refreshInterval;
} VkSwapchainTimingPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `refreshDuration` is zero or an indication of the duration of a refresh cycle.
- `refreshInterval` is zero or a duration in nanoseconds indicating the interval between refresh cycle durations.

If `refreshDuration` is zero, the presentation engine is not able to determine the duration of the refresh cycle. Similarly, if `refreshInterval` is zero, the presentation engine is not able to determine whether it is operating in VRR mode.

Otherwise, if `refreshInterval` is the same as `refreshDuration`, the presentation engine is operating in FRR mode. In this case, `refreshDuration` is the number of nanoseconds from the start of one refresh cycle to the start of the next refresh cycle.

If `refreshInterval` is `UINT64_MAX`, the presentation engine is operating in VRR mode, and `refreshDuration` is the minimum number of nanoseconds from the start of one refresh cycle to the start of the next refresh cycle.

If `refreshDuration` and `refreshInterval` are not zero, `refreshInterval` is a factor of `refreshDuration`.

Valid Usage (Implicit)

- VUID-VkSwapchainTimingPropertiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SWAPCHAIN_TIMING_PROPERTIES_EXT
- VUID-VkSwapchainTimingPropertiesEXT-pNext-pNext
  `pNext` **must** be `NULL`

NOTE

Some display technologies allow the vblank interval to be decoupled from the display's effective refresh rate. In other words, presented images are visible for a fixed duration (`refreshDuration`) that is a multiple of the vblank interval (`refreshInterval`) which **may** be much smaller. This allows such displays, for example, to effectively operate at a fixed refresh rate with the ability to quickly change that refresh rate to adapt to the current system's workload.

Note that `VkSwapchainTimingPropertiesEXT` only represents the current swapchain's view of those capabilities, which **may** differ from the actual hardware capabilities, based e.g. on the current surface properties. For example, swapchains that render to a windowed surface requiring complex composition **may** not be able to reach their display hardware's full potential.

NOTE

The rate at which an application renders and presents new images is known as the image present rate (IPR, a.k.a. frame rate). The inverse of IPR, or the duration between each image present, is the image present duration (IPD).

In order to provide a smooth, stutter-free animation on non-VRR displays, an application needs its IPD to be a multiple of `refreshInterval` that is at least equal to `refreshDuration`.

For example, if a FRR display has a set 60Hz refresh rate, where `refreshDuration` is equal to `refreshInterval`, `refreshDuration` will be a value in nanoseconds that is approximately equal to 16.67ms. In such a case, an application will want an IPD of 16.67ms (1X multiplier of `refreshInterval`), or 33.33ms (2X multiplier of `refreshInterval`), or 50.0ms (3X multiplier of `refreshInterval`), etc.

In order to determine a target IPD for a display (i.e. a multiple of `refreshInterval`), an application needs to determine when its images are actually displayed.

Consider an application that has an initial target IPD of 16.67ms (1X multiplier of `refreshDuration`). It will therefore position the geometry of a new image 16.67ms later than the previous image.

If this application is running on slower hardware, so that it actually takes 20ms to render each new image, the images will not be displayed to the user every 16.67ms, nor every 20ms, which will create visual anomalies. In this case, it is better for the application to adjust its target IPD to 33.33ms (i.e. a 2X multiplier of `refreshDuration`), and tell the presentation engine to not present images any sooner than every 33.33ms. This will allow the geometry to be correctly positioned for each

> presentable image.
>
> On VRR displays, where `refreshInterval` is `UINT64_MAX`, applications **should** target an IPD that is at least equal to `refreshDuration`.
>
> Adjustments to an application's IPD may be needed because different views of an application's geometry can take different amounts of time to render. For example, looking at the sky may take less time to render than looking at multiple, complex items in a room.
>
> In general, it is good to not frequently change IPD, as that can cause visual anomalies.
>
> Adjustments to a larger IPD because of late images should happen quickly, but adjustments to a smaller IPD should only happen if the periodic feedback of VkPastPresentationTimingEXT values indicates that the target IPD can be durably achieved.

The implementation maintains an internal monotonically increasing counter which updates when the presentation engine's list of supported time domains for a swapchain is modified.

To query the time domains supported by the presentation engine for a given swapchain, call:

```
// Provided by VK_EXT_present_timing
VkResult vkGetSwapchainTimeDomainPropertiesEXT(
    VkDevice                                    device,
    VkSwapchainKHR                              swapchain,
    VkSwapchainTimeDomainPropertiesEXT*         pSwapchainTimeDomainProperties,
    uint64_t*                                   pTimeDomainsCounter);
```

- `device` is the device associated with `swapchain`.
- `swapchain` is the swapchain to obtain time domain properties for.
- `pSwapchainTimeDomainProperties` is a pointer to an instance of the VkSwapchainTimeDomainPropertiesEXT structure.
- `pTimeDomainsCounter` is `NULL` or a pointer to a 64-bit unsigned integer set by the implementation to the current value of the swapchain's internal time domain properties counter.

If upon return VkSwapchainTimeDomainPropertiesEXT::`timeDomainCount` is smaller than the number of time domains supported for the given `swapchain`, VK_INCOMPLETE will be returned instead of VK_SUCCESS to indicate that not all the available values were returned.

Valid Usage (Implicit)

- VUID-vkGetSwapchainTimeDomainPropertiesEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetSwapchainTimeDomainPropertiesEXT-swapchain-parameter
  `swapchain` **must** be a valid VkSwapchainKHR handle

- VUID-vkGetSwapchainTimeDomainPropertiesEXT-pSwapchainTimeDomainProperties-parameter
  `pSwapchainTimeDomainProperties` **must** be a valid pointer to a VkSwapchainTimeDomainPropertiesEXT structure
- VUID-vkGetSwapchainTimeDomainPropertiesEXT-pTimeDomainsCounter-parameter
  If `pTimeDomainsCounter` is not `NULL`, `pTimeDomainsCounter` **must** be a valid pointer to a `uint64_t` value
- VUID-vkGetSwapchainTimeDomainPropertiesEXT-swapchain-parent
  `swapchain` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `swapchain` **must** be externally synchronized

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkSwapchainTimeDomainPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_present_timing
typedef struct VkSwapchainTimeDomainPropertiesEXT {
    VkStructureType     sType;
    void*               pNext;
    uint32_t            timeDomainCount;
    VkTimeDomainKHR*    pTimeDomains;
    uint64_t*           pTimeDomainIds;
} VkSwapchainTimeDomainPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `timeDomainCount` is an integer related to the number of time domains available or queried, as

described below.

- pTimeDomains is a pointer to an array of VkTimeDomainKHR values representing time domains that are available for the swapchain.
- pTimeDomainIds is a pointer to an array of unique identifiers for each time domain.

When calling vkGetSwapchainTimeDomainPropertiesEXT, if pTimeDomains is NULL and pTimeDomainIds is NULL, then the number of time domains supported for the given swapchain is returned in timeDomainCount. Otherwise, timeDomainCount **must** specify the number of elements in pTimeDomains and pTimeDomainIds, and on return is overwritten with the number of values actually written to each array.

Implementations **must** advertise support for at least one time domain of VK_TIME_DOMAIN_PRESENT_STAGE_LOCAL_EXT.

NOTE

Due to the dynamic nature of their underlying VkSurfaceKHR properties, swapchains may need to expose multiple swapchain-local opaque time domains using the same VkTimeDomainKHR value over time, for example when a surface is moved from one display hardware to another. Arbitrary identifiers, provided in pTimeDomainIds, are used by the implementation to differentiate opaque time domains of identical scopes.

Valid Usage

- VUID-VkSwapchainTimeDomainPropertiesEXT-pTimeDomains-12370
  pTimeDomains and pTimeDomainIds **must** both be NULL or both not be NULL
- VUID-VkSwapchainTimeDomainPropertiesEXT-pTimeDomains-12371
  If pTimeDomains and pTimeDomainIds are not NULL, then timeDomainCount **must** not be zero

Valid Usage (Implicit)

- VUID-VkSwapchainTimeDomainPropertiesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_SWAPCHAIN_TIME_DOMAIN_PROPERTIES_EXT
- VUID-VkSwapchainTimeDomainPropertiesEXT-pNext-pNext
  pNext **must** be NULL

Because of the asynchronous nature of the presentation engine, the timing information for a given vkQueuePresentKHR command **may** only become available some time after the presentation has occurred. These time values **should** be asynchronously queried, and are returned if available. All time values are in nanoseconds, according to the time-domain being used.

To asynchronously query the presentation engine for newly-available timing information about one or more previous presents to a given swapchain, call:

```
// Provided by VK_EXT_present_timing
VkResult vkGetPastPresentationTimingEXT(
    VkDevice                                    device,
    const VkPastPresentationTimingInfoEXT*      pPastPresentationTimingInfo,
    VkPastPresentationTimingPropertiesEXT*      pPastPresentationTimingProperties);
```

- `device` is the device associated with `swapchain`.
- `pPastPresentationTimingInfo` is a pointer to an instance of the VkPastPresentationTimingInfoEXT structure.
- `pPastPresentationTimingProperties` is a pointer to an instance of the VkPastPresentationTimingPropertiesEXT structure.

If upon return the value of `VkPastPresentationTimingPropertiesEXT::presentationTimingCount` is less than the number of available timing records for the given `VkPastPresentationTimingInfoEXT::swapchain`, VK_INCOMPLETE is returned instead of VK_SUCCESS to indicate that not all the available values were returned.

Upon return, zero or more slots of the `swapchain` internal timing results queue, equal to the number of entries written to `VkPastPresentationTimingPropertiesEXT::pPresentationTimings` for which `reportComplete` is VK_TRUE, are made available for future `vkQueuePresentKHR` calls. Elements of `pPresentationTimings` are arranged in ascending order of present ids.

Timing information **may** become available out of order with regards to their associated vkQueuePresentKHR order. VK_PAST_PRESENTATION_TIMING_ALLOW_OUT_OF_ORDER_RESULTS_BIT_EXT **can** be set in `VkPastPresentationTimingInfoEXT::flags` to allow `vkGetPastPresentationTimingEXT` to return results in that same order. Otherwise, results are returned in the order of their associated vkQueuePresentKHR calls.

There is no requirement for any precise timing relationship between the completion of a present stage and the availability of any associated timing information. However, results **must** be made available in finite time.

As an exception to the normal rules for objects which are externally synchronized, `swapchain` **may** be simultaneously used by other threads in calls to functions other than vkDestroySwapchainKHR and vkCreateSwapchainKHR with `swapchain` used as an `oldSwapchain`. Access to the swapchain timing information **must** be atomic within the implementation.

Valid Usage

- VUID-vkGetPastPresentationTimingEXT-flags-12230
  If VK_PAST_PRESENTATION_TIMING_ALLOW_OUT_OF_ORDER_RESULTS_BIT_EXT is set in `VkPastPresentationTimingInfoEXT::flags`, the `presentStageCount` value of each element of `VkPastPresentationTimingPropertiesEXT::pPresentationTimings` **must** be at least the maximum number of present stages set in VkPresentTimingInfoEXT::`presentStageQueries` among all vkQueuePresentKHR calls, with a non-zero `presentStageQueries`, for which complete results have not been returned yet by a previous call

- VUID-vkGetPastPresentationTimingEXT-flags-12231
  If VK_PAST_PRESENTATION_TIMING_ALLOW_OUT_OF_ORDER_RESULTS_BIT_EXT is not set in VkPastPresentationTimingInfoEXT::flags, the presentStageCount value of each element of VkPastPresentationTimingPropertiesEXT::pPresentationTimings **must** be at least the number of present stages set in VkPresentTimingInfoEXT::presentStageQueries for the earliest call to vkQueuePresentKHR, with a non-zero presentStageQueries, that corresponds to that element's index and for which complete results have not been returned yet by a previous call

Valid Usage (Implicit)

- VUID-vkGetPastPresentationTimingEXT-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkGetPastPresentationTimingEXT-pPastPresentationTimingInfo-parameter
  pPastPresentationTimingInfo **must** be a valid pointer to a valid VkPastPresentationTimingInfoEXT structure
- VUID-vkGetPastPresentationTimingEXT-pPastPresentationTimingProperties-parameter
  pPastPresentationTimingProperties **must** be a valid pointer to a VkPastPresentationTimingPropertiesEXT structure

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_OUT_OF_DATE_KHR
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkPastPresentationTimingInfoEXT structure is defined as:

```
// Provided by VK_EXT_present_timing
typedef struct VkPastPresentationTimingInfoEXT {
    VkStructureType                     sType;
    const void*                         pNext;
    VkPastPresentationTimingFlagsEXT    flags;
    VkSwapchainKHR                      swapchain;
} VkPastPresentationTimingInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkPastPresentationTimingFlagBitsEXT specifying options for queries of past presentation timing information.
- `swapchain` is the swapchain to obtain presentation timing information for.

Valid Usage (Implicit)

- VUID-VkPastPresentationTimingInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PAST_PRESENTATION_TIMING_INFO_EXT
- VUID-VkPastPresentationTimingInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkPastPresentationTimingInfoEXT-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkPastPresentationTimingFlagBitsEXT values
- VUID-VkPastPresentationTimingInfoEXT-swapchain-parameter
  `swapchain` **must** be a valid VkSwapchainKHR handle

Host Synchronization

- Host access to `swapchain` **must** be externally synchronized

```
// Provided by VK_EXT_present_timing
typedef VkFlags VkPastPresentationTimingFlagsEXT;
```

`VkPastPresentationTimingFlagsEXT` is a bitmask type for setting a mask of zero or more VkPastPresentationTimingFlagBitsEXT.

Bits which **can** be set in VkPastPresentationTimingInfoEXT::`flags`, specifying options for queries of past presentation timing information, are:

```
// Provided by VK_EXT_present_timing
typedef enum VkPastPresentationTimingFlagBitsEXT {
    VK_PAST_PRESENTATION_TIMING_ALLOW_PARTIAL_RESULTS_BIT_EXT = 0x00000001,
    VK_PAST_PRESENTATION_TIMING_ALLOW_OUT_OF_ORDER_RESULTS_BIT_EXT = 0x00000002,
} VkPastPresentationTimingFlagBitsEXT;
```

- VK_PAST_PRESENTATION_TIMING_ALLOW_PARTIAL_RESULTS_BIT_EXT specifies that vkGetPastPresentationTimingEXT **may** return partial results for presentation requests that have not completed all requested present stages.
- VK_PAST_PRESENTATION_TIMING_ALLOW_OUT_OF_ORDER_RESULTS_BIT_EXT specifies that vkGetPastPresentationTimingEXT **may** return results out of order with respect to the

presentation order.

The VkPastPresentationTimingPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_present_timing
typedef struct VkPastPresentationTimingPropertiesEXT {
    VkStructureType                 sType;
    void*                           pNext;
    uint64_t                        timingPropertiesCounter;
    uint64_t                        timeDomainsCounter;
    uint32_t                        presentationTimingCount;
    VkPastPresentationTimingEXT*    pPresentationTimings;
} VkPastPresentationTimingPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- timingPropertiesCounter is a 64-bit unsigned integer set by the implementation to the current value of the swapchain's internal timing properties counter.
- timeDomainsCounter is a 64-bit unsigned integer set by the implementation to the current value of the swapchain's internal time domains list counter.
- presentationTimingCount is an integer related to the number of VkPastPresentationTimingEXT structures available or queried, as described below.
- pPresentationTimings is NULL or a pointer to an array of VkPastPresentationTimingEXT structures.

When calling vkGetPastPresentationTimingEXT, if pPresentationTimings is NULL, then the number of available timing records for the given swapchain is returned in presentationTimingCount. Otherwise, presentationTimingCount **must** specify the number of elements in the pPresentationTimings array, and on return is overwritten with the number of structures actually written to pPresentationTimings.

If VK_PAST_PRESENTATION_TIMING_ALLOW_PARTIAL_RESULTS_BIT_EXT is specified in VkPastPresentationTimingInfoEXT::flags, vkGetPastPresentationTimingEXT **may** return incomplete results, containing only information for a subset of the requested present stages. Further calls to vkGetPastPresentationTimingEXT will keep providing all available results for a previously incomplete entry until it is complete.

The implementation **must** return a VkPastPresentationTimingEXT for every vkQueuePresentKHR referencing swapchain where a non-zero VkPresentTimingInfoEXT::presentStageQueries was specified and at least one present stage has available results.

Valid Usage (Implicit)

- VUID-VkPastPresentationTimingPropertiesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PAST_PRESENTATION_TIMING_PROPERTIES_EXT
- VUID-VkPastPresentationTimingPropertiesEXT-pNext-pNext

`pNext` **must** be `NULL`

NOTE

The presentation engine **may** change the timing properties of the `swapchain` for a variety of reasons.

This **may** occur, for example, if the window system changes its mode, including the refresh rate of the display. Another example is if an application's surface is being composited with other windows of a window system, and then the surface's window becomes a borderless, full-screen window. While composited, the timing properties **may** be FRR, and while full-screen, the timing properties **may** be VRR.

The available time domains for a swapchain **may** change for similar or identical reasons. Therefore, it is possible that the same event will cause both the swapchain's internal timing properties counter and time domains list counter to update.

The `VkPastPresentationTimingEXT` structure is defined as:

```
// Provided by VK_EXT_present_timing
typedef struct VkPastPresentationTimingEXT {
    VkStructureType           sType;
    void*                     pNext;
    uint64_t                  presentId;
    uint64_t                  targetTime;
    uint32_t                  presentStageCount;
    VkPresentStageTimeEXT*    pPresentStages;
    VkTimeDomainKHR           timeDomain;
    uint64_t                  timeDomainId;
    VkBool32                  reportComplete;
} VkPastPresentationTimingEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `presentId` is zero or a value that was given to a previous `vkQueuePresentKHR` command via VkPresentId2KHR::`pPresentIds`.
- `targetTime` is the application-provided target absolute time or duration of the associated presentation request in VkPresentTimingInfoEXT::`targetTime`.
- `presentStageCount` is a count of items contained in `pPresentStages`.
- `pPresentStages` is a pointer to an array of VkPresentStageTimeEXT providing timing information for the presentation request associated with `presentId`.
- `timeDomain` is the time domain used by the presentation engine to report times in `pPresentStages`.
- `timeDomainId` is the id associated with `timeDomain`.
- `reportComplete` is VK_TRUE if the presentation engine has reported all the requested results in `pPresentStages`.

When calling vkGetPastPresentationTimingEXT, the implementation sets `presentStageCount` to the number of present stages it has written results for. If VK_PAST_PRESENTATION_TIMING_ALLOW_PARTIAL_RESULTS_BIT_EXT was specified in VkPastPresentationTimingInfoEXT::`flags`, the implementation **may** return an incomplete report containing fewer present stage results than were queried by the associated presentation request. Otherwise, results for all the present stages queried by the presentation request are written by the implementation.

Timing information for some present stages **may** have a time value of 0, indicating that results for that present stage are not available.

For systems with multiple entities operating within the presentation engine, such as multiple displays, `pPresentStages` will return timing results for one entity which has been affected by the presentation.

`timeDomainId` **may** be different than the time domain that was specified in `VkPresentTimingInfoEXT` ::`timeDomainId` if the requirements for using this time domain could not be met at the time the presentation engine processed the presentation request. In such a case, the presentation engine **may** pick a time domain to fall back to, if one is available, and report results in that domain. Applications **can** continue to use this fallback time domain in future `vkQueuePresentKHR` calls, or they **can** call vkGetSwapchainTimeDomainPropertiesEXT to choose from the currently supported time domains.

Valid Usage (Implicit)

- VUID-VkPastPresentationTimingEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PAST_PRESENTATION_TIMING_EXT
- VUID-VkPastPresentationTimingEXT-pNext-pNext
  `pNext` **must** be `NULL`

The `VkPresentStageTimeEXT` structure is defined as:

```
// Provided by VK_EXT_present_timing
typedef struct VkPresentStageTimeEXT {
    VkPresentStageFlagsEXT    stage;
    uint64_t                  time;
} VkPresentStageTimeEXT;
```

- `stage` is a VkPresentStageFlagsEXT value specifying a present stage.
- `time` is a time in nanoseconds associated with the `stage`.

37.9. Present Wait

Applications wanting to control the pacing of the application by monitoring when presentation processes have completed to limit the number of outstanding images queued for presentation, need to have a method of being signaled during the presentation process.

Providing a mechanism which allows applications to block, waiting for a specific step of the presentation process to complete allows them to control the amount of outstanding work (and hence the potential lag in responding to user input or changes in the rendering environment).

The VK_KHR_present_wait2 extension allows applications to tell the presentation engine at the vkQueuePresentKHR call that it plans on waiting for presentation by passing a VkPresentId2KHR structure. The presentId passed in that structure may then be passed to a future vkWaitForPresent2KHR call to cause the application to block until that presentation is finished.

This functionality was originally provided by the VK_KHR_present_wait extension, which has been deprecated and replaced by VK_KHR_present_wait2.

The VkPresentWait2InfoKHR structure is defined as:

```
// Provided by VK_KHR_present_wait2
typedef struct VkPresentWait2InfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint64_t           presentId;
    uint64_t           timeout;
} VkPresentWait2InfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- presentId is the presentation presentId to wait for.
- timeout is the timeout period in units of nanoseconds. timeout is adjusted to the closest value allowed by the implementation-dependent timeout accuracy, which **may** be substantially longer than one nanosecond, and **may** be longer than the requested period.

Valid Usage (Implicit)

- VUID-VkPresentWait2InfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PRESENT_WAIT_2_INFO_KHR
- VUID-VkPresentWait2InfoKHR-pNext-pNext
  pNext **must** be NULL

37.10. WSI Swapchain

A swapchain object (a.k.a. swapchain) provides the ability to present rendering results to a surface. Swapchain objects are represented by VkSwapchainKHR handles:

```
// Provided by VK_KHR_swapchain
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkSwapchainKHR)
```

A swapchain is an abstraction for an array of presentable images that are associated with a surface. The presentable images are represented by `VkImage` objects created by the platform. One image (which **can** be an array image for multiview/stereoscopic-3D surfaces) is displayed at a time, but multiple images **can** be queued for presentation. An application renders to the image, and then queues the image for presentation to the surface.

A native window **cannot** be associated with more than one non-retired swapchain at a time. Further, swapchains **cannot** be created for native windows that have a non-Vulkan graphics API surface associated with them.

NOTE

The presentation engine is an abstraction for the platform's compositor or display engine.

The presentation engine **may** be synchronous or asynchronous with respect to the application and/or logical device.

Some implementations **may** use the device's graphics queue or dedicated presentation hardware to perform presentation.

The presentable images of a swapchain are owned by the presentation engine. An application **can** acquire use of a presentable image from the presentation engine. Use of a presentable image **must** occur only after the image is returned by vkAcquireNextImageKHR, and before it is released by vkQueuePresentKHR. This includes transitioning the image layout and rendering commands.

An application **can** acquire use of a presentable image with vkAcquireNextImageKHR. After acquiring a presentable image and before modifying it, the application **must** use a synchronization primitive to ensure that the presentation engine has finished reading from the image. The application **can** then transition the image's layout, queue rendering commands to it, etc. Finally, the application presents the image with vkQueuePresentKHR, which releases the acquisition of the image. The application **can** also release the acquisition of the image through vkReleaseSwapchainImagesKHR, if the image is not in use by the device, and skip the present operation.

The presentation engine controls the order in which presentable images are acquired for use by the application.

NOTE

This allows the platform to handle situations which require out-of-order return of images after presentation. At the same time, it allows the application to generate command buffers referencing all of the images in the swapchain at initialization time, rather than in its main loop.

How this all works is described below.

If a swapchain is created with `presentMode` set to either VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR or VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR, a single presentable image **can** be acquired, referred to as a shared presentable image. A shared presentable image **may** be concurrently accessed by the application and the presentation engine, without transitioning the image's layout after it is initially presented.

- With VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR, the presentation engine is only required to update to the latest contents of a shared presentable image after a present. The application **must** call vkQueuePresentKHR to guarantee an update. However, the presentation engine **may** update from it at any time.
- With VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR, the presentation engine will automatically present the latest contents of a shared presentable image during every refresh cycle. The application is only required to make one initial call to vkQueuePresentKHR, after which the presentation engine will update from it without any need for further present calls. The application **can** indicate the image contents have been updated by calling vkQueuePresentKHR, but this does not guarantee the timing of when updates will occur.

The presentation engine **may** access a shared presentable image at any time after it is first presented. To avoid tearing, an application **should** coordinate access with the presentation engine. This requires presentation engine timing information through platform-specific mechanisms and ensuring that color attachment writes are made available during the portion of the presentation engine's refresh cycle they are intended for.

NOTE The VK_KHR_shared_presentable_image extension does not provide functionality for determining the timing of the presentation engine's refresh cycles.

In order to query a swapchain's status when rendering to a shared presentable image, call:

```
// Provided by VK_KHR_shared_presentable_image
VkResult vkGetSwapchainStatusKHR(
    VkDevice                                    device,
    VkSwapchainKHR                              swapchain);
```

- device is the device associated with swapchain.
- swapchain is the swapchain to query.

Valid Usage (Implicit)

- VUID-vkGetSwapchainStatusKHR-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkGetSwapchainStatusKHR-swapchain-parameter
  swapchain **must** be a valid VkSwapchainKHR handle
- VUID-vkGetSwapchainStatusKHR-swapchain-parent
  swapchain **must** have been created, allocated, or retrieved from device

Host Synchronization

- Host access to swapchain **must** be externally synchronized

Return Codes

Success

- VK_SUBOPTIMAL_KHR
- VK_SUCCESS

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT
- VK_ERROR_OUT_OF_DATE_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The possible return values for `vkGetSwapchainStatusKHR` **should** be interpreted as follows:

- VK_SUCCESS specifies the presentation engine is presenting the contents of the shared presentable image, as per the swapchain's VkPresentModeKHR.
- VK_SUBOPTIMAL_KHR the swapchain no longer matches the surface properties exactly, but the presentation engine is presenting the contents of the shared presentable image, as per the swapchain's VkPresentModeKHR.
- VK_ERROR_OUT_OF_DATE_KHR the surface has changed in such a way that it is no longer compatible with the swapchain.
- VK_ERROR_SURFACE_LOST_KHR the surface is no longer available.

NOTE

The swapchain state **may** be cached by implementations, so applications **should** regularly call `vkGetSwapchainStatusKHR` when using a swapchain with VkPresentModeKHR equal to VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR.

To create a swapchain, call:

```
// Provided by VK_KHR_swapchain
VkResult vkCreateSwapchainKHR(
    VkDevice                                    device,
    const VkSwapchainCreateInfoKHR*             pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkSwapchainKHR*                             pSwapchain);
```

- `device` is the device to create the swapchain for.

- `pCreateInfo` is a pointer to a VkSwapchainCreateInfoKHR structure specifying the parameters of the created swapchain.
- `pAllocator` is the allocator used for host memory allocated for the swapchain object when there is no more specific allocator available (see Memory Allocation).
- `pSwapchain` is a pointer to a VkSwapchainKHR handle in which the created swapchain object will be returned.

As mentioned above, if `vkCreateSwapchainKHR` succeeds, it will return a handle to a swapchain containing an array of at least `pCreateInfo->minImageCount` presentable images.

While acquired by the application, presentable images **can** be used in any way that equivalent non-presentable images **can** be used. A presentable image is equivalent to a non-presentable image created with the following VkImageCreateInfo parameters:

`VkImageCreateInfo` **Field**	**Value**
`flags`	VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT is set if VK_SWAPCHAIN_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR is set VK_IMAGE_CREATE_PROTECTED_BIT is set if VK_SWAPCHAIN_CREATE_PROTECTED_BIT_KHR is set VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT and VK_IMAGE_CREATE_EXTENDED_USAGE_BIT_KHR are both set if VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR is set VK_IMAGE_CREATE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT is set if VK_SWAPCHAIN_CREATE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT is set all other bits are unset
`imageType`	VK_IMAGE_TYPE_2D
`format`	`pCreateInfo->imageFormat`
`extent`	{`pCreateInfo->imageExtent.width`, `pCreateInfo->imageExtent.height`, `1`}
`mipLevels`	1
`arrayLayers`	`pCreateInfo->imageArrayLayers`
`samples`	VK_SAMPLE_COUNT_1_BIT

VkImageCreateInfo Field	Value
tiling	VK_IMAGE_TILING_OPTIMAL
usage	pCreateInfo->imageUsage
sharingMode	pCreateInfo->imageSharingMode
queueFamilyIndexCount	pCreateInfo->queueFamilyIndexCount
pQueueFamilyIndices	pCreateInfo->pQueueFamilyIndices
initialLayout	VK_IMAGE_LAYOUT_UNDEFINED

The pCreateInfo->surface **must** not be destroyed until after the swapchain is destroyed.

If oldSwapchain is VK_NULL_HANDLE, and the native window referred to by pCreateInfo->surface is already associated with a Vulkan swapchain, VK_ERROR_NATIVE_WINDOW_IN_USE_KHR **must** be returned.

If oldSwapchain is a valid swapchain and there are outstanding calls to vkWaitForPresent2KHR, then vkCreateSwapchainKHR **may** block until those calls complete.

If the native window referred to by pCreateInfo->surface is already associated with a non-Vulkan graphics API surface, VK_ERROR_NATIVE_WINDOW_IN_USE_KHR **must** be returned.

The native window referred to by pCreateInfo->surface **must** not become associated with a non-Vulkan graphics API surface before all associated Vulkan swapchains have been destroyed.

vkCreateSwapchainKHR will return VK_ERROR_DEVICE_LOST if the logical device was lost. The VkSwapchainKHR is a child of the device, and **must** be destroyed before the device. However, VkSurfaceKHR is not a child of any VkDevice and is not affected by the lost device. After successfully recreating a VkDevice, the same VkSurfaceKHR **can** be used to create a new VkSwapchainKHR, provided the previous one was destroyed.

If the oldSwapchain parameter of pCreateInfo is a valid swapchain, which has exclusive full-screen access, that access is released from pCreateInfo->oldSwapchain. If the command succeeds in this case, the newly created swapchain will automatically acquire exclusive full-screen access from pCreateInfo->oldSwapchain.

NOTE This implicit transfer is intended to avoid exiting and entering full-screen exclusive mode, which may otherwise cause unwanted visual updates to the display.

In some cases, swapchain creation **may** fail if exclusive full-screen mode is requested for application control, but for some implementation-specific reason exclusive full-screen access is unavailable for the particular combination of parameters provided. If this occurs, VK_ERROR_INITIALIZATION_FAILED will be returned.

NOTE In particular, it will fail if the imageExtent member of pCreateInfo does not match the extents of the monitor. Other reasons for failure may include the application not being set as high-dpi aware, or if the physical device and monitor are not compatible in this mode.

Valid Usage (Implicit)

- VUID-vkCreateSwapchainKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateSwapchainKHR-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkSwapchainCreateInfoKHR structure
- VUID-vkCreateSwapchainKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateSwapchainKHR-pSwapchain-parameter
  `pSwapchain` **must** be a valid pointer to a VkSwapchainKHR handle
- VUID-vkCreateSwapchainKHR-device-queuecount
  The device **must** have been created with at least `1` queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_COMPRESSION_EXHAUSTED_EXT
- VK_ERROR_DEVICE_LOST
- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_NATIVE_WINDOW_IN_USE_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkSwapchainCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_swapchain
typedef struct VkSwapchainCreateInfoKHR {
    VkStructureType                  sType;
    const void*                      pNext;
    VkSwapchainCreateFlagsKHR        flags;
    VkSurfaceKHR                     surface;
    uint32_t                         minImageCount;
    VkFormat                         imageFormat;
    VkColorSpaceKHR                  imageColorSpace;
    VkExtent2D                       imageExtent;
    uint32_t                         imageArrayLayers;
    VkImageUsageFlags                imageUsage;
    VkSharingMode                    imageSharingMode;
    uint32_t                         queueFamilyIndexCount;
    const uint32_t*                  pQueueFamilyIndices;
    VkSurfaceTransformFlagBitsKHR    preTransform;
    VkCompositeAlphaFlagBitsKHR      compositeAlpha;
    VkPresentModeKHR                 presentMode;
    VkBool32                         clipped;
    VkSwapchainKHR                   oldSwapchain;
} VkSwapchainCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkSwapchainCreateFlagBitsKHR indicating parameters of the swapchain creation.
- `surface` is the surface onto which the swapchain will present images. If the creation succeeds, the swapchain becomes associated with `surface`.
- `minImageCount` is the minimum number of presentable images that the application needs. The implementation will either create the swapchain with at least that many images, or it will fail to create the swapchain.
- `imageFormat` is a VkFormat value specifying the format the swapchain image(s) will be created with.
- `imageColorSpace` is a VkColorSpaceKHR value specifying the way the swapchain interprets image data.
- `imageExtent` is the size (in pixels) of the swapchain image(s). The behavior is platform-dependent if the image extent does not match the surface's `currentExtent` as returned by `vkGetPhysicalDeviceSurfaceCapabilitiesKHR`.

> **NOTE**
>
> On some platforms, it is normal that `maxImageExtent` **may** become `(0, 0)`, for example when the window is minimized. In such a case, it is not possible to create a swapchain due to the Valid Usage requirements , unless scaling is selected through VkSwapchainPresentScalingCreateInfoKHR, if supported .

- `imageArrayLayers` is the number of views in a multiview/stereo surface. For non-stereoscopic-3D

applications, this value is 1.

- `imageUsage` is a bitmask of VkImageUsageFlagBits describing the intended usage of the (acquired) swapchain images.
- `imageSharingMode` is the sharing mode used for the image(s) of the swapchain.
- `queueFamilyIndexCount` is the number of queue families having access to the image(s) of the swapchain when `imageSharingMode` is VK_SHARING_MODE_CONCURRENT.
- `pQueueFamilyIndices` is a pointer to an array of queue family indices having access to the image(s) of the swapchain when `imageSharingMode` is VK_SHARING_MODE_CONCURRENT.
- `preTransform` is a VkSurfaceTransformFlagBitsKHR value describing the transform, relative to the presentation engine's natural orientation, applied to the image content prior to presentation. If it does not match the `currentTransform` value returned by `vkGetPhysicalDeviceSurfaceCapabilitiesKHR`, the presentation engine will transform the image content as part of the presentation operation.
- `compositeAlpha` is a VkCompositeAlphaFlagBitsKHR value indicating the alpha compositing mode to use when this surface is composited together with other surfaces on certain window systems.
- `presentMode` is the presentation mode the swapchain will use. A swapchain's present mode determines how incoming present requests will be processed and queued internally.
- `clipped` specifies whether the Vulkan implementation is allowed to discard rendering operations that affect regions of the surface that are not visible.
  - If `clipped` is VK_TRUE, the presentable images associated with the swapchain **may** not own all of their pixels. Pixels in the presentable images that correspond to regions of the target surface obscured by another window on the desktop, or subject to some other clipping mechanism will have undefined content when read back. Fragment shaders **may** not execute for these pixels, and thus any side effects they would have had will not occur. Setting VK_TRUE does not guarantee any clipping will occur, but allows more efficient presentation methods to be used on some platforms.
  - If `clipped` is VK_FALSE, presentable images associated with the swapchain will own all of the pixels they contain.

> **NOTE** Applications **should** set this value to VK_TRUE if they do not expect to read back the content of presentable images before presenting them or after reacquiring them, and if their fragment shaders do not have any side effects that require them to run for all pixels in the presentable image.

- `oldSwapchain` is VK_NULL_HANDLE, or the existing non-retired swapchain currently associated with `surface`. Providing a valid `oldSwapchain` **may** aid in the resource reuse, and also allows the application to still present any images that are already acquired from it.

Upon calling `vkCreateSwapchainKHR` with an `oldSwapchain` that is not VK_NULL_HANDLE, `oldSwapchain` is retired — even if creation of the new swapchain fails. The new swapchain is created in the non-retired state whether or not `oldSwapchain` is VK_NULL_HANDLE.

Upon calling `vkCreateSwapchainKHR` with an `oldSwapchain` that is not VK_NULL_HANDLE, any images from `oldSwapchain` that are not acquired by the application **may** be freed by the implementation,

which **may** occur even if creation of the new swapchain fails. The application **can** destroy `oldSwapchain` to free all memory associated with `oldSwapchain`.

NOTE

Multiple retired swapchains **can** be associated with the same `VkSurfaceKHR` through multiple uses of `oldSwapchain` that outnumber calls to vkDestroySwapchainKHR.

After `oldSwapchain` is retired, the application **can** pass to vkQueuePresentKHR any images it had already acquired from `oldSwapchain`. E.g., an application may present an image from the old swapchain before an image from the new swapchain is ready to be presented. As usual, vkQueuePresentKHR **may** fail if `oldSwapchain` has entered a state that causes VK_ERROR_OUT_OF_DATE_KHR to be returned.

The application **can** continue to use a shared presentable image obtained from `oldSwapchain` until a presentable image is acquired from the new swapchain, as long as it has not entered a state that causes it to return VK_ERROR_OUT_OF_DATE_KHR.

`imageUsage` defines the effective usage flags for the swapchain image(s). If the `pNext` chain includes a VkImageUsageFlags2CreateInfoKHR structure, `imageUsage` is ignored, and the effective usage flags are defined by VkImageUsageFlags2CreateInfoKHR::`usage`.

Valid Usage

- VUID-VkSwapchainCreateInfoKHR-surface-01270
  `surface` **must** be a surface that is supported by the device as determined using vkGetPhysicalDeviceSurfaceSupportKHR
- VUID-VkSwapchainCreateInfoKHR-minImageCount-01272
  `minImageCount` **must** be less than or equal to the value returned in the `maxImageCount` member of the `VkSurfaceCapabilitiesKHR` structure returned by `vkGetPhysicalDeviceSurfaceCapabilitiesKHR` for the surface if the returned `maxImageCount` is not zero
- VUID-VkSwapchainCreateInfoKHR-swapchainMaintenance1-10155
  If the `swapchainMaintenance1` feature is not enabled, then the `pNext` chain **must** not include a VkSwapchainPresentModesCreateInfoKHR structure
- VUID-VkSwapchainCreateInfoKHR-presentMode-02839
  If `presentMode` is not VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR nor VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR, then `minImageCount` **must** be greater than or equal to the value returned in the `minImageCount` member of the `VkSurfaceCapabilitiesKHR` structure returned by vkGetPhysicalDeviceSurfaceCapabilitiesKHR for the surface
- VUID-VkSwapchainCreateInfoKHR-minImageCount-01383
  `minImageCount` **must** be `1` if `presentMode` is either VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR or VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR
- VUID-VkSwapchainCreateInfoKHR-imageFormat-01273
  `imageFormat` and `imageColorSpace` **must** match the `format` and `colorSpace` members, respectively, of one of the `VkSurfaceFormatKHR` structures returned by

vkGetPhysicalDeviceSurfaceFormatsKHR for the surface

- VUID-VkSwapchainCreateInfoKHR-pNext-07781
  If a VkSwapchainPresentScalingCreateInfoKHR structure was not included in the pNext chain, or it is included and VkSwapchainPresentScalingCreateInfoKHR::scalingBehavior is zero then imageExtent **must** be between minImageExtent and maxImageExtent, inclusive, where minImageExtent and maxImageExtent are members of the VkSurfaceCapabilitiesKHR structure returned by vkGetPhysicalDeviceSurfaceCapabilitiesKHR for the surface
- VUID-VkSwapchainCreateInfoKHR-pNext-07782
  If a VkSwapchainPresentScalingCreateInfoKHR structure was included in the pNext chain and VkSwapchainPresentScalingCreateInfoKHR::scalingBehavior is not zero then imageExtent **must** be between minScaledImageExtent and maxScaledImageExtent, inclusive, where minScaledImageExtent and maxScaledImageExtent are members of the VkSurfacePresentScalingCapabilitiesKHR structure returned by vkGetPhysicalDeviceSurfaceCapabilities2KHR for the surface and presentMode
- VUID-VkSwapchainCreateInfoKHR-swapchainMaintenance1-10157
  If the swapchainMaintenance1 feature is not enabled, then flags **must** not include VK_SWAPCHAIN_CREATE_DEFERRED_MEMORY_ALLOCATION_BIT_KHR
- VUID-VkSwapchainCreateInfoKHR-imageExtent-01689
  imageExtent members width and height **must** both be non-zero
- VUID-VkSwapchainCreateInfoKHR-imageArrayLayers-01275
  imageArrayLayers **must** be greater than 0 and less than or equal to the maxImageArrayLayers member of the VkSurfaceCapabilitiesKHR structure returned by vkGetPhysicalDeviceSurfaceCapabilitiesKHR for the surface
- VUID-VkSwapchainCreateInfoKHR-presentMode-01427
  If presentMode is VK_PRESENT_MODE_FIFO_LATEST_READY_KHR, VK_PRESENT_MODE_IMMEDIATE_KHR, VK_PRESENT_MODE_MAILBOX_KHR, VK_PRESENT_MODE_FIFO_KHR or VK_PRESENT_MODE_FIFO_RELAXED_KHR, imageUsage **must** be a subset of the supported usage flags present in the supportedUsageFlags member of the VkSurfaceCapabilitiesKHR structure returned by vkGetPhysicalDeviceSurfaceCapabilitiesKHR for surface
- VUID-VkSwapchainCreateInfoKHR-imageUsage-01384
  If presentMode is VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR or VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR, imageUsage **must** be a subset of the supported usage flags present in the sharedPresentSupportedUsageFlags member of the VkSharedPresentSurfaceCapabilitiesKHR structure returned by vkGetPhysicalDeviceSurfaceCapabilities2KHR for surface
- VUID-VkSwapchainCreateInfoKHR-imageSharingMode-01277
  If imageSharingMode is VK_SHARING_MODE_CONCURRENT, pQueueFamilyIndices **must** be a valid pointer to an array of queueFamilyIndexCount uint32_t values
- VUID-VkSwapchainCreateInfoKHR-imageSharingMode-01278
  If imageSharingMode is VK_SHARING_MODE_CONCURRENT, queueFamilyIndexCount **must** be greater than 1
- VUID-VkSwapchainCreateInfoKHR-imageSharingMode-01428
  If imageSharingMode is VK_SHARING_MODE_CONCURRENT, each element of

pQueueFamilyIndices **must** be unique and **must** be less than pQueueFamilyPropertyCount returned by either vkGetPhysicalDeviceQueueFamilyProperties or vkGetPhysicalDeviceQueueFamilyProperties2 for the physicalDevice that was used to create device

- VUID-VkSwapchainCreateInfoKHR-preTransform-01279
  preTransform **must** be one of the bits present in the supportedTransforms member of the VkSurfaceCapabilitiesKHR structure returned by vkGetPhysicalDeviceSurfaceCapabilitiesKHR for the surface
- VUID-VkSwapchainCreateInfoKHR-compositeAlpha-01280
  compositeAlpha **must** be one of the bits present in the supportedCompositeAlpha member of the VkSurfaceCapabilitiesKHR structure returned by vkGetPhysicalDeviceSurfaceCapabilitiesKHR for the surface
- VUID-VkSwapchainCreateInfoKHR-presentMode-01281
  presentMode **must** be one of the VkPresentModeKHR values returned by vkGetPhysicalDeviceSurfacePresentModesKHR for the surface
- VUID-VkSwapchainCreateInfoKHR-presentModeFifoLatestReady-10161
  If the presentModeFifoLatestReady feature is not enabled, presentMode **must** not be VK_PRESENT_MODE_FIFO_LATEST_READY_KHR
- VUID-VkSwapchainCreateInfoKHR-physicalDeviceCount-01429
  If the logical device was created with VkDeviceGroupDeviceCreateInfo ::physicalDeviceCount equal to 1, flags **must** not contain VK_SWAPCHAIN_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR
- VUID-VkSwapchainCreateInfoKHR-oldSwapchain-01933
  If oldSwapchain is not VK_NULL_HANDLE, oldSwapchain **must** be a non-retired swapchain associated with the native window referred to by surface
- VUID-VkSwapchainCreateInfoKHR-imageFormat-01778
  The implied image creation parameters of the swapchain **must** be supported as reported by vkGetPhysicalDeviceImageFormatProperties or vkGetPhysicalDeviceImageFormatProperties2
- VUID-VkSwapchainCreateInfoKHR-flags-03168
  If flags contains VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR then the pNext chain **must** include a VkImageFormatListCreateInfo structure with a viewFormatCount greater than zero and pViewFormats **must** have an element equal to imageFormat
- VUID-VkSwapchainCreateInfoKHR-pNext-04099
  If a VkImageFormatListCreateInfo structure was included in the pNext chain and VkImageFormatListCreateInfo::viewFormatCount is not zero then all of the formats in VkImageFormatListCreateInfo::pViewFormats **must** be compatible with the format as described in the compatibility table
- VUID-VkSwapchainCreateInfoKHR-flags-04100
  If flags does not contain VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR and the pNext chain includes a VkImageFormatListCreateInfo structure then VkImageFormatListCreateInfo::viewFormatCount **must** be 0 or 1
- VUID-VkSwapchainCreateInfoKHR-flags-03187

If `flags` contains VK_SWAPCHAIN_CREATE_PROTECTED_BIT_KHR, then `VkSurfaceProtectedCapabilitiesKHR::supportsProtected` **must** be VK_TRUE in the VkSurfaceProtectedCapabilitiesKHR structure returned by vkGetPhysicalDeviceSurfaceCapabilities2KHR for `surface`

- VUID-VkSwapchainCreateInfoKHR-pNext-02679
  If the `pNext` chain includes a VkSurfaceFullScreenExclusiveInfoEXT structure with its `fullScreenExclusive` member set to VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT, and `surface` was created using vkCreateWin32SurfaceKHR, a VkSurfaceFullScreenExclusiveWin32InfoEXT structure **must** be included in the `pNext` chain
- VUID-VkSwapchainCreateInfoKHR-pNext-06752
  If the `imageCompressionControlSwapchain` feature is not enabled, the `pNext` chain **must** not include an VkImageCompressionControlEXT structure
- VUID-VkSwapchainCreateInfoKHR-multisampledRenderToSwapchain-12447
  If the `multisampledRenderToSwapchain` feature is not enabled, `flags` **must** not contain VK_SWAPCHAIN_CREATE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT
- VUID-VkSwapchainCreateInfoKHR-flags-12448
  `flags` **must** be a subset of flags returned in VkSwapchainFlagsSurfaceCapabilitiesEXT `::swapchainSupportedFlags`, given VkSwapchainCreateInfoKHR`::presentMode` in VkSurfacePresentModeKHR
- VUID-VkSwapchainCreateInfoKHR-presentTiming-12232
  If none of the `presentTiming`, `presentAtAbsoluteTime`, or `presentAtRelativeTime` features are enabled, `flags` **must** not contain VK_SWAPCHAIN_CREATE_PRESENT_TIMING_BIT_EXT

Valid Usage (Implicit)

- VUID-VkSwapchainCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SWAPCHAIN_CREATE_INFO_KHR
- VUID-VkSwapchainCreateInfoKHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkDeviceGroupSwapchainCreateInfoKHR, VkImageCompressionControlEXT, VkImageFormatListCreateInfo, VkImageUsageFlags2CreateInfoKHR, VkSurfaceFullScreenExclusiveInfoEXT, VkSurfaceFullScreenExclusiveWin32InfoEXT, VkSwapchainCounterCreateInfoEXT, VkSwapchainPresentModesCreateInfoKHR, or VkSwapchainPresentScalingCreateInfoKHR
- VUID-VkSwapchainCreateInfoKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkSwapchainCreateInfoKHR-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkSwapchainCreateFlagBitsKHR values
- VUID-VkSwapchainCreateInfoKHR-surface-parameter
  `surface` **must** be a valid VkSurfaceKHR handle
- VUID-VkSwapchainCreateInfoKHR-imageFormat-parameter

`imageFormat` **must** be a valid VkFormat value

- VUID-VkSwapchainCreateInfoKHR-imageColorSpace-parameter
  `imageColorSpace` **must** be a valid VkColorSpaceKHR value
- VUID-VkSwapchainCreateInfoKHR-imageUsage-parameter
  `imageUsage` **must** be a valid combination of VkImageUsageFlagBits values
- VUID-VkSwapchainCreateInfoKHR-imageSharingMode-parameter
  `imageSharingMode` **must** be a valid VkSharingMode value
- VUID-VkSwapchainCreateInfoKHR-preTransform-parameter
  `preTransform` **must** be a valid VkSurfaceTransformFlagBitsKHR value
- VUID-VkSwapchainCreateInfoKHR-compositeAlpha-parameter
  `compositeAlpha` **must** be a valid VkCompositeAlphaFlagBitsKHR value
- VUID-VkSwapchainCreateInfoKHR-presentMode-parameter
  `presentMode` **must** be a valid VkPresentModeKHR value
- VUID-VkSwapchainCreateInfoKHR-oldSwapchain-parameter
  If `oldSwapchain` is not VK_NULL_HANDLE, `oldSwapchain` **must** be a valid VkSwapchainKHR handle
- VUID-VkSwapchainCreateInfoKHR-commonparent
  Both of `oldSwapchain`, and `surface` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkInstance

Host Synchronization

- Host access to `surface` **must** be externally synchronized
- Host access to `oldSwapchain` **must** be externally synchronized

Bits which **can** be set in VkSwapchainCreateInfoKHR::`flags`, specifying parameters of swapchain creation, are:

```
// Provided by VK_KHR_swapchain
typedef enum VkSwapchainCreateFlagBitsKHR {
  // Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
    VK_SWAPCHAIN_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR = 0x00000001,
  // Provided by VK_VERSION_1_1 with VK_KHR_swapchain
    VK_SWAPCHAIN_CREATE_PROTECTED_BIT_KHR = 0x00000002,
  // Provided by VK_KHR_swapchain_mutable_format
    VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR = 0x00000004,
  // Provided by VK_EXT_present_timing
    VK_SWAPCHAIN_CREATE_PRESENT_TIMING_BIT_EXT = 0x00000200,
  // Provided by VK_KHR_present_id2
    VK_SWAPCHAIN_CREATE_PRESENT_ID_2_BIT_KHR = 0x00000040,
  // Provided by VK_KHR_present_wait2
    VK_SWAPCHAIN_CREATE_PRESENT_WAIT_2_BIT_KHR = 0x00000080,
  // Provided by VK_KHR_swapchain_maintenance1
    VK_SWAPCHAIN_CREATE_DEFERRED_MEMORY_ALLOCATION_BIT_KHR = 0x00000008,
  // Provided by VK_EXT_multisampled_render_to_swapchain
    VK_SWAPCHAIN_CREATE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT = 0x00000100,
  // Provided by VK_EXT_swapchain_maintenance1
    VK_SWAPCHAIN_CREATE_DEFERRED_MEMORY_ALLOCATION_BIT_EXT =
VK_SWAPCHAIN_CREATE_DEFERRED_MEMORY_ALLOCATION_BIT_KHR,
} VkSwapchainCreateFlagBitsKHR;
```

- VK_SWAPCHAIN_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR specifies that images created from the swapchain (i.e. with the `swapchain` member of VkImageSwapchainCreateInfoKHR set to this swapchain's handle) **must** use VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT.
- VK_SWAPCHAIN_CREATE_PROTECTED_BIT_KHR specifies that images created from the swapchain are protected images.
- VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR specifies that the images of the swapchain **can** be used to create a `VkImageView` with a different format than what the swapchain was created with. The list of allowed image view formats is specified by adding a VkImageFormatListCreateInfo structure to the `pNext` chain of VkSwapchainCreateInfoKHR. In addition, this flag also specifies that the swapchain **can** be created with usage flags that are not supported for the format the swapchain is created with but are supported for at least one of the allowed image view formats.
- VK_SWAPCHAIN_CREATE_DEFERRED_MEMORY_ALLOCATION_BIT_KHR specifies that the implementation **may** defer allocation of memory associated with each swapchain image until its index is to be returned from vkAcquireNextImageKHR or vkAcquireNextImage2KHR for the first time.
- VK_SWAPCHAIN_CREATE_PRESENT_ID_2_BIT_KHR specifies that applications **can** include the `VkPresentId2KHR` structure in the `pNext` chain of the VkPresentInfoKHR structure to associate an identifier with each presentation request.
- VK_SWAPCHAIN_CREATE_PRESENT_WAIT_2_BIT_KHR specifies that applications **can** use `vkWaitForPresent2KHR` to wait for the presentation engine to have begun presentation of the

presentation request associated with VkPresentWait2InfoKHR::presentId on swapchain.

- VK_SWAPCHAIN_CREATE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT specifies that an image **can** be used with multisampled rendering as a single-sampled render pass attachment.
- VK_SWAPCHAIN_CREATE_PRESENT_TIMING_BIT_EXT specifies that features supported by the swapchain device in VkPhysicalDevicePresentTimingFeaturesEXT **can** be used to collect timing information or schedule presentation requests at specific times.

```
// Provided by VK_KHR_swapchain
typedef VkFlags VkSwapchainCreateFlagsKHR;
```

VkSwapchainCreateFlagsKHR is a bitmask type for setting a mask of zero or more VkSwapchainCreateFlagBitsKHR.

If the pNext chain of VkSwapchainCreateInfoKHR includes a VkDeviceGroupSwapchainCreateInfoKHR structure, then that structure includes a set of device group present modes that the swapchain **can** be used with.

The VkDeviceGroupSwapchainCreateInfoKHR structure is defined as:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
typedef struct VkDeviceGroupSwapchainCreateInfoKHR {
    VkStructureType                  sType;
    const void*                      pNext;
    VkDeviceGroupPresentModeFlagsKHR modes;
} VkDeviceGroupSwapchainCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- modes is a bitfield of modes that the swapchain **can** be used with.

If this structure is not present, modes is considered to be VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR.

Valid Usage (Implicit)

- VUID-VkDeviceGroupSwapchainCreateInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DEVICE_GROUP_SWAPCHAIN_CREATE_INFO_KHR
- VUID-VkDeviceGroupSwapchainCreateInfoKHR-modes-parameter
  modes **must** be a valid combination of VkDeviceGroupPresentModeFlagBitsKHR values

Structure Chaining

Extends the structure

- VkSwapchainCreateInfoKHR

If the `pNext` chain of VkSwapchainCreateInfoKHR includes a VkSurfaceFullScreenExclusiveInfoEXT structure, then that structure specifies the application's preferred full-screen presentation behavior. If this structure is not present, `fullScreenExclusive` is considered to be VK_FULL_SCREEN_EXCLUSIVE_DEFAULT_EXT.

To enable surface counters when creating a swapchain, add a `VkSwapchainCounterCreateInfoEXT` structure to the `pNext` chain of VkSwapchainCreateInfoKHR. `VkSwapchainCounterCreateInfoEXT` is defined as:

```
// Provided by VK_EXT_display_control
typedef struct VkSwapchainCounterCreateInfoEXT {
    VkStructureType              sType;
    const void*                  pNext;
    VkSurfaceCounterFlagsEXT     surfaceCounters;
} VkSwapchainCounterCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `surfaceCounters` is a bitmask of VkSurfaceCounterFlagBitsEXT specifying surface counters to enable for the swapchain.

Valid Usage

- VUID-VkSwapchainCounterCreateInfoEXT-surfaceCounters-01244
  The bits in `surfaceCounters` **must** be supported by VkSwapchainCreateInfoKHR::`surface`, as reported by vkGetPhysicalDeviceSurfaceCapabilities2EXT

Valid Usage (Implicit)

- VUID-VkSwapchainCounterCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SWAPCHAIN_COUNTER_CREATE_INFO_EXT
- VUID-VkSwapchainCounterCreateInfoEXT-surfaceCounters-parameter
  If `surfaceCounters` is not `0`, `surfaceCounters` **must** be a valid combination of VkSurfaceCounterFlagBitsEXT values

Structure Chaining

Extends the structure

- VkSwapchainCreateInfoKHR

The requested counters become active when the first presentation command for the associated swapchain is processed by the presentation engine. To query the value of an active counter, use:

```
// Provided by VK_EXT_display_control
VkResult vkGetSwapchainCounterEXT(
    VkDevice                                    device,
    VkSwapchainKHR                              swapchain,
    VkSurfaceCounterFlagBitsEXT                 counter,
    uint64_t*                                   pCounterValue);
```

- `device` is the VkDevice associated with `swapchain`.
- `swapchain` is the swapchain from which to query the counter value.
- `counter` is a VkSurfaceCounterFlagBitsEXT value specifying the counter to query.
- `pCounterValue` will return the current value of the counter.

If a counter is not available because the swapchain is out of date, the implementation **may** return VK_ERROR_OUT_OF_DATE_KHR.

Valid Usage

- VUID-vkGetSwapchainCounterEXT-swapchain-01245
  One or more present commands on `swapchain` **must** have been processed by the presentation engine

Valid Usage (Implicit)

- VUID-vkGetSwapchainCounterEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetSwapchainCounterEXT-swapchain-parameter
  `swapchain` **must** be a valid VkSwapchainKHR handle
- VUID-vkGetSwapchainCounterEXT-counter-parameter
  `counter` **must** be a valid VkSurfaceCounterFlagBitsEXT value
- VUID-vkGetSwapchainCounterEXT-pCounterValue-parameter
  `pCounterValue` **must** be a valid pointer to a `uint64_t` value
- VUID-vkGetSwapchainCounterEXT-swapchain-parent
  `swapchain` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_OUT_OF_DATE_KHR
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To specify compression properties for the swapchain images in this swapchain, add a VkImageCompressionControlEXT structure to the `pNext` chain of the VkSwapchainCreateInfoKHR structure.

Applications **can** modify the presentation mode used by the swapchain on a per-presentation basis. However, all presentation modes the application intends to use with the swapchain **must** be specified at swapchain creation time. To specify more than one presentation mode when creating a swapchain, include the `VkSwapchainPresentModesCreateInfoKHR` structure in the `pNext` chain of the VkSwapchainCreateInfoKHR structure.

The `VkSwapchainPresentModesCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_swapchain_maintenance1
typedef struct VkSwapchainPresentModesCreateInfoKHR {
    VkStructureType            sType;
    const void*                pNext;
    uint32_t                   presentModeCount;
    const VkPresentModeKHR*    pPresentModes;
} VkSwapchainPresentModesCreateInfoKHR;
```

```
// Provided by VK_EXT_swapchain_maintenance1
// Equivalent to VkSwapchainPresentModesCreateInfoKHR
typedef VkSwapchainPresentModesCreateInfoKHR VkSwapchainPresentModesCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `presentModeCount` is the number of presentation modes provided.
- `pPresentModes` is a list of presentation modes with `presentModeCount` entries.

Valid Usage

- VUID-VkSwapchainPresentModesCreateInfoKHR-None-07762
  Each entry in pPresentModes **must** be one of the VkPresentModeKHR values returned by `vkGetPhysicalDeviceSurfacePresentModesKHR` for the surface
- VUID-VkSwapchainPresentModesCreateInfoKHR-presentModeFifoLatestReady-10160
  If the `presentModeFifoLatestReady` feature is not enabled, pPresentModes **must** not contain VK_PRESENT_MODE_FIFO_LATEST_READY_KHR
- VUID-VkSwapchainPresentModesCreateInfoKHR-pPresentModes-07763
  The entries in pPresentModes **must** be a subset of the present modes returned in VkSurfacePresentModeCompatibilityKHR::`pPresentModes`, given VkSwapchainCreateInfoKHR::`presentMode` in VkSurfacePresentModeKHR
- VUID-VkSwapchainPresentModesCreateInfoKHR-presentMode-07764
  VkSwapchainCreateInfoKHR::`presentMode` **must** be included in `pPresentModes`

Valid Usage (Implicit)

- VUID-VkSwapchainPresentModesCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_MODES_CREATE_INFO_KHR
- VUID-VkSwapchainPresentModesCreateInfoKHR-pPresentModes-parameter
  `pPresentModes` **must** be a valid pointer to an array of `presentModeCount` valid VkPresentModeKHR values
- VUID-VkSwapchainPresentModesCreateInfoKHR-presentModeCount-arraylength
  `presentModeCount` **must** be greater than `0`

Structure Chaining

Extends the structure

- VkSwapchainCreateInfoKHR

When an application presents a swapchain image with dimensions different than those of the target surface, different behavior is possible on different platforms per their respective specifications:

- Presentation fails and VK_ERROR_OUT_OF_DATE_KHR is returned
- Scaling is done and VK_SUCCESS or VK_SUBOPTIMAL_KHR is returned
- Unspecified scaling using an arbitrary combination of stretching, centering and/or clipping.

Applications **can** define specific behavior when creating a swapchain by including the `VkSwapchainPresentScalingCreateInfoKHR` structure in the `pNext` chain of the VkSwapchainCreateInfoKHR structure.

The VkSwapchainPresentScalingCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_swapchain_maintenance1
typedef struct VkSwapchainPresentScalingCreateInfoKHR {
    VkStructureType             sType;
    const void*                 pNext;
    VkPresentScalingFlagsKHR    scalingBehavior;
    VkPresentGravityFlagsKHR    presentGravityX;
    VkPresentGravityFlagsKHR    presentGravityY;
} VkSwapchainPresentScalingCreateInfoKHR;
```

```
// Provided by VK_EXT_swapchain_maintenance1
// Equivalent to VkSwapchainPresentScalingCreateInfoKHR
typedef VkSwapchainPresentScalingCreateInfoKHR VkSwapchainPresentScalingCreateInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- scalingBehavior is 0 or the scaling method to use when the dimensions of the surface and swapchain images differ.
- presentGravityX is 0 or the x-axis direction in which swapchain image pixels gravitate relative to the surface when scalingBehavior does not result in a one-to-one pixel mapping between the scaled swapchain image and the surface.
- presentGravityY is 0 or the y-axis direction in which swapchain image pixels gravitate relative to the surface when scalingBehavior does not result in a one-to-one pixel mapping between the scaled swapchain image and the surface.

If scalingBehavior is 0, the result of presenting a swapchain image with dimensions that do not match the surface dimensions is implementation and platform-dependent. If presentGravityX or presentGravityY are 0, the presentation gravity **must** match that defined by the native platform surface on platforms which define surface gravity.

Valid Usage

- VUID-VkSwapchainPresentScalingCreateInfoKHR-presentGravityX-07765
  If presentGravityX is 0, presentGravityY **must** be 0
- VUID-VkSwapchainPresentScalingCreateInfoKHR-presentGravityX-07766
  If presentGravityX is not 0, presentGravityY **must** not be 0
- VUID-VkSwapchainPresentScalingCreateInfoKHR-scalingBehavior-07767
  scalingBehavior **must** not have more than one bit set
- VUID-VkSwapchainPresentScalingCreateInfoKHR-presentGravityX-07768
  presentGravityX **must** not have more than one bit set
- VUID-VkSwapchainPresentScalingCreateInfoKHR-presentGravityY-07769
  presentGravityY **must** not have more than one bit set

- VUID-VkSwapchainPresentScalingCreateInfoKHR-scalingBehavior-07770
  `scalingBehavior` **must** be `0` or a valid scaling method for the surface as returned in VkSurfacePresentScalingCapabilitiesKHR::`supportedPresentScaling`, given VkSwapchainCreateInfoKHR::`presentMode` in VkSurfacePresentModeKHR
- VUID-VkSwapchainPresentScalingCreateInfoKHR-scalingBehavior-07771
  If the swapchain is created with VkSwapchainPresentModesCreateInfoKHR, `scalingBehavior` **must** be `0` or a valid scaling method for the surface as returned in VkSurfacePresentScalingCapabilitiesKHR::`supportedPresentScaling`, given each present mode in VkSwapchainPresentModesCreateInfoKHR::`pPresentModes` in VkSurfacePresentModeKHR
- VUID-VkSwapchainPresentScalingCreateInfoKHR-presentGravityX-07772
  `presentGravityX` **must** be `0` or a valid x-axis present gravity for the surface as returned in VkSurfacePresentScalingCapabilitiesKHR::`supportedPresentGravityX`, given VkSwapchainCreateInfoKHR::`presentMode` in VkSurfacePresentModeKHR
- VUID-VkSwapchainPresentScalingCreateInfoKHR-presentGravityX-07773
  If the swapchain is created with VkSwapchainPresentModesCreateInfoKHR, `presentGravityX` **must** be `0` or a valid x-axis present gravity for the surface as returned in VkSurfacePresentScalingCapabilitiesKHR::`supportedPresentGravityX`, given each present mode in VkSwapchainPresentModesCreateInfoKHR::`pPresentModes` in VkSurfacePresentModeKHR
- VUID-VkSwapchainPresentScalingCreateInfoKHR-presentGravityY-07774
  `presentGravityY` **must** be `0` or a valid y-axis present gravity for the surface as returned in VkSurfacePresentScalingCapabilitiesKHR::`supportedPresentGravityY`, given VkSwapchainCreateInfoKHR::`presentMode` in VkSurfacePresentModeKHR
- VUID-VkSwapchainPresentScalingCreateInfoKHR-presentGravityY-07775
  If the swapchain is created with VkSwapchainPresentModesCreateInfoKHR, `presentGravityY` **must** be `0` or a valid y-axis present gravity for the surface as returned in VkSurfacePresentScalingCapabilitiesKHR::`supportedPresentGravityY`, given each present mode in VkSwapchainPresentModesCreateInfoKHR::`pPresentModes` in VkSurfacePresentModeKHR
- VUID-VkSwapchainPresentScalingCreateInfoKHR-swapchainMaintenance1-10154
  If the `swapchainMaintenance1` feature is not enabled, then `scalingBehavior`, `presentGravityX`, and `presentGravityY` **must** be `0`

Valid Usage (Implicit)

- VUID-VkSwapchainPresentScalingCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_SCALING_CREATE_INFO_KHR
- VUID-VkSwapchainPresentScalingCreateInfoKHR-scalingBehavior-parameter
  If `scalingBehavior` is not `0`, `scalingBehavior` **must** be a valid combination of VkPresentScalingFlagBitsKHR values
- VUID-VkSwapchainPresentScalingCreateInfoKHR-presentGravityX-parameter

If `presentGravityX` is not `0`, `presentGravityX` **must** be a valid combination of VkPresentGravityFlagBitsKHR values

- VUID-VkSwapchainPresentScalingCreateInfoKHR-presentGravityY-parameter
  If `presentGravityY` is not `0`, `presentGravityY` **must** be a valid combination of VkPresentGravityFlagBitsKHR values

Structure Chaining

Extends the structure

- VkSwapchainCreateInfoKHR

To destroy a swapchain object call:

```
// Provided by VK_KHR_swapchain
void vkDestroySwapchainKHR(
    VkDevice                                    device,
    VkSwapchainKHR                              swapchain,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the VkDevice associated with `swapchain`.
- `swapchain` is the swapchain to destroy.
- `pAllocator` is the allocator used for host memory allocated for the swapchain object when there is no more specific allocator available (see Memory Allocation).

The application **must** not destroy a swapchain until after completion of all outstanding operations on images that were acquired from the swapchain. `swapchain` and all associated `VkImage` handles are destroyed, and **must** not be acquired or used any more by the application. The memory of each `VkImage` will only be freed after that image is no longer used by the presentation engine. For example, if one image of the swapchain is being displayed in a window, the memory for that image **may** not be freed until the window is destroyed, or another swapchain is created for the window. Destroying the swapchain does not invalidate the parent `VkSurfaceKHR`, and a new swapchain **can** be created with it.

When a swapchain associated with a display surface is destroyed, if the image most recently presented to the display surface is from the swapchain being destroyed, then either any display resources modified by presenting images from any swapchain associated with the display surface **must** be reverted by the implementation to their state prior to the first present performed on one of these swapchains, or such resources **must** be left in their current state.

If `swapchain` has exclusive full-screen access, it is released before the swapchain is destroyed.

Valid Usage

- VUID-vkDestroySwapchainKHR-swapchain-01282

All uses of presentable images acquired from `swapchain` **must** have completed execution

- VUID-vkDestroySwapchainKHR-swapchain-01283
  If `VkAllocationCallbacks` were provided when `swapchain` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroySwapchainKHR-swapchain-01284
  If no `VkAllocationCallbacks` were provided when `swapchain` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroySwapchainKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroySwapchainKHR-swapchain-parameter
  If `swapchain` is not VK_NULL_HANDLE, `swapchain` **must** be a valid VkSwapchainKHR handle
- VUID-vkDestroySwapchainKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroySwapchainKHR-swapchain-parent
  If `swapchain` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `swapchain` **must** be externally synchronized

When the `VK_KHR_display_swapchain` extension is enabled, multiple swapchains that share presentable images are created by calling:

```
// Provided by VK_KHR_display_swapchain
VkResult vkCreateSharedSwapchainsKHR(
    VkDevice                                    device,
    uint32_t                                    swapchainCount,
    const VkSwapchainCreateInfoKHR*             pCreateInfos,
    const VkAllocationCallbacks*                pAllocator,
    VkSwapchainKHR*                             pSwapchains);
```

- `device` is the device to create the swapchains for.
- `swapchainCount` is the number of swapchains to create.
- `pCreateInfos` is a pointer to an array of VkSwapchainCreateInfoKHR structures specifying the parameters of the created swapchains.

- `pAllocator` is the allocator used for host memory allocated for the swapchain objects when there is no more specific allocator available (see Memory Allocation).
- `pSwapchains` is a pointer to an array of VkSwapchainKHR handles in which the created swapchain objects will be returned.

`vkCreateSharedSwapchainsKHR` is similar to vkCreateSwapchainKHR, except that it takes an array of VkSwapchainCreateInfoKHR structures, and returns an array of swapchain objects.

The swapchain creation parameters that affect the properties and number of presentable images **must** match between all the swapchains. If the displays used by any of the swapchains do not use the same presentable image layout or are incompatible in a way that prevents sharing images, swapchain creation will fail with the result code VK_ERROR_INCOMPATIBLE_DISPLAY_KHR. If any error occurs, no swapchains will be created. Images presented to multiple swapchains **must** be re-acquired from all of them before being modified. After destroying one or more of the swapchains, the remaining swapchains and the presentable images **can** continue to be used.

Valid Usage (Implicit)

- VUID-vkCreateSharedSwapchainsKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateSharedSwapchainsKHR-pCreateInfos-parameter
  `pCreateInfos` **must** be a valid pointer to an array of `swapchainCount` valid VkSwapchainCreateInfoKHR structures
- VUID-vkCreateSharedSwapchainsKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateSharedSwapchainsKHR-pSwapchains-parameter
  `pSwapchains` **must** be a valid pointer to an array of `swapchainCount` VkSwapchainKHR handles
- VUID-vkCreateSharedSwapchainsKHR-device-queuecount
  The device **must** have been created with at least `1` queue
- VUID-vkCreateSharedSwapchainsKHR-swapchainCount-arraylength
  `swapchainCount` **must** be greater than `0`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_INCOMPATIBLE_DISPLAY_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To obtain the array of presentable images associated with a swapchain, call:

```
// Provided by VK_KHR_swapchain
VkResult vkGetSwapchainImagesKHR(
    VkDevice                                    device,
    VkSwapchainKHR                              swapchain,
    uint32_t*                                   pSwapchainImageCount,
    VkImage*                                    pSwapchainImages);
```

- `device` is the device associated with `swapchain`.
- `swapchain` is the swapchain to query.
- `pSwapchainImageCount` is a pointer to an integer related to the number of presentable images available or queried, as described below.
- `pSwapchainImages` is either `NULL` or a pointer to an array of `VkImage` handles.

If `pSwapchainImages` is `NULL`, then the number of presentable images for `swapchain` is returned in `pSwapchainImageCount`. Otherwise, `pSwapchainImageCount` **must** point to a variable set by the application to the number of elements in the `pSwapchainImages` array, and on return the variable is overwritten with the number of structures actually written to `pSwapchainImages`. If the value of `pSwapchainImageCount` is less than the number of presentable images for `swapchain`, at most `pSwapchainImageCount` structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available presentable images were returned.

Valid Usage (Implicit)

- VUID-vkGetSwapchainImagesKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetSwapchainImagesKHR-swapchain-parameter
  `swapchain` **must** be a valid VkSwapchainKHR handle
- VUID-vkGetSwapchainImagesKHR-pSwapchainImageCount-parameter
  `pSwapchainImageCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetSwapchainImagesKHR-pSwapchainImages-parameter
  If the value referenced by `pSwapchainImageCount` is not `0`, and `pSwapchainImages` is not `NULL`, `pSwapchainImages` **must** be a valid pointer to an array of `pSwapchainImageCount` VkImage handles
- VUID-vkGetSwapchainImagesKHR-swapchain-parent
  `swapchain` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

NOTE

By knowing all presentable images used in the swapchain, the application can create command buffers that reference these images prior to entering its main rendering loop. However, command buffers are not allowed to reference presentable images created with VK_SWAPCHAIN_CREATE_DEFERRED_MEMORY_ALLOCATION_BIT_KHR until their indices have been returned from vkAcquireNextImageKHR at least once.

Images returned by vkGetSwapchainImagesKHR are fully backed by memory before they are passed to the application, as if they are each bound completely and contiguously to a single `VkDeviceMemory` object , unless the swapchain is created with the VK_SWAPCHAIN_CREATE_DEFERRED_MEMORY_ALLOCATION_BIT_KHR flag . All presentable images are initially in the VK_IMAGE_LAYOUT_UNDEFINED layout, thus before using presentable images, the application **must** transition them to a valid layout for the intended use.

Further, the lifetime of presentable images is controlled by the implementation, so applications **must** not destroy a presentable image. See vkDestroySwapchainKHR for further details on the lifetime of presentable images.

Images **can** also be created by using vkCreateImage with VkImageSwapchainCreateInfoKHR and bound to swapchain memory using vkBindImageMemory2 with VkBindImageMemorySwapchainInfoKHR. These images **can** be used anywhere swapchain images are used, and are useful in logical devices with multiple physical devices to create peer memory bindings of swapchain memory. These images and bindings have no effect on what memory is presented. Unlike images retrieved from `vkGetSwapchainImagesKHR`, these images **must** be destroyed with vkDestroyImage.

To acquire an available presentable image to use, and retrieve the index of that image, call:

```
// Provided by VK_KHR_swapchain
VkResult vkAcquireNextImageKHR(
    VkDevice                                    device,
    VkSwapchainKHR                              swapchain,
    uint64_t                                    timeout,
    VkSemaphore                                 semaphore,
    VkFence                                     fence,
    uint32_t*                                   pImageIndex);
```

- `device` is the device associated with `swapchain`.
- `swapchain` is the non-retired swapchain from which an image is being acquired.
- `timeout` specifies how long the function waits, in nanoseconds, if no image is available.
- `semaphore` is VK_NULL_HANDLE or a semaphore defining a semaphore signal operation.
- `fence` is VK_NULL_HANDLE or a fence to signal.
- `pImageIndex` is a pointer to a `uint32_t` in which the index of the next image to use (i.e. an index into the array of images returned by `vkGetSwapchainImagesKHR`) is returned.

If the `swapchain` has been created with the VK_SWAPCHAIN_CREATE_DEFERRED_MEMORY_ALLOCATION_BIT_KHR flag, the image whose index is returned in `pImageIndex` will be fully backed by memory before this call returns to the application, as if it is bound completely and contiguously to a single `VkDeviceMemory` object.

If `semaphore` defines a semaphore signal operation, its first synchronization scope includes acquisition of the image.

Valid Usage

- VUID-vkAcquireNextImageKHR-swapchain-01285
  `swapchain` **must** not be in the retired state
- VUID-vkAcquireNextImageKHR-semaphore-01286
  If `semaphore` is not VK_NULL_HANDLE, it **must** be unsignaled
- VUID-vkAcquireNextImageKHR-semaphore-01779
  If `semaphore` is not VK_NULL_HANDLE, it **must** not have any uncompleted signal or wait operations pending
- VUID-vkAcquireNextImageKHR-fence-01287
  If `fence` is not VK_NULL_HANDLE, `fence` **must** be unsignaled
- VUID-vkAcquireNextImageKHR-fence-10066
  If `fence` is not VK_NULL_HANDLE, `fence` **must** not be associated with any other queue command that has not yet completed execution on that queue
- VUID-vkAcquireNextImageKHR-semaphore-01780
  `semaphore` and `fence` **must** not both be equal to VK_NULL_HANDLE
- VUID-vkAcquireNextImageKHR-surface-07783
  If forward progress cannot be guaranteed for the `surface` used to create the `swapchain`

member of pAcquireInfo, timeout **must** not be UINT64_MAX

- VUID-vkAcquireNextImageKHR-semaphore-03265
  semaphore **must** have a VkSemaphoreType of VK_SEMAPHORE_TYPE_BINARY

Valid Usage (Implicit)

- VUID-vkAcquireNextImageKHR-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkAcquireNextImageKHR-swapchain-parameter
  swapchain **must** be a valid VkSwapchainKHR handle
- VUID-vkAcquireNextImageKHR-semaphore-parameter
  If semaphore is not VK_NULL_HANDLE, semaphore **must** be a valid VkSemaphore handle
- VUID-vkAcquireNextImageKHR-fence-parameter
  If fence is not VK_NULL_HANDLE, fence **must** be a valid VkFence handle
- VUID-vkAcquireNextImageKHR-pImageIndex-parameter
  pImageIndex **must** be a valid pointer to a uint32_t value
- VUID-vkAcquireNextImageKHR-swapchain-parent
  swapchain **must** have been created, allocated, or retrieved from device
- VUID-vkAcquireNextImageKHR-semaphore-parent
  If semaphore is a valid handle, it **must** have been created, allocated, or retrieved from device
- VUID-vkAcquireNextImageKHR-fence-parent
  If fence is a valid handle, it **must** have been created, allocated, or retrieved from device

Host Synchronization

- Host access to swapchain **must** be externally synchronized
- Host access to semaphore **must** be externally synchronized
- Host access to fence **must** be externally synchronized

Return Codes

Success

- VK_NOT_READY
- VK_SUBOPTIMAL_KHR
- VK_SUCCESS
- VK_TIMEOUT

Failure

- VK_ERROR_DEVICE_LOST

- VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT
- VK_ERROR_OUT_OF_DATE_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

If an image is acquired successfully, vkAcquireNextImageKHR **must** either return VK_SUCCESS or VK_SUBOPTIMAL_KHR. The implementation **may** return VK_SUBOPTIMAL_KHR if the swapchain no longer matches the surface properties exactly, but **can** still be used for presentation.

When successful, vkAcquireNextImageKHR acquires a presentable image from swapchain that an application **can** use, and sets pImageIndex to the index of that image within the swapchain. The presentation engine **may** not have finished reading from the image at the time it is acquired, so the application **must** use semaphore and/or fence to ensure that the image layout and contents are not modified until the presentation engine reads have completed. Once vkAcquireNextImageKHR successfully acquires an image, the semaphore signal operation referenced by semaphore, if not VK_NULL_HANDLE, and the fence signal operation referenced by fence, if not VK_NULL_HANDLE, are submitted for execution. If vkAcquireNextImageKHR does not successfully acquire an image, semaphore and fence are unaffected. The order in which images are acquired is implementation-dependent, and **may** be different than the order the images were presented.

If timeout is zero, then vkAcquireNextImageKHR does not wait, and will either successfully acquire an image, or fail and return VK_NOT_READY if no image is available.

If the specified timeout period expires before an image is acquired, vkAcquireNextImageKHR returns VK_TIMEOUT. If timeout is UINT64_MAX, the timeout period is treated as infinite, and vkAcquireNextImageKHR will block until an image is acquired or an error occurs.

Let S be the number of images in swapchain. If swapchain is created with VkSwapchainPresentModesCreateInfoKHR, let M be the maximum of the values in VkSurfaceCapabilitiesKHR::minImageCount when queried with each present mode in VkSwapchainPresentModesCreateInfoKHR::pPresentModes in VkSurfacePresentModeKHR. Otherwise, let M be the value of VkSurfaceCapabilitiesKHR::minImageCount without a VkSurfacePresentModeKHR as part of the query input.

vkAcquireNextImageKHR **should** not be called if the number of images that the application has currently acquired is greater than S-M. If vkAcquireNextImageKHR is called when the number of images that the application has currently acquired is less than or equal to S-M, vkAcquireNextImageKHR **must** return in finite time with an allowed VkResult code.

NOTE

Returning a result in finite time guarantees that the implementation cannot deadlock an application, or suspend its execution indefinitely with correct API usage. Acquiring too many images at once may block indefinitely, which is covered by valid usage when attempting to use UINT64_MAX. For example, a scenario here is

when a compositor holds on to images which are currently being presented, and there are not any vacant images left to be acquired.

If the swapchain images no longer match native surface properties, either VK_SUBOPTIMAL_KHR or VK_ERROR_OUT_OF_DATE_KHR **must** be returned. If VK_ERROR_OUT_OF_DATE_KHR is returned, no image is acquired and attempts to present previously acquired images to the swapchain will also fail with VK_ERROR_OUT_OF_DATE_KHR. Applications need to create a new swapchain for the surface to continue presenting if VK_ERROR_OUT_OF_DATE_KHR is returned.

NOTE

VK_SUBOPTIMAL_KHR **may** happen, for example, if the platform surface has been resized but the platform is able to scale the presented images to the new size to produce valid surface updates. It is up to the application to decide whether it prefers to continue using the current swapchain in this state, or to re-create the swapchain to better match the platform surface properties.

If device loss occurs (see Lost Device) before the timeout has expired, `vkAcquireNextImageKHR` **must** return in finite time with either one of the allowed success codes, or VK_ERROR_DEVICE_LOST.

If `semaphore` is not VK_NULL_HANDLE, the semaphore **must** be unsignaled, with no signal or wait operations pending. It will become signaled when the application **can** use the image.

NOTE

Use of `semaphore` allows rendering operations to be recorded and submitted before the presentation engine has completed its use of the image.

If `fence` is not equal to VK_NULL_HANDLE, the fence **must** be unsignaled, with no signal operations pending. It will become signaled when the application **can** use the image.

NOTE

Applications **should** not rely on `vkAcquireNextImageKHR` blocking in order to meter their rendering speed. The implementation **may** return from this function immediately regardless of how many presentation requests are queued, and regardless of when queued presentation requests will complete relative to the call. Instead, applications **can** use `fence` to meter their frame generation work to match the presentation rate.

An application **must** wait until either the `semaphore` or `fence` is signaled before accessing the image's data.

NOTE

When the presentable image will be accessed by some stage S, the recommended idiom for ensuring correct synchronization is:

- The VkSubmitInfo used to submit the image layout transition for execution includes `vkAcquireNextImageKHR`::`semaphore` in its `pWaitSemaphores` member, with the corresponding element of `pWaitDstStageMask` including S.
- The synchronization command that performs any necessary image layout transition includes S in both the `srcStageMask` and `dstStageMask`.

After a successful return, the image indicated by `pImageIndex` and its data will be unmodified compared to when it was presented.

NOTE

Exclusive ownership of presentable images corresponding to a swapchain created with VK_SHARING_MODE_EXCLUSIVE as defined in Resource Sharing is not altered by a call to `vkAcquireNextImageKHR`. That means upon the first acquisition from such a swapchain presentable images are not owned by any queue family, while at subsequent acquisitions the presentable images remain owned by the queue family the image was previously presented on.

The possible return values for `vkAcquireNextImageKHR` depend on the `timeout` provided:

- VK_SUCCESS is returned if an image became available.
- VK_ERROR_SURFACE_LOST_KHR is returned if the surface becomes no longer available.
- VK_NOT_READY is returned if `timeout` is zero and no image was available.
- VK_TIMEOUT is returned if `timeout` is greater than zero and less than `UINT64_MAX`, and no image became available within the time allowed.
- VK_SUBOPTIMAL_KHR **may** be returned if an image became available, and the swapchain no longer matches the surface properties exactly, but **can** still be used to present to the surface successfully.

NOTE

This **may** happen, for example, if the platform surface has been resized but the platform is able to scale the presented images to the new size to produce valid surface updates. It is up to the application to decide whether it prefers to continue using the current swapchain indefinitely or temporarily in this state, or to re-create the swapchain to better match the platform surface properties.

- VK_ERROR_OUT_OF_DATE_KHR is returned if the surface has changed in such a way that it is no longer compatible with the swapchain, and further presentation requests using the swapchain will fail. Applications **must** query the new surface properties and recreate their swapchain if they wish to continue presenting to the surface.

If the native surface and presented image sizes no longer match, presentation **may** fail unless the swapchain is created with a non-zero value in VkSwapchainPresentScalingCreateInfoKHR ::`scalingBehavior` . If presentation does succeed, the mapping from the presented image to the native surface is defined by the VkSwapchainPresentScalingCreateInfoKHR structure if provided. Otherwise it is implementation-defined. It is the application's responsibility to detect surface size changes and react appropriately. If presentation fails because of a mismatch in the surface and presented image sizes, a VK_ERROR_OUT_OF_DATE_KHR error will be returned.

NOTE

For example, consider a 4x3 window/surface that gets resized to be 3x4 (taller than wider). On some window systems, the portion of the window/surface that was previously and still is visible (the 3x3 part) will contain the same contents as before, while the remaining parts of the window will have undefined contents. Other window systems **may** squash/stretch the image to fill the new window size without any undefined contents, or apply some other mapping.

To acquire an available presentable image to use, and retrieve the index of that image, call:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
VkResult vkAcquireNextImage2KHR(
    VkDevice                                    device,
    const VkAcquireNextImageInfoKHR*            pAcquireInfo,
    uint32_t*                                   pImageIndex);
```

- `device` is the device associated with `swapchain`.
- `pAcquireInfo` is a pointer to a VkAcquireNextImageInfoKHR structure containing parameters of the acquire.
- `pImageIndex` is a pointer to a `uint32_t` value specifying the index of the next image to use.

If the `swapchain` has been created with the VK_SWAPCHAIN_CREATE_DEFERRED_MEMORY_ALLOCATION_BIT_KHR flag, the image whose index is returned in `pImageIndex` will be fully backed by memory before this call returns to the application.

Valid Usage

- VUID-vkAcquireNextImage2KHR-surface-07784
  If forward progress cannot be guaranteed for the `surface` used to create `swapchain`, the `timeout` member of `pAcquireInfo` **must** not be `UINT64_MAX`

Valid Usage (Implicit)

- VUID-vkAcquireNextImage2KHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkAcquireNextImage2KHR-pAcquireInfo-parameter
  `pAcquireInfo` **must** be a valid pointer to a valid VkAcquireNextImageInfoKHR structure
- VUID-vkAcquireNextImage2KHR-pImageIndex-parameter
  `pImageIndex` **must** be a valid pointer to a `uint32_t` value

Return Codes

Success

- VK_NOT_READY
- VK_SUBOPTIMAL_KHR
- VK_SUCCESS
- VK_TIMEOUT

Failure

- VK_ERROR_DEVICE_LOST

- VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT
- VK_ERROR_OUT_OF_DATE_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkAcquireNextImageInfoKHR structure is defined as:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
typedef struct VkAcquireNextImageInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkSwapchainKHR     swapchain;
    uint64_t           timeout;
    VkSemaphore        semaphore;
    VkFence            fence;
    uint32_t           deviceMask;
} VkAcquireNextImageInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- swapchain is a non-retired swapchain from which an image is acquired.
- timeout specifies how long the function waits, in nanoseconds, if no image is available.
- semaphore is VK_NULL_HANDLE or a semaphore that defines a semaphore signal operation.
- fence is VK_NULL_HANDLE or a fence to signal.
- deviceMask is a mask of physical devices for which the swapchain image will be ready to use when the semaphore or fence is signaled.

If vkAcquireNextImageKHR is used, the device mask is considered to include all physical devices in the logical device.

NOTE

vkAcquireNextImage2KHR signals at most one semaphore, even if the application requests waiting for multiple physical devices to be ready via the deviceMask. However, only a single physical device **can** wait on that semaphore, since the semaphore becomes unsignaled when the wait succeeds. For other physical devices to wait for the image to be ready, it is necessary for the application to submit semaphore signal operation(s) to that first physical device to signal additional semaphore(s) after the wait succeeds, which the other physical device(s) **can** wait upon.

Valid Usage

- VUID-VkAcquireNextImageInfoKHR-swapchain-01675
  `swapchain` **must** not be in the retired state
- VUID-VkAcquireNextImageInfoKHR-semaphore-01288
  If `semaphore` is not VK_NULL_HANDLE, it **must** be unsignaled
- VUID-VkAcquireNextImageInfoKHR-semaphore-01781
  If `semaphore` is not VK_NULL_HANDLE, it **must** not have any uncompleted signal or wait operations pending
- VUID-VkAcquireNextImageInfoKHR-fence-01289
  If `fence` is not VK_NULL_HANDLE, `fence` **must** be unsignaled
- VUID-VkAcquireNextImageInfoKHR-fence-10067
  If `fence` is not VK_NULL_HANDLE, `fence` **must** not be associated with any other queue command that has not yet completed execution on that queue
- VUID-VkAcquireNextImageInfoKHR-semaphore-01782
  `semaphore` and `fence` **must** not both be equal to VK_NULL_HANDLE
- VUID-VkAcquireNextImageInfoKHR-deviceMask-01290
  `deviceMask` **must** be a valid device mask
- VUID-VkAcquireNextImageInfoKHR-deviceMask-01291
  `deviceMask` **must** not be zero
- VUID-VkAcquireNextImageInfoKHR-semaphore-03266
  `semaphore` **must** have a VkSemaphoreType of VK_SEMAPHORE_TYPE_BINARY

Valid Usage (Implicit)

- VUID-VkAcquireNextImageInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_ACQUIRE_NEXT_IMAGE_INFO_KHR
- VUID-VkAcquireNextImageInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkAcquireNextImageInfoKHR-swapchain-parameter
  `swapchain` **must** be a valid VkSwapchainKHR handle
- VUID-VkAcquireNextImageInfoKHR-semaphore-parameter
  If `semaphore` is not VK_NULL_HANDLE, `semaphore` **must** be a valid VkSemaphore handle
- VUID-VkAcquireNextImageInfoKHR-fence-parameter
  If `fence` is not VK_NULL_HANDLE, `fence` **must** be a valid VkFence handle
- VUID-VkAcquireNextImageInfoKHR-commonparent
  Each of `fence`, `semaphore`, and `swapchain` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `swapchain` **must** be externally synchronized
- Host access to `semaphore` **must** be externally synchronized
- Host access to `fence` **must** be externally synchronized

After queueing all rendering commands and transitioning the image to the correct layout, to queue an image for presentation, call:

```
// Provided by VK_KHR_swapchain
VkResult vkQueuePresentKHR(
    VkQueue                                     queue,
    const VkPresentInfoKHR*                     pPresentInfo);
```

- `queue` is a queue that is capable of presentation to the target surface's platform on the same device as the image's swapchain.
- `pPresentInfo` is a pointer to a VkPresentInfoKHR structure specifying parameters of the presentation.

NOTE There is no requirement for an application to present images in the same order that they were acquired - applications can arbitrarily present any image that is currently acquired.

NOTE The origin of the native orientation of the surface coordinate system is not specified in the Vulkan specification; it depends on the platform. For most platforms the origin is by default upper-left, meaning the pixel of the presented VkImage at coordinates (0,0) would appear at the upper left pixel of the platform surface (assuming VK_SURFACE_TRANSFORM_IDENTITY_BIT_KHR, and the display standing the right way up).

The result codes VK_ERROR_OUT_OF_DATE_KHR and VK_SUBOPTIMAL_KHR have the same meaning when returned by `vkQueuePresentKHR` as they do when returned by `vkAcquireNextImageKHR`. If any `swapchain` member of `pPresentInfo` was created with VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT, VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT will be returned if that swapchain does not have exclusive full-screen access, possibly for implementation-specific reasons outside of the application's control. If multiple swapchains are presented, the result code is determined by applying the following rules in order:

- If the device is lost, VK_ERROR_DEVICE_LOST is returned.
- If any of the target surfaces are no longer available the error VK_ERROR_SURFACE_LOST_KHR is returned.
- If any of the presents would have a result of VK_ERROR_OUT_OF_DATE_KHR if issued separately then VK_ERROR_OUT_OF_DATE_KHR is returned.

- If any of the presents would have a result of VK_ERROR_PRESENT_TIMING_QUEUE_FULL_EXT if issued separately then VK_ERROR_PRESENT_TIMING_QUEUE_FULL_EXT is returned.
- If any of the presents would have a result of VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT if issued separately then VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT is returned.
- If any of the presents would have a result of VK_SUBOPTIMAL_KHR if issued separately then VK_SUBOPTIMAL_KHR is returned.
- Otherwise VK_SUCCESS is returned.

Any writes to memory backing the images referenced by the `pImageIndices` and `pSwapchains` members of `pPresentInfo`, that are available before vkQueuePresentKHR is executed, are automatically made visible to the read access performed by the presentation engine. This automatic visibility operation for an image happens-after the semaphore wait operation, and happens-before the presentation engine accesses the image.

Presentation is a read-only operation that will not affect the content of the presentable images. Upon reacquiring the image and transitioning it away from the VK_IMAGE_LAYOUT_PRESENT_SRC_KHR layout, the contents will be the same as they were prior to transitioning the image to the present source layout and presenting it. However, if a mechanism other than Vulkan is used to modify the platform window associated with the swapchain, the content of all presentable images in the swapchain becomes undefined.

Calls to `vkQueuePresentKHR` **may** block, but **must** return in finite time. The processing of the presentation happens in issue order with other queue operations, but semaphores **must** be used to ensure that prior rendering and other commands in the specified queue complete before the presentation begins. The presentation command itself does not delay processing of subsequent commands on the queue. However, presentation requests sent to a particular queue are always performed in order. Exact presentation timing is controlled by the semantics of the presentation engine and native platform in use.

If an image is presented to a swapchain created from a display surface, the mode of the associated display will be updated, if necessary, to match the mode specified when creating the display surface. The mode switch and presentation of the specified image will be performed as one atomic operation.

Queueing an image for presentation defines a set of *queue operations*, including waiting on the semaphores and submitting a presentation request to the presentation engine. However, the scope of this set of queue operations does not include the actual processing of the image by the presentation engine.

If `vkQueuePresentKHR` fails to enqueue the corresponding set of queue operations, it **may** return VK_ERROR_OUT_OF_HOST_MEMORY or VK_ERROR_OUT_OF_DEVICE_MEMORY. If it does, the implementation **must** ensure that the state and contents of any resources or synchronization primitives referenced are unaffected by the call or its failure.

If `vkQueuePresentKHR` fails in such a way that the implementation is unable to make that guarantee, the implementation **must** return VK_ERROR_DEVICE_LOST.

However, if the presentation request is rejected by the presentation engine with an error VK_ERROR_OUT_OF_DATE_KHR, VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT, VK_ERROR_PRESENT_TIMING_QUEUE_FULL_EXT, or VK_ERROR_SURFACE_LOST_KHR, the set of queue operations is still considered to be enqueued and thus any semaphore wait operation specified in VkPresentInfoKHR will execute when the corresponding queue operation is complete.

vkQueuePresentKHR releases the acquisition of the images referenced by pImageIndices. The queue family corresponding to the queue vkQueuePresentKHR is executed on **must** have ownership of the presented images as defined in Resource Sharing. vkQueuePresentKHR does not alter the queue family ownership, but the presented images **must** not be used again before they have been reacquired using vkAcquireNextImageKHR.

NOTE The application **can** continue to present any acquired images from a retired swapchain as long as the swapchain has not entered a state that causes vkQueuePresentKHR to return VK_ERROR_OUT_OF_DATE_KHR.

Valid Usage

- VUID-vkQueuePresentKHR-pSwapchains-01292
  Each element of pSwapchains member of pPresentInfo **must** be a swapchain that is created for a surface for which presentation is supported from queue as determined using a call to vkGetPhysicalDeviceSurfaceSupportKHR
- VUID-vkQueuePresentKHR-pSwapchains-01293
  If more than one member of pSwapchains was created from a display surface, all display surfaces referenced that refer to the same display **must** use the same display mode
- VUID-vkQueuePresentKHR-pWaitSemaphores-01294
  When a semaphore wait operation referring to a binary semaphore defined by the elements of the pWaitSemaphores member of pPresentInfo executes on queue, there **must** be no other queues waiting on the same semaphore
- VUID-vkQueuePresentKHR-pWaitSemaphores-03267
  All elements of the pWaitSemaphores member of pPresentInfo **must** be created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_BINARY
- VUID-vkQueuePresentKHR-pWaitSemaphores-03268
  All elements of the pWaitSemaphores member of pPresentInfo **must** reference a semaphore signal operation that has been submitted for execution and any semaphore signal operations on which it depends **must** have also been submitted for execution

Valid Usage (Implicit)

- VUID-vkQueuePresentKHR-queue-parameter
  queue **must** be a valid VkQueue handle
- VUID-vkQueuePresentKHR-pPresentInfo-parameter
  pPresentInfo **must** be a valid pointer to a valid VkPresentInfoKHR structure

Host Synchronization

- Host access to `queue` **must** be externally synchronized if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	Any	-

Return Codes

Success

- VK_SUBOPTIMAL_KHR
- VK_SUCCESS

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT
- VK_ERROR_OUT_OF_DATE_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_PRESENT_TIMING_QUEUE_FULL_EXT
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkPresentInfoKHR` structure is defined as:

```
// Provided by VK_KHR_swapchain
typedef struct VkPresentInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    uint32_t                 waitSemaphoreCount;
    const VkSemaphore*       pWaitSemaphores;
    uint32_t                 swapchainCount;
    const VkSwapchainKHR*    pSwapchains;
    const uint32_t*          pImageIndices;
    VkResult*                pResults;
} VkPresentInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `waitSemaphoreCount` is the number of semaphores to wait for before issuing the present request. The number **may** be zero.
- `pWaitSemaphores` is `NULL` or a pointer to an array of VkSemaphore objects with `waitSemaphoreCount` entries, and specifies the semaphores to wait for before issuing the present request.
- `swapchainCount` is the number of swapchains being presented to by this command.
- `pSwapchains` is a pointer to an array of VkSwapchainKHR objects with `swapchainCount` entries.
- `pImageIndices` is a pointer to an array of indices into the array of each swapchain's presentable images, with `swapchainCount` entries. Each entry in this array identifies the image to present on the corresponding entry in the `pSwapchains` array.
- `pResults` is a pointer to an array of VkResult typed elements with `swapchainCount` entries. Applications that do not need per-swapchain results **can** use `NULL` for `pResults`. If non-`NULL`, each entry in `pResults` will be set to the VkResult for presenting the swapchain corresponding to the same index in `pSwapchains`.

Before an application **can** present an image, the image's layout **must** be transitioned to the VK_IMAGE_LAYOUT_PRESENT_SRC_KHR layout, or for a shared presentable image the VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR layout.

NOTE

When transitioning the image to the appropriate layout, there is no need to delay subsequent processing, or perform any visibility operations (as vkQueuePresentKHR performs automatic visibility operations). To achieve this, the `dstAccessMask` member of the VkImageMemoryBarrier **should** be `0`, and the `dstStageMask` parameter **should** be VK_PIPELINE_STAGE_BOTTOM_OF_PIPE_BIT.

The second synchronization scope of each semaphore wait operation defined by this structure includes presentation of each image indicated by `pSwapchains` and `pImageIndices`.

Valid Usage

- VUID-VkPresentInfoKHR-pSwapchain-09231

Elements of pSwapchains **must** be unique

- VUID-VkPresentInfoKHR-pImageIndices-01430
  Each element of pImageIndices **must** be the index of a presentable image acquired from the swapchain specified by the corresponding element of the pSwapchains array, and the presented image subresource **must** be in the VK_IMAGE_LAYOUT_PRESENT_SRC_KHR or VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR layout at the time the operation is executed on a VkDevice
- VUID-VkPresentInfoKHR-pNext-06235
  If a VkPresentIdKHR structure is included in the pNext chain, and the presentId feature is not enabled, each presentIds entry in that structure **must** be NULL
- VUID-VkPresentInfoKHR-swapchainMaintenance1-10158
  If the swapchainMaintenance1 feature is not enabled, then the pNext chain **must** not include a VkSwapchainPresentFenceInfoKHR structure
- VUID-VkPresentInfoKHR-pSwapchains-09199
  If any element of the pSwapchains array has been created with VkSwapchainPresentModesCreateInfoKHR, all of the elements of this array **must** be created with VkSwapchainPresentModesCreateInfoKHR
- VUID-VkPresentInfoKHR-pNext-10821
  If a VkPresentId2KHR structure is included in the pNext chain, and the presentId2 feature is not enabled, each presentIds entry in that structure **must** be zero
- VUID-VkPresentInfoKHR-presentId2Supported-10822
  If a VkPresentId2KHR structure is included and contains non-zero presentIds, presentId2Supported **must** be VK_TRUE in the VkSurfaceCapabilitiesPresentId2KHR structure returned by vkGetPhysicalDeviceSurfaceCapabilities2KHR for the surface

Valid Usage (Implicit)

- VUID-VkPresentInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PRESENT_INFO_KHR
- VUID-VkPresentInfoKHR-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkDeviceGroupPresentInfoKHR, VkDisplayPresentInfoKHR, VkFrameBoundaryEXT, VkPresentId2KHR, VkPresentIdKHR, VkPresentRegionsKHR, VkPresentTimingsInfoEXT, VkSwapchainPresentFenceInfoKHR, or VkSwapchainPresentModeInfoKHR
- VUID-VkPresentInfoKHR-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkPresentInfoKHR-pWaitSemaphores-parameter
  If waitSemaphoreCount is not 0, pWaitSemaphores **must** be a valid pointer to an array of waitSemaphoreCount valid VkSemaphore handles
- VUID-VkPresentInfoKHR-pSwapchains-parameter
  pSwapchains **must** be a valid pointer to an array of swapchainCount valid VkSwapchainKHR handles

- VUID-VkPresentInfoKHR-pImageIndices-parameter
  `pImageIndices` **must** be a valid pointer to an array of `swapchainCount` `uint32_t` values
- VUID-VkPresentInfoKHR-pResults-parameter
  If `pResults` is not `NULL`, `pResults` **must** be a valid pointer to an array of `swapchainCount` VkResult values
- VUID-VkPresentInfoKHR-swapchainCount-arraylength
  `swapchainCount` **must** be greater than `0`
- VUID-VkPresentInfoKHR-commonparent
  Both of the elements of `pSwapchains`, and the elements of `pWaitSemaphores` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to each member of `pWaitSemaphores` **must** be externally synchronized
- Host access to each member of `pSwapchains` **must** be externally synchronized

When the `VK_KHR_incremental_present` extension is enabled, additional fields **can** be specified that allow an application to specify that only certain rectangular regions of the presentable images of a swapchain are changed. This is an optimization hint that a presentation engine **may** use to only update the region of a surface that is actually changing. The application still **must** ensure that all pixels of a presented image contain the desired values, in case the presentation engine ignores this hint. An application **can** provide this hint by adding a `VkPresentRegionsKHR` structure to the `pNext` chain of the `VkPresentInfoKHR` structure.

The `VkPresentRegionsKHR` structure is defined as:

```
// Provided by VK_KHR_incremental_present
typedef struct VkPresentRegionsKHR {
    VkStructureType              sType;
    const void*                  pNext;
    uint32_t                     swapchainCount;
    const VkPresentRegionKHR*    pRegions;
} VkPresentRegionsKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `swapchainCount` is the number of swapchains being presented to by this command.
- `pRegions` is `NULL` or a pointer to an array of `VkPresentRegionKHR` elements with `swapchainCount` entries. If not `NULL`, each element of `pRegions` contains the region that has changed since the last present to the swapchain in the corresponding entry in the `VkPresentInfoKHR`::`pSwapchains` array.

Valid Usage

- VUID-VkPresentRegionsKHR-swapchainCount-01260
  `swapchainCount` **must** be the same value as `VkPresentInfoKHR`::`swapchainCount`, where `VkPresentInfoKHR` is included in the `pNext` chain of this `VkPresentRegionsKHR` structure

Valid Usage (Implicit)

- VUID-VkPresentRegionsKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PRESENT_REGIONS_KHR
- VUID-VkPresentRegionsKHR-pRegions-parameter
  If `pRegions` is not `NULL`, `pRegions` **must** be a valid pointer to an array of `swapchainCount` valid VkPresentRegionKHR structures
- VUID-VkPresentRegionsKHR-swapchainCount-arraylength
  `swapchainCount` **must** be greater than `0`

Structure Chaining

Extends the structure

- VkPresentInfoKHR

For a given image and swapchain, the region to present is specified by the `VkPresentRegionKHR` structure, which is defined as:

```
// Provided by VK_KHR_incremental_present
typedef struct VkPresentRegionKHR {
    uint32_t                rectangleCount;
    const VkRectLayerKHR*   pRectangles;
} VkPresentRegionKHR;
```

- `rectangleCount` is the number of rectangles in `pRectangles`, or zero if the entire image has changed and should be presented.
- `pRectangles` is either `NULL` or a pointer to an array of `VkRectLayerKHR` structures. The `VkRectLayerKHR` structure is the framebuffer coordinates, plus layer, of a portion of a presentable image that has changed and **must** be presented. If non-`NULL`, each entry in `pRectangles` is a rectangle of the given image that has changed since the last image was presented to the given swapchain. The rectangles **must** be specified relative to VkSurfaceCapabilitiesKHR ::`currentTransform`, regardless of the swapchain's `preTransform`. The presentation engine will apply the `preTransform` transformation to the rectangles, along with any further transformation it applies to the image content.

Valid Usage (Implicit)

- VUID-VkPresentRegionKHR-pRectangles-parameter
  If `rectangleCount` is not `0`, and `pRectangles` is not `NULL`, `pRectangles` **must** be a valid pointer to an array of `rectangleCount` valid VkRectLayerKHR structures

The `VkRectLayerKHR` structure is defined as:

```
// Provided by VK_KHR_incremental_present
typedef struct VkRectLayerKHR {
    VkOffset2D    offset;
    VkExtent2D    extent;
    uint32_t      layer;
} VkRectLayerKHR;
```

- `offset` is the origin of the rectangle, in pixels.
- `extent` is the size of the rectangle, in pixels.
- `layer` is the layer of the image. For images with only one layer, the value of `layer` **must** be 0.

Some platforms allow the size of a surface to change, and then scale the pixels of the image to fit the surface. `VkRectLayerKHR` specifies pixels of the swapchain's image(s), which will be constant for the life of the swapchain.

Valid Usage

- VUID-VkRectLayerKHR-offset-04864
  The sum of `offset` and `extent`, after being transformed according to the `preTransform` member of the VkSwapchainCreateInfoKHR structure, **must** be no greater than the `imageExtent` member of the VkSwapchainCreateInfoKHR structure passed to vkCreateSwapchainKHR
- VUID-VkRectLayerKHR-layer-01262
  `layer` **must** be less than the `imageArrayLayers` member of the VkSwapchainCreateInfoKHR structure passed to vkCreateSwapchainKHR

When the `VK_KHR_display_swapchain` extension is enabled, additional fields **can** be specified when presenting an image to a swapchain by setting VkPresentInfoKHR::`pNext` to point to a VkDisplayPresentInfoKHR structure.

The `VkDisplayPresentInfoKHR` structure is defined as:

```
// Provided by VK_KHR_display_swapchain
typedef struct VkDisplayPresentInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkRect2D           srcRect;
    VkRect2D           dstRect;
    VkBool32           persistent;
} VkDisplayPresentInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcRect` is a rectangular region of pixels to present. It **must** be a subset of the image being presented. If `VkDisplayPresentInfoKHR` is not specified, this region will be assumed to be the entire presentable image.
- `dstRect` is a rectangular region within the visible region of the swapchain's display mode. If `VkDisplayPresentInfoKHR` is not specified, this region will be assumed to be the entire visible region of the swapchain's mode. If the specified rectangle is a subset of the display mode's visible region, content from display planes below the swapchain's plane will be visible outside the rectangle. If there are no planes below the swapchain's, the area outside the specified rectangle will be black. If portions of the specified rectangle are outside of the display's visible region, pixels mapping only to those portions of the rectangle will be discarded.
- `persistent`: If this is VK_TRUE, the display engine will enable buffered mode on displays that support it. This allows the display engine to stop sending content to the display until a new image is presented. The display will instead maintain a copy of the last presented image. This allows less power to be used, but **may** increase presentation latency. If `VkDisplayPresentInfoKHR` is not specified, persistent mode will not be used.

If the extent of the `srcRect` and `dstRect` are not equal, the presented pixels will be scaled accordingly.

Valid Usage

- VUID-VkDisplayPresentInfoKHR-srcRect-01257
  `srcRect` **must** specify a rectangular region that is a subset of the image being presented
- VUID-VkDisplayPresentInfoKHR-dstRect-01258
  `dstRect` **must** specify a rectangular region that is a subset of the `visibleRegion` parameter of the display mode the swapchain being presented uses
- VUID-VkDisplayPresentInfoKHR-persistentContent-01259
  If the `persistentContent` member of the `VkDisplayPropertiesKHR` structure returned by `vkGetPhysicalDeviceDisplayPropertiesKHR` for the display the present operation targets is VK_FALSE, then `persistent` **must** be VK_FALSE

Valid Usage (Implicit)

- VUID-VkDisplayPresentInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DISPLAY_PRESENT_INFO_KHR

Structure Chaining

Extends the structure

- VkPresentInfoKHR

If the `pNext` chain of VkPresentInfoKHR includes a `VkDeviceGroupPresentInfoKHR` structure, then that structure includes an array of device masks and a device group present mode.

The `VkDeviceGroupPresentInfoKHR` structure is defined as:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
typedef struct VkDeviceGroupPresentInfoKHR {
    VkStructureType                          sType;
    const void*                              pNext;
    uint32_t                                 swapchainCount;
    const uint32_t*                          pDeviceMasks;
    VkDeviceGroupPresentModeFlagBitsKHR      mode;
} VkDeviceGroupPresentInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `swapchainCount` is zero or the number of elements in `pDeviceMasks`.
- `pDeviceMasks` is a pointer to an array of device masks, one for each element of VkPresentInfoKHR::`pSwapchains`.
- `mode` is a VkDeviceGroupPresentModeFlagBitsKHR value specifying the device group present mode that will be used for this present.

If `mode` is VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR, then each element of `pDeviceMasks` selects which instance of the swapchain image is presented. Each element of `pDeviceMasks` **must** have exactly one bit set, and the corresponding physical device **must** have a presentation engine as reported by VkDeviceGroupPresentCapabilitiesKHR.

If `mode` is VK_DEVICE_GROUP_PRESENT_MODE_REMOTE_BIT_KHR, then each element of `pDeviceMasks` selects which instance of the swapchain image is presented. Each element of `pDeviceMasks` **must** have exactly one bit set, and some physical device in the logical device **must** include that bit in its VkDeviceGroupPresentCapabilitiesKHR::`presentMask`.

If `mode` is VK_DEVICE_GROUP_PRESENT_MODE_SUM_BIT_KHR, then each element of `pDeviceMasks`

selects which instances of the swapchain image are component-wise summed and the sum of those images is presented. If the sum in any component is outside the representable range, the value of that component is undefined. Each element of pDeviceMasks **must** have a value for which all set bits are set in one of the elements of VkDeviceGroupPresentCapabilitiesKHR::presentMask.

If mode is VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_MULTI_DEVICE_BIT_KHR, then each element of pDeviceMasks selects which instance(s) of the swapchain images are presented. For each bit set in each element of pDeviceMasks, the corresponding physical device **must** have a presentation engine as reported by VkDeviceGroupPresentCapabilitiesKHR.

If VkDeviceGroupPresentInfoKHR is not provided or swapchainCount is zero then the masks are considered to be 1. If VkDeviceGroupPresentInfoKHR is not provided, mode is considered to be VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR.

Valid Usage

- VUID-VkDeviceGroupPresentInfoKHR-swapchainCount-01297
  swapchainCount **must** equal 0 or VkPresentInfoKHR::swapchainCount
- VUID-VkDeviceGroupPresentInfoKHR-mode-01298
  If mode is VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR, then each element of pDeviceMasks **must** have exactly one bit set, and the corresponding element of VkDeviceGroupPresentCapabilitiesKHR::presentMask **must** be non-zero
- VUID-VkDeviceGroupPresentInfoKHR-mode-01299
  If mode is VK_DEVICE_GROUP_PRESENT_MODE_REMOTE_BIT_KHR, then each element of pDeviceMasks **must** have exactly one bit set, and some physical device in the logical device **must** include that bit in its VkDeviceGroupPresentCapabilitiesKHR::presentMask
- VUID-VkDeviceGroupPresentInfoKHR-mode-01300
  If mode is VK_DEVICE_GROUP_PRESENT_MODE_SUM_BIT_KHR, then each element of pDeviceMasks **must** have a value for which all set bits are set in one of the elements of VkDeviceGroupPresentCapabilitiesKHR::presentMask
- VUID-VkDeviceGroupPresentInfoKHR-mode-01301
  If mode is VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_MULTI_DEVICE_BIT_KHR, then for each bit set in each element of pDeviceMasks, the corresponding element of VkDeviceGroupPresentCapabilitiesKHR::presentMask **must** be non-zero
- VUID-VkDeviceGroupPresentInfoKHR-pDeviceMasks-01302
  The value of each element of pDeviceMasks **must** be equal to the device mask passed in VkAcquireNextImageInfoKHR::deviceMask when the image index was last acquired
- VUID-VkDeviceGroupPresentInfoKHR-mode-01303
  mode **must** have exactly one bit set, and that bit **must** have been included in VkDeviceGroupSwapchainCreateInfoKHR::modes

Valid Usage (Implicit)

- VUID-VkDeviceGroupPresentInfoKHR-sType-sType

sType must be VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_INFO_KHR

- VUID-VkDeviceGroupPresentInfoKHR-pDeviceMasks-parameter
  If swapchainCount is not 0, pDeviceMasks must be a valid pointer to an array of swapchainCount uint32_t values
- VUID-VkDeviceGroupPresentInfoKHR-mode-parameter
  mode must be a valid VkDeviceGroupPresentModeFlagBitsKHR value

Structure Chaining

Extends the structure

- VkPresentInfoKHR

When the presentAtAbsoluteTime or presentAtRelativeTime feature is enabled, an application **can** instruct the presentation engine to attempt to display an image at a specified time, or for a minimum duration, by including the VkPresentTimingsInfoEXT structure in the pNext chain of the VkPresentInfoKHR structure.

The VkPresentTimingsInfoEXT structure is defined as:

```
// Provided by VK_EXT_present_timing
typedef struct VkPresentTimingsInfoEXT {
    VkStructureType                  sType;
    const void*                      pNext;
    uint32_t                         swapchainCount;
    const VkPresentTimingInfoEXT*    pTimingInfos;
} VkPresentTimingsInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- swapchainCount is the number of swapchains being presented to by this command.
- pTimingInfos is NULL or a pointer to an array of VkPresentTimingInfoEXT elements with swapchainCount entries. If not NULL, each element of pTimingInfos contains timing information for the presentation of the image corresponding to the entry in the VkPresentInfoKHR::pImageIndices array.

Valid Usage

- VUID-VkPresentTimingsInfoEXT-swapchainCount-12233
  swapchainCount must be equal to VkPresentInfoKHR::swapchainCount
- VUID-VkPresentTimingsInfoEXT-pSwapchains-12234
  All swapchains in VkPresentInfoKHR::pSwapchains must have been created with the VkSwapchainCreateInfoKHR::flags field containing VK_SWAPCHAIN_CREATE_PRESENT_TIMING_BIT_EXT

- VUID-VkPresentTimingsInfoEXT-pSwapchains-12235
  For each member of VkPresentInfoKHR::pSwapchains, if the associated VkPresentTimingInfoEXT::targetTime is not zero, the swapchain's current present mode **must** be VK_PRESENT_MODE_FIFO_LATEST_READY_KHR, VK_PRESENT_MODE_FIFO_KHR or VK_PRESENT_MODE_FIFO_RELAXED_KHR

Valid Usage (Implicit)

- VUID-VkPresentTimingsInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PRESENT_TIMINGS_INFO_EXT
- VUID-VkPresentTimingsInfoEXT-pTimingInfos-parameter
  If pTimingInfos is not NULL, pTimingInfos **must** be a valid pointer to an array of swapchainCount valid VkPresentTimingInfoEXT structures
- VUID-VkPresentTimingsInfoEXT-swapchainCount-arraylength
  swapchainCount **must** be greater than 0

Structure Chaining

Extends the structure

- VkPresentInfoKHR

The VkPresentTimingInfoEXT structure is defined as:

```
// Provided by VK_EXT_present_timing
typedef struct VkPresentTimingInfoEXT {
    VkStructureType                sType;
    const void*                    pNext;
    VkPresentTimingInfoFlagsEXT    flags;
    uint64_t                       targetTime;
    uint64_t                       timeDomainId;
    VkPresentStageFlagsEXT         presentStageQueries;
    VkPresentStageFlagsEXT         targetTimeDomainPresentStage;
} VkPresentTimingInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask of VkPresentTimingInfoFlagBitsEXT specifying options for how to interpret the timing information.
- targetTime is zero or a value specifying the target present time or duration, in nanoseconds, of the presentation request.
- timeDomainId is the id of the time domain used to specify the absolute target present time and the timing results obtained in a subsequent vkGetPastPresentationTimingEXT call for the

current presentation request.

- `presentStageQueries` is zero or a valid VkPresentStageFlagsEXT value indicating which present stages the presentation engine will collect timing information for.
- `targetTimeDomainPresentStage` is a valid VkPresentStageFlagsEXT specifying a single present stage used to interpret `targetTime`.

If `targetTime` is not zero, the implementation attempts to align the VK_PRESENT_STAGE_IMAGE_FIRST_PIXEL_VISIBLE_BIT_EXT present stage of that presentation request with the time specified in `targetTime` according to the time domain used. If VK_PRESENT_TIMING_INFO_PRESENT_AT_NEAREST_REFRESH_CYCLE_BIT_EXT is not set in `flags`, it indicates that the application would strictly prefer the image to not be visible before `targetTime` has lapsed.

If `targetTime` is not zero and `timeDomainId` is associated with a VK_TIME_DOMAIN_PRESENT_STAGE_LOCAL_EXT time domain, `targetTimeDomainPresentStage` is used to specify which present stage's time domain `targetTime` is specified for. Otherwise, `targetTimeDomainPresentStage` is ignored.

If `presentStageQueries` is not zero, a slot in the swapchain's internal results queue is reserved to hold the requested present timing data until the application retrieves them. If no slots are available, or if vkSetSwapchainPresentTimingQueueSizeEXT was never called, presentation fails and returns VK_ERROR_PRESENT_TIMING_QUEUE_FULL_EXT.

NOTE

Some platforms, due to hardware or system limitations, **may** not be able to accurately time `targetTime` with the actual physical event of the image becoming visible on the display. However, those timing capabilities **may** still be useful and result in improved animation quality.

As such, the `presentAtAbsoluteTime` and `presentAtRelativeTime` features do not provide a strict guarantee regarding the completion of the VK_PRESENT_STAGE_IMAGE_FIRST_PIXEL_VISIBLE_BIT_EXT present stage relative to the `targetTime`, and implementations **must** strive to make it as consistent and accurate as possible.

NOTE

Applications that specify an absolute present time **should** regularly rebase their calculations for their next target time on the feedback from vkGetPastPresentationTimingEXT to compensate for accumulated precision errors or potential clock drift. It is recommended that when targeting the time of a vertical blanking period, applications set VK_PRESENT_TIMING_INFO_PRESENT_AT_NEAREST_REFRESH_CYCLE_BIT_EXT to allow the implementation to compensate for small precision errors that may cause an image to be displayed one refresh cycle later than intended.

Some implementations **may** not provide timing data for the VK_PRESENT_STAGE_IMAGE_FIRST_PIXEL_VISIBLE_BIT_EXT present stage. Applications **should** use the supported present stage nearest to it to compute a desired target presentation time.

Valid Usage

- VUID-VkPresentTimingInfoEXT-targetTime-12236
  If `targetTime` is not zero and `flags` does not contain VK_PRESENT_TIMING_INFO_PRESENT_AT_RELATIVE_TIME_BIT_EXT, the `presentAtAbsoluteTime` feature **must** be enabled and the `presentAtAbsoluteTimeSupported` member of the `VkPresentTimingSurfaceCapabilitiesEXT` returned by `vkGetPhysicalDeviceSurfaceCapabilities2KHR` for the surface associated with the swapchain **must** be VK_TRUE
- VUID-VkPresentTimingInfoEXT-targetTime-12237
  If `targetTime` is not zero and `flags` contains VK_PRESENT_TIMING_INFO_PRESENT_AT_RELATIVE_TIME_BIT_EXT, the `presentAtRelativeTime` feature **must** be enabled and the `presentAtRelativeTimeSupported` member of the `VkPresentTimingSurfaceCapabilitiesEXT` returned by `vkGetPhysicalDeviceSurfaceCapabilities2KHR` for the surface associated with the swapchain **must** be VK_TRUE
- VUID-VkPresentTimingInfoEXT-timeDomainId-12238
  If `timeDomainId` is associated with a VK_TIME_DOMAIN_PRESENT_STAGE_LOCAL_EXT time domain, and `targetTime` is not zero, `targetTimeDomainPresentStage` **must** be a single `VkPresentStageFlagsEXT` value
- VUID-VkPresentTimingInfoEXT-timeDomainId-12400
  `timeDomainId` **must** be an id previously returned by vkGetSwapchainTimeDomainPropertiesEXT

Valid Usage (Implicit)

- VUID-VkPresentTimingInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PRESENT_TIMING_INFO_EXT
- VUID-VkPresentTimingInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkPresentTimingInfoEXT-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkPresentTimingInfoFlagBitsEXT values
- VUID-VkPresentTimingInfoEXT-presentStageQueries-parameter
  If `presentStageQueries` is not `0`, `presentStageQueries` **must** be a valid combination of VkPresentStageFlagBitsEXT values
- VUID-VkPresentTimingInfoEXT-targetTimeDomainPresentStage-parameter
  If `targetTimeDomainPresentStage` is not `0`, `targetTimeDomainPresentStage` **must** be a valid combination of VkPresentStageFlagBitsEXT values

Bits which **can** be set in VkPresentTimingInfoEXT::`flags`, specifying options for how to interpret timing information:

```
// Provided by VK_EXT_present_timing
typedef enum VkPresentTimingInfoFlagBitsEXT {
    VK_PRESENT_TIMING_INFO_PRESENT_AT_RELATIVE_TIME_BIT_EXT = 0x00000001,
    VK_PRESENT_TIMING_INFO_PRESENT_AT_NEAREST_REFRESH_CYCLE_BIT_EXT = 0x00000002,
} VkPresentTimingInfoFlagBitsEXT;
```

- VK_PRESENT_TIMING_INFO_PRESENT_AT_RELATIVE_TIME_BIT_EXT specifies that VkPresentTimingInfoEXT::targetTime is to be interpreted as a relative time from the previous presentation's VK_PRESENT_STAGE_IMAGE_FIRST_PIXEL_VISIBLE_BIT_EXT stage. If the swapchain has never been used to present an image, the provided targetTime is ignored.
- VK_PRESENT_TIMING_INFO_PRESENT_AT_NEAREST_REFRESH_CYCLE_BIT_EXT specifies that the application would prefer the image to be presented earlier than the time specified in VkPresentTimingInfoEXT::targetTime if that time falls within the first half of a refresh cycle. In that case, the presentation engine **may** choose to display the image at the start of that refresh cycle.

```
// Provided by VK_EXT_present_timing
typedef VkFlags VkPresentTimingInfoFlagsEXT;
```

VkPresentTimingInfoFlagsEXT is a bitmask type for setting a mask of zero or more VkPresentTimingInfoFlagBitsEXT.

Presenting an image to the user typically involves multiple stages. Bits which **can** be set to specify present stages are:

```
// Provided by VK_EXT_present_timing
typedef enum VkPresentStageFlagBitsEXT {
    VK_PRESENT_STAGE_QUEUE_OPERATIONS_END_BIT_EXT = 0x00000001,
    VK_PRESENT_STAGE_REQUEST_DEQUEUED_BIT_EXT = 0x00000002,
    VK_PRESENT_STAGE_IMAGE_FIRST_PIXEL_OUT_BIT_EXT = 0x00000004,
    VK_PRESENT_STAGE_IMAGE_FIRST_PIXEL_VISIBLE_BIT_EXT = 0x00000008,
} VkPresentStageFlagBitsEXT;
```

- VK_PRESENT_STAGE_QUEUE_OPERATIONS_END_BIT_EXT marks the end of the set of queue operations enqueued by vkQueuePresentKHR on the provided VkQueue for a presentation request.
- VK_PRESENT_STAGE_REQUEST_DEQUEUED_BIT_EXT is the stage after which the presentation request has been dequeued from the swapchain's internal presentation request queue, if any, as specified by the present mode associated with that request.
- VK_PRESENT_STAGE_IMAGE_FIRST_PIXEL_OUT_BIT_EXT is the stage after which data for the first pixel of the presentation request associated with the image has left the presentation engine for a display hardware.
- VK_PRESENT_STAGE_IMAGE_FIRST_PIXEL_VISIBLE_BIT_EXT is the stage after which a display hardware has made the first pixel visible for the presentation request associated with the image

to be presented.

NOTE	The set of queue operations delimited by VK_PRESENT_STAGE_QUEUE_OPERATIONS_END_BIT_EXT includes the wait for the semaphores specified in VkPresentInfoKHR::pWaitSemaphores, if any, and any work implicitly enqueued by the implementation.

```
// Provided by VK_EXT_present_timing
typedef VkFlags VkPresentStageFlagsEXT;
```

VkPresentStageFlagsEXT is a bitmask type for setting a mask of zero or more VkPresentStageFlagBitsEXT.

The VkPresentIdKHR structure is defined as:

```
// Provided by VK_KHR_present_id
typedef struct VkPresentIdKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           swapchainCount;
    const uint64_t*    pPresentIds;
} VkPresentIdKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- swapchainCount is the number of swapchains being presented to the vkQueuePresentKHR command.
- pPresentIds is NULL or a pointer to an array of uint64_t with swapchainCount entries. If not NULL, each non-zero value in pPresentIds specifies the present id to be associated with the presentation of the swapchain with the same index in the vkQueuePresentKHR call.

For applications to be able to reference specific presentation events queued by a call to vkQueuePresentKHR, an identifier needs to be associated with them. When the presentId feature is enabled, applications **can** include the VkPresentIdKHR structure in the pNext chain of the VkPresentInfoKHR structure to supply identifiers.

Each VkSwapchainKHR has a presentId associated with it. This value is initially zero when the VkSwapchainKHR is created.

When a VkPresentIdKHR structure with a non-NULL pPresentIds is included in the pNext chain of a VkPresentInfoKHR structure, each pSwapchains entry has a presentId associated in the pPresentIds array at the same index as the swapchain in the pSwapchains array. If this presentId is non-zero, then the application **can** later use this value to refer to that image presentation. A value of zero indicates that this presentation has no associated presentId. A non-zero presentId **must** be greater than any non-zero presentId passed previously by the application for the same swapchain.

There is no requirement for any precise timing relationship between the presentation of the image to the user and the update of the presentId value, but implementations **should** make this as close as possible to the presentation of the first pixel in the new image to the user.

Valid Usage

- VUID-VkPresentIdKHR-swapchainCount-04998
  `swapchainCount` **must** be the same value as VkPresentInfoKHR::`swapchainCount`, where this `VkPresentIdKHR` is in the `pNext` chain of the VkPresentInfoKHR structure
- VUID-VkPresentIdKHR-presentIds-04999
  Each non-zero entry in `pPresentIds` **must** be greater than all previously submitted present ids for the associated swapchain in VkPresentInfoKHR::`pSwapchains`

Valid Usage (Implicit)

- VUID-VkPresentIdKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PRESENT_ID_KHR
- VUID-VkPresentIdKHR-pPresentIds-parameter
  If `pPresentIds` is not `NULL`, `pPresentIds` **must** be a valid pointer to an array of `swapchainCount` `uint64_t` values
- VUID-VkPresentIdKHR-swapchainCount-arraylength
  `swapchainCount` **must** be greater than `0`

Structure Chaining

Extends the structure

- VkPresentInfoKHR

When the `presentWait` feature is enabled, an application **can** wait for an image to be presented to the user by first specifying a presentId for the target presentation by adding a VkPresentIdKHR structure to the `pNext` chain of the VkPresentInfoKHR structure and then waiting for that presentation to complete by calling:

```
// Provided by VK_KHR_present_wait
VkResult vkWaitForPresentKHR(
    VkDevice                                    device,
    VkSwapchainKHR                              swapchain,
    uint64_t                                    presentId,
    uint64_t                                    timeout);
```

- `device` is the device associated with `swapchain`.
- `swapchain` is the non-retired swapchain on which an image was queued for presentation.

- `presentId` is the presentation presentId to wait for.
- `timeout` is the timeout period in units of nanoseconds. `timeout` is adjusted to the closest value allowed by the implementation-dependent timeout accuracy, which **may** be substantially longer than one nanosecond, and **may** be longer than the requested period.

The call to `vkWaitForPresentKHR` will block until either the presentId associated with `swapchain` is greater than or equal to `presentId`, or `timeout` nanoseconds passes. When the swapchain becomes OUT_OF_DATE, the call will either return VK_SUCCESS (if the image was delivered to the presentation engine and may have been presented to the user) or will return early with status VK_ERROR_OUT_OF_DATE_KHR (if the image could not be presented to the user).

There is no requirement for any precise timing relationship between the presentation of the image to the user and the update of the presentId value, but implementations **should** make this as close as possible to the presentation of the first pixel in the next image being presented to the user.

For VK_PRESENT_MODE_MAILBOX_KHR (or other present mode where images may be replaced in the presentation queue) any wait of this type associated with such an image **must** be signaled no later than a wait associated with the replacing image would be signaled.

As an exception to the normal rules for objects which are externally synchronized, the `swapchain` passed to `vkWaitForPresentKHR` **may** be simultaneously used by other threads in calls to functions other than vkDestroySwapchainKHR. Access to the swapchain data associated with this extension **must** be atomic within the implementation.

Valid Usage

- VUID-vkWaitForPresentKHR-swapchain-04997
  `swapchain` **must** not be in the retired state
- VUID-vkWaitForPresentKHR-presentWait-06234
  The `presentWait` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkWaitForPresentKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkWaitForPresentKHR-swapchain-parameter
  `swapchain` **must** be a valid VkSwapchainKHR handle
- VUID-vkWaitForPresentKHR-swapchain-parent
  `swapchain` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `swapchain` **must** be externally synchronized

Return Codes

Success

- VK_SUBOPTIMAL_KHR
- VK_SUCCESS
- VK_TIMEOUT

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT
- VK_ERROR_OUT_OF_DATE_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkPresentId2KHR` structure is defined as:

```
// Provided by VK_KHR_present_id2
typedef struct VkPresentId2KHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           swapchainCount;
    const uint64_t*    pPresentIds;
} VkPresentId2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `swapchainCount` is the number of swapchains being presented to the vkQueuePresentKHR command.
- `pPresentIds` is `NULL` or a pointer to an array of uint64_t with `swapchainCount` entries. If not `NULL`, each non-zero value in `pPresentIds` specifies the present id to be associated with the presentation of the swapchain with the same index in the vkQueuePresentKHR call.

For applications to be able to reference specific presentation events queued by a call to vkQueuePresentKHR, an identifier needs to be associated with them.

When the VkSurfaceCapabilitiesPresentId2KHR surface capability is present for a surface, applications **can** include the `VkPresentId2KHR` structure in the `pNext` chain of the VkPresentInfoKHR structure to associate an identifier with each presentation request. The `pPresentIds` provides an identifier for the swapchain present at the corresponding index in VkPresentInfoKHR's `pSwapchains`

array.

If this presentId is non-zero, then the application **can** later use this value to refer to that image presentation. A value of zero indicates that this presentation has no associated presentId. A non-zero presentId **must** be greater than any non-zero presentId passed previously by the application for the same swapchain.

If a non-zero presentId was provided, this may be used with vkWaitForPresent2KHR for the application to synchronize against the presentation engine's processing of the presentation request.

NOTE

The ID namespace used by this extension **must** be shared with other extensions that allow the application to provide a 64-bit monotonically increasing presentation ID, such as the original VK_KHR_present_id.

This is to allow existing extensions that depend on VK_KHR_present_id to use VK_KHR_present_id2 provided IDs without change, as well as to simplify writing future extensions that require application provided presentation IDs.

Valid Usage

- VUID-VkPresentId2KHR-swapchainCount-10818
  `swapchainCount` **must** be the same value as VkPresentInfoKHR::`swapchainCount`, where this `VkPresentId2KHR` is in the `pNext` chain of the VkPresentInfoKHR structure
- VUID-VkPresentId2KHR-presentIds-10819
  Each non-zero entry in `pPresentIds` **must** be greater than all previously submitted present ids for the associated swapchain in VkPresentInfoKHR::`pSwapchains`
- VUID-VkPresentId2KHR-None-10820
  The swapchain **must** have been created with VK_SWAPCHAIN_CREATE_PRESENT_ID_2_BIT_KHR bit set in the VkSwapchainCreateFlagBitsKHR field

Valid Usage (Implicit)

- VUID-VkPresentId2KHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PRESENT_ID_2_KHR
- VUID-VkPresentId2KHR-pPresentIds-parameter
  If `pPresentIds` is not `NULL`, `pPresentIds` **must** be a valid pointer to an array of `swapchainCount` `uint64_t` values
- VUID-VkPresentId2KHR-swapchainCount-arraylength
  `swapchainCount` **must** be greater than `0`

Structure Chaining

Extends the structure

- VkPresentInfoKHR

When the VkSurfaceCapabilitiesPresentWait2KHR surface capability is present for a given surface, an application **can** wait for an image to be presented to the user by first specifying a presentId for the target presentation by adding a VkPresentId2KHR structure to the pNext chain of the VkPresentInfoKHR structure and then waiting for that presentation to complete by calling:

```
// Provided by VK_KHR_present_wait2
VkResult vkWaitForPresent2KHR(
    VkDevice                                    device,
    VkSwapchainKHR                              swapchain,
    const VkPresentWait2InfoKHR*                pPresentWait2Info);
```

- device is the device associated with swapchain.
- swapchain is the non-retired swapchain on which an image was queued for presentation.
- pPresentWait2Info is a pointer to a VkPresentWait2InfoKHR structure specifying the parameters of the wait.

vkWaitForPresent2KHR waits for the presentation engine to have begun presentation of the presentation request associated with the VkPresentWait2InfoKHR::presentId on swapchain, or for VkPresentWait2InfoKHR::timeout to have expired.

The wait request will complete when the timeout expires, or after the corresponding presentation request has either taken effect within the presentation engine or has been replaced without presentation.

The timing relationship between the presentation of the image to the user and the wait request completing is implementation-dependent due to variations in window system implementations.

If the swapchain becomes VK_ERROR_OUT_OF_DATE_KHR either before or during this call, the call **may** either return VK_SUCCESS (if the image was delivered to the presentation engine and **may** have been presented to the user) or return early with status VK_ERROR_OUT_OF_DATE_KHR (if the image could not be presented to the user).

As an exception to the normal rules for objects which are externally synchronized, the swapchain passed to vkWaitForPresent2KHR **may** be simultaneously used by other threads in calls to functions other than vkDestroySwapchainKHR. Access to the swapchain data associated with this extension **must** be atomic within the implementation.

Valid Usage

- VUID-vkWaitForPresent2KHR-presentWait2-10814
  The presentWait2 feature **must** be enabled

- VUID-vkWaitForPresent2KHR-None-10815
  The VkSurfaceCapabilitiesPresentWait2KHR surface capability **must** be present for the underlying surface
- VUID-vkWaitForPresent2KHR-None-10816
  The swapchain **must** have been created with VK_SWAPCHAIN_CREATE_PRESENT_WAIT_2_BIT_KHR bit set in the VkSwapchainCreateFlagBitsKHR field
- VUID-vkWaitForPresent2KHR-presentId-10817
  The `presentId` value **must** have been associated with a vkQueuePresentKHR request on the `swapchain` which returned a non-error value

Valid Usage (Implicit)

- VUID-vkWaitForPresent2KHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkWaitForPresent2KHR-swapchain-parameter
  `swapchain` **must** be a valid VkSwapchainKHR handle
- VUID-vkWaitForPresent2KHR-pPresentWait2Info-parameter
  `pPresentWait2Info` **must** be a valid pointer to a valid VkPresentWait2InfoKHR structure
- VUID-vkWaitForPresent2KHR-swapchain-parent
  `swapchain` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `swapchain` **must** be externally synchronized

Return Codes

Success

- VK_SUBOPTIMAL_KHR
- VK_SUCCESS
- VK_TIMEOUT

Failure

- VK_ERROR_DEVICE_LOST
- VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT
- VK_ERROR_OUT_OF_DATE_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_SURFACE_LOST_KHR

- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkSwapchainPresentModeInfoKHR structure is defined as:

```
// Provided by VK_KHR_swapchain_maintenance1
typedef struct VkSwapchainPresentModeInfoKHR {
    VkStructureType            sType;
    const void*                pNext;
    uint32_t                   swapchainCount;
    const VkPresentModeKHR*    pPresentModes;
} VkSwapchainPresentModeInfoKHR;
```

```
// Provided by VK_EXT_swapchain_maintenance1
// Equivalent to VkSwapchainPresentModeInfoKHR
typedef VkSwapchainPresentModeInfoKHR VkSwapchainPresentModeInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- swapchainCount is the number of swapchains being presented to by this command.
- pPresentModes is a list of presentation modes with swapchainCount entries.

If the pNext chain of VkPresentInfoKHR includes a VkSwapchainPresentModeInfoKHR structure, then that structure defines the presentation modes used for the current and subsequent presentation operations.

When the application changes present modes with VkSwapchainPresentModeInfoKHR, images that have already been queued for presentation will continue to be presented according to the previous present mode. The current image being queued for presentation and subsequent images will be presented according to the new present mode. The behavior during the transition between the two modes is defined as follows.

- Transition from VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR to VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR: the presentation engine updates the shared presentable image according to the behavior of VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR.
- Transition from VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR to VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR: the presentation engine **may** update the shared presentable image or defer that to its regular refresh cycle, according to the behavior of VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR.
- Transition between VK_PRESENT_MODE_FIFO_KHR and VK_PRESENT_MODE_FIFO_RELAXED_KHR: Images continue to be appended to the same FIFO queue, and the behavior with respect to waiting for vertical blanking period will follow the new mode for current and subsequent images.

- Transition from VK_PRESENT_MODE_IMMEDIATE_KHR to VK_PRESENT_MODE_FIFO_KHR or VK_PRESENT_MODE_FIFO_RELAXED_KHR or VK_PRESENT_MODE_FIFO_LATEST_READY_KHR : As all prior present requests in the VK_PRESENT_MODE_IMMEDIATE_KHR mode are applied immediately, there are no outstanding present operations in this mode, and current and subsequent images are appended to the FIFO queue and presented according to the new mode.
- Transition from VK_PRESENT_MODE_MAILBOX_KHR to VK_PRESENT_MODE_FIFO_KHR or VK_PRESENT_MODE_FIFO_RELAXED_KHR or VK_PRESENT_MODE_FIFO_LATEST_READY_KHR : Presentation in FIFO modes requires waiting for the next vertical blanking period, with VK_PRESENT_MODE_MAILBOX_KHR allowing the pending present operation to be replaced by a new one. In this case, the current present operation will replace the pending present operation and is applied according to the new mode.
- Transition from VK_PRESENT_MODE_FIFO_KHR or VK_PRESENT_MODE_FIFO_RELAXED_KHR or VK_PRESENT_MODE_FIFO_LATEST_READY_KHR to VK_PRESENT_MODE_IMMEDIATE_KHR or VK_PRESENT_MODE_MAILBOX_KHR: If the FIFO queue is empty, presentation is done according to the behavior of the new mode. If there are present operations in the FIFO queue, once the last present operation is performed based on the respective vertical blanking period, the current and subsequent updates are applied according to the new mode.
- Transition between VK_PRESENT_MODE_FIFO_KHR or VK_PRESENT_MODE_FIFO_RELAXED_KHR, and VK_PRESENT_MODE_FIFO_LATEST_READY_KHR: Images continue to be appended to the same FIFO queue, and the behavior with respect to waiting for vertical blanking period and dequeuing requests will follow the new mode for current and subsequent images.
- The behavior during transition between any other present modes, if possible, is implementation-defined.

Valid Usage

- VUID-VkSwapchainPresentModeInfoKHR-swapchainCount-07760
  `swapchainCount` **must** be equal to VkPresentInfoKHR::`swapchainCount`
- VUID-VkSwapchainPresentModeInfoKHR-pPresentModes-07761
  Each entry in `pPresentModes` **must** be a presentation mode specified in VkSwapchainPresentModesCreateInfoKHR::`pPresentModes` when creating the entry's corresponding swapchain

Valid Usage (Implicit)

- VUID-VkSwapchainPresentModeInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_MODE_INFO_KHR
- VUID-VkSwapchainPresentModeInfoKHR-pPresentModes-parameter
  `pPresentModes` **must** be a valid pointer to an array of `swapchainCount` valid VkPresentModeKHR values
- VUID-VkSwapchainPresentModeInfoKHR-swapchainCount-arraylength
  `swapchainCount` **must** be greater than `0`

Structure Chaining

Extends the structure

- VkPresentInfoKHR

The VkSwapchainPresentFenceInfoKHR structure is defined as:

```
// Provided by VK_KHR_swapchain_maintenance1
typedef struct VkSwapchainPresentFenceInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           swapchainCount;
    const VkFence*     pFences;
} VkSwapchainPresentFenceInfoKHR;
```

```
// Provided by VK_EXT_swapchain_maintenance1
// Equivalent to VkSwapchainPresentFenceInfoKHR
typedef VkSwapchainPresentFenceInfoKHR VkSwapchainPresentFenceInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- swapchainCount is the number of swapchains being presented to by this command.
- pFences is a list of fences with swapchainCount entries. Each entry **must** be VK_NULL_HANDLE or the handle of a fence to signal when the relevant operations on the associated swapchain have completed.

The set of *queue operations* defined by queuing an image for presentation, as well as operations performed by the presentation engine, access the payloads of objects associated with the presentation operation. The associated objects include:

- The swapchain image, its implicitly bound memory, and any other resources bound to that memory.
- The wait semaphores specified when queuing the image for presentation.

The application **can** provide a fence that the implementation will signal after all such queue operations have completed, and after the presentation engine has taken a reference to the payloads of all objects provided in VkPresentInfoKHR that the presentation engine accesses as part of the present operation. The fence **may** not wait for the present operation to complete. For all binary wait semaphores imported by the presentation engine using the equivalent of reference transference, as described in Importing Semaphore Payloads, this fence **must** not signal until all such semaphore payloads have been reset by the presentation engine.

The application **can** destroy the wait semaphores associated with a given presentation operation when at least one of the associated fences is signaled, and **can** destroy the swapchain when the

fences associated with all past presentation requests referring to that swapchain have signaled.

Fences associated with presentations to the same swapchain on the same VkQueue **must** be signaled in the same order as the present operations.

To specify a fence for each swapchain in a present operation, include the VkSwapchainPresentFenceInfoKHR structure in the pNext chain of the VkPresentInfoKHR structure.

Valid Usage

- VUID-VkSwapchainPresentFenceInfoKHR-swapchainCount-07757
  swapchainCount **must** be equal to VkPresentInfoKHR::swapchainCount
- VUID-VkSwapchainPresentFenceInfoKHR-pFences-07758
  Each element of pFences that is not VK_NULL_HANDLE **must** be unsignaled
- VUID-VkSwapchainPresentFenceInfoKHR-pFences-07759
  Each element of pFences that is not VK_NULL_HANDLE **must** not be associated with any other queue command that has not yet completed execution on that queue

Valid Usage (Implicit)

- VUID-VkSwapchainPresentFenceInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_FENCE_INFO_KHR
- VUID-VkSwapchainPresentFenceInfoKHR-pFences-parameter
  pFences **must** be a valid pointer to an array of swapchainCount valid or VK_NULL_HANDLE VkFence handles
- VUID-VkSwapchainPresentFenceInfoKHR-swapchainCount-arraylength
  swapchainCount **must** be greater than 0

Structure Chaining

Extends the structure

- VkPresentInfoKHR

To release images previously acquired through vkAcquireNextImage2KHR or vkAcquireNextImageKHR, call:

```
// Provided by VK_KHR_swapchain_maintenance1
VkResult vkReleaseSwapchainImagesKHR(
    VkDevice                                    device,
    const VkReleaseSwapchainImagesInfoKHR*      pReleaseInfo);
```

```
// Provided by VK_EXT_swapchain_maintenance1
// Equivalent to vkReleaseSwapchainImagesKHR
VkResult vkReleaseSwapchainImagesEXT(
    VkDevice                                    device,
    const VkReleaseSwapchainImagesInfoKHR*      pReleaseInfo);
```

- `device` is the device associated with VkReleaseSwapchainImagesInfoKHR::`swapchain`.
- `pReleaseInfo` is a pointer to a VkReleaseSwapchainImagesInfoKHR structure containing parameters of the release.

Only images that are not in use by the device **can** be released.

Releasing images is a read-only operation that will not affect the content of the released images. Upon reacquiring the image, the image contents and its layout will be the same as they were prior to releasing it. However, if a mechanism other than Vulkan is used to modify the platform window associated with the swapchain, the content of all presentable images in the swapchain becomes undefined.

NOTE This functionality is useful during swapchain recreation, where acquired images from the old swapchain can be released instead of presented.

Valid Usage

- VUID-vkReleaseSwapchainImagesKHR-swapchainMaintenance1-10159
  Feature `swapchainMaintenance1` **must** be enabled

Valid Usage (Implicit)

- VUID-vkReleaseSwapchainImagesKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkReleaseSwapchainImagesKHR-pReleaseInfo-parameter
  `pReleaseInfo` **must** be a valid pointer to a valid VkReleaseSwapchainImagesInfoKHR structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_SURFACE_LOST_KHR
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkReleaseSwapchainImagesInfoKHR structure is defined as:

```
// Provided by VK_KHR_swapchain_maintenance1
typedef struct VkReleaseSwapchainImagesInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkSwapchainKHR     swapchain;
    uint32_t           imageIndexCount;
    const uint32_t*    pImageIndices;
} VkReleaseSwapchainImagesInfoKHR;
```

```
// Provided by VK_EXT_swapchain_maintenance1
// Equivalent to VkReleaseSwapchainImagesInfoKHR
typedef VkReleaseSwapchainImagesInfoKHR VkReleaseSwapchainImagesInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- swapchain is a swapchain to which images are being released.
- imageIndexCount is the number of image indices to be released.
- pImageIndices is a pointer to an array of indices into the array of swapchain's presentable images, with imageIndexCount entries.

Valid Usage

- VUID-VkReleaseSwapchainImagesInfoKHR-pImageIndices-07785
  Each element of pImageIndices **must** be the index of a presentable image acquired from the swapchain specified by swapchain
- VUID-VkReleaseSwapchainImagesInfoKHR-pImageIndices-07786
  All uses of presentable images identified by elements of pImageIndices **must** have completed execution

Valid Usage (Implicit)

- VUID-VkReleaseSwapchainImagesInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_RELEASE_SWAPCHAIN_IMAGES_INFO_KHR
- VUID-VkReleaseSwapchainImagesInfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkReleaseSwapchainImagesInfoKHR-swapchain-parameter
  swapchain **must** be a valid VkSwapchainKHR handle
- VUID-VkReleaseSwapchainImagesInfoKHR-pImageIndices-parameter
  pImageIndices **must** be a valid pointer to an array of imageIndexCount uint32_t values

- VUID-VkReleaseSwapchainImagesInfoKHR-imageIndexCount-arraylength
  `imageIndexCount` **must** be greater than `0`

Host Synchronization

- Host access to `swapchain` **must** be externally synchronized

37.11. HDR Metadata

This section describes how to improve color reproduction of content to better reproduce colors as seen on the display used to originally optimize the content.

To provide HDR metadata to an implementation, call:

```
// Provided by VK_EXT_hdr_metadata
void vkSetHdrMetadataEXT(
    VkDevice                                    device,
    uint32_t                                    swapchainCount,
    const VkSwapchainKHR*                       pSwapchains,
    const VkHdrMetadataEXT*                     pMetadata);
```

- `device` is the logical device where the swapchain(s) were created.
- `swapchainCount` is the number of swapchains included in `pSwapchains`.
- `pSwapchains` is a pointer to an array of `swapchainCount` VkSwapchainKHR handles.
- `pMetadata` is a pointer to an array of `swapchainCount` VkHdrMetadataEXT structures.

The metadata will be applied to the specified VkSwapchainKHR objects at the next vkQueuePresentKHR call using that VkSwapchainKHR object. The metadata will persist until a subsequent `vkSetHdrMetadataEXT` changes it.

Valid Usage (Implicit)

- VUID-vkSetHdrMetadataEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkSetHdrMetadataEXT-pSwapchains-parameter
  `pSwapchains` **must** be a valid pointer to an array of `swapchainCount` valid VkSwapchainKHR handles
- VUID-vkSetHdrMetadataEXT-pMetadata-parameter
  `pMetadata` **must** be a valid pointer to an array of `swapchainCount` valid VkHdrMetadataEXT structures
- VUID-vkSetHdrMetadataEXT-swapchainCount-arraylength
  `swapchainCount` **must** be greater than `0`

- VUID-vkSetHdrMetadataEXT-pSwapchains-parent
  Each element of pSwapchains **must** have been created, allocated, or retrieved from device

Host Synchronization

- Host access to each member of pSwapchains **must** be externally synchronized

The VkHdrMetadataEXT structure is defined as:

```
// Provided by VK_EXT_hdr_metadata
typedef struct VkHdrMetadataEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkXYColorEXT       displayPrimaryRed;
    VkXYColorEXT       displayPrimaryGreen;
    VkXYColorEXT       displayPrimaryBlue;
    VkXYColorEXT       whitePoint;
    float              maxLuminance;
    float              minLuminance;
    float              maxContentLightLevel;
    float              maxFrameAverageLightLevel;
} VkHdrMetadataEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- displayPrimaryRed is a VkXYColorEXT structure specifying the red primary of the display used to optimize the content.
- displayPrimaryGreen is a VkXYColorEXT structure specifying the green primary of the display used to optimize the content.
- displayPrimaryBlue is a VkXYColorEXT structure specifying the blue primary of the display used to optimize the content.
- whitePoint is a VkXYColorEXT structure specifying the white-point of the display used to optimize the content.
- maxLuminance is the maximum luminance of the display used to optimize the content in nits.
- minLuminance is the minimum luminance of the display used to optimize the content in nits.
- maxContentLightLevel is the value in nits of the desired luminance for the brightest pixels in the displayed image.
- maxFrameAverageLightLevel is the value in nits of the average luminance of the frame which has the brightest average luminance anywhere in the content.

If any of the above values are unknown, they **can** be set to 0.

NOTE The meta-data provided here is intended to be used as defined in the SMPTE 2086,

> CTA 861.3 and CIE 15:2004 specifications. The validity and use of this data is outside the scope of Vulkan.

Valid Usage (Implicit)

- VUID-VkHdrMetadataEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_HDR_METADATA_EXT
- VUID-VkHdrMetadataEXT-pNext-pNext
  `pNext` **must** be `NULL`

The `VkXYColorEXT` structure is defined as:

```
// Provided by VK_EXT_hdr_metadata
typedef struct VkXYColorEXT {
    float    x;
    float    y;
} VkXYColorEXT;
```

- `x` is the x chromaticity coordinate.
- `y` is the y chromaticity coordinate.

Chromaticity coordinates are as specified in CIE 15:2004 “Calculation of chromaticity coordinates” (Section 7.3) and are limited to between 0 and 1 for real colors.

Chapter 38. Deferred Host Operations

Certain Vulkan commands are inherently expensive for the host CPU to execute. It is often desirable to offload such work onto background threads, and to parallelize the work across multiple CPUs. The concept of *deferred operations* allows applications and drivers to coordinate the execution of expensive host commands using an application-managed thread pool.

The `VK_KHR_deferred_host_operations` extension defines the infrastructure and usage patterns for *deferrable commands*, but does not specify any commands as deferrable. This is left to additional dependent extensions. Commands **must** not be deferred unless the deferral is specifically allowed by another extension which depends on `VK_KHR_deferred_host_operations`. This specification will refer to such extensions as *deferral extensions*.

38.1. Requesting Deferral

When an application requests an operation deferral, the implementation **may** defer the operation. When deferral is requested and the implementation defers any operation, the implementation **must** return VK_OPERATION_DEFERRED_KHR as the success code if no errors occurred. When deferral is requested, the implementation **should** defer the operation when the workload is significant, however if the implementation chooses not to defer any of the requested operations and instead executes all of them immediately, the implementation **must** return VK_OPERATION_NOT_DEFERRED_KHR as the success code if no errors occurred.

A deferred operation is created *complete* with an initial result value of VK_SUCCESS. The deferred operation becomes *pending* when an operation has been successfully deferred with that deferred operation object.

A deferred operation is considered pending until the deferred operation completes. A pending deferred operation becomes *complete* when it has been fully executed by one or more threads. Pending deferred operations will never complete until they are *joined* by an application thread, using vkDeferredOperationJoinKHR. Applications **can** join multiple threads to the same deferred operation, enabling concurrent execution of subtasks within that operation.

The application **can** query the status of a VkDeferredOperationKHR using the vkGetDeferredOperationMaxConcurrencyKHR or vkGetDeferredOperationResultKHR commands.

Parameters to the command requesting a deferred operation **may** be accessed by the implementation at any time until the deferred operation enters the complete state. The application **must** obey the following rules while a deferred operation is pending:

- Externally synchronized parameters **must** not be accessed.
- Pointer parameters **must** not be modified (e.g. reallocated/freed).
- The contents of pointer parameters which **may** be read by the command **must** not be modified.
- The contents of pointer parameters which **may** be written by the command **must** not be read.
- Vulkan object parameters **must** not be passed as externally synchronized parameters to any other command.

When the deferred operation is complete, the application **should** call vkGetDeferredOperationResultKHR to obtain the VkResult indicating success or failure of the operation. The VkResult value returned will be one of the values that the command requesting the deferred operation is able to return. Writes to output parameters of the requesting command will happen-before the deferred operation is complete.

When a deferral is requested for a command, the implementation **may** perform memory management operations on the allocator supplied to vkCreateDeferredOperationKHR for the deferred operation object, as described in the Memory Allocation chapter. Such allocations **must** occur on the thread which requests deferral.

If an allocator was supplied for the deferred command at the time of the deferral request, then the implementation **may** perform memory management operations on this allocator during the execution of vkDeferredOperationJoinKHR. These operations **may** occur concurrently and **may** be performed by any joined thread. The application **must** ensure that the supplied allocator is able to operate correctly under these conditions.

38.2. Deferred Host Operations API

The VkDeferredOperationKHR handle is defined as:

```
// Provided by VK_KHR_deferred_host_operations
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDeferredOperationKHR)
```

This handle refers to a tracking structure which manages the execution state for a deferred command.

To construct the tracking object for a deferred command, call:

```
// Provided by VK_KHR_deferred_host_operations
VkResult vkCreateDeferredOperationKHR(
    VkDevice                                    device,
    const VkAllocationCallbacks*                pAllocator,
    VkDeferredOperationKHR*                     pDeferredOperation);
```

- `device` is the device which owns `operation`.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pDeferredOperation` is a pointer to a handle in which the created VkDeferredOperationKHR is returned.

Valid Usage (Implicit)

- VUID-vkCreateDeferredOperationKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateDeferredOperationKHR-pAllocator-parameter

If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure

- VUID-vkCreateDeferredOperationKHR-pDeferredOperation-parameter
  `pDeferredOperation` **must** be a valid pointer to a VkDeferredOperationKHR handle
- VUID-vkCreateDeferredOperationKHR-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To assign a thread to a deferred operation, call:

```
// Provided by VK_KHR_deferred_host_operations
VkResult vkDeferredOperationJoinKHR(
    VkDevice                                    device,
    VkDeferredOperationKHR                      operation);
```

- `device` is the device which owns `operation`.
- `operation` is the deferred operation that the calling thread should work on.

The `vkDeferredOperationJoinKHR` command will execute a portion of the deferred operation on the calling thread.

The return value will be one of the following:

- A return value of VK_SUCCESS indicates that `operation` is complete. The application **should** use vkGetDeferredOperationResultKHR to retrieve the result of `operation`.
- A return value of VK_THREAD_DONE_KHR indicates that the deferred operation is not complete, but there is no work remaining to assign to threads. Future calls to vkDeferredOperationJoinKHR are not necessary and will simply harm performance. This situation **may** occur when other threads executing vkDeferredOperationJoinKHR are about to complete `operation`, and the implementation is unable to partition the workload any further.
- A return value of VK_THREAD_IDLE_KHR indicates that the deferred operation is not complete, and there is no work for the thread to do at the time of the call. This situation **may** occur if the operation encounters a temporary reduction in parallelism. By returning VK_THREAD_IDLE_KHR, the implementation is signaling that it expects that more opportunities for parallelism will emerge as execution progresses, and that future calls to

vkDeferredOperationJoinKHR **can** be beneficial. In the meantime, the application **can** perform other work on the calling thread.

Implementations **must** guarantee forward progress by enforcing the following invariants:

1. If only one thread has invoked vkDeferredOperationJoinKHR on a given operation, that thread **must** execute the operation to completion and return VK_SUCCESS.
2. If multiple threads have concurrently invoked vkDeferredOperationJoinKHR on the same operation, then at least one of them **must** complete the operation and return VK_SUCCESS.

Valid Usage (Implicit)

- VUID-vkDeferredOperationJoinKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDeferredOperationJoinKHR-operation-parameter
  `operation` **must** be a valid VkDeferredOperationKHR handle
- VUID-vkDeferredOperationJoinKHR-operation-parent
  `operation` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS
- VK_THREAD_DONE_KHR
- VK_THREAD_IDLE_KHR

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

When a deferred operation is completed, the application **can** destroy the tracking object by calling:

```
// Provided by VK_KHR_deferred_host_operations
void vkDestroyDeferredOperationKHR(
    VkDevice                                    device,
    VkDeferredOperationKHR                      operation,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the device which owns `operation`.
- `operation` is the completed operation to be destroyed.

- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyDeferredOperationKHR-operation-03434
  If `VkAllocationCallbacks` were provided when `operation` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyDeferredOperationKHR-operation-03435
  If no `VkAllocationCallbacks` were provided when `operation` was created, `pAllocator` **must** be `NULL`
- VUID-vkDestroyDeferredOperationKHR-operation-03436
  `operation` **must** be completed

Valid Usage (Implicit)

- VUID-vkDestroyDeferredOperationKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyDeferredOperationKHR-operation-parameter
  If `operation` is not VK_NULL_HANDLE, `operation` **must** be a valid VkDeferredOperationKHR handle
- VUID-vkDestroyDeferredOperationKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyDeferredOperationKHR-operation-parent
  If `operation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `operation` **must** be externally synchronized

To query the number of additional threads that can usefully be joined to a deferred operation, call:

```
// Provided by VK_KHR_deferred_host_operations
uint32_t vkGetDeferredOperationMaxConcurrencyKHR(
    VkDevice                                    device,
    VkDeferredOperationKHR                      operation);
```

- `device` is the device which owns `operation`.
- `operation` is the deferred operation to be queried.

The returned value is the maximum number of threads that can usefully execute a deferred

operation concurrently, reported for the state of the deferred operation at the point this command is called. This value is intended to be used to better schedule work onto available threads. Applications **can** join any number of threads to the deferred operation and expect it to eventually complete, though excessive joins **may** return VK_THREAD_DONE_KHR immediately, performing no useful work.

If `operation` is complete, `vkGetDeferredOperationMaxConcurrencyKHR` returns zero.

If `operation` is currently joined to any threads, the value returned by this command **may** immediately be out of date.

If `operation` is pending, implementations **must** not return zero unless at least one thread is currently executing vkDeferredOperationJoinKHR on `operation`. If there are such threads, the implementation **should** return an estimate of the number of additional threads which it could profitably use.

Implementations **may** return 2^{32}-1 to indicate that the maximum concurrency is unknown and cannot be easily derived. Implementations **may** return values larger than the maximum concurrency available on the host CPU. In these situations, an application **should** clamp the return value rather than oversubscribing the machine.

NOTE	The recommended usage pattern for applications is to query this value once, after deferral, and schedule no more than the specified number of threads to join the operation. Each time a joined thread receives VK_THREAD_IDLE_KHR, the application should schedule an additional join at some point in the future, but is not required to do so.

Valid Usage (Implicit)

- VUID-vkGetDeferredOperationMaxConcurrencyKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDeferredOperationMaxConcurrencyKHR-operation-parameter
  `operation` **must** be a valid VkDeferredOperationKHR handle
- VUID-vkGetDeferredOperationMaxConcurrencyKHR-operation-parent
  `operation` **must** have been created, allocated, or retrieved from `device`

The `vkGetDeferredOperationResultKHR` function is defined as:

```
// Provided by VK_KHR_deferred_host_operations
VkResult vkGetDeferredOperationResultKHR(
    VkDevice                                    device,
    VkDeferredOperationKHR                      operation);
```

- `device` is the device which owns `operation`.
- `operation` is the operation whose deferred result is being queried.

If no command has been deferred on `operation`, `vkGetDeferredOperationResultKHR` returns VK_SUCCESS.

If the deferred operation is pending, `vkGetDeferredOperationResultKHR` returns VK_NOT_READY.

If the deferred operation is complete, it returns the appropriate return value from the original command. This value **must** be one of the VkResult values which could have been returned by the original command if the operation had not been deferred.

Valid Usage (Implicit)

- VUID-vkGetDeferredOperationResultKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDeferredOperationResultKHR-operation-parameter
  `operation` **must** be a valid VkDeferredOperationKHR handle
- VUID-vkGetDeferredOperationResultKHR-operation-parent
  `operation` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_NOT_READY
- VK_SUCCESS

Failure

- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Chapter 39. Private Data

The private data extension provides a way for users to associate arbitrary application-defined data with Vulkan objects. This association is accomplished by storing 64-bit unsigned integers of application-defined data in private data slots. A private data slot represents a storage allocation for one data item for each child object of the device.

An application **can** reserve private data slots at device creation. To reserve private data slots, insert a VkDevicePrivateDataCreateInfo in the pNext chain in VkDeviceCreateInfo before device creation. Multiple VkDevicePrivateDataCreateInfo structures **can** be chained together, and the sum of the requested slots will be reserved. This is an exception to the specified valid usage for structure pointer chains. Reserving slots in this manner is not strictly necessary but it **may** improve performance.

Private data slots are represented by VkPrivateDataSlot handles:

```
// Provided by VK_VERSION_1_3
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkPrivateDataSlot)
```

To create a private data slot, call:

```
// Provided by VK_VERSION_1_3
VkResult vkCreatePrivateDataSlot(
    VkDevice                                    device,
    const VkPrivateDataSlotCreateInfo*          pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkPrivateDataSlot*                          pPrivateDataSlot);
```

- device is the logical device associated with the creation of the object(s) holding the private data slot.
- pCreateInfo is a pointer to a VkPrivateDataSlotCreateInfo structure.
- pAllocator controls host memory allocation as described in the Memory Allocation chapter.
- pPrivateDataSlot is a pointer to a VkPrivateDataSlot handle in which the resulting private data slot is returned.

Valid Usage

- VUID-vkCreatePrivateDataSlot-privateData-04564
  The privateData feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCreatePrivateDataSlot-device-parameter
  device **must** be a valid VkDevice handle

- VUID-vkCreatePrivateDataSlot-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkPrivateDataSlotCreateInfo structure
- VUID-vkCreatePrivateDataSlot-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreatePrivateDataSlot-pPrivateDataSlot-parameter
  `pPrivateDataSlot` **must** be a valid pointer to a VkPrivateDataSlot handle
- VUID-vkCreatePrivateDataSlot-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkPrivateDataSlotCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPrivateDataSlotCreateInfo {
    VkStructureType                 sType;
    const void*                     pNext;
    VkPrivateDataSlotCreateFlags    flags;
} VkPrivateDataSlotCreateInfo;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkPrivateDataSlotCreateFlagBits specifying additional parameters of the private data slot.

Valid Usage

Valid Usage (Implicit)

- VUID-VkPrivateDataSlotCreateInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PRIVATE_DATA_SLOT_CREATE_INFO

- VUID-VkPrivateDataSlotCreateInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkPrivateDataSlotCreateInfo-flags-zerobitmask
  `flags` **must** be `0`

```
// Provided by VK_VERSION_1_3
typedef VkFlags VkPrivateDataSlotCreateFlags;
```

`VkPrivateDataSlotCreateFlags` is a bitmask type for setting a mask of zero or more VkPrivateDataSlotCreateFlagBits.

To destroy a private data slot, call:

```
// Provided by VK_VERSION_1_3
void vkDestroyPrivateDataSlot(
    VkDevice                                    device,
    VkPrivateDataSlot                           privateDataSlot,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device associated with the creation of the object(s) holding the private data slot.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `privateDataSlot` is the private data slot to destroy.

Valid Usage

- VUID-vkDestroyPrivateDataSlot-privateDataSlot-04062
  If `VkAllocationCallbacks` were provided when `privateDataSlot` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyPrivateDataSlot-privateDataSlot-04063
  If no `VkAllocationCallbacks` were provided when `privateDataSlot` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyPrivateDataSlot-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyPrivateDataSlot-privateDataSlot-parameter
  If `privateDataSlot` is not VK_NULL_HANDLE, `privateDataSlot` **must** be a valid VkPrivateDataSlot handle
- VUID-vkDestroyPrivateDataSlot-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid

VkAllocationCallbacks structure

- VUID-vkDestroyPrivateDataSlot-privateDataSlot-parent
  If `privateDataSlot` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `privateDataSlot` **must** be externally synchronized

To store application-defined data in a slot associated with a Vulkan object, call:

```
// Provided by VK_VERSION_1_3
VkResult vkSetPrivateData(
    VkDevice                                    device,
    VkObjectType                                objectType,
    uint64_t                                    objectHandle,
    VkPrivateDataSlot                           privateDataSlot,
    uint64_t                                    data);
```

- `device` is the device that created the object.
- `objectType` is a VkObjectType specifying the type of object to associate data with.
- `objectHandle` is a handle to the object to associate data with.
- `privateDataSlot` is a handle to a VkPrivateDataSlot specifying location of private data storage.
- `data` is application-defined data to associate the object with. This data will be stored at `privateDataSlot`.

Valid Usage

- VUID-vkSetPrivateData-objectHandle-04016
  `objectHandle` **must** be `device` or a child of `device`
- VUID-vkSetPrivateData-objectHandle-04017
  `objectHandle` **must** be a valid handle to an object of type `objectType`

Valid Usage (Implicit)

- VUID-vkSetPrivateData-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkSetPrivateData-objectType-parameter
  `objectType` **must** be a valid VkObjectType value
- VUID-vkSetPrivateData-privateDataSlot-parameter
  `privateDataSlot` **must** be a valid VkPrivateDataSlot handle

- VUID-vkSetPrivateData-privateDataSlot-parent
  `privateDataSlot` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To retrieve application-defined data from a slot associated with a Vulkan object, call:

```
// Provided by VK_VERSION_1_3
void vkGetPrivateData(
    VkDevice                                    device,
    VkObjectType                                objectType,
    uint64_t                                    objectHandle,
    VkPrivateDataSlot                           privateDataSlot,
    uint64_t*                                   pData);
```

- `device` is the device that created the object.
- `objectType` is a VkObjectType specifying the type of object data is associated with.
- `objectHandle` is a handle to the object data is associated with.
- `privateDataSlot` is a handle to a VkPrivateDataSlot specifying location of private data pointer storage.
- `pData` is a pointer to specify where application-defined data is returned. `0` will be written in the absence of a previous call to `vkSetPrivateData` using the object specified by `objectHandle`.

NOTE

Due to platform details on Android, implementations might not be able to reliably return `0` from calls to `vkGetPrivateData` for VkSwapchainKHR objects on which `vkSetPrivateData` has not previously been called. This erratum is exclusive to the Android platform and objects of type VkSwapchainKHR.

Valid Usage

- VUID-vkGetPrivateData-objectType-04018
  `objectHandle` **must** be `device` or a child of `device`
- VUID-vkGetPrivateData-objectHandle-09498
  `objectHandle` **must** be a valid handle to an object of type `objectType`

Valid Usage (Implicit)

- VUID-vkGetPrivateData-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetPrivateData-objectType-parameter
  `objectType` **must** be a valid VkObjectType value
- VUID-vkGetPrivateData-privateDataSlot-parameter
  `privateDataSlot` **must** be a valid VkPrivateDataSlot handle
- VUID-vkGetPrivateData-pData-parameter
  `pData` **must** be a valid pointer to a `uint64_t` value
- VUID-vkGetPrivateData-privateDataSlot-parent
  `privateDataSlot` **must** have been created, allocated, or retrieved from `device`

Chapter 40. Acceleration Structures

40.1. Acceleration Structures

Acceleration structures are data structures used by the implementation to efficiently manage scene geometry as it is traversed during a ray tracing query. The application is responsible for managing acceleration structure objects (see Acceleration Structures), including allocation, destruction, executing builds or updates, and synchronizing resources used during ray tracing queries.

There are two types of acceleration structures, *top level acceleration structures* and *bottom level acceleration structures*.

An acceleration structure is considered to be constructed if an acceleration structure build command or copy command has been executed with the given acceleration structure as the destination.

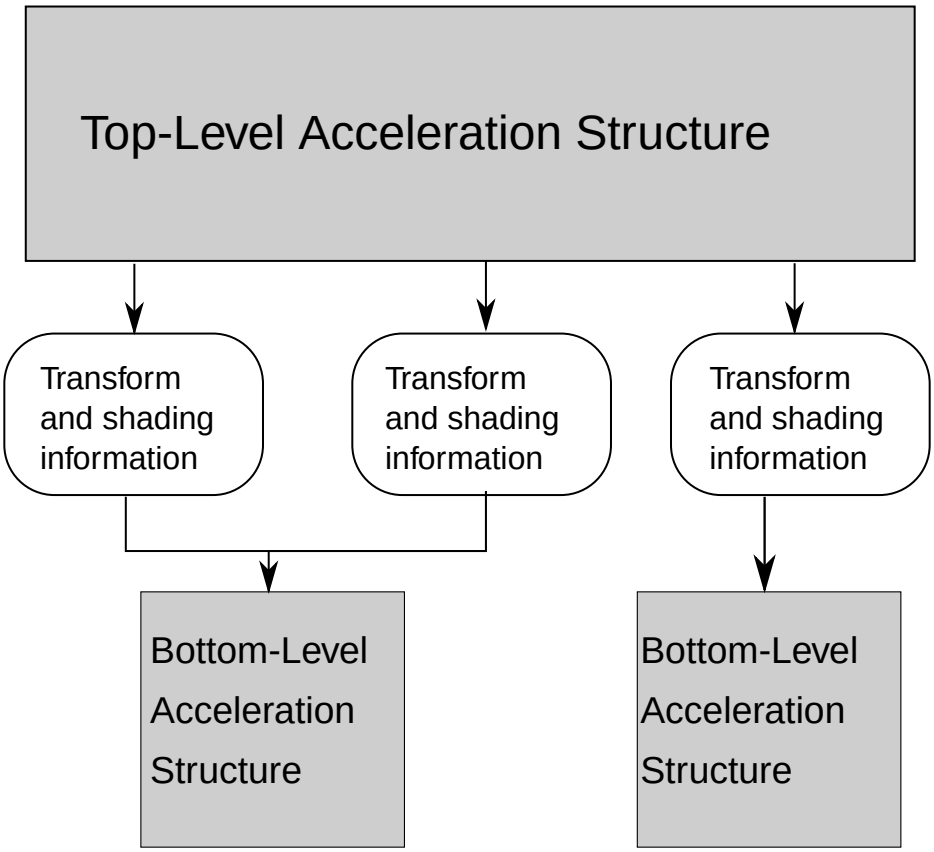


Figure 22. Acceleration Structure

Caption

The diagram shows the relationship between top and bottom level acceleration structures.

40.1.1. Geometry

Geometries refer to a triangle, or axis-aligned bounding box.

A triangle is a fundamental geometric primitive defined by three vertices in 3D space, forming a flat, planar surface.

An axis-aligned bounding box (AABB) is a rectangular box defined by two points (minimum and maximum corners) that encloses a 3D object or scene. Its faces are aligned with the coordinate axes, making intersection tests efficient for spatial partitioning and acceleration structures.

40.1.2. Top Level Acceleration Structures

Opaque acceleration structure for an array of instances. The descriptor or device address referencing this is the starting point for traversal.

The top level acceleration structure takes a reference to any bottom level acceleration structure referenced by its instances. Those bottom level acceleration structure objects **must** be valid when the top level acceleration structure is accessed.

40.1.3. Bottom Level Acceleration Structures

Opaque acceleration structure for an array of geometries.

40.1.4. Acceleration Structure Update Rules

The API defines two types of operations to produce acceleration structures from geometry:

- A *build* operation is used to construct an acceleration structure.
- An *update* operation is used to modify an existing acceleration structure.

An update operation imposes certain constraints on the input, in exchange for considerably faster execution. When performing an update, the application is required to provide a full description of the acceleration structure, but is prohibited from changing anything other than instance definitions, transform matrices, and vertex or AABB positions. All other aspects of the description **must** exactly match the one from the original build.

More precisely, the application **must** not use an update operation to do any of the following:

- Change primitives or instances from *active* to *inactive*, or vice versa (as defined in Inactive Primitives and Instances).
- Change the index or vertex formats of triangle geometry.
- Change triangle geometry transform pointers from null to non-null or vice versa.
- Change the number of geometries or instances in the structure.
- Change the geometry flags for any geometry in the structure.
- Change the number of vertices or primitives for any geometry in the structure.

If the original acceleration structure was built using opacity micromaps and VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_BIT_EXT was set in `flags`, the application **must** provide a micromap matching the original micromap in structure with only opacity values updated. The application is prohibited from changing anything

other than the specific opacity values assigned to the triangles.

More precisely, the application **must** not use an update operation to do any of the following:

- Remove micromaps or VkOpacityMicromapSpecialIndexKHR values from a geometry which previously had them, or vice versa.
- Change between use of VkOpacityMicromapSpecialIndexKHR values and explicit micro-map triangles.
- Change the subdivision level or format of the micromap triangle associated with any acceleration-structure triangle.

Acceleration structures constructed by deserialization with vkCmdCopyMemoryToAccelerationStructureKHR and originally built with the VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_BIT_EXT flag **must** provide a micromap deserialized using the exact serialized output of the micromap that was originally referenced by the serialized acceleration structure.

NOTE If applications need to update micromap data referenced by a deserialized acceleration structure, then they must originally build the acceleration structure with the VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_KHR flag and perform a full micromap update.

If the original acceleration structure was built using opacity micromaps and VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_KHR was set in `flags`, the application **must** provide a micromap to the update operation.

A VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_BIT_EXT or VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_KHR operation transfers the reference in the acceleration structure to the new micromap unless VkMicromapBuildSizesInfoEXT::`discardable` is VK_TRUE .

If the original acceleration structure was built using opacity micromaps and neither opacity micromap update flag is set, the application **must** provide the original micromap to the update operation or the deserialized micromap from the original micromap's exact serialized data if the original acceleration structure was deserialized prior.

40.1.5. Inactive Primitives and Instances

Acceleration structures allow the use of particular input values to signal *inactive* primitives or instances.

An *inactive* triangle is one for which the first (X) component of any vertex is NaN. If any other vertex component is NaN, and the first is not, the behavior is undefined. If the vertex format does not have a NaN representation, then all triangles are considered active.

An *inactive* instance is one whose acceleration structure reference is `0`.

An *inactive* AABB is one for which the minimum X coordinate is NaN. If any other component is

NaN, and the first is not, the behavior is undefined.

In the above definitions, “NaN” refers to any type of NaN. Signaling, non-signaling, quiet, loud, or otherwise.

An inactive object is considered invisible to all rays, and **should** not be represented in the acceleration structure. Implementations **should** ensure that the presence of inactive objects does not seriously degrade traversal performance.

Inactive objects are counted in the auto-generated index sequences which are provided to shaders via `InstanceId` and `PrimitiveId` SPIR-V decorations. This allows objects in the scene to change freely between the active and inactive states, without affecting the layout of any arrays which are being indexed using the ID values.

Any transition between the active and inactive states requires a full acceleration structure rebuild. Applications **must** not perform an acceleration structure update where an object is active in the source acceleration structure but would be inactive in the destination, or vice versa.

40.1.6. Degenerate Primitives and Instances

Degenerate primitives and instances behave differently to inactive primitives and instances, and are defined as:

- triangles that have one or more vertices whose respective (X), (Y), (Z) components are identical, or have three vertices that have at least two of the (X), (Y), or (Z) components identical, therefore forming a line or point. Degenerate triangles do not generate any intersections.
- AABBs whose `minX`=`maxX`, `minY`=`maxY`, and `minZ`=`maxZ`. Degenerate AABBs **may** invoke the intersection shader.
- instances that reference bottom level acceleration structures that contain no active primitives. When building an acceleration structure, implementations **should** treat degenerate instances as though they are a point at the instance origin, specified by VkAccelerationStructureInstanceKHR::`transform`.

Unlike *inactive* primitives and instances, *degenerate* primitives and instances **may** transition from the degenerate to the non-degenerate state, or vice versa, when performing an acceleration structure update.

If an acceleration structure is built without VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR set in VkAccelerationStructureBuildGeometryInfoKHR::`flags` , degenerate primitives **may** be discarded. Primitives that are defined with the same index value for more than one vertex **can** always be discarded.

40.1.7. Building Acceleration Structures

To build acceleration structures call:

```
// Provided by VK_KHR_acceleration_structure
void vkCmdBuildAccelerationStructuresKHR(
    VkCommandBuffer                                 commandBuffer,
    uint32_t                                        infoCount,
    const VkAccelerationStructureBuildGeometryInfoKHR* pInfos,
    const VkAccelerationStructureBuildRangeInfoKHR* const* ppBuildRangeInfos);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `infoCount` is the number of acceleration structures to build. It specifies the number of the `pInfos` structures and `ppBuildRangeInfos` pointers that **must** be provided.
- `pInfos` is a pointer to an array of `infoCount` VkAccelerationStructureBuildGeometryInfoKHR structures defining the geometry used to build each acceleration structure.
- `ppBuildRangeInfos` is a pointer to an array of `infoCount` pointers to arrays of VkAccelerationStructureBuildRangeInfoKHR structures. Each `ppBuildRangeInfos`[i] is a pointer to an array of `pInfos`[i].`geometryCount` VkAccelerationStructureBuildRangeInfoKHR structures defining dynamic offsets to the addresses where geometry data is stored, as defined by `pInfos`[i].

The `vkCmdBuildAccelerationStructuresKHR` command provides the ability to initiate multiple acceleration structure builds, however there is no ordering or synchronization implied between any of the individual acceleration structure builds.

NOTE This means that an application **cannot** build a top-level acceleration structure in the same vkCmdBuildAccelerationStructuresKHR call as the associated bottom-level or instance acceleration structures are being built. Similarly, the application **cannot** build a bottom-level acceleration structure in the same vkCmdBuildAccelerationStructuresKHR call as the associated micromaps are being built. There also **cannot** be any memory aliasing between any acceleration structure memories or scratch memories being used by any of the builds.

The input buffers passed to this command will be referenced by the implementation for the duration of this command's execution on the device.

After the command completes, the acceleration structure **may** hold a reference to any acceleration structure specified by an active instance contained therein. Apart from this referencing, acceleration structures **must** be fully self-contained. Micromaps **must** always be fully-self-contained.

The application **can** reuse or free any memory which was used by the command as an input or as scratch without affecting the results of future commands, provided proper synchronization is used as described below.

NOTE A VkAccelerationStructureBuildRangeInfoKHR structure is not used when building an acceleration structure with any of the following geometry types, and **must** be NULL: ** VK_GEOMETRY_TYPE_MICROMAP_KHR

NOTE The required alignment of the device addresses passed in to parameters below

> might not be provided by the base address of a VkBuffer created with the correct usage flags and must still be manually aligned by the application code.

Accesses to the acceleration structure scratch buffers as identified by the VkAccelerationStructureBuildGeometryInfoKHR::scratchData buffer device addresses **must** be synchronized with the VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR pipeline stage and an access type of (VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR | VK_ACCESS_ACCELERATION_STRUCTURE_WRITE_BIT_KHR). Accesses to each VkAccelerationStructureBuildGeometryInfoKHR::srcAccelerationStructure and VkAccelerationStructureBuildGeometryInfoKHR::dstAccelerationStructure **must** be synchronized with the VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR pipeline stage and an access type of VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR or VK_ACCESS_ACCELERATION_STRUCTURE_WRITE_BIT_KHR, as appropriate.

Accesses to other input buffers as identified by any used values of VkAccelerationStructureGeometryMicromapDataKHR::data, VkAccelerationStructureGeometryMicromapDataKHR::triangleArray, VkAccelerationStructureGeometryTrianglesDataKHR::vertexData, VkAccelerationStructureGeometryTrianglesDataKHR::indexData, VkAccelerationStructureGeometryTrianglesDataKHR::transformData, VkAccelerationStructureGeometryAabbsDataKHR::data, and VkAccelerationStructureGeometryInstancesDataKHR::data **must** be synchronized with the VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR pipeline stage and an access type of VK_ACCESS_SHADER_READ_BIT.

Valid Usage

- VUID-vkCmdBuildAccelerationStructuresKHR-accelerationStructure-08923
  The VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure feature **must** be enabled
- VUID-vkCmdBuildAccelerationStructuresKHR-ppBuildRangeInfos-11543
  Each element of ppBuildRangeInfos[i] **must** be a valid pointer to an array of pInfos[i].geometryCount VkAccelerationStructureBuildRangeInfoKHR structures if the pInfos[i].type is not one of:
  - VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR
- VUID-vkCmdBuildAccelerationStructuresKHR-ppBuildRangeInfos-11544
  For each element of pInfos[i] whose pGeometries or ppGeometries members have a geometryType of VK_GEOMETRY_TYPE_MICROMAP_KHR, ppBuildRangeInfos[i] **must** be NULL
- VUID-vkCmdBuildAccelerationStructuresKHR-mode-11545
  For each element of pInfos, if its type member is VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR, its mode **must** equal VK_BUILD_ACCELERATION_STRUCTURE_MODE_BUILD_KHR
- VUID-vkCmdBuildAccelerationStructuresKHR-srcAccelerationStructure-11546
  For each element of pInfos, if its type member is VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR, its srcAccelerationStructure **must** be VK_NULL_HANDLE

- VUID-vkCmdBuildAccelerationStructuresKHR-dstAccelerationStructure-11547
  For each element of pInfos, if its type member is VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR, its dstAccelerationStructure member **must** have been created with a value of VkAccelerationStructureCreateInfoKHR::type equal to either VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR or VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR
- VUID-vkCmdBuildAccelerationStructuresKHR-dstAccelerationStructure-11548
  The dstAccelerationStructure member of any element of pInfos **must** not be referenced by any VkAccelerationStructureTrianglesOpacityMicromapKHR::micromap member included in the pNext chain of any geometry.triangles member of any element of pGeometries or ppGeometries in any other element of pInfos
- VUID-vkCmdBuildAccelerationStructuresKHR-dstAccelerationStructure-11549
  The range of memory backing the dstAccelerationStructure member of any element of pInfos that is accessed by this command **must** not overlap the memory backing any VkAccelerationStructureTrianglesOpacityMicromapKHR::micromap member included in the pNext chain of any geometry.triangles member of any element of pGeometries or ppGeometries in any other element of pInfos, which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresKHR-micromap-11550
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries, if it includes a VkAccelerationStructureGeometryMicromapDataKHR structure in its pNext chain, its data and triangleArray members **must** be a valid VkDeviceAddress
- VUID-vkCmdBuildAccelerationStructuresKHR-micromap-11551
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries, if it includes a VkAccelerationStructureGeometryMicromapDataKHR structure in its pNext chain, the buffers from which the buffer device addresses for its data and triangleArray members are queried **must** have been created with the VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR usage flag
- VUID-vkCmdBuildAccelerationStructuresKHR-micromap-11552
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries, if it includes a VkAccelerationStructureGeometryMicromapDataKHR structure in its pNext chain, its data and triangleArray members **must** be a multiple of 128
- VUID-vkCmdBuildAccelerationStructuresKHR-micromap-11553
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries, if it includes a VkAccelerationStructureGeometryMicromapDataKHR structure in its pNext chain, each element of its triangleArray member **must** specify an access within the range of the VkBuffer that data was retrieved from, given by [dataOffset, dataOffset triangleSize] where triangleSize is calculated using its format and subdivisionLevel members rounded up to the nearest byte
- VUID-vkCmdBuildAccelerationStructuresKHR-micromap-11554
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries, if it includes a VkAccelerationStructureGeometryMicromapDataKHR structure in its pNext chain, the range of [triangleArray, triangleArray + (triangleArrayStride * numTriangles)] **must** be within the range of the VkBuffer from which the triangleArray member was retrieved,

where *numTriangles* is the sum of `count` of all elements of `pUsageCounts` or `ppUsageCounts`

- VUID-vkCmdBuildAccelerationStructuresKHR-micromap-11555
  For each element of `pInfos`[i].`pGeometries` or `pInfos`[i].`ppGeometries`, if it includes a VkAccelerationStructureGeometryMicromapDataKHR structure in its `pNext` chain, each element of its `geometry.triangleArray` member **must** have a `format` with a valid value from VkOpacityMicromapFormatKHR
- VUID-vkCmdBuildAccelerationStructuresKHR-micromap-11556
  For each element of `pInfos`[i].`pGeometries` or `pInfos`[i].`ppGeometries`, if it includes a VkAccelerationStructureGeometryMicromapDataKHR structure in its `pNext` chain, for each element of its `triangleArray` member with a `format` equal to VK_OPACITY_MICROMAP_FORMAT_2_STATE_KHR, `subdivisionLevel` **must** be less than or equal to `maxOpacity2StateSubdivisionLevel`
- VUID-vkCmdBuildAccelerationStructuresKHR-indexBuffer-11575
  For each element of `pInfos`[i].`pGeometries` or `pInfos`[i].`ppGeometries` with a `geometryType` equal to VK_GEOMETRY_TYPE_TRIANGLES_KHR, if its `geometry.triangles` member includes a VkAccelerationStructureTrianglesOpacityMicromapKHR structure in its `pNext` chain, the micromap's `indexType` member is not VK_INDEX_TYPE_NONE_KHR, and `micromap` is not VK_NULL_HANDLE, for each *index* in `indexBuffer`[*j*] that is not one of the VkOpacityMicromapSpecialIndexKHR values, where *j* is within the range [0, numTrianglesInGeometry - 1] where *numTrianglesInGeometry* is equal to `ppBuildRangeInfos`[i]→primitiveCount, *index* plus `baseTriangle` **must** be less than *numTriangles*, where *numTriangles* is given by the sum of all `count` parameters in the `pUsageCounts` or `ppUsageCounts` provided to the micromap's build command
- VUID-vkCmdBuildAccelerationStructuresKHR-indexBuffer-11577
  For each element of `pInfos`[i].`pGeometries` or `pInfos`[i].`ppGeometries` with a `geometryType` equal to VK_GEOMETRY_TYPE_TRIANGLES_KHR, if its `geometry.triangles` member includes a VkAccelerationStructureTrianglesOpacityMicromapKHR structure in its `pNext` chain, and the micromap's `indexType` member is not VK_INDEX_TYPE_NONE_KHR, `indexBuffer`
  (`indexStride` * numTrianglesInGeometry) **must** be within the range of the VkBuffer from which the `indexBuffer` device address was obtained where *numTrianglesInGeometry* is equal to `ppBuildRangeInfos`[i]→primitiveCount
- VUID-vkCmdBuildAccelerationStructuresKHR-indexBuffer-11578
  For each element of `pInfos`[i].`pGeometries` or `pInfos`[i].`ppGeometries` with a `geometryType` equal to VK_GEOMETRY_TYPE_TRIANGLES_KHR, if its `geometry.triangles` member includes a VkAccelerationStructureTrianglesOpacityMicromapKHR structure in its `pNext` chain, the micromap's `indexType` member is not VK_INDEX_TYPE_NONE_KHR, and `micromap` is VK_NULL_HANDLE, for each *index* in `indexBuffer`[*j*] where *j* is within the range from [0, numTrianglesInGeometry - 1], *index* **must** be a valid VkOpacityMicromapSpecialIndexKHR value where *numTrianglesInGeometry* is equal to `ppBuildRangeInfos`[i]→primitiveCount
- VUID-vkCmdBuildAccelerationStructuresKHR-mode-04628
  The `mode` member of each element of `pInfos` **must** be a valid VkBuildAccelerationStructureModeKHR value
- VUID-vkCmdBuildAccelerationStructuresKHR-srcAccelerationStructure-04629

If the srcAccelerationStructure member of any element of pInfos is not VK_NULL_HANDLE, the srcAccelerationStructure member **must** be a valid VkAccelerationStructureKHR handle

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-04630
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its srcAccelerationStructure member **must** not be VK_NULL_HANDLE
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03708
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, srcAccelerationStructure **must** be fully backed by physical memory
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03403
  The srcAccelerationStructure member of any element of pInfos **must** not be the same acceleration structure as the dstAccelerationStructure member of any other element of pInfos
- VUID-vkCmdBuildAccelerationStructuresKHR-dstAccelerationStructure-03698
  The dstAccelerationStructure member of any element of pInfos **must** not be the same acceleration structure as the dstAccelerationStructure member of any other element of pInfos
- VUID-vkCmdBuildAccelerationStructuresKHR-dstAccelerationStructure-03800
  The dstAccelerationStructure member of any element of pInfos **must** be a valid VkAccelerationStructureKHR handle
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03707
  For each element of pInfos, its dstAccelerationStructure member **must** be fully backed by physical memory
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03699
  For each element of pInfos, if its type member is VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR, its dstAccelerationStructure member **must** have been created with a value of VkAccelerationStructureCreateInfoKHR::type equal to either VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR or VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03700
  For each element of pInfos, if its type member is VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR, its dstAccelerationStructure member **must** have been created with a value of VkAccelerationStructureCreateInfoKHR::type equal to either VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR or VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03663
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, inactive primitives in its srcAccelerationStructure member **must** not be made active
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03664
  For each element of pInfos, if its mode member is

VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, active primitives in its srcAccelerationStructure member **must** not be made inactive

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03709
  For each element of pInfos, if an acceleration structure is referenced by the geometry.instances.data member of any element of pGeometries or ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, it **must** be fully backed by physical memory
- VUID-vkCmdBuildAccelerationStructuresKHR-None-03407
  The dstAccelerationStructure member of any element of pInfos **must** not be referenced by the geometry.instances.data member of any element of pGeometries or ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR in any other element of pInfos
- VUID-vkCmdBuildAccelerationStructuresKHR-dstAccelerationStructure-03701
  The range of memory backing the dstAccelerationStructure member of any element of pInfos that is accessed by this command **must** not overlap the memory backing the srcAccelerationStructure member of any other element of pInfos with a mode equal to VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresKHR-dstAccelerationStructure-03702
  The range of memory backing the dstAccelerationStructure member of any element of pInfos that is accessed by this command **must** not overlap the memory backing the dstAccelerationStructure member of any other element of pInfos, which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresKHR-dstAccelerationStructure-03703
  The range of memory backing the dstAccelerationStructure member of any element of pInfos that is accessed by this command **must** not overlap the memory backing the scratchData member of any element of pInfos (including the same element), which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresKHR-scratchData-03704
  The range of memory backing the scratchData member of any element of pInfos that is accessed by this command **must** not overlap the memory backing the scratchData member of any other element of pInfos, which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresKHR-scratchData-03705
  The range of memory backing the scratchData member of any element of pInfos that is accessed by this command **must** not overlap the memory backing the srcAccelerationStructure member of any element of pInfos with a mode equal to VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR (including the same element), which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresKHR-dstAccelerationStructure-03706
  The range of memory backing the dstAccelerationStructure member of any element of pInfos that is accessed by this command **must** not overlap the memory backing any acceleration structure referenced by the geometry.instances.data member of any element of pGeometries or ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR in any other element of pInfos, which is accessed by this command

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03667
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its srcAccelerationStructure member **must** have previously been constructed with VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR set in VkAccelerationStructureBuildGeometryInfoKHR::flags in the build
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03668
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its srcAccelerationStructure and dstAccelerationStructure members **must** either be the same VkAccelerationStructureKHR, or not have any memory aliasing
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03758
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its geometryCount member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03759
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its flags member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03760
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its type member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03761
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, its geometryType member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03762
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, its flags member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03763
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, its geometry.triangles.vertexFormat member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03764
  For each element of pInfos, if its mode member is

VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, its geometry.triangles.maxVertex member **must** have the same value which was specified when srcAccelerationStructure was last built

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03765
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, its geometry.triangles.indexType member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03766
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, if its geometry.triangles.transformData address was NULL when srcAccelerationStructure was last built, then it **must** be NULL
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03767
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, if its geometry.triangles.transformData address was not NULL when srcAccelerationStructure was last built, then it **must** not be NULL
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03768
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, and geometry.triangles.indexType is not VK_INDEX_TYPE_NONE_KHR, then the value of each index referenced **must** be the same as the corresponding index value when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresKHR-primitiveCount-03769
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, the primitiveCount member of its corresponding VkAccelerationStructureBuildRangeInfoKHR structure **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03801
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, the corresponding ppBuildRangeInfos[i][j].primitiveCount **must** be less than or equal to

VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxInstanceCount

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-12258
  If pInfos[i].mode is VK_BUILD_ACCELERATION_STRUCTURE_MODE_BUILD_KHR, and N is not 0, all addresses between pInfos[i].scratchData.deviceAddress and pInfos[i].scratchData.deviceAddress + N - 1 **must** be in the buffer device address range of the same buffer, where N is given by the buildScratchSize member of the VkAccelerationStructureBuildSizesInfoKHR structure returned from a call to vkGetAccelerationStructureBuildSizesKHR with an identical VkAccelerationStructureBuildGeometryInfoKHR structure and primitive count
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-12259
  If pInfos[i].mode is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, and N is not 0, all addresses between pInfos[i].scratchData.deviceAddress and pInfos[i].scratchData.deviceAddress + N - 1 **must** be in the buffer device address range of the same buffer, where N is given by the updateScratchSize member of the VkAccelerationStructureBuildSizesInfoKHR structure returned from a call to vkGetAccelerationStructureBuildSizesKHR with an identical VkAccelerationStructureBuildGeometryInfoKHR structure and primitive count
- VUID-vkCmdBuildAccelerationStructuresKHR-geometry-03673
  The buffers from which the buffer device addresses for all of the geometry.triangles.vertexData, geometry.triangles.indexData, geometry.triangles.transformData, geometry.aabbs.data, and geometry.instances.data members of all pInfos[i].pGeometries and pInfos[i].ppGeometries are queried **must** have been created with the VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR usage flag set
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-12260
  If pInfos[i].mode is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, and the size reported by the updateScratchSize member of the VkAccelerationStructureBuildSizesInfoKHR structure returned from a call to vkGetAccelerationStructureBuildSizesKHR with an identical VkAccelerationStructureBuildGeometryInfoKHR structure and primitive count is non-zero, pInfos[i].scratchData.deviceAddress **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-12261
  If pInfos[i].mode is VK_BUILD_ACCELERATION_STRUCTURE_MODE_BUILD_KHR, and the size reported by the buildScratchSize member of the VkAccelerationStructureBuildSizesInfoKHR structure returned from a call to vkGetAccelerationStructureBuildSizesKHR with an identical VkAccelerationStructureBuildGeometryInfoKHR structure and primitive count is non-zero, pInfos[i].scratchData.deviceAddress **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03710
  For each element of pInfos, its scratchData.deviceAddress member **must** be a multiple of

VkPhysicalDeviceAccelerationStructurePropertiesKHR::minAccelerationStructureScratchOffsetAlignment

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03804
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, geometry.triangles.vertexData.deviceAddress **must** be a valid VkDeviceAddress
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03711
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, geometry.triangles.vertexData.deviceAddress **must** be aligned to:
  - the size of the format specified in vertexFormat, in bytes, if that format is a packed format
  - the component size of the format specified in vertexFormat, in bytes, if that format is not a packed format
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03806
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if geometry.triangles.indexType is not VK_INDEX_TYPE_NONE_KHR, geometry.triangles.indexData.deviceAddress **must** be a valid VkDeviceAddress
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03712
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, and with geometry.triangles.indexType not equal to VK_INDEX_TYPE_NONE_KHR, geometry.triangles.indexData.deviceAddress **must** be aligned to the size in bytes of the type in indexType
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03808
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if geometry.triangles.transformData.deviceAddress is not 0, it **must** be a valid VkDeviceAddress
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03810
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if geometry.triangles.transformData.deviceAddress is not 0, it **must** be aligned to 16 bytes
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03811
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_AABBS_KHR, geometry.aabbs.data.deviceAddress **must** be a valid VkDeviceAddress
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03714
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_AABBS_KHR, geometry.aabbs.data.deviceAddress **must** be aligned to 8 bytes
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03715
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of

VK_GEOMETRY_TYPE_INSTANCES_KHR, if geometry.arrayOfPointers is VK_FALSE, geometry.instances.data.deviceAddress **must** be aligned to 16 bytes

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03716
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, if geometry.arrayOfPointers is VK_TRUE, geometry.instances.data.deviceAddress **must** be aligned to 8 bytes
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03717
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, if geometry.arrayOfPointers is VK_TRUE, each element of geometry.instances.data.deviceAddress in device memory **must** be aligned to 16 bytes
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03813
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, geometry.instances.data.deviceAddress **must** be a valid VkDeviceAddress
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-12281
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, each VkAccelerationStructureInstanceKHR ::accelerationStructureReference value in geometry.instances.data.deviceAddress **must** be 0 or a value obtained from vkGetAccelerationStructureDeviceAddressKHR for a valid bottom level acceleration structure
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-10607
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, if VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_BIT_KHR is set in VkAccelerationStructureInstanceKHR::flags then geometry.instances.data.deviceAddress **must** refer to an acceleration structure that was built with VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_BIT_KHR set in VkAccelerationStructureBuildGeometryInfoKHR::flags
- VUID-vkCmdBuildAccelerationStructuresKHR-commandBuffer-09547
  commandBuffer **must** not be a protected command buffer
- VUID-vkCmdBuildAccelerationStructuresKHR-micromap-11701
  For any element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if there is an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR in the geometry.triangles.pNext chain with a valid micromap member, then the micromap member **must** have been constructed prior to this command's execution on the device
- VUID-vkCmdBuildAccelerationStructuresKHR-micromap-11624
  For any element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if there is an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR in the geometry.triangles.pNext chain with a valid micromap member, then the micromap member **must** have been built with a type of VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR
- VUID-vkCmdBuildAccelerationStructuresKHR-micromap-11625

For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, and the same geometry in the original build of srcAccelerationStructure was built with an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR in its geometry.triangles.pNext chain with a valid micromap member, then an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR **must** be included in the geometry.triangles.pNext chain with a valid micromap member

- VUID-vkCmdBuildAccelerationStructuresKHR-micromap-11626
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, and the same geometry in the original build of srcAccelerationStructure was built with an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR in its geometry.triangles.pNext chain with its micromap member equal to VK_NULL_HANDLE, then an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR **must** be included in the geometry.triangles.pNext chain with its micromap member equal to VK_NULL_HANDLE
- VUID-vkCmdBuildAccelerationStructuresKHR-micromap-11627
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, and the same geometry in the original build of srcAccelerationStructure was not built with an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR in its geometry.triangles.pNext chain, then an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR **must** not be included in the geometry.triangles.pNext chain
- VUID-vkCmdBuildAccelerationStructuresKHR-micromap-11628
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its srcAccelerationStructure was originally built without the VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_KHR flag, and srcAccelerationStructure was constructed by deserialization with vkCmdCopyMemoryToAccelerationStructureKHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, and the same geometry in the original build of srcAccelerationStructure was built with an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR in its geometry.triangles.pNext chain with a valid micromap member, then an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR **must** be included in the geometry.triangles.pNext chain with a micromap that was deserialized from the serialized data of the micromap in the original build of srcAccelerationStructure
- VUID-vkCmdBuildAccelerationStructuresKHR-micromap-11629
  For each element of pInfos, if its mode member is

VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its srcAccelerationStructure was originally built without either the VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_BIT_EXT or the VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_KHR flags, and srcAccelerationStructure was not constructed by deserialization with vkCmdCopyMemoryToAccelerationStructureKHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, and the same geometry in the original build of srcAccelerationStructure was built with an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR in its geometry.triangles.pNext chain with a valid micromap member, then an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR **must** be included in the geometry.triangles.pNext chain with the same micromap handle as the original build

- VUID-vkCmdBuildAccelerationStructuresKHR-micromap-11632
  For any element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if there is an instance of VkAccelerationStructureTrianglesOpacityMicromapEXT in the geometry.triangles.pNext chain, then its micromap member **must** have been constructed prior to this command's execution on the device

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-10904
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if there is an instance of VkAccelerationStructureTrianglesOpacityMicromapEXT in the geometry.triangles.pNext chain, and its indexType is VK_INDEX_TYPE_NONE_KHR, then its indexBuffer.deviceAddress **must** be 0

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-10905
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if there is an instance of VkAccelerationStructureTrianglesOpacityMicromapEXT in the geometry.triangles.pNext chain, and its indexType is not VK_INDEX_TYPE_NONE_KHR, then its indexBuffer.deviceAddress **must** be a valid VkDeviceAddress

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-10126
  For each pInfos[i], dstAccelerationStructure **must** have been created with a value of VkAccelerationStructureCreateInfoKHR::size greater than or equal to either:
  - the memory size required by the build operation, as returned by vkGetAccelerationStructureBuildSizesKHR with pBuildInfo = pInfos[i] and with each element of the pMaxPrimitiveCounts array greater than or equal to the equivalent ppBuildRangeInfos[i][j].primitiveCount values for j in [0,pInfos[i].geometryCount) or,
  - the result of querying the corresponding VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR, if updating a compacted acceleration structure

Valid Usage (Implicit)

- VUID-vkCmdBuildAccelerationStructuresKHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-parameter
  `pInfos` **must** be a valid pointer to an array of `infoCount` valid VkAccelerationStructureBuildGeometryInfoKHR structures
- VUID-vkCmdBuildAccelerationStructuresKHR-ppBuildRangeInfos-parameter
  `ppBuildRangeInfos` **must** be a valid pointer to an array of `infoCount` VkAccelerationStructureBuildRangeInfoKHR structures
- VUID-vkCmdBuildAccelerationStructuresKHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBuildAccelerationStructuresKHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdBuildAccelerationStructuresKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdBuildAccelerationStructuresKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdBuildAccelerationStructuresKHR-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBuildAccelerationStructuresKHR-infoCount-arraylength
  `infoCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdBuildAccelerationStructuresKHR is not affected by conditional rendering

To build acceleration structures with some parameters sourced on the device call:

```
// Provided by VK_KHR_acceleration_structure
void vkCmdBuildAccelerationStructuresIndirectKHR(
    VkCommandBuffer                                 commandBuffer,
    uint32_t                                        infoCount,
    const VkAccelerationStructureBuildGeometryInfoKHR* pInfos,
    const VkDeviceAddress*                          pIndirectDeviceAddresses,
    const uint32_t*                                 pIndirectStrides,
    const uint32_t* const*                          ppMaxPrimitiveCounts);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `infoCount` is the number of acceleration structures to build.
- `pInfos` is a pointer to an array of `infoCount` VkAccelerationStructureBuildGeometryInfoKHR structures defining the geometry used to build each acceleration structure.
- `pIndirectDeviceAddresses` is a pointer to an array of `infoCount` buffer device addresses which point to `pInfos`[i].`geometryCount` VkAccelerationStructureBuildRangeInfoKHR structures defining dynamic offsets to the addresses where geometry data is stored, as defined by `pInfos`[i].
- `pIndirectStrides` is a pointer to an array of `infoCount` byte strides between elements of `pIndirectDeviceAddresses`.
- `ppMaxPrimitiveCounts` is a pointer to an array of `infoCount` pointers to arrays of `pInfos`[i].`geometryCount` values indicating the maximum number of primitives that will be built by this command for each geometry.

Accesses to any device memory referenced by this command **must** be synchronized as with vkCmdBuildAccelerationStructuresKHR.

Accesses to any element of `pIndirectDeviceAddresses` **must** be synchronized with the VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR pipeline stage and an access type of VK_ACCESS_INDIRECT_COMMAND_READ_BIT.

Valid Usage

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-accelerationStructureIndirectBuild-03650
  The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructureIndirectBuild` feature **must** be enabled

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-mode-04628
  The `mode` member of each element of `pInfos` **must** be a valid VkBuildAccelerationStructureModeKHR value

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-srcAccelerationStructure-04629
  If the srcAccelerationStructure member of any element of pInfos is not VK_NULL_HANDLE, the srcAccelerationStructure member **must** be a valid VkAccelerationStructureKHR handle
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-04630
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its srcAccelerationStructure member **must** not be VK_NULL_HANDLE
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03708
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, srcAccelerationStructure **must** be fully backed by physical memory
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03403
  The srcAccelerationStructure member of any element of pInfos **must** not be the same acceleration structure as the dstAccelerationStructure member of any other element of pInfos
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-dstAccelerationStructure-03698
  The dstAccelerationStructure member of any element of pInfos **must** not be the same acceleration structure as the dstAccelerationStructure member of any other element of pInfos
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-dstAccelerationStructure-03800
  The dstAccelerationStructure member of any element of pInfos **must** be a valid VkAccelerationStructureKHR handle
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03707
  For each element of pInfos, its dstAccelerationStructure member **must** be fully backed by physical memory
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03699
  For each element of pInfos, if its type member is VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR, its dstAccelerationStructure member **must** have been created with a value of VkAccelerationStructureCreateInfoKHR::type equal to either VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR or VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03700
  For each element of pInfos, if its type member is VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR, its dstAccelerationStructure member **must** have been created with a value of VkAccelerationStructureCreateInfoKHR::type equal to either VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR or VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03663
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, inactive primitives in its srcAccelerationStructure member **must** not be made active
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03664

For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, active primitives in its srcAccelerationStructure member **must** not be made inactive

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03709
  For each element of pInfos, if an acceleration structure is referenced by the geometry.instances.data member of any element of pGeometries or ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, it **must** be fully backed by physical memory
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-None-03407
  The dstAccelerationStructure member of any element of pInfos **must** not be referenced by the geometry.instances.data member of any element of pGeometries or ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR in any other element of pInfos
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-dstAccelerationStructure-03701
  The range of memory backing the dstAccelerationStructure member of any element of pInfos that is accessed by this command **must** not overlap the memory backing the srcAccelerationStructure member of any other element of pInfos with a mode equal to VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-dstAccelerationStructure-03702
  The range of memory backing the dstAccelerationStructure member of any element of pInfos that is accessed by this command **must** not overlap the memory backing the dstAccelerationStructure member of any other element of pInfos, which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-dstAccelerationStructure-03703
  The range of memory backing the dstAccelerationStructure member of any element of pInfos that is accessed by this command **must** not overlap the memory backing the scratchData member of any element of pInfos (including the same element), which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-scratchData-03704
  The range of memory backing the scratchData member of any element of pInfos that is accessed by this command **must** not overlap the memory backing the scratchData member of any other element of pInfos, which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-scratchData-03705
  The range of memory backing the scratchData member of any element of pInfos that is accessed by this command **must** not overlap the memory backing the srcAccelerationStructure member of any element of pInfos with a mode equal to VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR (including the same element), which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-dstAccelerationStructure-03706
  The range of memory backing the dstAccelerationStructure member of any element of pInfos that is accessed by this command **must** not overlap the memory backing any acceleration structure referenced by the geometry.instances.data member of any element of pGeometries or ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR in any other element of pInfos, which is accessed

by this command

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03667
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its srcAccelerationStructure member **must** have previously been constructed with VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR set in VkAccelerationStructureBuildGeometryInfoKHR::flags in the build
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03668
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its srcAccelerationStructure and dstAccelerationStructure members **must** either be the same VkAccelerationStructureKHR, or not have any memory aliasing
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03758
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its geometryCount member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03759
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its flags member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03760
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its type member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03761
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, its geometryType member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03762
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, its flags member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03763
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, its geometry.triangles.vertexFormat member **must** have the same value which was specified when srcAccelerationStructure was last built

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03764
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, its geometry.triangles.maxVertex member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03765
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, its geometry.triangles.indexType member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03766
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, if its geometry.triangles.transformData address was NULL when srcAccelerationStructure was last built, then it **must** be NULL
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03767
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, if its geometry.triangles.transformData address was not NULL when srcAccelerationStructure was last built, then it **must** not be NULL
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03768
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, and geometry.triangles.indexType is not VK_INDEX_TYPE_NONE_KHR, then the value of each index referenced **must** be the same as the corresponding index value when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-primitiveCount-03769
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, the primitiveCount member of its corresponding VkAccelerationStructureBuildRangeInfoKHR structure **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03801
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of

VK_GEOMETRY_TYPE_INSTANCES_KHR, the corresponding ppMaxPrimitiveCounts[i][j] **must** be less than or equal to VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxInstanceCount

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-12258
  If pInfos[i].mode is VK_BUILD_ACCELERATION_STRUCTURE_MODE_BUILD_KHR, and N is not 0, all addresses between pInfos[i].scratchData.deviceAddress and pInfos[i].scratchData.deviceAddress + N - 1 **must** be in the buffer device address range of the same buffer, where N is given by the buildScratchSize member of the VkAccelerationStructureBuildSizesInfoKHR structure returned from a call to vkGetAccelerationStructureBuildSizesKHR with an identical VkAccelerationStructureBuildGeometryInfoKHR structure and primitive count
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-12259
  If pInfos[i].mode is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, and N is not 0, all addresses between pInfos[i].scratchData.deviceAddress and pInfos[i].scratchData.deviceAddress + N - 1 **must** be in the buffer device address range of the same buffer, where N is given by the updateScratchSize member of the VkAccelerationStructureBuildSizesInfoKHR structure returned from a call to vkGetAccelerationStructureBuildSizesKHR with an identical VkAccelerationStructureBuildGeometryInfoKHR structure and primitive count
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-geometry-03673
  The buffers from which the buffer device addresses for all of the geometry.triangles.vertexData, geometry.triangles.indexData, geometry.triangles.transformData, geometry.aabbs.data, and geometry.instances.data members of all pInfos[i].pGeometries and pInfos[i].ppGeometries are queried **must** have been created with the VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR usage flag set
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-12260
  If pInfos[i].mode is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, and the size reported by the updateScratchSize member of the VkAccelerationStructureBuildSizesInfoKHR structure returned from a call to vkGetAccelerationStructureBuildSizesKHR with an identical VkAccelerationStructureBuildGeometryInfoKHR structure and primitive count is non-zero, pInfos[i].scratchData.deviceAddress **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-12261
  If pInfos[i].mode is VK_BUILD_ACCELERATION_STRUCTURE_MODE_BUILD_KHR, and the size reported by the buildScratchSize member of the VkAccelerationStructureBuildSizesInfoKHR structure returned from a call to vkGetAccelerationStructureBuildSizesKHR with an identical VkAccelerationStructureBuildGeometryInfoKHR structure and primitive count is non-zero, pInfos[i].scratchData.deviceAddress **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03710
  For each element of pInfos, its scratchData.deviceAddress member **must** be a multiple of VkPhysicalDeviceAccelerationStructurePropertiesKHR::minAccelerationStructureScratchOffsetAlignment
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03804
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, geometry.triangles.vertexData.deviceAddress **must** be a valid VkDeviceAddress
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03711
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, geometry.triangles.vertexData.deviceAddress **must** be aligned to:
  - the size of the format specified in vertexFormat, in bytes, if that format is a packed format
  - the component size of the format specified in vertexFormat, in bytes, if that format is not a packed format
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03806
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if geometry.triangles.indexType is not VK_INDEX_TYPE_NONE_KHR, geometry.triangles.indexData.deviceAddress **must** be a valid VkDeviceAddress
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03712
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, and with geometry.triangles.indexType not equal to VK_INDEX_TYPE_NONE_KHR, geometry.triangles.indexData.deviceAddress **must** be aligned to the size in bytes of the type in indexType
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03808
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if geometry.triangles.transformData.deviceAddress is not 0, it **must** be a valid VkDeviceAddress
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03810
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if geometry.triangles.transformData.deviceAddress is not 0, it **must** be aligned to 16 bytes
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03811
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_AABBS_KHR, geometry.aabbs.data.deviceAddress **must** be a valid VkDeviceAddress
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03714
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_AABBS_KHR, geometry.aabbs.data.deviceAddress **must** be aligned to 8 bytes

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03715
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, if geometry.arrayOfPointers is VK_FALSE, geometry.instances.data.deviceAddress **must** be aligned to 16 bytes
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03716
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, if geometry.arrayOfPointers is VK_TRUE, geometry.instances.data.deviceAddress **must** be aligned to 8 bytes
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03717
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, if geometry.arrayOfPointers is VK_TRUE, each element of geometry.instances.data.deviceAddress in device memory **must** be aligned to 16 bytes
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03813
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, geometry.instances.data.deviceAddress **must** be a valid VkDeviceAddress
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-12281
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, each VkAccelerationStructureInstanceKHR ::accelerationStructureReference value in geometry.instances.data.deviceAddress **must** be 0 or a value obtained from vkGetAccelerationStructureDeviceAddressKHR for a valid bottom level acceleration structure
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-10607
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, if VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_BIT_KHR is set in VkAccelerationStructureInstanceKHR::flags then geometry.instances.data.deviceAddress **must** refer to an acceleration structure that was built with VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_BIT_KHR set in VkAccelerationStructureBuildGeometryInfoKHR::flags
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-commandBuffer-09547
  commandBuffer **must** not be a protected command buffer
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-micromap-11701
  For any element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if there is an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR in the geometry.triangles.pNext chain with a valid micromap member, then the micromap member **must** have been constructed prior to this command's execution on the device
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-micromap-11624
  For any element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if there is an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR in the geometry.triangles.pNext chain with a valid micromap member, then the micromap member **must** have been built with a type of VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-micromap-11625
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, and the same geometry in the original build of srcAccelerationStructure was built with an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR in its geometry.triangles.pNext chain with a valid micromap member, then an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR **must** be included in the geometry.triangles.pNext chain with a valid micromap member
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-micromap-11626
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, and the same geometry in the original build of srcAccelerationStructure was built with an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR in its geometry.triangles.pNext chain with its micromap member equal to VK_NULL_HANDLE, then an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR **must** be included in the geometry.triangles.pNext chain with its micromap member equal to VK_NULL_HANDLE
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-micromap-11627
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, and the same geometry in the original build of srcAccelerationStructure was not built with an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR in its geometry.triangles.pNext chain, then an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR **must** not be included in the geometry.triangles.pNext chain
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-micromap-11628
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its srcAccelerationStructure was originally built without the VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_KHR flag, and srcAccelerationStructure was constructed by deserialization with vkCmdCopyMemoryToAccelerationStructureKHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, and the same geometry in the original build of srcAccelerationStructure was built with an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR in its geometry.triangles.pNext chain with a valid micromap member, then an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR **must** be included in the geometry.triangles.pNext chain with a micromap that was deserialized from the serialized data of the micromap in the original build of srcAccelerationStructure
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-micromap-11629

For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its srcAccelerationStructure was originally built without either the VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_BIT_EXT or the VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_KHR flags, and srcAccelerationStructure was not constructed by deserialization with vkCmdCopyMemoryToAccelerationStructureKHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, and the same geometry in the original build of srcAccelerationStructure was built with an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR in its geometry.triangles.pNext chain with a valid micromap member, then an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR **must** be included in the geometry.triangles.pNext chain with the same micromap handle as the original build

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-micromap-11632
  For any element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if there is an instance of VkAccelerationStructureTrianglesOpacityMicromapEXT in the geometry.triangles.pNext chain, then its micromap member **must** have been constructed prior to this command's execution on the device
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-10904
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if there is an instance of VkAccelerationStructureTrianglesOpacityMicromapEXT in the geometry.triangles.pNext chain, and its indexType is VK_INDEX_TYPE_NONE_KHR, then its indexBuffer.deviceAddress **must** be 0
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-10905
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if there is an instance of VkAccelerationStructureTrianglesOpacityMicromapEXT in the geometry.triangles.pNext chain, and its indexType is not VK_INDEX_TYPE_NONE_KHR, then its indexBuffer.deviceAddress **must** be a valid VkDeviceAddress
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pIndirectDeviceAddresses-03646
  For each element of pIndirectDeviceAddresses[i], all device addresses between pIndirectDeviceAddresses[i] and pIndirectDeviceAddresses[i] + (pInfos[i].geometryCount × pIndirectStrides[i]) - 1 **must** be in the buffer device address range of the same buffer
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pIndirectDeviceAddresses-03647
  For each element of pIndirectDeviceAddresses, the buffer from which it was queried **must** have been created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pIndirectDeviceAddresses-03648
  Each element of pIndirectDeviceAddresses **must** be a multiple of 4
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pIndirectStrides-03787
  Each element of pIndirectStrides **must** be a multiple of 4
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pIndirectDeviceAddresses-03651

Each VkAccelerationStructureBuildRangeInfoKHR structure referenced by any element of pIndirectDeviceAddresses **must** be a valid VkAccelerationStructureBuildRangeInfoKHR structure

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03652
  pInfos[i].dstAccelerationStructure **must** have been created with a value of VkAccelerationStructureCreateInfoKHR::size greater than or equal to the memory size required by the build operation, as returned by vkGetAccelerationStructureBuildSizesKHR with pBuildInfo = pInfos[i] and pMaxPrimitiveCounts = ppMaxPrimitiveCounts[i]
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-ppMaxPrimitiveCounts-03653
  Each ppMaxPrimitiveCounts[i][j] **must** be greater than or equal to the primitiveCount value specified by the VkAccelerationStructureBuildRangeInfoKHR structure located at pIndirectDeviceAddresses[i] + (j × pIndirectStrides[i])
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-type-11557
  For any element of pInfos, its type member **must** not be VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR

Valid Usage (Implicit)

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-parameter
  pInfos **must** be a valid pointer to an array of infoCount valid VkAccelerationStructureBuildGeometryInfoKHR structures
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pIndirectDeviceAddresses-parameter
  pIndirectDeviceAddresses **must** be a valid pointer to an array of infoCount VkDeviceAddress values
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pIndirectStrides-parameter
  pIndirectStrides **must** be a valid pointer to an array of infoCount uint32_t values
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-ppMaxPrimitiveCounts-parameter
  ppMaxPrimitiveCounts **must** be a valid pointer to an array of infoCount uint32_t values
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-videocoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-infoCount-arraylength
  `infoCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdBuildAccelerationStructuresIndirectKHR is not affected by conditional rendering

The `VkAccelerationStructureBuildGeometryInfoKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureBuildGeometryInfoKHR {
    VkStructureType                                    sType;
    const void*                                        pNext;
    VkAccelerationStructureTypeKHR                     type;
    VkBuildAccelerationStructureFlagsKHR               flags;
    VkBuildAccelerationStructureModeKHR                mode;
    VkAccelerationStructureKHR                         srcAccelerationStructure;
    VkAccelerationStructureKHR                         dstAccelerationStructure;
    uint32_t                                           geometryCount;
    const VkAccelerationStructureGeometryKHR*          pGeometries;
    const VkAccelerationStructureGeometryKHR* const*   ppGeometries;
    VkDeviceOrHostAddressKHR                           scratchData;
} VkAccelerationStructureBuildGeometryInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `type` is a VkAccelerationStructureTypeKHR value specifying the type of acceleration structure being built.

- `flags` is a bitmask of VkBuildAccelerationStructureFlagBitsKHR specifying additional parameters of the acceleration structure.
- `mode` is a VkBuildAccelerationStructureModeKHR value specifying the type of operation to perform.
- `srcAccelerationStructure` is a pointer to an existing acceleration structure that is to be used to update the `dstAccelerationStructure` acceleration structure when `mode` is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR.
- `dstAccelerationStructure` is a pointer to the target acceleration structure for the build.
- `geometryCount` specifies the number of geometries that will be built into `dstAccelerationStructure`.
- `pGeometries` is a pointer to an array of VkAccelerationStructureGeometryKHR structures.
- `ppGeometries` is a pointer to an array of pointers to VkAccelerationStructureGeometryKHR structures.
- `scratchData` is the device or host address of memory that will be used as scratch memory for the build.

Only one of `pGeometries` or `ppGeometries` **can** be a valid pointer, the other **must** be `NULL`. Each element of the non-`NULL` array describes the data used to build each acceleration structure geometry.

For bottom-level acceleration structures, the index of each element of the `pGeometries` or `ppGeometries` members of VkAccelerationStructureBuildGeometryInfoKHR is used as the *geometry index* during ray traversal. The geometry index is available in ray shaders via the `RayGeometryIndexKHR` built-in, and is used to determine hit and intersection shaders executed during traversal. The geometry index is available to ray queries via the `OpRayQueryGetIntersectionGeometryIndexKHR` instruction.

Members `srcAccelerationStructure` and `dstAccelerationStructure` **may** be the same or different for an update operation (when `mode` is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR). If they are the same, the update happens in-place. Otherwise, the target acceleration structure is updated and the source is not modified.

Valid Usage

- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03654
  `type` **must** not be VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-pGeometries-03788
  If `geometryCount` is not `0`, exactly one of `pGeometries` or `ppGeometries` **must** be a valid pointer, the other **must** be `NULL`
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03789
  If `type` is VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR, the `geometryType` member of elements of either `pGeometries` or `ppGeometries` **must** be VK_GEOMETRY_TYPE_INSTANCES_KHR
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03790

If type is VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR, geometryCount **must** be 1

- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03791
  If type is VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR the geometryType member of elements of either pGeometries or ppGeometries **must** not be VK_GEOMETRY_TYPE_INSTANCES_KHR
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03792
  If type is VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR then the geometryType member of each geometry in either pGeometries or ppGeometries **must** be the same
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03793
  If type is VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR then geometryCount **must** be less than or equal to VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxGeometryCount
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03794
  If type is VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR and the geometryType member of either pGeometries or ppGeometries is VK_GEOMETRY_TYPE_AABBS_KHR, the total number of AABBs in all geometries **must** be less than or equal to VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxPrimitiveCount
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03795
  If type is VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR and the geometryType member of either pGeometries or ppGeometries is VK_GEOMETRY_TYPE_TRIANGLES_KHR, the total number of triangles in all geometries **must** be less than or equal to VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxPrimitiveCount
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-flags-03796
  If flags has the VK_BUILD_ACCELERATION_STRUCTURE_PREFER_FAST_TRACE_BIT_KHR bit set, then it **must** not have the VK_BUILD_ACCELERATION_STRUCTURE_PREFER_FAST_BUILD_BIT_KHR bit set
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-flags-11558
  If flags has the VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_KHR bit set then it **must** not have the VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_BIT_EXT bit set
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-flags-11709
  If flags has the VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_BIT_EXT bit set, the VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap feature **must** be enabled
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-flags-11710
  If flags has the VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_KH

R bit set, the VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap or VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap feature **must** be enabled

- VUID-VkAccelerationStructureBuildGeometryInfoKHR-flags-11711
  If flags has the VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_BIT_KHR bit set, the VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap or VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap feature **must** be enabled
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-micromap-11559
  If type is VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR, the VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap feature **must** be enabled
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-geometryCount-11560
  If type is VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR, geometryCount **must** be 1
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-geometryType-11561
  If type is VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR, the geometryType member of elements of either pGeometries or ppGeometries **must** be VK_GEOMETRY_TYPE_MICROMAP_KHR
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-geometryType-11707
  If type is not VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR, the geometryType member of elements of either pGeometries or ppGeometries **must** not be VK_GEOMETRY_TYPE_MICROMAP_KHR
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-flags-11562
  If type is VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR, flags **must** only include the following bits:
  - VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_COMPACTION_BIT_KHR
  - VK_BUILD_ACCELERATION_STRUCTURE_PREFER_FAST_TRACE_BIT_KHR
  - VK_BUILD_ACCELERATION_STRUCTURE_PREFER_FAST_BUILD_BIT_KHR
  - VK_BUILD_ACCELERATION_STRUCTURE_MICROMAP_LOSSY_BIT_KHR
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-11563
  If flags includes VK_BUILD_ACCELERATION_STRUCTURE_MICROMAP_LOSSY_BIT_KHR, type **must** be VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-format-11712
  If flags includes VK_BUILD_ACCELERATION_STRUCTURE_MICROMAP_LOSSY_BIT_KHR, for each element of pGeometries or ppGeometries, if it includes a VkAccelerationStructureGeometryMicromapDataKHR structure in its pNext chain, each element of its pUsageCounts or ppUsageCounts members **must** have its format member equal to VK_OPACITY_MICROMAP_FORMAT_4_STATE_KHR
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-subdivisionLevel-11564
  If flags does not include VK_BUILD_ACCELERATION_STRUCTURE_MICROMAP_LOSSY_BIT_KHR, for each element of pGeometries or ppGeometries, if it includes a VkAccelerationStructureGeometryMicromapDataKHR structure in its pNext chain, for each element of its pUsageCounts or ppUsageCounts members, if its format member is

VK_OPACITY_MICROMAP_FORMAT_4_STATE_KHR, its subdivisionLevel member **must** be less than or equal to maxOpacity4StateSubdivisionLevel

- VUID-VkAccelerationStructureBuildGeometryInfoKHR-subdivisionLevel-11565
  If flags includes VK_BUILD_ACCELERATION_STRUCTURE_MICROMAP_LOSSY_BIT_KHR, for each element of pGeometries or ppGeometries, if it includes a VkAccelerationStructureGeometryMicromapDataKHR structure in its pNext chain, for each element of its pUsageCounts or ppUsageCounts members, its subdivisionLevel member **must** be less than or equal to maxOpacityLossy4StateSubdivisionLevel
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-format-11713
  If flags includes VK_BUILD_ACCELERATION_STRUCTURE_MICROMAP_LOSSY_BIT_KHR, for each element of pGeometries or ppGeometries, if it includes a VkAccelerationStructureGeometryMicromapDataKHR structure in its pNext chain, each element of its triangleArray member **must** have its format member equal to VK_OPACITY_MICROMAP_FORMAT_4_STATE_KHR
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-subdivisionLevel-11566
  If flags does not include VK_BUILD_ACCELERATION_STRUCTURE_MICROMAP_LOSSY_BIT_KHR, for each element of pGeometries or ppGeometries, if it includes a VkAccelerationStructureGeometryMicromapDataKHR structure in its pNext chain, for each element of its triangleArray member, if its format member is VK_OPACITY_MICROMAP_FORMAT_4_STATE_KHR, its subdivisionLevel member **must** be less than or equal to maxOpacity4StateSubdivisionLevel
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-subdivisionLevel-11567
  If flags includes VK_BUILD_ACCELERATION_STRUCTURE_MICROMAP_LOSSY_BIT_KHR, for each element of pGeometries or ppGeometries, if it includes a VkAccelerationStructureGeometryMicromapDataKHR structure in its pNext chain, for each element of its triangleArray member, its subdivisionLevel member **must** be less than or equal to maxOpacityLossy4StateSubdivisionLevel

Valid Usage (Implicit)

- VUID-VkAccelerationStructureBuildGeometryInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_BUILD_GEOMETRY_INFO_KHR
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-parameter
  type **must** be a valid VkAccelerationStructureTypeKHR value
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-flags-parameter
  If flags is not 0, flags **must** be a valid combination of VkBuildAccelerationStructureFlagBitsKHR values
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-pGeometries-parameter
  If geometryCount is not 0, and pGeometries is not NULL, pGeometries **must** be a valid pointer

to an array of geometryCount valid VkAccelerationStructureGeometryKHR structures

- VUID-VkAccelerationStructureBuildGeometryInfoKHR-ppGeometries-parameter
  If geometryCount is not 0, and ppGeometries is not NULL, ppGeometries **must** be a valid pointer to an array of geometryCount valid pointers to valid VkAccelerationStructureGeometryKHR structures
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-commonparent
  Both of dstAccelerationStructure, and srcAccelerationStructure that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

The VkBuildAccelerationStructureModeKHR enumeration is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkBuildAccelerationStructureModeKHR {
    VK_BUILD_ACCELERATION_STRUCTURE_MODE_BUILD_KHR = 0,
    VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR = 1,
} VkBuildAccelerationStructureModeKHR;
```

- VK_BUILD_ACCELERATION_STRUCTURE_MODE_BUILD_KHR specifies that the destination acceleration structure will be built using the specified geometries.
- VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR specifies that the destination acceleration structure will be built using data in a source acceleration structure, updated by the specified geometries.

The VkDeviceOrHostAddressKHR union is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef union VkDeviceOrHostAddressKHR {
    VkDeviceAddress    deviceAddress;
    void*              hostAddress;
} VkDeviceOrHostAddressKHR;
```

- deviceAddress is a buffer device address as returned by the vkGetBufferDeviceAddressKHR command.
- hostAddress is a host memory address.

The VkDeviceOrHostAddressConstKHR union is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef union VkDeviceOrHostAddressConstKHR {
    VkDeviceAddress    deviceAddress;
    const void*        hostAddress;
} VkDeviceOrHostAddressConstKHR;
```

- `deviceAddress` is a buffer device address as returned by the vkGetBufferDeviceAddressKHR command.
- `hostAddress` is a const host memory address.

The `VkAccelerationStructureGeometryKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureGeometryKHR {
    VkStructureType                              sType;
    const void*                                  pNext;
    VkGeometryTypeKHR                            geometryType;
    VkAccelerationStructureGeometryDataKHR       geometry;
    VkGeometryFlagsKHR                           flags;
} VkAccelerationStructureGeometryKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `geometryType` describes which type of geometry this `VkAccelerationStructureGeometryKHR` refers to.
- `geometry` is a VkAccelerationStructureGeometryDataKHR union describing the geometry data for the relevant geometry type.
- `flags` is a bitmask of VkGeometryFlagBitsKHR values describing additional properties of how the geometry should be built.

Valid Usage

- VUID-VkAccelerationStructureGeometryKHR-micromap-11568
  If `geometryType` is VK_GEOMETRY_TYPE_MICROMAP_KHR, `pNext` chain **must** include a single VkAccelerationStructureGeometryMicromapDataKHR structure
- VUID-VkAccelerationStructureGeometryKHR-flags-11569
  If `geometryType` is VK_GEOMETRY_TYPE_MICROMAP_KHR, `flags` **must** be `0`

Valid Usage (Implicit)

- VUID-VkAccelerationStructureGeometryKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_KHR
- VUID-VkAccelerationStructureGeometryKHR-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkAccelerationStructureGeometryMicromapDataKHR
- VUID-VkAccelerationStructureGeometryKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkAccelerationStructureGeometryKHR-geometryType-parameter
  `geometryType` **must** be a valid VkGeometryTypeKHR value

- VUID-VkAccelerationStructureGeometryKHR-triangles-parameter
  If `geometryType` is `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, the `triangles` member of `geometry` **must** be a valid `VkAccelerationStructureGeometryTrianglesDataKHR` structure
- VUID-VkAccelerationStructureGeometryKHR-aabbs-parameter
  If `geometryType` is `VK_GEOMETRY_TYPE_AABBS_KHR`, the `aabbs` member of `geometry` **must** be a valid `VkAccelerationStructureGeometryAabbsDataKHR` structure
- VUID-VkAccelerationStructureGeometryKHR-instances-parameter
  If `geometryType` is `VK_GEOMETRY_TYPE_INSTANCES_KHR`, the `instances` member of `geometry` **must** be a valid `VkAccelerationStructureGeometryInstancesDataKHR` structure
- VUID-VkAccelerationStructureGeometryKHR-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of `VkGeometryFlagBitsKHR` values

The `VkAccelerationStructureGeometryDataKHR` union is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef union VkAccelerationStructureGeometryDataKHR {
    VkAccelerationStructureGeometryTrianglesDataKHR    triangles;
    VkAccelerationStructureGeometryAabbsDataKHR        aabbs;
    VkAccelerationStructureGeometryInstancesDataKHR    instances;
} VkAccelerationStructureGeometryDataKHR;
```

- `triangles` is a `VkAccelerationStructureGeometryTrianglesDataKHR` structure.
- `aabbs` is a `VkAccelerationStructureGeometryAabbsDataKHR` structure.
- `instances` is a `VkAccelerationStructureGeometryInstancesDataKHR` structure.

The `VkAccelerationStructureGeometryTrianglesDataKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureGeometryTrianglesDataKHR {
    VkStructureType                  sType;
    const void*                      pNext;
    VkFormat                         vertexFormat;
    VkDeviceOrHostAddressConstKHR    vertexData;
    VkDeviceSize                     vertexStride;
    uint32_t                         maxVertex;
    VkIndexType                      indexType;
    VkDeviceOrHostAddressConstKHR    indexData;
    VkDeviceOrHostAddressConstKHR    transformData;
} VkAccelerationStructureGeometryTrianglesDataKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `vertexFormat` is the `VkFormat` of each vertex element.
- `vertexData` is a device or host address of memory containing vertex data for this geometry.

- `vertexStride` is the stride in bytes between each vertex.
- `maxVertex` is the number of vertices in `vertexData` minus one.
- `indexType` is the VkIndexType of each index element.
- `indexData` is a device or host address of memory containing index data for this geometry. When `indexType` is VK_INDEX_TYPE_NONE_KHR it **must** be `NULL`.
- `transformData` is a device or host address to memory containing an optional reference to a VkTransformMatrixKHR structure describing a transformation from the space in which the vertices in this geometry are described to the space in which the acceleration structure is defined.

NOTE Unlike the stride for vertex buffers in VkVertexInputBindingDescription for graphics pipelines which must not exceed `maxVertexInputBindingStride`, `vertexStride` for acceleration structure geometry is instead restricted to being a 32-bit value.

Valid Usage

- VUID-VkAccelerationStructureGeometryTrianglesDataKHR-vertexStride-03735
  `vertexStride` **must** be a multiple of the size in bytes of the smallest component of `vertexFormat`
- VUID-VkAccelerationStructureGeometryTrianglesDataKHR-vertexStride-03819
  `vertexStride` **must** be less than or equal to 2^{32}-1
- VUID-VkAccelerationStructureGeometryTrianglesDataKHR-vertexFormat-03797
  The format features of `vertexFormat` **must** contain VK_FORMAT_FEATURE_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR
- VUID-VkAccelerationStructureGeometryTrianglesDataKHR-indexType-03798
  `indexType` **must** be VK_INDEX_TYPE_UINT16, VK_INDEX_TYPE_UINT32, or VK_INDEX_TYPE_NONE_KHR

Valid Usage (Implicit)

- VUID-VkAccelerationStructureGeometryTrianglesDataKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_TRIANGLES_DATA_KHR
- VUID-VkAccelerationStructureGeometryTrianglesDataKHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkAccelerationStructureTrianglesOpacityMicromapEXT or VkAccelerationStructureTrianglesOpacityMicromapKHR
- VUID-VkAccelerationStructureGeometryTrianglesDataKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkAccelerationStructureGeometryTrianglesDataKHR-vertexFormat-parameter

vertexFormat **must** be a valid VkFormat value

- VUID-VkAccelerationStructureGeometryTrianglesDataKHR-indexType-parameter indexType **must** be a valid VkIndexType value

The VkAccelerationStructureTrianglesOpacityMicromapKHR structure is defined as:

```
// Provided by VK_KHR_opacity_micromap
typedef struct VkAccelerationStructureTrianglesOpacityMicromapKHR {
    VkStructureType                sType;
    void*                          pNext;
    VkIndexType                    indexType;
    VkDeviceAddress                indexBuffer;
    VkDeviceSize                   indexStride;
    uint32_t                       baseTriangle;
    VkAccelerationStructureKHR     micromap;
} VkAccelerationStructureTrianglesOpacityMicromapKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- indexType is the type of triangle indices used when indexing this micromap.
- indexBuffer is the address containing the triangle indices.
- indexStride is the byte stride between triangle indices.
- baseTriangle is the base value added to the non-negative triangle indices.
- micromap is the handle to the micromap object to include in this geometry.

If VkAccelerationStructureTrianglesOpacityMicromapKHR is included in the pNext chain of a VkAccelerationStructureGeometryTrianglesDataKHR structure, that geometry will reference that micromap.

For each triangle in the geometry, the acceleration structure build fetches an index from indexBuffer using indexType and indexStride if present. If indexType is VK_INDEX_TYPE_NONE_KHR, then the index used is the index of the triangle in the geometry.

If that value is the unsigned cast of one of the values from VkOpacityMicromapSpecialIndexKHR then that triangle behaves as described for that special value in Ray Opacity Micromap.

Otherwise that triangle uses the opacity micromap information from micromap at that index plus baseTriangle.

Valid Usage

- VUID-VkAccelerationStructureTrianglesOpacityMicromapKHR-indexType-11570 indexType **must** be VK_INDEX_TYPE_UINT8, VK_INDEX_TYPE_UINT16, VK_INDEX_TYPE_UINT32, or VK_INDEX_TYPE_NONE_KHR

- VUID-VkAccelerationStructureTrianglesOpacityMicromapKHR-indexBuffer-11571
  If `indexType` is VK_INDEX_TYPE_NONE_KHR, `indexBuffer` **must** be 0
- VUID-VkAccelerationStructureTrianglesOpacityMicromapKHR-indexBuffer-11572
  If `indexType` is not VK_INDEX_TYPE_NONE_KHR, `indexBuffer` **must** be a valid VkDeviceAddress
- VUID-VkAccelerationStructureTrianglesOpacityMicromapKHR-indexStride-11573
  If `indexType` is not VK_INDEX_TYPE_NONE_KHR, `indexStride` **must** be a multiple of the size in bytes of `indexType`
- VUID-VkAccelerationStructureTrianglesOpacityMicromapKHR-indexStride-11574
  If `indexType` is not VK_INDEX_TYPE_NONE_KHR, `indexStride` **must** be less than or equal to 2^{32}-1
- VUID-VkAccelerationStructureTrianglesOpacityMicromapKHR-geometry-11576
  If `indexType` is VK_INDEX_TYPE_NONE_KHR, for each triangle index in the geometry, index plus `baseTriangle` **must** be less than *numTriangles*, where *numTriangles* is given by the sum of all `count` parameters in the `pUsageCounts` or `ppUsageCounts` provided to the micromap's build command
- VUID-VkAccelerationStructureTrianglesOpacityMicromapKHR-micromap-11579
  If `indexType` is VK_INDEX_TYPE_NONE_KHR, `micromap` **must** not be VK_NULL_HANDLE

Valid Usage (Implicit)

- VUID-VkAccelerationStructureTrianglesOpacityMicromapKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_TRIANGLES_OPACITY_MICROMAP_KHR
- VUID-VkAccelerationStructureTrianglesOpacityMicromapKHR-indexType-parameter
  `indexType` **must** be a valid VkIndexType value
- VUID-VkAccelerationStructureTrianglesOpacityMicromapKHR-micromap-parameter
  If `micromap` is not VK_NULL_HANDLE, `micromap` **must** be a valid VkAccelerationStructureKHR handle

Structure Chaining

Extends the structure

- VkAccelerationStructureGeometryTrianglesDataKHR

The `VkAccelerationStructureTrianglesOpacityMicromapEXT` structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkAccelerationStructureTrianglesOpacityMicromapEXT {
    VkStructureType                       sType;
    void*                                 pNext;
    VkIndexType                           indexType;
    VkDeviceOrHostAddressConstKHR         indexBuffer;
    VkDeviceSize                          indexStride;
    uint32_t                              baseTriangle;
    uint32_t                              usageCountsCount;
    const VkMicromapUsageEXT*             pUsageCounts;
    const VkMicromapUsageEXT* const*      ppUsageCounts;
    VkMicromapEXT                         micromap;
} VkAccelerationStructureTrianglesOpacityMicromapEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `indexType` is the type of triangle indices used when indexing this micromap.
- `indexBuffer` is a device or host address of memory containing the triangle indices. When `indexType` is VK_INDEX_TYPE_NONE_KHR it **must** be `NULL`.
- `indexStride` is the byte stride between triangle indices.
- `baseTriangle` is the base value added to the non-negative triangle indices.
- `usageCountsCount` specifies the number of usage counts structures that will be used to determine the size of this micromap.
- `pUsageCounts` is a pointer to an array of VkMicromapUsageEXT structures.
- `ppUsageCounts` is a pointer to an array of pointers to VkMicromapUsageEXT structures.
- `micromap` is the handle to the micromap object to include in this geometry.

If `VkAccelerationStructureTrianglesOpacityMicromapEXT` is included in the `pNext` chain of a VkAccelerationStructureGeometryTrianglesDataKHR structure, that geometry will reference that micromap.

For each triangle in the geometry, the acceleration structure build fetches an index from `indexBuffer` using `indexType` and `indexStride` if present. If `indexType` is VK_INDEX_TYPE_NONE_KHR, then the index used is the index of the triangle in the geometry.

If that value is the unsigned cast of one of the values from VkOpacityMicromapSpecialIndexEXT then that triangle behaves as described for that special value in Ray Opacity Micromap.

Otherwise that triangle uses the opacity micromap information from micromap at that index plus `baseTriangle`.

Only one of `pUsageCounts` or `ppUsageCounts` **can** be a valid pointer, the other **must** be `NULL`. The elements of the non-`NULL` array describe the total count used to build this geometry. For a given `format` and `subdivisionLevel` the number of triangles in this geometry matching those values after indirection and special index handling **must** be equal to the sum of matching `count` provided.

If micromap is VK_NULL_HANDLE, then every value read from indexBuffer **must** be one of the values in VkOpacityMicromapSpecialIndexEXT.

Valid Usage

- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-pUsageCounts-07335
  Only one of pUsageCounts or ppUsageCounts **can** be a valid pointer, the other **must** be NULL
- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-indexType-10719
  indexType **must** be VK_INDEX_TYPE_UINT16, VK_INDEX_TYPE_UINT32, or VK_INDEX_TYPE_NONE_KHR
- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-indexType-10722
  If indexType is not VK_INDEX_TYPE_NONE_KHR, then indexStride **must** be a multiple of the size in bytes of indexType
- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-indexType-10723
  If indexType is not VK_INDEX_TYPE_NONE_KHR, then indexStride **must** be less than or equal to 2^{32}-1
- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-format-11679
  If micromap is not VK_NULL_HANDLE, the type used to construct micromap is VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT, the format member of each element of pUsageCounts or ppUsageCounts **must** be a valid value from VkOpacityMicromapFormatEXT
- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-subdivisionLevel-11680
  If micromap is not VK_NULL_HANDLE, the type used to build micromap is VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT, for each member of pUsageCounts or ppUsageCounts with format equal to VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT, subdivisionLevel **must** be less than or equal to VkPhysicalDeviceOpacityMicromapPropertiesEXT::maxOpacity2StateSubdivisionLevel
- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-subdivisionLevel-11681
  If micromap is not VK_NULL_HANDLE, the type used to build micromap is VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT, for each member of pUsageCounts or ppUsageCounts with format equal to VK_OPACITY_MICROMAP_FORMAT_4_STATE_EXT, subdivisionLevel **must** be less than or equal to VkPhysicalDeviceOpacityMicromapPropertiesEXT::maxOpacity4StateSubdivisionLevel

Valid Usage (Implicit)

- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_TRIANGLES_OPACITY_MICROMAP_EXT
- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-indexType-parameter
  indexType **must** be a valid VkIndexType value
- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-pUsageCounts-parameter
  If usageCountsCount is not 0, and pUsageCounts is not NULL, pUsageCounts **must** be a valid

pointer to an array of `usageCountsCount` VkMicromapUsageEXT structures

- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-ppUsageCounts-parameter
  If `usageCountsCount` is not `0`, and `ppUsageCounts` is not `NULL`, `ppUsageCounts` **must** be a valid pointer to an array of `usageCountsCount` valid pointers to VkMicromapUsageEXT structures
- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-micromap-parameter
  If `micromap` is not VK_NULL_HANDLE, `micromap` **must** be a valid VkMicromapEXT handle

Structure Chaining

Extends the structure

- VkAccelerationStructureGeometryTrianglesDataKHR

The VkOpacityMicromapSpecialIndexKHR enumeration is defined as:

```
// Provided by VK_KHR_opacity_micromap
typedef enum VkOpacityMicromapSpecialIndexKHR {
    VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_TRANSPARENT_KHR = -1,
    VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_OPAQUE_KHR = -2,
    VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_TRANSPARENT_KHR = -3,
    VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_OPAQUE_KHR = -4,
  // Provided by VK_EXT_opacity_micromap
    VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_TRANSPARENT_EXT =
VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_TRANSPARENT_KHR,
  // Provided by VK_EXT_opacity_micromap
    VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_OPAQUE_EXT =
VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_OPAQUE_KHR,
  // Provided by VK_EXT_opacity_micromap
    VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_TRANSPARENT_EXT =
VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_TRANSPARENT_KHR,
  // Provided by VK_EXT_opacity_micromap
    VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_OPAQUE_EXT =
VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_OPAQUE_KHR,
} VkOpacityMicromapSpecialIndexKHR;
```

```
// Provided by VK_EXT_opacity_micromap
// Equivalent to VkOpacityMicromapSpecialIndexKHR
typedef VkOpacityMicromapSpecialIndexKHR VkOpacityMicromapSpecialIndexEXT;
```

- VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_TRANSPARENT_KHR specifies that the entire triangle is fully transparent.
- VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_OPAQUE_KHR specifies that the entire triangle is fully opaque.
- VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_TRANSPARENT_KHR specifies that the entire triangle is unknown-transparent.

- VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_OPAQUE_KHR specifies that the entire triangle is unknown-opaque.

The VkAccelerationStructureSerializedBlockTypeKHR enumeration is defined as:

```
// Provided by VK_KHR_opacity_micromap
typedef enum VkAccelerationStructureSerializedBlockTypeKHR {
    VK_ACCELERATION_STRUCTURE_SERIALIZED_BLOCK_TYPE_OPACITY_MICROMAP_KHR = 0,
} VkAccelerationStructureSerializedBlockTypeKHR;
```

- VK_ACCELERATION_STRUCTURE_SERIALIZED_BLOCK_TYPE_OPACITY_MICROMAP_KHR specifies that the block contains opacity micromaps.

The VkTransformMatrixKHR structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkTransformMatrixKHR {
    float    matrix[3][4];
} VkTransformMatrixKHR;
```

- matrix is a 3x4 row-major affine transformation matrix.

Valid Usage

- VUID-VkTransformMatrixKHR-matrix-03799
  The first three columns of matrix **must** define an invertible 3x3 matrix

The VkAccelerationStructureGeometryAabbsDataKHR structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureGeometryAabbsDataKHR {
    VkStructureType                  sType;
    const void*                      pNext;
    VkDeviceOrHostAddressConstKHR    data;
    VkDeviceSize                     stride;
} VkAccelerationStructureGeometryAabbsDataKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- data is a device or host address of memory containing VkAabbPositionsKHR structures containing position data for each axis-aligned bounding box in the geometry.
- stride is the stride in bytes between each entry in data. The stride **must** be a multiple of 8.

Valid Usage

- VUID-VkAccelerationStructureGeometryAabbsDataKHR-stride-03545
  `stride` **must** be a multiple of 8
- VUID-VkAccelerationStructureGeometryAabbsDataKHR-stride-03820
  `stride` **must** be less than or equal to 2^{32}-1

Valid Usage (Implicit)

- VUID-VkAccelerationStructureGeometryAabbsDataKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_AABBS_DATA_KHR
- VUID-VkAccelerationStructureGeometryAabbsDataKHR-pNext-pNext
  `pNext` **must** be `NULL`

The `VkAabbPositionsKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAabbPositionsKHR {
    float    minX;
    float    minY;
    float    minZ;
    float    maxX;
    float    maxY;
    float    maxZ;
} VkAabbPositionsKHR;
```

- `minX` is the x position of one opposing corner of a bounding box.
- `minY` is the y position of one opposing corner of a bounding box.
- `minZ` is the z position of one opposing corner of a bounding box.
- `maxX` is the x position of the other opposing corner of a bounding box.
- `maxY` is the y position of the other opposing corner of a bounding box.
- `maxZ` is the z position of the other opposing corner of a bounding box.

Valid Usage

- VUID-VkAabbPositionsKHR-minX-03546
  `minX` **must** be less than or equal to `maxX`
- VUID-VkAabbPositionsKHR-minY-03547
  `minY` **must** be less than or equal to `maxY`
- VUID-VkAabbPositionsKHR-minZ-03548

`minZ` **must** be less than or equal to `maxZ`

The `VkAccelerationStructureGeometryInstancesDataKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureGeometryInstancesDataKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    VkBool32                        arrayOfPointers;
    VkDeviceOrHostAddressConstKHR   data;
} VkAccelerationStructureGeometryInstancesDataKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `arrayOfPointers` specifies whether `data` is used as an array of addresses or just an array.
- `data` is either the address of an array of device or host addresses referencing individual `VkAccelerationStructureInstanceKHR` structures if `arrayOfPointers` is `VK_TRUE`, or the address of an array of `VkAccelerationStructureInstanceKHR` structures. Addresses and `VkAccelerationStructureInstanceKHR` structures are tightly packed.

Valid Usage (Implicit)

- VUID-VkAccelerationStructureGeometryInstancesDataKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_INSTANCES_DATA_KHR`
- VUID-VkAccelerationStructureGeometryInstancesDataKHR-pNext-pNext
  `pNext` **must** be `NULL`

Acceleration structure instances **can** be built into top-level acceleration structures. Each acceleration structure instance is a separate entry in the top-level acceleration structure which includes all the geometry of a bottom-level acceleration structure at a transformed location. Multiple instances **can** point to the same bottom level acceleration structure.

An acceleration structure instance is defined by the structure:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureInstanceKHR {
    VkTransformMatrixKHR          transform;
    uint32_t                      instanceCustomIndex:24;
    uint32_t                      mask:8;
    uint32_t                      instanceShaderBindingTableRecordOffset:24;
    VkGeometryInstanceFlagsKHR    flags:8;
    uint64_t                      accelerationStructureReference;
} VkAccelerationStructureInstanceKHR;
```

- `transform` is a VkTransformMatrixKHR structure describing a transformation to be applied to the acceleration structure.
- `instanceCustomIndex` is a 24-bit application-specified index value accessible to ray shaders in the `InstanceCustomIndexKHR` built-in.
- `mask` is an 8-bit visibility mask for the geometry. The instance **may** only be hit if `Cull Mask & instance.mask != 0`
- `instanceShaderBindingTableRecordOffset` is a 24-bit offset used in calculating the hit shader binding table index.
- `flags` is an 8-bit mask of VkGeometryInstanceFlagBitsKHR values to apply to this instance.
- `accelerationStructureReference` is either :
  - a device address containing the value obtained from vkGetAccelerationStructureDeviceAddressKHR for a bottom level acceleration structure (used by device operations which reference acceleration structures)
  - a VkAccelerationStructureKHR object (used by host operations which reference acceleration structures).

The C language specification does not define the ordering of bit-fields, but in practice, this structure produces the correct layout with existing compilers. The intended bit pattern is for the following:

- `instanceCustomIndex` and `mask` occupy the same memory as if a single `uint32_t` was specified in their place
  - `instanceCustomIndex` occupies the 24 least significant bits of that memory
  - `mask` occupies the 8 most significant bits of that memory
- `instanceShaderBindingTableRecordOffset` and `flags` occupy the same memory as if a single `uint32_t` was specified in their place
  - `instanceShaderBindingTableRecordOffset` occupies the 24 least significant bits of that memory
  - `flags` occupies the 8 most significant bits of that memory

If a compiler produces code that diverges from that pattern, applications **must** employ another method to set values according to the correct bit pattern.

Valid Usage

- VUID-VkAccelerationStructureInstanceKHR-micromap-11580
  If `flags` includes VK_GEOMETRY_INSTANCE_FORCE_OPACITY_MICROMAP_2_STATE_BIT_KHR or VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_BIT_KHR, the `VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap` or `VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap` feature **must** be enabled
- VUID-VkAccelerationStructureInstanceKHR-flags-11581
  If `flags` includes VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_BIT_KHR, the `flags` used to build the acceleration structure referenced by `accelerationStructureReference` **must** have included

VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_BIT_KHR

Valid Usage (Implicit)

- VUID-VkAccelerationStructureInstanceKHR-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkGeometryInstanceFlagBitsKHR values

Possible values of `flags` in the instance modifying the behavior of that instance are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkGeometryInstanceFlagBitsKHR {
    VK_GEOMETRY_INSTANCE_TRIANGLE_FACING_CULL_DISABLE_BIT_KHR = 0x00000001,
    VK_GEOMETRY_INSTANCE_TRIANGLE_FLIP_FACING_BIT_KHR = 0x00000002,
    VK_GEOMETRY_INSTANCE_FORCE_OPAQUE_BIT_KHR = 0x00000004,
    VK_GEOMETRY_INSTANCE_FORCE_NO_OPAQUE_BIT_KHR = 0x00000008,
  // Provided by VK_KHR_opacity_micromap
    VK_GEOMETRY_INSTANCE_FORCE_OPACITY_MICROMAP_2_STATE_BIT_KHR = 0x00000010,
  // Provided by VK_KHR_opacity_micromap
    VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_BIT_KHR = 0x00000020,
    VK_GEOMETRY_INSTANCE_TRIANGLE_FRONT_COUNTERCLOCKWISE_BIT_KHR =
VK_GEOMETRY_INSTANCE_TRIANGLE_FLIP_FACING_BIT_KHR,
  // Provided by VK_EXT_opacity_micromap
    VK_GEOMETRY_INSTANCE_FORCE_OPACITY_MICROMAP_2_STATE_BIT_EXT =
VK_GEOMETRY_INSTANCE_FORCE_OPACITY_MICROMAP_2_STATE_BIT_KHR,
  // Provided by VK_EXT_opacity_micromap
  // VK_GEOMETRY_INSTANCE_FORCE_OPACITY_MICROMAP_2_STATE_EXT is a legacy alias
    VK_GEOMETRY_INSTANCE_FORCE_OPACITY_MICROMAP_2_STATE_EXT =
VK_GEOMETRY_INSTANCE_FORCE_OPACITY_MICROMAP_2_STATE_BIT_EXT,
  // Provided by VK_EXT_opacity_micromap
    VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_BIT_EXT =
VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_BIT_KHR,
  // Provided by VK_EXT_opacity_micromap
  // VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_EXT is a legacy alias
    VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_EXT =
VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_BIT_EXT,
} VkGeometryInstanceFlagBitsKHR;
```

- VK_GEOMETRY_INSTANCE_TRIANGLE_FACING_CULL_DISABLE_BIT_KHR disables face culling for this instance.
- VK_GEOMETRY_INSTANCE_TRIANGLE_FLIP_FACING_BIT_KHR specifies that the facing determination for geometry in this instance is inverted. Because the facing is determined in object space, an instance transform does not change the winding, but a geometry transform does.
- VK_GEOMETRY_INSTANCE_FORCE_OPAQUE_BIT_KHR causes this instance to act as though

VK_GEOMETRY_OPAQUE_BIT_KHR were specified on all geometries referenced by this instance. This behavior **can** be overridden by the SPIR-V NoOpaqueKHR ray flag.

- VK_GEOMETRY_INSTANCE_FORCE_NO_OPAQUE_BIT_KHR causes this instance to act as though VK_GEOMETRY_OPAQUE_BIT_KHR were not specified on all geometries referenced by this instance. This behavior **can** be overridden by the SPIR-V OpaqueKHR ray flag.
- VK_GEOMETRY_INSTANCE_FORCE_OPACITY_MICROMAP_2_STATE_BIT_KHR forces the micromap format to 2 state mode as described in traversal.
- VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_BIT_KHR disables any opacity micromaps referenced by acceleration structures referenced by this instance as described in traversal.

VK_GEOMETRY_INSTANCE_FORCE_NO_OPAQUE_BIT_KHR and VK_GEOMETRY_INSTANCE_FORCE_OPAQUE_BIT_KHR **must** not be used in the same flag.

```
// Provided by VK_KHR_acceleration_structure
typedef VkFlags VkGeometryInstanceFlagsKHR;
```

VkGeometryInstanceFlagsKHR is a bitmask type for setting a mask of zero or more VkGeometryInstanceFlagBitsKHR.

VkAccelerationStructureBuildRangeInfoKHR is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureBuildRangeInfoKHR {
    uint32_t    primitiveCount;
    uint32_t    primitiveOffset;
    uint32_t    firstVertex;
    uint32_t    transformOffset;
} VkAccelerationStructureBuildRangeInfoKHR;
```

- primitiveCount defines the number of primitives for a corresponding acceleration structure geometry.
- primitiveOffset defines an offset in bytes into the memory where primitive data is defined.
- firstVertex is the index of the first vertex to build from for triangle geometry.
- transformOffset defines an offset in bytes into the memory where a transform matrix is defined.

The primitive count and primitive offset are interpreted differently depending on the VkGeometryTypeKHR used:

- For geometries of type VK_GEOMETRY_TYPE_TRIANGLES_KHR, primitiveCount is the number of triangles to be built, where each triangle is treated as 3 vertices.
  - If the geometry uses indices, primitiveCount × 3 indices are consumed from VkAccelerationStructureGeometryTrianglesDataKHR::indexData, starting at an offset of primitiveOffset. The value of firstVertex is added to the index values before fetching vertices.

- If the geometry does not use indices, `primitiveCount` × 3 vertices are consumed from VkAccelerationStructureGeometryTrianglesDataKHR::`vertexData`, starting at an offset of `primitiveOffset` + VkAccelerationStructureGeometryTrianglesDataKHR::`vertexStride` × `firstVertex`.
- If VkAccelerationStructureGeometryTrianglesDataKHR::`transformData` is not `NULL`, a single VkTransformMatrixKHR structure is consumed from VkAccelerationStructureGeometryTrianglesDataKHR::`transformData`, at an offset of `transformOffset`. This matrix describes a transformation from the space in which the vertices for all triangles in this geometry are described to the space in which the acceleration structure is defined.

- For geometries of type VK_GEOMETRY_TYPE_AABBS_KHR, `primitiveCount` is the number of axis-aligned bounding boxes. `primitiveCount` VkAabbPositionsKHR structures are consumed from VkAccelerationStructureGeometryAabbsDataKHR::`data`, starting at an offset of `primitiveOffset`.
- For geometries of type VK_GEOMETRY_TYPE_INSTANCES_KHR, `primitiveCount` is the number of acceleration structures. `primitiveCount` VkAccelerationStructureInstanceKHR structures are consumed from VkAccelerationStructureGeometryInstancesDataKHR::`data`, starting at an offset of `primitiveOffset`.

Valid Usage

- VUID-VkAccelerationStructureBuildRangeInfoKHR-vertexData-10418
  The number of vertices consumed from VkAccelerationStructureGeometryTrianglesDataKHR::`vertexData` **must** be less than or equal to VkAccelerationStructureGeometryTrianglesDataKHR::`maxVertex` + 1
- VUID-VkAccelerationStructureBuildRangeInfoKHR-primitiveOffset-03656
  For geometries of type VK_GEOMETRY_TYPE_TRIANGLES_KHR, if the geometry uses indices, the offset `primitiveOffset` from VkAccelerationStructureGeometryTrianglesDataKHR::`indexData` **must** be a multiple of the element size of VkAccelerationStructureGeometryTrianglesDataKHR::`indexType`
- VUID-VkAccelerationStructureBuildRangeInfoKHR-primitiveOffset-03657
  For geometries of type VK_GEOMETRY_TYPE_TRIANGLES_KHR, if the geometry does not use indices, the offset `primitiveOffset` from VkAccelerationStructureGeometryTrianglesDataKHR::`vertexData` **must** be a multiple of:
  - the size of the format specified in VkAccelerationStructureGeometryTrianglesDataKHR::`vertexFormat`, if that format is a packed format
  - the component size of the VkAccelerationStructureGeometryTrianglesDataKHR::`vertexFormat`, if that format is not a packed format
- VUID-VkAccelerationStructureBuildRangeInfoKHR-maxVertex-10774
  For geometries of type VK_GEOMETRY_TYPE_TRIANGLES_KHR, if the geometry uses indices, then VkAccelerationStructureGeometryTrianglesDataKHR::`maxVertex` **must** be greater than or equal to `firstVertex` plus the maximum index value found in the VkAccelerationStructureGeometryTrianglesDataKHR::`indexData` in the range [`primitiveOffset`, `primitiveOffset`

primitiveCount x 3]

- VUID-VkAccelerationStructureBuildRangeInfoKHR-None-10775
  For geometries of type VK_GEOMETRY_TYPE_TRIANGLES_KHR, if the geometry does not use indices, then VkAccelerationStructureGeometryTrianglesDataKHR::maxVertex **must** be greater than or equal to firstVertex + primitiveCount x 3 - 1
- VUID-VkAccelerationStructureBuildRangeInfoKHR-transformOffset-03658
  For geometries of type VK_GEOMETRY_TYPE_TRIANGLES_KHR, the offset transformOffset from VkAccelerationStructureGeometryTrianglesDataKHR::transformData **must** be a multiple of 16
- VUID-VkAccelerationStructureBuildRangeInfoKHR-primitiveOffset-03659
  For geometries of type VK_GEOMETRY_TYPE_AABBS_KHR, the offset primitiveOffset from VkAccelerationStructureGeometryAabbsDataKHR::data **must** be a multiple of 8
- VUID-VkAccelerationStructureBuildRangeInfoKHR-primitiveOffset-03660
  For geometries of type VK_GEOMETRY_TYPE_INSTANCES_KHR, the offset primitiveOffset from VkAccelerationStructureGeometryInstancesDataKHR::data **must** be a multiple of 16

40.1.8. Copying Acceleration Structures

An additional command exists for copying acceleration structures without updating their contents. The acceleration structure object **can** be compacted in order to improve performance. Before copying, an application **must** query the size of the resulting acceleration structure.

To query acceleration structure size parameters call:

```
// Provided by VK_KHR_acceleration_structure
void vkCmdWriteAccelerationStructuresPropertiesKHR(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    accelerationStructureCount,
    const VkAccelerationStructureKHR*           pAccelerationStructures,
    VkQueryType                                 queryType,
    VkQueryPool                                 queryPool,
    uint32_t                                    firstQuery);
```

- commandBuffer is the command buffer into which the command will be recorded.
- accelerationStructureCount is the count of acceleration structures for which to query the property.
- pAccelerationStructures is a pointer to an array of existing previously built acceleration structures.
- queryType is a VkQueryType value specifying the type of queries managed by the pool.
- queryPool is the query pool that will manage the results of the query.
- firstQuery is the first query index within the query pool that will contain the accelerationStructureCount number of results.

Accesses to any of the acceleration structures listed in pAccelerationStructures **must** be

synchronized with the VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR pipeline stage or the VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR pipeline stage, and an access type of VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR.

- If queryType is VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR, then the value written out is the number of bytes required by a compacted acceleration structure.
- If queryType is VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR, then the value written out is the number of bytes required by a serialized acceleration structure.
- If queryType is VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR, then the value written out is the number of bytes required by the acceleration structure.
- VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR specifies a serialization acceleration structure pointer count query.

Valid Usage

- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-accelerationStructure-08924
  The VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure feature **must** be enabled
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-queryPool-02493
  queryPool **must** have been created with a queryType matching queryType
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-queryPool-02494
  The queries identified by queryPool and firstQuery **must** be *unavailable*
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-query-04880
  The sum of firstQuery plus accelerationStructureCount **must** be less than or equal to the number of queries in queryPool
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-pAccelerationStructures-11684
  All acceleration structures in pAccelerationStructures **must** have been constructed prior to the execution of this command on the device
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-accelerationStructures-03431
  All acceleration structures in pAccelerationStructures **must** have been built with VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_COMPACTION_BIT_KHR if queryType is VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-queryType-06742
  queryType **must** be VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR, VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR, VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR, or VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-pAccelerationStructures-12425
  All acceleration structures in pAccelerationStructures **must** have been built with VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR if queryType is VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-buffer-03736

Each element of `pAccelerationStructures` **must** be fully backed by physical memory

Valid Usage (Implicit)

- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-pAccelerationStructures-parameter
  `pAccelerationStructures` **must** be a valid pointer to an array of `accelerationStructureCount` valid VkAccelerationStructureKHR handles
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-queryType-parameter
  `queryType` **must** be a valid VkQueryType value
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-queryPool-parameter
  `queryPool` **must** be a valid VkQueryPool handle
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-accelerationStructureCount-arraylength
  `accelerationStructureCount` **must** be greater than `0`
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-commonparent
  Each of `commandBuffer`, `queryPool`, and the elements of `pAccelerationStructures` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdWriteAccelerationStructuresPropertiesKHR is not affected by conditional rendering

To copy an acceleration structure call:

```
// Provided by VK_KHR_acceleration_structure
void vkCmdCopyAccelerationStructureKHR(
    VkCommandBuffer                             commandBuffer,
    const VkCopyAccelerationStructureInfoKHR*   pInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pInfo` is a pointer to a VkCopyAccelerationStructureInfoKHR structure defining the copy operation.

This command copies the `pInfo->src` acceleration structure to the `pInfo->dst` acceleration structure in the manner specified by `pInfo->mode`.

Accesses to `pInfo->src` and `pInfo->dst` **must** be synchronized with the VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR pipeline stage or the VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR pipeline stage, and an access type of VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR or VK_ACCESS_ACCELERATION_STRUCTURE_WRITE_BIT_KHR as appropriate.

Valid Usage

- VUID-vkCmdCopyAccelerationStructureKHR-accelerationStructure-08925
  The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled
- VUID-vkCmdCopyAccelerationStructureKHR-src-11633
  The source acceleration structure `pInfo->src` **must** have been constructed prior to the execution of this command on the device
- VUID-vkCmdCopyAccelerationStructureKHR-src-11634
  If the source acceleration structure `pInfo->src` was constructed through deserialization, all micromap arrays the acceleration structure references that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized

data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device

Valid Usage (Implicit)

- VUID-vkCmdCopyAccelerationStructureKHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyAccelerationStructureKHR-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkCopyAccelerationStructureInfoKHR structure
- VUID-vkCmdCopyAccelerationStructureKHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdCopyAccelerationStructureKHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdCopyAccelerationStructureKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyAccelerationStructureKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyAccelerationStructureKHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdCopyAccelerationStructureKHR is not affected by conditional rendering

The `VkCopyAccelerationStructureInfoKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkCopyAccelerationStructureInfoKHR {
    VkStructureType                         sType;
    const void*                             pNext;
    VkAccelerationStructureKHR              src;
    VkAccelerationStructureKHR              dst;
    VkCopyAccelerationStructureModeKHR      mode;
} VkCopyAccelerationStructureInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `src` is the source acceleration structure for the copy.
- `dst` is the target acceleration structure for the copy.
- `mode` is a VkCopyAccelerationStructureModeKHR value specifying additional operations to perform during the copy.

Valid Usage

- VUID-VkCopyAccelerationStructureInfoKHR-mode-03410
  `mode` **must** be VK_COPY_ACCELERATION_STRUCTURE_MODE_COMPACT_KHR or VK_COPY_ACCELERATION_STRUCTURE_MODE_CLONE_KHR
- VUID-VkCopyAccelerationStructureInfoKHR-src-03411
  If `mode` is VK_COPY_ACCELERATION_STRUCTURE_MODE_COMPACT_KHR, `src` **must** have been constructed with VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_COMPACTION_BIT_KHR in the build
- VUID-VkCopyAccelerationStructureInfoKHR-buffer-03718
  The range of `src` accessed by this command **must** be fully backed by physical memory
- VUID-VkCopyAccelerationStructureInfoKHR-buffer-03719
  The range of `dst` accessed by this command **must** be fully backed by physical memory
- VUID-VkCopyAccelerationStructureInfoKHR-dst-07791
  The range of memory backing `dst` that is accessed by this command **must** not overlap the memory backing `src` that is accessed by this command
- VUID-VkCopyAccelerationStructureInfoKHR-mode-12418
  If `mode` is VK_COPY_ACCELERATION_STRUCTURE_MODE_CLONE_KHR, the memory pointed to by `dst` **must** be at least as large as the size of `src`, as reported by vkWriteAccelerationStructuresPropertiesKHR or vkCmdWriteAccelerationStructuresPropertiesKHR with a query type of VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR
- VUID-VkCopyAccelerationStructureInfoKHR-mode-12419
  If `mode` is VK_COPY_ACCELERATION_STRUCTURE_MODE_COMPACT_KHR, the memory pointed to by `dst` **must** be at least as large as the compacted size of `src`, as reported by vkWriteAccelerationStructuresPropertiesKHR or vkCmdWriteAccelerationStructuresPropertiesKHR with a query type of

VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR

Valid Usage (Implicit)

- VUID-VkCopyAccelerationStructureInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COPY_ACCELERATION_STRUCTURE_INFO_KHR
- VUID-VkCopyAccelerationStructureInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkCopyAccelerationStructureInfoKHR-src-parameter
  `src` **must** be a valid VkAccelerationStructureKHR handle
- VUID-VkCopyAccelerationStructureInfoKHR-dst-parameter
  `dst` **must** be a valid VkAccelerationStructureKHR handle
- VUID-VkCopyAccelerationStructureInfoKHR-mode-parameter
  `mode` **must** be a valid VkCopyAccelerationStructureModeKHR value
- VUID-VkCopyAccelerationStructureInfoKHR-commonparent
  Both of `dst`, and `src` **must** have been created, allocated, or retrieved from the same VkDevice

Possible values of `mode` specifying additional operations to perform during the copy, are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkCopyAccelerationStructureModeKHR {
    VK_COPY_ACCELERATION_STRUCTURE_MODE_CLONE_KHR = 0,
    VK_COPY_ACCELERATION_STRUCTURE_MODE_COMPACT_KHR = 1,
  // Provided by VK_KHR_acceleration_structure
    VK_COPY_ACCELERATION_STRUCTURE_MODE_SERIALIZE_KHR = 2,
  // Provided by VK_KHR_acceleration_structure
    VK_COPY_ACCELERATION_STRUCTURE_MODE_DESERIALIZE_KHR = 3,
} VkCopyAccelerationStructureModeKHR;
```

- VK_COPY_ACCELERATION_STRUCTURE_MODE_CLONE_KHR creates a direct copy of the acceleration structure specified in `src` into the one specified by `dst`. The `dst` acceleration structure **must** have been created with the same parameters as `src` with the exception of `size`, which **must** be at least as large as the queried VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR size . If `src` contains references to other acceleration structures, `dst` will reference the same acceleration structures.
- VK_COPY_ACCELERATION_STRUCTURE_MODE_COMPACT_KHR creates a more compact version of an acceleration structure `src` into `dst`. The acceleration structure `dst` **must** have been created with a size at least as large as that returned by vkCmdWriteAccelerationStructuresPropertiesKHR, or vkWriteAccelerationStructuresPropertiesKHR after the build of the acceleration structure specified by `src`. If `src` contains references to other acceleration structures, `dst` will reference the same acceleration structures.

- VK_COPY_ACCELERATION_STRUCTURE_MODE_SERIALIZE_KHR serializes the acceleration structure to a semi-opaque format which can be reloaded on a compatible implementation.
- VK_COPY_ACCELERATION_STRUCTURE_MODE_DESERIALIZE_KHR deserializes the semi-opaque serialization format in the buffer to the acceleration structure. After deserialization, the destination acceleration structure is treated as having the same acceleration structure type as the originally serialized acceleration structure.

To copy an acceleration structure to device memory call:

```
// Provided by VK_KHR_acceleration_structure
void vkCmdCopyAccelerationStructureToMemoryKHR(
    VkCommandBuffer                             commandBuffer,
    const VkCopyAccelerationStructureToMemoryInfoKHR* pInfo);
```

- commandBuffer is the command buffer into which the command will be recorded.
- pInfo is a pointer to a VkCopyAccelerationStructureToMemoryInfoKHR structure defining the copy operation.

Accesses to pInfo->src **must** be synchronized with the VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR pipeline stage or the VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR pipeline stage, and an access type of VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR. Accesses to the buffer indicated by pInfo->dst.deviceAddress **must** be synchronized with the VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR pipeline stage or the VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR pipeline stage, and an access type of VK_ACCESS_TRANSFER_WRITE_BIT.

This command produces the same results as vkCopyAccelerationStructureToMemoryKHR, but writes its result to a device address, and is executed on the device rather than the host. The output **may** not necessarily be bit-for-bit identical, but it can be equally used by either vkCmdCopyMemoryToAccelerationStructureKHR or vkCopyMemoryToAccelerationStructureKHR.

The defined header structure for the serialized data consists of:

- VK_UUID_SIZE bytes of data matching VkPhysicalDeviceIDProperties::driverUUID
- VK_UUID_SIZE bytes of data identifying the compatibility for comparison using vkGetDeviceAccelerationStructureCompatibilityKHR

For a top-level or bottom-level acceleration structure, this is followed by:

- A 64-bit integer of the total size matching the value queried using VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR
- A 64-bit integer of the deserialized size to be passed in to VkAccelerationStructureCreateInfoKHR::size

For a top-level acceleration structure, the remainder of the serialized header consists of:

- A 64-bit integer of the count of the number of acceleration structure handles following. This

value matches the value queried using VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR. For top-level acceleration structures this number is implementation-dependent; the number of and ordering of the handles may not match the instance descriptions which were used to build the acceleration structure.

The corresponding handles matching the values returned by vkGetAccelerationStructureDeviceAddressKHR are tightly packed in the buffer following the count. The application is expected to store a mapping between those handles and the original application-generated bottom-level acceleration structures to provide when deserializing.

For a bottom-level acceleration structure, the remainder of the serialized header consists of:

- A 64-bit integer consisting of two packed 32 bit values. The high 32 bits are 0xFFFFFFFF to indicate a block-based format, and the low 32 bits contain the number of serialized blocks that follow
- For each block:
  - A 32-bit unsigned integer indicating the block type corresponding to values in the VkAccelerationStructureSerializedBlockTypeKHR enumeration
  - A 32-bit reserved value for alignment
  - A 64-bit unsigned integer indicating the number of block buffer device addresses that follow the block header
  - An array of 64-bit buffer device addresses, with the count matching the previous value

If the block type is VK_ACCELERATION_STRUCTURE_SERIALIZED_BLOCK_TYPE_OPACITY_MICROMAP_KHR, the corresponding buffer device addresses are tightly packed in the buffer and match the values returned by vkGetBufferDeviceAddressKHR for the underlying buffer objects for the opacity micromaps. An implementation **may** either return `NULL` for a geometry with no opacity micromap or only return valid device address entries in the array. Similar to the above, the application **must** provide a mapping between the valid buffer device addresses and the original application-generated opacity micromaps when deserializing. If the implementation returned `NULL` addresses for geometry without micromaps, the application **must** provide `NULL` device addresses mapped to those when deserializing.

The addresses of micromaps provided to deserializing **must** be compatible with the micromaps that were used for building the acceleration structure that was serialized. They **must** be created before deserializing with identical `type` and `createFlags` except for the VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR flag set.

Before using the deserialized acceleration structure, the micromaps **must** be deserialized using the exact serialized output of the micromap that was originally referenced by the serialized acceleration structure, or, if the acceleration structure was originally built with the VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_KHR flag, the application **may** replace the micromap with a newly constructed micromap, using the normal update rules.

The serialized data is written to the buffer (or read from the buffer) according to the host endianness.

Valid Usage

- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-accelerationStructure-08926
  The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-pInfo-03739
  `pInfo->dst.deviceAddress` **must** be a valid VkDeviceAddress
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-pInfo-03740
  `pInfo->dst.deviceAddress` **must** be aligned to `256` bytes
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-pInfo-11699
  If `pInfo->src` was built with a `type` of VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR, `pInfo->dst.deviceAddress` **must** be aligned to `128` bytes

Valid Usage (Implicit)

- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkCopyAccelerationStructureToMemoryInfoKHR structure
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdCopyAccelerationStructureToMemoryKHR is not affected by conditional rendering

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkCopyAccelerationStructureToMemoryInfoKHR {
    VkStructureType                         sType;
    const void*                             pNext;
    VkAccelerationStructureKHR              src;
    VkDeviceOrHostAddressKHR                dst;
    VkCopyAccelerationStructureModeKHR      mode;
} VkCopyAccelerationStructureToMemoryInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `src` is the source acceleration structure for the copy.
- `dst` is the device or host address of memory which is the target for the copy.
- `mode` is a VkCopyAccelerationStructureModeKHR value specifying additional operations to perform during the copy.

Valid Usage

- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-src-04959
  The source acceleration structure `src` **must** have been constructed prior to the execution of this command
- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-src-11582
  If the source acceleration structure `src` was constructed through deserialization, all micromap arrays the acceleration structure references that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command
- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-dst-03561
  The memory pointed to by `dst` **must** be at least as large as the serialization size of `src`, as reported by vkWriteAccelerationStructuresPropertiesKHR or

vkCmdWriteAccelerationStructuresPropertiesKHR with a query type of VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR

- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-mode-03412
  `mode` **must** be VK_COPY_ACCELERATION_STRUCTURE_MODE_SERIALIZE_KHR
- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-mode-11714
  The range of `src` accessed by this command **must** be fully backed by physical memory
- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-mode-11715
  The range of `dst` accessed by this command **must** be fully backed by physical memory

Valid Usage (Implicit)

- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COPY_ACCELERATION_STRUCTURE_TO_MEMORY_INFO_KHR
- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-src-parameter
  `src` **must** be a valid VkAccelerationStructureKHR handle
- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-mode-parameter
  `mode` **must** be a valid VkCopyAccelerationStructureModeKHR value

To copy device memory to an acceleration structure call:

```
// Provided by VK_KHR_acceleration_structure
void vkCmdCopyMemoryToAccelerationStructureKHR(
    VkCommandBuffer                                 commandBuffer,
    const VkCopyMemoryToAccelerationStructureInfoKHR* pInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pInfo` is a pointer to a VkCopyMemoryToAccelerationStructureInfoKHR structure defining the copy operation.

Accesses to `pInfo->dst` **must** be synchronized with the VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR pipeline stage or the VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR pipeline stage, and an access type of VK_ACCESS_ACCELERATION_STRUCTURE_WRITE_BIT_KHR. Accesses to the buffer indicated by `pInfo->src.deviceAddress` **must** be synchronized with the VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR pipeline stage or the VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR pipeline stage, and an access type of VK_ACCESS_TRANSFER_READ_BIT.

This command can accept acceleration structures produced by either vkCmdCopyAccelerationStructureToMemoryKHR or vkCopyAccelerationStructureToMemoryKHR.

The structure provided as input to deserialize is as described in vkCmdCopyAccelerationStructureToMemoryKHR, with any acceleration structure handles filled in with the newly-queried handles to bottom level acceleration structures created before deserialization. These do not need to be built at deserialize time, but **must** be created.

The application **must** also provide any required micromap addresses in the header as described in `vkCmdCopyAccelerationStructureToMemoryKHR`.

Valid Usage

- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-accelerationStructure-08927
  The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-pInfo-03742
  `pInfo->src.deviceAddress` **must** be a valid VkDeviceAddress
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-pInfo-03743
  `pInfo->src.deviceAddress` **must** be aligned to `256` bytes
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-pInfo-11700
  If the serialized acceleration structure in `pInfo->src` was originally built with a `type` of VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR, `pInfo->src.deviceAddress` **must** be aligned to `128` bytes

Valid Usage (Implicit)

- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkCopyMemoryToAccelerationStructureInfoKHR structure
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdCopyMemoryToAccelerationStructureKHR is not affected by conditional rendering

The `VkCopyMemoryToAccelerationStructureInfoKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkCopyMemoryToAccelerationStructureInfoKHR {
    VkStructureType                         sType;
    const void*                             pNext;
    VkDeviceOrHostAddressConstKHR           src;
    VkAccelerationStructureKHR              dst;
    VkCopyAccelerationStructureModeKHR      mode;
} VkCopyMemoryToAccelerationStructureInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `src` is the device or host address of memory containing the source data for the copy.
- `dst` is the target acceleration structure for the copy.
- `mode` is a VkCopyAccelerationStructureModeKHR value specifying additional operations to perform during the copy.

If the serialized data in `src` represents a top-level acceleration structure, `dst` **must** have been created with a value of VkAccelerationStructureCreateInfoKHR::`type` equal to either VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR or VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR.

If the serialized data in `src` represents a bottom-level acceleration structure, `dst` **must** have been created with a value of VkAccelerationStructureCreateInfoKHR::`type` equal to either

VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR or VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR.

Valid Usage

- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-src-04960
  The source memory pointed to by `src` **must** contain data previously serialized using vkCmdCopyAccelerationStructureToMemoryKHR, potentially modified to relocate acceleration structure references as described in that command
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-mode-03413
  `mode` **must** be VK_COPY_ACCELERATION_STRUCTURE_MODE_DESERIALIZE_KHR
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-pInfo-03414
  The data in `src` **must** have a format compatible with the destination physical device as returned by vkGetDeviceAccelerationStructureCompatibilityKHR
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-dst-03746
  `dst` **must** have been created with a `size` greater than or equal to that used to serialize the data in `src`
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-dst-11716
  The range of `src` accessed by this command **must** be fully backed by physical memory
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-dst-11717
  The range of `dst` accessed by this command **must** be fully backed by physical memory
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-src-11583
  If the serialized acceleration structure in `src` is bottom-level, each block in the header **must** have a type that corresponds to a valid value in VkAccelerationStructureSerializedBlockTypeKHR
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-dst-11584
  If the serialized acceleration structure in `src` is bottom-level, and any block in the header has a type of VK_ACCELERATION_STRUCTURE_SERIALIZED_BLOCK_TYPE_OPACITY_MICROMAP_KHR the `VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap` feature **must** be enabled
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-dst-11585
  If the serialized acceleration structure in `src` is bottom-level, every device address in each block in the header with type VK_ACCELERATION_STRUCTURE_SERIALIZED_BLOCK_TYPE_OPACITY_MICROMAP_KHR **must** point to a valid VkDeviceAddress that was retrieved by vkGetBufferDeviceAddressKHR for the underlying buffer objects of compatible micromaps constructed on the VkDevice when the command is executed

Valid Usage (Implicit)

- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COPY_MEMORY_TO_ACCELERATION_STRUCTURE_INFO_KHR

- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-dst-parameter
  `dst` **must** be a valid VkAccelerationStructureKHR handle
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-mode-parameter
  `mode` **must** be a valid VkCopyAccelerationStructureModeKHR value

To check if a serialized acceleration structure is compatible with the current device call:

```
// Provided by VK_KHR_acceleration_structure
void vkGetDeviceAccelerationStructureCompatibilityKHR(
    VkDevice                                    device,
    const VkAccelerationStructureVersionInfoKHR* pVersionInfo,
    VkAccelerationStructureCompatibilityKHR*    pCompatibility);
```

- `device` is the device to check the version against.
- `pVersionInfo` is a pointer to a VkAccelerationStructureVersionInfoKHR structure specifying version information to check against the device.
- `pCompatibility` is a pointer to a VkAccelerationStructureCompatibilityKHR value in which compatibility information is returned.

Valid Usage

- VUID-vkGetDeviceAccelerationStructureCompatibilityKHR-accelerationStructure-08928
  The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetDeviceAccelerationStructureCompatibilityKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDeviceAccelerationStructureCompatibilityKHR-pVersionInfo-parameter
  `pVersionInfo` **must** be a valid pointer to a valid VkAccelerationStructureVersionInfoKHR structure
- VUID-vkGetDeviceAccelerationStructureCompatibilityKHR-pCompatibility-parameter
  `pCompatibility` **must** be a valid pointer to a VkAccelerationStructureCompatibilityKHR value

The `VkAccelerationStructureVersionInfoKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureVersionInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    const uint8_t*     pVersionData;
} VkAccelerationStructureVersionInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pVersionData` is a pointer to the version header of an acceleration structure as defined in vkCmdCopyAccelerationStructureToMemoryKHR.

NOTE

`pVersionData` is a *pointer* to an array of 2×VK_UUID_SIZE `uint8_t` values instead of two VK_UUID_SIZE arrays as the expected use case for this member is to be pointed at the header of a previously serialized acceleration structure (via vkCmdCopyAccelerationStructureToMemoryKHR or vkCopyAccelerationStructureToMemoryKHR) that is loaded in memory. Using arrays would necessitate extra memory copies of the UUIDs.

Valid Usage (Implicit)

- VUID-VkAccelerationStructureVersionInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_VERSION_INFO_KHR
- VUID-VkAccelerationStructureVersionInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkAccelerationStructureVersionInfoKHR-pVersionData-parameter
  `pVersionData` **must** be a valid pointer to an array of $2 \times VK_UUID_SIZE$ `uint8_t` values

Possible values of `pCompatibility` returned by vkGetDeviceAccelerationStructureCompatibilityKHR are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkAccelerationStructureCompatibilityKHR {
    VK_ACCELERATION_STRUCTURE_COMPATIBILITY_COMPATIBLE_KHR = 0,
    VK_ACCELERATION_STRUCTURE_COMPATIBILITY_INCOMPATIBLE_KHR = 1,
} VkAccelerationStructureCompatibilityKHR;
```

- VK_ACCELERATION_STRUCTURE_COMPATIBILITY_COMPATIBLE_KHR if the `pVersionData` version acceleration structure is compatible with `device`.
- VK_ACCELERATION_STRUCTURE_COMPATIBILITY_INCOMPATIBLE_KHR if the `pVersionData` version acceleration structure is not compatible with `device`.

40.2. Host Acceleration Structure Operations

Implementations are also required to provide host implementations of the acceleration structure operations if the `accelerationStructureHostCommands` feature is enabled:

- vkBuildAccelerationStructuresKHR corresponding to vkCmdBuildAccelerationStructuresKHR
- vkCopyAccelerationStructureKHR corresponding to vkCmdCopyAccelerationStructureKHR
- vkCopyAccelerationStructureToMemoryKHR corresponding to vkCmdCopyAccelerationStructureToMemoryKHR
- vkCopyMemoryToAccelerationStructureKHR corresponding to vkCmdCopyMemoryToAccelerationStructureKHR
- vkWriteAccelerationStructuresPropertiesKHR corresponding to vkCmdWriteAccelerationStructuresPropertiesKHR

These commands are functionally equivalent to their device counterparts, except that they are executed on the host timeline, rather than being enqueued into command buffers.

All acceleration structures used by the host commands **must** be bound to host-visible memory, and all input data for acceleration structure builds **must** be referenced using host addresses instead of device addresses. Applications are not required to map acceleration structure memory when using the host commands.

NOTE

The vkBuildAccelerationStructuresKHR and vkCmdBuildAccelerationStructuresKHR **may** use different algorithms, and thus are not required to produce identical structures. The structures produced by these two commands **may** exhibit different memory footprints or traversal performance, but should strive to be similar where possible.

Apart from these details, the host and device operations are interchangeable. For example, an application **can** use vkBuildAccelerationStructuresKHR to build a structure, compact it on the device using vkCmdCopyAccelerationStructureKHR, and serialize the result using vkCopyAccelerationStructureToMemoryKHR.

The exception to this are acceleration structures that are created with the vkCreateAccelerationStructure2KHR command as they are backed by a VkDeviceAddress and therefore are not valid to be used with host commands.

NOTE

For efficient execution, acceleration structures manipulated using these commands should always be bound to host cached memory, as the implementation may need to repeatedly read and write this memory during the execution of the command.

To build acceleration structures on the host, call:

WARNING

This functionality is superseded by vkCmdBuildAccelerationStructuresKHR. See Legacy Functionality for more information.

```
// Provided by VK_KHR_acceleration_structure
VkResult vkBuildAccelerationStructuresKHR(
    VkDevice                                    device,
    VkDeferredOperationKHR                      deferredOperation,
    uint32_t                                    infoCount,
    const VkAccelerationStructureBuildGeometryInfoKHR* pInfos,
    const VkAccelerationStructureBuildRangeInfoKHR* const* ppBuildRangeInfos);
```

- `device` is the `VkDevice` for which the acceleration structures are being built.
- `deferredOperation` is an optional VkDeferredOperationKHR to request deferral for this command.
- `infoCount` is the number of acceleration structures to build. It specifies the number of the `pInfos` structures and `ppBuildRangeInfos` pointers that **must** be provided.
- `pInfos` is a pointer to an array of `infoCount` VkAccelerationStructureBuildGeometryInfoKHR structures defining the geometry used to build each acceleration structure.
- `ppBuildRangeInfos` is a pointer to an array of `infoCount` pointers to arrays of VkAccelerationStructureBuildRangeInfoKHR structures. Each `ppBuildRangeInfos`[i] is a pointer to an array of `pInfos`[i].`geometryCount` VkAccelerationStructureBuildRangeInfoKHR structures defining dynamic offsets to the addresses where geometry data is stored, as defined by `pInfos`[i].

This command fulfills the same task as vkCmdBuildAccelerationStructuresKHR but is executed by the host.

The `vkBuildAccelerationStructuresKHR` command provides the ability to initiate multiple acceleration structure builds, however there is no ordering or synchronization implied between any of the individual acceleration structure builds.

NOTE This means that an application **cannot** build a top-level acceleration structure in the same vkBuildAccelerationStructuresKHR call as the associated bottom-level or instance acceleration structures are being built. There also **cannot** be any memory aliasing between any acceleration structure memories or scratch memories being used by any of the builds.

Valid Usage

- VUID-vkBuildAccelerationStructuresKHR-accelerationStructureHostCommands-03581
  The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructureHostCommands` feature **must** be enabled
- VUID-vkBuildAccelerationStructuresKHR-mode-04628
  The `mode` member of each element of `pInfos` **must** be a valid VkBuildAccelerationStructureModeKHR value
- VUID-vkBuildAccelerationStructuresKHR-srcAccelerationStructure-04629
  If the `srcAccelerationStructure` member of any element of `pInfos` is not VK_NULL_HANDLE, the `srcAccelerationStructure` member **must** be a valid

VkAccelerationStructureKHR handle

- VUID-vkBuildAccelerationStructuresKHR-pInfos-04630
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its srcAccelerationStructure member **must** not be VK_NULL_HANDLE
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03708
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, srcAccelerationStructure **must** be fully backed by physical memory
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03403
  The srcAccelerationStructure member of any element of pInfos **must** not be the same acceleration structure as the dstAccelerationStructure member of any other element of pInfos
- VUID-vkBuildAccelerationStructuresKHR-dstAccelerationStructure-03698
  The dstAccelerationStructure member of any element of pInfos **must** not be the same acceleration structure as the dstAccelerationStructure member of any other element of pInfos
- VUID-vkBuildAccelerationStructuresKHR-dstAccelerationStructure-03800
  The dstAccelerationStructure member of any element of pInfos **must** be a valid VkAccelerationStructureKHR handle
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03707
  For each element of pInfos, its dstAccelerationStructure member **must** be fully backed by physical memory
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03699
  For each element of pInfos, if its type member is VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR, its dstAccelerationStructure member **must** have been created with a value of VkAccelerationStructureCreateInfoKHR::type equal to either VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR or VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03700
  For each element of pInfos, if its type member is VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR, its dstAccelerationStructure member **must** have been created with a value of VkAccelerationStructureCreateInfoKHR::type equal to either VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR or VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03663
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, inactive primitives in its srcAccelerationStructure member **must** not be made active
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03664
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, active primitives in its srcAccelerationStructure member **must** not be made inactive

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03709
  For each element of `pInfos`, if an acceleration structure is referenced by the `geometry.instances.data` member of any element of `pGeometries` or `ppGeometries` with a `geometryType` of VK_GEOMETRY_TYPE_INSTANCES_KHR, it **must** be fully backed by physical memory
- VUID-vkBuildAccelerationStructuresKHR-None-03407
  The `dstAccelerationStructure` member of any element of `pInfos` **must** not be referenced by the `geometry.instances.data` member of any element of `pGeometries` or `ppGeometries` with a `geometryType` of VK_GEOMETRY_TYPE_INSTANCES_KHR in any other element of `pInfos`
- VUID-vkBuildAccelerationStructuresKHR-dstAccelerationStructure-03701
  The range of memory backing the `dstAccelerationStructure` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `srcAccelerationStructure` member of any other element of `pInfos` with a `mode` equal to VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, which is accessed by this command
- VUID-vkBuildAccelerationStructuresKHR-dstAccelerationStructure-03702
  The range of memory backing the `dstAccelerationStructure` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `dstAccelerationStructure` member of any other element of `pInfos`, which is accessed by this command
- VUID-vkBuildAccelerationStructuresKHR-dstAccelerationStructure-03703
  The range of memory backing the `dstAccelerationStructure` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `scratchData` member of any element of `pInfos` (including the same element), which is accessed by this command
- VUID-vkBuildAccelerationStructuresKHR-scratchData-03704
  The range of memory backing the `scratchData` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `scratchData` member of any other element of `pInfos`, which is accessed by this command
- VUID-vkBuildAccelerationStructuresKHR-scratchData-03705
  The range of memory backing the `scratchData` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `srcAccelerationStructure` member of any element of `pInfos` with a `mode` equal to VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR (including the same element), which is accessed by this command
- VUID-vkBuildAccelerationStructuresKHR-dstAccelerationStructure-03706
  The range of memory backing the `dstAccelerationStructure` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing any acceleration structure referenced by the `geometry.instances.data` member of any element of `pGeometries` or `ppGeometries` with a `geometryType` of VK_GEOMETRY_TYPE_INSTANCES_KHR in any other element of `pInfos`, which is accessed by this command
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03667
  For each element of `pInfos`, if its `mode` member is

VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its srcAccelerationStructure member **must** have previously been constructed with VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR set in VkAccelerationStructureBuildGeometryInfoKHR::flags in the build

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03668
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its srcAccelerationStructure and dstAccelerationStructure members **must** either be the same VkAccelerationStructureKHR, or not have any memory aliasing
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03758
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its geometryCount member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03759
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its flags member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03760
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its type member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03761
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, its geometryType member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03762
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, its flags member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03763
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, its geometry.triangles.vertexFormat member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03764
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or

ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, its geometry.triangles.maxVertex member **must** have the same value which was specified when srcAccelerationStructure was last built

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03765
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, its geometry.triangles.indexType member **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03766
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, if its geometry.triangles.transformData address was NULL when srcAccelerationStructure was last built, then it **must** be NULL
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03767
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, if its geometry.triangles.transformData address was not NULL when srcAccelerationStructure was last built, then it **must** not be NULL
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03768
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, and geometry.triangles.indexType is not VK_INDEX_TYPE_NONE_KHR, then the value of each index referenced **must** be the same as the corresponding index value when srcAccelerationStructure was last built
- VUID-vkBuildAccelerationStructuresKHR-primitiveCount-03769
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, the primitiveCount member of its corresponding VkAccelerationStructureBuildRangeInfoKHR structure **must** have the same value which was specified when srcAccelerationStructure was last built
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03801
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, the corresponding ppBuildRangeInfos[i][j].primitiveCount **must** be less than or equal to VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxInstanceCount
- VUID-vkBuildAccelerationStructuresKHR-pInfos-10126

For each pInfos[i], dstAccelerationStructure **must** have been created with a value of VkAccelerationStructureCreateInfoKHR::size greater than or equal to either:

  - the memory size required by the build operation, as returned by vkGetAccelerationStructureBuildSizesKHR with pBuildInfo = pInfos[i] and with each element of the pMaxPrimitiveCounts array greater than or equal to the equivalent ppBuildRangeInfos[i][j].primitiveCount values for j in [0,pInfos[i].geometryCount) or,
  - the result of querying the corresponding VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR, if updating a compacted acceleration structure

- VUID-vkBuildAccelerationStructuresKHR-deferredOperation-03678
  Any previous deferred operation that was associated with deferredOperation **must** be complete
- VUID-vkBuildAccelerationStructuresKHR-pInfos-11702
  Each element of ppBuildRangeInfos[i] **must** be a valid pointer to an array of pInfos[i].geometryCount VkAccelerationStructureBuildRangeInfoKHR structures
- VUID-vkBuildAccelerationStructuresKHR-pInfos-11703
  For each element of pInfos, its dstAccelerationStructure member **must** have been created with vkCreateAccelerationStructureKHR
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03722
  For each element of pInfos, its dstAccelerationStructure member **must** be bound to host-visible device memory
- VUID-vkBuildAccelerationStructuresKHR-pInfos-11706
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, its srcAccelerationStructure member **must** have been created with vkCreateAccelerationStructureKHR
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03723
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR the buffer used to create its srcAccelerationStructure member **must** be bound to host-visible device memory
- VUID-vkBuildAccelerationStructuresKHR-pInfos-11704
  For each element of pInfos, the acceleration structure referenced by the geometry.instances.data member of any element of pGeometries or ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR **must** have been created with vkCreateAccelerationStructureKHR
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03724
  For each element of pInfos, each acceleration structure referenced by the geometry.instances.data member of any element of pGeometries or ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR **must** be bound to host-visible device memory
- VUID-vkBuildAccelerationStructuresKHR-pInfos-12244
  If pInfos[i].mode is VK_BUILD_ACCELERATION_STRUCTURE_MODE_BUILD_KHR, and N is not 0, all addresses between pInfos[i].scratchData.hostAddress and pInfos

[i].scratchData.hostAddress + N - 1 **must** be valid host memory, where N is given by the buildScratchSize member of the VkAccelerationStructureBuildSizesInfoKHR structure returned from a call to vkGetAccelerationStructureBuildSizesKHR with an identical VkAccelerationStructureBuildGeometryInfoKHR structure and primitive count

- VUID-vkBuildAccelerationStructuresKHR-pInfos-12245
  If pInfos[i].mode is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, and N is not 0, all addresses between pInfos[i].scratchData.hostAddress and pInfos[i].scratchData.hostAddress + N - 1 **must** be valid host memory, where N is given by the updateScratchSize member of the VkAccelerationStructureBuildSizesInfoKHR structure returned from a call to vkGetAccelerationStructureBuildSizesKHR with an identical VkAccelerationStructureBuildGeometryInfoKHR structure and primitive count
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03771
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, geometry.triangles.vertexData.hostAddress **must** be a valid host address
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03772
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if geometry.triangles.indexType is not VK_INDEX_TYPE_NONE_KHR, geometry.triangles.indexData.hostAddress **must** be a valid host address
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03773
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if geometry.triangles.transformData.hostAddress is not 0, it **must** be a valid host address
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03774
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_AABBS_KHR, geometry.aabbs.data.hostAddress **must** be a valid host address
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03775
  For each element of pInfos, its dstAccelerationStructure member **must** be bound to memory that was not allocated with multiple instances
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03776
  For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR the buffer used to create its srcAccelerationStructure member **must** be bound to memory that was not allocated with multiple instances
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03777
  For each element of pInfos, each acceleration structure referenced by the geometry.instances.data member of any element of pGeometries or ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR **must** be bound to memory that was not allocated with multiple instances
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03778
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, geometry.instances.data.hostAddress **must** be a

valid host address

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03779
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_INSTANCES_KHR, each VkAccelerationStructureInstanceKHR ::accelerationStructureReference value in geometry.instances.data.hostAddress **must** be a valid VkAccelerationStructureKHR object
- VUID-vkBuildAccelerationStructuresKHR-pInfos-10892
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if there is an instance of VkAccelerationStructureTrianglesOpacityMicromapEXT in the geometry.triangles.pNext chain, and its indexType is VK_INDEX_TYPE_NONE_KHR, then its indexBuffer.hostAddress **must** be 0
- VUID-vkBuildAccelerationStructuresKHR-pInfos-11821
  For each element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if there is an instance of VkAccelerationStructureTrianglesOpacityMicromapEXT in the geometry.triangles.pNext chain, and its indexType is not VK_INDEX_TYPE_NONE_KHR, then its indexBuffer.hostAddress **must** not be 0
- VUID-vkBuildAccelerationStructuresKHR-pInfos-11705
  For any element of pInfos[i].pGeometries or pInfos[i].ppGeometries with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR, if there is an instance of VkAccelerationStructureTrianglesOpacityMicromapKHR in the geometry.triangles.pNext chain its micromap member **must** be VK_NULL_HANDLE

Valid Usage (Implicit)

- VUID-vkBuildAccelerationStructuresKHR-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkBuildAccelerationStructuresKHR-deferredOperation-parameter
  If deferredOperation is not VK_NULL_HANDLE, deferredOperation **must** be a valid VkDeferredOperationKHR handle
- VUID-vkBuildAccelerationStructuresKHR-pInfos-parameter
  pInfos **must** be a valid pointer to an array of infoCount valid VkAccelerationStructureBuildGeometryInfoKHR structures
- VUID-vkBuildAccelerationStructuresKHR-ppBuildRangeInfos-parameter
  ppBuildRangeInfos **must** be a valid pointer to an array of infoCount VkAccelerationStructureBuildRangeInfoKHR structures
- VUID-vkBuildAccelerationStructuresKHR-infoCount-arraylength
  infoCount **must** be greater than 0
- VUID-vkBuildAccelerationStructuresKHR-deferredOperation-parent
  If deferredOperation is a valid handle, it **must** have been created, allocated, or retrieved from device

Return Codes

Success

- VK_OPERATION_DEFERRED_KHR
- VK_OPERATION_NOT_DEFERRED_KHR
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To copy or compact an acceleration structure on the host, call:

WARNING This functionality is superseded by vkCmdCopyAccelerationStructureKHR. See Legacy Functionality for more information.

```
// Provided by VK_KHR_acceleration_structure
VkResult vkCopyAccelerationStructureKHR(
    VkDevice                                    device,
    VkDeferredOperationKHR                      deferredOperation,
    const VkCopyAccelerationStructureInfoKHR*   pInfo);
```

- `device` is the device which owns the acceleration structures.
- `deferredOperation` is an optional VkDeferredOperationKHR to request deferral for this command.
- `pInfo` is a pointer to a VkCopyAccelerationStructureInfoKHR structure defining the copy operation.

This command fulfills the same task as vkCmdCopyAccelerationStructureKHR but is executed by the host.

Valid Usage

- VUID-vkCopyAccelerationStructureKHR-accelerationStructureHostCommands-03582
  The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructureHostCommands` feature **must** be enabled
- VUID-vkCopyAccelerationStructureKHR-deferredOperation-03678
  Any previous deferred operation that was associated with `deferredOperation` **must** be complete
- VUID-vkCopyAccelerationStructureKHR-buffer-03727

`pInfo->src` **must** be bound to host-visible device memory

- VUID-vkCopyAccelerationStructureKHR-buffer-03728
  `pInfo->dst` **must** be bound to host-visible device memory
- VUID-vkCopyAccelerationStructureKHR-buffer-03780
  `pInfo->src` **must** be bound to memory that was not allocated with multiple instances
- VUID-vkCopyAccelerationStructureKHR-buffer-03781
  `pInfo->dst` **must** be bound to memory that was not allocated with multiple instances
- VUID-vkCopyAccelerationStructureKHR-src-11586
  The source acceleration structure `pInfo->src` **must** have been constructed prior to the execution of this command
- VUID-vkCopyAccelerationStructureKHR-src-11587
  If the source acceleration structure `pInfo->src` was constructed through deserialization, all micromap arrays the acceleration structure references that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command
- VUID-vkCopyAccelerationStructureKHR-src-11588
  `pInfo->src` **must** have been created with vkCreateAccelerationStructureKHR
- VUID-vkCopyAccelerationStructureKHR-dst-11589
  `pInfo->dst` **must** have been created with vkCreateAccelerationStructureKHR

Valid Usage (Implicit)

- VUID-vkCopyAccelerationStructureKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCopyAccelerationStructureKHR-deferredOperation-parameter
  If `deferredOperation` is not VK_NULL_HANDLE, `deferredOperation` **must** be a valid VkDeferredOperationKHR handle
- VUID-vkCopyAccelerationStructureKHR-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkCopyAccelerationStructureInfoKHR structure
- VUID-vkCopyAccelerationStructureKHR-deferredOperation-parent
  If `deferredOperation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_OPERATION_DEFERRED_KHR
- VK_OPERATION_NOT_DEFERRED_KHR
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To copy host accessible memory to an acceleration structure, call:

WARNING	This functionality is superseded by vkCmdCopyMemoryToAccelerationStructureKHR. See Legacy Functionality for more information.

```
// Provided by VK_KHR_acceleration_structure
VkResult vkCopyMemoryToAccelerationStructureKHR(
    VkDevice                                    device,
    VkDeferredOperationKHR                      deferredOperation,
    const VkCopyMemoryToAccelerationStructureInfoKHR* pInfo);
```

- `device` is the device which owns `pInfo->dst`.
- `deferredOperation` is an optional VkDeferredOperationKHR to request deferral for this command.
- `pInfo` is a pointer to a VkCopyMemoryToAccelerationStructureInfoKHR structure defining the copy operation.

This command fulfills the same task as vkCmdCopyMemoryToAccelerationStructureKHR but is executed by the host.

This command can accept acceleration structures produced by either vkCmdCopyAccelerationStructureToMemoryKHR or vkCopyAccelerationStructureToMemoryKHR.

Valid Usage

- VUID-vkCopyMemoryToAccelerationStructureKHR-accelerationStructureHostCommands-03583
  The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructureHostCommands` feature **must** be enabled
- VUID-vkCopyMemoryToAccelerationStructureKHR-deferredOperation-03678
  Any previous deferred operation that was associated with `deferredOperation` **must** be complete
- VUID-vkCopyMemoryToAccelerationStructureKHR-pInfo-03729
  `pInfo->src.hostAddress` **must** be a valid host pointer
- VUID-vkCopyMemoryToAccelerationStructureKHR-pInfo-03750
  `pInfo->src.hostAddress` **must** be aligned to 16 bytes

- VUID-vkCopyMemoryToAccelerationStructureKHR-buffer-03730
  `pInfo->dst` **must** be bound to host-visible device memory
- VUID-vkCopyMemoryToAccelerationStructureKHR-buffer-03782
  `pInfo->dst` **must** be bound to memory that was not allocated with multiple instances
- VUID-vkCopyMemoryToAccelerationStructureKHR-dst-11677
  `pInfo->dst` **must** have been created with vkCreateAccelerationStructureKHR

Valid Usage (Implicit)

- VUID-vkCopyMemoryToAccelerationStructureKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCopyMemoryToAccelerationStructureKHR-deferredOperation-parameter
  If `deferredOperation` is not VK_NULL_HANDLE, `deferredOperation` **must** be a valid VkDeferredOperationKHR handle
- VUID-vkCopyMemoryToAccelerationStructureKHR-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkCopyMemoryToAccelerationStructureInfoKHR structure
- VUID-vkCopyMemoryToAccelerationStructureKHR-deferredOperation-parent
  If `deferredOperation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_OPERATION_DEFERRED_KHR
- VK_OPERATION_NOT_DEFERRED_KHR
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To copy an acceleration structure to host accessible memory, call:

WARNING	This functionality is superseded by vkCmdCopyAccelerationStructureToMemoryKHR. See Legacy Functionality for more information.

```
// Provided by VK_KHR_acceleration_structure
VkResult vkCopyAccelerationStructureToMemoryKHR(
    VkDevice                                    device,
    VkDeferredOperationKHR                      deferredOperation,
    const VkCopyAccelerationStructureToMemoryInfoKHR* pInfo);
```

- `device` is the device which owns `pInfo->src`.
- `deferredOperation` is an optional VkDeferredOperationKHR to request deferral for this command.
- `pInfo` is a pointer to a VkCopyAccelerationStructureToMemoryInfoKHR structure defining the copy operation.

This command fulfills the same task as vkCmdCopyAccelerationStructureToMemoryKHR but is executed by the host.

This command produces the same results as vkCmdCopyAccelerationStructureToMemoryKHR, but writes its result directly to a host pointer, and is executed on the host rather than the device. The output **may** not necessarily be bit-for-bit identical, but it can be equally used by either vkCmdCopyMemoryToAccelerationStructureKHR or vkCopyMemoryToAccelerationStructureKHR.

Valid Usage

- VUID-vkCopyAccelerationStructureToMemoryKHR-accelerationStructureHostCommands-03584
  The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructureHostCommands` feature **must** be enabled
- VUID-vkCopyAccelerationStructureToMemoryKHR-deferredOperation-03678
  Any previous deferred operation that was associated with `deferredOperation` **must** be complete
- VUID-vkCopyAccelerationStructureToMemoryKHR-pInfo-03731
  `pInfo->src` **must** be bound to host-visible device memory
- VUID-vkCopyAccelerationStructureToMemoryKHR-pInfo-03732
  `pInfo->dst.hostAddress` **must** be a valid host pointer
- VUID-vkCopyAccelerationStructureToMemoryKHR-pInfo-03751
  `pInfo->dst.hostAddress` **must** be aligned to 16 bytes
- VUID-vkCopyAccelerationStructureToMemoryKHR-buffer-03783
  `pInfo->src` **must** not be bound to memory that was allocated with multiple instances
- VUID-vkCopyAccelerationStructureToMemoryKHR-src-11678
  `pInfo->src` **must** have been created with vkCreateAccelerationStructureKHR

Valid Usage (Implicit)

- VUID-vkCopyAccelerationStructureToMemoryKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCopyAccelerationStructureToMemoryKHR-deferredOperation-parameter
  If `deferredOperation` is not VK_NULL_HANDLE, `deferredOperation` **must** be a valid VkDeferredOperationKHR handle
- VUID-vkCopyAccelerationStructureToMemoryKHR-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkCopyAccelerationStructureToMemoryInfoKHR structure
- VUID-vkCopyAccelerationStructureToMemoryKHR-deferredOperation-parent
  If `deferredOperation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_OPERATION_DEFERRED_KHR
- VK_OPERATION_NOT_DEFERRED_KHR
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To query acceleration structure size parameters on the host, call:

WARNING	This functionality is superseded by vkCmdWriteAccelerationStructuresPropertiesKHR. See Legacy Functionality for more information.

```
// Provided by VK_KHR_acceleration_structure
VkResult vkWriteAccelerationStructuresPropertiesKHR(
    VkDevice                                    device,
    uint32_t                                    accelerationStructureCount,
    const VkAccelerationStructureKHR*           pAccelerationStructures,
    VkQueryType                                 queryType,
    size_t                                      dataSize,
    void*                                       pData,
    size_t                                      stride);
```

- `device` is the device which owns the acceleration structures in `pAccelerationStructures`.
- `accelerationStructureCount` is the count of acceleration structures for which to query the property.
- `pAccelerationStructures` is a pointer to an array of existing previously built acceleration structures.
- `queryType` is a VkQueryType value specifying the property to be queried.
- `dataSize` is the size in bytes of the buffer pointed to by `pData`.
- `pData` is a pointer to an application-allocated buffer where the results will be written.
- `stride` is the stride in bytes between results for individual queries within `pData`.

This command fulfills the same task as vkCmdWriteAccelerationStructuresPropertiesKHR but is executed by the host.

Valid Usage

- VUID-vkWriteAccelerationStructuresPropertiesKHR-accelerationStructureHostCommands-03585
  The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructureHostCommands` feature **must** be enabled
- VUID-vkWriteAccelerationStructuresPropertiesKHR-accelerationStructures-03431
  All acceleration structures in `pAccelerationStructures` **must** have been built with VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_COMPACTION_BIT_KHR if `queryType` is VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-06742
  `queryType` **must** be VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR, VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR, VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR, or VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR
- VUID-vkWriteAccelerationStructuresPropertiesKHR-pAccelerationStructures-12425
  All acceleration structures in `pAccelerationStructures` **must** have been built with VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR if `queryType` is VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR
- VUID-vkWriteAccelerationStructuresPropertiesKHR-buffer-03736
  Each element of `pAccelerationStructures` **must** be fully backed by physical memory
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-03448
  If `queryType` is VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR, then `stride` **must** be a multiple of the size of VkDeviceSize
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-03449
  If `queryType` is VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR, then `pData` **must** point to a VkDeviceSize
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-03450

If `queryType` is VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR, then `stride` **must** be a multiple of the size of VkDeviceSize

- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-03451
  If `queryType` is VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR, then `pData` **must** point to a VkDeviceSize
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-06731
  If `queryType` is VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR, then `stride` **must** be a multiple of the size of VkDeviceSize
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-06732
  If `queryType` is VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR, then `pData` **must** point to a VkDeviceSize
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-06733
  If `queryType` is VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR, then `stride` **must** be a multiple of the size of VkDeviceSize
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-06734
  If `queryType` is VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR, then `pData` **must** point to a VkDeviceSize
- VUID-vkWriteAccelerationStructuresPropertiesKHR-dataSize-03452
  `dataSize` **must** be greater than or equal to `accelerationStructureCount`*`stride`
- VUID-vkWriteAccelerationStructuresPropertiesKHR-buffer-03733
  Each element of `pAccelerationStructures` **must** be bound to host-visible device memory
- VUID-vkWriteAccelerationStructuresPropertiesKHR-pAccelerationStructures-11592
  Each acceleration structure in `pAccelerationStructures` **must** have been created with vkCreateAccelerationStructureKHR
- VUID-vkWriteAccelerationStructuresPropertiesKHR-buffer-03784
  Each element of `pAccelerationStructures` **must** be bound to memory that was not allocated with multiple instances
- VUID-vkWriteAccelerationStructuresPropertiesKHR-pAccelerationStructures-11591
  All acceleration structures in `pAccelerationStructures` **must** have been constructed prior to the execution of this command

Valid Usage (Implicit)

- VUID-vkWriteAccelerationStructuresPropertiesKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkWriteAccelerationStructuresPropertiesKHR-pAccelerationStructures-parameter
  `pAccelerationStructures` **must** be a valid pointer to an array of `accelerationStructureCount` valid VkAccelerationStructureKHR handles
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-parameter
  `queryType` **must** be a valid VkQueryType value

- VUID-vkWriteAccelerationStructuresPropertiesKHR-pData-parameter
  `pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-vkWriteAccelerationStructuresPropertiesKHR-accelerationStructureCount-arraylength
  `accelerationStructureCount` **must** be greater than `0`
- VUID-vkWriteAccelerationStructuresPropertiesKHR-dataSize-arraylength
  `dataSize` **must** be greater than `0`
- VUID-vkWriteAccelerationStructuresPropertiesKHR-pAccelerationStructures-parent
  Each element of `pAccelerationStructures` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Chapter 41. Micromap

41.1. Micromaps

Acceleration structures store and organize geometry for ray tracing, but in some cases it is beneficial to include some information within the geometry, particularly for triangles. A *micromap* organizes this data around a map of values corresponding to subdivided microtriangles which can be added to a triangle geometry when building a *bottom level acceleration structure*.

An *opacity micromap* is a type of micromap which stores information to control intersection opacity as described in Ray Opacity Micromap.

A micromap is considered to be constructed if a micromap build command or copy command has been executed with the given micromap as the destination.

41.1.1. Building Micromaps

Micromaps created with `VK_KHR_opacity_micromap`, which are represented by VkAccelerationStructureKHR objects, are built with device build commands.

When building micromaps with vkCmdBuildAccelerationStructuresKHR, a `VkAccelerationStructureGeometryMicromapDataKHR` structure can be added to the `pNext` chain of VkAccelerationStructureGeometryKHR to specify the data used to build the micromap.

The `VkAccelerationStructureGeometryMicromapDataKHR` structure is defined as:

```
// Provided by VK_KHR_opacity_micromap
typedef struct VkAccelerationStructureGeometryMicromapDataKHR {
    VkStructureType                       sType;
    const void*                           pNext;
    uint32_t                              usageCountsCount;
    const VkMicromapUsageKHR*             pUsageCounts;
    const VkMicromapUsageKHR* const*      ppUsageCounts;
    VkDeviceAddress                       data;
    VkDeviceAddress                       triangleArray;
    VkDeviceSize                          triangleArrayStride;
} VkAccelerationStructureGeometryMicromapDataKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `usageCountsCount` specifies the number of usage counts structures that will be used to determine the size of this micromap.
- `pUsageCounts` is a pointer to an array of VkMicromapUsageKHR structures.
- `ppUsageCounts` is a pointer to an array of pointers to VkMicromapUsageKHR structures.
- `data` is the device address to memory which contains the data for the micromap.

- `triangleArray` is the device address to memory containing the VkMicromapTriangleKHR data.
- `triangleArrayStride` is the stride in bytes between each element of `triangleArray`.

Only one of `pUsageCounts` or `ppUsageCounts` **can** be a valid pointer, the other **must** be `NULL`. The elements of the non-`NULL` array describe the total counts used to build each micromap. Each element contains a `count` which is the number of micromap triangles of that `format` and `subdivisionLevel` contained in the micromap. Multiple elements with the same `format` and `subdivisionLevel` are allowed and the total count for that `format` and `subdivisionLevel` is the sum of the `count` for each element.

Each micromap triangle refers to one element in `triangleArray` which contains the `format` and `subdivisionLevel` for that particular triangle as well as a `dataOffset` in bytes which is the location relative to `data` where that triangle's micromap data begins. The data at `triangleArray` is laid out as a 4 byte unsigned integer for the `dataOffset` followed by a 2 byte unsigned integer for the subdivision level then a 2 byte unsigned integer for the format. In practice, compilers compile VkMicromapTriangleKHR to match this pattern.

For opacity micromaps, the data at `data` is packed as either one bit per element for VK_OPACITY_MICROMAP_FORMAT_2_STATE_KHR or two bits per element for VK_OPACITY_MICROMAP_FORMAT_4_STATE_KHR and is packed from LSB to MSB in each byte. The data at each index in those bytes is interpreted as discussed in Ray Opacity Micromap.

Valid Usage

- VUID-VkAccelerationStructureGeometryMicromapDataKHR-pUsageCounts-11642
  Only one of `pUsageCounts` or `ppUsageCounts` **can** be a valid pointer, the other **must** be `NULL`
- VUID-VkAccelerationStructureGeometryMicromapDataKHR-count-11643
  The total sum of `count` from all elements of `pUsageCounts` and `ppUsageCounts` **must** be less than or equal to `maxMicromapTriangles`
- VUID-VkAccelerationStructureGeometryMicromapDataKHR-subdivisionLevel-11645
  For each member of `pUsageCounts` or `ppUsageCounts`, if `format` is VK_OPACITY_MICROMAP_FORMAT_2_STATE_KHR, `subdivisionLevel` **must** be less than or equal to `maxOpacity2StateSubdivisionLevel`
- VUID-VkAccelerationStructureGeometryMicromapDataKHR-triangleArrayStride-11646
  `triangleArrayStride` **must** be greater than or equal to `8` bytes
- VUID-VkAccelerationStructureGeometryMicromapDataKHR-triangleArrayStride-11647
  `triangleArrayStride` **must** be a multiple of `4` bytes

Valid Usage (Implicit)

- VUID-VkAccelerationStructureGeometryMicromapDataKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_MICROMAP_DATA_KHR
- VUID-VkAccelerationStructureGeometryMicromapDataKHR-pUsageCounts-parameter
  If `usageCountsCount` is not `0`, and `pUsageCounts` is not `NULL`, `pUsageCounts` **must** be a valid

pointer to an array of usageCountsCount valid VkMicromapUsageKHR structures

- VUID-VkAccelerationStructureGeometryMicromapDataKHR-ppUsageCounts-parameter
  If usageCountsCount is not 0, and ppUsageCounts is not NULL, ppUsageCounts **must** be a valid pointer to an array of usageCountsCount valid pointers to valid VkMicromapUsageKHR structures

Structure Chaining

Extends the structure

- VkAccelerationStructureGeometryKHR

The VkMicromapUsageKHR structure is defined as:

```
// Provided by VK_KHR_opacity_micromap
typedef struct VkMicromapUsageKHR {
    uint32_t                      count;
    uint32_t                      subdivisionLevel;
    VkOpacityMicromapFormatKHR    format;
} VkMicromapUsageKHR;
```

- count is the number of triangles in the usage format defined by the subdivisionLevel and format below in the micromap
- subdivisionLevel is the subdivision level of this usage format
- format is the format of this usage format

The format is interpreted based on the type of the micromap using it.

Valid Usage (Implicit)

- VUID-VkMicromapUsageKHR-format-parameter
  format **must** be a valid VkOpacityMicromapFormatKHR value

The VkMicromapUsageEXT structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkMicromapUsageEXT {
    uint32_t    count;
    uint32_t    subdivisionLevel;
    uint32_t    format;
} VkMicromapUsageEXT;
```

- count is the number of triangles in the usage format defined by the subdivisionLevel and format below in the micromap

- `subdivisionLevel` is the subdivision level of this usage format
- `format` is the format of this usage format

The `format` is interpreted based on the `type` of the micromap using it.

The `VkMicromapTriangleKHR` structure is defined as:

```
// Provided by VK_KHR_opacity_micromap
typedef struct VkMicromapTriangleKHR {
    uint32_t    dataOffset;
    uint16_t    subdivisionLevel;
    uint16_t    format;
} VkMicromapTriangleKHR;
```

```
// Provided by VK_EXT_opacity_micromap
// Equivalent to VkMicromapTriangleKHR
typedef VkMicromapTriangleKHR VkMicromapTriangleEXT;
```

- `dataOffset` is the offset in bytes of the start of the data for this triangle. This is a byte aligned value.
- `subdivisionLevel` is the subdivision level of this triangle.
- `format` is the format of this triangle.

The `format` is interpreted based on the `type` of the micromap using it.

Formats which **can** be set in VkMicromapUsageKHR::`format` and VkMicromapTriangleKHR::`format` for micromap builds, are:

```
// Provided by VK_KHR_opacity_micromap
typedef enum VkOpacityMicromapFormatKHR {
    VK_OPACITY_MICROMAP_FORMAT_2_STATE_KHR = 1,
    VK_OPACITY_MICROMAP_FORMAT_4_STATE_KHR = 2,
  // Provided by VK_EXT_opacity_micromap
    VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT = VK_OPACITY_MICROMAP_FORMAT_2_STATE_KHR,
  // Provided by VK_EXT_opacity_micromap
    VK_OPACITY_MICROMAP_FORMAT_4_STATE_EXT = VK_OPACITY_MICROMAP_FORMAT_4_STATE_KHR,
} VkOpacityMicromapFormatKHR;
```

```
// Provided by VK_EXT_opacity_micromap
// Equivalent to VkOpacityMicromapFormatKHR
typedef VkOpacityMicromapFormatKHR VkOpacityMicromapFormatEXT;
```

- VK_OPACITY_MICROMAP_FORMAT_2_STATE_KHR specifies that the given micromap format has one bit per subtriangle encoding either fully opaque or fully transparent.

- VK_OPACITY_MICROMAP_FORMAT_4_STATE_KHR specifies that the given micromap format has two bits per subtriangle encoding four modes which can be interpreted as described in Ray Opacity Micromap.

> **NOTE** For compactness, these values are stored as 16-bit in some structures.

To build micromaps represented as a VkMicromapEXT object call:

```
// Provided by VK_EXT_opacity_micromap
void vkCmdBuildMicromapsEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    infoCount,
    const VkMicromapBuildInfoEXT*               pInfos);
```

- commandBuffer is the command buffer into which the command will be recorded.
- infoCount is the number of micromaps to build. It specifies the number of the pInfos structures that **must** be provided.
- pInfos is a pointer to an array of infoCount VkMicromapBuildInfoEXT structures defining the data used to build each micromap.

The vkCmdBuildMicromapsEXT command provides the ability to initiate multiple micromap builds, however there is no ordering or synchronization implied between any of the individual micromap builds.

> **NOTE** This means that there **cannot** be any memory aliasing between any micromap memories or scratch memories being used by any of the builds.

The input buffers passed to this command will be referenced by the implementation for the duration of the command's execution on the device. Micromaps **must** be fully self-contained. The application **can** reuse or free any memory which was used by this command's execution on the device as an input or as scratch without affecting the results of a subsequent acceleration structure build using the micromap or traversal of that acceleration structure, provided proper synchronization is used as described below.

Accesses to the micromap scratch buffers as identified by the VkMicromapBuildInfoEXT::scratchData buffer device addresses **must** be synchronized with the VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT pipeline stage and an access type of (VK_ACCESS_2_MICROMAP_READ_BIT_EXT | VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT). Accesses to VkMicromapBuildInfoEXT::dstMicromap **must** be synchronized with the VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT pipeline stage and an access type of VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT.

Accesses to other input buffers as identified by any used values of VkMicromapBuildInfoEXT::data or VkMicromapBuildInfoEXT::triangleArray **must** be synchronized with the VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT pipeline stage and an access type of VK_ACCESS_SHADER_READ_BIT.

Valid Usage

- VUID-vkCmdBuildMicromapsEXT-pInfos-07461
  For each `pInfos`[i], `dstMicromap` **must** have been created with a value of VkMicromapCreateInfoEXT::`size` greater than or equal to the memory size required by the build operation, as returned by vkGetMicromapBuildSizesEXT with `pBuildInfo` = `pInfos`[i]
- VUID-vkCmdBuildMicromapsEXT-mode-07462
  The `mode` member of each element of `pInfos` **must** be a valid VkBuildMicromapModeEXT value
- VUID-vkCmdBuildMicromapsEXT-dstMicromap-07463
  The `dstMicromap` member of any element of `pInfos` **must** be a valid VkMicromapEXT handle
- VUID-vkCmdBuildMicromapsEXT-pInfos-07464
  For each element of `pInfos` its `type` member **must** match the value of VkMicromapCreateInfoEXT::`type` when its `dstMicromap` was created
- VUID-vkCmdBuildMicromapsEXT-micromap-11648
  The `VkPhysicalDeviceOpacityMicromapFeaturesEXT`::`micromap` feature **must** be enabled
- VUID-vkCmdBuildMicromapsEXT-pInfos-07508
  For each element of `pInfos`, the `buffer` used to create its `dstMicromap` member **must** be bound to device memory
- VUID-vkCmdBuildMicromapsEXT-pInfos-07509
  If `pInfos`[i].`mode` is VK_BUILD_MICROMAP_MODE_BUILD_EXT, all addresses between `pInfos`[i].`scratchData.deviceAddress` and `pInfos`[i].`scratchData.deviceAddress` + N - 1 **must** be in the buffer device address range of the same buffer, where N is given by the `buildScratchSize` member of the VkMicromapBuildSizesInfoEXT structure returned from a call to vkGetMicromapBuildSizesEXT with an identical VkMicromapBuildInfoEXT structure and primitive count
- VUID-vkCmdBuildMicromapsEXT-data-07510
  The buffers from which the buffer device addresses for all of the `data` and `triangleArray` members of all `pInfos`[i] are queried **must** have been created with the VK_BUFFER_USAGE_MICROMAP_BUILD_INPUT_READ_ONLY_BIT_EXT usage flag set
- VUID-vkCmdBuildMicromapsEXT-pInfos-07511
  For each element of `pInfos`[i] the buffer from which the buffer device address `pInfos`[i].`scratchData.deviceAddress` is queried **must** have been created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdBuildMicromapsEXT-pInfos-07512
  For each element of `pInfos`, if the value of VkMicromapBuildSizesInfoEXT::`buildScratchSize`, returned from a call to vkGetMicromapBuildSizesEXT with an identical VkMicromapBuildInfoEXT structure, is not `0`, `scratchData.deviceAddress` **must** be a valid VkDeviceAddress
- VUID-vkCmdBuildMicromapsEXT-pInfos-10896
  For each element of `pInfos`, `data.deviceAddress` **must** be a valid VkDeviceAddress

- VUID-vkCmdBuildMicromapsEXT-pInfos-10897
  For each element of pInfos, triangleArray.deviceAddress **must** be a valid VkDeviceAddress
- VUID-vkCmdBuildMicromapsEXT-pInfos-07514
  For each element of pInfos, its scratchData.deviceAddress member **must** be a multiple of VkPhysicalDeviceAccelerationStructurePropertiesKHR::minAccelerationStructureScratchOffsetAlignment
- VUID-vkCmdBuildMicromapsEXT-pInfos-07515
  For each element of pInfos, its triangleArray.deviceAddress and data.deviceAddress members **must** be a multiple of 256
- VUID-vkCmdBuildMicromapsEXT-dstMicromap-11649
  The range of memory backing the dstMicromap member of any element of pInfos that is accessed by the execution of this command on the device **must** not overlap the memory backing the dstMicromap member of any other element of pInfos, which is accessed by the execution of this command on the device
- VUID-vkCmdBuildMicromapsEXT-dstMicromap-11650
  The range of memory backing the dstMicromap member of any element of pInfos that is accessed by the execution of this command on the device **must** not overlap the memory backing the scratchData member of any element of pInfos (including the same element), which is accessed by the execution of this command on the device
- VUID-vkCmdBuildMicromapsEXT-scratchData-11651
  The range of memory backing the scratchData member of any element of pInfos that is accessed by the execution of this command on the device **must** not overlap the memory backing the scratchData member of any other element of pInfos, which is accessed by the execution of this command on the device

Valid Usage (Implicit)

- VUID-vkCmdBuildMicromapsEXT-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBuildMicromapsEXT-pInfos-parameter
  pInfos **must** be a valid pointer to an array of infoCount valid VkMicromapBuildInfoEXT structures
- VUID-vkCmdBuildMicromapsEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdBuildMicromapsEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdBuildMicromapsEXT-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdBuildMicromapsEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdBuildMicromapsEXT-videocoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdBuildMicromapsEXT-infoCount-arraylength
  `infoCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdBuildMicromapsEXT is not affected by conditional rendering

The `VkMicromapBuildInfoEXT` structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkMicromapBuildInfoEXT {
    VkStructureType                       sType;
    const void*                           pNext;
    VkMicromapTypeEXT                     type;
    VkBuildMicromapFlagsEXT               flags;
    VkBuildMicromapModeEXT                mode;
    VkMicromapEXT                         dstMicromap;
    uint32_t                              usageCountsCount;
    const VkMicromapUsageEXT*             pUsageCounts;
    const VkMicromapUsageEXT* const*      ppUsageCounts;
    VkDeviceOrHostAddressConstKHR         data;
    VkDeviceOrHostAddressKHR              scratchData;
    VkDeviceOrHostAddressConstKHR         triangleArray;
    VkDeviceSize                          triangleArrayStride;
} VkMicromapBuildInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `type` is a VkMicromapTypeEXT value specifying the type of micromap being built.
- `flags` is a bitmask of VkBuildMicromapFlagBitsEXT specifying additional parameters of the micromap.
- `mode` is a VkBuildMicromapModeEXT value specifying the type of operation to perform.
- `dstMicromap` is a pointer to the target micromap for the build.
- `usageCountsCount` specifies the number of usage counts structures that will be used to determine the size of this micromap.
- `pUsageCounts` is a pointer to an array of VkMicromapUsageEXT structures.
- `ppUsageCounts` is a pointer to an array of pointers to VkMicromapUsageEXT structures.
- `data` is the device or host address of memory which contains the data for the micromap.
- `scratchData` is the device or host address of memory that will be used as scratch memory for the build.
- `triangleArray` is the device or host address of memory containing the VkMicromapTriangleEXT data.
- `triangleArrayStride` is the stride in bytes between each element of `triangleArray`.

Only one of `pUsageCounts` or `ppUsageCounts` **can** be a valid pointer, the other **must** be `NULL`. The elements of the non-`NULL` array describe the total counts used to build each micromap. Each element contains a `count` which is the number of micromap triangles of that `format` and `subdivisionLevel` contained in the micromap. Multiple elements with the same `format` and `subdivisionLevel` are allowed and the total count for that `format` and `subdivisionLevel` is the sum of the `count` for each element.

Each micromap triangle refers to one element in `triangleArray` which contains the `format` and `subdivisionLevel` for that particular triangle as well as a `dataOffset` in bytes which is the location relative to `data` where that triangle's micromap data begins. The data at `triangleArray` is laid out as a 4 byte unsigned integer for the `dataOffset` followed by a 2 byte unsigned integer for the subdivision level then a 2 byte unsigned integer for the format. In practice, compilers compile VkMicromapTriangleEXT to match this pattern.

For opacity micromaps, the data at `data` is packed as either one bit per element for VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT or two bits per element for VK_OPACITY_MICROMAP_FORMAT_4_STATE_EXT and is packed from LSB to MSB in each byte. The data at each index in those bytes is interpreted as discussed in Ray Opacity Micromap.

Valid Usage

- VUID-VkMicromapBuildInfoEXT-pUsageCounts-07516
  Only one of `pUsageCounts` or `ppUsageCounts` **can** be a valid pointer, the other **must** be `NULL`
- VUID-VkMicromapBuildInfoEXT-format-11652
  If `type` is VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT, for each member of `pUsageCounts` or `ppUsageCounts`, `format` **must** be a valid value from VkOpacityMicromapFormatEXT

- VUID-VkMicromapBuildInfoEXT-subdivisionLevel-11654
  If `type` is VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT, for each member of `pUsageCounts` or `ppUsageCounts`, if `format` is VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT, `subdivisionLevel` **must** be less than or equal to VkPhysicalDeviceOpacityMicromapPropertiesEXT::`maxOpacity2StateSubdivisionLevel`
- VUID-VkMicromapBuildInfoEXT-subdivisionLevel-11655
  If `type` is VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT, for each member of `pUsageCounts` or `ppUsageCounts`, if `format` is VK_OPACITY_MICROMAP_FORMAT_4_STATE_EXT, `subdivisionLevel` **must** be less than or equal to VkPhysicalDeviceOpacityMicromapPropertiesEXT::`maxOpacity4StateSubdivisionLevel`
- VUID-VkMicromapBuildInfoEXT-format-11657
  If `type` is VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT, for each member of `triangleArray`, `format` **must** be a valid value from VkOpacityMicromapFormatEXT
- VUID-VkMicromapBuildInfoEXT-subdivisionLevel-11659
  If `type` is VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT, for each member of `triangleArray`, if `format` is VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT, `subdivisionLevel` **must** be less than or equal to VkPhysicalDeviceOpacityMicromapPropertiesEXT::`maxOpacity2StateSubdivisionLevel`
- VUID-VkMicromapBuildInfoEXT-subdivisionLevel-11660
  If `type` is VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT, for each member of `triangleArray`, if `format` is VK_OPACITY_MICROMAP_FORMAT_4_STATE_EXT, `subdivisionLevel` **must** be less than or equal to VkPhysicalDeviceOpacityMicromapPropertiesEXT::`maxOpacity4StateSubdivisionLevel`

Valid Usage (Implicit)

- VUID-VkMicromapBuildInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MICROMAP_BUILD_INFO_EXT
- VUID-VkMicromapBuildInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkMicromapBuildInfoEXT-type-parameter
  `type` **must** be a valid VkMicromapTypeEXT value
- VUID-VkMicromapBuildInfoEXT-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkBuildMicromapFlagBitsEXT values
- VUID-VkMicromapBuildInfoEXT-pUsageCounts-parameter
  If `usageCountsCount` is not `0`, and `pUsageCounts` is not `NULL`, `pUsageCounts` **must** be a valid pointer to an array of `usageCountsCount` VkMicromapUsageEXT structures
- VUID-VkMicromapBuildInfoEXT-ppUsageCounts-parameter
  If `usageCountsCount` is not `0`, and `ppUsageCounts` is not `NULL`, `ppUsageCounts` **must** be a valid pointer to an array of `usageCountsCount` valid pointers to VkMicromapUsageEXT structures

Bits which **can** be set in VkMicromapBuildInfoEXT::`flags` specifying additional parameters for micromap builds, are:

```
// Provided by VK_EXT_opacity_micromap
typedef enum VkBuildMicromapFlagBitsEXT {
    VK_BUILD_MICROMAP_PREFER_FAST_TRACE_BIT_EXT = 0x00000001,
    VK_BUILD_MICROMAP_PREFER_FAST_BUILD_BIT_EXT = 0x00000002,
    VK_BUILD_MICROMAP_ALLOW_COMPACTION_BIT_EXT = 0x00000004,
} VkBuildMicromapFlagBitsEXT;
```

- VK_BUILD_MICROMAP_PREFER_FAST_TRACE_BIT_EXT specifies that the given micromap build **should** prioritize trace performance over build time.
- VK_BUILD_MICROMAP_PREFER_FAST_BUILD_BIT_EXT specifies that the given micromap build **should** prioritize build time over trace performance.
- VK_BUILD_MICROMAP_ALLOW_COMPACTION_BIT_EXT specifies that the micromap **can** act as the source for a copy micromap command with `mode` of VK_COPY_MICROMAP_MODE_COMPACT_EXT to produce a compacted micromap.

```
// Provided by VK_EXT_opacity_micromap
typedef VkFlags VkBuildMicromapFlagsEXT;
```

`VkBuildMicromapFlagsEXT` is a bitmask type for setting a mask of zero or more VkBuildMicromapFlagBitsEXT.

The VkBuildMicromapModeEXT enumeration is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef enum VkBuildMicromapModeEXT {
    VK_BUILD_MICROMAP_MODE_BUILD_EXT = 0,
} VkBuildMicromapModeEXT;
```

- VK_BUILD_MICROMAP_MODE_BUILD_EXT specifies that the destination micromap will be built using the specified data.

41.1.2. Copying Micromaps

Micromaps created with `VK_KHR_opacity_micromap`, which are represented by VkAccelerationStructureKHR objects, are copied with device copy commands.

For VkMicromapEXT, an additional command exists for copying micromaps without updating their contents. Before copying, an application **must** query the size of the resulting micromap.

To query VkMicromapEXT size parameters call:

```
// Provided by VK_EXT_opacity_micromap
void vkCmdWriteMicromapsPropertiesEXT(
    VkCommandBuffer                             commandBuffer,
    uint32_t                                    micromapCount,
    const VkMicromapEXT*                        pMicromaps,
    VkQueryType                                 queryType,
    VkQueryPool                                 queryPool,
    uint32_t                                    firstQuery);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `micromapCount` is the count of micromaps for which to query the property.
- `pMicromaps` is a pointer to an array of existing previously built micromaps.
- `queryType` is a VkQueryType value specifying the type of queries managed by the pool.
- `queryPool` is the query pool that will manage the results of the query.
- `firstQuery` is the first query index within the query pool that will contain the `micromapCount` number of results.

Accesses to any of the micromaps listed in `pMicromaps` **must** be synchronized with the VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT pipeline stage and an access type of VK_ACCESS_2_MICROMAP_READ_BIT_EXT.

- If `queryType` is VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT, then the value written out is the number of bytes required by a serialized micromap.
- If `queryType` is VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT, then the value written out is the number of bytes required by a compacted micromap.

Valid Usage

- VUID-vkCmdWriteMicromapsPropertiesEXT-queryPool-07525
  `queryPool` **must** have been created with a `queryType` matching `queryType`
- VUID-vkCmdWriteMicromapsPropertiesEXT-None-12417
  All queries used by this command **must** be *unavailable*
- VUID-vkCmdWriteMicromapsPropertiesEXT-buffer-07527
  The `buffer` used to create each micromap in `pMicromaps` **must** be bound to device memory
- VUID-vkCmdWriteMicromapsPropertiesEXT-query-07528
  The sum of `firstQuery` plus `micromapCount` **must** be less than or equal to the number of queries in `queryPool`
- VUID-vkCmdWriteMicromapsPropertiesEXT-pMicromaps-11662
  All micromaps in `pMicromaps` **must** have been constructed prior to the execution of this command on the device
- VUID-vkCmdWriteMicromapsPropertiesEXT-pMicromaps-07502
  All micromaps in `pMicromaps` **must** have been constructed with VK_BUILD_MICROMAP_ALLOW_COMPACTION_BIT_EXT if `queryType` is

VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT

- VUID-vkCmdWriteMicromapsPropertiesEXT-queryType-07503
  `queryType` **must** be VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT, or VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT

Valid Usage (Implicit)

- VUID-vkCmdWriteMicromapsPropertiesEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdWriteMicromapsPropertiesEXT-pMicromaps-parameter
  `pMicromaps` **must** be a valid pointer to an array of `micromapCount` valid VkMicromapEXT handles
- VUID-vkCmdWriteMicromapsPropertiesEXT-queryPool-parameter
  `queryPool` **must** be a valid VkQueryPool handle
- VUID-vkCmdWriteMicromapsPropertiesEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdWriteMicromapsPropertiesEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdWriteMicromapsPropertiesEXT-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdWriteMicromapsPropertiesEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdWriteMicromapsPropertiesEXT-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdWriteMicromapsPropertiesEXT-micromapCount-arraylength
  `micromapCount` **must** be greater than `0`
- VUID-vkCmdWriteMicromapsPropertiesEXT-commonparent
  Each of `commandBuffer`, `queryPool`, and the elements of `pMicromaps` **must** have been created, allocated, or retrieved from the same VkDevice

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdWriteMicromapsPropertiesEXT is not affected by conditional rendering

To copy a VkMicromapEXT call:

```
// Provided by VK_EXT_opacity_micromap
void vkCmdCopyMicromapEXT(
    VkCommandBuffer                             commandBuffer,
    const VkCopyMicromapInfoEXT*                pInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pInfo` is a pointer to a VkCopyMicromapInfoEXT structure defining the copy operation.

This command copies the `pInfo->src` micromap to the `pInfo->dst` micromap in the manner specified by `pInfo->mode`.

Accesses to `pInfo->src` and `pInfo->dst` **must** be synchronized with the VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT pipeline stage and an access type of VK_ACCESS_2_MICROMAP_READ_BIT_EXT or VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT as appropriate.

Valid Usage

- VUID-vkCmdCopyMicromapEXT-buffer-07529
  The `buffer` used to create `pInfo->src` **must** be bound to device memory
- VUID-vkCmdCopyMicromapEXT-buffer-07530
  The `buffer` used to create `pInfo->dst` **must** be bound to device memory
- VUID-vkCmdCopyMicromapEXT-src-07532
  The source micromap `src` **must** have been constructed prior to the execution of this command on the device

Valid Usage (Implicit)

- VUID-vkCmdCopyMicromapEXT-commandBuffer-parameter

commandBuffer **must** be a valid VkCommandBuffer handle

- VUID-vkCmdCopyMicromapEXT-pInfo-parameter
  pInfo **must** be a valid pointer to a valid VkCopyMicromapInfoEXT structure
- VUID-vkCmdCopyMicromapEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdCopyMicromapEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdCopyMicromapEXT-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyMicromapEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyMicromapEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdCopyMicromapEXT is not affected by conditional rendering

The VkCopyMicromapInfoEXT structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkCopyMicromapInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkMicromapEXT            src;
    VkMicromapEXT            dst;
    VkCopyMicromapModeEXT    mode;
} VkCopyMicromapInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `src` is the source micromap for the copy.
- `dst` is the target micromap for the copy.
- `mode` is a VkCopyMicromapModeEXT value specifying additional operations to perform during the copy.

Valid Usage

- VUID-VkCopyMicromapInfoEXT-mode-07531
  `mode` **must** be VK_COPY_MICROMAP_MODE_COMPACT_EXT or VK_COPY_MICROMAP_MODE_CLONE_EXT
- VUID-VkCopyMicromapInfoEXT-mode-07533
  If `mode` is VK_COPY_MICROMAP_MODE_COMPACT_EXT, `src` **must** have been constructed with VK_BUILD_MICROMAP_ALLOW_COMPACTION_BIT_EXT in the build

Valid Usage (Implicit)

- VUID-VkCopyMicromapInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COPY_MICROMAP_INFO_EXT
- VUID-VkCopyMicromapInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkCopyMicromapInfoEXT-src-parameter
  `src` **must** be a valid VkMicromapEXT handle
- VUID-VkCopyMicromapInfoEXT-dst-parameter
  `dst` **must** be a valid VkMicromapEXT handle
- VUID-VkCopyMicromapInfoEXT-commonparent
  Both of `dst`, and `src` **must** have been created, allocated, or retrieved from the same VkDevice

Possible values of `mode` specifying additional operations to perform during the copy, are:

```
// Provided by VK_EXT_opacity_micromap
typedef enum VkCopyMicromapModeEXT {
    VK_COPY_MICROMAP_MODE_CLONE_EXT = 0,
    VK_COPY_MICROMAP_MODE_SERIALIZE_EXT = 1,
    VK_COPY_MICROMAP_MODE_DESERIALIZE_EXT = 2,
    VK_COPY_MICROMAP_MODE_COMPACT_EXT = 3,
} VkCopyMicromapModeEXT;
```

- VK_COPY_MICROMAP_MODE_CLONE_EXT creates a direct copy of the micromap specified in `src` into the one specified by `dst`. The `dst` micromap **must** have been created with the same parameters as `src`.
- VK_COPY_MICROMAP_MODE_SERIALIZE_EXT serializes the micromap to a semi-opaque format which can be reloaded on a compatible implementation.
- VK_COPY_MICROMAP_MODE_DESERIALIZE_EXT deserializes the semi-opaque serialization format in the buffer to the micromap.
- VK_COPY_MICROMAP_MODE_COMPACT_EXT creates a more compact version of a micromap `src` into `dst`. The micromap `dst` **must** have been created with a size at least as large as that returned by vkCmdWriteMicromapsPropertiesEXT after the build of the micromap specified by `src`.

To copy a VkMicromapEXT to device memory call:

```
// Provided by VK_EXT_opacity_micromap
void vkCmdCopyMicromapToMemoryEXT(
    VkCommandBuffer                             commandBuffer,
    const VkCopyMicromapToMemoryInfoEXT*        pInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pInfo` is a pointer to a VkCopyMicromapToMemoryInfoEXT structure defining the copy operation.

Accesses to `pInfo->src` **must** be synchronized with the VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT pipeline stage and an access type of VK_ACCESS_2_MICROMAP_READ_BIT_EXT. Accesses to the buffer indicated by `pInfo->dst` **must** be synchronized with the VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT pipeline stage and an access type of VK_ACCESS_TRANSFER_WRITE_BIT.

This command produces the same results as vkCopyMicromapToMemoryEXT, but writes its result to a device address, and is executed on the device rather than the host. The output **may** not necessarily be bit-for-bit identical, but is equally compatible for use with either vkCmdCopyMemoryToMicromapEXT or vkCopyMemoryToMicromapEXT.

The defined header structure for the serialized data consists of:

- VK_UUID_SIZE bytes of data matching `VkPhysicalDeviceIDProperties`::`driverUUID`
- VK_UUID_SIZE bytes of data identifying the compatibility for comparison using

vkGetDeviceMicromapCompatibilityEXT

The serialized data is written to the buffer (or read from the buffer) according to the host endianness.

Valid Usage

- VUID-vkCmdCopyMicromapToMemoryEXT-pInfo-07536
  `pInfo->dst.deviceAddress` **must** be a valid VkDeviceAddress
- VUID-vkCmdCopyMicromapToMemoryEXT-pInfo-07537
  `pInfo->dst.deviceAddress` **must** be aligned to `256` bytes
- VUID-vkCmdCopyMicromapToMemoryEXT-buffer-07539
  The `buffer` used to create `pInfo->src` **must** be bound to device memory

Valid Usage (Implicit)

- VUID-vkCmdCopyMicromapToMemoryEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyMicromapToMemoryEXT-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkCopyMicromapToMemoryInfoEXT structure
- VUID-vkCmdCopyMicromapToMemoryEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdCopyMicromapToMemoryEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdCopyMicromapToMemoryEXT-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyMicromapToMemoryEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyMicromapToMemoryEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdCopyMicromapToMemoryEXT is not affected by conditional rendering

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkCopyMicromapToMemoryInfoEXT {
    VkStructureType             sType;
    const void*                 pNext;
    VkMicromapEXT               src;
    VkDeviceOrHostAddressKHR    dst;
    VkCopyMicromapModeEXT       mode;
} VkCopyMicromapToMemoryInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `src` is the source micromap for the copy.
- `dst` is the device or host address of memory which is the target for the copy.
- `mode` is a VkCopyMicromapModeEXT value specifying additional operations to perform during the copy.

Valid Usage

- VUID-VkCopyMicromapToMemoryInfoEXT-src-07540
  The source micromap `src` **must** have been constructed prior to the execution of this command
- VUID-VkCopyMicromapToMemoryInfoEXT-dst-07541
  The memory pointed to by `dst` **must** be at least as large as the serialization size of `src`, as reported by vkWriteMicromapsPropertiesEXT or vkCmdWriteMicromapsPropertiesEXT with a query type of VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT
- VUID-VkCopyMicromapToMemoryInfoEXT-mode-07542
  `mode` **must** be VK_COPY_MICROMAP_MODE_SERIALIZE_EXT

Valid Usage (Implicit)

- VUID-VkCopyMicromapToMemoryInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COPY_MICROMAP_TO_MEMORY_INFO_EXT
- VUID-VkCopyMicromapToMemoryInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkCopyMicromapToMemoryInfoEXT-src-parameter
  `src` **must** be a valid VkMicromapEXT handle
- VUID-VkCopyMicromapToMemoryInfoEXT-mode-parameter
  `mode` **must** be a valid VkCopyMicromapModeEXT value

To copy device memory to a micromap call:

```
// Provided by VK_EXT_opacity_micromap
void vkCmdCopyMemoryToMicromapEXT(
    VkCommandBuffer                             commandBuffer,
    const VkCopyMemoryToMicromapInfoEXT*        pInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pInfo` is a pointer to a VkCopyMemoryToMicromapInfoEXT structure defining the copy operation.

Accesses to `pInfo->dst` **must** be synchronized with the VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT pipeline stage and an access type of VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT. Accesses to the buffer indicated by `pInfo->src` **must** be synchronized with the VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT pipeline stage and an access type of VK_ACCESS_TRANSFER_READ_BIT.

This command **can** accept micromaps produced by either vkCopyMicromapToMemoryEXT or vkCmdCopyMicromapToMemoryEXT.

Valid Usage

- VUID-vkCmdCopyMemoryToMicromapEXT-pInfo-07543
  `pInfo->src.deviceAddress` **must** be a valid VkDeviceAddress
- VUID-vkCmdCopyMemoryToMicromapEXT-pInfo-07544
  `pInfo->src.deviceAddress` **must** be aligned to `256` bytes
- VUID-vkCmdCopyMemoryToMicromapEXT-buffer-07546
  The `buffer` used to create `pInfo->dst` **must** be bound to device memory

Valid Usage (Implicit)

- VUID-vkCmdCopyMemoryToMicromapEXT-commandBuffer-parameter

commandBuffer **must** be a valid VkCommandBuffer handle

- VUID-vkCmdCopyMemoryToMicromapEXT-pInfo-parameter
  pInfo **must** be a valid pointer to a valid VkCopyMemoryToMicromapInfoEXT structure
- VUID-vkCmdCopyMemoryToMicromapEXT-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdCopyMemoryToMicromapEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdCopyMemoryToMicromapEXT-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyMemoryToMicromapEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdCopyMemoryToMicromapEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdCopyMemoryToMicromapEXT is not affected by conditional rendering

The VkCopyMemoryToMicromapInfoEXT structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkCopyMemoryToMicromapInfoEXT {
    VkStructureType                  sType;
    const void*                      pNext;
    VkDeviceOrHostAddressConstKHR    src;
    VkMicromapEXT                    dst;
    VkCopyMicromapModeEXT            mode;
} VkCopyMemoryToMicromapInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `src` is the device or host address of memory containing the source data for the copy.
- `dst` is the target micromap for the copy.
- `mode` is a VkCopyMicromapModeEXT value specifying additional operations to perform during the copy.

Valid Usage

- VUID-VkCopyMemoryToMicromapInfoEXT-src-07547
  The source memory pointed to by `src` **must** contain data previously serialized using vkCmdCopyMicromapToMemoryEXT
- VUID-VkCopyMemoryToMicromapInfoEXT-mode-07548
  `mode` **must** be VK_COPY_MICROMAP_MODE_DESERIALIZE_EXT
- VUID-VkCopyMemoryToMicromapInfoEXT-src-07549
  The data in `src` **must** have a format compatible with the destination physical device as returned by vkGetDeviceMicromapCompatibilityEXT
- VUID-VkCopyMemoryToMicromapInfoEXT-dst-07550
  `dst` **must** have been created with a `size` greater than or equal to that used to serialize the data in `src`

Valid Usage (Implicit)

- VUID-VkCopyMemoryToMicromapInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_COPY_MEMORY_TO_MICROMAP_INFO_EXT
- VUID-VkCopyMemoryToMicromapInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkCopyMemoryToMicromapInfoEXT-dst-parameter
  `dst` **must** be a valid VkMicromapEXT handle
- VUID-VkCopyMemoryToMicromapInfoEXT-mode-parameter
  `mode` **must** be a valid VkCopyMicromapModeEXT value

To check if a serialized VkMicromapEXT is compatible with the current device call:

```
// Provided by VK_EXT_opacity_micromap
void vkGetDeviceMicromapCompatibilityEXT(
    VkDevice                                    device,
    const VkMicromapVersionInfoEXT*             pVersionInfo,
    VkAccelerationStructureCompatibilityKHR*    pCompatibility);
```

- device is the device to check the version against.
- pVersionInfo is a pointer to a VkMicromapVersionInfoEXT structure specifying version information to check against the device.
- pCompatibility is a pointer to a VkAccelerationStructureCompatibilityKHR value in which compatibility information is returned.

Valid Usage

- VUID-vkGetDeviceMicromapCompatibilityEXT-micromap-11666
  The VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromap feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetDeviceMicromapCompatibilityEXT-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkGetDeviceMicromapCompatibilityEXT-pVersionInfo-parameter
  pVersionInfo **must** be a valid pointer to a valid VkMicromapVersionInfoEXT structure
- VUID-vkGetDeviceMicromapCompatibilityEXT-pCompatibility-parameter
  pCompatibility **must** be a valid pointer to a VkAccelerationStructureCompatibilityKHR value

The VkMicromapVersionInfoEXT structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkMicromapVersionInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    const uint8_t*     pVersionData;
} VkMicromapVersionInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pVersionData is a pointer to the version header of a micromap as defined in vkCmdCopyMicromapToMemoryEXT.

NOTE

`pVersionData` is a *pointer* to an array of 2×VK_UUID_SIZE `uint8_t` values instead of two VK_UUID_SIZE arrays as the expected use case for this member is to be pointed at the header of a previously serialized micromap that is loaded in memory. Using arrays would necessitate extra memory copies of the UUIDs.

Valid Usage (Implicit)

- VUID-VkMicromapVersionInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MICROMAP_VERSION_INFO_EXT
- VUID-VkMicromapVersionInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkMicromapVersionInfoEXT-pVersionData-parameter
  `pVersionData` **must** be a valid pointer to an array of $2 \times VK_UUID_SIZE$ `uint8_t` values

41.2. Host Micromap Operations

Implementations are also required to provide host implementations of the micromap operations if the `VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromapHostCommands` feature is enabled:

- vkBuildMicromapsEXT corresponding to vkCmdBuildMicromapsEXT
- vkCopyMicromapEXT corresponding to vkCmdCopyMicromapEXT
- vkCopyMicromapToMemoryEXT corresponding to vkCmdCopyMicromapToMemoryEXT
- vkCopyMemoryToMicromapEXT corresponding to vkCmdCopyMemoryToMicromapEXT
- vkWriteMicromapsPropertiesEXT corresponding to vkCmdWriteMicromapsPropertiesEXT

These commands are functionally equivalent to their device counterparts, except that they are executed on the host timeline, rather than being enqueued into command buffers.

All micromaps used by the host commands **must** be bound to host-visible memory, and all input data for micromap builds **must** be referenced using host addresses instead of device addresses. Applications are not required to map micromap memory when using the host commands.

NOTE

The vkBuildMicromapsEXT and vkCmdBuildMicromapsEXT **may** use different algorithms, and thus are not required to produce identical structures.

Apart from these details, the host and device operations are interchangeable.

NOTE

For efficient execution, micromaps manipulated using these commands should always be bound to host cached memory, as the implementation may need to repeatedly read and write this memory during the execution of the command.

To build VkMicromapEXT on the host, call:

WARNING

This functionality is superseded by vkCmdBuildMicromapsEXT. See Legacy

Functionality for more information.

```
// Provided by VK_EXT_opacity_micromap
VkResult vkBuildMicromapsEXT(
    VkDevice                                    device,
    VkDeferredOperationKHR                      deferredOperation,
    uint32_t                                    infoCount,
    const VkMicromapBuildInfoEXT*               pInfos);
```

- `device` is the `VkDevice` for which the micromaps are being built.
- `deferredOperation` is an optional VkDeferredOperationKHR to request deferral for this command.
- `infoCount` is the number of micromaps to build. It specifies the number of the `pInfos` that **must** be provided.
- `pInfos` is a pointer to an array of `infoCount` VkMicromapBuildInfoEXT structures defining the geometry used to build each micromap.

This command fulfills the same task as vkCmdBuildMicromapsEXT but is executed by the host.

The `vkBuildMicromapsEXT` command provides the ability to initiate multiple micromap builds, however there is no ordering or synchronization implied between any of the individual micromap builds.

NOTE This means that there **cannot** be any memory aliasing between any micromap memories or scratch memories being used by any of the builds.

The input buffers passed to this command will be referenced by the implementation for the duration of the command. Micromaps **must** be fully self-contained. The application **can** reuse or free any memory which was used by the command as an input or as scratch without affecting the results of a subsequent acceleration structure build using the micromap or traversal of that acceleration structure.

Valid Usage

- VUID-vkBuildMicromapsEXT-pInfos-07461
  For each `pInfos`[i], `dstMicromap` **must** have been created with a value of VkMicromapCreateInfoEXT::`size` greater than or equal to the memory size required by the build operation, as returned by vkGetMicromapBuildSizesEXT with `pBuildInfo` = `pInfos`[i]
- VUID-vkBuildMicromapsEXT-mode-07462
  The `mode` member of each element of `pInfos` **must** be a valid VkBuildMicromapModeEXT value
- VUID-vkBuildMicromapsEXT-dstMicromap-07463
  The `dstMicromap` member of any element of `pInfos` **must** be a valid VkMicromapEXT handle

- VUID-vkBuildMicromapsEXT-pInfos-07464
  For each element of pInfos its type member **must** match the value of VkMicromapCreateInfoEXT::type when its dstMicromap was created
- VUID-vkBuildMicromapsEXT-pInfos-07552
  For each element of pInfos, the buffer used to create its dstMicromap member **must** be bound to host-visible device memory
- VUID-vkBuildMicromapsEXT-pInfos-07553
  For each element of pInfos, all referenced addresses of pInfos[i].data.hostAddress **must** be valid host memory
- VUID-vkBuildMicromapsEXT-pInfos-07554
  For each element of pInfos, all referenced addresses of pInfos[i].triangleArray.hostAddress **must** be valid host memory
- VUID-vkBuildMicromapsEXT-micromapHostCommands-07555
  The VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromapHostCommands feature **must** be enabled
- VUID-vkBuildMicromapsEXT-pInfos-07556
  If pInfos[i].mode is VK_BUILD_MICROMAP_MODE_BUILD_EXT, and N is not 0, then all addresses between pInfos[i].scratchData.hostAddress and pInfos[i].scratchData.hostAddress + N - 1 **must** be valid host memory, where N is given by the value of VkMicromapBuildSizesInfoEXT::buildScratchSize returned from a call to vkGetMicromapBuildSizesEXT with an identical VkMicromapBuildInfoEXT structure
- VUID-vkBuildMicromapsEXT-pInfos-07557
  For each element of pInfos, the buffer used to create its dstMicromap member **must** be bound to memory that was not allocated with multiple instances
- VUID-vkBuildMicromapsEXT-dstMicromap-11667
  The range of memory backing the dstMicromap member of any element of pInfos that is accessed by this command **must** not overlap the memory backing the dstMicromap member of any other element of pInfos, which is accessed by this command
- VUID-vkBuildMicromapsEXT-dstMicromap-11668
  The range of memory backing the dstMicromap member of any element of pInfos that is accessed by this command **must** not overlap the memory backing the scratchData member of any element of pInfos (including the same element), which is accessed by this command
- VUID-vkBuildMicromapsEXT-scratchData-11669
  The range of memory backing the scratchData member of any element of pInfos that is accessed by this command **must** not overlap the memory backing the scratchData member of any other element of pInfos, which is accessed by this command

Valid Usage (Implicit)

- VUID-vkBuildMicromapsEXT-device-parameter
  device **must** be a valid VkDevice handle

- VUID-vkBuildMicromapsEXT-deferredOperation-parameter
  If `deferredOperation` is not VK_NULL_HANDLE, `deferredOperation` **must** be a valid VkDeferredOperationKHR handle
- VUID-vkBuildMicromapsEXT-pInfos-parameter
  `pInfos` **must** be a valid pointer to an array of `infoCount` valid VkMicromapBuildInfoEXT structures
- VUID-vkBuildMicromapsEXT-infoCount-arraylength
  `infoCount` **must** be greater than `0`
- VUID-vkBuildMicromapsEXT-deferredOperation-parent
  If `deferredOperation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_OPERATION_DEFERRED_KHR
- VK_OPERATION_NOT_DEFERRED_KHR
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To copy or compact a VkMicromapEXT on the host, call:

WARNING This functionality is superseded by vkCmdCopyMicromapEXT. See Legacy Functionality for more information.

```
// Provided by VK_EXT_opacity_micromap
VkResult vkCopyMicromapEXT(
    VkDevice                                    device,
    VkDeferredOperationKHR                      deferredOperation,
    const VkCopyMicromapInfoEXT*                pInfo);
```

- `device` is the device which owns the micromaps.
- `deferredOperation` is an optional VkDeferredOperationKHR to request deferral for this command.
- `pInfo` is a pointer to a VkCopyMicromapInfoEXT structure defining the copy operation.

This command fulfills the same task as vkCmdCopyMicromapEXT but is executed by the host.

Valid Usage

- VUID-vkCopyMicromapEXT-deferredOperation-03678
  Any previous deferred operation that was associated with `deferredOperation` **must** be complete
- VUID-vkCopyMicromapEXT-buffer-07558
  The `buffer` used to create `pInfo->src` **must** be bound to host-visible device memory
- VUID-vkCopyMicromapEXT-buffer-07559
  The `buffer` used to create `pInfo->dst` **must** be bound to host-visible device memory
- VUID-vkCopyMicromapEXT-micromapHostCommands-07560
  The `VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromapHostCommands` feature **must** be enabled
- VUID-vkCopyMicromapEXT-buffer-07561
  The `buffer` used to create `pInfo->src` **must** be bound to memory that was not allocated with multiple instances
- VUID-vkCopyMicromapEXT-buffer-07562
  The `buffer` used to create `pInfo->dst` **must** be bound to memory that was not allocated with multiple instances
- VUID-vkCopyMicromapEXT-src-11670
  The source micromap `src` **must** have been constructed prior to the execution of this command

Valid Usage (Implicit)

- VUID-vkCopyMicromapEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCopyMicromapEXT-deferredOperation-parameter
  If `deferredOperation` is not VK_NULL_HANDLE, `deferredOperation` **must** be a valid VkDeferredOperationKHR handle
- VUID-vkCopyMicromapEXT-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkCopyMicromapInfoEXT structure
- VUID-vkCopyMicromapEXT-deferredOperation-parent
  If `deferredOperation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_OPERATION_DEFERRED_KHR
- VK_OPERATION_NOT_DEFERRED_KHR

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To copy host accessible VkMicromapEXT to a micromap, call:

WARNING	This functionality is superseded by vkCmdCopyMemoryToMicromapEXT. See Legacy Functionality for more information.

```
// Provided by VK_EXT_opacity_micromap
VkResult vkCopyMemoryToMicromapEXT(
    VkDevice                                    device,
    VkDeferredOperationKHR                      deferredOperation,
    const VkCopyMemoryToMicromapInfoEXT*        pInfo);
```

- `device` is the device which owns `pInfo->dst`.
- `deferredOperation` is an optional VkDeferredOperationKHR to request deferral for this command.
- `pInfo` is a pointer to a VkCopyMemoryToMicromapInfoEXT structure defining the copy operation.

This command fulfills the same task as vkCmdCopyMemoryToMicromapEXT but is executed by the host.

This command **can** accept micromaps produced by either vkCmdCopyMicromapToMemoryEXT or vkCopyMicromapToMemoryEXT.

Valid Usage

- VUID-vkCopyMemoryToMicromapEXT-deferredOperation-03678
  Any previous deferred operation that was associated with `deferredOperation` **must** be complete
- VUID-vkCopyMemoryToMicromapEXT-pInfo-07563
  `pInfo->src.hostAddress` **must** be a valid host pointer
- VUID-vkCopyMemoryToMicromapEXT-pInfo-07564
  `pInfo->src.hostAddress` **must** be aligned to 16 bytes
- VUID-vkCopyMemoryToMicromapEXT-buffer-07565
  The `buffer` used to create `pInfo->dst` **must** be bound to host-visible device memory
- VUID-vkCopyMemoryToMicromapEXT-micromapHostCommands-07566

The VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromapHostCommands feature **must** be enabled

- VUID-vkCopyMemoryToMicromapEXT-buffer-07567
  The buffer used to create pInfo->dst **must** be bound to memory that was not allocated with multiple instances

Valid Usage (Implicit)

- VUID-vkCopyMemoryToMicromapEXT-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkCopyMemoryToMicromapEXT-deferredOperation-parameter
  If deferredOperation is not VK_NULL_HANDLE, deferredOperation **must** be a valid VkDeferredOperationKHR handle
- VUID-vkCopyMemoryToMicromapEXT-pInfo-parameter
  pInfo **must** be a valid pointer to a valid VkCopyMemoryToMicromapInfoEXT structure
- VUID-vkCopyMemoryToMicromapEXT-deferredOperation-parent
  If deferredOperation is a valid handle, it **must** have been created, allocated, or retrieved from device

Return Codes

Success

- VK_OPERATION_DEFERRED_KHR
- VK_OPERATION_NOT_DEFERRED_KHR
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To copy a VkMicromapEXT to host accessible memory, call:

WARNING	This functionality is superseded by vkCmdCopyMicromapToMemoryEXT. See Legacy Functionality for more information.

```
// Provided by VK_EXT_opacity_micromap
VkResult vkCopyMicromapToMemoryEXT(
    VkDevice                                    device,
    VkDeferredOperationKHR                      deferredOperation,
    const VkCopyMicromapToMemoryInfoEXT*        pInfo);
```

- `device` is the device which owns `pInfo->src`.
- `deferredOperation` is an optional VkDeferredOperationKHR to request deferral for this command.
- `pInfo` is a pointer to a VkCopyMicromapToMemoryInfoEXT structure defining the copy operation.

This command fulfills the same task as vkCmdCopyMicromapToMemoryEXT but is executed by the host.

This command produces the same results as vkCmdCopyMicromapToMemoryEXT, but writes its result directly to a host pointer, and is executed on the host rather than the device. The output **may** not necessarily be bit-for-bit identical, but is compatible for use with either vkCmdCopyMemoryToMicromapEXT or vkCopyMemoryToMicromapEXT.

Valid Usage

- VUID-vkCopyMicromapToMemoryEXT-deferredOperation-03678
  Any previous deferred operation that was associated with `deferredOperation` **must** be complete
- VUID-vkCopyMicromapToMemoryEXT-buffer-07568
  The `buffer` used to create `pInfo->src` **must** be bound to host-visible device memory
- VUID-vkCopyMicromapToMemoryEXT-pInfo-07569
  `pInfo->dst.hostAddress` **must** be a valid host pointer
- VUID-vkCopyMicromapToMemoryEXT-pInfo-07570
  `pInfo->dst.hostAddress` **must** be aligned to 16 bytes
- VUID-vkCopyMicromapToMemoryEXT-micromapHostCommands-07571
  The `VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromapHostCommands` feature **must** be enabled
- VUID-vkCopyMicromapToMemoryEXT-buffer-07572
  The `buffer` used to create `pInfo->src` **must** be bound to memory that was not allocated with multiple instances

Valid Usage (Implicit)

- VUID-vkCopyMicromapToMemoryEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCopyMicromapToMemoryEXT-deferredOperation-parameter

If `deferredOperation` is not VK_NULL_HANDLE, `deferredOperation` **must** be a valid VkDeferredOperationKHR handle

- VUID-vkCopyMicromapToMemoryEXT-pInfo-parameter
  `pInfo` **must** be a valid pointer to a valid VkCopyMicromapToMemoryInfoEXT structure
- VUID-vkCopyMicromapToMemoryEXT-deferredOperation-parent
  If `deferredOperation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_OPERATION_DEFERRED_KHR
- VK_OPERATION_NOT_DEFERRED_KHR
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

To query VkMicromapEXT size parameters on the host, call:

WARNING This functionality is superseded by vkCmdWriteMicromapsPropertiesEXT. See Legacy Functionality for more information.

```
// Provided by VK_EXT_opacity_micromap
VkResult vkWriteMicromapsPropertiesEXT(
    VkDevice                                    device,
    uint32_t                                    micromapCount,
    const VkMicromapEXT*                        pMicromaps,
    VkQueryType                                 queryType,
    size_t                                      dataSize,
    void*                                       pData,
    size_t                                      stride);
```

- `device` is the device which owns the micromaps in `pMicromaps`.
- `micromapCount` is the count of micromaps for which to query the property.
- `pMicromaps` is a pointer to an array of existing previously built micromaps.
- `queryType` is a VkQueryType value specifying the property to be queried.
- `dataSize` is the size in bytes of the buffer pointed to by `pData`.

- `pData` is a pointer to an application-allocated buffer where the results will be written.
- `stride` is the stride in bytes between results for individual queries within `pData`.

This command fulfills the same task as vkCmdWriteMicromapsPropertiesEXT but is executed by the host.

Valid Usage

- VUID-vkWriteMicromapsPropertiesEXT-pMicromaps-07502
  All micromaps in `pMicromaps` **must** have been constructed with VK_BUILD_MICROMAP_ALLOW_COMPACTION_BIT_EXT if `queryType` is VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT
- VUID-vkWriteMicromapsPropertiesEXT-queryType-07503
  `queryType` **must** be VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT, or VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT
- VUID-vkWriteMicromapsPropertiesEXT-queryType-10071
  If `queryType` is VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT or VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT then `stride` **must** be a multiple of the size of VkDeviceSize
- VUID-vkWriteMicromapsPropertiesEXT-queryType-10072
  If `queryType` is VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT or VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT then `pData` **must** point to a VkDeviceSize
- VUID-vkWriteMicromapsPropertiesEXT-dataSize-07576
  `dataSize` **must** be greater than or equal to `micromapCount`*`stride`
- VUID-vkWriteMicromapsPropertiesEXT-buffer-07577
  The `buffer` used to create each micromap in `pMicromaps` **must** be bound to host-visible device memory
- VUID-vkWriteMicromapsPropertiesEXT-micromapHostCommands-07578
  The `VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromapHostCommands` feature **must** be enabled
- VUID-vkWriteMicromapsPropertiesEXT-buffer-07579
  The `buffer` used to create each micromap in `pMicromaps` **must** be bound to memory that was not allocated with multiple instances
- VUID-vkWriteMicromapsPropertiesEXT-pMicromaps-11671
  All micromaps in `pMicromaps` **must** have been constructed prior to the execution of this command

Valid Usage (Implicit)

- VUID-vkWriteMicromapsPropertiesEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkWriteMicromapsPropertiesEXT-pMicromaps-parameter

`pMicromaps` **must** be a valid pointer to an array of `micromapCount` valid VkMicromapEXT handles

- VUID-vkWriteMicromapsPropertiesEXT-queryType-parameter
  `queryType` **must** be a valid VkQueryType value
- VUID-vkWriteMicromapsPropertiesEXT-pData-parameter
  `pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-vkWriteMicromapsPropertiesEXT-micromapCount-arraylength
  `micromapCount` **must** be greater than `0`
- VUID-vkWriteMicromapsPropertiesEXT-dataSize-arraylength
  `dataSize` **must** be greater than `0`
- VUID-vkWriteMicromapsPropertiesEXT-pMicromaps-parent
  Each element of `pMicromaps` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Chapter 42. Ray Traversal

The ray traversal process identifies and handles intersections between a ray and geometries in an acceleration structure.

Ray traversal cannot be started by a Vulkan API command directly - a shader **must** execute `OpRayQueryProceedKHR` or a pipeline trace ray instruction . When the `rayTracingPipeline` feature is enabled, `OpTraceRayKHR` **can** be used for ray tracing in a ray tracing pipeline. When the `rayQuery` feature is enabled, `OpRayQueryProceedKHR` **can** be used in any shader stage.

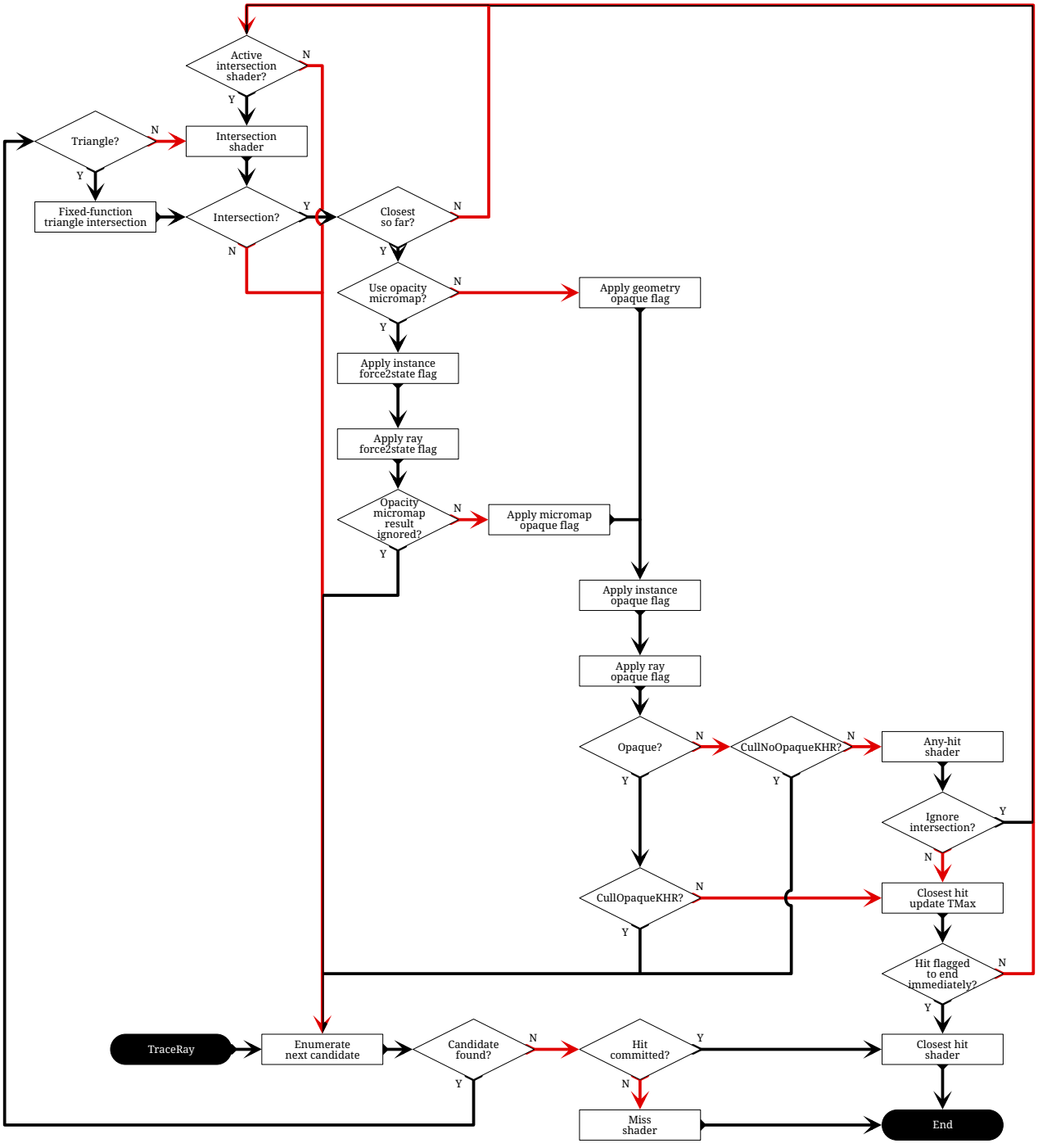


Figure 23. Ray tracing traversal

42.1. Ray Intersection Candidate Determination

Once tracing begins, rays are first tested against instances in a top-level acceleration structure. A ray that intersects an instance will be transformed into the space of the instance to continue traversal within that instance; therefore the transform matrix stored in the instance **must** be invertible.

In case multiple instances are intersected by a ray, the ray transformation into the space of the instance is invariant under the order in which these instances are encountered in the top-level acceleration structure.

NOTE	Applying multiple forward and reverse transforms to a ray to transition from one instance to another could result in accumulated errors. Thus an implementation should behave as if the ray is transformed from the origin for each instance independently.

Next, rays are tested against geometries in a bottom-level acceleration structure to determine if a hit occurred between them, initially based only on their geometric properties (i.e. their vertices). The implementation performs similar operations to that of rasterization, but with the effective viewport determined by the parameters of the ray, and the geometry transformed into a space determined by that viewport.

The vertices of each primitive are transformed from acceleration structure space `as` to ray space `r` according to the ray origin and direction as follows:

$$\begin{pmatrix} x_r \\ y_r \\ z_r \end{pmatrix} = \begin{pmatrix} a_x^2(1-c)+c & a_x a_y(1-c)-sa_z & a_x a_z(1-c)+sa_y \\ a_x a_y(1-c)+sa_z & a_y^2(1-c)+c & a_y a_z(1-c)-sa_x \\ a_x a_z(1-c)-sa_y & a_y a_z(1-c)+sa_x & a_z^2(1-c)+c \end{pmatrix} \begin{pmatrix} x_{as}-o_x \\ y_{as}-o_y \\ z_{as}-o_z \end{pmatrix}$$

$\mathbf{a}$ is the axis of rotation from the unnormalized ray direction vector $\mathbf{d}$ to the axis vector $\mathbf{k}$:

$$\mathbf{a} = \begin{cases} \dfrac{\mathbf{d}\times\mathbf{k}}{\|\mathbf{d}\times\mathbf{k}\|} & \text{if } \|\mathbf{d}\times\mathbf{k}\| \neq 0 \\ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} & \text{if } \|\mathbf{d}\times\mathbf{k}\| = 0 \end{cases}$$

s and c are the sine and cosine of the angle of rotation about $\mathbf{a}$ from $\mathbf{d}$ to $\mathbf{k}$:

$$c = \frac{\mathbf{d}\cdot\mathbf{k}}{\|\mathbf{d}\|}$$
$$s = \sqrt{1-c^2}$$

$\mathbf{k}$ is the unit vector:

$$\mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}$$

o and **d** are the ray origin and unnormalized direction, respectively; the vector described by x_{as}, y_{as}, and z_{as} is any position in acceleration structure space; and the vector described by x_r, y_r, and z_r is the same position in ray space.

An *intersection candidate* is a unique point of intersection between a ray and a geometric primitive. For any primitive that has within its bounds a position $\mathbf{xyz_{as}}$ such that

$$\begin{aligned} x_r &= 0 \\ y_r &= 0 \\ t_{min} < -\frac{z_r}{\|\mathbf{d}\|} &< t_{max} \qquad \text{if the primitive is a triangle,} \\ t_{min} \leq -\frac{z_r}{\|\mathbf{d}\|} &\leq t_{max} \qquad \text{otherwise} \end{aligned}$$

(where $t = -\frac{z_r}{\|\mathbf{d}\|}$), an intersection candidate exists.

Triangle primitive bounds consist of all points on the plane formed by the three vertices and within the bounds of the edges between the vertices, subject to the watertightness constraints below. AABB primitive bounds consist of all points within an implementation-defined bound which includes the specified box.

NOTE The bounds of the AABB including all points internal to the bound implies that a ray started within the AABB will hit that AABB.

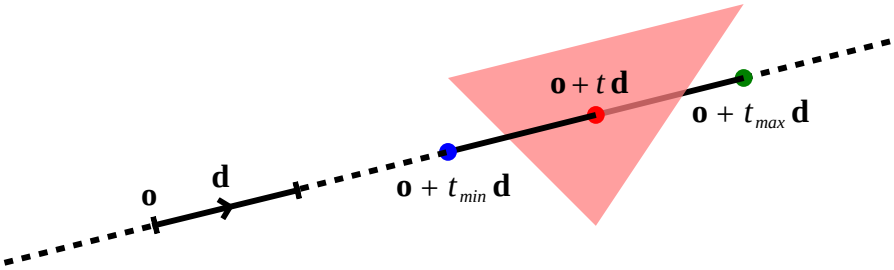


Figure 24. Ray intersection candidate

The determination of this condition is performed in an implementation specific manner, and **may** be performed with floating-point operations. Due to the complexity and number of operations involved, inaccuracies are expected, particularly as the scale of values involved begins to diverge. Implementations **should** take efforts to maintain as much precision as possible.

NOTE One very common case is when geometries are close to each other at some distance from the origin in acceleration structure space, where an effect similar to "z-fighting" is likely to be observed. Applications can mitigate this by ensuring their detailed geometries remain close to the origin.

Another likely case is when the origin of a ray is set to a position on a previously intersected surface, and its t_{min} is zero or near zero; an intersection may be detected

on the emitting surface. This case can usually be mitigated by offsetting t_{min} slightly.

In the case of AABB geometries, implementations **may** increase their size in an acceleration structure in order to mitigate precision issues. This **may** result in false positive intersections being reported to the application.

For triangle intersection candidates, the b and c barycentric coordinates on the triangle where the above condition is met are made available to future shading. If the ray was traced with a pipeline trace ray instruction, these values are available as a vector of 2 32-bit floating-point values in the `HitAttributeKHR` storage class. These values are included in the triangle intersection candidate.

Once an intersection candidate is determined, it proceeds through the following operations, in order:

1. Ray Intersection Culling
2. Ray Intersection Confirmation
3. Ray Closest Hit Determination
4. Ray Result Determination

The sections below describe the exact details of these tests. There is no ordering guarantee between operations performed on different intersection candidates.

42.1.1. Watertightness

For a set of triangles with identical transforms, within a single instance:

- Any set of two or more triangles where all triangles have one vertex with an identical position value, that vertex is a *shared vertex*.
- Any set of two triangles with two shared vertices that were specified in the same winding order in each triangle have a *shared edge* defined by those vertices.

A *closed fan* is a set of three or more triangles where:

- All triangles in the set have the same shared vertex as one of their vertices.
- All edges that include the above vertex are shared edges.
- All above shared edges are shared by exactly two triangles from the set.
- No two triangles in the set intersect, except at shared edges.
- Every triangle in the set is joined to every other triangle in the set by a series of the above shared edges.

Implementations **should** not double-hit or miss when a ray intersects a shared edge, or a shared vertex of a closed fan.

NOTE Because of the complexity of a definition of watertightness the language above does not list a **must** requirement but it is expected and tested that implementations will perform watertight intersection tests.

42.2. Ray Intersection Culling

Candidate intersections go through several phases of culling before confirmation as an actual hit. There is no particular ordering dependency between the different culling operations.

42.2.1. Ray Primitive Culling

If the `rayTraversalPrimitiveCulling` or `rayQuery` features are enabled, the `SkipTrianglesKHR` and `SkipAABBsKHR` ray flags **can** be specified when tracing a ray. `SkipTrianglesKHR` and `SkipAABBsKHR` are mutually exclusive. `SkipTrianglesKHR` is also mutually exclusive with `CullBackFacingTrianglesKHR` and `CullFrontFacingTrianglesKHR`.

If `SkipTrianglesKHR` was included in the `Ray Flags` operand of the ray trace instruction, and the intersection is with a triangle primitive, the intersection is dropped, and no further processing of this intersection occurs. If VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR was included in the pipeline, traversal with pipeline trace ray instructions will all behave as if `SkipTrianglesKHR` was included in their `Ray Flags` operand.

If `SkipAABBsKHR` was included in the `Ray Flags` operand of the ray trace instruction, and the intersection is with an AABB primitive, the intersection is dropped, and no further processing of this intersection occurs. If VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR was included in the pipeline, traversal with pipeline trace ray instructions will all behave as if `SkipAABBsKHR` was included in their `Ray Flags` operand.

42.2.2. Ray Mask Culling

Instances **can** be made invisible to particular rays based on the value of VkAccelerationStructureInstanceKHR::`mask` used to add that instance to a top-level acceleration structure, and the `Cull Mask` parameter used to trace the ray.

For the instance which is intersected, if `mask` & `Cull Mask` == 0, the intersection is dropped, and no further processing occurs.

42.2.3. Ray Face Culling

As in polygon rasterization, one of the stages of ray traversal is to determine if a triangle primitive is back- or front-facing, and primitives **can** be culled based on that facing.

If the intersection candidate is with an AABB primitive, this operation is skipped.

Determination

When a ray intersects a triangle primitive, the order that vertices are specified for the polygon affects whether the ray intersects the front or back face. Front or back facing is determined in the same way as they are for rasterization, based on the sign of the polygon's area but using the ray space coordinates instead of framebuffer coordinates. One way to compute this area is:

$$a = -\frac{1}{2}\sum_{i=0}^{n-1} x_r^i y_r^{i \oplus 1} - x_r^{i \oplus 1} y_r^i$$

where x_r^i and y_r^i are the x and y ray space coordinates of the ith vertex of the n-vertex polygon (vertices are numbered starting at zero for the purposes of this computation) and i ⊕ 1 is (i + 1) mod n.

By default, if a is negative then the intersection is with the front face of the triangle, otherwise it is with the back face. If VK_GEOMETRY_INSTANCE_TRIANGLE_FLIP_FACING_BIT_KHR is included in VkAccelerationStructureInstanceKHR::`flags` for the instance containing the intersected triangle, this determination is reversed. Additionally, if a is 0, the intersection candidate is treated as not intersecting with any face, irrespective of the sign.

NOTE

In a left-handed coordinate system, an intersection will be with the front face of a triangle if the vertices of the triangle, as defined in index order, appear from the ray's perspective in a clockwise rotation order. VK_GEOMETRY_INSTANCE_TRIANGLE_FLIP_FACING_BIT_KHR was previously annotated as VK_GEOMETRY_INSTANCE_TRIANGLE_FRONT_COUNTERCLOCKWISE_BIT_KHR because of this.

If the ray was traced with a pipeline trace ray instruction, the `HitKindKHR` built-in is set to `HitKindFrontFacingTriangleKHR` if the intersection is with front-facing geometry, and `HitKindBackFacingTriangleKHR` if the intersection is with back-facing geometry, for shader stages considering this intersection.

If the ray was traced with `OpRayQueryProceedKHR`, `OpRayQueryGetIntersectionFrontFaceKHR` will return true for intersection candidates with front faces, or false for back faces.

Culling

If `CullBackFacingTrianglesKHR` was included in the `Ray Flags` parameter of the ray trace instruction, and the intersection is determined as with the back face of a triangle primitive, the intersection is dropped, and no further processing of this intersection occurs.

If `CullFrontFacingTrianglesKHR` was included in the `Ray Flags` parameter of the ray trace instruction, and the intersection is determined as with the front face of a triangle primitive, the intersection is dropped, and no further processing of this intersection occurs.

This culling is disabled if VK_GEOMETRY_INSTANCE_TRIANGLE_FACING_CULL_DISABLE_BIT_KHR was included in VkAccelerationStructureInstanceKHR::`flags` for the instance which the intersected geometry belongs to.

Intersection candidates that have not intersected with any face (a == 0) are unconditionally culled, irrespective of ray flags and geometry instance flags.

The `CullBackFacingTrianglesKHR` and `CullFrontFacingTrianglesKHR` `Ray Flags` are mutually exclusive.

42.2.4. Ray Opacity Culling

Each geometry in the acceleration structure **may** be considered either opaque or non-opaque. Opaque geometries continue through traversal as normal, whereas non-opaque geometries need to

be either confirmed or discarded by shader code. Intersection candidates **can** also be culled based on their opacity.

Determination

If the geometry includes an opacity micromap, the intersection candidate is determined as opaque or non-opaque as described in Ray Opacity Micromap. If the result of the opacity micromap operation is *ignored*, the candidate is considered neither opaque nor non-opaque and execution instead proceeds with enumerating the next candidate. Otherwise, the intersection candidate is determined as opaque if VK_GEOMETRY_OPAQUE_BIT_KHR was included in the VkAccelerationStructureGeometryKHR::`flags` when the geometry it intersected with was built.

Setting VK_GEOMETRY_INSTANCE_FORCE_OPAQUE_BIT_KHR in VkAccelerationStructureInstanceKHR::`flags` will override all non-opaque geometries in the instance to be considered opaque. Similarly, setting VK_GEOMETRY_INSTANCE_FORCE_NO_OPAQUE_BIT_KHR will override all opaque geometries in the instance to be considered non-opaque.

Including `OpaqueKHR` in the `Ray Flags` parameter when tracing a ray or initializing a ray query object overrides all non-opaque geometries to behave as if they are opaque, ignoring any build instance flag overrides. Similarly, `NoOpaqueKHR` overrides all opaque geometries to behave as if they are non-opaque.

Culling

If `CullOpaqueKHR` is included in the `Ray Flags` parameter when tracing a ray or initializing a ray query object, an intersection with geometry considered to be opaque is dropped and execution proceeds with enumerating the next candidate. Similarly, `CullNoOpaqueKHR` drops intersections with geometry considered to be non-opaque.

The `OpaqueKHR`, `NoOpaqueKHR`, `CullOpaqueKHR`, and `CullNoOpaqueKHR` `Ray Flags` are mutually exclusive.

Ray Query

The opacity of geometries is determined as described above by `OpRayQueryProceedKHR` when using it to trace rays.

Intersection candidates with AABB geometry cause proceed to return `true`, incomplete traversal, in order for the shader to confirm the hit. To query the opacity that was determined by proceed, `OpRayQueryGetIntersectionCandidateAABBOpaqueKHR` **can** be used.

Intersection candidates with triangle geometry that cause proceed to return `true` have been determined to be non-opaque. Intersection candidates with opaque triangle geometry continue execution without causing proceed to return.

42.2.5. Ray Opacity Micromap

A VK_GEOMETRY_TYPE_TRIANGLES_KHR geometry in the acceleration structure **may** have an opacity micromap associated with it to give finer-grained opacity information.

If the intersection candidate is with a geometry with an associated opacity micromap and VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_BIT_KHR is not set in its instance then the micromap is used to determine geometry opacity instead of the VK_GEOMETRY_OPAQUE_BIT_KHR flag in the geometry.

Using an acceleration structure that contains opacity micromap data requires the shader or pipeline to declare that it does so. For traversals performed by pipeline trace ray instructions the pipeline **must** be created with VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR. For traversals performed by `OpRayQueryProceedKHR` where VK_EXT_opacity_micromap is not enabled, the shader **must** declare the `OpacityMicromapIdKHR` execution mode with its specialization-constant operand equal to VK_TRUE. If VK_EXT_opacity_micromap is enabled, `OpRayQueryProceedKHR` traversal always uses opacity micromap data without the `OpacityMicromapIdKHR` execution mode being declared. This applies whenever VK_EXT_opacity_micromap is enabled, including when VK_KHR_opacity_micromap is enabled as well. A shader executed in a ray tracing pipeline that itself performs ray query operations is subject to both requirements independently, each applying only to the traversal it governs.

NOTE These declarations are requirements, not a means of disabling opacity micromaps. Traversing an acceleration structure that contains opacity micromap data without the applicable declaration has undefined results, and is not equivalent to ignoring the opacity micromap data. To traverse such an acceleration structure without applying its opacity micromaps, set VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_BIT_KHR in the instance instead.

The opacity information in the micromap object is accessed using the candidate intersection u and v coordinates. The integer u and v are computed from □u□ + □v□, clamping □u□ as needed to keep the sum less than or equal to 1 << subdivisionlevel. These values are mapped into a linear index with a space filling curve which is defined recursively by traversing into the sub-triangle nearest vertex 0, then the middle triangle with ordering flipped, then nearest vertex 1, then nearest vertex 2 with ordering flipped.

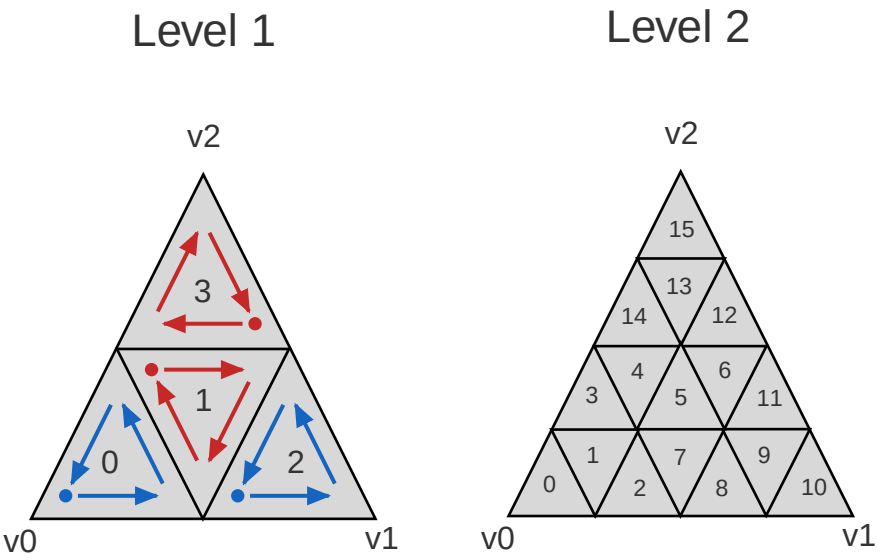


Figure 25. Example ordering for micromap data

Caption

In the ordering for micromap data diagram, the dot in each sub-triangle indicates the entry point of the curve and the arrows show the traversal order through its children at the next subdivision level.

NOTE

This encoding is spatially coherent, purely hierarchical, and allows a bit-parallel conversion between barycentric address and index values.

See the Reference Code implementing this mapping.

The result of the opacity micromap lookup and operations is to treat the intersection as opaque, non-opaque, or ignored.

If the associated opacity micromap has format VK_OPACITY_MICROMAP_FORMAT_2_STATE_KHR, each element of the micromap is represented by a single bit at the index derived above.

If the associated opacity micromap has format VK_OPACITY_MICROMAP_FORMAT_4_STATE_KHR, each element is represented by a two bit value at the index derived above.

If the micromap was built with VK_BUILD_ACCELERATION_STRUCTURE_MICROMAP_LOSSY_BIT_KHR and has a format of VK_OPACITY_MICROMAP_FORMAT_4_STATE_KHR, the implementation **may** substitute any element's state value with VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_TRANSPARENT_KHR or VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_OPAQUE_KHR. The application **should** accept the hit if the element was built with VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_OPAQUE_KHR and ignore the hit if the element was built with VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_TRANSPARENT_KHR.

The implementation, on an identically created instance and device, **must** perform any lossy substitutions invariantly with respect to acceleration structures and micromaps constructed with an equivalent shape and data.

The final result of the opacity micromap lookup operation is determined as shown in the following table based on the format and retrieved value from the micromap:

4 State value	2 State value	Special index value	Result
0	0	VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_TRANSPARENT_KHR	Ignored
1	1	VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_OPAQUE_KHR	Opaque

4 State value	2 State value	Special index value	Result
2		VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_TRANSPARENT_KHR	Non-opaque
3		VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_OPAQUE_KHR	Non-opaque

Unless the VK_GEOMETRY_INSTANCE_FORCE_OPACITY_MICROMAP_2_STATE_BIT_KHR flag is included in the instance flags of the candidate intersection or `ForceOpacityMicromap2StateEXT` is included in the ray flags on the ray, in which case the opacity micromap information is interpreted in 2 state override mode , including if lossy substitutions were made, and the final result of the opacity lookup operation is instead determined as shown in the following table:

4 State value	2 State value	Special index value	Result
0	0	VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_TRANSPARENT_KHR	Ignored
1	1	VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_OPAQUE_KHR	Opaque
2		VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_TRANSPARENT_KHR	Ignored
3		VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_OPAQUE_KHR	Opaque

NOTE The Force2state override never evaluates to a *non-opaque* result, even when lossy substitutions are made. While lossy substitutions can affect the final result of Force2state, this behavior preserves the invariance guarantees required for lossy substitutions.

If the result of the micromap lookup is to treat the intersection candidate as ignored, no further processing of that candidate is done and execution proceeds with enumerating the next candidate.

42.3. Ray Intersection Confirmation

Depending on the opacity of intersected geometry and whether it is a triangle or an AABB, candidate intersections are further processed to determine the eventual hit result. Candidates

generated from AABB intersections run through the same confirmation process as triangle hits.

For ray queries, the candidate intersection and committed intersection are distinct states. A candidate intersection is available only while OpRayQueryProceedKHR has returned true, and identifies the intersection currently requiring shader evaluation. Calling OpRayQueryGenerateIntersectionKHR or OpRayQueryConfirmIntersectionKHR confirms a hit for Ray Closest Hit Determination, but does not by itself complete traversal. If that confirmed hit becomes the current closest hit, it becomes the ray query's committed intersection. The committed intersection is the closest recorded hit so far, **can** be queried before traversal is complete, and **may** be replaced by a later confirmed hit that is closer.

42.3.1. AABB Intersection Candidates

For an intersection candidate with an AABB geometry generated by Ray Intersection Candidate Determination, shader code is executed to determine whether any hits should be reported to the traversal infrastructure; no further processing of this intersection candidate occurs. The occurrence of an AABB intersection candidate does not guarantee the ray intersects the application-provided AABB. To avoid propagating false intersections the application **should** verify the intersection candidate before reporting any hits and only report intersections within the bounds of the desired geometry. Reporting an intersection outside the AABB either through the implementation giving a conservative bound or reporting a t out of range is legal but **may** result in unpredictable closest hit results.

If the ray was traced with a pipeline trace ray instruction, an intersection shader is invoked from the Shader Binding Table according to the specified indexing for the intersected geometry. If this shader calls OpReportIntersectionKHR, a new intersection candidate is generated as described below. If the intersection shader is VK_SHADER_UNUSED_KHR (which is only allowed for a zero shader group) then no further processing of the intersection candidate occurs.

Each new candidate generated as a result of this processing is a generated intersection candidate from the intersection with AABB geometry, with a t value equal to the Hit parameter of the OpReportIntersectionKHR instruction. The generated intersection candidate includes the values stored in variables with the HitAttributeKHR storage class at the time OpReportIntersectionKHR is executed. The new generated candidate is then independently run through Ray Intersection Confirmation as a generated intersection.

If the ray was traced with OpRayQueryProceedKHR, control is returned to the shader which executed OpRayQueryProceedKHR, returning true. The resulting ray query has a candidate intersection type of RayQueryCandidateIntersectionAABBKHR. OpRayQueryGenerateIntersectionKHR **can** be called to report a generated hit for Ray Closest Hit Determination. If that hit becomes the current closest hit, the ray query's committed intersection type is RayQueryCommittedIntersectionGeneratedKHR. Further ray query processing **can** be continued by executing OpRayQueryProceedKHR with the same ray query, or intersection **can** be terminated with OpRayQueryTerminateKHR. Unlike rays traced with a pipeline trace ray instruction, candidates generated in this way skip generated intersection candidate confirmation; applications **should** make this determination before generating the intersection.

This operation **may** be executed multiple times for the same intersection candidate.

42.3.2. Triangle and Generated Intersection Candidates

For triangle and generated intersection candidates, additional shader code **may** be executed based on the intersection's opacity.

If the intersection is opaque, the candidate is immediately confirmed as a valid hit and passes to the next stage of processing.

For non-opaque intersection candidates, shader code is executed to determine whether a hit occurred or not.

If the ray was traced with a pipeline trace ray instruction, an any-hit shader is invoked from the Shader Binding Table according to the specified indexing. If this shader calls `OpIgnoreIntersectionKHR`, the candidate is dropped and no further processing of the candidate occurs. If the any-hit shader identified is VK_SHADER_UNUSED_KHR, the candidate is immediately confirmed as a valid hit and passes to the next stage of processing.

If the ray was traced with `OpRayQueryProceedKHR`, control is returned to the shader which executed `OpRayQueryProceedKHR`, returning `true`. As only triangle candidates participate in this operation with ray queries, the resulting candidate intersection type is always `RayQueryCandidateIntersectionTriangleKHR`. `OpRayQueryConfirmIntersectionKHR` **can** be called on the ray query to confirm the candidate as a hit for Ray Closest Hit Determination. If that hit becomes the current closest hit, the ray query's committed intersection type is `RayQueryCommittedIntersectionTriangleKHR`. Further ray query processing **can** be continued by executing `OpRayQueryProceedKHR` with the same ray query, or intersection **can** be terminated with `OpRayQueryTerminateKHR`. If `OpRayQueryConfirmIntersectionKHR` has not been executed, the candidate is dropped and no further processing of the candidate occurs.

This operation **may** be executed multiple times for the same intersection candidate unless VK_GEOMETRY_NO_DUPLICATE_ANY_HIT_INVOCATION_BIT_KHR was specified for the intersected geometry.

42.4. Ray Closest Hit Determination

Unless the ray was traced with the `TerminateOnFirstHitKHR` ray flag, the implementation **must** track the closest confirmed hit until all geometries have been tested and either confirmed or dropped.

After an intersection candidate is confirmed, its t value is compared to t_{max} to determine which intersection is closer, where t is the parametric distance along the ray at which the intersection occurred.

- If $t < t_{max}$, t_{max} is set to t and the candidate is set as the current closest hit.
- If $t > t_{max}$, the candidate is dropped and no further processing of that candidate occurs.
- If $t = t_{max}$, the candidate **may** be set as the current closest hit or dropped.

For ray queries, the current closest hit is the current committed intersection. When a confirmed hit is set as the current closest hit, values returned by `OpRayQueryGetIntersection*KHR` instructions with `Intersection` set to `RayQueryCommittedIntersectionKHR` are updated to describe that hit. If no closest hit has been identified, the committed intersection type is `RayQueryCommittedIntersectionNoneKHR`.

If `TerminateOnFirstHitKHR` was included in the `Ray Flags` used to trace the ray, once the first hit is confirmed, the ray trace is terminated.

42.5. Ray Result Determination

Once all candidates have finished processing the prior stages, or if the ray is forcibly terminated, the final result of the ray trace is determined.

If a closest hit result was identified by Ray Closest Hit Determination, a closest hit has occurred, otherwise the final result is a miss.

For rays traced with pipeline trace ray instructions which **can** invoke a closest hit shader, if a closest hit result was identified, a closest hit shader is invoked from the Shader Binding Table according to the specified indexing for the intersected geometry. Control returns to the shader that executed the pipeline trace ray instruction once this shader returns. This shader is skipped if either the ray flags included `SkipClosestHitShaderKHR`, or if the closest hit shader identified is VK_SHADER_UNUSED_KHR.

For rays traced with a pipeline trace ray instruction where no hit result was identified, the miss shader identified by the `Miss Index` parameter of the instruction is invoked. Control returns to the shader that executed the pipeline trace ray instruction once this shader returns. This shader is skipped if the miss shader identified is VK_SHADER_UNUSED_KHR.

If the ray was traced with `OpRayQueryProceedKHR`, control is returned to the shader which executed `OpRayQueryProceedKHR`, returning `false`. No candidate intersection is available after this result is returned. If a closest hit was identified by Ray Closest Hit Determination, the ray query's committed intersection is the final result and has a committed intersection type of `RayQueryCommittedIntersectionGeneratedKHR` or `RayQueryCommittedIntersectionTriangleKHR`. If no closest hit was identified, the committed intersection type will be `RayQueryCommittedIntersectionNoneKHR`.

No further processing of a ray query occurs after this result is determined.

Chapter 43. Ray Tracing

Ray tracing uses a separate rendering pipeline from both the graphics and compute pipelines (see Ray Tracing Pipeline).

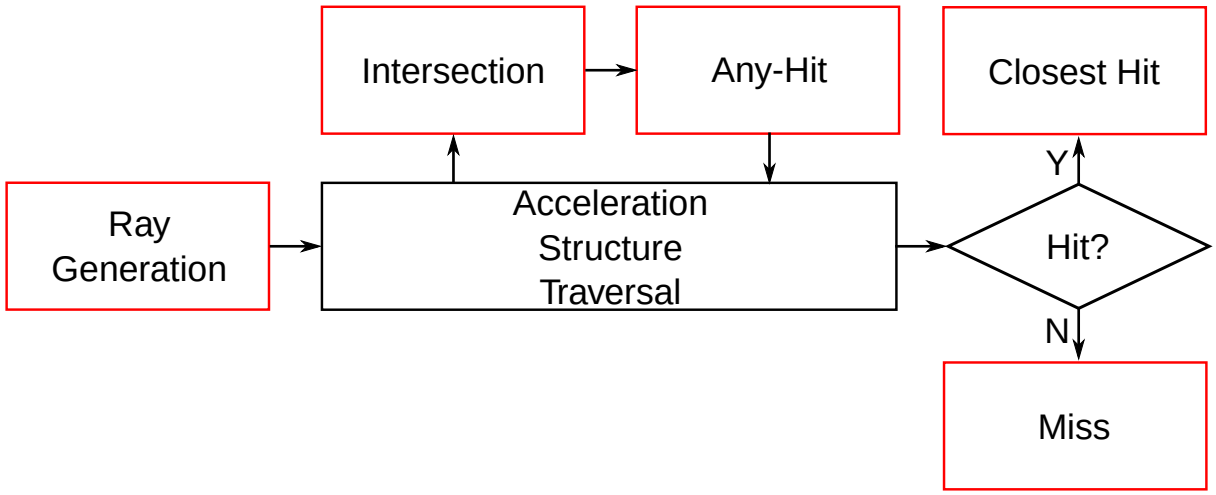


Figure 26. Ray tracing pipeline execution

Caption

Interaction between the different shader stages in the ray tracing pipeline

Within the ray tracing pipeline, a pipeline trace ray instruction **can** be called to perform a ray traversal that invokes the various ray tracing shader stages during its execution. The relationship between the ray tracing pipeline object and the geometries present in the acceleration structure traversed is passed into the ray tracing command in a VkBuffer object known as a *shader binding table*. `OpExecuteCallableKHR` can also be used in ray tracing pipelines to invoke a callable shader.

During execution, control alternates between scheduling and other operations. The scheduling functionality is implementation-specific and is responsible for workload execution. The shader stages are programmable. *Traversal*, which refers to the process of traversing acceleration structures to find potential intersections of rays with geometry, is fixed function.

The programmable portions of the pipeline are exposed in a single-ray programming model, with each invocation handling one ray at a time. Memory operations **can** be synchronized using standard memory barriers. The `Workgroup` scope and variables with a storage class of `Workgroup` **must** not be used in the ray tracing pipeline.

43.1. Shader Call Instructions

A *shader call* is an instruction which **may** cause execution to continue elsewhere by creating one or more invocations that execute a different shader stage.

The following table lists all shader call instructions and which stages each one **can** directly call.

Instruction	Intersection	Any-Hit	Closest Hit	Miss	Callable
OpTraceRayKHR	X	X	X	X	
OpReportIntersectionKHR		X			
OpExecuteCallableKHR					X
OpHitObjectExecuteShaderEXT			X	X	
OpHitObjectTraceRayEXT	X	X			
OpHitObjectReorderExecuteShaderEXT			X	X	
OpHitObjectTraceReorderExecuteEXT	X	X	X	X	

The invocations created by shader call instructions are grouped into subgroups by the implementation. Those subgroups **may** be unrelated to the subgroup of the parent invocation.

Pipeline trace ray instructions **can** be used recursively; invoked shaders **can** themselves execute pipeline trace ray instructions, to a maximum depth defined by the maxRayRecursionDepth limit.

Shaders directly invoked from the API always have a recursion depth of 0; each shader executed by a pipeline trace ray instruction has a recursion depth one higher than the recursion depth of the shader which invoked it. Applications **must** not invoke a shader with a recursion depth greater than the value of maxPipelineRayRecursionDepth specified in the pipeline.

There is no explicit recursion limit for other shader call instructions which may recurse (e.g. OpExecuteCallableKHR) but there is an upper bound determined by the stack size.

An *invocation repack instruction* is a ray tracing instruction where the implementation **may** change the set of invocations that are executing. When a repack instruction is encountered, the invocation is suspended and a new invocation begins and executes the instruction. After executing the repack instruction (which **may** result in other ray tracing shader stages executing) the new invocation ends and the original invocation is resumed, but it **may** be resumed in a different subgroup or at a different SubgroupLocalInvocationId within the same subgroup.

The OpTraceRayKHR, OpReorderThreadWithHintEXT, OpReorderThreadWithHitObjectEXT, OpHitObjectTraceRayEXT, OpHitObjectReorderExecuteShaderEXT, OpHitObjectTraceReorderExecuteEXT, OpReportIntersectionKHR, and OpExecuteCallableKHR instructions are invocation repack instructions.

The invocations that are executing before a shader call instruction, after the instruction, or are created by the instruction, are shader-call-related.

If the implementation changes the composition of subgroups, the values of SubgroupSize, SubgroupLocalInvocationId, and builtin variables that are derived from them (SubgroupEqMask, SubgroupGeMask, SubgroupGtMask, SubgroupLeMask, SubgroupLtMask) **must** be changed accordingly by the invocation repack instruction. The application **must** use Volatile semantics on these BuiltIn

variables when used in the ray generation, closest hit, miss, intersection, and callable shaders. Similarly, the application **must** use `Volatile` semantics on any `RayTmaxKHR` decorated `BuiltIn` used in an intersection shader.

> Subgroup operations are permitted in the programmable ray tracing shader stages. However, shader call instructions place a bound on where results of subgroup instructions or subgroup-scoped instructions that execute the dynamic instance of that instruction are potentially valid. For example, care **must** be taken when using the result of a ballot operation that was computed before an invocation repack instruction, after that repack instruction. The ballot **may** be incorrect as the set of invocations could have changed.

NOTE

While the `SubgroupSize` built-in is required to be declared `Volatile`, its value will never change unless VK_PIPELINE_SHADER_STAGE_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT is set on pipeline creation, as without that bit set, its value is required to match that of VkPhysicalDeviceSubgroupProperties::`subgroupSize`.

For clock operations, the value of a `Subgroup` scoped `OpReadClockKHR` read before the dynamic instance of a repack instruction **should** not be compared to the result of that clock instruction after the repack instruction.

When a ray tracing shader executes a dynamic instance of an invocation repack instruction which results in another ray tracing shader being invoked, their instructions are related by shader-call-order.

For ray tracing invocations that are shader-call-related:

- memory operations on `StorageBuffer`, `Image`, and `ShaderRecordBufferKHR` storage classes **can** be synchronized using the `ShaderCallKHR` scope.
- the `CallableDataKHR`, `IncomingCallableDataKHR`, `RayPayloadKHR`, `HitAttributeKHR`, and `IncomingRayPayloadKHR` storage classes are system-synchronized and no application availability and visibility operations are required.
- memory operations within a single invocation before and after the shader call instruction are ordered by program-order and do not require explicit synchronization.

43.2. Ray Tracing Commands

Ray tracing commands provoke work in the ray tracing pipeline. Ray tracing commands are recorded into a command buffer and when executed by a queue will produce work that executes according to the bound ray tracing pipeline. A ray tracing pipeline **must** be bound to a command buffer before any ray tracing commands are recorded in that command buffer.

To dispatch ray tracing use:

```
// Provided by VK_KHR_ray_tracing_pipeline
void vkCmdTraceRaysKHR(
    VkCommandBuffer                             commandBuffer,
    const VkStridedDeviceAddressRegionKHR*      pRaygenShaderBindingTable,
    const VkStridedDeviceAddressRegionKHR*      pMissShaderBindingTable,
    const VkStridedDeviceAddressRegionKHR*      pHitShaderBindingTable,
    const VkStridedDeviceAddressRegionKHR*      pCallableShaderBindingTable,
    uint32_t                                    width,
    uint32_t                                    height,
    uint32_t                                    depth);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pRaygenShaderBindingTable` is a VkStridedDeviceAddressRegionKHR that holds the shader binding table data for the ray generation shader stage.
- `pMissShaderBindingTable` is a VkStridedDeviceAddressRegionKHR that holds the shader binding table data for the miss shader stage.
- `pHitShaderBindingTable` is a VkStridedDeviceAddressRegionKHR that holds the shader binding table data for the hit shader stage.
- `pCallableShaderBindingTable` is a VkStridedDeviceAddressRegionKHR that holds the shader binding table data for the callable shader stage.
- `width` is the width of the ray trace query dimensions.
- `height` is the height of the ray trace query dimensions.
- `depth` is the depth of the ray trace query dimensions.

When the command is executed, a ray generation group of `width` × `height` × `depth` rays is assembled.

Valid Usage

- VUID-vkCmdTraceRaysKHR-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdTraceRaysKHR-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdTraceRaysKHR-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdTraceRaysKHR-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created

with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT

- VUID-vkCmdTraceRaysKHR-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdTraceRaysKHR-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdTraceRaysKHR-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdTraceRaysKHR-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdTraceRaysKHR-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdTraceRaysKHR-magFilter-09598
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdTraceRaysKHR-mipmapMode-04770
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdTraceRaysKHR-mipmapMode-09599
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT

- VUID-vkCmdTraceRaysKHR-unnormalizedCoordinates-09635
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdTraceRaysKHR-None-08609
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `viewType` **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdTraceRaysKHR-None-08610
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions with `ImplicitLod`, `Dref` or `Proj` in their name
- VUID-vkCmdTraceRaysKHR-None-08611
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions that include a LOD bias or any offset values
- VUID-vkCmdTraceRaysKHR-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdTraceRaysKHR-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdTraceRaysKHR-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdTraceRaysKHR-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdTraceRaysKHR-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdTraceRaysKHR-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdTraceRaysKHR-filterCubicMinmax-02695

Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2

- VUID-vkCmdTraceRaysKHR-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdTraceRaysKHR-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdTraceRaysKHR-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdTraceRaysKHR-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdTraceRaysKHR-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdTraceRaysKHR-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdTraceRaysKHR-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdTraceRaysKHR-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdTraceRaysKHR-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or

VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdTraceRaysKHR-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdTraceRaysKHR-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdTraceRaysKHR-None-08613
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdTraceRaysKHR-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdTraceRaysKHR-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match
- VUID-vkCmdTraceRaysKHR-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdTraceRaysKHR-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have four components
- VUID-vkCmdTraceRaysKHR-OpImageWrite-04469
  If a VkBufferView is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdTraceRaysKHR-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a

result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64

- VUID-vkCmdTraceRaysKHR-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdTraceRaysKHR-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdTraceRaysKHR-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdTraceRaysKHR-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdTraceRaysKHR-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdTraceRaysKHR-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdTraceRaysKHR-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdTraceRaysKHR-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdTraceRaysKHR-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding

micromaps used to originally build the acceleration structure prior to the execution of this command on the device

- VUID-vkCmdTraceRaysKHR-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with `flags` that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdTraceRaysKHR-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with `flags` that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdTraceRaysKHR-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdTraceRaysKHR-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`
- VUID-vkCmdTraceRaysKHR-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdTraceRaysKHR-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset

for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`

- VUID-vkCmdTraceRaysKHR-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdTraceRaysKHR-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdTraceRaysKHR-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdTraceRaysKHR-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdTraceRaysKHR-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdTraceRaysKHR-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdTraceRaysKHR-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdTraceRaysKHR-None-11309

If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands

- VUID-vkCmdTraceRaysKHR-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdTraceRaysKHR-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdTraceRaysKHR-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdTraceRaysKHR-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdTraceRaysKHR-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdTraceRaysKHR-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or

VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds

- VUID-vkCmdTraceRaysKHR-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdTraceRaysKHR-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdTraceRaysKHR-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdTraceRaysKHR-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdTraceRaysKHR-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`

- VUID-vkCmdTraceRaysKHR-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdTraceRaysKHR-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdTraceRaysKHR-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdTraceRaysKHR-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdTraceRaysKHR-None-03429
  Any shader group handle referenced by this call **must** have been queried from the bound ray tracing pipeline
- VUID-vkCmdTraceRaysKHR-None-09458
  If the bound ray tracing pipeline state was created with the VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR dynamic state enabled then vkCmdSetRayTracingPipelineStackSizeKHR **must** have been called in the current command buffer prior to this trace command
- VUID-vkCmdTraceRaysKHR-None-11319
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, the value of

the address at the expected location in shader record data **must** be a valid address

- VUID-vkCmdTraceRaysKHR-maxPipelineRayRecursionDepth-03679
  This command **must** not cause a shader call instruction to be executed from a shader invocation with a recursion depth greater than the value of `maxPipelineRayRecursionDepth` used to create the bound ray tracing pipeline
- VUID-vkCmdTraceRaysKHR-commandBuffer-03635
  `commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdTraceRaysKHR-micromap-11640
  If a VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR descriptor is accessed as a result of this command, the access is not through an `OpRayQueryProceedKHR` operation, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR or VkAccelerationStructureTrianglesOpacityMicromapEXT structure included in its `pNext` chain, the VkPipeline bound to the pipeline bind point used by this command **must** have been created with `flags` that includes VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR
- VUID-vkCmdTraceRaysKHR-size-04023
  The `size` member of `pRayGenShaderBindingTable` **must** be equal to its `stride` member
- VUID-vkCmdTraceRaysKHR-pRayGenShaderBindingTable-03681
  `pRaygenShaderBindingTable->deviceAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR usage flag set
- VUID-vkCmdTraceRaysKHR-pRayGenShaderBindingTable-03682
  `pRaygenShaderBindingTable->deviceAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR`::`shaderGroupBaseAlignment`
- VUID-vkCmdTraceRaysKHR-pMissShaderBindingTable-03684
  `pMissShaderBindingTable->deviceAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR usage flag set
- VUID-vkCmdTraceRaysKHR-pMissShaderBindingTable-03685
  `pMissShaderBindingTable->deviceAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR`::`shaderGroupBaseAlignment`
- VUID-vkCmdTraceRaysKHR-stride-03686
  `pMissShaderBindingTable->stride` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR`::`shaderGroupHandleAlignment`
- VUID-vkCmdTraceRaysKHR-stride-04029
  `pMissShaderBindingTable->stride` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR`::`maxShaderGroupStride`
- VUID-vkCmdTraceRaysKHR-pHitShaderBindingTable-03688
  `pHitShaderBindingTable->deviceAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR usage flag set

- VUID-vkCmdTraceRaysKHR-pHitShaderBindingTable-03689
  pHitShaderBindingTable->deviceAddress **must** be a multiple of VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment
- VUID-vkCmdTraceRaysKHR-stride-03690
  pHitShaderBindingTable->stride **must** be a multiple of VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleAlignment
- VUID-vkCmdTraceRaysKHR-stride-04035
  pHitShaderBindingTable->stride **must** be less than or equal to VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxShaderGroupStride
- VUID-vkCmdTraceRaysKHR-pCallableShaderBindingTable-03692
  pCallableShaderBindingTable->deviceAddress **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR usage flag set
- VUID-vkCmdTraceRaysKHR-pCallableShaderBindingTable-03693
  pCallableShaderBindingTable->deviceAddress **must** be a multiple of VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment
- VUID-vkCmdTraceRaysKHR-stride-03694
  pCallableShaderBindingTable->stride **must** be a multiple of VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleAlignment
- VUID-vkCmdTraceRaysKHR-stride-04041
  pCallableShaderBindingTable->stride **must** be less than or equal to VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxShaderGroupStride
- VUID-vkCmdTraceRaysKHR-flags-03511
  If the bound ray tracing pipeline was created with flags that included VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR, the entries in the table identified by pMissShaderBindingTable->deviceAddress accessed as a result of this command in order to execute a miss shader **must** not be zero
- VUID-vkCmdTraceRaysKHR-flags-03512
  If the bound ray tracing pipeline was created with flags that included VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR, entries in the table identified by pHitShaderBindingTable->deviceAddress accessed as a result of this command in order to execute an any-hit shader **must** not be zero
- VUID-vkCmdTraceRaysKHR-flags-03513
  If the bound ray tracing pipeline was created with flags that included VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR, entries in the table identified by pHitShaderBindingTable->deviceAddress accessed as a result of this command in order to execute a closest hit shader **must** not be zero
- VUID-vkCmdTraceRaysKHR-flags-03514
  If the bound ray tracing pipeline was created with flags that included VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR, entries in the table identified by pHitShaderBindingTable->deviceAddress accessed as a result of this command in order to execute an intersection shader **must** not be zero
- VUID-vkCmdTraceRaysKHR-pHitShaderBindingTable-04735

Any non-zero hit shader group entries in the table identified by pHitShaderBindingTable->deviceAddress accessed by this call from a geometry with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR **must** have been created with VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR

- VUID-vkCmdTraceRaysKHR-pHitShaderBindingTable-04736
  Any non-zero hit shader group entries in the table identified by pHitShaderBindingTable->deviceAddress accessed by this call from a geometry with a geometryType of VK_GEOMETRY_TYPE_AABBS_KHR **must** have been created with VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR

- VUID-vkCmdTraceRaysKHR-width-03638
  width **must** be less than or equal to VkPhysicalDeviceLimits::maxComputeWorkGroupCount[0] × VkPhysicalDeviceLimits::maxComputeWorkGroupSize[0]
- VUID-vkCmdTraceRaysKHR-height-03639
  height **must** be less than or equal to VkPhysicalDeviceLimits::maxComputeWorkGroupCount[1] × VkPhysicalDeviceLimits::maxComputeWorkGroupSize[1]
- VUID-vkCmdTraceRaysKHR-depth-03640
  depth **must** be less than or equal to VkPhysicalDeviceLimits::maxComputeWorkGroupCount[2] × VkPhysicalDeviceLimits::maxComputeWorkGroupSize[2]
- VUID-vkCmdTraceRaysKHR-width-03641
  width × height × depth **must** be less than or equal to VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxRayDispatchInvocationCount

Valid Usage (Implicit)

- VUID-vkCmdTraceRaysKHR-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdTraceRaysKHR-pRaygenShaderBindingTable-parameter
  pRaygenShaderBindingTable **must** be a valid pointer to a valid VkStridedDeviceAddressRegionKHR structure
- VUID-vkCmdTraceRaysKHR-pMissShaderBindingTable-parameter
  pMissShaderBindingTable **must** be a valid pointer to a valid VkStridedDeviceAddressRegionKHR structure
- VUID-vkCmdTraceRaysKHR-pHitShaderBindingTable-parameter
  pHitShaderBindingTable **must** be a valid pointer to a valid VkStridedDeviceAddressRegionKHR structure
- VUID-vkCmdTraceRaysKHR-pCallableShaderBindingTable-parameter
  pCallableShaderBindingTable **must** be a valid pointer to a valid VkStridedDeviceAddressRegionKHR structure
- VUID-vkCmdTraceRaysKHR-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdTraceRaysKHR-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support

VK_QUEUE_COMPUTE_BIT operations

- VUID-vkCmdTraceRaysKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdTraceRaysKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdTraceRaysKHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdTraceRaysKHR is not affected by conditional rendering

To dispatch ray tracing, with some parameters sourced on the device, use:

```
// Provided by VK_KHR_ray_tracing_pipeline
void vkCmdTraceRaysIndirectKHR(
    VkCommandBuffer                             commandBuffer,
    const VkStridedDeviceAddressRegionKHR*      pRaygenShaderBindingTable,
    const VkStridedDeviceAddressRegionKHR*      pMissShaderBindingTable,
    const VkStridedDeviceAddressRegionKHR*      pHitShaderBindingTable,
    const VkStridedDeviceAddressRegionKHR*      pCallableShaderBindingTable,
    VkDeviceAddress                             indirectDeviceAddress);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pRaygenShaderBindingTable` is a VkStridedDeviceAddressRegionKHR that holds the shader binding table data for the ray generation shader stage.
- `pMissShaderBindingTable` is a VkStridedDeviceAddressRegionKHR that holds the shader binding

table data for the miss shader stage.

- `pHitShaderBindingTable` is a VkStridedDeviceAddressRegionKHR that holds the shader binding table data for the hit shader stage.
- `pCallableShaderBindingTable` is a VkStridedDeviceAddressRegionKHR that holds the shader binding table data for the callable shader stage.
- `indirectDeviceAddress` is a buffer device address which is a pointer to a VkTraceRaysIndirectCommandKHR structure containing the trace ray parameters.

`vkCmdTraceRaysIndirectKHR` behaves similarly to vkCmdTraceRaysKHR except that the ray trace query dimensions are read by the device from `indirectDeviceAddress` during execution.

Valid Usage

- VUID-vkCmdTraceRaysIndirectKHR-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdTraceRaysIndirectKHR-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdTraceRaysIndirectKHR-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdTraceRaysIndirectKHR-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdTraceRaysIndirectKHR-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdTraceRaysIndirectKHR-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdTraceRaysIndirectKHR-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdTraceRaysIndirectKHR-None-08119
  If a descriptor is dynamically used with a VkPipeline created with

VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a VkDescriptorSetLayout that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident

- VUID-vkCmdTraceRaysIndirectKHR-magFilter-04553
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdTraceRaysIndirectKHR-magFilter-09598
  If a VkSampler created with magFilter or minFilter equal to VK_FILTER_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdTraceRaysIndirectKHR-mipmapMode-04770
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, reductionMode equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and compareEnable equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdTraceRaysIndirectKHR-mipmapMode-09599
  If a VkSampler created with mipmapMode equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and reductionMode equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdTraceRaysIndirectKHR-unnormalizedCoordinates-09635
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's levelCount and layerCount **must** be 1
- VUID-vkCmdTraceRaysIndirectKHR-None-08609
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's viewType **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdTraceRaysIndirectKHR-None-08610
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions with ImplicitLod, Dref or Proj in their name
- VUID-vkCmdTraceRaysIndirectKHR-None-08611
  If a VkSampler created with unnormalizedCoordinates equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V OpImageSample* or OpImageSparseSample* instructions that include a LOD bias

or any offset values

- VUID-vkCmdTraceRaysIndirectKHR-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdTraceRaysIndirectKHR-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdTraceRaysIndirectKHR-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdTraceRaysIndirectKHR-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdTraceRaysIndirectKHR-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdTraceRaysIndirectKHR-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdTraceRaysIndirectKHR-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdTraceRaysIndirectKHR-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdTraceRaysIndirectKHR-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdTraceRaysIndirectKHR-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain

VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT

- VUID-vkCmdTraceRaysIndirectKHR-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdTraceRaysIndirectKHR-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdTraceRaysIndirectKHR-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdTraceRaysIndirectKHR-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdTraceRaysIndirectKHR-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdTraceRaysIndirectKHR-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdTraceRaysIndirectKHR-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdTraceRaysIndirectKHR-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdTraceRaysIndirectKHR-None-08613
  If the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdTraceRaysIndirectKHR-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's viewType **must** match the Dim operand of the OpTypeImage as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdTraceRaysIndirectKHR-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's format and the Sampled Type operand of the OpTypeImage **must** match
- VUID-vkCmdTraceRaysIndirectKHR-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdTraceRaysIndirectKHR-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components
- VUID-vkCmdTraceRaysIndirectKHR-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdTraceRaysIndirectKHR-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdTraceRaysIndirectKHR-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdTraceRaysIndirectKHR-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdTraceRaysIndirectKHR-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdTraceRaysIndirectKHR-sparseImageInt64Atomics-04474
  If the sparseImageInt64Atomics feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic

instructions through an OpTypeImage with a SampledType with a Width of 64 by this command

- VUID-vkCmdTraceRaysIndirectKHR-sparseImageInt64Atomics-04475
  If the sparseImageInt64Atomics feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an OpTypeImage with a SampledType with a Width of 64 by this command
- VUID-vkCmdTraceRaysIndirectKHR-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdTraceRaysIndirectKHR-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdTraceRaysIndirectKHR-micromap-11636
  If an acceleration structure is accessed as a result of this command through an OpRayQueryProceedKHR operation , the VK_EXT_opacity_micromap extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its pNext chain, the shader **must** have specified a OpacityMicromapIdKHR execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdTraceRaysIndirectKHR-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdTraceRaysIndirectKHR-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with flags that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdTraceRaysIndirectKHR-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an indexBuffer containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkShaderEXT that accessed the descriptor **must** not have been created with flags that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT
- VUID-vkCmdTraceRaysIndirectKHR-None-12472

Descriptors dynamically accessed in bound descriptor heaps **must** be valid

- VUID-vkCmdTraceRaysIndirectKHR-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`
- VUID-vkCmdTraceRaysIndirectKHR-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdTraceRaysIndirectKHR-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdTraceRaysIndirectKHR-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdTraceRaysIndirectKHR-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdTraceRaysIndirectKHR-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound

to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress

- VUID-vkCmdTraceRaysIndirectKHR-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdTraceRaysIndirectKHR-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdTraceRaysIndirectKHR-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdTraceRaysIndirectKHR-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdTraceRaysIndirectKHR-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdTraceRaysIndirectKHR-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdTraceRaysIndirectKHR-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either

VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdTraceRaysIndirectKHR-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdTraceRaysIndirectKHR-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdTraceRaysIndirectKHR-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdTraceRaysIndirectKHR-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdTraceRaysIndirectKHR-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdTraceRaysIndirectKHR-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to

must have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set

- VUID-vkCmdTraceRaysIndirectKHR-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdTraceRaysIndirectKHR-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdTraceRaysIndirectKHR-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdTraceRaysIndirectKHR-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdTraceRaysIndirectKHR-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered before this command was recorded, and still be registered during the sampling operation, with an identically defined color
- VUID-vkCmdTraceRaysIndirectKHR-protectedNoFault-11455

If protectedNoFault is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set

- VUID-vkCmdTraceRaysIndirectKHR-protectedNoFault-11456
  If protectedNoFault is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdTraceRaysIndirectKHR-None-03429
  Any shader group handle referenced by this call **must** have been queried from the bound ray tracing pipeline
- VUID-vkCmdTraceRaysIndirectKHR-None-09458
  If the bound ray tracing pipeline state was created with the VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR dynamic state enabled then vkCmdSetRayTracingPipelineStackSizeKHR **must** have been called in the current command buffer prior to this trace command
- VUID-vkCmdTraceRaysIndirectKHR-None-11319
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, the value of the address at the expected location in shader record data **must** be a valid address
- VUID-vkCmdTraceRaysIndirectKHR-maxPipelineRayRecursionDepth-03679
  This command **must** not cause a shader call instruction to be executed from a shader invocation with a recursion depth greater than the value of maxPipelineRayRecursionDepth used to create the bound ray tracing pipeline
- VUID-vkCmdTraceRaysIndirectKHR-commandBuffer-03635
  commandBuffer **must** not be a protected command buffer
- VUID-vkCmdTraceRaysIndirectKHR-micromap-11640
  If a VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR descriptor is accessed as a result of this command, the access is not through an OpRayQueryProceedKHR operation, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR or VkAccelerationStructureTrianglesOpacityMicromapEXT structure included in its pNext chain, the VkPipeline bound to the pipeline bind point used by this command **must** have been created with flags that includes VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR

- VUID-vkCmdTraceRaysIndirectKHR-size-04023
  The `size` member of `pRayGenShaderBindingTable` **must** be equal to its `stride` member
- VUID-vkCmdTraceRaysIndirectKHR-pRayGenShaderBindingTable-03681
  `pRaygenShaderBindingTable->deviceAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR usage flag set
- VUID-vkCmdTraceRaysIndirectKHR-pRayGenShaderBindingTable-03682
  `pRaygenShaderBindingTable->deviceAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment`
- VUID-vkCmdTraceRaysIndirectKHR-pMissShaderBindingTable-03684
  `pMissShaderBindingTable->deviceAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR usage flag set
- VUID-vkCmdTraceRaysIndirectKHR-pMissShaderBindingTable-03685
  `pMissShaderBindingTable->deviceAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment`
- VUID-vkCmdTraceRaysIndirectKHR-stride-03686
  `pMissShaderBindingTable->stride` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleAlignment`
- VUID-vkCmdTraceRaysIndirectKHR-stride-04029
  `pMissShaderBindingTable->stride` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxShaderGroupStride`
- VUID-vkCmdTraceRaysIndirectKHR-pHitShaderBindingTable-03688
  `pHitShaderBindingTable->deviceAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR usage flag set
- VUID-vkCmdTraceRaysIndirectKHR-pHitShaderBindingTable-03689
  `pHitShaderBindingTable->deviceAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment`
- VUID-vkCmdTraceRaysIndirectKHR-stride-03690
  `pHitShaderBindingTable->stride` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleAlignment`
- VUID-vkCmdTraceRaysIndirectKHR-stride-04035
  `pHitShaderBindingTable->stride` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxShaderGroupStride`
- VUID-vkCmdTraceRaysIndirectKHR-pCallableShaderBindingTable-03692
  `pCallableShaderBindingTable->deviceAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR usage flag set
- VUID-vkCmdTraceRaysIndirectKHR-pCallableShaderBindingTable-03693
  `pCallableShaderBindingTable->deviceAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment`

- VUID-vkCmdTraceRaysIndirectKHR-stride-03694
  pCallableShaderBindingTable->stride **must** be a multiple of VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleAlignment
- VUID-vkCmdTraceRaysIndirectKHR-stride-04041
  pCallableShaderBindingTable->stride **must** be less than or equal to VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxShaderGroupStride
- VUID-vkCmdTraceRaysIndirectKHR-flags-03511
  If the bound ray tracing pipeline was created with flags that included VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR, the entries in the table identified by pMissShaderBindingTable->deviceAddress accessed as a result of this command in order to execute a miss shader **must** not be zero
- VUID-vkCmdTraceRaysIndirectKHR-flags-03512
  If the bound ray tracing pipeline was created with flags that included VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR, entries in the table identified by pHitShaderBindingTable->deviceAddress accessed as a result of this command in order to execute an any-hit shader **must** not be zero
- VUID-vkCmdTraceRaysIndirectKHR-flags-03513
  If the bound ray tracing pipeline was created with flags that included VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR, entries in the table identified by pHitShaderBindingTable->deviceAddress accessed as a result of this command in order to execute a closest hit shader **must** not be zero
- VUID-vkCmdTraceRaysIndirectKHR-flags-03514
  If the bound ray tracing pipeline was created with flags that included VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR, entries in the table identified by pHitShaderBindingTable->deviceAddress accessed as a result of this command in order to execute an intersection shader **must** not be zero
- VUID-vkCmdTraceRaysIndirectKHR-pHitShaderBindingTable-04735
  Any non-zero hit shader group entries in the table identified by pHitShaderBindingTable->deviceAddress accessed by this call from a geometry with a geometryType of VK_GEOMETRY_TYPE_TRIANGLES_KHR **must** have been created with VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR
- VUID-vkCmdTraceRaysIndirectKHR-pHitShaderBindingTable-04736
  Any non-zero hit shader group entries in the table identified by pHitShaderBindingTable->deviceAddress accessed by this call from a geometry with a geometryType of VK_GEOMETRY_TYPE_AABBS_KHR **must** have been created with VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR
- VUID-vkCmdTraceRaysIndirectKHR-indirectDeviceAddress-03633
  indirectDeviceAddress **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set
- VUID-vkCmdTraceRaysIndirectKHR-indirectDeviceAddress-03634
  indirectDeviceAddress **must** be a multiple of 4
- VUID-vkCmdTraceRaysIndirectKHR-indirectDeviceAddress-03636
  All device addresses between indirectDeviceAddress and indirectDeviceAddress + sizeof

(VkTraceRaysIndirectCommandKHR) - 1 **must** be in the buffer device address range of the same buffer

- VUID-vkCmdTraceRaysIndirectKHR-rayTracingPipelineTraceRaysIndirect-03637
  The rayTracingPipelineTraceRaysIndirect feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdTraceRaysIndirectKHR-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdTraceRaysIndirectKHR-pRaygenShaderBindingTable-parameter
  pRaygenShaderBindingTable **must** be a valid pointer to a valid VkStridedDeviceAddressRegionKHR structure
- VUID-vkCmdTraceRaysIndirectKHR-pMissShaderBindingTable-parameter
  pMissShaderBindingTable **must** be a valid pointer to a valid VkStridedDeviceAddressRegionKHR structure
- VUID-vkCmdTraceRaysIndirectKHR-pHitShaderBindingTable-parameter
  pHitShaderBindingTable **must** be a valid pointer to a valid VkStridedDeviceAddressRegionKHR structure
- VUID-vkCmdTraceRaysIndirectKHR-pCallableShaderBindingTable-parameter
  pCallableShaderBindingTable **must** be a valid pointer to a valid VkStridedDeviceAddressRegionKHR structure
- VUID-vkCmdTraceRaysIndirectKHR-indirectDeviceAddress-parameter
  indirectDeviceAddress **must** be a valid VkDeviceAddress value
- VUID-vkCmdTraceRaysIndirectKHR-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdTraceRaysIndirectKHR-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdTraceRaysIndirectKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdTraceRaysIndirectKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdTraceRaysIndirectKHR-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdTraceRaysIndirectKHR is not affected by conditional rendering

The VkTraceRaysIndirectCommandKHR structure is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef struct VkTraceRaysIndirectCommandKHR {
    uint32_t    width;
    uint32_t    height;
    uint32_t    depth;
} VkTraceRaysIndirectCommandKHR;
```

- width is the width of the ray trace query dimensions.
- height is the height of the ray trace query dimensions.
- depth is the depth of the ray trace query dimensions.

The members of VkTraceRaysIndirectCommandKHR have the same meaning as the similarly named parameters of vkCmdTraceRaysKHR.

Valid Usage

- VUID-VkTraceRaysIndirectCommandKHR-width-03638
  width **must** be less than or equal to VkPhysicalDeviceLimits::maxComputeWorkGroupCount[0] × VkPhysicalDeviceLimits::maxComputeWorkGroupSize[0]
- VUID-VkTraceRaysIndirectCommandKHR-height-03639
  height **must** be less than or equal to VkPhysicalDeviceLimits::maxComputeWorkGroupCount[1] × VkPhysicalDeviceLimits::maxComputeWorkGroupSize[1]
- VUID-VkTraceRaysIndirectCommandKHR-depth-03640
  depth **must** be less than or equal to VkPhysicalDeviceLimits::maxComputeWorkGroupCount[2] × VkPhysicalDeviceLimits::maxComputeWorkGroupSize[2]
- VUID-VkTraceRaysIndirectCommandKHR-width-03641
  width × height × depth **must** be less than or equal to VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxRayDispatchInvocationCount

To dispatch ray tracing, with some parameters sourced on the device, use:

```
// Provided by VK_KHR_ray_tracing_maintenance1 with VK_KHR_ray_tracing_pipeline
void vkCmdTraceRaysIndirect2KHR(
    VkCommandBuffer                             commandBuffer,
    VkDeviceAddress                             indirectDeviceAddress);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `indirectDeviceAddress` is a buffer device address which is a pointer to a VkTraceRaysIndirectCommand2KHR structure containing the trace ray parameters.

`vkCmdTraceRaysIndirect2KHR` behaves similarly to vkCmdTraceRaysIndirectKHR except that shader binding table parameters as well as dispatch dimensions are read by the device from `indirectDeviceAddress` during execution.

Valid Usage

- VUID-vkCmdTraceRaysIndirect2KHR-None-08606
  If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdTraceRaysIndirect2KHR-None-08600
  If a bound shader was created with a pipeline layout and statically uses a set *n*, a descriptor set **must** have been bound to *n* at the same pipeline bind point, with layouts compatible for set *n* with the layout(s) used to create the shader, as described in Pipeline Layout Compatibility
- VUID-vkCmdTraceRaysIndirect2KHR-commandBuffer-02707
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, any resource accessed by bound shaders **must** not be a protected resource
- VUID-vkCmdTraceRaysIndirect2KHR-None-08114
  Descriptors in each bound descriptor set, specified via vkCmdBindDescriptorSets, **must** be valid if they are accessed as described by descriptor validity by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was not created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdTraceRaysIndirect2KHR-None-08115
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified via vkCmdBindDescriptorSets, the bound VkPipeline **must** have been created without VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdTraceRaysIndirect2KHR-None-08116
  Descriptors in bound descriptor buffers, specified via vkCmdSetDescriptorBufferOffsetsEXT, **must** be valid if they are dynamically used by the VkPipeline bound to the pipeline bind point used by this command and the bound VkPipeline was created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- VUID-vkCmdTraceRaysIndirect2KHR-None-08117
  If the descriptors used by the VkPipeline bound to the pipeline bind point were specified

via vkCmdSetDescriptorBufferOffsetsEXT, the bound VkPipeline **must** have been created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT

- VUID-vkCmdTraceRaysIndirect2KHR-None-08119
  If a descriptor is dynamically used with a VkPipeline created with VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, or a VkShaderEXT created with a `VkDescriptorSetLayout` that was created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT, the descriptor memory **must** be resident
- VUID-vkCmdTraceRaysIndirect2KHR-magFilter-04553
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdTraceRaysIndirect2KHR-magFilter-09598
  If a VkSampler created with `magFilter` or `minFilter` equal to VK_FILTER_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdTraceRaysIndirect2KHR-mipmapMode-04770
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR, `reductionMode` equal to VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE, and `compareEnable` equal to VK_FALSE is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
- VUID-vkCmdTraceRaysIndirect2KHR-mipmapMode-09599
  If a VkSampler created with `mipmapMode` equal to VK_SAMPLER_MIPMAP_MODE_LINEAR and `reductionMode` equal to either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX is used to sample a VkImageView as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- VUID-vkCmdTraceRaysIndirect2KHR-unnormalizedCoordinates-09635
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdTraceRaysIndirect2KHR-None-08609
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the image view's `viewType` **must** be VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D
- VUID-vkCmdTraceRaysIndirect2KHR-None-08610
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions with `ImplicitLod`, `Dref` or `Proj` in their name

- VUID-vkCmdTraceRaysIndirect2KHR-None-08611
  If a VkSampler created with `unnormalizedCoordinates` equal to VK_TRUE is used to sample a VkImageView as a result of this command, then the sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions that include a LOD bias or any offset values
- VUID-vkCmdTraceRaysIndirect2KHR-None-06479
  If a VkImageView is sampled with depth comparison, the image view's format features **must** contain VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT
- VUID-vkCmdTraceRaysIndirect2KHR-None-02691
  If a VkImageView is accessed using atomic operations as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
- VUID-vkCmdTraceRaysIndirect2KHR-None-07888
  If a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's format features **must** contain VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT
- VUID-vkCmdTraceRaysIndirect2KHR-None-02692
  If a VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, then the image view's format features **must** contain VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- VUID-vkCmdTraceRaysIndirect2KHR-None-02693
  If the VK_EXT_filter_cubic extension is not enabled and any VkImageView is sampled with VK_FILTER_CUBIC_EXT as a result of this command, it **must** not have a VkImageViewType of VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY
- VUID-vkCmdTraceRaysIndirect2KHR-filterCubic-02694
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT as a result of this command **must** have a VkImageViewType and format that supports cubic filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubic` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdTraceRaysIndirect2KHR-filterCubicMinmax-02695
  Any VkImageView being sampled with VK_FILTER_CUBIC_EXT with a reduction mode of either VK_SAMPLER_REDUCTION_MODE_MIN or VK_SAMPLER_REDUCTION_MODE_MAX as a result of this command **must** have a VkImageViewType and format that supports cubic filtering together with minmax filtering, as specified by VkFilterCubicImageViewImageFormatPropertiesEXT::`filterCubicMinmax` returned by vkGetPhysicalDeviceImageFormatProperties2
- VUID-vkCmdTraceRaysIndirect2KHR-OpTypeImage-07027
  For any VkImageView being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdTraceRaysIndirect2KHR-OpTypeImage-07028
  For any VkImageView being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain

VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT

- VUID-vkCmdTraceRaysIndirect2KHR-OpTypeImage-07029
  For any VkBufferView being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT
- VUID-vkCmdTraceRaysIndirect2KHR-OpTypeImage-07030
  For any VkBufferView being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT
- VUID-vkCmdTraceRaysIndirect2KHR-None-08601
  If a bound shader was created with a pipeline layout or specified push constant ranges and statically uses a push constant value, that value **must** have been set for the same pipeline bind point, with push constant ranges that are compatible with the push constant range used to create the shader
- VUID-vkCmdTraceRaysIndirect2KHR-None-10068
  For each array of resources that is used by a bound shader, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdTraceRaysIndirect2KHR-imageLayout-00344
  If an image descriptor is accessed by a shader, the VkImageLayout **must** match the subresource accessible from the VkImageView as defined by the image layout matching rules
- VUID-vkCmdTraceRaysIndirect2KHR-None-08608
  If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the VkPipeline object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdTraceRaysIndirect2KHR-uniformBuffers-06935
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdTraceRaysIndirect2KHR-None-08612
  If the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdTraceRaysIndirect2KHR-storageBuffers-06936
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer, and that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or

VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for storageBuffers, and the robustBufferAccess feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdTraceRaysIndirect2KHR-None-08613
  If the robustBufferAccess feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdTraceRaysIndirect2KHR-viewType-07752
  If a VkImageView is accessed as a result of this command, then the image view's viewType **must** match the Dim operand of the OpTypeImage as described in Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types
- VUID-vkCmdTraceRaysIndirect2KHR-format-07753
  If a VkImageView or VkBufferView is accessed as a result of this command, then the numeric type of the view's format and the Sampled Type operand of the OpTypeImage **must** match
- VUID-vkCmdTraceRaysIndirect2KHR-OpImageWrite-08795
  If a VkImageView created with a format other than VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdTraceRaysIndirect2KHR-OpImageWrite-08796
  If a VkImageView created with the format VK_FORMAT_A8_UNORM is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have four components
- VUID-vkCmdTraceRaysIndirect2KHR-OpImageWrite-04469
  If a VkBufferView is accessed using OpImageWrite as a result of this command, then the Type of the Texel operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdTraceRaysIndirect2KHR-SampledType-04470
  If a VkImageView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdTraceRaysIndirect2KHR-SampledType-04471
  If a VkImageView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 32
- VUID-vkCmdTraceRaysIndirect2KHR-SampledType-04472
  If a VkBufferView with a VkFormat that has a 64-bit component width is accessed as a result of this command, the SampledType of the OpTypeImage operand of that instruction **must** have a Width of 64
- VUID-vkCmdTraceRaysIndirect2KHR-SampledType-04473
  If a VkBufferView with a VkFormat that has a component width less than 64-bit is accessed as a result of this command, the SampledType of the OpTypeImage operand of that

instruction **must** have a `Width` of 32

- VUID-vkCmdTraceRaysIndirect2KHR-sparseImageInt64Atomics-04474
  If the `sparseImageInt64Atomics` feature is not enabled, VkImage objects created with the VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdTraceRaysIndirect2KHR-sparseImageInt64Atomics-04475
  If the `sparseImageInt64Atomics` feature is not enabled, VkBuffer objects created with the VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdTraceRaysIndirect2KHR-None-07288
  Any shader invocation executed by this command **must** terminate
- VUID-vkCmdTraceRaysIndirect2KHR-None-09600
  If a descriptor with type equal to any of VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT is accessed as a result of this command, all image subresources identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdTraceRaysIndirect2KHR-micromap-11636
  If an acceleration structure is accessed as a result of this command through an `OpRayQueryProceedKHR` operation , the `VK_EXT_opacity_micromap` extension is not enabled, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR structure included in its `pNext` chain, the shader **must** have specified a `OpacityMicromapIdKHR` execution mode with its specialization constant operand equal to VK_TRUE
- VUID-vkCmdTraceRaysIndirect2KHR-micromap-11637
  If an acceleration structure is accessed as a result of this command and the acceleration structure was constructed through deserialization, all micromap arrays referenced by the acceleration structure that were not replaced by an acceleration structure update command **must** have been deserialized using the serialized data of the corresponding micromaps used to originally build the acceleration structure prior to the execution of this command on the device
- VUID-vkCmdTraceRaysIndirect2KHR-micromap-11638
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap array, any VkPipeline bound to the pipeline bind point used by this command **must** not have been created with `flags` that includes VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
- VUID-vkCmdTraceRaysIndirect2KHR-micromap-11639
  If an acceleration structure is accessed as a result of this command and the acceleration structure is built using geometry with an `indexBuffer` containing both VkOpacityMicromapSpecialIndexKHR values and indices into an associated micromap

array, any VkShaderEXT that accessed the descriptor **must** not have been created with `flags` that includes VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT

- VUID-vkCmdTraceRaysIndirect2KHR-None-12472
  Descriptors dynamically accessed in bound descriptor heaps **must** be valid
- VUID-vkCmdTraceRaysIndirect2KHR-None-11297
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a `OpTypeStruct` decorated with `Block` or `BufferBlock` using that mapping, the calculated offset for the resource heap **must** be a multiple of `bufferDescriptorAlignment`
- VUID-vkCmdTraceRaysIndirect2KHR-None-11298
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeImage` or `OpTypeSampledImage` using that mapping, the calculated offset for the resource heap **must** be a multiple of `imageDescriptorAlignment`
- VUID-vkCmdTraceRaysIndirect2KHR-None-11299
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_PUSH_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses an `OpTypeSampler` or `OpTypeSampledImage` using that mapping, the calculated offset for the sampler heap **must** be a multiple of `samplerDescriptorAlignment`
- VUID-vkCmdTraceRaysIndirect2KHR-None-11300
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 4
- VUID-vkCmdTraceRaysIndirect2KHR-None-11301
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location

in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping

- VUID-vkCmdTraceRaysIndirect2KHR-None-11302
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdTraceRaysIndirect2KHR-None-11304
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a multiple of 8
- VUID-vkCmdTraceRaysIndirect2KHR-None-11305
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address at the expected location in push data **must** be a valid VkDeviceAddress backed by physical memory at every offset specified by each mapping
- VUID-vkCmdTraceRaysIndirect2KHR-None-11306
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the value of the address pointed to by the address in push data **must** be a valid VkDeviceAddress
- VUID-vkCmdTraceRaysIndirect2KHR-None-11308
  For each descriptor heap that is statically used by a bound shader, either directly or via a descriptor mapping, a valid descriptor heap **must** be bound
- VUID-vkCmdTraceRaysIndirect2KHR-None-11309
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, execution of this command **must** not result in any descriptor read accessing data outside of the user range of the respective heap bound by `vkCmdBind*HeapEXT` commands
- VUID-vkCmdTraceRaysIndirect2KHR-None-11372
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a uniform buffer or uniform texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `uniformBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified via VkDeviceAddressRangeKHR when the descriptor was written

- VUID-vkCmdTraceRaysIndirect2KHR-None-11373
  If any stage of the VkPipeline object bound to the pipeline bind point used by this command accesses a storage buffer or storage texel buffer through a descriptor in the bound resource heap, that stage was created without enabling either VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS or VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2 for `storageBuffers`, the `robustBufferAccess2` feature is not enabled, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the descriptor specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdTraceRaysIndirect2KHR-None-11374
  If the `robustBufferAccess2` feature is not enabled, the `robustBufferAccess` feature is not enabled, and any VkShaderEXT bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, uniform texel buffer, storage buffer, or storage texel buffer, that shader **must** not access values outside of the range of the buffer as specified by VkDeviceAddressRangeKHR when the descriptor was written
- VUID-vkCmdTraceRaysIndirect2KHR-pBindInfo-11375
  If any bound shader uses an embedded sampler via a descriptor mapping, the value of `pBindInfo->reservedRangeSize` set for vkCmdBindSamplerHeapEXT **must** be greater than or equal to `minSamplerHeapReservedRangeWithEmbedded`
- VUID-vkCmdTraceRaysIndirect2KHR-None-11376
  If a bound shader was created as a VkShaderEXT with the VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT flag or as part of a pipeline with the VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT flag, and that shader statically uses a push constant value, that value **must** have been set by vkCmdPushDataEXT
- VUID-vkCmdTraceRaysIndirect2KHR-None-11398
  If a bound shader was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the access **must** not be out of bounds
- VUID-vkCmdTraceRaysIndirect2KHR-None-11437
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a resource using that mapping, the buffer from which the address in push data was queried **must** have been created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set
- VUID-vkCmdTraceRaysIndirect2KHR-None-11438
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping

using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT usage flag set

- VUID-vkCmdTraceRaysIndirect2KHR-None-11441
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT and a shader accesses a uniform buffer using that mapping, the address that the uniform buffer is mapped to **must** be aligned to `minUniformBufferOffsetAlignment`
- VUID-vkCmdTraceRaysIndirect2KHR-None-11439
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** have been queried from a buffer created with the VK_BUFFER_USAGE_STORAGE_BUFFER_BIT usage flag set
- VUID-vkCmdTraceRaysIndirect2KHR-None-11442
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses a storage buffer using that mapping, the address that the storage buffer is mapped to **must** be aligned to `minStorageBufferOffsetAlignment`
- VUID-vkCmdTraceRaysIndirect2KHR-None-11485
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, and a shader accesses an acceleration structure using that mapping, the address that the acceleration structure is mapped to **must** be an acceleration structure address retrieved from a VkAccelerationStructureKHR object via vkGetAccelerationStructureDeviceAddressKHR
- VUID-vkCmdTraceRaysIndirect2KHR-index-11450
  If a shader uses a sampler descriptor to sample an image as a result of this command, and that sampler descriptor uses a custom border color with an index defined by VkSamplerCustomBorderColorIndexCreateInfoEXT, the value of VkSamplerCustomBorderColorIndexCreateInfoEXT::`index` **must** have been registered

before this command was recorded, and still be registered during the sampling operation, with an identically defined color

- VUID-vkCmdTraceRaysIndirect2KHR-protectedNoFault-11455
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, the address that the resource is mapped to **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdTraceRaysIndirect2KHR-protectedNoFault-11456
  If `protectedNoFault` is not supported, a pipeline is bound to the pipeline bind point used by this command, or a shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_EXT or VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_INDIRECT_INDEX_ARRAY_EXT, the address of the indirect memory **must** have been queried from a buffer created without the VK_BUFFER_CREATE_PROTECTED_BIT create flag set
- VUID-vkCmdTraceRaysIndirect2KHR-None-03429
  Any shader group handle referenced by this call **must** have been queried from the bound ray tracing pipeline
- VUID-vkCmdTraceRaysIndirect2KHR-None-09458
  If the bound ray tracing pipeline state was created with the VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR dynamic state enabled then vkCmdSetRayTracingPipelineStackSizeKHR **must** have been called in the current command buffer prior to this trace command
- VUID-vkCmdTraceRaysIndirect2KHR-None-11319
  If a pipeline is bound to the pipeline bind point used by this command, or shader is bound to a shader stage used by this command, and it was created with a descriptor mapping using VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, the value of the address at the expected location in shader record data **must** be a valid address
- VUID-vkCmdTraceRaysIndirect2KHR-maxPipelineRayRecursionDepth-03679
  This command **must** not cause a shader call instruction to be executed from a shader invocation with a recursion depth greater than the value of `maxPipelineRayRecursionDepth` used to create the bound ray tracing pipeline
- VUID-vkCmdTraceRaysIndirect2KHR-commandBuffer-03635
  `commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdTraceRaysIndirect2KHR-micromap-11640
  If a VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR descriptor is accessed as a result of this command, the access is not through an `OpRayQueryProceedKHR` operation, and the acceleration structure is built using geometry with a VkAccelerationStructureTrianglesOpacityMicromapKHR or VkAccelerationStructureTrianglesOpacityMicromapEXT structure included in its `pNext`

chain, the VkPipeline bound to the pipeline bind point used by this command **must** have been created with `flags` that includes VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR

- VUID-vkCmdTraceRaysIndirect2KHR-indirectDeviceAddress-03633
`indirectDeviceAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set
- VUID-vkCmdTraceRaysIndirect2KHR-indirectDeviceAddress-03634
`indirectDeviceAddress` **must** be a multiple of `4`
- VUID-vkCmdTraceRaysIndirect2KHR-indirectDeviceAddress-03636
All device addresses between `indirectDeviceAddress` and `indirectDeviceAddress` + `sizeof` (`VkTraceRaysIndirectCommand2KHR`) - 1 **must** be in the buffer device address range of the same buffer
- VUID-vkCmdTraceRaysIndirect2KHR-rayTracingPipelineTraceRaysIndirect2-03637
The `rayTracingPipelineTraceRaysIndirect2` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdTraceRaysIndirect2KHR-commandBuffer-parameter
`commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdTraceRaysIndirect2KHR-indirectDeviceAddress-parameter
`indirectDeviceAddress` **must** be a valid VkDeviceAddress value
- VUID-vkCmdTraceRaysIndirect2KHR-commandBuffer-recording
`commandBuffer` **must** be in the recording state
- VUID-vkCmdTraceRaysIndirect2KHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT operations
- VUID-vkCmdTraceRaysIndirect2KHR-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdTraceRaysIndirect2KHR-suspended
This command **must** not be called between suspended render pass instances
- VUID-vkCmdTraceRaysIndirect2KHR-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT	Action

Conditional Rendering

vkCmdTraceRaysIndirect2KHR is not affected by conditional rendering

The VkTraceRaysIndirectCommand2KHR structure is defined as:

```
// Provided by VK_KHR_ray_tracing_maintenance1 with VK_KHR_ray_tracing_pipeline
typedef struct VkTraceRaysIndirectCommand2KHR {
    VkDeviceAddress    raygenShaderRecordAddress;
    VkDeviceSize       raygenShaderRecordSize;
    VkDeviceAddress    missShaderBindingTableAddress;
    VkDeviceSize       missShaderBindingTableSize;
    VkDeviceSize       missShaderBindingTableStride;
    VkDeviceAddress    hitShaderBindingTableAddress;
    VkDeviceSize       hitShaderBindingTableSize;
    VkDeviceSize       hitShaderBindingTableStride;
    VkDeviceAddress    callableShaderBindingTableAddress;
    VkDeviceSize       callableShaderBindingTableSize;
    VkDeviceSize       callableShaderBindingTableStride;
    uint32_t           width;
    uint32_t           height;
    uint32_t           depth;
} VkTraceRaysIndirectCommand2KHR;
```

- raygenShaderRecordAddress is a VkDeviceAddress of the ray generation shader binding table record used by this command.
- raygenShaderRecordSize is a VkDeviceSize number of bytes corresponding to the ray generation shader binding table record at base address raygenShaderRecordAddress.
- missShaderBindingTableAddress is a VkDeviceAddress of the first record in the miss shader binding table used by this command.
- missShaderBindingTableSize is a VkDeviceSize number of bytes corresponding to the total size of the miss shader binding table at missShaderBindingTableAddress that may be accessed by this command.
- missShaderBindingTableStride is a VkDeviceSize number of bytes between records of the miss shader binding table.
- hitShaderBindingTableAddress is a VkDeviceAddress of the first record in the hit shader binding

table used by this command.

- `hitShaderBindingTableSize` is a VkDeviceSize number of bytes corresponding to the total size of the hit shader binding table at `hitShaderBindingTableAddress` that may be accessed by this command.
- `hitShaderBindingTableStride` is a VkDeviceSize number of bytes between records of the hit shader binding table.
- `callableShaderBindingTableAddress` is a VkDeviceAddress of the first record in the callable shader binding table used by this command.
- `callableShaderBindingTableSize` is a VkDeviceSize number of bytes corresponding to the total size of the callable shader binding table at `callableShaderBindingTableAddress` that may be accessed by this command.
- `callableShaderBindingTableStride` is a VkDeviceSize number of bytes between records of the callable shader binding table.
- `width` is the width of the ray trace query dimensions.
- `height` is the height of the ray trace query dimensions.
- `depth` is the depth of the ray trace query dimensions.

The members of `VkTraceRaysIndirectCommand2KHR` have the same meaning as the similarly named parameters of vkCmdTraceRaysKHR.

Indirect shader binding table buffer parameters **must** satisfy the same memory alignment and binding requirements as their counterparts in vkCmdTraceRaysIndirectKHR and vkCmdTraceRaysKHR.

Valid Usage

- VUID-VkTraceRaysIndirectCommand2KHR-pRayGenShaderBindingTable-03681 `raygenShaderRecordAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR usage flag set
- VUID-VkTraceRaysIndirectCommand2KHR-pRayGenShaderBindingTable-03682 `raygenShaderRecordAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR`::`shaderGroupBaseAlignment`
- VUID-VkTraceRaysIndirectCommand2KHR-pMissShaderBindingTable-03684 `missShaderBindingTableAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR usage flag set
- VUID-VkTraceRaysIndirectCommand2KHR-pMissShaderBindingTable-03685 `missShaderBindingTableAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR`::`shaderGroupBaseAlignment`
- VUID-VkTraceRaysIndirectCommand2KHR-stride-03686 `missShaderBindingTableStride` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR`::`shaderGroupHandleAlignment`

- VUID-VkTraceRaysIndirectCommand2KHR-stride-04029
  `missShaderBindingTableStride` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR`::`maxShaderGroupStride`
- VUID-VkTraceRaysIndirectCommand2KHR-pHitShaderBindingTable-03688
  `hitShaderBindingTableAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR usage flag set
- VUID-VkTraceRaysIndirectCommand2KHR-pHitShaderBindingTable-03689
  `hitShaderBindingTableAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR`::`shaderGroupBaseAlignment`
- VUID-VkTraceRaysIndirectCommand2KHR-stride-03690
  `hitShaderBindingTableStride` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR`::`shaderGroupHandleAlignment`
- VUID-VkTraceRaysIndirectCommand2KHR-stride-04035
  `hitShaderBindingTableStride` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR`::`maxShaderGroupStride`
- VUID-VkTraceRaysIndirectCommand2KHR-pCallableShaderBindingTable-03692
  `callableShaderBindingTableAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR usage flag set
- VUID-VkTraceRaysIndirectCommand2KHR-pCallableShaderBindingTable-03693
  `callableShaderBindingTableAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR`::`shaderGroupBaseAlignment`
- VUID-VkTraceRaysIndirectCommand2KHR-stride-03694
  `callableShaderBindingTableStride` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR`::`shaderGroupHandleAlignment`
- VUID-VkTraceRaysIndirectCommand2KHR-stride-04041
  `callableShaderBindingTableStride` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR`::`maxShaderGroupStride`
- VUID-VkTraceRaysIndirectCommand2KHR-flags-03511
  If the bound ray tracing pipeline was created with `flags` that included VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR, the entries in the table identified by `missShaderBindingTableAddress` accessed as a result of this command in order to execute a miss shader **must** not be zero
- VUID-VkTraceRaysIndirectCommand2KHR-flags-03512
  If the bound ray tracing pipeline was created with `flags` that included VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR, entries in the table identified by `hitShaderBindingTableAddress` accessed as a result of this command in order to execute an any-hit shader **must** not be zero
- VUID-VkTraceRaysIndirectCommand2KHR-flags-03513
  If the bound ray tracing pipeline was created with `flags` that included VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR, entries in the table identified by `hitShaderBindingTableAddress` accessed as a result of this

command in order to execute a closest hit shader **must** not be zero

- VUID-VkTraceRaysIndirectCommand2KHR-flags-03514
  If the bound ray tracing pipeline was created with `flags` that included VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR, entries in the table identified by `hitShaderBindingTableAddress` accessed as a result of this command in order to execute an intersection shader **must** not be zero
- VUID-VkTraceRaysIndirectCommand2KHR-pHitShaderBindingTable-04735
  Any non-zero hit shader group entries in the table identified by `hitShaderBindingTableAddress` accessed by this call from a geometry with a `geometryType` of VK_GEOMETRY_TYPE_TRIANGLES_KHR **must** have been created with VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR
- VUID-VkTraceRaysIndirectCommand2KHR-pHitShaderBindingTable-04736
  Any non-zero hit shader group entries in the table identified by `hitShaderBindingTableAddress` accessed by this call from a geometry with a `geometryType` of VK_GEOMETRY_TYPE_AABBS_KHR **must** have been created with VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR
- VUID-VkTraceRaysIndirectCommand2KHR-width-03638
  `width` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxComputeWorkGroupCount`[0] × `VkPhysicalDeviceLimits`::`maxComputeWorkGroupSize`[0]
- VUID-VkTraceRaysIndirectCommand2KHR-height-03639
  `height` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxComputeWorkGroupCount`[1] × `VkPhysicalDeviceLimits`::`maxComputeWorkGroupSize`[1]
- VUID-VkTraceRaysIndirectCommand2KHR-depth-03640
  `depth` **must** be less than or equal to `VkPhysicalDeviceLimits`::`maxComputeWorkGroupCount`[2] × `VkPhysicalDeviceLimits`::`maxComputeWorkGroupSize`[2]
- VUID-VkTraceRaysIndirectCommand2KHR-width-03641
  `width` × `height` × `depth` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR`::`maxRayDispatchInvocationCount`

Valid Usage (Implicit)

- VUID-VkTraceRaysIndirectCommand2KHR-raygenShaderRecordAddress-parameter
  `raygenShaderRecordAddress` **must** be a valid VkDeviceAddress value
- VUID-VkTraceRaysIndirectCommand2KHR-missShaderBindingTableAddress-parameter
  `missShaderBindingTableAddress` **must** be a valid VkDeviceAddress value
- VUID-VkTraceRaysIndirectCommand2KHR-hitShaderBindingTableAddress-parameter
  `hitShaderBindingTableAddress` **must** be a valid VkDeviceAddress value
- VUID-VkTraceRaysIndirectCommand2KHR-callableShaderBindingTableAddress-parameter
  `callableShaderBindingTableAddress` **must** be a valid VkDeviceAddress value

43.3. Shader Binding Table

A *shader binding table* is a resource which establishes the relationship between the ray tracing pipeline and the acceleration structures that were built for the ray tracing pipeline. It indicates the shaders that operate on each geometry in an acceleration structure. In addition, it contains the resources accessed by each shader, including indices of textures, buffer device addresses, and constants. The application allocates and manages *shader binding tables* as VkBuffer objects.

Each entry in the shader binding table consists of `shaderGroupHandleSize` bytes of data, either as queried by vkGetRayTracingShaderGroupHandlesKHR to refer to those specified shaders, or all zeros to refer to a zero shader group. A zero shader group behaves as though it is a shader group consisting entirely of VK_SHADER_UNUSED_KHR. The remainder of the data specified by the stride is application-visible data that can be referenced by a `ShaderRecordBufferKHR` block in the shader.

The shader binding tables to use in a ray tracing pipeline are passed to the vkCmdTraceRaysKHR, or vkCmdTraceRaysIndirectKHR commands. Shader binding tables are read-only in shaders that are executing on the ray tracing pipeline.

Shader variables identified with the `ShaderRecordBufferKHR` storage class are used to access the provided shader binding table. Such variables **must** be:

- typed as `OpTypeStruct`, or an array of this type,
- identified with a `Block` decoration, and
- laid out explicitly using the `Offset`, `ArrayStride`, and `MatrixStride` decorations as specified in Offset and Stride Assignment.

The `Offset` decoration for any member of a `Block`-decorated variable in the `ShaderRecordBufferKHR` storage class **must** not cause the space required for that variable to extend outside the range [0, `maxStorageBufferRange`).

Accesses to the shader binding table from ray tracing pipelines **must** be synchronized with the VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR pipeline stage and an access type of VK_ACCESS_SHADER_READ_BIT.

NOTE

Because different shader record buffers can be associated with the same shader, a shader variable with `ShaderRecordBufferKHR` storage class will not be dynamically uniform if different invocations of the same shader can reference different data in the shader record buffer, such as if the same shader occurs twice in the shader binding table with a different shader record buffer. In this case, if the `DescriptorHeapEXT` capability is not declared, when indexing resources based on values in the `ShaderRecordBufferKHR` storage class, the index should be decorated as `NonUniform`.

43.3.1. Indexing Rules

In order to execute the correct shaders and access the correct resources during a ray tracing dispatch, the implementation **must** be able to locate shader binding table entries at various stages of execution. This is accomplished by defining a set of indexing rules that compute shader binding

table record positions relative to the buffer's base address in memory. The application **must** organize the contents of the shader binding table's memory in a way that application of the indexing rules will lead to correct records.

Ray Generation Shaders

Only one ray generation shader is executed per ray tracing dispatch.

For vkCmdTraceRaysKHR, the location of the ray generation shader is specified by the `pRaygenShaderBindingTable->deviceAddress` parameter — there is no indexing. All data accessed **must** be less than `pRaygenShaderBindingTable->size` bytes from `deviceAddress`. `pRaygenShaderBindingTable->stride` is unused, and **must** be equal to `pRaygenShaderBindingTable->size`.

Hit Shaders

The base for the computation of intersection, any-hit, and closest hit shader locations is the `instanceShaderBindingTableRecordOffset` value stored with each instance of a top-level acceleration structure (VkAccelerationStructureInstanceKHR). This value determines the beginning of the shader binding table records for a given instance.

In the following rule, `geometryIndex` refers to the geometry index of the intersected geometry within the instance.

The `sbtRecordOffset` and `sbtRecordStride` values are passed in as parameters to `traceRayEXT`() calls made in the shaders. See Section 8.19 (Ray Tracing Functions) of the OpenGL Shading Language Specification for more details. In SPIR-V, these correspond to the `SBTOffset` and `SBTStride` parameters to the pipeline trace ray instructions.

The result of this computation is then added to `pHitShaderBindingTable->deviceAddress`, a device address passed to vkCmdTraceRaysKHR .

For vkCmdTraceRaysKHR, the complete rule to compute a hit shader binding table record address in the `pHitShaderBindingTable` is:

```
pHitShaderBindingTable->deviceAddress + pHitShaderBindingTable->stride × ( instanceShaderBindingTableRecordOffset + geometryIndex × sbtRecordStride + sbtRecordOffset )
```

All data accessed **must** be less than `pHitShaderBindingTable->size` bytes from the base address.

Miss Shaders

A miss shader is executed whenever a ray query fails to find an intersection for the given scene geometry. Multiple miss shaders **may** be executed throughout a ray tracing dispatch.

The base for the computation of miss shader locations is `pMissShaderBindingTable->deviceAddress`, a device address passed into vkCmdTraceRaysKHR .

The `missIndex` value is passed in as a parameter to `traceRayEXT`() calls made in the shaders. See Section 8.19 (Ray Tracing Functions) of the OpenGL Shading Language Specification for more

details. In SPIR-V, this corresponds to the MissIndex parameter to the pipeline trace ray instructions.

For vkCmdTraceRaysKHR, the complete rule to compute a miss shader binding table record address in the pMissShaderBindingTable is:

pMissShaderBindingTable->deviceAddress + pMissShaderBindingTable->stride × missIndex

All data accessed **must** be less than pMissShaderBindingTable->size bytes from the base address.

Callable Shaders

A callable shader is executed when requested by a ray tracing shader. Multiple callable shaders **may** be executed throughout a ray tracing dispatch.

The base for the computation of callable shader locations is pCallableShaderBindingTable->deviceAddress, a device address passed into vkCmdTraceRaysKHR .

The sbtRecordIndex value is passed in as a parameter to executeCallableEXT() calls made in the shaders. See Section 8.19 (Ray Tracing Functions) of the OpenGL Shading Language Specification for more details. In SPIR-V, this corresponds to the SBTIndex parameter to the OpExecuteCallableKHR instruction.

For vkCmdTraceRaysKHR, the complete rule to compute a callable shader binding table record address in the pCallableShaderBindingTable is:

pCallableShaderBindingTable->deviceAddress + pCallableShaderBindingTable->stride × sbtRecordIndex

All data accessed **must** be less than pCallableShaderBindingTable->size bytes from the base address.

43.4. Ray Tracing Pipeline Stack

Ray tracing pipelines have a potentially large set of shaders which **may** be invoked in various call chain combinations to perform ray tracing. To store parameters for a given shader execution, an implementation **may** use a stack of data in memory. This stack **must** be sized to the sum of the stack sizes of all shaders in any call chain executed by the application.

If the stack size is not set explicitly, the stack size for a pipeline is:

rayGenStackMax + min(1, maxPipelineRayRecursionDepth) × max(closestHitStackMax, missStackMax, intersectionStackMax + anyHitStackMax) + max(0, maxPipelineRayRecursionDepth -1) × max(closestHitStackMax, missStackMax) + 2 × callableStackMax

where rayGenStackMax, closestHitStackMax, missStackMax, anyHitStackMax, intersectionStackMax, and callableStackMax are the maximum stack values queried by the respective shader stages for any shaders in any shader groups defined by the pipeline.

This stack size is potentially significant, so an application **may** want to provide a more accurate stack size after pipeline compilation. The value that the application provides is the maximum value of the sum of all shaders in a call chain across all possible call chains, taking into account any application specific knowledge about the properties of the call chains.

NOTE

For example, if an application has two types of closest hit and miss shaders that it can use but the first level of rays will only use the first kind (possibly reflection) and the second level will only use the second kind (occlusion or shadow ray, for example) then the application can compute the stack size by something similar to:

```
rayGenStack + max(closestHit1Stack, miss1Stack) + max(closestHit2Stack, miss2Stack)
```

This is guaranteed to be no larger than the default stack size computation which assumes that both call levels may be the larger of the two.

43.5. Ray Tracing Capture Replay

In a similar way to bufferDeviceAddressCaptureReplay, the `rayTracingPipelineShaderGroupHandleCaptureReplay` feature allows the querying of opaque data which **can** be used in a future replay.

During the capture phase, capture/replay tools are expected to query opaque data for shader group handle replay using vkGetRayTracingCaptureReplayShaderGroupHandlesKHR.

Providing the opaque data during replay, using VkRayTracingShaderGroupCreateInfoKHR ::`pShaderGroupCaptureReplayHandle` at pipeline creation time, causes the implementation to generate identical shader group handles to those in the capture phase, allowing capture/replay tools to reuse previously recorded shader binding table buffer contents or to obtain the same handles by calling vkGetRayTracingCaptureReplayShaderGroupHandlesKHR again.

Chapter 44. Memory Decompression

Commands that **can** be used to decompress memory between one or more memory regions are vkCmdDecompressMemoryEXT, and vkCmdDecompressMemoryIndirectCountEXT.

To decompress memory containing compressed data, call:

```
// Provided by VK_EXT_memory_decompression
void vkCmdDecompressMemoryEXT(
    VkCommandBuffer                             commandBuffer,
    const VkDecompressMemoryInfoEXT*            pDecompressMemoryInfoEXT);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pDecompressMemoryInfoEXT` is a pointer to a VkDecompressMemoryInfoEXT structure describing the decompression parameters.

Valid Usage

- VUID-vkCmdDecompressMemoryEXT-memoryDecompression-11761
  The `memoryDecompression` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdDecompressMemoryEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDecompressMemoryEXT-pDecompressMemoryInfoEXT-parameter
  `pDecompressMemoryInfoEXT` **must** be a valid pointer to a valid VkDecompressMemoryInfoEXT structure
- VUID-vkCmdDecompressMemoryEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdDecompressMemoryEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDecompressMemoryEXT-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdDecompressMemoryEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDecompressMemoryEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDecompressMemoryEXT is not affected by conditional rendering

The `VkDecompressMemoryInfoEXT` structure is defined as:

```
// Provided by VK_EXT_memory_decompression
typedef struct VkDecompressMemoryInfoEXT {
    VkStructureType                          sType;
    const void*                              pNext;
    VkMemoryDecompressionMethodFlagsEXT      decompressionMethod;
    uint32_t                                 regionCount;
    const VkDecompressMemoryRegionEXT*       pRegions;
} VkDecompressMemoryInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `decompressionMethod` is a bitmask of `VkMemoryDecompressionMethodFlagBitsEXT` with a single bit set specifying the method used to decompress data.
- `regionCount` is the number of regions to decompress.
- `pRegions` is a pointer to an array of `VkDecompressMemoryRegionEXT` structures specifying the regions to decompress.

Each memory region specified in `pRegions` is decompressed from the source to the destination address based on the decompression method specified in `decompressionMethod`. If any of the specified source and destination regions overlap in memory, then the results of decompression are

undefined.

Valid Usage

- VUID-VkDecompressMemoryInfoEXT-decompressionMethod-07690
  The `decompressionMethod` **must** have a single bit set
- VUID-VkDecompressMemoryInfoEXT-decompressionMethod-11762
  If `decompressionMethod` is VK_MEMORY_DECOMPRESSION_METHOD_GDEFLATE_1_0_BIT_EXT, then for each element of `pRegions`, `decompressedSize` **must** be less than or equal to 65536 bytes
- VUID-VkDecompressMemoryInfoEXT-decompressionMethod-11763
  `decompressionMethod` **must** be a valid bit specified in `VkPhysicalDeviceMemoryDecompressionPropertiesEXT`::`decompressionMethods`

Valid Usage (Implicit)

- VUID-VkDecompressMemoryInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DECOMPRESS_MEMORY_INFO_EXT
- VUID-VkDecompressMemoryInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDecompressMemoryInfoEXT-decompressionMethod-parameter
  `decompressionMethod` **must** be a valid combination of VkMemoryDecompressionMethodFlagBitsEXT values
- VUID-VkDecompressMemoryInfoEXT-pRegions-parameter
  `pRegions` **must** be a valid pointer to an array of `regionCount` VkDecompressMemoryRegionEXT structures
- VUID-VkDecompressMemoryInfoEXT-regionCount-arraylength
  `regionCount` **must** be greater than `0`

The `VkDecompressMemoryRegionEXT` structure is defined as:

```
// Provided by VK_EXT_memory_decompression
typedef struct VkDecompressMemoryRegionEXT {
    VkDeviceAddress    srcAddress;
    VkDeviceAddress    dstAddress;
    VkDeviceSize       compressedSize;
    VkDeviceSize       decompressedSize;
} VkDecompressMemoryRegionEXT;
```

- `srcAddress` is the address where compressed data is stored.
- `dstAddress` is the destination address where decompressed data will be written.
- `compressedSize` is the size of compressed data in bytes.

- `decompressedSize` is the size of decompressed data in bytes.

Accesses to compressed and decompressed data specified in `srcAddress` and `dstAddress` **must** be synchronized with the VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT pipeline stage with access type of VK_ACCESS_2_MEMORY_DECOMPRESSION_READ_BIT_EXT or VK_ACCESS_2_MEMORY_DECOMPRESSION_WRITE_BIT_EXT.

Valid Usage

- VUID-VkDecompressMemoryRegionEXT-srcAddress-07685
  `srcAddress` **must** be 4 byte aligned
- VUID-VkDecompressMemoryRegionEXT-srcAddress-07686
  The memory range defined by `srcAddress` and `compressedSize` **must** be contained within the size of the buffer bound to `srcAddress`, minus the offset of `srcAddress` from the base address of that buffer
- VUID-VkDecompressMemoryRegionEXT-dstAddress-07687
  `dstAddress` **must** be 4 byte aligned
- VUID-VkDecompressMemoryRegionEXT-dstAddress-07688
  The memory range defined by `dstAddress` and `decompressedSize` **must** be contained within the size of the buffer bound to `dstAddress`, minus the offset of `dstAddress` from the base address of that buffer
- VUID-VkDecompressMemoryRegionEXT-decompressedSize-07689
  `decompressedSize` **must** be large enough to hold the decompressed data based on the `decompressionMethod`
- VUID-VkDecompressMemoryRegionEXT-compressedSize-11795
  `compressedSize` **must** not be zero
- VUID-VkDecompressMemoryRegionEXT-decompressedSize-11796
  `decompressedSize` **must** not be zero
- VUID-VkDecompressMemoryRegionEXT-srcAddress-07691
  The memory range defined by `srcAddress` and `compressedSize` **must** not overlap the memory range defined by `dstAddress` and `decompressedSize`
- VUID-VkDecompressMemoryRegionEXT-srcAddress-11764
  `srcAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_2_MEMORY_DECOMPRESSION_BIT_EXT usage flag set
- VUID-VkDecompressMemoryRegionEXT-dstAddress-11765
  `dstAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_2_MEMORY_DECOMPRESSION_BIT_EXT usage flag set

Valid Usage (Implicit)

- VUID-VkDecompressMemoryRegionEXT-srcAddress-parameter
  `srcAddress` **must** be a valid VkDeviceAddress value

- VUID-VkDecompressMemoryRegionEXT-dstAddress-parameter
  `dstAddress` **must** be a valid VkDeviceAddress value

To decompress data between one or more memory regions by specifying decompression parameters indirectly in a buffer, call:

```
// Provided by VK_EXT_memory_decompression
void vkCmdDecompressMemoryIndirectCountEXT(
    VkCommandBuffer                             commandBuffer,
    VkMemoryDecompressionMethodFlagsEXT         decompressionMethod,
    VkDeviceAddress                             indirectCommandsAddress,
    VkDeviceAddress                             indirectCommandsCountAddress,
    uint32_t                                    maxDecompressionCount,
    uint32_t                                    stride);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `decompressionMethod` is a bitmask of VkMemoryDecompressionMethodFlagBitsEXT with a single bit set specifying the method used to decompress data.
- `indirectCommandsAddress` is the device address containing decompression parameters laid out as an array of VkDecompressMemoryRegionEXT structures.
- `indirectCommandsCountAddress` is the device address containing a 32-bit integer value specifying the decompression count.
- `maxDecompressionCount` is the maximum number of decompressions that will be executed. The actual number of executed decompressions is the minimum of the count specified in `indirectCommandsCountAddress` and `maxDecompressionCount`.
- `stride` is the byte stride between successive sets of decompression parameters located starting from `indirectCommandsAddress`.

Each region specified in `indirectCommandsAddress` is decompressed from the source to destination region based on the specified `decompressionMethod`.

Valid Usage

- VUID-vkCmdDecompressMemoryIndirectCountEXT-None-07692
  The `memoryDecompression` feature **must** be enabled
- VUID-vkCmdDecompressMemoryIndirectCountEXT-indirectCommandsAddress-07694
  `indirectCommandsAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set
- VUID-vkCmdDecompressMemoryIndirectCountEXT-indirectCommandsAddress-07695
  `indirectCommandsAddress` **must** be a multiple of 4
- VUID-vkCmdDecompressMemoryIndirectCountEXT-indirectCommandsCountAddress-07697
  `indirectCommandsCountAddress` **must** be a device address allocated to the application from a buffer created with the VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT usage flag set

- VUID-vkCmdDecompressMemoryIndirectCountEXT-indirectCommandsCountAddress-07698
  `indirectCommandsCountAddress` **must** be a multiple of `4`
- VUID-vkCmdDecompressMemoryIndirectCountEXT-indirectCommandsCountAddress-07699
  The count stored in `indirectCommandsCountAddress` **must** be less than or equal to `VkPhysicalDeviceMemoryDecompressionPropertiesEXT`::`maxDecompressionIndirectCount`
- VUID-vkCmdDecompressMemoryIndirectCountEXT-indirectCommandsAddress-11794
  All device addresses between `indirectCommandsAddress` and `indirectCommandsAddress` + (`stride` × (count stored in `indirectCommandsCountAddress`)) - 1 **must** be in the buffer device address range of the same buffer
- VUID-vkCmdDecompressMemoryIndirectCountEXT-decompressionMethod-07690
  The `decompressionMethod` **must** have a single bit set
- VUID-vkCmdDecompressMemoryIndirectCountEXT-stride-11767
  `stride` **must** be a multiple of `4` and **must** be greater than or equal to sizeof(VkDecompressMemoryRegionEXT)
- VUID-vkCmdDecompressMemoryIndirectCountEXT-maxDecompressionCount-11768
  `maxDecompressionCount` **must** be less than or equal to `VkPhysicalDeviceMemoryDecompressionPropertiesEXT`::`maxDecompressionIndirectCount`
- VUID-vkCmdDecompressMemoryIndirectCountEXT-decompressionMethod-11769
  If `decompressionMethod` is VK_MEMORY_DECOMPRESSION_METHOD_GDEFLATE_1_0_BIT_EXT, then all values in VkDecompressMemoryRegionEXT::`decompressedSize` **must** be less than or equal to 65536 bytes
- VUID-vkCmdDecompressMemoryIndirectCountEXT-decompressionMethod-11810
  `decompressionMethod` **must** be a valid bit specified in `VkPhysicalDeviceMemoryDecompressionPropertiesEXT`::`decompressionMethods`

Valid Usage (Implicit)

- VUID-vkCmdDecompressMemoryIndirectCountEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDecompressMemoryIndirectCountEXT-decompressionMethod-parameter
  `decompressionMethod` **must** be a valid combination of VkMemoryDecompressionMethodFlagBitsEXT values
- VUID-vkCmdDecompressMemoryIndirectCountEXT-indirectCommandsAddress-parameter
  `indirectCommandsAddress` **must** be a valid VkDeviceAddress value
- VUID-vkCmdDecompressMemoryIndirectCountEXT-indirectCommandsCountAddress-parameter
  `indirectCommandsCountAddress` **must** be a valid VkDeviceAddress value
- VUID-vkCmdDecompressMemoryIndirectCountEXT-commandBuffer-recording

commandBuffer **must** be in the recording state

- VUID-vkCmdDecompressMemoryIndirectCountEXT-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT operations
- VUID-vkCmdDecompressMemoryIndirectCountEXT-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdDecompressMemoryIndirectCountEXT-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDecompressMemoryIndirectCountEXT-videocoding
  This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT	Action

Conditional Rendering

vkCmdDecompressMemoryIndirectCountEXT is not affected by conditional rendering

Bits which **can** be set in VkDecompressMemoryRegionEXT::decompressionMethod specifying the decompression method to select, or returned in VkPhysicalDeviceMemoryDecompressionPropertiesEXT::decompressionMethods specifying the available decompression methods are:

```
// Provided by VK_EXT_memory_decompression
// Flag bits for VkMemoryDecompressionMethodFlagBitsEXT
typedef VkFlags64 VkMemoryDecompressionMethodFlagBitsEXT;
static const VkMemoryDecompressionMethodFlagBitsEXT
VK_MEMORY_DECOMPRESSION_METHOD_GDEFLATE_1_0_BIT_EXT = 0x00000001ULL;
static const VkMemoryDecompressionMethodFlagBitsEXT
VK_MEMORY_DECOMPRESSION_METHOD_GDEFLATE_1_0_BIT_NV = 0x00000001ULL;
```

- VK_MEMORY_DECOMPRESSION_METHOD_GDEFLATE_1_0_BIT_EXT specifies that the GDeflate 1.0 algorithm is used to decompress data.

```
// Provided by VK_EXT_memory_decompression
typedef VkFlags64 VkMemoryDecompressionMethodFlagsEXT;
```

VkMemoryDecompressionMethodFlagsEXT is a bitmask type for specifying a mask of one or more VkMemoryDecompressionMethodFlagBitsEXT.

Chapter 45. Video Coding

Vulkan implementations **may** expose one or more queue families supporting video coding operations. These operations are performed by recording them into a command buffer within a video coding scope, and submitting them to queues with compatible video coding capabilities.

The Vulkan video functionalities are designed to be made available through a set of APIs built on top of each other, consisting of:

- A core API providing common video coding functionalities,
- APIs providing codec-independent video decode and video encode related functionalities, respectively,
- Additional codec-specific APIs built on top of those.

This chapter details the fundamental components and operations of these.

45.1. Video Picture Resources

In the context of video coding, multidimensional arrays of image data that **can** be used as the source or target of video coding operations are referred to as *video picture resources*. They **may** store additional metadata that includes implementation-private information used during the execution of video coding operations, as discussed later.

Video picture resources are backed by VkImage objects. Individual subregions of VkImageView objects created from such resources **can** be used as decode output pictures, encode input pictures, reconstructed pictures, and/or reference pictures.

The parameters of a video picture resource are specified using a `VkVideoPictureResourceInfoKHR` structure.

The `VkVideoPictureResourceInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoPictureResourceInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkOffset2D         codedOffset;
    VkExtent2D         codedExtent;
    uint32_t           baseArrayLayer;
    VkImageView        imageViewBinding;
} VkVideoPictureResourceInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `codedOffset` is the offset in texels of the image subregion to use.
- `codedExtent` is the size in pixels of the coded image data.

- `baseArrayLayer` is the array layer of the image view specified in `imageViewBinding` to use as the video picture resource.
- `imageViewBinding` is an image view representing the video picture resource.

The image subresource referred to by such a structure is defined as the image array layer index specified in `baseArrayLayer` relative to the image subresource range the image view specified in `imageViewBinding` was created with.

The meaning of the `codedOffset` and `codedExtent` depends on the command and context the video picture resource is used in, as well as on the used video profile and corresponding codec-specific semantics, as described later.

A video picture resource is uniquely defined by the image subresource referred to by an instance of this structure, together with the `codedOffset` and `codedExtent` members that identify the image subregion within the image subresource referenced corresponding to the video picture resource according to the particular codec-specific semantics.

Accesses to image data within a video picture resource happen at the granularity indicated by VkVideoCapabilitiesKHR::`pictureAccessGranularity`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile. As a result, given an effective image subregion corresponding to a video picture resource, the actual image subregion accessed **may** be larger than that as it **may** include additional padding texels due to the picture access granularity. Any writes performed by video coding operations to such padding texels will result in undefined texel values.

Two video picture resources match if they refer to the same image subresource and they specify identical `codedOffset` and `codedExtent` values.

Valid Usage

- VUID-VkVideoPictureResourceInfoKHR-baseArrayLayer-07175
  `baseArrayLayer` **must** be less than the VkImageViewCreateInfo ::`subresourceRange.layerCount` specified when the image view `imageViewBinding` was created

Valid Usage (Implicit)

- VUID-VkVideoPictureResourceInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_PICTURE_RESOURCE_INFO_KHR
- VUID-VkVideoPictureResourceInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkVideoPictureResourceInfoKHR-imageViewBinding-parameter
  `imageViewBinding` **must** be a valid VkImageView handle

45.2. Decoded Picture Buffer

An integral part of video coding pipelines is the reconstruction of pictures from a compressed video bitstream. A *reconstructed picture* is a video picture resource resulting from this process.

Such reconstructed pictures **can** be used as *reference pictures* in subsequent video coding operations to provide predictions of the values of samples of subsequently decoded or encoded pictures. The correct use of such reconstructed pictures as reference pictures is driven by the video compression standard, the implementation, and the application-specific use cases.

The list of reference pictures used to provide such predictions within a single video coding operation is referred to as the list of *active reference pictures*.

The *decoded picture buffer (DPB)* is an indexed data structure that maintains the set of reference pictures available to be used in video coding operations. Individual indexed entries of the DPB are referred to as the *decoded picture buffer (DPB) slots*. The range of valid DPB slot indices is between zero and `N-1`, where `N` is the capacity of the DPB. Each DPB slot **can** refer to a reference picture containing a *video frame* or **can** refer to up to two reference pictures containing the top and/or bottom *fields* that, when both present, together represent a full *video frame* .

In Vulkan, the state and the backing store of the DPB is separated as follows:

- The state of individual DPB slots is maintained by video session objects.
- The backing store of DPB slots is provided by subregions of VkImage objects used as video picture resources.

In addition, the implementation **may** also maintain opaque metadata associated with DPB slots, including:

- *Reference picture metadata* corresponding to the video picture resource associated with the DPB slot.

Such metadata **may** be stored by the implementation as part of the DPB slot state maintained by the video session, or as part of the video picture resource backing the DPB slot.

Any metadata stored in the video picture resources backing DPB slots are independent of the video session used to store it, hence such video picture resources **can** be shared with other video sessions. Correspondingly, any metadata that is dependent on the video session will always be stored as part of the DPB slot state maintained by that video session.

The responsibility of managing the DPB is split between the application and the implementation as follows:

- The application maintains the association between DPB slot indices and corresponding video picture resources.
- The implementation maintains global and per-slot opaque reference picture metadata.

In addition, the application is also responsible for managing the mapping between the codec-specific picture IDs and DPB slots, and any other codec-specific states unless otherwise specified.

45.2.1. DPB Slot States

At a given time, each DPB slot is either in *active* or *inactive* state. Initially, all DPB slots managed by a video session are in *inactive* state.

A DPB slot **can** be *activated* by using it as the target of picture reconstruction in a video coding operation with the reconstructed picture requested to be set up as a reference picture, according to the codec-specific semantics, changing its state to *active* and associating it with a *picture reference* to the reconstructed pictures.

Some video coding standards allow multiple picture references to be associated with a single DPB slot. In this case the state of the individual picture references **can** be independently updated.

> **NOTE** As an example, H.264 decoding allows associating a separate top field and bottom field picture with the same DPB slot.

As part of reference picture setup, the implementation **may** also generate reference picture metadata. Such reference picture metadata is specific to each picture reference associated with the DPB slot.

If such a video coding operation completes successfully, the activated DPB slot will have a *valid picture reference* and the reconstructed picture is associated with the DPB slot. This is true even if the DPB slot is used as the target of a picture reconstruction that only sets up a top field or bottom field reference picture and thus does not yet refer to a complete frame. However, if any data provided as input to such a video coding operation is not compliant with the video compression standard used, that video coding operation **may** complete unsuccessfully, in which case the activated DPB slot will have an *invalid picture reference*. This is true even if the DPB slot previously had a valid picture reference to a top field or bottom field reference picture, but the reconstruction of the other field corresponding to the DPB slot failed.

The application **can** use queries to get feedback about the outcome of video coding operations and use the resulting VkQueryResultStatusKHR value to determine whether the video coding operation completed successfully (result status is positive) or unsuccessfully (result status is negative).

Using a reference picture associated with a DPB slot that has an *invalid picture reference* as an active reference picture in subsequent video coding operations is legal, however, the contents of the outputs of such operations are undefined, and any DPB slots activated by such video coding operations will also have an *invalid picture reference*. This is true even if such video coding operations **may** otherwise complete successfully.

A DPB slot **can** also be *deactivated* by the application, changing its state to *inactive* and invalidating any picture references and reference picture metadata associated with the DPB slot.

If an already active DPB slot is used as the target of picture reconstruction in a video coding operation, but the decoded picture is not requested to be set up as a reference picture, according to the codec-specific semantics, no reference picture setup happens and the corresponding picture reference and reference picture metadata is invalidated within the DPB slot. If the DPB slot no longer has any associated picture references after such an operation, the DPB slot is implicitly deactivated.

If an already active DPB slot is used as the target of picture reconstruction when decoding a field picture that is not marked as reference, then the behavior is as follows:

- If the DPB slot is currently associated with a frame, then the DPB slot is deactivated.
- If the DPB slot is not currently associated with a top field picture and the decoded picture is a top field picture, or if the DPB slot is not currently associated with a bottom field picture and the decoded picture is a bottom field picture, then the other field picture association of the DPB slot, if any, is not disturbed.
- If the DPB slot is currently associated with a top field picture and the decoded picture is a top field picture, or if the DPB slot is currently associated with a bottom field picture and the decoded picture is a bottom field picture, then that picture association is invalidated, without disturbing the other field picture association, if any. If the DPB slot no longer has any associated picture references after such an operation, the DPB slot is implicitly deactivated.

A DPB slot **can** be activated with a new frame even if it is already active. In this case all previous associations of the DPB slots with reference pictures are replaced with an association with the reconstructed picture used to activate it.

If an already active DPB slot is activated with a reconstructed field picture, then the behavior is as follows:

- If the DPB slot is currently associated with a frame, then that association is replaced with an association with the reconstructed field picture used to activate it.
- If the DPB slot is not currently associated with a top field picture and the DPB slot is activated with a top field picture, or if the DPB slot is not currently associated with a bottom field picture and the DPB slot is activated with a bottom field picture, then the DPB slot is associated with the reconstructed field picture used to activate it, without disturbing the other field picture association, if any.
- If the DPB slot is currently associated with a top field picture and the DPB slot is activated with a new top field picture, or if the DPB slot is currently associated with a bottom field picture and the DPB slot is activated with a new bottom field picture, then that association is replaced with an association with the reconstructed field picture used to activate it, without disturbing the other field picture association, if any.

45.3. Video Profiles

The `VkVideoProfileInfoKHR` structure is defined as follows:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoProfileInfoKHR {
    VkStructureType                      sType;
    const void*                          pNext;
    VkVideoCodecOperationFlagBitsKHR     videoCodecOperation;
    VkVideoChromaSubsamplingFlagsKHR     chromaSubsampling;
    VkVideoComponentBitDepthFlagsKHR     lumaBitDepth;
    VkVideoComponentBitDepthFlagsKHR     chromaBitDepth;
} VkVideoProfileInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `videoCodecOperation` is a VkVideoCodecOperationFlagBitsKHR value specifying a video codec operation.
- `chromaSubsampling` is a bitmask of VkVideoChromaSubsamplingFlagBitsKHR specifying video chroma subsampling information.
- `lumaBitDepth` is a bitmask of VkVideoComponentBitDepthFlagBitsKHR specifying video luma bit depth information.
- `chromaBitDepth` is a bitmask of VkVideoComponentBitDepthFlagBitsKHR specifying video chroma bit depth information.

Video profiles are provided as input to video capability queries such as vkGetPhysicalDeviceVideoCapabilitiesKHR or vkGetPhysicalDeviceVideoFormatPropertiesKHR, as well as when creating resources to be used by video coding operations such as images, buffers, query pools, and video sessions.

The full description of a video profile is specified by an instance of this structure, and the codec-specific and auxiliary structures provided in its `pNext` chain.

When this structure is specified as an input parameter to vkGetPhysicalDeviceVideoCapabilitiesKHR, or through the `pProfiles` member of a VkVideoProfileListInfoKHR structure in the `pNext` chain of the input parameter of a query command such as vkGetPhysicalDeviceVideoFormatPropertiesKHR or vkGetPhysicalDeviceImageFormatProperties2, the following error codes indicate specific causes of the failure of the query operation:

- VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR specifies that the requested video picture layout (e.g. through the `pictureLayout` member of a VkVideoDecodeH264ProfileInfoKHR structure included in the `pNext` chain of `VkVideoProfileInfoKHR`) is not supported.
- VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR specifies that a video profile operation specified by `videoCodecOperation` is not supported.
- VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR specifies that video format parameters specified by `chromaSubsampling`, `lumaBitDepth`, or `chromaBitDepth` are not supported.
- VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR specifies that the codec-specific parameters corresponding to the video codec operation are not supported.

Valid Usage

- VUID-VkVideoProfileInfoKHR-chromaSubsampling-07013
  `chromaSubsampling` **must** have a single bit set
- VUID-VkVideoProfileInfoKHR-lumaBitDepth-07014
  `lumaBitDepth` **must** have a single bit set
- VUID-VkVideoProfileInfoKHR-chromaSubsampling-07015
  If `chromaSubsampling` is not VK_VIDEO_CHROMA_SUBSAMPLING_MONOCHROME_BIT_KHR, then `chromaBitDepth` **must** have a single bit set
- VUID-VkVideoProfileInfoKHR-videoCodecOperation-07179
  If `videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the `pNext` chain **must** include a VkVideoDecodeH264ProfileInfoKHR structure
- VUID-VkVideoProfileInfoKHR-videoCodecOperation-07180
  If `videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the `pNext` chain **must** include a VkVideoDecodeH265ProfileInfoKHR structure
- VUID-VkVideoProfileInfoKHR-videoCodecOperation-10791
  If `videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR, then the `pNext` chain **must** include a VkVideoDecodeVP9ProfileInfoKHR structure
- VUID-VkVideoProfileInfoKHR-videoCodecOperation-09256
  If `videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then the `pNext` chain **must** include a VkVideoDecodeAV1ProfileInfoKHR structure
- VUID-VkVideoProfileInfoKHR-videoCodecOperation-07181
  If `videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the `pNext` chain **must** include a VkVideoEncodeH264ProfileInfoKHR structure
- VUID-VkVideoProfileInfoKHR-videoCodecOperation-07182
  If `videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the `pNext` chain **must** include a VkVideoEncodeH265ProfileInfoKHR structure
- VUID-VkVideoProfileInfoKHR-videoCodecOperation-10262
  If `videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then the `pNext` chain **must** include a VkVideoEncodeAV1ProfileInfoKHR structure

Valid Usage (Implicit)

- VUID-VkVideoProfileInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_PROFILE_INFO_KHR
- VUID-VkVideoProfileInfoKHR-videoCodecOperation-parameter
  `videoCodecOperation` **must** be a valid VkVideoCodecOperationFlagBitsKHR value
- VUID-VkVideoProfileInfoKHR-chromaSubsampling-parameter
  `chromaSubsampling` **must** be a valid combination of VkVideoChromaSubsamplingFlagBitsKHR values

- VUID-VkVideoProfileInfoKHR-lumaBitDepth-parameter
  `lumaBitDepth` **must** be a valid combination of VkVideoComponentBitDepthFlagBitsKHR values
- VUID-VkVideoProfileInfoKHR-chromaBitDepth-parameter
  If `chromaBitDepth` is not `0`, `chromaBitDepth` **must** be a valid combination of VkVideoComponentBitDepthFlagBitsKHR values

Structure Chaining

Extends the structure

- VkQueryPoolCreateInfo

Possible values of VkVideoProfileInfoKHR::`videoCodecOperation`, specifying the type of video coding operation and video compression standard used by a video profile, are:

```
// Provided by VK_KHR_video_queue
typedef enum VkVideoCodecOperationFlagBitsKHR {
    VK_VIDEO_CODEC_OPERATION_NONE_KHR = 0,
  // Provided by VK_KHR_video_encode_h264
    VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR = 0x00010000,
  // Provided by VK_KHR_video_encode_h265
    VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR = 0x00020000,
  // Provided by VK_KHR_video_decode_h264
    VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR = 0x00000001,
  // Provided by VK_KHR_video_decode_h265
    VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR = 0x00000002,
  // Provided by VK_KHR_video_decode_av1
    VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR = 0x00000004,
  // Provided by VK_KHR_video_encode_av1
    VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR = 0x00040000,
  // Provided by VK_KHR_video_decode_vp9
    VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR = 0x00000008,
} VkVideoCodecOperationFlagBitsKHR;
```

- VK_VIDEO_CODEC_OPERATION_NONE_KHR specifies that no video codec operations are supported.
- VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR specifies support for H.264 decode operations.
- VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR specifies support for H.265 decode operations.
- VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR specifies support for VP9 decode operations.
- VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR specifies support for AV1 decode operations.

- VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR specifies support for H.264 encode operations.
- VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR specifies support for H.265 encode operations.
- VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR specifies support for AV1 encode operations.

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoCodecOperationFlagsKHR;
```

VkVideoCodecOperationFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoCodecOperationFlagBitsKHR.

The video format chroma subsampling is defined with the following enums:

```
// Provided by VK_KHR_video_queue
typedef enum VkVideoChromaSubsamplingFlagBitsKHR {
    VK_VIDEO_CHROMA_SUBSAMPLING_INVALID_KHR = 0,
    VK_VIDEO_CHROMA_SUBSAMPLING_MONOCHROME_BIT_KHR = 0x00000001,
    VK_VIDEO_CHROMA_SUBSAMPLING_420_BIT_KHR = 0x00000002,
    VK_VIDEO_CHROMA_SUBSAMPLING_422_BIT_KHR = 0x00000004,
    VK_VIDEO_CHROMA_SUBSAMPLING_444_BIT_KHR = 0x00000008,
} VkVideoChromaSubsamplingFlagBitsKHR;
```

- VK_VIDEO_CHROMA_SUBSAMPLING_MONOCHROME_BIT_KHR specifies that the format is monochrome.
- VK_VIDEO_CHROMA_SUBSAMPLING_420_BIT_KHR specifies that the format is 4:2:0 chroma subsampled, i.e. the two chroma components are sampled horizontally and vertically at half the sample rate of the luma component.
- VK_VIDEO_CHROMA_SUBSAMPLING_422_BIT_KHR - the format is 4:2:2 chroma subsampled, i.e. the two chroma components are sampled horizontally at half the sample rate of the luma component.
- VK_VIDEO_CHROMA_SUBSAMPLING_444_BIT_KHR - the format is 4:4:4 chroma sampled, i.e. all three components of the $Y'C_BC_R$ format are sampled at the same rate, thus there is no chroma subsampling.

Chroma subsampling is described in more detail in the Chroma Reconstruction section.

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoChromaSubsamplingFlagsKHR;
```

VkVideoChromaSubsamplingFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoChromaSubsamplingFlagBitsKHR.

Possible values for the video format component bit depth are:

```
// Provided by VK_KHR_video_queue
typedef enum VkVideoComponentBitDepthFlagBitsKHR {
    VK_VIDEO_COMPONENT_BIT_DEPTH_INVALID_KHR = 0,
    VK_VIDEO_COMPONENT_BIT_DEPTH_8_BIT_KHR = 0x00000001,
    VK_VIDEO_COMPONENT_BIT_DEPTH_10_BIT_KHR = 0x00000004,
    VK_VIDEO_COMPONENT_BIT_DEPTH_12_BIT_KHR = 0x00000010,
} VkVideoComponentBitDepthFlagBitsKHR;
```

- VK_VIDEO_COMPONENT_BIT_DEPTH_8_BIT_KHR specifies a component bit depth of 8 bits.
- VK_VIDEO_COMPONENT_BIT_DEPTH_10_BIT_KHR specifies a component bit depth of 10 bits.
- VK_VIDEO_COMPONENT_BIT_DEPTH_12_BIT_KHR specifies a component bit depth of 12 bits.

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoComponentBitDepthFlagsKHR;
```

VkVideoComponentBitDepthFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoComponentBitDepthFlagBitsKHR.

Additional information about the video decode use case **can** be provided by adding a VkVideoDecodeUsageInfoKHR structure to the pNext chain of VkVideoProfileInfoKHR.

The VkVideoDecodeUsageInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_queue
typedef struct VkVideoDecodeUsageInfoKHR {
    VkStructureType                sType;
    const void*                    pNext;
    VkVideoDecodeUsageFlagsKHR     videoUsageHints;
} VkVideoDecodeUsageInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- videoUsageHints is a bitmask of VkVideoDecodeUsageFlagBitsKHR specifying hints about the intended use of the video decode profile.

Valid Usage (Implicit)

- VUID-VkVideoDecodeUsageInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_USAGE_INFO_KHR
- VUID-VkVideoDecodeUsageInfoKHR-videoUsageHints-parameter
  If videoUsageHints is not 0, videoUsageHints **must** be a valid combination of VkVideoDecodeUsageFlagBitsKHR values

Structure Chaining

Extends the structures

- VkQueryPoolCreateInfo
- VkVideoProfileInfoKHR

The following bits **can** be specified in VkVideoDecodeUsageInfoKHR::videoUsageHints as a hint about the video decode use case:

```
// Provided by VK_KHR_video_decode_queue
typedef enum VkVideoDecodeUsageFlagBitsKHR {
    VK_VIDEO_DECODE_USAGE_DEFAULT_KHR = 0,
    VK_VIDEO_DECODE_USAGE_TRANSCODING_BIT_KHR = 0x00000001,
    VK_VIDEO_DECODE_USAGE_OFFLINE_BIT_KHR = 0x00000002,
    VK_VIDEO_DECODE_USAGE_STREAMING_BIT_KHR = 0x00000004,
} VkVideoDecodeUsageFlagBitsKHR;
```

- VK_VIDEO_DECODE_USAGE_TRANSCODING_BIT_KHR specifies that video decoding is intended to be used in conjunction with video encoding to transcode a video bitstream with the same and/or different codecs.
- VK_VIDEO_DECODE_USAGE_OFFLINE_BIT_KHR specifies that video decoding is intended to be used to consume a local video bitstream.
- VK_VIDEO_DECODE_USAGE_STREAMING_BIT_KHR specifies that video decoding is intended to be used to consume a video bitstream received as a continuous flow over network.

NOTE There are no restrictions on the combination of bits that **can** be specified by the application. However, applications **should** use reasonable combinations in order for the implementation to be able to select the most appropriate mode of operation for the particular use case.

```
// Provided by VK_KHR_video_decode_queue
typedef VkFlags VkVideoDecodeUsageFlagsKHR;
```

VkVideoDecodeUsageFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoDecodeUsageFlagBitsKHR.

Additional information about the video encode use case **can** be provided by adding a VkVideoEncodeUsageInfoKHR structure to the pNext chain of VkVideoProfileInfoKHR.

The VkVideoEncodeUsageInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeUsageInfoKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    VkVideoEncodeUsageFlagsKHR      videoUsageHints;
    VkVideoEncodeContentFlagsKHR    videoContentHints;
    VkVideoEncodeTuningModeKHR      tuningMode;
} VkVideoEncodeUsageInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `videoUsageHints` is a bitmask of VkVideoEncodeUsageFlagBitsKHR specifying hints about the intended use of the video encode profile.
- `videoContentHints` is a bitmask of VkVideoEncodeContentFlagBitsKHR specifying hints about the content to be encoded using the video encode profile.
- `tuningMode` is a VkVideoEncodeTuningModeKHR value specifying the tuning mode to use when encoding with the video profile.

Valid Usage (Implicit)

- VUID-VkVideoEncodeUsageInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_USAGE_INFO_KHR
- VUID-VkVideoEncodeUsageInfoKHR-videoUsageHints-parameter
  If `videoUsageHints` is not `0`, `videoUsageHints` **must** be a valid combination of VkVideoEncodeUsageFlagBitsKHR values
- VUID-VkVideoEncodeUsageInfoKHR-videoContentHints-parameter
  If `videoContentHints` is not `0`, `videoContentHints` **must** be a valid combination of VkVideoEncodeContentFlagBitsKHR values
- VUID-VkVideoEncodeUsageInfoKHR-tuningMode-parameter
  If `tuningMode` is not `0`, `tuningMode` **must** be a valid VkVideoEncodeTuningModeKHR value

Structure Chaining

Extends the structures

- VkQueryPoolCreateInfo
- VkVideoProfileInfoKHR

The following bits **can** be specified in VkVideoEncodeUsageInfoKHR::`videoUsageHints` as a hint about the video encode use case:

```
// Provided by VK_KHR_video_encode_queue
typedef enum VkVideoEncodeUsageFlagBitsKHR {
    VK_VIDEO_ENCODE_USAGE_DEFAULT_KHR = 0,
    VK_VIDEO_ENCODE_USAGE_TRANSCODING_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_USAGE_STREAMING_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_USAGE_RECORDING_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_USAGE_CONFERENCING_BIT_KHR = 0x00000008,
} VkVideoEncodeUsageFlagBitsKHR;
```

- VK_VIDEO_ENCODE_USAGE_TRANSCODING_BIT_KHR specifies that video encoding is intended to be used in conjunction with video decoding to transcode a video bitstream with the same and/or different codecs.
- VK_VIDEO_ENCODE_USAGE_STREAMING_BIT_KHR specifies that video encoding is intended to be used to produce a video bitstream that is expected to be sent as a continuous flow over network.
- VK_VIDEO_ENCODE_USAGE_RECORDING_BIT_KHR specifies that video encoding is intended to be used for real-time recording for offline consumption.
- VK_VIDEO_ENCODE_USAGE_CONFERENCING_BIT_KHR specifies that video encoding is intended to be used in a video conferencing scenario.

NOTE

There are no restrictions on the combination of bits that **can** be specified by the application. However, applications **should** use reasonable combinations in order for the implementation to be able to select the most appropriate mode of operation for the particular use case.

```
// Provided by VK_KHR_video_encode_queue
typedef VkFlags VkVideoEncodeUsageFlagsKHR;
```

VkVideoEncodeUsageFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeUsageFlagBitsKHR.

The following bits **can** be specified in VkVideoEncodeUsageInfoKHR::videoContentHints as a hint about the encoded video content:

```
// Provided by VK_KHR_video_encode_queue
typedef enum VkVideoEncodeContentFlagBitsKHR {
    VK_VIDEO_ENCODE_CONTENT_DEFAULT_KHR = 0,
    VK_VIDEO_ENCODE_CONTENT_CAMERA_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_CONTENT_DESKTOP_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_CONTENT_RENDERED_BIT_KHR = 0x00000004,
} VkVideoEncodeContentFlagBitsKHR;
```

- VK_VIDEO_ENCODE_CONTENT_CAMERA_BIT_KHR specifies that video encoding is intended to be used to encode camera content.

- VK_VIDEO_ENCODE_CONTENT_DESKTOP_BIT_KHR specifies that video encoding is intended to be used to encode desktop content.
- VK_VIDEO_ENCODE_CONTENT_RENDERED_BIT_KHR specifies that video encoding is intended to be used to encode rendered (e.g. game) content.

NOTE	There are no restrictions on the combination of bits that **can** be specified by the application. However, applications **should** use reasonable combinations in order for the implementation to be able to select the most appropriate mode of operation for the particular content type.

```
// Provided by VK_KHR_video_encode_queue
typedef VkFlags VkVideoEncodeContentFlagsKHR;
```

VkVideoEncodeContentFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeContentFlagBitsKHR.

Possible video encode tuning mode values are as follows:

```
// Provided by VK_KHR_video_encode_queue
typedef enum VkVideoEncodeTuningModeKHR {
    VK_VIDEO_ENCODE_TUNING_MODE_DEFAULT_KHR = 0,
    VK_VIDEO_ENCODE_TUNING_MODE_HIGH_QUALITY_KHR = 1,
    VK_VIDEO_ENCODE_TUNING_MODE_LOW_LATENCY_KHR = 2,
    VK_VIDEO_ENCODE_TUNING_MODE_ULTRA_LOW_LATENCY_KHR = 3,
    VK_VIDEO_ENCODE_TUNING_MODE_LOSSLESS_KHR = 4,
} VkVideoEncodeTuningModeKHR;
```

- VK_VIDEO_ENCODE_TUNING_MODE_DEFAULT_KHR specifies the default tuning mode.
- VK_VIDEO_ENCODE_TUNING_MODE_HIGH_QUALITY_KHR specifies that video encoding is tuned for high quality. When using this tuning mode, the implementation **may** compromise the latency of video encoding operations to improve quality.
- VK_VIDEO_ENCODE_TUNING_MODE_LOW_LATENCY_KHR specifies that video encoding is tuned for low latency. When using this tuning mode, the implementation **may** compromise quality to increase the performance and lower the latency of video encode operations.
- VK_VIDEO_ENCODE_TUNING_MODE_ULTRA_LOW_LATENCY_KHR specifies that video encoding is tuned for ultra-low latency. When using this tuning mode, the implementation **may** compromise quality to maximize the performance and minimize the latency of video encoding operations.
- VK_VIDEO_ENCODE_TUNING_MODE_LOSSLESS_KHR specifies that video encoding is tuned for lossless encoding. When using this tuning mode, video encode operations produce lossless output.

The VkVideoProfileListInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoProfileListInfoKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    uint32_t                        profileCount;
    const VkVideoProfileInfoKHR*    pProfiles;
} VkVideoProfileListInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `profileCount` is the number of elements in the `pProfiles` array.
- `pProfiles` is a pointer to an array of VkVideoProfileInfoKHR structures.

NOTE	Video transcoding is an example of a use case that necessitates the specification of multiple profiles in various contexts.

When the application provides a video decode profile and one or more video encode profiles in the profile list, the implementation ensures that any capabilities returned or resources created are suitable for the video transcoding use cases without the need for manual data transformations.

Valid Usage

- VUID-VkVideoProfileListInfoKHR-pProfiles-06813
  `pProfiles` **must** not contain more than one element whose `videoCodecOperation` member specifies a decode operation

Valid Usage (Implicit)

- VUID-VkVideoProfileListInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_PROFILE_LIST_INFO_KHR
- VUID-VkVideoProfileListInfoKHR-pProfiles-parameter
  If `profileCount` is not `0`, `pProfiles` **must** be a valid pointer to an array of `profileCount` valid VkVideoProfileInfoKHR structures

Structure Chaining

Extends the structures

- VkBufferCreateInfo
- VkImageCreateInfo
- VkPhysicalDeviceImageFormatInfo2
- VkPhysicalDeviceVideoFormatInfoKHR

45.4. Video Capabilities

45.4.1. Video Coding Capabilities

To query video coding capabilities for a specific video profile, call:

```
// Provided by VK_KHR_video_queue
VkResult vkGetPhysicalDeviceVideoCapabilitiesKHR(
    VkPhysicalDevice                            physicalDevice,
    const VkVideoProfileInfoKHR*                pVideoProfile,
    VkVideoCapabilitiesKHR*                     pCapabilities);
```

- `physicalDevice` is the physical device from which to query the video decode or encode capabilities.
- `pVideoProfile` is a pointer to a VkVideoProfileInfoKHR structure.
- `pCapabilities` is a pointer to a VkVideoCapabilitiesKHR structure in which the capabilities are returned.

If the video profile described by `pVideoProfile` is supported by the implementation, then this command returns VK_SUCCESS and `pCapabilities` is filled with the capabilities supported with the specified video profile. Otherwise, one of the video-profile-specific error codes is returned.

Valid Usage

- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-07183
  If `pVideoProfile->videoCodecOperation` specifies a decode operation, then the `pNext` chain of `pCapabilities` **must** include a VkVideoDecodeCapabilitiesKHR structure
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-07184
  If `pVideoProfile->videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the `pNext` chain of `pCapabilities` **must** include a VkVideoDecodeH264CapabilitiesKHR structure
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-07185
  If `pVideoProfile->videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the `pNext` chain of `pCapabilities` **must** include a VkVideoDecodeH265CapabilitiesKHR structure
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-10792
  If `pVideoProfile->videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR, then the `pNext` chain of `pCapabilities` **must** include a VkVideoDecodeVP9CapabilitiesKHR structure
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-09257
  If `pVideoProfile->videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then the `pNext` chain of `pCapabilities` **must** include a VkVideoDecodeAV1CapabilitiesKHR structure
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-07186

If pVideoProfile->videoCodecOperation specifies an encode operation, then the pNext chain of pCapabilities **must** include a VkVideoEncodeCapabilitiesKHR structure

- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-07187
  If pVideoProfile->videoCodecOperation is VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the pNext chain of pCapabilities **must** include a VkVideoEncodeH264CapabilitiesKHR structure
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-07188
  If pVideoProfile->videoCodecOperation is VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the pNext chain of pCapabilities **must** include a VkVideoEncodeH265CapabilitiesKHR structure
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-10263
  If pVideoProfile->videoCodecOperation is VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then the pNext chain of pCapabilities **must** include a VkVideoEncodeAV1CapabilitiesKHR structure

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-physicalDevice-parameter
  physicalDevice **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-parameter
  pVideoProfile **must** be a valid pointer to a valid VkVideoProfileInfoKHR structure
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pCapabilities-parameter
  pCapabilities **must** be a valid pointer to a VkVideoCapabilitiesKHR structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED
- VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR
- VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR
- VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR
- VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR

The VkVideoCapabilitiesKHR structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoCapabilitiesKHR {
    VkStructureType              sType;
    void*                        pNext;
    VkVideoCapabilityFlagsKHR    flags;
    VkDeviceSize                 minBitstreamBufferOffsetAlignment;
    VkDeviceSize                 minBitstreamBufferSizeAlignment;
    VkExtent2D                   pictureAccessGranularity;
    VkExtent2D                   minCodedExtent;
    VkExtent2D                   maxCodedExtent;
    uint32_t                     maxDpbSlots;
    uint32_t                     maxActiveReferencePictures;
    VkExtensionProperties        stdHeaderVersion;
} VkVideoCapabilitiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkVideoCapabilityFlagBitsKHR specifying capability flags.
- `minBitstreamBufferOffsetAlignment` is the minimum alignment for bitstream buffer offsets.
- `minBitstreamBufferSizeAlignment` is the minimum alignment for bitstream buffer range sizes.
- `pictureAccessGranularity` is the granularity at which image access to video picture resources happens.
- `minCodedExtent` is the minimum width and height of the coded frames.
- `maxCodedExtent` is the maximum width and height of the coded frames.
- `maxDpbSlots` is the maximum number of DPB slots supported by a single video session.
- `maxActiveReferencePictures` is the maximum number of active reference pictures a single video coding operation **can** use.
- `stdHeaderVersion` is a VkExtensionProperties structure reporting the Video Std header name and version supported for the video profile.

NOTE It is common for video compression standards to allow using all reference pictures associated with active DPB slots as active reference pictures, hence for video decode profiles the values returned in `maxDpbSlots` and `maxActiveReferencePictures` are often equal. Similarly, in case of video decode profiles supporting field pictures the value of `maxActiveReferencePictures` often equals `maxDpbSlots` × 2.

Valid Usage (Implicit)

- VUID-VkVideoCapabilitiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_CAPABILITIES_KHR
- VUID-VkVideoCapabilitiesKHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkVideoDecodeAV1CapabilitiesKHR,

VkVideoDecodeCapabilitiesKHR, VkVideoDecodeH264CapabilitiesKHR, VkVideoDecodeH265CapabilitiesKHR, VkVideoDecodeVP9CapabilitiesKHR, VkVideoEncodeAV1CapabilitiesKHR, VkVideoEncodeAV1QuantizationMapCapabilitiesKHR, VkVideoEncodeCapabilitiesKHR, VkVideoEncodeFeedback2CapabilitiesKHR, VkVideoEncodeH264CapabilitiesKHR, VkVideoEncodeH264QuantizationMapCapabilitiesKHR, VkVideoEncodeH265CapabilitiesKHR, VkVideoEncodeH265QuantizationMapCapabilitiesKHR, VkVideoEncodeIntraRefreshCapabilitiesKHR, or VkVideoEncodeQuantizationMapCapabilitiesKHR

- VUID-VkVideoCapabilitiesKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

Bits which **can** be set in VkVideoCapabilitiesKHR::`flags` are:

```
// Provided by VK_KHR_video_queue
typedef enum VkVideoCapabilityFlagBitsKHR {
    VK_VIDEO_CAPABILITY_PROTECTED_CONTENT_BIT_KHR = 0x00000001,
    VK_VIDEO_CAPABILITY_SEPARATE_REFERENCE_IMAGES_BIT_KHR = 0x00000002,
} VkVideoCapabilityFlagBitsKHR;
```

- VK_VIDEO_CAPABILITY_PROTECTED_CONTENT_BIT_KHR specifies that video sessions support producing and consuming protected content.
- VK_VIDEO_CAPABILITY_SEPARATE_REFERENCE_IMAGES_BIT_KHR indicates that the video picture resources associated with the DPB slots of a video session **can** be backed by separate `VkImage` objects. If this capability flag is not present, then all DPB slots of a video session **must** be associated with video picture resources backed by the same `VkImage` object (e.g. using different layers of the same image).

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoCapabilityFlagsKHR;
```

`VkVideoCapabilityFlagsKHR` is a bitmask type for setting a mask of zero or more VkVideoCapabilityFlagBitsKHR.

45.4.2. Video Format Capabilities

To enumerate the supported video formats and corresponding capabilities for a specific video profile, call:

```
// Provided by VK_KHR_video_queue
VkResult vkGetPhysicalDeviceVideoFormatPropertiesKHR(
    VkPhysicalDevice                            physicalDevice,
    const VkPhysicalDeviceVideoFormatInfoKHR*   pVideoFormatInfo,
    uint32_t*                                   pVideoFormatPropertyCount,
    VkVideoFormatPropertiesKHR*                 pVideoFormatProperties);
```

- `physicalDevice` is the physical device from which to query the video format properties.
- `pVideoFormatInfo` is a pointer to a VkPhysicalDeviceVideoFormatInfoKHR structure specifying the usage and video profiles for which supported image formats and capabilities are returned.
- `pVideoFormatPropertyCount` is a pointer to an integer related to the number of video format properties available or queried, as described below.
- `pVideoFormatProperties` is a pointer to an array of VkVideoFormatPropertiesKHR structures in which supported image formats and capabilities are returned.

If `pVideoFormatProperties` is `NULL`, then the number of video format properties supported for the given `physicalDevice` is returned in `pVideoFormatPropertyCount`. Otherwise, `pVideoFormatPropertyCount` **must** point to a variable set by the application to the number of elements in the `pVideoFormatProperties` array, and on return the variable is overwritten with the number of values actually written to `pVideoFormatProperties`. If the value of `pVideoFormatPropertyCount` is less than the number of video format properties supported, at most `pVideoFormatPropertyCount` values will be written to `pVideoFormatProperties`, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available values were returned.

Video format properties are always queried with respect to a specific set of video profiles. These are specified by chaining the VkVideoProfileListInfoKHR structure to `pVideoFormatInfo`.

For most use cases, the images are used by a single video session and a single video profile is provided. For a use case such as video transcoding, where a decode session output image **can** be used as encode input in one or more encode sessions, multiple video profiles corresponding to the video sessions that will share the image **must** be provided.

If any of the video profiles specified via VkVideoProfileListInfoKHR::`pProfiles` are not supported, then this command returns one of the video-profile-specific error codes. Furthermore, if VkPhysicalDeviceVideoFormatInfoKHR::`imageUsage` includes any image usage flags not supported by the specified video profiles, then this command returns VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR.

If the decode capability flags include VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR, then querying video format properties that support both decode DPB and output usage **can** be done by including both VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR and VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR in VkPhysicalDeviceVideoFormatInfoKHR::`imageUsage`. However, even in this case, querying video format properties with VkPhysicalDeviceVideoFormatInfoKHR::`imageUsage` including only VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR or VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR will also return formats supporting both.

NOTE

This enables applications to be able to query all formats supporting VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR and/or VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR by just including one of the flags, respectively, regardless of whether the implementation supports VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR, VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR, or both. This makes enumerating decode DPB and output formats simpler and unified.

The `imageUsage` member of the VkPhysicalDeviceVideoFormatInfoKHR structure specifies the expected video usage flags that the returned video formats **must** support. Correspondingly, the `imageUsageFlags` member of each VkVideoFormatPropertiesKHR structure returned will contain at least the same set of image usage flags.

If the `pNext` chain includes a VkImageUsageFlags2CreateInfoKHR structure, `usage` from that structure specifies the expected video usage flags instead of VkPhysicalDeviceVideoFormatInfoKHR::`imageUsage`.

If the implementation supports using images of a particular format in operations other than video decode/encode then the `imageUsageFlags` member of the corresponding VkVideoFormatPropertiesKHR structure returned will include additional image usage flags indicating that.

NOTE

For most use cases, only decode or encode related usage flags are going to be specified. For a use case such as transcode, if the image were to be shared between decode and encode session(s), then both decode and encode related usage flags **can** be set.

Multiple `VkVideoFormatPropertiesKHR` entries **may** be returned with the same `format` member with different `componentMapping`, `imageType`, or `imageTiling` values, as described later.

If VkPhysicalDeviceVideoFormatInfoKHR::`imageUsage` includes VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR, multiple `VkVideoFormatPropertiesKHR` entries **may** be returned with the same `format`, `componentMapping`, `imageType`, and `imageTiling` member values, but different `quantizationMapTexelSize` returned in the VkVideoFormatQuantizationMapPropertiesKHR structure, if one is included in the VkVideoFormatPropertiesKHR::`pNext` chain, when the queried quantization map type supports multiple distinct quantization map texel sizes.

In addition, a different set of `VkVideoFormatPropertiesKHR` entries **may** be returned depending on the `imageUsage` member of the `VkPhysicalDeviceVideoFormatInfoKHR` structure, even for the same set of video profiles, for example, based on whether encode input, encode DPB, decode output, and/or decode DPB usage is requested.

The application **can** select the parameters returned in the `VkVideoFormatPropertiesKHR` entries and use compatible parameters when creating the input, output, and DPB images. The implementation will report all image creation and usage flags that are valid for images used with the requested video profiles but applications **should** create images only with those that are necessary for the

particular use case.

Before creating an image, the application **can** obtain the complete set of supported image format features by calling vkGetPhysicalDeviceImageFormatProperties2 using parameters derived from the members of one of the reported VkVideoFormatPropertiesKHR entries and adding the same VkVideoProfileListInfoKHR structure to the pNext chain of VkPhysicalDeviceImageFormatInfo2.

The following applies to all VkVideoFormatPropertiesKHR entries returned by vkGetPhysicalDeviceVideoFormatPropertiesKHR:

- vkGetPhysicalDeviceFormatProperties2 **must** succeed when called with VkVideoFormatPropertiesKHR::format
- If VkVideoFormatPropertiesKHR::imageTiling is VK_IMAGE_TILING_OPTIMAL, then the optimalTilingFeatures returned by vkGetPhysicalDeviceFormatProperties2 **must** include all format features required by the image usage flags reported in VkVideoFormatPropertiesKHR ::imageUsageFlags for the format, as indicated in the Format Feature Dependent Usage Flags section.
- If VkVideoFormatPropertiesKHR::imageTiling is VK_IMAGE_TILING_LINEAR, then the linearTilingFeatures returned by vkGetPhysicalDeviceFormatProperties2 **must** include all format features required by the image usage flags reported in VkVideoFormatPropertiesKHR ::imageUsageFlags for the format, as indicated in the Format Feature Dependent Usage Flags section.
- vkGetPhysicalDeviceImageFormatProperties2 **must** succeed when called with a VkPhysicalDeviceImageFormatInfo2 structure containing the following information:
  - The pNext chain including the same VkVideoProfileListInfoKHR structure used to call vkGetPhysicalDeviceVideoFormatPropertiesKHR.
  - format set to the value of VkVideoFormatPropertiesKHR::format.
  - type set to the value of VkVideoFormatPropertiesKHR::imageType.
  - tiling set to the value of VkVideoFormatPropertiesKHR::imageTiling.
  - usage set to the value of VkVideoFormatPropertiesKHR::imageUsageFlags.
  - flags set to the value of VkVideoFormatPropertiesKHR::imageCreateFlags.

The componentMapping member of VkVideoFormatPropertiesKHR defines the ordering of the $Y'C_BC_R$ color channels from the perspective of the video codec operations specified in VkVideoProfileListInfoKHR. For example, if the implementation produces video decode output with the format VK_FORMAT_G8_B8R8_2PLANE_420_UNORM where the blue and red chrominance channels are swapped then the componentMapping member of the corresponding VkVideoFormatPropertiesKHR structure will have the following member values:

```
components.r = VK_COMPONENT_SWIZZLE_B;        // Cb component
components.g = VK_COMPONENT_SWIZZLE_IDENTITY; // Y component
components.b = VK_COMPONENT_SWIZZLE_R;        // Cr component
components.a = VK_COMPONENT_SWIZZLE_IDENTITY; // unused, defaults to 1.0
```

Valid Usage

- VUID-vkGetPhysicalDeviceVideoFormatPropertiesKHR-pNext-06812
  The `pNext` chain of `pVideoFormatInfo` **must** include a VkVideoProfileListInfoKHR structure with `profileCount` greater than `0`

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceVideoFormatPropertiesKHR-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceVideoFormatPropertiesKHR-pVideoFormatInfo-parameter
  `pVideoFormatInfo` **must** be a valid pointer to a valid VkPhysicalDeviceVideoFormatInfoKHR structure
- VUID-vkGetPhysicalDeviceVideoFormatPropertiesKHR-pVideoFormatPropertyCount-parameter
  `pVideoFormatPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceVideoFormatPropertiesKHR-pVideoFormatProperties-parameter
  If the value referenced by `pVideoFormatPropertyCount` is not `0`, and `pVideoFormatProperties` is not `NULL`, `pVideoFormatProperties` **must** be a valid pointer to an array of `pVideoFormatPropertyCount` VkVideoFormatPropertiesKHR structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED
- VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR
- VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR
- VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR
- VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR

The `VkPhysicalDeviceVideoFormatInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkPhysicalDeviceVideoFormatInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    VkImageUsageFlags    imageUsage;
} VkPhysicalDeviceVideoFormatInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `imageUsage` is a bitmask of VkImageUsageFlagBits specifying the intended usage of the video images.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVideoFormatInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_FORMAT_INFO_KHR
- VUID-VkPhysicalDeviceVideoFormatInfoKHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkImageUsageFlags2CreateInfoKHR or VkVideoProfileListInfoKHR
- VUID-VkPhysicalDeviceVideoFormatInfoKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkPhysicalDeviceVideoFormatInfoKHR-imageUsage-parameter
  `imageUsage` **must** be a valid combination of VkImageUsageFlagBits values

The `VkVideoFormatPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoFormatPropertiesKHR {
    VkStructureType       sType;
    void*                 pNext;
    VkFormat              format;
    VkComponentMapping    componentMapping;
    VkImageCreateFlags    imageCreateFlags;
    VkImageType           imageType;
    VkImageTiling         imageTiling;
    VkImageUsageFlags     imageUsageFlags;
} VkVideoFormatPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `format` is a VkFormat that specifies the format that **can** be used with the specified video profiles and image usages.

- `componentMapping` defines the color channel order used for the format. `format` along with `componentMapping` describe how the color channels are ordered when producing video decoder output or are expected to be ordered in video encoder input, when applicable. If the `format` reported does not require component swizzling then all members of `componentMapping` will be set to VK_COMPONENT_SWIZZLE_IDENTITY.
- `imageCreateFlags` is a bitmask of VkImageCreateFlagBits specifying the supported image creation flags for the format.
- `imageType` is a VkImageType that specifies the image type the format **can** be used with.
- `imageTiling` is a VkImageTiling that specifies the image tiling the format **can** be used with.
- `imageUsageFlags` is a bitmask of VkImageUsageFlagBits specifying the supported image usage flags for the format.

Only create flags representable in VkImageCreateFlagBits are returned in `imageCreateFlags`. If the `pNext` chain includes a VkImageCreateFlags2CreateInfoKHR structure, all create flags are returned in VkImageCreateFlags2CreateInfoKHR::`flags`.

Only usage flags representable in VkImageUsageFlagBits are returned in `imageUsageFlags`. If the `pNext` chain includes a VkImageUsageFlags2CreateInfoKHR structure, all usage flags are returned in VkImageUsageFlags2CreateInfoKHR::`usage`.

Valid Usage (Implicit)

- VUID-VkVideoFormatPropertiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_FORMAT_PROPERTIES_KHR
- VUID-VkVideoFormatPropertiesKHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkImageCreateFlags2CreateInfoKHR, VkImageUsageFlags2CreateInfoKHR, VkVideoFormatAV1QuantizationMapPropertiesKHR, VkVideoFormatH265QuantizationMapPropertiesKHR, or VkVideoFormatQuantizationMapPropertiesKHR
- VUID-VkVideoFormatPropertiesKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

The *list of supported video format properties* for a set of image usage flags with respect to a video profile is defined as the list of VkVideoFormatPropertiesKHR structures and any structures included in its `pNext` chain, obtained by calling vkGetPhysicalDeviceVideoFormatPropertiesKHR with VkPhysicalDeviceVideoFormatInfoKHR::`imageUsage` equal to the VkImageUsageFlags in question and the VkPhysicalDeviceVideoFormatInfoKHR::`pNext` chain including a VkVideoProfileListInfoKHR structure with its `pProfiles` member containing a single array element specifying the VkVideoProfileInfoKHR structure chain describing the video profile in question.

45.5. Video Sessions

Video sessions are objects that represent and maintain the state needed to perform video decode or encode operations using a specific video profile.

In case of video encode profiles this includes the current rate control configuration and the currently set video encode quality level.

Video sessions are represented by VkVideoSessionKHR handles:

```
// Provided by VK_KHR_video_queue
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkVideoSessionKHR)
```

45.5.1. Creating a Video Session

To create a video session object, call:

```
// Provided by VK_KHR_video_queue
VkResult vkCreateVideoSessionKHR(
    VkDevice                                    device,
    const VkVideoSessionCreateInfoKHR*          pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkVideoSessionKHR*                          pVideoSession);
```

- device is the logical device that creates the video session.
- pCreateInfo is a pointer to a VkVideoSessionCreateInfoKHR structure containing parameters to be used to create the video session.
- pAllocator controls host memory allocation as described in the Memory Allocation chapter.
- pVideoSession is a pointer to a VkVideoSessionKHR handle in which the resulting video session object is returned.

The resulting video session object is said to be created with the video codec operation specified in pCreateInfo->pVideoProfile->videoCodecOperation.

The name and version of the codec-specific Video Std header to be used with the video session is specified by the VkExtensionProperties structure pointed to by pCreateInfo->pStdHeaderVersion. If a non-existent or unsupported Video Std header version is specified in pCreateInfo->pStdHeaderVersion->specVersion, then this command returns VK_ERROR_VIDEO_STD_VERSION_NOT_SUPPORTED_KHR.

Video session objects are created in *uninitialized* state. In order to transition the video session into *initial* state, the application **must** issue a vkCmdControlVideoCodingKHR command with VkVideoCodingControlInfoKHR::flags including VK_VIDEO_CODING_CONTROL_RESET_BIT_KHR.

Video session objects also maintain the state of the DPB. The number of DPB slots usable with the created video session is specified in pCreateInfo->maxDpbSlots, and each slot is initially in the inactive state.

Each DPB slot maintained by the created video session **can** refer to a reference picture representing a video frame.

In addition, if the videoCodecOperation member of the VkVideoProfileInfoKHR structure pointed to

by pCreateInfo->pVideoProfile is VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and the pictureLayout member of the VkVideoDecodeH264ProfileInfoKHR structure provided in the VkVideoProfileInfoKHR::pNext chain is not VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_PROGRESSIVE_KHR, then the created video session supports *interlaced* frames and each DPB slot maintained by the created video session **can** instead refer to separate top field and bottom field reference pictures that together **can** represent a full video frame. In this case, it is up to the application, driven by the video content, whether it associates any individual DPB slot with separate top and/or bottom field pictures or a single picture representing a full frame.

The created video session **can** be used to perform video coding operations using video frames up to the maximum size specified in pCreateInfo->maxCodedExtent. The minimum frame size allowed is implicitly derived from VkVideoCapabilitiesKHR::minCodedExtent, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified by pCreateInfo->pVideoProfile. Accordingly, the created video session is said to be created with a minCodedExtent equal to that.

In case of video session objects created with a video encode operation, implementations **may** return the VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR error if any of the specified Video Std parameters do not adhere to the syntactic or semantic requirements of the used video compression standard, or if values derived from parameters according to the rules defined by the used video compression standard do not adhere to the capabilities of the video compression standard or the implementation.

NOTE	Applications **should** not rely on the VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR error being returned by any command as a means to verify Video Std parameters, as implementations are not required to report the error in any specific set of cases.

Valid Usage (Implicit)

- VUID-vkCreateVideoSessionKHR-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkCreateVideoSessionKHR-pCreateInfo-parameter
  pCreateInfo **must** be a valid pointer to a valid VkVideoSessionCreateInfoKHR structure
- VUID-vkCreateVideoSessionKHR-pAllocator-parameter
  If pAllocator is not NULL, pAllocator **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateVideoSessionKHR-pVideoSession-parameter
  pVideoSession **must** be a valid pointer to a VkVideoSessionKHR handle
- VUID-vkCreateVideoSessionKHR-device-queuecount
  The device **must** have been created with at least 1 queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED
- VK_ERROR_VIDEO_STD_VERSION_NOT_SUPPORTED_KHR

The VkVideoSessionCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoSessionCreateInfoKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    uint32_t                        queueFamilyIndex;
    VkVideoSessionCreateFlagsKHR    flags;
    const VkVideoProfileInfoKHR*    pVideoProfile;
    VkFormat                        pictureFormat;
    VkExtent2D                      maxCodedExtent;
    VkFormat                        referencePictureFormat;
    uint32_t                        maxDpbSlots;
    uint32_t                        maxActiveReferencePictures;
    const VkExtensionProperties*    pStdHeaderVersion;
} VkVideoSessionCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- queueFamilyIndex is the index of the queue family the created video session will be used with.
- flags is a bitmask of VkVideoSessionCreateFlagBitsKHR specifying creation flags.
- pVideoProfile is a pointer to a VkVideoProfileInfoKHR structure specifying the video profile the created video session will be used with.
- pictureFormat is the image format the created video session will be used with. If pVideoProfile->videoCodecOperation specifies a decode operation, then pictureFormat is the image format of decode output pictures usable with the created video session. If pVideoProfile->videoCodecOperation specifies an encode operation, then pictureFormat is the image format of encode input pictures usable with the created video session.

- `maxCodedExtent` is the maximum width and height of the coded frames the created video session will be used with.
- `referencePictureFormat` is the image format of reference pictures stored in the DPB the created video session will be used with.
- `maxDpbSlots` is the maximum number of DPB Slots that **can** be used with the created video session.
- `maxActiveReferencePictures` is the maximum number of active reference pictures that **can** be used in a single video coding operation using the created video session.
- `pStdHeaderVersion` is a pointer to a VkExtensionProperties structure requesting the Video Std header version to use for the `videoCodecOperation` specified in `pVideoProfile`.

Valid Usage

- VUID-VkVideoSessionCreateInfoKHR-protectedMemory-07189
  If the `protectedMemory` feature is not enabled or if VkVideoCapabilitiesKHR::`flags` does not include VK_VIDEO_CAPABILITY_PROTECTED_CONTENT_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified by `pVideoProfile`, then `flags` **must** not include VK_VIDEO_SESSION_CREATE_PROTECTED_CONTENT_BIT_KHR
- VUID-VkVideoSessionCreateInfoKHR-flags-08371
  If `flags` includes VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR, then `videoMaintenance1` **must** be enabled
- VUID-VkVideoSessionCreateInfoKHR-flags-10398
  If `flags` includes VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR, then `videoMaintenance2` **must** be enabled
- VUID-VkVideoSessionCreateInfoKHR-flags-10399
  If `flags` includes VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR, then `pVideoProfile->videoCodecOperation` **must** specify a decode operation
- VUID-VkVideoSessionCreateInfoKHR-flags-10264
  If `flags` includes VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_EMPHASIS_MAP_BIT_KHR, then the `videoEncodeQuantizationMap` feature **must** be enabled
- VUID-VkVideoSessionCreateInfoKHR-flags-10265
  If `flags` includes VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_EMPHASIS_MAP_BIT_KHR, then `pVideoProfile->videoCodecOperation` **must** specify an encode operation
- VUID-VkVideoSessionCreateInfoKHR-flags-10266
  If `flags` includes VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR, then it **must** not also include VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_EMPHASIS_MAP_BIT_KHR

- VUID-VkVideoSessionCreateInfoKHR-flags-10267
  If VkVideoEncodeCapabilitiesKHR::flags does not include VK_VIDEO_ENCODE_CAPABILITY_QUANTIZATION_DELTA_MAP_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified by pVideoProfile, then flags **must** not include VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR
- VUID-VkVideoSessionCreateInfoKHR-flags-10268
  If VkVideoEncodeCapabilitiesKHR::flags does not include VK_VIDEO_ENCODE_CAPABILITY_EMPHASIS_MAP_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified by pVideoProfile, then flags **must** not include VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_EMPHASIS_MAP_BIT_KHR
- VUID-VkVideoSessionCreateInfoKHR-pVideoProfile-11759
  pVideoProfile->videoCodecOperation **must** be supported by the queue family index specified in queueFamilyIndex, as reported in VkQueueFamilyVideoPropertiesKHR::videoCodecOperations
- VUID-VkVideoSessionCreateInfoKHR-pVideoProfile-04845
  pVideoProfile **must** be a supported video profile
- VUID-VkVideoSessionCreateInfoKHR-maxDpbSlots-04847
  maxDpbSlots **must** be less than or equal to VkVideoCapabilitiesKHR::maxDpbSlots, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified by pVideoProfile
- VUID-VkVideoSessionCreateInfoKHR-maxActiveReferencePictures-04849
  maxActiveReferencePictures **must** be less than or equal to VkVideoCapabilitiesKHR::maxActiveReferencePictures, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified by pVideoProfile
- VUID-VkVideoSessionCreateInfoKHR-maxDpbSlots-04850
  If either maxDpbSlots or maxActiveReferencePictures is 0, then both **must** be 0
- VUID-VkVideoSessionCreateInfoKHR-maxCodedExtent-04851
  maxCodedExtent **must** be between VkVideoCapabilitiesKHR::minCodedExtent and VkVideoCapabilitiesKHR::maxCodedExtent, inclusive, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified by pVideoProfile
- VUID-VkVideoSessionCreateInfoKHR-referencePictureFormat-04852
  If pVideoProfile->videoCodecOperation specifies a decode operation and maxActiveReferencePictures is greater than 0, then referencePictureFormat **must** be one of the supported decode DPB formats, as returned by vkGetPhysicalDeviceVideoFormatPropertiesKHR in VkVideoFormatPropertiesKHR::format when called with the imageUsage member of its pVideoFormatInfo parameter containing VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR, and with a VkVideoProfileListInfoKHR structure specified in the pNext chain of its pVideoFormatInfo parameter whose pProfiles member contains an element matching pVideoProfile
- VUID-VkVideoSessionCreateInfoKHR-referencePictureFormat-06814
  If pVideoProfile->videoCodecOperation specifies an encode operation and maxActiveReferencePictures is greater than 0, then referencePictureFormat **must** be one of

the supported decode DPB formats, as returned by then `referencePictureFormat` **must** be one of the supported encode DPB formats, as returned by vkGetPhysicalDeviceVideoFormatPropertiesKHR in VkVideoFormatPropertiesKHR::`format` when called with the `imageUsage` member of its `pVideoFormatInfo` parameter containing VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR, and with a VkVideoProfileListInfoKHR structure specified in the `pNext` chain of its `pVideoFormatInfo` parameter whose `pProfiles` member contains an element matching `pVideoProfile`

- VUID-VkVideoSessionCreateInfoKHR-pictureFormat-04853
  If `pVideoProfile->videoCodecOperation` specifies a decode operation, then `pictureFormat` **must** be one of the supported decode output formats, as returned by vkGetPhysicalDeviceVideoFormatPropertiesKHR in VkVideoFormatPropertiesKHR::`format` when called with the `imageUsage` member of its `pVideoFormatInfo` parameter containing VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR, and with a VkVideoProfileListInfoKHR structure specified in the `pNext` chain of its `pVideoFormatInfo` parameter whose `pProfiles` member contains an element matching `pVideoProfile`
- VUID-VkVideoSessionCreateInfoKHR-pictureFormat-04854
  If `pVideoProfile->videoCodecOperation` specifies an encode operation, then `pictureFormat` **must** be one of the supported encode input formats, as returned by vkGetPhysicalDeviceVideoFormatPropertiesKHR in VkVideoFormatPropertiesKHR::`format` when called with the `imageUsage` member of its `pVideoFormatInfo` parameter containing VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR, and with a VkVideoProfileListInfoKHR structure specified in the `pNext` chain of its `pVideoFormatInfo` parameter whose `pProfiles` member contains an element matching `pVideoProfile`
- VUID-VkVideoSessionCreateInfoKHR-pVideoProfile-10835
  If `pVideoProfile->videoCodecOperation` specifies an encode operation, the `pNext` chain of this structure includes a VkVideoEncodeSessionIntraRefreshCreateInfoKHR structure, and its `intraRefreshMode` member is not VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_NONE_KHR, then the `videoEncodeIntraRefresh` feature **must** be enabled
- VUID-VkVideoSessionCreateInfoKHR-pVideoProfile-10836
  If `pVideoProfile->videoCodecOperation` specifies an encode operation, the `pNext` chain of this structure includes a VkVideoEncodeSessionIntraRefreshCreateInfoKHR structure, and its `intraRefreshMode` member is not VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_NONE_KHR, then `intraRefreshMode` **must** specify one of the bits included in VkVideoEncodeIntraRefreshCapabilitiesKHR::`intraRefreshModes`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified by `pVideoProfile`
- VUID-VkVideoSessionCreateInfoKHR-pStdHeaderVersion-07190
  `pStdHeaderVersion->extensionName` **must** match VkVideoCapabilitiesKHR::`stdHeaderVersion.extensionName`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified by `pVideoProfile`
- VUID-VkVideoSessionCreateInfoKHR-pStdHeaderVersion-07191
  `pStdHeaderVersion->specVersion` **must** be less than or equal to VkVideoCapabilitiesKHR::`stdHeaderVersion.specVersion`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified by `pVideoProfile`

- VUID-VkVideoSessionCreateInfoKHR-pVideoProfile-10793
  If `pVideoProfile->videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR, then the `videoDecodeVP9` feature **must** be enabled
- VUID-VkVideoSessionCreateInfoKHR-pVideoProfile-08251
  If `pVideoProfile->videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and the `pNext` chain of this structure includes a VkVideoEncodeH264SessionCreateInfoKHR structure, then its `maxLevelIdc` member **must** be less than or equal to VkVideoEncodeH264CapabilitiesKHR::`maxLevelIdc`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified in `pVideoProfile`
- VUID-VkVideoSessionCreateInfoKHR-pVideoProfile-08252
  If `pVideoProfile->videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and the `pNext` chain of this structure includes a VkVideoEncodeH265SessionCreateInfoKHR structure, then its `maxLevelIdc` member **must** be less than or equal to VkVideoEncodeH265CapabilitiesKHR::`maxLevelIdc`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified in `pVideoProfile`
- VUID-VkVideoSessionCreateInfoKHR-pVideoProfile-10269
  If `pVideoProfile->videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then the `videoEncodeAV1` feature **must** be enabled
- VUID-VkVideoSessionCreateInfoKHR-pVideoProfile-10270
  If `pVideoProfile->videoCodecOperation` is VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the `pNext` chain of this structure includes a VkVideoEncodeAV1SessionCreateInfoKHR structure, then its `maxLevel` member **must** be less than or equal to VkVideoEncodeAV1CapabilitiesKHR::`maxLevel`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified in `pVideoProfile`

Valid Usage (Implicit)

- VUID-VkVideoSessionCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_SESSION_CREATE_INFO_KHR
- VUID-VkVideoSessionCreateInfoKHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkVideoEncodeAV1SessionCreateInfoKHR, VkVideoEncodeH264SessionCreateInfoKHR, VkVideoEncodeH265SessionCreateInfoKHR, or VkVideoEncodeSessionIntraRefreshCreateInfoKHR
- VUID-VkVideoSessionCreateInfoKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkVideoSessionCreateInfoKHR-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkVideoSessionCreateFlagBitsKHR values

- VUID-VkVideoSessionCreateInfoKHR-pVideoProfile-parameter
  `pVideoProfile` **must** be a valid pointer to a valid VkVideoProfileInfoKHR structure
- VUID-VkVideoSessionCreateInfoKHR-pictureFormat-parameter
  `pictureFormat` **must** be a valid VkFormat value
- VUID-VkVideoSessionCreateInfoKHR-referencePictureFormat-parameter
  `referencePictureFormat` **must** be a valid VkFormat value
- VUID-VkVideoSessionCreateInfoKHR-pStdHeaderVersion-parameter
  `pStdHeaderVersion` **must** be a valid pointer to a valid VkExtensionProperties structure

Bits which **can** be set in VkVideoSessionCreateInfoKHR::`flags` are:

```
// Provided by VK_KHR_video_queue
typedef enum VkVideoSessionCreateFlagBitsKHR {
    VK_VIDEO_SESSION_CREATE_PROTECTED_CONTENT_BIT_KHR = 0x00000001,
  // Provided by VK_KHR_video_encode_queue
    VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_PARAMETER_OPTIMIZATIONS_BIT_KHR = 0x00000002,
  // Provided by VK_KHR_video_maintenance1
    VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR = 0x00000004,
  // Provided by VK_KHR_video_encode_quantization_map
    VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR = 0x00000008,
  // Provided by VK_KHR_video_encode_quantization_map
    VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_EMPHASIS_MAP_BIT_KHR = 0x00000010,
  // Provided by VK_KHR_video_decode_queue with VK_KHR_video_maintenance2
    VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR = 0x00000020,
} VkVideoSessionCreateFlagBitsKHR;
```

- VK_VIDEO_SESSION_CREATE_PROTECTED_CONTENT_BIT_KHR specifies that the video session uses protected video content.
- VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_PARAMETER_OPTIMIZATIONS_BIT_KHR specifies that the implementation is allowed to override video session parameters and other codec-specific encoding parameters to optimize video encode operations based on the use case information specified in the video profile and the used video encode quality level.

  NOTE Not specifying VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_PARAMETER_OPTIMIZATIONS_BIT_KHR does not guarantee that the implementation will not do any codec-specific parameter overrides, as certain overrides are necessary for the correct operation of the video encoder implementation due to limitations to the available encoding tools on that implementation. This flag, however, enables the implementation to apply further optimizing overrides.

- VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR specifies that queries within video coding scopes using the created video session are executed inline with video coding operations.
- VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR specifies that the video session **can** be used to encode pictures with quantization delta maps.

- VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_EMPHASIS_MAP_BIT_KHR specifies that the video session **can** be used to encode pictures with emphasis maps.
- VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR specifies that the application **can** specify video session parameters inline with video decode operations instead of sourcing them from the bound video session parameters object.

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoSessionCreateFlagsKHR;
```

VkVideoSessionCreateFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoSessionCreateFlagBitsKHR.

45.5.2. Destroying a Video Session

To destroy a video session, call:

```
// Provided by VK_KHR_video_queue
void vkDestroyVideoSessionKHR(
    VkDevice                                    device,
    VkVideoSessionKHR                           videoSession,
    const VkAllocationCallbacks*                pAllocator);
```

- device is the logical device that destroys the video session.
- videoSession is the video session to destroy.
- pAllocator controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyVideoSessionKHR-videoSession-07192
  All submitted commands that refer to videoSession **must** have completed execution
- VUID-vkDestroyVideoSessionKHR-videoSession-07193
  If VkAllocationCallbacks were provided when videoSession was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyVideoSessionKHR-videoSession-07194
  If no VkAllocationCallbacks were provided when videoSession was created, pAllocator **must** be NULL

Valid Usage (Implicit)

- VUID-vkDestroyVideoSessionKHR-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkDestroyVideoSessionKHR-videoSession-parameter
  If videoSession is not VK_NULL_HANDLE, videoSession **must** be a valid

VkVideoSessionKHR handle

- VUID-vkDestroyVideoSessionKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyVideoSessionKHR-videoSession-parent
  If `videoSession` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `videoSession` **must** be externally synchronized

45.5.3. Video Session Memory Association

After creating a video session object, and before the object **can** be used to record video coding operations into command buffers using it, the application **must** allocate and bind device memory to the video session. Device memory is allocated separately (see Device Memory) and then associated with the video session.

Video sessions **may** have multiple memory bindings identified by unique unsigned integer values. Appropriate device memory **must** be bound to each such memory binding before using the video session to record command buffer commands with it.

To determine the memory requirements for a video session object, call:

```
// Provided by VK_KHR_video_queue
VkResult vkGetVideoSessionMemoryRequirementsKHR(
    VkDevice                                    device,
    VkVideoSessionKHR                           videoSession,
    uint32_t*                                   pMemoryRequirementsCount,
    VkVideoSessionMemoryRequirementsKHR*        pMemoryRequirements);
```

- `device` is the logical device that owns the video session.
- `videoSession` is the video session to query.
- `pMemoryRequirementsCount` is a pointer to an integer related to the number of memory binding requirements available or queried, as described below.
- `pMemoryRequirements` is `NULL` or a pointer to an array of VkVideoSessionMemoryRequirementsKHR structures in which the memory binding requirements of the video session are returned.

If `pMemoryRequirements` is `NULL`, then the number of memory bindings required for the video session is returned in `pMemoryRequirementsCount`. Otherwise, `pMemoryRequirementsCount` **must** point to a variable set by the application to the number of elements in the `pMemoryRequirements` array, and on return the variable is overwritten with the number of memory binding requirements actually

written to pMemoryRequirements. If pMemoryRequirementsCount is less than the number of memory bindings required for the video session, then at most pMemoryRequirementsCount elements will be written to pMemoryRequirements, and VK_INCOMPLETE will be returned, instead of VK_SUCCESS, to indicate that not all required memory binding requirements were returned.

Valid Usage (Implicit)

- VUID-vkGetVideoSessionMemoryRequirementsKHR-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkGetVideoSessionMemoryRequirementsKHR-videoSession-parameter
  videoSession **must** be a valid VkVideoSessionKHR handle
- VUID-vkGetVideoSessionMemoryRequirementsKHR-pMemoryRequirementsCount-parameter
  pMemoryRequirementsCount **must** be a valid pointer to a uint32_t value
- VUID-vkGetVideoSessionMemoryRequirementsKHR-pMemoryRequirements-parameter
  If the value referenced by pMemoryRequirementsCount is not 0, and pMemoryRequirements is not NULL, pMemoryRequirements **must** be a valid pointer to an array of pMemoryRequirementsCount VkVideoSessionMemoryRequirementsKHR structures
- VUID-vkGetVideoSessionMemoryRequirementsKHR-videoSession-parent
  videoSession **must** have been created, allocated, or retrieved from device

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkVideoSessionMemoryRequirementsKHR structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoSessionMemoryRequirementsKHR {
    VkStructureType         sType;
    void*                   pNext;
    uint32_t                memoryBindIndex;
    VkMemoryRequirements    memoryRequirements;
} VkVideoSessionMemoryRequirementsKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.

- `memoryBindIndex` is the index of the memory binding.
- `memoryRequirements` is a VkMemoryRequirements structure in which the requested memory binding requirements for the binding index specified by `memoryBindIndex` are returned.

Valid Usage (Implicit)

- VUID-VkVideoSessionMemoryRequirementsKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_SESSION_MEMORY_REQUIREMENTS_KHR
- VUID-VkVideoSessionMemoryRequirementsKHR-pNext-pNext
  `pNext` **must** be `NULL`

To attach memory to a video session object, call:

```
// Provided by VK_KHR_video_queue
VkResult vkBindVideoSessionMemoryKHR(
    VkDevice                                    device,
    VkVideoSessionKHR                           videoSession,
    uint32_t                                    bindSessionMemoryInfoCount,
    const VkBindVideoSessionMemoryInfoKHR*      pBindSessionMemoryInfos);
```

- `device` is the logical device that owns the video session.
- `videoSession` is the video session to be bound with device memory.
- `bindSessionMemoryInfoCount` is the number of elements in `pBindSessionMemoryInfos`.
- `pBindSessionMemoryInfos` is a pointer to an array of `bindSessionMemoryInfoCount` VkBindVideoSessionMemoryInfoKHR structures specifying memory regions to be bound to specific memory bindings of the video session.

The valid usage statements below refer to the VkMemoryRequirements structure corresponding to a specific element of `pBindSessionMemoryInfos`, which is defined as follows:

- If the `memoryBindIndex` member of the element of `pBindSessionMemoryInfos` in question matches the `memoryBindIndex` member of one of the elements returned in `pMemoryRequirements` when vkGetVideoSessionMemoryRequirementsKHR is called with the same `videoSession` and with `pMemoryRequirementsCount` equal to `bindSessionMemoryInfoCount`, then the `memoryRequirements` member of that element of `pMemoryRequirements` is the VkMemoryRequirements structure corresponding to the element of `pBindSessionMemoryInfos` in question.
- Otherwise the element of `pBindSessionMemoryInfos` in question is said to not have a corresponding VkMemoryRequirements structure.

Valid Usage

- VUID-vkBindVideoSessionMemoryKHR-videoSession-07195
  The memory binding of `videoSession` identified by the `memoryBindIndex` member of any element of `pBindSessionMemoryInfos` **must** not already be backed by a memory object

- VUID-vkBindVideoSessionMemoryKHR-memoryBindIndex-07196
  The `memoryBindIndex` member of each element of `pBindSessionMemoryInfos` **must** be unique within `pBindSessionMemoryInfos`
- VUID-vkBindVideoSessionMemoryKHR-pBindSessionMemoryInfos-07197
  Each element of `pBindSessionMemoryInfos` **must** have a corresponding VkMemoryRequirements structure
- VUID-vkBindVideoSessionMemoryKHR-pBindSessionMemoryInfos-07198
  If an element of `pBindSessionMemoryInfos` has a corresponding VkMemoryRequirements structure, then the `memory` member of that element of `pBindSessionMemoryInfos` **must** have been allocated using one of the memory types allowed in the `memoryTypeBits` member of the corresponding VkMemoryRequirements structure
- VUID-vkBindVideoSessionMemoryKHR-pBindSessionMemoryInfos-07199
  If an element of `pBindSessionMemoryInfos` has a corresponding VkMemoryRequirements structure, then the `memoryOffset` member of that element of `pBindSessionMemoryInfos` **must** be an integer multiple of the `alignment` member of the corresponding VkMemoryRequirements structure
- VUID-vkBindVideoSessionMemoryKHR-pBindSessionMemoryInfos-07200
  If an element of `pBindSessionMemoryInfos` has a corresponding VkMemoryRequirements structure, then the `memorySize` member of that element of `pBindSessionMemoryInfos` **must** equal the `size` member of the corresponding VkMemoryRequirements structure

Valid Usage (Implicit)

- VUID-vkBindVideoSessionMemoryKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkBindVideoSessionMemoryKHR-videoSession-parameter
  `videoSession` **must** be a valid VkVideoSessionKHR handle
- VUID-vkBindVideoSessionMemoryKHR-pBindSessionMemoryInfos-parameter
  `pBindSessionMemoryInfos` **must** be a valid pointer to an array of `bindSessionMemoryInfoCount` valid VkBindVideoSessionMemoryInfoKHR structures
- VUID-vkBindVideoSessionMemoryKHR-bindSessionMemoryInfoCount-arraylength
  `bindSessionMemoryInfoCount` **must** be greater than `0`
- VUID-vkBindVideoSessionMemoryKHR-videoSession-parent
  `videoSession` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `videoSession` **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkBindVideoSessionMemoryInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkBindVideoSessionMemoryInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           memoryBindIndex;
    VkDeviceMemory     memory;
    VkDeviceSize       memoryOffset;
    VkDeviceSize       memorySize;
} VkBindVideoSessionMemoryInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryBindIndex` is the memory binding index to bind memory to.
- `memory` is the allocated device memory to be bound to the video session's memory binding with index `memoryBindIndex`.
- `memoryOffset` is the start offset of the region of `memory` which is to be bound.
- `memorySize` is the size in bytes of the region of `memory`, starting from `memoryOffset` bytes, to be bound.

Valid Usage

- VUID-VkBindVideoSessionMemoryInfoKHR-memoryOffset-07201
  `memoryOffset` **must** be less than the size of `memory`
- VUID-VkBindVideoSessionMemoryInfoKHR-memorySize-07202
  `memorySize` **must** be less than or equal to the size of `memory` minus `memoryOffset`

Valid Usage (Implicit)

- VUID-VkBindVideoSessionMemoryInfoKHR-sType-sType

`sType` **must** be VK_STRUCTURE_TYPE_BIND_VIDEO_SESSION_MEMORY_INFO_KHR

- VUID-VkBindVideoSessionMemoryInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkBindVideoSessionMemoryInfoKHR-memory-parameter
  `memory` **must** be a valid VkDeviceMemory handle

45.6. Video Profile Compatibility

Resources and query pools used with a particular video session **must** be compatible with the video profile the video session was created with.

A VkBuffer is compatible with a video profile if it was created with the VkBufferCreateInfo::`pNext` chain including a VkVideoProfileListInfoKHR structure with its `pProfiles` member containing an element matching the VkVideoProfileInfoKHR structure chain describing the video profile, and VkBufferCreateInfo::`usage` including at least one bit specific to video coding usage.

- VK_BUFFER_USAGE_VIDEO_DECODE_SRC_BIT_KHR
- VK_BUFFER_USAGE_VIDEO_DECODE_DST_BIT_KHR
- VK_BUFFER_USAGE_VIDEO_ENCODE_SRC_BIT_KHR
- VK_BUFFER_USAGE_VIDEO_ENCODE_DST_BIT_KHR

A VkBuffer is also compatible with a video profile if it was created with VkBufferCreateInfo::`flags` including VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR.

A VkImage is compatible with a video profile if it was created with the VkImageCreateInfo::`pNext` chain including a VkVideoProfileListInfoKHR structure with its `pProfiles` member containing an element matching the VkVideoProfileInfoKHR structure chain describing the video profile, and VkImageCreateInfo::`usage` including at least one bit specific to video coding usage.

- VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR
- VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR
- VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR
- VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR
- VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR
- VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR
- VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR
- VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR

A VkImage is also compatible with a video profile if all of the following conditions are true for the VkImageCreateInfo structure the image was created with:

- VkImageCreateInfo::`flags` included VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR.

- The list of VkVideoFormatPropertiesKHR structures, obtained by calling vkGetPhysicalDeviceVideoFormatPropertiesKHR with VkPhysicalDeviceVideoFormatInfoKHR::imageUsage equal to the VkImageCreateInfo::usage the image was created with and the VkPhysicalDeviceVideoFormatInfoKHR::pNext chain including a VkVideoProfileListInfoKHR structure with its pProfiles member containing a single array element specifying the VkVideoProfileInfoKHR structure chain describing the video profile in question, contains an element for which all of the following conditions are true with respect to the VkImageCreateInfo structure the image was created with:
  - VkImageCreateInfo::format equals VkVideoFormatPropertiesKHR::format.
  - VkImageCreateInfo::flags only contains VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR and/or bits also set in VkVideoFormatPropertiesKHR::imageCreateFlags.

    NOTE Specifying VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR when creating decode output pictures or encode input pictures is always supported when the videoMaintenance1 feature is enabled, regardless of the supported VkImageCreateFlags reported in VkVideoFormatPropertiesKHR::imageCreateFlags. Accordingly, implementations **should** not report VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR in VkVideoFormatPropertiesKHR::imageCreateFlags for any video format.
  - VkImageCreateInfo::imageType equals VkVideoFormatPropertiesKHR::imageType.
  - VkImageCreateInfo::tiling equals VkVideoFormatPropertiesKHR::imageTiling.
  - VkImageCreateInfo::usage only contains bits also set in VkVideoFormatPropertiesKHR::imageUsageFlags, or VkImageCreateInfo::flags includes VK_IMAGE_CREATE_EXTENDED_USAGE_BIT.

    NOTE While some of these rules allow creating buffer or image resources that **may** be compatible with any video profile, applications **should** still prefer to include the specific video profiles the buffer or image resource is expected to be used with (through a VkVideoProfileListInfoKHR structure included in the pNext chain of the corresponding create info structure) whenever the information about the complete set of video profiles is available at resource creation time, to enable the implementation to optimize the created resource for the specific use case. In the absence of that information, the implementation **may** have to make conservative decisions about the memory requirements or representation of the resource.

A VkImageView is compatible with a video profile if the VkImage it was created from is also compatible with that video profile.

A VkQueryPool is compatible with a video profile if it was created with the VkQueryPoolCreateInfo::pNext chain including a VkVideoProfileInfoKHR structure chain describing the same video profile, and VkQueryPoolCreateInfo::queryType having one of the following values:

- VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR
- VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR

45.7. Video Session Parameters

Video session parameters objects **can** store preprocessed codec-specific parameters used with a compatible video session, and enable reducing the number of parameters needed to be provided and processed by the implementation while recording video coding operations into command buffers.

Parameters stored in such objects are *immutable* to facilitate the concurrent use of the stored parameters in multiple threads. At the same time, new parameters **can** be added to existing objects using the vkUpdateVideoSessionParametersKHR command.

In order to support concurrent use of the stored immutable parameters while also allowing the video session parameters object to be extended with new parameters, each video session parameters object maintains an *update sequence counter* that is set to `0` at object creation time and **must** be incremented by each subsequent update operation.

Certain video sequences that adhere to particular video compression standards permit updating previously supplied parameters. If a parameter update is necessary, the application has the following options:

- Cache the set of parameters on the application side and create a new video session parameters object adding all the parameters with appropriate changes, as necessary; or
- Create a new video session parameters object providing only the updated parameters and the previously used object as the template, which ensures that parameters not specified at creation time will be copied unmodified from the template object.

The actual types of parameters that **can** be stored and the capacity for individual parameter types, and the methods of initializing, updating, and referring to individual parameters are specific to the video codec operation the video session parameters object was created with.

- For VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR these are defined in the H.264 Decode Parameter Sets section.
- For VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR these are defined in the H.265 Decode Parameter Sets section.
- For VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR there are no defined parameter sets and therefore do not use video session parameters objects.
- For VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR these are defined in the AV1 Decode Parameter Sets section.
- For VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR these are defined in the H.264 Encode Parameter Sets section.
- For VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR these are defined in the H.265 Encode Parameter Sets section.
- For VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR these are defined in the AV1 Encode Parameter Sets section.

Video session parameters objects created with an encode operation are further specialized based

on the video encode quality level the video session parameters are used with, as implementations **may** apply different sets of parameter overrides depending on the used quality level. This enables implementations to store the potentially optimized set of parameters in these objects, further limiting the necessary processing required while recording video encode operations into command buffers.

Video session parameters are represented by VkVideoSessionParametersKHR handles:

```
// Provided by VK_KHR_video_queue
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkVideoSessionParametersKHR)
```

45.7.1. Creating Video Session Parameters

To create a video session parameters object, call:

```
// Provided by VK_KHR_video_queue
VkResult vkCreateVideoSessionParametersKHR(
    VkDevice                                    device,
    const VkVideoSessionParametersCreateInfoKHR* pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkVideoSessionParametersKHR*                pVideoSessionParameters);
```

- device is the logical device that creates the video session parameters object.
- pCreateInfo is a pointer to a VkVideoSessionParametersCreateInfoKHR structure containing parameters to be used to create the video session parameters object.
- pAllocator controls host memory allocation as described in the Memory Allocation chapter.
- pVideoSessionParameters is a pointer to a VkVideoSessionParametersKHR handle in which the resulting video session parameters object is returned.

The resulting video session parameters object is said to be created with the video codec operation pCreateInfo->videoSession was created with.

Video session parameters objects created with an encode operation are always created with respect to a video encode quality level. By default, the created video session parameters objects are created with quality level zero, unless otherwise specified by including a VkVideoEncodeQualityLevelInfoKHR structure in the pCreateInfo->pNext chain, in which case the video session parameters object is created with the quality level specified in VkVideoEncodeQualityLevelInfoKHR::qualityLevel.

If pCreateInfo->videoSessionParametersTemplate is not VK_NULL_HANDLE, then it will be used as a template for constructing the new video session parameters object. This happens by first adding any parameters according to the additional creation parameters provided in the pCreateInfo->pNext chain, followed by adding any parameters from the template object that have a key that does not match the key of any of the already added parameters.

For video session parameters objects created with an encode operation, the template object

specified in pCreateInfo->videoSessionParametersTemplate **must** have been created with the same video encode quality level as the newly created object.

NOTE This means that codec-specific parameters stored in video session parameters objects **can** only be reused across different video encode quality levels by re-specifying them, as previously created video session parameters against other quality levels **cannot** be used as template because the original codec-specific parameters (before the implementation **may** have applied parameter overrides) **may** no longer be available in them for the purposes of constructing the derived object.

Video session parameters objects are only compatible with quantization maps if they are created with pCreateInfo->flags including VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR.

Video session parameters objects created with VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR against a video session object that was created with VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_EMPHASIS_MAP_BIT_KHR are created with a specific compatible quantization map texel size specified in the quantizationMapTexelSize member of the VkVideoEncodeQuantizationMapSessionParametersCreateInfoKHR structure included in the pNext chain of pCreateInfo.

NOTE This means that the quantization map texel size that such a video session parameters object is compatible with is fixed for the lifetime of the object. Applications have to create separate video session parameters objects to use different quantization map texel sizes with a single video session object. This is necessary because the used quantization map texel size may affect the parameter overrides the implementation has to perform and thus the final values of the used codec-specific parameters.

For video session parameters objects created with VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR, the template object specified in pCreateInfo->videoSessionParametersTemplate **must** also have been created with VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR and the same compatible quantization map texel size specified in VkVideoEncodeQuantizationMapSessionParametersCreateInfoKHR::quantizationMapTexelSize.

NOTE This means that codec-specific parameters stored in video session parameters objects **can** only be reused with different quantization map texel sizes by re-specifying them, as previously created video session parameters against other quantization map texel sizes **cannot** be used as template because the original codec-specific parameters (before the implementation **may** have applied parameter overrides) **may** no longer be available in them for the purposes of constructing the derived object.

For video session parameters objects created without VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR, the template object specified in pCreateInfo->videoSessionParametersTemplate **must** also have been created without VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR.

If pCreateInfo->videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the created video session parameters object will initially contain the following sets of parameter entries:

- StdVideoH264SequenceParameterSet structures representing H.264 SPS entries, as follows:
  - If the pParametersAddInfo member of the VkVideoDecodeH264SessionParametersCreateInfoKHR structure provided in the pCreateInfo->pNext chain is not NULL, then the set of StdVideoH264SequenceParameterSet entries specified in pParametersAddInfo->pStdSPSs are added first;
  - If pCreateInfo->videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH264SequenceParameterSet entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same seq_parameter_set_id.
- StdVideoH264PictureParameterSet structures representing H.264 PPS entries, as follows:
  - If the pParametersAddInfo member of the VkVideoDecodeH264SessionParametersCreateInfoKHR structure provided in the pCreateInfo->pNext chain is not NULL, then the set of StdVideoH264PictureParameterSet entries specified in pParametersAddInfo->pStdPPSs are added first;
  - If pCreateInfo->videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH264PictureParameterSet entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same seq_parameter_set_id and pic_parameter_set_id.

If pCreateInfo->videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the created video session parameters object will initially contain the following sets of parameter entries:

- StdVideoH265VideoParameterSet structures representing H.265 VPS entries, as follows:
  - If the pParametersAddInfo member of the VkVideoDecodeH265SessionParametersCreateInfoKHR structure provided in the pCreateInfo->pNext chain is not NULL, then the set of StdVideoH265VideoParameterSet entries specified in pParametersAddInfo->pStdVPSs are added first;
  - If pCreateInfo->videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH265VideoParameterSet entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same vps_video_parameter_set_id.
- StdVideoH265SequenceParameterSet structures representing H.265 SPS entries, as follows:
  - If the pParametersAddInfo member of the VkVideoDecodeH265SessionParametersCreateInfoKHR structure provided in the

pCreateInfo->pNext chain is not NULL, then the set of StdVideoH265SequenceParameterSet entries specified in pParametersAddInfo->pStdSPSs are added first;

- If pCreateInfo->videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH265SequenceParameterSet entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same sps_video_parameter_set_id and sps_seq_parameter_set_id.

- StdVideoH265PictureParameterSet structures representing H.265 PPS entries, as follows:
  - If the pParametersAddInfo member of the VkVideoDecodeH265SessionParametersCreateInfoKHR structure provided in the pCreateInfo->pNext chain is not NULL, then the set of StdVideoH265PictureParameterSet entries specified in pParametersAddInfo->pStdPPSs are added first;
  - If pCreateInfo->videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH265PictureParameterSet entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same sps_video_parameter_set_id, pps_seq_parameter_set_id, and pps_pic_parameter_set_id.

If pCreateInfo->videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then the created video session parameters object will contain a single AV1 sequence header represented by a StdVideoAV1SequenceHeader structure specified through the pStdSequenceHeader member of the VkVideoDecodeAV1SessionParametersCreateInfoKHR structure provided in the pCreateInfo->pNext chain. As such video session parameters objects **can** only contain a single AV1 sequence header, it is not possible to use a previously created object as a template or subsequently update the created video session parameters object.

If pCreateInfo->videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the created video session parameters object will initially contain the following sets of parameter entries:

- StdVideoH264SequenceParameterSet structures representing H.264 SPS entries, as follows:
  - If the pParametersAddInfo member of the VkVideoEncodeH264SessionParametersCreateInfoKHR structure provided in the pCreateInfo->pNext chain is not NULL, then the set of StdVideoH264SequenceParameterSet entries specified in pParametersAddInfo->pStdSPSs are added first;
  - If pCreateInfo->videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH264SequenceParameterSet entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same seq_parameter_set_id.
- StdVideoH264PictureParameterSet structures representing H.264 PPS entries, as follows:
  - If the pParametersAddInfo member of the VkVideoEncodeH264SessionParametersCreateInfoKHR structure provided in the pCreateInfo->pNext chain is not NULL, then the set of StdVideoH264PictureParameterSet entries specified in pParametersAddInfo->pStdPPSs are added first;
  - If pCreateInfo->videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH264PictureParameterSet entry stored in it is copied to the created video session

parameters object if the created object does not already contain such an entry with the same `seq_parameter_set_id` and `pic_parameter_set_id`.

If `pCreateInfo->videoSession` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the created video session parameters object will initially contain the following sets of parameter entries:

- `StdVideoH265VideoParameterSet` structures representing H.265 VPS entries, as follows:
  - If the `pParametersAddInfo` member of the VkVideoEncodeH265SessionParametersCreateInfoKHR structure provided in the `pCreateInfo->pNext` chain is not `NULL`, then the set of `StdVideoH265VideoParameterSet` entries specified in `pParametersAddInfo->pStdVPSs` are added first;
  - If `pCreateInfo->videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH265VideoParameterSet` entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same `vps_video_parameter_set_id`.
- `StdVideoH265SequenceParameterSet` structures representing H.265 SPS entries, as follows:
  - If the `pParametersAddInfo` member of the VkVideoEncodeH265SessionParametersCreateInfoKHR structure provided in the `pCreateInfo->pNext` chain is not `NULL`, then the set of `StdVideoH265SequenceParameterSet` entries specified in `pParametersAddInfo->pStdSPSs` are added first;
  - If `pCreateInfo->videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH265SequenceParameterSet` entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same `sps_video_parameter_set_id` and `sps_seq_parameter_set_id`.
- `StdVideoH265PictureParameterSet` structures representing H.265 PPS entries, as follows:
  - If the `pParametersAddInfo` member of the VkVideoEncodeH265SessionParametersCreateInfoKHR structure provided in the `pCreateInfo->pNext` chain is not `NULL`, then the set of `StdVideoH265PictureParameterSet` entries specified in `pParametersAddInfo->pStdPPSs` are added first;
  - If `pCreateInfo->videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH265PictureParameterSet` entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id`.

If `pCreateInfo->videoSession` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then the created video session parameters object will contain a single AV1 sequence header specified through the members of the VkVideoEncodeAV1SessionParametersCreateInfoKHR structure provided in the `pCreateInfo->pNext` chain. As such video session parameters objects **can** only contain a single AV1 sequence header, it is not possible to use a previously created object as a template or subsequently update the created video session parameters object.

In case of video session parameters objects created with a video encode operation, implementations **may** return the VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR error if any of the specified

Video Std parameters do not adhere to the syntactic or semantic requirements of the used video compression standard, or if values derived from parameters according to the rules defined by the used video compression standard do not adhere to the capabilities of the video compression standard or the implementation.

NOTE	Applications **should** not rely on the VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR error being returned by any command as a means to verify Video Std parameters, as implementations are not required to report the error in any specific set of cases.

Valid Usage (Implicit)

- VUID-vkCreateVideoSessionParametersKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkCreateVideoSessionParametersKHR-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkVideoSessionParametersCreateInfoKHR structure
- VUID-vkCreateVideoSessionParametersKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateVideoSessionParametersKHR-pVideoSessionParameters-parameter
  `pVideoSessionParameters` **must** be a valid pointer to a VkVideoSessionParametersKHR handle
- VUID-vkCreateVideoSessionParametersKHR-device-queuecount
  The device **must** have been created with at least `1` queue

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INITIALIZATION_FAILED
- VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkVideoSessionParametersCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoSessionParametersCreateInfoKHR {
    VkStructureType                              sType;
    const void*                                  pNext;
    VkVideoSessionParametersCreateFlagsKHR       flags;
    VkVideoSessionParametersKHR                  videoSessionParametersTemplate;
    VkVideoSessionKHR                            videoSession;
} VkVideoSessionParametersCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkVideoSessionParametersCreateFlagBitsKHR specifying create flags.
- `videoSessionParametersTemplate` is `VK_NULL_HANDLE` or a valid handle to a VkVideoSessionParametersKHR object used as a template for constructing the new video session parameters object.
- `videoSession` is the video session object against which the video session parameters object is going to be created.

Limiting values are defined below that are referenced by the relevant valid usage statements of this structure.

- If `videoSession` was created with the codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then let `StdVideoH264SequenceParameterSet spsAddList[]` be the list of H.264 SPS entries to add to the created video session parameters object, defined as follows:
  - If the `pParametersAddInfo` member of the VkVideoDecodeH264SessionParametersCreateInfoKHR structure provided in the `pNext` chain is not `NULL`, then the set of `StdVideoH264SequenceParameterSet` entries specified in `pParametersAddInfo->pStdSPSs` are added to `spsAddList`;
  - If `videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH264SequenceParameterSet` entry stored in it with `seq_parameter_set_id` not matching any of the entries already in `spsAddList` is added to `spsAddList`.
- If `videoSession` was created with the codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then let `StdVideoH264PictureParameterSet ppsAddList[]` be the list of H.264 PPS entries to add to the created video session parameters object, defined as follows:
  - If the `pParametersAddInfo` member of the VkVideoDecodeH264SessionParametersCreateInfoKHR structure provided in the `pNext` chain is not `NULL`, then the set of `StdVideoH264PictureParameterSet` entries specified in `pParametersAddInfo->pStdPPSs` are added to `ppsAddList`;
  - If `videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH264PictureParameterSet` entry stored in it with `seq_parameter_set_id` or `pic_parameter_set_id` not matching any of the entries already in `ppsAddList` is added to `ppsAddList`.

- If videoSession was created with the codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then let StdVideoH265VideoParameterSet vpsAddList[] be the list of H.265 VPS entries to add to the created video session parameters object, defined as follows:
  - If the pParametersAddInfo member of the VkVideoDecodeH265SessionParametersCreateInfoKHR structure provided in the pNext chain is not NULL, then the set of StdVideoH265VideoParameterSet entries specified in pParametersAddInfo->pStdVPSs are added to vpsAddList;
  - If videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH265VideoParameterSet entry stored in it with vps_video_parameter_set_id not matching any of the entries already in vpsAddList is added to vpsAddList.
- If videoSession was created with the codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then let StdVideoH265SequenceParameterSet spsAddList[] be the list of H.265 SPS entries to add to the created video session parameters object, defined as follows:
  - If the pParametersAddInfo member of the VkVideoDecodeH265SessionParametersCreateInfoKHR structure provided in the pNext chain is not NULL, then the set of StdVideoH265SequenceParameterSet entries specified in pParametersAddInfo->pStdSPSs are added to spsAddList;
  - If videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH265SequenceParameterSet entry stored in it with sps_video_parameter_set_id or sps_seq_parameter_set_id not matching any of the entries already in spsAddList is added to spsAddList.
- If videoSession was created with the codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then let StdVideoH265PictureParameterSet ppsAddList[] be the list of H.265 PPS entries to add to the created video session parameters object, defined as follows:
  - If the pParametersAddInfo member of the VkVideoDecodeH265SessionParametersCreateInfoKHR structure provided in the pNext chain is not NULL, then the set of StdVideoH265PictureParameterSet entries specified in pParametersAddInfo->pStdPPSs are added to ppsAddList;
  - If videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH265PictureParameterSet entry stored in it with sps_video_parameter_set_id, pps_seq_parameter_set_id, or pps_pic_parameter_set_id not matching any of the entries already in ppsAddList is added to ppsAddList.
- If videoSession was created with an encode operation, then let uint32_t qualityLevel be the video encode quality level of the created video session parameters object, defined as follows:
  - If the pNext chain of this structure includes a VkVideoEncodeQualityLevelInfoKHR structure, then qualityLevel is equal to VkVideoEncodeQualityLevelInfoKHR::qualityLevel.
  - Otherwise qualityLevel is 0.
- If videoSession was created with the codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then let StdVideoH264SequenceParameterSet spsAddList[] be the list of H.264 SPS entries to add to the

created video session parameters object, defined as follows:

  - If the pParametersAddInfo member of the VkVideoEncodeH264SessionParametersCreateInfoKHR structure provided in the pNext chain is not NULL, then the set of StdVideoH264SequenceParameterSet entries specified in pParametersAddInfo->pStdSPSs are added to spsAddList;
  - If videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH264SequenceParameterSet entry stored in it with seq_parameter_set_id not matching any of the entries already in spsAddList is added to spsAddList.

- If videoSession was created with the codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then let StdVideoH264PictureParameterSet ppsAddList[] be the list of H.264 PPS entries to add to the created video session parameters object, defined as follows:
  - If the pParametersAddInfo member of the VkVideoEncodeH264SessionParametersCreateInfoKHR structure provided in the pNext chain is not NULL, then the set of StdVideoH264PictureParameterSet entries specified in pParametersAddInfo->pStdPPSs are added to ppsAddList;
  - If videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH264PictureParameterSet entry stored in it with seq_parameter_set_id or pic_parameter_set_id not matching any of the entries already in ppsAddList is added to ppsAddList.
- If videoSession was created with the codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then let StdVideoH265VideoParameterSet vpsAddList[] be the list of H.265 VPS entries to add to the created video session parameters object, defined as follows:
  - If the pParametersAddInfo member of the VkVideoEncodeH265SessionParametersCreateInfoKHR structure provided in the pNext chain is not NULL, then the set of StdVideoH265VideoParameterSet entries specified in pParametersAddInfo->pStdVPSs are added to vpsAddList;
  - If videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH265VideoParameterSet entry stored in it with vps_video_parameter_set_id not matching any of the entries already in vpsAddList is added to vpsAddList.
- If videoSession was created with the codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then let StdVideoH265SequenceParameterSet spsAddList[] be the list of H.265 SPS entries to add to the created video session parameters object, defined as follows:
  - If the pParametersAddInfo member of the VkVideoEncodeH265SessionParametersCreateInfoKHR structure provided in the pNext chain is not NULL, then the set of StdVideoH265SequenceParameterSet entries specified in pParametersAddInfo->pStdSPSs are added to spsAddList;
  - If videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH265SequenceParameterSet entry stored in it with sps_video_parameter_set_id or sps_seq_parameter_set_id not matching any of the entries already in spsAddList is added to spsAddList.

- If videoSession was created with the codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then let StdVideoH265PictureParameterSet ppsAddList[] be the list of H.265 PPS entries to add to the created video session parameters object, defined as follows:
  - If the pParametersAddInfo member of the VkVideoEncodeH265SessionParametersCreateInfoKHR structure provided in the pNext chain is not NULL, then the set of StdVideoH265PictureParameterSet entries specified in pParametersAddInfo->pStdPPSs are added to ppsAddList;
  - If videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH265PictureParameterSet entry stored in it with sps_video_parameter_set_id, pps_seq_parameter_set_id, or pps_pic_parameter_set_id not matching any of the entries already in ppsAddList is added to ppsAddList.

Valid Usage

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSessionParametersTemplate-04855
  If videoSessionParametersTemplate is not VK_NULL_HANDLE, it **must** have been created against videoSession
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSessionParametersTemplate-08310
  If videoSessionParametersTemplate is not VK_NULL_HANDLE and videoSession was created with an encode operation, then qualityLevel **must** equal the video encode quality level videoSessionParametersTemplate was created with
- VUID-VkVideoSessionParametersCreateInfoKHR-flags-10271
  If flags includes VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR, then videoSession **must** have been created with VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_EMPHASIS_MAP_BIT_KHR
- VUID-VkVideoSessionParametersCreateInfoKHR-flags-10272
  If flags includes VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR, then the pNext chain **must** include a VkVideoEncodeQuantizationMapSessionParametersCreateInfoKHR structure
- VUID-VkVideoSessionParametersCreateInfoKHR-flags-10273
  If flags includes VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR and videoSession was created with VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR, then the list of video format properties supported for the image usage flag VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR **must** have an element for which VkVideoFormatQuantizationMapPropertiesKHR ::quantizationMapTexelSize equals the quantizationMapTexelSize member of the VkVideoEncodeQuantizationMapSessionParametersCreateInfoKHR structure included in the pNext chain
- VUID-VkVideoSessionParametersCreateInfoKHR-flags-10274

If flags includes VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR and videoSession was created with VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_EMPHASIS_MAP_BIT_KHR, then the list of video format properties supported for the image usage flag VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR **must** have an element for which VkVideoFormatQuantizationMapPropertiesKHR::quantizationMapTexelSize equals the quantizationMapTexelSize member of the VkVideoEncodeQuantizationMapSessionParametersCreateInfoKHR structure included in the pNext chain

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSessionParametersTemplate-10275
  If videoSessionParametersTemplate is not VK_NULL_HANDLE and flags includes VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR, then videoSessionParametersTemplate **must** have been created with VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSessionParametersTemplate-10276
  If videoSessionParametersTemplate is not VK_NULL_HANDLE and flags includes VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR, then videoSessionParametersTemplate **must** have been created with the same quantization map texel size as the one specified in the quantizationMapTexelSize member of the VkVideoEncodeQuantizationMapSessionParametersCreateInfoKHR structure included in the pNext chain
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSessionParametersTemplate-10277
  If videoSessionParametersTemplate is not VK_NULL_HANDLE and flags does not include VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR, then videoSessionParametersTemplate **must** have been created without VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07203
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the pNext chain **must** include a VkVideoDecodeH264SessionParametersCreateInfoKHR structure
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07204
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the number of elements of spsAddList **must** be less than or equal to the maxStdSPSCount specified in the VkVideoDecodeH264SessionParametersCreateInfoKHR structure included in the pNext chain
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07205
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the number of elements of ppsAddList **must** be less than or equal to the maxStdPPSCount specified in the VkVideoDecodeH264SessionParametersCreateInfoKHR structure included in the pNext chain

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07206
  If `videoSession` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the `pNext` chain **must** include a VkVideoDecodeH265SessionParametersCreateInfoKHR structure
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07207
  If `videoSession` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the number of elements of `vpsAddList` **must** be less than or equal to the `maxStdVPSCount` specified in the VkVideoDecodeH265SessionParametersCreateInfoKHR structure included in the `pNext` chain
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07208
  If `videoSession` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the number of elements of `spsAddList` **must** be less than or equal to the `maxStdSPSCount` specified in the VkVideoDecodeH265SessionParametersCreateInfoKHR structure included in the `pNext` chain
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07209
  If `videoSession` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the number of elements of `ppsAddList` **must** be less than or equal to the `maxStdPPSCount` specified in the VkVideoDecodeH265SessionParametersCreateInfoKHR structure included in the `pNext` chain
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-10794
  `videoSession` **must** not have been created with the codec operation VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-09258
  If `videoSession` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then `videoSessionParametersTemplate` **must** be `VK_NULL_HANDLE`
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-09259
  If `videoSession` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then the `pNext` chain **must** include a VkVideoDecodeAV1SessionParametersCreateInfoKHR structure
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07210
  If `videoSession` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the `pNext` chain **must** include a VkVideoEncodeH264SessionParametersCreateInfoKHR structure
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-04839
  If `videoSession` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the number of elements of `spsAddList` **must** be less than or equal to the `maxStdSPSCount` specified in the VkVideoEncodeH264SessionParametersCreateInfoKHR structure included in the `pNext` chain
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-04840

If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the number of elements of ppsAddList **must** be less than or equal to the maxStdPPSCount specified in the VkVideoEncodeH264SessionParametersCreateInfoKHR structure included in the pNext chain

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07211
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the pNext chain **must** include a VkVideoEncodeH265SessionParametersCreateInfoKHR structure
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-04841
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the number of elements of vpsAddList **must** be less than or equal to the maxStdVPSCount specified in the VkVideoEncodeH265SessionParametersCreateInfoKHR structure included in the pNext chain
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-04842
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the number of elements of spsAddList **must** be less than or equal to the maxStdSPSCount specified in the VkVideoEncodeH265SessionParametersCreateInfoKHR structure included in the pNext chain
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-04843
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the number of elements of ppsAddList **must** be less than or equal to the maxStdPPSCount specified in the VkVideoEncodeH265SessionParametersCreateInfoKHR structure included in the pNext chain
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-08319
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then num_tile_columns_minus1 **must** be less than VkVideoEncodeH265CapabilitiesKHR::maxTiles.width, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile videoSession was created with, for each element of ppsAddList
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-08320
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then num_tile_rows_minus1 **must** be less than VkVideoEncodeH265CapabilitiesKHR::maxTiles.height, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile videoSession was created with, for each element of ppsAddList
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-10278
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then videoSessionParametersTemplate **must** be VK_NULL_HANDLE
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-10279
  If videoSession was created with the video codec operation

VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then the pNext chain **must** include a VkVideoEncodeAV1SessionParametersCreateInfoKHR structure

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-10280
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then the stdOperatingPointCount member of the VkVideoEncodeAV1SessionParametersCreateInfoKHR structure included in the pNext chain **must** be less than or equal to VkVideoEncodeAV1CapabilitiesKHR::maxOperatingPoints, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile videoSession was created with

Valid Usage (Implicit)

- VUID-VkVideoSessionParametersCreateInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_SESSION_PARAMETERS_CREATE_INFO_KHR
- VUID-VkVideoSessionParametersCreateInfoKHR-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkVideoDecodeAV1SessionParametersCreateInfoKHR, VkVideoDecodeH264SessionParametersCreateInfoKHR, VkVideoDecodeH265SessionParametersCreateInfoKHR, VkVideoEncodeAV1SessionParametersCreateInfoKHR, VkVideoEncodeH264SessionParametersCreateInfoKHR, VkVideoEncodeH265SessionParametersCreateInfoKHR, VkVideoEncodeQualityLevelInfoKHR, or VkVideoEncodeQuantizationMapSessionParametersCreateInfoKHR
- VUID-VkVideoSessionParametersCreateInfoKHR-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkVideoSessionParametersCreateInfoKHR-flags-parameter
  If flags is not 0, flags **must** be a valid combination of VkVideoSessionParametersCreateFlagBitsKHR values
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSessionParametersTemplate-parameter
  If videoSessionParametersTemplate is not VK_NULL_HANDLE, videoSessionParametersTemplate **must** be a valid VkVideoSessionParametersKHR handle
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-parameter
  videoSession **must** be a valid VkVideoSessionKHR handle
- VUID-VkVideoSessionParametersCreateInfoKHR-commonparent
  Both of videoSession, and videoSessionParametersTemplate that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Bits which **can** be set in VkVideoSessionParametersCreateInfoKHR::flags are:

```
// Provided by VK_KHR_video_encode_quantization_map
typedef enum VkVideoSessionParametersCreateFlagBitsKHR {
  // Provided by VK_KHR_video_encode_quantization_map
    VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR =
0x00000001,
} VkVideoSessionParametersCreateFlagBitsKHR;
```

- VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR specifies that the created video session parameters object **can** be used with quantization maps.

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoSessionParametersCreateFlagsKHR;
```

VkVideoSessionParametersCreateFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoSessionParametersCreateFlagBitsKHR.

The VkVideoEncodeQuantizationMapSessionParametersCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_quantization_map
typedef struct VkVideoEncodeQuantizationMapSessionParametersCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkExtent2D         quantizationMapTexelSize;
} VkVideoEncodeQuantizationMapSessionParametersCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- quantizationMapTexelSize specifies the quantization map texel size a video session parameters object created with VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR is compatible with.

Valid Usage (Implicit)

- VUID-VkVideoEncodeQuantizationMapSessionParametersCreateInfoKHR-sType-sType sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUANTIZATION_MAP_SESSION_PARAMETERS_CREATE_INFO_KHR

Structure Chaining

Extends the structure

- VkVideoSessionParametersCreateInfoKHR

45.7.2. Destroying Video Session Parameters

To destroy a video session parameters object, call:

```
// Provided by VK_KHR_video_queue
void vkDestroyVideoSessionParametersKHR(
    VkDevice                                    device,
    VkVideoSessionParametersKHR                 videoSessionParameters,
    const VkAllocationCallbacks*                pAllocator);
```

- `device` is the logical device that destroys the video session parameters object.
- `videoSessionParameters` is the video session parameters object to destroy.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyVideoSessionParametersKHR-videoSessionParameters-07212
  All submitted commands that refer to `videoSessionParameters` **must** have completed execution
- VUID-vkDestroyVideoSessionParametersKHR-videoSessionParameters-07213
  If `VkAllocationCallbacks` were provided when `videoSessionParameters` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyVideoSessionParametersKHR-videoSessionParameters-07214
  If no `VkAllocationCallbacks` were provided when `videoSessionParameters` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyVideoSessionParametersKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkDestroyVideoSessionParametersKHR-videoSessionParameters-parameter
  If `videoSessionParameters` is not VK_NULL_HANDLE, `videoSessionParameters` **must** be a valid VkVideoSessionParametersKHR handle
- VUID-vkDestroyVideoSessionParametersKHR-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyVideoSessionParametersKHR-videoSessionParameters-parent
  If `videoSessionParameters` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `videoSessionParameters` **must** be externally synchronized

45.7.3. Updating Video Session Parameters

To update a video session parameters object with new parameters, call:

```
// Provided by VK_KHR_video_queue
VkResult vkUpdateVideoSessionParametersKHR(
    VkDevice                                    device,
    VkVideoSessionParametersKHR                 videoSessionParameters,
    const VkVideoSessionParametersUpdateInfoKHR* pUpdateInfo);
```

- `device` is the logical device that updates the video session parameters.
- `videoSessionParameters` is the video session parameters object to update.
- `pUpdateInfo` is a pointer to a VkVideoSessionParametersUpdateInfoKHR structure specifying the parameter update information.

After a successful call to this command, the update sequence counter of `videoSessionParameters` is changed to the value specified in `pUpdateInfo->updateSequenceCount`.

NOTE

As each update issued to a video session parameters object needs to specify the next available update sequence count value, concurrent updates of the same video session parameters object are inherently disallowed. However, recording video coding operations to command buffers referring to parameters previously added to the video session parameters object is allowed, even if there is a concurrent update in progress adding some new entries to the object.

If `videoSessionParameters` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and the `pUpdateInfo->pNext` chain includes a VkVideoDecodeH264SessionParametersAddInfoKHR structure, then this command adds the following parameter entries to `videoSessionParameters`:

- The H.264 SPS entries specified in VkVideoDecodeH264SessionParametersAddInfoKHR::`pStdSPSs`.
- The H.264 PPS entries specified in VkVideoDecodeH264SessionParametersAddInfoKHR::`pStdPPSs`.

If `videoSessionParameters` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR and the `pUpdateInfo->pNext` chain includes a VkVideoDecodeH265SessionParametersAddInfoKHR structure, then this command adds the following parameter entries to `videoSessionParameters`:

- The H.265 VPS entries specified in VkVideoDecodeH265SessionParametersAddInfoKHR::`pStdVPSs`.

- The H.265 SPS entries specified in VkVideoDecodeH265SessionParametersAddInfoKHR::pStdSPSs.
- The H.265 PPS entries specified in VkVideoDecodeH265SessionParametersAddInfoKHR::pStdPPSs.

If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and the pUpdateInfo->pNext chain includes a VkVideoEncodeH264SessionParametersAddInfoKHR structure, then this command adds the following parameter entries to videoSessionParameters:

- The H.264 SPS entries specified in VkVideoEncodeH264SessionParametersAddInfoKHR::pStdSPSs.
- The H.264 PPS entries specified in VkVideoEncodeH264SessionParametersAddInfoKHR::pStdPPSs.

If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and the pUpdateInfo->pNext chain includes a VkVideoEncodeH265SessionParametersAddInfoKHR structure, then this command adds the following parameter entries to videoSessionParameters:

- The H.265 VPS entries specified in VkVideoEncodeH265SessionParametersAddInfoKHR::pStdVPSs.
- The H.265 SPS entries specified in VkVideoEncodeH265SessionParametersAddInfoKHR::pStdSPSs.
- The H.265 PPS entries specified in VkVideoEncodeH265SessionParametersAddInfoKHR::pStdPPSs.

In case of video session parameters objects created with a video encode operation, implementations **may** return the VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR error if any of the specified Video Std parameters do not adhere to the syntactic or semantic requirements of the used video compression standard, or if values derived from parameters according to the rules defined by the used video compression standard do not adhere to the capabilities of the video compression standard or the implementation.

NOTE Applications **should** not rely on the VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR error being returned by any command as a means to verify Video Std parameters, as implementations are not required to report the error in any specific set of cases.

Valid Usage

- VUID-vkUpdateVideoSessionParametersKHR-pUpdateInfo-07215
  pUpdateInfo->updateSequenceCount **must** equal the current update sequence counter of videoSessionParameters plus one
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07216
  If videoSessionParameters was created with the video codec operation

VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and the pNext chain of pUpdateInfo includes a VkVideoDecodeH264SessionParametersAddInfoKHR structure, then videoSessionParameters **must** not already contain a StdVideoH264SequenceParameterSet entry with seq_parameter_set_id matching any of the elements of VkVideoDecodeH264SessionParametersAddInfoKHR::pStdSPSs

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07217
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the number of StdVideoH264SequenceParameterSet entries already stored in it plus the value of the stdSPSCount member of the VkVideoDecodeH264SessionParametersAddInfoKHR structure included in the pUpdateInfo->pNext chain **must** be less than or equal to the VkVideoDecodeH264SessionParametersCreateInfoKHR::maxStdSPSCount videoSessionParameters was created with
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07218
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and the pNext chain of pUpdateInfo includes a VkVideoDecodeH264SessionParametersAddInfoKHR structure, then videoSessionParameters **must** not already contain a StdVideoH264PictureParameterSet entry with both seq_parameter_set_id and pic_parameter_set_id matching any of the elements of VkVideoDecodeH264SessionParametersAddInfoKHR::pStdPPSs
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07219
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the number of StdVideoH264PictureParameterSet entries already stored in it plus the value of the stdPPSCount member of the VkVideoDecodeH264SessionParametersAddInfoKHR structure included in the pUpdateInfo->pNext chain **must** be less than or equal to the VkVideoDecodeH264SessionParametersCreateInfoKHR::maxStdPPSCount videoSessionParameters was created with
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07220
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR and the pNext chain of pUpdateInfo includes a VkVideoDecodeH265SessionParametersAddInfoKHR structure, then videoSessionParameters **must** not already contain a StdVideoH265VideoParameterSet entry with vps_video_parameter_set_id matching any of the elements of VkVideoDecodeH265SessionParametersAddInfoKHR::pStdVPSs
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07221
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the number of StdVideoH265VideoParameterSet entries already stored in it plus the value of the stdVPSCount member of the VkVideoDecodeH265SessionParametersAddInfoKHR structure included in the pUpdateInfo->pNext chain **must** be less than or equal to the VkVideoDecodeH265SessionParametersCreateInfoKHR::maxStdVPSCount videoSessionParameters was created with
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07222
  If videoSessionParameters was created with the video codec operation

VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR and the pNext chain of pUpdateInfo includes a VkVideoDecodeH265SessionParametersAddInfoKHR structure, then videoSessionParameters **must** not already contain a StdVideoH265SequenceParameterSet entry with both sps_video_parameter_set_id and sps_seq_parameter_set_id matching any of the elements of VkVideoDecodeH265SessionParametersAddInfoKHR::pStdSPSs

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07223
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the number of StdVideoH265SequenceParameterSet entries already stored in it plus the value of the stdSPSCount member of the VkVideoDecodeH265SessionParametersAddInfoKHR structure included in the pUpdateInfo->pNext chain **must** be less than or equal to the VkVideoDecodeH265SessionParametersCreateInfoKHR::maxStdSPSCount videoSessionParameters was created with
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07224
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR and the pNext chain of pUpdateInfo includes a VkVideoDecodeH265SessionParametersAddInfoKHR structure, then videoSessionParameters **must** not already contain a StdVideoH265PictureParameterSet entry with sps_video_parameter_set_id, pps_seq_parameter_set_id, and pps_pic_parameter_set_id all matching any of the elements of VkVideoDecodeH265SessionParametersAddInfoKHR::pStdPPSs
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07225
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the number of StdVideoH265PictureParameterSet entries already stored in it plus the value of the stdPPSCount member of the VkVideoDecodeH265SessionParametersAddInfoKHR structure included in the pUpdateInfo->pNext chain **must** be less than or equal to the VkVideoDecodeH265SessionParametersCreateInfoKHR::maxStdPPSCount videoSessionParameters was created with
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-09260
  videoSessionParameters **must** not have been created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07226
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and the pNext chain of pUpdateInfo includes a VkVideoEncodeH264SessionParametersAddInfoKHR structure, then videoSessionParameters **must** not already contain a StdVideoH264SequenceParameterSet entry with seq_parameter_set_id matching any of the elements of VkVideoEncodeH264SessionParametersAddInfoKHR::pStdSPSs
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-06441
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the number of StdVideoH264SequenceParameterSet entries already stored in it plus the value of the stdSPSCount member of the VkVideoEncodeH264SessionParametersAddInfoKHR structure included in the pUpdateInfo->pNext chain **must** be less than or equal to the

VkVideoEncodeH264SessionParametersCreateInfoKHR::maxStdSPSCount videoSessionParameters was created with

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07227
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and the pNext chain of pUpdateInfo includes a VkVideoEncodeH264SessionParametersAddInfoKHR structure, then videoSessionParameters **must** not already contain a StdVideoH264PictureParameterSet entry with both seq_parameter_set_id and pic_parameter_set_id matching any of the elements of VkVideoEncodeH264SessionParametersAddInfoKHR::pStdPPSs
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-06442
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the number of StdVideoH264PictureParameterSet entries already stored in it plus the value of the stdPPSCount member of the VkVideoEncodeH264SessionParametersAddInfoKHR structure included in the pUpdateInfo->pNext chain **must** be less than or equal to the VkVideoEncodeH264SessionParametersCreateInfoKHR::maxStdPPSCount videoSessionParameters was created with
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07228
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and the pNext chain of pUpdateInfo includes a VkVideoEncodeH265SessionParametersAddInfoKHR structure, then videoSessionParameters **must** not already contain a StdVideoH265VideoParameterSet entry with vps_video_parameter_set_id matching any of the elements of VkVideoEncodeH265SessionParametersAddInfoKHR::pStdVPSs
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-06443
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the number of StdVideoH265VideoParameterSet entries already stored in it plus the value of the stdVPSCount member of the VkVideoEncodeH265SessionParametersAddInfoKHR structure included in the pUpdateInfo->pNext chain **must** be less than or equal to the VkVideoEncodeH265SessionParametersCreateInfoKHR::maxStdVPSCount videoSessionParameters was created with
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07229
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and the pNext chain of pUpdateInfo includes a VkVideoEncodeH265SessionParametersAddInfoKHR structure, then videoSessionParameters **must** not already contain a StdVideoH265SequenceParameterSet entry with both sps_video_parameter_set_id and sps_seq_parameter_set_id matching any of the elements of VkVideoEncodeH265SessionParametersAddInfoKHR::pStdSPSs
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-06444
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the number of StdVideoH265SequenceParameterSet entries already stored in it plus the value of the stdSPSCount member of the VkVideoEncodeH265SessionParametersAddInfoKHR structure included in the pUpdateInfo->pNext chain **must** be less than or equal to the

VkVideoEncodeH265SessionParametersCreateInfoKHR::maxStdSPSCount videoSessionParameters was created with

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07230
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and the pNext chain of pUpdateInfo includes a VkVideoEncodeH265SessionParametersAddInfoKHR structure, then videoSessionParameters **must** not already contain a StdVideoH265PictureParameterSet entry with sps_video_parameter_set_id, pps_seq_parameter_set_id, and pps_pic_parameter_set_id all matching any of the elements of VkVideoEncodeH265SessionParametersAddInfoKHR::pStdPPSs
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-06445
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the number of StdVideoH265PictureParameterSet entries already stored in it plus the value of the stdPPSCount member of the VkVideoEncodeH265SessionParametersAddInfoKHR structure included in the pUpdateInfo->pNext chain **must** be less than or equal to the VkVideoEncodeH265SessionParametersCreateInfoKHR::maxStdPPSCount videoSessionParameters was created with
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-08321
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and the pNext chain of pUpdateInfo includes a VkVideoEncodeH265SessionParametersAddInfoKHR structure, then num_tile_columns_minus1 **must** be less than VkVideoEncodeH265CapabilitiesKHR::maxTiles.width, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile videoSessionParameters was created with, for each element of VkVideoEncodeH265SessionParametersAddInfoKHR::pStdPPSs
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-08322
  If videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and the pNext chain of pUpdateInfo includes a VkVideoEncodeH265SessionParametersAddInfoKHR structure, then num_tile_rows_minus1 **must** be less than VkVideoEncodeH265CapabilitiesKHR::maxTiles.height, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile videoSessionParameters was created with, for each element of VkVideoEncodeH265SessionParametersAddInfoKHR::pStdPPSs
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-10281
  videoSessionParameters **must** not have been created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR

Valid Usage (Implicit)

- VUID-vkUpdateVideoSessionParametersKHR-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-parameter
  videoSessionParameters **must** be a valid VkVideoSessionParametersKHR handle

- VUID-vkUpdateVideoSessionParametersKHR-pUpdateInfo-parameter
  `pUpdateInfo` **must** be a valid pointer to a valid VkVideoSessionParametersUpdateInfoKHR structure
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-parent
  `videoSessionParameters` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkVideoSessionParametersUpdateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoSessionParametersUpdateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           updateSequenceCount;
} VkVideoSessionParametersUpdateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `updateSequenceCount` is the new update sequence count to set for the video session parameters object.

Valid Usage (Implicit)

- VUID-VkVideoSessionParametersUpdateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_SESSION_PARAMETERS_UPDATE_INFO_KHR
- VUID-VkVideoSessionParametersUpdateInfoKHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkVideoDecodeH264SessionParametersAddInfoKHR, VkVideoDecodeH265SessionParametersAddInfoKHR, VkVideoEncodeH264SessionParametersAddInfoKHR, or

VkVideoEncodeH265SessionParametersAddInfoKHR

- VUID-VkVideoSessionParametersUpdateInfoKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

45.8. Video Coding Scope

Applications **can** record video coding commands for a video session only within a video coding scope.

To begin a video coding scope, call:

```
// Provided by VK_KHR_video_queue
void vkCmdBeginVideoCodingKHR(
    VkCommandBuffer                             commandBuffer,
    const VkVideoBeginCodingInfoKHR*            pBeginInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pBeginInfo` is a pointer to a VkVideoBeginCodingInfoKHR structure specifying the parameters of the video coding scope, including the video session and video session parameters object to use.

After beginning a video coding scope, the video session object specified in `pBeginInfo->videoSession` is *bound* to the command buffer, and the command buffer is ready to record video coding operations. Similarly, if `pBeginInfo->videoSessionParameters` is not `VK_NULL_HANDLE`, it is also *bound* to the command buffer, and video coding operations **can** refer to the codec-specific parameters stored in it.

This command also establishes the set of *bound reference picture resources* that **can** be used as reconstructed pictures or reference pictures within the video coding scope. Each element of this set consists of a video picture resource and the DPB slot index associated with it, if there is one.

The set of bound reference picture resources is immutable within a video coding scope, however, the DPB slot index associated with any of the bound reference picture resources **can** change during the video coding scope in response to video coding operations.

The VkVideoReferenceSlotInfoKHR structures provided as the elements of `pBeginInfo->pReferenceSlots` are interpreted by this command as follows:

- If `slotIndex` is non-negative and `pPictureResource` is not `NULL`, then the video picture resource defined by the VkVideoPictureResourceInfoKHR structure pointed to by `pPictureResource` is added to the set of bound reference picture resources and is associated with the DPB slot index specified in `slotIndex`.
- If `slotIndex` is non-negative and `pPictureResource` is `NULL`, then the DPB slot with index `slotIndex` is deactivated by this command.
- If `slotIndex` is negative and `pPictureResource` is not `NULL`, then the video picture resource defined by the VkVideoPictureResourceInfoKHR structure pointed to by `pPictureResource` is added to the set of bound reference picture resources without an associated DPB slot. Such a picture

resource **can** be subsequently used as a reconstructed picture to associate it with a DPB slot.

- If `slotIndex` is negative and `pPictureResource` is `NULL`, then the element is ignored.

NOTE

It is possible for multiple bound reference picture resources to be associated with the same DPB slot index, or for a single bound reference picture to refer to multiple separate reference pictures. For example, in case of an H.264 decode profile with interlaced frame support a single DPB slot can refer to two separate pictures for the top and bottom fields. Depending on the picture layout used by the H.264 decode profile, the following special cases **may** arise:

- If the picture layout is VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_INTERLEAVED_LINES_BIT_KHR, then the top and bottom field pictures are physically co-located in the same video picture resource with even scanlines corresponding to the top field and odd scanlines corresponding to the bottom field, respectively.
- If the picture layout is VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR, then the top and bottom field pictures are stored in separate video picture resources (in separate subregions of the same image layer, in separate layers of the same image, or in entirely separate images), hence two elements of VkVideoBeginCodingInfoKHR::`pReferenceSlots` **can** contain the same `slotIndex` but specify different video picture resources in their `pPictureResource` members.

All non-negative `slotIndex` values specified in the elements of `pBeginInfo->pReferenceSlots` **must** identify DPB slots of the video session that are in the active state at the time this command is executed on the device.

NOTE

The application does not have to specify an entry in `pBeginInfo->pReferenceSlots` corresponding to all active DPB slots of the video session, but only for those which are intended to be used in the video coding scope. This way the application can avoid any potential runtime cost associated with binding the corresponding picture resources to the command buffer.

In case of a video encode session, the application is also responsible for providing information about the current rate control state configured for the video session by including an instance of the VkVideoEncodeRateControlInfoKHR structure in the `pNext` chain of `pBeginInfo`. If no VkVideoEncodeRateControlInfoKHR is included, then the presence of an empty VkVideoEncodeRateControlInfoKHR structure is implied which indicates that the current rate control mode is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR. The specified state **must** match the effective rate control state configured for the video session at the time the recorded command is executed on the device.

NOTE

Including an instance of the VkVideoEncodeRateControlInfoKHR structure in the `pNext` chain of `pBeginInfo` does not change the rate control state configured for the video session, but only specifies the expected rate control state configured at the time the recorded command is executed on the device which allows the implementation to have information about the configured rate control state at

command buffer recording time. In order to change the current rate control state of a video session, the application has to issue an appropriate vkCmdControlVideoCodingKHR command as described in the Video Coding Control and Rate Control State sections.

Valid Usage

- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-11760
  The `VkCommandPool` that `commandBuffer` was allocated from **must** have been created with the same queue family index that `pBeginInfo->videoSession` was created with
- VUID-vkCmdBeginVideoCodingKHR-None-07232
  There **must** be no active queries
- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-07233
  If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, then `pBeginInfo->videoSession` **must** not have been created with VK_VIDEO_SESSION_CREATE_PROTECTED_CONTENT_BIT_KHR
- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-07234
  If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, then `pBeginInfo->videoSession` **must** have been created with VK_VIDEO_SESSION_CREATE_PROTECTED_CONTENT_BIT_KHR
- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-07235
  If `commandBuffer` is an unprotected command buffer, `protectedNoFault` is not supported, and the `pPictureResource` member of any element of `pBeginInfo->pReferenceSlots` is not `NULL`, then `pPictureResource->imageViewBinding` for that element **must** not specify an image view created from a protected image
- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-07236
  If `commandBuffer` is a protected command buffer, `protectedNoFault` is not supported, and the `pPictureResource` member of any element of `pBeginInfo->pReferenceSlots` is not `NULL`, then `pPictureResource->imageViewBinding` for that element **must** specify an image view created from a protected image
- VUID-vkCmdBeginVideoCodingKHR-slotIndex-07239
  If the `slotIndex` member of any element of `pBeginInfo->pReferenceSlots` is not negative, then it **must** specify the index of a DPB slot that is in the active state in `pBeginInfo->videoSession` at the time the command is executed on the device
- VUID-vkCmdBeginVideoCodingKHR-pPictureResource-07265
  Each video picture resource specified by any non-`NULL` `pPictureResource` member specified in the elements of `pBeginInfo->pReferenceSlots` for which `slotIndex` is not negative **must** match one of the video picture resources currently associated with the DPB slot index of `pBeginInfo->videoSession` specified by `slotIndex` at the time the command is executed on the device
- VUID-vkCmdBeginVideoCodingKHR-pBeginInfo-08253
  If `pBeginInfo->videoSession` was created with a video encode operation and the `pNext` chain of `pBeginInfo` does not include an instance of the VkVideoEncodeRateControlInfoKHR structure, then the rate control mode configured for

pBeginInfo->videoSession at the time the command is executed on the device **must** be VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR

- VUID-vkCmdBeginVideoCodingKHR-pBeginInfo-08254
  If pBeginInfo->videoSession was created with a video encode operation and the pNext chain of pBeginInfo includes an instance of the VkVideoEncodeRateControlInfoKHR structure, then it **must** match the rate control state configured for pBeginInfo->videoSession at the time the command is executed on the device
- VUID-vkCmdBeginVideoCodingKHR-pBeginInfo-08255
  If pBeginInfo->videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, the current rate control mode is not VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR or VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, and VkVideoEncodeH264CapabilitiesKHR::requiresGopRemainingFrames is VK_TRUE, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the pBeginInfo->videoSession was created with, then the pNext chain of pBeginInfo **must** include an instance of the VkVideoEncodeH264GopRemainingFrameInfoKHR with its useGopRemainingFrames member set to VK_TRUE
- VUID-vkCmdBeginVideoCodingKHR-pBeginInfo-08256
  If pBeginInfo->videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, the current rate control mode is not VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR or VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, and VkVideoEncodeH265CapabilitiesKHR::requiresGopRemainingFrames is VK_TRUE, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the pBeginInfo->videoSession was created with, then the pNext chain of pBeginInfo **must** include an instance of the VkVideoEncodeH265GopRemainingFrameInfoKHR with its useGopRemainingFrames member set to VK_TRUE
- VUID-vkCmdBeginVideoCodingKHR-pBeginInfo-10282
  If pBeginInfo->videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, the current rate control mode is not VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR or VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, and VkVideoEncodeAV1CapabilitiesKHR::requiresGopRemainingFrames is VK_TRUE, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the pBeginInfo->videoSession was created with, then the pNext chain of pBeginInfo **must** include an instance of the VkVideoEncodeAV1GopRemainingFrameInfoKHR with its useGopRemainingFrames member set to VK_TRUE

Valid Usage (Implicit)

- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBeginVideoCodingKHR-pBeginInfo-parameter
  pBeginInfo **must** be a valid pointer to a valid VkVideoBeginCodingInfoKHR structure

- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdBeginVideoCodingKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdBeginVideoCodingKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdBeginVideoCodingKHR-videocoding
  This command **must** only be called outside of a video coding scope
- VUID-vkCmdBeginVideoCodingKHR-bufferlevel
  `commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Outside	VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Action State

Conditional Rendering

vkCmdBeginVideoCodingKHR is not affected by conditional rendering

The VkVideoBeginCodingInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoBeginCodingInfoKHR {
    VkStructureType                       sType;
    const void*                           pNext;
    VkVideoBeginCodingFlagsKHR            flags;
    VkVideoSessionKHR                     videoSession;
    VkVideoSessionParametersKHR           videoSessionParameters;
    uint32_t                              referenceSlotCount;
    const VkVideoReferenceSlotInfoKHR*    pReferenceSlots;
} VkVideoBeginCodingInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `videoSession` is the video session object to be bound for the processing of the video commands.
- `videoSessionParameters` is `VK_NULL_HANDLE` or a handle of a VkVideoSessionParametersKHR object to be used for the processing of the video commands. If `VK_NULL_HANDLE`, then no video session parameters object is bound for the duration of the video coding scope.
- `referenceSlotCount` is the number of elements in the `pReferenceSlots` array.
- `pReferenceSlots` is a pointer to an array of VkVideoReferenceSlotInfoKHR structures specifying the information used to determine the set of bound reference picture resources for the video coding scope and their initial association with DPB slot indices.

Limiting values are defined below that are referenced by the relevant valid usage statements of this structure.

- Let `VkOffset2D codedOffsetGranularity` be the minimum alignment requirement for the coded offset of video picture resources. Unless otherwise defined, the value of the `x` and `y` members of `codedOffsetGranularity` are `0`.
  - If `videoSession` was created with an H.264 decode profile with a VkVideoDecodeH264ProfileInfoKHR::`pictureLayout` of VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR, then `codedOffsetGranularity` is equal to VkVideoDecodeH264CapabilitiesKHR::`fieldOffsetGranularity`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for that video profile.

Valid Usage

- VUID-VkVideoBeginCodingInfoKHR-videoSession-07237
  `videoSession` **must** have memory bound to all of its memory bindings returned by vkGetVideoSessionMemoryRequirementsKHR for `videoSession`
- VUID-VkVideoBeginCodingInfoKHR-slotIndex-04856
  Each non-negative VkVideoReferenceSlotInfoKHR::`slotIndex` specified in the elements of `pReferenceSlots` **must** be less than the VkVideoSessionCreateInfoKHR::`maxDpbSlots`

specified when videoSession was created

- VUID-VkVideoBeginCodingInfoKHR-pPictureResource-07238
  Each video picture resource corresponding to any non-NULL pPictureResource member specified in the elements of pReferenceSlots **must** be unique within pReferenceSlots
- VUID-VkVideoBeginCodingInfoKHR-pPictureResource-07240
  If the pPictureResource member of any element of pReferenceSlots is not NULL, then the image view specified in pPictureResource->imageViewBinding for that element **must** be compatible with the video profile videoSession was created with
- VUID-VkVideoBeginCodingInfoKHR-pPictureResource-07241
  If the pPictureResource member of any element of pReferenceSlots is not NULL, then the format of the image view specified in pPictureResource->imageViewBinding for that element **must** match the VkVideoSessionCreateInfoKHR::referencePictureFormat videoSession was created with
- VUID-VkVideoBeginCodingInfoKHR-pPictureResource-07242
  If the pPictureResource member of any element of pReferenceSlots is not NULL, then its codedOffset member **must** be an integer multiple of codedOffsetGranularity
- VUID-VkVideoBeginCodingInfoKHR-pPictureResource-07243
  If the pPictureResource member of any element of pReferenceSlots is not NULL, then its codedExtent member **must** be between minCodedExtent and maxCodedExtent, inclusive, videoSession was created with
- VUID-VkVideoBeginCodingInfoKHR-flags-07244
  If VkVideoCapabilitiesKHR::flags does not include VK_VIDEO_CAPABILITY_SEPARATE_REFERENCE_IMAGES_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile videoSession was created with, then pPictureResource->imageViewBinding of all elements of pReferenceSlots with a non-NULL pPictureResource member **must** specify image views created from the same image
- VUID-VkVideoBeginCodingInfoKHR-slotIndex-07245
  If videoSession was created with a decode operation and the slotIndex member of any element of pReferenceSlots is not negative, then the image view specified in pPictureResource->imageViewBinding for that element **must** have been created with the VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR usage flag set
- VUID-VkVideoBeginCodingInfoKHR-slotIndex-07246
  If videoSession was created with an encode operation and the slotIndex member of any element of pReferenceSlots is not negative, then the image view specified in pPictureResource->imageViewBinding for that element **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR usage flag set
- VUID-VkVideoBeginCodingInfoKHR-videoSession-07247
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then videoSessionParameters **must** not be VK_NULL_HANDLE , unless the videoMaintenance2 feature is enabled
- VUID-VkVideoBeginCodingInfoKHR-videoSession-07248
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then videoSessionParameters

must not be VK_NULL_HANDLE , unless the videoMaintenance2 feature is enabled

- VUID-VkVideoBeginCodingInfoKHR-videoSession-09261
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then videoSessionParameters **must** not be VK_NULL_HANDLE , unless the videoMaintenance2 feature is enabled
- VUID-VkVideoBeginCodingInfoKHR-videoSession-07249
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then videoSessionParameters **must** not be VK_NULL_HANDLE
- VUID-VkVideoBeginCodingInfoKHR-videoSession-07250
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then videoSessionParameters **must** not be VK_NULL_HANDLE
- VUID-VkVideoBeginCodingInfoKHR-videoSession-10283
  If videoSession was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then videoSessionParameters **must** not be VK_NULL_HANDLE
- VUID-VkVideoBeginCodingInfoKHR-videoSessionParameters-04857
  If videoSessionParameters is not VK_NULL_HANDLE, it **must** have been created with videoSession specified in VkVideoSessionParametersCreateInfoKHR::videoSession

Valid Usage (Implicit)

- VUID-VkVideoBeginCodingInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_BEGIN_CODING_INFO_KHR
- VUID-VkVideoBeginCodingInfoKHR-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkVideoEncodeAV1GopRemainingFrameInfoKHR, VkVideoEncodeAV1RateControlInfoKHR, VkVideoEncodeH264GopRemainingFrameInfoKHR, VkVideoEncodeH264RateControlInfoKHR, VkVideoEncodeH265GopRemainingFrameInfoKHR, VkVideoEncodeH265RateControlInfoKHR, or VkVideoEncodeRateControlInfoKHR
- VUID-VkVideoBeginCodingInfoKHR-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkVideoBeginCodingInfoKHR-flags-zerobitmask
  flags **must** be 0
- VUID-VkVideoBeginCodingInfoKHR-videoSession-parameter
  videoSession **must** be a valid VkVideoSessionKHR handle
- VUID-VkVideoBeginCodingInfoKHR-videoSessionParameters-parameter
  If videoSessionParameters is not VK_NULL_HANDLE, videoSessionParameters **must** be a valid VkVideoSessionParametersKHR handle

- VUID-VkVideoBeginCodingInfoKHR-pReferenceSlots-parameter
  If `referenceSlotCount` is not `0`, `pReferenceSlots` **must** be a valid pointer to an array of `referenceSlotCount` valid VkVideoReferenceSlotInfoKHR structures
- VUID-VkVideoBeginCodingInfoKHR-commonparent
  Both of `videoSession`, and `videoSessionParameters` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoBeginCodingFlagsKHR;
```

`VkVideoBeginCodingFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

The `VkVideoReferenceSlotInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoReferenceSlotInfoKHR {
    VkStructureType                          sType;
    const void*                              pNext;
    int32_t                                  slotIndex;
    const VkVideoPictureResourceInfoKHR*     pPictureResource;
} VkVideoReferenceSlotInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `slotIndex` is the index of the DPB slot or a negative integer value.
- `pPictureResource` is `NULL` or a pointer to a VkVideoPictureResourceInfoKHR structure describing the video picture resource associated with the DPB slot index specified by `slotIndex`.

Valid Usage (Implicit)

- VUID-VkVideoReferenceSlotInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_REFERENCE_SLOT_INFO_KHR
- VUID-VkVideoReferenceSlotInfoKHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkVideoDecodeAV1DpbSlotInfoKHR, VkVideoDecodeH264DpbSlotInfoKHR, VkVideoDecodeH265DpbSlotInfoKHR, VkVideoEncodeAV1DpbSlotInfoKHR, VkVideoEncodeH264DpbSlotInfoKHR, VkVideoEncodeH265DpbSlotInfoKHR, or VkVideoReferenceIntraRefreshInfoKHR
- VUID-VkVideoReferenceSlotInfoKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkVideoReferenceSlotInfoKHR-pPictureResource-parameter
  If `pPictureResource` is not `NULL`, `pPictureResource` **must** be a valid pointer to a valid

VkVideoPictureResourceInfoKHR structure

To end a video coding scope, call:

```
// Provided by VK_KHR_video_queue
void vkCmdEndVideoCodingKHR(
    VkCommandBuffer                             commandBuffer,
    const VkVideoEndCodingInfoKHR*              pEndCodingInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pEndCodingInfo` is a pointer to a VkVideoEndCodingInfoKHR structure specifying the parameters for ending the video coding scope.

After ending a video coding scope, the video session object, the optional video session parameters object, and all reference picture resources previously bound by the corresponding vkCmdBeginVideoCodingKHR command are *unbound*.

Valid Usage

- VUID-vkCmdEndVideoCodingKHR-None-07251
  There **must** be no active queries

Valid Usage (Implicit)

- VUID-vkCmdEndVideoCodingKHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdEndVideoCodingKHR-pEndCodingInfo-parameter
  `pEndCodingInfo` **must** be a valid pointer to a valid VkVideoEndCodingInfoKHR structure
- VUID-vkCmdEndVideoCodingKHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdEndVideoCodingKHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdEndVideoCodingKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdEndVideoCodingKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdEndVideoCodingKHR-videocoding
  This command **must** only be called inside of a video coding scope
- VUID-vkCmdEndVideoCodingKHR-bufferlevel
  `commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Inside	VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Action State

Conditional Rendering

vkCmdEndVideoCodingKHR is not affected by conditional rendering

The `VkVideoEndCodingInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoEndCodingInfoKHR {
    VkStructureType             sType;
    const void*                 pNext;
    VkVideoEndCodingFlagsKHR    flags;
} VkVideoEndCodingInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.

Valid Usage (Implicit)

- VUID-VkVideoEndCodingInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_END_CODING_INFO_KHR
- VUID-VkVideoEndCodingInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkVideoEndCodingInfoKHR-flags-zerobitmask

`flags` **must** be `0`

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoEndCodingFlagsKHR;
```

`VkVideoEndCodingFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

45.9. Video Coding Control

To apply dynamic controls to the bound video session object, call:

```
// Provided by VK_KHR_video_queue
void vkCmdControlVideoCodingKHR(
    VkCommandBuffer                             commandBuffer,
    const VkVideoCodingControlInfoKHR*          pCodingControlInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pCodingControlInfo` is a pointer to a VkVideoCodingControlInfoKHR structure specifying the control parameters.

The control parameters provided in this call are applied to the video session at the time the command executes on the device and are in effect until a subsequent call to this command with the same video session bound changes the corresponding control parameters.

A newly created video session **must** be reset before performing video coding operations using it by including VK_VIDEO_CODING_CONTROL_RESET_BIT_KHR in `pCodingControlInfo->flags`. The reset operation also returns all DPB slots of the video session to the inactive state. Correspondingly, any DPB slot index associated with the bound reference picture resources is removed.

For encode sessions, the reset operation returns rate control configuration to implementation default settings and sets the video encode quality level to zero.

After video coding operations are performed using a video session, the reset operation **can** be used to return the video session to the same *initial* state as after the reset of a newly created video session. This **can** be used, for example, when different video sequences are needed to be processed with the same video session object.

If `pCodingControlInfo->flags` includes VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR, then the command replaces the rate control configuration maintained by the video session with the configuration specified in the VkVideoEncodeRateControlInfoKHR structure included in the `pCodingControlInfo->pNext` chain.

If `pCodingControlInfo->flags` includes VK_VIDEO_CODING_CONTROL_ENCODE_QUALITY_LEVEL_BIT_KHR, then the command changes the current video encode quality level to the value specified in the `qualityLevel` member of the

VkVideoEncodeQualityLevelInfoKHR structure included in the pCodingControlInfo->pNext chain.

Valid Usage

- VUID-vkCmdControlVideoCodingKHR-flags-07017
  If pCodingControlInfo->flags does not include VK_VIDEO_CODING_CONTROL_RESET_BIT_KHR, then the bound video session **must** not be in uninitialized state at the time the command is executed on the device
- VUID-vkCmdControlVideoCodingKHR-pCodingControlInfo-08243
  If the bound video session was not created with an encode operation, then pCodingControlInfo->flags **must** not include VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR or VK_VIDEO_CODING_CONTROL_ENCODE_QUALITY_LEVEL_BIT_KHR

Valid Usage (Implicit)

- VUID-vkCmdControlVideoCodingKHR-commandBuffer-parameter
  commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdControlVideoCodingKHR-pCodingControlInfo-parameter
  pCodingControlInfo **must** be a valid pointer to a valid VkVideoCodingControlInfoKHR structure
- VUID-vkCmdControlVideoCodingKHR-commandBuffer-recording
  commandBuffer **must** be in the recording state
- VUID-vkCmdControlVideoCodingKHR-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from **must** support VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdControlVideoCodingKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdControlVideoCodingKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdControlVideoCodingKHR-videocoding
  This command **must** only be called inside of a video coding scope
- VUID-vkCmdControlVideoCodingKHR-bufferlevel
  commandBuffer **must** be a primary VkCommandBuffer

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Inside	VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Action

Conditional Rendering

vkCmdControlVideoCodingKHR is not affected by conditional rendering

The `VkVideoCodingControlInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoCodingControlInfoKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    VkVideoCodingControlFlagsKHR    flags;
} VkVideoCodingControlInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkVideoCodingControlFlagsKHR specifying control flags.

Valid Usage

- VUID-VkVideoCodingControlInfoKHR-flags-07018
  If `flags` includes VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR, then the `pNext` chain **must** include a VkVideoEncodeRateControlInfoKHR structure
- VUID-VkVideoCodingControlInfoKHR-flags-08349
  If `flags` includes VK_VIDEO_CODING_CONTROL_ENCODE_QUALITY_LEVEL_BIT_KHR, then the `pNext` chain **must** include a VkVideoEncodeQualityLevelInfoKHR structure

Valid Usage (Implicit)

- VUID-VkVideoCodingControlInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_CODING_CONTROL_INFO_KHR

- VUID-VkVideoCodingControlInfoKHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkVideoEncodeAV1RateControlInfoKHR, VkVideoEncodeH264RateControlInfoKHR, VkVideoEncodeH265RateControlInfoKHR, VkVideoEncodeQualityLevelInfoKHR, or VkVideoEncodeRateControlInfoKHR
- VUID-VkVideoCodingControlInfoKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkVideoCodingControlInfoKHR-flags-parameter
  `flags` **must** be a valid combination of VkVideoCodingControlFlagBitsKHR values

Bits which **can** be set in VkVideoCodingControlInfoKHR::`flags`, specifying the video coding control parameters to be modified, are:

```
// Provided by VK_KHR_video_queue
typedef enum VkVideoCodingControlFlagBitsKHR {
    VK_VIDEO_CODING_CONTROL_RESET_BIT_KHR = 0x00000001,
  // Provided by VK_KHR_video_encode_queue
    VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR = 0x00000002,
  // Provided by VK_KHR_video_encode_queue
    VK_VIDEO_CODING_CONTROL_ENCODE_QUALITY_LEVEL_BIT_KHR = 0x00000004,
} VkVideoCodingControlFlagBitsKHR;
```

- VK_VIDEO_CODING_CONTROL_RESET_BIT_KHR specifies a request for the bound video session to be reset before other coding control parameters are applied.
- VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR specifies that the coding control parameters include video encode rate control parameters (see VkVideoEncodeRateControlInfoKHR).
- VK_VIDEO_CODING_CONTROL_ENCODE_QUALITY_LEVEL_BIT_KHR specifies that the coding control parameters include video encode quality level parameters (see VkVideoEncodeQualityLevelInfoKHR).

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoCodingControlFlagsKHR;
```

`VkVideoCodingControlFlagsKHR` is a bitmask type for setting a mask of zero or more VkVideoCodingControlFlagBitsKHR.

45.10. Inline Queries

If a video session was created with VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR, beginning queries using commands such as vkCmdBeginQuery within a video coding scope is not allowed. Instead, queries are executed inline by including an instance of the VkVideoInlineQueryInfoKHR structure in the `pNext` chain of the parameters of one of the video coding commands, with its `queryPool` member set to a valid `VkQueryPool` handle.

The VkVideoInlineQueryInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_maintenance1
typedef struct VkVideoInlineQueryInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkQueryPool        queryPool;
    uint32_t           firstQuery;
    uint32_t           queryCount;
} VkVideoInlineQueryInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- queryPool is VK_NULL_HANDLE or a valid handle to a VkQueryPool object that will manage the results of the queries.
- firstQuery is the query index within the query pool that will contain the query results for the first video coding operation. The query results of subsequent video coding operations will be contained by subsequent query indices.
- queryCount is the number of queries to execute.

NOTE	In practice, if queryPool is not VK_NULL_HANDLE, then queryCount will always have to match the number of video coding operations issued by the video coding command this structure is specified to, meaning that using inline queries in a video coding command will always execute a query for each issued video coding operation.

This structure **can** be included in the pNext chain of the input parameter structure of video coding commands.

- In the pNext chain of the pDecodeInfo parameter of the vkCmdDecodeVideoKHR command to execute a query for each video decode operation issued by the command.
- In the pNext chain of the pEncodeInfo parameter of the vkCmdEncodeVideoKHR command to execute a query for each video encode operation issued by the command.

Valid Usage

- VUID-VkVideoInlineQueryInfoKHR-queryPool-08372
  If queryPool is not VK_NULL_HANDLE, then firstQuery **must** be less than the number of queries in queryPool
- VUID-VkVideoInlineQueryInfoKHR-queryPool-08373
  If queryPool is not VK_NULL_HANDLE, then the sum of firstQuery and queryCount **must** be less than or equal to the number of queries in queryPool

Valid Usage (Implicit)

- VUID-VkVideoInlineQueryInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_INLINE_QUERY_INFO_KHR
- VUID-VkVideoInlineQueryInfoKHR-queryPool-parameter
  If `queryPool` is not VK_NULL_HANDLE, `queryPool` **must** be a valid VkQueryPool handle

Structure Chaining

Extends the structures

- VkVideoDecodeInfoKHR
- VkVideoEncodeInfoKHR

45.11. Video Decode Operations

Video decode operations consume compressed video data from a video bitstream buffer and zero or more reference pictures, and produce a *decode output picture* and an optional reconstructed picture.

NOTE	Such decode output pictures can be shared with the Decoded Picture Buffer, and can also be used as the input of video encode operations, with graphics or compute operations, or with Window System Integration APIs, depending on the capabilities of the implementation.

Video decode operations **may** access the following resources in the VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR stage:

- The source video bitstream buffer range and the image subregions corresponding to the list of active reference pictures with access VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR.
- The image subregions corresponding to the target decode output picture and reconstructed picture with access VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR.

The image subresource of each video picture resource accessed by the video decode operation is specified using a corresponding VkVideoPictureResourceInfoKHR structure. Each such image subresource **must** be in the appropriate image layout as follows:

- If the image subresource is used in the video decode operation only as decode output picture, then it **must** be in the VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR layout , unless the `unifiedImageLayoutsVideo` feature is enabled, in which case it **may** be in the VK_IMAGE_LAYOUT_GENERAL layout .
- If the image subresource is used in the video decode operation both as decode output picture and reconstructed picture, then it **must** be in the VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR layout , unless the `unifiedImageLayoutsVideo` feature is enabled, in which case it **may** be in the VK_IMAGE_LAYOUT_GENERAL layout .

- If the image subresource is used in the video decode operation only as reconstructed picture, then it **must** be in the VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR layout , unless the `unifiedImageLayoutsVideo` feature is enabled, in which case it **may** be in the VK_IMAGE_LAYOUT_GENERAL layout .
- If the image subresource is used in the video decode operation as a reference picture, then it **must** be in the VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR layout , unless the `unifiedImageLayoutsVideo` feature is enabled, in which case it **may** be in the VK_IMAGE_LAYOUT_GENERAL layout .

A video decode operation **may** complete unsuccessfully. In this case the decode output picture will have undefined contents. Similarly, if reference picture setup is requested, the reconstructed picture will also have undefined contents, and the activated DPB slot will have an invalid picture reference.

45.11.1. Codec-Specific Semantics

The following aspects of video decode operations are codec-specific:

- The interpretation of the contents of the source video bitstream buffer range.
- The construction and interpretation of the list of active reference pictures and the interpretation of the picture data referred to by the corresponding image subregions.
- The construction and interpretation of information related to the decode output picture and the generation of picture data to the corresponding image subregion.
- The decision on reference picture setup.
- The construction and interpretation of information related to the optional reconstructed picture and the generation of picture data to the corresponding image subregion.

These codec-specific behaviors are defined for each video codec operation separately.

- If the used video codec operation is VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the codec-specific aspects of the video decoding process are performed as defined in the H.264 Decode Operations section.
- If the used video codec operation is VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the codec-specific aspects of the video decoding process are performed as defined in the H.265 Decode Operations section.
- If the used video codec operation is VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR, then the codec-specific aspects of the video decoding process are performed as defined in the VP9 Decode Operations section.
- If the used video codec operation is VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then the codec-specific aspects of the video decoding process are performed as defined in the AV1 Decode Operations section.

45.11.2. Video Decode Operation Steps

Each video decode operation performs the following steps in the VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR stage:

1. Reads the encoded video data from the source video bitstream buffer range.
2. Performs picture reconstruction of the encoded video data according to the codec-specific semantics, applying any prediction data read from the active reference pictures in the process;
3. Writes the decoded picture data to the decode output picture, and optionally to the reconstructed picture, if one is specified and is different from the decode output picture, according to the codec-specific semantics;
4. If reference picture setup is requested, the DPB slot index specified in the reconstructed picture information is activated with the reconstructed picture.

When reconstructed picture information is provided, the specified DPB slot index is associated with the corresponding bound reference picture resource, regardless of whether reference picture setup is requested.

45.11.3. Capabilities

When calling vkGetPhysicalDeviceVideoCapabilitiesKHR with `pVideoProfile->videoCodecOperation` specifying a decode operation, the `VkVideoDecodeCapabilitiesKHR` structure **must** be included in the `pNext` chain of the VkVideoCapabilitiesKHR structure to retrieve capabilities specific to video decoding.

The `VkVideoDecodeCapabilitiesKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_queue
typedef struct VkVideoDecodeCapabilitiesKHR {
    VkStructureType                     sType;
    void*                               pNext;
    VkVideoDecodeCapabilityFlagsKHR     flags;
} VkVideoDecodeCapabilitiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkVideoDecodeCapabilityFlagBitsKHR describing the supported video decoding capabilities.

Valid Usage (Implicit)

- VUID-VkVideoDecodeCapabilitiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkVideoCapabilitiesKHR

Bits which **may** be set in VkVideoDecodeCapabilitiesKHR::flags, indicating the decoding capabilities supported, are:

```
// Provided by VK_KHR_video_decode_queue
typedef enum VkVideoDecodeCapabilityFlagBitsKHR {
    VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR = 0x00000001,
    VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR = 0x00000002,
} VkVideoDecodeCapabilityFlagBitsKHR;
```

- VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR specifies support for using the same video picture resource as the reconstructed picture and decode output picture in a video decode operation.
- VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR specifies support for using distinct video picture resources as the reconstructed picture and decode output picture in a video decode operation.

NOTE Some video profiles allow using distinct video picture resources as the reconstructed picture and decode output picture in specific video decode operations even when the video decode profile does not support VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR. Even if the implementation only reports coincide, the decode output picture for film grain enabled frames must be a different video picture resource from the reconstructed picture because film grain is applied outside of the coding loop.

Implementations are only **required** to support one of VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR and VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR. Accordingly, applications **should** handle both cases to maximize portability.

NOTE If both VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR and VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR are supported, an application can choose to create separate images for decode DPB and decode output. E.g. in cases when linear tiling is preferred (and supported) for the decode output picture and the DPB requires optimal tiling, this avoids the need for a separate copy at the expense of additional memory bandwidth requirements during decoding.

```
// Provided by VK_KHR_video_decode_queue
typedef VkFlags VkVideoDecodeCapabilityFlagsKHR;
```

VkVideoDecodeCapabilityFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoDecodeCapabilityFlagBitsKHR.

45.11.4. Video Decode Commands

To launch video decode operations, call:

```
// Provided by VK_KHR_video_decode_queue
void vkCmdDecodeVideoKHR(
    VkCommandBuffer                             commandBuffer,
    const VkVideoDecodeInfoKHR*                 pDecodeInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pDecodeInfo` is a pointer to a VkVideoDecodeInfoKHR structure specifying the parameters of the video decode operations.

Each call issues one or more video decode operations. The implicit parameter `opCount` corresponds to the number of video decode operations issued by the command. After calling this command, the active query index of each active query is incremented by `opCount`.

Currently each call to this command results in the issue of a single video decode operation.

If the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR and the `pNext` chain of `pDecodeInfo` includes a VkVideoInlineQueryInfoKHR structure with its `queryPool` member specifying a valid `VkQueryPool` handle, then this command will execute a query for each video decode operation issued by it.

Active Reference Picture Information

The list of active reference pictures used by a video decode operation is a list of image subregions used as the source of reference picture data and related parameters, and is derived from the VkVideoReferenceSlotInfoKHR structures provided as the elements of the `pDecodeInfo->pReferenceSlots` array. For each element of `pDecodeInfo->pReferenceSlots`, one or more elements are added to the active reference picture list, as defined by the codec-specific semantics. Each element of this list contains the following information:

- The image subregion within the image subresource referred to by the video picture resource used as the reference picture.
- The DPB slot index the reference picture is associated with.
- The codec-specific reference information related to the reference picture.

Reconstructed Picture Information

Information related to the optional reconstructed picture used by a video decode operation is derived from the VkVideoReferenceSlotInfoKHR structure pointed to by `pDecodeInfo->pSetupReferenceSlot`, if not `NULL`, as defined by the codec-specific semantics, and consists of the following:

- The image subregion within the image subresource referred to by the video picture resource used as the reconstructed picture.
- The DPB slot index to use for picture reconstruction.

- The codec-specific reference information related to the reconstructed picture.

Specifying a valid VkVideoReferenceSlotInfoKHR structure in pDecodeInfo->pSetupReferenceSlot is always required, unless the video session was created with VkVideoSessionCreateInfoKHR::maxDpbSlots equal to zero. However, the DPB slot identified by pDecodeInfo->pSetupReferenceSlot->slotIndex is only activated with the reconstructed picture specified in pDecodeInfo->pSetupReferenceSlot->pPictureResource if reference picture setup is requested according to the codec-specific semantics.

If reconstructed picture information is specified, and pDecodeInfo->pSetupReferenceSlot->pPictureResource refers to a video picture resource different than that of the decode output picture, but reference picture setup is not requested, the contents of the video picture resource corresponding to the reconstructed picture will be undefined after the video decode operation.

NOTE	Some implementations may always output the reconstructed picture or use it as temporary storage during the video decode operation even when the reconstructed picture is not marked for future reference.

Decode Output Picture Information

Information related to the decode output picture used by a video decode operation is derived from pDecodeInfo->dstPictureResource and any codec-specific parameters provided in the pDecodeInfo->pNext chain, as defined by the codec-specific semantics, and consists of the following:

- The image subregion within the image subresource referred to by the video picture resource used as the decode output picture.
- The codec-specific picture information related to the decode output picture.

Several limiting values are defined below that are referenced by the relevant valid usage statements of this command.

- Let uint32_t activeReferencePictureCount be the size of the list of active reference pictures used by the video decode operation. Unless otherwise defined, activeReferencePictureCount is set to the value of pDecodeInfo->referenceSlotCount.
  - If the bound video session was created with an H.264 decode profile, then let activeReferencePictureCount be the value of pDecodeInfo->referenceSlotCount plus the number of elements of the pDecodeInfo->pReferenceSlots array that have a VkVideoDecodeH264DpbSlotInfoKHR structure included in their pNext chain with both pStdReferenceInfo->flags.top_field_flag and pStdReferenceInfo->flags.bottom_field_flag set.

NOTE	This means that the elements of pDecodeInfo->pReferenceSlots that include both a top and bottom field reference are counted as two separate active reference pictures, as described in the active reference picture list construction rules for H.264 decode operations.

- Let VkOffset2D codedOffsetGranularity be the minimum alignment requirement for the coded offset of video picture resources. Unless otherwise defined, the value of the x and y members of

codedOffsetGranularity are 0.

  - If the bound video session was created with an H.264 decode profile with a VkVideoDecodeH264ProfileInfoKHR::pictureLayout of VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR, then codedOffsetGranularity is equal to VkVideoDecodeH264CapabilitiesKHR::fieldOffsetGranularity, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for that video profile.

- Let uint32_t dpbFrameUseCount[] be an array of size maxDpbSlots, where maxDpbSlots is the VkVideoSessionCreateInfoKHR::maxDpbSlots the bound video session was created with, with each element indicating the number of times a frame associated with the corresponding DPB slot index is referred to by the video coding operation. Let the initial value of each element of the array be 0.
  - If pDecodeInfo->pSetupReferenceSlot is not NULL, then dpbFrameUseCount[i] is incremented by one, where i equals pDecodeInfo->pSetupReferenceSlot->slotIndex. If the bound video session object was created with an H.264 decode profile, then dpbFrameUseCount[i] is decremented by one if either pStdReferenceInfo->flags.top_field_flag or pStdReferenceInfo->flags.bottom_field_flag is set in the VkVideoDecodeH264DpbSlotInfoKHR structure in the pDecodeInfo->pSetupReferenceSlot->pNext chain.
  - For each element of pDecodeInfo->pReferenceSlots, dpbFrameUseCount[i] is incremented by one, where i equals the slotIndex member of the corresponding element. If the bound video session object was created with an H.264 decode profile, then dpbFrameUseCount[i] is decremented by one if either pStdReferenceInfo->flags.top_field_flag or pStdReferenceInfo->flags.bottom_field_flag is set in the VkVideoDecodeH264DpbSlotInfoKHR structure in the pNext chain of the corresponding element of pDecodeInfo->pReferenceSlots.
- Let uint32_t dpbTopFieldUseCount[] and uint32_t dpbBottomFieldUseCount[] be arrays of size maxDpbSlots, where maxDpbSlots is the VkVideoSessionCreateInfoKHR::maxDpbSlots the bound video session was created with, with each element indicating the number of times the top field or the bottom field, respectively, associated with the corresponding DPB slot index is referred to by the video coding operation. Let the initial value of each element of the arrays be 0.
  - If the bound video session object was created with an H.264 decode profile and pDecodeInfo->pSetupReferenceSlot is not NULL, then perform the following:
    - If pStdReferenceInfo->flags.top_field_flag is set in the VkVideoDecodeH264DpbSlotInfoKHR structure in the pDecodeInfo->pSetupReferenceSlot->pNext chain, then dpbTopFieldUseCount[i] is incremented by one, where i equals pDecodeInfo->pSetupReferenceSlot->slotIndex.
    - If pStdReferenceInfo->flags.bottom_field_flag is set in the VkVideoDecodeH264DpbSlotInfoKHR structure in the pDecodeInfo->pSetupReferenceSlot->pNext chain, then dpbBottomFieldUseCount[i] is incremented by one, where i equals pDecodeInfo->pSetupReferenceSlot->slotIndex.
  - If the bound video session object was created with an H.264 decode profile, then perform the following for each element of pDecodeInfo->pReferenceSlots:
    - If pStdReferenceInfo->flags.top_field_flag is set in the

VkVideoDecodeH264DpbSlotInfoKHR structure in the pNext chain of the element, then dpbTopFieldUseCount[i] is incremented by one, where i equals the slotIndex member of the element.

- If pStdReferenceInfo->flags.bottom_field_flag is set in the VkVideoDecodeH264DpbSlotInfoKHR structure in the pNext chain of the element, then dpbBottomFieldUseCount[i] is incremented by one, where i equals the slotIndex member of the element.

Valid Usage

- VUID-vkCmdDecodeVideoKHR-None-08249
  The bound video session **must** have been created with a decode operation
- VUID-vkCmdDecodeVideoKHR-None-07011
  The bound video session **must** not be in uninitialized state at the time the command is executed on the device
- VUID-vkCmdDecodeVideoKHR-opCount-07134
  For each active query, the active query index corresponding to the query type of that query plus opCount **must** be less than or equal to the last activatable query index corresponding to the query type of that query plus one
- VUID-vkCmdDecodeVideoKHR-pNext-08365
  If the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR, and the pNext chain of pDecodeInfo includes a VkVideoInlineQueryInfoKHR structure with its queryPool member specifying a valid VkQueryPool handle, then VkVideoInlineQueryInfoKHR::queryCount **must** equal opCount
- VUID-vkCmdDecodeVideoKHR-pNext-08366
  If the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR, and the pNext chain of pDecodeInfo includes a VkVideoInlineQueryInfoKHR structure with its queryPool member specifying a valid VkQueryPool handle, then all the queries used by the command, as specified by the VkVideoInlineQueryInfoKHR structure, **must** be *unavailable*
- VUID-vkCmdDecodeVideoKHR-queryType-08367
  If the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR, then the queryType used to create the queryPool specified in the VkVideoInlineQueryInfoKHR structure included in the pNext chain of pDecodeInfo **must** be VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR
- VUID-vkCmdDecodeVideoKHR-queryPool-08368
  If the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR, then the queryPool specified in the VkVideoInlineQueryInfoKHR structure included in the pNext chain of pDecodeInfo **must** have been created with a VkVideoProfileInfoKHR structure included in the pNext chain of VkQueryPoolCreateInfo identical to the one specified in VkVideoSessionCreateInfoKHR::pVideoProfile the bound video session was created with
- VUID-vkCmdDecodeVideoKHR-queryType-08369

If the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR, and the queryType used to create the queryPool specified in the VkVideoInlineQueryInfoKHR structure included in the pNext chain of pDecodeInfo is VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR, then the VkCommandPool that commandBuffer was allocated from **must** have been created with a queue family index that supports result status queries, as indicated by VkQueueFamilyQueryResultStatusPropertiesKHR::queryResultStatusSupport

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07135
  pDecodeInfo->srcBuffer **must** be compatible with the video profile the bound video session was created with
- VUID-vkCmdDecodeVideoKHR-commandBuffer-07136
  If commandBuffer is an unprotected command buffer and protectedNoFault is not supported, then pDecodeInfo->srcBuffer **must** not be a protected buffer
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07138
  pDecodeInfo->srcBufferOffset **must** be an integer multiple of VkVideoCapabilitiesKHR::minBitstreamBufferOffsetAlignment, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07139
  pDecodeInfo->srcBufferRange **must** be an integer multiple of VkVideoCapabilitiesKHR::minBitstreamBufferSizeAlignment, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07140
  If pDecodeInfo->pSetupReferenceSlot is not NULL and VkVideoDecodeCapabilitiesKHR::flags does not include VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then the video picture resources specified by pDecodeInfo->dstPictureResource and pDecodeInfo->pSetupReferenceSlot->pPictureResource **must** not match
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07141
  If pDecodeInfo->pSetupReferenceSlot is not NULL and none of the following is true:
  - VkVideoDecodeCapabilitiesKHR::flags includes VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with
  - the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR and VkVideoDecodeAV1ProfileInfoKHR::filmGrainSupport set to VK_TRUE, and film grain is enabled for the decoded picture

  then the video picture resources specified by pDecodeInfo->dstPictureResource and pDecodeInfo->pSetupReferenceSlot->pPictureResource **must** match

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07142
  pDecodeInfo->dstPictureResource.imageViewBinding **must** be compatible with the video profile the bound video session was created with
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07143
  The format of pDecodeInfo->dstPictureResource.imageViewBinding **must** match the VkVideoSessionCreateInfoKHR::pictureFormat the bound video session was created with
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07144
  pDecodeInfo->dstPictureResource.codedOffset **must** be an integer multiple of codedOffsetGranularity
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07145
  pDecodeInfo->dstPictureResource.codedExtent **must** be between minCodedExtent and maxCodedExtent, inclusive, the bound video session was created with
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07146
  pDecodeInfo->dstPictureResource.imageViewBinding **must** have been created with the VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR usage flag set
- VUID-vkCmdDecodeVideoKHR-commandBuffer-07147
  If commandBuffer is an unprotected command buffer and protectedNoFault is not supported, then pDecodeInfo->dstPictureResource.imageViewBinding **must** not have been created from a protected image
- VUID-vkCmdDecodeVideoKHR-commandBuffer-07148
  If commandBuffer is a protected command buffer and protectedNoFault is not supported, then pDecodeInfo->dstPictureResource.imageViewBinding **must** have been created from a protected image
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-08376
  pDecodeInfo->pSetupReferenceSlot **must** not be NULL unless the bound video session was created with VkVideoSessionCreateInfoKHR::maxDpbSlots equal to zero
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07170
  If pDecodeInfo->pSetupReferenceSlot is not NULL, then pDecodeInfo->pSetupReferenceSlot->slotIndex **must** be less than the VkVideoSessionCreateInfoKHR::maxDpbSlots specified when the bound video session was created
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07173
  If pDecodeInfo->pSetupReferenceSlot is not NULL, then pDecodeInfo->pSetupReferenceSlot->pPictureResource->codedOffset **must** be an integer multiple of codedOffsetGranularity
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07149
  If pDecodeInfo->pSetupReferenceSlot is not NULL, then pDecodeInfo->pSetupReferenceSlot->pPictureResource **must** match one of the bound reference picture resources
- VUID-vkCmdDecodeVideoKHR-activeReferencePictureCount-07150
  activeReferencePictureCount **must** be less than or equal to the VkVideoSessionCreateInfoKHR::maxActiveReferencePictures specified when the bound video session was created
- VUID-vkCmdDecodeVideoKHR-slotIndex-07256
  The slotIndex member of each element of pDecodeInfo->pReferenceSlots **must** be less than the VkVideoSessionCreateInfoKHR::maxDpbSlots specified when the bound video session

was created

- VUID-vkCmdDecodeVideoKHR-codedOffset-07257
  The codedOffset member of the VkVideoPictureResourceInfoKHR structure pointed to by the pPictureResource member of each element of pDecodeInfo->pReferenceSlots **must** be an integer multiple of codedOffsetGranularity
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07151
  The pPictureResource member of each element of pDecodeInfo->pReferenceSlots **must** match one of the bound reference picture resources associated with the DPB slot index specified in the slotIndex member of that element
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07264
  Each video picture resource corresponding to the pPictureResource member specified in the elements of pDecodeInfo->pReferenceSlots **must** be unique within pDecodeInfo->pReferenceSlots
- VUID-vkCmdDecodeVideoKHR-dpbFrameUseCount-07176
  All elements of dpbFrameUseCount **must** be less than or equal to 1
- VUID-vkCmdDecodeVideoKHR-dpbTopFieldUseCount-07177
  All elements of dpbTopFieldUseCount **must** be less than or equal to 1
- VUID-vkCmdDecodeVideoKHR-dpbBottomFieldUseCount-07178
  All elements of dpbBottomFieldUseCount **must** be less than or equal to 1
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-10801
  If pDecodeInfo->pSetupReferenceSlot is NULL or pDecodeInfo->pSetupReferenceSlot->pPictureResource does not refer to the same image subresource as pDecodeInfo->dstPictureResource, then the image subresource referred to by pDecodeInfo->dstPictureResource **must** be in the VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR layout at the time the video decode operation is executed on the device , unless the unifiedImageLayoutsVideo feature is enabled, in which case it **may** be in the VK_IMAGE_LAYOUT_GENERAL layout
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-10802
  If pDecodeInfo->pSetupReferenceSlot is not NULL and pDecodeInfo->pSetupReferenceSlot->pPictureResource refers to the same image subresource as pDecodeInfo->dstPictureResource, then the image subresource referred to by pDecodeInfo->dstPictureResource **must** be in the VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR layout at the time the video decode operation is executed on the device , unless the unifiedImageLayoutsVideo feature is enabled, in which case it **may** be in the VK_IMAGE_LAYOUT_GENERAL layout
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-10803
  If pDecodeInfo->pSetupReferenceSlot is not NULL, then the image subresource referred to by pDecodeInfo->pSetupReferenceSlot->pPictureResource **must** be in the VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR layout at the time the video decode operation is executed on the device , unless the unifiedImageLayoutsVideo feature is enabled, in which case it **may** be in the VK_IMAGE_LAYOUT_GENERAL layout
- VUID-vkCmdDecodeVideoKHR-pPictureResource-10804
  The image subresource referred to by the pPictureResource member of each element of pDecodeInfo->pReferenceSlots **must** be in the

VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR layout at the time the video decode operation is executed on the device , unless the unifiedImageLayoutsVideo feature is enabled, in which case it **may** be in the VK_IMAGE_LAYOUT_GENERAL layout

- VUID-vkCmdDecodeVideoKHR-pNext-07152
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the pNext chain of pDecodeInfo **must** include a VkVideoDecodeH264PictureInfoKHR structure
- VUID-vkCmdDecodeVideoKHR-None-07258
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR but was not created with interlaced frame support, then the decode output picture **must** represent a frame
- VUID-vkCmdDecodeVideoKHR-pSliceOffsets-07153
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then all elements of the pSliceOffsets member of the VkVideoDecodeH264PictureInfoKHR structure included in the pNext chain of pDecodeInfo **must** be less than pDecodeInfo->srcBufferRange
- VUID-vkCmdDecodeVideoKHR-None-10400
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then there **must** be a bound video session parameters object if any of the following conditions are not met:
  - the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR
  - the pNext chain of pDecodeInfo includes a VkVideoDecodeH264InlineSessionParametersInfoKHR structure
  - the pStdSPS member of the VkVideoDecodeH264InlineSessionParametersInfoKHR structure included in the pNext chain of pDecodeInfo is not NULL
  - the pStdPPS member of the VkVideoDecodeH264InlineSessionParametersInfoKHR structure included in the pNext chain of pDecodeInfo is not NULL
- VUID-vkCmdDecodeVideoKHR-StdVideoH264SequenceParameterSet-07154
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the bound video session parameters object **must** contain a StdVideoH264SequenceParameterSet entry with seq_parameter_set_id matching StdVideoDecodeH264PictureInfo::seq_parameter_set_id that is provided in the pStdPictureInfo member of the VkVideoDecodeH264PictureInfoKHR structure included in the pNext chain of pDecodeInfo , unless the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR and the pNext chain of pDecodeInfo includes a VkVideoDecodeH264InlineSessionParametersInfoKHR structure, and its pStdSPS member is not equal to NULL
- VUID-vkCmdDecodeVideoKHR-pNext-10401
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR, and the pNext chain of pDecodeInfo includes a VkVideoDecodeH264InlineSessionParametersInfoKHR

structure with a non-NULL pStdSPS member, then pStdSPS->seq_parameter_set_id **must** equal StdVideoDecodeH264PictureInfo::seq_parameter_set_id provided in the pStdPictureInfo member of the VkVideoDecodeH264PictureInfoKHR structure included in the pNext chain of pDecodeInfo

- VUID-vkCmdDecodeVideoKHR-StdVideoH264PictureParameterSet-07155
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the bound video session parameters object **must** contain a StdVideoH264PictureParameterSet entry with seq_parameter_set_id and pic_parameter_set_id matching StdVideoDecodeH264PictureInfo::seq_parameter_set_id and StdVideoDecodeH264PictureInfo::pic_parameter_set_id, respectively, that are provided in the pStdPictureInfo member of the VkVideoDecodeH264PictureInfoKHR structure included in the pNext chain of pDecodeInfo , unless the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR and the pNext chain of pDecodeInfo includes a VkVideoDecodeH264InlineSessionParametersInfoKHR structure, and its pStdPPS member is not equal to NULL
- VUID-vkCmdDecodeVideoKHR-pNext-10402
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR, and the pNext chain of pDecodeInfo includes a VkVideoDecodeH264InlineSessionParametersInfoKHR structure with a non-NULL pStdPPS member, then pStdPPS->seq_parameter_set_id and pStdPPS->pic_parameter_set_id **must** equal StdVideoDecodeH264PictureInfo::seq_parameter_set_id and StdVideoDecodeH264PictureInfo::pic_parameter_set_id, respectively, provided in the pStdPictureInfo member of the VkVideoDecodeH264PictureInfoKHR structure included in the pNext chain of pDecodeInfo
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07156
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and pDecodeInfo->pSetupReferenceSlot is not NULL, then the pNext chain of pDecodeInfo->pSetupReferenceSlot **must** include a VkVideoDecodeH264DpbSlotInfoKHR structure
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07259
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR but was not created with interlaced frame support, and pDecodeInfo->pSetupReferenceSlot is not NULL, then the reconstructed picture **must** represent a frame
- VUID-vkCmdDecodeVideoKHR-pNext-07157
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the pNext chain of each element of pDecodeInfo->pReferenceSlots **must** include a VkVideoDecodeH264DpbSlotInfoKHR structure
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07260
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR but was not created with interlaced frame support, then each active reference picture corresponding to the

elements of pDecodeInfo->pReferenceSlots **must** represent a frame

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07261
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, pDecodeInfo->pSetupReferenceSlot is not NULL, and the decode output picture represents a frame, then the reconstructed picture **must** also represent a frame
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07262
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, pDecodeInfo->pSetupReferenceSlot is not NULL, and the decode output picture represents a top field, then the reconstructed picture **must** also represent a top field
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07263
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, pDecodeInfo->pSetupReferenceSlot is not NULL, and the decode output picture represents a bottom field, then the reconstructed picture **must** also represent a bottom field
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07266
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and an active reference picture corresponding to any element of pDecodeInfo->pReferenceSlots represents a frame, then the DPB slot index of the bound video session specified by the slotIndex member of that element **must** be currently associated with a frame picture matching the video picture resource specified by the pPictureResource member of the same element at the time the command is executed on the device
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07267
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and an active reference picture corresponding to any element of pDecodeInfo->pReferenceSlots represents a top field, then the DPB slot index of the bound video session specified by the slotIndex member of that element **must** be currently associated with a top field picture matching the video picture resource specified by the pPictureResource member of the same element at the time the command is executed on the device
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07268
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and an active reference picture corresponding to any element of pDecodeInfo->pReferenceSlots represents a bottom field, then the DPB slot index of the bound video session specified by the slotIndex member of that element **must** be currently associated with a bottom field picture matching the video picture resource specified by the pPictureResource member of the same element at the time the command is executed on the device
- VUID-vkCmdDecodeVideoKHR-pNext-07158
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the pNext chain of pDecodeInfo **must** include a VkVideoDecodeH265PictureInfoKHR structure
- VUID-vkCmdDecodeVideoKHR-pSliceSegmentOffsets-07159

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then all elements of the pSliceSegmentOffsets member of the VkVideoDecodeH265PictureInfoKHR structure included in the pNext chain of pDecodeInfo **must** be less than pDecodeInfo->srcBufferRange

- VUID-vkCmdDecodeVideoKHR-None-10403
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then there **must** be a bound video session parameters object if any of the following conditions are not met:
  - the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR
  - the pNext chain of pDecodeInfo includes a VkVideoDecodeH265InlineSessionParametersInfoKHR structure
  - the pStdVPS member of the VkVideoDecodeH265InlineSessionParametersInfoKHR structure included in the pNext chain of pDecodeInfo is not NULL
  - the pStdSPS member of the VkVideoDecodeH265InlineSessionParametersInfoKHR structure included in the pNext chain of pDecodeInfo is not NULL
  - the pStdPPS member of the VkVideoDecodeH265InlineSessionParametersInfoKHR structure included in the pNext chain of pDecodeInfo is not NULL
- VUID-vkCmdDecodeVideoKHR-StdVideoH265VideoParameterSet-07160
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the bound video session parameters object **must** contain a StdVideoH265VideoParameterSet entry with vps_video_parameter_set_id matching StdVideoDecodeH265PictureInfo ::sps_video_parameter_set_id that is provided in the pStdPictureInfo member of the VkVideoDecodeH265PictureInfoKHR structure included in the pNext chain of pDecodeInfo , unless the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR and the pNext chain of pDecodeInfo includes a VkVideoDecodeH265InlineSessionParametersInfoKHR structure, and its pStdVPS member is not equal to NULL
- VUID-vkCmdDecodeVideoKHR-pNext-10404
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR and with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR, and the pNext chain of pDecodeInfo includes a VkVideoDecodeH265InlineSessionParametersInfoKHR structure with a non-NULL pStdVPS member, then pStdVPS->vps_video_parameter_set_id **must** equal StdVideoDecodeH265PictureInfo::sps_video_parameter_set_id provided in the pStdPictureInfo member of the VkVideoDecodeH265PictureInfoKHR structure included in the pNext chain of pDecodeInfo
- VUID-vkCmdDecodeVideoKHR-StdVideoH265SequenceParameterSet-07161
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the bound video session parameters object **must** contain a StdVideoH265SequenceParameterSet entry with sps_video_parameter_set_id and sps_seq_parameter_set_id matching StdVideoDecodeH265PictureInfo::sps_video_parameter_set_id and

StdVideoDecodeH265PictureInfo::pps_seq_parameter_set_id, respectively, that are provided in the pStdPictureInfo member of the VkVideoDecodeH265PictureInfoKHR structure included in the pNext chain of pDecodeInfo , unless the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR and the pNext chain of pDecodeInfo includes a VkVideoDecodeH265InlineSessionParametersInfoKHR structure, and its pStdSPS member is not equal to NULL

- VUID-vkCmdDecodeVideoKHR-pNext-10405
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR and with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR, and the pNext chain of pDecodeInfo includes a VkVideoDecodeH265InlineSessionParametersInfoKHR structure with a non-NULL pStdSPS member, then pStdSPS->sps_video_parameter_set_id and pStdSPS->sps_seq_parameter_set_id **must** equal StdVideoDecodeH265PictureInfo::sps_video_parameter_set_id and StdVideoDecodeH265PictureInfo::pps_seq_parameter_set_id, respectively, provided in the pStdPictureInfo member of the VkVideoDecodeH265PictureInfoKHR structure included in the pNext chain of pDecodeInfo
- VUID-vkCmdDecodeVideoKHR-StdVideoH265PictureParameterSet-07162
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the bound video session parameters object **must** contain a StdVideoH265PictureParameterSet entry with sps_video_parameter_set_id, pps_seq_parameter_set_id, and pps_pic_parameter_set_id matching StdVideoDecodeH265PictureInfo::sps_video_parameter_set_id, StdVideoDecodeH265PictureInfo::pps_seq_parameter_set_id, and StdVideoDecodeH265PictureInfo::pps_pic_parameter_set_id, respectively, that are provided in the pStdPictureInfo member of the VkVideoDecodeH265PictureInfoKHR structure included in the pNext chain of pDecodeInfo , unless the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR and the pNext chain of pDecodeInfo includes a VkVideoDecodeH265InlineSessionParametersInfoKHR structure, and its pStdPPS member is not equal to NULL
- VUID-vkCmdDecodeVideoKHR-pNext-10406
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR and with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR, and the pNext chain of pDecodeInfo includes a VkVideoDecodeH265InlineSessionParametersInfoKHR structure with a non-NULL pStdPPS member, then pStdPPS->sps_video_parameter_set_id, pStdPPS->pps_seq_parameter_set_id, and pStdPPS->pps_pic_parameter_set_id **must** equal StdVideoDecodeH265PictureInfo::sps_video_parameter_set_id, StdVideoDecodeH265PictureInfo::pps_seq_parameter_set_id, and StdVideoDecodeH265PictureInfo::pps_pic_parameter_set_id, respectively, provided in the pStdPictureInfo member of the VkVideoDecodeH265PictureInfoKHR structure included in the pNext chain of pDecodeInfo
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07163
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR and pDecodeInfo-

>pSetupReferenceSlot is not NULL, then the pNext chain of pDecodeInfo->pSetupReferenceSlot **must** include a VkVideoDecodeH265DpbSlotInfoKHR structure

- VUID-vkCmdDecodeVideoKHR-pNext-07164
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the pNext chain of each element of pDecodeInfo->pReferenceSlots **must** include a VkVideoDecodeH265DpbSlotInfoKHR structure
- VUID-vkCmdDecodeVideoKHR-pNext-10805
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR, then the pNext chain of pDecodeInfo **must** include a VkVideoDecodeVP9PictureInfoKHR structure
- VUID-vkCmdDecodeVideoKHR-uncompressedHeaderOffset-10806
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR, then the uncompressedHeaderOffset member of the VkVideoDecodeVP9PictureInfoKHR structure included in the pNext chain of pDecodeInfo **must** be less than pDecodeInfo->srcBufferRange
- VUID-vkCmdDecodeVideoKHR-compressedHeaderOffset-10807
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR, then the compressedHeaderOffset member of the VkVideoDecodeVP9PictureInfoKHR structure included in the pNext chain of pDecodeInfo **must** be less than pDecodeInfo->srcBufferRange
- VUID-vkCmdDecodeVideoKHR-tilesOffset-10808
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR, then the tilesOffset member of the VkVideoDecodeVP9PictureInfoKHR structure included in the pNext chain of pDecodeInfo **must** be less than pDecodeInfo->srcBufferRange
- VUID-vkCmdDecodeVideoKHR-referenceNameSlotIndices-10809
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR, then each element of the referenceNameSlotIndices array member of the VkVideoDecodeVP9PictureInfoKHR structure included in the pNext chain of pDecodeInfo **must** either be negative or **must** equal the slotIndex member of one of the elements of pDecodeInfo->pReferenceSlots
- VUID-vkCmdDecodeVideoKHR-slotIndex-10810
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR, then the slotIndex member of each element of pDecodeInfo->pReferenceSlots **must** equal one of the elements of the referenceNameSlotIndices array member of the VkVideoDecodeVP9PictureInfoKHR structure included in the pNext chain of pDecodeInfo
- VUID-vkCmdDecodeVideoKHR-filmGrainSupport-09248
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR and VkVideoDecodeAV1ProfileInfoKHR::filmGrainSupport set to VK_FALSE, then film grain **must** not be enabled for the decoded picture
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-09249

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, pDecodeInfo->pSetupReferenceSlot is not NULL, and film grain is enabled for the decoded picture, then the video picture resources specified by pDecodeInfo->dstPictureResource and pDecodeInfo->pSetupReferenceSlot->pPictureResource **must** not match

- VUID-vkCmdDecodeVideoKHR-pNext-09250
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then the pNext chain of pDecodeInfo **must** include a VkVideoDecodeAV1PictureInfoKHR structure
- VUID-vkCmdDecodeVideoKHR-frameHeaderOffset-09251
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then the frameHeaderOffset member of the VkVideoDecodeAV1PictureInfoKHR structure included in the pNext chain of pDecodeInfo **must** be less than pDecodeInfo->srcBufferRange
- VUID-vkCmdDecodeVideoKHR-pTileOffsets-09253
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then all elements of the pTileOffsets member of the VkVideoDecodeAV1PictureInfoKHR structure included in the pNext chain of pDecodeInfo **must** be less than pDecodeInfo->srcBufferRange
- VUID-vkCmdDecodeVideoKHR-pTileOffsets-09252
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then for each element i of the pTileOffsets and pTileSizes members of the VkVideoDecodeAV1PictureInfoKHR structure included in the pNext chain of pDecodeInfo, the sum of pTileOffsets[i] and pTileSizes[i] **must** be less than or equal to pDecodeInfo->srcBufferRange
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-09254
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR and pDecodeInfo->pSetupReferenceSlot is not NULL, then the pNext chain of pDecodeInfo->pSetupReferenceSlot **must** include a VkVideoDecodeAV1DpbSlotInfoKHR structure
- VUID-vkCmdDecodeVideoKHR-pNext-09255
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then the pNext chain of each element of pDecodeInfo->pReferenceSlots **must** include a VkVideoDecodeAV1DpbSlotInfoKHR structure
- VUID-vkCmdDecodeVideoKHR-referenceNameSlotIndices-09262
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then each element of the referenceNameSlotIndices array member of the VkVideoDecodeAV1PictureInfoKHR structure included in the pNext chain of pDecodeInfo **must** either be negative or **must** equal the slotIndex member of one of the elements of pDecodeInfo->pReferenceSlots
- VUID-vkCmdDecodeVideoKHR-slotIndex-09263
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then the slotIndex member of each element of pDecodeInfo->pReferenceSlots **must** equal one of the elements of the

`referenceNameSlotIndices` array member of the VkVideoDecodeAV1PictureInfoKHR structure included in the `pNext` chain of `pDecodeInfo`

- VUID-vkCmdDecodeVideoKHR-None-10407
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then there **must** be a bound video session parameters object if any of the following conditions are not met:
  - the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR
  - the `pNext` chain of `pDecodeInfo` includes a VkVideoDecodeAV1InlineSessionParametersInfoKHR structure
  - the `pStdSequenceHeader` member of the VkVideoDecodeAV1InlineSessionParametersInfoKHR structure included in the `pNext` chain of `pDecodeInfo` is not `NULL`

Valid Usage (Implicit)

- VUID-vkCmdDecodeVideoKHR-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-parameter
  `pDecodeInfo` **must** be a valid pointer to a valid VkVideoDecodeInfoKHR structure
- VUID-vkCmdDecodeVideoKHR-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdDecodeVideoKHR-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_VIDEO_DECODE_BIT_KHR operations
- VUID-vkCmdDecodeVideoKHR-renderpass
  This command **must** only be called outside of a render pass instance
- VUID-vkCmdDecodeVideoKHR-suspended
  This command **must** not be called between suspended render pass instances
- VUID-vkCmdDecodeVideoKHR-videocoding
  This command **must** only be called inside of a video coding scope
- VUID-vkCmdDecodeVideoKHR-bufferlevel
  `commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Inside	VK_QUEUE_VIDEO_DECODE_BIT_KHR	Action

Conditional Rendering

vkCmdDecodeVideoKHR is not affected by conditional rendering

The VkVideoDecodeInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_queue
typedef struct VkVideoDecodeInfoKHR {
    VkStructureType                          sType;
    const void*                              pNext;
    VkVideoDecodeFlagsKHR                    flags;
    VkBuffer                                 srcBuffer;
    VkDeviceSize                             srcBufferOffset;
    VkDeviceSize                             srcBufferRange;
    VkVideoPictureResourceInfoKHR            dstPictureResource;
    const VkVideoReferenceSlotInfoKHR*       pSetupReferenceSlot;
    uint32_t                                 referenceSlotCount;
    const VkVideoReferenceSlotInfoKHR*       pReferenceSlots;
} VkVideoDecodeInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is reserved for future use.
- srcBuffer is the source video bitstream buffer to read the encoded bitstream from.
- srcBufferOffset is the starting offset in bytes from the start of srcBuffer to read the encoded bitstream from.
- srcBufferRange is the size in bytes of the encoded bitstream to decode from srcBuffer, starting from srcBufferOffset.
- dstPictureResource is the video picture resource to use as the decode output picture.
- pSetupReferenceSlot is NULL or a pointer to a VkVideoReferenceSlotInfoKHR structure specifying the reconstructed picture information.
- referenceSlotCount is the number of elements in the pReferenceSlots array.
- pReferenceSlots is NULL or a pointer to an array of VkVideoReferenceSlotInfoKHR structures

describing the DPB slots and corresponding reference picture resources to use in this video decode operation (the set of active reference pictures).

Valid Usage

- VUID-VkVideoDecodeInfoKHR-srcBuffer-07165
  `srcBuffer` **must** have been created with the VK_BUFFER_USAGE_VIDEO_DECODE_SRC_BIT_KHR usage flag set
- VUID-VkVideoDecodeInfoKHR-srcBufferOffset-07166
  `srcBufferOffset` **must** be less than the size of `srcBuffer`
- VUID-VkVideoDecodeInfoKHR-srcBufferRange-07167
  `srcBufferRange` **must** be less than or equal to the size of `srcBuffer` minus `srcBufferOffset`
- VUID-VkVideoDecodeInfoKHR-pSetupReferenceSlot-07168
  If `pSetupReferenceSlot` is not `NULL`, then its `slotIndex` member **must** not be negative
- VUID-VkVideoDecodeInfoKHR-pSetupReferenceSlot-07169
  If `pSetupReferenceSlot` is not `NULL`, then its `pPictureResource` **must** not be `NULL`
- VUID-VkVideoDecodeInfoKHR-slotIndex-07171
  The `slotIndex` member of each element of `pReferenceSlots` **must** not be negative
- VUID-VkVideoDecodeInfoKHR-pPictureResource-07172
  The `pPictureResource` member of each element of `pReferenceSlots` **must** not be `NULL`

Valid Usage (Implicit)

- VUID-VkVideoDecodeInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_INFO_KHR
- VUID-VkVideoDecodeInfoKHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkVideoDecodeAV1InlineSessionParametersInfoKHR, VkVideoDecodeAV1PictureInfoKHR, VkVideoDecodeH264InlineSessionParametersInfoKHR, VkVideoDecodeH264PictureInfoKHR, VkVideoDecodeH265InlineSessionParametersInfoKHR, VkVideoDecodeH265PictureInfoKHR, VkVideoDecodeVP9PictureInfoKHR, or VkVideoInlineQueryInfoKHR
- VUID-VkVideoDecodeInfoKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkVideoDecodeInfoKHR-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkVideoDecodeInfoKHR-srcBuffer-parameter
  `srcBuffer` **must** be a valid VkBuffer handle
- VUID-VkVideoDecodeInfoKHR-dstPictureResource-parameter
  `dstPictureResource` **must** be a valid VkVideoPictureResourceInfoKHR structure

- VUID-VkVideoDecodeInfoKHR-pSetupReferenceSlot-parameter
  If `pSetupReferenceSlot` is not `NULL`, `pSetupReferenceSlot` **must** be a valid pointer to a valid VkVideoReferenceSlotInfoKHR structure
- VUID-VkVideoDecodeInfoKHR-pReferenceSlots-parameter
  If `referenceSlotCount` is not `0`, `pReferenceSlots` **must** be a valid pointer to an array of `referenceSlotCount` valid VkVideoReferenceSlotInfoKHR structures

```
// Provided by VK_KHR_video_decode_queue
typedef VkFlags VkVideoDecodeFlagsKHR;
```

`VkVideoDecodeFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

45.12. H.264 Decode Operations

Video decode operations using an H.264 decode profile **can** be used to decode elementary video stream sequences compliant to the ITU-T H.264 Specification.

NOTE	Refer to the Preamble for information on how the Khronos Intellectual Property Rights Policy relates to normative references to external materials not created by Khronos.

This process is performed according to the video decode operation steps with the codec-specific semantics defined in section 8 of the ITU-T H.264 Specification as follows:

- Syntax elements, derived values, and other parameters are applied from the following structures:
  - The `StdVideoH264SequenceParameterSet` structure corresponding to the active SPS specifying the H.264 sequence parameter set.
  - The `StdVideoH264PictureParameterSet` structure corresponding to the active PPS specifying the H.264 picture parameter set.
  - The `StdVideoDecodeH264PictureInfo` structure specifying the H.264 picture information.
  - The `StdVideoDecodeH264ReferenceInfo` structures specifying the H.264 reference information corresponding to the optional reconstructed picture and any active reference pictures.
- The contents of the provided video bitstream buffer range are interpreted as defined in the H.264 Decode Bitstream Data Access section.
- Picture data in the video picture resources corresponding to the used active reference pictures, decode output picture, and optional reconstructed picture is accessed as defined in the H.264 Decode Picture Data Access section.
- The decision on reference picture setup is made according to the parameters specified in the H.264 picture information.

If the parameters and the bitstream adhere to the syntactic and semantic requirements defined in the corresponding sections of the ITU-T H.264 Specification, as described above, and the DPB slots

associated with the active reference pictures all refer to valid picture references, then the video decode operation will complete successfully. Otherwise, the video decode operation **may** complete unsuccessfully.

45.12.1. H.264 Decode Bitstream Data Access

If the target decode output picture is a frame, then the video bitstream buffer range **should** contain a VCL NAL unit comprised of the slice headers and data of a picture representing an entire frame, as defined in sections 7.3.3 and 7.3.4, and this data is interpreted as defined in sections 7.4.3 and 7.4.4 of the ITU-T H.264 Specification, respectively.

If the target decode output picture is a field, then the video bitstream buffer range should contain a VCL NAL unit comprised of the slice headers and data of a picture representing a field, as defined in sections 7.3.3 and 7.3.4, and this data is interpreted as defined in sections 7.4.3 and 7.4.4 of the ITU-T H.264 Specification, respectively.

The offsets provided in VkVideoDecodeH264PictureInfoKHR::`pSliceOffsets` **should** specify the starting offsets corresponding to each slice header within the video bitstream buffer range.

45.12.2. H.264 Decode Picture Data Access

The effective `imageOffset` and `imageExtent` corresponding to a decode output picture, reference picture, or reconstructed picture used in video decode operations with an H.264 decode profile are defined as follows:

- `imageOffset` is (`codedOffset.x`,`codedOffset.y`) and `imageExtent` is (`codedExtent.width`, `codedExtent.height`), if the picture represents a frame.
- `imageOffset` is (`codedOffset.x`,`codedOffset.y`) and `imageExtent` is (`codedExtent.width`, `codedExtent.height`), if the picture represents a field and the picture layout of the used H.264 decode profile is VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_INTERLEAVED_LINES_BIT_KHR.
- `imageOffset` is (`codedOffset.x`,`codedOffset.y`) and `imageExtent` is (`codedExtent.width`, `codedExtent.height` / 2), if the picture represents a field and the picture layout of the used H.264 decode profile is VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR.

Where `codedOffset` and `codedExtent` are the members of the VkVideoPictureResourceInfoKHR structure corresponding to the picture.

However, accesses to image data within a video picture resource happen at the granularity indicated by VkVideoCapabilitiesKHR::`pictureAccessGranularity`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile. This means that the complete image subregion accessed by video coding operations using an H.264 decode profile for the video picture resource is defined as the set of texels within the coordinate range:

([`startX`,`endX`), [`startY`,`endY`))

Where:

- `startX` equals `imageOffset.x` rounded down to the nearest integer multiple of `pictureAccessGranularity.width`;
- `endX` equals `imageOffset.x` + `imageExtent.width` rounded up to the nearest integer multiple of `pictureAccessGranularity.width` and clamped to the width of the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure;
- startY equals `imageOffset.y` rounded down to the nearest integer multiple of `pictureAccessGranularity.height`;
- endY equals `imageOffset.y` + `imageExtent.height` rounded up to the nearest integer multiple of `pictureAccessGranularity.height` and clamped to the height of the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure.

In case of video decode operations using an H.264 decode profile, any access to a picture at the coordinates (x,y), as defined by the ITU-T H.264 Specification, is an access to the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure at the texel coordinates specified below:

- (x,y), if the accessed picture represents a frame.
- (x,y × 2), if the accessed picture represents a top field and the picture layout of the used H.264 decode profile is VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_INTERLEAVED_LINES_BIT_KHR.
- (x,y × 2 + 1), if the accessed picture represents a bottom field and the picture layout of the used H.264 decode profile is VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_INTERLEAVED_LINES_BIT_KHR.
- (x,y), if the accessed picture represents a top field and the picture layout of the used H.264 decode profile is VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR.
- (`codedOffset.x` + x,`codedOffset.y` + y), if the accessed picture represents a bottom field and the picture layout of the used H.264 decode profile is VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR.

Where `codedOffset` is the member of the corresponding VkVideoPictureResourceInfoKHR structure.

45.12.3. H.264 Decode Profile

A video profile supporting H.264 video decode operations is specified by setting VkVideoProfileInfoKHR::`videoCodecOperation` to VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and adding a `VkVideoDecodeH264ProfileInfoKHR` structure to the VkVideoProfileInfoKHR::`pNext` chain.

The `VkVideoDecodeH264ProfileInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_h264
typedef struct VkVideoDecodeH264ProfileInfoKHR {
    VkStructureType                              sType;
    const void*                                  pNext;
    StdVideoH264ProfileIdc                       stdProfileIdc;
    VkVideoDecodeH264PictureLayoutFlagBitsKHR    pictureLayout;
} VkVideoDecodeH264ProfileInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stdProfileIdc` is a `StdVideoH264ProfileIdc` value specifying the H.264 codec profile IDC, where enum constant `STD_VIDEO_H264_PROFILE_IDC_BASELINE` identifies the Constrained Baseline profile as defined in A.2.1.1 of the ITU-T H.264 Specification, and all other values correspond to profiles defined in section A.2 of the ITU-T H.264 Specification.
- `pictureLayout` is a VkVideoDecodeH264PictureLayoutFlagBitsKHR value specifying the picture layout used by the H.264 video sequence to be decoded.

Valid Usage (Implicit)

- VUID-VkVideoDecodeH264ProfileInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_PROFILE_INFO_KHR
- VUID-VkVideoDecodeH264ProfileInfoKHR-pictureLayout-parameter
  If `pictureLayout` is not `0`, `pictureLayout` **must** be a valid VkVideoDecodeH264PictureLayoutFlagBitsKHR value

Structure Chaining

Extends the structures

- VkQueryPoolCreateInfo
- VkVideoProfileInfoKHR

The H.264 video decode picture layout flags are defined as follows:

```
// Provided by VK_KHR_video_decode_h264
typedef enum VkVideoDecodeH264PictureLayoutFlagBitsKHR {
    VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_PROGRESSIVE_KHR = 0,
    VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_INTERLEAVED_LINES_BIT_KHR =
0x00000001,
    VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR =
0x00000002,
} VkVideoDecodeH264PictureLayoutFlagBitsKHR;
```

- VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_PROGRESSIVE_KHR specifies support for progressive content. This flag has the value 0.
- VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_INTERLEAVED_LINES_BIT_KHR specifies support for or use of a picture layout for interlaced content where all lines belonging to the top field are decoded to the even-numbered lines within the picture resource, and all lines belonging to the bottom field are decoded to the odd-numbered lines within the picture resource.
- VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR specifies support for or use of a picture layout for interlaced content where all lines belonging to a field are grouped together in a single image subregion, and the two fields comprising the frame **can** be stored in separate image subregions of the same image subresource or in separate image subresources.

```
// Provided by VK_KHR_video_decode_h264
typedef VkFlags VkVideoDecodeH264PictureLayoutFlagsKHR;
```

VkVideoDecodeH264PictureLayoutFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoDecodeH264PictureLayoutFlagBitsKHR.

45.12.4. H.264 Decode Capabilities

When calling vkGetPhysicalDeviceVideoCapabilitiesKHR to query the capabilities for an H.264 decode profile, the VkVideoCapabilitiesKHR::pNext chain **must** include a VkVideoDecodeH264CapabilitiesKHR structure that will be filled with the profile-specific capabilities.

The VkVideoDecodeH264CapabilitiesKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_h264
typedef struct VkVideoDecodeH264CapabilitiesKHR {
    VkStructureType         sType;
    void*                   pNext;
    StdVideoH264LevelIdc    maxLevelIdc;
    VkOffset2D              fieldOffsetGranularity;
} VkVideoDecodeH264CapabilitiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- maxLevelIdc is a StdVideoH264LevelIdc value indicating the maximum H.264 level supported by the profile, where enum constant STD_VIDEO_H264_LEVEL_IDC_<major>_<minor> identifies H.264 level <major>.<minor> as defined in section A.3 of the ITU-T H.264 Specification.
- fieldOffsetGranularity is the minimum alignment for VkVideoPictureResourceInfoKHR ::codedOffset specified for a video picture resource when using the picture layout VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR.

Valid Usage (Implicit)

- VUID-VkVideoDecodeH264CapabilitiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkVideoCapabilitiesKHR

45.12.5. H.264 Decode Parameter Sets

Video session parameters objects created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR **can** contain the following types of parameters:

H.264 Sequence Parameter Sets (SPS)

Represented by `StdVideoH264SequenceParameterSet` structures and interpreted as follows:

- `reserved1` and `reserved2` are used only for padding purposes and are otherwise ignored;
- `seq_parameter_set_id` is used as the key of the SPS entry;
- `level_idc` is one of the enum constants `STD_VIDEO_H264_LEVEL_IDC_<major>_<minor>` identifying the H.264 level `<major>.<minor>` as defined in section A.3 of the ITU-T H.264 Specification;
- `pOffsetForRefFrame` is a pointer to an array of `num_ref_frames_in_pic_order_cnt_cycle` number of signed integers specifying the elements of the `offset_for_ref_frame` array, as defined in section 7.4.2.1.1 of the ITU-T H.264 Specification;
- if `flags.seq_scaling_matrix_present_flag` is set, then the `StdVideoH264ScalingLists` structure pointed to by `pScalingLists` is interpreted as follows:
  - `scaling_list_present_mask` is a bitmask where bit index i corresponds to `seq_scaling_list_present_flag[i]` as defined in section 7.4.2.1 of the ITU-T H.264 Specification;
  - `use_default_scaling_matrix_mask` is a bitmask where bit index i corresponds to `UseDefaultScalingMatrix4x4Flag[i]`, when i < 6, or corresponds to `UseDefaultScalingMatrix8x8Flag[i-6]`, otherwise, as defined in section 7.3.2.1 of the ITU-T H.264 Specification;
  - `ScalingList4x4` and `ScalingList8x8` correspond to the identically named syntax elements defined in section 7.3.2.1 of the ITU-T H.264 Specification;
- if `flags.vui_parameters_present_flag` is set, then `pSequenceParameterSetVui` is a pointer to a `StdVideoH264SequenceParameterSetVui` structure that is interpreted as follows:
  - `reserved1` is used only for padding purposes and is otherwise ignored;

- `flags.color_description_present_flag` is interpreted as the value of `colour_description_present_flag`, as defined in section E.2.1 of the ITU-T H.264 Specification;

  > **NOTE** The name of `colour_description_present_flag` was misspelled in the Video Std header.

- if `flags.nal_hrd_parameters_present_flag` or `flags.vcl_hrd_parameters_present_flag` is set, then the `StdVideoH264HrdParameters` structure pointed to by `pHrdParameters` is interpreted as follows:
  - `reserved1` is used only for padding purposes and is otherwise ignored;
  - all other members of `StdVideoH264HrdParameters` are interpreted as defined in section E.2.2 of the ITU-T H.264 Specification;
- all other members of `StdVideoH264SequenceParameterSetVui` are interpreted as defined in section E.2.1 of the ITU-T H.264 Specification;

- all other members of `StdVideoH264SequenceParameterSet` are interpreted as defined in section 7.4.2.1 of the ITU-T H.264 Specification.

H.264 Picture Parameter Sets (PPS)

Represented by `StdVideoH264PictureParameterSet` structures and interpreted as follows:

- the pair constructed from `seq_parameter_set_id` and `pic_parameter_set_id` is used as the key of the PPS entry;
- if `flags.pic_scaling_matrix_present_flag` is set, then the `StdVideoH264ScalingLists` structure pointed to by `pScalingLists` is interpreted as follows:
  - `scaling_list_present_mask` is a bitmask where bit index i corresponds to `pic_scaling_list_present_flag[i]` as defined in section 7.4.2.2 of the ITU-T H.264 Specification;
  - `use_default_scaling_matrix_mask` is a bitmask where bit index i corresponds to `UseDefaultScalingMatrix4x4Flag[i]`, when i < 6, or corresponds to `UseDefaultScalingMatrix8x8Flag[i-6]`, otherwise, as defined in section 7.3.2.2 of the ITU-T H.264 Specification;
  - `ScalingList4x4` and `ScalingList8x8` correspond to the identically named syntax elements defined in section 7.3.2.2 of the ITU-T H.264 Specification;
- all other members of `StdVideoH264PictureParameterSet` are interpreted as defined in section 7.4.2.2 of the ITU-T H.264 Specification.

When a video session parameters object is created with the codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, the VkVideoSessionParametersCreateInfoKHR::`pNext` chain **must** include a `VkVideoDecodeH264SessionParametersCreateInfoKHR` structure specifying the capacity and initial contents of the object.

The VkVideoDecodeH264SessionParametersCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_h264
typedef struct VkVideoDecodeH264SessionParametersCreateInfoKHR {
    VkStructureType                                    sType;
    const void*                                        pNext;
    uint32_t                                           maxStdSPSCount;
    uint32_t                                           maxStdPPSCount;
    const VkVideoDecodeH264SessionParametersAddInfoKHR*    pParametersAddInfo;
} VkVideoDecodeH264SessionParametersCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- maxStdSPSCount is the maximum number of H.264 SPS entries the created VkVideoSessionParametersKHR **can** contain.
- maxStdPPSCount is the maximum number of H.264 PPS entries the created VkVideoSessionParametersKHR **can** contain.
- pParametersAddInfo is NULL or a pointer to a VkVideoDecodeH264SessionParametersAddInfoKHR structure specifying H.264 parameters to add upon object creation.

Valid Usage (Implicit)

- VUID-VkVideoDecodeH264SessionParametersCreateInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_SESSION_PARAMETERS_CREATE_INFO_KHR
- VUID-VkVideoDecodeH264SessionParametersCreateInfoKHR-pParametersAddInfo-parameter
  If pParametersAddInfo is not NULL, pParametersAddInfo **must** be a valid pointer to a valid VkVideoDecodeH264SessionParametersAddInfoKHR structure

Structure Chaining

Extends the structure

- VkVideoSessionParametersCreateInfoKHR

The VkVideoDecodeH264SessionParametersAddInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_h264
typedef struct VkVideoDecodeH264SessionParametersAddInfoKHR {
    VkStructureType                             sType;
    const void*                                 pNext;
    uint32_t                                    stdSPSCount;
    const StdVideoH264SequenceParameterSet*     pStdSPSs;
    uint32_t                                    stdPPSCount;
    const StdVideoH264PictureParameterSet*      pStdPPSs;
} VkVideoDecodeH264SessionParametersAddInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stdSPSCount` is the number of elements in the `pStdSPSs` array.
- `pStdSPSs` is a pointer to an array of `StdVideoH264SequenceParameterSet` structures describing the H.264 SPS entries to add.
- `stdPPSCount` is the number of elements in the `pStdPPSs` array.
- `pStdPPSs` is a pointer to an array of `StdVideoH264PictureParameterSet` structures describing the H.264 PPS entries to add.

This structure **can** be specified in the following places:

- In the `pParametersAddInfo` member of the VkVideoDecodeH264SessionParametersCreateInfoKHR structure specified in the `pNext` chain of VkVideoSessionParametersCreateInfoKHR used to create a video session parameters object. In this case, if the video codec operation the video session parameters object is created with is VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then it defines the set of initial parameters to add to the created object (see Creating Video Session Parameters).
- In the `pNext` chain of VkVideoSessionParametersUpdateInfoKHR. In this case, if the video codec operation the video session parameters object to be updated was created with is VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then it defines the set of parameters to add to it (see Updating Video Session Parameters).

Valid Usage

- VUID-VkVideoDecodeH264SessionParametersAddInfoKHR-None-04825
  The `seq_parameter_set_id` member of each `StdVideoH264SequenceParameterSet` structure specified in the elements of `pStdSPSs` **must** be unique within `pStdSPSs`
- VUID-VkVideoDecodeH264SessionParametersAddInfoKHR-None-04826
  The pair constructed from the `seq_parameter_set_id` and `pic_parameter_set_id` members of each `StdVideoH264PictureParameterSet` structure specified in the elements of `pStdPPSs` **must** be unique within `pStdPPSs`

Valid Usage (Implicit)

- VUID-VkVideoDecodeH264SessionParametersAddInfoKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_SESSION_PARAMETERS_ADD_INFO_KHR`
- VUID-VkVideoDecodeH264SessionParametersAddInfoKHR-pStdSPSs-parameter
  If `stdSPSCount` is not `0`, `pStdSPSs` **must** be a valid pointer to an array of `stdSPSCount` `StdVideoH264SequenceParameterSet` values
- VUID-VkVideoDecodeH264SessionParametersAddInfoKHR-pStdPPSs-parameter
  If `stdPPSCount` is not `0`, `pStdPPSs` **must** be a valid pointer to an array of `stdPPSCount` `StdVideoH264PictureParameterSet` values

Structure Chaining

Extends the structure

- VkVideoSessionParametersUpdateInfoKHR

Inline Parameter Sets

In case of video sessions created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR, the application **can** also specify the active parameter sets *inline* by including an instance of the VkVideoDecodeH264InlineSessionParametersInfoKHR structure in the `pNext` chain of VkVideoDecodeInfoKHR.

The `VkVideoDecodeH264InlineSessionParametersInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_h264 with VK_KHR_video_maintenance2
typedef struct VkVideoDecodeH264InlineSessionParametersInfoKHR {
    VkStructureType                             sType;
    const void*                                 pNext;
    const StdVideoH264SequenceParameterSet*     pStdSPS;
    const StdVideoH264PictureParameterSet*      pStdPPS;
} VkVideoDecodeH264InlineSessionParametersInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pStdSPS` is `NULL` or a pointer to an instance of the `StdVideoH264SequenceParameterSet` structure describing the active H.264 SPS.
- `pStdPPS` is `NULL` or a pointer to an instance of the `StdVideoH264PictureParameterSet` structure describing the active H.264 PPS.

If pStdSPS or pStdPPS is not NULL, the issued video decode operations will use the parameter sets specified by them, respectively, instead of the corresponding parameter sets being sourced from the bound video session parameters object.

Valid Usage (Implicit)

- VUID-VkVideoDecodeH264InlineSessionParametersInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_INLINE_SESSION_PARAMETERS_INFO_KHR
- VUID-VkVideoDecodeH264InlineSessionParametersInfoKHR-pStdSPS-parameter
  If pStdSPS is not NULL, pStdSPS **must** be a valid pointer to a valid StdVideoH264SequenceParameterSet value
- VUID-VkVideoDecodeH264InlineSessionParametersInfoKHR-pStdPPS-parameter
  If pStdPPS is not NULL, pStdPPS **must** be a valid pointer to a valid StdVideoH264PictureParameterSet value

Structure Chaining

Extends the structure

- VkVideoDecodeInfoKHR

45.12.6. H.264 Decoding Parameters

The VkVideoDecodeH264PictureInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_h264
typedef struct VkVideoDecodeH264PictureInfoKHR {
    VkStructureType                         sType;
    const void*                             pNext;
    const StdVideoDecodeH264PictureInfo*    pStdPictureInfo;
    uint32_t                                sliceCount;
    const uint32_t*                         pSliceOffsets;
} VkVideoDecodeH264PictureInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pStdPictureInfo is a pointer to a StdVideoDecodeH264PictureInfo structure specifying H.264 picture information.
- sliceCount is the number of elements in pSliceOffsets.
- pSliceOffsets is a pointer to an array of sliceCount offsets specifying the start offset of the slices of the picture within the video bitstream buffer range specified in VkVideoDecodeInfoKHR.

This structure is specified in the pNext chain of the VkVideoDecodeInfoKHR structure passed to

vkCmdDecodeVideoKHR to specify the codec-specific picture information for an H.264 decode operation.

Decode Output Picture Information

When this structure is specified in the pNext chain of the VkVideoDecodeInfoKHR structure passed to vkCmdDecodeVideoKHR, the information related to the decode output picture is defined as follows:

- If pStdPictureInfo->flags.field_pic_flag is not set, then the picture represents a frame.
- If pStdPictureInfo->flags.field_pic_flag is set, then the picture represents a field. Specifically:
  - If pStdPictureInfo->flags.bottom_field_flag is not set, then the picture represents the top field of the frame.
  - If pStdPictureInfo->flags.bottom_field_flag is set, then the picture represents the bottom field of the frame.
- The image subregion used is determined according to the H.264 Decode Picture Data Access section.
- The decode output picture is associated with the H.264 picture information provided in pStdPictureInfo.

Std Picture Information

The members of the StdVideoDecodeH264PictureInfo structure pointed to by pStdPictureInfo are interpreted as follows:

- reserved1 and reserved2 are used only for padding purposes and are otherwise ignored;
- flags.is_intra as defined in section 3.73 of the ITU-T H.264 Specification;
- flags.is_reference as defined in section 3.136 of the ITU-T H.264 Specification;
- flags.complementary_field_pair as defined in section 3.35 of the ITU-T H.264 Specification;
- seq_parameter_set_id and pic_parameter_set_id are used to identify the active parameter sets, as described below;
- all other members are interpreted as defined in section 7.4.3 of the ITU-T H.264 Specification.

Reference picture setup is controlled by the value of StdVideoDecodeH264PictureInfo::flags.is_reference. If it is set and a reconstructed picture is specified, then the latter is used as the target of picture reconstruction to activate the DPB slot specified in pDecodeInfo->pSetupReferenceSlot->slotIndex. If StdVideoDecodeH264PictureInfo::flags.is_reference is not set, but a reconstructed picture is specified, then the corresponding picture reference associated with the DPB slot is invalidated, as described in the DPB Slot States section.

Active Parameter Sets

The members of the StdVideoDecodeH264PictureInfo structure pointed to by pStdPictureInfo are used to select the active parameter sets to use from the specified inline parameter sets or from the bound video session parameters object, as follows:

- The *active SPS* is the SPS identified by the key specified in StdVideoDecodeH264PictureInfo

::seq_parameter_set_id.

- The *active PPS* is the PPS identified by the key specified by the pair constructed from StdVideoDecodeH264PictureInfo::seq_parameter_set_id and StdVideoDecodeH264PictureInfo::pic_parameter_set_id.

If the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR and any of the active parameter sets are specified inline, then the corresponding inline parameter set is used. Otherwise, the parameter set in question is sourced from the bound video session parameters object.

Valid Usage (Implicit)

- VUID-VkVideoDecodeH264PictureInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_PICTURE_INFO_KHR
- VUID-VkVideoDecodeH264PictureInfoKHR-pStdPictureInfo-parameter
  pStdPictureInfo **must** be a valid pointer to a valid StdVideoDecodeH264PictureInfo value
- VUID-VkVideoDecodeH264PictureInfoKHR-pSliceOffsets-parameter
  pSliceOffsets **must** be a valid pointer to an array of sliceCount uint32_t values
- VUID-VkVideoDecodeH264PictureInfoKHR-sliceCount-arraylength
  sliceCount **must** be greater than 0

Structure Chaining

Extends the structure

- VkVideoDecodeInfoKHR

The VkVideoDecodeH264DpbSlotInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_h264
typedef struct VkVideoDecodeH264DpbSlotInfoKHR {
    VkStructureType                          sType;
    const void*                              pNext;
    const StdVideoDecodeH264ReferenceInfo*   pStdReferenceInfo;
} VkVideoDecodeH264DpbSlotInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pStdReferenceInfo is a pointer to a StdVideoDecodeH264ReferenceInfo structure specifying H.264 reference information.

This structure is specified in the pNext chain of VkVideoDecodeInfoKHR::pSetupReferenceSlot, if not NULL, and the pNext chain of the elements of VkVideoDecodeInfoKHR::pReferenceSlots to specify the codec-specific reference picture information for an H.264 decode operation.

Active Reference Picture Information

When this structure is specified in the `pNext` chain of the elements of VkVideoDecodeInfoKHR::`pReferenceSlots`, one or two elements are added to the list of active reference pictures used by the video decode operation for each element of VkVideoDecodeInfoKHR::`pReferenceSlots` as follows:

- If neither `pStdReferenceInfo->flags.top_field_flag` nor `pStdReferenceInfo->flags.bottom_field_flag` is set, then the picture is added as a frame reference to the list of active reference pictures.
- If `pStdReferenceInfo->flags.top_field_flag` is set, then the picture is added as a top field reference to the list of active reference pictures.
- If `pStdReferenceInfo->flags.bottom_field_flag` is set, then the picture is added as a bottom field reference to the list of active reference pictures.
- For each added reference picture, the corresponding image subregion used is determined according to the H.264 Decode Picture Data Access section.
- Each added reference picture is associated with the DPB slot index specified in the `slotIndex` member of the corresponding element of VkVideoDecodeInfoKHR::`pReferenceSlots`.
- Each added reference picture is associated with the H.264 reference information provided in `pStdReferenceInfo`.

NOTE

When both the top and bottom field of an interlaced frame currently associated with a DPB slot is intended to be used as an active reference picture and both fields are stored in the same image subregion (which is the case when using VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_INTERLEAVED_LINES_BIT_KHR which stores the two fields at even and odd scanlines of the same image subregion), both references have to be provided through a single VkVideoReferenceSlotInfoKHR structure that has both `flags.top_field_flag` and `flags.bottom_field_flag` set in the `StdVideoDecodeH264ReferenceInfo` structure pointed to by the `pStdReferenceInfo` member of the VkVideoDecodeH264DpbSlotInfoKHR structure included in the corresponding VkVideoReferenceSlotInfoKHR structure's `pNext` chain. However, this approach can only be used when both fields are stored in the same image subregion. If that is not the case (e.g. when using VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR which requires separate `codedOffset` values for the two fields and also allows storing the two fields of a frame in separate image layers or entirely separate images), then a separate VkVideoReferenceSlotInfoKHR structure needs to be provided for referencing the two fields, each only setting one of `flags.top_field_flag` or `flags.bottom_field_flag`, and providing the appropriate video picture resource information in VkVideoReferenceSlotInfoKHR::`pPictureResource`.

Reconstructed Picture Information

When this structure is specified in the `pNext` chain of VkVideoDecodeInfoKHR::`pSetupReferenceSlot`, the information related to the reconstructed picture is defined as follows:

- If neither `pStdReferenceInfo->flags.top_field_flag` nor `pStdReferenceInfo->flags.bottom_field_flag` is set, then the picture represents a frame.
- If `pStdReferenceInfo->flags.top_field_flag` is set, then the picture represents a field, specifically, the top field of the frame.
- If `pStdReferenceInfo->flags.bottom_field_flag` is set, then the picture represents a field, specifically, the bottom field of the frame.
- The image subregion used is determined according to the H.264 Decode Picture Data Access section.
- If reference picture setup is requested, then the reconstructed picture is used to activate the DPB slot with the index specified in VkVideoDecodeInfoKHR::`pSetupReferenceSlot->slotIndex`.
- The reconstructed picture is associated with the H.264 reference information provided in `pStdReferenceInfo`.

Std Reference Information

The members of the `StdVideoDecodeH264ReferenceInfo` structure pointed to by `pStdReferenceInfo` are interpreted as follows:

- `flags.top_field_flag` is used to indicate whether the reference is used as top field reference;
- `flags.bottom_field_flag` is used to indicate whether the reference is used as bottom field reference;
- `flags.used_for_long_term_reference` is used to indicate whether the picture is marked as "used for long-term reference" as defined in section 8.2.5.1 of the ITU-T H.264 Specification;
- `flags.is_non_existing` is used to indicate whether the picture is marked as "non-existing" as defined in section 8.2.5.2 of the ITU-T H.264 Specification;
- all other members are interpreted as defined in section 8.2 of the ITU-T H.264 Specification.

Valid Usage (Implicit)

- VUID-VkVideoDecodeH264DpbSlotInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_DPB_SLOT_INFO_KHR
- VUID-VkVideoDecodeH264DpbSlotInfoKHR-pStdReferenceInfo-parameter
  `pStdReferenceInfo` **must** be a valid pointer to a valid `StdVideoDecodeH264ReferenceInfo` value

Structure Chaining

Extends the structure

- VkVideoReferenceSlotInfoKHR

45.12.7. H.264 Decode Requirements

This section describes the **required** H.264 decoding capabilities for physical devices that have at least one queue family that supports the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, as returned by vkGetPhysicalDeviceQueueFamilyProperties2 in VkQueueFamilyVideoPropertiesKHR ::videoCodecOperations.

Table 52. Required Video Std Header Versions

Video Std Header Name	Version
vulkan_video_codec_h264std_decode	1.0.0

Table 53. Required Video Capabilities

Video Capability	Requirement	Requirement Type[1]
VkVideoCapabilitiesKHR		
flags	-	min
minBitstreamBufferOffsetAlignment	4096	max
minBitstreamBufferSizeAlignment	4096	max
pictureAccessGranularity	(64,64)	max
minCodedExtent	-	max
maxCodedExtent	-	min
maxDpbSlots	0	min
maxActiveReferencePictures	0	min
VkVideoDecodeCapabilitiesKHR		
flags	VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR or VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR	min
VkVideoDecodeH264CapabilitiesKHR		
maxLevelIdc	STD_VIDEO_H264_LEVEL_IDC_1_0	min

Video Capability	Requirement	Requirement Type[1]
`fieldOffsetGranularity`	(0,0) except for profiles using VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR	implementation-dependent

1 The **Requirement Type** column specifies the requirement is either the minimum value all implementations **must** support, the maximum value all implementations **must** support, or the exact value all implementations **must** support. For bitmasks a minimum value is the least bits all implementations **must** set, but they **may** have additional bits set beyond this minimum.

45.13. H.265 Decode Operations

Video decode operations using an H.265 decode profile **can** be used to decode elementary video stream sequences compliant to the ITU-T H.265 Specification.

NOTE	Refer to the Preamble for information on how the Khronos Intellectual Property Rights Policy relates to normative references to external materials not created by Khronos.

This process is performed according to the video decode operation steps with the codec-specific semantics defined in section 8 of ITU-T H.265 Specification:

- Syntax elements, derived values, and other parameters are applied from the following structures:
  - The `StdVideoH265VideoParameterSet` structure corresponding to the active VPS specifying the H.265 video parameter set.
  - The `StdVideoH265SequenceParameterSet` structure corresponding to the active SPS specifying the H.265 sequence parameter set.
  - The `StdVideoH265PictureParameterSet` structure corresponding to the active PPS specifying the H.265 picture parameter set.
  - The `StdVideoDecodeH265PictureInfo` structure specifying the H.265 picture information.
  - The `StdVideoDecodeH265ReferenceInfo` structures specifying the H.265 reference information corresponding to the optional reconstructed picture and any active reference pictures.
- The contents of the provided video bitstream buffer range are interpreted as defined in the H.265 Decode Bitstream Data Access section.
- Picture data in the video picture resources corresponding to the used active reference pictures, decode output picture, and optional reconstructed picture is accessed as defined in the H.265 Decode Picture Data Access section.

- The decision on reference picture setup is made according to the parameters specified in the H.265 picture information.

If the parameters and the bitstream adhere to the syntactic and semantic requirements defined in the corresponding sections of the ITU-T H.265 Specification, as described above, and the DPB slots associated with the active reference pictures all refer to valid picture references, then the video decode operation will complete successfully. Otherwise, the video decode operation **may** complete unsuccessfully.

45.13.1. H.265 Decode Bitstream Data Access

The video bitstream buffer range **should** contain a VCL NAL unit comprised of the slice segment headers and data of a picture representing a frame, as defined in sections 7.3.6 and 7.3.8, and this data is interpreted as defined in sections 7.4.7 and 7.4.9 of the ITU-T H.265 Specification, respectively.

The offsets provided in VkVideoDecodeH265PictureInfoKHR::`pSliceSegmentOffsets` **should** specify the starting offsets corresponding to each slice segment header within the video bitstream buffer range.

45.13.2. H.265 Decode Picture Data Access

Accesses to image data within a video picture resource happen at the granularity indicated by VkVideoCapabilitiesKHR::`pictureAccessGranularity`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile. Accordingly, the complete image subregion of a decode output picture, reference picture, or reconstructed picture accessed by video coding operations using an H.265 decode profile is defined as the set of texels within the coordinate range:

([0,`endX`), [0,`endY`))

Where:

- `endX` equals `codedExtent.width` rounded up to the nearest integer multiple of `pictureAccessGranularity.width` and clamped to the width of the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure;
- endY equals `codedExtent.height` rounded up to the nearest integer multiple of `pictureAccessGranularity.height` and clamped to the height of the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure;

Where `codedExtent` is the member of the VkVideoPictureResourceInfoKHR structure corresponding to the picture.

In case of video decode operations using an H.265 decode profile, any access to a picture at the coordinates (`x`,`y`), as defined by the ITU-T H.265 Specification, is an access to the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure at the texel coordinates (`x`,`y`).

45.13.3. H.265 Decode Profile

A video profile supporting H.265 video decode operations is specified by setting VkVideoProfileInfoKHR::videoCodecOperation to VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR and adding a VkVideoDecodeH265ProfileInfoKHR structure to the VkVideoProfileInfoKHR::pNext chain.

The VkVideoDecodeH265ProfileInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_h265
typedef struct VkVideoDecodeH265ProfileInfoKHR {
    VkStructureType           sType;
    const void*               pNext;
    StdVideoH265ProfileIdc    stdProfileIdc;
} VkVideoDecodeH265ProfileInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- stdProfileIdc is a StdVideoH265ProfileIdc value specifying the H.265 codec profile IDC, as defined in section A.3 of the ITU-T H.265 Specification.

Valid Usage (Implicit)

- VUID-VkVideoDecodeH265ProfileInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_PROFILE_INFO_KHR

Structure Chaining

Extends the structures

- VkQueryPoolCreateInfo
- VkVideoProfileInfoKHR

45.13.4. H.265 Decode Capabilities

When calling vkGetPhysicalDeviceVideoCapabilitiesKHR to query the capabilities for an H.265 decode profile, the VkVideoCapabilitiesKHR::pNext chain **must** include a VkVideoDecodeH265CapabilitiesKHR structure that will be filled with the profile-specific capabilities.

The VkVideoDecodeH265CapabilitiesKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_h265
typedef struct VkVideoDecodeH265CapabilitiesKHR {
    VkStructureType         sType;
    void*                   pNext;
    StdVideoH265LevelIdc    maxLevelIdc;
} VkVideoDecodeH265CapabilitiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxLevelIdc` is a `StdVideoH265LevelIdc` value indicating the maximum H.265 level supported by the profile, where enum constant `STD_VIDEO_H265_LEVEL_IDC_<major>_<minor>` identifies H.265 level `<major>.<minor>` as defined in section A.4 of the ITU-T H.265 Specification.

Valid Usage (Implicit)

- VUID-VkVideoDecodeH265CapabilitiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkVideoCapabilitiesKHR

45.13.5. H.265 Decode Parameter Sets

Video session parameters objects created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR **can** contain the following types of parameters:

H.265 Video Parameter Sets (VPS)

Represented by `StdVideoH265VideoParameterSet` structures and interpreted as follows:

- `reserved1`, `reserved2`, and `reserved3` are used only for padding purposes and are otherwise ignored;
- `vps_video_parameter_set_id` is used as the key of the VPS entry;
- the `max_latency_increase_plus1`, `max_dec_pic_buffering_minus1`, and `max_num_reorder_pics` members of the `StdVideoH265DecPicBufMgr` structure pointed to by `pDecPicBufMgr` correspond to `vps_max_latency_increase_plus1`, `vps_max_dec_pic_buffering_minus1`, and `vps_max_num_reorder_pics`, respectively, as defined in section 7.4.3.1 of the ITU-T H.265 Specification;
- the `StdVideoH265HrdParameters` structure pointed to by `pHrdParameters` is interpreted as

follows:

  - `reserved` is used only for padding purposes and is otherwise ignored;
  - `flags.fixed_pic_rate_general_flag` is a bitmask where bit index i corresponds to `fixed_pic_rate_general_flag[i]` as defined in section E.3.2 of the ITU-T H.265 Specification;
  - `flags.fixed_pic_rate_within_cvs_flag` is a bitmask where bit index i corresponds to `fixed_pic_rate_within_cvs_flag[i]` as defined in section E.3.2 of the ITU-T H.265 Specification;
  - `flags.low_delay_hrd_flag` is a bitmask where bit index i corresponds to `low_delay_hrd_flag[i]` as defined in section E.3.2 of the ITU-T H.265 Specification;
  - if `flags.nal_hrd_parameters_present_flag` is set, then `pSubLayerHrdParametersNal` is a pointer to an array of `vps_max_sub_layers_minus1 + 1` number of `StdVideoH265SubLayerHrdParameters` structures where `vps_max_sub_layers_minus1` is the corresponding member of the encompassing `StdVideoH265VideoParameterSet` structure and each element is interpreted as follows:
    - `cbr_flag` is a bitmask where bit index i corresponds to `cbr_flag[i]` as defined in section E.3.3 of the ITU-T H.265 Specification;
    - all other members of the `StdVideoH265SubLayerHrdParameters` structure are interpreted as defined in section E.3.3 of the ITU-T H.265 Specification;
  - if `flags.vcl_hrd_parameters_present_flag` is set, then `pSubLayerHrdParametersVcl` is a pointer to an array of `vps_max_sub_layers_minus1 + 1` number of `StdVideoH265SubLayerHrdParameters` structures where `vps_max_sub_layers_minus1` is the corresponding member of the encompassing `StdVideoH265VideoParameterSet` structure and each element is interpreted as follows:
    - `cbr_flag` is a bitmask where bit index i corresponds to `cbr_flag[i]` as defined in section E.3.3 of the ITU-T H.265 Specification;
    - all other members of the `StdVideoH265SubLayerHrdParameters` structure are interpreted as defined in section E.3.3 of the ITU-T H.265 Specification;
  - all other members of `StdVideoH265HrdParameters` are interpreted as defined in section E.3.2 of the ITU-T H.265 Specification;
- the `StdVideoH265ProfileTierLevel` structure pointed to by `pProfileTierLevel` is interpreted as follows:
  - `general_level_idc` is one of the enum constants `STD_VIDEO_H265_LEVEL_IDC_<major>_<minor>` identifying the H.265 level `<major>.<minor>` as defined in section A.4 of the ITU-T H.265 Specification;
  - all other members of `StdVideoH265ProfileTierLevel` are interpreted as defined in section 7.4.4 of the ITU-T H.265 Specification;
- all other members of `StdVideoH265VideoParameterSet` are interpreted as defined in section 7.4.3.1 of the ITU-T H.265 Specification.

H.265 Sequence Parameter Sets (SPS)

Represented by `StdVideoH265SequenceParameterSet` structures and interpreted as follows:

- `reserved1` and `reserved2` are used only for padding purposes and are otherwise ignored;
- the pair constructed from `sps_video_parameter_set_id` and `sps_seq_parameter_set_id` is used as the key of the SPS entry;
- the `StdVideoH265ProfileTierLevel` structure pointed to by `pProfileTierLevel` is interpreted as follows:
  - `general_level_idc` is one of the enum constants `STD_VIDEO_H265_LEVEL_IDC_<major>_<minor>` identifying the H.265 level `<major>.<minor>` as defined in section A.4 of the ITU-T H.265 Specification;
  - all other members of `StdVideoH265ProfileTierLevel` are interpreted as defined in section 7.4.4 of the ITU-T H.265 Specification;
- the `max_latency_increase_plus1`, `max_dec_pic_buffering_minus1`, and `max_num_reorder_pics` members of the `StdVideoH265DecPicBufMgr` structure pointed to by `pDecPicBufMgr` correspond to `sps_max_latency_increase_plus1`, `sps_max_dec_pic_buffering_minus1`, and `sps_max_num_reorder_pics`, respectively, as defined in section 7.4.3.2 of the ITU-T H.265 Specification;
- if `flags.sps_scaling_list_data_present_flag` is set, then the `StdVideoH265ScalingLists` structure pointed to by `pScalingLists` is interpreted as follows:
  - `ScalingList4x4`, `ScalingList8x8`, `ScalingList16x16`, and `ScalingList32x32` correspond to `ScalingList[0]`, `ScalingList[1]`, `ScalingList[2]`, and `ScalingList[3]`, respectively, as defined in section 7.3.4 of the ITU-T H.265 Specification;
  - `ScalingListDCCoef16x16` and `ScalingListDCCoef32x32` correspond to `scaling_list_dc_coef_minus8[0]` and `scaling_list_dc_coef_minus8[1]`, respectively, as defined in section 7.3.4 of the ITU-T H.265 Specification;
- `pShortTermRefPicSet` is a pointer to an array of `num_short_term_ref_pic_sets` number of `StdVideoH265ShortTermRefPicSet` structures where each element is interpreted as follows:
  - `reserved1`, `reserved2`, and `reserved3` are used only for padding purposes and are otherwise ignored;
  - `used_by_curr_pic_flag` is a bitmask where bit index i corresponds to `used_by_curr_pic_flag[i]` as defined in section 7.4.8 of the ITU-T H.265 Specification;
  - `use_delta_flag` is a bitmask where bit index i corresponds to `use_delta_flag[i]` as defined in section 7.4.8 of the ITU-T H.265 Specification;
  - `used_by_curr_pic_s0_flag` is a bitmask where bit index i corresponds to `used_by_curr_pic_s0_flag[i]` as defined in section 7.4.8 of the ITU-T H.265 Specification;
  - `used_by_curr_pic_s1_flag` is a bitmask where bit index i corresponds to `used_by_curr_pic_s1_flag[i]` as defined in section 7.4.8 of the ITU-T H.265 Specification;
  - all other members of `StdVideoH265ShortTermRefPicSet` are interpreted as defined in section 7.4.8 of the ITU-T H.265 Specification;
- if `flags.long_term_ref_pics_present_flag` is set then the `StdVideoH265LongTermRefPicsSps` structure pointed to by `pLongTermRefPicsSps` is interpreted as follows:

    - used_by_curr_pic_lt_sps_flag is a bitmask where bit index i corresponds to used_by_curr_pic_lt_sps_flag[i] as defined in section 7.4.3.2 of the ITU-T H.265 Specification;
    - all other members of StdVideoH265LongTermRefPicsSps are interpreted as defined in section 7.4.3.2 of the ITU-T H.265 Specification;
- if flags.vui_parameters_present_flag is set, then the StdVideoH265SequenceParameterSetVui structure pointed to by pSequenceParameterSetVui is interpreted as follows:
    - reserved1, reserved2, and reserved3 are used only for padding purposes and are otherwise ignored;
    - the StdVideoH265HrdParameters structure pointed to by pHrdParameters is interpreted as follows:
        - flags.fixed_pic_rate_general_flag is a bitmask where bit index i corresponds to fixed_pic_rate_general_flag[i] as defined in section E.3.2 of the ITU-T H.265 Specification;
        - flags.fixed_pic_rate_within_cvs_flag is a bitmask where bit index i corresponds to fixed_pic_rate_within_cvs_flag[i] as defined in section E.3.2 of the ITU-T H.265 Specification;
        - flags.low_delay_hrd_flag is a bitmask where bit index i corresponds to low_delay_hrd_flag[i] as defined in section E.3.2 of the ITU-T H.265 Specification;
        - if flags.nal_hrd_parameters_present_flag is set, then pSubLayerHrdParametersNal is a pointer to an array of sps_max_sub_layers_minus1 + 1 number of StdVideoH265SubLayerHrdParameters structures where sps_max_sub_layers_minus1 is the corresponding member of the encompassing StdVideoH265SequenceParameterSet structure and each element is interpreted as follows:
            - cbr_flag is a bitmask where bit index i corresponds to cbr_flag[i] as defined in section E.3.3 of the ITU-T H.265 Specification;
            - all other members of the StdVideoH265SubLayerHrdParameters structure are interpreted as defined in section E.3.3 of the ITU-T H.265 Specification;
        - if flags.vcl_hrd_parameters_present_flag is set, then pSubLayerHrdParametersVcl is a pointer to an array of sps_max_sub_layers_minus1 + 1 number of StdVideoH265SubLayerHrdParameters structures where sps_max_sub_layers_minus1 is the corresponding member of the encompassing StdVideoH265SequenceParameterSet structure and each element is interpreted as follows:
            - cbr_flag is a bitmask where bit index i corresponds to cbr_flag[i] as defined in section E.3.3 of the ITU-T H.265 Specification;
            - all other members of the StdVideoH265SubLayerHrdParameters structure are interpreted as defined in section E.3.3 of the ITU-T H.265 Specification;
        - all other members of StdVideoH265HrdParameters are interpreted as defined in section E.3.2 of the ITU-T H.265 Specification;
    - all other members of pSequenceParameterSetVui are interpreted as defined in section E.3.1 of the ITU-T H.265 Specification;

- if flags.sps_palette_predictor_initializer_present_flag is set, then the PredictorPaletteEntries member of the StdVideoH265PredictorPaletteEntries structure pointed to by pPredictorPaletteEntries is interpreted as defined in section 7.4.9.13 of the ITU-T H.265 Specification;
- all other members of StdVideoH265SequenceParameterSet are interpreted as defined in section 7.4.3.1 of the ITU-T H.265 Specification.

H.265 Picture Parameter Sets (PPS)

Represented by StdVideoH265PictureParameterSet structures and interpreted as follows:

- reserved1, reserved2, and reserved3 are used only for padding purposes and are otherwise ignored;
- the triplet constructed from sps_video_parameter_set_id, pps_seq_parameter_set_id, and pps_pic_parameter_set_id is used as the key of the PPS entry;
- if flags.pps_scaling_list_data_present_flag is set, then the StdVideoH265ScalingLists structure pointed to by pScalingLists is interpreted as follows:
  - ScalingList4x4, ScalingList8x8, ScalingList16x16, and ScalingList32x32 correspond to ScalingList[0], ScalingList[1], ScalingList[2], and ScalingList[3], respectively, as defined in section 7.3.4 of the ITU-T H.265 Specification;
  - ScalingListDCCoef16x16 and ScalingListDCCoef32x32 correspond to scaling_list_dc_coef_minus8[0] and scaling_list_dc_coef_minus8[1], respectively, as defined in section 7.3.4 of the ITU-T H.265 Specification;
- if flags.pps_palette_predictor_initializer_present_flag is set, then the PredictorPaletteEntries member of the StdVideoH265PredictorPaletteEntries structure pointed to by pPredictorPaletteEntries is interpreted as defined in section 7.4.9.13 of the ITU-T H.265 Specification;
- all other members of StdVideoH265PictureParameterSet are interpreted as defined in section 7.4.3.3 of the ITU-T H.265 Specification.

When a video session parameters object is created with the codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, the VkVideoSessionParametersCreateInfoKHR::pNext chain **must** include a VkVideoDecodeH265SessionParametersCreateInfoKHR structure specifying the capacity and initial contents of the object.

The VkVideoDecodeH265SessionParametersCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_h265
typedef struct VkVideoDecodeH265SessionParametersCreateInfoKHR {
    VkStructureType                                    sType;
    const void*                                        pNext;
    uint32_t                                           maxStdVPSCount;
    uint32_t                                           maxStdSPSCount;
    uint32_t                                           maxStdPPSCount;
    const VkVideoDecodeH265SessionParametersAddInfoKHR*    pParametersAddInfo;
} VkVideoDecodeH265SessionParametersCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxStdVPSCount` is the maximum number of H.265 VPS entries the created `VkVideoSessionParametersKHR` **can** contain.
- `maxStdSPSCount` is the maximum number of H.265 SPS entries the created `VkVideoSessionParametersKHR` **can** contain.
- `maxStdPPSCount` is the maximum number of H.265 PPS entries the created `VkVideoSessionParametersKHR` **can** contain.
- `pParametersAddInfo` is `NULL` or a pointer to a VkVideoDecodeH265SessionParametersAddInfoKHR structure specifying H.265 parameters to add upon object creation.

Valid Usage (Implicit)

- VUID-VkVideoDecodeH265SessionParametersCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_SESSION_PARAMETERS_CREATE_INFO_KHR
- VUID-VkVideoDecodeH265SessionParametersCreateInfoKHR-pParametersAddInfo-parameter
  If `pParametersAddInfo` is not `NULL`, `pParametersAddInfo` **must** be a valid pointer to a valid VkVideoDecodeH265SessionParametersAddInfoKHR structure

Structure Chaining

Extends the structure

- VkVideoSessionParametersCreateInfoKHR

The `VkVideoDecodeH265SessionParametersAddInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_h265
typedef struct VkVideoDecodeH265SessionParametersAddInfoKHR {
    VkStructureType                              sType;
    const void*                                  pNext;
    uint32_t                                     stdVPSCount;
    const StdVideoH265VideoParameterSet*         pStdVPSs;
    uint32_t                                     stdSPSCount;
    const StdVideoH265SequenceParameterSet*      pStdSPSs;
    uint32_t                                     stdPPSCount;
    const StdVideoH265PictureParameterSet*       pStdPPSs;
} VkVideoDecodeH265SessionParametersAddInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stdVPSCount` is the number of elements in the `pStdVPSs` array.
- `pStdVPSs` is a pointer to an array of `StdVideoH265VideoParameterSet` structures describing the H.265 VPS entries to add.
- `stdSPSCount` is the number of elements in the `pStdSPSs` array.
- `pStdSPSs` is a pointer to an array of `StdVideoH265SequenceParameterSet` structures describing the H.265 SPS entries to add.
- `stdPPSCount` is the number of elements in the `pStdPPSs` array.
- `pStdPPSs` is a pointer to an array of `StdVideoH265PictureParameterSet` structures describing the H.265 PPS entries to add.

This structure **can** be specified in the following places:

- In the `pParametersAddInfo` member of the VkVideoDecodeH265SessionParametersCreateInfoKHR structure specified in the `pNext` chain of VkVideoSessionParametersCreateInfoKHR used to create a video session parameters object. In this case, if the video codec operation the video session parameters object is created with is VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then it defines the set of initial parameters to add to the created object (see Creating Video Session Parameters).
- In the `pNext` chain of VkVideoSessionParametersUpdateInfoKHR. In this case, if the video codec operation the video session parameters object to be updated was created with is VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then it defines the set of parameters to add to it (see Updating Video Session Parameters).

Valid Usage

- VUID-VkVideoDecodeH265SessionParametersAddInfoKHR-None-04833
  The `vps_video_parameter_set_id` member of each `StdVideoH265VideoParameterSet` structure specified in the elements of `pStdVPSs` **must** be unique within `pStdVPSs`
- VUID-VkVideoDecodeH265SessionParametersAddInfoKHR-None-04834
  The pair constructed from the `sps_video_parameter_set_id` and `sps_seq_parameter_set_id`

members of each `StdVideoH265SequenceParameterSet` structure specified in the elements of `pStdSPSs` **must** be unique within `pStdSPSs`

- VUID-VkVideoDecodeH265SessionParametersAddInfoKHR-None-04835
  The triplet constructed from the `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id` members of each `StdVideoH265PictureParameterSet` structure specified in the elements of `pStdPPSs` **must** be unique within `pStdPPSs`

Valid Usage (Implicit)

- VUID-VkVideoDecodeH265SessionParametersAddInfoKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_SESSION_PARAMETERS_ADD_INFO_KHR`
- VUID-VkVideoDecodeH265SessionParametersAddInfoKHR-pStdVPSs-parameter
  If `stdVPSCount` is not `0`, `pStdVPSs` **must** be a valid pointer to an array of `stdVPSCount` `StdVideoH265VideoParameterSet` values
- VUID-VkVideoDecodeH265SessionParametersAddInfoKHR-pStdSPSs-parameter
  If `stdSPSCount` is not `0`, `pStdSPSs` **must** be a valid pointer to an array of `stdSPSCount` `StdVideoH265SequenceParameterSet` values
- VUID-VkVideoDecodeH265SessionParametersAddInfoKHR-pStdPPSs-parameter
  If `stdPPSCount` is not `0`, `pStdPPSs` **must** be a valid pointer to an array of `stdPPSCount` `StdVideoH265PictureParameterSet` values

Structure Chaining

Extends the structure

- VkVideoSessionParametersUpdateInfoKHR

Inline Parameter Sets

In case of video sessions created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR` and with `VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR`, the application **can** also specify the active parameter sets *inline* by including an instance of the `VkVideoDecodeH265InlineSessionParametersInfoKHR` structure in the `pNext` chain of `VkVideoDecodeInfoKHR`.

The `VkVideoDecodeH265InlineSessionParametersInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_h265 with VK_KHR_video_maintenance2
typedef struct VkVideoDecodeH265InlineSessionParametersInfoKHR {
    VkStructureType                             sType;
    const void*                                 pNext;
    const StdVideoH265VideoParameterSet*        pStdVPS;
    const StdVideoH265SequenceParameterSet*     pStdSPS;
    const StdVideoH265PictureParameterSet*      pStdPPS;
} VkVideoDecodeH265InlineSessionParametersInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pStdVPS` is `NULL` or a pointer to an instance of the `StdVideoH265VideoParameterSet` structure describing the active H.265 VPS.
- `pStdSPS` is `NULL` or a pointer to an instance of the `StdVideoH265SequenceParameterSet` structure describing the active H.265 SPS.
- `pStdPPS` is `NULL` or a pointer to an instance of the `StdVideoH265PictureParameterSet` structure describing the active H.265 PPS.

If `pStdVPS`, `pStdSPS`, or `pStdPPS` is not `NULL`, the issued video decode operations will use the parameter sets specified by them, respectively, instead of the corresponding active parameter sets being sourced from the bound video session parameters object.

Valid Usage (Implicit)

- VUID-VkVideoDecodeH265InlineSessionParametersInfoKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_INLINE_SESSION_PARAMETERS_INFO_KHR`
- VUID-VkVideoDecodeH265InlineSessionParametersInfoKHR-pStdVPS-parameter
  If `pStdVPS` is not `NULL`, `pStdVPS` **must** be a valid pointer to a valid `StdVideoH265VideoParameterSet` value
- VUID-VkVideoDecodeH265InlineSessionParametersInfoKHR-pStdSPS-parameter
  If `pStdSPS` is not `NULL`, `pStdSPS` **must** be a valid pointer to a valid `StdVideoH265SequenceParameterSet` value
- VUID-VkVideoDecodeH265InlineSessionParametersInfoKHR-pStdPPS-parameter
  If `pStdPPS` is not `NULL`, `pStdPPS` **must** be a valid pointer to a valid `StdVideoH265PictureParameterSet` value

Structure Chaining

Extends the structure

- VkVideoDecodeInfoKHR

45.13.6. H.265 Decoding Parameters

The VkVideoDecodeH265PictureInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_h265
typedef struct VkVideoDecodeH265PictureInfoKHR {
    VkStructureType                          sType;
    const void*                              pNext;
    const StdVideoDecodeH265PictureInfo*     pStdPictureInfo;
    uint32_t                                 sliceSegmentCount;
    const uint32_t*                          pSliceSegmentOffsets;
} VkVideoDecodeH265PictureInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pStdPictureInfo is a pointer to a StdVideoDecodeH265PictureInfo structure specifying H.265 picture information.
- sliceSegmentCount is the number of elements in pSliceSegmentOffsets.
- pSliceSegmentOffsets is a pointer to an array of sliceSegmentCount offsets specifying the start offset of the slice segments of the picture within the video bitstream buffer range specified in VkVideoDecodeInfoKHR.

This structure is specified in the pNext chain of the VkVideoDecodeInfoKHR structure passed to vkCmdDecodeVideoKHR to specify the codec-specific picture information for an H.265 decode operation.

Decode Output Picture Information

When this structure is specified in the pNext chain of the VkVideoDecodeInfoKHR structure passed to vkCmdDecodeVideoKHR, the information related to the decode output picture is defined as follows:

- The image subregion used is determined according to the H.265 Decode Picture Data Access section.
- The decode output picture is associated with the H.265 picture information provided in pStdPictureInfo.

Std Picture Information

The members of the StdVideoDecodeH265PictureInfo structure pointed to by pStdPictureInfo are interpreted as follows:

- reserved is used only for padding purposes and is otherwise ignored;
- flags.IrapPicFlag as defined in section 3.73 of the ITU-T H.265 Specification;
- flags.IdrPicFlag as defined in section 3.67 of the ITU-T H.265 Specification;
- flags.IsReference as defined in section 3.132 of the ITU-T H.265 Specification;
- sps_video_parameter_set_id, pps_seq_parameter_set_id, and pps_pic_parameter_set_id are

used to identify the active parameter sets, as described below;

- `PicOrderCntVal` as defined in section 8.3.1 of the ITU-T H.265 Specification;
- `NumBitsForSTRefPicSetInSlice` is the number of bits used in `st_ref_pic_set` when `short_term_ref_pic_set_sps_flag` is `0`, or `0` otherwise, as defined in sections 7.4.7 and 7.4.8 of the ITU-T H.265 Specification;
- `NumDeltaPocsOfRefRpsIdx` is the value of `NumDeltaPocs[RefRpsIdx]` when `short_term_ref_pic_set_sps_flag` is `1`, or `0` otherwise, as defined in sections 7.4.7 and 7.4.8 of the ITU-T H.265 Specification;
- `RefPicSetStCurrBefore`, `RefPicSetStCurrAfter`, and `RefPicSetLtCurr` are interpreted as defined in section 8.3.2 of the ITU-T H.265 Specification where each element of these arrays either identifies an active reference picture using its DPB slot index or contains the value `STD_VIDEO_H265_NO_REFERENCE_PICTURE` to indicate “no reference picture”;
- all other members are interpreted as defined in section 8.3.2 of the ITU-T H.265 Specification.

Reference picture setup is controlled by the value of `StdVideoDecodeH265PictureInfo::flags.IsReference`. If it is set and a reconstructed picture is specified, then the latter is used as the target of picture reconstruction to activate the corresponding DPB slot. If `StdVideoDecodeH265PictureInfo::flags.IsReference` is not set, but a reconstructed picture is specified, then the corresponding picture reference associated with the DPB slot is invalidated, as described in the DPB Slot States section.

Active Parameter Sets

The members of the `StdVideoDecodeH265PictureInfo` structure pointed to by `pStdPictureInfo` are used to select the active parameter sets to use from the specified inline parameter sets or from the bound video session parameters object, as follows:

- The *active VPS* is the VPS identified by the key specified in `StdVideoDecodeH265PictureInfo::sps_video_parameter_set_id`.
- The *active SPS* is the SPS identified by the key specified by the pair constructed from `StdVideoDecodeH265PictureInfo::sps_video_parameter_set_id` and `StdVideoDecodeH265PictureInfo::pps_seq_parameter_set_id`.
- The *active PPS* is the PPS identified by the key specified by the triplet constructed from `StdVideoDecodeH265PictureInfo::sps_video_parameter_set_id`, `StdVideoDecodeH265PictureInfo::pps_seq_parameter_set_id`, and `StdVideoDecodeH265PictureInfo::pps_pic_parameter_set_id`.

If the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR and any of the active parameter sets are specified inline, then the corresponding inline parameter set is used. Otherwise, the parameter set in question is sourced from the bound video session parameters object.

Valid Usage (Implicit)

- VUID-VkVideoDecodeH265PictureInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_PICTURE_INFO_KHR
- VUID-VkVideoDecodeH265PictureInfoKHR-pStdPictureInfo-parameter

pStdPictureInfo **must** be a valid pointer to a valid StdVideoDecodeH265PictureInfo value

- VUID-VkVideoDecodeH265PictureInfoKHR-pSliceSegmentOffsets-parameter pSliceSegmentOffsets **must** be a valid pointer to an array of sliceSegmentCount uint32_t values
- VUID-VkVideoDecodeH265PictureInfoKHR-sliceSegmentCount-arraylength sliceSegmentCount **must** be greater than 0

Structure Chaining

Extends the structure

- VkVideoDecodeInfoKHR

The VkVideoDecodeH265DpbSlotInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_h265
typedef struct VkVideoDecodeH265DpbSlotInfoKHR {
    VkStructureType                          sType;
    const void*                              pNext;
    const StdVideoDecodeH265ReferenceInfo*   pStdReferenceInfo;
} VkVideoDecodeH265DpbSlotInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pStdReferenceInfo is a pointer to a StdVideoDecodeH265ReferenceInfo structure specifying reference picture information described in section 8.3 of the ITU-T H.265 Specification.

This structure is specified in the pNext chain of VkVideoDecodeInfoKHR::pSetupReferenceSlot, if not NULL, and the pNext chain of the elements of VkVideoDecodeInfoKHR::pReferenceSlots to specify the codec-specific reference picture information for an H.265 decode operation.

Active Reference Picture Information

When this structure is specified in the pNext chain of the elements of VkVideoDecodeInfoKHR ::pReferenceSlots, one element is added to the list of active reference pictures used by the video decode operation for each element of VkVideoDecodeInfoKHR::pReferenceSlots as follows:

- The image subregion used is determined according to the H.265 Decode Picture Data Access section.
- The reference picture is associated with the DPB slot index specified in the slotIndex member of the corresponding element of VkVideoDecodeInfoKHR::pReferenceSlots.
- The reference picture is associated with the H.265 reference information provided in pStdReferenceInfo.

Reconstructed Picture Information

When this structure is specified in the `pNext` chain of VkVideoDecodeInfoKHR ::`pSetupReferenceSlot`, the information related to the reconstructed picture is defined as follows:

- The image subregion used is determined according to the H.265 Decode Picture Data Access section.
- If reference picture setup is requested, then the reconstructed picture is used to activate the DPB slot with the index specified in VkVideoDecodeInfoKHR::`pSetupReferenceSlot->slotIndex`.
- The reconstructed picture is associated with the H.265 reference information provided in `pStdReferenceInfo`.

Std Reference Information

The members of the `StdVideoDecodeH265ReferenceInfo` structure pointed to by `pStdReferenceInfo` are interpreted as follows:

- `flags.used_for_long_term_reference` is used to indicate whether the picture is marked as "used for long-term reference" as defined in section 8.3.2 of the ITU-T H.265 Specification;
- `flags.unused_for_reference` is used to indicate whether the picture is marked as "unused for reference" as defined in section 8.3.2 of the ITU-T H.265 Specification;
- all other members are interpreted as defined in section 8.3 of the ITU-T H.265 Specification.

Valid Usage (Implicit)

- VUID-VkVideoDecodeH265DpbSlotInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_DPB_SLOT_INFO_KHR
- VUID-VkVideoDecodeH265DpbSlotInfoKHR-pStdReferenceInfo-parameter
  `pStdReferenceInfo` **must** be a valid pointer to a valid `StdVideoDecodeH265ReferenceInfo` value

Structure Chaining

Extends the structure

- VkVideoReferenceSlotInfoKHR

45.13.7. H.265 Decode Requirements

This section describes the **required** H.265 decoding capabilities for physical devices that have at least one queue family that supports the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, as returned by vkGetPhysicalDeviceQueueFamilyProperties2 in VkQueueFamilyVideoPropertiesKHR ::`videoCodecOperations`.

Table 54. Required Video Std Header Versions

Video Std Header Name	Version
vulkan_video_codec_h265std_decode	1.0.0

Table 55. Required Video Capabilities

Video Capability	Requirement	Requirement Type[1]
VkVideoCapabilitiesKHR		
flags	-	min
minBitstreamBufferOffsetAlignment	4096	max
minBitstreamBufferSizeAlignment	4096	max
pictureAccessGranularity	(64,64)	max
minCodedExtent	-	max
maxCodedExtent	-	min
maxDpbSlots	0	min
maxActiveReferencePictures	0	min
VkVideoDecodeCapabilitiesKHR		
flags	VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR or VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR	min
VkVideoDecodeH265CapabilitiesKHR		
maxLevelIdc	STD_VIDEO_H265_LEVEL_IDC_1_0	min

1

The **Requirement Type** column specifies the requirement is either the minimum value all implementations **must** support, the maximum value all implementations **must** support, or the exact value all implementations **must** support. For bitmasks a minimum value is the least bits all implementations **must** set, but they **may** have additional bits set beyond this minimum.

45.14. VP9 Decode Operations

Video decode operations using a VP9 decode profile **can** be used to decode elementary video stream sequences compliant with the VP9 Specification.

NOTE Refer to the Preamble for information on how the Khronos Intellectual Property Rights Policy relates to normative references to external materials not created by

> Khronos.

This process is performed according to the video decode operation steps with the codec-specific semantics defined in section 8 of the VP9 Specification:

- Syntax elements, derived values, and other parameters are applied from the `StdVideoDecodeVP9PictureInfo` structure specifying the VP9 picture information.
- The contents of the provided video bitstream buffer range are interpreted as defined in the VP9 Decode Bitstream Data Access section.
- Picture data in the video picture resources corresponding to the used active reference pictures, decode output picture, and optional reconstructed picture is accessed as defined in the VP9 Decode Picture Data Access section.
- The decision on reference picture setup is made according to the parameters specified in the VP9 picture information.

If the parameters and the bitstream adhere to the syntactic and semantic requirements defined in the corresponding sections of the VP9 Specification, as described above, and the DPB slots associated with the active reference pictures all refer to valid picture references, then the video decode operation will complete successfully. Otherwise, the video decode operation **may** complete unsuccessfully.

45.14.1. VP9 Decode Bitstream Data Access

The video bitstream buffer range **should** contain the bitstream that represents an entire frame, as defined in section 6.1, and this data is interpreted as defined in section 7.1 of the VP9 Specification, respectively.

The offset specified in VkVideoDecodeVP9PictureInfoKHR::`uncompressedHeaderOffset` **should** specify the starting offset of the uncompressed header of the frame, as defined in section 6.2 of the VP9 Specification.

The offset specified in VkVideoDecodeVP9PictureInfoKHR::`compressedHeaderOffset` **should** specify the starting offset of the compressed header of the frame, as defined in section 6.3 of the VP9 Specification.

The offset specified in VkVideoDecodeVP9PictureInfoKHR::`tilesOffset` **should** specify the starting offset of the frame tile data, as defined in section 6.4 of the VP9 Specification.

45.14.2. VP9 Decode Picture Data Access

Accesses to image data within a video picture resource happen at the granularity indicated by VkVideoCapabilitiesKHR::`pictureAccessGranularity`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile. Accordingly, the complete image subregion of a decode output picture, reference picture, or reconstructed picture accessed by video coding operations using a VP9 decode profile is defined as the set of texels within the coordinate range:

([0,endX), [0,endY))

Where:

- endX equals codedExtent.width rounded up to the nearest integer multiple of pictureAccessGranularity.width and clamped to the width of the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure;
- endY equals codedExtent.height rounded up to the nearest integer multiple of pictureAccessGranularity.height and clamped to the height of the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure;

Where codedExtent is the member of the VkVideoPictureResourceInfoKHR structure corresponding to the picture.

In case of video decode operations using a VP9 decode profile, any access to a picture at the coordinates (x,y), as defined by the VP9 Specification, is an access to the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure at the texel coordinates (x,y).

45.14.3. VP9 Reference Names and Semantics

Individual reference frames used in the decoding process have different semantics, as defined in section 7.4.12 of the VP9 Specification. The VP9 semantics associated with a reference picture are indicated by the corresponding enumeration constant defined in the Video Std enumeration type StdVideoVP9ReferenceName:

- STD_VIDEO_VP9_REFERENCE_NAME_INTRA_FRAME identifies the reference used for intra coding (INTRA_FRAME), as defined in sections 2 and 8.5.1 of the VP9 Specification.
- All other enumeration constants refer to backward or forward references used for inter coding, as defined in sections 2 and 8.5.2 of the VP9 Specification:
  - STD_VIDEO_VP9_REFERENCE_NAME_LAST_FRAME identifies the LAST_FRAME reference
  - STD_VIDEO_VP9_REFERENCE_NAME_GOLDEN_FRAME identifies the GOLDEN_FRAME reference
  - STD_VIDEO_VP9_REFERENCE_NAME_ALTREF_FRAME identifies the ALTREF_FRAME reference

These enumeration constants are not directly used in any APIs but are used to indirectly index into certain Video Std and Vulkan API parameter arrays.

45.14.4. VP9 Decode Profile

A video profile supporting VP9 video decode operations is specified by setting VkVideoProfileInfoKHR::videoCodecOperation to VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR and adding a VkVideoDecodeVP9ProfileInfoKHR structure to the VkVideoProfileInfoKHR::pNext chain.

The VkVideoDecodeVP9ProfileInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_vp9
typedef struct VkVideoDecodeVP9ProfileInfoKHR {
    VkStructureType       sType;
    const void*           pNext;
    StdVideoVP9Profile    stdProfile;
} VkVideoDecodeVP9ProfileInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stdProfile` is a `StdVideoVP9Profile` value specifying the VP9 codec profile, as defined in section 7.2 of the VP9 Specification.

Valid Usage (Implicit)

- VUID-VkVideoDecodeVP9ProfileInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_VP9_PROFILE_INFO_KHR

Structure Chaining

Extends the structures

- VkQueryPoolCreateInfo
- VkVideoProfileInfoKHR

45.14.5. VP9 Decode Capabilities

When calling vkGetPhysicalDeviceVideoCapabilitiesKHR to query the capabilities for a VP9 decode profile, the VkVideoCapabilitiesKHR::`pNext` chain **must** include a `VkVideoDecodeVP9CapabilitiesKHR` structure that will be filled with the profile-specific capabilities.

The `VkVideoDecodeVP9CapabilitiesKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_vp9
typedef struct VkVideoDecodeVP9CapabilitiesKHR {
    VkStructureType     sType;
    void*               pNext;
    StdVideoVP9Level    maxLevel;
} VkVideoDecodeVP9CapabilitiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxLevel` is a `StdVideoVP9Level` value specifying the maximum VP9 level supported by the profile, as defined in section A.1 of the VP9 Specification.

Valid Usage (Implicit)

- VUID-VkVideoDecodeVP9CapabilitiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_VP9_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkVideoCapabilitiesKHR

45.14.6. VP9 Decoding Parameters

The `VkVideoDecodeVP9PictureInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_vp9
typedef struct VkVideoDecodeVP9PictureInfoKHR {
    VkStructureType                         sType;
    const void*                             pNext;
    const StdVideoDecodeVP9PictureInfo*     pStdPictureInfo;
    int32_t                                 referenceNameSlotIndices
[VK_MAX_VIDEO_VP9_REFERENCES_PER_FRAME_KHR];
    uint32_t                                uncompressedHeaderOffset;
    uint32_t                                compressedHeaderOffset;
    uint32_t                                tilesOffset;
} VkVideoDecodeVP9PictureInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pStdPictureInfo` is a pointer to a `StdVideoDecodeVP9PictureInfo` structure specifying VP9 picture information.
- `referenceNameSlotIndices` is an array of three (VK_MAX_VIDEO_VP9_REFERENCES_PER_FRAME_KHR, which is equal to the Video Std definition `STD_VIDEO_VP9_REFS_PER_FRAME`) signed integer values specifying the index of the DPB slot or a negative integer value for each VP9 reference name used for inter coding. In particular, the DPB slot index for the VP9 reference name `frame` is specified in `referenceNameSlotIndices` [`frame` - `STD_VIDEO_VP9_REFERENCE_NAME_LAST_FRAME`].
- `uncompressedHeaderOffset` is the byte offset of the uncompressed frame header, as defined in section 6.2 of the VP9 Specification.
- `compressedHeaderOffset` is the byte offset of the compressed frame header, as defined in section 6.3 of the VP9 Specification.
- `tilesOffset` is the byte offset of the frame tile data, as defined in section 6.4 of the VP9 Specification.

This structure is specified in the pNext chain of the VkVideoDecodeInfoKHR structure passed to vkCmdDecodeVideoKHR to specify the codec-specific picture information for a VP9 decode operation.

Decode Output Picture Information

When this structure is specified in the pNext chain of the VkVideoDecodeInfoKHR structure passed to vkCmdDecodeVideoKHR, the information related to the decode output picture is defined as follows:

- The image subregion used is determined according to the VP9 Decode Picture Data Access section.
- The decode output picture is associated with the VP9 picture information provided in pStdPictureInfo.

Std Picture Information

The members of the StdVideoDecodeVP9PictureInfo structure pointed to by pStdPictureInfo are interpreted as follows:

- flags.reserved and reserved1 are used only for padding purposes and are otherwise ignored;
- ref_frame_sign_bias_mask is a bitmask where bit index i corresponds to ref_frame_sign_bias[i] as defined in section 7.2 of the VP9 Specification;
- the StdVideoVP9ColorConfig structure pointed to by pColorConfig is interpreted as follows:
  - flags.reserved and reserved1 are used only for padding purposes and are otherwise ignored;
  - all other members of StdVideoVP9ColorConfig are interpreted as defined in sections 6.2, 6.2.2, and 7.2.2 of the VP9 Specification;
- the StdVideoVP9LoopFilter structure pointed to by pLoopFilter is interpreted as follows:
  - flags.reserved is used only for padding purposes and is otherwise ignored;
  - update_ref_delta is a bitmask where bit index i is interpreted as the value of update_ref_delta corresponding to element i of loop_filter_ref_deltas as defined in section 7.2.8 of the VP9 Specification;
  - update_mode_delta is a bitmask where bit index i is interpreted as the value of update_mode_delta corresponding to element i of loop_filter_mode_deltas as defined in section 7.2.8 of the VP9 Specification;
  - all other members of StdVideoVP9LoopFilter are interpreted as defined in section 7.2.8 of the VP9 Specification;

> **NOTE** If the syntax elements corresponding to loop_filter_ref_deltas and loop_filter_mode_deltas are not present, the application should specify the previous values, as defined in section 7.2.8 of the VP9 Specification.

- if flags.segmentation_enabled is set, then the StdVideoVP9Segmentation structure pointed to by pSegmentation is interpreted as follows:
  - flags.reserved is used only for padding purposes and is otherwise ignored;

    - the elements of `FeatureEnabled` are bitmasks where bit index j of element i corresponds to `FeatureEnabled[i][j]` as defined in section 6.2.11 of the VP9 Specification;
    - `FeatureData` is interpreted as defined in section 6.2.11 of the VP9 Specification;
    - all other members of `StdVideoVP9Segmentation` are interpreted as defined in section 7.2.10 of the VP9 Specification;
- all other members are interpreted as defined in section 7.2 of the VP9 Specification.

Reference picture setup is controlled by the value of `StdVideoDecodeVP9PictureInfo::refresh_frame_flags`. If it is not zero and a reconstructed picture is specified, then the latter is used as the target of picture reconstruction to activate the DPB slot specified in `pDecodeInfo->pSetupReferenceSlot->slotIndex`. If `StdVideoDecodeVP9PictureInfo::refresh_frame_flags` is zero, but a reconstructed picture is specified, then the corresponding picture reference associated with the DPB slot is invalidated, as described in the DPB Slot States section.

Valid Usage (Implicit)

- VUID-VkVideoDecodeVP9PictureInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_VP9_PICTURE_INFO_KHR
- VUID-VkVideoDecodeVP9PictureInfoKHR-pStdPictureInfo-parameter
  `pStdPictureInfo` **must** be a valid pointer to a valid `StdVideoDecodeVP9PictureInfo` value

Structure Chaining

Extends the structure

- VkVideoDecodeInfoKHR

NOTE VP9 decode operations do not need any std reference information to be specified for the active reference pictures and the optional reconstructed picture. Accordingly, no DPB slot info structure exists in the API that would need to be chained to the corresponding VkVideoReferenceSlotInfoKHR structures and no `StdVideoDecodeVP9ReferenceInfo` structure exists in the VP9 Video Std Headers.

VK_MAX_VIDEO_VP9_REFERENCES_PER_FRAME_KHR is the length of the array of DPB slot or VP9 reference names passed in VkVideoDecodeVP9PictureInfoKHR::`referenceNameSlotIndices`.

```
#define VK_MAX_VIDEO_VP9_REFERENCES_PER_FRAME_KHR 3U
```

45.14.7. VP9 Decode Requirements

This section describes the **required** VP9 decoding capabilities for physical devices that have at least one queue family that supports the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR, as returned by vkGetPhysicalDeviceQueueFamilyProperties2 in VkQueueFamilyVideoPropertiesKHR

::videoCodecOperations.

Table 56. Required Video Std Header Versions

Video Std Header Name	Version
vulkan_video_codec_vp9std_decode	1.0.0

Table 57. Required Video Capabilities

Video Capability	Requirement	Requirement Type[1]
VkVideoCapabilitiesKHR		
flags	-	min
minBitstreamBufferOffsetAlignment	4096	max
minBitstreamBufferSizeAlignment	4096	max
pictureAccessGranularity	(64,64)	max
minCodedExtent	-	max
maxCodedExtent	-	min
maxDpbSlots	0	min
maxActiveReferencePictures	0	min
VkVideoDecodeCapabilitiesKHR		
flags	VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR or VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR	min
VkVideoDecodeVP9CapabilitiesKHR		
maxLevel	STD_VIDEO_VP9_LEVEL_2_0	min

1

The **Requirement Type** column specifies the requirement is either the minimum value all implementations **must** support, the maximum value all implementations **must** support, or the exact value all implementations **must** support. For bitmasks a minimum value is the least bits all implementations **must** set, but they **may** have additional bits set beyond this minimum.

45.15. AV1 Decode Operations

Video decode operations using an AV1 decode profile **can** be used to decode elementary video stream sequences compliant with the AV1 Specification.

NOTE	Refer to the Preamble for information on how the Khronos Intellectual Property Rights Policy relates to normative references to external materials not created by Khronos.

This process is performed according to the video decode operation steps with the codec-specific semantics defined in section 7 of the AV1 Specification:

- Syntax elements, derived values, and other parameters are applied from the following structures:
  - The `StdVideoAV1SequenceHeader` structure stored in the bound video session parameters object specifying the active sequence header.
  - The `StdVideoDecodeAV1PictureInfo` structure specifying the AV1 picture information.
  - The `StdVideoDecodeAV1ReferenceInfo` structures specifying the AV1 reference information corresponding to the optional reconstructed picture and any active reference pictures.
- The contents of the provided video bitstream buffer range are interpreted as defined in the AV1 Decode Bitstream Data Access section.
- Picture data in the video picture resources corresponding to the used active reference pictures, decode output picture, and optional reconstructed picture is accessed as defined in the AV1 Decode Picture Data Access section.
- The decision on reference picture setup is made according to the parameters specified in the AV1 picture information.

If the parameters and the bitstream adhere to the syntactic and semantic requirements defined in the corresponding sections of the AV1 Specification, as described above, and the DPB slots associated with the active reference pictures all refer to valid picture references, then the video decode operation will complete successfully. Otherwise, the video decode operation **may** complete unsuccessfully.

45.15.1. AV1 Decode Bitstream Data Access

The video bitstream buffer range **should** contain one or more frame OBUs, comprised of a frame header OBU and tile group OBU, that together represent an entire frame, as defined in sections 5.10, 5.9, and 5.11, and this data is interpreted as defined in sections 6.9, 6.8, and 6.10 of the AV1 Specification, respectively.

The offset specified in VkVideoDecodeAV1PictureInfoKHR::`frameHeaderOffset` **should** specify the starting offset of the frame header OBU of the frame.

NOTE	When the tiles of the frame are encoded into multiple tile groups, each frame OBU has a separate frame header OBU but their content is expected to match per the requirements of the AV1 Specification. Accordingly, the offset specified in `frameHeaderOffset` **can** be the offset of any of the otherwise identical frame header OBUs when multiple tile groups are present.

The offsets and sizes provided in VkVideoDecodeAV1PictureInfoKHR::`pTileOffsets` and VkVideoDecodeAV1PictureInfoKHR::`pTileSizes`, respectively, **should** specify the starting offsets and

sizes corresponding to each tile within the video bitstream buffer range.

45.15.2. AV1 Decode Picture Data Access

Accesses to image data within a video picture resource happen at the granularity indicated by VkVideoCapabilitiesKHR::`pictureAccessGranularity`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile. Accordingly, the complete image subregion of a decode output picture, reference picture, or reconstructed picture accessed by video coding operations using an AV1 decode profile is defined as the set of texels within the coordinate range:

([0,`endX`), [0,`endY`))

Where:

- `endX` equals `codedExtent.width` rounded up to the nearest integer multiple of `pictureAccessGranularity.width` and clamped to the width of the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure;
- endY equals `codedExtent.height` rounded up to the nearest integer multiple of `pictureAccessGranularity.height` and clamped to the height of the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure;

Where `codedExtent` is the member of the VkVideoPictureResourceInfoKHR structure corresponding to the picture.

In case of video decode operations using an AV1 decode profile, any access to a picture at the coordinates (`x`,`y`), as defined by the AV1 Specification, is an access to the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure at the texel coordinates (`x`,`y`).

45.15.3. AV1 Reference Names and Semantics

Individual reference frames used in the decoding process have different semantics, as defined in section 6.10.24 of the AV1 Specification. The AV1 semantics associated with a reference picture are indicated by the corresponding enumeration constant defined in the Video Std enumeration type `StdVideoAV1ReferenceName`:

- `STD_VIDEO_AV1_REFERENCE_NAME_INTRA_FRAME` identifies the reference used for intra coding (`INTRA_FRAME`), as defined in sections 2 and 7.11.2 of the AV1 Specification.
- All other enumeration constants refer to backward or forward references used for inter coding, as defined in sections 2 and 7.11.3 of the AV1 Specification:
  - `STD_VIDEO_AV1_REFERENCE_NAME_LAST_FRAME` identifies the `LAST_FRAME` reference
  - `STD_VIDEO_AV1_REFERENCE_NAME_LAST2_FRAME` identifies the `LAST2_FRAME` reference
  - `STD_VIDEO_AV1_REFERENCE_NAME_LAST3_FRAME` identifies the `LAST3_FRAME` reference
  - `STD_VIDEO_AV1_REFERENCE_NAME_GOLDEN_FRAME` identifies the `GOLDEN_FRAME` reference
  - `STD_VIDEO_AV1_REFERENCE_NAME_BWDREF_FRAME` identifies the `BWDREF_FRAME` reference

- STD_VIDEO_AV1_REFERENCE_NAME_ALTREF2_FRAME identifies the ALTREF2_FRAME reference
- STD_VIDEO_AV1_REFERENCE_NAME_ALTREF_FRAME identifies the ALTREF_FRAME reference

These enumeration constants are not directly used in any APIs but are used to indirectly index into certain Video Std and Vulkan API parameter arrays.

45.15.4. AV1 Decode Profile

A video profile supporting AV1 video decode operations is specified by setting VkVideoProfileInfoKHR::videoCodecOperation to VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR and adding a VkVideoDecodeAV1ProfileInfoKHR structure to the VkVideoProfileInfoKHR::pNext chain.

The VkVideoDecodeAV1ProfileInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_av1
typedef struct VkVideoDecodeAV1ProfileInfoKHR {
    VkStructureType       sType;
    const void*           pNext;
    StdVideoAV1Profile    stdProfile;
    VkBool32              filmGrainSupport;
} VkVideoDecodeAV1ProfileInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- stdProfile is a StdVideoAV1Profile value specifying the AV1 codec profile, as defined in section A.2 of the AV1 Specification.
- filmGrainSupport specifies whether AV1 film grain, as defined in section 7.8.3 of the AV1 Specification, **can** be used with the video profile. When this member is VK_TRUE, video session objects created against the video profile will be able to decode pictures that have film grain enabled.

NOTE	Enabling filmGrainSupport **may** increase the memory requirements of video sessions and/or video picture resources on some implementations.

Valid Usage (Implicit)

- VUID-VkVideoDecodeAV1ProfileInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_PROFILE_INFO_KHR

Structure Chaining

Extends the structures

- VkQueryPoolCreateInfo

- VkVideoProfileInfoKHR

45.15.5. AV1 Decode Capabilities

When calling vkGetPhysicalDeviceVideoCapabilitiesKHR to query the capabilities for an AV1 decode profile, the VkVideoCapabilitiesKHR::pNext chain **must** include a VkVideoDecodeAV1CapabilitiesKHR structure that will be filled with the profile-specific capabilities.

The VkVideoDecodeAV1CapabilitiesKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_av1
typedef struct VkVideoDecodeAV1CapabilitiesKHR {
    VkStructureType     sType;
    void*               pNext;
    StdVideoAV1Level    maxLevel;
} VkVideoDecodeAV1CapabilitiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- maxLevel is a StdVideoAV1Level value specifying the maximum AV1 level supported by the profile, as defined in section A.3 of the AV1 Specification.

Valid Usage (Implicit)

- VUID-VkVideoDecodeAV1CapabilitiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkVideoCapabilitiesKHR

45.15.6. AV1 Decode Parameter Sets

Video session parameters objects created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR contain a single instance of the following parameter set:

AV1 Sequence Header

Represented by StdVideoAV1SequenceHeader structures and interpreted as follows:

- flags.reserved and reserved1 are used only for padding purposes and are otherwise ignored;

- the StdVideoAV1ColorConfig structure pointed to by pColorConfig is interpreted as follows:
  - flags.reserved and reserved1 are used only for padding purposes and are otherwise ignored;
  - all other members of StdVideoAV1ColorConfig are interpreted as defined in section 6.4.2 of the AV1 Specification;
- if flags.timing_info_present_flag is set, then the StdVideoAV1TimingInfo structure pointed to by pTimingInfo is interpreted as follows:
  - flags.reserved is used only for padding purposes and is otherwise ignored;
  - all other members of StdVideoAV1TimingInfo are interpreted as defined in section 6.4.3 of the AV1 Specification;
- all other members of StdVideoAV1SequenceHeader are interpreted as defined in section 6.4 of the AV1 Specification.

When a video session parameters object is created with the codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, the VkVideoSessionParametersCreateInfoKHR::pNext chain **must** include a VkVideoDecodeAV1SessionParametersCreateInfoKHR structure specifying the contents of the object.

The VkVideoDecodeAV1SessionParametersCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_av1
typedef struct VkVideoDecodeAV1SessionParametersCreateInfoKHR {
    VkStructureType                    sType;
    const void*                        pNext;
    const StdVideoAV1SequenceHeader*   pStdSequenceHeader;
} VkVideoDecodeAV1SessionParametersCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pStdSequenceHeader is a pointer to a StdVideoAV1SequenceHeader structure describing the AV1 sequence header entry to store in the created object.

NOTE

As AV1 video session parameters objects will only ever contain a single AV1 sequence header, this has to be specified at object creation time and such video session parameters objects cannot be updated using the vkUpdateVideoSessionParametersKHR command. When a new AV1 sequence header is decoded from the input video bitstream the application needs to create a new video session parameters object to store it.

Valid Usage (Implicit)

- VUID-VkVideoDecodeAV1SessionParametersCreateInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_SESSION_PARAMETERS_CREATE_INFO_KHR

- VUID-VkVideoDecodeAV1SessionParametersCreateInfoKHR-pStdSequenceHeader-parameter
  `pStdSequenceHeader` **must** be a valid pointer to a valid `StdVideoAV1SequenceHeader` value

Structure Chaining

Extends the structure

- VkVideoSessionParametersCreateInfoKHR

Inline Parameter Sets

In case of video sessions created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR and with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR, the application **can** also specify the active sequence header *inline* by including an instance of the VkVideoDecodeAV1InlineSessionParametersInfoKHR structure in the `pNext` chain of VkVideoDecodeInfoKHR.

The `VkVideoDecodeAV1InlineSessionParametersInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_av1 with VK_KHR_video_maintenance2
typedef struct VkVideoDecodeAV1InlineSessionParametersInfoKHR {
    VkStructureType                    sType;
    const void*                        pNext;
    const StdVideoAV1SequenceHeader*   pStdSequenceHeader;
} VkVideoDecodeAV1InlineSessionParametersInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pStdSequenceHeader` is `NULL` or a pointer to an instance of the `StdVideoAV1SequenceHeader` structure describing the active AV1 sequence header.

If `pStdSequenceHeader` is not `NULL`, the issued video decode operations will use the specified sequence header parameters instead of the active sequence header being sourced from the bound video session parameters object.

Valid Usage (Implicit)

- VUID-VkVideoDecodeAV1InlineSessionParametersInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_INLINE_SESSION_PARAMETERS_INFO_KHR
- VUID-VkVideoDecodeAV1InlineSessionParametersInfoKHR-pStdSequenceHeader-parameter
  If `pStdSequenceHeader` is not `NULL`, `pStdSequenceHeader` **must** be a valid pointer to a valid

StdVideoAV1SequenceHeader value

Structure Chaining

Extends the structure

- VkVideoDecodeInfoKHR

45.15.7. AV1 Decoding Parameters

The VkVideoDecodeAV1PictureInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_av1
typedef struct VkVideoDecodeAV1PictureInfoKHR {
    VkStructureType                         sType;
    const void*                             pNext;
    const StdVideoDecodeAV1PictureInfo*     pStdPictureInfo;
    int32_t                                 referenceNameSlotIndices
[VK_MAX_VIDEO_AV1_REFERENCES_PER_FRAME_KHR];
    uint32_t                                frameHeaderOffset;
    uint32_t                                tileCount;
    const uint32_t*                         pTileOffsets;
    const uint32_t*                         pTileSizes;
} VkVideoDecodeAV1PictureInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pStdPictureInfo is a pointer to a StdVideoDecodeAV1PictureInfo structure specifying AV1 picture information.
- referenceNameSlotIndices is an array of seven (VK_MAX_VIDEO_AV1_REFERENCES_PER_FRAME_KHR, which is equal to the Video Std definition STD_VIDEO_AV1_REFS_PER_FRAME) signed integer values specifying the index of the DPB slot or a negative integer value for each AV1 reference name used for inter coding. In particular, the DPB slot index for the AV1 reference name frame is specified in referenceNameSlotIndices [frame - STD_VIDEO_AV1_REFERENCE_NAME_LAST_FRAME].
- frameHeaderOffset is the byte offset of the AV1 frame header OBU, as defined in section 5.9 of the AV1 Specification, within the video bitstream buffer range specified in VkVideoDecodeInfoKHR.
- tileCount is the number of elements in pTileOffsets and pTileSizes.
- pTileOffsets is a pointer to an array of tileCount integers specifying the byte offset of the tiles of the picture within the video bitstream buffer range specified in VkVideoDecodeInfoKHR.
- pTileSizes is a pointer to an array of tileCount integers specifying the byte size of the tiles of the picture within the video bitstream buffer range specified in VkVideoDecodeInfoKHR.

This structure is specified in the pNext chain of the VkVideoDecodeInfoKHR structure passed to

vkCmdDecodeVideoKHR to specify the codec-specific picture information for an AV1 decode operation.

Decode Output Picture Information

When this structure is specified in the pNext chain of the VkVideoDecodeInfoKHR structure passed to vkCmdDecodeVideoKHR, the information related to the decode output picture is defined as follows:

- The image subregion used is determined according to the AV1 Decode Picture Data Access section.
- The decode output picture is associated with the AV1 picture information provided in pStdPictureInfo.

Std Picture Information

The members of the StdVideoDecodeAV1PictureInfo structure pointed to by pStdPictureInfo are interpreted as follows:

- flags.reserved, reserved1, and reserved2 are used only for padding purposes and are otherwise ignored;
- flags.apply_grain indicates that film grain is enabled for the decoded picture, as defined in section 6.8.20 of the AV1 Specification;
- OrderHint, OrderHints, and expectedFrameId are interpreted as defined in section 6.8.2 of the AV1 Specification;
- the StdVideoAV1TileInfo structure pointed to by pTileInfo is interpreted as follows:
  - flags.reserved and reserved1 are used only for padding purposes and are otherwise ignored;
  - pMiColStarts is a pointer to an array of TileCols number of unsigned integers that corresponds to MiColStarts defined in section 6.8.14 of the AV1 Specification;
  - pMiRowStarts is a pointer to an array of TileRows number of unsigned integers that corresponds to MiRowStarts defined in section 6.8.14 of the AV1 Specification;

    NOTE Historically some applications incorrectly specified the values in pMiColStarts and pMiRowStarts in terms of superblocks instead of mode info blocks, therefore it is recommended for implementations to ignore the values specified in pMiColStarts and pMiRowStarts. Instead, the values should be calculated based on flags.uniform_tile_spacing_flag, pWidthInSbsMinus1, pHeightInSbsMinus1, the sequence header parameter flags.use_128x128_superblock, and the coded extent of the decoded picture.

  - pWidthInSbsMinus1 is a pointer to an array of TileCols number of unsigned integers that corresponds to width_in_sbs_minus_1 defined in section 6.8.14 of the AV1 Specification;
  - pHeightInSbsMinus1 is a pointer to an array of TileRows number of unsigned integers that corresponds to height_in_sbs_minus_1 defined in section 6.8.14 of the AV1 Specification;
  - all other members of StdVideoAV1TileInfo are interpreted as defined in section 6.8.14 of

the AV1 Specification;

- the `StdVideoAV1Quantization` structure pointed to by `pQuantization` is interpreted as follows:
  - `flags.reserved` is used only for padding purposes and is otherwise ignored;
  - all other members of `StdVideoAV1Quantization` are interpreted as defined in section 6.8.11 of the AV1 Specification;
- if `flags.segmentation_enabled` is set, then the `StdVideoAV1Segmentation` structure pointed to by `pSegmentation` is interpreted as follows:
  - the elements of `FeatureEnabled` are bitmasks where bit index j of element i corresponds to `FeatureEnabled[i][j]` as defined in section 5.9.14 of the AV1 Specification;
  - `FeatureData` is interpreted as defined in section 5.9.14 of the AV1 Specification;
- the `StdVideoAV1LoopFilter` structure pointed to by `pLoopFilter` is interpreted as follows:
  - `flags.reserved` is used only for padding purposes and is otherwise ignored;
  - `update_ref_delta` is a bitmask where bit index i is interpreted as the value of `update_ref_delta` corresponding to element i of `loop_filter_ref_deltas` as defined in section 6.8.10 of the AV1 Specification;
  - `update_mode_delta` is a bitmask where bit index i is interpreted as the value of `update_mode_delta` corresponding to element i of `loop_filter_mode_deltas` as defined in section 6.8.10 of the AV1 Specification;
  - all other members of `StdVideoAV1LoopFilter` are interpreted as defined in section 6.8.10 of the AV1 Specification;

> **NOTE** If the syntax elements corresponding to `loop_filter_ref_deltas` and `loop_filter_mode_deltas` are not present or otherwise defined according to section 5.9.11 of the AV1 Specification, the application should specify the previous values, as defined in section 6.8.10 of the AV1 Specification.

- if `flags.enable_cdef` is set in the active sequence header, then the members of the `StdVideoAV1CDEF` structure pointed to by `pCDEF` are interpreted as follows:
  - `cdef_y_sec_strength` and `cdef_uv_sec_strength` are the bitstream values of the corresponding syntax elements defined in section 5.9.19 of the AV1 Specification;
  - all other members of `StdVideoAV1CDEF` are interpreted as defined in section 6.10.14 of the AV1 Specification;
- if `flags.UsesLr` is set in the active sequence header, then the `StdVideoAV1LoopRestoration` structure pointed to by `pLoopRestoration` is interpreted as follows:
  - `LoopRestorationSize[plane]` is interpreted as log2(`size`) - 5, where `size` is the value of `LoopRestorationSize[plane]` as defined in section 6.10.15 of the AV1 Specification;
  - all other members of `StdVideoAV1LoopRestoration` are defined as in section 6.10.15 of the AV1 Specification;
- the members of the `StdVideoAV1GlobalMotion` structure pointed to by `pGlobalMotion` are interpreted as defined in section 7.10 of the AV1 Specification;
- if `flags.film_grain_params_present` is set in the active sequence header, then the

StdVideoAV1FilmGrain structure pointed to by pFilmGrain is interpreted as follows:

  - flags.reserved is used only for padding purposes and is otherwise ignored;
  - all other members of StdVideoAV1FilmGrain are interpreted as defined in section 6.8.20 of the AV1 Specification;

- all other members are interpreted as defined in section 6.8 of the AV1 Specification.

When film grain is enabled for the decoded frame, the flags.update_grain and film_grain_params_ref_idx values specified in StdVideoAV1FilmGrain are ignored by AV1 decode operations and the load_grain_params function, as defined in section 6.8.20 of the AV1 Specification, is not executed. Instead, the application is responsible for specifying the effective film grain parameters for the frame in StdVideoAV1FilmGrain.

When film grain is enabled for the decoded frame, the application is required to specify a different decode output picture resource in VkVideoDecodeInfoKHR::dstPictureResource compared to the reconstructed picture specified in VkVideoDecodeInfoKHR::pSetupReferenceSlot->pPictureResource even if the implementation does not report support for VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR in VkVideoDecodeCapabilitiesKHR::flags for the video decode profile.

Reference picture setup is controlled by the value of StdVideoDecodeAV1PictureInfo::refresh_frame_flags. If it is not zero and a reconstructed picture is specified, then the latter is used as the target of picture reconstruction to activate the DPB slot specified in pDecodeInfo->pSetupReferenceSlot->slotIndex. If StdVideoDecodeAV1PictureInfo::refresh_frame_flags is zero, but a reconstructed picture is specified, then the corresponding picture reference associated with the DPB slot is invalidated, as described in the DPB Slot States section.

Active Parameter Sets

The *active sequence header* is the AV1 sequence header stored in the bound video session parameters object , unless the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR and the active sequence header is specified inline. .

Valid Usage (Implicit)

- VUID-VkVideoDecodeAV1PictureInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_PICTURE_INFO_KHR
- VUID-VkVideoDecodeAV1PictureInfoKHR-pStdPictureInfo-parameter
  pStdPictureInfo **must** be a valid pointer to a valid StdVideoDecodeAV1PictureInfo value
- VUID-VkVideoDecodeAV1PictureInfoKHR-pTileOffsets-parameter
  pTileOffsets **must** be a valid pointer to an array of tileCount uint32_t values
- VUID-VkVideoDecodeAV1PictureInfoKHR-pTileSizes-parameter
  pTileSizes **must** be a valid pointer to an array of tileCount uint32_t values
- VUID-VkVideoDecodeAV1PictureInfoKHR-tileCount-arraylength
  tileCount **must** be greater than 0

Structure Chaining

Extends the structure

- VkVideoDecodeInfoKHR

VK_MAX_VIDEO_AV1_REFERENCES_PER_FRAME_KHR is the length of the array of DPB slot or AV1 reference names passed in VkVideoDecodeAV1PictureInfoKHR::referenceNameSlotIndices.

```
#define VK_MAX_VIDEO_AV1_REFERENCES_PER_FRAME_KHR 7U
```

The VkVideoDecodeAV1DpbSlotInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_decode_av1
typedef struct VkVideoDecodeAV1DpbSlotInfoKHR {
    VkStructureType                          sType;
    const void*                              pNext;
    const StdVideoDecodeAV1ReferenceInfo*    pStdReferenceInfo;
} VkVideoDecodeAV1DpbSlotInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pStdReferenceInfo is a pointer to a StdVideoDecodeAV1ReferenceInfo structure specifying AV1 reference information.

This structure is specified in the pNext chain of VkVideoDecodeInfoKHR::pSetupReferenceSlot, if not NULL, and the pNext chain of the elements of VkVideoDecodeInfoKHR::pReferenceSlots to specify the codec-specific reference picture information for an AV1 decode operation.

Active Reference Picture Information

When this structure is specified in the pNext chain of the elements of VkVideoDecodeInfoKHR::pReferenceSlots, one element is added to the list of active reference pictures used by the video decode operation for each element of VkVideoDecodeInfoKHR::pReferenceSlots as follows:

- The image subregion used is determined according to the AV1 Decode Picture Data Access section.
- The reference picture is associated with the DPB slot index specified in the slotIndex member of the corresponding element of VkVideoDecodeInfoKHR::pReferenceSlots.
- The reference picture is associated with the AV1 reference information provided in pStdReferenceInfo.

Reconstructed Picture Information

When this structure is specified in the pNext chain of VkVideoDecodeInfoKHR::pSetupReferenceSlot, the information related to the reconstructed picture is defined as follows:

- The image subregion used is determined according to the AV1 Decode Picture Data Access section.
- If reference picture setup is requested, then the reconstructed picture is used to activate the DPB slot with the index specified in VkVideoDecodeInfoKHR::pSetupReferenceSlot->slotIndex.
- The reconstructed picture is associated with the AV1 reference information provided in pStdReferenceInfo.

Std Reference Information

The members of the StdVideoDecodeAV1ReferenceInfo structure pointed to by pStdReferenceInfo are interpreted as follows:

- flags.reserved and reserved1 are used only for padding purposes and are otherwise ignored;
- flags.disable_frame_end_update_cdf is interpreted as defined in section 6.8.2 of the AV1 Specification;
- flags.segmentation_enabled is interpreted as defined in section 6.8.13 of the AV1 Specification;
- frame_type is interpreted as defined in section 6.8.2 of the AV1 Specification;

> **NOTE** The frame_type member is defined with the type uint8_t, but it takes the same values defined in the StdVideoAV1FrameType enumeration type as StdVideoDecodeAV1PictureInfo::frame_type.

- RefFrameSignBias is a bitmask where bit index i corresponds to RefFrameSignBias[i] as defined in section 6.8.2 of the AV1 Specification;
- OrderHint is interpreted as defined in section 6.8.2 of the AV1 Specification;
- SavedOrderHints is interpreted as defined in section 7.20 of the AV1 Specification.

> **NOTE** When the AV1 reference information is provided for the reconstructed picture, certain parameters (e.g. frame_type) are specified both in the AV1 picture information and in the AV1 reference information. This is necessary because unlike the AV1 picture information, which is only used for the purposes of the video decode operation in question, the AV1 reference information specified for the reconstructed picture **may** be associated with the activated DPB slot, meaning that some implementations **may** maintain it as part of the reference picture metadata corresponding to the video picture resource associated with the DPB slot. When the AV1 reference information is provided for an active reference picture, the specified parameters correspond to the parameters specified when the DPB slot was activated (set up) with the reference picture, as usual, in order to communicate these parameters for implementations that do not maintain any subset of these parameters as part of the DPB slot's reference picture metadata.

Valid Usage (Implicit)

- VUID-VkVideoDecodeAV1DpbSlotInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_DPB_SLOT_INFO_KHR
- VUID-VkVideoDecodeAV1DpbSlotInfoKHR-pStdReferenceInfo-parameter
  `pStdReferenceInfo` **must** be a valid pointer to a valid `StdVideoDecodeAV1ReferenceInfo` value

Structure Chaining

Extends the structure

- VkVideoReferenceSlotInfoKHR

45.15.8. AV1 Decode Requirements

This section describes the **required** AV1 decoding capabilities for physical devices that have at least one queue family that supports the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, as returned by vkGetPhysicalDeviceQueueFamilyProperties2 in VkQueueFamilyVideoPropertiesKHR ::`videoCodecOperations`.

Table 58. Required Video Std Header Versions

Video Std Header Name	Version
`vulkan_video_codec_av1std_decode`	1.0.0

Table 59. Required Video Capabilities

Video Capability	Requirement	Requirement Type[1]
VkVideoCapabilitiesKHR		
`flags`	-	min
`minBitstreamBufferOffsetAlignment`	4096	max
`minBitstreamBufferSizeAlignment`	4096	max
`pictureAccessGranularity`	(64,64)	max
`minCodedExtent`	-	max
`maxCodedExtent`	-	min
`maxDpbSlots`	0	min
`maxActiveReferencePictures`	0	min
VkVideoDecodeCapabilitiesKHR		

Video Capability	Requirement	Requirement Type[1]
flags	VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR or VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR	min
VkVideoDecodeAV1CapabilitiesKHR		
maxLevel	STD_VIDEO_AV1_LEVEL_2_0	min

1

The **Requirement Type** column specifies the requirement is either the minimum value all implementations **must** support, the maximum value all implementations **must** support, or the exact value all implementations **must** support. For bitmasks a minimum value is the least bits all implementations **must** set, but they **may** have additional bits set beyond this minimum.

45.16. Video Encode Operations

Video encode operations consume an *encode input picture* and zero or more reference pictures, and produce compressed video data to a video bitstream buffer and an optional reconstructed picture.

NOTE	Such encode input pictures can be used as the output of video decode operations, with graphics or compute operations, or with Window System Integration APIs, depending on the capabilities of the implementation.

Video encode operations **may** access the following resources in the VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR stage:

- The image subregions corresponding to the source encode input picture and active reference pictures with access VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR.
- The destination video bitstream buffer range and the optional reconstructed picture with access VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR.
- The image subregion corresponding to a quantization map used in the video encode operation with access VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR.

The image subresource of each video picture resource accessed by the video encode operation is specified using a corresponding VkVideoPictureResourceInfoKHR structure. Each such image subresource **must** be in the appropriate image layout as follows:

- If the image subresource is used in the video encode operation as an encode input picture, then it **must** be in the VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR layout , unless the unifiedImageLayoutsVideo feature is enabled, in which case it **may** be in the

VK_IMAGE_LAYOUT_GENERAL layout .

- If the image subresource is used in the video encode operation as a reconstructed picture or reference picture, then it **must** be in the VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR layout , unless the `unifiedImageLayoutsVideo` feature is enabled, in which case it **may** be in the VK_IMAGE_LAYOUT_GENERAL layout .
- If the image subresource is used in the video encode operation as a quantization map, then it **must** be in the VK_IMAGE_LAYOUT_VIDEO_ENCODE_QUANTIZATION_MAP_KHR layout , unless the `unifiedImageLayoutsVideo` feature is enabled, in which case it **may** be in the VK_IMAGE_LAYOUT_GENERAL layout .

A video encode operation **may** complete unsuccessfully. In this case the target video bitstream buffer will have undefined contents. Similarly, if reference picture setup is requested, the reconstructed picture will also have undefined contents, and the activated DPB slot will have an invalid picture reference.

If a video encode operation completes successfully and the codec-specific parameters provided by the application adhere to the syntactic and semantic requirements defined in the corresponding video compression standard, then the target video bitstream buffer will contain compressed video data after the execution of the video encode operation according to the respective codec-specific semantics.

45.16.1. Codec-Specific Semantics

The following aspects of video encode operations are codec-specific:

- The compressed video data written to the target video bitstream buffer range.
- The construction and interpretation of the list of active reference pictures and the interpretation of the picture data referred to by the corresponding image subregions.
- The construction and interpretation of information related to the encode input picture and the interpretation of the picture data referred to by the corresponding image subregion.
- The decision on reference picture setup.
- The construction and interpretation of information related to the optional reconstructed picture and the generation of picture data to the corresponding image subregion.
- Certain aspects of rate control.

These codec-specific behaviors are defined for each video codec operation separately.

- If the used video codec operation is VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the codec-specific aspects of the video encoding process are performed as defined in the H.264 Encode Operations section.
- If the used video codec operation is VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the codec-specific aspects of the video encoding process are performed as defined in the H.265 Encode Operations section.
- If the used video codec operation is VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then the codec-specific aspects of the video encoding process are performed as defined in the

AV1 Encode Operations section.

Video Encode Parameter Overrides

Implementations supporting video encode operations for any particular video codec operation often support only a subset of the available encoding tools defined by the corresponding video compression standards. Accordingly, certain implementation-dependent limitations **may** apply to codec-specific parameters provided through the structures defined in the Video Std headers corresponding to the used video codec operation.

Exposing all of these restrictions on particular codec-specific parameter values or combinations thereof in the form of application-queryable capabilities is impractical, hence this specification allows implementations to *override* the value of any of the codec-specific parameters, unless otherwise specified, as long as all of the following conditions are met:

- If the application-provided codec-specific parameters adhere to the syntactic and semantic requirements and rules defined by the used video compression standard, and thus would be usable to produce a video bitstream compliant with that standard, then the codec-specific parameters resulting from the process of implementation overrides **must** also adhere to the same requirements and rules, and any video bitstream produced using the overridden parameters **must** also be compliant.
- The overridden codec-specific parameter values **must** not have an impact on the codec-independent behaviors defined for video encode operations.
- The implementation **must** not override any codec-specific parameters specified to a command that **may** cause application-provided codec-specific parameters specified to subsequent commands to no longer adhere to the semantic requirements and rules defined by the used video compression standard, unless the implementation also overrides those parameters to adhere to any such requirements and rules.
- The overridden codec-specific parameter values **must** not have an impact on the codec-specific picture data access semantics.
- The overridden codec-specific parameter values **may** change the contents of the codec-specific bitstream elements produced by video encode operations or otherwise retrieved by the application (e.g. using the vkGetEncodedVideoSessionParametersKHR command) but **must** still adhere to the codec-specific semantics defined for that video codec operation, including, but not limited to, the number, type, and order of the encoded codec-specific bitstream elements.

Besides codec-specific parameter overrides performed for implementation-dependent reasons, applications **can** enable the implementation to apply additional optimizing overrides that **may** improve the efficiency or performance of video encoding operations. However, implementations **must** meet the conditions listed above even in case of such optimizing overrides.

NOTE	Unless the application opts in for optimizing overrides, implementations are not expected to override any of the codec-specific parameters, except when such overrides are necessary for the correct operation of video encoder implementation due to limitations to the available encoding tools on that implementation.

45.16.2. Video Encode Operation Steps

Each video encode operation performs the following steps in the VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR stage:

1. Reads the input picture data from the encode input picture;
2. Determines derived encoding quality parameters according to the codec-specific semantics and the current rate control state;
3. Compresses the input picture data according to the codec-specific semantics, applying any prediction data read from the active reference pictures and rate control restrictions in the process;
4. Writes the encoded bitstream data to the destination video bitstream buffer range;
5. Performs picture reconstruction of the encoded video data according to the codec-specific semantics, applying any prediction data read from the active reference pictures in the process, if a reconstructed picture is specified and reference picture setup is requested;
6. If reference picture setup is requested, the DPB slot index specified in the reconstructed picture information is activated with the reconstructed picture;
7. Writes the reconstructed picture data to the reconstructed picture, if one is specified, according to the codec-specific semantics.

When reconstructed picture information is provided, the specified DPB slot index is associated with the corresponding bound reference picture resource, regardless of whether reference picture setup is requested.

45.16.3. Capabilities

When calling vkGetPhysicalDeviceVideoCapabilitiesKHR with `pVideoProfile->videoCodecOperation` specifying an encode operation, the VkVideoEncodeCapabilitiesKHR structure **must** be included in the `pNext` chain of the VkVideoCapabilitiesKHR structure to retrieve capabilities specific to video encoding.

The `VkVideoEncodeCapabilitiesKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeCapabilitiesKHR {
    VkStructureType                          sType;
    void*                                    pNext;
    VkVideoEncodeCapabilityFlagsKHR          flags;
    VkVideoEncodeRateControlModeFlagsKHR     rateControlModes;
    uint32_t                                 maxRateControlLayers;
    uint64_t                                 maxBitrate;
    uint32_t                                 maxQualityLevels;
    VkExtent2D                               encodeInputPictureGranularity;
    VkVideoEncodeFeedbackFlagsKHR            supportedEncodeFeedbackFlags;
} VkVideoEncodeCapabilitiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkVideoEncodeCapabilityFlagBitsKHR describing supported encoding features.
- `rateControlModes` is a bitmask of VkVideoEncodeRateControlModeFlagBitsKHR indicating supported rate control modes.
- `maxRateControlLayers` indicates the maximum number of rate control layers supported.
- `maxBitrate` indicates the maximum supported bitrate.
- `maxQualityLevels` indicates the number of discrete video encode quality levels supported. Implementations **must** support at least one quality level.
- `encodeInputPictureGranularity` indicates the granularity at which encode input picture data is encoded and **may** indicate a texel granularity up to the size of the largest supported codec-specific coding block. This capability does not impose any valid usage constraints on the application, however, depending on the contents of the encode input picture, it **may** have effects on the encoded bitstream, as described in more detail below.
- `supportedEncodeFeedbackFlags` is a bitmask of VkVideoEncodeFeedbackFlagBitsKHR values specifying the supported flags for video encode feedback queries.

Implementations **must** include support for at least VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR and VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR in `supportedEncodeFeedbackFlags`.

`encodeInputPictureGranularity` provides information about the way encode input picture data is used as input to video encode operations. In particular, some implementations **may** not be able to limit the set of texels used to encode the output video bitstream to the image subregion specified in the VkVideoPictureResourceInfoKHR structure corresponding to the encode input picture (i.e. to the resolution of the image data to encode specified in its `codedExtent` member).

NOTE

For example, the application requests the coded extent to be 1920x1080, but the implementation is only able to source the encode input picture data at the granularity of the codec-specific coding block size which is 16x16 pixels (or as otherwise indicated in `encodeInputPictureGranularity`). In this example, the content is horizontally aligned with the coding block size, but not vertically aligned with it. Thus encoding of the last row of coding blocks will be impacted by the contents of the input image at texel rows 1080 to 1087 (the latter being the next row which is vertically aligned with the coding block size, assuming a zero-based texel row index).

If `codedExtent` rounded up to the next integer multiple of `encodeInputPictureGranularity` is greater than the extent of the image subresource specified for the encode input picture, then the texel values corresponding to texel coordinates outside of the bounds of the image subresource **may** be undefined. However, implementations **should** use well-defined default values for such texels in order to maximize the encoding efficiency for the last coding block row/column, and/or to ensure consistent encoding results across repeated encoding of the same input content. Nonetheless, the

values used for such texels **must** not have an effect on whether the video encode operation produces a compliant bitstream, and **must** not have any other effects on the encoded picture data beyond what **may** otherwise result from using these texel values as input to any compression algorithm, as defined in the used video compression standard.

NOTE While not required, it is generally a good practice for applications to make sure that the image subresource used for the encode input picture has an extent that is an integer multiple of the codec-specific coding block size (or at least `encodeInputPictureGranularity`) and that this padding area is filled with known values in order to improve encoding efficiency, portability, and reproducibility.

Valid Usage (Implicit)

- VUID-VkVideoEncodeCapabilitiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkVideoCapabilitiesKHR

Bits which **may** be set in VkVideoEncodeCapabilitiesKHR::`flags`, indicating the encoding tools supported, are:

```
// Provided by VK_KHR_video_encode_queue
typedef enum VkVideoEncodeCapabilityFlagBitsKHR {
    VK_VIDEO_ENCODE_CAPABILITY_PRECEDING_EXTERNALLY_ENCODED_BYTES_BIT_KHR =
0x00000001,
    VK_VIDEO_ENCODE_CAPABILITY_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_DETECTION_BIT_KHR =
0x00000002,
  // Provided by VK_KHR_video_encode_quantization_map
    VK_VIDEO_ENCODE_CAPABILITY_QUANTIZATION_DELTA_MAP_BIT_KHR = 0x00000004,
  // Provided by VK_KHR_video_encode_quantization_map
    VK_VIDEO_ENCODE_CAPABILITY_EMPHASIS_MAP_BIT_KHR = 0x00000008,
} VkVideoEncodeCapabilityFlagBitsKHR;
```

- VK_VIDEO_ENCODE_CAPABILITY_PRECEDING_EXTERNALLY_ENCODED_BYTES_BIT_KHR specifies that the implementation supports the use of VkVideoEncodeInfoKHR::`precedingExternallyEncodedBytes`.
- VK_VIDEO_ENCODE_CAPABILITY_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_DETECTION_BIT_KHR specifies that the implementation is able to detect and report when the destination video bitstream buffer range provided by the application is not sufficiently large to fit the encoded bitstream data produced by a video encode operation by reporting the VK_QUERY_RESULT_STATUS_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_KHR query result status code.

NOTE

Some implementations **may** not be able to reliably detect insufficient bitstream buffer range conditions in all situations. Such implementations will not report support for the VK_VIDEO_ENCODE_CAPABILITY_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_DETECTION_BIT_KHR encode capability flag for the video profile, but **may** still report the VK_QUERY_RESULT_STATUS_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_KHR query result status code in certain cases. Applications **should** always check for the specific query result status code VK_QUERY_RESULT_STATUS_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_KHR even when this encode capability flag is not supported by the implementation for the video profile in question. However, applications **must** not assume that a different negative query result status code indicating an unsuccessful completion of a video encode operation is not the result of an insufficient bitstream buffer condition unless this encode capability flag is supported.

- VK_VIDEO_ENCODE_CAPABILITY_QUANTIZATION_DELTA_MAP_BIT_KHR specifies support for using quantization delta maps.
- VK_VIDEO_ENCODE_CAPABILITY_EMPHASIS_MAP_BIT_KHR specifies support for using emphasis maps.

```
// Provided by VK_KHR_video_encode_queue
typedef VkFlags VkVideoEncodeCapabilityFlagsKHR;
```

VkVideoEncodeCapabilityFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeCapabilityFlagBitsKHR.

The VkVideoEncodeFeedback2CapabilitiesKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_feedback2
typedef struct VkVideoEncodeFeedback2CapabilitiesKHR {
    VkStructureType                                 sType;
    void*                                           pNext;
    uint32_t                                        maxPerPartitionFeedbackEntries;
    VkVideoEncodePerPartitionFeedbackFlagsKHR
supportedPerPartitionEncodeFeedbackFlags;
} VkVideoEncodeFeedback2CapabilitiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- maxPerPartitionFeedbackEntries specifies the maximum number of picture partitions for which the implementation supports capturing per picture partition video encode feedback values.
- supportedPerPartitionEncodeFeedbackFlags is a bitmask of VkVideoEncodePerPartitionFeedbackFlagBitsKHR values specifying the set of per picture partition video encode feedback values supported by video encode feedback queries.

Valid Usage (Implicit)

- VUID-VkVideoEncodeFeedback2CapabilitiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_FEEDBACK_2_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkVideoCapabilitiesKHR

45.16.4. Video Encode Quality Levels

Implementations **can** support more than one video encode quality level for a video encode profile, which control the number and type of implementation-specific encoding tools and algorithms utilized in the encoding process.

NOTE

Generally, using higher video encode quality levels **may** produce higher quality video streams at the cost of additional processing time. However, as the final quality of an encoded picture depends on the contents of the encode input picture, the contents of the active reference pictures, the codec-specific encode parameters, and the particular implementation-specific tools used corresponding to the individual video encode quality levels, there are no guarantees that using a higher video encode quality level will always produce a higher quality encoded picture for any given set of inputs.

To query properties for a specific video encode quality level supported by a video encode profile, call:

```
// Provided by VK_KHR_video_encode_queue
VkResult vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR(
    VkPhysicalDevice                                physicalDevice,
    const VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR* pQualityLevelInfo,
    VkVideoEncodeQualityLevelPropertiesKHR*         pQualityLevelProperties);
```

- `physicalDevice` is the physical device to query the video encode quality level properties for.
- `pQualityLevelInfo` is a pointer to a VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR structure specifying the video encode profile and quality level to query properties for.
- `pQualityLevelProperties` is a pointer to a VkVideoEncodeQualityLevelPropertiesKHR structure in which the properties are returned.

Valid Usage

- VUID-vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR-pQualityLevelInfo-

08257
If pQualityLevelInfo->pVideoProfile->videoCodecOperation is VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the pNext chain of pQualityLevelProperties **must** include a VkVideoEncodeH264QualityLevelPropertiesKHR structure

- VUID-vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR-pQualityLevelInfo-08258
If pQualityLevelInfo->pVideoProfile->videoCodecOperation is VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the pNext chain of pQualityLevelProperties **must** include a VkVideoEncodeH265QualityLevelPropertiesKHR structure

- VUID-vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR-pQualityLevelInfo-10305
If pQualityLevelInfo->pVideoProfile->videoCodecOperation is VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then the pNext chain of pQualityLevelProperties **must** include a VkVideoEncodeAV1QualityLevelPropertiesKHR structure

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR-physicalDevice-parameter
physicalDevice **must** be a valid VkPhysicalDevice handle

- VUID-vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR-pQualityLevelInfo-parameter
pQualityLevelInfo **must** be a valid pointer to a valid VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR structure

- VUID-vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR-pQualityLevelProperties-parameter
pQualityLevelProperties **must** be a valid pointer to a VkVideoEncodeQualityLevelPropertiesKHR structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED
- VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR

- VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR
- VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR
- VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR

The VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    const VkVideoProfileInfoKHR*    pVideoProfile;
    uint32_t                        qualityLevel;
} VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pVideoProfile is a pointer to a VkVideoProfileInfoKHR structure specifying the video profile to query the video encode quality level properties for.
- qualityLevel is the video encode quality level to query properties for.

Valid Usage

- VUID-VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR-pVideoProfile-08259
  pVideoProfile **must** be a supported video profile
- VUID-VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR-pVideoProfile-08260
  pVideoProfile->videoCodecOperation **must** specify an encode operation
- VUID-VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR-qualityLevel-08261
  qualityLevel **must** be less than VkVideoEncodeCapabilitiesKHR::maxQualityLevels, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile specified in pVideoProfile

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_QUALITY_LEVEL_INFO_KHR
- VUID-VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR-pVideoProfile-parameter
  pVideoProfile **must** be a valid pointer to a valid VkVideoProfileInfoKHR structure

The VkVideoEncodeQualityLevelPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeQualityLevelPropertiesKHR {
    VkStructureType                          sType;
    void*                                    pNext;
    VkVideoEncodeRateControlModeFlagBitsKHR  preferredRateControlMode;
    uint32_t                                 preferredRateControlLayerCount;
} VkVideoEncodeQualityLevelPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- preferredRateControlMode is a VkVideoEncodeRateControlModeFlagBitsKHR value indicating the preferred rate control mode to use with the video encode quality level.
- preferredRateControlLayerCount indicates the preferred number of rate control layers to use with the video encode quality level.

Valid Usage (Implicit)

- VUID-VkVideoEncodeQualityLevelPropertiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUALITY_LEVEL_PROPERTIES_KHR
- VUID-VkVideoEncodeQualityLevelPropertiesKHR-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkVideoEncodeAV1QualityLevelPropertiesKHR, VkVideoEncodeH264QualityLevelPropertiesKHR, or VkVideoEncodeH265QualityLevelPropertiesKHR
- VUID-VkVideoEncodeQualityLevelPropertiesKHR-sType-unique
  The sType value of each structure in the pNext chain **must** be unique

The VkVideoEncodeQualityLevelInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeQualityLevelInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           qualityLevel;
} VkVideoEncodeQualityLevelInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- qualityLevel is the used video encode quality level.

This structure **can** be specified in the following places:

- In the `pNext` chain of VkVideoSessionParametersCreateInfoKHR to specify the video encode quality level to use for a video session parameters object created for a video encode session. If no instance of this structure is included in the `pNext` chain of VkVideoSessionParametersCreateInfoKHR, then the video session parameters object is created with a video encode quality level of zero.
- In the `pNext` chain of VkVideoCodingControlInfoKHR to change the video encode quality level state of the bound video session.

Valid Usage

- VUID-VkVideoEncodeQualityLevelInfoKHR-qualityLevel-08311
  `qualityLevel` **must** be less than VkVideoEncodeCapabilitiesKHR::`maxQualityLevels`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile

Valid Usage (Implicit)

- VUID-VkVideoEncodeQualityLevelInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUALITY_LEVEL_INFO_KHR

Structure Chaining

Extends the structures

- VkVideoCodingControlInfoKHR
- VkVideoSessionParametersCreateInfoKHR

45.16.5. Retrieving Encoded Session Parameters

Any codec-specific parameters stored in video session parameters objects **may** need to be separately encoded and included in the final video bitstream data, depending on the used video compression standard. In such cases the application **must** call the vkGetEncodedVideoSessionParametersKHR command to retrieve the encoded parameter data from the used video session parameters object in order to be able to produce a compliant video bitstream.

NOTE

This is needed because implementations **may** have changed some of the codec-specific parameters stored in the video session parameters object, as defined in the Video Encode Parameter Overrides section. In addition, the vkGetEncodedVideoSessionParametersKHR command enables the application to retrieve the encoded parameter data without having to encode these codec-specific parameters manually.

Encoded parameter data **can** be retrieved from a video session parameters object created with a

video encode operation using the command:

```
// Provided by VK_KHR_video_encode_queue
VkResult vkGetEncodedVideoSessionParametersKHR(
    VkDevice                                    device,
    const VkVideoEncodeSessionParametersGetInfoKHR* pVideoSessionParametersInfo,
    VkVideoEncodeSessionParametersFeedbackInfoKHR* pFeedbackInfo,
    size_t*                                     pDataSize,
    void*                                       pData);
```

- `device` is the logical device that owns the video session parameters object.
- `pVideoSessionParametersInfo` is a pointer to a VkVideoEncodeSessionParametersGetInfoKHR structure specifying the parameters of the encoded parameter data to retrieve.
- `pFeedbackInfo` is either `NULL` or a pointer to a VkVideoEncodeSessionParametersFeedbackInfoKHR structure in which feedback about the requested parameter data is returned.
- `pDataSize` is a pointer to a `size_t` value related to the amount of encode parameter data returned, as described below.
- `pData` is either `NULL` or a pointer to a buffer to write the encoded parameter data to.

If `pData` is `NULL`, then the size of the encoded parameter data, in bytes, that **can** be retrieved is returned in `pDataSize`. Otherwise, `pDataSize` **must** point to a variable set by the application to the size of the buffer, in bytes, pointed to by `pData`, and on return the variable is overwritten with the number of bytes actually written to `pData`. If `pDataSize` is less than the size of the encoded parameter data that **can** be retrieved, then no data will be written to `pData`, zero will be written to `pDataSize`, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that no encoded parameter data was returned.

If `pFeedbackInfo` is not `NULL` then the members of the VkVideoEncodeSessionParametersFeedbackInfoKHR structure and any additional structures included in its `pNext` chain that are applicable to the video session parameters object specified in `pVideoSessionParametersInfo->videoSessionParameters` will be filled with feedback about the requested parameter data on all successful calls to this command.

NOTE

This includes the cases when `pData` is `NULL` or when VK_INCOMPLETE is returned by the command, and enables the application to determine whether the implementation overrode any of the requested video session parameters without actually needing to retrieve the encoded parameter data itself.

NOTE

This query does not behave consistently with the behavior described in Opaque Binary Data Results, for historical reasons.

If the amount of data available is larger than the passed `pDataSize`, the query returns a VK_INCOMPLETE success status instead of a VK_ERROR_NOT_ENOUGH_SPACE_KHR error status, and writes zero to `pDataSize`.

Valid Usage

- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08359
  `pVideoSessionParametersInfo->videoSessionParameters` **must** have been created with an encode operation
- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08262
  If `pVideoSessionParametersInfo->videoSessionParameters` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the `pNext` chain of `pVideoSessionParametersInfo` **must** include a VkVideoEncodeH264SessionParametersGetInfoKHR structure
- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08263
  If `pVideoSessionParametersInfo->videoSessionParameters` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then for the VkVideoEncodeH264SessionParametersGetInfoKHR structure included in the `pNext` chain of `pVideoSessionParametersInfo`, if its `writeStdSPS` member is VK_TRUE, then `pVideoSessionParametersInfo->videoSessionParameters` **must** contain a `StdVideoH264SequenceParameterSet` entry with `seq_parameter_set_id` matching VkVideoEncodeH264SessionParametersGetInfoKHR::`stdSPSId`
- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08264
  If `pVideoSessionParametersInfo->videoSessionParameters` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then for the VkVideoEncodeH264SessionParametersGetInfoKHR structure included in the `pNext` chain of `pVideoSessionParametersInfo`, if its `writeStdPPS` member is VK_TRUE, then `pVideoSessionParametersInfo->videoSessionParameters` **must** contain a `StdVideoH264PictureParameterSet` entry with `seq_parameter_set_id` and `pic_parameter_set_id` matching VkVideoEncodeH264SessionParametersGetInfoKHR::`stdSPSId` and VkVideoEncodeH264SessionParametersGetInfoKHR::`stdPPSId`, respectively
- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08265
  If `pVideoSessionParametersInfo->videoSessionParameters` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the `pNext` chain of `pVideoSessionParametersInfo` **must** include a VkVideoEncodeH265SessionParametersGetInfoKHR structure
- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08266
  If `pVideoSessionParametersInfo->videoSessionParameters` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then for the VkVideoEncodeH265SessionParametersGetInfoKHR structure included in the `pNext` chain of `pVideoSessionParametersInfo`, if its `writeStdVPS` member is VK_TRUE, then `pVideoSessionParametersInfo->videoSessionParameters` **must** contain a `StdVideoH265VideoParameterSet` entry with `vps_video_parameter_set_id` matching VkVideoEncodeH265SessionParametersGetInfoKHR::`stdVPSId`
- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08267
  If `pVideoSessionParametersInfo->videoSessionParameters` was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then for the VkVideoEncodeH265SessionParametersGetInfoKHR structure included in the `pNext` chain of `pVideoSessionParametersInfo`, if its `writeStdSPS` member is VK_TRUE, then

pVideoSessionParametersInfo->videoSessionParameters **must** contain a StdVideoH265SequenceParameterSet entry with sps_video_parameter_set_id and sps_seq_parameter_set_id matching VkVideoEncodeH265SessionParametersGetInfoKHR::stdVPSId and VkVideoEncodeH265SessionParametersGetInfoKHR::stdSPSId, respectively

- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08268
  If pVideoSessionParametersInfo->videoSessionParameters was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then for the VkVideoEncodeH265SessionParametersGetInfoKHR structure included in the pNext chain of pVideoSessionParametersInfo, if its writeStdPPS member is VK_TRUE, then pVideoSessionParametersInfo->videoSessionParameters **must** contain a StdVideoH265PictureParameterSet entry with sps_video_parameter_set_id, pps_seq_parameter_set_id, and pps_pic_parameter_set_id matching VkVideoEncodeH265SessionParametersGetInfoKHR::stdVPSId, VkVideoEncodeH265SessionParametersGetInfoKHR::stdSPSId, and VkVideoEncodeH265SessionParametersGetInfoKHR::stdPPSId, respectively

Valid Usage (Implicit)

- VUID-vkGetEncodedVideoSessionParametersKHR-device-parameter
  device **must** be a valid VkDevice handle
- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-parameter
  pVideoSessionParametersInfo **must** be a valid pointer to a valid VkVideoEncodeSessionParametersGetInfoKHR structure
- VUID-vkGetEncodedVideoSessionParametersKHR-pFeedbackInfo-parameter
  If pFeedbackInfo is not NULL, pFeedbackInfo **must** be a valid pointer to a VkVideoEncodeSessionParametersFeedbackInfoKHR structure
- VUID-vkGetEncodedVideoSessionParametersKHR-pDataSize-parameter
  pDataSize **must** be a valid pointer to a size_t value
- VUID-vkGetEncodedVideoSessionParametersKHR-pData-parameter
  If the value referenced by pDataSize is not 0, and pData is not NULL, pData **must** be a valid pointer to an array of pDataSize bytes

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN

- VK_ERROR_VALIDATION_FAILED

The VkVideoEncodeSessionParametersGetInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeSessionParametersGetInfoKHR {
    VkStructureType                sType;
    const void*                    pNext;
    VkVideoSessionParametersKHR    videoSessionParameters;
} VkVideoEncodeSessionParametersGetInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- videoSessionParameters is the VkVideoSessionParametersKHR object to retrieve encoded parameter data from.

Depending on the used video encode operation, additional codec-specific structures **may** need to be included in the pNext chain of this structure to identify the specific video session parameters to retrieve encoded parameter data for, as described in the corresponding sections.

Valid Usage (Implicit)

- VUID-VkVideoEncodeSessionParametersGetInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_SESSION_PARAMETERS_GET_INFO_KHR
- VUID-VkVideoEncodeSessionParametersGetInfoKHR-pNext-pNext
  Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkVideoEncodeH264SessionParametersGetInfoKHR or VkVideoEncodeH265SessionParametersGetInfoKHR
- VUID-VkVideoEncodeSessionParametersGetInfoKHR-sType-unique
  The sType value of each structure in the pNext chain **must** be unique
- VUID-VkVideoEncodeSessionParametersGetInfoKHR-videoSessionParameters-parameter
  videoSessionParameters **must** be a valid VkVideoSessionParametersKHR handle

The VkVideoEncodeSessionParametersFeedbackInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeSessionParametersFeedbackInfoKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           hasOverrides;
} VkVideoEncodeSessionParametersFeedbackInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `hasOverrides` indicates whether any of the requested parameter data were overridden by the implementation.

Depending on the used video encode operation, additional codec-specific structures **can** be included in the `pNext` chain of this structure to capture codec-specific feedback information about the requested parameter data, as described in the corresponding sections.

Valid Usage (Implicit)

- VUID-VkVideoEncodeSessionParametersFeedbackInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_SESSION_PARAMETERS_FEEDBACK_INFO_KHR
- VUID-VkVideoEncodeSessionParametersFeedbackInfoKHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkVideoEncodeH264SessionParametersFeedbackInfoKHR or VkVideoEncodeH265SessionParametersFeedbackInfoKHR
- VUID-VkVideoEncodeSessionParametersFeedbackInfoKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

45.16.6. Video Encode Commands

To launch video encode operations, call:

```
// Provided by VK_KHR_video_encode_queue
void vkCmdEncodeVideoKHR(
    VkCommandBuffer                             commandBuffer,
    const VkVideoEncodeInfoKHR*                 pEncodeInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pEncodeInfo` is a pointer to a VkVideoEncodeInfoKHR structure specifying the parameters of the video encode operations.

Each call issues one or more video encode operations. The implicit parameter `opCount` corresponds to the number of video encode operations issued by the command. After calling this command, the active query index of each active query is incremented by `opCount`.

Currently each call to this command results in the issue of a single video encode operation.

If the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR and the `pNext` chain of `pEncodeInfo` includes a VkVideoInlineQueryInfoKHR structure with its `queryPool` member specifying a valid `VkQueryPool` handle, then this command will execute a query for each video encode operation

issued by it.

Active Reference Picture Information

The list of active reference pictures used by a video encode operation is a list of image subregions used as the source of reference picture data and related parameters, and is derived from the VkVideoReferenceSlotInfoKHR structures provided as the elements of the `pEncodeInfo->pReferenceSlots` array. For each element of `pEncodeInfo->pReferenceSlots`, one or more elements are added to the active reference picture list, as defined by the codec-specific semantics. Each element of this list contains the following information:

- The image subregion within the image subresource referred to by the video picture resource used as the reference picture.
- The DPB slot index the reference picture is associated with.
- The codec-specific reference information related to the reference picture.

Reconstructed Picture Information

Information related to the optional reconstructed picture used by a video encode operation is derived from the VkVideoReferenceSlotInfoKHR structure pointed to by `pEncodeInfo->pSetupReferenceSlot`, if not `NULL`, as defined by the codec-specific semantics, and consists of the following:

- The image subregion within the image subresource referred to by the video picture resource used as the reconstructed picture.
- The DPB slot index to use for picture reconstruction.
- The codec-specific reference information related to the reconstructed picture.

Specifying a valid VkVideoReferenceSlotInfoKHR structure in `pEncodeInfo->pSetupReferenceSlot` is always required, unless the video session was created with VkVideoSessionCreateInfoKHR::`maxDpbSlots` equal to zero. However, the DPB slot identified by `pEncodeInfo->pSetupReferenceSlot->slotIndex` is only activated with the reconstructed picture specified in `pEncodeInfo->pSetupReferenceSlot->pPictureResource` if reference picture setup is requested according to the codec-specific semantics.

If reconstructed picture information is specified, but reference picture setup is not requested, according to the codec-specific semantics, the contents of the video picture resource corresponding to the reconstructed picture will be undefined after the video encode operation.

NOTE	Some implementations may always output the reconstructed picture or use it as temporary storage during the video encode operation even when the reconstructed picture is not marked for future reference.

Encode Input Picture Information

Information related to the encode input picture used by a video encode operation is derived from `pEncodeInfo->srcPictureResource` and any codec-specific parameters provided in the `pEncodeInfo->pNext` chain, as defined by the codec-specific semantics, and consists of the following:

- The image subregion within the image subresource referred to by the video picture resource used as the encode input picture.
- The codec-specific picture information related to the encoded picture.

Several limiting values are defined below that are referenced by the relevant valid usage statements of this command.

- Let `uint32_t activeReferencePictureCount` be the size of the list of active reference pictures used by the video encode operation. Unless otherwise defined, `activeReferencePictureCount` is set to the value of `pEncodeInfo->referenceSlotCount`.
- Let `VkOffset2D codedOffsetGranularity` be the minimum alignment requirement for the coded offset of video picture resources. Unless otherwise defined, the value of the `x` and `y` members of `codedOffsetGranularity` are `0`.
- Let `uint32_t dpbFrameUseCount[]` be an array of size `maxDpbSlots`, where `maxDpbSlots` is the VkVideoSessionCreateInfoKHR::`maxDpbSlots` the bound video session was created with, with each element indicating the number of times a frame associated with the corresponding DPB slot index is referred to by the video coding operation. Let the initial value of each element of the array be `0`.
  - If `pEncodeInfo->pSetupReferenceSlot` is not `NULL`, then `dpbFrameUseCount[i]` is incremented by one, where `i` equals `pEncodeInfo->pSetupReferenceSlot->slotIndex`.
  - For each element of `pEncodeInfo->pReferenceSlots`, `dpbFrameUseCount[i]` is incremented by one, where `i` equals the `slotIndex` member of the corresponding element.
- If there is a bound video session parameters object created with VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR, then let `VkExtent2D quantizationMapTexelSize` be the quantization map texel size the bound video session parameters object was created with.
- Let `VkExtent2D maxCodingBlockSize` be the maximum codec-specific coding block size that **may** be used by the video encode operation.
  - If the bound video session object was created with an H.264 encode profile, then let `maxCodingBlockSize` be equal to the size of an H.264 macroblock, i.e. `{16,16}`.
  - If the bound video session object was created with an H.265 encode profile, then let `maxCodingBlockSize` be equal to the maximum H.265 coding block size that **may** be used by the video encode operation derived as the maximum of the CTB sizes corresponding to the VkVideoEncodeH265CtbSizeFlagBitsKHR bits set in VkVideoEncodeH265CapabilitiesKHR::`ctbSizes`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with.
  - If the bound video session object was created with an AV1 encode profile, then let `maxCodingBlockSize` be equal to the maximum AV1 superblock size that **may** be used by the video encode operation derived as the maximum of the superblock sizes corresponding to the VkVideoEncodeAV1SuperblockSizeFlagBitsKHR bits set in VkVideoEncodeAV1CapabilitiesKHR::`superblockSizes`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with.
  - Otherwise, `maxCodingBlockSize` is undefined.

- If maxCodingBlockSize is defined, then let VkExtent2D minCodingBlockExtent be the coded extent of the encode input picture expressed in terms of codec-specific coding blocks, assuming the maximum size of such coding blocks, as defined by maxCodingBlockSize, calculated from the value of the codedExtent member of pEncodeInfo->srcPictureResource as follows:
  - minCodingBlockExtent.width = (codedExtent.width maxCodingBlockSize.width - 1) / maxCodingBlockSize.width
  - minCodingBlockExtent.height = (codedExtent.height maxCodingBlockSize.height - 1) / maxCodingBlockSize.height
- If the bound video session object was created with an H.264 encode profile, then:
  - Let StdVideoH264PictureType h264PictureType be the picture type of the encoded picture set to the value of pStdPictureInfo->primary_pic_type specified in the VkVideoEncodeH264PictureInfoKHR structure included in the pEncodeInfo->pNext chain.
  - Let StdVideoH264PictureType h264L0PictureTypes[] and StdVideoH264PictureType h264L1PictureTypes[] be the picture types of the reference pictures in the L0 and L1 reference lists, respectively. If pStdPictureInfo->pRefLists specified in the VkVideoEncodeH264PictureInfoKHR structure included in the pEncodeInfo->pNext chain is not NULL, then for each reference index specified in the elements of the pStdPictureInfo->pRefLists->RefPicList0 and pStdPictureInfo->pRefLists->RefPicList1 arrays, if the reference index is not STD_VIDEO_H264_NO_REFERENCE_PICTURE, pStdReferenceInfo->primary_pic_type is added to h264L0PictureTypes or h264L1PictureTypes, respectively, where pStdReferenceInfo is the member of the VkVideoEncodeH264DpbSlotInfoKHR structure included in the pNext chain of the element of pEncodeInfo->pReferenceSlots for which slotIndex equals the reference index in question.
- If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and with the intra refresh mode VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_PER_PICTURE_PARTITION_BIT_KHR, pEncodeInfo->flags includes VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR, and the pNext chain of pEncodeInfo includes a VkVideoEncodeIntraRefreshInfoKHR structure, then let uint32_t intraRefreshH264SliceIndex be the intra refresh index specified in VkVideoEncodeIntraRefreshInfoKHR::intraRefreshIndex. Otherwise intraRefreshH264SliceIndex is not defined.
- If the bound video session object was created with an H.265 encode profile, then:
  - Let StdVideoH265PictureType h265PictureType be the picture type of the encoded picture set to the value of pStdPictureInfo->pic_type specified in the VkVideoEncodeH265PictureInfoKHR structure included in the pEncodeInfo->pNext chain.
  - Let StdVideoH265PictureType h265L0PictureTypes[] and StdVideoH265PictureType h265L1PictureTypes[] be the picture types of the reference pictures in the L0 and L1 reference lists, respectively. If pStdPictureInfo->pRefLists specified in the VkVideoEncodeH265PictureInfoKHR structure included in the pEncodeInfo->pNext chain is not NULL, then for each reference index specified in the elements of the pStdPictureInfo->pRefLists->RefPicList0 and pStdPictureInfo->pRefLists->RefPicList1 arrays, if the reference index is not STD_VIDEO_H265_NO_REFERENCE_PICTURE, pStdReferenceInfo->pic_type is added to h265L0PictureTypes or h265L1PictureTypes, respectively, where pStdReferenceInfo is the member of the VkVideoEncodeH265DpbSlotInfoKHR structure included in the pNext

chain of the element of pEncodeInfo->pReferenceSlots for which slotIndex equals the reference index in question.

- If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and with the intra refresh mode VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_PER_PICTURE_PARTITION_BIT_KHR, pEncodeInfo->flags includes VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR, and the pNext chain of pEncodeInfo includes a VkVideoEncodeIntraRefreshInfoKHR structure, then let uint32_t intraRefreshH265SliceSegmentIndex be the intra refresh index specified in VkVideoEncodeIntraRefreshInfoKHR::intraRefreshIndex. Otherwise intraRefreshH265SliceSegmentIndex is not defined.
- If the bound video session object was created with an AV1 encode profile, then:
  - If the primaryReferenceCdfOnly member of the VkVideoEncodeAV1PictureInfoKHR structure included in the pEncodeInfo->pNext chain is set to VK_TRUE, then let int32_t cdfOnlyReferenceIndex be the value of VkVideoEncodeAV1PictureInfoKHR::pStdPictureInfo->primary_ref_frame.
  - Otherwise let int32_t cdfOnlyReferenceIndex be -1.

Valid Usage

- VUID-vkCmdEncodeVideoKHR-None-08250
  The bound video session **must** have been created with an encode operation
- VUID-vkCmdEncodeVideoKHR-None-07012
  The bound video session **must** not be in uninitialized state at the time the command is executed on the device
- VUID-vkCmdEncodeVideoKHR-None-08318
  If there is a bound video session parameters object, then it **must** have been created with the currently set video encode quality level for the bound video session at the time the command is executed on the device
- VUID-vkCmdEncodeVideoKHR-opCount-07174
  For each active query, the active query index corresponding to the query type of that query plus opCount **must** be less than or equal to the last activatable query index corresponding to the query type of that query plus one
- VUID-vkCmdEncodeVideoKHR-pNext-08360
  If the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR, and the pNext chain of pEncodeInfo includes a VkVideoInlineQueryInfoKHR structure with its queryPool member specifying a valid VkQueryPool handle, then VkVideoInlineQueryInfoKHR::queryCount **must** equal opCount
- VUID-vkCmdEncodeVideoKHR-pNext-08361
  If the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR, and the pNext chain of pEncodeInfo includes a VkVideoInlineQueryInfoKHR structure with its queryPool member specifying a valid VkQueryPool handle, then all the queries used by the command, as specified by the VkVideoInlineQueryInfoKHR structure, **must** be *unavailable*

- VUID-vkCmdEncodeVideoKHR-queryType-08362
  If the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR, then the queryType used to create the queryPool specified in the VkVideoInlineQueryInfoKHR structure included in the pNext chain of pEncodeInfo **must** be VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR or VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR
- VUID-vkCmdEncodeVideoKHR-queryPool-08363
  If the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR, then the queryPool specified in the VkVideoInlineQueryInfoKHR structure included in the pNext chain of pEncodeInfo **must** have been created with a VkVideoProfileInfoKHR structure included in the pNext chain of VkQueryPoolCreateInfo identical to the one specified in VkVideoSessionCreateInfoKHR::pVideoProfile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-queryType-08364
  If the bound video session was created with VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR, and the queryType used to create the queryPool specified in the VkVideoInlineQueryInfoKHR structure included in the pNext chain of pEncodeInfo is VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR, then the VkCommandPool that commandBuffer was allocated from **must** have been created with a queue family index that supports result status queries, as indicated by VkQueueFamilyQueryResultStatusPropertiesKHR::queryResultStatusSupport
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08201
  pEncodeInfo->dstBuffer **must** be compatible with the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-commandBuffer-08202
  If commandBuffer is an unprotected command buffer and protectedNoFault is not supported, then pEncodeInfo->dstBuffer **must** not be a protected buffer
- VUID-vkCmdEncodeVideoKHR-commandBuffer-08203
  If commandBuffer is a protected command buffer and protectedNoFault is not supported, then pEncodeInfo->dstBuffer **must** be a protected buffer
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08204
  pEncodeInfo->dstBufferOffset **must** be an integer multiple of VkVideoCapabilitiesKHR::minBitstreamBufferOffsetAlignment, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08205
  pEncodeInfo->dstBufferRange **must** be an integer multiple of VkVideoCapabilitiesKHR::minBitstreamBufferSizeAlignment, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08206
  pEncodeInfo->srcPictureResource.imageViewBinding **must** be compatible with the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08207

The format of pEncodeInfo->srcPictureResource.imageViewBinding **must** match the VkVideoSessionCreateInfoKHR::pictureFormat the bound video session was created with

- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08208
  pEncodeInfo->srcPictureResource.codedOffset **must** be an integer multiple of codedOffsetGranularity
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08209
  pEncodeInfo->srcPictureResource.codedExtent **must** be between minCodedExtent and maxCodedExtent, inclusive, the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08210
  pEncodeInfo->srcPictureResource.imageViewBinding **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR usage flag set
- VUID-vkCmdEncodeVideoKHR-commandBuffer-08211
  If commandBuffer is an unprotected command buffer and protectedNoFault is not supported, then pEncodeInfo->srcPictureResource.imageViewBinding **must** not have been created from a protected image
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08377
  pEncodeInfo->pSetupReferenceSlot **must** not be NULL unless the bound video session was created with VkVideoSessionCreateInfoKHR::maxDpbSlots equal to zero
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08213
  If pEncodeInfo->pSetupReferenceSlot is not NULL, then pEncodeInfo->pSetupReferenceSlot->slotIndex **must** be less than the VkVideoSessionCreateInfoKHR::maxDpbSlots specified when the bound video session was created
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08214
  If pEncodeInfo->pSetupReferenceSlot is not NULL, then pEncodeInfo->pSetupReferenceSlot->pPictureResource->codedOffset **must** be an integer multiple of codedOffsetGranularity
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08215
  If pEncodeInfo->pSetupReferenceSlot is not NULL, then pEncodeInfo->pSetupReferenceSlot->pPictureResource **must** match one of the bound reference picture resources
- VUID-vkCmdEncodeVideoKHR-activeReferencePictureCount-08216
  activeReferencePictureCount **must** be less than or equal to the VkVideoSessionCreateInfoKHR::maxActiveReferencePictures specified when the bound video session was created
- VUID-vkCmdEncodeVideoKHR-slotIndex-08217
  The slotIndex member of each element of pEncodeInfo->pReferenceSlots **must** be less than the VkVideoSessionCreateInfoKHR::maxDpbSlots specified when the bound video session was created
- VUID-vkCmdEncodeVideoKHR-codedOffset-08218
  The codedOffset member of the VkVideoPictureResourceInfoKHR structure pointed to by the pPictureResource member of each element of pEncodeInfo->pReferenceSlots **must** be an integer multiple of codedOffsetGranularity
- VUID-vkCmdEncodeVideoKHR-pPictureResource-08219
  The pPictureResource member of each element of pEncodeInfo->pReferenceSlots **must** match one of the bound reference picture resources associated with the DPB slot index

specified in the slotIndex member of that element

- VUID-vkCmdEncodeVideoKHR-pPictureResource-08220
  Each video picture resource corresponding to the pPictureResource member specified in the elements of pEncodeInfo->pReferenceSlots **must** be unique within pEncodeInfo->pReferenceSlots
- VUID-vkCmdEncodeVideoKHR-dpbFrameUseCount-08221
  All elements of dpbFrameUseCount **must** be less than or equal to 1
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10811
  The image subresource referred to by pEncodeInfo->srcPictureResource **must** be in the VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR layout at the time the video encode operation is executed on the device , unless the unifiedImageLayoutsVideo feature is enabled, in which case it **may** be in the VK_IMAGE_LAYOUT_GENERAL layout
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10812
  If pEncodeInfo->pSetupReferenceSlot is not NULL, then the image subresource referred to by pEncodeInfo->pSetupReferenceSlot->pPictureResource **must** be in the VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR layout at the time the video encode operation is executed on the device , unless the unifiedImageLayoutsVideo feature is enabled, in which case it **may** be in the VK_IMAGE_LAYOUT_GENERAL layout
- VUID-vkCmdEncodeVideoKHR-pPictureResource-10813
  The image subresource referred to by the pPictureResource member of each element of pEncodeInfo->pReferenceSlots **must** be in the VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR layout at the time the video encode operation is executed on the device , unless the unifiedImageLayoutsVideo feature is enabled, in which case it **may** be in the VK_IMAGE_LAYOUT_GENERAL layout
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10306
  If pEncodeInfo->flags includes VK_VIDEO_ENCODE_WITH_QUANTIZATION_DELTA_MAP_BIT_KHR, then the bound video session **must** have been created with VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10307
  If pEncodeInfo->flags includes VK_VIDEO_ENCODE_WITH_EMPHASIS_MAP_BIT_KHR, then the bound video session **must** have been created with VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_EMPHASIS_MAP_BIT_KHR
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10308
  If the current rate control mode is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR or VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, then pEncodeInfo->flags **must** not include VK_VIDEO_ENCODE_WITH_EMPHASIS_MAP_BIT_KHR
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10309
  If pEncodeInfo->flags includes VK_VIDEO_ENCODE_WITH_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_VIDEO_ENCODE_WITH_EMPHASIS_MAP_BIT_KHR, then the pNext chain of pEncodeInfo **must** include a VkVideoEncodeQuantizationMapInfoKHR structure with its quantizationMap member specifying a valid VkImageView handle

- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10310
  If pEncodeInfo->flags includes VK_VIDEO_ENCODE_WITH_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_VIDEO_ENCODE_WITH_EMPHASIS_MAP_BIT_KHR, then the VkImageView specified by the quantizationMap member of the VkVideoEncodeQuantizationMapInfoKHR structure included in the pNext chain **must** be compatible with the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10311
  If pEncodeInfo->flags includes VK_VIDEO_ENCODE_WITH_QUANTIZATION_DELTA_MAP_BIT_KHR, then the VkImageView specified by the quantizationMap member of the VkVideoEncodeQuantizationMapInfoKHR structure included in the pNext chain of pEncodeInfo **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR usage flag set
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10312
  If pEncodeInfo->flags includes VK_VIDEO_ENCODE_WITH_EMPHASIS_MAP_BIT_KHR, then the VkImageView specified by the quantizationMap member of the VkVideoEncodeQuantizationMapInfoKHR structure included in the pNext chain of pEncodeInfo **must** have been created with the VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR usage flag set
- VUID-vkCmdEncodeVideoKHR-pNext-10313
  If an instance of the VkVideoEncodeQuantizationMapInfoKHR structure is included in the pNext chain of pEncodeInfo, its quantizationMap member is not VK_NULL_HANDLE, commandBuffer is an unprotected command buffer, and protectedNoFault is not supported, then VkVideoEncodeQuantizationMapInfoKHR::quantizationMap **must** not have been created from a protected image
- VUID-vkCmdEncodeVideoKHR-pNext-10314
  If an instance of the VkVideoEncodeQuantizationMapInfoKHR structure is included in the pNext chain of pEncodeInfo and its quantizationMap member is not VK_NULL_HANDLE, then the image subresource range referenced by quantizationMap **must** be in the VK_IMAGE_LAYOUT_VIDEO_ENCODE_QUANTIZATION_MAP_KHR layout at the time the video encode operation is executed on the device , unless the unifiedImageLayoutsVideo feature is enabled, in which case it **may** be in the VK_IMAGE_LAYOUT_GENERAL layout
- VUID-vkCmdEncodeVideoKHR-pNext-10315
  If an instance of the VkVideoEncodeQuantizationMapInfoKHR structure is included in the pNext chain of pEncodeInfo and its quantizationMap member is not VK_NULL_HANDLE, and there is a bound video session parameters object, then the bound video session parameters object **must** have been created with VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR
- VUID-vkCmdEncodeVideoKHR-pNext-10316
  If an instance of the VkVideoEncodeQuantizationMapInfoKHR structure is included in the pNext chain of pEncodeInfo, its quantizationMap member is not VK_NULL_HANDLE, and there is a bound video session parameters object created with VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR, then quantizationMapExtent **must** equal □pEncodeInfo->srcPictureResource.codedExtent /

quantizationMapTexelSize

- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10837
  If pEncodeInfo->flags includes VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR, then the intra refresh mode the bound video session was created with **must** not be VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_NONE_KHR
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10838
  If pEncodeInfo->flags includes VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR, then activeReferencePictureCount **must** be less than or equal to VkVideoEncodeIntraRefreshCapabilitiesKHR::maxIntraRefreshActiveReferencePictures, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10839
  If pEncodeInfo->flags includes VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR, then the pNext chain of pEncodeInfo **must** include a VkVideoEncodeIntraRefreshInfoKHR structure
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10840
  If pEncodeInfo->flags includes VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR, then the intraRefreshCycleDuration member of the VkVideoEncodeIntraRefreshInfoKHR structure included in the pNext chain of pEncodeInfo **must** not be zero
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10841
  If pEncodeInfo->flags includes VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR, then the intraRefreshIndex member of the VkVideoEncodeIntraRefreshInfoKHR structure included in the pNext chain of pEncodeInfo **must** be less than VkVideoEncodeIntraRefreshInfoKHR::intraRefreshCycleDuration
- VUID-vkCmdEncodeVideoKHR-pNext-10842
  If the pNext chain of any element of pEncodeInfo->pReferenceSlots includes a VkVideoReferenceIntraRefreshInfoKHR structure with a non-zero dirtyIntraRefreshRegions member, then pEncodeInfo->flags **must** include VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR
- VUID-vkCmdEncodeVideoKHR-pNext-10843
  If the pNext chain of any element of pEncodeInfo->pReferenceSlots includes a VkVideoReferenceIntraRefreshInfoKHR structure with a non-zero dirtyIntraRefreshRegions member, then for each such element dirtyIntraRefreshRegions **must** equal intraRefreshCycleDuration minus intraRefreshIndex where intraRefreshCycleDuration and intraRefreshIndex are the members of the VkVideoEncodeIntraRefreshInfoKHR structure included in the pNext chain of pEncodeInfo
- VUID-vkCmdEncodeVideoKHR-pNext-10844
  If the pNext chain of pEncodeInfo includes a VkVideoEncodeIntraRefreshInfoKHR structure, then its intraRefreshCycleDuration member **must** be zero or **must** be between 2 and VkVideoEncodeIntraRefreshCapabilitiesKHR::maxIntraRefreshCycleDuration, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pNext-08225
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the pNext chain of

pEncodeInfo **must** include a VkVideoEncodeH264PictureInfoKHR structure

- VUID-vkCmdEncodeVideoKHR-StdVideoH264SequenceParameterSet-08226
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the bound video session parameters object **must** contain a StdVideoH264SequenceParameterSet entry with seq_parameter_set_id matching StdVideoEncodeH264PictureInfo::seq_parameter_set_id that is provided in the pStdPictureInfo member of the VkVideoEncodeH264PictureInfoKHR structure included in the pNext chain of pEncodeInfo
- VUID-vkCmdEncodeVideoKHR-StdVideoH264PictureParameterSet-08227
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the bound video session parameters object **must** contain a StdVideoH264PictureParameterSet entry with seq_parameter_set_id and pic_parameter_set_id matching StdVideoEncodeH264PictureInfo::seq_parameter_set_id and StdVideoEncodeH264PictureInfo::pic_parameter_set_id, respectively, that are provided in the pStdPictureInfo member of the VkVideoEncodeH264PictureInfoKHR structure included in the pNext chain of pEncodeInfo
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08228
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and pEncodeInfo->pSetupReferenceSlot is not NULL, then the pNext chain of pEncodeInfo->pSetupReferenceSlot **must** include a VkVideoEncodeH264DpbSlotInfoKHR structure
- VUID-vkCmdEncodeVideoKHR-pNext-08229
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the pNext chain of each element of pEncodeInfo->pReferenceSlots **must** include a VkVideoEncodeH264DpbSlotInfoKHR structure
- VUID-vkCmdEncodeVideoKHR-constantQp-08269
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and the current rate control mode is not VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, then VkVideoEncodeH264NaluSliceInfoKHR::constantQp **must** be zero for each element of the pNaluSliceEntries member of the VkVideoEncodeH264PictureInfoKHR structure included in the pNext chain of pEncodeInfo
- VUID-vkCmdEncodeVideoKHR-constantQp-08270
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and the current rate control mode is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, then VkVideoEncodeH264NaluSliceInfoKHR::constantQp **must** be between VkVideoEncodeH264CapabilitiesKHR::minQp and VkVideoEncodeH264CapabilitiesKHR::maxQp, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, for each element of the pNaluSliceEntries member of the VkVideoEncodeH264PictureInfoKHR structure included in the pNext chain of pEncodeInfo
- VUID-vkCmdEncodeVideoKHR-constantQp-08271
  If the bound video session was created with the video codec operation

VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and VkVideoEncodeH264CapabilitiesKHR::flags does not include VK_VIDEO_ENCODE_H264_CAPABILITY_PER_SLICE_CONSTANT_QP_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then VkVideoEncodeH264NaluSliceInfoKHR::constantQp **must** have the same value for each element of the pNaluSliceEntries member of the VkVideoEncodeH264PictureInfoKHR structure included in the pNext chain of pEncodeInfo

- VUID-vkCmdEncodeVideoKHR-naluSliceEntryCount-08302
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then the naluSliceEntryCount member of the VkVideoEncodeH264PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** be less than or equal to minCodingBlockExtent.width multiplied by minCodingBlockExtent.height
- VUID-vkCmdEncodeVideoKHR-naluSliceEntryCount-08312
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and VkVideoEncodeH264CapabilitiesKHR::flags does not include VK_VIDEO_ENCODE_H264_CAPABILITY_ROW_UNALIGNED_SLICE_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then the naluSliceEntryCount member of the VkVideoEncodeH264PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** be less than or equal to minCodingBlockExtent.height
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10845
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and with an intra refresh mode other than VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_PER_PICTURE_PARTITION_BIT_KHR, pEncodeInfo->flags includes VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR, and VkVideoEncodeIntraRefreshCapabilitiesKHR::partitionIndependentIntraRefreshRegions is VK_FALSE, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then the naluSliceEntryCount member of the VkVideoEncodeH264PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** equal 1
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10846
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and with the intra refresh mode VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_PER_PICTURE_PARTITION_BIT_KHR, and pEncodeInfo->flags includes VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR, then the naluSliceEntryCount member of the VkVideoEncodeH264PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** equal the intraRefreshCycleDuration member of the VkVideoEncodeIntraRefreshInfoKHR structure included in the pNext chain of pEncodeInfo
- VUID-vkCmdEncodeVideoKHR-intraRefreshH264SliceIndex-10847
  If intraRefreshH264SliceIndex is defined, then pNaluSliceEntries[intraRefreshH264SliceIndex].pStdSliceHeader->slice_type **must** be STD_VIDEO_H264_SLICE_TYPE_I in the VkVideoEncodeH264PictureInfoKHR structure

included in the pNext chain of pEncodeInfo

- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10848
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and with the intra refresh mode VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_PER_PICTURE_PARTITION_BIT_KHR, pEncodeInfo->flags includes VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR, and VkVideoEncodeIntraRefreshCapabilitiesKHR::nonRectangularIntraRefreshRegions is VK_FALSE, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then the naluSliceEntryCount member of the VkVideoEncodeH264PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** be less than or equal to minCodingBlockExtent.height
- VUID-vkCmdEncodeVideoKHR-h264PictureType-10849
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, h264PictureType is STD_VIDEO_H264_PICTURE_TYPE_B, and VkVideoEncodeH264CapabilitiesKHR::flags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, does not include VK_VIDEO_ENCODE_H264_CAPABILITY_B_PICTURE_INTRA_REFRESH_BIT_KHR, then pEncodeInfo->flags **must** not include VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR
- VUID-vkCmdEncodeVideoKHR-flags-10850
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and VkVideoEncodeH264CapabilitiesKHR::flags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, does not include VK_VIDEO_ENCODE_H264_CAPABILITY_DIFFERENT_SLICE_TYPE_BIT_KHR, then pNaluSliceEntries[i].pStdSliceHeader->slice_type **must** be identical for all elements i of the pNaluSliceEntries member of the VkVideoEncodeH264PictureInfoKHR structure included in the pNext chain of pEncodeInfo , except for element index i equal to intraRefreshH264SliceIndex, if intraRefreshH264SliceIndex is defined
- VUID-vkCmdEncodeVideoKHR-pNext-08352
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, the pNext chain of pEncodeInfo includes a VkVideoEncodeH264PictureInfoKHR structure, and pEncodeInfo->referenceSlotCount is greater than zero, then VkVideoEncodeH264PictureInfoKHR::pStdPictureInfo->pRefLists **must** not be NULL
- VUID-vkCmdEncodeVideoKHR-pNext-08339
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, the pNext chain of pEncodeInfo includes a VkVideoEncodeH264PictureInfoKHR structure, and VkVideoEncodeH264PictureInfoKHR::pStdPictureInfo->pRefLists is not NULL, then each element of the RefPicList0 and RefPicList1 array members of the StdVideoEncodeH264ReferenceListsInfo structure pointed to by VkVideoEncodeH264PictureInfoKHR::pStdPictureInfo->pRefLists **must** either be STD_VIDEO_H264_NO_REFERENCE_PICTURE or **must** equal the slotIndex member of one of the elements of pEncodeInfo->pReferenceSlots

- VUID-vkCmdEncodeVideoKHR-pNext-08353
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, the pNext chain of pEncodeInfo includes a VkVideoEncodeH264PictureInfoKHR structure, and pEncodeInfo->referenceSlotCount is greater than zero, then the slotIndex member of each element of pEncodeInfo->pReferenceSlots **must** equal one of the elements of the RefPicList0 or RefPicList1 array members of the StdVideoEncodeH264ReferenceListsInfo structure pointed to by VkVideoEncodeH264PictureInfoKHR::pStdPictureInfo->pRefLists
- VUID-vkCmdEncodeVideoKHR-maxPPictureL0ReferenceCount-08340
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and VkVideoEncodeH264CapabilitiesKHR::maxPPictureL0ReferenceCount is zero, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then h264PictureType and each element of h264L0PictureTypes and h264L1PictureTypes **must** not be STD_VIDEO_H264_PICTURE_TYPE_P
- VUID-vkCmdEncodeVideoKHR-maxBPictureL0ReferenceCount-08341
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and VkVideoEncodeH264CapabilitiesKHR::maxBPictureL0ReferenceCount and VkVideoEncodeH264CapabilitiesKHR::maxL1ReferenceCount are both zero, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then h264PictureType and each element of h264L0PictureTypes and h264L1PictureTypes **must** not be STD_VIDEO_H264_PICTURE_TYPE_B
- VUID-vkCmdEncodeVideoKHR-flags-08342
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and VkVideoEncodeH264CapabilitiesKHR::flags does not include VK_VIDEO_ENCODE_H264_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then each element of h264L0PictureTypes **must** not be STD_VIDEO_H264_PICTURE_TYPE_B
- VUID-vkCmdEncodeVideoKHR-flags-08343
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and VkVideoEncodeH264CapabilitiesKHR::flags does not include VK_VIDEO_ENCODE_H264_CAPABILITY_B_FRAME_IN_L1_LIST_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then each element of h264L1PictureTypes **must** not be STD_VIDEO_H264_PICTURE_TYPE_B
- VUID-vkCmdEncodeVideoKHR-pNext-08230
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the pNext chain of pEncodeInfo **must** include a VkVideoEncodeH265PictureInfoKHR structure
- VUID-vkCmdEncodeVideoKHR-StdVideoH265VideoParameterSet-08231
  If the bound video session was created with the video codec operation

VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the bound video session parameters object **must** contain a StdVideoH265VideoParameterSet entry with vps_video_parameter_set_id matching StdVideoEncodeH265PictureInfo::sps_video_parameter_set_id that is provided in the pStdPictureInfo member of the VkVideoEncodeH265PictureInfoKHR structure included in the pNext chain of pEncodeInfo

- VUID-vkCmdEncodeVideoKHR-StdVideoH265SequenceParameterSet-08232
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the bound video session parameters object **must** contain a StdVideoH265SequenceParameterSet entry with sps_video_parameter_set_id and sps_seq_parameter_set_id matching StdVideoEncodeH265PictureInfo::sps_video_parameter_set_id and StdVideoEncodeH265PictureInfo::pps_seq_parameter_set_id, respectively, that are provided in the pStdPictureInfo member of the VkVideoEncodeH265PictureInfoKHR structure included in the pNext chain of pEncodeInfo
- VUID-vkCmdEncodeVideoKHR-StdVideoH265PictureParameterSet-08233
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the bound video session parameters object **must** contain a StdVideoH265PictureParameterSet entry with sps_video_parameter_set_id, pps_seq_parameter_set_id, and pps_pic_parameter_set_id matching StdVideoEncodeH265PictureInfo::sps_video_parameter_set_id, StdVideoEncodeH265PictureInfo::pps_seq_parameter_set_id, and StdVideoEncodeH265PictureInfo::pps_pic_parameter_set_id, respectively, that are provided in the pStdPictureInfo member of the VkVideoEncodeH265PictureInfoKHR structure included in the pNext chain of pEncodeInfo
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08234
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and pEncodeInfo->pSetupReferenceSlot is not NULL, then the pNext chain of pEncodeInfo->pSetupReferenceSlot **must** include a VkVideoEncodeH265DpbSlotInfoKHR structure
- VUID-vkCmdEncodeVideoKHR-pNext-08235
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the pNext chain of each element of pEncodeInfo->pReferenceSlots **must** include a VkVideoEncodeH265DpbSlotInfoKHR structure
- VUID-vkCmdEncodeVideoKHR-constantQp-08272
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and the current rate control mode is not VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, then VkVideoEncodeH265NaluSliceSegmentInfoKHR::constantQp **must** be zero for each element of the pNaluSliceSegmentEntries member of the VkVideoEncodeH265PictureInfoKHR structure included in the pNext chain of pEncodeInfo
- VUID-vkCmdEncodeVideoKHR-constantQp-08273
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and the current rate control mode is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, then

VkVideoEncodeH265NaluSliceSegmentInfoKHR::constantQp **must** be between VkVideoEncodeH265CapabilitiesKHR::minQp and VkVideoEncodeH265CapabilitiesKHR::maxQp, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, for each element of the pNaluSliceSegmentEntries member of the VkVideoEncodeH265PictureInfoKHR structure included in the pNext chain of pEncodeInfo

- VUID-vkCmdEncodeVideoKHR-constantQp-08274
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and VkVideoEncodeH265CapabilitiesKHR::flags does not include VK_VIDEO_ENCODE_H265_CAPABILITY_PER_SLICE_SEGMENT_CONSTANT_QP_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then VkVideoEncodeH265NaluSliceSegmentInfoKHR::constantQp **must** have the same value for each element of the pNaluSliceSegmentEntries member of the VkVideoEncodeH265PictureInfoKHR structure included in the pNext chain of pEncodeInfo
- VUID-vkCmdEncodeVideoKHR-naluSliceSegmentEntryCount-08307
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then the naluSliceSegmentEntryCount member of the VkVideoEncodeH265PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** be less than or equal to minCodingBlockExtent.width multiplied by minCodingBlockExtent.height
- VUID-vkCmdEncodeVideoKHR-naluSliceSegmentEntryCount-08313
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and VkVideoEncodeH265CapabilitiesKHR::flags does not include VK_VIDEO_ENCODE_H265_CAPABILITY_ROW_UNALIGNED_SLICE_SEGMENT_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then the naluSliceSegmentEntryCount member of the VkVideoEncodeH265PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** be less than or equal to minCodingBlockExtent.height
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10851
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and with an intra refresh mode other than VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_PER_PICTURE_PARTITION_BIT_KHR, pEncodeInfo->flags includes VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR, and VkVideoEncodeIntraRefreshCapabilitiesKHR::partitionIndependentIntraRefreshRegions is VK_FALSE, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then the naluSliceSegmentEntryCount member of the VkVideoEncodeH265PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** equal 1
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10852
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and with the intra refresh mode VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_PER_PICTURE_PARTITION_BIT_KHR, and

pEncodeInfo->flags includes VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR, then the naluSliceSegmentEntryCount member of the VkVideoEncodeH265PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** equal VkVideoEncodeIntraRefreshInfoKHR::intraRefreshCycleDuration

- VUID-vkCmdEncodeVideoKHR-intraRefreshH265SliceSegmentIndex-10853
  If intraRefreshH265SliceSegmentIndex is defined, then pNaluSliceSegmentEntries[intraRefreshH265SliceSegmentIndex].pStdSliceSegmentHeader->slice_type **must** be STD_VIDEO_H265_SLICE_TYPE_I in the VkVideoEncodeH265PictureInfoKHR structure included in the pNext chain of pEncodeInfo
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10854
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and with the intra refresh mode VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_PER_PICTURE_PARTITION_BIT_KHR, pEncodeInfo->flags includes VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR, and VkVideoEncodeIntraRefreshCapabilitiesKHR::nonRectangularIntraRefreshRegions is VK_FALSE, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then the naluSliceSegmentEntryCount member of the VkVideoEncodeH265PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** be less than or equal to minCodingBlockExtent.height
- VUID-vkCmdEncodeVideoKHR-h265PictureType-10855
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, h265PictureType is STD_VIDEO_H265_PICTURE_TYPE_B, and VkVideoEncodeH265CapabilitiesKHR::flags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, does not include VK_VIDEO_ENCODE_H265_CAPABILITY_B_PICTURE_INTRA_REFRESH_BIT_KHR, then pEncodeInfo->flags **must** not include VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR
- VUID-vkCmdEncodeVideoKHR-flags-10856
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and VkVideoEncodeH265CapabilitiesKHR::flags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_H265_CAPABILITY_DIFFERENT_SLICE_SEGMENT_TYPE_BIT_KHR, then pNaluSliceSegmentEntries[i].pStdSliceSegmentHeader->slice_type **must** be identical for all elements i of the pNaluSliceSegmentEntries member of the VkVideoEncodeH265PictureInfoKHR structure included in the pNext chain of pEncodeInfo , except for element index i equal to intraRefreshH265SliceSegmentIndex, if intraRefreshH265SliceSegmentIndex is defined
- VUID-vkCmdEncodeVideoKHR-pNext-08354
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, the pNext chain of pEncodeInfo includes a VkVideoEncodeH265PictureInfoKHR structure, and pEncodeInfo->referenceSlotCount is greater than zero, then VkVideoEncodeH265PictureInfoKHR::pStdPictureInfo->pRefLists **must** not be NULL
- VUID-vkCmdEncodeVideoKHR-pNext-08344

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, the pNext chain of pEncodeInfo includes a VkVideoEncodeH265PictureInfoKHR structure, and VkVideoEncodeH265PictureInfoKHR::pStdPictureInfo->pRefLists is not NULL, then each element of the RefPicList0 and RefPicList1 array members of the StdVideoEncodeH265ReferenceListsInfo structure pointed to by VkVideoEncodeH265PictureInfoKHR::pStdPictureInfo->pRefLists **must** either be STD_VIDEO_H265_NO_REFERENCE_PICTURE or **must** equal the slotIndex member of one of the elements of pEncodeInfo->pReferenceSlots

- VUID-vkCmdEncodeVideoKHR-pNext-08355
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, the pNext chain of pEncodeInfo includes a VkVideoEncodeH265PictureInfoKHR structure, and pEncodeInfo->referenceSlotCount is greater than zero, then the slotIndex member of each element of pEncodeInfo->pReferenceSlots **must** equal one of the elements of the RefPicList0 or RefPicList1 array members of the StdVideoEncodeH265ReferenceListsInfo structure pointed to by VkVideoEncodeH265PictureInfoKHR::pStdPictureInfo->pRefLists
- VUID-vkCmdEncodeVideoKHR-maxPPictureL0ReferenceCount-08345
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and VkVideoEncodeH265CapabilitiesKHR::maxPPictureL0ReferenceCount is zero, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then h265PictureType and each element of h265L0PictureTypes and h265L1PictureTypes **must** not be STD_VIDEO_H265_PICTURE_TYPE_P
- VUID-vkCmdEncodeVideoKHR-maxBPictureL0ReferenceCount-08346
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and VkVideoEncodeH265CapabilitiesKHR::maxBPictureL0ReferenceCount and VkVideoEncodeH265CapabilitiesKHR::maxL1ReferenceCount are both zero, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then h265PictureType and each element of h265L0PictureTypes and h265L1PictureTypes **must** not be STD_VIDEO_H265_PICTURE_TYPE_B
- VUID-vkCmdEncodeVideoKHR-flags-08347
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and VkVideoEncodeH265CapabilitiesKHR::flags does not include VK_VIDEO_ENCODE_H265_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then each element of h265L0PictureTypes **must** not be STD_VIDEO_H265_PICTURE_TYPE_B
- VUID-vkCmdEncodeVideoKHR-flags-08348
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR and VkVideoEncodeH265CapabilitiesKHR::flags does not include VK_VIDEO_ENCODE_H265_CAPABILITY_B_FRAME_IN_L1_LIST_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session

was created with, then each element of h265L1PictureTypes **must** not be STD_VIDEO_H265_PICTURE_TYPE_B

- VUID-vkCmdEncodeVideoKHR-pNext-10317
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then the pNext chain of pEncodeInfo **must** include a VkVideoEncodeAV1PictureInfoKHR structure
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10318
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and pEncodeInfo->pSetupReferenceSlot is not NULL, then the pNext chain of pEncodeInfo->pSetupReferenceSlot **must** include a VkVideoEncodeAV1DpbSlotInfoKHR structure
- VUID-vkCmdEncodeVideoKHR-pNext-10319
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then the pNext chain of each element of pEncodeInfo->pReferenceSlots **must** include a VkVideoEncodeAV1DpbSlotInfoKHR structure
- VUID-vkCmdEncodeVideoKHR-constantQIndex-10320
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the current rate control mode is not VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, then the constantQIndex member of the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** be zero
- VUID-vkCmdEncodeVideoKHR-constantQIndex-10321
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the current rate control mode is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, then the constantQIndex member of the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** be between VkVideoEncodeAV1CapabilitiesKHR::minQIndex and VkVideoEncodeAV1CapabilitiesKHR::maxQIndex, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-flags-10322
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and VkVideoEncodeAV1CapabilitiesKHR::flags does not include VK_VIDEO_ENCODE_AV1_CAPABILITY_FRAME_SIZE_OVERRIDE_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then VkVideoEncodeAV1PictureInfoKHR::pStdPictureInfo->flags.frame_size_override_flag **must** be zero for the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo
- VUID-vkCmdEncodeVideoKHR-flags-10323
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and VkVideoEncodeAV1CapabilitiesKHR::flags does not include VK_VIDEO_ENCODE_AV1_CAPABILITY_FRAME_SIZE_OVERRIDE_BIT_KHR, as returned by

vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then pEncodeInfo->srcPictureResource.codedExtent.width **must** equal StdVideoAV1SequenceHeader::max_frame_width_minus_1 + 1 of the active AV1 sequence header

- VUID-vkCmdEncodeVideoKHR-flags-10324
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and VkVideoEncodeAV1CapabilitiesKHR::flags does not include VK_VIDEO_ENCODE_AV1_CAPABILITY_FRAME_SIZE_OVERRIDE_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then pEncodeInfo->srcPictureResource.codedExtent.height **must** equal StdVideoAV1SequenceHeader::max_frame_height_minus_1 + 1 of the active AV1 sequence header
- VUID-vkCmdEncodeVideoKHR-flags-10325
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and VkVideoEncodeAV1CapabilitiesKHR::flags does not include VK_VIDEO_ENCODE_AV1_CAPABILITY_MOTION_VECTOR_SCALING_BIT_KHR, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then for each element i of pEncodeInfo->pReferenceSlots pEncodeInfo->pReferenceSlots[i].pPictureResource->codedExtent **must** match pEncodeInfo->srcPictureResource.codedExtent
- VUID-vkCmdEncodeVideoKHR-predictionMode-10326
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the predictionMode member of the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo is VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_INTRA_ONLY_KHR, then VkVideoEncodeAV1PictureInfoKHR::pStdPictureInfo->frame_type **must** be STD_VIDEO_AV1_FRAME_TYPE_KEY or STD_VIDEO_AV1_FRAME_TYPE_INTRA_ONLY
- VUID-vkCmdEncodeVideoKHR-pStdPictureInfo-10327
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and pStdPictureInfo->frame_type for the pStdPictureInfo member of the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo is STD_VIDEO_AV1_FRAME_TYPE_KEY or STD_VIDEO_AV1_FRAME_TYPE_INTRA_ONLY, then VkVideoEncodeAV1PictureInfoKHR::predictionMode **must** be VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_INTRA_ONLY_KHR
- VUID-vkCmdEncodeVideoKHR-maxSingleReferenceCount-10328
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and VkVideoEncodeAV1CapabilitiesKHR::maxSingleReferenceCount is zero, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then the predictionMode member of the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** not be VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_SINGLE_REFERENCE_KHR
- VUID-vkCmdEncodeVideoKHR-predictionMode-10329

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the predictionMode member of the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo is VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_SINGLE_REFERENCE_KHR, then there **must** be at least one non-negative element of VkVideoEncodeAV1PictureInfoKHR::referenceNameSlotIndices with element index i that does not equal cdfOnlyReferenceIndex and for which bit index i is set in VkVideoEncodeAV1CapabilitiesKHR::singleReferenceNameMask, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with

- VUID-vkCmdEncodeVideoKHR-maxUnidirectionalCompoundReferenceCount-10330
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and VkVideoEncodeAV1CapabilitiesKHR::maxUnidirectionalCompoundReferenceCount is zero, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then the predictionMode member of the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** not be VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_UNIDIRECTIONAL_COMPOUND_KHR
- VUID-vkCmdEncodeVideoKHR-predictionMode-10331
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the predictionMode member of the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo is VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_UNIDIRECTIONAL_COMPOUND_KHR, then there **must** be at least two non-negative elements of VkVideoEncodeAV1PictureInfoKHR::referenceNameSlotIndices with element indices i and j where (i,j) □ {(0,1),(0,2),(0,3),(4,6)}, such that neither element equals cdfOnlyReferenceIndex and for which bit indices i and j are set in VkVideoEncodeAV1CapabilitiesKHR::unidirectionalCompoundReferenceNameMask, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-maxBidirectionalCompoundReferenceCount-10332
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and VkVideoEncodeAV1CapabilitiesKHR::maxBidirectionalCompoundReferenceCount is zero, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, then the predictionMode member of the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** not be VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_BIDIRECTIONAL_COMPOUND_KHR
- VUID-vkCmdEncodeVideoKHR-predictionMode-10333
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the predictionMode member of the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo is VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_BIDIRECTIONAL_COMPOUND_KHR, then

there **must** be at least two non-negative elements of VkVideoEncodeAV1PictureInfoKHR::referenceNameSlotIndices with element indices i ∈ {0,1,2,3} and j ∈ {4,5,6}, respectively, such that neither element equals cdfOnlyReferenceIndex, and for which bit indices i and j are set in VkVideoEncodeAV1CapabilitiesKHR::bidirectionalCompoundReferenceNameMask, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with

- VUID-vkCmdEncodeVideoKHR-predictionMode-10857
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, the predictionMode member of the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo is VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_UNIDIRECTIONAL_COMPOUND_KHR or VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_BIDIRECTIONAL_COMPOUND_KHR, and VkVideoEncodeAV1CapabilitiesKHR::flags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with, does not include VK_VIDEO_ENCODE_AV1_CAPABILITY_COMPOUND_PREDICTION_INTRA_REFRESH_BIT_KHR, then pEncodeInfo->flags **must** not include VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR
- VUID-vkCmdEncodeVideoKHR-referenceNameSlotIndices-10334
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then each element of the referenceNameSlotIndices array member of the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** either be negative or **must** equal the slotIndex member of one of the elements of pEncodeInfo->pReferenceSlots
- VUID-vkCmdEncodeVideoKHR-slotIndex-10335
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then the slotIndex member of each element of pEncodeInfo->pReferenceSlots **must** equal one of the elements of the referenceNameSlotIndices array member of the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo
- VUID-vkCmdEncodeVideoKHR-pExtensionHeader-10336
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the pExtensionHeader member of VkVideoEncodeAV1PictureInfoKHR::pStdPictureInfo for the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo is not NULL, then pExtensionHeader->temporal_id **must** be less than VkVideoEncodeAV1CapabilitiesKHR::maxTemporalLayerCount, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pExtensionHeader-10337
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the pExtensionHeader member of VkVideoEncodeAV1PictureInfoKHR::pStdPictureInfo for the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo is not NULL, then pExtensionHeader->spatial_id **must** be less than

VkVideoEncodeAV1CapabilitiesKHR::maxSpatialLayerCount, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with

- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10338
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and pEncodeInfo->pSetupReferenceSlot is not NULL, then the pExtensionHeader member of VkVideoEncodeAV1DpbSlotInfoKHR::pStdReferenceInfo for the VkVideoEncodeAV1DpbSlotInfoKHR structure included in the pNext chain of pEncodeInfo->pSetupReferenceSlot and the pExtensionHeader member of VkVideoEncodeAV1PictureInfoKHR::pStdPictureInfo for the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo **must** both be NULL or not NULL
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10339
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, pEncodeInfo->pSetupReferenceSlot is not NULL, and the pExtensionHeader member of VkVideoEncodeAV1DpbSlotInfoKHR::pStdReferenceInfo for the VkVideoEncodeAV1DpbSlotInfoKHR structure included in the pNext chain of pEncodeInfo->pSetupReferenceSlot is not NULL, then pExtensionHeader->temporal_id **must** equal pStdPictureInfo->pExtensionHeader->temporal_id in the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-10340
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, pEncodeInfo->pSetupReferenceSlot is not NULL, and the pExtensionHeader member of VkVideoEncodeAV1DpbSlotInfoKHR::pStdReferenceInfo for the VkVideoEncodeAV1DpbSlotInfoKHR structure included in the pNext chain of pEncodeInfo->pSetupReferenceSlot is not NULL, then pExtensionHeader->spatial_id **must** equal pStdPictureInfo->pExtensionHeader->spatial_id in the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo
- VUID-vkCmdEncodeVideoKHR-pExtensionHeader-10341
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the pExtensionHeader member of VkVideoEncodeAV1DpbSlotInfoKHR::pStdReferenceInfo for any of the VkVideoEncodeAV1DpbSlotInfoKHR structures included in the pNext chain of any element of pEncodeInfo->pReferenceSlots is not NULL, then pExtensionHeader->temporal_id **must** be less than VkVideoEncodeAV1CapabilitiesKHR::maxTemporalLayerCount, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pExtensionHeader-10342
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the pExtensionHeader member of VkVideoEncodeAV1DpbSlotInfoKHR::pStdReferenceInfo for any of the VkVideoEncodeAV1DpbSlotInfoKHR structures included in the pNext chain of any element of pEncodeInfo->pReferenceSlots is not NULL, then pExtensionHeader->spatial_id **must** be

less than VkVideoEncodeAV1CapabilitiesKHR::maxSpatialLayerCount, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with

- VUID-vkCmdEncodeVideoKHR-pTileInfo-10343
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the pTileInfo member of VkVideoEncodeAV1PictureInfoKHR::pStdPictureInfo for the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo is not NULL, then pTileInfo->TileCols **must** be greater than 0
- VUID-vkCmdEncodeVideoKHR-pTileInfo-10344
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the pTileInfo member of VkVideoEncodeAV1PictureInfoKHR::pStdPictureInfo for the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo is not NULL, then pTileInfo->TileRows **must** be greater than 0
- VUID-vkCmdEncodeVideoKHR-pTileInfo-10345
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the pTileInfo member of VkVideoEncodeAV1PictureInfoKHR::pStdPictureInfo for the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo is not NULL, then pTileInfo->TileCols **must** be less than or equal to VkVideoEncodeAV1CapabilitiesKHR::maxTiles.width, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pTileInfo-10346
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the pTileInfo member of VkVideoEncodeAV1PictureInfoKHR::pStdPictureInfo for the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo is not NULL, then pTileInfo->TileRows **must** be less than or equal to VkVideoEncodeAV1CapabilitiesKHR::maxTiles.height, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pTileInfo-10347
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the pTileInfo member of VkVideoEncodeAV1PictureInfoKHR::pStdPictureInfo for the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo is not NULL, then ⌊pEncodeInfo->srcPictureResource.codedExtent.width / pTileInfo->TileCols⌋ **must** be between VkVideoEncodeAV1CapabilitiesKHR::minTileSize.width and VkVideoEncodeAV1CapabilitiesKHR::maxTileSize.width, inclusive, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pTileInfo-10348
  If the bound video session was created with the video codec operation

VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and the pTileInfo member of VkVideoEncodeAV1PictureInfoKHR::pStdPictureInfo for the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo is not NULL, then pEncodeInfo->srcPictureResource.codedExtent.height / pTileInfo->TileRows must be between VkVideoEncodeAV1CapabilitiesKHR::minTileSize.height and VkVideoEncodeAV1CapabilitiesKHR::maxTileSize.height, inclusive, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile the bound video session was created with

- VUID-vkCmdEncodeVideoKHR-pStdPictureInfo-10349
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then VkVideoEncodeAV1PictureInfoKHR::pStdPictureInfo->flags.segmentation_enabled for the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo must be zero
- VUID-vkCmdEncodeVideoKHR-pStdPictureInfo-10350
  If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then VkVideoEncodeAV1PictureInfoKHR::pStdPictureInfo->pSegmentation for the VkVideoEncodeAV1PictureInfoKHR structure included in the pNext chain of pEncodeInfo must be NULL

Valid Usage (Implicit)

- VUID-vkCmdEncodeVideoKHR-commandBuffer-parameter
  commandBuffer must be a valid VkCommandBuffer handle
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-parameter
  pEncodeInfo must be a valid pointer to a valid VkVideoEncodeInfoKHR structure
- VUID-vkCmdEncodeVideoKHR-commandBuffer-recording
  commandBuffer must be in the recording state
- VUID-vkCmdEncodeVideoKHR-commandBuffer-cmdpool
  The VkCommandPool that commandBuffer was allocated from must support VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations
- VUID-vkCmdEncodeVideoKHR-renderpass
  This command must only be called outside of a render pass instance
- VUID-vkCmdEncodeVideoKHR-suspended
  This command must not be called between suspended render pass instances
- VUID-vkCmdEncodeVideoKHR-videocoding
  This command must only be called inside of a video coding scope
- VUID-vkCmdEncodeVideoKHR-bufferlevel
  commandBuffer must be a primary VkCommandBuffer

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Inside	VK_QUEUE_VIDEO_ENCODE_BIT_KHR	Action

Conditional Rendering

vkCmdEncodeVideoKHR is not affected by conditional rendering

The `VkVideoEncodeInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeInfoKHR {
    VkStructureType                       sType;
    const void*                           pNext;
    VkVideoEncodeFlagsKHR                 flags;
    VkBuffer                              dstBuffer;
    VkDeviceSize                          dstBufferOffset;
    VkDeviceSize                          dstBufferRange;
    VkVideoPictureResourceInfoKHR         srcPictureResource;
    const VkVideoReferenceSlotInfoKHR*    pSetupReferenceSlot;
    uint32_t                              referenceSlotCount;
    const VkVideoReferenceSlotInfoKHR*    pReferenceSlots;
    uint32_t                              precedingExternallyEncodedBytes;
} VkVideoEncodeInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is a pointer to a structure extending this structure.
- `flags` is a bitmask of VkVideoEncodeFlagBitsKHR indicating video encode command flags.
- `dstBuffer` is the destination video bitstream buffer to write the encoded bitstream to.
- `dstBufferOffset` is the starting offset in bytes from the start of `dstBuffer` to write the encoded bitstream to.

- `dstBufferRange` is the maximum bitstream size in bytes that **can** be written to `dstBuffer`, starting from `dstBufferOffset`.
- `srcPictureResource` is the video picture resource to use as the encode input picture.
- `pSetupReferenceSlot` is `NULL` or a pointer to a VkVideoReferenceSlotInfoKHR structure specifying the reconstructed picture information.
- `referenceSlotCount` is the number of elements in the `pReferenceSlots` array.
- `pReferenceSlots` is `NULL` or a pointer to an array of VkVideoReferenceSlotInfoKHR structures describing the DPB slots and corresponding reference picture resources to use in this video encode operation (the set of active reference pictures).
- `precedingExternallyEncodedBytes` is the number of bytes externally encoded by the application to the video bitstream and is used to update the internal state of the implementation's rate control algorithm to account for the bitrate budget consumed by these externally encoded bytes.

Valid Usage

- VUID-VkVideoEncodeInfoKHR-dstBuffer-08236
  `dstBuffer` **must** have been created with the VK_BUFFER_USAGE_VIDEO_ENCODE_DST_BIT_KHR usage flag set
- VUID-VkVideoEncodeInfoKHR-dstBufferOffset-08237
  `dstBufferOffset` **must** be less than the size of `dstBuffer`
- VUID-VkVideoEncodeInfoKHR-dstBufferRange-08238
  `dstBufferRange` **must** be less than or equal to the size of `dstBuffer` minus `dstBufferOffset`
- VUID-VkVideoEncodeInfoKHR-pSetupReferenceSlot-08239
  If `pSetupReferenceSlot` is not `NULL`, then its `slotIndex` member **must** not be negative
- VUID-VkVideoEncodeInfoKHR-pSetupReferenceSlot-08240
  If `pSetupReferenceSlot` is not `NULL`, then its `pPictureResource` **must** not be `NULL`
- VUID-VkVideoEncodeInfoKHR-slotIndex-08241
  The `slotIndex` member of each element of `pReferenceSlots` **must** not be negative
- VUID-VkVideoEncodeInfoKHR-pPictureResource-08242
  The `pPictureResource` member of each element of `pReferenceSlots` **must** not be `NULL`

Valid Usage (Implicit)

- VUID-VkVideoEncodeInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_INFO_KHR
- VUID-VkVideoEncodeInfoKHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkVideoEncodeAV1PictureInfoKHR, VkVideoEncodeH264PictureInfoKHR, VkVideoEncodeH265PictureInfoKHR, VkVideoEncodeIntraRefreshInfoKHR, VkVideoEncodeQuantizationMapInfoKHR, or VkVideoInlineQueryInfoKHR

- VUID-VkVideoEncodeInfoKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkVideoEncodeInfoKHR-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkVideoEncodeFlagBitsKHR values
- VUID-VkVideoEncodeInfoKHR-dstBuffer-parameter
  `dstBuffer` **must** be a valid VkBuffer handle
- VUID-VkVideoEncodeInfoKHR-srcPictureResource-parameter
  `srcPictureResource` **must** be a valid VkVideoPictureResourceInfoKHR structure
- VUID-VkVideoEncodeInfoKHR-pSetupReferenceSlot-parameter
  If `pSetupReferenceSlot` is not `NULL`, `pSetupReferenceSlot` **must** be a valid pointer to a valid VkVideoReferenceSlotInfoKHR structure
- VUID-VkVideoEncodeInfoKHR-pReferenceSlots-parameter
  If `referenceSlotCount` is not `0`, `pReferenceSlots` **must** be a valid pointer to an array of `referenceSlotCount` valid VkVideoReferenceSlotInfoKHR structures

Bits which **can** be set in VkVideoEncodeInfoKHR::`flags`, specifying video encode flags, are:

```
// Provided by VK_KHR_video_encode_quantization_map
typedef enum VkVideoEncodeFlagBitsKHR {
  // Provided by VK_KHR_video_encode_intra_refresh
    VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR = 0x00000004,
  // Provided by VK_KHR_video_encode_quantization_map
    VK_VIDEO_ENCODE_WITH_QUANTIZATION_DELTA_MAP_BIT_KHR = 0x00000001,
  // Provided by VK_KHR_video_encode_quantization_map
    VK_VIDEO_ENCODE_WITH_EMPHASIS_MAP_BIT_KHR = 0x00000002,
} VkVideoEncodeFlagBitsKHR;
```

- VK_VIDEO_ENCODE_WITH_QUANTIZATION_DELTA_MAP_BIT_KHR specifies the use of a quantization delta map in the issued video encode operations.
- VK_VIDEO_ENCODE_WITH_EMPHASIS_MAP_BIT_KHR specifies the use of an emphasis map in the issued video encode operations.
- VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR enables intra refresh for the encoded picture.

```
// Provided by VK_KHR_video_encode_queue
typedef VkFlags VkVideoEncodeFlagsKHR;
```

VkVideoEncodeFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeFlagBitsKHR.

45.17. Video Encode Intra Refresh

Encoding a picture with intra refresh enables encoding a subset of the coding blocks produced for an encode input picture with intra prediction.

NOTE This enables the application to perform a decoder refresh through a series of pictures instead of a single picture. In conjunction with restricting reference picture prediction, this enables avoiding error propagation across refresh cycles without introducing sudden spikes in the bitrate. This is achieved by amortizing the bitstream consumption of the refresh across a series of subsequent pictures.

An *intra refresh cycle* is a series of encode operations using intra refresh that contain at least one encode operation for any subregion of the coded extent that encoded the subregion with intra prediction.

The *intra refresh cycle duration* is the number of encode operations in an intra refresh cycle.

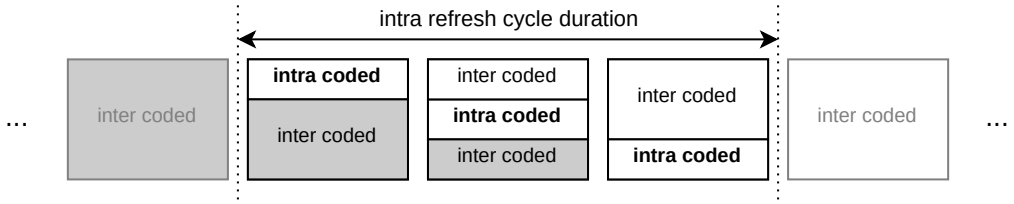


Figure 27. Intra refresh cycle duration

For the purposes of performing an intra refresh cycle, the coded extent is divided into a set of *intra refresh regions*. The number of intra refresh regions for a given intra refresh cycle equals the intra refresh cycle duration.

Implementations **may** partition the coded extent into intra refresh regions in an implementation-specific manner, unless otherwise specified. The following is true for the resulting set of intra refresh regions:

- Each intra refresh region encompasses entire coding blocks.
- Any intra refresh region **may** be non-rectangular.
- Some intra refresh regions **may** be empty.
- The union of intra refresh regions of a given partitioning covers the entire coded extent.

NOTE While intra refresh regions are generally disjoint, there may be overlap between distinct intra refresh regions of a given partitioning in order to accommodate for any filtering that may need to be applied across coding blocks of neighboring intra refresh regions.

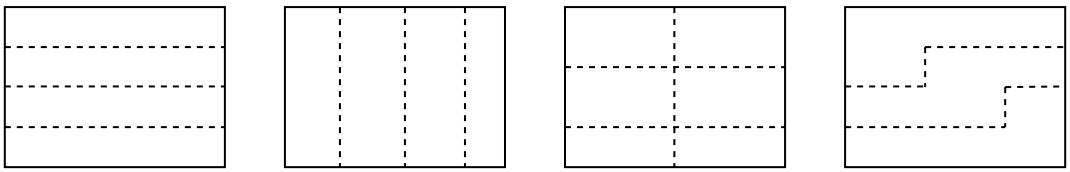


Figure 28. Examples of picture partitioning into intra refresh regions

An intra refresh cycle is performed by encoding subsequent pictures with intra refresh, each encoding subsequent intra refresh regions using intra prediction. This inherently defines an ordering of encode operations within an intra refresh cycle. Therefore each encode operation in an intra refresh cycle has an *intra refresh index* that provides the following information:

- The intra refresh index of an encode operation using intra refresh is the ordinal index of the intra refresh region being encoded using intra prediction.
- The intra refresh index of an encode operation using intra refresh inherently tells the number of intra refresh regions already refreshed by previous encode operations of the intra refresh cycle.

NOTE	For AV1 encoding, applications should consider setting the `error_resilient_mode` flag for the first frame encoded as part of the intra refresh cycle in order to avoid any error propagation of CDF data.

An intra refresh region of a picture encoded using intra refresh is considered *clean* once the encode operation within the intra refresh cycle that encoded that intra refresh region using intra prediction is complete. Otherwise, the intra refresh region is considered *dirty*.

The number of *dirty intra refresh regions* of a picture encoded using intra refresh is always one less than the difference between the intra refresh cycle duration and the intra refresh index used to encode that picture.

NOTE	As the intra refresh cycle progresses, each subsequent picture encoded with intra refresh will have fewer dirty intra refresh regions. Consequently, the final encode operation of the intra refresh cycle will produce an encoded picture with no more dirty intra refresh regions.

45.17.1. Intra Refresh Capabilities

When calling vkGetPhysicalDeviceVideoCapabilitiesKHR with `pVideoProfile->videoCodecOperation` specifying an encode operation, the VkVideoEncodeIntraRefreshCapabilitiesKHR structure **can** be included in the `pNext` chain of the VkVideoCapabilitiesKHR structure to retrieve capabilities specific to video encode intra refresh.

The `VkVideoEncodeIntraRefreshCapabilitiesKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_intra_refresh
typedef struct VkVideoEncodeIntraRefreshCapabilitiesKHR {
    VkStructureType                          sType;
    void*                                    pNext;
    VkVideoEncodeIntraRefreshModeFlagsKHR    intraRefreshModes;
    uint32_t                                 maxIntraRefreshCycleDuration;
    uint32_t                                 maxIntraRefreshActiveReferencePictures;
    VkBool32                                 partitionIndependentIntraRefreshRegions;
    VkBool32                                 nonRectangularIntraRefreshRegions;
} VkVideoEncodeIntraRefreshCapabilitiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `intraRefreshModes` is a bitmask of VkVideoEncodeIntraRefreshModeFlagBitsKHR values indicating the set of supported intra refresh modes.
- `maxIntraRefreshCycleDuration` specifies the maximum supported intra refresh cycle duration.
- `maxIntraRefreshActiveReferencePictures` is the maximum number of active reference pictures when encoding pictures with intra refresh enabled. This capability indicates additional restrictions beyond the maximum number of active reference pictures supported by the video profile, as reported in VkVideoCapabilitiesKHR::`maxActiveReferencePictures` and the maximum requested at video session creation time in VkVideoSessionCreateInfoKHR::`maxActiveReferencePictures`.
- `partitionIndependentIntraRefreshRegions` specifies whether the implementation supports intra refresh regions that are independent of the picture partitioning used during encoding. If it is VK_TRUE, then pictures **can** be encoded with multiple picture partitions, independent of the used intra refresh mode. Otherwise, pictures **cannot** be encoded with multiple picture partitions with any intra refresh mode other than VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_PER_PICTURE_PARTITION_BIT_KHR.

  NOTE This capability is only indicative for AV1 encode profiles and does not impose any restrictions on the application as implementations may change the application requested picture partitioning according to implementation-specific restrictions.

- `nonRectangularIntraRefreshRegions` specifies whether the implementation supports non-rectangular intra refresh regions.

  NOTE If this capability is not supported, then using per picture partition intra refresh may impose additional restrictions on the number of picture partitions a picture can be encoded with.

Valid Usage (Implicit)

- VUID-VkVideoEncodeIntraRefreshCapabilitiesKHR-sType-sType
  `sType` **must** be

VK_STRUCTURE_TYPE_VIDEO_ENCODE_INTRA_REFRESH_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkVideoCapabilitiesKHR

45.17.2. Intra Refresh Modes

The intra refresh modes are defined with the following enums:

```
// Provided by VK_KHR_video_encode_intra_refresh
typedef enum VkVideoEncodeIntraRefreshModeFlagBitsKHR {
    VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_NONE_KHR = 0,
    VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_PER_PICTURE_PARTITION_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_BLOCK_BASED_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_BLOCK_ROW_BASED_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_BLOCK_COLUMN_BASED_BIT_KHR = 0x00000008,
} VkVideoEncodeIntraRefreshModeFlagBitsKHR;
```

- VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_NONE_KHR specifies that intra refresh **must** not be used.
- VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_PER_PICTURE_PARTITION_BIT_KHR specifies the use of *per picture partition intra refresh*. In this mode each intra refresh region i corresponds to the encoded picture partition i.
- VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_BLOCK_BASED_BIT_KHR specifies the use of any *block-based intra refresh*. In this mode each intra refresh region encompasses a set of coding blocks, independent of encoded picture partitions but without any additional guarantees on the granularity at which the picture is split into intra refresh regions. When using this mode, the set of coding blocks comprising the intra refresh regions and the direction of intra refresh are implementation-defined.
- VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_BLOCK_ROW_BASED_BIT_KHR specifies the use of *block-row-based intra refresh*. This mode is a block-based intra refresh mode where each intra refresh region encompasses a set of coding block rows.
- VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_BLOCK_COLUMN_BASED_BIT_KHR specifies the use of *block-column-based intra refresh*. This mode is a block-based intra refresh mode where each intra refresh region encompasses a set of coding block columns.

Implementations reporting support for VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_BLOCK_ROW_BASED_BIT_KHR and/or VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_BLOCK_COLUMN_BASED_BIT_KHR in VkVideoEncodeIntraRefreshCapabilitiesKHR::`intraRefreshModes` are also **required** to report support for VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_BLOCK_BASED_BIT_KHR.

NOTE Both block-row-based and block-column-based intra refresh are just specific types of block-based intra refresh that provide additional guarantees about the granularity at which the picture is split into intra refresh regions, therefore implementations supporting either block-row-based or block-column-based intra refresh inherently support block-based intra refresh. Applications can use VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_BLOCK_BASED_BIT_KHR on such implementations when they do not have any preference for a more specific block-based mode and would rather leave the decision about the division into intra refresh regions to the implementation.

```
// Provided by VK_KHR_video_encode_intra_refresh
typedef VkFlags VkVideoEncodeIntraRefreshModeFlagsKHR;
```

VkVideoEncodeIntraRefreshModeFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeIntraRefreshModeFlagBitsKHR.

The used intra refresh mode is selected at video session creation time by including an instance of the VkVideoEncodeSessionIntraRefreshCreateInfoKHR structure in the pNext chain of the VkVideoSessionCreateInfoKHR structure and specifying one of the supported intra refresh modes, as returned in VkVideoEncodeIntraRefreshCapabilitiesKHR::intraRefreshModes, in its intraRefreshMode member.

The VkVideoEncodeSessionIntraRefreshCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_intra_refresh
typedef struct VkVideoEncodeSessionIntraRefreshCreateInfoKHR {
    VkStructureType                              sType;
    const void*                                  pNext;
    VkVideoEncodeIntraRefreshModeFlagBitsKHR     intraRefreshMode;
} VkVideoEncodeSessionIntraRefreshCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- intraRefreshMode is a VkVideoEncodeIntraRefreshModeFlagBitsKHR specifying the used intra refresh mode.

Valid Usage (Implicit)

- VUID-VkVideoEncodeSessionIntraRefreshCreateInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_SESSION_INTRA_REFRESH_CREATE_INFO_KHR
- VUID-VkVideoEncodeSessionIntraRefreshCreateInfoKHR-intraRefreshMode-parameter
  If intraRefreshMode is not 0, intraRefreshMode **must** be a valid VkVideoEncodeIntraRefreshModeFlagBitsKHR value

Structure Chaining

Extends the structure

- VkVideoSessionCreateInfoKHR

45.17.3. Encoding Pictures with Intra Refresh

If the video encode operation is issued with the VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR flag, the picture will be encoded with intra refresh, resulting in the intra refresh region identified by the specified intra refresh index to be encoded with intra prediction.

Intra refresh parameters are specified to video encode operations by including an instance of the VkVideoEncodeIntraRefreshInfoKHR structure in the `pNext` chain of the VkVideoEncodeInfoKHR structure specified to the video encode command.

The `VkVideoEncodeIntraRefreshInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_intra_refresh
typedef struct VkVideoEncodeIntraRefreshInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           intraRefreshCycleDuration;
    uint32_t           intraRefreshIndex;
} VkVideoEncodeIntraRefreshInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `intraRefreshCycleDuration` is the used intra refresh cycle duration.
- `intraRefreshIndex` is the intra refresh index of the encoded picture.

Valid Usage (Implicit)

- VUID-VkVideoEncodeIntraRefreshInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_INTRA_REFRESH_INFO_KHR

Structure Chaining

Extends the structure

- VkVideoEncodeInfoKHR

45.17.4. Referencing Pictures Encoded with Intra Refresh

When using pictures encoded with intra refresh as an active reference picture in a video encode

operation, applications **may** want to limit the set of intra refresh regions of the reference picture that are used to predict samples of the encoded picture to exclude dirty intra refresh regions of the reference picture in question.

NOTE This enables the application to avoid error propagation from previous refresh cycles by only using already refreshed (clean) intra refresh regions for sample prediction.

In order to limit the set of intra refresh regions of an active reference picture used for sample prediction, the application **must** specify an instance of the VkVideoReferenceIntraRefreshInfoKHR structure in the `pNext` chain of the VkVideoReferenceSlotInfoKHR structure specifying the active reference picture in question.

The `VkVideoReferenceIntraRefreshInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_intra_refresh
typedef struct VkVideoReferenceIntraRefreshInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           dirtyIntraRefreshRegions;
} VkVideoReferenceIntraRefreshInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `dirtyIntraRefreshRegions` is the number of dirty intra refresh regions in the reference picture.

Valid Usage (Implicit)

- VUID-VkVideoReferenceIntraRefreshInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_REFERENCE_INTRA_REFRESH_INFO_KHR

Structure Chaining

Extends the structure

- VkVideoReferenceSlotInfoKHR

In order to use active reference pictures with a non-zero number of dirty intra refresh regions and therefore limit sample prediction to clean intra refresh regions, the currently encoded picture **must** be encoded with intra refresh enabled.

When limiting sample prediction to clean intra refresh regions of a reference picture with a non-zero number of dirty intra refresh regions, the number of dirty intra refresh regions **must** equal the used intra refresh cycle duration minus the intra refresh index of the encoded picture.

NOTE This has the practical effect that pictures can only reference the previous picture

within the intra refresh cycle or pictures outside of the intra refresh cycle.

Given an active reference picture with `dirtyIntraRefreshRegions` number of dirty intra refresh regions and an intra refresh cycle duration of `intraRefreshCycleDuration`, the samples within intra refresh region index i of the encoded picture are allowed to be predicted by samples within intra refresh region index j of the active reference picture in question if and only if any of the following conditions are true:

- The picture is encoded with VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR and i is greater than the `intraRefreshIndex` of the encoded picture.
- The picture is encoded with VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR, i is less than the `intraRefreshIndex` of the encoded picture, and j is less than the difference of `intraRefreshCycleDuration` and `dirtyIntraRefreshRegions`.

NOTE

Even if a particular intra refresh region is allowed to be used for sample prediction, as defined above, implementations may not always support sample prediction from the given intra refresh region.

Some implementations may only support sample prediction from a single set of intra refresh regions across the entire set of active reference pictures. For example, when encoding a picture with two active reference pictures, one with no dirty intra refresh regions, and one with some dirty intra refresh regions, sample prediction may not happen from those intra refresh regions of the first active reference picture which are marked dirty in the second active reference picture.

Other implementations may have the additional restriction that the set of intra refresh regions to use for sample prediction across the entire set of active reference pictures is implied from the intra refresh index of the currently encoded picture, further restricting the set of intra refresh regions that such implementations can use for sample prediction. For example, when encoding a picture with intra refresh using an active reference picture that has no dirty intra refresh regions, sample prediction may not happen from intra refresh regions of the active reference picture which have an index greater than or equal to the currently encoded picture's intra refresh index, even though those intra refresh regions were not marked as dirty.

None of these implementation limitations will have an effect on application behavior from the perspective of correctness, but they may negatively impact the encoding efficiency.

45.18. Video Encode Rate Control

The size of the encoded bitstream data produced by video encode operations is a function of the following set of constraints:

- The capabilities of the compression algorithms defined and employed by the used video compression standard;

- Restrictions imposed by the selected video profile according to the rules defined by the used video compression standard;
- Further restrictions imposed by the capabilities supported by the implementation for the selected video profile;
- The image data in the encode input picture and the set of active reference pictures (as these affect the effectiveness of the compression algorithms employed by the video encode operations);
- The set of codec-specific and codec-independent encoding parameters provided by the application.

These also inherently define the set of decoder capabilities required for reconstructing and processing the picture data in the encoded bitstream.

Video coding uses *bitrate* as the quantitative metric associated with encoded bitstream data size which expresses the rate at which video bitstream data **can** be transferred or processed, measured in number of bits per second. This bitrate is both a function of the encoded bitstream data size of the encoded pictures as well as the *frame rate* used by the video sequence.

Rate control algorithms are used by video encode operations to enable adjusting encoding parameters to achieve a target bitrate, or otherwise directly or indirectly control the bitrate of the generated video bitstream data. These algorithms are usually not defined by the used video compression standard, although some video compression standards do provide non-normative guidelines for implementations.

Accordingly, this specification does not mandate implementations to produce identical encoded bitstream data outputs in response to video encode operations, however, it does define a set of codec-independent and codec-specific parameters that enable the application to control the behavior of the rate control algorithms supported by the implementation. Some of these parameters guarantee certain implementation behavior while others provide guidance for implementations to apply various rate control heuristics.

NOTE

Applications need to make sure that they configure rate control parameters appropriately and that they follow the promises made to the implementation through parameters providing guidance for the implementation's rate control algorithms and heuristics in order to be able to get the desired rate control behavior and to be able to hit the set bitrate targets. In addition, the behavior of rate control may also differ across implementations even if the capabilities of the used video profile match between those implementations. This may happen due to implementations applying different rate control algorithms or heuristics internally, and thus even the same set of guidance parameter values may have different effects on the rate control behavior across implementations.

45.18.1. Rate Control Modes

After a video session is reset to the initial state, the default behavior and parameters of video encode rate control are entirely implementation-dependent and the application **cannot** affect the bitrate or quality parameters of the encoded bitstream data produced by video encode operations

unless the application changes the rate control configuration of the video session, as described in the Video Coding Control section.

For each supported video profile, the implementation **may** expose a set of *rate control modes* that are available for use by the application when encoding bitstreams targeting that video profile. These modes allow using different rate control algorithms that fall into one of the following two categories:

1. Per-operation rate control
2. Stream-level rate control

In case of *per-operation rate control*, the bitrate of the generated video bitstream data is indirectly controlled by quality, size, or other encoding parameters specified by the application for each individual video encode operation.

In case of *stream-level rate control*, the application **can** directly specify target bitrates besides other encoding parameters to control the behavior of the rate control algorithm used by the implementation across multiple video encode operations.

The rate control modes are defined with the following enums:

```
// Provided by VK_KHR_video_encode_queue
typedef enum VkVideoEncodeRateControlModeFlagBitsKHR {
    VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR = 0,
    VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_RATE_CONTROL_MODE_CBR_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_RATE_CONTROL_MODE_VBR_BIT_KHR = 0x00000004,
} VkVideoEncodeRateControlModeFlagBitsKHR;
```

- VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR specifies the use of implementation-specific rate control.
- VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR specifies that rate control is disabled and the application will specify per-operation rate control parameters controlling the encoding quality. In this mode implementations will encode pictures independently of the output bitrate of prior video encode operations.
  - When using an H.264 encode profile, implementations will use the QP value specified in VkVideoEncodeH264NaluSliceInfoKHR::`constantQp` to control the quality of the encoded picture.
  - When using an H.265 encode profile, implementations will use the QP value specified in VkVideoEncodeH265NaluSliceSegmentInfoKHR::`constantQp` to control the quality of the encoded picture.
  - When using an AV1 encode profile, implementations will use the quantizer index value specified in VkVideoEncodeAV1PictureInfoKHR::`constantQIndex` to control the quality of the encoded picture.
- VK_VIDEO_ENCODE_RATE_CONTROL_MODE_CBR_BIT_KHR specifies the use of constant bitrate (CBR) rate control mode. In this mode the implementation will attempt to produce the encoded

bitstream at a constant bitrate while conforming to the constraints of other rate control parameters.

- VK_VIDEO_ENCODE_RATE_CONTROL_MODE_VBR_BIT_KHR specifies the use of variable bitrate (VBR) rate control mode. In this mode the implementation will produce the encoded bitstream at a variable bitrate according to the constraints of other rate control parameters.

```
// Provided by VK_KHR_video_encode_queue
typedef VkFlags VkVideoEncodeRateControlModeFlagsKHR;
```

VkVideoEncodeRateControlModeFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeRateControlModeFlagBitsKHR.

45.18.2. Leaky Bucket Model

Video encoding implementations use the *leaky bucket model* for stream-level rate control. The leaky bucket is a concept referring to the interface between the video encoder and the consumer (for example, a network connection), where the video encoder produces encoded bitstream data corresponding to the encoded pictures and adds them in the leaky bucket while its contents are drained by the consumer.

Analogously, a similar leaky bucket is considered to exist at the input interface of a video decoder, into which encoded bitstream data is continuously added and is subsequently consumed by the video decoder. It is desirable to avoid overflowing or underflowing this leaky bucket because:

- In case of an underflow, the video decoder will be unable to consume encoded bitstream data in order to decode pictures (and optionally display them).
- In case of an overflow, the leaky bucket will be unable to accommodate more encoded bitstream data and such data **may** need to be thrown away, leading to the loss of the corresponding encoded pictures.

These requirements **can** be satisfied by imposing various constraints on the encoder-side leaky bucket to avoid its overflow or underflow, depending on the used rate control algorithm and codec parameters. However, enumerating these constraints is outside the scope of this specification.

The term *virtual buffer* is often used as an alternative to refer to the leaky bucket.

This virtual buffer model is defined by the following parameters:

- The bitrate (R) at which the encoded bitstream is expected to be processed.
- The size (B) of the virtual buffer.
- The initial occupancy (F) of the virtual buffer.

In this model the virtual buffer is used to smooth out fluctuations in the bitrate of the encoded bitstream over time without experiencing buffer overflow or underflow, as long as the bitrate of the encoded stream does not diverge from the target bitrate for extended periods of time.

This buffering **may** inherently impose a processing delay, as the goal of the model is to enable

decoders to maintain a consistent processing rate of an encoded bitstream with varying data rate.

The initial or start-up delay (`D`) is computed as:

```
D = F / R
```

NOTE	Applications need to configure the virtual buffer with sufficient size to avoid or minimize buffer overflows and underflows while also keeping it small enough to meet their latency goals.

45.18.3. Rate Control Layers

Some video compression standards and video profiles allow associating encoded pictures with specific *video coding layers*. The name, identification, and semantics associated with such video coding layers are defined by the corresponding video compression standards.

Analogously, stream-level rate control **can** be configured to use one or more *rate control layers*:

- When a single rate control layer is configured, it is applied to all encoded pictures, regardless of the picture's video coding layer. In this case the distribution of the available bitrate budget across video coding layers is implementation-dependent.
- When multiple rate control layers are configured, each rate control layer is applied to the corresponding video coding layer, i.e. only across encoded pictures pertaining to the corresponding video coding layer.

Individual rate control layers are identified using *layer indices* between zero and `N-1`, where `N` is the number of active rate control layers.

Rate control layers are only applicable when using stream-level rate control modes.

45.18.4. Rate Control State

Rate control state is maintained by the implementation in the video session objects and its parameters are specified using an instance of the `VkVideoEncodeRateControlInfoKHR` structure. The complete rate control state of a video session is defined by the following set of parameters:

- The values of the members of the VkVideoEncodeRateControlInfoKHR structure used to configure the rate control state.
- The values of the members of any VkVideoEncodeRateControlLayerInfoKHR structures specified in VkVideoEncodeRateControlInfoKHR::`pLayers` used to configure the state of individual rate control layers.
- If the video session was created with an H.264 encode profile:
  - The values of the members of the VkVideoEncodeH264RateControlInfoKHR structure, if one is specified in the `pNext` chain of the VkVideoEncodeRateControlInfoKHR used to configure the rate control state.
  - The values of the members of any VkVideoEncodeH264RateControlLayerInfoKHR structures

included in the pNext chain of a VkVideoEncodeRateControlLayerInfoKHR structure used to configure the state of a rate control layer.

- If the video session was created with an H.265 encode profile:
  - The values of the members of the VkVideoEncodeH265RateControlInfoKHR structure, if one is specified in the pNext chain of the VkVideoEncodeRateControlInfoKHR used to configure the rate control state.
  - The values of the members of any VkVideoEncodeH265RateControlLayerInfoKHR structures included in the pNext chain of a VkVideoEncodeRateControlLayerInfoKHR structure used to configure the state of a rate control layer.
- If the video session was created with an AV1 encode profile:
  - The values of the members of the VkVideoEncodeAV1RateControlInfoKHR structure, if one is specified in the pNext chain of the VkVideoEncodeRateControlInfoKHR used to configure the rate control state.
  - The values of the members of any VkVideoEncodeAV1RateControlLayerInfoKHR structures included in the pNext chain of a VkVideoEncodeRateControlLayerInfoKHR structure used to configure the state of a rate control layer.

Two rate control states match if all the parameters listed above match between them.

The VkVideoEncodeRateControlInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeRateControlInfoKHR {
    VkStructureType                                 sType;
    const void*                                     pNext;
    VkVideoEncodeRateControlFlagsKHR                flags;
    VkVideoEncodeRateControlModeFlagBitsKHR         rateControlMode;
    uint32_t                                        layerCount;
    const VkVideoEncodeRateControlLayerInfoKHR*     pLayers;
    uint32_t                                        virtualBufferSizeInMs;
    uint32_t                                        initialVirtualBufferSizeInMs;
} VkVideoEncodeRateControlInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is reserved for future use.
- rateControlMode is a VkVideoEncodeRateControlModeFlagBitsKHR value specifying the rate control mode.
- layerCount specifies the number of rate control layers to use.
- pLayers is a pointer to an array of layerCount VkVideoEncodeRateControlLayerInfoKHR structures, each specifying the rate control configuration of the corresponding rate control layer.
- virtualBufferSizeInMs is the size in milliseconds of the virtual buffer used by the

implementation's rate control algorithm for the leaky bucket model, with respect to the average bitrate of the stream calculated by summing the values of the `averageBitrate` members of the elements of the `pLayers` array.

- `initialVirtualBufferSizeInMs` is the initial occupancy in milliseconds of the virtual buffer used by the implementation's rate control algorithm for the leaky bucket model.

If `layerCount` is zero then the values of `virtualBufferSizeInMs` and `initialVirtualBufferSizeInMs` are ignored.

This structure **can** be specified in the following places:

- In the `pNext` chain of VkVideoBeginCodingInfoKHR to specify the current rate control state expected to be configured when beginning a video coding scope.
- In the `pNext` chain of VkVideoCodingControlInfoKHR to change the rate control configuration of the bound video session.

Including this structure in the `pNext` chain of VkVideoCodingControlInfoKHR and including VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR in VkVideoCodingControlInfoKHR::`flags` enables updating the rate control configuration of the bound video session. This replaces the entire rate control configuration of the bound video session and **may** reset the state of all enabled rate control layers to an initial state according to the codec-specific rate control semantics defined in the corresponding sections listed below.

When `layerCount` is greater than one, multiple rate control layers are configured, and each rate control layer is applied to the corresponding video coding layer identified by the index of the corresponding element of `pLayers`.

- If the video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then this index specifies the H.264 temporal layer ID of the video coding layer the rate control layer is applied to.
- If the video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then this index specifies the H.265 temporal ID of the video coding layer the rate control layer is applied to.
- If the video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, then this index specifies the AV1 temporal ID of the temporal layer the rate control layer is applied to.

Additional structures providing codec-specific rate control parameters **can** be included in the `pNext` chain of `VkVideoCodingControlInfoKHR` depending on the video profile the bound video session was created with. For further details see:

- Video Coding Control
- H.264 Encode Rate Control
- H.265 Encode Rate Control
- AV1 Encode Rate Control

The new rate control configuration takes effect when the corresponding

vkCmdControlVideoCodingKHR is executed on the device, and only impacts video encode operations that follow in execution order.

Valid Usage

- VUID-VkVideoEncodeRateControlInfoKHR-rateControlMode-08248
  If `rateControlMode` is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR or VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, then `layerCount` **must** be `0`
- VUID-VkVideoEncodeRateControlInfoKHR-rateControlMode-08275
  If `rateControlMode` is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_CBR_BIT_KHR or VK_VIDEO_ENCODE_RATE_CONTROL_MODE_VBR_BIT_KHR, then `layerCount` **must** be greater than `0`
- VUID-VkVideoEncodeRateControlInfoKHR-rateControlMode-08244
  If `rateControlMode` is not VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR, then it **must** specify one of the bits included in VkVideoEncodeCapabilitiesKHR::`rateControlModes`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile
- VUID-VkVideoEncodeRateControlInfoKHR-layerCount-08245
  `layerCount` member **must** be less than or equal to VkVideoEncodeCapabilitiesKHR::`maxRateControlLayers`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile
- VUID-VkVideoEncodeRateControlInfoKHR-pLayers-08276
  For each element of `pLayers`, its `averageBitrate` member **must** be between `1` and VkVideoEncodeCapabilitiesKHR::`maxBitrate`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile
- VUID-VkVideoEncodeRateControlInfoKHR-pLayers-08277
  For each element of `pLayers`, its `maxBitrate` member **must** be between `1` and VkVideoEncodeCapabilitiesKHR::`maxBitrate`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile
- VUID-VkVideoEncodeRateControlInfoKHR-rateControlMode-08356
  If `rateControlMode` is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_CBR_BIT_KHR, then for each element of `pLayers`, its `averageBitrate` member **must** equal its `maxBitrate` member
- VUID-VkVideoEncodeRateControlInfoKHR-rateControlMode-08278
  If `rateControlMode` is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_VBR_BIT_KHR, then for each element of `pLayers`, its `averageBitrate` member **must** be less than or equal to its `maxBitrate` member
- VUID-VkVideoEncodeRateControlInfoKHR-layerCount-08357
  If `layerCount` is not zero, then `virtualBufferSizeInMs` **must** be greater than zero
- VUID-VkVideoEncodeRateControlInfoKHR-layerCount-08358
  If `layerCount` is not zero, then `initialVirtualBufferSizeInMs` **must** be less than or equal to `virtualBufferSizeInMs`
- VUID-VkVideoEncodeRateControlInfoKHR-videoCodecOperation-07022

If the videoCodecOperation of the used video profile is VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, the pNext chain this structure is included in also includes an instance of the VkVideoEncodeH264RateControlInfoKHR structure, and layerCount is greater than 1, then layerCount **must** equal VkVideoEncodeH264RateControlInfoKHR::temporalLayerCount

- VUID-VkVideoEncodeRateControlInfoKHR-videoCodecOperation-07025
  If the videoCodecOperation of the used video profile is VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, the pNext chain this structure is included in also includes an instance of the VkVideoEncodeH265RateControlInfoKHR structure, and layerCount is greater than 1, then layerCount **must** equal VkVideoEncodeH265RateControlInfoKHR::subLayerCount
- VUID-VkVideoEncodeRateControlInfoKHR-videoCodecOperation-10351
  If the videoCodecOperation of the used video profile is VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, the pNext chain this structure is included in also includes an instance of the VkVideoEncodeAV1RateControlInfoKHR structure, and layerCount is greater than 1, then layerCount **must** equal VkVideoEncodeAV1RateControlInfoKHR::temporalLayerCount

Valid Usage (Implicit)

- VUID-VkVideoEncodeRateControlInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_RATE_CONTROL_INFO_KHR
- VUID-VkVideoEncodeRateControlInfoKHR-flags-zerobitmask
  flags **must** be 0
- VUID-VkVideoEncodeRateControlInfoKHR-rateControlMode-parameter
  If rateControlMode is not 0, rateControlMode **must** be a valid VkVideoEncodeRateControlModeFlagBitsKHR value
- VUID-VkVideoEncodeRateControlInfoKHR-pLayers-parameter
  If layerCount is not 0, pLayers **must** be a valid pointer to an array of layerCount valid VkVideoEncodeRateControlLayerInfoKHR structures

Structure Chaining

Extends the structures

- VkVideoBeginCodingInfoKHR
- VkVideoCodingControlInfoKHR

```
// Provided by VK_KHR_video_encode_queue
typedef VkFlags VkVideoEncodeRateControlFlagsKHR;
```

VkVideoEncodeRateControlFlagsKHR is a bitmask type for setting a mask, but is currently reserved for

future use.

Rate Control Layer State

The configuration of individual rate control layers is specified using an instance of the VkVideoEncodeRateControlLayerInfoKHR structure.

The VkVideoEncodeRateControlLayerInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeRateControlLayerInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint64_t           averageBitrate;
    uint64_t           maxBitrate;
    uint32_t           frameRateNumerator;
    uint32_t           frameRateDenominator;
} VkVideoEncodeRateControlLayerInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is a pointer to a structure extending this structure.
- averageBitrate is the average bitrate to be targeted by the implementation's rate control algorithm.
- maxBitrate is the peak bitrate to be targeted by the implementation's rate control algorithm.
- frameRateNumerator is the numerator of the frame rate assumed by the implementation's rate control algorithm.
- frameRateDenominator is the denominator of the frame rate assumed by the implementation's rate control algorithm.

NOTE	The ability of the implementation's rate control algorithm to be able to match the requested average and/or peak bitrates **may** be limited by the set of other codec-independent and codec-specific rate control parameters specified by the application, the input content, as well as the application conforming to the rate control guidance provided to the implementation, as described earlier.

Additional structures providing codec-specific rate control parameters **can** be included in the pNext chain of VkVideoEncodeRateControlLayerInfoKHR depending on the video profile the bound video session was created with. For further details see:

- Video Coding Control
- H.264 Encode Rate Control
- H.265 Encode Rate Control
- AV1 Encode Rate Control

Valid Usage

- VUID-VkVideoEncodeRateControlLayerInfoKHR-frameRateNumerator-08350
  `frameRateNumerator` **must** be greater than zero
- VUID-VkVideoEncodeRateControlLayerInfoKHR-frameRateDenominator-08351
  `frameRateDenominator` **must** be greater than zero

Valid Usage (Implicit)

- VUID-VkVideoEncodeRateControlLayerInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_RATE_CONTROL_LAYER_INFO_KHR
- VUID-VkVideoEncodeRateControlLayerInfoKHR-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkVideoEncodeAV1RateControlLayerInfoKHR, VkVideoEncodeH264RateControlLayerInfoKHR, or VkVideoEncodeH265RateControlLayerInfoKHR
- VUID-VkVideoEncodeRateControlLayerInfoKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

45.19. Video Encode Quantization Maps

Quantization maps are VkImage objects that are used in video encode operations to control the relative quantization parameter values across the encoded picture. Each texel in the quantization map controls the relative quantization parameter values used to encode the corresponding rectangular block of texels in the encode input picture.

The size of the rectangular block of texels each quantization map texel covers is referred to as the *quantization map texel size*.

The extent of the image subresource used as a quantization map when encoding a picture with a coded extent of (`width`,`height`) thus has to be at least (⌈`width` / `texelSize.width`⌉, ⌈`height` / `texelSize.height`⌉), where `texelSize` is the used quantization map texel size.

In particular, the quantization map texel at location (`x`,`y`) contains relative quantization parameter values used when encoding the `texelSize` sized rectangular block of the encode input picture starting at the texel location (`x` × `texelSize.width`, `y` × `texelSize.height`).

The quantization map texel size does not always match the size of the codec-specific coding blocks used during encoding. Furthermore, some video compression standards allow the size of the codec-specific coding blocks to vary across the encoded picture. In order to accommodate for such mismatches between the granularity at which quantization parameters are stored in quantization maps and the granularity at which they are applied to codec-specific coding blocks during encoding, the following mapping rules are applied to define the quantization map texel value corresponding to a given codec-specific coding block with a size (`width`,`height`) at the texel location (`x`,`y`) in the encode input picture:

- If the size of the codec-specific coding block matches the used quantization map texel size, then the fetched quantization map value corresponding to the codec-specific coding block is the texel value at the texel location (`x` / `texelSize.width`, `y` / `texelSize.height`).
- If the size of the codec-specific coding block is smaller than the used quantization map texel size, then the fetched quantization map value corresponding to the codec-specific coding block is the texel value at the texel location (⌊`x` / `texelSize.width`⌋, ⌊`y` / `texelSize.height`⌋).
- If the size of the codec-specific coding block is larger than the used quantization map texel size, then the fetched quantization map value corresponding to the codec-specific coding block **may** be any value determined as the linear interpolation of the quantization map texel values in the subregion starting at texel location (`x` / `texelSize.width`,`y` / `texelSize.height`) with a size (⌊`width` / `texelSize.width`⌋, ⌊`height` / `texelSize.height`⌋).

The actual control parameters stored in the quantization map depend on its type. This specification supports the following types of quantization maps:

- Quantization delta maps
- Emphasis maps

45.19.1. Quantization Delta Maps

Quantization delta maps contain values that directly affect the codec-specific quantization parameter values used to encode the corresponding block of the encode input picture.

Quantization delta maps **can** be used in conjunction with any rate control mode, including VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR.

Due to their codec-specific nature, they are described in more detail in the corresponding codec-specific section for video encode operations that support them. In particular:

- The behavior of quantization delta maps used with an H.264 encode profile is described in the H.264 Encode Quantization and H.264 QP Delta Maps sections.
- The behavior of quantization delta maps used with an H.265 encode profile is described in the H.265 Encode Quantization and H.265 QP Delta Maps sections.
- The behavior of quantization delta maps used with an AV1 encode profile is described in the AV1 Encode Quantization and AV1 Quantizer Index Delta Maps sections.

This specification does not support quantization delta maps for any other video encode operation.

45.19.2. Emphasis Maps

Emphasis maps contain values that indirectly affect the codec-specific quantization parameter values used to encode the corresponding block of the encode input picture.

The texels of emphasis maps contain values that provide input to the encoder implementation about the relative importance (emphasis) of regions of the encoded pictures in order to enable the implementation's rate control algorithm to allocate more bitrate budget for regions of the encoded picture with higher emphasis values than to those with lower emphasis values.

Emphasis maps **can** only be used when the current rate control mode configured for the video session is not VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR or VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR.

As these emphasis values only control the otherwise implementation-specific behavior of the used rate control algorithm, this specification does not impose additional restrictions on implementations beyond the ones outlined in the corresponding codec-specific sections describing quantization behavior:

- The behavior of emphasis maps used with an H.264 encode profile is described in the H.264 Encode Quantization section.
- The behavior of emphasis maps used with an H.265 encode profile is described in the H.265 Encode Quantization section.
- The behavior of emphasis maps used with an AV1 encode profile is described in the AV1 Encode Quantization section.

This specification does not support emphasis maps for any other video encode operation.

Emphasis maps always have single channel unsigned normalized integer formats and implementations are **required** to support the VK_FORMAT_R8_UNORM format for emphasis maps, as reported in VkVideoFormatPropertiesKHR::`format`, when the video encode profile supports VK_VIDEO_ENCODE_CAPABILITY_EMPHASIS_MAP_BIT_KHR.

45.19.3. Quantization Map Capabilities

When calling vkGetPhysicalDeviceVideoCapabilitiesKHR with `pVideoProfile->videoCodecOperation` specifying an encode operation, the VkVideoEncodeQuantizationMapCapabilitiesKHR structure **can** be included in the `pNext` chain of the VkVideoCapabilitiesKHR structure to retrieve capabilities specific to video encode quantization maps.

The `VkVideoEncodeQuantizationMapCapabilitiesKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_quantization_map
typedef struct VkVideoEncodeQuantizationMapCapabilitiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkExtent2D         maxQuantizationMapExtent;
} VkVideoEncodeQuantizationMapCapabilitiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxQuantizationMapExtent` indicates the maximum supported width and height of quantization maps.

Valid Usage (Implicit)

- VUID-VkVideoEncodeQuantizationMapCapabilitiesKHR-sType-sType

sType must be VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUANTIZATION_MAP_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkVideoCapabilitiesKHR

When calling vkGetPhysicalDeviceVideoCapabilitiesKHR to query the capabilities of an H.264 encode profile, the VkVideoEncodeH264QuantizationMapCapabilitiesKHR structure **can** be included in the pNext chain of the VkVideoCapabilitiesKHR structure to retrieve additional video encode quantization map capabilities specific to H.264 encode profiles.

The VkVideoEncodeH264QuantizationMapCapabilitiesKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h264 with VK_KHR_video_encode_quantization_map
typedef struct VkVideoEncodeH264QuantizationMapCapabilitiesKHR {
    VkStructureType    sType;
    void*              pNext;
    int32_t            minQpDelta;
    int32_t            maxQpDelta;
} VkVideoEncodeH264QuantizationMapCapabilitiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- minQpDelta indicates the minimum QP delta value supported for H.264 QP delta maps.
- maxQpDelta indicates the maximum QP delta value supported for H.264 QP delta maps.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264QuantizationMapCapabilitiesKHR-sType-sType
  sType must be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_QUANTIZATION_MAP_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkVideoCapabilitiesKHR

When calling vkGetPhysicalDeviceVideoCapabilitiesKHR to query the capabilities of an H.265 encode profile, the VkVideoEncodeH265QuantizationMapCapabilitiesKHR structure **can** be included in the pNext chain of the VkVideoCapabilitiesKHR structure to retrieve additional video encode

quantization map capabilities specific to H.265 encode profiles.

The VkVideoEncodeH265QuantizationMapCapabilitiesKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h265 with VK_KHR_video_encode_quantization_map
typedef struct VkVideoEncodeH265QuantizationMapCapabilitiesKHR {
    VkStructureType    sType;
    void*              pNext;
    int32_t            minQpDelta;
    int32_t            maxQpDelta;
} VkVideoEncodeH265QuantizationMapCapabilitiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- minQpDelta indicates the minimum QP delta value supported for H.265 QP delta maps.
- maxQpDelta indicates the maximum QP delta value supported for H.265 QP delta maps.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265QuantizationMapCapabilitiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_QUANTIZATION_MAP_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkVideoCapabilitiesKHR

When calling vkGetPhysicalDeviceVideoCapabilitiesKHR to query the capabilities of an AV1 encode profile, the VkVideoEncodeAV1QuantizationMapCapabilitiesKHR structure **can** be included in the pNext chain of the VkVideoCapabilitiesKHR structure to retrieve additional video encode quantization map capabilities specific to AV1 encode profiles.

The VkVideoEncodeAV1QuantizationMapCapabilitiesKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_av1 with VK_KHR_video_encode_quantization_map
typedef struct VkVideoEncodeAV1QuantizationMapCapabilitiesKHR {
    VkStructureType    sType;
    void*              pNext;
    int32_t            minQIndexDelta;
    int32_t            maxQIndexDelta;
} VkVideoEncodeAV1QuantizationMapCapabilitiesKHR;
```

- sType is a VkStructureType value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `minQIndexDelta` indicates the minimum quantizer index delta value supported for AV1 quantizer index delta maps.
- `maxQIndexDelta` indicates the maximum quantizer index delta value supported for AV1 quantizer index delta maps.

Valid Usage (Implicit)

- VUID-VkVideoEncodeAV1QuantizationMapCapabilitiesKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_QUANTIZATION_MAP_CAPABILITIES_KHR`

Structure Chaining

Extends the structure

- VkVideoCapabilitiesKHR

45.19.4. Quantization Map Format Properties

When calling vkGetPhysicalDeviceVideoFormatPropertiesKHR, the VkVideoFormatQuantizationMapPropertiesKHR structure **can** be included in the `pNext` chain of the VkVideoFormatPropertiesKHR structure to retrieve video format properties specific to video encode quantization maps.

The `VkVideoFormatQuantizationMapPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_quantization_map
typedef struct VkVideoFormatQuantizationMapPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkExtent2D         quantizationMapTexelSize;
} VkVideoFormatQuantizationMapPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `quantizationMapTexelSize` indicates the quantization map texel size of the video format, i.e. the number of pixels corresponding to each quantization map texel.

The values returned in this structure are only defined if the allowed image usage flags returned in VkVideoFormatPropertiesKHR::`imageUsageFlags` for this video format include VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR.

Implementations **may** support multiple quantization map texel sizes for a particular video format

which is indicated by vkGetPhysicalDeviceVideoFormatPropertiesKHR returning multiple entries with different quantizationMapTexelSize values.

Valid Usage (Implicit)

- VUID-VkVideoFormatQuantizationMapPropertiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_FORMAT_QUANTIZATION_MAP_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkVideoFormatPropertiesKHR

When calling vkGetPhysicalDeviceVideoFormatPropertiesKHR, the VkVideoFormatH265QuantizationMapPropertiesKHR structure **can** be included in the pNext chain of the VkVideoFormatPropertiesKHR structure to retrieve video format properties specific to video encode quantization maps used with an H.265 encode profile.

The VkVideoFormatH265QuantizationMapPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h265 with VK_KHR_video_encode_quantization_map
typedef struct VkVideoFormatH265QuantizationMapPropertiesKHR {
    VkStructureType                    sType;
    void*                              pNext;
    VkVideoEncodeH265CtbSizeFlagsKHR   compatibleCtbSizes;
} VkVideoFormatH265QuantizationMapPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- compatibleCtbSizes is a bitmask of VkVideoEncodeH265CtbSizeFlagBitsKHR indicating the CTB sizes that quantization maps using this video format are compatible with.

NOTE

The value of compatibleCtbSizes does not limit the use of the specific quantization map format, but does limit the implementation in being able to encode pictures with CTB sizes not included in compatibleCtbSizes but otherwise supported by the used video profile, as indicated by VkVideoEncodeH265CapabilitiesKHR::ctbSizes. In particular, using smaller quantization map texel sizes may prevent implementations from encoding with larger CTB sizes which may have a negative impact on the efficiency of the encoder.

The values returned in this structure are only defined if the allowed image usage flags returned in VkVideoFormatPropertiesKHR::imageUsageFlags for this video format include

VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR.

Valid Usage (Implicit)

- VUID-VkVideoFormatH265QuantizationMapPropertiesKHR-sType-sType sType **must** be VK_STRUCTURE_TYPE_VIDEO_FORMAT_H265_QUANTIZATION_MAP_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkVideoFormatPropertiesKHR

When calling vkGetPhysicalDeviceVideoFormatPropertiesKHR, the VkVideoFormatAV1QuantizationMapPropertiesKHR structure **can** be included in the pNext chain of the VkVideoFormatPropertiesKHR structure to retrieve video format properties specific to video encode quantization maps used with an AV1 encode profile.

The VkVideoFormatAV1QuantizationMapPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_av1 with VK_KHR_video_encode_quantization_map
typedef struct VkVideoFormatAV1QuantizationMapPropertiesKHR {
    VkStructureType                          sType;
    void*                                    pNext;
    VkVideoEncodeAV1SuperblockSizeFlagsKHR   compatibleSuperblockSizes;
} VkVideoFormatAV1QuantizationMapPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- compatibleSuperblockSizes is a bitmask of VkVideoEncodeAV1SuperblockSizeFlagBitsKHR indicating the AV1 superblock sizes that quantization maps using this video format are compatible with.

NOTE

The value of compatibleSuperblockSizes does not limit the use of the specific quantization map format, but does limit the implementation in being able to encode pictures with superblock sizes not included in compatibleSuperblockSizes but otherwise supported by the used video profile, as indicated by VkVideoEncodeAV1CapabilitiesKHR::superblockSizes. In particular, using smaller quantization map texel sizes may prevent implementations from encoding with larger superblock sizes which may have a negative impact on the efficiency of the encoder.

The values returned in this structure are only defined if the allowed image usage flags returned in

VkVideoFormatPropertiesKHR::imageUsageFlags for this video format include VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR.

Valid Usage (Implicit)

- VUID-VkVideoFormatAV1QuantizationMapPropertiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_FORMAT_AV1_QUANTIZATION_MAP_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkVideoFormatPropertiesKHR

45.19.5. Encoding with Quantization Maps

The VkVideoEncodeQuantizationMapInfoKHR structure **can** be included in the pNext chain of the VkVideoEncodeInfoKHR structure passed to the vkCmdEncodeVideoKHR command to specify the quantization map used by the issued video encode operations.

The VkVideoEncodeQuantizationMapInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_quantization_map
typedef struct VkVideoEncodeQuantizationMapInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkImageView        quantizationMap;
    VkExtent2D         quantizationMapExtent;
} VkVideoEncodeQuantizationMapInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- quantizationMap specifies the image view to use as the quantization map.
- quantizationMapExtent specifies the extent of the image subregion of quantizationMap to use as the quantization map starting at offset (0,0).

Valid Usage

- VUID-VkVideoEncodeQuantizationMapInfoKHR-quantizationMapExtent-10352
  quantizationMapExtent.width **must** be less than or equal to the width of quantizationMap
- VUID-VkVideoEncodeQuantizationMapInfoKHR-quantizationMapExtent-10353
  quantizationMapExtent.height **must** be less than or equal to the height of quantizationMap

Valid Usage (Implicit)

- VUID-VkVideoEncodeQuantizationMapInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUANTIZATION_MAP_INFO_KHR
- VUID-VkVideoEncodeQuantizationMapInfoKHR-quantizationMap-parameter
  If `quantizationMap` is not VK_NULL_HANDLE, `quantizationMap` **must** be a valid VkImageView handle

Structure Chaining

Extends the structure

- VkVideoEncodeInfoKHR

45.20. H.264 Encode Operations

Video encode operations using an H.264 encode profile **can** be used to encode elementary video stream sequences compliant to the ITU-T H.264 Specification.

NOTE	Refer to the Preamble for information on how the Khronos Intellectual Property Rights Policy relates to normative references to external materials not created by Khronos.

This process is performed according to the video encode operation steps with the codec-specific semantics defined in section 8 of the ITU-T H.264 Specification as follows:

- Syntax elements, derived values, and other parameters are applied from the following structures:
  - The `StdVideoH264SequenceParameterSet` structure corresponding to the active SPS specifying the H.264 sequence parameter set.
  - The `StdVideoH264PictureParameterSet` structure corresponding to the active PPS specifying the H.264 picture parameter set.
  - The `StdVideoEncodeH264PictureInfo` structure specifying the H.264 picture information.
  - The `StdVideoEncodeH264SliceHeader` structures specifying the H.264 slice header parameters for each encoded H.264 slice.
  - The `StdVideoEncodeH264ReferenceInfo` structures specifying the H.264 reference information corresponding to the optional reconstructed picture and any active reference pictures.
- The encoded bitstream data is written to the destination video bitstream buffer range as defined in the H.264 Encode Bitstream Data Access section.
- Picture data in the video picture resources corresponding to the used encode input picture, active reference pictures, and optional reconstructed picture is accessed as defined in the H.264 Encode Picture Data Access section.

- The decision on reference picture setup is made according to the parameters specified in the H.264 picture information.

If the parameters adhere to the syntactic and semantic requirements defined in the corresponding sections of the ITU-T H.264 Specification, as described above, and the DPB slots associated with the active reference pictures all refer to valid picture references, then the video encode operation will complete successfully. Otherwise, the video encode operation **may** complete unsuccessfully.

45.20.1. H.264 Encode Parameter Overrides

Implementations **may** override, unless otherwise specified, any of the H.264 encode parameters specified in the following Video Std structures:

- `StdVideoH264SequenceParameterSet`
- `StdVideoH264PictureParameterSet`
- `StdVideoEncodeH264PictureInfo`
- `StdVideoEncodeH264SliceHeader`
- `StdVideoEncodeH264ReferenceInfo`

All such H.264 encode parameter overrides **must** fulfill the conditions defined in the Video Encode Parameter Overrides section.

In addition, implementations **must** not override any of the following H.264 encode parameters:

- `StdVideoEncodeH264PictureInfo`::`primary_pic_type`
- `StdVideoEncodeH264SliceHeader`::`slice_type`

If the `videoMaintenance2` feature is enabled, implementations **must** not override any of the following H.264 encode parameters:

- the following parameters specified in `StdVideoH264SequenceParameterSet`:
  - `flags.vui_parameters_present_flag`
  - `profile_idc`
  - `level_idc`
  - `chroma_format_idc`
- the following parameters specified in the `StdVideoH264SequenceParameterSetVui` structure pointed to by `StdVideoH264SequenceParameterSet`::`pSequenceParameterSetVui`:
  - `flags.aspect_ratio_info_present_flag`
  - `flags.overscan_info_present_flag`
  - `flags.overscan_appropriate_flag`
  - `flags.video_signal_type_present_flag`
  - `flags.video_full_range_flag`
  - `flags.color_description_present_flag`

- `flags.chroma_loc_info_present_flag`
- `flags.timing_info_present_flag`
- `flags.fixed_frame_rate_flag`
- `aspect_ratio_idc`
- `sar_width`
- `sar_height`
- `video_format`
- `colour_primaries`
- `transfer_characteristics`
- `matrix_coefficients`
- `num_units_in_tick`
- `time_scale`
- `chroma_sample_loc_type_top_field`
- `chroma_sample_loc_type_bottom_field`

In case of H.264 encode parameters stored in video session parameters objects, applications need to use the vkGetEncodedVideoSessionParametersKHR command to determine whether any implementation overrides happened. If the query indicates that implementation overrides were applied, then the application needs to retrieve and use the encoded H.264 parameter sets in the bitstream in order to be able to produce a compliant H.264 video bitstream using the H.264 encode parameters stored in the video session parameters object.

In case of any H.264 encode parameters stored in the encoded bitstream produced by video encode operations, if the implementation supports the VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR video encode feedback query flag, the application **can** use such queries to retrieve feedback about whether any implementation overrides have been applied to those H.264 encode parameters.

45.20.2. H.264 Encode Bitstream Data Access

Each video encode operation writes one or more VCL NAL units comprising of slice headers and data of the encoded picture, in the format defined in sections 7.3.3 and 7.3.4, according to the semantics defined in sections 7.4.3 and 7.4.4 of the ITU-T H.264 Specification, respectively. The number of VCL NAL units written is specified by VkVideoEncodeH264PictureInfoKHR ::`naluSliceEntryCount`.

In addition, if VkVideoEncodeH264PictureInfoKHR::`generatePrefixNalu` is VK_TRUE for the video encode operation, then an additional prefix NAL unit is written before each VCL NAL unit corresponding to individual slices in the format defined in section 7.3.2.12, according to the semantics defined in section 7.4.2.12 of the ITU-T H.264 Specification, respectively.

45.20.3. H.264 Encode Picture Data Access

Accesses to image data within a video picture resource happen at the granularity indicated by VkVideoCapabilitiesKHR::pictureAccessGranularity, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile. Accordingly, the complete image subregion of an encode input picture, reference picture, or reconstructed picture accessed by video coding operations using an H.264 encode profile is defined as the set of texels within the coordinate range:

([0,endX), [0,endY))

Where:

- endX equals codedExtent.width rounded up to the nearest integer multiple of pictureAccessGranularity.width and clamped to the width of the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure;
- endY equals codedExtent.height rounded up to the nearest integer multiple of pictureAccessGranularity.height and clamped to the height of the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure;

Where codedExtent is the member of the VkVideoPictureResourceInfoKHR structure corresponding to the picture.

In case of video encode operations using an H.264 encode profile, any access to a picture at the coordinates (x,y), as defined by the ITU-T H.264 Specification, is an access to the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure at the texel coordinates (x,y).

Implementations **may** choose not to access some or all texels within particular reference pictures available to a video encode operation (e.g. due to video encode parameter overrides restricting the effective set of used reference pictures, or if the encoding algorithm chooses not to use certain subregions of the reference picture data for sample prediction).

45.20.4. H.264 Frame, Picture, and Slice

H.264 pictures are partitioned into slices, as defined in section 6.3 of the ITU-T H.264 Specification.

For the purposes of this specification, the H.264 slices comprising a picture are referred to as the *picture partitions* of the picture.

Video encode operations using an H.264 encode profile **can** encode slices of different types, as defined in section 7.4.3 of the ITU-T H.264 Specification, by specifying the corresponding enumeration constant value in StdVideoEncodeH264SliceHeader::slice_type in the H.264 slice header parameters from the Video Std enumeration type StdVideoH264SliceType:

- STD_VIDEO_H264_SLICE_TYPE_P indicates that the slice is a *P slice* as defined in section 3.109 of the ITU-T H.264 Specification.
- STD_VIDEO_H264_SLICE_TYPE_B indicates that the slice is a *B slice* as defined in section 3.9 of the

ITU-T H.264 Specification.

- STD_VIDEO_H264_SLICE_TYPE_I indicates that the slice is an *I slice* as defined in section 3.66 of the ITU-T H.264 Specification.

Pictures constructed from such slices **can** be of different types, as defined in section 7.4.2.4 of the ITU-T H.264 Specification. Video encode operations using an H.264 encode profile **can** encode pictures of a specific type by specifying the corresponding enumeration constant value in StdVideoEncodeH264PictureInfo::primary_pic_type in the H.264 picture information from the Video Std enumeration type StdVideoH264PictureType:

- STD_VIDEO_H264_PICTURE_TYPE_P indicates that the picture is a *P picture*. A frame consisting of a P picture is also referred to as a *P frame*.
- STD_VIDEO_H264_PICTURE_TYPE_B indicates that the picture is a *B picture*. A frame consisting of a B picture is also referred to as a *B frame*.
- STD_VIDEO_H264_PICTURE_TYPE_I indicates that the picture is an *I picture*. A frame consisting of an I picture is also referred to as an *I frame*.
- STD_VIDEO_H264_PICTURE_TYPE_IDR indicates that the picture is a special type of I picture called an *IDR picture* as defined in section 3.69 of the ITU-T H.264 Specification. A frame consisting of an IDR picture is also referred to as an *IDR frame*.

45.20.5. H.264 Coding Blocks

H.264 encode supports a single type of coding block called a *macroblock*, as defined in section 3.84 of the ITU-T H.264 Specification.

An H.264 macroblock is considered *skipped* when the value of mb_type for the macroblock equals P_Skip or B_Skip, as defined in section 7.4.5 of the ITU-T H.264 Specification.

45.20.6. H.264 Encode Profile

A video profile supporting H.264 video encode operations is specified by setting VkVideoProfileInfoKHR::videoCodecOperation to VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR and adding a VkVideoEncodeH264ProfileInfoKHR structure to the VkVideoProfileInfoKHR::pNext chain.

The VkVideoEncodeH264ProfileInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264ProfileInfoKHR {
    VkStructureType           sType;
    const void*               pNext;
    StdVideoH264ProfileIdc    stdProfileIdc;
} VkVideoEncodeH264ProfileInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.

- `stdProfileIdc` is a `StdVideoH264ProfileIdc` value specifying the H.264 codec profile IDC, where enum constant `STD_VIDEO_H264_PROFILE_IDC_BASELINE` identifies the Constrained Baseline profile as defined in A.2.1.1 of the ITU-T H.264 Specification, and all other values correspond to profiles defined in section A.2 of the ITU-T H.264 Specification.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264ProfileInfoKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_PROFILE_INFO_KHR`

Structure Chaining

Extends the structures

- VkQueryPoolCreateInfo
- VkVideoProfileInfoKHR

45.20.7. H.264 Encode Capabilities

When calling vkGetPhysicalDeviceVideoCapabilitiesKHR to query the capabilities for an H.264 encode profile, the VkVideoCapabilitiesKHR::`pNext` chain **must** include a `VkVideoEncodeH264CapabilitiesKHR` structure that will be filled with the profile-specific capabilities.

The `VkVideoEncodeH264CapabilitiesKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264CapabilitiesKHR {
    VkStructureType                          sType;
    void*                                    pNext;
    VkVideoEncodeH264CapabilityFlagsKHR      flags;
    StdVideoH264LevelIdc                     maxLevelIdc;
    uint32_t                                 maxSliceCount;
    uint32_t                                 maxPPictureL0ReferenceCount;
    uint32_t                                 maxBPictureL0ReferenceCount;
    uint32_t                                 maxL1ReferenceCount;
    uint32_t                                 maxTemporalLayerCount;
    VkBool32                                 expectDyadicTemporalLayerPattern;
    int32_t                                  minQp;
    int32_t                                  maxQp;
    VkBool32                                 prefersGopRemainingFrames;
    VkBool32                                 requiresGopRemainingFrames;
    VkVideoEncodeH264StdFlagsKHR             stdSyntaxFlags;
} VkVideoEncodeH264CapabilitiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `flags` is a bitmask of VkVideoEncodeH264CapabilityFlagBitsKHR indicating supported H.264 encoding capabilities.
- `maxLevelIdc` is a `StdVideoH264LevelIdc` value indicating the maximum H.264 level supported by the profile, where enum constant `STD_VIDEO_H264_LEVEL_IDC_<major>_<minor>` identifies H.264 level `<major>.<minor>` as defined in section A.3 of the ITU-T H.264 Specification.
- `maxSliceCount` indicates the maximum number of slices that **can** be encoded for a single picture. Further restrictions **may** apply to the number of slices that **can** be encoded for a single picture depending on other capabilities and codec-specific rules.
- `maxPPictureL0ReferenceCount` indicates the maximum number of reference pictures the implementation supports in the reference list L0 for P pictures.

> **NOTE** As implementations **may** override the reference lists, `maxPPictureL0ReferenceCount` does not limit the number of elements that the application **can** specify in the L0 reference list for P pictures. However, if `maxPPictureL0ReferenceCount` is zero, then the use of P pictures is not allowed.

- `maxBPictureL0ReferenceCount` indicates the maximum number of reference pictures the implementation supports in the reference list L0 for B pictures.
- `maxL1ReferenceCount` indicates the maximum number of reference pictures the implementation supports in the reference list L1 if encoding of B pictures is supported.

> **NOTE** As implementations **may** override the reference lists, `maxBPictureL0ReferenceCount` and `maxL1ReferenceCount` do not limit the number of elements that the application **can** specify in the L0 and L1 reference lists for B pictures. However, if `maxBPictureL0ReferenceCount` and `maxL1ReferenceCount` are both zero, then the use of B pictures is not allowed.

- `maxTemporalLayerCount` indicates the maximum number of H.264 temporal layers supported by the implementation.
- `expectDyadicTemporalLayerPattern` indicates that the implementation's rate control algorithms expect the application to use a dyadic temporal layer pattern when encoding multiple temporal layers.
- `minQp` indicates the minimum QP value supported.
- `maxQp` indicates the maximum QP value supported.
- `prefersGopRemainingFrames` indicates that the implementation's rate control algorithm prefers the application to specify the number of frames of each type remaining in the current group of pictures when beginning a video coding scope.
- `requiresGopRemainingFrames` indicates that the implementation's rate control algorithm requires the application to specify the number of frames of each type remaining in the current group of pictures when beginning a video coding scope.
- `stdSyntaxFlags` is a bitmask of VkVideoEncodeH264StdFlagBitsKHR indicating capabilities related to H.264 syntax elements.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264CapabilitiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkVideoCapabilitiesKHR

Bits which **may** be set in VkVideoEncodeH264CapabilitiesKHR::`flags`, indicating the H.264 encoding capabilities supported, are:

```
// Provided by VK_KHR_video_encode_h264
typedef enum VkVideoEncodeH264CapabilityFlagBitsKHR {
    VK_VIDEO_ENCODE_H264_CAPABILITY_HRD_COMPLIANCE_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H264_CAPABILITY_PREDICTION_WEIGHT_TABLE_GENERATED_BIT_KHR =
0x00000002,
    VK_VIDEO_ENCODE_H264_CAPABILITY_ROW_UNALIGNED_SLICE_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_H264_CAPABILITY_DIFFERENT_SLICE_TYPE_BIT_KHR = 0x00000008,
    VK_VIDEO_ENCODE_H264_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_KHR = 0x00000010,
    VK_VIDEO_ENCODE_H264_CAPABILITY_B_FRAME_IN_L1_LIST_BIT_KHR = 0x00000020,
    VK_VIDEO_ENCODE_H264_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR = 0x00000040,
    VK_VIDEO_ENCODE_H264_CAPABILITY_PER_SLICE_CONSTANT_QP_BIT_KHR = 0x00000080,
    VK_VIDEO_ENCODE_H264_CAPABILITY_GENERATE_PREFIX_NALU_BIT_KHR = 0x00000100,
  // Provided by VK_KHR_video_encode_h264 with VK_KHR_video_encode_intra_refresh
    VK_VIDEO_ENCODE_H264_CAPABILITY_B_PICTURE_INTRA_REFRESH_BIT_KHR = 0x00000400,
  // Provided by VK_KHR_video_encode_h264 with VK_KHR_video_encode_quantization_map
    VK_VIDEO_ENCODE_H264_CAPABILITY_MB_QP_DIFF_WRAPAROUND_BIT_KHR = 0x00000200,
} VkVideoEncodeH264CapabilityFlagBitsKHR;
```

- VK_VIDEO_ENCODE_H264_CAPABILITY_HRD_COMPLIANCE_BIT_KHR specifies whether the implementation **may** be able to generate HRD compliant bitstreams if any of the `nal_hrd_parameters_present_flag` or `vcl_hrd_parameters_present_flag` members of `StdVideoH264SpsVuiFlags` are set to `1` in the active SPS.
- VK_VIDEO_ENCODE_H264_CAPABILITY_PREDICTION_WEIGHT_TABLE_GENERATED_BIT_KHR specifies that if `StdVideoH264PpsFlags`::`weighted_pred_flag` is set to `1` or `StdVideoH264PictureParameterSet`::`weighted_bipred_idc` is set to `STD_VIDEO_H264_WEIGHTED_BIPRED_IDC_EXPLICIT` in the active PPS when encoding a P picture or B picture, respectively, then the implementation is able to internally decide syntax for `pred_weight_table`, as defined in section 7.4.3.2 of the ITU-T H.264 Specification, and the application is not **required** to provide a weight table in the H.264 slice header parameters.
- VK_VIDEO_ENCODE_H264_CAPABILITY_ROW_UNALIGNED_SLICE_BIT_KHR specifies that each slice in a frame with multiple slices may begin or finish at any offset in a macroblock row. If not supported, all slices in the frame **must** begin at the start of a macroblock row (and hence each

slice **must** finish at the end of a macroblock row). When a picture is encoded with intra refresh, encoding non-rectangular slices also requires: support for the VkVideoEncodeIntraRefreshCapabilitiesKHR::nonRectangularIntraRefreshRegions capability.

- VK_VIDEO_ENCODE_H264_CAPABILITY_DIFFERENT_SLICE_TYPE_BIT_KHR specifies that when a frame is encoded with multiple slices, the implementation allows encoding each slice with a different StdVideoEncodeH264SliceHeader::slice_type specified in the H.264 slice header parameters. If not supported, all slices of the frame **must** be encoded with the same slice_type which corresponds to the picture type of the frame. There is one exception to this rule: if the picture is encoded with the intra refresh mode VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_PER_PICTURE_PARTITION_BIT_KHR, then the currently refreshed slice **must** specify the slice_type STD_VIDEO_H264_SLICE_TYPE_I and **can** differ from the slice_type of the other slices regardless of whether VK_VIDEO_ENCODE_H264_CAPABILITY_DIFFERENT_SLICE_TYPE_BIT_KHR is supported.
- VK_VIDEO_ENCODE_H264_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_KHR specifies support for using a B frame as L0 reference, as specified in StdVideoEncodeH264ReferenceListsInfo::RefPicList0 in the H.264 picture information.
- VK_VIDEO_ENCODE_H264_CAPABILITY_B_FRAME_IN_L1_LIST_BIT_KHR specifies support for using a B frame as L1 reference, as specified in StdVideoEncodeH264ReferenceListsInfo::RefPicList1 in the H.264 picture information.
- VK_VIDEO_ENCODE_H264_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR specifies support for specifying different QP values in the members of VkVideoEncodeH264QpKHR.
- VK_VIDEO_ENCODE_H264_CAPABILITY_PER_SLICE_CONSTANT_QP_BIT_KHR specifies support for specifying different constant QP values for each slice.
- VK_VIDEO_ENCODE_H264_CAPABILITY_GENERATE_PREFIX_NALU_BIT_KHR specifies support for generating prefix NAL units by setting VkVideoEncodeH264PictureInfoKHR::generatePrefixNalu to VK_TRUE.
- VK_VIDEO_ENCODE_H264_CAPABILITY_MB_QP_DIFF_WRAPAROUND_BIT_KHR specifies support for wraparound during the calculation of the QP values of subsequently encoded macroblocks, as defined in equation 7-37 of the ITU-T H.264 Specification. If not supported, equation 7-37 of the ITU-T H.264 Specification is effectively reduced to the following:

  $QP_Y = QP_{Y,PREV} + mb_qp_delta$

  NOTE The effect of this is that the maximum QP difference across subsequent macroblocks is limited to the $[-(26 + QpBdOffset_Y / 2), 25 + QpBdOffset_Y / 2]$ range.

- VK_VIDEO_ENCODE_H264_CAPABILITY_B_PICTURE_INTRA_REFRESH_BIT_KHR specifies support for encoding B pictures with intra refresh enabled.

```
// Provided by VK_KHR_video_encode_h264
typedef VkFlags VkVideoEncodeH264CapabilityFlagsKHR;
```

VkVideoEncodeH264CapabilityFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeH264CapabilityFlagBitsKHR.

Bits which **may** be set in VkVideoEncodeH264CapabilitiesKHR::stdSyntaxFlags, indicating the capabilities related to the H.264 syntax elements, are:

```
// Provided by VK_KHR_video_encode_h264
typedef enum VkVideoEncodeH264StdFlagBitsKHR {
    VK_VIDEO_ENCODE_H264_STD_SEPARATE_COLOR_PLANE_FLAG_SET_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H264_STD_QPPRIME_Y_ZERO_TRANSFORM_BYPASS_FLAG_SET_BIT_KHR =
0x00000002,
    VK_VIDEO_ENCODE_H264_STD_SCALING_MATRIX_PRESENT_FLAG_SET_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_H264_STD_CHROMA_QP_INDEX_OFFSET_BIT_KHR = 0x00000008,
    VK_VIDEO_ENCODE_H264_STD_SECOND_CHROMA_QP_INDEX_OFFSET_BIT_KHR = 0x00000010,
    VK_VIDEO_ENCODE_H264_STD_PIC_INIT_QP_MINUS26_BIT_KHR = 0x00000020,
    VK_VIDEO_ENCODE_H264_STD_WEIGHTED_PRED_FLAG_SET_BIT_KHR = 0x00000040,
    VK_VIDEO_ENCODE_H264_STD_WEIGHTED_BIPRED_IDC_EXPLICIT_BIT_KHR = 0x00000080,
    VK_VIDEO_ENCODE_H264_STD_WEIGHTED_BIPRED_IDC_IMPLICIT_BIT_KHR = 0x00000100,
    VK_VIDEO_ENCODE_H264_STD_TRANSFORM_8X8_MODE_FLAG_SET_BIT_KHR = 0x00000200,
    VK_VIDEO_ENCODE_H264_STD_DIRECT_SPATIAL_MV_PRED_FLAG_UNSET_BIT_KHR = 0x00000400,
    VK_VIDEO_ENCODE_H264_STD_ENTROPY_CODING_MODE_FLAG_UNSET_BIT_KHR = 0x00000800,
    VK_VIDEO_ENCODE_H264_STD_ENTROPY_CODING_MODE_FLAG_SET_BIT_KHR = 0x00001000,
    VK_VIDEO_ENCODE_H264_STD_DIRECT_8X8_INFERENCE_FLAG_UNSET_BIT_KHR = 0x00002000,
    VK_VIDEO_ENCODE_H264_STD_CONSTRAINED_INTRA_PRED_FLAG_SET_BIT_KHR = 0x00004000,
    VK_VIDEO_ENCODE_H264_STD_DEBLOCKING_FILTER_DISABLED_BIT_KHR = 0x00008000,
    VK_VIDEO_ENCODE_H264_STD_DEBLOCKING_FILTER_ENABLED_BIT_KHR = 0x00010000,
    VK_VIDEO_ENCODE_H264_STD_DEBLOCKING_FILTER_PARTIAL_BIT_KHR = 0x00020000,
    VK_VIDEO_ENCODE_H264_STD_SLICE_QP_DELTA_BIT_KHR = 0x00080000,
    VK_VIDEO_ENCODE_H264_STD_DIFFERENT_SLICE_QP_DELTA_BIT_KHR = 0x00100000,
} VkVideoEncodeH264StdFlagBitsKHR;
```

- VK_VIDEO_ENCODE_H264_STD_SEPARATE_COLOR_PLANE_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH264SpsFlags::separate_colour_plane_flag in the SPS when that value is 1.
- VK_VIDEO_ENCODE_H264_STD_QPPRIME_Y_ZERO_TRANSFORM_BYPASS_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH264SpsFlags::qpprime_y_zero_transform_bypass_flag in the SPS when that value is 1.
- VK_VIDEO_ENCODE_H264_STD_SCALING_MATRIX_PRESENT_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided values for StdVideoH264SpsFlags::seq_scaling_matrix_present_flag in the SPS and StdVideoH264PpsFlags ::pic_scaling_matrix_present_flag in the PPS when any of those values are 1.
- VK_VIDEO_ENCODE_H264_STD_CHROMA_QP_INDEX_OFFSET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH264PictureParameterSet::chroma_qp_index_offset in the PPS when that value is non-zero.
- VK_VIDEO_ENCODE_H264_STD_SECOND_CHROMA_QP_INDEX_OFFSET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH264PictureParameterSet::second_chroma_qp_index_offset in the PPS when that value is non-zero.

- VK_VIDEO_ENCODE_H264_STD_PIC_INIT_QP_MINUS26_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH264PictureParameterSet::pic_init_qp_minus26 in the PPS when that value is non-zero.
- VK_VIDEO_ENCODE_H264_STD_WEIGHTED_PRED_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH264PpsFlags ::weighted_pred_flag in the PPS when that value is 1.
- VK_VIDEO_ENCODE_H264_STD_WEIGHTED_BIPRED_IDC_EXPLICIT_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH264PictureParameterSet::weighted_bipred_idc in the PPS when that value is STD_VIDEO_H264_WEIGHTED_BIPRED_IDC_EXPLICIT.
- VK_VIDEO_ENCODE_H264_STD_WEIGHTED_BIPRED_IDC_IMPLICIT_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH264PictureParameterSet::weighted_bipred_idc in the PPS when that value is STD_VIDEO_H264_WEIGHTED_BIPRED_IDC_IMPLICIT.
- VK_VIDEO_ENCODE_H264_STD_TRANSFORM_8X8_MODE_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH264PpsFlags ::transform_8x8_mode_flag in the PPS when that value is 1.
- VK_VIDEO_ENCODE_H264_STD_DIRECT_SPATIAL_MV_PRED_FLAG_UNSET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoEncodeH264SliceHeaderFlags::direct_spatial_mv_pred_flag in the H.264 slice header parameters when that value is 0.
- VK_VIDEO_ENCODE_H264_STD_ENTROPY_CODING_MODE_FLAG_UNSET_BIT_KHR specifies whether the implementation supports CAVLC entropy coding, as defined in section 9.2 of the ITU-T H.264 Specification, and thus supports using the application-provided value for StdVideoH264PpsFlags::entropy_coding_mode_flag in the PPS when that value is 0.
- VK_VIDEO_ENCODE_H264_STD_ENTROPY_CODING_MODE_FLAG_SET_BIT_KHR specifies whether the implementation supports CABAC entropy coding, as defined in section 9.3 of the ITU-T H.264 Specification, and thus supports using the application-provided value for StdVideoH264PpsFlags::entropy_coding_mode_flag in the PPS when that value is 1.
- VK_VIDEO_ENCODE_H264_STD_DIRECT_8X8_INFERENCE_FLAG_UNSET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH264SpsFlags::direct_8x8_inference_flag in the SPS when that value is 0.
- VK_VIDEO_ENCODE_H264_STD_CONSTRAINED_INTRA_PRED_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH264PpsFlags::constrained_intra_pred_flag in the PPS when that value is 1.
- VK_VIDEO_ENCODE_H264_STD_DEBLOCKING_FILTER_DISABLED_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoEncodeH264SliceHeader::disable_deblocking_filter_idc in the H.264 slice header parameters when that value is STD_VIDEO_H264_DISABLE_DEBLOCKING_FILTER_IDC_DISABLED.
- VK_VIDEO_ENCODE_H264_STD_DEBLOCKING_FILTER_ENABLED_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoEncodeH264SliceHeader::disable_deblocking_filter_idc in the H.264 slice header parameters when that value is STD_VIDEO_H264_DISABLE_DEBLOCKING_FILTER_IDC_ENABLED.

- VK_VIDEO_ENCODE_H264_STD_DEBLOCKING_FILTER_PARTIAL_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoEncodeH264SliceHeader::disable_deblocking_filter_idc in the H.264 slice header parameters when that value is STD_VIDEO_H264_DISABLE_DEBLOCKING_FILTER_IDC_PARTIAL.
- VK_VIDEO_ENCODE_H264_STD_SLICE_QP_DELTA_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoEncodeH264SliceHeader::slice_qp_delta in the H.264 slice header parameters when that value is identical across the slices of the encoded frame.
- VK_VIDEO_ENCODE_H264_STD_DIFFERENT_SLICE_QP_DELTA_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoEncodeH264SliceHeader::slice_qp_delta in the H.264 slice header parameters when that value is different across the slices of the encoded frame.

These capability flags provide information to the application about specific H.264 syntax element values that the implementation supports without having to override them and do not otherwise restrict the values that the application **can** specify for any of the mentioned H.264 syntax elements.

```
// Provided by VK_KHR_video_encode_h264
typedef VkFlags VkVideoEncodeH264StdFlagsKHR;
```

VkVideoEncodeH264StdFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeH264StdFlagBitsKHR.

45.20.8. H.264 Encode Quality Level Properties

When calling vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR with pVideoProfile->videoCodecOperation specified as VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, the VkVideoEncodeH264QualityLevelPropertiesKHR structure **must** be included in the pNext chain of the VkVideoEncodeQualityLevelPropertiesKHR structure to retrieve additional video encode quality level properties specific to H.264 encoding.

The VkVideoEncodeH264QualityLevelPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264QualityLevelPropertiesKHR {
    VkStructureType                          sType;
    void*                                    pNext;
    VkVideoEncodeH264RateControlFlagsKHR     preferredRateControlFlags;
    uint32_t                                 preferredGopFrameCount;
    uint32_t                                 preferredIdrPeriod;
    uint32_t                                 preferredConsecutiveBFrameCount;
    uint32_t                                 preferredTemporalLayerCount;
    VkVideoEncodeH264QpKHR                   preferredConstantQp;
    uint32_t                                 preferredMaxL0ReferenceCount;
    uint32_t                                 preferredMaxL1ReferenceCount;
    VkBool32                                 preferredStdEntropyCodingModeFlag;
} VkVideoEncodeH264QualityLevelPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `preferredRateControlFlags` is a bitmask of VkVideoEncodeH264RateControlFlagBitsKHR values indicating the preferred flags to use for VkVideoEncodeH264RateControlInfoKHR::`flags`.
- `preferredGopFrameCount` indicates the preferred value to use for VkVideoEncodeH264RateControlInfoKHR::`gopFrameCount`.
- `preferredIdrPeriod` indicates the preferred value to use for VkVideoEncodeH264RateControlInfoKHR::`idrPeriod`.
- `preferredConsecutiveBFrameCount` indicates the preferred value to use for VkVideoEncodeH264RateControlInfoKHR::`consecutiveBFrameCount`.
- `preferredTemporalLayerCount` indicates the preferred value to use for VkVideoEncodeH264RateControlInfoKHR::`temporalLayerCount`.
- `preferredConstantQp` indicates the preferred values to use for VkVideoEncodeH264NaluSliceInfoKHR::`constantQp` for each picture type when using rate control mode VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR.
- `preferredMaxL0ReferenceCount` indicates the preferred maximum number of reference pictures to use in the reference list L0.
- `preferredMaxL1ReferenceCount` indicates the preferred maximum number of reference pictures to use in the reference list L1.
- `preferredStdEntropyCodingModeFlag` indicates the preferred value to use for `entropy_coding_mode_flag` in `StdVideoH264PpsFlags`.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264QualityLevelPropertiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_QUALITY_LEVEL_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkVideoEncodeQualityLevelPropertiesKHR

45.20.9. H.264 Encode Session

Additional parameters can be specified when creating a video session with an H.264 encode profile by including an instance of the VkVideoEncodeH264SessionCreateInfoKHR structure in the pNext chain of VkVideoSessionCreateInfoKHR.

The VkVideoEncodeH264SessionCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264SessionCreateInfoKHR {
    VkStructureType         sType;
    const void*             pNext;
    VkBool32                useMaxLevelIdc;
    StdVideoH264LevelIdc    maxLevelIdc;
} VkVideoEncodeH264SessionCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- useMaxLevelIdc indicates whether the value of maxLevelIdc should be used by the implementation. When it is VK_FALSE, the implementation ignores the value of maxLevelIdc and uses the value of VkVideoEncodeH264CapabilitiesKHR::maxLevelIdc, as reported by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile.
- maxLevelIdc is a StdVideoH264LevelIdc value specifying the upper bound on the H.264 level for the video bitstreams produced by the created video session, where enum constant STD_VIDEO_H264_LEVEL_IDC_<major>_<minor> identifies H.264 level <major>.<minor> as defined in section A.3 of the ITU-T H.264 Specification.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264SessionCreateInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_CREATE_INFO_KHR

Structure Chaining

Extends the structure

- VkVideoSessionCreateInfoKHR

45.20.10. H.264 Encode Parameter Sets

Video session parameters objects created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR **can** contain the following types of parameters:

H.264 Sequence Parameter Sets (SPS)

Represented by StdVideoH264SequenceParameterSet structures and interpreted as follows:

- reserved1 and reserved2 are used only for padding purposes and are otherwise ignored;
- seq_parameter_set_id is used as the key of the SPS entry;
- level_idc is one of the enum constants STD_VIDEO_H264_LEVEL_IDC_<major>_<minor> identifying the H.264 level <major>.<minor> as defined in section A.3 of the ITU-T H.264 Specification;
- pOffsetForRefFrame is a pointer to an array of num_ref_frames_in_pic_order_cnt_cycle number of signed integers specifying the elements of the offset_for_ref_frame array, as defined in section 7.4.2.1.1 of the ITU-T H.264 Specification;
- if flags.seq_scaling_matrix_present_flag is set, then the StdVideoH264ScalingLists structure pointed to by pScalingLists is interpreted as follows:
  - scaling_list_present_mask is a bitmask where bit index i corresponds to seq_scaling_list_present_flag[i] as defined in section 7.4.2.1 of the ITU-T H.264 Specification;
  - use_default_scaling_matrix_mask is a bitmask where bit index i corresponds to UseDefaultScalingMatrix4x4Flag[i], when i < 6, or corresponds to UseDefaultScalingMatrix8x8Flag[i-6], otherwise, as defined in section 7.3.2.1 of the ITU-T H.264 Specification;
  - ScalingList4x4 and ScalingList8x8 correspond to the identically named syntax elements defined in section 7.3.2.1 of the ITU-T H.264 Specification;
- if flags.vui_parameters_present_flag is set, then pSequenceParameterSetVui is a pointer to a StdVideoH264SequenceParameterSetVui structure that is interpreted as follows:
  - reserved1 is used only for padding purposes and is otherwise ignored;
  - flags.color_description_present_flag is interpreted as the value of colour_description_present_flag, as defined in section E.2.1 of the ITU-T H.264 Specification;

    NOTE The name of colour_description_present_flag was misspelled in the Video Std header.

  - if flags.nal_hrd_parameters_present_flag or flags.vcl_hrd_parameters_present_flag is set, then the StdVideoH264HrdParameters structure pointed to by pHrdParameters is interpreted as follows:
    - reserved1 is used only for padding purposes and is otherwise ignored;
    - all other members of StdVideoH264HrdParameters are interpreted as defined in section

E.2.2 of the ITU-T H.264 Specification;

- all other members of `StdVideoH264SequenceParameterSetVui` are interpreted as defined in section E.2.1 of the ITU-T H.264 Specification;

- all other members of `StdVideoH264SequenceParameterSet` are interpreted as defined in section 7.4.2.1 of the ITU-T H.264 Specification.

H.264 Picture Parameter Sets (PPS)

Represented by `StdVideoH264PictureParameterSet` structures and interpreted as follows:

- the pair constructed from `seq_parameter_set_id` and `pic_parameter_set_id` is used as the key of the PPS entry;
- if `flags.pic_scaling_matrix_present_flag` is set, then the `StdVideoH264ScalingLists` structure pointed to by `pScalingLists` is interpreted as follows:
  - `scaling_list_present_mask` is a bitmask where bit index i corresponds to `pic_scaling_list_present_flag[i]` as defined in section 7.4.2.2 of the ITU-T H.264 Specification;
  - `use_default_scaling_matrix_mask` is a bitmask where bit index i corresponds to `UseDefaultScalingMatrix4x4Flag[i]`, when i < 6, or corresponds to `UseDefaultScalingMatrix8x8Flag[i-6]`, otherwise, as defined in section 7.3.2.2 of the ITU-T H.264 Specification;
  - `ScalingList4x4` and `ScalingList8x8` correspond to the identically named syntax elements defined in section 7.3.2.2 of the ITU-T H.264 Specification;
- all other members of `StdVideoH264PictureParameterSet` are interpreted as defined in section 7.4.2.2 of the ITU-T H.264 Specification.

Implementations **may** override any of these parameters according to the semantics defined in the Video Encode Parameter Overrides section before storing the resulting H.264 parameter sets into the video session parameters object. Applications need to use the vkGetEncodedVideoSessionParametersKHR command to determine whether any implementation overrides happened and to retrieve the encoded H.264 parameter sets in order to be able to produce a compliant H.264 video bitstream.

Such H.264 parameter set overrides **may** also have cascading effects on the implementation overrides applied to the encoded bitstream produced by video encode operations. If the implementation supports the VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR video encode feedback query flag, then the application **can** use such queries to retrieve feedback about whether any implementation overrides have been applied to the encoded bitstream.

When a video session parameters object is created with the codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, the VkVideoSessionParametersCreateInfoKHR::`pNext` chain **must** include a `VkVideoEncodeH264SessionParametersCreateInfoKHR` structure specifying the capacity and initial contents of the object.

The VkVideoEncodeH264SessionParametersCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264SessionParametersCreateInfoKHR {
    VkStructureType                                       sType;
    const void*                                           pNext;
    uint32_t                                              maxStdSPSCount;
    uint32_t                                              maxStdPPSCount;
    const VkVideoEncodeH264SessionParametersAddInfoKHR*   pParametersAddInfo;
} VkVideoEncodeH264SessionParametersCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- maxStdSPSCount is the maximum number of H.264 SPS entries the created VkVideoSessionParametersKHR **can** contain.
- maxStdPPSCount is the maximum number of H.264 PPS entries the created VkVideoSessionParametersKHR **can** contain.
- pParametersAddInfo is NULL or a pointer to a VkVideoEncodeH264SessionParametersAddInfoKHR structure specifying H.264 parameters to add upon object creation.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264SessionParametersCreateInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_CREATE_INFO_KHR
- VUID-VkVideoEncodeH264SessionParametersCreateInfoKHR-pParametersAddInfo-parameter
  If pParametersAddInfo is not NULL, pParametersAddInfo **must** be a valid pointer to a valid VkVideoEncodeH264SessionParametersAddInfoKHR structure

Structure Chaining

Extends the structure

- VkVideoSessionParametersCreateInfoKHR

The VkVideoEncodeH264SessionParametersAddInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264SessionParametersAddInfoKHR {
    VkStructureType                             sType;
    const void*                                 pNext;
    uint32_t                                    stdSPSCount;
    const StdVideoH264SequenceParameterSet*     pStdSPSs;
    uint32_t                                    stdPPSCount;
    const StdVideoH264PictureParameterSet*      pStdPPSs;
} VkVideoEncodeH264SessionParametersAddInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stdSPSCount` is the number of elements in the `pStdSPSs` array.
- `pStdSPSs` is a pointer to an array of `StdVideoH264SequenceParameterSet` structures describing the H.264 SPS entries to add.
- `stdPPSCount` is the number of elements in the `pStdPPSs` array.
- `pStdPPSs` is a pointer to an array of `StdVideoH264PictureParameterSet` structures describing the H.264 PPS entries to add.

This structure **can** be specified in the following places:

- In the `pParametersAddInfo` member of the VkVideoEncodeH264SessionParametersCreateInfoKHR structure specified in the `pNext` chain of VkVideoSessionParametersCreateInfoKHR used to create a video session parameters object. In this case, if the video codec operation the video session parameters object is created with is VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then it defines the set of initial parameters to add to the created object (see Creating Video Session Parameters).
- In the `pNext` chain of VkVideoSessionParametersUpdateInfoKHR. In this case, if the video codec operation the video session parameters object to be updated was created with is VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, then it defines the set of parameters to add to it (see Updating Video Session Parameters).

Valid Usage

- VUID-VkVideoEncodeH264SessionParametersAddInfoKHR-None-04837
  The `seq_parameter_set_id` member of each `StdVideoH264SequenceParameterSet` structure specified in the elements of `pStdSPSs` **must** be unique within `pStdSPSs`
- VUID-VkVideoEncodeH264SessionParametersAddInfoKHR-None-04838
  The pair constructed from the `seq_parameter_set_id` and `pic_parameter_set_id` members of each `StdVideoH264PictureParameterSet` structure specified in the elements of `pStdPPSs` **must** be unique within `pStdPPSs`

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264SessionParametersAddInfoKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_ADD_INFO_KHR`
- VUID-VkVideoEncodeH264SessionParametersAddInfoKHR-pStdSPSs-parameter
  If `stdSPSCount` is not `0`, and `pStdSPSs` is not `NULL`, `pStdSPSs` **must** be a valid pointer to an array of `stdSPSCount` `StdVideoH264SequenceParameterSet` values
- VUID-VkVideoEncodeH264SessionParametersAddInfoKHR-pStdPPSs-parameter
  If `stdPPSCount` is not `0`, and `pStdPPSs` is not `NULL`, `pStdPPSs` **must** be a valid pointer to an array of `stdPPSCount` `StdVideoH264PictureParameterSet` values

Structure Chaining

Extends the structure

- VkVideoSessionParametersUpdateInfoKHR

The `VkVideoEncodeH264SessionParametersGetInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264SessionParametersGetInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           writeStdSPS;
    VkBool32           writeStdPPS;
    uint32_t           stdSPSId;
    uint32_t           stdPPSId;
} VkVideoEncodeH264SessionParametersGetInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `writeStdSPS` indicates whether the encoded H.264 sequence parameter set identified by `stdSPSId` is requested to be retrieved.
- `writeStdPPS` indicates whether the encoded H.264 picture parameter set identified by the pair constructed from `stdSPSId` and `stdPPSId` is requested to be retrieved.
- `stdSPSId` specifies the H.264 sequence parameter set ID used to identify the retrieved H.264 sequence and/or picture parameter set(s).
- `stdPPSId` specifies the H.264 picture parameter set ID used to identify the retrieved H.264 picture parameter set when `writeStdPPS` is VK_TRUE.

When this structure is specified in the `pNext` chain of the VkVideoEncodeSessionParametersGetInfoKHR structure passed to

vkGetEncodedVideoSessionParametersKHR, the command will write encoded parameter data to the output buffer in the following order:

1. The H.264 sequence parameter set identified by stdSPSId, if writeStdSPS is VK_TRUE.
2. The H.264 picture parameter set identified by the pair constructed from stdSPSId and stdPPSId, if writeStdPPS is VK_TRUE.

Valid Usage

- VUID-VkVideoEncodeH264SessionParametersGetInfoKHR-writeStdSPS-08279
  At least one of writeStdSPS and writeStdPPS **must** be VK_TRUE

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264SessionParametersGetInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_GET_INFO_KHR

Structure Chaining

Extends the structure

- VkVideoEncodeSessionParametersGetInfoKHR

The VkVideoEncodeH264SessionParametersFeedbackInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264SessionParametersFeedbackInfoKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           hasStdSPSOverrides;
    VkBool32           hasStdPPSOverrides;
} VkVideoEncodeH264SessionParametersFeedbackInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- hasStdSPSOverrides indicates whether any of the parameters of the requested H.264 sequence parameter set, if one was requested via VkVideoEncodeH264SessionParametersGetInfoKHR ::writeStdSPS, were overridden by the implementation.
- hasStdPPSOverrides indicates whether any of the parameters of the requested H.264 picture parameter set, if one was requested via VkVideoEncodeH264SessionParametersGetInfoKHR ::writeStdPPS, were overridden by the implementation.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264SessionParametersFeedbackInfoKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_FEEDBACK_INFO_KHR`

Structure Chaining

Extends the structure

- VkVideoEncodeSessionParametersFeedbackInfoKHR

45.20.11. H.264 Encoding Parameters

The VkVideoEncodeH264PictureInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264PictureInfoKHR {
    VkStructureType                                 sType;
    const void*                                     pNext;
    uint32_t                                        naluSliceEntryCount;
    const VkVideoEncodeH264NaluSliceInfoKHR*        pNaluSliceEntries;
    const StdVideoEncodeH264PictureInfo*            pStdPictureInfo;
    VkBool32                                        generatePrefixNalu;
} VkVideoEncodeH264PictureInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `naluSliceEntryCount` is the number of elements in `pNaluSliceEntries`.
- `pNaluSliceEntries` is a pointer to an array of `naluSliceEntryCount` VkVideoEncodeH264NaluSliceInfoKHR structures specifying the parameters of the individual H.264 slices to encode for the input picture.
- `pStdPictureInfo` is a pointer to a `StdVideoEncodeH264PictureInfo` structure specifying H.264 picture information.
- `generatePrefixNalu` controls whether prefix NALUs are generated before slice NALUs into the target bitstream, as defined in sections 7.3.2.12 and 7.4.2.12 of the ITU-T H.264 Specification.

This structure is specified in the `pNext` chain of the VkVideoEncodeInfoKHR structure passed to vkCmdEncodeVideoKHR to specify the codec-specific picture information for an H.264 encode operation.

Encode Input Picture Information

When this structure is specified in the `pNext` chain of the VkVideoEncodeInfoKHR structure

passed to vkCmdEncodeVideoKHR, the information related to the encode input picture is defined as follows:

- The image subregion used is determined according to the H.264 Encode Picture Data Access section.
- The encode input picture is associated with the H.264 picture information provided in pStdPictureInfo.

Std Picture Information

The members of the StdVideoEncodeH264PictureInfo structure pointed to by pStdPictureInfo are interpreted as follows:

- flags.reserved and reserved1 are used only for padding purposes and are otherwise ignored;
- flags.IdrPicFlag as defined in section 7.4.1 of the ITU-T H.264 Specification;
- flags.is_reference as defined in section 3.136 of the ITU-T H.264 Specification;
- seq_parameter_set_id and pic_parameter_set_id are used to identify the active parameter sets, as described below;
- primary_pic_type as defined in section 7.4.2 of the ITU-T H.264 Specification;
- PicOrderCnt as defined in section 8.2 of the ITU-T H.264 Specification;
- temporal_id as defined in section G.7.4.1.1 of the ITU-T H.264 Specification;
- if pRefLists is not NULL, then it is a pointer to a StdVideoEncodeH264ReferenceListsInfo structure that is interpreted as follows:
  - flags.reserved is used only for padding purposes and is otherwise ignored;
  - ref_pic_list_modification_flag_l0 and ref_pic_list_modification_flag_l1 as defined in section 7.4.3.1 of the ITU-T H.264 Specification;
  - num_ref_idx_l0_active_minus1 and num_ref_idx_l1_active_minus1 as defined in section 7.4.3 of the ITU-T H.264 Specification;
  - RefPicList0 and RefPicList1 as defined in section 8.2.4 of the ITU-T H.264 Specification where each element of these arrays either identifies an active reference picture using its DPB slot index or contains the value STD_VIDEO_H264_NO_REFERENCE_PICTURE to indicate “no reference picture”;
  - if refList0ModOpCount is not zero, then pRefList0ModOperations is a pointer to an array of refList0ModOpCount number of StdVideoEncodeH264RefListModEntry structures specifying the modification parameters for the reference list L0 as defined in section 7.4.3.1 of the ITU-T H.264 Specification;
  - if refList1ModOpCount is not zero, then pRefList1ModOperations is a pointer to an array of refList1ModOpCount number of StdVideoEncodeH264RefListModEntry structures specifying the modification parameters for the reference list L1 as defined in section 7.4.3.1 of the ITU-T H.264 Specification;
  - if refPicMarkingOpCount is not zero, then pRefPicMarkingOperations is a pointer to an array of refPicMarkingOpCount number of StdVideoEncodeH264RefPicMarkingEntry structures specifying the reference picture marking parameters as defined in section 7.4.3.3 of the

ITU-T H.264 Specification;

- all other members are interpreted as defined in section 7.4.3 of the ITU-T H.264 Specification.

Reference picture setup is controlled by the value of StdVideoEncodeH264PictureInfo::flags.is_reference. If it is set and a reconstructed picture is specified, then the latter is used as the target of picture reconstruction to activate the DPB slot specified in pEncodeInfo->pSetupReferenceSlot->slotIndex. If StdVideoEncodeH264PictureInfo::flags.is_reference is not set, but a reconstructed picture is specified, then the corresponding picture reference associated with the DPB slot is invalidated, as described in the DPB Slot States section.

Active Parameter Sets

The members of the StdVideoEncodeH264PictureInfo structure pointed to by pStdPictureInfo are used to select the active parameter sets to use from the bound video session parameters object, as follows:

- The *active SPS* is the SPS identified by the key specified in StdVideoEncodeH264PictureInfo::seq_parameter_set_id.
- The *active PPS* is the PPS identified by the key specified by the pair constructed from StdVideoEncodeH264PictureInfo::seq_parameter_set_id and StdVideoEncodeH264PictureInfo::pic_parameter_set_id.

H.264 encoding uses *explicit weighted sample prediction* for a slice, as defined in section 8.4.2.3 of the ITU-T H.264 Specification, if any of the following conditions are true for the active PPS and the pStdSliceHeader member of the corresponding element of pNaluSliceEntries:

- pStdSliceHeader->slice_type is STD_VIDEO_H264_SLICE_TYPE_P and weighted_pred_flag is enabled in the active PPS.
- pStdSliceHeader->slice_type is STD_VIDEO_H264_SLICE_TYPE_B and weighted_bipred_idc in the active PPS equals STD_VIDEO_H264_WEIGHTED_BIPRED_IDC_EXPLICIT.

Valid Usage

- VUID-VkVideoEncodeH264PictureInfoKHR-naluSliceEntryCount-08301
  naluSliceEntryCount **must** be between 1 and VkVideoEncodeH264CapabilitiesKHR::maxSliceCount, inclusive, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile
- VUID-VkVideoEncodeH264PictureInfoKHR-flags-08304
  If VkVideoEncodeH264CapabilitiesKHR::flags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_H264_CAPABILITY_GENERATE_PREFIX_NALU_BIT_KHR, then generatePrefixNalu **must** be VK_FALSE
- VUID-VkVideoEncodeH264PictureInfoKHR-flags-08314
  If VkVideoEncodeH264CapabilitiesKHR::flags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_H264_CAPABILITY_PREDICTION_WEIGHT_TABLE_GENERATED_BIT_KHR and the slice corresponding to any element of pNaluSliceEntries uses explicit

weighted sample prediction, then VkVideoEncodeH264NaluSliceInfoKHR ::pStdSliceHeader->pWeightTable **must** not be NULL for that element of pNaluSliceEntries

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264PictureInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_PICTURE_INFO_KHR
- VUID-VkVideoEncodeH264PictureInfoKHR-pNaluSliceEntries-parameter
  pNaluSliceEntries **must** be a valid pointer to an array of naluSliceEntryCount valid VkVideoEncodeH264NaluSliceInfoKHR structures
- VUID-VkVideoEncodeH264PictureInfoKHR-pStdPictureInfo-parameter
  pStdPictureInfo **must** be a valid pointer to a valid StdVideoEncodeH264PictureInfo value
- VUID-VkVideoEncodeH264PictureInfoKHR-naluSliceEntryCount-arraylength
  naluSliceEntryCount **must** be greater than 0

Structure Chaining

Extends the structure

- VkVideoEncodeInfoKHR

The VkVideoEncodeH264NaluSliceInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264NaluSliceInfoKHR {
    VkStructureType                         sType;
    const void*                             pNext;
    int32_t                                 constantQp;
    const StdVideoEncodeH264SliceHeader*    pStdSliceHeader;
} VkVideoEncodeH264NaluSliceInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- constantQp is the QP to use for the slice if the current rate control mode configured for the video session is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR.
- pStdSliceHeader is a pointer to a StdVideoEncodeH264SliceHeader structure specifying H.264 slice header parameters for the slice.

Std Slice Header Parameters

The members of the StdVideoEncodeH264SliceHeader structure pointed to by pStdSliceHeader are interpreted as follows:

- flags.reserved and reserved1 are used only for padding purposes and are otherwise ignored;

- if pWeightTable is not NULL, then it is a pointer to a StdVideoEncodeH264WeightTable that is interpreted as follows:
  - flags.reserved is used only for padding purposes and is otherwise ignored;
  - all other members of StdVideoEncodeH264WeightTable are interpreted as defined in section 7.4.3.2 of the ITU-T H.264 Specification;
- all other members are interpreted as defined in section 7.4.3 of the ITU-T H.264 Specification.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264NaluSliceInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_NALU_SLICE_INFO_KHR
- VUID-VkVideoEncodeH264NaluSliceInfoKHR-pNext-pNext
  pNext **must** be NULL
- VUID-VkVideoEncodeH264NaluSliceInfoKHR-pStdSliceHeader-parameter
  pStdSliceHeader **must** be a valid pointer to a valid StdVideoEncodeH264SliceHeader value

The VkVideoEncodeH264DpbSlotInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264DpbSlotInfoKHR {
    VkStructureType                          sType;
    const void*                              pNext;
    const StdVideoEncodeH264ReferenceInfo*   pStdReferenceInfo;
} VkVideoEncodeH264DpbSlotInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pStdReferenceInfo is a pointer to a StdVideoEncodeH264ReferenceInfo structure specifying H.264 reference information.

This structure is specified in the pNext chain of VkVideoEncodeInfoKHR::pSetupReferenceSlot, if not NULL, and the pNext chain of the elements of VkVideoEncodeInfoKHR::pReferenceSlots to specify the codec-specific reference picture information for an H.264 encode operation.

Active Reference Picture Information

When this structure is specified in the pNext chain of the elements of VkVideoEncodeInfoKHR::pReferenceSlots, one element is added to the list of active reference pictures used by the video encode operation for each element of VkVideoEncodeInfoKHR::pReferenceSlots as follows:

- The image subregion used is determined according to the H.264 Encode Picture Data Access section.
- The reference picture is associated with the DPB slot index specified in the slotIndex member of the corresponding element of VkVideoEncodeInfoKHR::pReferenceSlots.

- The reference picture is associated with the H.264 reference information provided in pStdReferenceInfo.

Reconstructed Picture Information

When this structure is specified in the pNext chain of VkVideoEncodeInfoKHR ::pSetupReferenceSlot, the information related to the reconstructed picture is defined as follows:

- The image subregion used is determined according to the H.264 Encode Picture Data Access section.
- If reference picture setup is requested, then the reconstructed picture is used to activate the DPB slot with the index specified in VkVideoEncodeInfoKHR::pSetupReferenceSlot->slotIndex.
- The reconstructed picture is associated with the H.264 reference information provided in pStdReferenceInfo.

Std Reference Information

The members of the StdVideoEncodeH264ReferenceInfo structure pointed to by pStdReferenceInfo are interpreted as follows:

- flags.reserved is used only for padding purposes and is otherwise ignored;
- flags.used_for_long_term_reference is used to indicate whether the picture is marked as "used for long-term reference" as defined in section 8.2.5.1 of the ITU-T H.264 Specification;
- primary_pic_type as defined in section 7.4.2 of the ITU-T H.264 Specification;
- long_term_pic_num and long_term_frame_idx as defined in section 7.4.3 of the ITU-T H.264 Specification;
- temporal_id as defined in section G.7.4.1.1 of the ITU-T H.264 Specification;
- all other members are interpreted as defined in section 8.2 of the ITU-T H.264 Specification.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264DpbSlotInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_DPB_SLOT_INFO_KHR
- VUID-VkVideoEncodeH264DpbSlotInfoKHR-pStdReferenceInfo-parameter
  pStdReferenceInfo **must** be a valid pointer to a valid StdVideoEncodeH264ReferenceInfo value

Structure Chaining

Extends the structure

- VkVideoReferenceSlotInfoKHR

45.20.12. H.264 Encode Rate Control

Group of Pictures

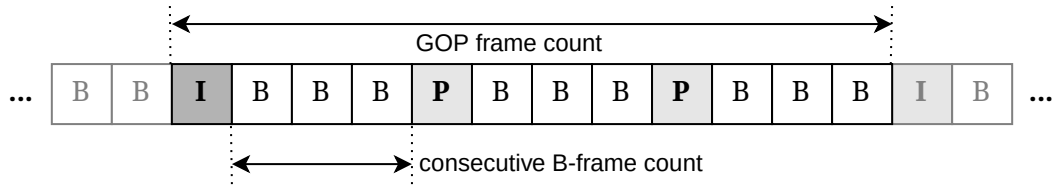
In case of H.264 encoding it is common practice to follow a regular pattern of different picture types in display order when encoding subsequent frames. This pattern is referred to as the *group of pictures* (GOP).

A regular GOP is defined by the following parameters:

- The number of frames in the GOP;
- The number of consecutive B frames between I and/or P frames in display order.

Frame types in a regular GOP follow each other in display order according to the following algorithm:

1. The first frame is always an I or an IDR frame.
2. This is followed by a number of consecutive B frames, as defined above.
3. If the number of frames in the GOP is not reached yet, then the next frame is a P frame and the algorithm continues from step 2.



Text is not SVG - cannot display

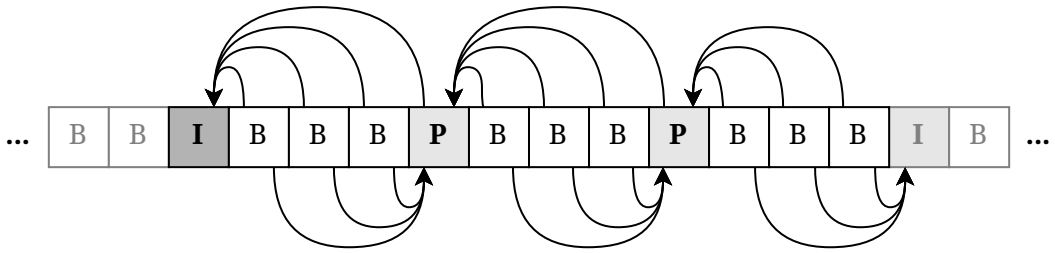
Figure 29. H.264 regular GOP starting with an I frame

When IDR frames are used in a video sequence, the *IDR period* refers to the distance between subsequent IDR frames.

It is also typical for H.264 encoding to use specific reference picture usage patterns across the frames of the GOP. The two most common reference patterns used are as follows:

Flat Reference Pattern

- Each P frame uses the last non-B frame, in display order, as reference.
- Each B frame uses the last non-B frame, in display order, as its forward reference, and uses the next non-B frame, in display order, as its backward reference.

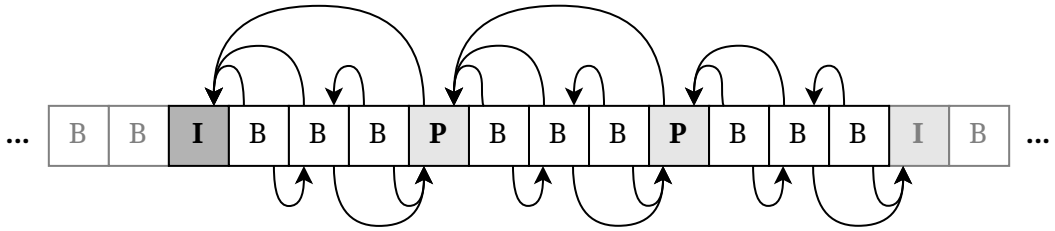


Text is not SVG - cannot display

Figure 30. H.264 flat reference pattern

Dyadic Reference Pattern

- Each P frame uses the last non-B frame, in display order, as reference.
- The following algorithm is applied to the sequence of consecutive B frames between I and/or P frames in display order:
  1. The B frame in the middle of this sequence uses the frame preceding the sequence as its forward reference, and uses the frame following the sequence as its backward reference.
  2. The algorithm is executed recursively for the following frame sequences:
     - The B frames of the original sequence preceding the frame in the middle, if any.
     - The B frames of the original sequence following the frame in the middle, if any.



Text is not SVG - cannot display

Figure 31. H.264 dyadic reference pattern

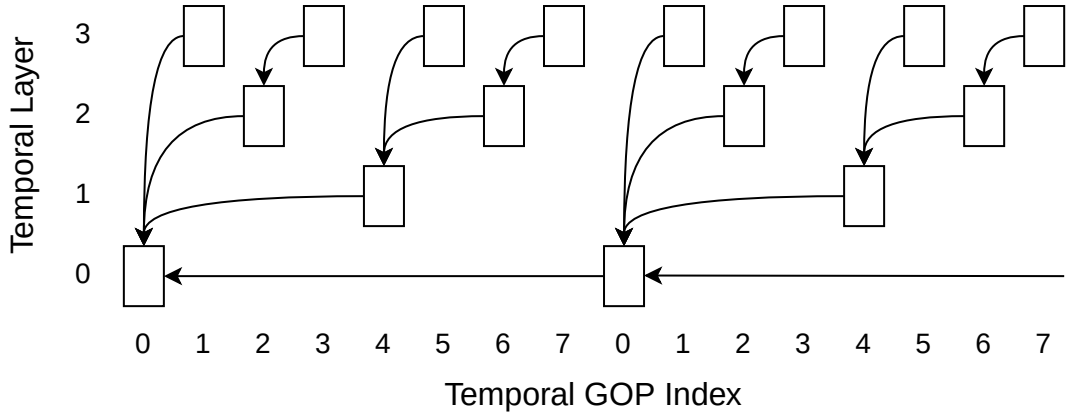
The application **can** provide guidance to the implementation's rate control algorithm about the structure of the GOP used by the application. Any such guidance about the GOP and its structure does not mandate that specific GOP structure to be used by the application, as the picture type of individual encoded pictures is still application-controlled. However, any deviation from the provided guidance **may** result in undesired rate control behavior including, but not limited to the implementation not being able to conform to the expected average or target bitrates, or other rate control parameters specified by the application.

When an H.264 encode session is used to encode multiple temporal layers, it is also common practice to follow a regular pattern for the H.264 temporal ID for the encoded pictures in display order when encoding subsequent frames. This pattern is referred to as the *temporal GOP*. The most

common temporal layer pattern used is as follows:

Dyadic Temporal Layer Pattern

- The number of frames in the temporal GOP is 2^{n-1}, where n is the number of temporal layers.
- The i^{th} frame in the temporal GOP uses temporal ID t, if and only if the index of the least significant bit set in i equals n-t-1, except for the first frame, which is the only frame in the temporal GOP using temporal ID zero.
- The i^{th} frame in the temporal GOP uses the r^{th} frame as reference, where r is calculated from i by clearing the least significant bit set in it, except for the first frame in the temporal GOP, which uses the first frame of the previous temporal GOP, if any, as reference.



Text is not SVG - cannot display

Figure 32. H.264 dyadic temporal layer pattern

NOTE Multi-layer rate control and multi-layer coding are typically used for streaming cases where low latency is expected, hence B pictures with backward prediction are usually not used.

The `VkVideoEncodeH264RateControlInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264RateControlInfoKHR {
    VkStructureType                          sType;
    const void*                              pNext;
    VkVideoEncodeH264RateControlFlagsKHR     flags;
    uint32_t                                 gopFrameCount;
    uint32_t                                 idrPeriod;
    uint32_t                                 consecutiveBFrameCount;
    uint32_t                                 temporalLayerCount;
} VkVideoEncodeH264RateControlInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkVideoEncodeH264RateControlFlagBitsKHR specifying H.264 rate control flags.
- `gopFrameCount` is the number of frames within a group of pictures (GOP) intended to be used by the application. If it is 0, the rate control algorithm **may** assume an implementation-dependent GOP length. If it is `UINT32_MAX`, the GOP length is treated as infinite.
- `idrPeriod` is the interval, in terms of number of frames, between two IDR frames (see IDR period). If it is 0, the rate control algorithm **may** assume an implementation-dependent IDR period. If it is `UINT32_MAX`, the IDR period is treated as infinite.
- `consecutiveBFrameCount` is the number of consecutive B frames between I and/or P frames within the GOP.
- `temporalLayerCount` specifies the number of H.264 temporal layers that the application intends to use.

When an instance of this structure is included in the `pNext` chain of the VkVideoCodingControlInfoKHR structure passed to the vkCmdControlVideoCodingKHR command, and VkVideoCodingControlInfoKHR::`flags` includes VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR, the parameters in this structure are used as guidance for the implementation's rate control algorithm (see Video Coding Control).

If `flags` includes VK_VIDEO_ENCODE_H264_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR, then the rate control state is reset to an initial state to meet HRD compliance requirements. Otherwise the new rate control state **may** be applied without a reset depending on the implementation and the specified rate control parameters.

NOTE It would be possible to infer the picture type to be used when encoding a frame, on the basis of the values provided for `consecutiveBFrameCount`, `idrPeriod`, and `gopFrameCount`, but this inferred picture type will not be used by implementations to override the picture type provided to the video encode operation.

Valid Usage

- VUID-VkVideoEncodeH264RateControlInfoKHR-flags-08280
  If VkVideoEncodeH264CapabilitiesKHR::`flags`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_H264_CAPABILITY_HRD_COMPLIANCE_BIT_KHR, then `flags` **must** not contain VK_VIDEO_ENCODE_H264_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR
- VUID-VkVideoEncodeH264RateControlInfoKHR-flags-08281
  If `flags` contains VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR or VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR, then it **must** also contain

VK_VIDEO_ENCODE_H264_RATE_CONTROL_REGULAR_GOP_BIT_KHR

- VUID-VkVideoEncodeH264RateControlInfoKHR-flags-08282
  If `flags` contains VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR, then it **must** not also contain VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR
- VUID-VkVideoEncodeH264RateControlInfoKHR-flags-08283
  If `flags` contains VK_VIDEO_ENCODE_H264_RATE_CONTROL_REGULAR_GOP_BIT_KHR, then `gopFrameCount` **must** be greater than `0`
- VUID-VkVideoEncodeH264RateControlInfoKHR-idrPeriod-08284
  If `idrPeriod` is not `0`, then it **must** be greater than or equal to `gopFrameCount`
- VUID-VkVideoEncodeH264RateControlInfoKHR-consecutiveBFrameCount-08285
  If `consecutiveBFrameCount` is not `0`, then it **must** be less than `gopFrameCount`

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264RateControlInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_RATE_CONTROL_INFO_KHR
- VUID-VkVideoEncodeH264RateControlInfoKHR-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkVideoEncodeH264RateControlFlagBitsKHR values

Structure Chaining

Extends the structures

- VkVideoBeginCodingInfoKHR
- VkVideoCodingControlInfoKHR

Bits which **can** be set in VkVideoEncodeH264RateControlInfoKHR::`flags`, specifying H.264 rate control flags, are:

```
// Provided by VK_KHR_video_encode_h264
typedef enum VkVideoEncodeH264RateControlFlagBitsKHR {
    VK_VIDEO_ENCODE_H264_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H264_RATE_CONTROL_REGULAR_GOP_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR = 0x00000008,
    VK_VIDEO_ENCODE_H264_RATE_CONTROL_TEMPORAL_LAYER_PATTERN_DYADIC_BIT_KHR =
0x00000010,
} VkVideoEncodeH264RateControlFlagBitsKHR;
```

- VK_VIDEO_ENCODE_H264_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR specifies

that rate control **should** attempt to produce an HRD compliant bitstream, as defined in annex C of the ITU-T H.264 Specification.

- VK_VIDEO_ENCODE_H264_RATE_CONTROL_REGULAR_GOP_BIT_KHR specifies that the application intends to use a regular GOP structure according to the parameters specified in the gopFrameCount, idrPeriod, and consecutiveBFrameCount members of the VkVideoEncodeH264RateControlInfoKHR structure.
- VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR specifies that the application intends to follow a flat reference pattern in the GOP.
- VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR specifies that the application intends to follow a dyadic reference pattern in the GOP.
- VK_VIDEO_ENCODE_H264_RATE_CONTROL_TEMPORAL_LAYER_PATTERN_DYADIC_BIT_KHR specifies that the application intends to follow a dyadic temporal layer pattern.

```
// Provided by VK_KHR_video_encode_h264
typedef VkFlags VkVideoEncodeH264RateControlFlagsKHR;
```

VkVideoEncodeH264RateControlFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeH264RateControlFlagBitsKHR.

Rate Control Layers

The VkVideoEncodeH264RateControlLayerInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264RateControlLayerInfoKHR {
    VkStructureType                  sType;
    const void*                      pNext;
    VkBool32                         useMinQp;
    VkVideoEncodeH264QpKHR           minQp;
    VkBool32                         useMaxQp;
    VkVideoEncodeH264QpKHR           maxQp;
    VkBool32                         useMaxFrameSize;
    VkVideoEncodeH264FrameSizeKHR    maxFrameSize;
} VkVideoEncodeH264RateControlLayerInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- useMinQp indicates whether the QP values determined by rate control will be clamped to the lower bounds on the QP values specified in minQp.
- minQp specifies the lower bounds on the QP values, for each picture type, that the implementation's rate control algorithm will use when useMinQp is VK_TRUE.
- useMaxQp indicates whether the QP values determined by rate control will be clamped to the upper bounds on the QP values specified in maxQp.
- maxQp specifies the upper bounds on the QP values, for each picture type, that the

implementation's rate control algorithm will use when useMaxQp is VK_TRUE.

- useMaxFrameSize indicates whether the implementation's rate control algorithm **should** use the values specified in maxFrameSize as the upper bounds on the encoded frame size for each picture type.
- maxFrameSize specifies the upper bounds on the encoded frame size, for each picture type, when useMaxFrameSize is VK_TRUE.

When used, the values in minQp and maxQp guarantee that the effective QP values used by the implementation will respect those lower and upper bounds, respectively. However, limiting the range of QP values that the implementation is able to use will also limit the capabilities of the implementation's rate control algorithm to comply to other constraints. In particular, the implementation **may** not be able to comply to the following:

- The average and/or peak bitrate values to be used for the encoded bitstream specified in the averageBitrate and maxBitrate members of the VkVideoEncodeRateControlLayerInfoKHR structure.
- The upper bounds on the encoded frame size, for each picture type, specified in the maxFrameSize member of VkVideoEncodeH264RateControlLayerInfoKHR.

NOTE	In general, applications need to configure rate control parameters appropriately in order to be able to get the desired rate control behavior, as described in the Video Encode Rate Control section.

When an instance of this structure is included in the pNext chain of a VkVideoEncodeRateControlLayerInfoKHR structure specified in one of the elements of the pLayers array member of the VkVideoEncodeRateControlInfoKHR structure passed to the vkCmdControlVideoCodingKHR command, VkVideoCodingControlInfoKHR::flags includes VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR, and the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, it specifies the H.264-specific rate control parameters of the rate control layer corresponding to that element of pLayers.

Valid Usage

- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-useMinQp-08286
  If useMinQp is VK_TRUE, then the qpI, qpP, and qpB members of minQp **must** all be between VkVideoEncodeH264CapabilitiesKHR::minQp and VkVideoEncodeH264CapabilitiesKHR::maxQp, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile
- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-useMaxQp-08287
  If useMaxQp is VK_TRUE, then the qpI, qpP, and qpB members of maxQp **must** all be between VkVideoEncodeH264CapabilitiesKHR::minQp and VkVideoEncodeH264CapabilitiesKHR::maxQp, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile
- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-useMinQp-08288
  If useMinQp is VK_TRUE and VkVideoEncodeH264CapabilitiesKHR::flags, as returned by

vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_H264_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR, then the qpI, qpP, and qpB members of minQp **must** all specify the same value

- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-useMaxQp-08289
  If useMaxQp is VK_TRUE and VkVideoEncodeH264CapabilitiesKHR::flags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_H264_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR, then the qpI, qpP, and qpB members of maxQp **must** all specify the same value
- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-useMinQp-08374
  If useMinQp and useMaxQp are both VK_TRUE, then the qpI, qpP, and qpB members of minQp **must** all be less than or equal to the respective members of maxQp

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_RATE_CONTROL_LAYER_INFO_KHR
- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-minQp-parameter
  minQp **must** be a valid VkVideoEncodeH264QpKHR structure
- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-maxQp-parameter
  maxQp **must** be a valid VkVideoEncodeH264QpKHR structure
- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-maxFrameSize-parameter
  maxFrameSize **must** be a valid VkVideoEncodeH264FrameSizeKHR structure

Structure Chaining

Extends the structure

- VkVideoEncodeRateControlLayerInfoKHR

The VkVideoEncodeH264QpKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264QpKHR {
    int32_t    qpI;
    int32_t    qpP;
    int32_t    qpB;
} VkVideoEncodeH264QpKHR;
```

- qpI is the QP to be used for I pictures.
- qpP is the QP to be used for P pictures.
- qpB is the QP to be used for B pictures.

The VkVideoEncodeH264FrameSizeKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264FrameSizeKHR {
    uint32_t    frameISize;
    uint32_t    framePSize;
    uint32_t    frameBSize;
} VkVideoEncodeH264FrameSizeKHR;
```

- frameISize is the size in bytes to be used for I pictures.
- framePSize is the size in bytes to be used for P pictures.
- frameBSize is the size in bytes to be used for B pictures.

GOP Remaining Frames

Besides session level rate control configuration, the application **can** specify the number of frames per frame type remaining in the group of pictures (GOP).

The VkVideoEncodeH264GopRemainingFrameInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264GopRemainingFrameInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           useGopRemainingFrames;
    uint32_t           gopRemainingI;
    uint32_t           gopRemainingP;
    uint32_t           gopRemainingB;
} VkVideoEncodeH264GopRemainingFrameInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- useGopRemainingFrames indicates whether the implementation's rate control algorithm **should** use the values specified in gopRemainingI, gopRemainingP, and gopRemainingB. If useGopRemainingFrames is VK_FALSE, then the values of gopRemainingI, gopRemainingP, and gopRemainingB are ignored.
- gopRemainingI specifies the number of I frames the implementation's rate control algorithm **should** assume to be remaining in the GOP prior to executing the video encode operation.
- gopRemainingP specifies the number of P frames the implementation's rate control algorithm **should** assume to be remaining in the GOP prior to executing the video encode operation.
- gopRemainingB specifies the number of B frames the implementation's rate control algorithm **should** assume to be remaining in the GOP prior to executing the video encode operation.

Setting useGopRemainingFrames to VK_TRUE and including this structure in the pNext chain of VkVideoBeginCodingInfoKHR is only mandatory if the VkVideoEncodeH264CapabilitiesKHR

::requiresGopRemainingFrames reported for the used video profile is VK_TRUE. However, implementations **may** use these remaining frame counts, when specified, even when it is not required. In particular, when the application does not use a regular GOP structure, these values **may** provide additional guidance for the implementation's rate control algorithm.

The VkVideoEncodeH264CapabilitiesKHR::prefersGopRemainingFrames capability is also used to indicate that the implementation's rate control algorithm **may** operate more accurately if the application specifies the remaining frame counts using this structure.

As with other rate control guidance values, if the effective order and number of frames encoded by the application are not in line with the remaining frame counts specified in this structure at any given point, then the behavior of the implementation's rate control algorithm **may** deviate from the one expected by the application.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264GopRemainingFrameInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_GOP_REMAINING_FRAME_INFO_KHR

Structure Chaining

Extends the structure

- VkVideoBeginCodingInfoKHR

H.264 QP Delta Maps

Quantization delta maps used with an H.264 encode profile are referred to as *QP delta maps* and their texels contain integer values representing QP delta values that are applied in the process of determining the quantization parameters of the encoded picture.

Accordingly, H.264 QP delta maps always have single channel integer formats, as reported in VkVideoFormatPropertiesKHR::format.

When the rate control mode is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, the QP delta values are added to the per slice constant QP values that, in effect, enable the application to explicitly control the used QP values at the granularity of the used quantization map texel size.

For all other rate control modes, the QP delta values **can** be used to offset the QP values that the rate control algorithm would otherwise produce.

H.264 Encode Quantization

Performing H.264 encode operations involves the process of assigning QP values to individual H.264 macroblocks. This process depends on the used rate control mode, as well as other encode and rate control parameters, as described below:

- If the configured rate control mode is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR, then the QP value is initialized by the implementation-specific default rate control algorithm.
  - If the video encode operation is issued with a quantization delta map, the QP delta value corresponding to the macroblock, as fetched from the quantization map, is added to the previously determined QP value. If the fetched QP delta value falls outside the supported QP delta value range reported in the `minQpDelta` and `maxQpDelta` members of VkVideoEncodeH264QuantizationMapCapabilitiesKHR, then the QP value used for the macroblock becomes undefined.
- If the configured rate control mode is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, then the QP value is initialized from the constant QP value specified for the H.264 slice the macroblock is part of.
  - If the video encode operation is issued with a quantization delta map, the QP delta value corresponding to the macroblock, as fetched from the quantization map, is added to the previously determined QP value. If the fetched QP delta value falls outside the supported QP delta value range reported in the `minQpDelta` and `maxQpDelta` members of VkVideoEncodeH264QuantizationMapCapabilitiesKHR, then the QP value used for the macroblock becomes undefined.
- If the configured rate control mode is not VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR or VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, then the QP value is initialized by the corresponding rate control algorithm.
  - If the video encode operation is issued with a quantization delta map, the QP delta value corresponding to the macroblock, as fetched from the quantization map, is added to the previously determined QP value. If the fetched QP delta value falls outside the supported QP delta value range reported in the `minQpDelta` and `maxQpDelta` members of VkVideoEncodeH264QuantizationMapCapabilitiesKHR, then the QP value used for the macroblock becomes undefined.
  - If the video encode operation is issued with an emphasis map, the rate control will adjust the QP value based on the emphasis value corresponding to the macroblock, as fetched from the quantization map, according to the following equation:

    $QP_{new} = f(QP_{prev},e)$

    Where QP_{new} is the resulting QP value, QP_{prev} is the previously determined QP value, e is the emphasis value corresponding to the macroblock, and f is an implementation-defined function for which the following implication is true:

    $e_1 < e_2 \Rightarrow f(QP,e_1) \geq f(QP,e_2)$

    > **NOTE** This means that lower emphasis values will result in higher QP values, whereas higher emphasis values will result in lower QP values, but the function is not strictly decreasing with respect to the input emphasis value for a given input QP value.

  - If clamping to minimum QP values is enabled in the applied rate control layer, then the QP

value is clamped to the corresponding minimum QP value.

- If clamping to maximum QP values is enabled in the applied rate control layer, then the QP value is clamped to the corresponding maximum QP value.

- If VK_VIDEO_ENCODE_H264_CAPABILITY_MB_QP_DIFF_WRAPAROUND_BIT_KHR is not supported, then the determined QP value is clamped in such a way that the `mb_qp_delta` value of the encoded macroblock complies to the modified version of equation 7-37 of the ITU-T H.264 Specification.

NOTE	The effect of this is that the maximum QP difference across subsequent macroblocks is limited to the [-(26 + $QpBdOffset_Y$ / 2), 25 + $QpBdOffset_Y$ / 2] range and only has an observable change in behavior when the video encode operation is issued with a QP delta map.

- In all cases, the final QP value is clamped to the QP value range supported by the video profile, as reported in the `minQp` and `maxQp` members of VkVideoEncodeH264CapabilitiesKHR.

45.20.13. H.264 Encode Requirements

This section describes the **required** H.264 encoding capabilities for physical devices that have at least one queue family that supports the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR, as returned by vkGetPhysicalDeviceQueueFamilyProperties2 in VkQueueFamilyVideoPropertiesKHR ::`videoCodecOperations`.

Table 60. Required Video Std Header Versions

Video Std Header Name	Version
`vulkan_video_codec_h264std_encode`	1.0.0

Table 61. Required Video Capabilities

Video Capability	Requirement	Requirement Type[1]
VkVideoCapabilitiesKHR		
`flags`	-	min
`minBitstreamBufferOffsetAlignment`	4096	max
`minBitstreamBufferSizeAlignment`	4096	max
`pictureAccessGranularity`	(64,64)	max
`minCodedExtent`	-	max
`maxCodedExtent`	-	min
`maxDpbSlots`	0	min
`maxActiveReferencePictures`	0	min
VkVideoEncodeCapabilitiesKHR		
`flags`	-	min

Video Capability	Requirement	Requirement Type[1]
rateControlModes	VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR [4]	min
maxBitrate	64000	min
maxQualityLevels	1	min
encodeInputPictureGranularity	(64,64)	max
supportedEncodeFeedbackFlags	VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR + VK_VIDEO_ENCODE_FEEDBACK_AVERAGE_QUANTIZATION_BIT_KHR [6] + VK_VIDEO_ENCODE_FEEDBACK_INTRA_PIXELS_BIT_KHR [6] + VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR [6] + VK_VIDEO_ENCODE_FEEDBACK_PICTURE_PARTITION_COUNT_BIT_KHR [6]	min
VkVideoEncodeFeedback2CapabilitiesKHR		
maxPerPartitionFeedbackEntries	-	min

Video Capability	Requirement	Requirement Type[1]
supportedPerPartitionEncodeFeedbackFlags	VK_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_STATUS_BIT_KHR[7] + VK_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR[7] + VK_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR[7]	min
VkVideoEncodeH264CapabilitiesKHR		
flags	-	min
maxLevelIdc	STD_VIDEO_H264_LEVEL_IDC_1_0	min
maxSliceCount	1	min
maxPPictureL0ReferenceCount	0	min
maxBPictureL0ReferenceCount	0	min
maxL1ReferenceCount	0	min
maxTemporalLayerCount	1	min
expectDyadicTemporalLayerPattern	-	implementation-dependent
minQp	-	max
maxQp	-	min
prefersGopRemainingFrames	-	implementation-dependent
requiresGopRemainingFrames	-	implementation-dependent
stdSyntaxFlags	-	min
VkVideoEncodeQuantizationMapCapabilitiesKHR		
maxQuantizationMapExtent	-[2]	min
VkVideoEncodeH264QuantizationMapCapabilitiesKHR		
minQpDelta	-[3]	max
maxQpDelta	-[3]	min

Video Capability	Requirement	Requirement Type[1]
VkVideoEncodeIntraRefreshCapabilitiesKHR		
intraRefreshModes	0	min
maxIntraRefreshCycleDuration	2 [5]	min
maxIntraRefreshActiveReferencePictures	1 [5]	min
partitionIndependentIntraRefreshRegions	-	implementation-dependent
nonRectangularIntraRefreshRegions	-	implementation-dependent

1

The **Requirement Type** column specifies the requirement is either the minimum value all implementations **must** support, the maximum value all implementations **must** support, or the exact value all implementations **must** support. For bitmasks a minimum value is the least bits all implementations **must** set, but they **may** have additional bits set beyond this minimum.

2

If VkVideoEncodeCapabilitiesKHR::flags includes VK_VIDEO_ENCODE_CAPABILITY_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_VIDEO_ENCODE_CAPABILITY_EMPHASIS_MAP_BIT_KHR, then the width and height members of maxQuantizationMapExtent **must** be greater than zero.

3

If VkVideoEncodeCapabilitiesKHR::flags includes VK_VIDEO_ENCODE_CAPABILITY_QUANTIZATION_DELTA_MAP_BIT_KHR, then maxQpDelta **must** be greater than minQpDelta.

4

If the videoMaintenance2 feature is supported.

5

If VkVideoEncodeIntraRefreshCapabilitiesKHR::intraRefreshModes is not zero.

6

If the videoEncodeFeedback2 feature is supported.

7

If maxPerPartitionFeedbackEntries is greater than zero.

45.21. H.265 Encode Operations

Video encode operations using an H.265 encode profile **can** be used to encode elementary video stream sequences compliant to the ITU-T H.265 Specification.

NOTE	Refer to the Preamble for information on how the Khronos Intellectual Property Rights Policy relates to normative references to external materials not created by Khronos.

This process is performed according to the video encode operation steps with the codec-specific semantics defined in section 8 of the ITU-T H.265 Specification as follows:

- Syntax elements, derived values, and other parameters are applied from the following structures:
  - The `StdVideoH265VideoParameterSet` structure corresponding to the active VPS specifying the H.265 video parameter set.
  - The `StdVideoH265SequenceParameterSet` structure corresponding to the active SPS specifying the H.265 sequence parameter set.
  - The `StdVideoH265PictureParameterSet` structure corresponding to the active PPS specifying the H.265 picture parameter set.
  - The `StdVideoEncodeH265PictureInfo` structure specifying the H.265 picture information.
  - The `StdVideoEncodeH265SliceSegmentHeader` structures specifying the H.265 slice segment header parameters for each encoded H.265 slice segment.
  - The `StdVideoEncodeH265ReferenceInfo` structures specifying the H.265 reference information corresponding to the optional reconstructed picture and any active reference pictures.
- The encoded bitstream data is written to the destination video bitstream buffer range as defined in the H.265 Encode Bitstream Data Access section.
- Picture data in the video picture resources corresponding to the used encode input picture, active reference pictures, and optional reconstructed picture is accessed as defined in the H.265 Encode Picture Data Access section.
- The decision on reference picture setup is made according to the parameters specified in the H.265 picture information.

If the parameters adhere to the syntactic and semantic requirements defined in the corresponding sections of the ITU-T H.265 Specification, as described above, and the DPB slots associated with the active reference pictures all refer to valid picture references, then the video encode operation will complete successfully. Otherwise, the video encode operation **may** complete unsuccessfully.

45.21.1. H.265 Encode Parameter Overrides

Implementations **may** override, unless otherwise specified, any of the H.265 encode parameters specified in the following Video Std structures:

- `StdVideoH265VideoParameterSet`
- `StdVideoH265SequenceParameterSet`
- `StdVideoH265PictureParameterSet`
- `StdVideoEncodeH265PictureInfo`
- `StdVideoEncodeH265SliceSegmentHeader`

- StdVideoEncodeH265ReferenceInfo

All such H.265 encode parameter overrides **must** fulfill the conditions defined in the Video Encode Parameter Overrides section.

In addition, implementations **must** not override any of the following H.265 encode parameters:

- StdVideoEncodeH265PictureInfo::pic_type
- StdVideoEncodeH265SliceSegmentHeader::slice_type

If the videoMaintenance2 feature is enabled, implementations **must** not override any of the following H.265 encode parameters:

- the following parameters specified in the StdVideoH265ProfileTierLevel structure pointed to by StdVideoH265VideoParameterSet::pProfileTierLevel and StdVideoH265SequenceParameterSet::pProfileTierLevel:
  - flags.general_tier_flag
  - flags.general_progressive_source_flag, if set to 1
  - flags.general_interlaced_source_flag, if set to 0
  - flags.general_frame_only_constraint_flag
  - general_profile_idc
  - general_level_idc
- the following parameters specified in StdVideoH265SequenceParameterSet:
  - chroma_format_idc
- the following parameters specified in the StdVideoH265SequenceParameterSetVui structure pointed to by StdVideoH265SequenceParameterSet::pSequenceParameterSetVui:
  - flags.aspect_ratio_info_present_flag
  - flags.overscan_info_present_flag
  - flags.overscan_appropriate_flag
  - flags.video_signal_type_present_flag
  - flags.video_full_range_flag
  - flags.colour_description_present_flag
  - flags.chroma_loc_info_present_flag
  - flags.neutral_chroma_indication_flag
  - flags.field_seq_flag
  - flags.frame_field_info_present_flag
  - flags.default_display_window_flag
  - flags.vui_timing_info_present_flag
  - flags.vui_poc_proportional_to_timing_flag
  - aspect_ratio_idc

- sar_width
- sar_height
- video_format
- colour_primaries
- transfer_characteristics
- matrix_coeffs
- chroma_sample_loc_type_top_field
- chroma_sample_loc_type_bottom_field
- def_disp_win_left_offset
- def_disp_win_right_offset
- def_disp_win_top_offset
- def_disp_win_bottom_offset
- vui_num_units_in_tick
- vui_time_scale
- vui_num_ticks_poc_diff_one_minus1

In case of a video session parameters object created with VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR, the following H.265 SPS and PPS parameters **may** be overridden by the implementation according to the quantization map texel size the video session parameters object was created with:

- StdVideoH265SequenceParameterSet::log2_min_luma_coding_block_size_minus3
- StdVideoH265SequenceParameterSet::log2_diff_max_min_luma_coding_block_size
- StdVideoH265SequenceParameterSet::log2_min_pcm_luma_coding_block_size_minus3
- StdVideoH265SequenceParameterSet::log2_diff_max_min_pcm_luma_coding_block_size
- StdVideoH265PictureParameterSet::diff_cu_qp_delta_depth

This **may** be necessary in order to limit the set of H.265 coding unit and coding tree unit sizes used during picture encoding to those that are supported by the implementation when using the specific quantization map texel size.

In case of H.265 encode parameters stored in video session parameters objects, applications need to use the vkGetEncodedVideoSessionParametersKHR command to determine whether any implementation overrides happened. If the query indicates that implementation overrides were applied, then the application needs to retrieve and use the encoded H.265 parameter sets in the bitstream in order to be able to produce a compliant H.265 video bitstream using the H.265 encode parameters stored in the video session parameters object.

In case of any H.265 encode parameters stored in the encoded bitstream produced by video encode operations, if the implementation supports the VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR video encode feedback query flag, the application **can** use such queries to retrieve feedback about whether any

implementation overrides have been applied to those H.265 encode parameters.

45.21.2. H.265 Encode Bitstream Data Access

Each video encode operation writes one or more VCL NAL units comprising of slice segment headers and data of the encoded picture, in the format defined in sections 7.3.6 and 7.3.8, according to the semantics defined in sections 7.4.7 and 7.4.9 of the ITU-T H.265 Specification, respectively. The number of VCL NAL units written is specified by VkVideoEncodeH265PictureInfoKHR ::`naluSliceSegmentEntryCount`.

45.21.3. H.265 Encode Picture Data Access

Accesses to image data within a video picture resource happen at the granularity indicated by VkVideoCapabilitiesKHR::`pictureAccessGranularity`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile. Accordingly, the complete image subregion of an encode input picture, reference picture, or reconstructed picture accessed by video coding operations using an H.265 encode profile is defined as the set of texels within the coordinate range:

([0,`endX`), [0,`endY`))

Where:

- `endX` equals `codedExtent.width` rounded up to the nearest integer multiple of `pictureAccessGranularity.width` and clamped to the width of the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure;
- endY equals `codedExtent.height` rounded up to the nearest integer multiple of `pictureAccessGranularity.height` and clamped to the height of the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure;

Where `codedExtent` is the member of the VkVideoPictureResourceInfoKHR structure corresponding to the picture.

In case of video encode operations using an H.265 encode profile, any access to a picture at the coordinates (`x`,`y`), as defined by the ITU-T H.265 Specification, is an access to the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure at the texel coordinates (`x`,`y`).

Implementations **may** choose not to access some or all texels within particular reference pictures available to a video encode operation (e.g. due to video encode parameter overrides restricting the effective set of used reference pictures, or if the encoding algorithm chooses not to use certain subregions of the reference picture data for sample prediction).

45.21.4. H.265 Frame, Picture, Slice Segments, and Tiles

H.265 pictures consist of one or more slices, slice segments, and tiles, as defined in section 6.3.1 of the ITU-T H.265 Specification.

For the purposes of this specification, the H.265 slice segments comprising a picture are referred to as the *picture partitions* of the picture.

Video encode operations using an H.265 encode profile **can** encode slice segments of different types, as defined in section 7.4.7.1 of the ITU-T H.265 Specification, by specifying the corresponding enumeration constant value in `StdVideoEncodeH265SliceSegmentHeader`::`slice_type` in the H.265 slice segment header parameters from the Video Std enumeration type `StdVideoH265SliceType`:

- `STD_VIDEO_H265_SLICE_TYPE_B` indicates that the slice segment is part of a *B slice* as defined in section 3.12 of the ITU-T H.265 Specification.
- `STD_VIDEO_H265_SLICE_TYPE_P` indicates that the slice segment is part of a *P slice* as defined in section 3.111 of the ITU-T H.265 Specification.
- `STD_VIDEO_H265_SLICE_TYPE_I` indicates that the slice segment is part of an *I slice* as defined in section 3.74 of the ITU-T H.265 Specification.

Pictures constructed from such slice segments **can** be of different types, as defined in section 7.4.3.5 of the ITU-T H.265 Specification. Video encode operations using an H.265 encode profile **can** encode pictures of a specific type by specifying the corresponding enumeration constant value in `StdVideoEncodeH265PictureInfo`::`pic_type` in the H.265 picture information from the Video Std enumeration type `StdVideoH265PictureType`:

- `STD_VIDEO_H265_PICTURE_TYPE_P` indicates that the picture is a *P picture*. A frame consisting of a P picture is also referred to as a *P frame*.
- `STD_VIDEO_H265_PICTURE_TYPE_B` indicates that the picture is a *B picture*. A frame consisting of a B picture is also referred to as a *B frame*.
- `STD_VIDEO_H265_PICTURE_TYPE_I` indicates that the picture is an *I picture*. A frame consisting of an I picture is also referred to as an *I frame*.
- `STD_VIDEO_H265_PICTURE_TYPE_IDR` indicates that the picture is a special type of I picture called an *IDR picture* as defined in section 3.67 of the ITU-T H.265 Specification. A frame consisting of an IDR picture is also referred to as an *IDR frame*.

45.21.5. H.265 Coding Blocks

H.265 encode supports two types of coding blocks:

- *Coding tree unit*, as defined in section 3.35 of the ITU-T H.265 Specification.
- *Coding unit*, as defined in section 3.36 of the ITU-T H.265 Specification.

An H.265 coding unit is considered *skipped* when the value of `cu_skip_flag` for the coding unit equals 1, as defined in section 7.4.95 of the ITU-T H.265 Specification.

45.21.6. H.265 Encode Profile

A video profile supporting H.265 video encode operations is specified by setting VkVideoProfileInfoKHR::`videoCodecOperation` to `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and adding a `VkVideoEncodeH265ProfileInfoKHR` structure to the VkVideoProfileInfoKHR::`pNext` chain.

The VkVideoEncodeH265ProfileInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265ProfileInfoKHR {
    VkStructureType           sType;
    const void*               pNext;
    StdVideoH265ProfileIdc    stdProfileIdc;
} VkVideoEncodeH265ProfileInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- stdProfileIdc is a StdVideoH265ProfileIdc value specifying the H.265 codec profile IDC, as defined in section A.3 of the ITU-T H.265 Specification.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265ProfileInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_PROFILE_INFO_KHR

Structure Chaining

Extends the structures

- VkQueryPoolCreateInfo
- VkVideoProfileInfoKHR

45.21.7. H.265 Encode Capabilities

When calling vkGetPhysicalDeviceVideoCapabilitiesKHR to query the capabilities for an H.265 encode profile, the VkVideoCapabilitiesKHR::pNext chain **must** include a VkVideoEncodeH265CapabilitiesKHR structure that will be filled with the profile-specific capabilities.

The VkVideoEncodeH265CapabilitiesKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265CapabilitiesKHR {
    VkStructureType                                 sType;
    void*                                           pNext;
    VkVideoEncodeH265CapabilityFlagsKHR             flags;
    StdVideoH265LevelIdc                            maxLevelIdc;
    uint32_t                                        maxSliceSegmentCount;
    VkExtent2D                                      maxTiles;
    VkVideoEncodeH265CtbSizeFlagsKHR                ctbSizes;
    VkVideoEncodeH265TransformBlockSizeFlagsKHR     transformBlockSizes;
    uint32_t                                        maxPPictureL0ReferenceCount;
    uint32_t                                        maxBPictureL0ReferenceCount;
    uint32_t                                        maxL1ReferenceCount;
    uint32_t                                        maxSubLayerCount;
    VkBool32
expectDyadicTemporalSubLayerPattern;
    int32_t                                         minQp;
    int32_t                                         maxQp;
    VkBool32                                        prefersGopRemainingFrames;
    VkBool32                                        requiresGopRemainingFrames;
    VkVideoEncodeH265StdFlagsKHR                    stdSyntaxFlags;
} VkVideoEncodeH265CapabilitiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkVideoEncodeH265CapabilityFlagBitsKHR indicating supported H.265 encoding capabilities.
- `maxLevelIdc` is a `StdVideoH265LevelIdc` value indicating the maximum H.265 level supported by the profile, where enum constant `STD_VIDEO_H265_LEVEL_IDC_<major>_<minor>` identifies H.265 level `<major>.<minor>` as defined in section A.4 of the ITU-T H.265 Specification.
- `maxSliceSegmentCount` indicates the maximum number of slice segments that **can** be encoded for a single picture. Further restrictions **may** apply to the number of slice segments that **can** be encoded for a single picture depending on other capabilities and codec-specific rules.
- `maxTiles` indicates the maximum number of H.265 tile columns and rows, as defined in sections 3.175 and 3.176 of the ITU-T H.265 Specification that **can** be encoded for a single picture. Further restrictions **may** apply to the number of H.265 tiles that **can** be encoded for a single picture depending on other capabilities and codec-specific rules.
- `ctbSizes` is a bitmask of VkVideoEncodeH265CtbSizeFlagBitsKHR describing the supported CTB sizes.
- `transformBlockSizes` is a bitmask of VkVideoEncodeH265TransformBlockSizeFlagBitsKHR describing the supported transform block sizes.
- `maxPPictureL0ReferenceCount` indicates the maximum number of reference pictures the implementation supports in the reference list L0 for P pictures.

> **NOTE** As implementations **may** override the reference lists,

> `maxPPictureL0ReferenceCount` does not limit the number of elements that the application **can** specify in the L0 reference list for P pictures. However, if `maxPPictureL0ReferenceCount` is zero, then the use of P pictures is not allowed. In case of H.265 encoding, pictures **can** be encoded using only forward prediction even if P pictures are not supported, as the ITU-T H.265 Specification supports *generalized P & B frames* (also known as low delay B frames) whereas B frames **can** refer to past frames through both the L0 and L1 reference lists.

- `maxBPictureL0ReferenceCount` indicates the maximum number of reference pictures the implementation supports in the reference list L0 for B pictures.
- `maxL1ReferenceCount` indicates the maximum number of reference pictures the implementation supports in the reference list L1 if encoding of B pictures is supported.

> **NOTE** As implementations **may** override the reference lists, `maxBPictureL0ReferenceCount` and `maxL1ReferenceCount` do not limit the number of elements that the application **can** specify in the L0 and L1 reference lists for B pictures. However, if `maxBPictureL0ReferenceCount` and `maxL1ReferenceCount` are both zero, then the use of B pictures is not allowed.

- `maxSubLayerCount` indicates the maximum number of H.265 sub-layers supported by the implementation.
- `expectDyadicTemporalSubLayerPattern` indicates that the implementation's rate control algorithms expect the application to use a dyadic temporal sub-layer pattern when encoding multiple temporal sub-layers.
- `minQp` indicates the minimum QP value supported.
- `maxQp` indicates the maximum QP value supported.
- `prefersGopRemainingFrames` indicates that the implementation's rate control algorithm prefers the application to specify the number of frames of each type remaining in the current group of pictures when beginning a video coding scope.
- `requiresGopRemainingFrames` indicates that the implementation's rate control algorithm requires the application to specify the number of frames of each type remaining in the current group of pictures when beginning a video coding scope.
- `stdSyntaxFlags` is a bitmask of VkVideoEncodeH265StdFlagBitsKHR indicating capabilities related to H.265 syntax elements.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265CapabilitiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkVideoCapabilitiesKHR

Bits which **may** be set in VkVideoEncodeH265CapabilitiesKHR::flags, indicating the H.265 encoding capabilities supported, are:

```
// Provided by VK_KHR_video_encode_h265
typedef enum VkVideoEncodeH265CapabilityFlagBitsKHR {
    VK_VIDEO_ENCODE_H265_CAPABILITY_HRD_COMPLIANCE_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H265_CAPABILITY_PREDICTION_WEIGHT_TABLE_GENERATED_BIT_KHR =
0x00000002,
    VK_VIDEO_ENCODE_H265_CAPABILITY_ROW_UNALIGNED_SLICE_SEGMENT_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_H265_CAPABILITY_DIFFERENT_SLICE_SEGMENT_TYPE_BIT_KHR = 0x00000008,
    VK_VIDEO_ENCODE_H265_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_KHR = 0x00000010,
    VK_VIDEO_ENCODE_H265_CAPABILITY_B_FRAME_IN_L1_LIST_BIT_KHR = 0x00000020,
    VK_VIDEO_ENCODE_H265_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR = 0x00000040,
    VK_VIDEO_ENCODE_H265_CAPABILITY_PER_SLICE_SEGMENT_CONSTANT_QP_BIT_KHR =
0x00000080,
    VK_VIDEO_ENCODE_H265_CAPABILITY_MULTIPLE_TILES_PER_SLICE_SEGMENT_BIT_KHR =
0x00000100,
    VK_VIDEO_ENCODE_H265_CAPABILITY_MULTIPLE_SLICE_SEGMENTS_PER_TILE_BIT_KHR =
0x00000200,
  // Provided by VK_KHR_video_encode_h265 with VK_KHR_video_encode_intra_refresh
    VK_VIDEO_ENCODE_H265_CAPABILITY_B_PICTURE_INTRA_REFRESH_BIT_KHR = 0x00000800,
  // Provided by VK_KHR_video_encode_h265 with VK_KHR_video_encode_quantization_map
    VK_VIDEO_ENCODE_H265_CAPABILITY_CU_QP_DIFF_WRAPAROUND_BIT_KHR = 0x00000400,
} VkVideoEncodeH265CapabilityFlagBitsKHR;
```

- VK_VIDEO_ENCODE_H265_CAPABILITY_HRD_COMPLIANCE_BIT_KHR specifies whether the implementation **may** be able to generate HRD compliant bitstreams if any of the nal_hrd_parameters_present_flag, vcl_hrd_parameters_present_flag, or sub_pic_hrd_params_present_flag members of StdVideoH265HrdFlags are set to 1 in the HRD parameters of the active VPS or active SPS, or if StdVideoH265SpsVuiFlags ::vui_hrd_parameters_present_flag is set to 1 in the active SPS.
- VK_VIDEO_ENCODE_H265_CAPABILITY_PREDICTION_WEIGHT_TABLE_GENERATED_BIT_KHR specifies that if the weighted_pred_flag or the weighted_bipred_flag member of StdVideoH265PpsFlags is set to 1 in the active PPS when encoding a P picture or B picture, respectively, then the implementation is able to internally decide syntax for pred_weight_table, as defined in section 7.4.7.3 of the ITU-T H.265 Specification, and the application is not **required** to provide a weight table in the H.265 slice segment header parameters.
- VK_VIDEO_ENCODE_H265_CAPABILITY_ROW_UNALIGNED_SLICE_SEGMENT_BIT_KHR specifies that each slice segment in a frame with a single or multiple tiles per slice may begin or finish at any offset in a CTB row. If not supported, all slice segments in such a frame **must** begin at the start of a CTB row (and hence each slice segment **must** finish at the end of a CTB row). Also

indicates that each slice segment in a frame with multiple slices per tile may begin or finish at any offset within the enclosing tile's CTB row. If not supported, slice segments in such a frame **must** begin at the start of the enclosing tile's CTB row (and hence each slice segment **must** finish at the end of the enclosing tile's CTB row). When a picture is encoded with intra refresh, encoding non-rectangular slice segments also requires: support for the VkVideoEncodeIntraRefreshCapabilitiesKHR::`nonRectangularIntraRefreshRegions` capability.

- VK_VIDEO_ENCODE_H265_CAPABILITY_DIFFERENT_SLICE_SEGMENT_TYPE_BIT_KHR specifies that when a frame is encoded with multiple slice segments, the implementation allows encoding each slice segment with a different `StdVideoEncodeH265SliceSegmentHeader`::`slice_type` specified in the H.265 slice segment header parameters. If not supported, all slice segments of the frame **must** be encoded with the same `slice_type` which corresponds to the picture type of the frame. There is one exception to this rule: if the picture is encoded with the intra refresh mode VK_VIDEO_ENCODE_INTRA_REFRESH_MODE_PER_PICTURE_PARTITION_BIT_KHR, then the currently refreshed slice segment **must** specify the `slice_type` `STD_VIDEO_H265_SLICE_TYPE_I` and **can** differ from the `slice_type` of the other slice segments regardless of whether VK_VIDEO_ENCODE_H265_CAPABILITY_DIFFERENT_SLICE_SEGMENT_TYPE_BIT_KHR is supported.
- VK_VIDEO_ENCODE_H265_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_KHR specifies support for using a B frame as L0 reference, as specified in `StdVideoEncodeH265ReferenceListsInfo`::`RefPicList0` in the H.265 picture information.
- VK_VIDEO_ENCODE_H265_CAPABILITY_B_FRAME_IN_L1_LIST_BIT_KHR specifies support for using a B frame as L1 reference, as specified in `StdVideoEncodeH265ReferenceListsInfo`::`RefPicList1` in the H.265 picture information.
- VK_VIDEO_ENCODE_H265_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR specifies support for specifying different QP values in the members of VkVideoEncodeH265QpKHR.
- VK_VIDEO_ENCODE_H265_CAPABILITY_PER_SLICE_SEGMENT_CONSTANT_QP_BIT_KHR specifies support for specifying different constant QP values for each slice segment.
- VK_VIDEO_ENCODE_H265_CAPABILITY_MULTIPLE_TILES_PER_SLICE_SEGMENT_BIT_KHR specifies whether encoding multiple tiles per slice segment, as defined in section 6.3.1 of the ITU-T H.265 Specification, is supported. If this capability flag is not present, then the implementation is only able to encode a single tile for each slice segment.
- VK_VIDEO_ENCODE_H265_CAPABILITY_MULTIPLE_SLICE_SEGMENTS_PER_TILE_BIT_KHR specifies whether encoding multiple slice segments per tile, as defined in section 6.3.1 of the ITU-T H.265 Specification, is supported. If this capability flag is not present, then the implementation is only able to encode a single slice segment for each tile.
- VK_VIDEO_ENCODE_H265_CAPABILITY_CU_QP_DIFF_WRAPAROUND_BIT_KHR indicates support for wraparound during the calculation of the QP values of subsequently encoded coding units, as defined in section 7.4.9.14 of the ITU-T H.265 Specification. If not supported, equation 8-283 of the ITU-T H.265 Specification is effectively reduced to the following:

$Qp_Y = qP_{Y_PRED} + \texttt{CuQpDeltaVal}$

NOTE	The effect of this is that the maximum QP difference across subsequent coding units is limited to the $[-(26 + QpBdOffset_Y / 2), 25 + QpBdOffset_Y / 2]$ range.

- VK_VIDEO_ENCODE_H265_CAPABILITY_B_PICTURE_INTRA_REFRESH_BIT_KHR specifies support for encoding B pictures with intra refresh enabled.

```
// Provided by VK_KHR_video_encode_h265
typedef VkFlags VkVideoEncodeH265CapabilityFlagsKHR;
```

VkVideoEncodeH265CapabilityFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeH265CapabilityFlagBitsKHR.

Bits which **may** be set in VkVideoEncodeH265CapabilitiesKHR::stdSyntaxFlags, indicating the capabilities related to the H.265 syntax elements, are:

```
// Provided by VK_KHR_video_encode_h265
typedef enum VkVideoEncodeH265StdFlagBitsKHR {
    VK_VIDEO_ENCODE_H265_STD_SEPARATE_COLOR_PLANE_FLAG_SET_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H265_STD_SAMPLE_ADAPTIVE_OFFSET_ENABLED_FLAG_SET_BIT_KHR =
0x00000002,
    VK_VIDEO_ENCODE_H265_STD_SCALING_LIST_DATA_PRESENT_FLAG_SET_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_H265_STD_PCM_ENABLED_FLAG_SET_BIT_KHR = 0x00000008,
    VK_VIDEO_ENCODE_H265_STD_SPS_TEMPORAL_MVP_ENABLED_FLAG_SET_BIT_KHR = 0x00000010,
    VK_VIDEO_ENCODE_H265_STD_INIT_QP_MINUS26_BIT_KHR = 0x00000020,
    VK_VIDEO_ENCODE_H265_STD_WEIGHTED_PRED_FLAG_SET_BIT_KHR = 0x00000040,
    VK_VIDEO_ENCODE_H265_STD_WEIGHTED_BIPRED_FLAG_SET_BIT_KHR = 0x00000080,
    VK_VIDEO_ENCODE_H265_STD_LOG2_PARALLEL_MERGE_LEVEL_MINUS2_BIT_KHR = 0x00000100,
    VK_VIDEO_ENCODE_H265_STD_SIGN_DATA_HIDING_ENABLED_FLAG_SET_BIT_KHR = 0x00000200,
    VK_VIDEO_ENCODE_H265_STD_TRANSFORM_SKIP_ENABLED_FLAG_SET_BIT_KHR = 0x00000400,
    VK_VIDEO_ENCODE_H265_STD_TRANSFORM_SKIP_ENABLED_FLAG_UNSET_BIT_KHR = 0x00000800,
    VK_VIDEO_ENCODE_H265_STD_PPS_SLICE_CHROMA_QP_OFFSETS_PRESENT_FLAG_SET_BIT_KHR =
0x00001000,
    VK_VIDEO_ENCODE_H265_STD_TRANSQUANT_BYPASS_ENABLED_FLAG_SET_BIT_KHR = 0x00002000,
    VK_VIDEO_ENCODE_H265_STD_CONSTRAINED_INTRA_PRED_FLAG_SET_BIT_KHR = 0x00004000,
    VK_VIDEO_ENCODE_H265_STD_ENTROPY_CODING_SYNC_ENABLED_FLAG_SET_BIT_KHR =
0x00008000,
    VK_VIDEO_ENCODE_H265_STD_DEBLOCKING_FILTER_OVERRIDE_ENABLED_FLAG_SET_BIT_KHR =
0x00010000,
    VK_VIDEO_ENCODE_H265_STD_DEPENDENT_SLICE_SEGMENTS_ENABLED_FLAG_SET_BIT_KHR =
0x00020000,
    VK_VIDEO_ENCODE_H265_STD_DEPENDENT_SLICE_SEGMENT_FLAG_SET_BIT_KHR = 0x00040000,
    VK_VIDEO_ENCODE_H265_STD_SLICE_QP_DELTA_BIT_KHR = 0x00080000,
    VK_VIDEO_ENCODE_H265_STD_DIFFERENT_SLICE_QP_DELTA_BIT_KHR = 0x00100000,
} VkVideoEncodeH265StdFlagBitsKHR;
```

- VK_VIDEO_ENCODE_H265_STD_SEPARATE_COLOR_PLANE_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265SpsFlags::separate_colour_plane_flag in the SPS when that value is 1.
- VK_VIDEO_ENCODE_H265_STD_SAMPLE_ADAPTIVE_OFFSET_ENABLED_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265SpsFlags::sample_adaptive_offset_enabled_flag in the SPS when that value is 1.

- VK_VIDEO_ENCODE_H265_STD_SCALING_LIST_DATA_PRESENT_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for the scaling_list_enabled_flag and sps_scaling_list_data_present_flag members of StdVideoH265SpsFlags in the SPS, and the application-provided value for StdVideoH265PpsFlags ::pps_scaling_list_data_present_flag in the PPS when those values are 1.
- VK_VIDEO_ENCODE_H265_STD_PCM_ENABLED_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265SpsFlags ::pcm_enabled_flag in the SPS when that value is 1.
- VK_VIDEO_ENCODE_H265_STD_SPS_TEMPORAL_MVP_ENABLED_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265SpsFlags::sps_temporal_mvp_enabled_flag in the SPS when that value is 1.
- VK_VIDEO_ENCODE_H265_STD_INIT_QP_MINUS26_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265PictureParameterSet::init_qp_minus26 in the PPS when that value is non-zero.
- VK_VIDEO_ENCODE_H265_STD_WEIGHTED_PRED_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265PpsFlags ::weighted_pred_flag in the PPS when that value is 1.
- VK_VIDEO_ENCODE_H265_STD_WEIGHTED_BIPRED_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265PpsFlags ::weighted_bipred_flag in the PPS when that value is 1.
- VK_VIDEO_ENCODE_H265_STD_LOG2_PARALLEL_MERGE_LEVEL_MINUS2_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265PictureParameterSet::log2_parallel_merge_level_minus2 in the PPS when that value is non-zero.
- VK_VIDEO_ENCODE_H265_STD_SIGN_DATA_HIDING_ENABLED_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265PpsFlags::sign_data_hiding_enabled_flag in the PPS when that value is 1.
- VK_VIDEO_ENCODE_H265_STD_TRANSFORM_SKIP_ENABLED_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265PpsFlags::transform_skip_enabled_flag in the PPS when that value is 1.
- VK_VIDEO_ENCODE_H265_STD_TRANSFORM_SKIP_ENABLED_FLAG_UNSET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265PpsFlags::transform_skip_enabled_flag in the PPS when that value is 0.
- VK_VIDEO_ENCODE_H265_STD_PPS_SLICE_CHROMA_QP_OFFSETS_PRESENT_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265PpsFlags::pps_slice_chroma_qp_offsets_present_flag in the PPS when that value is 1.
- VK_VIDEO_ENCODE_H265_STD_TRANSQUANT_BYPASS_ENABLED_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265PpsFlags::transquant_bypass_enabled_flag in the PPS when that value is 1.
- VK_VIDEO_ENCODE_H265_STD_CONSTRAINED_INTRA_PRED_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265PpsFlags::constrained_intra_pred_flag in the PPS when that value is 1.
- VK_VIDEO_ENCODE_H265_STD_ENTROPY_CODING_SYNC_ENABLED_FLAG_SET_BIT_KHR

specifies whether the implementation supports using the application-provided value for StdVideoH265PpsFlags::entropy_coding_sync_enabled_flag in the PPS when that value is 1.

- VK_VIDEO_ENCODE_H265_STD_DEBLOCKING_FILTER_OVERRIDE_ENABLED_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265PpsFlags::deblocking_filter_override_enabled_flag in the PPS when that value is 1.
- VK_VIDEO_ENCODE_H265_STD_DEPENDENT_SLICE_SEGMENTS_ENABLED_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoH265PpsFlags::dependent_slice_segments_enabled_flag in the PPS when that value is 1.
- VK_VIDEO_ENCODE_H265_STD_DEPENDENT_SLICE_SEGMENT_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoEncodeH265SliceSegmentHeader::dependent_slice_segment_flag in the H.265 slice segment header parameters when that value is 1.
- VK_VIDEO_ENCODE_H265_STD_SLICE_QP_DELTA_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoEncodeH265SliceSegmentHeader::slice_qp_delta in the H.265 slice segment header parameters when that value is identical across the slice segments of the encoded frame.
- VK_VIDEO_ENCODE_H265_STD_DIFFERENT_SLICE_QP_DELTA_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoEncodeH265SliceSegmentHeader::slice_qp_delta in the H.265 slice segment header parameters when that value is different across the slice segments of the encoded frame.

These capability flags provide information to the application about specific H.265 syntax element values that the implementation supports without having to override them and do not otherwise restrict the values that the application **can** specify for any of the mentioned H.265 syntax elements.

```
// Provided by VK_KHR_video_encode_h265
typedef VkFlags VkVideoEncodeH265StdFlagsKHR;
```

VkVideoEncodeH265StdFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeH265StdFlagBitsKHR.

Bits which **may** be set in VkVideoEncodeH265CapabilitiesKHR::ctbSizes, indicating the CTB sizes supported by the implementation, are:

```
// Provided by VK_KHR_video_encode_h265
typedef enum VkVideoEncodeH265CtbSizeFlagBitsKHR {
    VK_VIDEO_ENCODE_H265_CTB_SIZE_16_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H265_CTB_SIZE_32_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_H265_CTB_SIZE_64_BIT_KHR = 0x00000004,
} VkVideoEncodeH265CtbSizeFlagBitsKHR;
```

- VK_VIDEO_ENCODE_H265_CTB_SIZE_16_BIT_KHR specifies that a CTB size of 16x16 is supported.
- VK_VIDEO_ENCODE_H265_CTB_SIZE_32_BIT_KHR specifies that a CTB size of 32x32 is

supported.

- VK_VIDEO_ENCODE_H265_CTB_SIZE_64_BIT_KHR specifies that a CTB size of 64x64 is supported.

```
// Provided by VK_KHR_video_encode_h265
typedef VkFlags VkVideoEncodeH265CtbSizeFlagsKHR;
```

VkVideoEncodeH265CtbSizeFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeH265CtbSizeFlagBitsKHR.

Implementations **must** support at least one of VkVideoEncodeH265CtbSizeFlagBitsKHR.

Bits which **may** be set in VkVideoEncodeH265CapabilitiesKHR::transformBlockSizes, indicating the transform block sizes supported by the implementation, are:

```
// Provided by VK_KHR_video_encode_h265
typedef enum VkVideoEncodeH265TransformBlockSizeFlagBitsKHR {
    VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_4_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_8_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_16_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_32_BIT_KHR = 0x00000008,
} VkVideoEncodeH265TransformBlockSizeFlagBitsKHR;
```

- VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_4_BIT_KHR specifies that a transform block size of 4x4 is supported.
- VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_8_BIT_KHR specifies that a transform block size of 8x8 is supported.
- VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_16_BIT_KHR specifies that a transform block size of 16x16 is supported.
- VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_32_BIT_KHR specifies that a transform block size of 32x32 is supported.

```
// Provided by VK_KHR_video_encode_h265
typedef VkFlags VkVideoEncodeH265TransformBlockSizeFlagsKHR;
```

VkVideoEncodeH265TransformBlockSizeFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeH265TransformBlockSizeFlagBitsKHR.

Implementations **must** support at least one of VkVideoEncodeH265TransformBlockSizeFlagBitsKHR.

45.21.8. H.265 Encode Quality Level Properties

When calling vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR with pVideoProfile->videoCodecOperation specified as VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, the

VkVideoEncodeH265QualityLevelPropertiesKHR structure **must** be included in the pNext chain of the VkVideoEncodeQualityLevelPropertiesKHR structure to retrieve additional video encode quality level properties specific to H.265 encoding.

The VkVideoEncodeH265QualityLevelPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265QualityLevelPropertiesKHR {
    VkStructureType                         sType;
    void*                                   pNext;
    VkVideoEncodeH265RateControlFlagsKHR    preferredRateControlFlags;
    uint32_t                                preferredGopFrameCount;
    uint32_t                                preferredIdrPeriod;
    uint32_t                                preferredConsecutiveBFrameCount;
    uint32_t                                preferredSubLayerCount;
    VkVideoEncodeH265QpKHR                  preferredConstantQp;
    uint32_t                                preferredMaxL0ReferenceCount;
    uint32_t                                preferredMaxL1ReferenceCount;
} VkVideoEncodeH265QualityLevelPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- preferredRateControlFlags is a bitmask of VkVideoEncodeH265RateControlFlagBitsKHR values indicating the preferred flags to use for VkVideoEncodeH265RateControlInfoKHR::flags.
- preferredGopFrameCount indicates the preferred value to use for VkVideoEncodeH265RateControlInfoKHR::gopFrameCount.
- preferredIdrPeriod indicates the preferred value to use for VkVideoEncodeH265RateControlInfoKHR::idrPeriod.
- preferredConsecutiveBFrameCount indicates the preferred value to use for VkVideoEncodeH265RateControlInfoKHR::consecutiveBFrameCount.
- preferredSubLayerCount indicates the preferred value to use for VkVideoEncodeH265RateControlInfoKHR::subLayerCount.
- preferredConstantQp indicates the preferred values to use for VkVideoEncodeH265NaluSliceSegmentInfoKHR::constantQp for each picture type when using rate control mode VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR.
- preferredMaxL0ReferenceCount indicates the preferred maximum number of reference pictures to use in the reference list L0.
- preferredMaxL1ReferenceCount indicates the preferred maximum number of reference pictures to use in the reference list L1.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265QualityLevelPropertiesKHR-sType-sType
  sType **must** be

VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_QUALITY_LEVEL_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkVideoEncodeQualityLevelPropertiesKHR

45.21.9. H.265 Encode Session

Additional parameters can be specified when creating a video session with an H.265 encode profile by including an instance of the VkVideoEncodeH265SessionCreateInfoKHR structure in the `pNext` chain of VkVideoSessionCreateInfoKHR.

The `VkVideoEncodeH265SessionCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265SessionCreateInfoKHR {
    VkStructureType         sType;
    const void*             pNext;
    VkBool32                useMaxLevelIdc;
    StdVideoH265LevelIdc    maxLevelIdc;
} VkVideoEncodeH265SessionCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `useMaxLevelIdc` indicates whether the value of `maxLevelIdc` should be used by the implementation. When it is VK_FALSE, the implementation ignores the value of `maxLevelIdc` and uses the value of VkVideoEncodeH265CapabilitiesKHR::`maxLevelIdc`, as reported by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile.
- `maxLevelIdc` is a `StdVideoH265LevelIdc` value specifying the upper bound on the H.265 level for the video bitstreams produced by the created video session, where enum constant `STD_VIDEO_H265_LEVEL_IDC_<major>_<minor>` identifies H.265 level `<major>.<minor>` as defined in section A.4 of the ITU-T H.265 Specification.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265SessionCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_CREATE_INFO_KHR

Structure Chaining

Extends the structure

- VkVideoSessionCreateInfoKHR

45.21.10. H.265 Encode Parameter Sets

Video session parameters objects created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR **can** contain the following types of parameters:

H.265 Video Parameter Sets (VPS)

Represented by StdVideoH265VideoParameterSet structures and interpreted as follows:

- reserved1, reserved2, and reserved3 are used only for padding purposes and are otherwise ignored;
- vps_video_parameter_set_id is used as the key of the VPS entry;
- the max_latency_increase_plus1, max_dec_pic_buffering_minus1, and max_num_reorder_pics members of the StdVideoH265DecPicBufMgr structure pointed to by pDecPicBufMgr correspond to vps_max_latency_increase_plus1, vps_max_dec_pic_buffering_minus1, and vps_max_num_reorder_pics, respectively, as defined in section 7.4.3.1 of the ITU-T H.265 Specification;
- the StdVideoH265HrdParameters structure pointed to by pHrdParameters is interpreted as follows:
  - reserved is used only for padding purposes and is otherwise ignored;
  - flags.fixed_pic_rate_general_flag is a bitmask where bit index i corresponds to fixed_pic_rate_general_flag[i] as defined in section E.3.2 of the ITU-T H.265 Specification;
  - flags.fixed_pic_rate_within_cvs_flag is a bitmask where bit index i corresponds to fixed_pic_rate_within_cvs_flag[i] as defined in section E.3.2 of the ITU-T H.265 Specification;
  - flags.low_delay_hrd_flag is a bitmask where bit index i corresponds to low_delay_hrd_flag[i] as defined in section E.3.2 of the ITU-T H.265 Specification;
  - if flags.nal_hrd_parameters_present_flag is set, then pSubLayerHrdParametersNal is a pointer to an array of vps_max_sub_layers_minus1 + 1 number of StdVideoH265SubLayerHrdParameters structures where vps_max_sub_layers_minus1 is the corresponding member of the encompassing StdVideoH265VideoParameterSet structure and each element is interpreted as follows:
    - cbr_flag is a bitmask where bit index i corresponds to cbr_flag[i] as defined in section E.3.3 of the ITU-T H.265 Specification;
    - all other members of the StdVideoH265SubLayerHrdParameters structure are interpreted as defined in section E.3.3 of the ITU-T H.265 Specification;
  - if flags.vcl_hrd_parameters_present_flag is set, then pSubLayerHrdParametersVcl is a pointer to an array of vps_max_sub_layers_minus1 + 1 number of StdVideoH265SubLayerHrdParameters structures where vps_max_sub_layers_minus1 is the corresponding member of the encompassing StdVideoH265VideoParameterSet structure and each element is interpreted as follows:

        - `cbr_flag` is a bitmask where bit index i corresponds to `cbr_flag[i]` as defined in section E.3.3 of the ITU-T H.265 Specification;
        - all other members of the `StdVideoH265SubLayerHrdParameters` structure are interpreted as defined in section E.3.3 of the ITU-T H.265 Specification;
    - all other members of `StdVideoH265HrdParameters` are interpreted as defined in section E.3.2 of the ITU-T H.265 Specification;
- the `StdVideoH265ProfileTierLevel` structure pointed to by `pProfileTierLevel` is interpreted as follows:
    - `general_level_idc` is one of the enum constants `STD_VIDEO_H265_LEVEL_IDC_<major>_<minor>` identifying the H.265 level `<major>.<minor>` as defined in section A.4 of the ITU-T H.265 Specification;
    - all other members of `StdVideoH265ProfileTierLevel` are interpreted as defined in section 7.4.4 of the ITU-T H.265 Specification;
- all other members of `StdVideoH265VideoParameterSet` are interpreted as defined in section 7.4.3.1 of the ITU-T H.265 Specification.

H.265 Sequence Parameter Sets (SPS)

Represented by `StdVideoH265SequenceParameterSet` structures and interpreted as follows:

- `reserved1` and `reserved2` are used only for padding purposes and are otherwise ignored;
- the pair constructed from `sps_video_parameter_set_id` and `sps_seq_parameter_set_id` is used as the key of the SPS entry;
- the `StdVideoH265ProfileTierLevel` structure pointed to by `pProfileTierLevel` is interpreted as follows:
    - `general_level_idc` is one of the enum constants `STD_VIDEO_H265_LEVEL_IDC_<major>_<minor>` identifying the H.265 level `<major>.<minor>` as defined in section A.4 of the ITU-T H.265 Specification;
    - all other members of `StdVideoH265ProfileTierLevel` are interpreted as defined in section 7.4.4 of the ITU-T H.265 Specification;
- the `max_latency_increase_plus1`, `max_dec_pic_buffering_minus1`, and `max_num_reorder_pics` members of the `StdVideoH265DecPicBufMgr` structure pointed to by `pDecPicBufMgr` correspond to `sps_max_latency_increase_plus1`, `sps_max_dec_pic_buffering_minus1`, and `sps_max_num_reorder_pics`, respectively, as defined in section 7.4.3.2 of the ITU-T H.265 Specification;
- if `flags.sps_scaling_list_data_present_flag` is set, then the `StdVideoH265ScalingLists` structure pointed to by `pScalingLists` is interpreted as follows:
    - `ScalingList4x4`, `ScalingList8x8`, `ScalingList16x16`, and `ScalingList32x32` correspond to `ScalingList[0]`, `ScalingList[1]`, `ScalingList[2]`, and `ScalingList[3]`, respectively, as defined in section 7.3.4 of the ITU-T H.265 Specification;
    - `ScalingListDCCoef16x16` and `ScalingListDCCoef32x32` correspond to `scaling_list_dc_coef_minus8[0]` and `scaling_list_dc_coef_minus8[1]`, respectively, as

defined in section 7.3.4 of the ITU-T H.265 Specification;

- `pShortTermRefPicSet` is a pointer to an array of `num_short_term_ref_pic_sets` number of `StdVideoH265ShortTermRefPicSet` structures where each element is interpreted as follows:
  - `reserved1`, `reserved2`, and `reserved3` are used only for padding purposes and are otherwise ignored;
  - `used_by_curr_pic_flag` is a bitmask where bit index i corresponds to `used_by_curr_pic_flag[i]` as defined in section 7.4.8 of the ITU-T H.265 Specification;
  - `use_delta_flag` is a bitmask where bit index i corresponds to `use_delta_flag[i]` as defined in section 7.4.8 of the ITU-T H.265 Specification;
  - `used_by_curr_pic_s0_flag` is a bitmask where bit index i corresponds to `used_by_curr_pic_s0_flag[i]` as defined in section 7.4.8 of the ITU-T H.265 Specification;
  - `used_by_curr_pic_s1_flag` is a bitmask where bit index i corresponds to `used_by_curr_pic_s1_flag[i]` as defined in section 7.4.8 of the ITU-T H.265 Specification;
  - all other members of `StdVideoH265ShortTermRefPicSet` are interpreted as defined in section 7.4.8 of the ITU-T H.265 Specification;
- if `flags.long_term_ref_pics_present_flag` is set then the `StdVideoH265LongTermRefPicsSps` structure pointed to by `pLongTermRefPicsSps` is interpreted as follows:
  - `used_by_curr_pic_lt_sps_flag` is a bitmask where bit index i corresponds to `used_by_curr_pic_lt_sps_flag[i]` as defined in section 7.4.3.2 of the ITU-T H.265 Specification;
  - all other members of `StdVideoH265LongTermRefPicsSps` are interpreted as defined in section 7.4.3.2 of the ITU-T H.265 Specification;
- if `flags.vui_parameters_present_flag` is set, then the `StdVideoH265SequenceParameterSetVui` structure pointed to by `pSequenceParameterSetVui` is interpreted as follows:
  - `reserved1`, `reserved2`, and `reserved3` are used only for padding purposes and are otherwise ignored;
  - the `StdVideoH265HrdParameters` structure pointed to by `pHrdParameters` is interpreted as follows:
    - `flags.fixed_pic_rate_general_flag` is a bitmask where bit index i corresponds to `fixed_pic_rate_general_flag[i]` as defined in section E.3.2 of the ITU-T H.265 Specification;
    - `flags.fixed_pic_rate_within_cvs_flag` is a bitmask where bit index i corresponds to `fixed_pic_rate_within_cvs_flag[i]` as defined in section E.3.2 of the ITU-T H.265 Specification;
    - `flags.low_delay_hrd_flag` is a bitmask where bit index i corresponds to `low_delay_hrd_flag[i]` as defined in section E.3.2 of the ITU-T H.265 Specification;
    - if `flags.nal_hrd_parameters_present_flag` is set, then `pSubLayerHrdParametersNal` is a pointer to an array of `sps_max_sub_layers_minus1` + 1 number of `StdVideoH265SubLayerHrdParameters` structures where `sps_max_sub_layers_minus1` is the corresponding member of the encompassing `StdVideoH265SequenceParameterSet` structure and each element is interpreted as follows:

        - `cbr_flag` is a bitmask where bit index i corresponds to `cbr_flag[i]` as defined in section E.3.3 of the ITU-T H.265 Specification;
        - all other members of the `StdVideoH265SubLayerHrdParameters` structure are interpreted as defined in section E.3.3 of the ITU-T H.265 Specification;
      - if `flags.vcl_hrd_parameters_present_flag` is set, then `pSubLayerHrdParametersVcl` is a pointer to an array of `sps_max_sub_layers_minus1 + 1` number of `StdVideoH265SubLayerHrdParameters` structures where `sps_max_sub_layers_minus1` is the corresponding member of the encompassing `StdVideoH265SequenceParameterSet` structure and each element is interpreted as follows:
        - `cbr_flag` is a bitmask where bit index i corresponds to `cbr_flag[i]` as defined in section E.3.3 of the ITU-T H.265 Specification;
        - all other members of the `StdVideoH265SubLayerHrdParameters` structure are interpreted as defined in section E.3.3 of the ITU-T H.265 Specification;
      - all other members of `StdVideoH265HrdParameters` are interpreted as defined in section E.3.2 of the ITU-T H.265 Specification;
    - all other members of `pSequenceParameterSetVui` are interpreted as defined in section E.3.1 of the ITU-T H.265 Specification;
- if `flags.sps_palette_predictor_initializer_present_flag` is set, then the `PredictorPaletteEntries` member of the `StdVideoH265PredictorPaletteEntries` structure pointed to by `pPredictorPaletteEntries` is interpreted as defined in section 7.4.9.13 of the ITU-T H.265 Specification;
- all other members of `StdVideoH265SequenceParameterSet` are interpreted as defined in section 7.4.3.1 of the ITU-T H.265 Specification.

H.265 Picture Parameter Sets (PPS)

Represented by `StdVideoH265PictureParameterSet` structures and interpreted as follows:

- `reserved1`, `reserved2`, and `reserved3` are used only for padding purposes and are otherwise ignored;
- the triplet constructed from `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id` is used as the key of the PPS entry;
- if `flags.pps_scaling_list_data_present_flag` is set, then the `StdVideoH265ScalingLists` structure pointed to by `pScalingLists` is interpreted as follows:
  - `ScalingList4x4`, `ScalingList8x8`, `ScalingList16x16`, and `ScalingList32x32` correspond to `ScalingList[0]`, `ScalingList[1]`, `ScalingList[2]`, and `ScalingList[3]`, respectively, as defined in section 7.3.4 of the ITU-T H.265 Specification;
  - `ScalingListDCCoef16x16` and `ScalingListDCCoef32x32` correspond to `scaling_list_dc_coef_minus8[0]` and `scaling_list_dc_coef_minus8[1]`, respectively, as defined in section 7.3.4 of the ITU-T H.265 Specification;
- if `flags.pps_palette_predictor_initializer_present_flag` is set, then the `PredictorPaletteEntries` member of the `StdVideoH265PredictorPaletteEntries` structure

pointed to by pPredictorPaletteEntries is interpreted as defined in section 7.4.9.13 of the ITU-T H.265 Specification;

- all other members of StdVideoH265PictureParameterSet are interpreted as defined in section 7.4.3.3 of the ITU-T H.265 Specification.

Implementations **may** override any of these parameters according to the semantics defined in the Video Encode Parameter Overrides section before storing the resulting H.265 parameter sets into the video session parameters object. Applications need to use the vkGetEncodedVideoSessionParametersKHR command to determine whether any implementation overrides happened and to retrieve the encoded H.265 parameter sets in order to be able to produce a compliant H.265 video bitstream.

Such H.265 parameter set overrides **may** also have cascading effects on the implementation overrides applied to the encoded bitstream produced by video encode operations. If the implementation supports the VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR video encode feedback query flag, then the application **can** use such queries to retrieve feedback about whether any implementation overrides have been applied to the encoded bitstream.

When a video session parameters object is created with the codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, the VkVideoSessionParametersCreateInfoKHR::pNext chain **must** include a VkVideoEncodeH265SessionParametersCreateInfoKHR structure specifying the capacity and initial contents of the object.

The VkVideoEncodeH265SessionParametersCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265SessionParametersCreateInfoKHR {
    VkStructureType                                       sType;
    const void*                                           pNext;
    uint32_t                                              maxStdVPSCount;
    uint32_t                                              maxStdSPSCount;
    uint32_t                                              maxStdPPSCount;
    const VkVideoEncodeH265SessionParametersAddInfoKHR*   pParametersAddInfo;
} VkVideoEncodeH265SessionParametersCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- maxStdVPSCount is the maximum number of H.265 VPS entries the created VkVideoSessionParametersKHR **can** contain.
- maxStdSPSCount is the maximum number of H.265 SPS entries the created VkVideoSessionParametersKHR **can** contain.
- maxStdPPSCount is the maximum number of H.265 PPS entries the created VkVideoSessionParametersKHR **can** contain.
- pParametersAddInfo is NULL or a pointer to a VkVideoEncodeH265SessionParametersAddInfoKHR

structure specifying H.265 parameters to add upon object creation.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265SessionParametersCreateInfoKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_CREATE_INFO_KHR`
- VUID-VkVideoEncodeH265SessionParametersCreateInfoKHR-pParametersAddInfo-parameter
  If `pParametersAddInfo` is not `NULL`, `pParametersAddInfo` **must** be a valid pointer to a valid `VkVideoEncodeH265SessionParametersAddInfoKHR` structure

Structure Chaining

Extends the structure

- `VkVideoSessionParametersCreateInfoKHR`

The `VkVideoEncodeH265SessionParametersAddInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265SessionParametersAddInfoKHR {
    VkStructureType                              sType;
    const void*                                  pNext;
    uint32_t                                     stdVPSCount;
    const StdVideoH265VideoParameterSet*        pStdVPSs;
    uint32_t                                     stdSPSCount;
    const StdVideoH265SequenceParameterSet*     pStdSPSs;
    uint32_t                                     stdPPSCount;
    const StdVideoH265PictureParameterSet*      pStdPPSs;
} VkVideoEncodeH265SessionParametersAddInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stdVPSCount` is the number of elements in the `pStdVPSs` array.
- `pStdVPSs` is a pointer to an array of `StdVideoH265VideoParameterSet` structures describing the H.265 VPS entries to add.
- `stdSPSCount` is the number of elements in the `pStdSPSs` array.
- `pStdSPSs` is a pointer to an array of `StdVideoH265SequenceParameterSet` structures describing the H.265 SPS entries to add.
- `stdPPSCount` is the number of elements in the `pStdPPSs` array.
- `pStdPPSs` is a pointer to an array of `StdVideoH265PictureParameterSet` structures describing the

H.265 PPS entries to add.

This structure **can** be specified in the following places:

- In the `pParametersAddInfo` member of the VkVideoEncodeH265SessionParametersCreateInfoKHR structure specified in the `pNext` chain of VkVideoSessionParametersCreateInfoKHR used to create a video session parameters object. In this case, if the video codec operation the video session parameters object is created with is VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then it defines the set of initial parameters to add to the created object (see Creating Video Session Parameters).
- In the `pNext` chain of VkVideoSessionParametersUpdateInfoKHR. In this case, if the video codec operation the video session parameters object to be updated was created with is VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then it defines the set of parameters to add to it (see Updating Video Session Parameters).

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265SessionParametersAddInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_ADD_INFO_KHR
- VUID-VkVideoEncodeH265SessionParametersAddInfoKHR-pStdVPSs-parameter
  If `stdVPSCount` is not `0`, and `pStdVPSs` is not `NULL`, `pStdVPSs` **must** be a valid pointer to an array of `stdVPSCount` `StdVideoH265VideoParameterSet` values
- VUID-VkVideoEncodeH265SessionParametersAddInfoKHR-pStdSPSs-parameter
  If `stdSPSCount` is not `0`, and `pStdSPSs` is not `NULL`, `pStdSPSs` **must** be a valid pointer to an array of `stdSPSCount` `StdVideoH265SequenceParameterSet` values
- VUID-VkVideoEncodeH265SessionParametersAddInfoKHR-pStdPPSs-parameter
  If `stdPPSCount` is not `0`, and `pStdPPSs` is not `NULL`, `pStdPPSs` **must** be a valid pointer to an array of `stdPPSCount` `StdVideoH265PictureParameterSet` values

Structure Chaining

Extends the structure

- VkVideoSessionParametersUpdateInfoKHR

Valid Usage

- VUID-VkVideoEncodeH265SessionParametersAddInfoKHR-None-06438
  The `vps_video_parameter_set_id` member of each `StdVideoH265VideoParameterSet` structure specified in the elements of `pStdVPSs` **must** be unique within `pStdVPSs`
- VUID-VkVideoEncodeH265SessionParametersAddInfoKHR-None-06439
  The pair constructed from the `sps_video_parameter_set_id` and `sps_seq_parameter_set_id` members of each `StdVideoH265SequenceParameterSet` structure specified in the elements of

pStdSPSs **must** be unique within pStdSPSs

- VUID-VkVideoEncodeH265SessionParametersAddInfoKHR-None-06440
  The triplet constructed from the sps_video_parameter_set_id, pps_seq_parameter_set_id, and pps_pic_parameter_set_id members of each StdVideoH265PictureParameterSet structure specified in the elements of pStdPPSs **must** be unique within pStdPPSs

The VkVideoEncodeH265SessionParametersGetInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265SessionParametersGetInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           writeStdVPS;
    VkBool32           writeStdSPS;
    VkBool32           writeStdPPS;
    uint32_t           stdVPSId;
    uint32_t           stdSPSId;
    uint32_t           stdPPSId;
} VkVideoEncodeH265SessionParametersGetInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- writeStdVPS indicates whether the encoded H.265 video parameter set identified by stdVPSId is requested to be retrieved.
- writeStdSPS indicates whether the encoded H.265 sequence parameter set identified by the pair constructed from stdVPSId and stdSPSId is requested to be retrieved.
- writeStdPPS indicates whether the encoded H.265 picture parameter set identified by the triplet constructed from stdVPSId, stdSPSId, and stdPPSId is requested to be retrieved.
- stdVPSId specifies the H.265 video parameter set ID used to identify the retrieved H.265 video, sequence, and/or picture parameter set(s).
- stdSPSId specifies the H.265 sequence parameter set ID used to identify the retrieved H.265 sequence and/or picture parameter set(s) when writeStdSPS and/or writeStdPPS is VK_TRUE.
- stdPPSId specifies the H.265 picture parameter set ID used to identify the retrieved H.265 picture parameter set when writeStdPPS is VK_TRUE.

When this structure is specified in the pNext chain of the VkVideoEncodeSessionParametersGetInfoKHR structure passed to vkGetEncodedVideoSessionParametersKHR, the command will write encoded parameter data to the output buffer in the following order:

1. The H.265 video parameter set identified by stdVPSId, if writeStdVPS is VK_TRUE.
2. The H.265 sequence parameter set identified by the pair constructed from stdVPSId and stdSPSId, if writeStdSPS is VK_TRUE.
3. The H.265 picture parameter set identified by the triplet constructed from stdVPSId, stdSPSId,

and stdPPSId, if writeStdPPS is VK_TRUE.

Valid Usage

- VUID-VkVideoEncodeH265SessionParametersGetInfoKHR-writeStdVPS-08290
  At least one of writeStdVPS, writeStdSPS, and writeStdPPS **must** be VK_TRUE

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265SessionParametersGetInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_GET_INFO_KHR

Structure Chaining

Extends the structure

- VkVideoEncodeSessionParametersGetInfoKHR

The VkVideoEncodeH265SessionParametersFeedbackInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265SessionParametersFeedbackInfoKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           hasStdVPSOverrides;
    VkBool32           hasStdSPSOverrides;
    VkBool32           hasStdPPSOverrides;
} VkVideoEncodeH265SessionParametersFeedbackInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- hasStdVPSOverrides indicates whether any of the parameters of the requested H.265 video parameter set, if one was requested via VkVideoEncodeH265SessionParametersGetInfoKHR::writeStdVPS, were overridden by the implementation.
- hasStdSPSOverrides indicates whether any of the parameters of the requested H.265 sequence parameter set, if one was requested via VkVideoEncodeH265SessionParametersGetInfoKHR::writeStdSPS, were overridden by the implementation.
- hasStdPPSOverrides indicates whether any of the parameters of the requested H.265 picture parameter set, if one was requested via VkVideoEncodeH265SessionParametersGetInfoKHR::writeStdPPS, were overridden by the implementation.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265SessionParametersFeedbackInfoKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_FEEDBACK_INFO_KHR`

Structure Chaining

Extends the structure

- VkVideoEncodeSessionParametersFeedbackInfoKHR

45.21.11. H.265 Encoding Parameters

The VkVideoEncodeH265PictureInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265PictureInfoKHR {
    VkStructureType                                   sType;
    const void*                                       pNext;
    uint32_t                                          naluSliceSegmentEntryCount;
    const VkVideoEncodeH265NaluSliceSegmentInfoKHR*   pNaluSliceSegmentEntries;
    const StdVideoEncodeH265PictureInfo*              pStdPictureInfo;
} VkVideoEncodeH265PictureInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `naluSliceSegmentEntryCount` is the number of elements in `pNaluSliceSegmentEntries`.
- `pNaluSliceSegmentEntries` is a pointer to an array of `naluSliceSegmentEntryCount` VkVideoEncodeH265NaluSliceSegmentInfoKHR structures specifying the parameters of the individual H.265 slice segments to encode for the input picture.
- `pStdPictureInfo` is a pointer to a `StdVideoEncodeH265PictureInfo` structure specifying H.265 picture information.

This structure is specified in the `pNext` chain of the VkVideoEncodeInfoKHR structure passed to vkCmdEncodeVideoKHR to specify the codec-specific picture information for an H.265 encode operation.

Encode Input Picture Information

When this structure is specified in the `pNext` chain of the VkVideoEncodeInfoKHR structure passed to vkCmdEncodeVideoKHR, the information related to the encode input picture is defined as follows:

- The image subregion used is determined according to the H.265 Encode Picture Data Access

section.

- The encode input picture is associated with the H.265 picture information provided in `pStdPictureInfo`.

Std Picture Information

The members of the `StdVideoEncodeH265PictureInfo` structure pointed to by `pStdPictureInfo` are interpreted as follows:

- `flags.reserved` and `reserved1` are used only for padding purposes and are otherwise ignored;
- `flags.is_reference` as defined in section 3.132 of the ITU-T H.265 Specification;
- `flags.IrapPicFlag` as defined in section 3.73 of the ITU-T H.265 Specification;
- `flags.used_for_long_term_reference` is used to indicate whether the picture is marked as "used for long-term reference" as defined in section 8.3.2 of the ITU-T H.265 Specification;
- `flags.discardable_flag` and `cross_layer_bla_flag` as defined in section F.7.4.7.1 of the ITU-T H.265 Specification;
- `pic_type` as defined in section 7.4.3.5 of the ITU-T H.265 Specification;
- `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id` are used to identify the active parameter sets, as described below;
- `PicOrderCntVal` as defined in section 8.3.1 of the ITU-T H.265 Specification;
- `TemporalId` as defined in section 7.4.2.2 of the ITU-T H.265 Specification;
- if `pRefLists` is not `NULL`, then it is a pointer to a `StdVideoEncodeH265ReferenceListsInfo` structure that is interpreted as follows:
  - `flags.reserved` is used only for padding purposes and is otherwise ignored;
  - `ref_pic_list_modification_flag_l0` and `ref_pic_list_modification_flag_l1` as defined in section 7.4.7.2 of the ITU-T H.265 Specification;
  - `num_ref_idx_l0_active_minus1` and `num_ref_idx_l1_active_minus1` as defined in section 7.4.7.1 of the ITU-T H.265 Specification;
  - `RefPicList0` and `RefPicList1` as defined in section 8.3.4 of the ITU-T H.265 Specification where each element of these arrays either identifies an active reference picture using its DPB slot index or contains the value `STD_VIDEO_H265_NO_REFERENCE_PICTURE` to indicate "no reference picture";
  - `list_entry_l0` and `list_entry_l1` as defined in section 7.4.7.2 of the ITU-T H.265 Specification;
- if `flags.short_term_ref_pic_set_sps_flag` is set, then the `StdVideoH265ShortTermRefPicSet` structure pointed to by `pShortTermRefPicSet` is interpreted as defined for the elements of the `pShortTermRefPicSet` array specified in H.265 sequence parameter sets.
- if `flags.long_term_ref_pics_present_flag` is set in the active SPS, then the `StdVideoEncodeH265LongTermRefPics` structure pointed to by `pLongTermRefPics` is interpreted as follows:
  - `used_by_curr_pic_lt_flag` is a bitmask where bit index i corresponds to `used_by_curr_pic_lt_flag[i]` as defined in section 7.4.7.1 of the ITU-T H.265 Specification;

    - all other members of StdVideoEncodeH265LongTermRefPics are interpreted as defined in section 7.4.7.1 of the ITU-T H.265 Specification;
- all other members are interpreted as defined in section 7.4.7.1 of the ITU-T H.265 Specification.

Reference picture setup is controlled by the value of StdVideoEncodeH265PictureInfo::flags.is_reference. If it is set and a reconstructed picture is specified, then the latter is used as the target of picture reconstruction to activate the DPB slot specified in pEncodeInfo->pSetupReferenceSlot->slotIndex. If StdVideoEncodeH265PictureInfo::flags.is_reference is not set, but a reconstructed picture is specified, then the corresponding picture reference associated with the DPB slot is invalidated, as described in the DPB Slot States section.

Active Parameter Sets

The members of the StdVideoEncodeH265PictureInfo structure pointed to by pStdPictureInfo are used to select the active parameter sets to use from the bound video session parameters object, as follows:

- The *active VPS* is the VPS identified by the key specified in StdVideoEncodeH265PictureInfo::sps_video_parameter_set_id.
- The *active SPS* is the SPS identified by the key specified by the pair constructed from StdVideoEncodeH265PictureInfo::sps_video_parameter_set_id and StdVideoEncodeH265PictureInfo::pps_seq_parameter_set_id.
- The *active PPS* is the PPS identified by the key specified by the triplet constructed from StdVideoEncodeH265PictureInfo::sps_video_parameter_set_id, StdVideoEncodeH265PictureInfo::pps_seq_parameter_set_id, and StdVideoEncodeH265PictureInfo::pps_pic_parameter_set_id.

H.265 encoding uses *explicit weighted sample prediction* for a slice segment, as defined in section 8.5.3.3.4 of the ITU-T H.265 Specification, if any of the following conditions are true for the active PPS and the pStdSliceSegmentHeader member of the corresponding element of pNaluSliceSegmentEntries:

- pStdSliceSegmentHeader->slice_type is STD_VIDEO_H265_SLICE_TYPE_P and weighted_pred_flag is enabled in the active PPS.
- pStdSliceSegmentHeader->slice_type is STD_VIDEO_H265_SLICE_TYPE_B and weighted_bipred_flag is enabled in the active PPS.

The number of H.265 tiles, as defined in section 3.174 of the ITU-T H.265 Specification, is derived from the num_tile_columns_minus1 and num_tile_rows_minus1 members of the active PPS as follows:

(num_tile_columns_minus1 + 1) × (num_tile_rows_minus1 + 1)

Valid Usage

- VUID-VkVideoEncodeH265PictureInfoKHR-naluSliceSegmentEntryCount-08306
naluSliceSegmentEntryCount **must** be between 1 and VkVideoEncodeH265CapabilitiesKHR::maxSliceSegmentCount, inclusive, as returned by

vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile

- VUID-VkVideoEncodeH265PictureInfoKHR-flags-08323
  If VkVideoEncodeH265CapabilitiesKHR::flags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_H265_CAPABILITY_MULTIPLE_TILES_PER_SLICE_SEGMENT_BIT_KHR, then naluSliceSegmentEntryCount **must** be greater than or equal to the number of H.265 tiles in the picture
- VUID-VkVideoEncodeH265PictureInfoKHR-flags-08324
  If VkVideoEncodeH265CapabilitiesKHR::flags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_H265_CAPABILITY_MULTIPLE_SLICE_SEGMENTS_PER_TILE_BIT_KHR, then naluSliceSegmentEntryCount **must** be less than or equal to the number of H.265 tiles in the picture
- VUID-VkVideoEncodeH265PictureInfoKHR-flags-08316
  If VkVideoEncodeH265CapabilitiesKHR::flags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_H265_CAPABILITY_PREDICTION_WEIGHT_TABLE_GENERATED_BIT_KHR and the slice segment corresponding to any element of pNaluSliceSegmentEntries uses explicit weighted sample prediction, then VkVideoEncodeH265NaluSliceSegmentInfoKHR::pStdSliceSegmentHeader->pWeightTable **must** not be NULL for that element of pNaluSliceSegmentEntries

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265PictureInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_PICTURE_INFO_KHR
- VUID-VkVideoEncodeH265PictureInfoKHR-pNaluSliceSegmentEntries-parameter
  pNaluSliceSegmentEntries **must** be a valid pointer to an array of naluSliceSegmentEntryCount valid VkVideoEncodeH265NaluSliceSegmentInfoKHR structures
- VUID-VkVideoEncodeH265PictureInfoKHR-pStdPictureInfo-parameter
  pStdPictureInfo **must** be a valid pointer to a valid StdVideoEncodeH265PictureInfo value
- VUID-VkVideoEncodeH265PictureInfoKHR-naluSliceSegmentEntryCount-arraylength
  naluSliceSegmentEntryCount **must** be greater than 0

Structure Chaining

Extends the structure

- VkVideoEncodeInfoKHR

The VkVideoEncodeH265NaluSliceSegmentInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265NaluSliceSegmentInfoKHR {
    VkStructureType                                 sType;
    const void*                                     pNext;
    int32_t                                         constantQp;
    const StdVideoEncodeH265SliceSegmentHeader*     pStdSliceSegmentHeader;
} VkVideoEncodeH265NaluSliceSegmentInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `constantQp` is the QP to use for the slice segment if the current rate control mode configured for the video session is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR.
- `pStdSliceSegmentHeader` is a pointer to a `StdVideoEncodeH265SliceSegmentHeader` structure specifying H.265 slice segment header parameters for the slice segment.

Std Slice Segment Header Parameters

The members of the `StdVideoEncodeH265SliceSegmentHeader` structure pointed to by `pStdSliceSegmentHeader` are interpreted as follows:

- `flags.reserved` and `reserved1` are used only for padding purposes and are otherwise ignored;
- if `pWeightTable` is not `NULL`, then it is a pointer to a `StdVideoEncodeH265WeightTable` that is interpreted as follows:
  - `flags.luma_weight_l0_flag`, `flags.chroma_weight_l0_flag`, `flags.luma_weight_l1_flag`, and `flags.chroma_weight_l1_flag` are bitmasks where bit index i corresponds to `luma_weight_l0_flag[i]`, `chroma_weight_l0_flag[i]`, `luma_weight_l1_flag[i]`, and `chroma_weight_l1_flag[i]`, respectively, as defined in section 7.4.7.3 of the ITU-T H.265 Specification;
  - all other members of `StdVideoEncodeH265WeightTable` are interpreted as defined in section 7.4.7.3 of the ITU-T H.265 Specification;
- all other members are interpreted as defined in section 7.4.7.1 of the ITU-T H.265 Specification.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265NaluSliceSegmentInfoKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_NALU_SLICE_SEGMENT_INFO_KHR`
- VUID-VkVideoEncodeH265NaluSliceSegmentInfoKHR-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkVideoEncodeH265NaluSliceSegmentInfoKHR-pStdSliceSegmentHeader-parameter
  `pStdSliceSegmentHeader` **must** be a valid pointer to a valid `StdVideoEncodeH265SliceSegmentHeader` value

The VkVideoEncodeH265DpbSlotInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265DpbSlotInfoKHR {
    VkStructureType                          sType;
    const void*                              pNext;
    const StdVideoEncodeH265ReferenceInfo*   pStdReferenceInfo;
} VkVideoEncodeH265DpbSlotInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pStdReferenceInfo is a pointer to a StdVideoEncodeH265ReferenceInfo structure specifying H.265 reference information.

This structure is specified in the pNext chain of VkVideoEncodeInfoKHR::pSetupReferenceSlot, if not NULL, and the pNext chain of the elements of VkVideoEncodeInfoKHR::pReferenceSlots to specify the codec-specific reference picture information for an H.265 encode operation.

Active Reference Picture Information

When this structure is specified in the pNext chain of the elements of VkVideoEncodeInfoKHR::pReferenceSlots, one element is added to the list of active reference pictures used by the video encode operation for each element of VkVideoEncodeInfoKHR::pReferenceSlots as follows:

- The image subregion used is determined according to the H.265 Encode Picture Data Access section.
- The reference picture is associated with the DPB slot index specified in the slotIndex member of the corresponding element of VkVideoEncodeInfoKHR::pReferenceSlots.
- The reference picture is associated with the H.265 reference information provided in pStdReferenceInfo.

Reconstructed Picture Information

When this structure is specified in the pNext chain of VkVideoEncodeInfoKHR::pSetupReferenceSlot, the information related to the reconstructed picture is defined as follows:

- The image subregion used is determined according to the H.265 Encode Picture Data Access section.
- If reference picture setup is requested, then the reconstructed picture is used to activate the DPB slot with the index specified in VkVideoEncodeInfoKHR::pSetupReferenceSlot->slotIndex.
- The reconstructed picture is associated with the H.265 reference information provided in pStdReferenceInfo.

Std Reference Information

The members of the StdVideoEncodeH265ReferenceInfo structure pointed to by pStdReferenceInfo are interpreted as follows:

- `flags.reserved` is used only for padding purposes and is otherwise ignored;
- `flags.used_for_long_term_reference` is used to indicate whether the picture is marked as "used for long-term reference" as defined in section 8.3.2 of the ITU-T H.265 Specification;
- `flags.unused_for_reference` is used to indicate whether the picture is marked as "unused for reference" as defined in section 8.3.2 of the ITU-T H.265 Specification;
- `pic_type` as defined in section 7.4.3.5 of the ITU-T H.265 Specification;
- `PicOrderCntVal` as defined in section 8.3.1 of the ITU-T H.265 Specification;
- `TemporalId` as defined in section 7.4.2.2 of the ITU-T H.265 Specification.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265DpbSlotInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_DPB_SLOT_INFO_KHR
- VUID-VkVideoEncodeH265DpbSlotInfoKHR-pStdReferenceInfo-parameter
  `pStdReferenceInfo` **must** be a valid pointer to a valid `StdVideoEncodeH265ReferenceInfo` value

Structure Chaining

Extends the structure

- VkVideoReferenceSlotInfoKHR

45.21.12. H.265 Encode Rate Control

Group of Pictures

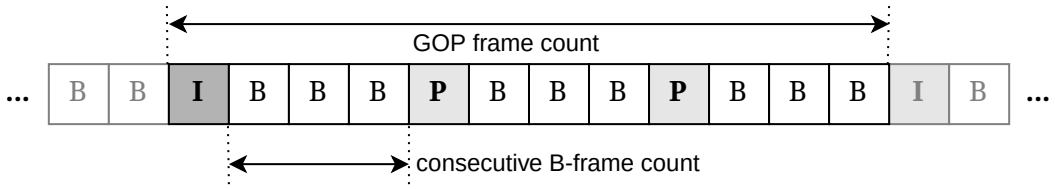
In case of H.265 encoding it is common practice to follow a regular pattern of different picture types in display order when encoding subsequent frames. This pattern is referred to as the *group of pictures* (GOP).

A regular GOP is defined by the following parameters:

- The number of frames in the GOP;
- The number of consecutive B frames between I and/or P frames in display order.

Frame types in a regular GOP follow each other in display order according to the following algorithm:

1. The first frame is always an I or an IDR frame.
2. This is followed by a number of consecutive B frames, as defined above.
3. If the number of frames in the GOP is not reached yet, then the next frame is a P frame and the algorithm continues from step 2.



Text is not SVG - cannot display

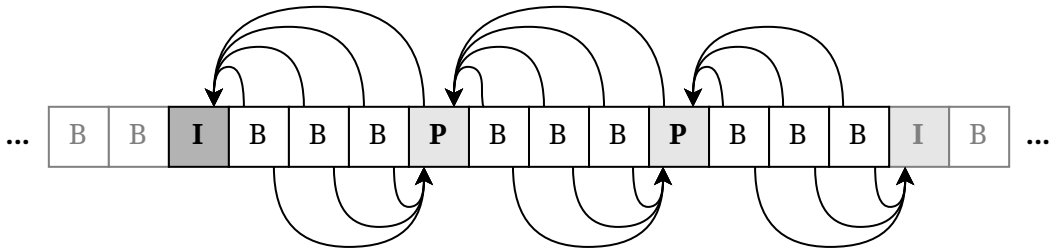
Figure 33. H.265 regular GOP starting with an I frame

When IDR frames are used in a video sequence, the *IDR period* refers to the distance between subsequent IDR frames.

It is also typical for H.265 encoding to use specific reference picture usage patterns across the frames of the GOP. The two most common reference patterns used are as follows:

Flat Reference Pattern

- Each P frame uses the last non-B frame, in display order, as reference.
- Each B frame uses the last non-B frame, in display order, as its forward reference, and uses the next non-B frame, in display order, as its backward reference.

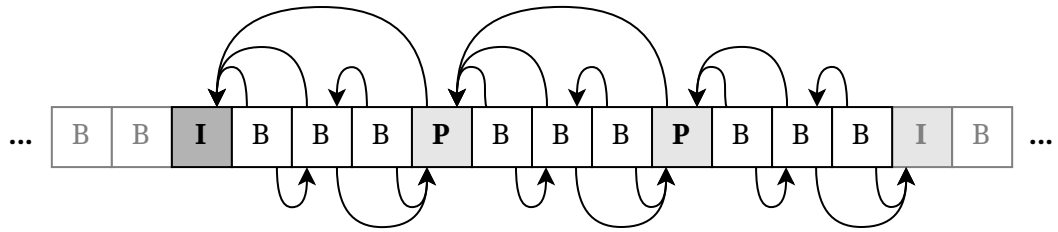


Text is not SVG - cannot display

Figure 34. H.265 flat reference pattern

Dyadic Reference Pattern

- Each P frame uses the last non-B frame, in display order, as reference.
- The following algorithm is applied to the sequence of consecutive B frames between I and/or P frames in display order:
  1. The B frame in the middle of this sequence uses the frame preceding the sequence as its forward reference, and uses the frame following the sequence as its backward reference.
  2. The algorithm is executed recursively for the following frame sequences:
     - The B frames of the original sequence preceding the frame in the middle, if any.
     - The B frames of the original sequence following the frame in the middle, if any.



Text is not SVG - cannot display

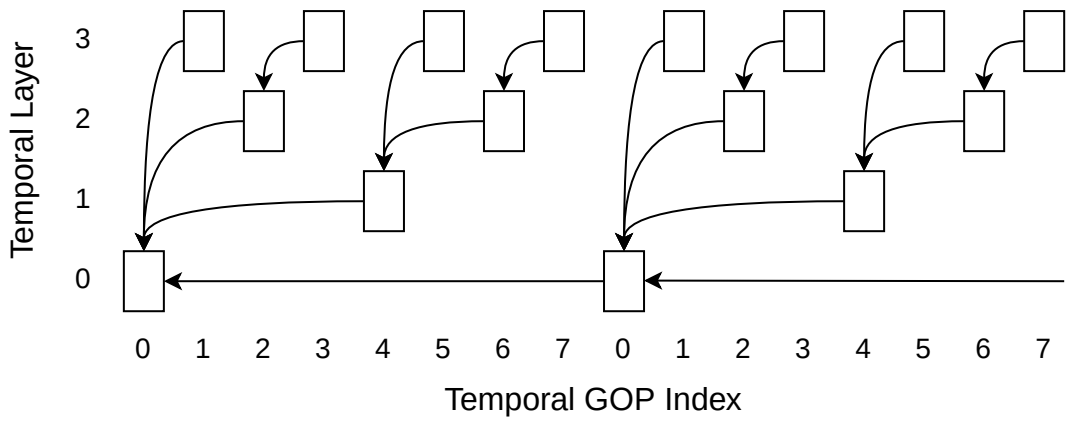
Figure 35. H.265 dyadic reference pattern

The application **can** provide guidance to the implementation's rate control algorithm about the structure of the GOP used by the application. Any such guidance about the GOP and its structure does not mandate that specific GOP structure to be used by the application, as the picture type of individual encoded pictures is still application-controlled. However, any deviation from the provided guidance **may** result in undesired rate control behavior including, but not limited to the implementation not being able to conform to the expected average or target bitrates, or other rate control parameters specified by the application.

When an H.265 encode session is used to encode multiple temporal sub-layers, it is also common practice to follow a regular pattern for the H.265 temporal ID for the encoded pictures in display order when encoding subsequent frames. This pattern is referred to as the *temporal GOP*. The most common temporal layer pattern used is as follows:

Dyadic Temporal Sub-Layer Pattern

- The number of frames in the temporal GOP is 2^{n-1}, where n is the number of temporal sub-layers.
- The i^{th} frame in the temporal GOP uses temporal ID t, if and only if the index of the least significant bit set in i equals n-t-1, except for the first frame, which is the only frame in the temporal GOP using temporal ID zero.
- The i^{th} frame in the temporal GOP uses the r^{th} frame as reference, where r is calculated from i by clearing the least significant bit set in it, except for the first frame in the temporal GOP, which uses the first frame of the previous temporal GOP, if any, as reference.



Text is not SVG - cannot display

Figure 36. H.265 dyadic temporal sub-layer pattern

NOTE	Multi-layer rate control and multi-layer coding are typically used for streaming cases where low latency is expected, hence B pictures with backward prediction are usually not used.

The `VkVideoEncodeH265RateControlInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265RateControlInfoKHR {
    VkStructureType                          sType;
    const void*                              pNext;
    VkVideoEncodeH265RateControlFlagsKHR     flags;
    uint32_t                                 gopFrameCount;
    uint32_t                                 idrPeriod;
    uint32_t                                 consecutiveBFrameCount;
    uint32_t                                 subLayerCount;
} VkVideoEncodeH265RateControlInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkVideoEncodeH265RateControlFlagBitsKHR specifying H.265 rate control flags.
- `gopFrameCount` is the number of frames within a group of pictures (GOP) intended to be used by the application. If it is 0, the rate control algorithm **may** assume an implementation-dependent GOP length. If it is `UINT32_MAX`, the GOP length is treated as infinite.
- `idrPeriod` is the interval, in terms of number of frames, between two IDR frames (see IDR period). If it is 0, the rate control algorithm **may** assume an implementation-dependent IDR period. If it is `UINT32_MAX`, the IDR period is treated as infinite.
- `consecutiveBFrameCount` is the number of consecutive B frames between I and/or P frames within

the GOP.

- `subLayerCount` specifies the number of H.265 sub-layers that the application intends to use.

When an instance of this structure is included in the `pNext` chain of the VkVideoCodingControlInfoKHR structure passed to the vkCmdControlVideoCodingKHR command, and VkVideoCodingControlInfoKHR::`flags` includes VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR, the parameters in this structure are used as guidance for the implementation's rate control algorithm (see Video Coding Control).

If `flags` includes VK_VIDEO_ENCODE_H265_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR, then the rate control state is reset to an initial state to meet HRD compliance requirements. Otherwise the new rate control state **may** be applied without a reset depending on the implementation and the specified rate control parameters.

NOTE

It would be possible to infer the picture type to be used when encoding a frame, on the basis of the values provided for `consecutiveBFrameCount`, `idrPeriod`, and `gopFrameCount`, but this inferred picture type will not be used by implementations to override the picture type provided to the video encode operation.

Valid Usage

- VUID-VkVideoEncodeH265RateControlInfoKHR-flags-08291
  If VkVideoEncodeH265CapabilitiesKHR::`flags`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_H265_CAPABILITY_HRD_COMPLIANCE_BIT_KHR, then `flags` **must** not contain VK_VIDEO_ENCODE_H265_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR
- VUID-VkVideoEncodeH265RateControlInfoKHR-flags-08292
  If `flags` contains VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR or VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR, then it **must** also contain VK_VIDEO_ENCODE_H265_RATE_CONTROL_REGULAR_GOP_BIT_KHR
- VUID-VkVideoEncodeH265RateControlInfoKHR-flags-08293
  If `flags` contains VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR, then it **must** not also contain VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR
- VUID-VkVideoEncodeH265RateControlInfoKHR-flags-08294
  If `flags` contains VK_VIDEO_ENCODE_H265_RATE_CONTROL_REGULAR_GOP_BIT_KHR, then `gopFrameCount` **must** be greater than `0`
- VUID-VkVideoEncodeH265RateControlInfoKHR-idrPeriod-08295
  If `idrPeriod` is not `0`, then it **must** be greater than or equal to `gopFrameCount`

- VUID-VkVideoEncodeH265RateControlInfoKHR-consecutiveBFrameCount-08296
  If `consecutiveBFrameCount` is not `0`, then it **must** be less than `gopFrameCount`

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265RateControlInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_RATE_CONTROL_INFO_KHR
- VUID-VkVideoEncodeH265RateControlInfoKHR-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkVideoEncodeH265RateControlFlagBitsKHR values

Structure Chaining

Extends the structures

- VkVideoBeginCodingInfoKHR
- VkVideoCodingControlInfoKHR

Bits which **can** be set in VkVideoEncodeH265RateControlInfoKHR::`flags`, specifying H.265 rate control flags, are:

```
// Provided by VK_KHR_video_encode_h265
typedef enum VkVideoEncodeH265RateControlFlagBitsKHR {
    VK_VIDEO_ENCODE_H265_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H265_RATE_CONTROL_REGULAR_GOP_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR = 0x00000008,
    VK_VIDEO_ENCODE_H265_RATE_CONTROL_TEMPORAL_SUB_LAYER_PATTERN_DYADIC_BIT_KHR =
0x00000010,
} VkVideoEncodeH265RateControlFlagBitsKHR;
```

- VK_VIDEO_ENCODE_H265_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR specifies that rate control **should** attempt to produce an HRD compliant bitstream, as defined in annex C of the ITU-T H.265 Specification.
- VK_VIDEO_ENCODE_H265_RATE_CONTROL_REGULAR_GOP_BIT_KHR specifies that the application intends to use a regular GOP structure according to the parameters specified in the `gopFrameCount`, `idrPeriod`, and `consecutiveBFrameCount` members of the VkVideoEncodeH265RateControlInfoKHR structure.
- VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR specifies that the application intends to follow a flat reference pattern in the GOP.
- VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR specifies that the application intends to follow a dyadic reference pattern in the GOP.
- VK_VIDEO_ENCODE_H265_RATE_CONTROL_TEMPORAL_SUB_LAYER_PATTERN_DYADIC_BIT_K

HR specifies that the application intends to follow a dyadic temporal sub-layer pattern.

```
// Provided by VK_KHR_video_encode_h265
typedef VkFlags VkVideoEncodeH265RateControlFlagsKHR;
```

VkVideoEncodeH265RateControlFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeH265RateControlFlagBitsKHR.

Rate Control Layers

The VkVideoEncodeH265RateControlLayerInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265RateControlLayerInfoKHR {
    VkStructureType                  sType;
    const void*                      pNext;
    VkBool32                         useMinQp;
    VkVideoEncodeH265QpKHR           minQp;
    VkBool32                         useMaxQp;
    VkVideoEncodeH265QpKHR           maxQp;
    VkBool32                         useMaxFrameSize;
    VkVideoEncodeH265FrameSizeKHR    maxFrameSize;
} VkVideoEncodeH265RateControlLayerInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- useMinQp indicates whether the QP values determined by rate control will be clamped to the lower bounds on the QP values specified in minQp.
- minQp specifies the lower bounds on the QP values, for each picture type, that the implementation's rate control algorithm will use when useMinQp is VK_TRUE.
- useMaxQp indicates whether the QP values determined by rate control will be clamped to the upper bounds on the QP values specified in maxQp.
- maxQp specifies the upper bounds on the QP values, for each picture type, that the implementation's rate control algorithm will use when useMaxQp is VK_TRUE.
- useMaxFrameSize indicates whether the implementation's rate control algorithm **should** use the values specified in maxFrameSize as the upper bounds on the encoded frame size for each picture type.
- maxFrameSize specifies the upper bounds on the encoded frame size, for each picture type, when useMaxFrameSize is VK_TRUE.

When used, the values in minQp and maxQp guarantee that the effective QP values used by the implementation will respect those lower and upper bounds, respectively. However, limiting the range of QP values that the implementation is able to use will also limit the capabilities of the implementation's rate control algorithm to comply to other constraints. In particular, the implementation **may** not be able to comply to the following:

- The average and/or peak bitrate values to be used for the encoded bitstream specified in the `averageBitrate` and `maxBitrate` members of the VkVideoEncodeRateControlLayerInfoKHR structure.
- The upper bounds on the encoded frame size, for each picture type, specified in the `maxFrameSize` member of `VkVideoEncodeH265RateControlLayerInfoKHR`.

NOTE	In general, applications need to configure rate control parameters appropriately in order to be able to get the desired rate control behavior, as described in the Video Encode Rate Control section.

When an instance of this structure is included in the `pNext` chain of a VkVideoEncodeRateControlLayerInfoKHR structure specified in one of the elements of the `pLayers` array member of the VkVideoEncodeRateControlInfoKHR structure passed to the vkCmdControlVideoCodingKHR command, VkVideoCodingControlInfoKHR::`flags` includes VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR, and the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, it specifies the H.265-specific rate control parameters of the rate control layer corresponding to that element of `pLayers`.

Valid Usage

- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-useMinQp-08297
  If `useMinQp` is VK_TRUE, then the `qpI`, `qpP`, and `qpB` members of `minQp` **must** all be between VkVideoEncodeH265CapabilitiesKHR::`minQp` and VkVideoEncodeH265CapabilitiesKHR::`maxQp`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile
- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-useMaxQp-08298
  If `useMaxQp` is VK_TRUE, then the `qpI`, `qpP`, and `qpB` members of `maxQp` **must** all be between VkVideoEncodeH265CapabilitiesKHR::`minQp` and VkVideoEncodeH265CapabilitiesKHR::`maxQp`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile
- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-useMinQp-08299
  If `useMinQp` is VK_TRUE and VkVideoEncodeH265CapabilitiesKHR::`flags`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_H265_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR, then the `qpI`, `qpP`, and `qpB` members of `minQp` **must** all specify the same value
- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-useMaxQp-08300
  If `useMaxQp` is VK_TRUE and VkVideoEncodeH265CapabilitiesKHR::`flags`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_H265_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR, then the `qpI`, `qpP`, and `qpB` members of `maxQp` **must** all specify the same value
- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-useMinQp-08375
  If `useMinQp` and `useMaxQp` are both VK_TRUE, then the `qpI`, `qpP`, and `qpB` members of `minQp` **must** all be less than or equal to the respective members of `maxQp`

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_RATE_CONTROL_LAYER_INFO_KHR`
- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-minQp-parameter
  `minQp` **must** be a valid VkVideoEncodeH265QpKHR structure
- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-maxQp-parameter
  `maxQp` **must** be a valid VkVideoEncodeH265QpKHR structure
- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-maxFrameSize-parameter
  `maxFrameSize` **must** be a valid VkVideoEncodeH265FrameSizeKHR structure

Structure Chaining

Extends the structure

- VkVideoEncodeRateControlLayerInfoKHR

The `VkVideoEncodeH265QpKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265QpKHR {
    int32_t    qpI;
    int32_t    qpP;
    int32_t    qpB;
} VkVideoEncodeH265QpKHR;
```

- `qpI` is the QP to be used for I pictures.
- `qpP` is the QP to be used for P pictures.
- `qpB` is the QP to be used for B pictures.

The `VkVideoEncodeH265FrameSizeKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265FrameSizeKHR {
    uint32_t    frameISize;
    uint32_t    framePSize;
    uint32_t    frameBSize;
} VkVideoEncodeH265FrameSizeKHR;
```

- `frameISize` is the size in bytes to be used for I frames.
- `framePSize` is the size in bytes to be used for P frames.
- `frameBSize` is the size in bytes to be used for B frames.

GOP Remaining Frames

Besides session level rate control configuration, the application **can** specify the number of frames per frame type remaining in the group of pictures (GOP).

The VkVideoEncodeH265GopRemainingFrameInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265GopRemainingFrameInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           useGopRemainingFrames;
    uint32_t           gopRemainingI;
    uint32_t           gopRemainingP;
    uint32_t           gopRemainingB;
} VkVideoEncodeH265GopRemainingFrameInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- useGopRemainingFrames indicates whether the implementation's rate control algorithm **should** use the values specified in gopRemainingI, gopRemainingP, and gopRemainingB. If useGopRemainingFrames is VK_FALSE, then the values of gopRemainingI, gopRemainingP, and gopRemainingB are ignored.
- gopRemainingI specifies the number of I frames the implementation's rate control algorithm **should** assume to be remaining in the GOP prior to executing the video encode operation.
- gopRemainingP specifies the number of P frames the implementation's rate control algorithm **should** assume to be remaining in the GOP prior to executing the video encode operation.
- gopRemainingB specifies the number of B frames the implementation's rate control algorithm **should** assume to be remaining in the GOP prior to executing the video encode operation.

Setting useGopRemainingFrames to VK_TRUE and including this structure in the pNext chain of VkVideoBeginCodingInfoKHR is only mandatory if the VkVideoEncodeH265CapabilitiesKHR ::requiresGopRemainingFrames reported for the used video profile is VK_TRUE. However, implementations **may** use these remaining frame counts, when specified, even when it is not required. In particular, when the application does not use a regular GOP structure, these values **may** provide additional guidance for the implementation's rate control algorithm.

The VkVideoEncodeH265CapabilitiesKHR::prefersGopRemainingFrames capability is also used to indicate that the implementation's rate control algorithm **may** operate more accurately if the application specifies the remaining frame counts using this structure.

As with other rate control guidance values, if the effective order and number of frames encoded by the application are not in line with the remaining frame counts specified in this structure at any given point, then the behavior of the implementation's rate control algorithm **may** deviate from the one expected by the application.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265GopRemainingFrameInfoKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_GOP_REMAINING_FRAME_INFO_KHR`

Structure Chaining

Extends the structure

- VkVideoBeginCodingInfoKHR

H.265 QP Delta Maps

Quantization delta maps used with an H.265 encode profile are referred to as *QP delta maps* and their texels contain integer values representing QP delta values that are applied in the process of determining the quantization parameters of the encoded picture.

Accordingly, H.265 QP delta maps always have single channel integer formats, as reported in VkVideoFormatPropertiesKHR::`format`.

When the rate control mode is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, the QP delta values are added to the per slice segment constant QP values that, in effect, enable the application to explicitly control the used QP values at the granularity of the used quantization map texel size.

For all other rate control modes, the QP delta values **can** be used to offset the QP values that the rate control algorithm would otherwise produce.

H.265 Encode Quantization

Performing H.265 encode operations involves the process of assigning QP values to individual H.265 coding units. This process depends on the used rate control mode, as well as other encode and rate control parameters, as described below:

- If the configured rate control mode is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR, then the QP value is initialized by the implementation-specific default rate control algorithm.
  - If the video encode operation is issued with a quantization delta map, the QP delta value corresponding to the coding unit, as fetched from the quantization map, is added to the previously determined QP value. If the fetched QP delta value falls outside the supported QP delta value range reported in the `minQpDelta` and `maxQpDelta` members of VkVideoEncodeH265QuantizationMapCapabilitiesKHR, then the QP value used for the coding unit becomes undefined.
- If the configured rate control mode is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, then the QP value is initialized from the constant QP value specified for the H.265 slice segment the coding unit is

part of.

  - If the video encode operation is issued with a quantization delta map, the QP delta value corresponding to the coding unit, as fetched from the quantization map, is added to the previously determined QP value. If the fetched QP delta value falls outside the supported QP delta value range reported in the `minQpDelta` and `maxQpDelta` members of VkVideoEncodeH265QuantizationMapCapabilitiesKHR, then the QP value used for the coding unit becomes undefined.

- If the configured rate control mode is not VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR or VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, then the QP value is initialized by the corresponding rate control algorithm.
  - If the video encode operation is issued with a quantization delta map, the QP delta value corresponding to the coding unit, as fetched from the quantization map, is added to the previously determined QP value. If the fetched QP delta value falls outside the supported QP delta value range reported in the `minQpDelta` and `maxQpDelta` members of VkVideoEncodeH265QuantizationMapCapabilitiesKHR, then the QP value used for the coding unit becomes undefined.
  - If the video encode operation is issued with an emphasis map, the rate control will adjust the QP value based on the emphasis value corresponding to the coding unit, as fetched from the quantization map, according to the following equation:

    $QP_{new} = f(QP_{prev},e)$

    Where QP_{new} is the resulting QP value, QP_{prev} is the previously determined QP value, e is the emphasis value corresponding to the coding unit, and f is an implementation-defined function for which the following implication is true:

    $e_1 < e_2 \Rightarrow f(QP,e_1) \geq f(QP,e_2)$

    > **NOTE** This means that lower emphasis values will result in higher QP values, whereas higher emphasis values will result in lower QP values, but the function is not strictly decreasing with respect to the input emphasis value for a given input QP value.

  - If clamping to minimum QP values is enabled in the applied rate control layer, then the QP value is clamped to the corresponding minimum QP value.
  - If clamping to maximum QP values is enabled in the applied rate control layer, then the QP value is clamped to the corresponding maximum QP value.

- If VK_VIDEO_ENCODE_H265_CAPABILITY_CU_QP_DIFF_WRAPAROUND_BIT_KHR is not supported, then the determined QP value is clamped in such a way that the `CuQpDeltaVal` value of the encoded coding unit complies to the modified version of equation 8-283 of the ITU-T H.265 Specification.

  > **NOTE** The effect of this is that the maximum QP difference across subsequent coding units is limited to the $[-(26 + QpBdOffset_Y / 2), 25 + QpBdOffset_Y / 2]$ range and only has an observable change in behavior when the video encode operation is

issued with a QP delta map.

- In all cases, the final QP value is clamped to the QP value range supported by the video profile, as reported in the `minQp` and `maxQp` members of VkVideoEncodeH265CapabilitiesKHR.

45.21.13. H.265 Encode Requirements

This section describes the **required** H.265 encoding capabilities for physical devices that have at least one queue family that supports the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, as returned by vkGetPhysicalDeviceQueueFamilyProperties2 in VkQueueFamilyVideoPropertiesKHR ::`videoCodecOperations`.

Table 62. Required Video Std Header Versions

Video Std Header Name	Version
`vulkan_video_codec_h265std_encode`	1.0.0

Table 63. Required Video Capabilities

Video Capability	Requirement	Requirement Type[1]
VkVideoCapabilitiesKHR		
`flags`	-	min
`minBitstreamBufferOffsetAlignment`	4096	max
`minBitstreamBufferSizeAlignment`	4096	max
`pictureAccessGranularity`	(64,64)	max
`minCodedExtent`	-	max
`maxCodedExtent`	-	min
`maxDpbSlots`	0	min
`maxActiveReferencePictures`	0	min
VkVideoEncodeCapabilitiesKHR		
`flags`	-	min
`rateControlModes`	VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR [4]	min
`maxBitrate`	128000	min
`maxQualityLevels`	1	min
`encodeInputPictureGranularity`	(64,64)	max

Video Capability	Requirement	Requirement Type[1]
supportedEncodeFeedbackFlags	VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR + VK_VIDEO_ENCODE_FEEDBACK_AVERAGE_QUANTIZATION_BIT_KHR [6] + VK_VIDEO_ENCODE_FEEDBACK_INTRA_PIXELS_BIT_KHR [6] + VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR [6] + VK_VIDEO_ENCODE_FEEDBACK_PICTURE_PARTITION_COUNT_BIT_KHR [6]	min
VkVideoEncodeFeedback2CapabilitiesKHR		
maxPerPartitionFeedbackEntries	-	min
supportedPerPartitionEncodeFeedbackFlags	VK_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_STATUS_BIT_KHR [7] + VK_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR [7] + VK_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR [7]	min
VkVideoEncodeH265CapabilitiesKHR		
flags	-	min
maxLevelIdc	STD_VIDEO_H265_LEVEL_IDC_1_0	min

Video Capability	Requirement	Requirement Type[1]
maxSliceSegmentCount	1	min
maxTiles	(1,1)	min
ctbSizes	at least one bit set	implementation-dependent
transformBlockSizes	at least one bit set	implementation-dependent
maxPPictureL0ReferenceCount	0	min
maxBPictureL0ReferenceCount	0	min
maxL1ReferenceCount	0	min
maxSubLayerCount	1	min
expectDyadicTemporalSubLayerPattern	-	implementation-dependent
minQp	-	max
maxQp	-	min
prefersGopRemainingFrames	-	implementation-dependent
requiresGopRemainingFrames	-	implementation-dependent
stdSyntaxFlags	-	min
VkVideoEncodeQuantizationMapCapabilitiesKHR		
maxQuantizationMapExtent	-[2]	min
VkVideoEncodeH265QuantizationMapCapabilitiesKHR		
minQpDelta	-[3]	max
maxQpDelta	-[3]	min
VkVideoEncodeIntraRefreshCapabilitiesKHR		
intraRefreshModes	0	min
maxIntraRefreshCycleDuration	2[5]	min
maxIntraRefreshActiveReferencePictures	1[5]	min
partitionIndependentIntraRefreshRegions	-	implementation-dependent
nonRectangularIntraRefreshRegions	-	implementation-dependent

1

The **Requirement Type** column specifies the requirement is either the minimum value all

implementations **must** support, the maximum value all implementations **must** support, or the exact value all implementations **must** support. For bitmasks a minimum value is the least bits all implementations **must** set, but they **may** have additional bits set beyond this minimum.

2

If VkVideoEncodeCapabilitiesKHR::`flags` includes VK_VIDEO_ENCODE_CAPABILITY_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_VIDEO_ENCODE_CAPABILITY_EMPHASIS_MAP_BIT_KHR, then the `width` and `height` members of `maxQuantizationMapExtent` **must** be greater than zero.

3

If VkVideoEncodeCapabilitiesKHR::`flags` includes VK_VIDEO_ENCODE_CAPABILITY_QUANTIZATION_DELTA_MAP_BIT_KHR, then `maxQpDelta` **must** be greater than `minQpDelta`.

4

If the `videoMaintenance2` feature is supported.

5

If VkVideoEncodeIntraRefreshCapabilitiesKHR::`intraRefreshModes` is not zero.

6

If the `videoEncodeFeedback2` feature is supported.

7

If `maxPerPartitionFeedbackEntries` is greater than zero.

45.22. AV1 Encode Operations

Video encode operations using an AV1 encode profile **can** be used to encode elementary video stream sequences compliant with the AV1 Specification.

> **NOTE** Refer to the Preamble for information on how the Khronos Intellectual Property Rights Policy relates to normative references to external materials not created by Khronos.

This process is performed according to the video encode operation steps with the codec-specific semantics defined in section 7 of the AV1 Specification:

- Syntax elements, derived values, and other parameters are applied from the following structures:
  - The `StdVideoAV1SequenceHeader` structure, the optional `StdVideoEncodeAV1DecoderModelInfo` structure, and the optional array of `StdVideoEncodeAV1OperatingPointInfo` structures stored in the bound video session parameters object specifying the active sequence header.
  - The `StdVideoEncodeAV1PictureInfo` structure specifying the AV1 picture information.
  - The `StdVideoEncodeAV1ReferenceInfo` structures specifying the AV1 reference information corresponding to the optional reconstructed picture and any active reference pictures.

    - The encoded bitstream data is written to the destination video bitstream buffer range as defined in the AV1 Encode Bitstream Data Access section.
    - Picture data in the video picture resources corresponding to the used encode input picture, active reference pictures, and optional reconstructed picture is accessed as defined in the AV1 Encode Picture Data Access section.
- The decision on reference picture setup is made according to the parameters specified in the AV1 picture information.

If the parameters adhere to the syntactic and semantic requirements defined in the corresponding sections of the AV1 Specification, as described above, and the DPB slots associated with the active reference pictures all refer to valid picture references, then the video encode operation will complete successfully. Otherwise, the video encode operation **may** complete unsuccessfully.

45.22.1. AV1 Encode Parameter Overrides

Implementations **may** override, unless otherwise specified, any of the AV1 encode parameters specified in the following Video Std structures:

- `StdVideoAV1SequenceHeader`
- `StdVideoEncodeAV1DecoderModelInfo`
- `StdVideoEncodeAV1OperatingPointInfo`
- `StdVideoEncodeAV1PictureInfo`
- `StdVideoEncodeAV1ReferenceInfo`

All such AV1 encode parameter overrides **must** fulfill the conditions defined in the Video Encode Parameter Overrides section.

In addition, implementations **must** not override any of the following AV1 encode parameters:

- the following parameters specified in `StdVideoAV1SequenceHeader`:
    - `flags.still_picture`
    - `flags.enable_order_hint`
    - `flags.frame_id_numbers_present_flag`
    - `flags.film_grain_params_present`
    - `flags.timing_info_present_flag`
    - `flags.initial_display_delay_present_flag`
    - `delta_frame_id_length_minus_2`
    - `additional_frame_id_length_minus_1`
    - `order_hint_bits_minus_1`
- the following parameters specified in the `StdVideoAV1ColorConfig` structure pointed to by `StdVideoAV1SequenceHeader`::`pColorConfig`:
    - `flags.mono_chrome`

    - flags.color_range
    - BitDepth
    - subsampling_x
    - subsampling_y
    - color_primaries
    - transfer_characteristics
    - matrix_coefficients
    - chroma_sample_position
- the following parameters specified in the StdVideoAV1TimingInfo structure pointed to by StdVideoAV1SequenceHeader::pTimingInfo:
    - flags.equal_picture_interval
    - num_units_in_display_tick
    - time_scale
    - num_ticks_per_picture_minus_1
- the parameters specified in StdVideoEncodeAV1DecoderModelInfo
- the parameters specified in StdVideoEncodeAV1OperatingPointInfo
- the following parameters specified in StdVideoEncodeAV1PictureInfo:
    - flags.show_frame
    - flags.showable_frame
    - frame_type
    - frame_presentation_time
    - current_frame_id
    - order_hint
    - refresh_frame_flags
    - render_width_minus_1
    - render_height_minus_1
    - ref_order_hint
    - ref_frame_idx
    - delta_frame_id_minus_1
- the following parameters specified in the StdVideoEncodeAV1ExtensionHeader structure pointed to by StdVideoEncodeAV1PictureInfo::pExtensionHeader when VkVideoEncodeAV1PictureInfoKHR::generateObuExtensionHeader is set to VK_TRUE:
    - temporal_id
    - spatial_id

If VkVideoEncodeAV1PictureInfoKHR::primaryReferenceCdfOnly is set to VK_TRUE for a video encode

operation, the implementation will not override StdVideoEncodeAV1PictureInfo::primary_ref_frame.

NOTE

Implementations supporting the VK_VIDEO_ENCODE_AV1_STD_PRIMARY_REF_FRAME_BIT_KHR AV1 syntax element capability reported in VkVideoEncodeAV1CapabilitiesKHR::stdSyntaxFlags also guarantee the use of the application-specified value for StdVideoEncodeAV1PictureInfo::primary_ref_frame. While support for VK_VIDEO_ENCODE_AV1_STD_PRIMARY_REF_FRAME_BIT_KHR guarantees that the implementation will not override the application-specified primary_ref_frame, it does not mandate the implementation to use the reference picture indicated by primary_ref_frame for sample prediction as implementations can always decide to use only a subset of the application-specified reference pictures for sample prediction. This means that implementations supporting VK_VIDEO_ENCODE_AV1_STD_PRIMARY_REF_FRAME_BIT_KHR may end up using the reference picture indicated by primary_ref_frame only for CDF data reference even if the application did not set VkVideoEncodeAV1PictureInfoKHR::primaryReferenceCdfOnly to VK_TRUE.

If VkVideoEncodeAV1CapabilitiesKHR::codedPictureAlignment is not equal to {8,8} for the used video profile, implementations will override the coded picture's resolution and parameters related to the width and height in the following manner:

- Let w and h be the codedExtent.width and codedExtent.height of the VkVideoPictureResourceInfoKHR structure corresponding to the encode input picture, rounded up to the nearest integer multiple of 8.
- Let aW and aH be w and h rounded up to the nearest integer multiple of codedPictureAlignment.width and codedPictureAlignment.height respectively.
- If w equals aW, no override will occur. Otherwise the coded width will be aW.
- If h equals aH, no override will occur. Otherwise the coded height will be aH.

NOTE

The AV1 specification codes all resolutions to an 8x8 alignment, but supports unaligned resolutions through implicit cropping. Thus, if the original coded extent, aligned to 8x8, meets the implementation required alignment, no override needs to occur. Otherwise, the implementation cannot code the requested coded extent, so the final resolution in the bitstream is overridden to be aligned to the implementation required alignment.

For example, consider an implementation that is only able to output bitstreams that are 16x16 aligned (as indicated by VkVideoEncodeAV1CapabilitiesKHR::codedPictureAlignment). If an application requests the coded extent to be 1920x1080, the resulting bitstream will be 1920x1088. On the other hand, a request of 1920x1082 will result in no override, since the 8x8 alignment of this resolution (1920x1088) is 16x16 aligned.

In case of a video session parameters object created with VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR, the following AV1 sequence header parameters **may** be overridden by the implementation according to

the quantization map texel size the video session parameters object was created with:

- `StdVideoAV1SequenceHeader`::`flags.use_128x128_superblock`

This **may** be necessary to change the AV1 superblock size used during encoding to be compatible with the used quantization map texel size.

In case of AV1 encode parameters stored in video session parameters objects, applications need to use the vkGetEncodedVideoSessionParametersKHR command to determine whether any implementation overrides happened. If the query indicates that implementation overrides were applied, then the application needs to retrieve and use the encoded AV1 sequence header in the bitstream in order to be able to produce a compliant AV1 video bitstream using the AV1 encode parameters stored in the video session parameters object.

In case of any AV1 encode parameters stored in the encoded bitstream produced by video encode operations, if the implementation supports the VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR video encode feedback query flag, the application **can** use such queries to retrieve feedback about whether any implementation overrides have been applied to those AV1 encode parameters.

45.22.2. AV1 Encode Bitstream Data Access

Each video encode operation writes either:

- A single OBU with `obu_type` equal to `OBU_FRAME` comprising of the frame header and tile data of the encoded picture, or
- An OBU with `obu_type` equal to `OBU_FRAME_HEADER` encapsulating the frame header of the encoded picture, followed by one or more OBUs with `obu_type` equal to `OBU_TILE_GROUP` comprising of the tile data of the encoded picture.

In addition, if VkVideoEncodeAV1PictureInfoKHR::`generateObuExtensionHeader` is set to VK_TRUE for the video encode operation, then OBU extension headers are included in the generated bitstream as defined in sections 5.3.1, 5.3.2, and 5.3.3 of the AV1 Specification.

45.22.3. AV1 Encode Picture Data Access

Accesses to image data within a video picture resource happen at the granularity indicated by VkVideoCapabilitiesKHR::`pictureAccessGranularity`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile. Accordingly, the complete image subregion of an encode input picture, reference picture, or reconstructed picture accessed by video coding operations using an AV1 encode profile is defined as the set of texels within the coordinate range:

([0,`endX`),[0,`endY`))

Where:

- `endX` equals `codedExtent.width` rounded up to the nearest integer multiple of

pictureAccessGranularity.width and clamped to the width of the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure;

- endY equals codedExtent.height rounded up to the nearest integer multiple of pictureAccessGranularity.height and clamped to the height of the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure;

Where codedExtent is the member of the VkVideoPictureResourceInfoKHR structure corresponding to the picture.

In case of video encode operations using an AV1 encode profile, any access to a picture at the coordinates (x,y), as defined by the AV1 Specification, is an access to the image subresource referred to by the corresponding VkVideoPictureResourceInfoKHR structure at the texel coordinates (x,y).

Implementations **may** choose not to access some or all texels within particular reference pictures available to a video encode operation (e.g. due to video encode parameter overrides restricting the effective set of used reference pictures, or if the encoding algorithm chooses not to use certain subregions of the reference picture data for sample prediction).

45.22.4. AV1 Reference Names and Semantics

Individual reference frames used in the encoding process have different semantics, as defined in section 6.10.24 of the AV1 Specification. The AV1 semantics associated with a reference picture is indicated by the corresponding enumeration constant defined in the Video Std enumeration type StdVideoAV1ReferenceName:

- STD_VIDEO_AV1_REFERENCE_NAME_INTRA_FRAME identifies the reference used for intra coding (INTRA_FRAME), as defined in sections 2 and 7.11.2 of the AV1 Specification.
- All other enumeration constants refer to backward or forward references used for inter coding, as defined in sections 2 and 7.11.3 of the AV1 Specification:
  - STD_VIDEO_AV1_REFERENCE_NAME_LAST_FRAME identifies the LAST_FRAME reference
  - STD_VIDEO_AV1_REFERENCE_NAME_LAST2_FRAME identifies the LAST2_FRAME reference
  - STD_VIDEO_AV1_REFERENCE_NAME_LAST3_FRAME identifies the LAST3_FRAME reference
  - STD_VIDEO_AV1_REFERENCE_NAME_GOLDEN_FRAME identifies the GOLDEN_FRAME reference
  - STD_VIDEO_AV1_REFERENCE_NAME_BWDREF_FRAME identifies the BWDREF_FRAME reference
  - STD_VIDEO_AV1_REFERENCE_NAME_ALTREF2_FRAME identifies the ALTREF2_FRAME reference
  - STD_VIDEO_AV1_REFERENCE_NAME_ALTREF_FRAME identifies the ALTREF_FRAME reference

These enumeration constants are not directly used in any APIs but are used to indirectly index into certain Video Std and Vulkan API parameter arrays.

45.22.5. AV1 Prediction Modes

AV1 encoding supports multiple types of prediction modes, as described in section 6.10.24 of the AV1 Specification.

Possible AV1 encode prediction modes are as follows:

```
// Provided by VK_KHR_video_encode_av1
typedef enum VkVideoEncodeAV1PredictionModeKHR {
    VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_INTRA_ONLY_KHR = 0,
    VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_SINGLE_REFERENCE_KHR = 1,
    VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_UNIDIRECTIONAL_COMPOUND_KHR = 2,
    VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_BIDIRECTIONAL_COMPOUND_KHR = 3,
} VkVideoEncodeAV1PredictionModeKHR;
```

- VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_INTRA_ONLY_KHR specifies the use of *intra-only prediction mode*, used when encoding AV1 frames of type STD_VIDEO_AV1_FRAME_TYPE_KEY or STD_VIDEO_AV1_FRAME_TYPE_INTRA_ONLY.
- VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_SINGLE_REFERENCE_KHR specifies the use of *single reference prediction mode*, used when encoding AV1 frames of type STD_VIDEO_AV1_FRAME_TYPE_INTER or STD_VIDEO_AV1_FRAME_TYPE_SWITCH with reference_select, as defined in section 6.8.23 of the AV1 Specification, equal to 0. When using this prediction mode, the application **must** specify a reference picture for at least one AV1 reference name in VkVideoEncodeAV1PictureInfoKHR::referenceNameSlotIndices that is supported by the implementation, as reported in VkVideoEncodeAV1CapabilitiesKHR::singleReferenceNameMask.
- VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_UNIDIRECTIONAL_COMPOUND_KHR specifies the use of *unidirectional compound prediction mode*, used when encoding AV1 frames of type STD_VIDEO_AV1_FRAME_TYPE_INTER or STD_VIDEO_AV1_FRAME_TYPE_SWITCH with reference_select, as defined in section 6.8.23 of the AV1 Specification, equal to 1, and both reference names used for prediction are from the same reference frame group, as defined in section 6.10.24 of the AV1 Specification. When using this prediction mode, the application **must** specify a reference picture for at least two AV1 reference names in VkVideoEncodeAV1PictureInfoKHR::referenceNameSlotIndices that is supported by the implementation, as reported in VkVideoEncodeAV1CapabilitiesKHR::unidirectionalCompoundReferenceNameMask, where those two reference names are one of the allowed pairs of reference names, as defined in section 5.11.25 of the AV1 Specification, listed below:
  - LAST_FRAME and LAST2_FRAME,
  - LAST_FRAME and LAST3_FRAME,
  - LAST_FRAME and GOLDEN_FRAME, or
  - BWDREF_FRAME and ALTREF_FRAME.
- VK_VIDEO_ENCODE_AV1_PREDICTION_MODE_BIDIRECTIONAL_COMPOUND_KHR specifies the use of *bidirectional compound prediction mode*, used when encoding AV1 frames of type STD_VIDEO_AV1_FRAME_TYPE_INTER or STD_VIDEO_AV1_FRAME_TYPE_SWITCH with reference_select, as defined in section 6.8.23 of the AV1 Specification, equal to 1, and the two reference names used for prediction are from different reference frame groups, as defined in section 6.10.24 of the AV1 Specification. When using this prediction mode, the application **must** specify a reference picture for at least one AV1 reference name from each reference frame group in VkVideoEncodeAV1PictureInfoKHR::referenceNameSlotIndices that is supported by the implementation, as reported in VkVideoEncodeAV1CapabilitiesKHR::bidirectionalCompoundReferenceNameMask.

The effective prediction mode used to encode individual AV1 mode info blocks **may** use simpler prediction modes than the one set by the application for the frame, as allowed by the AV1 Specification, in particular:

- Frames encoded with single reference prediction mode **may** contain mode info blocks encoded with intra-only prediction mode.
- Frames encoded with unidirectional compound prediction mode **may** contain mode info blocks encoded with intra-only or single reference prediction mode.
- Frames encoded with bidirectional compound prediction mode **may** contain mode info blocks encoded with intra-only, single reference, or unidirectional compound prediction mode.

45.22.6. AV1 Frame and Tile

AV1 frames are partitioned into tiles, as defined in section 2 of the AV1 Specification.

For the purposes of this specification, the AV1 tiles comprising a frame are referred to as the *picture partitions* of the frame.

45.22.7. AV1 Coding Blocks

AV1 encode supports two types of coding blocks, as defined in section 2 of the AV1 Specification:

- *Superblock*.
- *Mode info block*.

An AV1 mode info block is considered *skipped* when the value of `skip` for the mode info block is set to 1, as defined in section 5.11.11 of the AV1 Specification.

45.22.8. AV1 Encode Profile

A video profile supporting AV1 video encode operations is specified by setting VkVideoProfileInfoKHR::`videoCodecOperation` to VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR and adding a `VkVideoEncodeAV1ProfileInfoKHR` structure to the VkVideoProfileInfoKHR::`pNext` chain.

The `VkVideoEncodeAV1ProfileInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_av1
typedef struct VkVideoEncodeAV1ProfileInfoKHR {
    VkStructureType       sType;
    const void*           pNext;
    StdVideoAV1Profile    stdProfile;
} VkVideoEncodeAV1ProfileInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stdProfile` is a `StdVideoAV1Profile` value specifying the AV1 codec profile, as defined in section

A.2 of the AV1 Specification.

Valid Usage (Implicit)

- VUID-VkVideoEncodeAV1ProfileInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_PROFILE_INFO_KHR

Structure Chaining

Extends the structures

- VkQueryPoolCreateInfo
- VkVideoProfileInfoKHR

45.22.9. AV1 Encode Capabilities

When calling vkGetPhysicalDeviceVideoCapabilitiesKHR to query the capabilities for an AV1 encode profile, the VkVideoCapabilitiesKHR::`pNext` chain **must** include a `VkVideoEncodeAV1CapabilitiesKHR` structure that will be filled with the profile-specific capabilities.

The `VkVideoEncodeAV1CapabilitiesKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_av1
typedef struct VkVideoEncodeAV1CapabilitiesKHR {
    VkStructureType                          sType;
    void*                                    pNext;
    VkVideoEncodeAV1CapabilityFlagsKHR       flags;
    StdVideoAV1Level                         maxLevel;
    VkExtent2D                               codedPictureAlignment;
    VkExtent2D                               maxTiles;
    VkExtent2D                               minTileSize;
    VkExtent2D                               maxTileSize;
    VkVideoEncodeAV1SuperblockSizeFlagsKHR   superblockSizes;
    uint32_t                                 maxSingleReferenceCount;
    uint32_t                                 singleReferenceNameMask;
    uint32_t                                 maxUnidirectionalCompoundReferenceCount;
    uint32_t
maxUnidirectionalCompoundGroup1ReferenceCount;
    uint32_t                                 unidirectionalCompoundReferenceNameMask;
    uint32_t                                 maxBidirectionalCompoundReferenceCount;
    uint32_t
maxBidirectionalCompoundGroup1ReferenceCount;
    uint32_t
maxBidirectionalCompoundGroup2ReferenceCount;
    uint32_t                                 bidirectionalCompoundReferenceNameMask;
    uint32_t                                 maxTemporalLayerCount;
    uint32_t                                 maxSpatialLayerCount;
    uint32_t                                 maxOperatingPoints;
    uint32_t                                 minQIndex;
    uint32_t                                 maxQIndex;
    VkBool32                                 prefersGopRemainingFrames;
    VkBool32                                 requiresGopRemainingFrames;
    VkVideoEncodeAV1StdFlagsKHR              stdSyntaxFlags;
} VkVideoEncodeAV1CapabilitiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkVideoEncodeAV1CapabilityFlagBitsKHR indicating supported AV1 encoding capabilities.
- `maxLevel` is a `StdVideoAV1Level` value indicating the maximum AV1 level supported by the profile, as defined in section A.3 of the AV1 Specification.
- `codedPictureAlignment` indicates the alignment at which the implementation will code pictures. This capability does not impose any valid usage constraints on the application. However, depending on the `codedExtent` of the encode input picture resource, this capability **may** result in a change of the resolution of the encoded picture, as described in more detail below.
- `maxTiles` indicates the maximum number of AV1 tile columns and rows the implementation supports.
- `minTileSize` indicates the minimum extent of individual AV1 tiles the implementation supports.

- `maxTileSize` indicates the maximum extent of individual AV1 tiles the implementation supports.
- `superblockSizes` is a bitmask of VkVideoEncodeAV1SuperblockSizeFlagBitsKHR values indicating the supported AV1 superblock sizes.
- `maxSingleReferenceCount` indicates the maximum number of reference pictures the implementation supports when using single reference prediction mode.
- `singleReferenceNameMask` is a bitmask of supported AV1 reference names when using single reference prediction mode.
- `maxUnidirectionalCompoundReferenceCount` indicates the maximum number of reference pictures the implementation supports when using unidirectional compound prediction mode.
- `maxUnidirectionalCompoundGroup1ReferenceCount` indicates the maximum number of reference pictures the implementation supports when using unidirectional compound prediction mode from reference frame group 1, as defined in section 6.10.24 of the AV1 Specification.
- `unidirectionalCompoundReferenceNameMask` is a bitmask of supported AV1 reference names when using unidirectional compound prediction mode.
- `maxBidirectionalCompoundReferenceCount` indicates the maximum number of reference pictures the implementation supports when using bidirectional compound prediction mode.
- `maxBidirectionalCompoundGroup1ReferenceCount` indicates the maximum number of reference pictures the implementation supports when using bidirectional compound prediction mode from reference frame group 1, as defined in section 6.10.24 of the AV1 Specification.
- `maxBidirectionalCompoundGroup2ReferenceCount` indicates the maximum number of reference pictures the implementation supports when using bidirectional compound prediction mode from reference frame group 2, as defined in section 6.10.24 of the AV1 Specification.
- `bidirectionalCompoundReferenceNameMask` is a bitmask of supported AV1 reference names when using bidirectional compound prediction mode.
- `maxTemporalLayerCount` indicates the maximum number of AV1 temporal layers supported by the implementation.
- `maxSpatialLayerCount` indicates the maximum number of AV1 spatial layers supported by the implementation.
- `maxOperatingPoints` indicates the maximum number of AV1 operating points supported by the implementation.
- `minQIndex` indicates the minimum quantizer index value supported.
- `maxQIndex` indicates the maximum quantizer index value supported.
- `prefersGopRemainingFrames` indicates that the implementation's rate control algorithm prefers the application to specify the number of frames in each AV1 rate control group remaining in the current group of pictures when beginning a video coding scope.
- `requiresGopRemainingFrames` indicates that the implementation's rate control algorithm requires the application to specify the number of frames in each AV1 rate control group remaining in the current group of pictures when beginning a video coding scope.
- `stdSyntaxFlags` is a bitmask of VkVideoEncodeAV1StdFlagBitsKHR indicating capabilities related to AV1 syntax elements.

singleReferenceNameMask, unidirectionalCompoundReferenceNameMask, and bidirectionalCompoundReferenceNameMask are encoded such that when bit index i is set, it indicates support for the AV1 reference name STD_VIDEO_AV1_REFERENCE_NAME_LAST_FRAME + i.

NOTE These masks indicate which elements of the referenceNameSlotIndices member of VkVideoEncodeAV1PictureInfoKHR are supported to be used by the implementation. It is important to note that both the bits of these masks and the elements of referenceNameSlotIndices are indexed such that the first value specifies the support bit and DPB slot index, respectively, for the AV1 reference name STD_VIDEO_AV1_REFERENCE_NAME_LAST_FRAME (i.e. there is no bit or element for STD_VIDEO_AV1_REFERENCE_NAME_INTRA_FRAME).

codedPictureAlignment provides information about implementation limitations to encode arbitrary resolutions. In particular, some implementations **may** not be able to generate bitstreams aligned to the requirements of the AV1 Specification (8x8). In such cases, the implementation **may** override the width and height of the bitstream, in order to produce a bitstream compliant to the AV1 Specification. If such an override occurs, the encoded resolution of the coded picture is enlarged, with the texel values used for the texel coordinates outside of the bounds of the codedExtent of the encode input picture resource being first governed by the rules regarding the encode input picture granularity. Any texel values outside of the region described by the encode input picture granularity are implementation-defined. Implementations **should** use well-defined values to minimize impact on the produced encoded content.

NOTE This capability does not impose additional application requirements. However, these overrides change the effective resolution of the bitstream and add padding pixels. Applications sensitive to such overrides **can** use this capability and the corresponding override behavior to compute the cropping needed to reproduce the original input of the encoding and transmit it in a side channel (i.e. by using cropping fields available in a container). Additionally, applications **can** explicitly consider this alignment in their coded extent, to avoid implementation-defined texel values being included in the encoded content.

Valid Usage (Implicit)

- VUID-VkVideoEncodeAV1CapabilitiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_CAPABILITIES_KHR

Structure Chaining

Extends the structure

- VkVideoCapabilitiesKHR

Bits which **may** be set in VkVideoEncodeAV1CapabilitiesKHR::flags, indicating the AV1 encoding capabilities supported, are:

```
// Provided by VK_KHR_video_encode_av1
typedef enum VkVideoEncodeAV1CapabilityFlagBitsKHR {
    VK_VIDEO_ENCODE_AV1_CAPABILITY_PER_RATE_CONTROL_GROUP_MIN_MAX_Q_INDEX_BIT_KHR =
0x00000001,
    VK_VIDEO_ENCODE_AV1_CAPABILITY_GENERATE_OBU_EXTENSION_HEADER_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_AV1_CAPABILITY_PRIMARY_REFERENCE_CDF_ONLY_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_AV1_CAPABILITY_FRAME_SIZE_OVERRIDE_BIT_KHR = 0x00000008,
    VK_VIDEO_ENCODE_AV1_CAPABILITY_MOTION_VECTOR_SCALING_BIT_KHR = 0x00000010,
  // Provided by VK_KHR_video_encode_av1 with VK_KHR_video_encode_intra_refresh
    VK_VIDEO_ENCODE_AV1_CAPABILITY_COMPOUND_PREDICTION_INTRA_REFRESH_BIT_KHR =
0x00000020,
} VkVideoEncodeAV1CapabilityFlagBitsKHR;
```

- VK_VIDEO_ENCODE_AV1_CAPABILITY_PER_RATE_CONTROL_GROUP_MIN_MAX_Q_INDEX_BIT_KHR specifies support for specifying different quantizer index values in the members of VkVideoEncodeAV1QIndexKHR.
- VK_VIDEO_ENCODE_AV1_CAPABILITY_GENERATE_OBU_EXTENSION_HEADER_BIT_KHR specifies support for generating OBU extension headers, as defined in section 5.3.3 of the AV1 Specification.
- VK_VIDEO_ENCODE_AV1_CAPABILITY_PRIMARY_REFERENCE_CDF_ONLY_BIT_KHR specifies support for using the primary reference frame indicated by the value of StdVideoEncodeAV1PictureInfo::primary_ref_frame in the AV1 picture information only for CDF data reference, as defined in section 6.8.2 of the AV1 Specification.
- VK_VIDEO_ENCODE_AV1_CAPABILITY_FRAME_SIZE_OVERRIDE_BIT_KHR specifies support for encoding a picture with a frame size different from the maximum frame size defined in the active AV1 sequence header. If this capability is not supported, then frame_size_override_flag **must** not be set in the AV1 picture information of the encoded frame and the coded extent of the encode input picture **must** match the maximum coded extent allowed by the active AV1 sequence header, i.e. (max_frame_width_minus_1 + 1, max_frame_height_minus_1 + 1).
- VK_VIDEO_ENCODE_AV1_CAPABILITY_MOTION_VECTOR_SCALING_BIT_KHR specifies support for motion vector scaling, as defined in section 7.11.3.3 of the AV1 Specification. If this capability is not supported, then the coded extent of all active reference pictures **must** match the coded extent of the encode input picture. This capability **may** only be supported by a video profile when VK_VIDEO_ENCODE_AV1_CAPABILITY_FRAME_SIZE_OVERRIDE_BIT_KHR is also supported.
- VK_VIDEO_ENCODE_AV1_CAPABILITY_COMPOUND_PREDICTION_INTRA_REFRESH_BIT_KHR specifies support for encoding frames using unidirectional or bidirectional compound prediction mode with intra refresh enabled.

```
// Provided by VK_KHR_video_encode_av1
typedef VkFlags VkVideoEncodeAV1CapabilityFlagsKHR;
```

VkVideoEncodeAV1CapabilityFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeAV1CapabilityFlagBitsKHR.

Bits which **may** be set in VkVideoEncodeAV1CapabilitiesKHR::stdSyntaxFlags, indicating the capabilities related to the AV1 syntax elements, are:

```
// Provided by VK_KHR_video_encode_av1
typedef enum VkVideoEncodeAV1StdFlagBitsKHR {
    VK_VIDEO_ENCODE_AV1_STD_UNIFORM_TILE_SPACING_FLAG_SET_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_AV1_STD_SKIP_MODE_PRESENT_UNSET_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_AV1_STD_PRIMARY_REF_FRAME_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_AV1_STD_DELTA_Q_BIT_KHR = 0x00000008,
} VkVideoEncodeAV1StdFlagBitsKHR;
```

- VK_VIDEO_ENCODE_AV1_STD_UNIFORM_TILE_SPACING_FLAG_SET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoAV1TileInfoFlags::uniform_tile_spacing_flag in the AV1 tile parameters when that value is 1, regardless of the coded extent of the encode input picture and the number of tile columns and rows requested in the TileCols and TileRows members of StdVideoAV1TileInfo.
- VK_VIDEO_ENCODE_AV1_STD_SKIP_MODE_PRESENT_UNSET_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoEncodeAV1PictureInfoFlags::skip_mode_present when that value is 0.
- VK_VIDEO_ENCODE_AV1_STD_PRIMARY_REF_FRAME_BIT_KHR specifies whether the implementation supports using the application-provided value for StdVideoEncodeAV1PictureInfo::primary_ref_frame.
- VK_VIDEO_ENCODE_AV1_STD_DELTA_Q_BIT_KHR specifies whether the implementation supports using the application-provided values for the DeltaQYDc, DeltaQUDc, DeltaQUAc, DeltaQVDc, and DeltaQVAc members of StdVideoAV1Quantization.

These capability flags provide information to the application about specific AV1 syntax element values that the implementation supports without having to override them and do not otherwise restrict the values that the application **can** specify for any of the mentioned AV1 syntax elements.

```
// Provided by VK_KHR_video_encode_av1
typedef VkFlags VkVideoEncodeAV1StdFlagsKHR;
```

VkVideoEncodeAV1StdFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeAV1StdFlagBitsKHR.

Bits which **may** be set in VkVideoEncodeAV1CapabilitiesKHR::superblockSizes, indicating the superblock sizes supported by the implementation, are:

```
// Provided by VK_KHR_video_encode_av1
typedef enum VkVideoEncodeAV1SuperblockSizeFlagBitsKHR {
    VK_VIDEO_ENCODE_AV1_SUPERBLOCK_SIZE_64_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_AV1_SUPERBLOCK_SIZE_128_BIT_KHR = 0x00000002,
} VkVideoEncodeAV1SuperblockSizeFlagBitsKHR;
```

- VK_VIDEO_ENCODE_AV1_SUPERBLOCK_SIZE_64_BIT_KHR specifies that a superblock size of 64x64 is supported.
- VK_VIDEO_ENCODE_AV1_SUPERBLOCK_SIZE_128_BIT_KHR specifies that a superblock size of 128x128 is supported.

```
// Provided by VK_KHR_video_encode_av1
typedef VkFlags VkVideoEncodeAV1SuperblockSizeFlagsKHR;
```

VkVideoEncodeAV1SuperblockSizeFlagsKHR is a bitmask type for setting a mask of zero or more VkVideoEncodeAV1SuperblockSizeFlagBitsKHR.

Implementations **must** support at least one of VkVideoEncodeAV1SuperblockSizeFlagBitsKHR.

45.22.10. AV1 Encode Quality Level Properties

When calling vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR with pVideoProfile->videoCodecOperation specified as VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, the VkVideoEncodeAV1QualityLevelPropertiesKHR structure **must** be included in the pNext chain of the VkVideoEncodeQualityLevelPropertiesKHR structure to retrieve additional video encode quality level properties specific to AV1 encoding.

The VkVideoEncodeAV1QualityLevelPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_av1
typedef struct VkVideoEncodeAV1QualityLevelPropertiesKHR {
    VkStructureType                          sType;
    void*                                    pNext;
    VkVideoEncodeAV1RateControlFlagsKHR      preferredRateControlFlags;
    uint32_t                                 preferredGopFrameCount;
    uint32_t                                 preferredKeyFramePeriod;
    uint32_t                                 preferredConsecutiveBipredictiveFrameCount;
    uint32_t                                 preferredTemporalLayerCount;
    VkVideoEncodeAV1QIndexKHR                preferredConstantQIndex;
    uint32_t                                 preferredMaxSingleReferenceCount;
    uint32_t                                 preferredSingleReferenceNameMask;
    uint32_t
preferredMaxUnidirectionalCompoundReferenceCount;
    uint32_t
preferredMaxUnidirectionalCompoundGroup1ReferenceCount;
    uint32_t
preferredUnidirectionalCompoundReferenceNameMask;
    uint32_t
preferredMaxBidirectionalCompoundReferenceCount;
    uint32_t
preferredMaxBidirectionalCompoundGroup1ReferenceCount;
    uint32_t
preferredMaxBidirectionalCompoundGroup2ReferenceCount;
    uint32_t
preferredBidirectionalCompoundReferenceNameMask;
} VkVideoEncodeAV1QualityLevelPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `preferredRateControlFlags` is a bitmask of VkVideoEncodeAV1RateControlFlagBitsKHR values indicating the preferred flags to use for VkVideoEncodeAV1RateControlInfoKHR::`flags`.
- `preferredGopFrameCount` indicates the preferred value to use for VkVideoEncodeAV1RateControlInfoKHR::`gopFrameCount`.
- `preferredKeyFramePeriod` indicates the preferred value to use for VkVideoEncodeAV1RateControlInfoKHR::`keyFramePeriod`.
- `preferredConsecutiveBipredictiveFrameCount` indicates the preferred value to use for VkVideoEncodeAV1RateControlInfoKHR::`consecutiveBipredictiveFrameCount`.
- `preferredTemporalLayerCount` indicates the preferred value to use for VkVideoEncodeAV1RateControlInfoKHR::`temporalLayerCount`.
- `preferredConstantQIndex` indicates the preferred value to use for VkVideoEncodeAV1PictureInfoKHR::`constantQIndex` when using rate control mode VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR.
- `preferredMaxSingleReferenceCount` indicates the preferred maximum number of reference pictures to use with single reference prediction mode.

- preferredSingleReferenceNameMask is a bitmask of preferred AV1 reference names when using single reference prediction mode.
- preferredMaxUnidirectionalCompoundReferenceCount indicates the preferred maximum number of reference pictures to use with unidirectional compound prediction mode.
- preferredMaxUnidirectionalCompoundGroup1ReferenceCount indicates the preferred maximum number of reference pictures to use with unidirectional compound prediction mode from reference frame group 1, as defined in section 6.10.24 of the AV1 Specification.
- preferredUnidirectionalCompoundReferenceNameMask is a bitmask of preferred AV1 reference names when using unidirectional compound prediction mode.
- preferredMaxBidirectionalCompoundReferenceCount indicates the preferred maximum number of reference pictures to use with bidirectional compound prediction mode.
- preferredMaxBidirectionalCompoundGroup1ReferenceCount indicates the preferred maximum number of reference pictures to use with bidirectional compound prediction mode from reference frame group 1, as defined in section 6.10.24 of the AV1 Specification.
- preferredMaxBidirectionalCompoundGroup2ReferenceCount indicates the preferred maximum number of reference pictures to use with bidirectional compound prediction mode from reference frame group 2, as defined in section 6.10.24 of the AV1 Specification.
- preferredBidirectionalCompoundReferenceNameMask is a bitmask of preferred AV1 reference names when using bidirectional compound prediction mode.

preferredSingleReferenceNameMask, preferredUnidirectionalCompoundReferenceNameMask, and preferredBidirectionalCompoundReferenceNameMask are encoded such that when bit index i is set, it indicates preference for using the AV1 reference name STD_VIDEO_AV1_REFERENCE_NAME_LAST_FRAME + i.

Valid Usage (Implicit)

- VUID-VkVideoEncodeAV1QualityLevelPropertiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_QUALITY_LEVEL_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkVideoEncodeQualityLevelPropertiesKHR

45.22.11. AV1 Encode Session

Additional parameters can be specified when creating a video session with an AV1 encode profile by including an instance of the VkVideoEncodeAV1SessionCreateInfoKHR structure in the pNext chain of VkVideoSessionCreateInfoKHR.

The VkVideoEncodeAV1SessionCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_av1
typedef struct VkVideoEncodeAV1SessionCreateInfoKHR {
    VkStructureType     sType;
    const void*         pNext;
    VkBool32            useMaxLevel;
    StdVideoAV1Level    maxLevel;
} VkVideoEncodeAV1SessionCreateInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `useMaxLevel` indicates whether the value of `maxLevel` should be used by the implementation. When it is set to VK_FALSE, the implementation ignores the value of `maxLevel` and uses the value of VkVideoEncodeAV1CapabilitiesKHR::`maxLevel`, as reported by vkGetPhysicalDeviceVideoCapabilitiesKHR for the video profile.
- `maxLevel` is a `StdVideoAV1Level` value specifying the upper bound on the AV1 level for the video bitstreams produced by the created video session.

Valid Usage (Implicit)

- VUID-VkVideoEncodeAV1SessionCreateInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_SESSION_CREATE_INFO_KHR

Structure Chaining

Extends the structure

- VkVideoSessionCreateInfoKHR

45.22.12. AV1 Encode Parameter Sets

Video session parameters objects created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR contain a single instance of the following parameter set:

AV1 Sequence Header

Represented by `StdVideoAV1SequenceHeader` structures and interpreted as follows:

- `flags.reserved` and `reserved1` are used only for padding purposes and are otherwise ignored;
- the `StdVideoAV1ColorConfig` structure pointed to by `pColorConfig` is interpreted as follows:
  - `flags.reserved` and `reserved1` are used only for padding purposes and are otherwise ignored;
  - all other members of `StdVideoAV1ColorConfig` are interpreted as defined in section 6.4.2 of

the AV1 Specification;

- if `flags.timing_info_present_flag` is set, then the `StdVideoAV1TimingInfo` structure pointed to by `pTimingInfo` is interpreted as follows:
  - `flags.reserved` is used only for padding purposes and is otherwise ignored;
  - all other members of `StdVideoAV1TimingInfo` are interpreted as defined in section 6.4.3 of the AV1 Specification;
- all other members of `StdVideoAV1SequenceHeader` are interpreted as defined in section 6.4 of the AV1 Specification.

When `StdVideoAV1SequenceHeader`::`flags.timing_info_present_flag` is set, the AV1 sequence header **can** be amended with AV1 decoder model information, represented by a `StdVideoEncodeAV1DecoderModelInfo` structure and interpreted as follows:

- `reserved1` is used only for padding purposes and is otherwise ignored;
- all other members of `StdVideoEncodeAV1DecoderModelInfo` are interpreted as defined in section 6.4.4 of the AV1 Specification.

When `StdVideoAV1SequenceHeader`::`flags.reduced_still_picture_header` is not set, the AV1 sequence header **can** be amended with AV1 operating point information, represented by an array of `StdVideoEncodeAV1OperatingPointInfo` structures and interpreted as follows:

- `flags.reserved` is used only for padding purposes and is otherwise ignored;
- all other members of `StdVideoEncodeAV1OperatingPointInfo` are interpreted as the corresponding element of the respective arrays defined in section 6.4 of the AV1 Specification.

Implementations **may** override any of these parameters according to the semantics defined in the Video Encode Parameter Overrides section before storing the resulting AV1 sequence header into the video session parameters object. Applications need to use the vkGetEncodedVideoSessionParametersKHR command to determine whether any implementation overrides happened and to retrieve the encoded AV1 sequence header in order to be able to produce a compliant AV1 video bitstream.

The encoded AV1 sequence header retrieved using the vkGetEncodedVideoSessionParametersKHR command is encoded as a single OBU with `obu_type` equal to `OBU_SEQUENCE_HEADER`, as defined in section 5.3 of the AV1 Specification.

Such AV1 sequence header overrides **may** also have cascading effects on the implementation overrides applied to the encoded bitstream produced by video encode operations. If the implementation supports the VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR video encode feedback query flag, then the application **can** use such queries to retrieve feedback about whether any implementation overrides have been applied to the encoded bitstream.

When a video session parameters object is created with the codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, the VkVideoSessionParametersCreateInfoKHR::`pNext` chain **must** include a `VkVideoEncodeAV1SessionParametersCreateInfoKHR` structure specifying the contents of the object.

The VkVideoEncodeAV1SessionParametersCreateInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_av1
typedef struct VkVideoEncodeAV1SessionParametersCreateInfoKHR {
    VkStructureType                                 sType;
    const void*                                     pNext;
    const StdVideoAV1SequenceHeader*                pStdSequenceHeader;
    const StdVideoEncodeAV1DecoderModelInfo*        pStdDecoderModelInfo;
    uint32_t                                        stdOperatingPointCount;
    const StdVideoEncodeAV1OperatingPointInfo*      pStdOperatingPoints;
} VkVideoEncodeAV1SessionParametersCreateInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pStdSequenceHeader is a pointer to a StdVideoAV1SequenceHeader structure describing parameters of the AV1 sequence header entry to store in the created object.
- pStdDecoderModelInfo is NULL or a pointer to a StdVideoEncodeAV1DecoderModelInfo structure specifying the AV1 decoder model information to store in the created object.
- stdOperatingPointCount is the number of elements in the pStdOperatingPoints array.
- pStdOperatingPoints is NULL or a pointer to an array of stdOperatingPointCount number of StdVideoEncodeAV1OperatingPointInfo structures specifying the AV1 operating point information to store in the created object. Each element i specifies the parameter values corresponding to element i of the syntax elements defined in section 6.4 of the AV1 Specification.

Valid Usage

- VUID-VkVideoEncodeAV1SessionParametersCreateInfoKHR-pStdSequenceHeader-10288
  pStdSequenceHeader->flags.film_grain_params_present **must** be zero

Valid Usage (Implicit)

- VUID-VkVideoEncodeAV1SessionParametersCreateInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_SESSION_PARAMETERS_CREATE_INFO_KHR
- VUID-VkVideoEncodeAV1SessionParametersCreateInfoKHR-pStdSequenceHeader-parameter
  pStdSequenceHeader **must** be a valid pointer to a valid StdVideoAV1SequenceHeader value
- VUID-VkVideoEncodeAV1SessionParametersCreateInfoKHR-pStdDecoderModelInfo-parameter
  If pStdDecoderModelInfo is not NULL, pStdDecoderModelInfo **must** be a valid pointer to a valid StdVideoEncodeAV1DecoderModelInfo value
- VUID-VkVideoEncodeAV1SessionParametersCreateInfoKHR-pStdOperatingPoints-parameter

If `stdOperatingPointCount` is not `0`, and `pStdOperatingPoints` is not `NULL`, `pStdOperatingPoints` **must** be a valid pointer to an array of `stdOperatingPointCount` `StdVideoEncodeAV1OperatingPointInfo` values

Structure Chaining

Extends the structure

- VkVideoSessionParametersCreateInfoKHR

45.22.13. AV1 Encoding Parameters

The VkVideoEncodeAV1PictureInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_av1
typedef struct VkVideoEncodeAV1PictureInfoKHR {
    VkStructureType                         sType;
    const void*                             pNext;
    VkVideoEncodeAV1PredictionModeKHR       predictionMode;
    VkVideoEncodeAV1RateControlGroupKHR     rateControlGroup;
    uint32_t                                constantQIndex;
    const StdVideoEncodeAV1PictureInfo*     pStdPictureInfo;
    int32_t                                 referenceNameSlotIndices
[VK_MAX_VIDEO_AV1_REFERENCES_PER_FRAME_KHR];
    VkBool32                                primaryReferenceCdfOnly;
    VkBool32                                generateObuExtensionHeader;
} VkVideoEncodeAV1PictureInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `predictionMode` specifies the AV1 prediction mode to use for the encoded frame.
- `rateControlGroup` specifies the AV1 rate control group to use for the encoded frame when the current rate control mode is not VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR. Otherwise it is ignored.
- `constantQIndex` is the quantizer index to use for the encoded frame if the current rate control mode configured for the video session is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR.
- `pStdPictureInfo` is a pointer to a `StdVideoEncodeAV1PictureInfo` structure specifying AV1 picture information.
- `referenceNameSlotIndices` is an array of seven (VK_MAX_VIDEO_AV1_REFERENCES_PER_FRAME_KHR, which is equal to the Video Std definition `STD_VIDEO_AV1_REFS_PER_FRAME`) signed integer values specifying the index of the DPB slot or a negative integer value for each AV1 reference name used for inter coding. In particular, the DPB slot index for the AV1 reference name `frame` is specified in `referenceNameSlotIndices`

[frame - STD_VIDEO_AV1_REFERENCE_NAME_LAST_FRAME].

- primaryReferenceCdfOnly controls whether the primary reference frame indicated by the value of pStdPictureInfo->primary_ref_frame is used only for CDF data reference, as defined in section 6.8.2 of the AV1 Specification. If set to VK_TRUE, then the primary reference frame's picture data will not be used for sample prediction.
- generateObuExtensionHeader controls whether OBU extension headers are generated into the target bitstream, as defined in sections 5.3.1, 5.3.2, and 5.3.3 of the AV1 Specification.

This structure is specified in the pNext chain of the VkVideoEncodeInfoKHR structure passed to vkCmdEncodeVideoKHR to specify the codec-specific picture information for an AV1 encode operation.

Encode Input Picture Information

When this structure is specified in the pNext chain of the VkVideoEncodeInfoKHR structure passed to vkCmdEncodeVideoKHR, the information related to the encode input picture is defined as follows:

- The image subregion used is determined according to the AV1 Encode Picture Data Access section.
- The encode input picture is associated with the AV1 picture information provided in pStdPictureInfo.

Std Picture Information

The members of the StdVideoEncodeAV1PictureInfo structure pointed to by pStdPictureInfo are interpreted as follows:

- flags.reserved and reserved1 are used only for padding purposes and are otherwise ignored;
- pSegmentation **must** be NULL

> **NOTE** AV1 segmentation is currently not supported in video encode operations. Accordingly, the application needs to set flags.segmentation_enabled to 0 and pSegmentation to NULL.

- pTileInfo is NULL or a pointer to a StdVideoAV1TileInfo structure specifying AV1 tile parameters;
- the StdVideoAV1Quantization structure pointed to by pQuantization is interpreted as follows:
  - flags.reserved is used only for padding purposes and is otherwise ignored;
  - all other members of StdVideoAV1Quantization are interpreted as defined in section 6.8.11 of the AV1 Specification;
- the StdVideoAV1LoopFilter structure pointed to by pLoopFilter is interpreted as follows:
  - flags.reserved is used only for padding purposes and is otherwise ignored;
  - update_ref_delta is a bitmask where bit index i is interpreted as the value of update_ref_delta corresponding to element i of loop_filter_ref_deltas as defined in section 6.8.10 of the AV1 Specification;

    - update_mode_delta is a bitmask where bit index i is interpreted as the value of update_mode_delta corresponding to element i of loop_filter_mode_deltas as defined in section 6.8.10 of the AV1 Specification;
    - all other members of StdVideoAV1LoopFilter are interpreted as defined in section 6.8.10 of the AV1 Specification;
- if flags.enable_cdef is set in the active sequence header, then the members of the StdVideoAV1CDEF structure pointed to by pCDEF are interpreted as follows:
    - cdef_y_sec_strength and cdef_uv_sec_strength are the bitstream values of the corresponding syntax elements defined in section 5.9.19 of the AV1 Specification;
    - all other members of StdVideoAV1CDEF are interpreted as defined in section 6.10.14 of the AV1 Specification;
- if flags.UsesLr is set in the active sequence header, then the StdVideoAV1LoopRestoration structure pointed to by pLoopRestoration is interpreted as follows:
    - LoopRestorationSize[plane] is interpreted as log2(size) - 5, where size is the value of LoopRestorationSize[plane] as defined in section 6.10.15 of the AV1 Specification;
    - all other members of StdVideoAV1LoopRestoration are defined as in section 6.10.15 of the AV1 Specification;
- the members of the StdVideoAV1GlobalMotion structure pointed to by pGlobalMotion are interpreted as defined in section 7.10 of the AV1 Specification;
- pExtensionHeader is NULL or a pointer to a StdVideoEncodeAV1ExtensionHeader structure whose temporal_id and spatial_id members specify the temporal and spatial layer ID of the reference frame, respectively (these IDs are encoded into the OBU extension header if VkVideoEncodeAV1PictureInfoKHR::generateObuExtensionHeader is set to VK_TRUE for the encode operation);
- if flags.buffer_removal_time_present_flag is set, then pBufferRemovalTimes is a pointer to an array of N number of unsigned integer values specifying the elements of the buffer_removal_time array, as defined in section 6.8.2 of the AV1 Specification, where N is the number of operating points specified for the active sequence header through VkVideoEncodeAV1SessionParametersCreateInfoKHR::stdOperatingPointCount;
- all other members are interpreted as defined in section 6.8 of the AV1 Specification.

Reference picture setup is controlled by the value of StdVideoEncodeAV1PictureInfo::refresh_frame_flags. If it is not zero and a reconstructed picture is specified, then the latter is used as the target of picture reconstruction to activate the DPB slot specified in pEncodeInfo->pSetupReferenceSlot->slotIndex. If StdVideoEncodeAV1PictureInfo::refresh_frame_flags is zero, but a reconstructed picture is specified, then the corresponding picture reference associated with the DPB slot is invalidated, as described in the DPB Slot States section.

Std Tile Parameters

Specifying AV1 tile parameters is optional. If StdVideoEncodeAV1PictureInfo::pTileInfo is NULL, then the implementation determines the values of AV1 tile parameters defined in section 6.8.14 of the AV1 Specification in an implementation-dependent manner. If StdVideoEncodeAV1PictureInfo::pTileInfo is not NULL, then the members of the StdVideoAV1TileInfo structure pointed to by StdVideoEncodeAV1PictureInfo::pTileInfo are

interpreted as follows:

- `flags.reserved` and `reserved1` are used only for padding purposes and are otherwise ignored;
- `TileCols` and `TileRows` specify the number of tile columns and tile rows as defined in section 6.8.14 of the AV1 Specification;
- `tile_size_bytes_minus_1` is ignored, as its value, as defined in section 6.8.14 of the AV1 Specification, is determined as the result of the encoding process;
- `pMiColStarts` and `pMiRowStarts` are ignored, as the elements of the `MiColStarts` and `MiRowStarts` arrays defined in section 6.8.14 of the AV1 Specification are determined by the implementation based on the tile widths and heights determined by the implementation or specified through the `pWidthInSbsMinus1` and `pHeightInSbsMinus1` arrays, respectively;
- `pWidthInSbsMinus1` is `NULL` or a pointer to an array of `TileCols` number of unsigned integers that corresponds to `width_in_sbs_minus_1` defined in section 6.8.14 of the AV1 Specification;
- `pHeightInSbsMinus1` is `NULL` or is a pointer to an array of `TileRows` number of unsigned integers that corresponds to `height_in_sbs_minus_1` defined in section 6.8.14 of the AV1 Specification;
- all other members of `StdVideoAV1TileInfo` are interpreted as defined in section 6.8.14 of the AV1 Specification.

If `flags.uniform_tile_spacing_flag` is set, then `pWidthInSbsMinus1` and `pHeightInSbsMinus1` are ignored.

If `flags.uniform_tile_spacing_flag` is not set and `pWidthInSbsMinus1` is `NULL`, then the width of individual tile columns is determined in an implementation-dependent manner.

If `flags.uniform_tile_spacing_flag` is not set and `pHeightInSbsMinus1` is `NULL`, then the height of individual tile rows is determined in an implementation-dependent manner.

NOTE

In general, implementations are expected to respect the application-specified AV1 tile parameters. However, as implementations may have restrictions on the combination of tile column and row counts, and tile widths and heights with respect to the extent of the encoded frame beyond the restrictions specified in the AV1 Specification and this specification (through video profile capabilities), certain parameter combinations may require the implementation to override them in order to conform to such implementation-specific limitations.

Active Parameter Sets

The *active sequence header* is the AV1 sequence header stored in the bound video session parameters object.

Valid Usage

- VUID-VkVideoEncodeAV1PictureInfoKHR-flags-10289
  If VkVideoEncodeAV1CapabilitiesKHR::`flags`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_AV1_CAPABILITY_PRIMARY_REFERENCE_CDF_ONLY_BIT_KHR, then `primaryReferenceCdfOnly` **must** be VK_FALSE

- VUID-VkVideoEncodeAV1PictureInfoKHR-primaryReferenceCdfOnly-10290
  If `primaryReferenceCdfOnly` is set to VK_TRUE, then `pStdPictureInfo->primary_ref_frame` **must** be less than VK_MAX_VIDEO_AV1_REFERENCES_PER_FRAME_KHR
- VUID-VkVideoEncodeAV1PictureInfoKHR-pStdPictureInfo-10291
  If `pStdPictureInfo->primary_ref_frame` is less than VK_MAX_VIDEO_AV1_REFERENCES_PER_FRAME_KHR, then `referenceNameSlotIndices`[`pStdPictureInfo->primary_ref_frame`] **must** not be negative
- VUID-VkVideoEncodeAV1PictureInfoKHR-flags-10292
  If VkVideoEncodeAV1CapabilitiesKHR::`flags`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_AV1_CAPABILITY_GENERATE_OBU_EXTENSION_HEADER_BIT_KHR, then `generateObuExtensionHeader` **must** be VK_FALSE
- VUID-VkVideoEncodeAV1PictureInfoKHR-generateObuExtensionHeader-10293
  If `generateObuExtensionHeader` is set to VK_TRUE, then `pStdPictureInfo->pExtensionHeader` **must** not be `NULL`

Valid Usage (Implicit)

- VUID-VkVideoEncodeAV1PictureInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_PICTURE_INFO_KHR
- VUID-VkVideoEncodeAV1PictureInfoKHR-predictionMode-parameter
  `predictionMode` **must** be a valid VkVideoEncodeAV1PredictionModeKHR value
- VUID-VkVideoEncodeAV1PictureInfoKHR-rateControlGroup-parameter
  `rateControlGroup` **must** be a valid VkVideoEncodeAV1RateControlGroupKHR value
- VUID-VkVideoEncodeAV1PictureInfoKHR-pStdPictureInfo-parameter
  `pStdPictureInfo` **must** be a valid pointer to a valid `StdVideoEncodeAV1PictureInfo` value

Structure Chaining

Extends the structure

- VkVideoEncodeInfoKHR

The VkVideoEncodeAV1DpbSlotInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_av1
typedef struct VkVideoEncodeAV1DpbSlotInfoKHR {
    VkStructureType                          sType;
    const void*                              pNext;
    const StdVideoEncodeAV1ReferenceInfo*    pStdReferenceInfo;
} VkVideoEncodeAV1DpbSlotInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.

- pNext is NULL or a pointer to a structure extending this structure.
- pStdReferenceInfo is a pointer to a StdVideoEncodeAV1ReferenceInfo structure specifying AV1 reference information.

This structure is specified in the pNext chain of VkVideoEncodeInfoKHR::pSetupReferenceSlot, if not NULL, and the pNext chain of the elements of VkVideoEncodeInfoKHR::pReferenceSlots to specify the codec-specific reference picture information for an AV1 encode operation.

Active Reference Picture Information

When this structure is specified in the pNext chain of the elements of VkVideoEncodeInfoKHR::pReferenceSlots, one element is added to the list of active reference pictures used by the video encode operation for each element of VkVideoEncodeInfoKHR::pReferenceSlots as follows:

- The image subregion used is determined according to the AV1 Encode Picture Data Access section.
- The reference picture is associated with the DPB slot index specified in the slotIndex member of the corresponding element of VkVideoEncodeInfoKHR::pReferenceSlots.
- The reference picture is associated with the AV1 reference information provided in pStdReferenceInfo.

Reconstructed Picture Information

When this structure is specified in the pNext chain of VkVideoEncodeInfoKHR::pSetupReferenceSlot, the information related to the reconstructed picture is defined as follows:

- The image subregion used is determined according to the AV1 Encode Picture Data Access section.
- If reference picture setup is requested, then the reconstructed picture is used to activate the DPB slot with the index specified in VkVideoEncodeInfoKHR::pSetupReferenceSlot->slotIndex.
- The reconstructed picture is associated with the AV1 reference information provided in pStdReferenceInfo.

Std Reference Information

The members of the StdVideoEncodeAV1ReferenceInfo structure pointed to by pStdReferenceInfo are interpreted as follows:

- flags.reserved and reserved1 are used only for padding purposes and are otherwise ignored;
- flags.disable_frame_end_update_cdf is interpreted as defined in section 6.8.2 of the AV1 Specification;
- flags.segmentation_enabled is interpreted as defined in section 6.8.13 of the AV1 Specification;
- RefFrameId is interpreted as the element of the RefFrameId array defined in section 6.8.2 of the AV1 Specification corresponding to the reference frame;
- frame_type is interpreted as defined in section 6.8.2 of the AV1 Specification;
- OrderHint is interpreted as defined in section 6.8.2 of the AV1 Specification;

- `pExtensionHeader` is `NULL` or a pointer to a `StdVideoEncodeAV1ExtensionHeader` structure whose `temporal_id` and `spatial_id` members specify the temporal and spatial layer ID of the reference frame, respectively.

Valid Usage (Implicit)

- VUID-VkVideoEncodeAV1DpbSlotInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_DPB_SLOT_INFO_KHR
- VUID-VkVideoEncodeAV1DpbSlotInfoKHR-pStdReferenceInfo-parameter
  `pStdReferenceInfo` **must** be a valid pointer to a valid `StdVideoEncodeAV1ReferenceInfo` value

Structure Chaining

Extends the structure

- VkVideoReferenceSlotInfoKHR

45.22.14. AV1 Encode Rate Control

Group of Pictures

In case of AV1 encoding it is common practice to follow a regular pattern of frame types and prediction directions in display order when encoding subsequent frames. This pattern is referred to as the *group of pictures* (GOP).

The AV1 Specification, unlike some other video compression standards, does not restrict the direction in display order of the referenced frames based on the used frame type or AV1 prediction mode. Accordingly, this specification introduces the concept of *rate control groups* for which the application **can** specify separate rate control configuration parameters. When encoding a frame, the application specifies the rate control group the encoded frame belongs to through a VkVideoEncodeAV1RateControlGroupKHR value in VkVideoEncodeAV1PictureInfoKHR ::`rateControlGroup`. This value is then used by the implementation's rate control algorithm to determine which rate control configuration parameters apply to it.

Possible AV1 encode rate control groups are as follows:

```
// Provided by VK_KHR_video_encode_av1
typedef enum VkVideoEncodeAV1RateControlGroupKHR {
    VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_INTRA_KHR = 0,
    VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_PREDICTIVE_KHR = 1,
    VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_BIPREDICTIVE_KHR = 2,
} VkVideoEncodeAV1RateControlGroupKHR;
```

- VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_INTRA_KHR **should** be specified when encoding AV1 frames that use intra-only prediction (e.g. when encoding AV1 frames of type `STD_VIDEO_AV1_FRAME_TYPE_KEY` or `STD_VIDEO_AV1_FRAME_TYPE_INTRA_ONLY`).

- VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_PREDICTIVE_KHR **should** be specified when encoding AV1 frames that only have forward references in display order.
- VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_BIPREDICTIVE_KHR **should** be specified when encoding AV1 frames that have backward references in display order.

NOTE	While the application can specify any rate control group for any frame, regardless of the frame type, prediction mode, or prediction direction, specifying a rate control group that does not reflect the prediction direction used by the encoded frame may result in unexpected behavior of the implementation's rate control algorithm.

A regular GOP is defined by the following parameters:

- The number of frames in the GOP;
- The number of consecutive frames encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_BIPREDICTIVE_KHR between frames encoded with other rate control groups in display order.

Frame types in a regular GOP follow each other in display order according to the following algorithm:

1. The first frame is always a frame encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_INTRA_KHR.
2. This is followed by a number of consecutive frames encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_BIPREDICTIVE_KHR.
3. If the number of frames in the GOP is not reached yet, then the next frame is a frame encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_PREDICTIVE_KHR and the algorithm continues from step 2.

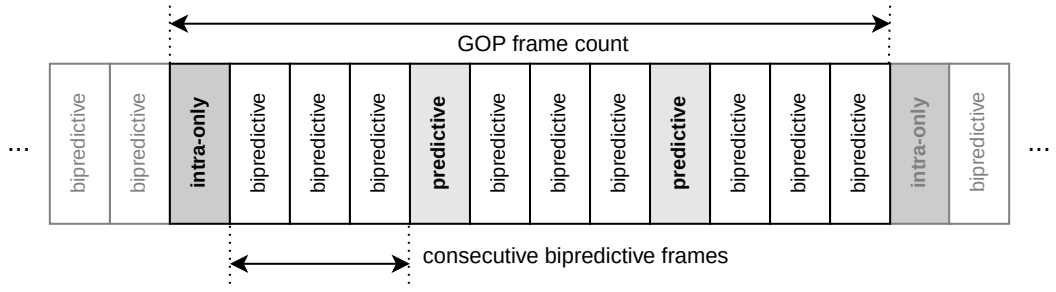


Figure 37. AV1 regular GOP starting with an intra-only frame

When frames with the AV1 frame type `STD_VIDEO_AV1_FRAME_TYPE_KEY` are used in a video sequence, the *key frame period* refers to the distance between subsequent frames with the AV1 frame type `STD_VIDEO_AV1_FRAME_TYPE_KEY`.

It is also typical for AV1 encoding to use specific reference picture usage patterns across the frames of the GOP. The two most common reference patterns used are as follows:

Flat Reference Pattern

- Each frame encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_PREDICTIVE_KHR refers to the last frame that was not encoded using VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_BIPREDICTIVE_KHR, in display order, as its forward reference.
- Each frame encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_BIPREDICTIVE_KHR refers to the last frame that was not encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_BIPREDICTIVE_KHR, in display order, as its forward reference, and refers to the next frame that was not encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_BIPREDICTIVE_KHR, in display order, as its backward reference.

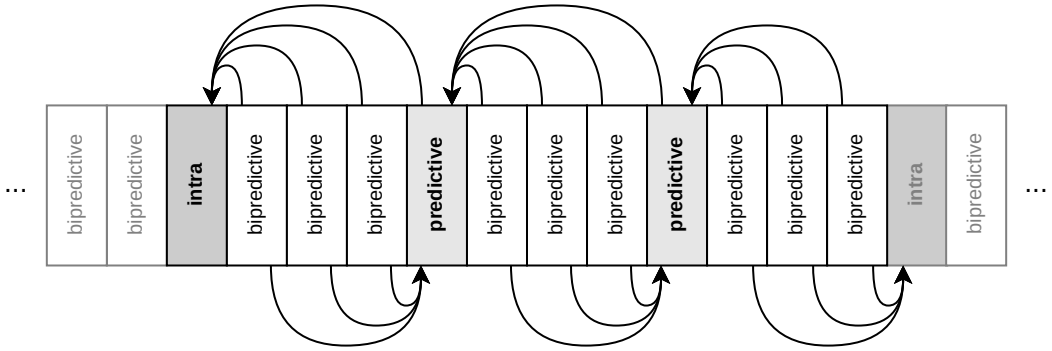


Figure 38. AV1 flat reference pattern

Dyadic Reference Pattern

- Each frame encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_PREDICTIVE_KHR refers to the last frame that was not encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_BIPREDICTIVE_KHR, in display order, as its forward reference.
- The following algorithm is applied to the sequence of consecutive frames encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_BIPREDICTIVE_KHR between frames using other rate control groups in display order:
  1. The frame in the middle of this sequence uses the frame preceding the sequence as its forward reference, and uses the frame following the sequence as its backward reference.
  2. The algorithm is executed recursively for the following frame sequences:
     - The frames of the original sequence preceding the frame in the middle, if any.
     - The frames of the original sequence following the frame in the middle, if any.

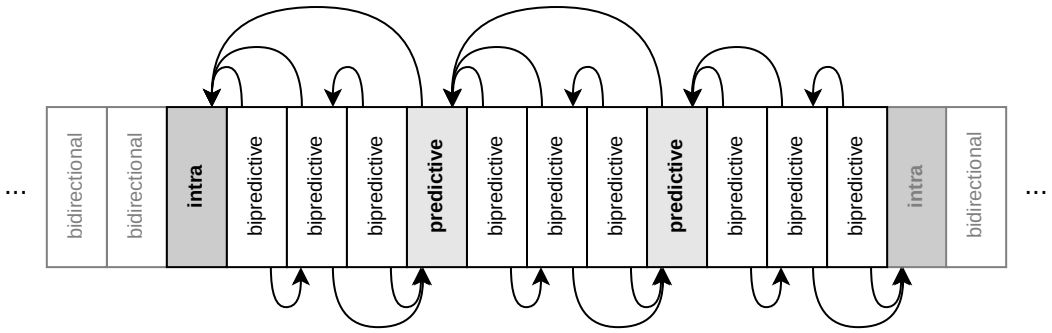


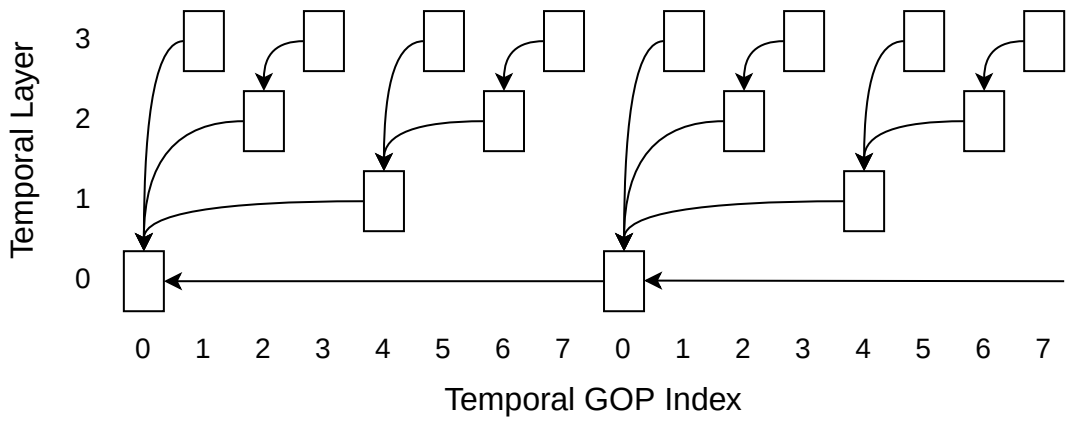
Figure 39. AV1 dyadic reference pattern

The application **can** provide guidance to the implementation's rate control algorithm about the structure of the GOP used by the application. Any such guidance about the GOP and its structure does not mandate that specific GOP structure to be used by the application, as the frame type and the selected rate control group is still application-controlled, however, any deviation from the provided guidance **may** result in undesired rate control behavior including, but not limited to, the implementation not being able to conform to the expected average or target bitrates, or other rate control parameters specified by the application.

When an AV1 encode session is used to encode multiple temporal layers, it is also common practice to follow a regular pattern for the AV1 temporal ID for the encoded frames in display order when encoding subsequent frames. This pattern is referred to as the *temporal GOP*. The most common temporal layer pattern used is as follows:

Dyadic Temporal Layer Pattern

- The number of frames in the temporal GOP is 2^{n-1}, where n is the number of temporal layers.
- The i^{th} frame in the temporal GOP uses temporal ID t, if and only if the index of the least significant bit set in i equals n-t-1, except for the first frame, which is the only frame in the temporal GOP using temporal ID zero.
- The i^{th} frame in the temporal GOP uses the r^{th} frame as reference, where r is calculated from i by clearing the least significant bit set in it, except for the first frame in the temporal GOP, which uses the first frame of the previous temporal GOP, if any, as reference.



Text is not SVG - cannot display

Figure 40. AV1 dyadic temporal layer pattern

NOTE	Multi-layer rate control and multi-layer coding are typically used for streaming cases where low latency is expected, hence frames usually do not use backward references in display order.

The `VkVideoEncodeAV1RateControlInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_av1
typedef struct VkVideoEncodeAV1RateControlInfoKHR {
    VkStructureType                          sType;
    const void*                              pNext;
    VkVideoEncodeAV1RateControlFlagsKHR      flags;
    uint32_t                                 gopFrameCount;
    uint32_t                                 keyFramePeriod;
    uint32_t                                 consecutiveBipredictiveFrameCount;
    uint32_t                                 temporalLayerCount;
} VkVideoEncodeAV1RateControlInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkVideoEncodeAV1RateControlFlagBitsKHR specifying AV1 rate control flags.
- `gopFrameCount` is the number of frames within a group of pictures (GOP) intended to be used by the application. If it is set to 0, the rate control algorithm **may** assume an implementation-dependent GOP length. If it is set to `UINT32_MAX`, the GOP length is treated as infinite.
- `keyFramePeriod` is the interval, in terms of number of frames, between two frames with the AV1 frame type `STD_VIDEO_AV1_FRAME_TYPE_KEY` (see key frame period). If it is set to 0, the rate control algorithm **may** assume an implementation-dependent key frame period. If it is set to `UINT32_MAX`, the key frame period is treated as infinite.

- `consecutiveBipredictiveFrameCount` is the number of consecutive frames encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_BIPREDICTIVE_KHR between frames encoded with other rate control groups within the GOP.
- `temporalLayerCount` specifies the number of AV1 temporal layers that the application intends to use.

When an instance of this structure is included in the `pNext` chain of the VkVideoCodingControlInfoKHR structure passed to the vkCmdControlVideoCodingKHR command, and VkVideoCodingControlInfoKHR::`flags` includes VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR, the parameters in this structure are used as guidance for the implementation's rate control algorithm (see Video Coding Control).

Valid Usage

- VUID-VkVideoEncodeAV1RateControlInfoKHR-flags-10294
  If `flags` contains VK_VIDEO_ENCODE_AV1_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR or VK_VIDEO_ENCODE_AV1_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR, then it **must** also contain VK_VIDEO_ENCODE_AV1_RATE_CONTROL_REGULAR_GOP_BIT_KHR
- VUID-VkVideoEncodeAV1RateControlInfoKHR-flags-10295
  If `flags` contains VK_VIDEO_ENCODE_AV1_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR, then it **must** not also contain VK_VIDEO_ENCODE_AV1_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR
- VUID-VkVideoEncodeAV1RateControlInfoKHR-flags-10296
  If `flags` contains VK_VIDEO_ENCODE_AV1_RATE_CONTROL_REGULAR_GOP_BIT_KHR, then `gopFrameCount` **must** be greater than `0`
- VUID-VkVideoEncodeAV1RateControlInfoKHR-keyFramePeriod-10297
  If `keyFramePeriod` is not `0`, then it **must** be greater than or equal to `gopFrameCount`
- VUID-VkVideoEncodeAV1RateControlInfoKHR-consecutiveBipredictiveFrameCount-10298
  If `consecutiveBipredictiveFrameCount` is not `0`, then it **must** be less than `gopFrameCount`
- VUID-VkVideoEncodeAV1RateControlInfoKHR-temporalLayerCount-10299
  `temporalLayerCount` **must** be less than or equal to VkVideoEncodeAV1CapabilitiesKHR::`maxTemporalLayerCount`, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile

Valid Usage (Implicit)

- VUID-VkVideoEncodeAV1RateControlInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_RATE_CONTROL_INFO_KHR
- VUID-VkVideoEncodeAV1RateControlInfoKHR-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkVideoEncodeAV1RateControlFlagBitsKHR values

Structure Chaining

Extends the structures

- VkVideoBeginCodingInfoKHR
- VkVideoCodingControlInfoKHR

Bits which **can** be set in VkVideoEncodeAV1RateControlInfoKHR::`flags`, specifying AV1 rate control flags, are:

```
// Provided by VK_KHR_video_encode_av1
typedef enum VkVideoEncodeAV1RateControlFlagBitsKHR {
    VK_VIDEO_ENCODE_AV1_RATE_CONTROL_REGULAR_GOP_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_AV1_RATE_CONTROL_TEMPORAL_LAYER_PATTERN_DYADIC_BIT_KHR =
0x00000002,
    VK_VIDEO_ENCODE_AV1_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_AV1_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR = 0x00000008,
} VkVideoEncodeAV1RateControlFlagBitsKHR;
```

- VK_VIDEO_ENCODE_AV1_RATE_CONTROL_REGULAR_GOP_BIT_KHR specifies that the application intends to use a regular GOP structure according to the parameters specified in the `gopFrameCount` and `keyFramePeriod` members of the VkVideoEncodeAV1RateControlInfoKHR structure.
- VK_VIDEO_ENCODE_AV1_RATE_CONTROL_TEMPORAL_LAYER_PATTERN_DYADIC_BIT_KHR specifies that the application intends to follow a dyadic temporal layer pattern.
- VK_VIDEO_ENCODE_AV1_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR specifies that the application intends to follow a flat reference pattern in the GOP.
- VK_VIDEO_ENCODE_AV1_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR specifies that the application intends to follow a dyadic reference pattern in the GOP.

```
// Provided by VK_KHR_video_encode_av1
typedef VkFlags VkVideoEncodeAV1RateControlFlagsKHR;
```

`VkVideoEncodeAV1RateControlFlagsKHR` is a bitmask type for setting a mask of zero or more VkVideoEncodeAV1RateControlFlagBitsKHR.

Rate Control Layers

The `VkVideoEncodeAV1RateControlLayerInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_av1
typedef struct VkVideoEncodeAV1RateControlLayerInfoKHR {
    VkStructureType                 sType;
    const void*                     pNext;
    VkBool32                        useMinQIndex;
    VkVideoEncodeAV1QIndexKHR       minQIndex;
    VkBool32                        useMaxQIndex;
    VkVideoEncodeAV1QIndexKHR       maxQIndex;
    VkBool32                        useMaxFrameSize;
    VkVideoEncodeAV1FrameSizeKHR    maxFrameSize;
} VkVideoEncodeAV1RateControlLayerInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `useMinQIndex` indicates whether the quantizer index values determined by rate control will be clamped to the lower bounds on the quantizer index values specified in `minQIndex`.
- `minQIndex` specifies the lower bounds on the quantizer index values, for each rate control group, that the implementation's rate control algorithm will use when `useMinQIndex` is set to VK_TRUE.
- `useMaxQIndex` indicates whether the quantizer index values determined by rate control will be clamped to the upper bounds on the quantizer index values specified in `maxQIndex`.
- `maxQIndex` specifies the upper bounds on the quantizer index values, for each rate control group, that the implementation's rate control algorithm will use when `useMaxQIndex` is set to VK_TRUE.
- `useMaxFrameSize` indicates whether the implementation's rate control algorithm **should** use the values specified in `maxFrameSize` as the upper bounds on the encoded frame size for each rate control group.
- `maxFrameSize` specifies the upper bounds on the encoded frame size, for each rate control group, when `useMaxFrameSize` is set to VK_TRUE.

When used, the values in `minQIndex` and `maxQIndex` guarantee that the effective quantizer index values used by the implementation will respect those lower and upper bounds, respectively. However, limiting the range of quantizer index values that the implementation is able to use will also limit the capabilities of the implementation's rate control algorithm to comply to other constraints. In particular, the implementation **may** not be able to comply to the following:

- The average and/or peak bitrate values to be used for the encoded bitstream specified in the `averageBitrate` and `maxBitrate` members of the VkVideoEncodeRateControlLayerInfoKHR structure.
- The upper bounds on the encoded frame size, for each rate control group, specified in the `maxFrameSize` member of `VkVideoEncodeAV1RateControlLayerInfoKHR`.

NOTE	In general, applications need to configure rate control parameters appropriately in order to be able to get the desired rate control behavior, as described in the Video Encode Rate Control section.

When an instance of this structure is included in the `pNext` chain of a

VkVideoEncodeRateControlLayerInfoKHR structure specified in one of the elements of the pLayers array member of the VkVideoEncodeRateControlInfoKHR structure passed to the vkCmdControlVideoCodingKHR command, VkVideoCodingControlInfoKHR::flags includes VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR, and the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, it specifies the AV1-specific rate control parameters of the rate control layer corresponding to that element of pLayers.

Valid Usage

- VUID-VkVideoEncodeAV1RateControlLayerInfoKHR-useMinQIndex-10300
  If useMinQIndex is VK_TRUE, then the intraQIndex, predictiveQIndex, and bipredictiveQIndex members of minQIndex **must** all be between VkVideoEncodeAV1CapabilitiesKHR::minQIndex and VkVideoEncodeAV1CapabilitiesKHR::maxQIndex, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile
- VUID-VkVideoEncodeAV1RateControlLayerInfoKHR-useMinQIndex-10301
  If useMinQIndex is VK_TRUE and VkVideoEncodeAV1CapabilitiesKHR::flags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_AV1_CAPABILITY_PER_RATE_CONTROL_GROUP_MIN_MAX_Q_INDEX_BIT_KHR, then the intraQIndex, predictiveQIndex, and bipredictiveQIndex members of minQIndex **must** all specify the same value
- VUID-VkVideoEncodeAV1RateControlLayerInfoKHR-useMaxQIndex-10302
  If useMaxQIndex is VK_TRUE, then the intraQIndex, predictiveQIndex, and bipredictiveQIndex members of maxQIndex **must** all be between VkVideoEncodeAV1CapabilitiesKHR::minQIndex and VkVideoEncodeAV1CapabilitiesKHR::maxQIndex, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile
- VUID-VkVideoEncodeAV1RateControlLayerInfoKHR-useMaxQIndex-10303
  If useMaxQIndex is VK_TRUE and VkVideoEncodeAV1CapabilitiesKHR::flags, as returned by vkGetPhysicalDeviceVideoCapabilitiesKHR for the used video profile, does not include VK_VIDEO_ENCODE_AV1_CAPABILITY_PER_RATE_CONTROL_GROUP_MIN_MAX_Q_INDEX_BIT_KHR, then the intraQIndex, predictiveQIndex, and bipredictiveQIndex members of maxQIndex **must** all specify the same value
- VUID-VkVideoEncodeAV1RateControlLayerInfoKHR-useMinQIndex-10304
  If useMinQIndex and useMaxQIndex are both VK_TRUE, then the intraQIndex, predictiveQIndex, and bipredictiveQIndex members of minQIndex **must** all be less than or equal to the respective members of maxQIndex

Valid Usage (Implicit)

- VUID-VkVideoEncodeAV1RateControlLayerInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_RATE_CONTROL_LAYER_INFO_KHR

- VUID-VkVideoEncodeAV1RateControlLayerInfoKHR-minQIndex-parameter
  `minQIndex` **must** be a valid VkVideoEncodeAV1QIndexKHR structure
- VUID-VkVideoEncodeAV1RateControlLayerInfoKHR-maxQIndex-parameter
  `maxQIndex` **must** be a valid VkVideoEncodeAV1QIndexKHR structure
- VUID-VkVideoEncodeAV1RateControlLayerInfoKHR-maxFrameSize-parameter
  `maxFrameSize` **must** be a valid VkVideoEncodeAV1FrameSizeKHR structure

Structure Chaining

Extends the structure

- VkVideoEncodeRateControlLayerInfoKHR

The `VkVideoEncodeAV1QIndexKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_av1
typedef struct VkVideoEncodeAV1QIndexKHR {
    uint32_t    intraQIndex;
    uint32_t    predictiveQIndex;
    uint32_t    bipredictiveQIndex;
} VkVideoEncodeAV1QIndexKHR;
```

- `intraQIndex` is the quantizer index to be used for frames encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_INTRA_KHR.
- `predictiveQIndex` is the quantizer index to be used for frames encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_PREDICTIVE_KHR.
- `bipredictiveQIndex` is the quantizer index to be used for frames encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_BIPREDICTIVE_KHR.

The `VkVideoEncodeAV1FrameSizeKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_av1
typedef struct VkVideoEncodeAV1FrameSizeKHR {
    uint32_t    intraFrameSize;
    uint32_t    predictiveFrameSize;
    uint32_t    bipredictiveFrameSize;
} VkVideoEncodeAV1FrameSizeKHR;
```

- `intraFrameSize` is the size in bytes to be used for frames encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_INTRA_KHR.
- `predictiveFrameSize` is the size in bytes to be used for frames encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_PREDICTIVE_KHR.
- `bipredictiveFrameSize` is the size in bytes to be used for frames encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_BIPREDICTIVE_KHR.

GOP Remaining Frames

Besides session level rate control configuration, the application **can** specify the number of frames per frame type remaining in the group of pictures (GOP).

The VkVideoEncodeAV1GopRemainingFrameInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_av1
typedef struct VkVideoEncodeAV1GopRemainingFrameInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           useGopRemainingFrames;
    uint32_t           gopRemainingIntra;
    uint32_t           gopRemainingPredictive;
    uint32_t           gopRemainingBipredictive;
} VkVideoEncodeAV1GopRemainingFrameInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- useGopRemainingFrames indicates whether the implementation's rate control algorithm **should** use the values specified in gopRemainingIntra, gopRemainingPredictive, and gopRemainingBipredictive. If useGopRemainingFrames is VK_FALSE, then the values of gopRemainingIntra, gopRemainingPredictive, and gopRemainingBipredictive are ignored.
- gopRemainingIntra specifies the number of frames encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_INTRA_KHR the implementation's rate control algorithm **should** assume to be remaining in the GOP prior to executing the next video encode operation.
- gopRemainingPredictive specifies the number of frames encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_PREDICTIVE_KHR the implementation's rate control algorithm **should** assume to be remaining in the GOP prior to executing the next video encode operation.
- gopRemainingBipredictive specifies the number of frames encoded with VK_VIDEO_ENCODE_AV1_RATE_CONTROL_GROUP_BIPREDICTIVE_KHR the implementation's rate control algorithm **should** assume to be remaining in the GOP prior to executing the next video encode operation.

Setting useGopRemainingFrames to VK_TRUE and including this structure in the pNext chain of VkVideoBeginCodingInfoKHR is only mandatory if the VkVideoEncodeAV1CapabilitiesKHR ::requiresGopRemainingFrames reported for the used video profile is VK_TRUE. However, implementations **may** use these remaining frame counts, when specified, even when it is not required. In particular, when the application does not use a regular GOP structure, these values **may** provide additional guidance for the implementation's rate control algorithm.

The VkVideoEncodeAV1CapabilitiesKHR::prefersGopRemainingFrames capability is also used to indicate that the implementation's rate control algorithm **may** operate more accurately if the application specifies the remaining frame counts using this structure.

As with other rate control guidance values, if the effective order and number of frames encoded by the application are not in line with the remaining frame counts specified in this structure at any given point, then the behavior of the implementation's rate control algorithm **may** deviate from the one expected by the application.

Valid Usage (Implicit)

- VUID-VkVideoEncodeAV1GopRemainingFrameInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_GOP_REMAINING_FRAME_INFO_KHR

Structure Chaining

Extends the structure

- VkVideoBeginCodingInfoKHR

AV1 Quantizer Index Delta Maps

Quantization delta maps used with an AV1 encode profile are referred to as *quantizer index delta maps* and their texels contain integer values representing quantizer index delta values that are applied in the process of determining the quantizer indices of the encoded picture.

Accordingly, AV1 quantizer index delta maps always have single channel integer formats, as reported in VkVideoFormatPropertiesKHR::`format`.

When the rate control mode is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, the quantizer index delta values are added to the constant quantizer index value that, in effect, enable the application to explicitly control the used quantizer index values at the granularity of the used quantization map texel size.

For all other rate control modes, the quantizer index delta values **can** be used to offset the quantizer index values that the rate control algorithm would otherwise produce.

AV1 Encode Quantization

Performing AV1 encode operations involves the process of assigning quantizer index values to individual AV1 mode info blocks. This process depends on the used rate control mode, as well as other encode and rate control parameters, as described below:

- If the configured rate control mode is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR, then the quantizer index value is initialized by the implementation-specific default rate control algorithm.
  - If the video encode operation is issued with a quantization delta map, the quantizer index delta value corresponding to the mode info block, as fetched from the quantization map, is added to the previously determined quantizer index value. If the fetched quantizer index delta value falls outside the supported quantizer index delta value range reported in the

`minQIndexDelta` and `maxQIndexDelta` members of VkVideoEncodeAV1QuantizationMapCapabilitiesKHR, then the quantizer index value used for the mode info block becomes undefined.

- If the configured rate control mode is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, then the quantizer index value is initialized from the constant quantizer index value specified for the encoded frame.
  - If the video encode operation is issued with a quantization delta map, the quantizer index delta value corresponding to the mode info block, as fetched from the quantization map, is added to the previously determined quantizer index value. If the fetched quantizer index delta value falls outside the supported quantizer index delta value range reported in the `minQIndexDelta` and `maxQIndexDelta` members of VkVideoEncodeAV1QuantizationMapCapabilitiesKHR, then the quantizer index value used for the mode info block becomes undefined.
- If the configured rate control mode is not VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR or VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR, then the quantizer index value is initialized by the corresponding rate control algorithm.
  - If the video encode operation is issued with a quantization delta map, the quantizer index delta value corresponding to the mode info block, as fetched from the quantization map, is added to the previously determined quantizer index value. If the fetched quantizer index delta value falls outside the supported quantizer index delta value range reported in the `minQIndexDelta` and `maxQIndexDelta` members of VkVideoEncodeAV1QuantizationMapCapabilitiesKHR, then the quantizer index value used for the mode info block becomes undefined.
  - If the video encode operation is issued with an emphasis map, the rate control will adjust the quantizer index value based on the emphasis value corresponding to the mode info block, as fetched from the quantization map, according to the following equation:

    $QIndex_{new} = f(QIndex_{prev}, e)$

    Where $QIndex_{new}$ is the resulting quantizer index value, $QIndex_{prev}$ is the previously determined quantizer index value, e is the emphasis value corresponding to the macroblock, and f is an implementation-defined function for which the following implication is true:

    $e_1 < e_2 \Rightarrow f(QIndex, e_1) \geq f(QIndex, e_2)$

    > **NOTE** This means that lower emphasis values will result in higher quantizer index values, whereas higher emphasis values will result in lower quantizer index values, but the function is not strictly decreasing with respect to the input emphasis value for a given input quantizer index value.

  - If clamping to minimum quantizer index values is enabled in the applied rate control layer, then the quantizer index value is clamped to the corresponding minimum quantizer index value.
  - If clamping to maximum quantizer index values is enabled in the applied rate control layer,

then the quantizer index value is clamped to the corresponding maximum quantizer index value.

- In all cases, the final quantizer index value is clamped to the minimum and maximum quantizer index values supported by the video profile.

45.22.15. AV1 Encode Requirements

This section describes the **required** AV1 encoding capabilities for physical devices that have at least one queue family that supports the video codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR, as returned by vkGetPhysicalDeviceQueueFamilyProperties2 in VkQueueFamilyVideoPropertiesKHR ::videoCodecOperations.

Table 64. Required Video Std Header Versions

Video Std Header Name	Version
vulkan_video_codec_av1std_encode	1.0.0

Table 65. Required Video Capabilities

Video Capability	Requirement	Requirement Type[1]
VkVideoCapabilitiesKHR		
flags	-	min
minBitstreamBufferOffsetAlignment	4096	max
minBitstreamBufferSizeAlignment	4096	max
pictureAccessGranularity	(64,64)	max
minCodedExtent	-	max
maxCodedExtent	-	min
maxDpbSlots	0	min
maxActiveReferencePictures	0	min
VkVideoEncodeCapabilitiesKHR		
flags	-	min
rateControlModes	VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR [8]	min
maxBitrate	5529600	min
maxQualityLevels	1	min
encodeInputPictureGranularity	(64,64)	max

Video Capability	Requirement	Requirement Type[1]
supportedEncodeFeedbackFlags	VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR + VK_VIDEO_ENCODE_FEEDBACK_AVERAGE_QUANTIZATION_BIT_KHR [10] + VK_VIDEO_ENCODE_FEEDBACK_INTRA_PIXELS_BIT_KHR [10] + VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR [10] + VK_VIDEO_ENCODE_FEEDBACK_PICTURE_PARTITION_COUNT_BIT_KHR [10]	min
VkVideoEncodeFeedback2CapabilitiesKHR		
maxPerPartitionFeedbackEntries	-	min
supportedPerPartitionEncodeFeedbackFlags	VK_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_STATUS_BIT_KHR [11] + VK_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR [11] + VK_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR [11]	min
VkVideoEncodeAV1CapabilitiesKHR		
flags	-	min
codedPictureAlignment	(8,8)	min
maxTiles	1	min

Video Capability	Requirement	Requirement Type[1]
minTileSize	-	max
maxTileSize	-	min
maxLevel	STD_VIDEO_AV1_LEVEL_2_0	min
superblockSizes	at least one bit set	implementation-dependent
maxSingleReferenceCount	0	min
singleReferenceNameMask	- [2]	min
maxUnidirectionalCompoundReferenceCount	0 [3]	min
maxUnidirectionalCompoundGroup1ReferenceCount	0 [3,4]	min
unidirectionalCompoundReferenceNameMask	- [2]	min
maxBidirectionalCompoundReferenceCount	0 [3]	min
maxBidirectionalCompoundGroup1ReferenceCount	0 [3,5]	min
maxBidirectionalCompoundGroup2ReferenceCount	0 [3,5]	min
bidirectionalCompoundReferenceNameMask	- [2]	min
maxTemporalLayerCount	1	min
maxSpatialLayerCount	1	min
maxOperatingPoints	0	min
minQIndex	-	max
maxQIndex	-	min
prefersGopRemainingFrames	-	implementation-dependent
requiresGopRemainingFrames	-	implementation-dependent
stdSyntaxFlags	-	min
VkVideoEncodeQuantizationMapCapabilitiesKHR		
maxQuantizationMapExtent	- [6]	min
VkVideoEncodeAV1QuantizationMapCapabilitiesKHR		
minQIndexDelta	- [7]	max
maxQIndexDelta	- [7]	min
VkVideoEncodeIntraRefreshCapabilitiesKHR		
intraRefreshModes	0	min
maxIntraRefreshCycleDuration	2 [9]	min
maxIntraRefreshActiveReferencePictures	1 [9]	min

Video Capability	Requirement	Requirement Type[1]
partitionIndependentIntraRefreshRegions	-	implementation-dependent
nonRectangularIntraRefreshRegions	-	implementation-dependent

1

The **Requirement Type** column specifies the requirement is either the minimum value all implementations **must** support, the maximum value all implementations **must** support, or the exact value all implementations **must** support. For bitmasks a minimum value is the least bits all implementations **must** set, but they **may** have additional bits set beyond this minimum.

2

These masks **must** only have bits set in the least significant VK_MAX_VIDEO_AV1_REFERENCES_PER_FRAME_KHR bits (bit index i indicates support for the AV1 reference name STD_VIDEO_AV1_REFERENCE_NAME_LAST_FRAME + i when using the corresponding AV1 prediction mode), and **must** have at least as many bits set in any *ReferenceNameMask capability as the value of the corresponding max*ReferenceCount capability.

3

If greater than zero, it **must** be at least 2.

4

maxUnidirectionalCompoundGroup1ReferenceCount **must** be less than or equal to maxUnidirectionalCompoundReferenceCount

5

The sum of maxBidirectionalCompoundGroup1ReferenceCount and maxBidirectionalCompoundGroup2ReferenceCount **must** be greater than or equal to maxBidirectionalCompoundReferenceCount

6

If VkVideoEncodeCapabilitiesKHR::flags includes VK_VIDEO_ENCODE_CAPABILITY_QUANTIZATION_DELTA_MAP_BIT_KHR or VK_VIDEO_ENCODE_CAPABILITY_EMPHASIS_MAP_BIT_KHR, then the width and height members of maxQuantizationMapExtent **must** be greater than zero.

7

If VkVideoEncodeCapabilitiesKHR::flags includes VK_VIDEO_ENCODE_CAPABILITY_QUANTIZATION_DELTA_MAP_BIT_KHR, then maxQIndexDelta **must** be greater than minQIndexDelta.

8

If the videoMaintenance2 feature is supported.

9

If VkVideoEncodeIntraRefreshCapabilitiesKHR::intraRefreshModes is not zero.

10

If the videoEncodeFeedback2 feature is supported.

11

If maxPerPartitionFeedbackEntries is greater than zero.

Chapter 46. Extending Vulkan

New functionality **may** be added to Vulkan via either new extensions or new versions of the core, or new versions of an extension in some cases.

This chapter describes how Vulkan is versioned, how compatibility is affected between different versions, and compatibility rules that are followed by the Vulkan Working Group.

46.1. Functionality Levels

Functionality in Vulkan is divided into several different levels; global, instance-level, physical-device-level, and device-level.

- VkInstance and any objects created from a VkInstance other than VkPhysicalDevice are instance-level.
- VkPhysicalDevice is the only physical-device-level object.
- VkDevice and any objects created from a VkDevice are device-level.

The level of a command is the same as the level of its first parameter - if the first parameter is not a dispatchable handle, it is a global command. Different levels of functionality **may** be advertised in different ways.

46.2. Instance and Device Versions

Starting with Vulkan 1.1, there are separate versions advertised for the Vulkan instance, and for each device supported on the system. This allows a system with multiple devices to advertise all devices at their full capabilities, even if those devices do not support the same version of Vulkan.

The instance version indicates which global and instance-level functionality is supported, while each device version indicates the physical-device-level and device-level functionality supported.

The instance version **can** be queried by calling vkEnumerateInstanceVersion. Querying for this function via vkGetInstanceProcAddr will return `NULL` on implementations that only support Vulkan 1.0 functionality.

The device version **can** be queried by calling vkGetPhysicalDeviceProperties or vkGetPhysicalDeviceProperties2, and is returned in VkPhysicalDeviceProperties::`apiVersion`, encoded as described in Version Numbers.

46.3. Core Versions

The Vulkan Specification is regularly updated with bug fixes and clarifications. Occasionally new functionality is added to the core and at some point it is expected that there will be a desire to perform a large, breaking change to the API. In order to indicate to developers how and when these changes are made to the specification, and to provide a way to identify each set of changes, the Vulkan API maintains a version number.

46.3.1. Version Numbers

The Vulkan version number comprises four parts indicating the variant, major, minor and patch version of the Vulkan API Specification.

The *variant* indicates the variant of the Vulkan API supported by the implementation. This is always 0 for the Vulkan API.

NOTE A non-zero variant indicates the API is a variant of the Vulkan API and applications will typically need to be modified to run against it. The variant field was a later addition to the version number, added in version 1.2.175 of the Specification. As Vulkan uses variant 0, this change is fully backwards compatible with the previous version number format for Vulkan implementations. New version number macros have been added for this change and the old macros marked as legacy. For existing applications using the older format and macros, an implementation with non-zero variant will decode as a very high Vulkan version. The high version number should be detectable by applications performing suitable version checking.

The *major version* indicates a significant change in the API, which will encompass a wholly new version of the specification.

The *minor version* indicates the incorporation of new functionality into the core specification.

The *patch version* indicates bug fixes, clarifications, and language improvements have been incorporated into the specification.

Compatibility guarantees made about versions of the API sharing any of the same version numbers are documented in Core Versions.

The version number is used in several places in the API. In each such use, the version numbers are packed into a 32-bit integer as follows:

- The variant is a 3-bit integer packed into bits 31-29.
- The major version is a 7-bit integer packed into bits 28-22.
- The minor version number is a 10-bit integer packed into bits 21-12.
- The patch version number is a 12-bit integer packed into bits 11-0.

`VK_API_VERSION_VARIANT` extracts the API variant number from a packed version number:

```
// Provided by VK_VERSION_1_0
#define VK_API_VERSION_VARIANT(version) ((uint32_t)(version) >> 29U)
```

`VK_API_VERSION_MAJOR` extracts the API major version number from a packed version number:

```
// Provided by VK_VERSION_1_0
#define VK_API_VERSION_MAJOR(version) (((uint32_t)(version) >> 22U) & 0x7FU)
```

VK_VERSION_MAJOR extracts the API major version number from a packed version number:

WARNING	This functionality is superseded by Vulkan Version 1.0. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
#define VK_VERSION_MAJOR(version) ((uint32_t)(version) >> 22U)
```

VK_API_VERSION_MINOR extracts the API minor version number from a packed version number:

```
// Provided by VK_VERSION_1_0
#define VK_API_VERSION_MINOR(version) (((uint32_t)(version) >> 12U) & 0x3FFU)
```

VK_VERSION_MINOR extracts the API minor version number from a packed version number:

WARNING	This functionality is superseded by Vulkan Version 1.0. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
#define VK_VERSION_MINOR(version) (((uint32_t)(version) >> 12U) & 0x3FFU)
```

VK_API_VERSION_PATCH extracts the API patch version number from a packed version number:

```
// Provided by VK_VERSION_1_0
#define VK_API_VERSION_PATCH(version) ((uint32_t)(version) & 0xFFFU)
```

VK_VERSION_PATCH extracts the API patch version number from a packed version number:

WARNING	This functionality is superseded by Vulkan Version 1.0. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
#define VK_VERSION_PATCH(version) ((uint32_t)(version) & 0xFFFU)
```

VK_MAKE_API_VERSION constructs an API version number.

```
// Provided by VK_VERSION_1_0
#define VK_MAKE_API_VERSION(variant, major, minor, patch) \
    ((((uint32_t)(variant)) << 29U) | (((uint32_t)(major)) << 22U) |
(((uint32_t)(minor)) << 12U) | ((uint32_t)(patch)))
```

- variant is the variant number.

- `major` is the major version number.
- `minor` is the minor version number.
- `patch` is the patch version number.

`VK_MAKE_VERSION` constructs an API version number.

WARNING	This functionality is superseded by Vulkan Version 1.0. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
#define VK_MAKE_VERSION(major, minor, patch) \
    ((((uint32_t)(major)) << 22U) | (((uint32_t)(minor)) << 12U) |
((uint32_t)(patch)))
```

- `major` is the major version number.
- `minor` is the minor version number.
- `patch` is the patch version number.

`VK_API_VERSION_1_0` returns the API version number for Vulkan 1.0.0.

```
// Provided by VK_VERSION_1_0
// Vulkan 1.0 version number
#define VK_API_VERSION_1_0 VK_MAKE_API_VERSION(0, 1, 0, 0)// Patch version should
always be set to 0
```

`VK_API_VERSION_1_1` returns the API version number for Vulkan 1.1.0.

```
// Provided by VK_VERSION_1_1
// Vulkan 1.1 version number
#define VK_API_VERSION_1_1 VK_MAKE_API_VERSION(0, 1, 1, 0)// Patch version should
always be set to 0
```

`VK_API_VERSION_1_2` returns the API version number for Vulkan 1.2.0.

```
// Provided by VK_VERSION_1_2
// Vulkan 1.2 version number
#define VK_API_VERSION_1_2 VK_MAKE_API_VERSION(0, 1, 2, 0)// Patch version should
always be set to 0
```

`VK_API_VERSION_1_3` returns the API version number for Vulkan 1.3.0.

```
// Provided by VK_VERSION_1_3
// Vulkan 1.3 version number
#define VK_API_VERSION_1_3 VK_MAKE_API_VERSION(0, 1, 3, 0)// Patch version should
always be set to 0
```

`VK_API_VERSION_1_4` returns the API version number for Vulkan 1.4.0.

```
// Provided by VK_VERSION_1_4
// Vulkan 1.4 version number
#define VK_API_VERSION_1_4 VK_MAKE_API_VERSION(0, 1, 4, 0)// Patch version should
always be set to 0
```

46.4. Layers

When a layer is enabled, it inserts itself into the call chain for Vulkan commands the layer is interested in. Layers **can** be used for a variety of tasks that extend the base behavior of Vulkan beyond what is required by the specification - such as call logging, tracing, validation, or providing additional extensions.

NOTE For example, an implementation is not expected to check that the values of enums used by the application fall within allowed ranges. Instead, a validation layer would do those checks and flag issues. This avoids a performance penalty during production use of the application because those layers would not be enabled in production.

NOTE Vulkan layers **may** wrap object handles (i.e. return a different handle value to the application than that generated by the implementation). This is generally discouraged, as it increases the probability of incompatibilities with new extensions. The validation layers wrap handles in order to track the proper use and destruction of each object. See the "Architecture of the Vulkan Loader Interfaces" document for additional information.

To query the available layers, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEnumerateInstanceLayerProperties(
    uint32_t*                                   pPropertyCount,
    VkLayerProperties*                          pProperties);
```

- `pPropertyCount` is a pointer to an integer related to the number of layer properties available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of VkLayerProperties structures.

If `pProperties` is `NULL`, then the number of layer properties available is returned in `pPropertyCount`.

Otherwise, pPropertyCount **must** point to a variable set by the application to the number of elements in the pProperties array, and on return the variable is overwritten with the number of structures actually written to pProperties. If pPropertyCount is less than the number of layer properties available, at most pPropertyCount structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available properties were returned.

The list of available layers may change at any time due to actions outside of the Vulkan implementation, so two calls to vkEnumerateInstanceLayerProperties with the same parameters **may** return different results, or retrieve different pPropertyCount values or pProperties contents. Once an instance has been created, the layers enabled for that instance will continue to be enabled and valid for the lifetime of that instance, even if some of them become unavailable for future instances.

Valid Usage (Implicit)

- VUID-vkEnumerateInstanceLayerProperties-pPropertyCount-parameter
  pPropertyCount **must** be a valid pointer to a uint32_t value
- VUID-vkEnumerateInstanceLayerProperties-pProperties-parameter
  If the value referenced by pPropertyCount is not 0, and pProperties is not NULL, pProperties **must** be a valid pointer to an array of pPropertyCount VkLayerProperties structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkLayerProperties structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkLayerProperties {
    char        layerName[VK_MAX_EXTENSION_NAME_SIZE];
    uint32_t    specVersion;
    uint32_t    implementationVersion;
    char        description[VK_MAX_DESCRIPTION_SIZE];
} VkLayerProperties;
```

- layerName is an array of VK_MAX_EXTENSION_NAME_SIZE char containing a null-terminated

UTF-8 string which is the name of the layer. Use this name in the ppEnabledLayerNames array passed in the VkInstanceCreateInfo structure to enable this layer for an instance.

- specVersion is the Vulkan version the layer was written to, encoded as described in Version Numbers.
- implementationVersion is the version of this layer. It is an integer, increasing with backward compatible changes.
- description is an array of VK_MAX_DESCRIPTION_SIZE char containing a null-terminated UTF-8 string which provides additional details that **can** be used by the application to identify the layer.

VK_MAX_EXTENSION_NAME_SIZE is the length in char values of an array containing a layer or extension name string, as returned in VkLayerProperties::layerName, VkExtensionProperties::extensionName, and other queries.

```
#define VK_MAX_EXTENSION_NAME_SIZE        256U
```

VK_MAX_DESCRIPTION_SIZE is the length in char values of an array containing a string with additional descriptive information about a query, as returned in VkLayerProperties::description and other queries.

```
#define VK_MAX_DESCRIPTION_SIZE           256U
```

To enable a layer, the name of the layer **should** be added to the ppEnabledLayerNames member of VkInstanceCreateInfo when creating a VkInstance.

Loader implementations **may** provide mechanisms outside the Vulkan API for enabling specific layers. Layers enabled through such a mechanism are *implicitly enabled*, while layers enabled by including the layer name in the ppEnabledLayerNames member of VkInstanceCreateInfo are *explicitly enabled*. Implicitly enabled layers are loaded before explicitly enabled layers, such that implicitly enabled layers are closer to the application, and explicitly enabled layers are closer to the driver. Except where otherwise specified, implicitly enabled and explicitly enabled layers differ only in the way they are enabled, and the order in which they are loaded. Explicitly enabling a layer that is implicitly enabled results in this layer being loaded as an implicitly enabled layer; it has no additional effect.

To enumerate device layers, call:

WARNING	This functionality is superseded by Vulkan Version 1.0. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
VkResult vkEnumerateDeviceLayerProperties(
    VkPhysicalDevice                            physicalDevice,
    uint32_t*                                   pPropertyCount,
    VkLayerProperties*                          pProperties);
```

- `physicalDevice` is the physical device that will be queried.
- `pPropertyCount` is a pointer to an integer related to the number of layer properties available or queried.
- `pProperties` is either `NULL` or a pointer to an array of VkLayerProperties structures.

If `pProperties` is `NULL`, then the number of layer properties available is returned in `pPropertyCount`. Otherwise, `pPropertyCount` **must** point to a variable set by the application to the number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If `pPropertyCount` is less than the number of layer properties available, at most `pPropertyCount` structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available properties were returned.

The list of layers enumerated by `vkEnumerateDeviceLayerProperties` **must** be exactly the sequence of layers enabled for the instance. The members of `VkLayerProperties` for each enumerated layer **must** be the same as the properties when the layer was enumerated by `vkEnumerateInstanceLayerProperties`.

NOTE	Due to platform details on Android, `vkEnumerateDeviceLayerProperties` may be called with `physicalDevice` equal to `NULL` during layer discovery. This behavior will only be observed by layer implementations, and not the underlying Vulkan driver.

Valid Usage (Implicit)

- VUID-vkEnumerateDeviceLayerProperties-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkEnumerateDeviceLayerProperties-pPropertyCount-parameter
  `pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkEnumerateDeviceLayerProperties-pProperties-parameter
  If the value referenced by `pPropertyCount` is not `0`, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` VkLayerProperties structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The ppEnabledLayerNames and enabledLayerCount members of VkDeviceCreateInfo are legacy and their values **must** be ignored by implementations. However, for compatibility, only an empty list of layers or a list that exactly matches the sequence enabled at instance creation time are valid, and validation layers **should** issue diagnostics for other cases.

Regardless of the enabled layer list provided in VkDeviceCreateInfo, the sequence of layers active for a device will be exactly the sequence of layers enabled when the parent instance was created.

46.5. Extensions

Extensions **may** define new Vulkan commands, structures, and enumerants. For compilation purposes, the interfaces defined by registered extensions, including new structures and enumerants as well as function pointer types for new commands, are defined in the Khronos-supplied vulkan_core.h together with the core API. However, commands defined by extensions **may** not be available for static linking - in which case function pointers to these commands **should** be queried at runtime as described in Command Function Pointers. Extensions **may** be provided by layers as well as by a Vulkan implementation.

Because extensions **may** extend or change the behavior of the Vulkan API, extension authors **should** add support for their extensions to the Khronos validation layers. This is especially important for new commands whose parameters have been wrapped by the validation layers. See the "Architecture of the Vulkan Loader Interfaces" document for additional information.

NOTE

To enable an instance extension, the name of the extension **can** be added to the ppEnabledExtensionNames member of VkInstanceCreateInfo when creating a VkInstance.

To enable a device extension, the name of the extension **can** be added to the ppEnabledExtensionNames member of VkDeviceCreateInfo when creating a VkDevice.

Physical-Device-Level functionality does not have any enabling mechanism and **can** be used as long as the VkPhysicalDevice supports the device extension as determined by vkEnumerateDeviceExtensionProperties.

Enabling an extension (with no further use of that extension) does not change the behavior of functionality exposed by the core Vulkan API or any other extension, other than making valid the use of the commands, enums and structures defined by that extension.

Valid Usage sections for individual commands and structures do not currently contain which extensions have to be enabled in order to make their use valid, although they might do so in the future. It is defined only in the Valid Usage for Extensions section.

46.5.1. Instance Extensions

Instance extensions add new global or instance-level functionality to the API, outside of the core specification. Instance extensions **may** also add physical-device-level functionality.

To query the available instance extensions, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEnumerateInstanceExtensionProperties(
    const char*                                 pLayerName,
    uint32_t*                                   pPropertyCount,
    VkExtensionProperties*                      pProperties);
```

- `pLayerName` is either `NULL` or a pointer to a null-terminated UTF-8 string naming the layer to retrieve extensions from.
- `pPropertyCount` is a pointer to an integer related to the number of extension properties available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of VkExtensionProperties structures.

When the `pLayerName` parameter is `NULL`, only extensions provided by the Vulkan implementation or by implicitly enabled layers are returned. When `pLayerName` is the name of a layer, the instance extensions provided by that layer are returned.

If `pProperties` is `NULL`, then the number of extension properties available is returned in `pPropertyCount`. Otherwise, `pPropertyCount` **must** point to a variable set by the application to the number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If `pPropertyCount` is less than the number of extension properties available, at most `pPropertyCount` structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available properties were returned.

Because the list of available layers may change externally between calls to vkEnumerateInstanceExtensionProperties, two calls may retrieve different results if a `pLayerName` is available in one call but not in another. The extensions supported by a layer may also change between two calls, e.g. if the layer implementation is replaced by a different version between those calls.

Implementations **must** not advertise any pair of extensions that cannot be enabled together due to behavioral differences, or any extension that cannot be enabled against the advertised version.

Valid Usage (Implicit)

- VUID-vkEnumerateInstanceExtensionProperties-pLayerName-parameter
  If `pLayerName` is not `NULL`, `pLayerName` **must** be a null-terminated UTF-8 string
- VUID-vkEnumerateInstanceExtensionProperties-pPropertyCount-parameter
  `pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkEnumerateInstanceExtensionProperties-pProperties-parameter
  If the value referenced by `pPropertyCount` is not `0`, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` VkExtensionProperties structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_LAYER_NOT_PRESENT
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

46.5.2. Device Extensions

Device extensions add new device-level functionality to the API, outside of the core specification. If Vulkan 1.1 or VK_KHR_get_physical_device_properties2 is required by the extension, it **may** also add new physical-device-level functionality.

To query the extensions available to a given physical device, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEnumerateDeviceExtensionProperties(
    VkPhysicalDevice                            physicalDevice,
    const char*                                 pLayerName,
    uint32_t*                                   pPropertyCount,
    VkExtensionProperties*                      pProperties);
```

- `physicalDevice` is the physical device that will be queried.
- `pLayerName` is either `NULL` or a pointer to a null-terminated UTF-8 string naming the layer to retrieve extensions from.
- `pPropertyCount` is a pointer to an integer related to the number of extension properties available or queried, and is treated in the same fashion as the vkEnumerateInstanceExtensionProperties ::`pPropertyCount` parameter.
- `pProperties` is either `NULL` or a pointer to an array of VkExtensionProperties structures.

When the `pLayerName` parameter is `NULL`, only extensions provided by the Vulkan implementation or by implicitly enabled layers are returned. When `pLayerName` is the name of a layer, the device extensions provided by that layer are returned.

Implementations **must** not advertise any pair of extensions that cannot be enabled together due to behavioral differences, or any extension that cannot be enabled against the advertised version.

If the `VK_KHR_ray_tracing_pipeline` extension is advertised as supported by this query, the

VK_KHR_pipeline_library extension **must** also be supported.

If the VK_EXT_descriptor_heap extension is advertised as supported by this query, the VK_KHR_shader_untyped_pointers extension **must** also be supported.

NOTE	Due to platform details on Android, vkEnumerateDeviceExtensionProperties may be called with physicalDevice equal to NULL during layer discovery. This behavior will only be observed by layer implementations, and not the underlying Vulkan driver.

Valid Usage (Implicit)

- VUID-vkEnumerateDeviceExtensionProperties-physicalDevice-parameter
  physicalDevice **must** be a valid VkPhysicalDevice handle
- VUID-vkEnumerateDeviceExtensionProperties-pLayerName-parameter
  If pLayerName is not NULL, pLayerName **must** be a null-terminated UTF-8 string
- VUID-vkEnumerateDeviceExtensionProperties-pPropertyCount-parameter
  pPropertyCount **must** be a valid pointer to a uint32_t value
- VUID-vkEnumerateDeviceExtensionProperties-pProperties-parameter
  If the value referenced by pPropertyCount is not 0, and pProperties is not NULL, pProperties **must** be a valid pointer to an array of pPropertyCount VkExtensionProperties structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_LAYER_NOT_PRESENT
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkExtensionProperties structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkExtensionProperties {
    char        extensionName[VK_MAX_EXTENSION_NAME_SIZE];
    uint32_t    specVersion;
} VkExtensionProperties;
```

- `extensionName` is an array of VK_MAX_EXTENSION_NAME_SIZE `char` containing a null-terminated UTF-8 string which is the name of the extension.
- `specVersion` is the version of this extension. It is an integer, incremented with backward compatible changes.

46.6. Extension Dependencies

Some extensions are dependent on other extensions, or on specific core API versions, to function. To enable extensions with dependencies, any *required extensions* **must** also be enabled through the same API mechanisms when creating an instance with vkCreateInstance or a device with vkCreateDevice. Each extension which has such dependencies documents them in the appendix summarizing that extension.

If an extension is supported (as queried by vkEnumerateInstanceExtensionProperties or vkEnumerateDeviceExtensionProperties), then *required extensions* of that extension **must** also be supported for the same instance or physical device.

Any device extension that has an instance extension dependency that is not enabled by vkCreateInstance is considered to be unsupported, hence it **must** not be returned by vkEnumerateDeviceExtensionProperties for any VkPhysicalDevice child of the instance. Instance extensions do not have dependencies on device extensions.

If a required extension has been promoted to another extension or to a core API version, then as a *general* rule, the dependency is also satisfied by the promoted extension or core version. This will be true so long as any features required by the original extension are also required or enabled by the promoted extension or core version. However, in some cases an extension is promoted while making some of its features optional in the promoted extension or core version. In this case, the dependency **may** not be satisfied. The only way to be certain is to look at the descriptions of the original dependency and the promoted version in the Layers & Extensions and Core Revisions appendices.

NOTE

There is metadata in `vk.xml` describing some aspects of promotion, especially `requires`, `promotedto` and `deprecatedby` attributes of `<extension>` tags. However, the metadata does not yet fully describe this scenario. In the future, we may extend the XML schema to describe the full set of extensions and versions satisfying a dependency. As discussed in more detail for Promotion below, when an extension is promoted it does not mean that a mechanical substitution of an extension API by the corresponding promoted API will work in exactly the same fashion; be supported at runtime; or even exist.

46.7. Compatibility Guarantees (Informative)

This section is marked as informal as there is no binding responsibility on implementations of the Vulkan API - these guarantees are however a contract between the Vulkan Working Group and developers using this Specification.

46.7.1. Core Versions

Each of the major, minor, and patch versions of the Vulkan specification provides different compatibility guarantees.

Patch Versions

A difference in the patch version indicates that a set of bug fixes or clarifications have been made to the Specification. Informative enums returned by Vulkan commands that will not affect the runtime behavior of a valid application may be added in a patch version (e.g. VkVendorId).

The specification's patch version is strictly increasing for a given major version of the specification; any change to a specification as described above will result in the patch version being increased by 1. Patch versions are applied to all minor versions, even if a given minor version is not affected by the provoking change.

Specifications with different patch versions but the same major and minor version are *fully compatible* with each other - such that a valid application written against one will work with an implementation of another.

NOTE	If a patch version includes a bug fix or clarification that could have a significant impact on developer expectations, these will be highlighted in the change log. Generally the Vulkan Working Group tries to avoid these kinds of changes, instead fixing them in either an extension or core version.

Minor Versions

Changes in the minor version of the specification indicate that new functionality has been added to the core specification. This will usually include new interfaces in the header, and **may** also include behavior changes and bug fixes. Core functionality **may** be marked as legacy in a minor version, but will not be obsoleted or removed.

The specification's minor version is strictly increasing for a given major version of the specification; any change to a specification as described above will result in the minor version being increased by 1. Changes that can be accommodated in a patch version will not increase the minor version.

Specifications with a lower minor version are *backwards compatible* with an implementation of a specification with a higher minor version for core functionality and extensions issued with the KHR vendor tag. Vendor and multi-vendor extensions are not guaranteed to remain functional across minor versions, though in general they are with few exceptions - see Obsoletion for more information.

Major Versions

A difference in the major version of specifications indicates a large set of changes which will likely include interface changes, behavioral changes, removal of legacy functionality, and the modification, addition, or replacement of other functionality.

The specification's major version is monotonically increasing; any change to the specification as

described above will result in the major version being increased. Changes that can be accommodated in a patch or minor version will not increase the major version.

The Vulkan Working Group intends to only issue a new major version of the Specification in order to realize significant improvements to the Vulkan API that will necessarily require breaking compatibility.

A new major version will likely include a wholly new version of the specification to be issued - which could include an overhaul of the versioning semantics for the minor and patch versions. The patch and minor versions of a specification are therefore not meaningful across major versions. If a major version of the specification includes similar versioning semantics, it is expected that the patch and the minor version will be reset to 0 for that major version.

46.7.2. Extensions

A KHR extension **must** be able to be enabled alongside any other KHR extension, and for any minor or patch version of the core Specification beyond the minimum version it requires. A multi-vendor extension **should** be able to be enabled alongside any KHR extension or other multi-vendor extension, and for any minor or patch version of the core Specification beyond the minimum version it requires. A vendor extension **should** be able to be enabled alongside any KHR extension, multi-vendor extension, or other vendor extension from the same vendor, and for any minor or patch version of the core Specification beyond the minimum version it requires. A vendor extension **may** be able to be enabled alongside vendor extensions from another vendor.

The one other exception to this is if a vendor or multi-vendor extension is made obsolete by either a core version or another extension, which will be highlighted in the extension appendix.

Promotion

Extensions, or features of an extension, **may** be promoted to a new core version of the API, or a newer extension which an equal or greater number of implementors are in favor of.

NOTE	Structure and enumerated types defined in extensions that were promoted to core in a later version of Vulkan are defined in terms of the equivalent Vulkan core interfaces in that version and its successors. This affects the Vulkan Specification, the Vulkan header files, and the corresponding XML Registry.

When extension functionality is promoted, minor changes **may** be introduced, limited to the following:

- Naming
- Non-intrusive parameter changes
- Feature advertisement/enablement
- Combining structure parameters into larger structures
- Author ID suffixes changed or removed

NOTE	If extension functionality is promoted, there is no guarantee of direct compatibility,

however it should require little effort to port code from the original feature to the promoted one.

The Vulkan Working Group endeavors to ensure that larger changes are marked as either deprecated or obsoleted as appropriate, and can do so retroactively if necessary.

Extensions that are promoted are listed as being promoted in their extension appendices, with reference to where they were promoted to.

When an extension is promoted, any backwards compatibility aliases which exist in the extension will **not** be promoted.

NOTE As a hypothetical example, if the `VK_KHR_surface` extension were promoted to part of a future core version, the `VK_COLOR_SPACE_SRGB_NONLINEAR_KHR` token defined by that extension would be promoted to `VK_COLOR_SPACE_SRGB_NONLINEAR`. However, the `VK_COLORSPACE_SRGB_NONLINEAR_KHR` token aliases `VK_COLOR_SPACE_SRGB_NONLINEAR_KHR`. The `VK_COLORSPACE_SRGB_NONLINEAR_KHR` would not be promoted, because it is a backwards compatibility alias that exists only due to a naming mistake when the extension was initially published.

Deprecation

Extensions **may** be marked as deprecated when they are no longer intended to be used by applications under certain conditions. Generally, a new feature will become available to solve the use case in another extension or core version of the API, but it is not guaranteed.

NOTE Features that are intended to replace deprecated functionality have no guarantees of compatibility, and applications may require drastic modification in order to make use of the new features.

Extensions that are deprecated are listed as being deprecated in their extension appendices, with an explanation of the deprecation and any related features that are relevant.

Obsoletion

Occasionally, an extension will be marked as obsolete if a new version of the core API or a new extension is fundamentally incompatible with it. An obsoleted extension **must** not be used with the extension or core version that obsoleted it.

Extensions that are obsoleted are listed as being obsoleted in their extension appendices, with reference to what they were obsoleted by.

Aliases

When an extension is promoted or deprecated by a newer feature, some or all of its functionality **may** be replicated into the newer feature. Rather than duplication of all the documentation and definitions, the specification instead identifies the identical commands and types as *aliases* of one another. Each alias is mentioned together with the definition it aliases, with the older aliases

marked as “equivalents”. Each alias of the same command has identical behavior, and each alias of the same type has identical meaning - they can be used interchangeably in an application with no compatibility issues.

NOTE

For promoted types, the aliased extension type is semantically identical to the new core type. The C99 headers simply `typedef` the older aliases to the promoted types.

For promoted command aliases, however, there are two separate command definitions, due to the fact that the C99 ABI has no way to alias command definitions without resorting to macros. Calling either command will produce identical behavior within the bounds of the specification, and should still invoke the same path in the implementation. Debug tools may use separate commands with different debug behavior; to write the appropriate command name to an output log, for instance.

Special Use Extensions

Some extensions exist only to support a specific purpose or specific class of application. These are referred to as “special use extensions”. Use of these extensions in applications not meeting the special use criteria is not recommended.

Special use cases are restricted, and only those defined below are used to describe extensions:

Table 66. Extension Special Use Cases

Special Use	XML Tag	Full Description
CAD support	cadsupport	Extension is intended to support specialized functionality used by CAD/CAM applications.
D3D support	d3demulation	Extension is intended to support D3D emulation layers, and applications ported from D3D, by adding functionality specific to D3D.
Developer tools	devtools	Extension is intended to support developer tools such as capture-replay libraries.
Debugging tools	debugging	Extension is intended for use by applications when debugging.
OpenGL / ES support	glemulation	Extension is intended to support OpenGL and/or OpenGL ES emulation layers, and applications ported from those APIs, by adding functionality specific to those APIs.

Special use extensions are identified in the metadata for each such extension in the Layers & Extensions appendix, using the name in the “Special Use” column above.

Special use extensions are also identified in `vk.xml` with the short name in the “XML Tag” column above, as described in the “API Extensions (`extension` tag)” section of the registry schema documentation.

Chapter 47. Features

Features advertise additional functionality which **can** be enabled in the API. If a feature is *supported* by a physical device, it **must** be *enabled* when creating a VkDevice in order to use its associated functionality with that VkDevice. If a feature is *unsupported*, that functionality **cannot** be used with that VkDevice.

NOTE Features are reported via the extensible structure `VkPhysicalDeviceFeatures2`, which was added in the `VK_KHR_get_physical_device_properties2` extension and included in Vulkan 1.1. When new features are added in future Vulkan versions or extensions, each extension **should** introduce one new feature structure, if needed. This structure **can** be added to the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure.

For convenience, new core versions of Vulkan **may** introduce new unified feature structures for features promoted from extensions. At the same time, the extension's original feature structure (if any) is also promoted to the core API, and is an alias of the extension's structure. This results in multiple names for the same feature: in the original extension's feature structure and the promoted structure alias, in the unified feature structure. When a feature was implicitly supported and enabled in the extension, but an explicit name was added during promotion, then the extension itself acts as an alias for the feature as listed in the table below.

All aliases of the same feature in the core API **must** be reported consistently: either all **must** be reported as supported, or none of them. When a promoted extension is available, any corresponding feature aliases **must** be supported.

Table 67. Extension Feature Aliases

Extension	Feature(s)
`VK_KHR_shader_draw_parameters`	`shaderDrawParameters`
`VK_KHR_draw_indirect_count`	`drawIndirectCount`
`VK_KHR_sampler_mirror_clamp_to_edge`	`samplerMirrorClampToEdge`
`VK_EXT_descriptor_indexing`	`descriptorIndexing`
`VK_KHR_push_descriptor`	`pushDescriptor`

To query supported features, call:

WARNING This functionality is superseded by vkGetPhysicalDeviceFeatures2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceFeatures(
    VkPhysicalDevice                            physicalDevice,
    VkPhysicalDeviceFeatures*                   pFeatures);
```

- `physicalDevice` is the physical device from which to query the supported features.

- `pFeatures` is a pointer to a VkPhysicalDeviceFeatures structure in which the physical device features are returned. For each feature, a value of VK_TRUE specifies that the feature is supported on this physical device, and VK_FALSE specifies that the feature is not supported.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceFeatures-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceFeatures-pFeatures-parameter
  `pFeatures` **must** be a valid pointer to a VkPhysicalDeviceFeatures structure

Fine-grained features used by a logical device **must** be enabled at `VkDevice` creation time. If a feature is enabled that the physical device does not support, `VkDevice` creation will fail and return VK_ERROR_FEATURE_NOT_PRESENT.

The fine-grained features are enabled by passing a pointer to the `VkPhysicalDeviceFeatures` structure via the `pEnabledFeatures` member of the VkDeviceCreateInfo structure that is passed into the `vkCreateDevice` call. If a member of `pEnabledFeatures` is VK_TRUE or VK_FALSE, then the device will be created with the indicated feature enabled or disabled, respectively. Features **can** also be enabled by using the VkPhysicalDeviceFeatures2 structure.

If an application wishes to enable all features supported by a device, it **can** simply pass in the `VkPhysicalDeviceFeatures` structure that was previously returned by `vkGetPhysicalDeviceFeatures`. To disable an individual feature, the application **can** set the desired member to VK_FALSE in the same structure. Setting `pEnabledFeatures` to `NULL` and not including a VkPhysicalDeviceFeatures2 in the `pNext` chain of VkDeviceCreateInfo is equivalent to setting all members of the structure to VK_FALSE.

NOTE	Some features, such as `robustBufferAccess`, **may** incur a runtime performance cost. Application writers **should** carefully consider the implications of enabling all supported features.

To query supported features defined by the core or extensions, call:

```
// Provided by VK_VERSION_1_1
void vkGetPhysicalDeviceFeatures2(
    VkPhysicalDevice                            physicalDevice,
    VkPhysicalDeviceFeatures2*                  pFeatures);
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to vkGetPhysicalDeviceFeatures2
void vkGetPhysicalDeviceFeatures2KHR(
    VkPhysicalDevice                            physicalDevice,
    VkPhysicalDeviceFeatures2*                  pFeatures);
```

- `physicalDevice` is the physical device from which to query the supported features.
- `pFeatures` is a pointer to a VkPhysicalDeviceFeatures2 structure in which the physical device features are returned.

Each structure in `pFeatures` and its `pNext` chain contains members corresponding to fine-grained features. Each structure in `pFeatures` and its `pNext` chain contains VkBool32 members corresponding to fine-grained features. Each such member is returned with a VK_TRUE value indicating that feature is supported on this physical device, or a VK_FALSE value indicating it is unsupported.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceFeatures2-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceFeatures2-pFeatures-parameter
  `pFeatures` **must** be a valid pointer to a VkPhysicalDeviceFeatures2 structure

The `VkPhysicalDeviceFeatures2` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceFeatures2 {
    VkStructureType             sType;
    void*                       pNext;
    VkPhysicalDeviceFeatures    features;
} VkPhysicalDeviceFeatures2;
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to VkPhysicalDeviceFeatures2
typedef VkPhysicalDeviceFeatures2 VkPhysicalDeviceFeatures2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `features` is a VkPhysicalDeviceFeatures structure describing the fine-grained features of the Vulkan 1.0 API.

The `pNext` chain of this structure is used to extend the structure with features defined by extensions. This structure **can** be used in vkGetPhysicalDeviceFeatures2 or **can** be included in the `pNext` chain of a VkDeviceCreateInfo structure, in which case it controls which features are enabled in lieu of `pEnabledFeatures`.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFeatures2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FEATURES_2

Structure Chaining

Extends the structure

- VkDeviceCreateInfo

The VkPhysicalDeviceFeatures structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPhysicalDeviceFeatures {
    VkBool32    robustBufferAccess;
    VkBool32    fullDrawIndexUint32;
    VkBool32    imageCubeArray;
    VkBool32    independentBlend;
    VkBool32    geometryShader;
    VkBool32    tessellationShader;
    VkBool32    sampleRateShading;
    VkBool32    dualSrcBlend;
    VkBool32    logicOp;
    VkBool32    multiDrawIndirect;
    VkBool32    drawIndirectFirstInstance;
    VkBool32    depthClamp;
    VkBool32    depthBiasClamp;
    VkBool32    fillModeNonSolid;
    VkBool32    depthBounds;
    VkBool32    wideLines;
    VkBool32    largePoints;
    VkBool32    alphaToOne;
    VkBool32    multiViewport;
    VkBool32    samplerAnisotropy;
    VkBool32    textureCompressionETC2;
    VkBool32    textureCompressionASTC_LDR;
    VkBool32    textureCompressionBC;
    VkBool32    occlusionQueryPrecise;
    VkBool32    pipelineStatisticsQuery;
    VkBool32    vertexPipelineStoresAndAtomics;
    VkBool32    fragmentStoresAndAtomics;
    VkBool32    shaderTessellationAndGeometryPointSize;
    VkBool32    shaderImageGatherExtended;
    VkBool32    shaderStorageImageExtendedFormats;
    VkBool32    shaderStorageImageMultisample;
    VkBool32    shaderStorageImageReadWithoutFormat;
    VkBool32    shaderStorageImageWriteWithoutFormat;
    VkBool32    shaderUniformBufferArrayDynamicIndexing;
    VkBool32    shaderSampledImageArrayDynamicIndexing;
    VkBool32    shaderStorageBufferArrayDynamicIndexing;
    VkBool32    shaderStorageImageArrayDynamicIndexing;
    VkBool32    shaderClipDistance;
    VkBool32    shaderCullDistance;
    VkBool32    shaderFloat64;
```

```
    VkBool32    shaderInt64;
    VkBool32    shaderInt16;
    VkBool32    shaderResourceResidency;
    VkBool32    shaderResourceMinLod;
    VkBool32    sparseBinding;
    VkBool32    sparseResidencyBuffer;
    VkBool32    sparseResidencyImage2D;
    VkBool32    sparseResidencyImage3D;
    VkBool32    sparseResidency2Samples;
    VkBool32    sparseResidency4Samples;
    VkBool32    sparseResidency8Samples;
    VkBool32    sparseResidency16Samples;
    VkBool32    sparseResidencyAliased;
    VkBool32    variableMultisampleRate;
    VkBool32    inheritedQueries;
} VkPhysicalDeviceFeatures;
```

This structure describes the following features:

- `robustBufferAccess` enables Robust Buffer Access guarantees for shader buffer accesses.
- `fullDrawIndexUint32` specifies the full 32-bit range of indices is supported for indexed draw calls when using a VkIndexType of VK_INDEX_TYPE_UINT32. `maxDrawIndexedIndexValue` is the maximum index value that **may** be used (aside from the primitive restart index, which is always 2^{32}-1 when the VkIndexType is VK_INDEX_TYPE_UINT32). If this feature is supported, `maxDrawIndexedIndexValue` **must** be 2^{32}-1; otherwise it **must** be no smaller than 2^{24}-1. See `maxDrawIndexedIndexValue`.
- `imageCubeArray` specifies whether image views with a VkImageViewType of VK_IMAGE_VIEW_TYPE_CUBE_ARRAY **can** be created, and that the corresponding `SampledCubeArray` and `ImageCubeArray` SPIR-V capabilities **can** be used in shader code.
- `independentBlend` specifies whether the `VkPipelineColorBlendAttachmentState` settings are controlled independently per-attachment. If this feature is not enabled, the `VkPipelineColorBlendAttachmentState` settings for all color attachments **must** be identical. Otherwise, a different `VkPipelineColorBlendAttachmentState` **can** be provided for each bound color attachment.
- `geometryShader` specifies whether geometry shaders are supported. If this feature is not enabled, the VK_SHADER_STAGE_GEOMETRY_BIT and VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT enum values **must** not be used. This also specifies whether shader modules **can** declare the `Geometry` capability.
- `tessellationShader` specifies whether tessellation control and evaluation shaders are supported. If this feature is not enabled, the VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT, VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT, VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT, and VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_STATE_CREATE_INFO enum values **must** not be used. This also specifies whether shader modules **can** declare the `Tessellation` capability.
- `sampleRateShading` specifies whether Sample Shading and multisample interpolation are

supported. If this feature is not enabled, the sampleShadingEnable member of the VkPipelineMultisampleStateCreateInfo structure **must** be VK_FALSE and the minSampleShading member is ignored. This also specifies whether shader modules **can** declare the SampleRateShading capability.

- dualSrcBlend specifies whether blend operations which take two sources are supported. If this feature is not enabled, the VK_BLEND_FACTOR_SRC1_COLOR, VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR, VK_BLEND_FACTOR_SRC1_ALPHA, and VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA enum values **must** not be used as source or destination blending factors. See Dual-Source Blending.
- logicOp specifies whether logic operations are supported. If this feature is not enabled, the logicOpEnable member of the VkPipelineColorBlendStateCreateInfo structure **must** be VK_FALSE, and the logicOp member is ignored.
- multiDrawIndirect specifies whether multiple draw indirect is supported. If this feature is not enabled, the drawCount parameter to the vkCmdDrawIndirect and vkCmdDrawIndexedIndirect commands **must** be 0 or 1. The maxDrawIndirectCount member of the VkPhysicalDeviceLimits structure **must** also be 1 if this feature is not supported. See maxDrawIndirectCount.
- drawIndirectFirstInstance specifies whether indirect drawing calls support the firstInstance parameter. If this feature is not enabled, the firstInstance member of all VkDrawIndirectCommand and VkDrawIndexedIndirectCommand structures that are provided to the vkCmdDrawIndirect and vkCmdDrawIndexedIndirect commands **must** be 0.
- depthClamp specifies whether depth clamping is supported. If this feature is not enabled, the depthClampEnable member of the VkPipelineRasterizationStateCreateInfo structure **must** be VK_FALSE. Otherwise, setting depthClampEnable to VK_TRUE will enable depth clamping.
- depthBiasClamp specifies whether depth bias clamping is supported. If this feature is not enabled, the depthBiasClamp member of the VkPipelineRasterizationStateCreateInfo structure **must** be 0.0 unless the VK_DYNAMIC_STATE_DEPTH_BIAS dynamic state is enabled, in which case the depthBiasClamp parameter to vkCmdSetDepthBias **must** be 0.0.
- fillModeNonSolid specifies whether point and wireframe fill modes are supported. If this feature is not enabled, the VK_POLYGON_MODE_POINT and VK_POLYGON_MODE_LINE enum values **must** not be used.
- depthBounds specifies whether depth bounds tests are supported. If this feature is not enabled, the depthBoundsTestEnable member of the VkPipelineDepthStencilStateCreateInfo structure **must** be VK_FALSE unless the VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE dynamic state is enabled, in which case the depthBoundsTestEnable parameter to vkCmdSetDepthBoundsTestEnable **must** be VK_FALSE. When depthBoundsTestEnable is VK_FALSE, the minDepthBounds and maxDepthBounds members of the VkPipelineDepthStencilStateCreateInfo structure are ignored.
- wideLines specifies whether lines with width other than 1.0 are supported. If this feature is not enabled, the lineWidth member of the VkPipelineRasterizationStateCreateInfo structure **must** be 1.0 unless the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state is enabled, in which case the lineWidth parameter to vkCmdSetLineWidth **must** be 1.0. When this feature is supported, the range and granularity of supported line widths are indicated by the lineWidthRange and lineWidthGranularity members of the VkPhysicalDeviceLimits structure, respectively.
- largePoints specifies whether points with size greater than 1.0 are supported. If this feature is

not enabled, only a point size of 1.0 written by a shader is supported. The range and granularity of supported point sizes are indicated by the `pointSizeRange` and `pointSizeGranularity` members of the `VkPhysicalDeviceLimits` structure, respectively.

- `alphaToOne` specifies whether the implementation is able to replace the alpha value of the fragment shader color output in the Multisample Coverage fragment operation. If this feature is not enabled, then the `alphaToOneEnable` member of the VkPipelineMultisampleStateCreateInfo structure **must** be VK_FALSE. Otherwise setting `alphaToOneEnable` to VK_TRUE will enable alpha-to-one behavior.
- `multiViewport` specifies whether more than one viewport is supported. If this feature is not enabled:
  - The `viewportCount` and `scissorCount` members of the VkPipelineViewportStateCreateInfo structure **must** be 1.
  - The `firstViewport` and `viewportCount` parameters to the vkCmdSetViewport command **must** be 0 and 1, respectively.
  - The `firstScissor` and `scissorCount` parameters to the vkCmdSetScissor command **must** be 0 and 1, respectively.
- `samplerAnisotropy` specifies whether anisotropic filtering is supported. If this feature is not enabled, the `anisotropyEnable` member of the VkSamplerCreateInfo structure **must** be VK_FALSE.
- `textureCompressionETC2` specifies whether all of the ETC2 and EAC compressed texture formats are supported. If this feature is enabled, then the VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT, VK_FORMAT_FEATURE_BLIT_SRC_BIT and VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT features **must** be supported in `optimalTilingFeatures` for the following formats:
  - VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK
  - VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK
  - VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK
  - VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK
  - VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK
  - VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK
  - VK_FORMAT_EAC_R11_UNORM_BLOCK
  - VK_FORMAT_EAC_R11_SNORM_BLOCK
  - VK_FORMAT_EAC_R11G11_UNORM_BLOCK
  - VK_FORMAT_EAC_R11G11_SNORM_BLOCK

  To query for additional properties, or if the feature is not enabled, vkGetPhysicalDeviceFormatProperties and vkGetPhysicalDeviceImageFormatProperties **can** be used to check for supported properties of individual formats as normal.
- `textureCompressionASTC_LDR` specifies whether all of the ASTC LDR compressed texture formats are supported. If this feature is enabled, then the VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT, VK_FORMAT_FEATURE_BLIT_SRC_BIT and

VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT features **must** be supported in `optimalTilingFeatures` for the following formats:

- VK_FORMAT_ASTC_4x4_UNORM_BLOCK
- VK_FORMAT_ASTC_4x4_SRGB_BLOCK
- VK_FORMAT_ASTC_5x4_UNORM_BLOCK
- VK_FORMAT_ASTC_5x4_SRGB_BLOCK
- VK_FORMAT_ASTC_5x5_UNORM_BLOCK
- VK_FORMAT_ASTC_5x5_SRGB_BLOCK
- VK_FORMAT_ASTC_6x5_UNORM_BLOCK
- VK_FORMAT_ASTC_6x5_SRGB_BLOCK
- VK_FORMAT_ASTC_6x6_UNORM_BLOCK
- VK_FORMAT_ASTC_6x6_SRGB_BLOCK
- VK_FORMAT_ASTC_8x5_UNORM_BLOCK
- VK_FORMAT_ASTC_8x5_SRGB_BLOCK
- VK_FORMAT_ASTC_8x6_UNORM_BLOCK
- VK_FORMAT_ASTC_8x6_SRGB_BLOCK
- VK_FORMAT_ASTC_8x8_UNORM_BLOCK
- VK_FORMAT_ASTC_8x8_SRGB_BLOCK
- VK_FORMAT_ASTC_10x5_UNORM_BLOCK
- VK_FORMAT_ASTC_10x5_SRGB_BLOCK
- VK_FORMAT_ASTC_10x6_UNORM_BLOCK
- VK_FORMAT_ASTC_10x6_SRGB_BLOCK
- VK_FORMAT_ASTC_10x8_UNORM_BLOCK
- VK_FORMAT_ASTC_10x8_SRGB_BLOCK
- VK_FORMAT_ASTC_10x10_UNORM_BLOCK
- VK_FORMAT_ASTC_10x10_SRGB_BLOCK
- VK_FORMAT_ASTC_12x10_UNORM_BLOCK
- VK_FORMAT_ASTC_12x10_SRGB_BLOCK
- VK_FORMAT_ASTC_12x12_UNORM_BLOCK
- VK_FORMAT_ASTC_12x12_SRGB_BLOCK

To query for additional properties, or if the feature is not enabled, vkGetPhysicalDeviceFormatProperties and vkGetPhysicalDeviceImageFormatProperties **can** be used to check for supported properties of individual formats as normal.

- `textureCompressionBC` specifies whether all of the BC compressed texture formats are supported. If this feature is enabled, then the VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT,

VK_FORMAT_FEATURE_BLIT_SRC_BIT and VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT features **must** be supported in `optimalTilingFeatures` for the following formats:

- VK_FORMAT_BC1_RGB_UNORM_BLOCK
- VK_FORMAT_BC1_RGB_SRGB_BLOCK
- VK_FORMAT_BC1_RGBA_UNORM_BLOCK
- VK_FORMAT_BC1_RGBA_SRGB_BLOCK
- VK_FORMAT_BC2_UNORM_BLOCK
- VK_FORMAT_BC2_SRGB_BLOCK
- VK_FORMAT_BC3_UNORM_BLOCK
- VK_FORMAT_BC3_SRGB_BLOCK
- VK_FORMAT_BC4_UNORM_BLOCK
- VK_FORMAT_BC4_SNORM_BLOCK
- VK_FORMAT_BC5_UNORM_BLOCK
- VK_FORMAT_BC5_SNORM_BLOCK
- VK_FORMAT_BC6H_UFLOAT_BLOCK
- VK_FORMAT_BC6H_SFLOAT_BLOCK
- VK_FORMAT_BC7_UNORM_BLOCK
- VK_FORMAT_BC7_SRGB_BLOCK

  To query for additional properties, or if the feature is not enabled, vkGetPhysicalDeviceFormatProperties and vkGetPhysicalDeviceImageFormatProperties **can** be used to check for supported properties of individual formats as normal.

- `occlusionQueryPrecise` specifies whether occlusion queries returning actual sample counts are supported. Occlusion queries are created in a `VkQueryPool` by specifying the `queryType` of VK_QUERY_TYPE_OCCLUSION in the VkQueryPoolCreateInfo structure which is passed to vkCreateQueryPool. If this feature is enabled, queries of this type **can** enable VK_QUERY_CONTROL_PRECISE_BIT in the `flags` parameter to vkCmdBeginQuery. If this feature is not supported, the implementation supports only boolean occlusion queries. When any samples are passed, boolean queries will return a non-zero result value, otherwise a result value of zero is returned. When this feature is enabled and VK_QUERY_CONTROL_PRECISE_BIT is set, occlusion queries will report the actual number of samples passed.
- `pipelineStatisticsQuery` specifies whether the pipeline statistics queries are supported. If this feature is not enabled, queries of type VK_QUERY_TYPE_PIPELINE_STATISTICS **cannot** be created, and none of the VkQueryPipelineStatisticFlagBits bits **can** be set in the `pipelineStatistics` member of the VkQueryPoolCreateInfo structure.
- `vertexPipelineStoresAndAtomics` specifies whether storage buffers and images support stores and atomic operations in the vertex, tessellation, and geometry shader stages. If this feature is not enabled, all storage image, storage texel buffer, and storage buffer variables used by these stages in shader modules **must** be decorated with the `NonWritable` decoration (or the `readonly`

memory qualifier in GLSL).

- `fragmentStoresAndAtomics` specifies whether storage buffers and images support stores and atomic operations in the fragment shader stage. If this feature is not enabled, all storage image, storage texel buffer, and storage buffer variables used by the fragment stage in shader modules **must** be decorated with the `NonWritable` decoration (or the `readonly` memory qualifier in GLSL).
- `shaderTessellationAndGeometryPointSize` specifies whether the `PointSize` built-in decoration is available in the tessellation control, tessellation evaluation, and geometry shader stages. If this feature is not enabled, members decorated with the `PointSize` built-in decoration **must** not be read from or written to and all points written from a tessellation or geometry shader will have a size of 1.0. This also specifies whether shader modules **can** declare the `TessellationPointSize` capability for tessellation control and evaluation shaders, or if the shader modules **can** declare the `GeometryPointSize` capability for geometry shaders. An implementation supporting this feature **must** also support one or both of the `tessellationShader` or `geometryShader` features.
- `shaderImageGatherExtended` specifies whether the extended set of image gather instructions are available in shader code. If this feature is not enabled, the `OpImage*Gather` instructions do not support the `Offset` and `ConstOffsets` operands. This also specifies whether shader modules **can** declare the `ImageGatherExtended` capability.
- `shaderStorageImageExtendedFormats` specifies whether all the “storage image extended formats” below are supported; if this feature is supported, then the VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT **must** be supported in `optimalTilingFeatures` for the following formats:
  - VK_FORMAT_R16G16_SFLOAT
  - VK_FORMAT_B10G11R11_UFLOAT_PACK32
  - VK_FORMAT_R16_SFLOAT
  - VK_FORMAT_R16G16B16A16_UNORM
  - VK_FORMAT_A2B10G10R10_UNORM_PACK32
  - VK_FORMAT_R16G16_UNORM
  - VK_FORMAT_R8G8_UNORM
  - VK_FORMAT_R16_UNORM
  - VK_FORMAT_R8_UNORM
  - VK_FORMAT_R16G16B16A16_SNORM
  - VK_FORMAT_R16G16_SNORM
  - VK_FORMAT_R8G8_SNORM
  - VK_FORMAT_R16_SNORM
  - VK_FORMAT_R8_SNORM
  - VK_FORMAT_R16G16_SINT
  - VK_FORMAT_R8G8_SINT
  - VK_FORMAT_R16_SINT
  - VK_FORMAT_R8_SINT

- VK_FORMAT_A2B10G10R10_UINT_PACK32
- VK_FORMAT_R16G16_UINT
- VK_FORMAT_R8G8_UINT
- VK_FORMAT_R16_UINT
- VK_FORMAT_R8_UINT

NOTE

`shaderStorageImageExtendedFormats` feature only adds a guarantee of format support, which is specified for the whole physical device. Therefore enabling or disabling the feature via vkCreateDevice has no practical effect.

To query for additional properties, or if the feature is not supported, vkGetPhysicalDeviceFormatProperties and vkGetPhysicalDeviceImageFormatProperties **can** be used to check for supported properties of individual formats, as usual rules allow.

VK_FORMAT_R32G32_UINT, VK_FORMAT_R32G32_SINT, and VK_FORMAT_R32G32_SFLOAT from `StorageImageExtendedFormats` SPIR-V capability, are already covered by core Vulkan mandatory format support.

- `shaderStorageImageMultisample` specifies whether multisampled storage images are supported. If this feature is not enabled, images that are created with the VK_IMAGE_USAGE_STORAGE_BIT usage flag set **must** be created with `samples` equal to VK_SAMPLE_COUNT_1_BIT. This also specifies whether shader modules **can** declare the `StorageImageMultisample` and `ImageMSArray` capabilities.
- `shaderStorageImageReadWithoutFormat` specifies whether storage images and storage texel buffers require a format qualifier to be specified when reading. `shaderStorageImageReadWithoutFormat` applies only to formats listed in the storage without format list.
- `shaderStorageImageWriteWithoutFormat` specifies whether storage images and storage texel buffers require a format qualifier to be specified when writing. `shaderStorageImageWriteWithoutFormat` applies only to formats listed in the storage without format list.
- `shaderUniformBufferArrayDynamicIndexing` specifies whether arrays of uniform buffers **can** be indexed by integer expressions that are dynamically uniform within either the subgroup or the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also specifies whether shader modules **can** declare the `UniformBufferArrayDynamicIndexing` capability.
- `shaderSampledImageArrayDynamicIndexing` specifies whether arrays of samplers or sampled images **can** be indexed by integer expressions that are dynamically uniform within either the subgroup or the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_SAMPLER, VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, or VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also specifies whether shader

modules **can** declare the SampledImageArrayDynamicIndexing capability.

- shaderStorageBufferArrayDynamicIndexing specifies whether arrays of storage buffers **can** be indexed by integer expressions that are dynamically uniform within either the subgroup or the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also specifies whether shader modules **can** declare the StorageBufferArrayDynamicIndexing capability.
- shaderStorageImageArrayDynamicIndexing specifies whether arrays of storage images **can** be indexed by integer expressions that are dynamically uniform within either the subgroup or the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_STORAGE_IMAGE **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also specifies whether shader modules **can** declare the StorageImageArrayDynamicIndexing capability.
- shaderClipDistance specifies whether clip distances are supported in shader code. If this feature is not enabled, any members decorated with the ClipDistance built-in decoration **must** not be read from or written to in shader modules. This also specifies whether shader modules **can** declare the ClipDistance capability.
- shaderCullDistance specifies whether cull distances are supported in shader code. If this feature is not enabled, any members decorated with the CullDistance built-in decoration **must** not be read from or written to in shader modules. This also specifies whether shader modules **can** declare the CullDistance capability.
- shaderFloat64 specifies whether 64-bit floats (doubles) are supported in shader code. If this feature is not enabled, 64-bit floating-point types **must** not be used in shader code. This also specifies whether shader modules **can** declare the Float64 capability. Declaring and using 64-bit floats is enabled for all storage classes that SPIR-V allows with the Float64 capability.
- shaderInt64 specifies whether 64-bit integers (signed and unsigned) are supported in shader code. If this feature is not enabled, 64-bit integer types **must** not be used in shader code. This also specifies whether shader modules **can** declare the Int64 capability. Declaring and using 64-bit integers is enabled for all storage classes that SPIR-V allows with the Int64 capability.
- shaderInt16 specifies whether 16-bit integers (signed and unsigned) are supported in shader code. If this feature is not enabled, 16-bit integer types **must** not be used in shader code. This also specifies whether shader modules **can** declare the Int16 capability. However, this only enables a subset of the storage classes that SPIR-V allows for the Int16 SPIR-V capability: Declaring and using 16-bit integers in the Private, Workgroup (for non-Block variables), and Function storage classes is enabled, while declaring them in the interface storage classes (e.g., UniformConstant, Uniform, StorageBuffer, Input, Output, and PushConstant) is not enabled.
- shaderResourceResidency specifies whether image operations that return resource residency information are supported in shader code. If this feature is not enabled, the OpImageSparse* instructions **must** not be used in shader code. This also specifies whether shader modules **can** declare the SparseResidency capability. The feature requires at least one of the sparseResidency* features to be supported.
- shaderResourceMinLod specifies whether image operations specifying the minimum resource LOD are supported in shader code. If this feature is not enabled, the MinLod image operand **must** not

be used in shader code. This also specifies whether shader modules **can** declare the `MinLod` capability.

- `sparseBinding` specifies whether resource memory **can** be managed at opaque sparse block level instead of at the object level. If this feature is not enabled, resource memory **must** be bound only on a per-object basis using the vkBindBufferMemory and vkBindImageMemory commands. In this case, buffers and images **must** not be created with VK_BUFFER_CREATE_SPARSE_BINDING_BIT and VK_IMAGE_CREATE_SPARSE_BINDING_BIT set in the `flags` member of the VkBufferCreateInfo and VkImageCreateInfo structures, respectively. Otherwise resource memory **can** be managed as described in Sparse Resource Features.
- `sparseResidencyBuffer` specifies whether the device **can** access partially resident buffers. If this feature is not enabled, buffers **must** not be created with VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT set in the `flags` member of the VkBufferCreateInfo structure.
- `sparseResidencyImage2D` specifies whether the device **can** access partially resident 2D images with 1 sample per pixel. If this feature is not enabled, images with an `imageType` of VK_IMAGE_TYPE_2D and `samples` set to VK_SAMPLE_COUNT_1_BIT **must** not be created with VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT set in the `flags` member of the VkImageCreateInfo structure.
- `sparseResidencyImage3D` specifies whether the device **can** access partially resident 3D images. If this feature is not enabled, images with an `imageType` of VK_IMAGE_TYPE_3D **must** not be created with VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT set in the `flags` member of the VkImageCreateInfo structure.
- `sparseResidency2Samples` specifies whether the physical device **can** access partially resident 2D images with 2 samples per pixel. If this feature is not enabled, images with an `imageType` of VK_IMAGE_TYPE_2D and `samples` set to VK_SAMPLE_COUNT_2_BIT **must** not be created with VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT set in the `flags` member of the VkImageCreateInfo structure.
- `sparseResidency4Samples` specifies whether the physical device **can** access partially resident 2D images with 4 samples per pixel. If this feature is not enabled, images with an `imageType` of VK_IMAGE_TYPE_2D and `samples` set to VK_SAMPLE_COUNT_4_BIT **must** not be created with VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT set in the `flags` member of the VkImageCreateInfo structure.
- `sparseResidency8Samples` specifies whether the physical device **can** access partially resident 2D images with 8 samples per pixel. If this feature is not enabled, images with an `imageType` of VK_IMAGE_TYPE_2D and `samples` set to VK_SAMPLE_COUNT_8_BIT **must** not be created with VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT set in the `flags` member of the VkImageCreateInfo structure.
- `sparseResidency16Samples` specifies whether the physical device **can** access partially resident 2D images with 16 samples per pixel. If this feature is not enabled, images with an `imageType` of VK_IMAGE_TYPE_2D and `samples` set to VK_SAMPLE_COUNT_16_BIT **must** not be created with VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT set in the `flags` member of the VkImageCreateInfo structure.
- `sparseResidencyAliased` specifies whether the physical device **can** correctly access data aliased into multiple locations. If this feature is not enabled, the

VK_BUFFER_CREATE_SPARSE_ALIASED_BIT and VK_IMAGE_CREATE_SPARSE_ALIASED_BIT enum values **must** not be used in flags members of the VkBufferCreateInfo and VkImageCreateInfo structures, respectively.

- variableMultisampleRate specifies whether all pipelines that will be bound to a command buffer during a subpass which uses no attachments **must** have the same value for VkPipelineMultisampleStateCreateInfo::rasterizationSamples. If set to VK_TRUE, the implementation supports variable multisample rates in a subpass which uses no attachments. If set to VK_FALSE, then all pipelines bound in such a subpass **must** have the same multisample rate. This has no effect in situations where a subpass uses any attachments.
- inheritedQueries specifies whether a secondary command buffer **may** be executed while a query is active.

The VkPhysicalDeviceVulkan11Features structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceVulkan11Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           storageBuffer16BitAccess;
    VkBool32           uniformAndStorageBuffer16BitAccess;
    VkBool32           storagePushConstant16;
    VkBool32           storageInputOutput16;
    VkBool32           multiview;
    VkBool32           multiviewGeometryShader;
    VkBool32           multiviewTessellationShader;
    VkBool32           variablePointersStorageBuffer;
    VkBool32           variablePointers;
    VkBool32           protectedMemory;
    VkBool32           samplerYcbcrConversion;
    VkBool32           shaderDrawParameters;
} VkPhysicalDeviceVulkan11Features;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- storageBuffer16BitAccess specifies whether objects in the StorageBuffer, ShaderRecordBufferKHR, or PhysicalStorageBuffer storage class with the Block decoration **can** have 16-bit integer and 16-bit floating-point members. If this feature is not enabled, 16-bit integer or 16-bit floating-point members **must** not be used in such objects unless the storageBuffer8BitAccess or uniformAndStorageBuffer8BitAccess features are enabled or they are accessed in 32-bit multiples if the shaderUntypedPointers feature is enabled. This also specifies whether shader modules **can** declare the StorageBuffer16BitAccess capability.
- uniformAndStorageBuffer16BitAccess specifies whether objects in the Uniform storage class with the Block decoration **can** have 16-bit integer and 16-bit floating-point members. If this feature is not enabled, 16-bit integer or 16-bit floating-point members **must** not be used in such objects

unless the uniformAndStorageBuffer8BitAccess are enabled or they are accessed in 32-bit multiples if the shaderUntypedPointers feature is enabled. This also specifies whether shader modules **can** declare the UniformAndStorageBuffer16BitAccess capability.

- storagePushConstant16 specifies whether objects in the PushConstant storage class **can** have 16-bit integer and 16-bit floating-point members. If this feature is not enabled, 16-bit integer or floating-point members **must** not be used in such objects unless the storagePushConstant8 feature is enabled or they are accessed in 32-bit multiples if the shaderUntypedPointers feature is enabled. This also specifies whether shader modules **can** declare the StoragePushConstant16 capability.
- storageInputOutput16 specifies whether objects in the Input and Output storage classes **can** have 16-bit integer and 16-bit floating-point members. If this feature is not enabled, 16-bit integer or 16-bit floating-point members **must** not be used in such objects. This also specifies whether shader modules **can** declare the StorageInputOutput16 capability.
- multiview specifies whether the implementation supports multiview rendering within a render pass. If this feature is not enabled, the view mask of each subpass **must** always be zero.
- multiviewGeometryShader specifies whether the implementation supports multiview rendering within a render pass, with geometry shaders. If this feature is not enabled, then a pipeline compiled against a subpass with a non-zero view mask **must** not include a geometry shader.
- multiviewTessellationShader specifies whether the implementation supports multiview rendering within a render pass, with tessellation shaders. If this feature is not enabled, then a pipeline compiled against a subpass with a non-zero view mask **must** not include any tessellation shaders.
- variablePointersStorageBuffer specifies whether the implementation supports the SPIR-V VariablePointersStorageBuffer capability. When this feature is not enabled, shader modules **must** not declare the SPV_KHR_variable_pointers extension or the VariablePointersStorageBuffer capability.
- variablePointers specifies whether the implementation supports the SPIR-V VariablePointers capability. When this feature is not enabled, shader modules **must** not declare the VariablePointers capability.
- protectedMemory specifies whether protected memory is supported.
- samplerYcbcrConversion specifies whether the implementation supports sampler $Y'C_BC_R$ conversion. If samplerYcbcrConversion is VK_FALSE, sampler $Y'C_BC_R$ conversion is not supported, and samplers using sampler $Y'C_BC_R$ conversion **must** not be used.
- shaderDrawParameters specifies whether the implementation supports the SPIR-V DrawParameters capability. When this feature is not enabled, shader modules **must** not declare the SPV_KHR_shader_draw_parameters extension or the DrawParameters capability.

If the VkPhysicalDeviceVulkan11Features structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceVulkan11Features, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVulkan11Features-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_1_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceVulkan12Features` structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceVulkan12Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           samplerMirrorClampToEdge;
    VkBool32           drawIndirectCount;
    VkBool32           storageBuffer8BitAccess;
    VkBool32           uniformAndStorageBuffer8BitAccess;
    VkBool32           storagePushConstant8;
    VkBool32           shaderBufferInt64Atomics;
    VkBool32           shaderSharedInt64Atomics;
    VkBool32           shaderFloat16;
    VkBool32           shaderInt8;
    VkBool32           descriptorIndexing;
    VkBool32           shaderInputAttachmentArrayDynamicIndexing;
    VkBool32           shaderUniformTexelBufferArrayDynamicIndexing;
    VkBool32           shaderStorageTexelBufferArrayDynamicIndexing;
    VkBool32           shaderUniformBufferArrayNonUniformIndexing;
    VkBool32           shaderSampledImageArrayNonUniformIndexing;
    VkBool32           shaderStorageBufferArrayNonUniformIndexing;
    VkBool32           shaderStorageImageArrayNonUniformIndexing;
    VkBool32           shaderInputAttachmentArrayNonUniformIndexing;
    VkBool32           shaderUniformTexelBufferArrayNonUniformIndexing;
    VkBool32           shaderStorageTexelBufferArrayNonUniformIndexing;
    VkBool32           descriptorBindingUniformBufferUpdateAfterBind;
    VkBool32           descriptorBindingSampledImageUpdateAfterBind;
    VkBool32           descriptorBindingStorageImageUpdateAfterBind;
    VkBool32           descriptorBindingStorageBufferUpdateAfterBind;
    VkBool32           descriptorBindingUniformTexelBufferUpdateAfterBind;
    VkBool32           descriptorBindingStorageTexelBufferUpdateAfterBind;
    VkBool32           descriptorBindingUpdateUnusedWhilePending;
    VkBool32           descriptorBindingPartiallyBound;
    VkBool32           descriptorBindingVariableDescriptorCount;
    VkBool32           runtimeDescriptorArray;
```

```
    VkBool32           samplerFilterMinmax;
    VkBool32           scalarBlockLayout;
    VkBool32           imagelessFramebuffer;
    VkBool32           uniformBufferStandardLayout;
    VkBool32           shaderSubgroupExtendedTypes;
    VkBool32           separateDepthStencilLayouts;
    VkBool32           hostQueryReset;
    VkBool32           timelineSemaphore;
    VkBool32           bufferDeviceAddress;
    VkBool32           bufferDeviceAddressCaptureReplay;
    VkBool32           bufferDeviceAddressMultiDevice;
    VkBool32           vulkanMemoryModel;
    VkBool32           vulkanMemoryModelDeviceScope;
    VkBool32           vulkanMemoryModelAvailabilityVisibilityChains;
    VkBool32           shaderOutputViewportIndex;
    VkBool32           shaderOutputLayer;
    VkBool32           subgroupBroadcastDynamicId;
} VkPhysicalDeviceVulkan12Features;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `samplerMirrorClampToEdge` indicates whether the implementation supports the VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE sampler address mode. If this feature is not enabled, the VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE sampler address mode **must** not be used.
- `drawIndirectCount` indicates whether the implementation supports the vkCmdDrawIndirectCount and vkCmdDrawIndexedIndirectCount functions. If this feature is not enabled, these functions **must** not be used.
- `storageBuffer8BitAccess` indicates whether objects in the `StorageBuffer`, `ShaderRecordBufferKHR`, or `PhysicalStorageBuffer` storage class with the `Block` decoration **can** have 8-bit integer members. If this feature is not enabled, 8-bit integer members **must** not be used in such objects unless the `shaderUntypedPointers` feature is enabled and they are accessed in 32-bit multiples or 16-bit multiples if the `storageBuffer16BitAccess` feature is enabled. This also indicates whether shader modules **can** declare the `StorageBuffer8BitAccess` capability.
- `uniformAndStorageBuffer8BitAccess` indicates whether objects in the `Uniform` storage class with the `Block` decoration **can** have 8-bit integer members. If this feature is not enabled, 8-bit integer members **must** not be used in such objects unless the `shaderUntypedPointers` feature is enabled and they are accessed in 32-bit multiples or 16-bit multiples if the `uniformAndStorageBuffer16BitAccess` feature is enabled. This also indicates whether shader modules **can** declare the `UniformAndStorageBuffer8BitAccess` capability.
- `storagePushConstant8` indicates whether objects in the `PushConstant` storage class **can** have 8-bit integer members. If this feature is not enabled, 8-bit integer members **must** not be used in such objects unless the `shaderUntypedPointers` feature is enabled and they are accessed in 32-bit multiples or 16-bit multiples if the `storagePushConstant16` feature is enabled. This also indicates

whether shader modules **can** declare the StoragePushConstant8 capability.

- shaderBufferInt64Atomics indicates whether shaders **can** perform 64-bit unsigned and signed integer atomic operations on buffers.
- shaderSharedInt64Atomics indicates whether shaders **can** perform 64-bit unsigned and signed integer atomic operations on shared and payload memory.
- shaderFloat16 indicates whether 16-bit floats (halfs) are supported in shader code. This also indicates whether shader modules **can** declare the Float16 capability. However, this only enables a subset of the storage classes that SPIR-V allows for the Float16 SPIR-V capability: Declaring and using 16-bit floats in the Private, Workgroup (for non-Block variables), and Function storage classes is enabled, while declaring them in the interface storage classes (e.g., UniformConstant, Uniform, StorageBuffer, Input, Output, and PushConstant) is not enabled.
- shaderInt8 indicates whether 8-bit integers (signed and unsigned) are supported in shader code. This also indicates whether shader modules **can** declare the Int8 capability. However, this only enables a subset of the storage classes that SPIR-V allows for the Int8 SPIR-V capability: Declaring and using 8-bit integers in the Private, Workgroup (for non-Block variables), and Function storage classes is enabled, while declaring them in the interface storage classes (e.g., UniformConstant, Uniform, StorageBuffer, Input, Output, and PushConstant) is not enabled.
- descriptorIndexing indicates whether the implementation supports the minimum set of descriptor indexing features as described in the Feature Requirements section. Enabling this feature when vkCreateDevice is called does not imply the other minimum descriptor indexing features are also enabled. Those other descriptor indexing features **must** be enabled individually as needed by the application.
- shaderInputAttachmentArrayDynamicIndexing indicates whether arrays of input attachments **can** be indexed by integer expressions that are dynamically uniform within either the subgroup or the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the InputAttachmentArrayDynamicIndexing capability.
- shaderUniformTexelBufferArrayDynamicIndexing indicates whether arrays of uniform texel buffers **can** be indexed by integer expressions that are dynamically uniform within either the subgroup or the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the UniformTexelBufferArrayDynamicIndexing capability.
- shaderStorageTexelBufferArrayDynamicIndexing indicates whether arrays of storage texel buffers **can** be indexed by integer expressions that are dynamically uniform within either the subgroup or the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the StorageTexelBufferArrayDynamicIndexing capability.
- shaderUniformBufferArrayNonUniformIndexing indicates whether arrays of uniform buffers **can** be indexed by non-uniform integer expressions in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC **must** not be indexed by non-uniform

integer expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `UniformBufferArrayNonUniformIndexing` capability.

- `shaderSampledImageArrayNonUniformIndexing` indicates whether arrays of samplers or sampled images **can** be indexed by non-uniform integer expressions in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_SAMPLER, VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, or VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE **must** not be indexed by non-uniform integer expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `SampledImageArrayNonUniformIndexing` capability.
- `shaderStorageBufferArrayNonUniformIndexing` indicates whether arrays of storage buffers **can** be indexed by non-uniform integer expressions in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC **must** not be indexed by non-uniform integer expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `StorageBufferArrayNonUniformIndexing` capability.
- `shaderStorageImageArrayNonUniformIndexing` indicates whether arrays of storage images **can** be indexed by non-uniform integer expressions in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_STORAGE_IMAGE **must** not be indexed by non-uniform integer expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `StorageImageArrayNonUniformIndexing` capability.
- `shaderInputAttachmentArrayNonUniformIndexing` indicates whether arrays of input attachments **can** be indexed by non-uniform integer expressions in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT **must** not be indexed by non-uniform integer expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `InputAttachmentArrayNonUniformIndexing` capability.
- `shaderUniformTexelBufferArrayNonUniformIndexing` indicates whether arrays of uniform texel buffers **can** be indexed by non-uniform integer expressions in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER **must** not be indexed by non-uniform integer expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `UniformTexelBufferArrayNonUniformIndexing` capability.
- `shaderStorageTexelBufferArrayNonUniformIndexing` indicates whether arrays of storage texel buffers **can** be indexed by non-uniform integer expressions in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER **must** not be indexed by non-uniform integer expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `StorageTexelBufferArrayNonUniformIndexing` capability.
- `descriptorBindingUniformBufferUpdateAfterBind` indicates whether the implementation supports updating uniform buffer descriptors after a set is bound. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER.
- `descriptorBindingSampledImageUpdateAfterBind` indicates whether the implementation supports updating sampled image descriptors after a set is bound. If this feature is not enabled,

VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with VK_DESCRIPTOR_TYPE_SAMPLER, VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, or VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE.

- `descriptorBindingStorageImageUpdateAfterBind` indicates whether the implementation supports updating storage image descriptors after a set is bound. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with VK_DESCRIPTOR_TYPE_STORAGE_IMAGE.
- `descriptorBindingStorageBufferUpdateAfterBind` indicates whether the implementation supports updating storage buffer descriptors after a set is bound. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with VK_DESCRIPTOR_TYPE_STORAGE_BUFFER.
- `descriptorBindingUniformTexelBufferUpdateAfterBind` indicates whether the implementation supports updating uniform texel buffer descriptors after a set is bound. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER.
- `descriptorBindingStorageTexelBufferUpdateAfterBind` indicates whether the implementation supports updating storage texel buffer descriptors after a set is bound. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER.
- `descriptorBindingUpdateUnusedWhilePending` indicates whether the implementation supports updating descriptors while the set is in use. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_UNUSED_WHILE_PENDING_BIT **must** not be used.
- `descriptorBindingPartiallyBound` indicates whether the implementation supports statically using a descriptor set binding in which some descriptors are not valid. If this feature is not enabled, VK_DESCRIPTOR_BINDING_PARTIALLY_BOUND_BIT **must** not be used.
- `descriptorBindingVariableDescriptorCount` indicates whether the implementation supports descriptor sets with a variable-sized last binding. If this feature is not enabled, VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT **must** not be used.
- `runtimeDescriptorArray` indicates whether the implementation supports the SPIR-V `RuntimeDescriptorArray` capability. If this feature is not enabled, descriptors **must** not be declared in runtime arrays.
- `samplerFilterMinmax` indicates whether the implementation supports a minimum set of required formats supporting min/max filtering as defined by the `filterMinmaxSingleComponentFormats` property minimum requirements. If this feature is not enabled, then VkSamplerReductionModeCreateInfo **must** only use VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE.
- `scalarBlockLayout` indicates that the implementation supports the layout of resource blocks in shaders using scalar alignment.
- `imagelessFramebuffer` indicates that the implementation supports specifying the image view for attachments at render pass begin time via VkRenderPassAttachmentBeginInfo.
- `uniformBufferStandardLayout` indicates that the implementation supports the same layouts for uniform buffers as for storage and other kinds of buffers. See Standard Buffer Layout.
- `shaderSubgroupExtendedTypes` is a boolean specifying whether subgroup operations can use 8-bit

integer, 16-bit integer, 64-bit integer, 16-bit floating-point, and vectors of these types in group operations with subgroup scope, if the implementation supports the types.

- `separateDepthStencilLayouts` indicates whether the implementation supports a `VkImageMemoryBarrier` for a depth/stencil image with only one of VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT set, and whether VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL can be used.
- `hostQueryReset` indicates that the implementation supports resetting queries from the host with vkResetQueryPool.
- `timelineSemaphore` indicates whether semaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE are supported.
- `bufferDeviceAddress` indicates that the implementation supports accessing buffer memory in shaders as storage buffers via an address queried from vkGetBufferDeviceAddress.
- `bufferDeviceAddressCaptureReplay` indicates that the implementation supports saving and reusing buffer and device addresses, e.g. for trace capture and replay.
- `bufferDeviceAddressMultiDevice` indicates that the implementation supports the `bufferDeviceAddress` , `rayTracingPipeline` and `rayQuery` features for logical devices created with multiple physical devices. If this feature is not supported, buffer and acceleration structure addresses **must** not be queried on a logical device created with more than one physical device.
- `vulkanMemoryModel` indicates whether shader modules **can** declare the `VulkanMemoryModel` capability.
- `vulkanMemoryModelDeviceScope` indicates whether the Vulkan Memory Model can use `Device` scope synchronization. This also indicates whether shader modules **can** declare the `VulkanMemoryModelDeviceScope` capability.
- `vulkanMemoryModelAvailabilityVisibilityChains` indicates whether the Vulkan Memory Model can use availability and visibility chains with more than one element.
- `shaderOutputViewportIndex` indicates whether the implementation supports the `ShaderViewportIndex` SPIR-V capability enabling variables decorated with the `ViewportIndex` built-in to be exported from mesh, vertex or tessellation evaluation shaders. If this feature is not enabled, the `ViewportIndex` built-in decoration **must** not be used on outputs in mesh, vertex or tessellation evaluation shaders.
- `shaderOutputLayer` indicates whether the implementation supports the `ShaderLayer` SPIR-V capability enabling variables decorated with the `Layer` built-in to be exported from mesh, vertex or tessellation evaluation shaders. If this feature is not enabled, the `Layer` built-in decoration **must** not be used on outputs in mesh, vertex or tessellation evaluation shaders.
- If `subgroupBroadcastDynamicId` is VK_TRUE, the "Id" operand of `OpGroupNonUniformBroadcast` **can** be dynamically uniform within a subgroup, and the "Index" operand of `OpGroupNonUniformQuadBroadcast` **can** be dynamically uniform within the derivative group. If it is VK_FALSE, these operands **must** be constants.

If the `VkPhysicalDeviceVulkan12Features` structure is included in the `pNext` chain of the

VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceVulkan12Features, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVulkan12Features-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_2_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceVulkan13Features structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceVulkan13Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           robustImageAccess;
    VkBool32           inlineUniformBlock;
    VkBool32           descriptorBindingInlineUniformBlockUpdateAfterBind;
    VkBool32           pipelineCreationCacheControl;
    VkBool32           privateData;
    VkBool32           shaderDemoteToHelperInvocation;
    VkBool32           shaderTerminateInvocation;
    VkBool32           subgroupSizeControl;
    VkBool32           computeFullSubgroups;
    VkBool32           synchronization2;
    VkBool32           textureCompressionASTC_HDR;
    VkBool32           shaderZeroInitializeWorkgroupMemory;
    VkBool32           dynamicRendering;
    VkBool32           shaderIntegerDotProduct;
    VkBool32           maintenance4;
} VkPhysicalDeviceVulkan13Features;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- robustImageAccess enables Robust Image Access guarantees for shader image accesses.

- `inlineUniformBlock` indicates whether the implementation supports inline uniform block descriptors. If this feature is not enabled, VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK **must** not be used.
- `descriptorBindingInlineUniformBlockUpdateAfterBind` indicates whether the implementation supports updating inline uniform block descriptors after a set is bound. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK.
- `pipelineCreationCacheControl` indicates that the implementation supports:
  - The following **can** be used in `Vk*PipelineCreateInfo`::`flags`:
    - VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT
    - VK_PIPELINE_CREATE_EARLY_RETURN_ON_FAILURE_BIT
  - The following **can** be used in VkPipelineCacheCreateInfo::`flags`:
    - VK_PIPELINE_CACHE_CREATE_EXTERNALLY_SYNCHRONIZED_BIT
- `privateData` indicates whether the implementation supports private data. See Private Data.
- `shaderDemoteToHelperInvocation` indicates whether the implementation supports the SPIR-V `DemoteToHelperInvocationEXT` capability.
- `shaderTerminateInvocation` specifies whether the implementation supports SPIR-V modules that use the `SPV_KHR_terminate_invocation` extension.
- `subgroupSizeControl` indicates whether the implementation supports controlling shader subgroup sizes via the VK_PIPELINE_SHADER_STAGE_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT flag and the VkPipelineShaderStageRequiredSubgroupSizeCreateInfo structure.
- `computeFullSubgroups` indicates whether the implementation supports requiring full subgroups in compute shaders via the VK_PIPELINE_SHADER_STAGE_CREATE_REQUIRE_FULL_SUBGROUPS_BIT flag.
- `synchronization2` indicates whether the implementation supports the new set of synchronization commands introduced in `VK_KHR_synchronization2`.
- `textureCompressionASTC_HDR` indicates whether all of the ASTC HDR compressed texture formats are supported. If this feature is enabled, then the VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT, VK_FORMAT_FEATURE_BLIT_SRC_BIT and VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT features **must** be supported in `optimalTilingFeatures` for the following formats:
  - VK_FORMAT_ASTC_4x4_SFLOAT_BLOCK
  - VK_FORMAT_ASTC_5x4_SFLOAT_BLOCK
  - VK_FORMAT_ASTC_5x5_SFLOAT_BLOCK
  - VK_FORMAT_ASTC_6x5_SFLOAT_BLOCK
  - VK_FORMAT_ASTC_6x6_SFLOAT_BLOCK
  - VK_FORMAT_ASTC_8x5_SFLOAT_BLOCK
  - VK_FORMAT_ASTC_8x6_SFLOAT_BLOCK

    - VK_FORMAT_ASTC_8x8_SFLOAT_BLOCK
    - VK_FORMAT_ASTC_10x5_SFLOAT_BLOCK
    - VK_FORMAT_ASTC_10x6_SFLOAT_BLOCK
    - VK_FORMAT_ASTC_10x8_SFLOAT_BLOCK
    - VK_FORMAT_ASTC_10x10_SFLOAT_BLOCK
    - VK_FORMAT_ASTC_12x10_SFLOAT_BLOCK
    - VK_FORMAT_ASTC_12x12_SFLOAT_BLOCK

    To query for additional properties, or if the feature is not enabled, vkGetPhysicalDeviceFormatProperties and vkGetPhysicalDeviceImageFormatProperties **can** be used to check for supported properties of individual formats as normal.

- `shaderZeroInitializeWorkgroupMemory` specifies whether the implementation supports initializing a variable in Workgroup storage class.
- `dynamicRendering` specifies that the implementation supports dynamic render pass instances using the vkCmdBeginRendering command.
- `shaderIntegerDotProduct` specifies whether shader modules **can** declare the `DotProductInputAllKHR`, `DotProductInput4x8BitKHR`, `DotProductInput4x8BitPackedKHR` and `DotProductKHR` capabilities.
- `maintenance4` indicates that the implementation supports the following:
    - The application **may** destroy a VkPipelineLayout object immediately after using it to create another object.
    - `LocalSizeId` **can** be used as an alternative to `LocalSize` to specify the local workgroup size with specialization constants.
    - Images created with identical creation parameters will always have the same alignment requirements.
    - The size memory requirement of a buffer or image is never greater than that of another buffer or image created with a greater or equal size.
    - Push constants do not have to be initialized before they are dynamically accessed.
    - The interface matching rules allow a larger output vector to match with a smaller input vector, with additional values being discarded.

If the `VkPhysicalDeviceVulkan13Features` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceVulkan13Features`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVulkan13Features-sType-sType

sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_3_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceVulkan14Features structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceVulkan14Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           globalPriorityQuery;
    VkBool32           shaderSubgroupRotate;
    VkBool32           shaderSubgroupRotateClustered;
    VkBool32           shaderFloatControls2;
    VkBool32           shaderExpectAssume;
    VkBool32           rectangularLines;
    VkBool32           bresenhamLines;
    VkBool32           smoothLines;
    VkBool32           stippledRectangularLines;
    VkBool32           stippledBresenhamLines;
    VkBool32           stippledSmoothLines;
    VkBool32           vertexAttributeInstanceRateDivisor;
    VkBool32           vertexAttributeInstanceRateZeroDivisor;
    VkBool32           indexTypeUint8;
    VkBool32           dynamicRenderingLocalRead;
    VkBool32           maintenance5;
    VkBool32           maintenance6;
    VkBool32           pipelineProtectedAccess;
    VkBool32           pipelineRobustness;
    VkBool32           hostImageCopy;
    VkBool32           pushDescriptor;
} VkPhysicalDeviceVulkan14Features;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- globalPriorityQuery indicates whether the implementation supports the ability to query global queue priorities.
- shaderSubgroupRotate specifies whether shader modules **can** declare the GroupNonUniformRotateKHR capability.

- `shaderSubgroupRotateClustered` specifies whether shader modules **can** use the `ClusterSize` operand to `OpGroupNonUniformRotateKHR`.
- `shaderFloatControls2` specifies whether shader modules **can** declare the `FloatControls2` capability.
- `shaderExpectAssume` specifies whether shader modules **can** declare the `ExpectAssumeKHR` capability.
- `rectangularLines` indicates whether the implementation supports rectangular line rasterization.
- `bresenhamLines` indicates whether the implementation supports Bresenham-style line rasterization.
- `smoothLines` indicates whether the implementation supports smooth line rasterization.
- `stippledRectangularLines` indicates whether the implementation supports stippled line rasterization with VK_LINE_RASTERIZATION_MODE_RECTANGULAR lines.
- `stippledBresenhamLines` indicates whether the implementation supports stippled line rasterization with VK_LINE_RASTERIZATION_MODE_BRESENHAM lines.
- `stippledSmoothLines` indicates whether the implementation supports stippled line rasterization with VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH lines.
- `vertexAttributeInstanceRateDivisor` specifies whether vertex attribute fetching may be repeated in the case of instanced rendering.
- `vertexAttributeInstanceRateZeroDivisor` specifies whether a zero value for VkVertexInputBindingDivisorDescription::`divisor` is supported.
- `indexTypeUint8` indicates that VK_INDEX_TYPE_UINT8 can be used with vkCmdBindIndexBuffer2 and vkCmdBindIndexBuffer.
- `dynamicRenderingLocalRead` specifies that the implementation supports local reads inside dynamic render pass instances using the vkCmdBeginRendering command.
- `maintenance5` indicates that the implementation supports the following:
  - The ability to expose support for the optional format VK_FORMAT_A1B5G5R5_UNORM_PACK16.
  - The ability to expose support for the optional format VK_FORMAT_A8_UNORM.
  - A property to indicate that multisample coverage operations are performed after sample counting in EarlyFragmentTests mode.
  - Creating a `VkBufferView` with a subset of the associated `VkBuffer` usage using VkBufferUsageFlags2CreateInfo.
  - A new function vkCmdBindIndexBuffer2, allowing a range of memory to be bound as an index buffer.
  - vkGetDeviceProcAddr will return `NULL` for function pointers of core functions for versions higher than the version requested by the application.
  - vkCmdBindVertexBuffers2 supports using VK_WHOLE_SIZE in the `pSizes` parameter.
  - If `PointSize` is not written, a default value of `1.0` is used for the size of points.
  - VkShaderModuleCreateInfo **can** be added as a chained structure to pipeline creation via

VkPipelineShaderStageCreateInfo, rather than having to create a shader module.

- A function vkGetRenderingAreaGranularity to query the optimal render area for a dynamic rendering instance.
- A property to indicate that depth/stencil texturing operations with VK_COMPONENT_SWIZZLE_ONE have defined behavior.
- vkGetDeviceImageSubresourceLayout allows an application to perform a vkGetImageSubresourceLayout query without having to create an image.
- VK_REMAINING_ARRAY_LAYERS as the `layerCount` member of VkImageSubresourceLayers.
- A property to indicate whether `PointSize` controls the final rasterization of polygons if polygon mode is VK_POLYGON_MODE_POINT.
- Two properties to indicate the non-strict line rasterization algorithm used.
- Two new flags words VkPipelineCreateFlagBits2 and VkBufferUsageFlagBits2.
- Physical-device-level functions **can** now be called with any value in the valid range for a type beyond the defined enumerants, such that applications can avoid checking individual features, extensions, or versions before querying supported properties of a particular enumerant.
- Copies between images of any type are allowed, with 1D images treated as 2D images with a height of 1.

- `maintenance6` indicates that the implementation supports the following:
  - VK_NULL_HANDLE **can** be used when binding an index buffer.
  - VkBindMemoryStatus **can** be included in the `pNext` chain of the VkBindBufferMemoryInfo and VkBindImageMemoryInfo structures, enabling applications to retrieve VkResult values for individual memory binding operations.
  - VkPhysicalDeviceMaintenance6Properties::`blockTexelViewCompatibleMultipleLayers` property to indicate that the implementation supports creating image views with VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT where the `layerCount` member of `subresourceRange` is greater than 1.
  - VkPhysicalDeviceMaintenance6Properties::`maxCombinedImageSamplerDescriptorCount` property which indicates the maximum number of descriptors required for sampled images with $Y'C_BC_R$ conversion.
  - A VkPhysicalDeviceMaintenance6Properties::`fragmentShadingRateClampCombinerInputs` property which indicates whether the implementation clamps the inputs to fragment shading rate combiner operations.
- `pipelineProtectedAccess` indicates whether the implementation supports specifying protected access on individual pipelines.
- `pipelineRobustness` indicates that robustness **can** be requested on a per-pipeline-stage granularity.
- `hostImageCopy` indicates that the implementation supports copying from host memory to images using the vkCopyMemoryToImage command, copying from images to host memory using the vkCopyImageToMemory command, and copying between images using the vkCopyImageToImage command.

- pushDescriptor indicates that the implementation supports push descriptors.

If the VkPhysicalDeviceVulkan14Features structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceVulkan14Features, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVulkan14Features-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_4_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceVariablePointersFeatures structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceVariablePointersFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           variablePointersStorageBuffer;
    VkBool32           variablePointers;
} VkPhysicalDeviceVariablePointersFeatures;
```

```
// Provided by VK_VERSION_1_1
// Equivalent to VkPhysicalDeviceVariablePointersFeatures
typedef VkPhysicalDeviceVariablePointersFeatures
VkPhysicalDeviceVariablePointerFeatures;
```

```
// Provided by VK_KHR_variable_pointers
// Equivalent to VkPhysicalDeviceVariablePointersFeatures
typedef VkPhysicalDeviceVariablePointersFeatures
VkPhysicalDeviceVariablePointersFeaturesKHR;
```

```
// Provided by VK_KHR_variable_pointers
// Equivalent to VkPhysicalDeviceVariablePointersFeatures
typedef VkPhysicalDeviceVariablePointersFeatures
VkPhysicalDeviceVariablePointerFeaturesKHR;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `variablePointersStorageBuffer` specifies whether the implementation supports the SPIR-V `VariablePointersStorageBuffer` capability. When this feature is not enabled, shader modules **must** not declare the `SPV_KHR_variable_pointers` extension or the `VariablePointersStorageBuffer` capability.
- `variablePointers` specifies whether the implementation supports the SPIR-V `VariablePointers` capability. When this feature is not enabled, shader modules **must** not declare the `VariablePointers` capability.

If the `VkPhysicalDeviceVariablePointersFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceVariablePointersFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage

- VUID-VkPhysicalDeviceVariablePointersFeatures-variablePointers-01431
  If `variablePointers` is enabled then `variablePointersStorageBuffer` **must** also be enabled

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVariablePointersFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTERS_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceMultiviewFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceMultiviewFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           multiview;
    VkBool32           multiviewGeometryShader;
    VkBool32           multiviewTessellationShader;
} VkPhysicalDeviceMultiviewFeatures;
```

```
// Provided by VK_KHR_multiview
// Equivalent to VkPhysicalDeviceMultiviewFeatures
typedef VkPhysicalDeviceMultiviewFeatures VkPhysicalDeviceMultiviewFeaturesKHR;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `multiview` specifies whether the implementation supports multiview rendering within a render pass. If this feature is not enabled, the view mask of each subpass **must** always be zero.
- `multiviewGeometryShader` specifies whether the implementation supports multiview rendering within a render pass, with geometry shaders. If this feature is not enabled, then a pipeline compiled against a subpass with a non-zero view mask **must** not include a geometry shader.
- `multiviewTessellationShader` specifies whether the implementation supports multiview rendering within a render pass, with tessellation shaders. If this feature is not enabled, then a pipeline compiled against a subpass with a non-zero view mask **must** not include any tessellation shaders.

If the `VkPhysicalDeviceMultiviewFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceMultiviewFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage

- VUID-VkPhysicalDeviceMultiviewFeatures-multiviewGeometryShader-00580
  If `multiviewGeometryShader` is enabled then `multiview` **must** also be enabled
- VUID-VkPhysicalDeviceMultiviewFeatures-multiviewTessellationShader-00581
  If `multiviewTessellationShader` is enabled then `multiview` **must** also be enabled

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMultiviewFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderAtomicFloatFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_shader_atomic_float
typedef struct VkPhysicalDeviceShaderAtomicFloatFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderBufferFloat32Atomics;
    VkBool32           shaderBufferFloat32AtomicAdd;
    VkBool32           shaderBufferFloat64Atomics;
    VkBool32           shaderBufferFloat64AtomicAdd;
    VkBool32           shaderSharedFloat32Atomics;
    VkBool32           shaderSharedFloat32AtomicAdd;
    VkBool32           shaderSharedFloat64Atomics;
    VkBool32           shaderSharedFloat64AtomicAdd;
    VkBool32           shaderImageFloat32Atomics;
    VkBool32           shaderImageFloat32AtomicAdd;
    VkBool32           sparseImageFloat32Atomics;
    VkBool32           sparseImageFloat32AtomicAdd;
} VkPhysicalDeviceShaderAtomicFloatFeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderBufferFloat32Atomics` indicates whether shaders **can** perform 32-bit floating-point load, store, and exchange atomic operations on storage buffers.
- `shaderBufferFloat32AtomicAdd` indicates whether shaders **can** perform 32-bit floating-point add atomic operations on storage buffers.
- `shaderBufferFloat64Atomics` indicates whether shaders **can** perform 64-bit floating-point load, store, and exchange atomic operations on storage buffers.
- `shaderBufferFloat64AtomicAdd` indicates whether shaders **can** perform 64-bit floating-point add atomic operations on storage buffers.

- shaderSharedFloat32Atomics indicates whether shaders **can** perform 32-bit floating-point load, store, and exchange atomic operations on shared and payload memory.
- shaderSharedFloat32AtomicAdd indicates whether shaders **can** perform 32-bit floating-point add atomic operations on shared and payload memory.
- shaderSharedFloat64Atomics indicates whether shaders **can** perform 64-bit floating-point load, store, and exchange atomic operations on shared and payload memory.
- shaderSharedFloat64AtomicAdd indicates whether shaders **can** perform 64-bit floating-point add atomic operations on shared and payload memory.
- shaderImageFloat32Atomics indicates whether shaders **can** perform 32-bit floating-point load, store, and exchange atomic image operations.
- shaderImageFloat32AtomicAdd indicates whether shaders **can** perform 32-bit floating-point add atomic image operations.
- sparseImageFloat32Atomics indicates whether 32-bit floating-point load, store, and exchange atomic operations **can** be used on sparse images.
- sparseImageFloat32AtomicAdd indicates whether 32-bit floating-point add atomic operations **can** be used on sparse images.

If the VkPhysicalDeviceShaderAtomicFloatFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderAtomicFloatFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderAtomicFloatFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_FLOAT_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT structure is defined as:

```
// Provided by VK_EXT_shader_atomic_float2
typedef struct VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderBufferFloat16Atomics;
    VkBool32           shaderBufferFloat16AtomicAdd;
    VkBool32           shaderBufferFloat16AtomicMinMax;
    VkBool32           shaderBufferFloat32AtomicMinMax;
    VkBool32           shaderBufferFloat64AtomicMinMax;
    VkBool32           shaderSharedFloat16Atomics;
    VkBool32           shaderSharedFloat16AtomicAdd;
    VkBool32           shaderSharedFloat16AtomicMinMax;
    VkBool32           shaderSharedFloat32AtomicMinMax;
    VkBool32           shaderSharedFloat64AtomicMinMax;
    VkBool32           shaderImageFloat32AtomicMinMax;
    VkBool32           sparseImageFloat32AtomicMinMax;
} VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderBufferFloat16Atomics` indicates whether shaders **can** perform 16-bit floating-point load, store, and exchange atomic operations on storage buffers.
- `shaderBufferFloat16AtomicAdd` indicates whether shaders **can** perform 16-bit floating-point add atomic operations on storage buffers.
- `shaderBufferFloat16AtomicMinMax` indicates whether shaders **can** perform 16-bit floating-point min and max atomic operations on storage buffers.
- `shaderBufferFloat32AtomicMinMax` indicates whether shaders **can** perform 32-bit floating-point min and max atomic operations on storage buffers.
- `shaderBufferFloat64AtomicMinMax` indicates whether shaders **can** perform 64-bit floating-point min and max atomic operations on storage buffers.
- `shaderSharedFloat16Atomics` indicates whether shaders **can** perform 16-bit floating-point load, store, and exchange atomic operations on shared and payload memory.
- `shaderSharedFloat16AtomicAdd` indicates whether shaders **can** perform 16-bit floating-point add atomic operations on shared and payload memory.
- `shaderSharedFloat16AtomicMinMax` indicates whether shaders **can** perform 16-bit floating-point min and max atomic operations on shared and payload memory.
- `shaderSharedFloat32AtomicMinMax` indicates whether shaders **can** perform 32-bit floating-point min and max atomic operations on shared and payload memory.
- `shaderSharedFloat64AtomicMinMax` indicates whether shaders **can** perform 64-bit floating-point min and max atomic operations on shared and payload memory.
- `shaderImageFloat32AtomicMinMax` indicates whether shaders **can** perform 32-bit floating-point

min and max atomic image operations.

- `sparseImageFloat32AtomicMinMax` indicates whether 32-bit floating-point min and max atomic operations **can** be used on sparse images.

If the `VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_FLOAT_2_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderAtomicInt64Features structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceShaderAtomicInt64Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderBufferInt64Atomics;
    VkBool32           shaderSharedInt64Atomics;
} VkPhysicalDeviceShaderAtomicInt64Features;
```

```
// Provided by VK_KHR_shader_atomic_int64
// Equivalent to VkPhysicalDeviceShaderAtomicInt64Features
typedef VkPhysicalDeviceShaderAtomicInt64Features
VkPhysicalDeviceShaderAtomicInt64FeaturesKHR;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderBufferInt64Atomics` indicates whether shaders **can** perform 64-bit unsigned and signed

integer atomic operations on buffers.

- shaderSharedInt64Atomics indicates whether shaders **can** perform 64-bit unsigned and signed integer atomic operations on shared and payload memory.

If the VkPhysicalDeviceShaderAtomicInt64Features structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderAtomicInt64Features, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderAtomicInt64Features-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_INT64_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT structure is defined as:

```
// Provided by VK_EXT_shader_image_atomic_int64
typedef struct VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderImageInt64Atomics;
    VkBool32           sparseImageInt64Atomics;
} VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderImageInt64Atomics indicates whether shaders **can** support 64-bit unsigned and signed integer atomic operations on images.
- sparseImageInt64Atomics indicates whether 64-bit integer atomics **can** be used on sparse images.

If the VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a

VkDevice with any features described by VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_IMAGE_ATOMIC_INT64_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevice8BitStorageFeatures structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDevice8BitStorageFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           storageBuffer8BitAccess;
    VkBool32           uniformAndStorageBuffer8BitAccess;
    VkBool32           storagePushConstant8;
} VkPhysicalDevice8BitStorageFeatures;
```

```
// Provided by VK_KHR_8bit_storage
// Equivalent to VkPhysicalDevice8BitStorageFeatures
typedef VkPhysicalDevice8BitStorageFeatures VkPhysicalDevice8BitStorageFeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- storageBuffer8BitAccess indicates whether objects in the StorageBuffer, ShaderRecordBufferKHR, or PhysicalStorageBuffer storage class with the Block decoration **can** have 8-bit integer members. If this feature is not enabled, 8-bit integer members **must** not be used in such objects unless the shaderUntypedPointers feature is enabled and they are accessed in 32-bit multiples or 16-bit multiples if the storageBuffer16BitAccess feature is enabled. This also indicates whether shader modules **can** declare the StorageBuffer8BitAccess capability.

- `uniformAndStorageBuffer8BitAccess` indicates whether objects in the `Uniform` storage class with the `Block` decoration **can** have 8-bit integer members. If this feature is not enabled, 8-bit integer members **must** not be used in such objects unless the `shaderUntypedPointers` feature is enabled and they are accessed in 32-bit multiples or 16-bit multiples if the `uniformAndStorageBuffer16BitAccess` feature is enabled. This also indicates whether shader modules **can** declare the `UniformAndStorageBuffer8BitAccess` capability.
- `storagePushConstant8` indicates whether objects in the `PushConstant` storage class **can** have 8-bit integer members. If this feature is not enabled, 8-bit integer members **must** not be used in such objects unless the `shaderUntypedPointers` feature is enabled and they are accessed in 32-bit multiples or 16-bit multiples if the `storagePushConstant16` feature is enabled. This also indicates whether shader modules **can** declare the `StoragePushConstant8` capability.

If the `VkPhysicalDevice8BitStorageFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDevice8BitStorageFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevice8BitStorageFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_8BIT_STORAGE_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevice16BitStorageFeatures structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDevice16BitStorageFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           storageBuffer16BitAccess;
    VkBool32           uniformAndStorageBuffer16BitAccess;
    VkBool32           storagePushConstant16;
    VkBool32           storageInputOutput16;
} VkPhysicalDevice16BitStorageFeatures;
```

```
// Provided by VK_KHR_16bit_storage
// Equivalent to VkPhysicalDevice16BitStorageFeatures
typedef VkPhysicalDevice16BitStorageFeatures VkPhysicalDevice16BitStorageFeaturesKHR;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `storageBuffer16BitAccess` specifies whether objects in the `StorageBuffer`, `ShaderRecordBufferKHR`, or `PhysicalStorageBuffer` storage class with the `Block` decoration **can** have 16-bit integer and 16-bit floating-point members. If this feature is not enabled, 16-bit integer or 16-bit floating-point members **must** not be used in such objects unless the `storageBuffer8BitAccess` or `uniformAndStorageBuffer8BitAccess` features are enabled or they are accessed in 32-bit multiples if the `shaderUntypedPointers` feature is enabled. This also specifies whether shader modules **can** declare the `StorageBuffer16BitAccess` capability.
- `uniformAndStorageBuffer16BitAccess` specifies whether objects in the `Uniform` storage class with the `Block` decoration **can** have 16-bit integer and 16-bit floating-point members. If this feature is not enabled, 16-bit integer or 16-bit floating-point members **must** not be used in such objects unless the `uniformAndStorageBuffer8BitAccess` are enabled or they are accessed in 32-bit multiples if the `shaderUntypedPointers` feature is enabled. This also specifies whether shader modules **can** declare the `UniformAndStorageBuffer16BitAccess` capability.
- `storagePushConstant16` specifies whether objects in the `PushConstant` storage class **can** have 16-bit integer and 16-bit floating-point members. If this feature is not enabled, 16-bit integer or floating-point members **must** not be used in such objects unless the `storagePushConstant8` feature is enabled or they are accessed in 32-bit multiples if the `shaderUntypedPointers` feature is enabled. This also specifies whether shader modules **can** declare the `StoragePushConstant16` capability.
- `storageInputOutput16` specifies whether objects in the `Input` and `Output` storage classes **can** have 16-bit integer and 16-bit floating-point members. If this feature is not enabled, 16-bit integer or 16-bit floating-point members **must** not be used in such objects. This also specifies whether shader modules **can** declare the `StorageInputOutput16` capability.

If the `VkPhysicalDevice16BitStorageFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDevice16BitStorageFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevice16BitStorageFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_16BIT_STORAGE_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderFloat16Int8Features structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceShaderFloat16Int8Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderFloat16;
    VkBool32           shaderInt8;
} VkPhysicalDeviceShaderFloat16Int8Features;
```

```
// Provided by VK_KHR_shader_float16_int8
// Equivalent to VkPhysicalDeviceShaderFloat16Int8Features
typedef VkPhysicalDeviceShaderFloat16Int8Features
VkPhysicalDeviceShaderFloat16Int8FeaturesKHR;
```

```
// Provided by VK_KHR_shader_float16_int8
// Equivalent to VkPhysicalDeviceShaderFloat16Int8Features
typedef VkPhysicalDeviceShaderFloat16Int8Features
VkPhysicalDeviceFloat16Int8FeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderFloat16 indicates whether 16-bit floats (halfs) are supported in shader code. This also indicates whether shader modules **can** declare the Float16 capability. However, this only enables a subset of the storage classes that SPIR-V allows for the Float16 SPIR-V capability: Declaring and using 16-bit floats in the Private, Workgroup (for non-Block variables), and Function storage classes is enabled, while declaring them in the interface storage classes (e.g., UniformConstant, Uniform, StorageBuffer, Input, Output, and PushConstant) is not enabled.
- shaderInt8 indicates whether 8-bit integers (signed and unsigned) are supported in shader code. This also indicates whether shader modules **can** declare the Int8 capability. However, this only enables a subset of the storage classes that SPIR-V allows for the Int8 SPIR-V capability: Declaring and using 8-bit integers in the Private, Workgroup (for non-Block variables), and Function storage classes is enabled, while declaring them in the interface storage classes (e.g., UniformConstant, Uniform, StorageBuffer, Input, Output, and PushConstant) is not enabled.

If the VkPhysicalDeviceShaderFloat16Int8Features structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderFloat16Int8Features, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderFloat16Int8Features-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT16_INT8_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderClockFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_shader_clock
typedef struct VkPhysicalDeviceShaderClockFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderSubgroupClock;
    VkBool32           shaderDeviceClock;
} VkPhysicalDeviceShaderClockFeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderSubgroupClock indicates whether shaders **can** perform Subgroup scoped clock reads.
- shaderDeviceClock indicates whether shaders **can** perform Device scoped clock reads.

If the VkPhysicalDeviceShaderClockFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderClockFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderClockFeaturesKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_CLOCK_FEATURES_KHR`

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceSamplerYcbcrConversionFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceSamplerYcbcrConversionFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           samplerYcbcrConversion;
} VkPhysicalDeviceSamplerYcbcrConversionFeatures;
```

```
// Provided by VK_KHR_sampler_ycbcr_conversion
// Equivalent to VkPhysicalDeviceSamplerYcbcrConversionFeatures
typedef VkPhysicalDeviceSamplerYcbcrConversionFeatures
VkPhysicalDeviceSamplerYcbcrConversionFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `samplerYcbcrConversion` specifies whether the implementation supports sampler $Y'C_BC_R$ conversion. If `samplerYcbcrConversion` is VK_FALSE, sampler $Y'C_BC_R$ conversion is not supported, and samplers using sampler $Y'C_BC_R$ conversion **must** not be used.

If the `VkPhysicalDeviceSamplerYcbcrConversionFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceSamplerYcbcrConversionFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSamplerYcbcrConversionFeatures-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLER_YCBCR_CONVERSION_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceProtectedMemoryFeatures structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceProtectedMemoryFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           protectedMemory;
} VkPhysicalDeviceProtectedMemoryFeatures;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- protectedMemory specifies whether protected memory is supported.

If the VkPhysicalDeviceProtectedMemoryFeatures structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceProtectedMemoryFeatures, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceProtectedMemoryFeatures-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROTECTED_MEMORY_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_blend_operation_advanced
typedef struct VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           advancedBlendCoherentOperations;
} VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- advancedBlendCoherentOperations specifies whether blending using advanced blend operations is guaranteed to execute atomically and in primitive order. If this is VK_TRUE, VK_ACCESS_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT is treated the same as VK_ACCESS_COLOR_ATTACHMENT_READ_BIT, and advanced blending needs no additional synchronization over basic blending. If this is VK_FALSE, then memory dependencies are required to guarantee order between two advanced blending operations that occur on the same sample.

If the VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BLEND_OPERATION_ADVANCED_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceConditionalRenderingFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_conditional_rendering
typedef struct VkPhysicalDeviceConditionalRenderingFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           conditionalRendering;
    VkBool32           inheritedConditionalRendering;
} VkPhysicalDeviceConditionalRenderingFeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- conditionalRendering specifies whether conditional rendering is supported.
- inheritedConditionalRendering specifies whether a secondary command buffer **can** be executed while conditional rendering is active in the primary command buffer.

If the VkPhysicalDeviceConditionalRenderingFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceConditionalRenderingFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceConditionalRenderingFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CONDITIONAL_RENDERING_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderDrawParametersFeatures structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceShaderDrawParametersFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderDrawParameters;
} VkPhysicalDeviceShaderDrawParametersFeatures;
```

```
// Provided by VK_VERSION_1_1
// Equivalent to VkPhysicalDeviceShaderDrawParametersFeatures
typedef VkPhysicalDeviceShaderDrawParametersFeatures
VkPhysicalDeviceShaderDrawParameterFeatures;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderDrawParameters specifies whether the implementation supports the SPIR-V DrawParameters capability. When this feature is not enabled, shader modules **must** not declare the SPV_KHR_shader_draw_parameters extension or the DrawParameters capability.

If the VkPhysicalDeviceShaderDrawParametersFeatures structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderDrawParametersFeatures, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderDrawParametersFeatures-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_DRAW_PARAMETERS_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceMeshShaderFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_mesh_shader
typedef struct VkPhysicalDeviceMeshShaderFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           taskShader;
    VkBool32           meshShader;
    VkBool32           multiviewMeshShader;
    VkBool32           primitiveFragmentShadingRateMeshShader;
    VkBool32           meshShaderQueries;
} VkPhysicalDeviceMeshShaderFeaturesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `taskShader` specifies whether task shaders are supported. If this feature is not enabled, the VK_SHADER_STAGE_TASK_BIT_EXT and VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT enum values **must** not be used.
- `meshShader` specifies whether mesh shaders are supported. If this feature is not enabled, the VK_SHADER_STAGE_MESH_BIT_EXT and VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT enum values **must** not be used.
- `multiviewMeshShader` specifies whether the implementation supports `multiview` rendering within a render pass, with mesh shaders. If this feature is not enabled, then a pipeline compiled against a subpass with a non-zero view mask **must** not include a mesh shader.
- `primitiveFragmentShadingRateMeshShader` indicates that the implementation supports the primitive fragment shading rate in mesh shaders.
- `meshShaderQueries` indicates that the implementation supports creating query pools using the VK_QUERY_TYPE_MESH_PRIMITIVES_GENERATED_EXT query type and statistic queries containing the VK_QUERY_PIPELINE_STATISTIC_TASK_SHADER_INVOCATIONS_BIT_EXT and VK_QUERY_PIPELINE_STATISTIC_MESH_SHADER_INVOCATIONS_BIT_EXT flags.

If the `VkPhysicalDeviceMeshShaderFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceMeshShaderFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage

- VUID-VkPhysicalDeviceMeshShaderFeaturesEXT-multiviewMeshShader-07032
  If `multiviewMeshShader` is enabled then `VkPhysicalDeviceMultiviewFeaturesKHR`::`multiview` **must** also be enabled
- VUID-VkPhysicalDeviceMeshShaderFeaturesEXT-primitiveFragmentShadingRateMeshShader-07033
  If `primitiveFragmentShadingRateMeshShader` is enabled then

VkPhysicalDeviceFragmentShadingRateFeaturesKHR::primitiveFragmentShadingRate **must** also be enabled

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMeshShaderFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MESH_SHADER_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceMemoryDecompressionFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_memory_decompression
typedef struct VkPhysicalDeviceMemoryDecompressionFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           memoryDecompression;
} VkPhysicalDeviceMemoryDecompressionFeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- memoryDecompression indicates whether memory decompression is supported.

If the VkPhysicalDeviceMemoryDecompressionFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceMemoryDecompressionFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMemoryDecompressionFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_DECOMPRESSION_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceDescriptorIndexingFeatures structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceDescriptorIndexingFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderInputAttachmentArrayDynamicIndexing;
    VkBool32           shaderUniformTexelBufferArrayDynamicIndexing;
    VkBool32           shaderStorageTexelBufferArrayDynamicIndexing;
    VkBool32           shaderUniformBufferArrayNonUniformIndexing;
    VkBool32           shaderSampledImageArrayNonUniformIndexing;
    VkBool32           shaderStorageBufferArrayNonUniformIndexing;
    VkBool32           shaderStorageImageArrayNonUniformIndexing;
    VkBool32           shaderInputAttachmentArrayNonUniformIndexing;
    VkBool32           shaderUniformTexelBufferArrayNonUniformIndexing;
    VkBool32           shaderStorageTexelBufferArrayNonUniformIndexing;
    VkBool32           descriptorBindingUniformBufferUpdateAfterBind;
    VkBool32           descriptorBindingSampledImageUpdateAfterBind;
    VkBool32           descriptorBindingStorageImageUpdateAfterBind;
    VkBool32           descriptorBindingStorageBufferUpdateAfterBind;
    VkBool32           descriptorBindingUniformTexelBufferUpdateAfterBind;
    VkBool32           descriptorBindingStorageTexelBufferUpdateAfterBind;
    VkBool32           descriptorBindingUpdateUnusedWhilePending;
    VkBool32           descriptorBindingPartiallyBound;
    VkBool32           descriptorBindingVariableDescriptorCount;
    VkBool32           runtimeDescriptorArray;
} VkPhysicalDeviceDescriptorIndexingFeatures;
```

```
// Provided by VK_EXT_descriptor_indexing
// Equivalent to VkPhysicalDeviceDescriptorIndexingFeatures
typedef VkPhysicalDeviceDescriptorIndexingFeatures
VkPhysicalDeviceDescriptorIndexingFeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderInputAttachmentArrayDynamicIndexing indicates whether arrays of input attachments **can** be indexed by integer expressions that are dynamically uniform within either the subgroup or

the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `InputAttachmentArrayDynamicIndexing` capability.

- `shaderUniformTexelBufferArrayDynamicIndexing` indicates whether arrays of uniform texel buffers **can** be indexed by integer expressions that are dynamically uniform within either the subgroup or the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `UniformTexelBufferArrayDynamicIndexing` capability.
- `shaderStorageTexelBufferArrayDynamicIndexing` indicates whether arrays of storage texel buffers **can** be indexed by integer expressions that are dynamically uniform within either the subgroup or the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `StorageTexelBufferArrayDynamicIndexing` capability.
- `shaderUniformBufferArrayNonUniformIndexing` indicates whether arrays of uniform buffers **can** be indexed by non-uniform integer expressions in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC **must** not be indexed by non-uniform integer expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `UniformBufferArrayNonUniformIndexing` capability.
- `shaderSampledImageArrayNonUniformIndexing` indicates whether arrays of samplers or sampled images **can** be indexed by non-uniform integer expressions in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_SAMPLER, VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, or VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE **must** not be indexed by non-uniform integer expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `SampledImageArrayNonUniformIndexing` capability.
- `shaderStorageBufferArrayNonUniformIndexing` indicates whether arrays of storage buffers **can** be indexed by non-uniform integer expressions in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC **must** not be indexed by non-uniform integer expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `StorageBufferArrayNonUniformIndexing` capability.
- `shaderStorageImageArrayNonUniformIndexing` indicates whether arrays of storage images **can** be indexed by non-uniform integer expressions in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_STORAGE_IMAGE **must** not be indexed by non-uniform integer expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `StorageImageArrayNonUniformIndexing` capability.
- `shaderInputAttachmentArrayNonUniformIndexing` indicates whether arrays of input attachments **can** be indexed by non-uniform integer expressions in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT **must** not be indexed by non-uniform integer expressions when aggregated into arrays in shader

code. This also indicates whether shader modules **can** declare the `InputAttachmentArrayNonUniformIndexing` capability.

- `shaderUniformTexelBufferArrayNonUniformIndexing` indicates whether arrays of uniform texel buffers **can** be indexed by non-uniform integer expressions in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER **must** not be indexed by non-uniform integer expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `UniformTexelBufferArrayNonUniformIndexing` capability.
- `shaderStorageTexelBufferArrayNonUniformIndexing` indicates whether arrays of storage texel buffers **can** be indexed by non-uniform integer expressions in shader code. If this feature is not enabled, resources with a descriptor type of VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER **must** not be indexed by non-uniform integer expressions when aggregated into arrays in shader code. This also indicates whether shader modules **can** declare the `StorageTexelBufferArrayNonUniformIndexing` capability.
- `descriptorBindingUniformBufferUpdateAfterBind` indicates whether the implementation supports updating uniform buffer descriptors after a set is bound. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER.
- `descriptorBindingSampledImageUpdateAfterBind` indicates whether the implementation supports updating sampled image descriptors after a set is bound. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with VK_DESCRIPTOR_TYPE_SAMPLER, VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, or VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE.
- `descriptorBindingStorageImageUpdateAfterBind` indicates whether the implementation supports updating storage image descriptors after a set is bound. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with VK_DESCRIPTOR_TYPE_STORAGE_IMAGE.
- `descriptorBindingStorageBufferUpdateAfterBind` indicates whether the implementation supports updating storage buffer descriptors after a set is bound. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with VK_DESCRIPTOR_TYPE_STORAGE_BUFFER.
- `descriptorBindingUniformTexelBufferUpdateAfterBind` indicates whether the implementation supports updating uniform texel buffer descriptors after a set is bound. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER.
- `descriptorBindingStorageTexelBufferUpdateAfterBind` indicates whether the implementation supports updating storage texel buffer descriptors after a set is bound. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER.
- `descriptorBindingUpdateUnusedWhilePending` indicates whether the implementation supports updating descriptors while the set is in use. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_UNUSED_WHILE_PENDING_BIT **must** not be used.
- `descriptorBindingPartiallyBound` indicates whether the implementation supports statically using a descriptor set binding in which some descriptors are not valid. If this feature is not

enabled, VK_DESCRIPTOR_BINDING_PARTIALLY_BOUND_BIT **must** not be used.

- `descriptorBindingVariableDescriptorCount` indicates whether the implementation supports descriptor sets with a variable-sized last binding. If this feature is not enabled, VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT **must** not be used.
- `runtimeDescriptorArray` indicates whether the implementation supports the SPIR-V `RuntimeDescriptorArray` capability. If this feature is not enabled, descriptors **must** not be declared in runtime arrays.

If the `VkPhysicalDeviceDescriptorIndexingFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceDescriptorIndexingFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDescriptorIndexingFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_INDEXING_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceCopyMemoryIndirectFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_copy_memory_indirect
typedef struct VkPhysicalDeviceCopyMemoryIndirectFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           indirectMemoryCopy;
    VkBool32           indirectMemoryToImageCopy;
} VkPhysicalDeviceCopyMemoryIndirectFeaturesKHR;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `indirectMemoryCopy` indicates whether indirect memory to memory copies are supported.
- `indirectMemoryToImageCopy` indicates whether indirect memory to image copies are supported.

If the VkPhysicalDeviceCopyMemoryIndirectFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceCopyMemoryIndirectFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceCopyMemoryIndirectFeaturesKHR-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COPY_MEMORY_INDIRECT_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceVertexAttributeDivisorFeatures structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceVertexAttributeDivisorFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           vertexAttributeInstanceRateDivisor;
    VkBool32           vertexAttributeInstanceRateZeroDivisor;
} VkPhysicalDeviceVertexAttributeDivisorFeatures;
```

```
// Provided by VK_KHR_vertex_attribute_divisor
// Equivalent to VkPhysicalDeviceVertexAttributeDivisorFeatures
typedef VkPhysicalDeviceVertexAttributeDivisorFeatures
VkPhysicalDeviceVertexAttributeDivisorFeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- vertexAttributeInstanceRateDivisor specifies whether vertex attribute fetching may be repeated in the case of instanced rendering.
- vertexAttributeInstanceRateZeroDivisor specifies whether a zero value for VkVertexInputBindingDivisorDescription::divisor is supported.

If the VkPhysicalDeviceVertexAttributeDivisorFeatures structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceVertexAttributeDivisorFeatures, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVertexAttributeDivisorFeatures-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceASTCDecodeFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_astc_decode_mode
typedef struct VkPhysicalDeviceASTCDecodeFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           decodeModeSharedExponent;
} VkPhysicalDeviceASTCDecodeFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- decodeModeSharedExponent indicates whether the implementation supports decoding ASTC compressed formats to VK_FORMAT_E5B9G9R9_UFLOAT_PACK32 internal precision.

If the VkPhysicalDeviceASTCDecodeFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceASTCDecodeFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceASTCDecodeFeaturesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ASTC_DECODE_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceTransformFeedbackFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_transform_feedback
typedef struct VkPhysicalDeviceTransformFeedbackFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           transformFeedback;
    VkBool32           geometryStreams;
} VkPhysicalDeviceTransformFeedbackFeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `transformFeedback` indicates whether the implementation supports transform feedback and shader modules **can** declare the `TransformFeedback` capability.
- `geometryStreams` indicates whether the implementation supports the `GeometryStreams` SPIR-V capability.

If the `VkPhysicalDeviceTransformFeedbackFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceTransformFeedbackFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceTransformFeedbackFeaturesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TRANSFORM_FEEDBACK_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceVulkanMemoryModelFeatures structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceVulkanMemoryModelFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           vulkanMemoryModel;
    VkBool32           vulkanMemoryModelDeviceScope;
    VkBool32           vulkanMemoryModelAvailabilityVisibilityChains;
} VkPhysicalDeviceVulkanMemoryModelFeatures;
```

```
// Provided by VK_KHR_vulkan_memory_model
// Equivalent to VkPhysicalDeviceVulkanMemoryModelFeatures
typedef VkPhysicalDeviceVulkanMemoryModelFeatures
VkPhysicalDeviceVulkanMemoryModelFeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- vulkanMemoryModel indicates whether shader modules **can** declare the VulkanMemoryModel capability.
- vulkanMemoryModelDeviceScope indicates whether the Vulkan Memory Model can use Device scope synchronization. This also indicates whether shader modules **can** declare the VulkanMemoryModelDeviceScope capability.
- vulkanMemoryModelAvailabilityVisibilityChains indicates whether the Vulkan Memory Model can use availability and visibility chains with more than one element.

If the VkPhysicalDeviceVulkanMemoryModelFeatures structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceVulkanMemoryModelFeatures, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVulkanMemoryModelFeatures-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_MEMORY_MODEL_FEATURES`

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceInlineUniformBlockFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceInlineUniformBlockFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           inlineUniformBlock;
    VkBool32           descriptorBindingInlineUniformBlockUpdateAfterBind;
} VkPhysicalDeviceInlineUniformBlockFeatures;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `inlineUniformBlock` indicates whether the implementation supports inline uniform block descriptors. If this feature is not enabled, VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK **must** not be used.
- `descriptorBindingInlineUniformBlockUpdateAfterBind` indicates whether the implementation supports updating inline uniform block descriptors after a set is bound. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK.

If the `VkPhysicalDeviceInlineUniformBlockFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceInlineUniformBlockFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceInlineUniformBlockFeatures-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INLINE_UNIFORM_BLOCK_FEATURES`

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_compute_shader_derivatives
typedef struct VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           computeDerivativeGroupQuads;
    VkBool32           computeDerivativeGroupLinear;
} VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `computeDerivativeGroupQuads` indicates that the implementation supports the `ComputeDerivativeGroupQuadsKHR` SPIR-V capability.
- `computeDerivativeGroupLinear` indicates that the implementation supports the `ComputeDerivativeGroupLinearKHR` SPIR-V capability.

See Quad shader scope for more information.

If the `VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR-sType-sType
  `sType` **must** be

VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COMPUTE_SHADER_DERIVATIVES_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_fragment_shader_barycentric
typedef struct VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           fragmentShaderBarycentric;
} VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- fragmentShaderBarycentric indicates that the implementation supports the BaryCoordKHR and BaryCoordNoPerspKHR SPIR-V fragment shader built-ins and supports the PerVertexKHR SPIR-V decoration on fragment shader input variables.

See Barycentric Interpolation for more information.

If the VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_BARYCENTRIC_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceFragmentDensityMapFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_fragment_density_map
typedef struct VkPhysicalDeviceFragmentDensityMapFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           fragmentDensityMap;
    VkBool32           fragmentDensityMapDynamic;
    VkBool32           fragmentDensityMapNonSubsampledImages;
} VkPhysicalDeviceFragmentDensityMapFeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- fragmentDensityMap specifies whether the implementation supports render passes with a fragment density map attachment. If this feature is not enabled and the pNext chain of VkRenderPassCreateInfo includes a VkRenderPassFragmentDensityMapCreateInfoEXT structure, fragmentDensityMapAttachment **must** be VK_ATTACHMENT_UNUSED.
- fragmentDensityMapDynamic specifies whether the implementation supports dynamic fragment density map image views. If this feature is not enabled, VK_IMAGE_VIEW_CREATE_FRAGMENT_DENSITY_MAP_DYNAMIC_BIT_EXT **must** not be included in VkImageViewCreateInfo::flags.
- fragmentDensityMapNonSubsampledImages specifies whether the implementation supports regular non-subsampled image attachments with fragment density map render passes. If this feature is not enabled, render passes with a fragment density map attachment **must** only have subsampled attachments bound.

If the VkPhysicalDeviceFragmentDensityMapFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceFragmentDensityMapFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentDensityMapFeaturesEXT-sType-sType

sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceFragmentDensityMap2FeaturesEXT structure is defined as:

```
// Provided by VK_EXT_fragment_density_map2
typedef struct VkPhysicalDeviceFragmentDensityMap2FeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           fragmentDensityMapDeferred;
} VkPhysicalDeviceFragmentDensityMap2FeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- fragmentDensityMapDeferred specifies whether the implementation supports deferred reads of fragment density map image views. If this feature is not enabled, VK_IMAGE_VIEW_CREATE_FRAGMENT_DENSITY_MAP_DEFERRED_BIT_EXT **must** not be included in VkImageViewCreateInfo::flags.

If the VkPhysicalDeviceFragmentDensityMap2FeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceFragmentDensityMap2FeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentDensityMap2FeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_2_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceFragmentDensityMapOffsetFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_fragment_density_map_offset
typedef struct VkPhysicalDeviceFragmentDensityMapOffsetFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           fragmentDensityMapOffset;
} VkPhysicalDeviceFragmentDensityMapOffsetFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- fragmentDensityMapOffset specifies whether the implementation supports fragment density map offsets.

If the VkPhysicalDeviceFragmentDensityMapOffsetFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceFragmentDensityMapOffsetFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentDensityMapOffsetFeaturesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_OFFSET_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceScalarBlockLayoutFeatures structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceScalarBlockLayoutFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           scalarBlockLayout;
} VkPhysicalDeviceScalarBlockLayoutFeatures;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `scalarBlockLayout` indicates that the implementation supports the layout of resource blocks in shaders using scalar alignment.

If the `VkPhysicalDeviceScalarBlockLayoutFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceScalarBlockLayoutFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceScalarBlockLayoutFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SCALAR_BLOCK_LAYOUT_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceUniformBufferStandardLayoutFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceUniformBufferStandardLayoutFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           uniformBufferStandardLayout;
} VkPhysicalDeviceUniformBufferStandardLayoutFeatures;
```

```
// Provided by VK_KHR_uniform_buffer_standard_layout
// Equivalent to VkPhysicalDeviceUniformBufferStandardLayoutFeatures
typedef VkPhysicalDeviceUniformBufferStandardLayoutFeatures
VkPhysicalDeviceUniformBufferStandardLayoutFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `uniformBufferStandardLayout` indicates that the implementation supports the same layouts for uniform buffers as for storage and other kinds of buffers. See Standard Buffer Layout.

If the `VkPhysicalDeviceUniformBufferStandardLayoutFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceUniformBufferStandardLayoutFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceUniformBufferStandardLayoutFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_UNIFORM_BUFFER_STANDARD_LAYOUT_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceDepthClipEnableFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_depth_clip_enable
typedef struct VkPhysicalDeviceDepthClipEnableFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           depthClipEnable;
} VkPhysicalDeviceDepthClipEnableFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `depthClipEnable` indicates that the implementation supports setting the depth clipping operation explicitly via the VkPipelineRasterizationDepthClipStateCreateInfoEXT pipeline state. Otherwise depth clipping is only enabled when VkPipelineRasterizationStateCreateInfo::`depthClampEnable` is VK_FALSE.

If the `VkPhysicalDeviceDepthClipEnableFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceDepthClipEnableFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDepthClipEnableFeaturesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLIP_ENABLE_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceMemoryPriorityFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_memory_priority
typedef struct VkPhysicalDeviceMemoryPriorityFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           memoryPriority;
} VkPhysicalDeviceMemoryPriorityFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryPriority` indicates that the implementation supports memory priorities specified at memory allocation time via VkMemoryPriorityAllocateInfoEXT.

If the `VkPhysicalDeviceMemoryPriorityFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to

indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceMemoryPriorityFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMemoryPriorityFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PRIORITY_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceBufferDeviceAddressFeatures structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceBufferDeviceAddressFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           bufferDeviceAddress;
    VkBool32           bufferDeviceAddressCaptureReplay;
    VkBool32           bufferDeviceAddressMultiDevice;
} VkPhysicalDeviceBufferDeviceAddressFeatures;
```

```
// Provided by VK_KHR_buffer_device_address
// Equivalent to VkPhysicalDeviceBufferDeviceAddressFeatures
typedef VkPhysicalDeviceBufferDeviceAddressFeatures
VkPhysicalDeviceBufferDeviceAddressFeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- bufferDeviceAddress indicates that the implementation supports accessing buffer memory in shaders as storage buffers via an address queried from vkGetBufferDeviceAddress.
- bufferDeviceAddressCaptureReplay indicates that the implementation supports saving and reusing buffer and device addresses, e.g. for trace capture and replay.
- bufferDeviceAddressMultiDevice indicates that the implementation supports the

`bufferDeviceAddress` , `rayTracingPipeline` and `rayQuery` features for logical devices created with multiple physical devices. If this feature is not supported, buffer and acceleration structure addresses **must** not be queried on a logical device created with more than one physical device.

NOTE `bufferDeviceAddressMultiDevice` exists to allow certain legacy platforms to be able to support `bufferDeviceAddress` without needing to support shared GPU virtual addresses for multi-device configurations.

See vkGetBufferDeviceAddress for more information.

If the `VkPhysicalDeviceBufferDeviceAddressFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceBufferDeviceAddressFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceBufferDeviceAddressFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BUFFER_DEVICE_ADDRESS_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceImagelessFramebufferFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceImagelessFramebufferFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           imagelessFramebuffer;
} VkPhysicalDeviceImagelessFramebufferFeatures;
```

```
// Provided by VK_KHR_imageless_framebuffer
// Equivalent to VkPhysicalDeviceImagelessFramebufferFeatures
typedef VkPhysicalDeviceImagelessFramebufferFeatures
VkPhysicalDeviceImagelessFramebufferFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `imagelessFramebuffer` indicates that the implementation supports specifying the image view for attachments at render pass begin time via VkRenderPassAttachmentBeginInfo.

If the `VkPhysicalDeviceImagelessFramebufferFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceImagelessFramebufferFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImagelessFramebufferFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGELESS_FRAMEBUFFER_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_fragment_shader_interlock
typedef struct VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           fragmentShaderSampleInterlock;
    VkBool32           fragmentShaderPixelInterlock;
    VkBool32           fragmentShaderShadingRateInterlock;
} VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `fragmentShaderSampleInterlock` indicates that the implementation supports the `FragmentShaderSampleInterlockEXT` SPIR-V capability.
- `fragmentShaderPixelInterlock` indicates that the implementation supports the `FragmentShaderPixelInterlockEXT` SPIR-V capability.

- `fragmentShaderShadingRateInterlock` indicates that the implementation supports the `FragmentShaderShadingRateInterlockEXT` SPIR-V capability.

If the `VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_INTERLOCK_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceCooperativeMatrixFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_cooperative_matrix
typedef struct VkPhysicalDeviceCooperativeMatrixFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           cooperativeMatrix;
    VkBool32           cooperativeMatrixRobustBufferAccess;
} VkPhysicalDeviceCooperativeMatrixFeaturesKHR;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `cooperativeMatrix` indicates that the implementation supports the `CooperativeMatrixKHR` SPIR-V capability.
- `cooperativeMatrixRobustBufferAccess` indicates that the implementation supports robust buffer access for SPIR-V `OpCooperativeMatrixLoadKHR` and `OpCooperativeMatrixStoreKHR` instructions.

If the `VkPhysicalDeviceCooperativeMatrixFeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to

indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceCooperativeMatrixFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceCooperativeMatrixFeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT structure is defined as:

```
// Provided by VK_EXT_cooperative_matrix_maintenance1
typedef struct VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           cooperativeMatrixProperties2;
    VkBool32           cooperativeMatrixReductions;
    VkBool32           cooperativeMatrixConversions;
    VkBool32           cooperativeMatrixPerElementOperations;
    VkBool32           cooperativeMatrixGetCoordinate;
} VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- cooperativeMatrixProperties2 indicates that the implementation supports the vkGetPhysicalDeviceCooperativeMatrixProperties2EXT command.
- cooperativeMatrixReductions indicates that the implementation supports the CooperativeMatrixReductionsEXT SPIR-V capability. This allows performing (row, column, 2x2, or all element) reductions on matrices.
- cooperativeMatrixConversions indicates that the implementation supports the CooperativeMatrixConversionsEXT SPIR-V capability. This allows converting accumulator matrices to A or B matrices.
- cooperativeMatrixPerElementOperations indicates that the implementation supports the

CooperativeMatrixPerElementOperationsEXT SPIR-V capability. This allows performing element-wise operations on matrix elements using a callback function.

- cooperativeMatrixGetCoordinate indicates that the implementation supports the CooperativeMatrixGetCoordinateEXT SPIR-V capability. This allows querying locations for a matrix element.

If the VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_MAINTENANCE_1_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderLongVectorFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_shader_long_vector
typedef struct VkPhysicalDeviceShaderLongVectorFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           longVector;
} VkPhysicalDeviceShaderLongVectorFeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- longVector indicates that the implementation supports the LongVectorEXT SPIR-V capability.

If the VkPhysicalDeviceShaderLongVectorFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to

indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderLongVectorFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderLongVectorFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_LONG_VECTOR_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceYcbcrImageArraysFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_ycbcr_image_arrays
typedef struct VkPhysicalDeviceYcbcrImageArraysFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           ycbcrImageArrays;
} VkPhysicalDeviceYcbcrImageArraysFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- ycbcrImageArrays indicates that the implementation supports creating images with a format that requires $Y'C_BC_R$ conversion and has multiple array layers.

If the VkPhysicalDeviceYcbcrImageArraysFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceYcbcrImageArraysFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceYcbcrImageArraysFeaturesEXT-sType-sType
  sType **must** be

VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_YCBCR_IMAGE_ARRAYS_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderSubgroupExtendedTypesFeatures structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceShaderSubgroupExtendedTypesFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderSubgroupExtendedTypes;
} VkPhysicalDeviceShaderSubgroupExtendedTypesFeatures;
```

```
// Provided by VK_KHR_shader_subgroup_extended_types
// Equivalent to VkPhysicalDeviceShaderSubgroupExtendedTypesFeatures
typedef VkPhysicalDeviceShaderSubgroupExtendedTypesFeatures
VkPhysicalDeviceShaderSubgroupExtendedTypesFeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderSubgroupExtendedTypes is a boolean specifying whether subgroup operations can use 8-bit integer, 16-bit integer, 64-bit integer, 16-bit floating-point, and vectors of these types in group operations with subgroup scope, if the implementation supports the types.

If the VkPhysicalDeviceShaderSubgroupExtendedTypesFeatures structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderSubgroupExtendedTypesFeatures, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderSubgroupExtendedTypesFeatures-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_EXTENDED_TYPES_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceHostQueryResetFeatures structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceHostQueryResetFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           hostQueryReset;
} VkPhysicalDeviceHostQueryResetFeatures;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- hostQueryReset indicates that the implementation supports resetting queries from the host with vkResetQueryPool.

If the VkPhysicalDeviceHostQueryResetFeatures structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceHostQueryResetFeatures, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceHostQueryResetFeatures-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_QUERY_RESET_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceTimelineSemaphoreFeatures structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceTimelineSemaphoreFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           timelineSemaphore;
} VkPhysicalDeviceTimelineSemaphoreFeatures;
```

```
// Provided by VK_KHR_timeline_semaphore
// Equivalent to VkPhysicalDeviceTimelineSemaphoreFeatures
typedef VkPhysicalDeviceTimelineSemaphoreFeatures
VkPhysicalDeviceTimelineSemaphoreFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `timelineSemaphore` indicates whether semaphores created with a VkSemaphoreType of VK_SEMAPHORE_TYPE_TIMELINE are supported.

If the `VkPhysicalDeviceTimelineSemaphoreFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceTimelineSemaphoreFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceTimelineSemaphoreFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceInternallySynchronizedQueuesFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_internally_synchronized_queues
typedef struct VkPhysicalDeviceInternallySynchronizedQueuesFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           internallySynchronizedQueues;
} VkPhysicalDeviceInternallySynchronizedQueuesFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `internallySynchronizedQueues` indicates that VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR can be used to make queues internally synchronized.

If the `VkPhysicalDeviceInternallySynchronizedQueuesFeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceInternallySynchronizedQueuesFeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceInternallySynchronizedQueuesFeaturesKHR-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INTERNALLY_SYNCHRONIZED_QUEUES_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceIndexTypeUint8Features` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceIndexTypeUint8Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           indexTypeUint8;
} VkPhysicalDeviceIndexTypeUint8Features;
```

```
// Provided by VK_KHR_index_type_uint8
// Equivalent to VkPhysicalDeviceIndexTypeUint8Features
typedef VkPhysicalDeviceIndexTypeUint8Features
VkPhysicalDeviceIndexTypeUint8FeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `indexTypeUint8` indicates that VK_INDEX_TYPE_UINT8 can be used with vkCmdBindIndexBuffer2 and vkCmdBindIndexBuffer.

If the `VkPhysicalDeviceIndexTypeUint8Features` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceIndexTypeUint8Features`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceIndexTypeUint8Features-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INDEX_TYPE_UINT8_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_primitive_topology_list_restart
typedef struct VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           primitiveTopologyListRestart;
    VkBool32           primitiveTopologyPatchListRestart;
} VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `primitiveTopologyListRestart` indicates that list type primitives, VK_PRIMITIVE_TOPOLOGY_POINT_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY and VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, **can** use the primitive restart index value in index buffers.
- `primitiveTopologyPatchListRestart` indicates that the VK_PRIMITIVE_TOPOLOGY_PATCH_LIST topology **can** use the primitive restart index value in index buffers.

If the `VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIMITIVE_TOPOLOGY_LIST_RESTART_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           separateDepthStencilLayouts;
} VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures;
```

```
// Provided by VK_KHR_separate_depth_stencil_layouts
// Equivalent to VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures
typedef VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures
VkPhysicalDeviceSeparateDepthStencilLayoutsFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `separateDepthStencilLayouts` indicates whether the implementation supports a `VkImageMemoryBarrier` for a depth/stencil image with only one of VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT set, and whether VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL can be used.

If the `VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SEPARATE_DEPTH_STENCIL_LAYOUTS_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef struct VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           pipelineExecutableInfo;
} VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pipelineExecutableInfo indicates that the implementation supports reporting properties and statistics about the pipeline executables associated with a compiled pipeline.

If the VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_EXECUTABLE_PROPERTIES_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderDemoteToHelperInvocationFeatures structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceShaderDemoteToHelperInvocationFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderDemoteToHelperInvocation;
} VkPhysicalDeviceShaderDemoteToHelperInvocationFeatures;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderDemoteToHelperInvocation` indicates whether the implementation supports the SPIR-V `DemoteToHelperInvocationEXT` capability.

If the `VkPhysicalDeviceShaderDemoteToHelperInvocationFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceShaderDemoteToHelperInvocationFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderDemoteToHelperInvocationFeatures-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_DEMOTE_TO_HELPER_INVOCATION_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_attachment_feedback_loop_dynamic_state
typedef struct VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           attachmentFeedbackLoopDynamicState;
} VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `attachmentFeedbackLoopDynamicState` specifies whether dynamic feedback loops are supported.

If the `VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ATTACHMENT_FEEDBACK_LOOP_DYNAMIC_STATE_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_legacy_vertex_attributes
typedef struct VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           legacyVertexAttributes;
} VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `legacyVertexAttributes` specifies whether compatibility features for vertex attributes are supported when using dynamic vertex input state.

If the `VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT` structure is included in the `pNext` chain of

the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LEGACY_VERTEX_ATTRIBUTES_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceTextureCompressionASTCHDRFeatures structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceTextureCompressionASTCHDRFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           textureCompressionASTC_HDR;
} VkPhysicalDeviceTextureCompressionASTCHDRFeatures;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- textureCompressionASTC_HDR indicates whether all of the ASTC HDR compressed texture formats are supported. If this feature is enabled, then the VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT, VK_FORMAT_FEATURE_BLIT_SRC_BIT and VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT features **must** be supported in optimalTilingFeatures for the following formats:
  - VK_FORMAT_ASTC_4x4_SFLOAT_BLOCK
  - VK_FORMAT_ASTC_5x4_SFLOAT_BLOCK
  - VK_FORMAT_ASTC_5x5_SFLOAT_BLOCK
  - VK_FORMAT_ASTC_6x5_SFLOAT_BLOCK
  - VK_FORMAT_ASTC_6x6_SFLOAT_BLOCK

- VK_FORMAT_ASTC_8x5_SFLOAT_BLOCK
- VK_FORMAT_ASTC_8x6_SFLOAT_BLOCK
- VK_FORMAT_ASTC_8x8_SFLOAT_BLOCK
- VK_FORMAT_ASTC_10x5_SFLOAT_BLOCK
- VK_FORMAT_ASTC_10x6_SFLOAT_BLOCK
- VK_FORMAT_ASTC_10x8_SFLOAT_BLOCK
- VK_FORMAT_ASTC_10x10_SFLOAT_BLOCK
- VK_FORMAT_ASTC_12x10_SFLOAT_BLOCK
- VK_FORMAT_ASTC_12x12_SFLOAT_BLOCK

To query for additional properties, or if the feature is not enabled, vkGetPhysicalDeviceFormatProperties and vkGetPhysicalDeviceImageFormatProperties **can** be used to check for supported properties of individual formats as normal.

If the VkPhysicalDeviceTextureCompressionASTCHDRFeatures structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceTextureCompressionASTCHDRFeatures, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceTextureCompressionASTCHDRFeatures-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TEXTURE_COMPRESSION_ASTC_HDR_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceTextureCompressionASTC3DFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_texture_compression_astc_3d
typedef struct VkPhysicalDeviceTextureCompressionASTC3DFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           textureCompressionASTC_3D;
} VkPhysicalDeviceTextureCompressionASTC3DFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `textureCompressionASTC_3D` indicates whether all of the ASTC 3D compressed texture formats are supported. If this feature is enabled, then the VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT, VK_FORMAT_FEATURE_BLIT_SRC_BIT and VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT features **must** be supported in `optimalTilingFeatures` for the following formats:
  - VK_FORMAT_ASTC_3x3x3_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_3x3x3_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_3x3x3_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_4x3x3_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_4x3x3_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_4x3x3_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_4x4x3_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_4x4x3_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_4x4x3_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_4x4x4_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_4x4x4_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_4x4x4_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_5x4x4_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_5x4x4_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_5x4x4_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_5x5x4_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_5x5x4_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_5x5x4_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_5x5x5_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_5x5x5_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_5x5x5_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_6x5x5_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_6x5x5_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_6x5x5_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_6x6x5_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_6x6x5_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_6x6x5_SFLOAT_BLOCK_EXT

- VK_FORMAT_ASTC_6x6x6_UNORM_BLOCK_EXT
- VK_FORMAT_ASTC_6x6x6_SRGB_BLOCK_EXT
- VK_FORMAT_ASTC_6x6x6_SFLOAT_BLOCK_EXT

If the VkPhysicalDeviceTextureCompressionASTC3DFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceTextureCompressionASTC3DFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceTextureCompressionASTC3DFeaturesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TEXTURE_COMPRESSION_ASTC_3D_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceLineRasterizationFeatures structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceLineRasterizationFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           rectangularLines;
    VkBool32           bresenhamLines;
    VkBool32           smoothLines;
    VkBool32           stippledRectangularLines;
    VkBool32           stippledBresenhamLines;
    VkBool32           stippledSmoothLines;
} VkPhysicalDeviceLineRasterizationFeatures;
```

```
// Provided by VK_KHR_line_rasterization
// Equivalent to VkPhysicalDeviceLineRasterizationFeatures
typedef VkPhysicalDeviceLineRasterizationFeatures
VkPhysicalDeviceLineRasterizationFeaturesKHR;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `rectangularLines` indicates whether the implementation supports rectangular line rasterization.
- `bresenhamLines` indicates whether the implementation supports Bresenham-style line rasterization.
- `smoothLines` indicates whether the implementation supports smooth line rasterization.
- `stippledRectangularLines` indicates whether the implementation supports stippled line rasterization with VK_LINE_RASTERIZATION_MODE_RECTANGULAR lines.
- `stippledBresenhamLines` indicates whether the implementation supports stippled line rasterization with VK_LINE_RASTERIZATION_MODE_BRESENHAM lines.
- `stippledSmoothLines` indicates whether the implementation supports stippled line rasterization with VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH lines.

If the `VkPhysicalDeviceLineRasterizationFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceLineRasterizationFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceLineRasterizationFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceSubgroupSizeControlFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceSubgroupSizeControlFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           subgroupSizeControl;
    VkBool32           computeFullSubgroups;
} VkPhysicalDeviceSubgroupSizeControlFeatures;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `subgroupSizeControl` indicates whether the implementation supports controlling shader subgroup sizes via the VK_PIPELINE_SHADER_STAGE_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT flag and the VkPipelineShaderStageRequiredSubgroupSizeCreateInfo structure.
- `computeFullSubgroups` indicates whether the implementation supports requiring full subgroups in compute shaders via the VK_PIPELINE_SHADER_STAGE_CREATE_REQUIRE_FULL_SUBGROUPS_BIT flag.

If the `VkPhysicalDeviceSubgroupSizeControlFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceSubgroupSizeControlFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

NOTE

The `VkPhysicalDeviceSubgroupSizeControlFeaturesEXT` structure was added in version 2 of the `VK_EXT_subgroup_size_control` extension. Version 1 implementations of this extension will not fill out the features structure but applications may assume that both `subgroupSizeControl` and `computeFullSubgroups` are supported if the extension is supported. (See also the Feature Requirements section.) Applications are advised to add a `VkPhysicalDeviceSubgroupSizeControlFeaturesEXT` structure to the `pNext` chain of VkDeviceCreateInfo to enable the features regardless of the version of the extension supported by the implementation. If the implementation only supports version 1, it will safely ignore the `VkPhysicalDeviceSubgroupSizeControlFeaturesEXT` structure.

Vulkan 1.3 implementations always support the features structure.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSubgroupSizeControlFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SUBGROUP_SIZE_CONTROL_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceAccelerationStructureFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkPhysicalDeviceAccelerationStructureFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           accelerationStructure;
    VkBool32           accelerationStructureCaptureReplay;
    VkBool32           accelerationStructureIndirectBuild;
    // accelerationStructureHostCommands is legacy, but no reason was given in the API
XML
    VkBool32           accelerationStructureHostCommands;
    VkBool32           descriptorBindingAccelerationStructureUpdateAfterBind;
} VkPhysicalDeviceAccelerationStructureFeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- accelerationStructure indicates whether the implementation supports the acceleration structure functionality. See Acceleration Structures.
- accelerationStructureCaptureReplay indicates whether the implementation supports saving and reusing acceleration structure device addresses, e.g. for trace capture and replay.
- accelerationStructureIndirectBuild indicates whether the implementation supports indirect acceleration structure build commands, e.g. vkCmdBuildAccelerationStructuresIndirectKHR.
- accelerationStructureHostCommands indicates whether the implementation supports host side acceleration structure commands, e.g. vkBuildAccelerationStructuresKHR, vkCopyAccelerationStructureKHR, vkCopyAccelerationStructureToMemoryKHR, vkCopyMemoryToAccelerationStructureKHR, vkWriteAccelerationStructuresPropertiesKHR.
- descriptorBindingAccelerationStructureUpdateAfterBind indicates whether the implementation supports updating acceleration structure descriptors after a set is bound. If this feature is not enabled, VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR.

If the VkPhysicalDeviceAccelerationStructureFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceAccelerationStructureFeaturesKHR, it

must add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceAccelerationStructureFeaturesKHR-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ACCELERATION_STRUCTURE_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceRayTracingPipelineFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef struct VkPhysicalDeviceRayTracingPipelineFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           rayTracingPipeline;
    VkBool32           rayTracingPipelineShaderGroupHandleCaptureReplay;
    VkBool32           rayTracingPipelineShaderGroupHandleCaptureReplayMixed;
    VkBool32           rayTracingPipelineTraceRaysIndirect;
    VkBool32           rayTraversalPrimitiveCulling;
} VkPhysicalDeviceRayTracingPipelineFeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- rayTracingPipeline indicates whether the implementation supports the ray tracing pipeline functionality. See Ray Tracing.
- rayTracingPipelineShaderGroupHandleCaptureReplay indicates whether the implementation supports saving and reusing shader group handles, e.g. for trace capture and replay.
- rayTracingPipelineShaderGroupHandleCaptureReplayMixed indicates whether the implementation supports reuse of shader group handles being arbitrarily mixed with creation of non-reused shader group handles. If this is VK_FALSE, all reused shader group handles **must** be specified before any non-reused handles **may** be created.
- rayTracingPipelineTraceRaysIndirect indicates whether the implementation supports indirect ray tracing commands, e.g. vkCmdTraceRaysIndirectKHR.
- rayTraversalPrimitiveCulling indicates whether the implementation supports primitive culling

during ray traversal.

If the VkPhysicalDeviceRayTracingPipelineFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceRayTracingPipelineFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage

- VUID-VkPhysicalDeviceRayTracingPipelineFeaturesKHR-rayTracingPipelineShaderGroupHandleCaptureReplayMixed-03575
  If rayTracingPipelineShaderGroupHandleCaptureReplayMixed is VK_TRUE, rayTracingPipelineShaderGroupHandleCaptureReplay **must** also be VK_TRUE

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRayTracingPipelineFeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_PIPELINE_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceRayQueryFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_ray_query
typedef struct VkPhysicalDeviceRayQueryFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           rayQuery;
} VkPhysicalDeviceRayQueryFeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- rayQuery indicates whether the implementation supports ray query (OpRayQueryProceedKHR) functionality.

If the VkPhysicalDeviceRayQueryFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceRayQueryFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRayQueryFeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_QUERY_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR structure is defined as:

```
// Provided by VK_KHR_ray_tracing_maintenance1
typedef struct VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           rayTracingMaintenance1;
    VkBool32           rayTracingPipelineTraceRaysIndirect2;
} VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- rayTracingMaintenance1 indicates that the implementation supports the following:
  - The CullMaskKHR SPIR-V builtin using the SPV_KHR_ray_cull_mask SPIR-V extension.
  - Additional acceleration structure property queries: VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR and VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR.
  - A new access flag VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR.
  - A new pipeline stage flag bit VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR.
- rayTracingPipelineTraceRaysIndirect2 indicates whether the implementation supports the extended indirect ray tracing command vkCmdTraceRaysIndirect2KHR.

If the VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_MAINTENANCE_1_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceVideoEncodeFeedback2FeaturesKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_feedback2
typedef struct VkPhysicalDeviceVideoEncodeFeedback2FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           videoEncodeFeedback2;
} VkPhysicalDeviceVideoEncodeFeedback2FeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- videoEncodeFeedback2 specifies that the implementation supports per picture partition feedback and the following additional video encode feedback flags, as reported in VkVideoEncodeFeedback2CapabilitiesKHR and/or VkVideoEncodeCapabilitiesKHR ::supportedEncodeFeedbackFlags, respectively:
  - VK_VIDEO_ENCODE_FEEDBACK_AVERAGE_QUANTIZATION_BIT_KHR.
  - VK_VIDEO_ENCODE_FEEDBACK_MIN_QUANTIZATION_BIT_KHR.
  - VK_VIDEO_ENCODE_FEEDBACK_MAX_QUANTIZATION_BIT_KHR.
  - VK_VIDEO_ENCODE_FEEDBACK_INTRA_PIXELS_BIT_KHR.
  - VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR.

- VK_VIDEO_ENCODE_FEEDBACK_SKIPPED_PIXELS_BIT_KHR.
- VK_VIDEO_ENCODE_FEEDBACK_PICTURE_PARTITION_COUNT_BIT_KHR.

NOTE	Support for videoEncodeFeedback2 does not indicate that all video encode profiles support per picture partition feedback and the new video encode feedback flags. Support for individual video encode profiles is subject to video-profile-specific capabilities.

If the VkPhysicalDeviceVideoEncodeFeedback2FeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceVideoEncodeFeedback2FeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVideoEncodeFeedback2FeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_FEEDBACK_2_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceVideoEncodeQuantizationMapFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_quantization_map
typedef struct VkPhysicalDeviceVideoEncodeQuantizationMapFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           videoEncodeQuantizationMap;
} VkPhysicalDeviceVideoEncodeQuantizationMapFeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- videoEncodeQuantizationMap indicates that the implementation supports video encode quantization maps.

NOTE	Support for videoEncodeQuantizationMap does not indicate that all video encode

profiles support quantization maps. Support for quantization maps for any specific video encode profile is subject to video-profile-specific capabilities.

If the VkPhysicalDeviceVideoEncodeQuantizationMapFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceVideoEncodeQuantizationMapFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVideoEncodeQuantizationMapFeaturesKHR-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_QUANTIZATION_MAP_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceVideoEncodeIntraRefreshFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_intra_refresh
typedef struct VkPhysicalDeviceVideoEncodeIntraRefreshFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           videoEncodeIntraRefresh;
} VkPhysicalDeviceVideoEncodeIntraRefreshFeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- videoEncodeIntraRefresh specifies that the implementation supports video encode intra refresh.

NOTE Support for videoEncodeIntraRefresh does not indicate that all video encode profiles support intra refresh. Support for intra refresh for any specific video encode profile is subject to video-profile-specific capabilities.

If the VkPhysicalDeviceVideoEncodeIntraRefreshFeaturesKHR structure is included in the pNext chain

of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceVideoEncodeIntraRefreshFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVideoEncodeIntraRefreshFeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_INTRA_REFRESH_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceVideoMaintenance1FeaturesKHR structure is defined as:

```
// Provided by VK_KHR_video_maintenance1
typedef struct VkPhysicalDeviceVideoMaintenance1FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           videoMaintenance1;
} VkPhysicalDeviceVideoMaintenance1FeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- videoMaintenance1 indicates that the implementation supports the following:
  - The new buffer creation flag VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR.
  - The new image creation flag VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR.
  - The new video session creation flag VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR.

If the VkPhysicalDeviceVideoMaintenance1FeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceVideoMaintenance1FeaturesKHR, it **must**

add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVideoMaintenance1FeaturesKHR-sType-sType sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_MAINTENANCE_1_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceVideoMaintenance2FeaturesKHR structure is defined as:

```
// Provided by VK_KHR_video_maintenance2
typedef struct VkPhysicalDeviceVideoMaintenance2FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           videoMaintenance2;
} VkPhysicalDeviceVideoMaintenance2FeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- videoMaintenance2 specifies that the implementation supports the following:
  - Support for issuing video coding control commands against video decode sessions without a bound video session parameters object.
  - The new video session creation flag VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR for video decode sessions.
  - Required support for the rate control mode VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR for the following video encode profiles:
    - H.264 encode profiles;
    - H.265 encode profiles;
    - AV1 encode profiles.
  - Additional guarantees on Video Std parameters used with video encode profiles that the

implementations support without the need to override them.

If the VkPhysicalDeviceVideoMaintenance2FeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceVideoMaintenance2FeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVideoMaintenance2FeaturesKHR-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_MAINTENANCE_2_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceVideoEncodeAV1FeaturesKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_av1
typedef struct VkPhysicalDeviceVideoEncodeAV1FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           videoEncodeAV1;
} VkPhysicalDeviceVideoEncodeAV1FeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- videoEncodeAV1 indicates that the implementation supports AV1 encode operations.

If the VkPhysicalDeviceVideoEncodeAV1FeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceVideoEncodeAV1FeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVideoEncodeAV1FeaturesKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_AV1_FEATURES_KHR`

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceVideoDecodeVP9FeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_vp9
typedef struct VkPhysicalDeviceVideoDecodeVP9FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           videoDecodeVP9;
} VkPhysicalDeviceVideoDecodeVP9FeaturesKHR;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `videoDecodeVP9` specifies that the implementation supports VP9 decode operations.

If the `VkPhysicalDeviceVideoDecodeVP9FeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceVideoDecodeVP9FeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVideoDecodeVP9FeaturesKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_DECODE_VP9_FEATURES_KHR`

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceExtendedDynamicStateFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_extended_dynamic_state
typedef struct VkPhysicalDeviceExtendedDynamicStateFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           extendedDynamicState;
} VkPhysicalDeviceExtendedDynamicStateFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- extendedDynamicState indicates that the implementation supports the following dynamic states:
  - VK_DYNAMIC_STATE_CULL_MODE
  - VK_DYNAMIC_STATE_FRONT_FACE
  - VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY
  - VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT
  - VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT
  - VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE
  - VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE
  - VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE
  - VK_DYNAMIC_STATE_DEPTH_COMPARE_OP
  - VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE
  - VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE
  - VK_DYNAMIC_STATE_STENCIL_OP

If the VkPhysicalDeviceExtendedDynamicStateFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceExtendedDynamicStateFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExtendedDynamicStateFeaturesEXT-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_FEATURES_EXT`

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceExtendedDynamicState2FeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_extended_dynamic_state2
typedef struct VkPhysicalDeviceExtendedDynamicState2FeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           extendedDynamicState2;
    VkBool32           extendedDynamicState2LogicOp;
    VkBool32           extendedDynamicState2PatchControlPoints;
} VkPhysicalDeviceExtendedDynamicState2FeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `extendedDynamicState2` indicates that the implementation supports the following dynamic states:
  - VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE
  - VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE
  - VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE
- `extendedDynamicState2LogicOp` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_LOGIC_OP_EXT
- `extendedDynamicState2PatchControlPoints` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT

If the `VkPhysicalDeviceExtendedDynamicState2FeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a

VkDevice with any features described by VkPhysicalDeviceExtendedDynamicState2FeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExtendedDynamicState2FeaturesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_2_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceExtendedDynamicState3FeaturesEXT structure is defined as:

```
// Provided by VK_EXT_extended_dynamic_state3
typedef struct VkPhysicalDeviceExtendedDynamicState3FeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           extendedDynamicState3TessellationDomainOrigin;
    VkBool32           extendedDynamicState3DepthClampEnable;
    VkBool32           extendedDynamicState3PolygonMode;
    VkBool32           extendedDynamicState3RasterizationSamples;
    VkBool32           extendedDynamicState3SampleMask;
    VkBool32           extendedDynamicState3AlphaToCoverageEnable;
    VkBool32           extendedDynamicState3AlphaToOneEnable;
    VkBool32           extendedDynamicState3LogicOpEnable;
    VkBool32           extendedDynamicState3ColorBlendEnable;
    VkBool32           extendedDynamicState3ColorBlendEquation;
    VkBool32           extendedDynamicState3ColorWriteMask;
    VkBool32           extendedDynamicState3RasterizationStream;
    VkBool32           extendedDynamicState3ConservativeRasterizationMode;
    VkBool32           extendedDynamicState3ExtraPrimitiveOverestimationSize;
    VkBool32           extendedDynamicState3DepthClipEnable;
    VkBool32           extendedDynamicState3SampleLocationsEnable;
    VkBool32           extendedDynamicState3ColorBlendAdvanced;
    VkBool32           extendedDynamicState3ProvokingVertexMode;
    VkBool32           extendedDynamicState3LineRasterizationMode;
    VkBool32           extendedDynamicState3LineStippleEnable;
    VkBool32           extendedDynamicState3DepthClipNegativeOneToOne;
    VkBool32           extendedDynamicState3ViewportWScalingEnable;
    VkBool32           extendedDynamicState3ViewportSwizzle;
    VkBool32           extendedDynamicState3CoverageToColorEnable;
    VkBool32           extendedDynamicState3CoverageToColorLocation;
    VkBool32           extendedDynamicState3CoverageModulationMode;
    VkBool32           extendedDynamicState3CoverageModulationTableEnable;
    VkBool32           extendedDynamicState3CoverageModulationTable;
    VkBool32           extendedDynamicState3CoverageReductionMode;
    VkBool32           extendedDynamicState3RepresentativeFragmentTestEnable;
    VkBool32           extendedDynamicState3ShadingRateImageEnable;
} VkPhysicalDeviceExtendedDynamicState3FeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `extendedDynamicState3TessellationDomainOrigin` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT
- `extendedDynamicState3DepthClampEnable` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT

- `extendedDynamicState3PolygonMode` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_POLYGON_MODE_EXT
- `extendedDynamicState3RasterizationSamples` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT
- `extendedDynamicState3SampleMask` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_SAMPLE_MASK_EXT
- `extendedDynamicState3AlphaToCoverageEnable` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT
- `extendedDynamicState3AlphaToOneEnable` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT
- `extendedDynamicState3LogicOpEnable` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT
- `extendedDynamicState3ColorBlendEnable` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT
- `extendedDynamicState3ColorBlendEquation` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT
- `extendedDynamicState3ColorWriteMask` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT
- `extendedDynamicState3RasterizationStream` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT
- `extendedDynamicState3ConservativeRasterizationMode` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT
- `extendedDynamicState3ExtraPrimitiveOverestimationSize` indicates that the implementation supports the following dynamic state:
  - VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT
- `extendedDynamicState3DepthClipEnable` indicates that the implementation supports the following dynamic state:

    - VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT
- `extendedDynamicState3SampleLocationsEnable` indicates that the implementation supports the following dynamic state:
    - VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT
- `extendedDynamicState3ColorBlendAdvanced` indicates that the implementation supports the following dynamic state:
    - VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT
- `extendedDynamicState3ProvokingVertexMode` indicates that the implementation supports the following dynamic state:
    - VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT
- `extendedDynamicState3LineRasterizationMode` indicates that the implementation supports the following dynamic state:
    - VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT
- `extendedDynamicState3LineStippleEnable` indicates that the implementation supports the following dynamic state:
    - VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT
- `extendedDynamicState3DepthClipNegativeOneToOne` indicates that the implementation supports the following dynamic state:
    - VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT
- `extendedDynamicState3ViewportWScalingEnable` indicates that the implementation supports the following dynamic state:
    - VK_DYNAMIC_STATE_VIEWPORT_W_SCALING_ENABLE_NV
- `extendedDynamicState3ViewportSwizzle` indicates that the implementation supports the following dynamic state:
    - VK_DYNAMIC_STATE_VIEWPORT_SWIZZLE_NV
- `extendedDynamicState3CoverageToColorEnable` indicates that the implementation supports the following dynamic state:
    - VK_DYNAMIC_STATE_COVERAGE_TO_COLOR_ENABLE_NV
- `extendedDynamicState3CoverageToColorLocation` indicates that the implementation supports the following dynamic state:
    - VK_DYNAMIC_STATE_COVERAGE_TO_COLOR_LOCATION_NV
- `extendedDynamicState3CoverageModulationMode` indicates that the implementation supports the following dynamic state:
    - VK_DYNAMIC_STATE_COVERAGE_MODULATION_MODE_NV
- `extendedDynamicState3CoverageModulationTableEnable` indicates that the implementation supports the following dynamic state:
    - VK_DYNAMIC_STATE_COVERAGE_MODULATION_TABLE_ENABLE_NV
- `extendedDynamicState3CoverageModulationTable` indicates that the implementation supports the

following dynamic state:

  - VK_DYNAMIC_STATE_COVERAGE_MODULATION_TABLE_NV

- extendedDynamicState3CoverageReductionMode indicates that the implementation supports the following dynamic state:

  - VK_DYNAMIC_STATE_COVERAGE_REDUCTION_MODE_NV

- extendedDynamicState3RepresentativeFragmentTestEnable indicates that the implementation supports the following dynamic state:

  - VK_DYNAMIC_STATE_REPRESENTATIVE_FRAGMENT_TEST_ENABLE_NV

- extendedDynamicState3ShadingRateImageEnable indicates that the implementation supports the following dynamic state:

  - VK_DYNAMIC_STATE_SHADING_RATE_IMAGE_ENABLE_NV

If the VkPhysicalDeviceExtendedDynamicState3FeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceExtendedDynamicState3FeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExtendedDynamicState3FeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_3_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceExtendedFlagsFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_extended_flags
typedef struct VkPhysicalDeviceExtendedFlagsFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           extendedFlags;
} VkPhysicalDeviceExtendedFlagsFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `extendedFlags` indicates that the implementation supports the following:
- VkFormatProperties4KHR, extending VkFormatProperties2, and corresponding VkFormatFeatureFlagBits4KHR and VkFormatFeatureFlags4KHR types providing up to 64 new format feature flags.
- VkImageCreateFlags2CreateInfoKHR, extending structures passed to image create and query functions, and corresponding VkImageCreateFlagBits2KHR and VkImageCreateFlags2KHR types providing up to 32 new image create flags.
- VkImageUsageFlags2CreateInfoKHR, extending structures passed to image create and query functions, and corresponding VkImageUsageFlagBits2KHR and VkImageUsageFlags2KHR types providing up to 32 new image usage flags.
- The VkPipelineCreateFlags2CreateInfoKHR, VkPipelineCreateFlags2KHR, VkPipelineCreateFlagBits2KHR, VkBufferUsageFlags2CreateInfoKHR, VkBufferUsageFlags2KHR, and VkBufferUsageFlagBits2KHR types originally introduced by VK_KHR_maintenance5.

If the `VkPhysicalDeviceExtendedFlagsFeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceExtendedFlagsFeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExtendedFlagsFeaturesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_FLAGS_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           deviceGeneratedCommands;
    VkBool32           dynamicGeneratedPipelineLayout;
} VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `deviceGeneratedCommands` indicates whether the implementation supports functionality to generate commands on the device.
- `dynamicGeneratedPipelineLayout` indicates the implementation allows the `pipelineLayout` member of VkIndirectCommandsLayoutCreateInfoEXT to be VK_NULL_HANDLE and VkPipelineLayoutCreateInfo **can** be chained off those structures' `pNext` instead.

If the `VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_GENERATED_COMMANDS_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceDeviceMemoryReportFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_device_memory_report
typedef struct VkPhysicalDeviceDeviceMemoryReportFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           deviceMemoryReport;
} VkPhysicalDeviceDeviceMemoryReportFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `deviceMemoryReport` indicates whether the implementation supports the ability to register device memory report callbacks.

If the `VkPhysicalDeviceDeviceMemoryReportFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceDeviceMemoryReportFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDeviceMemoryReportFeaturesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_MEMORY_REPORT_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceGlobalPriorityQueryFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceGlobalPriorityQueryFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           globalPriorityQuery;
} VkPhysicalDeviceGlobalPriorityQueryFeatures;
```

```
// Provided by VK_KHR_global_priority
// Equivalent to VkPhysicalDeviceGlobalPriorityQueryFeatures
typedef VkPhysicalDeviceGlobalPriorityQueryFeatures
VkPhysicalDeviceGlobalPriorityQueryFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `globalPriorityQuery` indicates whether the implementation supports the ability to query global queue priorities.

If the `VkPhysicalDeviceGlobalPriorityQueryFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceGlobalPriorityQueryFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceGlobalPriorityQueryFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GLOBAL_PRIORITY_QUERY_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDevicePipelineCreationCacheControlFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDevicePipelineCreationCacheControlFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           pipelineCreationCacheControl;
} VkPhysicalDevicePipelineCreationCacheControlFeatures;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.

- pNext is NULL or a pointer to a structure extending this structure.
- pipelineCreationCacheControl indicates that the implementation supports:
  - The following **can** be used in Vk*PipelineCreateInfo::flags:
    - VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT
    - VK_PIPELINE_CREATE_EARLY_RETURN_ON_FAILURE_BIT
  - The following **can** be used in VkPipelineCacheCreateInfo::flags:
    - VK_PIPELINE_CACHE_CREATE_EXTERNALLY_SYNCHRONIZED_BIT

If the VkPhysicalDevicePipelineCreationCacheControlFeatures structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDevicePipelineCreationCacheControlFeatures, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePipelineCreationCacheControlFeatures-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_CREATION_CACHE_CONTROL_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderZeroInitializeWorkgroupMemory;
} VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures;
```

```
// Provided by VK_KHR_zero_initialize_workgroup_memory
// Equivalent to VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures
typedef VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures
VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderZeroInitializeWorkgroupMemory` specifies whether the implementation supports initializing a variable in Workgroup storage class.

If the `VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ZERO_INITIALIZE_WORKGROUP_MEMORY_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDevicePrivateDataFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDevicePrivateDataFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           privateData;
} VkPhysicalDevicePrivateDataFeatures;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- privateData indicates whether the implementation supports private data. See Private Data.

If the VkPhysicalDevicePrivateDataFeatures structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDevicePrivateDataFeatures, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePrivateDataFeatures-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIVATE_DATA_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_shader_subgroup_uniform_control_flow
typedef struct VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderSubgroupUniformControlFlow;
} VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderSubgroupUniformControlFlow specifies whether the implementation supports the shader execution mode SubgroupUniformControlFlowKHR.

If the VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR, it **must** add an instance of the

structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_UNIFORM_CONTROL_FLOW_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceRobustness2FeaturesKHR structure is defined as:

```
// Provided by VK_KHR_robustness2
typedef struct VkPhysicalDeviceRobustness2FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           robustBufferAccess2;
    VkBool32           robustImageAccess2;
    VkBool32           nullDescriptor;
} VkPhysicalDeviceRobustness2FeaturesKHR;
```

```
// Provided by VK_EXT_robustness2
// Equivalent to VkPhysicalDeviceRobustness2FeaturesKHR
typedef VkPhysicalDeviceRobustness2FeaturesKHR VkPhysicalDeviceRobustness2FeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- robustBufferAccess2 enables Robust Buffer Access 2 guarantees for shader buffer accesses.
- robustImageAccess2 enables Robust Image Access 2 guarantees for shader image accesses.
- nullDescriptor indicates whether descriptors **can** be written with a VK_NULL_HANDLE resource or view, which are considered valid to access and act as if the descriptor were bound to nothing.

If the VkPhysicalDeviceRobustness2FeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to

indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceRobustness2FeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage

- VUID-VkPhysicalDeviceRobustness2FeaturesKHR-robustBufferAccess2-04000
  If robustBufferAccess2 is enabled then robustBufferAccess **must** also be enabled

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRobustness2FeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceImageRobustnessFeatures structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceImageRobustnessFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           robustImageAccess;
} VkPhysicalDeviceImageRobustnessFeatures;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- robustImageAccess enables Robust Image Access guarantees for shader image accesses.

If the VkPhysicalDeviceImageRobustnessFeatures structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceImageRobustnessFeatures, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImageRobustnessFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_ROBUSTNESS_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceShaderTerminateInvocationFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceShaderTerminateInvocationFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderTerminateInvocation;
} VkPhysicalDeviceShaderTerminateInvocationFeatures;
```

```
// Provided by VK_KHR_shader_terminate_invocation
// Equivalent to VkPhysicalDeviceShaderTerminateInvocationFeatures
typedef VkPhysicalDeviceShaderTerminateInvocationFeatures
VkPhysicalDeviceShaderTerminateInvocationFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderTerminateInvocation` specifies whether the implementation supports SPIR-V modules that use the `SPV_KHR_terminate_invocation` extension.

If the `VkPhysicalDeviceShaderTerminateInvocationFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceShaderTerminateInvocationFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderTerminateInvocationFeatures-sType-sType

sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TERMINATE_INVOCATION_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceCustomBorderColorFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_custom_border_color
typedef struct VkPhysicalDeviceCustomBorderColorFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           customBorderColors;
    VkBool32           customBorderColorWithoutFormat;
} VkPhysicalDeviceCustomBorderColorFeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- customBorderColors indicates that the implementation supports providing a borderColor value with one of the following values at sampler creation time:
  - VK_BORDER_COLOR_FLOAT_CUSTOM_EXT
  - VK_BORDER_COLOR_INT_CUSTOM_EXT
- customBorderColorWithoutFormat indicates that explicit formats are not required for custom border colors and the value of the format member of the VkSamplerCustomBorderColorCreateInfoEXT structure **may** be VK_FORMAT_UNDEFINED. If this feature bit is not set, applications **must** provide the VkFormat of the image view(s) being sampled by this sampler in the format member of the VkSamplerCustomBorderColorCreateInfoEXT structure.

If the VkPhysicalDeviceCustomBorderColorFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceCustomBorderColorFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceCustomBorderColorFeaturesEXT-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CUSTOM_BORDER_COLOR_FEATURES_EXT`

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceBorderColorSwizzleFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_border_color_swizzle
typedef struct VkPhysicalDeviceBorderColorSwizzleFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           borderColorSwizzle;
    VkBool32           borderColorSwizzleFromImage;
} VkPhysicalDeviceBorderColorSwizzleFeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `borderColorSwizzle` indicates that defined values are returned by sampled image operations when used with a sampler that uses a VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK, VK_BORDER_COLOR_INT_OPAQUE_BLACK, VK_BORDER_COLOR_FLOAT_CUSTOM_EXT, or VK_BORDER_COLOR_INT_CUSTOM_EXT `borderColor` and an image view that uses a non-identity component mapping, when either `borderColorSwizzleFromImage` is enabled or the VkSamplerBorderColorComponentMappingCreateInfoEXT is specified.
- `borderColorSwizzleFromImage` indicates that the implementation will return the correct border color values from sampled image operations under the conditions expressed above, without the application having to specify the border color component mapping when creating the sampler object. If this feature bit is not set, applications **can** chain a VkSamplerBorderColorComponentMappingCreateInfoEXT structure when creating samplers for use with image views that do not have an identity swizzle and, when those samplers are combined with image views using the same component mapping, sampled image operations that use opaque black or custom border colors will return the correct border color values.

If the `VkPhysicalDeviceBorderColorSwizzleFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a

VkDevice with any features described by VkPhysicalDeviceBorderColorSwizzleFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceBorderColorSwizzleFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BORDER_COLOR_SWIZZLE_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevicePortabilitySubsetFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_portability_subset
typedef struct VkPhysicalDevicePortabilitySubsetFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           constantAlphaColorBlendFactors;
    VkBool32           events;
    VkBool32           imageViewFormatReinterpretation;
    VkBool32           imageViewFormatSwizzle;
    VkBool32           imageView2DOn3DImage;
    VkBool32           multisampleArrayImage;
    VkBool32           mutableComparisonSamplers;
    VkBool32           pointPolygons;
    VkBool32           samplerMipLodBias;
    VkBool32           separateStencilMaskRef;
    VkBool32           shaderSampleRateInterpolationFunctions;
    VkBool32           tessellationIsolines;
    VkBool32           tessellationPointMode;
    VkBool32           triangleFans;
    VkBool32           vertexAttributeAccessBeyondStride;
} VkPhysicalDevicePortabilitySubsetFeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- constantAlphaColorBlendFactors indicates whether this implementation supports constant *alpha* Blend Factors used as source or destination *color* Blending.

- `events` indicates whether this implementation supports synchronization using Events.
- `imageViewFormatReinterpretation` indicates whether this implementation supports a `VkImageView` being created with a texel format containing a different number of components, or a different number of bits in each component, than the texel format of the underlying `VkImage`.
- `imageViewFormatSwizzle` indicates whether this implementation supports remapping format components using VkImageViewCreateInfo::`components`.
- `imageView2DOn3DImage` indicates whether this implementation supports a `VkImage` being created with the VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT flag set, permitting a 2D or 2D array image view to be created on a 3D `VkImage`.
- `multisampleArrayImage` indicates whether this implementation supports a `VkImage` being created as a 2D array with multiple samples per texel.
- `mutableComparisonSamplers` indicates whether this implementation allows descriptors with comparison samplers to be updated.
- `pointPolygons` indicates whether this implementation supports Rasterization using a *point* Polygon Mode.
- `samplerMipLodBias` indicates whether this implementation supports setting a mipmap LOD bias value when creating a sampler.
- `separateStencilMaskRef` indicates whether this implementation supports separate front and back Stencil Test reference values.
- `shaderSampleRateInterpolationFunctions` indicates whether this implementation supports fragment shaders which use the `InterpolationFunction` capability and the extended instructions `InterpolateAtCentroid`, `InterpolateAtOffset`, and `InterpolateAtSample` from the `GLSL.std.450` extended instruction set. This member is only meaningful if the `sampleRateShading` feature is supported.
- `tessellationIsolines` indicates whether this implementation supports isoline output from the Tessellation stage of a graphics pipeline. This member is only meaningful if the `tessellationShader` feature is supported.
- `tessellationPointMode` indicates whether this implementation supports point output from the Tessellation stage of a graphics pipeline. This member is only meaningful if the `tessellationShader` feature is supported.
- `triangleFans` indicates whether this implementation supports Triangle Fans primitive topology.
- `vertexAttributeAccessBeyondStride` indicates whether this implementation supports accessing a vertex input attribute beyond the stride of the corresponding vertex input binding.

If the `VkPhysicalDevicePortabilitySubsetFeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDevicePortabilitySubsetFeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePortabilitySubsetFeaturesKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PORTABILITY_SUBSET_FEATURES_KHR`

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDevicePerformanceQueryFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_performance_query
typedef struct VkPhysicalDevicePerformanceQueryFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           performanceCounterQueryPools;
    VkBool32           performanceCounterMultipleQueryPools;
} VkPhysicalDevicePerformanceQueryFeaturesKHR;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `performanceCounterQueryPools` indicates whether the implementation supports performance counter query pools.
- `performanceCounterMultipleQueryPools` indicates whether the implementation supports using multiple performance query pools in a primary command buffer and secondary command buffers executed within it.

If the `VkPhysicalDevicePerformanceQueryFeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDevicePerformanceQueryFeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePerformanceQueryFeaturesKHR-sType-sType
  `sType` **must** be

VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PERFORMANCE_QUERY_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceMutableDescriptorTypeFeaturesEXT structure is defined as:

WARNING	This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_mutable_descriptor_type
typedef struct VkPhysicalDeviceMutableDescriptorTypeFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           mutableDescriptorType;
} VkPhysicalDeviceMutableDescriptorTypeFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- mutableDescriptorType indicates that the implementation **must** support using the VkDescriptorType of VK_DESCRIPTOR_TYPE_MUTABLE_EXT with at least the following descriptor types, where any combination of the types **must** be supported:
  - VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE
  - VK_DESCRIPTOR_TYPE_STORAGE_IMAGE
  - VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER
  - VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER
  - VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER
  - VK_DESCRIPTOR_TYPE_STORAGE_BUFFER
- Additionally, mutableDescriptorType indicates that:
  - Non-uniform descriptor indexing **must** be supported if all descriptor types in a VkMutableDescriptorTypeListEXT for VK_DESCRIPTOR_TYPE_MUTABLE_EXT have the corresponding non-uniform indexing features enabled in VkPhysicalDeviceDescriptorIndexingFeatures.
  - VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT with descriptorType of VK_DESCRIPTOR_TYPE_MUTABLE_EXT relaxes the list of required descriptor types to the descriptor types which have the corresponding update-after-bind feature enabled in

VkPhysicalDeviceDescriptorIndexingFeatures.

- Dynamically uniform descriptor indexing **must** be supported if all descriptor types in a VkMutableDescriptorTypeListEXT for VK_DESCRIPTOR_TYPE_MUTABLE_EXT have the corresponding dynamic indexing features enabled.
- VK_DESCRIPTOR_SET_LAYOUT_CREATE_HOST_ONLY_POOL_BIT_EXT **must** be supported.
- VK_DESCRIPTOR_POOL_CREATE_HOST_ONLY_BIT_EXT **must** be supported.

If the VkPhysicalDeviceMutableDescriptorTypeFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceMutableDescriptorTypeFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMutableDescriptorTypeFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MUTABLE_DESCRIPTOR_TYPE_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceDepthClipControlFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_depth_clip_control
typedef struct VkPhysicalDeviceDepthClipControlFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           depthClipControl;
} VkPhysicalDeviceDepthClipControlFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- depthClipControl indicates that the implementation supports setting VkPipelineViewportDepthClipControlCreateInfoEXT::negativeOneToOne to VK_TRUE.

If the VkPhysicalDeviceDepthClipControlFeaturesEXT structure is included in the pNext chain of the

VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceDepthClipControlFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDepthClipControlFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLIP_CONTROL_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceDepthClampControlFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_depth_clamp_control
typedef struct VkPhysicalDeviceDepthClampControlFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           depthClampControl;
} VkPhysicalDeviceDepthClampControlFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- depthClampControl indicates that the implementation supports setting VkPipelineViewportDepthClampControlCreateInfoEXT::depthClampMode to VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT.

If the VkPhysicalDeviceDepthClampControlFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceDepthClampControlFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDepthClampControlFeaturesEXT-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLAMP_CONTROL_FEATURES_EXT`

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_workgroup_memory_explicit_layout
typedef struct VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           workgroupMemoryExplicitLayout;
    VkBool32           workgroupMemoryExplicitLayoutScalarBlockLayout;
    VkBool32           workgroupMemoryExplicitLayout8BitAccess;
    VkBool32           workgroupMemoryExplicitLayout16BitAccess;
} VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `workgroupMemoryExplicitLayout` indicates whether the implementation supports the SPIR-V `WorkgroupMemoryExplicitLayoutKHR` capability.
- `workgroupMemoryExplicitLayoutScalarBlockLayout` indicates whether the implementation supports scalar alignment for laying out Workgroup Blocks.
- `workgroupMemoryExplicitLayout8BitAccess` indicates whether objects in the `Workgroup` storage class with the `Block` decoration **can** have 8-bit integer members. If this feature is not enabled, 8-bit integer members **must** not be used in such objects. This also indicates whether shader modules **can** declare the `WorkgroupMemoryExplicitLayout8BitAccessKHR` capability.
- `workgroupMemoryExplicitLayout16BitAccess` indicates whether objects in the `Workgroup` storage class with the `Block` decoration **can** have 16-bit integer and 16-bit floating-point members. If this feature is not enabled, 16-bit integer or 16-bit floating-point members **must** not be used in such objects. This also indicates whether shader modules **can** declare the `WorkgroupMemoryExplicitLayout16BitAccessKHR` capability.

If the `VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR` structure is included in the `pNext`

chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_WORKGROUP_MEMORY_EXPLICIT_LAYOUT_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceSynchronization2Features structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceSynchronization2Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           synchronization2;
} VkPhysicalDeviceSynchronization2Features;
```

```
// Provided by VK_KHR_synchronization2
// Equivalent to VkPhysicalDeviceSynchronization2Features
typedef VkPhysicalDeviceSynchronization2Features
VkPhysicalDeviceSynchronization2FeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- synchronization2 indicates whether the implementation supports the new set of synchronization commands introduced in VK_KHR_synchronization2.

If the VkPhysicalDeviceSynchronization2Features structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to

indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceSynchronization2Features, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSynchronization2Features-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SYNCHRONIZATION_2_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_vertex_input_dynamic_state
typedef struct VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           vertexInputDynamicState;
} VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- vertexInputDynamicState indicates that the implementation supports the following dynamic states:
  - VK_DYNAMIC_STATE_VERTEX_INPUT_EXT

If the VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT-sType-sType

sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_INPUT_DYNAMIC_STATE_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_primitives_generated_query
typedef struct VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           primitivesGeneratedQuery;
    VkBool32           primitivesGeneratedQueryWithRasterizerDiscard;
    VkBool32           primitivesGeneratedQueryWithNonZeroStreams;
} VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- primitivesGeneratedQuery indicates whether the implementation supports the VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT query type.
- primitivesGeneratedQueryWithRasterizerDiscard indicates whether the implementation supports this query when rasterization discard is enabled.
- primitivesGeneratedQueryWithNonZeroStreams indicates whether the implementation supports this query with a non-zero index in vkCmdBeginQueryIndexedEXT.

If the VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT-sType-sType
sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIMITIVES_GENERATED_QUERY_FEATURES_

EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceFragmentShadingRateFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_fragment_shading_rate
typedef struct VkPhysicalDeviceFragmentShadingRateFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           pipelineFragmentShadingRate;
    VkBool32           primitiveFragmentShadingRate;
    VkBool32           attachmentFragmentShadingRate;
} VkPhysicalDeviceFragmentShadingRateFeaturesKHR;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pipelineFragmentShadingRate indicates that the implementation supports the pipeline fragment shading rate.
- primitiveFragmentShadingRate indicates that the implementation supports the primitive fragment shading rate.
- attachmentFragmentShadingRate indicates that the implementation supports the attachment fragment shading rate.

If the VkPhysicalDeviceFragmentShadingRateFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceFragmentShadingRateFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentShadingRateFeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceLegacyDitheringFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_legacy_dithering
typedef struct VkPhysicalDeviceLegacyDitheringFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           legacyDithering;
} VkPhysicalDeviceLegacyDitheringFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- legacyDithering indicates whether the implementation supports Legacy Dithering.

If the VkPhysicalDeviceLegacyDitheringFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceLegacyDitheringFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceLegacyDitheringFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LEGACY_DITHERING_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderUntypedPointersFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_shader_untyped_pointers
typedef struct VkPhysicalDeviceShaderUntypedPointersFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderUntypedPointers;
} VkPhysicalDeviceShaderUntypedPointersFeaturesKHR;
```

The members of VkPhysicalDeviceShaderUntypedPointersFeaturesKHR describe the following features:

- shaderUntypedPointers specifies whether shader modules **can** declare the UntypedPointersKHR capability and untyped pointers in any explicitly laid out storage class.

If the VkPhysicalDeviceShaderUntypedPointersFeaturesKHR structure is included in the pNext chain of VkPhysicalDeviceFeatures2, it is filled with values indicating whether the features are supported. VkPhysicalDeviceShaderUntypedPointersFeaturesKHR **can** also be included in the pNext chain of VkDeviceCreateInfo to enable the features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderUntypedPointersFeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_UNTYPED_POINTERS_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevicePipelineProtectedAccessFeatures structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDevicePipelineProtectedAccessFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           pipelineProtectedAccess;
} VkPhysicalDevicePipelineProtectedAccessFeatures;
```

```
// Provided by VK_EXT_pipeline_protected_access
// Equivalent to VkPhysicalDevicePipelineProtectedAccessFeatures
typedef VkPhysicalDevicePipelineProtectedAccessFeatures
VkPhysicalDevicePipelineProtectedAccessFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pipelineProtectedAccess` indicates whether the implementation supports specifying protected access on individual pipelines.

If the `VkPhysicalDevicePipelineProtectedAccessFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDevicePipelineProtectedAccessFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePipelineProtectedAccessFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_PROTECTED_ACCESS_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceColorWriteEnableFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_color_write_enable
typedef struct VkPhysicalDeviceColorWriteEnableFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           colorWriteEnable;
} VkPhysicalDeviceColorWriteEnableFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `colorWriteEnable` indicates that the implementation supports the dynamic state VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT.

If the `VkPhysicalDeviceColorWriteEnableFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to

indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceColorWriteEnableFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceColorWriteEnableFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COLOR_WRITE_ENABLE_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevicePipelinePropertiesFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_pipeline_properties
typedef struct VkPhysicalDevicePipelinePropertiesFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           pipelinePropertiesIdentifier;
} VkPhysicalDevicePipelinePropertiesFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pipelinePropertiesIdentifier indicates that the implementation supports querying a unique pipeline identifier.

If the VkPhysicalDevicePipelinePropertiesFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDevicePipelinePropertiesFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePipelinePropertiesFeaturesEXT-sType-sType
  sType **must** be

VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_PROPERTIES_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceProvokingVertexFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_provoking_vertex
typedef struct VkPhysicalDeviceProvokingVertexFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           provokingVertexLast;
    VkBool32           transformFeedbackPreservesProvokingVertex;
} VkPhysicalDeviceProvokingVertexFeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- provokingVertexLast indicates whether the implementation supports the VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT provoking vertex mode for flat shading.
- transformFeedbackPreservesProvokingVertex indicates that the order of vertices within each primitive written by transform feedback will preserve the provoking vertex. This does not apply to triangle fan primitives when transformFeedbackPreservesTriangleFanProvokingVertex is VK_FALSE. transformFeedbackPreservesProvokingVertex **must** be VK_FALSE when the VK_EXT_transform_feedback extension is not supported.

If the VkPhysicalDeviceProvokingVertexFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceProvokingVertexFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

When VkPhysicalDeviceProvokingVertexFeaturesEXT is in the pNext chain of VkDeviceCreateInfo but the transformFeedback feature is not enabled, the value of transformFeedbackPreservesProvokingVertex is ignored.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceProvokingVertexFeaturesEXT-sType-sType

sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROVOKING_VERTEX_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceDescriptorBufferFeaturesEXT structure is defined as:

WARNING This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
typedef struct VkPhysicalDeviceDescriptorBufferFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           descriptorBuffer;
    VkBool32           descriptorBufferCaptureReplay;
    VkBool32           descriptorBufferImageLayoutIgnored;
    VkBool32           descriptorBufferPushDescriptors;
} VkPhysicalDeviceDescriptorBufferFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- descriptorBuffer indicates that the implementation supports putting shader-accessible descriptors directly in memory.
- descriptorBufferCaptureReplay indicates that the implementation supports capture and replay when using descriptor buffers. If this is VK_TRUE, all resources created with VK_BUFFER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT, VK_IMAGE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT, VK_IMAGE_VIEW_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT, VK_SAMPLER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT, or VK_ACCELERATION_STRUCTURE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT **must** be created before resources of the same types without those flags.
- descriptorBufferImageLayoutIgnored indicates that the implementation will ignore imageLayout in VkDescriptorImageInfo when calling vkGetDescriptorEXT.
- descriptorBufferPushDescriptors indicates that the implementation supports using push descriptors with descriptor buffers.

If the VkPhysicalDeviceDescriptorBufferFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceDescriptorBufferFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDescriptorBufferFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_BUFFER_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevicePageableDeviceLocalMemoryFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_pageable_device_local_memory
typedef struct VkPhysicalDevicePageableDeviceLocalMemoryFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           pageableDeviceLocalMemory;
} VkPhysicalDevicePageableDeviceLocalMemoryFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pageableDeviceLocalMemory indicates that the implementation supports pageable device-local memory and **may** transparently move device-local memory allocations to host-local memory to better share device-local memory with other applications.

If the VkPhysicalDevicePageableDeviceLocalMemoryFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDevicePageableDeviceLocalMemoryFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePageableDeviceLocalMemoryFeaturesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PAGEABLE_DEVICE_LOCAL_MEMORY_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceMultiDrawFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_multi_draw
typedef struct VkPhysicalDeviceMultiDrawFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           multiDraw;
} VkPhysicalDeviceMultiDrawFeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `multiDraw` indicates that the implementation supports vkCmdDrawMultiEXT and vkCmdDrawMultiIndexedEXT.

If the `VkPhysicalDeviceMultiDrawFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceMultiDrawFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMultiDrawFeaturesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTI_DRAW_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceOpacityMicromapFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_opacity_micromap
typedef struct VkPhysicalDeviceOpacityMicromapFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           micromap;
} VkPhysicalDeviceOpacityMicromapFeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- micromap indicates whether the implementation supports the micromap array feature.

If the VkPhysicalDeviceOpacityMicromapFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceOpacityMicromapFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceOpacityMicromapFeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceOpacityMicromapFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkPhysicalDeviceOpacityMicromapFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           micromap;
    VkBool32           micromapCaptureReplay;
    // micromapHostCommands is legacy, but no reason was given in the API XML
    VkBool32           micromapHostCommands;
} VkPhysicalDeviceOpacityMicromapFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `micromap` indicates whether the implementation supports the micromap array feature.
- `micromapCaptureReplay` indicates whether the implementation supports capture and replay of addresses for micromap arrays.
- `micromapHostCommands` indicates whether the implementation supports host side micromap array commands.

If the `VkPhysicalDeviceOpacityMicromapFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceOpacityMicromapFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceOpacityMicromapFeaturesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceZeroInitializeDeviceMemoryFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_zero_initialize_device_memory
typedef struct VkPhysicalDeviceZeroInitializeDeviceMemoryFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           zeroInitializeDeviceMemory;
} VkPhysicalDeviceZeroInitializeDeviceMemoryFeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `zeroInitializeDeviceMemory` indicates that the implementation supports zeroing memory allocations using a user-specified flag.

If the `VkPhysicalDeviceZeroInitializeDeviceMemoryFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceZeroInitializeDeviceMemoryFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceZeroInitializeDeviceMemoryFeaturesEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ZERO_INITIALIZE_DEVICE_MEMORY_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDevicePresentIdFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_present_id
typedef struct VkPhysicalDevicePresentIdFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           presentId;
} VkPhysicalDevicePresentIdFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `presentId` indicates that the implementation supports specifying present ID values in the `VkPresentIdKHR` extension to the `VkPresentInfoKHR` struct.

If the `VkPhysicalDevicePresentIdFeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDevicePresentIdFeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePresentIdFeaturesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_ID_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDevicePresentId2FeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_present_id2
typedef struct VkPhysicalDevicePresentId2FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           presentId2;
} VkPhysicalDevicePresentId2FeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `presentId2` indicates that the implementation supports specifying present ID values in the `VkPresentId2KHR` extension to the `VkPresentInfoKHR` struct.

If the `VkPhysicalDevicePresentId2FeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a

VkDevice with any features described by VkPhysicalDevicePresentId2FeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePresentId2FeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_ID_2_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevicePresentWaitFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_present_wait
typedef struct VkPhysicalDevicePresentWaitFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           presentWait;
} VkPhysicalDevicePresentWaitFeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- presentWait indicates that the implementation supports vkWaitForPresentKHR.

If the VkPhysicalDevicePresentWaitFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDevicePresentWaitFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePresentWaitFeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_WAIT_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevicePresentWait2FeaturesKHR structure is defined as:

```
// Provided by VK_KHR_present_wait2
typedef struct VkPhysicalDevicePresentWait2FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           presentWait2;
} VkPhysicalDevicePresentWait2FeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- presentWait2 indicates that the implementation supports vkWaitForPresent2KHR.

If the VkPhysicalDevicePresentWait2FeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDevicePresentWait2FeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePresentWait2FeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_WAIT_2_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceUnifiedImageLayoutsFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_unified_image_layouts
typedef struct VkPhysicalDeviceUnifiedImageLayoutsFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           unifiedImageLayouts;
    VkBool32           unifiedImageLayoutsVideo;
} VkPhysicalDeviceUnifiedImageLayoutsFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `unifiedImageLayouts` specifies whether usage of VK_IMAGE_LAYOUT_GENERAL, where valid, incurs no loss in efficiency. Additionally, it indicates whether it **can** be used in place of VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT.
- `unifiedImageLayoutsVideo` specifies whether VK_IMAGE_LAYOUT_GENERAL can be used in place of any of the following image layouts with no loss in efficiency:
  - VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR
  - VK_IMAGE_LAYOUT_VIDEO_DECODE_SRC_KHR
  - VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR
  - VK_IMAGE_LAYOUT_VIDEO_ENCODE_DST_KHR
  - VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR
  - VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR
  - VK_IMAGE_LAYOUT_VIDEO_ENCODE_QUANTIZATION_MAP_KHR

If the `VkPhysicalDeviceUnifiedImageLayoutsFeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceUnifiedImageLayoutsFeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceUnifiedImageLayoutsFeaturesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_UNIFIED_IMAGE_LAYOUTS_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo

- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceHostImageCopyFeatures structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceHostImageCopyFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           hostImageCopy;
} VkPhysicalDeviceHostImageCopyFeatures;
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to VkPhysicalDeviceHostImageCopyFeatures
typedef VkPhysicalDeviceHostImageCopyFeatures
VkPhysicalDeviceHostImageCopyFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- hostImageCopy indicates that the implementation supports copying from host memory to images using the vkCopyMemoryToImage command, copying from images to host memory using the vkCopyImageToMemory command, and copying between images using the vkCopyImageToImage command.

If the VkPhysicalDeviceHostImageCopyFeatures structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceHostImageCopyFeatures, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceHostImageCopyFeatures-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevicePresentTimingFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_present_timing
typedef struct VkPhysicalDevicePresentTimingFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           presentTiming;
    VkBool32           presentAtAbsoluteTime;
    VkBool32           presentAtRelativeTime;
} VkPhysicalDevicePresentTimingFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- presentTiming indicates that the implementation supports vkGetPastPresentationTimingEXT.
- presentAtAbsoluteTime indicates that the implementation supports specifying absolute target present times.
- presentAtRelativeTime indicates that the implementation supports specifying relative target present times.

If the VkPhysicalDevicePresentTimingFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDevicePresentTimingFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePresentTimingFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_TIMING_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderFmaFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_shader_fma
typedef struct VkPhysicalDeviceShaderFmaFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderFmaFloat16;
    VkBool32           shaderFmaFloat32;
    VkBool32           shaderFmaFloat64;
} VkPhysicalDeviceShaderFmaFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderFmaFloat16` indicates whether the implementation supports `OpFmaKHR` for Float16 types in shaders.
- `shaderFmaFloat32` indicates whether the implementation supports `OpFmaKHR` for Float32 types in shaders.
- `shaderFmaFloat64` indicates whether the implementation supports `OpFmaKHR` for Float64 types in shaders.

If the `VkPhysicalDeviceShaderFmaFeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceShaderFmaFeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderFmaFeaturesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FMA_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceShaderIntegerDotProductFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceShaderIntegerDotProductFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderIntegerDotProduct;
} VkPhysicalDeviceShaderIntegerDotProductFeatures;
```

```
// Provided by VK_KHR_shader_integer_dot_product
// Equivalent to VkPhysicalDeviceShaderIntegerDotProductFeatures
typedef VkPhysicalDeviceShaderIntegerDotProductFeatures
VkPhysicalDeviceShaderIntegerDotProductFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderIntegerDotProduct` specifies whether shader modules **can** declare the `DotProductInputAllKHR`, `DotProductInput4x8BitKHR`, `DotProductInput4x8BitPackedKHR` and `DotProductKHR` capabilities.

If the `VkPhysicalDeviceShaderIntegerDotProductFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceShaderIntegerDotProductFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderIntegerDotProductFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceMaintenance4Features` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceMaintenance4Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           maintenance4;
} VkPhysicalDeviceMaintenance4Features;
```

```
// Provided by VK_KHR_maintenance4
// Equivalent to VkPhysicalDeviceMaintenance4Features
typedef VkPhysicalDeviceMaintenance4Features VkPhysicalDeviceMaintenance4FeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maintenance4` indicates that the implementation supports the following:
  - The application **may** destroy a VkPipelineLayout object immediately after using it to create another object.
  - `LocalSizeId` **can** be used as an alternative to `LocalSize` to specify the local workgroup size with specialization constants.
  - Images created with identical creation parameters will always have the same alignment requirements.
  - The size memory requirement of a buffer or image is never greater than that of another buffer or image created with a greater or equal size.
  - Push constants do not have to be initialized before they are dynamically accessed.
  - The interface matching rules allow a larger output vector to match with a smaller input vector, with additional values being discarded.

If the `VkPhysicalDeviceMaintenance4Features` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceMaintenance4Features`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance4Features-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceMaintenance5Features structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceMaintenance5Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           maintenance5;
} VkPhysicalDeviceMaintenance5Features;
```

```
// Provided by VK_KHR_maintenance5
// Equivalent to VkPhysicalDeviceMaintenance5Features
typedef VkPhysicalDeviceMaintenance5Features VkPhysicalDeviceMaintenance5FeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- maintenance5 indicates that the implementation supports the following:
  - The ability to expose support for the optional format VK_FORMAT_A1B5G5R5_UNORM_PACK16.
  - The ability to expose support for the optional format VK_FORMAT_A8_UNORM.
  - A property to indicate that multisample coverage operations are performed after sample counting in EarlyFragmentTests mode.
  - Creating a VkBufferView with a subset of the associated VkBuffer usage using VkBufferUsageFlags2CreateInfo.
  - A new function vkCmdBindIndexBuffer2, allowing a range of memory to be bound as an index buffer.
  - vkGetDeviceProcAddr will return NULL for function pointers of core functions for versions higher than the version requested by the application.
  - vkCmdBindVertexBuffers2 supports using VK_WHOLE_SIZE in the pSizes parameter.
  - If PointSize is not written, a default value of 1.0 is used for the size of points.
  - VkShaderModuleCreateInfo **can** be added as a chained structure to pipeline creation via VkPipelineShaderStageCreateInfo, rather than having to create a shader module.
  - A function vkGetRenderingAreaGranularity to query the optimal render area for a dynamic

rendering instance.

- A property to indicate that depth/stencil texturing operations with VK_COMPONENT_SWIZZLE_ONE have defined behavior.
- vkGetDeviceImageSubresourceLayout allows an application to perform a vkGetImageSubresourceLayout query without having to create an image.
- VK_REMAINING_ARRAY_LAYERS as the `layerCount` member of VkImageSubresourceLayers.
- A property to indicate whether `PointSize` controls the final rasterization of polygons if polygon mode is VK_POLYGON_MODE_POINT.
- Two properties to indicate the non-strict line rasterization algorithm used.
- Two new flags words VkPipelineCreateFlagBits2 and VkBufferUsageFlagBits2.
- Physical-device-level functions **can** now be called with any value in the valid range for a type beyond the defined enumerants, such that applications can avoid checking individual features, extensions, or versions before querying supported properties of a particular enumerant.
- Copies between images of any type are allowed, with 1D images treated as 2D images with a height of 1.

If the `VkPhysicalDeviceMaintenance5Features` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceMaintenance5Features`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance5Features-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceMaintenance6Features` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceMaintenance6Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           maintenance6;
} VkPhysicalDeviceMaintenance6Features;
```

```
// Provided by VK_KHR_maintenance6
// Equivalent to VkPhysicalDeviceMaintenance6Features
typedef VkPhysicalDeviceMaintenance6Features VkPhysicalDeviceMaintenance6FeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maintenance6` indicates that the implementation supports the following:
  - VK_NULL_HANDLE **can** be used when binding an index buffer.
  - VkBindMemoryStatus **can** be included in the `pNext` chain of the VkBindBufferMemoryInfo and VkBindImageMemoryInfo structures, enabling applications to retrieve VkResult values for individual memory binding operations.
  - VkPhysicalDeviceMaintenance6Properties::`blockTexelViewCompatibleMultipleLayers` property to indicate that the implementation supports creating image views with VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT where the `layerCount` member of `subresourceRange` is greater than 1.
  - VkPhysicalDeviceMaintenance6Properties::`maxCombinedImageSamplerDescriptorCount` property which indicates the maximum number of descriptors required for sampled images with $Y'C_BC_R$ conversion.
  - A VkPhysicalDeviceMaintenance6Properties::`fragmentShadingRateClampCombinerInputs` property which indicates whether the implementation clamps the inputs to fragment shading rate combiner operations.

If the `VkPhysicalDeviceMaintenance6Features` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceMaintenance6Features`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance6Features-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceMaintenance7FeaturesKHR structure is defined as:

```
// Provided by VK_KHR_maintenance7
typedef struct VkPhysicalDeviceMaintenance7FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           maintenance7;
} VkPhysicalDeviceMaintenance7FeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- maintenance7 indicates that the implementation supports the following:
  - The VK_RENDERING_CONTENTS_INLINE_BIT_KHR and VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR flags **can** be used to record commands in render pass instances both inline and in secondary command buffers executed with vkCmdExecuteCommands for dynamic rendering and legacy render passes respectively.
  - Querying information regarding the underlying devices in environments where the Vulkan implementation is provided through layered implementations. This is done by chaining VkPhysicalDeviceLayeredApiPropertiesListKHR to VkPhysicalDeviceProperties2.
  - New limits which indicate the maximum total count of dynamic uniform buffers and dynamic storage buffers that **can** be included in a pipeline layout.
  - 32-bit timestamp queries **must** wrap on overflow.
  - A property that indicates whether a fragment shading rate attachment can have a size that is too small to cover a specified render area.
  - A property that indicates support for writing to one aspect of a depth/stencil attachment without performing a read-modify-write operation on the other aspect.

If the VkPhysicalDeviceMaintenance7FeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceMaintenance7FeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance7FeaturesKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_7_FEATURES_KHR`

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceMaintenance8FeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance8
typedef struct VkPhysicalDeviceMaintenance8FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           maintenance8;
} VkPhysicalDeviceMaintenance8FeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maintenance8` indicates that the implementation supports the following:
  - Allow copies between depth/stencil and “matching” color attachments
  - Allow `dstCache` in vkMergePipelineCaches to be implicitly synchronized.
  - Require src/dst sync scopes to work when doing queue family ownership transfers
  - Support `Offset` (as an alternative to `ConstOffset`) image operand in texture sampling and fetch operations
  - Use the SPIR-V definition of `OpSRem` and `OpSMod`, making these operations produce well-defined results for negative operands
  - Loosen layer restrictions when blitting from 3D images to other image types
  - Add space for an additional 64 access flags for use with VkMemoryBarrier2, VkBufferMemoryBarrier2, and VkImageMemoryBarrier2

If the `VkPhysicalDeviceMaintenance8FeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceMaintenance8FeaturesKHR`, it **must** add an

instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance8FeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_8_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceMaintenance9FeaturesKHR structure is defined as:

```
// Provided by VK_KHR_maintenance9
typedef struct VkPhysicalDeviceMaintenance9FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           maintenance9;
} VkPhysicalDeviceMaintenance9FeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- maintenance9 indicates that the implementation supports the following:
  - The restriction that certain bitfield SPIR-V instructions only operate on 32-bit integers is relaxed.
  - The value returned when a vertex shader reads an unbound vertex attribute is defined by way of the defaultVertexAttributeValue property.
  - A new VkQueryPoolCreateFlagBits::VK_QUERY_POOL_CREATE_RESET_BIT_KHR flag **can** be used to initialize all queries in the query pool to the reset state on creation.
  - vkCmdSetEvent2 **may** not provide a dependency other than the event src stage mask.
  - The effects of image memory barriers and image layout transitions on 3D images created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT are limited to only those slices specified in VkImageSubresourceRange.
  - A device **can** be created with no queues. This can be used for compiling pipelines or shaders for the purpose of filling pipeline caches.

- Queue family ownership transfers are no longer required for buffers and linear images. For optimally tiled images, a new physical device query is added to check if resources created with VK_SHARING_MODE_EXCLUSIVE **can** automatically transfer ownership between two queue families.
- `image2DViewOf3DSparse` enables 2D views of 3D sparse images.

If the `VkPhysicalDeviceMaintenance9FeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceMaintenance9FeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance9FeaturesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_9_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceMaintenance10FeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance10
typedef struct VkPhysicalDeviceMaintenance10FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           maintenance10;
} VkPhysicalDeviceMaintenance10FeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maintenance10` indicates that the implementation supports the following:
  - New image format feature bits that indicate support for copying depth or stencil aspects using non-graphics queue families
  - If vkCmdSetSampleMaskEXT is called with `pSampleMask` set to `NULL`, it is treated as if the mask has all bits set to `1`.

- Add vkCmdEndRendering2KHR as an extensible version of vkCmdEndRendering
- Add input attachment information to dynamic rendering
- Require that vertex inputs follow sRGB encoding when those formats are used, instead of being underspecified.
- Add a query to determine if sRGB images are resolved in nonlinear or linear space by default
- Add an optional feature to allow applications to override the default sRGB resolve behavior
- Add resolve mode and depth-stencil resolve support to vkCmdResolveImage2 to bring it in-line with render pass attachment resolves

If the VkPhysicalDeviceMaintenance10FeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceMaintenance10FeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance10FeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_10_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceMaintenance11FeaturesKHR structure is defined as:

```
// Provided by VK_KHR_maintenance11
typedef struct VkPhysicalDeviceMaintenance11FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           maintenance11;
} VkPhysicalDeviceMaintenance11FeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.

- `maintenance11` indicates that the implementation supports the following:
  - VK_IMAGE_CREATE_ALIAS_SINGLE_LAYER_DESCRIPTOR_BIT_KHR
  - The depth clipping state of the pipeline is well defined when the pipeline is created without VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT being set and the VkPipelineRasterizationDepthClipStateCreateInfoEXT structure is not present
  - VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR
  - Allow `queueFamilyIndexCount` of 1 in VkBufferCreateInfo, VkImageCreateInfo, VkPhysicalDeviceImageDrmFormatModifierInfoEXT, when `sharingMode` is VK_SHARING_MODE_CONCURRENT
  - Require `minImageTransferGranularity` to be (1,1,1) even on transfer-only queues and add `optimalImageTransferGranularity` queue family property to communicate the performance bump for copies not aligned to the optimal granularity
  - When copying between a buffer and an image on a transfer-only queue, do not require `bufferOffset` to be a multiple of 4

If the `VkPhysicalDeviceMaintenance11FeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceMaintenance11FeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance11FeaturesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_11_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceDynamicRenderingFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceDynamicRenderingFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           dynamicRendering;
} VkPhysicalDeviceDynamicRenderingFeatures;
```

```
// Provided by VK_KHR_dynamic_rendering
// Equivalent to VkPhysicalDeviceDynamicRenderingFeatures
typedef VkPhysicalDeviceDynamicRenderingFeatures
VkPhysicalDeviceDynamicRenderingFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `dynamicRendering` specifies that the implementation supports dynamic render pass instances using the vkCmdBeginRendering command.

If the `VkPhysicalDeviceDynamicRenderingFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceDynamicRenderingFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDynamicRenderingFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceCustomResolveFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_custom_resolve
typedef struct VkPhysicalDeviceCustomResolveFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           customResolve;
} VkPhysicalDeviceCustomResolveFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.

- pNext is NULL or a pointer to a structure extending this structure.
- customResolve specifies that the implementation supports render pass resolves using shaders.

If the VkPhysicalDeviceCustomResolveFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceCustomResolveFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceCustomResolveFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CUSTOM_RESOLVE_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceRGBA10X6FormatsFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_rgba10x6_formats
typedef struct VkPhysicalDeviceRGBA10X6FormatsFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           formatRgba10x6WithoutYCbCrSampler;
} VkPhysicalDeviceRGBA10X6FormatsFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- formatRgba10x6WithoutYCbCrSampler indicates that VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16 **can** be used with a VkImageView with subresourceRange.aspectMask equal to VK_IMAGE_ASPECT_COLOR_BIT without a sampler $Y'C_BC_R$ conversion enabled.

If the VkPhysicalDeviceRGBA10X6FormatsFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceRGBA10X6FormatsFeaturesEXT, it **must** add

an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRGBA10X6FormatsFeaturesEXT-sType-sType sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RGBA10X6_FORMATS_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevicePipelineRobustnessFeatures structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDevicePipelineRobustnessFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           pipelineRobustness;
} VkPhysicalDevicePipelineRobustnessFeatures;
```

```
// Provided by VK_EXT_pipeline_robustness
// Equivalent to VkPhysicalDevicePipelineRobustnessFeatures
typedef VkPhysicalDevicePipelineRobustnessFeatures
VkPhysicalDevicePipelineRobustnessFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pipelineRobustness indicates that robustness **can** be requested on a per-pipeline-stage granularity.

NOTE Enabling the pipelineRobustness feature may, on some platforms, incur a minor performance cost when the robustBufferAccess feature is not enabled, even for pipelines which do not make use of any robustness features. If robustness is not needed, the pipelineRobustness feature should not be enabled by an application.

If the VkPhysicalDevicePipelineRobustnessFeatures structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to

indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDevicePipelineRobustnessFeatures, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePipelineRobustnessFeatures-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_ROBUSTNESS_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceImageViewMinLodFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_image_view_min_lod
typedef struct VkPhysicalDeviceImageViewMinLodFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           minLod;
} VkPhysicalDeviceImageViewMinLodFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- minLod indicates whether the implementation supports clamping the minimum LOD value during Image Level(s) Selection, Texel Gathering and Integer Texel Coordinate Operations with a given VkImageView by VkImageViewMinLodCreateInfoEXT::minLod.

If the VkPhysicalDeviceImageViewMinLodFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceImageViewMinLodFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImageViewMinLodFeaturesEXT-sType-sType

sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_VIEW_MIN_LOD_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceRasterizationOrderAttachmentAccessFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_rasterization_order_attachment_access
typedef struct VkPhysicalDeviceRasterizationOrderAttachmentAccessFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           rasterizationOrderColorAttachmentAccess;
    VkBool32           rasterizationOrderDepthAttachmentAccess;
    VkBool32           rasterizationOrderStencilAttachmentAccess;
} VkPhysicalDeviceRasterizationOrderAttachmentAccessFeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- rasterizationOrderColorAttachmentAccess indicates that rasterization order access to color and input attachments is supported by the implementation.
- rasterizationOrderDepthAttachmentAccess indicates that rasterization order access to the depth aspect of depth/stencil and input attachments is supported by the implementation.
- rasterizationOrderStencilAttachmentAccess indicates that rasterization order access to the stencil aspect of depth/stencil and input attachments is supported by the implementation.

If the VkPhysicalDeviceRasterizationOrderAttachmentAccessFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceRasterizationOrderAttachmentAccessFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRasterizationOrderAttachmentAccessFeaturesEXT-sType-sType sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RASTERIZATION_ORDER_ATTACHMENT_ACCE

SS_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceSubpassMergeFeedbackFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_subpass_merge_feedback
typedef struct VkPhysicalDeviceSubpassMergeFeedbackFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           subpassMergeFeedback;
} VkPhysicalDeviceSubpassMergeFeedbackFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- subpassMergeFeedback indicates whether the implementation supports feedback of subpass merging.

If the VkPhysicalDeviceSubpassMergeFeedbackFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceSubpassMergeFeedbackFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSubpassMergeFeedbackFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SUBPASS_MERGE_FEEDBACK_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceAttachmentFeedbackLoopLayoutFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_attachment_feedback_loop_layout
typedef struct VkPhysicalDeviceAttachmentFeedbackLoopLayoutFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           attachmentFeedbackLoopLayout;
} VkPhysicalDeviceAttachmentFeedbackLoopLayoutFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- attachmentFeedbackLoopLayout indicates whether the implementation supports using VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT image layout for images created with the VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT usage flag set.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceAttachmentFeedbackLoopLayoutFeaturesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ATTACHMENT_FEEDBACK_LOOP_LAYOUT_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceNestedCommandBufferFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_nested_command_buffer
typedef struct VkPhysicalDeviceNestedCommandBufferFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           nestedCommandBuffer;
    VkBool32           nestedCommandBufferRendering;
    VkBool32           nestedCommandBufferSimultaneousUse;
} VkPhysicalDeviceNestedCommandBufferFeaturesEXT;
```

This structure describes the following features:

- nestedCommandBuffer indicates the implementation supports nested command buffers, which

allows Secondary Command Buffers to execute other Secondary Command Buffers.

- `nestedCommandBufferRendering` indicates that it is valid to call vkCmdExecuteCommands inside a Secondary Command Buffer recorded with VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT.
- `nestedCommandBufferSimultaneousUse` indicates that the implementation supports nested command buffers with command buffers that are recorded with VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT.

If the `VkPhysicalDeviceNestedCommandBufferFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceNestedCommandBufferFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceNestedCommandBufferFeaturesEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NESTED_COMMAND_BUFFER_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceGraphicsPipelineLibraryFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_graphics_pipeline_library
typedef struct VkPhysicalDeviceGraphicsPipelineLibraryFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           graphicsPipelineLibrary;
} VkPhysicalDeviceGraphicsPipelineLibraryFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `graphicsPipelineLibrary` indicates that the implementation supports graphics pipeline libraries.

If the `VkPhysicalDeviceGraphicsPipelineLibraryFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to

indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceGraphicsPipelineLibraryFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceGraphicsPipelineLibraryFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GRAPHICS_PIPELINE_LIBRARY_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevicePipelineBinaryFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPhysicalDevicePipelineBinaryFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           pipelineBinaries;
} VkPhysicalDevicePipelineBinaryFeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pipelineBinaries indicates that the implementation supports Pipeline Binaries.

If the VkPhysicalDevicePipelineBinaryFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDevicePipelineBinaryFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePipelineBinaryFeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_BINARY_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceMultisampledRenderToSingleSampledFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_multisampled_render_to_single_sampled
typedef struct VkPhysicalDeviceMultisampledRenderToSingleSampledFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           multisampledRenderToSingleSampled;
} VkPhysicalDeviceMultisampledRenderToSingleSampledFeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- multisampledRenderToSingleSampled indicates that the implementation supports multisampled rendering to single-sampled render pass attachments.

If the VkPhysicalDeviceMultisampledRenderToSingleSampledFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceMultisampledRenderToSingleSampledFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMultisampledRenderToSingleSampledFeaturesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceMultisampledRenderToSwapchainFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_multisampled_render_to_swapchain
typedef struct VkPhysicalDeviceMultisampledRenderToSwapchainFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           multisampledRenderToSwapchain;
} VkPhysicalDeviceMultisampledRenderToSwapchainFeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- multisampledRenderToSwapchain indicates that the implementation supports multisampled rendering to single-sampled render pass attachments referencing swapchain images.

If the VkPhysicalDeviceMultisampledRenderToSwapchainFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceMultisampledRenderToSwapchainFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMultisampledRenderToSwapchainFeaturesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTISAMPLED_RENDER_TO_SWAPCHAIN_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceImage2DViewOf3DFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_image_2d_view_of_3d
typedef struct VkPhysicalDeviceImage2DViewOf3DFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           image2DViewOf3D;
    VkBool32           sampler2DViewOf3D;
} VkPhysicalDeviceImage2DViewOf3DFeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `image2DViewOf3D` indicates that the implementation supports using a 2D view of a 3D image in a descriptor of type VK_DESCRIPTOR_TYPE_STORAGE_IMAGE if the image is created using VK_IMAGE_CREATE_2D_VIEW_COMPATIBLE_BIT_EXT.
- `sampler2DViewOf3D` indicates that the implementation supports using a 2D view of a 3D image in a descriptor of type VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER if the image is created using VK_IMAGE_CREATE_2D_VIEW_COMPATIBLE_BIT_EXT.

If the `VkPhysicalDeviceImage2DViewOf3DFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceImage2DViewOf3DFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImage2DViewOf3DFeaturesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_2D_VIEW_OF_3D_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceImageSlicedViewOf3DFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_image_sliced_view_of_3d
typedef struct VkPhysicalDeviceImageSlicedViewOf3DFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           imageSlicedViewOf3D;
} VkPhysicalDeviceImageSlicedViewOf3DFeaturesEXT;
```

The members of the VkPhysicalDeviceImageSlicedViewOf3DFeaturesEXT structure describe the following features:

- imageSlicedViewOf3D indicates that the implementation supports using a sliced view of a 3D image in a descriptor of type VK_DESCRIPTOR_TYPE_STORAGE_IMAGE by using a VkImageViewSlicedCreateInfoEXT structure when creating the view.

If the VkPhysicalDeviceImageSlicedViewOf3DFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceImageSlicedViewOf3DFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImageSlicedViewOf3DFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_SLICED_VIEW_OF_3D_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceImageCompressionControlFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_image_compression_control
typedef struct VkPhysicalDeviceImageCompressionControlFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           imageCompressionControl;
} VkPhysicalDeviceImageCompressionControlFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `imageCompressionControl` indicates that the implementation supports providing controls for image compression at image creation time.

If the `VkPhysicalDeviceImageCompressionControlFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceImageCompressionControlFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImageCompressionControlFeaturesEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_COMPRESSION_CONTROL_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceImageCompressionControlSwapchainFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_image_compression_control_swapchain
typedef struct VkPhysicalDeviceImageCompressionControlSwapchainFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           imageCompressionControlSwapchain;
} VkPhysicalDeviceImageCompressionControlSwapchainFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `imageCompressionControlSwapchain` indicates that the implementation supports controlling image controls per swapchain and querying image compression properties per surface.

If the `VkPhysicalDeviceImageCompressionControlSwapchainFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to

use a VkDevice with any features described by VkPhysicalDeviceImageCompressionControlSwapchainFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImageCompressionControlSwapchainFeaturesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_COMPRESSION_CONTROL_SWAPCHAIN_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_ray_tracing_position_fetch
typedef struct VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           rayTracingPositionFetch;
} VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- rayTracingPositionFetch indicates that the implementation supports fetching the object space vertex positions of a hit triangle.

If the VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR-sType-sType

sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_POSITION_FETCH_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderFloatControls2Features structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceShaderFloatControls2Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderFloatControls2;
} VkPhysicalDeviceShaderFloatControls2Features;
```

```
// Provided by VK_KHR_shader_float_controls2
// Equivalent to VkPhysicalDeviceShaderFloatControls2Features
typedef VkPhysicalDeviceShaderFloatControls2Features
VkPhysicalDeviceShaderFloatControls2FeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderFloatControls2 specifies whether shader modules **can** declare the FloatControls2 capability.

If the VkPhysicalDeviceShaderFloatControls2Features structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderFloatControls2Features, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderFloatControls2Features-sType-sType
  sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT_CONTROLS_2_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_non_seamless_cube_map
typedef struct VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           nonSeamlessCubeMap;
} VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- nonSeamlessCubeMap indicates that the implementation supports VK_SAMPLER_CREATE_NON_SEAMLESS_CUBE_MAP_BIT_EXT.

If the VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NON_SEAMLESS_CUBE_MAP_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderModuleIdentifierFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_shader_module_identifier
typedef struct VkPhysicalDeviceShaderModuleIdentifierFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderModuleIdentifier;
} VkPhysicalDeviceShaderModuleIdentifierFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderModuleIdentifier` indicates whether the implementation supports querying an identifier of a VkShaderModule or VkShaderModuleCreateInfo structure, and creating pipelines from identifiers only.

If the `VkPhysicalDeviceShaderModuleIdentifierFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceShaderModuleIdentifierFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderModuleIdentifierFeaturesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_MODULE_IDENTIFIER_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceDepthClampZeroOneFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_depth_clamp_zero_one
typedef struct VkPhysicalDeviceDepthClampZeroOneFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           depthClampZeroOne;
} VkPhysicalDeviceDepthClampZeroOneFeaturesKHR;
```

```
// Provided by VK_EXT_depth_clamp_zero_one
// Equivalent to VkPhysicalDeviceDepthClampZeroOneFeaturesKHR
typedef VkPhysicalDeviceDepthClampZeroOneFeaturesKHR
VkPhysicalDeviceDepthClampZeroOneFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `depthClampZeroOne` indicates that the implementation supports clamping the depth to a range of `0` to `1`.

If the `VkPhysicalDeviceDepthClampZeroOneFeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceDepthClampZeroOneFeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDepthClampZeroOneFeaturesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLAMP_ZERO_ONE_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceShaderTileImageFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_shader_tile_image
typedef struct VkPhysicalDeviceShaderTileImageFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderTileImageColorReadAccess;
    VkBool32           shaderTileImageDepthReadAccess;
    VkBool32           shaderTileImageStencilReadAccess;
} VkPhysicalDeviceShaderTileImageFeaturesEXT;
```

The members of the `VkPhysicalDeviceShaderTileImageFeaturesEXT` structure describe the following

features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderTileImageColorReadAccess` indicates that the implementation supports the `TileImageColorReadAccessEXT` SPIR-V capability.
- `shaderTileImageDepthReadAccess` indicates that the implementation supports the `TileImageDepthReadAccessEXT` SPIR-V capability.
- `shaderTileImageStencilReadAccess` indicates that the implementation supports the `TileImageStencilReadAccessEXT` SPIR-V capability.

If the `VkPhysicalDeviceShaderTileImageFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceShaderTileImageFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderTileImageFeaturesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TILE_IMAGE_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceDepthBiasControlFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_depth_bias_control
typedef struct VkPhysicalDeviceDepthBiasControlFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           depthBiasControl;
    VkBool32           leastRepresentableValueForceUnormRepresentation;
    VkBool32           floatRepresentation;
    VkBool32           depthBiasExact;
} VkPhysicalDeviceDepthBiasControlFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- depthBiasControl indicates whether the implementation supports the vkCmdSetDepthBias2EXT command and the VkDepthBiasRepresentationInfoEXT structure.
- leastRepresentableValueForceUnormRepresentation indicates whether the implementation supports using the VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORCE_UNORM_EXT depth bias representation.
- floatRepresentation indicates whether the implementation supports using the VK_DEPTH_BIAS_REPRESENTATION_FLOAT_EXT depth bias representation.
- depthBiasExact indicates whether the implementation supports forcing depth bias to not be scaled to ensure a minimum resolvable difference using VkDepthBiasRepresentationInfoEXT ::depthBiasExact.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDepthBiasControlFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_BIAS_CONTROL_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceAddressBindingReportFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_device_address_binding_report
typedef struct VkPhysicalDeviceAddressBindingReportFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           reportAddressBinding;
} VkPhysicalDeviceAddressBindingReportFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- reportAddressBinding indicates whether this implementation supports reporting the binding of GPU virtual address ranges to Vulkan objects.

If the VkPhysicalDeviceAddressBindingReportFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceAddressBindingReportFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceAddressBindingReportFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ADDRESS_BINDING_REPORT_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceFaultFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_device_fault
typedef struct VkPhysicalDeviceFaultFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           deviceFault;
    VkBool32           deviceFaultVendorBinary;
} VkPhysicalDeviceFaultFeaturesEXT;
```

This structure describes the following features:

- deviceFault indicates that the implementation supports the reporting of device fault information.
- deviceFaultVendorBinary indicates that the implementation supports the generation of vendor-specific binary crash dumps. These may provide additional information when imported into vendor-specific external tools.

If the VkPhysicalDeviceFaultFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceFaultFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFaultFeaturesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FAULT_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceFaultFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_device_fault
typedef struct VkPhysicalDeviceFaultFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           deviceFault;
    VkBool32           deviceFaultVendorBinary;
    VkBool32           deviceFaultReportMasked;
    VkBool32           deviceFaultDeviceLostOnMasked;
} VkPhysicalDeviceFaultFeaturesKHR;
```

This structure describes the following features:

- `deviceFault` indicates that the implementation supports the reporting of device fault information.
- `deviceFaultVendorBinary` indicates that the implementation supports the generation of vendor-specific binary crash dumps. These **may** provide additional information when imported into vendor-specific external tools.
- `deviceFaultReportMasked` indicates that the implementation supports masked faults in normal operation (i.e. automatically recovered by the driver internally without the application receiving a VK_ERROR_DEVICE_LOST error) which **may** be reported via this extension even if they did not result in a VK_ERROR_DEVICE_LOST condition being returned to the application.
- `deviceFaultDeviceLostOnMasked` indicates that the implementation supports returning VK_ERROR_DEVICE_LOST for faults that would normally be masked.

In exceptional circumstances, some implementations **may** mask faults and attempt to recover from an error. In such circumstances, the device is not lost and further work can be submitted to the device. When such faults occur, the contents of all resources being written to at the time of the fault are undefined.

If the `VkPhysicalDeviceFaultFeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to

indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceFaultFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFaultFeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FAULT_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevicePipelineLibraryGroupHandlesFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_pipeline_library_group_handles
typedef struct VkPhysicalDevicePipelineLibraryGroupHandlesFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           pipelineLibraryGroupHandles;
} VkPhysicalDevicePipelineLibraryGroupHandlesFeaturesKHR;
```

```
// Provided by VK_EXT_pipeline_library_group_handles
// Equivalent to VkPhysicalDevicePipelineLibraryGroupHandlesFeaturesKHR
typedef VkPhysicalDevicePipelineLibraryGroupHandlesFeaturesKHR
VkPhysicalDevicePipelineLibraryGroupHandlesFeaturesEXT;
```

This structure describes the following features:

- pipelineLibraryGroupHandles indicates whether the implementation supports querying group handles directly from a ray tracing pipeline library, and guarantees bitwise identical group handles for such libraries when linked into other pipelines.

If the VkPhysicalDevicePipelineLibraryGroupHandlesFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDevicePipelineLibraryGroupHandlesFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePipelineLibraryGroupHandlesFeaturesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_LIBRARY_GROUP_HANDLES_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceShaderObjectFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_shader_object
typedef struct VkPhysicalDeviceShaderObjectFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderObject;
} VkPhysicalDeviceShaderObjectFeaturesEXT;
```

This structure describes the following feature:

- `shaderObject` indicates whether the implementation supports shader objects.

If the `VkPhysicalDeviceShaderObjectFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceShaderObjectFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderObjectFeaturesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_OBJECT_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo

- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceFrameBoundaryFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_frame_boundary
typedef struct VkPhysicalDeviceFrameBoundaryFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           frameBoundary;
} VkPhysicalDeviceFrameBoundaryFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- frameBoundary indicates whether the implementation supports frame boundary information.

If the VkPhysicalDeviceFrameBoundaryFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceFrameBoundaryFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFrameBoundaryFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAME_BOUNDARY_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceSwapchainMaintenance1FeaturesKHR structure is defined as:

```
// Provided by VK_KHR_swapchain_maintenance1
typedef struct VkPhysicalDeviceSwapchainMaintenance1FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           swapchainMaintenance1;
} VkPhysicalDeviceSwapchainMaintenance1FeaturesKHR;
```

```
// Provided by VK_EXT_swapchain_maintenance1
// Equivalent to VkPhysicalDeviceSwapchainMaintenance1FeaturesKHR
typedef VkPhysicalDeviceSwapchainMaintenance1FeaturesKHR
VkPhysicalDeviceSwapchainMaintenance1FeaturesEXT;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `swapchainMaintenance1` indicates that the implementation supports the following:
  - VkSwapchainPresentFenceInfoKHR, specifying a fence that is signaled when the resources associated with a present operation **can** be safely destroyed.
  - VkSwapchainPresentModesCreateInfoKHR and VkSwapchainPresentModeInfoKHR, allowing the swapchain to switch present modes without a need for recreation.
  - VkSwapchainPresentScalingCreateInfoKHR, specifying the scaling behavior of the swapchain in presence of window resizing.
  - The VK_SWAPCHAIN_CREATE_DEFERRED_MEMORY_ALLOCATION_BIT_KHR flag, allowing the implementation to defer the allocation of swapchain image memory until first acquisition.
  - vkReleaseSwapchainImagesKHR, allowing acquired swapchain images to be released without presenting them.

If the `VkPhysicalDeviceSwapchainMaintenance1FeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceSwapchainMaintenance1FeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSwapchainMaintenance1FeaturesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SWAPCHAIN_MAINTENANCE_1_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevicePresentModeFifoLatestReadyFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_present_mode_fifo_latest_ready
typedef struct VkPhysicalDevicePresentModeFifoLatestReadyFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           presentModeFifoLatestReady;
} VkPhysicalDevicePresentModeFifoLatestReadyFeaturesKHR;
```

```
// Provided by VK_EXT_present_mode_fifo_latest_ready
// Equivalent to VkPhysicalDevicePresentModeFifoLatestReadyFeaturesKHR
typedef VkPhysicalDevicePresentModeFifoLatestReadyFeaturesKHR
VkPhysicalDevicePresentModeFifoLatestReadyFeaturesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- presentModeFifoLatestReady specifies whether the implementation supports the VK_PRESENT_MODE_FIFO_LATEST_READY_KHR present mode.

If the VkPhysicalDevicePresentModeFifoLatestReadyFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDevicePresentModeFifoLatestReadyFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePresentModeFifoLatestReadyFeaturesKHR-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_MODE_FIFO_LATEST_READY_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_dynamic_rendering_unused_attachments
typedef struct VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           dynamicRenderingUnusedAttachments;
} VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- dynamicRenderingUnusedAttachments indicates that the implementation supports binding graphics pipelines within a render pass instance where any pipeline VkPipelineRenderingCreateInfo::pColorAttachmentFormats element with a format other than VK_FORMAT_UNDEFINED is allowed with a corresponding VkRenderingInfo::pColorAttachments element with an imageView equal to VK_NULL_HANDLE, or any pipeline VkPipelineRenderingCreateInfo::pColorAttachmentFormats element with a VK_FORMAT_UNDEFINED format is allowed with a corresponding VkRenderingInfo::pColorAttachments element with a non-VK_NULL_HANDLE imageView. Also a VkPipelineRenderingCreateInfo::depthAttachmentFormat other than VK_FORMAT_UNDEFINED is allowed with a VK_NULL_HANDLE VkRenderingInfo::pDepthAttachment, or a VkPipelineRenderingCreateInfo::depthAttachmentFormat of VK_FORMAT_UNDEFINED is allowed with a non-VK_NULL_HANDLE VkRenderingInfo::pDepthAttachment. Also a VkPipelineRenderingCreateInfo::stencilAttachmentFormat other than VK_FORMAT_UNDEFINED is allowed with a VK_NULL_HANDLE VkRenderingInfo::pStencilAttachment, or a VkPipelineRenderingCreateInfo::stencilAttachmentFormat of VK_FORMAT_UNDEFINED is allowed with a non-VK_NULL_HANDLE VkRenderingInfo::pStencilAttachment. Any writes to a VkRenderingInfo::pColorAttachments, VkRenderingInfo::pDepthAttachment, or VkRenderingInfo::pStencilAttachment with VK_NULL_HANDLE are discarded.

If the VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_UNUSED_ATTACHMENTS_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceRayTracingInvocationReorderFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_ray_tracing_invocation_reorder
typedef struct VkPhysicalDeviceRayTracingInvocationReorderFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           rayTracingInvocationReorder;
} VkPhysicalDeviceRayTracingInvocationReorderFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- rayTracingInvocationReorder indicates that the implementation supports SPV_EXT_shader_invocation_reorder.

If the VkPhysicalDeviceRayTracingInvocationReorderFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceRayTracingInvocationReorderFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRayTracingInvocationReorderFeaturesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_INVOCATION_REORDER_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderMaximalReconvergenceFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_shader_maximal_reconvergence
typedef struct VkPhysicalDeviceShaderMaximalReconvergenceFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderMaximalReconvergence;
} VkPhysicalDeviceShaderMaximalReconvergenceFeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderMaximalReconvergence specifies whether the implementation supports the shader execution mode MaximallyReconvergesKHR.

If the VkPhysicalDeviceShaderMaximalReconvergenceFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderMaximalReconvergenceFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderMaximalReconvergenceFeaturesKHR-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_MAXIMAL_RECONVERGENCE_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderSubgroupRotateFeatures structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceShaderSubgroupRotateFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderSubgroupRotate;
    VkBool32           shaderSubgroupRotateClustered;
} VkPhysicalDeviceShaderSubgroupRotateFeatures;
```

```
// Provided by VK_KHR_shader_subgroup_rotate
// Equivalent to VkPhysicalDeviceShaderSubgroupRotateFeatures
typedef VkPhysicalDeviceShaderSubgroupRotateFeatures
VkPhysicalDeviceShaderSubgroupRotateFeaturesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderSubgroupRotate specifies whether shader modules **can** declare the GroupNonUniformRotateKHR capability.
- shaderSubgroupRotateClustered specifies whether shader modules **can** use the ClusterSize operand to OpGroupNonUniformRotateKHR.

If the VkPhysicalDeviceShaderSubgroupRotateFeatures structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderSubgroupRotateFeatures, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderSubgroupRotateFeatures-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_ROTATE_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderExpectAssumeFeatures structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceShaderExpectAssumeFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderExpectAssume;
} VkPhysicalDeviceShaderExpectAssumeFeatures;
```

```
// Provided by VK_KHR_shader_expect_assume
// Equivalent to VkPhysicalDeviceShaderExpectAssumeFeatures
typedef VkPhysicalDeviceShaderExpectAssumeFeatures
VkPhysicalDeviceShaderExpectAssumeFeaturesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderExpectAssume` specifies whether shader modules **can** declare the `ExpectAssumeKHR` capability.

If the `VkPhysicalDeviceShaderExpectAssumeFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceShaderExpectAssumeFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderExpectAssumeFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_EXPECT_ASSUME_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceDynamicRenderingLocalReadFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceDynamicRenderingLocalReadFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           dynamicRenderingLocalRead;
} VkPhysicalDeviceDynamicRenderingLocalReadFeatures;
```

```
// Provided by VK_KHR_dynamic_rendering_local_read
// Equivalent to VkPhysicalDeviceDynamicRenderingLocalReadFeatures
typedef VkPhysicalDeviceDynamicRenderingLocalReadFeatures
VkPhysicalDeviceDynamicRenderingLocalReadFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `dynamicRenderingLocalRead` specifies that the implementation supports local reads inside dynamic render pass instances using the vkCmdBeginRendering command.

If the `VkPhysicalDeviceDynamicRenderingLocalReadFeatures` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceDynamicRenderingLocalReadFeatures`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDynamicRenderingLocalReadFeatures-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_LOCAL_READ_FEATURES

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceShaderQuadControlFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_shader_quad_control
typedef struct VkPhysicalDeviceShaderQuadControlFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderQuadControl;
} VkPhysicalDeviceShaderQuadControlFeaturesKHR;
```

This structure describes the following features:

- `shaderQuadControl` indicates whether the implementation supports shaders with the `QuadControlKHR` capability.

If the `VkPhysicalDeviceShaderQuadControlFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2`, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a `VkDevice` with any features described by `VkPhysicalDeviceShaderQuadControlFeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to `VK_TRUE`, to the `pNext` chain of `VkDeviceCreateInfo` when creating the `VkDevice`.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderQuadControlFeaturesKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_QUAD_CONTROL_FEATURES_KHR`

Structure Chaining

Extends the structures

- `VkDeviceCreateInfo`
- `VkPhysicalDeviceFeatures2`

The `VkPhysicalDeviceShaderBfloat16FeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_shader_bfloat16
typedef struct VkPhysicalDeviceShaderBfloat16FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderBFloat16Type;
    VkBool32           shaderBFloat16DotProduct;
    VkBool32           shaderBFloat16CooperativeMatrix;
} VkPhysicalDeviceShaderBfloat16FeaturesKHR;
```

This structure describes the following features:

- `shaderBFloat16Type` indicates whether the implementation supports shaders with the

BFloat16TypeKHR capability.

- shaderBFloat16DotProduct indicates whether the implementation supports shaders with the BFloat16DotProductKHR capability.
- shaderBFloat16CooperativeMatrix indicates whether the implementation supports shaders with the BFloat16CooperativeMatrixKHR capability.

If the VkPhysicalDeviceShaderBfloat16FeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderBfloat16FeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderBfloat16FeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_BFLOAT16_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceMapMemoryPlacedFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_map_memory_placed
typedef struct VkPhysicalDeviceMapMemoryPlacedFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           memoryMapPlaced;
    VkBool32           memoryMapRangePlaced;
    VkBool32           memoryUnmapReserve;
} VkPhysicalDeviceMapMemoryPlacedFeaturesEXT;
```

This structure describes the following features:

- memoryMapPlaced indicates that the implementation supports placing memory maps at application-specified virtual addresses.
- memoryMapRangePlaced indicates that the implementation supports placing memory maps of a subrange of a memory object at application-specified virtual addresses.
- memoryUnmapReserve indicates that the implementation supports leaving the memory range reserved when unmapping a memory object.

If the VkPhysicalDeviceMapMemoryPlacedFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceMapMemoryPlacedFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMapMemoryPlacedFeaturesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAP_MEMORY_PLACED_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_shader_replicated_composites
typedef struct VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderReplicatedComposites;
} VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderReplicatedComposites specifies whether shader modules **can** declare the ReplicatedCompositesEXT capability.

If the VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT-sType-sType

sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_REPLICATED_COMPOSITES_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_shader_relaxed_extended_instruction
typedef struct VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderRelaxedExtendedInstruction;
} VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderRelaxedExtendedInstruction specifies whether the implementation supports SPIR-V modules that use the SPV_KHR_relaxed_extended_instruction extension.

If the VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR-sType-sType sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_RELAXED_EXTENDED_INSTRUCTION_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderUniformBufferUnsizedArrayFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_shader_uniform_buffer_unsized_array
typedef struct VkPhysicalDeviceShaderUniformBufferUnsizedArrayFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderUniformBufferUnsizedArray;
} VkPhysicalDeviceShaderUniformBufferUnsizedArrayFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderUniformBufferUnsizedArray indicates that the implementation supports declaring the last member of a uniform buffer block as an unsized array.

If the VkPhysicalDeviceShaderUniformBufferUnsizedArrayFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderUniformBufferUnsizedArrayFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderUniformBufferUnsizedArrayFeaturesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_UNIFORM_BUFFER_UNSIZED_ARRAY_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDevicePrimitiveRestartIndexFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_primitive_restart_index
typedef struct VkPhysicalDevicePrimitiveRestartIndexFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           primitiveRestartIndex;
} VkPhysicalDevicePrimitiveRestartIndexFeaturesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- primitiveRestartIndex specifies whether the index for primitive restart can be set.

If the VkPhysicalDevicePrimitiveRestartIndexFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDevicePrimitiveRestartIndexFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePrimitiveRestartIndexFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIMITIVE_RESTART_INDEX_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderFloat8FeaturesEXT structure is defined as:

```
// Provided by VK_EXT_shader_float8
typedef struct VkPhysicalDeviceShaderFloat8FeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderFloat8;
    VkBool32           shaderFloat8CooperativeMatrix;
} VkPhysicalDeviceShaderFloat8FeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderFloat8` indicates whether the implementation supports shaders with the `Float8EXT` capability.
- `shaderFloat8CooperativeMatrix` indicates whether the implementation supports shaders with the `Float8CooperativeMatrixEXT` capability.

If the `VkPhysicalDeviceShaderFloat8FeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceShaderFloat8FeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderFloat8FeaturesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT8_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceShader64BitIndexingFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_shader_64bit_indexing
typedef struct VkPhysicalDeviceShader64BitIndexingFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shader64BitIndexing;
} VkPhysicalDeviceShader64BitIndexingFeaturesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shader64BitIndexing` indicates that the implementation supports using 64-bit address calculations for indexing cooperative matrices, storage buffers, and physical storage buffers.

If the `VkPhysicalDeviceShader64BitIndexingFeaturesEXT` structure is included in the `pNext` chain of

the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShader64BitIndexingFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShader64BitIndexingFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_64_BIT_INDEXING_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderSubgroupPartitionedFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_shader_subgroup_partitioned
typedef struct VkPhysicalDeviceShaderSubgroupPartitionedFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderSubgroupPartitioned;
} VkPhysicalDeviceShaderSubgroupPartitionedFeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderSubgroupPartitioned indicates that the implementation supports VK_SUBGROUP_FEATURE_PARTITIONED_BIT_EXT.

If the VkPhysicalDeviceShaderSubgroupPartitionedFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderSubgroupPartitionedFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderSubgroupPartitionedFeaturesEXT-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_PARTITIONED_FEATURES_EXT`

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceShaderSplitBarrierFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_shader_split_barrier
typedef struct VkPhysicalDeviceShaderSplitBarrierFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderSplitBarrier;
} VkPhysicalDeviceShaderSplitBarrierFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderSplitBarrier` indicates that the implementation supports the `SplitBarrierEXT` SPIR-V capability.

If the `VkPhysicalDeviceShaderSplitBarrierFeaturesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceShaderSplitBarrierFeaturesEXT`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderSplitBarrierFeaturesEXT-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SPLIT_BARRIER_FEATURES_EXT`

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceDescriptorHeapFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkPhysicalDeviceDescriptorHeapFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           descriptorHeap;
    VkBool32           descriptorHeapCaptureReplay;
} VkPhysicalDeviceDescriptorHeapFeaturesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- descriptorHeap specifies whether descriptor heaps **can** be used.
- descriptorHeapCaptureReplay specifies whether heap descriptors **can** be captured and replayed.

If the VkPhysicalDeviceDescriptorHeapFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceDescriptorHeapFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDescriptorHeapFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_HEAP_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceDeviceAddressCommandsFeaturesKHR structure is defined as:

```
// Provided by VK_KHR_device_address_commands
typedef struct VkPhysicalDeviceDeviceAddressCommandsFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           deviceAddressCommands;
} VkPhysicalDeviceDeviceAddressCommandsFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `deviceAddressCommands` specifies whether the implementation supports commands taking a VkDeviceAddress that would otherwise take a VkBuffer object.

If the `VkPhysicalDeviceDeviceAddressCommandsFeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceDeviceAddressCommandsFeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDeviceAddressCommandsFeaturesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_ADDRESS_COMMANDS_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceShaderConstantDataFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_shader_constant_data
typedef struct VkPhysicalDeviceShaderConstantDataFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderConstantData;
} VkPhysicalDeviceShaderConstantDataFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderConstantData` specifies whether the implementation supports SPIR-V modules that use the `SPV_KHR_constant_data` extension.

If the `VkPhysicalDeviceShaderConstantDataFeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by `VkPhysicalDeviceShaderConstantDataFeaturesKHR`, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the `pNext` chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderConstantDataFeaturesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_CONSTANT_DATA_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The `VkPhysicalDeviceShaderAbortFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_shader_abort
typedef struct VkPhysicalDeviceShaderAbortFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderAbort;
} VkPhysicalDeviceShaderAbortFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderAbort` specifies whether shaders **can** declare the `AbortKHR` capability.

If the `VkPhysicalDeviceShaderAbortFeaturesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a

VkDevice with any features described by VkPhysicalDeviceShaderAbortFeaturesKHR, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderAbortFeaturesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ABORT_FEATURES_KHR

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceImageTilingControlFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_image_tiling_control
typedef struct VkPhysicalDeviceImageTilingControlFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           imageTilingControl;
} VkPhysicalDeviceImageTilingControlFeaturesEXT;
```

This structure describes the following feature:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- imageTilingControl indicates that the implementation supports control of optimal tiling arrangements at image creation time.

If the VkPhysicalDeviceImageTilingControlFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceImageTilingControlFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImageTilingControlFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_TILING_CONTROL_FEATURES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

The VkPhysicalDeviceShaderOCPMicroscalingTypesFeaturesEXT structure is defined as:

```
// Provided by VK_EXT_shader_ocp_microscaling_types
typedef struct VkPhysicalDeviceShaderOCPMicroscalingTypesFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderFloat4;
    VkBool32           shaderFloat6;
    VkBool32           shaderFloat8UnsignedE8M0;
    VkBool32           shaderMXInt8;
} VkPhysicalDeviceShaderOCPMicroscalingTypesFeaturesEXT;
```

This structure describes the following features:

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderFloat4 indicates whether the implementation supports shaders with the Float4EXT or BitcastExtractEXT capabilities.
- shaderFloat6 indicates whether the implementation supports shaders with the Float6EXT or BitcastExtractEXT capabilities.
- shaderFloat8UnsignedE8M0 indicates whether the implementation supports shaders with the Float8UnsignedE8M0EXT capability.
- shaderMXInt8 indicates whether the implementation supports shaders with the MXInt8EXT capability.

If the VkPhysicalDeviceShaderOCPMicroscalingTypesFeaturesEXT structure is included in the pNext chain of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceShaderOCPMicroscalingTypesFeaturesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderOCPMicroscalingTypesFeaturesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_OCP_MICROSCALING_TYPES_FEATUR

ES_EXT

Structure Chaining

Extends the structures

- VkDeviceCreateInfo
- VkPhysicalDeviceFeatures2

47.1. Feature Requirements

All Vulkan graphics implementations **must** support the following features:

- If Vulkan 1.0 is supported, `robustBufferAccess` **must** be supported if `VK_KHR_portability_subset` is not advertised
- If Vulkan 1.1 is supported, the following features **must** be supported:
  - `multiview`
  - `storageBuffer16BitAccess` if `uniformAndStorageBuffer16BitAccess` is supported
  - `shaderDrawParameters` if `VK_KHR_shader_draw_parameters` is supported
- If Vulkan 1.2 is supported, the following features **must** be supported:
  - `subgroupBroadcastDynamicId`
  - `imagelessFramebuffer`
  - `uniformBufferStandardLayout`
  - `shaderSubgroupExtendedTypes`
  - `separateDepthStencilLayouts`
  - `hostQueryReset`
  - `timelineSemaphore`
  - `samplerMirrorClampToEdge` if `VK_KHR_sampler_mirror_clamp_to_edge` is supported
  - `drawIndirectCount` if `VK_KHR_draw_indirect_count` is supported
  - `storageBuffer8BitAccess` if `uniformAndStorageBuffer8BitAccess` is supported
  - `shaderInt64` if `shaderSharedInt64Atomics` or `shaderBufferInt64Atomics` are supported
  - `descriptorIndexing` if `VK_EXT_descriptor_indexing` is supported
  - `shaderSampledImageArrayDynamicIndexing` if `descriptorIndexing` is supported
  - `shaderStorageBufferArrayDynamicIndexing` if `descriptorIndexing` is supported
  - `shaderUniformTexelBufferArrayDynamicIndexing` if `descriptorIndexing` is supported
  - `shaderStorageTexelBufferArrayDynamicIndexing` if `descriptorIndexing` is supported
  - `shaderSampledImageArrayNonUniformIndexing` if `descriptorIndexing` is supported

    - shaderStorageBufferArrayNonUniformIndexing if descriptorIndexing is supported
    - shaderUniformTexelBufferArrayNonUniformIndexing if descriptorIndexing is supported
    - descriptorBindingSampledImageUpdateAfterBind if descriptorIndexing is supported
    - descriptorBindingStorageImageUpdateAfterBind if descriptorIndexing is supported
    - descriptorBindingStorageBufferUpdateAfterBind if descriptorIndexing is supported
    - descriptorBindingUniformTexelBufferUpdateAfterBind if descriptorIndexing is supported
    - descriptorBindingStorageTexelBufferUpdateAfterBind if descriptorIndexing is supported
    - descriptorBindingUpdateUnusedWhilePending if descriptorIndexing is supported
    - descriptorBindingPartiallyBound if descriptorIndexing is supported
    - runtimeDescriptorArray if descriptorIndexing is supported
    - shaderOutputViewportIndex if VK_EXT_shader_viewport_index_layer is supported
    - shaderOutputLayer if VK_EXT_shader_viewport_index_layer is supported
    - samplerFilterMinmax if VK_EXT_sampler_filter_minmax is supported
- If Vulkan 1.3 is supported, the following features **must** be supported:
    - shaderTerminateInvocation
    - shaderDemoteToHelperInvocation
    - privateData
    - pipelineCreationCacheControl
    - synchronization2
    - shaderZeroInitializeWorkgroupMemory
    - robustImageAccess
    - subgroupSizeControl
    - computeFullSubgroups
    - dynamicRendering
    - shaderIntegerDotProduct
    - maintenance4
    - vulkanMemoryModel
    - vulkanMemoryModelDeviceScope
    - inlineUniformBlock
    - bufferDeviceAddress
    - descriptorBindingInlineUniformBlockUpdateAfterBind if VK_EXT_descriptor_indexing or descriptorIndexing are supported
- If Vulkan 1.4 is supported, the following features **must** be supported:
    - fullDrawIndexUint32
    - imageCubeArray

- independentBlend
- sampleRateShading
- drawIndirectFirstInstance
- depthClamp
- depthBiasClamp
- samplerAnisotropy
- fragmentStoresAndAtomics
- shaderStorageImageExtendedFormats
- shaderUniformBufferArrayDynamicIndexing
- shaderSampledImageArrayDynamicIndexing
- shaderStorageBufferArrayDynamicIndexing
- shaderStorageImageArrayDynamicIndexing
- shaderImageGatherExtended
- shaderInt16
- largePoints
- samplerYcbcrConversion
- storageBuffer16BitAccess
- variablePointers
- variablePointersStorageBuffer
- samplerMirrorClampToEdge
- scalarBlockLayout
- shaderUniformTexelBufferArrayDynamicIndexing
- shaderStorageTexelBufferArrayDynamicIndexing
- shaderInt8
- storageBuffer8BitAccess
- globalPriorityQuery
- shaderSubgroupRotate
- shaderSubgroupRotateClustered
- shaderFloatControls2
- shaderExpectAssume
- bresenhamLines
- vertexAttributeInstanceRateDivisor
- indexTypeUint8
- maintenance5
- pushDescriptor

    - dynamicRenderingLocalRead
    - maintenance6
    - pipelineRobustness
    - pipelineProtectedAccess if protectedMemory is supported
- If VK_KHR_dynamic_rendering is supported, dynamicRendering **must** be supported
- If VK_KHR_multiview is supported, multiview **must** be supported
- If VK_KHR_shader_float16_int8 is supported, at least one of shaderFloat16 or shaderInt8 **must** be supported
- If VK_KHR_16bit_storage is supported, storageBuffer16BitAccess **must** be supported
- If VK_KHR_imageless_framebuffer is supported, imagelessFramebuffer **must** be supported
- If VK_KHR_performance_query is supported, performanceCounterQueryPools **must** be supported
- If VK_KHR_variable_pointers is supported, variablePointersStorageBuffer **must** be supported
- If VK_KHR_shader_bfloat16 is supported, the following features **must** be supported:
    - shaderBFloat16Type
    - at least one of shaderBFloat16CooperativeMatrix or shaderBFloat16DotProduct
- If VK_KHR_sampler_ycbcr_conversion is supported, samplerYcbcrConversion **must** be supported
- If VK_KHR_shader_subgroup_extended_types is supported, shaderSubgroupExtendedTypes **must** be supported
- If VK_KHR_8bit_storage is supported, storageBuffer8BitAccess **must** be supported
- If VK_KHR_shader_atomic_int64 is supported, shaderBufferInt64Atomics **must** be supported
- If VK_KHR_shader_clock is supported, shaderSubgroupClock **must** be supported
- If VK_KHR_global_priority is supported, globalPriorityQuery **must** be supported
- If VK_KHR_timeline_semaphore is supported, timelineSemaphore **must** be supported
- If VK_KHR_vulkan_memory_model is supported, vulkanMemoryModel **must** be supported
- If VK_KHR_shader_terminate_invocation is supported, shaderTerminateInvocation **must** be supported
- If VK_KHR_fragment_shading_rate is supported, pipelineFragmentShadingRate **must** be supported
- If VK_KHR_shader_constant_data is supported, shaderConstantData **must** be supported
- If VK_KHR_dynamic_rendering_local_read is supported, dynamicRenderingLocalRead **must** be supported
- If VK_KHR_shader_abort is supported, shaderAbort **must** be supported
- If VK_KHR_shader_quad_control is supported, shaderQuadControl **must** be supported
- If VK_KHR_separate_depth_stencil_layouts is supported, separateDepthStencilLayouts **must** be supported
- If VK_KHR_present_wait is supported, presentWait **must** be supported
- If VK_KHR_uniform_buffer_standard_layout is supported, uniformBufferStandardLayout **must** be

supported

- If VK_KHR_buffer_device_address is supported, bufferDeviceAddress **must** be supported
- If VK_KHR_pipeline_executable_properties is supported, pipelineExecutableInfo **must** be supported
- If VK_KHR_shader_integer_dot_product is supported, shaderIntegerDotProduct **must** be supported
- If VK_KHR_present_id is supported, presentId **must** be supported
- If VK_KHR_synchronization2 is supported, synchronization2 **must** be supported
- If VK_KHR_device_address_commands is supported, deviceAddressCommands **must** be supported
- If VK_KHR_fragment_shader_barycentric is supported, fragmentShaderBarycentric **must** be supported
- If VK_KHR_shader_subgroup_uniform_control_flow is supported, shaderSubgroupUniformControlFlow **must** be supported
- If VK_KHR_zero_initialize_workgroup_memory is supported, shaderZeroInitializeWorkgroupMemory **must** be supported
- If VK_KHR_workgroup_memory_explicit_layout is supported, workgroupMemoryExplicitLayout **must** be supported
- If VK_KHR_ray_tracing_maintenance1 is supported, rayTracingMaintenance1 **must** be supported
- If VK_KHR_shader_untyped_pointers is supported, shaderUntypedPointers **must** be supported
- If VK_KHR_maintenance4 is supported, maintenance4 **must** be supported
- If VK_KHR_shader_subgroup_rotate is supported, shaderSubgroupRotate **must** be supported
- If VK_KHR_shader_maximal_reconvergence is supported, shaderMaximalReconvergence **must** be supported
- If VK_KHR_maintenance5 is supported, maintenance5 **must** be supported
- If VK_KHR_present_id2 is supported, presentId2 **must** be supported
- If VK_KHR_present_wait2 is supported, presentWait2 **must** be supported
- If VK_KHR_ray_tracing_position_fetch is supported, rayTracingPositionFetch **must** be supported
- If VK_KHR_pipeline_binary is supported, pipelineBinaries **must** be supported
- If VK_KHR_swapchain_maintenance1 is supported, swapchainMaintenance1 **must** be supported
- If VK_KHR_internally_synchronized_queues is supported, internallySynchronizedQueues **must** be supported
- If VK_KHR_cooperative_matrix is supported, cooperativeMatrix **must** be supported
- If VK_KHR_compute_shader_derivatives is supported, computeDerivativeGroupLinear **must** be supported
- If VK_KHR_video_encode_av1 is supported, videoEncodeAV1 **must** be supported
- If VK_KHR_video_decode_vp9 is supported, videoDecodeVP9 **must** be supported
- If VK_KHR_video_maintenance1 is supported, videoMaintenance1 **must** be supported
- If VK_KHR_vertex_attribute_divisor is supported, vertexAttributeInstanceRateDivisor **must** be

supported

- If VK_KHR_unified_image_layouts is supported, unifiedImageLayouts **must** be supported
- If VK_KHR_shader_float_controls2 is supported, shaderFloatControls2 **must** be supported
- If VK_KHR_index_type_uint8 is supported, indexTypeUint8 **must** be supported
- If VK_KHR_line_rasterization is supported, at least one of rectangularLines, bresenhamLines, smoothLines, stippledRectangularLines, stippledBresenhamLines, or stippledSmoothLines **must** be supported
- If VK_KHR_shader_expect_assume is supported, shaderExpectAssume **must** be supported
- If VK_KHR_maintenance6 is supported, maintenance6 **must** be supported
- If VK_KHR_copy_memory_indirect is supported, indirectMemoryCopy **must** be supported
- If VK_KHR_video_encode_intra_refresh is supported, videoEncodeIntraRefresh **must** be supported
- If VK_KHR_video_encode_quantization_map is supported, videoEncodeQuantizationMap **must** be supported
- If VK_KHR_shader_relaxed_extended_instruction is supported, shaderRelaxedExtendedInstruction **must** be supported
- If VK_KHR_maintenance7 is supported, maintenance7 **must** be supported
- If VK_KHR_device_fault is supported, deviceFault **must** be supported
- If VK_KHR_maintenance8 is supported, maintenance8 **must** be supported
- If VK_KHR_shader_fma is supported, shaderFmaFloat32 **must** be supported
- If VK_KHR_maintenance9 is supported, maintenance9 **must** be supported
- If VK_KHR_video_maintenance2 is supported, videoMaintenance2 **must** be supported
- If VK_KHR_video_encode_feedback2 is supported, videoEncodeFeedback2 **must** be supported
- If VK_KHR_depth_clamp_zero_one is supported, depthClampZeroOne **must** be supported
- If VK_KHR_robustness2 is supported, at least one of robustBufferAccess2, robustImageAccess2, or nullDescriptor **must** be supported
- If VK_KHR_present_mode_fifo_latest_ready is supported, presentModeFifoLatestReady **must** be supported
- If VK_KHR_opacity_micromap is supported, micromap **must** be supported
- If VK_KHR_maintenance10 is supported, maintenance10 **must** be supported
- If VK_KHR_pipeline_library_group_handles is supported, pipelineLibraryGroupHandles **must** be supported
- If VK_KHR_maintenance11 is supported, maintenance11 **must** be supported
- If VK_KHR_extended_flags is supported, extendedFlags **must** be supported
- If VK_EXT_transform_feedback is supported, transformFeedback **must** be supported
- If VK_EXT_pipeline_robustness is supported, pipelineRobustness **must** be supported
- If VK_EXT_conditional_rendering is supported, conditionalRendering **must** be supported
- If VK_EXT_depth_clip_enable is supported, depthClipEnable **must** be supported

- If VK_EXT_descriptor_heap is supported, descriptorHeap **must** be supported
- If VK_EXT_descriptor_indexing is supported, the following features **must** be supported:
  - shaderSampledImageArrayDynamicIndexing
  - shaderStorageBufferArrayDynamicIndexing
  - shaderUniformTexelBufferArrayDynamicIndexing
  - shaderStorageTexelBufferArrayDynamicIndexing
  - shaderSampledImageArrayNonUniformIndexing
  - shaderStorageBufferArrayNonUniformIndexing
  - shaderUniformTexelBufferArrayNonUniformIndexing
  - descriptorBindingSampledImageUpdateAfterBind
  - descriptorBindingStorageImageUpdateAfterBind
  - descriptorBindingStorageBufferUpdateAfterBind
  - descriptorBindingUniformTexelBufferUpdateAfterBind
  - descriptorBindingStorageTexelBufferUpdateAfterBind
  - descriptorBindingUpdateUnusedWhilePending
  - descriptorBindingPartiallyBound
  - runtimeDescriptorArray
- If VK_EXT_present_timing is supported, presentTiming **must** be supported
- If VK_EXT_fragment_density_map is supported, fragmentDensityMap **must** be supported
- If VK_EXT_shader_image_atomic_int64 is supported, the following features **must** be supported:
  - shaderInt64
  - shaderImageInt64Atomics
- If VK_EXT_memory_priority is supported, memoryPriority **must** be supported
- If VK_EXT_fragment_shader_interlock is supported, at least one of fragmentShaderSampleInterlock, fragmentShaderPixelInterlock, or fragmentShaderShadingRateInterlock **must** be supported
- If VK_EXT_ycbcr_image_arrays is supported, ycbcrImageArrays **must** be supported
- If VK_EXT_provoking_vertex is supported, provokingVertexLast **must** be supported
- If VK_EXT_shader_atomic_float is supported, the following features **must** be supported:
  - at least one of shaderBufferFloat32Atomics, shaderBufferFloat32AtomicAdd, shaderBufferFloat64Atomics, shaderBufferFloat64AtomicAdd, shaderSharedFloat32Atomics, shaderSharedFloat32AtomicAdd, shaderSharedFloat64Atomics, shaderSharedFloat64AtomicAdd, shaderImageFloat32Atomics, or shaderImageFloat32AtomicAdd
  - shaderImageFloat32Atomics if sparseImageFloat32Atomics is supported
  - shaderImageFloat32AtomicAdd if sparseImageFloat32AtomicAdd is supported
- If VK_EXT_extended_dynamic_state is supported, extendedDynamicState **must** be supported
- If VK_EXT_host_image_copy is supported, hostImageCopy **must** be supported

- If VK_EXT_map_memory_placed is supported, memoryMapPlaced **must** be supported
- If VK_EXT_shader_atomic_float2 is supported, the following features **must** be supported:
  - at least one of shaderBufferFloat16Atomics, shaderBufferFloat16AtomicAdd, shaderBufferFloat16AtomicMinMax, shaderBufferFloat32AtomicMinMax, shaderBufferFloat64AtomicMinMax, shaderSharedFloat16Atomics, shaderSharedFloat16AtomicAdd, shaderSharedFloat16AtomicMinMax, shaderSharedFloat32AtomicMinMax, shaderSharedFloat64AtomicMinMax, or shaderImageFloat32AtomicMinMax
  - shaderImageFloat32AtomicMinMax if sparseImageFloat32AtomicMinMax is supported
- If VK_EXT_swapchain_maintenance1 is supported, swapchainMaintenance1 **must** be supported
- If VK_EXT_depth_bias_control is supported, depthBiasControl **must** be supported
- If VK_EXT_device_memory_report is supported, deviceMemoryReport **must** be supported
- If VK_EXT_robustness2 is supported, at least one of robustBufferAccess2, robustImageAccess2, or nullDescriptor **must** be supported
- If VK_EXT_custom_border_color is supported, customBorderColors **must** be supported
- If VK_EXT_texture_compression_astc_3d is supported, textureCompressionASTC_3D **must** be supported
- If VK_EXT_shader_split_barrier is supported, shaderSplitBarrier **must** be supported
- If VK_EXT_descriptor_buffer is supported, descriptorBuffer **must** be supported
- If VK_EXT_graphics_pipeline_library is supported, graphicsPipelineLibrary **must** be supported
- If VK_EXT_fragment_density_map2 is supported, fragmentDensityMapDeferred **must** be supported
- If VK_EXT_image_compression_control is supported, imageCompressionControl **must** be supported
- If VK_EXT_attachment_feedback_loop_layout is supported, attachmentFeedbackLoopLayout **must** be supported
- If VK_EXT_device_fault is supported, deviceFault **must** be supported
- If VK_EXT_rgba10x6_formats is supported, formatRgba10x6WithoutYCbCrSampler **must** be supported
- If VK_EXT_vertex_input_dynamic_state is supported, vertexInputDynamicState **must** be supported
- If VK_EXT_device_address_binding_report is supported, reportAddressBinding **must** be supported
- If VK_EXT_depth_clip_control is supported, depthClipControl **must** be supported
- If VK_EXT_primitive_topology_list_restart is supported, primitiveTopologyListRestart **must** be supported
- If VK_EXT_present_mode_fifo_latest_ready is supported, presentModeFifoLatestReady **must** be supported
- If VK_EXT_pipeline_properties is supported, pipelinePropertiesIdentifier **must** be supported
- If VK_EXT_frame_boundary is supported, frameBoundary **must** be supported
- If VK_EXT_multisampled_render_to_single_sampled is supported, multisampledRenderToSingleSampled **must** be supported
- If VK_EXT_extended_dynamic_state2 is supported, extendedDynamicState2 **must** be supported

- If VK_EXT_color_write_enable is supported, colorWriteEnable **must** be supported
- If VK_EXT_primitives_generated_query is supported, primitivesGeneratedQuery **must** be supported
- If VK_EXT_image_view_min_lod is supported, minLod **must** be supported
- If VK_EXT_multi_draw is supported, multiDraw **must** be supported
- If VK_EXT_image_2d_view_of_3d is supported, image2DViewOf3D **must** be supported
- If VK_EXT_shader_tile_image is supported, shaderTileImageColorReadAccess **must** be supported
- If VK_EXT_opacity_micromap is supported, micromap **must** be supported
- If VK_EXT_border_color_swizzle is supported, borderColorSwizzle **must** be supported
- If VK_EXT_pageable_device_local_memory is supported, pageableDeviceLocalMemory **must** be supported
- If VK_EXT_image_sliced_view_of_3d is supported, imageSlicedViewOf3D **must** be supported
- If VK_EXT_depth_clamp_zero_one is supported, depthClampZeroOne **must** be supported
- If VK_EXT_non_seamless_cube_map is supported, nonSeamlessCubeMap **must** be supported
- If VK_EXT_image_compression_control_swapchain is supported, imageCompressionControlSwapchain **must** be supported
- If VK_EXT_nested_command_buffer is supported, nestedCommandBuffer **must** be supported
- If VK_EXT_extended_dynamic_state3 is supported, the following features **must** be supported:
  - tessellationShader if extendedDynamicState3TessellationDomainOrigin is supported
  - depthClamp if extendedDynamicState3DepthClampEnable is supported
  - fillModeNonSolid if extendedDynamicState3PolygonMode is supported
  - alphaToOne if extendedDynamicState3AlphaToOneEnable is supported
  - logicOp if extendedDynamicState3LogicOpEnable is supported
  - geometryStreams if VK_EXT_transform_feedback and extendedDynamicState3RasterizationStream are supported
- If VK_EXT_subpass_merge_feedback is supported, subpassMergeFeedback **must** be supported
- If VK_EXT_shader_module_identifier is supported, shaderModuleIdentifier **must** be supported
- If VK_EXT_legacy_dithering is supported, legacyDithering **must** be supported
- If VK_EXT_pipeline_protected_access is supported, pipelineProtectedAccess **must** be supported
- If VK_EXT_shader_object is supported, shaderObject **must** be supported
- If VK_EXT_mutable_descriptor_type is supported, mutableDescriptorType **must** be supported
- If VK_EXT_legacy_vertex_attributes is supported, legacyVertexAttributes **must** be supported
- If VK_EXT_pipeline_library_group_handles is supported, pipelineLibraryGroupHandles **must** be supported
- If VK_EXT_dynamic_rendering_unused_attachments is supported, dynamicRenderingUnusedAttachments **must** be supported
- If VK_EXT_attachment_feedback_loop_dynamic_state is supported,

attachmentFeedbackLoopDynamicState **must** be supported

- If VK_EXT_memory_decompression is supported, memoryDecompression **must** be supported
- If VK_EXT_shader_replicated_composites is supported, shaderReplicatedComposites **must** be supported
- If VK_EXT_shader_float8 is supported, shaderFloat8 **must** be supported
- If VK_EXT_device_generated_commands is supported, deviceGeneratedCommands **must** be supported
- If VK_EXT_ray_tracing_invocation_reorder is supported, rayTracingInvocationReorder **must** be supported
- If VK_EXT_depth_clamp_control is supported, depthClampControl **must** be supported
- If VK_EXT_multisampled_render_to_swapchain is supported, multisampledRenderToSwapchain **must** be supported
- If VK_EXT_fragment_density_map_offset is supported, fragmentDensityMapOffset **must** be supported
- If VK_EXT_zero_initialize_device_memory is supported, zeroInitializeDeviceMemory **must** be supported
- If VK_EXT_shader_64bit_indexing is supported, shader64BitIndexing **must** be supported
- If VK_EXT_custom_resolve is supported, customResolve **must** be supported
- If VK_EXT_shader_long_vector is supported, the following features **must** be supported:
  - longVector
  - scalarBlockLayout
  - workgroupMemoryExplicitLayoutScalarBlockLayout if VK_KHR_workgroup_memory_explicit_layout is supported
- If VK_EXT_shader_uniform_buffer_unsized_array is supported, shaderUniformBufferUnsizedArray **must** be supported
- If VK_EXT_cooperative_matrix_maintenance1 is supported, VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT::cooperativeMatrixProperties2 **must** be supported
- If VK_EXT_shader_subgroup_partitioned is supported, shaderSubgroupPartitioned **must** be supported
- If VK_EXT_shader_ocp_microscaling_types is supported, at least one of shaderFloat4, shaderFloat6, shaderFloat8UnsignedE8M0, or shaderMXInt8 **must** be supported
- If VK_EXT_primitive_restart_index is supported, primitiveRestartIndex **must** be supported
- If VK_EXT_image_tiling_control is supported, imageTilingControl **must** be supported
- If VK_KHR_acceleration_structure is supported, the following features **must** be supported:
  - accelerationStructure
  - bufferDeviceAddress
  - descriptorBindingAccelerationStructureUpdateAfterBind
  - descriptorIndexing if Vulkan 1.2 is supported

- If VK_KHR_ray_tracing_pipeline is supported, the following features **must** be supported:
  - rayTracingPipeline
  - rayTracingPipelineTraceRaysIndirect
  - rayTraversalPrimitiveCulling if VK_KHR_ray_query is supported
- If VK_KHR_ray_query is supported, rayQuery **must** be supported
- If VK_EXT_mesh_shader is supported, the following features **must** be supported:
  - taskShader
  - meshShader
  - primitiveFragmentShadingRate if VK_KHR_fragment_shading_rate and primitiveFragmentShadingRateMeshShader are supported
- VK_KHR_fragment_shading_rate if the primitiveFragmentShadingRateMeshShader feature is supported.
- VK_EXT_transform_feedback, if the extendedDynamicState3RasterizationStream feature is supported.
- VK_EXT_conservative_rasterization extension, if the extendedDynamicState3ConservativeRasterizationMode feature is supported.
- VK_EXT_conservative_rasterization extension, if the extendedDynamicState3ExtraPrimitiveOverestimationSize feature is supported.
- VK_EXT_sample_locations extension, if the extendedDynamicState3SampleLocationsEnable feature is supported.
- VK_EXT_blend_operation_advanced extension, if the extendedDynamicState3ColorBlendAdvanced feature is supported.
- VK_EXT_provoking_vertex, if the extendedDynamicState3ProvokingVertexMode feature is supported.
- Vulkan 1.4 or the VK_KHR_line_rasterization extension or the VK_EXT_line_rasterization extension, if the extendedDynamicState3LineRasterizationMode feature is supported.
- VK_KHR_line_rasterization or VK_EXT_line_rasterization extension, if the extendedDynamicState3LineStippleEnable feature is supported.
- VK_EXT_depth_clip_control, if the extendedDynamicState3DepthClipNegativeOneToOne feature is supported.
- If Vulkan 1.4 is supported, at least one queue family is reported by vkGetPhysicalDeviceQueueFamilyProperties that includes the VK_QUEUE_GRAPHICS_BIT flag, and no additional queue families are reported that support the VK_QUEUE_TRANSFER_BIT flag, the hostImageCopy feature **must** be supported.

All other features defined in the Specification are **optional**.

NOTE	Applications running on Vulkan implementations advertising a VkPhysicalDeviceDriverProperties::conformanceVersion less than 1.4.4.0 should be aware that the formatRgba10x6WithoutYCbCrSampler feature may not be supported despite VK_EXT_rgba10x6_formats being advertised.

Chapter 48. Limits

Limits are implementation-dependent minimums, maximums, and other device characteristics that an application **may** need to be aware of.

NOTE

Limits are reported via the basic VkPhysicalDeviceLimits structure as well as the extensible structure VkPhysicalDeviceProperties2, which was added in VK_KHR_get_physical_device_properties2 and included in Vulkan 1.1. When limits are added in future Vulkan versions or extensions, each extension **should** introduce one new limit structure, if needed. This structure **can** be added to the pNext chain of the VkPhysicalDeviceProperties2 structure.

The VkPhysicalDeviceLimits structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPhysicalDeviceLimits {
    uint32_t              maxImageDimension1D;
    uint32_t              maxImageDimension2D;
    uint32_t              maxImageDimension3D;
    uint32_t              maxImageDimensionCube;
    uint32_t              maxImageArrayLayers;
    uint32_t              maxTexelBufferElements;
    uint32_t              maxUniformBufferRange;
    uint32_t              maxStorageBufferRange;
    uint32_t              maxPushConstantsSize;
    uint32_t              maxMemoryAllocationCount;
    uint32_t              maxSamplerAllocationCount;
    VkDeviceSize          bufferImageGranularity;
    VkDeviceSize          sparseAddressSpaceSize;
    uint32_t              maxBoundDescriptorSets;
    uint32_t              maxPerStageDescriptorSamplers;
    uint32_t              maxPerStageDescriptorUniformBuffers;
    uint32_t              maxPerStageDescriptorStorageBuffers;
    uint32_t              maxPerStageDescriptorSampledImages;
    uint32_t              maxPerStageDescriptorStorageImages;
    uint32_t              maxPerStageDescriptorInputAttachments;
    uint32_t              maxPerStageResources;
    uint32_t              maxDescriptorSetSamplers;
    uint32_t              maxDescriptorSetUniformBuffers;
    uint32_t              maxDescriptorSetUniformBuffersDynamic;
    uint32_t              maxDescriptorSetStorageBuffers;
    uint32_t              maxDescriptorSetStorageBuffersDynamic;
    uint32_t              maxDescriptorSetSampledImages;
    uint32_t              maxDescriptorSetStorageImages;
    uint32_t              maxDescriptorSetInputAttachments;
    uint32_t              maxVertexInputAttributes;
    uint32_t              maxVertexInputBindings;
    uint32_t              maxVertexInputAttributeOffset;
    uint32_t              maxVertexInputBindingStride;
```

```
    uint32_t              maxVertexOutputComponents;
    uint32_t              maxTessellationGenerationLevel;
    uint32_t              maxTessellationPatchSize;
    uint32_t              maxTessellationControlPerVertexInputComponents;
    uint32_t              maxTessellationControlPerVertexOutputComponents;
    uint32_t              maxTessellationControlPerPatchOutputComponents;
    uint32_t              maxTessellationControlTotalOutputComponents;
    uint32_t              maxTessellationEvaluationInputComponents;
    uint32_t              maxTessellationEvaluationOutputComponents;
    uint32_t              maxGeometryShaderInvocations;
    uint32_t              maxGeometryInputComponents;
    uint32_t              maxGeometryOutputComponents;
    uint32_t              maxGeometryOutputVertices;
    uint32_t              maxGeometryTotalOutputComponents;
    uint32_t              maxFragmentInputComponents;
    uint32_t              maxFragmentOutputAttachments;
    uint32_t              maxFragmentDualSrcAttachments;
    uint32_t              maxFragmentCombinedOutputResources;
    uint32_t              maxComputeSharedMemorySize;
    uint32_t              maxComputeWorkGroupCount[3];
    uint32_t              maxComputeWorkGroupInvocations;
    uint32_t              maxComputeWorkGroupSize[3];
    uint32_t              subPixelPrecisionBits;
    uint32_t              subTexelPrecisionBits;
    uint32_t              mipmapPrecisionBits;
    uint32_t              maxDrawIndexedIndexValue;
    uint32_t              maxDrawIndirectCount;
    float                 maxSamplerLodBias;
    float                 maxSamplerAnisotropy;
    uint32_t              maxViewports;
    uint32_t              maxViewportDimensions[2];
    float                 viewportBoundsRange[2];
    uint32_t              viewportSubPixelBits;
    size_t                minMemoryMapAlignment;
    VkDeviceSize          minTexelBufferOffsetAlignment;
    VkDeviceSize          minUniformBufferOffsetAlignment;
    VkDeviceSize          minStorageBufferOffsetAlignment;
    int32_t               minTexelOffset;
    uint32_t              maxTexelOffset;
    int32_t               minTexelGatherOffset;
    uint32_t              maxTexelGatherOffset;
    float                 minInterpolationOffset;
    float                 maxInterpolationOffset;
    uint32_t              subPixelInterpolationOffsetBits;
    uint32_t              maxFramebufferWidth;
    uint32_t              maxFramebufferHeight;
    uint32_t              maxFramebufferLayers;
    VkSampleCountFlags    framebufferColorSampleCounts;
    VkSampleCountFlags    framebufferDepthSampleCounts;
    VkSampleCountFlags    framebufferStencilSampleCounts;
    VkSampleCountFlags    framebufferNoAttachmentsSampleCounts;
```

```
    uint32_t              maxColorAttachments;
    VkSampleCountFlags    sampledImageColorSampleCounts;
    VkSampleCountFlags    sampledImageIntegerSampleCounts;
    VkSampleCountFlags    sampledImageDepthSampleCounts;
    VkSampleCountFlags    sampledImageStencilSampleCounts;
    VkSampleCountFlags    storageImageSampleCounts;
    uint32_t              maxSampleMaskWords;
    VkBool32              timestampComputeAndGraphics;
    float                 timestampPeriod;
    uint32_t              maxClipDistances;
    uint32_t              maxCullDistances;
    uint32_t              maxCombinedClipAndCullDistances;
    uint32_t              discreteQueuePriorities;
    float                 pointSizeRange[2];
    float                 lineWidthRange[2];
    float                 pointSizeGranularity;
    float                 lineWidthGranularity;
    VkBool32              strictLines;
    VkBool32              standardSampleLocations;
    VkDeviceSize          optimalBufferCopyOffsetAlignment;
    VkDeviceSize          optimalBufferCopyRowPitchAlignment;
    VkDeviceSize          nonCoherentAtomSize;
} VkPhysicalDeviceLimits;
```

The VkPhysicalDeviceLimits are properties of the physical device. These are available in the limits member of the VkPhysicalDeviceProperties structure which is returned from vkGetPhysicalDeviceProperties.

- maxImageDimension1D is the largest dimension (width) that is guaranteed to be supported for all images created with an imageType of VK_IMAGE_TYPE_1D. Some combinations of image parameters (format, usage, etc.) **may** allow support for larger dimensions, which **can** be queried using vkGetPhysicalDeviceImageFormatProperties.
- maxImageDimension2D is the largest dimension (width or height) that is guaranteed to be supported for all images created with an imageType of VK_IMAGE_TYPE_2D and without VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT set in flags. Some combinations of image parameters (format, usage, etc.) **may** allow support for larger dimensions, which **can** be queried using vkGetPhysicalDeviceImageFormatProperties.
- maxImageDimension3D is the largest dimension (width, height, or depth) that is guaranteed to be supported for all images created with an imageType of VK_IMAGE_TYPE_3D. Some combinations of image parameters (format, usage, etc.) **may** allow support for larger dimensions, which **can** be queried using vkGetPhysicalDeviceImageFormatProperties.
- maxImageDimensionCube is the largest dimension (width or height) that is guaranteed to be supported for all images created with an imageType of VK_IMAGE_TYPE_2D and with VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT set in flags. Some combinations of image parameters (format, usage, etc.) **may** allow support for larger dimensions, which **can** be queried using vkGetPhysicalDeviceImageFormatProperties.
- maxImageArrayLayers is the maximum number of layers (arrayLayers) for an image.

- `maxTexelBufferElements` is the maximum number of addressable texels for a buffer view created on a buffer which was created with the VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT or VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT usage flag set.
- `maxUniformBufferRange` is the maximum value that **can** be specified in the `range` member of a VkDescriptorBufferInfo structure passed to vkUpdateDescriptorSets for descriptors of type VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC.
- `maxStorageBufferRange` is the maximum value that **can** be specified in the `range` member of a VkDescriptorBufferInfo structure passed to vkUpdateDescriptorSets for descriptors of type VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC. If the `shader64BitIndexing` feature is enabled, this limit does not apply.
- `maxPushConstantsSize` is the maximum size, in bytes, of the pool of push constant memory. For each of the push constant ranges indicated by the `pPushConstantRanges` member of the VkPipelineLayoutCreateInfo structure, (`offset` + `size`) **must** be less than or equal to this limit.
- `maxMemoryAllocationCount` is the maximum number of device memory allocations, as created by vkAllocateMemory, which **can** simultaneously exist.
- `maxSamplerAllocationCount` is the maximum number of sampler objects, as created by vkCreateSampler, which **can** simultaneously exist on a device. If the `descriptorHeap` feature is enabled and the application intends to use embedded samplers, the number advertised here is effectively reduced by the quotient of `minSamplerHeapReservedRangeWithEmbedded` divided by `samplerDescriptorSize`, to provide storage for embedded samplers when switching to heaps. If embedded samplers are not used, this can be ignored.
- `bufferImageGranularity` is the granularity, in bytes, at which buffer or linear image resources, and optimal image resources **can** be bound to adjacent offsets in the same `VkDeviceMemory` object without aliasing. See Buffer-Image Granularity for more details.
- `sparseAddressSpaceSize` is the total amount of address space available, in bytes, for sparse memory resources. This is an upper bound on the sum of the sizes of all sparse resources, regardless of whether any memory is bound to them.
- `maxBoundDescriptorSets` is the maximum number of descriptor sets that **can** be simultaneously used by a pipeline. See Descriptor Sets.
- `maxPerStageDescriptorSamplers` is the maximum number of samplers that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_SAMPLER or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. A descriptor is accessible to a shader stage when the `stageFlags` member of the `VkDescriptorSetLayoutBinding` structure has the bit for that shader stage set. See Sampler and Combined Image Sampler.
- `maxPerStageDescriptorUniformBuffers` is the maximum number of uniform buffers that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC count against this limit. Only descriptors in descriptor set layouts created without the

VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. A descriptor is accessible to a shader stage when the stageFlags member of the VkDescriptorSetLayoutBinding structure has the bit for that shader stage set. See Uniform Buffer and Dynamic Uniform Buffer.

- maxPerStageDescriptorStorageBuffers is the maximum number of storage buffers that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. A descriptor is accessible to a pipeline shader stage when the stageFlags member of the VkDescriptorSetLayoutBinding structure has the bit for that shader stage set. See Storage Buffer and Dynamic Storage Buffer.
- maxPerStageDescriptorSampledImages is the maximum number of sampled images that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, or VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. A descriptor is accessible to a pipeline shader stage when the stageFlags member of the VkDescriptorSetLayoutBinding structure has the bit for that shader stage set. See Combined Image Sampler, Sampled Image, and Uniform Texel Buffer.
- maxPerStageDescriptorStorageImages is the maximum number of storage images that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. A descriptor is accessible to a pipeline shader stage when the stageFlags member of the VkDescriptorSetLayoutBinding structure has the bit for that shader stage set. See Storage Image, and Storage Texel Buffer.
- maxPerStageDescriptorInputAttachments is the maximum number of input attachments that **can** be accessible to a single shader stage in a pipeline layout, as well as the maximum usable input attachment index. Descriptors with a type of VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. A descriptor is accessible to a pipeline shader stage when the stageFlags member of the VkDescriptorSetLayoutBinding structure has the bit for that shader stage set. These are only supported for the fragment stage. See Input Attachment.

NOTE

maxPerStageDescriptorInputAttachments was originally only intended to limit the number of attachments per stage, not the number of available input indices across all shaders in a render pass. The input indices were allowed to be semi arbitrary for render pass objects, or fully arbitrary for dynamic rendering, however some implementations have fixed limits for them. Applications already exist that exceed this limit, and they will continue to work where they already did, but will fail to render on some platforms. For forward looking applications,

> this can be worked around by either making careful use of index remapping with VkRenderingInputAttachmentIndexInfo for dynamic rendering, or splitting a subpass with too many input attachments into multiple subpasses.

- `maxPerStageResources` is the maximum number of resources that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER, VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER, VK_DESCRIPTOR_TYPE_STORAGE_BUFFER, VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC, VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC, or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. For the fragment shader stage the framebuffer color attachments also count against this limit.
- `maxDescriptorSetSamplers` is the maximum number of samplers that **can** be included in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_SAMPLER or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. See Sampler and Combined Image Sampler.
- `maxDescriptorSetUniformBuffers` is the maximum number of uniform buffers that **can** be included in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. See Uniform Buffer and Dynamic Uniform Buffer.
- `maxDescriptorSetUniformBuffersDynamic` is the maximum number of dynamic uniform buffers that **can** be included in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. See Dynamic Uniform Buffer.
- `maxDescriptorSetStorageBuffers` is the maximum number of storage buffers that **can** be included in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. See Storage Buffer and Dynamic Storage Buffer.
- `maxDescriptorSetStorageBuffersDynamic` is the maximum number of dynamic storage buffers that **can** be included in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC count against this limit. Only descriptors

in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. See Dynamic Storage Buffer.

- `maxDescriptorSetSampledImages` is the maximum number of sampled images that **can** be included in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE, or VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. See Combined Image Sampler, Sampled Image, and Uniform Texel Buffer.
- `maxDescriptorSetStorageImages` is the maximum number of storage images that **can** be included in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, or VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. See Storage Image, and Storage Texel Buffer.
- `maxDescriptorSetInputAttachments` is the maximum number of input attachments that **can** be included in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. See Input Attachment.
- `maxVertexInputAttributes` is the maximum number of vertex input attributes that **can** be specified for a graphics pipeline. These are described in the array of `VkVertexInputAttributeDescription` structures that are provided at graphics pipeline creation time via the `pVertexAttributeDescriptions` member of the VkPipelineVertexInputStateCreateInfo structure. See Vertex Attributes and Vertex Input Description.
- `maxVertexInputBindings` is the maximum number of vertex buffers that **can** be specified for providing vertex attributes to a graphics pipeline. These are described in the array of `VkVertexInputBindingDescription` structures that are provided at graphics pipeline creation time via the `pVertexBindingDescriptions` member of the VkPipelineVertexInputStateCreateInfo structure. The `binding` member of `VkVertexInputBindingDescription` **must** be less than this limit. See Vertex Input Description.
- `maxVertexInputAttributeOffset` is the maximum vertex input attribute offset that **can** be added to the vertex input binding stride. The `offset` member of the `VkVertexInputAttributeDescription` structure **must** be less than or equal to this limit. See Vertex Input Description.
- `maxVertexInputBindingStride` is the maximum vertex input binding stride that **can** be specified in a vertex input binding. The `stride` member of the `VkVertexInputBindingDescription` structure **must** be less than or equal to this limit. See Vertex Input Description.
- `maxVertexOutputComponents` is the maximum number of components of output variables which **can** be output by a vertex shader. See Vertex Shaders.
- `maxTessellationGenerationLevel` is the maximum tessellation generation level supported by the fixed-function tessellation primitive generator. See Tessellation.
- `maxTessellationPatchSize` is the maximum patch size, in vertices, of patches that **can** be

processed by the tessellation control shader and tessellation primitive generator. The patchControlPoints member of the VkPipelineTessellationStateCreateInfo structure specified at pipeline creation time and the value provided in the OutputVertices execution mode of shader modules **must** be less than or equal to this limit. See Tessellation.

- maxTessellationControlPerVertexInputComponents is the maximum number of components of input variables which **can** be provided as per-vertex inputs to the tessellation control shader stage.
- maxTessellationControlPerVertexOutputComponents is the maximum number of components of per-vertex output variables which **can** be output from the tessellation control shader stage.
- maxTessellationControlPerPatchOutputComponents is the maximum number of components of per-patch output variables which **can** be output from the tessellation control shader stage.
- maxTessellationControlTotalOutputComponents is the maximum total number of components of per-vertex and per-patch output variables which **can** be output from the tessellation control shader stage.
- maxTessellationEvaluationInputComponents is the maximum number of components of input variables which **can** be provided as per-vertex inputs to the tessellation evaluation shader stage.
- maxTessellationEvaluationOutputComponents is the maximum number of components of per-vertex output variables which **can** be output from the tessellation evaluation shader stage.
- maxGeometryShaderInvocations is the maximum invocation count supported for instanced geometry shaders. The value provided in the Invocations execution mode of shader modules **must** be less than or equal to this limit. See Geometry Shading.
- maxGeometryInputComponents is the maximum number of components of input variables which **can** be provided as inputs to the geometry shader stage.
- maxGeometryOutputComponents is the maximum number of components of output variables which **can** be output from the geometry shader stage.
- maxGeometryOutputVertices is the maximum number of vertices which **can** be emitted by any geometry shader.
- maxGeometryTotalOutputComponents is the maximum total number of components of output variables, across all emitted vertices, which **can** be output from the geometry shader stage.
- maxFragmentInputComponents is the maximum number of components of input variables which **can** be provided as inputs to the fragment shader stage.
- maxFragmentOutputAttachments is the maximum number of output attachments which **can** be written to by the fragment shader stage.
- maxFragmentDualSrcAttachments is the maximum number of output attachments which **can** be written to by the fragment shader stage when blending is enabled and one of the dual source blend modes is in use. See Dual-Source Blending and dualSrcBlend.
- maxFragmentCombinedOutputResources is the total number of storage buffers, storage images, and output Location decorated color attachments (described in Fragment Output Interface) which **can** be used in the fragment shader stage.
- maxComputeSharedMemorySize is the maximum total storage size, in bytes, available for variables

declared with the Workgroup storage class in shader modules (or with the shared storage qualifier in GLSL) in the compute shader stage.

- maxComputeWorkGroupCount[3] is the maximum number of local workgroups that **can** be dispatched by a single dispatching command. These three values represent the maximum number of local workgroups for the X, Y, and Z dimensions, respectively. The workgroup count parameters to the dispatching commands **must** be less than or equal to the corresponding limit. See Dispatching Commands.
- maxComputeWorkGroupInvocations is the maximum total number of compute shader invocations in a single local workgroup. The product of the X, Y, and Z sizes, as specified by the LocalSize or LocalSizeId execution mode in shader modules or by the object decorated by the WorkgroupSize decoration, **must** be less than or equal to this limit.
- maxComputeWorkGroupSize[3] is the maximum size of a local compute workgroup, per dimension. These three values represent the maximum local workgroup size in the X, Y, and Z dimensions, respectively. The x, y, and z sizes, as specified by the LocalSize or LocalSizeId execution mode or by the object decorated by the WorkgroupSize decoration in shader modules, **must** be less than or equal to the corresponding limit.
- subPixelPrecisionBits is the number of bits of subpixel precision in framebuffer coordinates x_f and y_f. See Rasterization.
- subTexelPrecisionBits is the number of bits of precision in the division along an axis of an image used for minification and magnification filters. $2^{subTexelPrecisionBits}$ is the actual number of divisions along each axis of the image represented. Sub-texel values calculated during image sampling will snap to these locations when generating the filtered results.
- mipmapPrecisionBits is the number of bits of division that the LOD calculation for mipmap fetching get snapped to when determining the contribution from each mip level to the mip filtered results. $2^{mipmapPrecisionBits}$ is the actual number of divisions.
- maxDrawIndexedIndexValue is the maximum index value that **can** be used for indexed draw calls when using 32-bit indices. This excludes the primitive restart index value of 0xFFFFFFFF. See fullDrawIndexUint32.
- maxDrawIndirectCount is the maximum draw count that is supported for indirect drawing calls. See multiDrawIndirect.
- maxSamplerLodBias is the maximum absolute sampler LOD bias. The sum of the mipLodBias member of the VkSamplerCreateInfo structure and the Bias operand of image sampling operations in shader modules (or 0 if no Bias operand is provided to an image sampling operation) is clamped to the range [-maxSamplerLodBias,+maxSamplerLodBias]. See [samplers-mipLodBias].
- maxSamplerAnisotropy is the maximum degree of sampler anisotropy. The maximum degree of anisotropic filtering used for an image sampling operation is the minimum of the maxAnisotropy member of the VkSamplerCreateInfo structure and this limit. See [samplers-maxAnisotropy].
- maxViewports is the maximum number of active viewports. The viewportCount member of the VkPipelineViewportStateCreateInfo structure that is provided at pipeline creation **must** be less than or equal to this limit.
- maxViewportDimensions[2] are the maximum viewport dimensions in the X (width) and Y (height) dimensions, respectively. The maximum viewport dimensions **must** be greater than or equal to

the largest image which **can** be created and used as a framebuffer attachment. See Controlling the Viewport.

- `viewportBoundsRange`[2] is the [minimum, maximum] range that the corners of a viewport **must** be contained in. This range **must** be at least [-2 × `size`, 2 × `size` - 1], where `size` = max(`maxViewportDimensions`[0], `maxViewportDimensions`[1]). See Controlling the Viewport.

> **NOTE**
>
> The wide range of values required for `viewportBoundsRange` allows the viewport to be arbitrarily shifted relative to the output render target while still partially overlapping. However, the minimum range required to achieve this would actually be [-`size` + 1, 2 × `size` - 1]. As these limits in implementations are typically simple power-of-two values, the specification reflects this convention, rounding the lower bound accordingly.

- `viewportSubPixelBits` is the number of bits of subpixel precision for viewport bounds. The subpixel precision that floating-point viewport bounds are interpreted at is given by this limit.
- `minMemoryMapAlignment` is the minimum **required** alignment, in bytes, of host visible memory allocations within the host address space. When mapping a memory allocation with vkMapMemory, subtracting `offset` bytes from the returned pointer will always produce an integer multiple of this limit. See Host Access to Device Memory Objects. The value **must** be a power of two.
- `minTexelBufferOffsetAlignment` is the minimum **required** alignment, in bytes, for the `offset` member of the VkBufferViewCreateInfo structure for texel buffers. The value **must** be a power of two. This limit is equivalent to the maximum of the `uniformTexelBufferOffsetAlignmentBytes` and `storageTexelBufferOffsetAlignmentBytes` members of VkPhysicalDeviceTexelBufferAlignmentProperties, but smaller alignment is **optionally** allowed by `storageTexelBufferOffsetSingleTexelAlignment` and `uniformTexelBufferOffsetSingleTexelAlignment`. For single texel alignment, a format has an alignment requirement which is the size of a single component if the size of the format is a multiple of three bytes, otherwise, it is the size of the format itself. The effective alignment requirement is the minimum of the per-format alignment and `uniformTexelBufferOffsetAlignmentBytes` or `storageTexelBufferOffsetAlignmentBytes` depending on the descriptor type. If the `texelBufferAlignment` feature is not enabled, the effective alignment requirement for any format is `minTexelBufferOffsetAlignment`. VkBufferViewCreateInfo::`offset` **must** be a multiple of this value.
- `minUniformBufferOffsetAlignment` is the minimum **required** alignment, in bytes, for the `offset` member of the `VkDescriptorBufferInfo` structure for uniform buffers. When a descriptor of type VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC is updated, the `offset` **must** be an integer multiple of this limit. Similarly, dynamic offsets for uniform buffers **must** be multiples of this limit. The value **must** be a power of two.
- `minStorageBufferOffsetAlignment` is the minimum **required** alignment, in bytes, for the `offset` member of the `VkDescriptorBufferInfo` structure for storage buffers. When a descriptor of type VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC is updated, the `offset` **must** be an integer multiple of this limit. Similarly, dynamic offsets for storage buffers **must** be multiples of this limit. The value **must** be a power of two.

- `minTexelOffset` is the minimum offset value for the `Offset` or `ConstOffset` image operand of any of the `OpImageSample*` or `OpImageFetch*` image instructions.
- `maxTexelOffset` is the maximum offset value for the `Offset` or `ConstOffset` image operand of any of the `OpImageSample*` or `OpImageFetch*` image instructions.
- `minTexelGatherOffset` is the minimum offset value for the `Offset`, `ConstOffset`, or `ConstOffsets` image operands of any of the `OpImage*Gather` image instructions.
- `maxTexelGatherOffset` is the maximum offset value for the `Offset`, `ConstOffset`, or `ConstOffsets` image operands of any of the `OpImage*Gather` image instructions.
- `minInterpolationOffset` is the base minimum (inclusive) negative offset value for the `Offset` operand of the `InterpolateAtOffset` extended instruction.
- `maxInterpolationOffset` is the base maximum (inclusive) positive offset value for the `Offset` operand of the `InterpolateAtOffset` extended instruction.
- `subPixelInterpolationOffsetBits` is the number of fractional bits that the `x` and `y` offsets to the `InterpolateAtOffset` extended instruction **may** be rounded to as fixed-point values.
- `maxFramebufferWidth` is the maximum width for a framebuffer. The `width` member of the VkFramebufferCreateInfo structure **must** be less than or equal to this limit.
- `maxFramebufferHeight` is the maximum height for a framebuffer. The `height` member of the VkFramebufferCreateInfo structure **must** be less than or equal to this limit.
- `maxFramebufferLayers` is the maximum layer count for a layered framebuffer. The `layers` member of the VkFramebufferCreateInfo structure **must** be less than or equal to this limit.
- `framebufferColorSampleCounts` is a bitmask[1] of VkSampleCountFlagBits indicating the color sample counts that are supported for all framebuffer color attachments with floating- or fixed-point formats. For color attachments with integer formats, see `framebufferIntegerColorSampleCounts`.
- `framebufferDepthSampleCounts` is a bitmask[1] of VkSampleCountFlagBits indicating the supported depth sample counts for all framebuffer depth/stencil attachments, when the format includes a depth component.
- `framebufferStencilSampleCounts` is a bitmask[1] of VkSampleCountFlagBits indicating the supported stencil sample counts for all framebuffer depth/stencil attachments, when the format includes a stencil component.
- `framebufferNoAttachmentsSampleCounts` is a bitmask[1] of VkSampleCountFlagBits indicating the supported sample counts for a subpass which uses no attachments.
- `maxColorAttachments` is the maximum number of color attachments that **can** be used by a subpass in a render pass. The `colorAttachmentCount` member of the `VkSubpassDescription` or `VkSubpassDescription2` structure **must** be less than or equal to this limit.
- `sampledImageColorSampleCounts` is a bitmask[1] of VkSampleCountFlagBits indicating the sample counts supported for all 2D images created with VK_IMAGE_TILING_OPTIMAL, the VK_IMAGE_USAGE_SAMPLED_BIT usage flag set, and a non-integer color format.
- `sampledImageIntegerSampleCounts` is a bitmask[1] of VkSampleCountFlagBits indicating the sample counts supported for all 2D images created with VK_IMAGE_TILING_OPTIMAL, the VK_IMAGE_USAGE_SAMPLED_BIT usage flag set, and an integer color format.

- `sampledImageDepthSampleCounts` is a bitmask[1] of VkSampleCountFlagBits indicating the sample counts supported for all 2D images created with VK_IMAGE_TILING_OPTIMAL, the VK_IMAGE_USAGE_SAMPLED_BIT usage flag set, and a depth format.
- `sampledImageStencilSampleCounts` is a bitmask[1] of VkSampleCountFlagBits indicating the sample counts supported for all 2D images created with VK_IMAGE_TILING_OPTIMAL, the VK_IMAGE_USAGE_SAMPLED_BIT usage flag set, and a stencil format.
- `storageImageSampleCounts` is a bitmask[1] of VkSampleCountFlagBits indicating the sample counts supported for all 2D images created with VK_IMAGE_TILING_OPTIMAL, and the VK_IMAGE_USAGE_STORAGE_BIT usage flag set.
- `maxSampleMaskWords` is the maximum number of array elements of a variable decorated with the `SampleMask` built-in decoration.
- `timestampComputeAndGraphics` specifies support for timestamps on all graphics and compute queues. If this limit is VK_TRUE, all queues that advertise the VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT in the `VkQueueFamilyProperties`::`queueFlags` support `VkQueueFamilyProperties`::`timestampValidBits` of at least 36. See Timestamp Queries.
- `timestampPeriod` is the number of nanoseconds **required** for a timestamp query to be incremented by 1. See Timestamp Queries.
- `maxClipDistances` is the maximum number of clip distances that **can** be used in a single shader stage. The size of any array declared with the `ClipDistance` built-in decoration in a shader module **must** be less than or equal to this limit.
- `maxCullDistances` is the maximum number of cull distances that **can** be used in a single shader stage. The size of any array declared with the `CullDistance` built-in decoration in a shader module **must** be less than or equal to this limit.
- `maxCombinedClipAndCullDistances` is the maximum combined number of clip and cull distances that **can** be used in a single shader stage. The sum of the sizes of all arrays declared with the `ClipDistance` and `CullDistance` built-in decoration used by a single shader stage in a shader module **must** be less than or equal to this limit.
- `discreteQueuePriorities` is the number of discrete priorities that **can** be assigned to a queue based on the value of each member of VkDeviceQueueCreateInfo::`pQueuePriorities`. This **must** be at least 2, and levels **must** be spread evenly over the range, with at least one level at 1.0, and another at 0.0. See Queue Priority.
- `pointSizeRange`[2] is the range [`minimum`,`maximum`] of supported sizes for points. Values written to variables decorated with the `PointSize` built-in decoration are clamped to this range.
- `lineWidthRange`[2] is the range [`minimum`,`maximum`] of supported widths for lines. Values specified by the `lineWidth` member of the VkPipelineRasterizationStateCreateInfo or the `lineWidth` parameter to `vkCmdSetLineWidth` are clamped to this range.
- `pointSizeGranularity` is the granularity of supported point sizes. Not all point sizes in the range defined by `pointSizeRange` are supported. This limit specifies the granularity (or increment) between successive supported point sizes.
- `lineWidthGranularity` is the granularity of supported line widths. Not all line widths in the range defined by `lineWidthRange` are supported. This limit specifies the granularity (or increment) between successive supported line widths.

- `strictLines` specifies whether lines are rasterized according to the preferred method of rasterization. If set to VK_FALSE, lines **may** be rasterized under a relaxed set of rules. If set to VK_TRUE, lines are rasterized as per the strict definition. See Basic Line Segment Rasterization.
- `standardSampleLocations` specifies whether rasterization uses the standard sample locations as documented in Multisampling. If set to VK_TRUE, the implementation uses the documented sample locations. If set to VK_FALSE, the implementation **may** use different sample locations.
- `optimalBufferCopyOffsetAlignment` is the optimal buffer offset alignment in bytes for vkCmdCopyBufferToImage2, vkCmdCopyBufferToImage, vkCmdCopyImageToBuffer2, and vkCmdCopyImageToBuffer. This value is also the optimal host memory offset alignment in bytes for vkCopyMemoryToImage and vkCopyImageToMemory. The per texel alignment requirements are enforced, but applications **should** use the optimal alignment for optimal performance and power use. The value **must** be a power of two.
- `optimalBufferCopyRowPitchAlignment` is the optimal buffer row pitch alignment in bytes for vkCmdCopyBufferToImage2, vkCmdCopyBufferToImage, vkCmdCopyImageToBuffer2, and vkCmdCopyImageToBuffer. This value is also the optimal host memory row pitch alignment in bytes for vkCopyMemoryToImage and vkCopyImageToMemory. Row pitch is the number of bytes between texels with the same X coordinate in adjacent rows (Y coordinates differ by one). The per texel alignment requirements are enforced, but applications **should** use the optimal alignment for optimal performance and power use. The value **must** be a power of two.
- `nonCoherentAtomSize` is the size and alignment in bytes that bounds concurrent access to host-mapped device memory. The value **must** be a power of two.

  1

  For all bitmasks of VkSampleCountFlagBits, the sample count limits defined above represent the minimum supported sample counts for each image type. Individual images **may** support additional sample counts, which are queried using vkGetPhysicalDeviceImageFormatProperties as described in Supported Sample Counts.

Bits which **may** be set in the sample count limits returned by VkPhysicalDeviceLimits, as well as in other queries and structures representing image sample counts, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSampleCountFlagBits {
    VK_SAMPLE_COUNT_1_BIT = 0x00000001,
    VK_SAMPLE_COUNT_2_BIT = 0x00000002,
    VK_SAMPLE_COUNT_4_BIT = 0x00000004,
    VK_SAMPLE_COUNT_8_BIT = 0x00000008,
    VK_SAMPLE_COUNT_16_BIT = 0x00000010,
    VK_SAMPLE_COUNT_32_BIT = 0x00000020,
    VK_SAMPLE_COUNT_64_BIT = 0x00000040,
} VkSampleCountFlagBits;
```

- VK_SAMPLE_COUNT_1_BIT specifies an image with one sample per pixel.
- VK_SAMPLE_COUNT_2_BIT specifies an image with 2 samples per pixel.
- VK_SAMPLE_COUNT_4_BIT specifies an image with 4 samples per pixel.

- VK_SAMPLE_COUNT_8_BIT specifies an image with 8 samples per pixel.
- VK_SAMPLE_COUNT_16_BIT specifies an image with 16 samples per pixel.
- VK_SAMPLE_COUNT_32_BIT specifies an image with 32 samples per pixel.
- VK_SAMPLE_COUNT_64_BIT specifies an image with 64 samples per pixel.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSampleCountFlags;
```

VkSampleCountFlags is a bitmask type for setting a mask of zero or more VkSampleCountFlagBits.

The VkPhysicalDevicePushDescriptorProperties structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDevicePushDescriptorProperties {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxPushDescriptors;
} VkPhysicalDevicePushDescriptorProperties;
```

```
// Provided by VK_KHR_push_descriptor
// Equivalent to VkPhysicalDevicePushDescriptorProperties
typedef VkPhysicalDevicePushDescriptorProperties
VkPhysicalDevicePushDescriptorPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- maxPushDescriptors is the maximum number of descriptors that **can** be used in a descriptor set layout created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT set.

If the VkPhysicalDevicePushDescriptorProperties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePushDescriptorProperties-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PUSH_DESCRIPTOR_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceMultiviewProperties structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceMultiviewProperties {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxMultiviewViewCount;
    uint32_t           maxMultiviewInstanceIndex;
} VkPhysicalDeviceMultiviewProperties;
```

```
// Provided by VK_KHR_multiview
// Equivalent to VkPhysicalDeviceMultiviewProperties
typedef VkPhysicalDeviceMultiviewProperties VkPhysicalDeviceMultiviewPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- maxMultiviewViewCount is one greater than the maximum view index that **can** be used in a subpass.
- maxMultiviewInstanceIndex is the maximum valid value of instance index allowed to be generated by a drawing command recorded within a subpass of a multiview render pass instance.

If the VkPhysicalDeviceMultiviewProperties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMultiviewProperties-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceFloatControlsProperties structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceFloatControlsProperties {
    VkStructureType                      sType;
    void*                                pNext;
    VkShaderFloatControlsIndependence    denormBehaviorIndependence;
    VkShaderFloatControlsIndependence    roundingModeIndependence;
    VkBool32                             shaderSignedZeroInfNanPreserveFloat16;
    VkBool32                             shaderSignedZeroInfNanPreserveFloat32;
    VkBool32                             shaderSignedZeroInfNanPreserveFloat64;
    VkBool32                             shaderDenormPreserveFloat16;
    VkBool32                             shaderDenormPreserveFloat32;
    VkBool32                             shaderDenormPreserveFloat64;
    VkBool32                             shaderDenormFlushToZeroFloat16;
    VkBool32                             shaderDenormFlushToZeroFloat32;
    VkBool32                             shaderDenormFlushToZeroFloat64;
    VkBool32                             shaderRoundingModeRTEFloat16;
    VkBool32                             shaderRoundingModeRTEFloat32;
    VkBool32                             shaderRoundingModeRTEFloat64;
    VkBool32                             shaderRoundingModeRTZFloat16;
    VkBool32                             shaderRoundingModeRTZFloat32;
    VkBool32                             shaderRoundingModeRTZFloat64;
} VkPhysicalDeviceFloatControlsProperties;
```

```
// Provided by VK_KHR_shader_float_controls
// Equivalent to VkPhysicalDeviceFloatControlsProperties
typedef VkPhysicalDeviceFloatControlsProperties
VkPhysicalDeviceFloatControlsPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- denormBehaviorIndependence is a VkShaderFloatControlsIndependence value indicating whether, and how, denorm behavior can be set independently for different bit widths.
- roundingModeIndependence is a VkShaderFloatControlsIndependence value indicating whether, and how, rounding modes can be set independently for different bit widths.
- shaderSignedZeroInfNanPreserveFloat16 is a boolean value indicating whether sign of a zero, Nans and ±∞ **can** be preserved in 16-bit floating-point computations. It also indicates whether the SignedZeroInfNanPreserve execution mode **can** be used for 16-bit floating-point types.
- shaderSignedZeroInfNanPreserveFloat32 is a boolean value indicating whether sign of a zero, Nans and ±∞ **can** be preserved in 32-bit floating-point computations. It also indicates whether the SignedZeroInfNanPreserve execution mode **can** be used for 32-bit floating-point types.
- shaderSignedZeroInfNanPreserveFloat64 is a boolean value indicating whether sign of a zero, Nans and ±∞ **can** be preserved in 64-bit floating-point computations. It also indicates whether the SignedZeroInfNanPreserve execution mode **can** be used for 64-bit floating-point types.

- `shaderDenormPreserveFloat16` is a boolean value indicating whether denormals **can** be preserved in 16-bit floating-point computations. It also indicates whether the `DenormPreserve` execution mode **can** be used for 16-bit floating-point types.
- `shaderDenormPreserveFloat32` is a boolean value indicating whether denormals **can** be preserved in 32-bit floating-point computations. It also indicates whether the `DenormPreserve` execution mode **can** be used for 32-bit floating-point types.
- `shaderDenormPreserveFloat64` is a boolean value indicating whether denormals **can** be preserved in 64-bit floating-point computations. It also indicates whether the `DenormPreserve` execution mode **can** be used for 64-bit floating-point types.
- `shaderDenormFlushToZeroFloat16` is a boolean value indicating whether denormals **can** be flushed to zero in 16-bit floating-point computations. It also indicates whether the `DenormFlushToZero` execution mode **can** be used for 16-bit floating-point types.
- `shaderDenormFlushToZeroFloat32` is a boolean value indicating whether denormals **can** be flushed to zero in 32-bit floating-point computations. It also indicates whether the `DenormFlushToZero` execution mode **can** be used for 32-bit floating-point types.
- `shaderDenormFlushToZeroFloat64` is a boolean value indicating whether denormals **can** be flushed to zero in 64-bit floating-point computations. It also indicates whether the `DenormFlushToZero` execution mode **can** be used for 64-bit floating-point types.
- `shaderRoundingModeRTEFloat16` is a boolean value indicating whether an implementation supports the round-to-nearest-even rounding mode for 16-bit floating-point arithmetic and conversion instructions. It also indicates whether the `RoundingModeRTE` execution mode **can** be used for 16-bit floating-point types.
- `shaderRoundingModeRTEFloat32` is a boolean value indicating whether an implementation supports the round-to-nearest-even rounding mode for 32-bit floating-point arithmetic and conversion instructions. It also indicates whether the `RoundingModeRTE` execution mode **can** be used for 32-bit floating-point types.
- `shaderRoundingModeRTEFloat64` is a boolean value indicating whether an implementation supports the round-to-nearest-even rounding mode for 64-bit floating-point arithmetic and conversion instructions. It also indicates whether the `RoundingModeRTE` execution mode **can** be used for 64-bit floating-point types.
- `shaderRoundingModeRTZFloat16` is a boolean value indicating whether an implementation supports the round-towards-zero rounding mode for 16-bit floating-point arithmetic and conversion instructions. It also indicates whether the `RoundingModeRTZ` execution mode **can** be used for 16-bit floating-point types.
- `shaderRoundingModeRTZFloat32` is a boolean value indicating whether an implementation supports the round-towards-zero rounding mode for 32-bit floating-point arithmetic and conversion instructions. It also indicates whether the `RoundingModeRTZ` execution mode **can** be used for 32-bit floating-point types.
- `shaderRoundingModeRTZFloat64` is a boolean value indicating whether an implementation supports the round-towards-zero rounding mode for 64-bit floating-point arithmetic and conversion instructions. It also indicates whether the `RoundingModeRTZ` execution mode **can** be used for 64-bit floating-point types.

If the `VkPhysicalDeviceFloatControlsProperties` structure is included in the `pNext` chain of the

VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFloatControlsProperties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FLOAT_CONTROLS_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

Values which **may** be returned in the `denormBehaviorIndependence` and `roundingModeIndependence` fields of `VkPhysicalDeviceFloatControlsProperties` are:

```
// Provided by VK_VERSION_1_2
typedef enum VkShaderFloatControlsIndependence {
    VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_32_BIT_ONLY = 0,
    VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_ALL = 1,
    VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_NONE = 2,
  // Provided by VK_KHR_shader_float_controls
    VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_32_BIT_ONLY_KHR =
VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_32_BIT_ONLY,
  // Provided by VK_KHR_shader_float_controls
    VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_ALL_KHR =
VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_ALL,
  // Provided by VK_KHR_shader_float_controls
    VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_NONE_KHR =
VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_NONE,
} VkShaderFloatControlsIndependence;
```

```
// Provided by VK_KHR_shader_float_controls
// Equivalent to VkShaderFloatControlsIndependence
typedef VkShaderFloatControlsIndependence VkShaderFloatControlsIndependenceKHR;
```

- VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_32_BIT_ONLY specifies that shader float controls for 32-bit floating-point **can** be set independently; other bit widths **must** be set identically to each other.
- VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_ALL specifies that shader float controls for all bit widths **can** be set independently.
- VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_NONE specifies that shader float controls for all bit widths **must** be set identically.

The VkPhysicalDeviceDiscardRectanglePropertiesEXT structure is defined as:

```
// Provided by VK_EXT_discard_rectangles
typedef struct VkPhysicalDeviceDiscardRectanglePropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxDiscardRectangles;
} VkPhysicalDeviceDiscardRectanglePropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- maxDiscardRectangles is the maximum number of active discard rectangles that **can** be specified.

If the VkPhysicalDeviceDiscardRectanglePropertiesEXT structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDiscardRectanglePropertiesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DISCARD_RECTANGLE_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceSampleLocationsPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_sample_locations
typedef struct VkPhysicalDeviceSampleLocationsPropertiesEXT {
    VkStructureType       sType;
    void*                 pNext;
    VkSampleCountFlags    sampleLocationSampleCounts;
    VkExtent2D            maxSampleLocationGridSize;
    float                 sampleLocationCoordinateRange[2];
    uint32_t              sampleLocationSubPixelBits;
    VkBool32              variableSampleLocations;
} VkPhysicalDeviceSampleLocationsPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.

- `sampleLocationSampleCounts` is a bitmask of VkSampleCountFlagBits indicating the sample counts supporting custom sample locations.
- `maxSampleLocationGridSize` is the maximum size of the pixel grid in which sample locations **can** vary that is supported for all sample counts in `sampleLocationSampleCounts`.
- `sampleLocationCoordinateRange`[2] is the range of supported sample location coordinates.
- `sampleLocationSubPixelBits` is the number of bits of subpixel precision for sample locations.
- `variableSampleLocations` specifies whether the sample locations used by all pipelines that will be bound to a command buffer during a subpass **must** match. If set to VK_TRUE, the implementation supports variable sample locations in a subpass. If set to VK_FALSE, then the sample locations **must** stay constant in each subpass.

If the `VkPhysicalDeviceSampleLocationsPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSampleLocationsPropertiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLE_LOCATIONS_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_legacy_vertex_attributes
typedef struct VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           nativeUnalignedPerformance;
} VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT;
```

This structure describes the following features:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `nativeUnalignedPerformance` specifies whether unaligned vertex fetches do not incur significant performance penalties as compared to aligned fetches.

If the `VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT` structure is included in the `pNext` chain

of the VkPhysicalDeviceFeatures2 structure passed to vkGetPhysicalDeviceFeatures2, it is filled in to indicate whether each corresponding feature is supported. If the application wishes to use a VkDevice with any features described by VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT, it **must** add an instance of the structure, with the desired feature members set to VK_TRUE, to the pNext chain of VkDeviceCreateInfo when creating the VkDevice.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LEGACY_VERTEX_ATTRIBUTES_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceExternalMemoryHostPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_external_memory_host
typedef struct VkPhysicalDeviceExternalMemoryHostPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceSize       minImportedHostPointerAlignment;
} VkPhysicalDeviceExternalMemoryHostPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- minImportedHostPointerAlignment is the minimum **required** alignment, in bytes, for the base address and size of host pointers that **can** be imported to a Vulkan memory object. The value **must** be a power of two.

If the VkPhysicalDeviceExternalMemoryHostPropertiesEXT structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExternalMemoryHostPropertiesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_MEMORY_HOST_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDevicePointClippingProperties structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDevicePointClippingProperties {
    VkStructureType            sType;
    void*                      pNext;
    VkPointClippingBehavior    pointClippingBehavior;
} VkPhysicalDevicePointClippingProperties;
```

```
// Provided by VK_KHR_maintenance2
// Equivalent to VkPhysicalDevicePointClippingProperties
typedef VkPhysicalDevicePointClippingProperties
VkPhysicalDevicePointClippingPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- pointClippingBehavior is a VkPointClippingBehavior value specifying the point clipping behavior supported by the implementation.

If the VkPhysicalDevicePointClippingProperties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePointClippingProperties-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_POINT_CLIPPING_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceSubgroupProperties structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceSubgroupProperties {
    VkStructureType           sType;
    void*                     pNext;
    uint32_t                  subgroupSize;
    VkShaderStageFlags        supportedStages;
    VkSubgroupFeatureFlags    supportedOperations;
    VkBool32                  quadOperationsInAllStages;
} VkPhysicalDeviceSubgroupProperties;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `subgroupSize` is the default number of invocations in each subgroup. `subgroupSize` is at least 1 if any of the physical device's queues support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT. `subgroupSize` is a power-of-two.
- `supportedStages` is a bitfield of VkShaderStageFlagBits describing the shader stages that group operations with subgroup scope are supported in. `supportedStages` will have the VK_SHADER_STAGE_COMPUTE_BIT bit set if any of the physical device's queues support VK_QUEUE_COMPUTE_BIT.
- `supportedOperations` is a bitmask of VkSubgroupFeatureFlagBits specifying the sets of group operations with subgroup scope supported on this device. `supportedOperations` will have the VK_SUBGROUP_FEATURE_BASIC_BIT bit set if any of the physical device's queues support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT.
- `quadOperationsInAllStages` is a boolean specifying whether quad group operations are available in all stages, or are restricted to fragment and compute stages.

If the `VkPhysicalDeviceSubgroupProperties` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

If `supportedOperations` includes VK_SUBGROUP_FEATURE_QUAD_BIT, or the `shaderSubgroupUniformControlFlow` feature is enabled, `subgroupSize` **must** be greater than or equal to 4.

If the `shaderQuadControl` feature is supported, `supportedOperations` **must** include VK_SUBGROUP_FEATURE_QUAD_BIT.

If VK_KHR_shader_subgroup_rotate is supported, and the implementation advertises support with a VkExtensionProperties::`specVersion` greater than or equal to 2, and the `shaderSubgroupRotate` feature is supported, VK_SUBGROUP_FEATURE_ROTATE_BIT **must** be returned in VkPhysicalDeviceVulkan11Properties::`subgroupSupportedOperations` and VkPhysicalDeviceSubgroupProperties::`supportedOperations`. If VK_KHR_shader_subgroup_rotate is supported, and the implementation advertises support with a VkExtensionProperties::`specVersion` greater than or equal to 2, and the `shaderSubgroupRotateClustered` feature is supported, VK_SUBGROUP_FEATURE_ROTATE_CLUSTERED_BIT **must** be returned in VkPhysicalDeviceVulkan11Properties::`subgroupSupportedOperations` and

VkPhysicalDeviceSubgroupProperties::supportedOperations.

If Vulkan 1.4 is supported, VK_SUBGROUP_FEATURE_ROTATE_BIT and VK_SUBGROUP_FEATURE_ROTATE_CLUSTERED_BIT **must** be returned in VkPhysicalDeviceSubgroupProperties::supportedOperations and VkPhysicalDeviceVulkan11Properties::subgroupSupportedOperations.

NOTE VK_SUBGROUP_FEATURE_ROTATE_BIT and VK_SUBGROUP_FEATURE_ROTATE_CLUSTERED_BIT were added in version 2 of the VK_KHR_shader_subgroup_rotate extension, after the initial release, so there are implementations that do not advertise these bits. Applications should use the shaderSubgroupRotate and shaderSubgroupRotateClustered features to determine and enable support. These bits are advertised here for consistency and for future dependencies.

If the shaderSubgroupPartitioned feature is supported, supportedOperations **must** include VK_SUBGROUP_FEATURE_PARTITIONED_BIT_EXT.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSubgroupProperties-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SUBGROUP_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

Bits which **can** be set in VkPhysicalDeviceSubgroupProperties::supportedOperations and VkPhysicalDeviceVulkan11Properties::subgroupSupportedOperations to specify supported group operations with subgroup scope are:

```
// Provided by VK_VERSION_1_1
typedef enum VkSubgroupFeatureFlagBits {
    VK_SUBGROUP_FEATURE_BASIC_BIT = 0x00000001,
    VK_SUBGROUP_FEATURE_VOTE_BIT = 0x00000002,
    VK_SUBGROUP_FEATURE_ARITHMETIC_BIT = 0x00000004,
    VK_SUBGROUP_FEATURE_BALLOT_BIT = 0x00000008,
    VK_SUBGROUP_FEATURE_SHUFFLE_BIT = 0x00000010,
    VK_SUBGROUP_FEATURE_SHUFFLE_RELATIVE_BIT = 0x00000020,
    VK_SUBGROUP_FEATURE_CLUSTERED_BIT = 0x00000040,
    VK_SUBGROUP_FEATURE_QUAD_BIT = 0x00000080,
  // Provided by VK_VERSION_1_4
    VK_SUBGROUP_FEATURE_ROTATE_BIT = 0x00000200,
  // Provided by VK_VERSION_1_4
    VK_SUBGROUP_FEATURE_ROTATE_CLUSTERED_BIT = 0x00000400,
  // Provided by VK_EXT_shader_subgroup_partitioned
    VK_SUBGROUP_FEATURE_PARTITIONED_BIT_EXT = 0x00000100,
  // Provided by VK_KHR_shader_subgroup_rotate
    VK_SUBGROUP_FEATURE_ROTATE_BIT_KHR = VK_SUBGROUP_FEATURE_ROTATE_BIT,
  // Provided by VK_KHR_shader_subgroup_rotate
    VK_SUBGROUP_FEATURE_ROTATE_CLUSTERED_BIT_KHR =
VK_SUBGROUP_FEATURE_ROTATE_CLUSTERED_BIT,
} VkSubgroupFeatureFlagBits;
```

- VK_SUBGROUP_FEATURE_BASIC_BIT specifies the device will accept SPIR-V shader modules containing the GroupNonUniform capability.
- VK_SUBGROUP_FEATURE_VOTE_BIT specifies the device will accept SPIR-V shader modules containing the GroupNonUniformVote capability.
- VK_SUBGROUP_FEATURE_ARITHMETIC_BIT specifies the device will accept SPIR-V shader modules containing the GroupNonUniformArithmetic capability.
- VK_SUBGROUP_FEATURE_BALLOT_BIT specifies the device will accept SPIR-V shader modules containing the GroupNonUniformBallot capability.
- VK_SUBGROUP_FEATURE_SHUFFLE_BIT specifies the device will accept SPIR-V shader modules containing the GroupNonUniformShuffle capability.
- VK_SUBGROUP_FEATURE_SHUFFLE_RELATIVE_BIT specifies the device will accept SPIR-V shader modules containing the GroupNonUniformShuffleRelative capability.
- VK_SUBGROUP_FEATURE_CLUSTERED_BIT specifies the device will accept SPIR-V shader modules containing the GroupNonUniformClustered capability.
- VK_SUBGROUP_FEATURE_QUAD_BIT specifies the device will accept SPIR-V shader modules containing the GroupNonUniformQuad capability.
- VK_SUBGROUP_FEATURE_PARTITIONED_BIT_EXT specifies the device will accept SPIR-V shader modules containing the GroupNonUniformPartitionedEXT capability.
- VK_SUBGROUP_FEATURE_ROTATE_BIT specifies the device will accept SPIR-V shader modules containing the GroupNonUniformRotateKHR capability.
- VK_SUBGROUP_FEATURE_ROTATE_CLUSTERED_BIT specifies the device will accept SPIR-V

shader modules that use the ClusterSize operand to OpGroupNonUniformRotateKHR.

```
// Provided by VK_VERSION_1_1
typedef VkFlags VkSubgroupFeatureFlags;
```

VkSubgroupFeatureFlags is a bitmask type for setting a mask of zero or more VkSubgroupFeatureFlagBits.

The VkPhysicalDeviceSubgroupSizeControlProperties structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceSubgroupSizeControlProperties {
    VkStructureType       sType;
    void*                 pNext;
    uint32_t              minSubgroupSize;
    uint32_t              maxSubgroupSize;
    uint32_t              maxComputeWorkgroupSubgroups;
    VkShaderStageFlags    requiredSubgroupSizeStages;
} VkPhysicalDeviceSubgroupSizeControlProperties;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- minSubgroupSize is the minimum subgroup size supported by this device. minSubgroupSize is at least one if any of the physical device's queues support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT. minSubgroupSize is a power-of-two. minSubgroupSize is less than or equal to maxSubgroupSize. minSubgroupSize is less than or equal to subgroupSize.
- maxSubgroupSize is the maximum subgroup size supported by this device. maxSubgroupSize is at least one if any of the physical device's queues support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT. maxSubgroupSize is a power-of-two. maxSubgroupSize is greater than or equal to minSubgroupSize. maxSubgroupSize is greater than or equal to subgroupSize.
- maxComputeWorkgroupSubgroups is the maximum number of subgroups supported by the implementation within a workgroup.
- requiredSubgroupSizeStages is a bitfield of what shader stages support having a required subgroup size specified.

If the VkPhysicalDeviceSubgroupSizeControlProperties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

If VkPhysicalDeviceSubgroupProperties::supportedOperations includes VK_SUBGROUP_FEATURE_QUAD_BIT, minSubgroupSize **must** be greater than or equal to 4.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSubgroupSizeControlProperties-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SUBGROUP_SIZE_CONTROL_PROPERTIES`

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_blend_operation_advanced
typedef struct VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           advancedBlendMaxColorAttachments;
    VkBool32           advancedBlendIndependentBlend;
    VkBool32           advancedBlendNonPremultipliedSrcColor;
    VkBool32           advancedBlendNonPremultipliedDstColor;
    VkBool32           advancedBlendCorrelatedOverlap;
    VkBool32           advancedBlendAllOperations;
} VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `advancedBlendMaxColorAttachments` is one greater than the highest color attachment index that **can** be used in a render pass instance, for a pipeline that uses an advanced blend operation.
- `advancedBlendIndependentBlend` specifies whether advanced blend operations **can** vary per-attachment.
- `advancedBlendNonPremultipliedSrcColor` specifies whether the source color **can** be treated as non-premultiplied. If this is VK_FALSE, then VkPipelineColorBlendAdvancedStateCreateInfoEXT::`srcPremultiplied` **must** be VK_TRUE.
- `advancedBlendNonPremultipliedDstColor` specifies whether the destination color **can** be treated as non-premultiplied. If this is VK_FALSE, then VkPipelineColorBlendAdvancedStateCreateInfoEXT::`dstPremultiplied` **must** be VK_TRUE.
- `advancedBlendCorrelatedOverlap` specifies whether the overlap mode **can** be treated as correlated. If this is VK_FALSE, then VkPipelineColorBlendAdvancedStateCreateInfoEXT::`blendOverlap` **must** be VK_BLEND_OVERLAP_UNCORRELATED_EXT.
- `advancedBlendAllOperations` specifies whether all advanced blend operation enums are supported. See the valid usage of VkPipelineColorBlendAttachmentState.

If the `VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BLEND_OPERATION_ADVANCED_PROPERTIES_EXT`

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceVertexAttributeDivisorProperties` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceVertexAttributeDivisorProperties {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxVertexAttribDivisor;
    VkBool32           supportsNonZeroFirstInstance;
} VkPhysicalDeviceVertexAttributeDivisorProperties;
```

```
// Provided by VK_KHR_vertex_attribute_divisor
// Equivalent to VkPhysicalDeviceVertexAttributeDivisorProperties
typedef VkPhysicalDeviceVertexAttributeDivisorProperties
VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxVertexAttribDivisor` is the maximum value of the number of instances that will repeat the value of vertex attribute data when instanced rendering is enabled.
- `supportsNonZeroFirstInstance` specifies whether a non-zero value for the `firstInstance` parameter of drawing commands is supported when VkVertexInputBindingDivisorDescription ::`divisor` is not `1`.

If the `VkPhysicalDeviceVertexAttributeDivisorProperties` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVertexAttributeDivisorProperties-sType-sType

sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceSamplerFilterMinmaxProperties structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceSamplerFilterMinmaxProperties {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           filterMinmaxSingleComponentFormats;
    VkBool32           filterMinmaxImageComponentMapping;
} VkPhysicalDeviceSamplerFilterMinmaxProperties;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- filterMinmaxSingleComponentFormats is a boolean value indicating whether a minimum set of required formats support min/max filtering.
- filterMinmaxImageComponentMapping is a boolean value indicating whether the implementation supports non-identity component mapping of the image when doing min/max filtering.

If the VkPhysicalDeviceSamplerFilterMinmaxProperties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

If filterMinmaxSingleComponentFormats is VK_TRUE, the following formats **must** support the VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT feature with VK_IMAGE_TILING_OPTIMAL, if they support VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT:

- VK_FORMAT_R8_UNORM
- VK_FORMAT_R8_SNORM
- VK_FORMAT_R16_UNORM
- VK_FORMAT_R16_SNORM
- VK_FORMAT_R16_SFLOAT
- VK_FORMAT_R32_SFLOAT
- VK_FORMAT_D16_UNORM
- VK_FORMAT_X8_D24_UNORM_PACK32

- VK_FORMAT_D32_SFLOAT
- VK_FORMAT_D16_UNORM_S8_UINT
- VK_FORMAT_D24_UNORM_S8_UINT
- VK_FORMAT_D32_SFLOAT_S8_UINT

If the format is a depth/stencil format, this bit only specifies that the depth aspect (not the stencil aspect) of an image of this format supports min/max filtering, and that min/max filtering of the depth aspect is supported when depth compare is disabled in the sampler.

If `filterMinmaxImageComponentMapping` is VK_FALSE the component mapping of the image view used with min/max filtering **must** have been created with the `r` component set to the identity swizzle. Only the `r` component of the sampled image value is defined; reading other component values results in poison. If `filterMinmaxImageComponentMapping` is VK_TRUE this restriction does not apply and image component mapping works as normal.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSamplerFilterMinmaxProperties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLER_FILTER_MINMAX_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceProtectedMemoryProperties` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceProtectedMemoryProperties {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           protectedNoFault;
} VkPhysicalDeviceProtectedMemoryProperties;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `protectedNoFault` specifies how an implementation behaves when an application attempts to write to unprotected memory in a protected queue operation, read from protected memory in an unprotected queue operation, or perform a query in a protected queue operation. If this limit is VK_TRUE, such writes will be discarded or have undefined values written; reads and queries will return poison. If this limit is VK_FALSE, applications **must** not perform these operations. See Protected Memory Access Rules for more information.

If the VkPhysicalDeviceProtectedMemoryProperties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceProtectedMemoryProperties-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROTECTED_MEMORY_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceMaintenance3Properties structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceMaintenance3Properties {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxPerSetDescriptors;
    VkDeviceSize       maxMemoryAllocationSize;
} VkPhysicalDeviceMaintenance3Properties;
```

```
// Provided by VK_KHR_maintenance3
// Equivalent to VkPhysicalDeviceMaintenance3Properties
typedef VkPhysicalDeviceMaintenance3Properties
VkPhysicalDeviceMaintenance3PropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- maxPerSetDescriptors is a maximum number of descriptors (summed over all descriptor types) in a single descriptor set that is guaranteed to satisfy any implementation-dependent constraints on the size of a descriptor set itself. Applications **can** query whether a descriptor set that goes beyond this limit is supported using vkGetDescriptorSetLayoutSupport.
- maxMemoryAllocationSize is the maximum size of a memory allocation that **can** be created, even if there is more space available in the heap. If VkMemoryAllocateInfo::allocationSize is larger the error VK_ERROR_OUT_OF_DEVICE_MEMORY **may** be returned.

If the VkPhysicalDeviceMaintenance3Properties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance3Properties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_3_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceMaintenance4Properties` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceMaintenance4Properties {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceSize       maxBufferSize;
} VkPhysicalDeviceMaintenance4Properties;
```

```
// Provided by VK_KHR_maintenance4
// Equivalent to VkPhysicalDeviceMaintenance4Properties
typedef VkPhysicalDeviceMaintenance4Properties
VkPhysicalDeviceMaintenance4PropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxBufferSize` is the maximum size of a `VkBuffer` that **can** be created.

If the `VkPhysicalDeviceMaintenance4Properties` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance4Properties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceMaintenance5Properties structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceMaintenance5Properties {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           earlyFragmentMultisampleCoverageAfterSampleCounting;
    VkBool32           earlyFragmentSampleMaskTestBeforeSampleCounting;
    VkBool32           depthStencilSwizzleOneSupport;
    VkBool32           polygonModePointSize;
    VkBool32           nonStrictSinglePixelWideLinesUseParallelogram;
    VkBool32           nonStrictWideLinesUseParallelogram;
} VkPhysicalDeviceMaintenance5Properties;
```

```
// Provided by VK_KHR_maintenance5
// Equivalent to VkPhysicalDeviceMaintenance5Properties
typedef VkPhysicalDeviceMaintenance5Properties
VkPhysicalDeviceMaintenance5PropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- earlyFragmentMultisampleCoverageAfterSampleCounting is a boolean value indicating whether the fragment shading and multisample coverage operations are performed after sample counting for fragment shaders with EarlyFragmentTests execution mode.
- earlyFragmentSampleMaskTestBeforeSampleCounting is a boolean value indicating whether the sample mask test operation is performed before sample counting for fragment shaders using the EarlyFragmentTests execution mode.
- depthStencilSwizzleOneSupport is a boolean indicating that depth/stencil texturing operations with VK_COMPONENT_SWIZZLE_ONE have defined behavior.
- polygonModePointSize is a boolean value indicating whether the point size of the final rasterization of polygons with VK_POLYGON_MODE_POINT is controlled by PointSize.
- nonStrictSinglePixelWideLinesUseParallelogram is a boolean value indicating whether non-strict lines with a width of 1.0 are rasterized as parallelograms or using Bresenham's algorithm.
- nonStrictWideLinesUseParallelogram is a boolean value indicating whether non-strict lines with a width greater than 1.0 are rasterized as parallelograms or using Bresenham's algorithm.

If the VkPhysicalDeviceMaintenance5Properties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance5Properties-sType-sType

sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceMaintenance6Properties structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceMaintenance6Properties {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           blockTexelViewCompatibleMultipleLayers;
    uint32_t           maxCombinedImageSamplerDescriptorCount;
    VkBool32           fragmentShadingRateClampCombinerInputs;
} VkPhysicalDeviceMaintenance6Properties;
```

```
// Provided by VK_KHR_maintenance6
// Equivalent to VkPhysicalDeviceMaintenance6Properties
typedef VkPhysicalDeviceMaintenance6Properties
VkPhysicalDeviceMaintenance6PropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- blockTexelViewCompatibleMultipleLayers is a boolean value indicating that an implementation supports creating image views with VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT where the layerCount member of subresourceRange is greater than 1.
- maxCombinedImageSamplerDescriptorCount is the maximum number of descriptors that the implementation uses for sampled images with $Y'C_BC_R$ conversion.
- fragmentShadingRateClampCombinerInputs is a boolean value indicating that an implementation clamps the inputs to combiner operations.

If the VkPhysicalDeviceMaintenance6Properties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance6Properties-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceMaintenance7PropertiesKHR structure is defined as:

```
// Provided by VK_KHR_maintenance7
typedef struct VkPhysicalDeviceMaintenance7PropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           robustFragmentShadingRateAttachmentAccess;
    VkBool32           separateDepthStencilAttachmentAccess;
    uint32_t           maxDescriptorSetTotalUniformBuffersDynamic;
    uint32_t           maxDescriptorSetTotalStorageBuffersDynamic;
    uint32_t           maxDescriptorSetTotalBuffersDynamic;
    uint32_t           maxDescriptorSetUpdateAfterBindTotalUniformBuffersDynamic;
    uint32_t           maxDescriptorSetUpdateAfterBindTotalStorageBuffersDynamic;
    uint32_t           maxDescriptorSetUpdateAfterBindTotalBuffersDynamic;
} VkPhysicalDeviceMaintenance7PropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- robustFragmentShadingRateAttachmentAccess indicates whether the scaled size of a fragment shading rate attachment **can** be less than the size of the render area. If robustFragmentShadingRateAttachmentAccess is VK_FALSE, the size of the attachment multiplied by the texel size **must** be greater than or equal to the size of the render area. If it is VK_TRUE and the fragment shading rate attachment was created with VkImageSubresourceRange ::baseMipLevel equal to 0, the scaled size **can** be smaller than the render area, with missing shading rates defined by out of bounds behavior.
- separateDepthStencilAttachmentAccess indicates support for writing to one aspect of a depth/stencil attachment without performing read-modify-write operations on the other aspect. If this property is VK_TRUE, writes to one aspect **must** not result in read-modify-write operations on the other aspect. If VK_FALSE, writes to one aspect **may** result in writes to the other aspect as defined by render pass load operations, render pass store operations and render pass resolve operations.
- maxDescriptorSetTotalUniformBuffersDynamic is the maximum total count of dynamic uniform buffers that **can** be included in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. See Dynamic Uniform Buffer.
- maxDescriptorSetTotalStorageBuffersDynamic is the maximum total count of dynamic storage buffers that **can** be included in a pipeline layout. Descriptors with a type of

VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit. See Dynamic Storage Buffer.

- maxDescriptorSetTotalBuffersDynamic is the maximum total count of dynamic uniform buffers and storage buffers that **can** be included in a pipeline layout. Descriptors with a type of VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC count against this limit. Only descriptors in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit.
- maxDescriptorSetUpdateAfterBindTotalUniformBuffersDynamic is similar to maxDescriptorSetTotalUniformBuffersDynamic but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set. While an application **can** allocate dynamic uniform buffer descriptors from a pool created with VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT, and **can** include dynamic uniform buffer descriptors in descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT, bindings for these descriptors **must** not be present in any descriptor set layout that includes bindings created with VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT. Therefore this limit has no practical effect and **may** be zero.
- maxDescriptorSetUpdateAfterBindTotalStorageBuffersDynamic is similar to maxDescriptorSetTotalStorageBuffersDynamic but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set. While an application **can** allocate dynamic storage buffer descriptors from a pool created with VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT, and **can** include dynamic storage buffer descriptors in descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT, bindings for these descriptors **must** not be present in any descriptor set layout that includes bindings created with VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT. Therefore this limit has no practical effect and **may** be zero.
- maxDescriptorSetUpdateAfterBindTotalBuffersDynamic is similar to maxDescriptorSetTotalBuffersDynamic but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set. While an application **can** allocate dynamic uniform/storage buffer descriptors from a pool created with VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT, and **can** include dynamic uniform/storage buffer descriptors in descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT, bindings for these descriptors **must** not be present in any descriptor set layout that includes bindings created with VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT. Therefore this limit has no practical effect and **may** be zero.

If the VkPhysicalDeviceMaintenance7PropertiesKHR structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance7PropertiesKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_7_PROPERTIES_KHR`

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceLayeredApiPropertiesListKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance7
typedef struct VkPhysicalDeviceLayeredApiPropertiesListKHR {
    VkStructureType                               sType;
    void*                                         pNext;
    uint32_t                                      layeredApiCount;
    VkPhysicalDeviceLayeredApiPropertiesKHR*      pLayeredApis;
} VkPhysicalDeviceLayeredApiPropertiesListKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `layeredApiCount` is an integer related to the number of layered implementations underneath the Vulkan physical device, as described below.
- `pLayeredApis` is a pointer to an array of VkPhysicalDeviceLayeredApiPropertiesKHR in which information regarding the layered implementations underneath the Vulkan physical device is returned.

If `pLayeredApis` is `NULL`, then the number of layered implementations that are underneath the top-most Vulkan physical device (i.e. the one returned by vkGetPhysicalDeviceProperties2) is returned in `layeredApiCount`. Otherwise, `layeredApiCount` **must** be set by the application to the number of elements in the `pLayeredApis` array, and on return is overwritten with the number of values actually written to `pLayeredApis`. If the value of `layeredApiCount` is less than the number of layered implementations underneath the Vulkan physical device, at most `layeredApiCount` values will be written to `pLayeredApis`. An implementation that is not a layer will return 0 in `layeredApiCount`.

In the presence of multiple layered implementations, each element of `pLayeredApis` corresponds to an API implementation that is implemented on top of the API at the previous index. If there are layered implementations underneath a non-Vulkan implementation, they may not be visible in this query as the corresponding APIs may lack such a query.

If the `VkPhysicalDeviceLayeredApiPropertiesListKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with

each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceLayeredApiPropertiesListKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_PROPERTIES_LIST_KHR`
- VUID-VkPhysicalDeviceLayeredApiPropertiesListKHR-pLayeredApis-parameter
  If `layeredApiCount` is not `0`, and `pLayeredApis` is not `NULL`, `pLayeredApis` **must** be a valid pointer to an array of `layeredApiCount` VkPhysicalDeviceLayeredApiPropertiesKHR structures

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceLayeredApiPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance7
typedef struct VkPhysicalDeviceLayeredApiPropertiesKHR {
    VkStructureType                 sType;
    void*                           pNext;
    uint32_t                        vendorID;
    uint32_t                        deviceID;
    VkPhysicalDeviceLayeredApiKHR   layeredAPI;
    char                            deviceName[VK_MAX_PHYSICAL_DEVICE_NAME_SIZE];
} VkPhysicalDeviceLayeredApiPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `vendorID` is a unique identifier for the vendor of the layered implementation.
- `deviceID` is a unique identifier for the layered implementation among devices available from the vendor.
- `layeredAPI` is a VkPhysicalDeviceLayeredApiKHR specifying the API implemented by the layered implementation.
- `deviceName` is an array of `VK_MAX_PHYSICAL_DEVICE_NAME_SIZE` `char` containing a null-terminated UTF-8 string which is the name of the device.

If `layeredAPI` is `VK_PHYSICAL_DEVICE_LAYERED_API_VULKAN_KHR`, additional Vulkan-specific information can be queried by including the VkPhysicalDeviceLayeredApiVulkanPropertiesKHR structure in the `pNext` chain. Otherwise if such a structure is included in the `pNext` chain, it is ignored.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceLayeredApiPropertiesKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_PROPERTIES_KHR`
- VUID-VkPhysicalDeviceLayeredApiPropertiesKHR-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of `VkPhysicalDeviceLayeredApiVulkanPropertiesKHR`
- VUID-VkPhysicalDeviceLayeredApiPropertiesKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

The list of possible API implementations of a layered implementation underneath the Vulkan physical device, as returned in `VkPhysicalDeviceLayeredApiPropertiesKHR`::`layeredAPI`, are:

```
// Provided by VK_KHR_maintenance7
typedef enum VkPhysicalDeviceLayeredApiKHR {
    VK_PHYSICAL_DEVICE_LAYERED_API_VULKAN_KHR = 0,
    VK_PHYSICAL_DEVICE_LAYERED_API_D3D12_KHR = 1,
    VK_PHYSICAL_DEVICE_LAYERED_API_METAL_KHR = 2,
    VK_PHYSICAL_DEVICE_LAYERED_API_OPENGL_KHR = 3,
    VK_PHYSICAL_DEVICE_LAYERED_API_OPENGLES_KHR = 4,
} VkPhysicalDeviceLayeredApiKHR;
```

- `VK_PHYSICAL_DEVICE_LAYERED_API_VULKAN_KHR` - the device implements the Vulkan API.
- `VK_PHYSICAL_DEVICE_LAYERED_API_D3D12_KHR` - the device implements the D3D12 API.
- `VK_PHYSICAL_DEVICE_LAYERED_API_METAL_KHR` - the device implements the Metal API.
- `VK_PHYSICAL_DEVICE_LAYERED_API_OPENGL_KHR` - the device implements the OpenGL API.
- `VK_PHYSICAL_DEVICE_LAYERED_API_OPENGLES_KHR` - the device implements the OpenGL ES API.

The `VkPhysicalDeviceLayeredApiVulkanPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance7
typedef struct VkPhysicalDeviceLayeredApiVulkanPropertiesKHR {
    VkStructureType                sType;
    void*                          pNext;
    VkPhysicalDeviceProperties2    properties;
} VkPhysicalDeviceLayeredApiVulkanPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `properties` is a `VkPhysicalDeviceProperties2` in which properties of the underlying layered Vulkan implementation are returned.

The implementation **must** zero-fill the contents of `properties.properties.limits` and `properties.properties.sparseProperties`.

Valid Usage

- VUID-VkPhysicalDeviceLayeredApiVulkanPropertiesKHR-pNext-10011
  Only VkPhysicalDeviceDriverProperties and VkPhysicalDeviceIDProperties are allowed in the `pNext` chain of `properties`

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceLayeredApiVulkanPropertiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_VULKAN_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkPhysicalDeviceLayeredApiPropertiesKHR

The `VkPhysicalDeviceMaintenance9PropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance9
typedef struct VkPhysicalDeviceMaintenance9PropertiesKHR {
    VkStructureType                    sType;
    void*                              pNext;
    VkBool32                           image2DViewOf3DSparse;
    VkDefaultVertexAttributeValueKHR   defaultVertexAttributeValue;
} VkPhysicalDeviceMaintenance9PropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- If the `image2DViewOf3D` feature is enabled, `image2DViewOf3DSparse` indicates whether the implementation supports binding a slice of a sparse 3D image to a 2D image view.
- `defaultVertexAttributeValue` is a VkDefaultVertexAttributeValueKHR that indicates what value the implementation will return when the vertex shader reads unbound vertex attributes. Unbound attributes are those that have no corresponding VkVertexInputAttributeDescription ::`location` defined in the bound graphics pipeline or no corresponding VkVertexInputAttributeDescription2EXT::`location` set by the most recent call to vkCmdSetVertexInputEXT when the state is dynamic .

If the `VkPhysicalDeviceMaintenance9PropertiesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with

each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance9PropertiesKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_9_PROPERTIES_KHR`

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The possible values returned by the implementation when the vertex shader reads an unbound vertex attribute are:

```
// Provided by VK_KHR_maintenance9
typedef enum VkDefaultVertexAttributeValueKHR {
    VK_DEFAULT_VERTEX_ATTRIBUTE_VALUE_ZERO_ZERO_ZERO_ZERO_KHR = 0,
    VK_DEFAULT_VERTEX_ATTRIBUTE_VALUE_ZERO_ZERO_ZERO_ONE_KHR = 1,
} VkDefaultVertexAttributeValueKHR;
```

- VK_DEFAULT_VERTEX_ATTRIBUTE_VALUE_ZERO_ZERO_ZERO_ZERO_KHR - the value read for an unbound vertex attribute is (0,0,0,0).
- VK_DEFAULT_VERTEX_ATTRIBUTE_VALUE_ZERO_ZERO_ZERO_ONE_KHR - the value read for an unbound vertex attribute is (0,0,0,1).

The `VkPhysicalDeviceMaintenance10PropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance10
typedef struct VkPhysicalDeviceMaintenance10PropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           rgba4OpaqueBlackSwizzled;
    VkBool32           resolveSrgbFormatAppliesTransferFunction;
    VkBool32           resolveSrgbFormatSupportsTransferFunctionControl;
} VkPhysicalDeviceMaintenance10PropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `rgba4OpaqueBlackSwizzled` indicates whether correct swizzling is applied to the opaque black border color when using either the VK_FORMAT_B4G4R4A4_UNORM_PACK16 or VK_FORMAT_R4G4B4A4_UNORM_PACK16 format. If it is VK_TRUE, the implementation will correctly produce an opaque black border color with these formats. If it is VK_FALSE, the

implementation **may** swap the first channel with the alpha channel for the border color when sampling.

- `resolveSrgbFormatAppliesTransferFunction` indicates whether resolving a multi-sampled sRGB format to single-sampled sRGB by a weighted average converts the samples to linear before averaging. This applies to both attachment resolves in a render pass and standalone resolve commands. If VK_TRUE, implementation always converts to linear before averaging unless overridden. If VK_FALSE, implementation never converts to linear before averaging unless overridden.
- `resolveSrgbFormatSupportsTransferFunctionControl` indicates whether the implementation supports overriding the default behavior in `resolveSrgbFormatAppliesTransferFunction` in render passes and vkCmdResolveImage2.

Implementations supporting `maintenance10` **should** set `resolveSrgbFormatAppliesTransferFunction` to VK_TRUE.

If the `VkPhysicalDeviceMaintenance10PropertiesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance10PropertiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_10_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceMeshShaderPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_mesh_shader
typedef struct VkPhysicalDeviceMeshShaderPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxTaskWorkGroupTotalCount;
    uint32_t           maxTaskWorkGroupCount[3];
    uint32_t           maxTaskWorkGroupInvocations;
    uint32_t           maxTaskWorkGroupSize[3];
    uint32_t           maxTaskPayloadSize;
    uint32_t           maxTaskSharedMemorySize;
    uint32_t           maxTaskPayloadAndSharedMemorySize;
    uint32_t           maxMeshWorkGroupTotalCount;
    uint32_t           maxMeshWorkGroupCount[3];
    uint32_t           maxMeshWorkGroupInvocations;
    uint32_t           maxMeshWorkGroupSize[3];
    uint32_t           maxMeshSharedMemorySize;
    uint32_t           maxMeshPayloadAndSharedMemorySize;
    uint32_t           maxMeshOutputMemorySize;
    uint32_t           maxMeshPayloadAndOutputMemorySize;
    uint32_t           maxMeshOutputComponents;
    uint32_t           maxMeshOutputVertices;
    uint32_t           maxMeshOutputPrimitives;
    uint32_t           maxMeshOutputLayers;
    uint32_t           maxMeshMultiviewViewCount;
    uint32_t           meshOutputPerVertexGranularity;
    uint32_t           meshOutputPerPrimitiveGranularity;
    uint32_t           maxPreferredTaskWorkGroupInvocations;
    uint32_t           maxPreferredMeshWorkGroupInvocations;
    VkBool32           prefersLocalInvocationVertexOutput;
    VkBool32           prefersLocalInvocationPrimitiveOutput;
    VkBool32           prefersCompactVertexOutput;
    VkBool32           prefersCompactPrimitiveOutput;
} VkPhysicalDeviceMeshShaderPropertiesEXT;
```

The members of the `VkPhysicalDeviceMeshShaderPropertiesEXT` structure describe the following implementation-dependent limits:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxTaskWorkGroupTotalCount` is the maximum number of local workgroups that **can** be launched for a task shader by a mesh tasks drawing command.
- `maxTaskWorkGroupCount`[3] is the maximum number of local workgroups that **can** be launched for a task shader in each dimension by a mesh tasks drawing command. These three values represent the maximum number of local workgroups for the X, Y, and Z dimensions, respectively. The workgroup count parameters to the drawing commands **must** be less than or equal to the corresponding limit. The product of these dimensions **must** be less than or equal to `maxTaskWorkGroupTotalCount`.

- maxTaskWorkGroupInvocations is the maximum total number of task shader invocations in a single local workgroup. The product of the X, Y, and Z sizes, as specified by the LocalSize or LocalSizeId execution mode in shader modules or by the object decorated by the WorkgroupSize decoration, **must** be less than or equal to this limit.
- maxTaskWorkGroupSize[3] is the maximum size of a local workgroup for a task shader in each dimension. These three values represent the maximum local workgroup size in the X, Y, and Z dimensions, respectively. The x, y, and z sizes, as specified by the LocalSize or LocalSizeId execution mode or by the object decorated by the WorkgroupSize decoration in shader modules, **must** be less than or equal to the corresponding limit.
- maxTaskPayloadSize is the maximum total storage size, in bytes, available for variables declared with the TaskPayloadWorkgroupEXT storage class in shader modules in the task shader stage.
- maxTaskSharedMemorySize is the maximum total storage size, in bytes, available for variables declared with the Workgroup storage class in shader modules in the task shader stage.
- maxTaskPayloadAndSharedMemorySize is the maximum total storage size, in bytes, available for variables that are declared with the TaskPayloadWorkgroupEXT or Workgroup storage class, in shader modules in the task shader stage.
- maxMeshWorkGroupTotalCount is the maximum number of local workgroups that **can** be launched for a mesh shader, either directly by a mesh tasks drawing command, or emitted by a single task shader workgroup.
- maxMeshWorkGroupCount[3] is the maximum number of local workgroups that **can** be launched for a mesh shader in each dimension, either directly by a mesh tasks drawing command, or emitted by a single task shader workgroup. These three values represent the maximum number of local output tasks for the X, Y, and Z dimensions, respectively. The workgroup count parameters to the OpEmitMeshTasksEXT **must** be less than or equal to the corresponding limit. The product of these dimensions **must** be less than or equal to maxMeshWorkGroupTotalCount.
- maxMeshWorkGroupInvocations is the maximum total number of mesh shader invocations in a single local workgroup. The product of the X, Y, and Z sizes, as specified by the LocalSize or LocalSizeId execution mode in shader modules or by the object decorated by the WorkgroupSize decoration, **must** be less than or equal to this limit.
- maxMeshWorkGroupSize[3] is the maximum size of a local workgroup for a mesh shader in each dimension. These three values represent the maximum local workgroup size in the X, Y, and Z dimensions, respectively. The x, y, and z sizes, as specified by the LocalSize or LocalSizeId execution mode or by the object decorated by the WorkgroupSize decoration in shader modules, **must** be less than or equal to the corresponding limit.
- maxMeshSharedMemorySize is the maximum total storage size, in bytes, available for variables declared with the Workgroup storage class in shader modules in the mesh shader stage.
- maxMeshPayloadAndSharedMemorySize is the maximum total storage size, in bytes, available for variables that are declared with the TaskPayloadWorkgroupEXT or Workgroup storage class in shader modules in the mesh shader stage.
- maxMeshOutputMemorySize is the maximum total storage size, in bytes, available for output variables in shader modules in the mesh shader stage, according to the formula in Mesh Shader Output.
- maxMeshPayloadAndOutputMemorySize is the maximum total storage size, in bytes, available for

variables that are declared with the TaskPayloadWorkgroupEXT storage class, or output variables in shader modules in the mesh shader stage, according to the formula in Mesh Shader Output.

- maxMeshOutputComponents is the maximum number of components of output variables which **can** be output from the mesh shader stage.
- maxMeshOutputVertices is the maximum number of vertices which **can** be emitted by a single mesh shader workgroup.
- maxMeshOutputPrimitives is the maximum number of primitives which **can** be emitted by a single mesh shader workgroup.
- maxMeshOutputLayers is one greater than the maximum layer index that **can** be output from the mesh shader stage.
- maxMeshMultiviewViewCount is one greater than the maximum view index that **can** be used by any mesh shader.
- meshOutputPerVertexGranularity is the granularity of vertex allocation. The number of output vertices allocated for the mesh shader stage is padded to a multiple of this number. The value can be used to calculate the required storage size for output variables in shader modules in the mesh shader stage, which **must** be less than or equal to maxMeshOutputMemorySize.
- meshOutputPerPrimitiveGranularity is the granularity of primitive allocation. The number of output primitives allocated for the mesh shader stage is padded to a multiple of this number. The value can be used to calculate the required storage size for output variables in shader modules in the mesh shader stage, which **must** be less than or equal to maxMeshOutputMemorySize.
- maxPreferredTaskWorkGroupInvocations is the maximum number of task shader invocations in a single workgroup that is preferred by the implementation for optimal performance. The value is guaranteed to be a multiple of a supported subgroup size for the task shader stage.
- maxPreferredMeshWorkGroupInvocations is the maximum number of mesh shader invocations in a single workgroup that is preferred by the implementation for optimal performance. The value is guaranteed to be a multiple of a supported subgroup size for the mesh shader stage.
- prefersLocalInvocationVertexOutput specifies whether writes to the vertex output array in a mesh shader yield best performance when the array index matches LocalInvocationIndex.
- prefersLocalInvocationPrimitiveOutput specifies whether writes to the primitive output array in a mesh shader yield best performance when the array index matches LocalInvocationIndex.
- prefersCompactVertexOutput specifies whether output vertices should be compacted after custom culling in the mesh shader for best performance, otherwise keeping the vertices at their original location may be better.
- prefersCompactPrimitiveOutput specifies whether output primitives should be compacted after custom culling in the mesh shader for best performance, otherwise the use of CullPrimitiveEXT may be better.

If the VkPhysicalDeviceMeshShaderPropertiesEXT structure is included in the pNext chain of VkPhysicalDeviceProperties2, it is filled with the implementation-dependent limits.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMeshShaderPropertiesEXT-sType-sType

sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MESH_SHADER_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceDescriptorIndexingProperties structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceDescriptorIndexingProperties {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxUpdateAfterBindDescriptorsInAllPools;
    VkBool32           shaderUniformBufferArrayNonUniformIndexingNative;
    VkBool32           shaderSampledImageArrayNonUniformIndexingNative;
    VkBool32           shaderStorageBufferArrayNonUniformIndexingNative;
    VkBool32           shaderStorageImageArrayNonUniformIndexingNative;
    VkBool32           shaderInputAttachmentArrayNonUniformIndexingNative;
    VkBool32           robustBufferAccessUpdateAfterBind;
    VkBool32           quadDivergentImplicitLod;
    uint32_t           maxPerStageDescriptorUpdateAfterBindSamplers;
    uint32_t           maxPerStageDescriptorUpdateAfterBindUniformBuffers;
    uint32_t           maxPerStageDescriptorUpdateAfterBindStorageBuffers;
    uint32_t           maxPerStageDescriptorUpdateAfterBindSampledImages;
    uint32_t           maxPerStageDescriptorUpdateAfterBindStorageImages;
    uint32_t           maxPerStageDescriptorUpdateAfterBindInputAttachments;
    uint32_t           maxPerStageUpdateAfterBindResources;
    uint32_t           maxDescriptorSetUpdateAfterBindSamplers;
    uint32_t           maxDescriptorSetUpdateAfterBindUniformBuffers;
    uint32_t           maxDescriptorSetUpdateAfterBindUniformBuffersDynamic;
    uint32_t           maxDescriptorSetUpdateAfterBindStorageBuffers;
    uint32_t           maxDescriptorSetUpdateAfterBindStorageBuffersDynamic;
    uint32_t           maxDescriptorSetUpdateAfterBindSampledImages;
    uint32_t           maxDescriptorSetUpdateAfterBindStorageImages;
    uint32_t           maxDescriptorSetUpdateAfterBindInputAttachments;
} VkPhysicalDeviceDescriptorIndexingProperties;
```

```
// Provided by VK_EXT_descriptor_indexing
// Equivalent to VkPhysicalDeviceDescriptorIndexingProperties
typedef VkPhysicalDeviceDescriptorIndexingProperties
VkPhysicalDeviceDescriptorIndexingPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxUpdateAfterBindDescriptorsInAllPools` is the maximum number of descriptors (summed over all descriptor types) that **can** be created across all pools that are created with the VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT bit set. Pool creation **may** fail when this limit is exceeded, or when the space this limit represents is unable to satisfy a pool creation due to fragmentation.
- `shaderUniformBufferArrayNonUniformIndexingNative` is a boolean value indicating whether uniform buffer descriptors natively support non-uniform indexing. If this is VK_FALSE, then a single dynamic instance of an instruction that non-uniformly indexes an array of uniform buffers **may** execute multiple times in order to access all the descriptors.
- `shaderSampledImageArrayNonUniformIndexingNative` is a boolean value indicating whether sampler and image descriptors natively support non-uniform indexing. If this is VK_FALSE, then a single dynamic instance of an instruction that non-uniformly indexes an array of samplers or images **may** execute multiple times in order to access all the descriptors.
- `shaderStorageBufferArrayNonUniformIndexingNative` is a boolean value indicating whether storage buffer descriptors natively support non-uniform indexing. If this is VK_FALSE, then a single dynamic instance of an instruction that non-uniformly indexes an array of storage buffers **may** execute multiple times in order to access all the descriptors.
- `shaderStorageImageArrayNonUniformIndexingNative` is a boolean value indicating whether storage image descriptors natively support non-uniform indexing. If this is VK_FALSE, then a single dynamic instance of an instruction that non-uniformly indexes an array of storage images **may** execute multiple times in order to access all the descriptors.
- `shaderInputAttachmentArrayNonUniformIndexingNative` is a boolean value indicating whether input attachment descriptors natively support non-uniform indexing. If this is VK_FALSE, then a single dynamic instance of an instruction that non-uniformly indexes an array of input attachments **may** execute multiple times in order to access all the descriptors.
- `robustBufferAccessUpdateAfterBind` is a boolean value indicating whether `robustBufferAccess` **can** be enabled on a device simultaneously with `descriptorBindingUniformBufferUpdateAfterBind`, `descriptorBindingStorageBufferUpdateAfterBind`, `descriptorBindingUniformTexelBufferUpdateAfterBind`, and/or `descriptorBindingStorageTexelBufferUpdateAfterBind`. If this is VK_FALSE, then either `robustBufferAccess` **must** be disabled or all of these update-after-bind features **must** be disabled. Similarly, if this property is VK_FALSE, robustness **must** not be enabled through the VkPipelineRobustnessCreateInfo mechanism.
- `quadDivergentImplicitLod` is a boolean value indicating whether implicit LOD calculations for image operations have well-defined results when the image and/or sampler objects used for the instruction are not uniform within a quad. See Derivative Image Operations.
- `maxPerStageDescriptorUpdateAfterBindSamplers` is similar to `maxPerStageDescriptorSamplers` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxPerStageDescriptorUpdateAfterBindUniformBuffers` is similar to `maxPerStageDescriptorUniformBuffers` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.

- `maxPerStageDescriptorUpdateAfterBindStorageBuffers` is similar to `maxPerStageDescriptorStorageBuffers` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxPerStageDescriptorUpdateAfterBindSampledImages` is similar to `maxPerStageDescriptorSampledImages` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxPerStageDescriptorUpdateAfterBindStorageImages` is similar to `maxPerStageDescriptorStorageImages` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxPerStageDescriptorUpdateAfterBindInputAttachments` is similar to `maxPerStageDescriptorInputAttachments` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxPerStageUpdateAfterBindResources` is similar to `maxPerStageResources` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxDescriptorSetUpdateAfterBindSamplers` is similar to `maxDescriptorSetSamplers` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxDescriptorSetUpdateAfterBindUniformBuffers` is similar to `maxDescriptorSetUniformBuffers` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxDescriptorSetUpdateAfterBindUniformBuffersDynamic` is similar to `maxDescriptorSetUniformBuffersDynamic` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set. While an application **can** allocate dynamic uniform buffer descriptors from a pool created with VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT, and **can** include dynamic uniform buffer descriptors in descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT, bindings for these descriptors **must** not be present in any descriptor set layout that includes bindings created with VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT. Therefore this limit has no practical effect and **may** be zero.
- `maxDescriptorSetUpdateAfterBindStorageBuffers` is similar to `maxDescriptorSetStorageBuffers` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxDescriptorSetUpdateAfterBindStorageBuffersDynamic` is similar to `maxDescriptorSetStorageBuffersDynamic` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set. While an application **can** allocate dynamic storage buffer descriptors from a pool created with VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT, and **can** include dynamic storage buffer descriptors in descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT, bindings for these descriptors **must** not be present in any descriptor set layout that includes bindings created with VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT. Therefore this limit has no practical effect and **may** be zero.

- `maxDescriptorSetUpdateAfterBindSampledImages` is similar to `maxDescriptorSetSampledImages` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxDescriptorSetUpdateAfterBindStorageImages` is similar to `maxDescriptorSetStorageImages` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxDescriptorSetUpdateAfterBindInputAttachments` is similar to `maxDescriptorSetInputAttachments` but counts descriptors from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set. While an application **can** allocate input attachment descriptors from a pool created with VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT, and **can** include input attachment descriptors in descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT, bindings for these descriptors **must** not be created with VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT. Therefore this limit only exists to allow implementations exposing a higher limit for input attachment descriptors in a pipeline layout including any descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT than in a pipeline layout not including any descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT and **may** be zero to indicate that only the `maxDescriptorSetInputAttachments` applies.

If the `VkPhysicalDeviceDescriptorIndexingProperties` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDescriptorIndexingProperties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_INDEXING_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceInlineUniformBlockProperties` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceInlineUniformBlockProperties {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxInlineUniformBlockSize;
    uint32_t           maxPerStageDescriptorInlineUniformBlocks;
    uint32_t           maxPerStageDescriptorUpdateAfterBindInlineUniformBlocks;
    uint32_t           maxDescriptorSetInlineUniformBlocks;
    uint32_t           maxDescriptorSetUpdateAfterBindInlineUniformBlocks;
} VkPhysicalDeviceInlineUniformBlockProperties;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxInlineUniformBlockSize` is the maximum size in bytes of an inline uniform block binding.
- `maxPerStageDescriptorInlineUniformBlocks` is the maximum number of inline uniform block bindings that **can** be accessible to a single shader stage in a pipeline layout. Descriptor bindings with a descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK count against this limit. Only descriptor bindings in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit.
- `maxPerStageDescriptorUpdateAfterBindInlineUniformBlocks` is similar to `maxPerStageDescriptorInlineUniformBlocks` but counts descriptor bindings from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- `maxDescriptorSetInlineUniformBlocks` is the maximum number of inline uniform block bindings that **can** be included in descriptor bindings in a pipeline layout across all pipeline shader stages and descriptor set numbers. Descriptor bindings with a descriptor type of VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK count against this limit. Only descriptor bindings in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit.
- `maxDescriptorSetUpdateAfterBindInlineUniformBlocks` is similar to `maxDescriptorSetInlineUniformBlocks` but counts descriptor bindings from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.

If the `VkPhysicalDeviceInlineUniformBlockProperties` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceInlineUniformBlockProperties-sType-sType
  `sType` **must** be

VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INLINE_UNIFORM_BLOCK_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceConservativeRasterizationPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_conservative_rasterization
typedef struct VkPhysicalDeviceConservativeRasterizationPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    float              primitiveOverestimationSize;
    float              maxExtraPrimitiveOverestimationSize;
    float              extraPrimitiveOverestimationSizeGranularity;
    VkBool32           primitiveUnderestimation;
    VkBool32           conservativePointAndLineRasterization;
    VkBool32           degenerateTrianglesRasterized;
    VkBool32           degenerateLinesRasterized;
    VkBool32           fullyCoveredFragmentShaderInputVariable;
    VkBool32           conservativeRasterizationPostDepthCoverage;
} VkPhysicalDeviceConservativeRasterizationPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- primitiveOverestimationSize is the size in pixels the generating primitive is increased at each of its edges during conservative rasterization overestimation mode. Even with a size of 0.0, conservative rasterization overestimation rules still apply and if any part of the pixel rectangle is covered by the generating primitive, fragments are generated for the entire pixel. However implementations **may** make the pixel coverage area even more conservative by increasing the size of the generating primitive.
- maxExtraPrimitiveOverestimationSize is the maximum size in pixels of extra overestimation the implementation supports in the pipeline state. A value of 0.0 means the implementation does not support any additional overestimation of the generating primitive during conservative rasterization. A value above 0.0 allows the application to further increase the size of the generating primitive during conservative rasterization overestimation.
- extraPrimitiveOverestimationSizeGranularity is the granularity of extra overestimation that can be specified in the pipeline state between 0.0 and maxExtraPrimitiveOverestimationSize inclusive. A value of 0.0 means the implementation can use the smallest representable non-zero value in the screen space pixel fixed-point grid.
- primitiveUnderestimation is VK_TRUE if the implementation supports the VK_CONSERVATIVE_RASTERIZATION_MODE_UNDERESTIMATE_EXT conservative rasterization mode in addition to VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT.

Otherwise the implementation only supports VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT.

- `conservativePointAndLineRasterization` is VK_TRUE if the implementation supports conservative rasterization of point and line primitives as well as triangle primitives. Otherwise the implementation only supports triangle primitives.
- `degenerateTrianglesRasterized` is VK_FALSE if the implementation culls primitives generated from triangles that become zero area after they are quantized to the fixed-point rasterization pixel grid. `degenerateTrianglesRasterized` is VK_TRUE if these primitives are not culled and the provoking vertex attributes and depth value are used for the fragments. The primitive area calculation is done on the primitive generated from the clipped triangle if applicable. Zero area primitives are backfacing and the application **can** enable backface culling if desired.
- `degenerateLinesRasterized` is VK_FALSE if the implementation culls lines that become zero length after they are quantized to the fixed-point rasterization pixel grid. `degenerateLinesRasterized` is VK_TRUE if zero length lines are not culled and the provoking vertex attributes and depth value are used for the fragments.
- `fullyCoveredFragmentShaderInputVariable` is VK_TRUE if the implementation supports the SPIR-V builtin fragment shader input variable `FullyCoveredEXT` specifying that conservative rasterization is enabled and the fragment area is fully covered by the generating primitive.
- `conservativeRasterizationPostDepthCoverage` is VK_TRUE if the implementation supports conservative rasterization with the `PostDepthCoverage` execution mode enabled. Otherwise the `PostDepthCoverage` execution mode **must** not be used when conservative rasterization is enabled.

If the `VkPhysicalDeviceConservativeRasterizationPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceConservativeRasterizationPropertiesEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CONSERVATIVE_RASTERIZATION_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceShaderSplitBarrierPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_shader_split_barrier
typedef struct VkPhysicalDeviceShaderSplitBarrierPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           splitBarrierReservedSharedMemory;
} VkPhysicalDeviceShaderSplitBarrierPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `splitBarrierReservedSharedMemory` is the number of bytes of shared memory reserved for the implementation when the module executes `OpControlBarrierArriveEXT` or `OpControlBarrierWaitEXT` instructions.

If the `VkPhysicalDeviceShaderSplitBarrierPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderSplitBarrierPropertiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SPLIT_BARRIER_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceFragmentDensityMapPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_fragment_density_map
typedef struct VkPhysicalDeviceFragmentDensityMapPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkExtent2D         minFragmentDensityTexelSize;
    VkExtent2D         maxFragmentDensityTexelSize;
    VkBool32           fragmentDensityInvocations;
} VkPhysicalDeviceFragmentDensityMapPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `minFragmentDensityTexelSize` is the minimum fragment density texel size.
- `maxFragmentDensityTexelSize` is the maximum fragment density texel size.

- `fragmentDensityInvocations` specifies whether the implementation **may** invoke additional fragment shader invocations for each covered sample.

If the `VkPhysicalDeviceFragmentDensityMapPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentDensityMapPropertiesEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceFragmentDensityMap2PropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_fragment_density_map2
typedef struct VkPhysicalDeviceFragmentDensityMap2PropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           subsampledLoads;
    VkBool32           subsampledCoarseReconstructionEarlyAccess;
    uint32_t           maxSubsampledArrayLayers;
    uint32_t           maxDescriptorSetSubsampledSamplers;
} VkPhysicalDeviceFragmentDensityMap2PropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `subsampledLoads` specifies if performing image data read with load operations on subsampled attachments will be resampled to the fragment density of the render pass.
- `subsampledCoarseReconstructionEarlyAccess` specifies if performing image data read with samplers created with `flags` containing VK_SAMPLER_CREATE_SUBSAMPLED_COARSE_RECONSTRUCTION_BIT_EXT in fragment shader will trigger additional reads during VK_PIPELINE_STAGE_VERTEX_SHADER_BIT.
- `maxSubsampledArrayLayers` is the maximum number of VkImageView array layers for usages supporting subsampled samplers.
- `maxDescriptorSetSubsampledSamplers` is the maximum number of subsampled samplers that **can** be included in a VkPipelineLayout.

If the `VkPhysicalDeviceFragmentDensityMap2PropertiesEXT` structure is included in the `pNext` chain of

the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentDensityMap2PropertiesEXT-sType-sType
  sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_2_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceFragmentDensityMapOffsetPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_fragment_density_map_offset
typedef struct VkPhysicalDeviceFragmentDensityMapOffsetPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkExtent2D         fragmentDensityOffsetGranularity;
} VkPhysicalDeviceFragmentDensityMapOffsetPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- fragmentDensityOffsetGranularity is the granularity for fragment density offsets.

If the VkPhysicalDeviceFragmentDensityMapOffsetPropertiesEXT structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentDensityMapOffsetPropertiesEXT-sType-sType
  sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_OFFSET_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceDepthStencilResolveProperties structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceDepthStencilResolveProperties {
    VkStructureType       sType;
    void*                 pNext;
    VkResolveModeFlags    supportedDepthResolveModes;
    VkResolveModeFlags    supportedStencilResolveModes;
    VkBool32              independentResolveNone;
    VkBool32              independentResolve;
} VkPhysicalDeviceDepthStencilResolveProperties;
```

```
// Provided by VK_KHR_depth_stencil_resolve
// Equivalent to VkPhysicalDeviceDepthStencilResolveProperties
typedef VkPhysicalDeviceDepthStencilResolveProperties
VkPhysicalDeviceDepthStencilResolvePropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- supportedDepthResolveModes is a bitmask of VkResolveModeFlagBits indicating the set of supported depth resolve modes. VK_RESOLVE_MODE_SAMPLE_ZERO_BIT **must** be included in the set but implementations **may** support additional modes.
- supportedStencilResolveModes is a bitmask of VkResolveModeFlagBits indicating the set of supported stencil resolve modes. VK_RESOLVE_MODE_SAMPLE_ZERO_BIT **must** be included in the set but implementations **may** support additional modes. VK_RESOLVE_MODE_AVERAGE_BIT **must** not be included in the set.
- independentResolveNone is VK_TRUE if the implementation supports setting the depth and stencil resolve modes to different values when one of those modes is VK_RESOLVE_MODE_NONE. Otherwise the implementation only supports setting both modes to the same value.
- independentResolve is VK_TRUE if the implementation supports all combinations of the supported depth and stencil resolve modes, including setting either depth or stencil resolve mode to VK_RESOLVE_MODE_NONE. An implementation that supports independentResolve **must** also support independentResolveNone.

If the VkPhysicalDeviceDepthStencilResolveProperties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDepthStencilResolveProperties-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_STENCIL_RESOLVE_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDevicePerformanceQueryPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_performance_query
typedef struct VkPhysicalDevicePerformanceQueryPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           allowCommandBufferQueryCopies;
} VkPhysicalDevicePerformanceQueryPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- allowCommandBufferQueryCopies is VK_TRUE if the performance query pools are allowed to be used with vkCmdCopyQueryPoolResults.

If the VkPhysicalDevicePerformanceQueryPropertiesKHR structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePerformanceQueryPropertiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PERFORMANCE_QUERY_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceMemoryDecompressionPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_memory_decompression
typedef struct VkPhysicalDeviceMemoryDecompressionPropertiesEXT {
    VkStructureType                       sType;
    void*                                 pNext;
    VkMemoryDecompressionMethodFlagsEXT   decompressionMethods;
    uint64_t                              maxDecompressionIndirectCount;
} VkPhysicalDeviceMemoryDecompressionPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `decompressionMethods` is a bitmask of VkMemoryDecompressionMethodFlagBitsEXT specifying memory decompression methods supported by the implementation.
- `maxDecompressionIndirectCount` specifies the maximum supported count value identified by either vkCmdDecompressMemoryIndirectCountEXT::`maxDecompressionCount` or the value specified in vkCmdDecompressMemoryIndirectCountEXT::`indirectCommandsCountAddress`.

If the `memoryDecompression` feature is supported, `decompressionMethods` **must** have at least one bit set.

If the `VkPhysicalDeviceMemoryDecompressionPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMemoryDecompressionPropertiesEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_DECOMPRESSION_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceTransformFeedbackPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_transform_feedback
typedef struct VkPhysicalDeviceTransformFeedbackPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxTransformFeedbackStreams;
    uint32_t           maxTransformFeedbackBuffers;
    VkDeviceSize       maxTransformFeedbackBufferSize;
    uint32_t           maxTransformFeedbackStreamDataSize;
    uint32_t           maxTransformFeedbackBufferDataSize;
    uint32_t           maxTransformFeedbackBufferDataStride;
    VkBool32           transformFeedbackQueries;
    VkBool32           transformFeedbackStreamsLinesTriangles;
    VkBool32           transformFeedbackRasterizationStreamSelect;
    VkBool32           transformFeedbackDraw;
} VkPhysicalDeviceTransformFeedbackPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxTransformFeedbackStreams` is the maximum number of vertex streams that can be output from geometry shaders declared with the `GeometryStreams` capability. If the implementation does not support `VkPhysicalDeviceTransformFeedbackFeaturesEXT::geometryStreams` then `maxTransformFeedbackStreams` **must** be 1.
- `maxTransformFeedbackBuffers` is the maximum number of transform feedback buffers that can be bound for capturing shader outputs from the last pre-rasterization shader stage.
- `maxTransformFeedbackBufferSize` is the maximum size that can be specified when binding a buffer for transform feedback in vkCmdBindTransformFeedbackBuffersEXT.
- `maxTransformFeedbackStreamDataSize` is the maximum amount of data in bytes for each vertex that is captured to one or more transform feedback buffers associated with a specific vertex stream.
- `maxTransformFeedbackBufferDataSize` is the maximum amount of data in bytes for each vertex that can be captured to a specific transform feedback buffer.
- `maxTransformFeedbackBufferDataStride` is the maximum stride between each capture of vertex data to the buffer.
- `transformFeedbackQueries` is VK_TRUE if the implementation supports the VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT query type. `transformFeedbackQueries` is VK_FALSE if queries of this type **cannot** be created.
- `transformFeedbackStreamsLinesTriangles` is VK_TRUE if the implementation supports the geometry shader `OpExecutionMode` of `OutputLineStrip` and `OutputTriangleStrip` in addition to `OutputPoints` when more than one vertex stream is output. If `transformFeedbackStreamsLinesTriangles` is VK_FALSE the implementation only supports an `OpExecutionMode` of `OutputPoints` when more than one vertex stream is output from the geometry shader.
- `transformFeedbackRasterizationStreamSelect` is VK_TRUE if the implementation supports the `GeometryStreams` SPIR-V capability and the application can use

VkPipelineRasterizationStateStreamCreateInfoEXT to modify which vertex stream output is used for rasterization. Otherwise vertex stream `0` **must** always be used for rasterization.

- `transformFeedbackDraw` is VK_TRUE if the implementation supports the vkCmdDrawIndirectByteCountEXT function otherwise the function **must** not be called.

If the `VkPhysicalDeviceTransformFeedbackPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceTransformFeedbackPropertiesEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TRANSFORM_FEEDBACK_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceCopyMemoryIndirectPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_copy_memory_indirect
typedef struct VkPhysicalDeviceCopyMemoryIndirectPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkQueueFlags       supportedQueues;
} VkPhysicalDeviceCopyMemoryIndirectPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `supportedQueues` is a bitmask of VkQueueFlagBits indicating the types of queues on which indirect copy commands are supported. If a queue family supports any of the bits set in `supportedQueues`, then it **must** support at least one indirect copy command.

If the `indirectMemoryCopy` or `indirectMemoryToImageCopy` feature is supported, `supportedQueues` **must** return at least one supported queue type.

If the `VkPhysicalDeviceCopyMemoryIndirectPropertiesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceCopyMemoryIndirectPropertiesKHR-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COPY_MEMORY_INDIRECT_PROPERTIES_KHR`

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceAccelerationStructurePropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkPhysicalDeviceAccelerationStructurePropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint64_t           maxGeometryCount;
    uint64_t           maxInstanceCount;
    uint64_t           maxPrimitiveCount;
    uint32_t           maxPerStageDescriptorAccelerationStructures;
    uint32_t           maxPerStageDescriptorUpdateAfterBindAccelerationStructures;
    uint32_t           maxDescriptorSetAccelerationStructures;
    uint32_t           maxDescriptorSetUpdateAfterBindAccelerationStructures;
    uint32_t           minAccelerationStructureScratchOffsetAlignment;
} VkPhysicalDeviceAccelerationStructurePropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxGeometryCount` is the maximum number of geometries in the bottom level acceleration structure.
- `maxInstanceCount` is the maximum number of instances in the top level acceleration structure.
- `maxPrimitiveCount` is the maximum number of triangles or AABBs in all geometries in the bottom level acceleration structure.
- `maxPerStageDescriptorAccelerationStructures` is the maximum number of acceleration structure bindings that **can** be accessible to a single shader stage in a pipeline layout. Descriptor bindings with a descriptor type of VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR count against this limit. Only descriptor bindings in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit.
- `maxPerStageDescriptorUpdateAfterBindAccelerationStructures` is similar to `maxPerStageDescriptorAccelerationStructures` but counts descriptor bindings from descriptor sets created with or without the

VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.

- maxDescriptorSetAccelerationStructures is the maximum number of acceleration structure descriptors that **can** be included in descriptor bindings in a pipeline layout across all pipeline shader stages and descriptor set numbers. Descriptor bindings with a descriptor type of VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR count against this limit. Only descriptor bindings in descriptor set layouts created without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set count against this limit.
- maxDescriptorSetUpdateAfterBindAccelerationStructures is similar to maxDescriptorSetAccelerationStructures but counts descriptor bindings from descriptor sets created with or without the VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT bit set.
- minAccelerationStructureScratchOffsetAlignment is the minimum **required** alignment, in bytes, for scratch data passed in to an acceleration structure build command. The value **must** be a power of two.

Due to the fact that the geometry, instance, and primitive counts are specified at acceleration structure creation as 32-bit values, maxGeometryCount, maxInstanceCount, and maxPrimitiveCount **must** not exceed 2^{32}-1.

If the VkPhysicalDeviceAccelerationStructurePropertiesKHR structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceAccelerationStructurePropertiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ACCELERATION_STRUCTURE_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceRayTracingPipelinePropertiesKHR structure is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef struct VkPhysicalDeviceRayTracingPipelinePropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           shaderGroupHandleSize;
    uint32_t           maxRayRecursionDepth;
    uint32_t           maxShaderGroupStride;
    uint32_t           shaderGroupBaseAlignment;
    uint32_t           shaderGroupHandleCaptureReplaySize;
    uint32_t           maxRayDispatchInvocationCount;
    uint32_t           shaderGroupHandleAlignment;
    uint32_t           maxRayHitAttributeSize;
} VkPhysicalDeviceRayTracingPipelinePropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderGroupHandleSize` is the size in bytes of the shader header.
- `maxRayRecursionDepth` is the maximum number of levels of ray recursion allowed in a trace command.
- `maxShaderGroupStride` is the maximum stride in bytes allowed between shader groups in the shader binding table.
- `shaderGroupBaseAlignment` is the **required** alignment in bytes for the base of the shader binding table.
- `shaderGroupHandleCaptureReplaySize` is the number of bytes for the information required to do capture and replay for shader group handles.
- `maxRayDispatchInvocationCount` is the maximum number of ray generation shader invocations which **may** be produced by a single vkCmdTraceRaysIndirectKHR or vkCmdTraceRaysKHR command.
- `shaderGroupHandleAlignment` is the **required** alignment in bytes for each entry in a shader binding table. The value **must** be a power of two.
- `maxRayHitAttributeSize` is the maximum size in bytes for a ray attribute structure.

If the `VkPhysicalDeviceRayTracingPipelinePropertiesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRayTracingPipelinePropertiesKHR-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_PIPELINE_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceCooperativeMatrixPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_cooperative_matrix
typedef struct VkPhysicalDeviceCooperativeMatrixPropertiesKHR {
    VkStructureType       sType;
    void*                 pNext;
    VkShaderStageFlags    cooperativeMatrixSupportedStages;
} VkPhysicalDeviceCooperativeMatrixPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- cooperativeMatrixSupportedStages is a bitfield of VkShaderStageFlagBits describing the shader stages that cooperative matrix instructions are supported in. cooperativeMatrixSupportedStages will have the VK_SHADER_STAGE_COMPUTE_BIT bit set if any of the physical device's queues support VK_QUEUE_COMPUTE_BIT.

cooperativeMatrixSupportedStages **must** not have any bits other than VK_SHADER_STAGE_COMPUTE_BIT set.

If the VkPhysicalDeviceCooperativeMatrixPropertiesKHR structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceCooperativeMatrixPropertiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceShaderLongVectorPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_shader_long_vector
typedef struct VkPhysicalDeviceShaderLongVectorPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxVectorComponents;
} VkPhysicalDeviceShaderLongVectorPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxVectorComponents` indicates the maximum number of components that **can** be in a vector type.

If the `VkPhysicalDeviceShaderLongVectorPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderLongVectorPropertiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_LONG_VECTOR_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceTexelBufferAlignmentProperties` structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceTexelBufferAlignmentProperties {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceSize       storageTexelBufferOffsetAlignmentBytes;
    VkBool32           storageTexelBufferOffsetSingleTexelAlignment;
    VkDeviceSize       uniformTexelBufferOffsetAlignmentBytes;
    VkBool32           uniformTexelBufferOffsetSingleTexelAlignment;
} VkPhysicalDeviceTexelBufferAlignmentProperties;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `storageTexelBufferOffsetAlignmentBytes` is a byte alignment that is sufficient for a storage texel buffer of any format. The value **must** be a power of two.

- `storageTexelBufferOffsetSingleTexelAlignment` indicates whether single texel alignment is sufficient for a storage texel buffer of any format.
- `uniformTexelBufferOffsetAlignmentBytes` is a byte alignment that is sufficient for a uniform texel buffer of any format. The value **must** be a power of two.
- `uniformTexelBufferOffsetSingleTexelAlignment` indicates whether single texel alignment is sufficient for a uniform texel buffer of any format.

If the `VkPhysicalDeviceTexelBufferAlignmentProperties` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

If the single texel alignment property is VK_FALSE, then the buffer view's offset **must** be aligned to the corresponding byte alignment value. If the single texel alignment property is VK_TRUE, then the buffer view's offset **must** be aligned to the lesser of the corresponding byte alignment value or the size of a single texel, based on VkBufferViewCreateInfo::`format`. If the size of a single texel is a multiple of three bytes, then the size of a single component of the format is used instead.

These limits **must** not advertise a larger alignment than the required maximum minimum value of VkPhysicalDeviceLimits::`minTexelBufferOffsetAlignment`, for any format that supports use as a texel buffer.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceTexelBufferAlignmentProperties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TEXEL_BUFFER_ALIGNMENT_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceTimelineSemaphoreProperties` structure is defined as:

```
// Provided by VK_VERSION_1_2
typedef struct VkPhysicalDeviceTimelineSemaphoreProperties {
    VkStructureType    sType;
    void*              pNext;
    uint64_t           maxTimelineSemaphoreValueDifference;
} VkPhysicalDeviceTimelineSemaphoreProperties;
```

```
// Provided by VK_KHR_timeline_semaphore
// Equivalent to VkPhysicalDeviceTimelineSemaphoreProperties
typedef VkPhysicalDeviceTimelineSemaphoreProperties
VkPhysicalDeviceTimelineSemaphorePropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxTimelineSemaphoreValueDifference` indicates the maximum difference allowed by the implementation between the current value of a timeline semaphore and any pending signal or wait operations.

If the `VkPhysicalDeviceTimelineSemaphoreProperties` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceTimelineSemaphoreProperties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceLineRasterizationProperties` structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceLineRasterizationProperties {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           lineSubPixelPrecisionBits;
} VkPhysicalDeviceLineRasterizationProperties;
```

```
// Provided by VK_KHR_line_rasterization
// Equivalent to VkPhysicalDeviceLineRasterizationProperties
typedef VkPhysicalDeviceLineRasterizationProperties
VkPhysicalDeviceLineRasterizationPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `lineSubPixelPrecisionBits` is the number of bits of subpixel precision in framebuffer coordinates x_f and y_f when rasterizing line segments.

If the `VkPhysicalDeviceLineRasterizationProperties` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceLineRasterizationProperties-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_PROPERTIES`

Structure Chaining

Extends the structure

- `VkPhysicalDeviceProperties2`

The `VkPhysicalDeviceRobustness2PropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_robustness2
typedef struct VkPhysicalDeviceRobustness2PropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceSize       robustStorageBufferAccessSizeAlignment;
    VkDeviceSize       robustUniformBufferAccessSizeAlignment;
} VkPhysicalDeviceRobustness2PropertiesKHR;
```

```
// Provided by VK_EXT_robustness2
// Equivalent to VkPhysicalDeviceRobustness2PropertiesKHR
typedef VkPhysicalDeviceRobustness2PropertiesKHR
VkPhysicalDeviceRobustness2PropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `robustStorageBufferAccessSizeAlignment` is the number of bytes that the range of a storage buffer descriptor is rounded up to when used for bounds-checking when the `robustBufferAccess2` feature is enabled. This value **must** be either 1 or 4.
- `robustUniformBufferAccessSizeAlignment` is the number of bytes that the range of a uniform buffer descriptor is rounded up to when used for bounds-checking when the `robustBufferAccess2` feature is enabled. This value **must** be a power of two in the range [1, 256].

If the `VkPhysicalDeviceRobustness2PropertiesKHR` structure is included in the `pNext` chain of the

VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRobustness2PropertiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT {
    VkStructureType                       sType;
    void*                                 pNext;
    uint32_t                              maxIndirectPipelineCount;
    uint32_t                              maxIndirectShaderObjectCount;
    uint32_t                              maxIndirectSequenceCount;
    uint32_t                              maxIndirectCommandsTokenCount;
    uint32_t                              maxIndirectCommandsTokenOffset;
    uint32_t                              maxIndirectCommandsIndirectStride;
    VkIndirectCommandsInputModeFlagsEXT   supportedIndirectCommandsInputModes;
    VkShaderStageFlags                    supportedIndirectCommandsShaderStages;
    VkShaderStageFlags
supportedIndirectCommandsShaderStagesPipelineBinding;
    VkShaderStageFlags
supportedIndirectCommandsShaderStagesShaderBinding;
    VkBool32                              deviceGeneratedCommandsTransformFeedback;
    VkBool32
deviceGeneratedCommandsMultiDrawIndirectCount;
} VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- maxIndirectPipelineCount is the maximum number of pipelines passed to vkCreateIndirectExecutionSetEXT.
- maxIndirectShaderObjectCount is the maximum number of shader objects passed to vkCreateIndirectExecutionSetEXT. If this value is zero, binding shader objects indirectly is not supported.

- `maxIndirectSequenceCount` is the maximum number of sequences in VkGeneratedCommandsInfoEXT and in VkGeneratedCommandsMemoryRequirementsInfoEXT.
- `maxIndirectCommandsTokenCount` is the maximum number of tokens in VkIndirectCommandsLayoutCreateInfoEXT.
- `maxIndirectCommandsTokenOffset` is the maximum offset in VkIndirectCommandsLayoutTokenEXT.
- `maxIndirectCommandsIndirectStride` is the maximum stream stride in VkIndirectCommandsLayoutCreateInfoEXT.
- `supportedIndirectCommandsInputModes` indicates the supported input modes.
- `supportedIndirectCommandsShaderStages` indicates the stages which **can** be used to generate indirect commands. Implementations are required to support, at minimum: VK_SHADER_STAGE_VERTEX_BIT, VK_SHADER_STAGE_FRAGMENT_BIT, VK_SHADER_STAGE_COMPUTE_BIT.
- `supportedIndirectCommandsShaderStagesPipelineBinding` indicates the stages which **can** be used within indirect execution sets for indirectly binding shader stages using pipelines.
- `supportedIndirectCommandsShaderStagesShaderBinding` indicates the stages which **can** be used within indirect execution sets for indirectly binding shader stages using shader objects.
- `deviceGeneratedCommandsTransformFeedback` indicates whether the implementation supports interactions with `VK_EXT_transform_feedback` for pipelines not created with VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT.
- `deviceGeneratedCommandsMultiDrawIndirectCount` indicates whether the implementation supports COUNT variants of multi-draw indirect tokens.

If the `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_GENERATED_COMMANDS_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDevicePortabilitySubsetPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_portability_subset
typedef struct VkPhysicalDevicePortabilitySubsetPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           minVertexInputBindingStrideAlignment;
} VkPhysicalDevicePortabilitySubsetPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `minVertexInputBindingStrideAlignment` indicates the minimum alignment for vertex input strides. VkVertexInputBindingDescription::`stride` **must** be a multiple of, and at least as large as, this value. The value **must** be a power of two.

If the `VkPhysicalDevicePortabilitySubsetPropertiesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePortabilitySubsetPropertiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PORTABILITY_SUBSET_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceFragmentShadingRatePropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_fragment_shading_rate
typedef struct VkPhysicalDeviceFragmentShadingRatePropertiesKHR {
    VkStructureType          sType;
    void*                    pNext;
    VkExtent2D               minFragmentShadingRateAttachmentTexelSize;
    VkExtent2D               maxFragmentShadingRateAttachmentTexelSize;
    uint32_t                 maxFragmentShadingRateAttachmentTexelSizeAspectRatio;
    VkBool32                 primitiveFragmentShadingRateWithMultipleViewports;
    VkBool32                 layeredShadingRateAttachments;
    VkBool32                 fragmentShadingRateNonTrivialCombinerOps;
    VkExtent2D               maxFragmentSize;
    uint32_t                 maxFragmentSizeAspectRatio;
    uint32_t                 maxFragmentShadingRateCoverageSamples;
    VkSampleCountFlagBits    maxFragmentShadingRateRasterizationSamples;
    VkBool32                 fragmentShadingRateWithShaderDepthStencilWrites;
    VkBool32                 fragmentShadingRateWithSampleMask;
    VkBool32                 fragmentShadingRateWithShaderSampleMask;
    VkBool32                 fragmentShadingRateWithConservativeRasterization;
    VkBool32                 fragmentShadingRateWithFragmentShaderInterlock;
    VkBool32                 fragmentShadingRateWithCustomSampleLocations;
    VkBool32                 fragmentShadingRateStrictMultiplyCombiner;
} VkPhysicalDeviceFragmentShadingRatePropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `minFragmentShadingRateAttachmentTexelSize` indicates minimum supported width and height of the portion of the framebuffer corresponding to each texel in a fragment shading rate attachment. Each value **must** be less than or equal to the values in `maxFragmentShadingRateAttachmentTexelSize`. Each value **must** be a power-of-two. It **must** be (0,0) if the `attachmentFragmentShadingRate` feature is not supported.
- `maxFragmentShadingRateAttachmentTexelSize` indicates maximum supported width and height of the portion of the framebuffer corresponding to each texel in a fragment shading rate attachment. Each value **must** be greater than or equal to the values in `minFragmentShadingRateAttachmentTexelSize`. Each value **must** be a power-of-two. It **must** be (0,0) if the `attachmentFragmentShadingRate` feature is not supported.
- `maxFragmentShadingRateAttachmentTexelSizeAspectRatio` indicates the maximum ratio between the width and height of the portion of the framebuffer corresponding to each texel in a fragment shading rate attachment. `maxFragmentShadingRateAttachmentTexelSizeAspectRatio` **must** be a power-of-two value, and **must** be less than or equal to max(`maxFragmentShadingRateAttachmentTexelSize.width` / `minFragmentShadingRateAttachmentTexelSize.height`, `maxFragmentShadingRateAttachmentTexelSize.height` / `minFragmentShadingRateAttachmentTexelSize.width`). It **must** be 0 if the `attachmentFragmentShadingRate` feature is not supported.
- `primitiveFragmentShadingRateWithMultipleViewports` specifies whether the primitive fragment shading rate **can** be used when multiple viewports are used. If this value is VK_FALSE, only a

single viewport **must** be used, and applications **must** not write to the ViewportIndex built-in when setting PrimitiveShadingRateKHR. It **must** be VK_FALSE if the shaderOutputViewportIndex feature, or the geometryShader feature is not supported, or if the primitiveFragmentShadingRate feature is not supported.

- layeredShadingRateAttachments specifies whether a shading rate attachment image view **can** be created with multiple layers. If this value is VK_FALSE, when creating an image view with a usage that includes VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR, layerCount **must** be 1. It **must** be VK_FALSE if the multiview feature, the shaderOutputViewportIndex feature, or the geometryShader feature is not supported, or if the attachmentFragmentShadingRate feature is not supported.
- fragmentShadingRateNonTrivialCombinerOps specifies whether VkFragmentShadingRateCombinerOpKHR enums other than VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR or VK_FRAGMENT_SHADING_RATE_COMBINER_OP_REPLACE_KHR **can** be used. It **must** be VK_FALSE unless either the primitiveFragmentShadingRate or attachmentFragmentShadingRate feature is supported.
- maxFragmentSize indicates the maximum supported width and height of a fragment. Its width and height members **must** both be power-of-two values. This limit is purely informational, and is not validated.
- maxFragmentSizeAspectRatio indicates the maximum ratio between the width and height of a fragment. maxFragmentSizeAspectRatio **must** be a power-of-two value, and **must** be less than or equal to the maximum of the width and height members of maxFragmentSize. This limit is purely informational, and is not validated.
- maxFragmentShadingRateCoverageSamples specifies the maximum number of coverage samples supported in a single fragment. maxFragmentShadingRateCoverageSamples **must** be less than or equal to the product of the width and height members of maxFragmentSize, and the sample count reported by maxFragmentShadingRateRasterizationSamples. maxFragmentShadingRateCoverageSamples **must** be less than or equal to maxSampleMaskWords × 32 if fragmentShadingRateWithShaderSampleMask is supported. This limit is purely informational, and is not validated.
- maxFragmentShadingRateRasterizationSamples is a VkSampleCountFlagBits value specifying the maximum sample rate supported when a fragment covers multiple pixels. This limit is purely informational, and is not validated.
- fragmentShadingRateWithShaderDepthStencilWrites specifies whether the implementation supports writing FragDepth or FragStencilRefEXT from a fragment shader for multi-pixel fragments. If this value is VK_FALSE, writing to those built-ins will clamp the fragment shading rate to (1,1).
- fragmentShadingRateWithSampleMask specifies whether the implementation supports setting valid bits of VkPipelineMultisampleStateCreateInfo::pSampleMask to 0 for multi-pixel fragments. If this value is VK_FALSE, zeroing valid bits in the sample mask will clamp the fragment shading rate to (1,1).
- fragmentShadingRateWithShaderSampleMask specifies whether the implementation supports reading or writing SampleMask for multi-pixel fragments. If this value is VK_FALSE, using that built-in will clamp the fragment shading rate to (1,1).

- `fragmentShadingRateWithConservativeRasterization` specifies whether conservative rasterization is supported for multi-pixel fragments. It **must** be VK_FALSE if `VK_EXT_conservative_rasterization` is not supported. If this value is VK_FALSE, using conservative rasterization will clamp the fragment shading rate to (1,1).
- `fragmentShadingRateWithFragmentShaderInterlock` specifies whether fragment shader interlock is supported for multi-pixel fragments. It **must** be VK_FALSE if `VK_EXT_fragment_shader_interlock` is not supported. If this value is VK_FALSE, using fragment shader interlock will clamp the fragment shading rate to (1,1).
- `fragmentShadingRateWithCustomSampleLocations` specifies whether custom sample locations are supported for multi-pixel fragments. It **must** be VK_FALSE if `VK_EXT_sample_locations` is not supported. If this value is VK_FALSE, using custom sample locations will clamp the fragment shading rate to (1,1).
- `fragmentShadingRateStrictMultiplyCombiner` specifies whether VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MUL_KHR accurately performs a multiplication or not. Implementations where this value is VK_FALSE will instead combine rates with an addition. If `fragmentShadingRateNonTrivialCombinerOps` is VK_FALSE, implementations **must** report this as VK_FALSE. If `fragmentShadingRateNonTrivialCombinerOps` is VK_TRUE, implementations **should** report this as VK_TRUE.

NOTE

Multiplication of the combiner rates using the fragment width/height in linear space is equivalent to an addition of those values in log2 space. Some implementations inadvertently implemented an addition in linear space due to unclear requirements originating outside of this specification. This resulted in `fragmentShadingRateStrictMultiplyCombiner` being added. Fortunately, this only affects situations where a rate of 1 in either dimension is combined with another rate of 1. All other combinations result in the exact same result as if multiplication was performed in linear space due to the clamping logic, and the fact that both the sum and product of 2 and 2 are equal. In many cases, this limit will not affect the correct operation of applications.

If the `VkPhysicalDeviceFragmentShadingRatePropertiesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

These properties are related to fragment shading rates.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentShadingRatePropertiesKHR-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceCustomBorderColorPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_custom_border_color
typedef struct VkPhysicalDeviceCustomBorderColorPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxCustomBorderColorSamplers;
} VkPhysicalDeviceCustomBorderColorPropertiesEXT;
```

- maxCustomBorderColorSamplers indicates the maximum number of samplers with custom border colors which **can** simultaneously exist on a device.

If the VkPhysicalDeviceCustomBorderColorPropertiesEXT structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceCustomBorderColorPropertiesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CUSTOM_BORDER_COLOR_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceProvokingVertexPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_provoking_vertex
typedef struct VkPhysicalDeviceProvokingVertexPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           provokingVertexModePerPipeline;
    VkBool32           transformFeedbackPreservesTriangleFanProvokingVertex;
} VkPhysicalDeviceProvokingVertexPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `provokingVertexModePerPipeline` indicates whether the implementation supports graphics pipelines with different provoking vertex modes within the same render pass instance.
- `transformFeedbackPreservesTriangleFanProvokingVertex` indicates whether the implementation can preserve the provoking vertex order when writing triangle fan vertices to transform feedback.

If the `VkPhysicalDeviceProvokingVertexPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceProvokingVertexPropertiesEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROVOKING_VERTEX_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceDescriptorBufferPropertiesEXT` structure is defined as:

WARNING	This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer
typedef struct VkPhysicalDeviceDescriptorBufferPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           combinedImageSamplerDescriptorSingleArray;
    VkBool32           bufferlessPushDescriptors;
    VkBool32           allowSamplerImageViewPostSubmitCreation;
    VkDeviceSize       descriptorBufferOffsetAlignment;
    uint32_t           maxDescriptorBufferBindings;
    uint32_t           maxResourceDescriptorBufferBindings;
    uint32_t           maxSamplerDescriptorBufferBindings;
    uint32_t           maxEmbeddedImmutableSamplerBindings;
    uint32_t           maxEmbeddedImmutableSamplers;
    size_t             bufferCaptureReplayDescriptorDataSize;
    size_t             imageCaptureReplayDescriptorDataSize;
    size_t             imageViewCaptureReplayDescriptorDataSize;
    size_t             samplerCaptureReplayDescriptorDataSize;
    size_t             accelerationStructureCaptureReplayDescriptorDataSize;
    size_t             samplerDescriptorSize;
    size_t             combinedImageSamplerDescriptorSize;
    size_t             sampledImageDescriptorSize;
    size_t             storageImageDescriptorSize;
    size_t             uniformTexelBufferDescriptorSize;
    size_t             robustUniformTexelBufferDescriptorSize;
    size_t             storageTexelBufferDescriptorSize;
    size_t             robustStorageTexelBufferDescriptorSize;
    size_t             uniformBufferDescriptorSize;
    size_t             robustUniformBufferDescriptorSize;
    size_t             storageBufferDescriptorSize;
    size_t             robustStorageBufferDescriptorSize;
    size_t             inputAttachmentDescriptorSize;
    size_t             accelerationStructureDescriptorSize;
    VkDeviceSize       maxSamplerDescriptorBufferRange;
    VkDeviceSize       maxResourceDescriptorBufferRange;
    VkDeviceSize       samplerDescriptorBufferAddressSpaceSize;
    VkDeviceSize       resourceDescriptorBufferAddressSpaceSize;
    VkDeviceSize       descriptorBufferAddressSpaceSize;
} VkPhysicalDeviceDescriptorBufferPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `combinedImageSamplerDescriptorSingleArray` indicates that the implementation does not require an array of VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER descriptors to be written into a descriptor buffer as an array of image descriptors, immediately followed by an array of sampler descriptors.
- `bufferlessPushDescriptors` indicates that the implementation does not require a buffer created with the VK_BUFFER_USAGE_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT usage flag set to be bound when using push descriptors.

- `allowSamplerImageViewPostSubmitCreation` indicates that the implementation does not restrict when the VkSampler or VkImageView objects used to retrieve descriptor data **can** be created in relation to command buffer submission. If this value is VK_FALSE, then the application **must** create any VkSampler or VkImageView objects whose descriptor data is accessed during the execution of a command buffer, before the vkQueueSubmit , or vkQueueSubmit2, call that submits that command buffer.
- `descriptorBufferOffsetAlignment` indicates the **required** alignment in bytes when setting offsets into the descriptor buffer.
- `maxDescriptorBufferBindings` indicates the maximum number of descriptor buffer bindings.
- `maxResourceDescriptorBufferBindings` indicates the maximum number of descriptor buffer bindings with VK_BUFFER_USAGE_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT that **can** be used.
- `maxSamplerDescriptorBufferBindings` indicates the maximum number of descriptor buffer bindings with VK_BUFFER_USAGE_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT that **can** be used.
- `maxEmbeddedImmutableSamplerBindings` indicates the maximum number of embedded immutable sampler sets that **can** be bound.
- `maxEmbeddedImmutableSamplers` indicates the maximum number of unique immutable samplers in descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_EMBEDDED_IMMUTABLE_SAMPLERS_BIT_EXT, and pipeline layouts created from them, which **can** simultaneously exist on a device.
- `bufferCaptureReplayDescriptorDataSize` indicates the maximum size in bytes of the opaque data used for capture and replay with buffers.
- `imageCaptureReplayDescriptorDataSize` indicates the maximum size in bytes of the opaque data used for capture and replay with images.
- `imageViewCaptureReplayDescriptorDataSize` indicates the maximum size in bytes of the opaque data used for capture and replay with image views.
- `samplerCaptureReplayDescriptorDataSize` indicates the maximum size in bytes of the opaque data used for capture and replay with samplers.
- `accelerationStructureCaptureReplayDescriptorDataSize` indicates the maximum size in bytes of the opaque data used for capture and replay with acceleration structures.
- `samplerDescriptorSize` indicates the size in bytes of a VK_DESCRIPTOR_TYPE_SAMPLER descriptor.
- `combinedImageSamplerDescriptorSize` indicates the size in bytes of a VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER descriptor.
- `sampledImageDescriptorSize` indicates the size in bytes of a VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE descriptor.
- `storageImageDescriptorSize` indicates the size in bytes of a VK_DESCRIPTOR_TYPE_STORAGE_IMAGE descriptor.
- `uniformTexelBufferDescriptorSize` indicates the size in bytes of a VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER descriptor if the `robustBufferAccess` feature is not enabled.
- `robustUniformTexelBufferDescriptorSize` indicates the size in bytes of a

VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER descriptor if the robustBufferAccess feature is enabled.

- storageTexelBufferDescriptorSize indicates the size in bytes of a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor if the robustBufferAccess feature is not enabled.
- robustStorageTexelBufferDescriptorSize indicates the size in bytes of a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor if the robustBufferAccess feature is enabled.
- uniformBufferDescriptorSize indicates the size in bytes of a VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER descriptor.
- robustUniformBufferDescriptorSize indicates the size in bytes of a VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER descriptor if the robustBufferAccess feature is enabled.
- storageBufferDescriptorSize indicates the size in bytes of a VK_DESCRIPTOR_TYPE_STORAGE_BUFFER descriptor.
- robustStorageBufferDescriptorSize indicates the size in bytes of a VK_DESCRIPTOR_TYPE_STORAGE_BUFFER descriptor if the robustBufferAccess feature is enabled.
- inputAttachmentDescriptorSize indicates the size in bytes of a VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT descriptor.
- accelerationStructureDescriptorSize indicates the size in bytes of a VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR or VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_NV descriptor.
- maxSamplerDescriptorBufferRange indicates the maximum range in bytes from the address of a sampler descriptor buffer binding that is accessible to a shader.
- maxResourceDescriptorBufferRange indicates the maximum range in bytes from the address of a resource descriptor buffer binding that is accessible to a shader.
- samplerDescriptorBufferAddressSpaceSize indicates the total size in bytes of the address space available for descriptor buffers created with the VK_BUFFER_USAGE_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT usage flag set.
- resourceDescriptorBufferAddressSpaceSize indicates the total size in bytes of the address space available for descriptor buffers created with the VK_BUFFER_USAGE_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT usage flag set.
- descriptorBufferAddressSpaceSize indicates the total size in bytes of the address space available for descriptor buffers created with both the VK_BUFFER_USAGE_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT and VK_BUFFER_USAGE_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT usage flags set.

A descriptor binding with type VK_DESCRIPTOR_TYPE_MUTABLE_EXT has a descriptor size which is implied by the descriptor types included in the VkMutableDescriptorTypeCreateInfoEXT ::pDescriptorTypes list. The descriptor size is equal to the maximum size of any descriptor type included in the pDescriptorTypes list.

As there is no way to request robust and non-robust descriptors separately, or specify robust/non-robust descriptors in the set layout, if the `robustBufferAccess` feature is enabled then robust descriptors are always used.

If the `VkPhysicalDeviceDescriptorBufferPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDescriptorBufferPropertiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_BUFFER_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceDescriptorBufferDensityMapPropertiesEXT` structure is defined as:

WARNING	This functionality is superseded by VK_EXT_descriptor_heap. See Legacy Functionality for more information.

```
// Provided by VK_EXT_descriptor_buffer with VK_EXT_fragment_density_map
typedef struct VkPhysicalDeviceDescriptorBufferDensityMapPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    size_t             combinedImageSamplerDensityMapDescriptorSize;
} VkPhysicalDeviceDescriptorBufferDensityMapPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `combinedImageSamplerDensityMapDescriptorSize` indicates the size in bytes of a VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER descriptor when creating the descriptor with VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT set.

If the `VkPhysicalDeviceDescriptorBufferDensityMapPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDescriptorBufferDensityMapPropertiesEXT-sType-sType

sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_BUFFER_DENSITY_MAP_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceHostImageCopyProperties structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDeviceHostImageCopyProperties {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           copySrcLayoutCount;
    VkImageLayout*     pCopySrcLayouts;
    uint32_t           copyDstLayoutCount;
    VkImageLayout*     pCopyDstLayouts;
    uint8_t            optimalTilingLayoutUUID[VK_UUID_SIZE];
    VkBool32           identicalMemoryTypeRequirements;
} VkPhysicalDeviceHostImageCopyProperties;
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to VkPhysicalDeviceHostImageCopyProperties
typedef VkPhysicalDeviceHostImageCopyProperties
VkPhysicalDeviceHostImageCopyPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- copySrcLayoutCount is an integer related to the number of image layouts for host copies from images available or queried, as described below.
- pCopySrcLayouts is a pointer to an array of VkImageLayout in which supported image layouts for use with host copy operations from images are returned.
- copyDstLayoutCount is an integer related to the number of image layouts for host copies to images available or queried, as described below.
- pCopyDstLayouts is a pointer to an array of VkImageLayout in which supported image layouts for use with host copy operations to images are returned.
- optimalTilingLayoutUUID is an array of VK_UUID_SIZE uint8_t values representing a universally unique identifier for the implementation's swizzling layout of images created with VK_IMAGE_TILING_OPTIMAL.
- identicalMemoryTypeRequirements indicates that specifying the

VK_IMAGE_USAGE_HOST_TRANSFER_BIT flag in VkImageCreateInfo::usage does not affect the memory type requirements of the image.

If the VkPhysicalDeviceHostImageCopyProperties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

If pCopyDstLayouts is NULL, then the number of image layouts that are supported in VkCopyMemoryToImageInfo::dstImageLayout and VkCopyImageToImageInfo::dstImageLayout is returned in copyDstLayoutCount. Otherwise, copyDstLayoutCount **must** be set by the application to the number of elements in the pCopyDstLayouts array, and on return is overwritten with the number of values actually written to pCopyDstLayouts. If the value of copyDstLayoutCount is less than the number of image layouts that are supported, at most copyDstLayoutCount values will be written to pCopyDstLayouts. The implementation **must** include the VK_IMAGE_LAYOUT_GENERAL image layout in pCopyDstLayouts. If the unifiedImageLayouts feature is supported, the implementation **must** include all the image layouts that are interchangeable with VK_IMAGE_LAYOUT_GENERAL in pCopyDstLayouts.

If pCopySrcLayouts is NULL, then the number of image layouts that are supported in VkCopyImageToMemoryInfo::srcImageLayout and VkCopyImageToImageInfo::srcImageLayout is returned in copySrcLayoutCount. Otherwise, copySrcLayoutCount **must** be set by the application to the number of elements in the pCopySrcLayouts array, and on return is overwritten with the number of values actually written to pCopySrcLayouts. If the value of copySrcLayoutCount is less than the number of image layouts that are supported, at most copySrcLayoutCount values will be written to pCopySrcLayouts. The implementation **must** include the VK_IMAGE_LAYOUT_GENERAL image layout in pCopySrcLayouts. If the unifiedImageLayouts feature is supported, the implementation **must** include all the image layouts that are interchangeable with VK_IMAGE_LAYOUT_GENERAL in pCopySrcLayouts.

The optimalTilingLayoutUUID value can be used to ensure compatible data layouts when using the VK_HOST_IMAGE_COPY_MEMCPY_BIT flag in vkCopyMemoryToImage and vkCopyImageToMemory.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceHostImageCopyProperties-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_PROPERTIES
- VUID-VkPhysicalDeviceHostImageCopyProperties-pCopySrcLayouts-parameter
  If copySrcLayoutCount is not 0, and pCopySrcLayouts is not NULL, pCopySrcLayouts **must** be a valid pointer to an array of copySrcLayoutCount VkImageLayout values
- VUID-VkPhysicalDeviceHostImageCopyProperties-pCopyDstLayouts-parameter
  If copyDstLayoutCount is not 0, and pCopyDstLayouts is not NULL, pCopyDstLayouts **must** be a valid pointer to an array of copyDstLayoutCount VkImageLayout values

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceMultiDrawPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_multi_draw
typedef struct VkPhysicalDeviceMultiDrawPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxMultiDrawCount;
} VkPhysicalDeviceMultiDrawPropertiesEXT;
```

The members of the VkPhysicalDeviceMultiDrawPropertiesEXT structure describe the following features:

- maxMultiDrawCount indicates the maximum number of draw calls which **can** be batched into a single multidraw.

If the VkPhysicalDeviceMultiDrawPropertiesEXT structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMultiDrawPropertiesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTI_DRAW_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceNestedCommandBufferPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_nested_command_buffer
typedef struct VkPhysicalDeviceNestedCommandBufferPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxCommandBufferNestingLevel;
} VkPhysicalDeviceNestedCommandBufferPropertiesEXT;
```

The members of the VkPhysicalDeviceNestedCommandBufferPropertiesEXT structure describe the

following features:

- `maxCommandBufferNestingLevel` indicates the maximum nesting level of calls to vkCmdExecuteCommands from Secondary Command Buffers. A `maxCommandBufferNestingLevel` of `UINT32_MAX` means there is no limit to the nesting level.

If the `VkPhysicalDeviceNestedCommandBufferPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceNestedCommandBufferPropertiesEXT-sType-sType `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NESTED_COMMAND_BUFFER_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceGraphicsPipelineLibraryPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_graphics_pipeline_library
typedef struct VkPhysicalDeviceGraphicsPipelineLibraryPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           graphicsPipelineLibraryFastLinking;
    VkBool32           graphicsPipelineLibraryIndependentInterpolationDecoration;
} VkPhysicalDeviceGraphicsPipelineLibraryPropertiesEXT;
```

- `graphicsPipelineLibraryFastLinking` indicates whether fast linking of graphics pipelines is supported. If it is VK_TRUE, creating a graphics pipeline entirely from pipeline libraries without VK_PIPELINE_CREATE_LINK_TIME_OPTIMIZATION_BIT_EXT is comparable in cost to recording a command in a command buffer.
- `graphicsPipelineLibraryIndependentInterpolationDecoration` indicates whether `NoPerspective` and `Flat` interpolation decorations in the last vertex processing stage and the fragment shader are required to match when using graphics pipeline libraries. If it is VK_TRUE, the interpolation decorations do not need to match. If it is VK_FALSE, these decorations **must** either be present in both stages or neither stage in order for a given interface variable to match.

If the `VkPhysicalDeviceGraphicsPipelineLibraryPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceGraphicsPipelineLibraryPropertiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GRAPHICS_PIPELINE_LIBRARY_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_fragment_shader_barycentric with VK_EXT_provoking_vertex
typedef struct VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           triStripVertexOrderIndependentOfProvokingVertex;
} VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR;
```

- When the provoking vertex mode is VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT, and the primitive order is odd in a triangle strip, the ordering of vertices is defined in last vertex table. `triStripVertexOrderIndependentOfProvokingVertex` equal to VK_TRUE indicates that the implementation ignores this and uses the vertex order defined by VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT instead.

If the `VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_BARYCENTRIC_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceShaderModuleIdentifierPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_shader_module_identifier
typedef struct VkPhysicalDeviceShaderModuleIdentifierPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint8_t            shaderModuleIdentifierAlgorithmUUID[VK_UUID_SIZE];
} VkPhysicalDeviceShaderModuleIdentifierPropertiesEXT;
```

The members of the VkPhysicalDeviceShaderModuleIdentifierPropertiesEXT structure describe the following:

- shaderModuleIdentifierAlgorithmUUID is an array of VK_UUID_SIZE uint8_t values which uniquely represents the algorithm used to compute an identifier in vkGetShaderModuleIdentifierEXT and vkGetShaderModuleCreateInfoIdentifierEXT. Implementations **should** not change this value in different driver versions if the algorithm used to compute an identifier is the same.

NOTE

The algorithm UUID may be the same in different ICDs if the algorithms are guaranteed to produce the same results. This may happen in driver stacks which support different kinds of hardware with shared code.

Khronos' conformance testing can not guarantee that shaderModuleIdentifierAlgorithmUUID values are actually unique, so implementors should make their own best efforts to ensure that their UUID is unlikely to conflict with other implementations which may use a different algorithm. In particular, hard-coded values which easily conflict, such as all-0 bits, **should** never be used. Hard-coded values are acceptable if best effort is ensured that the value will not accidentally conflict.

If the VkPhysicalDeviceShaderModuleIdentifierPropertiesEXT structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderModuleIdentifierPropertiesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_MODULE_IDENTIFIER_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDevicePipelineRobustnessProperties structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkPhysicalDevicePipelineRobustnessProperties {
    VkStructureType                       sType;
    void*                                 pNext;
    VkPipelineRobustnessBufferBehavior    defaultRobustnessStorageBuffers;
    VkPipelineRobustnessBufferBehavior    defaultRobustnessUniformBuffers;
    VkPipelineRobustnessBufferBehavior    defaultRobustnessVertexInputs;
    VkPipelineRobustnessImageBehavior     defaultRobustnessImages;
} VkPhysicalDevicePipelineRobustnessProperties;
```

```
// Provided by VK_EXT_pipeline_robustness
// Equivalent to VkPhysicalDevicePipelineRobustnessProperties
typedef VkPhysicalDevicePipelineRobustnessProperties
VkPhysicalDevicePipelineRobustnessPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- defaultRobustnessStorageBuffers describes the behavior of out of bounds accesses made to storage buffers when no robustness features are enabled.
- defaultRobustnessUniformBuffers describes the behavior of out of bounds accesses made to uniform buffers when no robustness features are enabled.
- defaultRobustnessVertexInputs describes the behavior of out of bounds accesses made to vertex input attributes when no robustness features are enabled.
- defaultRobustnessImages describes the behavior of out of bounds accesses made to images when no robustness features are enabled.

Some implementations of Vulkan may be able to guarantee that certain types of accesses are always performed with robustness even when the Vulkan API's robustness features are not explicitly enabled.

Even when an implementation reports that accesses to a given resource type are robust by default, it remains invalid to make an out of bounds access without requesting the appropriate robustness feature.

If the VkPhysicalDevicePipelineRobustnessProperties structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePipelineRobustnessProperties-sType-sType
  sType **must** be

VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_ROBUSTNESS_PROPERTIES

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceExtendedDynamicState3PropertiesEXT structure is defined as:

```
// Provided by VK_EXT_extended_dynamic_state3
typedef struct VkPhysicalDeviceExtendedDynamicState3PropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           dynamicPrimitiveTopologyUnrestricted;
} VkPhysicalDeviceExtendedDynamicState3PropertiesEXT;
```

- dynamicPrimitiveTopologyUnrestricted indicates that the implementation allows vkCmdSetPrimitiveTopology to use a different primitive topology class to the one specified in the active graphics pipeline.

If the VkPhysicalDeviceExtendedDynamicState3PropertiesEXT structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExtendedDynamicState3PropertiesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_3_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceOpacityMicromapPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_opacity_micromap
typedef struct VkPhysicalDeviceOpacityMicromapPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxOpacity2StateSubdivisionLevel;
    uint32_t           maxOpacity4StateSubdivisionLevel;
    uint32_t           maxOpacityLossy4StateSubdivisionLevel;
    uint64_t           maxMicromapTriangles;
} VkPhysicalDeviceOpacityMicromapPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxOpacity2StateSubdivisionLevel` is the maximum allowed `subdivisionLevel` when `format` is VK_OPACITY_MICROMAP_FORMAT_2_STATE_KHR.
- `maxOpacity4StateSubdivisionLevel` is the maximum allowed `subdivisionLevel` when `format` is VK_OPACITY_MICROMAP_FORMAT_4_STATE_KHR and the micromap is not built with VK_BUILD_ACCELERATION_STRUCTURE_MICROMAP_LOSSY_BIT_KHR.
- `maxOpacityLossy4StateSubdivisionLevel` is the maximum allowed `subdivisionLevel` when `format` is VK_OPACITY_MICROMAP_FORMAT_4_STATE_KHR and the micromap is built with VK_BUILD_ACCELERATION_STRUCTURE_MICROMAP_LOSSY_BIT_KHR.
- `maxMicromapTriangles` limits the number of triangles allowed to be specified when building a micromap.

If the `VkPhysicalDeviceOpacityMicromapPropertiesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceOpacityMicromapPropertiesKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceOpacityMicromapPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkPhysicalDeviceOpacityMicromapPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxOpacity2StateSubdivisionLevel;
    uint32_t           maxOpacity4StateSubdivisionLevel;
} VkPhysicalDeviceOpacityMicromapPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxOpacity2StateSubdivisionLevel` is the maximum allowed `subdivisionLevel` when `format` is VK_OPACITY_MICROMAP_FORMAT_2_STATE_KHR.
- `maxOpacity4StateSubdivisionLevel` is the maximum allowed `subdivisionLevel` when `format` is VK_OPACITY_MICROMAP_FORMAT_4_STATE_KHR.

If the `VkPhysicalDeviceOpacityMicromapPropertiesEXT` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceOpacityMicromapPropertiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

Values which **may** be returned in the `rayTracingInvocationReorderReorderingHint` field of `VkPhysicalDeviceRayTracingInvocationReorderPropertiesEXT` or are:

```
// Provided by VK_EXT_ray_tracing_invocation_reorder
typedef enum VkRayTracingInvocationReorderModeEXT {
    VK_RAY_TRACING_INVOCATION_REORDER_MODE_NONE_EXT = 0,
    VK_RAY_TRACING_INVOCATION_REORDER_MODE_REORDER_EXT = 1,
} VkRayTracingInvocationReorderModeEXT;
```

- VK_RAY_TRACING_INVOCATION_REORDER_MODE_NONE_EXT specifies that the implementation does not reorder at reorder calls.
- VK_RAY_TRACING_INVOCATION_REORDER_MODE_REORDER_EXT specifies that the implementation **may** reorder at reorder calls.

The VkPhysicalDeviceRayTracingInvocationReorderPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_ray_tracing_invocation_reorder
typedef struct VkPhysicalDeviceRayTracingInvocationReorderPropertiesEXT {
    VkStructureType                         sType;
    void*                                   pNext;
    VkRayTracingInvocationReorderModeEXT    rayTracingInvocationReorderReorderingHint;
    uint32_t                                maxShaderBindingTableRecordIndex;
} VkPhysicalDeviceRayTracingInvocationReorderPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- rayTracingInvocationReorderReorderingHint is a hint indicating if the implementation **may** reorder at the reorder calls.
- maxShaderBindingTableRecordIndex is the maximum shader binding table record index allowed to be passed in to OpHitObjectSetShaderBindingTableRecordIndexEXT.

If rayTracingInvocationReorderReorderingHint is VK_RAY_TRACING_INVOCATION_REORDER_MODE_REORDER_EXT there **must** exist conditions under which the ordered set of invocations before a reorder instruction is different than the ordered set of invocations after the reorder instruction. The ordering of a set of invocations is determined by the SubgroupId of an invocation's subgroup and the SubgroupLocalInvocationId of an invocation within that subgroup.

The reorder instructions are:

- OpReorderThreadWithHintEXT
- OpReorderThreadWithHitObjectEXT
- OpHitObjectReorderExecuteShaderEXT
- OpHitObjectTraceReorderExecuteEXT

NOTE Because the extension changes how hits are managed there is a compatibility reason to expose the extension even when an implementation does not have sorting active.

If the VkPhysicalDeviceRayTracingInvocationReorderPropertiesEXT structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRayTracingInvocationReorderPropertiesEXT-sType-sType sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_INVOCATION_REORDER_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceShaderObjectPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_shader_object
typedef struct VkPhysicalDeviceShaderObjectPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint8_t            shaderBinaryUUID[VK_UUID_SIZE];
    uint32_t           shaderBinaryVersion;
} VkPhysicalDeviceShaderObjectPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- shaderBinaryUUID is an array of VK_UUID_SIZE uint8_t values representing a universally unique identifier for one or more implementations whose shader binaries are guaranteed to be compatible with each other.
- shaderBinaryVersion is an unsigned integer incremented to represent backwards compatible differences between implementations with the same shaderBinaryUUID.

The purpose and usage of the values of this structure are described in greater detail in Binary Shader Compatibility.

If the VkPhysicalDeviceShaderObjectPropertiesEXT structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderObjectPropertiesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_OBJECT_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDevicePipelineBinaryPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPhysicalDevicePipelineBinaryPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           pipelineBinaryInternalCache;
    VkBool32           pipelineBinaryInternalCacheControl;
    VkBool32           pipelineBinaryPrefersInternalCache;
    VkBool32           pipelineBinaryPrecompiledInternalCache;
    VkBool32           pipelineBinaryCompressedData;
} VkPhysicalDevicePipelineBinaryPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pipelineBinaryInternalCache` specifies that the implementation maintains a pipeline cache internal to the implementation. If this is VK_TRUE, applications **can** create pipeline binaries with only a pipeline create info, and in this case, an implementation **may** be able to create a pipeline binary directly without application needing to capture the binary itself.
- `pipelineBinaryInternalCacheControl` specifies whether the driver's internal cache **can** be disabled. If this property is VK_TRUE VkDevicePipelineBinaryInternalCacheControlKHR ::`disableInternalCache` **can** be used to disable the driver's internal cache, allowing an application to take full control of both memory and disk usage.
- `pipelineBinaryPrefersInternalCache` specifies that the implementation prefers to maintain an internal cache, and applications **should** not store pipeline binaries in their own on-disk caches to avoid increased on-disk storage requirements. Applications are encouraged to only store pipeline keys instead, and aim to create pipeline binaries from key alone on subsequent runs of the application.
- `pipelineBinaryPrecompiledInternalCache` specifies that the implementation **may** have pipeline binaries in its internal cache, which is populated without the application ever having generated that pipeline itself. Applications **can** attempt to create binaries without extracting pipeline binary data from the pipeline prior for a set of pipeline keys, including from previous runs of the application.
- `pipelineBinaryCompressedData` specifies that the binary data is already compressed and so applications **should** not attempt to compress it.

NOTE

These properties tend to be platform specific and may change depending on external configuration which is outside the scope of this specification. These properties are intended to guide applications when implementations have dedicated caching solutions available. In particular, if the `pipelineBinaryPrefersInternalCache` limit is exposed, relying on the internal cache may provide some advantage compared to an application-specific solution. An application with its own dedicated solution may still use its own caching system even with this limit exposed.

If the `VkPhysicalDevicePipelineBinaryPropertiesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with

each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePipelineBinaryPropertiesKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_BINARY_PROPERTIES_KHR`

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The `VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_compute_shader_derivatives
typedef struct VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           meshAndTaskShaderDerivatives;
} VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR;
```

The members of the `VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR` structure describe the following:

- `meshAndTaskShaderDerivatives` indicates whether the mesh and task shader stages support the `ComputeDerivativeGroupQuadsKHR` and `ComputeDerivativeGroupLinearKHR` SPIR-V capabilities.

If the `VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR` structure is included in the `pNext` chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COMPUTE_SHADER_DERIVATIVES_PROPERTIES_KHR`

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceMapMemoryPlacedPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_map_memory_placed
typedef struct VkPhysicalDeviceMapMemoryPlacedPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceSize       minPlacedMemoryMapAlignment;
} VkPhysicalDeviceMapMemoryPlacedPropertiesEXT;
```

The members of the VkPhysicalDeviceMapMemoryPlacedPropertiesEXT structure describe the following:

- minPlacedMemoryMapAlignment is the minimum alignment required for memory object offsets and virtual address ranges when using placed memory mapping.

If the VkPhysicalDeviceMapMemoryPlacedPropertiesEXT structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMapMemoryPlacedPropertiesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAP_MEMORY_PLACED_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceFaultPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_device_fault
typedef struct VkPhysicalDeviceFaultPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxDeviceFaultCount;
} VkPhysicalDeviceFaultPropertiesKHR;
```

The members of the VkPhysicalDeviceFaultPropertiesKHR structure describe the following:

- maxDeviceFaultCount is the maximum number of instances of VkDeviceFaultInfoKHR that will be retained by the implementation. This **must** be greater than or equal to 1. If the application does not retrieve fault reports and overflow occurs, the oldest fault reports will be overwritten by the most recent record.

If the VkPhysicalDeviceFaultPropertiesKHR structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFaultPropertiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FAULT_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceDescriptorHeapPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_descriptor_heap
typedef struct VkPhysicalDeviceDescriptorHeapPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceSize       samplerHeapAlignment;
    VkDeviceSize       resourceHeapAlignment;
    VkDeviceSize       maxSamplerHeapSize;
    VkDeviceSize       maxResourceHeapSize;
    VkDeviceSize       minSamplerHeapReservedRange;
    VkDeviceSize       minSamplerHeapReservedRangeWithEmbedded;
    VkDeviceSize       minResourceHeapReservedRange;
    VkDeviceSize       samplerDescriptorSize;
    VkDeviceSize       imageDescriptorSize;
    VkDeviceSize       bufferDescriptorSize;
    VkDeviceSize       samplerDescriptorAlignment;
    VkDeviceSize       imageDescriptorAlignment;
    VkDeviceSize       bufferDescriptorAlignment;
    VkDeviceSize       maxPushDataSize;
    size_t             imageCaptureReplayOpaqueDataSize;
    uint32_t           maxDescriptorHeapEmbeddedSamplers;
    uint32_t           samplerYcbcrConversionCount;
    VkBool32           sparseDescriptorHeaps;
    VkBool32           protectedDescriptorHeaps;
} VkPhysicalDeviceDescriptorHeapPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- samplerHeapAlignment specifies the required alignment of the heapRange->address member of VkBindHeapInfoEXT for binding sampler heaps. It must be a power-of-two value.

- `resourceHeapAlignment` specifies the required alignment of the `heapRange->address` member of VkBindHeapInfoEXT for binding resource heaps. It must be a power-of-two value.
- `maxSamplerHeapSize` describes maximum value of the `size` member of VkDeviceAddressRangeKHR for binding sampler heaps, including the reservation, when embedded samplers are used.
- `maxResourceHeapSize` describes maximum value of the `size` member of VkDeviceAddressRangeKHR for binding resource heaps, including the reservation.
- `minSamplerHeapReservedRange` specifies the minimum amount of data that the implementation needs to be reserved within the bound sampler heap range when embedded samplers are not used.
- `minSamplerHeapReservedRangeWithEmbedded` specifies the minimum amount of data that the implementation needs to be reserved within the bound sampler heap range when embedded samplers are used.
- `minResourceHeapReservedRange` specifies the minimum amount of data that the implementation needs to be reserved within the bound resource heap range.
- `samplerDescriptorSize` specifies the size of sampler descriptors written by vkWriteSamplerDescriptorsEXT. It **must** be a power-of-two value.
- `imageDescriptorSize` specifies the maximum size of image and texel buffer descriptors written by vkWriteResourceDescriptorsEXT. It **must** be a power-of-two value.
- `bufferDescriptorSize` specifies the maximum size of unformatted buffer descriptors or acceleration structures written by vkWriteResourceDescriptorsEXT. It **must** be a power-of-two value.
- `samplerDescriptorAlignment` specifies the required alignment of sampler descriptors within a sampler heap. It must be a power-of-two value, and less than or equal to `samplerDescriptorSize`.
- `imageDescriptorAlignment` specifies the required alignment of image descriptors within a resource heap. It must be a power-of-two value, and less than or equal to `imageDescriptorSize`.
- `bufferDescriptorAlignment` specifies the required alignment of unformatted buffer descriptors or acceleration structures within a resource heap. It must be a power-of-two value, and less than or equal to `bufferDescriptorSize`.
- `maxPushDataSize` specifies the maximum total size of all push data. It must be greater than or equal to maxPushConstantsSize.
- `imageCaptureReplayOpaqueDataSize` specifies the size of the opaque capture/replay data for an image.
- `maxDescriptorHeapEmbeddedSamplers` specifies the maximum number of unique embedded samplers across all pipelines.
- `samplerYcbcrConversionCount` specifies the number of sampler descriptors required for any sampler using YC_BC_R conversion.
- `sparseDescriptorHeaps` specifies whether descriptor heaps can be backed by sparse memory or not. If this value is VK_FALSE, buffers cannot be specified as both sparse and having descriptor heap usage.
- `protectedDescriptorHeaps` specifies whether descriptor heaps can be used with protected

submissions or not. If this value is VK_FALSE, buffers cannot be specified as both protected and having descriptor heap usage.

If the VkPhysicalDeviceDescriptorHeapPropertiesEXT structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDescriptorHeapPropertiesEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_HEAP_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

The VkPhysicalDeviceShaderAbortPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_shader_abort
typedef struct VkPhysicalDeviceShaderAbortPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint64_t           maxShaderAbortMessageSize;
} VkPhysicalDeviceShaderAbortPropertiesKHR;
```

The VkPhysicalDeviceShaderAbortPropertiesKHR structure describes the following:

- maxShaderAbortMessageSize indicates the maximum size of a shader abort message that the application **can** specify.

If the VkPhysicalDeviceShaderAbortPropertiesKHR structure is included in the pNext chain of the VkPhysicalDeviceProperties2 structure passed to vkGetPhysicalDeviceProperties2, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderAbortPropertiesKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ABORT_PROPERTIES_KHR

Structure Chaining

Extends the structure

- VkPhysicalDeviceProperties2

48.1. Limit Requirements

The following table specifies the **required** minimum/maximum for all Vulkan graphics implementations. Where a limit corresponds to a fine-grained device feature which is **optional**, the feature name is listed with two **required** limits, one when the feature is supported and one when it is not supported. If an implementation supports a feature, the limits reported are the same whether or not the feature is enabled.

Table 68. Required Limit Types

Type	Limit	Feature
uint32_t	maxImageDimension1D	-
uint32_t	maxImageDimension2D	-
uint32_t	maxImageDimension3D	-
uint32_t	maxImageDimensionCube	-
uint32_t	maxImageArrayLayers	-
uint32_t	maxTexelBufferElements	-
uint32_t	maxUniformBufferRange	-
uint32_t	maxStorageBufferRange	-
uint32_t	maxPushConstantsSize	-
uint32_t	maxMemoryAllocationCount	-
uint32_t	maxSamplerAllocationCount	-
VkDeviceSize	bufferImageGranularity	-
VkDeviceSize	sparseAddressSpaceSize	sparseBinding
uint32_t	maxBoundDescriptorSets	-
uint32_t	maxPerStageDescriptorSamplers	-
uint32_t	maxPerStageDescriptorUniformBuffers	-
uint32_t	maxPerStageDescriptorStorageBuffers	-
uint32_t	maxPerStageDescriptorSampledImages	-
uint32_t	maxPerStageDescriptorStorageImages	-
uint32_t	maxPerStageDescriptorInputAttachments	-
uint32_t	maxPerStageResources	-
uint32_t	maxDescriptorSetSamplers	-

Type	Limit	Feature
uint32_t	maxDescriptorSetUniformBuffers	-
uint32_t	maxDescriptorSetUniformBuffersDynamic	-
uint32_t	maxDescriptorSetStorageBuffers	-
uint32_t	maxDescriptorSetStorageBuffersDynamic	-
uint32_t	maxDescriptorSetSampledImages	-
uint32_t	maxDescriptorSetStorageImages	-
uint32_t	maxDescriptorSetInputAttachments	-
uint32_t	maxVertexInputAttributes	-
uint32_t	maxVertexInputBindings	-
uint32_t	maxVertexInputAttributeOffset	-
uint32_t	maxVertexInputBindingStride	-
uint32_t	maxVertexOutputComponents	-
uint32_t	maxTessellationGenerationLevel	tessellationShader
uint32_t	maxTessellationPatchSize	tessellationShader
uint32_t	maxTessellationControlPerVertexInputComponents	tessellationShader
uint32_t	maxTessellationControlPerVertexOutputComponents	tessellationShader
uint32_t	maxTessellationControlPerPatchOutputComponents	tessellationShader
uint32_t	maxTessellationControlTotalOutputComponents	tessellationShader
uint32_t	maxTessellationEvaluationInputComponents	tessellationShader
uint32_t	maxTessellationEvaluationOutputComponents	tessellationShader
uint32_t	maxGeometryShaderInvocations	geometryShader
uint32_t	maxGeometryInputComponents	geometryShader
uint32_t	maxGeometryOutputComponents	geometryShader
uint32_t	maxGeometryOutputVertices	geometryShader
uint32_t	maxGeometryTotalOutputComponents	geometryShader
uint32_t	maxFragmentInputComponents	-
uint32_t	maxFragmentOutputAttachments	-
uint32_t	maxFragmentDualSrcAttachments	dualSrcBlend
uint32_t	maxFragmentCombinedOutputResources	-
uint32_t	maxComputeSharedMemorySize	-
3 × uint32_t	maxComputeWorkGroupCount	-
uint32_t	maxComputeWorkGroupInvocations	-
3 × uint32_t	maxComputeWorkGroupSize	-
uint32_t	subPixelPrecisionBits	-
uint32_t	subTexelPrecisionBits	-

Type	Limit	Feature
uint32_t	mipmapPrecisionBits	-
uint32_t	maxDrawIndexedIndexValue	fullDrawIndexUint32
uint32_t	maxDrawIndirectCount	multiDrawIndirect
float	maxSamplerLodBias	-
float	maxSamplerAnisotropy	samplerAnisotropy
uint32_t	maxViewports	multiViewport
2 × uint32_t	maxViewportDimensions	-
2 × float	viewportBoundsRange	-
uint32_t	viewportSubPixelBits	-
size_t	minMemoryMapAlignment	-
VkDeviceSize	minTexelBufferOffsetAlignment	-
VkDeviceSize	minUniformBufferOffsetAlignment	-
VkDeviceSize	minStorageBufferOffsetAlignment	-
int32_t	minTexelOffset	-
uint32_t	maxTexelOffset	-
int32_t	minTexelGatherOffset	shaderImageGatherExtended
uint32_t	maxTexelGatherOffset	shaderImageGatherExtended
float	minInterpolationOffset	sampleRateShading
float	maxInterpolationOffset	sampleRateShading
uint32_t	subPixelInterpolationOffsetBits	sampleRateShading
uint32_t	maxFramebufferWidth	-
uint32_t	maxFramebufferHeight	-
uint32_t	maxFramebufferLayers	-
VkSampleCountFlags	framebufferColorSampleCounts	-
VkSampleCountFlags	framebufferIntegerColorSampleCounts	-
VkSampleCountFlags	framebufferDepthSampleCounts	-
VkSampleCountFlags	framebufferStencilSampleCounts	-
VkSampleCountFlags	framebufferNoAttachmentsSampleCounts	-
uint32_t	maxColorAttachments	-
VkSampleCountFlags	sampledImageColorSampleCounts	-

Type	Limit	Feature
VkSampleCountFlags	sampledImageIntegerSampleCounts	-
VkSampleCountFlags	sampledImageDepthSampleCounts	-
VkSampleCountFlags	sampledImageStencilSampleCounts	-
VkSampleCountFlags	storageImageSampleCounts	shaderStorageImageMultisample
uint32_t	maxSampleMaskWords	-
VkBool32	timestampComputeAndGraphics	-
float	timestampPeriod	-
uint32_t	maxClipDistances	shaderClipDistance
uint32_t	maxCullDistances	shaderCullDistance
uint32_t	maxCombinedClipAndCullDistances	shaderCullDistance
uint32_t	discreteQueuePriorities	-
2 × float	pointSizeRange	largePoints
2 × float	lineWidthRange	wideLines
float	pointSizeGranularity	largePoints
float	lineWidthGranularity	wideLines
VkBool32	strictLines	-
VkBool32	standardSampleLocations	-
VkDeviceSize	optimalBufferCopyOffsetAlignment	-
VkDeviceSize	optimalBufferCopyRowPitchAlignment	-
VkDeviceSize	nonCoherentAtomSize	-
uint32_t	maxDiscardRectangles	VK_EXT_discard_rectangles
VkBool32	filterMinmaxSingleComponentFormats	samplerFilterMinmax
VkBool32	filterMinmaxImageComponentMapping	samplerFilterMinmax
VkDeviceSize	maxBufferSize	maintenance4
float	primitiveOverestimationSize	VK_EXT_conservative_rasterization
float	maxExtraPrimitiveOverestimationSize	VK_EXT_conservative_rasterization
float	extraPrimitiveOverestimationSizeGranularity	VK_EXT_conservative_rasterization
VkBool32	degenerateTrianglesRasterized	VK_EXT_conservative_rasterization
VkBool32	degenerateLinesRasterized	VK_EXT_conservative_rasterization

Type	Limit	Feature
VkBool32	fullyCoveredFragmentShaderInputVariable	VK_EXT_conservative_rasterization
VkBool32	conservativeRasterizationPostDepthCoverage	VK_EXT_conservative_rasterization
uint32_t	maxUpdateAfterBindDescriptorsInAllPools	descriptorIndexing
VkBool32	shaderUniformBufferArrayNonUniformIndexingNative	-
VkBool32	shaderSampledImageArrayNonUniformIndexingNative	-
VkBool32	shaderStorageBufferArrayNonUniformIndexingNative	-
VkBool32	shaderStorageImageArrayNonUniformIndexingNative	-
VkBool32	shaderInputAttachmentArrayNonUniformIndexingNative	-
uint32_t	maxPerStageDescriptorUpdateAfterBindSamplers	descriptorIndexing
uint32_t	maxPerStageDescriptorUpdateAfterBindUniformBuffers	descriptorIndexing
uint32_t	maxPerStageDescriptorUpdateAfterBindStorageBuffers	descriptorIndexing
uint32_t	maxPerStageDescriptorUpdateAfterBindSampledImages	descriptorIndexing
uint32_t	maxPerStageDescriptorUpdateAfterBindStorageImages	descriptorIndexing
uint32_t	maxPerStageDescriptorUpdateAfterBindInputAttachments	descriptorIndexing
uint32_t	maxPerStageUpdateAfterBindResources	descriptorIndexing
uint32_t	maxDescriptorSetUpdateAfterBindSamplers	descriptorIndexing
uint32_t	maxDescriptorSetUpdateAfterBindUniformBuffers	descriptorIndexing
uint32_t	maxDescriptorSetUpdateAfterBindUniformBuffersDynamic	descriptorIndexing
uint32_t	maxDescriptorSetUpdateAfterBindStorageBuffers	descriptorIndexing
uint32_t	maxDescriptorSetUpdateAfterBindStorageBuffersDynamic	descriptorIndexing
uint32_t	maxDescriptorSetUpdateAfterBindSampledImages	descriptorIndexing
uint32_t	maxDescriptorSetUpdateAfterBindStorageImages	descriptorIndexing
uint32_t	maxDescriptorSetUpdateAfterBindInputAttachments	descriptorIndexing
uint32_t	maxInlineUniformBlockSize	inlineUniformBlock
uint32_t	maxPerStageDescriptorInlineUniformBlocks	inlineUniformBlock
uint32_t	maxPerStageDescriptorUpdateAfterBindInlineUniformBlocks	inlineUniformBlock
uint32_t	maxDescriptorSetInlineUniformBlocks	inlineUniformBlock

Type	Limit	Feature
uint32_t	maxDescriptorSetUpdateAfterBindInlineUniformBlocks	inlineUniformBlock
uint32_t	maxInlineUniformTotalSize	inlineUniformBlock
uint32_t	maxVertexAttribDivisor	Vulkan 1.4, VK_KHR_vertex_attribute_divisor, VK_EXT_vertex_attribute_divisor
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskWorkGroupTotalCount	VK_EXT_mesh_shader
3 × uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskWorkGroupCount	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskWorkGroupInvocations	VK_EXT_mesh_shader
3 × uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskWorkGroupSize	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskPayloadSize	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskSharedMemorySize	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskPayloadAndSharedMemorySize	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupTotalCount	VK_EXT_mesh_shader
3 × uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupCount	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupInvocations	VK_EXT_mesh_shader
3 × uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupSize	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshSharedMemorySize	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshPayloadAndSharedMemorySize	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshOutputMemorySize	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshPayloadAndOutputMemorySize	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshOutputComponents	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshOutputVertices	VK_EXT_mesh_shader

Type	Limit	Feature
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshOutputPrimitives	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshOutputLayers	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshMultiviewViewCount	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::meshOutputPerVertexGranularity	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::meshOutputPerPrimitiveGranularity	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxPreferredTaskWorkGroupInvocations	VK_EXT_mesh_shader
uint32_t	VkPhysicalDeviceMeshShaderPropertiesEXT::maxPreferredMeshWorkGroupInvocations	VK_EXT_mesh_shader
VkBool32	VkPhysicalDeviceMeshShaderPropertiesEXT::prefersLocalInvocationVertexOutput	VK_EXT_mesh_shader
VkBool32	VkPhysicalDeviceMeshShaderPropertiesEXT::prefersLocalInvocationPrimitiveOutput	VK_EXT_mesh_shader
VkBool32	VkPhysicalDeviceMeshShaderPropertiesEXT::prefersCompactVertexOutput	VK_EXT_mesh_shader
VkBool32	VkPhysicalDeviceMeshShaderPropertiesEXT::prefersCompactPrimitiveOutput	VK_EXT_mesh_shader
uint32_t	maxTransformFeedbackStreams	VK_EXT_transform_feedback
uint32_t	maxTransformFeedbackBuffers	VK_EXT_transform_feedback
VkDeviceSize	maxTransformFeedbackBufferSize	VK_EXT_transform_feedback
uint32_t	maxTransformFeedbackStreamDataSize	VK_EXT_transform_feedback
uint32_t	maxTransformFeedbackBufferDataSize	VK_EXT_transform_feedback
uint32_t	maxTransformFeedbackBufferDataStride	VK_EXT_transform_feedback
VkBool32	transformFeedbackQueries	VK_EXT_transform_feedback
VkBool32	transformFeedbackStreamsLinesTriangles	VK_EXT_transform_feedback
VkBool32	transformFeedbackRasterizationStreamSelect	VK_EXT_transform_feedback
VkBool32	transformFeedbackDraw	VK_EXT_transform_feedback
uint32_t	splitBarrierReservedSharedMemory	shaderSplitBarrier
VkExtent2D	minFragmentDensityTexelSize	fragmentDensityMap
VkExtent2D	maxFragmentDensityTexelSize	fragmentDensityMap
VkBool32	fragmentDensityInvocations	fragmentDensityMap
VkBool32	subsampledLoads	VK_EXT_fragment_density_map2
VkBool32	subsampledCoarseReconstructionEarlyAccess	VK_EXT_fragment_density_map2

Type	Limit	Feature
uint32_t	maxSubsampledArrayLayers	VK_EXT_fragment_density_map2
uint32_t	maxDescriptorSetSubsampledSamplers	VK_EXT_fragment_density_map2
uint32_t	maxGeometryCount	VK_NV_ray_tracing, VK_KHR_acceleration_structure
uint32_t	maxInstanceCount	VK_NV_ray_tracing, VK_KHR_acceleration_structure
uint32_t	maxVerticesPerCluster	clusterAccelerationStructure
uint32_t	maxTrianglesPerCluster	clusterAccelerationStructure
uint32_t	clusterScratchByteAlignment	clusterAccelerationStructure
uint32_t	clusterByteAlignment	clusterAccelerationStructure
uint32_t	clusterTemplateByteAlignment	clusterAccelerationStructure
uint32_t	clusterBottomLevelByteAlignment	clusterAccelerationStructure
uint32_t	clusterTemplateBoundsByteAlignment	clusterAccelerationStructure
uint32_t	maxClusterGeometryIndex	clusterAccelerationStructure
uint32_t	shaderGroupHandleSize	VK_NV_ray_tracing, VK_KHR_ray_tracing_pipeline
uint32_t	maxShaderGroupStride	VK_NV_ray_tracing, VK_KHR_ray_tracing_pipeline
uint32_t	shaderGroupBaseAlignment	VK_NV_ray_tracing, VK_KHR_ray_tracing_pipeline
uint32_t	maxPerStageDescriptorAccelerationStructures	VK_KHR_acceleration_structure
uint32_t	maxPerStageDescriptorUpdateAfterBindAccelerationStructures	VK_KHR_acceleration_structure
uint32_t	maxDescriptorSetAccelerationStructures	VK_NV_ray_tracing, VK_KHR_acceleration_structure
uint32_t	maxDescriptorSetUpdateAfterBindAccelerationStructures	VK_KHR_acceleration_structure
uint32_t	minAccelerationStructureScratchOffsetAlignment	VK_KHR_acceleration_structure
uint32_t	maxRayRecursionDepth	VK_KHR_ray_tracing_pipeline
uint32_t	shaderGroupHandleCaptureReplaySize	VK_KHR_ray_tracing_pipeline

Type	Limit	Feature
uint32_t	maxRayDispatchInvocationCount	VK_KHR_ray_tracing_pipeline
uint32_t	shaderGroupHandleAlignment	VK_KHR_ray_tracing_pipeline
uint32_t	maxRayHitAttributeSize	VK_KHR_ray_tracing_pipeline
uint32_t	maxPartitionCount	partitionedAccelerationStructure
uint64_t	maxTimelineSemaphoreValueDifference	timelineSemaphore
uint32_t	lineSubPixelPrecisionBits	Vulkan 1.4, VK_KHR_line_rasterization, VK_EXT_line_rasterization
uint32_t	maxCustomBorderColorSamplers	VK_EXT_custom_border_color
VkDeviceSize	robustStorageBufferAccessSizeAlignment	VK_EXT_robustness2, VK_KHR_robustness2
VkDeviceSize	robustUniformBufferAccessSizeAlignment	VK_EXT_robustness2, VK_KHR_robustness2
2 × uint32_t	minFragmentShadingRateAttachmentTexelSize	attachmentFragmentShadingRate
2 × uint32_t	maxFragmentShadingRateAttachmentTexelSize	attachmentFragmentShadingRate
uint32_t	maxFragmentShadingRateAttachmentTexelSizeAspectRatio	attachmentFragmentShadingRate
VkBool32	primitiveFragmentShadingRateWithMultipleViewports	primitiveFragmentShadingRate
VkBool32	layeredShadingRateAttachments	attachmentFragmentShadingRate
VkBool32	fragmentShadingRateNonTrivialCombinerOps	pipelineFragmentShadingRate
2 × uint32_t	maxFragmentSize	pipelineFragmentShadingRate
uint32_t	maxFragmentSizeAspectRatio	pipelineFragmentShadingRate
uint32_t	maxFragmentShadingRateCoverageSamples	pipelineFragmentShadingRate
VkSampleCountFlagBits	maxFragmentShadingRateRasterizationSamples	pipelineFragmentShadingRate
VkBool32	fragmentShadingRateWithShaderDepthStencilWrites	pipelineFragmentShadingRate
VkBool32	fragmentShadingRateWithSampleMask	pipelineFragmentShadingRate
VkBool32	fragmentShadingRateWithShaderSampleMask	pipelineFragmentShadingRate
VkBool32	fragmentShadingRateWithConservativeRasterization	pipelineFragmentShadingRate
VkBool32	fragmentShadingRateWithFragmentShaderInterlock	pipelineFragmentShadingRate
VkBool32	fragmentShadingRateWithCustomSampleLocations	pipelineFragmentShadingRate
VkBool32	fragmentShadingRateStrictMultiplyCombiner	pipelineFragmentShadingRate
VkBool32	combinedImageSamplerDescriptorSingleArray	VK_EXT_descriptor_buffer

Type	Limit	Feature
VkBool32	bufferlessPushDescriptors	VK_EXT_descriptor_buffer
VkBool32	allowSamplerImageViewPostSubmitCreation	VK_EXT_descriptor_buffer
VkDeviceSize	descriptorBufferOffsetAlignment	VK_EXT_descriptor_buffer
uint32_t	maxDescriptorBufferBindings	VK_EXT_descriptor_buffer
uint32_t	maxResourceDescriptorBufferBindings	VK_EXT_descriptor_buffer
uint32_t	maxSamplerDescriptorBufferBindings	VK_EXT_descriptor_buffer
uint32_t	maxEmbeddedImmutableSamplerBindings	VK_EXT_descriptor_buffer
uint32_t	maxEmbeddedImmutableSamplers	VK_EXT_descriptor_buffer
size_t	bufferCaptureReplayDescriptorDataSize	VK_EXT_descriptor_buffer
size_t	imageCaptureReplayDescriptorDataSize	VK_EXT_descriptor_buffer
size_t	imageViewCaptureReplayDescriptorDataSize	VK_EXT_descriptor_buffer
size_t	samplerCaptureReplayDescriptorDataSize	VK_EXT_descriptor_buffer
size_t	accelerationStructureCaptureReplayDescriptorDataSize	VK_EXT_descriptor_buffer
size_t	samplerDescriptorSize	VK_EXT_descriptor_buffer
size_t	combinedImageSamplerDescriptorSize	VK_EXT_descriptor_buffer
size_t	sampledImageDescriptorSize	VK_EXT_descriptor_buffer
size_t	storageImageDescriptorSize	VK_EXT_descriptor_buffer
size_t	uniformTexelBufferDescriptorSize	VK_EXT_descriptor_buffer
size_t	robustUniformTexelBufferDescriptorSize	VK_EXT_descriptor_buffer
size_t	storageTexelBufferDescriptorSize	VK_EXT_descriptor_buffer
size_t	robustStorageTexelBufferDescriptorSize	VK_EXT_descriptor_buffer
size_t	uniformBufferDescriptorSize	VK_EXT_descriptor_buffer
size_t	robustUniformBufferDescriptorSize	VK_EXT_descriptor_buffer
size_t	storageBufferDescriptorSize	VK_EXT_descriptor_buffer
size_t	robustStorageBufferDescriptorSize	VK_EXT_descriptor_buffer
size_t	inputAttachmentDescriptorSize	VK_EXT_descriptor_buffer
size_t	accelerationStructureDescriptorSize	VK_EXT_descriptor_buffer
VkDeviceSize	maxSamplerDescriptorBufferRange	VK_EXT_descriptor_buffer
VkDeviceSize	maxResourceDescriptorBufferRange	VK_EXT_descriptor_buffer
VkDeviceSize	samplerDescriptorBufferAddressSpaceSize	VK_EXT_descriptor_buffer
VkDeviceSize	resourceDescriptorBufferAddressSpaceSize	VK_EXT_descriptor_buffer
VkDeviceSize	descriptorBufferAddressSpaceSize	VK_EXT_descriptor_buffer
size_t	combinedImageSamplerDensityMapDescriptorSize	VK_EXT_descriptor_buffer
VkBool32	graphicsPipelineLibraryFastLinking	graphicsPipelineLibrary
VkBool32	graphicsPipelineLibraryIndependentInterpolationDecoration	graphicsPipelineLibrary

Type	Limit	Feature
VkBool32	triStripVertexOrderIndependentOfProvokingVertex	-
VkBool32	dynamicPrimitiveTopologyUnrestricted	VK_EXT_extended_dynamic_state3
uint32_t	VkPhysicalDeviceOpacityMicromapPropertiesKHR::maxOpacity2StateSubdivisionLevel	VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap
uint32_t	VkPhysicalDeviceOpacityMicromapPropertiesKHR::maxOpacity4StateSubdivisionLevel	VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap
uint32_t	VkPhysicalDeviceOpacityMicromapPropertiesKHR::maxOpacityLossy4StateSubdivisionLevel	VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap
uint64_t	VkPhysicalDeviceOpacityMicromapPropertiesKHR::maxMicromapTriangles	VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap
uint32_t	VkPhysicalDeviceOpacityMicromapPropertiesEXT::maxOpacity2StateSubdivisionLevel	VK_EXT_opacity_micromap
uint32_t	VkPhysicalDeviceOpacityMicromapPropertiesEXT::maxOpacity4StateSubdivisionLevel	VK_EXT_opacity_micromap
uint64_t	maxDecompressionIndirectCount	VK_NV_memory_decompression, VK_EXT_memory_decompression
uint32_t	maxIndirectShaderObjectCount	shaderObject
VkBool32	separateDepthStencilAttachmentAccess	maintenance7
uint32_t	maxDescriptorSetTotalUniformBuffersDynamic	maintenance7
uint32_t	maxDescriptorSetTotalStorageBuffersDynamic	maintenance7
uint32_t	maxDescriptorSetTotalBuffersDynamic	maintenance7
uint32_t	maxDescriptorSetUpdateAfterBindTotalUniformBuffersDynamic	maintenance7
uint32_t	maxDescriptorSetUpdateAfterBindTotalStorageBuffersDynamic	maintenance7
uint32_t	maxDescriptorSetUpdateAfterBindTotalBuffersDynamic	maintenance7
VkBool32	shaderSignedZeroInfNanPreserveFloat16	shaderFloat16
uint32_t	maxShaderBindingTableRecordIndex	VK_EXT_ray_tracing_invocation_reorder
VkBool32	resolveSrgbFormatAppliesTransferFunction	maintenance10
VkBool32	resolveSrgbFormatSupportsTransferFunctionControl	maintenance10
VkDeviceSize	samplerHeapAlignment	descriptorHeap
VkDeviceSize	resourceHeapAlignment	descriptorHeap
VkDeviceSize	maxSamplerHeapSize	descriptorHeap
VkDeviceSize	maxResourceHeapSize	descriptorHeap
VkDeviceSize	minSamplerHeapReservedRange	descriptorHeap

Type	Limit	Feature
VkDeviceSize	minSamplerHeapReservedRangeWithEmbedded	descriptorHeap
VkDeviceSize	minResourceHeapReservedRange	descriptorHeap
VkDeviceSize	samplerDescriptorSize	descriptorHeap
VkDeviceSize	imageDescriptorSize	descriptorHeap
VkDeviceSize	bufferDescriptorSize	descriptorHeap
VkDeviceSize	samplerDescriptorAlignment	descriptorHeap
VkDeviceSize	imageDescriptorAlignment	descriptorHeap
VkDeviceSize	bufferDescriptorAlignment	descriptorHeap
VkDeviceSize	maxPushDataSize	descriptorHeap
size_t	imageCaptureReplayOpaqueDataSize	descriptorHeap
uint32_t	maxDescriptorHeapEmbeddedSamplers	descriptorHeap
uint32_t	samplerYcbcrConversionCount	descriptorHeap
VkBool32	sparseDescriptorHeaps	descriptorHeap
VkBool32	protectedDescriptorHeaps	descriptorHeap
uint64_t	maxShaderAbortMessageSize	shaderAbort
uint32_t	maxDeviceFaultCount	VK_KHR_device_fault

Table 69. Required Limits

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
maxImageDimension1D	-	4096 (Vulkan Core) 8192 (Vulkan Roadmap 2022, Vulkan 1.4)	min
maxImageDimension2D	-	4096 (Vulkan Core) 8192 (Vulkan Roadmap 2022, Vulkan 1.4)	min
maxImageDimension3D	-	256 (Vulkan Core) 512 (Vulkan 1.4)	min
maxImageDimensionCube	-	4096 (Vulkan Core) 8192 (Vulkan Roadmap 2022, Vulkan 1.4)	min
maxImageArrayLayers	-	256 (Vulkan Core) 2048 (Vulkan Roadmap 2022, Vulkan 1.4)	min

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
maxTexelBufferElements	-	65536	min
maxUniformBufferRange	-	16384 (Vulkan Core) 65536 (Vulkan Roadmap 2022, Vulkan 1.4)	min
maxStorageBufferRange	-	2^{27}	min
maxPushConstantsSize	-	128 (Vulkan Core) 256 (Vulkan 1.4)	min
maxMemoryAllocationCount	-	4096	min
maxSamplerAllocationCount	-	4000	min
bufferImageGranularity	-	131072 (Vulkan Core) 4096 (Vulkan Roadmap 2022, Vulkan 1.4)	max
sparseAddressSpaceSize	0	2^{31}	min
maxBoundDescriptorSets	-	4 (Vulkan Core) 7 (Vulkan Roadmap 2024, Vulkan 1.4)	min
maxPerStageDescriptorSamplers	-	16 (Vulkan Core) 64 (Vulkan Roadmap 2022)	min
maxPerStageDescriptorUniformBuffers	-	12 (Vulkan Core) 15 (Vulkan Roadmap 2022, Vulkan 1.4) 200 (Vulkan Roadmap 2026)	min
maxPerStageDescriptorStorageBuffers	-	4 (Vulkan Core) 30 (Vulkan Roadmap 2022) 200 (Vulkan Roadmap 2026)	min
maxPerStageDescriptorSampledImages	-	16 (Vulkan Core) 200 (Vulkan Roadmap 2022)	min

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
maxPerStageDescriptorStorageImages	-	4 (Vulkan Core) 16 (Vulkan Roadmap 2022)	min
maxPerStageDescriptorInputAttachments	-	4 8 (Vulkan Roadmap 2026)	min
maxPerStageResources	-	128 [2] (Vulkan Core) 200 (Vulkan Roadmap 2022, Vulkan 1.4)	min
maxDescriptorSetSamplers	-	96 [8] (Vulkan Core) 576 (Vulkan Roadmap 2022)	min, *n* × PerStage
maxDescriptorSetUniformBuffers	-	72 [8] (Vulkan Core) 90 (Vulkan Roadmap 2022, Vulkan 1.4) 1800 (Vulkan Roadmap 2026)	min, *n* × PerStage
maxDescriptorSetUniformBuffersDynamic	-	8	min
maxDescriptorSetStorageBuffers	-	24 [8] (Vulkan Core) 96 (Vulkan Roadmap 2022, Vulkan 1.4) 1800 (Vulkan Roadmap 2026)	min, *n* × PerStage
maxDescriptorSetStorageBuffersDynamic	-	4	min
maxDescriptorSetTotalUniformBuffersDynamic	-	maxDescriptorSetUniformBuffersDynamic	min
maxDescriptorSetTotalStorageBuffersDynamic	-	maxDescriptorSetStorageBuffersDynamic	min
maxDescriptorSetTotalBuffersDynamic	-	maxDescriptorSetUniformBuffersDynamic + maxDescriptorSetStorageBuffersDynamic	min
maxDescriptorSetSampledImages	-	96 [8] (Vulkan Core) 1800 (Vulkan Roadmap 2022)	min, *n* × PerStage

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
maxDescriptorSetStorageImages	-	24 [8] (Vulkan Core) 144 (Vulkan Roadmap 2022, Vulkan 1.4)	min, $n \times$ PerStage
maxDescriptorSetInputAttachments	-	4 8 (Vulkan Roadmap 2026)	min
maxVertexInputAttributes	-	16	min
maxVertexInputBindings	-	16 [10]	min
maxVertexInputAttributeOffset	-	2047	min
maxVertexInputBindingStride	-	2048	min
maxVertexOutputComponents	-	64 124 (Vulkan Roadmap 2026)	min
maxTessellationGenerationLevel	0	64	min
maxTessellationPatchSize	0	32	min
maxTessellationControlPerVertexInputComponents	0	64 128 (Vulkan Roadmap 2026)	min
maxTessellationControlPerVertexOutputComponents	0	64 128 (Vulkan Roadmap 2026)	min
maxTessellationControlPerPatchOutputComponents	0	120	min
maxTessellationControlTotalOutputComponents	0	2048 4096 (Vulkan Roadmap 2026)	min
maxTessellationEvaluationInputComponents	0	64 128 (Vulkan Roadmap 2026)	min
maxTessellationEvaluationOutputComponents	0	64 128 (Vulkan Roadmap 2026)	min
maxGeometryShaderInvocations	0	32	min
maxGeometryInputComponents	0	64	min
maxGeometryOutputComponents	0	64 128 (Vulkan Roadmap 2026)	min
maxGeometryOutputVertices	0	256	min

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
maxGeometryTotalOutputComponents	0	1024	min
maxFragmentInputComponents	-	64 112 (Vulkan Roadmap 2026)	min
maxFragmentOutputAttachments	-	4 8 (Vulkan Roadmap 2026)	min
maxFragmentDualSrcAttachments	0	1	min
maxFragmentCombinedOutputResources	-	4 (Vulkan Core) 16 (Vulkan Roadmap 2022, Vulkan 1.4)	min
maxComputeSharedMemorySize	-	16384 32768 (Vulkan Roadmap 2026)	min
maxComputeWorkGroupCount	-	(65535,65535,65535)	min
maxComputeWorkGroupInvocations	-	128 (Vulkan Core) 256 (Vulkan Roadmap 2022, Vulkan 1.4)	min
maxComputeWorkGroupSize	-	(128,128,64) (Vulkan Core) (256,256,64) (Vulkan Roadmap 2022, Vulkan 1.4)	min
subgroupSize	-	1/4 (Vulkan Core) 4 (Vulkan Roadmap 2022)	min
subgroupSupportedStages	-	VK_SHADER_STAGE_COMPUTE_BIT (Vulkan Core) VK_SHADER_STAGE_COMPUTE_BIT | VK_SHADER_STAGE_FRAGMENT_BIT (Vulkan Roadmap 2022)	bitfield

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
subgroupSupportedOperations	-	VK_SUBGROUP_FEATURE_BASIC_BIT (Vulkan Core) VK_SUBGROUP_FEATURE_BASIC_BIT | VK_SUBGROUP_FEATURE_VOTE_BIT | VK_SUBGROUP_FEATURE_ARITHMETIC_BIT | VK_SUBGROUP_FEATURE_BALLOT_BIT | VK_SUBGROUP_FEATURE_SHUFFLE_BIT | VK_SUBGROUP_FEATURE_SHUFFLE_RELATIVE_BIT | VK_SUBGROUP_FEATURE_QUAD_BIT (Vulkan Roadmap 2022)	bitfield
shaderSignedZeroInfNanPreserveFloat16	-	- (Vulkan Core) VK_TRUE (Vulkan Roadmap 2022, Vulkan 1.4)	Boolean
shaderSignedZeroInfNanPreserveFloat32	-	- (Vulkan Core) VK_TRUE (Vulkan Roadmap 2022, Vulkan 1.4)	Boolean
shaderRoundingModeRTEFloat16	-	VK_FALSE (Vulkan Core) VK_TRUE (Vulkan Roadmap 2024)	Boolean
shaderRoundingModeRTEFloat32	-	VK_FALSE (Vulkan Core) VK_TRUE (Vulkan Roadmap 2024)	Boolean

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
maxSubgroupSize	-	- (Vulkan Core) 4 (Vulkan Roadmap 2022)	min
subPixelPrecisionBits	-	4 8 (Vulkan Roadmap 2026)	min
subTexelPrecisionBits	-	4 (Vulkan Core) 8 (Vulkan Roadmap 2022, Vulkan 1.4)	min
mipmapPrecisionBits	-	4 (Vulkan Core) 6 (Vulkan Roadmap 2022, Vulkan 1.4)	min
maxDrawIndexedIndexValue	2^{24}-1	2^{32}-1	min
maxDrawIndirectCount	1	2^{16}-1	min
maxSamplerLodBias	-	2 (Vulkan Core) 14 (Vulkan Roadmap 2022, Vulkan 1.4)	min
maxSamplerAnisotropy	1	16	min
maxViewports	1	16	min
maxViewportDimensions [3]	-	(4096,4096) (Vulkan Core) (7680,7680) (Vulkan 1.4) (8192,8192) (Vulkan Roadmap 2026)	min
viewportBoundsRange [4]	-	(-8192,8191) (Vulkan Core) (-15360,15359) (Vulkan 1.4)	(max,min)
viewportSubPixelBits	-	0	min
minMemoryMapAlignment	-	64	min
minTexelBufferOffsetAlignment	-	256	max
minUniformBufferOffsetAlignment	-	256	max
minStorageBufferOffsetAlignment	-	256	max
minTexelOffset	-	-8	max

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
maxTexelOffset	-	7	min
minTexelGatherOffset	0	-8	max
maxTexelGatherOffset	0	7	min
minInterpolationOffset	0.0	-0.5 [5]	max
maxInterpolationOffset	0.0	0.5 - (1 ULP) [5]	min
subPixelInterpolationOffsetBits	0	4 [5]	min
maxFramebufferWidth	-	4096 (Vulkan Core) 7680 (Vulkan 1.4) 8192 (Vulkan Roadmap 2026)	min
maxFramebufferHeight	-	4096 (Vulkan Core) 7680 (Vulkan 1.4) 8192 (Vulkan Roadmap 2026)	min
maxFramebufferLayers	-	256	min
framebufferColorSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT | VK_SAMPLE_COUNT_4_BIT)	min
framebufferIntegerColorSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT)	min
framebufferDepthSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT | VK_SAMPLE_COUNT_4_BIT)	min
framebufferStencilSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT | VK_SAMPLE_COUNT_4_BIT)	min
framebufferNoAttachmentsSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT | VK_SAMPLE_COUNT_4_BIT)	min
maxColorAttachments	-	4 (Vulkan Core) 7 (Vulkan Roadmap 2022) 8 (Vulkan Roadmap 2024, Vulkan 1.4)	min

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
sampledImageColorSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT | VK_SAMPLE_COUNT_4_BIT)	min
sampledImageIntegerSampleCounts	-	VK_SAMPLE_COUNT_1_BIT	min
sampledImageDepthSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT | VK_SAMPLE_COUNT_4_BIT)	min
sampledImageStencilSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT | VK_SAMPLE_COUNT_4_BIT)	min
storageImageSampleCounts	VK_SAMPLE_COUNT_1_BIT	(VK_SAMPLE_COUNT_1_BIT | VK_SAMPLE_COUNT_4_BIT)	min
maxSampleMaskWords	-	1	min
timestampComputeAndGraphics	-	- (Vulkan Core) VK_TRUE (Vulkan Roadmap 2024, Vulkan 1.4)	Boolean
timestampPeriod	-	-	duration
maxClipDistances	0	8	min
maxCullDistances	0	8	min
maxCombinedClipAndCullDistances	0	8	min
discreteQueuePriorities	-	2	min
pointSizeRange	(1.0,1.0)	(1.0,64.0 - ULP) [6] (Vulkan Core) (1.0,256.0 - pointSizeGranularity) (Vulkan 1.4)	(max,min)
lineWidthRange	(1.0,1.0)	(1.0,8.0 - ULP) [7]	(max,min)
pointSizeGranularity	0.0	1.0 [6] (Vulkan Core) 0.125 (Vulkan Roadmap 2022, Vulkan 1.4)	max, fixed point increment

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
lineWidthGranularity	0.0	1.0 [7] (Vulkan Core) 0.5 (Vulkan Roadmap 2022, Vulkan 1.4)	max, fixed point increment
strictLines	-	-	implementation-dependent
standardSampleLocations	-	- (Vulkan Core) VK_TRUE (Vulkan Roadmap 2022, Vulkan 1.4)	Boolean
optimalBufferCopyOffsetAlignment	-	-	recommendation
optimalBufferCopyRowPitchAlignment	-	-	recommendation
nonCoherentAtomSize	-	256	max
maxPushDescriptors	-	32	min
maxMultiviewViewCount	-	6	min
maxMultiviewInstanceIndex	-	2^{27}-1	min
maxDiscardRectangles	0	4	min
sampleLocationSampleCounts	-	VK_SAMPLE_COUNT_4_BIT	min
maxSampleLocationGridSize	-	(1,1)	min
sampleLocationCoordinateRange	-	(0.0, 0.9375)	(max,min)
sampleLocationSubPixelBits	-	4	min
variableSampleLocations	-	VK_FALSE	implementation-dependent
nativeUnalignedPerformance	-	VK_FALSE	implementation-dependent
minImportedHostPointerAlignment	-	65536	max
filterMinmaxSingleComponentFormats	-	-	implementation-dependent
filterMinmaxImageComponentMapping	-	-	implementation-dependent
advancedBlendMaxColorAttachments	-	1	min
advancedBlendIndependentBlend	-	VK_FALSE	implementation-dependent

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
advancedBlendNonPremultipliedSrcColor	-	VK_FALSE	implementation-dependent
advancedBlendNonPremultipliedDstColor	-	VK_FALSE	implementation-dependent
advancedBlendCorrelatedOverlap	-	VK_FALSE	implementation-dependent
advancedBlendAllOperations	-	VK_FALSE	implementation-dependent
maxPerSetDescriptors	-	1024	min
maxMemoryAllocationSize	-	2^{30}	min
maxBufferSize	-	2^{30}	min
primitiveOverestimationSize	-	0.0	min
maxExtraPrimitiveOverestimationSize	-	0.0	min
extraPrimitiveOverestimationSizeGranularity	-	0.0	min
primitiveUnderestimation	-	VK_FALSE	implementation-dependent
conservativePointAndLineRasterization	-	VK_FALSE	implementation-dependent
degenerateTrianglesRasterized	-	VK_FALSE	implementation-dependent
degenerateLinesRasterized	-	VK_FALSE	implementation-dependent
fullyCoveredFragmentShaderInputVariable	-	VK_FALSE	implementation-dependent
conservativeRasterizationPostDepthCoverage	-	VK_FALSE	implementation-dependent
maxUpdateAfterBindDescriptorsInAllPools	-	500000	min
shaderUniformBufferArrayNonUniformIndexingNative	-	VK_FALSE	implementation-dependent
shaderSampledImageArrayNonUniformIndexingNative	-	VK_FALSE	implementation-dependent
shaderStorageBufferArrayNonUniformIndexingNative	-	VK_FALSE	implementation-dependent
shaderStorageImageArrayNonUniformIndexingNative	-	VK_FALSE	implementation-dependent
shaderInputAttachmentArrayNonUniformIndexingNative	-	VK_FALSE	implementation-dependent

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
maxPerStageDescriptorUpdateAfterBindSamplers	-	500000 [9]	min
maxPerStageDescriptorUpdateAfterBindUniformBuffers	-	12 [9]	min
maxPerStageDescriptorUpdateAfterBindStorageBuffers	-	500000 [9]	min
maxPerStageDescriptorUpdateAfterBindSampledImages	-	500000 [9]	min
maxPerStageDescriptorUpdateAfterBindStorageImages	-	500000 [9]	min
maxPerStageDescriptorUpdateAfterBindInputAttachments	-	4 (Vulkan Core) 7 (Vulkan Roadmap 2022) [14]	min
maxPerStageUpdateAfterBindResources	-	500000	min
maxDescriptorSetUpdateAfterBindSamplers	-	500000 [9]	min
maxDescriptorSetUpdateAfterBindUniformBuffers	-	72 [8] [9]	min, $n \times$ PerStage
maxDescriptorSetUpdateAfterBindUniformBuffersDynamic	-	8 [13]	min
maxDescriptorSetUpdateAfterBindStorageBuffers	-	500000 [9]	min
maxDescriptorSetUpdateAfterBindStorageBuffersDynamic	-	4 [13]	min
maxDescriptorSetUpdateAfterBindTotalUniformBuffersDynamic	-	maxDescriptorSetUpdateAfterBindUniformBuffersDynamic [13]	min
maxDescriptorSetUpdateAfterBindTotalStorageBuffersDynamic	-	maxDescriptorSetUpdateAfterBindStorageBuffersDynamic [13]	min
maxDescriptorSetUpdateAfterBindTotalBuffersDynamic	-	maxDescriptorSetUpdateAfterBindUniformBuffersDynamic + maxDescriptorSetUpdateAfterBindStorageBuffersDynamic [13]	min
maxDescriptorSetUpdateAfterBindSampledImages	-	500000 [9]	min
maxDescriptorSetUpdateAfterBindStorageImages	-	500000 [9]	min
maxDescriptorSetUpdateAfterBindInputAttachments	-	4 [14]	min
maxInlineUniformBlockSize	-	256	min
maxPerStageDescriptorInlineUniformBlocks	-	4	min
maxPerStageDescriptorUpdateAfterBindInlineUniformBlocks	-	4 [9]	min
maxDescriptorSetInlineUniformBlocks	-	4	min

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
maxDescriptorSetUpdateAfterBindInlineUniformBlocks	-	4 [9]	min
maxInlineUniformTotalSize	-	256	min
maxVertexAttribDivisor	-	2^{16}-1	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskWorkGroupTotalCount	-	2^{22}	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskWorkGroupCount	-	(65535,65535,65535)	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskWorkGroupInvocations	-	128	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskWorkGroupSize	-	(128,128,128)	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskPayloadSize	-	16384	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskSharedMemorySize	-	32768	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskPayloadAndSharedMemorySize	-	32768	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupTotalCount	-	2^{22}	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupCount	-	(65535,65535,65535)	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupInvocations	-	128	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupSize	-	(128,128,128)	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshSharedMemorySize	-	28672	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshPayloadAndSharedMemorySize	-	28672	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshOutputMemorySize	-	32768	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshPayloadAndOutputMemorySize	-	48128	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshOutputComponents	-	128	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshOutputVertices	-	256	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshOutputPrimitives	-	256	min

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshOutputLayers	-	8	min
VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshMultiviewViewCount	-	1	min
VkPhysicalDeviceMeshShaderPropertiesEXT::meshOutputPerVertexGranularity	0	32	max
VkPhysicalDeviceMeshShaderPropertiesEXT::meshOutputPerPrimitiveGranularity	0	32	max
VkPhysicalDeviceMeshShaderPropertiesEXT::maxPreferredTaskWorkGroupInvocations	-	-	implementation-dependent
VkPhysicalDeviceMeshShaderPropertiesEXT::maxPreferredMeshWorkGroupInvocations	-	-	implementation-dependent
VkPhysicalDeviceMeshShaderPropertiesEXT::prefersLocalInvocationVertexOutput	-	-	implementation-dependent
VkPhysicalDeviceMeshShaderPropertiesEXT::prefersLocalInvocationPrimitiveOutput	-	-	implementation-dependent
VkPhysicalDeviceMeshShaderPropertiesEXT::prefersCompactVertexOutput	-	-	implementation-dependent
VkPhysicalDeviceMeshShaderPropertiesEXT::prefersCompactPrimitiveOutput	-	-	implementation-dependent
maxTransformFeedbackStreams	-	1	min
maxTransformFeedbackBuffers	-	1	min
maxTransformFeedbackBufferSize	-	2^{27}	min
maxTransformFeedbackStreamDataSize	-	512	min
maxTransformFeedbackBufferDataSize	-	512	min
maxTransformFeedbackBufferDataStride	-	512	min
transformFeedbackQueries	-	VK_FALSE	implementation-dependent
transformFeedbackStreamsLinesTriangles	-	VK_FALSE	implementation-dependent
transformFeedbackRasterizationStreamSelect	-	VK_FALSE	implementation-dependent
transformFeedbackDraw	-	VK_FALSE	implementation-dependent
splitBarrierReservedSharedMemory	-	8	max
minFragmentDensityTexelSize	-	(1,1)	min
maxFragmentDensityTexelSize	-	(1,1)	min

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
fragmentDensityInvocations	-	-	implementation-dependent
subsampledLoads	VK_TRUE	VK_FALSE	implementation-dependent
subsampledCoarseReconstructionEarlyAccess	VK_FALSE	VK_FALSE	implementation-dependent
maxSubsampledArrayLayers	2	2	min
maxDescriptorSetSubsampledSamplers	1	1	min
VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleSize	-	32	exact
VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxRayRecursionDepth	-	1	min
maxShaderGroupStride	-	4096	min
shaderGroupBaseAlignment	-	64	max
maxGeometryCount	-	2^{24}-1	min
maxInstanceCount	-	2^{24}-1	min
maxPrimitiveCount	-	2^{29}-1	min
maxPerStageDescriptorAccelerationStructures	-	16	min
maxPerStageDescriptorUpdateAfterBindAccelerationStructures	-	500000 [9]	min
maxDescriptorSetAccelerationStructures	-	16	min
maxDescriptorSetUpdateAfterBindAccelerationStructures	-	500000 [9]	min
minAccelerationStructureScratchOffsetAlignment	-	256	max
shaderGroupHandleCaptureReplaySize	-	64	max
maxRayDispatchInvocationCount	-	2^{30}	min
shaderGroupHandleAlignment	-	32	max
maxRayHitAttributeSize	-	32	min
maxTimelineSemaphoreValueDifference	-	2^{31}-1	min
lineSubPixelPrecisionBits	-	4	min
maxIndirectSequenceCount	-	2^{20}	min
maxIndirectCommandsTokenCount	-	16	min
maxIndirectCommandsTokenOffset	-	2047	min
maxIndirectPipelineCount	-	2^{12}	min

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
deviceGeneratedCommandsTransformFeedback	-	false	implementation-dependent
deviceGeneratedCommandsMultiDrawIndirectCount	-	false	implementation-dependent
maxIndirectShaderObjectCount	0	2^{12}	implementation-dependent
maxIndirectCommandsIndirectStride	-	2048	min
supportedIndirectCommandsInputModes	-	VK_INDIRECT_COMMANDS_INPUT_MODE_VULKAN_INDEX_BUFFER_EXT	min
supportedIndirectCommandsShaderStages	-	(VK_SHADER_STAGE_COMPUTE_BIT | VK_SHADER_STAGE_VERTEX_BIT | VK_SHADER_STAGE_FRAGMENT_BIT)	min
supportedIndirectCommandsShaderStagesPipelineBinding	-	0	min
supportedIndirectCommandsShaderStagesShaderBinding	-	0	min
maxCustomBorderColorSamplers	-	32	min
robustStorageBufferAccessSizeAlignment	-	4	max
robustUniformBufferAccessSizeAlignment	-	256	max
minFragmentShadingRateAttachmentTexelSize	(0,0)	(32,32)	max
maxFragmentShadingRateAttachmentTexelSize	(0,0)	(8,8)	min
maxFragmentShadingRateAttachmentTexelSizeAspectRatio	0	1	min
primitiveFragmentShadingRateWithMultipleViewports	VK_FALSE	VK_FALSE	implementation-dependent
layeredShadingRateAttachments	VK_FALSE	VK_FALSE	implementation-dependent
fragmentShadingRateNonTrivialCombinerOps	-	VK_FALSE	implementation-dependent
maxFragmentSize	-	(2,2)	min
maxFragmentSizeAspectRatio	-	2	min
maxFragmentShadingRateCoverageSamples	-	16	min

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
maxFragmentShadingRateRasterizationSamples	-	VK_SAMPLE_COUNT_4_BIT	min
fragmentShadingRateWithShaderDepthStencilWrites	-	VK_FALSE	implementation-dependent
fragmentShadingRateWithSampleMask	-	VK_FALSE	implementation-dependent
fragmentShadingRateWithShaderSampleMask	-	VK_FALSE	implementation-dependent
fragmentShadingRateWithConservativeRasterization	-	VK_FALSE	implementation-dependent
fragmentShadingRateWithFragmentShaderInterlock	-	VK_FALSE	implementation-dependent
fragmentShadingRateWithCustomSampleLocations	-	VK_FALSE	implementation-dependent
fragmentShadingRateStrictMultiplyCombiner	-	VK_FALSE	implementation-dependent
combinedImageSamplerDescriptorSingleArray	-	VK_FALSE	implementation-dependent
bufferlessPushDescriptors	-	VK_FALSE	implementation-dependent
allowSamplerImageViewPostSubmitCreation	-	VK_FALSE	implementation-dependent
descriptorBufferOffsetAlignment	-	256	max
maxDescriptorBufferBindings	-	3	min
maxResourceDescriptorBufferBindings	-	1	min
maxSamplerDescriptorBufferBindings	-	1	min
maxEmbeddedImmutableSamplerBindings	-	1	min
maxEmbeddedImmutableSamplers	-	2032	min
bufferCaptureReplayDescriptorDataSize	-	64	max
imageCaptureReplayDescriptorDataSize	-	64	max
imageViewCaptureReplayDescriptorDataSize	-	64	max
samplerCaptureReplayDescriptorDataSize	-	64	max
accelerationStructureCaptureReplayDescriptorDataSize	-	64	max
samplerDescriptorSize	-	256	max
combinedImageSamplerDescriptorSize	-	256	max

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
sampledImageDescriptorSize	-	256	max
storageImageDescriptorSize	-	256	max
uniformTexelBufferDescriptorSize	-	256	max
robustUniformTexelBufferDescriptorSize	-	256	max
storageTexelBufferDescriptorSize	-	256	max
robustStorageTexelBufferDescriptorSize	-	256	max
uniformBufferDescriptorSize	-	256	max
robustUniformBufferDescriptorSize	-	256	max
storageBufferDescriptorSize	-	256	max
robustStorageBufferDescriptorSize	-	256	max
inputAttachmentDescriptorSize	-	256	max
accelerationStructureDescriptorSize	-	256	max
maxSamplerDescriptorBufferRange	-	2^{11} × samplerDescriptorSize	min
maxResourceDescriptorBufferRange	-	(2^{20} - 2^{15}) × maxResourceDescriptorSize [12]	min
samplerDescriptorBufferAddressSpaceSize	-	2^{27}	min
resourceDescriptorBufferAddressSpaceSize	-	2^{27}	min
descriptorBufferAddressSpaceSize	-	2^{27}	min
combinedImageSamplerDensityMapDescriptorSize	-	256	max
maxMultiDrawCount	-	1024	min
maxCommandBufferNestingLevel	-	1	min
graphicsPipelineLibraryFastLinking	-	VK_FALSE	implementation-dependent
graphicsPipelineLibraryIndependentInterpolationDecoration	-	VK_FALSE	implementation-dependent
triStripVertexOrderIndependentOfProvokingVertex	-	VK_FALSE	implementation-dependent
dynamicPrimitiveTopologyUnrestricted	-	-	implementation-dependent
VkPhysicalDeviceOpacityMicromapPropertiesKHR::maxOpacity2StateSubdivisionLevel	-	3	min
VkPhysicalDeviceOpacityMicromapPropertiesKHR::maxOpacity4StateSubdivisionLevel	-	3	min

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
VkPhysicalDeviceOpacityMicromapPropertiesKHR::maxOpacityLossy4StateSubdivisionLevel	-	3	min
VkPhysicalDeviceOpacityMicromapPropertiesKHR::maxMicromapTriangles	-	2^{27}	min
VkPhysicalDeviceOpacityMicromapPropertiesEXT::maxOpacity2StateSubdivisionLevel	-	3	min
VkPhysicalDeviceOpacityMicromapPropertiesEXT::maxOpacity4StateSubdivisionLevel	-	3	min
maxDecompressionIndirectCount	1	2^{16}-1	min
minPlacedMemoryMapAlignment	-	65536	max
separateDepthStencilAttachmentAccess	VK_FALSE	-	implementation-dependent
maxVectorComponents	-	1024	min
maxShaderBindingTableRecordIndex	-	2^{28}-1	min
resolveSrgbFormatAppliesTransferFunction	-	-	implementation-dependent
resolveSrgbFormatSupportsTransferFunctionControl	VK_FALSE	VK_FALSE	min
samplerHeapAlignment	-	65536	max
resourceHeapAlignment	-	65536	max
maxSamplerHeapSize	-	max(4000 × samplerDescriptorSize + minSamplerHeapReservedRange, 2048 × samplerDescriptorSize + minSamplerHeapReservedRangeWithEmbedded)	min
maxResourceHeapSize	-	(2^{20} - 2^{15}) × max(imageDescriptorSize, bufferDescriptorSize) + minResourceHeapReservedRange	min
minSamplerHeapReservedRange	-	96 × samplerDescriptorSize	max

Limit	Unsupported Limit	Supported Limit	Limit Type[1]
minSamplerHeapReservedRangeWithEmbedded	-	2048 × samplerDescriptorSize	max
minResourceHeapReservedRange	-	2^{15} × max(imageDescriptorSize, bufferDescriptorSize)	max
samplerDescriptorSize	-	32	max
imageDescriptorSize	-	64	max
bufferDescriptorSize	-	128	max
samplerDescriptorAlignment	-	min(32, samplerDescriptorSize)	max
imageDescriptorAlignment	-	min(64, imageDescriptorSize)	max
bufferDescriptorAlignment	-	min(128, bufferDescriptorSize)	max
maxPushDataSize	-	256	min
maxDescriptorHeapEmbeddedSamplers	-	2032	min
samplerYcbcrConversionCount	-	3	max
imageCaptureReplayOpaqueDataSize	-	-	implementation-dependent
maxShaderAbortMessageSize	0	65536	min
maxDeviceFaultCount	0	1	min

1

The **Limit Type** column specifies the limit is either the minimum limit all implementations **must** support, the maximum limit all implementations **must** support, or the exact value all implementations **must** support. For bitmasks a minimum limit is the least bits all implementations **must** set, but they **may** have additional bits set beyond this minimum.

2

The maxPerStageResources **must** be at least the smallest of the following:

- the sum of the maxPerStageDescriptorUniformBuffers, maxPerStageDescriptorStorageBuffers, maxPerStageDescriptorSampledImages, maxPerStageDescriptorStorageImages, maxPerStageDescriptorInputAttachments, maxColorAttachments limits, or
- 128.

It **may** not be possible to reach this limit in every stage.

3

See `maxViewportDimensions` for the **required** relationship to other limits.

4

See `viewportBoundsRange` for the **required** relationship to other limits.

5

The values `minInterpolationOffset` and `maxInterpolationOffset` describe the closed interval of supported interpolation offsets: [`minInterpolationOffset`, `maxInterpolationOffset`]. The ULP is determined by `subPixelInterpolationOffsetBits`. If `subPixelInterpolationOffsetBits` is 4, this provides increments of $(1/2^4) = 0.0625$, and thus the range of supported interpolation offsets would be [-0.5, 0.4375].

6

The point size ULP is determined by `pointSizeGranularity`. If the `pointSizeGranularity` is 0.125, the range of supported point sizes **must** be at least [1.0, 63.875].

7

The line width ULP is determined by `lineWidthGranularity`. If the `lineWidthGranularity` is 0.0625, the range of supported line widths **must** be at least [1.0, 7.9375].

8

The minimum `maxDescriptorSet*` limit is *n* times the corresponding *specification* minimum `maxPerStageDescriptor*` limit, where *n* is the number of shader stages supported by the VkPhysicalDevice. If all shader stages are supported, *n* = 6 (vertex, tessellation control, tessellation evaluation, geometry, fragment, compute).

9

This `UpdateAfterBind` descriptor limit **must** be greater than or equal to the corresponding `non`-UpdateAfterBind limit if the corresponding `UpdateAfterBind` feature is supported. Otherwise it **can** be zero.

10

If the `VK_KHR_portability_subset` extension is enabled, the required minimum value of `maxVertexInputBindings` is `8`.

12

`maxResourceDescriptorSize` is defined as the maximum value of `storageImageDescriptorSize`, `sampledImageDescriptorSize`, `robustUniformTexelBufferDescriptorSize`, `robustStorageTexelBufferDescriptorSize`, `robustUniformBufferDescriptorSize`, `robustStorageBufferDescriptorSize`, `inputAttachmentDescriptorSize`, and `accelerationStructureDescriptorSize`.

13

This `UpdateAfterBind` descriptor limit exists and is defined to have a non-zero supported limit for historical reasons. As VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used

with the corresponding descriptor type, and descriptors of this type **must** not be included in any descriptor set layout that includes bindings created with VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT, the limit has no practical effect and **may** be zero.

14

As VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT **must** not be used with the corresponding descriptor type, this limit only exists to allow implementations exposing a higher limit for descriptors of this type in a pipeline layout including any descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT than in a pipeline layout not including any descriptor set layouts created with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT and **may** be zero to indicate that only the corresponding non-UpdateAfterBind limit applies.

48.2. Additional Multisampling Capabilities

To query additional multisampling capabilities which **may** be supported for a specific sample count, beyond the minimum capabilities described for Limits above, call:

```
// Provided by VK_EXT_sample_locations
void vkGetPhysicalDeviceMultisamplePropertiesEXT(
    VkPhysicalDevice                            physicalDevice,
    VkSampleCountFlagBits                       samples,
    VkMultisamplePropertiesEXT*                 pMultisampleProperties);
```

- physicalDevice is the physical device from which to query the additional multisampling capabilities.
- samples is a VkSampleCountFlagBits value specifying the sample count to query capabilities for.
- pMultisampleProperties is a pointer to a VkMultisamplePropertiesEXT structure in which information about additional multisampling capabilities specific to the sample count is returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceMultisamplePropertiesEXT-physicalDevice-parameter
  physicalDevice **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceMultisamplePropertiesEXT-samples-parameter
  samples **must** be a valid VkSampleCountFlagBits value
- VUID-vkGetPhysicalDeviceMultisamplePropertiesEXT-pMultisampleProperties-parameter
  pMultisampleProperties **must** be a valid pointer to a VkMultisamplePropertiesEXT structure

The VkMultisamplePropertiesEXT structure is defined as:

```
// Provided by VK_EXT_sample_locations
typedef struct VkMultisamplePropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkExtent2D         maxSampleLocationGridSize;
} VkMultisamplePropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxSampleLocationGridSize` is the maximum size of the pixel grid in which sample locations **can** vary.

Valid Usage (Implicit)

- VUID-VkMultisamplePropertiesEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_MULTISAMPLE_PROPERTIES_EXT
- VUID-VkMultisamplePropertiesEXT-pNext-pNext
  `pNext` **must** be `NULL`

If the sample count for which additional multisampling capabilities are requested using `vkGetPhysicalDeviceMultisamplePropertiesEXT` is set in `sampleLocationSampleCounts` the `width` and `height` members of `VkMultisamplePropertiesEXT`::`maxSampleLocationGridSize` **must** be greater than or equal to the corresponding members of `maxSampleLocationGridSize`, respectively, otherwise both members **must** be `0`.

48.3. Milestone Limits

48.3.1. Roadmap 2022

Implementations that claim support for the Roadmap 2022 milestone **must** satisfy the following additional limit requirements:

Limit	Supported Limit	Limit Type[1]
`maxImageDimension1D`	8192	min
`maxImageDimension2D`	8192	min
`maxImageDimensionCube`	8192	min
`maxImageArrayLayers`	2048	min
`maxUniformBufferRange`	65536	min
`bufferImageGranularity`	4096	max
`maxPerStageDescriptorSamplers`	64	min
`maxPerStageDescriptorUniformBuffers`	15	min

Limit	Supported Limit	Limit Type[1]
maxPerStageDescriptorStorageBuffers	30	min
maxPerStageDescriptorSampledImages	200	min
maxPerStageDescriptorStorageImages	16	min
maxPerStageResources	200	min
maxDescriptorSetSamplers	576	min
maxDescriptorSetUniformBuffers	90	min
maxDescriptorSetStorageBuffers	96	min
maxDescriptorSetSampledImages	1800	min
maxDescriptorSetStorageImages	144	min
maxFragmentCombinedOutputResources	16	min
maxComputeWorkGroupInvocations	256	min
maxComputeWorkGroupSize	(256,256,64)	min
subTexelPrecisionBits	8	min
mipmapPrecisionBits	6	min
maxSamplerLodBias	14	min
pointSizeGranularity	0.125	max
lineWidthGranularity	0.5	max
standardSampleLocations	VK_TRUE	Boolean
maxColorAttachments	7	min
subgroupSize	4	min
subgroupSupportedStages	VK_SHADER_STAGE_COMPUTE_BIT VK_SHADER_STAGE_FRAGMENT_BIT	bitfield

Limit	Supported Limit	Limit Type[1]
subgroupSupportedOperations	VK_SUBGROUP_FEATURE_BASIC_BIT VK_SUBGROUP_FEATURE_VOTE_BIT VK_SUBGROUP_FEATURE_ARITHMETIC_BIT VK_SUBGROUP_FEATURE_BALLOT_BIT VK_SUBGROUP_FEATURE_SHUFFLE_BIT VK_SUBGROUP_FEATURE_SHUFFLE_RELATIVE_BIT VK_SUBGROUP_FEATURE_QUAD_BIT	bitfield
shaderSignedZeroInfNanPreserveFloat16	VK_TRUE	Boolean
shaderSignedZeroInfNanPreserveFloat32	VK_TRUE	Boolean
maxSubgroupSize	4	min
maxPerStageDescriptorUpdateAfterBindInputAttachments	7	min

48.3.2. Roadmap 2024

Implementations that claim support for the Roadmap 2024 milestone **must** satisfy the following additional limit requirements:

Limit	Supported Limit	Limit Type[1]
shaderRoundingModeRTEFloat16	VK_TRUE	Boolean
shaderRoundingModeRTEFloat32	VK_TRUE	Boolean
timestampComputeAndGraphics	VK_TRUE	Boolean
maxColorAttachments	8	min
maxBoundDescriptorSets	7	min

48.3.3. Roadmap 2026

Implementations that claim support for the Roadmap 2026 milestone **must** satisfy the following additional limit requirements:

Limit	Supported Limit	Limit Type[1]
maxPerStageDescriptorUniformBuffers	200	min
maxPerStageDescriptorStorageBuffers	200	min

Limit	Supported Limit	Limit Type[1]
maxPerStageDescriptorInputAttachments	8	min
maxDescriptorSetUniformBuffers	1800	min
maxDescriptorSetStorageBuffers	1800	min
maxDescriptorSetInputAttachments	8	min
maxVertexOutputComponents	124	min
maxTessellationControlPerVertexInputComponents	128	min
maxTessellationControlPerVertexOutputComponents	128	min
maxTessellationControlTotalOutputComponents	4096	min
maxTessellationEvaluationInputComponents	128	min
maxTessellationEvaluationOutputComponents	128	min
maxGeometryOutputComponents	128	min
maxFragmentInputComponents	112	min
maxFragmentOutputAttachments	8	min
maxComputeSharedMemorySize	32768	min
subPixelPrecisionBits	8	min
maxViewportDimensions.width	8192	min
maxViewportDimensions.height	8192	min
maxFramebufferWidth	8192	min
maxFramebufferHeight	8192	min

Chapter 49. Formats

Supported buffer and image formats **may** vary across implementations. A minimum set of format features are guaranteed, but others **must** be explicitly queried before use to ensure they are supported by the implementation.

The features for the set of formats (VkFormat) supported by the implementation are queried individually using the vkGetPhysicalDeviceFormatProperties command.

49.1. Format Definition

The following image formats **can** be passed to, and **may** be returned from Vulkan commands. The memory required to store each format is discussed with that format, and also summarized in the Representation and Texel Block Size section and the Compatible formats table.

```
// Provided by VK_VERSION_1_0
typedef enum VkFormat {
    VK_FORMAT_UNDEFINED = 0,
    VK_FORMAT_R4G4_UNORM_PACK8 = 1,
    VK_FORMAT_R4G4B4A4_UNORM_PACK16 = 2,
    VK_FORMAT_B4G4R4A4_UNORM_PACK16 = 3,
    VK_FORMAT_R5G6B5_UNORM_PACK16 = 4,
    VK_FORMAT_B5G6R5_UNORM_PACK16 = 5,
    VK_FORMAT_R5G5B5A1_UNORM_PACK16 = 6,
    VK_FORMAT_B5G5R5A1_UNORM_PACK16 = 7,
    VK_FORMAT_A1R5G5B5_UNORM_PACK16 = 8,
    VK_FORMAT_R8_UNORM = 9,
    VK_FORMAT_R8_SNORM = 10,
    VK_FORMAT_R8_USCALED = 11,
    VK_FORMAT_R8_SSCALED = 12,
    VK_FORMAT_R8_UINT = 13,
    VK_FORMAT_R8_SINT = 14,
    VK_FORMAT_R8_SRGB = 15,
    VK_FORMAT_R8G8_UNORM = 16,
    VK_FORMAT_R8G8_SNORM = 17,
    VK_FORMAT_R8G8_USCALED = 18,
    VK_FORMAT_R8G8_SSCALED = 19,
    VK_FORMAT_R8G8_UINT = 20,
    VK_FORMAT_R8G8_SINT = 21,
    VK_FORMAT_R8G8_SRGB = 22,
    VK_FORMAT_R8G8B8_UNORM = 23,
    VK_FORMAT_R8G8B8_SNORM = 24,
    VK_FORMAT_R8G8B8_USCALED = 25,
    VK_FORMAT_R8G8B8_SSCALED = 26,
    VK_FORMAT_R8G8B8_UINT = 27,
    VK_FORMAT_R8G8B8_SINT = 28,
    VK_FORMAT_R8G8B8_SRGB = 29,
    VK_FORMAT_B8G8R8_UNORM = 30,
    VK_FORMAT_B8G8R8_SNORM = 31,
```

```
    VK_FORMAT_B8G8R8_USCALED = 32,
    VK_FORMAT_B8G8R8_SSCALED = 33,
    VK_FORMAT_B8G8R8_UINT = 34,
    VK_FORMAT_B8G8R8_SINT = 35,
    VK_FORMAT_B8G8R8_SRGB = 36,
    VK_FORMAT_R8G8B8A8_UNORM = 37,
    VK_FORMAT_R8G8B8A8_SNORM = 38,
    VK_FORMAT_R8G8B8A8_USCALED = 39,
    VK_FORMAT_R8G8B8A8_SSCALED = 40,
    VK_FORMAT_R8G8B8A8_UINT = 41,
    VK_FORMAT_R8G8B8A8_SINT = 42,
    VK_FORMAT_R8G8B8A8_SRGB = 43,
    VK_FORMAT_B8G8R8A8_UNORM = 44,
    VK_FORMAT_B8G8R8A8_SNORM = 45,
    VK_FORMAT_B8G8R8A8_USCALED = 46,
    VK_FORMAT_B8G8R8A8_SSCALED = 47,
    VK_FORMAT_B8G8R8A8_UINT = 48,
    VK_FORMAT_B8G8R8A8_SINT = 49,
    VK_FORMAT_B8G8R8A8_SRGB = 50,
    VK_FORMAT_A8B8G8R8_UNORM_PACK32 = 51,
    VK_FORMAT_A8B8G8R8_SNORM_PACK32 = 52,
    VK_FORMAT_A8B8G8R8_USCALED_PACK32 = 53,
    VK_FORMAT_A8B8G8R8_SSCALED_PACK32 = 54,
    VK_FORMAT_A8B8G8R8_UINT_PACK32 = 55,
    VK_FORMAT_A8B8G8R8_SINT_PACK32 = 56,
    VK_FORMAT_A8B8G8R8_SRGB_PACK32 = 57,
    VK_FORMAT_A2R10G10B10_UNORM_PACK32 = 58,
    VK_FORMAT_A2R10G10B10_SNORM_PACK32 = 59,
    VK_FORMAT_A2R10G10B10_USCALED_PACK32 = 60,
    VK_FORMAT_A2R10G10B10_SSCALED_PACK32 = 61,
    VK_FORMAT_A2R10G10B10_UINT_PACK32 = 62,
    VK_FORMAT_A2R10G10B10_SINT_PACK32 = 63,
    VK_FORMAT_A2B10G10R10_UNORM_PACK32 = 64,
    VK_FORMAT_A2B10G10R10_SNORM_PACK32 = 65,
    VK_FORMAT_A2B10G10R10_USCALED_PACK32 = 66,
    VK_FORMAT_A2B10G10R10_SSCALED_PACK32 = 67,
    VK_FORMAT_A2B10G10R10_UINT_PACK32 = 68,
    VK_FORMAT_A2B10G10R10_SINT_PACK32 = 69,
    VK_FORMAT_R16_UNORM = 70,
    VK_FORMAT_R16_SNORM = 71,
    VK_FORMAT_R16_USCALED = 72,
    VK_FORMAT_R16_SSCALED = 73,
    VK_FORMAT_R16_UINT = 74,
    VK_FORMAT_R16_SINT = 75,
    VK_FORMAT_R16_SFLOAT = 76,
    VK_FORMAT_R16G16_UNORM = 77,
    VK_FORMAT_R16G16_SNORM = 78,
    VK_FORMAT_R16G16_USCALED = 79,
    VK_FORMAT_R16G16_SSCALED = 80,
    VK_FORMAT_R16G16_UINT = 81,
    VK_FORMAT_R16G16_SINT = 82,
```

```
    VK_FORMAT_R16G16_SFLOAT = 83,
    VK_FORMAT_R16G16B16_UNORM = 84,
    VK_FORMAT_R16G16B16_SNORM = 85,
    VK_FORMAT_R16G16B16_USCALED = 86,
    VK_FORMAT_R16G16B16_SSCALED = 87,
    VK_FORMAT_R16G16B16_UINT = 88,
    VK_FORMAT_R16G16B16_SINT = 89,
    VK_FORMAT_R16G16B16_SFLOAT = 90,
    VK_FORMAT_R16G16B16A16_UNORM = 91,
    VK_FORMAT_R16G16B16A16_SNORM = 92,
    VK_FORMAT_R16G16B16A16_USCALED = 93,
    VK_FORMAT_R16G16B16A16_SSCALED = 94,
    VK_FORMAT_R16G16B16A16_UINT = 95,
    VK_FORMAT_R16G16B16A16_SINT = 96,
    VK_FORMAT_R16G16B16A16_SFLOAT = 97,
    VK_FORMAT_R32_UINT = 98,
    VK_FORMAT_R32_SINT = 99,
    VK_FORMAT_R32_SFLOAT = 100,
    VK_FORMAT_R32G32_UINT = 101,
    VK_FORMAT_R32G32_SINT = 102,
    VK_FORMAT_R32G32_SFLOAT = 103,
    VK_FORMAT_R32G32B32_UINT = 104,
    VK_FORMAT_R32G32B32_SINT = 105,
    VK_FORMAT_R32G32B32_SFLOAT = 106,
    VK_FORMAT_R32G32B32A32_UINT = 107,
    VK_FORMAT_R32G32B32A32_SINT = 108,
    VK_FORMAT_R32G32B32A32_SFLOAT = 109,
    VK_FORMAT_R64_UINT = 110,
    VK_FORMAT_R64_SINT = 111,
    VK_FORMAT_R64_SFLOAT = 112,
    VK_FORMAT_R64G64_UINT = 113,
    VK_FORMAT_R64G64_SINT = 114,
    VK_FORMAT_R64G64_SFLOAT = 115,
    VK_FORMAT_R64G64B64_UINT = 116,
    VK_FORMAT_R64G64B64_SINT = 117,
    VK_FORMAT_R64G64B64_SFLOAT = 118,
    VK_FORMAT_R64G64B64A64_UINT = 119,
    VK_FORMAT_R64G64B64A64_SINT = 120,
    VK_FORMAT_R64G64B64A64_SFLOAT = 121,
    VK_FORMAT_B10G11R11_UFLOAT_PACK32 = 122,
    VK_FORMAT_E5B9G9R9_UFLOAT_PACK32 = 123,
    VK_FORMAT_D16_UNORM = 124,
    VK_FORMAT_X8_D24_UNORM_PACK32 = 125,
    VK_FORMAT_D32_SFLOAT = 126,
    VK_FORMAT_S8_UINT = 127,
    VK_FORMAT_D16_UNORM_S8_UINT = 128,
    VK_FORMAT_D24_UNORM_S8_UINT = 129,
    VK_FORMAT_D32_SFLOAT_S8_UINT = 130,
    VK_FORMAT_BC1_RGB_UNORM_BLOCK = 131,
    VK_FORMAT_BC1_RGB_SRGB_BLOCK = 132,
    VK_FORMAT_BC1_RGBA_UNORM_BLOCK = 133,
```

```
    VK_FORMAT_BC1_RGBA_SRGB_BLOCK = 134,
    VK_FORMAT_BC2_UNORM_BLOCK = 135,
    VK_FORMAT_BC2_SRGB_BLOCK = 136,
    VK_FORMAT_BC3_UNORM_BLOCK = 137,
    VK_FORMAT_BC3_SRGB_BLOCK = 138,
    VK_FORMAT_BC4_UNORM_BLOCK = 139,
    VK_FORMAT_BC4_SNORM_BLOCK = 140,
    VK_FORMAT_BC5_UNORM_BLOCK = 141,
    VK_FORMAT_BC5_SNORM_BLOCK = 142,
    VK_FORMAT_BC6H_UFLOAT_BLOCK = 143,
    VK_FORMAT_BC6H_SFLOAT_BLOCK = 144,
    VK_FORMAT_BC7_UNORM_BLOCK = 145,
    VK_FORMAT_BC7_SRGB_BLOCK = 146,
    VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK = 147,
    VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK = 148,
    VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK = 149,
    VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK = 150,
    VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK = 151,
    VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK = 152,
    VK_FORMAT_EAC_R11_UNORM_BLOCK = 153,
    VK_FORMAT_EAC_R11_SNORM_BLOCK = 154,
    VK_FORMAT_EAC_R11G11_UNORM_BLOCK = 155,
    VK_FORMAT_EAC_R11G11_SNORM_BLOCK = 156,
    VK_FORMAT_ASTC_4x4_UNORM_BLOCK = 157,
    VK_FORMAT_ASTC_4x4_SRGB_BLOCK = 158,
    VK_FORMAT_ASTC_5x4_UNORM_BLOCK = 159,
    VK_FORMAT_ASTC_5x4_SRGB_BLOCK = 160,
    VK_FORMAT_ASTC_5x5_UNORM_BLOCK = 161,
    VK_FORMAT_ASTC_5x5_SRGB_BLOCK = 162,
    VK_FORMAT_ASTC_6x5_UNORM_BLOCK = 163,
    VK_FORMAT_ASTC_6x5_SRGB_BLOCK = 164,
    VK_FORMAT_ASTC_6x6_UNORM_BLOCK = 165,
    VK_FORMAT_ASTC_6x6_SRGB_BLOCK = 166,
    VK_FORMAT_ASTC_8x5_UNORM_BLOCK = 167,
    VK_FORMAT_ASTC_8x5_SRGB_BLOCK = 168,
    VK_FORMAT_ASTC_8x6_UNORM_BLOCK = 169,
    VK_FORMAT_ASTC_8x6_SRGB_BLOCK = 170,
    VK_FORMAT_ASTC_8x8_UNORM_BLOCK = 171,
    VK_FORMAT_ASTC_8x8_SRGB_BLOCK = 172,
    VK_FORMAT_ASTC_10x5_UNORM_BLOCK = 173,
    VK_FORMAT_ASTC_10x5_SRGB_BLOCK = 174,
    VK_FORMAT_ASTC_10x6_UNORM_BLOCK = 175,
    VK_FORMAT_ASTC_10x6_SRGB_BLOCK = 176,
    VK_FORMAT_ASTC_10x8_UNORM_BLOCK = 177,
    VK_FORMAT_ASTC_10x8_SRGB_BLOCK = 178,
    VK_FORMAT_ASTC_10x10_UNORM_BLOCK = 179,
    VK_FORMAT_ASTC_10x10_SRGB_BLOCK = 180,
    VK_FORMAT_ASTC_12x10_UNORM_BLOCK = 181,
    VK_FORMAT_ASTC_12x10_SRGB_BLOCK = 182,
    VK_FORMAT_ASTC_12x12_UNORM_BLOCK = 183,
    VK_FORMAT_ASTC_12x12_SRGB_BLOCK = 184,
```

```
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G8B8G8R8_422_UNORM = 1000156000,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_B8G8R8G8_422_UNORM = 1000156001,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM = 1000156002,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G8_B8R8_2PLANE_420_UNORM = 1000156003,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM = 1000156004,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G8_B8R8_2PLANE_422_UNORM = 1000156005,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM = 1000156006,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_R10X6_UNORM_PACK16 = 1000156007,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_R10X6G10X6_UNORM_2PACK16 = 1000156008,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16 = 1000156009,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16 = 1000156010,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16 = 1000156011,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16 = 1000156012,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16 = 1000156013,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16 = 1000156014,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16 = 1000156015,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16 = 1000156016,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_R12X4_UNORM_PACK16 = 1000156017,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_R12X4G12X4_UNORM_2PACK16 = 1000156018,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16 = 1000156019,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16 = 1000156020,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16 = 1000156021,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16 = 1000156022,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16 = 1000156023,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16 = 1000156024,
  // Provided by VK_VERSION_1_1
```

```
    VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16 = 1000156025,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16 = 1000156026,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G16B16G16R16_422_UNORM = 1000156027,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_B16G16R16G16_422_UNORM = 1000156028,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM = 1000156029,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G16_B16R16_2PLANE_420_UNORM = 1000156030,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM = 1000156031,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G16_B16R16_2PLANE_422_UNORM = 1000156032,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM = 1000156033,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_G8_B8R8_2PLANE_444_UNORM = 1000330000,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_G10X6_B10X6R10X6_2PLANE_444_UNORM_3PACK16 = 1000330001,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_G12X4_B12X4R12X4_2PLANE_444_UNORM_3PACK16 = 1000330002,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_G16_B16R16_2PLANE_444_UNORM = 1000330003,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_A4R4G4B4_UNORM_PACK16 = 1000340000,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_A4B4G4R4_UNORM_PACK16 = 1000340001,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_ASTC_4x4_SFLOAT_BLOCK = 1000066000,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_ASTC_5x4_SFLOAT_BLOCK = 1000066001,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_ASTC_5x5_SFLOAT_BLOCK = 1000066002,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_ASTC_6x5_SFLOAT_BLOCK = 1000066003,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_ASTC_6x6_SFLOAT_BLOCK = 1000066004,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_ASTC_8x5_SFLOAT_BLOCK = 1000066005,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_ASTC_8x6_SFLOAT_BLOCK = 1000066006,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_ASTC_8x8_SFLOAT_BLOCK = 1000066007,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_ASTC_10x5_SFLOAT_BLOCK = 1000066008,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_ASTC_10x6_SFLOAT_BLOCK = 1000066009,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_ASTC_10x8_SFLOAT_BLOCK = 1000066010,
```

```
  // Provided by VK_VERSION_1_3
    VK_FORMAT_ASTC_10x10_SFLOAT_BLOCK = 1000066011,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_ASTC_12x10_SFLOAT_BLOCK = 1000066012,
  // Provided by VK_VERSION_1_3
    VK_FORMAT_ASTC_12x12_SFLOAT_BLOCK = 1000066013,
  // Provided by VK_VERSION_1_4
    VK_FORMAT_A1B5G5R5_UNORM_PACK16 = 1000470000,
  // Provided by VK_VERSION_1_4
    VK_FORMAT_A8_UNORM = 1000470001,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_3x3x3_UNORM_BLOCK_EXT = 1000288000,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_3x3x3_SRGB_BLOCK_EXT = 1000288001,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_3x3x3_SFLOAT_BLOCK_EXT = 1000288002,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_4x3x3_UNORM_BLOCK_EXT = 1000288003,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_4x3x3_SRGB_BLOCK_EXT = 1000288004,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_4x3x3_SFLOAT_BLOCK_EXT = 1000288005,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_4x4x3_UNORM_BLOCK_EXT = 1000288006,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_4x4x3_SRGB_BLOCK_EXT = 1000288007,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_4x4x3_SFLOAT_BLOCK_EXT = 1000288008,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_4x4x4_UNORM_BLOCK_EXT = 1000288009,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_4x4x4_SRGB_BLOCK_EXT = 1000288010,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_4x4x4_SFLOAT_BLOCK_EXT = 1000288011,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_5x4x4_UNORM_BLOCK_EXT = 1000288012,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_5x4x4_SRGB_BLOCK_EXT = 1000288013,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_5x4x4_SFLOAT_BLOCK_EXT = 1000288014,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_5x5x4_UNORM_BLOCK_EXT = 1000288015,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_5x5x4_SRGB_BLOCK_EXT = 1000288016,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_5x5x4_SFLOAT_BLOCK_EXT = 1000288017,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_5x5x5_UNORM_BLOCK_EXT = 1000288018,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_5x5x5_SRGB_BLOCK_EXT = 1000288019,
  // Provided by VK_EXT_texture_compression_astc_3d
```

```
    VK_FORMAT_ASTC_5x5x5_SFLOAT_BLOCK_EXT = 1000288020,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_6x5x5_UNORM_BLOCK_EXT = 1000288021,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_6x5x5_SRGB_BLOCK_EXT = 1000288022,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_6x5x5_SFLOAT_BLOCK_EXT = 1000288023,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_6x6x5_UNORM_BLOCK_EXT = 1000288024,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_6x6x5_SRGB_BLOCK_EXT = 1000288025,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_6x6x5_SFLOAT_BLOCK_EXT = 1000288026,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_6x6x6_UNORM_BLOCK_EXT = 1000288027,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_6x6x6_SRGB_BLOCK_EXT = 1000288028,
  // Provided by VK_EXT_texture_compression_astc_3d
    VK_FORMAT_ASTC_6x6x6_SFLOAT_BLOCK_EXT = 1000288029,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G8B8G8R8_422_UNORM_KHR = VK_FORMAT_G8B8G8R8_422_UNORM,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_B8G8R8G8_422_UNORM_KHR = VK_FORMAT_B8G8R8G8_422_UNORM,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM_KHR = VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G8_B8R8_2PLANE_420_UNORM_KHR = VK_FORMAT_G8_B8R8_2PLANE_420_UNORM,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM_KHR = VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G8_B8R8_2PLANE_422_UNORM_KHR = VK_FORMAT_G8_B8R8_2PLANE_422_UNORM,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM_KHR = VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_R10X6_UNORM_PACK16_KHR = VK_FORMAT_R10X6_UNORM_PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_R10X6G10X6_UNORM_2PACK16_KHR = VK_FORMAT_R10X6G10X6_UNORM_2PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16_KHR =
VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16_KHR =
VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16_KHR =
VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16_KHR =
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16_KHR =
```

```
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16_KHR =
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16_KHR =
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16_KHR =
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_R12X4_UNORM_PACK16_KHR = VK_FORMAT_R12X4_UNORM_PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_R12X4G12X4_UNORM_2PACK16_KHR = VK_FORMAT_R12X4G12X4_UNORM_2PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16_KHR =
VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16_KHR =
VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16_KHR =
VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16_KHR =
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16_KHR =
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16_KHR =
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16_KHR =
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16_KHR =
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G16B16G16R16_422_UNORM_KHR = VK_FORMAT_G16B16G16R16_422_UNORM,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_B16G16R16G16_422_UNORM_KHR = VK_FORMAT_B16G16R16G16_422_UNORM,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM_KHR =
VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G16_B16R16_2PLANE_420_UNORM_KHR = VK_FORMAT_G16_B16R16_2PLANE_420_UNORM,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM_KHR =
VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM,
  // Provided by VK_KHR_sampler_ycbcr_conversion
```

```
    VK_FORMAT_G16_B16R16_2PLANE_422_UNORM_KHR = VK_FORMAT_G16_B16R16_2PLANE_422_UNORM,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM_KHR =
VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM,
  // Provided by VK_KHR_maintenance5
    VK_FORMAT_A1B5G5R5_UNORM_PACK16_KHR = VK_FORMAT_A1B5G5R5_UNORM_PACK16,
  // Provided by VK_KHR_maintenance5
    VK_FORMAT_A8_UNORM_KHR = VK_FORMAT_A8_UNORM,
} VkFormat;
```

- VK_FORMAT_UNDEFINED specifies that the format is not specified.
- VK_FORMAT_R4G4_UNORM_PACK8 specifies a two-component, 8-bit packed unsigned normalized format that has a 4-bit R component in bits 4..7, and a 4-bit G component in bits 0..3.
- VK_FORMAT_R4G4B4A4_UNORM_PACK16 specifies a four-component, 16-bit packed unsigned normalized format that has a 4-bit R component in bits 12..15, a 4-bit G component in bits 8..11, a 4-bit B component in bits 4..7, and a 4-bit A component in bits 0..3.
- VK_FORMAT_B4G4R4A4_UNORM_PACK16 specifies a four-component, 16-bit packed unsigned normalized format that has a 4-bit B component in bits 12..15, a 4-bit G component in bits 8..11, a 4-bit R component in bits 4..7, and a 4-bit A component in bits 0..3.
- VK_FORMAT_A4R4G4B4_UNORM_PACK16 specifies a four-component, 16-bit packed unsigned normalized format that has a 4-bit A component in bits 12..15, a 4-bit R component in bits 8..11, a 4-bit G component in bits 4..7, and a 4-bit B component in bits 0..3.
- VK_FORMAT_A4B4G4R4_UNORM_PACK16 specifies a four-component, 16-bit packed unsigned normalized format that has a 4-bit A component in bits 12..15, a 4-bit B component in bits 8..11, a 4-bit G component in bits 4..7, and a 4-bit R component in bits 0..3.
- VK_FORMAT_R5G6B5_UNORM_PACK16 specifies a three-component, 16-bit packed unsigned normalized format that has a 5-bit R component in bits 11..15, a 6-bit G component in bits 5..10, and a 5-bit B component in bits 0..4.
- VK_FORMAT_B5G6R5_UNORM_PACK16 specifies a three-component, 16-bit packed unsigned normalized format that has a 5-bit B component in bits 11..15, a 6-bit G component in bits 5..10, and a 5-bit R component in bits 0..4.
- VK_FORMAT_R5G5B5A1_UNORM_PACK16 specifies a four-component, 16-bit packed unsigned normalized format that has a 5-bit R component in bits 11..15, a 5-bit G component in bits 6..10, a 5-bit B component in bits 1..5, and a 1-bit A component in bit 0.
- VK_FORMAT_B5G5R5A1_UNORM_PACK16 specifies a four-component, 16-bit packed unsigned normalized format that has a 5-bit B component in bits 11..15, a 5-bit G component in bits 6..10, a 5-bit R component in bits 1..5, and a 1-bit A component in bit 0.
- VK_FORMAT_A1R5G5B5_UNORM_PACK16 specifies a four-component, 16-bit packed unsigned normalized format that has a 1-bit A component in bit 15, a 5-bit R component in bits 10..14, a 5-bit G component in bits 5..9, and a 5-bit B component in bits 0..4.
- VK_FORMAT_A1B5G5R5_UNORM_PACK16 specifies a four-component, 16-bit packed unsigned normalized format that has a 1-bit A component in bit 15, a 5-bit B component in bits 10..14, a 5-bit G component in bits 5..9, and a 5-bit R component in bits 0..4.

- VK_FORMAT_A8_UNORM specifies a one-component, 8-bit unsigned normalized format that has a single 8-bit A component.
- VK_FORMAT_R8_UNORM specifies a one-component, 8-bit unsigned normalized format that has a single 8-bit R component.
- VK_FORMAT_R8_SNORM specifies a one-component, 8-bit signed normalized format that has a single 8-bit R component.
- VK_FORMAT_R8_USCALED specifies a one-component, 8-bit unsigned scaled integer format that has a single 8-bit R component.
- VK_FORMAT_R8_SSCALED specifies a one-component, 8-bit signed scaled integer format that has a single 8-bit R component.
- VK_FORMAT_R8_UINT specifies a one-component, 8-bit unsigned integer format that has a single 8-bit R component.
- VK_FORMAT_R8_SINT specifies a one-component, 8-bit signed integer format that has a single 8-bit R component.
- VK_FORMAT_R8_SRGB specifies a one-component, 8-bit unsigned normalized format that has a single 8-bit R component stored with sRGB nonlinear encoding.
- VK_FORMAT_R8G8_UNORM specifies a two-component, 16-bit unsigned normalized format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- VK_FORMAT_R8G8_SNORM specifies a two-component, 16-bit signed normalized format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- VK_FORMAT_R8G8_USCALED specifies a two-component, 16-bit unsigned scaled integer format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- VK_FORMAT_R8G8_SSCALED specifies a two-component, 16-bit signed scaled integer format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- VK_FORMAT_R8G8_UINT specifies a two-component, 16-bit unsigned integer format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- VK_FORMAT_R8G8_SINT specifies a two-component, 16-bit signed integer format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- VK_FORMAT_R8G8_SRGB specifies a two-component, 16-bit unsigned normalized format that has an 8-bit R component stored with sRGB nonlinear encoding in byte 0, and an 8-bit G component stored with sRGB nonlinear encoding in byte 1.
- VK_FORMAT_R8G8B8_UNORM specifies a three-component, 24-bit unsigned normalized format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- VK_FORMAT_R8G8B8_SNORM specifies a three-component, 24-bit signed normalized format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- VK_FORMAT_R8G8B8_USCALED specifies a three-component, 24-bit unsigned scaled format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- VK_FORMAT_R8G8B8_SSCALED specifies a three-component, 24-bit signed scaled format that

has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.

- VK_FORMAT_R8G8B8_UINT specifies a three-component, 24-bit unsigned integer format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- VK_FORMAT_R8G8B8_SINT specifies a three-component, 24-bit signed integer format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- VK_FORMAT_R8G8B8_SRGB specifies a three-component, 24-bit unsigned normalized format that has an 8-bit R component stored with sRGB nonlinear encoding in byte 0, an 8-bit G component stored with sRGB nonlinear encoding in byte 1, and an 8-bit B component stored with sRGB nonlinear encoding in byte 2.
- VK_FORMAT_B8G8R8_UNORM specifies a three-component, 24-bit unsigned normalized format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- VK_FORMAT_B8G8R8_SNORM specifies a three-component, 24-bit signed normalized format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- VK_FORMAT_B8G8R8_USCALED specifies a three-component, 24-bit unsigned scaled format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- VK_FORMAT_B8G8R8_SSCALED specifies a three-component, 24-bit signed scaled format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- VK_FORMAT_B8G8R8_UINT specifies a three-component, 24-bit unsigned integer format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- VK_FORMAT_B8G8R8_SINT specifies a three-component, 24-bit signed integer format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- VK_FORMAT_B8G8R8_SRGB specifies a three-component, 24-bit unsigned normalized format that has an 8-bit B component stored with sRGB nonlinear encoding in byte 0, an 8-bit G component stored with sRGB nonlinear encoding in byte 1, and an 8-bit R component stored with sRGB nonlinear encoding in byte 2.
- VK_FORMAT_R8G8B8A8_UNORM specifies a four-component, 32-bit unsigned normalized format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- VK_FORMAT_R8G8B8A8_SNORM specifies a four-component, 32-bit signed normalized format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- VK_FORMAT_R8G8B8A8_USCALED specifies a four-component, 32-bit unsigned scaled format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.

- VK_FORMAT_R8G8B8A8_SSCALED specifies a four-component, 32-bit signed scaled format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- VK_FORMAT_R8G8B8A8_UINT specifies a four-component, 32-bit unsigned integer format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- VK_FORMAT_R8G8B8A8_SINT specifies a four-component, 32-bit signed integer format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- VK_FORMAT_R8G8B8A8_SRGB specifies a four-component, 32-bit unsigned normalized format that has an 8-bit R component stored with sRGB nonlinear encoding in byte 0, an 8-bit G component stored with sRGB nonlinear encoding in byte 1, an 8-bit B component stored with sRGB nonlinear encoding in byte 2, and an 8-bit A component in byte 3.
- VK_FORMAT_B8G8R8A8_UNORM specifies a four-component, 32-bit unsigned normalized format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- VK_FORMAT_B8G8R8A8_SNORM specifies a four-component, 32-bit signed normalized format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- VK_FORMAT_B8G8R8A8_USCALED specifies a four-component, 32-bit unsigned scaled format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- VK_FORMAT_B8G8R8A8_SSCALED specifies a four-component, 32-bit signed scaled format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- VK_FORMAT_B8G8R8A8_UINT specifies a four-component, 32-bit unsigned integer format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- VK_FORMAT_B8G8R8A8_SINT specifies a four-component, 32-bit signed integer format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- VK_FORMAT_B8G8R8A8_SRGB specifies a four-component, 32-bit unsigned normalized format that has an 8-bit B component stored with sRGB nonlinear encoding in byte 0, an 8-bit G component stored with sRGB nonlinear encoding in byte 1, an 8-bit R component stored with sRGB nonlinear encoding in byte 2, and an 8-bit A component in byte 3.
- VK_FORMAT_A8B8G8R8_UNORM_PACK32 specifies a four-component, 32-bit packed unsigned normalized format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- VK_FORMAT_A8B8G8R8_SNORM_PACK32 specifies a four-component, 32-bit packed signed normalized format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- VK_FORMAT_A8B8G8R8_USCALED_PACK32 specifies a four-component, 32-bit packed unsigned scaled integer format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits

16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.

- VK_FORMAT_A8B8G8R8_SSCALED_PACK32 specifies a four-component, 32-bit packed signed scaled integer format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- VK_FORMAT_A8B8G8R8_UINT_PACK32 specifies a four-component, 32-bit packed unsigned integer format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- VK_FORMAT_A8B8G8R8_SINT_PACK32 specifies a four-component, 32-bit packed signed integer format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- VK_FORMAT_A8B8G8R8_SRGB_PACK32 specifies a four-component, 32-bit packed unsigned normalized format that has an 8-bit A component in bits 24..31, an 8-bit B component stored with sRGB nonlinear encoding in bits 16..23, an 8-bit G component stored with sRGB nonlinear encoding in bits 8..15, and an 8-bit R component stored with sRGB nonlinear encoding in bits 0..7.
- VK_FORMAT_A2R10G10B10_UNORM_PACK32 specifies a four-component, 32-bit packed unsigned normalized format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- VK_FORMAT_A2R10G10B10_SNORM_PACK32 specifies a four-component, 32-bit packed signed normalized format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- VK_FORMAT_A2R10G10B10_USCALED_PACK32 specifies a four-component, 32-bit packed unsigned scaled integer format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- VK_FORMAT_A2R10G10B10_SSCALED_PACK32 specifies a four-component, 32-bit packed signed scaled integer format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- VK_FORMAT_A2R10G10B10_UINT_PACK32 specifies a four-component, 32-bit packed unsigned integer format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- VK_FORMAT_A2R10G10B10_SINT_PACK32 specifies a four-component, 32-bit packed signed integer format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- VK_FORMAT_A2B10G10R10_UNORM_PACK32 specifies a four-component, 32-bit packed unsigned normalized format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- VK_FORMAT_A2B10G10R10_SNORM_PACK32 specifies a four-component, 32-bit packed signed normalized format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- VK_FORMAT_A2B10G10R10_USCALED_PACK32 specifies a four-component, 32-bit packed unsigned scaled integer format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.

- VK_FORMAT_A2B10G10R10_SSCALED_PACK32 specifies a four-component, 32-bit packed signed scaled integer format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- VK_FORMAT_A2B10G10R10_UINT_PACK32 specifies a four-component, 32-bit packed unsigned integer format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- VK_FORMAT_A2B10G10R10_SINT_PACK32 specifies a four-component, 32-bit packed signed integer format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- VK_FORMAT_R16_UNORM specifies a one-component, 16-bit unsigned normalized format that has a single 16-bit R component.
- VK_FORMAT_R16_SNORM specifies a one-component, 16-bit signed normalized format that has a single 16-bit R component.
- VK_FORMAT_R16_USCALED specifies a one-component, 16-bit unsigned scaled integer format that has a single 16-bit R component.
- VK_FORMAT_R16_SSCALED specifies a one-component, 16-bit signed scaled integer format that has a single 16-bit R component.
- VK_FORMAT_R16_UINT specifies a one-component, 16-bit unsigned integer format that has a single 16-bit R component.
- VK_FORMAT_R16_SINT specifies a one-component, 16-bit signed integer format that has a single 16-bit R component.
- VK_FORMAT_R16_SFLOAT specifies a one-component, 16-bit signed floating-point format that has a single 16-bit R component.
- VK_FORMAT_R16G16_UNORM specifies a two-component, 32-bit unsigned normalized format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- VK_FORMAT_R16G16_SNORM specifies a two-component, 32-bit signed normalized format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- VK_FORMAT_R16G16_USCALED specifies a two-component, 32-bit unsigned scaled integer format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- VK_FORMAT_R16G16_SSCALED specifies a two-component, 32-bit signed scaled integer format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- VK_FORMAT_R16G16_UINT specifies a two-component, 32-bit unsigned integer format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- VK_FORMAT_R16G16_SINT specifies a two-component, 32-bit signed integer format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- VK_FORMAT_R16G16_SFLOAT specifies a two-component, 32-bit signed floating-point format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- VK_FORMAT_R16G16B16_UNORM specifies a three-component, 48-bit unsigned normalized format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- VK_FORMAT_R16G16B16_SNORM specifies a three-component, 48-bit signed normalized format

that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.

- VK_FORMAT_R16G16B16_USCALED specifies a three-component, 48-bit unsigned scaled integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- VK_FORMAT_R16G16B16_SSCALED specifies a three-component, 48-bit signed scaled integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- VK_FORMAT_R16G16B16_UINT specifies a three-component, 48-bit unsigned integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- VK_FORMAT_R16G16B16_SINT specifies a three-component, 48-bit signed integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- VK_FORMAT_R16G16B16_SFLOAT specifies a three-component, 48-bit signed floating-point format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- VK_FORMAT_R16G16B16A16_UNORM specifies a four-component, 64-bit unsigned normalized format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- VK_FORMAT_R16G16B16A16_SNORM specifies a four-component, 64-bit signed normalized format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- VK_FORMAT_R16G16B16A16_USCALED specifies a four-component, 64-bit unsigned scaled integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- VK_FORMAT_R16G16B16A16_SSCALED specifies a four-component, 64-bit signed scaled integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- VK_FORMAT_R16G16B16A16_UINT specifies a four-component, 64-bit unsigned integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- VK_FORMAT_R16G16B16A16_SINT specifies a four-component, 64-bit signed integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- VK_FORMAT_R16G16B16A16_SFLOAT specifies a four-component, 64-bit signed floating-point format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- VK_FORMAT_R32_UINT specifies a one-component, 32-bit unsigned integer format that has a single 32-bit R component.
- VK_FORMAT_R32_SINT specifies a one-component, 32-bit signed integer format that has a single 32-bit R component.

- VK_FORMAT_R32_SFLOAT specifies a one-component, 32-bit signed floating-point format that has a single 32-bit R component.
- VK_FORMAT_R32G32_UINT specifies a two-component, 64-bit unsigned integer format that has a 32-bit R component in bytes 0..3, and a 32-bit G component in bytes 4..7.
- VK_FORMAT_R32G32_SINT specifies a two-component, 64-bit signed integer format that has a 32-bit R component in bytes 0..3, and a 32-bit G component in bytes 4..7.
- VK_FORMAT_R32G32_SFLOAT specifies a two-component, 64-bit signed floating-point format that has a 32-bit R component in bytes 0..3, and a 32-bit G component in bytes 4..7.
- VK_FORMAT_R32G32B32_UINT specifies a three-component, 96-bit unsigned integer format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, and a 32-bit B component in bytes 8..11.
- VK_FORMAT_R32G32B32_SINT specifies a three-component, 96-bit signed integer format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, and a 32-bit B component in bytes 8..11.
- VK_FORMAT_R32G32B32_SFLOAT specifies a three-component, 96-bit signed floating-point format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, and a 32-bit B component in bytes 8..11.
- VK_FORMAT_R32G32B32A32_UINT specifies a four-component, 128-bit unsigned integer format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, a 32-bit B component in bytes 8..11, and a 32-bit A component in bytes 12..15.
- VK_FORMAT_R32G32B32A32_SINT specifies a four-component, 128-bit signed integer format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, a 32-bit B component in bytes 8..11, and a 32-bit A component in bytes 12..15.
- VK_FORMAT_R32G32B32A32_SFLOAT specifies a four-component, 128-bit signed floating-point format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, a 32-bit B component in bytes 8..11, and a 32-bit A component in bytes 12..15.
- VK_FORMAT_R64_UINT specifies a one-component, 64-bit unsigned integer format that has a single 64-bit R component.
- VK_FORMAT_R64_SINT specifies a one-component, 64-bit signed integer format that has a single 64-bit R component.
- VK_FORMAT_R64_SFLOAT specifies a one-component, 64-bit signed floating-point format that has a single 64-bit R component.
- VK_FORMAT_R64G64_UINT specifies a two-component, 128-bit unsigned integer format that has a 64-bit R component in bytes 0..7, and a 64-bit G component in bytes 8..15.
- VK_FORMAT_R64G64_SINT specifies a two-component, 128-bit signed integer format that has a 64-bit R component in bytes 0..7, and a 64-bit G component in bytes 8..15.
- VK_FORMAT_R64G64_SFLOAT specifies a two-component, 128-bit signed floating-point format that has a 64-bit R component in bytes 0..7, and a 64-bit G component in bytes 8..15.
- VK_FORMAT_R64G64B64_UINT specifies a three-component, 192-bit unsigned integer format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, and a 64-bit B component in bytes 16..23.

- VK_FORMAT_R64G64B64_SINT specifies a three-component, 192-bit signed integer format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, and a 64-bit B component in bytes 16..23.
- VK_FORMAT_R64G64B64_SFLOAT specifies a three-component, 192-bit signed floating-point format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, and a 64-bit B component in bytes 16..23.
- VK_FORMAT_R64G64B64A64_UINT specifies a four-component, 256-bit unsigned integer format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, a 64-bit B component in bytes 16..23, and a 64-bit A component in bytes 24..31.
- VK_FORMAT_R64G64B64A64_SINT specifies a four-component, 256-bit signed integer format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, a 64-bit B component in bytes 16..23, and a 64-bit A component in bytes 24..31.
- VK_FORMAT_R64G64B64A64_SFLOAT specifies a four-component, 256-bit signed floating-point format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, a 64-bit B component in bytes 16..23, and a 64-bit A component in bytes 24..31.
- VK_FORMAT_B10G11R11_UFLOAT_PACK32 specifies a three-component, 32-bit packed unsigned floating-point format that has a 10-bit B component in bits 22..31, an 11-bit G component in bits 11..21, and an 11-bit R component in bits 0..10. See Unsigned 10-Bit Floating-Point Numbers and Unsigned 11-Bit Floating-Point Numbers.
- VK_FORMAT_E5B9G9R9_UFLOAT_PACK32 specifies a three-component, 32-bit packed unsigned floating-point format that has a 5-bit shared exponent in bits 27..31, a 9-bit B component mantissa in bits 18..26, a 9-bit G component mantissa in bits 9..17, and a 9-bit R component mantissa in bits 0..8.
- VK_FORMAT_D16_UNORM specifies a one-component, 16-bit unsigned normalized format that has a single 16-bit depth component.
- VK_FORMAT_X8_D24_UNORM_PACK32 specifies a two-component, 32-bit format that has 24 unsigned normalized bits in the depth component and, **optionally**, 8 bits that are unused.
- VK_FORMAT_D32_SFLOAT specifies a one-component, 32-bit signed floating-point format that has 32 bits in the depth component.
- VK_FORMAT_S8_UINT specifies a one-component, 8-bit unsigned integer format that has 8 bits in the stencil component.
- VK_FORMAT_D16_UNORM_S8_UINT specifies a two-component, 24-bit format that has 16 unsigned normalized bits in the depth component and 8 unsigned integer bits in the stencil component.
- VK_FORMAT_D24_UNORM_S8_UINT specifies a two-component, 32-bit packed format that has 8 unsigned integer bits in the stencil component, and 24 unsigned normalized bits in the depth component.
- VK_FORMAT_D32_SFLOAT_S8_UINT specifies a two-component format that has 32 signed float bits in the depth component and 8 unsigned integer bits in the stencil component. There are **optionally** 24 bits that are unused.
- VK_FORMAT_BC1_RGB_UNORM_BLOCK specifies a three-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB

texel data. This format has no alpha and is considered opaque.

- VK_FORMAT_BC1_RGB_SRGB_BLOCK specifies a three-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data with sRGB nonlinear encoding. This format has no alpha and is considered opaque.
- VK_FORMAT_BC1_RGBA_UNORM_BLOCK specifies a four-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data, and provides 1 bit of alpha.
- VK_FORMAT_BC1_RGBA_SRGB_BLOCK specifies a four-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data with sRGB nonlinear encoding, and provides 1 bit of alpha.
- VK_FORMAT_BC2_UNORM_BLOCK specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values.
- VK_FORMAT_BC2_SRGB_BLOCK specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values with sRGB nonlinear encoding.
- VK_FORMAT_BC3_UNORM_BLOCK specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values.
- VK_FORMAT_BC3_SRGB_BLOCK specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values with sRGB nonlinear encoding.
- VK_FORMAT_BC4_UNORM_BLOCK specifies a one-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized red texel data.
- VK_FORMAT_BC4_SNORM_BLOCK specifies a one-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of signed normalized red texel data.
- VK_FORMAT_BC5_UNORM_BLOCK specifies a two-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RG texel data with the first 64 bits encoding red values followed by 64 bits encoding green values.
- VK_FORMAT_BC5_SNORM_BLOCK specifies a two-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of signed normalized RG texel data with the first 64 bits encoding red values followed by 64 bits encoding green values.
- VK_FORMAT_BC6H_UFLOAT_BLOCK specifies a three-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned floating-point RGB texel data.
- VK_FORMAT_BC6H_SFLOAT_BLOCK specifies a three-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of signed floating-point RGB texel data.
- VK_FORMAT_BC7_UNORM_BLOCK specifies a four-component, block-compressed format where

each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data.

- VK_FORMAT_BC7_SRGB_BLOCK specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK specifies a three-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data. This format has no alpha and is considered opaque.
- VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK specifies a three-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data with sRGB nonlinear encoding. This format has no alpha and is considered opaque.
- VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK specifies a four-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data, and provides 1 bit of alpha.
- VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK specifies a four-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data with sRGB nonlinear encoding, and provides 1 bit of alpha.
- VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK specifies a four-component, ETC2 compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values.
- VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK specifies a four-component, ETC2 compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values with sRGB nonlinear encoding applied.
- VK_FORMAT_EAC_R11_UNORM_BLOCK specifies a one-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized red texel data.
- VK_FORMAT_EAC_R11_SNORM_BLOCK specifies a one-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of signed normalized red texel data.
- VK_FORMAT_EAC_R11G11_UNORM_BLOCK specifies a two-component, ETC2 compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RG texel data with the first 64 bits encoding red values followed by 64 bits encoding green values.
- VK_FORMAT_EAC_R11G11_SNORM_BLOCK specifies a two-component, ETC2 compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of signed normalized RG texel data with the first 64 bits encoding red values followed by 64 bits encoding green values.
- VK_FORMAT_ASTC_4x4_UNORM_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_4x4_SRGB_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized

RGBA texel data with sRGB nonlinear encoding applied to the RGB components.

- VK_FORMAT_ASTC_4x4_SFLOAT_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_5x4_UNORM_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×4 rectangle of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_5x4_SRGB_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×4 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_5x4_SFLOAT_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×4 rectangle of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_5x5_UNORM_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×5 rectangle of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_5x5_SRGB_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×5 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_5x5_SFLOAT_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×5 rectangle of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_6x5_UNORM_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×5 rectangle of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_6x5_SRGB_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×5 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_6x5_SFLOAT_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×5 rectangle of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_6x6_UNORM_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×6 rectangle of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_6x6_SRGB_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×6 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_6x6_SFLOAT_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×6 rectangle of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_8x5_UNORM_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×5 rectangle of unsigned normalized

RGBA texel data.

- VK_FORMAT_ASTC_8x5_SRGB_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×5 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_8x5_SFLOAT_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×5 rectangle of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_8x6_UNORM_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×6 rectangle of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_8x6_SRGB_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×6 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_8x6_SFLOAT_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×6 rectangle of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_8x8_UNORM_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×8 rectangle of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_8x8_SRGB_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×8 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_8x8_SFLOAT_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×8 rectangle of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_10x5_UNORM_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×5 rectangle of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_10x5_SRGB_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×5 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_10x5_SFLOAT_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×5 rectangle of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_10x6_UNORM_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×6 rectangle of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_10x6_SRGB_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×6 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_10x6_SFLOAT_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×6 rectangle of signed floating-point

RGBA texel data.

- VK_FORMAT_ASTC_10x8_UNORM_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×8 rectangle of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_10x8_SRGB_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×8 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_10x8_SFLOAT_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×8 rectangle of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_10x10_UNORM_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×10 rectangle of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_10x10_SRGB_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×10 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_10x10_SFLOAT_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×10 rectangle of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_12x10_UNORM_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 12×10 rectangle of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_12x10_SRGB_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 12×10 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_12x10_SFLOAT_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 12×10 rectangle of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_12x12_UNORM_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 12×12 rectangle of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_12x12_SRGB_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 12×12 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_12x12_SFLOAT_BLOCK specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 12×12 rectangle of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_3x3x3_UNORM_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 3×3×3 cuboid of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_3x3x3_SRGB_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 3×3×3 cuboid of unsigned

normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.

- VK_FORMAT_ASTC_3x3x3_SFLOAT_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 3×3×3 cuboid of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_4x3x3_UNORM_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×3×3 cuboid of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_4x3x3_SRGB_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×3×3 cuboid of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_4x3x3_SFLOAT_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×3×3 cuboid of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_4x4x3_UNORM_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×4×3 cuboid of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_4x4x3_SRGB_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×4×3 cuboid of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_4x4x3_SFLOAT_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×4×3 cuboid of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_4x4x4_UNORM_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×4×4 cuboid of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_4x4x4_SRGB_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×4×4 cuboid of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_4x4x4_SFLOAT_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×4×4 cuboid of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_5x4x4_UNORM_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×4×4 cuboid of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_5x4x4_SRGB_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×4×4 cuboid of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_5x4x4_SFLOAT_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×4×4 cuboid of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_5x5x4_UNORM_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×5×4 cuboid of unsigned

normalized RGBA texel data.

- VK_FORMAT_ASTC_5x5x4_SRGB_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×5×4 cuboid of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_5x5x4_SFLOAT_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×5×4 cuboid of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_5x5x5_UNORM_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×5×5 cuboid of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_5x5x5_SRGB_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×5×5 cuboid of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_5x5x5_SFLOAT_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×5×5 cuboid of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_6x5x5_UNORM_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×5×5 cuboid of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_6x5x5_SRGB_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×5×5 cuboid of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_6x5x5_SFLOAT_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×5×5 cuboid of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_6x6x5_UNORM_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×6×5 cuboid of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_6x6x5_SRGB_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×6×5 cuboid of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_6x6x5_SFLOAT_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×6×5 cuboid of signed floating-point RGBA texel data.
- VK_FORMAT_ASTC_6x6x6_UNORM_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×6×6 cuboid of unsigned normalized RGBA texel data.
- VK_FORMAT_ASTC_6x6x6_SRGB_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×6×6 cuboid of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- VK_FORMAT_ASTC_6x6x6_SFLOAT_BLOCK_EXT specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×6×6 cuboid of signed floating-

point RGBA texel data.

- VK_FORMAT_G8B8G8R8_422_UNORM specifies a four-component, 32-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each i coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the image. This format has an 8-bit G component for the even i coordinate in byte 0, an 8-bit B component in byte 1, an 8-bit G component for the odd i coordinate in byte 2, and an 8-bit R component in byte 3. This format only supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.
- VK_FORMAT_B8G8R8G8_422_UNORM specifies a four-component, 32-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each i coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the image. This format has an 8-bit B component in byte 0, an 8-bit G component for the even i coordinate in byte 1, an 8-bit R component in byte 2, and an 8-bit G component for the odd i coordinate in byte 3. This format only supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.
- VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM specifies an unsigned normalized *multi-planar format* that has an 8-bit G component in plane 0, an 8-bit B component in plane 1, and an 8-bit R component in plane 2. The horizontal and vertical dimensions of the R and B planes are halved relative to the image dimensions, and each R and B component is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, VK_IMAGE_ASPECT_PLANE_1_BIT for the B plane, and VK_IMAGE_ASPECT_PLANE_2_BIT for the R plane. This format only supports images with a width and height that is a multiple of two.
- VK_FORMAT_G8_B8R8_2PLANE_420_UNORM specifies an unsigned normalized *multi-planar format* that has an 8-bit G component in plane 0, and a two-component, 16-bit BR plane 1 consisting of an 8-bit B component in byte 0 and an 8-bit R component in byte 1. The horizontal and vertical dimensions of the BR plane are halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, and VK_IMAGE_ASPECT_PLANE_1_BIT for the BR plane. This format only supports images with a width and height that is a multiple of two.
- VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM specifies an unsigned normalized *multi-planar format* that has an 8-bit G component in plane 0, an 8-bit B component in plane 1, and an 8-bit R component in plane 2. The horizontal dimension of the R and B plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, VK_IMAGE_ASPECT_PLANE_1_BIT for the B plane, and VK_IMAGE_ASPECT_PLANE_2_BIT for the R plane. This format only supports images with a width that is a multiple of two.

- VK_FORMAT_G8_B8R8_2PLANE_422_UNORM specifies an unsigned normalized *multi-planar format* that has an 8-bit G component in plane 0, and a two-component, 16-bit BR plane 1 consisting of an 8-bit B component in byte 0 and an 8-bit R component in byte 1. The horizontal dimension of the BR plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, and VK_IMAGE_ASPECT_PLANE_1_BIT for the BR plane. This format only supports images with a width that is a multiple of two.
- VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM specifies an unsigned normalized *multi-planar format* that has an 8-bit G component in plane 0, an 8-bit B component in plane 1, and an 8-bit R component in plane 2. Each plane has the same dimensions and each R, G, and B component contributes to a single texel. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, VK_IMAGE_ASPECT_PLANE_1_BIT for the B plane, and VK_IMAGE_ASPECT_PLANE_2_BIT for the R plane.
- VK_FORMAT_R10X6_UNORM_PACK16 specifies a one-component, 16-bit unsigned normalized format that has a single 10-bit R component in the top 10 bits of a 16-bit word, with the bottom 6 bits unused.
- VK_FORMAT_R10X6G10X6_UNORM_2PACK16 specifies a two-component, 32-bit unsigned normalized format that has a 10-bit R component in the top 10 bits of the word in bytes 0..1, and a 10-bit G component in the top 10 bits of the word in bytes 2..3, with the bottom 6 bits of each word unused.
- VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16 specifies a four-component, 64-bit unsigned normalized format that has a 10-bit R component in the top 10 bits of the word in bytes 0..1, a 10-bit G component in the top 10 bits of the word in bytes 2..3, a 10-bit B component in the top 10 bits of the word in bytes 4..5, and a 10-bit A component in the top 10 bits of the word in bytes 6..7, with the bottom 6 bits of each word unused.
- VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16 specifies a four-component, 64-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each i coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the image. This format has a 10-bit G component for the even i coordinate in the top 10 bits of the word in bytes 0..1, a 10-bit B component in the top 10 bits of the word in bytes 2..3, a 10-bit G component for the odd i coordinate in the top 10 bits of the word in bytes 4..5, and a 10-bit R component in the top 10 bits of the word in bytes 6..7, with the bottom 6 bits of each word unused. This format only supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.
- VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16 specifies a four-component, 64-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each i coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the image. This format has a 10-bit B component in the top 10 bits of the word in bytes 0..1, a 10-bit G component for the even i coordinate in the top 10 bits of the word in bytes 2..3, a 10-bit R component in the top 10 bits of the word in bytes 4..5, and a 10-bit G

component for the odd *i* coordinate in the top 10 bits of the word in bytes 6..7, with the bottom 6 bits of each word unused. This format only supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.

- VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16 specifies an unsigned normalized *multi-planar format* that has a 10-bit G component in the top 10 bits of each 16-bit word of plane 0, a 10-bit B component in the top 10 bits of each 16-bit word of plane 1, and a 10-bit R component in the top 10 bits of each 16-bit word of plane 2, with the bottom 6 bits of each word unused. The horizontal and vertical dimensions of the R and B planes are halved relative to the image dimensions, and each R and B component is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, VK_IMAGE_ASPECT_PLANE_1_BIT for the B plane, and VK_IMAGE_ASPECT_PLANE_2_BIT for the R plane. This format only supports images with a width and height that is a multiple of two.
- VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16 specifies an unsigned normalized *multi-planar format* that has a 10-bit G component in the top 10 bits of each 16-bit word of plane 0, and a two-component, 32-bit BR plane 1 consisting of a 10-bit B component in the top 10 bits of the word in bytes 0..1, and a 10-bit R component in the top 10 bits of the word in bytes 2..3, with the bottom 6 bits of each word unused. The horizontal and vertical dimensions of the BR plane are halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, and VK_IMAGE_ASPECT_PLANE_1_BIT for the BR plane. This format only supports images with a width and height that is a multiple of two.
- VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16 specifies an unsigned normalized *multi-planar format* that has a 10-bit G component in the top 10 bits of each 16-bit word of plane 0, a 10-bit B component in the top 10 bits of each 16-bit word of plane 1, and a 10-bit R component in the top 10 bits of each 16-bit word of plane 2, with the bottom 6 bits of each word unused. The horizontal dimension of the R and B plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, VK_IMAGE_ASPECT_PLANE_1_BIT for the B plane, and VK_IMAGE_ASPECT_PLANE_2_BIT for the R plane. This format only supports images with a width that is a multiple of two.
- VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16 specifies an unsigned normalized *multi-planar format* that has a 10-bit G component in the top 10 bits of each 16-bit word of plane 0, and a two-component, 32-bit BR plane 1 consisting of a 10-bit B component in the top 10 bits of the word in bytes 0..1, and a 10-bit R component in the top 10 bits of the word in bytes 2..3, with the bottom 6 bits of each word unused. The horizontal dimension of the BR plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, and VK_IMAGE_ASPECT_PLANE_1_BIT for

the BR plane. This format only supports images with a width that is a multiple of two.

- VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16 specifies an unsigned normalized *multi-planar format* that has a 10-bit G component in the top 10 bits of each 16-bit word of plane 0, a 10-bit B component in the top 10 bits of each 16-bit word of plane 1, and a 10-bit R component in the top 10 bits of each 16-bit word of plane 2, with the bottom 6 bits of each word unused. Each plane has the same dimensions and each R, G, and B component contributes to a single texel. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, VK_IMAGE_ASPECT_PLANE_1_BIT for the B plane, and VK_IMAGE_ASPECT_PLANE_2_BIT for the R plane.
- VK_FORMAT_R12X4_UNORM_PACK16 specifies a one-component, 16-bit unsigned normalized format that has a single 12-bit R component in the top 12 bits of a 16-bit word, with the bottom 4 bits unused.
- VK_FORMAT_R12X4G12X4_UNORM_2PACK16 specifies a two-component, 32-bit unsigned normalized format that has a 12-bit R component in the top 12 bits of the word in bytes 0..1, and a 12-bit G component in the top 12 bits of the word in bytes 2..3, with the bottom 4 bits of each word unused.
- VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16 specifies a four-component, 64-bit unsigned normalized format that has a 12-bit R component in the top 12 bits of the word in bytes 0..1, a 12-bit G component in the top 12 bits of the word in bytes 2..3, a 12-bit B component in the top 12 bits of the word in bytes 4..5, and a 12-bit A component in the top 12 bits of the word in bytes 6..7, with the bottom 4 bits of each word unused.
- VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16 specifies a four-component, 64-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each i coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the image. This format has a 12-bit G component for the even i coordinate in the top 12 bits of the word in bytes 0..1, a 12-bit B component in the top 12 bits of the word in bytes 2..3, a 12-bit G component for the odd i coordinate in the top 12 bits of the word in bytes 4..5, and a 12-bit R component in the top 12 bits of the word in bytes 6..7, with the bottom 4 bits of each word unused. This format only supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.
- VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16 specifies a four-component, 64-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each i coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the image. This format has a 12-bit B component in the top 12 bits of the word in bytes 0..1, a 12-bit G component for the even i coordinate in the top 12 bits of the word in bytes 2..3, a 12-bit R component in the top 12 bits of the word in bytes 4..5, and a 12-bit G component for the odd i coordinate in the top 12 bits of the word in bytes 6..7, with the bottom 4 bits of each word unused. This format only supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.
- VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16 specifies an unsigned

normalized *multi-planar format* that has a 12-bit G component in the top 12 bits of each 16-bit word of plane 0, a 12-bit B component in the top 12 bits of each 16-bit word of plane 1, and a 12-bit R component in the top 12 bits of each 16-bit word of plane 2, with the bottom 4 bits of each word unused. The horizontal and vertical dimensions of the R and B planes are halved relative to the image dimensions, and each R and B component is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, VK_IMAGE_ASPECT_PLANE_1_BIT for the B plane, and VK_IMAGE_ASPECT_PLANE_2_BIT for the R plane. This format only supports images with a width and height that is a multiple of two.

- VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16 specifies an unsigned normalized *multi-planar format* that has a 12-bit G component in the top 12 bits of each 16-bit word of plane 0, and a two-component, 32-bit BR plane 1 consisting of a 12-bit B component in the top 12 bits of the word in bytes 0..1, and a 12-bit R component in the top 12 bits of the word in bytes 2..3, with the bottom 4 bits of each word unused. The horizontal and vertical dimensions of the BR plane are halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, and VK_IMAGE_ASPECT_PLANE_1_BIT for the BR plane. This format only supports images with a width and height that is a multiple of two.
- VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16 specifies an unsigned normalized *multi-planar format* that has a 12-bit G component in the top 12 bits of each 16-bit word of plane 0, a 12-bit B component in the top 12 bits of each 16-bit word of plane 1, and a 12-bit R component in the top 12 bits of each 16-bit word of plane 2, with the bottom 4 bits of each word unused. The horizontal dimension of the R and B plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, VK_IMAGE_ASPECT_PLANE_1_BIT for the B plane, and VK_IMAGE_ASPECT_PLANE_2_BIT for the R plane. This format only supports images with a width that is a multiple of two.
- VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16 specifies an unsigned normalized *multi-planar format* that has a 12-bit G component in the top 12 bits of each 16-bit word of plane 0, and a two-component, 32-bit BR plane 1 consisting of a 12-bit B component in the top 12 bits of the word in bytes 0..1, and a 12-bit R component in the top 12 bits of the word in bytes 2..3, with the bottom 4 bits of each word unused. The horizontal dimension of the BR plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, and VK_IMAGE_ASPECT_PLANE_1_BIT for the BR plane. This format only supports images with a width that is a multiple of two.
- VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16 specifies an unsigned normalized *multi-planar format* that has a 12-bit G component in the top 12 bits of each 16-bit word of plane 0, a 12-bit B component in the top 12 bits of each 16-bit word of plane 1, and a 12-bit R component in the top 12 bits of each 16-bit word of plane 2, with the bottom 4 bits of each

word unused. Each plane has the same dimensions and each R, G, and B component contributes to a single texel. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, VK_IMAGE_ASPECT_PLANE_1_BIT for the B plane, and VK_IMAGE_ASPECT_PLANE_2_BIT for the R plane.

- VK_FORMAT_G16B16G16R16_422_UNORM specifies a four-component, 64-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each *i* coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the image. This format has a 16-bit G component for the even *i* coordinate in the word in bytes 0..1, a 16-bit B component in the word in bytes 2..3, a 16-bit G component for the odd *i* coordinate in the word in bytes 4..5, and a 16-bit R component in the word in bytes 6..7. This format only supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.
- VK_FORMAT_B16G16R16G16_422_UNORM specifies a four-component, 64-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each *i* coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the image. This format has a 16-bit B component in the word in bytes 0..1, a 16-bit G component for the even *i* coordinate in the word in bytes 2..3, a 16-bit R component in the word in bytes 4..5, and a 16-bit G component for the odd *i* coordinate in the word in bytes 6..7. This format only supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.
- VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM specifies an unsigned normalized *multi-planar format* that has a 16-bit G component in each 16-bit word of plane 0, a 16-bit B component in each 16-bit word of plane 1, and a 16-bit R component in each 16-bit word of plane 2. The horizontal and vertical dimensions of the R and B planes are halved relative to the image dimensions, and each R and B component is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, VK_IMAGE_ASPECT_PLANE_1_BIT for the B plane, and VK_IMAGE_ASPECT_PLANE_2_BIT for the R plane. This format only supports images with a width and height that is a multiple of two.
- VK_FORMAT_G16_B16R16_2PLANE_420_UNORM specifies an unsigned normalized *multi-planar format* that has a 16-bit G component in each 16-bit word of plane 0, and a two-component, 32-bit BR plane 1 consisting of a 16-bit B component in the word in bytes 0..1, and a 16-bit R component in the word in bytes 2..3. The horizontal and vertical dimensions of the BR plane are halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, and VK_IMAGE_ASPECT_PLANE_1_BIT for the BR plane. This format only supports images with a width and height that is a multiple of two.
- VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM specifies an unsigned normalized *multi-*

planar format that has a 16-bit G component in each 16-bit word of plane 0, a 16-bit B component in each 16-bit word of plane 1, and a 16-bit R component in each 16-bit word of plane 2. The horizontal dimension of the R and B plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, VK_IMAGE_ASPECT_PLANE_1_BIT for the B plane, and VK_IMAGE_ASPECT_PLANE_2_BIT for the R plane. This format only supports images with a width that is a multiple of two.

- VK_FORMAT_G16_B16R16_2PLANE_422_UNORM specifies an unsigned normalized *multi-planar format* that has a 16-bit G component in each 16-bit word of plane 0, and a two-component, 32-bit BR plane 1 consisting of a 16-bit B component in the word in bytes 0..1, and a 16-bit R component in the word in bytes 2..3. The horizontal dimension of the BR plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, and VK_IMAGE_ASPECT_PLANE_1_BIT for the BR plane. This format only supports images with a width that is a multiple of two.
- VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM specifies an unsigned normalized *multi-planar format* that has a 16-bit G component in each 16-bit word of plane 0, a 16-bit B component in each 16-bit word of plane 1, and a 16-bit R component in each 16-bit word of plane 2. Each plane has the same dimensions and each R, G, and B component contributes to a single texel. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, VK_IMAGE_ASPECT_PLANE_1_BIT for the B plane, and VK_IMAGE_ASPECT_PLANE_2_BIT for the R plane.
- VK_FORMAT_G8_B8R8_2PLANE_444_UNORM specifies an unsigned normalized *multi-planar format* that has an 8-bit G component in plane 0, and a two-component, 16-bit BR plane 1 consisting of an 8-bit B component in byte 0 and an 8-bit R component in byte 1. Both planes have the same dimensions and each R, G, and B component contributes to a single texel. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, and VK_IMAGE_ASPECT_PLANE_1_BIT for the BR plane.
- VK_FORMAT_G10X6_B10X6R10X6_2PLANE_444_UNORM_3PACK16 specifies an unsigned normalized *multi-planar format* that has a 10-bit G component in the top 10 bits of each 16-bit word of plane 0, and a two-component, 32-bit BR plane 1 consisting of a 10-bit B component in the top 10 bits of the word in bytes 0..1, and a 10-bit R component in the top 10 bits of the word in bytes 2..3, with the bottom 6 bits of each word unused. Both planes have the same dimensions and each R, G, and B component contributes to a single texel. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, and VK_IMAGE_ASPECT_PLANE_1_BIT for the BR plane.
- VK_FORMAT_G12X4_B12X4R12X4_2PLANE_444_UNORM_3PACK16 specifies an unsigned normalized *multi-planar format* that has a 12-bit G component in the top 12 bits of each 16-bit word of plane 0, and a two-component, 32-bit BR plane 1 consisting of a 12-bit B component in the top 12 bits of the word in bytes 0..1, and a 12-bit R component in the top 12 bits of the word

in bytes 2..3, with the bottom 4 bits of each word unused. Both planes have the same dimensions and each R, G, and B component contributes to a single texel. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, and VK_IMAGE_ASPECT_PLANE_1_BIT for the BR plane.

- VK_FORMAT_G16_B16R16_2PLANE_444_UNORM specifies an unsigned normalized *multi-planar format* that has a 16-bit G component in each 16-bit word of plane 0, and a two-component, 32-bit BR plane 1 consisting of a 16-bit B component in the word in bytes 0..1, and a 16-bit R component in the word in bytes 2..3. Both planes have the same dimensions and each R, G, and B component contributes to a single texel. The location of each plane when this image is in linear layout can be determined via vkGetImageSubresourceLayout, using VK_IMAGE_ASPECT_PLANE_0_BIT for the G plane, and VK_IMAGE_ASPECT_PLANE_1_BIT for the BR plane.

49.1.1. Compatible Formats of Planes of Multi-Planar Formats

Individual planes of multi-planar formats are size-compatible with single-plane color formats if they occupy the same number of bits per texel block, and are compatible with those formats if they have the same block extent.

In the following table, individual planes of a *multi-planar* format are compatible with the format listed against the relevant plane index for that multi-planar format, and any format compatible with the listed single-plane format according to Format Compatibility Classes. These planes are also size-compatible with any format that is size-compatible with the listed single-plane format.

Table 70. Plane Format Compatibility

Plane	Compatible format for plane	Width relative to the width *w* of the plane with the largest dimensions	Height relative to the height *h* of the plane with the largest dimensions
VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM			
0	VK_FORMAT_R8_UNORM	w	h
1	VK_FORMAT_R8_UNORM	w/2	h/2
2	VK_FORMAT_R8_UNORM	w/2	h/2
VK_FORMAT_G8_B8R8_2PLANE_420_UNORM			
0	VK_FORMAT_R8_UNORM	w	h
1	VK_FORMAT_R8G8_UNORM	w/2	h/2
VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM			
0	VK_FORMAT_R8_UNORM	w	h
1	VK_FORMAT_R8_UNORM	w/2	h
2	VK_FORMAT_R8_UNORM	w/2	h
VK_FORMAT_G8_B8R8_2PLANE_422_UNORM			

Plane	Compatible format for plane	Width relative to the width *w* of the plane with the largest dimensions	Height relative to the height *h* of the plane with the largest dimensions
0	VK_FORMAT_R8_UNORM	w	h
1	VK_FORMAT_R8G8_UNORM	w/2	h
VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM			
0	VK_FORMAT_R8_UNORM	w	h
1	VK_FORMAT_R8_UNORM	w	h
2	VK_FORMAT_R8_UNORM	w	h
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16			
0	VK_FORMAT_R10X6_UNORM_PACK16	w	h
1	VK_FORMAT_R10X6_UNORM_PACK16	w/2	h/2
2	VK_FORMAT_R10X6_UNORM_PACK16	w/2	h/2
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16			
0	VK_FORMAT_R10X6_UNORM_PACK16	w	h
1	VK_FORMAT_R10X6G10X6_UNORM_2PACK16	w/2	h/2
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16			
0	VK_FORMAT_R10X6_UNORM_PACK16	w	h
1	VK_FORMAT_R10X6_UNORM_PACK16	w/2	h
2	VK_FORMAT_R10X6_UNORM_PACK16	w/2	h
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16			
0	VK_FORMAT_R10X6_UNORM_PACK16	w	h
1	VK_FORMAT_R10X6G10X6_UNORM_2PACK16	w/2	h
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16			
0	VK_FORMAT_R10X6_UNORM_PACK16	w	h
1	VK_FORMAT_R10X6_UNORM_PACK16	w	h
2	VK_FORMAT_R10X6_UNORM_PACK16	w	h
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16			
0	VK_FORMAT_R12X4_UNORM_PACK16	w	h
1	VK_FORMAT_R12X4_UNORM_PACK16	w/2	h/2
2	VK_FORMAT_R12X4_UNORM_PACK16	w/2	h/2
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16			
0	VK_FORMAT_R12X4_UNORM_PACK16	w	h

Plane	Compatible format for plane	Width relative to the width *w* of the plane with the largest dimensions	Height relative to the height *h* of the plane with the largest dimensions
1	VK_FORMAT_R12X4G12X4_UNORM_2PACK16	w/2	h/2
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16			
0	VK_FORMAT_R12X4_UNORM_PACK16	w	h
1	VK_FORMAT_R12X4_UNORM_PACK16	w/2	h
2	VK_FORMAT_R12X4_UNORM_PACK16	w/2	h
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16			
0	VK_FORMAT_R12X4_UNORM_PACK16	w	h
1	VK_FORMAT_R12X4G12X4_UNORM_2PACK16	w/2	h
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16			
0	VK_FORMAT_R12X4_UNORM_PACK16	w	h
1	VK_FORMAT_R12X4_UNORM_PACK16	w	h
2	VK_FORMAT_R12X4_UNORM_PACK16	w	h
VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM			
0	VK_FORMAT_R16_UNORM	w	h
1	VK_FORMAT_R16_UNORM	w/2	h/2
2	VK_FORMAT_R16_UNORM	w/2	h/2
VK_FORMAT_G16_B16R16_2PLANE_420_UNORM			
0	VK_FORMAT_R16_UNORM	w	h
1	VK_FORMAT_R16G16_UNORM	w/2	h/2
VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM			
0	VK_FORMAT_R16_UNORM	w	h
1	VK_FORMAT_R16_UNORM	w/2	h
2	VK_FORMAT_R16_UNORM	w/2	h
VK_FORMAT_G16_B16R16_2PLANE_422_UNORM			
0	VK_FORMAT_R16_UNORM	w	h
1	VK_FORMAT_R16G16_UNORM	w/2	h
VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM			
0	VK_FORMAT_R16_UNORM	w	h
1	VK_FORMAT_R16_UNORM	w	h
2	VK_FORMAT_R16_UNORM	w	h

Plane	Compatible format for plane	Width relative to the width w of the plane with the largest dimensions	Height relative to the height h of the plane with the largest dimensions
VK_FORMAT_G8_B8R8_2PLANE_444_UNORM			
0	VK_FORMAT_R8_UNORM	w	h
1	VK_FORMAT_R8G8_UNORM	w	h
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_444_UNORM_3PACK16			
0	VK_FORMAT_R10X6_UNORM_PACK16	w	h
1	VK_FORMAT_R10X6G10X6_UNORM_2PACK16	w	h
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_444_UNORM_3PACK16			
0	VK_FORMAT_R12X4_UNORM_PACK16	w	h
1	VK_FORMAT_R12X4G12X4_UNORM_2PACK16	w	h
VK_FORMAT_G16_B16R16_2PLANE_444_UNORM			
0	VK_FORMAT_R16_UNORM	w	h
1	VK_FORMAT_R16G16_UNORM	w	h

49.1.2. Compatible Formats for Depth-Stencil to/from Color Copies

If the `maintenance8` feature is enabled, individual aspects of depth-stencil formats are size-compatible with certain single-component color formats when performing bit-wise copy operations. The following is a complete list of all per-aspect format compatibility relations by size:

- 32-bit depth (VK_FORMAT_D32_SFLOAT, VK_FORMAT_D32_SFLOAT_S8_UINT)
  - VK_FORMAT_R32_SFLOAT, VK_FORMAT_R32_SINT, VK_FORMAT_R32_UINT
- 24-bit depth (VK_FORMAT_X8_D24_UNORM_PACK32, VK_FORMAT_D24_UNORM_S8_UINT)
  - VK_FORMAT_R32_SFLOAT, VK_FORMAT_R32_SINT, VK_FORMAT_R32_UINT
- 16-bit depth (VK_FORMAT_D16_UNORM, VK_FORMAT_D16_UNORM_S8_UINT)
  - VK_FORMAT_R16_SFLOAT, VK_FORMAT_R16_UNORM, VK_FORMAT_R16_SNORM, VK_FORMAT_R16_UINT, VK_FORMAT_R16_SINT
- 8-bit stencil (VK_FORMAT_S8_UINT, VK_FORMAT_D16_UNORM_S8_UINT, VK_FORMAT_D24_UNORM_S8_UINT, VK_FORMAT_D32_SFLOAT_S8_UINT)
  - VK_FORMAT_R8_UINT, VK_FORMAT_R8_SINT, VK_FORMAT_R8_UNORM, VK_FORMAT_R8_SNORM

49.1.3. Multi-Planar Format Image Aspect

When using VkImageAspectFlagBits to select a plane of a multi-planar format, the following are the

valid options:

- Two planes
  - VK_IMAGE_ASPECT_PLANE_0_BIT
  - VK_IMAGE_ASPECT_PLANE_1_BIT
- Three planes
  - VK_IMAGE_ASPECT_PLANE_0_BIT
  - VK_IMAGE_ASPECT_PLANE_1_BIT
  - VK_IMAGE_ASPECT_PLANE_2_BIT

49.1.4. Packed Formats

For the purposes of address alignment when accessing buffer memory containing vertex attribute or texel data, the following formats are considered *packed* - components of the texels or attributes are stored in bitfields packed into one or more 8-, 16-, or 32-bit fundamental data types.

- Packed into 8-bit data types:
  - VK_FORMAT_R4G4_UNORM_PACK8
- Packed into 16-bit data types:
  - VK_FORMAT_R4G4B4A4_UNORM_PACK16
  - VK_FORMAT_B4G4R4A4_UNORM_PACK16
  - VK_FORMAT_R5G6B5_UNORM_PACK16
  - VK_FORMAT_B5G6R5_UNORM_PACK16
  - VK_FORMAT_R5G5B5A1_UNORM_PACK16
  - VK_FORMAT_B5G5R5A1_UNORM_PACK16
  - VK_FORMAT_A1R5G5B5_UNORM_PACK16
  - VK_FORMAT_A1B5G5R5_UNORM_PACK16
  - VK_FORMAT_R10X6_UNORM_PACK16
  - VK_FORMAT_R10X6G10X6_UNORM_2PACK16
  - VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16
  - VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16
  - VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16
  - VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16
  - VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16
  - VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16
  - VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16
  - VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16
  - VK_FORMAT_R12X4_UNORM_PACK16

    - VK_FORMAT_R12X4G12X4_UNORM_2PACK16
    - VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16
    - VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16
    - VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16
    - VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16
    - VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16
    - VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16
    - VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16
    - VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16
    - VK_FORMAT_G10X6_B10X6R10X6_2PLANE_444_UNORM_3PACK16
    - VK_FORMAT_G12X4_B12X4R12X4_2PLANE_444_UNORM_3PACK16
    - VK_FORMAT_A4R4G4B4_UNORM_PACK16
    - VK_FORMAT_A4B4G4R4_UNORM_PACK16
- Packed into 32-bit data types:
    - VK_FORMAT_A8B8G8R8_UNORM_PACK32
    - VK_FORMAT_A8B8G8R8_SNORM_PACK32
    - VK_FORMAT_A8B8G8R8_USCALED_PACK32
    - VK_FORMAT_A8B8G8R8_SSCALED_PACK32
    - VK_FORMAT_A8B8G8R8_UINT_PACK32
    - VK_FORMAT_A8B8G8R8_SINT_PACK32
    - VK_FORMAT_A8B8G8R8_SRGB_PACK32
    - VK_FORMAT_A2R10G10B10_UNORM_PACK32
    - VK_FORMAT_A2R10G10B10_SNORM_PACK32
    - VK_FORMAT_A2R10G10B10_USCALED_PACK32
    - VK_FORMAT_A2R10G10B10_SSCALED_PACK32
    - VK_FORMAT_A2R10G10B10_UINT_PACK32
    - VK_FORMAT_A2R10G10B10_SINT_PACK32
    - VK_FORMAT_A2B10G10R10_UNORM_PACK32
    - VK_FORMAT_A2B10G10R10_SNORM_PACK32
    - VK_FORMAT_A2B10G10R10_USCALED_PACK32
    - VK_FORMAT_A2B10G10R10_SSCALED_PACK32
    - VK_FORMAT_A2B10G10R10_UINT_PACK32
    - VK_FORMAT_A2B10G10R10_SINT_PACK32
    - VK_FORMAT_B10G11R11_UFLOAT_PACK32
    - VK_FORMAT_E5B9G9R9_UFLOAT_PACK32

- VK_FORMAT_X8_D24_UNORM_PACK32

49.1.5. Identification of Formats

A “format” is represented by a single enum value. The name of a format is usually built up by using the following pattern:

```
VK_FORMAT_{component-format|compression-scheme}_{numeric-format}
```

The component-format indicates either the size of the R, G, B, and A components (if they are present) in the case of a color format, or the size of the depth (D) and stencil (S) components (if they are present) in the case of a depth/stencil format (see below). An X indicates a component that is unused, but **may** be present for padding.

Table 71. Interpretation of Numeric Format

Numeric format	Type-Declaration instructions	Numeric type	Description
`UNORM`	OpTypeFloat	floating-point	The components are unsigned normalized values in the range [0,1]
`SNORM`	OpTypeFloat	floating-point	The components are signed normalized values in the range [-1,1]
`USCALED`	OpTypeFloat	floating-point	The components are unsigned integer values that get converted to floating-point in the range [0,2^n-1]
`SSCALED`	OpTypeFloat	floating-point	The components are signed integer values that get converted to floating-point in the range [-2^{n-1},2^{n-1}-1]
`UINT`	OpTypeInt	unsigned integer	The components are unsigned integer values in the range [0,2^n-1]
`SINT`	OpTypeInt	signed integer	The components are signed integer values in the range [-2^{n-1},2^{n-1}-1]
`UFLOAT`	OpTypeFloat	floating-point	The components are unsigned floating-point numbers (used by packed, shared exponent, and some compressed formats)
`SFLOAT`	OpTypeFloat	floating-point	The components are signed floating-point numbers
`SRGB`	OpTypeFloat	floating-point	The R, G, and B components are unsigned normalized values that represent values using sRGB nonlinear encoding, while the A component (if one exists) is a regular unsigned normalized value
n is the number of bits in the component.			

The suffix `_PACKnn` indicates that the format is packed into an underlying type with `nn` bits. The suffix `_mPACKnn` is a short-hand that indicates that the format has `m` groups of components (which may or may not be stored in separate *planes*) that are each packed into an underlying type with `nn` bits.

The suffix `_BLOCK` indicates that the format is a block-compressed format, with the representation of multiple texels encoded interdependently within a region.

Table 72. Interpretation of Compression Scheme

Compression scheme	Description
`BC`	Block Compression. See Block-Compressed Image Formats.
`ETC2`	Ericsson Texture Compression. See ETC Compressed Image Formats.
`EAC`	ETC2 Alpha Compression. See ETC Compressed Image Formats.

Compression scheme	Description
ASTC	Adaptive Scalable Texture Compression (LDR Profile). See ASTC Compressed Image Formats.

For *multi-planar* images, the components in separate *planes* are separated by underscores, and the number of planes is indicated by the addition of a `_2PLANE` or `_3PLANE` suffix. Similarly, the separate aspects of depth-stencil formats are separated by underscores, although these are not considered separate planes. Formats are suffixed by `_422` to indicate that planes other than the first are reduced in size by a factor of two horizontally or that the R and B values appear at half the horizontal frequency of the G values, `_420` to indicate that planes other than the first are reduced in size by a factor of two both horizontally and vertically, and `_444` for consistency to indicate that all three planes of a three-planar image are the same size.

NOTE	No common format has a single plane containing both R and B components but does not store these components at reduced horizontal resolution.

49.1.6. Representation and Texel Block Size

Color formats **must** be represented in memory in exactly the form indicated by the format's name. This means that promoting one format to another with more bits per component and/or additional components **must** not occur for color formats. Depth/stencil formats have more relaxed requirements as discussed below.

Each format has a *texel block size*, the number of bytes used to store one *texel block* (a single addressable element of an uncompressed image, or a single compressed block of a compressed image). The texel block size for each format is shown in the Compatible formats table.

The representation of non-packed formats is that the first component specified in the name of the format is in the lowest memory addresses and the last component specified is in the highest memory addresses. See Byte Mappings for Non-Packed/Compressed Color Formats. The in-memory ordering of bytes within a component is determined by the host endianness.

Table 73. Byte Mappings for Non-Packed/Compressed Color Formats

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	← Byte
R																VK_FORMAT_R8_*
R	G															VK_FORMAT_R8G8_*
R	G	B														VK_FORMAT_R8G8B8_*
B	G	R														VK_FORMAT_B8G8R8_*
R	G	B	A													VK_FORMAT_R8G8B8A8_*
B	G	R	A													VK_FORMAT_B8G8R8A8_*
A																**VK_FORMAT_A8_UNORM**
G_0	B	G_1	R													**VK_FORMAT_G8B8G8R8_422_UNORM**
B	G_0	R	G_1													**VK_FORMAT_B8G8R8G8_422_UNORM**

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	← Byte
R																VK_FORMAT_R16_*
R		G														VK_FORMAT_R16G16_*
R		G		B												VK_FORMAT_R16G16B16_*
R		G		B		A										VK_FORMAT_R16G16B16A16_*
G_0		B		G_1		R										VK_FORMAT_G10X6B10X6G10X6R10X6_4PACK16_422_UNORM VK_FORMAT_G12X4B12X4G12X4R12X4_4PACK16_422_UNORM VK_FORMAT_G16B16G16R16_UNORM
B		G_0		R		G_1										VK_FORMAT_B10X6G10X6R10X6G10X6_4PACK16_422_UNORM VK_FORMAT_B12X4G12X4R12X4G12X4_4PACK16_422_UNORM **VK_FORMAT_B16G16R16G16_422_UNORM**
R																VK_FORMAT_R32_*
R				G												VK_FORMAT_R32G32_*
R				G				B								VK_FORMAT_R32G32B32_*
R				G				B				A				VK_FORMAT_R32G32B32A32_*
R																VK_FORMAT_R64_*
R								G								VK_FORMAT_R64G64_*
VK_FORMAT_R64G64B64_* **as** VK_FORMAT_R64G64_* **but with B in bytes 16-23**																
VK_FORMAT_R64G64B64A64_* **as** VK_FORMAT_R64G64B64_* **but with A in bytes 24-31**																

Packed formats store multiple components within one underlying type. The bit representation is that the first component specified in the name of the format is in the most-significant bits and the last component specified is in the least-significant bits of the underlying type. The in-memory ordering of bytes comprising the underlying type is determined by the host endianness.

Table 74. Bit Mappings for Packed 8-Bit Formats

Bit							
7	6	5	4	3	2	1	0
VK_FORMAT_R4G4_UNORM_PACK8							
R				**G**			
3	2	1	0	3	2	1	0

Table 75. Bit Mappings for Packed 16-Bit Formats

Bit															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VK_FORMAT_R4G4B4A4_UNORM_PACK16															
R				**G**				**B**				**A**			
3	2	1	0	3	2	1	0	3	2	1	0	3	2	1	0
VK_FORMAT_B4G4R4A4_UNORM_PACK16															

Bit															
B				G				R				A			
3	2	1	0	3	2	1	0	3	2	1	0	3	2	1	0
VK_FORMAT_A4R4G4B4_UNORM_PACK16															
A				R				G				B			
3	2	1	0	3	2	1	0	3	2	1	0	3	2	1	0
VK_FORMAT_A4B4G4R4_UNORM_PACK16															
A				B				G				R			
3	2	1	0	3	2	1	0	3	2	1	0	3	2	1	0
VK_FORMAT_R5G6B5_UNORM_PACK16															
R					G						B				
4	3	2	1	0	5	4	3	2	1	0	4	3	2	1	0
VK_FORMAT_B5G6R5_UNORM_PACK16															
B					G						R				
4	3	2	1	0	5	4	3	2	1	0	4	3	2	1	0
VK_FORMAT_R5G5B5A1_UNORM_PACK16															
R					G					B					A
4	3	2	1	0	4	3	2	1	0	4	3	2	1	0	0
VK_FORMAT_B5G5R5A1_UNORM_PACK16															
B					G					R					A
4	3	2	1	0	4	3	2	1	0	4	3	2	1	0	0
VK_FORMAT_A1R5G5B5_UNORM_PACK16															
A		R				G					B				
0	4	3	2	1	0	4	3	2	1	0	4	3	2	1	0
VK_FORMAT_A1B5G5R5_UNORM_PACK16															
A		B				G					R				
0	4	3	2	1	0	4	3	2	1	0	4	3	2	1	0
VK_FORMAT_R10X6_UNORM_PACK16															
R										X					
9	8	7	6	5	4	3	2	1	0	5	4	3	2	1	0
VK_FORMAT_R12X4_UNORM_PACK16															
R												X			
11	10	9	8	7	6	5	4	3	2	1	0	3	2	1	0

Table 76. Bit Mappings for Packed 32-Bit Formats

Bit																													
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VK_FORMAT_A8B8G8R8_*_PACK32																															

Bit																															
A								B								G								R							
7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
VK_FORMAT_A2R10G10B10_*_PACK32																															
A								R								G								B							
1	0	9	8	7	6	5	4	3	2	1	0	9	8	7	6	5	4	3	2	1	0	9	8	7	6	5	4	3	2	1	0
VK_FORMAT_A2B10G10R10_*_PACK32																															
A								B								G								R							
1	0	9	8	7	6	5	4	3	2	1	0	9	8	7	6	5	4	3	2	1	0	9	8	7	6	5	4	3	2	1	0
VK_FORMAT_B10G11R11_UFLOAT_PACK32																															
B										G										R											
9	8	7	6	5	4	3	2	1	0	10	9	8	7	6	5	4	3	2	1	0	10	9	8	7	6	5	4	3	2	1	0
VK_FORMAT_E5B9G9R9_UFLOAT_PACK32																															
E					B							G								R											
4	3	2	1	0	8	7	6	5	4	3	2	1	0	8	7	6	5	4	3	2	1	0	8	7	6	5	4	3	2	1	0
VK_FORMAT_X8_D24_UNORM_PACK32																															
X								D																							
7	6	5	4	3	2	1	0	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

49.1.7. Depth/Stencil Formats

Depth/stencil formats are considered opaque and need not be stored in the exact number of bits per texel or component ordering indicated by the format enum. However, implementations **must** not substitute a different depth or stencil precision than is described in the format (e.g. D16 **must** not be implemented as D24 or D32).

49.1.8. Format Compatibility Classes

Uncompressed color formats are *compatible* with each other if they occupy the same number of bits per texel block as long as neither or both are alpha formats (e.g., VK_FORMAT_A8_UNORM) . Compressed color formats are compatible with each other if the only difference between them is the numeric format of the uncompressed texels. Each depth/stencil format is only compatible with itself. In the following table, all the formats in the same row are compatible. Each format has a defined *texel block extent* specifying how many texels each texel block represents in each dimension.

Table 77. Compatible Formats

Class, Texel Block Size, Texel Block Extent, # Texels/Block	**Formats**
8-bit Block size 1 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R4G4_UNORM_PACK8, VK_FORMAT_R8_UNORM, VK_FORMAT_R8_SNORM, VK_FORMAT_R8_USCALED, VK_FORMAT_R8_SSCALED, VK_FORMAT_R8_UINT, VK_FORMAT_R8_SINT, VK_FORMAT_R8_SRGB
16-bit Block size 2 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_A1B5G5R5_UNORM_PACK16, VK_FORMAT_R10X6_UNORM_PACK16, VK_FORMAT_R12X4_UNORM_PACK16, VK_FORMAT_A4R4G4B4_UNORM_PACK16, VK_FORMAT_A4B4G4R4_UNORM_PACK16, VK_FORMAT_R4G4B4A4_UNORM_PACK16, VK_FORMAT_B4G4R4A4_UNORM_PACK16, VK_FORMAT_R5G6B5_UNORM_PACK16, VK_FORMAT_B5G6R5_UNORM_PACK16, VK_FORMAT_R5G5B5A1_UNORM_PACK16, VK_FORMAT_B5G5R5A1_UNORM_PACK16, VK_FORMAT_A1R5G5B5_UNORM_PACK16, VK_FORMAT_R8G8_UNORM, VK_FORMAT_R8G8_SNORM, VK_FORMAT_R8G8_USCALED, VK_FORMAT_R8G8_SSCALED, VK_FORMAT_R8G8_UINT, VK_FORMAT_R8G8_SINT, VK_FORMAT_R8G8_SRGB, VK_FORMAT_R16_UNORM, VK_FORMAT_R16_SNORM, VK_FORMAT_R16_USCALED, VK_FORMAT_R16_SSCALED, VK_FORMAT_R16_UINT, VK_FORMAT_R16_SINT, VK_FORMAT_R16_SFLOAT
8-bit alpha Block size 1 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_A8_UNORM

Class, Texel Block Size, Texel Block Extent, # Texels/Block	**Formats**
24-bit Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R8G8B8_UNORM, VK_FORMAT_R8G8B8_SNORM, VK_FORMAT_R8G8B8_USCALED, VK_FORMAT_R8G8B8_SSCALED, VK_FORMAT_R8G8B8_UINT, VK_FORMAT_R8G8B8_SINT, VK_FORMAT_R8G8B8_SRGB, VK_FORMAT_B8G8R8_UNORM, VK_FORMAT_B8G8R8_SNORM, VK_FORMAT_B8G8R8_USCALED, VK_FORMAT_B8G8R8_SSCALED, VK_FORMAT_B8G8R8_UINT, VK_FORMAT_B8G8R8_SINT, VK_FORMAT_B8G8R8_SRGB

Class, Texel Block Size, Texel Block Extent, # Texels/Block	**Formats**
32-bit Block size 4 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R10X6G10X6_UNORM_2PACK16, VK_FORMAT_R12X4G12X4_UNORM_2PACK16, VK_FORMAT_R8G8B8A8_UNORM, VK_FORMAT_R8G8B8A8_SNORM, VK_FORMAT_R8G8B8A8_USCALED, VK_FORMAT_R8G8B8A8_SSCALED, VK_FORMAT_R8G8B8A8_UINT, VK_FORMAT_R8G8B8A8_SINT, VK_FORMAT_R8G8B8A8_SRGB, VK_FORMAT_B8G8R8A8_UNORM, VK_FORMAT_B8G8R8A8_SNORM, VK_FORMAT_B8G8R8A8_USCALED, VK_FORMAT_B8G8R8A8_SSCALED, VK_FORMAT_B8G8R8A8_UINT, VK_FORMAT_B8G8R8A8_SINT, VK_FORMAT_B8G8R8A8_SRGB, VK_FORMAT_A8B8G8R8_UNORM_PACK32, VK_FORMAT_A8B8G8R8_SNORM_PACK32, VK_FORMAT_A8B8G8R8_USCALED_PACK32, VK_FORMAT_A8B8G8R8_SSCALED_PACK32, VK_FORMAT_A8B8G8R8_UINT_PACK32, VK_FORMAT_A8B8G8R8_SINT_PACK32, VK_FORMAT_A8B8G8R8_SRGB_PACK32, VK_FORMAT_A2R10G10B10_UNORM_PACK32, VK_FORMAT_A2R10G10B10_SNORM_PACK32, VK_FORMAT_A2R10G10B10_USCALED_PACK32, VK_FORMAT_A2R10G10B10_SSCALED_PACK32, VK_FORMAT_A2R10G10B10_UINT_PACK32, VK_FORMAT_A2R10G10B10_SINT_PACK32, VK_FORMAT_A2B10G10R10_UNORM_PACK32, VK_FORMAT_A2B10G10R10_SNORM_PACK32, VK_FORMAT_A2B10G10R10_USCALED_PACK32, VK_FORMAT_A2B10G10R10_SSCALED_PACK32, VK_FORMAT_A2B10G10R10_UINT_PACK32, VK_FORMAT_A2B10G10R10_SINT_PACK32, VK_FORMAT_R16G16_UNORM, VK_FORMAT_R16G16_SNORM, VK_FORMAT_R16G16_USCALED, VK_FORMAT_R16G16_SSCALED, VK_FORMAT_R16G16_UINT, VK_FORMAT_R16G16_SINT, VK_FORMAT_R16G16_SFLOAT, VK_FORMAT_R32_UINT, VK_FORMAT_R32_SINT

Class, Texel Block Size, Texel Block Extent, # Texels/Block	**Formats**
32-bit (continued) Block size 4 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R32_SFLOAT, VK_FORMAT_B10G11R11_UFLOAT_PACK32, VK_FORMAT_E5B9G9R9_UFLOAT_PACK32
48-bit Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R16G16B16_UNORM, VK_FORMAT_R16G16B16_SNORM, VK_FORMAT_R16G16B16_USCALED, VK_FORMAT_R16G16B16_SSCALED, VK_FORMAT_R16G16B16_UINT, VK_FORMAT_R16G16B16_SINT, VK_FORMAT_R16G16B16_SFLOAT
64-bit Block size 8 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R16G16B16A16_UNORM, VK_FORMAT_R16G16B16A16_SNORM, VK_FORMAT_R16G16B16A16_USCALED, VK_FORMAT_R16G16B16A16_SSCALED, VK_FORMAT_R16G16B16A16_UINT, VK_FORMAT_R16G16B16A16_SINT, VK_FORMAT_R16G16B16A16_SFLOAT, VK_FORMAT_R32G32_UINT, VK_FORMAT_R32G32_SINT, VK_FORMAT_R32G32_SFLOAT, VK_FORMAT_R64_UINT, VK_FORMAT_R64_SINT, VK_FORMAT_R64_SFLOAT
96-bit Block size 12 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R32G32B32_UINT, VK_FORMAT_R32G32B32_SINT, VK_FORMAT_R32G32B32_SFLOAT
128-bit Block size 16 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R32G32B32A32_UINT, VK_FORMAT_R32G32B32A32_SINT, VK_FORMAT_R32G32B32A32_SFLOAT, VK_FORMAT_R64G64_UINT, VK_FORMAT_R64G64_SINT, VK_FORMAT_R64G64_SFLOAT
192-bit Block size 24 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R64G64B64_UINT, VK_FORMAT_R64G64B64_SINT, VK_FORMAT_R64G64B64_SFLOAT

Class, Texel Block Size, Texel Block Extent, # Texels/Block	**Formats**
256-bit Block size 32 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R64G64B64A64_UINT, VK_FORMAT_R64G64B64A64_SINT, VK_FORMAT_R64G64B64A64_SFLOAT
D16 Block size 2 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_D16_UNORM
D24 Block size 4 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_X8_D24_UNORM_PACK32
D32 Block size 4 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_D32_SFLOAT
S8 Block size 1 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_S8_UINT
D16S8 Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_D16_UNORM_S8_UINT
D24S8 Block size 4 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_D24_UNORM_S8_UINT
D32S8 Block size 5 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_D32_SFLOAT_S8_UINT
BC1_RGB Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC1_RGB_UNORM_BLOCK, VK_FORMAT_BC1_RGB_SRGB_BLOCK
BC1_RGBA Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC1_RGBA_UNORM_BLOCK, VK_FORMAT_BC1_RGBA_SRGB_BLOCK

Class, Texel Block Size, Texel Block Extent, # Texels/Block	**Formats**
BC2 Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC2_UNORM_BLOCK, VK_FORMAT_BC2_SRGB_BLOCK
BC3 Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC3_UNORM_BLOCK, VK_FORMAT_BC3_SRGB_BLOCK
BC4 Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC4_UNORM_BLOCK, VK_FORMAT_BC4_SNORM_BLOCK
BC5 Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC5_UNORM_BLOCK, VK_FORMAT_BC5_SNORM_BLOCK
BC6H Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC6H_UFLOAT_BLOCK, VK_FORMAT_BC6H_SFLOAT_BLOCK
BC7 Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC7_UNORM_BLOCK, VK_FORMAT_BC7_SRGB_BLOCK
ETC2_RGB Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK, VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK
ETC2_RGBA Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK, VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK
ETC2_EAC_RGBA Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK, VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK
EAC_R Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_EAC_R11_UNORM_BLOCK, VK_FORMAT_EAC_R11_SNORM_BLOCK

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
EAC_RG Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_EAC_R11G11_UNORM_BLOCK, VK_FORMAT_EAC_R11G11_SNORM_BLOCK
ASTC_4x4 Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_ASTC_4x4_SFLOAT_BLOCK, VK_FORMAT_ASTC_4x4_UNORM_BLOCK, VK_FORMAT_ASTC_4x4_SRGB_BLOCK
ASTC_5x4 Block size 16 byte 5x4x1 block extent 20 texel/block	VK_FORMAT_ASTC_5x4_SFLOAT_BLOCK, VK_FORMAT_ASTC_5x4_UNORM_BLOCK, VK_FORMAT_ASTC_5x4_SRGB_BLOCK
ASTC_5x5 Block size 16 byte 5x5x1 block extent 25 texel/block	VK_FORMAT_ASTC_5x5_SFLOAT_BLOCK, VK_FORMAT_ASTC_5x5_UNORM_BLOCK, VK_FORMAT_ASTC_5x5_SRGB_BLOCK
ASTC_6x5 Block size 16 byte 6x5x1 block extent 30 texel/block	VK_FORMAT_ASTC_6x5_SFLOAT_BLOCK, VK_FORMAT_ASTC_6x5_UNORM_BLOCK, VK_FORMAT_ASTC_6x5_SRGB_BLOCK
ASTC_6x6 Block size 16 byte 6x6x1 block extent 36 texel/block	VK_FORMAT_ASTC_6x6_SFLOAT_BLOCK, VK_FORMAT_ASTC_6x6_UNORM_BLOCK, VK_FORMAT_ASTC_6x6_SRGB_BLOCK
ASTC_8x5 Block size 16 byte 8x5x1 block extent 40 texel/block	VK_FORMAT_ASTC_8x5_SFLOAT_BLOCK, VK_FORMAT_ASTC_8x5_UNORM_BLOCK, VK_FORMAT_ASTC_8x5_SRGB_BLOCK
ASTC_8x6 Block size 16 byte 8x6x1 block extent 48 texel/block	VK_FORMAT_ASTC_8x6_SFLOAT_BLOCK, VK_FORMAT_ASTC_8x6_UNORM_BLOCK, VK_FORMAT_ASTC_8x6_SRGB_BLOCK
ASTC_8x8 Block size 16 byte 8x8x1 block extent 64 texel/block	VK_FORMAT_ASTC_8x8_SFLOAT_BLOCK, VK_FORMAT_ASTC_8x8_UNORM_BLOCK, VK_FORMAT_ASTC_8x8_SRGB_BLOCK
ASTC_10x5 Block size 16 byte 10x5x1 block extent 50 texel/block	VK_FORMAT_ASTC_10x5_SFLOAT_BLOCK, VK_FORMAT_ASTC_10x5_UNORM_BLOCK, VK_FORMAT_ASTC_10x5_SRGB_BLOCK

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
ASTC_10x6 Block size 16 byte 10x6x1 block extent 60 texel/block	VK_FORMAT_ASTC_10x6_SFLOAT_BLOCK, VK_FORMAT_ASTC_10x6_UNORM_BLOCK, VK_FORMAT_ASTC_10x6_SRGB_BLOCK
ASTC_10x8 Block size 16 byte 10x8x1 block extent 80 texel/block	VK_FORMAT_ASTC_10x8_SFLOAT_BLOCK, VK_FORMAT_ASTC_10x8_UNORM_BLOCK, VK_FORMAT_ASTC_10x8_SRGB_BLOCK
ASTC_10x10 Block size 16 byte 10x10x1 block extent 100 texel/block	VK_FORMAT_ASTC_10x10_SFLOAT_BLOCK, VK_FORMAT_ASTC_10x10_UNORM_BLOCK, VK_FORMAT_ASTC_10x10_SRGB_BLOCK
ASTC_12x10 Block size 16 byte 12x10x1 block extent 120 texel/block	VK_FORMAT_ASTC_12x10_SFLOAT_BLOCK, VK_FORMAT_ASTC_12x10_UNORM_BLOCK, VK_FORMAT_ASTC_12x10_SRGB_BLOCK
ASTC_12x12 Block size 16 byte 12x12x1 block extent 144 texel/block	VK_FORMAT_ASTC_12x12_SFLOAT_BLOCK, VK_FORMAT_ASTC_12x12_UNORM_BLOCK, VK_FORMAT_ASTC_12x12_SRGB_BLOCK
32-bit G8B8G8R8 Block size 4 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_G8B8G8R8_422_UNORM
32-bit B8G8R8G8 Block size 4 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_B8G8R8G8_422_UNORM
8-bit 3-plane 420 Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM
8-bit 2-plane 420 Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G8_B8R8_2PLANE_420_UNORM
8-bit 3-plane 422 Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
8-bit 2-plane 422 Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G8_B8R8_2PLANE_422_UNORM
8-bit 3-plane 444 Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM
64-bit R10G10B10A10 Block size 8 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16
64-bit G10B10G10R10 Block size 8 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16
64-bit B10G10R10G10 Block size 8 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16
10-bit 3-plane 420 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16
10-bit 2-plane 420 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16
10-bit 3-plane 422 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16
10-bit 2-plane 422 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16
10-bit 3-plane 444 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
64-bit R12G12B12A12 Block size 8 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16
64-bit G12B12G12R12 Block size 8 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16
64-bit B12G12R12G12 Block size 8 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16
12-bit 3-plane 420 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16
12-bit 2-plane 420 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16
12-bit 3-plane 422 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16
12-bit 2-plane 422 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16
12-bit 3-plane 444 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16
64-bit G16B16G16R16 Block size 8 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_G16B16G16R16_422_UNORM
64-bit B16G16R16G16 Block size 8 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_B16G16R16G16_422_UNORM

Class, Texel Block Size, Texel Block Extent, # Texels/Block	**Formats**
16-bit 3-plane 420 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM
16-bit 2-plane 420 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G16_B16R16_2PLANE_420_UNORM
16-bit 3-plane 422 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM
16-bit 2-plane 422 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G16_B16R16_2PLANE_422_UNORM
16-bit 3-plane 444 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM
ASTC_3x3x3 Block size 16 byte 3x3x3 block extent 27 texel/block	VK_FORMAT_ASTC_3x3x3_UNORM_BLOCK_EXT, VK_FORMAT_ASTC_3x3x3_SRGB_BLOCK_EXT, VK_FORMAT_ASTC_3x3x3_SFLOAT_BLOCK_EXT
ASTC_4x3x3 Block size 16 byte 4x3x3 block extent 36 texel/block	VK_FORMAT_ASTC_4x3x3_UNORM_BLOCK_EXT, VK_FORMAT_ASTC_4x3x3_SRGB_BLOCK_EXT, VK_FORMAT_ASTC_4x3x3_SFLOAT_BLOCK_EXT
ASTC_4x4x3 Block size 16 byte 4x4x3 block extent 48 texel/block	VK_FORMAT_ASTC_4x4x3_UNORM_BLOCK_EXT, VK_FORMAT_ASTC_4x4x3_SRGB_BLOCK_EXT, VK_FORMAT_ASTC_4x4x3_SFLOAT_BLOCK_EXT
ASTC_4x4x4 Block size 16 byte 4x4x4 block extent 64 texel/block	VK_FORMAT_ASTC_4x4x4_UNORM_BLOCK_EXT, VK_FORMAT_ASTC_4x4x4_SRGB_BLOCK_EXT, VK_FORMAT_ASTC_4x4x4_SFLOAT_BLOCK_EXT
ASTC_5x4x4 Block size 16 byte 5x4x4 block extent 80 texel/block	VK_FORMAT_ASTC_5x4x4_UNORM_BLOCK_EXT, VK_FORMAT_ASTC_5x4x4_SRGB_BLOCK_EXT, VK_FORMAT_ASTC_5x4x4_SFLOAT_BLOCK_EXT

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
ASTC_5x5x4 Block size 16 byte 5x5x4 block extent 100 texel/block	VK_FORMAT_ASTC_5x5x4_UNORM_BLOCK_EXT, VK_FORMAT_ASTC_5x5x4_SRGB_BLOCK_EXT, VK_FORMAT_ASTC_5x5x4_SFLOAT_BLOCK_EXT
ASTC_5x5x5 Block size 16 byte 5x5x5 block extent 125 texel/block	VK_FORMAT_ASTC_5x5x5_UNORM_BLOCK_EXT, VK_FORMAT_ASTC_5x5x5_SRGB_BLOCK_EXT, VK_FORMAT_ASTC_5x5x5_SFLOAT_BLOCK_EXT
ASTC_6x5x5 Block size 16 byte 6x5x5 block extent 150 texel/block	VK_FORMAT_ASTC_6x5x5_UNORM_BLOCK_EXT, VK_FORMAT_ASTC_6x5x5_SRGB_BLOCK_EXT, VK_FORMAT_ASTC_6x5x5_SFLOAT_BLOCK_EXT
ASTC_6x6x5 Block size 16 byte 6x6x5 block extent 180 texel/block	VK_FORMAT_ASTC_6x6x5_UNORM_BLOCK_EXT, VK_FORMAT_ASTC_6x6x5_SRGB_BLOCK_EXT, VK_FORMAT_ASTC_6x6x5_SFLOAT_BLOCK_EXT
ASTC_6x6x6 Block size 16 byte 6x6x6 block extent 216 texel/block	VK_FORMAT_ASTC_6x6x6_UNORM_BLOCK_EXT, VK_FORMAT_ASTC_6x6x6_SRGB_BLOCK_EXT, VK_FORMAT_ASTC_6x6x6_SFLOAT_BLOCK_EXT
8-bit 2-plane 444 Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G8_B8R8_2PLANE_444_UNORM
10-bit 2-plane 444 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G10X6_B10X6R10X6_2PLANE_444_UNORM_3PACK16
12-bit 2-plane 444 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G12X4_B12X4R12X4_2PLANE_444_UNORM_3PACK16
16-bit 2-plane 444 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G16_B16R16_2PLANE_444_UNORM

Size Compatibility

Color formats with the same texel block size are considered *size-compatible* as long as neither or both are alpha formats (e.g., VK_FORMAT_A8_UNORM). If two size-compatible formats have

different block extents (i.e. for compressed formats), then an image with size A × B × C in one format with a block extent of a × b × c can be represented as an image with size X × Y × Z in the other format with block extent x × y × z at the ratio between the block extents for each format, where

$$\lceil A/a \rceil = \lceil X/x \rceil$$

$$\lceil B/b \rceil = \lceil Y/y \rceil$$

$$\lceil C/c \rceil = \lceil Z/z \rceil$$

NOTE	For example, a 7x3 image in the VK_FORMAT_ASTC_8x5_UNORM_BLOCK format can be represented as a 1x1 VK_FORMAT_R64G64_UINT image.

Images created with the VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT flag can have size-compatible views created from them to enable access via different size-compatible formats. Image views created in this way will be sized to match the expectations of the block extents noted above.

Copy operations are able to copy between size-compatible formats in different resources to enable manipulation of data in different formats. The extent used in these copy operations always matches the source image, and is resized to the expectations of the block extents noted above for the destination image.

Copy operations between color formats and a depth-stencil are size-compatible as defined by the list of compatible depth-stencil and color formats.

49.2. Format Properties

To query supported format features which are properties of the physical device, call:

WARNING	This functionality is superseded by vkGetPhysicalDeviceFormatProperties2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceFormatProperties(
    VkPhysicalDevice                            physicalDevice,
    VkFormat                                    format,
    VkFormatProperties*                         pFormatProperties);
```

- physicalDevice is the physical device from which to query the format properties.
- format is the format whose properties are queried.
- pFormatProperties is a pointer to a VkFormatProperties structure in which physical device

properties for format are returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceFormatProperties-physicalDevice-parameter
  physicalDevice **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceFormatProperties-format-parameter
  format **must** be a valid VkFormat value
- VUID-vkGetPhysicalDeviceFormatProperties-pFormatProperties-parameter
  pFormatProperties **must** be a valid pointer to a VkFormatProperties structure

The VkFormatProperties structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkFormatProperties {
    VkFormatFeatureFlags    linearTilingFeatures;
    VkFormatFeatureFlags    optimalTilingFeatures;
    VkFormatFeatureFlags    bufferFeatures;
} VkFormatProperties;
```

- linearTilingFeatures is a bitmask of VkFormatFeatureFlagBits specifying features supported by images created with a tiling parameter of VK_IMAGE_TILING_LINEAR.
- optimalTilingFeatures is a bitmask of VkFormatFeatureFlagBits specifying features supported by images created with a tiling parameter of VK_IMAGE_TILING_OPTIMAL.
- bufferFeatures is a bitmask of VkFormatFeatureFlagBits specifying features supported by buffers.

NOTE If no format feature flags are supported, the format itself is not supported, and images of that format cannot be created.

If format is block-compressed, requires sampler $Y'C_BC_R$ conversion, or is a depth/stencil format then bufferFeatures **must** not support any features for the format.

If format is not a multi-plane format then linearTilingFeatures and optimalTilingFeatures **must** not contain VK_FORMAT_FEATURE_DISJOINT_BIT.

Bits which **can** be set in the VkFormatProperties features linearTilingFeatures, optimalTilingFeatures, VkDrmFormatModifierPropertiesEXT::drmFormatModifierTilingFeatures, and bufferFeatures are:

WARNING This functionality is superseded by VkFormatFeatureFlagBits2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef enum VkFormatFeatureFlagBits {
```

```
    VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT = 0x00000001,
    VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT = 0x00000002,
    VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT = 0x00000004,
    VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT = 0x00000008,
    VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT = 0x00000010,
    VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT = 0x00000020,
    VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT = 0x00000040,
    VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT = 0x00000080,
    VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT = 0x00000100,
    VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT = 0x00000200,
    VK_FORMAT_FEATURE_BLIT_SRC_BIT = 0x00000400,
    VK_FORMAT_FEATURE_BLIT_DST_BIT = 0x00000800,
    VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT = 0x00001000,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_FEATURE_TRANSFER_SRC_BIT = 0x00004000,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_FEATURE_TRANSFER_DST_BIT = 0x00008000,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT = 0x00020000,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT = 0x00040000,
  // Provided by VK_VERSION_1_1

VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT =
0x00080000,
  // Provided by VK_VERSION_1_1

VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT =
0x00100000,
  // Provided by VK_VERSION_1_1

VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEA
BLE_BIT = 0x00200000,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_FEATURE_DISJOINT_BIT = 0x00400000,
  // Provided by VK_VERSION_1_1
    VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT = 0x00800000,
  // Provided by VK_VERSION_1_2
    VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT = 0x00010000,
  // Provided by VK_KHR_video_decode_queue
    VK_FORMAT_FEATURE_VIDEO_DECODE_OUTPUT_BIT_KHR = 0x02000000,
  // Provided by VK_KHR_video_decode_queue
    VK_FORMAT_FEATURE_VIDEO_DECODE_DPB_BIT_KHR = 0x04000000,
  // Provided by VK_KHR_acceleration_structure
    VK_FORMAT_FEATURE_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR = 0x20000000,
  // Provided by VK_EXT_filter_cubic
    VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT = 0x00002000,
  // Provided by VK_EXT_fragment_density_map
    VK_FORMAT_FEATURE_FRAGMENT_DENSITY_MAP_BIT_EXT = 0x01000000,
  // Provided by VK_KHR_fragment_shading_rate
    VK_FORMAT_FEATURE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR = 0x40000000,
```

```
  // Provided by VK_KHR_video_encode_queue
    VK_FORMAT_FEATURE_VIDEO_ENCODE_INPUT_BIT_KHR = 0x08000000,
  // Provided by VK_KHR_video_encode_queue
    VK_FORMAT_FEATURE_VIDEO_ENCODE_DPB_BIT_KHR = 0x10000000,
  // Provided by VK_KHR_maintenance1
    VK_FORMAT_FEATURE_TRANSFER_SRC_BIT_KHR = VK_FORMAT_FEATURE_TRANSFER_SRC_BIT,
  // Provided by VK_KHR_maintenance1
    VK_FORMAT_FEATURE_TRANSFER_DST_BIT_KHR = VK_FORMAT_FEATURE_TRANSFER_DST_BIT,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT_KHR =
VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT_KHR =
VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT,
  // Provided by VK_KHR_sampler_ycbcr_conversion

VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT_KH
R =
VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT,
  // Provided by VK_KHR_sampler_ycbcr_conversion

VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT_KH
R =
VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT,
  // Provided by VK_KHR_sampler_ycbcr_conversion

VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEA
BLE_BIT_KHR =
VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEA
BLE_BIT,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_FEATURE_DISJOINT_BIT_KHR = VK_FORMAT_FEATURE_DISJOINT_BIT,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT_KHR =
VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT,
} VkFormatFeatureFlagBits;
```

These values all have the same meaning as the equivalently named values for VkFormatFeatureFlags2 and **may** be set in `linearTilingFeatures`, `optimalTilingFeatures`, and VkDrmFormatModifierPropertiesEXT::`drmFormatModifierTilingFeatures`, specifying that the features are supported by images or image views or sampler $Y'C_BC_R$ conversion objects created with the queried vkGetPhysicalDeviceFormatProperties::`format`:

- VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT specifies that an image view **can** be sampled from.
- VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT specifies that an image view **can** be used as a storage image.
- VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT specifies that an image view **can** be used as a storage image that supports atomic operations.

- VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT specifies that an image view **can** be used as a framebuffer color attachment and as an input attachment.
- VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT specifies that an image view **can** be used as a framebuffer color attachment that supports blending.
- VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT specifies that an image view **can** be used as a framebuffer depth/stencil attachment and as an input attachment.
- VK_FORMAT_FEATURE_BLIT_SRC_BIT specifies that an image **can** be used as `srcImage` for the `vkCmdBlitImage2` and `vkCmdBlitImage` commands.
- VK_FORMAT_FEATURE_BLIT_DST_BIT specifies that an image **can** be used as `dstImage` for the `vkCmdBlitImage2` and `vkCmdBlitImage` commands.
- VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT specifies that if VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT is also set, an image view **can** be used with a sampler that has either of `magFilter` or `minFilter` set to VK_FILTER_LINEAR, or `mipmapMode` set to VK_SAMPLER_MIPMAP_MODE_LINEAR. If VK_FORMAT_FEATURE_BLIT_SRC_BIT is also set, an image can be used as the `srcImage` to `vkCmdBlitImage2` and `vkCmdBlitImage` with a `filter` of VK_FILTER_LINEAR. This bit **must** only be exposed for formats that also support the VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT or VK_FORMAT_FEATURE_BLIT_SRC_BIT.

  If the format being queried is a depth/stencil format, this bit only specifies that the depth aspect (not the stencil aspect) of an image of this format supports linear filtering, and that linear filtering of the depth aspect is supported whether depth compare is enabled in the sampler or not. Where depth comparison is supported it **may** be linear filtered whether this bit is present or not, but where this bit is not present the filtered value **may** be computed in an implementation-dependent manner which differs from the normal rules of linear filtering. The resulting value **must** be in the range [0,1] and **should** be proportional to, or a weighted average of, the number of comparison passes or failures.

- VK_FORMAT_FEATURE_TRANSFER_SRC_BIT specifies that an image **can** be used as a source image for copy commands. If the application `apiVersion` is Vulkan 1.0 and `VK_KHR_maintenance1` is not supported, VK_FORMAT_FEATURE_TRANSFER_SRC_BIT is implied to be set when the format feature flag is not 0.
- VK_FORMAT_FEATURE_TRANSFER_DST_BIT specifies that an image **can** be used as a destination image for copy commands and clear commands. If the application `apiVersion` is Vulkan 1.0 and `VK_KHR_maintenance1` is not supported, VK_FORMAT_FEATURE_TRANSFER_DST_BIT is implied to be set when the format feature flag is not 0.
- VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT specifies that `VkImage` **can** be used as a sampled image with a min or max VkSamplerReductionMode. This bit **must** only be exposed for formats that also support the VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT.
- VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT specifies that `VkImage` **can** be used with a sampler that has either of `magFilter` or `minFilter` set to VK_FILTER_CUBIC_EXT, or be the source image for a blit with `filter` set to VK_FILTER_CUBIC_EXT. This bit **must** only be exposed for formats that also support the VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT. If the format being queried is a depth/stencil format, this only specifies that the depth aspect is cubic filterable.
- VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT specifies that an application **can**

define a sampler $Y'C_BC_R$ conversion using this format as a source, and that an image of this format **can** be used with a VkSamplerYcbcrConversionCreateInfo `xChromaOffset` and/or `yChromaOffset` of VK_CHROMA_LOCATION_MIDPOINT. Otherwise both `xChromaOffset` and `yChromaOffset` **must** be VK_CHROMA_LOCATION_COSITED_EVEN. If a format does not incorporate chroma downsampling (it is not a "422" or "420" format) but the implementation supports sampler $Y'C_BC_R$ conversion for this format, the implementation **must** set VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT.

- VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT specifies that an application **can** define a sampler $Y'C_BC_R$ conversion using this format as a source, and that an image of this format **can** be used with a VkSamplerYcbcrConversionCreateInfo `xChromaOffset` and/or `yChromaOffset` of VK_CHROMA_LOCATION_COSITED_EVEN. Otherwise both `xChromaOffset` and `yChromaOffset` **must** be VK_CHROMA_LOCATION_MIDPOINT. If neither VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT nor VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT is set, the application **must** not define a sampler $Y'C_BC_R$ conversion using this format as a source.
- VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT specifies that an application **can** define a sampler $Y'C_BC_R$ conversion using this format as a source with `chromaFilter` set to VK_FILTER_LINEAR.
- VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT specifies that the format can have different chroma, min, and mag filters.
- VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT specifies that reconstruction is explicit, as described in Chroma Reconstruction. If this bit is not present, reconstruction is implicit by default.
- VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT specifies that reconstruction **can** be forcibly made explicit by setting VkSamplerYcbcrConversionCreateInfo::`forceExplicitReconstruction` to VK_TRUE. If the format being queried supports VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT it **must** also support VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT.
- VK_FORMAT_FEATURE_DISJOINT_BIT specifies that a multi-planar image **can** have the VK_IMAGE_CREATE_DISJOINT_BIT set during image creation. An implementation **must** not set VK_FORMAT_FEATURE_DISJOINT_BIT for *single-plane formats*.
- VK_FORMAT_FEATURE_FRAGMENT_DENSITY_MAP_BIT_EXT specifies that an image view **can** be used as a fragment density map attachment.
- VK_FORMAT_FEATURE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR specifies that an image view **can** be used as a fragment shading rate attachment. An implementation **must** not set this feature for formats with a numeric format other than `UINT`, or set it as a buffer feature.
- VK_FORMAT_FEATURE_VIDEO_DECODE_OUTPUT_BIT_KHR specifies that an image view with this format **can** be used as a decode output picture in video decode operations.
- VK_FORMAT_FEATURE_VIDEO_DECODE_DPB_BIT_KHR specifies that an image view with this format **can** be used as an output reconstructed picture or an input reference picture in video decode operations.

- VK_FORMAT_FEATURE_VIDEO_ENCODE_INPUT_BIT_KHR specifies that an image view with this format **can** be used as an encode input picture in video encode operations.
- VK_FORMAT_FEATURE_VIDEO_ENCODE_DPB_BIT_KHR specifies that an image view with this format **can** be used as an output reconstructed picture or an input reference picture in video encode operations.

> **NOTE** Specific video profiles **may** have additional restrictions on the format and other image creation parameters corresponding to image views used by video coding operations that **can** be enumerated using the vkGetPhysicalDeviceVideoFormatPropertiesKHR command.

The following bits **may** be set in `bufferFeatures`, specifying that the features are supported by buffers or buffer views created with the queried vkGetPhysicalDeviceFormatProperties::`format`:

- VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT specifies that the format **can** be used to create a buffer view that **can** be bound to a VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER descriptor.
- VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT specifies that the format **can** be used to create a buffer view that **can** be bound to a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor.
- VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT specifies that atomic operations are supported on VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER with this format.
- VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT specifies that the format **can** be used as a vertex attribute format (`VkVertexInputAttributeDescription`::`format`).
- VK_FORMAT_FEATURE_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR specifies that the format **can** be used as the vertex format when creating an acceleration structure (`VkAccelerationStructureGeometryTrianglesDataKHR`::`vertexFormat`). This format **can** also be used as the vertex format in host memory when doing host acceleration structure builds.

> **NOTE** VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT and VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT are only intended to be advertised for single-component formats, since SPIR-V atomic operations require a scalar type.

> **WARNING** This functionality is superseded by VkFormatFeatureFlags2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkFormatFeatureFlags;
```

`VkFormatFeatureFlags` is a bitmask type for setting a mask of zero or more VkFormatFeatureFlagBits.

To query supported format features which are properties of the physical device, call:

```
// Provided by VK_VERSION_1_1
void vkGetPhysicalDeviceFormatProperties2(
    VkPhysicalDevice                            physicalDevice,
    VkFormat                                    format,
    VkFormatProperties2*                        pFormatProperties);
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to vkGetPhysicalDeviceFormatProperties2
void vkGetPhysicalDeviceFormatProperties2KHR(
    VkPhysicalDevice                            physicalDevice,
    VkFormat                                    format,
    VkFormatProperties2*                        pFormatProperties);
```

- `physicalDevice` is the physical device from which to query the format properties.
- `format` is the format whose properties are queried.
- `pFormatProperties` is a pointer to a VkFormatProperties2 structure in which physical device properties for `format` are returned.

`vkGetPhysicalDeviceFormatProperties2` behaves similarly to vkGetPhysicalDeviceFormatProperties, with the ability to return extended information in a `pNext` chain of output structures.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceFormatProperties2-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceFormatProperties2-format-parameter
  `format` **must** be a valid VkFormat value
- VUID-vkGetPhysicalDeviceFormatProperties2-pFormatProperties-parameter
  `pFormatProperties` **must** be a valid pointer to a VkFormatProperties2 structure

The `VkFormatProperties2` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkFormatProperties2 {
    VkStructureType       sType;
    void*                 pNext;
    VkFormatProperties    formatProperties;
} VkFormatProperties2;
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to VkFormatProperties2
typedef VkFormatProperties2 VkFormatProperties2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `formatProperties` is a VkFormatProperties structure describing features supported by the requested format.

Valid Usage (Implicit)

- VUID-VkFormatProperties2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_2
- VUID-VkFormatProperties2-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkDrmFormatModifierPropertiesList2EXT, VkDrmFormatModifierPropertiesListEXT, VkFormatProperties3, VkFormatProperties4KHR, or VkSubpassResolvePerformanceQueryEXT
- VUID-VkFormatProperties2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

To obtain the list of Linux DRM format modifiers compatible with a VkFormat, add a VkDrmFormatModifierPropertiesListEXT structure to the `pNext` chain of VkFormatProperties2.

The VkDrmFormatModifierPropertiesListEXT structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier
typedef struct VkDrmFormatModifierPropertiesListEXT {
    VkStructureType                      sType;
    void*                                pNext;
    uint32_t                             drmFormatModifierCount;
    VkDrmFormatModifierPropertiesEXT*    pDrmFormatModifierProperties;
} VkDrmFormatModifierPropertiesListEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `drmFormatModifierCount` is an inout parameter related to the number of modifiers compatible with the `format`, as described below.
- `pDrmFormatModifierProperties` is either `NULL` or a pointer to an array of VkDrmFormatModifierPropertiesEXT structures.

If `pDrmFormatModifierProperties` is `NULL`, then the function returns in `drmFormatModifierCount` the number of modifiers compatible with the queried `format`. Otherwise, the application **must** set `drmFormatModifierCount` to the length of the array `pDrmFormatModifierProperties`; the function will write at most `drmFormatModifierCount` elements to the array, and will return in `drmFormatModifierCount` the number of elements written.

Among the elements in array `pDrmFormatModifierProperties`, each returned `drmFormatModifier` **must**

be unique.

Valid Usage (Implicit)

- VUID-VkDrmFormatModifierPropertiesListEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DRM_FORMAT_MODIFIER_PROPERTIES_LIST_EXT

Structure Chaining

Extends the structure

- VkFormatProperties2

The VkDrmFormatModifierPropertiesEXT structure describes properties of a VkFormat when that format is combined with a Linux DRM format modifier. These properties, like those of VkFormatProperties2, are independent of any particular image.

The VkDrmFormatModifierPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier
typedef struct VkDrmFormatModifierPropertiesEXT {
    uint64_t                drmFormatModifier;
    uint32_t                drmFormatModifierPlaneCount;
    VkFormatFeatureFlags    drmFormatModifierTilingFeatures;
} VkDrmFormatModifierPropertiesEXT;
```

- `drmFormatModifier` is a *Linux DRM format modifier*.
- `drmFormatModifierPlaneCount` is the number of *memory planes* in any image created with `format` and `drmFormatModifier`. An image's *memory planecount* is distinct from its *format planecount*, as explained below.
- `drmFormatModifierTilingFeatures` is a bitmask of VkFormatFeatureFlagBits that are supported by any image created with `format` and `drmFormatModifier`.

The returned `drmFormatModifierTilingFeatures` **must** contain at least one bit.

The implementation **must** not return `DRM_FORMAT_MOD_INVALID` in `drmFormatModifier`.

An image's *memory planecount* (as returned by `drmFormatModifierPlaneCount`) is distinct from its *format planecount* (in the sense of multi-planar format). In VkImageAspectFlags, each `VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT` represents a *memory plane* and each `VK_IMAGE_ASPECT_PLANE_i_BIT` a *format plane*.

An image's set of *format planes* is an ordered partition of the image's **content** into separable groups of format components. The ordered partition is encoded in the name of each VkFormat. For example, VK_FORMAT_G8_B8R8_2PLANE_420_UNORM contains two *format planes*; the first plane contains the green component and the second plane contains the blue component and red

component. If the format name does not contain PLANE, then the format contains a single plane; for example, VK_FORMAT_R8G8B8A8_UNORM. Some commands, such as vkCmdCopyBufferToImage, do not operate on all format components in the image, but instead operate only on the *format planes* explicitly chosen by the application and operate on each *format plane* independently.

An image's set of *memory planes* is an ordered partition of the image's **memory** rather than the image's **content**. Each *memory plane* is a contiguous range of memory. The union of an image's *memory planes* is not necessarily contiguous.

If an image is linear, then the partition is the same for *memory planes* and for *format planes*. Therefore, if the returned drmFormatModifier is DRM_FORMAT_MOD_LINEAR, then drmFormatModifierPlaneCount **must** equal the *format planecount*, and drmFormatModifierTilingFeatures **must** be identical to the VkFormatProperties2 ::formatProperties.linearTilingFeatures returned in the same pNext chain.

If an image is non-linear, then the partition of the image's **memory** into *memory planes* is implementation-specific and **may** be unrelated to the partition of the image's **content** into *format planes*. For example, consider an image whose format is VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM, tiling is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, whose drmFormatModifier is not DRM_FORMAT_MOD_LINEAR, and flags lacks VK_IMAGE_CREATE_DISJOINT_BIT. The image has 3 *format planes*, and commands such vkCmdCopyBufferToImage act on each *format plane* independently as if the data of each *format plane* were separable from the data of the other planes. In a straightforward implementation, the implementation **may** store the image's content in 3 adjacent *memory planes* where each *memory plane* corresponds exactly to a *format plane*. However, the implementation **may** also store the image's content in a single *memory plane* where all format components are combined using an implementation-private block-compressed format; or the implementation **may** store the image's content in a collection of 7 adjacent *memory planes* using an implementation-private sharding technique. Because the image is non-linear and non-disjoint, the implementation has much freedom when choosing the image's placement in memory.

The *memory planecount* applies to function parameters and structures only when the API specifies an explicit requirement on drmFormatModifierPlaneCount. In all other cases, the *memory planecount* is ignored.

The list of Linux DRM format modifiers compatible with a VkFormat **can** be obtained by adding a VkDrmFormatModifierPropertiesList2EXT structure to the pNext chain of VkFormatProperties2.

The VkDrmFormatModifierPropertiesList2EXT structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
typedef struct VkDrmFormatModifierPropertiesList2EXT {
    VkStructureType                      sType;
    void*                                pNext;
    uint32_t                             drmFormatModifierCount;
    VkDrmFormatModifierProperties2EXT*   pDrmFormatModifierProperties;
} VkDrmFormatModifierPropertiesList2EXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `drmFormatModifierCount` is an inout parameter related to the number of modifiers compatible with the `format`, as described below.
- `pDrmFormatModifierProperties` is either `NULL` or a pointer to an array of VkDrmFormatModifierProperties2EXT structures.

If `pDrmFormatModifierProperties` is `NULL`, the number of modifiers compatible with the queried `format` is returned in `drmFormatModifierCount`. Otherwise, the application **must** set `drmFormatModifierCount` to the length of the array `pDrmFormatModifierProperties`; the function will write at most `drmFormatModifierCount` elements to the array, and will return in `drmFormatModifierCount` the number of elements written.

Among the elements in array `pDrmFormatModifierProperties`, each returned `drmFormatModifier` **must** be unique.

Among the elements in array `pDrmFormatModifierProperties`, the bits reported in `drmFormatModifierTilingFeatures` **must** include the bits reported in the corresponding element of `VkDrmFormatModifierPropertiesListEXT`::`pDrmFormatModifierProperties`.

Valid Usage (Implicit)

- VUID-VkDrmFormatModifierPropertiesList2EXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DRM_FORMAT_MODIFIER_PROPERTIES_LIST_2_EXT

Structure Chaining

Extends the structure

- VkFormatProperties2

The VkDrmFormatModifierProperties2EXT structure describes properties of a VkFormat when that format is combined with a Linux DRM format modifier. These properties, like those of VkFormatProperties2, are independent of any particular image.

The VkDrmFormatModifierProperties2EXT structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
typedef struct VkDrmFormatModifierProperties2EXT {
    uint64_t                drmFormatModifier;
    uint32_t                drmFormatModifierPlaneCount;
    VkFormatFeatureFlags2   drmFormatModifierTilingFeatures;
} VkDrmFormatModifierProperties2EXT;
```

- `drmFormatModifier` is a *Linux DRM format modifier*.

- drmFormatModifierPlaneCount is the number of *memory planes* in any image created with format and drmFormatModifier. An image's *memory planecount* is distinct from its *format planecount*, as explained below.
- drmFormatModifierTilingFeatures is a bitmask of VkFormatFeatureFlagBits2 that are supported by any image created with format and drmFormatModifier.

To query supported format extended features which are properties of the physical device, add VkFormatProperties3 structure to the pNext chain of VkFormatProperties2.

The VkFormatProperties3 structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkFormatProperties3 {
    VkStructureType          sType;
    void*                    pNext;
    VkFormatFeatureFlags2    linearTilingFeatures;
    VkFormatFeatureFlags2    optimalTilingFeatures;
    VkFormatFeatureFlags2    bufferFeatures;
} VkFormatProperties3;
```

```
// Provided by VK_KHR_format_feature_flags2
// Equivalent to VkFormatProperties3
typedef VkFormatProperties3 VkFormatProperties3KHR;
```

- linearTilingFeatures is a bitmask of VkFormatFeatureFlagBits2 specifying features supported by images created with a tiling parameter of VK_IMAGE_TILING_LINEAR.
- optimalTilingFeatures is a bitmask of VkFormatFeatureFlagBits2 specifying features supported by images created with a tiling parameter of VK_IMAGE_TILING_OPTIMAL.
- bufferFeatures is a bitmask of VkFormatFeatureFlagBits2 specifying features supported by buffers.

The bits reported in linearTilingFeatures, optimalTilingFeatures and bufferFeatures **must** include the bits reported in the corresponding fields of VkFormatProperties2::formatProperties.

Valid Usage (Implicit)

- VUID-VkFormatProperties3-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_3

Structure Chaining

Extends the structure

- VkFormatProperties2

Bits which **can** be set in the VkFormatProperties3 features `linearTilingFeatures`, `optimalTilingFeatures`, and `bufferFeatures` are:

```
// Provided by VK_VERSION_1_3
// Flag bits for VkFormatFeatureFlagBits2
typedef VkFlags64 VkFormatFeatureFlagBits2;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT =
0x00000001ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_STORAGE_IMAGE_BIT =
0x00000002ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_STORAGE_IMAGE_ATOMIC_BIT =
0x00000004ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_UNIFORM_TEXEL_BUFFER_BIT =
0x00000008ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_STORAGE_TEXEL_BUFFER_BIT =
0x00000010ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_STORAGE_TEXEL_BUFFER_ATOMIC_BIT = 0x00000020ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_VERTEX_BUFFER_BIT =
0x00000040ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_COLOR_ATTACHMENT_BIT =
0x00000080ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_COLOR_ATTACHMENT_BLEND_BIT =
0x00000100ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_DEPTH_STENCIL_ATTACHMENT_BIT
= 0x00000200ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_BLIT_SRC_BIT =
0x00000400ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_BLIT_DST_BIT =
0x00000800ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_LINEAR_BIT = 0x00001000ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_TRANSFER_SRC_BIT =
0x00004000ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_TRANSFER_DST_BIT =
0x00008000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_MINMAX_BIT = 0x00010000ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_MIDPOINT_CHROMA_SAMPLES_BIT
= 0x00020000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT = 0x00040000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT
= 0x00080000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT
= 0x00100000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORC
```

```
EABLE_BIT = 0x00200000ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_DISJOINT_BIT =
0x00400000ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_COSITED_CHROMA_SAMPLES_BIT =
0x00800000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT = 0x80000000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT = 0x100000000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT = 0x200000000ULL;
// Provided by VK_VERSION_1_3
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_CUBIC_BIT = 0x00002000ULL;
// Provided by VK_VERSION_1_4
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_HOST_IMAGE_TRANSFER_BIT =
0x400000000000ULL;
// Provided by VK_KHR_video_decode_queue with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_VIDEO_DECODE_OUTPUT_BIT_KHR
= 0x02000000ULL;
// Provided by VK_KHR_video_decode_queue with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_VIDEO_DECODE_DPB_BIT_KHR =
0x04000000ULL;
// Provided by VK_KHR_acceleration_structure with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR = 0x20000000ULL;
// Provided by VK_EXT_fragment_density_map with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_FRAGMENT_DENSITY_MAP_BIT_EXT
= 0x01000000ULL;
// Provided by VK_KHR_fragment_shading_rate with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR = 0x40000000ULL;
// Provided by VK_EXT_host_image_copy
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_HOST_IMAGE_TRANSFER_BIT_EXT
= 0x400000000000ULL;
// Provided by VK_KHR_video_encode_queue with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_VIDEO_ENCODE_INPUT_BIT_KHR =
0x08000000ULL;
// Provided by VK_KHR_video_encode_queue with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_VIDEO_ENCODE_DPB_BIT_KHR =
0x10000000ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT_KHR =
0x00000001ULL;
```

```
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_STORAGE_IMAGE_BIT_KHR =
0x00000002ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_STORAGE_IMAGE_ATOMIC_BIT_KHR
= 0x00000004ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_UNIFORM_TEXEL_BUFFER_BIT_KHR
= 0x00000008ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_STORAGE_TEXEL_BUFFER_BIT_KHR
= 0x00000010ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_STORAGE_TEXEL_BUFFER_ATOMIC_BIT_KHR = 0x00000020ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_VERTEX_BUFFER_BIT_KHR =
0x00000040ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_COLOR_ATTACHMENT_BIT_KHR =
0x00000080ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_COLOR_ATTACHMENT_BLEND_BIT_KHR = 0x00000100ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_DEPTH_STENCIL_ATTACHMENT_BIT_KHR = 0x00000200ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_BLIT_SRC_BIT_KHR =
0x00000400ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_BLIT_DST_BIT_KHR =
0x00000800ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_LINEAR_BIT_KHR = 0x00001000ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_TRANSFER_SRC_BIT_KHR =
0x00004000ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_TRANSFER_DST_BIT_KHR =
0x00008000ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_MIDPOINT_CHROMA_SAMPLES_BIT_KHR = 0x00020000ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT_KHR =
0x00040000ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2
```

```
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT_
KHR = 0x00080000ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT_
KHR = 0x00100000ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORC
EABLE_BIT_KHR = 0x00200000ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_DISJOINT_BIT_KHR =
0x00400000ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_COSITED_CHROMA_SAMPLES_BIT_KHR = 0x00800000ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT_KHR = 0x80000000ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT_KHR = 0x100000000ULL;
// Provided by VK_KHR_format_feature_flags2
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT_KHR = 0x200000000ULL;
// Provided by VK_KHR_format_feature_flags2 with VK_VERSION_1_2 or
VK_EXT_sampler_filter_minmax
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_MINMAX_BIT_KHR = 0x00010000ULL;
// Provided by VK_KHR_format_feature_flags2 with VK_EXT_filter_cubic or
VK_IMG_filter_cubic
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT = 0x00002000ULL;
// Provided by VK_KHR_copy_memory_indirect
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR = 0x8000000000000000ULL;
// Provided by VK_KHR_video_encode_quantization_map
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR = 0x2000000000000ULL;
// Provided by VK_KHR_video_encode_quantization_map
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR = 0x4000000000000ULL;
// Provided by VK_KHR_maintenance10 with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_COMPUTE_QUEUE_BIT_KHR = 0x1000000000000ULL;
// Provided by VK_KHR_maintenance10 with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_TRANSFER_QUEUE_BIT_KHR = 0x2000000000000ULL;
// Provided by VK_KHR_maintenance10 with VK_KHR_format_feature_flags2 or
```

```
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_COMPUTE_QUEUE_BIT_KHR = 0x4000000000000ULL;
// Provided by VK_KHR_maintenance10 with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_TRANSFER_QUEUE_BIT_KHR = 0x8000000000000ULL;
```

```
// Provided by VK_KHR_format_feature_flags2
// Equivalent to VkFormatFeatureFlagBits2
typedef VkFormatFeatureFlagBits2 VkFormatFeatureFlagBits2KHR;
```

The following bits **may** be set in `linearTilingFeatures` and `optimalTilingFeatures`, specifying that the features are supported by images or image views or sampler $Y'C_BC_R$ conversion objects created with the queried vkGetPhysicalDeviceFormatProperties2::`format`:

- VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT specifies that an image view **can** be sampled from.
- VK_FORMAT_FEATURE_2_STORAGE_IMAGE_BIT specifies that an image view **can** be used as a storage image.
- VK_FORMAT_FEATURE_2_STORAGE_IMAGE_ATOMIC_BIT specifies that an image view **can** be used as a storage image that supports atomic operations.
- VK_FORMAT_FEATURE_2_COLOR_ATTACHMENT_BIT specifies that an image view **can** be used as a framebuffer color attachment and as an input attachment.
- VK_FORMAT_FEATURE_2_COLOR_ATTACHMENT_BLEND_BIT specifies that an image view **can** be used as a framebuffer color attachment that supports blending.
- VK_FORMAT_FEATURE_2_DEPTH_STENCIL_ATTACHMENT_BIT specifies that an image view **can** be used as a framebuffer depth/stencil attachment and as an input attachment.
- VK_FORMAT_FEATURE_2_BLIT_SRC_BIT specifies that an image **can** be used as the `srcImage` for vkCmdBlitImage2 and vkCmdBlitImage.
- VK_FORMAT_FEATURE_2_BLIT_DST_BIT specifies that an image **can** be used as the `dstImage` for vkCmdBlitImage2 and vkCmdBlitImage.
- VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_LINEAR_BIT specifies that if VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT is also set, an image view **can** be used with a sampler that has either of `magFilter` or `minFilter` set to VK_FILTER_LINEAR, or `mipmapMode` set to VK_SAMPLER_MIPMAP_MODE_LINEAR. If VK_FORMAT_FEATURE_2_BLIT_SRC_BIT is also set, an image can be used as the `srcImage` for vkCmdBlitImage2 and `vkCmdBlitImage` with a `filter` of VK_FILTER_LINEAR. This bit **must** only be exposed for formats that also support the VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT or VK_FORMAT_FEATURE_2_BLIT_SRC_BIT.

  If the format being queried is a depth/stencil format, this bit only specifies that the depth aspect (not the stencil aspect) of an image of this format supports linear filtering. Where depth comparison is supported it **may** be linear filtered whether this bit is present or not, but where this bit is not present the filtered value **may** be computed in an implementation-dependent

manner which differs from the normal rules of linear filtering. The resulting value **must** be in the range [0,1] and **should** be proportional to, or a weighted average of, the number of comparison passes or failures.

- VK_FORMAT_FEATURE_2_TRANSFER_SRC_BIT specifies that an image **can** be used as a source image for copy commands.
- VK_FORMAT_FEATURE_2_TRANSFER_DST_BIT specifies that an image **can** be used as a destination image for copy commands and clear commands.
- VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_MINMAX_BIT specifies that `VkImage` **can** be used as a sampled image with a min or max VkSamplerReductionMode. This bit **must** only be exposed for formats that also support the VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT.
- VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_CUBIC_BIT specifies that `VkImage` **can** be used with a sampler that has either of `magFilter` or `minFilter` set to VK_FILTER_CUBIC_EXT, or be the source image for a blit with `filter` set to VK_FILTER_CUBIC_EXT. This bit **must** only be exposed for formats that also support the VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT. If the format being queried is a depth/stencil format, this only specifies that the depth aspect is cubic filterable.
- VK_FORMAT_FEATURE_2_MIDPOINT_CHROMA_SAMPLES_BIT specifies that an application **can** define a sampler $Y'C_BC_R$ conversion using this format as a source, and that an image of this format **can** be used with a VkSamplerYcbcrConversionCreateInfo `xChromaOffset` and/or `yChromaOffset` of VK_CHROMA_LOCATION_MIDPOINT. Otherwise both `xChromaOffset` and `yChromaOffset` **must** be VK_CHROMA_LOCATION_COSITED_EVEN. If a format does not incorporate chroma downsampling (it is not a “422” or “420” format) but the implementation supports sampler $Y'C_BC_R$ conversion for this format, the implementation **must** set VK_FORMAT_FEATURE_2_MIDPOINT_CHROMA_SAMPLES_BIT.
- VK_FORMAT_FEATURE_2_COSITED_CHROMA_SAMPLES_BIT specifies that an application **can** define a sampler $Y'C_BC_R$ conversion using this format as a source, and that an image of this format **can** be used with a VkSamplerYcbcrConversionCreateInfo `xChromaOffset` and/or `yChromaOffset` of VK_CHROMA_LOCATION_COSITED_EVEN. Otherwise both `xChromaOffset` and `yChromaOffset` **must** be VK_CHROMA_LOCATION_MIDPOINT. If neither VK_FORMAT_FEATURE_2_COSITED_CHROMA_SAMPLES_BIT nor VK_FORMAT_FEATURE_2_MIDPOINT_CHROMA_SAMPLES_BIT is set, the application **must** not define a sampler $Y'C_BC_R$ conversion using this format as a source.
- VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT specifies that an application **can** define a sampler $Y'C_BC_R$ conversion using this format as a source with `chromaFilter` set to VK_FILTER_LINEAR.
- VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT specifies that the format can have different chroma, min, and mag filters.
- VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT specifies that reconstruction is explicit, as described in Chroma Reconstruction. If this bit is not present, reconstruction is implicit by default.
- VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT specifies that reconstruction **can** be forcibly made explicit by setting VkSamplerYcbcrConversionCreateInfo::`forceExplicitReconstruction` to VK_TRUE. If the

format being queried supports VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT it **must** also support VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT.

- VK_FORMAT_FEATURE_2_DISJOINT_BIT specifies that a multi-planar image **can** have the VK_IMAGE_CREATE_DISJOINT_BIT set during image creation. An implementation **must** not set VK_FORMAT_FEATURE_2_DISJOINT_BIT for *single-plane formats*.
- VK_FORMAT_FEATURE_2_FRAGMENT_DENSITY_MAP_BIT_EXT specifies that an image view **can** be used as a fragment density map attachment.
- VK_FORMAT_FEATURE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR specifies that an image view **can** be used as a fragment shading rate attachment. An implementation **must** not set this feature for formats with a numeric format other than `UINT`, or set it as a buffer feature.
- VK_FORMAT_FEATURE_2_VIDEO_DECODE_OUTPUT_BIT_KHR specifies that an image view with this format **can** be used as a decode output picture in video decode operations.
- VK_FORMAT_FEATURE_2_VIDEO_DECODE_DPB_BIT_KHR specifies that an image view with this format **can** be used as an output reconstructed picture or an input reference picture in video decode operations.
- VK_FORMAT_FEATURE_2_VIDEO_ENCODE_INPUT_BIT_KHR specifies that an image view with this format **can** be used as an encode input picture in video encode operations.
- VK_FORMAT_FEATURE_2_VIDEO_ENCODE_DPB_BIT_KHR specifies that an image view with this format **can** be used as an output reconstructed picture or an input reference picture in video encode operations.
- VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_COMPUTE_QUEUE_BIT_KHR specifies that the depth aspect of this format can be copied using a queue family that supports VK_QUEUE_COMPUTE_BIT but does not support VK_QUEUE_GRAPHICS_BIT.
- VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_TRANSFER_QUEUE_BIT_KHR specifies that the depth aspect of this format can be copied using a queue family that supports VK_QUEUE_TRANSFER_BIT but does not support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT.
- VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_COMPUTE_QUEUE_BIT_KHR specifies that the stencil aspect of this format can be copied using a queue family that supports VK_QUEUE_COMPUTE_BIT but does not support VK_QUEUE_GRAPHICS_BIT.
- VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_TRANSFER_QUEUE_BIT_KHR specifies that the stencil aspect of this format can be copied using a queue family that supports VK_QUEUE_TRANSFER_BIT but does not support VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT.

NOTE	Specific video profiles **may** have additional restrictions on the format and other image creation parameters corresponding to image views used by video coding operations that **can** be enumerated using the vkGetPhysicalDeviceVideoFormatPropertiesKHR command.

- VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT specifies that image views or buffer views created with this format **can** be used as storage images or storage texel buffers respectively for read operations without specifying a format.
- VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT specifies that image views or buffer views created with this format **can** be used as storage images or storage texel buffers respectively for write operations without specifying a format.
- VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT specifies that image views created with this format **can** be used for depth comparison performed by `OpImage*Dref*` instructions.
- VK_FORMAT_FEATURE_2_HOST_IMAGE_TRANSFER_BIT specifies that an image **can** be created with the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag set.
- VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR specifies that a VkImage **can** be used as a destination for indirect copies.
- VK_FORMAT_FEATURE_2_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR specifies that an image view with this format **can** be used as a quantization delta map in video encode operations.
- VK_FORMAT_FEATURE_2_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR specifies that an image view with this format **can** be used as an emphasis map in video encode operations.

> **NOTE** Video encode quantization maps have additional restrictions specific to the video profile they are used with that **can** be enumerated using the vkGetPhysicalDeviceVideoFormatPropertiesKHR command.

The following bits **may** be set in `bufferFeatures`, specifying that the features are supported by buffers or buffer views created with the queried vkGetPhysicalDeviceFormatProperties2::`format`:

- VK_FORMAT_FEATURE_2_UNIFORM_TEXEL_BUFFER_BIT specifies that the format **can** be used to create a buffer view that **can** be bound to a VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER descriptor.
- VK_FORMAT_FEATURE_2_STORAGE_TEXEL_BUFFER_BIT specifies that the format **can** be used to create a buffer view that **can** be bound to a VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER descriptor.
- VK_FORMAT_FEATURE_2_STORAGE_TEXEL_BUFFER_ATOMIC_BIT specifies that atomic operations are supported on VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER with this format.
- VK_FORMAT_FEATURE_2_VERTEX_BUFFER_BIT specifies that the format **can** be used as a vertex attribute format (`VkVertexInputAttributeDescription`::`format`).
- VK_FORMAT_FEATURE_2_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR specifies that the format **can** be used as the vertex format when creating an acceleration structure (`VkAccelerationStructureGeometryTrianglesDataKHR`::`vertexFormat`). This format **can** also be used as the vertex format in host memory when doing host acceleration structure builds.
- VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT specifies that buffer views created with this format **can** be used as storage texel buffers for read operations without specifying a format.

- VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT specifies that buffer views created with this format **can** be used as storage texel buffers for write operations without specifying a format.

```
// Provided by VK_VERSION_1_3
typedef VkFlags64 VkFormatFeatureFlags2;
```

```
// Provided by VK_KHR_format_feature_flags2
// Equivalent to VkFormatFeatureFlags2
typedef VkFormatFeatureFlags2 VkFormatFeatureFlags2KHR;
```

VkFormatFeatureFlags2 is a bitmask type for setting a mask of zero or more VkFormatFeatureFlagBits2.

The VkFormatProperties4KHR structure is defined as:

```
// Provided by VK_KHR_extended_flags
typedef struct VkFormatProperties4KHR {
    VkStructureType             sType;
    void*                       pNext;
    VkFormatFeatureFlags4KHR    linearTilingFeatures;
    VkFormatFeatureFlags4KHR    optimalTilingFeatures;
    VkFormatFeatureFlags4KHR    bufferFeatures;
} VkFormatProperties4KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- linearTilingFeatures is a bitmask of VkFormatFeatureFlagBits4KHR specifying features supported by images created with a tiling parameter of VK_IMAGE_TILING_LINEAR.
- optimalTilingFeatures is a bitmask of VkFormatFeatureFlagBits4KHR specifying features supported by images created with a tiling parameter of VK_IMAGE_TILING_OPTIMAL.
- bufferFeatures is a bitmask of VkFormatFeatureFlagBits4KHR specifying features supported by buffers.

To query supported format extended features which are properties of the physical device, add a VkFormatProperties4KHR structure to the pNext chain of VkFormatProperties2.

NOTE VkFormatProperties4KHR supports additional format feature bits beyond those in VkFormatProperties3.

Valid Usage (Implicit)

- VUID-VkFormatProperties4KHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_4_KHR

Structure Chaining

Extends the structure

- VkFormatProperties2

Bits which **can** be set in the VkFormatProperties4KHR features `linearTilingFeatures`, `optimalTilingFeatures`, and `bufferFeatures` are:

```
// Provided by VK_KHR_extended_flags
// Flag bits for VkFormatFeatureFlagBits4KHR
typedef VkFlags64 VkFormatFeatureFlagBits4KHR;
```

```
// Provided by VK_KHR_extended_flags
typedef VkFlags64 VkFormatFeatureFlags4KHR;
```

`VkFormatFeatureFlags4KHR` is a bitmask type for setting a mask of zero or more VkFormatFeatureFlagBits4KHR.

To query the performance characteristics of a subpass resolve operation for an attachment with a VkFormat, add a VkSubpassResolvePerformanceQueryEXT structure to the `pNext` chain of VkFormatProperties2.

The VkSubpassResolvePerformanceQueryEXT structure is defined as:

```
// Provided by VK_EXT_multisampled_render_to_single_sampled
typedef struct VkSubpassResolvePerformanceQueryEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           optimal;
} VkSubpassResolvePerformanceQueryEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `optimal` specifies that a subpass resolve operation is optimally performed.

If `optimal` is VK_FALSE for a VkFormat, using a subpass resolve operation on a multisampled attachment with this format can incur additional costs, including additional memory bandwidth usage and a higher memory footprint. If an attachment with such a format is used in a multisampled-render-to-single-sampled subpass, the additional memory and memory bandwidth usage can nullify the benefits of using the `VK_EXT_multisampled_render_to_single_sampled` extension.

Valid Usage (Implicit)

- VUID-VkSubpassResolvePerformanceQueryEXT-sType-sType

sType must be VK_STRUCTURE_TYPE_SUBPASS_RESOLVE_PERFORMANCE_QUERY_EXT

Structure Chaining

Extends the structure

- VkFormatProperties2

49.2.1. Potential Format Features

Some valid usage conditions depend on the format features supported by a VkImage whose VkImageTiling is unknown. In such cases the exact VkFormatFeatureFlagBits supported by the VkImage cannot be determined, so the valid usage conditions are expressed in terms of the *potential format features* of the VkImage format.

The *potential format features* of a VkFormat are defined as follows:

- The union of VkFormatFeatureFlagBits and VkFormatFeatureFlagBits2, supported when the VkImageTiling is VK_IMAGE_TILING_OPTIMAL, VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, or VK_IMAGE_TILING_LINEAR

49.3. Required Format Support

Implementations **must** support at least the following set of features on the listed formats. For images, these features **must** be supported for every VkImageType (including arrayed and cube variants) unless otherwise noted. These features are supported on existing formats without needing to advertise an extension or needing to explicitly enable them. Support for additional functionality beyond the requirements listed here is queried using the vkGetPhysicalDeviceFormatProperties command.

NOTE Unless otherwise excluded below, the required formats are supported for all VkImageCreateFlags values as long as those flag values are otherwise allowed.

The following tables show which feature bits **must** be supported for each format. Formats that are required to support VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT **must** also support VK_FORMAT_FEATURE_TRANSFER_SRC_BIT and VK_FORMAT_FEATURE_TRANSFER_DST_BIT.

Table 78. Key for Format Feature Tables

✔	This feature **must** be supported on the named format
†	This feature **must** be supported on at least some of the named formats, with more information in the table where the symbol appears
‡	This feature **must** be supported with some caveats or preconditions, with more information in the table where the symbol appears

§	This feature **must** be supported with some caveats or preconditions, with more information in the table where the symbol appears

Table 79. Feature Bits in `optimalTilingFeatures`

VK_FORMAT_FEATURE_TRANSFER_SRC_BIT
VK_FORMAT_FEATURE_TRANSFER_DST_BIT
VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT
VK_FORMAT_FEATURE_BLIT_SRC_BIT
VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT
VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT
VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT
VK_FORMAT_FEATURE_BLIT_DST_BIT
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT
VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR

Table 80. Feature Bits in `bufferFeatures`

VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT
VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT
VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT
VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT

Table 81. Mandatory Format Support: Sub-Byte Components

Format	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT	VK_FORMAT_FEATURE_BLIT_SRC_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT	VK_FORMAT_FEATURE_BLIT_DST_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT	VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR
VK_FORMAT_UNDEFINED														
VK_FORMAT_R4G4_UNORM_PACK8														
VK_FORMAT_R4G4B4A4_UNORM_PACK16														
VK_FORMAT_B4G4R4A4_UNORM_PACK16	✔	✔	✔											
VK_FORMAT_R5G6B5_UNORM_PACK16	✔	✔	✔			✔	✔	✔						
VK_FORMAT_B5G6R5_UNORM_PACK16														
VK_FORMAT_R5G5B5A1_UNORM_PACK16														
VK_FORMAT_B5G5R5A1_UNORM_PACK16														
VK_FORMAT_A1R5G5B5_UNORM_PACK16	✔	✔	✔			✔	✔	✔						
VK_FORMAT_A1B5G5R5_UNORM_PACK16														
VK_FORMAT_A4R4G4B4_UNORM_PACK16	†	†	†											
VK_FORMAT_A4B4G4R4_UNORM_PACK16	‡	‡	‡											

Format features marked † **must** be supported for `optimalTilingFeatures` if the `VK_EXT_4444_formats` extension is supported

Format features marked ‡ **must** be supported for `optimalTilingFeatures` if the `VK_EXT_4444_formats` extension is supported

Table 82. Mandatory Format Support: 1-3 Byte-Sized Components

Format	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT	VK_FORMAT_FEATURE_BLIT_SRC_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT	VK_FORMAT_FEATURE_BLIT_DST_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT	VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR
VK_FORMAT_R8_UNORM	✔	✔	✔	‡		✔	✔	✔		✔	✔			✔
VK_FORMAT_R8_SNORM	✔	✔	✔	‡						✔	✔			✔
VK_FORMAT_R8_USCALED														
VK_FORMAT_R8_SSCALED														
VK_FORMAT_R8_UINT	✔	✔		‡		✔	✔			✔	✔			✔
VK_FORMAT_R8_SINT	✔	✔		‡		✔	✔			✔	✔			✔
VK_FORMAT_R8_SRGB														
VK_FORMAT_R8G8_UNORM	✔	✔	✔	‡		✔	✔	✔		✔	✔			✔
VK_FORMAT_R8G8_SNORM	✔	✔	✔	‡						✔	✔			✔
VK_FORMAT_R8G8_USCALED														
VK_FORMAT_R8G8_SSCALED														
VK_FORMAT_R8G8_UINT	✔	✔		‡		✔	✔			✔	✔			✔
VK_FORMAT_R8G8_SINT	✔	✔		‡		✔	✔			✔	✔			✔
VK_FORMAT_R8G8_SRGB														
VK_FORMAT_R8G8B8_UNORM														
VK_FORMAT_R8G8B8_SNORM														
VK_FORMAT_R8G8B8_USCALED														
VK_FORMAT_R8G8B8_SSCALED														

VK_FORMAT_R8G8B8_UINT														
VK_FORMAT_R8G8B8_SINT														
VK_FORMAT_R8G8B8_SRGB														
VK_FORMAT_B8G8R8_UNORM														
VK_FORMAT_B8G8R8_SNORM														
VK_FORMAT_B8G8R8_USCALED														
VK_FORMAT_B8G8R8_SSCALED														
VK_FORMAT_B8G8R8_UINT														
VK_FORMAT_B8G8R8_SINT														
VK_FORMAT_B8G8R8_SRGB														
VK_FORMAT_A8_UNORM														
Format features marked with ‡ **must** be supported for `optimalTilingFeatures` if the `VkPhysicalDevice` supports the `shaderStorageImageExtendedFormats` feature.														

Table 83. Mandatory Format Support: 4 Byte-Sized Components

Format	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT	VK_FORMAT_FEATURE_BLIT_SRC_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT	VK_FORMAT_FEATURE_BLIT_DST_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT	VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR
VK_FORMAT_R8G8B8A8_UNORM	✔	✔	✔	✔		✔	✔	✔		✔	✔	✔		✔
VK_FORMAT_R8G8B8A8_SNORM	✔	✔	✔	✔						✔	✔	✔		✔
VK_FORMAT_R8G8B8A8_USCALED														
VK_FORMAT_R8G8B8A8_SSCALED														
VK_FORMAT_R8G8B8A8_UINT	✔	✔		✔		✔	✔			✔	✔	✔		✔
VK_FORMAT_R8G8B8A8_SINT	✔	✔		✔		✔	✔			✔	✔	✔		✔
VK_FORMAT_R8G8B8A8_SRGB	✔	✔	✔			✔	✔	✔						
VK_FORMAT_B8G8R8A8_UNORM	✔	✔	✔			✔	✔	✔		✔	✔			
VK_FORMAT_B8G8R8A8_SNORM														
VK_FORMAT_B8G8R8A8_USCALED														
VK_FORMAT_B8G8R8A8_SSCALED														
VK_FORMAT_B8G8R8A8_UINT														
VK_FORMAT_B8G8R8A8_SINT														
VK_FORMAT_B8G8R8A8_SRGB	✔	✔	✔			✔	✔	✔						
VK_FORMAT_A8B8G8R8_UNORM_PACK32	✔	✔	✔			✔	✔	✔		✔	✔	✔		
VK_FORMAT_A8B8G8R8_SNORM_PACK32	✔	✔	✔							✔	✔	✔		
VK_FORMAT_A8B8G8R8_USCALED_PACK32														
VK_FORMAT_A8B8G8R8_SSCALED_PACK32														

VK_FORMAT_A8B8G8R8_UINT_PACK32	✔	✔				✔	✔			✔	✔	✔		
VK_FORMAT_A8B8G8R8_SINT_PACK32	✔	✔				✔	✔			✔	✔	✔		
VK_FORMAT_A8B8G8R8_SRGB_PACK32	✔	✔	✔			✔	✔	✔						

Table 84. Mandatory Format Support: 10-, 12-, and 14-Bit Components

Format	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT	VK_FORMAT_FEATURE_BLIT_SRC_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT	VK_FORMAT_FEATURE_BLIT_DST_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT	VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR
VK_FORMAT_A2R10G10B10_UNORM_PACK32														
VK_FORMAT_A2R10G10B10_SNORM_PACK32														
VK_FORMAT_A2R10G10B10_USCALED_PACK32														
VK_FORMAT_A2R10G10B10_SSCALED_PACK32														
VK_FORMAT_A2R10G10B10_UINT_PACK32														
VK_FORMAT_A2R10G10B10_SINT_PACK32														
VK_FORMAT_A2B10G10R10_UNORM_PACK32	✔	✔	✔	‡		✔	✔	✔		✔	✔			✔
VK_FORMAT_A2B10G10R10_SNORM_PACK32														
VK_FORMAT_A2B10G10R10_USCALED_PACK32														
VK_FORMAT_A2B10G10R10_SSCALED_PACK32														
VK_FORMAT_A2B10G10R10_UINT_PACK32	✔	✔		‡		✔	✔				✔			✔
VK_FORMAT_A2B10G10R10_SINT_PACK32														
VK_FORMAT_R10X6_UNORM_PACK16														
VK_FORMAT_R10X6G10X6_UNORM_2PACK16														
VK_FORMAT_R12X4_UNORM_PACK16														
VK_FORMAT_R12X4G12X4_UNORM_2PACK16														

Format features marked with ‡ **must** be supported for `optimalTilingFeatures` if the `VkPhysicalDevice` supports the `shaderStorageImageExtendedFormats` feature.

Table 85. Mandatory Format Support: 16-bit Components

Format	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT	VK_FORMAT_FEATURE_BLIT_SRC_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT	VK_FORMAT_FEATURE_BLIT_DST_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT	VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR
VK_FORMAT_R16_UNORM				‡						✔				✔
VK_FORMAT_R16_SNORM				‡						✔				✔
VK_FORMAT_R16_USCALED														
VK_FORMAT_R16_SSCALED														
VK_FORMAT_R16_UINT	✔	✔		‡		✔	✔			✔	✔			✔
VK_FORMAT_R16_SINT	✔	✔		‡		✔	✔			✔	✔			✔
VK_FORMAT_R16_SFLOAT	✔	✔	✔	‡		✔	✔	✔		✔	✔			✔
VK_FORMAT_R16G16_UNORM				‡						✔				✔
VK_FORMAT_R16G16_SNORM				‡						✔				✔
VK_FORMAT_R16G16_USCALED														
VK_FORMAT_R16G16_SSCALED														
VK_FORMAT_R16G16_UINT	✔	✔		‡		✔	✔			✔	✔			✔
VK_FORMAT_R16G16_SINT	✔	✔		‡		✔	✔			✔	✔			✔
VK_FORMAT_R16G16_SFLOAT	✔	✔	✔	‡	§	✔	✔	✔		✔	✔	§		✔
VK_FORMAT_R16G16B16_UNORM														
VK_FORMAT_R16G16B16_SNORM														
VK_FORMAT_R16G16B16_USCALED														
VK_FORMAT_R16G16B16_SSCALED														

VK_FORMAT_R16G16B16_UINT													
VK_FORMAT_R16G16B16_SINT													
VK_FORMAT_R16G16B16_SFLOAT													
VK_FORMAT_R16G16B16A16_UNORM				‡					✔				✔
VK_FORMAT_R16G16B16A16_SNORM				‡					✔				✔
VK_FORMAT_R16G16B16A16_USCALED													
VK_FORMAT_R16G16B16A16_SSCALED													
VK_FORMAT_R16G16B16A16_UINT	✔	✔		✔		✔	✔			✔	✔	✔		✔
VK_FORMAT_R16G16B16A16_SINT	✔	✔		✔		✔	✔			✔	✔	✔		✔
VK_FORMAT_R16G16B16A16_SFLOAT	✔	✔	✔	✔	§	✔	✔	✔		✔	✔	✔		✔

Format features marked with ‡ **must** be supported for `optimalTilingFeatures` if the `VkPhysicalDevice` supports the `shaderStorageImageExtendedFormats` feature.

Table 86. Mandatory Format Support: 32-bit Components

Format	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT	VK_FORMAT_FEATURE_BLIT_SRC_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT	VK_FORMAT_FEATURE_BLIT_DST_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT	VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR
VK_FORMAT_R32_UINT	✔	✔		✔	✔	✔	✔			✔	✔	✔		✔
VK_FORMAT_R32_SINT	✔	✔		✔	✔	✔	✔			✔	✔	✔		✔
VK_FORMAT_R32_SFLOAT	✔	✔		✔	†	✔	✔			✔	✔	✔		✔
VK_FORMAT_R32G32_UINT	✔	✔		✔		✔	✔			✔	✔	✔		✔
VK_FORMAT_R32G32_SINT	✔	✔		✔		✔	✔			✔	✔	✔		✔
VK_FORMAT_R32G32_SFLOAT	✔	✔		✔		✔	✔			✔	✔	✔		✔
VK_FORMAT_R32G32B32_UINT										✔				
VK_FORMAT_R32G32B32_SINT										✔				
VK_FORMAT_R32G32B32_SFLOAT										✔				
VK_FORMAT_R32G32B32A32_UINT	✔	✔		✔		✔	✔			✔	✔	✔		✔
VK_FORMAT_R32G32B32A32_SINT	✔	✔		✔		✔	✔			✔	✔	✔		✔
VK_FORMAT_R32G32B32A32_SFLOAT	✔	✔		✔		✔	✔			✔	✔	✔		✔

Format features marked with † **must** be supported for `optimalTilingFeatures` if the `VkPhysicalDevice` supports the `shaderImageFloat32Atomics` or the `shaderImageFloat32AtomicAdd` or the `shaderImageFloat32AtomicMinMax` feature.

Table 87. Mandatory Format Support: 64-bit/uneven Components

Format	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT	VK_FORMAT_FEATURE_BLIT_SRC_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT	VK_FORMAT_FEATURE_BLIT_DST_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT	VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR
VK_FORMAT_R64_UINT				†	†									
VK_FORMAT_R64_SINT				†	†									
VK_FORMAT_R64_SFLOAT														
VK_FORMAT_R64G64_UINT														
VK_FORMAT_R64G64_SINT														
VK_FORMAT_R64G64_SFLOAT														
VK_FORMAT_R64G64B64_UINT														
VK_FORMAT_R64G64B64_SINT														
VK_FORMAT_R64G64B64_SFLOAT														
VK_FORMAT_R64G64B64A64_UINT														
VK_FORMAT_R64G64B64A64_SINT														
VK_FORMAT_R64G64B64A64_SFLOAT														
VK_FORMAT_B10G11R11_UFLOAT_PACK32	✔	✔	✔	‡						✔				✔
VK_FORMAT_E5B9G9R9_UFLOAT_PACK32	✔	✔	✔											

Format features marked with ‡ **must** be supported for `optimalTilingFeatures` if the `VkPhysicalDevice` supports the `shaderStorageImageExtendedFormats` feature.

If the `shaderImageInt64Atomics` feature is supported, VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT and VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT **must** be advertised in `optimalTilingFeatures` for both VK_FORMAT_R64_UINT and VK_FORMAT_R64_SINT.

Table 88. Mandatory Format Support: Depth/Stencil With `VkImageType` VK_IMAGE_TYPE_2D

Format	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT	VK_FORMAT_FEATURE_BLIT_SRC_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT	VK_FORMAT_FEATURE_BLIT_DST_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT	VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR	
VK_FORMAT_D16_UNORM	✔	✔							✔						
VK_FORMAT_X8_D24_UNORM_PACK32									†						
VK_FORMAT_D32_SFLOAT	✔	✔							†						
VK_FORMAT_S8_UINT															
VK_FORMAT_D16_UNORM_S8_UINT															
VK_FORMAT_D24_UNORM_S8_UINT									†						
VK_FORMAT_D32_SFLOAT_S8_UINT									†						
VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT feature **must** be supported for at least one of VK_FORMAT_X8_D24_UNORM_PACK32 and VK_FORMAT_D32_SFLOAT, and **must** be supported for at least one of VK_FORMAT_D24_UNORM_S8_UINT and VK_FORMAT_D32_SFLOAT_S8_UINT.															
`bufferFeatures` **must** not support any features for these formats															

Table 89. Mandatory Format Support: BC Compressed Formats With `VkImageType` *VK_IMAGE_TYPE_2D and VK_IMAGE_TYPE_3D*

Format	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT	VK_FORMAT_FEATURE_BLIT_SRC_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT	VK_FORMAT_FEATURE_BLIT_DST_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT	VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR	
VK_FORMAT_BC1_RGB_UNORM_BLOCK	†	†	†												
VK_FORMAT_BC1_RGB_SRGB_BLOCK	†	†	†												
VK_FORMAT_BC1_RGBA_UNORM_BLOCK	†	†	†												
VK_FORMAT_BC1_RGBA_SRGB_BLOCK	†	†	†												
VK_FORMAT_BC2_UNORM_BLOCK	†	†	†												
VK_FORMAT_BC2_SRGB_BLOCK	†	†	†												
VK_FORMAT_BC3_UNORM_BLOCK	†	†	†												
VK_FORMAT_BC3_SRGB_BLOCK	†	†	†												
VK_FORMAT_BC4_UNORM_BLOCK	†	†	†												
VK_FORMAT_BC4_SNORM_BLOCK	†	†	†												
VK_FORMAT_BC5_UNORM_BLOCK	†	†	†												
VK_FORMAT_BC5_SNORM_BLOCK	†	†	†												
VK_FORMAT_BC6H_UFLOAT_BLOCK	†	†	†												
VK_FORMAT_BC6H_SFLOAT_BLOCK	†	†	†												
VK_FORMAT_BC7_UNORM_BLOCK	†	†	†												
VK_FORMAT_BC7_SRGB_BLOCK	†	†	†												

The VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT, VK_FORMAT_FEATURE_BLIT_SRC_BIT and VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT features **must** be supported in `optimalTilingFeatures` for all the formats in at least one of: this table, Mandatory Format Support: ETC2 and EAC Compressed Formats With `VkImageType` VK_IMAGE_TYPE_2D, or Mandatory Format Support: ASTC LDR Compressed Formats With `VkImageType` VK_IMAGE_TYPE_2D.

Table 90. Mandatory Format Support: ETC2 and EAC Compressed Formats With `VkImageType` VK_IMAGE_TYPE_2D

Format	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT	VK_FORMAT_FEATURE_BLIT_SRC_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT	VK_FORMAT_FEATURE_BLIT_DST_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT	VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR	
VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK	†	†	†												
VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK	†	†	†												
VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK	†	†	†												
VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK	†	†	†												
VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK	†	†	†												
VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK	†	†	†												
VK_FORMAT_EAC_R11_UNORM_BLOCK	†	†	†												
VK_FORMAT_EAC_R11_SNORM_BLOCK	†	†	†												
VK_FORMAT_EAC_R11G11_UNORM_BLOCK	†	†	†												
VK_FORMAT_EAC_R11G11_SNORM_BLOCK	†	†	†												

The VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT, VK_FORMAT_FEATURE_BLIT_SRC_BIT and VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT features **must** be supported in `optimalTilingFeatures` for all the formats in at least one of: this table, Mandatory Format Support: BC Compressed Formats With `VkImageType` VK_IMAGE_TYPE_2D and VK_IMAGE_TYPE_3D, or Mandatory Format Support: ASTC LDR Compressed Formats With `VkImageType` VK_IMAGE_TYPE_2D.

Table 91. Mandatory Format Support: ASTC LDR Compressed Formats With `VkImageType` *VK_IMAGE_TYPE_2D*

Format	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT	VK_FORMAT_FEATURE_BLIT_SRC_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT	VK_FORMAT_FEATURE_BLIT_DST_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT	VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR	
VK_FORMAT_ASTC_4x4_UNORM_BLOCK	†	†	†												
VK_FORMAT_ASTC_4x4_SRGB_BLOCK	†	†	†												
VK_FORMAT_ASTC_5x4_UNORM_BLOCK	†	†	†												
VK_FORMAT_ASTC_5x4_SRGB_BLOCK	†	†	†												
VK_FORMAT_ASTC_5x5_UNORM_BLOCK	†	†	†												
VK_FORMAT_ASTC_5x5_SRGB_BLOCK	†	†	†												
VK_FORMAT_ASTC_6x5_UNORM_BLOCK	†	†	†												
VK_FORMAT_ASTC_6x5_SRGB_BLOCK	†	†	†												
VK_FORMAT_ASTC_6x6_UNORM_BLOCK	†	†	†												
VK_FORMAT_ASTC_6x6_SRGB_BLOCK	†	†	†												
VK_FORMAT_ASTC_8x5_UNORM_BLOCK	†	†	†												
VK_FORMAT_ASTC_8x5_SRGB_BLOCK	†	†	†												
VK_FORMAT_ASTC_8x6_UNORM_BLOCK	†	†	†												
VK_FORMAT_ASTC_8x6_SRGB_BLOCK	†	†	†												
VK_FORMAT_ASTC_8x8_UNORM_BLOCK	†	†	†												
VK_FORMAT_ASTC_8x8_SRGB_BLOCK	†	†	†												
VK_FORMAT_ASTC_10x5_UNORM_BLOCK	†	†	†												

VK_FORMAT_ASTC_10x5_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x6_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x6_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x8_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x8_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x10_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x10_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_12x10_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_12x10_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_12x12_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_12x12_SRGB_BLOCK	†	†	†										
The VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT, VK_FORMAT_FEATURE_BLIT_SRC_BIT and VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT features **must** be supported in `optimalTilingFeatures` for all the formats in at least one of: this table, Mandatory Format Support: BC Compressed Formats With `VkImageType` VK_IMAGE_TYPE_2D and VK_IMAGE_TYPE_3D, or Mandatory Format Support: ETC2 and EAC Compressed Formats With `VkImageType` VK_IMAGE_TYPE_2D.													

If cubic filtering is supported, VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT **must** be supported for the following image view types:

- VK_IMAGE_VIEW_TYPE_2D
- VK_IMAGE_VIEW_TYPE_2D_ARRAY

for the following formats:

- VK_FORMAT_R4G4_UNORM_PACK8
- VK_FORMAT_R4G4B4A4_UNORM_PACK16
- VK_FORMAT_B4G4R4A4_UNORM_PACK16
- VK_FORMAT_R5G6B5_UNORM_PACK16
- VK_FORMAT_B5G6R5_UNORM_PACK16
- VK_FORMAT_R5G5B5A1_UNORM_PACK16
- VK_FORMAT_B5G5R5A1_UNORM_PACK16
- VK_FORMAT_A1R5G5B5_UNORM_PACK16
- VK_FORMAT_R8_UNORM
- VK_FORMAT_R8_SNORM
- VK_FORMAT_R8_SRGB
- VK_FORMAT_R8G8_UNORM
- VK_FORMAT_R8G8_SNORM

- VK_FORMAT_R8G8_SRGB
- VK_FORMAT_R8G8B8_UNORM
- VK_FORMAT_R8G8B8_SNORM
- VK_FORMAT_R8G8B8_SRGB
- VK_FORMAT_B8G8R8_UNORM
- VK_FORMAT_B8G8R8_SNORM
- VK_FORMAT_B8G8R8_SRGB
- VK_FORMAT_R8G8B8A8_UNORM
- VK_FORMAT_R8G8B8A8_SNORM
- VK_FORMAT_R8G8B8A8_SRGB
- VK_FORMAT_B8G8R8A8_UNORM
- VK_FORMAT_B8G8R8A8_SNORM
- VK_FORMAT_B8G8R8A8_SRGB
- VK_FORMAT_A8B8G8R8_UNORM_PACK32
- VK_FORMAT_A8B8G8R8_SNORM_PACK32
- VK_FORMAT_A8B8G8R8_SRGB_PACK32

If ETC compressed formats are supported, VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT **must** be supported for the following image view types:

- VK_IMAGE_VIEW_TYPE_2D
- VK_IMAGE_VIEW_TYPE_2D_ARRAY

for the following additional formats:

- VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK
- VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK
- VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK
- VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK
- VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK
- VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK

If cubic filtering is supported for any other formats, the following image view types **must** be supported for those formats:

- VK_IMAGE_VIEW_TYPE_2D
- VK_IMAGE_VIEW_TYPE_2D_ARRAY

To be used with `VkImageView` with `subresourceRange.aspectMask` equal to VK_IMAGE_ASPECT_COLOR_BIT, sampler $Y'C_BC_R$ conversion **must** be enabled for the following

formats:

Table 92. Formats Requiring Sampler $Y'C_BC_R$ Conversion for VK_IMAGE_ASPECT_COLOR_BIT Image Views

Format	Planes	VK_FORMAT_FEATURE_DISJOINT_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT	VK_FORMAT_FEATURE_TRANSFER_SRC_BIT	VK_FORMAT_FEATURE_TRANSFER_DST_BIT	VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT	VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT
VK_FORMAT_G8B8G8R8_422_UNORM	1										
VK_FORMAT_B8G8R8G8_422_UNORM	1										
VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM	3		†	†	†	†					
VK_FORMAT_G8_B8R8_2PLANE_420_UNORM	2		†	†	†	†					
VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM	3										
VK_FORMAT_G8_B8R8_2PLANE_422_UNORM	2										
VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM	3										
VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16 ‡	1										
VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16	1										
VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16	1										
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16	3										
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16	2										
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16	3										
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16	2										

VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16	3										
VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16	1										
VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16	1										
VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16	1										
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16	3										
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16	2										
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16	3										
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16	2										
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16	3										
VK_FORMAT_G16B16G16R16_422_UNORM	1										
VK_FORMAT_B16G16R16G16_422_UNORM	1										
VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM	3										
VK_FORMAT_G16_B16R16_2PLANE_420_UNORM	2										
VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM	3										
VK_FORMAT_G16_B16R16_2PLANE_422_UNORM	2										
VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM	3										
VK_FORMAT_G8_B8R8_2PLANE_444_UNORM	2										
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_444_UNORM_3PACK16	2										
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_444_UNORM_3PACK16	2										
VK_FORMAT_G16_B16R16_2PLANE_444_UNORM	2										
Format features marked † **must** be supported for `optimalTilingFeatures` with VkImageType VK_IMAGE_TYPE_2D if the `VkPhysicalDevice` supports the VkPhysicalDeviceSamplerYcbcrConversionFeatures feature.											
Formats marked ‡ do not require a sampler $Y'C_BC_R$ conversion for VK_IMAGE_ASPECT_COLOR_BIT image views if the VkPhysicalDeviceRGBA10X6FormatsFeaturesEXT ::`formatRgba10x6WithoutYCbCrSampler` feature is enabled.											

Implementations are not required to support the VK_IMAGE_CREATE_SPARSE_BINDING_BIT,

VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT, or VK_IMAGE_CREATE_SPARSE_ALIASED_BIT VkImageCreateFlags for the above formats that require sampler $Y'C_BC_R$ conversion. To determine whether the implementation supports sparse image creation flags with these formats use vkGetPhysicalDeviceImageFormatProperties or vkGetPhysicalDeviceImageFormatProperties2.

VK_FORMAT_FEATURE_FRAGMENT_DENSITY_MAP_BIT_EXT **must** be supported for the following formats if the `fragmentDensityMap` feature is enabled:

- VK_FORMAT_R8G8_UNORM

VK_FORMAT_FEATURE_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR **must** be supported in `bufferFeatures` for the following formats if the `accelerationStructure` feature is supported:

- VK_FORMAT_R32G32_SFLOAT
- VK_FORMAT_R32G32B32_SFLOAT
- VK_FORMAT_R16G16_SFLOAT
- VK_FORMAT_R16G16B16A16_SFLOAT
- VK_FORMAT_R16G16_SNORM
- VK_FORMAT_R16G16B16A16_SNORM

VK_FORMAT_FEATURE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR **must** be supported for the following formats if the `attachmentFragmentShadingRate` feature is supported:

- VK_FORMAT_R8_UINT

If the `hostImageCopy` feature is supported and VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT is supported in `optimalTilingFeatures` or `linearTilingFeatures` for a color format, VK_FORMAT_FEATURE_2_HOST_IMAGE_TRANSFER_BIT **must** also be supported in `optimalTilingFeatures` or `linearTilingFeatures` respectively.

49.3.1. Formats Without Shader Storage Format

The device-level features for using a storage image or a storage texel buffer with an image format of `Unknown`, `shaderStorageImageReadWithoutFormat` and `shaderStorageImageWriteWithoutFormat`, only apply to the following formats:

- VK_FORMAT_R8G8B8A8_UNORM
- VK_FORMAT_R8G8B8A8_SNORM
- VK_FORMAT_R8G8B8A8_UINT
- VK_FORMAT_R8G8B8A8_SINT
- VK_FORMAT_R32_UINT
- VK_FORMAT_R32_SINT
- VK_FORMAT_R32_SFLOAT
- VK_FORMAT_R32G32_UINT

- VK_FORMAT_R32G32_SINT
- VK_FORMAT_R32G32_SFLOAT
- VK_FORMAT_R32G32B32A32_UINT
- VK_FORMAT_R32G32B32A32_SINT
- VK_FORMAT_R32G32B32A32_SFLOAT
- VK_FORMAT_R16G16B16A16_UINT
- VK_FORMAT_R16G16B16A16_SINT
- VK_FORMAT_R16G16B16A16_SFLOAT
- VK_FORMAT_R16G16_SFLOAT
- VK_FORMAT_B10G11R11_UFLOAT_PACK32
- VK_FORMAT_R16_SFLOAT
- VK_FORMAT_R16G16B16A16_UNORM
- VK_FORMAT_A2B10G10R10_UNORM_PACK32
- VK_FORMAT_R16G16_UNORM
- VK_FORMAT_R8G8_UNORM
- VK_FORMAT_R16_UNORM
- VK_FORMAT_R8_UNORM
- VK_FORMAT_R16G16B16A16_SNORM
- VK_FORMAT_R16G16_SNORM
- VK_FORMAT_R8G8_SNORM
- VK_FORMAT_R16_SNORM
- VK_FORMAT_R8_SNORM
- VK_FORMAT_R16G16_SINT
- VK_FORMAT_R8G8_SINT
- VK_FORMAT_R16_SINT
- VK_FORMAT_R8_SINT
- VK_FORMAT_A2B10G10R10_UINT_PACK32
- VK_FORMAT_R16G16_UINT
- VK_FORMAT_R8G8_UINT
- VK_FORMAT_R16_UINT
- VK_FORMAT_R8_UINT
- VK_FORMAT_A8_UNORM

NOTE This list of formats is the union of required storage formats from Required Format Support section and formats listed in `shaderStorageImageExtendedFormats`.

An implementation that supports VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT for any format from the given list of formats and supports `shaderStorageImageReadWithoutFormat` **must** support VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT for that same format if Vulkan 1.3 or the `VK_KHR_format_feature_flags2` extension is supported.

An implementation that supports VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT for any format from the given list of formats and supports `shaderStorageImageWriteWithoutFormat` **must** support VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT for that same format if Vulkan 1.3 or the `VK_KHR_format_feature_flags2` extension is supported.

An implementation that does not support either of VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT or VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT for a format **must** not report support for VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT or VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT for that format if it is not listed in the SPIR-V and Vulkan Image Format Compatibility table.

NOTE

Some older implementations do not follow this restriction. They report support for formats as storage images even though they do not support access without the `Format` qualifier and there is no matching `Format` token. Such images cannot be either read from or written to.

Drivers which pass Vulkan conformance test suite version 1.3.9.0, or any subsequent version will conform to the requirement above.

49.3.2. Depth Comparison Format Support

If Vulkan 1.3 or the `VK_KHR_format_feature_flags2` extension is supported, a depth/stencil format with a depth component supporting VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT **must** support VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT.

49.3.3. Format Feature Dependent Usage Flags

Certain resource usage flags depend on support for the corresponding format feature flag for the format in question. The following tables list the VkBufferUsageFlagBits and VkImageUsageFlagBits that have such dependencies, and the format feature flags they depend on. Additional restrictions, including, but not limited to, further required format feature flags specific to the particular use of the resource **may** apply, as described in the respective sections of this specification.

Table 93. Format Feature Dependent Buffer Usage Flags

Buffer usage flag	Required format feature flag
VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT
VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT
VK_BUFFER_USAGE_VERTEX_BUFFER_BIT	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT

Table 94. Format Feature Dependent Image Usage Flags

Image usage flag	Required format feature flag
VK_IMAGE_USAGE_SAMPLED_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT
VK_IMAGE_USAGE_STORAGE_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT
VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT
VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT or VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR	VK_FORMAT_FEATURE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR	VK_FORMAT_FEATURE_VIDEO_DECODE_OUTPUT_BIT_KHR
VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR	VK_FORMAT_FEATURE_VIDEO_DECODE_DPB_BIT_KHR
VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR	VK_FORMAT_FEATURE_VIDEO_ENCODE_INPUT_BIT_KHR
VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR	VK_FORMAT_FEATURE_VIDEO_ENCODE_DPB_BIT_KHR
VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR	VK_FORMAT_FEATURE_2_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR
VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR	VK_FORMAT_FEATURE_2_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR

Chapter 50. Additional Capabilities

This chapter describes additional capabilities beyond the minimum capabilities described in the Limits and Formats chapters, including:

- Additional Image Capabilities
- Additional Buffer Capabilities
- Optional Semaphore Capabilities
- Optional Fence Capabilities
- Timestamp Calibration Capabilities

50.1. Additional Image Capabilities

Additional image capabilities, such as larger dimensions or additional sample counts for certain image types, or additional capabilities for *linear* tiling format images, are described in this section.

To query additional capabilities specific to image types, call:

WARNING	This functionality is superseded by vkGetPhysicalDeviceImageFormatProperties2. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
VkResult vkGetPhysicalDeviceImageFormatProperties(
    VkPhysicalDevice                            physicalDevice,
    VkFormat                                    format,
    VkImageType                                 type,
    VkImageTiling                               tiling,
    VkImageUsageFlags                           usage,
    VkImageCreateFlags                          flags,
    VkImageFormatProperties*                    pImageFormatProperties);
```

- `physicalDevice` is the physical device from which to query the image capabilities.
- `format` is a VkFormat value specifying the image format, corresponding to VkImageCreateInfo::`format`.
- `type` is a VkImageType value specifying the image type, corresponding to VkImageCreateInfo::`imageType`.
- `tiling` is a VkImageTiling value specifying the image tiling, corresponding to VkImageCreateInfo::`tiling`.
- `usage` is a bitmask of VkImageUsageFlagBits specifying the intended usage of the image, corresponding to VkImageCreateInfo::`usage`.
- `flags` is a bitmask of VkImageCreateFlagBits specifying additional parameters of the image, corresponding to VkImageCreateInfo::`flags`.

- `pImageFormatProperties` is a pointer to a VkImageFormatProperties structure in which capabilities are returned.

The `format`, `type`, `tiling`, `usage`, and `flags` parameters correspond to parameters that would be consumed by vkCreateImage (as members of VkImageCreateInfo).

If `format` is not a supported image format, or if the combination of `format`, `type`, `tiling`, `usage`, and `flags` is not supported for images, then `vkGetPhysicalDeviceImageFormatProperties` returns VK_ERROR_FORMAT_NOT_SUPPORTED.

The limitations on an image format that are reported by `vkGetPhysicalDeviceImageFormatProperties` have the following property: if `usage1` and `usage2` of type VkImageUsageFlags are such that the bits set in `usage1` are a subset of the bits set in `usage2`, and `flags1` and `flags2` of type VkImageCreateFlags are such that the bits set in `flags1` are a subset of the bits set in `flags2`, then the limitations for `usage1` and `flags1` **must** be no more strict than the limitations for `usage2` and `flags2`, for all values of `format`, `type`, and `tiling`.

If the `hostImageCopy` feature is supported, and:

- `usage` includes VK_IMAGE_USAGE_SAMPLED_BIT, and
- `flags` does not include any of VK_IMAGE_CREATE_SPARSE_BINDING_BIT, VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT, or VK_IMAGE_CREATE_SPARSE_ALIASED_BIT

Then the result of calls to `vkGetPhysicalDeviceImageFormatProperties` with identical parameters except for the inclusion of VK_IMAGE_USAGE_HOST_TRANSFER_BIT in `usage` **must** be identical.

Valid Usage

- VUID-vkGetPhysicalDeviceImageFormatProperties-tiling-02248
  `tiling` **must** not be VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceImageFormatProperties-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceImageFormatProperties-format-parameter
  `format` **must** be a valid VkFormat value
- VUID-vkGetPhysicalDeviceImageFormatProperties-type-parameter
  `type` **must** be a valid VkImageType value
- VUID-vkGetPhysicalDeviceImageFormatProperties-tiling-parameter
  `tiling` **must** be a valid VkImageTiling value
- VUID-vkGetPhysicalDeviceImageFormatProperties-usage-parameter
  `usage` **must** be a valid combination of VkImageUsageFlagBits values
- VUID-vkGetPhysicalDeviceImageFormatProperties-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkImageCreateFlagBits values

- VUID-vkGetPhysicalDeviceImageFormatProperties-pImageFormatProperties-parameter `pImageFormatProperties` **must** be a valid pointer to a VkImageFormatProperties structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_FORMAT_NOT_SUPPORTED
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkImageFormatProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageFormatProperties {
    VkExtent3D            maxExtent;
    uint32_t              maxMipLevels;
    uint32_t              maxArrayLayers;
    VkSampleCountFlags    sampleCounts;
    VkDeviceSize          maxResourceSize;
} VkImageFormatProperties;
```

- `maxExtent` are the maximum image dimensions. See the Allowed Extent Values section below for how these values are constrained by `type`.
- `maxMipLevels` is the maximum number of mipmap levels. `maxMipLevels` **must** be equal to the number of levels in the complete mipmap chain based on the `maxExtent.width`, `maxExtent.height`, and `maxExtent.depth`, except when one of the following conditions is true, in which case it **may** instead be `1`:
  - `vkGetPhysicalDeviceImageFormatProperties`::`tiling` was VK_IMAGE_TILING_LINEAR
  - VkPhysicalDeviceImageFormatInfo2::`tiling` was VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT
  - the VkPhysicalDeviceImageFormatInfo2::`pNext` chain included a VkPhysicalDeviceExternalImageFormatInfo structure with a handle type included in the `handleTypes` member for which mipmap image support is not required
  - image `format` is one of the formats that require a sampler $Y'C_BC_R$ conversion
  - `flags` contains VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- `maxArrayLayers` is the maximum number of array layers. `maxArrayLayers` **must** be no less than VkPhysicalDeviceLimits::`maxImageArrayLayers`, except when one of the following conditions is

true, in which case it **may** instead be 1:

  - `tiling` is VK_IMAGE_TILING_LINEAR
  - `tiling` is VK_IMAGE_TILING_OPTIMAL and `type` is VK_IMAGE_TYPE_3D
  - `format` is one of the formats that require a sampler $Y'C_BC_R$ conversion

- If `tiling` is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT, then `maxArrayLayers` **must** not be 0.
- `sampleCounts` is a bitmask of VkSampleCountFlagBits specifying all the supported sample counts for this image as described below.
- `maxResourceSize` is an upper bound on the total image size in bytes, inclusive of all image subresources. Implementations **may** have an address space limit on total size of a resource, which is advertised by this property. `maxResourceSize` **must** be at least 2^{31}.

NOTE

There is no mechanism to query the size of an image before creating it, to compare that size against `maxResourceSize`. If an application attempts to create an image that exceeds this limit, the creation will fail and vkCreateImage will return VK_ERROR_OUT_OF_DEVICE_MEMORY. While the advertised limit **must** be at least 2^{31}, it **may** not be possible to create an image that approaches that size, particularly for VK_IMAGE_TYPE_1D.

If the combination of parameters to `vkGetPhysicalDeviceImageFormatProperties` is not supported by the implementation for use in vkCreateImage, then all members of `VkImageFormatProperties` will be filled with zero.

NOTE

Filling `VkImageFormatProperties` with zero for unsupported formats is an exception to the usual rule that output structures have undefined contents on error. This exception was unintentional, but is preserved for backwards compatibility.

To query additional capabilities specific to image types, call:

```
// Provided by VK_VERSION_1_1
VkResult vkGetPhysicalDeviceImageFormatProperties2(
    VkPhysicalDevice                            physicalDevice,
    const VkPhysicalDeviceImageFormatInfo2*     pImageFormatInfo,
    VkImageFormatProperties2*                   pImageFormatProperties);
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to vkGetPhysicalDeviceImageFormatProperties2
VkResult vkGetPhysicalDeviceImageFormatProperties2KHR(
    VkPhysicalDevice                            physicalDevice,
    const VkPhysicalDeviceImageFormatInfo2*     pImageFormatInfo,
    VkImageFormatProperties2*                   pImageFormatProperties);
```

- `physicalDevice` is the physical device from which to query the image capabilities.

- `pImageFormatInfo` is a pointer to a VkPhysicalDeviceImageFormatInfo2 structure describing the parameters that would be consumed by vkCreateImage.
- `pImageFormatProperties` is a pointer to a VkImageFormatProperties2 structure in which capabilities are returned.

`vkGetPhysicalDeviceImageFormatProperties2` behaves similarly to vkGetPhysicalDeviceImageFormatProperties, with the ability to return extended information in a `pNext` chain of output structures.

If the `pNext` chain of `pImageFormatInfo` includes a VkVideoProfileListInfoKHR structure with a `profileCount` member greater than `0`, then this command returns format capabilities specific to image types used in conjunction with the specified video profiles. In this case, this command will return one of the video-profile-specific error codes if any of the profiles specified via VkVideoProfileListInfoKHR::`pProfiles` are not supported. Furthermore, if VkPhysicalDeviceImageFormatInfo2::`usage` includes any image usage flag not supported by the specified video profiles, then this command returns VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR.

If the `hostImageCopy` feature is supported, and:

- `pImageFormatInfo->usage` includes VK_IMAGE_USAGE_SAMPLED_BIT
- `pImageFormatInfo->flags` does not include any of VK_IMAGE_CREATE_SPARSE_BINDING_BIT, VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT, or VK_IMAGE_CREATE_SPARSE_ALIASED_BIT
- The `pNext` chain of `pImageFormatInfo` does not include a VkPhysicalDeviceExternalImageFormatInfo structure with non-zero `handleType`
- `pImageFormatInfo->tiling` is not VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT

Then the result of calls to `vkGetPhysicalDeviceImageFormatProperties2` with identical parameters except for the inclusion of VK_IMAGE_USAGE_HOST_TRANSFER_BIT in `pImageFormatInfo->usage` **must** be identical.

Valid Usage

- VUID-vkGetPhysicalDeviceImageFormatProperties2-pNext-09004
  If the `pNext` chain of `pImageFormatProperties` includes a VkHostImageCopyDevicePerformanceQuery structure, `pImageFormatInfo->usage` **must** contain VK_IMAGE_USAGE_HOST_TRANSFER_BIT

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceImageFormatProperties2-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceImageFormatProperties2-pImageFormatInfo-parameter
  `pImageFormatInfo` **must** be a valid pointer to a valid VkPhysicalDeviceImageFormatInfo2 structure

- VUID-vkGetPhysicalDeviceImageFormatProperties2-pImageFormatProperties-parameter `pImageFormatProperties` **must** be a valid pointer to a VkImageFormatProperties2 structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_FORMAT_NOT_SUPPORTED
- VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR
- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED
- VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR
- VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR
- VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR
- VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR

The `VkPhysicalDeviceImageFormatInfo2` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceImageFormatInfo2 {
    VkStructureType       sType;
    const void*           pNext;
    VkFormat              format;
    VkImageType           type;
    VkImageTiling         tiling;
    VkImageUsageFlags     usage;
    VkImageCreateFlags    flags;
} VkPhysicalDeviceImageFormatInfo2;
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to VkPhysicalDeviceImageFormatInfo2
typedef VkPhysicalDeviceImageFormatInfo2 VkPhysicalDeviceImageFormatInfo2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure. The `pNext` chain of `VkPhysicalDeviceImageFormatInfo2` is used to provide additional image parameters to `vkGetPhysicalDeviceImageFormatProperties2`.

- `format` is a VkFormat value indicating the image format, corresponding to VkImageCreateInfo::`format`.
- `type` is a VkImageType value indicating the image type, corresponding to VkImageCreateInfo::`imageType`.
- `tiling` is a VkImageTiling value indicating the image tiling, corresponding to VkImageCreateInfo::`tiling`.
- `usage` is a bitmask of VkImageUsageFlagBits indicating the intended usage of the image, corresponding to VkImageCreateInfo::`usage`.
- `flags` is a bitmask of VkImageCreateFlagBits indicating additional parameters of the image, corresponding to VkImageCreateInfo::`flags`.

The members of `VkPhysicalDeviceImageFormatInfo2` correspond to the arguments to vkGetPhysicalDeviceImageFormatProperties, with `sType` and `pNext` added for extensibility.

Only create flags representable in VkImageCreateFlagBits are returned in `flags`. If the `pNext` chain includes a VkImageCreateFlags2CreateInfoKHR structure, all create flags are returned in VkImageCreateFlags2CreateInfoKHR::`flags`.

Only usage flags representable in VkImageUsageFlagBits are returned in `usage`. If the `pNext` chain includes a VkImageUsageFlags2CreateInfoKHR structure, all usage flags are returned in VkImageUsageFlags2CreateInfoKHR::`usage`.

Valid Usage

- VUID-VkPhysicalDeviceImageFormatInfo2-tiling-02249
  `tiling` **must** be VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT if and only if the `pNext` chain includes VkPhysicalDeviceImageDrmFormatModifierInfoEXT
- VUID-VkPhysicalDeviceImageFormatInfo2-tiling-02313
  If `tiling` is VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT and `flags` contains VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT, then the `pNext` chain **must** include a VkImageFormatListCreateInfo structure with non-zero `viewFormatCount`

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImageFormatInfo2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_FORMAT_INFO_2
- VUID-VkPhysicalDeviceImageFormatInfo2-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkImageCompressionControlEXT, VkImageCreateFlags2CreateInfoKHR, VkImageFormatListCreateInfo, VkImageStencilUsage2CreateInfoKHR, VkImageStencilUsageCreateInfo, VkImageUsageFlags2CreateInfoKHR, VkPhysicalDeviceExternalImageFormatInfo, VkPhysicalDeviceImageDrmFormatModifierInfoEXT, VkPhysicalDeviceImageViewImageFormatInfoEXT, or VkVideoProfileListInfoKHR

- VUID-VkPhysicalDeviceImageFormatInfo2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkPhysicalDeviceImageFormatInfo2-format-parameter
  `format` **must** be a valid VkFormat value
- VUID-VkPhysicalDeviceImageFormatInfo2-type-parameter
  `type` **must** be a valid VkImageType value
- VUID-VkPhysicalDeviceImageFormatInfo2-tiling-parameter
  `tiling` **must** be a valid VkImageTiling value
- VUID-VkPhysicalDeviceImageFormatInfo2-usage-parameter
  `usage` **must** be a valid combination of VkImageUsageFlagBits values
- VUID-VkPhysicalDeviceImageFormatInfo2-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkImageCreateFlagBits values

The `VkImageFormatProperties2` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkImageFormatProperties2 {
    VkStructureType             sType;
    void*                       pNext;
    VkImageFormatProperties     imageFormatProperties;
} VkImageFormatProperties2;
```

```
// Provided by VK_KHR_get_physical_device_properties2
// Equivalent to VkImageFormatProperties2
typedef VkImageFormatProperties2 VkImageFormatProperties2KHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure. The `pNext` chain of `VkImageFormatProperties2` is used to allow the specification of additional capabilities to be returned from `vkGetPhysicalDeviceImageFormatProperties2`.
- `imageFormatProperties` is a VkImageFormatProperties structure in which capabilities are returned.

If the combination of parameters to `vkGetPhysicalDeviceImageFormatProperties2` is not supported by the implementation for use in vkCreateImage, then all members of `imageFormatProperties` will be filled with zero.

NOTE

Filling `imageFormatProperties` with zero for unsupported formats is an exception to the usual rule that output structures have undefined contents on error. This exception was unintentional, but is preserved for backwards compatibility. This exception only applies to `imageFormatProperties`, not `sType`, `pNext`, or any structures chained from `pNext`.

Valid Usage (Implicit)

- VUID-VkImageFormatProperties2-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_IMAGE_FORMAT_PROPERTIES_2
- VUID-VkImageFormatProperties2-pNext-pNext
  Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of VkExternalImageFormatProperties, VkFilterCubicImageViewImageFormatPropertiesEXT, VkHostImageCopyDevicePerformanceQuery, VkImageCompressionPropertiesEXT, VkSamplerYcbcrConversionImageFormatProperties, or VkSubsampledImageFormatPropertiesEXT
- VUID-VkImageFormatProperties2-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

To determine the image capabilities compatible with an external memory handle type, add a VkPhysicalDeviceExternalImageFormatInfo structure to the `pNext` chain of the VkPhysicalDeviceImageFormatInfo2 structure and a `VkExternalImageFormatProperties` structure to the `pNext` chain of the VkImageFormatProperties2 structure.

The `VkPhysicalDeviceExternalImageFormatInfo` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceExternalImageFormatInfo {
    VkStructureType                       sType;
    const void*                           pNext;
    VkExternalMemoryHandleTypeFlagBits    handleType;
} VkPhysicalDeviceExternalImageFormatInfo;
```

```
// Provided by VK_KHR_external_memory_capabilities
// Equivalent to VkPhysicalDeviceExternalImageFormatInfo
typedef VkPhysicalDeviceExternalImageFormatInfo
VkPhysicalDeviceExternalImageFormatInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `handleType` is a VkExternalMemoryHandleTypeFlagBits value specifying the memory handle type that will be used with the memory associated with the image.

If `handleType` is 0, vkGetPhysicalDeviceImageFormatProperties2 will behave as if VkPhysicalDeviceExternalImageFormatInfo was not present, and VkExternalImageFormatProperties will be ignored.

If `handleType` is not compatible with the `format`, `type`, `tiling`, `usage`, and `flags` specified in VkPhysicalDeviceImageFormatInfo2, then vkGetPhysicalDeviceImageFormatProperties2 returns

VK_ERROR_FORMAT_NOT_SUPPORTED.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExternalImageFormatInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_IMAGE_FORMAT_INFO
- VUID-VkPhysicalDeviceExternalImageFormatInfo-handleType-parameter
  If `handleType` is not `0`, `handleType` **must** be a valid VkExternalMemoryHandleTypeFlagBits value

Structure Chaining

Extends the structure

- VkPhysicalDeviceImageFormatInfo2

Possible values of VkPhysicalDeviceExternalImageFormatInfo::`handleType`, specifying an external memory handle type, are:

```
// Provided by VK_VERSION_1_1
typedef enum VkExternalMemoryHandleTypeFlagBits {
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT = 0x00000001,
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT = 0x00000002,
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT = 0x00000004,
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT = 0x00000008,
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT = 0x00000010,
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT = 0x00000020,
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT = 0x00000040,
  // Provided by VK_EXT_external_memory_dma_buf
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT = 0x00000200,
  // Provided by VK_EXT_external_memory_host
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT = 0x00000080,
  // Provided by VK_EXT_external_memory_host
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT = 0x00000100,
  // Provided by VK_EXT_external_memory_metal
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLBUFFER_BIT_EXT = 0x00010000,
  // Provided by VK_EXT_external_memory_metal
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLTEXTURE_BIT_EXT = 0x00020000,
  // Provided by VK_EXT_external_memory_metal
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLHEAP_BIT_EXT = 0x00040000,
  // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT_KHR =
VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT,
  // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR =
VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT,
  // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT_KHR =
VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT,
  // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT_KHR =
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT,
  // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT_KHR =
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT,
  // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT_KHR =
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT,
  // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT_KHR =
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT,
} VkExternalMemoryHandleTypeFlagBits;
```

```
// Provided by VK_KHR_external_memory_capabilities
// Equivalent to VkExternalMemoryHandleTypeFlagBits
typedef VkExternalMemoryHandleTypeFlagBits VkExternalMemoryHandleTypeFlagBitsKHR;
```

- VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT specifies a POSIX file descriptor handle that has only limited valid usage outside of Vulkan and other compatible APIs. It **must** be compatible with the POSIX system calls `dup`, `dup2`, `close`, and the non-standard system call `dup3`. Additionally, it **must** be transportable over a socket using an `SCM_RIGHTS` control message. It owns a reference to the underlying memory resource represented by its Vulkan memory object.
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT specifies an NT handle that has only limited valid usage outside of Vulkan and other compatible APIs. It **must** be compatible with the functions `DuplicateHandle`, `CloseHandle`, `CompareObjectHandles`, `GetHandleInformation`, and `SetHandleInformation`. It owns a reference to the underlying memory resource represented by its Vulkan memory object.
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT specifies a global share handle that has only limited valid usage outside of Vulkan and other compatible APIs. It is not compatible with any native APIs. It does not own a reference to the underlying memory resource represented by its Vulkan memory object, and will therefore become invalid when all Vulkan memory objects associated with it are destroyed.
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT specifies an NT handle returned by `IDXGIResource1`::`CreateSharedHandle` referring to a Direct3D 10 or 11 texture resource. It owns a reference to the memory used by the Direct3D resource.
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT specifies a global share handle returned by `IDXGIResource`::`GetSharedHandle` referring to a Direct3D 10 or 11 texture resource. It does not own a reference to the underlying Direct3D resource, and will therefore become invalid when all Vulkan memory objects and Direct3D resources associated with it are destroyed.
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT specifies an NT handle returned by `ID3D12Device`::`CreateSharedHandle` referring to a Direct3D 12 heap resource. It owns a reference to the resources used by the Direct3D heap.
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT specifies an NT handle returned by `ID3D12Device`::`CreateSharedHandle` referring to a Direct3D 12 committed resource. It owns a reference to the memory used by the Direct3D resource.
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT specifies a host pointer returned by a host memory allocation command. It does not own a reference to the underlying memory resource, and will therefore become invalid if the host memory is freed.
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT specifies a host pointer to *host mapped foreign memory*. It does not own a reference to the underlying memory resource, and will therefore become invalid if the foreign memory is unmapped or otherwise becomes no longer available.
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT is a file descriptor for a Linux dma_buf. It owns a reference to the underlying memory resource represented by its Vulkan memory object.
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLBUFFER_BIT_EXT is a handle to a `MTLResource` holding a `MTLBuffer`.
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLTEXTURE_BIT_EXT is a handle to a `MTLResource` holding a `MTLTexture`.

- VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLHEAP_BIT_EXT is a handle to a `MTLResource` holding a `MTLHeap`.

Some external memory handle types can only be shared within the same underlying physical device and/or the same driver version, as defined in the following table:

Table 95. External Memory Handle Types Compatibility

Handle type	VkPhysicalDeviceIDProperties::driverUUID	VkPhysicalDeviceIDProperties::deviceUUID
VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT	Must match	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT	Must match	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT	Must match	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT	Must match	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT	Must match	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT	Must match	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT	Must match	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT	No restriction	No restriction
VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT	No restriction	No restriction
VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT	No restriction	No restriction
VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLBUFFER_BIT_EXT	No restriction	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLTEXTURE_BIT_EXT	No restriction	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLHEAP_BIT_EXT	No restriction	Must match

NOTE The above table does not restrict the drivers and devices with which

VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT and VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT **may** be shared, as these handle types inherently mean memory that does not come from the same device, as they import memory from the host or a foreign device, respectively.

NOTE

Even though the above table does not restrict the drivers and devices with which VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT **may** be shared, query mechanisms exist in the Vulkan API that prevent the import of incompatible dma-bufs (such as vkGetMemoryFdPropertiesKHR) and that prevent incompatible usage of dma-bufs (such as VkPhysicalDeviceExternalBufferInfo and VkPhysicalDeviceExternalImageFormatInfo).

```
// Provided by VK_VERSION_1_1
typedef VkFlags VkExternalMemoryHandleTypeFlags;
```

```
// Provided by VK_KHR_external_memory_capabilities
// Equivalent to VkExternalMemoryHandleTypeFlags
typedef VkExternalMemoryHandleTypeFlags VkExternalMemoryHandleTypeFlagsKHR;
```

VkExternalMemoryHandleTypeFlags is a bitmask type for setting a mask of zero or more VkExternalMemoryHandleTypeFlagBits.

The VkExternalImageFormatProperties structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkExternalImageFormatProperties {
    VkStructureType               sType;
    void*                         pNext;
    VkExternalMemoryProperties    externalMemoryProperties;
} VkExternalImageFormatProperties;
```

```
// Provided by VK_KHR_external_memory_capabilities
// Equivalent to VkExternalImageFormatProperties
typedef VkExternalImageFormatProperties VkExternalImageFormatPropertiesKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- externalMemoryProperties is a VkExternalMemoryProperties structure specifying various capabilities of the external handle type when used with the specified image creation parameters.

Valid Usage (Implicit)

- VUID-VkExternalImageFormatProperties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_EXTERNAL_IMAGE_FORMAT_PROPERTIES

Structure Chaining

Extends the structure

- VkImageFormatProperties2

The `VkExternalMemoryProperties` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkExternalMemoryProperties {
    VkExternalMemoryFeatureFlags       externalMemoryFeatures;
    VkExternalMemoryHandleTypeFlags    exportFromImportedHandleTypes;
    VkExternalMemoryHandleTypeFlags    compatibleHandleTypes;
} VkExternalMemoryProperties;
```

```
// Provided by VK_KHR_external_memory_capabilities
// Equivalent to VkExternalMemoryProperties
typedef VkExternalMemoryProperties VkExternalMemoryPropertiesKHR;
```

- `externalMemoryFeatures` is a bitmask of VkExternalMemoryFeatureFlagBits specifying the features of `handleType`.
- `exportFromImportedHandleTypes` is a bitmask of VkExternalMemoryHandleTypeFlagBits specifying which types of imported handle `handleType` **can** be exported from.
- `compatibleHandleTypes` is a bitmask of VkExternalMemoryHandleTypeFlagBits specifying handle types which **can** be specified at the same time as `handleType` when creating an image compatible with external memory.

`compatibleHandleTypes` **must** include at least `handleType`. Inclusion of a handle type in `compatibleHandleTypes` does not imply the values returned in VkImageFormatProperties2 will be the same when VkPhysicalDeviceExternalImageFormatInfo::`handleType` is set to that type. The application is responsible for querying the capabilities of all handle types intended for concurrent use in a single image and intersecting them to obtain the compatible set of capabilities.

Bits which **may** be set in VkExternalMemoryProperties::`externalMemoryFeatures`, specifying features of an external memory handle type, are:

```
// Provided by VK_VERSION_1_1
typedef enum VkExternalMemoryFeatureFlagBits {
    VK_EXTERNAL_MEMORY_FEATURE_DEDICATED_ONLY_BIT = 0x00000001,
    VK_EXTERNAL_MEMORY_FEATURE_EXPORTABLE_BIT = 0x00000002,
    VK_EXTERNAL_MEMORY_FEATURE_IMPORTABLE_BIT = 0x00000004,
  // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_FEATURE_DEDICATED_ONLY_BIT_KHR =
VK_EXTERNAL_MEMORY_FEATURE_DEDICATED_ONLY_BIT,
  // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_FEATURE_EXPORTABLE_BIT_KHR =
VK_EXTERNAL_MEMORY_FEATURE_EXPORTABLE_BIT,
  // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_FEATURE_IMPORTABLE_BIT_KHR =
VK_EXTERNAL_MEMORY_FEATURE_IMPORTABLE_BIT,
} VkExternalMemoryFeatureFlagBits;
```

```
// Provided by VK_KHR_external_memory_capabilities
// Equivalent to VkExternalMemoryFeatureFlagBits
typedef VkExternalMemoryFeatureFlagBits VkExternalMemoryFeatureFlagBitsKHR;
```

- VK_EXTERNAL_MEMORY_FEATURE_DEDICATED_ONLY_BIT specifies that images or buffers created with the specified parameters and handle type **must** use the mechanisms defined by VkMemoryDedicatedRequirements and VkMemoryDedicatedAllocateInfo to create (or import) a dedicated allocation for the image or buffer.
- VK_EXTERNAL_MEMORY_FEATURE_EXPORTABLE_BIT specifies that handles of this type **can** be exported from Vulkan memory objects.
- VK_EXTERNAL_MEMORY_FEATURE_IMPORTABLE_BIT specifies that handles of this type **can** be imported as Vulkan memory objects.

Because their semantics in external APIs roughly align with that of an image or buffer with a dedicated allocation in Vulkan, implementations are **required** to report VK_EXTERNAL_MEMORY_FEATURE_DEDICATED_ONLY_BIT for the following external handle types:

- VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT
- VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLTEXTURE_BIT_EXT

Implementations **must** not report VK_EXTERNAL_MEMORY_FEATURE_DEDICATED_ONLY_BIT for images or buffers with external handle type VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT, or VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT.

```
// Provided by VK_VERSION_1_1
typedef VkFlags VkExternalMemoryFeatureFlags;
```

```
// Provided by VK_KHR_external_memory_capabilities
// Equivalent to VkExternalMemoryFeatureFlags
typedef VkExternalMemoryFeatureFlags VkExternalMemoryFeatureFlagsKHR;
```

VkExternalMemoryFeatureFlags is a bitmask type for setting a mask of zero or more VkExternalMemoryFeatureFlagBits.

To query the image capabilities that are compatible with a Linux DRM format modifier, set VkPhysicalDeviceImageFormatInfo2::tiling to VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT and add a VkPhysicalDeviceImageDrmFormatModifierInfoEXT structure to the pNext chain of VkPhysicalDeviceImageFormatInfo2.

The VkPhysicalDeviceImageDrmFormatModifierInfoEXT structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier
typedef struct VkPhysicalDeviceImageDrmFormatModifierInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    uint64_t           drmFormatModifier;
    VkSharingMode      sharingMode;
    uint32_t           queueFamilyIndexCount;
    const uint32_t*    pQueueFamilyIndices;
} VkPhysicalDeviceImageDrmFormatModifierInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- drmFormatModifier is the image's *Linux DRM format modifier*, corresponding to VkImageDrmFormatModifierExplicitCreateInfoEXT::drmFormatModifier or to VkImageDrmFormatModifierListCreateInfoEXT::pDrmFormatModifiers.
- sharingMode specifies how the image will be accessed by multiple queue families.
- queueFamilyIndexCount is the number of entries in the pQueueFamilyIndices array.
- pQueueFamilyIndices is a pointer to an array of queue families that will access the image. It is ignored if sharingMode is not VK_SHARING_MODE_CONCURRENT.

If the drmFormatModifier is incompatible with the parameters specified in VkPhysicalDeviceImageFormatInfo2 and its pNext chain, then vkGetPhysicalDeviceImageFormatProperties2 returns VK_ERROR_FORMAT_NOT_SUPPORTED. The implementation **must** support the query of any drmFormatModifier, including unknown and invalid modifier values.

Valid Usage

- VUID-VkPhysicalDeviceImageDrmFormatModifierInfoEXT-sharingMode-02314
  If `sharingMode` is VK_SHARING_MODE_CONCURRENT, then `pQueueFamilyIndices` **must** be a valid pointer to an array of `queueFamilyIndexCount` `uint32_t` values
- VUID-VkPhysicalDeviceImageDrmFormatModifierInfoEXT-maintenance11-13351
  If the `maintenance11` feature is supported and `sharingMode` is VK_SHARING_MODE_CONCURRENT, then `queueFamilyIndexCount` **must** be greater than `0`
- VUID-VkPhysicalDeviceImageDrmFormatModifierInfoEXT-sharingMode-02315
  If the `maintenance11` feature is not supported and `sharingMode` is VK_SHARING_MODE_CONCURRENT, then `queueFamilyIndexCount` **must** be greater than `1`
- VUID-VkPhysicalDeviceImageDrmFormatModifierInfoEXT-sharingMode-02316
  If `sharingMode` is VK_SHARING_MODE_CONCURRENT, each element of `pQueueFamilyIndices` **must** be unique and **must** be less than the `pQueueFamilyPropertyCount` returned by vkGetPhysicalDeviceQueueFamilyProperties2 for the `physicalDevice` that was used to create `device`

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImageDrmFormatModifierInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_DRM_FORMAT_MODIFIER_INFO_EXT
- VUID-VkPhysicalDeviceImageDrmFormatModifierInfoEXT-sharingMode-parameter
  `sharingMode` **must** be a valid VkSharingMode value

Structure Chaining

Extends the structure

- VkPhysicalDeviceImageFormatInfo2

To determine the number of descriptors required to support an image format for sampled images with $Y'C_BC_R$ conversion, add VkSamplerYcbcrConversionImageFormatProperties to the `pNext` chain of the VkImageFormatProperties2 structure in a call to `vkGetPhysicalDeviceImageFormatProperties2`.

The `VkSamplerYcbcrConversionImageFormatProperties` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkSamplerYcbcrConversionImageFormatProperties {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           combinedImageSamplerDescriptorCount;
} VkSamplerYcbcrConversionImageFormatProperties;
```

```
// Provided by VK_KHR_sampler_ycbcr_conversion
// Equivalent to VkSamplerYcbcrConversionImageFormatProperties
typedef VkSamplerYcbcrConversionImageFormatProperties
VkSamplerYcbcrConversionImageFormatPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `combinedImageSamplerDescriptorCount` is the number of descriptors that the implementation uses to access a sampled image with $Y'C_BC_R$ conversion.

`combinedImageSamplerDescriptorCount` is a number between 1 and the number of planes in the format.

The size an image descriptor with a VkSamplerYcbcrConversionInfo structure consumes in the resource heap is multiplied by the `combinedImageSamplerDescriptorCount` of its format. These images are sampled with embedded samplers and therefore do not consume space in the sampler heap.

For descriptor buffers or descriptor sets, each combined image sampler descriptor binding with a layout that includes at least one immutable $Y'C_BC_R$ conversion sampler will internally use a number of descriptors equal to the maximum `combinedImageSamplerDescriptorCount` of the formats of all the immutable $Y'C_BC_R$ conversion samplers in that binding.

This expanded number of descriptors will be consumed from the descriptor pool when a descriptor set is allocated, and counts towards the `maxDescriptorSetSamplers`, `maxDescriptorSetSampledImages`, `maxPerStageDescriptorSamplers`, and `maxPerStageDescriptorSampledImages` limits.

NOTE

All descriptors in a binding use the same expanded number of descriptors to allow implementations to use a uniform stride for dynamic indexing.

For example, consider a descriptor set layout binding with two immutable $Y'C_BC_R$ samplers with formats that have `VkSamplerYcbcrConversionImageFormatProperties::combinedImageSamplerDescriptorCount` values of `2` and `3` respectively. The maximum `combinedImageSamplerDescriptorCount` for this binding layout is therefore `3`.

If there are `3` descriptors in a descriptor set binding with this layout, including one descriptor without an immutable $Y'C_BC_R$ sampler, the binding will consume `9` combined image sampler descriptors from the descriptor pool.

For descriptor heaps, the application explicitly maps the descriptor binding to a heap offset with VkDescriptorSetAndBindingMappingEXT and **should** make similar calculations as it **must** offset each element in the binding correctly.

Instead of querying all the potential formats that an application might use, it **can** use VkPhysicalDeviceMaintenance6Properties::`maxCombinedImageSamplerDescriptorCount` property to determine the maximum `combinedImageSamplerDescriptorCount` for all formats the implementation supports.

Valid Usage (Implicit)

- VUID-VkSamplerYcbcrConversionImageFormatProperties-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_IMAGE_FORMAT_PROPERTIES`

Structure Chaining

Extends the structure

- VkImageFormatProperties2

To determine the number of image descriptors required to support a subsampled image created with VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT, add VkSubsampledImageFormatPropertiesEXT to the `pNext` chain of the VkImageFormatProperties2 structure in a call to `vkGetPhysicalDeviceImageFormatProperties2`.

The `VkSubsampledImageFormatPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_descriptor_heap with VK_EXT_fragment_density_map
typedef struct VkSubsampledImageFormatPropertiesEXT {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           subsampledImageDescriptorCount;
} VkSubsampledImageFormatPropertiesEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `subsampledImageDescriptorCount` is the number of image descriptors that the implementation uses to access the image.

Valid Usage (Implicit)

- VUID-VkSubsampledImageFormatPropertiesEXT-sType-sType
  `sType` **must** be `VK_STRUCTURE_TYPE_SUBSAMPLED_IMAGE_FORMAT_PROPERTIES_EXT`

Structure Chaining

Extends the structure

- VkImageFormatProperties2

To query if using VK_IMAGE_USAGE_HOST_TRANSFER_BIT has a negative impact on device performance when accessing an image, add VK_IMAGE_USAGE_HOST_TRANSFER_BIT to

VkPhysicalDeviceImageFormatInfo2::usage, and add a VkHostImageCopyDevicePerformanceQuery structure to the pNext chain of a VkImageFormatProperties2 structure passed to vkGetPhysicalDeviceImageFormatProperties2. This structure is defined as:

```
// Provided by VK_VERSION_1_4
typedef struct VkHostImageCopyDevicePerformanceQuery {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           optimalDeviceAccess;
    VkBool32           identicalMemoryLayout;
} VkHostImageCopyDevicePerformanceQuery;
```

```
// Provided by VK_EXT_host_image_copy
// Equivalent to VkHostImageCopyDevicePerformanceQuery
typedef VkHostImageCopyDevicePerformanceQuery
VkHostImageCopyDevicePerformanceQueryEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- optimalDeviceAccess returns VK_TRUE if use of host image copy has no adverse effect on device access performance, compared to an image that is created with exact same creation parameters, and bound to the same VkDeviceMemory, except that the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag is replaced with VK_IMAGE_USAGE_TRANSFER_SRC_BIT and VK_IMAGE_USAGE_TRANSFER_DST_BIT.
- identicalMemoryLayout returns VK_TRUE if use of host image copy has no impact on memory layout compared to an image that is created with exact same creation parameters, and bound to the same VkDeviceMemory, except that the VK_IMAGE_USAGE_HOST_TRANSFER_BIT usage flag is replaced with VK_IMAGE_USAGE_TRANSFER_SRC_BIT and VK_IMAGE_USAGE_TRANSFER_DST_BIT.

The implementation **may** return VK_FALSE in optimalDeviceAccess if identicalMemoryLayout is VK_FALSE. If identicalMemoryLayout is VK_TRUE, optimalDeviceAccess **must** be VK_TRUE.

The implementation **may** return VK_TRUE in optimalDeviceAccess while identicalMemoryLayout is VK_FALSE. In this situation, any device performance impact **should** not be measurable.

If VkPhysicalDeviceImageFormatInfo2::format is a block-compressed format and vkGetPhysicalDeviceImageFormatProperties2 returns VK_SUCCESS, the implementation **must** return VK_TRUE in optimalDeviceAccess.

NOTE Applications can make use of optimalDeviceAccess to determine their resource copying strategy. If a resource is expected to be accessed more on device than on the host, and the implementation considers the resource sub-optimally accessed, it is likely better to use device copies instead.

NOTE Layout not being identical yet still considered optimal for device access could

happen if the implementation has different memory layout patterns, some of which are easier to access on the host.

NOTE

The most practical reason for `optimalDeviceAccess` to be VK_FALSE is that host image access may disable framebuffer compression where it would otherwise have been enabled. This represents far more efficient host image access since no compression algorithm is required to read or write to the image, but it would impact device access performance. Some implementations may only set `optimalDeviceAccess` to VK_FALSE if certain conditions are met, such as specific image usage flags or creation flags.

Valid Usage (Implicit)

- VUID-VkHostImageCopyDevicePerformanceQuery-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_HOST_IMAGE_COPY_DEVICE_PERFORMANCE_QUERY

Structure Chaining

Extends the structure

- VkImageFormatProperties2

To determine if cubic filtering can be used with a given image format and a given image view type add a VkPhysicalDeviceImageViewImageFormatInfoEXT structure to the `pNext` chain of the VkPhysicalDeviceImageFormatInfo2 structure, and a VkFilterCubicImageViewImageFormatPropertiesEXT structure to the `pNext` chain of the VkImageFormatProperties2 structure.

The `VkPhysicalDeviceImageViewImageFormatInfoEXT` structure is defined as:

```
// Provided by VK_EXT_filter_cubic
typedef struct VkPhysicalDeviceImageViewImageFormatInfoEXT {
    VkStructureType    sType;
    void*              pNext;
    VkImageViewType    imageViewType;
} VkPhysicalDeviceImageViewImageFormatInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `imageViewType` is a VkImageViewType value specifying the type of the image view.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImageViewImageFormatInfoEXT-sType-sType

sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_VIEW_IMAGE_FORMAT_INFO_EXT

- VUID-VkPhysicalDeviceImageViewImageFormatInfoEXT-imageViewType-parameter imageViewType **must** be a valid VkImageViewType value

Structure Chaining

Extends the structure

- VkPhysicalDeviceImageFormatInfo2

The VkFilterCubicImageViewImageFormatPropertiesEXT structure is defined as:

```
// Provided by VK_EXT_filter_cubic
typedef struct VkFilterCubicImageViewImageFormatPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           filterCubic;
    VkBool32           filterCubicMinmax;
} VkFilterCubicImageViewImageFormatPropertiesEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- filterCubic tells if image format, image type and image view type **can** be used with cubic filtering. This field is set by the implementation. An application-specified value is ignored.
- filterCubicMinmax tells if image format, image type and image view type **can** be used with cubic filtering and minmax filtering. This field is set by the implementation. An application-specified value is ignored.

Valid Usage (Implicit)

- VUID-VkFilterCubicImageViewImageFormatPropertiesEXT-sType-sType sType must be VK_STRUCTURE_TYPE_FILTER_CUBIC_IMAGE_VIEW_IMAGE_FORMAT_PROPERTIES_EXT

Structure Chaining

Extends the structure

- VkImageFormatProperties2

Valid Usage

- VUID-VkFilterCubicImageViewImageFormatPropertiesEXT-pNext-02627
  If the pNext chain of the VkImageFormatProperties2 structure includes a VkFilterCubicImageViewImageFormatPropertiesEXT structure, the pNext chain of the VkPhysicalDeviceImageFormatInfo2 structure **must** include a VkPhysicalDeviceImageViewImageFormatInfoEXT structure with an imageViewType that is compatible with imageType

50.1.1. Supported Sample Counts

vkGetPhysicalDeviceImageFormatProperties returns a bitmask of VkSampleCountFlagBits in sampleCounts specifying the supported sample counts for the image parameters.

sampleCounts will be VK_SAMPLE_COUNT_1_BIT if at least one of the following conditions is true:

- tiling is VK_IMAGE_TILING_LINEAR
- type is not VK_IMAGE_TYPE_2D
- flags contains VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT and does not contain VK_IMAGE_CREATE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT
- Neither the VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT flag nor the VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT flag in VkFormatProperties::optimalTilingFeatures returned by vkGetPhysicalDeviceFormatProperties is set
- VkPhysicalDeviceExternalImageFormatInfo::handleType is an external handle type for which multisampled image support is not required.
- format is one of the formats that require a sampler $Y'C_BC_R$ conversion
- usage contains VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- usage contains VK_IMAGE_USAGE_FRAGMENT_DENSITY_MAP_BIT_EXT

Otherwise, the bits set in sampleCounts will be the sample counts supported for the specified values of usage and format. For each bit set in usage, the supported sample counts relate to the limits in VkPhysicalDeviceLimits as follows:

- If usage includes VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT and format is a floating- or fixed-point color format, a superset of VkPhysicalDeviceLimits::framebufferColorSampleCounts
- If usage includes VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT and format is an integer format, a superset of VkPhysicalDeviceVulkan12Properties::framebufferIntegerColorSampleCounts
- If usage includes VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, and format includes a depth component, a superset of VkPhysicalDeviceLimits::framebufferDepthSampleCounts
- If usage includes VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, and format includes a stencil component, a superset of VkPhysicalDeviceLimits::framebufferStencilSampleCounts
- If usage includes VK_IMAGE_USAGE_SAMPLED_BIT, and format includes a color component, a superset of VkPhysicalDeviceLimits::sampledImageColorSampleCounts

- If `usage` includes VK_IMAGE_USAGE_SAMPLED_BIT, and `format` includes a depth component, a superset of `VkPhysicalDeviceLimits`::`sampledImageDepthSampleCounts`
- If `usage` includes VK_IMAGE_USAGE_SAMPLED_BIT, and `format` is an integer format, a superset of `VkPhysicalDeviceLimits`::`sampledImageIntegerSampleCounts`
- If `usage` includes VK_IMAGE_USAGE_STORAGE_BIT, a superset of `VkPhysicalDeviceLimits`::`storageImageSampleCounts`

If multiple bits are set in `usage`, `sampleCounts` will be the intersection of the per-usage values described above.

If none of the bits described above are set in `usage`, then there is no corresponding limit in `VkPhysicalDeviceLimits`. In this case, `sampleCounts` **must** include at least VK_SAMPLE_COUNT_1_BIT.

50.1.2. Allowed Extent Values Based on Image Type

Implementations **may** support extent values larger than the required minimum/maximum values for certain types of images. VkImageFormatProperties::`maxExtent` for each type is subject to the constraints below.

NOTE	Implementations **must** support images with dimensions up to the required minimum/maximum values for all types of images. It follows that the query for additional capabilities **must** return extent values that are at least as large as the required values.

For VK_IMAGE_TYPE_1D:

- `maxExtent.width` ≥ VkPhysicalDeviceLimits::`maxImageDimension1D`
- `maxExtent.height` = 1
- `maxExtent.depth` = 1

For VK_IMAGE_TYPE_2D when `flags` does not contain VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT:

- `maxExtent.width` ≥ VkPhysicalDeviceLimits::`maxImageDimension2D`
- `maxExtent.height` ≥ VkPhysicalDeviceLimits::`maxImageDimension2D`
- `maxExtent.depth` = 1

For VK_IMAGE_TYPE_2D when `flags` contains VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT:

- `maxExtent.width` ≥ VkPhysicalDeviceLimits::`maxImageDimensionCube`
- `maxExtent.height` ≥ VkPhysicalDeviceLimits::`maxImageDimensionCube`
- `maxExtent.depth` = 1

For VK_IMAGE_TYPE_3D:

- `maxExtent.width` ≥ VkPhysicalDeviceLimits::`maxImageDimension3D`
- `maxExtent.height` ≥ VkPhysicalDeviceLimits::`maxImageDimension3D`

- `maxExtent.depth` ≥ VkPhysicalDeviceLimits::`maxImageDimension3D`

50.2. Additional Buffer Capabilities

To query the external handle types supported by buffers, call:

```
// Provided by VK_VERSION_1_1
void vkGetPhysicalDeviceExternalBufferProperties(
    VkPhysicalDevice                            physicalDevice,
    const VkPhysicalDeviceExternalBufferInfo*   pExternalBufferInfo,
    VkExternalBufferProperties*                 pExternalBufferProperties);
```

```
// Provided by VK_KHR_external_memory_capabilities
// Equivalent to vkGetPhysicalDeviceExternalBufferProperties
void vkGetPhysicalDeviceExternalBufferPropertiesKHR(
    VkPhysicalDevice                            physicalDevice,
    const VkPhysicalDeviceExternalBufferInfo*   pExternalBufferInfo,
    VkExternalBufferProperties*                 pExternalBufferProperties);
```

- `physicalDevice` is the physical device from which to query the buffer capabilities.
- `pExternalBufferInfo` is a pointer to a VkPhysicalDeviceExternalBufferInfo structure describing the parameters that would be consumed by vkCreateBuffer.
- `pExternalBufferProperties` is a pointer to a VkExternalBufferProperties structure in which capabilities are returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceExternalBufferProperties-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceExternalBufferProperties-pExternalBufferInfo-parameter
  `pExternalBufferInfo` **must** be a valid pointer to a valid VkPhysicalDeviceExternalBufferInfo structure
- VUID-vkGetPhysicalDeviceExternalBufferProperties-pExternalBufferProperties-parameter
  `pExternalBufferProperties` **must** be a valid pointer to a VkExternalBufferProperties structure

The `VkPhysicalDeviceExternalBufferInfo` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceExternalBufferInfo {
    VkStructureType                       sType;
    const void*                           pNext;
    VkBufferCreateFlags                   flags;
    VkBufferUsageFlags                    usage;
    VkExternalMemoryHandleTypeFlagBits    handleType;
} VkPhysicalDeviceExternalBufferInfo;
```

```
// Provided by VK_KHR_external_memory_capabilities
// Equivalent to VkPhysicalDeviceExternalBufferInfo
typedef VkPhysicalDeviceExternalBufferInfo VkPhysicalDeviceExternalBufferInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkBufferCreateFlagBits describing additional parameters of the buffer, corresponding to VkBufferCreateInfo::`flags`.
- `usage` is a bitmask of VkBufferUsageFlagBits describing the intended usage of the buffer, corresponding to VkBufferCreateInfo::`usage`.
- `handleType` is a VkExternalMemoryHandleTypeFlagBits value specifying the memory handle type that will be used with the memory associated with the buffer.

Only usage flags representable in VkBufferUsageFlagBits are returned in this structure's `usage`. If the `pNext` chain includes a VkBufferUsageFlags2CreateInfo structure, all usage flags of the buffer are returned in VkBufferUsageFlags2CreateInfo::`usage`.

Valid Usage

- VUID-VkPhysicalDeviceExternalBufferInfo-None-09499
  If the `pNext` chain does not include a VkBufferUsageFlags2CreateInfo structure, `usage` **must** be a valid combination of VkBufferUsageFlagBits values
- VUID-VkPhysicalDeviceExternalBufferInfo-None-09500
  If the `pNext` chain does not include a VkBufferUsageFlags2CreateInfo structure, `usage` **must** not be 0

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExternalBufferInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_BUFFER_INFO
- VUID-VkPhysicalDeviceExternalBufferInfo-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkBufferUsageFlags2CreateInfo
- VUID-VkPhysicalDeviceExternalBufferInfo-sType-unique

The `sType` value of each structure in the `pNext` chain **must** be unique

- VUID-VkPhysicalDeviceExternalBufferInfo-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkBufferCreateFlagBits values
- VUID-VkPhysicalDeviceExternalBufferInfo-handleType-parameter
  `handleType` **must** be a valid VkExternalMemoryHandleTypeFlagBits value

The `VkExternalBufferProperties` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkExternalBufferProperties {
    VkStructureType              sType;
    void*                        pNext;
    VkExternalMemoryProperties   externalMemoryProperties;
} VkExternalBufferProperties;
```

```
// Provided by VK_KHR_external_memory_capabilities
// Equivalent to VkExternalBufferProperties
typedef VkExternalBufferProperties VkExternalBufferPropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `externalMemoryProperties` is a VkExternalMemoryProperties structure specifying various capabilities of the external handle type when used with the specified buffer creation parameters.

Valid Usage (Implicit)

- VUID-VkExternalBufferProperties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_EXTERNAL_BUFFER_PROPERTIES
- VUID-VkExternalBufferProperties-pNext-pNext
  `pNext` **must** be `NULL`

50.3. Optional Semaphore Capabilities

Semaphores **may** support import and export of their payload to external handles. To query the external handle types supported by semaphores, call:

```
// Provided by VK_VERSION_1_1
void vkGetPhysicalDeviceExternalSemaphoreProperties(
    VkPhysicalDevice                            physicalDevice,
    const VkPhysicalDeviceExternalSemaphoreInfo* pExternalSemaphoreInfo,
    VkExternalSemaphoreProperties*              pExternalSemaphoreProperties);
```

```
// Provided by VK_KHR_external_semaphore_capabilities
// Equivalent to vkGetPhysicalDeviceExternalSemaphoreProperties
void vkGetPhysicalDeviceExternalSemaphorePropertiesKHR(
    VkPhysicalDevice                            physicalDevice,
    const VkPhysicalDeviceExternalSemaphoreInfo* pExternalSemaphoreInfo,
    VkExternalSemaphoreProperties*              pExternalSemaphoreProperties);
```

- `physicalDevice` is the physical device from which to query the semaphore capabilities.
- `pExternalSemaphoreInfo` is a pointer to a VkPhysicalDeviceExternalSemaphoreInfo structure describing the parameters that would be consumed by vkCreateSemaphore.
- `pExternalSemaphoreProperties` is a pointer to a VkExternalSemaphoreProperties structure in which capabilities are returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceExternalSemaphoreProperties-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceExternalSemaphoreProperties-pExternalSemaphoreInfo-parameter
  `pExternalSemaphoreInfo` **must** be a valid pointer to a valid VkPhysicalDeviceExternalSemaphoreInfo structure
- VUID-vkGetPhysicalDeviceExternalSemaphoreProperties-pExternalSemaphoreProperties-parameter
  `pExternalSemaphoreProperties` **must** be a valid pointer to a VkExternalSemaphoreProperties structure

The `VkPhysicalDeviceExternalSemaphoreInfo` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceExternalSemaphoreInfo {
    VkStructureType                         sType;
    const void*                             pNext;
    VkExternalSemaphoreHandleTypeFlagBits   handleType;
} VkPhysicalDeviceExternalSemaphoreInfo;
```

```
// Provided by VK_KHR_external_semaphore_capabilities
// Equivalent to VkPhysicalDeviceExternalSemaphoreInfo
typedef VkPhysicalDeviceExternalSemaphoreInfo
VkPhysicalDeviceExternalSemaphoreInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `handleType` is a VkExternalSemaphoreHandleTypeFlagBits value specifying the external semaphore handle type for which capabilities will be returned.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExternalSemaphoreInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_SEMAPHORE_INFO
- VUID-VkPhysicalDeviceExternalSemaphoreInfo-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkSemaphoreTypeCreateInfo
- VUID-VkPhysicalDeviceExternalSemaphoreInfo-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkPhysicalDeviceExternalSemaphoreInfo-handleType-parameter
  `handleType` **must** be a valid VkExternalSemaphoreHandleTypeFlagBits value

Bits which **may** be set in VkPhysicalDeviceExternalSemaphoreInfo::`handleType`, specifying an external semaphore handle type, are:

```
// Provided by VK_VERSION_1_1
typedef enum VkExternalSemaphoreHandleTypeFlagBits {
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT = 0x00000001,
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT = 0x00000002,
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT = 0x00000004,
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT = 0x00000008,
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT = 0x00000010,
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D11_FENCE_BIT =
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT,
  // Provided by VK_KHR_external_semaphore_capabilities
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT_KHR =
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT,
  // Provided by VK_KHR_external_semaphore_capabilities
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR =
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT,
  // Provided by VK_KHR_external_semaphore_capabilities
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT_KHR =
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT,
  // Provided by VK_KHR_external_semaphore_capabilities
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT_KHR =
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT,
  // Provided by VK_KHR_external_semaphore_capabilities
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT_KHR =
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT,
} VkExternalSemaphoreHandleTypeFlagBits;
```

```
// Provided by VK_KHR_external_semaphore_capabilities
// Equivalent to VkExternalSemaphoreHandleTypeFlagBits
typedef VkExternalSemaphoreHandleTypeFlagBits
VkExternalSemaphoreHandleTypeFlagBitsKHR;
```

- VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT specifies a POSIX file descriptor handle that has only limited valid usage outside of Vulkan and other compatible APIs. It **must** be compatible with the POSIX system calls `dup`, `dup2`, `close`, and the non-standard system call `dup3`. Additionally, it **must** be transportable over a socket using an `SCM_RIGHTS` control message. It owns a reference to the underlying synchronization primitive represented by its Vulkan semaphore object.
- VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT specifies an NT handle that has only limited valid usage outside of Vulkan and other compatible APIs. It **must** be compatible with the functions `DuplicateHandle`, `CloseHandle`, `CompareObjectHandles`, `GetHandleInformation`, and `SetHandleInformation`. It owns a reference to the underlying synchronization primitive represented by its Vulkan semaphore object.
- VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT specifies a global share handle that has only limited valid usage outside of Vulkan and other compatible APIs. It is not compatible with any native APIs. It does not own a reference to the underlying synchronization primitive represented by its Vulkan semaphore object, and will therefore become invalid when

all Vulkan semaphore objects associated with it are destroyed.

- VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT specifies an NT handle returned by `ID3D12Device`::`CreateSharedHandle` referring to a Direct3D 12 fence, or `ID3D11Device5` ::`CreateFence` referring to a Direct3D 11 fence. It owns a reference to the underlying synchronization primitive associated with the Direct3D fence.
- VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D11_FENCE_BIT is an alias of VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT with the same meaning. It is provided for convenience and code clarity when interacting with D3D11 fences.
- VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT specifies a POSIX file descriptor handle to a Linux Sync File or Android Fence object. It can be used with any native API accepting a valid sync file or fence as input. It owns a reference to the underlying synchronization primitive associated with the file descriptor. Implementations which support importing this handle type **must** accept any type of sync or fence FD supported by the native system they are running on.

NOTE

Handles of type VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT generated by the implementation may represent either Linux Sync Files or Android Fences at the implementation's discretion. Applications **should** only use operations defined for both types of file descriptors, unless they know via means external to Vulkan the type of the file descriptor, or are prepared to deal with the system-defined operation failures resulting from using the wrong type.

Some external semaphore handle types can only be shared within the same underlying physical device and/or the same driver version, as defined in the following table:

Table 96. External Semaphore Handle Types Compatibility

Handle type	VkPhysicalDeviceIDProperties::driverUUID	VkPhysicalDeviceIDProperties::deviceUUID
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT	Must match	Must match
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT	Must match	Must match
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT	Must match	Must match
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT	Must match	Must match
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT	No restriction	No restriction

```
// Provided by VK_VERSION_1_1
typedef VkFlags VkExternalSemaphoreHandleTypeFlags;
```

```
// Provided by VK_KHR_external_semaphore_capabilities
// Equivalent to VkExternalSemaphoreHandleTypeFlags
typedef VkExternalSemaphoreHandleTypeFlags VkExternalSemaphoreHandleTypeFlagsKHR;
```

VkExternalSemaphoreHandleTypeFlags is a bitmask type for setting a mask of zero or more VkExternalSemaphoreHandleTypeFlagBits.

The VkExternalSemaphoreProperties structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkExternalSemaphoreProperties {
    VkStructureType                       sType;
    void*                                 pNext;
    VkExternalSemaphoreHandleTypeFlags    exportFromImportedHandleTypes;
    VkExternalSemaphoreHandleTypeFlags    compatibleHandleTypes;
    VkExternalSemaphoreFeatureFlags       externalSemaphoreFeatures;
} VkExternalSemaphoreProperties;
```

```
// Provided by VK_KHR_external_semaphore_capabilities
// Equivalent to VkExternalSemaphoreProperties
typedef VkExternalSemaphoreProperties VkExternalSemaphorePropertiesKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `exportFromImportedHandleTypes` is a bitmask of VkExternalSemaphoreHandleTypeFlagBits specifying which types of imported handle `handleType` **can** be exported from.
- `compatibleHandleTypes` is a bitmask of VkExternalSemaphoreHandleTypeFlagBits specifying handle types which **can** be specified at the same time as `handleType` when creating a semaphore.
- `externalSemaphoreFeatures` is a bitmask of VkExternalSemaphoreFeatureFlagBits describing the features of `handleType`.

If `handleType` is not supported by the implementation, then VkExternalSemaphoreProperties ::`externalSemaphoreFeatures` will be zero.

Valid Usage (Implicit)

- VUID-VkExternalSemaphoreProperties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_EXTERNAL_SEMAPHORE_PROPERTIES
- VUID-VkExternalSemaphoreProperties-pNext-pNext
  `pNext` **must** be `NULL`

Bits which **may** be set in VkExternalSemaphoreProperties::`externalSemaphoreFeatures`, specifying the features of an external semaphore handle type, are:

```
// Provided by VK_VERSION_1_1
typedef enum VkExternalSemaphoreFeatureFlagBits {
    VK_EXTERNAL_SEMAPHORE_FEATURE_EXPORTABLE_BIT = 0x00000001,
    VK_EXTERNAL_SEMAPHORE_FEATURE_IMPORTABLE_BIT = 0x00000002,
  // Provided by VK_KHR_external_semaphore_capabilities
    VK_EXTERNAL_SEMAPHORE_FEATURE_EXPORTABLE_BIT_KHR =
VK_EXTERNAL_SEMAPHORE_FEATURE_EXPORTABLE_BIT,
  // Provided by VK_KHR_external_semaphore_capabilities
    VK_EXTERNAL_SEMAPHORE_FEATURE_IMPORTABLE_BIT_KHR =
VK_EXTERNAL_SEMAPHORE_FEATURE_IMPORTABLE_BIT,
} VkExternalSemaphoreFeatureFlagBits;
```

```
// Provided by VK_KHR_external_semaphore_capabilities
// Equivalent to VkExternalSemaphoreFeatureFlagBits
typedef VkExternalSemaphoreFeatureFlagBits VkExternalSemaphoreFeatureFlagBitsKHR;
```

- VK_EXTERNAL_SEMAPHORE_FEATURE_EXPORTABLE_BIT specifies that handles of this type **can**

be exported from Vulkan semaphore objects.

- VK_EXTERNAL_SEMAPHORE_FEATURE_IMPORTABLE_BIT specifies that handles of this type **can** be imported as Vulkan semaphore objects.

```
// Provided by VK_VERSION_1_1
typedef VkFlags VkExternalSemaphoreFeatureFlags;
```

```
// Provided by VK_KHR_external_semaphore_capabilities
// Equivalent to VkExternalSemaphoreFeatureFlags
typedef VkExternalSemaphoreFeatureFlags VkExternalSemaphoreFeatureFlagsKHR;
```

VkExternalSemaphoreFeatureFlags is a bitmask type for setting a mask of zero or more VkExternalSemaphoreFeatureFlagBits.

50.4. Optional Fence Capabilities

Fences **may** support import and export of their payload to external handles. To query the external handle types supported by fences, call:

```
// Provided by VK_VERSION_1_1
void vkGetPhysicalDeviceExternalFenceProperties(
    VkPhysicalDevice                            physicalDevice,
    const VkPhysicalDeviceExternalFenceInfo*    pExternalFenceInfo,
    VkExternalFenceProperties*                  pExternalFenceProperties);
```

```
// Provided by VK_KHR_external_fence_capabilities
// Equivalent to vkGetPhysicalDeviceExternalFenceProperties
void vkGetPhysicalDeviceExternalFencePropertiesKHR(
    VkPhysicalDevice                            physicalDevice,
    const VkPhysicalDeviceExternalFenceInfo*    pExternalFenceInfo,
    VkExternalFenceProperties*                  pExternalFenceProperties);
```

- physicalDevice is the physical device from which to query the fence capabilities.
- pExternalFenceInfo is a pointer to a VkPhysicalDeviceExternalFenceInfo structure describing the parameters that would be consumed by vkCreateFence.
- pExternalFenceProperties is a pointer to a VkExternalFenceProperties structure in which capabilities are returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceExternalFenceProperties-physicalDevice-parameter
  physicalDevice **must** be a valid VkPhysicalDevice handle

- VUID-vkGetPhysicalDeviceExternalFenceProperties-pExternalFenceInfo-parameter
  `pExternalFenceInfo` **must** be a valid pointer to a valid VkPhysicalDeviceExternalFenceInfo structure
- VUID-vkGetPhysicalDeviceExternalFenceProperties-pExternalFenceProperties-parameter
  `pExternalFenceProperties` **must** be a valid pointer to a VkExternalFenceProperties structure

The `VkPhysicalDeviceExternalFenceInfo` structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkPhysicalDeviceExternalFenceInfo {
    VkStructureType                       sType;
    const void*                           pNext;
    VkExternalFenceHandleTypeFlagBits     handleType;
} VkPhysicalDeviceExternalFenceInfo;
```

```
// Provided by VK_KHR_external_fence_capabilities
// Equivalent to VkPhysicalDeviceExternalFenceInfo
typedef VkPhysicalDeviceExternalFenceInfo VkPhysicalDeviceExternalFenceInfoKHR;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `handleType` is a VkExternalFenceHandleTypeFlagBits value specifying an external fence handle type for which capabilities will be returned.

NOTE

Handles of type VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT generated by the implementation may represent either Linux Sync Files or Android Fences at the implementation's discretion. Applications **should** only use operations defined for both types of file descriptors, unless they know via means external to Vulkan the type of the file descriptor, or are prepared to deal with the system-defined operation failures resulting from using the wrong type.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExternalFenceInfo-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_FENCE_INFO
- VUID-VkPhysicalDeviceExternalFenceInfo-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkPhysicalDeviceExternalFenceInfo-handleType-parameter
  `handleType` **must** be a valid VkExternalFenceHandleTypeFlagBits value

Bits which **may** be set in

- VkPhysicalDeviceExternalFenceInfo::handleType
- VkExternalFenceProperties::exportFromImportedHandleTypes
- VkExternalFenceProperties::compatibleHandleTypes

indicate external fence handle types, and are:

```
// Provided by VK_VERSION_1_1
typedef enum VkExternalFenceHandleTypeFlagBits {
    VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_FD_BIT = 0x00000001,
    VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT = 0x00000002,
    VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT = 0x00000004,
    VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT = 0x00000008,
  // Provided by VK_KHR_external_fence_capabilities
    VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_FD_BIT_KHR =
VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_FD_BIT,
  // Provided by VK_KHR_external_fence_capabilities
    VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR =
VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT,
  // Provided by VK_KHR_external_fence_capabilities
    VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT_KHR =
VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT,
  // Provided by VK_KHR_external_fence_capabilities
    VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT_KHR =
VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT,
} VkExternalFenceHandleTypeFlagBits;
```

```
// Provided by VK_KHR_external_fence_capabilities
// Equivalent to VkExternalFenceHandleTypeFlagBits
typedef VkExternalFenceHandleTypeFlagBits VkExternalFenceHandleTypeFlagBitsKHR;
```

- VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_FD_BIT specifies a POSIX file descriptor handle that has only limited valid usage outside of Vulkan and other compatible APIs. It **must** be compatible with the POSIX system calls dup, dup2, close, and the non-standard system call dup3. Additionally, it **must** be transportable over a socket using an SCM_RIGHTS control message. It owns a reference to the underlying synchronization primitive represented by its Vulkan fence object.
- VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT specifies an NT handle that has only limited valid usage outside of Vulkan and other compatible APIs. It **must** be compatible with the functions DuplicateHandle, CloseHandle, CompareObjectHandles, GetHandleInformation, and SetHandleInformation. It owns a reference to the underlying synchronization primitive represented by its Vulkan fence object.
- VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT specifies a global share handle that has only limited valid usage outside of Vulkan and other compatible APIs. It is not compatible with any native APIs. It does not own a reference to the underlying synchronization primitive represented by its Vulkan fence object, and will therefore become invalid when all Vulkan fence objects associated with it are destroyed.

- VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT specifies a POSIX file descriptor handle to a Linux Sync File or Android Fence. It can be used with any native API accepting a valid sync file or fence as input. It owns a reference to the underlying synchronization primitive associated with the file descriptor. Implementations which support importing this handle type **must** accept any type of sync or fence FD supported by the native system they are running on.

Some external fence handle types can only be shared within the same underlying physical device and/or the same driver version, as defined in the following table:

Table 97. External Fence Handle Types Compatibility

Handle type	VkPhysicalDeviceIDProperties::driverUUID	VkPhysicalDeviceIDProperties::deviceUUID
VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_FD_BIT	Must match	Must match
VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT	Must match	Must match
VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT	Must match	Must match
VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT	No restriction	No restriction

```
// Provided by VK_VERSION_1_1
typedef VkFlags VkExternalFenceHandleTypeFlags;
```

```
// Provided by VK_KHR_external_fence_capabilities
// Equivalent to VkExternalFenceHandleTypeFlags
typedef VkExternalFenceHandleTypeFlags VkExternalFenceHandleTypeFlagsKHR;
```

VkExternalFenceHandleTypeFlags is a bitmask type for setting a mask of zero or more VkExternalFenceHandleTypeFlagBits.

The VkExternalFenceProperties structure is defined as:

```
// Provided by VK_VERSION_1_1
typedef struct VkExternalFenceProperties {
    VkStructureType                 sType;
    void*                           pNext;
    VkExternalFenceHandleTypeFlags  exportFromImportedHandleTypes;
    VkExternalFenceHandleTypeFlags  compatibleHandleTypes;
    VkExternalFenceFeatureFlags     externalFenceFeatures;
} VkExternalFenceProperties;
```

```
// Provided by VK_KHR_external_fence_capabilities
// Equivalent to VkExternalFenceProperties
typedef VkExternalFenceProperties VkExternalFencePropertiesKHR;
```

- exportFromImportedHandleTypes is a bitmask of VkExternalFenceHandleTypeFlagBits indicating which types of imported handle handleType **can** be exported from.

- `compatibleHandleTypes` is a bitmask of VkExternalFenceHandleTypeFlagBits specifying handle types which **can** be specified at the same time as `handleType` when creating a fence.
- `externalFenceFeatures` is a bitmask of VkExternalFenceFeatureFlagBits indicating the features of `handleType`.

If `handleType` is not supported by the implementation, then VkExternalFenceProperties ::`externalFenceFeatures` will be zero.

Valid Usage (Implicit)

- VUID-VkExternalFenceProperties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_EXTERNAL_FENCE_PROPERTIES
- VUID-VkExternalFenceProperties-pNext-pNext
  `pNext` **must** be `NULL`

Bits which **may** be set in VkExternalFenceProperties::`externalFenceFeatures`, indicating features of a fence external handle type, are:

```
// Provided by VK_VERSION_1_1
typedef enum VkExternalFenceFeatureFlagBits {
    VK_EXTERNAL_FENCE_FEATURE_EXPORTABLE_BIT = 0x00000001,
    VK_EXTERNAL_FENCE_FEATURE_IMPORTABLE_BIT = 0x00000002,
  // Provided by VK_KHR_external_fence_capabilities
    VK_EXTERNAL_FENCE_FEATURE_EXPORTABLE_BIT_KHR =
VK_EXTERNAL_FENCE_FEATURE_EXPORTABLE_BIT,
  // Provided by VK_KHR_external_fence_capabilities
    VK_EXTERNAL_FENCE_FEATURE_IMPORTABLE_BIT_KHR =
VK_EXTERNAL_FENCE_FEATURE_IMPORTABLE_BIT,
} VkExternalFenceFeatureFlagBits;
```

```
// Provided by VK_KHR_external_fence_capabilities
// Equivalent to VkExternalFenceFeatureFlagBits
typedef VkExternalFenceFeatureFlagBits VkExternalFenceFeatureFlagBitsKHR;
```

- VK_EXTERNAL_FENCE_FEATURE_EXPORTABLE_BIT specifies handles of this type **can** be exported from Vulkan fence objects.
- VK_EXTERNAL_FENCE_FEATURE_IMPORTABLE_BIT specifies handles of this type **can** be imported to Vulkan fence objects.

```
// Provided by VK_VERSION_1_1
typedef VkFlags VkExternalFenceFeatureFlags;
```

```
// Provided by VK_KHR_external_fence_capabilities
// Equivalent to VkExternalFenceFeatureFlags
typedef VkExternalFenceFeatureFlags VkExternalFenceFeatureFlagsKHR;
```

VkExternalFenceFeatureFlags is a bitmask type for setting a mask of zero or more VkExternalFenceFeatureFlagBits.

50.5. Timestamp Calibration Capabilities

To query the set of time domains for which a physical device supports timestamp calibration, call:

```
// Provided by VK_KHR_calibrated_timestamps
VkResult vkGetPhysicalDeviceCalibrateableTimeDomainsKHR(
    VkPhysicalDevice                            physicalDevice,
    uint32_t*                                   pTimeDomainCount,
    VkTimeDomainKHR*                            pTimeDomains);
```

- physicalDevice is the physical device from which to query the set of calibrateable time domains.
- pTimeDomainCount is a pointer to an integer related to the number of calibrateable time domains available or queried, as described below.
- pTimeDomains is either NULL or a pointer to an array of VkTimeDomainKHR values, indicating the supported calibrateable time domains.

If pTimeDomains is NULL, then the number of calibrateable time domains supported for the given physicalDevice is returned in pTimeDomainCount. Otherwise, pTimeDomainCount **must** point to a variable set by the application to the number of elements in the pTimeDomains array, and on return the variable is overwritten with the number of values actually written to pTimeDomains. If the value of pTimeDomainCount is less than the number of calibrateable time domains supported, at most pTimeDomainCount values will be written to pTimeDomains, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available time domains were returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceCalibrateableTimeDomainsKHR-physicalDevice-parameter
  physicalDevice **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceCalibrateableTimeDomainsKHR-pTimeDomainCount-parameter
  pTimeDomainCount **must** be a valid pointer to a uint32_t value
- VUID-vkGetPhysicalDeviceCalibrateableTimeDomainsKHR-pTimeDomains-parameter
  If the value referenced by pTimeDomainCount is not 0, and pTimeDomains is not NULL, pTimeDomains **must** be a valid pointer to an array of pTimeDomainCount VkTimeDomainKHR values

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

Chapter 51. Debugging

To aid developers in tracking down errors in the application's use of Vulkan, particularly in combination with an external debugger or profiler, *debugging extensions* may be available.

The VkObjectType enumeration defines values, each of which corresponds to a specific Vulkan handle type. These values **can** be used to associate debug information with a particular type of object through one or more extensions.

```
// Provided by VK_VERSION_1_0
typedef enum VkObjectType {
    VK_OBJECT_TYPE_UNKNOWN = 0,
    VK_OBJECT_TYPE_INSTANCE = 1,
    VK_OBJECT_TYPE_PHYSICAL_DEVICE = 2,
    VK_OBJECT_TYPE_DEVICE = 3,
    VK_OBJECT_TYPE_QUEUE = 4,
    VK_OBJECT_TYPE_SEMAPHORE = 5,
    VK_OBJECT_TYPE_COMMAND_BUFFER = 6,
    VK_OBJECT_TYPE_FENCE = 7,
    VK_OBJECT_TYPE_DEVICE_MEMORY = 8,
    VK_OBJECT_TYPE_BUFFER = 9,
    VK_OBJECT_TYPE_IMAGE = 10,
    VK_OBJECT_TYPE_EVENT = 11,
    VK_OBJECT_TYPE_QUERY_POOL = 12,
    VK_OBJECT_TYPE_BUFFER_VIEW = 13,
    VK_OBJECT_TYPE_IMAGE_VIEW = 14,
    VK_OBJECT_TYPE_SHADER_MODULE = 15,
    VK_OBJECT_TYPE_PIPELINE_CACHE = 16,
    VK_OBJECT_TYPE_PIPELINE_LAYOUT = 17,
    VK_OBJECT_TYPE_RENDER_PASS = 18,
    VK_OBJECT_TYPE_PIPELINE = 19,
    VK_OBJECT_TYPE_DESCRIPTOR_SET_LAYOUT = 20,
    VK_OBJECT_TYPE_SAMPLER = 21,
    VK_OBJECT_TYPE_DESCRIPTOR_POOL = 22,
    VK_OBJECT_TYPE_DESCRIPTOR_SET = 23,
    VK_OBJECT_TYPE_FRAMEBUFFER = 24,
    VK_OBJECT_TYPE_COMMAND_POOL = 25,
  // Provided by VK_VERSION_1_1
    VK_OBJECT_TYPE_DESCRIPTOR_UPDATE_TEMPLATE = 1000085000,
  // Provided by VK_VERSION_1_1
    VK_OBJECT_TYPE_SAMPLER_YCBCR_CONVERSION = 1000156000,
  // Provided by VK_VERSION_1_3
    VK_OBJECT_TYPE_PRIVATE_DATA_SLOT = 1000295000,
  // Provided by VK_KHR_surface
    VK_OBJECT_TYPE_SURFACE_KHR = 1000000000,
  // Provided by VK_KHR_swapchain
    VK_OBJECT_TYPE_SWAPCHAIN_KHR = 1000001000,
  // Provided by VK_KHR_display
    VK_OBJECT_TYPE_DISPLAY_KHR = 1000002000,
  // Provided by VK_KHR_display
```

```
    VK_OBJECT_TYPE_DISPLAY_MODE_KHR = 1000002001,
  // Provided by VK_KHR_video_queue
    VK_OBJECT_TYPE_VIDEO_SESSION_KHR = 1000023000,
  // Provided by VK_KHR_video_queue
    VK_OBJECT_TYPE_VIDEO_SESSION_PARAMETERS_KHR = 1000023001,
  // Provided by VK_EXT_debug_utils
    VK_OBJECT_TYPE_DEBUG_UTILS_MESSENGER_EXT = 1000128000,
  // Provided by VK_KHR_acceleration_structure
    VK_OBJECT_TYPE_ACCELERATION_STRUCTURE_KHR = 1000150000,
  // Provided by VK_EXT_validation_cache
    VK_OBJECT_TYPE_VALIDATION_CACHE_EXT = 1000160000,
  // Provided by VK_KHR_deferred_host_operations
    VK_OBJECT_TYPE_DEFERRED_OPERATION_KHR = 1000268000,
  // Provided by VK_EXT_opacity_micromap
    VK_OBJECT_TYPE_MICROMAP_EXT = 1000396000,
  // Provided by VK_EXT_shader_object
    VK_OBJECT_TYPE_SHADER_EXT = 1000482000,
  // Provided by VK_KHR_pipeline_binary
    VK_OBJECT_TYPE_PIPELINE_BINARY_KHR = 1000483000,
  // Provided by VK_EXT_device_generated_commands
    VK_OBJECT_TYPE_INDIRECT_COMMANDS_LAYOUT_EXT = 1000572000,
  // Provided by VK_EXT_device_generated_commands
    VK_OBJECT_TYPE_INDIRECT_EXECUTION_SET_EXT = 1000572001,
  // Provided by VK_KHR_descriptor_update_template
    VK_OBJECT_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_KHR =
VK_OBJECT_TYPE_DESCRIPTOR_UPDATE_TEMPLATE,
  // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_OBJECT_TYPE_SAMPLER_YCBCR_CONVERSION_KHR =
VK_OBJECT_TYPE_SAMPLER_YCBCR_CONVERSION,
} VkObjectType;
```

Table 98. `VkObjectType` and Vulkan Handle Relationship

VkObjectType	**Vulkan Handle Type**
VK_OBJECT_TYPE_UNKNOWN	Unknown/Undefined Handle
VK_OBJECT_TYPE_INSTANCE	VkInstance
VK_OBJECT_TYPE_PHYSICAL_DEVICE	VkPhysicalDevice
VK_OBJECT_TYPE_DEVICE	VkDevice
VK_OBJECT_TYPE_QUEUE	VkQueue
VK_OBJECT_TYPE_SEMAPHORE	VkSemaphore
VK_OBJECT_TYPE_COMMAND_BUFFER	VkCommandBuffer
VK_OBJECT_TYPE_FENCE	VkFence
VK_OBJECT_TYPE_DEVICE_MEMORY	VkDeviceMemory
VK_OBJECT_TYPE_BUFFER	VkBuffer
VK_OBJECT_TYPE_IMAGE	VkImage

VkObjectType	Vulkan Handle Type
VK_OBJECT_TYPE_EVENT	VkEvent
VK_OBJECT_TYPE_QUERY_POOL	VkQueryPool
VK_OBJECT_TYPE_BUFFER_VIEW	VkBufferView
VK_OBJECT_TYPE_IMAGE_VIEW	VkImageView
VK_OBJECT_TYPE_SHADER_MODULE	VkShaderModule
VK_OBJECT_TYPE_PIPELINE_CACHE	VkPipelineCache
VK_OBJECT_TYPE_PIPELINE_LAYOUT	VkPipelineLayout
VK_OBJECT_TYPE_RENDER_PASS	VkRenderPass
VK_OBJECT_TYPE_PIPELINE	VkPipeline
VK_OBJECT_TYPE_DESCRIPTOR_SET_LAYOUT	VkDescriptorSetLayout
VK_OBJECT_TYPE_SAMPLER	VkSampler
VK_OBJECT_TYPE_DESCRIPTOR_POOL	VkDescriptorPool
VK_OBJECT_TYPE_DESCRIPTOR_SET	VkDescriptorSet
VK_OBJECT_TYPE_FRAMEBUFFER	VkFramebuffer
VK_OBJECT_TYPE_COMMAND_POOL	VkCommandPool
VK_OBJECT_TYPE_SAMPLER_YCBCR_CONVERSION	VkSamplerYcbcrConversion
VK_OBJECT_TYPE_DESCRIPTOR_UPDATE_TEMPLATE	VkDescriptorUpdateTemplate
VK_OBJECT_TYPE_PRIVATE_DATA_SLOT	VkPrivateDataSlot
VK_OBJECT_TYPE_SURFACE_KHR	VkSurfaceKHR
VK_OBJECT_TYPE_SWAPCHAIN_KHR	VkSwapchainKHR
VK_OBJECT_TYPE_DISPLAY_KHR	VkDisplayKHR
VK_OBJECT_TYPE_DISPLAY_MODE_KHR	VkDisplayModeKHR
VK_OBJECT_TYPE_VIDEO_SESSION_KHR	VkVideoSessionKHR
VK_OBJECT_TYPE_VIDEO_SESSION_PARAMETERS_KHR	VkVideoSessionParametersKHR
VK_OBJECT_TYPE_DEBUG_UTILS_MESSENGER_EXT	VkDebugUtilsMessengerEXT
VK_OBJECT_TYPE_ACCELERATION_STRUCTURE_KHR	VkAccelerationStructureKHR
VK_OBJECT_TYPE_VALIDATION_CACHE_EXT	VkValidationCacheEXT
VK_OBJECT_TYPE_DEFERRED_OPERATION_KHR	VkDeferredOperationKHR

VkObjectType	Vulkan Handle Type
VK_OBJECT_TYPE_INDIRECT_COMMANDS_LAYOUT_EXT	VkIndirectCommandsLayoutEXT
VK_OBJECT_TYPE_INDIRECT_EXECUTION_SET_EXT	VkIndirectExecutionSetEXT
VK_OBJECT_TYPE_MICROMAP_EXT	VkMicromapEXT
VK_OBJECT_TYPE_SHADER_EXT	VkShaderEXT

If this Specification was generated with any such extensions included, they will be described in the remainder of this chapter.

51.1. Debug Utilities

Vulkan provides flexible debugging utilities for debugging an application.

The Object Debug Annotation section describes how to associate either a name or binary data with a specific Vulkan object.

The Queue Labels section describes how to annotate and group the work submitted to a queue.

The Command Buffer Labels section describes how to associate logical elements of the scene with commands in a VkCommandBuffer.

The Debug Messengers section describes how to create debug messenger objects associated with an application supplied callback to capture debug messages from a variety of Vulkan components.

51.1.1. Object Debug Annotation

It can be useful for an application to provide its own content relative to a specific Vulkan object.

The following commands allow application developers to associate application-defined information with Vulkan objects. These commands are device-level commands but they **may** reference instance-level objects (such as VkInstance) and physical device-level objects (such as VkPhysicalDevice) with a few restrictions:

- The data for the corresponding object **may** still be available after the VkDevice used in the corresponding API call to set it is destroyed, but access to this data is not guaranteed and should be avoided.
- Subsequent calls to change the data of the same object across multiple `VkDevice` objects, **may** result in the data being changed to the most recent version for all `VkDevice` objects and not just the `VkDevice` used in the most recent API call.

Object Naming

An object can be given an application-defined name by calling:

```
// Provided by VK_EXT_debug_utils
VkResult vkSetDebugUtilsObjectNameEXT(
    VkDevice                                    device,
    const VkDebugUtilsObjectNameInfoEXT*        pNameInfo);
```

- `device` is the device that is associated with the named object passed in via `objectHandle`.
- `pNameInfo` is a pointer to a VkDebugUtilsObjectNameInfoEXT structure specifying parameters of the name to set on the object.

Valid Usage

- VUID-vkSetDebugUtilsObjectNameEXT-pNameInfo-02587
  `pNameInfo->objectType` **must** not be VK_OBJECT_TYPE_UNKNOWN
- VUID-vkSetDebugUtilsObjectNameEXT-pNameInfo-02588
  `pNameInfo->objectHandle` **must** not be VK_NULL_HANDLE
- VUID-vkSetDebugUtilsObjectNameEXT-pNameInfo-07872
  If `pNameInfo->objectHandle` is the valid handle of an instance-level object, the VkDevice identified by `device` **must** be a descendant of the same VkInstance as the object identified by `pNameInfo->objectHandle`
- VUID-vkSetDebugUtilsObjectNameEXT-pNameInfo-07873
  If `pNameInfo->objectHandle` is the valid handle of a physical-device-level object, the VkDevice identified by `device` **must** be a descendant of the same VkPhysicalDevice as the object identified by `pNameInfo->objectHandle`
- VUID-vkSetDebugUtilsObjectNameEXT-pNameInfo-07874
  If `pNameInfo->objectHandle` is the valid handle of a device-level object, that object **must** be a descendant of the VkDevice identified by `device`

Valid Usage (Implicit)

- VUID-vkSetDebugUtilsObjectNameEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkSetDebugUtilsObjectNameEXT-pNameInfo-parameter
  `pNameInfo` **must** be a valid pointer to a valid VkDebugUtilsObjectNameInfoEXT structure

Host Synchronization

- Host access to `pNameInfo->objectHandle` **must** be externally synchronized

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkDebugUtilsObjectNameInfoEXT structure is defined as:

```
// Provided by VK_EXT_debug_utils
typedef struct VkDebugUtilsObjectNameInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkObjectType       objectType;
    uint64_t           objectHandle;
    const char*        pObjectName;
} VkDebugUtilsObjectNameInfoEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- objectType is a VkObjectType specifying the type of the object to be named.
- objectHandle is the object to be named.
- pObjectName is either NULL or a null-terminated UTF-8 string specifying the name to apply to objectHandle.

Applications **may** change the name associated with an object simply by calling vkSetDebugUtilsObjectNameEXT again with a new string. If pObjectName is either NULL or an empty string, then any previously set name is removed.

The graphicsPipelineLibrary feature allows the specification of pipelines without the creation of VkShaderModule objects beforehand. In order to continue to allow naming these shaders independently, VkDebugUtilsObjectNameInfoEXT **can** be included in the pNext chain of VkPipelineShaderStageCreateInfo, which associates a static name with that particular shader.

This structure **can** be included in the pNext chain of VkResourceDescriptorInfoEXT or VkSamplerCreateInfo to label a descriptor or embedded sampler. This structure **may** be ignored when included in the pNext chain of VkSamplerCreateInfo when creating a sampler object. The label **must** remain valid while the descriptor is valid; it **may** be discarded if it becomes invalid.

Valid Usage

- VUID-VkDebugUtilsObjectNameInfoEXT-objectType-02589
  If `objectType` is VK_OBJECT_TYPE_UNKNOWN, `objectHandle` **must** not be VK_NULL_HANDLE
- VUID-VkDebugUtilsObjectNameInfoEXT-objectType-02590
  If `objectType` is not VK_OBJECT_TYPE_UNKNOWN, `objectHandle` **must** be VK_NULL_HANDLE or a valid Vulkan handle of the type associated with `objectType` as defined in the `VkObjectType` and Vulkan Handle Relationship table

Valid Usage (Implicit)

- VUID-VkDebugUtilsObjectNameInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEBUG_UTILS_OBJECT_NAME_INFO_EXT
- VUID-VkDebugUtilsObjectNameInfoEXT-objectType-parameter
  `objectType` **must** be a valid VkObjectType value
- VUID-VkDebugUtilsObjectNameInfoEXT-pObjectName-parameter
  If `pObjectName` is not `NULL`, `pObjectName` **must** be a null-terminated UTF-8 string

Structure Chaining

Extends the structures

- VkPipelineShaderStageCreateInfo
- VkResourceDescriptorInfoEXT
- VkSamplerCreateInfo

Object Data Association

In addition to setting a name for an object, debugging and validation layers **may** have uses for additional binary data on a per-object basis that have no other place in the Vulkan API.

For example, a `VkShaderModule` could have additional debugging data attached to it to aid in offline shader tracing.

Additional data can be attached to an object by calling `vkSetDebugUtilsObjectTagEXT` as defined below.

```
// Provided by VK_EXT_debug_utils
VkResult vkSetDebugUtilsObjectTagEXT(
    VkDevice                                    device,
    const VkDebugUtilsObjectTagInfoEXT*         pTagInfo);
```

- `device` is the device that created the object.
- `pTagInfo` is a pointer to a VkDebugUtilsObjectTagInfoEXT structure specifying parameters of the tag to attach to the object.

Valid Usage

- VUID-vkSetDebugUtilsObjectTagEXT-pNameInfo-07875
  If `pTagInfo->objectHandle` is the valid handle of an instance-level object, the VkDevice identified by `device` **must** be a descendent of the same VkInstance as the object identified by `pTagInfo->objectHandle`
- VUID-vkSetDebugUtilsObjectTagEXT-pNameInfo-07876
  If `pTagInfo->objectHandle` is the valid handle of a physical-device-level object, the VkDevice identified by `device` **must** be a descendant of the same VkPhysicalDevice as the object identified by `pTagInfo->objectHandle`
- VUID-vkSetDebugUtilsObjectTagEXT-pNameInfo-07877
  If `pTagInfo->objectHandle` is the valid handle of a device-level object, that object **must** be a descendent of the VkDevice identified by `device`

Valid Usage (Implicit)

- VUID-vkSetDebugUtilsObjectTagEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkSetDebugUtilsObjectTagEXT-pTagInfo-parameter
  `pTagInfo` **must** be a valid pointer to a valid VkDebugUtilsObjectTagInfoEXT structure

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_DEVICE_MEMORY
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkDebugUtilsObjectTagInfoEXT` structure is defined as:

```
// Provided by VK_EXT_debug_utils
typedef struct VkDebugUtilsObjectTagInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkObjectType       objectType;
    uint64_t           objectHandle;
    uint64_t           tagName;
    size_t             tagSize;
    const void*        pTag;
} VkDebugUtilsObjectTagInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `objectType` is a VkObjectType specifying the type of the object to be tagged.
- `objectHandle` is the object to be tagged.
- `tagName` is a numerical identifier of the tag.
- `tagSize` is the number of bytes of data to attach to the object.
- `pTag` is a pointer to an array of `tagSize` bytes containing the data to be associated with the object.

The `tagName` parameter gives a name or identifier to the type of data being tagged. This can be used by debugging layers to easily filter for only data that can be used by that implementation.

Valid Usage

- VUID-VkDebugUtilsObjectTagInfoEXT-objectType-01908
  `objectType` **must** not be VK_OBJECT_TYPE_UNKNOWN
- VUID-VkDebugUtilsObjectTagInfoEXT-objectHandle-01910
  `objectHandle` **must** be a valid Vulkan handle of the type associated with `objectType` as defined in the `VkObjectType` and Vulkan Handle Relationship table

Valid Usage (Implicit)

- VUID-VkDebugUtilsObjectTagInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEBUG_UTILS_OBJECT_TAG_INFO_EXT
- VUID-VkDebugUtilsObjectTagInfoEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDebugUtilsObjectTagInfoEXT-objectType-parameter
  `objectType` **must** be a valid VkObjectType value
- VUID-VkDebugUtilsObjectTagInfoEXT-pTag-parameter
  `pTag` **must** be a valid pointer to an array of `tagSize` bytes
- VUID-VkDebugUtilsObjectTagInfoEXT-tagSize-arraylength
  `tagSize` **must** be greater than `0`

Host Synchronization

- Host access to `objectHandle` **must** be externally synchronized

51.1.2. Queue Labels

All Vulkan work is submitted using queues. It is possible for an application to use multiple queues, each containing multiple command buffers, when performing work. It can be useful to identify which queue, or even where in a queue, something has occurred.

To begin identifying a region using a debug label inside a queue, you may use the vkQueueBeginDebugUtilsLabelEXT command.

Then, when the region of interest has passed, you may end the label region using vkQueueEndDebugUtilsLabelEXT.

Additionally, a single debug label may be inserted at any time using vkQueueInsertDebugUtilsLabelEXT.

A queue debug label region is opened by calling:

```
// Provided by VK_EXT_debug_utils
void vkQueueBeginDebugUtilsLabelEXT(
    VkQueue                                     queue,
    const VkDebugUtilsLabelEXT*                 pLabelInfo);
```

- `queue` is the queue in which to start a debug label region.
- `pLabelInfo` is a pointer to a VkDebugUtilsLabelEXT structure specifying parameters of the label region to open.

Valid Usage (Implicit)

- VUID-vkQueueBeginDebugUtilsLabelEXT-queue-parameter
  `queue` **must** be a valid VkQueue handle
- VUID-vkQueueBeginDebugUtilsLabelEXT-pLabelInfo-parameter
  `pLabelInfo` **must** be a valid pointer to a valid VkDebugUtilsLabelEXT structure

Host Synchronization

- Host access to `queue` **must** be externally synchronized if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	Any	-

The `VkDebugUtilsLabelEXT` structure is defined as:

```
// Provided by VK_EXT_debug_utils
typedef struct VkDebugUtilsLabelEXT {
    VkStructureType    sType;
    const void*        pNext;
    const char*        pLabelName;
    float              color[4];
} VkDebugUtilsLabelEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pLabelName` is a pointer to a null-terminated UTF-8 string containing the name of the label.
- `color` is an optional RGBA color value that can be associated with the label. A particular implementation **may** choose to ignore this color value. The values contain RGBA values in order, in the range 0.0 to 1.0. If all elements in `color` are 0.0, then it is ignored.

Valid Usage (Implicit)

- VUID-VkDebugUtilsLabelEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEBUG_UTILS_LABEL_EXT
- VUID-VkDebugUtilsLabelEXT-pNext-pNext
  `pNext` **must** be `NULL`
- VUID-VkDebugUtilsLabelEXT-pLabelName-parameter
  `pLabelName` **must** be a null-terminated UTF-8 string

A queue debug label region is closed by calling:

```
// Provided by VK_EXT_debug_utils
void vkQueueEndDebugUtilsLabelEXT(
    VkQueue                                     queue);
```

- `queue` is the queue in which a debug label region should be closed.

The calls to vkQueueBeginDebugUtilsLabelEXT and vkQueueEndDebugUtilsLabelEXT **must** be matched and balanced.

Valid Usage

- VUID-vkQueueEndDebugUtilsLabelEXT-None-01911
  There **must** be an outstanding `vkQueueBeginDebugUtilsLabelEXT` command prior to the `vkQueueEndDebugUtilsLabelEXT` on the queue

Valid Usage (Implicit)

- VUID-vkQueueEndDebugUtilsLabelEXT-queue-parameter
  `queue` **must** be a valid VkQueue handle

Host Synchronization

- Host access to `queue` **must** be externally synchronized if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	Any	-

A single label can be inserted into a queue by calling:

```
// Provided by VK_EXT_debug_utils
void vkQueueInsertDebugUtilsLabelEXT(
    VkQueue                                     queue,
    const VkDebugUtilsLabelEXT*                 pLabelInfo);
```

- `queue` is the queue into which a debug label will be inserted.
- `pLabelInfo` is a pointer to a VkDebugUtilsLabelEXT structure specifying parameters of the label to insert.

Valid Usage (Implicit)

- VUID-vkQueueInsertDebugUtilsLabelEXT-queue-parameter
  `queue` **must** be a valid VkQueue handle
- VUID-vkQueueInsertDebugUtilsLabelEXT-pLabelInfo-parameter
  `pLabelInfo` **must** be a valid pointer to a valid VkDebugUtilsLabelEXT structure

Host Synchronization

- Host access to `queue` **must** be externally synchronized if it was not created with VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	Any	-

51.1.3. Command Buffer Labels

Typical Vulkan applications will submit many command buffers in each frame, with each command buffer containing a large number of individual commands. Being able to logically annotate regions of command buffers that belong together as well as hierarchically subdivide the frame is important to a developer's ability to navigate the commands viewed holistically.

To identify the beginning of a debug label region in a command buffer, vkCmdBeginDebugUtilsLabelEXT **can** be used as defined below.

To indicate the end of a debug label region in a command buffer, vkCmdEndDebugUtilsLabelEXT **can** be used.

To insert a single command buffer debug label inside of a command buffer, vkCmdInsertDebugUtilsLabelEXT **can** be used as defined below.

A command buffer debug label region can be opened by calling:

```
// Provided by VK_EXT_debug_utils
void vkCmdBeginDebugUtilsLabelEXT(
    VkCommandBuffer                             commandBuffer,
    const VkDebugUtilsLabelEXT*                 pLabelInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `pLabelInfo` is a pointer to a VkDebugUtilsLabelEXT structure specifying parameters of the label region to open.

Valid Usage (Implicit)

- VUID-vkCmdBeginDebugUtilsLabelEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdBeginDebugUtilsLabelEXT-pLabelInfo-parameter
  `pLabelInfo` **must** be a valid pointer to a valid VkDebugUtilsLabelEXT structure

- VUID-vkCmdBeginDebugUtilsLabelEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdBeginDebugUtilsLabelEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_TRANSFER_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	State

Conditional Rendering

vkCmdBeginDebugUtilsLabelEXT is not affected by conditional rendering

A command buffer label region can be closed by calling:

```
// Provided by VK_EXT_debug_utils
void vkCmdEndDebugUtilsLabelEXT(
    VkCommandBuffer                             commandBuffer);
```

- `commandBuffer` is the command buffer into which the command is recorded.

An application **may** open a debug label region in one command buffer and close it in another, or otherwise split debug label regions across multiple command buffers or multiple queue submissions. When viewed from the linear series of submissions to a single queue, the calls to vkCmdBeginDebugUtilsLabelEXT and vkCmdEndDebugUtilsLabelEXT **must** be matched and balanced.

There **can** be problems reporting command buffer debug labels during the recording process because command buffers **may** be recorded out of sequence with the resulting execution order. Since the recording order **may** be different, a solitary command buffer **may** have an inconsistent view of the debug label regions by itself. Therefore, if an issue occurs during the recording of a command buffer, and the environment requires returning debug labels, the implementation **may** return only those labels it is aware of. This is true even if the implementation is aware of only the debug labels within the command buffer being actively recorded.

Valid Usage

- VUID-vkCmdEndDebugUtilsLabelEXT-commandBuffer-01912
  There **must** be an outstanding `vkCmdBeginDebugUtilsLabelEXT` command prior to the `vkCmdEndDebugUtilsLabelEXT` on the queue that `commandBuffer` is submitted to
- VUID-vkCmdEndDebugUtilsLabelEXT-commandBuffer-01913
  If `commandBuffer` is a secondary command buffer, there **must** be an outstanding `vkCmdBeginDebugUtilsLabelEXT` command recorded to `commandBuffer` that has not previously been ended by a call to `vkCmdEndDebugUtilsLabelEXT`

Valid Usage (Implicit)

- VUID-vkCmdEndDebugUtilsLabelEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdEndDebugUtilsLabelEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdEndDebugUtilsLabelEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_TRANSFER_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	State

Conditional Rendering

vkCmdEndDebugUtilsLabelEXT is not affected by conditional rendering

A single debug label can be inserted into a command buffer by calling:

```
// Provided by VK_EXT_debug_utils
void vkCmdInsertDebugUtilsLabelEXT(
    VkCommandBuffer                             commandBuffer,
    const VkDebugUtilsLabelEXT*                 pLabelInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `pLabelInfo` is a pointer to a VkDebugUtilsLabelEXT structure specifying parameters of the label to insert.

Valid Usage (Implicit)

- VUID-vkCmdInsertDebugUtilsLabelEXT-commandBuffer-parameter
  `commandBuffer` **must** be a valid VkCommandBuffer handle
- VUID-vkCmdInsertDebugUtilsLabelEXT-pLabelInfo-parameter
  `pLabelInfo` **must** be a valid pointer to a valid VkDebugUtilsLabelEXT structure
- VUID-vkCmdInsertDebugUtilsLabelEXT-commandBuffer-recording
  `commandBuffer` **must** be in the recording state
- VUID-vkCmdInsertDebugUtilsLabelEXT-commandBuffer-cmdpool
  The `VkCommandPool` that `commandBuffer` was allocated from **must** support

VK_QUEUE_COMPUTE_BIT, VK_QUEUE_GRAPHICS_BIT, VK_QUEUE_TRANSFER_BIT, VK_QUEUE_VIDEO_DECODE_BIT_KHR, or VK_QUEUE_VIDEO_ENCODE_BIT_KHR operations

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	VK_QUEUE_COMPUTE_BIT VK_QUEUE_GRAPHICS_BIT VK_QUEUE_TRANSFER_BIT VK_QUEUE_VIDEO_DECODE_BIT_KHR VK_QUEUE_VIDEO_ENCODE_BIT_KHR	State

Conditional Rendering

vkCmdInsertDebugUtilsLabelEXT is not affected by conditional rendering

51.1.4. Debug Messengers

Vulkan allows an application to register multiple callbacks with any Vulkan component wishing to report debug information. Some callbacks may log the information to a file, others may cause a debug break point or other application-defined behavior. A primary producer of callback messages are the validation layers. An application **can** register callbacks even when no validation layers are enabled, but they will only be called for the Vulkan loader and, if implemented, other layer and driver events.

A `VkDebugUtilsMessengerEXT` is a messenger object which handles passing along debug messages to a provided debug callback.

```
// Provided by VK_EXT_debug_utils
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDebugUtilsMessengerEXT)
```

The debug messenger will provide detailed feedback on the application's use of Vulkan when events of interest occur. When an event of interest does occur, the debug messenger will submit a debug message to the debug callback that was provided during its creation. Additionally, the debug messenger is responsible for filtering out debug messages that the callback is not interested in and will only provide desired debug messages.

A debug messenger triggers a debug callback with a debug message when an event of interest occurs. To create a debug messenger which will trigger a debug callback, call:

```
// Provided by VK_EXT_debug_utils
VkResult vkCreateDebugUtilsMessengerEXT(
    VkInstance                                  instance,
    const VkDebugUtilsMessengerCreateInfoEXT*   pCreateInfo,
    const VkAllocationCallbacks*                pAllocator,
    VkDebugUtilsMessengerEXT*                   pMessenger);
```

- `instance` is the instance the messenger will be used with.
- `pCreateInfo` is a pointer to a VkDebugUtilsMessengerCreateInfoEXT structure containing the callback pointer, as well as defining conditions under which this messenger will trigger the callback.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.
- `pMessenger` is a pointer to a VkDebugUtilsMessengerEXT handle in which the created object is returned.

Valid Usage (Implicit)

- VUID-vkCreateDebugUtilsMessengerEXT-instance-parameter
  `instance` **must** be a valid VkInstance handle
- VUID-vkCreateDebugUtilsMessengerEXT-pCreateInfo-parameter
  `pCreateInfo` **must** be a valid pointer to a valid VkDebugUtilsMessengerCreateInfoEXT structure
- VUID-vkCreateDebugUtilsMessengerEXT-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkCreateDebugUtilsMessengerEXT-pMessenger-parameter
  `pMessenger` **must** be a valid pointer to a VkDebugUtilsMessengerEXT handle

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The application **must** ensure that vkCreateDebugUtilsMessengerEXT is not executed in parallel with any Vulkan command that is also called with `instance` or child of `instance` as the dispatchable argument.

The definition of `VkDebugUtilsMessengerCreateInfoEXT` is:

```
// Provided by VK_EXT_debug_utils
typedef struct VkDebugUtilsMessengerCreateInfoEXT {
    VkStructureType                         sType;
    const void*                             pNext;
    VkDebugUtilsMessengerCreateFlagsEXT     flags;
    VkDebugUtilsMessageSeverityFlagsEXT     messageSeverity;
    VkDebugUtilsMessageTypeFlagsEXT         messageType;
    PFN_vkDebugUtilsMessengerCallbackEXT    pfnUserCallback;
    void*                                   pUserData;
} VkDebugUtilsMessengerCreateInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is `0` and is reserved for future use.
- `messageSeverity` is a bitmask of VkDebugUtilsMessageSeverityFlagBitsEXT specifying which severity of event(s) will cause this callback to be called.
- `messageType` is a bitmask of VkDebugUtilsMessageTypeFlagBitsEXT specifying which type of event(s) will cause this callback to be called.
- `pfnUserCallback` is the application callback function to call.
- `pUserData` is `NULL` or an application-defined user data pointer to be passed to the callback.

For each `VkDebugUtilsMessengerEXT` that is created the `VkDebugUtilsMessengerCreateInfoEXT`::`messageSeverity` and `VkDebugUtilsMessengerCreateInfoEXT`::`messageType` determine when that `VkDebugUtilsMessengerCreateInfoEXT`::`pfnUserCallback` is called. The process to determine if the user's `pfnUserCallback` is triggered when an event occurs is as follows:

1. The implementation will perform a bitwise AND of the event's VkDebugUtilsMessageSeverityFlagBitsEXT with the `messageSeverity` provided during creation of

the VkDebugUtilsMessengerEXT object.

   a. If the value is 0, the message is skipped.

2. The implementation will perform a bitwise AND of the event's VkDebugUtilsMessageTypeFlagBitsEXT with the `messageType` provided during the creation of the VkDebugUtilsMessengerEXT object.

   a. If the value is 0, the message is skipped.

3. The callback will trigger a debug message for the current event.

The callback will come directly from the component that detected the event, unless some other layer intercepts the calls for its own purposes (filter them in a different way, log to a system error log, etc.).

An application **can** receive multiple callbacks if multiple `VkDebugUtilsMessengerEXT` objects are created. A callback will always be executed in the same thread as the originating Vulkan call.

A callback **can** be called from multiple threads simultaneously (if the application is making Vulkan calls from multiple threads).

Valid Usage (Implicit)

- VUID-VkDebugUtilsMessengerCreateInfoEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CREATE_INFO_EXT
- VUID-VkDebugUtilsMessengerCreateInfoEXT-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkDebugUtilsMessengerCreateInfoEXT-messageSeverity-parameter
  `messageSeverity` **must** be a valid combination of VkDebugUtilsMessageSeverityFlagBitsEXT values
- VUID-VkDebugUtilsMessengerCreateInfoEXT-messageType-parameter
  `messageType` **must** be a valid combination of VkDebugUtilsMessageTypeFlagBitsEXT values
- VUID-VkDebugUtilsMessengerCreateInfoEXT-pfnUserCallback-parameter
  `pfnUserCallback` **must** be a valid PFN_vkDebugUtilsMessengerCallbackEXT value

Structure Chaining

Extends the structure

- VkInstanceCreateInfo

```
// Provided by VK_EXT_debug_utils
typedef VkFlags VkDebugUtilsMessengerCreateFlagsEXT;
```

`VkDebugUtilsMessengerCreateFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

Bits which **can** be set in VkDebugUtilsMessengerCreateInfoEXT::messageSeverity, specifying event severities which cause a debug messenger to call the callback, are:

```
// Provided by VK_EXT_debug_utils
typedef enum VkDebugUtilsMessageSeverityFlagBitsEXT {
    VK_DEBUG_UTILS_MESSAGE_SEVERITY_VERBOSE_BIT_EXT = 0x00000001,
    VK_DEBUG_UTILS_MESSAGE_SEVERITY_INFO_BIT_EXT = 0x00000010,
    VK_DEBUG_UTILS_MESSAGE_SEVERITY_WARNING_BIT_EXT = 0x00000100,
    VK_DEBUG_UTILS_MESSAGE_SEVERITY_ERROR_BIT_EXT = 0x00001000,
} VkDebugUtilsMessageSeverityFlagBitsEXT;
```

- VK_DEBUG_UTILS_MESSAGE_SEVERITY_VERBOSE_BIT_EXT specifies the most verbose output indicating all diagnostic messages from the Vulkan loader, layers, and drivers should be captured.
- VK_DEBUG_UTILS_MESSAGE_SEVERITY_INFO_BIT_EXT specifies an informational message such as resource details that may be handy when debugging an application.
- VK_DEBUG_UTILS_MESSAGE_SEVERITY_WARNING_BIT_EXT specifies use of Vulkan that **may** expose an application bug. Such cases may not be immediately harmful, such as a fragment shader outputting to a location with no attachment. Other cases **may** point to behavior that is almost certainly bad when unintended such as using an image whose memory has not been filled. In general if you see a warning but you know that the behavior is intended/desired, then simply ignore the warning.
- VK_DEBUG_UTILS_MESSAGE_SEVERITY_ERROR_BIT_EXT specifies that the application has violated a valid usage condition of the specification.

NOTE

The values of VkDebugUtilsMessageSeverityFlagBitsEXT are sorted based on severity. The higher the flag value, the more severe the message. This allows for simple boolean operation comparisons when looking at VkDebugUtilsMessageSeverityFlagBitsEXT values.

For example:

```
    if (messageSeverity >=
VK_DEBUG_UTILS_MESSAGE_SEVERITY_WARNING_BIT_EXT) {
        // Do something for warnings and errors
    }
```

In addition, space has been left between the enums to allow for later addition of new severities in between the existing values.

```
// Provided by VK_EXT_debug_utils
typedef VkFlags VkDebugUtilsMessageSeverityFlagsEXT;
```

VkDebugUtilsMessageSeverityFlagsEXT is a bitmask type for setting a mask of zero or more VkDebugUtilsMessageSeverityFlagBitsEXT.

Bits which **can** be set in VkDebugUtilsMessengerCreateInfoEXT::messageType, specifying event types which cause a debug messenger to call the callback, are:

```
// Provided by VK_EXT_debug_utils
typedef enum VkDebugUtilsMessageTypeFlagBitsEXT {
    VK_DEBUG_UTILS_MESSAGE_TYPE_GENERAL_BIT_EXT = 0x00000001,
    VK_DEBUG_UTILS_MESSAGE_TYPE_VALIDATION_BIT_EXT = 0x00000002,
    VK_DEBUG_UTILS_MESSAGE_TYPE_PERFORMANCE_BIT_EXT = 0x00000004,
  // Provided by VK_EXT_device_address_binding_report
    VK_DEBUG_UTILS_MESSAGE_TYPE_DEVICE_ADDRESS_BINDING_BIT_EXT = 0x00000008,
} VkDebugUtilsMessageTypeFlagBitsEXT;
```

- VK_DEBUG_UTILS_MESSAGE_TYPE_GENERAL_BIT_EXT specifies that some general event has occurred. This is typically a non-specification, non-performance event.
- VK_DEBUG_UTILS_MESSAGE_TYPE_VALIDATION_BIT_EXT specifies that something has occurred during validation against the Vulkan specification that may indicate invalid behavior.
- VK_DEBUG_UTILS_MESSAGE_TYPE_PERFORMANCE_BIT_EXT specifies a potentially non-optimal use of Vulkan, e.g. using vkCmdClearColorImage when setting VkAttachmentDescription::loadOp to VK_ATTACHMENT_LOAD_OP_CLEAR would have worked.
- VK_DEBUG_UTILS_MESSAGE_TYPE_DEVICE_ADDRESS_BINDING_BIT_EXT specifies that the implementation has modified the set of GPU-visible virtual addresses associated with a Vulkan object.

```
// Provided by VK_EXT_debug_utils
typedef VkFlags VkDebugUtilsMessageTypeFlagsEXT;
```

VkDebugUtilsMessageTypeFlagsEXT is a bitmask type for setting a mask of zero or more VkDebugUtilsMessageTypeFlagBitsEXT.

The prototype for the VkDebugUtilsMessengerCreateInfoEXT::pfnUserCallback function implemented by the application is:

```
// Provided by VK_EXT_debug_utils
typedef VkBool32 (*PFN_vkDebugUtilsMessengerCallbackEXT)(
    VkDebugUtilsMessageSeverityFlagBitsEXT      messageSeverity,
    VkDebugUtilsMessageTypeFlagsEXT             messageTypes,
    const VkDebugUtilsMessengerCallbackDataEXT* pCallbackData,
    void*                                       pUserData);
```

- messageSeverity specifies the VkDebugUtilsMessageSeverityFlagBitsEXT that triggered this callback.
- messageTypes is a bitmask of VkDebugUtilsMessageTypeFlagBitsEXT specifying which type of event(s) triggered this callback.
- pCallbackData contains all the callback related data in the

VkDebugUtilsMessengerCallbackDataEXT structure.

- pUserData is the application-defined user data pointer, equal to the value of VkDebugUtilsMessengerCreateInfoEXT::pUserData specified when the VkDebugUtilsMessengerEXT object was created.

The callback returns a VkBool32, which is interpreted in a layer-specified manner. The application **should** always return VK_FALSE. The VK_TRUE value is reserved for use in layer development.

Valid Usage

- VUID-PFN_vkDebugUtilsMessengerCallbackEXT-None-04769
  The callback **must** not make calls to any Vulkan commands

The definition of VkDebugUtilsMessengerCallbackDataEXT is:

```
// Provided by VK_EXT_debug_utils
typedef struct VkDebugUtilsMessengerCallbackDataEXT {
    VkStructureType                                 sType;
    const void*                                     pNext;
    VkDebugUtilsMessengerCallbackDataFlagsEXT       flags;
    const char*                                     pMessageIdName;
    int32_t                                         messageIdNumber;
    const char*                                     pMessage;
    uint32_t                                        queueLabelCount;
    const VkDebugUtilsLabelEXT*                     pQueueLabels;
    uint32_t                                        cmdBufLabelCount;
    const VkDebugUtilsLabelEXT*                     pCmdBufLabels;
    uint32_t                                        objectCount;
    const VkDebugUtilsObjectNameInfoEXT*            pObjects;
} VkDebugUtilsMessengerCallbackDataEXT;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is 0 and is reserved for future use.
- pMessageIdName is NULL or a null-terminated UTF-8 string that identifies the particular message ID that is associated with the provided message. If the message corresponds to a validation layer message, then this string will be the VUID.
- messageIdNumber is the ID number of the triggering message. If the message corresponds to a validation layer message, then this number is an internal hash of the VUID.
- pMessage is NULL if messageTypes is equal to VK_DEBUG_UTILS_MESSAGE_TYPE_DEVICE_ADDRESS_BINDING_BIT_EXT, or a null-terminated UTF-8 string detailing the trigger conditions. If the message corresponds to a validation layer message, then this will contain the main message with the specification text and link.
- queueLabelCount is a count of items contained in the pQueueLabels array.

- `pQueueLabels` is `NULL` or a pointer to an array of VkDebugUtilsLabelEXT active in the current `VkQueue` at the time the callback was triggered. Refer to Queue Labels for more information.
- `cmdBufLabelCount` is a count of items contained in the `pCmdBufLabels` array.
- `pCmdBufLabels` is `NULL` or a pointer to an array of VkDebugUtilsLabelEXT active in the current `VkCommandBuffer` at the time the callback was triggered. Refer to Command Buffer Labels for more information.
- `objectCount` is a count of items contained in the `pObjects` array.
- `pObjects` is a pointer to an array of VkDebugUtilsObjectNameInfoEXT objects related to the detected issue. The array is roughly in order of importance, but the 0th element is always guaranteed to be the most important object for this message.

NOTE This structure should only be considered valid during the lifetime of the triggered callback.

Since adding queue and command buffer labels behaves like pushing and popping onto a stack, the order of both `pQueueLabels` and `pCmdBufLabels` is based on the order the labels were defined. The result is that the first label in either `pQueueLabels` or `pCmdBufLabels` will be the first defined (and therefore the oldest) while the last label in each list will be the most recent.

NOTE

`pQueueLabels` will only be non-`NULL` if one of the objects in `pObjects` can be related directly to a defined `VkQueue` which has had one or more labels associated with it.

Likewise, `pCmdBufLabels` will only be non-`NULL` if one of the objects in `pObjects` can be related directly to a defined `VkCommandBuffer` which has had one or more labels associated with it. Additionally, while command buffer labels allow for beginning and ending across different command buffers, the debug messaging framework **cannot** guarantee that labels in `pCmdBufLabels` will contain those defined outside of the associated command buffer. This is partially due to the fact that the association of one command buffer with another may not have been defined at the time the debug message is triggered.

Valid Usage (Implicit)

- VUID-VkDebugUtilsMessengerCallbackDataEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CALLBACK_DATA_EXT
- VUID-VkDebugUtilsMessengerCallbackDataEXT-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkDeviceAddressBindingCallbackDataEXT
- VUID-VkDebugUtilsMessengerCallbackDataEXT-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique
- VUID-VkDebugUtilsMessengerCallbackDataEXT-flags-zerobitmask
  `flags` **must** be `0`
- VUID-VkDebugUtilsMessengerCallbackDataEXT-pMessageIdName-parameter
  If `pMessageIdName` is not `NULL`, `pMessageIdName` **must** be a null-terminated UTF-8 string

- VUID-VkDebugUtilsMessengerCallbackDataEXT-pMessage-parameter
  If `pMessage` is not `NULL`, `pMessage` **must** be a null-terminated UTF-8 string
- VUID-VkDebugUtilsMessengerCallbackDataEXT-pQueueLabels-parameter
  If `queueLabelCount` is not `0`, `pQueueLabels` **must** be a valid pointer to an array of `queueLabelCount` valid VkDebugUtilsLabelEXT structures
- VUID-VkDebugUtilsMessengerCallbackDataEXT-pCmdBufLabels-parameter
  If `cmdBufLabelCount` is not `0`, `pCmdBufLabels` **must** be a valid pointer to an array of `cmdBufLabelCount` valid VkDebugUtilsLabelEXT structures
- VUID-VkDebugUtilsMessengerCallbackDataEXT-pObjects-parameter
  If `objectCount` is not `0`, `pObjects` **must** be a valid pointer to an array of `objectCount` valid VkDebugUtilsObjectNameInfoEXT structures

```
// Provided by VK_EXT_debug_utils
typedef VkFlags VkDebugUtilsMessengerCallbackDataFlagsEXT;
```

`VkDebugUtilsMessengerCallbackDataFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

The definition of `VkDeviceAddressBindingCallbackDataEXT` is:

```
// Provided by VK_EXT_device_address_binding_report
typedef struct VkDeviceAddressBindingCallbackDataEXT {
    VkStructureType                   sType;
    void*                             pNext;
    VkDeviceAddressBindingFlagsEXT    flags;
    VkDeviceAddress                   baseAddress;
    VkDeviceSize                      size;
    VkDeviceAddressBindingTypeEXT     bindingType;
} VkDeviceAddressBindingCallbackDataEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkDeviceAddressBindingFlagBitsEXT specifying additional information about the binding event that caused the callback to be called.
- `baseAddress` is a GPU-accessible virtual address identifying the start of a region of the virtual address space associated with a Vulkan object, as identified by the `pObjects` member of VkDebugUtilsMessengerCallbackDataEXT.
- `size` is the size in bytes of a region of GPU-accessible virtual address space.
- `bindingType` is a VkDeviceAddressBindingTypeEXT specifying the type of binding event that caused the callback to be called.

If the `reportAddressBinding` feature is enabled and the implementation binds or unbinds a region of virtual address space associated with a Vulkan object, the implementation **must** submit a debug

message with the following properties:

- `messageSeverity` equal to VK_DEBUG_UTILS_MESSAGE_SEVERITY_INFO_BIT_EXT
- `messageTypes` equal to VK_DEBUG_UTILS_MESSAGE_TYPE_DEVICE_ADDRESS_BINDING_BIT_EXT
- `VkDebugUtilsMessengerCallbackDataEXT`::`pObjects` **must** identify the associated Vulkan object
- `VkDeviceAddressBindingCallbackDataEXT` **must** be included in the `pNext` chain of `VkDebugUtilsMessengerCallbackDataEXT`

These debug messages **must** be emitted both for GPU virtual address space regions that are explicitly bound to a Vulkan object via the `vkBind`*Memory/`vkBind`*Memory2 functions, and for those that are implicitly generated via memory allocation or importing external memory.

An implementation **may** report binding events associated with a Vulkan object via `VkDebugUtilsMessengerEXT` prior to the object becoming visible to an application via other Vulkan commands. For example, object creation functions **may** report binding events that occur during an object's creation. In such cases, `VkDeviceAddressBindingCallbackDataEXT`::`flags` **must** include VK_DEVICE_ADDRESS_BINDING_INTERNAL_OBJECT_BIT_EXT.

Object handles reported in this manner are not valid object handles, and **must** not be used as an input parameter to any Vulkan command.

Any valid object handle returned by an object creation function **must** match the handle specified via any previously reported binding events associated with the object's creation.

Valid Usage (Implicit)

- VUID-VkDeviceAddressBindingCallbackDataEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_ADDRESS_BINDING_CALLBACK_DATA_EXT
- VUID-VkDeviceAddressBindingCallbackDataEXT-flags-parameter
  If `flags` is not `0`, `flags` **must** be a valid combination of VkDeviceAddressBindingFlagBitsEXT values
- VUID-VkDeviceAddressBindingCallbackDataEXT-baseAddress-parameter
  `baseAddress` **must** be a valid VkDeviceAddress value
- VUID-VkDeviceAddressBindingCallbackDataEXT-bindingType-parameter
  `bindingType` **must** be a valid VkDeviceAddressBindingTypeEXT value

Structure Chaining

Extends the structure

- VkDebugUtilsMessengerCallbackDataEXT

Bits which **can** be set in VkDeviceAddressBindingCallbackDataEXT::`flags` specifying additional information about a binding event are:

```
// Provided by VK_EXT_device_address_binding_report
typedef enum VkDeviceAddressBindingFlagBitsEXT {
    VK_DEVICE_ADDRESS_BINDING_INTERNAL_OBJECT_BIT_EXT = 0x00000001,
} VkDeviceAddressBindingFlagBitsEXT;
```

- VK_DEVICE_ADDRESS_BINDING_INTERNAL_OBJECT_BIT_EXT specifies that VkDeviceAddressBindingCallbackDataEXT describes a Vulkan object that has not been made visible to the application via a Vulkan command.

```
// Provided by VK_EXT_device_address_binding_report
typedef VkFlags VkDeviceAddressBindingFlagsEXT;
```

VkDeviceAddressBindingFlagsEXT is a bitmask type for setting a mask of zero or more VkDeviceAddressBindingFlagBitsEXT.

The VkDeviceAddressBindingTypeEXT enum is defined as:

```
// Provided by VK_EXT_device_address_binding_report
typedef enum VkDeviceAddressBindingTypeEXT {
    VK_DEVICE_ADDRESS_BINDING_TYPE_BIND_EXT = 0,
    VK_DEVICE_ADDRESS_BINDING_TYPE_UNBIND_EXT = 1,
} VkDeviceAddressBindingTypeEXT;
```

- VK_DEVICE_ADDRESS_BINDING_TYPE_BIND_EXT specifies that a new GPU-accessible virtual address range has been bound.
- VK_DEVICE_ADDRESS_BINDING_TYPE_UNBIND_EXT specifies that a GPU-accessible virtual address range has been unbound.

To intentionally submit a debug message, call:

```
// Provided by VK_EXT_debug_utils
void vkSubmitDebugUtilsMessageEXT(
    VkInstance                                  instance,
    VkDebugUtilsMessageSeverityFlagBitsEXT      messageSeverity,
    VkDebugUtilsMessageTypeFlagsEXT             messageTypes,
    const VkDebugUtilsMessengerCallbackDataEXT* pCallbackData);
```

- `instance` is the debug stream's VkInstance.
- `messageSeverity` is a VkDebugUtilsMessageSeverityFlagBitsEXT value specifying the severity of this event/message.
- `messageTypes` is a bitmask of VkDebugUtilsMessageTypeFlagBitsEXT specifying which type of event(s) to identify with this message.
- `pCallbackData` contains all the callback related data in the VkDebugUtilsMessengerCallbackDataEXT structure.

The call will propagate through the layers and generate callback(s) as indicated by the message's flags. The parameters are passed on to the callback in addition to the `pUserData` value that was defined at the time the messenger was registered.

Valid Usage

- VUID-vkSubmitDebugUtilsMessageEXT-objectType-02591
  The `objectType` member of each element of `pCallbackData->pObjects` **must** not be VK_OBJECT_TYPE_UNKNOWN

Valid Usage (Implicit)

- VUID-vkSubmitDebugUtilsMessageEXT-instance-parameter
  `instance` **must** be a valid VkInstance handle
- VUID-vkSubmitDebugUtilsMessageEXT-messageSeverity-parameter
  `messageSeverity` **must** be a valid VkDebugUtilsMessageSeverityFlagBitsEXT value
- VUID-vkSubmitDebugUtilsMessageEXT-messageTypes-parameter
  `messageTypes` **must** be a valid combination of VkDebugUtilsMessageTypeFlagBitsEXT values
- VUID-vkSubmitDebugUtilsMessageEXT-pCallbackData-parameter
  `pCallbackData` **must** be a valid pointer to a valid VkDebugUtilsMessengerCallbackDataEXT structure

To destroy a `VkDebugUtilsMessengerEXT` object, call:

```
// Provided by VK_EXT_debug_utils
void vkDestroyDebugUtilsMessengerEXT(
    VkInstance                                  instance,
    VkDebugUtilsMessengerEXT                    messenger,
    const VkAllocationCallbacks*                pAllocator);
```

- `instance` is the instance where the callback was created.
- `messenger` is the VkDebugUtilsMessengerEXT object to destroy. `messenger` is an externally synchronized object and **must** not be used on more than one thread at a time. This means that `vkDestroyDebugUtilsMessengerEXT` **must** not be called when a callback is active.
- `pAllocator` controls host memory allocation as described in the Memory Allocation chapter.

Valid Usage

- VUID-vkDestroyDebugUtilsMessengerEXT-messenger-01915
  If `VkAllocationCallbacks` were provided when `messenger` was created, a compatible set of callbacks **must** be provided here

- VUID-vkDestroyDebugUtilsMessengerEXT-messenger-01916
  If no `VkAllocationCallbacks` were provided when `messenger` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyDebugUtilsMessengerEXT-instance-parameter
  `instance` **must** be a valid VkInstance handle
- VUID-vkDestroyDebugUtilsMessengerEXT-messenger-parameter
  If `messenger` is not VK_NULL_HANDLE, `messenger` **must** be a valid VkDebugUtilsMessengerEXT handle
- VUID-vkDestroyDebugUtilsMessengerEXT-pAllocator-parameter
  If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid VkAllocationCallbacks structure
- VUID-vkDestroyDebugUtilsMessengerEXT-messenger-parent
  If `messenger` is a valid handle, it **must** have been created, allocated, or retrieved from `instance`

Host Synchronization

- Host access to `messenger` **must** be externally synchronized

The application **must** ensure that vkDestroyDebugUtilsMessengerEXT is not executed in parallel with any Vulkan command that is also called with `instance` or child of `instance` as the dispatchable argument.

51.2. Device Loss Debugging

51.2.1. Device Fault Diagnosis

To retrieve diagnostic information about faults call:

```
// Provided by VK_KHR_device_fault
VkResult vkGetDeviceFaultReportsKHR(
    VkDevice                                    device,
    uint64_t                                    timeout,
    uint32_t*                                   pFaultCounts,
    VkDeviceFaultInfoKHR*                       pFaultInfo);
```

- `device` is the logical device from which to query the diagnostic fault information.
- `timeout` is the timeout period in units of nanoseconds. `timeout` is adjusted to the closest value allowed by the implementation-dependent timeout accuracy, which **may** be substantially longer

than one nanosecond, and **may** be longer than the requested period.

- `pFaultCount` is a pointer to an integer related to the number of fault reports available or queried, as described below.
- `pFaultInfo` is `NULL` or a pointer to an array of VkDeviceFaultInfoKHR structures in which fault information is returned.

If at least one fault report is available when `vkGetDeviceFaultReportsKHR` is called, then `vkGetDeviceFaultReportsKHR` returns immediately. If no faults have occurred at the time `vkGetDeviceFaultReportsKHR` is called, then `vkGetDeviceFaultReportsKHR` will block and wait until a fault occurs and a report becomes available or until the `timeout` has expired, whichever is sooner.

If `timeout` is zero, then `vkGetDeviceFaultReportsKHR` does not wait, but simply returns the current fault count and reports. VK_TIMEOUT will be returned in this case if no faults have occurred, even though no actual wait was performed.

If a fault has occurred before the `timeout` has expired, `vkGetDeviceFaultReportsKHR` returns VK_SUCCESS. Otherwise, `vkGetDeviceFaultReportsKHR` returns VK_TIMEOUT after the `timeout` has expired.

If `pFaultInfo` is `NULL`, then the number of fault reports available is returned in `pFaultCount`. Otherwise, `pFaultCount` **must** point to a variable set by the application to the number of elements in the `pFaultInfo` array, and on return the variable is overwritten with the number of handles actually written to `pFaultInfo`. If `pFaultCount` is less than the number of fault reports available, at most `pFaultCount` structures will be written, and VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available fault reports were returned.

If more than one fault report is available, they will be returned in order of occurrence via `pFaultInfo`.

Each individual fault report is returned exactly once.

`vkGetDeviceFaultReportsKHR`() **may** be invoked in parallel on different threads, in which case each invocation for a given `device` will return a unique set of reports. No fault report is returned to more than one invocation.

Valid Usage (Implicit)

- VUID-vkGetDeviceFaultReportsKHR-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDeviceFaultReportsKHR-pFaultCounts-parameter
  `pFaultCounts` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetDeviceFaultReportsKHR-pFaultInfo-parameter
  If `pFaultInfo` is not `NULL`, `pFaultInfo` **must** be a valid pointer to an array of `pFaultCounts` VkDeviceFaultInfoKHR structures
- VUID-vkGetDeviceFaultReportsKHR-pFaultCounts-arraylength
  If `pFaultInfo` is not `NULL`, the value referenced by `pFaultCounts` **must** be greater than `0`

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS
- VK_TIMEOUT

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkDeviceFaultInfoKHR structure is defined as:

```
// Provided by VK_KHR_device_fault
typedef struct VkDeviceFaultInfoKHR {
    VkStructureType                 sType;
    void*                           pNext;
    VkDeviceFaultFlagsKHR           flags;
    uint64_t                        groupId;
    char                            description[VK_MAX_DESCRIPTION_SIZE];
    VkDeviceFaultAddressInfoKHR     faultAddressInfo;
    VkDeviceFaultAddressInfoKHR     instructionAddressInfo;
    VkDeviceFaultVendorInfoKHR      vendorInfo;
} VkDeviceFaultInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask of VkDeviceFaultFlagBitsKHR values giving information as to the type of fault and which additional fields have been populated by the driver provide further information.
- groupId is a monotonically increasing value for associating multiple entries with a single fault.
- description is an array of VK_MAX_DESCRIPTION_SIZE char containing a null-terminated UTF-8 string which is a human readable description of the fault.
- faultAddressInfo is a VkDeviceFaultAddressInfoKHR structure providing details of the memory access which caused a fault.
- instructionAddressInfo is a VkDeviceFaultAddressInfoKHR structure providing details of the GPU instruction which caused a fault.
- vendorInfo is a VkDeviceFaultVendorInfoKHR structure providing vendor specific fault information.

An implementation **should** populate as many members of VkDeviceFaultInfoKHR as possible, given the information available at the time of the fault and the constraints of the implementation itself.

A single error **may** generate multiple instances of VkDeviceFaultInfoKHR. For example, multiple page faults **may** be generated, with the faultAddressInfo member of each instance identifying a different memory address. In this case, all instances of VkDeviceFaultInfoKHR **should** share the same groupId where the implementation can reasonably ascertain this association.

Due to hardware limitations, faultAddressInfo and instructionAddressInfo describe ranges of GPU virtual address space, rather than precise addresses. The precise memory address accessed or the precise value of the instruction pointer **must** lie within the region described.

NOTE

Each VkDeviceFaultInfoKHR reported may (depending on the flags set) provide:

- A memory access which may have triggered a page fault and may have contributed to device loss
- The value of an active instruction pointer at the time a fault occurred. This value may be indicative of the active pipeline or shader at the time of device loss

Comparison of the GPU virtual addresses described by faultAddressInfo or instructionAddressInfo to GPU virtual address ranges reported by the VK_EXT_device_address_binding_report extension may allow applications to correlate between these addresses and Vulkan objects. Applications should be aware that these addresses may also correspond to resources internal to an implementation, which will not be reported via the VK_EXT_device_address_binding_report extension.

Valid Usage (Implicit)

- VUID-VkDeviceFaultInfoKHR-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DEVICE_FAULT_INFO_KHR
- VUID-VkDeviceFaultInfoKHR-pNext-pNext
  pNext **must** be NULL

```
// Provided by VK_KHR_device_fault
typedef VkFlags VkDeviceFaultFlagsKHR;
```

VkDeviceFaultFlagsKHR is a bitmask type for receiving a mask of zero or more VkDeviceFaultFlagBitsKHR.

Bits which **can** be set in VkDeviceFaultInfoKHR::flags providing information of the status of the fault reported and which additional fields have been populated by the driver, are:

```
// Provided by VK_KHR_device_fault
typedef enum VkDeviceFaultFlagBitsKHR {
    VK_DEVICE_FAULT_FLAG_DEVICE_LOST_KHR = 0x00000001,
    VK_DEVICE_FAULT_FLAG_MEMORY_ADDRESS_KHR = 0x00000002,
    VK_DEVICE_FAULT_FLAG_INSTRUCTION_ADDRESS_KHR = 0x00000004,
    VK_DEVICE_FAULT_FLAG_VENDOR_KHR = 0x00000008,
    VK_DEVICE_FAULT_FLAG_WATCHDOG_TIMEOUT_KHR = 0x00000010,
    VK_DEVICE_FAULT_FLAG_OVERFLOW_KHR = 0x00000020,
} VkDeviceFaultFlagBitsKHR;
```

- VK_DEVICE_FAULT_FLAG_DEVICE_LOST_KHR specifies that the fault has resulted in a device lost condition. No subsequent entries will be returned for this device.
- VK_DEVICE_FAULT_FLAG_MEMORY_ADDRESS_KHR specifies that the fault has associated memory access address information which is stored in the faultAddressInfo field (see VkDeviceFaultAddressInfoKHR).
- VK_DEVICE_FAULT_FLAG_INSTRUCTION_ADDRESS_KHR specifies that the fault has an associated instruction address which is stored in the instructionAddressInfo field (see VkDeviceFaultAddressInfoKHR).
- VK_DEVICE_FAULT_FLAG_VENDOR_KHR specifies that the fault has associated vendor information stored in the vendorInfo field (see VkDeviceFaultVendorInfoKHR).
- VK_DEVICE_FAULT_FLAG_WATCHDOG_TIMEOUT_KHR specifies that the fault was the result of a GPU timeout. Further information **may** be made available using other platform specific extensions via the pNext chain of the VkDeviceFaultInfoKHR structure.
- VK_DEVICE_FAULT_FLAG_OVERFLOW_KHR specifies that prior faults have occurred, but information about them is no longer available. This typically indicates that faults are occurring more rapidly than the calling application is able to read them back.

To retrieve diagnostic information about faults that **may** have caused device loss, call:

```
// Provided by VK_EXT_device_fault
VkResult vkGetDeviceFaultInfoEXT(
    VkDevice                                    device,
    VkDeviceFaultCountsEXT*                     pFaultCounts,
    VkDeviceFaultInfoEXT*                       pFaultInfo);
```

- device is the logical device from which to query the diagnostic fault information.
- pFaultCounts is a pointer to a VkDeviceFaultCountsEXT structure in which counts for structures describing additional fault information are returned.
- pFaultInfo is NULL or a pointer to a VkDeviceFaultInfoEXT structure in which fault information is returned.

If pFaultInfo is NULL, then the counts of corresponding additional fault information structures available are returned in the addressInfoCount and vendorInfoCount members of pFaultCounts. Additionally, the size of any vendor-specific binary crash dump is returned in the vendorBinarySize

member of pFaultCounts.

If pFaultInfo is not NULL, pFaultCounts **must** point to a VkDeviceFaultCountsEXT structure with each structure count or size member (addressInfoCount, vendorInfoCount, vendorBinarySize) set by the application to the number of elements in the corresponding output array member of pFaultInfo (pAddressInfos and pVendorInfos), or to the size of the output buffer in bytes (pVendorBinaryData). On return, each structure count member is overwritten with the number of structures actually written to the corresponding output array member of pFaultInfo. Similarly, vendorBinarySize is overwritten with the number of bytes actually written to the pVendorBinaryData member of pFaultInfo.

If the vendor-specific crash dumps feature is not enabled, then implementations **must** set pFaultCounts->vendorBinarySize to zero and **must** not modify pFaultInfo->pVendorBinaryData.

If any pFaultCounts structure count member is less than the number of corresponding fault properties available, at most structure count (addressInfoCount, vendorInfoCount) elements will be written to the associated pFaultInfo output array. Similarly, if vendorBinarySize is less than the size in bytes of the available crash dump data, at most vendorBinarySize elements will be written to pVendorBinaryData.

If pFaultInfo is NULL, then subsequent calls to vkGetDeviceFaultInfoEXT for the same device **must** return identical values in the addressInfoCount, vendorInfoCount and vendorBinarySize members of pFaultCounts.

If pFaultInfo is not NULL, then subsequent calls to vkGetDeviceFaultInfoEXT for the same device **must** return identical values in the output members of pFaultInfo (pAddressInfos, pVendorInfos, pVendorBinaryData), up to the limits described by the structure count and buffer size members of pFaultCounts (addressInfoCount, vendorInfoCount, vendorBinarySize). If the sizes of the output members of pFaultInfo increase for a subsequent call to vkGetDeviceFaultInfoEXT, then supplementary information **may** be returned in the additional available space.

If any pFaultCounts structure count member is smaller than the number of corresponding fault properties available, or if pFaultCounts->vendorBinarySize is smaller than the size in bytes of the generated binary crash dump data, VK_INCOMPLETE will be returned instead of VK_SUCCESS, to indicate that not all the available properties were returned.

If pFaultCounts->vendorBinarySize is less than what is necessary to store the binary crash dump header, nothing will be written to pFaultInfo->pVendorBinaryData and zero will be written to pFaultCounts->vendorBinarySize.

Valid Usage

- VUID-vkGetDeviceFaultInfoEXT-device-07336
  device **must** be in the *lost* state
- VUID-vkGetDeviceFaultInfoEXT-pFaultCounts-07337
  If the value referenced by pFaultCounts->addressInfoCount is not 0, and pFaultInfo->pAddressInfos is not NULL, pFaultInfo->pAddressInfos **must** be a valid pointer to an array of pFaultCounts->addressInfoCount VkDeviceFaultAddressInfoEXT structures
- VUID-vkGetDeviceFaultInfoEXT-pFaultCounts-07338

If the value referenced by `pFaultCounts->vendorInfoCount` is not `0`, and `pFaultInfo->pVendorInfos` is not `NULL`, `pFaultInfo->pVendorInfos` **must** be a valid pointer to an array of `pFaultCounts->vendorInfoCount` VkDeviceFaultVendorInfoEXT structures

- VUID-vkGetDeviceFaultInfoEXT-pFaultCounts-07339
  If the value referenced by `pFaultCounts->vendorBinarySize` is not `0`, and `pFaultInfo->pVendorBinaryData` is not `NULL`, `pFaultInfo->pVendorBinaryData` **must** be a valid pointer to an array of `pFaultCounts->vendorBinarySize` bytes

Valid Usage (Implicit)

- VUID-vkGetDeviceFaultInfoEXT-device-parameter
  `device` **must** be a valid VkDevice handle
- VUID-vkGetDeviceFaultInfoEXT-pFaultCounts-parameter
  `pFaultCounts` **must** be a valid pointer to a VkDeviceFaultCountsEXT structure
- VUID-vkGetDeviceFaultInfoEXT-pFaultInfo-parameter
  If `pFaultInfo` is not `NULL`, `pFaultInfo` **must** be a valid pointer to a VkDeviceFaultInfoEXT structure

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkDeviceFaultCountsEXT` structure is defined as:

```
// Provided by VK_EXT_device_fault
typedef struct VkDeviceFaultCountsEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           addressInfoCount;
    uint32_t           vendorInfoCount;
    VkDeviceSize       vendorBinarySize;
} VkDeviceFaultCountsEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `addressInfoCount` is the number of VkDeviceFaultAddressInfoEXT structures describing either memory accesses which **may** have caused a page fault, or the addresses of active instructions at the time of the fault.
- `vendorInfoCount` is the number of VkDeviceFaultVendorInfoEXT structures describing vendor-specific fault information.
- `vendorBinarySize` is the size in bytes of a vendor-specific binary crash dump, which may provide additional information when imported into external tools.

Valid Usage (Implicit)

- VUID-VkDeviceFaultCountsEXT-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_FAULT_COUNTS_EXT
- VUID-VkDeviceFaultCountsEXT-pNext-pNext
  `pNext` **must** be `NULL`

The `VkDeviceFaultInfoEXT` structure is defined as:

```
// Provided by VK_EXT_device_fault
typedef struct VkDeviceFaultInfoEXT {
    VkStructureType                 sType;
    void*                           pNext;
    char                            description[VK_MAX_DESCRIPTION_SIZE];
    VkDeviceFaultAddressInfoKHR*    pAddressInfos;
    VkDeviceFaultVendorInfoKHR*     pVendorInfos;
    void*                           pVendorBinaryData;
} VkDeviceFaultInfoEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `description` is an array of VK_MAX_DESCRIPTION_SIZE `char` containing a null-terminated UTF-8 string which is a human readable description of the fault.
- `pAddressInfos` is `NULL` or a pointer to an array of VkDeviceFaultAddressInfoEXT structures describing either memory accesses which **may** have caused a page fault, or describing active instruction pointers at the time of the fault. The length of `pAddressInfos` is specified by the VkDeviceFaultCountsEXT::`addressInfoCount` value passed to vkGetDeviceFaultInfoEXT. If not `NULL`, each element of `pAddressInfos` describes a bounded region of GPU virtual address space containing either the GPU virtual address accessed, or the value of an active instruction pointer.
- `pVendorInfos` is `NULL` or a pointer to an array of VkDeviceFaultVendorInfoEXT structures describing vendor-specific fault information. The length of `pVendorInfos` is specified by the VkDeviceFaultCountsEXT::`vendorInfoCount` value passed to vkGetDeviceFaultInfoEXT.
- `pVendorBinaryData` is `NULL` or a pointer to memory which will be populated with a vendor-specific binary crash dump, as described in Vendor Binary Crash Dumps. The size of `pVendorBinaryData` is specified by the VkDeviceFaultCountsEXT::`vendorBinarySize` value passed to

vkGetDeviceFaultInfoEXT.

An implementation **should** populate as many members of VkDeviceFaultInfoEXT as possible, given the information available at the time of the fault and the constraints of the implementation itself.

Due to hardware limitations, pAddressInfos describes ranges of GPU virtual address space, rather than precise addresses. The precise memory address accessed or the precise value of the instruction pointer **must** lie within the region described.

NOTE

Each element of pAddressInfos describes either:

- A memory access which may have triggered a page fault and may have contributed to device loss
- The value of an active instruction pointer at the time a fault occurred. This value may be indicative of the active pipeline or shader at the time of device loss

Comparison of the GPU virtual addresses described by pAddressInfos to GPU virtual address ranges reported by the VK_EXT_device_address_binding_report extension may allow applications to correlate between these addresses and Vulkan objects. Applications should be aware that these addresses may also correspond to resources internal to an implementation, which will not be reported via the VK_EXT_device_address_binding_report extension.

Valid Usage (Implicit)

- VUID-VkDeviceFaultInfoEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_DEVICE_FAULT_INFO_EXT
- VUID-VkDeviceFaultInfoEXT-pNext-pNext
  pNext **must** be NULL

The VkDeviceFaultAddressInfoKHR structure is defined as:

```
// Provided by VK_KHR_device_fault
typedef struct VkDeviceFaultAddressInfoKHR {
    VkDeviceFaultAddressTypeKHR    addressType;
    VkDeviceAddress                reportedAddress;
    VkDeviceSize                   addressPrecision;
} VkDeviceFaultAddressInfoKHR;
```

```
// Provided by VK_EXT_device_fault
// Equivalent to VkDeviceFaultAddressInfoKHR
typedef VkDeviceFaultAddressInfoKHR VkDeviceFaultAddressInfoEXT;
```

- addressType is either the type of memory operation that triggered a page fault, or the type of association between an instruction pointer and a fault.

- `reportedAddress` is the GPU virtual address recorded by the device.
- `addressPrecision` is a power of two value that specifies how precisely the device can report the address.

The combination of `reportedAddress` and `addressPrecision` allow the possible range of addresses to be calculated, such that:

```
lower_address = (pInfo->reportedAddress & ~(pInfo->addressPrecision-1))
upper_address = (pInfo->reportedAddress |  (pInfo->addressPrecision-1))
```

NOTE

It is valid for the `reportedAddress` to contain a more precise address than indicated by `addressPrecision`. In this case, the value of `reportedAddress` should be treated as an additional hint as to the value of the address that triggered the page fault, or to the value of an instruction pointer.

Possible values of VkDeviceFaultAddressInfoKHR::`addressType` are:

```
// Provided by VK_KHR_device_fault
typedef enum VkDeviceFaultAddressTypeKHR {
    VK_DEVICE_FAULT_ADDRESS_TYPE_NONE_KHR = 0,
    VK_DEVICE_FAULT_ADDRESS_TYPE_READ_INVALID_KHR = 1,
    VK_DEVICE_FAULT_ADDRESS_TYPE_WRITE_INVALID_KHR = 2,
    VK_DEVICE_FAULT_ADDRESS_TYPE_EXECUTE_INVALID_KHR = 3,
    VK_DEVICE_FAULT_ADDRESS_TYPE_INSTRUCTION_POINTER_UNKNOWN_KHR = 4,
    VK_DEVICE_FAULT_ADDRESS_TYPE_INSTRUCTION_POINTER_INVALID_KHR = 5,
    VK_DEVICE_FAULT_ADDRESS_TYPE_INSTRUCTION_POINTER_FAULT_KHR = 6,
  // Provided by VK_EXT_device_fault
    VK_DEVICE_FAULT_ADDRESS_TYPE_NONE_EXT = VK_DEVICE_FAULT_ADDRESS_TYPE_NONE_KHR,
  // Provided by VK_EXT_device_fault
    VK_DEVICE_FAULT_ADDRESS_TYPE_READ_INVALID_EXT =
VK_DEVICE_FAULT_ADDRESS_TYPE_READ_INVALID_KHR,
  // Provided by VK_EXT_device_fault
    VK_DEVICE_FAULT_ADDRESS_TYPE_WRITE_INVALID_EXT =
VK_DEVICE_FAULT_ADDRESS_TYPE_WRITE_INVALID_KHR,
  // Provided by VK_EXT_device_fault
    VK_DEVICE_FAULT_ADDRESS_TYPE_EXECUTE_INVALID_EXT =
VK_DEVICE_FAULT_ADDRESS_TYPE_EXECUTE_INVALID_KHR,
  // Provided by VK_EXT_device_fault
    VK_DEVICE_FAULT_ADDRESS_TYPE_INSTRUCTION_POINTER_UNKNOWN_EXT =
VK_DEVICE_FAULT_ADDRESS_TYPE_INSTRUCTION_POINTER_UNKNOWN_KHR,
  // Provided by VK_EXT_device_fault
    VK_DEVICE_FAULT_ADDRESS_TYPE_INSTRUCTION_POINTER_INVALID_EXT =
VK_DEVICE_FAULT_ADDRESS_TYPE_INSTRUCTION_POINTER_INVALID_KHR,
  // Provided by VK_EXT_device_fault
    VK_DEVICE_FAULT_ADDRESS_TYPE_INSTRUCTION_POINTER_FAULT_EXT =
VK_DEVICE_FAULT_ADDRESS_TYPE_INSTRUCTION_POINTER_FAULT_KHR,
} VkDeviceFaultAddressTypeKHR;
```

```
// Provided by VK_EXT_device_fault
// Equivalent to VkDeviceFaultAddressTypeKHR
typedef VkDeviceFaultAddressTypeKHR VkDeviceFaultAddressTypeEXT;
```

- VK_DEVICE_FAULT_ADDRESS_TYPE_NONE_KHR specifies that VkDeviceFaultAddressInfoKHR does not describe a page fault, or an instruction address.
- VK_DEVICE_FAULT_ADDRESS_TYPE_READ_INVALID_KHR specifies that VkDeviceFaultAddressInfoKHR describes a page fault triggered by an invalid read operation.
- VK_DEVICE_FAULT_ADDRESS_TYPE_WRITE_INVALID_KHR specifies that VkDeviceFaultAddressInfoKHR describes a page fault triggered by an invalid write operation.
- VK_DEVICE_FAULT_ADDRESS_TYPE_EXECUTE_INVALID_KHR specifies that VkDeviceFaultAddressInfoKHR describes a page fault triggered by an attempt to execute non-executable memory.
- VK_DEVICE_FAULT_ADDRESS_TYPE_INSTRUCTION_POINTER_UNKNOWN_KHR specifies an instruction pointer value at the time the fault occurred. This may or may not be related to a fault.
- VK_DEVICE_FAULT_ADDRESS_TYPE_INSTRUCTION_POINTER_INVALID_KHR specifies an instruction pointer value associated with an invalid instruction fault.
- VK_DEVICE_FAULT_ADDRESS_TYPE_INSTRUCTION_POINTER_FAULT_KHR specifies an instruction pointer value associated with a fault.

NOTE The instruction pointer values recorded may not identify the specific instruction(s) that triggered the fault. The relationship between the instruction pointer reported and triggering instruction will be vendor-specific.

The VkDeviceFaultVendorInfoKHR structure is defined as:

```
// Provided by VK_KHR_device_fault
typedef struct VkDeviceFaultVendorInfoKHR {
    char        description[VK_MAX_DESCRIPTION_SIZE];
    uint64_t    vendorFaultCode;
    uint64_t    vendorFaultData;
} VkDeviceFaultVendorInfoKHR;
```

```
// Provided by VK_EXT_device_fault
// Equivalent to VkDeviceFaultVendorInfoKHR
typedef VkDeviceFaultVendorInfoKHR VkDeviceFaultVendorInfoEXT;
```

- description is an array of VK_MAX_DESCRIPTION_SIZE char containing a null-terminated UTF-8 string which is a human readable description of the fault.
- vendorFaultCode is the vendor-specific fault code for this fault.
- vendorFaultData is the vendor-specific fault data associated with this fault.

Vendor Binary Crash Dumps

Applications **can** retrieve extended vendor-specific binary crash dump data for later analysis using external tools.

When VK_KHR_device_fault is enabled, crash dump data can be retrieved by calling vkGetDeviceFaultDebugInfoKHR.

When VK_EXT_device_fault is enabled, crash dump data can be retrieved by calling vkGetDeviceFaultInfoEXT.

To retrieve a vendor-specific binary crash dump call:

```
// Provided by VK_KHR_device_fault
VkResult vkGetDeviceFaultDebugInfoKHR(
    VkDevice                                    device,
    VkDeviceFaultDebugInfoKHR*                  pDebugInfo);
```

- `device` is the logical device from which to query the crash dump.
- `pDebugInfo` is a pointer to a `VkDeviceFaultDebugInfoKHR` structure which will be populated with the vendor binary data crash dump binary, as described below.

If `pDebugInfo->pVendorBinaryData` is `NULL`, then the size of any vendor-specific binary crash dump is returned in `pDebugInfo->vendorBinarySize`.

If the `pDebugInfo->pVendorBinaryData` is not `NULL` then it **must** point to a buffer of size `pDebugInfo->vendorBinarySize` bytes.

On return, `pDebugInfo->vendorBinarySize` will be overwritten with the number of bytes actually written to the `pDebugInfo->pVendorBinaryData`.

If `pDebugInfo->pVendorBinaryData` is not `NULL`, then subsequent calls to vkGetDeviceFaultDebugInfoKHR for the same `device` **must** return identical binary values in the `pDebugInfo->pVendorBinaryData` buffer up to the limit defined by `pDebugInfo->vendorBinarySize`.

If the vendor-specific crash dumps feature is not enabled, then implementations **must** set `pDebugInfo`->vendorBinarySize to zero and **must** not modify `pDebugInfo`->pVendorBinaryData.

Valid Usage

- VUID-vkGetDeviceFaultDebugInfoKHR-device-12383
  `device` **must** be lost

Valid Usage (Implicit)

- VUID-vkGetDeviceFaultDebugInfoKHR-device-parameter
  `device` **must** be a valid VkDevice handle

- VUID-vkGetDeviceFaultDebugInfoKHR-pDebugInfo-parameter
  `pDebugInfo` **must** be a valid pointer to a VkDeviceFaultDebugInfoKHR structure

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_NOT_ENOUGH_SPACE_KHR
- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The `VkDeviceFaultDebugInfoKHR` structure is defined as:

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `vendorBinarySize` is the size in bytes of a vendor-specific binary crash dump, which **may** provide additional information when imported into external tools.
- `pVendorBinaryData` is `NULL` or a pointer to `vendorBinarySize` number of bytes of data, which will be populated with a vendor-specific binary crash dump, as described in Vendor Binary Crash Dumps.

```
// Provided by VK_KHR_device_fault
typedef struct VkDeviceFaultDebugInfoKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           vendorBinarySize;
    void*              pVendorBinaryData;
} VkDeviceFaultDebugInfoKHR;
```

Valid Usage (Implicit)

- VUID-VkDeviceFaultDebugInfoKHR-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_DEVICE_FAULT_DEBUG_INFO_KHR
- VUID-VkDeviceFaultDebugInfoKHR-pNext-pNext
  `pNext` **must** be `NULL` or a pointer to a valid instance of VkDeviceFaultShaderAbortMessageInfoKHR
- VUID-VkDeviceFaultDebugInfoKHR-sType-unique
  The `sType` value of each structure in the `pNext` chain **must** be unique

The format of this crash dump data **may** depend on the vendor ID, device ID, driver version, and other details of the device. To enable external applications to identify the original device from which a crash dump was generated, the initial bytes written to VkDeviceFaultInfoEXT::pVendorBinaryData, or to VkDeviceFaultDebugInfoKHR::pVendorBinaryData **must** begin with a valid crash dump header.

Version one of the crash dump header is defined as:

```
// Provided by VK_KHR_device_fault
typedef struct VkDeviceFaultVendorBinaryHeaderVersionOneKHR {
    uint32_t                                    headerSize;
    VkDeviceFaultVendorBinaryHeaderVersionKHR    headerVersion;
    uint32_t                                    vendorID;
    uint32_t                                    deviceID;
    uint32_t                                    driverVersion;
    uint8_t                                     pipelineCacheUUID[VK_UUID_SIZE];
    uint32_t                                    applicationNameOffset;
    uint32_t                                    applicationVersion;
    uint32_t                                    engineNameOffset;
    uint32_t                                    engineVersion;
    uint32_t                                    apiVersion;
} VkDeviceFaultVendorBinaryHeaderVersionOneKHR;
```

```
// Provided by VK_EXT_device_fault
// Equivalent to VkDeviceFaultVendorBinaryHeaderVersionOneKHR
typedef VkDeviceFaultVendorBinaryHeaderVersionOneKHR
VkDeviceFaultVendorBinaryHeaderVersionOneEXT;
```

- headerSize is the length in bytes of the crash dump header.
- headerVersion is a VkDeviceFaultVendorBinaryHeaderVersionKHR enum value specifying the version of the header. A consumer of the crash dump **should** use the header version to interpret the remainder of the header. headerVersion **must** be written as exactly 4 bytes.
- vendorID is the VkPhysicalDeviceProperties::vendorID of the implementation.
- deviceID is the VkPhysicalDeviceProperties::deviceID of the implementation.
- driverVersion is the VkPhysicalDeviceProperties::driverVersion of the implementation.
- pipelineCacheUUID is an array of VK_UUID_SIZE uint8_t values matching the VkPhysicalDeviceProperties::pipelineCacheUUID property of the implementation.
- applicationNameOffset is zero, or an offset from the base address of the crash dump header to a null-terminated UTF-8 string containing the name of the application. If applicationNameOffset is non-zero, this string **must** match the application name specified via VkApplicationInfo::pApplicationName during instance creation.
- applicationVersion **must** be zero or the value specified by VkApplicationInfo::applicationVersion during instance creation.
- engineNameOffset is zero, or an offset from the base address of the crash dump header to a null-

terminated UTF-8 string containing the name of the engine (if any) used to create the application. If `engineNameOffset` is non-zero, this string **must** match the engine name specified via VkApplicationInfo::`pEngineName` during instance creation.

- `engineVersion` **must** be zero or the value specified by VkApplicationInfo::`engineVersion` during instance creation.
- `apiVersion` **must** be zero or the value specified by VkApplicationInfo::`apiVersion` during instance creation.

Unlike most structures declared by the Vulkan API, all fields of this structure are written with the least significant byte first, regardless of host byte-order.

The C language specification does not define the packing of structure members. This layout assumes tight structure member packing, with members laid out in the order listed in the structure, and the intended size of the structure is 56 bytes. If a compiler produces code that diverges from that pattern, applications **must** employ another method to set values at the correct offsets.

Valid Usage

- VUID-VkDeviceFaultVendorBinaryHeaderVersionOneEXT-headerSize-07340
  `headerSize` **must** be 56
- VUID-VkDeviceFaultVendorBinaryHeaderVersionOneEXT-headerVersion-07341
  `headerVersion` **must** be VK_DEVICE_FAULT_VENDOR_BINARY_HEADER_VERSION_ONE_KHR

Valid Usage (Implicit)

- VUID-VkDeviceFaultVendorBinaryHeaderVersionOneKHR-headerVersion-parameter
  `headerVersion` **must** be a valid VkDeviceFaultVendorBinaryHeaderVersionKHR value

Possible values of the `headerVersion` value of the crash dump header are:

```
// Provided by VK_KHR_device_fault
typedef enum VkDeviceFaultVendorBinaryHeaderVersionKHR {
    VK_DEVICE_FAULT_VENDOR_BINARY_HEADER_VERSION_ONE_KHR = 1,
    VK_DEVICE_FAULT_VENDOR_BINARY_HEADER_VERSION_ONE_EXT =
VK_DEVICE_FAULT_VENDOR_BINARY_HEADER_VERSION_ONE_KHR,
} VkDeviceFaultVendorBinaryHeaderVersionKHR;
```

```
// Provided by VK_EXT_device_fault
// Equivalent to VkDeviceFaultVendorBinaryHeaderVersionKHR
typedef VkDeviceFaultVendorBinaryHeaderVersionKHR
VkDeviceFaultVendorBinaryHeaderVersionEXT;
```

- VK_DEVICE_FAULT_VENDOR_BINARY_HEADER_VERSION_ONE_KHR specifies version one of the

binary crash dump header.

Shader Abort Messages

Applications making use of the OpAbortKHR instruction in their shaders **can** retrieve the messages passed through those instructions with the following structures:

The VkDeviceFaultShaderAbortMessageInfoKHR structure is defined as:

```
// Provided by VK_KHR_shader_abort
typedef struct VkDeviceFaultShaderAbortMessageInfoKHR {
    VkStructureType    sType;
    void*              pNext;
    uint64_t           messageDataSize;
    void*              pMessageData;
} VkDeviceFaultShaderAbortMessageInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- messageDataSize is the size of pMessageData in bytes. If pMessageData is NULL, this value is populated by the implementation.
- pMessageData is NULL or a pointer to messageDataSize bytes of data, which will be populated with data for messages reported via OpAbortKHR. If pMessageData is NULL then messageDataSize will be populated with the required size.

This structure **can** be included in the pNext chain of VkDeviceFaultDebugInfoKHR to retrieve messages returned by OpAbortKHR instructions.

pMessageData is populated with a series of (size,payload) pairs, each aligned to 8-byte boundaries. The size in each pair is a 64-bit integer indicating the number of bytes in the subsequent payload. The data in payload is laid out in the exact manner specified in the OpAbortKHR instruction by the message type, with no modifications. If multiple messages are present, the next message size will always be at the following 8-byte aligned offset after the payload ends.

Implementations **must** report the message reported by the first OpAbortKHR instruction executed in an invocation for this device. They **may** report additional messages if other invocations continued to execute after the abort instruction was executed.

NOTE Even though shading languages may provide definitions for printf-style abort commands, no formatting is performed by the Vulkan implementation. Applications should consult documentation for the shader language they are using on how abort messages are packed, so that they can unpack them after they are queried.

Valid Usage (Implicit)

- VUID-VkDeviceFaultShaderAbortMessageInfoKHR-sType-sType
  sType **must** be

VK_STRUCTURE_TYPE_DEVICE_FAULT_SHADER_ABORT_MESSAGE_INFO_KHR

Structure Chaining

Extends the structure

- VkDeviceFaultDebugInfoKHR

51.3. Active Tooling Information

Information about tools providing debugging, profiling, or similar services, active for a given physical device, can be obtained by calling:

```
// Provided by VK_VERSION_1_3
VkResult vkGetPhysicalDeviceToolProperties(
    VkPhysicalDevice                            physicalDevice,
    uint32_t*                                   pToolCount,
    VkPhysicalDeviceToolProperties*             pToolProperties);
```

- `physicalDevice` is the handle to the physical device to query for active tools.
- `pToolCount` is a pointer to an integer describing the number of tools active on `physicalDevice`.
- `pToolProperties` is either `NULL` or a pointer to an array of VkPhysicalDeviceToolProperties structures.

If `pToolProperties` is `NULL`, then the number of tools currently active on `physicalDevice` is returned in `pToolCount`. Otherwise, `pToolCount` **must** point to a variable set by the application to the number of elements in the `pToolProperties` array, and on return the variable is overwritten with the number of structures actually written to `pToolProperties`. If `pToolCount` is less than the number of currently active tools, at most `pToolCount` structures will be written.

The count and properties of active tools **may** change in response to events outside the scope of the specification. An application **should** assume these properties might change at any given time.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceToolProperties-physicalDevice-parameter
  `physicalDevice` **must** be a valid VkPhysicalDevice handle
- VUID-vkGetPhysicalDeviceToolProperties-pToolCount-parameter
  `pToolCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceToolProperties-pToolProperties-parameter
  If the value referenced by `pToolCount` is not `0`, and `pToolProperties` is not `NULL`, `pToolProperties` **must** be a valid pointer to an array of `pToolCount` VkPhysicalDeviceToolProperties structures

Return Codes

Success

- VK_INCOMPLETE
- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_UNKNOWN
- VK_ERROR_VALIDATION_FAILED

The VkPhysicalDeviceToolProperties structure is defined as:

```
// Provided by VK_VERSION_1_3
typedef struct VkPhysicalDeviceToolProperties {
    VkStructureType       sType;
    void*                 pNext;
    char                  name[VK_MAX_EXTENSION_NAME_SIZE];
    char                  version[VK_MAX_EXTENSION_NAME_SIZE];
    VkToolPurposeFlags    purposes;
    char                  description[VK_MAX_DESCRIPTION_SIZE];
    char                  layer[VK_MAX_EXTENSION_NAME_SIZE];
} VkPhysicalDeviceToolProperties;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `name` is a null-terminated UTF-8 string containing the name of the tool.
- `version` is a null-terminated UTF-8 string containing the version of the tool.
- `purposes` is a bitmask of VkToolPurposeFlagBits which is populated with purposes supported by the tool.
- `description` is a null-terminated UTF-8 string containing a description of the tool.
- `layer` is a null-terminated UTF-8 string containing the name of the layer implementing the tool, if the tool is implemented in a layer - otherwise it **may** be an empty string.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceToolProperties-sType-sType
  `sType` **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TOOL_PROPERTIES
- VUID-VkPhysicalDeviceToolProperties-pNext-pNext
  `pNext` **must** be `NULL`

Bits which **can** be set in VkPhysicalDeviceToolProperties::`purposes`, specifying the purposes of an

active tool, are:

```
// Provided by VK_VERSION_1_3
typedef enum VkToolPurposeFlagBits {
    VK_TOOL_PURPOSE_VALIDATION_BIT = 0x00000001,
    VK_TOOL_PURPOSE_PROFILING_BIT = 0x00000002,
    VK_TOOL_PURPOSE_TRACING_BIT = 0x00000004,
    VK_TOOL_PURPOSE_ADDITIONAL_FEATURES_BIT = 0x00000008,
    VK_TOOL_PURPOSE_MODIFYING_FEATURES_BIT = 0x00000010,
} VkToolPurposeFlagBits;
```

- VK_TOOL_PURPOSE_VALIDATION_BIT specifies that the tool provides validation of API usage.
- VK_TOOL_PURPOSE_PROFILING_BIT specifies that the tool provides profiling of API usage.
- VK_TOOL_PURPOSE_TRACING_BIT specifies that the tool is capturing data about the application's API usage, including anything from simple logging to capturing data for later replay.
- VK_TOOL_PURPOSE_ADDITIONAL_FEATURES_BIT specifies that the tool provides additional API features/extensions on top of the underlying implementation.
- VK_TOOL_PURPOSE_MODIFYING_FEATURES_BIT specifies that the tool modifies the API features/limits/extensions presented to the application.
- VK_TOOL_PURPOSE_DEBUG_REPORTING_BIT_EXT specifies that the tool reports additional information to the application via callbacks specified by vkCreateDebugUtilsMessengerEXT
- VK_TOOL_PURPOSE_DEBUG_MARKERS_BIT_EXT specifies that the tool consumes object debug annotation, queue labels, or command buffer labels

```
// Provided by VK_VERSION_1_3
typedef VkFlags VkToolPurposeFlags;
```

VkToolPurposeFlags is a bitmask type for setting a mask of zero or more VkToolPurposeFlagBits.

51.4. Frame Boundary

The VkFrameBoundaryEXT structure is defined as:

```
// Provided by VK_EXT_frame_boundary
typedef struct VkFrameBoundaryEXT {
    VkStructureType            sType;
    const void*                pNext;
    VkFrameBoundaryFlagsEXT    flags;
    uint64_t                   frameID;
    uint32_t                   imageCount;
    const VkImage*             pImages;
    uint32_t                   bufferCount;
    const VkBuffer*            pBuffers;
    uint64_t                   tagName;
    size_t                     tagSize;
    const void*                pTag;
} VkFrameBoundaryEXT;
```

- `sType` is a VkStructureType value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of VkFrameBoundaryFlagBitsEXT that can flag the last submission of a frame identifier.
- `frameID` is the frame identifier.
- `imageCount` is the number of images that store frame results.
- `pImages` is a pointer to an array of VkImage objects with imageCount entries.
- `bufferCount` is the number of buffers that store the frame results.
- `pBuffers` is a pointer to an array of VkBuffer objects with bufferCount entries.
- `tagName` is a numerical identifier for tag data.
- `tagSize` is the number of bytes of tag data.
- `pTag` is a pointer to an array of `tagSize` bytes containing tag data.

The application **can** associate frame boundary information to a queue submission call by adding a `VkFrameBoundaryEXT` structure to the `pNext` chain of queue submission, VkPresentInfoKHR, or VkBindSparseInfo.

NOTE

The frame identifier is used to associate one or more queue submissions to a frame. It is meant to be unique within a frame lifetime, i.e. it is possible (though not recommended) to reuse frame identifiers, as long as any two frames that may have overlapping queue submissions (as in the example above) use different frame identifiers.

Since the concept of frame is application-dependent, there is no way to validate the use of frame identifier. It is good practice to use a monotonically increasing counter as the frame identifier and not reuse identifiers between frames.

The `pImages` and `pBuffers` arrays contain a list of images and buffers which store the “end result” of the frame. As the concept of frame is application-dependent, not all frames **may** produce their

results in images or buffers, yet this is a sufficiently common case to be handled by VkFrameBoundaryEXT. Note that no extra information, such as image layout is being provided, since the images are meant to be used by tools which would already be tracking this required information. Having the possibility of passing a list of end-result images makes VkFrameBoundaryEXT as expressive as vkQueuePresentKHR, which is often the default frame boundary delimiter.

The application **can** also associate arbitrary extra information via tag data using tagName, tagSize and pTag. This extra information is typically tool-specific.

Valid Usage (Implicit)

- VUID-VkFrameBoundaryEXT-sType-sType
  sType **must** be VK_STRUCTURE_TYPE_FRAME_BOUNDARY_EXT
- VUID-VkFrameBoundaryEXT-flags-parameter
  If flags is not 0, flags **must** be a valid combination of VkFrameBoundaryFlagBitsEXT values
- VUID-VkFrameBoundaryEXT-pImages-parameter
  If imageCount is not 0, and pImages is not NULL, pImages **must** be a valid pointer to an array of imageCount valid VkImage handles
- VUID-VkFrameBoundaryEXT-pBuffers-parameter
  If bufferCount is not 0, and pBuffers is not NULL, pBuffers **must** be a valid pointer to an array of bufferCount valid VkBuffer handles
- VUID-VkFrameBoundaryEXT-pTag-parameter
  If tagSize is not 0, and pTag is not NULL, pTag **must** be a valid pointer to an array of tagSize bytes
- VUID-VkFrameBoundaryEXT-commonparent
  Both of the elements of pBuffers, and the elements of pImages that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

Structure Chaining

Extends the structures

- VkBindSparseInfo
- VkPresentInfoKHR
- VkSubmitInfo
- VkSubmitInfo2

The bit which **can** be set in VkFrameBoundaryEXT::flags is:

```
// Provided by VK_EXT_frame_boundary
typedef enum VkFrameBoundaryFlagBitsEXT {
    VK_FRAME_BOUNDARY_FRAME_END_BIT_EXT = 0x00000001,
} VkFrameBoundaryFlagBitsEXT;
```

- VK_FRAME_BOUNDARY_FRAME_END_BIT_EXT specifies that this queue submission is the last one for this frame, i.e. once this queue submission has terminated, then the work for this frame is completed.

Note that in the presence of timeline semaphores, the last queue submission might not be the last one to be submitted, as timeline semaphores allow for wait-before-signal submissions. In the context of frame boundary, the queue submission that should be flagged as the last one is the one that is meant to be executed last, even if it **may** not be the last one to be submitted.

```
// Provided by VK_EXT_frame_boundary
typedef VkFlags VkFrameBoundaryFlagsEXT;
```

VkFrameBoundaryFlagsEXT is a bitmask type for setting a mask of zero or more VkFrameBoundaryFlagBitsEXT.

Appendix A: Vulkan Environment for SPIR-V

Shaders for Vulkan are defined by the Khronos SPIR-V Specification as well as the Khronos SPIR-V Extended Instructions for GLSL Specification. This appendix defines additional SPIR-V requirements applying to Vulkan shaders.

Versions and Formats

A Vulkan 1.4 implementation **must** support the 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, and 1.6 versions of SPIR-V and the 1.0 version of the SPIR-V Extended Instructions for GLSL.

A SPIR-V module passed into vkCreateShaderModule is interpreted as a series of 32-bit words in host endianness, with literal strings packed as described in section 2.2 of the SPIR-V Specification. The first few words of the SPIR-V module **must** be a magic number and a SPIR-V version number, as described in section 2.3 of the SPIR-V Specification.

Capabilities

The table below lists the set of SPIR-V capabilities that **may** be supported in Vulkan implementations. The application **must** not use any of these capabilities in SPIR-V passed to vkCreateShaderModule unless one of the following conditions is met for the VkDevice specified in the `device` parameter of vkCreateShaderModule:

- The corresponding field in the table is blank.
- Any corresponding Vulkan feature is enabled.
- Any corresponding Vulkan extension is enabled.
- Any corresponding Vulkan property is supported.
- The corresponding core version is supported (as returned by VkPhysicalDeviceProperties ::`apiVersion`).

Table 99. List of SPIR-V Capabilities and Corresponding Vulkan Features, Extensions, or Core Version

SPIR-V `OpCapability` **Vulkan feature, extension, or core version**
`Matrix` VK_VERSION_1_0
`Shader` VK_VERSION_1_0
`InputAttachment` VK_VERSION_1_0
`Sampled1D` VK_VERSION_1_0
`Image1D` VK_VERSION_1_0

SPIR-V `OpCapability` **Vulkan feature, extension, or core version**
`SampledBuffer` VK_VERSION_1_0
`ImageBuffer` VK_VERSION_1_0
`ImageQuery` VK_VERSION_1_0
`DerivativeControl` VK_VERSION_1_0
`Geometry` `VkPhysicalDeviceFeatures::geometryShader`
`Tessellation` `VkPhysicalDeviceFeatures::tessellationShader`
`Float64` `VkPhysicalDeviceFeatures::shaderFloat64`
`Int64` `VkPhysicalDeviceFeatures::shaderInt64`

`Int64Atomics`
`VkPhysicalDeviceVulkan12Features::shaderBufferInt64Atomics`
`VkPhysicalDeviceVulkan12Features::shaderSharedInt64Atomics`
`VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT::shaderImageInt64Atomics`

`AtomicFloat16AddEXT`
`VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderBufferFloat16AtomicAdd`
`VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderSharedFloat16AtomicAdd`

`AtomicFloat32AddEXT`
`VkPhysicalDeviceShaderAtomicFloatFeaturesEXT::shaderBufferFloat32AtomicAdd`
`VkPhysicalDeviceShaderAtomicFloatFeaturesEXT::shaderSharedFloat32AtomicAdd`
`VkPhysicalDeviceShaderAtomicFloatFeaturesEXT::shaderImageFloat32AtomicAdd`

`AtomicFloat64AddEXT`
`VkPhysicalDeviceShaderAtomicFloatFeaturesEXT::shaderBufferFloat64AtomicAdd`
`VkPhysicalDeviceShaderAtomicFloatFeaturesEXT::shaderSharedFloat64AtomicAdd`

`AtomicFloat16MinMaxEXT`
`VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderBufferFloat16AtomicMinMax`
`VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderSharedFloat16AtomicMinMax`

`AtomicFloat32MinMaxEXT`
`VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderBufferFloat32AtomicMinMax`
`VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderSharedFloat32AtomicMinMax`
`VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderImageFloat32AtomicMinMax`

SPIR-V `OpCapability` **Vulkan feature, extension, or core version**

`AtomicFloat64MinMaxEXT`
`VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderBufferFloat64AtomicMinMax`
`VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderSharedFloat64AtomicMinMax`

`Int64ImageEXT` `VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT::shaderImageInt64Atomics`
`Int16` `VkPhysicalDeviceFeatures::shaderInt16`
`TessellationPointSize` `VkPhysicalDeviceFeatures::shaderTessellationAndGeometryPointSize`
`GeometryPointSize` `VkPhysicalDeviceFeatures::shaderTessellationAndGeometryPointSize`
`ImageGatherExtended` `VkPhysicalDeviceFeatures::shaderImageGatherExtended`
`StorageImageMultisample` `VkPhysicalDeviceFeatures::shaderStorageImageMultisample`
`UniformBufferArrayDynamicIndexing` `VkPhysicalDeviceFeatures::shaderUniformBufferArrayDynamicIndexing`
`SampledImageArrayDynamicIndexing` `VkPhysicalDeviceFeatures::shaderSampledImageArrayDynamicIndexing`
`StorageBufferArrayDynamicIndexing` `VkPhysicalDeviceFeatures::shaderStorageBufferArrayDynamicIndexing`
`StorageImageArrayDynamicIndexing` `VkPhysicalDeviceFeatures::shaderStorageImageArrayDynamicIndexing`
`ClipDistance` `VkPhysicalDeviceFeatures::shaderClipDistance`
`CullDistance` `VkPhysicalDeviceFeatures::shaderCullDistance`
`ImageCubeArray` `VkPhysicalDeviceFeatures::imageCubeArray`
`SampleRateShading` `VkPhysicalDeviceFeatures::sampleRateShading`
`SparseResidency` `VkPhysicalDeviceFeatures::shaderResourceResidency`
`MinLod` `VkPhysicalDeviceFeatures::shaderResourceMinLod`
`SampledCubeArray` `VkPhysicalDeviceFeatures::imageCubeArray`

SPIR-V `OpCapability` **Vulkan feature, extension, or core version**
`ImageMSArray` `VkPhysicalDeviceFeatures::shaderStorageImageMultisample`
`StorageImageExtendedFormats` `VK_VERSION_1_0`
`InterpolationFunction` `VkPhysicalDeviceFeatures::sampleRateShading`

`StorageImageReadWithoutFormat`
`VkPhysicalDeviceFeatures::shaderStorageImageReadWithoutFormat`
`VK_VERSION_1_3`
`VK_KHR_format_feature_flags2`

`StorageImageWriteWithoutFormat`
`VkPhysicalDeviceFeatures::shaderStorageImageWriteWithoutFormat`
`VK_VERSION_1_3`
`VK_KHR_format_feature_flags2`

`MultiViewport` `VkPhysicalDeviceFeatures::multiViewport`

`DrawParameters`
`VkPhysicalDeviceVulkan11Features::shaderDrawParameters`
`VkPhysicalDeviceShaderDrawParametersFeatures::shaderDrawParameters`
`VK_KHR_shader_draw_parameters`

`MultiView`
`VkPhysicalDeviceVulkan11Features::multiview`
`VkPhysicalDeviceMultiviewFeatures::multiview`

`DeviceGroup`
`VK_VERSION_1_1`
`VK_KHR_device_group`

`VariablePointersStorageBuffer`
`VkPhysicalDeviceVulkan11Features::variablePointersStorageBuffer`
`VkPhysicalDeviceVariablePointersFeatures::variablePointersStorageBuffer`

`VariablePointers`
`VkPhysicalDeviceVulkan11Features::variablePointers`
`VkPhysicalDeviceVariablePointersFeatures::variablePointers`

`ShaderClockKHR` `VK_KHR_shader_clock`
`StencilExportEXT` `VK_EXT_shader_stencil_export`
`ShaderViewportIndex` `VkPhysicalDeviceVulkan12Features::shaderOutputViewportIndex`

SPIR-V `OpCapability` **Vulkan feature, extension, or core version**
`ShaderLayer` `VkPhysicalDeviceVulkan12Features::shaderOutputLayer`

`StorageBuffer16BitAccess`
`VkPhysicalDeviceVulkan11Features::storageBuffer16BitAccess`
`VkPhysicalDevice16BitStorageFeatures::storageBuffer16BitAccess`

`UniformAndStorageBuffer16BitAccess`
`VkPhysicalDeviceVulkan11Features::uniformAndStorageBuffer16BitAccess`
`VkPhysicalDevice16BitStorageFeatures::uniformAndStorageBuffer16BitAccess`

`StoragePushConstant16`
`VkPhysicalDeviceVulkan11Features::storagePushConstant16`
`VkPhysicalDevice16BitStorageFeatures::storagePushConstant16`

`StorageInputOutput16`
`VkPhysicalDeviceVulkan11Features::storageInputOutput16`
`VkPhysicalDevice16BitStorageFeatures::storageInputOutput16`

`GroupNonUniform` `VK_SUBGROUP_FEATURE_BASIC_BIT`
`GroupNonUniformVote` `VK_SUBGROUP_FEATURE_VOTE_BIT`
`GroupNonUniformArithmetic` `VK_SUBGROUP_FEATURE_ARITHMETIC_BIT`
`GroupNonUniformBallot` `VK_SUBGROUP_FEATURE_BALLOT_BIT`
`GroupNonUniformShuffle` `VK_SUBGROUP_FEATURE_SHUFFLE_BIT`
`GroupNonUniformShuffleRelative` `VK_SUBGROUP_FEATURE_SHUFFLE_RELATIVE_BIT`
`GroupNonUniformClustered` `VK_SUBGROUP_FEATURE_CLUSTERED_BIT`
`GroupNonUniformQuad` `VK_SUBGROUP_FEATURE_QUAD_BIT`
`GroupNonUniformPartitionedEXT` `VkPhysicalDeviceShaderSubgroupPartitionedFeaturesEXT::shaderSubgroupPartitioned`
`SampleMaskPostDepthCoverage` `VK_EXT_post_depth_coverage`

`ShaderNonUniform`
`VK_VERSION_1_2`
`VK_EXT_descriptor_indexing`

`RuntimeDescriptorArray` `VkPhysicalDeviceVulkan12Features::runtimeDescriptorArray`

SPIR-V OpCapability **Vulkan feature, extension, or core version**
InputAttachmentArrayDynamicIndexing VkPhysicalDeviceVulkan12Features::shaderInputAttachmentArrayDynamicIndexing
UniformTexelBufferArrayDynamicIndexing VkPhysicalDeviceVulkan12Features::shaderUniformTexelBufferArrayDynamicIndexing
StorageTexelBufferArrayDynamicIndexing VkPhysicalDeviceVulkan12Features::shaderStorageTexelBufferArrayDynamicIndexing
UniformBufferArrayNonUniformIndexing VkPhysicalDeviceVulkan12Features::shaderUniformBufferArrayNonUniformIndexing
SampledImageArrayNonUniformIndexing VkPhysicalDeviceVulkan12Features::shaderSampledImageArrayNonUniformIndexing
StorageBufferArrayNonUniformIndexing VkPhysicalDeviceVulkan12Features::shaderStorageBufferArrayNonUniformIndexing
StorageImageArrayNonUniformIndexing VkPhysicalDeviceVulkan12Features::shaderStorageImageArrayNonUniformIndexing
InputAttachmentArrayNonUniformIndexing VkPhysicalDeviceVulkan12Features::shaderInputAttachmentArrayNonUniformIndexing
UniformTexelBufferArrayNonUniformIndexing VkPhysicalDeviceVulkan12Features::shaderUniformTexelBufferArrayNonUniformIndexing
StorageTexelBufferArrayNonUniformIndexing VkPhysicalDeviceVulkan12Features::shaderStorageTexelBufferArrayNonUniformIndexing
FragmentFullyCoveredEXT VK_EXT_conservative_rasterization
Float16 VkPhysicalDeviceVulkan12Features::shaderFloat16
Int8 VkPhysicalDeviceVulkan12Features::shaderInt8
StorageBuffer8BitAccess VkPhysicalDeviceVulkan12Features::storageBuffer8BitAccess
UniformAndStorageBuffer8BitAccess VkPhysicalDeviceVulkan12Features::uniformAndStorageBuffer8BitAccess
StoragePushConstant8 VkPhysicalDeviceVulkan12Features::storagePushConstant8
VulkanMemoryModel VkPhysicalDeviceVulkan12Features::vulkanMemoryModel
VulkanMemoryModelDeviceScope VkPhysicalDeviceVulkan12Features::vulkanMemoryModelDeviceScope

SPIR-V `OpCapability` **Vulkan feature, extension, or core version**

`DenormPreserve`
`VkPhysicalDeviceVulkan12Properties::shaderDenormPreserveFloat16`
`VkPhysicalDeviceVulkan12Properties::shaderDenormPreserveFloat32`
`VkPhysicalDeviceVulkan12Properties::shaderDenormPreserveFloat64`

`DenormFlushToZero`
`VkPhysicalDeviceVulkan12Properties::shaderDenormFlushToZeroFloat16`
`VkPhysicalDeviceVulkan12Properties::shaderDenormFlushToZeroFloat32`
`VkPhysicalDeviceVulkan12Properties::shaderDenormFlushToZeroFloat64`

`SignedZeroInfNanPreserve`
`VkPhysicalDeviceVulkan12Properties::shaderSignedZeroInfNanPreserveFloat16`
`VkPhysicalDeviceVulkan12Properties::shaderSignedZeroInfNanPreserveFloat32`
`VkPhysicalDeviceVulkan12Properties::shaderSignedZeroInfNanPreserveFloat64`

`RoundingModeRTE`
`VkPhysicalDeviceVulkan12Properties::shaderRoundingModeRTEFloat16`
`VkPhysicalDeviceVulkan12Properties::shaderRoundingModeRTEFloat32`
`VkPhysicalDeviceVulkan12Properties::shaderRoundingModeRTEFloat64`

`RoundingModeRTZ`
`VkPhysicalDeviceVulkan12Properties::shaderRoundingModeRTZFloat16`
`VkPhysicalDeviceVulkan12Properties::shaderRoundingModeRTZFloat32`
`VkPhysicalDeviceVulkan12Properties::shaderRoundingModeRTZFloat64`

`ComputeDerivativeGroupQuadsKHR` `VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR::computeDerivativeGroupQuads`
`ComputeDerivativeGroupLinearKHR` `VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR::computeDerivativeGroupLinear`
`RayTracingKHR` `VkPhysicalDeviceRayTracingPipelineFeaturesKHR::rayTracingPipeline`
`RayQueryKHR` `VkPhysicalDeviceRayQueryFeaturesKHR::rayQuery`

`RayTraversalPrimitiveCullingKHR`
`VkPhysicalDeviceRayTracingPipelineFeaturesKHR::rayTraversalPrimitiveCulling`
`VkPhysicalDeviceRayQueryFeaturesKHR::rayQuery`

`RayCullMaskKHR` `VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR::rayTracingMaintenance1`
`TransformFeedback` `VkPhysicalDeviceTransformFeedbackFeaturesEXT::transformFeedback`
`GeometryStreams` `VkPhysicalDeviceTransformFeedbackFeaturesEXT::geometryStreams`
`FragmentDensityEXT` `VkPhysicalDeviceFragmentDensityMapFeaturesEXT::fragmentDensityMap`

SPIR-V OpCapability **Vulkan feature, extension, or core version**
PhysicalStorageBufferAddresses VkPhysicalDeviceVulkan12Features::bufferDeviceAddress
FragmentShaderSampleInterlockEXT VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT::fragmentShaderSampleInterlock
FragmentShaderPixelInterlockEXT VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT::fragmentShaderPixelInterlock
FragmentShaderShadingRateInterlockEXT VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT::fragmentShaderShadingRateInterlock
DemoteToHelperInvocation VkPhysicalDeviceVulkan13Features::shaderDemoteToHelperInvocation
FragmentShadingRateKHR VkPhysicalDeviceFragmentShadingRateFeaturesKHR::pipelineFragmentShadingRate VkPhysicalDeviceFragmentShadingRateFeaturesKHR::primitiveFragmentShadingRate VkPhysicalDeviceFragmentShadingRateFeaturesKHR::attachmentFragmentShadingRate
WorkgroupMemoryExplicitLayoutKHR VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR ::workgroupMemoryExplicitLayout
WorkgroupMemoryExplicitLayout8BitAccessKHR VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR ::workgroupMemoryExplicitLayout8BitAccess
WorkgroupMemoryExplicitLayout16BitAccessKHR VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR ::workgroupMemoryExplicitLayout16BitAccess
DotProductInputAll VkPhysicalDeviceVulkan13Features::shaderIntegerDotProduct VkPhysicalDeviceShaderIntegerDotProductFeaturesKHR::shaderIntegerDotProduct
DotProductInput4x8Bit VkPhysicalDeviceVulkan13Features::shaderIntegerDotProduct VkPhysicalDeviceShaderIntegerDotProductFeaturesKHR::shaderIntegerDotProduct
DotProductInput4x8BitPacked VkPhysicalDeviceVulkan13Features::shaderIntegerDotProduct VkPhysicalDeviceShaderIntegerDotProductFeaturesKHR::shaderIntegerDotProduct
DotProduct VkPhysicalDeviceVulkan13Features::shaderIntegerDotProduct VkPhysicalDeviceShaderIntegerDotProductFeaturesKHR::shaderIntegerDotProduct
FragmentBarycentricKHR VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR::fragmentShaderBarycentric
MeshShadingEXT VK_EXT_mesh_shader

SPIR-V OpCapability **Vulkan feature, extension, or core version**
RayTracingOpacityMicromapKHR VK_EXT_opacity_micromap VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap
RayTracingOpacityMicromapExecutionModeKHR VkPhysicalDeviceOpacityMicromapFeaturesKHR::micromap
RayTracingPositionFetchKHR VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR::rayTracingPositionFetch
RayQueryPositionFetchKHR VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR::rayTracingPositionFetch
TileImageColorReadAccessEXT VkPhysicalDeviceShaderTileImageFeaturesEXT::shaderTileImageColorReadAccess
TileImageDepthReadAccessEXT VkPhysicalDeviceShaderTileImageFeaturesEXT::shaderTileImageDepthReadAccess
TileImageStencilReadAccessEXT VkPhysicalDeviceShaderTileImageFeaturesEXT::shaderTileImageStencilReadAccess
CooperativeMatrixKHR VkPhysicalDeviceCooperativeMatrixFeaturesKHR::cooperativeMatrix
GroupNonUniformRotateKHR VkPhysicalDeviceVulkan14Features::shaderSubgroupRotate VkPhysicalDeviceShaderSubgroupRotateFeatures::shaderSubgroupRotate
ExpectAssumeKHR VkPhysicalDeviceVulkan14Features::shaderExpectAssume VkPhysicalDeviceShaderExpectAssumeFeatures::shaderExpectAssume
FloatControls2 VkPhysicalDeviceVulkan14Features::shaderFloatControls2 VkPhysicalDeviceShaderFloatControls2Features::shaderFloatControls2
QuadControlKHR VkPhysicalDeviceShaderQuadControlFeaturesKHR::shaderQuadControl
BFloat16TypeKHR VkPhysicalDeviceShaderBfloat16FeaturesKHR::shaderBFloat16Type
BFloat16DotProductKHR VkPhysicalDeviceShaderBfloat16FeaturesKHR::shaderBFloat16DotProduct
BFloat16CooperativeMatrixKHR VkPhysicalDeviceShaderBfloat16FeaturesKHR::shaderBFloat16CooperativeMatrix
ReplicatedCompositesEXT VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT::shaderReplicatedComposites
ShaderInvocationReorderEXT VK_EXT_ray_tracing_invocation_reorder

SPIR-V OpCapability **Vulkan feature, extension, or core version**
SplitBarrierEXT VkPhysicalDeviceShaderSplitBarrierFeaturesEXT::shaderSplitBarrier
Float8EXT VkPhysicalDeviceShaderFloat8FeaturesEXT::shaderFloat8
Float8CooperativeMatrixEXT VkPhysicalDeviceShaderFloat8FeaturesEXT::shaderFloat8CooperativeMatrix
UntypedPointersKHR VkPhysicalDeviceShaderUntypedPointersFeaturesKHR::shaderUntypedPointers
FMAKHR VkPhysicalDeviceShaderFmaFeaturesKHR::shaderFmaFloat16 VkPhysicalDeviceShaderFmaFeaturesKHR::shaderFmaFloat32 VkPhysicalDeviceShaderFmaFeaturesKHR::shaderFmaFloat64
Shader64BitIndexingEXT VkPhysicalDeviceShader64BitIndexingFeaturesEXT::shader64BitIndexing
LongVectorEXT VkPhysicalDeviceShaderLongVectorFeaturesEXT::longVector
DescriptorHeapEXT VkPhysicalDeviceDescriptorHeapFeaturesEXT::descriptorHeap
ConstantDataKHR VkPhysicalDeviceShaderConstantDataFeaturesKHR::shaderConstantData
AbortKHR VkPhysicalDeviceShaderAbortFeaturesKHR::shaderAbort
Float4EXT VkPhysicalDeviceShaderOCPMicroscalingTypesFeaturesEXT::shaderFloat4
Float6EXT VkPhysicalDeviceShaderOCPMicroscalingTypesFeaturesEXT::shaderFloat6
Float8UnsignedE8M0EXT VkPhysicalDeviceShaderOCPMicroscalingTypesFeaturesEXT::shaderFloat8UnsignedE8M0
MXInt8EXT VkPhysicalDeviceShaderOCPMicroscalingTypesFeaturesEXT::shaderMXInt8
BitcastExtractEXT VkPhysicalDeviceShaderOCPMicroscalingTypesFeaturesEXT::shaderFloat4 VkPhysicalDeviceShaderOCPMicroscalingTypesFeaturesEXT::shaderFloat6
CooperativeMatrixReductionsEXT VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT::cooperativeMatrixReductions
CooperativeMatrixConversionsEXT VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT::cooperativeMatrixConversions

SPIR-V `OpCapability` **Vulkan feature, extension, or core version**

`CooperativeMatrixPerElementOperationsEXT`
`VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT`
`::cooperativeMatrixPerElementOperations`

`CooperativeMatrixGetCoordinateEXT`
`VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT`
`::cooperativeMatrixGetCoordinate`

The application **must** not pass a SPIR-V module containing any of the following to vkCreateShaderModule:

- any `OpCapability` not listed above,
- an unsupported capability, or
- a capability which corresponds to a Vulkan feature or extension which has not been enabled.

SPIR-V Extensions

The following table lists SPIR-V extensions that implementations **may** support. The application **must** not pass a SPIR-V module to vkCreateShaderModule that uses the following SPIR-V extensions unless one of the following conditions is met for the VkDevice specified in the `device` parameter of vkCreateShaderModule:

- Any corresponding Vulkan extension is enabled.
- The corresponding core version is supported (as returned by VkPhysicalDeviceProperties `::apiVersion`).

Table 100. List of SPIR-V Extensions and Corresponding Vulkan Extensions or Core Version

SPIR-V `OpExtension` **Vulkan extension or core version**

`SPV_KHR_variable_pointers`
`VK_VERSION_1_1`
`VK_KHR_variable_pointers`

`SPV_KHR_shader_draw_parameters`
`VK_VERSION_1_1`
`VK_KHR_shader_draw_parameters`

`SPV_KHR_8bit_storage`
`VK_VERSION_1_2`
`VK_KHR_8bit_storage`

`SPV_KHR_16bit_storage`
`VK_VERSION_1_1`
`VK_KHR_16bit_storage`

`SPV_KHR_shader_clock` `VK_KHR_shader_clock`

SPIR-V `OpExtension` **Vulkan extension or core version**

`SPV_KHR_float_controls`
`VK_VERSION_1_2`
`VK_KHR_shader_float_controls`

`SPV_KHR_storage_buffer_storage_class`
`VK_VERSION_1_1`
`VK_KHR_storage_buffer_storage_class`

`SPV_KHR_post_depth_coverage` `VK_EXT_post_depth_coverage`
`SPV_EXT_shader_stencil_export` `VK_EXT_shader_stencil_export`
`SPV_NV_shader_subgroup_partitioned` `VK_EXT_shader_subgroup_partitioned`
`SPV_EXT_shader_subgroup_partitioned` `VK_EXT_shader_subgroup_partitioned`
`SPV_EXT_shader_viewport_index_layer` `VK_VERSION_1_2`

`SPV_EXT_descriptor_indexing`
`VK_VERSION_1_2`
`VK_EXT_descriptor_indexing`

`SPV_KHR_vulkan_memory_model`
`VK_VERSION_1_2`
`VK_KHR_vulkan_memory_model`

`SPV_KHR_ray_tracing` `VK_KHR_ray_tracing_pipeline`
`SPV_KHR_ray_query` `VK_KHR_ray_query`
`SPV_KHR_ray_cull_mask` `VK_KHR_ray_tracing_maintenance1`
`SPV_EXT_fragment_invocation_density` `VK_EXT_fragment_density_map`

`SPV_KHR_physical_storage_buffer`
`VK_VERSION_1_2`
`VK_KHR_buffer_device_address`

`SPV_EXT_fragment_shader_interlock` `VK_EXT_fragment_shader_interlock`
`SPV_EXT_demote_to_helper_invocation` `VK_VERSION_1_3`
`SPV_KHR_fragment_shading_rate` `VK_KHR_fragment_shading_rate`

SPIR-V `OpExtension` **Vulkan extension or core version**

`SPV_KHR_non_semantic_info`
`VK_VERSION_1_3`
`VK_KHR_shader_non_semantic_info`

`SPV_EXT_shader_image_int64` `VK_EXT_shader_image_atomic_int64`

`SPV_KHR_terminate_invocation`
`VK_VERSION_1_3`
`VK_KHR_shader_terminate_invocation`

`SPV_KHR_multiview`
`VK_VERSION_1_1`
`VK_KHR_multiview`

`SPV_KHR_workgroup_memory_explicit_layout` `VK_KHR_workgroup_memory_explicit_layout`
`SPV_EXT_shader_atomic_float_add` `VK_EXT_shader_atomic_float`
`SPV_KHR_fragment_shader_barycentric` `VK_KHR_fragment_shader_barycentric`
`SPV_KHR_subgroup_uniform_control_flow` `VK_KHR_shader_subgroup_uniform_control_flow`
`SPV_EXT_shader_atomic_float_min_max` `VK_EXT_shader_atomic_float2`
`SPV_EXT_shader_atomic_float16_add` `VK_EXT_shader_atomic_float2`
`SPV_EXT_fragment_fully_covered` `VK_EXT_conservative_rasterization`

`SPV_KHR_integer_dot_product`
`VK_VERSION_1_3`
`VK_KHR_shader_integer_dot_product`

`SPV_KHR_device_group`
`VK_VERSION_1_1`
`VK_KHR_device_group`

`SPV_EXT_mesh_shader` `VK_EXT_mesh_shader`
`SPV_KHR_ray_tracing_position_fetch` `VK_KHR_ray_tracing_position_fetch`
`SPV_EXT_shader_tile_image` `VK_EXT_shader_tile_image`

SPIR-V `OpExtension` **Vulkan extension or core version**

`SPV_EXT_opacity_micromap`
`VK_EXT_opacity_micromap`
`VK_KHR_opacity_micromap`

`SPV_KHR_opacity_micromap` `VK_KHR_opacity_micromap`
`SPV_EXT_split_barrier` `VK_EXT_shader_split_barrier`
`SPV_KHR_cooperative_matrix` `VK_KHR_cooperative_matrix`
`SPV_KHR_maximal_reconvergence` `VK_KHR_shader_maximal_reconvergence`

`SPV_KHR_subgroup_rotate`
`VK_VERSION_1_4`
`VK_KHR_shader_subgroup_rotate`

`SPV_KHR_expect_assume`
`VK_VERSION_1_4`
`VK_KHR_shader_expect_assume`

`SPV_KHR_float_controls2`
`VK_VERSION_1_4`
`VK_KHR_shader_float_controls2`

`SPV_KHR_fma` `VK_KHR_shader_fma`
`SPV_KHR_quad_control` `VK_KHR_shader_quad_control`
`SPV_KHR_bfloat16` `VK_KHR_shader_bfloat16`
`SPV_KHR_compute_shader_derivatives` `VK_KHR_compute_shader_derivatives`
`SPV_EXT_replicated_composites` `VK_EXT_shader_replicated_composites`
`SPV_KHR_relaxed_extended_instruction` `VK_KHR_shader_relaxed_extended_instruction`
`SPV_EXT_shader_invocation_reorder` `VK_EXT_ray_tracing_invocation_reorder`
`SPV_EXT_float8` `VK_EXT_shader_float8`
`SPV_KHR_untyped_pointers` `VK_KHR_shader_untyped_pointers`

SPIR-V OpExtension Vulkan extension or core version
SPV_EXT_shader_64bit_indexing VK_EXT_shader_64bit_indexing
SPV_EXT_long_vector VK_EXT_shader_long_vector
SPV_EXT_descriptor_heap VK_EXT_descriptor_heap
SPV_KHR_abort VK_KHR_shader_abort
SPV_KHR_constant_data VK_KHR_shader_constant_data
SPV_EXT_ocp_microscaling_types VK_EXT_shader_ocp_microscaling_types
SPV_EXT_cooperative_matrix_maintenance1 VK_EXT_cooperative_matrix_maintenance1

Validation Rules Within a Module

A SPIR-V module passed to vkCreateShaderModule **must** conform to the following rules:

Standalone SPIR-V Validation

The following rules **can** be validated with only the SPIR-V module itself. They do not depend on knowledge of the implementation and its capabilities or knowledge of runtime information, such as enabled features.

Valid Usage

- VUID-StandaloneSpirv-None-04633
  Every entry point **must** have no return value and accept no arguments
- VUID-StandaloneSpirv-None-04634
  The static function-call graph for an entry point **must** not contain cycles; that is, static recursion is not allowed
- VUID-StandaloneSpirv-None-04635
  The Logical or PhysicalStorageBuffer64 addressing model **must** be selected
- VUID-StandaloneSpirv-None-04636
  Scope for execution **must** be limited to Workgroup or Subgroup
- VUID-StandaloneSpirv-Scope-12243
  The Scope operand of OpTypeCooperativeMatrixKHR **must** be limited to Workgroup or Subgroup
- VUID-StandaloneSpirv-None-04637
  If the Scope for execution is Workgroup, then it **must** only be used in the task, mesh,

tessellation control, or compute Execution Model

- VUID-StandaloneSpirv-None-04638
  Scope for memory **must** be limited to Device, QueueFamily, Workgroup, ShaderCallKHR, Subgroup, or Invocation
- VUID-StandaloneSpirv-ExecutionModel-07320
  If the Execution Model is TessellationControl, and the MemoryModel is GLSL450, the Scope for memory **must** not be Workgroup
- VUID-StandaloneSpirv-None-07321
  If the Scope for memory is Workgroup, then it **must** only be used in the task, mesh, tessellation control, or compute Execution Model
- VUID-StandaloneSpirv-None-04640
  If the Scope for memory is ShaderCallKHR, then it **must** only be used in ray generation, intersection, closest hit, any-hit, miss, and callable Execution Model
- VUID-StandaloneSpirv-None-04641
  If the Scope for memory is Invocation, then MemorySemantics **must** use Relaxed memory order
- VUID-StandaloneSpirv-None-04642
  Scope for group operations **must** be limited to Subgroup
- VUID-StandaloneSpirv-SubgroupVoteKHR-07951
  If none of the SubgroupVoteKHR, GroupNonUniform, or SubgroupBallotKHR capabilities are declared, Scope for memory **must** not be Subgroup
- VUID-StandaloneSpirv-None-04643
  Storage Class **must** be limited to UniformConstant, Input, Uniform, Output, Workgroup, Private, Function, PushConstant, Image, StorageBuffer, RayPayloadKHR, IncomingRayPayloadKHR, HitAttributeKHR, CallableDataKHR, IncomingCallableDataKHR, ShaderRecordBufferKHR, PhysicalStorageBuffer, or TileImageEXT
- VUID-StandaloneSpirv-None-04644
  If the Storage Class is Output, then it **must** not be used in the GlCompute, RayGenerationKHR, IntersectionKHR, AnyHitKHR, ClosestHitKHR, MissKHR, or CallableKHR Execution Model
- VUID-StandaloneSpirv-None-04645
  If the Storage Class is Workgroup, then it **must** only be used in the task, mesh, or compute Execution Model
- VUID-StandaloneSpirv-None-08720
  If the Storage Class is TileImageEXT, then it **must** only be used in the fragment execution model
- VUID-StandaloneSpirv-MemorySemantics-10865
  MemorySemantics **must** have at most one non-relaxed memory order bit set (Acquire, Release, or AcquireRelease)
- VUID-StandaloneSpirv-MemorySemantics-10866
  MemorySemantics with SequentiallyConsistent memory order **must** not be used in the Vulkan API
- VUID-StandaloneSpirv-MemorySemantics-10867

MemorySemantics **must** not use Acquire or AcquireRelease memory order with OpAtomicStore

- VUID-StandaloneSpirv-MemorySemantics-10868
  MemorySemantics **must** not use Release or AcquireRelease memory order with OpAtomicLoad
- VUID-StandaloneSpirv-MemorySemantics-10869
  MemorySemantics **must** not use Relaxed memory order with OpMemoryBarrier
- VUID-StandaloneSpirv-MemorySemantics-10870
  MemorySemantics with a non-relaxed memory order (Acquire, Release, or AcquireRelease) **must** have at least one Vulkan-supported storage class semantics bit set (UniformMemory, WorkgroupMemory, ImageMemory, or OutputMemory)
- VUID-StandaloneSpirv-MemorySemantics-10871
  MemorySemantics with at least one Vulkan-supported storage class semantics bit set (UniformMemory, WorkgroupMemory, ImageMemory, or OutputMemory) **must** use a non-relaxed memory order (Acquire, Release, or AcquireRelease)
- VUID-StandaloneSpirv-MemorySemantics-10872
  MemorySemantics with MakeAvailable bit set **must** use Release or AcquireRelease memory order
- VUID-StandaloneSpirv-MemorySemantics-10873
  MemorySemantics with MakeVisible bit set **must** use Acquire or AcquireRelease memory order
- VUID-StandaloneSpirv-MemorySemantics-13551
  MemorySemantics with Volatile bit set **must** not be used with barrier instructions (OpControlBarrier, OpControlBarrierArriveEXT, OpControlBarrierWaitEXT, or OpMemoryBarrier)
- VUID-StandaloneSpirv-UnequalMemorySemantics-10875
  UnequalMemorySemantics of OpAtomicCompareExchange **must** not use Release or AcquireRelease memory order
- VUID-StandaloneSpirv-UnequalMemorySemantics-10876
  UnequalMemorySemantics of OpAtomicCompareExchange **must** not use a stronger memory order than the corresponding EqualMemorySemantics
- VUID-StandaloneSpirv-UnequalMemorySemantics-10877
  UnequalMemorySemantics of OpAtomicCompareExchange **must** not have any Vulkan-supported storage class semantics bit set (UniformMemory, WorkgroupMemory, ImageMemory, or OutputMemory) unless this bit is also set in the corresponding EqualMemorySemantics
- VUID-StandaloneSpirv-UnequalMemorySemantics-10878
  UnequalMemorySemantics of OpAtomicCompareExchange **must** not have MakeVisible bit set unless this bit is also set in the corresponding EqualMemorySemantics
- VUID-StandaloneSpirv-UnequalMemorySemantics-10879
  UnequalMemorySemantics of OpAtomicCompareExchange **must** have Volatile bit set if and only if this bit is also set in the corresponding EqualMemorySemantics
- VUID-StandaloneSpirv-OpVariable-04651
  Any variable with an Initializer operand **must** have Output, Private, Function, or Workgroup as its Storage Class operand

- VUID-StandaloneSpirv-OpVariable-04734
  Any variable with an `Initializer` operand and `Workgroup` as its `Storage Class` operand **must** use `OpConstantNull` as the initializer
- VUID-StandaloneSpirv-OpReadClockKHR-04652
  `Scope` for `OpReadClockKHR` **must** be limited to `Subgroup` or `Device`
- VUID-StandaloneSpirv-OriginLowerLeft-04653
  The `OriginLowerLeft Execution Mode` **must** not be used; fragment entry points **must** declare `OriginUpperLeft`
- VUID-StandaloneSpirv-PixelCenterInteger-04654
  The `PixelCenterInteger Execution Mode` **must** not be used (pixels are always centered at half-integer coordinates)
- VUID-StandaloneSpirv-UniformConstant-04655
  Any variable in the `UniformConstant Storage Class` **must** be typed as either `OpTypeImage`, `OpTypeSampler`, `OpTypeSampledImage`, `OpTypeAccelerationStructureKHR`, `OpTypeTensorARM` or an array of one of these types
- VUID-StandaloneSpirv-Uniform-06807
  Any variable in the `Uniform` or `StorageBuffer Storage Class` **must** be typed as `OpTypeStruct` or an array of this type
- VUID-StandaloneSpirv-PushConstant-06808
  Any variable in the `PushConstant Storage Class` **must** be typed as `OpTypeStruct`
- VUID-StandaloneSpirv-OpTypeImage-04656
  `OpTypeImage` **must** declare a scalar 32-bit float, 64-bit integer, or 32-bit integer type for the "Sampled Type" (`RelaxedPrecision` **can** be applied to a sampling instruction and to the variable holding the result of a sampling instruction)
- VUID-StandaloneSpirv-OpTypeImage-04657
  `OpTypeImage` **must** have a "Sampled" operand of 1 (sampled image) or 2 (storage image)
- VUID-StandaloneSpirv-OpTypeSampledImage-06671
  `OpTypeSampledImage` **must** have a `OpTypeImage` with a "Sampled" operand of 1 (sampled image)
- VUID-StandaloneSpirv-Image-04965
  The SPIR-V Type of the `Image Format` operand of an `OpTypeImage` **must** match the `Sampled Type`, as defined in Image Format and Type Matching
- VUID-StandaloneSpirv-OpImageTexelPointer-04658
  If an `OpImageTexelPointer` is used in an atomic operation, the image type of the `image` parameter to `OpImageTexelPointer` **must** have an image format of `R64i`, `R64ui`, `R32f`, `R32i`, or `R32ui`
- VUID-StandaloneSpirv-OpUntypedImageTexelPointerEXT-11416
  If an `OpUntypedImageTexelPointerEXT` instruction is used in an atomic operation, the image type operand **must** have an image format of `R64i`, `R64ui`, `R32f`, `R32i`, or `R32ui`
- VUID-StandaloneSpirv-OpImageQuerySizeLod-04659
  `OpImageQuerySizeLod`, `OpImageQueryLod`, and `OpImageQueryLevels` **must** only consume an "Image" operand whose type has its "Sampled" operand set to 1

- VUID-StandaloneSpirv-OpTypeImage-09638
  An `OpTypeImage` **must** not have a “Dim” operand of `Rect`
- VUID-StandaloneSpirv-OpTypeImage-06214
  An `OpTypeImage` with a “Dim” operand of `SubpassData` **must** have an “Arrayed” operand of 0 (non-arrayed) and a “Sampled” operand of 2 (storage image)
- VUID-StandaloneSpirv-SubpassData-04660
  The (u,v) coordinates used for a `SubpassData` **must** be the <id> of a constant vector (0,0)
- VUID-StandaloneSpirv-OpTypeImage-06924
  Objects of types `OpTypeImage`, `OpTypeSampler`, `OpTypeSampledImage`, `OpTypeAccelerationStructureKHR`, `OpTypeTensorARM`, and arrays of these types **must** not be stored to or modified
- VUID-StandaloneSpirv-Uniform-06925
  Any variable in the `Uniform Storage Class` decorated as `Block` **must** not be stored to or modified
- VUID-StandaloneSpirv-Offset-04865
  Any image instruction which uses an `Offset`, `ConstOffset`, or `ConstOffsets` image operand, **must** only consume a “Sampled Image” operand whose type has its “Sampled” operand set to 1
- VUID-StandaloneSpirv-OpImageGather-04664
  The “Component” operand of `OpImageGather`, and `OpImageSparseGather` **must** be the <id> of a constant instruction
- VUID-StandaloneSpirv-OpImage-04777
  `OpImage*Dref*` instructions **must** not consume an image whose `Dim` is 3D
- VUID-StandaloneSpirv-None-04667
  If the `DescriptorHeapEXT` capability is not declared, structure types **must** not contain opaque types
- VUID-StandaloneSpirv-DescriptorHeapEXT-11482
  If the `DescriptorHeapEXT` capability is declared, structure types **must** not contain opaque types other than descriptors
- VUID-StandaloneSpirv-BuiltIn-04668
  Any `BuiltIn` decoration not listed in Built-In Variables **must** not be used
- VUID-StandaloneSpirv-OpEntryPoint-09658
  For a given `OpEntryPoint`, any `BuiltIn` decoration **must** not be used more than once by the `Input` interface
- VUID-StandaloneSpirv-OpEntryPoint-09659
  For a given `OpEntryPoint`, any `BuiltIn` decoration **must** not be used more than once by the `Output` interface
- VUID-StandaloneSpirv-Location-06672
  The `Location` or `Component` decorations **must** only be used with the `Input`, `Output`, `RayPayloadKHR`, `IncomingRayPayloadKHR`, `HitAttributeKHR`, `HitObjectAttributeNV`, `CallableDataKHR`, `IncomingCallableDataKHR`, or `ShaderRecordBufferKHR` storage classes
- VUID-StandaloneSpirv-Location-04915

The Location or Component decorations **must** not be used with BuiltIn

- VUID-StandaloneSpirv-Location-04916
  The Location decorations **must** be used on user-defined variables
- VUID-StandaloneSpirv-Location-04917
  If a user-defined variable is not a pointer to a Block decorated OpTypeStruct, then the variable **must** have a Location decoration
- VUID-StandaloneSpirv-Location-04918
  If a user-defined variable has a Location decoration, and the variable is a pointer to a OpTypeStruct, then the members of that structure **must** not have Location decorations
- VUID-StandaloneSpirv-Location-04919
  If a user-defined variable does not have a Location decoration, and the variable is a pointer to a Block decorated OpTypeStruct, then each member of the struct **must** have a Location decoration
- VUID-StandaloneSpirv-Component-04920
  The Component decoration value **must** not be greater than 3
- VUID-StandaloneSpirv-Component-04921
  If the Component decoration is used on a variable that has a vector type with a Component Type with a Width that is less than or equal to 32, the sum of its Component Count and the Component decoration value **must** be less than or equal to 4
- VUID-StandaloneSpirv-Component-04922
  If the Component decoration is used on a variable that has a vector type with a Component Type with a Width that is equal to 64, the sum of two times its Component Count and the Component decoration value **must** be less than or equal to 4
- VUID-StandaloneSpirv-Component-04923
  The Component decorations value **must** not be 1 or 3 for scalar or two-component 64-bit data types
- VUID-StandaloneSpirv-Component-07703
  The Component decorations **must** not be used for a 64-bit vector type with more than two components
- VUID-StandaloneSpirv-Component-10583
  The Component decorations **must** not be used with any type that is not a scalar, vector, array of scalars or vectors, or an array of arrays of scalars or vectors
- VUID-StandaloneSpirv-Execution-10584
  If the Execution Model of an entry point is not TessellationControl or Geometry, its interface definition **must** not include any variables in the Input storage class decorated with Component that are arrays of arrays of scalars or vectors
- VUID-StandaloneSpirv-Execution-10585
  If the Execution Model of an entry point is not MeshNV, its interface definition **must** not include any variables in the Output storage class decorated with Component that are arrays of arrays of scalars or vectors
- VUID-StandaloneSpirv-Output-10586
  Variables in the Output storage class in the Vertex, TessellationControl, TessellationEvaluation, or Geometry execution model **must** not have overlapping

Component and Location decorations as defined by Location and Component Assignment

- VUID-StandaloneSpirv-Output-10587
  Variables in the Output storage class in the Fragment execution model **must** not have both identical Index decorations and overlapping Component and Location decorations as defined by Location and Component Assignment
- VUID-StandaloneSpirv-Input-10588
  Variables in the Input storage class in the TessellationControl, TessellationEvaluation, Geometry, or Fragment execution model **must** not have overlapping Component and Location decorations as defined by Location and Component Assignment
- VUID-StandaloneSpirv-Input-10604
  Any variable in the Input storage class in the Fragment execution model **must** not have an assigned Location shared with another variable with different decorations for PerVertexKHR, Flat, NoPerspective, Sample, or Centroid
- VUID-StandaloneSpirv-Input-09557
  The pointers of any Input or Output Interface user-defined variables **must** not contain any PhysicalStorageBuffer Storage Class pointers
- VUID-StandaloneSpirv-None-10684
  All variables **must** have valid explicit layout decorations as described in Shader Interfaces
- VUID-StandaloneSpirv-GLSLShared-04669
  The GLSLShared and GLSLPacked decorations **must** not be used
- VUID-StandaloneSpirv-TessLevelInner-10880
  Any variable decorated with TessLevelInner or TessLevelOuter **must** also be decorated with Patch
- VUID-StandaloneSpirv-Flat-04670
  The Flat, NoPerspective, Sample, and Centroid decorations **must** only be used on variables with the Output or Input Storage Class
- VUID-StandaloneSpirv-Flat-06201
  The Flat, NoPerspective, Sample, and Centroid decorations **must** not be used on variables with the Output storage class in a fragment shader
- VUID-StandaloneSpirv-Flat-06202
  The Flat, NoPerspective, Sample, and Centroid decorations **must** not be used on variables with the Input storage class in a vertex shader
- VUID-StandaloneSpirv-PerVertexKHR-06777
  The PerVertexKHR decoration **must** only be used on variables with the Input Storage Class in a fragment shader
- VUID-StandaloneSpirv-Flat-04744
  Any variable with integer or double-precision floating-point type and with Input Storage Class in a fragment shader, **must** be decorated Flat
- VUID-StandaloneSpirv-ViewportRelativeNV-04672
  The ViewportRelativeNV decoration **must** only be used on a variable decorated with Layer in the vertex, tessellation evaluation, or geometry shader stages
- VUID-StandaloneSpirv-ViewportRelativeNV-04673

The ViewportRelativeNV decoration **must** not be used unless a variable decorated with one of ViewportIndex or ViewportMaskNV is also statically used by the same OpEntryPoint

- VUID-StandaloneSpirv-ViewportMaskNV-04674
  The ViewportMaskNV and ViewportIndex decorations **must** not both be statically used by one or more OpEntryPoint's that form the pre-rasterization shader stages of a graphics pipeline
- VUID-StandaloneSpirv-FPRoundingMode-04675
  Rounding modes other than round-to-nearest-even and round-towards-zero **must** not be used for the FPRoundingMode decoration
- VUID-StandaloneSpirv-Invariant-04677
  Variables decorated with Invariant and variables with structure types that have any members decorated with Invariant **must** be in the Output or Input Storage Class, Invariant used on an Input Storage Class variable or structure member has no effect
- VUID-StandaloneSpirv-VulkanMemoryModel-04678
  If the VulkanMemoryModel capability is not declared, the Volatile decoration **must** be used on any variable declaration that includes one of the SMIDNV, WarpIDNV, SubgroupSize, SubgroupLocalInvocationId, SubgroupEqMask, SubgroupGeMask, SubgroupGtMask, SubgroupLeMask, or SubgroupLtMask BuiltIn decorations when used in the ray generation, closest hit, miss, intersection, or callable shaders, or with the RayTmaxKHR Builtin decoration when used in an intersection shader
- VUID-StandaloneSpirv-VulkanMemoryModel-04679
  If the VulkanMemoryModel capability is declared, the OpLoad instruction **must** use MemorySemantics with the Volatile flag when it accesses into any variable that includes one of the SMIDNV, WarpIDNV, SubgroupSize, SubgroupLocalInvocationId, SubgroupEqMask, SubgroupGeMask, SubgroupGtMask, SubgroupLeMask, or SubgroupLtMask BuiltIn decorations when used in the ray generation, closest hit, miss, intersection, or callable shaders, or with the RayTmaxKHR Builtin decoration when used in an intersection shader
- VUID-StandaloneSpirv-OpTypeRuntimeArray-04680
  OpTypeRuntimeArray **must** only be instantiated by a variable as:
  - the last member of a Block-decorated OpTypeStruct in StorageBuffer or PhysicalStorageBuffer storage Storage Class
  - BufferBlock-decorated OpTypeStruct in the Uniform storage Storage Class
  - the last member of a Block-decorated OpTypeStruct in the Uniform storage Storage Class
  - the outermost dimension of an arrayed variable in the StorageBuffer, Uniform, or UniformConstant storage Storage Class
  - the NodePayloadAMDX storage Storage Class when the CoalescingAMDX Execution Mode is specified
- VUID-StandaloneSpirv-OpArrayLength-11805
  OpArrayLength and OpUntypedArrayLengthKHR **must** not be used with an OpTypeRuntimeArray that is the last member of a Block-decorated OpTypeStruct in the Uniform storage Storage Class
- VUID-StandaloneSpirv-Function-04681
  A type *T* that is an array sized with a specialization constant **must** neither be, nor be contained in, the type *T2* of a variable *V*, unless either: a) *T* is equal to *T2*, b) *V* is declared

in the Function, or Private Storage Class, c) *V* is a non-Block variable in the Workgroup Storage Class, or d) *V* is an interface variable with an additional level of arrayness, as described in interface matching, and *T* is the member type of the array type *T2*

- VUID-StandaloneSpirv-Function-12294
  A type *T* that is a vector sized with a specialization constant **must** neither be, nor be contained in, the type *T2* of a variable *V*, unless either: a) *T* is equal to *T2*, b) *V* is declared in the Function, or Private Storage Class, c) *V* is a non-Block variable in the Workgroup Storage Class, or d) *V* is an interface variable with an additional level of arrayness, as described in interface matching, and *T* is the member type of the array type *T2*
- VUID-StandaloneSpirv-OpControlBarrier-04682
  If OpControlBarrier is used in ray generation, intersection, any-hit, closest hit, miss, fragment, vertex, tessellation evaluation, or geometry shaders, the execution Scope **must** be Subgroup
- VUID-StandaloneSpirv-SplitBarrierEXT-13552
  The SplitBarrierEXT capability **must** not be enabled in any stage other than compute
- VUID-StandaloneSpirv-OpControlBarrierArriveEXT-13553
  The execution Scope for OpControlBarrierArriveEXT and OpControlBarrierWaitEXT **must** be Workgroup or Subgroup
- VUID-StandaloneSpirv-MemorySemantics-13556
  The MemorySemantics for OpControlBarrierArriveEXT **must** not have any non-relaxed memory order set other than Release
- VUID-StandaloneSpirv-MemorySemantics-13557
  The MemorySemantics for OpControlBarrierWaitEXT **must** not have any non-relaxed memory order set other than Acquire
- VUID-StandaloneSpirv-None-10685
  Either a TileShadingRateQCOM, LocalSize, or LocalSizeId Execution Mode, or an object decorated with the WorkgroupSize decoration **must** be specified for each entry point with a task, mesh, or compute Execution Model
- VUID-StandaloneSpirv-DerivativeGroupQuadsNV-04684
  For compute shaders using the DerivativeGroupQuadsNV execution mode, the first two dimensions of the local workgroup size **must** be a multiple of two
- VUID-StandaloneSpirv-DerivativeGroupLinearNV-04778
  For compute shaders using the DerivativeGroupLinearNV execution mode, the product of the dimensions of the local workgroup size **must** be a multiple of four
- VUID-StandaloneSpirv-DerivativeGroupQuadsKHR-10151
  For compute, mesh, or task shaders using the DerivativeGroupQuadsKHR execution mode, the first two dimensions of the local workgroup size **must** be a multiple of two
- VUID-StandaloneSpirv-DerivativeGroupLinearKHR-10152
  For compute, mesh, or task shaders using the DerivativeGroupLinearKHR execution mode, the product of the dimensions of the local workgroup size **must** be a multiple of four
- VUID-StandaloneSpirv-OpGroupNonUniformBallotBitCount-04685
  If OpGroupNonUniformBallotBitCount is used, the group operation **must** be limited to Reduce, InclusiveScan, or ExclusiveScan

- VUID-StandaloneSpirv-None-04686
  The *Pointer* operand of all atomic instructions **must** have a `Storage Class` limited to `Uniform`, `Workgroup`, `Image`, `StorageBuffer`, `PhysicalStorageBuffer`, or `TaskPayloadWorkgroupEXT`
- VUID-StandaloneSpirv-Offset-04687
  Output variables or block members decorated with `Offset` that have a 64-bit type, or a composite type containing a 64-bit type, **must** specify an `Offset` value aligned to a 8 byte boundary
- VUID-StandaloneSpirv-Offset-04689
  The size of any output block containing any member decorated with `Offset` that is a 64-bit type **must** be a multiple of 8
- VUID-StandaloneSpirv-Offset-04690
  The first member of an output block specifying a `Offset` decoration **must** specify a `Offset` value that is aligned to an 8 byte boundary if that block contains any member decorated with `Offset` and is a 64-bit type
- VUID-StandaloneSpirv-Offset-04691
  Output variables or block members decorated with `Offset` that have a 32-bit type, or a composite type contains a 32-bit type, **must** specify an `Offset` value aligned to a 4 byte boundary
- VUID-StandaloneSpirv-Offset-04692
  Output variables, blocks, or block members decorated with `Offset` **must** only contain base types that have components that are either 32-bit or 64-bit in size
- VUID-StandaloneSpirv-Offset-12509
  Variables or block members in the input interface decorated with `Offset` **must** also be decorated with `PerTaskNV`
- VUID-StandaloneSpirv-Offset-04716
  Variables or block members in the output interface decorated with `Offset` **must** also be decorated with `XfbBuffer` and `XfbStride`, inherit `XfbBuffer` and `XfbStride` decorations from a block containing them, or be decorated with `PerTaskNV`
- VUID-StandaloneSpirv-XfbBuffer-04693
  All variables or block members in the output interface of the entry point being compiled decorated with a specific `XfbBuffer` value **must** all be decorated with identical `XfbStride` values
- VUID-StandaloneSpirv-Stream-04694
  If any variables or block members in the output interface of the entry point being compiled are decorated with `Stream`, then all variables belonging to the same `XfbBuffer` **must** specify the same `Stream` value
- VUID-StandaloneSpirv-XfbBuffer-04696
  For any two variables or block members in the output interface of the entry point being compiled with the same `XfbBuffer` value, the ranges determined by the `Offset` decoration and the size of the type **must** not overlap
- VUID-StandaloneSpirv-XfbBuffer-04697
  All block members in the output interface of the entry point being compiled that are in the same block and have a declared or inherited `XfbBuffer` decoration **must** specify the

same XfbBuffer value

- VUID-StandaloneSpirv-RayPayloadKHR-04698
  RayPayloadKHR Storage Class **must** only be used in ray generation, closest hit or miss shaders
- VUID-StandaloneSpirv-IncomingRayPayloadKHR-04699
  IncomingRayPayloadKHR Storage Class **must** only be used in closest hit, any-hit, or miss shaders
- VUID-StandaloneSpirv-IncomingRayPayloadKHR-04700
  There **must** be at most one variable with the IncomingRayPayloadKHR Storage Class in the input interface of an entry point
- VUID-StandaloneSpirv-HitAttributeKHR-04701
  HitAttributeKHR Storage Class **must** only be used in intersection, any-hit, or closest hit shaders
- VUID-StandaloneSpirv-HitAttributeKHR-04702
  There **must** be at most one variable with the HitAttributeKHR Storage Class in the input interface of an entry point
- VUID-StandaloneSpirv-HitAttributeKHR-04703
  A variable with HitAttributeKHR Storage Class **must** only be written to in an intersection shader
- VUID-StandaloneSpirv-CallableDataKHR-04704
  CallableDataKHR Storage Class **must** only be used in ray generation, closest hit, miss, and callable shaders
- VUID-StandaloneSpirv-IncomingCallableDataKHR-04705
  IncomingCallableDataKHR Storage Class **must** only be used in callable shaders
- VUID-StandaloneSpirv-IncomingCallableDataKHR-04706
  There **must** be at most one variable with the IncomingCallableDataKHR Storage Class in the input interface of an entry point
- VUID-StandaloneSpirv-ShaderRecordBufferKHR-07119
  ShaderRecordBufferKHR Storage Class **must** only be used in ray generation, intersection, any-hit, closest hit, callable, or miss shaders
- VUID-StandaloneSpirv-Base-07650
  The Base operand of OpPtrAccessChain or OpUntypedPtrAccessChainKHR **must** have a storage class of Workgroup, StorageBuffer, or PhysicalStorageBuffer
- VUID-StandaloneSpirv-Base-07651
  If the Base operand of OpPtrAccessChain or OpUntypedPtrAccessChainKHR has a Workgroup Storage Class, then the VariablePointers capability **must** be declared
- VUID-StandaloneSpirv-Base-07652
  If the Base operand of OpPtrAccessChain or OpUntypedPtrAccessChainKHR has a StorageBuffer Storage Class, then the VariablePointers or VariablePointersStorageBuffer capability **must** be declared
- VUID-StandaloneSpirv-OpUntypedVariableKHR-11167
  Any OpUntypedVariableKHR with a Storage Class other than UniformConstant **must** have a

Data Type operand specified

- VUID-StandaloneSpirv-OpUntypedVariableKHR-11347
  Any OpUntypedVariableKHR in the UniformConstant storage class without a Data Type **must** be decorated with SamplerHeapEXT or ResourceHeapEXT
- VUID-StandaloneSpirv-PhysicalStorageBuffer64-04708
  If the PhysicalStorageBuffer64 addressing model is enabled, all instructions that support memory access operands and that use a physical pointer **must** include the Aligned operand
- VUID-StandaloneSpirv-PhysicalStorageBuffer64-04709
  If the PhysicalStorageBuffer64 addressing model is enabled, any access chain instruction that accesses into a RowMajor matrix **must** only be used as the Pointer operand to OpLoad or OpStore
- VUID-StandaloneSpirv-PhysicalStorageBuffer64-04710
  If the PhysicalStorageBuffer64 addressing model is enabled, OpConvertUToPtr and OpConvertPtrToU **must** use an integer type whose Width is 64
- VUID-StandaloneSpirv-PhysicalStorageBuffer64-06314
  If the PhysicalStorageBuffer64 addressing model is enabled any load or store through a physical pointer type **must** have an Aligned operand which is a multiple of the size of the largest scalar type in the pointed-to type
- VUID-StandaloneSpirv-OpTypeForwardPointer-04711
  OpTypeForwardPointer **must** have a Storage Class of PhysicalStorageBuffer
- VUID-StandaloneSpirv-OpVariable-06673
  There **must** be at most one variable in the PushConstant Storage Class listed in the Interface for each OpEntryPoint unless the PushConstantBanksNV capability is declared
- VUID-StandaloneSpirv-OpEntryPoint-06674
  Each OpEntryPoint **must** statically use at most one variable in the PushConstant Storage Class unless the PushConstantBanksNV capability is declared
- VUID-StandaloneSpirv-OpEntryPoint-08721
  Each OpEntryPoint **must** not have more than one Input variable assigned the same Component word inside a Location slot, either explicitly or implicitly
- VUID-StandaloneSpirv-OpEntryPoint-08722
  Each OpEntryPoint **must** not have more than one Output variable assigned the same Component word inside a Location slot, either explicitly or implicitly
- VUID-StandaloneSpirv-Result-04780
  The Result Type operand of any OpImageRead or OpImageSparseRead instruction **must** be a vector of four components
- VUID-StandaloneSpirv-PushConstant-06675
  Any variable in the PushConstant or StorageBuffer storage class **must** be decorated as Block
- VUID-StandaloneSpirv-Uniform-06676
  Any variable in the Uniform Storage Class **must** be decorated as Block or BufferBlock
- VUID-StandaloneSpirv-UniformConstant-06677
  Any variable in the UniformConstant, StorageBuffer, or Uniform Storage Class **must** be

decorated with either DescriptorSet and Binding, or BuiltIn with SamplerHeapEXT or ResourceHeapEXT

- VUID-StandaloneSpirv-InputAttachmentIndex-06678
  Variables decorated with InputAttachmentIndex **must** be in the UniformConstant Storage Class
- VUID-StandaloneSpirv-DescriptorSet-06491
  If a variable is decorated by DescriptorSet or Binding, the Storage Class **must** correspond to an entry in Shader Resource and Storage Class Correspondence
- VUID-StandaloneSpirv-Input-06778
  Variables with a Storage Class of Input in a fragment shader stage that are decorated with PerVertexKHR **must** be declared as arrays
- VUID-StandaloneSpirv-MeshEXT-07102
  The module **must** not contain both an entry point that uses the TaskEXT or MeshEXT Execution Model and an entry point that uses the TaskNV or MeshNV Execution Model
- VUID-StandaloneSpirv-MeshEXT-07106
  In mesh shaders using the MeshEXT Execution Model OpSetMeshOutputsEXT **must** be called before any outputs are written
- VUID-StandaloneSpirv-MeshEXT-07107
  In mesh shaders using the MeshEXT Execution Model all variables declared in the Output Storage Class **must** not be read
- VUID-StandaloneSpirv-MeshEXT-07108
  In mesh shaders using the MeshEXT Execution Model for OpSetMeshOutputsEXT instructions, the “Vertex Count” and “Primitive Count” operands **must** not depend on ViewIndex
- VUID-StandaloneSpirv-MeshEXT-07109
  In mesh shaders using the MeshEXT Execution Model variables decorated with PrimitivePointIndicesEXT, PrimitiveLineIndicesEXT, or PrimitiveTriangleIndicesEXT declared as an array **must** not be accessed by indices that depend on ViewIndex
- VUID-StandaloneSpirv-MeshEXT-07110
  In mesh shaders using the MeshEXT Execution Model any values stored in variables decorated with PrimitivePointIndicesEXT, PrimitiveLineIndicesEXT, or PrimitiveTriangleIndicesEXT **must** not depend on ViewIndex
- VUID-StandaloneSpirv-MeshEXT-07111
  In mesh shaders using the MeshEXT Execution Model variables in workgroup or private Storage Class declared as or containing a composite type **must** not be accessed by indices that depend on ViewIndex
- VUID-StandaloneSpirv-MeshEXT-07330
  In mesh shaders using the MeshEXT Execution Model the OutputVertices Execution Mode **must** be greater than 0
- VUID-StandaloneSpirv-MeshEXT-07331
  In mesh shaders using the MeshEXT Execution Model the OutputPrimitivesEXT Execution Mode **must** be greater than 0
- VUID-StandaloneSpirv-Input-07290
  Variables with a Storage Class of Input or Output and a type of OpTypeBool **must** be

decorated with the BuiltIn decoration

- VUID-StandaloneSpirv-TileImageEXT-08723
  The tile image variable declarations **must** obey the constraints on the TileImageEXT Storage Class and the Location decoration described in Fragment Tile Image Interface
- VUID-StandaloneSpirv-None-08724
  The TileImageEXT Storage Class **must** only be used for declaring tile image variables
- VUID-StandaloneSpirv-Pointer-08973
  The Storage Class of the Pointer operand to OpCooperativeMatrixLoadKHR or OpCooperativeMatrixStoreKHR **must** be limited to Workgroup, StorageBuffer, or PhysicalStorageBuffer
- VUID-StandaloneSpirv-OpTypeFloat-10370
  Variables with a type of OpTypeFloat and an encoding of BFloat16KHR **must** not be declared with a Storage Class of Input or Output
- VUID-StandaloneSpirv-UniformBufferArrayDynamicIndexing-10127
  If the UniformBufferArrayDynamicIndexing capability is not declared, and an instruction accesses memory through a uniform buffer, the uniform buffer through which that memory is accessed **must** be determined by constant integral expressions
- VUID-StandaloneSpirv-SampledImageArrayDynamicIndexing-10128
  If the SampledImageArrayDynamicIndexing capability is not declared, and an instruction accesses memory through a sampled image or sampler, the sampled image or sampler through which that memory is accessed **must** be determined by constant integral expressions
- VUID-StandaloneSpirv-StorageBufferArrayDynamicIndexing-10129
  If the StorageBufferArrayDynamicIndexing capability is not declared, and an instruction accesses memory through a storage buffer, the storage buffer through which that memory is accessed **must** be determined by constant integral expressions
- VUID-StandaloneSpirv-StorageImageArrayDynamicIndexing-10130
  If the StorageImageArrayDynamicIndexing capability is not declared, and an instruction accesses memory through a storage image, the storage image through which that memory is accessed **must** be determined by constant integral expressions
- VUID-StandaloneSpirv-InputAttachmentArrayDynamicIndexing-10131
  If the InputAttachmentArrayDynamicIndexing capability is not declared, and an instruction accesses memory through an input attachment, the input attachment through which that memory is accessed **must** be determined by constant integral expressions
- VUID-StandaloneSpirv-UniformTexelBufferArrayDynamicIndexing-10132
  If the UniformTexelBufferArrayDynamicIndexing capability is not declared, and an instruction accesses memory through a uniform texel buffer, the uniform texel buffer through which that memory is accessed **must** be determined by constant integral expressions
- VUID-StandaloneSpirv-StorageTexelBufferArrayDynamicIndexing-10133
  If the StorageTexelBufferArrayDynamicIndexing capability is not declared, and an instruction accesses memory through a storage texel buffer, the storage texel buffer through which that memory is accessed **must** be determined by constant integral

expressions

- VUID-StandaloneSpirv-MaximallyReconvergesKHR-09565
  The `MaximallyReconvergesKHR Execution Mode` **must** not be applied to an entry point if an *invocation repack instruction* is statically used
- VUID-StandaloneSpirv-TileShadingQCOM-10686
  `TileShadingQCOM` capability **must** not be enabled in any stage other than compute or fragment
- VUID-StandaloneSpirv-Execution-10687
  `Execution Mode NonCoherentTileAttachmentReadQCOM` **must** not be used in any stage other than fragment
- VUID-StandaloneSpirv-Execution-10688
  `Execution Mode TileShadingRateQCOM` **must** not be used in any stage other than compute
- VUID-StandaloneSpirv-TileAttachmentQCOM-10689
  If the `TileAttachmentQCOM Storage Class` is used, the `TileShadingQCOM` capability **must** be enabled
- VUID-StandaloneSpirv-NonCoherentTileAttachmentReadQCOM-10690
  If the `NonCoherentTileAttachmentReadQCOM Execution Mode` is used, the `TileShadingQCOM` capability **must** be enabled
- VUID-StandaloneSpirv-TileShadingRateQCOM-10691
  If the `TileShadingRateQCOM Execution Mode` is used, the `TileShadingQCOM` capability **must** be enabled
- VUID-StandaloneSpirv-TileShadingRateQCOM-10692
  If the `TileShadingRateQCOM Execution Mode` is used, `LocalSize` and `LocalSizeId Execution Mode` **must** not be specified
- VUID-StandaloneSpirv-OpTypeImage-10693
  `OpTypeImage` variables in the `TileAttachmentQCOM Storage Class` **must** have `Dim` equal to `2D`
- VUID-StandaloneSpirv-OpTypeImage-10694
  `OpTypeImage` variables in the `TileAttachmentQCOM Storage Class` **must** `Sampled` equal to `1` or `2`
- VUID-StandaloneSpirv-TileAttachmentQCOM-10695
  Any variable in the `TileAttachmentQCOM Storage Class` **must** be decorated with `DescriptorSet` and `Binding`
- VUID-StandaloneSpirv-TileAttachmentQCOM-10696
  Any variable in the `TileAttachmentQCOM Storage Class` **must** not be decorated with `Component` decoration
- VUID-StandaloneSpirv-TileAttachmentQCOM-10697
  An OpTypeImage variables in the `TileAttachmentQCOM Storage Class` **must** not be consumed by an `OpImageQuery*` instruction
- VUID-StandaloneSpirv-OpTypeFloat-10823
  Variables with a type of `OpTypeFloat` and an encoding of `Float8E4M3EXT` or `Float8E5M2EXT` **must** not be declared with a `Storage Class` of `Input` or `Output`
- VUID-StandaloneSpirv-OpGraphInputARM-09931
  The `InputIndex` and `ElementIndex` operands to `OpGraphInputARM` **must** be the <id> of a

constant instruction

- VUID-StandaloneSpirv-OpGraphSetOutputARM-09932
  The OutputIndex and ElementIndex operands to OpGraphSetOutputARM **must** be the <id> of a constant instruction
- VUID-StandaloneSpirv-Result-11336
  If the Result Type operand of OpLoad is OpTypeSampler, Pointer **must** be derived from a variable decorated with Binding and DescriptorSet, or decorated with BuiltIn and SamplerHeapEXT
- VUID-StandaloneSpirv-Result-11337
  If the Result Type operand of OpLoad is OpTypeImage, Pointer **must** be derived from a variable decorated with Binding and DescriptorSet, or decorated with BuiltIn and ResourceHeapEXT
- VUID-StandaloneSpirv-Result-11339
  If the Result Type operand of OpLoad is OpTypeAccelerationStructureKHR, and Pointer is not in the Private or Function Storage Class, Pointer **must** be derived from a variable decorated with Binding and DescriptorSet, or decorated with BuiltIn and ResourceHeapEXT
- VUID-StandaloneSpirv-OpTypeUntypedPointerKHR-11417
  The Storage Class of OpTypeUntypedPointerKHR **must** be Image, UniformConstant,Workgroup, StorageBuffer, Uniform, PushConstant, or PhysicalStorageBuffer
- VUID-StandaloneSpirv-None-12295
  If the LongVectorEXT capability is not declared, the Component Count of any vector type **must** be less than or equal to 4 and greater than 1
- VUID-StandaloneSpirv-Type-12297
  Any pointer type whose Type parameter is a vector type with more than four components (or an aggregate containing such a type) **must** have Storage Class of Function, Private, Uniform, Workgroup, StorageBuffer, PhysicalStorageBuffer, PushConstant, or ShaderRecordBufferKHR
- VUID-StandaloneSpirv-OpUndef-12464
  Any OpUndef instruction **must** not have a Result Type of OpTypeImage, OpTypeSampler, OpTypeSampledImage, OpTypeAccelerationStructureKHR, or OpTypeTensorARM
- VUID-StandaloneSpirv-OpConvertFToU-12465
  Floating-point objects consumed or produced by OpConvertFToU, OpConvertFToS, OpConvertSToF, or OpConvertUToF **must** not have a Float6E2M3EXT, Float6E3M2EXT, Float4E2M1EXT, Float8UnsignedE8M0EXT, or MXInt8EXT encoding
- VUID-StandaloneSpirv-OpFConvert-12466
  Floating-point objects produced by OpFConvert **must** not have a Float6E2M3EXT, Float6E3M2EXT, Float4E2M1EXT, Float8UnsignedE8M0EXT, or MXInt8EXT encoding
- VUID-StandaloneSpirv-OpFConvert-12467
  If the object consumed by OpFConvert has a Float6E2M3EXT, Float6E3M2EXT, Float4E2M1EXT, Float8UnsignedE8M0EXT, or MXInt8EXT encoding, then the Result Type **must** have an IEEE 754, Float8E4M3EXT, Float8E5M2EXT, or BFloat16KHR encoding
- VUID-StandaloneSpirv-Result-12468
  The Result Type operand of OpBitcastExtractEXT **must** be OpTypeFloat with a Float6E2M3EXT,

`Float6E3M2EXT`, or `Float4E2M1EXT` encoding

- VUID-StandaloneSpirv-Base-12469
  The `Base` operand of `OpBitcastExtractEXT` **must** be a scalar or vector integer type

Runtime SPIR-V Validation

The following rules **must** be validated at runtime. These rules depend on knowledge of the implementation and its capabilities and knowledge of runtime information, such as enabled features.

Valid Usage

- VUID-RuntimeSpirv-vulkanMemoryModel-06265
  If the `vulkanMemoryModel` feature is enabled and the `vulkanMemoryModelDeviceScope` feature is not enabled, `Device` memory scope **must** not be used
- VUID-RuntimeSpirv-vulkanMemoryModel-06266
  If the `vulkanMemoryModel` feature is not enabled, `QueueFamily` memory scope **must** not be used
- VUID-RuntimeSpirv-shaderSubgroupClock-06267
  If the `shaderSubgroupClock` feature is not enabled, the `Subgroup` scope **must** not be used for `OpReadClockKHR`
- VUID-RuntimeSpirv-shaderDeviceClock-06268
  If the `shaderDeviceClock` feature is not enabled, the `Device` scope **must** not be used for `OpReadClockKHR`
- VUID-RuntimeSpirv-shaderRelaxedExtendedInstruction-10773
  If the `shaderRelaxedExtendedInstruction` feature is not enabled, the `OpExtInstWithForwardRefsKHR` **must** not be used
- VUID-RuntimeSpirv-None-09558
  If the `dynamicRenderingLocalRead` feature is not enabled, any variable created with a "Type" of `OpTypeImage` that has a "Dim" operand of `SubpassData` **must** be decorated with `InputAttachmentIndex`
- VUID-RuntimeSpirv-OpTypeImage-09644
  Any variable declared as an `OpTypeArray` where the `Element Type` is an `OpTypeImage` with a "Dim" operand of `SubpassData` **must** be decorated with `InputAttachmentIndex`
- VUID-RuntimeSpirv-apiVersion-07954
  If VkPhysicalDeviceProperties::`apiVersion` is less than Vulkan 1.3, the VK_KHR_format_feature_flags2 extension is not supported, and the `shaderStorageImageWriteWithoutFormat` feature is not enabled, any variable created with a "Type" of `OpTypeImage` that has a "Sampled" operand of 2 and an "Image Format" operand of `Unknown` **must** be decorated with `NonWritable`
- VUID-RuntimeSpirv-apiVersion-07955
  If VkPhysicalDeviceProperties::`apiVersion` is less than Vulkan 1.3, the VK_KHR_format_feature_flags2 extension is not supported, and the

shaderStorageImageReadWithoutFormat feature is not enabled, any variable created with a "Type" of OpTypeImage that has a "Sampled" operand of 2 and an "Image Format" operand of Unknown **must** be decorated with NonReadable

- VUID-RuntimeSpirv-OpImageWrite-07112
  OpImageWrite to any Image whose Image Format is not Unknown **must** have the Texel operand contain at least as many components as the corresponding VkFormat as given in the SPIR-V Image Format compatibility table
- VUID-RuntimeSpirv-Location-06272
  The sum of Location and the number of locations the variable it decorates consumes **must** be less than or equal to the value for the matching Execution Model defined in Shader Input and Output Locations
- VUID-RuntimeSpirv-Location-06428
  The maximum number of storage buffers, storage images, and output Location decorated color attachments written to in the Fragment Execution Model **must** be less than or equal to maxFragmentCombinedOutputResources
- VUID-RuntimeSpirv-UniformBufferArrayNonUniformIndexing-10134
  If the UniformBufferArrayNonUniformIndexing capability is not declared, and an instruction accesses memory through a uniform buffer, the uniform buffer through which that memory is accessed **must** be dynamically uniform within the invocation group or subgroup
- VUID-RuntimeSpirv-SampledImageArrayNonUniformIndexing-10135
  If the SampledImageArrayNonUniformIndexing capability is not declared, and an instruction accesses memory through a sampled image or sampler, the sampled image or sampler through which that memory is accessed **must** be dynamically uniform within the invocation group or subgroup
- VUID-RuntimeSpirv-StorageBufferArrayNonUniformIndexing-10136
  If the StorageBufferArrayNonUniformIndexing capability is not declared, and an instruction accesses memory through a storage buffer, the storage buffer through which that memory is accessed **must** be dynamically uniform within the invocation group or subgroup
- VUID-RuntimeSpirv-StorageImageArrayNonUniformIndexing-10137
  If the StorageImageArrayNonUniformIndexing capability is not declared, and an instruction accesses memory through a storage image, the storage image through which that memory is accessed **must** be dynamically uniform within the invocation group or subgroup
- VUID-RuntimeSpirv-InputAttachmentArrayNonUniformIndexing-10138
  If the InputAttachmentArrayNonUniformIndexing capability is not declared, and an instruction accesses memory through an input attachment, the input attachment through which that memory is accessed **must** be dynamically uniform within the invocation group or subgroup
- VUID-RuntimeSpirv-UniformTexelBufferArrayNonUniformIndexing-10139
  If the UniformTexelBufferArrayNonUniformIndexing capability is not declared, and an instruction accesses memory through a uniform texel buffer, the uniform texel buffer through which that memory is accessed **must** be dynamically uniform within the invocation group or subgroup
- VUID-RuntimeSpirv-StorageTexelBufferArrayNonUniformIndexing-10140

If the StorageTexelBufferArrayNonUniformIndexing capability is not declared, and an instruction accesses memory through a storage texel buffer, the storage texel buffer through which that memory is accessed **must** be dynamically uniform within the invocation group or subgroup

- VUID-RuntimeSpirv-subgroupSize-10141
  If the effective subgroup size is 1, the UniformBufferArrayNonUniformIndexing capability is not declared, and an instruction accesses memory through a uniform buffer, the uniform buffer through which that memory is accessed **must** be dynamically uniform within the invocation group
- VUID-RuntimeSpirv-subgroupSize-10142
  If the effective subgroup size is 1, the SampledImageArrayNonUniformIndexing capability is not declared, and an instruction accesses memory through a sampled image or sampler, the sampled image or sampler through which that memory is accessed **must** be dynamically uniform within the invocation group
- VUID-RuntimeSpirv-subgroupSize-10143
  If the effective subgroup size is 1, the StorageBufferArrayNonUniformIndexing capability is not declared, and an instruction accesses memory through a storage buffer, the storage buffer through which that memory is accessed **must** be dynamically uniform within the invocation group
- VUID-RuntimeSpirv-subgroupSize-10144
  If the effective subgroup size is 1, the StorageImageArrayNonUniformIndexing capability is not declared, and an instruction accesses memory through a storage image, the storage image through which that memory is accessed **must** be dynamically uniform within the invocation group
- VUID-RuntimeSpirv-subgroupSize-10145
  If the effective subgroup size is 1, the InputAttachmentArrayNonUniformIndexing capability is not declared, and an instruction accesses memory through an input attachment, the input attachment through which that memory is accessed **must** be dynamically uniform within the invocation group
- VUID-RuntimeSpirv-subgroupSize-10146
  If the effective subgroup size is 1, the UniformTexelBufferArrayNonUniformIndexing capability is not declared, and an instruction accesses memory through a uniform texel buffer, the uniform texel buffer through which that memory is accessed **must** be dynamically uniform within the invocation group
- VUID-RuntimeSpirv-subgroupSize-10147
  If the effective subgroup size is 1, the StorageTexelBufferArrayNonUniformIndexing capability is not is not declared, and an instruction accesses memory through a storage texel buffer, the storage texel buffer through which that memory is accessed **must** be dynamically uniform within the invocation group
- VUID-RuntimeSpirv-maxClipDistances-00708
  If VkPhysicalDeviceLimits::maxClipDistances is not zero, then any entry point interface variable declared with the ClipDistance BuiltIn decoration, **must** not have an array size greater than VkPhysicalDeviceLimits::maxClipDistances
- VUID-RuntimeSpirv-maxCullDistances-00709

If VkPhysicalDeviceLimits::maxCullDistances is not zero, then any entry point interface variable declared with the CullDistance BuiltIn decoration, **must** not have an array size greater than VkPhysicalDeviceLimits::maxCullDistances

- VUID-RuntimeSpirv-maxCombinedClipAndCullDistances-00710
  If VkPhysicalDeviceLimits::maxCombinedClipAndCullDistances is not zero, then any entry point interface variable declared with the ClipDistance BuiltIn and CullDistance BuiltIn decoration **must** not have array sizes which sum to more than VkPhysicalDeviceLimits::maxCombinedClipAndCullDistances
- VUID-RuntimeSpirv-maxClipDistances-08448
  If VkPhysicalDeviceLimits::maxClipDistances is zero, then entry point interface variables declared with the ClipDistance BuiltIn decoration **must** not be statically accessed
- VUID-RuntimeSpirv-maxCullDistances-08449
  If VkPhysicalDeviceLimits::maxCullDistances is zero, then entry point interface variables declared with the CullDistance BuiltIn decoration **must** not be statically accessed
- VUID-RuntimeSpirv-None-04745
  All block members in a variable with a Storage Class of PushConstant declared as an array **must** only be accessed by dynamically uniform indices
- VUID-RuntimeSpirv-None-10148
  If an instruction accesses memory through any resource, the effective subgroup size is 1, the DescriptorHeapEXT capability is not declared, and the resource through which that memory is accessed is not uniform within the invocation group, then the operand corresponding to that resource (e.g. the pointer or sampled image operand) **must** be decorated with NonUniform
- VUID-RuntimeSpirv-subgroupSize-10149
  If an instruction accesses memory through any resource, the effective subgroup size is greater than 1, the DescriptorHeapEXT capability is not declared, and the resource through which that memory is accessed is not uniform within the invocation group, and not uniform within the subgroup, then the operand corresponding to that resource (e.g. the pointer or sampled image operand) **must** be decorated with NonUniform
- VUID-RuntimeSpirv-None-06275
  shaderSubgroupExtendedTypes **must** be enabled for group operations to use 8-bit integer, 16-bit integer, 64-bit integer, 16-bit floating-point, and vectors of these types
- VUID-RuntimeSpirv-subgroupBroadcastDynamicId-06276
  If subgroupBroadcastDynamicId is VK_TRUE, and the shader module version is 1.5 or higher, the “Index” for OpGroupNonUniformQuadBroadcast **must** be dynamically uniform within the derivative group. Otherwise, “Index” **must** be a constant
- VUID-RuntimeSpirv-subgroupBroadcastDynamicId-06277
  If subgroupBroadcastDynamicId is VK_TRUE, and the shader module version is 1.5 or higher, the “Id” for OpGroupNonUniformBroadcast **must** be dynamically uniform within the subgroup. Otherwise, “Id” **must** be a constant
- VUID-RuntimeSpirv-None-06278
  shaderBufferInt64Atomics **must** be enabled for 64-bit integer atomic operations to be supported on a *Pointer* with a Storage Class of StorageBuffer or Uniform

- VUID-RuntimeSpirv-None-06279
  shaderSharedInt64Atomics **must** be enabled for 64-bit integer atomic operations to be supported on a *Pointer* with a Storage Class of Workgroup
- VUID-RuntimeSpirv-None-06284
  shaderBufferFloat32Atomics, or shaderBufferFloat32AtomicAdd, or shaderBufferFloat64Atomics, or shaderBufferFloat64AtomicAdd, or shaderBufferFloat16Atomics, or shaderBufferFloat16AtomicAdd, or shaderBufferFloat16AtomicMinMax, or shaderBufferFloat32AtomicMinMax, or shaderBufferFloat64AtomicMinMax, **must** be enabled for floating-point atomic operations to be supported on a *Pointer* with a Storage Class of StorageBuffer
- VUID-RuntimeSpirv-None-06285
  shaderSharedFloat32Atomics, or shaderSharedFloat32AtomicAdd, or shaderSharedFloat64Atomics, or shaderSharedFloat64AtomicAdd, or shaderSharedFloat16Atomics, or shaderSharedFloat16AtomicAdd, or shaderSharedFloat16AtomicMinMax, or shaderSharedFloat32AtomicMinMax, or shaderSharedFloat64AtomicMinMax, **must** be enabled for floating-point atomic operations to be supported on a *Pointer* with a Storage Class of Workgroup
- VUID-RuntimeSpirv-None-06286
  shaderImageFloat32Atomics, or shaderImageFloat32AtomicAdd, or shaderImageFloat32AtomicMinMax, **must** be enabled for 32-bit floating-point atomic operations to be supported on a *Pointer* with a Storage Class of Image
- VUID-RuntimeSpirv-None-06287
  sparseImageFloat32Atomics, or sparseImageFloat32AtomicAdd, or sparseImageFloat32AtomicMinMax, **must** be enabled for 32-bit floating-point atomics to be supported on sparse images
- VUID-RuntimeSpirv-None-06288
  shaderImageInt64Atomics **must** be enabled for 64-bit integer atomic operations to be supported on a *Pointer* with a Storage Class of Image
- VUID-RuntimeSpirv-denormBehaviorIndependence-06289
  If denormBehaviorIndependence is VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_32_BIT_ONLY, then the entry point **must** use the same denormals Execution Mode for both 16-bit and 64-bit floating-point types
- VUID-RuntimeSpirv-denormBehaviorIndependence-06290
  If denormBehaviorIndependence is VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_NONE, then the entry point **must** use the same denormals Execution Mode for all floating-point types
- VUID-RuntimeSpirv-roundingModeIndependence-06291
  If roundingModeIndependence is VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_32_BIT_ONLY, then the entry point **must** use the same rounding Execution Mode for both 16-bit and 64-bit floating-point types
- VUID-RuntimeSpirv-roundingModeIndependence-06292
  If roundingModeIndependence is VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_NONE, then the entry point **must** use the same rounding Execution Mode for all floating-point

types

- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat16-06293
  If `shaderSignedZeroInfNanPreserveFloat16` is VK_FALSE, then `SignedZeroInfNanPreserve` for 16-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat32-06294
  If `shaderSignedZeroInfNanPreserveFloat32` is VK_FALSE, then `SignedZeroInfNanPreserve` for 32-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat64-06295
  If `shaderSignedZeroInfNanPreserveFloat64` is VK_FALSE, then `SignedZeroInfNanPreserve` for 64-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderDenormPreserveFloat16-06296
  If `shaderDenormPreserveFloat16` is VK_FALSE, then `DenormPreserve` for 16-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderDenormPreserveFloat32-06297
  If `shaderDenormPreserveFloat32` is VK_FALSE, then `DenormPreserve` for 32-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderDenormPreserveFloat64-06298
  If `shaderDenormPreserveFloat64` is VK_FALSE, then `DenormPreserve` for 64-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderDenormFlushToZeroFloat16-06299
  If `shaderDenormFlushToZeroFloat16` is VK_FALSE, then `DenormFlushToZero` for 16-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderDenormFlushToZeroFloat32-06300
  If `shaderDenormFlushToZeroFloat32` is VK_FALSE, then `DenormFlushToZero` for 32-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderDenormFlushToZeroFloat64-06301
  If `shaderDenormFlushToZeroFloat64` is VK_FALSE, then `DenormFlushToZero` for 64-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderRoundingModeRTEFloat16-06302
  If `shaderRoundingModeRTEFloat16` is VK_FALSE, then `RoundingModeRTE` for 16-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderRoundingModeRTEFloat32-06303
  If `shaderRoundingModeRTEFloat32` is VK_FALSE, then `RoundingModeRTE` for 32-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderRoundingModeRTEFloat64-06304
  If `shaderRoundingModeRTEFloat64` is VK_FALSE, then `RoundingModeRTE` for 64-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderRoundingModeRTZFloat16-06305
  If `shaderRoundingModeRTZFloat16` is VK_FALSE, then `RoundingModeRTZ` for 16-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderRoundingModeRTZFloat32-06306
  If `shaderRoundingModeRTZFloat32` is VK_FALSE, then `RoundingModeRTZ` for 32-bit floating-

point type **must** not be used

- VUID-RuntimeSpirv-shaderRoundingModeRTZFloat64-06307
  If `shaderRoundingModeRTZFloat64` is VK_FALSE, then `RoundingModeRTZ` for 64-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat16-09559
  If `shaderSignedZeroInfNanPreserveFloat16` is VK_FALSE then any `FPFastMathDefault` execution mode with a type of 16-bit float **must** include the `NSZ`, `NotInf`, and `NotNaN` flags
- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat16-09560
  If `shaderSignedZeroInfNanPreserveFloat16` is VK_FALSE then any `FPFastMathMode` decoration on an instruction with result type or any operand type that includes a 16-bit float **must** include the `NSZ`, `NotInf`, and `NotNaN` flags
- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat32-09561
  If `shaderSignedZeroInfNanPreserveFloat32` is VK_FALSE then any `FPFastMathDefault` execution mode with a type of 32-bit float **must** include the `NSZ`, `NotInf`, and `NotNaN` flags
- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat32-09562
  If `shaderSignedZeroInfNanPreserveFloat32` is VK_FALSE then any `FPFastMathMode` decoration on an instruction with result type or any operand type that includes a 32-bit float **must** include the `NSZ`, `NotInf`, and `NotNaN` flags
- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat64-09563
  If `shaderSignedZeroInfNanPreserveFloat64` is VK_FALSE then any `FPFastMathDefault` execution mode with a type of 64-bit float **must** include the `NSZ`, `NotInf`, and `NotNaN` flags
- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat64-09564
  If `shaderSignedZeroInfNanPreserveFloat64` is VK_FALSE then any `FPFastMathMode` decoration on an instruction with result type or any operand type that includes a 64-bit float **must** include the `NSZ`, `NotInf`, and `NotNaN` flags
- VUID-RuntimeSpirv-shaderFmaFloat16-10977
  `shaderFmaFloat16` **must** be enabled for `OpFmaKHR` to be supported with a result type that includes a 16-bit float
- VUID-RuntimeSpirv-shaderFmaFloat32-10978
  `shaderFmaFloat32` **must** be enabled for `OpFmaKHR` to be supported with a result type that includes a 32-bit float
- VUID-RuntimeSpirv-shaderFmaFloat64-10979
  `shaderFmaFloat64` **must** be enabled for `OpFmaKHR` to be supported with a result type that includes a 64-bit float
- VUID-RuntimeSpirv-Offset-06308
  The `Offset` plus size of the type of each variable, in the output interface of the entry point being compiled, decorated with `XfbBuffer` **must** not be greater than VkPhysicalDeviceTransformFeedbackPropertiesEXT::`maxTransformFeedbackBufferDataSize`
- VUID-RuntimeSpirv-XfbBuffer-06309
  For any given `XfbBuffer` value, define the buffer data size to be smallest number of bytes such that, for all outputs decorated with the same `XfbBuffer` value, the size of the output interface variable plus the `Offset` is less than or equal to the buffer data size. For a given `Stream`, the sum of all the buffer data sizes for all buffers writing to that stream the **must**

not exceed VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreamDataSize

- VUID-RuntimeSpirv-OpEmitStreamVertex-06310
  The Stream value to OpEmitStreamVertex and OpEndStreamPrimitive **must** be less than VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams
- VUID-RuntimeSpirv-transformFeedbackStreamsLinesTriangles-06311
  If the geometry shader emits to more than one vertex stream and VkPhysicalDeviceTransformFeedbackPropertiesEXT::transformFeedbackStreamsLinesTriangles is VK_FALSE, then Execution Mode **must** be OutputPoints
- VUID-RuntimeSpirv-Stream-06312
  The stream number value to Stream **must** be less than VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams
- VUID-RuntimeSpirv-XfbStride-06313
  The XFB Stride value to XfbStride **must** be less than or equal to VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBufferDataStride
- VUID-RuntimeSpirv-PhysicalStorageBuffer64-06315
  If the PhysicalStorageBuffer64 addressing model is enabled the pointer value of a memory access instruction **must** be at least as aligned as specified by the Aligned memory access operand
- VUID-RuntimeSpirv-PhysicalStorageBuffer64-11819
  If the PhysicalStorageBuffer64 addressing model is enabled the pointer value of a memory access instruction in the PhysicalStorageBuffer Storage Class **must** reference a buffer created with the VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT usage flag set
- VUID-RuntimeSpirv-OpTypeCooperativeMatrixKHR-10163
  For OpTypeCooperativeMatrixKHR, if the VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT::cooperativeMatrixProperties2 feature is not enabled, the type **must** match one of the supported VkCooperativeMatrixPropertiesKHR structures as defined in VkCooperativeMatrixPropertiesKHR Type Matching
- VUID-RuntimeSpirv-OpCooperativeMatrixMulAddKHR-10060
  For OpCooperativeMatrixMulAddKHR, if the VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT::cooperativeMatrixProperties2 feature is not enabled, the operands **must** match a supported VkCooperativeMatrixPropertiesKHR structure as defined in VkCooperativeMatrixPropertiesKHR Multiply-Add Matching
- VUID-RuntimeSpirv-cooperativeMatrixWorkgroupScope-10164
  The scope of all OpTypeCooperativeMatrixKHR **must** not be VkScopeKHR::VK_SCOPE_WORKGROUP_KHR
- VUID-RuntimeSpirv-cooperativeMatrixProperties2-13382
  For OpTypeCooperativeMatrixKHR, if the VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT::cooperativeMatrixProperties2 feature is enabled then the type **must** either match one of the supported VkCooperativeMatrixPropertiesKHR structures as defined in

VkCooperativeMatrixPropertiesKHR Type Matching, or match one of the supported VkCooperativeMatrixProperties2EXT structures as defined in VkCooperativeMatrixProperties2EXT Type Matching

- VUID-RuntimeSpirv-cooperativeMatrixProperties2-13383
  For OpCooperativeMatrixMulAddKHR, if the VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT::cooperativeMatrixProperties2 feature is enabled then the operands **must** either match one of the supported VkCooperativeMatrixPropertiesKHR structures as defined in VkCooperativeMatrixPropertiesKHR Multiply-Add Matching, or match one of the supported VkCooperativeMatrixProperties2EXT structures as defined in VkCooperativeMatrixProperties2EXT Multiply-Add Matching
- VUID-RuntimeSpirv-cooperativeMatrixSupportedStages-08985
  OpTypeCooperativeMatrixKHR and OpCooperativeMatrix* instructions **must** not be used in shader stages not included in VkPhysicalDeviceCooperativeMatrixPropertiesKHR::cooperativeMatrixSupportedStages
- VUID-RuntimeSpirv-OpTypeCooperativeMatrixKHR-10770
  Any pipeline containing a shader with OpTypeCooperativeMatrixKHR or OpCooperativeMatrix*KHR instructions **must** be created with the VK_PIPELINE_SHADER_STAGE_CREATE_REQUIRE_FULL_SUBGROUPS_BIT flag or the shader module must be version 1.6 or greater
- VUID-RuntimeSpirv-OpTypeCooperativeMatrixKHR-10771
  Any shader object containing OpTypeCooperativeMatrixKHR or OpCooperativeMatrix*KHR instructions **must** be created with the VK_SHADER_CREATE_REQUIRE_FULL_SUBGROUPS_BIT_EXT flag or the shader module must be version 1.6 or greater
- VUID-RuntimeSpirv-OpCooperativeMatrix-12379
  If any OpCooperativeMatrix* instructions accesses a descriptor, it **must** not be a null descriptor
- VUID-RuntimeSpirv-MeshEXT-07115
  For mesh shaders using the MeshEXT Execution Model the OutputVertices OpExecutionMode **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshOutputVertices
- VUID-RuntimeSpirv-MeshEXT-07332
  For mesh shaders using the MeshEXT Execution Model the “Vertex Count” operand of OpSetMeshOutputsEXT **must** be less than or equal to OutputVertices OpExecutionMode
- VUID-RuntimeSpirv-MeshEXT-07116
  For mesh shaders using the MeshEXT Execution Model the OutputPrimitivesEXT OpExecutionMode **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshOutputPrimitives
- VUID-RuntimeSpirv-MeshEXT-07333
  For mesh shaders using the MeshEXT Execution Model the “Primitive Count” operand of OpSetMeshOutputsEXT **must** be less than or equal to OutputPrimitivesEXT OpExecutionMode
- VUID-RuntimeSpirv-MeshEXT-12333
  For mesh shaders using the MeshEXT Execution Model the index into the array of any

variable decorated with the PerPrimitiveEXT decoration **must** be less than the "Primitive Count" operand of OpSetMeshOutputsEXT

- VUID-RuntimeSpirv-TaskEXT-07117
  In task shaders using the TaskEXT Execution Model OpEmitMeshTasksEXT **must** be called exactly once under dynamically uniform conditions
- VUID-RuntimeSpirv-MeshEXT-07118
  In mesh shaders using the MeshEXT Execution Model OpSetMeshOutputsEXT **must** be called at most once under dynamically uniform conditions
- VUID-RuntimeSpirv-TaskEXT-07291
  In task shaders using the TaskEXT Execution Model the x size in LocalSize or LocalSizeId **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT ::maxTaskWorkGroupSize[0]
- VUID-RuntimeSpirv-TaskEXT-07292
  In task shaders using the TaskEXT Execution Model the y size in LocalSize or LocalSizeId **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT ::maxTaskWorkGroupSize[1]
- VUID-RuntimeSpirv-TaskEXT-07293
  In task shaders using the TaskEXT Execution Model the z size in LocalSize or LocalSizeId **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT ::maxTaskWorkGroupSize[2]
- VUID-RuntimeSpirv-TaskEXT-07294
  In task shaders using the TaskEXT Execution Model the product of x size, y size, and z size in LocalSize or LocalSizeId **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT::maxTaskWorkGroupInvocations
- VUID-RuntimeSpirv-MeshEXT-07295
  For mesh shaders using the MeshEXT Execution Model the x size in LocalSize or LocalSizeId **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT ::maxMeshWorkGroupSize[0]
- VUID-RuntimeSpirv-MeshEXT-07296
  For mesh shaders using the MeshEXT Execution Model the y size in LocalSize or LocalSizeId **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT ::maxMeshWorkGroupSize[1]
- VUID-RuntimeSpirv-MeshEXT-07297
  For mesh shaders using the MeshEXT Execution Model the z size in LocalSize or LocalSizeId **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT ::maxMeshWorkGroupSize[2]
- VUID-RuntimeSpirv-MeshEXT-07298
  For mesh shaders using the MeshEXT Execution Model the product of x size, y size, and z size in LocalSize or LocalSizeId **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupInvocations
- VUID-RuntimeSpirv-TaskEXT-07299
  In task shaders using the TaskEXT Execution Model the value of the "Group Count X" operand of OpEmitMeshTasksEXT **must** be less than or equal to

VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupCount[0]

- VUID-RuntimeSpirv-TaskEXT-07300
  In task shaders using the TaskEXT Execution Model the value of the "Group Count Y" operand of OpEmitMeshTasksEXT **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupCount[1]
- VUID-RuntimeSpirv-TaskEXT-07301
  In task shaders using the TaskEXT Execution Model the value of the "Group Count Z" operand of OpEmitMeshTasksEXT **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupCount[2]
- VUID-RuntimeSpirv-TaskEXT-07302
  In task shaders using the TaskEXT Execution Model the product of the "Group Count" operands of OpEmitMeshTasksEXT **must** be less than or equal to VkPhysicalDeviceMeshShaderPropertiesEXT::maxMeshWorkGroupTotalCount
- VUID-RuntimeSpirv-maxMeshSharedMemorySize-08754
  The sum of size in bytes for variables and padding in the Workgroup Storage Class in the MeshEXT Execution Model **must** be less than or equal to maxMeshSharedMemorySize
- VUID-RuntimeSpirv-maxMeshPayloadAndSharedMemorySize-08755
  The sum of size in bytes for variables and padding in the TaskPayloadWorkgroupEXT or Workgroup Storage Class in the MeshEXT Execution Model **must** be less than or equal to maxMeshPayloadAndSharedMemorySize
- VUID-RuntimeSpirv-maxMeshOutputMemorySize-08756
  The sum of size in bytes for variables in the Output Storage Class in the MeshEXT Execution Model **must** be less than or equal to maxMeshOutputMemorySize according to the formula in Mesh Shader Output
- VUID-RuntimeSpirv-maxMeshPayloadAndOutputMemorySize-08757
  The sum of size in bytes for variables and in the TaskPayloadWorkgroupEXT or Output Storage Class in the MeshEXT Execution Model **must** be less than or equal to maxMeshPayloadAndOutputMemorySize according to the formula in Mesh Shader Output
- VUID-RuntimeSpirv-maxTaskPayloadSize-08758
  The sum of size in bytes for variables and in the TaskPayloadWorkgroupEXT Storage Class in the TaskEXT Execution Model **must** be less than or equal to maxTaskPayloadSize
- VUID-RuntimeSpirv-maxTaskSharedMemorySize-08759
  The sum of size in bytes for variables and padding in the Workgroup Storage Class in the TaskEXT Execution Model **must** be less than or equal to maxTaskSharedMemorySize
- VUID-RuntimeSpirv-maxTaskPayloadAndSharedMemorySize-08760
  The sum of size in bytes for variables and padding in the TaskPayloadWorkgroupEXT or Workgroup Storage Class in the TaskEXT Execution Model **must** be less than or equal to maxTaskPayloadAndSharedMemorySize
- VUID-RuntimeSpirv-MeshEXT-12380
  If the MeshEXT Execution Model dynamically accesses a variable with the TaskPayloadWorkgroupEXT Storage Class, there must be a matching TaskPayloadWorkgroupEXT Storage Class variable in the TaskEXT Execution Model passed as an argument to OpEmitMeshTasksEXT

- VUID-RuntimeSpirv-OpCooperativeMatrixLoadKHR-08986
  For OpCooperativeMatrixLoadKHR and OpCooperativeMatrixStoreKHR instructions, the Pointer and Stride operands **must** be aligned to at least the lesser of 16 bytes or the natural alignment of a row or column (depending on ColumnMajor) of the matrix (where the natural alignment is the number of columns/rows multiplied by the component size)
- VUID-RuntimeSpirv-shaderSampleRateInterpolationFunctions-06325
  If the VK_KHR_portability_subset extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::shaderSampleRateInterpolationFunctions is VK_FALSE, then GLSL.std.450 fragment interpolation functions are not supported by the implementation and OpCapability **must** not be InterpolationFunction
- VUID-RuntimeSpirv-tessellationShader-06326
  If the tessellationShader feature is enabled, and the VK_KHR_portability_subset extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::tessellationIsolines is VK_FALSE, then OpExecutionMode **must** not be IsoLines
- VUID-RuntimeSpirv-tessellationShader-06327
  If the tessellationShader feature is enabled, and the VK_KHR_portability_subset extension is enabled, and VkPhysicalDevicePortabilitySubsetFeaturesKHR::tessellationPointMode is VK_FALSE, then OpExecutionMode **must** not be PointMode
- VUID-RuntimeSpirv-storageBuffer8BitAccess-06328
  If storageBuffer8BitAccess is VK_FALSE, then objects containing an 8-bit integer element **must** not have Storage Class of StorageBuffer, ShaderRecordBufferKHR, or PhysicalStorageBuffer unless shaderUntypedPointers is VK_TRUE and they are accessed in:
  - 32-bit multiples, or
  - 16-bit multiples if storageBuffer16BitAccess is VK_TRUE.
- VUID-RuntimeSpirv-uniformAndStorageBuffer8BitAccess-06329
  If uniformAndStorageBuffer8BitAccess is VK_FALSE, then objects in the Uniform Storage Class with the Block decoration **must** not have an 8-bit integer member unless shaderUntypedPointers is VK_TRUE and they are accessed in:
  - 32-bit multiples, or
  - 16-bit multiples if uniformAndStorageBuffer16BitAccess is VK_TRUE.
- VUID-RuntimeSpirv-storagePushConstant8-06330
  If storagePushConstant8 is VK_FALSE, then objects containing an 8-bit integer element **must** not have Storage Class of PushConstant unless shaderUntypedPointers is VK_TRUE and they are accessed in:
  - 32-bit multiples, or
  - 16-bit multiples if storagePushConstant16 is VK_TRUE.
- VUID-RuntimeSpirv-workgroupMemoryExplicitLayout8BitAccess-10756
  If workgroupMemoryExplicitLayout8BitAccess is VK_FALSE, then objects in the Workgroup Storage Class with the Block decoration **must** not have an 8-bit integer element unless shaderUntypedPointers is VK_TRUE and they are accessed in:
  - 32-bit multiples, or
  - 16-bit multiples if workgroupMemoryExplicitLayout16BitAccess is VK_TRUE

- VUID-RuntimeSpirv-storageBuffer16BitAccess-11161
  If storageBuffer16BitAccess is VK_FALSE, then objects containing 16-bit integer or 16-bit floating-point elements **must** not have Storage Class of StorageBuffer, ShaderRecordBufferKHR, or PhysicalStorageBuffer unless:
  - storageBuffer8BitAccess is VK_TRUE, or
  - the elements are accessed in 32-bit multiples if shaderUntypedPointers is VK_TRUE.
- VUID-RuntimeSpirv-uniformAndStorageBuffer16BitAccess-06332
  If uniformAndStorageBuffer16BitAccess is VK_FALSE, then objects in the Uniform Storage Class with the Block decoration **must** not have 16-bit integer or 16-bit floating-point members unless:
  - uniformAndStorageBuffer8BitAccess is VK_TRUE, or
  - members are accessed in 32-bit multiples and shaderUntypedPointers is VK_TRUE
- VUID-RuntimeSpirv-storagePushConstant16-06333
  If storagePushConstant16 is VK_FALSE, then objects containing 16-bit integer or 16-bit floating-point elements **must** not have Storage Class of PushConstant unless:
  - StoragePushConstant8 is VK_TRUE, or
  - elements are accessed in 32-bit multiples if shaderUntypedPointers is VK_TRUE
- VUID-RuntimeSpirv-storageInputOutput16-11162
  If storageInputOutput16 is VK_FALSE, then objects containing 16-bit integer or 16-bit floating-point elements **must** not have storage class of **Input** or **Output**
- VUID-RuntimeSpirv-None-10980
  Objects containing 8-bit integer or 8-bit floating-point elements **must** not have storage class of **Input** or **Output**
- VUID-RuntimeSpirv-workgroupMemoryExplicitLayout16BitAccess-10757
  If workgroupMemoryExplicitLayout16BitAccess is VK_FALSE, then objects in the Workgroup Storage Class with the Block decoration **must** not have an 16-bit integer or 16-bit floating-point elements unless:
  - workgroupMemoryExplicitLayout8BitAccess is VK_TRUE, or
  - elements are accessed in 32-bit multiples if shaderUntypedPointers is VK_TRUE
- VUID-RuntimeSpirv-None-06337
  shaderBufferFloat16Atomics, or shaderBufferFloat16AtomicAdd, or shaderBufferFloat16AtomicMinMax, or shaderSharedFloat16Atomics, or shaderSharedFloat16AtomicAdd, or shaderSharedFloat16AtomicMinMax **must** be enabled for 16-bit floating-point atomic operations
- VUID-RuntimeSpirv-None-06338
  shaderBufferFloat32Atomics, or shaderBufferFloat32AtomicAdd, or shaderSharedFloat32Atomics, or shaderSharedFloat32AtomicAdd, or shaderImageFloat32Atomics, or shaderImageFloat32AtomicAdd or shaderBufferFloat32AtomicMinMax, or shaderSharedFloat32AtomicMinMax, or shaderImageFloat32AtomicMinMax **must** be enabled for 32-bit floating-point atomic operations

- VUID-RuntimeSpirv-None-06339
  `shaderBufferFloat64Atomics`, or `shaderBufferFloat64AtomicAdd`, or `shaderSharedFloat64Atomics`, or `shaderSharedFloat64AtomicAdd`, or `shaderBufferFloat64AtomicMinMax`, or `shaderSharedFloat64AtomicMinMax`, **must** be enabled for 64-bit floating-point atomic operations
- VUID-RuntimeSpirv-NonWritable-06340
  If the `fragmentStoresAndAtomics` feature is not enabled, then all storage image, storage texel buffer, and storage buffer variables in the fragment stage **must** be decorated with the `NonWritable` decoration
- VUID-RuntimeSpirv-NonWritable-06341
  If the `vertexPipelineStoresAndAtomics` feature is not enabled, then all storage image, storage texel buffer, and storage buffer variables in the vertex, tessellation, and geometry stages **must** be decorated with the `NonWritable` decoration
- VUID-RuntimeSpirv-None-06342
  If `subgroupQuadOperationsInAllStages` is VK_FALSE, then quad subgroup operations **must** not be used except for in fragment and compute stages
- VUID-RuntimeSpirv-None-06343
  Group operations with subgroup scope **must** not be used if the shader stage is not in `subgroupSupportedStages`
- VUID-RuntimeSpirv-Offset-06344
  The first element of the `Offset` operand of `InterpolateAtOffset` **must** be greater than or equal to:
  $frag_{width}$ × `minInterpolationOffset`
  where $frag_{width}$ is the width of the current fragment in pixels
- VUID-RuntimeSpirv-Offset-06345
  The first element of the `Offset` operand of `InterpolateAtOffset` **must** be less than or equal to
  $frag_{width}$ × (`maxInterpolationOffset` + ULP) - ULP
  where $frag_{width}$ is the width of the current fragment in pixels and ULP = 1 / 2^`subPixelInterpolationOffsetBits`^
- VUID-RuntimeSpirv-Offset-06346
  The second element of the `Offset` operand of `InterpolateAtOffset` **must** be greater than or equal to
  $frag_{height}$ × `minInterpolationOffset`
  where $frag_{height}$ is the height of the current fragment in pixels
- VUID-RuntimeSpirv-Offset-06347
  The second element of the `Offset` operand of `InterpolateAtOffset` **must** be less than or equal to
  $frag_{height}$ × (`maxInterpolationOffset` + ULP) - ULP
  where $frag_{height}$ is the height of the current fragment in pixels and ULP = 1 / 2^`subPixelInterpolationOffsetBits`^
- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06348
  For `OpRayQueryInitializeKHR` instructions, all components of the `RayOrigin` and `RayDirection` operands **must** be finite floating-point values

- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06349
  For OpRayQueryInitializeKHR instructions, the RayTmin and RayTmax operands **must** be non-negative floating-point values
- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06350
  For OpRayQueryInitializeKHR instructions, the RayTmin operand **must** be less than or equal to the RayTmax operand
- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06351
  For OpRayQueryInitializeKHR instructions, RayOrigin, RayDirection, RayTmin, and RayTmax operands **must** not contain NaNs
- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06352
  For OpRayQueryInitializeKHR instructions, Acceleration Structure **must** be an acceleration structure built as a top-level acceleration structure
- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06889
  For OpRayQueryInitializeKHR instructions, the Rayflags operand **must** not contain both SkipTrianglesKHR and SkipAABBsKHR
- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06890
  For OpRayQueryInitializeKHR instructions, the Rayflags operand **must** not contain more than one of SkipTrianglesKHR, CullBackFacingTrianglesKHR, and CullFrontFacingTrianglesKHR
- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06891
  For OpRayQueryInitializeKHR instructions, the Rayflags operand **must** not contain more than one of OpaqueKHR, NoOpaqueKHR, CullOpaqueKHR, and CullNoOpaqueKHR
- VUID-RuntimeSpirv-OpRayQueryGenerateIntersectionKHR-06353
  For OpRayQueryGenerateIntersectionKHR instructions, Hit T **must** satisfy the condition RayTmin ≤ Hit T ≤ RayTmax, where RayTmin is equal to the value returned by OpRayQueryGetRayTMinKHR with the same ray query object, and RayTmax is equal to the value of OpRayQueryGetIntersectionTKHR for the current committed intersection with the same ray query object
- VUID-RuntimeSpirv-RayTracingOpacityMicromapKHR-11542
  If the RayTracingOpacityMicromapKHR capability is not enabled, any Rayflags operand provided to an instruction **must** not contain ForceOpacityMicromap2StateKHR
- VUID-RuntimeSpirv-flags-08761
  For OpRayQueryGetIntersectionTriangleVertexPositionsKHR instructions, Acceleration Structure **must** have been built with VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_BIT_KHR in flags
- VUID-RuntimeSpirv-OpTraceRayKHR-06355
  For OpTraceRayKHR instructions, all components of the RayOrigin and RayDirection operands **must** be finite floating-point values
- VUID-RuntimeSpirv-OpTraceRayKHR-06356
  For OpTraceRayKHR instructions, the RayTmin and RayTmax operands **must** be non-negative floating-point values
- VUID-RuntimeSpirv-OpTraceRayKHR-06552
  For OpTraceRayKHR instructions, the Rayflags operand **must** not contain both

SkipTrianglesKHR and SkipAABBsKHR

- VUID-RuntimeSpirv-OpTraceRayKHR-06892
  For OpTraceRayKHR instructions, the Rayflags operand **must** not contain more than one of SkipTrianglesKHR, CullBackFacingTrianglesKHR, and CullFrontFacingTrianglesKHR
- VUID-RuntimeSpirv-OpTraceRayKHR-06893
  For OpTraceRayKHR instructions, the Rayflags operand **must** not contain more than one of OpaqueKHR, NoOpaqueKHR, CullOpaqueKHR, and CullNoOpaqueKHR
- VUID-RuntimeSpirv-OpTraceRayKHR-06553
  For OpTraceRayKHR instructions, if the Rayflags operand contains SkipTrianglesKHR, the pipeline **must** not have been created with VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR set
- VUID-RuntimeSpirv-OpTraceRayKHR-06554
  For OpTraceRayKHR instructions, if the Rayflags operand contains SkipAABBsKHR, the pipeline **must** not have been created with VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR set
- VUID-RuntimeSpirv-OpTraceRayKHR-06357
  For OpTraceRayKHR instructions, the RayTmin operand **must** be less than or equal to the RayTmax operand
- VUID-RuntimeSpirv-OpTraceRayKHR-06358
  For OpTraceRayKHR instructions, RayOrigin, RayDirection, RayTmin, and RayTmax operands **must** not contain NaNs
- VUID-RuntimeSpirv-OpTraceRayKHR-06359
  For OpTraceRayKHR instructions, Acceleration Structure **must** be an acceleration structure built as a top-level acceleration structure
- VUID-RuntimeSpirv-OpReportIntersectionKHR-06998
  The value of the “Hit Kind” operand of OpReportIntersectionKHR **must** be in the range [0,127]
- VUID-RuntimeSpirv-OpTraceRayKHR-11855
  For modules which contain OpHitObjectTraceRayEXT, OpHitObjectTraceReorderExecuteEXT, OpHitObjectExecuteShaderEXT, OpHitObjectReorderExecuteShaderEXT, or OpTraceRayKHR instructions that declare a variable in the RayPayloadKHR Storage Class, all shaders which may be invoked as part of that *shader call* **must** declare an identical variable in the IncomingRayPayloadKHR Storage Class
- VUID-RuntimeSpirv-OpHitObjectTraceRayEXT-11878
  For OpHitObjectTraceRayEXT, and OpHitObjectTraceReorderExecuteEXT instructions, all components of the RayOrigin and RayDirection operands **must** be finite floating-point values
- VUID-RuntimeSpirv-OpHitObjectTraceRayEXT-11879
  For OpHitObjectTraceRayEXT, and OpHitObjectTraceReorderExecuteEXT instructions, the RayTmin and RayTmax operands **must** be non-negative floating-point values
- VUID-RuntimeSpirv-OpHitObjectTraceRayEXT-11880
  For OpHitObjectTraceRayEXT, and OpHitObjectTraceReorderExecuteEXT instructions, the RayTmin operand **must** be less than or equal to the RayTmax operand

- VUID-RuntimeSpirv-OpHitObjectTraceRayEXT-11881
  For OpHitObjectTraceRayEXT, and OpHitObjectTraceReorderExecuteEXT instructions, RayOrigin, RayDirection, RayTmin, and RayTmax operands **must** not contain NaNs
- VUID-RuntimeSpirv-OpHitObjectTraceRayEXT-11882
  For OpHitObjectTraceRayEXT, and OpHitObjectTraceReorderExecuteEXT instructions the time operand **must** be between 0.0 and 1.0
- VUID-RuntimeSpirv-OpHitObjectTraceRayEXT-11883
  For OpHitObjectTraceRayEXT, and OpHitObjectTraceReorderExecuteEXT instructions, the Rayflags operand **must** not contain both SkipTrianglesKHR and SkipAABBsKHR
- VUID-RuntimeSpirv-OpHitObjectTraceRayEXT-11884
  For OpHitObjectTraceRayEXT, and OpHitObjectTraceReorderExecuteEXT instructions, the Rayflags operand **must** not contain more than one of SkipTrianglesKHR, CullBackFacingTrianglesKHR, and CullFrontFacingTrianglesKHR
- VUID-RuntimeSpirv-OpHitObjectTraceRayEXT-11885
  For OpHitObjectTraceRayEXT, and OpHitObjectTraceReorderExecuteEXT instructions, the Rayflags operand **must** not contain more than one of OpaqueKHR, NoOpaqueKHR, CullOpaqueKHR, and CullNoOpaqueKHR
- VUID-RuntimeSpirv-OpHitObjectTraceRayEXT-11886
  For OpHitObjectTraceRayEXT, and OpHitObjectTraceReorderExecuteEXT instructions, if the Rayflags operand contains SkipTrianglesKHR, the pipeline **must** not have been created with VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR set
- VUID-RuntimeSpirv-OpHitObjectTraceRayEXT-11887
  For OpHitObjectTraceRayEXT, and OpHitObjectTraceReorderExecuteEXT instructions, if the Rayflags operand contains SkipAABBsKHR, the pipeline **must** not have been created with VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR set
- VUID-RuntimeSpirv-x-06429
  In compute shaders using the GLCompute Execution Model the x size in LocalSize or LocalSizeId **must** be less than or equal to VkPhysicalDeviceLimits ::maxComputeWorkGroupSize[0]
- VUID-RuntimeSpirv-y-06430
  In compute shaders using the GLCompute Execution Model the y size in LocalSize or LocalSizeId **must** be less than or equal to VkPhysicalDeviceLimits ::maxComputeWorkGroupSize[1]
- VUID-RuntimeSpirv-z-06431
  In compute shaders using the GLCompute Execution Model the z size in LocalSize or LocalSizeId **must** be less than or equal to VkPhysicalDeviceLimits ::maxComputeWorkGroupSize[2]
- VUID-RuntimeSpirv-x-06432
  In compute shaders using the GLCompute Execution Model the product of x size, y size, and z size in LocalSize or LocalSizeId **must** be less than or equal to VkPhysicalDeviceLimits ::maxComputeWorkGroupInvocations
- VUID-RuntimeSpirv-LocalSizeId-06434
  If Execution Mode LocalSizeId is used, maintenance4 **must** be enabled

- VUID-RuntimeSpirv-maintenance4-06817
  If the maintenance4 feature is not enabled, any vector type output interface variables **must** not have a higher Component Count than a matching vector type input interface variable
- VUID-RuntimeSpirv-OpEntryPoint-08743
  Any user-defined variables shared between the OpEntryPoint of two shader stages, and declared with Input as its Storage Class for the subsequent shader stage, **must** have all Location slots and Component words declared in the preceding shader stage's OpEntryPoint with Output as the Storage Class
- VUID-RuntimeSpirv-OpEntryPoint-07754
  Any user-defined variables between the OpEntryPoint of two shader stages **must** have the same type and width for each Component
- VUID-RuntimeSpirv-OpVariable-08746
  Any variable, Block-decorated OpTypeStruct, or Block-decorated OpTypeStruct members shared between the OpEntryPoint of two shader stages **must** have matching decorations as defined in interface matching
- VUID-RuntimeSpirv-Workgroup-06530
  The sum of size in bytes for variables and padding in the Workgroup Storage Class in the GLCompute Execution Model **must** be less than or equal to maxComputeSharedMemorySize
- VUID-RuntimeSpirv-shaderZeroInitializeWorkgroupMemory-06372
  If the shaderZeroInitializeWorkgroupMemory feature is not enabled, any variable with Workgroup as its Storage Class **must** not have an Initializer operand
- VUID-RuntimeSpirv-Offset-10213
  If the maintenance8 feature is not enabled, image operand Offset **must** only be used with OpImage*Gather instructions
- VUID-RuntimeSpirv-Size-11165
  The Size operand of OpCopyMemorySized **must** be a multiple of 4
  - If 16-bit storage is enabled for the storage classes of both the Target and Source operands the Size operand may instead be a multiple of 2
  - If 8-bit storage is enabled for the storage classes of both the Target and Source operands the Size operand may instead be any value
- VUID-RuntimeSpirv-OpTypeUntypedPointerKHR-11166
  Any memory access made using an OpTypeUntypedPointerKHR must have an alignment that satisfies Offset and Stride Assignment
- VUID-RuntimeSpirv-OpImage-06376
  If an OpImage*Gather operation has an image operand of Offset, ConstOffset, or ConstOffsets the offset value **must** be greater than or equal to minTexelGatherOffset
- VUID-RuntimeSpirv-OpImage-06377
  If an OpImage*Gather operation has an image operand of Offset, ConstOffset, or ConstOffsets the offset value **must** be less than or equal to maxTexelGatherOffset
- VUID-RuntimeSpirv-OpImageSample-06435
  If an OpImageSample* or OpImageFetch* operation has an image operand of Offset or ConstOffset then the offset value **must** be greater than or equal to minTexelOffset

- VUID-RuntimeSpirv-OpImageSample-06436
  If an `OpImageSample*` or `OpImageFetch*` operation has an image operand of `Offset` or `ConstOffset` then the offset value **must** be less than or equal to `maxTexelOffset`
- VUID-RuntimeSpirv-samples-08725
  If an `OpTypeImage` has an `MS` operand 0, its bound image **must** have been created with VkImageCreateInfo::`samples` as VK_SAMPLE_COUNT_1_BIT
- VUID-RuntimeSpirv-samples-08726
  If an `OpTypeImage` has an `MS` operand 1, its bound image **must** not have been created with VkImageCreateInfo::`samples` as VK_SAMPLE_COUNT_1_BIT
- VUID-RuntimeSpirv-SampleRateShading-06378
  If the subpass description contains VK_SUBPASS_DESCRIPTION_FRAGMENT_REGION_BIT_EXT, then the SPIR-V fragment shader Capability `SampleRateShading` **must** not be enabled
- VUID-RuntimeSpirv-SubgroupUniformControlFlowKHR-06379
  The `Execution Mode SubgroupUniformControlFlowKHR` **must** not be applied to an entry point unless the `shaderSubgroupUniformControlFlow` feature is enabled, the corresponding shader stage bit is set in `subgroupSupportedStages`, and the entry point does not execute any *invocation repack instructions*
- VUID-RuntimeSpirv-OpEntryPoint-08727
  Each `OpEntryPoint` **must** not have more than one variable decorated with `InputAttachmentIndex` per image aspect of the attachment image bound to it, either explicitly or implicitly as described by input attachment interface
- VUID-RuntimeSpirv-minSampleShading-08731
  If sample shading is enabled and `minSampleShading` is 1.0, the `sample` operand of any `OpColorAttachmentReadEXT`, `OpDepthAttachmentReadEXT`, or `OpStencilAttachmentReadEXT` operation **must** evaluate to the value of the coverage index for any given fragment invocation
- VUID-RuntimeSpirv-minSampleShading-08732
  If sample shading is enabled and any of the `OpColorAttachmentReadEXT`, `OpDepthAttachmentReadEXT`, or `OpStencilAttachmentReadEXT` operations are used, then `minSampleShading` **must** be 1.0
- VUID-RuntimeSpirv-MeshEXT-09218
  In mesh shaders using the `MeshEXT` or `MeshNV Execution Model` and the `OutputPoints Execution Mode`, if the `maintenance5` feature is not enabled, and if the number of output points is greater than 0, a `PointSize` decorated variable **must** be written to for each output point
- VUID-RuntimeSpirv-maintenance5-09190
  If the `maintenance5` feature is enabled and a `PointSize` decorated variable is written to, all execution paths **must** write to a `PointSize` decorated variable
- VUID-RuntimeSpirv-maintenance5-10934
  If the `maintenance5` feature is enabled and a `PointSize` decorated variable is written to for any output vertex in the `Geometry Execution Model`, all execution paths for all output vertices **must** write to a `PointSize` decorated variable

- VUID-RuntimeSpirv-OpControlBarrierArriveEXT-13558
  If a OpControlBarrierArriveEXT or OpControlBarrierWaitEXT instruction is executed with an execution scope equal to Workgroup, its execution **must** be inside *uniform control flow* within the invocation group
- VUID-RuntimeSpirv-OpControlBarrierWaitEXT-13559
  If a OpControlBarrierWaitEXT instruction is executed, a OpControlBarrierArriveEXT instruction with the same execution scope **must** have been executed previously
- VUID-RuntimeSpirv-OpControlBarrierWaitEXT-13560
  If a OpControlBarrierWaitEXT instruction is executed, another OpControlBarrierWaitEXT instruction with the same execution scope **must** not have been previously executed prior to the last executed OpControlBarrierArriveEXT instruction with the same execution scope
- VUID-RuntimeSpirv-OpControlBarrierArriveEXT-13561
  If a OpControlBarrierArriveEXT instruction is executed, another OpControlBarrierArriveEXT instruction with the same execution scope **must** not have been previously executed unless a OpControlBarrierWaitEXT instruction with the same execution scope has been executed after it
- VUID-RuntimeSpirv-OpControlBarrierArriveEXT-13562
  If the module uses OpControlBarrierArriveEXT or OpControlBarrierWaitEXT, the sum of size in bytes for variables and padding in the Workgroup Storage Class in the GLCompute Execution Model **must** be less than or equal to maxComputeSharedMemorySize minus splitBarrierReservedSharedMemory
- VUID-RuntimeSpirv-shaderSubgroupRotateClustered-09566
  If shaderSubgroupRotateClustered is VK_FALSE, then the ClusterSize operand to OpGroupNonUniformRotateKHR **must** not be used
- VUID-RuntimeSpirv-protectedNoFault-09645
  If protectedNoFault is not supported, the Storage Class of the PhysicalStorageBuffer **must** not be used if the buffer being accessed is protected
- VUID-RuntimeSpirv-meshAndTaskShaderDerivatives-10153
  If meshAndTaskShaderDerivatives is VK_FALSE, the DerivativeGroupLinearKHR and DerivativeGroupQuadsKHR execution modes **must** not be used in the MeshEXT, MeshNV, TaskEXT, or TaskNV Execution Model
- VUID-RuntimeSpirv-OpTypeSampler-12203
  If a variable with type OpTypeSampler is declared in the Shader Resource Interface, it **must** not be backed by a sampler that requires sampler $Y'C_BC_R$ conversion
- VUID-RuntimeSpirv-OpTypeImage-12204
  If a variable with type OpTypeImage is declared in the Shader Resource Interface, it **must** not be backed by an image view that requires sampler $Y'C_BC_R$ conversion
- VUID-RuntimeSpirv-None-12205
  If an image view or sampler that requires sampler $Y'C_BC_R$ conversion is accessed in a shader, it **must** be determined by constant integral expressions
- VUID-RuntimeSpirv-OpTypeSampledImage-12206
  If an OpTypeSampledImage variable backed by an image view and sampler that require sampler $Y'C_BC_R$ conversion is statically used in a shader, it **must** only be used with

OpImageSample*, OpImageSparseSample*, or OpImage instructions

- VUID-RuntimeSpirv-OpTypeImage-12207
  If a OpTypeImage variable backed by an image view that requires sampler $Y'C_BC_R$ conversion is statically used in a shader, it **must** only be used with OpImageQueryLevels or OpImageQuerySizeLod instructions
- VUID-RuntimeSpirv-ConstOffset-10718
  If an OpTypeSampledImage variable backed by an image view and sampler that require sampler $Y'C_BC_R$ conversion is statically used in a shader with a sampling instruction, it **must** not use the ConstOffset or Offset operands
- VUID-RuntimeSpirv-OpTypeSampledImage-12208
  If an OpTypeSampledImage variable backed by an image view and sampler that require sampler $Y'C_BC_R$ conversion is statically used in a shader, the image view and sampler **must** have been created with an *identically defined* VkSamplerYcbcrConversion set via VkSamplerYcbcrConversionInfo
- VUID-RuntimeSpirv-shaderUniformBufferUnsizedArray-11806
  If the shaderUniformBufferUnsizedArray feature is not enabled, OpTypeRuntimeArray **must** not be used for the last member of a Block-decorated OpTypeStruct in the Uniform storage Storage Class
- VUID-RuntimeSpirv-None-10824
  If the maintenance9 feature is not enabled, the Base operand of any OpBitCount, OpBitReverse, OpBitFieldInsert, OpBitFieldSExtract, or OpBitFieldUExtract instruction **must** be a 32-bit integer scalar or a vector of 32-bit integers
- VUID-RuntimeSpirv-maxShaderBindingTableRecordIndex-11888
  The SBT Index passed in to OpHitObjectSetShaderBindingTableRecordIndexEXT **must** be less than or equal to VkPhysicalDeviceRayTracingInvocationReorderPropertiesEXT ::maxShaderBindingTableRecordIndex
- VUID-RuntimeSpirv-None-10834
  Buffer indexing calculations **must** not wrap 32 bits if the pipeline or shader was not created with the VK_PIPELINE_CREATE_2_64_BIT_INDEXING_BIT_EXT or VK_SHADER_CREATE_64_BIT_INDEXING_BIT_EXT flags and the entry point does not set the Shader64BitIndexingEXT execution mode
- VUID-RuntimeSpirv-OpArrayLength-11807
  OpArrayLength and OpUntypedArrayLengthKHR result type **must** be a 32-bit integer type if the pipeline or shader was not compiled with the VK_PIPELINE_CREATE_2_64_BIT_INDEXING_BIT_EXT or VK_SHADER_CREATE_64_BIT_INDEXING_BIT_EXT flags and the entry point does not set the Shader64BitIndexingEXT execution mode
- VUID-RuntimeSpirv-OpConstantSizeOfEXT-11475
  The result type of OpConstantSizeOfEXT **must** not be a 64-bit integer type if the pipeline or shader was not compiled with the VK_PIPELINE_CREATE_2_64_BIT_INDEXING_BIT_EXT or VK_SHADER_CREATE_64_BIT_INDEXING_BIT_EXT flags and the entry point does not set the Shader64BitIndexingEXT execution mode
- VUID-RuntimeSpirv-longVector-12296
  If the longVector feature is enabled, the Component Count of any vector type **must** be less

than or equal to maxVectorComponents

- VUID-RuntimeSpirv-samplerDescriptorAlignment-11348
  If the Result Type operand of OpLoad is OpTypeSampler, and Pointer is derived from a variable decorated with SamplerHeapEXT, it **must** be at an offset from the base that is a multiple of samplerDescriptorAlignment
- VUID-RuntimeSpirv-imageDescriptorAlignment-11349
  If the Result Type operand of OpLoad is OpTypeImage, and Pointer is derived from a variable decorated with ResourceHeapEXT, it **must** be at an offset from the base that is a multiple of imageDescriptorAlignment
- VUID-RuntimeSpirv-imageDescriptorAlignment-11383
  If the Image operand of OpImageTexelPointer or OpUntypedImageTexelPointerEXT is derived from a variable decorated with ResourceHeapEXT, it **must** be at an offset from the base that is a multiple of imageDescriptorAlignment
- VUID-RuntimeSpirv-Result-11350
  If the Result Type operand of OpLoad is OpTypeAccelerationStructureKHR and Pointer is derived from a variable decorated with ResourceHeapEXT, it **must** be at an offset from the base that is a multiple of bufferDescriptorAlignment
- VUID-RuntimeSpirv-bufferDescriptorAlignment-11384
  The Buffer operand of OpBufferPointerEXT **must** be at an offset from the base of the ResourceHeapEXT that is a multiple of bufferDescriptorAlignment
- VUID-RuntimeSpirv-samplerDescriptorAlignment-11476
  If a OpTypeSampler member of a struct is decorated with Offset or OffsetIdEXT, the Byte Offset value must be a multiple of samplerDescriptorAlignment
- VUID-RuntimeSpirv-imageDescriptorAlignment-11477
  If a OpTypeImage member of a struct is decorated with Offset or OffsetIdEXT, the Byte Offset value must be a multiple of imageDescriptorAlignment
- VUID-RuntimeSpirv-bufferDescriptorAlignment-11478
  If a OpTypeBufferEXT member of a struct is decorated with Offset or OffsetIdEXT, the Byte Offset value must be a multiple of bufferDescriptorAlignment
- VUID-RuntimeSpirv-bufferDescriptorAlignment-11479
  If a OpTypeAccelerationStructureKHR member of a struct is decorated with Offset or OffsetIdEXT, the Byte Offset value must be a multiple of bufferDescriptorAlignment
- VUID-RuntimeSpirv-DescriptorSet-11385
  If an instruction accesses memory through any resource with a DescriptorSet and Binding that are mapped using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT, the resource through which that memory is accessed **must** be determined by constant integral expressions
- VUID-RuntimeSpirv-source-11387
  All possible values of a variable pointer to a resource that are not OpConstantNull **must**

either all be pointers to resources with DescriptorSet and Binding decorations with the same mapping types as specified by VkDescriptorSetAndBindingMappingEXT::source, or all be pointers to resources without DescriptorSet and Binding decorations

- VUID-RuntimeSpirv-DescriptorSet-11388
  All possible values of a variable pointer to a resource **must** not be resources with a DescriptorSet and Binding that are mapped using VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_PUSH_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT, VK_DESCRIPTOR_MAPPING_SOURCE_INDIRECT_ADDRESS_EXT, or VK_DESCRIPTOR_MAPPING_SOURCE_RESOURCE_HEAP_DATA_EXT
- VUID-RuntimeSpirv-Result-11340
  If the Result Type operand of OpLoad is OpTypeSampler, and Pointer is derived from a variable decorated with SamplerHeapEXT, it **must** correspond to loading a descriptor matching one of the descriptor types listed in Heap Resource Type Correspondence for OpTypeSampler
- VUID-RuntimeSpirv-Result-11341
  If the Result Type operand of OpLoad is OpTypeImage, and Pointer is derived from a variable decorated with ResourceHeapEXT, it **must** correspond to loading a descriptor matching one of the descriptor types listed in Heap Resource Type Correspondence for OpTypeImage
- VUID-RuntimeSpirv-Result-11342
  The Buffer operand of OpBufferPointerEXT **must** correspond to accessing a descriptor matching one of the descriptor types listed in Heap Resource Type Correspondence for OpTypePointer
- VUID-RuntimeSpirv-Result-11343
  If the Result Type operand of OpLoad is OpTypeImage, and Pointer is derived from a variable decorated with ResourceHeapEXT, the Image Format and Sampled Type of that OpTypeImage **must** correspond to a descriptor with a VkFormat that matches as described by Compatibility Between SPIR-V Image Formats and Vulkan Formats and Image Format and Type Matching Between SPIR-V Image Formats and Vulkan Formats
- VUID-RuntimeSpirv-Result-11345
  If the Result Type operand of OpLoad is OpTypeImage, and Pointer is derived from a variable decorated with ResourceHeapEXT, the operands of that OpTypeImage **must** correspond to a descriptor with operands that match as described by Heap Image Operand Correspondence and Heap Image View Type Dimensionality
- VUID-RuntimeSpirv-Image-11379
  If the Image operand of OpImageTexelPointer or OpUntypedImageTexelPointerEXT is derived from a variable decorated with ResourceHeapEXT, it **must** correspond to accessing a descriptor matching one of the descriptor types listed in Heap Resource Type Correspondence for the OpTypeImage pointed to
- VUID-RuntimeSpirv-Image-11380
  If the Image operand of OpImageTexelPointer or OpUntypedImageTexelPointerEXT is derived from a variable decorated with ResourceHeapEXT, the Image Format and Sampled Type of the OpTypeImage pointed to **must** correspond to a descriptor with a VkFormat that matches as

described by Compatibility Between SPIR-V Image Formats and Vulkan Formats and Image Format and Type Matching

- VUID-RuntimeSpirv-Image-11382
  If the Image operand of OpImageTexelPointer or OpUntypedImageTexelPointerEXT is derived from a variable decorated with ResourceHeapEXT, the operands of that OpTypeImage **must** correspond to a descriptor with operands that match as described by Heap Image Operand Correspondence and Heap Image View Type Dimensionality
- VUID-RuntimeSpirv-maxPushDataSize-12455
  The size of the PushConstant Storage Class block **must** be less than or equal to maxPushDataSize

Precision and Operation of SPIR-V Instructions

The following rules apply to operations on all floating-point values:

- Positive and negative infinities and positive and negative zeros are generated as dictated by IEEE 754 for the specified encoding, but subject to the precisions allowed by each operation for a given encoding, as specified in Precision of Individual Operations.
- Signaling NaNs are not required to be generated and exceptions are never raised. Signaling NaN **may** be converted to quiet NaNs values by any floating-point instruction.
- The set of operations OpPhi, OpSelect, OpFunctionCall, OpReturnValue, OpVectorExtractDynamic, OpVectorInsertDynamic, OpVectorShuffle, OpCompositeConstruct, OpCompositeExtract, OpCompositeInsert, OpTranspose, OpCopyObject, OpCopyLogical, OpCopyMemory, OpGroupNonUniformBroadcast, OpGroupNonUniformBroadcastFirst, OpGroupNonUniformShuffle, OpGroupNonUniformShuffleXor, OpGroupNonUniformShuffleUp, OpGroupNonUniformShuffleDown, OpGroupNonUniformQuadBroadcast, OpGroupNonUniformQuadSwap, OpGroupNonUniformRotateKHR, OpCooperativeMatrixLoadKHR, OpCooperativeMatrixStoreKHR, OpAtomicLoad, OpAtomicStore, OpAtomicExchange, OpStore, and OpLoad are referred to as *bit-preserving operations*.
- The floating-point environment used for an instruction can be determined as follows:
  - If the SPIR-V specifies it explicitly using the FPFastMath decoration or FPFastMathDefault Execution Mode then that is used.
  - If the environment is not specified in the SPIR-V then it is determined as follows:
    - If the operation is not decorated NoContraction then the flags AllowContract, AllowReassoc, AllowRecip, and AllowTransform are assumed.
    - If any of the following conditions are true then the flags NSZ, NotInf, and NotNaN are assumed:
      - The entry point does not use the Execution Mode SignedZeroInfNanPreserve with a bit-width corresponding to one of the operands or to the result type.
      - The operation is not a bit-preserving operation and is not one of OpFConvert, OpFNegate, OpFAdd, OpFSub, OpFMul, OpFDiv, OpIsNan, OpIsInf, OpVectorTimesScalar, OpMatrixTimesScalar, OpVectorTimesMatrix, OpMatrixTimesVector, OpMatrixTimesMatrix, OpOuterProduct, OpDot, OpFOrdEqual, OpFUnordEqual, OpFOrdNotEqual, OpFUnordNotEqual, OpFOrdLessThan, OpFUnordLessThan, OpFOrdGreaterThan, OpFUnordGreaterThan,

OpFOrdLessThanEqual, OpFUnordLessThanEqual, OpFOrdGreaterThanEqual, OpFUnordGreaterThanEqual, OpGroupNonUniformAllEqual, OpGroupNonUniformFMin, OpGroupNonUniformFMax, OpAtomicCompareExchange, OpAtomicCompareExchangeWeak, OpDPdx, OpDPdy, OpFwidth, OpDPdxFine, OpDPdyFine, OpFwidthFine, OpDPdxCoarse, OpDPdyCoarse, or OpFwidthCoarse.

        - The operation is an OpLoad from the Input Storage Class in the fragment shader stage.

- All bit-preserving operations and the following instructions **must** not flush denormalized values: OpConstant, OpConstantComposite, OpSpecConstant, OpSpecConstantComposite, and OpBitcast.
- Denormalized values are supported.
    - By default any denormalized floating-point value input into a shader or potentially generated by any instruction (except those listed above) or any extended instructions for GLSL in a shader **may** be flushed to zero.
    - Denormalized floating-point values whose type use the Float8E4M3EXT or Float8E5M2EXT FP Encoding **must** be preserved during conversions to IEEE 754 binary 16 floating-point values.
    - If the entry point is declared with the DenormFlushToZero Execution Mode then for the affected instructions the denormalized result **must** be flushed to zero and the denormalized operands **may** be flushed to zero. Denormalized values obtained via unpacking an integer into a vector of values with smaller bit width and interpreting those values as floating-point numbers **must** be flushed to zero.
    - If the entry point is declared with the DenormPreserve Execution Mode then for the affected instructions the denormalized result and any denormalized operands **must** be preserved. Denormalized values obtained via unpacking an integer into a vector of values with smaller bit width and interpreting those values as floating-point numbers **must** be preserved.
    - When denormal values are being flushed, the result of an operation **may** be considered denormal whenever the infinitely precise result is non-zero and of smaller magnitude than the smallest normal value, even if rounding would otherwise give a normal result.
    - The following core SPIR-V instructions **must** respect the DenormFlushToZero Execution Mode: OpSpecConstantOp (with opcode OpFConvert), OpFConvert, OpFNegate, OpFAdd, OpFSub, OpFMul, OpFmaKHR, OpFDiv, OpFRem, OpFMod, OpVectorTimesScalar, OpMatrixTimesScalar, OpVectorTimesMatrix, OpMatrixTimesVector, OpMatrixTimesMatrix, OpOuterProduct, OpDot, OpGroupNonUniformFMin, OpGroupNonUniformFMax, OpAtomicFAddEXT, OpAtomicFMinEXT, OpAtomicFMaxEXT, OpDPdx, OpDPdy, OpFwidth, OpDPdxFine, OpDPdyFine OpFwidthFine, OpDPdxCoarse, OpDPdyCoarse, OpFwidthCoarse; and the following extended instructions for GLSL: Round, RoundEven, Trunc, FAbs, Floor, Ceil, Fract, Radians, Degrees, Sin, Cos, Tan, Asin, Acos, Atan, Sinh, Cosh, Tanh, Asinh, Acosh, Atanh, Atan2, Pow, Exp, Log, Exp2, Log2, Sqrt, InverseSqrt, Determinant, MatrixInverse, Modf, ModfStruct, FMin, FMax, FClamp, FMix, Step, SmoothStep, Fma, UnpackHalf2x16, Length, Distance, Cross, Normalize, FaceForward, Reflect, Refract, NMin, NMax, and NClamp.
    - The following core SPIR-V instructions **must** respect the DenormPreserve Execution Mode for floating-point values with an IEEE 754 encoding: OpSpecConstantOp, OpFConvert, OpFNegate, OpFAdd, OpFSub, OpFMul, OpFmaKHR, OpVectorTimesScalar, OpMatrixTimesScalar, OpVectorTimesMatrix, OpMatrixTimesVector, OpMatrixTimesMatrix, OpOuterProduct, OpDot, OpFOrdEqual, OpFUnordEqual, OpFOrdNotEqual, OpFUnordNotEqual, OpFOrdLessThan, OpFUnordLessThan, OpFOrdGreaterThan, OpFUnordGreaterThan, OpFOrdLessThanEqual, OpFUnordLessThanEqual, OpFOrdGreaterThanEqual, OpFUnordGreaterThanEqual,

OpGroupNonUniformAllEqual, OpGroupNonUniformFMin, OpGroupNonUniformFMax, OpAtomicCompareExchange, OpAtomicCompareExchangeWeak, OpAtomicFAddEXT, OpAtomicFMinEXT, OpAtomicFMaxEXT, OpDPdx, OpDPdy, OpFwidth, OpDPdxFine, OpDPdyFine OpFwidthFine, OpDPdxCoarse, OpDPdyCoarse, OpFwidthCoarse; and the following extended instructions for GLSL: FAbs, FSign, Radians, Degrees, FMin, FMax, FClamp, FMix, Fma, PackHalf2x16, PackDouble2x32, UnpackHalf2x16, UnpackDouble2x32, NMin, NMax, and NClamp.

The precision of double-precision instructions is at least that of single precision.

The precision of individual operations is defined in Precision of Individual Operations. Subject to the constraints below, however, implementations **may** reorder or combine operations, resulting in expressions exhibiting different precisions than might be expected from the constituent operations.

NOTE The interpretation of objects with a OpTypeFloat type with MXInt8EXT encoding that have the bit pattern 0x80 is undefined.

Evaluation of Expressions

Implementations **may** rearrange floating-point operations using any of the mathematical properties governing the expressions in precise arithmetic, even where the floating- point operations do not share these properties. This includes, but is not limited to associativity and distributivity, and **may** involve a different number of rounding steps than would occur if the operations were not rearranged. In shaders that use the SignedZeroInfNanPreserve Execution Mode the values **must** be preserved if they are generated after any rearrangement but the Execution Mode does not change which rearrangements are valid. This rearrangement **can** be prevented for particular operations by using the NoContraction decoration.

NOTE For example, in the absence of the NoContraction decoration implementations are allowed to implement a + b - a and $\frac{a \times b}{a}$ as b. The SignedZeroInfNanPreserve does not prevent these transformations, even though they may overflow to infinity or NaN when evaluated in floating-point.

NOTE If the NoContraction decoration is applied then operations may not be rearranged, so, for example, a + a - a must account for possible overflow to infinity. If infinities are not preserved then the expression may be replaced with a despite the NoContraction decoration, since the replacement is exact when overflow does not occur. If both NoContraction and SignedZeroInfNanPreserve are used then the result must be infinity for sufficiently large a.

Precision of Individual Operations

The precision of individual operations is defined either in terms of rounding (correctly rounded), as an error bound in ULP, or as inherited from a formula as follows:

Correct Result

Operations that are described as returning the “correct result” will return the infinitely precise result which, due to the nature of the operation, will not need rounding.

Correctly Rounded

Operations described as “correctly rounded” will return the infinitely precise result, x, rounded so as to be representable in floating-point. If the entry point is declared with the `RoundingModeRTE` or the `RoundingModeRTZ Execution Mode` then this is done according to IEEE 754 “roundTiesToEven” or “roundTowardZero” rounding directions, respectively. These execution modes do not affect operations on floating-point values with the following encodings:

- `BFloat16KHR`
- `Float8E4M3EXT`
- `Float8E5M2EXT`
- `Float6E2M3EXT`
- `Float6E3M2EXT`
- `Float4E2M1EXT`
- `Float8UnsignedE8M0EXT`
- `MXInt8EXT`

Otherwise, they are rounded with implementation-defined rounding mode.

Correctly Rounded with Implementation-Defined Rounding Mode

Operations described as “correctly rounded with implementation-defined rounding mode” will return the infinitely precise result, x, rounded so as to be representable in floating-point. If x is exactly representable then x will be returned. Otherwise, either the floating-point value closest to and no less than x or the value closest to and no greater than x will be returned. Which value is chosen is implementation-defined.

ULP

Where an error bound of n ULP (units in the last place) is given, for an operation with infinitely precise result x the value returned **must** be in the range [x - n × ulp(x), x + n × ulp(x)]. The function ulp(x) is defined as follows:

> If there exist non-equal, finite floating-point numbers a and b such that a ≤ x ≤ b then ulp(x) is the minimum possible distance between such numbers, $ulp(x) = \min_{a,\ b} |b - a|$. If such numbers do not exist then ulp(x) is defined to be the difference between the two non-equal, finite floating-point numbers nearest to x.

Where the range of allowed return values includes any value of magnitude larger than that of the largest representable finite floating-point number, operations **may**, additionally, return either an infinity of the appropriate sign or the finite number with the largest magnitude of the appropriate sign. If the infinitely precise result of the operation is not mathematically defined then the value returned is poison.

Inherited From ...

Where an operation's precision is described as being inherited from a formula, the result returned **must** be at least as accurate as the result of computing an approximation to x using a formula equivalent to the given formula applied to the supplied inputs. Specifically, the formula given may be transformed using the mathematical associativity, commutativity, and distributivity of the operators involved to yield an equivalent formula. The SPIR-V precision rules, when applied to each such formula and the given input values, define a range of permitted values. If NaN is one of the permitted values then the operation may return any result, otherwise let the largest permitted value in any of the ranges be F_{max} and the smallest be F_{min}. The operation **must** return a value in the range [x - E, x + E] where $E = \max(|x - F_{min}|, |x - F_{max}|)$. If the entry point is declared with the `DenormPreserve` `Execution Mode`, then denormals **must** be preserved throughout the formula. Otherwise, any intermediate denormal value(s) while evaluating the formula **may** be flushed to zero. If the entry point is declared with the `DenormFlushToZero` execution mode, then denormal final results **must** be flushed to zero.

The precision of all instructions on float values with the `BFloat16KHR` encoding is required to be correctly rounded with implementation defined rounding mode.

For IEEE754 half- (16 bit) and single- (32 bit) precision instructions, precisions are **required** to be at least as follows:

Table 101. Precision of Core SPIR-V Instructions

Instruction	Single precision, unless decorated with RelaxedPrecision	Half precision
`OpFNegate`	Correct result.	
`OpFAdd`	Correctly rounded.	
`OpFSub`	Correctly rounded.	
`OpFMul`, `OpVectorTimesScalar`, `OpMatrixTimesScalar`	Correctly rounded.	
`OpMatrixTimesVector`	Inherited from $\sum_{j=0}^{n-1} M_{ij} \times v_j$.	
`OpVectorTimesMatrix`	Inherited from $\sum_{j=0}^{n-1} v_j \times M_{ji}$.	
`OpMatrixTimesMatrix`	Inherited from $\sum_{k=0}^{n-1} M_{ik} \times N_{kj}$.	
`OpOuterProduct`	Correctly rounded.	
`OpFmaKHR`	Correctly rounded.	
`OpDot`(x, y)	Inherited from $\sum_{i=0}^{n-1} x_i \times y_i$.	
`OpIsNan`, `OpIsInf`	Correct result.	
`OpFOrdEqual`, `OpFUnordEqual`	Correct result.	
`OpFOrdNotEqual`, `OpFUnordNotEqual`	Correct result.	
`OpFOrdLessThan`, `OpFUnordLessThan`	Correct result.	

Instruction	Single precision, unless decorated with RelaxedPrecision	Half precision
OpFOrdGreaterThan, OpFUnordGreaterThan	Correct result.	
OpFOrdLessThanEqual, OpFUnordLessThanEqual	Correct result.	
OpFOrdGreaterThanEqual, OpFUnordGreaterThanEqual	Correct result.	
OpGroupNonUniformAllEqual	Correct result.	
OpGroupNonUniformFMin, OpGroupNonUniformFMax	Correct result.	
OpFDiv(x,y)	2.5 ULP for $\|y\| = 0$ or $\|y\|$ in the range $[2^{-126}, 2^{126}]$.	2.5 ULP for $\|y\| = 0$ or $\|y\|$ in the range $[2^{-14}, 2^{14}]$.
OpFRem(x,y)	Inherited from x - y × trunc(x/y).	
OpFMod(x,y)	Inherited from x - y × floor(x/y).	
OpQuantizeToF16	Correctly rounded with implementation defined rounding mode.	
conversions between types	Correctly rounded.	
OpAtomicCompareExchange, OpAtomicCompareExchangeWeak	Correct result.	
OpAtomicFAddEXT	Return value correct result, value in memory correctly rounded.	
OpAtomicFMinEXT, OpAtomicFMaxEXT	Correct result.	
OpDPdx, OpDPdy, OpFwidth	Correctly rounded.	
OpDPdxCoarse, OpDPdyCoarse, OpFwidthCoarse	Correctly rounded.	
OpDPdxFine, OpDPdyFine, OpFwidthFine	Correctly rounded.	

NOTE The OpFRem and OpFMod instructions use cheap approximations of remainder, and the error can be large due to the discontinuity in trunc() and floor(). This can produce mathematically unexpected results in some cases, such as FMod(x,x) computing x rather than 0, and can also cause the result to have a different sign than the infinitely precise result.

Table 102. Precision of GLSL.std.450 Instructions

Instruction	Single precision, unless decorated with RelaxedPrecision	Half precision
fma()	Inherited from OpFMul followed by OpFAdd.	
exp(x), exp2(x)	$3 + 2 \times \|x\|$ ULP.	$1 + 2 \times \|x\|$ ULP.

Instruction	Single precision, unless decorated with RelaxedPrecision	Half precision
log(), log2()	3 ULP outside the range $[0.5, 2.0]$. Absolute error $< 2^{-21}$ inside the range $[0.5, 2.0]$.	3 ULP outside the range $[0.5, 2.0]$. Absolute error $< 2^{-7}$ inside the range $[0.5, 2.0]$.
pow(x, y)	Inherited from exp2(y × log2(x)).	
sqrt()	Inherited from 1.0 / inversesqrt().	
inversesqrt()	2 ULP.	
radians(x)	Inherited from $x \times C_{\pi_180}$, where C_{π_180} is a correctly rounded approximation to $\frac{\pi}{180}$.	
degrees(x)	Inherited from $x \times C_{180_\pi}$, where C_{180_π} is a correctly rounded approximation to $\frac{180}{\pi}$.	
sin()	Absolute error $\leq 2^{-11}$ inside the range $[-\pi, \pi]$.	Absolute error $\leq 2^{-7}$ inside the range $[-\pi, \pi]$.
cos()	Absolute error $\leq 2^{-11}$ inside the range $[-\pi, \pi]$.	Absolute error $\leq 2^{-7}$ inside the range $[-\pi, \pi]$.
tan()	Inherited from $\frac{\sin()}{\cos()}$.	
asin(x)	Inherited from $\text{atan2}(x, sqrt(1.0 - x \times x))$.	
acos(x)	Inherited from $\text{atan2}(sqrt(1.0 - x \times x), x)$.	
atan(), atan2()	4096 ULP	5 ULP.
sinh(x)	Inherited from $(\exp(x) - \exp(-x)) \times 0.5$.	
cosh(x)	Inherited from $(\exp(x) + \exp(-x)) \times 0.5$.	
tanh()	Inherited from $\frac{\sinh()}{\cosh()}$.	
asinh(x)	Inherited from $\log(x + sqrt(x \times x + 1.0))$.	
acosh(x)	Inherited from $\log(x + sqrt(x \times x - 1.0))$.	
atanh(x)	Inherited from $\log(\frac{1.0 + x}{1.0 - x}) \times 0.5$.	
frexp()	Correct result.	
ldexp()	Correctly rounded.	
length(x)	Inherited from $sqrt(dot(x, x))$.	
distance(x, y)	Inherited from $length(x - y)$.	
cross()	Inherited from OpFSub(OpFMul, OpFMul).	
normalize(x)	Inherited from $x \times inversesqrt(dot(x, x))$.	
faceforward(N, I, NRef)	Inherited from dot(NRef, I) < 0.0 ? N : -N.	
reflect(x, y)	Inherited from x - 2.0 × dot(y, x) × y.	
refract(I, N, eta)	Inherited from k < 0.0 ? 0.0 : eta × I - (eta × dot(N, I) + sqrt(k)) × N, where k = 1 - eta × eta × (1.0 - dot(N, I) × dot(N, I)).	

Instruction	Single precision, unless decorated with RelaxedPrecision	Half precision
round	Correctly rounded.	
roundEven	Correctly rounded.	
trunc	Correctly rounded.	
fabs	Correct result.	
fsign	Correct result.	
floor	Correctly rounded.	
ceil	Correctly rounded.	
fract	Correctly rounded.	
modf	Correctly rounded.	
fmin	Correct result.	
fmax	Correct result.	
fclamp	Correct result.	
fmix(x, y, a)	Inherited from $x \times (1.0 - a) + y \times a$.	
step	Correctly rounded.	
smoothStep(edge0, edge1, x)	Inherited from $t \times t \times (3.0 - 2.0 \times t)$, where $t = clamp(\frac{x - edge0}{edge1 - edge0}, 0.0, 1.0)$.	
nmin	Correct result.	
nmax	Correct result.	
nclamp	Correct result.	
packHalf2x16	Correctly rounded with implementation defined rounding mode.	

GLSL.std.450 extended instructions specifically defined in terms of the above instructions inherit the above errors. GLSL.std.450 extended instructions not listed above and not defined in terms of the above have implementation-defined precision.

If the maintenance8 feature is not enabled and if either operand to OpSRem and OpSMod instructions is negative the result is poison.

NOTE While the OpSRem and OpSMod instructions are supported by the Vulkan environment, If the maintenance8 feature is not enabled, they require non-negative values and thus do not enable additional functionality beyond what OpUMod provides.

OpCooperativeMatrixMulAddKHR performs its operations in an implementation-dependent order and internal precision.

Buffer Indexing Calculations

Pipelines and shaders **may** be compiled to use 32-bit integer arithmetic to compute byte offsets (e.g. based on OpAccessChain instructions) within a physical storage buffer or buffer, limiting the accessible range to 4GB. For physical storage buffers, the base address is a 64-bit value and the implicit base plus offset calculation uses 64-bit addition.

NOTE	Note that OpAccessChain indices are always treated as signed, so a 32-bit index can only address 2GB if the ArrayStride is 1.

The offset calculations that have the 4GB limit include:

- OpAccessChain and OpPtrAccessChain - the total offset summed over all indices multiplied by strides, including spanning multiple access chain instructions that lead to a given memory access.
- OpCooperativeMatrixLoadKHR and OpCooperativeMatrixStoreKHR instructions computing an offset based on element, row, and stride parameters.

64-Bit Indexing

The application **can** enable 64-bit indexing either by setting the flags VK_PIPELINE_CREATE_2_64_BIT_INDEXING_BIT_EXT or VK_SHADER_CREATE_64_BIT_INDEXING_BIT_EXT or by setting the Execution Mode Shader64BitIndexingEXT on the entry point, which will cause the implementation to use 64 bits of range for those addressing calculations.

If 64-bit indexing is enabled, the calculation of index multiplied by ArrayStride for a runtime-sized array when evaluating an access chain in StorageBuffer or PhysicalStorageBuffer will have 64 bits of range. Similarly, the Element times ArrayStride calculation for OpPtrAccessChain and OpUntypedPtrAccessChainKHR will have 64 bits of range.

NOTE	The size of a structure type or sized array type in a storage buffer is still limited to 32 bits for various practical reasons, such as that Offset and ArrayStride decorations use 32-bit literals. Only runtime-sized arrays can access beyond the 32-bit limit.

NOTE	Enabling 64-bit indexing may incur a runtime cost.

If 64-bit indexing is enabled, OpArrayLength and OpUntypedArrayLengthKHR **can** have a 64-bit result type.

If 64-bit indexing is enabled, the row/col times stride calculations in OpCooperativeMatrixLoadKHR, and OpCooperativeMatrixStoreKHR are changed to have 64 bits of range. i.e. Pointer[row*(uint64_t)Stride] and Pointer[col*(uint64_t)Stride]. The stride itself is still limited to 32 bits.

Signedness of SPIR-V Image Accesses

SPIR-V associates a signedness with all integer image accesses. This is required in certain parts of the SPIR-V and the Vulkan image access pipeline to ensure defined results. The signedness is determined from a combination of the access instruction's `Image Operands` and the underlying image's `Sampled Type` as follows:

1. If the instruction's `Image Operands` contains the `SignExtend` operand then the access is signed.
2. If the instruction's `Image Operands` contains the `ZeroExtend` operand then the access is unsigned.
3. Otherwise, the image accesses signedness matches that of the `Sampled Type` of the `OpTypeImage` being accessed.

Image Format and Type Matching

When specifying the `Image Format` of an `OpTypeImage`, the converted bit width and type, as shown in the table below, **must** match the `Sampled Type`. The signedness **must** match the signedness of any access to the image.

> **NOTE** Formatted accesses are always converted from a shader readable type to the resource's format or vice versa via Texel Decode for reads and Texel Encode for writes. As such, the bit width and format below do not necessarily match 1:1 with what might be expected for some formats.

For a given `Image Format`, the `Sampled Type` **must** be the type described in the *Type* column of the below table, with its `Literal Width` set to that in the *Bit Width* column. Every access that is made to the image **must** have a signedness equal to that in the *Signedness* column (where applicable).

Image Format	Type-Declaration instructions	Bit Width	Signedness
`Unknown`	Any	Any	Any

Image Format	Type-Declaration instructions	Bit Width	Signedness
Rgba32f	OpTypeFloat	32	N/A
Rg32f			
R32f			
Rgba16f			
Rg16f			
R16f			
Rgba16			
Rg16			
R16			
Rgba16Snorm			
Rg16Snorm			
R16Snorm			
Rgb10A2			
R11fG11fB10f			
Rgba8			
Rg8			
R8			
Rgba8Snorm			
Rg8Snorm			
R8Snorm			

Image Format	Type-Declaration instructions	Bit Width	Signedness
Rgba32i	OpTypeInt	32	1
Rg32i			
R32i			
Rgba16i			
Rg16i			
R16i			
Rgba8i			
Rg8i			
R8i			
Rgba32ui			0
Rg32ui			
R32ui			
Rgba16ui			
Rg16ui			
R16ui			
Rgb10a2ui			
Rgba8ui			
Rg8ui			
R8ui			
R64i	OpTypeInt	64	1
R64ui			0

The *SPIR-V Type* is defined by an instruction in SPIR-V, declared with the Type-Declaration Instruction, Bit Width, and Signedness from above.

Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types

SPIR-V `Image Dim` values are compatible with VkImageView `viewType` values as defined below:

Table 103. SPIR-V and Vulkan ImageView Dimension Compatibility

SPIR-V Image Dim	Compatible Vulkan ImageView viewTypes
1D	VK_IMAGE_VIEW_TYPE_1D, VK_IMAGE_VIEW_TYPE_1D_ARRAY
2D	VK_IMAGE_VIEW_TYPE_2D, VK_IMAGE_VIEW_TYPE_2D_ARRAY
3D	VK_IMAGE_VIEW_TYPE_3D
Cube	VK_IMAGE_VIEW_TYPE_CUBE, VK_IMAGE_VIEW_TYPE_CUBE_ARRAY

Compatibility Between SPIR-V Image Formats and Vulkan Formats

SPIR-V `Image Format` values are compatible with VkFormat values as defined below:

Table 104. SPIR-V and Vulkan Image Format Compatibility

SPIR-V Image Format	Compatible Vulkan Format
`Unknown`	Any
`R8`	VK_FORMAT_R8_UNORM
`R8Snorm`	VK_FORMAT_R8_SNORM
`R8ui`	VK_FORMAT_R8_UINT
`R8i`	VK_FORMAT_R8_SINT
`Rg8`	VK_FORMAT_R8G8_UNORM
`Rg8Snorm`	VK_FORMAT_R8G8_SNORM
`Rg8ui`	VK_FORMAT_R8G8_UINT
`Rg8i`	VK_FORMAT_R8G8_SINT
`Rgba8`	VK_FORMAT_R8G8B8A8_UNORM
`Rgba8Snorm`	VK_FORMAT_R8G8B8A8_SNORM
`Rgba8ui`	VK_FORMAT_R8G8B8A8_UINT
`Rgba8i`	VK_FORMAT_R8G8B8A8_SINT
`Rgb10A2`	VK_FORMAT_A2B10G10R10_UNORM_PACK32
`Rgb10a2ui`	VK_FORMAT_A2B10G10R10_UINT_PACK32
`R16`	VK_FORMAT_R16_UNORM
`R16Snorm`	VK_FORMAT_R16_SNORM
`R16ui`	VK_FORMAT_R16_UINT
`R16i`	VK_FORMAT_R16_SINT
`R16f`	VK_FORMAT_R16_SFLOAT
`Rg16`	VK_FORMAT_R16G16_UNORM
`Rg16Snorm`	VK_FORMAT_R16G16_SNORM
`Rg16ui`	VK_FORMAT_R16G16_UINT
`Rg16i`	VK_FORMAT_R16G16_SINT
`Rg16f`	VK_FORMAT_R16G16_SFLOAT
`Rgba16`	VK_FORMAT_R16G16B16A16_UNORM
`Rgba16Snorm`	VK_FORMAT_R16G16B16A16_SNORM
`Rgba16ui`	VK_FORMAT_R16G16B16A16_UINT

SPIR-V Image Format	Compatible Vulkan Format
Rgba16i	VK_FORMAT_R16G16B16A16_SINT
Rgba16f	VK_FORMAT_R16G16B16A16_SFLOAT
R32ui	VK_FORMAT_R32_UINT
R32i	VK_FORMAT_R32_SINT
R32f	VK_FORMAT_R32_SFLOAT
Rg32ui	VK_FORMAT_R32G32_UINT
Rg32i	VK_FORMAT_R32G32_SINT
Rg32f	VK_FORMAT_R32G32_SFLOAT
Rgba32ui	VK_FORMAT_R32G32B32A32_UINT
Rgba32i	VK_FORMAT_R32G32B32A32_SINT
Rgba32f	VK_FORMAT_R32G32B32A32_SFLOAT
R64ui	VK_FORMAT_R64_UINT
R64i	VK_FORMAT_R64_SINT
R11fG11fB10f	VK_FORMAT_B10G11R11_UFLOAT_PACK32

Ray Query Precision and Operation

The values returned by OpRayQueryGetIntersectionTriangleVertexPositionsKHR are transformed by the geometry transform, which is performed at standard floating-point precision, but without a specifically defined order of floating-point operations to perform the matrix multiplication.

Appendix B: Memory Model

NOTE	This memory model describes synchronizations provided by all implementations; however, some of the synchronizations defined require extra features to be supported by the implementation. See VkPhysicalDeviceVulkanMemoryModelFeatures.

Agent

Operation is a general term for any task that is executed on the system.

NOTE	An operation is by definition something that is executed. Thus if an instruction is skipped due to control flow, it does not constitute an operation.

Each operation is executed by a particular *agent*. Possible agents include each:

- fixed-function stage of the pipeline,
- shader invocation, or
- host thread.

Memory Location

A *memory location* identifies unique storage for 8 bits of data. Memory operations access a *set of memory locations* consisting of one or more memory locations at a time, e.g. an operation accessing a 32-bit integer in memory would read/write a set of four memory locations. Memory operations that access whole aggregates **may** access any padding bytes between elements or members, but no padding bytes at the end of the aggregate. Two sets of memory locations *overlap* if the intersection of their sets of memory locations is non-empty. A memory operation **must** not affect memory at a memory location not within its set of memory locations.

Memory locations are implicitly allocated for SPIR-V variables by the implementation. Buffers and images are explicitly mapped to memory locations allocated in VkDeviceMemory objects.

Variables with `Workgroup` storage class that are decorated with `Aliased` and that point to a block-decorated type share a set of memory locations.

Allocation

The values stored in memory locations newly allocated by the API are set to 0 if allocated with the VK_MEMORY_ALLOCATE_ZERO_INITIALIZE_BIT_EXT flag, otherwise they are undefined. For SPIR-V variables, the values are determined according to the SPIR-V specification.

At the time an allocation is created there have been no memory operations to any of its memory locations. Using an initializer for a SPIR-V variable is not considered to be a memory operation.

NOTE	For tessellation control shader output variables, a consequence of initialization not

> being considered a memory operation is that some implementations may need to insert a barrier between the initialization of the output variables and any reads of those variables.

Memory Operation

For an operation A and memory location M:

- A *reads* M if and only if the data stored in M is an input to A.
- A *writes* M if and only if the data output from A is stored to M.
- A *accesses* M if and only if it either reads or writes (or both) M.

> **NOTE** A write whose value is the same as what was already in those memory locations is still considered to be a write and has all the same effects.

Reference

A *reference* is an object that a particular agent **can** use to access a set of memory locations. On the host, a reference is a host virtual address. On the device, a reference is:

- The descriptor that a variable is bound to, for variables in Image, Uniform, or StorageBuffer storage classes. If the variable is an array (or array of arrays, etc.) then each element of the array **may** be a unique reference.
- The address range for a buffer in `PhysicalStorageBuffer` storage class, where the base of the address range is queried with vkGetBufferDeviceAddress and the length of the range is the size of the buffer.
- A single common reference for all variables with `Workgroup` storage class that are decorated with `Aliased` and that point to a block-decorated type.
- The variable itself for all other variables in the `Workgroup` storage class.
- A single common reference for all variables with `Workgroup` storage class that point to a block-decorated type.
- The variable itself for non-block-decorated type variables in `Workgroup` storage class.
- The variable itself for variables in other storage classes.

Two memory accesses through distinct references **may** require availability and visibility operations as defined below.

Program-Order

A *dynamic instance* of an instruction is defined in SPIR-V (https://registry.khronos.org/spir-v/specs/unified1/SPIRV.html#DynamicInstance) as a way of referring to a particular execution of a static instruction. Program-order is an ordering on dynamic instances of instructions executed by a single shader invocation:

- (Basic block): If instructions A and B are in the same basic block, and A is listed in the module before B, then the n'th dynamic instance of A is program-ordered before the n'th dynamic instance of B.
- (Branch): The dynamic instance of a branch or switch instruction is program-ordered before the dynamic instance of the OpLabel instruction to which it transfers control.
- (Call entry): The dynamic instance of an `OpFunctionCall` instruction is program-ordered before the dynamic instances of the `OpFunctionParameter` instructions and the body of the called function.
- (Call exit): The dynamic instance of the instruction following an `OpFunctionCall` instruction is program-ordered after the dynamic instance of the return instruction executed by the called function.
- (Transitive Closure): If dynamic instance A of any instruction is program-ordered before dynamic instance B of any instruction and B is program-ordered before dynamic instance C of any instruction then A is program-ordered before C.
- (Complete definition): No other dynamic instances are program-ordered.

For instructions executed on the host, the source language defines the program-order relation (e.g. as "sequenced-before").

Shader Call Related

Shader-call-related is an equivalence relation on invocations defined as the symmetric and transitive closure of:

- A is shader-call-related to B if A is created by an shader call instruction executed by B.

Shader Call Order

Shader-call-order is a partial order on dynamic instances of instructions executed by invocations that are shader-call-related:

- (Program order): If dynamic instance A is program-ordered before B, then A is shader-call-ordered before B.
- (Shader call entry): If A is a dynamic instance of an shader call instruction and B is a dynamic instance executed by an invocation that is created by A, then A is shader-call-ordered before B.
- (Shader call exit): If A is a dynamic instance of an shader call instruction, B is the next dynamic instance executed by the same invocation, and C is a dynamic instance executed by an invocation that is created by A, then C is shader-call-ordered before B.
- (Transitive closure): If A is shader-call-ordered-before B and B is shader-call-ordered-before C, then A is shader-call-ordered-before C.
- (Complete definition): No other dynamic instances are shader-call-ordered.

Scope

Atomic and barrier instructions include scopes which identify sets of shader invocations that **must** obey the requested ordering and atomicity rules of the operation, as defined below.

The various scopes are described in detail in the Shaders chapter.

Atomic Operation

An *atomic operation* on the device is any SPIR-V operation whose name begins with `OpAtomic`. An atomic operation on the host is any operation performed with an std::atomic typed object.

Each atomic operation has a memory scope and a semantics. Informally, the scope determines which other agents it is atomic with respect to, and the semantics constrains its ordering against other memory accesses.

Device atomic operations have explicit scopes and semantics. Each host atomic operation implicitly uses the `CrossDevice` scope, and uses a memory semantics equivalent to a C++ std::memory_order value of relaxed, acquire, release, acq_rel, or seq_cst.

Two atomic operations A and B are *potentially-mutually-ordered* if and only if all of the following are true:

- They access the same set of memory locations.
- They use the same reference.
- A is in the instance of B's memory scope.
- B is in the instance of A's memory scope.
- A and B are not the same operation (irreflexive).

Two atomic operations A and B are *mutually-ordered* if and only if they are potentially-mutually-ordered and any of the following are true:

- A and B are both device operations.
- A and B are both host operations.

NOTE	If two atomic operations are not mutually-ordered, and if their sets of memory locations overlap, then each **must** be synchronized against the other as if they were non-atomic operations.

Scoped Modification Order

For a given atomic write A, all atomic writes that are mutually-ordered with A occur in an order known as A's *scoped modification order*. A's scoped modification order relates no other operations.

NOTE	Invocations outside the instance of A's memory scope **may** observe the values at A's set of memory locations becoming visible to it in an order that disagrees with the

scoped modification order.

NOTE It is valid to have non-atomic operations or atomics in a different scope instance to the same set of memory locations, as long as they are synchronized against each other as if they were non-atomic (if they are not, it is treated as a data race). That means this definition of A's scoped modification order could include atomic operations that occur much later, after intervening non-atomics. That is a bit non-intuitive, but it helps to keep this definition simple and non-circular.

Memory Semantics

Non-atomic memory operations, by default, **may** be observed by one agent in a different order than they were written by another agent.

Atomics and some synchronization operations include *memory semantics*, which are flags that constrain the order in which other memory accesses (including non-atomic memory accesses and availability and visibility operations) performed by the same agent **can** be observed by other agents, or **can** observe accesses by other agents.

Host instructions that include semantics are some std::atomic methods and memory fences. Device instructions that include semantics are `OpAtomic*`, `OpControlBarrier`, `OpControlBarrierArriveEXT`, `OpControlBarrierWaitEXT`, `OpMemoryBarrier`, and `OpMemoryNamedBarrier`.

Vulkan supports the following memory semantics:

- Relaxed: No constraints on order of other memory accesses.
- Acquire: A memory read with this semantic performs an *acquire operation*. A memory barrier with this semantic is an *acquire barrier*.
- Release: A memory write with this semantic performs a *release operation*. A memory barrier with this semantic is a *release barrier*.
- AcquireRelease: A memory read-modify-write operation with this semantic performs both an acquire operation and a release operation, and inherits the limitations on ordering from both of those operations. A memory barrier with this semantic is both a release and acquire barrier.

SequentiallyConsistent memory semantics is not supported and **must** not be used.

NOTE SPIR-V does not support "consume" semantics on the device.

The memory semantics operand **can** also include *storage class semantics* flags which indicate the storage classes constrained by the synchronization. Vulkan supports the following SPIR-V storage class semantics flags:

- UniformMemory
- WorkgroupMemory
- ImageMemory
- OutputMemory

The remaining storage class semantics flags (SubgroupMemory, CrossWorkgroupMemory, and AtomicCounterMemory) are ignored.

Each SPIR-V memory operation accesses a single storage class. Semantics in synchronization operations can include a combination of storage classes.

The UniformMemory storage class semantic applies to accesses to memory in the PhysicalStorageBuffer, `ShaderRecordBufferKHR`, Uniform and StorageBuffer storage classes. The WorkgroupMemory storage class semantic applies to accesses to memory in the Workgroup storage class. The ImageMemory storage class semantic applies to accesses to memory in the Image storage class. The OutputMemory storage class semantic applies to accesses to memory in the Output storage class.

NOTE	Informally, these constraints limit how memory operations can be reordered, and these limits apply not only to the order of accesses as performed in the agent that executes the instruction, but also to the order the effects of writes become visible to all other agents within the same instance of the instruction's memory scope.
NOTE	Release and acquire operations in different threads **can** act as synchronization operations, to guarantee that writes that happened before the release are visible after the acquire. (This is not a formal definition, just an Informative forward reference.)
NOTE	The OutputMemory storage class semantic is only useful in tessellation control shaders, which is the only execution model where output variables are shared between invocations.

The memory semantics operand **can** also include availability and visibility flags, which apply availability and visibility operations as described in availability and visibility. The availability/visibility flags are:

- MakeAvailable: Semantics **must** be Release or AcquireRelease. Performs an availability operation before the release operation or barrier.
- MakeVisible: Semantics **must** be Acquire or AcquireRelease. Performs a visibility operation after the acquire operation or barrier.

The specifics of these operations are defined in Availability and Visibility Semantics.

Host atomic operations **may** support a different list of memory semantics and synchronization operations, depending on the host architecture and source language.

Release Sequence

After an atomic operation A performs a release operation on a set of memory locations M, the *release sequence headed by A* is the longest continuous subsequence of A's scoped modification order that consists of:

- the atomic operation A as its first element

- atomic read-modify-write operations on M by any agent

NOTE	The atomics in the last bullet **must** be mutually-ordered with A by virtue of being in A's scoped modification order.

NOTE	This intentionally omits "atomic writes to M performed by the same agent that performed A", which is present in the corresponding C++ definition.

Synchronizes-With

Synchronizes-with is a relation between operations, where each operation is either an atomic operation or a memory barrier (aka fence on the host).

If A and B are atomic operations, then A synchronizes-with B if and only if all of the following are true:

- A performs a release operation
- B performs an acquire operation
- A and B are mutually-ordered
- B reads a value written by A or by an operation in the release sequence headed by A

`OpControlBarrier`, `OpControlBarrierArriveEXT`, `OpControlBarrierWaitEXT`, `OpMemoryBarrier`, and `OpMemoryNamedBarrier` are *memory barrier* instructions in SPIR-V. `OpControlBarrierArriveEXT` is a *control barrier arrival* and `OpControlBarrierWaitEXT` is a *control barrier wait*.

If A is a release barrier and B is an atomic operation that performs an acquire operation, then A synchronizes-with B if and only if all of the following are true:

- there exists an atomic write X (with any memory semantics)
- A is program-ordered before X
- X and B are mutually-ordered
- B reads a value written by X or by an operation in the release sequence headed by X
  - If X is relaxed, it is still considered to head a hypothetical release sequence for this rule
- A and B are in the instance of each other's memory scopes
- X's storage class is in A's semantics.

If A is an atomic operation that performs a release operation and B is an acquire barrier, then A synchronizes-with B if and only if all of the following are true:

- there exists an atomic read X (with any memory semantics)
- X is program-ordered before B
- X and A are mutually-ordered
- X reads a value written by A or by an operation in the release sequence headed by A

- A and B are in the instance of each other's memory scopes
- X's storage class is in B's semantics.

If A is a release barrier and B is an acquire barrier, then A synchronizes-with B if all of the following are true:

- there exists an atomic write X (with any memory semantics)
- A is program-ordered before X
- there exists an atomic read Y (with any memory semantics)
- Y is program-ordered before B
- X and Y are mutually-ordered
- Y reads the value written by X or by an operation in the release sequence headed by X
  - If X is relaxed, it is still considered to head a hypothetical release sequence for this rule
- A and B are in the instance of each other's memory scopes
- X's and Y's storage class is in A's and B's semantics.
  - NOTE: X and Y **must** have the same storage class, because they are mutually ordered.

If A is a release barrier, B is an acquire barrier, and C is a control barrier (where A **can** equal C, and B **can** equal C), then A synchronizes-with B if all of the following are true:

- A is program-ordered before (or equals) C
- C is program-ordered before (or equals) B
- A and B are in the instance of each other's memory scopes
- A and B are in the instance of C's execution scope

NOTE	This is similar to the barrier-barrier synchronization above, but with a control barrier filling the role of the relaxed atomics.

If A is a release barrier, B is an acquire barrier, C is a control barrier arrival, and D is a control barrier wait (where A **can** equal C, and B **can** equal D), then A synchronizes-with B if all of the following are true:

- A is program-ordered before (or equals) C
- D is program-ordered before (or equals) B
- C and D have the same execution scope
- A and B are in the instance of each other's memory scopes
- A and B are in the instance of C's execution scope

Let F be an ordering of fragment shader invocations, such that invocation F_1 is ordered before invocation F_2 if and only if F_1 and F_2 overlap as described in Fragment Shader Interlock and F_1 executes the interlocked code before F_2.

If A is an `OpEndInvocationInterlockEXT` instruction and B is an `OpBeginInvocationInterlockEXT`

instruction, then A synchronizes-with B if the agent that executes A is ordered before the agent that executes B in F. A and B are both considered to have `FragmentInterlock` memory scope and semantics of UniformMemory and ImageMemory, and A is considered to have Release semantics and B is considered to have Acquire semantics.

NOTE `OpBeginInvocationInterlockEXT` and `OpBeginInvocationInterlockEXT` do not perform implicit availability or visibility operations. Usually, shaders using fragment shader interlock will declare the relevant resources as `coherent` to get implicit per-instruction availability and visibility operations.

If A is a release barrier and B is an acquire barrier, then A synchronizes-with B if all of the following are true:

- A is shader-call-ordered-before B
- A and B are in the instance of each other's memory scopes

No other release and acquire barriers synchronize-with each other.

System-Synchronizes-With

System-synchronizes-with is a relation between arbitrary operations on the device or host. Certain operations system-synchronize-with each other, which informally means the first operation occurs before the second and that the synchronization is performed without using application-visible memory accesses.

If there is an execution dependency between two operations A and B, then the operation in the first synchronization scope system-synchronizes-with the operation in the second synchronization scope.

NOTE This covers all Vulkan synchronization primitives, including device operations executing before a synchronization primitive is signaled, wait operations happening before subsequent device operations, signal operations happening before host operations that wait on them, and host operations happening before vkQueueSubmit. The list is spread throughout the synchronization chapter, and is not repeated here.

System-synchronizes-with implicitly includes all storage class semantics and has `CrossDevice` scope.

If A system-synchronizes-with B, we also say A is *system-synchronized-before* B and B is *system-synchronized-after* A.

Private vs. Non-Private

By default, non-atomic memory operations are treated as *private*, meaning such a memory operation is not intended to be used for communication with other agents.

Non-private memory operations are intended to be used for communication with other agents. Atomic operations are always considered *non-private*.

Memory operations with the `NonPrivatePointer`, `NonPrivateTexel` bit set are treated as *non-private*

More precisely, for private memory operations to be Location-Ordered between distinct agents requires using system-synchronizes-with rather than shader-based synchronization. Private memory operations still obey program-order.

Inter-Thread-Happens-Before

Let SC be a non-empty set of storage class semantics. Then (using template syntax) operation A *inter-thread-happens-before*<SC> operation B if and only if any of the following is true:

- A system-synchronizes-with B
- A synchronizes-with B, and both A and B have all of SC in their semantics
- A is an operation on memory in a storage class in SC or that has all of SC in its semantics, B is a release barrier or release atomic with all of SC in its semantics, and A is program-ordered before B
- A is an acquire barrier or acquire atomic with all of SC in its semantics, B is an operation on memory in a storage class in SC or that has all of SC in its semantics, and A is program-ordered before B
- A and B are both host operations and A inter-thread-happens-before B as defined in the host language specification
- A inter-thread-happens-before<SC> some X and X inter-thread-happens-before<SC> B

Happens-Before

Operation A *happens-before* operation B if and only if any of the following is true:

- A is program-ordered before B
- A inter-thread-happens-before<SC> B for some set of storage classes SC

Happens-after is defined similarly.

NOTE	Unlike C++, happens-before is not always sufficient for a write to be visible to a read. Additional availability and visibility operations **may** be required for writes to be visible-to other memory accesses.
NOTE	Happens-before is not transitive, but each of program-order and inter-thread-happens-before<SC> are transitive. These can be thought of as covering the “single-threaded” case and the “multi-threaded” case, and it is not necessary (and not valid) to form chains between the two.

Availability and Visibility

Availability and *visibility* are states of a write operation, which (informally) track how far the write has permeated the system, i.e. which agents and references are able to observe the write.

Availability state is per *memory domain*. Visibility state is per (agent,reference) pair. Availability and visibility states are per-memory location for each write.

Memory domains are named according to the agents whose memory accesses use the domain. Domains used by shader invocations are organized hierarchically into multiple smaller memory domains which correspond to the different scopes. Each memory domain is considered the *dual* of a scope, and vice versa. The memory domains defined in Vulkan include:

- *host* - accessible by host agents
- *device* - accessible by all device agents for a particular device
- *shader* - accessible by shader agents for a particular device, corresponding to the `Device` scope
- *queue family instance* - accessible by shader agents in a single queue family, corresponding to the `QueueFamily` scope.
- *fragment interlock instance* - accessible by fragment shader agents that overlap, corresponding to the `FragmentInterlock` scope.
- *shader call instance* - accessible by shader agents that are shader-call-related, corresponding to the `ShaderCallKHR` scope.
- *workgroup instance* - accessible by shader agents in the same workgroup, corresponding to the `Workgroup` scope.
- *subgroup instance* - accessible by shader agents in the same subgroup, corresponding to the `Subgroup` scope.

The memory domains are nested in the order listed above, except for shader call instance domain, with memory domains later in the list nested in the domains earlier in the list. The shader call instance domain is at an implementation-dependent location in the list, and is nested according to that location. The shader call instance domain is not broader than the queue family instance domain.

NOTE

Memory domains do not correspond to storage classes or device-local and host-local VkDeviceMemory allocations, rather they indicate whether a write can be made visible only to agents in the same subgroup, same workgroup, in overlapping fragment shader invocation, in shader-call-related ray tracing invocation, in any shader invocation, or anywhere on the device, or host. The shader, queue family instance, fragment interlock instance, shader call instance, workgroup instance, and subgroup instance domains are only used for shader-based availability/visibility operations, in other cases writes can be made available from/visible to the shader via the device domain.

Availability operations, *visibility operations*, and *memory domain operations* alter the state of the write operations that happen-before them, and which are included in their *source scope* to be available or visible to their *destination scope*.

- For an availability operation, the source scope is a set of (agent,reference,memory location) tuples, and the destination scope is a set of memory domains.
- For a memory domain operation, the source scope is a memory domain and the destination

scope is a memory domain.

- For a visibility operation, the source scope is a set of memory domains and the destination scope is a set of (agent,reference,memory location) tuples.

How the scopes are determined depends on the specific operation. Availability and memory domain operations expand the set of memory domains to which the write is available. Visibility operations expand the set of (agent,reference,memory location) tuples to which the write is visible.

Recall that availability and visibility states are per-memory location, and let W be a write operation to one or more locations performed by agent A via reference R. Let L be one of the locations written. (W,L) (the write W to L), is initially not available to any memory domain and only visible to (A,R,L). An availability operation AV that happens-after W and that includes (A,R,L) in its source scope makes (W,L) *available* to the memory domains in its destination scope.

A memory domain operation DOM that happens-after AV and for which (W,L) is available in the source scope makes (W,L) available in the destination memory domain.

A visibility operation VIS that happens-after AV (or DOM) and for which (W,L) is available in any domain in the source scope makes (W,L) *visible* to all (agent,reference,L) tuples included in its destination scope.

If write W_2 happens-after W, and their sets of memory locations overlap, then W will not be available/visible to all agents/references for those memory locations that overlap (and future AV/DOM/VIS ops cannot revive W's write to those locations).

Availability, memory domain, and visibility operations are treated like other non-atomic memory accesses for the purpose of memory semantics, meaning they can be ordered by release-acquire sequences or memory barriers.

An *availability chain* is a sequence of availability operations to increasingly broad memory domains, where element N+1 of the chain is performed in the dual scope instance of the destination memory domain of element N and element N happens-before element N+1. An example is an availability operation with destination scope of the workgroup instance domain that happens-before an availability operation to the shader domain performed by an invocation in the same workgroup. An availability chain AVC that happens-after W and that includes (A,R,L) in the source scope makes (W,L) *available* to the memory domains in its final destination scope. An availability chain with a single element is just the availability operation.

Similarly, a *visibility chain* is a sequence of visibility operations from increasingly narrow memory domains, where element N of the chain is performed in the dual scope instance of the source memory domain of element N+1 and element N happens-before element N+1. An example is a visibility operation with source scope of the shader domain that happens-before a visibility operation with source scope of the workgroup instance domain performed by an invocation in the same workgroup. A visibility chain VISC that happens-after AVC (or DOM) and for which (W,L) is available in any domain in the source scope makes (W,L) *visible* to all (agent,reference,L) tuples included in its final destination scope. A visibility chain with a single element is just the visibility operation.

Availability, Visibility, and Domain Operations

The following operations generate availability, visibility, and domain operations. When multiple availability/visibility/domain operations are described, they are system-synchronized-with each other in the order listed.

An operation that performs a memory dependency generates:

- If the source access mask includes VK_ACCESS_HOST_WRITE_BIT, then the dependency includes a memory domain operation from host domain to device domain.
- An availability operation with source scope of all writes in the first access scope of the dependency and a destination scope of the device domain.
- A visibility operation with source scope of the device domain and destination scope of the second access scope of the dependency.
- If the destination access mask includes VK_ACCESS_HOST_READ_BIT or VK_ACCESS_HOST_WRITE_BIT, then the dependency includes a memory domain operation from device domain to host domain.

vkFlushMappedMemoryRanges performs an availability operation, with a source scope of (agents,references) = (all host threads, all mapped memory ranges passed to the command), and destination scope of the host domain.

vkInvalidateMappedMemoryRanges performs a visibility operation, with a source scope of the host domain and a destination scope of (agents,references) = (all host threads, all mapped memory ranges passed to the command).

vkQueueSubmit performs a memory domain operation from host to device, and a visibility operation with source scope of the device domain and destination scope of all agents and references on the device.

Availability and Visibility Semantics

A memory barrier or atomic operation via agent A that includes MakeAvailable in its semantics performs an availability operation whose source scope includes agent A and all references in the storage classes in that instruction's storage class semantics, and all memory locations, and whose destination scope is a set of memory domains selected as specified below. The implicit availability operation is program-ordered between the barrier or atomic and all other operations program-ordered before the barrier or atomic.

A memory barrier or atomic operation via agent A that includes MakeVisible in its semantics performs a visibility operation whose source scope is a set of memory domains selected as specified below, and whose destination scope includes agent A and all references in the storage classes in that instruction's storage class semantics, and all memory locations. The implicit visibility operation is program-ordered between the barrier or atomic and all other operations program-ordered after the barrier or atomic.

The memory domains are selected based on the memory scope of the instruction as follows:

- `Device` scope uses the shader domain
- `QueueFamily` scope uses the queue family instance domain
- `FragmentInterlock` scope uses the fragment interlock instance domain
- `ShaderCallKHR` scope uses the shader call instance domain
- `Workgroup` scope uses the workgroup instance domain
- `Subgroup` uses the subgroup instance domain
- `Invocation` perform no availability/visibility operations.

When an availability operation performed by an agent A includes a memory domain D in its destination scope, where D corresponds to scope instance S, it also includes the memory domains that correspond to each smaller scope instance S' that is a subset of S and that includes A. Similarly for visibility operations.

Per-Instruction Availability and Visibility Semantics

A memory write instruction that includes `MakePointerAvailable`, or an image write instruction that includes `MakeTexelAvailable`, performs an availability operation whose source scope includes the agent and reference used to perform the write and the memory locations written by the instruction, and whose destination scope is a set of memory domains selected by the Scope operand specified in Availability and Visibility Semantics. The implicit availability operation is program-ordered between the write and all other operations program-ordered after the write.

A memory read instruction that includes `MakePointerVisible`, or an image read instruction that includes `MakeTexelVisible`, performs a visibility operation whose source scope is a set of memory domains selected by the Scope operand as specified in Availability and Visibility Semantics, and whose destination scope includes the agent and reference used to perform the read and the memory locations read by the instruction. The implicit visibility operation is program-ordered between read and all other operations program-ordered before the read.

NOTE	Although reads with per-instruction visibility only perform visibility ops from the shader or fragment interlock instance or shader call instance or workgroup instance or subgroup instance domain, they will also see writes that were made visible via the device domain, i.e. those writes previously performed by non-shader agents and made visible via API commands.
NOTE	It is expected that all invocations in a subgroup execute on the same processor with the same path to memory, and thus availability and visibility operations with subgroup scope can be expected to be “free”.

Location-Ordered

Let X and Y be memory accesses to overlapping sets of memory locations M, where X != Y. Let (A_X,R_X) be the agent and reference used for X, and (A_Y,R_Y) be the agent and reference used for Y. For now, let “→” denote happens-before and “$\rightarrow^{rcpo}$” denote the reflexive closure of program-ordered before.

If D_1 and D_2 are different memory domains, then let $DOM(D_1,D_2)$ be a memory domain operation from D_1 to D_2. Otherwise, let DOM(D,D) be a placeholder such that $X \rightarrow DOM(D,D) \rightarrow Y$ if and only if $X \rightarrow Y$.

X is *location-ordered* before Y for a location L in M if and only if any of the following is true:

- $A_X == A_Y$ and $R_X == R_Y$ and $X \rightarrow Y$
  - NOTE: this case means no availability/visibility ops are required when it is the same (agent,reference).
- X is a read, both X and Y are non-private, and $X \rightarrow Y$
- X is a read, and X (transitively) system-synchronizes with Y
- If $R_X == R_Y$ and A_X and A_Y access a common memory domain D (e.g. are in the same workgroup instance if D is the workgroup instance domain), and both X and Y are non-private:
  - X is a write, Y is a write, $AVC(A_X,R_X,D,L)$ is an availability chain making (X,L) available to domain D, and $X \rightarrow^{rcpo} AVC(A_X,R_X,D,L) \rightarrow Y$
  - X is a write, Y is a read, $AVC(A_X,R_X,D,L)$ is an availability chain making (X,L) available to domain D, $VISC(A_Y,R_Y,D,L)$ is a visibility chain making writes to L available in domain D visible to Y, and $X \rightarrow^{rcpo} AVC(A_X,R_X,D,L) \rightarrow VISC(A_Y,R_Y,D,L) \rightarrow^{rcpo} Y$
  - If VkPhysicalDeviceVulkanMemoryModelFeatures::`vulkanMemoryModelAvailabilityVisibilityChains` is VK_FALSE, then AVC and VISC **must** each only have a single element in the chain, in each sub-bullet above.
- Let D_X and D_Y each be either the device domain or the host domain, depending on whether A_X and A_Y execute on the device or host:
  - X is a write and Y is a write, and $X \rightarrow AV(A_X,R_X,D_X,L) \rightarrow DOM(D_X,D_Y) \rightarrow Y$
  - X is a write and Y is a read, and $X \rightarrow AV(A_X,R_X,D_X,L) \rightarrow DOM(D_X,D_Y) \rightarrow VIS(A_Y,R_Y,D_Y,L) \rightarrow Y$

NOTE	The final bullet (synchronization through device/host domain) requires API-level synchronization operations, since the device/host domains are not accessible via shader instructions. And “device domain” is not to be confused with “device scope”, which synchronizes through the “shader domain”.

Data Race

Let X and Y be operations that access overlapping sets of memory locations M, where X != Y, and at least one of X and Y is a write, and X and Y are not mutually-ordered atomic operations. If there does not exist a location-ordered relation between X and Y for each location in M, then there is a *data race*.

Applications **must** ensure that no data races occur during the execution of their application.

NOTE	Data races can only occur due to instructions that are actually executed. For example, an instruction skipped due to control flow must not contribute to a data race.

Visible-To

Let X be a write and Y be a read whose sets of memory locations overlap, and let M be the set of memory locations that overlap. Let M_2 be a non-empty subset of M. Then X is *visible-to* Y for memory locations M_2 if and only if all of the following are true:

- X is location-ordered before Y for each location L in M_2.
- There does not exist another write Z to any location L in M_2 such that X is location-ordered before Z for location L and Z is location-ordered before Y for location L.

If X is visible-to Y, then Y reads the value written by X for locations M_2.

NOTE	It is possible for there to be a write between X and Y that overwrites a subset of the memory locations, but the remaining memory locations (M_2) will still be visible-to Y.

Acyclicity

Reads-from is a relation between operations, where the first operation is a write, the second operation is a read, and the second operation reads the value written by the first operation. *From-reads* is a relation between operations, where the first operation is a read, the second operation is a write, and the first operation reads a value written earlier than the second operation in the second operation's scoped modification order or location order (or the first operation reads from the initial value, and the second operation is any write to the same locations).

Then the implementation **must** guarantee that no cycles exist in the union of the following relations:

- location-ordered
- scoped modification order (over all atomic writes)
- reads-from
- from-reads

NOTE	This is a "consistency" axiom, which informally guarantees that sequences of operations cannot violate causality.

Scoped Modification Order Coherence

Let A and B be mutually-ordered atomic operations, where A is location-ordered before B. Then the following rules are a consequence of acyclicity:

- If A and B are both reads and A does not read the initial value, then the write that A takes its value from **must** be earlier in its own scoped modification order than (or the same as) the write that B takes its value from (no cycles between location-order, reads-from, and from-reads).
- If A is a read and B is a write and A does not read the initial value, then A **must** take its value from a write earlier than B in B's scoped modification order (no cycles between location-order, scope modification order, and reads-from).

- If A is a write and B is a read, then B **must** take its value from A or a write later than A in A's scoped modification order (no cycles between location-order, scoped modification order, and from-reads).
- If A and B are both writes, then A **must** be earlier than B in A's scoped modification order (no cycles between location-order and scoped modification order).
- If A is a write and B is a read-modify-write and B reads the value written by A, then B comes immediately after A in A's scoped modification order (no cycles between scoped modification order and from-reads).

Shader I/O

If a shader invocation A in a shader stage other than `Vertex` performs a memory read operation X from an object in storage class `CallableDataKHR`, `IncomingCallableDataKHR`, `RayPayloadKHR`, `HitAttributeKHR`, `IncomingRayPayloadKHR`, or `Input`, then X is system-synchronized-after all writes to the corresponding `CallableDataKHR`, `IncomingCallableDataKHR`, `RayPayloadKHR`, `HitAttributeKHR`, `IncomingRayPayloadKHR`, or `Output` storage variable(s) in the shader invocation(s) that contribute to generating invocation A, and those writes are all visible-to X.

NOTE It is not necessary for the upstream shader invocations to have completed execution, they only need to have generated the output that is being read.

Deallocation

A call to vkFreeMemory **must** happen-after all memory operations on all memory locations in that VkDeviceMemory object.

NOTE Normally, device memory operations in a given queue are synchronized with vkFreeMemory by having a host thread wait on a fence signaled by that queue, and the wait happens-before the call to vkFreeMemory on the host.

The deallocation of SPIR-V variables is managed by the system and happens-after all operations on those variables.

Descriptions (Informative)

This subsection offers more easily understandable consequences of the memory model for app/compiler developers.

Let SC be the storage class(es) specified by a release or acquire operation or barrier.

- An atomic write with release semantics must not be reordered against any read or write to SC that is program-ordered before it (regardless of the storage class the atomic is in).
- An atomic read with acquire semantics must not be reordered against any read or write to SC that is program-ordered after it (regardless of the storage class the atomic is in).
- Any write to SC program-ordered after a release barrier must not be reordered against any read

or write to SC program-ordered before that barrier.

- Any read from SC program-ordered before an acquire barrier must not be reordered against any read or write to SC program-ordered after the barrier.

A control barrier (even if it has no memory semantics) must not be reordered against any memory barriers.

This memory model allows memory accesses with and without availability and visibility operations, as well as atomic operations, all to be performed on the same memory location. This is critical to allow it to reason about memory that is reused in multiple ways, e.g. across the lifetime of different shader invocations or draw calls. While GLSL (and legacy SPIR-V) applies the "coherent" decoration to variables (for historical reasons), this model treats each memory access instruction as having optional implicit availability/visibility operations. GLSL to SPIR-V compilers should map all (non-atomic) operations on a coherent variable to Make{Pointer,Texel}{Available}{Visible} flags in this model.

Atomic operations implicitly have availability/visibility operations, and the scope of those operations is taken from the atomic operation's scope.

Tessellation Output Ordering

For SPIR-V that uses the Vulkan Memory Model, the `OutputMemory` storage class is used to synchronize accesses to tessellation control output variables. For legacy SPIR-V that does not enable the Vulkan Memory Model via `OpMemoryModel`, tessellation outputs can be ordered using a control barrier with no particular memory scope or semantics, as defined below.

Let X and Y be memory operations performed by shader invocations A_X and A_Y. Operation X is *tessellation-output-ordered* before operation Y if and only if all of the following are true:

- There is a dynamic instance of an `OpControlBarrier` instruction C such that X is program-ordered before C in A_X and C is program-ordered before Y in A_Y.
- A_X and A_Y are in the same instance of C's execution scope.

If shader invocations A_X and A_Y in the `TessellationControl` execution model execute memory operations X and Y, respectively, on the `Output` storage class, and X is tessellation-output-ordered before Y with a scope of `Workgroup`, then X is location-ordered before Y, and if X is a write and Y is a read then X is visible-to Y.

Cooperative Matrix Memory Access

For each dynamic instance of a cooperative matrix load instruction (`OpCooperativeMatrixLoadKHR`), some implementation-dependent invocation(s) within the instance of the matrix's scope perform a non-atomic load from each memory location that is defined to be accessed by the instruction.

For each memory location accessed by a dynamic instance of a cooperative matrix store instruction (`OpCooperativeMatrixStoreKHR`), a single implementation-dependent invocation within the instance of the matrix's scope performs a non-atomic store to that memory location.

Appendix C: Compressed Image Formats

The compressed texture formats used by Vulkan are described in the specifically identified sections of the Khronos Data Format Specification, version 1.3.

Unless otherwise described, the quantities encoded in these compressed formats are treated as normalized, unsigned values.

Those formats listed as sRGB-encoded have in-memory representations of R, G and B components which are nonlinearly-encoded as R', G', and B'; any alpha component is unchanged. As part of filtering, the nonlinear R', G', and B' values are converted to linear R, G, and B components; any alpha component is unchanged. The conversion between linear and nonlinear encoding is performed as described in the “KHR_DF_TRANSFER_SRGB” section of the Khronos Data Format Specification.

Block-Compressed Image Formats

BC1, BC2, and BC3 formats are described in “S3TC Compressed Texture Image Formats” chapter of the Khronos Data Format Specification. BC4 and BC5 are described in the “RGTC Compressed Texture Image Formats” chapter. BC6H and BC7 are described in the “BPTC Compressed Texture Image Formats” chapter.

Table 105. Mapping of Vulkan BC Formats to Descriptions

VkFormat	**Khronos Data Format Specification description**
Formats described in the “S3TC Compressed Texture Image Formats” chapter	
VK_FORMAT_BC1_RGB_UNORM_BLOCK	BC1 with no alpha
VK_FORMAT_BC1_RGB_SRGB_BLOCK	BC1 with no alpha, sRGB-encoded
VK_FORMAT_BC1_RGBA_UNORM_BLOCK	BC1 with alpha
VK_FORMAT_BC1_RGBA_SRGB_BLOCK	BC1 with alpha, sRGB-encoded
VK_FORMAT_BC2_UNORM_BLOCK	BC2
VK_FORMAT_BC2_SRGB_BLOCK	BC2, sRGB-encoded
VK_FORMAT_BC3_UNORM_BLOCK	BC3
VK_FORMAT_BC3_SRGB_BLOCK	BC3, sRGB-encoded
Formats described in the “RGTC Compressed Texture Image Formats” chapter	
VK_FORMAT_BC4_UNORM_BLOCK	BC4 unsigned
VK_FORMAT_BC4_SNORM_BLOCK	BC4 signed
VK_FORMAT_BC5_UNORM_BLOCK	BC5 unsigned
VK_FORMAT_BC5_SNORM_BLOCK	BC5 signed
Formats described in the “BPTC Compressed Texture Image Formats” chapter	
VK_FORMAT_BC6H_UFLOAT_BLOCK	BC6H (unsigned version)
VK_FORMAT_BC6H_SFLOAT_BLOCK	BC6H (signed version)
VK_FORMAT_BC7_UNORM_BLOCK	BC7
VK_FORMAT_BC7_SRGB_BLOCK	BC7, sRGB-encoded

ETC Compressed Image Formats

The following formats are described in the “ETC2 Compressed Texture Image Formats” chapter of the Khronos Data Format Specification.

Table 106. Mapping of Vulkan ETC Formats to Descriptions

VkFormat	**Khronos Data Format Specification description**
VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK	RGB ETC2
VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK	RGB ETC2 with sRGB encoding
VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK	RGB ETC2 with punch-through alpha
VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK	RGB ETC2 with punch-through alpha and sRGB
VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK	RGBA ETC2
VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK	RGBA ETC2 with sRGB encoding
VK_FORMAT_EAC_R11_UNORM_BLOCK	Unsigned R11 EAC
VK_FORMAT_EAC_R11_SNORM_BLOCK	Signed R11 EAC
VK_FORMAT_EAC_R11G11_UNORM_BLOCK	Unsigned RG11 EAC
VK_FORMAT_EAC_R11G11_SNORM_BLOCK	Signed RG11 EAC

ASTC Compressed Image Formats

ASTC formats are described in the “ASTC Compressed Texture Image Formats” chapter of the Khronos Data Format Specification.

Table 107. Mapping of Vulkan ASTC Formats to Descriptions

VkFormat	Compressed texel block dimensions	Requested mode
VK_FORMAT_ASTC_4x4_UNORM_BLOCK	4 × 4	Linear LDR
VK_FORMAT_ASTC_4x4_SRGB_BLOCK	4 × 4	sRGB
VK_FORMAT_ASTC_5x4_UNORM_BLOCK	5 × 4	Linear LDR
VK_FORMAT_ASTC_5x4_SRGB_BLOCK	5 × 4	sRGB
VK_FORMAT_ASTC_5x5_UNORM_BLOCK	5 × 5	Linear LDR
VK_FORMAT_ASTC_5x5_SRGB_BLOCK	5 × 5	sRGB
VK_FORMAT_ASTC_6x5_UNORM_BLOCK	6 × 5	Linear LDR
VK_FORMAT_ASTC_6x5_SRGB_BLOCK	6 × 5	sRGB
VK_FORMAT_ASTC_6x6_UNORM_BLOCK	6 × 6	Linear LDR
VK_FORMAT_ASTC_6x6_SRGB_BLOCK	6 × 6	sRGB
VK_FORMAT_ASTC_8x5_UNORM_BLOCK	8 × 5	Linear LDR
VK_FORMAT_ASTC_8x5_SRGB_BLOCK	8 × 5	sRGB
VK_FORMAT_ASTC_8x6_UNORM_BLOCK	8 × 6	Linear LDR
VK_FORMAT_ASTC_8x6_SRGB_BLOCK	8 × 6	sRGB
VK_FORMAT_ASTC_8x8_UNORM_BLOCK	8 × 8	Linear LDR
VK_FORMAT_ASTC_8x8_SRGB_BLOCK	8 × 8	sRGB
VK_FORMAT_ASTC_10x5_UNORM_BLOCK	10 × 5	Linear LDR
VK_FORMAT_ASTC_10x5_SRGB_BLOCK	10 × 5	sRGB
VK_FORMAT_ASTC_10x6_UNORM_BLOCK	10 × 6	Linear LDR
VK_FORMAT_ASTC_10x6_SRGB_BLOCK	10 × 6	sRGB
VK_FORMAT_ASTC_10x8_UNORM_BLOCK	10 × 8	Linear LDR
VK_FORMAT_ASTC_10x8_SRGB_BLOCK	10 × 8	sRGB
VK_FORMAT_ASTC_10x10_UNORM_BLOCK	10 × 10	Linear LDR
VK_FORMAT_ASTC_10x10_SRGB_BLOCK	10 × 10	sRGB
VK_FORMAT_ASTC_12x10_UNORM_BLOCK	12 × 10	Linear LDR
VK_FORMAT_ASTC_12x10_SRGB_BLOCK	12 × 10	sRGB
VK_FORMAT_ASTC_12x12_UNORM_BLOCK	12 × 12	Linear LDR

VkFormat	Compressed texel block dimensions	Requested mode
VK_FORMAT_ASTC_12x12_SRGB_BLOCK	12 × 12	sRGB
VK_FORMAT_ASTC_4x4_SFLOAT_BLOCK	4 × 4	HDR
VK_FORMAT_ASTC_5x4_SFLOAT_BLOCK	5 × 4	HDR
VK_FORMAT_ASTC_5x5_SFLOAT_BLOCK	5 × 5	HDR
VK_FORMAT_ASTC_6x5_SFLOAT_BLOCK	6 × 5	HDR
VK_FORMAT_ASTC_6x6_SFLOAT_BLOCK	6 × 6	HDR
VK_FORMAT_ASTC_8x5_SFLOAT_BLOCK	8 × 5	HDR
VK_FORMAT_ASTC_8x6_SFLOAT_BLOCK	8 × 6	HDR
VK_FORMAT_ASTC_8x8_SFLOAT_BLOCK	8 × 8	HDR
VK_FORMAT_ASTC_10x5_SFLOAT_BLOCK	10 × 5	HDR
VK_FORMAT_ASTC_10x6_SFLOAT_BLOCK	10 × 6	HDR
VK_FORMAT_ASTC_10x8_SFLOAT_BLOCK	10 × 8	HDR
VK_FORMAT_ASTC_10x10_SFLOAT_BLOCK	10 × 10	HDR
VK_FORMAT_ASTC_12x10_SFLOAT_BLOCK	12 × 10	HDR
VK_FORMAT_ASTC_12x12_SFLOAT_BLOCK	12 × 12	HDR
VK_FORMAT_ASTC_3x3x3_UNORM_BLOCK_EXT	3 × 3 × 3	Linear LDR
VK_FORMAT_ASTC_3x3x3_SRGB_BLOCK_EXT	3 × 3 × 3	sRGB
VK_FORMAT_ASTC_3x3x3_SFLOAT_BLOCK_EXT	3 × 3 × 3	HDR
VK_FORMAT_ASTC_4x3x3_UNORM_BLOCK_EXT	4 × 3 × 3	Linear LDR
VK_FORMAT_ASTC_4x3x3_SRGB_BLOCK_EXT	4 × 3 × 3	sRGB
VK_FORMAT_ASTC_4x3x3_SFLOAT_BLOCK_EXT	4 × 3 × 3	HDR
VK_FORMAT_ASTC_4x4x3_UNORM_BLOCK_EXT	4 × 4 × 3	Linear LDR
VK_FORMAT_ASTC_4x4x3_SRGB_BLOCK_EXT	4 × 4 × 3	sRGB
VK_FORMAT_ASTC_4x4x3_SFLOAT_BLOCK_EXT	4 × 4 × 3	HDR
VK_FORMAT_ASTC_4x4x4_UNORM_BLOCK_EXT	4 × 4 × 4	Linear LDR
VK_FORMAT_ASTC_4x4x4_SRGB_BLOCK_EXT	4 × 4 × 4	sRGB
VK_FORMAT_ASTC_4x4x4_SFLOAT_BLOCK_EXT	4 × 4 × 4	HDR
VK_FORMAT_ASTC_5x4x4_UNORM_BLOCK_EXT	5 × 4 × 4	Linear LDR
VK_FORMAT_ASTC_5x4x4_SRGB_BLOCK_EXT	5 × 4 × 4	sRGB
VK_FORMAT_ASTC_5x4x4_SFLOAT_BLOCK_EXT	5 × 4 × 4	HDR
VK_FORMAT_ASTC_5x5x4_UNORM_BLOCK_EXT	5 × 5 × 4	Linear LDR
VK_FORMAT_ASTC_5x5x4_SRGB_BLOCK_EXT	5 × 5 × 4	sRGB

VkFormat	Compressed texel block dimensions	Requested mode
VK_FORMAT_ASTC_5x5x4_SFLOAT_BLOCK_EXT	5 × 5 × 4	HDR
VK_FORMAT_ASTC_5x5x5_UNORM_BLOCK_EXT	5 × 5 × 5	Linear LDR
VK_FORMAT_ASTC_5x5x5_SRGB_BLOCK_EXT	5 × 5 × 5	sRGB
VK_FORMAT_ASTC_5x5x5_SFLOAT_BLOCK_EXT	5 × 5 × 5	HDR
VK_FORMAT_ASTC_6x5x5_UNORM_BLOCK_EXT	6 × 5 × 5	Linear LDR
VK_FORMAT_ASTC_6x5x5_SRGB_BLOCK_EXT	6 × 5 × 5	sRGB
VK_FORMAT_ASTC_6x5x5_SFLOAT_BLOCK_EXT	6 × 5 × 5	HDR
VK_FORMAT_ASTC_6x6x5_UNORM_BLOCK_EXT	6 × 6 × 5	Linear LDR
VK_FORMAT_ASTC_6x6x5_SRGB_BLOCK_EXT	6 × 6 × 5	sRGB
VK_FORMAT_ASTC_6x6x5_SFLOAT_BLOCK_EXT	6 × 6 × 5	HDR
VK_FORMAT_ASTC_6x6x6_UNORM_BLOCK_EXT	6 × 6 × 6	Linear LDR
VK_FORMAT_ASTC_6x6x6_SRGB_BLOCK_EXT	6 × 6 × 6	sRGB
VK_FORMAT_ASTC_6x6x6_SFLOAT_BLOCK_EXT	6 × 6 × 6	HDR

ASTC textures containing HDR block encodings **should** be passed to the API using an ASTC SFLOAT texture format.

NOTE An HDR block in a texture passed using a LDR UNORM format will return the appropriate ASTC error color if the implementation supports only the ASTC LDR profile, but may result in either the error color or a decompressed HDR color if the implementation supports HDR decoding.

ASTC Decode Mode

If the `VK_EXT_astc_decode_mode` extension is enabled, the decode mode is determined as follows:

Table 108. Mapping of Vulkan ASTC Decoding Format to ASTC Decoding Modes

VkFormat	Decoding mode
VK_FORMAT_R16G16B16A16_SFLOAT	decode_float16
VK_FORMAT_R8G8B8A8_UNORM	decode_unorm8
VK_FORMAT_E5B9G9R9_UFLOAT_PACK32	decode_rgb9e5

Otherwise, the ASTC decode mode is decode_float16.

Note that an implementation **may** use HDR mode when linear LDR mode is requested unless the decode mode is decode_unorm8.

Appendix D: Core Revisions (Informative)

New minor versions of the Vulkan API are defined periodically by the Khronos Vulkan Working Group. These consist of some amount of additional functionality added to the core API, potentially including both new functionality and functionality promoted from extensions.

It is possible to build the specification for earlier versions, but to aid readability of the latest versions, this appendix gives an overview of the changes as compared to earlier versions.

Vulkan Version 1.4

Vulkan Version 1.4 promoted a number of key extensions into the core API:

- VK_KHR_dynamic_rendering_local_read
- VK_KHR_global_priority
- VK_KHR_index_type_uint8
- VK_KHR_line_rasterization
- VK_KHR_load_store_op_none
- VK_KHR_maintenance5
- VK_KHR_maintenance6
- VK_KHR_map_memory2
- VK_KHR_push_descriptor
- VK_KHR_shader_expect_assume
- VK_KHR_shader_float_controls2
- VK_KHR_shader_subgroup_rotate
- VK_KHR_vertex_attribute_divisor
- VK_EXT_host_image_copy
- VK_EXT_pipeline_protected_access
- VK_EXT_pipeline_robustness

All differences in behavior between these extensions and the corresponding Vulkan 1.4 functionality are summarized below.

Differences Relative to `VK_KHR_dynamic_rendering_local_read`

If the VK_KHR_dynamic_rendering_local_read extension is not supported, Vulkan 1.4 implementations **must** support local read only for storage resources and single sampled color

attachments.

Support for reading depth/stencil attachments and multi-sampled attachments are respectively gated behind the new boolean `dynamicRenderingLocalReadDepthStencilAttachments` and `dynamicRenderingLocalReadMultisampledAttachments` properties.

- If `dynamicRenderingLocalReadDepthStencilAttachments` is VK_FALSE, implementations do not support depth/stencil attachment access within dynamic rendering.
- If `dynamicRenderingLocalReadMultisampledAttachments` is VK_FALSE, implementations do not support multisampled attachment access within dynamic rendering.
- If both properties are VK_TRUE, the full functionality of the extension is supported.

Differences Relative to `VK_EXT_host_image_copy`

If the VK_EXT_host_image_copy extension is not supported, support for it is optional in Vulkan 1.4.

- An implementation that has a VK_QUEUE_GRAPHICS_BIT queue must support either:
  - the `hostImageCopy` feature; or
  - an additional queue that supports VK_QUEUE_TRANSFER_BIT.

Differences Relative to `VK_KHR_push_descriptor`

VK_KHR_push_descriptor did not include a feature bit, so a new feature bit has been added to VkPhysicalDeviceVulkan14Features to gate its functionality: `pushDescriptor`. Enabling this new feature has the same effect as enabling the extension.

Differences Relative to `VK_EXT_pipeline_protected_access`

VK_EXT_pipeline_protected_access is only useful when the `protectedMemory` feature is supported. As the `protectedMemory` feature is optional in core Vulkan, the `pipelineProtectedAccess` feature is only required when the `protectedMemory` feature is supported.

Differences Relative to `VK_KHR_line_rasterization`

The `bresenhamLines` feature is required, rather than just any one of the line style features.

Differences Relative to `VK_KHR_shader_subgroup_rotate`

The `shaderSubgroupRotateClustered` feature is required in addition to `shaderSubgroupRotate`.

Additional Vulkan 1.4 Feature Support

In addition to the promoted extensions described above, Vulkan 1.4 added required support for:

- All queues supporting VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT **must** also advertise VK_QUEUE_TRANSFER_BIT.
- Clustered subgroup operations **must** be advertised in Vulkan 1.4 via setting both VK_SUBGROUP_FEATURE_CLUSTERED_BIT and VK_SUBGROUP_FEATURE_ROTATE_CLUSTERED_BIT (as an interaction with the promoted VK_KHR_shader_subgroup_rotate functionality) in `supportedOperations`.
- The following features that were optional in earlier versions:

    - `fullDrawIndexUint32`
    - `imageCubeArray`
    - `independentBlend`
    - `sampleRateShading`
    - `drawIndirectFirstInstance`
    - `depthClamp`
    - `depthBiasClamp`
    - `samplerAnisotropy`
    - `fragmentStoresAndAtomics`
    - `shaderStorageImageExtendedFormats`
    - `shaderUniformBufferArrayDynamicIndexing`
    - `shaderSampledImageArrayDynamicIndexing`
    - `shaderStorageBufferArrayDynamicIndexing`
    - `shaderStorageImageArrayDynamicIndexing`
    - `shaderImageGatherExtended`
    - `shaderInt16`
    - `largePoints`
    - `samplerYcbcrConversion`
    - `storageBuffer16BitAccess`
    - `variablePointers`
    - `variablePointersStorageBuffer`
    - `samplerMirrorClampToEdge`
    - `scalarBlockLayout`
    - `shaderUniformTexelBufferArrayDynamicIndexing`
    - `shaderStorageTexelBufferArrayDynamicIndexing`
    - `shaderInt8`
    - `storageBuffer8BitAccess`

Updated Vulkan 1.4 Limit Support

Vulkan 1.4 also requires support for the following updated limits:

- `maxImageDimension1D` is increased from 4096 to 8192
- `maxImageDimension2D` is increased from 4096 to 8192
- `maxImageDimension3D` is increased from 256 to 512
- `maxImageDimensionCube` is increased from 4096 to 8192
- `maxImageArrayLayers` is increased from 256 to 2048

- `maxUniformBufferRange` is increased from 16384 to 65536
- `maxPushConstantsSize` is increased from 128 to 256
- `bufferImageGranularity` is decreased from 131072 to 4096
- `maxBoundDescriptorSets` is increased from 4 to 7
- `maxPerStageDescriptorUniformBuffers` is increased from 12 to 15
- `maxPerStageResources` is increased from 128 to 200
- `maxDescriptorSetUniformBuffers` is increased from 72 to 90
- `maxDescriptorSetStorageBuffers` is increased from 24 to 96
- `maxDescriptorSetStorageImages` is increased from 24 to 144
- `maxFragmentCombinedOutputResources` is increased from 4 to 16
- `maxComputeWorkGroupInvocations` is increased from 128 to 256
- `maxComputeWorkGroupSize` is increased from (128,128,64) to (256,256,64)
- `shaderSignedZeroInfNanPreserveFloat16` is changed from unspecified to VK_TRUE
- `shaderSignedZeroInfNanPreserveFloat32` is changed from unspecified to VK_TRUE
- `subTexelPrecisionBits` is increased from 4 to 8
- `mipmapPrecisionBits` is increased from 4 to 6
- `maxSamplerLodBias` is increased from 2 to 14
- `maxViewportDimensions` is increased from (4096,4096) to (7680,7680)
- `viewportBoundsRange` is increased from (-8192,8191) to (-15360,15359)
- `maxFramebufferWidth` is increased from 4096 to 7680
- `maxFramebufferHeight` is increased from 4096 to 7680
- `maxColorAttachments` is increased from 7 to 8
- `timestampComputeAndGraphics` is changed from unspecified to VK_TRUE
- `pointSizeRange` is increased from (1.0,64.0 - ULP) to (1.0,256.0 - `pointSizeGranularity`)
- `pointSizeGranularity` is decreased from 1.0 to 0.125
- `lineWidthGranularity` is decreased from 1.0 to 0.5
- `maxPushDescriptors` is increased from 16 to 32
- `standardSampleLocations` is changed from unspecified to VK_TRUE

New Macros

- VK_API_VERSION_1_4

New Commands

- vkCmdBindDescriptorSets2
- vkCmdBindIndexBuffer2

- vkCmdPushConstants2
- vkCmdPushDescriptorSet
- vkCmdPushDescriptorSet2
- vkCmdPushDescriptorSetWithTemplate
- vkCmdPushDescriptorSetWithTemplate2
- vkCmdSetLineStipple
- vkCmdSetRenderingAttachmentLocations
- vkCmdSetRenderingInputAttachmentIndices
- vkCopyImageToImage
- vkCopyImageToMemory
- vkCopyMemoryToImage
- vkGetDeviceImageSubresourceLayout
- vkGetImageSubresourceLayout2
- vkGetRenderingAreaGranularity
- vkMapMemory2
- vkTransitionImageLayout
- vkUnmapMemory2

New Structures

- VkBindDescriptorSetsInfo
- VkCopyImageToImageInfo
- VkCopyImageToMemoryInfo
- VkCopyMemoryToImageInfo
- VkDeviceImageSubresourceInfo
- VkHostImageLayoutTransitionInfo
- VkImageSubresource2
- VkImageToMemoryCopy
- VkMemoryMapInfo
- VkMemoryToImageCopy
- VkMemoryUnmapInfo
- VkPushConstantsInfo
- VkPushDescriptorSetInfo
- VkPushDescriptorSetWithTemplateInfo
- VkRenderingAreaInfo
- VkSubresourceLayout2

- VkVertexInputBindingDivisorDescription
- Extending VkBindBufferMemoryInfo, VkBindImageMemoryInfo:
  - VkBindMemoryStatus
- Extending VkBufferViewCreateInfo, VkBufferCreateInfo, VkPhysicalDeviceExternalBufferInfo, VkDescriptorBufferBindingInfoEXT:
  - VkBufferUsageFlags2CreateInfo
- Extending VkComputePipelineCreateInfo, VkGraphicsPipelineCreateInfo, VkRayTracingPipelineCreateInfoKHR:
  - VkPipelineCreateFlags2CreateInfo
- Extending VkDeviceQueueCreateInfo:
  - VkDeviceQueueGlobalPriorityCreateInfo
- Extending VkGraphicsPipelineCreateInfo, VkCommandBufferInheritanceInfo:
  - VkRenderingAttachmentLocationInfo
  - VkRenderingInputAttachmentIndexInfo
- Extending VkGraphicsPipelineCreateInfo, VkComputePipelineCreateInfo, VkPipelineShaderStageCreateInfo, VkRayTracingPipelineCreateInfoKHR:
  - VkPipelineRobustnessCreateInfo
- Extending VkImageFormatProperties2:
  - VkHostImageCopyDevicePerformanceQuery
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDynamicRenderingLocalReadFeatures
  - VkPhysicalDeviceGlobalPriorityQueryFeatures
  - VkPhysicalDeviceHostImageCopyFeatures
  - VkPhysicalDeviceIndexTypeUint8Features
  - VkPhysicalDeviceLineRasterizationFeatures
  - VkPhysicalDeviceMaintenance5Features
  - VkPhysicalDeviceMaintenance6Features
  - VkPhysicalDevicePipelineProtectedAccessFeatures
  - VkPhysicalDevicePipelineRobustnessFeatures
  - VkPhysicalDeviceShaderExpectAssumeFeatures
  - VkPhysicalDeviceShaderFloatControls2Features
  - VkPhysicalDeviceShaderSubgroupRotateFeatures
  - VkPhysicalDeviceVertexAttributeDivisorFeatures
  - VkPhysicalDeviceVulkan14Features
- Extending VkPhysicalDeviceProperties2:

    - VkPhysicalDeviceHostImageCopyProperties
    - VkPhysicalDeviceLineRasterizationProperties
    - VkPhysicalDeviceMaintenance5Properties
    - VkPhysicalDeviceMaintenance6Properties
    - VkPhysicalDevicePipelineRobustnessProperties
    - VkPhysicalDevicePushDescriptorProperties
    - VkPhysicalDeviceVertexAttributeDivisorProperties
    - VkPhysicalDeviceVulkan14Properties
- Extending VkPipelineRasterizationStateCreateInfo:
    - VkPipelineRasterizationLineStateCreateInfo
- Extending VkPipelineVertexInputStateCreateInfo:
    - VkPipelineVertexInputDivisorStateCreateInfo
- Extending VkQueueFamilyProperties2:
    - VkQueueFamilyGlobalPriorityProperties
- Extending VkSubresourceLayout2:
    - VkSubresourceHostMemcpySize

New Enums

- VkBufferUsageFlagBits2
- VkHostImageCopyFlagBits
- VkLineRasterizationMode
- VkMemoryUnmapFlagBits
- VkPipelineCreateFlagBits2
- VkPipelineRobustnessBufferBehavior
- VkPipelineRobustnessImageBehavior
- VkQueueGlobalPriority

New Bitmasks

- VkBufferUsageFlags2
- VkHostImageCopyFlags
- VkMemoryUnmapFlags
- VkPipelineCreateFlags2

New Enum Constants

- VK_MAX_GLOBAL_PRIORITY_SIZE

- Extending VkAttachmentLoadOp:
  - VK_ATTACHMENT_LOAD_OP_NONE
- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_SHADER_DEVICE_ADDRESS_BIT
- Extending VkDescriptorSetLayoutCreateFlagBits:
  - VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT
- Extending VkDescriptorUpdateTemplateType:
  - VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS
- Extending VkDynamicState:
  - VK_DYNAMIC_STATE_LINE_STIPPLE
- Extending VkFormat:
  - VK_FORMAT_A1B5G5R5_UNORM_PACK16
  - VK_FORMAT_A8_UNORM
- Extending VkFormatFeatureFlagBits2:
  - VK_FORMAT_FEATURE_2_HOST_IMAGE_TRANSFER_BIT
- Extending VkImageLayout:
  - VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ
- Extending VkImageUsageFlagBits:
  - VK_IMAGE_USAGE_HOST_TRANSFER_BIT
- Extending VkIndexType:
  - VK_INDEX_TYPE_UINT8
- Extending VkPipelineCreateFlagBits:
  - VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT
  - VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT
- Extending VkResult:
  - VK_ERROR_NOT_PERMITTED
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_BIND_DESCRIPTOR_SETS_INFO
  - VK_STRUCTURE_TYPE_BIND_MEMORY_STATUS
  - VK_STRUCTURE_TYPE_BUFFER_USAGE_FLAGS_2_CREATE_INFO
  - VK_STRUCTURE_TYPE_COPY_IMAGE_TO_IMAGE_INFO
  - VK_STRUCTURE_TYPE_COPY_IMAGE_TO_MEMORY_INFO
  - VK_STRUCTURE_TYPE_COPY_MEMORY_TO_IMAGE_INFO
  - VK_STRUCTURE_TYPE_DEVICE_IMAGE_SUBRESOURCE_INFO
  - VK_STRUCTURE_TYPE_DEVICE_QUEUE_GLOBAL_PRIORITY_CREATE_INFO

- VK_STRUCTURE_TYPE_HOST_IMAGE_COPY_DEVICE_PERFORMANCE_QUERY
- VK_STRUCTURE_TYPE_HOST_IMAGE_LAYOUT_TRANSITION_INFO
- VK_STRUCTURE_TYPE_IMAGE_SUBRESOURCE_2
- VK_STRUCTURE_TYPE_IMAGE_TO_MEMORY_COPY
- VK_STRUCTURE_TYPE_MEMORY_MAP_INFO
- VK_STRUCTURE_TYPE_MEMORY_TO_IMAGE_COPY
- VK_STRUCTURE_TYPE_MEMORY_UNMAP_INFO
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_LOCAL_READ_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GLOBAL_PRIORITY_QUERY_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INDEX_TYPE_UINT8_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_PROTECTED_ACCESS_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_ROBUSTNESS_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_ROBUSTNESS_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PUSH_DESCRIPTOR_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_EXPECT_ASSUME_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT_CONTROLS_2_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_ROTATE_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_4_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_4_PROPERTIES
- VK_STRUCTURE_TYPE_PIPELINE_CREATE_FLAGS_2_CREATE_INFO
- VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_LINE_STATE_CREATE_INFO
- VK_STRUCTURE_TYPE_PIPELINE_ROBUSTNESS_CREATE_INFO
- VK_STRUCTURE_TYPE_PIPELINE_VERTEX_INPUT_DIVISOR_STATE_CREATE_INFO
- VK_STRUCTURE_TYPE_PUSH_CONSTANTS_INFO
- VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_INFO

    - VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_WITH_TEMPLATE_INFO
    - VK_STRUCTURE_TYPE_QUEUE_FAMILY_GLOBAL_PRIORITY_PROPERTIES
    - VK_STRUCTURE_TYPE_RENDERING_AREA_INFO
    - VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_LOCATION_INFO
    - VK_STRUCTURE_TYPE_RENDERING_INPUT_ATTACHMENT_INDEX_INFO
    - VK_STRUCTURE_TYPE_SUBRESOURCE_HOST_MEMCPY_SIZE
    - VK_STRUCTURE_TYPE_SUBRESOURCE_LAYOUT_2
- Extending VkSubgroupFeatureFlagBits:
    - VK_SUBGROUP_FEATURE_ROTATE_BIT
    - VK_SUBGROUP_FEATURE_ROTATE_CLUSTERED_BIT

Vulkan Version 1.3

Vulkan Version 1.3 promoted a number of key extensions into the core API:

- VK_KHR_copy_commands2
- VK_KHR_dynamic_rendering
- VK_KHR_format_feature_flags2
- VK_KHR_maintenance4
- VK_KHR_shader_integer_dot_product
- VK_KHR_shader_non_semantic_info
- VK_KHR_shader_terminate_invocation
- VK_KHR_synchronization2
- VK_KHR_zero_initialize_workgroup_memory
- `VK_EXT_4444_formats`
- VK_EXT_extended_dynamic_state
- VK_EXT_extended_dynamic_state2
- `VK_EXT_image_robustness`
- `VK_EXT_inline_uniform_block`
- `VK_EXT_pipeline_creation_cache_control`
- `VK_EXT_pipeline_creation_feedback`
- `VK_EXT_private_data`
- `VK_EXT_shader_demote_to_helper_invocation`

- VK_EXT_subgroup_size_control
- VK_EXT_texel_buffer_alignment
- VK_EXT_texture_compression_astc_hdr
- VK_EXT_tooling_info
- VK_EXT_ycbcr_2plane_444_formats

All differences in behavior between these extensions and the corresponding Vulkan 1.3 functionality are summarized below.

Differences Relative to VK_EXT_4444_formats

If the VK_EXT_4444_formats extension is not supported, support for all formats defined by it are optional in Vulkan 1.3. There are no members in the VkPhysicalDeviceVulkan13Features structure corresponding to the VkPhysicalDevice4444FormatsFeaturesEXT structure.

Differences Relative to VK_EXT_extended_dynamic_state

All dynamic state enumerants and commands defined by VK_EXT_extended_dynamic_state are required in Vulkan 1.3. There are no members in the VkPhysicalDeviceVulkan13Features structure corresponding to the VkPhysicalDeviceExtendedDynamicStateFeaturesEXT structure.

Differences Relative to VK_EXT_extended_dynamic_state2

The optional dynamic state enumerants and commands defined by VK_EXT_extended_dynamic_state2 for patch control points and logic op are not promoted in Vulkan 1.3. There are no members in the VkPhysicalDeviceVulkan13Features structure corresponding to the VkPhysicalDeviceExtendedDynamicState2FeaturesEXT structure.

Differences Relative to VK_EXT_texel_buffer_alignment

The more specific alignment requirements defined by VkPhysicalDeviceTexelBufferAlignmentProperties are required in Vulkan 1.3. There are no members in the VkPhysicalDeviceVulkan13Features structure corresponding to the VkPhysicalDeviceTexelBufferAlignmentFeaturesEXT structure. The texelBufferAlignment feature is enabled if using a Vulkan 1.3 instance.

Differences Relative to VK_EXT_texture_compression_astc_hdr

If the VK_EXT_texture_compression_astc_hdr extension is not supported, support for all formats defined by it are optional in Vulkan 1.3. The textureCompressionASTC_HDR member of VkPhysicalDeviceVulkan13Features indicates whether a Vulkan 1.3 implementation supports these formats.

Differences Relative to VK_EXT_ycbcr_2plane_444_formats

If the VK_EXT_ycbcr_2plane_444_formats extension is not supported, support for all formats defined by it are optional in Vulkan 1.3. There are no members in the VkPhysicalDeviceVulkan13Features structure corresponding to the VkPhysicalDeviceYcbcr2Plane444FormatsFeaturesEXT structure.

Additional Vulkan 1.3 Feature Support

In addition to the promoted extensions described above, Vulkan 1.3 added required support for:

- SPIR-V version 1.6
  - SPIR-V 1.6 deprecates (but does not remove) the `WorkgroupSize` decoration.
- The `bufferDeviceAddress` feature which indicates support for accessing memory in shaders as storage buffers via vkGetBufferDeviceAddress.
- The `vulkanMemoryModel` and `vulkanMemoryModelDeviceScope` features, which indicate support for the corresponding Vulkan Memory Model capabilities.
- The `maxInlineUniformTotalSize` limit is added to provide the total size of all inline uniform block bindings in a pipeline layout.

New Macros

- VK_API_VERSION_1_3

New Base Types

- VkFlags64

New Object Types

- VkPrivateDataSlot

New Commands

- vkCmdBeginRendering
- vkCmdBindVertexBuffers2
- vkCmdBlitImage2
- vkCmdCopyBuffer2
- vkCmdCopyBufferToImage2
- vkCmdCopyImage2
- vkCmdCopyImageToBuffer2
- vkCmdEndRendering
- vkCmdPipelineBarrier2
- vkCmdResetEvent2
- vkCmdResolveImage2
- vkCmdSetCullMode
- vkCmdSetDepthBiasEnable
- vkCmdSetDepthBoundsTestEnable
- vkCmdSetDepthCompareOp
- vkCmdSetDepthTestEnable
- vkCmdSetDepthWriteEnable

- vkCmdSetEvent2
- vkCmdSetFrontFace
- vkCmdSetPrimitiveRestartEnable
- vkCmdSetPrimitiveTopology
- vkCmdSetRasterizerDiscardEnable
- vkCmdSetScissorWithCount
- vkCmdSetStencilOp
- vkCmdSetStencilTestEnable
- vkCmdSetViewportWithCount
- vkCmdWaitEvents2
- vkCmdWriteTimestamp2
- vkCreatePrivateDataSlot
- vkDestroyPrivateDataSlot
- vkGetDeviceBufferMemoryRequirements
- vkGetDeviceImageMemoryRequirements
- vkGetDeviceImageSparseMemoryRequirements
- vkGetPhysicalDeviceToolProperties
- vkGetPrivateData
- vkQueueSubmit2
- vkSetPrivateData

New Structures

- VkBlitImageInfo2
- VkBufferCopy2
- VkBufferImageCopy2
- VkBufferMemoryBarrier2
- VkCommandBufferSubmitInfo
- VkCopyBufferInfo2
- VkCopyBufferToImageInfo2
- VkCopyImageInfo2
- VkCopyImageToBufferInfo2
- VkDependencyInfo
- VkDeviceBufferMemoryRequirements
- VkDeviceImageMemoryRequirements
- VkImageBlit2

- VkImageCopy2
- VkImageMemoryBarrier2
- VkImageResolve2
- VkPhysicalDeviceToolProperties
- VkPipelineCreationFeedback
- VkPrivateDataSlotCreateInfo
- VkRenderingAttachmentInfo
- VkRenderingInfo
- VkResolveImageInfo2
- VkSemaphoreSubmitInfo
- VkSubmitInfo2
- Extending VkCommandBufferInheritanceInfo:
  - VkCommandBufferInheritanceRenderingInfo
- Extending VkDescriptorPoolCreateInfo:
  - VkDescriptorPoolInlineUniformBlockCreateInfo
- Extending VkDeviceCreateInfo:
  - VkDevicePrivateDataCreateInfo
- Extending VkFormatProperties2:
  - VkFormatProperties3
- Extending VkGraphicsPipelineCreateInfo:
  - VkPipelineRenderingCreateInfo
- Extending VkGraphicsPipelineCreateInfo, VkComputePipelineCreateInfo, VkRayTracingPipelineCreateInfoKHR:
  - VkPipelineCreationFeedbackCreateInfo
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDynamicRenderingFeatures
  - VkPhysicalDeviceImageRobustnessFeatures
  - VkPhysicalDeviceInlineUniformBlockFeatures
  - VkPhysicalDeviceMaintenance4Features
  - VkPhysicalDevicePipelineCreationCacheControlFeatures
  - VkPhysicalDevicePrivateDataFeatures
  - VkPhysicalDeviceShaderDemoteToHelperInvocationFeatures
  - VkPhysicalDeviceShaderIntegerDotProductFeatures
  - VkPhysicalDeviceShaderTerminateInvocationFeatures
  - VkPhysicalDeviceSubgroupSizeControlFeatures

    - VkPhysicalDeviceSynchronization2Features
    - VkPhysicalDeviceTextureCompressionASTCHDRFeatures
    - VkPhysicalDeviceVulkan13Features
    - VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures
- Extending VkPhysicalDeviceProperties2:
    - VkPhysicalDeviceInlineUniformBlockProperties
    - VkPhysicalDeviceMaintenance4Properties
    - VkPhysicalDeviceShaderIntegerDotProductProperties
    - VkPhysicalDeviceSubgroupSizeControlProperties
    - VkPhysicalDeviceTexelBufferAlignmentProperties
    - VkPhysicalDeviceVulkan13Properties
- Extending VkPipelineShaderStageCreateInfo, VkShaderCreateInfoEXT:
    - VkPipelineShaderStageRequiredSubgroupSizeCreateInfo
- Extending VkSubpassDependency2:
    - VkMemoryBarrier2
- Extending VkWriteDescriptorSet:
    - VkWriteDescriptorSetInlineUniformBlock

New Enums

- VkAccessFlagBits2
- VkFormatFeatureFlagBits2
- VkPipelineCreationFeedbackFlagBits
- VkPipelineStageFlagBits2
- VkPrivateDataSlotCreateFlagBits
- VkRenderingFlagBits
- VkSubmitFlagBits
- VkToolPurposeFlagBits

New Bitmasks

- VkAccessFlags2
- VkFormatFeatureFlags2
- VkPipelineCreationFeedbackFlags
- VkPipelineStageFlags2
- VkPrivateDataSlotCreateFlags
- VkRenderingFlags

- VkSubmitFlags
- VkToolPurposeFlags

New Enum Constants

- Extending VkAccessFlagBits:
  - VK_ACCESS_NONE
- Extending VkAttachmentStoreOp:
  - VK_ATTACHMENT_STORE_OP_NONE
- Extending VkDescriptorType:
  - VK_DESCRIPTOR_TYPE_INLINE_UNIFORM_BLOCK
- Extending VkDynamicState:
  - VK_DYNAMIC_STATE_CULL_MODE
  - VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE
  - VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE
  - VK_DYNAMIC_STATE_DEPTH_COMPARE_OP
  - VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE
  - VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE
  - VK_DYNAMIC_STATE_FRONT_FACE
  - VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE
  - VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY
  - VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE
  - VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT
  - VK_DYNAMIC_STATE_STENCIL_OP
  - VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE
  - VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE
  - VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT
- Extending VkEventCreateFlagBits:
  - VK_EVENT_CREATE_DEVICE_ONLY_BIT
- Extending VkFormat:
  - VK_FORMAT_A4B4G4R4_UNORM_PACK16
  - VK_FORMAT_A4R4G4B4_UNORM_PACK16
  - VK_FORMAT_ASTC_10x10_SFLOAT_BLOCK
  - VK_FORMAT_ASTC_10x5_SFLOAT_BLOCK
  - VK_FORMAT_ASTC_10x6_SFLOAT_BLOCK
  - VK_FORMAT_ASTC_10x8_SFLOAT_BLOCK

    - VK_FORMAT_ASTC_12x10_SFLOAT_BLOCK
    - VK_FORMAT_ASTC_12x12_SFLOAT_BLOCK
    - VK_FORMAT_ASTC_4x4_SFLOAT_BLOCK
    - VK_FORMAT_ASTC_5x4_SFLOAT_BLOCK
    - VK_FORMAT_ASTC_5x5_SFLOAT_BLOCK
    - VK_FORMAT_ASTC_6x5_SFLOAT_BLOCK
    - VK_FORMAT_ASTC_6x6_SFLOAT_BLOCK
    - VK_FORMAT_ASTC_8x5_SFLOAT_BLOCK
    - VK_FORMAT_ASTC_8x6_SFLOAT_BLOCK
    - VK_FORMAT_ASTC_8x8_SFLOAT_BLOCK
    - VK_FORMAT_G10X6_B10X6R10X6_2PLANE_444_UNORM_3PACK16
    - VK_FORMAT_G12X4_B12X4R12X4_2PLANE_444_UNORM_3PACK16
    - VK_FORMAT_G16_B16R16_2PLANE_444_UNORM
    - VK_FORMAT_G8_B8R8_2PLANE_444_UNORM
- Extending VkFormatFeatureFlagBits2:
    - VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_CUBIC_BIT
- Extending VkImageAspectFlagBits:
    - VK_IMAGE_ASPECT_NONE
- Extending VkImageLayout:
    - VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL
    - VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL
- Extending VkObjectType:
    - VK_OBJECT_TYPE_PRIVATE_DATA_SLOT
- Extending VkPipelineCacheCreateFlagBits:
    - VK_PIPELINE_CACHE_CREATE_EXTERNALLY_SYNCHRONIZED_BIT
- Extending VkPipelineCreateFlagBits:
    - VK_PIPELINE_CREATE_EARLY_RETURN_ON_FAILURE_BIT
    - VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT
- Extending VkPipelineShaderStageCreateFlagBits:
    - VK_PIPELINE_SHADER_STAGE_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT
    - VK_PIPELINE_SHADER_STAGE_CREATE_REQUIRE_FULL_SUBGROUPS_BIT
- Extending VkPipelineStageFlagBits:
    - VK_PIPELINE_STAGE_NONE
- Extending VkResult:
    - VK_PIPELINE_COMPILE_REQUIRED

- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_BLIT_IMAGE_INFO_2
  - VK_STRUCTURE_TYPE_BUFFER_COPY_2
  - VK_STRUCTURE_TYPE_BUFFER_IMAGE_COPY_2
  - VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER_2
  - VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_RENDERING_INFO
  - VK_STRUCTURE_TYPE_COMMAND_BUFFER_SUBMIT_INFO
  - VK_STRUCTURE_TYPE_COPY_BUFFER_INFO_2
  - VK_STRUCTURE_TYPE_COPY_BUFFER_TO_IMAGE_INFO_2
  - VK_STRUCTURE_TYPE_COPY_IMAGE_INFO_2
  - VK_STRUCTURE_TYPE_COPY_IMAGE_TO_BUFFER_INFO_2
  - VK_STRUCTURE_TYPE_DEPENDENCY_INFO
  - VK_STRUCTURE_TYPE_DESCRIPTOR_POOL_INLINE_UNIFORM_BLOCK_CREATE_INFO
  - VK_STRUCTURE_TYPE_DEVICE_BUFFER_MEMORY_REQUIREMENTS
  - VK_STRUCTURE_TYPE_DEVICE_IMAGE_MEMORY_REQUIREMENTS
  - VK_STRUCTURE_TYPE_DEVICE_PRIVATE_DATA_CREATE_INFO
  - VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_3
  - VK_STRUCTURE_TYPE_IMAGE_BLIT_2
  - VK_STRUCTURE_TYPE_IMAGE_COPY_2
  - VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER_2
  - VK_STRUCTURE_TYPE_IMAGE_RESOLVE_2
  - VK_STRUCTURE_TYPE_MEMORY_BARRIER_2
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_FEATURES
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_ROBUSTNESS_FEATURES
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INLINE_UNIFORM_BLOCK_FEATURES
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INLINE_UNIFORM_BLOCK_PROPERTIES
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_FEATURES
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_PROPERTIES
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_CREATION_CACHE_CONTROL_FEATURES
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIVATE_DATA_FEATURES
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_DEMOTE_TO_HELPER_INVOCATION_FEATURES
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_FEATURES
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_PROPERTIES

- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TERMINATE_INVOCATION_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SUBGROUP_SIZE_CONTROL_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SUBGROUP_SIZE_CONTROL_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SYNCHRONIZATION_2_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TEXEL_BUFFER_ALIGNMENT_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TEXTURE_COMPRESSION_ASTC_HDR_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TOOL_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_3_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_3_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ZERO_INITIALIZE_WORKGROUP_MEMORY_FEATURES
- VK_STRUCTURE_TYPE_PIPELINE_CREATION_FEEDBACK_CREATE_INFO
- VK_STRUCTURE_TYPE_PIPELINE_RENDERING_CREATE_INFO
- VK_STRUCTURE_TYPE_PIPELINE_SHADER_STAGE_REQUIRED_SUBGROUP_SIZE_CREATE_INFO
- VK_STRUCTURE_TYPE_PRIVATE_DATA_SLOT_CREATE_INFO
- VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_INFO
- VK_STRUCTURE_TYPE_RENDERING_INFO
- VK_STRUCTURE_TYPE_RESOLVE_IMAGE_INFO_2
- VK_STRUCTURE_TYPE_SEMAPHORE_SUBMIT_INFO
- VK_STRUCTURE_TYPE_SUBMIT_INFO_2
- VK_STRUCTURE_TYPE_WRITE_DESCRIPTOR_SET_INLINE_UNIFORM_BLOCK

Vulkan Version 1.2

Vulkan Version 1.2 promoted a number of key extensions into the core API:

- VK_KHR_8bit_storage
- VK_KHR_buffer_device_address
- VK_KHR_create_renderpass2
- VK_KHR_depth_stencil_resolve
- VK_KHR_draw_indirect_count
- VK_KHR_driver_properties
- VK_KHR_image_format_list
- VK_KHR_imageless_framebuffer

- VK_KHR_sampler_mirror_clamp_to_edge
- VK_KHR_separate_depth_stencil_layouts
- VK_KHR_shader_atomic_int64
- VK_KHR_shader_float16_int8
- VK_KHR_shader_float_controls
- VK_KHR_shader_subgroup_extended_types
- VK_KHR_spirv_1_4
- VK_KHR_timeline_semaphore
- VK_KHR_uniform_buffer_standard_layout
- VK_KHR_vulkan_memory_model
- VK_EXT_descriptor_indexing
- VK_EXT_host_query_reset
- VK_EXT_sampler_filter_minmax
- VK_EXT_scalar_block_layout
- VK_EXT_separate_stencil_usage
- VK_EXT_shader_viewport_index_layer

All differences in behavior between these extensions and the corresponding Vulkan 1.2 functionality are summarized below.

Differences Relative to VK_KHR_8bit_storage

If the VK_KHR_8bit_storage extension is not supported, support for the SPIR-V storageBuffer8BitAccess capability in shader modules is optional. Support for this feature is defined by VkPhysicalDeviceVulkan12Features::storageBuffer8BitAccess when queried via vkGetPhysicalDeviceFeatures2.

Differences Relative to VK_KHR_draw_indirect_count

If the VK_KHR_draw_indirect_count extension is not supported, support for the commands vkCmdDrawIndirectCount and vkCmdDrawIndexedIndirectCount is optional. Support for this feature is defined by VkPhysicalDeviceVulkan12Features::drawIndirectCount when queried via vkGetPhysicalDeviceFeatures2.

Differences Relative to VK_KHR_sampler_mirror_clamp_to_edge

If the VK_KHR_sampler_mirror_clamp_to_edge extension is not supported, support for the VkSamplerAddressMode VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE is optional. Support for this feature is defined by VkPhysicalDeviceVulkan12Features ::samplerMirrorClampToEdge when queried via vkGetPhysicalDeviceFeatures2.

Differences Relative to VK_EXT_descriptor_indexing

If the VK_EXT_descriptor_indexing extension is not supported, support for the descriptorIndexing feature is optional. Support for this feature is defined by VkPhysicalDeviceVulkan12Features::descriptorIndexing when queried via vkGetPhysicalDeviceFeatures2.

Differences Relative to VK_EXT_scalar_block_layout

If the VK_EXT_scalar_block_layout extension is not supported, support for the scalarBlockLayout feature is optional. Support for this feature is defined by VkPhysicalDeviceVulkan12Features::scalarBlockLayout when queried via vkGetPhysicalDeviceFeatures2.

Differences Relative to VK_EXT_shader_viewport_index_layer

The ShaderViewportIndexLayerEXT SPIR-V capability was replaced with the ShaderViewportIndex and ShaderLayer capabilities. Declaring both is equivalent to declaring ShaderViewportIndexLayerEXT. If the VK_EXT_shader_viewport_index_layer extension is not supported, support for the ShaderViewportIndexLayerEXT SPIR-V capability is optional. Support for this feature is defined by VkPhysicalDeviceVulkan12Features::shaderOutputViewportIndex and VkPhysicalDeviceVulkan12Features::shaderOutputLayer when queried via vkGetPhysicalDeviceFeatures2.

Differences Relative to VK_KHR_buffer_device_address

If the VK_KHR_buffer_device_address extension is not supported, support for the bufferDeviceAddress feature is optional. Support for this feature is defined by VkPhysicalDeviceVulkan12Features::bufferDeviceAddress when queried via vkGetPhysicalDeviceFeatures2.

Differences Relative to VK_KHR_shader_atomic_int64

If the VK_KHR_shader_atomic_int64 extension is not supported, support for the shaderBufferInt64Atomics feature is optional. Support for this feature is defined by VkPhysicalDeviceVulkan12Features::shaderBufferInt64Atomics when queried via vkGetPhysicalDeviceFeatures2.

Differences Relative to VK_KHR_shader_float16_int8

If the VK_KHR_shader_float16_int8 extension is not supported, support for the shaderFloat16 and shaderInt8 features is optional. Support for these features are defined by VkPhysicalDeviceVulkan12Features::shaderFloat16 and VkPhysicalDeviceVulkan12Features::shaderInt8 when queried via vkGetPhysicalDeviceFeatures2.

Differences Relative to VK_KHR_vulkan_memory_model

If the VK_KHR_vulkan_memory_model extension is not supported, support for the vulkanMemoryModel feature is optional. Support for this feature is defined by VkPhysicalDeviceVulkan12Features::vulkanMemoryModel when queried via vkGetPhysicalDeviceFeatures2.

Additional Vulkan 1.2 Feature Support

In addition to the promoted extensions described above, Vulkan 1.2 added support for:

- SPIR-V version 1.4.
- SPIR-V version 1.5.

- The `samplerMirrorClampToEdge` feature which indicates whether the implementation supports the VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE sampler address mode.
- The `ShaderNonUniform` capability in SPIR-V version 1.5.
- The `shaderOutputViewportIndex` feature which indicates that the `ShaderViewportIndex` capability can be used.
- The `shaderOutputLayer` feature which indicates that the `ShaderLayer` capability can be used.
- The `subgroupBroadcastDynamicId` feature which allows the "Id" operand of `OpGroupNonUniformBroadcast` to be dynamically uniform within a subgroup, and the "Index" operand of `OpGroupNonUniformQuadBroadcast` to be dynamically uniform within a derivative group, in shader modules of version 1.5 or higher.
- The `drawIndirectCount` feature which indicates whether the vkCmdDrawIndirectCount and vkCmdDrawIndexedIndirectCount functions can be used.
- The `descriptorIndexing` feature which indicates the implementation supports the minimum number of descriptor indexing features as defined in the Feature Requirements section.
- The `samplerFilterMinmax` feature which indicates whether the implementation supports the minimum number of image formats that support the VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT feature bit as defined by the `filterMinmaxSingleComponentFormats` property minimum requirements.
- The `framebufferIntegerColorSampleCounts` limit which indicates the color sample counts that are supported for all framebuffer color attachments with integer formats.

New Macros

- VK_API_VERSION_1_2

New Commands

- vkCmdBeginRenderPass2
- vkCmdDrawIndexedIndirectCount
- vkCmdDrawIndirectCount
- vkCmdEndRenderPass2
- vkCmdNextSubpass2
- vkCreateRenderPass2
- vkGetBufferDeviceAddress
- vkGetBufferOpaqueCaptureAddress
- vkGetDeviceMemoryOpaqueCaptureAddress
- vkGetSemaphoreCounterValue
- vkResetQueryPool
- vkSignalSemaphore
- vkWaitSemaphores

New Structures

- VkAttachmentDescription2
- VkAttachmentReference2
- VkBufferDeviceAddressInfo
- VkConformanceVersion
- VkDeviceMemoryOpaqueCaptureAddressInfo
- VkFramebufferAttachmentImageInfo
- VkRenderPassCreateInfo2
- VkSemaphoreSignalInfo
- VkSemaphoreWaitInfo
- VkSubpassBeginInfo
- VkSubpassDependency2
- VkSubpassDescription2
- VkSubpassEndInfo
- Extending VkAttachmentDescription2:
  - VkAttachmentDescriptionStencilLayout
- Extending VkAttachmentReference2:
  - VkAttachmentReferenceStencilLayout
- Extending VkBufferCreateInfo:
  - VkBufferOpaqueCaptureAddressCreateInfo
- Extending VkDescriptorSetAllocateInfo:
  - VkDescriptorSetVariableDescriptorCountAllocateInfo
- Extending VkDescriptorSetLayoutCreateInfo:
  - VkDescriptorSetLayoutBindingFlagsCreateInfo
- Extending VkDescriptorSetLayoutSupport:
  - VkDescriptorSetVariableDescriptorCountLayoutSupport
- Extending VkFramebufferCreateInfo:
  - VkFramebufferAttachmentsCreateInfo
- Extending VkImageCreateInfo, VkPhysicalDeviceImageFormatInfo2:
  - VkImageStencilUsageCreateInfo
- Extending VkImageCreateInfo, VkSwapchainCreateInfoKHR, VkPhysicalDeviceImageFormatInfo2:
  - VkImageFormatListCreateInfo
- Extending VkMemoryAllocateInfo:
  - VkMemoryOpaqueCaptureAddressAllocateInfo

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevice8BitStorageFeatures
  - VkPhysicalDeviceBufferDeviceAddressFeatures
  - VkPhysicalDeviceDescriptorIndexingFeatures
  - VkPhysicalDeviceHostQueryResetFeatures
  - VkPhysicalDeviceImagelessFramebufferFeatures
  - VkPhysicalDeviceScalarBlockLayoutFeatures
  - VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures
  - VkPhysicalDeviceShaderAtomicInt64Features
  - VkPhysicalDeviceShaderFloat16Int8Features
  - VkPhysicalDeviceShaderSubgroupExtendedTypesFeatures
  - VkPhysicalDeviceTimelineSemaphoreFeatures
  - VkPhysicalDeviceUniformBufferStandardLayoutFeatures
  - VkPhysicalDeviceVulkan11Features
  - VkPhysicalDeviceVulkan12Features
  - VkPhysicalDeviceVulkanMemoryModelFeatures
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceDepthStencilResolveProperties
  - VkPhysicalDeviceDescriptorIndexingProperties
  - VkPhysicalDeviceDriverProperties
  - VkPhysicalDeviceFloatControlsProperties
  - VkPhysicalDeviceSamplerFilterMinmaxProperties
  - VkPhysicalDeviceTimelineSemaphoreProperties
  - VkPhysicalDeviceVulkan11Properties
  - VkPhysicalDeviceVulkan12Properties
- Extending VkRenderPassBeginInfo:
  - VkRenderPassAttachmentBeginInfo
- Extending VkSamplerCreateInfo:
  - VkSamplerReductionModeCreateInfo
- Extending VkSemaphoreCreateInfo, VkPhysicalDeviceExternalSemaphoreInfo:
  - VkSemaphoreTypeCreateInfo
- Extending VkSubmitInfo, VkBindSparseInfo:
  - VkTimelineSemaphoreSubmitInfo
- Extending VkSubpassDescription2:
  - VkSubpassDescriptionDepthStencilResolve

New Enums

- VkDescriptorBindingFlagBits
- VkDriverId
- VkResolveModeFlagBits
- VkSamplerReductionMode
- VkSemaphoreType
- VkSemaphoreWaitFlagBits
- VkShaderFloatControlsIndependence

New Bitmasks

- VkDescriptorBindingFlags
- VkResolveModeFlags
- VkSemaphoreWaitFlags

New Enum Constants

- VK_MAX_DRIVER_INFO_SIZE
- VK_MAX_DRIVER_NAME_SIZE
- Extending VkBufferCreateFlagBits:
  - VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT
- Extending VkBufferUsageFlagBits:
  - VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT
- Extending VkDescriptorPoolCreateFlagBits:
  - VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT
- Extending VkDescriptorSetLayoutCreateFlagBits:
  - VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT
- Extending VkFormatFeatureFlagBits:
  - VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_MINMAX_BIT
- Extending VkFramebufferCreateFlagBits:
  - VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT
- Extending VkImageLayout:
  - VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL
  - VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL
  - VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL
  - VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL
- Extending VkMemoryAllocateFlagBits:

    - VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT
    - VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT
- Extending VkResult:
    - VK_ERROR_FRAGMENTATION
    - VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS
- Extending VkSamplerAddressMode:
    - VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_2
    - VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_STENCIL_LAYOUT
    - VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_2
    - VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_STENCIL_LAYOUT
    - VK_STRUCTURE_TYPE_BUFFER_DEVICE_ADDRESS_INFO
    - VK_STRUCTURE_TYPE_BUFFER_OPAQUE_CAPTURE_ADDRESS_CREATE_INFO
    - VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_BINDING_FLAGS_CREATE_INFO
    - VK_STRUCTURE_TYPE_DESCRIPTOR_SET_VARIABLE_DESCRIPTOR_COUNT_ALLOCATE_INFO
    - VK_STRUCTURE_TYPE_DESCRIPTOR_SET_VARIABLE_DESCRIPTOR_COUNT_LAYOUT_SUPPORT
    - VK_STRUCTURE_TYPE_DEVICE_MEMORY_OPAQUE_CAPTURE_ADDRESS_INFO
    - VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENTS_CREATE_INFO
    - VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENT_IMAGE_INFO
    - VK_STRUCTURE_TYPE_IMAGE_FORMAT_LIST_CREATE_INFO
    - VK_STRUCTURE_TYPE_IMAGE_STENCIL_USAGE_CREATE_INFO
    - VK_STRUCTURE_TYPE_MEMORY_OPAQUE_CAPTURE_ADDRESS_ALLOCATE_INFO
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_8BIT_STORAGE_FEATURES
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BUFFER_DEVICE_ADDRESS_FEATURES
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_STENCIL_RESOLVE_PROPERTIES
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_INDEXING_FEATURES
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_INDEXING_PROPERTIES
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRIVER_PROPERTIES
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FLOAT_CONTROLS_PROPERTIES
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_QUERY_RESET_FEATURES
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGELESS_FRAMEBUFFER_FEATURES
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLER_FILTER_MINMAX_PROPERTIES
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SCALAR_BLOCK_LAYOUT_FEATURES

- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SEPARATE_DEPTH_STENCIL_LAYOUTS_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_INT64_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT16_INT8_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_EXTENDED_TYPES_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_UNIFORM_BUFFER_STANDARD_LAYOUT_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_1_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_1_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_2_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_1_2_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_MEMORY_MODEL_FEATURES
- VK_STRUCTURE_TYPE_RENDER_PASS_ATTACHMENT_BEGIN_INFO
- VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO_2
- VK_STRUCTURE_TYPE_SAMPLER_REDUCTION_MODE_CREATE_INFO
- VK_STRUCTURE_TYPE_SEMAPHORE_SIGNAL_INFO
- VK_STRUCTURE_TYPE_SEMAPHORE_TYPE_CREATE_INFO
- VK_STRUCTURE_TYPE_SEMAPHORE_WAIT_INFO
- VK_STRUCTURE_TYPE_SUBPASS_BEGIN_INFO
- VK_STRUCTURE_TYPE_SUBPASS_DEPENDENCY_2
- VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_2
- VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_DEPTH_STENCIL_RESOLVE
- VK_STRUCTURE_TYPE_SUBPASS_END_INFO
- VK_STRUCTURE_TYPE_TIMELINE_SEMAPHORE_SUBMIT_INFO

Vulkan Version 1.1

Vulkan Version 1.1 promoted a number of key extensions into the core API:

- VK_KHR_16bit_storage
- VK_KHR_bind_memory2
- VK_KHR_dedicated_allocation
- VK_KHR_descriptor_update_template
- VK_KHR_device_group

- VK_KHR_device_group_creation
- VK_KHR_external_fence
- VK_KHR_external_fence_capabilities
- VK_KHR_external_memory
- VK_KHR_external_memory_capabilities
- VK_KHR_external_semaphore
- VK_KHR_external_semaphore_capabilities
- VK_KHR_get_memory_requirements2
- VK_KHR_get_physical_device_properties2
- VK_KHR_maintenance1
- VK_KHR_maintenance2
- VK_KHR_maintenance3
- VK_KHR_multiview
- VK_KHR_relaxed_block_layout
- VK_KHR_sampler_ycbcr_conversion
- VK_KHR_shader_draw_parameters
- VK_KHR_storage_buffer_storage_class
- VK_KHR_variable_pointers

All differences in behavior between these extensions and the corresponding Vulkan 1.1 functionality are summarized below.

Differences Relative to `VK_KHR_16bit_storage`

If the `VK_KHR_16bit_storage` extension is not supported, support for the `storageBuffer16BitAccess` feature is optional. Support for this feature is defined by VkPhysicalDevice16BitStorageFeatures::`storageBuffer16BitAccess` or VkPhysicalDeviceVulkan11Features::`storageBuffer16BitAccess` when queried via vkGetPhysicalDeviceFeatures2.

Differences Relative to `VK_KHR_sampler_ycbcr_conversion`

If the `VK_KHR_sampler_ycbcr_conversion` extension is not supported, support for the `samplerYcbcrConversion` feature is optional. Support for this feature is defined by VkPhysicalDeviceSamplerYcbcrConversionFeatures::`samplerYcbcrConversion` or VkPhysicalDeviceVulkan11Features::`samplerYcbcrConversion` when queried via vkGetPhysicalDeviceFeatures2.

Differences Relative to VK_KHR_shader_draw_parameters

If the VK_KHR_shader_draw_parameters extension is not supported, support for the SPV_KHR_shader_draw_parameters SPIR-V extension is optional. Support for this feature is defined by VkPhysicalDeviceShaderDrawParametersFeatures::shaderDrawParameters or VkPhysicalDeviceVulkan11Features::shaderDrawParameters when queried via vkGetPhysicalDeviceFeatures2.

Differences Relative to VK_KHR_variable_pointers

If the VK_KHR_variable_pointers extension is not supported, support for the variablePointersStorageBuffer feature is optional. Support for this feature is defined by VkPhysicalDeviceVariablePointersFeatures::variablePointersStorageBuffer or VkPhysicalDeviceVulkan11Features::variablePointersStorageBuffer when queried via vkGetPhysicalDeviceFeatures2.

Additional Vulkan 1.1 Feature Support

In addition to the promoted extensions described above, Vulkan 1.1 added support for:

- SPIR-V version 1.1
- SPIR-V version 1.2
- SPIR-V version 1.3
- The group operations and subgroup scope.
- The protected memory feature.
- A new command to enumerate the instance version: vkEnumerateInstanceVersion.
- The VkPhysicalDeviceShaderDrawParametersFeatures feature query structure (where the VK_KHR_shader_draw_parameters extension did not have one).

New Macros

- VK_API_VERSION_1_1

New Object Types

- VkDescriptorUpdateTemplate
- VkSamplerYcbcrConversion

New Commands

- vkBindBufferMemory2
- vkBindImageMemory2
- vkCmdDispatchBase
- vkCmdSetDeviceMask
- vkCreateDescriptorUpdateTemplate
- vkCreateSamplerYcbcrConversion

- vkDestroyDescriptorUpdateTemplate
- vkDestroySamplerYcbcrConversion
- vkEnumerateInstanceVersion
- vkEnumeratePhysicalDeviceGroups
- vkGetBufferMemoryRequirements2
- vkGetDescriptorSetLayoutSupport
- vkGetDeviceGroupPeerMemoryFeatures
- vkGetDeviceQueue2
- vkGetImageMemoryRequirements2
- vkGetImageSparseMemoryRequirements2
- vkGetPhysicalDeviceExternalBufferProperties
- vkGetPhysicalDeviceExternalFenceProperties
- vkGetPhysicalDeviceExternalSemaphoreProperties
- vkGetPhysicalDeviceFeatures2
- vkGetPhysicalDeviceFormatProperties2
- vkGetPhysicalDeviceImageFormatProperties2
- vkGetPhysicalDeviceMemoryProperties2
- vkGetPhysicalDeviceProperties2
- vkGetPhysicalDeviceQueueFamilyProperties2
- vkGetPhysicalDeviceSparseImageFormatProperties2
- vkTrimCommandPool
- vkUpdateDescriptorSetWithTemplate

New Structures

- VkBindBufferMemoryInfo
- VkBindImageMemoryInfo
- VkBufferMemoryRequirementsInfo2
- VkDescriptorSetLayoutSupport
- VkDescriptorUpdateTemplateCreateInfo
- VkDescriptorUpdateTemplateEntry
- VkDeviceQueueInfo2
- VkExternalBufferProperties
- VkExternalFenceProperties
- VkExternalMemoryProperties
- VkExternalSemaphoreProperties

- VkFormatProperties2
- VkImageFormatProperties2
- VkImageMemoryRequirementsInfo2
- VkImageSparseMemoryRequirementsInfo2
- VkInputAttachmentAspectReference
- VkMemoryRequirements2
- VkPhysicalDeviceExternalBufferInfo
- VkPhysicalDeviceExternalFenceInfo
- VkPhysicalDeviceExternalSemaphoreInfo
- VkPhysicalDeviceGroupProperties
- VkPhysicalDeviceImageFormatInfo2
- VkPhysicalDeviceMemoryProperties2
- VkPhysicalDeviceProperties2
- VkPhysicalDeviceSparseImageFormatInfo2
- VkQueueFamilyProperties2
- VkSamplerYcbcrConversionCreateInfo
- VkSparseImageFormatProperties2
- VkSparseImageMemoryRequirements2
- Extending VkBindBufferMemoryInfo:
  - VkBindBufferMemoryDeviceGroupInfo
- Extending VkBindImageMemoryInfo:
  - VkBindImageMemoryDeviceGroupInfo
  - VkBindImagePlaneMemoryInfo
- Extending VkBindSparseInfo:
  - VkDeviceGroupBindSparseInfo
- Extending VkBufferCreateInfo:
  - VkExternalMemoryBufferCreateInfo
- Extending VkCommandBufferBeginInfo:
  - VkDeviceGroupCommandBufferBeginInfo
- Extending VkDeviceCreateInfo:
  - VkDeviceGroupDeviceCreateInfo
  - VkPhysicalDeviceFeatures2
- Extending VkFenceCreateInfo:
  - VkExportFenceCreateInfo
- Extending VkImageCreateInfo:

    - VkExternalMemoryImageCreateInfo
- Extending VkImageFormatProperties2:
    - VkExternalImageFormatProperties
    - VkSamplerYcbcrConversionImageFormatProperties
- Extending VkImageMemoryRequirementsInfo2:
    - VkImagePlaneMemoryRequirementsInfo
- Extending VkImageViewCreateInfo:
    - VkImageViewUsageCreateInfo
- Extending VkMemoryAllocateInfo:
    - VkExportMemoryAllocateInfo
    - VkMemoryAllocateFlagsInfo
    - VkMemoryDedicatedAllocateInfo
- Extending VkMemoryRequirements2:
    - VkMemoryDedicatedRequirements
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
    - VkPhysicalDevice16BitStorageFeatures
    - VkPhysicalDeviceMultiviewFeatures
    - VkPhysicalDeviceProtectedMemoryFeatures
    - VkPhysicalDeviceSamplerYcbcrConversionFeatures
    - VkPhysicalDeviceShaderDrawParameterFeatures
    - VkPhysicalDeviceShaderDrawParametersFeatures
    - VkPhysicalDeviceVariablePointerFeatures
    - VkPhysicalDeviceVariablePointersFeatures
- Extending VkPhysicalDeviceImageFormatInfo2:
    - VkPhysicalDeviceExternalImageFormatInfo
- Extending VkPhysicalDeviceProperties2:
    - VkPhysicalDeviceIDProperties
    - VkPhysicalDeviceMaintenance3Properties
    - VkPhysicalDeviceMultiviewProperties
    - VkPhysicalDevicePointClippingProperties
    - VkPhysicalDeviceProtectedMemoryProperties
    - VkPhysicalDeviceSubgroupProperties
- Extending VkPipelineTessellationStateCreateInfo:
    - VkPipelineTessellationDomainOriginStateCreateInfo
- Extending VkRenderPassBeginInfo, VkRenderingInfo:

    - VkDeviceGroupRenderPassBeginInfo
- Extending VkRenderPassCreateInfo:
    - VkRenderPassInputAttachmentAspectCreateInfo
    - VkRenderPassMultiviewCreateInfo
- Extending VkSamplerCreateInfo, VkImageViewCreateInfo:
    - VkSamplerYcbcrConversionInfo
- Extending VkSemaphoreCreateInfo:
    - VkExportSemaphoreCreateInfo
- Extending VkSubmitInfo:
    - VkDeviceGroupSubmitInfo
    - VkProtectedSubmitInfo

New Enums

- VkChromaLocation
- VkDescriptorUpdateTemplateType
- VkDeviceQueueCreateFlagBits
- VkExternalFenceFeatureFlagBits
- VkExternalFenceHandleTypeFlagBits
- VkExternalMemoryFeatureFlagBits
- VkExternalMemoryHandleTypeFlagBits
- VkExternalSemaphoreFeatureFlagBits
- VkExternalSemaphoreHandleTypeFlagBits
- VkFenceImportFlagBits
- VkMemoryAllocateFlagBits
- VkPeerMemoryFeatureFlagBits
- VkPointClippingBehavior
- VkSamplerYcbcrModelConversion
- VkSamplerYcbcrRange
- VkSemaphoreImportFlagBits
- VkSubgroupFeatureFlagBits
- VkTessellationDomainOrigin

New Bitmasks

- VkCommandPoolTrimFlags
- VkDescriptorUpdateTemplateCreateFlags

- VkExternalFenceFeatureFlags
- VkExternalFenceHandleTypeFlags
- VkExternalMemoryFeatureFlags
- VkExternalMemoryHandleTypeFlags
- VkExternalSemaphoreFeatureFlags
- VkExternalSemaphoreHandleTypeFlags
- VkFenceImportFlags
- VkMemoryAllocateFlags
- VkPeerMemoryFeatureFlags
- VkSemaphoreImportFlags
- VkSubgroupFeatureFlags

New Enum Constants

- VK_LUID_SIZE
- VK_MAX_DEVICE_GROUP_SIZE
- VK_QUEUE_FAMILY_EXTERNAL
- Extending VkBufferCreateFlagBits:
  - VK_BUFFER_CREATE_PROTECTED_BIT
- Extending VkCommandPoolCreateFlagBits:
  - VK_COMMAND_POOL_CREATE_PROTECTED_BIT
- Extending VkDependencyFlagBits:
  - VK_DEPENDENCY_DEVICE_GROUP_BIT
  - VK_DEPENDENCY_VIEW_LOCAL_BIT
- Extending VkDeviceQueueCreateFlagBits:
  - VK_DEVICE_QUEUE_CREATE_PROTECTED_BIT
- Extending VkFormat:
  - VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16
  - VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16
  - VK_FORMAT_B16G16R16G16_422_UNORM
  - VK_FORMAT_B8G8R8G8_422_UNORM
  - VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16
  - VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16
  - VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16
  - VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16
  - VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16

    - VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16
    - VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16
    - VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16
    - VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16
    - VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16
    - VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16
    - VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16
    - VK_FORMAT_G16B16G16R16_422_UNORM
    - VK_FORMAT_G16_B16R16_2PLANE_420_UNORM
    - VK_FORMAT_G16_B16R16_2PLANE_422_UNORM
    - VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM
    - VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM
    - VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM
    - VK_FORMAT_G8B8G8R8_422_UNORM
    - VK_FORMAT_G8_B8R8_2PLANE_420_UNORM
    - VK_FORMAT_G8_B8R8_2PLANE_422_UNORM
    - VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM
    - VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM
    - VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM
    - VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16
    - VK_FORMAT_R10X6G10X6_UNORM_2PACK16
    - VK_FORMAT_R10X6_UNORM_PACK16
    - VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16
    - VK_FORMAT_R12X4G12X4_UNORM_2PACK16
    - VK_FORMAT_R12X4_UNORM_PACK16
- Extending VkFormatFeatureFlagBits:
    - VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT
    - VK_FORMAT_FEATURE_DISJOINT_BIT
    - VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT
    - VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT
    - VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT
    - VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT
    - VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCT

ION_FILTER_BIT
    - VK_FORMAT_FEATURE_TRANSFER_DST_BIT
    - VK_FORMAT_FEATURE_TRANSFER_SRC_BIT
- Extending VkImageAspectFlagBits:
    - VK_IMAGE_ASPECT_PLANE_0_BIT
    - VK_IMAGE_ASPECT_PLANE_1_BIT
    - VK_IMAGE_ASPECT_PLANE_2_BIT
- Extending VkImageCreateFlagBits:
    - VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT
    - VK_IMAGE_CREATE_ALIAS_BIT
    - VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT
    - VK_IMAGE_CREATE_DISJOINT_BIT
    - VK_IMAGE_CREATE_EXTENDED_USAGE_BIT
    - VK_IMAGE_CREATE_PROTECTED_BIT
    - VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT
- Extending VkImageLayout:
    - VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL
    - VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL
- Extending VkMemoryHeapFlagBits:
    - VK_MEMORY_HEAP_MULTI_INSTANCE_BIT
- Extending VkMemoryPropertyFlagBits:
    - VK_MEMORY_PROPERTY_PROTECTED_BIT
- Extending VkObjectType:
    - VK_OBJECT_TYPE_DESCRIPTOR_UPDATE_TEMPLATE
    - VK_OBJECT_TYPE_SAMPLER_YCBCR_CONVERSION
- Extending VkPipelineCreateFlagBits:
    - VK_PIPELINE_CREATE_DISPATCH_BASE
    - VK_PIPELINE_CREATE_DISPATCH_BASE_BIT
    - VK_PIPELINE_CREATE_VIEW_INDEX_FROM_DEVICE_INDEX_BIT
- Extending VkQueueFlagBits:
    - VK_QUEUE_PROTECTED_BIT
- Extending VkResult:
    - VK_ERROR_INVALID_EXTERNAL_HANDLE
    - VK_ERROR_OUT_OF_POOL_MEMORY
- Extending VkStructureType:

- VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_DEVICE_GROUP_INFO
- VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_INFO
- VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_DEVICE_GROUP_INFO
- VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_INFO
- VK_STRUCTURE_TYPE_BIND_IMAGE_PLANE_MEMORY_INFO
- VK_STRUCTURE_TYPE_BUFFER_MEMORY_REQUIREMENTS_INFO_2
- VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_SUPPORT
- VK_STRUCTURE_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_CREATE_INFO
- VK_STRUCTURE_TYPE_DEVICE_GROUP_BIND_SPARSE_INFO
- VK_STRUCTURE_TYPE_DEVICE_GROUP_COMMAND_BUFFER_BEGIN_INFO
- VK_STRUCTURE_TYPE_DEVICE_GROUP_DEVICE_CREATE_INFO
- VK_STRUCTURE_TYPE_DEVICE_GROUP_RENDER_PASS_BEGIN_INFO
- VK_STRUCTURE_TYPE_DEVICE_GROUP_SUBMIT_INFO
- VK_STRUCTURE_TYPE_DEVICE_QUEUE_INFO_2
- VK_STRUCTURE_TYPE_EXPORT_FENCE_CREATE_INFO
- VK_STRUCTURE_TYPE_EXPORT_MEMORY_ALLOCATE_INFO
- VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_CREATE_INFO
- VK_STRUCTURE_TYPE_EXTERNAL_BUFFER_PROPERTIES
- VK_STRUCTURE_TYPE_EXTERNAL_FENCE_PROPERTIES
- VK_STRUCTURE_TYPE_EXTERNAL_IMAGE_FORMAT_PROPERTIES
- VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_BUFFER_CREATE_INFO
- VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_IMAGE_CREATE_INFO
- VK_STRUCTURE_TYPE_EXTERNAL_SEMAPHORE_PROPERTIES
- VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_2
- VK_STRUCTURE_TYPE_IMAGE_FORMAT_PROPERTIES_2
- VK_STRUCTURE_TYPE_IMAGE_MEMORY_REQUIREMENTS_INFO_2
- VK_STRUCTURE_TYPE_IMAGE_PLANE_MEMORY_REQUIREMENTS_INFO
- VK_STRUCTURE_TYPE_IMAGE_SPARSE_MEMORY_REQUIREMENTS_INFO_2
- VK_STRUCTURE_TYPE_IMAGE_VIEW_USAGE_CREATE_INFO
- VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_FLAGS_INFO
- VK_STRUCTURE_TYPE_MEMORY_DEDICATED_ALLOCATE_INFO
- VK_STRUCTURE_TYPE_MEMORY_DEDICATED_REQUIREMENTS
- VK_STRUCTURE_TYPE_MEMORY_REQUIREMENTS_2
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_16BIT_STORAGE_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_BUFFER_INFO

- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_FENCE_INFO
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_IMAGE_FORMAT_INFO
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_SEMAPHORE_INFO
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FEATURES_2
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GROUP_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ID_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_FORMAT_INFO_2
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_3_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PROPERTIES_2
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_POINT_CLIPPING_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROPERTIES_2
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROTECTED_MEMORY_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROTECTED_MEMORY_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLER_YCBCR_CONVERSION_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_DRAW_PARAMETERS_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_DRAW_PARAMETER_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SPARSE_IMAGE_FORMAT_INFO_2
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SUBGROUP_PROPERTIES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTERS_FEATURES
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTER_FEATURES
- VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_DOMAIN_ORIGIN_STATE_CREATE_INFO
- VK_STRUCTURE_TYPE_PROTECTED_SUBMIT_INFO
- VK_STRUCTURE_TYPE_QUEUE_FAMILY_PROPERTIES_2
- VK_STRUCTURE_TYPE_RENDER_PASS_INPUT_ATTACHMENT_ASPECT_CREATE_INFO
- VK_STRUCTURE_TYPE_RENDER_PASS_MULTIVIEW_CREATE_INFO
- VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_CREATE_INFO
- VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_IMAGE_FORMAT_PROPERTIES
- VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_INFO
- VK_STRUCTURE_TYPE_SPARSE_IMAGE_FORMAT_PROPERTIES_2
- VK_STRUCTURE_TYPE_SPARSE_IMAGE_MEMORY_REQUIREMENTS_2

Vulkan Version 1.0

Vulkan Version 1.0 was the initial release of the Vulkan API.

New Macros

- VK_API_VERSION
- VK_API_VERSION_1_0
- VK_API_VERSION_MAJOR
- VK_API_VERSION_MINOR
- VK_API_VERSION_PATCH
- VK_API_VERSION_VARIANT
- VK_DEFINE_HANDLE
- VK_DEFINE_NON_DISPATCHABLE_HANDLE
- VK_HEADER_VERSION
- VK_HEADER_VERSION_COMPLETE
- VK_MAKE_API_VERSION
- VK_MAKE_VERSION
- VK_NULL_HANDLE
- VK_USE_64_BIT_PTR_DEFINES
- VK_VERSION_MAJOR
- VK_VERSION_MINOR
- VK_VERSION_PATCH

New Base Types

- VkBool32
- VkDeviceAddress
- VkDeviceSize
- VkFlags
- VkSampleMask

New Object Types

- VkBuffer
- VkBufferView
- VkCommandBuffer
- VkCommandPool
- VkDescriptorPool

- VkDescriptorSet
- VkDescriptorSetLayout
- VkDevice
- VkDeviceMemory
- VkEvent
- VkFence
- VkFramebuffer
- VkImage
- VkImageView
- VkInstance
- VkPhysicalDevice
- VkPipeline
- VkPipelineCache
- VkPipelineLayout
- VkQueryPool
- VkQueue
- VkRenderPass
- VkSampler
- VkSemaphore
- VkShaderModule

New Commands

- vkAllocateCommandBuffers
- vkAllocateDescriptorSets
- vkAllocateMemory
- vkBeginCommandBuffer
- vkBindBufferMemory
- vkBindImageMemory
- vkCmdBeginQuery
- vkCmdBeginRenderPass
- vkCmdBindDescriptorSets
- vkCmdBindIndexBuffer
- vkCmdBindPipeline
- vkCmdBindVertexBuffers
- vkCmdBlitImage

- vkCmdClearAttachments
- vkCmdClearColorImage
- vkCmdClearDepthStencilImage
- vkCmdCopyBuffer
- vkCmdCopyBufferToImage
- vkCmdCopyImage
- vkCmdCopyImageToBuffer
- vkCmdCopyQueryPoolResults
- vkCmdDispatch
- vkCmdDispatchIndirect
- vkCmdDraw
- vkCmdDrawIndexed
- vkCmdDrawIndexedIndirect
- vkCmdDrawIndirect
- vkCmdEndQuery
- vkCmdEndRenderPass
- vkCmdExecuteCommands
- vkCmdFillBuffer
- vkCmdNextSubpass
- vkCmdPipelineBarrier
- vkCmdPushConstants
- vkCmdResetEvent
- vkCmdResetQueryPool
- vkCmdResolveImage
- vkCmdSetBlendConstants
- vkCmdSetDepthBias
- vkCmdSetDepthBounds
- vkCmdSetEvent
- vkCmdSetLineWidth
- vkCmdSetScissor
- vkCmdSetStencilCompareMask
- vkCmdSetStencilReference
- vkCmdSetStencilWriteMask
- vkCmdSetViewport
- vkCmdUpdateBuffer

- vkCmdWaitEvents
- vkCmdWriteTimestamp
- vkCreateBuffer
- vkCreateBufferView
- vkCreateCommandPool
- vkCreateComputePipelines
- vkCreateDescriptorPool
- vkCreateDescriptorSetLayout
- vkCreateDevice
- vkCreateEvent
- vkCreateFence
- vkCreateFramebuffer
- vkCreateGraphicsPipelines
- vkCreateImage
- vkCreateImageView
- vkCreateInstance
- vkCreatePipelineCache
- vkCreatePipelineLayout
- vkCreateQueryPool
- vkCreateRenderPass
- vkCreateSampler
- vkCreateSemaphore
- vkCreateShaderModule
- vkDestroyBuffer
- vkDestroyBufferView
- vkDestroyCommandPool
- vkDestroyDescriptorPool
- vkDestroyDescriptorSetLayout
- vkDestroyDevice
- vkDestroyEvent
- vkDestroyFence
- vkDestroyFramebuffer
- vkDestroyImage
- vkDestroyImageView
- vkDestroyInstance

- vkDestroyPipeline
- vkDestroyPipelineCache
- vkDestroyPipelineLayout
- vkDestroyQueryPool
- vkDestroyRenderPass
- vkDestroySampler
- vkDestroySemaphore
- vkDestroyShaderModule
- vkDeviceWaitIdle
- vkEndCommandBuffer
- vkEnumerateDeviceExtensionProperties
- vkEnumerateDeviceLayerProperties
- vkEnumerateInstanceExtensionProperties
- vkEnumerateInstanceLayerProperties
- vkEnumeratePhysicalDevices
- vkFlushMappedMemoryRanges
- vkFreeCommandBuffers
- vkFreeDescriptorSets
- vkFreeMemory
- vkGetBufferMemoryRequirements
- vkGetDeviceMemoryCommitment
- vkGetDeviceProcAddr
- vkGetDeviceQueue
- vkGetEventStatus
- vkGetFenceStatus
- vkGetImageMemoryRequirements
- vkGetImageSparseMemoryRequirements
- vkGetImageSubresourceLayout
- vkGetInstanceProcAddr
- vkGetPhysicalDeviceFeatures
- vkGetPhysicalDeviceFormatProperties
- vkGetPhysicalDeviceImageFormatProperties
- vkGetPhysicalDeviceMemoryProperties
- vkGetPhysicalDeviceProperties
- vkGetPhysicalDeviceQueueFamilyProperties

- vkGetPhysicalDeviceSparseImageFormatProperties
- vkGetPipelineCacheData
- vkGetQueryPoolResults
- vkGetRenderAreaGranularity
- vkInvalidateMappedMemoryRanges
- vkMapMemory
- vkMergePipelineCaches
- vkQueueBindSparse
- vkQueueSubmit
- vkQueueWaitIdle
- vkResetCommandBuffer
- vkResetCommandPool
- vkResetDescriptorPool
- vkResetEvent
- vkResetFences
- vkSetEvent
- vkUnmapMemory
- vkUpdateDescriptorSets
- vkWaitForFences

New Structures

- VkAllocationCallbacks
- VkApplicationInfo
- VkAttachmentDescription
- VkAttachmentReference
- VkBaseInStructure
- VkBaseOutStructure
- VkBindSparseInfo
- VkBufferCopy
- VkBufferCreateInfo
- VkBufferImageCopy
- VkBufferMemoryBarrier
- VkBufferViewCreateInfo
- VkClearAttachment
- VkClearDepthStencilValue

- VkClearRect
- VkCommandBufferAllocateInfo
- VkCommandBufferBeginInfo
- VkCommandBufferInheritanceInfo
- VkCommandPoolCreateInfo
- VkComponentMapping
- VkCopyDescriptorSet
- VkDescriptorBufferInfo
- VkDescriptorImageInfo
- VkDescriptorPoolCreateInfo
- VkDescriptorPoolSize
- VkDescriptorSetAllocateInfo
- VkDescriptorSetLayoutBinding
- VkDescriptorSetLayoutCreateInfo
- VkDeviceCreateInfo
- VkDeviceQueueCreateInfo
- VkDispatchIndirectCommand
- VkDrawIndexedIndirectCommand
- VkDrawIndirectCommand
- VkEventCreateInfo
- VkExtensionProperties
- VkExtent2D
- VkExtent3D
- VkFenceCreateInfo
- VkFormatProperties
- VkFramebufferCreateInfo
- VkImageBlit
- VkImageCopy
- VkImageCreateInfo
- VkImageFormatProperties
- VkImageMemoryBarrier
- VkImageResolve
- VkImageSubresource
- VkImageSubresourceLayers
- VkImageSubresourceRange

- VkImageViewCreateInfo
- VkInstanceCreateInfo
- VkLayerProperties
- VkMappedMemoryRange
- VkMemoryAllocateInfo
- VkMemoryBarrier
- VkMemoryHeap
- VkMemoryRequirements
- VkMemoryType
- VkOffset2D
- VkOffset3D
- VkPhysicalDeviceFeatures
- VkPhysicalDeviceLimits
- VkPhysicalDeviceMemoryProperties
- VkPhysicalDeviceProperties
- VkPhysicalDeviceSparseProperties
- VkPipelineCacheCreateInfo
- VkPipelineCacheHeaderVersionOne
- VkPipelineColorBlendAttachmentState
- VkPipelineColorBlendStateCreateInfo
- VkPipelineDepthStencilStateCreateInfo
- VkPipelineDynamicStateCreateInfo
- VkPipelineInputAssemblyStateCreateInfo
- VkPipelineMultisampleStateCreateInfo
- VkPipelineRasterizationStateCreateInfo
- VkPipelineShaderStageCreateInfo
- VkPipelineTessellationStateCreateInfo
- VkPipelineVertexInputStateCreateInfo
- VkPipelineViewportStateCreateInfo
- VkPushConstantRange
- VkQueryPoolCreateInfo
- VkQueueFamilyProperties
- VkRect2D
- VkRenderPassBeginInfo
- VkRenderPassCreateInfo

- VkSamplerCreateInfo
- VkSemaphoreCreateInfo
- VkSparseBufferMemoryBindInfo
- VkSparseImageFormatProperties
- VkSparseImageMemoryBind
- VkSparseImageMemoryBindInfo
- VkSparseImageMemoryRequirements
- VkSparseImageOpaqueMemoryBindInfo
- VkSparseMemoryBind
- VkSpecializationInfo
- VkSpecializationMapEntry
- VkStencilOpState
- VkSubmitInfo
- VkSubpassDependency
- VkSubpassDescription
- VkSubresourceLayout
- VkVertexInputAttributeDescription
- VkVertexInputBindingDescription
- VkViewport
- VkWriteDescriptorSet
- Extending VkBindDescriptorSetsInfo, VkPushConstantsInfo, VkPushDescriptorSetInfo, VkPushDescriptorSetWithTemplateInfo, VkSetDescriptorBufferOffsetsInfoEXT, VkBindDescriptorBufferEmbeddedSamplersInfoEXT, VkIndirectCommandsLayoutCreateInfoEXT:
  - VkPipelineLayoutCreateInfo
- Extending VkPipelineCreateInfoKHR:
  - VkComputePipelineCreateInfo
  - VkGraphicsPipelineCreateInfo
- Extending VkPipelineShaderStageCreateInfo:
  - VkShaderModuleCreateInfo

New Unions

- VkClearColorValue
- VkClearValue

New Function Pointers

- PFN_vkAllocationFunction
- PFN_vkFreeFunction
- PFN_vkInternalAllocationNotification
- PFN_vkInternalFreeNotification
- PFN_vkReallocationFunction
- PFN_vkVoidFunction

New Enums

- VkAccessFlagBits
- VkAttachmentDescriptionFlagBits
- VkAttachmentLoadOp
- VkAttachmentStoreOp
- VkBlendFactor
- VkBlendOp
- VkBorderColor
- VkBufferCreateFlagBits
- VkBufferUsageFlagBits
- VkColorComponentFlagBits
- VkCommandBufferLevel
- VkCommandBufferResetFlagBits
- VkCommandBufferUsageFlagBits
- VkCommandPoolCreateFlagBits
- VkCommandPoolResetFlagBits
- VkCompareOp
- VkComponentSwizzle
- VkCullModeFlagBits
- VkDependencyFlagBits
- VkDescriptorPoolCreateFlagBits
- VkDescriptorSetLayoutCreateFlagBits
- VkDescriptorType
- VkDynamicState
- VkEventCreateFlagBits
- VkFenceCreateFlagBits

- VkFilter
- VkFormat
- VkFormatFeatureFlagBits
- VkFramebufferCreateFlagBits
- VkFrontFace
- VkImageAspectFlagBits
- VkImageCreateFlagBits
- VkImageLayout
- VkImageTiling
- VkImageType
- VkImageUsageFlagBits
- VkImageViewCreateFlagBits
- VkImageViewType
- VkIndexType
- VkInstanceCreateFlagBits
- VkInternalAllocationType
- VkLogicOp
- VkMemoryHeapFlagBits
- VkMemoryMapFlagBits
- VkMemoryPropertyFlagBits
- VkObjectType
- VkPhysicalDeviceType
- VkPipelineBindPoint
- VkPipelineCacheCreateFlagBits
- VkPipelineCacheHeaderVersion
- VkPipelineColorBlendStateCreateFlagBits
- VkPipelineCreateFlagBits
- VkPipelineDepthStencilStateCreateFlagBits
- VkPipelineLayoutCreateFlagBits
- VkPipelineShaderStageCreateFlagBits
- VkPipelineStageFlagBits
- VkPolygonMode
- VkPrimitiveTopology
- VkQueryControlFlagBits
- VkQueryPipelineStatisticFlagBits

- VkQueryPoolCreateFlagBits
- VkQueryResultFlagBits
- VkQueryType
- VkQueueFlagBits
- VkRenderPassCreateFlagBits
- VkResult
- VkSampleCountFlagBits
- VkSamplerAddressMode
- VkSamplerCreateFlagBits
- VkSamplerMipmapMode
- VkShaderStageFlagBits
- VkSharingMode
- VkSparseImageFormatFlagBits
- VkSparseMemoryBindFlagBits
- VkStencilFaceFlagBits
- VkStencilOp
- VkStructureType
- VkSubpassContents
- VkSubpassDescriptionFlagBits
- VkSystemAllocationScope
- VkVendorId
- VkVertexInputRate

New Bitmasks

- VkAccessFlags
- VkAttachmentDescriptionFlags
- VkBufferCreateFlags
- VkBufferUsageFlags
- VkBufferViewCreateFlags
- VkColorComponentFlags
- VkCommandBufferResetFlags
- VkCommandBufferUsageFlags
- VkCommandPoolCreateFlags
- VkCommandPoolResetFlags
- VkCullModeFlags

- VkDependencyFlags
- VkDescriptorPoolCreateFlags
- VkDescriptorPoolResetFlags
- VkDescriptorSetLayoutCreateFlags
- VkDeviceCreateFlags
- VkDeviceQueueCreateFlags
- VkEventCreateFlags
- VkFenceCreateFlags
- VkFormatFeatureFlags
- VkFramebufferCreateFlags
- VkImageAspectFlags
- VkImageCreateFlags
- VkImageUsageFlags
- VkImageViewCreateFlags
- VkInstanceCreateFlags
- VkMemoryHeapFlags
- VkMemoryMapFlags
- VkMemoryPropertyFlags
- VkPipelineCacheCreateFlags
- VkPipelineColorBlendStateCreateFlags
- VkPipelineCreateFlags
- VkPipelineDepthStencilStateCreateFlags
- VkPipelineDynamicStateCreateFlags
- VkPipelineInputAssemblyStateCreateFlags
- VkPipelineLayoutCreateFlags
- VkPipelineMultisampleStateCreateFlags
- VkPipelineRasterizationStateCreateFlags
- VkPipelineShaderStageCreateFlags
- VkPipelineStageFlags
- VkPipelineTessellationStateCreateFlags
- VkPipelineVertexInputStateCreateFlags
- VkPipelineViewportStateCreateFlags
- VkQueryControlFlags
- VkQueryPipelineStatisticFlags
- VkQueryPoolCreateFlags

- VkQueryResultFlags
- VkQueueFlags
- VkRenderPassCreateFlags
- VkSampleCountFlags
- VkSamplerCreateFlags
- VkSemaphoreCreateFlags
- VkShaderModuleCreateFlags
- VkShaderStageFlags
- VkSparseImageFormatFlags
- VkSparseMemoryBindFlags
- VkStencilFaceFlags
- VkSubpassDescriptionFlags

New Headers

- `vk_platform`

New Enum Constants

- VK_ATTACHMENT_UNUSED
- VK_FALSE
- VK_LOD_CLAMP_NONE
- VK_MAX_DESCRIPTION_SIZE
- VK_MAX_EXTENSION_NAME_SIZE
- VK_MAX_MEMORY_HEAPS
- VK_MAX_MEMORY_TYPES
- VK_MAX_PHYSICAL_DEVICE_NAME_SIZE
- VK_QUEUE_FAMILY_IGNORED
- VK_REMAINING_ARRAY_LAYERS
- VK_REMAINING_MIP_LEVELS
- VK_SUBPASS_EXTERNAL
- VK_TRUE
- VK_UUID_SIZE
- VK_WHOLE_SIZE
- Extending VkResult:
  - VK_ERROR_VALIDATION_FAILED

Appendix E: Layers & Extensions (Informative)

Extensions to the Vulkan API **can** be defined by authors, groups of authors, and the Khronos Vulkan Working Group. The online Registry of extensions is available at URL

https://registry.khronos.org/vulkan/

Authors creating extensions and layers **must** follow the mandatory procedures described in the Vulkan Documentation and Extensions document when creating extensions and layers.

The version of the Vulkan specification published on the Vulkan Documentation site and in the Registry includes all registered and published extensions. It is also possible to build the specification with an arbitrary set of extensions included, and the specification you are viewing may not include all extensions.

In the remainder of this appendix, extensions are grouped as Khronos `KHR`, multivendor `EXT`, and then alphabetically by author ID. Within each group, extensions are listed in alphabetical order by their name.

Extension Dependencies

Extensions which have dependencies on specific core versions or on other extensions will list such dependencies.

For core versions, the specified version **must** be supported at runtime. All extensions implicitly require support for Vulkan 1.0.

For a device extension, use of any device-level functionality defined by that extension requires that any extensions that extension depends on be enabled.

For any extension, use of any instance-level functionality defined by that extension requires only that any extensions that extension depends on be supported at runtime.

Extension Interactions

Some extensions define APIs which are only supported when other extensions or core versions are supported at runtime. Such interactions are noted as “API Interactions”.

List of Current Extensions

- VK_KHR_acceleration_structure
- VK_KHR_android_surface
- VK_KHR_calibrated_timestamps
- VK_KHR_compute_shader_derivatives
- VK_KHR_cooperative_matrix
- VK_KHR_copy_memory_indirect
- VK_KHR_deferred_host_operations
- VK_KHR_depth_clamp_zero_one
- VK_KHR_device_address_commands
- VK_KHR_device_fault
- VK_KHR_display
- VK_KHR_display_swapchain
- VK_KHR_extended_flags
- VK_KHR_external_fence_fd
- VK_KHR_external_fence_win32
- VK_KHR_external_memory_fd
- VK_KHR_external_memory_win32
- VK_KHR_external_semaphore_fd
- VK_KHR_external_semaphore_win32
- VK_KHR_fragment_shader_barycentric
- VK_KHR_fragment_shading_rate
- VK_KHR_get_display_properties2
- VK_KHR_get_surface_capabilities2
- VK_KHR_incremental_present
- VK_KHR_internally_synchronized_queues
- VK_KHR_maintenance10
- VK_KHR_maintenance11
- VK_KHR_maintenance7
- VK_KHR_maintenance8
- VK_KHR_maintenance9
- VK_KHR_opacity_micromap
- VK_KHR_performance_query
- VK_KHR_pipeline_binary

- VK_KHR_pipeline_executable_properties
- VK_KHR_pipeline_library
- VK_KHR_pipeline_library_group_handles
- VK_KHR_portability_enumeration
- VK_KHR_present_id
- VK_KHR_present_id2
- VK_KHR_present_mode_fifo_latest_ready
- VK_KHR_present_wait
- VK_KHR_present_wait2
- VK_KHR_ray_query
- VK_KHR_ray_tracing_maintenance1
- VK_KHR_ray_tracing_pipeline
- VK_KHR_ray_tracing_position_fetch
- VK_KHR_robustness2
- VK_KHR_shader_abort
- VK_KHR_shader_bfloat16
- VK_KHR_shader_clock
- VK_KHR_shader_constant_data
- VK_KHR_shader_fma
- VK_KHR_shader_maximal_reconvergence
- VK_KHR_shader_quad_control
- VK_KHR_shader_relaxed_extended_instruction
- VK_KHR_shader_subgroup_uniform_control_flow
- VK_KHR_shader_untyped_pointers
- VK_KHR_shared_presentable_image
- VK_KHR_surface
- VK_KHR_surface_maintenance1
- VK_KHR_surface_protected_capabilities
- VK_KHR_swapchain
- VK_KHR_swapchain_maintenance1
- VK_KHR_swapchain_mutable_format
- VK_KHR_unified_image_layouts
- VK_KHR_video_decode_av1
- VK_KHR_video_decode_h264
- VK_KHR_video_decode_h265

- VK_KHR_video_decode_queue
- VK_KHR_video_decode_vp9
- VK_KHR_video_encode_av1
- VK_KHR_video_encode_feedback2
- VK_KHR_video_encode_h264
- VK_KHR_video_encode_h265
- VK_KHR_video_encode_intra_refresh
- VK_KHR_video_encode_quantization_map
- VK_KHR_video_encode_queue
- VK_KHR_video_maintenance1
- VK_KHR_video_maintenance2
- VK_KHR_video_queue
- VK_KHR_wayland_surface
- VK_KHR_win32_keyed_mutex
- VK_KHR_win32_surface
- VK_KHR_workgroup_memory_explicit_layout
- VK_KHR_xcb_surface
- VK_KHR_xlib_surface
- VK_EXT_acquire_drm_display
- VK_EXT_acquire_xlib_display
- VK_EXT_astc_decode_mode
- VK_EXT_attachment_feedback_loop_dynamic_state
- VK_EXT_attachment_feedback_loop_layout
- VK_EXT_blend_operation_advanced
- VK_EXT_border_color_swizzle
- VK_EXT_color_write_enable
- VK_EXT_conditional_rendering
- VK_EXT_conservative_rasterization
- VK_EXT_cooperative_matrix_maintenance1
- VK_EXT_custom_border_color
- VK_EXT_custom_resolve
- VK_EXT_debug_utils
- VK_EXT_depth_bias_control
- VK_EXT_depth_clamp_control
- VK_EXT_depth_clip_control

- VK_EXT_depth_clip_enable
- VK_EXT_depth_range_unrestricted
- VK_EXT_descriptor_heap
- VK_EXT_device_address_binding_report
- VK_EXT_device_generated_commands
- VK_EXT_device_memory_report
- VK_EXT_direct_mode_display
- VK_EXT_directfb_surface
- VK_EXT_discard_rectangles
- VK_EXT_display_control
- VK_EXT_display_surface_counter
- VK_EXT_dynamic_rendering_unused_attachments
- VK_EXT_extended_dynamic_state3
- VK_EXT_external_memory_acquire_unmodified
- VK_EXT_external_memory_dma_buf
- VK_EXT_external_memory_host
- VK_EXT_external_memory_metal
- VK_EXT_filter_cubic
- VK_EXT_fragment_density_map
- VK_EXT_fragment_density_map2
- VK_EXT_fragment_density_map_offset
- VK_EXT_fragment_shader_interlock
- VK_EXT_frame_boundary
- VK_EXT_full_screen_exclusive
- VK_EXT_graphics_pipeline_library
- VK_EXT_hdr_metadata
- VK_EXT_headless_surface
- VK_EXT_image_2d_view_of_3d
- VK_EXT_image_compression_control
- VK_EXT_image_compression_control_swapchain
- VK_EXT_image_drm_format_modifier
- VK_EXT_image_sliced_view_of_3d
- VK_EXT_image_tiling_control
- VK_EXT_image_view_min_lod
- VK_EXT_layer_settings

- VK_EXT_legacy_dithering
- VK_EXT_legacy_vertex_attributes
- VK_EXT_map_memory_placed
- VK_EXT_memory_budget
- VK_EXT_memory_decompression
- VK_EXT_memory_priority
- VK_EXT_mesh_shader
- VK_EXT_metal_objects
- VK_EXT_metal_surface
- VK_EXT_multi_draw
- VK_EXT_multisampled_render_to_single_sampled
- VK_EXT_multisampled_render_to_swapchain
- VK_EXT_mutable_descriptor_type
- VK_EXT_nested_command_buffer
- VK_EXT_non_seamless_cube_map
- VK_EXT_pageable_device_local_memory
- VK_EXT_pci_bus_info
- VK_EXT_physical_device_drm
- VK_EXT_pipeline_properties
- VK_EXT_post_depth_coverage
- VK_EXT_present_timing
- VK_EXT_primitive_restart_index
- VK_EXT_primitive_topology_list_restart
- VK_EXT_primitives_generated_query
- VK_EXT_provoking_vertex
- VK_EXT_queue_family_foreign
- VK_EXT_rasterization_order_attachment_access
- VK_EXT_ray_tracing_invocation_reorder
- VK_EXT_rgba10x6_formats
- VK_EXT_sample_locations
- VK_EXT_shader_64bit_indexing
- VK_EXT_shader_atomic_float
- VK_EXT_shader_atomic_float2
- VK_EXT_shader_float8
- VK_EXT_shader_image_atomic_int64

- VK_EXT_shader_long_vector
- VK_EXT_shader_module_identifier
- VK_EXT_shader_object
- VK_EXT_shader_ocp_microscaling_types
- VK_EXT_shader_replicated_composites
- VK_EXT_shader_split_barrier
- VK_EXT_shader_stencil_export
- VK_EXT_shader_subgroup_partitioned
- VK_EXT_shader_tile_image
- VK_EXT_shader_uniform_buffer_unsized_array
- VK_EXT_subpass_merge_feedback
- VK_EXT_swapchain_colorspace
- VK_EXT_texture_compression_astc_3d
- VK_EXT_transform_feedback
- VK_EXT_validation_cache
- VK_EXT_vertex_input_dynamic_state
- VK_EXT_ycbcr_image_arrays
- VK_EXT_zero_initialize_device_memory

VK_KHR_acceleration_structure

Name String

`VK_KHR_acceleration_structure`

Extension Type

Device extension

Registered Extension Number

151

Revision

13

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.1
and
VK_EXT_descriptor_indexing
and
VK_KHR_buffer_device_address
or
Vulkan Version 1.2
and
VK_KHR_deferred_host_operations

API Interactions

- Interacts with VK_VERSION_1_2
- Interacts with VK_VERSION_1_3
- Interacts with VK_EXT_debug_report
- Interacts with VK_KHR_format_feature_flags2

Contact

- Daniel Koch [GitHub]dgkoch

Other Extension Metadata

Last Modified Date

2021-09-30

Contributors

- Samuel Bourasseau, Adobe
- Matthäus Chajdas, AMD
- Greg Grebe, AMD

- Nicolai Hähnle, AMD
- Tobias Hector, AMD
- Dave Oldcorn, AMD
- Skyler Saleh, AMD
- Mathieu Robart, Arm
- Marius Bjorge, Arm
- Tom Olson, Arm
- Sebastian Tafuri, EA
- Henrik Rydgard, Embark
- Juan Cañada, Epic Games
- Patrick Kelly, Epic Games
- Yuriy O'Donnell, Epic Games
- Michael Doggett, Facebook/Oculus
- Ricardo Garcia, Igalia
- Andrew Garrard, Imagination
- Don Scorgie, Imagination
- Dae Kim, Imagination
- Joshua Barczak, Intel
- Slawek Grajewski, Intel
- Jeff Bolz, NVIDIA
- Pascal Gautron, NVIDIA
- Daniel Koch, NVIDIA
- Christoph Kubisch, NVIDIA
- Ashwin Lele, NVIDIA
- Robert Stepinski, NVIDIA
- Martin Stich, NVIDIA
- Nuno Subtil, NVIDIA
- Eric Werness, NVIDIA
- Jon Leech, Khronos
- Jeroen van Schijndel, OTOY
- Juul Joosten, OTOY
- Alex Bourd, Qualcomm
- Roman Larionov, Qualcomm
- David McAllister, Qualcomm
- Lewis Gordon, Samsung

- Ralph Potter, Samsung
- Jasper Bekkers, Traverse Research
- Jesse Barker, Unity
- Baldur Karlsson, Valve

Description

In order to be efficient, rendering techniques such as ray tracing need a quick way to identify which primitives may be intersected by a ray traversing the geometries. Acceleration structures are the most common way to represent the geometry spatially sorted, in order to quickly identify such potential intersections.

This extension adds new functionalities:

- Acceleration structure objects and build commands
- Structures to describe geometry inputs to acceleration structure builds
- Acceleration structure copy commands

New Object Types

- VkAccelerationStructureKHR

New Commands

- vkBuildAccelerationStructuresKHR
- vkCmdBuildAccelerationStructuresIndirectKHR
- vkCmdBuildAccelerationStructuresKHR
- vkCmdCopyAccelerationStructureKHR
- vkCmdCopyAccelerationStructureToMemoryKHR
- vkCmdCopyMemoryToAccelerationStructureKHR
- vkCmdWriteAccelerationStructuresPropertiesKHR
- vkCopyAccelerationStructureKHR
- vkCopyAccelerationStructureToMemoryKHR
- vkCopyMemoryToAccelerationStructureKHR
- vkCreateAccelerationStructureKHR
- vkDestroyAccelerationStructureKHR
- vkGetAccelerationStructureBuildSizesKHR
- vkGetAccelerationStructureDeviceAddressKHR
- vkGetDeviceAccelerationStructureCompatibilityKHR
- vkWriteAccelerationStructuresPropertiesKHR

New Structures

- VkAabbPositionsKHR
- VkAccelerationStructureBuildGeometryInfoKHR
- VkAccelerationStructureBuildRangeInfoKHR
- VkAccelerationStructureBuildSizesInfoKHR
- VkAccelerationStructureCreateInfoKHR
- VkAccelerationStructureDeviceAddressInfoKHR
- VkAccelerationStructureGeometryAabbsDataKHR
- VkAccelerationStructureGeometryInstancesDataKHR
- VkAccelerationStructureGeometryKHR
- VkAccelerationStructureGeometryTrianglesDataKHR
- VkAccelerationStructureInstanceKHR
- VkAccelerationStructureVersionInfoKHR
- VkCopyAccelerationStructureInfoKHR
- VkCopyAccelerationStructureToMemoryInfoKHR
- VkCopyMemoryToAccelerationStructureInfoKHR
- VkTransformMatrixKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceAccelerationStructureFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceAccelerationStructurePropertiesKHR
- Extending VkWriteDescriptorSet:
  - VkWriteDescriptorSetAccelerationStructureKHR

New Unions

- VkAccelerationStructureGeometryDataKHR
- VkDeviceOrHostAddressConstKHR
- VkDeviceOrHostAddressKHR

New Enums

- VkAccelerationStructureBuildTypeKHR
- VkAccelerationStructureCompatibilityKHR
- VkAccelerationStructureCreateFlagBitsKHR
- VkAccelerationStructureTypeKHR
- VkBuildAccelerationStructureFlagBitsKHR

- VkBuildAccelerationStructureModeKHR
- VkCopyAccelerationStructureModeKHR
- VkGeometryFlagBitsKHR
- VkGeometryInstanceFlagBitsKHR
- VkGeometryTypeKHR

New Bitmasks

- VkAccelerationStructureCreateFlagsKHR
- VkBuildAccelerationStructureFlagsKHR
- VkGeometryFlagsKHR
- VkGeometryInstanceFlagsKHR

New Enum Constants

- `VK_KHR_ACCELERATION_STRUCTURE_EXTENSION_NAME`
- `VK_KHR_ACCELERATION_STRUCTURE_SPEC_VERSION`
- Extending VkAccessFlagBits:
  - VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR
  - VK_ACCESS_ACCELERATION_STRUCTURE_WRITE_BIT_KHR
- Extending VkBufferUsageFlagBits:
  - VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR
  - VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR
- Extending VkCopyAccelerationStructureModeKHR:
  - VK_COPY_ACCELERATION_STRUCTURE_MODE_DESERIALIZE_KHR
  - VK_COPY_ACCELERATION_STRUCTURE_MODE_SERIALIZE_KHR
- Extending VkDescriptorType:
  - VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR
- Extending VkFormatFeatureFlagBits:
  - VK_FORMAT_FEATURE_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR
- Extending VkIndexType:
  - VK_INDEX_TYPE_NONE_KHR
- Extending VkObjectType:
  - VK_OBJECT_TYPE_ACCELERATION_STRUCTURE_KHR
- Extending VkPipelineStageFlagBits:
  - VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
- Extending VkQueryType:

    - VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR
    - VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_BUILD_GEOMETRY_INFO_KHR
    - VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_BUILD_SIZES_INFO_KHR
    - VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CREATE_INFO_KHR
    - VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_DEVICE_ADDRESS_INFO_KHR
    - VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_AABBS_DATA_KHR
    - VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_INSTANCES_DATA_KHR
    - VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_KHR
    - VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_TRIANGLES_DATA_KHR
    - VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_VERSION_INFO_KHR
    - VK_STRUCTURE_TYPE_COPY_ACCELERATION_STRUCTURE_INFO_KHR
    - VK_STRUCTURE_TYPE_COPY_ACCELERATION_STRUCTURE_TO_MEMORY_INFO_KHR
    - VK_STRUCTURE_TYPE_COPY_MEMORY_TO_ACCELERATION_STRUCTURE_INFO_KHR
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ACCELERATION_STRUCTURE_FEATURES_KHR
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ACCELERATION_STRUCTURE_PROPERTIES_KHR
    - VK_STRUCTURE_TYPE_WRITE_DESCRIPTOR_SET_ACCELERATION_STRUCTURE_KHR

If VK_KHR_format_feature_flags2 or Vulkan Version 1.3 is supported:

- Extending VkFormatFeatureFlagBits2:
    - VK_FORMAT_FEATURE_2_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR

Issues

(1) How does this extension differ from VK_NV_ray_tracing?

DISCUSSION:

The following is a summary of the main functional differences between VK_KHR_acceleration_structure and VK_NV_ray_tracing:

- added acceleration structure serialization / deserialization (VK_COPY_ACCELERATION_STRUCTURE_MODE_SERIALIZE_KHR, VK_COPY_ACCELERATION_STRUCTURE_MODE_DESERIALIZE_KHR, vkCmdCopyAccelerationStructureToMemoryKHR, vkCmdCopyMemoryToAccelerationStructureKHR)
- document inactive primitives and instances
- added VkPhysicalDeviceAccelerationStructureFeaturesKHR structure
- added indirect and batched acceleration structure builds

(vkCmdBuildAccelerationStructuresIndirectKHR)

- added host acceleration structure commands
- reworked geometry structures so they could be better shared between device, host, and indirect builds
- explicitly made VkAccelerationStructureKHR use device addresses
- added acceleration structure compatibility check function (vkGetDeviceAccelerationStructureCompatibilityKHR)
- add parameter for requesting memory requirements for host and/or device build
- added format feature for acceleration structure build vertex formats (VK_FORMAT_FEATURE_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR)

(3) What are the changes between the public provisional (VK_KHR_ray_tracing v8) release and the internal provisional (VK_KHR_ray_tracing v9) release?

- added `geometryFlags` to `VkAccelerationStructureCreateGeometryTypeInfoKHR` (later reworked to obsolete this)
- added `minAccelerationStructureScratchOffsetAlignment` property to VkPhysicalDeviceRayTracingPropertiesKHR
- fix naming and return enum from vkGetDeviceAccelerationStructureCompatibilityKHR
  - renamed `VkAccelerationStructureVersionKHR` to VkAccelerationStructureVersionInfoKHR
  - renamed `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_VERSION_KHR` to VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_VERSION_INFO_KHR
  - removed `VK_ERROR_INCOMPATIBLE_VERSION_KHR`
  - added VkAccelerationStructureCompatibilityKHR enum
  - remove return value from vkGetDeviceAccelerationStructureCompatibilityKHR and added return enum parameter
- Require Vulkan 1.1
- added creation time capture and replay flags
  - added VkAccelerationStructureCreateFlagBitsKHR and VkAccelerationStructureCreateFlagsKHR
  - renamed the `flags` member of VkAccelerationStructureCreateInfoKHR to `buildFlags` (later removed) and added the `createFlags` member
- change vkCmdBuildAccelerationStructuresIndirectKHR to use buffer device address for indirect parameter
- make `VK_KHR_deferred_host_operations` an interaction instead of a required extension (later went back on this)
- renamed `VkAccelerationStructureBuildOffsetInfoKHR` to VkAccelerationStructureBuildRangeInfoKHR
  - renamed the `ppOffsetInfos` parameter of vkCmdBuildAccelerationStructuresKHR to `ppBuildRangeInfos`

- Re-unify geometry description between build and create
  - remove VkAccelerationStructureCreateGeometryTypeInfoKHR and VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CREATE_GEOMETRY_TYPE_INFO_KHR
  - added VkAccelerationStructureCreateSizeInfoKHR structure (later removed)
  - change type of the pGeometryInfos member of VkAccelerationStructureCreateInfoKHR from VkAccelerationStructureCreateGeometryTypeInfoKHR to VkAccelerationStructureGeometryKHR (later removed)
  - added pCreateSizeInfos member to VkAccelerationStructureCreateInfoKHR (later removed)
- Fix ppGeometries ambiguity, add pGeometries
  - remove geometryArrayOfPointers member of VkAccelerationStructureBuildGeometryInfoKHR
  - disambiguate two meanings of ppGeometries by explicitly adding pGeometries to the VkAccelerationStructureBuildGeometryInfoKHR structure and require one of them be NULL
- added nullDescriptor support for acceleration structures
- changed the update member of VkAccelerationStructureBuildGeometryInfoKHR from a bool to the mode VkBuildAccelerationStructureModeKHR enum which allows future extensibility in update types
- Clarify deferred host ops for pipeline creation
  - VkDeferredOperationKHR is now a top-level parameter for vkBuildAccelerationStructuresKHR, vkCreateRayTracingPipelinesKHR, vkCopyAccelerationStructureToMemoryKHR, vkCopyAccelerationStructureKHR, and vkCopyMemoryToAccelerationStructureKHR
  - removed VkDeferredOperationInfoKHR structure
  - change deferred host creation/return parameter behavior such that the implementation can modify such parameters until the deferred host operation completes
  - VK_KHR_deferred_host_operations is required again
- Change acceleration structure build to always be sized
  - de-alias VkAccelerationStructureMemoryRequirementsTypeNV and VkAccelerationStructureMemoryRequirementsTypeKHR, and remove VkAccelerationStructureMemoryRequirementsTypeKHR
  - add vkGetAccelerationStructureBuildSizesKHR command and VkAccelerationStructureBuildSizesInfoKHR structure and VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_BUILD_SIZES_INFO_KHR enum to query sizes for acceleration structures and scratch storage
  - move size queries for scratch space to vkGetAccelerationStructureBuildSizesKHR
  - remove compactedSize, buildFlags, maxGeometryCount, pGeometryInfos, pCreateSizeInfos members of VkAccelerationStructureCreateInfoKHR and add the size member
  - add maxVertex member to VkAccelerationStructureGeometryTrianglesDataKHR structure
  - remove VkAccelerationStructureCreateSizeInfoKHR structure

(4) What are the changes between the internal provisional (VK_KHR_ray_tracing v9) release and the final (VK_KHR_acceleration_structure v11) release?

- refactor VK_KHR_ray_tracing into 3 extensions, enabling implementation flexibility and decoupling ray query support from ray pipelines:
  - `VK_KHR_acceleration_structure` (for acceleration structure operations)
  - `VK_KHR_ray_tracing_pipeline` (for ray tracing pipeline and shader stages)
  - `VK_KHR_ray_query` (for ray queries in existing shader stages)
- clarify buffer usage flags for ray tracing
  - VK_BUFFER_USAGE_RAY_TRACING_BIT_NV is left alone in `VK_NV_ray_tracing` (required on `scratch` and `instanceData`)
  - VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR is added as an alias of VK_BUFFER_USAGE_RAY_TRACING_BIT_NV in `VK_KHR_ray_tracing_pipeline` and is required on shader binding table buffers
  - VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR is added in `VK_KHR_acceleration_structure` for all vertex, index, transform, aabb, and instance buffer data referenced by device build commands
  - VK_BUFFER_USAGE_STORAGE_BUFFER_BIT is used for `scratchData`
- add max primitive counts (`ppMaxPrimitiveCounts`) to vkCmdBuildAccelerationStructuresIndirectKHR
- Allocate acceleration structures from `VkBuffers` and add a mode to constrain the device address
  - de-alias `VkBindAccelerationStructureMemoryInfoNV` and `vkBindAccelerationStructureMemoryNV`, and remove `VkBindAccelerationStructureMemoryInfoKHR`, `VkAccelerationStructureMemoryRequirementsInfoKHR`, and `vkGetAccelerationStructureMemoryRequirementsKHR`
  - acceleration structures now take a VkBuffer and offset at creation time for memory placement
  - add a new VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR buffer usage for such buffers
  - add a new VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR acceleration structure type for layering
- move VK_GEOMETRY_TYPE_INSTANCES_KHR to main enum instead of being added via extension
- make build commands more consistent - all now build multiple acceleration structures and are named plurally (vkCmdBuildAccelerationStructuresIndirectKHR, vkCmdBuildAccelerationStructuresKHR, vkBuildAccelerationStructuresKHR)
- add interactions with VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT for acceleration structures, including a new feature (`descriptorBindingAccelerationStructureUpdateAfterBind`) and 3 new properties (`maxPerStageDescriptorAccelerationStructures`, `maxPerStageDescriptorUpdateAfterBindAccelerationStructures`,

maxDescriptorSetUpdateAfterBindAccelerationStructures)

- extension is no longer provisional
- define synchronization requirements for builds, traces, and copies
- define synchronization requirements for AS build inputs and indirect build buffer

(5) What is VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR for?

RESOLVED: It is primarily intended for API layering. In DXR, the acceleration structure is basically just a buffer in a special layout, and you do not know at creation time whether it will be used as a top or bottom level acceleration structure. We thus added a generic acceleration structure type whose type is unknown at creation time, but is specified at build time instead. Applications which are written directly for Vulkan should not use it.

Version History

- Revision 1, 2019-12-05 (Members of the Vulkan Ray Tracing TSG)
  - Internal revisions (forked from VK_NV_ray_tracing)
- Revision 2, 2019-12-20 (Daniel Koch, Eric Werness)
  - Add const version of DeviceOrHostAddress (!3515)
  - Add VU to clarify that only handles in the current pipeline are valid (!3518)
  - Restore some missing VUs and add in-place update language (#1902, !3522)
  - rename VkAccelerationStructureInstanceKHR member from accelerationStructure to accelerationStructureReference to better match its type (!3523)
  - Allow VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS for pipeline creation if shader group handles cannot be reused (!3523)
  - update documentation for the VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS error code and add missing documentation for new return codes from VK_KHR_deferred_host_operations (!3523)
  - list new query types for VK_KHR_ray_tracing (!3523)
  - Fix VU statements for VkAccelerationStructureGeometryKHR referring to correct union members and update to use more current wording (!3523)
- Revision 3, 2020-01-10 (Daniel Koch, Jon Leech, Christoph Kubisch)
  - Fix 'instance of' and 'that/which contains/defines' markup issues (!3528)
  - factor out VK_KHR_pipeline_library as stand-alone extension (!3540)
  - Resolve Vulkan-hpp issues (!3543)
    - add missing require for VkGeometryInstanceFlagsKHR
    - de-alias VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CREATE_INFO_NV since the KHR structure is no longer equivalent
    - add len to pDataSize attribute for vkWriteAccelerationStructuresPropertiesKHR
- Revision 4, 2020-01-23 (Daniel Koch, Eric Werness)

- Improve vkWriteAccelerationStructuresPropertiesKHR, add return value and VUs (#1947)
- Clarify language to allow multiple raygen shaders (#1959)
- Various editorial feedback (!3556)
- Add language to help deal with looped self-intersecting fans (#1901)
- Change vkCmdTraceRays{,Indirect}KHR args to pointers (!3559)
- Add scratch address validation language (#1941, !3551)
- Fix definition and add hierarchy information for shader call scope (#1977, !3571)

- Revision 5, 2020-02-04 (Eric Werness, Jeff Bolz, Daniel Koch)
  - remove vestigial accelerationStructureUUID (!3582)
  - update definition of repack instructions and improve memory model interactions (#1910, #1913, !3584)
  - Fix wrong sType for VkPhysicalDeviceRayTracingFeaturesKHR (#1988)
  - Use provisional SPIR-V capabilities (#1987)
  - require rayTraversalPrimitiveCulling if rayQuery is supported (#1927)
  - Miss shaders do not have object parameters (!3592)
  - Fix missing required types in XML (!3592)
  - clarify matching conditions for update (!3592)
  - add goal that host and device builds be similar (!3592)
  - clarify that `maxPrimitiveCount` limit should apply to triangles and AABBs (!3592)
  - Require alignment for instance arrayOfPointers (!3592)
  - Zero is a valid value for instance flags (!3592)
  - Add some alignment VUs that got lost in refactoring (!3592)
  - Recommend TMin epsilon rather than culling (!3592)
  - Get angle from dot product not cross product (!3592)
  - Clarify that AH can access the payload and attributes (!3592)
  - Match DXR behavior for inactive primitive definition (!3592)
  - Use a more generic term than degenerate for inactive to avoid confusion (!3592)
- Revision 6, 2020-02-20 (Daniel Koch)
  - fix some dangling NV references (#1996)
  - rename VkCmdTraceRaysIndirectCommandKHR to VkTraceRaysIndirectCommandKHR (!3607)
  - update contributor list (!3611)
  - use uint64_t instead of VkAccelerationStructureReferenceKHR in VkAccelerationStructureInstanceKHR (#2004)
- Revision 7, 2020-02-28 (Tobias Hector)

- remove HitTKHR SPIR-V builtin (spirv/spirv-extensions#7)

- Revision 8, 2020-03-06 (Tobias Hector, Dae Kim, Daniel Koch, Jeff Bolz, Eric Werness)
  - explicitly state that Tmax is updated when new closest intersection is accepted (#2020,!3536)
  - Made references to min and max t values consistent (!3644)
  - finish enumerating differences relative to VK_NV_ray_tracing in issues (1) and (2) (#1974,!3642)
  - fix formatting in some math equations (!3642)
  - Restrict the Hit Kind operand of `OpReportIntersectionKHR` to 7-bits (spirv/spirv-extensions#8,!3646)
  - Say ray tracing '**should**' be watertight (#2008,!3631)
  - Clarify memory requirements for ray tracing buffers (#2005,!3649)
  - Add callable size limits (#1997,!3652)
- Revision 9, 2020-04-15 (Eric Werness, Daniel Koch, Tobias Hector, Joshua Barczak)
  - Add geometry flags to acceleration structure creation (!3672)
  - add build scratch memory alignment (minAccelerationStructureScratchOffsetAlignment) (#2065,!3725)
  - fix naming and return enum from vkGetDeviceAccelerationStructureCompatibilityKHR (#2051,!3726)
  - require SPIR-V 1.4 (#2096,!3777)
  - added creation time capture/replay flags (#2104,!3774)
  - require Vulkan 1.1 (#2133,!3806)
  - use device addresses instead of VkBuffers for ray tracing commands (#2074,!3815)
  - add interactions with Vulkan 1.2 and VK_KHR_vulkan_memory_model (#2133,!3830)
  - make VK_KHR_pipeline_library an interaction instead of required (#2045,#2108,!3830)
  - make VK_KHR_deferred_host_operations an interaction instead of required (#2045,!3830)
  - removed maxCallableSize and added explicit stack size management for ray pipelines (#1997,!3817,!3772,!3844)
  - improved documentation for VkAccelerationStructureVersionInfoKHR (#2135,3835)
  - rename VkAccelerationStructureBuildOffsetInfoKHR to VkAccelerationStructureBuildRangeInfoKHR (#2058,!3754)
  - Re-unify geometry description between build and create (!3754)
  - Fix ppGeometries ambiguity, add pGeometries (#2032,!3811)
  - add interactions with VK_EXT_robustness2 and allow nullDescriptor support for acceleration structures (#1920,!3848)
  - added future extensibility for AS updates (#2114,!3849)
  - Fix VU for dispatchrays and add a limit on the size of the full grid (#2160,!3851)
  - Add shaderGroupHandleAlignment property (#2180,!3875)

- Clarify deferred host ops for pipeline creation (#2067,!3813)
- Change acceleration structure build to always be sized (#2131,#2197,#2198,!3854,!3883,!3880)

- Revision 10, 2020-07-03 (Mathieu Robart, Daniel Koch, Eric Werness, Tobias Hector)
  - Decomposition of the specification, from VK_KHR_ray_tracing to VK_KHR_acceleration_structure (#1918,!3912)
  - clarify buffer usage flags for ray tracing (#2181,!3939)
  - add max primitive counts to build indirect command (#2233,!3944)
  - Allocate acceleration structures from VkBuffers and add a mode to constrain the device address (#2131,!3936)
  - Move VK_GEOMETRY_TYPE_INSTANCES_KHR to main enum (#2243,!3952)
  - make build commands more consistent (#2247,!3958)
  - add interactions with UPDATE_AFTER_BIND (#2128,!3986)
  - correct and expand build command VUs (!4020)
  - fix copy command VUs (!4018)
  - added various alignment requirements (#2229,!3943)
  - fix valid usage for arrays of geometryCount items (#2198,!4010)
  - define what is allowed to change on RTAS updates and relevant VUs (#2177,!3961)
- Revision 11, 2020-11-12 (Eric Werness, Josh Barczak, Daniel Koch, Tobias Hector)
  - de-alias NV and KHR acceleration structure types and associated commands (#2271,!4035)
  - specify alignment for host copy commands (#2273,!4037)
  - document VK_FORMAT_FEATURE_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR
  - specify that acceleration structures are non-linear (#2289,!4068)
  - add several missing VUs for strides, vertexFormat, and indexType (#2315,!4069)
  - restore VUs for VkAccelerationStructureBuildGeometryInfoKHR (#2337,!4098)
  - ban multi-instance memory for host operations (#2324,!4102)
  - allow dstAccelerationStructure to be null for vkGetAccelerationStructureBuildSizesKHR (#2330,!4111)
  - more build VU cleanup (#2138,#4130)
  - specify host endianness for AS serialization (#2261,!4136)
  - add invertible transform matrix VU (#1710,!4140)
  - require geometryCount to be 1 for TLAS builds (!4145)
  - improved validity conditions for build addresses (#4142)
  - add single statement SPIR-V VUs, build limit VUs (!4158)
  - document limits for vertex and aabb strides (#2390,!4184)
  - specify that VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR applies to

AS copies (#2382,#4173)

- define sync for AS build inputs and indirect buffer (#2407,!4208)

- Revision 12, 2021-08-06 (Samuel Bourasseau)
  - rename VK_GEOMETRY_INSTANCE_TRIANGLE_FRONT_COUNTERCLOCKWISE_BIT_KHR to VK_GEOMETRY_INSTANCE_TRIANGLE_FLIP_FACING_BIT_KHR (keep previous as alias).
  - Clarify description and add note.
- Revision 13, 2021-09-30 (Jon Leech)
  - Add interaction with `VK_KHR_format_feature_flags2` to `vk.xml`

VK_KHR_android_surface

Name String

`VK_KHR_android_surface`

Extension Type

Instance extension

Registered Extension Number

9

Revision

6

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_surface

Contact

- Jesse Hall [GitHub]critsec

Other Extension Metadata

Last Modified Date

2016-01-14

IP Status

No known IP claims.

Contributors

- Patrick Doane, Blizzard
- Faith Ekstrand, Intel
- Ian Elliott, LunarG
- Courtney Goeltzenleuchter, LunarG
- Jesse Hall, Google
- James Jones, NVIDIA
- Antoine Labour, Google
- Jon Leech, Khronos
- David Mao, AMD
- Norbert Nopper, Freescale
- Alon Or-bach, Samsung
- Daniel Rakos, AMD

- Graham Sellers, AMD
- Ray Smith, ARM
- Jeff Vigil, Qualcomm
- Chia-I Wu, LunarG

Description

The VK_KHR_android_surface extension is an instance extension. It provides a mechanism to create a VkSurfaceKHR object (defined by the VK_KHR_surface extension) that refers to an ANativeWindow, Android's native surface type. The ANativeWindow represents the producer endpoint of any buffer queue, regardless of consumer endpoint. Common consumer endpoints for ANativeWindows are the system window compositor, video encoders, and application-specific compositors importing the images through a SurfaceTexture.

New Base Types

- ANativeWindow

New Commands

- vkCreateAndroidSurfaceKHR

New Structures

- VkAndroidSurfaceCreateInfoKHR

New Bitmasks

- VkAndroidSurfaceCreateFlagsKHR

New Enum Constants

- VK_KHR_ANDROID_SURFACE_EXTENSION_NAME
- VK_KHR_ANDROID_SURFACE_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_ANDROID_SURFACE_CREATE_INFO_KHR

Issues

1) Does Android need a way to query for compatibility between a particular physical device (and queue family?) and a specific Android display?

RESOLVED: No. Currently on Android, any physical device is expected to be able to present to the system compositor, and all queue families must support the necessary image layout transitions and synchronization operations.

Version History

- Revision 1, 2015-09-23 (Jesse Hall)
  - Initial draft.
- Revision 2, 2015-10-26 (Ian Elliott)
  - Renamed from VK_EXT_KHR_android_surface to VK_KHR_android_surface.
- Revision 3, 2015-11-03 (Daniel Rakos)
  - Added allocation callbacks to surface creation function.
- Revision 4, 2015-11-10 (Jesse Hall)
  - Removed VK_ERROR_INVALID_ANDROID_WINDOW_KHR.
- Revision 5, 2015-11-28 (Daniel Rakos)
  - Updated the surface create function to take a pCreateInfo structure.
- Revision 6, 2016-01-14 (James Jones)
  - Moved VK_ERROR_NATIVE_WINDOW_IN_USE_KHR from the VK_KHR_android_surface to the VK_KHR_surface extension.

VK_KHR_calibrated_timestamps

Name String

`VK_KHR_calibrated_timestamps`

Extension Type

Device extension

Registered Extension Number

544

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Daniel Rakos [GitHub]aqnuep

Extension Proposal

VK_EXT_calibrated_timestamps

Other Extension Metadata

Last Modified Date

2023-07-12

IP Status

No known IP claims.

Contributors

- Matthaeus G. Chajdas, AMD
- Alan Harrison, AMD
- Derrick Owens, AMD
- Daniel Rakos, RasterGrid
- Faith Ekstrand, Intel
- Keith Packard, Valve

Description

This extension provides an interface to query calibrated timestamps obtained quasi simultaneously

from two time domains.

New Commands

- vkGetCalibratedTimestampsKHR
- vkGetPhysicalDeviceCalibrateableTimeDomainsKHR

New Structures

- VkCalibratedTimestampInfoKHR

New Enums

- VkTimeDomainKHR

New Enum Constants

- `VK_KHR_CALIBRATED_TIMESTAMPS_EXTENSION_NAME`
- `VK_KHR_CALIBRATED_TIMESTAMPS_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_CALIBRATED_TIMESTAMP_INFO_KHR

Version History

- Revision 1, 2023-07-12 (Daniel Rakos)
  - Initial draft.

VK_KHR_compute_shader_derivatives

Name String

`VK_KHR_compute_shader_derivatives`

Extension Type

Device extension

Registered Extension Number

512

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_compute_shader_derivatives

Contact

- Jean-Noe Morissette [GitHub]MagicPoncho

Extension Proposal

VK_KHR_compute_shader_derivatives

Other Extension Metadata

Last Modified Date

2024-06-26

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension requires `SPV_KHR_compute_shader_derivatives`
- This extension provides API support for `GL_KHR_compute_shader_derivatives`

Contributors

- Jean-Noe Morissette, Epic Games
- Daniel Koch, NVIDIA
- Pat Brown, NVIDIA
- Stu Smith, AMD

- Jan-Harald Fredriksen, Arm
- Tobias Hector, AMD
- Ralph Potter, Samsung
- Pan Gao, Huawei
- Samuel (Sheng-Wen) Huang, MediaTek
- Graeme Leese, Broadcom
- Hans-Kristian Arntzen, Valve
- Matthew Netsch, Qualcomm

Description

This extension adds Vulkan support for the `SPV_KHR_compute_shader_derivatives` SPIR-V extension.

The SPIR-V extension provides two new execution modes, both of which allow execution models with defined workgroups to use built-ins that evaluate derivatives explicitly or implicitly. Derivatives will be computed via differencing over a 2x2 group of shader invocations. The `DerivativeGroupQuadsKHR` execution mode assembles shader invocations into 2x2 groups, where each group has x and y coordinates of the local invocation ID of the form (2m+{0,1}, 2n+{0,1}). The `DerivativeGroupLinearKHR` execution mode assembles shader invocations into 2x2 groups, where each group has local invocation index values of the form 4m+{0,1,2,3}.

The new execution modes are supported in compute shaders and optionally (see meshAndTaskShaderDerivatives) in mesh and task shaders.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR

New Enum Constants

- `VK_KHR_COMPUTE_SHADER_DERIVATIVES_EXTENSION_NAME`
- `VK_KHR_COMPUTE_SHADER_DERIVATIVES_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COMPUTE_SHADER_DERIVATIVES_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COMPUTE_SHADER_DERIVATIVES_PROPERTIES_KHR

New SPIR-V Capability

- `ComputeDerivativeGroupQuadsKHR`

- `ComputeDerivativeGroupLinearKHR`

Examples

None.

Version History

- Revision 1, 2023-02-27 (Jean-Noe Morissette)
  - Initial draft
  - Add properties and clarify mesh and task support (Daniel Koch)

VK_KHR_cooperative_matrix

Name String

`VK_KHR_cooperative_matrix`

Extension Type

Device extension

Registered Extension Number

507

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_cooperative_matrix

Contact

- Kevin Petit [GitHub]kpet

Extension Proposal

VK_KHR_cooperative_matrix

Other Extension Metadata

Last Modified Date

2023-05-03

Interactions and External Dependencies

- This extension provides API support for `GLSL_KHR_cooperative_matrix`

Contributors

- Jeff Bolz, NVIDIA
- Markus Tavenrath, NVIDIA
- Daniel Koch, NVIDIA
- Kevin Petit, Arm Ltd.
- Boris Zanin, AMD

Description

This extension adds support for using cooperative matrix types in SPIR-V. Cooperative matrix types are medium-sized matrices that are primarily supported in compute shaders, where the storage for the matrix is spread across all invocations in some scope (usually a subgroup) and those invocations cooperate to efficiently perform matrix multiplies.

Cooperative matrix types are defined by the `SPV_KHR_cooperative_matrix` SPIR-V extension and can be used with the `GLSL_KHR_cooperative_matrix` GLSL extension.

This extension includes support for enumerating the matrix types and dimensions that are supported by the implementation.

New Commands

- vkGetPhysicalDeviceCooperativeMatrixPropertiesKHR

New Structures

- VkCooperativeMatrixPropertiesKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceCooperativeMatrixFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceCooperativeMatrixPropertiesKHR

New Enums

- VkComponentTypeKHR
- VkScopeKHR

New Enum Constants

- `VK_KHR_COOPERATIVE_MATRIX_EXTENSION_NAME`
- `VK_KHR_COOPERATIVE_MATRIX_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_COOPERATIVE_MATRIX_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_PROPERTIES_KHR

New SPIR-V Capabilities

- CooperativeMatrixKHR

Issues

1.) How does cooperativeMatrixRobustBufferAccess interact with nullDescriptor?

RESOLVED: nullDescriptor is not allowed here While drivers can handle going OOB into a buffer, if there is not buffer, via nullDescriptor, it is invalid to load/store from the cooperative matrix.

Version History

- Revision 2, 2023-05-03 (Kevin Petit)
  - First KHR revision
- Revision 1, 2019-02-05 (Jeff Bolz)
  - NVIDIA vendor extension

VK_KHR_copy_memory_indirect

Name String

`VK_KHR_copy_memory_indirect`

Extension Type

Device extension

Registered Extension Number

550

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
and
VK_KHR_buffer_device_address
or
Vulkan Version 1.2

Contact

- Vikram Kushwaha [GitHub]vkushwaha-nv

Extension Proposal

VK_KHR_copy_memory_indirect

Other Extension Metadata

Last Modified Date

2025-01-25

Contributors

- Daniel Koch, NVIDIA
- Vikram Kushwaha, NVIDIA
- Jeff Bolz, NVIDIA
- Christoph Kubisch, NVIDIA
- Stuart Smith, AMD
- Faith Ekstrand, Collabora
- Caterina Shablia, Collabora
- Spencer Fricke, LunarG
- Matthew Netsch, Qualcomm Technologies, Inc

- Mike Blumenkrantz, Valve
- Alyssa Rosenzweig, Valve

Description

This extension adds support for performing copies between memory and image regions using indirect parameters that are read by the device from a buffer during execution. This functionality may be useful for performing copies where the copy parameters are not known during the command buffer creation time.

New Commands

- vkCmdCopyMemoryIndirectKHR
- vkCmdCopyMemoryToImageIndirectKHR

New Structures

- VkCopyMemoryIndirectCommandKHR
- VkCopyMemoryIndirectInfoKHR
- VkCopyMemoryToImageIndirectCommandKHR
- VkCopyMemoryToImageIndirectInfoKHR
- VkStridedDeviceAddressRangeKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceCopyMemoryIndirectFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceCopyMemoryIndirectPropertiesKHR

New Enums

- VkAddressCopyFlagBitsKHR

New Bitmasks

- VkAddressCopyFlagsKHR

New Enum Constants

- `VK_KHR_COPY_MEMORY_INDIRECT_EXTENSION_NAME`
- `VK_KHR_COPY_MEMORY_INDIRECT_SPEC_VERSION`
- Extending VkFormatFeatureFlagBits2:
  - VK_FORMAT_FEATURE_2_COPY_IMAGE_INDIRECT_DST_BIT_KHR
- Extending VkPipelineStageFlagBits2:
  - VK_PIPELINE_STAGE_2_COPY_INDIRECT_BIT_KHR
- Extending VkStructureType:

- VK_STRUCTURE_TYPE_COPY_MEMORY_INDIRECT_INFO_KHR
- VK_STRUCTURE_TYPE_COPY_MEMORY_TO_IMAGE_INDIRECT_INFO_KHR
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COPY_MEMORY_INDIRECT_FEATURES_KHR
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COPY_MEMORY_INDIRECT_PROPERTIES_KHR

Version History

- Revision 1, 2025-01-25 (Daniel Koch, Vikram Kushwaha)
  - Initial external release

VK_KHR_deferred_host_operations

Name String

`VK_KHR_deferred_host_operations`

Extension Type

Device extension

Registered Extension Number

269

Revision

4

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- Josh Barczak [GitHub]jbarczak

Other Extension Metadata

Last Modified Date

2020-11-12

IP Status

No known IP claims.

Contributors

- Joshua Barczak, Intel
- Jeff Bolz, NVIDIA
- Daniel Koch, NVIDIA
- Slawek Grajewski, Intel
- Tobias Hector, AMD
- Yuriy O’Donnell, Epic
- Eric Werness, NVIDIA
- Baldur Karlsson, Valve
- Jesse Barker, Unity
- Contributors to VK_KHR_acceleration_structure, VK_KHR_ray_tracing_pipeline

Description

The VK_KHR_deferred_host_operations extension defines the infrastructure and usage patterns for deferrable commands, but does not specify any commands as deferrable. This is left to additional dependent extensions. Commands **must** not be deferred unless the deferral is specifically allowed by another extension which depends on VK_KHR_deferred_host_operations.

New Object Types

- VkDeferredOperationKHR

New Commands

- vkCreateDeferredOperationKHR
- vkDeferredOperationJoinKHR
- vkDestroyDeferredOperationKHR
- vkGetDeferredOperationMaxConcurrencyKHR
- vkGetDeferredOperationResultKHR

New Enum Constants

- VK_KHR_DEFERRED_HOST_OPERATIONS_EXTENSION_NAME
- VK_KHR_DEFERRED_HOST_OPERATIONS_SPEC_VERSION
- Extending VkObjectType:
  - VK_OBJECT_TYPE_DEFERRED_OPERATION_KHR
- Extending VkResult:
  - VK_OPERATION_DEFERRED_KHR
  - VK_OPERATION_NOT_DEFERRED_KHR
  - VK_THREAD_DONE_KHR
  - VK_THREAD_IDLE_KHR

Code Examples

The following examples will illustrate the concept of deferrable operations using a hypothetical example. The command vkDoSomethingExpensive denotes a deferrable command.

The following example illustrates how a vulkan application might request deferral of an expensive operation:

```
// create a deferred operation
VkDeferredOperationKHR hOp;
VkResult result = vkCreateDeferredOperationKHR(device, pCallbacks, &hOp);
assert(result == VK_SUCCESS);

result = vkDoSomethingExpensive(device, hOp, ...);
```

```
assert( result == VK_OPERATION_DEFERRED_KHR );

// operation was deferred.  Execute it asynchronously
std::async::launch(
    [ hOp ] ( )
    {
        vkDeferredOperationJoinKHR(device, hOp);

        result = vkGetDeferredOperationResultKHR(device, hOp);

        // deferred operation is now complete.  'result' indicates success or failure

        vkDestroyDeferredOperationKHR(device, hOp, pCallbacks);
    }
);
```

The following example illustrates extracting concurrency from a single deferred operation:

```
// create a deferred operation
VkDeferredOperationKHR hOp;
VkResult result = vkCreateDeferredOperationKHR(device, pCallbacks, &hOp);
assert(result == VK_SUCCESS);

result = vkDoSomethingExpensive(device, hOp, ...);
assert( result == VK_OPERATION_DEFERRED_KHR );

// Query the maximum amount of concurrency and clamp to the desired maximum
uint32_t numLaunches = std::min(vkGetDeferredOperationMaxConcurrencyKHR(device, hOp),
maxThreads);

std::vector<std::future<void> > joins;

for (uint32_t i = 0; i < numLaunches; i++) {
  joins.emplace_back(std::async::launch(
    [ hOp ] ( )
    {
        vkDeferredOperationJoinKHR(device, hOp);
                // in a job system, a return of VK_THREAD_IDLE_KHR should queue
another
                // job, but it is not functionally required
    }
  ));
}

for (auto &f : joins) {
  f.get();
}

result = vkGetDeferredOperationResultKHR(device, hOp);
```

```
// deferred operation is now complete.  'result' indicates success or failure

vkDestroyDeferredOperationKHR(device, hOp, pCallbacks);
```

The following example shows a subroutine which guarantees completion of a deferred operation, in the presence of multiple worker threads, and returns the result of the operation.

```
VkResult FinishDeferredOperation(VkDeferredOperationKHR hOp)
{
    // Attempt to join the operation until the implementation indicates that we should
stop

    VkResult result = vkDeferredOperationJoinKHR(device, hOp);
    while( result == VK_THREAD_IDLE_KHR )
    {
        std::this_thread::yield();
        result = vkDeferredOperationJoinKHR(device, hOp);
    }

    switch( result )
    {
    case VK_SUCCESS:
        {
            // deferred operation has finished.  Query its result.
            result = vkGetDeferredOperationResultKHR(device, hOp);
        }
        break;

    case VK_THREAD_DONE_KHR:
        {
            // deferred operation is being wrapped up by another thread
            //  wait for that thread to finish
            do
            {
                std::this_thread::yield();
                result = vkGetDeferredOperationResultKHR(device, hOp);
            } while( result == VK_NOT_READY );
        }
        break;

    default:
        assert(false); // other conditions are illegal.
        break;
    }

    return result;
}
```

Issues

1. Should this extension have a VkPhysicalDevice*FeaturesKHR structure?

RESOLVED: No. This extension does not add any functionality on its own and requires a dependent extension to actually enable functionality and thus there is no value in adding a feature structure. If necessary, any dependent extension could add a feature boolean if it wanted to indicate that it is adding optional deferral support.

Version History

- Revision 1, 2019-12-05 (Josh Barczak, Daniel Koch)
  - Initial draft.
- Revision 2, 2020-03-06 (Daniel Koch, Tobias Hector)
  - Add missing VK_OBJECT_TYPE_DEFERRED_OPERATION_KHR enum
  - fix sample code
  - Clarified deferred operation parameter lifetimes (#2018,!3647)
- Revision 3, 2020-05-15 (Josh Barczak)
  - Clarify behavior of vkGetDeferredOperationMaxConcurrencyKHR, allowing it to return 0 if the operation is complete (#2036,!3850)
- Revision 4, 2020-11-12 (Tobias Hector, Daniel Koch)
  - Remove VkDeferredOperationInfoKHR and change return value semantics when deferred host operations are in use (#2067,3813)
  - clarify return value of vkGetDeferredOperationResultKHR (#2339,!4110)

VK_KHR_depth_clamp_zero_one

Name String

VK_KHR_depth_clamp_zero_one

Extension Type

Device extension

Registered Extension Number

605

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Graeme Leese [GitHub]gnl21

Other Extension Metadata

Last Modified Date

2024-09-10

Contributors

- Graeme Leese, Broadcom

Description

This extension is based on the VK_EXT_depth_clamp_zero_one extension. This extension gives defined behavior to fragment depth values which end up outside the conventional [0, 1] range. It can be used to ensure portability in edge cases of features like depthBias. The particular behavior is chosen to match OpenGL to aid porting or emulation.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDepthClampZeroOneFeaturesKHR

New Enum Constants

- VK_KHR_DEPTH_CLAMP_ZERO_ONE_EXTENSION_NAME

- VK_KHR_DEPTH_CLAMP_ZERO_ONE_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLAMP_ZERO_ONE_FEATURES_KHR

Version History

- Revision 1, 2024-09-10 (Graeme Leese)
  - Internal revisions

VK_KHR_device_address_commands

Name String

VK_KHR_device_address_commands

Extension Type

Device extension

Registered Extension Number

319

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1
and
VK_KHR_buffer_device_address
or
Vulkan Version 1.2
and
VK_KHR_synchronization2
and
VK_EXT_extended_dynamic_state
or
Vulkan Version 1.3

API Interactions

- Interacts with VK_VERSION_1_2
- Interacts with VK_AMD_buffer_marker
- Interacts with VK_EXT_conditional_rendering
- Interacts with VK_EXT_mesh_shader
- Interacts with VK_EXT_transform_feedback
- Interacts with VK_KHR_acceleration_structure
- Interacts with VK_KHR_draw_indirect_count

Contact

- Tobias Hector [GitHub]tobski

Extension Proposal

VK_KHR_device_address_commands

Other Extension Metadata

Last Modified Date

2026-03-10

Contributors

- Tobias Hector, AMD
- Matthew Netsch, Qualcomm Technologies, Inc
- Jan-Harald Fredriksen, Arm
- Mike Blumenkrantz, Valve
- Allan MacKinnon, Google
- Daniel Koch, NVIDIA
- Autumn Ashton, Valve
- Piers Daniell, NVIDIA
- Stuart Smith, AMD
- Caterina Shablia, Collabora
- Piotr Byszewski, Cognizant
- James Fitzpatrick, Imagination
- Daniel Story, Nintendo
- Baldur Karlsson, Valve
- Jon Leech, Khronos
- Samuel Pitoiset, Valve
- Lionel Landwerlin, Intel
- Hans-Kristian Arntzen, Valve
- Žiga Markuš, LunarG
- Spencer Fricke, Lunarg
- Per Inge Mathisen, Arm
- Vikram Kushwaha, NVIDIA
- Ralph Potter, Samsung
- Marijn Suijten, Traverse Research
- Ricard Garcia, Igalia
- Mengyang Liu, AMD
- Artem Kharytoniuk, LunarG
- Marty Johnson, Khronos
- Alyssa Rosenzweig, Valve

Description

This extension enables applications to use device addresses in place of buffer objects for most functionality.

New Commands

- vkCmdBindIndexBuffer3KHR
- vkCmdBindVertexBuffers3KHR
- vkCmdCopyImageToMemoryKHR
- vkCmdCopyMemoryKHR
- vkCmdCopyMemoryToImageKHR
- vkCmdCopyQueryPoolResultsToMemoryKHR
- vkCmdDispatchIndirect2KHR
- vkCmdDrawIndexedIndirect2KHR
- vkCmdDrawIndirect2KHR
- vkCmdFillMemoryKHR
- vkCmdUpdateMemoryKHR

If VK_KHR_draw_indirect_count or Vulkan Version 1.2 and VK_EXT_mesh_shader is supported:

- vkCmdDrawMeshTasksIndirectCount2EXT

If VK_EXT_conditional_rendering is supported:

- vkCmdBeginConditionalRendering2EXT

If VK_EXT_mesh_shader is supported:

- vkCmdDrawMeshTasksIndirect2EXT

If VK_EXT_transform_feedback is supported:

- vkCmdBeginTransformFeedback2EXT
- vkCmdBindTransformFeedbackBuffers2EXT
- vkCmdDrawIndirectByteCount2EXT
- vkCmdEndTransformFeedback2EXT

If VK_KHR_acceleration_structure is supported:

- vkCreateAccelerationStructure2KHR

If VK_KHR_draw_indirect_count or Vulkan Version 1.2 is supported:

- vkCmdDrawIndexedIndirectCount2KHR
- vkCmdDrawIndirectCount2KHR

New Structures

- VkBindIndexBuffer3InfoKHR
- VkBindVertexBuffer3InfoKHR
- VkCopyDeviceMemoryImageInfoKHR
- VkCopyDeviceMemoryInfoKHR
- VkDeviceAddressRangeKHR
- VkDeviceMemoryCopyKHR
- VkDeviceMemoryImageCopyKHR
- VkDispatchIndirect2InfoKHR
- VkDrawIndirect2InfoKHR
- VkDrawIndirectCount2InfoKHR
- VkMemoryRangeBarrierKHR
- VkStridedDeviceAddressRangeKHR
- Extending VkDependencyInfo:
  - VkMemoryRangeBarriersInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDeviceAddressCommandsFeaturesKHR

If VK_EXT_conditional_rendering is supported:

- VkConditionalRenderingBeginInfo2EXT

If VK_EXT_transform_feedback is supported:

- VkBindTransformFeedbackBuffer2InfoEXT

If VK_KHR_acceleration_structure is supported:

- VkAccelerationStructureCreateInfo2KHR

New Enums

- VkAddressCommandFlagBitsKHR

New Bitmasks

- VkAddressCommandFlagsKHR

New Enum Constants

- `VK_KHR_DEVICE_ADDRESS_COMMANDS_EXTENSION_NAME`
- `VK_KHR_DEVICE_ADDRESS_COMMANDS_SPEC_VERSION`
- Extending VkStructureType:

    - VK_STRUCTURE_TYPE_BIND_INDEX_BUFFER_3_INFO_KHR
    - VK_STRUCTURE_TYPE_BIND_VERTEX_BUFFER_3_INFO_KHR
    - VK_STRUCTURE_TYPE_COPY_DEVICE_MEMORY_IMAGE_INFO_KHR
    - VK_STRUCTURE_TYPE_COPY_DEVICE_MEMORY_INFO_KHR
    - VK_STRUCTURE_TYPE_DEVICE_MEMORY_COPY_KHR
    - VK_STRUCTURE_TYPE_DEVICE_MEMORY_IMAGE_COPY_KHR
    - VK_STRUCTURE_TYPE_DISPATCH_INDIRECT_2_INFO_KHR
    - VK_STRUCTURE_TYPE_DRAW_INDIRECT_2_INFO_KHR
    - VK_STRUCTURE_TYPE_DRAW_INDIRECT_COUNT_2_INFO_KHR
    - VK_STRUCTURE_TYPE_MEMORY_RANGE_BARRIERS_INFO_KHR
    - VK_STRUCTURE_TYPE_MEMORY_RANGE_BARRIER_KHR
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_ADDRESS_COMMANDS_FEATURES_KHR

If VK_EXT_conditional_rendering is supported:

- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_CONDITIONAL_RENDERING_BEGIN_INFO_2_EXT

If VK_EXT_transform_feedback is supported:

- Extending VkAddressCommandFlagBitsKHR:
    - VK_ADDRESS_COMMAND_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
    - VK_ADDRESS_COMMAND_UNKNOWN_TRANSFORM_FEEDBACK_BUFFER_USAGE_BIT_KHR
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_BIND_TRANSFORM_FEEDBACK_BUFFER_2_INFO_EXT

If VK_KHR_acceleration_structure is supported:

- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CREATE_INFO_2_KHR

Version History

- Revision 1, 2026-03-10 (Tobias Hector)
    - Initial revision

VK_KHR_device_fault

Name String

`VK_KHR_device_fault`

Extension Type

Device extension

Registered Extension Number

574

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Ralph Potter [GitLab]r_potter

Extension Proposal

VK_KHR_device_fault

Other Extension Metadata

Last Modified Date

2026-03-18

IP Status

No known IP claims.

Contributors

- Ralph Potter, Samsung
- Craig Graham, Samsung
- Stuart Smith, AMD
- Jan-Harald Fredriksen, ARM
- Mark Bellamy, ARM
- Andrew Ellem, Google
- Alex Walters, IMG
- Jeff Bolz, NVIDIA
- Baldur Karlsson, Valve

- Adrian Ravai, Samsung
- Peter Gal, Samsung
- Matthew Netsch, QUALCOMM
- Tobias Hector, AMD
- Alan Harrison, AMD
- Vikram Tarikere, IMG
- Jon Leech, Khronos
- Samuel Pitoiset, Valve

Description

Device loss can be triggered by a variety of issues, including invalid API usage, implementation errors, or hardware failures. This extension introduces two new commands:

- vkGetDeviceFaultReportsKHR
- vkGetDeviceFaultDebugInfoKHR

vkGetDeviceFaultReportsKHR allows developers to query for additional information on GPU faults which may have caused device loss, and to generate binary crash dumps, which may be loaded into external tools for further diagnosis. Additionally this command allows developers to query for additional information on GPU faults which were internally recovered by the implementation.

vkGetDeviceFaultReportsKHR differs from vkGetDeviceFaultInfoEXT in that it can be called at any time, is able to report faults which do not result in VK_ERROR_DEVICE_LOST and allows the caller to wait for such an error to occur.

vkGetDeviceFaultDebugInfoKHR provides a separate interface which **must** only be called when a device loss has occurred to provide extended GPU vendor specific crash post-mortem information.

New Commands

- vkGetDeviceFaultDebugInfoKHR
- vkGetDeviceFaultReportsKHR

New Structures

- VkDeviceFaultAddressInfoKHR
- VkDeviceFaultDebugInfoKHR
- VkDeviceFaultInfoKHR
- VkDeviceFaultVendorBinaryHeaderVersionOneKHR
- VkDeviceFaultVendorInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceFaultFeaturesKHR

- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceFaultPropertiesKHR

New Enums

- VkDeviceFaultAddressTypeKHR
- VkDeviceFaultFlagBitsKHR
- VkDeviceFaultVendorBinaryHeaderVersionKHR

New Bitmasks

- VkDeviceFaultFlagsKHR

New Enum Constants

- `VK_KHR_DEVICE_FAULT_EXTENSION_NAME`
- `VK_KHR_DEVICE_FAULT_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DEVICE_FAULT_DEBUG_INFO_KHR
  - VK_STRUCTURE_TYPE_DEVICE_FAULT_INFO_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FAULT_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FAULT_PROPERTIES_KHR

Version History

- Revision 0, 2024-03-01 (Ralph Potter)
  - Internal Revision
- Revision 1, 2025-06-10 (Craig Graham)
  - Revised API to support async fault queries.

VK_KHR_display

Name String

`VK_KHR_display`

Extension Type

Instance extension

Registered Extension Number

3

Revision

23

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_surface

Contact

- James Jones [GitHub]cubanismo
- Norbert Nopper [GitHub]FslNopper

Other Extension Metadata

Last Modified Date

2017-03-13

IP Status

No known IP claims.

Contributors

- James Jones, NVIDIA
- Norbert Nopper, Freescale
- Jeff Vigil, Qualcomm
- Daniel Rakos, AMD

Description

This extension provides the API to enumerate displays and available modes on a given device.

New Object Types

- VkDisplayKHR
- VkDisplayModeKHR

New Commands

- vkCreateDisplayModeKHR
- vkCreateDisplayPlaneSurfaceKHR
- vkGetDisplayModePropertiesKHR
- vkGetDisplayPlaneCapabilitiesKHR
- vkGetDisplayPlaneSupportedDisplaysKHR
- vkGetPhysicalDeviceDisplayPlanePropertiesKHR
- vkGetPhysicalDeviceDisplayPropertiesKHR

New Structures

- VkDisplayModeCreateInfoKHR
- VkDisplayModeParametersKHR
- VkDisplayModePropertiesKHR
- VkDisplayPlaneCapabilitiesKHR
- VkDisplayPlanePropertiesKHR
- VkDisplayPropertiesKHR
- VkDisplaySurfaceCreateInfoKHR

New Enums

- VkDisplayPlaneAlphaFlagBitsKHR

New Bitmasks

- VkDisplayModeCreateFlagsKHR
- VkDisplayPlaneAlphaFlagsKHR
- VkDisplaySurfaceCreateFlagsKHR
- VkSurfaceTransformFlagsKHR

New Enum Constants

- `VK_KHR_DISPLAY_EXTENSION_NAME`
- `VK_KHR_DISPLAY_SPEC_VERSION`
- Extending VkObjectType:
  - VK_OBJECT_TYPE_DISPLAY_KHR
  - VK_OBJECT_TYPE_DISPLAY_MODE_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DISPLAY_MODE_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_DISPLAY_SURFACE_CREATE_INFO_KHR

Issues

1) Which properties of a mode should be fixed in the mode information vs. settable in some other function when setting the mode? E.g., do we need to double the size of the mode pool to include both stereo and non-stereo modes? YUV and RGB scanout even if they both take RGB input images? BGR vs. RGB input? etc.

RESOLVED: Many modern displays support at most a handful of resolutions and timings natively. Other "modes" are expected to be supported using scaling hardware on the display engine or GPU. Other properties, such as rotation and mirroring should not require duplicating hardware modes just to express all combinations. Further, these properties may be implemented on a per-display or per-overlay granularity.

To avoid the exponential growth of modes as mutable properties are added, as was the case with `EGLConfig`/WGL pixel formats/`GLXFBConfig`, this specification should separate out hardware properties and configurable state into separate objects. Modes and overlay planes will express capabilities of the hardware, while a separate structure will allow applications to configure scaling, rotation, mirroring, color keys, LUT values, alpha masks, etc. for a given swapchain independent of the mode in use. Constraints on these settings will be established by properties of the immutable objects.

Note the resolution of this issue may affect issue 5 as well.

2) What properties of a display itself are useful?

RESOLVED: This issue is too broad. It was meant to prompt general discussion, but resolving this issue amounts to completing this specification. All interesting properties should be included. The issue will remain as a placeholder since removing it would make it hard to parse existing discussion notes that refer to issues by number.

3) How are multiple overlay planes within a display or mode enumerated?

RESOLVED: They are referred to by an index. Each display will report the number of overlay planes it contains.

4) Should swapchains be created relative to a mode or a display?

RESOLVED: When using this extension, swapchains are created relative to a mode and a plane. The mode implies the display object the swapchain will present to. If the specified mode is not the display's current mode, the new mode will be applied when the first image is presented to the swapchain, and the default operating system mode, if any, will be restored when the swapchain is destroyed.

5) Should users query generic ranges from displays and construct their own modes explicitly using those constraints rather than querying a fixed set of modes (Most monitors only have one real "mode" these days, even though many support relatively arbitrary scaling, either on the monitor side or in the GPU display engine, making "modes" something of a relic/compatibility construct).

RESOLVED: Expose both. Display information structures will expose a set of predefined modes, as well as any attributes necessary to construct a customized mode.

6) Is it fine if we return the display and display mode handles in the structure used to query their properties?

RESOLVED: Yes.

7) Is there a possibility that not all displays of a device work with all of the present queues of a device? If yes, how do we determine which displays work with which present queues?

RESOLVED: No known hardware has such limitations, but determining such limitations is supported automatically using the existing `VK_KHR_surface` and `VK_KHR_swapchain` query mechanisms.

8) Should all presentation need to be done relative to an overlay plane, or can a display mode + display be used alone to target an output?

RESOLVED: Require specifying a plane explicitly.

9) Should displays have an associated window system display, such as an `HDC` or `Display*`?

RESOLVED: No. Displays are independent of any windowing system in use on the system. Further, neither `HDC` nor `Display*` refer to a physical display object.

10) Are displays queried from a physical GPU or from a device instance?

RESOLVED: Developers prefer to query modes directly from the physical GPU so they can use display information as an input to their device selection algorithms prior to device creation. This avoids the need to create placeholder device instances to enumerate displays.

This preference must be weighed against the extra initialization that must be done by driver vendors prior to device instance creation to support this usage.

11) Should displays and/or modes be dispatchable objects? If functions are to take displays, overlays, or modes as their first parameter, they must be dispatchable objects as defined in Khronos bug 13529. If they are not added to the list of dispatchable objects, functions operating on them must take some higher-level object as their first parameter. There is no performance case against making them dispatchable objects, but they would be the first extension objects to be dispatchable.

RESOLVED: Do not make displays or modes dispatchable. They will dispatch based on their associated physical device.

12) Should hardware cursor capabilities be exposed?

RESOLVED: Defer. This could be a separate extension on top of the base WSI specs.

13) How many display objects should be enumerated for “tiled” display devices? There are ongoing design discussions among lower-level display API authors regarding how to expose displays if they are one physical display device to an end user, but may internally be implemented as two side-by-side displays using the same display engine (and sometimes cabling) resources as two physically separate display devices.

RESOLVED: Tiled displays will appear as a single display object in this API.

14) Should the raw EDID data be included in the display information?

RESOLVED: No. A future extension could be added which reports the EDID if necessary. This may be complicated by the outcome of issue 13.

15) Should min and max scaling factor capabilities of overlays be exposed?

RESOLVED: Yes. This is exposed indirectly by allowing applications to query the min/max position and extent of the source and destination regions from which image contents are fetched by the display engine when using a particular mode and overlay pair.

16) Should devices be able to expose planes that can be moved between displays? If so, how?

RESOLVED: Yes. Applications can determine which displays a given plane supports using vkGetDisplayPlaneSupportedDisplaysKHR.

17) Should there be a way to destroy display modes? If so, does it support destroying "built in" modes?

RESOLVED: Not in this extension. A future extension could add this functionality.

18) What should the lifetime of display and built-in display mode objects be?

RESOLVED: The lifetime of the instance. These objects cannot be destroyed. A future extension may be added to expose a way to destroy these objects and/or support display hotplug.

19) Should persistent mode for smart panels be enabled/disabled at swapchain creation time, or on a per-present basis.

RESOLVED: On a per-present basis.

Examples

NOTE	The example code for the `VK_KHR_display` and `VK_KHR_display_swapchain` extensions was removed from the appendix after revision 1.0.43. The display enumeration example code was ported to the cube demo that is shipped with the official Khronos SDK, and is being kept up-to-date in that location (see: https://github.com/KhronosGroup/Vulkan-Tools/blob/main/cube/cube.c).

Version History

- Revision 1, 2015-02-24 (James Jones)
  - Initial draft
- Revision 2, 2015-03-12 (Norbert Nopper)
  - Added overlay enumeration for a display.
- Revision 3, 2015-03-17 (Norbert Nopper)
  - Fixed typos and namings as discussed in Bugzilla.
  - Reordered and grouped functions.

    - Added functions to query count of display, mode, and overlay.
    - Added native display handle, which may be needed on some platforms to create a native Window.
- Revision 4, 2015-03-18 (Norbert Nopper)
    - Removed primary and virtualPostion members (see comment of James Jones in Bugzilla).
    - Added native overlay handle to information structure.
    - Replaced , with ; in struct.
- Revision 6, 2015-03-18 (Daniel Rakos)
    - Added WSI extension suffix to all items.
    - Made the whole API more “Vulkanish”.
    - Replaced all functions with a single vkGetDisplayInfoKHR function to better match the rest of the API.
    - Made the display, display mode, and overlay objects be top level objects, not subclasses of VkBaseObject as they do not support the common functions anyways.
    - Renamed *Info structures to *Properties.
    - Removed overlayIndex field from VkOverlayProperties as there is an implicit index already as a result of moving to a “Vulkanish” API.
    - Displays are not get through device, but through physical GPU to match the rest of the Vulkan API. Also this is something ISVs explicitly requested.
    - Added issue (6) and (7).
- Revision 7, 2015-03-25 (James Jones)
    - Added an issues section
    - Added rotation and mirroring flags
- Revision 8, 2015-03-25 (James Jones)
    - Combined the duplicate issues sections introduced in last change.
    - Added proposed resolutions to several issues.
- Revision 9, 2015-04-01 (Daniel Rakos)
    - Rebased extension against Vulkan 0.82.0
- Revision 10, 2015-04-01 (James Jones)
    - Added issues (10) and (11).
    - Added more straw-man issue resolutions, and cleaned up the proposed resolution for issue (4).
    - Updated the rotation and mirroring enums to have proper bitmask semantics.
- Revision 11, 2015-04-15 (James Jones)
    - Added proposed resolution for issues (1) and (2).
    - Added issues (12), (13), (14), and (15)

    - Removed pNativeHandle field from overlay structure.
    - Fixed small compilation errors in example code.
- Revision 12, 2015-07-29 (James Jones)
    - Rewrote the guts of the extension against the latest WSI swapchain specifications and the latest Vulkan API.
    - Address overlay planes by their index rather than an object handle and refer to them as “planes” rather than “overlays” to make it slightly clearer that even a display with no “overlays” still has at least one base “plane” that images can be displayed on.
    - Updated most of the issues.
    - Added an “extension type” section to the specification header.
    - Reused the VK_EXT_KHR_surface surface transform enumerations rather than redefining them here.
    - Updated the example code to use the new semantics.
- Revision 13, 2015-08-21 (Ian Elliott)
    - Renamed this extension and all of its enumerations, types, functions, etc. This makes it compliant with the proposed standard for Vulkan extensions.
    - Switched from “revision” to “version”, including use of the VK_MAKE_VERSION macro in the header file.
- Revision 14, 2015-09-01 (James Jones)
    - Restore single-field revision number.
- Revision 15, 2015-09-08 (James Jones)
    - Added alpha flags enum.
    - Added premultiplied alpha support.
- Revision 16, 2015-09-08 (James Jones)
    - Added description section to the spec.
    - Added issues 16 - 18.
- Revision 17, 2015-10-02 (James Jones)
    - Planes are now a property of the entire device rather than individual displays. This allows planes to be moved between multiple displays on devices that support it.
    - Added a function to create a VkSurfaceKHR object describing a display plane and mode to align with the new per-platform surface creation conventions.
    - Removed detailed mode timing data. It was agreed that the mode extents and refresh rate are sufficient for current use cases. Other information could be added back in as an extension if it is needed in the future.
    - Added support for smart/persistent/buffered display devices.
- Revision 18, 2015-10-26 (Ian Elliott)
    - Renamed from VK_EXT_KHR_display to VK_KHR_display.

- Revision 19, 2015-11-02 (James Jones)
  - Updated example code to match revision 17 changes.
- Revision 20, 2015-11-03 (Daniel Rakos)
  - Added allocation callbacks to creation functions.
- Revision 21, 2015-11-10 (Jesse Hall)
  - Added VK_DISPLAY_PLANE_ALPHA_OPAQUE_BIT_KHR, and use VkDisplayPlaneAlphaFlagBitsKHR for VkDisplayPlanePropertiesKHR::alphaMode instead of VkDisplayPlaneAlphaFlagsKHR, since it only represents one mode.
  - Added reserved flags bitmask to VkDisplayPlanePropertiesKHR.
  - Use VkSurfaceTransformFlagBitsKHR instead of obsolete VkSurfaceTransformKHR.
  - Renamed vkGetDisplayPlaneSupportedDisplaysKHR parameters for clarity.
- Revision 22, 2015-12-18 (James Jones)
  - Added missing “planeIndex” parameter to vkGetDisplayPlaneSupportedDisplaysKHR()
- Revision 23, 2017-03-13 (James Jones)
  - Closed all remaining issues. The specification and implementations have been shipping with the proposed resolutions for some time now.
  - Removed the sample code and noted it has been integrated into the official Vulkan SDK cube demo.

VK_KHR_display_swapchain

Name String

`VK_KHR_display_swapchain`

Extension Type

Device extension

Registered Extension Number

4

Revision

10

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_swapchain
and
VK_KHR_display

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2017-03-13

IP Status

No known IP claims.

Contributors

- James Jones, NVIDIA
- Jeff Vigil, Qualcomm
- Jesse Hall, Google

Description

This extension provides an API to create a swapchain directly on a device's display without any underlying window system.

New Commands

- vkCreateSharedSwapchainsKHR

New Structures

- Extending VkPresentInfoKHR:
  - VkDisplayPresentInfoKHR

New Enum Constants

- `VK_KHR_DISPLAY_SWAPCHAIN_EXTENSION_NAME`
- `VK_KHR_DISPLAY_SWAPCHAIN_SPEC_VERSION`
- Extending VkResult:
  - VK_ERROR_INCOMPATIBLE_DISPLAY_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DISPLAY_PRESENT_INFO_KHR

Issues

1) Should swapchains sharing images each hold a reference to the images, or should it be up to the application to destroy the swapchains and images in an order that avoids the need for reference counting?

RESOLVED: Take a reference. The lifetime of presentable images is already complex enough.

2) Should the `srcRect` and `dstRect` parameters be specified as part of the presentation command, or at swapchain creation time?

RESOLVED: As part of the presentation command. This allows moving and scaling the image on the screen without the need to respecify the mode or create a new swapchain and presentable images.

3) Should `srcRect` and `dstRect` be specified as rects, or separate offset/extent values?

RESOLVED: As rects. Specifying them separately might make it easier for hardware to expose support for one but not the other, but in such cases applications must just take care to obey the reported capabilities and not use non-zero offsets or extents that require scaling, as appropriate.

4) How can applications create multiple swapchains that use the same images?

RESOLVED: By calling vkCreateSharedSwapchainsKHR.

An earlier resolution used vkCreateSwapchainKHR, chaining multiple VkSwapchainCreateInfoKHR structures through `pNext`. In order to allow each swapchain to also allow other extension structs, a level of indirection was used: VkSwapchainCreateInfoKHR::`pNext` pointed to a different structure, which had both `sType` and `pNext` members for additional extensions, and also had a pointer to the next VkSwapchainCreateInfoKHR structure. The number of swapchains to be created could only be found by walking this linked list of alternating structures, and the `pSwapchains` out parameter was reinterpreted to be an array of VkSwapchainKHR handles.

Another option considered was a method to specify a “shared” swapchain when creating a new swapchain, such that groups of swapchains using the same images could be built up one at a time. This was deemed unusable because drivers need to know all of the displays an image will be used

on when determining which internal formats and layouts to use for that image.

5) Can VkDisplayPresentInfoKHR be used without calling vkCreateSharedSwapchainsKHR?

RESOLVED: Yes.

The two bits of functionality were meant to be independent. If the extension was created today, there would likely be two feature bits for VkDisplayPresentInfoKHR and vkCreateSharedSwapchainsKHR.

Examples

NOTE	The example code for the `VK_KHR_display` and `VK_KHR_display_swapchain` extensions was removed from the appendix after revision 1.0.43. The display swapchain creation example code was ported to the cube demo that is shipped with the official Khronos SDK, and is being kept up-to-date in that location (see: https://github.com/KhronosGroup/Vulkan-Tools/blob/main/cube/cube.c).

Version History

- Revision 1, 2015-07-29 (James Jones)
  - Initial draft
- Revision 2, 2015-08-21 (Ian Elliott)
  - Renamed this extension and all of its enumerations, types, functions, etc. This makes it compliant with the proposed standard for Vulkan extensions.
  - Switched from “revision” to “version”, including use of the VK_MAKE_VERSION macro in the header file.
- Revision 3, 2015-09-01 (James Jones)
  - Restore single-field revision number.
- Revision 4, 2015-09-08 (James Jones)
  - Allow creating multiple swapchains that share the same images using a single call to vkCreateSwapchainKHR().
- Revision 5, 2015-09-10 (Alon Or-bach)
  - Removed underscores from SWAP_CHAIN in two enums.
- Revision 6, 2015-10-02 (James Jones)
  - Added support for smart panels/buffered displays.
- Revision 7, 2015-10-26 (Ian Elliott)
  - Renamed from VK_EXT_KHR_display_swapchain to VK_KHR_display_swapchain.
- Revision 8, 2015-11-03 (Daniel Rakos)
  - Updated sample code based on the changes to VK_KHR_swapchain.
- Revision 9, 2015-11-10 (Jesse Hall)
  - Replaced VkDisplaySwapchainCreateInfoKHR with vkCreateSharedSwapchainsKHR,

changing resolution of issue #4.

- Revision 10, 2017-03-13 (James Jones)
  - Closed all remaining issues. The specification and implementations have been shipping with the proposed resolutions for some time now.
  - Removed the sample code and noted it has been integrated into the official Vulkan SDK cube demo.

VK_KHR_extended_flags

Name String

VK_KHR_extended_flags

Extension Type

Device extension

Registered Extension Number

669

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_VERSION_1_0
- Interacts with VK_VERSION_1_2
- Interacts with VK_VERSION_1_3
- Interacts with VK_VERSION_1_4
- Interacts with VK_ARM_pipeline_opacity_micromap
- Interacts with VK_ARM_tensors
- Interacts with VK_EXT_attachment_feedback_loop_layout
- Interacts with VK_EXT_buffer_device_address
- Interacts with VK_EXT_conditional_rendering
- Interacts with VK_EXT_descriptor_buffer
- Interacts with VK_EXT_fragment_density_map
- Interacts with VK_EXT_fragment_density_map_offset
- Interacts with VK_EXT_graphics_pipeline_library
- Interacts with VK_EXT_image_2d_view_of_3d
- Interacts with VK_EXT_multisampled_render_to_single_sampled
- Interacts with VK_EXT_opacity_micromap
- Interacts with VK_EXT_pipeline_creation_cache_control
- Interacts with VK_EXT_sample_locations

- Interacts with VK_EXT_separate_stencil_usage
- Interacts with VK_EXT_transform_feedback
- Interacts with VK_HUAWEI_invocation_mask
- Interacts with VK_KHR_acceleration_structure
- Interacts with VK_KHR_buffer_device_address
- Interacts with VK_KHR_dynamic_rendering
- Interacts with VK_KHR_fragment_shading_rate
- Interacts with VK_KHR_pipeline_executable_properties
- Interacts with VK_KHR_pipeline_library
- Interacts with VK_KHR_ray_tracing_pipeline
- Interacts with VK_KHR_shared_presentable_image
- Interacts with VK_KHR_video_decode_queue
- Interacts with VK_KHR_video_encode_quantization_map
- Interacts with VK_KHR_video_encode_queue
- Interacts with VK_KHR_video_maintenance1
- Interacts with VK_NV_corner_sampled_image
- Interacts with VK_NV_device_generated_commands
- Interacts with VK_NV_displacement_micromap
- Interacts with VK_NV_ray_tracing
- Interacts with VK_NV_ray_tracing_motion_blur
- Interacts with VK_QCOM_image_processing
- Interacts with VK_QCOM_tile_memory_heap

Contact

- Jon Leech [GitHub]oddhack

Extension Proposal

VK_KHR_extended_flags

Other Extension Metadata

Last Modified Date

2025-12-17

IP Status

No known IP claims.

Contributors

- Andreas Süßenbach, NVIDIA
- Daniel Koch, NVIDIA

- Daniel Rakos, RasterGrid
- Jon Leech, Independent
- Matthew Netsch, QUALCOMM
- Noah Fredriks, AMD
- Spencer Fricke, LunarG
- Ting Wei, ARM
- Žiga Markuš, LunarG

Description

This extension adds a new VkFormatFeatureFlagBits4KHR format feature flag type, supporting up to 64 additional flags beyond those defined by VkFormatFeatureFlagBits2; and a new VkFormatProperties4KHR structure to extend VkFormatProperties2. The new structure does not replace format feature flag bits specified by existing commands and structures, unlike VkFormatProperties3. Instead, it adds new flags to the existing APIs. Initially, no new format flag bits are defined.

This extension also adds a new VkImageCreateFlagBits2KHR image create flag type, supporting up to 32 additional flags, and a new VkImageCreateFlags2CreateInfoKHR structure to extend VkImageCreateInfo and other commands that specify image create flag bits. The new structure duplicates image create flag bits specified by existing commands and structures, while also adding up to 32 new flags. All the existing VkImageCreateFlagBits are given corresponding VkImageCreateFlagBits2KHR bits with the same meaning. Initially, no new image create flag bits are defined.

This extension also adds a new VkImageUsageFlagBits2KHR image usage flag type, supporting up to 32 additional flags, and a new VkImageUsageFlags2CreateInfoKHR structure to extend VkImageCreateInfo and other commands that specify image usage flag bits. The new structure duplicates image usage flag bits specified by existing commands and structures, while also adding up to 32 new flags. All the existing VkImageUsageFlagBits are given corresponding VkImageUsageFlagBits2KHR bits with the same meaning. Initially, no new image usage flag bits are defined.

The extension is infrastructure used because the API is running out of available bits in the VkFormatFeatureFlagBits2 and VkImageUsageFlagBits types, and will soon thereafter run out of bits in VkImageCreateFlagBits.

Finally, this extension adds all the same 64-bit flag types first introduced by VK_KHR_maintenance5. This will allow API variants like Vulkan SC to support the extended flags without requiring other, unrelated functionality of VK_KHR_maintenance5.

New Structures

- Extending VkBufferViewCreateInfo, VkBufferCreateInfo, VkPhysicalDeviceExternalBufferInfo, VkDescriptorBufferBindingInfoEXT:
  - VkBufferUsageFlags2CreateInfoKHR

- Extending VkComputePipelineCreateInfo, VkGraphicsPipelineCreateInfo, VkRayTracingPipelineCreateInfoKHR:
  - VkPipelineCreateFlags2CreateInfoKHR
- Extending VkFormatProperties2:
  - VkFormatProperties4KHR
- Extending VkFramebufferAttachmentImageInfo, VkImageCreateInfo, VkPhysicalDeviceImageFormatInfo2, VkPhysicalDeviceSparseImageFormatInfo2, VkPhysicalDeviceVideoFormatInfoKHR, VkSurfaceCapabilities2KHR, VkSwapchainCreateInfoKHR, VkVideoFormatPropertiesKHR:
  - VkImageUsageFlags2CreateInfoKHR
- Extending VkImageCreateInfo, VkPhysicalDeviceImageFormatInfo2, VkFramebufferAttachmentImageInfo, VkVideoFormatPropertiesKHR:
  - VkImageCreateFlags2CreateInfoKHR
- Extending VkImageViewCreateInfo:
  - VkImageViewUsage2CreateInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceExtendedFlagsFeaturesKHR

If VK_KHR_shared_presentable_image is supported:

- Extending VkSurfaceCapabilities2KHR:
  - VkSharedPresentSurfaceCapabilities2KHR

If Vulkan Version 1.2 or `VK_EXT_separate_stencil_usage` is supported:

- Extending VkImageCreateInfo, VkPhysicalDeviceImageFormatInfo2:
  - VkImageStencilUsage2CreateInfoKHR

New Enums

- VkBufferUsageFlagBits2KHR
- VkFormatFeatureFlagBits4KHR
- VkImageCreateFlagBits2KHR
- VkImageUsageFlagBits2KHR
- VkPipelineCreateFlagBits2KHR

New Bitmasks

- VkBufferUsageFlags2KHR
- VkFormatFeatureFlags4KHR
- VkImageCreateFlags2KHR
- VkImageUsageFlags2KHR

- VkPipelineCreateFlags2KHR

New Enum Constants

- `VK_KHR_EXTENDED_FLAGS_EXTENSION_NAME`
- `VK_KHR_EXTENDED_FLAGS_SPEC_VERSION`
- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_INDEX_BUFFER_BIT_KHR
  - VK_BUFFER_USAGE_2_INDIRECT_BUFFER_BIT_KHR
  - VK_BUFFER_USAGE_2_STORAGE_BUFFER_BIT_KHR
  - VK_BUFFER_USAGE_2_STORAGE_TEXEL_BUFFER_BIT_KHR
  - VK_BUFFER_USAGE_2_TRANSFER_DST_BIT_KHR
  - VK_BUFFER_USAGE_2_TRANSFER_SRC_BIT_KHR
  - VK_BUFFER_USAGE_2_UNIFORM_BUFFER_BIT_KHR
  - VK_BUFFER_USAGE_2_UNIFORM_TEXEL_BUFFER_BIT_KHR
  - VK_BUFFER_USAGE_2_VERTEX_BUFFER_BIT_KHR
- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_ALLOW_DERIVATIVES_BIT_KHR
  - VK_PIPELINE_CREATE_2_DERIVATIVE_BIT_KHR
  - VK_PIPELINE_CREATE_2_DISABLE_OPTIMIZATION_BIT_KHR
  - VK_PIPELINE_CREATE_2_DISPATCH_BASE_BIT_KHR
  - VK_PIPELINE_CREATE_2_VIEW_INDEX_FROM_DEVICE_INDEX_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_BUFFER_USAGE_FLAGS_2_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_4_KHR
  - VK_STRUCTURE_TYPE_IMAGE_CREATE_FLAGS_2_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_IMAGE_USAGE_FLAGS_2_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_IMAGE_VIEW_USAGE_2_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_FLAGS_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PIPELINE_CREATE_FLAGS_2_CREATE_INFO_KHR

If VK_KHR_dynamic_rendering or Vulkan Version 1.3 and VK_EXT_fragment_density_map is supported:

- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT

If VK_KHR_dynamic_rendering or Vulkan Version 1.3 and VK_KHR_fragment_shading_rate is supported:

- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR

If VK_EXT_attachment_feedback_loop_layout is supported:

- Extending VkImageUsageFlagBits2KHR:
  - VK_IMAGE_USAGE_2_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT
- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT
  - VK_PIPELINE_CREATE_2_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT

If VK_EXT_conditional_rendering is supported:

- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_CONDITIONAL_RENDERING_BIT_EXT

If VK_EXT_descriptor_buffer is supported:

- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT
  - VK_BUFFER_USAGE_2_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT
  - VK_BUFFER_USAGE_2_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT
- Extending VkImageCreateFlagBits2KHR:
  - VK_IMAGE_CREATE_2_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT
- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_DESCRIPTOR_BUFFER_BIT_EXT

If VK_EXT_fragment_density_map is supported:

- Extending VkImageCreateFlagBits2KHR:
  - VK_IMAGE_CREATE_2_SUBSAMPLED_BIT_EXT
- Extending VkImageUsageFlagBits2KHR:
  - VK_IMAGE_USAGE_2_FRAGMENT_DENSITY_MAP_BIT_EXT

If VK_EXT_fragment_density_map_offset is supported:

- Extending VkImageCreateFlagBits2KHR:
  - VK_IMAGE_CREATE_2_FRAGMENT_DENSITY_MAP_OFFSET_BIT_EXT

If VK_EXT_graphics_pipeline_library is supported:

- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_LINK_TIME_OPTIMIZATION_BIT_EXT

    - VK_PIPELINE_CREATE_2_RETAIN_LINK_TIME_OPTIMIZATION_INFO_BIT_EXT

If VK_EXT_image_2d_view_of_3d is supported:

- Extending VkImageCreateFlagBits2KHR:
    - VK_IMAGE_CREATE_2_2D_VIEW_COMPATIBLE_BIT_EXT

If VK_EXT_multisampled_render_to_single_sampled is supported:

- Extending VkImageCreateFlagBits2KHR:
    - VK_IMAGE_CREATE_2_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT

If VK_EXT_opacity_micromap is supported:

- Extending VkBufferUsageFlagBits2:
    - VK_BUFFER_USAGE_2_MICROMAP_BUILD_INPUT_READ_ONLY_BIT_EXT
    - VK_BUFFER_USAGE_2_MICROMAP_STORAGE_BIT_EXT
- Extending VkPipelineCreateFlagBits2:
    - VK_PIPELINE_CREATE_2_RAY_TRACING_OPACITY_MICROMAP_BIT_EXT

If VK_EXT_sample_locations is supported:

- Extending VkImageCreateFlagBits2KHR:
    - VK_IMAGE_CREATE_2_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT

If VK_EXT_transform_feedback is supported:

- Extending VkBufferUsageFlagBits2:
    - VK_BUFFER_USAGE_2_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT
    - VK_BUFFER_USAGE_2_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT

If VK_KHR_acceleration_structure is supported:

- Extending VkBufferUsageFlagBits2:
    - VK_BUFFER_USAGE_2_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR
    - VK_BUFFER_USAGE_2_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR

If VK_KHR_fragment_shading_rate is supported:

- Extending VkImageUsageFlagBits2KHR:
    - VK_IMAGE_USAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR

If VK_KHR_pipeline_executable_properties is supported:

- Extending VkPipelineCreateFlagBits2:
    - VK_PIPELINE_CREATE_2_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR

    - VK_PIPELINE_CREATE_2_CAPTURE_STATISTICS_BIT_KHR

If VK_KHR_pipeline_library is supported:

- Extending VkPipelineCreateFlagBits2:
    - VK_PIPELINE_CREATE_2_LIBRARY_BIT_KHR

If VK_KHR_ray_tracing_pipeline is supported:

- Extending VkBufferUsageFlagBits2:
    - VK_BUFFER_USAGE_2_SHADER_BINDING_TABLE_BIT_KHR
- Extending VkPipelineCreateFlagBits2:
    - VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR
    - VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR
    - VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR
    - VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR
    - VK_PIPELINE_CREATE_2_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR
    - VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_AABBS_BIT_KHR
    - VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR

If VK_KHR_shared_presentable_image is supported:

- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_SHARED_PRESENT_SURFACE_CAPABILITIES_2_KHR

If VK_KHR_video_decode_queue is supported:

- Extending VkBufferUsageFlagBits2:
    - VK_BUFFER_USAGE_2_VIDEO_DECODE_DST_BIT_KHR
    - VK_BUFFER_USAGE_2_VIDEO_DECODE_SRC_BIT_KHR
- Extending VkImageUsageFlagBits2KHR:
    - VK_IMAGE_USAGE_2_VIDEO_DECODE_DPB_BIT_KHR
    - VK_IMAGE_USAGE_2_VIDEO_DECODE_DST_BIT_KHR
    - VK_IMAGE_USAGE_2_VIDEO_DECODE_SRC_BIT_KHR

If VK_KHR_video_encode_quantization_map is supported:

- Extending VkImageUsageFlagBits2KHR:
    - VK_IMAGE_USAGE_2_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR
    - VK_IMAGE_USAGE_2_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR

If VK_KHR_video_encode_queue is supported:

- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_VIDEO_ENCODE_DST_BIT_KHR
  - VK_BUFFER_USAGE_2_VIDEO_ENCODE_SRC_BIT_KHR
- Extending VkImageUsageFlagBits2KHR:
  - VK_IMAGE_USAGE_2_VIDEO_ENCODE_DPB_BIT_KHR
  - VK_IMAGE_USAGE_2_VIDEO_ENCODE_DST_BIT_KHR
  - VK_IMAGE_USAGE_2_VIDEO_ENCODE_SRC_BIT_KHR

If VK_KHR_video_maintenance1 is supported:

- Extending VkImageCreateFlagBits2KHR:
  - VK_IMAGE_CREATE_2_VIDEO_PROFILE_INDEPENDENT_BIT_KHR

If Vulkan Version 1.0 is supported:

- Extending VkImageCreateFlagBits2KHR:
  - VK_IMAGE_CREATE_2_2D_ARRAY_COMPATIBLE_BIT_KHR
  - VK_IMAGE_CREATE_2_ALIAS_BIT_KHR
  - VK_IMAGE_CREATE_2_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT_KHR
  - VK_IMAGE_CREATE_2_DISJOINT_BIT_KHR
  - VK_IMAGE_CREATE_2_EXTENDED_USAGE_BIT_KHR
  - VK_IMAGE_CREATE_2_PROTECTED_BIT_KHR
  - VK_IMAGE_CREATE_2_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR

If Vulkan Version 1.2 or `VK_EXT_separate_stencil_usage` is supported:

- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_IMAGE_STENCIL_USAGE_2_CREATE_INFO_KHR

If Vulkan Version 1.2 or VK_KHR_buffer_device_address or `VK_EXT_buffer_device_address` is supported:

- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_SHADER_DEVICE_ADDRESS_BIT_KHR

If Vulkan Version 1.3 or `VK_EXT_pipeline_creation_cache_control` is supported:

- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_EARLY_RETURN_ON_FAILURE_BIT_KHR
  - VK_PIPELINE_CREATE_2_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT_KHR

If Vulkan Version 1.4 is supported:

- Extending VkImageUsageFlagBits2KHR:

- VK_IMAGE_USAGE_2_HOST_TRANSFER_BIT_KHR

Version History

- Revision 0.1, 2025-12-17 (Jon Leech)
  - Initial draft
- Revision 0.2, 2026-01-19 (Jon Leech)
  - Add VkImageCreateFlags2KHR and import 64-bit flags types from VK_KHR_maintenance5.
- Revision 0.3, 2026-02-14 (Jon Leech)
  - Add a feature structure.

VK_KHR_external_fence_fd

Name String

VK_KHR_external_fence_fd

Extension Type

Device extension

Registered Extension Number

116

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_external_fence
or
Vulkan Version 1.1

Contact

- Jesse Hall [GitHub]critsec

Other Extension Metadata

Last Modified Date

2017-05-08

IP Status

No known IP claims.

Contributors

- Jesse Hall, Google
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Cass Everitt, Oculus
- Contributors to VK_KHR_external_semaphore_fd

Description

An application using external memory may wish to synchronize access to that memory using fences. This extension enables an application to export fence payload to and import fence payload from POSIX file descriptors.

New Commands

- vkGetFenceFdKHR
- vkImportFenceFdKHR

New Structures

- VkFenceGetFdInfoKHR
- VkImportFenceFdInfoKHR

New Enum Constants

- `VK_KHR_EXTERNAL_FENCE_FD_EXTENSION_NAME`
- `VK_KHR_EXTERNAL_FENCE_FD_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_FENCE_GET_FD_INFO_KHR
  - VK_STRUCTURE_TYPE_IMPORT_FENCE_FD_INFO_KHR

Issues

This extension borrows concepts, semantics, and language from `VK_KHR_external_semaphore_fd`. That extension's issues apply equally to this extension.

Version History

- Revision 1, 2017-05-08 (Jesse Hall)
  - Initial revision

VK_KHR_external_fence_win32

Name String

VK_KHR_external_fence_win32

Extension Type

Device extension

Registered Extension Number

115

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_external_fence

Contact

- Jesse Hall [GitHub]critsec

Other Extension Metadata

Last Modified Date

2017-05-08

IP Status

No known IP claims.

Contributors

- Jesse Hall, Google
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Cass Everitt, Oculus
- Contributors to VK_KHR_external_semaphore_win32

Description

An application using external memory may wish to synchronize access to that memory using fences. This extension enables an application to export fence payload to and import fence payload from Windows handles.

New Commands

- vkGetFenceWin32HandleKHR

- vkImportFenceWin32HandleKHR

New Structures

- VkFenceGetWin32HandleInfoKHR
- VkImportFenceWin32HandleInfoKHR
- Extending VkFenceCreateInfo:
  - VkExportFenceWin32HandleInfoKHR

New Enum Constants

- `VK_KHR_EXTERNAL_FENCE_WIN32_EXTENSION_NAME`
- `VK_KHR_EXTERNAL_FENCE_WIN32_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_EXPORT_FENCE_WIN32_HANDLE_INFO_KHR
  - VK_STRUCTURE_TYPE_FENCE_GET_WIN32_HANDLE_INFO_KHR
  - VK_STRUCTURE_TYPE_IMPORT_FENCE_WIN32_HANDLE_INFO_KHR

Issues

This extension borrows concepts, semantics, and language from `VK_KHR_external_semaphore_win32`. That extension's issues apply equally to this extension.

1) Should D3D12 fence handle types be supported, like they are for semaphores?

RESOLVED: No. Doing so would require extending the fence signal and wait operations to provide values to signal / wait for, like `VkD3D12FenceSubmitInfoKHR` does. A D3D12 fence can be signaled by importing it into a VkSemaphore instead of a VkFence, and applications can check status or wait on the D3D12 fence using non-Vulkan APIs. The convenience of being able to do these operations on `VkFence` objects does not justify the extra API complexity.

Version History

- Revision 1, 2017-05-08 (Jesse Hall)
  - Initial revision

VK_KHR_external_memory_fd

Name String

`VK_KHR_external_memory_fd`

Extension Type

Device extension

Registered Extension Number

75

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_external_memory
or
Vulkan Version 1.1

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2016-10-21

IP Status

No known IP claims.

Contributors

- James Jones, NVIDIA
- Jeff Juliano, NVIDIA

Description

An application may wish to reference device memory in multiple Vulkan logical devices or instances, in multiple processes, and/or in multiple APIs. This extension enables an application to export POSIX file descriptor handles from Vulkan memory objects and to import Vulkan memory objects from POSIX file descriptor handles exported from other Vulkan memory objects or from similar resources in other APIs.

New Commands

- vkGetMemoryFdKHR

- vkGetMemoryFdPropertiesKHR

New Structures

- VkMemoryFdPropertiesKHR
- VkMemoryGetFdInfoKHR
- Extending VkMemoryAllocateInfo:
  - VkImportMemoryFdInfoKHR

New Enum Constants

- `VK_KHR_EXTERNAL_MEMORY_FD_EXTENSION_NAME`
- `VK_KHR_EXTERNAL_MEMORY_FD_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_IMPORT_MEMORY_FD_INFO_KHR
  - VK_STRUCTURE_TYPE_MEMORY_FD_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_MEMORY_GET_FD_INFO_KHR

Issues

1) Does the application need to close the file descriptor returned by vkGetMemoryFdKHR?

RESOLVED: Yes, unless it is passed back in to a driver instance to import the memory. A successful get call transfers ownership of the file descriptor to the application, and a successful import transfers it back to the driver. Destroying the original memory object will not close the file descriptor or remove its reference to the underlying memory resource associated with it.

2) Do drivers ever need to expose multiple file descriptors per memory object?

RESOLVED: No. This would indicate there are actually multiple memory objects, rather than a single memory object.

3) How should the valid size and memory type for POSIX file descriptor memory handles created outside of Vulkan be specified?

RESOLVED: The valid memory types are queried directly from the external handle. The size will be specified by future extensions that introduce such external memory handle types.

Version History

- Revision 1, 2016-10-21 (James Jones)
  - Initial revision

VK_KHR_external_memory_win32

Name String

VK_KHR_external_memory_win32

Extension Type

Device extension

Registered Extension Number

74

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_external_memory
or
Vulkan Version 1.1

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2016-10-21

IP Status

No known IP claims.

Contributors

- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Carsten Rohde, NVIDIA

Description

An application may wish to reference device memory in multiple Vulkan logical devices or instances, in multiple processes, and/or in multiple APIs. This extension enables an application to export Windows handles from Vulkan memory objects and to import Vulkan memory objects from Windows handles exported from other Vulkan memory objects or from similar resources in other APIs.

New Commands

- vkGetMemoryWin32HandleKHR
- vkGetMemoryWin32HandlePropertiesKHR

New Structures

- VkMemoryGetWin32HandleInfoKHR
- VkMemoryWin32HandlePropertiesKHR
- Extending VkMemoryAllocateInfo:
  - VkExportMemoryWin32HandleInfoKHR
  - VkImportMemoryWin32HandleInfoKHR

New Enum Constants

- `VK_KHR_EXTERNAL_MEMORY_WIN32_EXTENSION_NAME`
- `VK_KHR_EXTERNAL_MEMORY_WIN32_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_EXPORT_MEMORY_WIN32_HANDLE_INFO_KHR
  - VK_STRUCTURE_TYPE_IMPORT_MEMORY_WIN32_HANDLE_INFO_KHR
  - VK_STRUCTURE_TYPE_MEMORY_GET_WIN32_HANDLE_INFO_KHR
  - VK_STRUCTURE_TYPE_MEMORY_WIN32_HANDLE_PROPERTIES_KHR

Issues

1) Do applications need to call `CloseHandle`() on the values returned from vkGetMemoryWin32HandleKHR when `handleType` is VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR?

RESOLVED: Yes. A successful get call transfers ownership of the handle to the application. Destroying the memory object will not destroy the handle or the handle's reference to the underlying memory resource. Unlike file descriptor opaque handles, win32 opaque handle ownership can not be transferred back to a driver by an import operation.

2) Should the language regarding KMT/Windows 7 handles be moved to a separate extension so that it can be deprecated over time?

RESOLVED: No. Support for them can be deprecated by drivers if they choose, by no longer returning them in the supported handle types of the instance level queries.

3) How should the valid size and memory type for windows memory handles created outside of Vulkan be specified?

RESOLVED: The valid memory types are queried directly from the external handle. The size is determined by the associated image or buffer memory requirements for external handle types that require dedicated allocations, and by the size specified when creating the object from which the

handle was exported for other external handle types.

Version History

- Revision 1, 2016-10-21 (James Jones)
  - Initial revision

VK_KHR_external_semaphore_fd

Name String

`VK_KHR_external_semaphore_fd`

Extension Type

Device extension

Registered Extension Number

80

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_external_semaphore
or
Vulkan Version 1.1

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2016-10-21

IP Status

No known IP claims.

Contributors

- Jesse Hall, Google
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Carsten Rohde, NVIDIA

Description

An application using external memory may wish to synchronize access to that memory using semaphores. This extension enables an application to export semaphore payload to and import semaphore payload from POSIX file descriptors.

New Commands

- vkGetSemaphoreFdKHR
- vkImportSemaphoreFdKHR

New Structures

- VkImportSemaphoreFdInfoKHR
- VkSemaphoreGetFdInfoKHR

New Enum Constants

- `VK_KHR_EXTERNAL_SEMAPHORE_FD_EXTENSION_NAME`
- `VK_KHR_EXTERNAL_SEMAPHORE_FD_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_IMPORT_SEMAPHORE_FD_INFO_KHR
  - VK_STRUCTURE_TYPE_SEMAPHORE_GET_FD_INFO_KHR

Issues

1) Does the application need to close the file descriptor returned by vkGetSemaphoreFdKHR?

RESOLVED: Yes, unless it is passed back in to a driver instance to import the semaphore. A successful get call transfers ownership of the file descriptor to the application, and a successful import transfers it back to the driver. Destroying the original semaphore object will not close the file descriptor or remove its reference to the underlying semaphore resource associated with it.

Version History

- Revision 1, 2016-10-21 (Jesse Hall)
  - Initial revision

VK_KHR_external_semaphore_win32

Name String

VK_KHR_external_semaphore_win32

Extension Type

Device extension

Registered Extension Number

79

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_external_semaphore

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2016-10-21

IP Status

No known IP claims.

Contributors

- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Carsten Rohde, NVIDIA

Description

An application using external memory may wish to synchronize access to that memory using semaphores. This extension enables an application to export semaphore payload to and import semaphore payload from Windows handles.

New Commands

- vkGetSemaphoreWin32HandleKHR
- vkImportSemaphoreWin32HandleKHR

New Structures

- VkImportSemaphoreWin32HandleInfoKHR
- VkSemaphoreGetWin32HandleInfoKHR
- Extending VkSemaphoreCreateInfo:
  - VkExportSemaphoreWin32HandleInfoKHR
- Extending VkSubmitInfo:
  - VkD3D12FenceSubmitInfoKHR

New Enum Constants

- VK_KHR_EXTERNAL_SEMAPHORE_WIN32_EXTENSION_NAME
- VK_KHR_EXTERNAL_SEMAPHORE_WIN32_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_D3D12_FENCE_SUBMIT_INFO_KHR
  - VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_WIN32_HANDLE_INFO_KHR
  - VK_STRUCTURE_TYPE_IMPORT_SEMAPHORE_WIN32_HANDLE_INFO_KHR
  - VK_STRUCTURE_TYPE_SEMAPHORE_GET_WIN32_HANDLE_INFO_KHR

Issues

1) Do applications need to call `CloseHandle`() on the values returned from vkGetSemaphoreWin32HandleKHR when `handleType` is VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR?

RESOLVED: Yes. A successful get call transfers ownership of the handle to the application. Destroying the semaphore object will not destroy the handle or the handle's reference to the underlying semaphore resource. Unlike file descriptor opaque handles, win32 opaque handle ownership can not be transferred back to a driver by an import operation.

2) Should the language regarding KMT/Windows 7 handles be moved to a separate extension so that it can be deprecated over time?

RESOLVED: No. Support for them can be deprecated by drivers if they choose, by no longer returning them in the supported handle types of the instance level queries.

3) Should applications be allowed to specify additional object attributes for shared handles?

RESOLVED: Yes. Applications will be allowed to provide similar attributes to those they would to any other handle creation API.

4) How do applications communicate the desired fence values to use with `D3D12_FENCE`-based Vulkan semaphores?

RESOLVED: There are a couple of options. The values for the signaled and reset states could be communicated up front when creating the object and remain static for the life of the Vulkan

semaphore, or they could be specified using auxiliary structures when submitting semaphore signal and wait operations, similar to what is done with the keyed mutex extensions. The latter is more flexible and consistent with the keyed mutex usage, but the former is a much simpler API.

Since Vulkan tends to favor flexibility and consistency over simplicity, a new structure specifying D3D12 fence acquire and release values is added to the vkQueueSubmit function.

Version History

- Revision 1, 2016-10-21 (James Jones)
  - Initial revision

VK_KHR_fragment_shader_barycentric

Name String

`VK_KHR_fragment_shader_barycentric`

Extension Type

Device extension

Registered Extension Number

323

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_EXT_provoking_vertex

SPIR-V Dependencies

- SPV_KHR_fragment_shader_barycentric

Contact

- Stu Smith

Extension Proposal

VK_KHR_fragment_shader_barycentric

Other Extension Metadata

Last Modified Date

2022-03-10

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for `GL_EXT_fragment_shader_barycentric`

Contributors

- Stu Smith, AMD
- Tobias Hector, AMD
- Graeme Leese, Broadcom

- Jan-Harald Fredriksen, Arm
- Slawek Grajewski, Intel
- Pat Brown, NVIDIA
- Hans-Kristian Arntzen, Valve
- Contributors to the VK_NV_fragment_shader_barycentric specification

Description

This extension is based on the `VK_NV_fragment_shader_barycentric` extension, and adds support for the following SPIR-V extension in Vulkan:

- `SPV_KHR_fragment_shader_barycentric`

The extension provides access to three additional fragment shader variable decorations in SPIR-V:

- `PerVertexKHR`, which indicates that a fragment shader input will not have interpolated values, but instead must be accessed with an extra array index that identifies one of the vertices of the primitive producing the fragment
- `BaryCoordKHR`, which indicates that the variable is a three-component floating-point vector holding barycentric weights for the fragment produced using perspective interpolation
- `BaryCoordNoPerspKHR`, which indicates that the variable is a three-component floating-point vector holding barycentric weights for the fragment produced using linear interpolation

When using GLSL source-based shader languages, the following variables from `GL_EXT_fragment_shader_barycentric` map to these SPIR-V built-in decorations:

- `in vec3 gl_BaryCoordEXT;` → `BaryCoordKHR`
- `in vec3 gl_BaryCoordNoPerspEXT;` → `BaryCoordNoPerspKHR`

GLSL variables declared using the `pervertexEXT` GLSL qualifier are expected to be decorated with `PerVertexKHR` in SPIR-V.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR

If VK_EXT_provoking_vertex is supported:

- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR

New Enum Constants

- `VK_KHR_FRAGMENT_SHADER_BARYCENTRIC_EXTENSION_NAME`
- `VK_KHR_FRAGMENT_SHADER_BARYCENTRIC_SPEC_VERSION`

- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_BARYCENTRIC_FEATURES_KHR

If VK_EXT_provoking_vertex is supported:

- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_BARYCENTRIC_PROPERTIES_KHR

New Built-In Variables

- `BaryCoordKHR`
- `BaryCoordNoPerspKHR`

New SPIR-V Decorations

- `PerVertexKHR`

New SPIR-V Capabilities

- `FragmentBarycentricKHR`

Issues

1) What are the interactions with MSAA and how are `BaryCoordKHR` and `BaryCoordNoPerspKHR` interpolated?

RESOLVED: The inputs decorated with `BaryCoordKHR` or `BaryCoordNoPerspKHR` **may** also be decorated with the `Centroid` or `Sample` qualifiers to specify interpolation, like any other fragment shader input. If the `shaderSampleRateInterpolationFunctions` feature is enabled, the extended instructions InterpolateAtCentroid, InterpolateAtOffset, and InterpolateAtSample from the GLSL.std.450 **may** also be used with inputs decorated with `BaryCoordKHR` or `BaryCoordNoPerspKHR`.

Version History

- Revision 1, 2022-03-10 (Stu Smith)
  - Initial revision

VK_KHR_fragment_shading_rate

Name String

`VK_KHR_fragment_shading_rate`

Extension Type

Device extension

Registered Extension Number

227

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1
and
VK_KHR_create_renderpass2
or
Vulkan Version 1.2

API Interactions

- Interacts with VK_VERSION_1_3
- Interacts with VK_KHR_dynamic_rendering
- Interacts with VK_KHR_format_feature_flags2

SPIR-V Dependencies

- SPV_KHR_fragment_shading_rate

Contact

- Tobias Hector [GitHub]tobski

Extension Proposal

VK_KHR_fragment_shading_rate

Other Extension Metadata

Last Modified Date

2021-09-30

Interactions and External Dependencies

- This extension provides API support for `GL_EXT_fragment_shading_rate`

Contributors

- Tobias Hector, AMD
- Guennadi Riguer, AMD
- Matthaeus Chajdas, AMD
- Pat Brown, Nvidia
- Matthew Netsch, Qualcomm
- Slawomir Grajewski, Intel
- Jan-Harald Fredriksen, Arm
- Jeff Bolz, Nvidia
- Arseny Kapoulkine, Roblox
- Contributors to the VK_NV_shading_rate_image specification
- Contributors to the VK_EXT_fragment_density_map specification

Description

This extension adds the ability to change the rate at which fragments are shaded. Rather than the usual single fragment invocation for each pixel covered by a primitive, multiple pixels can be shaded by a single fragment shader invocation.

Up to three methods are available to the application to change the fragment shading rate:

- Pipeline Fragment Shading Rate, which allows the specification of a rate per-draw.
- Primitive Fragment Shading Rate, which allows the specification of a rate per primitive, specified during shading.
- Attachment Fragment Shading Rate, which allows the specification of a rate per-region of the framebuffer, specified in a specialized image attachment.

Additionally, these rates can all be specified and combined in order to adjust the overall detail in the image at each point.

This functionality can be used to focus shading efforts where higher levels of detail are needed in some parts of a scene compared to others. This can be particularly useful in high resolution rendering, or for XR contexts.

This extension also adds support for the `SPV_KHR_fragment_shading_rate` extension which enables setting the primitive fragment shading rate, and allows querying the final shading rate from a fragment shader.

New Commands

- vkCmdSetFragmentShadingRateKHR
- vkGetPhysicalDeviceFragmentShadingRatesKHR

New Structures

- VkPhysicalDeviceFragmentShadingRateKHR
- Extending VkGraphicsPipelineCreateInfo:
  - VkPipelineFragmentShadingRateStateCreateInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceFragmentShadingRateFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceFragmentShadingRatePropertiesKHR
- Extending VkSubpassDescription2:
  - VkFragmentShadingRateAttachmentInfoKHR

If Vulkan Version 1.3 or VK_KHR_dynamic_rendering is supported:

- Extending VkRenderingInfo:
  - VkRenderingFragmentShadingRateAttachmentInfoKHR

New Enums

- VkFragmentShadingRateCombinerOpKHR

New Enum Constants

- `VK_KHR_FRAGMENT_SHADING_RATE_EXTENSION_NAME`
- `VK_KHR_FRAGMENT_SHADING_RATE_SPEC_VERSION`
- Extending VkAccessFlagBits:
  - VK_ACCESS_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR
- Extending VkDynamicState:
  - VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR
- Extending VkFormatFeatureFlagBits:
  - VK_FORMAT_FEATURE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- Extending VkImageLayout:
  - VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR
- Extending VkImageUsageFlagBits:
  - VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- Extending VkPipelineStageFlagBits:
  - VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_FRAGMENT_SHADING_RATE_ATTACHMENT_INFO_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_FEATURES_KHR

    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_KHR
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_PROPERTIES_KHR
    - VK_STRUCTURE_TYPE_PIPELINE_FRAGMENT_SHADING_RATE_STATE_CREATE_INFO_KHR

If VK_KHR_format_feature_flags2 or Vulkan Version 1.3 is supported:

- Extending VkFormatFeatureFlagBits2:
    - VK_FORMAT_FEATURE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR

If Vulkan Version 1.3 or VK_KHR_dynamic_rendering is supported:

- Extending VkPipelineCreateFlagBits:
    - VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
    - VK_PIPELINE_RASTERIZATION_STATE_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_INFO_KHR

Version History

- Revision 1, 2020-05-06 (Tobias Hector)
    - Initial revision
- Revision 2, 2021-09-30 (Jon Leech)
    - Add interaction with `VK_KHR_format_feature_flags2` to `vk.xml`

VK_KHR_get_display_properties2

Name String

VK_KHR_get_display_properties2

Extension Type

Instance extension

Registered Extension Number

122

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_display

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2017-02-21

IP Status

No known IP claims.

Contributors

- Ian Elliott, Google
- James Jones, NVIDIA

Description

This extension provides new queries for device display properties and capabilities that can be easily extended by other extensions, without introducing any further queries. This extension can be considered the VK_KHR_display equivalent of the VK_KHR_get_physical_device_properties2 extension.

New Commands

- vkGetDisplayModeProperties2KHR
- vkGetDisplayPlaneCapabilities2KHR
- vkGetPhysicalDeviceDisplayPlaneProperties2KHR
- vkGetPhysicalDeviceDisplayProperties2KHR

New Structures

- VkDisplayModeProperties2KHR
- VkDisplayPlaneCapabilities2KHR
- VkDisplayPlaneInfo2KHR
- VkDisplayPlaneProperties2KHR
- VkDisplayProperties2KHR

New Enum Constants

- `VK_KHR_GET_DISPLAY_PROPERTIES_2_EXTENSION_NAME`
- `VK_KHR_GET_DISPLAY_PROPERTIES_2_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DISPLAY_MODE_PROPERTIES_2_KHR
  - VK_STRUCTURE_TYPE_DISPLAY_PLANE_CAPABILITIES_2_KHR
  - VK_STRUCTURE_TYPE_DISPLAY_PLANE_INFO_2_KHR
  - VK_STRUCTURE_TYPE_DISPLAY_PLANE_PROPERTIES_2_KHR
  - VK_STRUCTURE_TYPE_DISPLAY_PROPERTIES_2_KHR

Issues

1) What should this extension be named?

RESOLVED: `VK_KHR_get_display_properties2`. Other alternatives:

- `VK_KHR_display2`
- One extension, combined with `VK_KHR_surface_capabilites2`.

2) Should extensible input structs be added for these new functions:

RESOLVED:

- vkGetPhysicalDeviceDisplayProperties2KHR: No. The only current input is a VkPhysicalDevice. Other inputs would not make sense.
- vkGetPhysicalDeviceDisplayPlaneProperties2KHR: No. The only current input is a VkPhysicalDevice. Other inputs would not make sense.
- vkGetDisplayModeProperties2KHR: No. The only current inputs are a VkPhysicalDevice and a VkDisplayModeKHR. Other inputs would not make sense.

3) Should additional display query functions be extended?

RESOLVED:

- vkGetDisplayPlaneSupportedDisplaysKHR: No. Extensions should instead extend vkGetDisplayPlaneCapabilitiesKHR().

Version History

- Revision 1, 2017-02-21 (James Jones)
  - Initial draft.

VK_KHR_get_surface_capabilities2

Name String

`VK_KHR_get_surface_capabilities2`

Extension Type

Instance extension

Registered Extension Number

120

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_surface

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2017-02-27

IP Status

No known IP claims.

Contributors

- Ian Elliott, Google
- James Jones, NVIDIA
- Alon Or-bach, Samsung

Description

This extension provides new queries for device surface capabilities that can be easily extended by other extensions, without introducing any further queries. This extension can be considered the `VK_KHR_surface` equivalent of the `VK_KHR_get_physical_device_properties2` extension.

New Commands

- vkGetPhysicalDeviceSurfaceCapabilities2KHR
- vkGetPhysicalDeviceSurfaceFormats2KHR

New Structures

- VkPhysicalDeviceSurfaceInfo2KHR
- VkSurfaceCapabilities2KHR
- VkSurfaceFormat2KHR

New Enum Constants

- `VK_KHR_GET_SURFACE_CAPABILITIES_2_EXTENSION_NAME`
- `VK_KHR_GET_SURFACE_CAPABILITIES_2_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SURFACE_INFO_2_KHR
  - VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_2_KHR
  - VK_STRUCTURE_TYPE_SURFACE_FORMAT_2_KHR

Issues

1) What should this extension be named?

RESOLVED: `VK_KHR_get_surface_capabilities2`. Other alternatives:

- `VK_KHR_surface2`
- One extension, combining a separate display-specific query extension.

2) Should additional WSI query functions be extended?

RESOLVED:

- vkGetPhysicalDeviceSurfaceCapabilitiesKHR: Yes. The need for this motivated the extension.
- vkGetPhysicalDeviceSurfaceSupportKHR: No. Currently only has boolean output. Extensions should instead extend vkGetPhysicalDeviceSurfaceCapabilities2KHR.
- vkGetPhysicalDeviceSurfaceFormatsKHR: Yes.
- vkGetPhysicalDeviceSurfacePresentModesKHR: No. Recent discussion concluded this introduced too much variability for applications to deal with. Extensions should instead extend vkGetPhysicalDeviceSurfaceCapabilities2KHR.
- vkGetPhysicalDeviceXlibPresentationSupportKHR: Not in this extension.
- vkGetPhysicalDeviceXcbPresentationSupportKHR: Not in this extension.
- vkGetPhysicalDeviceWaylandPresentationSupportKHR: Not in this extension.
- vkGetPhysicalDeviceWin32PresentationSupportKHR: Not in this extension.

Version History

- Revision 1, 2017-02-27 (James Jones)
  - Initial draft.

VK_KHR_incremental_present

Name String

`VK_KHR_incremental_present`

Extension Type

Device extension

Registered Extension Number

85

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_swapchain

Contact

- Ian Elliott [GitHub]ianelliottus

Other Extension Metadata

Last Modified Date

2016-11-02

IP Status

No known IP claims.

Contributors

- Ian Elliott, Google
- Jesse Hall, Google
- Alon Or-bach, Samsung
- James Jones, NVIDIA
- Daniel Rakos, AMD
- Ray Smith, ARM
- Mika Isojarvi, Google
- Jeff Juliano, NVIDIA
- Jeff Bolz, NVIDIA

Description

This device extension extends vkQueuePresentKHR, from the `VK_KHR_swapchain` extension, allowing

an application to specify a list of rectangular, modified regions of each image to present. This should be used in situations where an application is only changing a small portion of the presentable images within a swapchain, since it enables the presentation engine to avoid wasting time presenting parts of the surface that have not changed.

This extension is leveraged from the `EGL_KHR_swap_buffers_with_damage` extension.

New Structures

- VkPresentRegionKHR
- VkRectLayerKHR
- Extending VkPresentInfoKHR:
  - VkPresentRegionsKHR

New Enum Constants

- `VK_KHR_INCREMENTAL_PRESENT_EXTENSION_NAME`
- `VK_KHR_INCREMENTAL_PRESENT_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PRESENT_REGIONS_KHR

Issues

1) How should we handle steroescopic-3D swapchains? We need to add a layer for each rectangle. One approach is to create another structure containing the VkRect2D plus layer, and have VkPresentRegionsKHR point to an array of that struct. Another approach is to have two parallel arrays, `pRectangles` and `pLayers`, where `pRectangles`[i] and `pLayers`[i] must be used together. Which approach should we use, and if the array of a new structure, what should that be called?

RESOLVED: Create a new structure, which is a VkRect2D plus a layer, and will be called VkRectLayerKHR.

2) Where is the origin of the VkRectLayerKHR?

RESOLVED: The upper left corner of the presentable image(s) of the swapchain, per the definition of framebuffer coordinates.

3) Does the rectangular region, VkRectLayerKHR, specify pixels of the swapchain's image(s), or of the surface?

RESOLVED: Of the image(s). Some presentation engines may scale the pixels of a swapchain's image(s) to the size of the surface. The size of the swapchain's image(s) will be consistent, where the size of the surface may vary over time.

4) What if all of the rectangles for a given swapchain contain a width and/or height of zero?

RESOLVED: The application is indicating that no pixels changed since the last present. The presentation engine may use such a hint and not update any pixels for the swapchain. However, all

other semantics of vkQueuePresentKHR must still be honored, including waiting for semaphores to signal.

5) When the swapchain is created with `VkSwapchainCreateInfoKHR`::`preTransform` set to a value other than VK_SURFACE_TRANSFORM_IDENTITY_BIT_KHR, should the rectangular region, VkRectLayerKHR, be transformed to align with the `preTransform`?

RESOLVED: No. The rectangular region in VkRectLayerKHR should not be transformed. As such, it may not align with the extents of the swapchain's image(s). It is the responsibility of the presentation engine to transform the rectangular region. This matches the behavior of the Android presentation engine, which set the precedent.

Version History

- Revision 1, 2016-11-02 (Ian Elliott)
  - Internal revisions
- Revision 2, 2021-03-18 (Ian Elliott)
  - Clarified alignment of rectangles for presentation engines that support transformed swapchains.

VK_KHR_internally_synchronized_queues

Name String

`VK_KHR_internally_synchronized_queues`

Extension Type

Device extension

Registered Extension Number

505

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.1

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Extension Proposal

VK_KHR_internally_synchronized_queues

Other Extension Metadata

Last Modified Date

2025-02-04

Contributors

- Shahbaz Youssefi, Google
- Daniel Rakos, RasterGrid
- Jeff Bolz, Nvidia
- Hans-Kristian Arntzen, Valve

Description

VK_KHR_internally_synchronized_queues allows queues to opt into being internally synchronized via the VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR flag.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceInternallySynchronizedQueuesFeaturesKHR

New Enum Constants

- `VK_KHR_INTERNALLY_SYNCHRONIZED_QUEUES_EXTENSION_NAME`
- `VK_KHR_INTERNALLY_SYNCHRONIZED_QUEUES_SPEC_VERSION`
- Extending VkDeviceQueueCreateFlagBits:
  - VK_DEVICE_QUEUE_CREATE_INTERNALLY_SYNCHRONIZED_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INTERNALLY_SYNCHRONIZED_QUEUES_FEATURES_KHR

Issues

None.

Version History

- Revision 0, 2025-01-07 (Daniel Rakos)
  - Initial proposal
- Revision 1, 2025-02-04 (Shahbaz Youssefi)
  - Initial revision

VK_KHR_maintenance10

Name String

VK_KHR_maintenance10

Extension Type

Device extension

Registered Extension Number

631

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_VERSION_1_3
- Interacts with VK_VERSION_1_4
- Interacts with VK_KHR_copy_commands2
- Interacts with VK_KHR_dynamic_rendering
- Interacts with VK_KHR_dynamic_rendering_local_read
- Interacts with VK_KHR_format_feature_flags2

Contact

- Mike Blumenkrantz [GitHub]zmike

Extension Proposal

VK_KHR_maintenance10

Other Extension Metadata

Last Modified Date

2025-05-13

Interactions and External Dependencies

- This extension interacts with VK_KHR_format_feature_flags2
- This extension interacts with VK_EXT_extended_dynamic_state3
- This extension interacts with VK_KHR_dynamic_rendering_local_read
- This extension interacts with VK_KHR_depth_stencil_resolve

Contributors

- Mike Blumenkrantz, Valve
- Piers Daniell, NVIDIA
- Hans-Kristian Arntzen, Valve

Description

VK_KHR_maintenance10 adds a collection of minor features, none of which would warrant an entire extension of their own.

The new features are as follows:

- New image format feature bits that indicate support for copying depth or stencil aspects using non-graphics queue families
- If vkCmdSetSampleMaskEXT is called with `pSampleMask` set to `NULL`, it is treated as if the mask has all bits set to `1`.
- Add vkCmdEndRendering2KHR as an extensible version of vkCmdEndRendering
- Add input attachment information to dynamic rendering
- Require that vertex inputs follow sRGB encoding when those formats are used, instead of being underspecified.
- Add a query to determine if sRGB images are resolved in nonlinear or linear space by default
- Add an optional feature to allow applications to override the default sRGB resolve behavior
- Add resolve mode and depth-stencil resolve support to `vkCmdResolveImage2` to bring it in-line with render pass attachment resolves

New Commands

- vkCmdEndRendering2KHR

New Structures

- VkRenderingEndInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMaintenance10FeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceMaintenance10PropertiesKHR
- Extending VkRenderingAttachmentInfo:
  - VkRenderingAttachmentFlagsInfoKHR
- Extending VkResolveImageInfo2:
  - VkResolveImageModeInfoKHR

New Enums

- VkRenderingAttachmentFlagBitsKHR
- VkResolveImageFlagBitsKHR

New Bitmasks

- VkRenderingAttachmentFlagsKHR
- VkResolveImageFlagsKHR

New Enum Constants

- `VK_KHR_MAINTENANCE_10_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE_10_SPEC_VERSION`
- Extending VkAttachmentDescriptionFlagBits:
  - VK_ATTACHMENT_DESCRIPTION_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR
  - VK_ATTACHMENT_DESCRIPTION_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_10_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_10_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_FLAGS_INFO_KHR
  - VK_STRUCTURE_TYPE_RENDERING_END_INFO_KHR
  - VK_STRUCTURE_TYPE_RESOLVE_IMAGE_MODE_INFO_KHR

If Vulkan Version 1.4 or VK_KHR_dynamic_rendering_local_read and Vulkan Version 1.3 or VK_KHR_dynamic_rendering is supported:

- Extending VkRenderingAttachmentFlagBitsKHR:
  - VK_RENDERING_ATTACHMENT_INPUT_ATTACHMENT_FEEDBACK_BIT_KHR
- Extending VkRenderingFlagBits:
  - VK_RENDERING_LOCAL_READ_CONCURRENT_ACCESS_CONTROL_BIT_KHR

If VK_KHR_format_feature_flags2 or Vulkan Version 1.3 is supported:

- Extending VkFormatFeatureFlagBits2:
  - VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_COMPUTE_QUEUE_BIT_KHR
  - VK_FORMAT_FEATURE_2_DEPTH_COPY_ON_TRANSFER_QUEUE_BIT_KHR
  - VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_COMPUTE_QUEUE_BIT_KHR
  - VK_FORMAT_FEATURE_2_STENCIL_COPY_ON_TRANSFER_QUEUE_BIT_KHR

If Vulkan Version 1.3 or VK_KHR_copy_commands2 is supported:

- Extending VkResolveImageFlagBitsKHR:

    - VK_RESOLVE_IMAGE_ENABLE_TRANSFER_FUNCTION_BIT_KHR
    - VK_RESOLVE_IMAGE_SKIP_TRANSFER_FUNCTION_BIT_KHR

If Vulkan Version 1.3 or VK_KHR_dynamic_rendering is supported:

- Extending VkRenderingAttachmentFlagBitsKHR:
    - VK_RENDERING_ATTACHMENT_RESOLVE_ENABLE_TRANSFER_FUNCTION_BIT_KHR
    - VK_RENDERING_ATTACHMENT_RESOLVE_SKIP_TRANSFER_FUNCTION_BIT_KHR

Issues

None.

Version History

- Revision 1, 2025-05-13 (Mike Blumenkrantz)
    - Initial revision

VK_KHR_maintenance11

Name String

VK_KHR_maintenance11

Extension Type

Device extension

Registered Extension Number

658

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_EXT_mesh_shader
- Interacts with VK_EXT_shader_object
- Interacts with VK_KHR_extended_flags
- Interacts with VK_NV_mesh_shader

Contact

- Mike Blumenkrantz [GitHub]zmike

Extension Proposal

VK_KHR_maintenance11

Other Extension Metadata

Last Modified Date

2025-09-09

Interactions and External Dependencies

Contributors

- Mike Blumenkrantz, Valve
- Piers Daniell, NVIDIA
- Hans-Kristian Arntzen, Valve
- Caterina Shablia, Collabora

Description

VK_KHR_maintenance11 adds a collection of minor features, none of which would warrant an entire extension of their own.

The new features are as follows:

- Add D3D compatibility for mismatch between `Arrayed` in shaders and the arrayness of the underlying descriptor when the descriptor contains a single array layer
- Clarify the pipeline depth clipping state when the pipeline is created without VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT being set and the VkPipelineRasterizationDepthClipStateCreateInfoEXT struct is not present
- Add VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR to enable shader object functionality to mimic VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT used for graphics pipeline libraries, including a new pipeline layout creation flag VK_PIPELINE_LAYOUT_CREATE_NO_TASK_SHADER_BIT_KHR to ensure pipeline layouts used with shader objects also created with VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT to be compatible
- Allow `queueFamilyIndexCount` of 1 in VkBufferCreateInfo, VkPhysicalDeviceImageDrmFormatModifierInfoEXT, and VkImageCreateInfo, when `sharingMode` is VK_SHARING_MODE_CONCURRENT.
- Require `minImageTransferGranularity` to be (1,1,1) even on transfer-only queues and add `optimalImageTransferGranularity` queue family property to communicate the performance bump for copies not aligned to the optimal granularity.
- When copying between a buffer and an image on a transfer-only queue, do not require `bufferOffset` to be a multiple of 4.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMaintenance11FeaturesKHR
- Extending VkQueueFamilyProperties2:
  - VkQueueFamilyOptimalImageTransferGranularityPropertiesKHR

New Enum Constants

- `VK_KHR_MAINTENANCE_11_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE_11_SPEC_VERSION`
- Extending VkImageCreateFlagBits:
  - VK_IMAGE_CREATE_ALIAS_SINGLE_LAYER_DESCRIPTOR_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_11_FEATURES_KHR
  - VK_STRUCTURE_TYPE_QUEUE_FAMILY_OPTIMAL_IMAGE_TRANSFER_GRANULARITY_PROPERTIES_KHR

If VK_EXT_shader_object is supported:

- Extending VkPipelineLayoutCreateFlagBits:
  - VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT
- Extending VkShaderCreateFlagBitsEXT:
  - VK_SHADER_CREATE_INDEPENDENT_SETS_BIT_KHR

If VK_EXT_shader_object and VK_EXT_mesh_shader or `VK_NV_mesh_shader` is supported:

- Extending VkPipelineLayoutCreateFlagBits:
  - VK_PIPELINE_LAYOUT_CREATE_NO_TASK_SHADER_BIT_KHR

If VK_KHR_extended_flags is supported:

- Extending VkImageCreateFlagBits2KHR:
  - VK_IMAGE_CREATE_2_ALIAS_SINGLE_LAYER_DESCRIPTOR_BIT_KHR

Issues

None.

Version History

- Revision 1, 2025-09-09 (Mike Blumenkrantz)
  - Initial revision

VK_KHR_maintenance7

Name String

VK_KHR_maintenance7

Extension Type

Device extension

Registered Extension Number

563

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.1

Contact

- Mike Blumenkrantz [GitHub]zmike

Extension Proposal

VK_KHR_maintenance7

Other Extension Metadata

Last Modified Date

2024-01-30

Interactions and External Dependencies

Contributors

- Mike Blumenkrantz, Valve
- Hans-Kristian Arntzen, Valve
- Pan Gao, Huawei
- Tobias Hector, AMD
- Jon Leech, Khronos
- Daniel Story, Nintendo
- Shahbaz Youssefi, Google
- Yiwei Zhang, Google
- Matthew Netsch, Qualcomm

Description

VK_KHR_maintenance7 adds a collection of minor features, none of which would warrant an entire extension of their own.

The proposed new features are as follows:

- Add a property query to determine if a framebuffer writes to depth or stencil aspect does not trigger a write access in the sibling aspect. For example, this allows sampling stencil aspect as a texture while rendering to the sibling depth attachment and vice-versa given appropriate image layouts.
- Add a way to query information regarding the underlying devices in environments where the Vulkan implementation is provided through layered implementations. For example, running on Mesa/Venus, driver ID is returned as VK_DRIVER_ID_MESA_VENUS, but it can be necessary to know what the real driver under the hood is. The new VkPhysicalDeviceLayeredApiPropertiesKHR structure can be used to gather information regarding layers underneath the top-level physical device.
- Promote VK_RENDERING_CONTENTS_INLINE_BIT_EXT and VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_EXT to KHR
- Add a limit to report the maximum total count of dynamic uniform buffers and dynamic storage buffers that can be included in a pipeline layout.
- Require that for an unsigned integer query, the 32-bit result value **must** be equal to the 32 least significant bits of the equivalent 64-bit result value.
- Add query for robust access support when using fragment shading rate attachments

New Structures

- VkPhysicalDeviceLayeredApiPropertiesKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMaintenance7FeaturesKHR
- Extending VkPhysicalDeviceLayeredApiPropertiesKHR:
  - VkPhysicalDeviceLayeredApiVulkanPropertiesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceLayeredApiPropertiesListKHR
  - VkPhysicalDeviceMaintenance7PropertiesKHR

New Enums

- VkPhysicalDeviceLayeredApiKHR

New Enum Constants

- `VK_KHR_MAINTENANCE_7_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE_7_SPEC_VERSION`

- Extending VkRenderingFlagBits:
  - VK_RENDERING_CONTENTS_INLINE_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_PROPERTIES_LIST_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_VULKAN_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_7_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_7_PROPERTIES_KHR
- Extending VkSubpassContents:
  - VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR

Issues

None.

Version History

- Revision 1, 2024-01-30 (Jon Leech)
  - Initial revision

VK_KHR_maintenance8

Name String

VK_KHR_maintenance8

Extension Type

Device extension

Registered Extension Number

575

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.1

Contact

- Mike Blumenkrantz [GitHub]zmike

Extension Proposal

VK_KHR_maintenance8

Other Extension Metadata

Last Modified Date

2025-01-07

Interactions and External Dependencies

Contributors

- Jon Leech, Khronos
- Mike Blumenkrantz, Valve
- Spencer Fricke, LunarG
- Jan-Harald Fredriksen, ARM
- Piers Daniell, NVIDIA
- Matthew Netsch, Qualcomm
- Ricardo Garcia, Igalia
- Lionel Landwerlin, Intel
- Rick Hammerstone, Qualcomm
- Daniel Story, Nintendo
- Hans-Kristian Arntzen, Valve

- Caterina Shablia, Collabora
- Georg Lehmann, Valve
- Shahbaz Youssefi, Google
- Tobias Hector, AMD

Description

VK_KHR_maintenance8 adds a collection of minor features, none of which would warrant an entire extension of their own.

The new features are as follows:

- Allow copies between depth/stencil and “matching” color attachments
- Allow `dstCache` in `vkMergePipelineCaches` to be implicitly synchronized.
- Require src/dst sync scopes to work when doing queue family ownership transfers
- Support `Offset` (as an alternative to `ConstOffset`) image operand in texture sampling and fetch operations
- Use the SPIR-V definition of `OpSRem` and `OpSMod`, making these operations produce well-defined results for negative operands
- Loosen layer restrictions when blitting from 3D images to other image types
- Add space for an additional 64 access flags for use with VkMemoryBarrier2, VkBufferMemoryBarrier2, and VkImageMemoryBarrier2

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMaintenance8FeaturesKHR
- Extending VkSubpassDependency2, VkBufferMemoryBarrier2, VkImageMemoryBarrier2, VkMemoryBarrier2, VkMemoryRangeBarrierKHR:
  - VkMemoryBarrierAccessFlags3KHR

New Enums

- VkAccessFlagBits3KHR
- VkPipelineCacheCreateFlagBits

New Bitmasks

- VkAccessFlags3KHR

New Enum Constants

- `VK_KHR_MAINTENANCE_8_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE_8_SPEC_VERSION`

- Extending VkDependencyFlagBits:
  - VK_DEPENDENCY_QUEUE_FAMILY_OWNERSHIP_TRANSFER_USE_ALL_STAGES_BIT_KHR
- Extending VkPipelineCacheCreateFlagBits:
  - VK_PIPELINE_CACHE_CREATE_INTERNALLY_SYNCHRONIZED_MERGE_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_MEMORY_BARRIER_ACCESS_FLAGS_3_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_8_FEATURES_KHR

Issues

None.

Version History

- Revision 1, 2024-06-20 (Jon Leech)
  - Initial revision

VK_KHR_maintenance9

Name String

VK_KHR_maintenance9

Extension Type

Device extension

Registered Extension Number

585

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Mike Blumenkrantz [GitHub]zmike

Extension Proposal

VK_KHR_maintenance9

Other Extension Metadata

Last Modified Date

2025-05-29

Interactions and External Dependencies

Contributors

- Mike Blumenkrantz, Valve
- Shahbaz Youssefi, Google
- Hans-Kristian Arntzen, Valve
- Piers Daniell, NVIDIA
- Daniel Story, Nintendo
- Jeff Bolz, NVIDIA

Description

VK_KHR_maintenance9 adds a collection of minor features, none of which would warrant an entire extension of their own.

The new features are as follows:

- Support VkDevice with no queues. These can be used as effectively an offline compiler to prepopulate pipeline caches, without expensive queue creation or internal memory allocations.
- Allow `vkCmdSetEvent2` to not provide a dependency, providing `vkCmdSetEvent`-style usage using enums from `VK_KHR_synchronization2`
- Add a VkQueryPoolCreateFlagBits::VK_QUERY_POOL_CREATE_RESET_BIT_KHR flag to create a query pool with all queries initialized to the reset state.
- Allow any integer bit width for specific bit-wise operations.
- Add a property to enable sparse support with `VK_EXT_image_2d_view_of_3d`.
- Add a property to indicate the implementation will return (0,0,0,0) or (0,0,0,1) to vertex shaders that read unassigned attributes.
- The effects of image memory barriers and image layout transitions on 3D images created with VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT are scoped to the slices specified by the user-provided VkImageSubresourceRange.
- Queue family ownership transfers are no longer required for buffers and linear images, and a new physical device queue family property is exposed to indicate whether queue family ownership transfers are required for optimal images.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMaintenance9FeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceMaintenance9PropertiesKHR
- Extending VkQueueFamilyProperties2:
  - VkQueueFamilyOwnershipTransferPropertiesKHR

New Enums

- VkDefaultVertexAttributeValueKHR

New Enum Constants

- `VK_KHR_MAINTENANCE_9_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE_9_SPEC_VERSION`
- Extending VkDependencyFlagBits:
  - VK_DEPENDENCY_ASYMMETRIC_EVENT_BIT_KHR
- Extending VkQueryPoolCreateFlagBits:
  - VK_QUERY_POOL_CREATE_RESET_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_9_FEATURES_KHR

- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_9_PROPERTIES_KHR
- VK_STRUCTURE_TYPE_QUEUE_FAMILY_OWNERSHIP_TRANSFER_PROPERTIES_KHR

Issues

None.

Version History

- Revision 1, 2025-05-29 (Contributors)
  - Internal revisions

VK_KHR_opacity_micromap

Name String

VK_KHR_opacity_micromap

Extension Type

Device extension

Registered Extension Number

624

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_acceleration_structure
and
VK_KHR_device_address_commands

API Interactions

- Interacts with VK_VERSION_1_4
- Interacts with VK_EXT_shader_object
- Interacts with VK_KHR_extended_flags
- Interacts with VK_KHR_maintenance5

SPIR-V Dependencies

- SPV_EXT_opacity_micromap
- SPV_KHR_opacity_micromap

Contact

- Matthew Netsch [GitHub]mnetsch

Extension Proposal

VK_KHR_opacity_micromap

Other Extension Metadata

Last Modified Date

2026-05-08

Interactions and External Dependencies

- This extension provides API support for GLSL_EXT_opacity_micromap
- This extension provides API support for GLSL_EXT_opacity_micromap_ray_query_mode

Contributors

- Matthew Netsch, Qualcomm Technologies, Inc
- Aleksandra Krstic, Qualcomm Technologies, Inc
- Eric Werness, NVIDIA
- Daniel Koch, NVIDIA
- Stu Smith, AMD
- Sven Woop, Intel
- Anton Berko, MediaTek
- Radoslaw Drabinski, Intel
- Simon Fenney, Imagination Technologies
- Andrew Garrard, Imagination Technologies
- Dae Kim, Imagination Technologies
- Fred Saucedo, Qualcomm Technologies, Inc
- Ramesh babu Admimula, Qualcomm Technologies, Inc
- Zedian Zhang, Qualcomm Technologies, Inc
- Hans-Kristian Arntzen, Valve
- Vikram Kushwaha, NVIDIA
- Spencer Fricke, LunarG
- Revanth Ponna, Qualcomm Technologies, Inc
- Contributors to `VK_EXT_opacity_micromap`

Description

When adding transparency to a ray traced scene, an application can choose between further tessellating the geometry or using an any-hit shader to allow the ray through specific parts of the geometry. These options have the downside of either significantly increasing memory consumption or adding runtime overhead to run shader code in the middle of traversal, respectively.

This extension adds the ability to add an *opacity micromap* to geometry when building an acceleration structure. The opacity micromap compactly encodes opacity information which can be read by the implementation to mark parts of triangles as opaque or transparent. The format is externally visible to allow the application to compress its internal geometry and surface representations into the compressed format ahead of time. The compressed format subdivides each triangle into a set of subtriangles, each of which can be assigned either two or four opacity values. These opacity values can control if a ray hitting that subtriangle is treated as an opaque hit, complete miss, or possible hit, depending on the controls described in Ray Opacity Micromap.

This extension provides:

- the ability to create micromaps as VkAccelerationStructureKHR structures
- the ability to build, copy, and query micromaps with the acceleration structure functions, and

- a structure to extend VkAccelerationStructureGeometryTrianglesDataKHR to attach a micromap to the geometry of the acceleration structure.

Traversing an acceleration structure that contains opacity micromap data requires the shader or pipeline to declare that it does so. Traversal performed by pipeline trace ray instructions requires the pipeline to have been created with VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR. Traversal performed by `OpRayQueryProceedKHR` instead requires the shader to declare the `OpacityMicromapIdKHR` execution mode, added by `SPV_KHR_opacity_micromap`, with its specialization-constant operand equal to VK_TRUE. See Ray Opacity Micromap for details.

New Structures

- VkMicromapTriangleKHR
- VkMicromapUsageKHR
- Extending VkAccelerationStructureGeometryKHR:
  - VkAccelerationStructureGeometryMicromapDataKHR
- Extending VkAccelerationStructureGeometryTrianglesDataKHR:
  - VkAccelerationStructureTrianglesOpacityMicromapKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceOpacityMicromapFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceOpacityMicromapPropertiesKHR

New Enums

- VkAccelerationStructureSerializedBlockTypeKHR
- VkOpacityMicromapFormatKHR
- VkOpacityMicromapSpecialIndexKHR

New Enum Constants

- `VK_KHR_OPACITY_MICROMAP_EXTENSION_NAME`
- `VK_KHR_OPACITY_MICROMAP_SPEC_VERSION`
- Extending VkAccelerationStructureTypeKHR:
  - VK_ACCELERATION_STRUCTURE_TYPE_OPACITY_MICROMAP_KHR
- Extending VkBuildAccelerationStructureFlagBitsKHR:
  - VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_BIT_KHR
  - VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_KHR
  - VK_BUILD_ACCELERATION_STRUCTURE_MICROMAP_LOSSY_BIT_KHR
- Extending VkGeometryInstanceFlagBitsKHR:

    - VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_BIT_KHR
    - VK_GEOMETRY_INSTANCE_FORCE_OPACITY_MICROMAP_2_STATE_BIT_KHR
- Extending VkGeometryTypeKHR:
    - VK_GEOMETRY_TYPE_MICROMAP_KHR
- Extending VkPipelineCreateFlagBits:
    - VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_MICROMAP_DATA_KHR
    - VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_TRIANGLES_OPACITY_MICROMAP_KHR
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_FEATURES_KHR
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_PROPERTIES_KHR

If VK_EXT_shader_object is supported:

- Extending VkShaderCreateFlagBitsEXT:
    - VK_SHADER_CREATE_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_EXT

If Vulkan Version 1.4 or VK_KHR_extended_flags or VK_KHR_maintenance5 is supported:

- Extending VkPipelineCreateFlagBits2:
    - VK_PIPELINE_CREATE_2_OPACITY_MICROMAP_DISALLOW_MIXED_SPECIAL_INDEX_BIT_KHR
    - VK_PIPELINE_CREATE_2_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR

Reference Code

The following code illustrates an algorithm that converts the barycentric coordinates inside a triangle into an OMM array index:

```
uint32_t BarycentricsToSpaceFillingCurveIndex(float u, float v, uint32_t level)
{
    u = clamp(u, 0.0f, 1.0f);
    v = clamp(v, 0.0f, 1.0f);

    uint32_t iu, iv, iw;

    // Quantize barycentric coordinates
    float fu = u * (1u << level);
    float fv = v * (1u << level);

    iu = (uint32_t)fu;
    iv = (uint32_t)fv;

    float uf = fu - float(iu);
    float vf = fv - float(iv);
```

```
    if (iu >= (1u << level)) iu = (1u << level) - 1u;
    if (iv >= (1u << level)) iv = (1u << level) - 1u;

    uint32_t iuv = iu + iv;

    if (iuv >= (1u << level))
        iu -= iuv - (1u << level) + 1u;

    iw = ~(iu + iv);

    if (uf + vf >= 1.0f && iuv < (1u << level) - 1u) --iw;

    uint32_t b0 = ~(iu ^ iw);
    b0 &= ((1u << level) - 1u);
    uint32_t t = (iu ^ iv) & b0;

    uint32_t f = t;
    f ^= f >> 1u;
    f ^= f >> 2u;
    f ^= f >> 4u;
    f ^= f >> 8u;
    uint32_t b1 = ((f ^ iu) & ~b0) | t;

    // Interleave bits
    b0 = (b0 | (b0 << 8u)) & 0x00ff00ffu;
    b0 = (b0 | (b0 << 4u)) & 0x0f0f0f0fu;
    b0 = (b0 | (b0 << 2u)) & 0x33333333u;
    b0 = (b0 | (b0 << 1u)) & 0x55555555u;
    b1 = (b1 | (b1 << 8u)) & 0x00ff00ffu;
    b1 = (b1 | (b1 << 4u)) & 0x0f0f0f0fu;
    b1 = (b1 | (b1 << 2u)) & 0x33333333u;
    b1 = (b1 | (b1 << 1u)) & 0x55555555u;

    return b0 | (b1 << 1u);
}
```

Issues

(1) Is the build actually similar to an acceleration structure build?

- Resolved: The build should be much lighter-weight than an acceleration structure build

(2) Why does VkMicromapUsageKHR not have type/pNext?

- Resolved: There can be a very large number of these structures, so doubling the size of these can be significant memory consumption. Also, an application may be loading these directly from a file which is more compatible with it being a flat structure. The including structures are extensible and are probably a more suitable place to add extensibility.

(3) Why is there a SPIR-V extension?

- Resolved: There is a ray flag and an execution mode. To be consistent with how the existing ray tracing extensions work that these needs its own extension.

(4) Should there be indirect micromap build?

- Resolved: Not for now. There is more in-depth usage metadata required and it seems less likely that something like a GPU culling system would need to change the counts for a micromap.

(5) Should the feature struct be aliased with VkPhysicalDeviceOpacityMicromapFeaturesEXT?

- Resolved: No. This extension is not an exact promotion of `VK_EXT_opacity_micromap` and provides significantly different functionality.

(6) Should micromaps API be similar to the `VK_EXT_opacity_micromap`?

`VK_EXT_opacity_micromap` introduced almost an identical set of functionality for micromaps as acceleration structures. Should this promotion fold in micromaps as an acceleration structure type?

- Resolved: Yes. While this is significant API breakage from the EXT, it is a better design choice going forward and can eliminate significant API surface area if promoted in the future.

(7) Should micromaps support host commands?

- Resolved: No. Host commands are deprecated and not widely supported.

(8) Should vkCreateAccelerationStructureKHR be used to create micromaps?

This interface that uses VkBuffer to back micromaps is deprecated, but is still available for acceleration structure object creation.

- Resolved: No. Make a new entry point that uses a device address to back micromaps and other acceleration structures instead of a buffer. Eliminate the ability to provide a separate capture/replay address as well. Top-level acceleration structures **must** reference the device address that backs the bottom-level acceleration structures and implementations **must** not expose an indirect handle.

(9) Should VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_BIT_EXT be promoted from `VK_EXT_opacity_micromap` to this extension?

VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_BIT_EXT distinguished between replacing a micromap with one of a different shape (VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_EXT) and replacing one where the shape is identical and only the opacity values have changed (DATA_UPDATE). The DATA_UPDATE flag was correlated with the discardable micromap feature, where an implementation might embed micromap state directly in the acceleration structure and exploit the tighter constraint to avoid a full rebuild. However, the spec never required DATA_UPDATE to be restricted to discardable micromaps, and it was already legal to treat DATA_UPDATE identically to UPDATE.

- Resolved: No. The discardable micromap feature is not promoted to this extension, removing

the primary motivation for DATA_UPDATE. No implementation identified a meaningful optimization from the stricter constraint beyond what UPDATE already provides, and no CTS coverage existed for either opacity micromap update flag in the EXT. Removing it is a pure API simplification with no functional loss. Applications must still perform a BLAS build update when opacity micromap data changes, even if only values and not structure have changed.

Version History

- Revision 1, 2026-05-08 (Matthew Netsch)
  - Initial draft

VK_KHR_performance_query

Name String

`VK_KHR_performance_query`

Extension Type

Device extension

Registered Extension Number

117

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Special Use

- Developer tools

Contact

- Alon Or-bach [GitHub]alonorbach

Other Extension Metadata

Last Modified Date

2019-10-08

IP Status

No known IP claims.

Contributors

- Jesse Barker, Unity Technologies
- Kenneth Benzie, Codeplay
- Jan-Harald Fredriksen, ARM
- Jeff Leger, Qualcomm
- Jesse Hall, Google
- Tobias Hector, AMD
- Neil Henning, Codeplay
- Baldur Karlsson
- Lionel Landwerlin, Intel

- Peter Lohrmann, AMD
- Alon Or-bach, Samsung
- Daniel Rakos, AMD
- Niklas Smedberg, Unity Technologies
- Igor Ostrowski, Intel

Description

The `VK_KHR_performance_query` extension adds a mechanism to allow querying of performance counters for use in applications and by profiling tools.

Each queue family **may** expose counters that **can** be enabled on a queue of that family. We extend VkQueryType to add a new query type for performance queries, and chain a structure on VkQueryPoolCreateInfo to specify the performance queries to enable.

New Commands

- vkAcquireProfilingLockKHR
- vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR
- vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR
- vkReleaseProfilingLockKHR

New Structures

- VkAcquireProfilingLockInfoKHR
- VkPerformanceCounterDescriptionKHR
- VkPerformanceCounterKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePerformanceQueryFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDevicePerformanceQueryPropertiesKHR
- Extending VkQueryPoolCreateInfo:
  - VkQueryPoolPerformanceCreateInfoKHR
- Extending VkSubmitInfo, VkSubmitInfo2:
  - VkPerformanceQuerySubmitInfoKHR

New Unions

- VkPerformanceCounterResultKHR

New Enums

- VkAcquireProfilingLockFlagBitsKHR

- VkPerformanceCounterDescriptionFlagBitsKHR
- VkPerformanceCounterScopeKHR
- VkPerformanceCounterStorageKHR
- VkPerformanceCounterUnitKHR

New Bitmasks

- VkAcquireProfilingLockFlagsKHR
- VkPerformanceCounterDescriptionFlagsKHR

New Enum Constants

- `VK_KHR_PERFORMANCE_QUERY_EXTENSION_NAME`
- `VK_KHR_PERFORMANCE_QUERY_SPEC_VERSION`
- Extending VkQueryType:
  - VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_ACQUIRE_PROFILING_LOCK_INFO_KHR
  - VK_STRUCTURE_TYPE_PERFORMANCE_COUNTER_DESCRIPTION_KHR
  - VK_STRUCTURE_TYPE_PERFORMANCE_COUNTER_KHR
  - VK_STRUCTURE_TYPE_PERFORMANCE_QUERY_SUBMIT_INFO_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PERFORMANCE_QUERY_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PERFORMANCE_QUERY_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_QUERY_POOL_PERFORMANCE_CREATE_INFO_KHR

Issues

1) Should this extension include a mechanism to begin a query in command buffer *A* and end the query in command buffer *B*?

RESOLVED No - queries are tied to command buffer creation and thus have to be encapsulated within a single command buffer.

2) Should this extension include a mechanism to begin and end queries globally on the queue, not using the existing command buffer commands?

RESOLVED No - for the same reasoning as the resolution of 1).

3) Should this extension expose counters that require multiple passes?

RESOLVED Yes - users should re-submit a command buffer with the same commands in it multiple times, specifying the pass to count as the query parameter in VkPerformanceQuerySubmitInfoKHR.

4) How to handle counters across parallel workloads?

RESOLVED In the spirit of Vulkan, a counter description flag VK_PERFORMANCE_COUNTER_DESCRIPTION_CONCURRENTLY_IMPACTED_BIT_KHR denotes that the accuracy of a counter result is affected by parallel workloads.

5) How to handle secondary command buffers?

RESOLVED Secondary command buffers inherit any counter pass index specified in the parent primary command buffer. Note: this is no longer an issue after change from issue 10 resolution

6) What commands does the profiling lock have to be held for?

RESOLVED For any command buffer that is being queried with a performance query pool, the profiling lock **must** be held while that command buffer is in the *recording*, *executable*, or *pending state*.

7) Should we support vkCmdCopyQueryPoolResults?

RESOLVED Yes.

8) Should we allow performance queries to interact with multiview?

RESOLVED Yes, but the performance queries must be performed once for each pass per view.

9) Should a `queryCount > 1` be usable for performance queries?

RESOLVED Yes. Some vendors will have costly performance counter query pool creation, and would rather if a certain set of counters were to be used multiple times that a `queryCount > 1` can be used to amortize the instantiation cost.

10) Should we introduce an indirect mechanism to set the counter pass index?

RESOLVED Specify the counter pass index at submit time instead, to avoid requiring re-recording of command buffers when multiple counter passes are needed.

Examples

The following example shows how to find what performance counters a queue family supports, setup a query pool to record these performance counters, how to add the query pool to the command buffer to record information, and how to get the results from the query pool.

```
// A previously created physical device
VkPhysicalDevice physicalDevice;

// One of the queue families our device supports
uint32_t queueFamilyIndex;

uint32_t counterCount;

// Get the count of counters supported
vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR(
  physicalDevice,
```

```
  queueFamilyIndex,
  &counterCount,
  NULL,
  NULL);

VkPerformanceCounterKHR* counters =
  malloc(sizeof(VkPerformanceCounterKHR) * counterCount);
VkPerformanceCounterDescriptionKHR* counterDescriptions =
  malloc(sizeof(VkPerformanceCounterDescriptionKHR) * counterCount);

// Get the counters supported
vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR(
  physicalDevice,
  queueFamilyIndex,
  &counterCount,
  counters,
  counterDescriptions);

// Try to enable the first 8 counters
uint32_t enabledCounters[8];

const uint32_t enabledCounterCount = min(counterCount, 8));

for (uint32_t i = 0; i < enabledCounterCount; i++) {
  enabledCounters[i] = i;
}

// A previously created device that had the performanceCounterQueryPools feature
// set to VK_TRUE
VkDevice device;

VkQueryPoolPerformanceCreateInfoKHR performanceQueryCreateInfo = {
  .sType = VK_STRUCTURE_TYPE_QUERY_POOL_PERFORMANCE_CREATE_INFO_KHR,
  .pNext = NULL,

  // Specify the queue family that this performance query is performed on
  .queueFamilyIndex = queueFamilyIndex,

  // The number of counters to enable
  .counterIndexCount = enabledCounterCount,

  // The array of indices of counters to enable
  .pCounterIndices = enabledCounters
};


// Get the number of passes our counters will require.
uint32_t numPasses;

vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR(
  physicalDevice,
```

```
  &performanceQueryCreateInfo,
  &numPasses);

VkQueryPoolCreateInfo queryPoolCreateInfo = {
  .sType = VK_STRUCTURE_TYPE_QUERY_POOL_CREATE_INFO,
  .pNext = &performanceQueryCreateInfo,
  .flags = 0,
  // Using our new query type here
  .queryType = VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR,
  .queryCount = 1,
  .pipelineStatistics = 0
};

VkQueryPool queryPool;

VkResult result = vkCreateQueryPool(
  device,
  &queryPoolCreateInfo,
  NULL,
  &queryPool);

assert(VK_SUCCESS == result);

// A queue from queueFamilyIndex
VkQueue queue;

// A command buffer we want to record counters on
VkCommandBuffer commandBuffer;

VkCommandBufferBeginInfo commandBufferBeginInfo = {
  .sType = VK_STRUCTURE_TYPE_COMMAND_BUFFER_BEGIN_INFO,
  .pNext = NULL,
  .flags = 0,
  .pInheritanceInfo = NULL
};

VkAcquireProfilingLockInfoKHR lockInfo = {
  .sType = VK_STRUCTURE_TYPE_ACQUIRE_PROFILING_LOCK_INFO_KHR,
  .pNext = NULL,
  .flags = 0,
  .timeout = UINT64_MAX // Wait forever for the lock
};

// Acquire the profiling lock before we record command buffers
// that will use performance queries

result = vkAcquireProfilingLockKHR(device, &lockInfo);

assert(VK_SUCCESS == result);

result = vkBeginCommandBuffer(commandBuffer, &commandBufferBeginInfo);
```

```
assert(VK_SUCCESS == result);

vkCmdResetQueryPool(
  commandBuffer,
  queryPool,
  0,
  1);

vkCmdBeginQuery(
  commandBuffer,
  queryPool,
  0,
  0);

// Perform the commands you want to get performance information on
// ...

// Perform a barrier to ensure all previous commands were complete before
// ending the query
vkCmdPipelineBarrier(commandBuffer,
  VK_PIPELINE_STAGE_BOTTOM_OF_PIPE_BIT,
  VK_PIPELINE_STAGE_BOTTOM_OF_PIPE_BIT,
  0,
  0,
  NULL,
  0,
  NULL,
  0,
  NULL);

vkCmdEndQuery(
  commandBuffer,
  queryPool,
  0);

result = vkEndCommandBuffer(commandBuffer);

assert(VK_SUCCESS == result);

for (uint32_t counterPass = 0; counterPass < numPasses; counterPass++) {

  VkPerformanceQuerySubmitInfoKHR performanceQuerySubmitInfo = {
    VK_STRUCTURE_TYPE_PERFORMANCE_QUERY_SUBMIT_INFO_KHR,
    NULL,
    counterPass
  };


  // Submit the command buffer and wait for its completion
  // ...
```

```
}

// Release the profiling lock after the command buffer is no longer in the
// pending state.
vkReleaseProfilingLockKHR(device);

result = vkResetCommandBuffer(commandBuffer, 0);

assert(VK_SUCCESS == result);

// Create an array to hold the results of all counters
VkPerformanceCounterResultKHR* recordedCounters = malloc(
  sizeof(VkPerformanceCounterResultKHR) * enabledCounterCount);

result = vkGetQueryPoolResults(
  device,
  queryPool,
  0,
  1,
  sizeof(VkPerformanceCounterResultKHR) * enabledCounterCount,
  recordedCounters,
  sizeof(VkPerformanceCounterResultKHR) * enabledCounterCount,
  NULL);

// recordedCounters is filled with our counters, we will look at one for posterity
switch (counters[0].storage) {
  case VK_PERFORMANCE_COUNTER_STORAGE_INT32:
    // use recordCounters[0].int32 to get at the counter result!
    break;
  case VK_PERFORMANCE_COUNTER_STORAGE_INT64:
    // use recordCounters[0].int64 to get at the counter result!
    break;
  case VK_PERFORMANCE_COUNTER_STORAGE_UINT32:
    // use recordCounters[0].uint32 to get at the counter result!
    break;
  case VK_PERFORMANCE_COUNTER_STORAGE_UINT64:
    // use recordCounters[0].uint64 to get at the counter result!
    break;
  case VK_PERFORMANCE_COUNTER_STORAGE_FLOAT32:
    // use recordCounters[0].float32 to get at the counter result!
    break;
  case VK_PERFORMANCE_COUNTER_STORAGE_FLOAT64:
    // use recordCounters[0].float64 to get at the counter result!
    break;
}
```

Version History

- Revision 1, 2019-10-08

VK_KHR_pipeline_binary

Name String

`VK_KHR_pipeline_binary`

Extension Type

Device extension

Registered Extension Number

484

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.4
or
VK_KHR_extended_flags
or
VK_KHR_maintenance5

Contact

- Stu Smith [GitHub]stu-s

Extension Proposal

VK_KHR_pipeline_binary

Other Extension Metadata

Last Modified Date

2024-07-01

Contributors

- Stu Smith, AMD
- Tobias Hector, AMD
- Alan Harrison, AMD
- Maciej Jesionowski, AMD
- Younggwan Kim, Arm
- Jan-Harald Fredriksen, Arm
- Ting Wei, Arm
- Chris Glover, Google
- Shahbaz Youssefi, Google

- Jakub Kuderski, Google
- Piotr Byszewski, Mobica
- Piers Daniell, NVIDIA
- Ralph Potter, Samsung
- Matthew Netsch, Qualcomm
- Hans-Kristian Arntzen, Valve
- Samuel Pitoiset, Valve
- Tatsuyuki Ishi, Valve

Description

This extension provides a method to obtain binary data associated with individual pipelines such that applications can manage caching themselves instead of using VkPipelineCache objects.

New Object Types

- VkPipelineBinaryKHR

New Commands

- vkCreatePipelineBinariesKHR
- vkDestroyPipelineBinaryKHR
- vkGetPipelineBinaryDataKHR
- vkGetPipelineKeyKHR
- vkReleaseCapturedPipelineDataKHR

New Structures

- VkPipelineBinaryCreateInfoKHR
- VkPipelineBinaryDataInfoKHR
- VkPipelineBinaryDataKHR
- VkPipelineBinaryHandlesInfoKHR
- VkPipelineBinaryKeyKHR
- VkPipelineBinaryKeysAndDataKHR
- VkPipelineCreateInfoKHR
- VkReleaseCapturedPipelineDataInfoKHR
- Extending VkDeviceCreateInfo:
  - VkDevicePipelineBinaryInternalCacheControlKHR
- Extending VkGraphicsPipelineCreateInfo, VkComputePipelineCreateInfo, VkRayTracingPipelineCreateInfoKHR:
  - VkPipelineBinaryInfoKHR

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePipelineBinaryFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDevicePipelineBinaryPropertiesKHR

New Enum Constants

- `VK_KHR_PIPELINE_BINARY_EXTENSION_NAME`
- `VK_KHR_PIPELINE_BINARY_SPEC_VERSION`
- VK_MAX_PIPELINE_BINARY_KEY_SIZE_KHR
- Extending VkObjectType:
  - VK_OBJECT_TYPE_PIPELINE_BINARY_KHR
- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR
- Extending VkResult:
  - VK_ERROR_NOT_ENOUGH_SPACE_KHR
  - VK_PIPELINE_BINARY_MISSING_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DEVICE_PIPELINE_BINARY_INTERNAL_CACHE_CONTROL_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_BINARY_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_BINARY_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_PIPELINE_BINARY_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_PIPELINE_BINARY_DATA_INFO_KHR
  - VK_STRUCTURE_TYPE_PIPELINE_BINARY_HANDLES_INFO_KHR
  - VK_STRUCTURE_TYPE_PIPELINE_BINARY_INFO_KHR
  - VK_STRUCTURE_TYPE_PIPELINE_BINARY_KEY_KHR
  - VK_STRUCTURE_TYPE_PIPELINE_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_RELEASE_CAPTURED_PIPELINE_DATA_INFO_KHR

Version History

- Revision 1, 2021-12-10 (Chris Glover)
  - Initial draft.

VK_KHR_pipeline_executable_properties

Name String

`VK_KHR_pipeline_executable_properties`

Extension Type

Device extension

Registered Extension Number

270

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Special Use

- Developer tools

Contact

- Faith Ekstrand [GitHub]gfxstrand

Other Extension Metadata

Last Modified Date

2019-05-28

IP Status

No known IP claims.

Interactions and External Dependencies

Contributors

- Faith Ekstrand, Intel
- Ian Romanick, Intel
- Kenneth Graunke, Intel
- Baldur Karlsson, Valve
- Jesse Hall, Google
- Jeff Bolz, Nvidia
- Piers Daniel, Nvidia
- Tobias Hector, AMD

- Jan-Harald Fredriksen, ARM
- Tom Olson, ARM
- Daniel Koch, Nvidia
- Spencer Fricke, Samsung

Description

When a pipeline is created, its state and shaders are compiled into zero or more device-specific executables, which are used when executing commands against that pipeline. This extension adds a mechanism to query properties and statistics about the different executables produced by the pipeline compilation process. This is intended to be used by debugging and performance tools to allow them to provide more detailed information to the user. Certain compile time shader statistics provided through this extension may be useful to developers for debugging or performance analysis.

New Commands

- vkGetPipelineExecutableInternalRepresentationsKHR
- vkGetPipelineExecutablePropertiesKHR
- vkGetPipelineExecutableStatisticsKHR

New Structures

- VkPipelineExecutableInfoKHR
- VkPipelineExecutableInternalRepresentationKHR
- VkPipelineExecutablePropertiesKHR
- VkPipelineExecutableStatisticKHR
- VkPipelineInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR

New Unions

- VkPipelineExecutableStatisticValueKHR

New Enums

- VkPipelineExecutableStatisticFormatKHR

New Enum Constants

- `VK_KHR_PIPELINE_EXECUTABLE_PROPERTIES_EXTENSION_NAME`
- `VK_KHR_PIPELINE_EXECUTABLE_PROPERTIES_SPEC_VERSION`
- Extending VkPipelineCreateFlagBits:

    - VK_PIPELINE_CREATE_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR
    - VK_PIPELINE_CREATE_CAPTURE_STATISTICS_BIT_KHR
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_EXECUTABLE_PROPERTIES_FEATURES_KHR
    - VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_INFO_KHR
    - VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_INTERNAL_REPRESENTATION_KHR
    - VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_PROPERTIES_KHR
    - VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_STATISTIC_KHR
    - VK_STRUCTURE_TYPE_PIPELINE_INFO_KHR

Issues

1) What should we call the pieces of the pipeline which are produced by the compilation process and about which you can query properties and statistics?

RESOLVED: Call them “executables”. The name “binary” was used in early drafts of the extension but it was determined that “pipeline binary” could have a fairly broad meaning (such as a binary serialized form of an entire pipeline) and was too big of a namespace for the very specific needs of this extension.

Version History

- Revision 1, 2019-05-28 (Faith Ekstrand)
    - Initial draft

VK_KHR_pipeline_library

Name String

`VK_KHR_pipeline_library`

Extension Type

Device extension

Registered Extension Number

291

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- Christoph Kubisch [GitHub]pixeljetstream

Other Extension Metadata

Last Modified Date

2020-01-08

IP Status

No known IP claims.

Contributors

- See contributors to `VK_KHR_ray_tracing_pipeline`

Description

A pipeline library is a special pipeline that cannot be bound, instead it defines a set of shaders and shader groups which can be linked into other pipelines. This extension defines the infrastructure for pipeline libraries, but does not specify the creation or usage of pipeline libraries. This is left to additional dependent extensions.

New Structures

- Extending VkGraphicsPipelineCreateInfo:
  - VkPipelineLibraryCreateInfoKHR

New Enum Constants

- `VK_KHR_PIPELINE_LIBRARY_EXTENSION_NAME`

- `VK_KHR_PIPELINE_LIBRARY_SPEC_VERSION`
- Extending VkPipelineCreateFlagBits:
  - VK_PIPELINE_CREATE_LIBRARY_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PIPELINE_LIBRARY_CREATE_INFO_KHR

Version History

- Revision 1, 2020-01-08 (Christoph Kubisch)
  - Initial draft.

VK_KHR_pipeline_library_group_handles

Name String

`VK_KHR_pipeline_library_group_handles`

Extension Type

Device extension

Registered Extension Number

651

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_ray_tracing_pipeline
and
VK_KHR_pipeline_library

Contact

- Stu Smith [GitHub]stu-s

Extension Proposal

VK_KHR_pipeline_library_group_handles

Other Extension Metadata

Last Modified Date

2026-03-09

IP Status

No known IP claims.

Contributors

- Hans-Kristian Arntzen, Valve
- Stu Smith, AMD
- Ricardo Garcia, Igalia
- Lionel Landwerlin, Intel
- Eric Werness, NVIDIA
- Daniel Koch, NVIDIA

Description

This extension is based on the `VK_EXT_pipeline_library_group_handles` extension.

When using pipeline libraries in ray tracing pipelines, a library might get linked into different pipelines in an incremental way. An application can have a strategy where a ray tracing pipeline is comprised of N pipeline libraries and is later augmented by creating a new pipeline with N + 1 libraries. Without this extension, all group handles must be re-queried as the group handle is tied to the pipeline, not the library. This is problematic for applications which aim to decouple construction of record buffers and the linkage of ray tracing pipelines.

To aid in this, this extension enables support for querying group handles directly from pipeline libraries. Group handles obtained from a library **must** remain bitwise identical in any `VkPipeline` that links to the library.

With this feature, the extension also improves compatibility with DXR 1.1 AddToStateObject(), which guarantees that group handles returned remain bitwise identical between parent and child pipelines. In addition, querying group handles from COLLECTION objects is also supported with that API.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePipelineLibraryGroupHandlesFeaturesKHR

New Enum Constants

- `VK_KHR_PIPELINE_LIBRARY_GROUP_HANDLES_EXTENSION_NAME`
- `VK_KHR_PIPELINE_LIBRARY_GROUP_HANDLES_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_LIBRARY_GROUP_HANDLES_FEATURES_KHR

Version History

- Revision 1, 2026-03-09 (Stu Smith)
  - Initial draft

VK_KHR_portability_enumeration

Name String

`VK_KHR_portability_enumeration`

Extension Type

Instance extension

Registered Extension Number

395

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- Charles Giessen [GitHub]charles-lunarg

Other Extension Metadata

Last Modified Date

2021-06-02

IP Status

No known IP claims.

Interactions and External Dependencies

- Interacts with `VK_KHR_portability_subset`

Contributors

- Lenny Komow, LunarG
- Charles Giessen, LunarG

Description

This extension allows applications to control whether devices that expose the `VK_KHR_portability_subset` extension are included in the results of physical device enumeration. Since devices which support the `VK_KHR_portability_subset` extension are not fully conformant Vulkan implementations, the Vulkan loader does not report those devices unless the application explicitly asks for them. This prevents applications which may not be aware of non-conformant devices from accidentally using them, as any device which supports the `VK_KHR_portability_subset` extension mandates that the extension must be enabled if that device is used.

This extension is implemented in the loader.

New Enum Constants

- `VK_KHR_PORTABILITY_ENUMERATION_EXTENSION_NAME`
- `VK_KHR_PORTABILITY_ENUMERATION_SPEC_VERSION`
- Extending VkInstanceCreateFlagBits:
  - VK_INSTANCE_CREATE_ENUMERATE_PORTABILITY_BIT_KHR

Version History

- Revision 1, 2021-06-02 (Lenny Komow)
  - Initial version

VK_KHR_present_id

Name String

`VK_KHR_present_id`

Extension Type

Device extension

Registered Extension Number

295

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_swapchain
and
VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Keith Packard [GitHub]keithp

Other Extension Metadata

Last Modified Date

2019-05-15

IP Status

No known IP claims.

Contributors

- Keith Packard, Valve
- Ian Elliott, Google
- Alon Or-bach, Samsung

Description

This device extension allows an application that uses the `VK_KHR_swapchain` extension to provide an identifier for present operations on a swapchain. An application **can** use this to reference specific present operations in other extensions.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePresentIdFeaturesKHR
- Extending VkPresentInfoKHR:
  - VkPresentIdKHR

New Enum Constants

- `VK_KHR_PRESENT_ID_EXTENSION_NAME`
- `VK_KHR_PRESENT_ID_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_ID_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PRESENT_ID_KHR

Issues

None.

Examples

Version History

- Revision 1, 2019-05-15 (Keith Packard)
  - Initial version

VK_KHR_present_id2

Name String

VK_KHR_present_id2

Extension Type

Device extension

Registered Extension Number

480

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_surface_capabilities2
and
VK_KHR_surface
and
VK_KHR_swapchain

Contact

- Daniel Stone

Extension Proposal

VK_KHR_present_id2

Other Extension Metadata

Last Modified Date

2025-01-06

IP Status

No known IP claims.

Contributors

- Hans-Kristian Arntzen, Valve
- James Jones, NVIDIA
- Daniel Stone, Collabora
- Derek Foreman, Collabora
- *contributors to `VK_KHR_present_id`*

Description

This device extension allows an application that uses the VK_KHR_swapchain extension to provide an identifier for present operations on a swapchain. An application **can** use this to reference specific present operations in other extensions.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePresentId2FeaturesKHR
- Extending VkPresentInfoKHR:
  - VkPresentId2KHR
- Extending VkSurfaceCapabilities2KHR:
  - VkSurfaceCapabilitiesPresentId2KHR

New Enum Constants

- VK_KHR_PRESENT_ID_2_EXTENSION_NAME
- VK_KHR_PRESENT_ID_2_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_ID_2_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PRESENT_ID_2_KHR
  - VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_PRESENT_ID_2_KHR
- Extending VkSwapchainCreateFlagBitsKHR:
  - VK_SWAPCHAIN_CREATE_PRESENT_ID_2_BIT_KHR

Issues

None.

Examples

Version History

- Revision 1, 2022-05-10 (Daniel Stone)
  - Repurposed VK_KHR_present_id to be driven by surface capabilities

VK_KHR_present_mode_fifo_latest_ready

Name String

VK_KHR_present_mode_fifo_latest_ready

Extension Type

Device extension

Registered Extension Number

622

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_swapchain

Contact

- Lionel Duc [GitHub]nvlduc

Extension Proposal

VK_KHR_present_mode_fifo_latest_ready

Other Extension Metadata

Last Modified Date

2025-03-18

IP Status

No known IP claims.

Contributors

- James Jones, NVIDIA
- Lionel Duc, NVIDIA
- Lina Versace, Google

Description

This extension is based on VK_EXT_present_mode_fifo_latest_ready and provides equivalent functionality.

This extension adds a new present mode, VK_PRESENT_MODE_FIFO_LATEST_READY_KHR.

This tear-free present mode behaves much like VK_PRESENT_MODE_FIFO_KHR, except that each vertical blanking period dequeues consecutive present requests until the latest ready is found to update the current image.

While this seems similar in concept to VK_PRESENT_MODE_MAILBOX_KHR, the fundamental difference is that the processing of the present requests is done during vblank. From the application perspective, this means for example, that in a flip-based model, a single vblank **may** cause multiple swapchain images to be released at once, while VK_PRESENT_MODE_MAILBOX_KHR **may** continuously be releasing images as new requests become ready.

This additional present mode is useful when using a time-based present API.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePresentModeFifoLatestReadyFeaturesKHR

New Enum Constants

- `VK_KHR_PRESENT_MODE_FIFO_LATEST_READY_EXTENSION_NAME`
- `VK_KHR_PRESENT_MODE_FIFO_LATEST_READY_SPEC_VERSION`
- Extending VkPresentModeKHR:
  - VK_PRESENT_MODE_FIFO_LATEST_READY_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_MODE_FIFO_LATEST_READY_FEATURES_KHR

Version History

- Revision 1, 2025-03-18 (Lina Versace)
  - Internal revisions

VK_KHR_present_wait

Name String

`VK_KHR_present_wait`

Extension Type

Device extension

Registered Extension Number

249

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_swapchain
and
VK_KHR_present_id

Contact

- Keith Packard [GitHub]keithp

Other Extension Metadata

Last Modified Date

2019-05-15

IP Status

No known IP claims.

Contributors

- Keith Packard, Valve
- Ian Elliott, Google
- Tobias Hector, AMD
- Daniel Stone, Collabora

Description

This device extension allows an application that uses the `VK_KHR_swapchain` extension to wait for present operations to complete. An application can use this to monitor and control the pacing of the application by managing the number of outstanding images yet to be presented.

New Commands

- vkWaitForPresentKHR

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePresentWaitFeaturesKHR

New Enum Constants

- `VK_KHR_PRESENT_WAIT_EXTENSION_NAME`
- `VK_KHR_PRESENT_WAIT_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_WAIT_FEATURES_KHR

Issues

1) When does the wait finish?

RESOLVED. The wait will finish when the present is visible to the user. There is no requirement for any precise timing relationship between the presentation of the image to the user, but implementations **should** signal the wait as close as possible to the presentation of the first pixel in the new image to the user.

2) Should this use fences or other existing synchronization mechanism.

RESOLVED. Because display and rendering are often implemented in separate drivers, this extension will provide a separate synchronization API.

3) Should this extension share present identification with other extensions?

RESOLVED. Yes. A new extension, VK_KHR_present_id, should be created to provide a shared structure for presentation identifiers.

4) What happens when presentations complete out of order wrt calls to vkQueuePresent? This could happen if the semaphores for the presentations were ready out of order.

OPTION A: Require that when a PresentId is set that the driver ensure that images are always presented in the order of calls to vkQueuePresent.

OPTION B: Finish both waits when the earliest present completes. This will complete the later present wait earlier than the actual presentation. This should be the easiest to implement as the driver need only track the largest present ID completed. This is also the 'natural' consequence of interpreting the existing vkWaitForPresentKHR specification.

OPTION C: Finish both waits when both have completed. This will complete the earlier presentation later than the actual presentation time. This is allowed by the current specification as there is no precise timing requirement for when the presentId value is updated. This requires slightly more complexity in the driver as it will need to track all outstanding presentId values.

Examples

Version History

- Revision 1, 2019-02-19 (Keith Packard)
  - Initial version

VK_KHR_present_wait2

Name String

VK_KHR_present_wait2

Extension Type

Device extension

Registered Extension Number

481

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_surface_capabilities2
and
VK_KHR_surface
and
VK_KHR_swapchain
and
VK_KHR_present_id2

Contact

- Daniel Stone

Extension Proposal

VK_KHR_present_wait2

Other Extension Metadata

Last Modified Date

2025-05-30

IP Status

No known IP claims.

Contributors

- Hans-Kristian Arntzen, Valve
- James Jones, NVIDIA
- Daniel Stone, Collabora
- Derek Foreman, Collabora
- *contributors to `VK_KHR_present_wait`*

Description

This device extension allows an application that uses the VK_KHR_swapchain extension to wait for present operations to complete. An application can use this to monitor and control the pacing of the application by managing the number of outstanding images yet to be presented.

New Commands

- vkWaitForPresent2KHR

New Structures

- VkPresentWait2InfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePresentWait2FeaturesKHR
- Extending VkSurfaceCapabilities2KHR:
  - VkSurfaceCapabilitiesPresentWait2KHR

New Enum Constants

- VK_KHR_PRESENT_WAIT_2_EXTENSION_NAME
- VK_KHR_PRESENT_WAIT_2_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_WAIT_2_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PRESENT_WAIT_2_INFO_KHR
  - VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_PRESENT_WAIT_2_KHR
- Extending VkSwapchainCreateFlagBitsKHR:
  - VK_SWAPCHAIN_CREATE_PRESENT_WAIT_2_BIT_KHR

Issues

1) When does the wait finish?

RESOLVED. The wait request will complete when the timeout expires, or after the corresponding presentation request has either taken effect within the presentation engine or has been replaced without presentation. Additionally, a wait may complete immediately if the swapchain becomes out of date.

In circumstances outside the application's control, this wait may be particularly long. For example, a user session may have the display locked and switched off for multiple days. During this time, the latest image presented through the WSI will never be presented to the user (because nothing is being presented), or replaced (because nothing newer has been queued by the application). Each operating system may have a separate mechanism to inform the application of states such as these, however it is out of scope of the Vulkan WSI.

There is no requirement for any precise timing relationship between the presentation of the image

to the user and the end of the wait.

This extension is not intended for time-to-light estimation, which is better solved by a separate extension dedicated to present-timing feedback for audio/visual/input synchronization.

2) Should this use fences or other existing synchronization mechanism?

RESOLVED. VkFence is a legacy primitive. Building a new API around a legacy primitive is undesirable.

Other existing synchronization mechanisms may lack a platform-provided framework for sharing synchronization objects between display and render drivers.

For these reasons, this extension will provide a separate synchronization API.

3) Should this extension share present identification with other extensions?

RESOLVED. Yes. A new extension, `VK_KHR_present_id2`, should be created to provide a shared structure for presentation identifiers.

4) What happens when presentations complete out of order with respect to calls to vkQueuePresent? This could happen if the semaphores for the presentations were ready out of order.

OPTION A: Require that when a PresentId is set that the driver ensure that images are always presented in the order of calls to vkQueuePresent.

OPTION B: Finish both waits when the earliest present completes. This will complete the later present wait earlier than the actual presentation. This should be the easiest to implement as the driver need only track the largest present ID completed. This is also the 'natural' consequence of interpreting the existing vkWaitForPresentKHR specification.

OPTION C: Finish both waits when both have completed. This will complete the earlier presentation later than the actual presentation time. This is allowed by the current specification as there is no precise timing requirement for when the presentId value is updated. This requires slightly more complexity in the driver as it will need to track all outstanding presentId values.

OPTION D: The order of completion between outstanding `vkWaitForPresent2KHR` calls is always undefined. However, a `VK_SUCCESS` return value in `VkPresentWait2InfoKHR`::`presentId` implies that future calls to `vkWaitForPresent2KHR` where `VkPresentWait2InfoKHR`::`presentId` is less than or equal to N will complete immediately.

RESOLVED. **OPTION D**: This option ensures implementations do not need to create complex internal queues to generate signals in the right order.

5) Should this extension deviate from `VK_KHR_present_wait` and require the presentation engine to provide the presentId values?

RESOLVED. No. This extension is intended to be a bugfix of `VK_KHR_present_wait`, and existing low-latency apis require an application provided id. At least on some platforms, a mapping mechanism would be required to translate between presentation engine and application ids. This exceeds the

intended scope of this extension.

When needed in the future, we can introduce an independent presentation engine driven id and a mechanism for mapping presentation engine ids to application provided ids.

Examples

Version History

- Revision 1, 2022-10-05 (Daniel Stone)
  - Repurposed from VK_KHR_present_wait to be based on surface capabilities
  - Reworded wait finish section to avoid time-to-light

VK_KHR_ray_query

Name String

`VK_KHR_ray_query`

Extension Type

Device extension

Registered Extension Number

349

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_spirv_1_4
or
Vulkan Version 1.2
and
VK_KHR_acceleration_structure

SPIR-V Dependencies

- SPV_KHR_ray_query

Contact

- Daniel Koch [GitHub]dgkoch

Other Extension Metadata

Last Modified Date

2020-11-12

Interactions and External Dependencies

- This extension provides API support for `GLSL_EXT_ray_query`

Contributors

- Matthäus Chajdas, AMD
- Greg Grebe, AMD
- Nicolai Hähnle, AMD
- Tobias Hector, AMD
- Dave Oldcorn, AMD
- Skyler Saleh, AMD
- Mathieu Robart, Arm

- Marius Bjorge, Arm
- Tom Olson, Arm
- Sebastian Tafuri, EA
- Henrik Rydgard, Embark
- Juan Cañada, Epic Games
- Patrick Kelly, Epic Games
- Yuriy O'Donnell, Epic Games
- Michael Doggett, Facebook/Oculus
- Andrew Garrard, Imagination
- Don Scorgie, Imagination
- Dae Kim, Imagination
- Joshua Barczak, Intel
- Slawek Grajewski, Intel
- Jeff Bolz, NVIDIA
- Pascal Gautron, NVIDIA
- Daniel Koch, NVIDIA
- Christoph Kubisch, NVIDIA
- Ashwin Lele, NVIDIA
- Robert Stepinski, NVIDIA
- Martin Stich, NVIDIA
- Nuno Subtil, NVIDIA
- Eric Werness, NVIDIA
- Jon Leech, Khronos
- Jeroen van Schijndel, OTOY
- Juul Joosten, OTOY
- Alex Bourd, Qualcomm
- Roman Larionov, Qualcomm
- David McAllister, Qualcomm
- Spencer Fricke, Samsung
- Lewis Gordon, Samsung
- Ralph Potter, Samsung
- Jasper Bekkers, Traverse Research
- Jesse Barker, Unity
- Baldur Karlsson, Valve

Description

Rasterization has been the dominant method to produce interactive graphics, but increasing performance of graphics hardware has made ray tracing a viable option for interactive rendering. Being able to integrate ray tracing with traditional rasterization makes it easier for applications to incrementally add ray traced effects to existing applications or to do hybrid approaches with rasterization for primary visibility and ray tracing for secondary queries.

Ray queries are available to all shader types, including graphics, compute, and ray tracing pipelines. Ray queries are not able to launch additional shaders, instead returning traversal results to the calling shader.

This extension adds support for the following SPIR-V extension in Vulkan:

- `SPV_KHR_ray_query`

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceRayQueryFeaturesKHR

New Enum Constants

- `VK_KHR_RAY_QUERY_EXTENSION_NAME`
- `VK_KHR_RAY_QUERY_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_QUERY_FEATURES_KHR

New SPIR-V Capabilities

- `RayQueryKHR`
- `RayTraversalPrimitiveCullingKHR`

Sample Code

Example of ray query in a GLSL shader, illustrating how to use ray queries to determine whether a given position (at ray origin) is in shadow or not, by tracing a ray towards the light, and checking for any intersections with geometry occluding the light.

```
rayQueryEXT rq;

rayQueryInitializeEXT(rq, accStruct, gl_RayFlagsTerminateOnFirstHitEXT, cullMask,
origin, tMin, direction, tMax);

// Traverse the acceleration structure and store information about the first
intersection (if any)
rayQueryProceedEXT(rq);
```

```
if (rayQueryGetIntersectionTypeEXT(rq, true) ==
gl_RayQueryCommittedIntersectionNoneEXT) {
    // Not in shadow
}
```

Issues

(1) What are the changes between the public provisional (VK_KHR_ray_tracing v8) release and the final (VK_KHR_acceleration_structure v11 / VK_KHR_ray_query v1) release?

- refactor VK_KHR_ray_tracing into 3 extensions, enabling implementation flexibility and decoupling ray query support from ray pipelines:
  - `VK_KHR_acceleration_structure` (for acceleration structure operations)
  - `VK_KHR_ray_tracing_pipeline` (for ray tracing pipeline and shader stages)
  - `VK_KHR_ray_query` (for ray queries in existing shader stages)
- Update SPIRV capabilities to use `RayQueryKHR`
- extension is no longer provisional

Version History

- Revision 1, 2020-11-12 (Mathieu Robart, Daniel Koch, Andrew Garrard)
  - Decomposition of the specification, from VK_KHR_ray_tracing to VK_KHR_ray_query (#1918,!3912)
  - update to use `RayQueryKHR` SPIR-V capability
  - add numerical limits for ray parameters (#2235,!3960)
  - relax formula for ray intersection candidate determination (#2322,!4080)
  - restrict traces to TLAS (#2239,!4141)
  - require `HitT` to be in ray interval for `OpRayQueryGenerateIntersectionKHR` (#2359,!4146)
  - add ray query shader stages for AS read bit (#2407,!4203)

VK_KHR_ray_tracing_maintenance1

Name String

`VK_KHR_ray_tracing_maintenance1`

Extension Type

Device extension

Registered Extension Number

387

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_acceleration_structure

API Interactions

- Interacts with VK_VERSION_1_3
- Interacts with VK_EXT_device_generated_commands
- Interacts with VK_KHR_ray_tracing_pipeline
- Interacts with VK_KHR_synchronization2

SPIR-V Dependencies

- SPV_KHR_ray_cull_mask

Contact

- Daniel Koch [GitHub]dgkoch

Other Extension Metadata

Last Modified Date

2022-02-21

Interactions and External Dependencies

- This extension provides API support for `GLSL_EXT_ray_cull_mask`
- Interacts with `VK_KHR_ray_tracing_pipeline`
- Interacts with `VK_KHR_synchronization2`

Contributors

- Stu Smith, AMD
- Tobias Hector, AMD
- Marius Bjorge, Arm

- Tom Olson, Arm
- Yuriy O'Donnell, Epic Games
- Yunpeng Zhu, Huawei
- Andrew Garrard, Imagination
- Dae Kim, Imagination
- Joshua Barczak, Intel
- Lionel Landwerlin, Intel
- Daniel Koch, NVIDIA
- Eric Werness, NVIDIA
- Spencer Fricke, Samsung

Description

`VK_KHR_ray_tracing_maintenance1` adds a collection of minor ray tracing features, none of which would warrant an entire extension of their own.

The new features are as follows:

- Adds support for the `SPV_KHR_ray_cull_mask` SPIR-V extension in Vulkan. This extension provides access to built-in `CullMaskKHR` shader variable which contains the value of the `OpTrace* Cull Mask` parameter. This new shader variable is accessible in the intersection, any-hit, closest hit and miss shader stages.
- Adds support for a new pipeline stage and access mask built on top of `VK_KHR_synchronization2`:
  - VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR to specify execution of acceleration structure copy commands
  - VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR to specify read access to a shader binding table in any shader pipeline stage
- Adds two new acceleration structure query parameters:
  - VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR to query the acceleration structure size on the device timeline
  - VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR to query the number of bottom level acceleration structure pointers for serialization
- Adds an optional new indirect ray tracing dispatch command, vkCmdTraceRaysIndirect2KHR, which sources the shader binding table parameters as well as the dispatch dimensions from the device. The `rayTracingPipelineTraceRaysIndirect2` feature indicates whether this functionality is supported.

New Commands

If VK_KHR_ray_tracing_pipeline is supported:

- vkCmdTraceRaysIndirect2KHR

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR

If VK_KHR_ray_tracing_pipeline is supported:

- VkTraceRaysIndirectCommand2KHR

New Enum Constants

- `VK_KHR_RAY_TRACING_MAINTENANCE_1_EXTENSION_NAME`
- `VK_KHR_RAY_TRACING_MAINTENANCE_1_SPEC_VERSION`
- Extending VkQueryType:
  - VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR
  - VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_MAINTENANCE_1_FEATURES_KHR

If VK_KHR_synchronization2 or Vulkan Version 1.3 and VK_KHR_ray_tracing_pipeline is supported:

- Extending VkAccessFlagBits2:
  - VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR

If VK_EXT_device_generated_commands is supported:

- Extending VkIndirectCommandsTokenTypeEXT:
  - VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT

If VK_KHR_synchronization2 or Vulkan Version 1.3 is supported:

- Extending VkPipelineStageFlagBits2:
  - VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR

New Built-In Variables

- `CullMaskKHR`

New SPIR-V Capabilities

- `RayCullMaskKHR`

Issues

None Yet!

Version History

- Revision 1, 2022-02-21 (Members of the Vulkan Ray Tracing TSG)
    - internal revisions

VK_KHR_ray_tracing_pipeline

Name String

`VK_KHR_ray_tracing_pipeline`

Extension Type

Device extension

Registered Extension Number

348

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_spirv_1_4
or
Vulkan Version 1.2
and
VK_KHR_acceleration_structure

API Interactions

- Interacts with VK_KHR_ray_query

SPIR-V Dependencies

- SPV_KHR_ray_tracing

Contact

- Daniel Koch [GitHub]dgkoch

Other Extension Metadata

Last Modified Date

2020-11-12

Interactions and External Dependencies

- This extension provides API support for `GLSL_EXT_ray_tracing`
- This extension interacts with Vulkan 1.2 and `VK_KHR_vulkan_memory_model`, adding the shader-call-related relation of invocations, shader-call-order partial order of dynamic instances of instructions, and the `ShaderCallKHR` scope.
- This extension interacts with `VK_KHR_pipeline_library`, enabling pipeline libraries to be used with ray tracing pipelines and enabling usage of VkRayTracingPipelineInterfaceCreateInfoKHR.

Contributors

- Matthäus Chajdas, AMD

- Greg Grebe, AMD
- Nicolai Hähnle, AMD
- Tobias Hector, AMD
- Dave Oldcorn, AMD
- Skyler Saleh, AMD
- Mathieu Robart, Arm
- Marius Bjorge, Arm
- Tom Olson, Arm
- Sebastian Tafuri, EA
- Henrik Rydgard, Embark
- Juan Cañada, Epic Games
- Patrick Kelly, Epic Games
- Yuriy O'Donnell, Epic Games
- Michael Doggett, Facebook/Oculus
- Andrew Garrard, Imagination
- Don Scorgie, Imagination
- Dae Kim, Imagination
- Joshua Barczak, Intel
- Slawek Grajewski, Intel
- Jeff Bolz, NVIDIA
- Pascal Gautron, NVIDIA
- Daniel Koch, NVIDIA
- Christoph Kubisch, NVIDIA
- Ashwin Lele, NVIDIA
- Robert Stepinski, NVIDIA
- Martin Stich, NVIDIA
- Nuno Subtil, NVIDIA
- Eric Werness, NVIDIA
- Jon Leech, Khronos
- Jeroen van Schijndel, OTOY
- Juul Joosten, OTOY
- Alex Bourd, Qualcomm
- Roman Larionov, Qualcomm
- David McAllister, Qualcomm
- Spencer Fricke, Samsung

- Lewis Gordon, Samsung
- Ralph Potter, Samsung
- Jasper Bekkers, Traverse Research
- Jesse Barker, Unity
- Baldur Karlsson, Valve

Description

Rasterization has been the dominant method to produce interactive graphics, but increasing performance of graphics hardware has made ray tracing a viable option for interactive rendering. Being able to integrate ray tracing with traditional rasterization makes it easier for applications to incrementally add ray traced effects to existing applications or to do hybrid approaches with rasterization for primary visibility and ray tracing for secondary queries.

To enable ray tracing, this extension adds a few different categories of new functionality:

- A new ray tracing pipeline type with new shader domains: ray generation, intersection, any-hit, closest hit, miss, and callable
- A shader binding indirection table to link shader groups with acceleration structure items
- Ray tracing commands which initiate the ray pipeline traversal and invocation of the various new shader domains depending on which traversal conditions are met

This extension adds support for the following SPIR-V extension in Vulkan:

- `SPV_KHR_ray_tracing`

New Commands

- vkCmdSetRayTracingPipelineStackSizeKHR
- vkCmdTraceRaysIndirectKHR
- vkCmdTraceRaysKHR
- vkCreateRayTracingPipelinesKHR
- vkGetRayTracingCaptureReplayShaderGroupHandlesKHR
- vkGetRayTracingShaderGroupHandlesKHR
- vkGetRayTracingShaderGroupStackSizeKHR

New Structures

- VkRayTracingPipelineInterfaceCreateInfoKHR
- VkRayTracingShaderGroupCreateInfoKHR
- VkStridedDeviceAddressRegionKHR
- VkTraceRaysIndirectCommandKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:

    - VkPhysicalDeviceRayTracingPipelineFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
    - VkPhysicalDeviceRayTracingPipelinePropertiesKHR
- Extending VkPipelineCreateInfoKHR:
    - VkRayTracingPipelineCreateInfoKHR

New Enums

- VkRayTracingShaderGroupTypeKHR
- VkShaderGroupShaderKHR

New Enum Constants

- `VK_KHR_RAY_TRACING_PIPELINE_EXTENSION_NAME`
- `VK_KHR_RAY_TRACING_PIPELINE_SPEC_VERSION`
- VK_SHADER_UNUSED_KHR
- Extending VkBufferUsageFlagBits:
    - VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR
- Extending VkDynamicState:
    - VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR
- Extending VkPipelineBindPoint:
    - VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR
- Extending VkPipelineCreateFlagBits:
    - VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR
    - VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR
    - VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR
    - VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR
    - VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR
    - VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR
    - VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR
- Extending VkPipelineCreateFlagBits2:
    - VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_BUILT_IN_PRIMITIVES_BIT_KHR
- Extending VkPipelineStageFlagBits:
    - VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR
- Extending VkShaderStageFlagBits:
    - VK_SHADER_STAGE_ANY_HIT_BIT_KHR
    - VK_SHADER_STAGE_CALLABLE_BIT_KHR

    - VK_SHADER_STAGE_CLOSEST_HIT_BIT_KHR
    - VK_SHADER_STAGE_INTERSECTION_BIT_KHR
    - VK_SHADER_STAGE_MISS_BIT_KHR
    - VK_SHADER_STAGE_RAYGEN_BIT_KHR
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_PIPELINE_FEATURES_KHR
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_PIPELINE_PROPERTIES_KHR
    - VK_STRUCTURE_TYPE_RAY_TRACING_PIPELINE_CREATE_INFO_KHR
    - VK_STRUCTURE_TYPE_RAY_TRACING_PIPELINE_INTERFACE_CREATE_INFO_KHR
    - VK_STRUCTURE_TYPE_RAY_TRACING_SHADER_GROUP_CREATE_INFO_KHR

New or Modified Built-In Variables

- `LaunchIdKHR`
- `LaunchSizeKHR`
- `WorldRayOriginKHR`
- `WorldRayDirectionKHR`
- `ObjectRayOriginKHR`
- `ObjectRayDirectionKHR`
- `RayTminKHR`
- `RayTmaxKHR`
- `InstanceCustomIndexKHR`
- `InstanceId`
- `ObjectToWorldKHR`
- `WorldToObjectKHR`
- `HitKindKHR`
- `IncomingRayFlagsKHR`
- `RayGeometryIndexKHR`
- (modified)`PrimitiveId`

New SPIR-V Capabilities

- `RayTracingKHR`
- `RayTraversalPrimitiveCullingKHR`

Issues

(1) How does this extension differ from VK_NV_ray_tracing?

DISCUSSION:

The following is a summary of the main functional differences between VK_KHR_ray_tracing_pipeline and VK_NV_ray_tracing:

- added support for indirect ray tracing (vkCmdTraceRaysIndirectKHR)
- uses SPV_KHR_ray_tracing instead of SPV_NV_ray_tracing
  - refer to KHR SPIR-V enums instead of NV SPIR-V enums (which are functionally equivalent and aliased to the same values).
  - added `RayGeometryIndexKHR` built-in
- removed vkCompileDeferredNV compilation functionality and replaced with deferred host operations interactions for ray tracing
- added VkPhysicalDeviceRayTracingPipelineFeaturesKHR structure
- extended VkPhysicalDeviceRayTracingPipelinePropertiesKHR structure
  - renamed `maxRecursionDepth` to `maxRayRecursionDepth` and it has a minimum of 1 instead of 31
  - require `shaderGroupHandleSize` to be 32 bytes
  - added `maxRayDispatchInvocationCount`, `shaderGroupHandleAlignment` and `maxRayHitAttributeSize`
- reworked geometry structures so they could be better shared between device, host, and indirect builds
- changed SBT parameters to a structure and added size (VkStridedDeviceAddressRegionKHR)
- add parameter for requesting memory requirements for host and/or device build
- added pipeline library support for ray tracing
- added watertightness guarantees
- added no-null-shader pipeline flags (`VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_*_SHADERS_BIT_KHR`)
- added memory model interactions with ray tracing and define how subgroups work and can be repacked

(3) What are the changes between the public provisional (VK_KHR_ray_tracing v8) release and the internal provisional (VK_KHR_ray_tracing v9) release?

- Require Vulkan 1.1 and SPIR-V 1.4
- Added interactions with Vulkan 1.2 and `VK_KHR_vulkan_memory_model`
- added creation time capture and replay flags
  - added VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR to VkPipelineCreateFlagBits
- replace `VkStridedBufferRegionKHR` with VkStridedDeviceAddressRegionKHR and change vkCmdTraceRaysKHR, vkCmdTraceRaysIndirectKHR, to take these for the shader binding table and use device addresses instead of buffers.
- require the shader binding table buffers to have the `VK_BUFFER_USAGE_RAY_TRACING_BIT_KHR` set

- make VK_KHR_pipeline_library an interaction instead of required extension
- rename the libraries member of VkRayTracingPipelineCreateInfoKHR to pLibraryInfo and make it a pointer
- make VK_KHR_deferred_host_operations an interaction instead of a required extension (later went back on this)
- added explicit stack size management for ray tracing pipelines
  - removed the maxCallableSize member of VkRayTracingPipelineInterfaceCreateInfoKHR
  - added the pDynamicState member to VkRayTracingPipelineCreateInfoKHR
  - added VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR dynamic state for ray tracing pipelines
  - added vkGetRayTracingShaderGroupStackSizeKHR and vkCmdSetRayTracingPipelineStackSizeKHR commands
  - added VkShaderGroupShaderKHR enum
- Added maxRayDispatchInvocationCount limit to VkPhysicalDeviceRayTracingPipelinePropertiesKHR
- Added shaderGroupHandleAlignment property to VkPhysicalDeviceRayTracingPipelinePropertiesKHR
- Added maxRayHitAttributeSize property to VkPhysicalDeviceRayTracingPipelinePropertiesKHR
- Clarify deferred host ops for pipeline creation
  - VkDeferredOperationKHR is now a top-level parameter for vkCreateRayTracingPipelinesKHR
  - removed VkDeferredOperationInfoKHR structure
  - change deferred host creation/return parameter behavior such that the implementation can modify such parameters until the deferred host operation completes
  - VK_KHR_deferred_host_operations is required again

(4) What are the changes between the internal provisional (VK_KHR_ray_tracing v9) release and the final (VK_KHR_acceleration_structure v11 / VK_KHR_ray_tracing_pipeline v1) release?

- refactor VK_KHR_ray_tracing into 3 extensions, enabling implementation flexibility and decoupling ray query support from ray pipelines:
  - VK_KHR_acceleration_structure (for acceleration structure operations)
  - VK_KHR_ray_tracing_pipeline (for ray tracing pipeline and shader stages)
  - VK_KHR_ray_query (for ray queries in existing shader stages)
- Require Volatile for the following builtins in the ray generation, closest hit, miss, intersection, and callable shader stages:
  - SubgroupSize, SubgroupLocalInvocationId, SubgroupEqMask, SubgroupGeMask, SubgroupGtMask, SubgroupLeMask, SubgroupLtMask
  - SMIDNV, WarpIDNV

- clarify buffer usage flags for ray tracing
  - VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR is added as an alias of VK_BUFFER_USAGE_RAY_TRACING_BIT_NV and is required on shader binding table buffers
  - VK_BUFFER_USAGE_STORAGE_BUFFER_BIT is used in VK_KHR_acceleration_structure for scratchData
- rename maxRecursionDepth to maxRayPipelineRecursionDepth (pipeline creation) and maxRayRecursionDepth (limit) to reduce confusion
- Add queryable maxRayHitAttributeSize limit and rename members of VkRayTracingPipelineInterfaceCreateInfoKHR to maxPipelineRayPayloadSize and maxPipelineRayHitAttributeSize for clarity
- Update SPIRV capabilities to use RayTracingKHR
- extension is no longer provisional
- define synchronization requirements for indirect trace rays and indirect buffer

(5) This extension adds gl_InstanceID for the intersection, any-hit, and closest hit shaders, but in KHR_vulkan_glsl, gl_InstanceID is replaced with gl_InstanceIndex. Which should be used for Vulkan in this extension?

RESOLVED: This extension uses gl_InstanceID and maps it to InstanceId in SPIR-V. It is acknowledged that this is different than other shader stages in Vulkan. There are two main reasons for the difference here:

- symmetry with gl_PrimitiveID which is also available in these shaders
- there is no "baseInstance" relevant for these shaders, and so ID makes it more obvious that this is zero-based.

(6) Why is VK_KHR_pipeline_library an interaction instead of a required dependency, particularly when the "Device Extensions" section says it is required to be supported anyhow?

RESOLVED: If the VK_KHR_pipeline_library extension were a required dependency, then every application would need to enable the extension whether or not they actually want to use the pipeline library functionality. Developers found this to be annoying and unfriendly behavior. We do wish to require all **implementations** to support it though, and thus it is listed as a requirement in device extensions section.

Sample Code

Example ray generation GLSL shader

```
#version 450 core
#extension GL_EXT_ray_tracing : require
layout(set = 0, binding = 0, rgba8) uniform image2D image;
layout(set = 0, binding = 1) uniform accelerationStructureEXT as;
layout(location = 0) rayPayloadEXT float payload;

void main()
```

```
{
   vec4 col = vec4(0, 0, 0, 1);

   vec3 origin = vec3(float(gl_LaunchIDEXT.x)/float(gl_LaunchSizeEXT.x), float
(gl_LaunchIDEXT.y)/float(gl_LaunchSizeEXT.y), 1.0);
   vec3 dir = vec3(0.0, 0.0, -1.0);

   traceRayEXT(as, 0, 0xff, 0, 1, 0, origin, 0.0, dir, 1000.0, 0);

   col.y = payload;

   imageStore(image, ivec2(gl_LaunchIDEXT.xy), col);
}
```

Version History

- Revision 1, 2020-11-12 (Mathieu Robart, Daniel Koch, Eric Werness, Tobias Hector)
  - Decomposition of the specification, from VK_KHR_ray_tracing to VK_KHR_ray_tracing_pipeline (#1918,!3912)
  - require certain subgroup and sm_shader_builtin shader builtins to be decorated as volatile in the ray generation, closest hit, miss, intersection, and callable stages (#1924,!3903,!3954)
  - clarify buffer usage flags for ray tracing (#2181,!3939)
  - rename maxRecursionDepth to maxRayPipelineRecursionDepth and maxRayRecursionDepth (#2203,!3937)
  - add queryable maxRayHitAttributeSize and rename members of VkRayTracingPipelineInterfaceCreateInfoKHR (#2102,!3966)
  - update to use `RayTracingKHR` SPIR-V capability
  - add VUs for matching hit group type against geometry type (#2245,!3994)
  - require `RayTMaxKHR` be volatile in intersection shaders (#2268,!4030)
  - add numerical limits for ray parameters (#2235,!3960)
  - fix SBT indexing rules for device addresses (#2308,!4079)
  - relax formula for ray intersection candidate determination (#2322,!4080)
  - add more details on `ShaderRecordBufferKHR` variables (#2230,!4083)
  - clarify valid bits for `InstanceCustomIndexKHR` (GLSL/GLSL#19,!4128)
  - allow at most one `IncomingRayPayloadKHR`, `IncomingCallableDataKHR`, and `HitAttributeKHR` (!4129)
  - add minimum for maxShaderGroupStride (#2353,!4131)
  - require VK_KHR_pipeline_library extension to be supported (#2348,!4135)
  - clarify meaning of 'geometry index' (#2272,!4137)
  - restrict traces to TLAS (#2239,!4141)
  - add note about maxPipelineRayPayloadSize (#2383,!4172)

- do not require raygen shader in pipeline libraries (!4185)
- define sync for indirect trace rays and indirect buffer (#2407,!4208)

VK_KHR_ray_tracing_position_fetch

Name String

`VK_KHR_ray_tracing_position_fetch`

Extension Type

Device extension

Registered Extension Number

482

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_acceleration_structure

SPIR-V Dependencies

- SPV_KHR_ray_tracing_position_fetch

Contact

- Eric Werness

Extension Proposal

VK_KHR_ray_tracing_position_fetch

Other Extension Metadata

Last Modified Date

2023-02-17

Interactions and External Dependencies

- This extension provides API support for `GLSL_EXT_ray_tracing_position_fetch`
- Interacts with `VK_KHR_ray_tracing_pipeline`
- Interacts with `VK_KHR_ray_query`

Contributors

- Eric Werness, NVIDIA
- Stu Smith, AMD
- Yuriy O’Donnell, Epic Games
- Ralph Potter, Samsung
- Joshua Barczak, Intel
- Lionel Landwerlin, Intel

- Andrew Garrard, Imagination Technologies
- Alex Bourd, Qualcomm
- Yunpeng Zhu, Huawei Technologies
- Marius Bjorge, Arm
- Daniel Koch, NVIDIA

Description

`VK_KHR_ray_tracing_position_fetch` adds the ability to fetch the vertex positions in the shader from a hit triangle as stored in the acceleration structure.

An application adds VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_BIT_KHR to the acceleration structure at build time. Then, if the hit is a triangle geometry, the shader (any-hit or closest hit for ray pipelines or using ray query) **can** fetch the three, three-component vertex positions in object space, of the triangle which was hit.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR

New Enum Constants

- `VK_KHR_RAY_TRACING_POSITION_FETCH_EXTENSION_NAME`
- `VK_KHR_RAY_TRACING_POSITION_FETCH_SPEC_VERSION`
- Extending VkBuildAccelerationStructureFlagBitsKHR:
  - VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_BIT_KHR
  - VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_POSITION_FETCH_FEATURES_KHR

New Built-In Variables

- `HitTriangleVertexPositionsKHR`

New SPIR-V Capabilities

- RayTracingPositionFetchKHR
- RayQueryPositionFetchKHR

Issues

None Yet!

Version History

- Revision 1, 2023-02-17 (Eric Werness)
    - internal revisions

VK_KHR_robustness2

Name String

`VK_KHR_robustness2`

Extension Type

Device extension

Registered Extension Number

613

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2025-01-10

IP Status

No known IP claims.

Contributors

- Liam Middlebrook, NVIDIA
- Jeff Bolz, NVIDIA

Description

This extension is based on the `VK_EXT_robustness2` extension. This extension adds stricter requirements for how out of bounds reads and writes are handled. Most accesses **must** be tightly bounds-checked, out of bounds writes **must** be discarded, out of bound reads **must** return zero. Rather than allowing multiple possible (0,0,0,x) vectors, the out of bounds values are treated as zero, and then missing components are inserted based on the format as described in Component Substitution and vertex input attribute extraction.

These additional requirements **may** be expensive on some implementations, and should only be enabled when truly necessary.

This extension also adds support for “null descriptors”, where VK_NULL_HANDLE **can** be used instead of a valid handle. Accesses to null descriptors have well-defined behavior, and do not rely on robustness.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceRobustness2FeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceRobustness2PropertiesKHR

New Enum Constants

- `VK_KHR_ROBUSTNESS_2_EXTENSION_NAME`
- `VK_KHR_ROBUSTNESS_2_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_PROPERTIES_KHR

Issues

1. Why do VkPhysicalDeviceRobustness2PropertiesKHR::`robustUniformBufferAccessSizeAlignment` and VkPhysicalDeviceRobustness2PropertiesKHR::`robustStorageBufferAccessSizeAlignment` exist?

RESOLVED: Some implementations cannot efficiently tightly bounds-check all buffer accesses. Rather, the size of the bound range is padded to some power of two multiple, up to 256 bytes for uniform buffers and up to 4 bytes for storage buffers, and that padded size is bounds-checked. This is sufficient to implement D3D-like behavior, because D3D only allows binding whole uniform buffers or ranges that are a multiple of 256 bytes, and D3D raw and structured buffers only support 32-bit accesses.

Version History

- Revision 1, 2025-01-10 (Piers Daniell)
  - Internal revisions

VK_KHR_shader_abort

Name String

`VK_KHR_shader_abort`

Extension Type

Device extension

Registered Extension Number

234

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_device_fault
and
VK_KHR_shader_constant_data

SPIR-V Dependencies

- SPV_KHR_abort

Contact

- Tobias Hector [GitHub]tobski

Extension Proposal

VK_KHR_shader_abort

Other Extension Metadata

Last Modified Date

2026-03-18

IP Status

No known IP claims.

Contributors

- Tobias Hector, AMD
- Piers Daniell, Nvidia
- Craig Graham, Samsung
- Erik Hogeman, ARM
- Ralph Potter, Samsung
- Vikram Tarikere, IMG

Description

This extension enables the use of the `OpAbortKHR` instruction in shaders.

New Structures

- Extending VkDeviceFaultDebugInfoKHR:
  - VkDeviceFaultShaderAbortMessageInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderAbortFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceShaderAbortPropertiesKHR

New Enum Constants

- `VK_KHR_SHADER_ABORT_EXTENSION_NAME`
- `VK_KHR_SHADER_ABORT_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DEVICE_FAULT_SHADER_ABORT_MESSAGE_INFO_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ABORT_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ABORT_PROPERTIES_KHR

Version History

- Revision 1, 2024-08-22 (Tobias Hector)
  - Initial revision

VK_KHR_shader_bfloat16

Name String

`VK_KHR_shader_bfloat16`

Extension Type

Device extension

Registered Extension Number

142

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_KHR_cooperative_matrix

SPIR-V Dependencies

- SPV_KHR_bfloat16

Contact

- Tobias Hector [GitHub]tobski

Extension Proposal

VK_KHR_shader_bfloat16

Other Extension Metadata

Last Modified Date

2024-04-09

IP Status

No known IP claims.

Contributors

- Tobias Hector, AMD
- Stu Smith, AMD
- Jeff Bolz, Nvidia
- Kévin Petit, Arm
- David Neto, Google

- Graeme Leese, Broadcom
- Ruihao Zhang, Qualcomm
- Mark Sheppard, Imagination
- Ben Ashbaugh, Intel
- Dmitry Sidorov, Intel
- Victor Mustya, Intel

Description

This extension enables support for bfloat16 (“brain float”) operations in shaders as defined in `SPV_KHR_bfloat16`.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderBfloat16FeaturesKHR

New Enum Constants

- `VK_KHR_SHADER_BFLOAT16_EXTENSION_NAME`
- `VK_KHR_SHADER_BFLOAT16_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_BFLOAT16_FEATURES_KHR

If VK_KHR_cooperative_matrix is supported:

- Extending VkComponentTypeKHR:
  - VK_COMPONENT_TYPE_BFLOAT16_KHR

New SPIR-V Capabilities

- BFloat16TypeKHR
- BFloat16DotProductKHR
- BFloat16CooperativeMatrixKHR

Version History

- Revision 1, 2024-04-09 (Stu Smith)
  - Initial draft

VK_KHR_shader_clock

Name String

`VK_KHR_shader_clock`

Extension Type

Device extension

Registered Extension Number

182

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_shader_clock

Contact

- Aaron Hagan [GitHub]ahagan

Other Extension Metadata

Last Modified Date

2019-4-25

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for `GL_ARB_shader_clock` and `GL_EXT_shader_realtime_clock`

Contributors

- Aaron Hagan, AMD
- Daniel Koch, NVIDIA

Description

This extension advertises the SPIR-V `ShaderClockKHR` capability for Vulkan, which allows a shader to query a real-time or monotonically incrementing counter at the subgroup level or across the device level. The two valid SPIR-V scopes for `OpReadClockKHR` are `Subgroup` and `Device`.

When using GLSL source-based shading languages, the `clockRealtime*EXT`() timing functions map to the `OpReadClockKHR` instruction with a scope of `Device`, and the `clock*ARB`() timing functions map to the `OpReadClockKHR` instruction with a scope of `Subgroup`.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderClockFeaturesKHR

New Enum Constants

- `VK_KHR_SHADER_CLOCK_EXTENSION_NAME`
- `VK_KHR_SHADER_CLOCK_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_CLOCK_FEATURES_KHR

New SPIR-V Capabilities

- `ShaderClockKHR`

Version History

- Revision 1, 2019-4-25 (Aaron Hagan)
  - Initial revision

VK_KHR_shader_constant_data

Name String

`VK_KHR_shader_constant_data`

Extension Type

Device extension

Registered Extension Number

232

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_constant_data

Contact

- Tobias Hector [GitHub]tobski

Extension Proposal

VK_KHR_shader_constant_data

Other Extension Metadata

Last Modified Date

2026-03-18

IP Status

No known IP claims.

Contributors

- Tobias Hector, AMD
- Piers Daniell, Nvidia
- Craig Graham, Samsung
- Vikram Tarikere, IMG

Description

This extension allows the use of the `SPV_KHR_constant_data` extension in SPIR-V shader modules

which enables the specification and specialization of arrays of constant data.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderConstantDataFeaturesKHR

New Enum Constants

- `VK_KHR_SHADER_CONSTANT_DATA_EXTENSION_NAME`
- `VK_KHR_SHADER_CONSTANT_DATA_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_CONSTANT_DATA_FEATURES_KHR

New SPIR-V Capabilities

- ConstantDataKHR

Version History

- Revision 1, 2024-10-30 (Tobias Hector)
  - Initial revision

VK_KHR_shader_fma

Name String

`VK_KHR_shader_fma`

Extension Type

Device extension

Registered Extension Number

580

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_fma

Contact

- Graeme Leese [GitHub]gnl21

Extension Proposal

VK_KHR_shader_fma

Other Extension Metadata

Last Modified Date

2025-06-10

Contributors

- Graeme Leese, Broadcom

Description

This extension allows applications to use the SPV_KHR_fma extension to obtain correctly-rounded results for fused-multiply add (fma) operations.

Fused-multiply add is a building block of many high-precision numerical functions. It provides better accuracy than separate operations, because of the removal of the intermediate rounding step, and often costs less than the pair of separate operations.

Vulkan currently exposes an fma primitive that can give the reduced cost, but it is not guaranteed to be a fused operation, so the accuracy cannot be relied on. For applications which require the

high accuracy, therefore, the operation must be emulated or the algorithm changed so as not to require fma. This is often vastly more costly, even though fma is supported in much of the underlying hardware.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderFmaFeaturesKHR

New Enum Constants

- `VK_KHR_SHADER_FMA_EXTENSION_NAME`
- `VK_KHR_SHADER_FMA_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FMA_FEATURES_KHR

Version History

- Revision 1, 2025-06-10 (Graeme Leese)
  - Initial draft

VK_KHR_shader_maximal_reconvergence

Name String

VK_KHR_shader_maximal_reconvergence

Extension Type

Device extension

Registered Extension Number

435

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_maximal_reconvergence

Contact

- Alan Baker [GitHub]alan-baker

Extension Proposal

VK_KHR_shader_maximal_reconvergence

Other Extension Metadata

Last Modified Date

2021-11-12

IP Status

No known IP claims.

Interactions and External Dependencies

- Requires SPIR-V 1.3.
- This extension requires SPV_KHR_maximal_reconvergence

Contributors

- Alan Baker, Google

Description

This extension allows the use of the SPV_KHR_maximal_reconvergence SPIR-V extension in shader modules. SPV_KHR_maximal_reconvergence provides stronger guarantees that diverged subgroups will reconverge. These guarantees should match shader author intuition about divergence and

reconvergence of invocations based on the structure of the code in the HLL.

Developers should utilize this extension if they require stronger guarantees about reconvergence than either the core spec or SPV_KHR_subgroup_uniform_control_flow. This extension will define the rules that govern how invocations diverge and reconverge in a way that should match developer intuition. It allows robust programs to be written relying on subgroup operations and other tangled instructions.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderMaximalReconvergenceFeaturesKHR

New Enum Constants

- `VK_KHR_SHADER_MAXIMAL_RECONVERGENCE_EXTENSION_NAME`
- `VK_KHR_SHADER_MAXIMAL_RECONVERGENCE_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_MAXIMAL_RECONVERGENCE_FEATURES_KHR

Version History

- Revision 1, 2021-11-12 (Alan Baker)
  - Internal draft version

VK_KHR_shader_quad_control

Name String

`VK_KHR_shader_quad_control`

Extension Type

Device extension

Registered Extension Number

236

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.1
and
VK_KHR_vulkan_memory_model
or
Vulkan Version 1.2
and
VK_KHR_shader_maximal_reconvergence

SPIR-V Dependencies

- SPV_KHR_quad_control

Contact

- Tobias Hector [GitHub]tobski

Extension Proposal

VK_KHR_shader_quad_control

Other Extension Metadata

Last Modified Date

2023-11-01

IP Status

No known IP claims.

Contributors

- Tobias Hector, AMD
- Bill Licea-Kane, Qualcomm
- Graeme Leese, Broadcom
- Jan-Harald Fredriksen, Arm

- Nicolai Hähnle, AMD
- Jeff Bolz, NVidia
- Alan Baker, Google
- Hans-Kristian Arntzen, Valve

Description

This extension adds new quad any/all operations, requires that derivatives are well-defined in quad-uniform control flow, and adds the ability to require helper invocations participate in group operations.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderQuadControlFeaturesKHR

New Enum Constants

- `VK_KHR_SHADER_QUAD_CONTROL_EXTENSION_NAME`
- `VK_KHR_SHADER_QUAD_CONTROL_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_QUAD_CONTROL_FEATURES_KHR

New SPIR-V Capabilities

- QuadControlKHR

Version History

- Revision 1, 2023-11-01 (Tobias Hector)
  - Initial draft

VK_KHR_shader_relaxed_extended_instruction

Name String

`VK_KHR_shader_relaxed_extended_instruction`

Extension Type

Device extension

Registered Extension Number

559

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_relaxed_extended_instruction

Contact

- Nathan Gauër [GitHub]Keenuts

Extension Proposal

VK_KHR_shader_relaxed_extended_instruction

Other Extension Metadata

Last Modified Date

2024-01-24

IP Status

No known IP claims.

Contributors

- Alan Baker, Google LLC
- Nathan Gauër, Google LLC

Description

This extension allows the use of the `SPV_KHR_relaxed_extended_instruction` extension in SPIR-V shader modules.

It adds a new SPIR-V instruction, which allows some usage of forward references in non-semantic

instruction sets. This extensions interacts with the `SPV_KHR_non_semantic_info` extension, hence with `VK_KHR_shader_non_semantic_info`.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR

New Enum Constants

- `VK_KHR_SHADER_RELAXED_EXTENDED_INSTRUCTION_EXTENSION_NAME`
- `VK_KHR_SHADER_RELAXED_EXTENDED_INSTRUCTION_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_RELAXED_EXTENDED_INSTRUCTION_FEATURES_KHR

Version History

- Revision 1, 2024-01-24 (Nathan Gauër)
  - Initial revision

VK_KHR_shader_subgroup_uniform_control_flow

Name String

`VK_KHR_shader_subgroup_uniform_control_flow`

Extension Type

Device extension

Registered Extension Number

324

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_subgroup_uniform_control_flow

Contact

- Alan Baker [GitHub]alan-baker

Other Extension Metadata

Last Modified Date

2020-08-27

IP Status

No known IP claims.

Interactions and External Dependencies

- Requires SPIR-V 1.3.
- This extension provides API support for `GL_EXT_subgroupuniform_qualifier`

Contributors

- Alan Baker, Google
- Jeff Bolz, NVIDIA

Description

This extension allows the use of the `SPV_KHR_subgroup_uniform_control_flow` SPIR-V extension in shader modules. `SPV_KHR_subgroup_uniform_control_flow` provides stronger guarantees that diverged subgroups will reconverge.

Developers should utilize this extension if they use subgroup operations to reduce the work

performed by a uniform subgroup. This extension will guarantee that uniform subgroup will reconverge in the same manner as invocation groups (see “Uniform Control Flow” in the Khronos SPIR-V Specification).

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR

New Enum Constants

- `VK_KHR_SHADER_SUBGROUP_UNIFORM_CONTROL_FLOW_EXTENSION_NAME`
- `VK_KHR_SHADER_SUBGROUP_UNIFORM_CONTROL_FLOW_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_UNIFORM_CONTROL_FLOW_FEATURES_KHR

Version History

- Revision 1, 2020-08-27 (Alan Baker)
  - Internal draft version

VK_KHR_shader_untyped_pointers

Name String

`VK_KHR_shader_untyped_pointers`

Extension Type

Device extension

Registered Extension Number

388

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2

SPIR-V Dependencies

- SPV_KHR_untyped_pointers

Contact

- Alan Baker [GitHub]alan-baker

Extension Proposal

VK_KHR_shader_untyped_pointers

Last Modified Date

2024-03-26

IP Status

No known IP claims.

Interactions and External Dependencies

- Requires the `SPV_KHR_untyped_pointers` SPIR-V extension.

Contributors

- Alan Baker, Google
- Jan-Harald Fredriksen, Arm
- Tom Olson, Arm
- Spencer Fricke, LunarG
- Shahbaz Youssefi, Google
- Tobias Hector, AMD

This extension adds Vulkan support for the `SPV_KHR_untyped_pointers` SPIR-V extension. It provides an alternative to strongly-typed pointers. Untyped pointers allow shader authors to reinterpret data

accessed through memory and atomic instructions versus the data type declared in the variable without extra conversion instructions. Untyped pointers also provide an efficient translation from templated load/store operations in high-level languages and simplify shaders that support operations, but not storage, on smaller data types (e.g. 16-bit floating-point types).

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderUntypedPointersFeaturesKHR

New Enum Constants

- `VK_KHR_SHADER_UNTYPED_POINTERS_EXTENSION_NAME`
- `VK_KHR_SHADER_UNTYPED_POINTERS_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_UNTYPED_POINTERS_FEATURES_KHR

Version History

- Revision 1, 2024-03-26 (Alan Baker)
  - Internal draft version

VK_KHR_shared_presentable_image

Name String

`VK_KHR_shared_presentable_image`

Extension Type

Device extension

Registered Extension Number

112

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_swapchain
and
VK_KHR_get_surface_capabilities2
and
VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Alon Or-bach [GitHub]alonorbach

Other Extension Metadata

Last Modified Date

2017-03-20

IP Status

No known IP claims.

Contributors

- Alon Or-bach, Samsung Electronics
- Ian Elliott, Google
- Jesse Hall, Google
- Pablo Ceballos, Google
- Chris Forbes, Google
- Jeff Juliano, NVIDIA
- James Jones, NVIDIA

- Daniel Rakos, AMD
- Tobias Hector, Imagination Technologies
- Graham Connor, Imagination Technologies
- Michael Worcester, Imagination Technologies
- Cass Everitt, Oculus
- Johannes Van Waveren, Oculus

Description

This extension extends `VK_KHR_swapchain` to enable creation of a shared presentable image. This allows the application to use the image while the presention engine is accessing it, in order to reduce the latency between rendering and presentation.

New Commands

- vkGetSwapchainStatusKHR

New Structures

- Extending VkSurfaceCapabilities2KHR:
  - VkSharedPresentSurfaceCapabilitiesKHR

New Enum Constants

- `VK_KHR_SHARED_PRESENTABLE_IMAGE_EXTENSION_NAME`
- `VK_KHR_SHARED_PRESENTABLE_IMAGE_SPEC_VERSION`
- Extending VkImageLayout:
  - VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR
- Extending VkPresentModeKHR:
  - VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR
  - VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_SHARED_PRESENT_SURFACE_CAPABILITIES_KHR

Issues

1) Should we allow a Vulkan WSI swapchain to toggle between normal usage and shared presentation usage?

RESOLVED: No. WSI swapchains are typically recreated with new properties instead of having their properties changed. This can also save resources, assuming that fewer images are needed for shared presentation, and assuming that most VR applications do not need to switch between normal and shared usage.

2) Should we have a query for determining how the presentation engine refresh is triggered?

RESOLVED: Yes. This is done via which presentation modes a surface supports.

3) Should the object representing a shared presentable image be an extension of a VkSwapchainKHR or a separate object?

RESOLVED: Extension of a swapchain due to overlap in creation properties and to allow common functionality between shared and normal presentable images and swapchains.

4) What should we call the extension and the new structures it creates?

RESOLVED: Shared presentable image / shared present.

5) Should the `minImageCount` and `presentMode` values of the VkSwapchainCreateInfoKHR be ignored, or required to be compatible values?

RESOLVED: `minImageCount` must be 1, and `presentMode` should be set to either VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR or VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR.

6) What should the layout of the shared presentable image be?

RESOLVED: After acquiring the shared presentable image, the application must transition it to the VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR layout prior to it being used. After this initial transition, any image usage that was requested during swapchain creation **can** be performed on the image without layout transitions being performed.

7) Do we need a new API for the trigger to refresh new content?

RESOLVED: vkQueuePresentKHR to act as API to trigger a refresh, as will allow combination with other compatible extensions to vkQueuePresentKHR.

8) How should an application detect a VK_ERROR_OUT_OF_DATE_KHR error on a swapchain using the VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR present mode?

RESOLVED: Introduce vkGetSwapchainStatusKHR to allow applications to query the status of a swapchain using a shared presentation mode.

9) What should subsequent calls to vkQueuePresentKHR for VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR swapchains be defined to do?

RESOLVED: State that implementations may use it as a hint for updated content.

10) Can the ownership of a shared presentable image be transferred to a different queue?

RESOLVED: No. It is not possible to transfer ownership of a shared presentable image obtained from a swapchain created using VK_SHARING_MODE_EXCLUSIVE after it has been presented.

11) How should vkQueueSubmit behave if a command buffer uses an image from a VK_ERROR_OUT_OF_DATE_KHR swapchain?

RESOLVED: vkQueueSubmit is expected to return the VK_ERROR_DEVICE_LOST error.

12) Can Vulkan provide any guarantee on the order of rendering, to enable beam chasing?

RESOLVED: This could be achieved via use of render passes to ensure strip rendering.

Version History

- Revision 1, 2017-03-20 (Alon Or-bach)
  - Internal revisions

VK_KHR_surface

Name String

`VK_KHR_surface`

Extension Type

Instance extension

Registered Extension Number

1

Revision

25

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- James Jones [GitHub]cubanismo
- Ian Elliott [GitHub]ianelliottus

Other Extension Metadata

Last Modified Date

2016-08-25

IP Status

No known IP claims.

Contributors

- Patrick Doane, Blizzard
- Ian Elliott, LunarG
- Jesse Hall, Google
- James Jones, NVIDIA
- David Mao, AMD
- Norbert Nopper, Freescale
- Alon Or-bach, Samsung
- Daniel Rakos, AMD
- Graham Sellers, AMD
- Jeff Vigil, Qualcomm
- Chia-I Wu, LunarG

- Faith Ekstrand, Intel

Description

The VK_KHR_surface extension is an instance extension. It introduces VkSurfaceKHR objects, which abstract native platform surface or window objects for use with Vulkan. It also provides a way to determine whether a queue family in a physical device supports presenting to particular surface.

Separate extensions for each platform provide the mechanisms for creating VkSurfaceKHR objects, but once created they may be used in this and other platform-independent extensions, in particular the VK_KHR_swapchain extension.

New Object Types

- VkSurfaceKHR

New Commands

- vkDestroySurfaceKHR
- vkGetPhysicalDeviceSurfaceCapabilitiesKHR
- vkGetPhysicalDeviceSurfaceFormatsKHR
- vkGetPhysicalDeviceSurfacePresentModesKHR
- vkGetPhysicalDeviceSurfaceSupportKHR

New Structures

- VkSurfaceCapabilitiesKHR
- VkSurfaceFormatKHR

New Enums

- VkColorSpaceKHR
- VkCompositeAlphaFlagBitsKHR
- VkPresentModeKHR
- VkSurfaceTransformFlagBitsKHR

New Bitmasks

- VkCompositeAlphaFlagsKHR
- VkSurfaceTransformFlagsKHR

New Enum Constants

- VK_KHR_SURFACE_EXTENSION_NAME
- VK_KHR_SURFACE_SPEC_VERSION
- Extending VkObjectType:

    - VK_OBJECT_TYPE_SURFACE_KHR
- Extending VkResult:
    - VK_ERROR_NATIVE_WINDOW_IN_USE_KHR
    - VK_ERROR_SURFACE_LOST_KHR

Examples

NOTE	The example code for the `VK_KHR_surface` and `VK_KHR_swapchain` extensions was removed from the appendix after revision 1.0.29. This WSI example code was ported to the cube demo that is shipped with the official Khronos SDK, and is being kept up-to-date in that location (see: https://github.com/KhronosGroup/Vulkan-Tools/blob/main/cube/cube.c).

Issues

1) Should this extension include a method to query whether a physical device supports presenting to a specific window or native surface on a given platform?

RESOLVED: Yes. Without this, applications would need to create a device instance to determine whether a particular window can be presented to. Knowing that a device supports presentation to a platform in general is not sufficient, as a single machine might support multiple seats, or instances of the platform that each use different underlying physical devices. Additionally, on some platforms, such as the X Window System, different drivers and devices might be used for different windows depending on which section of the desktop they exist on.

2) Should the vkGetPhysicalDeviceSurfaceCapabilitiesKHR, vkGetPhysicalDeviceSurfaceFormatsKHR, and vkGetPhysicalDeviceSurfacePresentModesKHR functions be in this extension and operate on physical devices, rather than being in `VK_KHR_swapchain` (i.e. device extension) and being dependent on VkDevice?

RESOLVED: Yes. While it might be useful to depend on `VkDevice` (and therefore on enabled extensions and features) for the queries, Vulkan was released only with the VkPhysicalDevice versions. Many cases can be resolved by a Valid Usage statement. The following extensible queries are also available to allow additional input parameters:

- vkGetPhysicalDeviceSurfacePresentModes2EXT
- vkGetPhysicalDeviceSurfaceCapabilities2KHR
- vkGetPhysicalDeviceSurfaceFormats2KHR

3) Should Vulkan support Xlib or XCB as the API for accessing the X Window System platform?

RESOLVED: Both. XCB is a more modern and efficient API, but Xlib usage is deeply ingrained in many applications and likely will remain in use for the foreseeable future. Not all drivers necessarily need to support both, but including both as options in the core specification will probably encourage support, which should in turn ease adoption of the Vulkan API in older codebases. Additionally, the performance improvements possible with XCB likely will not have a measurable impact on the performance of Vulkan presentation and other minimal window system

interactions defined here.

4) Should the GBM platform be included in the list of platform enums?

RESOLVED: Deferred, and will be addressed with a platform-specific extension to be written in the future.

Version History

- Revision 1, 2015-05-20 (James Jones)
  - Initial draft, based on LunarG KHR spec, other KHR specs, patches attached to bugs.
- Revision 2, 2015-05-22 (Ian Elliott)
  - Created initial Description section.
  - Removed query for whether a platform requires the use of a queue for presentation, since it was decided that presentation will always be modeled as being part of the queue.
  - Fixed typos and other minor mistakes.
- Revision 3, 2015-05-26 (Ian Elliott)
  - Improved the Description section.
- Revision 4, 2015-05-27 (James Jones)
  - Fixed compilation errors in example code.
- Revision 5, 2015-06-01 (James Jones)
  - Added issues 1 and 2 and made related spec updates.
- Revision 6, 2015-06-01 (James Jones)
  - Merged the platform type mappings table previously removed from VK_KHR_swapchain with the platform description table in this spec.
  - Added issues 3 and 4 documenting choices made when building the initial list of native platforms supported.
- Revision 7, 2015-06-11 (Ian Elliott)
  - Updated table 1 per input from the KHR TSG.
  - Updated issue 4 (GBM) per discussion with Daniel Stone. He will create a platform-specific extension sometime in the future.
- Revision 8, 2015-06-17 (James Jones)
  - Updated enum-extending values using new convention.
  - Fixed the value of VK_SURFACE_PLATFORM_INFO_TYPE_SUPPORTED_KHR.
- Revision 9, 2015-06-17 (James Jones)
  - Rebased on Vulkan API version 126.
- Revision 10, 2015-06-18 (James Jones)
  - Marked issues 2 and 3 resolved.
- Revision 11, 2015-06-23 (Ian Elliott)

    - Examples now show use of function pointers for extension functions.
    - Eliminated extraneous whitespace.
- Revision 12, 2015-07-07 (Daniel Rakos)
    - Added error section describing when each error is expected to be reported.
    - Replaced the term “queue node index” with “queue family index” in the spec as that is the agreed term to be used in the latest version of the core header and spec.
    - Replaced bool32_t with VkBool32.
- Revision 13, 2015-08-06 (Daniel Rakos)
    - Updated spec against latest core API header version.
- Revision 14, 2015-08-20 (Ian Elliott)
    - Renamed this extension and all of its enumerations, types, functions, etc. This makes it compliant with the proposed standard for Vulkan extensions.
    - Switched from “revision” to “version”, including use of the VK_MAKE_VERSION macro in the header file.
    - Did miscellaneous cleanup, etc.
- Revision 15, 2015-08-20 (Ian Elliott—porting a 2015-07-29 change from James Jones)
    - Moved the surface transform enums here from VK_WSI_swapchain so they could be reused by VK_WSI_display.
- Revision 16, 2015-09-01 (James Jones)
    - Restore single-field revision number.
- Revision 17, 2015-09-01 (James Jones)
    - Fix example code compilation errors.
- Revision 18, 2015-09-26 (Jesse Hall)
    - Replaced VkSurfaceDescriptionKHR with the VkSurfaceKHR object, which is created via layered extensions. Added VkDestroySurfaceKHR.
- Revision 19, 2015-09-28 (Jesse Hall)
    - Renamed from VK_EXT_KHR_swapchain to VK_EXT_KHR_surface.
- Revision 20, 2015-09-30 (Jeff Vigil)
    - Add error result VK_ERROR_SURFACE_LOST_KHR.
- Revision 21, 2015-10-15 (Daniel Rakos)
    - Updated the resolution of issue #2 and include the surface capability queries in this extension.
    - Renamed SurfaceProperties to SurfaceCapabilities as it better reflects that the values returned are the capabilities of the surface on a particular device.
    - Other minor cleanup and consistency changes.
- Revision 22, 2015-10-26 (Ian Elliott)
    - Renamed from VK_EXT_KHR_surface to VK_KHR_surface.

- Revision 23, 2015-11-03 (Daniel Rakos)
  - Added allocation callbacks to vkDestroySurfaceKHR.
- Revision 24, 2015-11-10 (Jesse Hall)
  - Removed VkSurfaceTransformKHR. Use VkSurfaceTransformFlagBitsKHR instead.
  - Rename VkSurfaceCapabilitiesKHR member maxImageArraySize to maxImageArrayLayers.
- Revision 25, 2016-01-14 (James Jones)
  - Moved VK_ERROR_NATIVE_WINDOW_IN_USE_KHR from the VK_KHR_android_surface to the VK_KHR_surface extension.
- 2016-08-23 (Ian Elliott)
  - Update the example code, to not have so many characters per line, and to split out a new example to show how to obtain function pointers.
- 2016-08-25 (Ian Elliott)
  - A note was added at the beginning of the example code, stating that it will be removed from future versions of the appendix.

VK_KHR_surface_maintenance1

Name String

VK_KHR_surface_maintenance1

Extension Type

Instance extension

Registered Extension Number

487

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_surface
and
VK_KHR_get_surface_capabilities2

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Extension Proposal

VK_KHR_surface_maintenance1

Other Extension Metadata

Last Modified Date

2025-03-31

Contributors

- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Lionel Landwerlin, Intel
- Shahbaz Youssefi, Google
- Chris Forbes, Google
- Ian Elliott, Google
- Hans-Kristian Arntzen, Valve
- Daniel Stone, Collabora

Description

This extension is based off the VK_EXT_surface_maintenance1 extension.

VK_KHR_surface_maintenance1 adds a collection of window system integration features that were intentionally left out or overlooked in the original VK_KHR_surface extension.

The new features are as follows:

- Allow querying number of min/max images from a surface for a particular presentation mode.
- Allow querying a surface's scaled presentation capabilities.
- Allow querying a surface for the set of presentation modes which can be easily switched between without requiring swapchain recreation.

New Structures

- Extending VkPhysicalDeviceSurfaceInfo2KHR:
  - VkSurfacePresentModeKHR
- Extending VkSurfaceCapabilities2KHR:
  - VkSurfacePresentModeCompatibilityKHR
  - VkSurfacePresentScalingCapabilitiesKHR

New Enums

- VkPresentGravityFlagBitsKHR
- VkPresentScalingFlagBitsKHR

New Bitmasks

- VkPresentGravityFlagsKHR
- VkPresentScalingFlagsKHR

New Enum Constants

- VK_KHR_SURFACE_MAINTENANCE_1_EXTENSION_NAME
- VK_KHR_SURFACE_MAINTENANCE_1_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_SURFACE_PRESENT_MODE_COMPATIBILITY_KHR
  - VK_STRUCTURE_TYPE_SURFACE_PRESENT_MODE_KHR
  - VK_STRUCTURE_TYPE_SURFACE_PRESENT_SCALING_CAPABILITIES_KHR

Version History

- Revision 1, 2025-03-31 (Shahbaz Youssefi)
  - Based on VK_EXT_surface_maintenance1

VK_KHR_surface_protected_capabilities

Name String

VK_KHR_surface_protected_capabilities

Extension Type

Instance extension

Registered Extension Number

240

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.1
and
VK_KHR_get_surface_capabilities2

Contact

- Sandeep Shinde [GitHub]sashinde

Other Extension Metadata

Last Modified Date

2018-12-18

IP Status

No known IP claims.

Contributors

- Sandeep Shinde, NVIDIA
- James Jones, NVIDIA
- Daniel Koch, NVIDIA

Description

This extension extends VkSurfaceCapabilities2KHR, providing applications a way to query whether swapchains **can** be created with the VK_SWAPCHAIN_CREATE_PROTECTED_BIT_KHR flag set.

Vulkan 1.1 added (optional) support for protect memory and protected resources including buffers (VK_BUFFER_CREATE_PROTECTED_BIT), images (VK_IMAGE_CREATE_PROTECTED_BIT), and swapchains (VK_SWAPCHAIN_CREATE_PROTECTED_BIT_KHR). However, on implementations which support multiple windowing systems, not all window systems **may** be able to provide a protected display path.

This extension provides a way to query if a protected swapchain created for a surface (and thus a specific windowing system) **can** be displayed on screen. It extends the existing VkSurfaceCapabilities2KHR structure with a new VkSurfaceProtectedCapabilitiesKHR structure from which the application **can** obtain information about support for protected swapchain creation through vkGetPhysicalDeviceSurfaceCapabilities2KHR.

New Structures

- Extending VkSurfaceCapabilities2KHR:
  - VkSurfaceProtectedCapabilitiesKHR

New Enum Constants

- `VK_KHR_SURFACE_PROTECTED_CAPABILITIES_EXTENSION_NAME`
- `VK_KHR_SURFACE_PROTECTED_CAPABILITIES_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_SURFACE_PROTECTED_CAPABILITIES_KHR

Version History

- Revision 1, 2018-12-18 (Sandeep Shinde, Daniel Koch)
  - Internal revisions.

VK_KHR_swapchain

Name String

`VK_KHR_swapchain`

Extension Type

Device extension

Registered Extension Number

2

Revision

70

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_surface

API Interactions

- Interacts with VK_VERSION_1_1

Contact

- James Jones [GitHub]cubanismo
- Ian Elliott [GitHub]ianelliottus

Other Extension Metadata

Last Modified Date

2017-10-06

IP Status

No known IP claims.

Interactions and External Dependencies

- Interacts with Vulkan 1.1

Contributors

- Patrick Doane, Blizzard
- Ian Elliott, LunarG
- Jesse Hall, Google
- Mathias Heyer, NVIDIA
- James Jones, NVIDIA
- David Mao, AMD
- Norbert Nopper, Freescale

- Alon Or-bach, Samsung
- Daniel Rakos, AMD
- Graham Sellers, AMD
- Jeff Vigil, Qualcomm
- Chia-I Wu, LunarG
- Faith Ekstrand, Intel
- Matthaeus G. Chajdas, AMD
- Ray Smith, ARM

Description

The VK_KHR_swapchain extension is the device-level companion to the VK_KHR_surface extension. It introduces VkSwapchainKHR objects, which provide the ability to present rendering results to a surface.

New Object Types

- VkSwapchainKHR

New Commands

- vkAcquireNextImageKHR
- vkCreateSwapchainKHR
- vkDestroySwapchainKHR
- vkGetSwapchainImagesKHR
- vkQueuePresentKHR

If Vulkan Version 1.1 is supported:

- vkAcquireNextImage2KHR
- vkGetDeviceGroupPresentCapabilitiesKHR
- vkGetDeviceGroupSurfacePresentModesKHR
- vkGetPhysicalDevicePresentRectanglesKHR

New Structures

- VkPresentInfoKHR
- VkSwapchainCreateInfoKHR

If Vulkan Version 1.1 is supported:

- VkAcquireNextImageInfoKHR
- VkDeviceGroupPresentCapabilitiesKHR

- Extending VkBindImageMemoryInfo:
  - VkBindImageMemorySwapchainInfoKHR
- Extending VkImageCreateInfo:
  - VkImageSwapchainCreateInfoKHR
- Extending VkPresentInfoKHR:
  - VkDeviceGroupPresentInfoKHR
- Extending VkSwapchainCreateInfoKHR:
  - VkDeviceGroupSwapchainCreateInfoKHR

New Enums

- VkSwapchainCreateFlagBitsKHR

If Vulkan Version 1.1 is supported:

- VkDeviceGroupPresentModeFlagBitsKHR

New Bitmasks

- VkSwapchainCreateFlagsKHR

If Vulkan Version 1.1 is supported:

- VkDeviceGroupPresentModeFlagsKHR

New Enum Constants

- `VK_KHR_SWAPCHAIN_EXTENSION_NAME`
- `VK_KHR_SWAPCHAIN_SPEC_VERSION`
- Extending VkImageLayout:
  - VK_IMAGE_LAYOUT_PRESENT_SRC_KHR
- Extending VkObjectType:
  - VK_OBJECT_TYPE_SWAPCHAIN_KHR
- Extending VkResult:
  - VK_ERROR_OUT_OF_DATE_KHR
  - VK_SUBOPTIMAL_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PRESENT_INFO_KHR
  - VK_STRUCTURE_TYPE_SWAPCHAIN_CREATE_INFO_KHR

If Vulkan Version 1.1 is supported:

- Extending VkStructureType:

    - VK_STRUCTURE_TYPE_ACQUIRE_NEXT_IMAGE_INFO_KHR
    - VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_SWAPCHAIN_INFO_KHR
    - VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_CAPABILITIES_KHR
    - VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_INFO_KHR
    - VK_STRUCTURE_TYPE_DEVICE_GROUP_SWAPCHAIN_CREATE_INFO_KHR
    - VK_STRUCTURE_TYPE_IMAGE_SWAPCHAIN_CREATE_INFO_KHR
- Extending VkSwapchainCreateFlagBitsKHR:
    - VK_SWAPCHAIN_CREATE_PROTECTED_BIT_KHR
    - VK_SWAPCHAIN_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR

Issues

1) Does this extension allow the application to specify the memory backing of the presentable images?

RESOLVED: No. Unlike standard images, the implementation will allocate the memory backing of the presentable image.

2) What operations are allowed on presentable images?

RESOLVED: This is determined by the image usage flags specified when creating the presentable image's swapchain.

3) Does this extension support MSAA presentable images?

RESOLVED: No. Presentable images are always single-sampled. Multi-sampled rendering must use regular images. To present the rendering results the application must manually resolve the multi-sampled image to a single-sampled presentable image prior to presentation.

4) Does this extension support stereo/multi-view presentable images?

RESOLVED: Yes. The number of views associated with a presentable image is determined by the `imageArrayLayers` specified when creating a swapchain. All presentable images in a given swapchain use the same array size.

5) Are the layers of stereo presentable images half-sized?

RESOLVED: No. The image extents always match those requested by the application.

6) Do the "present" and "acquire next image" commands operate on a queue? If not, do they need to include explicit semaphore objects to interlock them with queue operations?

RESOLVED: The present command operates on a queue. The image ownership operation it represents happens in order with other operations on the queue, so no explicit semaphore object is required to synchronize its actions.

Applications may want to acquire the next image in separate threads from those in which they manage their queue, or in multiple threads. To make such usage easier, the acquire next image

command takes a semaphore to signal as a method of explicit synchronization. The application must later queue a wait for this semaphore before queuing execution of any commands using the image.

7) Does vkAcquireNextImageKHR block if no images are available?

RESOLVED: The command takes a timeout parameter. Special values for the timeout are 0, which makes the call a non-blocking operation, and `UINT64_MAX`, which blocks indefinitely. Values in between will block for up to the specified time. The call will return when an image becomes available or an error occurs. It may, but is not required to, return before the specified timeout expires if the swapchain becomes out of date.

8) Can multiple presents be queued using one vkQueuePresentKHR call?

RESOLVED: Yes. VkPresentInfoKHR contains a list of swapchains and corresponding image indices that will be presented. When supported, all presentations queued with a single vkQueuePresentKHR call will be applied atomically as one operation. The same swapchain must not appear in the list more than once. Later extensions may provide applications stronger guarantees of atomicity for such present operations, and/or allow them to query whether atomic presentation of a particular group of swapchains is possible.

9) How do the presentation and acquire next image functions notify the application the targeted surface has changed?

RESOLVED: Two new result codes are introduced for this purpose:

- VK_SUBOPTIMAL_KHR - Presentation will still succeed, subject to the window resize behavior, but the swapchain is no longer configured optimally for the surface it targets. Applications should query updated surface information and recreate their swapchain at the next convenient opportunity.
- VK_ERROR_OUT_OF_DATE_KHR - Failure. The swapchain is no longer compatible with the surface it targets. The application must query updated surface information and recreate the swapchain before presentation will succeed.

These can be returned by both vkAcquireNextImageKHR and vkQueuePresentKHR.

10) Does the vkAcquireNextImageKHR command return a semaphore to the application via an output parameter, or accept a semaphore to signal from the application as an object handle parameter?

RESOLVED: Accept a semaphore to signal as an object handle. This avoids the need to specify whether the application must destroy the semaphore or whether it is owned by the swapchain, and if the latter, what its lifetime is and whether it can be reused for other operations once it is received from vkAcquireNextImageKHR.

11) What types of swapchain queuing behavior should be exposed? Options include swap interval specification, mailbox/most recent vs. FIFO queue management, targeting specific vertical blank intervals or absolute times for a given present operation, and probably others. For some of these, whether they are specified at swapchain creation time or as per-present parameters needs to be decided as well.

RESOLVED: The base swapchain extension will expose 3 possible behaviors (of which, FIFO will always be supported):

- Immediate present: Does not wait for vertical blanking period to update the current image, likely resulting in visible tearing. No internal queue is used. Present requests are applied immediately.
- Mailbox queue: Waits for the next vertical blanking period to update the current image. No tearing should be observed. An internal single-entry queue is used to hold pending presentation requests. If the queue is full when a new presentation request is received, the new request replaces the existing entry, and any images associated with the prior entry become available for reuse by the application.
- FIFO queue: Waits for the next vertical blanking period to update the current image. No tearing should be observed. An internal queue containing `numSwapchainImages` - 1 entries is used to hold pending presentation requests. New requests are appended to the end of the queue, and one request is removed from the beginning of the queue and processed during each vertical blanking period in which the queue is non-empty

Not all surfaces will support all of these modes, so the modes supported will be returned using a surface information query. All surfaces must support the FIFO queue mode. Applications must choose one of these modes up front when creating a swapchain. Switching modes can be accomplished by recreating the swapchain.

12) Can VK_PRESENT_MODE_MAILBOX_KHR provide non-blocking guarantees for vkAcquireNextImageKHR? If so, what is the proper criteria?

RESOLVED: Yes. The difficulty is not immediately obvious here. Naively, if at least 3 images are requested, mailbox mode should always have an image available for the application if the application does not own any images when the call to vkAcquireNextImageKHR was made. However, some presentation engines may have more than one “current” image, and would still need to block in some cases. The right requirement appears to be that if the application allocates the surface’s minimum number of images + 1 then it is guaranteed non-blocking behavior when it does not currently own any images.

13) Is there a way to create and initialize a new swapchain for a surface that has generated a VK_SUBOPTIMAL_KHR return code while still using the old swapchain?

RESOLVED: Not as part of this specification. This could be useful to allow the application to create an “optimal” replacement swapchain and rebuild all its command buffers using it in a background thread at a low priority while continuing to use the “suboptimal” swapchain in the main thread. It could probably use the same “atomic replace” semantics proposed for recreating direct-to-device swapchains without incurring a mode switch. However, after discussion, it was determined some platforms probably could not support concurrent swapchains for the same surface though, so this will be left out of the base KHR extensions. A future extension could add this for platforms where it is supported.

14) Should there be a special value for VkSurfaceCapabilitiesKHR::`maxImageCount` to indicate there are no practical limits on the number of images in a swapchain?

RESOLVED: Yes. There will often be cases where there is no practical limit to the number of images

in a swapchain other than the amount of available resources (i.e., memory) in the system. Trying to derive a hard limit from things like memory size is prone to failure. It is better in such cases to leave it to applications to figure such soft limits out via trial/failure iterations.

15) Should there be a special value for VkSurfaceCapabilitiesKHR::`currentExtent` to indicate the size of the platform surface is undefined?

RESOLVED: Yes. On some platforms (Wayland, for example), the surface size is defined by the images presented to it rather than the other way around.

16) Should there be a special value for VkSurfaceCapabilitiesKHR::`maxImageExtent` to indicate there is no practical limit on the surface size?

RESOLVED: No. It seems unlikely such a system would exist. 0 could be used to indicate the platform places no limits on the extents beyond those imposed by Vulkan for normal images, but this query could just as easily return those same limits, so a special "unlimited" value does not seem useful for this field.

17) How should surface rotation and mirroring be exposed to applications? How do they specify rotation and mirroring transforms applied prior to presentation?

RESOLVED: Applications can query both the supported and current transforms of a surface. Both are specified relative to the device's "natural" display rotation and direction. The supported transforms indicate which orientations the presentation engine accepts images in. For example, a presentation engine that does not support transforming surfaces as part of presentation, and which is presenting to a surface that is displayed with a 90-degree rotation, would return only one supported transform bit: VK_SURFACE_TRANSFORM_ROTATE_90_BIT_KHR. Applications must transform their rendering by the transform they specify when creating the swapchain in `preTransform` field.

18) Can surfaces ever not support `VK_MIRROR_NONE`? Can they support vertical and horizontal mirroring simultaneously? Relatedly, should `VK_MIRROR_NONE`[_BIT] be zero, or bit one, and should applications be allowed to specify multiple pre and current mirror transform bits, or exactly one?

RESOLVED: Since some platforms may not support presenting with a transform other than the native window's current transform, and prerotation/mirroring are specified relative to the device's natural rotation and direction, rather than relative to the surface's current rotation and direction, it is necessary to express lack of support for no mirroring. To allow this, the `MIRROR_NONE` enum must occupy a bit in the flags. Since `MIRROR_NONE` must be a bit in the bitmask rather than a bitmask with no values set, allowing more than one bit to be set in the bitmask would make it possible to describe undefined transforms such as `VK_MIRROR_NONE_BIT | VK_MIRROR_HORIZONTAL_BIT`, or a transform that includes both "no mirroring" and "horizontal mirroring" simultaneously. Therefore, it is desirable to allow specifying all supported mirroring transforms using only one bit. The question then becomes, should there be a `VK_MIRROR_HORIZONTAL_AND_VERTICAL_BIT` to represent a simultaneous horizontal and vertical mirror transform? However, such a transform is equivalent to a 180 degree rotation, so presentation engines and applications that wish to support or use such a transform can express it through rotation instead. Therefore, 3 exclusive bits are sufficient to express all needed mirroring transforms.

19) Should support for sRGB be required?

RESOLVED: In the advent of UHD and HDR display devices, proper color space information is vital to the display pipeline represented by the swapchain. The application can discover the supported format/color-space pairs and select a pair most suited to its rendering needs. Currently only the sRGB color space is supported, future extensions may provide support for more color spaces. See issues 23 and 24.

20) Is there a mechanism to modify or replace an existing swapchain with one targeting the same surface?

RESOLVED: Yes. This is described above in the text.

21) Should there be a way to set prerotation and mirroring using native APIs when presenting using a Vulkan swapchain?

RESOLVED: Yes. The transforms that can be expressed in this extension are a subset of those possible on native platforms. If a platform exposes a method to specify the transform of presented images for a given surface using native methods and exposes more transforms or other properties for surfaces than Vulkan supports, it might be impossible, difficult, or inconvenient to set some of those properties using Vulkan KHR extensions and some using the native interfaces. To avoid overwriting properties set using native commands when presenting using a Vulkan swapchain, the application can set the pretransform to “inherit”, in which case the current native properties will be used, or if none are available, a platform-specific default will be used. Platforms that do not specify a reasonable default or do not provide native mechanisms to specify such transforms should not include the inherit bits in the `supportedTransforms` bitmask they return in VkSurfaceCapabilitiesKHR.

22) Should the content of presentable images be clipped by objects obscuring their target surface?

RESOLVED: Applications can choose which behavior they prefer. Allowing the content to be clipped could enable more efficient presentation methods on some platforms, but some applications might rely on the content of presentable images to perform techniques such as partial updates or motion blurs.

23) What is the purpose of specifying a VkColorSpaceKHR along with VkFormat when creating a swapchain?

RESOLVED: While Vulkan itself is color space agnostic (e.g. even the meaning of R, G, B, and A can be freely defined by the rendering application), the swapchain eventually will have to present the images on a display device with specific color reproduction characteristics. If any color space transformations are necessary before an image can be displayed, the color space of the presented image must be known to the swapchain. A swapchain will only support a restricted set of color format and -space pairs. This set can be discovered via vkGetPhysicalDeviceSurfaceFormatsKHR. As it can be expected that most display devices support the sRGB color space, at least one format/color-space pair has to be exposed, where the color space is VK_COLOR_SPACE_SRGB_NONLINEAR_KHR.

24) How are sRGB formats and the sRGB color space related?

RESOLVED: While Vulkan exposes a number of SRGB texture formats, using such formats does not guarantee working in a specific color space. It merely means that the hardware can directly support applying the non-linear transfer functions defined by the sRGB standard color space when

reading from or writing to images of those formats. Still, it is unlikely that a swapchain will expose a *_SRGB format along with any color space other than VK_COLOR_SPACE_SRGB_NONLINEAR_KHR.

On the other hand, non-*_SRGB formats will be very likely exposed in pair with a SRGB color space. This means, the hardware will not apply any transfer function when reading from or writing to such images, yet they will still be presented on a device with sRGB display characteristics. In this case the application is responsible for applying the transfer function, for instance by using shader math.

25) How are the lifetimes of surfaces and swapchains targeting them related?

RESOLVED: A surface must outlive any swapchains targeting it. A VkSurfaceKHR owns the binding of the native window to the Vulkan driver.

26) How can the application control the way the alpha component of swapchain images is treated by the presentation engine during compositing?

RESOLVED: We should add new enum values to allow the application to negotiate with the presentation engine on how to treat image alpha values during the compositing process. Since not all platforms can practically control this through the Vulkan driver, a value of VK_COMPOSITE_ALPHA_INHERIT_BIT_KHR is provided like for surface transforms.

27) Is vkCreateSwapchainKHR the right function to return VK_ERROR_NATIVE_WINDOW_IN_USE_KHR, or should the various platform-specific VkSurfaceKHR factory functions catch this error earlier?

RESOLVED: For most platforms, the VkSurfaceKHR structure is a simple container holding the data that identifies a native window or other object representing a surface on a particular platform. For the surface factory functions to return this error, they would likely need to register a reference on the native objects with the native display server somehow, and ensure no other such references exist. Surfaces were not intended to be that heavyweight.

Swapchains are intended to be the objects that directly manipulate native windows and communicate with the native presentation mechanisms. Swapchains will already need to communicate with the native display server to negotiate allocation and/or presentation of presentable images for a native surface. Therefore, it makes more sense for swapchain creation to be the point at which native object exclusivity is enforced. Platforms may choose to enforce further restrictions on the number of VkSurfaceKHR objects that may be created for the same native window if such a requirement makes sense on a particular platform, but a global requirement is only sensible at the swapchain level.

Examples

NOTE	The example code for the VK_KHR_surface and VK_KHR_swapchain extensions was removed from the appendix after revision 1.0.29. This WSI example code was ported to the cube demo that is shipped with the official Khronos SDK, and is being kept up-to-date in that location (see: https://github.com/KhronosGroup/Vulkan-Tools/blob/main/cube/cube.c).

Version History

- Revision 1, 2015-05-20 (James Jones)
  - Initial draft, based on LunarG KHR spec, other KHR specs, patches attached to bugs.
- Revision 2, 2015-05-22 (Ian Elliott)
  - Made many agreed-upon changes from 2015-05-21 KHR TSG meeting. This includes using only a queue for presentation, and having an explicit function to acquire the next image.
  - Fixed typos and other minor mistakes.
- Revision 3, 2015-05-26 (Ian Elliott)
  - Improved the Description section.
  - Added or resolved issues that were found in improving the Description. For example, pSurfaceDescription is used consistently, instead of sometimes using pSurface.
- Revision 4, 2015-05-27 (James Jones)
  - Fixed some grammatical errors and typos
  - Filled in the description of imageUseFlags when creating a swapchain.
  - Added a description of swapInterval.
  - Replaced the paragraph describing the order of operations on a queue for image ownership and presentation.
- Revision 5, 2015-05-27 (James Jones)
  - Imported relevant issues from the (abandoned) vk_wsi_persistent_swapchain_images extension.
  - Added issues 6 and 7, regarding behavior of the acquire next image and present commands with respect to queues.
  - Updated spec language and examples to align with proposed resolutions to issues 6 and 7.
- Revision 6, 2015-05-27 (James Jones)
  - Added issue 8, regarding atomic presentation of multiple swapchains
  - Updated spec language and examples to align with proposed resolution to issue 8.
- Revision 7, 2015-05-27 (James Jones)
  - Fixed compilation errors in example code, and made related spec fixes.
- Revision 8, 2015-05-27 (James Jones)
  - Added issue 9, and the related VK_SUBOPTIMAL_KHR result code.
  - Renamed VK_OUT_OF_DATE_KHR to VK_ERROR_OUT_OF_DATE_KHR.
- Revision 9, 2015-05-27 (James Jones)
  - Added inline proposed resolutions (marked with [JRJ]) to some XXX questions/issues. These should be moved to the issues section in a subsequent update if the proposals are adopted.
- Revision 10, 2015-05-28 (James Jones)
  - Converted vkAcquireNextImageKHR back to a non-queue operation that uses a

VkSemaphore object for explicit synchronization.

  - Added issue 10 to determine whether vkAcquireNextImageKHR generates or returns semaphores, or whether it operates on a semaphore provided by the application.

- Revision 11, 2015-05-28 (James Jones)
  - Marked issues 6, 7, and 8 resolved.
  - Renamed VkSurfaceCapabilityPropertiesKHR to VkSurfacePropertiesKHR to better convey the mutable nature of the information it contains.
- Revision 12, 2015-05-28 (James Jones)
  - Added issue 11 with a proposed resolution, and the related issue 12.
  - Updated various sections of the spec to match the proposed resolution to issue 11.
- Revision 13, 2015-06-01 (James Jones)
  - Moved some structures to VK_EXT_KHR_swap_chain to resolve the specification's issues 1 and 2.
- Revision 14, 2015-06-01 (James Jones)
  - Added code for example 4 demonstrating how an application might make use of the two different present and acquire next image KHR result codes.
  - Added issue 13.
- Revision 15, 2015-06-01 (James Jones)
  - Added issues 14 - 16 and related spec language.
  - Fixed some spelling errors.
  - Added language describing the meaningful return values for vkAcquireNextImageKHR and vkQueuePresentKHR.
- Revision 16, 2015-06-02 (James Jones)
  - Added issues 17 and 18, as well as related spec language.
  - Removed some erroneous text added by mistake in the last update.
- Revision 17, 2015-06-15 (Ian Elliott)
  - Changed special value from "-1" to "0" so that the data types can be unsigned.
- Revision 18, 2015-06-15 (Ian Elliott)
  - Clarified the values of VkSurfacePropertiesKHR::minImageCount and the timeout parameter of the vkAcquireNextImageKHR function.
- Revision 19, 2015-06-17 (James Jones)
  - Misc. cleanup. Removed resolved inline issues and fixed typos.
  - Fixed clarification of VkSurfacePropertiesKHR::minImageCount made in version 18.
  - Added a brief "Image Ownership" definition to the list of terms used in the spec.
- Revision 20, 2015-06-17 (James Jones)
  - Updated enum-extending values using new convention.

- Revision 21, 2015-06-17 (James Jones)
  - Added language describing how to use VK_IMAGE_LAYOUT_PRESENT_SOURCE_KHR.
  - Cleaned up an XXX comment regarding the description of which queues vkQueuePresentKHR can be used on.
- Revision 22, 2015-06-17 (James Jones)
  - Rebased on Vulkan API version 126.
- Revision 23, 2015-06-18 (James Jones)
  - Updated language for issue 12 to read as a proposed resolution.
  - Marked issues 11, 12, 13, 16, and 17 resolved.
  - Temporarily added links to the relevant bugs under the remaining unresolved issues.
  - Added issues 19 and 20 as well as proposed resolutions.
- Revision 24, 2015-06-19 (Ian Elliott)
  - Changed special value for VkSurfacePropertiesKHR::currentExtent back to “-1” from “0”. This value will never need to be unsigned, and “0” is actually a legal value.
- Revision 25, 2015-06-23 (Ian Elliott)
  - Examples now show use of function pointers for extension functions.
  - Eliminated extraneous whitespace.
- Revision 26, 2015-06-25 (Ian Elliott)
  - Resolved Issues 9 & 10 per KHR TSG meeting.
- Revision 27, 2015-06-25 (James Jones)
  - Added oldSwapchain member to VkSwapchainCreateInfoKHR.
- Revision 28, 2015-06-25 (James Jones)
  - Added the “inherit” bits to the rotation and mirroring flags and the associated issue 21.
- Revision 29, 2015-06-25 (James Jones)
  - Added the “clipped” flag to VkSwapchainCreateInfoKHR, and the associated issue 22.
  - Specified that presenting an image does not modify it.
- Revision 30, 2015-06-25 (James Jones)
  - Added language to the spec that clarifies the behavior of vkCreateSwapchainKHR() when the oldSwapchain field of VkSwapchainCreateInfoKHR is not NULL.
- Revision 31, 2015-06-26 (Ian Elliott)
  - Example of new VkSwapchainCreateInfoKHR members, “oldSwapchain” and “clipped”.
  - Example of using VkSurfacePropertiesKHR::{min|max}ImageCount to set VkSwapchainCreateInfoKHR::minImageCount.
  - Rename vkGetSurfaceInfoKHR()'s 4th parameter to “pDataSize”, for consistency with other functions.
  - Add macro with C-string name of extension (just to header file).

- Revision 32, 2015-06-26 (James Jones)
  - Minor adjustments to the language describing the behavior of “oldSwapchain”
  - Fixed the version date on my previous two updates.
- Revision 33, 2015-06-26 (Jesse Hall)
  - Add usage flags to VkSwapchainCreateInfoKHR
- Revision 34, 2015-06-26 (Ian Elliott)
  - Rename vkQueuePresentKHR()'s 2nd parameter to “pPresentInfo”, for consistency with other functions.
- Revision 35, 2015-06-26 (Faith Ekstrand)
  - Merged the VkRotationFlagBitsKHR and VkMirrorFlagBitsKHR enums into a single VkSurfaceTransformFlagBitsKHR enum.
- Revision 36, 2015-06-26 (Faith Ekstrand)
  - Added a VkSurfaceTransformKHR enum that is not a bitmask. Each value in VkSurfaceTransformKHR corresponds directly to one of the bits in VkSurfaceTransformFlagBitsKHR so transforming from one to the other is easy. Having a separate enum means that currentTransform and preTransform are now unambiguous by definition.
- Revision 37, 2015-06-29 (Ian Elliott)
  - Corrected one of the signatures of vkAcquireNextImageKHR, which had the last two parameters switched from what it is elsewhere in the specification and header files.
- Revision 38, 2015-06-30 (Ian Elliott)
  - Corrected a typo in description of the vkGetSwapchainInfoKHR() function.
  - Corrected a typo in header file comment for VkPresentInfoKHR::sType.
- Revision 39, 2015-07-07 (Daniel Rakos)
  - Added error section describing when each error is expected to be reported.
  - Replaced bool32_t with VkBool32.
- Revision 40, 2015-07-10 (Ian Elliott)
  - Updated to work with version 138 of the `vulkan.h` header. This includes declaring the VkSwapchainKHR type using the new VK_DEFINE_NONDISP_HANDLE macro, and no longer extending VkObjectType (which was eliminated).
- Revision 41 2015-07-09 (Mathias Heyer)
  - Added color space language.
- Revision 42, 2015-07-10 (Daniel Rakos)
  - Updated query mechanism to reflect the convention changes done in the core spec.
  - Removed “queue” from the name of VK_STRUCTURE_TYPE_QUEUE_PRESENT_INFO_KHR to be consistent with the established naming convention.
  - Removed reference to the no longer existing VkObjectType enum.

- Revision 43, 2015-07-17 (Daniel Rakos)
  - Added support for concurrent sharing of swapchain images across queue families.
  - Updated sample code based on recent changes
- Revision 44, 2015-07-27 (Ian Elliott)
  - Noted that support for VK_PRESENT_MODE_FIFO_KHR is required. That is ICDs may optionally support IMMEDIATE and MAILBOX, but must support FIFO.
- Revision 45, 2015-08-07 (Ian Elliott)
  - Corrected a typo in spec file (type and variable name had wrong case for the imageColorSpace member of the VkSwapchainCreateInfoKHR struct).
  - Corrected a typo in header file (last parameter in PFN_vkGetSurfacePropertiesKHR was missing “KHR” at the end of type: VkSurfacePropertiesKHR).
- Revision 46, 2015-08-20 (Ian Elliott)
  - Renamed this extension and all of its enumerations, types, functions, etc. This makes it compliant with the proposed standard for Vulkan extensions.
  - Switched from “revision” to “version”, including use of the VK_MAKE_VERSION macro in the header file.
  - Made improvements to several descriptions.
  - Changed the status of several issues, leaving no unresolved issues.
  - Resolved several TODOs, did miscellaneous cleanup, etc.
- Revision 47, 2015-08-20 (Ian Elliott—porting a 2015-07-29 change from James Jones)
  - Moved the surface transform enums to VK_WSI_swapchain so they could be reused by VK_WSI_display.
- Revision 48, 2015-09-01 (James Jones)
  - Various minor cleanups.
- Revision 49, 2015-09-01 (James Jones)
  - Restore single-field revision number.
- Revision 50, 2015-09-01 (James Jones)
  - Update Example #4 to include code that illustrates how to use the oldSwapchain field.
- Revision 51, 2015-09-01 (James Jones)
  - Fix example code compilation errors.
- Revision 52, 2015-09-08 (Matthaeus G. Chajdas)
  - Corrected a typo.
- Revision 53, 2015-09-10 (Alon Or-bach)
  - Removed underscore from SWAP_CHAIN left in VK_STRUCTURE_TYPE_SWAPCHAIN_CREATE_INFO_KHR.
- Revision 54, 2015-09-11 (Jesse Hall)
  - Described the execution and memory coherence requirements for image transitions to and

from VK_IMAGE_LAYOUT_PRESENT_SOURCE_KHR.

- Revision 55, 2015-09-11 (Ray Smith)
  - Added errors for destroying and binding memory to presentable images
- Revision 56, 2015-09-18 (James Jones)
  - Added fence argument to vkAcquireNextImageKHR
  - Added example of how to meter a host thread based on presentation rate.
- Revision 57, 2015-09-26 (Jesse Hall)
  - Replace VkSurfaceDescriptionKHR with VkSurfaceKHR.
  - Added issue 25 with agreed resolution.
- Revision 58, 2015-09-28 (Jesse Hall)
  - Renamed from VK_EXT_KHR_device_swapchain to VK_EXT_KHR_swapchain.
- Revision 59, 2015-09-29 (Ian Elliott)
  - Changed vkDestroySwapchainKHR() to return void.
- Revision 60, 2015-10-01 (Jeff Vigil)
  - Added error result VK_ERROR_SURFACE_LOST_KHR.
- Revision 61, 2015-10-05 (Faith Ekstrand)
  - Added the VkCompositeAlpha enum and corresponding structure fields.
- Revision 62, 2015-10-12 (Daniel Rakos)
  - Added VK_PRESENT_MODE_FIFO_RELAXED_KHR.
- Revision 63, 2015-10-15 (Daniel Rakos)
  - Moved surface capability queries to VK_EXT_KHR_surface.
- Revision 64, 2015-10-26 (Ian Elliott)
  - Renamed from VK_EXT_KHR_swapchain to VK_KHR_swapchain.
- Revision 65, 2015-10-28 (Ian Elliott)
  - Added optional pResult member to VkPresentInfoKHR, so that per-swapchain results can be obtained from vkQueuePresentKHR().
- Revision 66, 2015-11-03 (Daniel Rakos)
  - Added allocation callbacks to create and destroy functions.
  - Updated resource transition language.
  - Updated sample code.
- Revision 67, 2015-11-10 (Jesse Hall)
  - Add reserved flags bitmask to VkSwapchainCreateInfoKHR.
  - Modify naming and member ordering to match API style conventions, and so the VkSwapchainCreateInfoKHR image property members mirror corresponding VkImageCreateInfo members but with an 'image' prefix.

    - Make VkPresentInfoKHR::pResults non-const; it is an output array parameter.
    - Make pPresentInfo parameter to vkQueuePresentKHR const.
- Revision 68, 2016-04-05 (Ian Elliott)
    - Moved the “validity” include for vkAcquireNextImage to be in its proper place, after the prototype and list of parameters.
    - Clarified language about presentable images, including how they are acquired, when applications can and cannot use them, etc. As part of this, removed language about “ownership” of presentable images, and replaced it with more-consistent language about presentable images being “acquired” by the application.
- 2016-08-23 (Ian Elliott)
    - Update the example code, to use the final API command names, to not have so many characters per line, and to split out a new example to show how to obtain function pointers. This code is more similar to the LunarG “cube” demo program.
- 2016-08-25 (Ian Elliott)
    - A note was added at the beginning of the example code, stating that it will be removed from future versions of the appendix.
- Revision 69, 2017-09-07 (Tobias Hector)
    - Added interactions with Vulkan 1.1
- Revision 70, 2017-10-06 (Ian Elliott)
    - Corrected interactions with Vulkan 1.1

VK_KHR_swapchain_maintenance1

Name String

`VK_KHR_swapchain_maintenance1`

Extension Type

Device extension

Registered Extension Number

488

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_swapchain
and
VK_KHR_surface_maintenance1
and
VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Extension Proposal

VK_KHR_swapchain_maintenance1

Other Extension Metadata

Last Modified Date

2025-03-31

Contributors

- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Shahbaz Youssefi, Google
- Chris Forbes, Google
- Ian Elliott, Google
- Yiwei Zhang, Google
- Charlie Lao, Google
- Lina Versace, Google

- Ralph Potter, Samsung
- Igor Nazarov, Samsung
- Hyunchang Kim, Samsung
- Suenghwan Lee, Samsung
- Munseong Kang, Samsung
- Joonyong Park, Samsung
- Hans-Kristian Arntzen, Valve
- Lisa Wu, Arm
- Daniel Stone, Collabora
- Pan Gao, Huawei

Description

This extension is based off the VK_EXT_swapchain_maintenance1 extension.

VK_KHR_swapchain_maintenance1 adds a collection of window system integration features that were intentionally left out or overlooked in the original VK_KHR_swapchain extension.

The new features are as follows:

- Specify a fence that will be signaled when the resources associated with a present operation **can** be safely destroyed.
- Allow changing the present mode a swapchain is using at per-present granularity.
- Allow applications to define the behavior when presenting a swapchain image to a surface with different dimensions than the image. Using this feature **may** allow implementations to avoid returning VK_ERROR_OUT_OF_DATE_KHR in this situation.
- Allow applications to defer swapchain memory allocation for improved startup time and memory footprint.
- Allow applications to release previously acquired images without presenting them.

New Commands

- vkReleaseSwapchainImagesKHR

New Structures

- VkReleaseSwapchainImagesInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceSwapchainMaintenance1FeaturesKHR
- Extending VkPresentInfoKHR:
  - VkSwapchainPresentFenceInfoKHR
  - VkSwapchainPresentModeInfoKHR

- Extending VkSwapchainCreateInfoKHR:
  - VkSwapchainPresentModesCreateInfoKHR
  - VkSwapchainPresentScalingCreateInfoKHR

New Enum Constants

- VK_KHR_SWAPCHAIN_MAINTENANCE_1_EXTENSION_NAME
- VK_KHR_SWAPCHAIN_MAINTENANCE_1_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SWAPCHAIN_MAINTENANCE_1_FEATURES_KHR
  - VK_STRUCTURE_TYPE_RELEASE_SWAPCHAIN_IMAGES_INFO_KHR
  - VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_FENCE_INFO_KHR
  - VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_MODES_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_MODE_INFO_KHR
  - VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_SCALING_CREATE_INFO_KHR
- Extending VkSwapchainCreateFlagBitsKHR:
  - VK_SWAPCHAIN_CREATE_DEFERRED_MEMORY_ALLOCATION_BIT_KHR

Version History

- Revision 1, 2025-03-31 (Shahbaz Youssefi)
  - Based on VK_EXT_swapchain_maintenance1

VK_KHR_swapchain_mutable_format

Name String

VK_KHR_swapchain_mutable_format

Extension Type

Device extension

Registered Extension Number

201

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_swapchain
and
  VK_KHR_maintenance2
  or
  Vulkan Version 1.1
and
  VK_KHR_image_format_list
  or
  Vulkan Version 1.2

Contact

- Daniel Rakos [GitHub]drakos-amd

Other Extension Metadata

Last Modified Date

2018-03-28

IP Status

No known IP claims.

Contributors

- Faith Ekstrand, Intel
- Jan-Harald Fredriksen, ARM
- Jesse Hall, Google
- Daniel Rakos, AMD
- Ray Smith, ARM

Description

This extension allows processing of swapchain images as different formats to that used by the window system, which is particularly useful for switching between sRGB and linear RGB formats.

It adds a new swapchain creation flag that enables creating image views from presentable images with a different format than the one used to create the swapchain.

New Enum Constants

- VK_KHR_SWAPCHAIN_MUTABLE_FORMAT_EXTENSION_NAME
- VK_KHR_SWAPCHAIN_MUTABLE_FORMAT_SPEC_VERSION
- Extending VkSwapchainCreateFlagBitsKHR:
  - VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR

Issues

1) Are there any new capabilities needed?

RESOLVED: No. It is expected that all implementations exposing this extension support swapchain image format mutability.

2) Do we need a separate VK_SWAPCHAIN_CREATE_EXTENDED_USAGE_BIT_KHR?

RESOLVED: No. This extension requires VK_KHR_maintenance2 and presentable images of swapchains created with VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR are created internally in a way equivalent to specifying both VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT and VK_IMAGE_CREATE_EXTENDED_USAGE_BIT.

3) Do we need a separate structure to allow specifying an image format list for swapchains?

RESOLVED: No. We use the VkImageFormatListCreateInfo structure introduced by VK_KHR_image_format_list. The structure is required to be included in the pNext chain of VkSwapchainCreateInfoKHR for swapchains created with VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR.

Version History

- Revision 1, 2018-03-28 (Daniel Rakos)
  - Internal revisions.

VK_KHR_unified_image_layouts

Name String

VK_KHR_unified_image_layouts

Extension Type

Device extension

Registered Extension Number

528

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_VERSION_1_3
- Interacts with VK_EXT_attachment_feedback_loop_layout
- Interacts with VK_KHR_dynamic_rendering

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Extension Proposal

VK_KHR_unified_image_layouts

Other Extension Metadata

Last Modified Date

2024-10-15

Interactions and External Dependencies

- This extension interacts with VK_EXT_attachment_feedback_loop_layout
- This extension interacts with VK_KHR_video_decode_queue
- This extension interacts with VK_KHR_video_encode_queue
- This extension interacts with VK_KHR_video_encode_quantization_map

Contributors

- Ahmed Abdelkhalek, AMD
- Tobias Hector, AMD

- Jan-Harald Fredriksen, ARM
- Ting Wei, ARM
- Faith Ekstrand, Collabora
- Lina Versace, Google
- Shahbaz Youssefi, Google
- James Fitzpatrick, Imagination
- Daniel Story, Nintendo
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Piers Daniell, NVIDIA
- Tony Zlatinski, NVIDIA
- Matthew Netsch, Qualcomm
- Patrick Boyle, Qualcomm
- Daniel Rakos, RasterGrid
- Ralph Potter, Samsung
- Hans-Kristian Arntzen, VALVE
- Samuel Pitoiset, VALVE

Description

This extension significantly simplifies synchronization in Vulkan by removing the need for image layout transitions in most cases. In particular, it guarantees that using the `VK_IMAGE_LAYOUT_GENERAL` layout everywhere possible is just as efficient as using the other layouts.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceUnifiedImageLayoutsFeaturesKHR

If VK_EXT_attachment_feedback_loop_layout and Vulkan Version 1.3 or VK_KHR_dynamic_rendering is supported:

- Extending VkRenderingAttachmentInfo:
  - VkAttachmentFeedbackLoopInfoEXT

New Enum Constants

- `VK_KHR_UNIFIED_IMAGE_LAYOUTS_EXTENSION_NAME`
- `VK_KHR_UNIFIED_IMAGE_LAYOUTS_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_UNIFIED_IMAGE_LAYOUTS_FEATURES_KHR

If VK_EXT_attachment_feedback_loop_layout and Vulkan Version 1.3 or VK_KHR_dynamic_rendering is supported:

- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_ATTACHMENT_FEEDBACK_LOOP_INFO_EXT

Version History

- Revision 1, 2024-10-15 (Shahbaz Youssefi)
  - Initial revision

VK_KHR_video_decode_av1

Name String

`VK_KHR_video_decode_av1`

Extension Type

Device extension

Registered Extension Number

513

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_video_decode_queue

Contact

- Daniel Rakos [GitHub]aqnuep

Extension Proposal

VK_KHR_video_decode_av1

Other Extension Metadata

Last Modified Date

2024-01-02

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- Benjamin Cheng, AMD
- Ho Hin Lau, AMD
- Lynne Iribarren, Independent
- David Airlie, Red Hat, Inc.
- Ping Liu, Intel
- Srinath Kumarapuram, NVIDIA
- Vassili Nikolaev, NVIDIA
- Tony Zlatinski, NVIDIA
- Charlie Turner, Igalia

- Daniel Almeida, Collabora
- Nicolas Dufresne, Collabora
- Daniel Rakos, RasterGrid

Description

This extension builds upon the `VK_KHR_video_decode_queue` extension by adding support for decoding elementary video stream sequences compliant with the AV1 video compression standard.

New Structures

- Extending VkVideoCapabilitiesKHR:
  - VkVideoDecodeAV1CapabilitiesKHR
- Extending VkVideoDecodeInfoKHR:
  - VkVideoDecodeAV1PictureInfoKHR
- Extending VkVideoProfileInfoKHR, VkQueryPoolCreateInfo:
  - VkVideoDecodeAV1ProfileInfoKHR
- Extending VkVideoReferenceSlotInfoKHR:
  - VkVideoDecodeAV1DpbSlotInfoKHR
- Extending VkVideoSessionParametersCreateInfoKHR:
  - VkVideoDecodeAV1SessionParametersCreateInfoKHR

New Enum Constants

- `VK_KHR_VIDEO_DECODE_AV1_EXTENSION_NAME`
- `VK_KHR_VIDEO_DECODE_AV1_SPEC_VERSION`
- VK_MAX_VIDEO_AV1_REFERENCES_PER_FRAME_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_CAPABILITIES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_DPB_SLOT_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_PICTURE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_PROFILE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_SESSION_PARAMETERS_CREATE_INFO_KHR
- Extending VkVideoCodecOperationFlagBitsKHR:
  - VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR

Version History

- Revision 1, 2024-01-02 (Daniel Rakos)
  - Internal revisions

VK_KHR_video_decode_h264

Name String

`VK_KHR_video_decode_h264`

Extension Type

Device extension

Registered Extension Number

41

Revision

9

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_video_decode_queue

Contact

- peter.fang@amd.com

Extension Proposal

VK_KHR_video_decode_h264

Other Extension Metadata

Last Modified Date

2023-12-05

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- Chunbo Chen, Intel
- HoHin Lau, AMD
- Jake Beju, AMD
- Peter Fang, AMD
- Ping Liu, Intel
- Srinath Kumarapuram, NVIDIA
- Tony Zlatinski, NVIDIA
- Daniel Rakos, RasterGrid

Description

This extension builds upon the VK_KHR_video_decode_queue extension by adding support for decoding elementary video stream sequences compliant with the H.264/AVC video compression standard.

NOTE	This extension was promoted to KHR from the provisional extension VK_EXT_video_decode_h264.

New Structures

- Extending VkVideoCapabilitiesKHR:
  - VkVideoDecodeH264CapabilitiesKHR
- Extending VkVideoDecodeInfoKHR:
  - VkVideoDecodeH264PictureInfoKHR
- Extending VkVideoProfileInfoKHR, VkQueryPoolCreateInfo:
  - VkVideoDecodeH264ProfileInfoKHR
- Extending VkVideoReferenceSlotInfoKHR:
  - VkVideoDecodeH264DpbSlotInfoKHR
- Extending VkVideoSessionParametersCreateInfoKHR:
  - VkVideoDecodeH264SessionParametersCreateInfoKHR
- Extending VkVideoSessionParametersUpdateInfoKHR:
  - VkVideoDecodeH264SessionParametersAddInfoKHR

New Enums

- VkVideoDecodeH264PictureLayoutFlagBitsKHR

New Bitmasks

- VkVideoDecodeH264PictureLayoutFlagsKHR

New Enum Constants

- VK_KHR_VIDEO_DECODE_H264_EXTENSION_NAME
- VK_KHR_VIDEO_DECODE_H264_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_CAPABILITIES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_DPB_SLOT_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_PICTURE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_PROFILE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_SESSION_PARAMETERS_ADD_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_SESSION_PARAMETERS_CREATE_INFO_KHR

- Extending VkVideoCodecOperationFlagBitsKHR:
  - VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR

Version History

- Revision 1, 2018-6-11 (Peter Fang)
  - Initial draft
- Revision 2, March 29 2021 (Tony Zlatinski)
  - Spec and API Updates
- Revision 3, August 1 2021 (Srinath Kumarapuram)
  - Rename `VkVideoDecodeH264FieldLayoutFlagsEXT` to `VkVideoDecodeH264PictureLayoutFlagsEXT`, `VkVideoDecodeH264FieldLayoutFlagBitsEXT` to `VkVideoDecodeH264PictureLayoutFlagBitsEXT` (along with the names of enumerants it defines), and `VkVideoDecodeH264ProfileEXT.fieldLayout` to `VkVideoDecodeH264ProfileEXT.pictureLayout`, following Vulkan naming conventions.
- Revision 4, 2022-03-16 (Ahmed Abdelkhalek)
  - Relocate Std header version reporting/requesting from this extension to VK_KHR_video_queue extension.
  - Remove the now empty VkVideoDecodeH264SessionCreateInfoEXT.
- Revision 5, 2022-03-31 (Ahmed Abdelkhalek)
  - Use type StdVideoH264Level for VkVideoDecodeH264Capabilities.maxLevel
- Revision 6, 2022-08-09 (Daniel Rakos)
  - Rename `VkVideoDecodeH264ProfileEXT` to `VkVideoDecodeH264ProfileInfoEXT`
  - Rename `VkVideoDecodeH264MvcEXT` to `VkVideoDecodeH264MvcInfoEXT`
- Revision 7, 2022-09-18 (Daniel Rakos)
  - Change type of `VkVideoDecodeH264ProfileInfoEXT::pictureLayout` to `VkVideoDecodeH264PictureLayoutFlagBitsEXT`
  - Remove MVC support and related `VkVideoDecodeH264MvcInfoEXT` structure
  - Rename `spsStdCount`, `pSpsStd`, `ppsStdCount`, and `pPpsStd` to `stdSPSCount`, `pStdSPSs`, `stdPPSCount`, and `pStdPPSs`, respectively, in `VkVideoDecodeH264SessionParametersAddInfoEXT`
  - Rename `maxSpsStdCount` and `maxPpsStdCount` to `maxStdSPSCount` and `maxStdPPSCount`, respectively, in `VkVideoDecodeH264SessionParametersCreateInfoEXT`
  - Rename `slicesCount` and `pSlicesDataOffsets` to `sliceCount` and `pSliceOffsets`, respectively, in `VkVideoDecodeH264PictureInfoEXT`
- Revision 8, 2022-09-29 (Daniel Rakos)
  - Change extension from `EXT` to `KHR`
  - Extension is no longer provisional
- Revision 9, 2023-12-05 (Daniel Rakos)

- Condition reference picture setup based on the value of `StdVideoDecodeH264PictureInfo::flags.is_reference`

VK_KHR_video_decode_h265

Name String

`VK_KHR_video_decode_h265`

Extension Type

Device extension

Registered Extension Number

188

Revision

8

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_video_decode_queue

Contact

- peter.fang@amd.com

Extension Proposal

VK_KHR_video_decode_h265

Other Extension Metadata

Last Modified Date

2023-12-05

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- HoHin Lau, AMD
- Jake Beju, AMD
- Peter Fang, AMD
- Ping Liu, Intel
- Srinath Kumarapuram, NVIDIA
- Tony Zlatinski, NVIDIA
- Daniel Rakos, RasterGrid

Description

This extension builds upon the VK_KHR_video_decode_queue extension by adding support for decoding elementary video stream sequences compliant with the H.265/HEVC video compression standard.

NOTE	This extension was promoted to KHR from the provisional extension VK_EXT_video_decode_h265.

New Structures

- Extending VkVideoCapabilitiesKHR:
  - VkVideoDecodeH265CapabilitiesKHR
- Extending VkVideoDecodeInfoKHR:
  - VkVideoDecodeH265PictureInfoKHR
- Extending VkVideoProfileInfoKHR, VkQueryPoolCreateInfo:
  - VkVideoDecodeH265ProfileInfoKHR
- Extending VkVideoReferenceSlotInfoKHR:
  - VkVideoDecodeH265DpbSlotInfoKHR
- Extending VkVideoSessionParametersCreateInfoKHR:
  - VkVideoDecodeH265SessionParametersCreateInfoKHR
- Extending VkVideoSessionParametersUpdateInfoKHR:
  - VkVideoDecodeH265SessionParametersAddInfoKHR

New Enum Constants

- VK_KHR_VIDEO_DECODE_H265_EXTENSION_NAME
- VK_KHR_VIDEO_DECODE_H265_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_CAPABILITIES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_DPB_SLOT_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_PICTURE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_PROFILE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_SESSION_PARAMETERS_ADD_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_SESSION_PARAMETERS_CREATE_INFO_KHR
- Extending VkVideoCodecOperationFlagBitsKHR:
  - VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR

Version History

- Revision 1, 2018-6-11 (Peter Fang)

    - Initial draft
- Revision 1.6, March 29 2021 (Tony Zlatinski)
    - Spec and API updates.
- Revision 2, 2022-03-16 (Ahmed Abdelkhalek)
    - Relocate Std header version reporting/requesting from this extension to VK_KHR_video_queue extension.
    - Remove the now empty VkVideoDecodeH265SessionCreateInfoEXT.
- Revision 3, 2022-03-31 (Ahmed Abdelkhalek)
    - Use type StdVideoH265Level for VkVideoDecodeH265Capabilities.maxLevel
- Revision 4, 2022-08-09 (Daniel Rakos)
    - Rename `VkVideoDecodeH265ProfileEXT` to `VkVideoDecodeH265ProfileInfoEXT`
- Revision 5, 2022-09-18 (Daniel Rakos)
    - Rename `vpsStdCount`, `pVpsStd`, `spsStdCount`, `pSpsStd`, `ppsStdCount`, and `pPpsStd` to `stdVPSCount`, `pStdVPSs`, `stdSPSCount`, `pStdSPSs`, `stdPPSCount`, and `pStdPPSs`, respectively, in `VkVideoDecodeH265SessionParametersAddInfoEXT`
    - Rename `maxVpsStdCount`, `maxSpsStdCount`, and `maxPpsStdCount` to `maxStdVPSCount`, `maxStdSPSCount` and `maxStdPPSCount`, respectively, in `VkVideoDecodeH265SessionParametersCreateInfoEXT`
    - Rename `slicesCount` and `pSlicesDataOffsets` to `sliceCount` and `pSliceOffsets`, respectively, in `VkVideoDecodeH265PictureInfoEXT`
- Revision 6, 2022-11-14 (Daniel Rakos)
    - Rename `slice` to `sliceSegment` in the APIs for better clarity
- Revision 7, 2022-11-14 (Daniel Rakos)
    - Change extension from `EXT` to `KHR`
    - Extension is no longer provisional
- Revision 8, 2023-12-05 (Daniel Rakos)
    - Condition reference picture setup based on the value of `StdVideoDecodeH265PictureInfo::flags.IsReference`

VK_KHR_video_decode_queue

Name String

VK_KHR_video_decode_queue

Extension Type

Device extension

Registered Extension Number

25

Revision

8

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_video_queue
and
VK_KHR_synchronization2
or
Vulkan Version 1.3

API Interactions

- Interacts with VK_VERSION_1_3
- Interacts with VK_KHR_format_feature_flags2

Contact

- jake.beju@amd.com

Extension Proposal

VK_KHR_video_decode_queue

Other Extension Metadata

Last Modified Date

2023-12-05

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- Jake Beju, AMD
- Olivier Lapicque, NVIDIA
- Peter Fang, AMD

- Piers Daniell, NVIDIA
- Srinath Kumarapuram, NVIDIA
- Tony Zlatinski, NVIDIA
- Daniel Rakos, RasterGrid

Description

This extension builds upon the `VK_KHR_video_queue` extension by adding common APIs specific to video decoding and thus enabling implementations to expose queue families supporting video decode operations.

More specifically, it adds video decode specific capabilities and a new command buffer command that allows recording video decode operations against a video session.

This extension is to be used in conjunction with other codec specific video decode extensions that enable decoding video sequences of specific video compression standards.

New Commands

- vkCmdDecodeVideoKHR

New Structures

- VkVideoDecodeInfoKHR
- Extending VkVideoCapabilitiesKHR:
  - VkVideoDecodeCapabilitiesKHR
- Extending VkVideoProfileInfoKHR, VkQueryPoolCreateInfo:
  - VkVideoDecodeUsageInfoKHR

New Enums

- VkVideoDecodeCapabilityFlagBitsKHR
- VkVideoDecodeUsageFlagBitsKHR

New Bitmasks

- VkVideoDecodeCapabilityFlagsKHR
- VkVideoDecodeFlagsKHR
- VkVideoDecodeUsageFlagsKHR

New Enum Constants

- `VK_KHR_VIDEO_DECODE_QUEUE_EXTENSION_NAME`
- `VK_KHR_VIDEO_DECODE_QUEUE_SPEC_VERSION`
- Extending VkAccessFlagBits2:

    - VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR
    - VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR
- Extending VkBufferUsageFlagBits:
    - VK_BUFFER_USAGE_VIDEO_DECODE_DST_BIT_KHR
    - VK_BUFFER_USAGE_VIDEO_DECODE_SRC_BIT_KHR
- Extending VkFormatFeatureFlagBits:
    - VK_FORMAT_FEATURE_VIDEO_DECODE_DPB_BIT_KHR
    - VK_FORMAT_FEATURE_VIDEO_DECODE_OUTPUT_BIT_KHR
- Extending VkImageLayout:
    - VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR
    - VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR
    - VK_IMAGE_LAYOUT_VIDEO_DECODE_SRC_KHR
- Extending VkImageUsageFlagBits:
    - VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR
    - VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR
    - VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR
- Extending VkPipelineStageFlagBits2:
    - VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
- Extending VkQueueFlagBits:
    - VK_QUEUE_VIDEO_DECODE_BIT_KHR
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_VIDEO_DECODE_CAPABILITIES_KHR
    - VK_STRUCTURE_TYPE_VIDEO_DECODE_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_DECODE_USAGE_INFO_KHR

If VK_KHR_format_feature_flags2 or Vulkan Version 1.3 is supported:

- Extending VkFormatFeatureFlagBits2:
    - VK_FORMAT_FEATURE_2_VIDEO_DECODE_DPB_BIT_KHR
    - VK_FORMAT_FEATURE_2_VIDEO_DECODE_OUTPUT_BIT_KHR

Version History

- Revision 1, 2018-6-11 (Peter Fang)
    - Initial draft
- Revision 1.5, Nov 09 2018 (Tony Zlatinski)
    - API Updates

- Revision 1.6, Jan 08 2020 (Tony Zlatinski)
  - API unify with the video_encode_queue spec
- Revision 1.7, March 29 2021 (Tony Zlatinski)
  - Spec and API updates.
- Revision 2, September 30 2021 (Jon Leech)
  - Add interaction with `VK_KHR_format_feature_flags2` to `vk.xml`
- Revision 3, 2022-02-25 (Ahmed Abdelkhalek)
  - Add VkVideoDecodeCapabilitiesKHR with new flags to report support for decode DPB and output coinciding in the same image, or in distinct images.
- Revision 4, 2022-03-31 (Ahmed Abdelkhalek)
  - Remove redundant VkVideoDecodeInfoKHR.coded{Offset|Extent}
- Revision 5, 2022-07-18 (Daniel Rakos)
  - Remove `VkVideoDecodeFlagBitsKHR` as it contains no defined flags for now
- Revision 6, 2022-08-12 (Daniel Rakos)
  - Add VkVideoDecodeUsageInfoKHR structure and related flags
- Revision 7, 2022-09-29 (Daniel Rakos)
  - Extension is no longer provisional
- Revision 8, 2023-12-05 (Daniel Rakos)
  - Require the specification of a reconstructed picture in all cases, except when the video session was created with no DPB slots to match shipping implementations
  - Make DPB slot activation behavior codec-specific to continue allowing application control over reference picture setup now that a reconstructed picture is always mandatory

VK_KHR_video_decode_vp9

Name String

`VK_KHR_video_decode_vp9`

Extension Type

Device extension

Registered Extension Number

515

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_video_decode_queue

Contact

- Ahmed Abdelkhalek [GitHub]aabdelkh

Extension Proposal

VK_KHR_video_decode_vp9

Other Extension Metadata

Last Modified Date

2025-04-11

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- Benjamin Cheng, AMD
- Lynne Iribarren, Independent
- David Airlie, Red Hat, Inc.
- Ping Liu, Intel
- Srinath Kumarapuram, NVIDIA
- Vassili Nikolaev, NVIDIA
- Tony Zlatinski, NVIDIA
- Konda Raju, NVIDIA
- Daniel Almeida, Collabora

- Nicolas Dufresne, Collabora
- Daniel Rakos, RasterGrid

Description

This extension builds upon the `VK_KHR_video_decode_queue` extension by adding support for decoding elementary video stream sequences compliant with the VP9 video compression standard.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceVideoDecodeVP9FeaturesKHR
- Extending VkVideoCapabilitiesKHR:
  - VkVideoDecodeVP9CapabilitiesKHR
- Extending VkVideoDecodeInfoKHR:
  - VkVideoDecodeVP9PictureInfoKHR
- Extending VkVideoProfileInfoKHR, VkQueryPoolCreateInfo:
  - VkVideoDecodeVP9ProfileInfoKHR

New Enum Constants

- `VK_KHR_VIDEO_DECODE_VP9_EXTENSION_NAME`
- `VK_KHR_VIDEO_DECODE_VP9_SPEC_VERSION`
- VK_MAX_VIDEO_VP9_REFERENCES_PER_FRAME_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_DECODE_VP9_FEATURES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_VP9_CAPABILITIES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_VP9_PICTURE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_DECODE_VP9_PROFILE_INFO_KHR
- Extending VkVideoCodecOperationFlagBitsKHR:
  - VK_VIDEO_CODEC_OPERATION_DECODE_VP9_BIT_KHR

Version History

- Revision 1, 2025-04-11 (Daniel Rakos)
  - Internal revisions

VK_KHR_video_encode_av1

Name String

VK_KHR_video_encode_av1

Extension Type

Device extension

Registered Extension Number

514

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_video_encode_queue

Contact

- Daniel Rakos [GitHub]aqnuep

Extension Proposal

VK_KHR_video_encode_av1

Other Extension Metadata

Last Modified Date

2024-09-23

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- Benjamin Cheng, AMD
- Ho Hin Lau, AMD
- Lynne Iribarren, Independent
- David Airlie, Red Hat, Inc.
- Ping Liu, Intel
- Srinath Kumarapuram, NVIDIA
- Vassili Nikolaev, NVIDIA
- Tony Zlatinski, NVIDIA
- Konda Raju, NVIDIA

- Charlie Turner, Igalia
- Daniel Almeida, Collabora
- Nicolas Dufresne, Collabora
- Daniel Rakos, RasterGrid

Description

This extension builds upon the `VK_KHR_video_encode_queue` extension by adding support for encoding elementary video stream sequences compliant with the AV1 video compression standard.

New Structures

- VkVideoEncodeAV1FrameSizeKHR
- VkVideoEncodeAV1QIndexKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceVideoEncodeAV1FeaturesKHR
- Extending VkVideoBeginCodingInfoKHR:
  - VkVideoEncodeAV1GopRemainingFrameInfoKHR
- Extending VkVideoCapabilitiesKHR:
  - VkVideoEncodeAV1CapabilitiesKHR
- Extending VkVideoCodingControlInfoKHR, VkVideoBeginCodingInfoKHR:
  - VkVideoEncodeAV1RateControlInfoKHR
- Extending VkVideoEncodeInfoKHR:
  - VkVideoEncodeAV1PictureInfoKHR
- Extending VkVideoEncodeQualityLevelPropertiesKHR:
  - VkVideoEncodeAV1QualityLevelPropertiesKHR
- Extending VkVideoEncodeRateControlLayerInfoKHR:
  - VkVideoEncodeAV1RateControlLayerInfoKHR
- Extending VkVideoProfileInfoKHR, VkQueryPoolCreateInfo:
  - VkVideoEncodeAV1ProfileInfoKHR
- Extending VkVideoReferenceSlotInfoKHR:
  - VkVideoEncodeAV1DpbSlotInfoKHR
- Extending VkVideoSessionCreateInfoKHR:
  - VkVideoEncodeAV1SessionCreateInfoKHR
- Extending VkVideoSessionParametersCreateInfoKHR:
  - VkVideoEncodeAV1SessionParametersCreateInfoKHR

New Enums

- VkVideoEncodeAV1CapabilityFlagBitsKHR
- VkVideoEncodeAV1PredictionModeKHR
- VkVideoEncodeAV1RateControlFlagBitsKHR
- VkVideoEncodeAV1RateControlGroupKHR
- VkVideoEncodeAV1StdFlagBitsKHR
- VkVideoEncodeAV1SuperblockSizeFlagBitsKHR

New Bitmasks

- VkVideoEncodeAV1CapabilityFlagsKHR
- VkVideoEncodeAV1RateControlFlagsKHR
- VkVideoEncodeAV1StdFlagsKHR
- VkVideoEncodeAV1SuperblockSizeFlagsKHR

New Enum Constants

- `VK_KHR_VIDEO_ENCODE_AV1_EXTENSION_NAME`
- `VK_KHR_VIDEO_ENCODE_AV1_SPEC_VERSION`
- VK_MAX_VIDEO_AV1_REFERENCES_PER_FRAME_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_AV1_FEATURES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_CAPABILITIES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_DPB_SLOT_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_GOP_REMAINING_FRAME_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_PICTURE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_PROFILE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_QUALITY_LEVEL_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_RATE_CONTROL_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_RATE_CONTROL_LAYER_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_SESSION_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_SESSION_PARAMETERS_CREATE_INFO_KHR
- Extending VkVideoCodecOperationFlagBitsKHR:
  - VK_VIDEO_CODEC_OPERATION_ENCODE_AV1_BIT_KHR

Version History

- Revision 1, 2024-09-23 (Daniel Rakos)
  - Internal revisions

VK_KHR_video_encode_feedback2

Name String

`VK_KHR_video_encode_feedback2`

Extension Type

Device extension

Registered Extension Number

599

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_video_encode_queue

Contact

- Ahmed Abdelkhalek [GitHub]aabdelkh

Extension Proposal

VK_KHR_video_encode_feedback2

Other Extension Metadata

Last Modified Date

2026-02-04

IP Status

No known IP claims.

Contributors

- Sreerenj Balachandran, Amazon
- Ahmed Abdelkhalek, AMD
- Benjamin Cheng, AMD
- Aidan Fabius, Core Avionics & Industrial Inc.
- Ping Liu, Intel
- Srinath Kumarapuram, NVIDIA
- Tony Zlatinski, NVIDIA
- Daniel Rakos, RasterGrid

Description

This extension builds upon the VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR queries introduced by the VK_KHR_video_encode_queue extension and extends the set of supported video encode feedback values with additional statistics about the encoded picture, including per picture partition feedback values.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceVideoEncodeFeedback2FeaturesKHR
- Extending VkQueryPoolCreateInfo:
  - VkQueryPoolVideoEncodePerPartitionFeedbackCreateInfoKHR
- Extending VkVideoCapabilitiesKHR:
  - VkVideoEncodeFeedback2CapabilitiesKHR

New Enums

- VkVideoEncodePerPartitionFeedbackFlagBitsKHR

New Bitmasks

- VkVideoEncodePerPartitionFeedbackFlagsKHR

New Enum Constants

- VK_KHR_VIDEO_ENCODE_FEEDBACK_2_EXTENSION_NAME
- VK_KHR_VIDEO_ENCODE_FEEDBACK_2_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_FEEDBACK_2_FEATURES_KHR
  - VK_STRUCTURE_TYPE_QUERY_POOL_VIDEO_ENCODE_PER_PARTITION_FEEDBACK_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_FEEDBACK_2_CAPABILITIES_KHR
- Extending VkVideoEncodeFeedbackFlagBitsKHR:
  - VK_VIDEO_ENCODE_FEEDBACK_AVERAGE_QUANTIZATION_BIT_KHR
  - VK_VIDEO_ENCODE_FEEDBACK_INTER_PIXELS_BIT_KHR
  - VK_VIDEO_ENCODE_FEEDBACK_INTRA_PIXELS_BIT_KHR
  - VK_VIDEO_ENCODE_FEEDBACK_MAX_QUANTIZATION_BIT_KHR
  - VK_VIDEO_ENCODE_FEEDBACK_MIN_QUANTIZATION_BIT_KHR
  - VK_VIDEO_ENCODE_FEEDBACK_PICTURE_PARTITION_COUNT_BIT_KHR
  - VK_VIDEO_ENCODE_FEEDBACK_SKIPPED_PIXELS_BIT_KHR

Version History

- Revision 1, 2026-02-04 (Daniel Rakos)
  - Internal revisions

VK_KHR_video_encode_h264

Name String

`VK_KHR_video_encode_h264`

Extension Type

Device extension

Registered Extension Number

39

Revision

14

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_video_encode_queue

Contact

- Ahmed Abdelkhalek [GitHub]aabdelkh

Extension Proposal

VK_KHR_video_encode_h264

Other Extension Metadata

Last Modified Date

2023-12-05

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- George Hao, AMD
- Jake Beju, AMD
- Peter Fang, AMD
- Ping Liu, Intel
- Srinath Kumarapuram, NVIDIA
- Tony Zlatinski, NVIDIA
- Ravi Chaudhary, NVIDIA
- Yang Liu, AMD
- Daniel Rakos, RasterGrid

- Aidan Fabius, Lynx
- Lynne Iribarren, Independent

Description

This extension builds upon the VK_KHR_video_encode_queue extension by adding support for encoding elementary video stream sequences compliant with the H.264/AVC video compression standard.

NOTE	This extension was promoted to KHR from the provisional extension VK_EXT_video_encode_h264.

New Structures

- VkVideoEncodeH264FrameSizeKHR
- VkVideoEncodeH264NaluSliceInfoKHR
- VkVideoEncodeH264QpKHR
- Extending VkVideoBeginCodingInfoKHR:
  - VkVideoEncodeH264GopRemainingFrameInfoKHR
- Extending VkVideoCapabilitiesKHR:
  - VkVideoEncodeH264CapabilitiesKHR
- Extending VkVideoCodingControlInfoKHR, VkVideoBeginCodingInfoKHR:
  - VkVideoEncodeH264RateControlInfoKHR
- Extending VkVideoEncodeInfoKHR:
  - VkVideoEncodeH264PictureInfoKHR
- Extending VkVideoEncodeQualityLevelPropertiesKHR:
  - VkVideoEncodeH264QualityLevelPropertiesKHR
- Extending VkVideoEncodeRateControlLayerInfoKHR:
  - VkVideoEncodeH264RateControlLayerInfoKHR
- Extending VkVideoEncodeSessionParametersFeedbackInfoKHR:
  - VkVideoEncodeH264SessionParametersFeedbackInfoKHR
- Extending VkVideoEncodeSessionParametersGetInfoKHR:
  - VkVideoEncodeH264SessionParametersGetInfoKHR
- Extending VkVideoProfileInfoKHR, VkQueryPoolCreateInfo:
  - VkVideoEncodeH264ProfileInfoKHR
- Extending VkVideoReferenceSlotInfoKHR:
  - VkVideoEncodeH264DpbSlotInfoKHR
- Extending VkVideoSessionCreateInfoKHR:

    - VkVideoEncodeH264SessionCreateInfoKHR
- Extending VkVideoSessionParametersCreateInfoKHR:
    - VkVideoEncodeH264SessionParametersCreateInfoKHR
- Extending VkVideoSessionParametersUpdateInfoKHR:
    - VkVideoEncodeH264SessionParametersAddInfoKHR

New Enums

- VkVideoEncodeH264CapabilityFlagBitsKHR
- VkVideoEncodeH264RateControlFlagBitsKHR
- VkVideoEncodeH264StdFlagBitsKHR

New Bitmasks

- VkVideoEncodeH264CapabilityFlagsKHR
- VkVideoEncodeH264RateControlFlagsKHR
- VkVideoEncodeH264StdFlagsKHR

New Enum Constants

- `VK_KHR_VIDEO_ENCODE_H264_EXTENSION_NAME`
- `VK_KHR_VIDEO_ENCODE_H264_SPEC_VERSION`
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_CAPABILITIES_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_DPB_SLOT_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_GOP_REMAINING_FRAME_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_NALU_SLICE_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_PICTURE_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_PROFILE_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_QUALITY_LEVEL_PROPERTIES_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_RATE_CONTROL_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_RATE_CONTROL_LAYER_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_CREATE_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_ADD_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_CREATE_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_FEEDBACK_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_GET_INFO_KHR
- Extending VkVideoCodecOperationFlagBitsKHR:

- VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR

Version History

- Revision 0, 2018-7-23 (Ahmed Abdelkhalek)
  - Initial draft
- Revision 0.5, 2020-02-13 (Tony Zlatinski)
  - General Spec cleanup
  - Added DPB structures
  - Change the VCL frame encode structure
  - Added a common Non-VCL Picture Paramarameters structure
- Revision 1, 2021-03-29 (Tony Zlatinski)
  - Spec and API updates
- Revision 2, August 1 2021 (Srinath Kumarapuram)
  - Rename `VkVideoEncodeH264CapabilitiesFlagsEXT` to `VkVideoEncodeH264CapabilityFlagsEXT` and `VkVideoEncodeH264CapabilitiesFlagsEXT` to `VkVideoEncodeH264CapabilityFlagsEXT`, following Vulkan naming conventions.
- Revision 3, 2021-12-08 (Ahmed Abdelkhalek)
  - Rate control updates
- Revision 4, 2022-02-04 (Ahmed Abdelkhalek)
  - Align VkVideoEncodeH264VclFrameInfoEXT structure to similar one in VK_EXT_video_encode_h265 extension
- Revision 5, 2022-02-10 (Ahmed Abdelkhalek)
  - Updates to encode capability interface
- Revision 6, 2022-03-16 (Ahmed Abdelkhalek)
  - Relocate Std header version reporting/requesting from this extension to VK_KHR_video_queue extension.
  - Remove redundant maxPictureSizeInMbs from VkVideoEncodeH264SessionCreateInfoEXT.
  - Remove the now empty VkVideoEncodeH264SessionCreateInfoEXT.
- Revision 7, 2022-04-06 (Ahmed Abdelkhalek)
  - Add capability flag to report support to use B frame in L1 reference list.
  - Add capability flag to report support for disabling SPS direct_8x8_inference_flag.
- Revision 8, 2022-07-18 (Daniel Rakos)
  - Replace `VkVideoEncodeH264RateControlStructureFlagBitsEXT` bit enum with `VkVideoEncodeH264RateControlStructureEXT` enum
  - Rename `VkVideoEncodeH264ProfileEXT` to `VkVideoEncodeH264ProfileInfoEXT`
  - Rename `VkVideoEncodeH264ReferenceListsEXT` to `VkVideoEncodeH264ReferenceListsInfoEXT`

- Rename `VkVideoEncodeH264EmitPictureParametersEXT` to `VkVideoEncodeH264EmitPictureParametersInfoEXT`
- Rename `VkVideoEncodeH264NaluSliceEXT` to `VkVideoEncodeH264NaluSliceInfoEXT`

- Revision 9, 2022-09-18 (Daniel Rakos)
  - Rename `spsStdCount`, `pSpsStd`, `ppsStdCount`, and `pPpsStd` to `stdSPSCount`, `pStdSPSs`, `stdPPSCount`, and `pStdPPSs`, respectively, in `VkVideoEncodeH264SessionParametersAddInfoEXT`
  - Rename `maxSpsStdCount` and `maxPpsStdCount` to `maxStdSPSCount` and `maxStdPPSCount`, respectively, in `VkVideoEncodeH264SessionParametersCreateInfoEXT`
- Revision 10, 2023-03-06 (Daniel Rakos)
  - Removed `VkVideoEncodeH264EmitPictureParametersInfoEXT`
  - Changed member types in `VkVideoEncodeH264CapabilitiesEXT` and `VkVideoEncodeH264ReferenceListsInfoEXT` from `uint8_t` to `uint32_t`
  - Changed the type of `VkVideoEncodeH264RateControlInfoEXT::temporalLayerCount` and `VkVideoEncodeH264RateControlLayerInfoEXT::temporalLayerId` from `uint8_t` to `uint32_t`
  - Removed `VkVideoEncodeH264InputModeFlagsEXT` and `VkVideoEncodeH264OutputModeFlagsEXT` as we only support frame-in-frame-out mode for now
  - Rename `pCurrentPictureInfo` in `VkVideoEncodeH264VclFrameInfoEXT` to `pStdPictureInfo`
  - Rename `pSliceHeaderStd` in `VkVideoEncodeH264NaluSliceInfoEXT` to `pStdSliceHeader`
  - Rename `pReferenceFinalLists` in `VkVideoEncodeH264VclFrameInfoEXT` and `VkVideoEncodeH264NaluSliceInfoEXT` to `pStdReferenceFinalLists`
  - Removed the `slotIndex` member of `VkVideoEncodeH264DpbSlotInfoEXT` and changed it to be chained to `VkVideoReferenceSlotInfoKHR`
  - Replaced `VkVideoEncodeH264ReferenceListsInfoEXT` with the new Video Std header structure `StdVideoEncodeH264ReferenceLists` that also includes data previously part of the now removed `StdVideoEncodeH264RefMemMgmtCtrlOperations` structure
  - Added new capability flag `VK_VIDEO_ENCODE_H264_CAPABILITY_DIFFERENT_REFERENCE_FINAL_LISTS_BIT_EXT`
- Revision 11, 2023-05-22 (Daniel Rakos)
  - Renamed `VkVideoEncodeH264VclFrameInfoEXT` to `VkVideoEncodeH264PictureInfoEXT`
  - Added `VkVideoEncodeH264PictureInfoEXT::generatePrefixNalu` and `VK_VIDEO_ENCODE_H264_CAPABILITY_GENERATE_PREFIX_NALU_BIT_EXT` to enable the generation of H.264 prefix NALUs when supported by the implementation
  - Removed `VkVideoEncodeH264RateControlLayerInfoEXT::temporalLayerId`
  - Added `expectDyadicTemporalLayerPattern` capability
  - Added the `VkVideoEncodeH264SessionParametersGetInfoEXT` structure to identify the H.264 parameter sets to retrieve encoded parameter data for, and the `VkVideoEncodeH264SessionParametersFeedbackInfoEXT` structure to retrieve H.264 parameter set override information when using the new `vkGetEncodedVideoSessionParametersKHR` command

- Added VkVideoEncodeH264NaluSliceInfoEXT::constantQp to specify per-slice constant QP when rate control mode is VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR
- Added VkVideoEncodeH264QualityLevelPropertiesEXT for retrieving H.264 specific quality level recommendations
- Replaced VkVideoEncodeH264RateControlStructureEXT enum with the flags type VkVideoEncodeH264RateControlFlagsEXT and bits defined in VkVideoEncodeH264RateControlFlagBitsEXT and added HRD compliance flag
- Removed useInitialRcQp and initialRcQp members of VkVideoEncodeH264RateControlLayerInfoEXT
- Added prefersGopRemainingFrames and requiresGopRemainingFrames, and the new VkVideoEncodeH264GopRemainingFrameInfoEXT structure to allow specifying remaining frames of each type in the rate control GOP
- Added maxTemporalLayers, maxQp, and minQp capabilities
- Added maxLevelIdc capability and new VkVideoEncodeH264SessionCreateInfoEXT structure to specify upper bounds on the H.264 level of the produced video bitstream
- Moved capability flags specific to codec syntax restrictions from VkVideoEncodeH264CapabilityFlagsEXT to the new VkVideoEncodeH264StdFlagsEXT which is now included as a separate stdSyntaxFlags member in VkVideoEncodeH264CapabilitiesEXT
- Removed codec syntax override values from VkVideoEncodeH264CapabilitiesEXT
- Removed VkVideoEncodeH264NaluSliceInfoEXT::mbCount and VK_VIDEO_ENCODE_H264_CAPABILITY_SLICE_MB_COUNT_BIT_EXT
- Replaced VK_VIDEO_ENCODE_H264_CAPABILITY_MULTIPLE_SLICES_PER_FRAME_BIT_EXT with the new maxSliceCount capability
- Removed capability flag VK_VIDEO_ENCODE_H264_CAPABILITY_DIFFERENT_REFERENCE_FINAL_LISTS_BIT_EXT and removed pStdReferenceFinalLists members from the VkVideoEncodeH264PictureInfoEXT and VkVideoEncodeH264NaluSliceInfoEXT structures as reference lists info is now included in pStdPictureInfo
- Added capability flag VK_VIDEO_ENCODE_H264_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_EXT

- Revision 12, 2023-07-19 (Daniel Rakos)
  - Added video std capability flags VK_VIDEO_ENCODE_H264_STD_SLICE_QP_DELTA_BIT_EXT and VK_VIDEO_ENCODE_H264_STD_DIFFERENT_SLICE_QP_DELTA_BIT_EXT
  - Fixed optionality of the array members of VkVideoEncodeH264SessionParametersAddInfoEXT
  - Fixed optionality of VkVideoEncodeH264RateControlInfoEXT::flags
- Revision 13, 2023-09-04 (Daniel Rakos)
  - Change extension from EXT to KHR
  - Extension is no longer provisional
- Revision 14, 2023-12-05 (Daniel Rakos)
  - Condition reference picture setup based on the value of StdVideoEncodeH264PictureInfo::flags.is_reference

VK_KHR_video_encode_h265

Name String

`VK_KHR_video_encode_h265`

Extension Type

Device extension

Registered Extension Number

40

Revision

14

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_video_encode_queue

Contact

- Ahmed Abdelkhalek [GitHub]aabdelkh

Extension Proposal

VK_KHR_video_encode_h265

Other Extension Metadata

Last Modified Date

2023-12-05

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- George Hao, AMD
- Jake Beju, AMD
- Chunbo Chen, Intel
- Ping Liu, Intel
- Srinath Kumarapuram, NVIDIA
- Tony Zlatinski, NVIDIA
- Ravi Chaudhary, NVIDIA
- Daniel Rakos, RasterGrid
- Aidan Fabius, Lynx

- Lynne Iribarren, Independent

Description

This extension builds upon the VK_KHR_video_encode_queue extension by adding support for encoding elementary video stream sequences compliant with the H.265/HEVC video compression standard.

NOTE	This extension was promoted to KHR from the provisional extension VK_EXT_video_encode_h265.

New Structures

- VkVideoEncodeH265FrameSizeKHR
- VkVideoEncodeH265NaluSliceSegmentInfoKHR
- VkVideoEncodeH265QpKHR
- Extending VkVideoBeginCodingInfoKHR:
  - VkVideoEncodeH265GopRemainingFrameInfoKHR
- Extending VkVideoCapabilitiesKHR:
  - VkVideoEncodeH265CapabilitiesKHR
- Extending VkVideoCodingControlInfoKHR, VkVideoBeginCodingInfoKHR:
  - VkVideoEncodeH265RateControlInfoKHR
- Extending VkVideoEncodeInfoKHR:
  - VkVideoEncodeH265PictureInfoKHR
- Extending VkVideoEncodeQualityLevelPropertiesKHR:
  - VkVideoEncodeH265QualityLevelPropertiesKHR
- Extending VkVideoEncodeRateControlLayerInfoKHR:
  - VkVideoEncodeH265RateControlLayerInfoKHR
- Extending VkVideoEncodeSessionParametersFeedbackInfoKHR:
  - VkVideoEncodeH265SessionParametersFeedbackInfoKHR
- Extending VkVideoEncodeSessionParametersGetInfoKHR:
  - VkVideoEncodeH265SessionParametersGetInfoKHR
- Extending VkVideoProfileInfoKHR, VkQueryPoolCreateInfo:
  - VkVideoEncodeH265ProfileInfoKHR
- Extending VkVideoReferenceSlotInfoKHR:
  - VkVideoEncodeH265DpbSlotInfoKHR
- Extending VkVideoSessionCreateInfoKHR:
  - VkVideoEncodeH265SessionCreateInfoKHR

- Extending VkVideoSessionParametersCreateInfoKHR:
  - VkVideoEncodeH265SessionParametersCreateInfoKHR
- Extending VkVideoSessionParametersUpdateInfoKHR:
  - VkVideoEncodeH265SessionParametersAddInfoKHR

New Enums

- VkVideoEncodeH265CapabilityFlagBitsKHR
- VkVideoEncodeH265CtbSizeFlagBitsKHR
- VkVideoEncodeH265RateControlFlagBitsKHR
- VkVideoEncodeH265StdFlagBitsKHR
- VkVideoEncodeH265TransformBlockSizeFlagBitsKHR

New Bitmasks

- VkVideoEncodeH265CapabilityFlagsKHR
- VkVideoEncodeH265CtbSizeFlagsKHR
- VkVideoEncodeH265RateControlFlagsKHR
- VkVideoEncodeH265StdFlagsKHR
- VkVideoEncodeH265TransformBlockSizeFlagsKHR

New Enum Constants

- `VK_KHR_VIDEO_ENCODE_H265_EXTENSION_NAME`
- `VK_KHR_VIDEO_ENCODE_H265_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_CAPABILITIES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_DPB_SLOT_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_GOP_REMAINING_FRAME_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_NALU_SLICE_SEGMENT_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_PICTURE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_PROFILE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_QUALITY_LEVEL_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_RATE_CONTROL_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_RATE_CONTROL_LAYER_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_ADD_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_FEEDBACK_INFO_KH

R

- VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_GET_INFO_KHR

- Extending VkVideoCodecOperationFlagBitsKHR:
  - VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR

Version History

- Revision 0, 2019-11-14 (Ahmed Abdelkhalek)
  - Initial draft
- Revision 0.5, 2020-02-13 (Tony Zlatinski)
  - General Spec cleanup
  - Added DPB structures
  - Change the VCL frame encode structure
  - Added a common Non-VCL Picture Paramarameters structure
- Revision 2, Oct 10 2021 (Srinath Kumarapuram)
  - Vulkan Video Encode h.265 update and spec edits
- Revision 3, 2021-12-08 (Ahmed Abdelkhalek)
  - Rate control updates
- Revision 4, 2022-01-11 (Ahmed Abdelkhalek)
  - Replace occurrences of “slice” by “slice segment” and rename structures/enums to reflect this.
- Revision 5, 2022-02-10 (Ahmed Abdelkhalek)
  - Updates to encode capability interface
- Revision 6, 2022-03-16 (Ahmed Abdelkhalek)
  - Relocate Std header version reporting/requesting from this extension to VK_KHR_video_queue extension.
  - Remove the now empty VkVideoEncodeH265SessionCreateInfoEXT.
- Revision 7, 2022-03-24 (Ahmed Abdelkhalek)
  - Add capability flags to report support to disable transform skip and support to use B frame in L1 reference list.
- Revision 8, 2022-07-18 (Daniel Rakos)
  - Replace `VkVideoEncodeH265RateControlStructureFlagBitsEXT` bit enum with `VkVideoEncodeH265RateControlStructureEXT` enum
  - Rename `VkVideoEncodeH265ProfileEXT` to `VkVideoEncodeH265ProfileInfoEXT`
  - Rename `VkVideoEncodeH265ReferenceListsEXT` to `VkVideoEncodeH265ReferenceListsInfoEXT`
  - Rename `VkVideoEncodeH265EmitPictureParametersEXT` to `VkVideoEncodeH265EmitPictureParametersInfoEXT`

- Rename `VkVideoEncodeH265NaluSliceSegmentEXT` to `VkVideoEncodeH265NaluSliceSegmentInfoEXT`

- Revision 9, 2022-09-18 (Daniel Rakos)
  - Rename `vpsStdCount`, `pVpsStd`, `spsStdCount`, `pSpsStd`, `ppsStdCount`, and `pPpsStd` to `stdVPSCount`, `pStdVPSs`, `stdSPSCount`, `pStdSPSs`, `stdPPSCount`, and `pStdPPSs`, respectively, in `VkVideoEncodeH265SessionParametersAddInfoEXT`
  - Rename `maxVpsStdCount`, `maxSpsStdCount`, and `maxPpsStdCount` to `maxStdVPSCount`, `maxStdSPSCount` and `maxStdPPSCount`, respectively, in `VkVideoEncodeH265SessionParametersCreateInfoEXT`
- Revision 10, 2023-03-06 (Daniel Rakos)
  - Removed `VkVideoEncodeH265EmitPictureParametersInfoEXT`
  - Changed member types in `VkVideoEncodeH265CapabilitiesEXT` and `VkVideoEncodeH265ReferenceListsInfoEXT` from `uint8_t` to `uint32_t`
  - Changed the type of `VkVideoEncodeH265RateControlInfoEXT::subLayerCount` and `VkVideoEncodeH265RateControlLayerInfoEXT::temporalId` from `uint8_t` to `uint32_t`
  - Removed `VkVideoEncodeH265InputModeFlagsEXT` and `VkVideoEncodeH265OutputModeFlagsEXT` as we only support frame-in-frame-out mode for now
  - Rename `pCurrentPictureInfo` in `VkVideoEncodeH265VclFrameInfoEXT` to `pStdPictureInfo`
  - Rename `pSliceSegmentHeaderStd` in `VkVideoEncodeH265NaluSliceSegmentInfoEXT` to `pStdSliceSegmentHeader`
  - Rename `pReferenceFinalLists` in `VkVideoEncodeH265VclFrameInfoEXT` and `VkVideoEncodeH265NaluSliceSegmentInfoEXT` to `pStdReferenceFinalLists`
  - Removed the `slotIndex` member of `VkVideoEncodeH265DpbSlotInfoEXT` and changed it to be chained to `VkVideoReferenceSlotInfoKHR`
  - Replaced `VkVideoEncodeH265ReferenceListsInfoEXT` with the new Video Std header structure `StdVideoEncodeH265ReferenceLists`
  - Added new capability flag `VK_VIDEO_ENCODE_H265_CAPABILITY_DIFFERENT_REFERENCE_FINAL_LISTS_BIT_EXT`
- Revision 11, 2023-05-26 (Daniel Rakos)
  - Renamed `VkVideoEncodeH265VclFrameInfoEXT` to `VkVideoEncodeH265PictureInfoEXT`
  - Removed `VkVideoEncodeH265RateControlLayerInfoEXT::temporalId`
  - Added `expectDyadicTemporalSubLayerPattern` capability
  - Added the `VkVideoEncodeH265SessionParametersGetInfoEXT` structure to identify the H.265 parameter sets to retrieve encoded parameter data for, and the `VkVideoEncodeH265SessionParametersFeedbackInfoEXT` structure to retrieve H.265 parameter set override information when using the new `vkGetEncodedVideoSessionParametersKHR` command
  - Added `VkVideoEncodeH265NaluSliceSegmentInfoEXT::constantQp` to specify per-slice segment constant QP when rate control mode is `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR`
  - Added `VkVideoEncodeH265QualityLevelPropertiesEXT` for retrieving H.265 specific quality level recommendations

- Replaced VkVideoEncodeH265RateControlStructureEXT enum with the flags type VkVideoEncodeH265RateControlFlagsEXT and bits defined in VkVideoEncodeH265RateControlFlagBitsEXT and added HRD compliance flag
- Removed useInitialRcQp and initialRcQp members of VkVideoEncodeH265RateControlLayerInfoEXT
- Added prefersGopRemainingFrames and requiresGopRemainingFrames, and the new VkVideoEncodeH265GopRemainingFrameInfoEXT structure to allow specifying remaining frames of each type in the rate control GOP
- Renamed maxSubLayersCount capability to maxSubLayerCount
- Added maxQp, and minQp capabilities
- Added maxLevelIdc capability and new VkVideoEncodeH265SessionCreateInfoEXT structure to specify upper bounds on the H.265 level of the produced video bitstream
- Moved capability flags specific to codec syntax restrictions from VkVideoEncodeH265CapabilityFlagsEXT to the new VkVideoEncodeH265StdFlagsEXT which is now included as a separate stdSyntaxFlags member in VkVideoEncodeH265CapabilitiesEXT
- Added std prefix to codec syntax capabilities in VkVideoEncodeH265CapabilitiesEXT
- Removed VkVideoEncodeH265NaluSliceSegmentInfoEXT::ctbCount and VK_VIDEO_ENCODE_H265_CAPABILITY_SLICE_SEGMENT_CTB_COUNT_BIT_EXT
- Replaced VK_VIDEO_ENCODE_H265_CAPABILITY_MULTIPLE_SLICE_SEGMENTS_PER_FRAME_BIT_EXT with the new maxSliceSegmentCount capability
- Added maxTiles capability
- Removed codec syntax min/max capabilities from VkVideoEncodeH265CapabilitiesEXT
- Removed capability flag VK_VIDEO_ENCODE_H265_CAPABILITY_DIFFERENT_REFERENCE_FINAL_LISTS_BIT_EXT and removed pStdReferenceFinalLists members from the VkVideoEncodeH265PictureInfoEXT and VkVideoEncodeH265NaluSliceSegmentInfoEXT structures as reference lists info is now included in pStdPictureInfo
- Added capability flag VK_VIDEO_ENCODE_H265_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_EXT

- Revision 12, 2023-07-19 (Daniel Rakos)
  - Added video std capability flags VK_VIDEO_ENCODE_H265_STD_SLICE_QP_DELTA_BIT_EXT and VK_VIDEO_ENCODE_H265_STD_DIFFERENT_SLICE_QP_DELTA_BIT_EXT
  - Fixed optionality of the array members of VkVideoEncodeH265SessionParametersAddInfoEXT
  - Fixed optionality of VkVideoEncodeH265RateControlInfoEXT::flags
- Revision 13, 2023-09-04 (Daniel Rakos)
  - Change extension from EXT to KHR
  - Extension is no longer provisional
- Revision 14, 2023-12-05 (Daniel Rakos)
  - Condition reference picture setup based on the value of StdVideoEncodeH265PictureInfo::flags.is_reference

VK_KHR_video_encode_intra_refresh

Name String

`VK_KHR_video_encode_intra_refresh`

Extension Type

Device extension

Registered Extension Number

553

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_video_encode_queue

API Interactions

- Interacts with VK_KHR_video_encode_av1
- Interacts with VK_KHR_video_encode_h264
- Interacts with VK_KHR_video_encode_h265

Contact

- Ahmed Abdelkhalek [GitHub]aabdelkh

Extension Proposal

VK_KHR_video_encode_intra_refresh

Other Extension Metadata

Last Modified Date

2025-03-28

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- Benjamin Cheng, AMD
- Srinath Kumarapuram, NVIDIA
- Tony Zlatinski, NVIDIA
- Ping Liu, Intel
- Daniel Rakos, RasterGrid

- Lynne Iribarren, Independent

Description

This extension builds upon the `VK_KHR_video_encode_queue` extension by enabling the application to perform intra refresh in video encode operations.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceVideoEncodeIntraRefreshFeaturesKHR
- Extending VkVideoCapabilitiesKHR:
  - VkVideoEncodeIntraRefreshCapabilitiesKHR
- Extending VkVideoEncodeInfoKHR:
  - VkVideoEncodeIntraRefreshInfoKHR
- Extending VkVideoReferenceSlotInfoKHR:
  - VkVideoReferenceIntraRefreshInfoKHR
- Extending VkVideoSessionCreateInfoKHR:
  - VkVideoEncodeSessionIntraRefreshCreateInfoKHR

New Enums

- VkVideoEncodeIntraRefreshModeFlagBitsKHR

New Bitmasks

- VkVideoEncodeIntraRefreshModeFlagsKHR

New Enum Constants

- `VK_KHR_VIDEO_ENCODE_INTRA_REFRESH_EXTENSION_NAME`
- `VK_KHR_VIDEO_ENCODE_INTRA_REFRESH_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_INTRA_REFRESH_FEATURES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_INTRA_REFRESH_CAPABILITIES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_INTRA_REFRESH_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_ENCODE_SESSION_INTRA_REFRESH_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_REFERENCE_INTRA_REFRESH_INFO_KHR
- Extending VkVideoEncodeFlagBitsKHR:
  - VK_VIDEO_ENCODE_INTRA_REFRESH_BIT_KHR

If VK_KHR_video_encode_av1 is supported:

- Extending VkVideoEncodeAV1CapabilityFlagBitsKHR:
  - VK_VIDEO_ENCODE_AV1_CAPABILITY_COMPOUND_PREDICTION_INTRA_REFRESH_BIT_KHR

If VK_KHR_video_encode_h264 is supported:

- Extending VkVideoEncodeH264CapabilityFlagBitsKHR:
  - VK_VIDEO_ENCODE_H264_CAPABILITY_B_PICTURE_INTRA_REFRESH_BIT_KHR

If VK_KHR_video_encode_h265 is supported:

- Extending VkVideoEncodeH265CapabilityFlagBitsKHR:
  - VK_VIDEO_ENCODE_H265_CAPABILITY_B_PICTURE_INTRA_REFRESH_BIT_KHR

Version History

- Revision 1, 2025-03-28 (Daniel Rakos)
  - Internal revisions

VK_KHR_video_encode_quantization_map

Name String

VK_KHR_video_encode_quantization_map

Extension Type

Device extension

Registered Extension Number

554

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_video_encode_queue
and
VK_KHR_format_feature_flags2
or
Vulkan Version 1.3

API Interactions

- Interacts with VK_KHR_video_encode_av1
- Interacts with VK_KHR_video_encode_h264
- Interacts with VK_KHR_video_encode_h265

Contact

- Ahmed Abdelkhalek [GitHub]aabdelkh

Extension Proposal

VK_KHR_video_encode_quantization_map

Other Extension Metadata

Last Modified Date

2024-09-23

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- Benjamin Cheng, AMD
- Srinath Kumarapuram, NVIDIA

- Tony Zlatinski, NVIDIA
- Ping Liu, Intel
- Daniel Rakos, RasterGrid
- Lynne Iribarren, Independent

Description

This extension builds upon the `VK_KHR_video_encode_queue` extension by enabling fine grained control of codec-specific quantization parameters in video encode operations.

More specifically, it adds support for quantization maps:

- Quantization delta maps to directly control the relative value of quantization parameter values on a per-block basis for all rate control modes (including when rate control is disabled).
- Emphasis maps to indirectly control the relative quantization parameter values on a per-block basis when rate control is not disabled and the rate control mode is not configured to the implementation-defined default mode.

This extension is to be used in conjunction with other codec specific video encode extensions that specify the codec specific quantization parameters these maps control.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceVideoEncodeQuantizationMapFeaturesKHR
- Extending VkVideoCapabilitiesKHR:
  - VkVideoEncodeQuantizationMapCapabilitiesKHR
- Extending VkVideoEncodeInfoKHR:
  - VkVideoEncodeQuantizationMapInfoKHR
- Extending VkVideoFormatPropertiesKHR:
  - VkVideoFormatQuantizationMapPropertiesKHR
- Extending VkVideoSessionParametersCreateInfoKHR:
  - VkVideoEncodeQuantizationMapSessionParametersCreateInfoKHR

If VK_KHR_video_encode_av1 is supported:

- Extending VkVideoCapabilitiesKHR:
  - VkVideoEncodeAV1QuantizationMapCapabilitiesKHR
- Extending VkVideoFormatPropertiesKHR:
  - VkVideoFormatAV1QuantizationMapPropertiesKHR

If VK_KHR_video_encode_h264 is supported:

- Extending VkVideoCapabilitiesKHR:

    - VkVideoEncodeH264QuantizationMapCapabilitiesKHR

If VK_KHR_video_encode_h265 is supported:

- Extending VkVideoCapabilitiesKHR:
    - VkVideoEncodeH265QuantizationMapCapabilitiesKHR
- Extending VkVideoFormatPropertiesKHR:
    - VkVideoFormatH265QuantizationMapPropertiesKHR

New Enums

- VkVideoEncodeFlagBitsKHR
- VkVideoSessionParametersCreateFlagBitsKHR

New Enum Constants

- `VK_KHR_VIDEO_ENCODE_QUANTIZATION_MAP_EXTENSION_NAME`
- `VK_KHR_VIDEO_ENCODE_QUANTIZATION_MAP_SPEC_VERSION`
- Extending VkFormatFeatureFlagBits2:
    - VK_FORMAT_FEATURE_2_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR
    - VK_FORMAT_FEATURE_2_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR
- Extending VkImageLayout:
    - VK_IMAGE_LAYOUT_VIDEO_ENCODE_QUANTIZATION_MAP_KHR
- Extending VkImageUsageFlagBits:
    - VK_IMAGE_USAGE_VIDEO_ENCODE_EMPHASIS_MAP_BIT_KHR
    - VK_IMAGE_USAGE_VIDEO_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_QUANTIZATION_MAP_FEATURES_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUANTIZATION_MAP_CAPABILITIES_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUANTIZATION_MAP_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUANTIZATION_MAP_SESSION_PARAMETERS_CREATE_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_FORMAT_QUANTIZATION_MAP_PROPERTIES_KHR
- Extending VkVideoEncodeCapabilityFlagBitsKHR:
    - VK_VIDEO_ENCODE_CAPABILITY_EMPHASIS_MAP_BIT_KHR
    - VK_VIDEO_ENCODE_CAPABILITY_QUANTIZATION_DELTA_MAP_BIT_KHR
- Extending VkVideoEncodeFlagBitsKHR:
    - VK_VIDEO_ENCODE_WITH_EMPHASIS_MAP_BIT_KHR

    - VK_VIDEO_ENCODE_WITH_QUANTIZATION_DELTA_MAP_BIT_KHR
- Extending VkVideoSessionCreateFlagBitsKHR:
    - VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_EMPHASIS_MAP_BIT_KHR
    - VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_QUANTIZATION_DELTA_MAP_BIT_KHR
- Extending VkVideoSessionParametersCreateFlagBitsKHR:
    - VK_VIDEO_SESSION_PARAMETERS_CREATE_QUANTIZATION_MAP_COMPATIBLE_BIT_KHR

If VK_KHR_video_encode_av1 is supported:

- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_AV1_QUANTIZATION_MAP_CAPABILITIES_KHR
    - VK_STRUCTURE_TYPE_VIDEO_FORMAT_AV1_QUANTIZATION_MAP_PROPERTIES_KHR

If VK_KHR_video_encode_h264 is supported:

- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_QUANTIZATION_MAP_CAPABILITIES_KHR
- Extending VkVideoEncodeH264CapabilityFlagBitsKHR:
    - VK_VIDEO_ENCODE_H264_CAPABILITY_MB_QP_DIFF_WRAPAROUND_BIT_KHR

If VK_KHR_video_encode_h265 is supported:

- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_QUANTIZATION_MAP_CAPABILITIES_KHR
    - VK_STRUCTURE_TYPE_VIDEO_FORMAT_H265_QUANTIZATION_MAP_PROPERTIES_KHR
- Extending VkVideoEncodeH265CapabilityFlagBitsKHR:
    - VK_VIDEO_ENCODE_H265_CAPABILITY_CU_QP_DIFF_WRAPAROUND_BIT_KHR

Version History

- Revision 1, 2024-08-21 (Daniel Rakos)
    - Internal revisions
- Revision 2, 2024-09-23 (Daniel Rakos)
    - Added interactions with VK_KHR_video_encode_av1

VK_KHR_video_encode_queue

Name String

VK_KHR_video_encode_queue

Extension Type

Device extension

Registered Extension Number

300

Revision

12

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_video_queue
and
VK_KHR_synchronization2
or
Vulkan Version 1.3

API Interactions

- Interacts with VK_VERSION_1_3
- Interacts with VK_KHR_format_feature_flags2

Contact

- Ahmed Abdelkhalek [GitHub]aabdelkh

Extension Proposal

VK_KHR_video_encode_queue

Other Extension Metadata

Last Modified Date

2023-12-05

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- Damien Kessler, NVIDIA
- George Hao, AMD
- Jake Beju, AMD

- Peter Fang, AMD
- Piers Daniell, NVIDIA
- Srinath Kumarapuram, NVIDIA
- Thomas J. Meier, NVIDIA
- Tony Zlatinski, NVIDIA
- Ravi Chaudhary, NVIDIA
- Yang Liu, AMD
- Daniel Rakos, RasterGrid
- Ping Liu, Intel
- Aidan Fabius, Lynx
- Lynne Iribarren, Independent

Description

This extension builds upon the `VK_KHR_video_queue` extension by adding common APIs specific to video encoding and thus enabling implementations to expose queue families supporting video encode operations.

More specifically, it adds video encode specific capabilities and a new command buffer command that allows recording video encode operations against a video session.

This extension is to be used in conjunction with other codec specific video encode extensions that enable encoding video sequences of specific video compression standards.

New Commands

- vkCmdEncodeVideoKHR
- vkGetEncodedVideoSessionParametersKHR
- vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR

New Structures

- VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR
- VkVideoEncodeInfoKHR
- VkVideoEncodeQualityLevelPropertiesKHR
- VkVideoEncodeRateControlLayerInfoKHR
- VkVideoEncodeSessionParametersFeedbackInfoKHR
- VkVideoEncodeSessionParametersGetInfoKHR
- Extending VkQueryPoolCreateInfo:
  - VkQueryPoolVideoEncodeFeedbackCreateInfoKHR
- Extending VkVideoCapabilitiesKHR:

    - VkVideoEncodeCapabilitiesKHR
- Extending VkVideoCodingControlInfoKHR, VkVideoBeginCodingInfoKHR:
    - VkVideoEncodeRateControlInfoKHR
- Extending VkVideoCodingControlInfoKHR, VkVideoSessionParametersCreateInfoKHR:
    - VkVideoEncodeQualityLevelInfoKHR
- Extending VkVideoProfileInfoKHR, VkQueryPoolCreateInfo:
    - VkVideoEncodeUsageInfoKHR

New Enums

- VkVideoEncodeCapabilityFlagBitsKHR
- VkVideoEncodeContentFlagBitsKHR
- VkVideoEncodeFeedbackFlagBitsKHR
- VkVideoEncodeRateControlModeFlagBitsKHR
- VkVideoEncodeTuningModeKHR
- VkVideoEncodeUsageFlagBitsKHR

New Bitmasks

- VkVideoEncodeCapabilityFlagsKHR
- VkVideoEncodeContentFlagsKHR
- VkVideoEncodeFeedbackFlagsKHR
- VkVideoEncodeFlagsKHR
- VkVideoEncodeRateControlFlagsKHR
- VkVideoEncodeRateControlModeFlagsKHR
- VkVideoEncodeUsageFlagsKHR

New Enum Constants

- `VK_KHR_VIDEO_ENCODE_QUEUE_EXTENSION_NAME`
- `VK_KHR_VIDEO_ENCODE_QUEUE_SPEC_VERSION`
- Extending VkAccessFlagBits2:
    - VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR
    - VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR
- Extending VkBufferUsageFlagBits:
    - VK_BUFFER_USAGE_VIDEO_ENCODE_DST_BIT_KHR
    - VK_BUFFER_USAGE_VIDEO_ENCODE_SRC_BIT_KHR
- Extending VkFormatFeatureFlagBits:
    - VK_FORMAT_FEATURE_VIDEO_ENCODE_DPB_BIT_KHR

    - VK_FORMAT_FEATURE_VIDEO_ENCODE_INPUT_BIT_KHR
- Extending VkImageLayout:
    - VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR
    - VK_IMAGE_LAYOUT_VIDEO_ENCODE_DST_KHR
    - VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR
- Extending VkImageUsageFlagBits:
    - VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR
    - VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR
    - VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR
- Extending VkPipelineStageFlagBits2:
    - VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
- Extending VkQueryResultStatusKHR:
    - VK_QUERY_RESULT_STATUS_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_KHR
- Extending VkQueryType:
    - VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR
- Extending VkQueueFlagBits:
    - VK_QUEUE_VIDEO_ENCODE_BIT_KHR
- Extending VkResult:
    - VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_QUALITY_LEVEL_INFO_KHR
    - VK_STRUCTURE_TYPE_QUERY_POOL_VIDEO_ENCODE_FEEDBACK_CREATE_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_CAPABILITIES_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUALITY_LEVEL_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUALITY_LEVEL_PROPERTIES_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_RATE_CONTROL_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_RATE_CONTROL_LAYER_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_SESSION_PARAMETERS_FEEDBACK_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_SESSION_PARAMETERS_GET_INFO_KHR
    - VK_STRUCTURE_TYPE_VIDEO_ENCODE_USAGE_INFO_KHR
- Extending VkVideoCodingControlFlagBitsKHR:
    - VK_VIDEO_CODING_CONTROL_ENCODE_QUALITY_LEVEL_BIT_KHR
    - VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR
- Extending VkVideoSessionCreateFlagBitsKHR:

    - VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_PARAMETER_OPTIMIZATIONS_BIT_KHR

If VK_KHR_format_feature_flags2 or Vulkan Version 1.3 is supported:

- Extending VkFormatFeatureFlagBits2:
    - VK_FORMAT_FEATURE_2_VIDEO_ENCODE_DPB_BIT_KHR
    - VK_FORMAT_FEATURE_2_VIDEO_ENCODE_INPUT_BIT_KHR

Version History

- Revision 1, 2018-07-23 (Ahmed Abdelkhalek)
    - Initial draft
- Revision 1.1, 10/29/2019 (Tony Zlatinski)
    - Updated the reserved spec tokens and renamed VkVideoEncoderKHR to VkVideoSessionKHR
- Revision 1.6, Jan 08 2020 (Tony Zlatinski)
    - API unify with the video_decode_queue spec
- Revision 2, March 29 2021 (Tony Zlatinski)
    - Spec and API updates.
- Revision 3, 2021-09-30 (Jon Leech)
    - Add interaction with `VK_KHR_format_feature_flags2` to `vk.xml`
- Revision 4, 2022-02-10 (Ahmed Abdelkhalek)
    - Updates to encode capability interface
- Revision 5, 2022-03-31 (Ahmed Abdelkhalek)
    - Remove redundant VkVideoEncodeInfoKHR.codedExtent
- Revision 6, 2022-07-18 (Daniel Rakos)
    - Remove `VkVideoEncodeRateControlFlagBitsKHR` and `VkVideoEncodeFlagBitsKHR` as they contain no defined flags for now
    - Add `VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR` and `VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_LAYER_BIT_KHR` to indicate rate control and rate control layer change requests, respectively, in video coding control operations
- Revision 7, 2022-08-12 (Daniel Rakos)
    - Add VkVideoEncodeUsageInfoKHR structure and related flags
- Revision 8, 2023-03-06 (Daniel Rakos)
    - Replace `VK_QUERY_TYPE_VIDEO_ENCODE_BITSTREAM_BUFFER_RANGE_KHR` queries with more generic `VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR` queries that can be extended in the future with more feedback values
    - Rename `dstBitstreamBuffer`, `dstBitstreamBufferOffset`, and `dstBitstreamBufferMaxRange` in `VkVideoEncodeInfoKHR` to `dstBuffer`, `dstBufferOffset`, and `dstBufferRange`, respectively, for consistency with the naming convention in the video decode extensions

- Change the type of `rateControlLayerCount` and `qualityLevelCount` in `VkVideoEncodeCapabilitiesKHR` from `uint8_t` to `uint32_t` and rename them to `maxRateControlLayers` and `maxQualityLevels`, respectively
- Change the type of `averageBitrate` and `maxBitrate` in `VkVideoEncodeRateControlLayerInfoKHR` from `uint32_t` to `uint64_t`
- Fixed the definition of rate control flag bits and added the new `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR` constant to indicate implementation-specific automatic rate control
- Change the type of `VkVideoEncodeRateControlInfoKHR::layerCount` from `uint8_t` to `uint32_t`
- Rename `pLayerConfigs` to `pLayers` in `VkVideoEncodeRateControlInfoKHR`

- Revision 9, 2023-03-28 (Daniel Rakos)
  - Removed `VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_LAYER_BIT_KHR` and the ability to change the state of individual rate control layers
  - Added new `VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR` flag to video encode feedback queries
  - Added new video session create flag `VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_PARAMETER_OPTIMIZATIONS_BIT_KHR` to opt-in to video session and encoding parameter optimizations
  - Added the `vkGetEncodedVideoSessionParametersKHR` command to enable retrieving encoded video session parameter data
  - Moved `virtualBufferSizeInMs` and `initialVirtualBufferSizeInMs` from `VkVideoEncodeRateControlLayerInfoKHR` to `VkVideoEncodeRateControlInfoKHR`
  - Added `maxBitrate` capability
  - Renamed `inputImageDataFillAlignment` capability to `encodeInputPictureGranularity` to better reflect its purpose
  - Added new `vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR` command and related structures to enable querying recommended settings for video encode quality levels
  - Added `VK_VIDEO_CODING_CONTROL_ENCODE_QUALITY_LEVEL_BIT_KHR` flag and `VkVideoEncodeQualityLevelInfoKHR` structure to allow controlling video encode quality level and removed `qualityLevel` from the encode operation parameters
- Revision 10, 2023-07-19 (Daniel Rakos)
  - Added `VK_QUERY_RESULT_STATUS_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_KHR` query result status code and the related capability flag `VK_VIDEO_ENCODE_CAPABILITY_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_DETECTION_BIT_KHR`
- Revision 11, 2023-09-04 (Daniel Rakos)
  - Extension is no longer provisional
- Revision 12, 2023-12-05 (Daniel Rakos)
  - Require the specification of a reconstructed picture in all cases, except when the video session was created with no DPB slots to match shipping implementations
  - Make DPB slot activation behavior codec-specific to continue allowing application control

over reference picture setup now that a reconstructed picture is always mandatory

VK_KHR_video_maintenance1

Name String

VK_KHR_video_maintenance1

Extension Type

Device extension

Registered Extension Number

516

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_video_queue

Contact

- Daniel Rakos [GitHub]aqnuep

Extension Proposal

VK_KHR_video_maintenance1

Other Extension Metadata

Last Modified Date

2023-07-27

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- Aidan Fabius, Lynx
- Ping Liu, Intel
- Lynne Iribarren, Independent
- Srinath Kumarapuram, NVIDIA
- Tony Zlatinski, NVIDIA
- Daniel Rakos, RasterGrid

Description

VK_KHR_video_maintenance1 adds a collection of minor video coding features, none of which would warrant an entire extension of their own.

The new features are as follows:

- Allow creating buffers that can be used in video coding operations, independent of the used video profile, using the new buffer creation flag VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR.
- Allow creating images that can be used as decode output or encode input pictures, independent of the used video profile, using the new image creation flag VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR.
- Allow specifying queries used by video coding operations as part of the video coding command parameters, instead of using begin/end query when the video session is created using the new video session creation flag VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceVideoMaintenance1FeaturesKHR
- Extending VkVideoDecodeInfoKHR, VkVideoEncodeInfoKHR:
  - VkVideoInlineQueryInfoKHR

New Enum Constants

- `VK_KHR_VIDEO_MAINTENANCE_1_EXTENSION_NAME`
- `VK_KHR_VIDEO_MAINTENANCE_1_SPEC_VERSION`
- Extending VkBufferCreateFlagBits:
  - VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR
- Extending VkImageCreateFlagBits:
  - VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_MAINTENANCE_1_FEATURES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_INLINE_QUERY_INFO_KHR
- Extending VkVideoSessionCreateFlagBitsKHR:
  - VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR

Version History

- Revision 1, 2023-07-27 (Daniel Rakos)
  - internal revisions

VK_KHR_video_maintenance2

Name String

`VK_KHR_video_maintenance2`

Extension Type

Device extension

Registered Extension Number

587

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_video_queue

API Interactions

- Interacts with VK_KHR_video_decode_av1
- Interacts with VK_KHR_video_decode_h264
- Interacts with VK_KHR_video_decode_h265
- Interacts with VK_KHR_video_decode_queue

Contact

- Daniel Rakos [GitHub]aqnuep

Extension Proposal

VK_KHR_video_maintenance2

Other Extension Metadata

Last Modified Date

2024-10-14

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- Benjamin Cheng, AMD
- Aidan Fabius, Lynx
- Ping Liu, Intel
- Lynne Iribarren, Independent

- Srinath Kumarapuram, NVIDIA
- Tony Zlatinski, NVIDIA
- Daniel Rakos, RasterGrid

Description

VK_KHR_video_maintenance2 adds a collection of minor video coding features, none of which would warrant an entire extension of their own.

The new features are as follows:

- Allow video coding control commands (such as video session reset) to be issued without the need for a bound video session parameters object for video decode operations that would otherwise require the use of video session parameters objects.
- Allow applications to specify codec-specific parameter sets inline for each decode operation instead of having to construct video session parameters objects.
- Require support for VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR in all applicable video encode profiles.
- Provide additional guarantees on Video Std parameters that the encoder implementation will not override.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceVideoMaintenance2FeaturesKHR

If VK_KHR_video_decode_av1 is supported:

- Extending VkVideoDecodeInfoKHR:
  - VkVideoDecodeAV1InlineSessionParametersInfoKHR

If VK_KHR_video_decode_h264 is supported:

- Extending VkVideoDecodeInfoKHR:
  - VkVideoDecodeH264InlineSessionParametersInfoKHR

If VK_KHR_video_decode_h265 is supported:

- Extending VkVideoDecodeInfoKHR:
  - VkVideoDecodeH265InlineSessionParametersInfoKHR

New Enum Constants

- VK_KHR_VIDEO_MAINTENANCE_2_EXTENSION_NAME
- VK_KHR_VIDEO_MAINTENANCE_2_SPEC_VERSION
- Extending VkStructureType:

    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_MAINTENANCE_2_FEATURES_KHR

If VK_KHR_video_decode_av1 is supported:

- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_INLINE_SESSION_PARAMETERS_INFO_KHR

If VK_KHR_video_decode_h264 is supported:

- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_INLINE_SESSION_PARAMETERS_INFO_KHR

If VK_KHR_video_decode_h265 is supported:

- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_INLINE_SESSION_PARAMETERS_INFO_KHR

If VK_KHR_video_decode_queue is supported:

- Extending VkVideoSessionCreateFlagBitsKHR:
    - VK_VIDEO_SESSION_CREATE_INLINE_SESSION_PARAMETERS_BIT_KHR

Version History

- Revision 1, 2024-10-14 (Daniel Rakos)
    - internal revisions

VK_KHR_video_queue

Name String

VK_KHR_video_queue

Extension Type

Device extension

Registered Extension Number

24

Revision

8

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.1
and
VK_KHR_synchronization2
or
Vulkan Version 1.3

Contact

- Tony Zlatinski [GitHub]tzlatinski

Extension Proposal

VK_KHR_video_queue

Other Extension Metadata

Last Modified Date

2022-09-29

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- George Hao, AMD
- Jake Beju, AMD
- Piers Daniell, NVIDIA
- Srinath Kumarapuram, NVIDIA
- Tobias Hector, AMD
- Tony Zlatinski, NVIDIA

- Daniel Rakos, RasterGrid

Description

This extension provides common APIs to enable exposing queue families with support for video codec operations by introducing the following new object types and related functionalities:

- Video session objects that represent and maintain the state needed to perform video codec operations.
- Video session parameters objects that act as a container for codec specific parameters.

In addition, it also introduces query commands that allow applications to determine video coding related capabilities, and command buffer commands that enable recording video coding operations against a video session.

This extension is to be used in conjunction with other extensions that enable specific video coding operations.

New Object Types

- VkVideoSessionKHR
- VkVideoSessionParametersKHR

New Commands

- vkBindVideoSessionMemoryKHR
- vkCmdBeginVideoCodingKHR
- vkCmdControlVideoCodingKHR
- vkCmdEndVideoCodingKHR
- vkCreateVideoSessionKHR
- vkCreateVideoSessionParametersKHR
- vkDestroyVideoSessionKHR
- vkDestroyVideoSessionParametersKHR
- vkGetPhysicalDeviceVideoCapabilitiesKHR
- vkGetPhysicalDeviceVideoFormatPropertiesKHR
- vkGetVideoSessionMemoryRequirementsKHR
- vkUpdateVideoSessionParametersKHR

New Structures

- VkBindVideoSessionMemoryInfoKHR
- VkPhysicalDeviceVideoFormatInfoKHR
- VkVideoBeginCodingInfoKHR
- VkVideoCapabilitiesKHR

- VkVideoCodingControlInfoKHR
- VkVideoEndCodingInfoKHR
- VkVideoFormatPropertiesKHR
- VkVideoPictureResourceInfoKHR
- VkVideoReferenceSlotInfoKHR
- VkVideoSessionCreateInfoKHR
- VkVideoSessionMemoryRequirementsKHR
- VkVideoSessionParametersCreateInfoKHR
- VkVideoSessionParametersUpdateInfoKHR
- Extending VkPhysicalDeviceImageFormatInfo2, VkPhysicalDeviceVideoFormatInfoKHR, VkImageCreateInfo, VkBufferCreateInfo:
  - VkVideoProfileListInfoKHR
- Extending VkQueryPoolCreateInfo:
  - VkVideoProfileInfoKHR
- Extending VkQueueFamilyProperties2:
  - VkQueueFamilyQueryResultStatusPropertiesKHR
  - VkQueueFamilyVideoPropertiesKHR

New Enums

- VkQueryResultStatusKHR
- VkVideoCapabilityFlagBitsKHR
- VkVideoChromaSubsamplingFlagBitsKHR
- VkVideoCodecOperationFlagBitsKHR
- VkVideoCodingControlFlagBitsKHR
- VkVideoComponentBitDepthFlagBitsKHR
- VkVideoSessionCreateFlagBitsKHR

New Bitmasks

- VkVideoBeginCodingFlagsKHR
- VkVideoCapabilityFlagsKHR
- VkVideoChromaSubsamplingFlagsKHR
- VkVideoCodecOperationFlagsKHR
- VkVideoCodingControlFlagsKHR
- VkVideoComponentBitDepthFlagsKHR
- VkVideoEndCodingFlagsKHR
- VkVideoSessionCreateFlagsKHR

- VkVideoSessionParametersCreateFlagsKHR

New Enum Constants

- `VK_KHR_VIDEO_QUEUE_EXTENSION_NAME`
- `VK_KHR_VIDEO_QUEUE_SPEC_VERSION`
- Extending VkObjectType:
  - VK_OBJECT_TYPE_VIDEO_SESSION_KHR
  - VK_OBJECT_TYPE_VIDEO_SESSION_PARAMETERS_KHR
- Extending VkQueryResultFlagBits:
  - VK_QUERY_RESULT_WITH_STATUS_BIT_KHR
- Extending VkQueryType:
  - VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR
- Extending VkResult:
  - VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR
  - VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR
  - VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR
  - VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR
  - VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR
  - VK_ERROR_VIDEO_STD_VERSION_NOT_SUPPORTED_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_BIND_VIDEO_SESSION_MEMORY_INFO_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_FORMAT_INFO_KHR
  - VK_STRUCTURE_TYPE_QUEUE_FAMILY_QUERY_RESULT_STATUS_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_QUEUE_FAMILY_VIDEO_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_BEGIN_CODING_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_CAPABILITIES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_CODING_CONTROL_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_END_CODING_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_FORMAT_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_VIDEO_PICTURE_RESOURCE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_PROFILE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_PROFILE_LIST_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_REFERENCE_SLOT_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_SESSION_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_VIDEO_SESSION_MEMORY_REQUIREMENTS_KHR

- VK_STRUCTURE_TYPE_VIDEO_SESSION_PARAMETERS_CREATE_INFO_KHR
- VK_STRUCTURE_TYPE_VIDEO_SESSION_PARAMETERS_UPDATE_INFO_KHR

Version History

- Revision 0.1, 2019-11-21 (Tony Zlatinski)
  - Initial draft
- Revision 0.2, 2019-11-27 (Tony Zlatinski)
  - Make vulkan video core common between decode and encode
- Revision 1, March 29 2021 (Tony Zlatinski)
  - Spec and API updates.
- Revision 2, August 1 2021 (Srinath Kumarapuram)
  - Rename `VkVideoCapabilitiesFlagBitsKHR` to `VkVideoCapabilityFlagBitsKHR` (along with the names of enumerants it defines) and `VkVideoCapabilitiesFlagsKHR` to `VkVideoCapabilityFlagsKHR`, following Vulkan naming conventions.
- Revision 3, 2022-03-16 (Ahmed Abdelkhalek)
  - Relocate Std header version reporting/requesting from codec-operation specific extensions to this extension.
  - Make Std header versions codec-operation specific instead of only codec-specific.
- Revision 4, 2022-05-30 (Daniel Rakos)
  - Refactor the video format query APIs and related language
  - Extend VkResult with video-specific error codes
- Revision 5, 2022-08-11 (Daniel Rakos)
  - Add `VkVideoSessionParametersCreateFlagsKHR`
  - Remove `VkVideoCodingQualityPresetFlagsKHR`
  - Rename `VkQueueFamilyQueryResultStatusProperties2KHR` to `VkQueueFamilyQueryResultStatusPropertiesKHR`
  - Rename `VkVideoQueueFamilyProperties2KHR` to `VkQueueFamilyVideoPropertiesKHR`
  - Rename `VkVideoProfileKHR` to `VkVideoProfileInfoKHR`
  - Rename `VkVideoProfilesKHR` to `VkVideoProfileListInfoKHR`
  - Rename `VkVideoGetMemoryPropertiesKHR` to `VkVideoSessionMemoryRequirementsKHR`
  - Rename `VkVideoBindMemoryKHR` to `VkBindVideoSessionMemoryInfoKHR`
  - Fix `pNext` constness of `VkPhysicalDeviceVideoFormatInfoKHR` and `VkVideoSessionMemoryRequirementsKHR`
  - Fix incorrectly named value enums in bit enum types `VkVideoCodecOperationFlagBitsKHR` and `VkVideoChromaSubsamplingFlagBitsKHR`
  - Remove unnecessary default values from `VkVideoSessionCreateFlagBitsKHR` and `VkVideoCodingControlFlagBitsKHR`

- Eliminate nested pointer in `VkVideoSessionMemoryRequirementsKHR`
- Rename `VkVideoPictureResourceKHR` to `VkVideoPictureResourceInfoKHR`
- Rename `VkVideoReferenceSlotKHR` to `VkVideoReferenceSlotInfoKHR`

- Revision 6, 2022-09-18 (Daniel Rakos)
  - Rename the `maxReferencePicturesSlotsCount` and `maxReferencePicturesActiveCount` fields of `VkVideoCapabilitiesKHR` and `VkVideoSessionCreateInfoKHR` to `maxDpbSlots` and `maxActiveReferencePictures`, respectively, to clarify their meaning
  - Rename `capabilityFlags` to `flags` in `VkVideoCapabilitiesKHR`
  - Rename `videoPictureExtentGranularity` to `pictureAccessGranularity` in `VkVideoCapabilitiesKHR`
  - Rename `minExtent` and `maxExtent` to `minCodedExtent` and `maxCodedExtent`, respectively, in `VkVideoCapabilitiesKHR`
  - Rename `referencePicturesFormat` to `referencePictureFormat` in `VkVideoSessionCreateInfoKHR`
- Revision 7, 2022-09-26 (Daniel Rakos)
  - Change type of `VkVideoReferenceSlotInfoKHR::slotIndex` from `int8_t` to `int32_t`
- Revision 8, 2022-09-29 (Daniel Rakos)
  - Extension is no longer provisional

VK_KHR_wayland_surface

Name String

`VK_KHR_wayland_surface`

Extension Type

Instance extension

Registered Extension Number

7

Revision

6

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_surface

Contact

- Jesse Hall [GitHub]critsec
- Ian Elliott [GitHub]ianelliottus

Other Extension Metadata

Last Modified Date

2015-11-28

IP Status

No known IP claims.

Contributors

- Patrick Doane, Blizzard
- Faith Ekstrand, Intel
- Ian Elliott, LunarG
- Courtney Goeltzenleuchter, LunarG
- Jesse Hall, Google
- James Jones, NVIDIA
- Antoine Labour, Google
- Jon Leech, Khronos
- David Mao, AMD
- Norbert Nopper, Freescale
- Alon Or-bach, Samsung

- Daniel Rakos, AMD
- Graham Sellers, AMD
- Ray Smith, ARM
- Jeff Vigil, Qualcomm
- Chia-I Wu, LunarG

Description

The VK_KHR_wayland_surface extension is an instance extension. It provides a mechanism to create a VkSurfaceKHR object (defined by the VK_KHR_surface extension) that refers to a Wayland wl_surface, as well as a query to determine support for rendering to a Wayland compositor.

New Commands

- vkCreateWaylandSurfaceKHR
- vkGetPhysicalDeviceWaylandPresentationSupportKHR

New Structures

- VkWaylandSurfaceCreateInfoKHR

New Bitmasks

- VkWaylandSurfaceCreateFlagsKHR

New Enum Constants

- VK_KHR_WAYLAND_SURFACE_EXTENSION_NAME
- VK_KHR_WAYLAND_SURFACE_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_WAYLAND_SURFACE_CREATE_INFO_KHR

Issues

1) Does Wayland need a way to query for compatibility between a particular physical device and a specific Wayland display? This would be a more general query than vkGetPhysicalDeviceSurfaceSupportKHR: if the Wayland-specific query returned VK_TRUE for a (VkPhysicalDevice, struct wl_display*) pair, then the physical device could be assumed to support presentation to any VkSurfaceKHR for surfaces on the display.

RESOLVED: Yes. vkGetPhysicalDeviceWaylandPresentationSupportKHR was added to address this issue.

2) Should we require surfaces created with vkCreateWaylandSurfaceKHR to support the VK_PRESENT_MODE_MAILBOX_KHR present mode?

RESOLVED: Yes. Wayland is an inherently mailbox window system and mailbox support is

required for some Wayland compositor interactions to work as expected. While handling these interactions may be possible with VK_PRESENT_MODE_FIFO_KHR, it is much more difficult to do without deadlock and requiring all Wayland applications to be able to support implementations which only support VK_PRESENT_MODE_FIFO_KHR would be an onerous restriction on application developers.

Version History

- Revision 1, 2015-09-23 (Jesse Hall)
  - Initial draft, based on the previous contents of VK_EXT_KHR_swapchain (later renamed VK_EXT_KHR_surface).
- Revision 2, 2015-10-02 (James Jones)
  - Added vkGetPhysicalDeviceWaylandPresentationSupportKHR() to resolve issue #1.
  - Adjusted wording of issue #1 to match the agreed-upon solution.
  - Renamed “window” parameters to “surface” to match Wayland conventions.
- Revision 3, 2015-10-26 (Ian Elliott)
  - Renamed from VK_EXT_KHR_wayland_surface to VK_KHR_wayland_surface.
- Revision 4, 2015-11-03 (Daniel Rakos)
  - Added allocation callbacks to vkCreateWaylandSurfaceKHR.
- Revision 5, 2015-11-28 (Daniel Rakos)
  - Updated the surface create function to take a pCreateInfo structure.
- Revision 6, 2017-02-08 (Faith Ekstrand)
  - Added the requirement that implementations support VK_PRESENT_MODE_MAILBOX_KHR.
  - Added wording about interactions between vkQueuePresentKHR and the Wayland requests sent to the compositor.

VK_KHR_win32_keyed_mutex

Name String

VK_KHR_win32_keyed_mutex

Extension Type

Device extension

Registered Extension Number

76

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_external_memory_win32

Contact

- Carsten Rohde [GitHub]crohde

Other Extension Metadata

Last Modified Date

2016-10-21

IP Status

No known IP claims.

Contributors

- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Carsten Rohde, NVIDIA

Description

Applications that wish to import Direct3D 11 memory objects into the Vulkan API may wish to use the native keyed mutex mechanism to synchronize access to the memory between Vulkan and Direct3D. This extension provides a way for an application to access the keyed mutex associated with an imported Vulkan memory object when submitting command buffers to a queue.

New Structures

- Extending VkSubmitInfo, VkSubmitInfo2:
  - VkWin32KeyedMutexAcquireReleaseInfoKHR

New Enum Constants

- `VK_KHR_WIN32_KEYED_MUTEX_EXTENSION_NAME`
- `VK_KHR_WIN32_KEYED_MUTEX_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_WIN32_KEYED_MUTEX_ACQUIRE_RELEASE_INFO_KHR

Version History

- Revision 1, 2016-10-21 (James Jones)
  - Initial revision

VK_KHR_win32_surface

Name String

`VK_KHR_win32_surface`

Extension Type

Instance extension

Registered Extension Number

10

Revision

6

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_surface

Contact

- Jesse Hall [GitHub]critsec
- Ian Elliott [GitHub]ianelliottus

Other Extension Metadata

Last Modified Date

2017-04-24

IP Status

No known IP claims.

Contributors

- Patrick Doane, Blizzard
- Faith Ekstrand, Intel
- Ian Elliott, LunarG
- Courtney Goeltzenleuchter, LunarG
- Jesse Hall, Google
- James Jones, NVIDIA
- Antoine Labour, Google
- Jon Leech, Khronos
- David Mao, AMD
- Norbert Nopper, Freescale
- Alon Or-bach, Samsung

- Daniel Rakos, AMD
- Graham Sellers, AMD
- Ray Smith, ARM
- Jeff Vigil, Qualcomm
- Chia-I Wu, LunarG

Description

The `VK_KHR_win32_surface` extension is an instance extension. It provides a mechanism to create a VkSurfaceKHR object (defined by the `VK_KHR_surface` extension) that refers to a Win32 `HWND`, as well as a query to determine support for rendering to the windows desktop.

New Commands

- vkCreateWin32SurfaceKHR
- vkGetPhysicalDeviceWin32PresentationSupportKHR

New Structures

- VkWin32SurfaceCreateInfoKHR

New Bitmasks

- VkWin32SurfaceCreateFlagsKHR

New Enum Constants

- `VK_KHR_WIN32_SURFACE_EXTENSION_NAME`
- `VK_KHR_WIN32_SURFACE_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_WIN32_SURFACE_CREATE_INFO_KHR

Issues

1) Does Win32 need a way to query for compatibility between a particular physical device and a specific screen? Compatibility between a physical device and a window generally only depends on what screen the window is on. However, there is not an obvious way to identify a screen without already having a window on the screen.

RESOLVED: No. While it may be useful, there is not a clear way to do this on Win32. However, a method was added to query support for presenting to the windows desktop as a whole.

2) If a native window object (`HWND`) is used by one graphics API, and then is later used by a different graphics API (one of which is Vulkan), can these uses interfere with each other?

RESOLVED: Yes.

Uses of a window object by multiple graphics APIs results in undefined behavior. Such behavior

may succeed when using one Vulkan implementation but fail when using a different Vulkan implementation. Potential failures include:

- Creating then destroying a flip presentation model DXGI swapchain on a window object can prevent vkCreateSwapchainKHR from succeeding on the same window object.
- Creating then destroying a VkSwapchainKHR on a window object can prevent creation of a bitblt model DXGI swapchain on the same window object.
- Creating then destroying a VkSwapchainKHR on a window object can effectively `SetPixelFormat` to a different format than the format chosen by an OpenGL application.
- Creating then destroying a VkSwapchainKHR on a window object on one VkPhysicalDevice can prevent vkCreateSwapchainKHR from succeeding on the same window object, but on a different VkPhysicalDevice that is associated with a different Vulkan ICD.

In all cases the problem can be worked around by creating a new window object.

Technical details include:

- Creating a DXGI swapchain over a window object can alter the object for the remainder of its lifetime. The alteration persists even after the DXGI swapchain has been destroyed. This alteration can make it impossible for a conformant Vulkan implementation to create a VkSwapchainKHR over the same window object. Mention of this alteration can be found in the remarks section of the MSDN documentation for `DXGI_SWAP_EFFECT`.
- Calling GDI's `SetPixelFormat` (needed by OpenGL's WGL layer) on a window object alters the object for the remainder of its lifetime. The MSDN documentation for `SetPixelFormat` explains that a window object's pixel format can be set only one time.
- Creating a VkSwapchainKHR over a window object can alter the object for its remaining lifetime. Either of the above alterations may occur as a side effect of vkCreateSwapchainKHR.

Version History

- Revision 1, 2015-09-23 (Jesse Hall)
  - Initial draft, based on the previous contents of VK_EXT_KHR_swapchain (later renamed VK_EXT_KHR_surface).
- Revision 2, 2015-10-02 (James Jones)
  - Added presentation support query for win32 desktops.
- Revision 3, 2015-10-26 (Ian Elliott)
  - Renamed from VK_EXT_KHR_win32_surface to VK_KHR_win32_surface.
- Revision 4, 2015-11-03 (Daniel Rakos)
  - Added allocation callbacks to vkCreateWin32SurfaceKHR.
- Revision 5, 2015-11-28 (Daniel Rakos)
  - Updated the surface create function to take a pCreateInfo structure.
- Revision 6, 2017-04-24 (Jeff Juliano)
  - Add issue 2 addressing reuse of a native window object in a different Graphics API, or by a

different Vulkan ICD.

VK_KHR_workgroup_memory_explicit_layout

Name String

`VK_KHR_workgroup_memory_explicit_layout`

Extension Type

Device extension

Registered Extension Number

337

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_workgroup_memory_explicit_layout

Contact

- Caio Marcelo de Oliveira Filho [GitHub]cmarcelo

Other Extension Metadata

Last Modified Date

2020-06-01

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for `GL_EXT_shared_memory_block`

Contributors

- Caio Marcelo de Oliveira Filho, Intel
- Jeff Bolz, NVIDIA
- Graeme Leese, Broadcom
- Faith Ekstrand, Intel
- Daniel Koch, NVIDIA

Description

This extension adds Vulkan support for the `SPV_KHR_workgroup_memory_explicit_layout` SPIR-V extension, which allows shaders to explicitly define the layout of `Workgroup` storage class memory and create aliases between variables from that storage class in a compute shader.

The aliasing feature allows different "views" on the same data, so the shader can bulk copy data from another storage class using one type (e.g. an array of large vectors), and then use the data with a more specific type. It also enables reducing the amount of workgroup memory consumed by allowing the shader to alias data whose lifetimes do not overlap.

The explicit layout support and some form of aliasing is also required for layering OpenCL on top of Vulkan.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR

New Enum Constants

- `VK_KHR_WORKGROUP_MEMORY_EXPLICIT_LAYOUT_EXTENSION_NAME`
- `VK_KHR_WORKGROUP_MEMORY_EXPLICIT_LAYOUT_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_WORKGROUP_MEMORY_EXPLICIT_LAYOUT_FEATURES_KHR

New SPIR-V Capabilities

- `WorkgroupMemoryExplicitLayoutKHR`
- `WorkgroupMemoryExplicitLayout8BitAccessKHR`
- `WorkgroupMemoryExplicitLayout16BitAccessKHR`

Version History

- Revision 1, 2020-06-01 (Caio Marcelo de Oliveira Filho)
  - Initial version

VK_KHR_xcb_surface

Name String

`VK_KHR_xcb_surface`

Extension Type

Instance extension

Registered Extension Number

6

Revision

6

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_surface

Contact

- Jesse Hall [GitHub]critsec
- Ian Elliott [GitHub]ianelliottus

Other Extension Metadata

Last Modified Date

2015-11-28

IP Status

No known IP claims.

Contributors

- Patrick Doane, Blizzard
- Faith Ekstrand, Intel
- Ian Elliott, LunarG
- Courtney Goeltzenleuchter, LunarG
- Jesse Hall, Google
- James Jones, NVIDIA
- Antoine Labour, Google
- Jon Leech, Khronos
- David Mao, AMD
- Norbert Nopper, Freescale
- Alon Or-bach, Samsung

- Daniel Rakos, AMD
- Graham Sellers, AMD
- Ray Smith, ARM
- Jeff Vigil, Qualcomm
- Chia-I Wu, LunarG

Description

The `VK_KHR_xcb_surface` extension is an instance extension. It provides a mechanism to create a VkSurfaceKHR object (defined by the `VK_KHR_surface` extension) that refers to an X11 `Window`, using the XCB client-side library, as well as a query to determine support for rendering via XCB.

New Commands

- vkCreateXcbSurfaceKHR
- vkGetPhysicalDeviceXcbPresentationSupportKHR

New Structures

- VkXcbSurfaceCreateInfoKHR

New Bitmasks

- VkXcbSurfaceCreateFlagsKHR

New Enum Constants

- `VK_KHR_XCB_SURFACE_EXTENSION_NAME`
- `VK_KHR_XCB_SURFACE_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_XCB_SURFACE_CREATE_INFO_KHR

Issues

1) Does XCB need a way to query for compatibility between a particular physical device and a specific screen? This would be a more general query than vkGetPhysicalDeviceSurfaceSupportKHR: If it returned VK_TRUE, then the physical device could be assumed to support presentation to any window on that screen.

RESOLVED: Yes, this is needed for toolkits that want to create a VkDevice before creating a window. To ensure the query is reliable, it must be made against a particular X visual rather than the screen in general.

Version History

- Revision 1, 2015-09-23 (Jesse Hall)
  - Initial draft, based on the previous contents of VK_EXT_KHR_swapchain (later renamed

VK_EXT_KHR_surface).

- Revision 2, 2015-10-02 (James Jones)
  - Added presentation support query for an (xcb_connection_t*, xcb_visualid_t) pair.
  - Removed “root” parameter from CreateXcbSurfaceKHR(), as it is redundant when a window on the same screen is specified as well.
  - Adjusted wording of issue #1 and added agreed upon resolution.
- Revision 3, 2015-10-14 (Ian Elliott)
  - Removed “root” parameter from CreateXcbSurfaceKHR() in one more place.
- Revision 4, 2015-10-26 (Ian Elliott)
  - Renamed from VK_EXT_KHR_xcb_surface to VK_KHR_xcb_surface.
- Revision 5, 2015-10-23 (Daniel Rakos)
  - Added allocation callbacks to vkCreateXcbSurfaceKHR.
- Revision 6, 2015-11-28 (Daniel Rakos)
  - Updated the surface create function to take a pCreateInfo structure.

VK_KHR_xlib_surface

Name String

`VK_KHR_xlib_surface`

Extension Type

Instance extension

Registered Extension Number

5

Revision

6

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_surface

Contact

- Jesse Hall [GitHub]critsec
- Ian Elliott [GitHub]ianelliottus

Other Extension Metadata

Last Modified Date

2015-11-28

IP Status

No known IP claims.

Contributors

- Patrick Doane, Blizzard
- Faith Ekstrand, Intel
- Ian Elliott, LunarG
- Courtney Goeltzenleuchter, LunarG
- Jesse Hall, Google
- James Jones, NVIDIA
- Antoine Labour, Google
- Jon Leech, Khronos
- David Mao, AMD
- Norbert Nopper, Freescale
- Alon Or-bach, Samsung

- Daniel Rakos, AMD
- Graham Sellers, AMD
- Ray Smith, ARM
- Jeff Vigil, Qualcomm
- Chia-I Wu, LunarG

Description

The VK_KHR_xlib_surface extension is an instance extension. It provides a mechanism to create a VkSurfaceKHR object (defined by the VK_KHR_surface extension) that refers to an X11 Window, using the Xlib client-side library, as well as a query to determine support for rendering via Xlib.

New Commands

- vkCreateXlibSurfaceKHR
- vkGetPhysicalDeviceXlibPresentationSupportKHR

New Structures

- VkXlibSurfaceCreateInfoKHR

New Bitmasks

- VkXlibSurfaceCreateFlagsKHR

New Enum Constants

- VK_KHR_XLIB_SURFACE_EXTENSION_NAME
- VK_KHR_XLIB_SURFACE_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_XLIB_SURFACE_CREATE_INFO_KHR

Issues

1) Does X11 need a way to query for compatibility between a particular physical device and a specific screen? This would be a more general query than vkGetPhysicalDeviceSurfaceSupportKHR; if it returned VK_TRUE, then the physical device could be assumed to support presentation to any window on that screen.

RESOLVED: Yes, this is needed for toolkits that want to create a VkDevice before creating a window. To ensure the query is reliable, it must be made against a particular X visual rather than the screen in general.

Version History

- Revision 1, 2015-09-23 (Jesse Hall)
  - Initial draft, based on the previous contents of VK_EXT_KHR_swapchain (later renamed

VK_EXT_KHR_surface).

- Revision 2, 2015-10-02 (James Jones)
  - Added presentation support query for (Display*, VisualID) pair.
  - Removed “root” parameter from CreateXlibSurfaceKHR(), as it is redundant when a window on the same screen is specified as well.
  - Added appropriate X errors.
  - Adjusted wording of issue #1 and added agreed upon resolution.
- Revision 3, 2015-10-14 (Ian Elliott)
  - Renamed this extension from VK_EXT_KHR_x11_surface to VK_EXT_KHR_xlib_surface.
- Revision 4, 2015-10-26 (Ian Elliott)
  - Renamed from VK_EXT_KHR_xlib_surface to VK_KHR_xlib_surface.
- Revision 5, 2015-11-03 (Daniel Rakos)
  - Added allocation callbacks to vkCreateXlibSurfaceKHR.
- Revision 6, 2015-11-28 (Daniel Rakos)
  - Updated the surface create function to take a pCreateInfo structure.

VK_EXT_acquire_drm_display

Name String

`VK_EXT_acquire_drm_display`

Extension Type

Instance extension

Registered Extension Number

286

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_EXT_direct_mode_display

Contact

- Drew DeVault sir@cmpwn.com

Other Extension Metadata

Last Modified Date

2021-06-09

IP Status

No known IP claims.

Contributors

- Simon Zeni, Status Holdings, Ltd.

Description

This extension allows an application to take exclusive control of a display using the Direct Rendering Manager (DRM) interface. When acquired, the display will be under full control of the application until the display is either released or the connector is unplugged.

New Commands

- vkAcquireDrmDisplayEXT
- vkGetDrmDisplayEXT

New Enum Constants

- `VK_EXT_ACQUIRE_DRM_DISPLAY_EXTENSION_NAME`

- `VK_EXT_ACQUIRE_DRM_DISPLAY_SPEC_VERSION`

Issues

None.

Version History

- Revision 1, 2021-05-11 (Simon Zeni)
  - Initial draft

VK_EXT_acquire_xlib_display

Name String

`VK_EXT_acquire_xlib_display`

Extension Type

Instance extension

Registered Extension Number

90

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_EXT_direct_mode_display

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2016-12-13

IP Status

No known IP claims.

Contributors

- Dave Airlie, Red Hat
- Pierre Boudier, NVIDIA
- James Jones, NVIDIA
- Damien Leone, NVIDIA
- Pierre-Loup Griffais, Valve
- Liam Middlebrook, NVIDIA
- Daniel Vetter, Intel

Description

This extension allows an application to take exclusive control on a display currently associated with an X11 screen. When control is acquired, the display will be deassociated from the X11 screen until control is released or the specified display connection is closed. Essentially, the X11 screen will behave as if the monitor has been unplugged until control is released.

New Commands

- vkAcquireXlibDisplayEXT
- vkGetRandROutputDisplayEXT

New Enum Constants

- `VK_EXT_ACQUIRE_XLIB_DISPLAY_EXTENSION_NAME`
- `VK_EXT_ACQUIRE_XLIB_DISPLAY_SPEC_VERSION`

Issues

1) Should vkAcquireXlibDisplayEXT take an RandR display ID, or a Vulkan display handle as input?

RESOLVED: A Vulkan display handle. Otherwise there would be no way to specify handles to displays that had been prevented from being included in the X11 display list by some native platform or vendor-specific mechanism.

2) How does an application figure out which RandR display corresponds to a Vulkan display?

RESOLVED: A new function, vkGetRandROutputDisplayEXT, is introduced for this purpose.

3) Should vkGetRandROutputDisplayEXT be part of this extension, or a general Vulkan / RandR or Vulkan / Xlib extension?

RESOLVED: To avoid yet another extension, include it in this extension.

Version History

- Revision 1, 2016-12-13 (James Jones)
  - Initial draft

VK_EXT_astc_decode_mode

Name String

VK_EXT_astc_decode_mode

Extension Type

Device extension

Registered Extension Number

68

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Jan-Harald Fredriksen [GitHub]janharaldfredriksen-arm

Other Extension Metadata

Last Modified Date

2018-08-07

Contributors

- Jan-Harald Fredriksen, Arm

Description

The existing specification requires that low dynamic range (LDR) ASTC textures are decompressed to FP16 values per component. In many cases, decompressing LDR textures to a lower precision intermediate result gives acceptable image quality. Source material for LDR textures is typically authored as 8-bit UNORM values, so decoding to FP16 values adds little value. On the other hand, reducing precision of the decoded result reduces the size of the decompressed data, potentially improving texture cache performance and saving power.

The goal of this extension is to enable this efficiency gain on existing ASTC texture data. This is achieved by giving the application the ability to select the intermediate decoding precision.

Three decoding options are provided:

- Decode to VK_FORMAT_R16G16B16A16_SFLOAT precision: This is the default, and matches the required behavior in the core API.

- Decode to VK_FORMAT_R8G8B8A8_UNORM precision: This is provided as an option in LDR mode.
- Decode to VK_FORMAT_E5B9G9R9_UFLOAT_PACK32 precision: This is provided as an option in both LDR and HDR mode. In this mode, negative values cannot be represented and are clamped to zero. The alpha component is ignored, and the results are as if alpha was 1.0. This decode mode is optional and support can be queried via the physical device properties.

New Structures

- Extending VkImageViewCreateInfo:
  - VkImageViewASTCDecodeModeEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceASTCDecodeFeaturesEXT

New Enum Constants

- VK_EXT_ASTC_DECODE_MODE_EXTENSION_NAME
- VK_EXT_ASTC_DECODE_MODE_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_IMAGE_VIEW_ASTC_DECODE_MODE_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ASTC_DECODE_FEATURES_EXT

Issues

1) Are implementations allowed to decode at a higher precision than what is requested?

```
RESOLUTION: No.
If we allow this, then this extension could be exposed on all
implementations that support ASTC.
But developers would have no way of knowing what precision was actually
used, and thus whether the image quality is sufficient at reduced
precision.
```

2) Should the decode mode be image view state and/or sampler state?

```
RESOLUTION: Image view state only.
Some implementations treat the different decode modes as different
texture formats.
```

Example

Create an image view that decodes to VK_FORMAT_R8G8B8A8_UNORM precision:

```
    VkImageViewASTCDecodeModeEXT decodeMode =
```

```
    {
        .sType = VK_STRUCTURE_TYPE_IMAGE_VIEW_ASTC_DECODE_MODE_EXT,
        .pNext = NULL,
        .decodeMode = VK_FORMAT_R8G8B8A8_UNORM
    };

    VkImageViewCreateInfo createInfo =
    {
        .sType = VK_STRUCTURE_TYPE_IMAGE_VIEW_CREATE_INFO,
        .pNext = &decodeMode,
        // flags, image, viewType set to application-desired values
        .format = VK_FORMAT_ASTC_8x8_UNORM_BLOCK,
        // components, subresourceRange set to application-desired values
    };

    VkImageView imageView;
    VkResult result = vkCreateImageView(
        device,
        &createInfo,
        NULL,
        &imageView);
```

Version History

- Revision 1, 2018-08-07 (Jan-Harald Fredriksen)
    - Initial revision

VK_EXT_attachment_feedback_loop_dynamic_state

Name String

VK_EXT_attachment_feedback_loop_dynamic_state

Extension Type

Device extension

Registered Extension Number

525

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1
and
VK_EXT_attachment_feedback_loop_layout

Special Uses

- OpenGL / ES support
- D3D support

Contact

- Mike Blumenkrantz [GitHub]zmike

Extension Proposal

VK_EXT_attachment_feedback_loop_dynamic_state

Other Extension Metadata

Last Modified Date

2023-04-28

IP Status

No known IP claims.

Contributors

- Mike Blumenkrantz, Valve
- Daniel Story, Nintendo
- Stu Smith, AMD
- Samuel Pitoiset, Valve

- Ricardo Garcia, Igalia

Description

This extension adds support for setting attachment feedback loops dynamically on command buffers.

New Commands

- vkCmdSetAttachmentFeedbackLoopEnableEXT

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT

New Enum Constants

- `VK_EXT_ATTACHMENT_FEEDBACK_LOOP_DYNAMIC_STATE_EXTENSION_NAME`
- `VK_EXT_ATTACHMENT_FEEDBACK_LOOP_DYNAMIC_STATE_SPEC_VERSION`
- Extending VkDynamicState:
  - VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ATTACHMENT_FEEDBACK_LOOP_DYNAMIC_STATE_FEATURES_EXT

Version History

- Revision 1, 2023-04-28 (Mike Blumenkrantz)
  - Initial revision

VK_EXT_attachment_feedback_loop_layout

Name String

`VK_EXT_attachment_feedback_loop_layout`

Extension Type

Device extension

Registered Extension Number

340

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Special Uses

- OpenGL / ES support
- D3D support

Contact

- Joshua Ashton [GitHub]Joshua-Ashton

Extension Proposal

VK_EXT_attachment_feedback_loop_layout

Other Extension Metadata

Last Modified Date

2022-04-04

IP Status

No known IP claims.

Contributors

- Joshua Ashton, Valve
- Faith Ekstrand, Collabora
- Bas Nieuwenhuizen, Google
- Samuel Iglesias Gonsálvez, Igalia
- Ralph Potter, Samsung
- Jan-Harald Fredriksen, Arm

- Ricardo Garcia, Igalia

Description

This extension adds a new image layout, VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT, which allows applications to have an image layout in which they are able to both render to and sample/fetch from the same subresource of an image in a given render pass.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceAttachmentFeedbackLoopLayoutFeaturesEXT

New Enum Constants

- `VK_EXT_ATTACHMENT_FEEDBACK_LOOP_LAYOUT_EXTENSION_NAME`
- `VK_EXT_ATTACHMENT_FEEDBACK_LOOP_LAYOUT_SPEC_VERSION`
- Extending VkDependencyFlagBits:
  - VK_DEPENDENCY_FEEDBACK_LOOP_BIT_EXT
- Extending VkImageLayout:
  - VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT
- Extending VkImageUsageFlagBits:
  - VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT
- Extending VkPipelineCreateFlagBits:
  - VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT
  - VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ATTACHMENT_FEEDBACK_LOOP_LAYOUT_FEATURES_EXT

Version History

- Revision 2, 2022-04-04 (Joshua Ashton)
  - Renamed from VALVE to EXT.
- Revision 1, 2021-03-09 (Joshua Ashton)
  - Initial draft.

VK_EXT_blend_operation_advanced

Name String

`VK_EXT_blend_operation_advanced`

Extension Type

Device extension

Registered Extension Number

149

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Jeff Bolz [GitHub]jeffbolznv

Other Extension Metadata

Last Modified Date

2017-06-12

Contributors

- Jeff Bolz, NVIDIA

Description

This extension adds a number of “advanced” blending operations that **can** be used to perform new color blending operations, many of which are more complex than the standard blend modes provided by unextended Vulkan. This extension requires different styles of usage, depending on the level of hardware support and the enabled features:

- If VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT::`advancedBlendCoherentOperations` is VK_FALSE, the new blending operations are supported, but a memory dependency **must** separate each advanced blend operation on a given sample. VK_ACCESS_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT is used to synchronize reads using advanced blend operations.
- If VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT::`advancedBlendCoherentOperations` is VK_TRUE, advanced blend operations obey primitive order just like basic blend operations.

In unextended Vulkan, the set of blending operations is limited, and **can** be expressed very simply.

The VK_BLEND_OP_MIN and VK_BLEND_OP_MAX blend operations simply compute component-wise minimums or maximums of source and destination color components. The VK_BLEND_OP_ADD, VK_BLEND_OP_SUBTRACT, and VK_BLEND_OP_REVERSE_SUBTRACT modes multiply the source and destination colors by source and destination factors and either add the two products together or subtract one from the other. This limited set of operations supports many common blending operations but precludes the use of more sophisticated transparency and blending operations commonly available in many dedicated imaging APIs.

This extension provides a number of new "advanced" blending operations. Unlike traditional blending operations using VK_BLEND_OP_ADD, these blending equations do not use source and destination factors specified by VkBlendFactor. Instead, each blend operation specifies a complete equation based on the source and destination colors. These new blend operations are used for both RGB and alpha components; they **must** not be used to perform separate RGB and alpha blending (via different values of color and alpha VkBlendOp).

These blending operations are performed using premultiplied colors, where RGB colors **can** be considered premultiplied or non-premultiplied by alpha, according to the `srcPremultiplied` and `dstPremultiplied` members of VkPipelineColorBlendAdvancedStateCreateInfoEXT. If a color is considered non-premultiplied, the (R,G,B) color components are multiplied by the alpha component prior to blending. For non-premultiplied color components in the range [0,1], the corresponding premultiplied color component would have values in the range [0 × A, 1 × A].

Many of these advanced blending equations are formulated where the result of blending source and destination colors with partial coverage have three separate contributions: from the portions covered by both the source and the destination, from the portion covered only by the source, and from the portion covered only by the destination. The blend parameter VkPipelineColorBlendAdvancedStateCreateInfoEXT::`blendOverlap` **can** be used to specify a correlation between source and destination pixel coverage. If set to VK_BLEND_OVERLAP_CONJOINT_EXT, the source and destination are considered to have maximal overlap, as would be the case if drawing two objects on top of each other. If set to VK_BLEND_OVERLAP_DISJOINT_EXT, the source and destination are considered to have minimal overlap, as would be the case when rendering a complex polygon tessellated into individual non-intersecting triangles. If set to VK_BLEND_OVERLAP_UNCORRELATED_EXT, the source and destination coverage are assumed to have no spatial correlation within the pixel.

In addition to the coherency issues on implementations not supporting `advancedBlendCoherentOperations`, this extension has several limitations worth noting. First, the new blend operations have a limit on the number of color attachments they **can** be used with, as indicated by VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT::`advancedBlendMaxColorAttachments`. Additionally, blending precision **may** be limited to 16-bit floating-point, which **may** result in a loss of precision and dynamic range for framebuffer formats with 32-bit floating-point components, and in a loss of precision for formats with 12- and 16-bit signed or unsigned normalized integer components.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT

- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT
- Extending VkPipelineColorBlendStateCreateInfo:
  - VkPipelineColorBlendAdvancedStateCreateInfoEXT

New Enums

- VkBlendOverlapEXT

New Enum Constants

- `VK_EXT_BLEND_OPERATION_ADVANCED_EXTENSION_NAME`
- `VK_EXT_BLEND_OPERATION_ADVANCED_SPEC_VERSION`
- Extending VkAccessFlagBits:
  - VK_ACCESS_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT
- Extending VkBlendOp:
  - VK_BLEND_OP_BLUE_EXT
  - VK_BLEND_OP_COLORBURN_EXT
  - VK_BLEND_OP_COLORDODGE_EXT
  - VK_BLEND_OP_CONTRAST_EXT
  - VK_BLEND_OP_DARKEN_EXT
  - VK_BLEND_OP_DIFFERENCE_EXT
  - VK_BLEND_OP_DST_ATOP_EXT
  - VK_BLEND_OP_DST_EXT
  - VK_BLEND_OP_DST_IN_EXT
  - VK_BLEND_OP_DST_OUT_EXT
  - VK_BLEND_OP_DST_OVER_EXT
  - VK_BLEND_OP_EXCLUSION_EXT
  - VK_BLEND_OP_GREEN_EXT
  - VK_BLEND_OP_HARDLIGHT_EXT
  - VK_BLEND_OP_HARDMIX_EXT
  - VK_BLEND_OP_HSL_COLOR_EXT
  - VK_BLEND_OP_HSL_HUE_EXT
  - VK_BLEND_OP_HSL_LUMINOSITY_EXT
  - VK_BLEND_OP_HSL_SATURATION_EXT
  - VK_BLEND_OP_INVERT_EXT
  - VK_BLEND_OP_INVERT_OVG_EXT

    - VK_BLEND_OP_INVERT_RGB_EXT
    - VK_BLEND_OP_LIGHTEN_EXT
    - VK_BLEND_OP_LINEARBURN_EXT
    - VK_BLEND_OP_LINEARDODGE_EXT
    - VK_BLEND_OP_LINEARLIGHT_EXT
    - VK_BLEND_OP_MINUS_CLAMPED_EXT
    - VK_BLEND_OP_MINUS_EXT
    - VK_BLEND_OP_MULTIPLY_EXT
    - VK_BLEND_OP_OVERLAY_EXT
    - VK_BLEND_OP_PINLIGHT_EXT
    - VK_BLEND_OP_PLUS_CLAMPED_ALPHA_EXT
    - VK_BLEND_OP_PLUS_CLAMPED_EXT
    - VK_BLEND_OP_PLUS_DARKER_EXT
    - VK_BLEND_OP_PLUS_EXT
    - VK_BLEND_OP_RED_EXT
    - VK_BLEND_OP_SCREEN_EXT
    - VK_BLEND_OP_SOFTLIGHT_EXT
    - VK_BLEND_OP_SRC_ATOP_EXT
    - VK_BLEND_OP_SRC_EXT
    - VK_BLEND_OP_SRC_IN_EXT
    - VK_BLEND_OP_SRC_OUT_EXT
    - VK_BLEND_OP_SRC_OVER_EXT
    - VK_BLEND_OP_VIVIDLIGHT_EXT
    - VK_BLEND_OP_XOR_EXT
    - VK_BLEND_OP_ZERO_EXT
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BLEND_OPERATION_ADVANCED_FEATURES_EXT
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BLEND_OPERATION_ADVANCED_PROPERTIES_EXT
    - VK_STRUCTURE_TYPE_PIPELINE_COLOR_BLEND_ADVANCED_STATE_CREATE_INFO_EXT

Issues

None.

Version History

- Revision 1, 2017-06-12 (Jeff Bolz)
  - Internal revisions
- Revision 2, 2017-06-12 (Jeff Bolz)
  - Internal revisions

VK_EXT_border_color_swizzle

Name String

`VK_EXT_border_color_swizzle`

Extension Type

Device extension

Registered Extension Number

412

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_EXT_custom_border_color

Special Uses

- OpenGL / ES support
- D3D support

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2021-10-12

IP Status

No known IP claims.

Contributors

- Graeme Leese, Broadcom
- Jan-Harald Fredriksen, Arm
- Ricardo Garcia, Igalia
- Shahbaz Youssefi, Google
- Stu Smith, AMD

Description

After the publication of VK_EXT_custom_border_color, it was discovered that some implementations had undefined behavior when combining a sampler that uses a custom border color with image views whose component mapping is not the identity mapping.

Since VK_EXT_custom_border_color has already shipped, this new extension VK_EXT_border_color_swizzle was created to define the interaction between custom border colors and non-identity image view swizzles, and provide a work-around for implementations that must pre-swizzle the sampler border color to match the image view component mapping it is combined with.

This extension also defines the behavior between samplers with an opaque black border color and image views with a non-identity component swizzle, which was previously left undefined.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceBorderColorSwizzleFeaturesEXT
- Extending VkSamplerCreateInfo:
  - VkSamplerBorderColorComponentMappingCreateInfoEXT

New Enum Constants

- `VK_EXT_BORDER_COLOR_SWIZZLE_EXTENSION_NAME`
- `VK_EXT_BORDER_COLOR_SWIZZLE_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BORDER_COLOR_SWIZZLE_FEATURES_EXT
  - VK_STRUCTURE_TYPE_SAMPLER_BORDER_COLOR_COMPONENT_MAPPING_CREATE_INFO_EXT

Issues

None.

Version History

- Revision 1, 2021-10-12 (Piers Daniell)
  - Internal revisions.

VK_EXT_color_write_enable

Name String

`VK_EXT_color_write_enable`

Extension Type

Device extension

Registered Extension Number

382

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Sharif Elcott [GitHub]selcott

Other Extension Metadata

Last Modified Date

2020-02-25

IP Status

No known IP claims.

Contributors

- Sharif Elcott, Google
- Tobias Hector, AMD
- Piers Daniell, NVIDIA

Description

This extension allows for selectively enabling and disabling writes to output color attachments via a pipeline dynamic state.

The intended use cases for this new state are mostly identical to those of colorWriteMask, such as selectively disabling writes to avoid feedback loops between subpasses or bandwidth savings for unused outputs. By making the state dynamic, one additional benefit is the ability to reduce pipeline counts and pipeline switching via shaders that write a superset of the desired data of which subsets are selected dynamically. The reason for a new state, colorWriteEnable, rather than

making colorWriteMask dynamic is that, on many implementations, the more flexible per-component semantics of the colorWriteMask state cannot be made dynamic in a performant manner.

New Commands

- vkCmdSetColorWriteEnableEXT

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceColorWriteEnableFeaturesEXT
- Extending VkPipelineColorBlendStateCreateInfo:
  - VkPipelineColorWriteCreateInfoEXT

New Enum Constants

- `VK_EXT_COLOR_WRITE_ENABLE_EXTENSION_NAME`
- `VK_EXT_COLOR_WRITE_ENABLE_SPEC_VERSION`
- Extending VkDynamicState:
  - VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COLOR_WRITE_ENABLE_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PIPELINE_COLOR_WRITE_CREATE_INFO_EXT

Version History

- Revision 1, 2020-01-25 (Sharif Elcott)
  - Internal revisions

VK_EXT_conditional_rendering

Name String

`VK_EXT_conditional_rendering`

Extension Type

Device extension

Registered Extension Number

82

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Vikram Kushwaha [GitHub]vkushwaha

Other Extension Metadata

Last Modified Date

2018-05-21

IP Status

No known IP claims.

Contributors

- Vikram Kushwaha, NVIDIA
- Daniel Rakos, AMD
- Jesse Hall, Google
- Jeff Bolz, NVIDIA
- Piers Daniell, NVIDIA
- Stuart Smith, Imagination Technologies

Description

This extension allows the execution of one or more rendering commands to be conditional on a value in buffer memory. This may help an application reduce the latency by conditionally discarding rendering commands without application intervention. The conditional rendering commands are limited to draws, compute dispatches and clearing attachments within a conditional

rendering block.

New Commands

- vkCmdBeginConditionalRenderingEXT
- vkCmdEndConditionalRenderingEXT

New Structures

- VkConditionalRenderingBeginInfoEXT
- Extending VkCommandBufferInheritanceInfo:
  - VkCommandBufferInheritanceConditionalRenderingInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceConditionalRenderingFeaturesEXT

New Enums

- VkConditionalRenderingFlagBitsEXT

New Bitmasks

- VkConditionalRenderingFlagsEXT

New Enum Constants

- `VK_EXT_CONDITIONAL_RENDERING_EXTENSION_NAME`
- `VK_EXT_CONDITIONAL_RENDERING_SPEC_VERSION`
- Extending VkAccessFlagBits:
  - VK_ACCESS_CONDITIONAL_RENDERING_READ_BIT_EXT
- Extending VkBufferUsageFlagBits:
  - VK_BUFFER_USAGE_CONDITIONAL_RENDERING_BIT_EXT
- Extending VkPipelineStageFlagBits:
  - VK_PIPELINE_STAGE_CONDITIONAL_RENDERING_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_CONDITIONAL_RENDERING_INFO_EXT
  - VK_STRUCTURE_TYPE_CONDITIONAL_RENDERING_BEGIN_INFO_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CONDITIONAL_RENDERING_FEATURES_EXT

Issues

1) Should conditional rendering affect copy and blit commands?

RESOLVED: Conditional rendering should not affect copies and blits.

2) Should secondary command buffers be allowed to execute while conditional rendering is active in the primary command buffer?

RESOLVED: The rendering commands in secondary command buffer will be affected by an active conditional rendering in primary command buffer if the `conditionalRenderingEnable` is set to VK_TRUE. Conditional rendering **must** not be active in the primary command buffer if `conditionalRenderingEnable` is VK_FALSE.

Examples

None.

Version History

- Revision 1, 2018-04-19 (Vikram Kushwaha)
  - First Version
- Revision 2, 2018-05-21 (Vikram Kushwaha)
  - Add new pipeline stage, access flags and limit conditional rendering to a subpass or entire render pass.

VK_EXT_conservative_rasterization

Name String

VK_EXT_conservative_rasterization

Extension Type

Device extension

Registered Extension Number

102

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_EXT_fragment_fully_covered

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2020-06-09

Interactions and External Dependencies

- This extension requires SPV_EXT_fragment_fully_covered if the VkPhysicalDeviceConservativeRasterizationPropertiesEXT::fullyCoveredFragmentShaderInputVariable feature is used.
- This extension requires SPV_KHR_post_depth_coverageif the VkPhysicalDeviceConservativeRasterizationPropertiesEXT::conservativeRasterizationPostDepthCoverage feature is used.
- This extension provides API support for GL_NV_conservative_raster_underestimation if the VkPhysicalDeviceConservativeRasterizationPropertiesEXT::fullyCoveredFragmentShaderInputVariable feature is used.

Contributors

- Daniel Koch, NVIDIA
- Daniel Rakos, AMD

- Jeff Bolz, NVIDIA
- Slawomir Grajewski, Intel
- Stu Smith, Imagination Technologies

Description

This extension adds a new rasterization mode called conservative rasterization. There are two modes of conservative rasterization; overestimation and underestimation.

When overestimation is enabled, if any part of the primitive, including its edges, covers any part of the rectangular pixel area, including its sides, then a fragment is generated with all coverage samples turned on. This extension allows for some variation in implementations by accounting for differences in overestimation, where the generating primitive size is increased at each of its edges by some sub-pixel amount to further increase conservative pixel coverage. Implementations can allow the application to specify an extra overestimation beyond the base overestimation the implementation already does. It also allows implementations to either cull degenerate primitives or rasterize them.

When underestimation is enabled, fragments are only generated if the rectangular pixel area is fully covered by the generating primitive. If supported by the implementation, when a pixel rectangle is fully covered the fragment shader input variable builtin called FullyCoveredEXT is set to true. The shader variable works in either overestimation or underestimation mode.

Implementations can process degenerate triangles and lines by either discarding them or generating conservative fragments for them. Degenerate triangles are those that end up with zero area after the rasterizer quantizes them to the fixed-point pixel grid. Degenerate lines are those with zero length after quantization.

New Structures

- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceConservativeRasterizationPropertiesEXT
- Extending VkPipelineRasterizationStateCreateInfo:
  - VkPipelineRasterizationConservativeStateCreateInfoEXT

New Enums

- VkConservativeRasterizationModeEXT

New Bitmasks

- VkPipelineRasterizationConservativeStateCreateFlagsEXT

New Enum Constants

- `VK_EXT_CONSERVATIVE_RASTERIZATION_EXTENSION_NAME`
- `VK_EXT_CONSERVATIVE_RASTERIZATION_SPEC_VERSION`

- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CONSERVATIVE_RASTERIZATION_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_CONSERVATIVE_STATE_CREATE_INFO_EXT

New Built-In Variables

- FullyCoveredEXT

New SPIR-V Capabilities

- FragmentFullyCoveredEXT

Version History

- Revision 1.1, 2020-09-06 (Piers Daniell)
  - Add missing SPIR-V and GLSL dependencies.
- Revision 1, 2017-08-28 (Piers Daniell)
  - Internal revisions

VK_EXT_cooperative_matrix_maintenance1

Name String

VK_EXT_cooperative_matrix_maintenance1

Extension Type

Device extension

Registered Extension Number

660

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_cooperative_matrix

SPIR-V Dependencies

- SPV_EXT_cooperative_matrix_maintenance1

Contact

- Jeff Bolz [GitHub]jeffbolznv

Extension Proposal

VK_EXT_cooperative_matrix_maintenance1

Other Extension Metadata

Last Modified Date

2026-06-05

Interactions and External Dependencies

- This extension provides API support for GLSL_EXT_cooperative_matrix_maintenance1

Contributors

- Jeff Bolz, NVIDIA
- Karthik Vaidyanathan, NVIDIA
- Matthew Netsch, Qualcomm Technologies, Inc
- Mariusz Merecki, Intel
- Kévin Petit, Arm Ltd.

Description

This extension adds several new features building on the cooperative matrix types added in VK_KHR_cooperative_matrix. The goal is to add and accelerate features beyond just simple GEMM

kernels, including adding support for type/use conversions, reductions, per-element operations, and conversion of an element index to a matrix coordinate.

The new functionality is defined by the `SPV_EXT_cooperative_matrix_maintenance1` SPIR-V extension and can be used with the `GLSL_EXT_cooperative_matrix_maintenance1` GLSL extension.

This extension also provides a new command to query the properties of cooperative matrices, which can be easily extended by other extensions in the future. The new query also includes inputs for the desired subgroup size and (when used together with extensions that support workgroup scope cooperative matrices, such as VK_NV_cooperative_matrix2) the number of invocations in the local workgroup, allowing implementations to expose properties that are optimal or compatible only with specific subgroup sizes or local workgroup sizes.

New Commands

- vkGetPhysicalDeviceCooperativeMatrixProperties2EXT

New Structures

- VkCooperativeMatrixProperties2EXT
- VkPhysicalDeviceCooperativeMatrixInfo2EXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceCooperativeMatrixMaintenance1FeaturesEXT

New Enums

- VkCooperativeMatrixFlagBitsEXT

New Bitmasks

- VkCooperativeMatrixFlagsEXT

New Enum Constants

- `VK_EXT_COOPERATIVE_MATRIX_MAINTENANCE_1_EXTENSION_NAME`
- `VK_EXT_COOPERATIVE_MATRIX_MAINTENANCE_1_SPEC_VERSION`
- Extending VkCooperativeMatrixFlagBitsEXT:
  - VK_COOPERATIVE_MATRIX_SATURATING_ACCUMULATION_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_COOPERATIVE_MATRIX_PROPERTIES_2_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_INFO_2_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_MAINTENANCE_1_FEATURES_EXT

New SPIR-V Capabilities

- CooperativeMatrixReductionsEXT
- CooperativeMatrixConversionsEXT
- CooperativeMatrixPerElementOperationsEXT
- CooperativeMatrixGetCoordinateEXT

Issues

Version History

- Revision 1, 2025-10-14 (Jeff Bolz)
  - Initial revision

VK_EXT_custom_border_color

Name String

`VK_EXT_custom_border_color`

Extension Type

Device extension

Registered Extension Number

288

Revision

12

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Special Uses

- OpenGL / ES support
- D3D support

Contact

- Liam Middlebrook [GitHub]liam-middlebrook

Other Extension Metadata

Last Modified Date

2020-04-16

IP Status

No known IP claims.

Contributors

- Joshua Ashton, Valve
- Hans-Kristian Arntzen, Valve
- Philip Rebohle, Valve
- Liam Middlebrook, NVIDIA
- Jeff Bolz, NVIDIA
- Tobias Hector, AMD
- Faith Ekstrand, Intel
- Spencer Fricke, Samsung Electronics

- Graeme Leese, Broadcom
- Jesse Hall, Google
- Jan-Harald Fredriksen, ARM
- Tom Olson, ARM
- Stuart Smith, Imagination Technologies
- Donald Scorgie, Imagination Technologies
- Alex Walters, Imagination Technologies
- Peter Quayle, Imagination Technologies

Description

This extension provides cross-vendor functionality to specify a custom border color for use when the sampler address mode VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_BORDER is used.

To create a sampler which uses a custom border color set VkSamplerCreateInfo::`borderColor` to one of:

- VK_BORDER_COLOR_FLOAT_CUSTOM_EXT
- VK_BORDER_COLOR_INT_CUSTOM_EXT

When VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT is used, applications must provide a VkSamplerCustomBorderColorCreateInfoEXT in the `pNext` chain for VkSamplerCreateInfo.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceCustomBorderColorFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceCustomBorderColorPropertiesEXT
- Extending VkSamplerCreateInfo:
  - VkSamplerCustomBorderColorCreateInfoEXT

New Enum Constants

- `VK_EXT_CUSTOM_BORDER_COLOR_EXTENSION_NAME`
- `VK_EXT_CUSTOM_BORDER_COLOR_SPEC_VERSION`
- Extending VkBorderColor:
  - VK_BORDER_COLOR_FLOAT_CUSTOM_EXT
  - VK_BORDER_COLOR_INT_CUSTOM_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CUSTOM_BORDER_COLOR_FEATURES_EXT

- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CUSTOM_BORDER_COLOR_PROPERTIES_EXT
- VK_STRUCTURE_TYPE_SAMPLER_CUSTOM_BORDER_COLOR_CREATE_INFO_EXT

Issues

1) Should VkClearColorValue be used for the border color value, or should we have our own struct/union? Do we need to specify the type of the input values for the components? This is more of a concern if VkClearColorValue is used here because it provides a union of float,int,uint types.

RESOLVED: Will reuse existing VkClearColorValue structure in order to easily take advantage of float,int,uint borderColor types.

2) For hardware which supports a limited number of border colors what happens if that number is exceeded? Should this be handled by the driver unbeknownst to the application? In Revision 1 we had solved this issue using a new Object type, however that may have lead to additional system resource consumption which would otherwise not be required.

RESOLVED: Added `VkPhysicalDeviceCustomBorderColorPropertiesEXT::maxCustomBorderColorSamplers` for tracking implementation-specific limit, and Valid Usage statement handling overflow.

3) Should this be supported for immutable samplers at all, or by a feature bit? Some implementations may not be able to support custom border colors on immutable samplers — is it worthwhile enabling this to work on them for implementations that can support it, or forbidding it entirely.

RESOLVED: Samplers created with a custom border color are forbidden from being immutable. This resolves concerns for implementations where the custom border color is an index to a LUT instead of being directly embedded into sampler state.

4) Should UINT and SINT (unsigned integer and signed integer) border color types be separated or should they be combined into one generic INT (integer) type?

RESOLVED: Separating these does not make much sense as the existing fixed border color types do not have this distinction, and there is no reason in hardware to do so. This separation would also create unnecessary work and considerations for the application.

Version History

- Revision 1, 2019-10-10 (Joshua Ashton)
  - Internal revisions.
- Revision 2, 2019-10-11 (Liam Middlebrook)
  - Remove VkCustomBorderColor object and associated functions
  - Add issues concerning HW limitations for custom border color count
- Revision 3, 2019-10-12 (Joshua Ashton)
  - Re-expose the limits for the maximum number of unique border colors
  - Add extra details about border color tracking
  - Fix typos

- Revision 4, 2019-10-12 (Joshua Ashton)
  - Changed maxUniqueCustomBorderColors to a uint32_t from a VkDeviceSize
- Revision 5, 2019-10-14 (Liam Middlebrook)
  - Added features bit
- Revision 6, 2019-10-15 (Joshua Ashton)
  - Type-ize VK_BORDER_COLOR_CUSTOM
  - Fix const-ness on `pNext` of VkSamplerCustomBorderColorCreateInfoEXT
- Revision 7, 2019-11-26 (Liam Middlebrook)
  - Renamed maxUniqueCustomBorderColors to maxCustomBorderColors
- Revision 8, 2019-11-29 (Joshua Ashton)
  - Renamed borderColor member of VkSamplerCustomBorderColorCreateInfoEXT to customBorderColor
- Revision 9, 2020-02-19 (Joshua Ashton)
  - Renamed maxCustomBorderColors to maxCustomBorderColorSamplers
- Revision 10, 2020-02-21 (Joshua Ashton)
  - Added format to VkSamplerCustomBorderColorCreateInfoEXT and feature bit
- Revision 11, 2020-04-07 (Joshua Ashton)
  - Dropped UINT/SINT border color differences, consolidated types
- Revision 12, 2020-04-16 (Joshua Ashton)
  - Renamed VK_BORDER_COLOR_CUSTOM_FLOAT_EXT to VK_BORDER_COLOR_FLOAT_CUSTOM_EXT for consistency

VK_EXT_custom_resolve

Name String

`VK_EXT_custom_resolve`

Extension Type

Device extension

Registered Extension Number

629

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_VERSION_1_3
- Interacts with VK_KHR_dynamic_rendering

Contact

- Mike Blumenkrantz [GitHub]zmike

Extension Proposal

VK_EXT_custom_resolve

Other Extension Metadata

Last Modified Date

2025-05-13

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension interacts with `VK_KHR_dynamic_rendering`
- This extension interacts with `VK_EXT_dynamic_rendering_unused_attachments`
- This extension interacts with `VK_EXT_fragment_density_map`
- This extension interacts with `VK_EXT_graphics_pipeline_library`
- This extension interacts with `VK_EXT_shader_object`

Contributors

- Mike Blumenkrantz, Valve
- Connor Abbott, Valve
- Samuel Pitoiset, Valve
- Matthew Netsch, Qualcomm
- Jan-Harald Fredriksen, ARM
- Ting Wei, ARM
- Ricardo Garcia, Igalia
- Spencer Fricke, LunarG
- Piers Daniell, Nvidia

Description

This extension provides functionality for using shaders to resolve multisample rendering attachments.

It builds upon mechanics introduced by VK_QCOM_render_pass_shader_resolve, additionally adding support for dynamic rendering.

New Commands

If VK_KHR_dynamic_rendering or Vulkan Version 1.3 is supported:

- vkCmdBeginCustomResolveEXT

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceCustomResolveFeaturesEXT

If VK_KHR_dynamic_rendering or Vulkan Version 1.3 is supported:

- VkBeginCustomResolveInfoEXT
- Extending VkGraphicsPipelineCreateInfo, VkCommandBufferInheritanceInfo, VkShaderCreateInfoEXT:
  - VkCustomResolveCreateInfoEXT

New Enum Constants

- `VK_EXT_CUSTOM_RESOLVE_EXTENSION_NAME`
- `VK_EXT_CUSTOM_RESOLVE_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CUSTOM_RESOLVE_FEATURES_EXT
- Extending VkSubpassDescriptionFlagBits:

    - VK_SUBPASS_DESCRIPTION_CUSTOM_RESOLVE_BIT_EXT
    - VK_SUBPASS_DESCRIPTION_FRAGMENT_REGION_BIT_EXT

If VK_KHR_dynamic_rendering or Vulkan Version 1.3 is supported:

- Extending VkRenderingFlagBits:
    - VK_RENDERING_CUSTOM_RESOLVE_BIT_EXT
    - VK_RENDERING_FRAGMENT_REGION_BIT_EXT
- Extending VkResolveModeFlagBits:
    - VK_RESOLVE_MODE_CUSTOM_BIT_EXT
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_BEGIN_CUSTOM_RESOLVE_INFO_EXT
    - VK_STRUCTURE_TYPE_CUSTOM_RESOLVE_CREATE_INFO_EXT

Issues

1) How will this work with shader objects?

Some vendors emit an epilog at the end of the FS that stores each color/depth/stencil attachment to the appropriate tilebuffer location, and to do that they need to know the layout of the tilebuffer which depends on the attachment formats/sample counts. We agreed that for shader object the FS epilog is emitted dynamically when the draw happens.

Version History

- Revision 1, 2025-05-13 (Mike Blumenkrantz)
    - Initial draft

VK_EXT_debug_utils

Name String

`VK_EXT_debug_utils`

Extension Type

Instance extension

Registered Extension Number

129

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

None

Special Use

- Debugging tools

Contact

- Mark Young [GitHub]marky-lunarg

Other Extension Metadata

Last Modified Date

2020-04-03

Revision

2

IP Status

No known IP claims.

Dependencies

- This extension is written against version 1.0 of the Vulkan API.
- Requires VkObjectType

Contributors

- Mark Young, LunarG
- Baldur Karlsson
- Ian Elliott, Google
- Courtney Goeltzenleuchter, Google
- Karl Schultz, LunarG

- Mark Lobodzinski, LunarG
- Mike Schuchardt, LunarG
- Jaakko Konttinen, AMD
- Dan Ginsburg, Valve Software
- Rolando Olivares, Epic Games
- Dan Baker, Oxide Games
- Kyle Spagnoli, NVIDIA
- Jon Ashburn, LunarG
- Piers Daniell, NVIDIA

Description

Due to the nature of the Vulkan interface, there is very little error information available to the developer and application. By using the `VK_EXT_debug_utils` extension, developers **can** obtain more information. When combined with validation layers, even more detailed feedback on the application's use of Vulkan will be provided.

This extension provides the following capabilities:

- The ability to create a debug messenger which will pass along debug messages to an application supplied callback.
- The ability to identify specific Vulkan objects using a name or tag to improve tracking.
- The ability to identify specific sections within a `VkQueue` or `VkCommandBuffer` using labels to aid organization and offline analysis in external tools.

New Object Types

- VkDebugUtilsMessengerEXT

New Commands

- vkCmdBeginDebugUtilsLabelEXT
- vkCmdEndDebugUtilsLabelEXT
- vkCmdInsertDebugUtilsLabelEXT
- vkCreateDebugUtilsMessengerEXT
- vkDestroyDebugUtilsMessengerEXT
- vkQueueBeginDebugUtilsLabelEXT
- vkQueueEndDebugUtilsLabelEXT
- vkQueueInsertDebugUtilsLabelEXT
- vkSetDebugUtilsObjectNameEXT
- vkSetDebugUtilsObjectTagEXT

- vkSubmitDebugUtilsMessageEXT

New Structures

- VkDebugUtilsLabelEXT
- VkDebugUtilsMessengerCallbackDataEXT
- VkDebugUtilsObjectTagInfoEXT
- Extending VkInstanceCreateInfo:
  - VkDebugUtilsMessengerCreateInfoEXT
- Extending VkPipelineShaderStageCreateInfo, VkResourceDescriptorInfoEXT, VkSamplerCreateInfo:
  - VkDebugUtilsObjectNameInfoEXT

New Function Pointers

- PFN_vkDebugUtilsMessengerCallbackEXT

New Enums

- VkDebugUtilsMessageSeverityFlagBitsEXT
- VkDebugUtilsMessageTypeFlagBitsEXT

New Bitmasks

- VkDebugUtilsMessageSeverityFlagsEXT
- VkDebugUtilsMessageTypeFlagsEXT
- VkDebugUtilsMessengerCallbackDataFlagsEXT
- VkDebugUtilsMessengerCreateFlagsEXT

New Enum Constants

- `VK_EXT_DEBUG_UTILS_EXTENSION_NAME`
- `VK_EXT_DEBUG_UTILS_SPEC_VERSION`
- Extending VkObjectType:
  - VK_OBJECT_TYPE_DEBUG_UTILS_MESSENGER_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DEBUG_UTILS_LABEL_EXT
  - VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CALLBACK_DATA_EXT
  - VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_DEBUG_UTILS_OBJECT_NAME_INFO_EXT
  - VK_STRUCTURE_TYPE_DEBUG_UTILS_OBJECT_TAG_INFO_EXT

Examples

Example 1

VK_EXT_debug_utils allows an application to register multiple callbacks with any Vulkan component wishing to report debug information. Some callbacks may log the information to a file, others may cause a debug break point or other application-defined behavior. An application **can** register callbacks even when no validation layers are enabled, but they will only be called for loader and, if implemented, driver events.

To capture events that occur while creating or destroying an instance an application **can** link a VkDebugUtilsMessengerCreateInfoEXT structure to the pNext chain of the VkInstanceCreateInfo structure passed to vkCreateInstance.

Example uses: Create three callback objects. One will log errors and warnings to the debug console using Windows OutputDebugString. The second will cause the debugger to break at that callback when an error happens and the third will log warnings to stdout.

```
    extern VkInstance instance;
    VkResult res;
    VkDebugUtilsMessengerEXT cb1, cb2, cb3;

    // Must call extension functions through a function pointer:
    PFN_vkCreateDebugUtilsMessengerEXT pfnCreateDebugUtilsMessengerEXT =
(PFN_vkCreateDebugUtilsMessengerEXT)vkGetInstanceProcAddr(instance,
"vkCreateDebugUtilsMessengerEXT");
    PFN_vkDestroyDebugUtilsMessengerEXT pfnDestroyDebugUtilsMessengerEXT =
(PFN_vkDestroyDebugUtilsMessengerEXT)vkGetInstanceProcAddr(instance,
"vkDestroyDebugUtilsMessengerEXT");

    VkDebugUtilsMessengerCreateInfoEXT callback1 = {
        .sType = VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = 0,
        .messageSeverity = VK_DEBUG_UTILS_MESSAGE_SEVERITY_ERROR_BIT_EXT |
                           VK_DEBUG_UTILS_MESSAGE_SEVERITY_WARNING_BIT_EXT,
        .messageType= VK_DEBUG_UTILS_MESSAGE_TYPE_GENERAL_BIT_EXT |
                      VK_DEBUG_UTILS_MESSAGE_TYPE_VALIDATION_BIT_EXT,
        .pfnUserCallback = myOutputDebugString,
        .pUserData = NULL
    };
    res = pfnCreateDebugUtilsMessengerEXT(instance, &callback1, NULL, &cb1);
    if (res != VK_SUCCESS) {
       // Do error handling for VK_ERROR_OUT_OF_MEMORY
    }

    callback1.messageSeverity = VK_DEBUG_UTILS_MESSAGE_SEVERITY_ERROR_BIT_EXT;
    callback1.pfnUserCallback = myDebugBreak;
    callback1.pUserData = NULL;
    res = pfnCreateDebugUtilsMessengerEXT(instance, &callback1, NULL, &cb2);
```

```
    if (res != VK_SUCCESS) {
       // Do error handling for VK_ERROR_OUT_OF_MEMORY
    }

    VkDebugUtilsMessengerCreateInfoEXT callback3 = {
        .sType = VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = 0,
        .messageSeverity = VK_DEBUG_UTILS_MESSAGE_SEVERITY_WARNING_BIT_EXT,
        .messageType = VK_DEBUG_UTILS_MESSAGE_TYPE_GENERAL_BIT_EXT |
                       VK_DEBUG_UTILS_MESSAGE_TYPE_VALIDATION_BIT_EXT,
        .pfnUserCallback = mystdOutLogger,
        .pUserData = NULL
    };
    res = pfnCreateDebugUtilsMessengerEXT(instance, &callback3, NULL, &cb3);
    if (res != VK_SUCCESS) {
       // Do error handling for VK_ERROR_OUT_OF_MEMORY
    }

    ...

    // Remove callbacks when cleaning up
    pfnDestroyDebugUtilsMessengerEXT(instance, cb1, NULL);
    pfnDestroyDebugUtilsMessengerEXT(instance, cb2, NULL);
    pfnDestroyDebugUtilsMessengerEXT(instance, cb3, NULL);
```

Example 2

Associate a name with an image, for easier debugging in external tools or with validation layers that can print a friendly name when referring to objects in error messages.

```
    extern VkInstance instance;
    extern VkDevice device;
    extern VkImage image;

    // Must call extension functions through a function pointer:
    PFN_vkSetDebugUtilsObjectNameEXT pfnSetDebugUtilsObjectNameEXT =
(PFN_vkSetDebugUtilsObjectNameEXT)vkGetInstanceProcAddr(instance,
"vkSetDebugUtilsObjectNameEXT");

    // Set a name on the image
    const VkDebugUtilsObjectNameInfoEXT imageNameInfo =
    {
        .sType = VK_STRUCTURE_TYPE_DEBUG_UTILS_OBJECT_NAME_INFO_EXT,
        .pNext = NULL,
        .objectType = VK_OBJECT_TYPE_IMAGE,
        .objectHandle = (uint64_t)image,
        .pObjectName = "Brick Diffuse Texture",
    };
```

```
    pfnSetDebugUtilsObjectNameEXT(device, &imageNameInfo);

    // A subsequent error might print:
    //   Image 'Brick Diffuse Texture' (0xc0dec0dedeadbeef) is used in a
    //   command buffer with no memory bound to it.
```

Example 3

Annotating regions of a workload with naming information so that offline analysis tools can display a more usable visualization of the commands submitted.

```
    extern VkInstance instance;
    extern VkCommandBuffer commandBuffer;

    // Must call extension functions through a function pointer:
    PFN_vkQueueBeginDebugUtilsLabelEXT pfnQueueBeginDebugUtilsLabelEXT =
(PFN_vkQueueBeginDebugUtilsLabelEXT)vkGetInstanceProcAddr(instance,
"vkQueueBeginDebugUtilsLabelEXT");
    PFN_vkQueueEndDebugUtilsLabelEXT pfnQueueEndDebugUtilsLabelEXT =
(PFN_vkQueueEndDebugUtilsLabelEXT)vkGetInstanceProcAddr(instance,
"vkQueueEndDebugUtilsLabelEXT");
    PFN_vkCmdBeginDebugUtilsLabelEXT pfnCmdBeginDebugUtilsLabelEXT =
(PFN_vkCmdBeginDebugUtilsLabelEXT)vkGetInstanceProcAddr(instance,
"vkCmdBeginDebugUtilsLabelEXT");
    PFN_vkCmdEndDebugUtilsLabelEXT pfnCmdEndDebugUtilsLabelEXT =
(PFN_vkCmdEndDebugUtilsLabelEXT)vkGetInstanceProcAddr(instance,
"vkCmdEndDebugUtilsLabelEXT");
    PFN_vkCmdInsertDebugUtilsLabelEXT pfnCmdInsertDebugUtilsLabelEXT =
(PFN_vkCmdInsertDebugUtilsLabelEXT)vkGetInstanceProcAddr(instance,
"vkCmdInsertDebugUtilsLabelEXT");

    // Describe the area being rendered
    const VkDebugUtilsLabelEXT houseLabel =
    {
        .sType = VK_STRUCTURE_TYPE_DEBUG_UTILS_LABEL_EXT,
        .pNext = NULL,
        .pLabelName = "Brick House",
        .color = { 1.0f, 0.0f, 0.0f, 1.0f },
    };

    // Start an annotated group of calls under the 'Brick House' name
    pfnCmdBeginDebugUtilsLabelEXT(commandBuffer, &houseLabel);
    {
        // A mutable structure for each part being rendered
        VkDebugUtilsLabelEXT housePartLabel =
        {
            .sType = VK_STRUCTURE_TYPE_DEBUG_UTILS_LABEL_EXT,
            .pNext = NULL,
            .pLabelName = NULL,
            .color = { 0.0f, 0.0f, 0.0f, 0.0f },
```

```
        };

        // Set the name and insert the marker
        housePartLabel.pLabelName = "Walls";
        pfnCmdInsertDebugUtilsLabelEXT(commandBuffer, &housePartLabel);

        // Insert the drawcall for the walls
        vkCmdDrawIndexed(commandBuffer, 1000, 1, 0, 0, 0);

        // Insert a recursive region for two sets of windows
        housePartLabel.pLabelName = "Windows";
        pfnCmdBeginDebugUtilsLabelEXT(commandBuffer, &housePartLabel);
        {
            vkCmdDrawIndexed(commandBuffer, 75, 6, 1000, 0, 0);
            vkCmdDrawIndexed(commandBuffer, 100, 2, 1450, 0, 0);
        }
        pfnCmdEndDebugUtilsLabelEXT(commandBuffer);

        housePartLabel.pLabelName = "Front Door";
        pfnCmdInsertDebugUtilsLabelEXT(commandBuffer, &housePartLabel);

        vkCmdDrawIndexed(commandBuffer, 350, 1, 1650, 0, 0);

        housePartLabel.pLabelName = "Roof";
        pfnCmdInsertDebugUtilsLabelEXT(commandBuffer, &housePartLabel);

        vkCmdDrawIndexed(commandBuffer, 500, 1, 2000, 0, 0);
    }
    // End the house annotation started above
    pfnCmdEndDebugUtilsLabelEXT(commandBuffer);

    // Do other work

    vkEndCommandBuffer(commandBuffer);

    // Describe the queue being used
    const VkDebugUtilsLabelEXT queueLabel =
    {
        .sType = VK_STRUCTURE_TYPE_DEBUG_UTILS_LABEL_EXT,
        .pNext = NULL,
        .pLabelName = "Main Render Work",
        .color = { 0.0f, 1.0f, 0.0f, 1.0f },
    };

    // Identify the queue label region
    pfnQueueBeginDebugUtilsLabelEXT(queue, &queueLabel);

    // Submit the work for the main render thread
    const VkCommandBuffer cmd_bufs[] = {commandBuffer};
    VkSubmitInfo submit_info =
    {
```

```
        .sType = VK_STRUCTURE_TYPE_SUBMIT_INFO,
        .pNext = NULL,
        .waitSemaphoreCount = 0,
        .pWaitSemaphores = NULL,
        .pWaitDstStageMask = NULL,
        .commandBufferCount = 1,
        .pCommandBuffers = cmd_bufs,
        .signalSemaphoreCount = 0,
        .pSignalSemaphores = NULL
    };
    vkQueueSubmit(queue, 1, &submit_info, fence);

    // End the queue label region
    pfnQueueEndDebugUtilsLabelEXT(queue);
```

Issues

1) Should we just name this extension `VK_EXT_debug_report2`

RESOLVED: No. There is enough additional changes to the structures to break backwards compatibility. So, a new name was decided that would not indicate any interaction with the previous extension.

2) Will validation layers immediately support all the new features.

RESOLVED: Not immediately. As one can imagine, there is a lot of work involved with converting the validation layer logging over to the new functionality. Basic logging, as seen in the origin `VK_EXT_debug_report` extension will be made available immediately. However, adding the labels and object names will take time. Since the priority for Khronos at this time is to continue focusing on Valid Usage statements, it may take a while before the new functionality is fully exposed.

3) If the validation layers will not expose the new functionality immediately, then what is the point of this extension?

RESOLVED: We needed a replacement for `VK_EXT_debug_report` because the `VkDebugReportObjectTypeEXT` enumeration will no longer be updated and any new objects will need to be debugged using the new functionality provided by this extension.

4) Should this extension be split into two separate parts (1 extension that is an instance extension providing the callback functionality, and another device extension providing the general debug marker and annotation functionality)?

RESOLVED: No, the functionality for this extension is too closely related. If we did split up the extension, where would the structures and enums live, and how would you define that the device behavior in the instance extension is really only valid if the device extension is enabled, and the functionality is passed in. It is cleaner to just define this all as an instance extension, plus it allows the application to enable all debug functionality provided with one enable string during vkCreateInstance.

Version History

- Revision 1, 2017-09-14 (Mark Young and all listed Contributors)
  - Initial draft, based on `VK_EXT_debug_report` and `VK_EXT_debug_marker` in addition to previous feedback supplied from various companies including Valve, Epic, and Oxide games.
- Revision 2, 2020-04-03 (Mark Young and Piers Daniell)
  - Updated to allow either `NULL` or an empty string to be passed in for `pObjectName` in `VkDebugUtilsObjectNameInfoEXT`, because the loader and various drivers support `NULL` already.

VK_EXT_depth_bias_control

Name String

`VK_EXT_depth_bias_control`

Extension Type

Device extension

Registered Extension Number

284

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Special Use

- D3D support

Contact

- Joshua Ashton [GitHub]Joshua-Ashton

Extension Proposal

VK_EXT_depth_bias_control

Other Extension Metadata

Last Modified Date

2023-02-15

IP Status

No known IP claims.

Contributors

- Joshua Ashton, VALVE
- Hans-Kristian Arntzen, VALVE
- Mike Blumenkrantz, VALVE
- Georg Lehmann, VALVE
- Piers Daniell, NVIDIA
- Lionel Landwerlin, INTEL
- Tobias Hector, AMD

- Ricardo Garcia, IGALIA
- Jan-Harald Fredriksen, ARM
- Shahbaz Youssefi, GOOGLE
- Tom Olson, ARM

Description

This extension adds a new structure, `VkDepthBiasRepresentationInfoEXT`, that can be added to a `pNext` chain of `VkPipelineRasterizationStateCreateInfo` and allows setting the scaling and representation of depth bias for a pipeline.

This state can also be set dynamically by using the new structure mentioned above in combination with the new `vkCmdSetDepthBias2EXT` command.

New Commands

- vkCmdSetDepthBias2EXT

New Structures

- VkDepthBiasInfoEXT
- Extending VkDepthBiasInfoEXT, VkPipelineRasterizationStateCreateInfo:
  - VkDepthBiasRepresentationInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDepthBiasControlFeaturesEXT

New Enums

- VkDepthBiasRepresentationEXT

New Enum Constants

- `VK_EXT_DEPTH_BIAS_CONTROL_EXTENSION_NAME`
- `VK_EXT_DEPTH_BIAS_CONTROL_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DEPTH_BIAS_INFO_EXT
  - VK_STRUCTURE_TYPE_DEPTH_BIAS_REPRESENTATION_INFO_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_BIAS_CONTROL_FEATURES_EXT

Version History

- Revision 1, 2022-09-22 (Joshua Ashton)
  - Initial draft.

VK_EXT_depth_clamp_control

Name String

VK_EXT_depth_clamp_control

Extension Type

Device extension

Registered Extension Number

583

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Jules Blok [GitHub]jules

Extension Proposal

VK_EXT_depth_clamp_control

Other Extension Metadata

Last Modified Date

2024-07-15

Contributors

- Jules Blok, Independent

Description

This extension allows the application to control the viewport depth clamp range separately from the viewport `minDepth` and `maxDepth`. This gives the ability for the application to restrict depth values to an application-defined range rather than the viewport depth range or the range defined in the VK_EXT_depth_clamp_zero_one extension.

It can be used to set a smaller or larger clamping range than the viewport depth range without affecting the depth mapping of the viewport transform. Another possible use of this extension is to restrict depth values beyond the viewport depth range to a clamping range other than the [0, 1] range defined in the VK_EXT_depth_clamp_zero_one extension.

New Commands

- vkCmdSetDepthClampRangeEXT

New Structures

- VkDepthClampRangeEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDepthClampControlFeaturesEXT
- Extending VkPipelineViewportStateCreateInfo:
  - VkPipelineViewportDepthClampControlCreateInfoEXT

New Enums

- VkDepthClampModeEXT

New Enum Constants

- `VK_EXT_DEPTH_CLAMP_CONTROL_EXTENSION_NAME`
- `VK_EXT_DEPTH_CLAMP_CONTROL_SPEC_VERSION`
- Extending VkDynamicState:
  - VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLAMP_CONTROL_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_DEPTH_CLAMP_CONTROL_CREATE_INFO_EXT

Issues

1) Should the depth clamp range be a per-viewport parameter?

RESOLVED: No. Because the depth clamp range was previously defined to be equal to the viewport depth range, conformant runtimes are already handling the depth clamp range as a per-viewport parameter. However because of complexities from interactions with multiple viewports a per-viewport clamp range is left to a future extensions if a use case arises.

2) Should this pipeline state be dynamic?

RESOLVED: Yes. Since the viewport depth range can already be a dynamic state conformant runtimes are already able to handle the depth clamp range as a dynamic state.

3) Can the depth clamp range be ignored when depth clamping is disabled?

RESOLVED: Yes. This extension overrides the clamping range used only when depth clamping is enabled. The alternative would be highly unintuitive. As a consequence the VK_EXT_depth_clip_enable extension is required if depth clipping is desired in combination with this extension.

Version History

- Revision 1, 2024-02-13 (Jules Blok)
  - Initial draft

VK_EXT_depth_clip_control

Name String

`VK_EXT_depth_clip_control`

Extension Type

Device extension

Registered Extension Number

356

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Special Use

- OpenGL / ES support

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Other Extension Metadata

Last Modified Date

2021-11-09

Contributors

- Spencer Fricke, Samsung Electronics
- Shahbaz Youssefi, Google
- Ralph Potter, Samsung Electronics

Description

This extension allows the application to use the OpenGL depth range in NDC, i.e. with depth in range [-1, 1], as opposed to Vulkan's default of [0, 1]. The purpose of this extension is to allow efficient layering of OpenGL over Vulkan, by avoiding emulation in the pre-rasterization shader stages. This emulation, which effectively duplicates gl_Position but with a different depth value, costs ALU and consumes shader output components that the implementation may not have to spare to meet OpenGL minimum requirements.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDepthClipControlFeaturesEXT
- Extending VkPipelineViewportStateCreateInfo:
  - VkPipelineViewportDepthClipControlCreateInfoEXT

New Enum Constants

- `VK_EXT_DEPTH_CLIP_CONTROL_EXTENSION_NAME`
- `VK_EXT_DEPTH_CLIP_CONTROL_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLIP_CONTROL_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_DEPTH_CLIP_CONTROL_CREATE_INFO_EXT

Issues

1) Should this extension include an origin control option to match GL_LOWER_LEFT found in ARB_clip_control?

RESOLVED: No. The fix for porting over the origin is a simple y-axis flip. The depth clip control is a much harder problem to solve than what this extension is aimed to solve. Adding an equivalent to GL_LOWER_LEFT would require more testing.

2) Should this pipeline state be dynamic?

RESOLVED: Yes. The purpose of this extension is to emulate the OpenGL depth range, which is expected to be globally fixed (in case of OpenGL ES) or very infrequently changed (with `glClipControl` in OpenGL).

3) Should the control provided in this extension be an enum that could be extended in the future?

RESOLVED: No. It is highly unlikely that the depth range is changed to anything other than [0, 1] in the future. Should that happen a new extension will be required to extend such an enum, and that extension might as well add a new structure to chain to VkPipelineViewportStateCreateInfo::`pNext` instead.

Version History

- Revision 0, 2020-10-01 (Spencer Fricke)
  - Internal revisions
- Revision 1, 2020-11-26 (Shahbaz Youssefi)
  - Language fixes

VK_EXT_depth_clip_enable

Name String

`VK_EXT_depth_clip_enable`

Extension Type

Device extension

Registered Extension Number

103

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Special Use

- D3D support

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2018-12-20

Contributors

- Daniel Rakos, AMD
- Henri Verbeet, CodeWeavers
- Jeff Bolz, NVIDIA
- Philip Rebohle, DXVK
- Tobias Hector, AMD

Description

This extension allows the depth clipping operation, that is normally implicitly controlled by VkPipelineRasterizationStateCreateInfo::`depthClampEnable`, to instead be controlled explicitly by VkPipelineRasterizationDepthClipStateCreateInfoEXT::`depthClipEnable`.

This is useful for translating DX content which assumes depth clamping is always enabled, but depth clip can be controlled by the DepthClipEnable rasterization state (D3D12_RASTERIZER_DESC).

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDepthClipEnableFeaturesEXT
- Extending VkPipelineRasterizationStateCreateInfo:
  - VkPipelineRasterizationDepthClipStateCreateInfoEXT

New Bitmasks

- VkPipelineRasterizationDepthClipStateCreateFlagsEXT

New Enum Constants

- `VK_EXT_DEPTH_CLIP_ENABLE_EXTENSION_NAME`
- `VK_EXT_DEPTH_CLIP_ENABLE_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLIP_ENABLE_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_DEPTH_CLIP_STATE_CREATE_INFO_EXT

Version History

- Revision 1, 2018-12-20 (Piers Daniell)
  - Internal revisions

VK_EXT_depth_range_unrestricted

Name String

`VK_EXT_depth_range_unrestricted`

Extension Type

Device extension

Registered Extension Number

14

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2017-06-22

Contributors

- Daniel Koch, NVIDIA
- Jeff Bolz, NVIDIA

Description

This extension removes the VkViewport `minDepth` and `maxDepth` restrictions that the values must be between `0.0` and `1.0`, inclusive. It also removes the same restriction on VkPipelineDepthStencilStateCreateInfo `minDepthBounds` and `maxDepthBounds`. Finally it removes the restriction on the `depth` value in VkClearDepthStencilValue.

New Enum Constants

- `VK_EXT_DEPTH_RANGE_UNRESTRICTED_EXTENSION_NAME`
- `VK_EXT_DEPTH_RANGE_UNRESTRICTED_SPEC_VERSION`

Issues

1) How do VkViewport `minDepth` and `maxDepth` values outside of the `0.0` to `1.0` range interact with Primitive Clipping?

RESOLVED: The behavior described in Primitive Clipping still applies. If depth clamping is disabled the depth values are still clipped to $0 \le z_c \le w_c$ before the viewport transform. If depth clamping is enabled the above equation is ignored and the depth values are instead clamped to the VkViewport `minDepth` and `maxDepth` values, which in the case of this extension can be outside of the `0.0` to `1.0` range.

2) What happens if a resulting depth fragment is outside of the `0.0` to `1.0` range and the depth buffer is fixed-point rather than floating-point?

RESOLVED: This situation can also arise without this extension (when fragment shaders replace depth values, for example), and this extension does not change the behavior, which is defined in the Depth Test section of the Fragment Operations chapter.

Version History

- Revision 1, 2017-06-22 (Piers Daniell)
  - Internal revisions

VK_EXT_descriptor_heap

Name String

VK_EXT_descriptor_heap

Extension Type

Device extension

Registered Extension Number

136

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_extended_flags
or
VK_KHR_maintenance5
and
VK_KHR_buffer_device_address
or
Vulkan Version 1.2
or
Vulkan Version 1.4

API Interactions

- Interacts with VK_ARM_tensors
- Interacts with VK_EXT_custom_border_color
- Interacts with VK_EXT_device_generated_commands
- Interacts with VK_EXT_fragment_density_map
- Interacts with VK_EXT_shader_object
- Interacts with VK_KHR_ray_tracing_pipeline
- Interacts with VK_NV_device_generated_commands
- Interacts with VK_NV_ray_tracing

SPIR-V Dependencies

- SPV_EXT_descriptor_heap

Contact

- Tobias Hector [GitHub]tobski

Extension Proposal

VK_EXT_descriptor_heap

Other Extension Metadata

Last Modified Date

2024-06-12

IP Status

No known IP claims.

Contributors

- Tobias Hector, AMD
- Jan-Harald Fredriksen, Arm
- Daniel Story, Nintendo
- Connor Abbot, Valve
- Hans-Kristian Arntzen, Valve
- Matthew Netsch, Qualcomm
- Jeff Bolz, NVIDIA
- Alyssa Rosenzweig, Valve
- Lionel Landerwerlin, Intel
- Baldur Karlsson, Valve
- Faith Ekstrand, Collabora
- Slawomir Grajewski, Intel
- Mike Blumenkrantz, Valve
- Yiwei Zhang, Google
- Stu Smith, AMD
- Piers Daniell, NVIDIA
- Jon Leech, Khronos
- Rodrigo Locatti, NVIDIA
- Krzysztof Niski, NVIDIA
- Alan Baker, Google
- James Fitzpatrick, Imagination
- Samuel (Sheng-Wen) Huang, Mediatek
- Hai Nguyen, Google
- Jeff Leger, Qualcomm
- Marty Johnson, Khronos
- Tom Olson, Arm
- Chris Glover, Google
- Daniel Koch, NVIDIA

- Robert Simpson, Qualcomm
- Dan Ginsburg, Valve
- Andreas Süßenbach, NVIDIA
- Anna Maniscalco, Valve
- Artem Kharytoniuk, LunarG
- Younggwan Kim, Arm
- Konstantin Seurer, Valve
- Catarina Shablia, Collabora
- Spencer Fricke, LunarG
- Chris Bieneman, Microsoft
- Ting Wei, Arm
- Boris Zanin, AMD
- Samuel Pitoiset, Valve
- Erik Hogeman, Arm
- Jesse Natalie, Microsoft
- Guang Xu, AMD
- Embla Flatlandsmo, Arm

Interactions and External Dependencies

- VK_KHR_shader_untyped_pointers must be supported, but it does not need to be enabled for applications using only the binding interface.

Description

This extension allows explicit management of descriptors, and the memory used to store descriptors, conceptualised as descriptor heaps. Descriptor heap memory can be accessed as any other memory, enabling management of descriptors on both CPU and the GPU.

This extension was developed based on issues discovered with VK_EXT_descriptor_buffer. There are more constraints on how it is implemented, to provide more portable guarantees and more predictable performance characteristics. For instance, rather than an arbitrary number of sampler or resource buffers, there is exactly one sampler heap and exactly one resource heap.

This extension also eliminates descriptor sets and pipeline layouts completely; instead applications can look descriptors up solely by their offset into a heap. Shaders still using descriptor set and binding decorations can still be mapped to heaps through an API that maps set and binding decorations to heap offsets, either as constants or through push data. This new mapping API also enables a much more straightforward mapping to HLSL shaders using the root signature and descriptor table interfaces.

The push constant API has also been replaced with the *push data* interface, which aims to provide much more clarity about how data is passed to the shader, without requiring any mapping information to be provided during pipeline or shader creation. Mappings are still available for

shaders written for the legacy interface.

There is also a much clearer path for mapping shader constant data, with two recommended options for mapping constant data through push data; either directly in push data, or through a device address stored in push data, both of which can be mapped to shaders with set and binding interfaces.

New Object Types

- VkTensorARM

New Commands

- vkCmdBindResourceHeapEXT
- vkCmdBindSamplerHeapEXT
- vkCmdPushDataEXT
- vkGetImageOpaqueCaptureDataEXT
- vkGetPhysicalDeviceDescriptorSizeEXT
- vkWriteResourceDescriptorsEXT
- vkWriteSamplerDescriptorsEXT

If VK_EXT_custom_border_color is supported:

- vkRegisterCustomBorderColorEXT
- vkUnregisterCustomBorderColorEXT

New Structures

- VkBindHeapInfoEXT
- VkDescriptorMappingSourceConstantOffsetEXT
- VkDescriptorMappingSourceHeapDataEXT
- VkDescriptorMappingSourceIndirectAddressEXT
- VkDescriptorMappingSourceIndirectIndexArrayEXT
- VkDescriptorMappingSourceIndirectIndexEXT
- VkDescriptorMappingSourcePushIndexEXT
- VkDescriptorMappingSourceShaderRecordIndexEXT
- VkDescriptorSetAndBindingMappingEXT
- VkDeviceAddressRangeEXT
- VkHostAddressRangeConstEXT
- VkHostAddressRangeEXT
- VkImageDescriptorInfoEXT
- VkPushDataInfoEXT

- VkResourceDescriptorInfoEXT
- VkTensorViewCreateInfoARM
- VkTexelBufferDescriptorInfoEXT
- Extending VkCommandBufferInheritanceInfo:
  - VkCommandBufferInheritanceDescriptorHeapInfoEXT
- Extending VkImageCreateInfo:
  - VkOpaqueCaptureDataCreateInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDescriptorHeapFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceDescriptorHeapPropertiesEXT
- Extending VkPipelineShaderStageCreateInfo, VkShaderCreateInfoEXT:
  - VkShaderDescriptorSetAndBindingMappingInfoEXT

If VK_EXT_custom_border_color is supported:

- Extending VkSamplerCreateInfo:
  - VkSamplerCustomBorderColorIndexCreateInfoEXT

If VK_EXT_fragment_density_map is supported:

- Extending VkImageFormatProperties2:
  - VkSubsampledImageFormatPropertiesEXT

New Unions

- VkDescriptorMappingSourceDataEXT
- VkResourceDescriptorDataEXT

New Enums

- VkDescriptorMappingSourceEXT
- VkSpirvResourceTypeFlagBitsEXT

New Bitmasks

- VkSpirvResourceTypeFlagsEXT
- VkTensorViewCreateFlagsARM

New Enum Constants

- `VK_EXT_DESCRIPTOR_HEAP_EXTENSION_NAME`
- `VK_EXT_DESCRIPTOR_HEAP_SPEC_VERSION`

- Extending VkAccessFlagBits2:
  - VK_ACCESS_2_RESOURCE_HEAP_READ_BIT_EXT
  - VK_ACCESS_2_SAMPLER_HEAP_READ_BIT_EXT
- Extending VkBufferUsageFlagBits:
  - VK_BUFFER_USAGE_DESCRIPTOR_HEAP_BIT_EXT
- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_DESCRIPTOR_HEAP_BIT_EXT
- Extending VkImageCreateFlagBits:
  - VK_IMAGE_CREATE_DESCRIPTOR_HEAP_CAPTURE_REPLAY_BIT_EXT
- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_DESCRIPTOR_HEAP_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_BIND_HEAP_INFO_EXT
  - VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_DESCRIPTOR_HEAP_INFO_EXT
  - VK_STRUCTURE_TYPE_DESCRIPTOR_SET_AND_BINDING_MAPPING_EXT
  - VK_STRUCTURE_TYPE_IMAGE_DESCRIPTOR_INFO_EXT
  - VK_STRUCTURE_TYPE_OPAQUE_CAPTURE_DATA_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_HEAP_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_HEAP_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_PUSH_DATA_INFO_EXT
  - VK_STRUCTURE_TYPE_RESOURCE_DESCRIPTOR_INFO_EXT
  - VK_STRUCTURE_TYPE_SHADER_DESCRIPTOR_SET_AND_BINDING_MAPPING_INFO_EXT
  - VK_STRUCTURE_TYPE_TEXEL_BUFFER_DESCRIPTOR_INFO_EXT

If VK_EXT_custom_border_color is supported:

- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_SAMPLER_CUSTOM_BORDER_COLOR_INDEX_CREATE_INFO_EXT

If VK_EXT_device_generated_commands is supported:

- Extending VkIndirectCommandsTokenTypeEXT:
  - VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_EXT
  - VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_DATA_SEQUENCE_INDEX_EXT

If VK_EXT_fragment_density_map is supported:

- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_SUBSAMPLED_IMAGE_FORMAT_PROPERTIES_EXT

If VK_EXT_shader_object is supported:

- Extending VkShaderCreateFlagBitsEXT:
  - VK_SHADER_CREATE_DESCRIPTOR_HEAP_BIT_EXT

If VK_KHR_ray_tracing_pipeline or `VK_NV_ray_tracing` is supported:

- Extending VkDescriptorMappingSourceEXT:
  - VK_DESCRIPTOR_MAPPING_SOURCE_HEAP_WITH_SHADER_RECORD_INDEX_EXT
  - VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_ADDRESS_EXT
  - VK_DESCRIPTOR_MAPPING_SOURCE_SHADER_RECORD_DATA_EXT
- Extending VkSpirvResourceTypeFlagBitsEXT:
  - VK_SPIRV_RESOURCE_TYPE_ACCELERATION_STRUCTURE_BIT_EXT

Version History

- Revision 1, 2024-06-12 (Tobias Hector)
  - Initial revision.

VK_EXT_device_address_binding_report

Name String

`VK_EXT_device_address_binding_report`

Extension Type

Device extension

Registered Extension Number

355

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1
and
VK_EXT_debug_utils

Special Uses

- Debugging tools
- Developer tools

Contact

- Ralph Potter [GitLab]r_potter

Other Extension Metadata

Last Modified Date

2020-11-23

Interactions and External Dependencies

- This extension requires `VK_EXT_debug_utils`

Contributors

- Ralph Potter, Samsung
- Spencer Fricke, Samsung
- Jan-Harald Fredriksen, ARM
- Andrew Ellem, Google
- Alex Walters, IMG
- Jeff Bolz, NVIDIA

Description

This extension enables applications to track the binding of regions of the GPU virtual address space, and to associate those regions with Vulkan objects. This extension is primarily intended to aid in crash postmortem, where applications may wish to map a faulting GPU address to a Vulkan object.

For example, a page fault triggered by accessing an address located within a region of the GPU virtual address space that was previously reported as bound and then unbound may indicate a use-after-free error. Similarly, faults generated by accessing virtual addresses outside the limits of a bound region of GPU virtual address space may indicate indexing beyond the bounds of a resource.

New Structures

- Extending VkDebugUtilsMessengerCallbackDataEXT:
  - VkDeviceAddressBindingCallbackDataEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceAddressBindingReportFeaturesEXT

New Enums

- VkDeviceAddressBindingFlagBitsEXT
- VkDeviceAddressBindingTypeEXT

New Bitmasks

- VkDeviceAddressBindingFlagsEXT

New Enum Constants

- `VK_EXT_DEVICE_ADDRESS_BINDING_REPORT_EXTENSION_NAME`
- `VK_EXT_DEVICE_ADDRESS_BINDING_REPORT_SPEC_VERSION`
- Extending VkDebugUtilsMessageTypeFlagBitsEXT:
  - VK_DEBUG_UTILS_MESSAGE_TYPE_DEVICE_ADDRESS_BINDING_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DEVICE_ADDRESS_BINDING_CALLBACK_DATA_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ADDRESS_BINDING_REPORT_FEATURES_EXT

Issues

1.) Should this extend VK_EXT_debug_utils or VK_EXT_device_memory_report?

RESOLVED: Extend VK_EXT_debug_utils. VK_EXT_device_memory_report focuses on memory allocations and would not normally trigger callbacks in all of the situations where VK_EXT_device_address_binding_report is expected to.

2.) Should this extension cover all Vulkan object types, or only resources such as buffers and images?

RESOLVED: The extension covers all Vulkan objects, and is not restricted to objects backed by VkDeviceMemory objects.

3.) Should reallocation be identified explicitly, or as an unbind/bind pair?

RESOLVED: Reallocation should be represented as an unbind/bind pair.

4.) Can multiple Vulkan objects share an overlapping virtual address range?

RESOLVED: Yes. This can be expected to occur due to resources aliasing.

5.) Can a single Vulkan object be associated with multiple virtual address ranges concurrently?

RESOLVED: Yes. These should be reported via multiple calls to the reporting callback.

6.) Should the virtual address ranges associated with internal allocations such as memory pools be reported?

RESOLVED: Virtual address ranges associated with internal allocations should only be reported when they become associated with a specific Vulkan object. In the case of internal pool allocations, a bind event should be reported when resources from the pool are assigned to a Vulkan object, and an unbind event should be reported when those resources are returned to the pool. Implementations should not report the binding or unbinding of virtual address ranges for which there are no related API objects visible to the application developer.

7.) Can an implementation report binding a virtual address range at VkImage or VkImageView creation, rather than in response to vkBindImageMemory?

RESOLVED: Yes. Virtual address range binding should be reported at the appropriate point at which it occurs within the implementation. This extension does not mandate when that should occur, and applications should anticipate receiving callback events at any point after registering callbacks.

8.) Can reporting of binding/unbinding be deferred until a resource is referenced by an executing command buffer?

RESOLVED: Changes to the virtual address ranges associated with a Vulkan object should be reported as close as possible to where they occur within the implementation. If virtual address binding is deferred, then the callback should also be deferred to match.

9.) Do bind/unbind callbacks have to form matched pairs? Can a large region be bound, and then subregions unbound, resulting in fragmentation?

RESOLVED: Splitting of virtual address regions, and unmatched bind/unbind callbacks may occur. Developers should anticipate that sparse memory may exhibit this behavior.

10.) The specification mandates that a callback **must** be triggered whenever a GPU virtual address range associated with any Vulkan object is bound or unbound. Do we need queries or properties indicating which Vulkan objects will report binding modifications?

RESOLVED: No. This extension is not intended to mandate how and when implementations bind

virtual ranges to objects. Adding queries or properties would constrain implementations, which might otherwise vary how virtual address bindings occur based on usage.

11.) Should vkAllocateMemory and vkFreeMemory trigger reporting callbacks?

RESOLVED: If an implementation binds a GPU virtual address range when vkAllocateMemory is called, then the callbacks must be triggered associating the virtual address range with the VkDeviceMemory object. If the device memory is subsequently bound to a buffer or image via vkBind*Memory, the callbacks should be triggered a second time, reporting the association between virtual address range and the buffer/image.

12.) Should binding reports be emitted for vkQueueBindSparse() operations?

RESOLVED: binding reports for sparse objects are best effort, some implementations might report events on vkQueueBindSparse() other implementations might report events on sparse object creation/destruction.

Version History

- Revision 1, 2020-09-23 (Ralph Potter)
    - Internal revisions

VK_EXT_device_generated_commands

Name String

VK_EXT_device_generated_commands

Extension Type

Device extension

Registered Extension Number

573

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_buffer_device_address
or
Vulkan Version 1.2
and
VK_KHR_extended_flags
or
VK_KHR_maintenance5
or
Vulkan Version 1.3

API Interactions

- Interacts with VK_EXT_shader_object

Contact

- Mike Blumenkrantz [GitHub]zmike

Extension Proposal

VK_EXT_device_generated_commands

Other Extension Metadata

Last Modified Date

2024-02-23

Interactions and External Dependencies

- This extension requires Vulkan 1.1
- This extension requires VK_EXT_buffer_device_address or VK_KHR_buffer_device_address or Vulkan 1.2 for the ability to bind vertex and index buffers on the device.
- This extension requires VK_KHR_extended_flags or VK_KHR_maintenance5 for the ability to use VkPipelineCreateFlags2KHR.

- This extension interacts with VK_NV_mesh_shader. If the latter extension is not supported, remove the command tokens to initiate NV mesh tasks drawing in this extension.
- This extension interacts with VK_EXT_mesh_shader. If the latter extension is not supported, remove the command tokens to initiate EXT mesh tasks drawing in this extension.
- This extension interacts with VK_KHR_ray_tracing_pipeline. If the latter extension is not supported, remove the command tokens to initiate ray tracing in this extension.
- This extension interacts with VK_EXT_shader_object. If the latter extension is not supported, remove references to shader objects in this extension.

Contributors

- Mike Blumenkrantz, VALVE
- Hans-Kristian Arntzen, VALVE
- Jan-Harald Fredriksen, ARM
- Spencer Fricke, LunarG
- Ricardo Garcia, Igalia
- Tobias Hector, AMD
- Baldur Karlsson, VALVE
- Christoph Kubisch, NVIDIA
- Lionel Landwerlin, INTEL
- Jon Leech, Khronos
- Ting Wei, ARM
- Ken Shanyi Zhang, AMD
- Faith Ekstrand, Collabora
- Vikram Kushwaha, NVIDIA
- Connor Abbott, VALVE
- Samuel Pitoiset, VALVE

Description

This extension allows the device to generate a number of commands for command buffers. It provides a subset of functionality from both VK_NV_device_generated_commands and VK_NV_device_generated_commands_compute as well as some new features.

When rendering a large number of objects, the device can be leveraged to implement a number of critical functions, like updating matrices, or implementing occlusion culling, frustum culling, front to back sorting, etc. Implementing those on the device does not require any special extension, since an application is free to define its own data structures, and just process them using shaders.

To render objects which have been processed on the device, Vulkan has several ways to perform indirect rendering, from the most basic vkCmdDrawIndirect with one indirect draw to vkCmdDrawIndirectCount which supports multiple indirect draws batched together, with a way to determine number of draws at device execution time.

However, if rendering state needs to change between the indirect draws, then unextended Vulkan forces the application to speculatively record a prohibitive number of redundant indirect commands covering all possible state combinations - which could end up processing nothing after culling - or read back the processed stream and issue graphics command from the host. For very large scenes, the synchronization overhead and cost to generate the command buffer can become the bottleneck. This extension allows an application to generate a device side stream of state changes and commands, and convert it efficiently into a command buffer without having to read it back to the host.

Furthermore, it allows incremental changes to such command buffers by manipulating only partial sections of a command stream — for example pipeline and shader object bindings. Unextended Vulkan requires re-creation of entire command buffers in such a scenario, or updates synchronized on the host.

The intended usage for this extension is for the application to:

- create `VkBuffer` objects and retrieve physical addresses from them via vkGetBufferDeviceAddress
- create a `VkIndirectExecutionSetEXT` for the ability to change shaders on the device.
- create a VkIndirectCommandsLayoutEXT, which lists the VkIndirectCommandsTokenTypeEXT it wants to dynamically execute as an atomic command sequence. This step likely involves some internal device code compilation, since the intent is for the GPU to generate the command buffer based on the layout.
- fill the input stream buffers with the data for each of the inputs it needs. Each input is an array that will be filled with token-dependent data.
- set up a preprocess `VkBuffer` that uses memory according to the information retrieved via vkGetGeneratedCommandsMemoryRequirementsEXT.
- optionally preprocess the generated content using vkCmdPreprocessGeneratedCommandsEXT, for example on an asynchronous compute queue, or for the purpose of reusing the data in multiple executions.
- call vkCmdExecuteGeneratedCommandsEXT to create and execute the actual device commands for all sequences based on the inputs provided.

For each draw in a sequence, the following can be specified:

- a number of vertex buffer bindings
- a different index buffer, with an optional dynamic offset and index type
- a number of different push constants
- updates to bound shader stages

For each dispatch in a sequence, the following can be specified:

- a number of different push constants
- updates to bound shader stages

For each trace rays in a sequence, the following can be specified:

- a number of different push constants
- updates to bound shader stages

While the GPU can be faster than a CPU to generate the commands, it will not happen asynchronously to the device, therefore the primary use case is generating “less” total work (occlusion culling, classification to use specialized shaders, etc.).

New Object Types

- VkIndirectCommandsLayoutEXT
- VkIndirectExecutionSetEXT

New Commands

- vkCmdExecuteGeneratedCommandsEXT
- vkCmdPreprocessGeneratedCommandsEXT
- vkCreateIndirectCommandsLayoutEXT
- vkCreateIndirectExecutionSetEXT
- vkDestroyIndirectCommandsLayoutEXT
- vkDestroyIndirectExecutionSetEXT
- vkGetGeneratedCommandsMemoryRequirementsEXT
- vkUpdateIndirectExecutionSetPipelineEXT
- vkUpdateIndirectExecutionSetShaderEXT

New Structures

- VkBindIndexBufferIndirectCommandEXT
- VkBindVertexBufferIndirectCommandEXT
- VkDrawIndirectCountIndirectCommandEXT
- VkGeneratedCommandsInfoEXT
- VkGeneratedCommandsMemoryRequirementsInfoEXT
- VkIndirectCommandsExecutionSetTokenEXT
- VkIndirectCommandsIndexBufferTokenEXT
- VkIndirectCommandsLayoutCreateInfoEXT
- VkIndirectCommandsLayoutTokenEXT
- VkIndirectCommandsPushConstantTokenEXT
- VkIndirectCommandsVertexBufferTokenEXT
- VkIndirectExecutionSetCreateInfoEXT
- VkIndirectExecutionSetPipelineInfoEXT
- VkIndirectExecutionSetShaderInfoEXT

- VkIndirectExecutionSetShaderLayoutInfoEXT
- VkWriteIndirectExecutionSetPipelineEXT
- Extending VkGeneratedCommandsInfoEXT, VkGeneratedCommandsMemoryRequirementsInfoEXT:
  - VkGeneratedCommandsPipelineInfoEXT
  - VkGeneratedCommandsShaderInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT

If VK_EXT_shader_object is supported:

- VkWriteIndirectExecutionSetShaderEXT

New Unions

- VkIndirectCommandsTokenDataEXT
- VkIndirectExecutionSetInfoEXT

New Enums

- VkIndirectCommandsInputModeFlagBitsEXT
- VkIndirectCommandsLayoutUsageFlagBitsEXT
- VkIndirectCommandsTokenTypeEXT
- VkIndirectExecutionSetInfoTypeEXT

New Bitmasks

- VkIndirectCommandsInputModeFlagsEXT
- VkIndirectCommandsLayoutUsageFlagsEXT

New Enum Constants

- `VK_EXT_DEVICE_GENERATED_COMMANDS_EXTENSION_NAME`
- `VK_EXT_DEVICE_GENERATED_COMMANDS_SPEC_VERSION`
- Extending VkAccessFlagBits:
  - VK_ACCESS_COMMAND_PREPROCESS_READ_BIT_EXT
  - VK_ACCESS_COMMAND_PREPROCESS_WRITE_BIT_EXT
- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_PREPROCESS_BUFFER_BIT_EXT
- Extending VkObjectType:

    - VK_OBJECT_TYPE_INDIRECT_COMMANDS_LAYOUT_EXT
    - VK_OBJECT_TYPE_INDIRECT_EXECUTION_SET_EXT
- Extending VkPipelineCreateFlagBits2:
    - VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT
- Extending VkPipelineStageFlagBits:
    - VK_PIPELINE_STAGE_COMMAND_PREPROCESS_BIT_EXT
- Extending VkShaderCreateFlagBitsEXT:
    - VK_SHADER_CREATE_INDIRECT_BINDABLE_BIT_EXT
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_GENERATED_COMMANDS_INFO_EXT
    - VK_STRUCTURE_TYPE_GENERATED_COMMANDS_MEMORY_REQUIREMENTS_INFO_EXT
    - VK_STRUCTURE_TYPE_GENERATED_COMMANDS_PIPELINE_INFO_EXT
    - VK_STRUCTURE_TYPE_GENERATED_COMMANDS_SHADER_INFO_EXT
    - VK_STRUCTURE_TYPE_INDIRECT_COMMANDS_LAYOUT_CREATE_INFO_EXT
    - VK_STRUCTURE_TYPE_INDIRECT_COMMANDS_LAYOUT_TOKEN_EXT
    - VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_CREATE_INFO_EXT
    - VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_PIPELINE_INFO_EXT
    - VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_SHADER_INFO_EXT
    - VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_SHADER_LAYOUT_INFO_EXT
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_GENERATED_COMMANDS_FEATURES_EXT
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_GENERATED_COMMANDS_PROPERTIES_EXT
    - VK_STRUCTURE_TYPE_WRITE_INDIRECT_EXECUTION_SET_PIPELINE_EXT
    - VK_STRUCTURE_TYPE_WRITE_INDIRECT_EXECUTION_SET_SHADER_EXT

Example Code

TODO

Version History

- Revision 1, 2024-02-23 (Mike Blumenkrantz)
    - Initial version

VK_EXT_device_memory_report

Name String

`VK_EXT_device_memory_report`

Extension Type

Device extension

Registered Extension Number

285

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Special Use

- Developer tools

Contact

- Yiwei Zhang [GitHub]zzyiwei

Other Extension Metadata

Last Modified Date

2021-01-06

IP Status

No known IP claims.

Contributors

- Yiwei Zhang, Google
- Jesse Hall, Google

Description

This device extension allows registration of device memory event callbacks upon device creation, so that applications or middleware can obtain detailed information about memory usage and how memory is associated with Vulkan objects. This extension exposes the actual underlying device memory usage, including allocations that are not normally visible to the application, such as memory consumed by vkCreateGraphicsPipelines. It is intended primarily for use by debug tooling rather than for production applications.

New Structures

- VkDeviceMemoryReportCallbackDataEXT
- Extending VkDeviceCreateInfo:
  - VkDeviceDeviceMemoryReportCreateInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDeviceMemoryReportFeaturesEXT

New Function Pointers

- PFN_vkDeviceMemoryReportCallbackEXT

New Enums

- VkDeviceMemoryReportEventTypeEXT

New Bitmasks

- VkDeviceMemoryReportFlagsEXT

New Enum Constants

- `VK_EXT_DEVICE_MEMORY_REPORT_EXTENSION_NAME`
- `VK_EXT_DEVICE_MEMORY_REPORT_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DEVICE_DEVICE_MEMORY_REPORT_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_DEVICE_MEMORY_REPORT_CALLBACK_DATA_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_MEMORY_REPORT_FEATURES_EXT

Issues

1) Should this be better expressed as an extension to VK_EXT_debug_utils and its general-purpose messenger construct?

RESOLVED: No. The intended lifecycle is quite different. We want to make this extension tied to the device's lifecycle. Each ICD just handles its own implementation of this extension, and this extension will only be directly exposed from the ICD. So we can avoid the extra implementation complexity used to accommodate the flexibility of `VK_EXT_debug_utils` extension.

2) Can we extend and use the existing internal allocation callbacks instead of adding the new callback structure in this extension?

RESOLVED: No. Our memory reporting layer that combines this information with other memory information it collects directly (e.g. bindings of resources to VkDeviceMemory) would have to intercept all entry points that take a VkAllocationCallbacks parameter and inject its own `pfnInternalAllocation` and `pfnInternalFree`. That may be doable for the extensions we know about, but not for ones we do not. The proposal would work fine in the face of most unknown extensions.

But even for ones we know about, since apps can provide a different set of callbacks and userdata and those can be retained by the driver and used later (esp. for pool object, but not just those), we would have to dynamically allocate the interception trampoline every time. That is getting to be an unreasonably large amount of complexity and (possibly) overhead.

We are interested in both alloc/free and import/unimport. The latter is fairly important for tracking (and avoiding double-counting) of swapchain images (still true with "native swapchains" based on external memory) and media/camera interop. Though we might be able to handle this with additional VkInternalAllocationType values, for import/export we do want to be able to tie this to the external resource, which is one thing that the `memoryObjectId` is for.

The internal alloc/free callbacks are not extensible except via new VkInternalAllocationType values. The VkDeviceMemoryReportCallbackDataEXT in this extension is extensible. That was deliberate: there is a real possibility we will want to get extra information in the future. As one example, currently this reports only physical allocations, but we believe there are interesting cases for tracking how populated that VA region is.

The callbacks are clearly specified as only callable within the context of a call from the application into Vulkan. We believe there are some cases where drivers can allocate device memory asynchronously. This was one of the sticky issues that derailed the internal device memory allocation reporting design (which is essentially what this extension is trying to do) leading up to 1.0.

VkAllocationCallbacks is described in a section called "Host memory" and the intro to it is very explicitly about host memory. The other callbacks are all inherently about host memory. But this extension is very focused on device memory.

3) Should the callback be reporting which heap is used?

RESOLVED: Yes. It is important for non-UMA systems to have all the device memory allocations attributed to the corresponding device memory heaps. For internally-allocated device memory, `heapIndex` will always correspond to an advertised heap, rather than having a magic value indicating a non-advertised heap. Drivers can advertise heaps that do not have any corresponding memory types if they need to.

4) Should we use an array of callback for the layers to intercept instead of chaining multiple of the VkDeviceDeviceMemoryReportCreateInfoEXT structures in the `pNext` of VkDeviceCreateInfo?

RESOLVED No. The pointer to the VkDeviceDeviceMemoryReportCreateInfoEXT structure itself is const and you cannot just cast it away. Thus we cannot update the callback array inside the structure. In addition, we cannot drop this `pNext` chain either, so making a copy of this whole structure does not work either.

5) Should we track bulk allocations shared among multiple objects?

RESOLVED No. Take the shader heap as an example. Some implementations will let multiple VkPipeline objects share the same shader heap. We are not asking the implementation to report VK_OBJECT_TYPE_PIPELINE along with a VK_NULL_HANDLE for this bulk allocation. Instead, this bulk allocation is considered as a layer below what this extension is interested in. Later, when the actual VkPipeline objects are created by suballocating from the bulk allocation, we ask the

implementation to report the valid handles of the VkPipeline objects along with the actual suballocated sizes and different `memoryObjectId`.

6) Can we require the callbacks to be always called in the same thread with the Vulkan commands?

RESOLVED No. Some implementations might choose to multiplex work from multiple application threads into a single backend thread and perform JIT allocations as a part of that flow. Since this behavior is theoretically legit, we cannot require the callbacks to be always called in the same thread with the Vulkan commands, and the note is to remind the applications to handle this case properly.

7) Should we add an additional “allocation failed” event type with things like size and heap index reported?

RESOLVED Yes. This fits in well with the callback infrastructure added in this extension, and implementation touches the same code and has the same overheads as the rest of the extension. It could help debugging things like getting a VK_ERROR_OUT_OF_HOST_MEMORY error when ending a command buffer. Right now the allocation failure could have happened anywhere during recording, and a callback would be really useful to understand where and why.

Version History

- Revision 1, 2020-08-26 (Yiwei Zhang)
  - Initial version
- Revision 2, 2021-01-06 (Yiwei Zhang)
  - Minor description update

VK_EXT_direct_mode_display

Name String

`VK_EXT_direct_mode_display`

Extension Type

Instance extension

Registered Extension Number

89

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_display

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2016-12-13

IP Status

No known IP claims.

Contributors

- Pierre Boudier, NVIDIA
- James Jones, NVIDIA
- Damien Leone, NVIDIA
- Pierre-Loup Griffais, Valve
- Liam Middlebrook, NVIDIA

Description

This is extension, along with related platform extensions, allows applications to take exclusive control of displays associated with a native windowing system. This is especially useful for virtual reality applications that wish to hide HMDs (head mounted displays) from the native platform's display management system, desktop, and/or other applications.

New Commands

- vkReleaseDisplayEXT

New Enum Constants

- VK_EXT_DIRECT_MODE_DISPLAY_EXTENSION_NAME
- VK_EXT_DIRECT_MODE_DISPLAY_SPEC_VERSION

Issues

1) Should this extension and its related platform-specific extensions leverage VK_KHR_display, or provide separate equivalent interfaces.

RESOLVED: Use VK_KHR_display concepts and objects. VK_KHR_display can be used to enumerate all displays on the system, including those attached to/in use by a window system or native platform, but VK_KHR_display_swapchain will fail to create a swapchain on in-use displays. This extension and its platform-specific children will allow applications to grab in-use displays away from window systems and/or native platforms, allowing them to be used with VK_KHR_display_swapchain.

2) Are separate calls needed to acquire displays and enable direct mode?

RESOLVED: No, these operations happen in one combined command. Acquiring a display puts it into direct mode.

Version History

- Revision 1, 2016-12-13 (James Jones)
  - Initial draft

VK_EXT_directfb_surface

Name String

`VK_EXT_directfb_surface`

Extension Type

Instance extension

Registered Extension Number

347

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_surface

Contact

- Nicolas Caramelli [GitHub]caramelli

Other Extension Metadata

Last Modified Date

2020-06-16

IP Status

No known IP claims.

Contributors

- Nicolas Caramelli

Description

The `VK_EXT_directfb_surface` extension is an instance extension. It provides a mechanism to create a VkSurfaceKHR object (defined by the `VK_KHR_surface` extension) that refers to a DirectFB `IDirectFBSurface`, as well as a query to determine support for rendering via DirectFB.

New Commands

- vkCreateDirectFBSurfaceEXT
- vkGetPhysicalDeviceDirectFBPresentationSupportEXT

New Structures

- VkDirectFBSurfaceCreateInfoEXT

New Bitmasks

- VkDirectFBSurfaceCreateFlagsEXT

New Enum Constants

- `VK_EXT_DIRECTFB_SURFACE_EXTENSION_NAME`
- `VK_EXT_DIRECTFB_SURFACE_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DIRECTFB_SURFACE_CREATE_INFO_EXT

Version History

- Revision 1, 2020-06-16 (Nicolas Caramelli)
  - Initial version

VK_EXT_discard_rectangles

Name String

`VK_EXT_discard_rectangles`

Extension Type

Device extension

Registered Extension Number

100

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2023-01-18

Interactions and External Dependencies

- Interacts with `VK_KHR_device_group`
- Interacts with Vulkan 1.1

Contributors

- Daniel Koch, NVIDIA
- Jeff Bolz, NVIDIA

Description

This extension provides additional orthogonally aligned “discard rectangles” specified in framebuffer-space coordinates that restrict rasterization of all points, lines, and triangles.

From zero to an implementation-dependent limit (specified by `maxDiscardRectangles`) number of discard rectangles can be operational at once. When one or more discard rectangles are active, rasterized fragments can either survive if the fragment is within any of the operational discard rectangles (VK_DISCARD_RECTANGLE_MODE_INCLUSIVE_EXT mode) or be rejected if the fragment is within any of the operational discard rectangles

(VK_DISCARD_RECTANGLE_MODE_EXCLUSIVE_EXT mode).

These discard rectangles operate orthogonally to the existing scissor test functionality. The discard rectangles can be different for each physical device in a device group by specifying the device mask and setting discard rectangle dynamic state.

Version 2 of this extension introduces new dynamic states VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT and VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT, and the corresponding functions vkCmdSetDiscardRectangleEnableEXT and vkCmdSetDiscardRectangleModeEXT. Applications that use these dynamic states must ensure the implementation advertises at least `specVersion 2` of this extension.

New Commands

- vkCmdSetDiscardRectangleEXT
- vkCmdSetDiscardRectangleEnableEXT
- vkCmdSetDiscardRectangleModeEXT

New Structures

- Extending VkGraphicsPipelineCreateInfo:
  - VkPipelineDiscardRectangleStateCreateInfoEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceDiscardRectanglePropertiesEXT

New Enums

- VkDiscardRectangleModeEXT

New Bitmasks

- VkPipelineDiscardRectangleStateCreateFlagsEXT

New Enum Constants

- `VK_EXT_DISCARD_RECTANGLES_EXTENSION_NAME`
- `VK_EXT_DISCARD_RECTANGLES_SPEC_VERSION`
- Extending VkDynamicState:
  - VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT
  - VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT
  - VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DISCARD_RECTANGLE_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_PIPELINE_DISCARD_RECTANGLE_STATE_CREATE_INFO_EXT

Version History

- Revision 2, 2023-01-18 (Piers Daniell)
  - Add dynamic states for discard rectangle enable/disable and mode.
- Revision 1, 2016-12-22 (Piers Daniell)
  - Internal revisions

VK_EXT_display_control

Name String

VK_EXT_display_control

Extension Type

Device extension

Registered Extension Number

92

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_EXT_display_surface_counter
and
VK_KHR_swapchain

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2016-12-13

IP Status

No known IP claims.

Contributors

- Pierre Boudier, NVIDIA
- James Jones, NVIDIA
- Damien Leone, NVIDIA
- Pierre-Loup Griffais, Valve
- Daniel Vetter, Intel

Description

This extension defines a set of utility functions for use with the VK_KHR_display and VK_KHR_display_swapchain extensions.

New Commands

- vkDisplayPowerControlEXT
- vkGetSwapchainCounterEXT
- vkRegisterDeviceEventEXT
- vkRegisterDisplayEventEXT

New Structures

- VkDeviceEventInfoEXT
- VkDisplayEventInfoEXT
- VkDisplayPowerInfoEXT
- Extending VkSwapchainCreateInfoKHR:
  - VkSwapchainCounterCreateInfoEXT

New Enums

- VkDeviceEventTypeEXT
- VkDisplayEventTypeEXT
- VkDisplayPowerStateEXT

New Enum Constants

- `VK_EXT_DISPLAY_CONTROL_EXTENSION_NAME`
- `VK_EXT_DISPLAY_CONTROL_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DEVICE_EVENT_INFO_EXT
  - VK_STRUCTURE_TYPE_DISPLAY_EVENT_INFO_EXT
  - VK_STRUCTURE_TYPE_DISPLAY_POWER_INFO_EXT
  - VK_STRUCTURE_TYPE_SWAPCHAIN_COUNTER_CREATE_INFO_EXT

Issues

1) Should this extension add an explicit “WaitForVsync” API or a fence signaled at vsync that the application can wait on?

RESOLVED: A fence. A separate API could later be provided that allows exporting the fence to a native object that could be inserted into standard run loops on POSIX and Windows systems.

2) Should callbacks be added for a vsync event, or in general to monitor events in Vulkan?

RESOLVED: No, fences should be used. Some events are generated by interrupts which are managed in the kernel. In order to use a callback provided by the application, drivers would need to have the userspace driver spawn threads that would wait on the kernel event, and hence the

callbacks could be difficult for the application to synchronize with its other work given they would arrive on a foreign thread.

3) Should vblank or scanline events be exposed?

RESOLVED: Vblank events. Scanline events could be added by a separate extension, but the latency of processing an interrupt and waking up a userspace event is high enough that the accuracy of a scanline event would be rather low. Further, per-scanline interrupts are not supported by all hardware.

Version History

- Revision 1, 2016-12-13 (James Jones)
    - Initial draft

VK_EXT_display_surface_counter

Name String

`VK_EXT_display_surface_counter`

Extension Type

Instance extension

Registered Extension Number

91

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_display

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2016-12-13

IP Status

No known IP claims.

Contributors

- Pierre Boudier, NVIDIA
- James Jones, NVIDIA
- Damien Leone, NVIDIA
- Pierre-Loup Griffais, Valve
- Daniel Vetter, Intel

Description

This extension defines a vertical blanking period counter associated with display surfaces. It provides a mechanism to query support for such a counter from a VkSurfaceKHR object.

New Commands

- vkGetPhysicalDeviceSurfaceCapabilities2EXT

New Structures

- VkSurfaceCapabilities2EXT

New Enums

- VkSurfaceCounterFlagBitsEXT

New Bitmasks

- VkSurfaceCounterFlagsEXT

New Enum Constants

- `VK_EXT_DISPLAY_SURFACE_COUNTER_EXTENSION_NAME`
- `VK_EXT_DISPLAY_SURFACE_COUNTER_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES2_EXT
  - VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_2_EXT

Version History

- Revision 1, 2016-12-13 (James Jones)
  - Initial draft

VK_EXT_dynamic_rendering_unused_attachments

Name String

`VK_EXT_dynamic_rendering_unused_attachments`

Extension Type

Device extension

Registered Extension Number

500

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1
and
VK_KHR_dynamic_rendering
or
Vulkan Version 1.3

Contact

- Piers Daniell [GitHub]pdaniell-nv

Extension Proposal

VK_EXT_dynamic_rendering_unused_attachments

Other Extension Metadata

Last Modified Date

2023-05-22

IP Status

No known IP claims.

Contributors

- Daniel Story, Nintendo
- Hans-Kristian Arntzen, Valve
- Jan-Harald Fredriksen, Arm
- James Fitzpatrick, Imagination Technologies
- Pan Gao, Huawei Technologies
- Ricardo Garcia, Igalia

- Stu Smith, AMD

Description

This extension lifts some restrictions in the `VK_KHR_dynamic_rendering` extension to allow render pass instances and bound pipelines within those render pass instances to have an unused attachment specified in one but not the other. It also allows pipelines to use different formats in a render pass as long the attachment is NULL.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT

New Enum Constants

- `VK_EXT_DYNAMIC_RENDERING_UNUSED_ATTACHMENTS_EXTENSION_NAME`
- `VK_EXT_DYNAMIC_RENDERING_UNUSED_ATTACHMENTS_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_UNUSED_ATTACHMENTS_FEATURES_EXT

Issues

None.

Version History

- Revision 1, 2023-05-22 (Piers Daniell)
  - Internal revisions

VK_EXT_extended_dynamic_state3

Name String

VK_EXT_extended_dynamic_state3

Extension Type

Device extension

Registered Extension Number

456

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_VERSION_1_1
- Interacts with VK_VERSION_1_4
- Interacts with VK_EXT_blend_operation_advanced
- Interacts with VK_EXT_conservative_rasterization
- Interacts with VK_EXT_depth_clip_control
- Interacts with VK_EXT_depth_clip_enable
- Interacts with VK_EXT_line_rasterization
- Interacts with VK_EXT_provoking_vertex
- Interacts with VK_EXT_sample_locations
- Interacts with VK_EXT_transform_feedback
- Interacts with VK_KHR_line_rasterization
- Interacts with VK_KHR_maintenance2
- Interacts with VK_NV_clip_space_w_scaling
- Interacts with VK_NV_coverage_reduction_mode
- Interacts with VK_NV_fragment_coverage_to_color
- Interacts with VK_NV_framebuffer_mixed_samples
- Interacts with VK_NV_representative_fragment_test
- Interacts with VK_NV_shading_rate_image

- Interacts with VK_NV_viewport_swizzle
- Interacts with VkPhysicalDeviceExtendedDynamicState3FeaturesEXT::extendedDynamicState3AlphaToOneEnable
- Interacts with VkPhysicalDeviceExtendedDynamicState3FeaturesEXT::extendedDynamicState3DepthClampEnable
- Interacts with VkPhysicalDeviceExtendedDynamicState3FeaturesEXT::extendedDynamicState3LogicOpEnable
- Interacts with VkPhysicalDeviceExtendedDynamicState3FeaturesEXT::extendedDynamicState3PolygonMode
- Interacts with VkPhysicalDeviceExtendedDynamicState3FeaturesEXT::extendedDynamicState3RasterizationStream
- Interacts with VkPhysicalDeviceExtendedDynamicState3FeaturesEXT::extendedDynamicState3TessellationDomainOrigin

Contact

- Piers Daniell [GitHub]pdaniell-nv

Extension Proposal

VK_EXT_extended_dynamic_state3

Other Extension Metadata

Last Modified Date

2022-09-02

IP Status

No known IP claims.

Contributors

- Daniel Story, Nintendo
- Jamie Madill, Google
- Jan-Harald Fredriksen, Arm
- Faith Ekstrand, Collabora
- Mike Blumenkrantz, Valve
- Ricardo Garcia, Igalia
- Samuel Pitoiset, Valve
- Shahbaz Youssefi, Google

- Stu Smith, AMD
- Tapani Pälli, Intel

Description

This extension adds almost all of the remaining pipeline state as dynamic state to help applications further reduce the number of monolithic pipelines they need to create and bind.

New Commands

- vkCmdSetAlphaToCoverageEnableEXT
- vkCmdSetAlphaToOneEnableEXT
- vkCmdSetColorBlendEnableEXT
- vkCmdSetColorBlendEquationEXT
- vkCmdSetColorWriteMaskEXT
- vkCmdSetDepthClampEnableEXT
- vkCmdSetLogicOpEnableEXT
- vkCmdSetPolygonModeEXT
- vkCmdSetRasterizationSamplesEXT
- vkCmdSetSampleMaskEXT

If VK_EXT_blend_operation_advanced is supported:

- vkCmdSetColorBlendAdvancedEXT

If VK_EXT_conservative_rasterization is supported:

- vkCmdSetConservativeRasterizationModeEXT
- vkCmdSetExtraPrimitiveOverestimationSizeEXT

If VK_EXT_depth_clip_control is supported:

- vkCmdSetDepthClipNegativeOneToOneEXT

If VK_EXT_depth_clip_enable is supported:

- vkCmdSetDepthClipEnableEXT

If VK_EXT_provoking_vertex is supported:

- vkCmdSetProvokingVertexModeEXT

If VK_EXT_sample_locations is supported:

- vkCmdSetSampleLocationsEnableEXT

If VK_EXT_transform_feedback is supported:

- vkCmdSetRasterizationStreamEXT

If VK_KHR_maintenance2 or Vulkan Version 1.1 is supported:

- vkCmdSetTessellationDomainOriginEXT

If Vulkan Version 1.4 or VK_KHR_line_rasterization or `VK_EXT_line_rasterization` is supported:

- vkCmdSetLineRasterizationModeEXT
- vkCmdSetLineStippleEnableEXT

New Structures

- VkColorBlendAdvancedEXT
- VkColorBlendEquationEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceExtendedDynamicState3FeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceExtendedDynamicState3PropertiesEXT

New Enum Constants

- `VK_EXT_EXTENDED_DYNAMIC_STATE_3_EXTENSION_NAME`
- `VK_EXT_EXTENDED_DYNAMIC_STATE_3_SPEC_VERSION`
- Extending VkDynamicState:
  - VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT
  - VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT
  - VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT
  - VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT
  - VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT
  - VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT
  - VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT
  - VK_DYNAMIC_STATE_POLYGON_MODE_EXT
  - VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT
  - VK_DYNAMIC_STATE_SAMPLE_MASK_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_3_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_3_PROPERTIES_EXT

If VK_EXT_blend_operation_advanced is supported:

- Extending VkDynamicState:

    - VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT

If VK_EXT_conservative_rasterization is supported:

- Extending VkDynamicState:
    - VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT
    - VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT

If VK_EXT_depth_clip_control is supported:

- Extending VkDynamicState:
    - VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT

If VK_EXT_depth_clip_enable is supported:

- Extending VkDynamicState:
    - VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT

If VK_EXT_provoking_vertex is supported:

- Extending VkDynamicState:
    - VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT

If VK_EXT_sample_locations is supported:

- Extending VkDynamicState:
    - VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT

If VK_EXT_transform_feedback is supported:

- Extending VkDynamicState:
    - VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT

If VK_KHR_maintenance2 or Vulkan Version 1.1 is supported:

- Extending VkDynamicState:
    - VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT

If Vulkan Version 1.4 or VK_KHR_line_rasterization or `VK_EXT_line_rasterization` is supported:

- Extending VkDynamicState:
    - VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT
    - VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT

Issues

1) What about the VkPipelineMultisampleStateCreateInfo state `sampleShadingEnable` and `minSampleShading`?

`sampleShadingEnable` and `minSampleShading` are required when compiling the fragment shader, and it is not meaningful to set them dynamically since they always need to match the fragment shader state, so this hardware state may as well just come from the pipeline with the fragment shader.

Version History

- Revision 2, 2022-07-18 (Piers Daniell)
  - Added rasterizationSamples
- Revision 1, 2022-05-18 (Piers Daniell)
  - Internal revisions

VK_EXT_external_memory_acquire_unmodified

Name String

`VK_EXT_external_memory_acquire_unmodified`

Extension Type

Device extension

Registered Extension Number

454

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_external_memory
or
Vulkan Version 1.1

Contact

- James Jones [GitHub]cubanismo

Extension Proposal

VK_EXT_external_memory_acquire_unmodified

Other Extension Metadata

Last Modified Date

2023-03-09

Contributors

- Lina Versace, Google
- Chia-I Wu, Google
- James Jones, NVIDIA
- Yiwei Zhang, Google

Description

A memory barrier **may** have a performance penalty when acquiring ownership of a subresource range from an external queue family. This extension provides API that **may** reduce the performance penalty if ownership of the subresource range was previously released to the external queue family and if the resource's memory has remained unmodified between the release and acquire operations.

New Structures

- Extending VkBufferMemoryBarrier, VkBufferMemoryBarrier2, VkImageMemoryBarrier, VkImageMemoryBarrier2:
  - VkExternalMemoryAcquireUnmodifiedEXT

New Enum Constants

- `VK_EXT_EXTERNAL_MEMORY_ACQUIRE_UNMODIFIED_EXTENSION_NAME`
- `VK_EXT_EXTERNAL_MEMORY_ACQUIRE_UNMODIFIED_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_ACQUIRE_UNMODIFIED_EXT

Version History

- Revision 1, 2023-03-09 (Lina Versace)
  - Initial revision

VK_EXT_external_memory_dma_buf

Name String

`VK_EXT_external_memory_dma_buf`

Extension Type

Device extension

Registered Extension Number

126

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_external_memory_fd

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2017-10-10

IP Status

No known IP claims.

Contributors

- Lina Versace, Google
- James Jones, NVIDIA
- Faith Ekstrand, Intel

Description

A `dma_buf` is a type of file descriptor, defined by the Linux kernel, that allows sharing memory across kernel device drivers and across processes. This extension enables applications to import a `dma_buf` as VkDeviceMemory, to export VkDeviceMemory as a `dma_buf`, and to create VkBuffer objects that **can** be bound to that memory.

New Enum Constants

- `VK_EXT_EXTERNAL_MEMORY_DMA_BUF_EXTENSION_NAME`
- `VK_EXT_EXTERNAL_MEMORY_DMA_BUF_SPEC_VERSION`

- Extending VkExternalMemoryHandleTypeFlagBits:
  - VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT

Issues

1) How does the application, when creating a VkImage that it intends to bind to `dma_buf` VkDeviceMemory containing an externally produced image, specify the memory layout (such as row pitch and DRM format modifier) of the VkImage? In other words, how does the application achieve behavior comparable to that provided by `EGL_EXT_image_dma_buf_import` and `EGL_EXT_image_dma_buf_import_modifiers` ?

RESOLVED: Features comparable to those in `EGL_EXT_image_dma_buf_import` and `EGL_EXT_image_dma_buf_import_modifiers` will be provided by an extension layered atop this one.

2) Without the ability to specify the memory layout of external `dma_buf` images, how is this extension useful?

RESOLVED: This extension provides exactly one new feature: the ability to import/export between `dma_buf` and VkDeviceMemory. This feature, together with features provided by `VK_KHR_external_memory_fd`, is sufficient to bind a VkBuffer to `dma_buf`.

Version History

- Revision 1, 2017-10-10 (Lina Versace)
  - Squashed internal revisions

VK_EXT_external_memory_host

Name String

`VK_EXT_external_memory_host`

Extension Type

Device extension

Registered Extension Number

179

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_external_memory
or
Vulkan Version 1.1

Contact

- Daniel Rakos [GitHub]drakos-amd

Other Extension Metadata

Last Modified Date

2017-11-10

IP Status

No known IP claims.

Contributors

- Jaakko Konttinen, AMD
- David Mao, AMD
- Daniel Rakos, AMD
- Tobias Hector, Imagination Technologies
- Faith Ekstrand, Intel
- James Jones, NVIDIA

Description

This extension enables an application to import host allocations and host mapped foreign device memory to Vulkan memory objects.

New Commands

- vkGetMemoryHostPointerPropertiesEXT

New Structures

- VkMemoryHostPointerPropertiesEXT
- Extending VkMemoryAllocateInfo:
  - VkImportMemoryHostPointerInfoEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceExternalMemoryHostPropertiesEXT

New Enum Constants

- `VK_EXT_EXTERNAL_MEMORY_HOST_EXTENSION_NAME`
- `VK_EXT_EXTERNAL_MEMORY_HOST_SPEC_VERSION`
- Extending VkExternalMemoryHandleTypeFlagBits:
  - VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT
  - VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_IMPORT_MEMORY_HOST_POINTER_INFO_EXT
  - VK_STRUCTURE_TYPE_MEMORY_HOST_POINTER_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_MEMORY_HOST_PROPERTIES_EXT

Issues

1) What memory type has to be used to import host pointers?

RESOLVED: Depends on the implementation. Applications have to use the new vkGetMemoryHostPointerPropertiesEXT command to query the supported memory types for a particular host pointer. The reported memory types may include memory types that come from a memory heap that is otherwise not usable for regular memory object allocation and thus such a heap's size may be zero.

2) Can the application still access the contents of the host allocation after importing?

RESOLVED: Yes. However, usual synchronization requirements apply.

3) Can the application free the host allocation?

RESOLVED: No, it violates valid usage conditions. Using the memory object imported from a host allocation that is already freed thus results in undefined behavior.

4) Is vkMapMemory expected to return the same host address which was specified when importing it to the memory object?

RESOLVED: No. Implementations are allowed to return the same address but it is not required. Some implementations might return a different virtual mapping of the allocation, although the same physical pages will be used.

5) Is there any limitation on the alignment of the host pointer and/or size?

RESOLVED: Yes. Both the address and the size have to be an integer multiple of `minImportedHostPointerAlignment`. In addition, some platforms and foreign devices may have additional restrictions.

6) Can the same host allocation be imported multiple times into a given physical device?

RESOLVED: No, at least not guaranteed by this extension. Some platforms do not allow locking the same physical pages for device access multiple times, so attempting to do it may result in undefined behavior.

7) Does this extension support exporting the new handle type?

RESOLVED: No.

8) Should we include the possibility to import host mapped foreign device memory using this API?

RESOLVED: Yes, through a separate handle type. Implementations are still allowed to support only one of the handle types introduced by this extension by not returning import support for a particular handle type as returned in VkExternalMemoryProperties.

Version History

- Revision 1, 2017-11-10 (Daniel Rakos)
  - Internal revisions

VK_EXT_external_memory_metal

Name String

VK_EXT_external_memory_metal

Extension Type

Device extension

Registered Extension Number

603

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_external_memory
or
Vulkan Version 1.1

Contact

- Aitor Camacho Larrondo [GitHub]aitor-lunarg

Extension Proposal

VK_EXT_external_memory_metal

Other Extension Metadata

Last Modified Date

2024-07-18

IP Status

No known IP claims.

Contributors

- Aitor Camacho Larrondo, LunarG Inc.

Description

An application may wish to reference device memory in multiple Vulkan device instances, in multiple processes, and/or in Metal API. This extension enables an application to export and import Metal handles from Vulkan memory objects such that the underlying resources can be referenced outside the scope of the Vulkan device instance that created them.

New Commands

- vkGetMemoryMetalHandleEXT

- vkGetMemoryMetalHandlePropertiesEXT

New Structures

- VkMemoryGetMetalHandleInfoEXT
- VkMemoryMetalHandlePropertiesEXT
- Extending VkMemoryAllocateInfo:
  - VkImportMemoryMetalHandleInfoEXT

New Enum Constants

- `VK_EXT_EXTERNAL_MEMORY_METAL_EXTENSION_NAME`
- `VK_EXT_EXTERNAL_MEMORY_METAL_SPEC_VERSION`
- Extending VkExternalMemoryHandleTypeFlagBits:
  - VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLBUFFER_BIT_EXT
  - VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLHEAP_BIT_EXT
  - VK_EXTERNAL_MEMORY_HANDLE_TYPE_MTLTEXTURE_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_IMPORT_MEMORY_METAL_HANDLE_INFO_EXT
  - VK_STRUCTURE_TYPE_MEMORY_GET_METAL_HANDLE_INFO_EXT
  - VK_STRUCTURE_TYPE_MEMORY_METAL_HANDLE_PROPERTIES_EXT

Version History

- Revision 1, 2024-07-18 (Aitor Camacho Larrondo)
  - Initial revision

VK_EXT_filter_cubic

Name String

VK_EXT_filter_cubic

Extension Type

Device extension

Registered Extension Number

171

Revision

3

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- Matthew Netsch [GitHub]mnetsch

Other Extension Metadata

Last Modified Date

2019-12-13

Contributors

- Bill Licea-Kane, Qualcomm Technologies, Inc.
- Andrew Garrard, Samsung
- Daniel Koch, NVIDIA
- Donald Scorgie, Imagination Technologies
- Graeme Leese, Broadcom
- Jan-Harald Fredriksen, ARM
- Jeff Leger, Qualcomm Technologies, Inc.
- Tobias Hector, AMD
- Tom Olson, ARM
- Stuart Smith, Imagination Technologies

Description

VK_EXT_filter_cubic extends VK_IMG_filter_cubic.

It documents cubic filtering of other image view types. It adds new structures that **can** be added to

the pNext chain of VkPhysicalDeviceImageFormatInfo2 and VkImageFormatProperties2 that **can** be used to determine which image types and which image view types support cubic filtering.

New Structures

- Extending VkImageFormatProperties2:
  - VkFilterCubicImageViewImageFormatPropertiesEXT
- Extending VkPhysicalDeviceImageFormatInfo2:
  - VkPhysicalDeviceImageViewImageFormatInfoEXT

New Enum Constants

- VK_EXT_FILTER_CUBIC_EXTENSION_NAME
- VK_EXT_FILTER_CUBIC_SPEC_VERSION
- Extending VkFilter:
  - VK_FILTER_CUBIC_EXT
- Extending VkFormatFeatureFlagBits:
  - VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_FILTER_CUBIC_IMAGE_VIEW_IMAGE_FORMAT_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_VIEW_IMAGE_FORMAT_INFO_EXT

Version History

- Revision 3, 2019-12-13 (wwlk)
  - Delete requirement to cubic filter the formats USCALED_PACKED32, SSCALED_PACKED32, UINT_PACK32, and SINT_PACK32 (cut/paste error)
- Revision 2, 2019-06-05 (wwlk)
  - Clarify 1D optional
- Revision 1, 2019-01-24 (wwlk)
  - Initial version

VK_EXT_fragment_density_map

Name String

`VK_EXT_fragment_density_map`

Extension Type

Device extension

Registered Extension Number

219

Revision

3

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_VERSION_1_3
- Interacts with VK_KHR_dynamic_rendering
- Interacts with VK_KHR_format_feature_flags2

SPIR-V Dependencies

- SPV_EXT_fragment_invocation_density

Contact

- Matthew Netsch [GitHub]mnetsch

Other Extension Metadata

Last Modified Date

2025-05-20

Interactions and External Dependencies

- This extension provides API support for `GL_EXT_fragment_invocation_density`

Contributors

- Matthew Netsch, Qualcomm Technologies, Inc.
- Robert VanReenen, Qualcomm Technologies, Inc.
- Jonathan Wicks, Qualcomm Technologies, Inc.
- Tate Hornbeck, Qualcomm Technologies, Inc.
- Sam Holmes, Qualcomm Technologies, Inc.

- Jeff Leger, Qualcomm Technologies, Inc.
- Jan-Harald Fredriksen, ARM
- Jeff Bolz, NVIDIA
- Pat Brown, NVIDIA
- Daniel Rakos, AMD
- Piers Daniell, NVIDIA

Description

This extension allows an application to specify areas of the render target where the fragment shader may be invoked fewer times. These fragments are broadcasted out to multiple pixels to cover the render target.

The primary use of this extension is to reduce workloads in areas where lower quality may not be perceived such as the distorted edges of a lens or the periphery of a user's gaze.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceFragmentDensityMapFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceFragmentDensityMapPropertiesEXT
- Extending VkRenderPassCreateInfo, VkRenderPassCreateInfo2:
  - VkRenderPassFragmentDensityMapCreateInfoEXT

If Vulkan Version 1.3 or VK_KHR_dynamic_rendering is supported:

- Extending VkRenderingInfo:
  - VkRenderingFragmentDensityMapAttachmentInfoEXT

New Enum Constants

- `VK_EXT_FRAGMENT_DENSITY_MAP_EXTENSION_NAME`
- `VK_EXT_FRAGMENT_DENSITY_MAP_SPEC_VERSION`
- Extending VkAccessFlagBits:
  - VK_ACCESS_FRAGMENT_DENSITY_MAP_READ_BIT_EXT
- Extending VkFormatFeatureFlagBits:
  - VK_FORMAT_FEATURE_FRAGMENT_DENSITY_MAP_BIT_EXT
- Extending VkImageCreateFlagBits:
  - VK_IMAGE_CREATE_SUBSAMPLED_BIT_EXT
- Extending VkImageLayout:
  - VK_IMAGE_LAYOUT_FRAGMENT_DENSITY_MAP_OPTIMAL_EXT

- Extending VkImageUsageFlagBits:
  - VK_IMAGE_USAGE_FRAGMENT_DENSITY_MAP_BIT_EXT
- Extending VkImageViewCreateFlagBits:
  - VK_IMAGE_VIEW_CREATE_FRAGMENT_DENSITY_MAP_DYNAMIC_BIT_EXT
- Extending VkPipelineStageFlagBits:
  - VK_PIPELINE_STAGE_FRAGMENT_DENSITY_PROCESS_BIT_EXT
- Extending VkSamplerCreateFlagBits:
  - VK_SAMPLER_CREATE_SUBSAMPLED_BIT_EXT
  - VK_SAMPLER_CREATE_SUBSAMPLED_COARSE_RECONSTRUCTION_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_RENDER_PASS_FRAGMENT_DENSITY_MAP_CREATE_INFO_EXT

If VK_KHR_format_feature_flags2 or Vulkan Version 1.3 is supported:

- Extending VkFormatFeatureFlagBits2:
  - VK_FORMAT_FEATURE_2_FRAGMENT_DENSITY_MAP_BIT_EXT

If Vulkan Version 1.3 or VK_KHR_dynamic_rendering is supported:

- Extending VkPipelineCreateFlagBits:
  - VK_PIPELINE_CREATE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
  - VK_PIPELINE_RASTERIZATION_STATE_CREATE_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_INFO_EXT

New or Modified Built-In Variables

- `FragInvocationCountEXT`
- `FragSizeEXT`

New SPIR-V Capabilities

- `FragmentDensityEXT`

Version History

- Revision 1, 2018-09-25 (Matthew Netsch)
  - Initial version
- Revision 2, 2021-09-30 (Jon Leech)

    - Add interaction with `VK_KHR_format_feature_flags2` to `vk.xml`
- Revision 3, 2025-05-20 (Matthew Netsch)
    - Fixes fragmentDensityTexelSize calculation equation

VK_EXT_fragment_density_map2

Name String

VK_EXT_fragment_density_map2

Extension Type

Device extension

Registered Extension Number

333

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_EXT_fragment_density_map

Contact

- Matthew Netsch [GitHub]mnetsch

Other Extension Metadata

Last Modified Date

2020-06-16

Interactions and External Dependencies

- Interacts with Vulkan 1.1

Contributors

- Matthew Netsch, Qualcomm Technologies, Inc.
- Jonathan Tinkham, Qualcomm Technologies, Inc.
- Jonathan Wicks, Qualcomm Technologies, Inc.
- Jan-Harald Fredriksen, ARM

Description

This extension adds additional features and properties to VK_EXT_fragment_density_map in order to reduce fragment density map host latency as well as improved queries for subsampled sampler implementation-dependent behavior.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceFragmentDensityMap2FeaturesEXT

- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceFragmentDensityMap2PropertiesEXT

New Enum Constants

- VK_EXT_FRAGMENT_DENSITY_MAP_2_EXTENSION_NAME
- VK_EXT_FRAGMENT_DENSITY_MAP_2_SPEC_VERSION
- Extending VkImageViewCreateFlagBits:
  - VK_IMAGE_VIEW_CREATE_FRAGMENT_DENSITY_MAP_DEFERRED_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_2_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_2_PROPERTIES_EXT

Version History

- Revision 1, 2020-06-16 (Matthew Netsch)
  - Initial version

VK_EXT_fragment_density_map_offset

Name String

VK_EXT_fragment_density_map_offset

Extension Type

Device extension

Registered Extension Number

620

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1
and
VK_EXT_fragment_density_map
and
VK_KHR_create_renderpass2
or
Vulkan Version 1.2
and
Vulkan Version 1.3
or
VK_KHR_dynamic_rendering

Contact

- Connor Abbott [GitHub]cwabbott0

Extension Proposal

VK_EXT_fragment_density_map_offset

Other Extension Metadata

Last Modified Date

2025-02-14

Contributors

- Connor Abbott, Valve Corporation
- Matthew Netsch, Qualcomm Technologies, Inc.
- Jonathan Wicks, Qualcomm Technologies, Inc.

- Jonathan Tinkham, Qualcomm Technologies, Inc.
- Jeff Leger, Qualcomm Technologies, Inc.
- Manan Katwala, Qualcomm Technologies, Inc.
- Mike Blumenkrantz, Valve Corporation

Description

This extension allows an application to specify offsets to a fragment density map attachment, changing the framebuffer location where density values are applied to without having to regenerate the fragment density map.

New Commands

- vkCmdEndRendering2EXT

New Structures

- VkRenderingEndInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceFragmentDensityMapOffsetFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceFragmentDensityMapOffsetPropertiesEXT
- Extending VkSubpassEndInfo, VkRenderingEndInfoKHR:
  - VkRenderPassFragmentDensityMapOffsetEndInfoEXT

New Enum Constants

- `VK_EXT_FRAGMENT_DENSITY_MAP_OFFSET_EXTENSION_NAME`
- `VK_EXT_FRAGMENT_DENSITY_MAP_OFFSET_SPEC_VERSION`
- Extending VkImageCreateFlagBits:
  - VK_IMAGE_CREATE_FRAGMENT_DENSITY_MAP_OFFSET_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_OFFSET_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_DENSITY_MAP_OFFSET_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_RENDERING_END_INFO_EXT
  - VK_STRUCTURE_TYPE_RENDER_PASS_FRAGMENT_DENSITY_MAP_OFFSET_END_INFO_EXT

Version History

- Revision 1, 2025-02-14 (Connor Abbott)

- Initial version

VK_EXT_fragment_shader_interlock

Name String

`VK_EXT_fragment_shader_interlock`

Extension Type

Device extension

Registered Extension Number

252

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_EXT_fragment_shader_interlock

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2019-05-02

Interactions and External Dependencies

- This extension provides API support for `GL_ARB_fragment_shader_interlock`

Contributors

- Daniel Koch, NVIDIA
- Graeme Leese, Broadcom
- Jan-Harald Fredriksen, Arm
- Faith Ekstrand, Intel
- Jeff Bolz, NVIDIA
- Ruihao Zhang, Qualcomm
- Slawomir Grajewski, Intel
- Spencer Fricke, Samsung

Description

This extension adds support for the `FragmentShaderPixelInterlockEXT`, `FragmentShaderSampleInterlockEXT`, and `FragmentShaderShadingRateInterlockEXT` capabilities from the `SPV_EXT_fragment_shader_interlock` extension to Vulkan.

Enabling these capabilities provides a critical section for fragment shaders to avoid overlapping pixels being processed at the same time, and certain guarantees about the ordering of fragment shader invocations of fragments of overlapping pixels.

This extension can be useful for algorithms that need to access per-pixel data structures via shader loads and stores. Algorithms using this extension can access per-pixel data structures in critical sections without other invocations accessing the same per-pixel data. Additionally, the ordering guarantees are useful for cases where the API ordering of fragments is meaningful. For example, applications may be able to execute programmable blending operations in the fragment shader, where the destination buffer is read via image loads and the final value is written via image stores.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT

New Enum Constants

- `VK_EXT_FRAGMENT_SHADER_INTERLOCK_EXTENSION_NAME`
- `VK_EXT_FRAGMENT_SHADER_INTERLOCK_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_INTERLOCK_FEATURES_EXT

New SPIR-V Capabilities

- `FragmentShaderInterlockEXT`
- `FragmentShaderPixelInterlockEXT`
- `FragmentShaderShadingRateInterlockEXT`

Version History

- Revision 1, 2019-05-24 (Piers Daniell)
  - Internal revisions

VK_EXT_frame_boundary

Name String

VK_EXT_frame_boundary

Extension Type

Device extension

Registered Extension Number

376

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- James Fitzpatrick [GitHub]jamesfitzpatrick

Extension Proposal

VK_EXT_frame_boundary

Other Extension Metadata

Last Modified Date

2023-06-14

Contributors

- James Fitzpatrick, Imagination Technologies
- Hugues Evrard, Google
- Melih Yasin Yalcin, Google
- Andrew Garrard, Imagination Technologies
- Jan-Harald Fredriksen, Arm
- Vassili Nikolaev, NVIDIA
- Ting Wei, Huawei

Description

VK_EXT_frame_boundary is a device extension that helps **tools** (such as debuggers) to group queue submissions per frames in non-trivial scenarios, typically when vkQueuePresentKHR is not a relevant frame boundary delimiter.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceFrameBoundaryFeaturesEXT
- Extending VkSubmitInfo, VkSubmitInfo2, VkPresentInfoKHR, VkBindSparseInfo:
  - VkFrameBoundaryEXT

New Enums

- VkFrameBoundaryFlagBitsEXT

New Bitmasks

- VkFrameBoundaryFlagsEXT

New Enum Constants

- `VK_EXT_FRAME_BOUNDARY_EXTENSION_NAME`
- `VK_EXT_FRAME_BOUNDARY_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_FRAME_BOUNDARY_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAME_BOUNDARY_FEATURES_EXT

Version History

- Revision 0, 2022-01-14 (Hugues Evard)
  - Initial proposal
- Revision 1, 2023-06-14 (James Fitzpatrick)
  - Initial draft

VK_EXT_full_screen_exclusive

Name String

`VK_EXT_full_screen_exclusive`

Extension Type

Device extension

Registered Extension Number

256

Revision

4

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1
and
VK_KHR_surface
and
VK_KHR_get_surface_capabilities2
and
VK_KHR_swapchain

API Interactions

- Interacts with VK_VERSION_1_1
- Interacts with VK_KHR_device_group
- Interacts with VK_KHR_win32_surface

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2019-03-12

IP Status

No known IP claims.

Interactions and External Dependencies

- Interacts with Vulkan 1.1
- Interacts with `VK_KHR_device_group`

- Interacts with VK_KHR_win32_surface

Contributors

- Hans-Kristian Arntzen, ARM
- Slawomir Grajewski, Intel
- Tobias Hector, AMD
- James Jones, NVIDIA
- Daniel Rakos, AMD
- Jeff Juliano, NVIDIA
- Joshua Schnarr, NVIDIA
- Aaron Hagan, AMD

Description

This extension allows applications to set the policy for swapchain creation and presentation mechanisms relating to full-screen access. Implementations may be able to acquire exclusive access to a particular display for an application window that covers the whole screen. This can increase performance on some systems by bypassing composition, however it can also result in disruptive or expensive transitions in the underlying windowing system when a change occurs.

Applications can choose between explicitly disallowing or allowing this behavior, letting the implementation decide, or managing this mode of operation directly using the new vkAcquireFullScreenExclusiveModeEXT and vkReleaseFullScreenExclusiveModeEXT commands.

New Commands

- vkAcquireFullScreenExclusiveModeEXT
- vkGetPhysicalDeviceSurfacePresentModes2EXT
- vkReleaseFullScreenExclusiveModeEXT

If VK_KHR_device_group or Vulkan Version 1.1 is supported:

- vkGetDeviceGroupSurfacePresentModes2EXT

New Structures

- Extending VkPhysicalDeviceSurfaceInfo2KHR, VkSwapchainCreateInfoKHR:
  - VkSurfaceFullScreenExclusiveInfoEXT
- Extending VkSurfaceCapabilities2KHR:
  - VkSurfaceCapabilitiesFullScreenExclusiveEXT

If VK_KHR_win32_surface is supported:

- Extending VkPhysicalDeviceSurfaceInfo2KHR, VkSwapchainCreateInfoKHR:
  - VkSurfaceFullScreenExclusiveWin32InfoEXT

New Enums

- VkFullScreenExclusiveEXT

New Enum Constants

- `VK_EXT_FULL_SCREEN_EXCLUSIVE_EXTENSION_NAME`
- `VK_EXT_FULL_SCREEN_EXCLUSIVE_SPEC_VERSION`
- Extending VkResult:
  - VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_FULL_SCREEN_EXCLUSIVE_EXT
  - VK_STRUCTURE_TYPE_SURFACE_FULL_SCREEN_EXCLUSIVE_INFO_EXT

If VK_KHR_win32_surface is supported:

- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_SURFACE_FULL_SCREEN_EXCLUSIVE_WIN32_INFO_EXT

Issues

1) What should the extension & flag be called?

RESOLVED: VK_EXT_full_screen_exclusive.

Other options considered (prior to the app-controlled mode) were:

- VK_EXT_smooth_fullscreen_transition
- VK_EXT_fullscreen_behavior
- VK_EXT_fullscreen_preference
- VK_EXT_fullscreen_hint
- VK_EXT_fast_fullscreen_transition
- VK_EXT_avoid_fullscreen_exclusive

2) Do we need more than a boolean toggle?

RESOLVED: Yes.

Using an enum with default/allowed/disallowed/app-controlled enables applications to accept driver default behavior, specifically override it in either direction without implying the driver is ever required to use full-screen exclusive mechanisms, or manage this mode explicitly.

3) Should this be a KHR or EXT extension?

RESOLVED: EXT, in order to allow it to be shipped faster.

4) Can the full-screen hint affect the surface capabilities, and if so, should the hint also be specified

as input when querying the surface capabilities?

RESOLVED: Yes on both accounts.

While the hint does not guarantee a particular full-screen mode will be used when the swapchain is created, it can sometimes imply particular modes will NOT be used. If the driver determines that it will opt-out of using a particular mode based on the policy, and knows it can only support certain capabilities if that mode is used, it would be confusing at best to the application to report those capabilities in such cases. Not allowing implementations to report this state to applications could result in situations where applications are unable to determine why swapchain creation fails when they specify certain hint values, which could result in never- terminating surface creation loops.

5) Should full-screen be one word or two?

RESOLVED: Two words.

"Fullscreen" is not in my dictionary, and web searches did not turn up definitive proof that it is a colloquially accepted compound word. Documentation for the corresponding Windows API mechanisms dithers. The text consistently uses a hyphen, but none-the-less, there is a SetFullscreenState method in the DXGI swapchain object. Given this inconclusive external guidance, it is best to adhere to the Vulkan style guidelines and avoid inventing new compound words.

Version History

- Revision 4, 2019-03-12 (Tobias Hector)
  - Added application-controlled mode, and related functions
  - Tidied up appendix
- Revision 3, 2019-01-03 (James Jones)
  - Renamed to VK_EXT_full_screen_exclusive
  - Made related adjustments to the tri-state enumerant names.
- Revision 2, 2018-11-27 (James Jones)
  - Renamed to VK_KHR_fullscreen_behavior
  - Switched from boolean flag to tri-state enum
- Revision 1, 2018-11-06 (James Jones)
  - Internal revision

VK_EXT_graphics_pipeline_library

Name String

VK_EXT_graphics_pipeline_library

Extension Type

Device extension

Registered Extension Number

321

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1
and
VK_KHR_pipeline_library

Contact

- Tobias Hector [GitHub]tobski

Extension Proposal

VK_EXT_graphics_pipeline_library

Other Extension Metadata

Last Modified Date

2021-08-17

Contributors

- Tobias Hector, AMD
- Chris Glover, Google
- Jeff Leger, Qualcomm
- Jan-Harald Fredriksen, Arm
- Piers Daniell, NVidia
- Boris Zanin, Mobica
- Krzysztof Niski, NVidia
- Dan Ginsburg, Valve
- Sebastian Aaltonen, Unity

- Arseny Kapoulkine, Roblox
- Calle Lejdfors, Ubisoft
- Tiago Rodrigues, Ubisoft
- Francois Duranleau, Gameloft

Description

This extension allows the separate compilation of four distinct parts of graphics pipelines, with the intent of allowing faster pipeline loading for applications reusing the same shaders or state in multiple pipelines. Each part can be independently compiled into a graphics pipeline library, with a final link step required to create an executable pipeline that can be bound to a command buffer.

New Structures

- Extending VkGraphicsPipelineCreateInfo:
  - VkGraphicsPipelineLibraryCreateInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceGraphicsPipelineLibraryFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceGraphicsPipelineLibraryPropertiesEXT

New Enums

- VkGraphicsPipelineLibraryFlagBitsEXT
- VkPipelineLayoutCreateFlagBits

New Bitmasks

- VkGraphicsPipelineLibraryFlagsEXT

New Enum Constants

- `VK_EXT_GRAPHICS_PIPELINE_LIBRARY_EXTENSION_NAME`
- `VK_EXT_GRAPHICS_PIPELINE_LIBRARY_SPEC_VERSION`
- Extending VkPipelineCreateFlagBits:
  - VK_PIPELINE_CREATE_LINK_TIME_OPTIMIZATION_BIT_EXT
  - VK_PIPELINE_CREATE_RETAIN_LINK_TIME_OPTIMIZATION_INFO_BIT_EXT
- Extending VkPipelineLayoutCreateFlagBits:
  - VK_PIPELINE_LAYOUT_CREATE_INDEPENDENT_SETS_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_GRAPHICS_PIPELINE_LIBRARY_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GRAPHICS_PIPELINE_LIBRARY_FEATURES_EXT

- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GRAPHICS_PIPELINE_LIBRARY_PROPERTIES_EXT

Version History

- Revision 1, 2021-08-17 (Tobias Hector)
  - Initial draft.

VK_EXT_hdr_metadata

Name String

`VK_EXT_hdr_metadata`

Extension Type

Device extension

Registered Extension Number

106

Revision

3

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_swapchain

Contact

- Courtney Goeltzenleuchter [GitHub]courtney-g

Other Extension Metadata

Last Modified Date

2024-03-26

IP Status

No known IP claims.

Contributors

- Courtney Goeltzenleuchter, Google
- Sebastian Wick, Red Hat Inc.
- Tobias Hector, AMD

Description

This extension defines two new structures and a function to assign SMPTE (the Society of Motion Picture and Television Engineers) 2086 metadata and CTA (Consumer Technology Association) 861.3 metadata to a swapchain.

SMPTE 2086 metadata defines the color volume of the display on which the content was optimized for viewing and includes the color primaries, white point, and luminance range. When such content is reproduced on another display, this metadata can be used by the presentation engine to improve processing of images. For instance, values in the image can first be clamped to the color volume described in the metadata, and then what remains can be remapped to the color volume of the presentation engine.

CTA 861.3 metadata additionally includes the maximum intended luminance for the content and the maximum average light level across frames.

This extension does not define exactly how this metadata is used, however, it simply provides a mechanism to provide it to the presentation engine. Presentation engines may process the image based on the metadata before displaying it, resulting in the image being modified outside of Vulkan. For example, the clamping of colors in the image to the color volume may change those values in the image itself.

The metadata does not override or otherwise influence the color space and color encoding.

New Commands

- vkSetHdrMetadataEXT

New Structures

- VkHdrMetadataEXT
- VkXYColorEXT

New Enum Constants

- `VK_EXT_HDR_METADATA_EXTENSION_NAME`
- `VK_EXT_HDR_METADATA_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_HDR_METADATA_EXT

Issues

1) Do we need a query function for the currently specified metadata?

No, Vulkan does not provide queries for state that the application can track on its own.

2) Should we specify default metadata if not specified by the application?

No, the metadata is optional and the absence of the metadata is well-defined.

Version History

- Revision 1, 2016-12-27 (Courtney Goeltzenleuchter)
  - Initial version
- Revision 2, 2018-12-19 (Courtney Goeltzenleuchter)
  - Correct implicit validity for VkHdrMetadataEXT structure
- Revision 3, 2024-03-26 (Tobias Hector & Sebastian Wick)
  - Clarifications and removal of erroneous “reference monitor” term

VK_EXT_headless_surface

Name String

VK_EXT_headless_surface

Extension Type

Instance extension

Registered Extension Number

257

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_surface

Contact

- Lisa Wu [GitHub]chengtianww

Other Extension Metadata

Last Modified Date

2019-03-21

IP Status

No known IP claims.

Contributors

- Ray Smith, Arm

Description

The VK_EXT_headless_surface extension is an instance extension. It provides a mechanism to create VkSurfaceKHR objects independently of any window system or display device. The presentation operation for a swapchain created from a headless surface is by default a no-op, resulting in no externally-visible result.

Because there is no real presentation target, future extensions can layer on top of the headless surface to introduce arbitrary or customizable sets of restrictions or features. These could include features like saving to a file or restrictions to emulate a particular presentation target.

This functionality is expected to be useful for application and driver development because it allows any platform to expose an arbitrary or customizable set of restrictions and features of a presentation engine. This makes it a useful portable test target for applications targeting a wide range of presentation engines where the actual target presentation engines might be scarce,

unavailable, or otherwise undesirable or inconvenient to use for general Vulkan application development.

New Commands

- vkCreateHeadlessSurfaceEXT

New Structures

- VkHeadlessSurfaceCreateInfoEXT

New Bitmasks

- VkHeadlessSurfaceCreateFlagsEXT

New Enum Constants

- `VK_EXT_HEADLESS_SURFACE_EXTENSION_NAME`
- `VK_EXT_HEADLESS_SURFACE_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_HEADLESS_SURFACE_CREATE_INFO_EXT

Version History

- Revision 1, 2019-03-21 (Ray Smith)
  - Initial draft

VK_EXT_image_2d_view_of_3d

Name String

`VK_EXT_image_2d_view_of_3d`

Extension Type

Device extension

Registered Extension Number

394

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_maintenance1
and
VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Special Use

- OpenGL / ES support

Contact

- Mike Blumenkrantz [GitHub]zmike

Extension Proposal

VK_EXT_image_2d_view_of_3d

Other Extension Metadata

Last Modified Date

2022-02-22

IP Status

No known IP claims.

Contributors

- Mike Blumenkrantz, Valve
- Piers Daniell, NVIDIA
- Spencer Fricke, Samsung
- Ricardo Garcia, Igalia
- Graeme Leese, Broadcom

- Ralph Potter, Samsung
- Stu Smith, AMD
- Shahbaz Youssefi, Google
- Alex Walters, Imagination

Description

This extension allows a single slice of a 3D image to be used as a 2D view in image descriptors, matching both the functionality of glBindImageTexture in OpenGL with the `layer` parameter set to true and 2D view binding provided by the extension EGL_KHR_gl_texture_3D_image. It is primarily intended to support GL emulation.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceImage2DViewOf3DFeaturesEXT

New Enum Constants

- `VK_EXT_IMAGE_2D_VIEW_OF_3D_EXTENSION_NAME`
- `VK_EXT_IMAGE_2D_VIEW_OF_3D_SPEC_VERSION`
- Extending VkImageCreateFlagBits:
  - VK_IMAGE_CREATE_2D_VIEW_COMPATIBLE_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_2D_VIEW_OF_3D_FEATURES_EXT

Version History

- Revision 1, 2022-03-25 (Mike Blumenkrantz)
  - Internal revisions

VK_EXT_image_compression_control

Name String

`VK_EXT_image_compression_control`

Extension Type

Device extension

Registered Extension Number

339

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Jan-Harald Fredriksen [GitHub]janharaldfredriksen-arm

Extension Proposal

VK_EXT_image_compression_control

Other Extension Metadata

Last Modified Date

2022-05-02

IP Status

No known IP claims.

Contributors

- Jan-Harald Fredriksen, Arm
- Graeme Leese, Broadcom
- Andrew Garrard, Imagination
- Lisa Wu, Arm
- Peter Kohaut, Arm

Description

This extension enables fixed-rate image compression and adds the ability to control when this kind of compression can be applied. Many implementations support some form of framebuffer

compression. This is typically transparent to applications as lossless compression schemes are used. With fixed-rate compression, the compression is done at a defined bitrate. Such compression algorithms generally produce results that are visually lossless, but the results are typically not bit-exact when compared to a non-compressed result. The implementation may not be able to use the requested compression rate in all cases. This extension adds a query that can be used to determine the compression scheme and rate that was applied to an image.

New Commands

- vkGetImageSubresourceLayout2EXT

New Structures

- VkImageSubresource2EXT
- VkSubresourceLayout2EXT
- Extending VkImageCreateInfo, VkSwapchainCreateInfoKHR, VkPhysicalDeviceImageFormatInfo2:
  - VkImageCompressionControlEXT
- Extending VkImageFormatProperties2, VkSurfaceFormat2KHR, VkSubresourceLayout2:
  - VkImageCompressionPropertiesEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceImageCompressionControlFeaturesEXT

New Enums

- VkImageCompressionFixedRateFlagBitsEXT
- VkImageCompressionFlagBitsEXT

New Bitmasks

- VkImageCompressionFixedRateFlagsEXT
- VkImageCompressionFlagsEXT

New Enum Constants

- `VK_EXT_IMAGE_COMPRESSION_CONTROL_EXTENSION_NAME`
- `VK_EXT_IMAGE_COMPRESSION_CONTROL_SPEC_VERSION`
- Extending VkResult:
  - VK_ERROR_COMPRESSION_EXHAUSTED_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_IMAGE_COMPRESSION_CONTROL_EXT
  - VK_STRUCTURE_TYPE_IMAGE_COMPRESSION_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_IMAGE_SUBRESOURCE_2_EXT

- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_COMPRESSION_CONTROL_FEATURES_EXT
- VK_STRUCTURE_TYPE_SUBRESOURCE_LAYOUT_2_EXT

Version History

- Revision 1, 2022-05-02 (Jan-Harald Fredriksen)
  - Initial draft

VK_EXT_image_compression_control_swapchain

Name String

VK_EXT_image_compression_control_swapchain

Extension Type

Device extension

Registered Extension Number

438

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_EXT_image_compression_control
and
VK_KHR_swapchain

Contact

- Jan-Harald Fredriksen [GitHub]janharaldfredriksen-arm

Other Extension Metadata

Last Modified Date

2022-05-02

IP Status

No known IP claims.

Contributors

- Jan-Harald Fredriksen, Arm
- Graeme Leese, Broadcom
- Andrew Garrard, Imagination
- Lisa Wu, Arm
- Peter Kohaut, Arm
- Ian Elliott, Google

Description

This extension enables fixed-rate image compression and adds the ability to control when this kind of compression can be applied to swapchain images.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceImageCompressionControlSwapchainFeaturesEXT

New Enum Constants

- VK_EXT_IMAGE_COMPRESSION_CONTROL_SWAPCHAIN_EXTENSION_NAME
- VK_EXT_IMAGE_COMPRESSION_CONTROL_SWAPCHAIN_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_COMPRESSION_CONTROL_SWAPCHAIN_FEATURES_EXT

Version History

- Revision 1, 2022-05-02 (Jan-Harald Fredriksen)
  - Initial draft

VK_EXT_image_drm_format_modifier

Name String

`VK_EXT_image_drm_format_modifier`

Extension Type

Device extension

Registered Extension Number

159

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_bind_memory2
and
VK_KHR_get_physical_device_properties2
and
VK_KHR_sampler_ycbcr_conversion
or
Vulkan Version 1.1
and
VK_KHR_image_format_list
or
Vulkan Version 1.2

API Interactions

- Interacts with VK_VERSION_1_3
- Interacts with VK_KHR_format_feature_flags2

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2021-09-30

IP Status

No known IP claims.

Contributors

- Antoine Labour, Google
- Bas Nieuwenhuizen, Google

- Lina Versace, Google
- James Jones, NVIDIA
- Faith Ekstrand, Intel
- Jőrg Wagner, ARM
- Kristian Høgsberg Kristensen, Google
- Ray Smith, ARM

Description

This extension provides the ability to use *DRM format modifiers* with images, enabling Vulkan to better integrate with the Linux ecosystem of graphics, video, and display APIs.

Its functionality closely overlaps with `EGL_EXT_image_dma_buf_import_modifiers`[2] and `EGL_MESA_image_dma_buf_export`[3]. Unlike the EGL extensions, this extension does not require the use of a specific handle type (such as a dma_buf) for external memory and provides more explicit control of image creation.

Introduction to DRM Format Modifiers

A *DRM format modifier* is a 64-bit, vendor-prefixed, semi-opaque unsigned integer. Most *modifiers* represent a concrete, vendor-specific tiling format for images. Some exceptions are `DRM_FORMAT_MOD_LINEAR` (which is not vendor-specific); `DRM_FORMAT_MOD_NONE` (which is an alias of `DRM_FORMAT_MOD_LINEAR` due to historical accident); and `DRM_FORMAT_MOD_INVALID` (which does not represent a tiling format). The *modifier's* vendor prefix consists of the 8 most significant bits. The canonical list of *modifiers* and vendor prefixes is found in `drm_fourcc.h` in the Linux kernel source. The other dominant source of *modifiers* are vendor kernel trees.

One goal of *modifiers* in the Linux ecosystem is to enumerate for each vendor a reasonably sized set of tiling formats that are appropriate for images shared across processes, APIs, and/or devices, where each participating component may possibly be from different vendors. A non-goal is to enumerate all tiling formats supported by all vendors. Some tiling formats used internally by vendors are inappropriate for sharing; no *modifiers* should be assigned to such tiling formats.

Modifier values typically do not *describe* memory layouts. More precisely, a *modifier*'s lower 56 bits usually have no structure. Instead, modifiers *name* memory layouts; they name a small set of vendor-preferred layouts for image sharing. As a consequence, in each vendor namespace the modifier values are often sequentially allocated starting at 1.

Each *modifier* is usually supported by a single vendor and its name matches the pattern `{VENDOR}_FORMAT_MOD_*` or `DRM_FORMAT_MOD_{VENDOR}_*`. Examples are `I915_FORMAT_MOD_X_TILED` and `DRM_FORMAT_MOD_BROADCOM_VC4_T_TILED`. An exception is `DRM_FORMAT_MOD_LINEAR`, which is supported by most vendors.

Many APIs in Linux use *modifiers* to negotiate and specify the memory layout of shared images. For example, a Wayland compositor and Wayland client may, by relaying *modifiers* over the Wayland protocol `zwp_linux_dmabuf_v1`, negotiate a vendor-specific tiling format for a shared `wl_buffer`. The client may allocate the underlying memory for the `wl_buffer` with GBM, providing the chosen *modifier* to `gbm_bo_create_with_modifiers`. The client may then import the `wl_buffer` into Vulkan for

producing image content, providing the resource's dma_buf to VkImportMemoryFdInfoKHR and its *modifier* to VkImageDrmFormatModifierExplicitCreateInfoEXT. The compositor may then import the `wl_buffer` into OpenGL for sampling, providing the resource's dma_buf and *modifier* to `eglCreateImage`. The compositor may also bypass OpenGL and submit the `wl_buffer` directly to the kernel's display API, providing the dma_buf and *modifier* through `drm_mode_fb_cmd2`.

Format Translation

Modifier-capable APIs often pair *modifiers* with DRM formats, which are defined in `drm_fourcc.h`. However, `VK_EXT_image_drm_format_modifier` uses VkFormat instead of DRM formats. The application must convert between VkFormat and DRM format when it sends or receives a DRM format to or from an external API.

The mapping from VkFormat to DRM format is lossy. Therefore, when receiving a DRM format from an external API, often the application must use information from the external API to accurately map the DRM format to a VkFormat. For example, DRM formats do not distinguish between RGB and sRGB (as of 2018-03-28); external information is required to identify the image's color space.

The mapping between VkFormat and DRM format is also incomplete. For some DRM formats there exist no corresponding Vulkan format, and for some Vulkan formats there exist no corresponding DRM format.

Usage Patterns

Three primary usage patterns are intended for this extension:

- **Negotiation.** The application negotiates with *modifier*-aware, external components to determine sets of image creation parameters supported among all components.

  In the Linux ecosystem, the negotiation usually assumes the image is a 2D, single-sampled, non-mipmapped, non-array image; this extension permits that assumption but does not require it. The result of the negotiation usually resembles a set of tuples such as *(drmFormat, drmFormatModifier)*, where each participating component supports all tuples in the set.

  Many details of this negotiation - such as the protocol used during negotiation, the set of image creation parameters expressible in the protocol, and how the protocol chooses which process and which API will create the image - are outside the scope of this specification.

  In this extension, vkGetPhysicalDeviceFormatProperties2 with VkDrmFormatModifierPropertiesListEXT serves a primary role during the negotiation, and vkGetPhysicalDeviceImageFormatProperties2 with VkPhysicalDeviceImageDrmFormatModifierInfoEXT serves a secondary role.

- **Import.** The application imports an image with a *modifier*.

  In this pattern, the application receives from an external source the image's memory and its creation parameters, which are often the result of the negotiation described above. Some image creation parameters are implicitly defined by the external source; for example, VK_IMAGE_TYPE_2D is often assumed. Some image creation parameters are usually explicit, such as the image's `format`, `drmFormatModifier`, and `extent`; and each plane's `offset` and `rowPitch`.

Before creating the image, the application first verifies that the physical device supports the received creation parameters by querying vkGetPhysicalDeviceFormatProperties2 with VkDrmFormatModifierPropertiesListEXT and vkGetPhysicalDeviceImageFormatProperties2 with VkPhysicalDeviceImageDrmFormatModifierInfoEXT. Then the application creates the image by chaining VkImageDrmFormatModifierExplicitCreateInfoEXT and VkExternalMemoryImageCreateInfo onto VkImageCreateInfo.

- **Export.** The application creates an image and allocates its memory. Then the application exports to *modifier*-aware consumers the image's memory handles; its creation parameters; its *modifier*; and the `offset`, `size`, and `rowPitch` of each *memory plane*.

  In this pattern, the Vulkan device is the authority for the image; it is the allocator of the image's memory and the decider of the image's creation parameters. When choosing the image's creation parameters, the application usually chooses a tuple *(format, drmFormatModifier)* from the result of the negotiation described above. The negotiation's result often contains multiple tuples that share the same format but differ in their *modifier*. In this case, the application should defer the choice of the image's *modifier* to the Vulkan implementation by providing all such *modifiers* to VkImageDrmFormatModifierListCreateInfoEXT::`pDrmFormatModifiers`; and the implementation should choose from `pDrmFormatModifiers` the optimal *modifier* in consideration with the other image parameters.

  The application creates the image by chaining VkImageDrmFormatModifierListCreateInfoEXT and VkExternalMemoryImageCreateInfo onto VkImageCreateInfo. The protocol and APIs by which the application will share the image with external consumers will likely determine the value of VkExternalMemoryImageCreateInfo::`handleTypes`. The implementation chooses for the image an optimal *modifier* from VkImageDrmFormatModifierListCreateInfoEXT ::`pDrmFormatModifiers`. The application then queries the implementation-chosen *modifier* with vkGetImageDrmFormatModifierPropertiesEXT, and queries the memory layout of each plane with vkGetImageSubresourceLayout.

  The application then allocates the image's memory with VkMemoryAllocateInfo, adding chained extending structures for external memory; binds it to the image; and exports the memory, for example, with vkGetMemoryFdKHR.

  Finally, the application sends the image's creation parameters, its *modifier*, its per-plane memory layout, and the exported memory handle to the external consumers. The details of how the application transmits this information to external consumers is outside the scope of this specification.

Prior Art

Extension `EGL_EXT_image_dma_buf_import`[1] introduced the ability to create an `EGLImage` by importing for each plane a dma_buf, offset, and row pitch.

Later, extension `EGL_EXT_image_dma_buf_import_modifiers`[2] introduced the ability to query which combination of formats and *modifiers* the implementation supports and to specify *modifiers* during creation of the `EGLImage`.

Extension `EGL_MESA_image_dma_buf_export`[3] is the inverse of `EGL_EXT_image_dma_buf_import_modifiers`.

The Linux kernel modesetting API (KMS), when configuring the display's framebuffer with `struct drm_mode_fb_cmd2`[4], allows one to specify the framebuffer's *modifier* as well as a per-plane memory handle, offset, and row pitch.

GBM, a graphics buffer manager for Linux, allows creation of a `gbm_bo` (that is, a graphics *buffer object*) by importing data similar to that in `EGL_EXT_image_dma_buf_import_modifiers`[1]; and symmetrically allows exporting the same data from the `gbm_bo`. See the references to *modifier* and *plane* in `gbm.h`[5].

New Commands

- vkGetImageDrmFormatModifierPropertiesEXT

New Structures

- VkDrmFormatModifierPropertiesEXT
- VkImageDrmFormatModifierPropertiesEXT
- Extending VkFormatProperties2:
  - VkDrmFormatModifierPropertiesListEXT
- Extending VkImageCreateInfo:
  - VkImageDrmFormatModifierExplicitCreateInfoEXT
  - VkImageDrmFormatModifierListCreateInfoEXT
- Extending VkPhysicalDeviceImageFormatInfo2:
  - VkPhysicalDeviceImageDrmFormatModifierInfoEXT

If VK_KHR_format_feature_flags2 or Vulkan Version 1.3 is supported:

- VkDrmFormatModifierProperties2EXT
- Extending VkFormatProperties2:
  - VkDrmFormatModifierPropertiesList2EXT

New Enum Constants

- `VK_EXT_IMAGE_DRM_FORMAT_MODIFIER_EXTENSION_NAME`
- `VK_EXT_IMAGE_DRM_FORMAT_MODIFIER_SPEC_VERSION`
- Extending VkImageAspectFlagBits:
  - VK_IMAGE_ASPECT_MEMORY_PLANE_0_BIT_EXT
  - VK_IMAGE_ASPECT_MEMORY_PLANE_1_BIT_EXT
  - VK_IMAGE_ASPECT_MEMORY_PLANE_2_BIT_EXT
  - VK_IMAGE_ASPECT_MEMORY_PLANE_3_BIT_EXT
- Extending VkImageTiling:
  - VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT

- Extending VkResult:
  - VK_ERROR_INVALID_DRM_FORMAT_MODIFIER_PLANE_LAYOUT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DRM_FORMAT_MODIFIER_PROPERTIES_LIST_EXT
  - VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_EXPLICIT_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_LIST_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_DRM_FORMAT_MODIFIER_INFO_EXT

If VK_KHR_format_feature_flags2 or Vulkan Version 1.3 is supported:

- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DRM_FORMAT_MODIFIER_PROPERTIES_LIST_2_EXT

Issues

1) Should this extension define a single DRM format modifier per `VkImage`? Or define one per plane?

+

RESOLVED: There exists a single DRM format modifier per `VkImage`.

DISCUSSION: Prior art, such as `EGL_EXT_image_dma_buf_import_modifiers`[2], `struct drm_mode_fb_cmd2`[4], and `struct gbm_import_fd_modifier_data`[5], allows defining one *modifier* per plane. However, developers of the GBM and kernel APIs concede it was a mistake. Beginning in Linux 4.10, the kernel requires that the application provide the same DRM format *modifier* for each plane. (See Linux commit bae781b259269590109e8a4a8227331362b88212). And GBM provides an entry point, `gbm_bo_get_modifier`, for querying the *modifier* of the image but does not provide one to query the modifier of individual planes.

2) When creating an image with VkImageDrmFormatModifierExplicitCreateInfoEXT, which is typically used when *importing* an image, should the application explicitly provide the size of each plane?

+

RESOLVED: No. The application **must** not provide the size. To enforce this, the API requires that VkImageDrmFormatModifierExplicitCreateInfoEXT::`pPlaneLayouts->size` **must** be 0.

DISCUSSION: Prior art, such as `EGL_EXT_image_dma_buf_import_modifiers`[2], `struct drm_mode_fb_cmd2`[4], and `struct gbm_import_fd_modifier_data`[5], omits from the API the size of each plane. Instead, the APIs infer each plane's size from the import parameters, which include the image's pixel format and a dma_buf, offset, and row pitch for each plane.

However, Vulkan differs from EGL and GBM with regards to image creation in the following ways:

Differences in Image Creation

- **Undedicated allocation by default.** When importing or exporting a set of dma_bufs as an `EGLImage` or `gbm_bo`, common practice mandates that each dma_buf's memory be dedicated (in the sense of `VK_KHR_dedicated_allocation`) to the image (though not necessarily dedicated to a single plane). In particular, neither the GBM documentation nor the EGL extension specifications explicitly state this requirement, but in light of common practice this is likely due to under-specification rather than intentional omission. In contrast, `VK_EXT_image_drm_format_modifier` permits, but does not require, the implementation to require dedicated allocations for images created with VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT.
- **Separation of image creation and memory allocation.** When importing a set of dma_bufs as an `EGLImage` or `gbm_bo`, EGL and GBM create the image resource and bind it to memory (the dma_bufs) simultaneously. This allows EGL and GBM to query each dma_buf's size during image creation. In Vulkan, image creation and memory allocation are independent unless a dedicated allocation is used (as in `VK_KHR_dedicated_allocation`). Therefore, without requiring dedicated allocation, Vulkan cannot query the size of each dma_buf (or other external handle) when calculating the image's memory layout. Even if dedication allocation were required, Vulkan cannot calculate the image's memory layout until after the image is bound to its dma_ufs.

The above differences complicate the potential inference of plane size in Vulkan. Consider the following problematic cases:

Problematic Plane Size Calculations

- **Padding.** Some plane of the image may require implementation-dependent padding.
- **Metadata.** For some *modifiers*, the image may have a metadata plane which requires a non-trivial calculation to determine its size.
- **Mipmapped, array, and 3D images.** The implementation may support VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT for images whose `mipLevels`, `arrayLayers`, or `depth` is greater than 1. For such images with certain *modifiers*, the calculation of each plane's size may be non-trivial.

However, an application-provided plane size solves none of the above problems.

For simplicity, consider an external image with a single memory plane. The implementation is obviously capable calculating the image's size when its tiling is VK_IMAGE_TILING_OPTIMAL. Likewise, any reasonable implementation is capable of calculating the image's size when its tiling uses a supported *modifier*.

Suppose that the external image's size is smaller than the implementation-calculated size. If the application provided the external image's size to vkCreateImage, the implementation would observe the mismatched size and recognize its inability to comprehend the external image's layout (unless the implementation used the application-provided size to select a refinement of the tiling layout indicated by the *modifier*, which is strongly discouraged). The implementation would observe the conflict, and reject image creation with VK_ERROR_INVALID_DRM_FORMAT_MODIFIER_PLANE_LAYOUT_EXT. On the other hand, if the application did not provide the external image's size to vkCreateImage, then the application would observe after calling vkGetImageMemoryRequirements that the external image's size is less than

the size required by the implementation. The application would observe the conflict and refuse to bind the VkImage to the external memory. In both cases, the result is explicit failure.

Suppose that the external image's size is larger than the implementation-calculated size. If the application provided the external image's size to vkCreateImage, for reasons similar to above the implementation would observe the mismatched size and recognize its inability to comprehend the image data residing in the extra size. The implementation, however, must assume that image data resides in the entire size provided by the application. The implementation would observe the conflict and reject image creation with VK_ERROR_INVALID_DRM_FORMAT_MODIFIER_PLANE_LAYOUT_EXT. On the other hand, if the application did not provide the external image's size to vkCreateImage, then the application would observe after calling vkGetImageMemoryRequirements that the external image's size is larger than the implementation-usable size. The application would observe the conflict and refuse to bind the VkImage to the external memory. In both cases, the result is explicit failure.

Therefore, an application-provided size provides no benefit, and this extension should not require it. This decision renders VkSubresourceLayout::size an unused field during image creation, and thus introduces a risk that implementations may require applications to submit sideband creation parameters in the unused field. To prevent implementations from relying on sideband data, this extension *requires* the application to set size to 0.

References

1. EGL_EXT_image_dma_buf_import
2. EGL_EXT_image_dma_buf_import_modifiers
3. EGL_MESA_image_dma_buf_export
4. struct drm_mode_fb_cmd2
5. gbm.h

Version History

- Revision 1, 2018-08-29 (Lina Versace)
  - First stable revision
- Revision 2, 2021-09-30 (Jon Leech)
  - Add interaction with VK_KHR_format_feature_flags2 to vk.xml

VK_EXT_image_sliced_view_of_3d

Name String

VK_EXT_image_sliced_view_of_3d

Extension Type

Device extension

Registered Extension Number

419

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_maintenance1
and
VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Special Use

- D3D support

Contact

- Mike Blumenkrantz [GitHub]zmike

Extension Proposal

VK_EXT_image_sliced_view_of_3d

Other Extension Metadata

Last Modified Date

2023-01-24

IP Status

No known IP claims.

Contributors

- Mike Blumenkrantz, Valve
- Hans-Kristian Arntzen, Valve
- Ricardo Garcia, Igalia
- Shahbaz Youssefi, Google
- Piers Daniell, NVIDIA

Description

This extension allows creating 3D views of 3D images such that the views contain a subset of the slices in the image, using a Z offset and range, for the purpose of using the views as storage image descriptors. This matches functionality in D3D12 and is primarily intended to support D3D12 emulation.

New Structures

- Extending VkImageViewCreateInfo:
  - VkImageViewSlicedCreateInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceImageSlicedViewOf3DFeaturesEXT

New Enum Constants

- `VK_EXT_IMAGE_SLICED_VIEW_OF_3D_EXTENSION_NAME`
- `VK_EXT_IMAGE_SLICED_VIEW_OF_3D_SPEC_VERSION`
- VK_REMAINING_3D_SLICES_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_IMAGE_VIEW_SLICED_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_SLICED_VIEW_OF_3D_FEATURES_EXT

Version History

- Revision 1, 2022-10-21 (Mike Blumenkrantz)
  - Initial revision

VK_EXT_image_tiling_control

Name String

`VK_EXT_image_tiling_control`

Extension Type

Device extension

Registered Extension Number

688

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Noah Fredriks [GitHub]noahf

Extension Proposal

VK_EXT_image_tiling_control

Other Extension Metadata

Last Modified Date

2026-06-19

Interactions and External Dependencies

- Interacts with VK_KHR_maintenance4
- Interacts with VK_KHR_maintenance5
- Interacts with VK_EXT_image_drm_format_modifier

IP Status

No known IP claims.

Contributors

- Noah Fredriks, AMD
- Stu Smith, AMD
- Tobias Hector, AMD
- Hans-Kristian Arntzen, Valve
- Ralph Potter, Samsung

- James Jones, NVIDIA
- Lionel Landwerlin, Intel
- Daniel Story, Nintendo

Description

This extension provides per-image selection of optimal tiling arrangements by extending VkImageCreateInfo with VkImageTilingControlCreateInfoEXT when multiple arrangements are supported by the implementation.

VK_IMAGE_TILING_CONTROL_DEFAULT_EXT **can** be set to use the tiling arrangement that is the same selection as with the VkImageTilingControlCreateInfoEXT structure omitted.

VK_IMAGE_TILING_CONTROL_MIN_SIZE_EXT **can** be set to minimize memory size requirements with reduced consideration for memory access performance.

VK_IMAGE_TILING_CONTROL_MAX_PERFORMANCE_EXT **can** be set to maximize memory access performance with reduced consideration for increases in memory size requirements.

New Structures

- Extending VkImageCreateInfo:
  - VkImageTilingControlCreateInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceImageTilingControlFeaturesEXT

New Enums

- VkImageTilingControlEXT

New Enum Constants

- `VK_EXT_IMAGE_TILING_CONTROL_EXTENSION_NAME`
- `VK_EXT_IMAGE_TILING_CONTROL_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_IMAGE_TILING_CONTROL_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_TILING_CONTROL_FEATURES_EXT

Version History

- Revision 1, 2026-06-19 (Noah Fredriks)
  - Initial revision

VK_EXT_image_view_min_lod

Name String

`VK_EXT_image_view_min_lod`

Extension Type

Device extension

Registered Extension Number

392

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Joshua Ashton [GitHub]Joshua-Ashton

Other Extension Metadata

Last Modified Date

2021-11-09

IP Status

No known IP claims.

Contributors

- Joshua Ashton, Valve
- Hans-Kristian Arntzen, Valve
- Samuel Iglesias Gonsalvez, Igalia
- Tobias Hector, AMD
- Faith Ekstrand, Intel
- Tom Olson, ARM

Description

This extension allows applications to clamp the minimum LOD value during Image Level(s) Selection, Texel Gathering and Integer Texel Coordinate Operations with a given VkImageView by VkImageViewMinLodCreateInfoEXT::`minLod`.

This extension may be useful to restrict a VkImageView to only mips which have been uploaded, and the use of fractional `minLod` can be useful for smoothly introducing new mip levels when using linear mipmap filtering.

New Structures

- Extending VkImageViewCreateInfo:
  - VkImageViewMinLodCreateInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceImageViewMinLodFeaturesEXT

New Enum Constants

- `VK_EXT_IMAGE_VIEW_MIN_LOD_EXTENSION_NAME`
- `VK_EXT_IMAGE_VIEW_MIN_LOD_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_IMAGE_VIEW_MIN_LOD_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_VIEW_MIN_LOD_FEATURES_EXT

Version History

- Revision 1, 2021-07-06 (Joshua Ashton)
  - Initial version

VK_EXT_layer_settings

Name String

`VK_EXT_layer_settings`

Extension Type

Instance extension

Registered Extension Number

497

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- Christophe Riccio [GitHub]christophe

Extension Proposal

VK_EXT_layer_settings

Other Extension Metadata

Last Modified Date

2023-09-23

IP Status

No known IP claims.

Contributors

- Christophe Riccio, LunarG
- Mark Lobodzinski, LunarG
- Charles Giessen, LunarG
- Spencer Fricke, LunarG
- Juan Ramos, LunarG
- Daniel Rakos, RasterGrid
- Shahbaz Youssefi, Google
- Lina Versace, Google
- Bill Hollings, The Brenwill Workshop
- Jon Leech, Khronos

- Tom Olson, Arm

Description

This extension provides a mechanism for configuring programmatically through the Vulkan API the behavior of layers.

This extension provides the VkLayerSettingsCreateInfoEXT structure that can be included in the pNext chain of the VkInstanceCreateInfo structure passed as the pCreateInfo parameter of vkCreateInstance.

The structure contains an array of VkLayerSettingEXT structure values that configure specific features of layers.

NOTE	The VK_EXT_layer_settings extension subsumes all the functionality provided in the VK_EXT_validation_flags extension and the VK_EXT_validation_features extension.

New Structures

- VkLayerSettingEXT
- Extending VkInstanceCreateInfo:
  - VkLayerSettingsCreateInfoEXT

New Enums

- VkLayerSettingTypeEXT

New Enum Constants

- VK_EXT_LAYER_SETTINGS_EXTENSION_NAME
- VK_EXT_LAYER_SETTINGS_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_LAYER_SETTINGS_CREATE_INFO_EXT

Example

One example usage of VK_EXT_layer_settings is as implemented by the Vulkan Profiles layer.

It allows the profiles layer tests used by the profiles layer C.I. to programmatically configure the layer for each test without affecting the C.I. environment, allowing to run multiple tests concurrently.

```
const char* profile_file_data = JSON_TEST_FILES_PATH "VP_KHR_roadmap_2022.json";
const char* profile_name_data = "VP_KHR_roadmap_2022";
VkBool32 emulate_portability_data = VK_TRUE;
const char* simulate_capabilities[] = {
    "SIMULATE_API_VERSION_BIT",
    "SIMULATE_FEATURES_BIT",
```

```
    "SIMULATE_PROPERTIES_BIT",
    "SIMULATE_EXTENSIONS_BIT",
    "SIMULATE_FORMATS_BIT",
    "SIMULATE_QUEUE_FAMILY_PROPERTIES_BIT"
};
const char* debug_reports[] = {
    "DEBUG_REPORT_ERROR_BIT",
    "DEBUG_REPORT_WARNING_BIT",
    "DEBUG_REPORT_NOTIFICATION_BIT",
    "DEBUG_REPORT_DEBUG_BIT"
};

const VkLayerSettingEXT settings[] = {
     {kLayerName, kLayerSettingsProfileFile, VK_LAYER_SETTING_TYPE_STRING_EXT, 1,
&profile_file_data},
     {kLayerName, kLayerSettingsProfileName, VK_LAYER_SETTING_TYPE_STRING_EXT, 1,
&profile_name_data},
     {kLayerName, kLayerSettingsEmulatePortability, VK_LAYER_SETTING_TYPE_BOOL32_EXT,
1, &emulate_portability_data},
     {kLayerName, kLayerSettingsSimulateCapabilities,
VK_LAYER_SETTING_TYPE_STRING_EXT,
        static_cast<uint32_t>(std::size(simulate_capabilities)),
simulate_capabilities},
     {kLayerName, kLayerSettingsDebugReports, VK_LAYER_SETTING_TYPE_STRING_EXT,
        static_cast<uint32_t>(std::size(debug_reports)), debug_reports}
};

const VkLayerSettingsCreateInfoEXT layer_settings_create_info{
    VK_STRUCTURE_TYPE_LAYER_SETTINGS_CREATE_INFO_EXT, nullptr,
    static_cast<uint32_t>(std::size(settings)), settings};

VkInstanceCreateInfo inst_create_info = {};
...
inst_create_info.pNext = &layer_settings_create_info;
vkCreateInstance(&inst_create_info, nullptr, &_instances);
```

Issues

- How should application developers figure out the list of available settings?

This extension does not provide a reflection API for layer settings. Layer settings are described in each layer JSON manifest and the documentation of each layer which implements this extension.

Version History

- Revision 1, 2020-06-17 (Mark Lobodzinski)
  - Initial revision for Validation layer internal usages
- Revision 2, 2023-09-26 (Christophe Riccio)
  - Refactor APIs for any layer usages and public release

VK_EXT_legacy_dithering

Name String

`VK_EXT_legacy_dithering`

Extension Type

Device extension

Registered Extension Number

466

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_VERSION_1_3
- Interacts with VK_VERSION_1_4
- Interacts with VK_KHR_dynamic_rendering
- Interacts with VK_KHR_extended_flags
- Interacts with VK_KHR_maintenance5

Special Use

- OpenGL / ES support

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Extension Proposal

VK_EXT_legacy_dithering

Other Extension Metadata

Last Modified Date

2024-02-22

Contributors

- Shahbaz Youssefi, Google
- Graeme Leese, Broadcom
- Jan-Harald Fredriksen, Arm

Description

This extension exposes a hardware feature used by some vendors to implement OpenGL's dithering. The purpose of this extension is to support layering OpenGL over Vulkan, by allowing the layer to take advantage of the same hardware feature and provide equivalent dithering to OpenGL applications.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceLegacyDitheringFeaturesEXT

New Enum Constants

- `VK_EXT_LEGACY_DITHERING_EXTENSION_NAME`
- `VK_EXT_LEGACY_DITHERING_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LEGACY_DITHERING_FEATURES_EXT
- Extending VkSubpassDescriptionFlagBits:
  - VK_SUBPASS_DESCRIPTION_ENABLE_LEGACY_DITHERING_BIT_EXT

If VK_KHR_dynamic_rendering or Vulkan Version 1.3 and Vulkan Version 1.4 or VK_KHR_extended_flags or VK_KHR_maintenance5 is supported:

- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_ENABLE_LEGACY_DITHERING_BIT_EXT
- Extending VkRenderingFlagBits:
  - VK_RENDERING_ENABLE_LEGACY_DITHERING_BIT_EXT

Version History

- Revision 1, 2022-03-31 (Shahbaz Youssefi)
  - Internal revisions
- Revision 2, 2024-02-22 (Shahbaz Youssefi)
  - Added pipeline create flag to support dynamic rendering

Issues

1) In OpenGL, the dither state can change dynamically. Should this extension add a pipeline state for dither?

RESOLVED: No. Changing dither state is rarely, if ever, done during rendering. Every surveyed Android application either entirely disables dither, explicitly enables it, or uses the default state (which is enabled). Additionally, on some hardware dither can only be specified in a render pass granularity, so a change in dither state would necessarily need to cause a render pass break. This extension considers dynamic changes in OpenGL dither state a theoretical situation, and expects

the layer to break the render pass in such a situation without any practical downsides.

VK_EXT_legacy_vertex_attributes

Name String

`VK_EXT_legacy_vertex_attributes`

Extension Type

Device extension

Registered Extension Number

496

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_EXT_vertex_input_dynamic_state

Special Use

- OpenGL / ES support

Contact

- Mike Blumenkrantz [GitHub]zmike

Extension Proposal

VK_EXT_legacy_vertex_attributes

Other Extension Metadata

Last Modified Date

2024-02-23

IP Status

No known IP claims.

Contributors

- Mike Blumenkrantz, Valve
- Piers Daniell, NVIDIA
- Spencer Fricke, LunarG
- Alyssa Rosenzweig, Valve

Description

This extension adds support for legacy features of (non-64-bit) vertex attributes as found in OpenGL:

- Vertex attributes loaded from arbitrary buffer alignments
- Vertex attributes using arbitrary strides
- Vertex attributes where the component data type of the binding does not match the component numeric type of the shader input

These features are only usable with dynamic vertex input. Unaligned loads of vertex attributes may incur performance penalties, indicated with a property.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT

New Enum Constants

- `VK_EXT_LEGACY_VERTEX_ATTRIBUTES_EXTENSION_NAME`
- `VK_EXT_LEGACY_VERTEX_ATTRIBUTES_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LEGACY_VERTEX_ATTRIBUTES_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LEGACY_VERTEX_ATTRIBUTES_PROPERTIES_EXT

Issues

1.) Should implementations convert float/integer values?

RESOLVED: No. When fetching an integer data type from float values or float data types from integer values, the resulting shader values are implementation-dependent.

Version History

- Revision 1, 2024-02-16 (Mike Blumenkrantz)
  - Initial revision

VK_EXT_map_memory_placed

Name String

`VK_EXT_map_memory_placed`

Extension Type

Device extension

Registered Extension Number

273

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_map_memory2
or
Vulkan Version 1.4

Contact

- Faith Ekstrand [GitHub]gfxstrand

Extension Proposal

VK_EXT_map_memory_placed

Other Extension Metadata

Last Modified Date

2023-03-21

IP Status

No known IP claims.

Interactions and External Dependencies

- Depends on apitext:VK_KHR_map_memory2
- Interacts with apitext:VK_EXT_external_memory_host

Contributors

- Faith Ekstrand, Collabora
- Tobias Hector, AMD
- James Jones, NVIDIA
- Georg Lehmann, Valve
- Derek Lesho, Codeweavers

Description

This extension allows an application to request that vkMapMemory2KHR attempt to place the memory map at a particular virtual address.

New Structures

- Extending VkMemoryMapInfo:
  - VkMemoryMapPlacedInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMapMemoryPlacedFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceMapMemoryPlacedPropertiesEXT

New Enum Constants

- `VK_EXT_MAP_MEMORY_PLACED_EXTENSION_NAME`
- `VK_EXT_MAP_MEMORY_PLACED_SPEC_VERSION`
- Extending VkMemoryMapFlagBits:
  - VK_MEMORY_MAP_PLACED_BIT_EXT
- Extending VkMemoryUnmapFlagBits:
  - VK_MEMORY_UNMAP_RESERVE_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_MEMORY_MAP_PLACED_INFO_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAP_MEMORY_PLACED_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAP_MEMORY_PLACED_PROPERTIES_EXT

Version History

- Revision 1, 2024-01-14 (Faith Ekstrand)
  - Internal revisions

VK_EXT_memory_budget

Name String

`VK_EXT_memory_budget`

Extension Type

Device extension

Registered Extension Number

238

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Jeff Bolz [GitHub]jeffbolznv

Other Extension Metadata

Last Modified Date

2018-10-08

Contributors

- Jeff Bolz, NVIDIA
- Jeff Juliano, NVIDIA

Description

While running a Vulkan application, other processes on the machine might also be attempting to use the same device memory, which can pose problems. This extension adds support for querying the amount of memory used and the total memory budget for a memory heap. The values returned by this query are implementation-dependent and can depend on a variety of factors including operating system and system load.

The VkPhysicalDeviceMemoryBudgetPropertiesEXT::`heapBudget` values can be used as a guideline for how much total memory from each heap the **current process** can use at any given time, before allocations may start failing or causing performance degradation. The values may change based on other activity in the system that is outside the scope and control of the Vulkan implementation.

The VkPhysicalDeviceMemoryBudgetPropertiesEXT::`heapUsage` will display the **current process** estimated heap usage.

With this information, the idea is for an application at some interval (once per frame, per few seconds, etc) to query `heapBudget` and `heapUsage`. From here the application can notice if it is over budget and decide how it wants to handle the memory situation (free it, move to host memory, changing mipmap levels, etc). This extension is designed to be used in concert with `VK_EXT_memory_priority` to help with this part of memory management.

New Structures

- Extending VkPhysicalDeviceMemoryProperties2:
  - VkPhysicalDeviceMemoryBudgetPropertiesEXT

New Enum Constants

- `VK_EXT_MEMORY_BUDGET_EXTENSION_NAME`
- `VK_EXT_MEMORY_BUDGET_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_BUDGET_PROPERTIES_EXT

Version History

- Revision 1, 2018-10-08 (Jeff Bolz)
  - Initial revision

VK_EXT_memory_decompression

Name String

`VK_EXT_memory_decompression`

Extension Type

Device extension

Registered Extension Number

551

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
and
VK_KHR_buffer_device_address

Contact

- Vikram Kushwaha [GitHub]vkushwaha-nv

Extension Proposal

VK_EXT_memory_decompression

Other Extension Metadata

Last Modified Date

2025-01-23

Contributors

- Vikram Kushwaha, NVIDIA
- Daniel Koch, NVIDIA
- Jeff Bolz, NVIDIA
- Christoph Kubisch, NVIDIA
- Piers Daniell, NVIDIA
- Spencer Fricke, LunarG

Description

This extension adds support for performing memory to memory decompression.

New Commands

- vkCmdDecompressMemoryEXT
- vkCmdDecompressMemoryIndirectCountEXT

New Structures

- VkDecompressMemoryInfoEXT
- VkDecompressMemoryRegionEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMemoryDecompressionFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceMemoryDecompressionPropertiesEXT

New Enums

- VkMemoryDecompressionMethodFlagBitsEXT

New Bitmasks

- VkMemoryDecompressionMethodFlagsEXT

New Enum Constants

- `VK_EXT_MEMORY_DECOMPRESSION_EXTENSION_NAME`
- `VK_EXT_MEMORY_DECOMPRESSION_SPEC_VERSION`
- Extending VkAccessFlagBits2:
  - VK_ACCESS_2_MEMORY_DECOMPRESSION_READ_BIT_EXT
  - VK_ACCESS_2_MEMORY_DECOMPRESSION_WRITE_BIT_EXT
- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- Extending VkPipelineStageFlagBits2:
  - VK_PIPELINE_STAGE_2_MEMORY_DECOMPRESSION_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DECOMPRESS_MEMORY_INFO_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_DECOMPRESSION_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_DECOMPRESSION_PROPERTIES_EXT

Issues

1) How does an app know the minimum size that `decompressedSize` should be set to?

RESOLVED: When decompressing, data is typically processed in chunks. For example, with

GDeflate 1.0, data is streamed in 64 KB blocks, but the final block may be smaller. The exact size of this last block depends on the compression method and original data size and so it must be stored in the compressed bitstream so that the decompressor can set `decompressedSize` correctly. It is still ok for the last block to take up all 64 KB, but setting it too low will cause issues and is undefined behavior. It is a known limitation that the validation layers will not be able to detect the minimum size of `decompressedSize` unless it decides to implement each decompression method specification.

Version History

- Revision 1, 2025-01-23 (Daniel Koch)
  - Initial draft

VK_EXT_memory_priority

Name String

`VK_EXT_memory_priority`

Extension Type

Device extension

Registered Extension Number

239

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Jeff Bolz [GitHub]jeffbolznv

Other Extension Metadata

Last Modified Date

2018-10-08

Contributors

- Jeff Bolz, NVIDIA
- Jeff Juliano, NVIDIA

Description

This extension adds a `priority` value specified at memory allocation time. On some systems with both device-local and non-device-local memory heaps, the implementation may transparently move memory from one heap to another when a heap becomes full (for example, when the total memory used across all processes exceeds the size of the heap). In such a case, this priority value may be used to determine which allocations are more likely to remain in device-local memory.

New Structures

- Extending VkMemoryAllocateInfo:
  - VkMemoryPriorityAllocateInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:

    - VkPhysicalDeviceMemoryPriorityFeaturesEXT

New Enum Constants

- VK_EXT_MEMORY_PRIORITY_EXTENSION_NAME
- VK_EXT_MEMORY_PRIORITY_SPEC_VERSION
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_MEMORY_PRIORITY_ALLOCATE_INFO_EXT
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PRIORITY_FEATURES_EXT

Version History

- Revision 1, 2018-10-08 (Jeff Bolz)
    - Initial revision

VK_EXT_mesh_shader

Name String

`VK_EXT_mesh_shader`

Extension Type

Device extension

Registered Extension Number

329

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_spirv_1_4
or
Vulkan Version 1.2

API Interactions

- Interacts with VK_VERSION_1_2
- Interacts with VK_AMD_draw_indirect_count
- Interacts with VK_EXT_device_generated_commands
- Interacts with VK_KHR_draw_indirect_count
- Interacts with VK_KHR_fragment_shading_rate
- Interacts with VK_NV_device_generated_commands
- Interacts with VkPhysicalDeviceMeshShaderFeaturesEXT::primitiveFragmentShadingRateMeshShader

SPIR-V Dependencies

- SPV_EXT_mesh_shader

Contact

- Christoph Kubisch [GitHub]pixeljetstream

Extension Proposal

VK_EXT_mesh_shader

Other Extension Metadata

Last Modified Date

2022-01-20

Interactions and External Dependencies

- This extension provides API support for `GLSL_EXT_mesh_shader`
- Interacts with Vulkan 1.1
- Interacts with `VK_KHR_multiview`
- Interacts with `VK_KHR_fragment_shading_rate`

Contributors

- Christoph Kubisch, NVIDIA
- Pat Brown, NVIDIA
- Jeff Bolz, NVIDIA
- Daniel Koch, NVIDIA
- Piers Daniell, NVIDIA
- Pierre Boudier, NVIDIA
- Patrick Mours, NVIDIA
- David Zhao Akeley, NVIDIA
- Kedarnath Thangudu, NVIDIA
- Timur Kristóf, Valve
- Hans-Kristian Arntzen, Valve
- Philip Rebohle, Valve
- Mike Blumenkrantz, Valve
- Slawomir Grajewski, Intel
- Michal Pietrasiuk, Intel
- Mariusz Merecki, Intel
- Tom Olson, ARM
- Jan-Harald Fredriksen, ARM
- Sandeep Kakarlapudi, ARM
- Ruihao Zhang, QUALCOMM
- Ricardo Garcia, Igalia, S.L.
- Tobias Hector, AMD
- Stu Smith, AMD

Description

This extension provides a new mechanism allowing applications to generate collections of geometric primitives via programmable mesh shading. It is an alternative to the existing programmable primitive shading pipeline, which relied on generating input primitives by a fixed function assembler as well as fixed function vertex fetch.

This extension also adds support for the following SPIR-V extension in Vulkan:

- `SPV_EXT_mesh_shader`

New Commands

- vkCmdDrawMeshTasksEXT
- vkCmdDrawMeshTasksIndirectEXT

If Vulkan Version 1.2 or VK_KHR_draw_indirect_count or `VK_AMD_draw_indirect_count` is supported:

- vkCmdDrawMeshTasksIndirectCountEXT

New Structures

- VkDrawMeshTasksIndirectCommandEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMeshShaderFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceMeshShaderPropertiesEXT

New Enum Constants

- `VK_EXT_MESH_SHADER_EXTENSION_NAME`
- `VK_EXT_MESH_SHADER_SPEC_VERSION`
- Extending VkPipelineStageFlagBits:
  - VK_PIPELINE_STAGE_MESH_SHADER_BIT_EXT
  - VK_PIPELINE_STAGE_TASK_SHADER_BIT_EXT
- Extending VkQueryPipelineStatisticFlagBits:
  - VK_QUERY_PIPELINE_STATISTIC_MESH_SHADER_INVOCATIONS_BIT_EXT
  - VK_QUERY_PIPELINE_STATISTIC_TASK_SHADER_INVOCATIONS_BIT_EXT
- Extending VkQueryType:
  - VK_QUERY_TYPE_MESH_PRIMITIVES_GENERATED_EXT
- Extending VkShaderStageFlagBits:
  - VK_SHADER_STAGE_MESH_BIT_EXT
  - VK_SHADER_STAGE_TASK_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MESH_SHADER_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MESH_SHADER_PROPERTIES_EXT

If VK_EXT_device_generated_commands is supported:

- Extending VkIndirectCommandsTokenTypeEXT:
  - VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_COUNT_EXT

    - VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_EXT

New or Modified Built-In Variables

- CullPrimitiveEXT
- PrimitivePointIndicesEXT
- PrimitiveLineIndicesEXT
- PrimitiveTriangleIndicesEXT
- (modified)`Position`
- (modified)`PointSize`
- (modified)`ClipDistance`
- (modified)`CullDistance`
- (modified)`PrimitiveId`
- (modified)`Layer`
- (modified)`ViewportIndex`
- (modified)`NumWorkgroups`
- (modified)`WorkgroupSize`
- (modified)`WorkgroupId`
- (modified)`LocalInvocationId`
- (modified)`GlobalInvocationId`
- (modified)`LocalInvocationIndex`
- (modified)`NumSubgroups`
- (modified)`SubgroupId`
- (modified)`DrawIndex`
- (modified)`PrimitiveShadingRateKHR`
- (modified)`ViewIndex`

New SPIR-V Capability

- `MeshShadingEXT`

Version History

- Revision 1, 2022-03-08 (Christoph Kubisch, Daniel Koch, Patrick Mours)
    - Initial revision

VK_EXT_metal_objects

Name String

VK_EXT_metal_objects

Extension Type

Device extension

Registered Extension Number

312

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- Bill Hollings [GitHub]billhollings

Extension Proposal

VK_EXT_metal_objects

Other Extension Metadata

Last Modified Date

2024-04-04

IP Status

No known IP claims.

Contributors

- Bill Hollings, The Brenwill Workshop Ltd.
- Dzmitry Malyshau, Mozilla Corp.

Description

In a Vulkan implementation that is layered on top of Metal on Apple device platforms, this extension provides the ability to import and export the underlying Metal objects associated with specific Vulkan objects.

As detailed in the extension proposal document, this extension adds one new Vulkan command, vkExportMetalObjectsEXT, to export underlying Metal objects from Vulkan objects, and supports importing the appropriate existing Metal objects when creating Vulkan objects of types VkDeviceMemory, VkImage, VkSemaphore, and VkEvent,

The intent is that this extension will be advertised and supported only on implementations that are layered on top of Metal on Apple device platforms.

New Base Types

- IOSurfaceRef
- MTLBuffer_id
- MTLCommandQueue_id
- MTLDevice_id
- MTLSharedEvent_id
- MTLTexture_id

New Commands

- vkExportMetalObjectsEXT

New Structures

- VkExportMetalObjectsInfoEXT
- Extending VkExportMetalObjectsInfoEXT:
  - VkExportMetalBufferInfoEXT
  - VkExportMetalCommandQueueInfoEXT
  - VkExportMetalDeviceInfoEXT
  - VkExportMetalIOSurfaceInfoEXT
  - VkExportMetalSharedEventInfoEXT
  - VkExportMetalTextureInfoEXT
- Extending VkImageCreateInfo:
  - VkImportMetalIOSurfaceInfoEXT
  - VkImportMetalTextureInfoEXT
- Extending VkInstanceCreateInfo, VkMemoryAllocateInfo, VkImageCreateInfo, VkImageViewCreateInfo, VkBufferViewCreateInfo, VkSemaphoreCreateInfo, VkEventCreateInfo:
  - VkExportMetalObjectCreateInfoEXT
- Extending VkMemoryAllocateInfo:
  - VkImportMetalBufferInfoEXT
- Extending VkSemaphoreCreateInfo, VkEventCreateInfo:
  - VkImportMetalSharedEventInfoEXT

New Enums

- VkExportMetalObjectTypeFlagBitsEXT

New Bitmasks

- VkExportMetalObjectTypeFlagsEXT

New Enum Constants

- `VK_EXT_METAL_OBJECTS_EXTENSION_NAME`
- `VK_EXT_METAL_OBJECTS_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_EXPORT_METAL_BUFFER_INFO_EXT
  - VK_STRUCTURE_TYPE_EXPORT_METAL_COMMAND_QUEUE_INFO_EXT
  - VK_STRUCTURE_TYPE_EXPORT_METAL_DEVICE_INFO_EXT
  - VK_STRUCTURE_TYPE_EXPORT_METAL_IO_SURFACE_INFO_EXT
  - VK_STRUCTURE_TYPE_EXPORT_METAL_OBJECTS_INFO_EXT
  - VK_STRUCTURE_TYPE_EXPORT_METAL_OBJECT_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_EXPORT_METAL_SHARED_EVENT_INFO_EXT
  - VK_STRUCTURE_TYPE_EXPORT_METAL_TEXTURE_INFO_EXT
  - VK_STRUCTURE_TYPE_IMPORT_METAL_BUFFER_INFO_EXT
  - VK_STRUCTURE_TYPE_IMPORT_METAL_IO_SURFACE_INFO_EXT
  - VK_STRUCTURE_TYPE_IMPORT_METAL_SHARED_EVENT_INFO_EXT
  - VK_STRUCTURE_TYPE_IMPORT_METAL_TEXTURE_INFO_EXT

Issues

None.

Version History

- Revision 1, 2022-05-28 (Bill Hollings)
  - Initial draft.
  - Incorporated feedback from review by the Vulkan Working Group. Renamed many structures, moved import/export of MTLBuffer to VkDeviceMemory, added export of MTLSharedEvent, added import of MTLSharedEvent for VkSemaphore and VkEvent, and changed used bit mask fields to individual bit fields to simplify Valid Usage rules.
- Revision 2, 2024-04-04 (Bill Hollings)
  - Add an `__unsafe_unretained` ownership qualifier to all Metal object declarations, to support Automatic Reference Counting (ARC) on Apple devices.

VK_EXT_metal_surface

Name String

VK_EXT_metal_surface

Extension Type

Instance extension

Registered Extension Number

218

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_surface

Contact

- Dzmitry Malyshau [GitHub]kvark

Other Extension Metadata

Last Modified Date

2018-10-01

IP Status

No known IP claims.

Contributors

- Dzmitry Malyshau, Mozilla Corp.

Description

The VK_EXT_metal_surface extension is an instance extension. It provides a mechanism to create a VkSurfaceKHR object (defined by the VK_KHR_surface extension) from CAMetalLayer, which is the native rendering surface of Apple's Metal framework.

New Base Types

- CAMetalLayer

New Commands

- vkCreateMetalSurfaceEXT

New Structures

- VkMetalSurfaceCreateInfoEXT

New Bitmasks

- VkMetalSurfaceCreateFlagsEXT

New Enum Constants

- `VK_EXT_METAL_SURFACE_EXTENSION_NAME`
- `VK_EXT_METAL_SURFACE_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_METAL_SURFACE_CREATE_INFO_EXT

Version History

- Revision 1, 2018-10-01 (Dzmitry Malyshau)
  - Initial version

VK_EXT_multi_draw

Name String

`VK_EXT_multi_draw`

Extension Type

Device extension

Registered Extension Number

393

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Mike Blumenkrantz [GitHub]zmike

Other Extension Metadata

Last Modified Date

2021-05-19

Interactions and External Dependencies

- Interacts with Vulkan 1.1.
- Interacts with `VK_KHR_shader_draw_parameters`.

IP Status

No known IP claims.

Contributors

- Mike Blumenkrantz, VALVE
- Piers Daniell, NVIDIA
- Faith Ekstrand, INTEL
- Spencer Fricke, SAMSUNG
- Ricardo Garcia, IGALIA
- Jon Leech, KHRONOS
- Stu Smith, AMD

Description

Processing multiple draw commands in sequence incurs measurable overhead within drivers due to repeated state checks and updates during dispatch. This extension enables passing the entire sequence of draws directly to the driver in order to avoid any such overhead, using an array of VkMultiDrawInfoEXT or VkMultiDrawIndexedInfoEXT structs with `vkCmdDrawMultiEXT` or `vkCmdDrawMultiIndexedEXT`, respectively. These functions could be used any time multiple draw commands are being recorded without any state changes between them in order to maximize performance.

New Commands

- vkCmdDrawMultiEXT
- vkCmdDrawMultiIndexedEXT

New Structures

- VkMultiDrawIndexedInfoEXT
- VkMultiDrawInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMultiDrawFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceMultiDrawPropertiesEXT

New Enum Constants

- `VK_EXT_MULTI_DRAW_EXTENSION_NAME`
- `VK_EXT_MULTI_DRAW_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTI_DRAW_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTI_DRAW_PROPERTIES_EXT

New or Modified Built-In Variables

- (modified)`DrawIndex`

Version History

- Revision 1, 2021-01-20 (Mike Blumenkrantz)
  - Initial version

VK_EXT_multisampled_render_to_single_sampled

Name String

`VK_EXT_multisampled_render_to_single_sampled`

Extension Type

Device extension

Registered Extension Number

377

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_create_renderpass2
and
VK_KHR_depth_stencil_resolve
or
Vulkan Version 1.2

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Extension Proposal

VK_EXT_multisampled_render_to_single_sampled

Other Extension Metadata

Last Modified Date

2021-04-16

IP Status

No known IP claims.

Contributors

- Shahbaz Youssefi, Google
- Jan-Harald Fredriksen, Arm
- Jörg Wagner, Arm
- Matthew Netsch, Qualcomm Technologies, Inc.
- Jarred Davies, Imagination Technologies

Description

With careful usage of resolve attachments, multisampled image memory allocated with VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT, `loadOp` not equal to VK_ATTACHMENT_LOAD_OP_LOAD and `storeOp` not equal to VK_ATTACHMENT_STORE_OP_STORE, a Vulkan application is able to efficiently perform multisampled rendering without incurring any additional memory penalty on some implementations.

Under certain circumstances however, the application may not be able to complete its multisampled rendering within a single render pass; for example if it does partial rasterization from frame to frame, blending on an image from a previous frame, or in emulation of GL_EXT_multisampled_render_to_texture. In such cases, the application can use an initial subpass to effectively load single-sampled data from the next subpass's resolve attachment and fill in the multisampled attachment which otherwise uses `loadOp` equal to VK_ATTACHMENT_LOAD_OP_DONT_CARE. However, this is not always possible (for example for stencil in the absence of VK_EXT_shader_stencil_export) and has multiple drawbacks.

Some implementations are able to perform said operation efficiently in hardware, effectively loading a multisampled attachment from the contents of a single sampled one. Together with the ability to perform a resolve operation at the end of a subpass, these implementations are able to perform multisampled rendering on single-sampled attachments with no extra memory or bandwidth overhead. This extension exposes this capability by allowing a framebuffer and render pass to include single-sampled attachments while rendering is done with a specified number of samples.

New Structures

- Extending VkFormatProperties2:
  - VkSubpassResolvePerformanceQueryEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMultisampledRenderToSingleSampledFeaturesEXT
- Extending VkSubpassDescription2, VkRenderingInfo:
  - VkMultisampledRenderToSingleSampledInfoEXT

New Enum Constants

- `VK_EXT_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_EXTENSION_NAME`
- `VK_EXT_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_SPEC_VERSION`
- Extending VkImageCreateFlagBits:
  - VK_IMAGE_CREATE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_INFO_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_FEATURES_EXT
  - VK_STRUCTURE_TYPE_SUBPASS_RESOLVE_PERFORMANCE_QUERY_EXT

Issues

1) Could the multisampled attachment be initialized through some form of copy?

RESOLVED: No. Some implementations do not support copying between attachments in general, and find expressing this operation through a copy unnatural.

2) Another way to achieve this is by introducing a new `loadOp` to load the contents of the multisampled image from a single-sampled one. Why is this extension preferred?

RESOLVED: Using this extension simplifies the application, as it does not need to manage a secondary lazily-allocated image. Additionally, using this extension leaves less room for error; for example a single mistake in `loadOp` or `storeOp` would result in the lazily-allocated image to actually take up memory, and remain so until destruction.

3) There is no guarantee that multisampled data between two subpasses with the same number of samples will be retained as the implementation may be forced to split the render pass implicitly for various reasons. Should this extension require that every subpass that uses multisampled-render-to-single-sampled end in an implicit render pass split (which results in a resolve operation)?

RESOLVED: No. Not requiring this allows render passes with multiple multisampled-render-to-single-sampled subpasses to potentially execute more efficiently (though there is no guarantee).

Version History

- Revision 1, 2021-04-12 (Shahbaz Youssefi)

VK_EXT_multisampled_render_to_swapchain

Name String

`VK_EXT_multisampled_render_to_swapchain`

Extension Type

Device extension

Registered Extension Number

617

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_swapchain
and
VK_EXT_multisampled_render_to_single_sampled

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Extension Proposal

VK_EXT_multisampled_render_to_swapchain

Other Extension Metadata

Last Modified Date

2025-05-20

IP Status

No known IP claims.

Contributors

- Shahbaz Youssefi, Google
- Mike Blumenkrantz, Valve
- James Jones, NVIDIA

Description

This extension adds the equivalent of VK_IMAGE_CREATE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT to swapchain create flags such that the functionality of the `VK_EXT_multisampled_render_to_single_sampled` extension can be used with its images.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMultisampledRenderToSwapchainFeaturesEXT
- Extending VkSurfaceCapabilities2KHR:
  - VkSwapchainFlagsSurfaceCapabilitiesEXT

New Enum Constants

- `VK_EXT_MULTISAMPLED_RENDER_TO_SWAPCHAIN_EXTENSION_NAME`
- `VK_EXT_MULTISAMPLED_RENDER_TO_SWAPCHAIN_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTISAMPLED_RENDER_TO_SWAPCHAIN_FEATURES_EXT
  - VK_STRUCTURE_TYPE_SWAPCHAIN_FLAGS_SURFACE_CAPABILITIES_EXT
- Extending VkSwapchainCreateFlagBitsKHR:
  - VK_SWAPCHAIN_CREATE_MULTISAMPLED_RENDER_TO_SINGLE_SAMPLED_BIT_EXT

Version History

- Revision 1, 2025-05-20 (Shahbaz Youssefi)

VK_EXT_mutable_descriptor_type

Name String

`VK_EXT_mutable_descriptor_type`

Extension Type

Device extension

Registered Extension Number

495

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_maintenance3
or
Vulkan Version 1.1

Special Use

- D3D support

Contact

- Joshua Ashton [GitHub]Joshua-Ashton
- Hans-Kristian Arntzen [GitHub]HansKristian-Work

Extension Proposal

VK_EXT_mutable_descriptor_type

Other Extension Metadata

Last Modified Date

2022-08-22

IP Status

No known IP claims.

Contributors

- Joshua Ashton, Valve
- Hans-Kristian Arntzen, Valve

Description

This extension allows applications to reduce descriptor memory footprint by allowing a descriptor to be able to mutate to a given list of descriptor types depending on which descriptor types are written into, or copied into a descriptor set.

The main use case this extension intends to address is descriptor indexing with VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT where the descriptor types are completely generic, as this means applications can allocate one large descriptor set, rather than having one large descriptor set per descriptor type, which significantly bloats descriptor memory usage and causes performance issues.

This extension also adds a mechanism to declare that a descriptor pool, and therefore the descriptor sets that are allocated from it, reside only in host memory; as such these descriptors can only be updated/copied, but not bound.

These features together allow much more efficient emulation of the raw D3D12 binding model. This extension is primarily intended to be useful for API layering efforts.

New Structures

- VkMutableDescriptorTypeListEXT
- Extending VkDescriptorSetLayoutCreateInfo, VkDescriptorPoolCreateInfo:
  - VkMutableDescriptorTypeCreateInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMutableDescriptorTypeFeaturesEXT

New Enum Constants

- `VK_EXT_MUTABLE_DESCRIPTOR_TYPE_EXTENSION_NAME`
- `VK_EXT_MUTABLE_DESCRIPTOR_TYPE_SPEC_VERSION`
- Extending VkDescriptorPoolCreateFlagBits:
  - VK_DESCRIPTOR_POOL_CREATE_HOST_ONLY_BIT_EXT
- Extending VkDescriptorSetLayoutCreateFlagBits:
  - VK_DESCRIPTOR_SET_LAYOUT_CREATE_HOST_ONLY_POOL_BIT_EXT
- Extending VkDescriptorType:
  - VK_DESCRIPTOR_TYPE_MUTABLE_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_MUTABLE_DESCRIPTOR_TYPE_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MUTABLE_DESCRIPTOR_TYPE_FEATURES_EXT

Version History

- Revision 1, 2022-08-22 (Jon Leech)
  - Initial version, promoted from `VK_VALVE_mutable_descriptor_type`.

VK_EXT_nested_command_buffer

Name String

`VK_EXT_nested_command_buffer`

Extension Type

Device extension

Registered Extension Number

452

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2023-09-18

Contributors

- Daniel Story, Nintendo
- Peter Kohaut, NVIDIA
- Shahbaz Youssefi, Google
- Slawomir Grajewski, Intel
- Stu Smith, AMD

Description

With core Vulkan it is not legal to call vkCmdExecuteCommands when recording a secondary command buffer. This extension relaxes that restriction, allowing secondary command buffers to execute other secondary command buffers.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceNestedCommandBufferFeaturesEXT

- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceNestedCommandBufferPropertiesEXT

New Enum Constants

- VK_EXT_NESTED_COMMAND_BUFFER_EXTENSION_NAME
- VK_EXT_NESTED_COMMAND_BUFFER_SPEC_VERSION
- Extending VkRenderingFlagBits:
  - VK_RENDERING_CONTENTS_INLINE_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NESTED_COMMAND_BUFFER_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NESTED_COMMAND_BUFFER_PROPERTIES_EXT
- Extending VkSubpassContents:
  - VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_EXT

Issues

1) The Command Buffer Levels property for the Vulkan commands comes from the `cmdbufferlevel` attribute in `vk.xml` for the command, and it is currently not possible to modify this attribute based on whether an extension is enabled. For this extension we want the `cmdbufferlevel` attribute for vkCmdExecuteCommands to be `primary,secondary` when this extension is enabled and `primary` otherwise.

RESOLVED: The `cmdbufferlevel` attribute for vkCmdExecuteCommands has been changed to `primary,secondary` and a new VUID added to prohibit recording this command in a secondary command buffer unless this extension is enabled.

Version History

- Revision 1, 2023-09-18 (Piers Daniell)
  - Internal revisions

VK_EXT_non_seamless_cube_map

Name String

`VK_EXT_non_seamless_cube_map`

Extension Type

Device extension

Registered Extension Number

423

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Special Uses

- D3D support
- OpenGL / ES support

Contact

- Georg Lehmann [GitHub]DadSchoorse

Extension Proposal

VK_EXT_non_seamless_cube_map

Other Extension Metadata

Last Modified Date

2021-09-04

IP Status

No known IP claims.

Contributors

- Georg Lehmann

Description

This extension provides functionality to disable cube map edge handling on a per sampler level which matches the behavior of other graphics APIs.

This extension may be useful for building translation layers for those APIs or for porting

applications that rely on this cube map behavior.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT

New Enum Constants

- `VK_EXT_NON_SEAMLESS_CUBE_MAP_EXTENSION_NAME`
- `VK_EXT_NON_SEAMLESS_CUBE_MAP_SPEC_VERSION`
- Extending VkSamplerCreateFlagBits:
  - VK_SAMPLER_CREATE_NON_SEAMLESS_CUBE_MAP_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NON_SEAMLESS_CUBE_MAP_FEATURES_EXT

Version History

- Revision 1, 2021-09-04 (Georg Lehmann)
  - First Version

VK_EXT_pageable_device_local_memory

Name String

`VK_EXT_pageable_device_local_memory`

Extension Type

Device extension

Registered Extension Number

413

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_EXT_memory_priority

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2021-08-24

Contributors

- Hans-Kristian Arntzen, Valve
- Axel Gneiting, id Software
- Billy Khan, id Software
- Daniel Koch, NVIDIA
- Chris Lentini, NVIDIA
- Joshua Schnarr, NVIDIA
- Stu Smith, AMD

Description

Vulkan is frequently implemented on multi-user and multi-process operating systems where the device-local memory can be shared by more than one process. On such systems the size of the device-local memory available to the application may not be the full size of the memory heap at all times. In order for these operating systems to support multiple applications the device-local memory is virtualized and paging is used to move memory between device-local and host-local memory heaps, transparent to the application.

The current Vulkan specification does not expose this behavior well and may cause applications to make suboptimal memory choices when allocating memory. For example, in a system with multiple applications running, the application may think that device-local memory is full and revert to making performance-sensitive allocations from host-local memory. In reality the memory heap might not have been full, it just appeared to be at the time memory consumption was queried, and a device-local allocation would have succeeded. A well designed operating system that implements pageable device-local memory will try to have all memory allocations for the foreground application paged into device-local memory, and paged out for other applications as needed when not in use.

When this extension is exposed by the Vulkan implementation it indicates to the application that the operating system implements pageable device-local memory and the application should adjust its memory allocation strategy accordingly. The extension also exposes a new vkSetDeviceMemoryPriorityEXT function to allow the application to dynamically adjust the priority of existing memory allocations based on its current needs. This will help the operating system page out lower priority memory allocations before higher priority allocations when needed. It will also help the operating system decide which memory allocations to page back into device-local memory first.

To take best advantage of pageable device-local memory the application must create the Vulkan device with the VkPhysicalDevicePageableDeviceLocalMemoryFeaturesEXT ::`pageableDeviceLocalMemory` feature enabled. When enabled the Vulkan implementation will allow device-local memory allocations to be paged in and out by the operating system, and **may** not return VK_ERROR_OUT_OF_DEVICE_MEMORY even if device-local memory appears to be full, but will instead page this, or other allocations, out to make room. The Vulkan implementation will also ensure that host-local memory allocations will never be promoted to device-local memory by the operating system, or consume device-local memory.

New Commands

- vkSetDeviceMemoryPriorityEXT

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePageableDeviceLocalMemoryFeaturesEXT

New Enum Constants

- `VK_EXT_PAGEABLE_DEVICE_LOCAL_MEMORY_EXTENSION_NAME`
- `VK_EXT_PAGEABLE_DEVICE_LOCAL_MEMORY_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PAGEABLE_DEVICE_LOCAL_MEMORY_FEATURES_EXT

Version History

- Revision 1, 2021-08-24 (Piers Daniell)

- Initial revision

VK_EXT_pci_bus_info

Name String

`VK_EXT_pci_bus_info`

Extension Type

Device extension

Registered Extension Number

213

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Matthaeus G. Chajdas [GitHub]anteru

Other Extension Metadata

Last Modified Date

2018-12-10

IP Status

No known IP claims.

Contributors

- Matthaeus G. Chajdas, AMD
- Daniel Rakos, AMD

Description

This extension adds a new query to obtain PCI bus information about a physical device.

Not all physical devices have PCI bus information, either due to the device not being connected to the system through a PCI interface or due to platform specific restrictions and policies. Thus this extension is only expected to be supported by physical devices which can provide the information.

As a consequence, applications should always check for the presence of the extension string for each individual physical device for which they intend to issue the new query for and should not have any assumptions about the availability of the extension on any given platform.

New Structures

- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDevicePCIBusInfoPropertiesEXT

New Enum Constants

- `VK_EXT_PCI_BUS_INFO_EXTENSION_NAME`
- `VK_EXT_PCI_BUS_INFO_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PCI_BUS_INFO_PROPERTIES_EXT

Version History

- Revision 2, 2018-12-10 (Daniel Rakos)
  - Changed all members of the new structure to have the uint32_t type
- Revision 1, 2018-10-11 (Daniel Rakos)
  - Initial revision

VK_EXT_physical_device_drm

Name String

VK_EXT_physical_device_drm

Extension Type

Device extension

Registered Extension Number

354

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Simon Ser [GitHub]emersion

Other Extension Metadata

Last Modified Date

2021-06-09

IP Status

No known IP claims.

Contributors

- Simon Ser

Description

This extension provides new facilities to query DRM properties for physical devices, enabling users to match Vulkan physical devices with DRM nodes on Linux.

Its functionality closely overlaps with EGL_EXT_device_drm[1]. Unlike the EGL extension, this extension does not expose a string containing the name of the device file and instead exposes device minor numbers.

DRM defines multiple device node types. Each physical device may have one primary node and one render node associated. Physical devices may have no primary node (e.g. if the device does not have a display subsystem), may have no render node (e.g. if it is a software rendering engine), or may have neither (e.g. if it is a software rendering engine without a display subsystem).

To query DRM properties for a physical device, chain VkPhysicalDeviceDrmPropertiesEXT to VkPhysicalDeviceProperties2.

New Structures

- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceDrmPropertiesEXT

New Enum Constants

- VK_EXT_PHYSICAL_DEVICE_DRM_EXTENSION_NAME
- VK_EXT_PHYSICAL_DEVICE_DRM_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRM_PROPERTIES_EXT

References

1. EGL_EXT_device_drm

Version History

- Revision 1, 2021-06-09
  - First stable revision

VK_EXT_pipeline_properties

Name String

`VK_EXT_pipeline_properties`

Extension Type

Device extension

Registered Extension Number

373

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Mukund Keshava [GitHub]mkeshavanv

Other Extension Metadata

Last Modified Date

2022-04-19

IP Status

No known IP claims.

Contributors

- Mukund Keshava, NVIDIA
- Daniel Koch, NVIDIA
- Mark Bellamy, Arm

Description

Vulkan SC requires offline compilation of pipelines. In order to support this, the pipeline state is represented in a JSON schema that is read by an offline tool for compilation.

One method of developing a Vulkan SC application is to author a Vulkan application and use a layer to record and serialize the pipeline state and shaders for offline compilation. Each pipeline is represented by a separate JSON file, and can be identified with a `pipelineIdentifier`.

Once the pipelines have been compiled by the offline pipeline cache compiler, the Vulkan SC

application can then use this `pipelineIdentifier` for identifying the pipeline via Vulkan SC's `VkPipelineIdentifierInfo` structure.

This extension allows the Vulkan application to query the `pipelineIdentifier` associated with each pipeline so that the application can store this with its pipeline metadata and the Vulkan SC application will then use to map the same state to an entry in the Vulkan SC pipeline cache.

It is expected that this extension will initially be implemented in the json generation layer, although we can envision that there might be future uses for it in native Vulkan drivers as well.

New Commands

- vkGetPipelinePropertiesEXT

New Structures

- VkPipelineInfoEXT
- VkPipelinePropertiesIdentifierEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePipelinePropertiesFeaturesEXT

New Enum Constants

- `VK_EXT_PIPELINE_PROPERTIES_EXTENSION_NAME`
- `VK_EXT_PIPELINE_PROPERTIES_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_PROPERTIES_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PIPELINE_INFO_EXT
  - VK_STRUCTURE_TYPE_PIPELINE_PROPERTIES_IDENTIFIER_EXT

Issues

(1) This extension does not make sense on a strict Vulkan SC implementation. It may however be of potential use in a non-strict Vulkan SC implementation. Should this extension be enabled as part of Vulkan SC as well?

RESOLVED: No. This extension will not be enabled for Vulkan SC.

(2) This is intended to be a general pipeline properties query, but is currently only retrieving the pipeline identifier. Should the pipeline identifier query be mandatory for this extension and for all queries using this command?

RESOLVED: Use VkBaseOutStructure for the return parameter. Currently this is required to actually be a VkPipelinePropertiesIdentifierEXT structure, but that could be relaxed in the future to allow other structure types or to allow other structures to be chained in along with this one.

(3) Should there be a feature structure? Should it be required?

RESOLVED: Add a feature structure, and a feature for querying pipeline identifier, but allow it to be optional so that this extension can be used as the basis for other pipeline property queries without requiring the pipeline identifier to be supported.

Version History

- Revision 1, 2022-04-19 (Mukund Keshava, Daniel Koch)
    - Initial draft

VK_EXT_post_depth_coverage

Name String

VK_EXT_post_depth_coverage

Extension Type

Device extension

Registered Extension Number

156

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

SPIR-V Dependencies

- SPV_KHR_post_depth_coverage

Contact

- Daniel Koch [GitHub]dgkoch

Other Extension Metadata

Last Modified Date

2017-07-17

Interactions and External Dependencies

- This extension provides API support for GL_ARB_post_depth_coverage and GL_EXT_post_depth_coverage

Contributors

- Jeff Bolz, NVIDIA

Description

This extension adds support for the following SPIR-V extension in Vulkan:

- SPV_KHR_post_depth_coverage

which allows the fragment shader to control whether values in the SampleMask built-in input variable reflect the coverage after early depth and stencil tests are applied.

This extension adds a new PostDepthCoverage execution mode under the SampleMaskPostDepthCoverage capability. When this mode is specified along with EarlyFragmentTests, the value of an input variable decorated with the SampleMask built-in reflects the coverage after the

early fragment tests are applied. Otherwise, it reflects the coverage before the depth and stencil tests.

When using GLSL source-based shading languages, the `post_depth_coverage` layout qualifier from GL_ARB_post_depth_coverage or GL_EXT_post_depth_coverage maps to the `PostDepthCoverage` execution mode.

New Enum Constants

- `VK_EXT_POST_DEPTH_COVERAGE_EXTENSION_NAME`
- `VK_EXT_POST_DEPTH_COVERAGE_SPEC_VERSION`

New SPIR-V Capabilities

- `SampleMaskPostDepthCoverage`

Version History

- Revision 1, 2017-07-17 (Daniel Koch)
  - Internal revisions

VK_EXT_present_timing

Name String

`VK_EXT_present_timing`

Extension Type

Device extension

Registered Extension Number

209

Revision

3

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_swapchain
and
VK_KHR_present_id2
and
VK_KHR_get_surface_capabilities2
and
VK_KHR_calibrated_timestamps

Contact

- Lionel Duc [GitHub]nvlduc

Extension Proposal

VK_EXT_present_timing

Other Extension Metadata

Last Modified Date

2024-10-09

IP Status

No known IP claims.

Contributors

- Ian Elliott, Google
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Daniel Rakos, AMD
- Daniel Stone, Collabora
- Daniel Vetter, Intel

- Aric Cyr, AMD
- Faith Ekstrand, Intel
- Nicolai Hähnle, AMD
- Alon Or-Bach, Samsung
- Niklas Smedberg, Unity Technologies
- Tobias Hector, AMD
- Lionel Duc, NVIDIA
- Lina Versace, Google
- Sebastian Wick, Red Hat
- Jakob Bornecrantz, Collabora
- David Kvasnica, NVIDIA

Description

This device extension allows an application that uses the `VK_KHR_swapchain` extension to obtain information about the presentation engine's display, to obtain timing information about each present, and to schedule a present to happen no earlier than a desired time. An application can use this to minimize various visual anomalies (e.g. stuttering).

Traditional game and real-time animation applications need to correctly position their geometry for when the presentable image will be presented to the user. To accomplish this, applications need various timing information about the presentation engine's display. They need to know when presentable images were actually presented, and when they could have been presented. Applications also need to tell the presentation engine to display an image no sooner than a given time. This allows the application to avoid stuttering, so the animation looks smooth to the user.

New Commands

- vkGetPastPresentationTimingEXT
- vkGetSwapchainTimeDomainPropertiesEXT
- vkGetSwapchainTimingPropertiesEXT
- vkSetSwapchainPresentTimingQueueSizeEXT

New Structures

- VkPastPresentationTimingEXT
- VkPastPresentationTimingInfoEXT
- VkPastPresentationTimingPropertiesEXT
- VkPresentStageTimeEXT
- VkPresentTimingInfoEXT
- VkSwapchainTimeDomainPropertiesEXT
- VkSwapchainTimingPropertiesEXT

- Extending VkCalibratedTimestampInfoKHR:
  - VkSwapchainCalibratedTimestampInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePresentTimingFeaturesEXT
- Extending VkPresentInfoKHR:
  - VkPresentTimingsInfoEXT
- Extending VkSurfaceCapabilities2KHR:
  - VkPresentTimingSurfaceCapabilitiesEXT

New Enums

- VkPastPresentationTimingFlagBitsEXT
- VkPresentStageFlagBitsEXT
- VkPresentTimingInfoFlagBitsEXT

New Bitmasks

- VkPastPresentationTimingFlagsEXT
- VkPresentStageFlagsEXT
- VkPresentTimingInfoFlagsEXT

New Enum Constants

- `VK_EXT_PRESENT_TIMING_EXTENSION_NAME`
- `VK_EXT_PRESENT_TIMING_SPEC_VERSION`
- Extending VkResult:
  - VK_ERROR_PRESENT_TIMING_QUEUE_FULL_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PAST_PRESENTATION_TIMING_EXT
  - VK_STRUCTURE_TYPE_PAST_PRESENTATION_TIMING_INFO_EXT
  - VK_STRUCTURE_TYPE_PAST_PRESENTATION_TIMING_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_TIMING_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PRESENT_TIMINGS_INFO_EXT
  - VK_STRUCTURE_TYPE_PRESENT_TIMING_INFO_EXT
  - VK_STRUCTURE_TYPE_PRESENT_TIMING_SURFACE_CAPABILITIES_EXT
  - VK_STRUCTURE_TYPE_SWAPCHAIN_CALIBRATED_TIMESTAMP_INFO_EXT
  - VK_STRUCTURE_TYPE_SWAPCHAIN_TIME_DOMAIN_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_SWAPCHAIN_TIMING_PROPERTIES_EXT
- Extending VkSwapchainCreateFlagBitsKHR:

    - VK_SWAPCHAIN_CREATE_PRESENT_TIMING_BIT_EXT
- Extending VkTimeDomainKHR:
    - VK_TIME_DOMAIN_PRESENT_STAGE_LOCAL_EXT
    - VK_TIME_DOMAIN_SWAPCHAIN_LOCAL_EXT

Issues

1) How does the application determine refresh duration, quanta for change, whether FRR vs. VRR, etc.

The query returns two values: 1) a refresh-cycle duration (`refreshDuration`), and 2) an indication whether the timing is currently fixed (FRR) or variable (VRR). If `refreshDuration` is zero, the platform cannot supply these values until after at least one vkQueuePresentKHR has been done, from this time (e.g. if vkQueuePresentKHR has been previously called for this swapchain, at least one additional call must be made). After calling vkQueuePresentKHR, the query can be repeated until `refreshDuration` is non-zero, at which point the FRR vs. VRR indication will also be valid.

If the presentation engine's `refreshDuration` is a fixed value, the application's image present duration (IPD) should be a multiple of `refreshDuration`. That is, the quanta for changing the IPD is `refreshDuration`. For example, if `refreshDuration` is 16.67ms, the IPD can be 16.67ms, 33.33ms, 50.0ms, etc.

If the presentation engine's `refreshDuration` is variable, `refreshDuration` is the minimum value of the application's IPD, and the IPD can be larger by any quanta that is meaningful to the application. For example, if the `refreshDuration` is 10ms (i.e. the maximum refresh rate is 100Hz), the application can choose an IPD of 11ms, 13.33ms, 13.5ms, or 66.0ms; any value greater than or equal to 10ms is valid. There may be negative consequences for choosing an IPD that is too high, as the presentation engine may actually have a practical maximum `refreshDuration`, where it needs to display the previous image again, and during this time the presentation engine might delay displaying a newly-presented image.

FRR displays on at least one platform (Wayland) are not necessarily fixed; but can change over time. For example, if a full-screen video player application is visible, the display may operate at a 24Hz refresh cycle; and then later switch to 60Hz when multiple windows are visible.

VRR displays on some platforms can also be seen as having different characteristics over time. For example, if an application's window is full-screen-exclusive (i.e. no other window or window system component is visible), the display can look like a VRR display (however that is defined). If the application's window is not full-screen-exclusive (e.g. a normal multi-window case), the display can look like an FRR display (i.e. because the compositor is trying to treat all windows in a consistent manner). A different issue will deal with how the timing characteristics can change over time.

2) Do we return min/max values for refresh duration for VRR?

Return only the minimum value of refreshDuration for a VRR.

VRR displays have a minimum and maximum refresh rate, and therefore a minimum and maximum refreshDuration. It has been asserted that the display effectively does not have a

minimum refresh rate. That is because if an application does not present soon enough, the display hardware will automatically re-display the previous image. However, when the display does that, an application cannot present a new image for a certain period of time. It is unclear about whether that period is large enough to cause visual artifacts.

3) How to deal with changes in timing properties?

RESOLVED: The VkPastPresentationTimingPropertiesEXT structure that is returned by vkGetPastPresentationTimingEXT contains `timeDomainsCounter`, which is incremented if the time domain enabled for the swapchain is not currently available.

An example of why display timing properties can change is if a surface changes from being a window that's a subset of the display size, to becoming full-screen-exclusive. While the surface was a subset of the display, a compositor might enforce fixed timings on the surface (e.g. FRR of 60Hz), where the presentation engine might be free to allow VRR behavior of a full-screen-exclusive surface.

It is possible that a full-screen-exclusive window can become temporarily obscured (e.g. when a short-term dialog pops up). In this case, the surface might use FRR timings while the dialog is visible and VRR otherwise.

4) One Query for all Timing info vs. an initial query to determine FRR vs. VRR, and then FRR-specific vs VRR-specific queries?

RESOLVED: Have one query, as described in issue 1, that can be called whenever the application needs to obtain the timing properties of the surface.

5) Query to determine time domain?

RESOLVED: Have a query to determine the time domain. This extension defines a basic swapchain-local time domain. Other extensions can add other platform-specific time domains.

6) What time to use for targetPresentTime for early images?

RESOLVED: Have no query for determining the current time in the PE's time domain; and do allow the special value of zero for targetPresentTime, meaning that there is no target.

On some platforms, there is no way to determine the current time, nor to determine surface timing properties until after at least one image has been presented.

In such cases, the special value of zero allows the application to indicate that timing feedback is desired, but that no targetPresentTime is requested. Later, once the application has obtained feedback, it can specify targetPresentTime by using the result's actualPresentTime.

7) How long before an application's request for new image duration is honored?

Apparently, changes to some vendors' display hardware settings do not take effect immediately. It is not clear what settings, and therefore, it is not clear how to address this issue.

8) Do we have a query for the anticipated latency from present to feedback?

RESOLVED: Do not provide a query for the feedback latency.

There is some amount of latency from when an application calls vkQueuePresentKHR to when the image is displayed to the user, to when feedback is available to the application on when the image was actually displayed to the user. The first time (from the call till the image is presented) generally doesn't matter, because the application will likely be providing a targetPresentTime (i.e. the application may have some indication for how long this will be). However, the latency between targetPresentTime until feedback is available may be much longer. For example, on Android on the 1st-generation Pixel phone (60Hz FRR display), the latency was approximately 5 refresh cycles (83.33ms). For higher-frequency displays, the latency may have a larger number of refresh cycles.

9) Do we have a query(s) about the number of VkPastPresentationTimingEXT structs to keep?

RESOLVED: Do not provide a query for the number of results the swapchain is allowed to store before querying them with vkGetPastPresentationTimingEXT. Let the application specify that value with a dedicated API.

10) How is the SWAPCHAIN_LOCAL and STAGE_LOCAL time domain used with the calibrated timestamps extension?

RESOLVED: Define a struct to chain into VkCalibratedTimestampInfoEXT::pNext that specifies a swapchain and present stage.

11) Should VK_PRESENT_MODE_FIFO_LATEST_READY_EXT be part of this extension, or split out into its own extension?

RESOLVED: It is only tangentially related. Split it out into its own extension and define the interaction here.

Version History

- Revision 1, 2018-05-11 (Ian Elliott)
  - Internal revisions.
- Revision 2, 2022-11-30 (Lionel Duc)
  - Rebase for public discussions.
- Revision 3, 2024-10-09 (Lionel Duc)
  - Public revisions.

VK_EXT_primitive_restart_index

Name String

`VK_EXT_primitive_restart_index`

Extension Type

Device extension

Registered Extension Number

679

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Special Use

- OpenGL / ES support

Contact

- Mike Blumenkrantz [GitHub]zmike

Extension Proposal

VK_EXT_primitive_restart_index

Other Extension Metadata

Last Modified Date

2026-03-10

IP Status

No known IP claims.

Contributors

- Mike Blumenkrantz, Valve
- Spencer Fricke, LunarG
- Ricardo Garcia, Igalia
- Piers Daniell, NVIDIA

Description

This extension allows setting a custom primitive restart index. It is primarily intended to support

GL emulation.

New Commands

- vkCmdSetPrimitiveRestartIndexEXT

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePrimitiveRestartIndexFeaturesEXT

New Enum Constants

- `VK_EXT_PRIMITIVE_RESTART_INDEX_EXTENSION_NAME`
- `VK_EXT_PRIMITIVE_RESTART_INDEX_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIMITIVE_RESTART_INDEX_FEATURES_EXT

Version History

- Revision 1, 2026-03-10 (Mike Blumenkrantz)
  - Internal revisions

VK_EXT_primitive_topology_list_restart

Name String

`VK_EXT_primitive_topology_list_restart`

Extension Type

Device extension

Registered Extension Number

357

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Special Use

- OpenGL / ES support

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Other Extension Metadata

Last Modified Date

2021-01-11

IP Status

No known IP claims.

Contributors

- Courtney Goeltzenleuchter, Google
- Shahbaz Youssefi, Google

Description

This extension allows list primitives to use the primitive restart index value. This provides a more efficient implementation when layering OpenGL functionality on Vulkan by avoiding emulation which incurs data copies.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:

    - VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT

New Enum Constants

- `VK_EXT_PRIMITIVE_TOPOLOGY_LIST_RESTART_EXTENSION_NAME`
- `VK_EXT_PRIMITIVE_TOPOLOGY_LIST_RESTART_SPEC_VERSION`
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIMITIVE_TOPOLOGY_LIST_RESTART_FEATURES_EXT

Version History

- Revision 0, 2020-09-14 (Courtney Goeltzenleuchter)
    - Internal revisions
- Revision 1, 2021-01-11 (Shahbaz Youssefi)
    - Add the `primitiveTopologyPatchListRestart` feature
    - Internal revisions

VK_EXT_primitives_generated_query

Name String

`VK_EXT_primitives_generated_query`

Extension Type

Device extension

Registered Extension Number

383

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_EXT_transform_feedback

Special Use

- OpenGL / ES support

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Extension Proposal

VK_EXT_primitives_generated_query

Other Extension Metadata

Last Modified Date

2022-01-24

Contributors

- Shahbaz Youssefi, Google
- Piers Daniell, NVIDIA
- Faith Ekstrand, Collabora
- Jan-Harald Fredriksen, Arm

Description

This extension adds support for a new query type to match OpenGL's `GL_PRIMITIVES_GENERATED` to support layering.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:

    - VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT

New Enum Constants

- VK_EXT_PRIMITIVES_GENERATED_QUERY_EXTENSION_NAME
- VK_EXT_PRIMITIVES_GENERATED_QUERY_SPEC_VERSION
- Extending VkQueryType:
    - VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIMITIVES_GENERATED_QUERY_FEATURES_EXT

Issues

1) Can the query from VK_EXT_transform_feedback be used instead?

RESOLVED: No. While the query from VK_EXT_transform_feedback can produce the same results as in this extension, it is only available while transform feedback is active. The OpenGL GL_PRIMITIVES_GENERATED query is independent from transform feedback. Emulation through artificial transform feedback is unnecessarily inefficient.

2) Can VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT be used instead?

RESOLVED: It could, but we prefer the extension for simplicity. Vulkan requires that only one query be active at a time. If both the GL_PRIMITIVES_GENERATED and the GL_CLIPPING_INPUT_PRIMITIVES_ARB queries need to be simultaneously enabled, emulation of both through VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT is inconvenient.

3) On some hardware, this query cannot be implemented if VkPipelineRasterizationStateCreateInfo::rasterizerDiscardEnable is enabled. How will this be handled?

RESOLVED: A feature flag is exposed by this extension for this. On said hardware, the GL implementation disables rasterizer-discard and achieves the same effect through other means. It will not be able to do the same in Vulkan due to lack of state information. A feature flag is exposed by this extension so the OpenGL implementation on top of Vulkan would be able to implement a similar workaround.

4) On some hardware, this query cannot be implemented for non-zero query indices. How will this be handled?

RESOLVED: A feature flag is exposed by this extension for this. If this feature is not present, the query from VK_EXT_transform_feedback can be used to the same effect.

5) How is the interaction of this extension with transformFeedbackRasterizationStreamSelect handled?

RESOLVED: Disallowed for non-zero streams. In OpenGL, the rasterization stream is always stream zero.

Version History

- Revision 1, 2021-06-23 (Shahbaz Youssefi)
  - Internal revisions

VK_EXT_provoking_vertex

Name String

`VK_EXT_provoking_vertex`

Extension Type

Device extension

Registered Extension Number

255

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Special Use

- OpenGL / ES support

Contact

- Jesse Hall [GitHub]jessehall

Other Extension Metadata

Last Modified Date

2021-02-22

IP Status

No known IP claims.

Contributors

- Alexis Hétu, Google
- Bill Licea-Kane, Qualcomm
- Daniel Koch, Nvidia
- Jamie Madill, Google
- Jan-Harald Fredriksen, Arm
- Faith Ekstrand, Intel
- Jeff Bolz, Nvidia
- Jeff Leger, Qualcomm
- Jesse Hall, Google

- Jörg Wagner, Arm
- Matthew Netsch, Qualcomm
- Mike Blumenkrantz, Valve
- Piers Daniell, Nvidia
- Tobias Hector, AMD

Description

This extension allows changing the provoking vertex convention between Vulkan's default convention (first vertex) and OpenGL's convention (last vertex).

This extension is intended for use by API-translation layers that implement APIs like OpenGL on top of Vulkan, and need to match the source API's provoking vertex convention. Applications using Vulkan directly should use Vulkan's default convention.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceProvokingVertexFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceProvokingVertexPropertiesEXT
- Extending VkPipelineRasterizationStateCreateInfo:
  - VkPipelineRasterizationProvokingVertexStateCreateInfoEXT

New Enums

- VkProvokingVertexModeEXT

New Enum Constants

- `VK_EXT_PROVOKING_VERTEX_EXTENSION_NAME`
- `VK_EXT_PROVOKING_VERTEX_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROVOKING_VERTEX_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROVOKING_VERTEX_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_PROVOKING_VERTEX_STATE_CREATE_INFO_EXT

Issues

1) At what granularity should this state be set?

RESOLVED: At pipeline bind, with an optional per-render pass restriction.

The most natural place to put this state is in the graphics pipeline object. Some implementations

require it to be known when creating the pipeline, and pipeline state is convenient for implementing OpenGL 3.2's glProvokingVertex, which can change the state between draw calls. However, some implementations can only change it approximately render pass granularity. To accommodate both, provoking vertex will be pipeline state, but implementations can require that only one mode is used within a render pass instance; the render pass's mode is chosen implicitly when the first pipeline is bound.

2) Does the provoking vertex mode affect the order that vertices are written to transform feedback buffers?

RESOLVED: Yes, to enable layered implementations of OpenGL and D3D.

All of OpenGL, OpenGL ES, and Direct3D 11 require that vertices are written to transform feedback buffers such that flat-shaded attributes have the same value when drawing the contents of the transform feedback buffer as they did in the original drawing when the transform feedback buffer was written (assuming the provoking vertex mode has not changed, in APIs that support more than one mode).

Version History

- Revision 1, (1c) 2021-02-22 (Jesse Hall)
  - Added VkPhysicalDeviceProvokingVertexPropertiesEXT::transformFeedbackPreservesTriangleFanProvokingVertex to accommodate implementations that cannot change the transform feedback vertex order for triangle fans.
- Revision 1, (1b) 2020-06-14 (Jesse Hall)
  - Added VkPhysicalDeviceProvokingVertexFeaturesEXT::transformFeedbackPreservesProvokingVertex and required that transform feedback write vertices so as to preserve the provoking vertex of each primitive.
- Revision 1, (1a) 2019-10-23 (Jesse Hall)
  - Initial draft, based on a proposal by Alexis Hétu

VK_EXT_queue_family_foreign

Name String

`VK_EXT_queue_family_foreign`

Extension Type

Device extension

Registered Extension Number

127

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_external_memory
or
Vulkan Version 1.1

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2017-11-01

IP Status

No known IP claims.

Contributors

- Lina Versace, Google
- James Jones, NVIDIA
- Faith Ekstrand, Intel
- Jesse Hall, Google
- Daniel Rakos, AMD
- Ray Smith, ARM

Description

This extension defines a special queue family, VK_QUEUE_FAMILY_FOREIGN_EXT, which can be used to transfer ownership of resources backed by external memory to foreign, external queues. This is similar to VK_QUEUE_FAMILY_EXTERNAL_KHR, defined in `VK_KHR_external_memory`. The key differences between the two are:

- The queues represented by VK_QUEUE_FAMILY_EXTERNAL_KHR must share the same physical device and the same driver version as the current VkInstance. VK_QUEUE_FAMILY_FOREIGN_EXT has no such restrictions. It can represent devices and drivers from other vendors, and can even represent non-Vulkan-capable devices.
- All resources backed by external memory support VK_QUEUE_FAMILY_EXTERNAL_KHR. Support for VK_QUEUE_FAMILY_FOREIGN_EXT is more restrictive.
- Applications should expect transitions to/from VK_QUEUE_FAMILY_FOREIGN_EXT to be more expensive than transitions to/from VK_QUEUE_FAMILY_EXTERNAL_KHR.

New Enum Constants

- `VK_EXT_QUEUE_FAMILY_FOREIGN_EXTENSION_NAME`
- `VK_EXT_QUEUE_FAMILY_FOREIGN_SPEC_VERSION`
- VK_QUEUE_FAMILY_FOREIGN_EXT

Version History

- Revision 1, 2017-11-01 (Lina Versace)
  - Squashed internal revisions

VK_EXT_rasterization_order_attachment_access

Name String

`VK_EXT_rasterization_order_attachment_access`

Extension Type

Device extension

Registered Extension Number

464

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Jan-Harald Fredriksen [GitHub]janharaldfredriksen-arm

Extension Proposal

VK_EXT_rasterization_order_attachment_access

Other Extension Metadata

Last Modified Date

2026-01-16

IP Status

No known IP claims.

Contributors

- Tobias Hector, AMD
- Jan-Harald Fredriksen, Arm

Description

This extension extends the mechanism of input attachments to allow access to framebuffer attachments that are used both as input and as color or depth/stencil attachments from one fragment to the next, in rasterization order, without explicit synchronization.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:

- VkPhysicalDeviceRasterizationOrderAttachmentAccessFeaturesEXT

New Enums

- VkPipelineColorBlendStateCreateFlagBits
- VkPipelineDepthStencilStateCreateFlagBits

New Enum Constants

- `VK_EXT_RASTERIZATION_ORDER_ATTACHMENT_ACCESS_EXTENSION_NAME`
- `VK_EXT_RASTERIZATION_ORDER_ATTACHMENT_ACCESS_SPEC_VERSION`
- Extending VkPipelineColorBlendStateCreateFlagBits:
  - VK_PIPELINE_COLOR_BLEND_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_ACCESS_BIT_EXT
- Extending VkPipelineDepthStencilStateCreateFlagBits:
  - VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT_EXT
  - VK_PIPELINE_DEPTH_STENCIL_STATE_CREATE_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RASTERIZATION_ORDER_ATTACHMENT_ACCESS_FEATURES_EXT
- Extending VkSubpassDescriptionFlagBits:
  - VK_SUBPASS_DESCRIPTION_RASTERIZATION_ORDER_ATTACHMENT_COLOR_ACCESS_BIT_EXT
  - VK_SUBPASS_DESCRIPTION_RASTERIZATION_ORDER_ATTACHMENT_DEPTH_ACCESS_BIT_EXT
  - VK_SUBPASS_DESCRIPTION_RASTERIZATION_ORDER_ATTACHMENT_STENCIL_ACCESS_BIT_EXT

Examples

None.

Version History

- Revision 1, 2022-07-04 (Jan-Harald Fredriksen)
  - Initial draft
  - Updated Issues section 2026-01-16

VK_EXT_ray_tracing_invocation_reorder

Name String

`VK_EXT_ray_tracing_invocation_reorder`

Extension Type

Device extension

Registered Extension Number

582

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_ray_tracing_pipeline

SPIR-V Dependencies

- SPV_EXT_shader_invocation_reorder

Contact

- Eric Werness [GitHub]ewerness-nv

Extension Proposal

VK_EXT_ray_tracing_invocation_reorder

Other Extension Metadata

Last Modified Date

2026-04-02

Interactions and External Dependencies

- This extension provides API support for `GL_EXT_shader_invocation_reorder`

Contributors

- Eric Werness, NVIDIA
- Ashwin Lele, NVIDIA
- Daniel Koch, NVIDIA
- Vikram Kushwaha, NVIDIA
- Piers Daniell, NVIDIA
- Stu Smith, AMD
- Aaron Hagan, AMD
- Tyler Nowicki, AMD

- Sebastian Neubauer, AMD
- Radoslaw Drabinski, Intel
- Sven Woop, Intel
- Aleksandra Krstic, QUALCOMM
- Andrew Garrard, Imagination Technologies
- Mathieu Robart, Arm Limited
- Tom Olson, Khronos
- Ralph Potter, Samsung Electronics
- Antonio Caggiano, LunarG

Description

The ray tracing pipeline API provides some ability to reorder for locality, but it is useful to have more control over how the reordering happens and what information is included in the reordering. The shader API provides a hit object to contain result information from the hit which can be used as part of the explicit sorting plus options that contain an integer for hint bits to use to add more coherency.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceRayTracingInvocationReorderFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceRayTracingInvocationReorderPropertiesEXT

New Enums

- VkRayTracingInvocationReorderModeEXT

New Enum Constants

- `VK_EXT_RAY_TRACING_INVOCATION_REORDER_EXTENSION_NAME`
- `VK_EXT_RAY_TRACING_INVOCATION_REORDER_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_INVOCATION_REORDER_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_INVOCATION_REORDER_PROPERTIES_EXT

Version History

- Revision 2, 2026-04-02 (Vikram Kushwaha)
  - Add `hitKind` overload for `hitObjectRecordFromQueryEXT` in `GL_EXT_shader_invocation_reorder`,

and fix for https://github.com/KhronosGroup/SPIRV-Registry/issues/373

- Revision 1, 2025-11-12 (Eric Werness)
  - Internal development - forked from NV

VK_EXT_rgba10x6_formats

Name String

VK_EXT_rgba10x6_formats

Extension Type

Device extension

Registered Extension Number

345

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_sampler_ycbcr_conversion
or
Vulkan Version 1.1

Contact

- Jan-Harald Fredriksen [GitHub]janharaldfredriksen-arm

Other Extension Metadata

Last Modified Date

2021-09-29

IP Status

No known IP claims.

Contributors

- Jan-Harald Fredriksen, Arm
- Graeme Leese, Broadcom
- Spencer Fricke, Samsung

Description

This extension enables the VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16 format to be used without a sampler $Y'C_BC_R$ conversion enabled.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceRGBA10X6FormatsFeaturesEXT

New Enum Constants

- `VK_EXT_RGBA10X6_FORMATS_EXTENSION_NAME`
- `VK_EXT_RGBA10X6_FORMATS_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RGBA10X6_FORMATS_FEATURES_EXT

Issues

1) Should we reuse the existing format enumeration or introduce a new one?

RESOLVED: We reuse an existing format enumeration, VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16, that was previously exclusively used for YCbCr and therefore had a set of limitations related to that usage. The alternative was to introduce a new format token with exactly the same bit representation as the existing token, but without the limitations.

2) Should we only introduce VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16 or also 1-3 component variations?

RESOLVED: Only the 4-component format is introduced because the 1- and 2- component variations are already not exclusive to YCbCr, and the 3-component variation is not a good match for hardware capabilities.

Version History

- Revision 1, 2021-09-29 (Jan-Harald Fredriksen)
  - Initial EXT version

VK_EXT_sample_locations

Name String

`VK_EXT_sample_locations`

Extension Type

Device extension

Registered Extension Number

144

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Daniel Rakos [GitHub]drakos-amd

Other Extension Metadata

Last Modified Date

2017-08-02

Contributors

- Mais Alnasser, AMD
- Matthaeus G. Chajdas, AMD
- Maciej Jesionowski, AMD
- Daniel Rakos, AMD
- Slawomir Grajewski, Intel
- Jeff Bolz, NVIDIA
- Bill Licea-Kane, Qualcomm

Description

This extension allows an application to modify the locations of samples within a pixel used in rasterization. Additionally, it allows applications to specify different sample locations for each pixel in a group of adjacent pixels, which **can** increase antialiasing quality (particularly if a custom resolve shader is used that takes advantage of these different locations).

It is common for implementations to optimize the storage of depth values by storing values that **can** be used to reconstruct depth at each sample location, rather than storing separate depth values for each sample. For example, the depth values from a single triangle **may** be represented using plane equations. When the depth value for a sample is needed, it is automatically evaluated at the sample location. Modifying the sample locations causes the reconstruction to no longer evaluate the same depth values as when the samples were originally generated, thus the depth aspect of a depth/stencil attachment **must** be cleared before rendering to it using different sample locations.

Some implementations **may** need to evaluate depth image values while performing image layout transitions. To accommodate this, instances of the VkSampleLocationsInfoEXT structure **can** be specified for each situation where an explicit or automatic layout transition has to take place. VkSampleLocationsInfoEXT **can** be chained from VkImageMemoryBarrier structures to provide sample locations for layout transitions performed by vkCmdWaitEvents and vkCmdPipelineBarrier calls, and VkRenderPassSampleLocationsBeginInfoEXT **can** be chained from VkRenderPassBeginInfo to provide sample locations for layout transitions performed implicitly by a render pass instance.

New Commands

- vkCmdSetSampleLocationsEXT
- vkGetPhysicalDeviceMultisamplePropertiesEXT

New Structures

- VkAttachmentSampleLocationsEXT
- VkMultisamplePropertiesEXT
- VkSampleLocationEXT
- VkSubpassSampleLocationsEXT
- Extending VkImageMemoryBarrier, VkImageMemoryBarrier2:
  - VkSampleLocationsInfoEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceSampleLocationsPropertiesEXT
- Extending VkPipelineMultisampleStateCreateInfo:
  - VkPipelineSampleLocationsStateCreateInfoEXT
- Extending VkRenderPassBeginInfo:
  - VkRenderPassSampleLocationsBeginInfoEXT

New Enum Constants

- `VK_EXT_SAMPLE_LOCATIONS_EXTENSION_NAME`
- `VK_EXT_SAMPLE_LOCATIONS_SPEC_VERSION`
- Extending VkDynamicState:
  - VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT

- Extending VkImageCreateFlagBits:
  - VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_MULTISAMPLE_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLE_LOCATIONS_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_PIPELINE_SAMPLE_LOCATIONS_STATE_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_RENDER_PASS_SAMPLE_LOCATIONS_BEGIN_INFO_EXT
  - VK_STRUCTURE_TYPE_SAMPLE_LOCATIONS_INFO_EXT

Issues

1) When using with Dynamic Rendering, is there a VkRenderPassSampleLocationsBeginInfoEXT equivalent struct

RESOLVED: No, there are no subpasses that need to have a sample location set.

Version History

- Revision 1, 2017-08-02 (Daniel Rakos)
  - Internal revisions

VK_EXT_shader_64bit_indexing

Name String

`VK_EXT_shader_64bit_indexing`

Extension Type

Device extension

Registered Extension Number

628

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_EXT_shader_64bit_indexing

Contact

- Jeff Bolz [GitHub]jeffbolznv

Extension Proposal

VK_EXT_shader_64bit_indexing

Other Extension Metadata

Last Modified Date

2025-05-02

Interactions and External Dependencies

- This extension provides API support for `GL_EXT_shader_64bit_indexing`

Contributors

- Jeff Bolz, NVIDIA

Description

This extension relaxes the maxStorageBufferRange limit, allowing more than 4GB to be accessed through a buffer binding (or through a buffer device address). It adds pipeline and shader creation flags that request 64-bit addressing support, and defines which addressing calculations use 64 bits of range.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShader64BitIndexingFeaturesEXT

New Enum Constants

- `VK_EXT_SHADER_64BIT_INDEXING_EXTENSION_NAME`
- `VK_EXT_SHADER_64BIT_INDEXING_SPEC_VERSION`
- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_64_BIT_INDEXING_BIT_EXT
- Extending VkShaderCreateFlagBitsEXT:
  - VK_SHADER_CREATE_64_BIT_INDEXING_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_64_BIT_INDEXING_FEATURES_EXT

Issues

Version History

- Revision 1, 2025-05-02 (Jeff Bolz)
  - Initial revisions

VK_EXT_shader_atomic_float

Name String

`VK_EXT_shader_atomic_float`

Extension Type

Device extension

Registered Extension Number

261

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VkPhysicalDeviceShaderAtomicFloatFeaturesEXT::sparseImageFloat32AtomicAdd
- Interacts with VkPhysicalDeviceShaderAtomicFloatFeaturesEXT::sparseImageFloat32Atomics

SPIR-V Dependencies

- SPV_EXT_shader_atomic_float_add

Contact

- Vikram Kushwaha [GitHub]vkushwaha-nv

Other Extension Metadata

Last Modified Date

2020-07-15

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for `GL_EXT_shader_atomic_float`

Contributors

- Vikram Kushwaha, NVIDIA
- Jeff Bolz, NVIDIA

Description

This extension allows a shader to contain floating-point atomic operations on buffer, workgroup, and image memory. It also advertises the SPIR-V `AtomicFloat32AddEXT` and `AtomicFloat64AddEXT` capabilities that allows atomic addition on floating-points numbers. The supported operations include `OpAtomicFAddEXT`, `OpAtomicExchange`, `OpAtomicLoad` and `OpAtomicStore`.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderAtomicFloatFeaturesEXT

New Enum Constants

- `VK_EXT_SHADER_ATOMIC_FLOAT_EXTENSION_NAME`
- `VK_EXT_SHADER_ATOMIC_FLOAT_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_FLOAT_FEATURES_EXT

New SPIR-V Capabilities

- `AtomicFloat32AddEXT`
- `AtomicFloat64AddEXT`

Version History

- Revision 1, 2020-07-15 (Vikram Kushwaha)
  - Internal revisions

VK_EXT_shader_atomic_float2

Name String

`VK_EXT_shader_atomic_float2`

Extension Type

Device extension

Registered Extension Number

274

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_EXT_shader_atomic_float

API Interactions

- Interacts with VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::sparseImageFloat32AtomicMinMax

SPIR-V Dependencies

- SPV_EXT_shader_atomic_float16_add
- SPV_EXT_shader_atomic_float_min_max

Contact

- Faith Ekstrand [GitHub]gfxstrand

Other Extension Metadata

Last Modified Date

2020-08-14

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for `GLSL_EXT_shader_atomic_float2`

Contributors

- Faith Ekstrand, Intel

Description

This extension allows a shader to perform 16-bit floating-point atomic operations on buffer and workgroup memory as well as floating-point atomic minimum and maximum operations on buffer,

workgroup, and image memory. It advertises the SPIR-V `AtomicFloat16AddEXT` capability which allows atomic add operations on 16-bit floating-point numbers and the SPIR-V `AtomicFloat16MinMaxEXT`, `AtomicFloat32MinMaxEXT` and `AtomicFloat64MinMaxEXT` capabilities which allow atomic minimum and maximum operations on floating-point numbers. The supported operations include `OpAtomicFAddEXT`, `OpAtomicFMinEXT` and `OpAtomicFMaxEXT`.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT

New Enum Constants

- `VK_EXT_SHADER_ATOMIC_FLOAT_2_EXTENSION_NAME`
- `VK_EXT_SHADER_ATOMIC_FLOAT_2_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_FLOAT_2_FEATURES_EXT

Issues

1) Should this extension add support for 16-bit image atomics?

RESOLVED: No. While Vulkan supports creating storage images with VK_FORMAT_R16_SFLOAT and doing load and store on them, the data in the shader has a 32-bit representation. Vulkan currently has no facility for even basic reading or writing such images using 16-bit float values in the shader. Adding such functionality would be required before 16-bit image atomics would make sense and is outside the scope of this extension.

New SPIR-V Capabilities

- `AtomicFloat32MinMaxEXT`
- `AtomicFloat32MinMaxEXT`
- `AtomicFloat32MinMaxEXT`
- `AtomicFloat64MinMaxEXT`

Version History

- Revision 1, 2020-08-14 (Faith Ekstrand)
  - Internal revisions

VK_EXT_shader_float8

Name String

`VK_EXT_shader_float8`

Extension Type

Device extension

Registered Extension Number

568

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_KHR_cooperative_matrix

SPIR-V Dependencies

- SPV_EXT_float8

Contact

- Kevin Petit [GitHub]kpet

Extension Proposal

VK_EXT_shader_float8

Other Extension Metadata

Last Modified Date

2025-04-16

IP Status

No known IP claims.

Contributors

- Kévin Petit, Arm Ltd.
- Stu Smith, AMD
- Jeff Bolz, NVIDIA
- Craig Graham, Samsung

Description

This extension enables support for 8-bit floating-point data types as defined in SPV_EXT_float8.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderFloat8FeaturesEXT

New Enum Constants

- `VK_EXT_SHADER_FLOAT8_EXTENSION_NAME`
- `VK_EXT_SHADER_FLOAT8_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT8_FEATURES_EXT

If VK_KHR_cooperative_matrix is supported:

- Extending VkComponentTypeKHR:
  - VK_COMPONENT_TYPE_FLOAT8_E4M3_EXT
  - VK_COMPONENT_TYPE_FLOAT8_E5M2_EXT

New SPIR-V Capabilities

- Float8EXT
- Float8CooperativeMatrixEXT

Issues

1) Resolve interactions with the changes VK_KHR_shader_float16 makes to rules for denorm flushing (always allowed by default for all FP formats). How to describe the requirement to preserve subnormals?

+ **RESOLVED**: Subnormals are always preserved when converting FP8 values to IEEE 754 binary 16. In all other cases, subnormals may be flushed to zero.

+

Version History

- Revision 1, 2025-04-16 (Kévin Petit)
  - Initial draft

VK_EXT_shader_image_atomic_int64

Name String

`VK_EXT_shader_image_atomic_int64`

Extension Type

Device extension

Registered Extension Number

235

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_EXT_shader_image_int64

Contact

- Tobias Hector [GitHub]tobski

Other Extension Metadata

Last Modified Date

2020-07-14

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for `GLSL_EXT_shader_image_int64`

Contributors

- Matthaeus Chajdas, AMD
- Graham Wihlidal, Epic Games
- Tobias Hector, AMD
- Jeff Bolz, Nvidia
- Faith Ekstrand, Intel

Description

This extension extends existing 64-bit integer atomic support to enable these operations on images as well.

When working with large 2- or 3-dimensional data sets (e.g. rasterization or screen-space effects), image accesses are generally more efficient than equivalent buffer accesses. This extension allows applications relying on 64-bit integer atomics in this manner to quickly improve performance with only relatively minor code changes.

64-bit integer atomic support is guaranteed for optimally tiled images with the VK_FORMAT_R64_UINT and VK_FORMAT_R64_SINT formats.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT

New Enum Constants

- VK_EXT_SHADER_IMAGE_ATOMIC_INT64_EXTENSION_NAME
- VK_EXT_SHADER_IMAGE_ATOMIC_INT64_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_IMAGE_ATOMIC_INT64_FEATURES_EXT

Version History

- Revision 1, 2020-07-14 (Tobias Hector)
  - Initial draft

VK_EXT_shader_long_vector

Name String

`VK_EXT_shader_long_vector`

Extension Type

Device extension

Registered Extension Number

636

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.2

API Interactions

- Interacts with VK_KHR_workgroup_memory_explicit_layout

SPIR-V Dependencies

- SPV_EXT_long_vector

Contact

- Jeff Bolz [GitHub]jeffbolznv

Extension Proposal

VK_EXT_shader_long_vector

Other Extension Metadata

Last Modified Date

2025-06-24

Interactions and External Dependencies

- This extension requires `SPV_EXT_long_vector`
- This extension provides API support for `GL_EXT_long_vector`

Contributors

- Jeff Bolz, NVIDIA
- Spencer Fricke, LunarG

Description

This extension adds support for using vector types with more than four components in SPIR-V.

Long vector types are defined by the `SPV_EXT_long_vector` SPIR-V extension and can be used with

the GL_EXT_long_vector GLSL extension.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderLongVectorFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceShaderLongVectorPropertiesEXT

New Enum Constants

- VK_EXT_SHADER_LONG_VECTOR_EXTENSION_NAME
- VK_EXT_SHADER_LONG_VECTOR_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_LONG_VECTOR_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_LONG_VECTOR_PROPERTIES_EXT

New SPIR-V Capabilities

- LongVectorEXT

Issues

Version History

- Revision 1, 2025-06-24 (Jeff Bolz)
  - Initial revisions

VK_EXT_shader_module_identifier

Name String

`VK_EXT_shader_module_identifier`

Extension Type

Device extension

Registered Extension Number

463

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1
and
`VK_EXT_pipeline_creation_cache_control`
or
Vulkan Version 1.3

Contact

- Hans-Kristian Arntzen [GitHub]HansKristian-Work

Extension Proposal

VK_EXT_shader_module_identifier

Other Extension Metadata

Last Modified Date

2022-05-16

IP Status

No known IP claims.

Contributors

- Hans-Kristian Arntzen, Valve
- Ricardo Garcia, Igalia
- Piers Daniell, NVIDIA
- Jan-Harald Fredriksen, Arm
- Tom Olson, Arm
- Faith Ekstrand, Collabora

Description

Some applications generate SPIR-V code at runtime. When pipeline caches are primed, either explicitly through e.g. VkPipelineCache mechanisms, or implicitly through driver managed caches, having to re-generate SPIR-V modules is redundant. SPIR-V modules could be cached on disk by an application, but the extra disk size requirement might be prohibitive in some use cases.

This extension adds the ability for an application to query a small identifier associated with a VkShaderModule. On subsequent runs of the application, the same identifier **can** be provided in lieu of a VkShaderModule object. A pipeline creation call with such a module **may** succeed if a pipeline could be created without invoking compilation, and information inside the SPIR-V module is not required by the implementation.

VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT **must** be used if only the identifier is provided, and this use case is intended to work like a non-blocking, speculative compile. Applications **can** fallback as necessary.

The main motivation for identifying the module itself and not the entire pipeline is that pipeline identifiers change when a driver is updated, but module identifiers are expected to be stable for any particular driver implementation. This approach is helpful for shader pre-compilation systems which can prime pipeline caches ahead of time. When on-disk pipeline caches are updated, the same shader identifiers could lead to a pipeline cache hit.

New Commands

- vkGetShaderModuleCreateInfoIdentifierEXT
- vkGetShaderModuleIdentifierEXT

New Structures

- VkShaderModuleIdentifierEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderModuleIdentifierFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceShaderModuleIdentifierPropertiesEXT
- Extending VkPipelineShaderStageCreateInfo:
  - VkPipelineShaderStageModuleIdentifierCreateInfoEXT

New Enum Constants

- `VK_EXT_SHADER_MODULE_IDENTIFIER_EXTENSION_NAME`
- `VK_EXT_SHADER_MODULE_IDENTIFIER_SPEC_VERSION`
- VK_MAX_SHADER_MODULE_IDENTIFIER_SIZE_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_MODULE_IDENTIFIER_FEATURES_EXT

- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_MODULE_IDENTIFIER_PROPERTIES_EXT
- VK_STRUCTURE_TYPE_PIPELINE_SHADER_STAGE_MODULE_IDENTIFIER_CREATE_INFO_EXT
- VK_STRUCTURE_TYPE_SHADER_MODULE_IDENTIFIER_EXT

Version History

- Revision 1, 2022-03-16 (Hans-Kristian Arntzen)
  - Initial draft

VK_EXT_shader_object

Name String

`VK_EXT_shader_object`

Extension Type

Device extension

Registered Extension Number

483

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1
and
VK_KHR_dynamic_rendering
or
Vulkan Version 1.3

API Interactions

- Interacts with VK_VERSION_1_1
- Interacts with VK_VERSION_1_3
- Interacts with VK_VERSION_1_4
- Interacts with VK_EXT_blend_operation_advanced
- Interacts with VK_EXT_conservative_rasterization
- Interacts with VK_EXT_depth_clamp_control
- Interacts with VK_EXT_depth_clip_control
- Interacts with VK_EXT_depth_clip_enable
- Interacts with VK_EXT_fragment_density_map
- Interacts with VK_EXT_line_rasterization
- Interacts with VK_EXT_mesh_shader
- Interacts with VK_EXT_provoking_vertex
- Interacts with VK_EXT_sample_locations
- Interacts with VK_EXT_subgroup_size_control
- Interacts with VK_EXT_transform_feedback

- Interacts with VK_KHR_device_group
- Interacts with VK_KHR_fragment_shading_rate
- Interacts with VK_KHR_line_rasterization
- Interacts with VK_NV_clip_space_w_scaling
- Interacts with VK_NV_coverage_reduction_mode
- Interacts with VK_NV_fragment_coverage_to_color
- Interacts with VK_NV_framebuffer_mixed_samples
- Interacts with VK_NV_mesh_shader
- Interacts with VK_NV_representative_fragment_test
- Interacts with VK_NV_shading_rate_image
- Interacts with VK_NV_viewport_swizzle

Contact

- Daniel Story [GitHub]daniel-story

Extension Proposal

VK_EXT_shader_object

Other Extension Metadata

Last Modified Date

2023-03-30

Interactions and External Dependencies

- Interacts with `VK_EXT_extended_dynamic_state`
- Interacts with `VK_EXT_extended_dynamic_state2`
- Interacts with `VK_EXT_extended_dynamic_state3`
- Interacts with `VK_EXT_vertex_input_dynamic_state`

IP Status

No known IP claims.

Contributors

- Piers Daniell, NVIDIA
- Sandy Jamieson, Nintendo
- Žiga Markuš, LunarG
- Tobias Hector, AMD
- Alex Walters, Imagination
- Shahbaz Youssefi, Google
- Ralph Potter, Samsung
- Jan-Harald Fredriksen, ARM

- Connor Abott, Valve
- Arseny Kapoulkine, Roblox
- Patrick Doane, Activision
- Jeff Leger, Qualcomm
- Stu Smith, AMD
- Chris Glover, Google
- Ricardo Garcia, Igalia
- Faith Ekstrand, Collabora
- Timur Kristóf, Valve
- Caterina Shablia, Collabora
- Daniel Koch, NVIDIA
- Alyssa Rosenzweig, Collabora
- Mike Blumenkrantz, Valve
- Samuel Pitoiset, Valve
- Qun Lin, AMD
- Spencer Fricke, LunarG
- Soroush Faghihi Kashani, Imagination

Description

This extension introduces a new VkShaderEXT object type which represents a single compiled shader stage. Shader objects provide a more flexible alternative to VkPipeline objects, which may be helpful in certain use cases.

New Object Types

- VkShaderEXT

New Commands

- vkCmdBindShadersEXT
- vkCmdBindVertexBuffers2EXT
- vkCmdSetAlphaToCoverageEnableEXT
- vkCmdSetAlphaToOneEnableEXT
- vkCmdSetColorBlendEnableEXT
- vkCmdSetColorBlendEquationEXT
- vkCmdSetColorWriteMaskEXT
- vkCmdSetCullModeEXT
- vkCmdSetDepthBiasEnableEXT

- vkCmdSetDepthBoundsTestEnableEXT
- vkCmdSetDepthClampEnableEXT
- vkCmdSetDepthCompareOpEXT
- vkCmdSetDepthTestEnableEXT
- vkCmdSetDepthWriteEnableEXT
- vkCmdSetFrontFaceEXT
- vkCmdSetLogicOpEXT
- vkCmdSetLogicOpEnableEXT
- vkCmdSetPatchControlPointsEXT
- vkCmdSetPolygonModeEXT
- vkCmdSetPrimitiveRestartEnableEXT
- vkCmdSetPrimitiveTopologyEXT
- vkCmdSetRasterizationSamplesEXT
- vkCmdSetRasterizerDiscardEnableEXT
- vkCmdSetSampleMaskEXT
- vkCmdSetScissorWithCountEXT
- vkCmdSetStencilOpEXT
- vkCmdSetStencilTestEnableEXT
- vkCmdSetTessellationDomainOriginEXT
- vkCmdSetVertexInputEXT
- vkCmdSetViewportWithCountEXT
- vkCreateShadersEXT
- vkDestroyShaderEXT
- vkGetShaderBinaryDataEXT

If VK_EXT_blend_operation_advanced is supported:

- vkCmdSetColorBlendAdvancedEXT

If VK_EXT_conservative_rasterization is supported:

- vkCmdSetConservativeRasterizationModeEXT
- vkCmdSetExtraPrimitiveOverestimationSizeEXT

If VK_EXT_depth_clamp_control is supported:

- vkCmdSetDepthClampRangeEXT

If VK_EXT_depth_clip_control is supported:

- vkCmdSetDepthClipNegativeOneToOneEXT

If VK_EXT_depth_clip_enable is supported:

- vkCmdSetDepthClipEnableEXT

If VK_EXT_provoking_vertex is supported:

- vkCmdSetProvokingVertexModeEXT

If VK_EXT_sample_locations is supported:

- vkCmdSetSampleLocationsEnableEXT

If VK_EXT_transform_feedback is supported:

- vkCmdSetRasterizationStreamEXT

If Vulkan Version 1.4 or VK_KHR_line_rasterization or `VK_EXT_line_rasterization` is supported:

- vkCmdSetLineRasterizationModeEXT
- vkCmdSetLineStippleEnableEXT

New Structures

- VkColorBlendAdvancedEXT
- VkColorBlendEquationEXT
- VkShaderCreateInfoEXT
- VkVertexInputAttributeDescription2EXT
- VkVertexInputBindingDescription2EXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderObjectFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceShaderObjectPropertiesEXT
- Extending VkPipelineShaderStageCreateInfo, VkShaderCreateInfoEXT:
  - VkShaderRequiredSubgroupSizeCreateInfoEXT

New Enums

- VkShaderCodeTypeEXT
- VkShaderCreateFlagBitsEXT

New Bitmasks

- VkShaderCreateFlagsEXT

New Enum Constants

- `VK_EXT_SHADER_OBJECT_EXTENSION_NAME`
- `VK_EXT_SHADER_OBJECT_SPEC_VERSION`
- Extending VkObjectType:
  - VK_OBJECT_TYPE_SHADER_EXT
- Extending VkResult:
  - VK_ERROR_INCOMPATIBLE_SHADER_BINARY_EXT
  - VK_INCOMPATIBLE_SHADER_BINARY_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_OBJECT_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_OBJECT_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_SHADER_REQUIRED_SUBGROUP_SIZE_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_VERTEX_INPUT_ATTRIBUTE_DESCRIPTION_2_EXT
  - VK_STRUCTURE_TYPE_VERTEX_INPUT_BINDING_DESCRIPTION_2_EXT

If VK_EXT_fragment_density_map is supported:

- Extending VkShaderCreateFlagBitsEXT:
  - VK_SHADER_CREATE_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT

If VK_EXT_mesh_shader or `VK_NV_mesh_shader` is supported:

- Extending VkShaderCreateFlagBitsEXT:
  - VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT

If `VK_EXT_subgroup_size_control` or Vulkan Version 1.3 is supported:

- Extending VkShaderCreateFlagBitsEXT:
  - VK_SHADER_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT_EXT
  - VK_SHADER_CREATE_REQUIRE_FULL_SUBGROUPS_BIT_EXT

If VK_KHR_device_group or Vulkan Version 1.1 is supported:

- Extending VkShaderCreateFlagBitsEXT:
  - VK_SHADER_CREATE_DISPATCH_BASE_BIT_EXT

If VK_KHR_fragment_shading_rate is supported:

- Extending VkShaderCreateFlagBitsEXT:
  - VK_SHADER_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_EXT

Examples

Example 1

Create linked pair of vertex and fragment shaders.

```
// Logical device created with the shaderObject feature enabled
VkDevice device;

// SPIR-V shader code for a vertex shader, along with its size in bytes
void* pVertexSpirv;
size_t vertexSpirvSize;

// SPIR-V shader code for a fragment shader, along with its size in bytes
void* pFragmentSpirv;
size_t fragmentSpirvSize;

// Descriptor set layout compatible with the shaders
VkDescriptorSetLayout descriptorSetLayout;

VkShaderCreateInfoEXT shaderCreateInfos[2] =
{
    {
        .sType = VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = VK_SHADER_CREATE_LINK_STAGE_BIT_EXT,
        .stage = VK_SHADER_STAGE_VERTEX_BIT,
        .nextStage = VK_SHADER_STAGE_FRAGMENT_BIT,
        .codeType = VK_SHADER_CODE_TYPE_SPIRV_EXT,
        .codeSize = vertexSpirvSize,
        .pCode = pVertexSpirv,
        .pName = "main",
        .setLayoutCount = 1,
        .pSetLayouts = &descriptorSetLayout;
        .pushConstantRangeCount = 0,
        .pPushConstantRanges = NULL,
        .pSpecializationInfo = NULL
    },
    {
        .sType = VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = VK_SHADER_CREATE_LINK_STAGE_BIT_EXT,
        .stage = VK_SHADER_STAGE_FRAGMENT_BIT,
        .nextStage = 0,
        .codeType = VK_SHADER_CODE_TYPE_SPIRV_EXT,
        .codeSize = fragmentSpirvSize,
        .pCode = pFragmentSpirv,
        .pName = "main",
        .setLayoutCount = 1,
        .pSetLayouts = &descriptorSetLayout;
        .pushConstantRangeCount = 0,
```

```
        .pPushConstantRanges = NULL,
        .pSpecializationInfo = NULL
    }
};

VkResult result;
VkShaderEXT shaders[2];

result = vkCreateShadersEXT(device, 2, &shaderCreateInfos, NULL, shaders);
if (result != VK_SUCCESS)
{
    // Handle error
}
```

Later, during command buffer recording, bind the linked shaders and draw.

```
// Command buffer in the recording state
VkCommandBuffer commandBuffer;

// Vertex and fragment shader objects created above
VkShaderEXT shaders[2];

// Assume vertex buffers, descriptor sets, etc. have been bound, and existing
// state setting commands have been called to set all required state

const VkShaderStageFlagBits stages[2] =
{
    VK_SHADER_STAGE_VERTEX_BIT,
    VK_SHADER_STAGE_FRAGMENT_BIT
};

// Bind linked shaders
vkCmdBindShadersEXT(commandBuffer, 2, stages, shaders);

// Equivalent to the previous line. Linked shaders can be bound one at a time,
// in any order:
// vkCmdBindShadersEXT(commandBuffer, 1, &stages[1], &shaders[1]);
// vkCmdBindShadersEXT(commandBuffer, 1, &stages[0], &shaders[0]);

// The above is sufficient to draw if the device was created with the
// tessellationShader and geometryShader features disabled. Otherwise, since
// those stages should not execute, vkCmdBindShadersEXT() must be called at
// least once with each of their stages in pStages before drawing:

const VkShaderStageFlagBits unusedStages[3] =
{
    VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT,
    VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT,
    VK_SHADER_STAGE_GEOMETRY_BIT
};
```

```
// NULL pShaders is equivalent to an array of stageCount VK_NULL_HANDLE values,
// meaning no shaders are bound to those stages, and that any previously bound
// shaders are unbound
vkCmdBindShadersEXT(commandBuffer, 3, unusedStages, NULL);

// Graphics shader objects may only be used to draw inside dynamic render pass
// instances begun with vkCmdBeginRendering(), assume one has already been begun

// Draw a triangle
vkCmdDraw(commandBuffer, 3, 1, 0, 0);
```

Example 2

Create unlinked vertex, geometry, and fragment shaders.

```
// Logical device created with the shaderObject feature enabled
VkDevice device;

// SPIR-V shader code for vertex shaders, along with their sizes in bytes
void* pVertexSpirv[2];
size_t vertexSpirvSize[2];

// SPIR-V shader code for a geometry shader, along with its size in bytes
void pGeometrySpirv;
size_t geometrySpirvSize;

// SPIR-V shader code for fragment shaders, along with their sizes in bytes
void* pFragmentSpirv[2];
size_t fragmentSpirvSize[2];

// Descriptor set layout compatible with the shaders
VkDescriptorSetLayout descriptorSetLayout;

VkShaderCreateInfoEXT shaderCreateInfos[5] =
{
    // Stage order does not matter
    {
        .sType = VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = 0,
        .stage = VK_SHADER_STAGE_GEOMETRY_BIT,
        .nextStage = VK_SHADER_STAGE_FRAGMENT_BIT,
        .codeType = VK_SHADER_CODE_TYPE_SPIRV_EXT,
        .codeSize = pGeometrySpirv,
        .pCode = geometrySpirvSize,
        .pName = "main",
        .setLayoutCount = 1,
        .pSetLayouts = &descriptorSetLayout;
        .pushConstantRangeCount = 0,
```

```
        .pPushConstantRanges = NULL,
        .pSpecializationInfo = NULL
    },
    {
        .sType = VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = 0,
        .stage = VK_SHADER_STAGE_VERTEX_BIT,
        .nextStage = VK_SHADER_STAGE_GEOMETRY_BIT,
        .codeType = VK_SHADER_CODE_TYPE_SPIRV_EXT,
        .codeSize = vertexSpirvSize[0],
        .pCode = pVertexSpirv[0],
        .pName = "main",
        .setLayoutCount = 1,
        .pSetLayouts = &descriptorSetLayout;
        .pushConstantRangeCount = 0,
        .pPushConstantRanges = NULL,
        .pSpecializationInfo = NULL
    },
    {
        .sType = VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = 0,
        .stage = VK_SHADER_STAGE_FRAGMENT_BIT,
        .nextStage = 0,
        .codeType = VK_SHADER_CODE_TYPE_SPIRV_EXT,
        .codeSize = fragmentSpirvSize[0],
        .pCode = pFragmentSpirv[0],
        .pName = "main",
        .setLayoutCount = 1,
        .pSetLayouts = &descriptorSetLayout;
        .pushConstantRangeCount = 0,
        .pPushConstantRanges = NULL,
        .pSpecializationInfo = NULL
    },
    {
        .sType = VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = 0,
        .stage = VK_SHADER_STAGE_FRAGMENT_BIT,
        .nextStage = 0,
        .codeType = VK_SHADER_CODE_TYPE_SPIRV_EXT,
        .codeSize = fragmentSpirvSize[1],
        .pCode = pFragmentSpirv[1],
        .pName = "main",
        .setLayoutCount = 1,
        .pSetLayouts = &descriptorSetLayout;
        .pushConstantRangeCount = 0,
        .pPushConstantRanges = NULL,
        .pSpecializationInfo = NULL
    },
```

```
    {
        .sType = VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = 0,
        .stage = VK_SHADER_STAGE_VERTEX_BIT,
        // Suppose we want this vertex shader to be able to be followed by
        // either a geometry shader or fragment shader:
        .nextStage = VK_SHADER_STAGE_GEOMETRY_BIT | VK_SHADER_STAGE_FRAGMENT_BIT,
        .codeType = VK_SHADER_CODE_TYPE_SPIRV_EXT,
        .codeSize = vertexSpirvSize[1],
        .pCode = pVertexSpirv[1],
        .pName = "main",
        .setLayoutCount = 1,
        .pSetLayouts = &descriptorSetLayout;
        .pushConstantRangeCount = 0,
        .pPushConstantRanges = NULL,
        .pSpecializationInfo = NULL
    }
};

VkResult result;
VkShaderEXT shaders[5];

result = vkCreateShadersEXT(device, 5, &shaderCreateInfos, NULL, shaders);
if (result != VK_SUCCESS)
{
    // Handle error
}
```

Later, during command buffer recording, bind the linked shaders in different combinations and draw.

```
// Command buffer in the recording state
VkCommandBuffer commandBuffer;

// Vertex, geometry, and fragment shader objects created above
VkShaderEXT shaders[5];

// Assume vertex buffers, descriptor sets, etc. have been bound, and existing
// state setting commands have been called to set all required state

const VkShaderStageFlagBits stages[3] =
{
    // Any order is allowed
    VK_SHADER_STAGE_FRAGMENT_BIT,
    VK_SHADER_STAGE_VERTEX_BIT,
    VK_SHADER_STAGE_GEOMETRY_BIT,
};

VkShaderEXT bindShaders[3] =
```

```
{
    shaders[2], // FS
    shaders[1], // VS
    shaders[0]  // GS
};

// Bind unlinked shaders
vkCmdBindShadersEXT(commandBuffer, 3, stages, bindShaders);

// Assume the tessellationShader feature is disabled, so vkCmdBindShadersEXT()
// need not have been called with either tessellation stage

// Graphics shader objects may only be used to draw inside dynamic render pass
// instances begun with vkCmdBeginRendering(), assume one has already been begun

// Draw a triangle
vkCmdDraw(commandBuffer, 3, 1, 0, 0);

// Bind a different unlinked fragment shader
const VkShaderStageFlagBits fragmentStage = VK_SHADER_STAGE_FRAGMENT_BIT;
vkCmdBindShadersEXT(commandBuffer, 1, &fragmentStage, &shaders[3]);

// Draw another triangle
vkCmdDraw(commandBuffer, 3, 1, 0, 0);

// Bind a different unlinked vertex shader
const VkShaderStageFlagBits vertexStage = VK_SHADER_STAGE_VERTEX_BIT;
vkCmdBindShadersEXT(commandBuffer, 1, &vertexStage, &shaders[4]);

// Draw another triangle
vkCmdDraw(commandBuffer, 3, 1, 0, 0);
```

Version History

- Revision 1, 2023-03-30 (Daniel Story)
    - Initial draft

VK_EXT_shader_ocp_microscaling_types

Name String

`VK_EXT_shader_ocp_microscaling_types`

Extension Type

Device extension

Registered Extension Number

673

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_KHR_cooperative_matrix

SPIR-V Dependencies

- SPV_EXT_ocp_microscaling_types

Contact

- Kevin Petit [GitHub]kpet

Extension Proposal

VK_EXT_shader_ocp_microscaling_types

Other Extension Metadata

Last Modified Date

2026-04-29

IP Status

No known IP claims.

Contributors

- Kévin Petit, Arm Ltd.
- Vikram Tarikere, Imagination Technologies
- Jeff Bolz, NVIDIA
- Craig Graham, Samsung

Description

This extension enables support for the floating-point data types defined in the Open Compute Project (OCP) Microscaling Formats specification.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderOCPMicroscalingTypesFeaturesEXT

New Enum Constants

- `VK_EXT_SHADER_OCP_MICROSCALING_TYPES_EXTENSION_NAME`
- `VK_EXT_SHADER_OCP_MICROSCALING_TYPES_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_OCP_MICROSCALING_TYPES_FEATURES_EXT

If VK_KHR_cooperative_matrix is supported:

- Extending VkComponentTypeKHR:
  - VK_COMPONENT_TYPE_FLOAT4_E2M1_EXT
  - VK_COMPONENT_TYPE_FLOAT6_E2M3_EXT
  - VK_COMPONENT_TYPE_FLOAT6_E3M2_EXT
  - VK_COMPONENT_TYPE_FLOAT8_UNSIGNED_E8M0_EXT
  - VK_COMPONENT_TYPE_MXINT8_EXT

New SPIR-V Capabilities

- Float4EXT
- Float6EXT
- Float8UnsignedE8M0EXT
- MXInt8EXT
- BitcastExtractEXT

Issues

1) How does this extension add support for microscaling?

RESOLVED: It does not directly do so. This extension just introduces basic support for the data types used for microscaling but it does not add any support for microscaled tensors or operations on microscaled data. Reading miscroscaled data and converting it to other formats such as FP16 or FP32 can be implemented in shaders with the new types provided by this extension but these new types cannot for example be used directly in matrix multiplication operations.

2) What conversions to/from the newly-added types should be supported?

RESOLVED: This extension will only add conversions from the newly-added types to a restricted list of existing larger floating-point types. Conversions from existing types to the newly-added types would require complex logic and are not expected to be as useful. Conversions between integer types and the newly-added types are not expected to be generally useful either.

Version History

- Revision 1, 2026-04-29 (Kévin Petit)
  - Initial draft

VK_EXT_shader_replicated_composites

Name String

`VK_EXT_shader_replicated_composites`

Extension Type

Device extension

Registered Extension Number

565

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_EXT_replicated_composites

Contact

- Kevin Petit [GitHub]kpet

Extension Proposal

VK_EXT_shader_replicated_composites

Last Modified Date

2024-02-08

IP Status

No known IP claims.

Contributors

- Kévin Petit, Arm Ltd.
- Jeff Bolz, NVIDIA
- Piers Daniell, NVIDIA

This extension adds support for creating composites from a single value in SPIR-V modules, as defined by SPV_EXT_replicated_composites.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:

    - VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT

New Enum Constants

- VK_EXT_SHADER_REPLICATED_COMPOSITES_EXTENSION_NAME
- VK_EXT_SHADER_REPLICATED_COMPOSITES_SPEC_VERSION
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_REPLICATED_COMPOSITES_FEATURES_EXT

New SPIR-V Capabilities

- ReplicatedCompositesEXT

Version History

- Revision 1, 2024-02-08 (Kévin Petit)
    - Initial revision

VK_EXT_shader_split_barrier

Name String

`VK_EXT_shader_split_barrier`

Extension Type

Device extension

Registered Extension Number

306

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_EXT_split_barrier

Contact

- Matthew Netsch [GitHub]mnetsch

Extension Proposal

VK_EXT_shader_split_barrier

Other Extension Metadata

Last Modified Date

2026-05-08

Interactions and External Dependencies

- This extension provides API support for `GLSL_EXT_split_barrier`

Contributors

- Matthew Netsch, Qualcomm Technologies, Inc.
- Elina Kamenetskaya, Qualcomm Technologies, Inc.
- Wooyoung Kim, Qualcomm Technologies, Inc.
- John Li, Qualcomm Technologies, Inc.
- Jeff Bolz, Nvidia
- Ben Ashbaugh, Intel

Description

This extension splits `OpControlBarrier` by exposing two new barrier operations with `SPV_EXT_split_barrier`:

- `OpControlBarrierArriveEXT` - notifies that invocation has arrived here
- `OpControlBarrierWaitEXT` - waits on all invocations before proceeding execution

In the Vulkan context, this allows apps to synchronize Subgroup execution flow within a Workgroup without requiring all Subgroups to wait at the arrival condition before proceeding to execute independent work. It also permits synchronizing memory access like other control barriers.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderSplitBarrierFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceShaderSplitBarrierPropertiesEXT

New Enum Constants

- `VK_EXT_SHADER_SPLIT_BARRIER_EXTENSION_NAME`
- `VK_EXT_SHADER_SPLIT_BARRIER_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SPLIT_BARRIER_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SPLIT_BARRIER_PROPERTIES_EXT

Version History

- Revision 1, 2026-05-08 (Matthew Netsch)
  - Initial version

VK_EXT_shader_stencil_export

Name String

`VK_EXT_shader_stencil_export`

Extension Type

Device extension

Registered Extension Number

141

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

SPIR-V Dependencies

- SPV_EXT_shader_stencil_export

Contact

- Dominik Witczak [GitHub]dominikwitczakamd

Other Extension Metadata

Last Modified Date

2017-07-19

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for `GL_ARB_shader_stencil_export`

Contributors

- Dominik Witczak, AMD
- Daniel Rakos, AMD
- Rex Xu, AMD

Description

This extension adds support for the SPIR-V extension `SPV_EXT_shader_stencil_export`, providing a mechanism whereby a shader may generate the stencil reference value per invocation. When stencil testing is enabled, this allows the test to be performed against the value generated in the shader.

New Enum Constants

- `VK_EXT_SHADER_STENCIL_EXPORT_EXTENSION_NAME`
- `VK_EXT_SHADER_STENCIL_EXPORT_SPEC_VERSION`

Version History

- Revision 1, 2017-07-19 (Dominik Witczak)
  - Initial draft

VK_EXT_shader_subgroup_partitioned

Name String

VK_EXT_shader_subgroup_partitioned

Extension Type

Device extension

Registered Extension Number

663

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_EXT_shader_subgroup_partitioned
- SPV_NV_shader_subgroup_partitioned

Contact

- Jeff Bolz [GitHub]jeffbolznv

Other Extension Metadata

Last Modified Date

2025-11-12

Interactions and External Dependencies

- This extension provides API support for GL_NV_shader_subgroup_partitioned

Contributors

- Jeff Bolz, NVIDIA

Description

This extension enables support for a new class of group operations on subgroups via the GL_NV_shader_subgroup_partitioned GLSL extension and SPV_EXT_shader_subgroup_partitioned SPIR-V extension. Support for these new operations is advertised via the VK_SUBGROUP_FEATURE_PARTITIONED_BIT_EXT bit. Note that there is not a new GLSL extension since SPV_EXT_shader_subgroup_partitioned is compatible with SPV_NV_shader_subgroup_partitioned, so the GL_NV_shader_subgroup_partitioned GLSL extension can still be used.

This extension requires Vulkan 1.1, for general subgroup support.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderSubgroupPartitionedFeaturesEXT

New Enum Constants

- `VK_EXT_SHADER_SUBGROUP_PARTITIONED_EXTENSION_NAME`
- `VK_EXT_SHADER_SUBGROUP_PARTITIONED_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_PARTITIONED_FEATURES_EXT
- Extending VkSubgroupFeatureFlagBits:
  - VK_SUBGROUP_FEATURE_PARTITIONED_BIT_EXT

Version History

- Revision 1, 2025-11-12 (Jeff Bolz)
  - Promoted from `VK_NV_shader_subgroup_partitioned`

VK_EXT_shader_tile_image

Name String

`VK_EXT_shader_tile_image`

Extension Type

Device extension

Registered Extension Number

396

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.3

SPIR-V Dependencies

- SPV_EXT_shader_tile_image

Contact

- Jan-Harald Fredriksen [GitHub]janharaldfredriksen-arm

Extension Proposal

VK_EXT_shader_tile_image

Other Extension Metadata

Last Modified Date

2023-03-23

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for `GL_EXT_shader_tile_image`

Contributors

- Sandeep Kakarlapudi, Arm
- Jan-Harald Fredriksen, Arm
- James Fitzpatrick, Imagination
- Andrew Garrard, Imagination
- Jeff Leger, Qualcomm
- Huilong Wang, Huawei

- Graeme Leese, Broadcom
- Hans-Kristian Arntzen, Valve
- Tobias Hector, AMD
- Jeff Bolz, NVIDIA
- Shahbaz Youssefi, Google

Description

This extension allows fragment shader invocations to read color, depth, and stencil values at their pixel location in rasterization order. The functionality is only available when using dynamic render passes introduced by VK_KHR_dynamic_rendering. Example use cases are programmable blending and deferred shading.

See fragment shader tile image reads for more information.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderTileImageFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceShaderTileImagePropertiesEXT

New Enum Constants

- VK_EXT_SHADER_TILE_IMAGE_EXTENSION_NAME
- VK_EXT_SHADER_TILE_IMAGE_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TILE_IMAGE_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TILE_IMAGE_PROPERTIES_EXT

Issues

None.

Examples

Color read example.

```
layout( location = 0 /* aliased to color attachment 0 */ ) tileImageEXT highp
attachmentEXT color0;
layout( location = 1 /* aliased to color attachment 1 */ ) tileImageEXT highp
attachmentEXT color1;

layout( location = 0 ) out vec4 fragColor;
```

```
void main()
{
    vec4 value = colorAttachmentReadEXT(color0) + colorAttachmentReadEXT(color1);
    fragColor = value;
}
```

Depth & Stencil read example.

```
void main()
{
    // read sample 0: works for non-MSAA or MSAA targets
    highp float last_depth = depthAttachmentReadEXT();
    lowp uint last_stencil = stencilAttachmentReadEXT();

    //..
}
```

Version History

- Revision 1, 2023-03-23 (Sandeep Kakarlapudi)
    - Initial version

VK_EXT_shader_uniform_buffer_unsized_array

Name String

`VK_EXT_shader_uniform_buffer_unsized_array`

Extension Type

Device extension

Registered Extension Number

643

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Piers Daniell [GitHub]pdaniell-nv

Extension Proposal

VK_EXT_shader_uniform_buffer_unsized_array

Other Extension Metadata

Last Modified Date

2025-05-28

Contributors

- Pat Brown, NVIDIA
- Jeff Bolz, NVIDIA
- Pranjal Dubey, NVIDIA

Description

This extension allows the last member of a uniform buffer block to be declared as an unsized array. This capability enables applications to create flexible buffer layouts where the array size can vary based on runtime requirements.

When the last member of a uniform buffer block is declared as an unsized array, the effective array size is inferred at runtime from the size of the buffer object backing the uniform buffer block. Such unsized arrays can be indexed with general integer expressions, but may not be passed as arguments to functions or indexed with negative constant expressions.

This extension leverages existing SPIR-V capabilities, allowing use of `OpTypeRuntimeArray` as the last member of a uniform buffer block structure while prohibiting `OpArrayLength`.

Applications needing to know array sizes at runtime should calculate the length and pass it to shaders via a separate uniform. This calculation can be performed using the formula: max((buffer_object_size - offset_of_array) / stride_of_array, 0), where buffer_object_size is the size of the bound buffer, offset_of_array is the byte offset of the array in the block, and stride_of_array is the byte stride between consecutive array elements.

Uniform buffers have traditionally required explicit sizes for all arrays which limits flexibility. With this extension, developers can create a single shader that adapts to different data set sizes at runtime by binding differently sized buffers.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderUniformBufferUnsizedArrayFeaturesEXT

New Enum Constants

- `VK_EXT_SHADER_UNIFORM_BUFFER_UNSIZED_ARRAY_EXTENSION_NAME`
- `VK_EXT_SHADER_UNIFORM_BUFFER_UNSIZED_ARRAY_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_UNIFORM_BUFFER_UNSIZED_ARRAY_FEATURES_EXT

Version History

- Revision 1, 2025-05-28 (Pranjal Dubey)
  - Initial draft

VK_EXT_subpass_merge_feedback

Name String

`VK_EXT_subpass_merge_feedback`

Extension Type

Device extension

Registered Extension Number

459

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Ting Wei [GitHub]catweiting

Extension Proposal

VK_EXT_subpass_merge_feedback

Other Extension Metadata

Last Modified Date

2022-05-24

IP Status

No known IP claims.

Contributors

- Jan-Harald Fredriksen, Arm
- Jorg Wagner, Arm
- Ting Wei, Arm

Description

This extension adds a mechanism to provide feedback to an application about whether the subpasses specified on render pass creation are merged by the implementation. Additionally, it provides a control to enable or disable subpass merging in the render pass.

New Structures

- VkRenderPassCreationFeedbackInfoEXT
- VkRenderPassSubpassFeedbackInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceSubpassMergeFeedbackFeaturesEXT
- Extending VkRenderPassCreateInfo2:
  - VkRenderPassCreationFeedbackCreateInfoEXT
- Extending VkRenderPassCreateInfo2, VkSubpassDescription2:
  - VkRenderPassCreationControlEXT
- Extending VkSubpassDescription2:
  - VkRenderPassSubpassFeedbackCreateInfoEXT

New Enums

- VkSubpassMergeStatusEXT

New Enum Constants

- `VK_EXT_SUBPASS_MERGE_FEEDBACK_EXTENSION_NAME`
- `VK_EXT_SUBPASS_MERGE_FEEDBACK_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SUBPASS_MERGE_FEEDBACK_FEATURES_EXT
  - VK_STRUCTURE_TYPE_RENDER_PASS_CREATION_CONTROL_EXT
  - VK_STRUCTURE_TYPE_RENDER_PASS_CREATION_FEEDBACK_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_RENDER_PASS_SUBPASS_FEEDBACK_CREATE_INFO_EXT

Version History

- Revision 1, 2022-03-10
  - Initial draft.
- Revision 2, 2022-05-24
  - Fix structextends and constness issues.

VK_EXT_swapchain_colorspace

Name String

VK_EXT_swapchain_colorspace

Extension Type

Instance extension

Registered Extension Number

105

Revision

5

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_surface

Contact

- Courtney Goeltzenleuchter [GitHub]courtney-g

Other Extension Metadata

Last Modified Date

2019-04-26

IP Status

No known IP claims.

Contributors

- Courtney Goeltzenleuchter, Google

Description

This extension expands VkColorSpaceKHR to add support for most standard color spaces beyond VK_COLOR_SPACE_SRGB_NONLINEAR_KHR. This extension also adds support for VK_COLOR_SPACE_PASS_THROUGH_EXT which allows applications to use color spaces not explicitly enumerated in VkColorSpaceKHR.

New Enum Constants

- VK_EXT_SWAPCHAIN_COLOR_SPACE_EXTENSION_NAME
- VK_EXT_SWAPCHAIN_COLOR_SPACE_SPEC_VERSION
- Extending VkColorSpaceKHR:
  - VK_COLOR_SPACE_ADOBERGB_LINEAR_EXT

- VK_COLOR_SPACE_ADOBERGB_NONLINEAR_EXT
- VK_COLOR_SPACE_BT2020_LINEAR_EXT
- VK_COLOR_SPACE_BT709_LINEAR_EXT
- VK_COLOR_SPACE_BT709_NONLINEAR_EXT
- VK_COLOR_SPACE_DCI_P3_LINEAR_EXT
- VK_COLOR_SPACE_DCI_P3_NONLINEAR_EXT
- VK_COLOR_SPACE_DISPLAY_P3_LINEAR_EXT
- VK_COLOR_SPACE_DISPLAY_P3_NONLINEAR_EXT
- VK_COLOR_SPACE_DOLBYVISION_EXT
- VK_COLOR_SPACE_EXTENDED_SRGB_LINEAR_EXT
- VK_COLOR_SPACE_EXTENDED_SRGB_NONLINEAR_EXT
- VK_COLOR_SPACE_HDR10_HLG_EXT
- VK_COLOR_SPACE_HDR10_ST2084_EXT
- VK_COLOR_SPACE_PASS_THROUGH_EXT

Issues

1) Does the spec need to specify which kinds of image formats support the color spaces?

RESOLVED: Pixel format is independent of color space (though some color spaces really want / need floating-point color components to be useful). Therefore, do not plan on documenting what formats support which color spaces. An application **can** call vkGetPhysicalDeviceSurfaceFormatsKHR to query what a particular implementation supports.

2) How does application determine if HW supports appropriate transfer function for a color space?

RESOLVED: Extension indicates that implementation **must** not do the OETF encoding if it is not sRGB. That responsibility falls to the application shaders. Any other native OETF / EOTF functions supported by an implementation can be described by separate extension.

Version History

- Revision 1, 2016-12-27 (Courtney Goeltzenleuchter)
  - Initial version
- Revision 2, 2017-01-19 (Courtney Goeltzenleuchter)
  - Add pass through and multiple options for BT2020.
  - Clean up some issues with equations not displaying properly.
- Revision 3, 2017-06-23 (Courtney Goeltzenleuchter)
  - Add extended sRGB non-linear enum.
- Revision 4, 2019-04-26 (Graeme Leese)
  - Clarify color space transfer function usage.

    - Refer to normative definitions in the Data Format Specification.
    - Clarify DCI-P3 and Display P3 usage.
- Revision 5, 2024-03-16 (Zehui Lin)
    - Fix interchanged concepts of EOTF and OETF.
    - Clarify that the presentation engine can accept the color spaces.

VK_EXT_texture_compression_astc_3d

Name String

`VK_EXT_texture_compression_astc_3d`

Extension Type

Device extension

Registered Extension Number

289

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Jan-Harald Fredriksen [GitHub]janharaldfredriksen-arm

Extension Proposal

VK_EXT_texture_compression_astc_3d

Other Extension Metadata

Last Modified Date

2025-06-09

IP Status

No known issues.

Contributors

- Jan-Harald Fredriksen, Arm

Description

This extension adds support for 3D textures compressed using the Adaptive Scalable Texture Compression (ASTC) format.

These formats are compressed in 3D. As such, each slice will contain data for a Width x Height x Depth block of the texture. All transfer operations are done at the granularity of block dimensions.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceTextureCompressionASTC3DFeaturesEXT

New Enum Constants

- `VK_EXT_TEXTURE_COMPRESSION_ASTC_3D_EXTENSION_NAME`
- `VK_EXT_TEXTURE_COMPRESSION_ASTC_3D_SPEC_VERSION`
- Extending VkFormat:
  - VK_FORMAT_ASTC_3x3x3_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_3x3x3_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_3x3x3_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_4x3x3_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_4x3x3_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_4x3x3_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_4x4x3_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_4x4x3_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_4x4x3_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_4x4x4_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_4x4x4_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_4x4x4_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_5x4x4_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_5x4x4_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_5x4x4_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_5x5x4_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_5x5x4_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_5x5x4_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_5x5x5_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_5x5x5_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_5x5x5_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_6x5x5_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_6x5x5_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_6x5x5_UNORM_BLOCK_EXT
  - VK_FORMAT_ASTC_6x6x5_SFLOAT_BLOCK_EXT
  - VK_FORMAT_ASTC_6x6x5_SRGB_BLOCK_EXT
  - VK_FORMAT_ASTC_6x6x5_UNORM_BLOCK_EXT

    - VK_FORMAT_ASTC_6x6x6_SFLOAT_BLOCK_EXT
    - VK_FORMAT_ASTC_6x6x6_SRGB_BLOCK_EXT
    - VK_FORMAT_ASTC_6x6x6_UNORM_BLOCK_EXT
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TEXTURE_COMPRESSION_ASTC_3D_FEATURES_EXT

Issues

None.

Version History

- Revision 1, 2025-06-09 (Jan-Harald Fredriksen)
    - Initial version

VK_EXT_transform_feedback

Name String

`VK_EXT_transform_feedback`

Extension Type

Device extension

Registered Extension Number

29

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Special Uses

- OpenGL / ES support
- D3D support
- Developer tools

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2018-10-09

Contributors

- Baldur Karlsson, Valve
- Boris Zanin, Mobica
- Daniel Rakos, AMD
- Donald Scorgie, Imagination
- Henri Verbeet, CodeWeavers
- Jan-Harald Fredriksen, Arm
- Faith Ekstrand, Intel
- Jeff Bolz, NVIDIA
- Jesse Barker, Unity

- Jesse Hall, Google
- Pierre-Loup Griffais, Valve
- Philip Rebohle, DXVK
- Ruihao Zhang, Qualcomm
- Samuel Pitoiset, Valve
- Slawomir Grajewski, Intel
- Stu Smith, Imagination Technologies

Description

This extension adds transform feedback to the Vulkan API by exposing the SPIR-V `TransformFeedback` and `GeometryStreams` capabilities to capture vertex, tessellation, or geometry shader outputs to one or more buffers. It adds API functionality to bind transform feedback buffers to capture the primitives emitted by the graphics pipeline from SPIR-V outputs decorated for transform feedback. The transform feedback capture can be paused and resumed by way of storing and retrieving a byte counter. The captured data can be drawn again where the vertex count is derived from the byte counter without CPU intervention. If the implementation is capable, a vertex stream other than zero can be rasterized.

All these features are designed to match the full capabilities of OpenGL core transform feedback functionality and beyond. Many of the features are optional to allow base OpenGL ES GPUs to also implement this extension.

The primary purpose of the functionality exposed by this extension is to support translation layers from other 3D APIs. This functionality is not considered forward looking, and is not expected to be promoted to a KHR extension or to core Vulkan. Unless this is needed for translation, it is recommended that developers use alternative techniques of using the GPU to process and capture vertex data.

New Commands

- vkCmdBeginQueryIndexedEXT
- vkCmdBeginTransformFeedbackEXT
- vkCmdBindTransformFeedbackBuffersEXT
- vkCmdDrawIndirectByteCountEXT
- vkCmdEndQueryIndexedEXT
- vkCmdEndTransformFeedbackEXT

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceTransformFeedbackFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceTransformFeedbackPropertiesEXT

- Extending VkPipelineRasterizationStateCreateInfo:
  - VkPipelineRasterizationStateStreamCreateInfoEXT

New Bitmasks

- VkPipelineRasterizationStateStreamCreateFlagsEXT

New Enum Constants

- VK_EXT_TRANSFORM_FEEDBACK_EXTENSION_NAME
- VK_EXT_TRANSFORM_FEEDBACK_SPEC_VERSION
- Extending VkAccessFlagBits:
  - VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT
  - VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT
  - VK_ACCESS_TRANSFORM_FEEDBACK_WRITE_BIT_EXT
- Extending VkBufferUsageFlagBits:
  - VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT
  - VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT
- Extending VkPipelineStageFlagBits:
  - VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT
- Extending VkQueryType:
  - VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TRANSFORM_FEEDBACK_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TRANSFORM_FEEDBACK_PROPERTIES_EXT
  - VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_STATE_STREAM_CREATE_INFO_EXT

Issues

1) Should we include pause/resume functionality?

RESOLVED: Yes, this is needed to ease layering other APIs which have this functionality. To pause use `vkCmdEndTransformFeedbackEXT` and provide valid buffer handles in the `pCounterBuffers` array and offsets in the `pCounterBufferOffsets` array for the implementation to save the resume points. Then to resume use `vkCmdBeginTransformFeedbackEXT` with the previous `pCounterBuffers` and `pCounterBufferOffsets` values. Between the pause and resume there needs to be a memory barrier for the counter buffers with a source access of VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT at pipeline stage VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT to a destination access of VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT at pipeline stage VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT.

2) How does this interact with multiview?

RESOLVED: Transform feedback cannot be made active in a render pass with multiview enabled.

3) How should queries be done?

RESOLVED: There is a new query type VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT. A query pool created with this type will capture 2 integers - numPrimitivesWritten and numPrimitivesNeeded - for the specified vertex stream output from the last pre-rasterization shader stage. The vertex stream output queried is zero by default, but can be specified with the new `vkCmdBeginQueryIndexedEXT` and `vkCmdEndQueryIndexedEXT` commands.

Version History

- Revision 1, 2018-10-09 (Piers Daniell)
  - Internal revisions

VK_EXT_validation_cache

Name String

`VK_EXT_validation_cache`

Extension Type

Device extension

Registered Extension Number

161

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- Cort Stratton [GitHub]cdwfs

Other Extension Metadata

Last Modified Date

2017-08-29

IP Status

No known IP claims.

Contributors

- Cort Stratton, Google
- Chris Forbes, Google

Description

This extension provides a mechanism for caching the results of potentially expensive internal validation operations across multiple runs of a Vulkan application. At the core is the VkValidationCacheEXT object type, which is managed similarly to the existing VkPipelineCache.

The new structure VkShaderModuleValidationCacheCreateInfoEXT can be included in the `pNext` chain at vkCreateShaderModule time. It contains a VkValidationCacheEXT to use when validating the VkShaderModule.

New Object Types

- VkValidationCacheEXT

New Commands

- vkCreateValidationCacheEXT
- vkDestroyValidationCacheEXT
- vkGetValidationCacheDataEXT
- vkMergeValidationCachesEXT

New Structures

- VkValidationCacheCreateInfoEXT
- Extending VkShaderModuleCreateInfo, VkPipelineShaderStageCreateInfo:
  - VkShaderModuleValidationCacheCreateInfoEXT

New Enums

- VkValidationCacheHeaderVersionEXT

New Bitmasks

- VkValidationCacheCreateFlagsEXT

New Enum Constants

- `VK_EXT_VALIDATION_CACHE_EXTENSION_NAME`
- `VK_EXT_VALIDATION_CACHE_SPEC_VERSION`
- Extending VkObjectType:
  - VK_OBJECT_TYPE_VALIDATION_CACHE_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_SHADER_MODULE_VALIDATION_CACHE_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_VALIDATION_CACHE_CREATE_INFO_EXT

Version History

- Revision 1, 2017-08-29 (Cort Stratton)
  - Initial draft

VK_EXT_vertex_input_dynamic_state

Name String

VK_EXT_vertex_input_dynamic_state

Extension Type

Device extension

Registered Extension Number

353

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2020-08-21

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Spencer Fricke, Samsung
- Stu Smith, AMD

Description

One of the states that contributes to the combinatorial explosion of pipeline state objects that need to be created, is the vertex input binding and attribute descriptions. By allowing them to be dynamic applications may reduce the number of pipeline objects they need to create.

This extension adds dynamic state support for what is normally static state in VkPipelineVertexInputStateCreateInfo.

New Commands

- vkCmdSetVertexInputEXT

New Structures

- VkVertexInputAttributeDescription2EXT
- VkVertexInputBindingDescription2EXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT

New Enum Constants

- `VK_EXT_VERTEX_INPUT_DYNAMIC_STATE_EXTENSION_NAME`
- `VK_EXT_VERTEX_INPUT_DYNAMIC_STATE_SPEC_VERSION`
- Extending VkDynamicState:
  - VK_DYNAMIC_STATE_VERTEX_INPUT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_INPUT_DYNAMIC_STATE_FEATURES_EXT
  - VK_STRUCTURE_TYPE_VERTEX_INPUT_ATTRIBUTE_DESCRIPTION_2_EXT
  - VK_STRUCTURE_TYPE_VERTEX_INPUT_BINDING_DESCRIPTION_2_EXT

Version History

- Revision 2, 2020-11-05 (Piers Daniell)
  - Make VkVertexInputBindingDescription2EXT extensible
  - Add new VkVertexInputAttributeDescription2EXT structure for the `pVertexAttributeDescriptions` parameter to vkCmdSetVertexInputEXT so it is also extensible
- Revision 1, 2020-08-21 (Piers Daniell)
  - Internal revisions

VK_EXT_ycbcr_image_arrays

Name String

VK_EXT_ycbcr_image_arrays

Extension Type

Device extension

Registered Extension Number

253

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_sampler_ycbcr_conversion
or
Vulkan Version 1.1

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2019-01-15

Contributors

- Piers Daniell, NVIDIA

Description

This extension allows images of a format that requires $Y'C_BC_R$ conversion to be created with multiple array layers, which is otherwise restricted.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceYcbcrImageArraysFeaturesEXT

New Enum Constants

- VK_EXT_YCBCR_IMAGE_ARRAYS_EXTENSION_NAME
- VK_EXT_YCBCR_IMAGE_ARRAYS_SPEC_VERSION
- Extending VkStructureType:

- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_YCBCR_IMAGE_ARRAYS_FEATURES_EXT

Version History

- Revision 1, 2019-01-15 (Piers Daniell)
  - Initial revision

VK_EXT_zero_initialize_device_memory

Name String

`VK_EXT_zero_initialize_device_memory`

Extension Type

Device extension

Registered Extension Number

621

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Contact

- Mike Blumenkrantz [GitHub]zmike

Extension Proposal

VK_EXT_zero_initialize_device_memory

Other Extension Metadata

Last Modified Date

2025-04-09

Interactions and External Dependencies

- Interacts with Vulkan 1.1.
- Interacts with `VK_KHR_get_physical_device_properties2`.

IP Status

No known IP claims.

Contributors

- Hans-Kristian Arntzen, VALVE
- Mike Blumenkrantz, VALVE
- Tobias Hector, AMD
- Faith Ekstrand, Collabora
- Ricardo Garcia, Igalia
- Jan-Harald Fredriksen, ARM

- Spencer Fricke, LunarG

Description

By default, Vulkan provides no guarantees that device memory allocated through vkAllocateMemory is cleared to zero. This means that applications wanting resources to be zero-initialized must execute a command such as vkCmdFillBuffer or vkCmdClearColorImage on the device to ensure a deterministic result. This can be wasteful if the underlying platform either:

- Already performs that zero clear anyway, due to e.g. security concerns.
- Can be performed more efficiently in implementation, by e.g. clearing pages to zero in the background after device memory is freed.

This extension also has uses in API layering and porting efforts, where zero memory behavior may be more strict than Vulkan. Different OS platforms also have wildly different behaviors here, which leads to implementations needing to apply workarounds to paper over these issues in the wild. If an extension exists to make allocation behavior explicit, we hopefully achieve a more robust ecosystem for Vulkan.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceZeroInitializeDeviceMemoryFeaturesEXT

New Enum Constants

- `VK_EXT_ZERO_INITIALIZE_DEVICE_MEMORY_EXTENSION_NAME`
- `VK_EXT_ZERO_INITIALIZE_DEVICE_MEMORY_SPEC_VERSION`
- Extending VkImageLayout:
  - VK_IMAGE_LAYOUT_ZERO_INITIALIZED_EXT
- Extending VkMemoryAllocateFlagBits:
  - VK_MEMORY_ALLOCATE_ZERO_INITIALIZE_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ZERO_INITIALIZE_DEVICE_MEMORY_FEATURES_EXT

Version History

- Revision 1, 2025-03-10 (Mike Blumenkrantz)
  - Initial version

List of Provisional Extensions

- VK_KHR_portability_subset

VK_KHR_portability_subset

Name String

`VK_KHR_portability_subset`

Extension Type

Device extension

Registered Extension Number

164

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

- **This is a *provisional* extension and must be used with caution. See the description of provisional header files for enablement and stability details.**

Contact

- Bill Hollings [GitHub]billhollings

Other Extension Metadata

Last Modified Date

2020-07-21

IP Status

No known IP claims.

Contributors

- Bill Hollings, The Brenwill Workshop Ltd.
- Daniel Koch, NVIDIA
- Dzmitry Malyshau, Mozilla
- Chip Davis, CodeWeavers
- Dan Ginsburg, Valve
- Mike Weiblen, LunarG
- Neil Trevett, NVIDIA
- Alexey Knyazev, Independent

Description

The VK_KHR_portability_subset extension allows a non-conformant Vulkan implementation to be built on top of another non-Vulkan graphics API, and identifies differences between that implementation and a fully-conformant native Vulkan implementation.

This extension provides Vulkan implementations with the ability to mark otherwise-required capabilities as unsupported, or to establish additional properties and limits that the application should adhere to in order to guarantee portable behavior and operation across platforms, including platforms where Vulkan is not natively supported.

The goal of this specification is to document, and make queryable, capabilities which are required to be supported by a fully-conformant Vulkan 1.0 implementation, but may be optional for an implementation of the Vulkan 1.0 Portability Subset.

The intent is that this extension will be advertised only on implementations of the Vulkan 1.0 Portability Subset, and not on conformant implementations of Vulkan 1.0. Fully-conformant Vulkan implementations provide all the required capabilities, and so will not provide this extension. Therefore, the existence of this extension can be used to determine that an implementation is likely not fully conformant with the Vulkan spec.

If this extension is supported by the Vulkan implementation, the application must enable this extension.

This extension defines several new structures that can be chained to the existing structures used by certain standard Vulkan calls, in order to query for non-conformant portable behavior.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePortabilitySubsetFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDevicePortabilitySubsetPropertiesKHR

New Enum Constants

- VK_KHR_PORTABILITY_SUBSET_EXTENSION_NAME
- VK_KHR_PORTABILITY_SUBSET_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PORTABILITY_SUBSET_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PORTABILITY_SUBSET_PROPERTIES_KHR

Issues

None.

Version History

- Revision 1, 2020-07-21 (Bill Hollings)
  - Initial draft.

List of Deprecated Extensions

- VK_KHR_16bit_storage (promoted to core)
- VK_KHR_8bit_storage (promoted to core)
- VK_KHR_bind_memory2 (promoted to core)
- VK_KHR_buffer_device_address (promoted to core)
- VK_KHR_copy_commands2 (promoted to core)
- VK_KHR_create_renderpass2 (promoted to core)
- VK_KHR_dedicated_allocation (promoted to core)
- VK_KHR_depth_stencil_resolve (promoted to core)
- VK_KHR_descriptor_update_template (promoted to core)
- VK_KHR_device_group (promoted to core)
- VK_KHR_device_group_creation (promoted to core)
- VK_KHR_draw_indirect_count (promoted to core)
- VK_KHR_driver_properties (promoted to core)
- VK_KHR_dynamic_rendering (promoted to core)
- VK_KHR_dynamic_rendering_local_read (promoted to core)
- VK_KHR_external_fence (promoted to core)
- VK_KHR_external_fence_capabilities (promoted to core)
- VK_KHR_external_memory (promoted to core)
- VK_KHR_external_memory_capabilities (promoted to core)
- VK_KHR_external_semaphore (promoted to core)
- VK_KHR_external_semaphore_capabilities (promoted to core)
- VK_KHR_format_feature_flags2 (promoted to core)
- VK_KHR_get_memory_requirements2 (promoted to core)
- VK_KHR_get_physical_device_properties2 (promoted to core)
- VK_KHR_global_priority (promoted to core)
- VK_KHR_image_format_list (promoted to core)
- VK_KHR_imageless_framebuffer (promoted to core)
- VK_KHR_index_type_uint8 (promoted to core)
- VK_KHR_line_rasterization (promoted to core)
- VK_KHR_load_store_op_none (promoted to core)
- VK_KHR_maintenance1 (promoted to core)
- VK_KHR_maintenance2 (promoted to core)
- VK_KHR_maintenance3 (promoted to core)

- VK_KHR_maintenance4 (promoted to core)
- VK_KHR_maintenance5 (promoted to core)
- VK_KHR_maintenance6 (promoted to core)
- VK_KHR_map_memory2 (promoted to core)
- VK_KHR_multiview (promoted to core)
- VK_KHR_push_descriptor (promoted to core)
- VK_KHR_relaxed_block_layout (promoted to core)
- VK_KHR_sampler_mirror_clamp_to_edge (promoted to core)
- VK_KHR_sampler_ycbcr_conversion (promoted to core)
- VK_KHR_separate_depth_stencil_layouts (promoted to core)
- VK_KHR_shader_atomic_int64 (promoted to core)
- VK_KHR_shader_draw_parameters (promoted to core)
- VK_KHR_shader_expect_assume (promoted to core)
- VK_KHR_shader_float16_int8 (promoted to core)
- VK_KHR_shader_float_controls (promoted to core)
- VK_KHR_shader_float_controls2 (promoted to core)
- VK_KHR_shader_integer_dot_product (promoted to core)
- VK_KHR_shader_non_semantic_info (promoted to core)
- VK_KHR_shader_subgroup_extended_types (promoted to core)
- VK_KHR_shader_subgroup_rotate (promoted to core)
- VK_KHR_shader_terminate_invocation (promoted to core)
- VK_KHR_spirv_1_4 (promoted to core)
- VK_KHR_storage_buffer_storage_class (promoted to core)
- VK_KHR_synchronization2 (promoted to core)
- VK_KHR_timeline_semaphore (promoted to core)
- VK_KHR_uniform_buffer_standard_layout (promoted to core)
- VK_KHR_variable_pointers (promoted to core)
- VK_KHR_vertex_attribute_divisor (promoted to core)
- VK_KHR_vulkan_memory_model (promoted to core)
- VK_KHR_zero_initialize_workgroup_memory (promoted to core)
- VK_EXT_depth_clamp_zero_one
- VK_EXT_descriptor_buffer
- VK_EXT_descriptor_indexing (promoted to core)
- VK_EXT_device_fault
- VK_EXT_extended_dynamic_state (promoted to core)

- VK_EXT_extended_dynamic_state2 (promoted to core)
- VK_EXT_host_image_copy (promoted to core)
- VK_EXT_load_store_op_none
- VK_EXT_opacity_micromap
- VK_EXT_pipeline_library_group_handles
- VK_EXT_pipeline_protected_access (promoted to core)
- VK_EXT_pipeline_robustness (promoted to core)
- VK_EXT_present_mode_fifo_latest_ready
- VK_EXT_robustness2
- VK_EXT_surface_maintenance1
- VK_EXT_swapchain_maintenance1
- VK_EXT_validation_features

VK_KHR_16bit_storage

Name String

`VK_KHR_16bit_storage`

Extension Type

Device extension

Registered Extension Number

84

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
and
VK_KHR_storage_buffer_storage_class
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_16bit_storage

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jan-Harald Fredriksen [GitHub]janharaldfredriksen-arm

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for `GL_EXT_shader_16bit_storage`

Contributors

- Alexander Galazin, ARM
- Jan-Harald Fredriksen, ARM
- Joerg Wagner, ARM

- Neil Henning, Codeplay
- Jeff Bolz, Nvidia
- Daniel Koch, Nvidia
- David Neto, Google
- John Kessenich, Google

Description

The VK_KHR_16bit_storage extension allows use of 16-bit types in shader input and output interfaces, and push constant blocks. This extension introduces several new optional features which map to SPIR-V capabilities and allow access to 16-bit data in Block-decorated objects in the Uniform and the StorageBuffer storage classes, and objects in the PushConstant storage class. This extension allows 16-bit variables to be declared and used as user-defined shader inputs and outputs but does not change location assignment and component assignment rules.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. However, if Vulkan 1.1 is supported and this extension is not, the storageBuffer16BitAccess capability is optional. The original type, enum, and command names are still available as aliases of the core functionality.

Promotion to Vulkan 1.4

If Vulkan 1.4 is supported, support for the storageBuffer16BitAccess capability is required.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevice16BitStorageFeaturesKHR

New Enum Constants

- VK_KHR_16BIT_STORAGE_EXTENSION_NAME
- VK_KHR_16BIT_STORAGE_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_16BIT_STORAGE_FEATURES_KHR

New SPIR-V Capabilities

- StorageBuffer16BitAccess
- UniformAndStorageBuffer16BitAccess
- StoragePushConstant16
- StorageInputOutput16

Version History

- Revision 1, 2017-03-23 (Alexander Galazin)
  - Initial draft

VK_KHR_8bit_storage

Name String

`VK_KHR_8bit_storage`

Extension Type

Device extension

Registered Extension Number

178

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
and
VK_KHR_storage_buffer_storage_class
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_8bit_storage

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Alexander Galazin [GitHub]alegal-arm

Other Extension Metadata

Last Modified Date

2018-02-05

Interactions and External Dependencies

- This extension provides API support for `GL_EXT_shader_16bit_storage`

IP Status

No known IP claims.

Contributors

- Alexander Galazin, Arm

Description

The `VK_KHR_8bit_storage` extension allows use of 8-bit types in uniform and storage buffers, and

push constant blocks. This extension introduces several new optional features which map to SPIR-V capabilities and allow access to 8-bit data in `Block`-decorated objects in the `Uniform` and the `StorageBuffer` storage classes, and objects in the `PushConstant` storage class.

The `StorageBuffer8BitAccess` capability **must** be supported by all implementations of this extension. The other capabilities are optional.

Promotion to Vulkan 1.2

Vulkan APIs in this extension are included in core Vulkan 1.2, with the KHR suffix omitted. However, if Vulkan 1.2 is supported and this extension is not, the `StorageBuffer8BitAccess` capability is optional. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Promotion to Vulkan 1.4

If Vulkan 1.4 is supported, support for the `storageBuffer8BitAccess` capability is required.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevice8BitStorageFeaturesKHR

New Enum Constants

- `VK_KHR_8BIT_STORAGE_EXTENSION_NAME`
- `VK_KHR_8BIT_STORAGE_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_8BIT_STORAGE_FEATURES_KHR

New SPIR-V Capabilities

- `StorageBuffer8BitAccess`
- `UniformAndStorageBuffer8BitAccess`
- `StoragePushConstant8`

Version History

- Revision 1, 2018-02-05 (Alexander Galazin)
  - Initial draft

VK_KHR_bind_memory2

Name String

`VK_KHR_bind_memory2`

Extension Type

Device extension

Registered Extension Number

158

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Tobias Hector [GitHub]tobski

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Tobias Hector, Imagination Technologies

Description

This extension provides versions of vkBindBufferMemory and vkBindImageMemory that allow multiple bindings to be performed at once, and are extensible.

This extension also introduces VK_IMAGE_CREATE_ALIAS_BIT_KHR, which allows “identical” images that alias the same memory to interpret the contents consistently, even across image layout changes.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- vkBindBufferMemory2KHR
- vkBindImageMemory2KHR

New Structures

- VkBindBufferMemoryInfoKHR
- VkBindImageMemoryInfoKHR

New Enum Constants

- `VK_KHR_BIND_MEMORY_2_EXTENSION_NAME`
- `VK_KHR_BIND_MEMORY_2_SPEC_VERSION`
- Extending VkImageCreateFlagBits:
  - VK_IMAGE_CREATE_ALIAS_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_INFO_KHR
  - VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_INFO_KHR

Version History

- Revision 1, 2017-05-19 (Tobias Hector)
  - Pulled bind memory functions into their own extension

VK_KHR_buffer_device_address

Name String

`VK_KHR_buffer_device_address`

Extension Type

Device extension

Registered Extension Number

258

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
and
VK_KHR_device_group
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_physical_storage_buffer

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Jeff Bolz [GitHub]jeffbolznv

Other Extension Metadata

Last Modified Date

2019-06-24

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for `GL_EXT_buffer_reference` and `GL_EXT_buffer_reference2` and `GL_EXT_buffer_reference_uvec2`

Contributors

- Jeff Bolz, NVIDIA
- Neil Henning, AMD
- Tobias Hector, AMD

- Faith Ekstrand, Intel
- Baldur Karlsson, Valve
- Jan-Harald Fredriksen, Arm

Description

This extension allows the application to query a 64-bit buffer device address value for a buffer, which can be used to access the buffer memory via the `PhysicalStorageBuffer` storage class in the `GL_EXT_buffer_reference` GLSL extension and `SPV_KHR_physical_storage_buffer` SPIR-V extension.

Another way to describe this extension is that it adds "pointers to buffer memory in shaders". By calling vkGetBufferDeviceAddress with a `VkBuffer`, it will return a VkDeviceAddress value which represents the address of the start of the buffer.

vkGetBufferOpaqueCaptureAddress and vkGetDeviceMemoryOpaqueCaptureAddress allow opaque addresses for buffers and memory objects to be queried for the current process. A trace capture and replay tool can then supply these addresses to be used at replay time to match the addresses used when the trace was captured. To enable tools to insert these queries, new memory allocation flags must be specified for memory objects that will be bound to buffers accessed via the `PhysicalStorageBuffer` storage class. **Note that this mechanism is intended only to support capture/replay tools, and is not recommended for use in other applications.**

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. However, if Vulkan 1.2 is supported and this extension is not, the `bufferDeviceAddress` feature is optional. The original type, enum, and command names are still available as aliases of the core functionality.

Promotion to Vulkan 1.3

If Vulkan 1.3 is supported, support for the `bufferDeviceAddress` capability is required.

New Commands

- vkGetBufferDeviceAddressKHR
- vkGetBufferOpaqueCaptureAddressKHR
- vkGetDeviceMemoryOpaqueCaptureAddressKHR

New Structures

- VkBufferDeviceAddressInfoKHR
- VkDeviceMemoryOpaqueCaptureAddressInfoKHR
- Extending VkBufferCreateInfo:
  - VkBufferOpaqueCaptureAddressCreateInfoKHR
- Extending VkMemoryAllocateInfo:

    - VkMemoryOpaqueCaptureAddressAllocateInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
    - VkPhysicalDeviceBufferDeviceAddressFeaturesKHR

New Enum Constants

- VK_KHR_BUFFER_DEVICE_ADDRESS_EXTENSION_NAME
- VK_KHR_BUFFER_DEVICE_ADDRESS_SPEC_VERSION
- Extending VkBufferCreateFlagBits:
    - VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR
- Extending VkBufferUsageFlagBits:
    - VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT_KHR
- Extending VkMemoryAllocateFlagBits:
    - VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT_KHR
    - VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR
- Extending VkResult:
    - VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_BUFFER_DEVICE_ADDRESS_INFO_KHR
    - VK_STRUCTURE_TYPE_BUFFER_OPAQUE_CAPTURE_ADDRESS_CREATE_INFO_KHR
    - VK_STRUCTURE_TYPE_DEVICE_MEMORY_OPAQUE_CAPTURE_ADDRESS_INFO_KHR
    - VK_STRUCTURE_TYPE_MEMORY_OPAQUE_CAPTURE_ADDRESS_ALLOCATE_INFO_KHR
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BUFFER_DEVICE_ADDRESS_FEATURES_KHR

New SPIR-V Capabilities

- PhysicalStorageBufferAddresses

Version History

- Revision 1, 2019-06-24 (Jan-Harald Fredriksen)
    - Internal revisions based on VK_EXT_buffer_device_address

VK_KHR_copy_commands2

Name String

VK_KHR_copy_commands2

Extension Type

Device extension

Registered Extension Number

338

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Matthew Netsch [GitHub]mnetsch

Other Extension Metadata

Last Modified Date

2020-07-06

Contributors

- Jeff Leger, Qualcomm
- Tobias Hector, AMD
- Jan-Harald Fredriksen, ARM
- Tom Olson, ARM

Description

This extension provides extensible versions of the Vulkan buffer and image copy commands. The new commands are functionally identical to the core commands, except that their copy parameters are specified using extensible structures that can be used to pass extension-specific information.

The following extensible copy commands are introduced with this extension: vkCmdCopyBuffer2KHR, vkCmdCopyImage2KHR, vkCmdCopyBufferToImage2KHR, vkCmdCopyImageToBuffer2KHR, vkCmdBlitImage2KHR, and vkCmdResolveImage2KHR. Each

command contains an *Info2KHR structure parameter that includes sType/pNext members. Lower level structures describing each region to be copied are also extended with sType/pNext members.

New Commands

- vkCmdBlitImage2KHR
- vkCmdCopyBuffer2KHR
- vkCmdCopyBufferToImage2KHR
- vkCmdCopyImage2KHR
- vkCmdCopyImageToBuffer2KHR
- vkCmdResolveImage2KHR

New Structures

- VkBlitImageInfo2KHR
- VkBufferCopy2KHR
- VkBufferImageCopy2KHR
- VkCopyBufferInfo2KHR
- VkCopyBufferToImageInfo2KHR
- VkCopyImageInfo2KHR
- VkCopyImageToBufferInfo2KHR
- VkImageBlit2KHR
- VkImageCopy2KHR
- VkImageResolve2KHR
- VkResolveImageInfo2KHR

New Enum Constants

- VK_KHR_COPY_COMMANDS_2_EXTENSION_NAME
- VK_KHR_COPY_COMMANDS_2_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_BLIT_IMAGE_INFO_2_KHR
  - VK_STRUCTURE_TYPE_BUFFER_COPY_2_KHR
  - VK_STRUCTURE_TYPE_BUFFER_IMAGE_COPY_2_KHR
  - VK_STRUCTURE_TYPE_COPY_BUFFER_INFO_2_KHR
  - VK_STRUCTURE_TYPE_COPY_BUFFER_TO_IMAGE_INFO_2_KHR
  - VK_STRUCTURE_TYPE_COPY_IMAGE_INFO_2_KHR
  - VK_STRUCTURE_TYPE_COPY_IMAGE_TO_BUFFER_INFO_2_KHR
  - VK_STRUCTURE_TYPE_IMAGE_BLIT_2_KHR

- VK_STRUCTURE_TYPE_IMAGE_COPY_2_KHR
- VK_STRUCTURE_TYPE_IMAGE_RESOLVE_2_KHR
- VK_STRUCTURE_TYPE_RESOLVE_IMAGE_INFO_2_KHR

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Version History

- Revision 1, 2020-07-06 (Jeff Leger)
  - Internal revisions

VK_KHR_create_renderpass2

Name String

VK_KHR_create_renderpass2

Extension Type

Device extension

Registered Extension Number

110

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_multiview
and
VK_KHR_maintenance2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Tobias Hector [GitHub]tobias

Other Extension Metadata

Last Modified Date

2018-02-07

Contributors

- Tobias Hector
- Jeff Bolz

Description

This extension provides a new command to create render passes in a way that can be easily extended by other extensions through the substructures of render pass creation. The Vulkan 1.0 render pass creation sub-structures do not include `sType`/`pNext` members. Additionally, the render pass begin/next/end commands have been augmented with new extensible structures for passing additional subpass information.

The VkRenderPassMultiviewCreateInfo and VkInputAttachmentAspectReference structures that extended the original VkRenderPassCreateInfo are not accepted into the new creation functions,

and instead their parameters are folded into this extension as follows:

- Elements of VkRenderPassMultiviewCreateInfo::pViewMasks are now specified in VkSubpassDescription2KHR::viewMask.
- Elements of VkRenderPassMultiviewCreateInfo::pViewOffsets are now specified in VkSubpassDependency2KHR::viewOffset.
- VkRenderPassMultiviewCreateInfo::correlationMaskCount and VkRenderPassMultiviewCreateInfo::pCorrelationMasks are directly specified in VkRenderPassCreateInfo2KHR.
- VkInputAttachmentAspectReference::aspectMask is now specified in the relevant input attachment reference in VkAttachmentReference2KHR::aspectMask

The details of these mappings are explained fully in the new structures.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- vkCmdBeginRenderPass2KHR
- vkCmdEndRenderPass2KHR
- vkCmdNextSubpass2KHR
- vkCreateRenderPass2KHR

New Structures

- VkAttachmentDescription2KHR
- VkAttachmentReference2KHR
- VkRenderPassCreateInfo2KHR
- VkSubpassBeginInfoKHR
- VkSubpassDependency2KHR
- VkSubpassDescription2KHR
- VkSubpassEndInfoKHR

New Enum Constants

- VK_KHR_CREATE_RENDERPASS_2_EXTENSION_NAME
- VK_KHR_CREATE_RENDERPASS_2_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_2_KHR
  - VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_2_KHR

- VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO_2_KHR
- VK_STRUCTURE_TYPE_SUBPASS_BEGIN_INFO_KHR
- VK_STRUCTURE_TYPE_SUBPASS_DEPENDENCY_2_KHR
- VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_2_KHR
- VK_STRUCTURE_TYPE_SUBPASS_END_INFO_KHR

Version History

- Revision 1, 2018-02-07 (Tobias Hector)
  - Internal revisions

VK_KHR_dedicated_allocation

Name String

`VK_KHR_dedicated_allocation`

Extension Type

Device extension

Registered Extension Number

128

Revision

3

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_memory_requirements2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Faith Ekstrand, Intel

Description

This extension enables resources to be bound to a dedicated allocation, rather than suballocated. For any particular resource, applications **can** query whether a dedicated allocation is recommended, in which case using a dedicated allocation **may** improve the performance of access to that resource. Normal device memory allocations must support multiple resources per allocation, memory aliasing and sparse binding, which could interfere with some optimizations. Applications should query the implementation for when a dedicated allocation **may** be beneficial by adding a `VkMemoryDedicatedRequirementsKHR` structure to the `pNext` chain of the

VkMemoryRequirements2 structure passed as the pMemoryRequirements parameter of a call to vkGetBufferMemoryRequirements2 or vkGetImageMemoryRequirements2. Certain external handle types and external images or buffers **may** also depend on dedicated allocations on implementations that associate image or buffer metadata with OS-level memory objects.

This extension adds a two small structures to memory requirements querying and memory allocation: a new structure that flags whether an image/buffer should have a dedicated allocation, and a structure indicating the image or buffer that an allocation will be bound to.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending VkMemoryAllocateInfo:
  - VkMemoryDedicatedAllocateInfoKHR
- Extending VkMemoryRequirements2:
  - VkMemoryDedicatedRequirementsKHR

New Enum Constants

- VK_KHR_DEDICATED_ALLOCATION_EXTENSION_NAME
- VK_KHR_DEDICATED_ALLOCATION_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_MEMORY_DEDICATED_ALLOCATE_INFO_KHR
  - VK_STRUCTURE_TYPE_MEMORY_DEDICATED_REQUIREMENTS_KHR

Examples

```
    // Create an image with a dedicated allocation based on the
    // implementation's preference

    VkImageCreateInfo imageCreateInfo =
    {
        // Image creation parameters
    };

    VkImage image;
    VkResult result = vkCreateImage(
        device,
        &imageCreateInfo,
        NULL,               // pAllocator
        &image);

    VkMemoryDedicatedRequirementsKHR dedicatedRequirements =
```

```
    {
        .sType = VK_STRUCTURE_TYPE_MEMORY_DEDICATED_REQUIREMENTS_KHR,
        .pNext = NULL,
    };

    VkMemoryRequirements2 memoryRequirements =
    {
        .sType = VK_STRUCTURE_TYPE_MEMORY_REQUIREMENTS_2,
        .pNext = &dedicatedRequirements,
    };

    const VkImageMemoryRequirementsInfo2 imageRequirementsInfo =
    {
        .sType = VK_STRUCTURE_TYPE_IMAGE_MEMORY_REQUIREMENTS_INFO_2,
        .pNext = NULL,
        .image = image
    };

    vkGetImageMemoryRequirements2(
        device,
        &imageRequirementsInfo,
        &memoryRequirements);

    if (dedicatedRequirements.prefersDedicatedAllocation) {
        // Allocate memory with VkMemoryDedicatedAllocateInfoKHR::image
        // pointing to the image we are allocating the memory for

        VkMemoryDedicatedAllocateInfoKHR dedicatedInfo =
        {
            .sType = VK_STRUCTURE_TYPE_MEMORY_DEDICATED_ALLOCATE_INFO_KHR,
            .pNext = NULL,
            .image = image,
            .buffer = VK_NULL_HANDLE,
        };

        VkMemoryAllocateInfo memoryAllocateInfo =
        {
            .sType = VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_INFO,
            .pNext = &dedicatedInfo,
            .allocationSize = memoryRequirements.size,
            .memoryTypeIndex = FindMemoryTypeIndex(memoryRequirements.memoryTypeBits),
        };

        VkDeviceMemory memory;
        vkAllocateMemory(
            device,
            &memoryAllocateInfo,
            NULL,               // pAllocator
            &memory);

        // Bind the image to the memory
```

```
        vkBindImageMemory(
            device,
            image,
            memory,
            0);
    } else {
        // Take the normal memory sub-allocation path
    }
```

Version History

- Revision 1, 2017-02-27 (James Jones)
  - Copy content from VK_NV_dedicated_allocation
  - Add some references to external object interactions to the overview.
- Revision 2, 2017-03-27 (Faith Ekstrand)
  - Rework the extension to be query-based
- Revision 3, 2017-07-31 (Faith Ekstrand)
  - Clarify that memory objects allocated with VkMemoryDedicatedAllocateInfoKHR can only have the specified resource bound and no others.

VK_KHR_depth_stencil_resolve

Name String

`VK_KHR_depth_stencil_resolve`

Extension Type

Device extension

Registered Extension Number

200

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_create_renderpass2
or
Vulkan Version 1.2

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Jan-Harald Fredriksen [GitHub]janharald

Other Extension Metadata

Last Modified Date

2018-04-09

Contributors

- Jan-Harald Fredriksen, Arm
- Andrew Garrard, Samsung Electronics
- Soowan Park, Samsung Electronics
- Jeff Bolz, NVIDIA
- Daniel Rakos, AMD

Description

This extension adds support for automatically resolving multisampled depth/stencil attachments in a subpass in a similar manner as for color attachments.

Multisampled color attachments can be resolved at the end of a subpass by specifying `pResolveAttachments` entries corresponding to the `pColorAttachments` array entries. This does not allow for a way to map the resolve attachments to the depth/stencil attachment. The

vkCmdResolveImage command does not allow for depth/stencil images. While there are other ways to resolve the depth/stencil attachment, they can give sub-optimal performance. Extending the VkSubpassDescription2 in this extension allows an application to add a pDepthStencilResolveAttachment, that is similar to the color pResolveAttachments, that the pDepthStencilAttachment can be resolved into.

Depth and stencil samples are resolved to a single value based on the resolve mode. The set of possible resolve modes is defined in the VkResolveModeFlagBits enum. The VK_RESOLVE_MODE_SAMPLE_ZERO_BIT mode is the only mode that is required of all implementations (that support the extension or support Vulkan 1.2 or higher). Some implementations may also support averaging (the same as color sample resolve) or taking the minimum or maximum sample, which may be more suitable for depth/stencil resolve.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceDepthStencilResolvePropertiesKHR
- Extending VkSubpassDescription2:
  - VkSubpassDescriptionDepthStencilResolveKHR

New Enums

- VkResolveModeFlagBitsKHR

New Bitmasks

- VkResolveModeFlagsKHR

New Enum Constants

- VK_KHR_DEPTH_STENCIL_RESOLVE_EXTENSION_NAME
- VK_KHR_DEPTH_STENCIL_RESOLVE_SPEC_VERSION
- Extending VkResolveModeFlagBits:
  - VK_RESOLVE_MODE_AVERAGE_BIT_KHR
  - VK_RESOLVE_MODE_MAX_BIT_KHR
  - VK_RESOLVE_MODE_MIN_BIT_KHR
  - VK_RESOLVE_MODE_NONE_KHR
  - VK_RESOLVE_MODE_SAMPLE_ZERO_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_STENCIL_RESOLVE_PROPERTIES_KHR

- VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_DEPTH_STENCIL_RESOLVE_KHR

Version History

- Revision 1, 2018-04-09 (Jan-Harald Fredriksen)
  - Initial revision

VK_KHR_descriptor_update_template

Name String

`VK_KHR_descriptor_update_template`

Extension Type

Device extension

Registered Extension Number

86

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

API Interactions

- Interacts with VK_EXT_debug_report
- Interacts with VK_KHR_push_descriptor

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Markus Tavenrath [GitHub]mtavenrath

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Interactions and External Dependencies

- Interacts with `VK_KHR_push_descriptor`

Contributors

- Jeff Bolz, NVIDIA
- Michael Worcester, Imagination Technologies

Description

Applications may wish to update a fixed set of descriptors in a large number of descriptor sets very frequently, i.e. during initialization phase or if it is required to rebuild descriptor sets for each

frame. For those cases it is also not unlikely that all information required to update a single descriptor set is stored in a single struct. This extension provides a way to update a fixed set of descriptors in a single VkDescriptorSet with a pointer to an application-defined data structure describing the new descriptors.

Promotion to Vulkan 1.1

vkCmdPushDescriptorSetWithTemplateKHR is included as an interaction with `VK_KHR_push_descriptor`. If Vulkan 1.1 and `VK_KHR_push_descriptor` are supported, this is included by `VK_KHR_push_descriptor`.

The base functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Object Types

- VkDescriptorUpdateTemplateKHR

New Commands

- vkCreateDescriptorUpdateTemplateKHR
- vkDestroyDescriptorUpdateTemplateKHR
- vkUpdateDescriptorSetWithTemplateKHR

If VK_KHR_push_descriptor is supported:

- vkCmdPushDescriptorSetWithTemplateKHR

New Structures

- VkDescriptorUpdateTemplateCreateInfoKHR
- VkDescriptorUpdateTemplateEntryKHR

New Enums

- VkDescriptorUpdateTemplateTypeKHR

New Bitmasks

- VkDescriptorUpdateTemplateCreateFlagsKHR

New Enum Constants

- `VK_KHR_DESCRIPTOR_UPDATE_TEMPLATE_EXTENSION_NAME`
- `VK_KHR_DESCRIPTOR_UPDATE_TEMPLATE_SPEC_VERSION`
- Extending VkDescriptorUpdateTemplateType:
  - VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET_KHR
- Extending VkObjectType:

    - VK_OBJECT_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_KHR
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_CREATE_INFO_KHR

If VK_KHR_push_descriptor is supported:

- Extending VkDescriptorUpdateTemplateType:
    - VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS_KHR

Version History

- Revision 1, 2016-01-11 (Markus Tavenrath)
    - Initial draft

VK_KHR_device_group

Name String

`VK_KHR_device_group`

Extension Type

Device extension

Registered Extension Number

61

Revision

4

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_device_group_creation

API Interactions

- Interacts with VK_KHR_bind_memory2
- Interacts with VK_KHR_surface
- Interacts with VK_KHR_swapchain

SPIR-V Dependencies

- SPV_KHR_device_group

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jeff Bolz [GitHub]jeffbolznv

Other Extension Metadata

Last Modified Date

2017-10-10

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Tobias Hector, Imagination Technologies

Description

This extension provides functionality to use a logical device that consists of multiple physical devices, as created with the VK_KHR_device_group_creation extension. A device group can allocate memory across the subdevices, bind memory from one subdevice to a resource on another subdevice, record command buffers where some work executes on an arbitrary subset of the subdevices, and potentially present a swapchain image from one or more subdevices.

Promotion to Vulkan 1.1

The following enums, types, and commands are included as interactions with VK_KHR_swapchain:

- VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_CAPABILITIES_KHR
- VK_STRUCTURE_TYPE_IMAGE_SWAPCHAIN_CREATE_INFO_KHR
- VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_SWAPCHAIN_INFO_KHR
- VK_STRUCTURE_TYPE_ACQUIRE_NEXT_IMAGE_INFO_KHR
- VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_INFO_KHR
- VK_STRUCTURE_TYPE_DEVICE_GROUP_SWAPCHAIN_CREATE_INFO_KHR
- VK_SWAPCHAIN_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR
- VkDeviceGroupPresentModeFlagBitsKHR
- VkDeviceGroupPresentCapabilitiesKHR
- VkImageSwapchainCreateInfoKHR
- VkBindImageMemorySwapchainInfoKHR
- VkAcquireNextImageInfoKHR
- VkDeviceGroupPresentInfoKHR
- VkDeviceGroupSwapchainCreateInfoKHR
- vkGetDeviceGroupPresentCapabilitiesKHR
- vkGetDeviceGroupSurfacePresentModesKHR
- vkGetPhysicalDevicePresentRectanglesKHR
- vkAcquireNextImage2KHR

If Vulkan 1.1 and VK_KHR_swapchain are supported, these are included by VK_KHR_swapchain.

The base functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- vkCmdDispatchBaseKHR
- vkCmdSetDeviceMaskKHR
- vkGetDeviceGroupPeerMemoryFeaturesKHR

If VK_KHR_surface is supported:

- vkGetDeviceGroupPresentCapabilitiesKHR
- vkGetDeviceGroupSurfacePresentModesKHR
- vkGetPhysicalDevicePresentRectanglesKHR

If VK_KHR_swapchain is supported:

- vkAcquireNextImage2KHR

New Structures

- Extending VkBindSparseInfo:
  - VkDeviceGroupBindSparseInfoKHR
- Extending VkCommandBufferBeginInfo:
  - VkDeviceGroupCommandBufferBeginInfoKHR
- Extending VkMemoryAllocateInfo:
  - VkMemoryAllocateFlagsInfoKHR
- Extending VkRenderPassBeginInfo, VkRenderingInfo:
  - VkDeviceGroupRenderPassBeginInfoKHR
- Extending VkSubmitInfo:
  - VkDeviceGroupSubmitInfoKHR

If VK_KHR_bind_memory2 is supported:

- Extending VkBindBufferMemoryInfo:
  - VkBindBufferMemoryDeviceGroupInfoKHR
- Extending VkBindImageMemoryInfo:
  - VkBindImageMemoryDeviceGroupInfoKHR

If VK_KHR_surface is supported:

- VkDeviceGroupPresentCapabilitiesKHR

If VK_KHR_swapchain is supported:

- VkAcquireNextImageInfoKHR
- Extending VkBindImageMemoryInfo:
  - VkBindImageMemorySwapchainInfoKHR
- Extending VkImageCreateInfo:
  - VkImageSwapchainCreateInfoKHR
- Extending VkPresentInfoKHR:

    - VkDeviceGroupPresentInfoKHR
- Extending VkSwapchainCreateInfoKHR:
    - VkDeviceGroupSwapchainCreateInfoKHR

New Enums

- VkMemoryAllocateFlagBitsKHR
- VkPeerMemoryFeatureFlagBitsKHR

If VK_KHR_surface is supported:

- VkDeviceGroupPresentModeFlagBitsKHR

New Bitmasks

- VkMemoryAllocateFlagsKHR
- VkPeerMemoryFeatureFlagsKHR

If VK_KHR_surface is supported:

- VkDeviceGroupPresentModeFlagsKHR

New Enum Constants

- `VK_KHR_DEVICE_GROUP_EXTENSION_NAME`
- `VK_KHR_DEVICE_GROUP_SPEC_VERSION`
- Extending VkDependencyFlagBits:
    - VK_DEPENDENCY_DEVICE_GROUP_BIT_KHR
- Extending VkMemoryAllocateFlagBits:
    - VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT_KHR
- Extending VkPeerMemoryFeatureFlagBits:
    - VK_PEER_MEMORY_FEATURE_COPY_DST_BIT_KHR
    - VK_PEER_MEMORY_FEATURE_COPY_SRC_BIT_KHR
    - VK_PEER_MEMORY_FEATURE_GENERIC_DST_BIT_KHR
    - VK_PEER_MEMORY_FEATURE_GENERIC_SRC_BIT_KHR
- Extending VkPipelineCreateFlagBits:
    - VK_PIPELINE_CREATE_DISPATCH_BASE_BIT_KHR
    - VK_PIPELINE_CREATE_DISPATCH_BASE_KHR
    - VK_PIPELINE_CREATE_VIEW_INDEX_FROM_DEVICE_INDEX_BIT_KHR
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_DEVICE_GROUP_BIND_SPARSE_INFO_KHR
    - VK_STRUCTURE_TYPE_DEVICE_GROUP_COMMAND_BUFFER_BEGIN_INFO_KHR

    - VK_STRUCTURE_TYPE_DEVICE_GROUP_RENDER_PASS_BEGIN_INFO_KHR
    - VK_STRUCTURE_TYPE_DEVICE_GROUP_SUBMIT_INFO_KHR
    - VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_FLAGS_INFO_KHR

If VK_KHR_bind_memory2 is supported:

- Extending VkImageCreateFlagBits:
    - VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_DEVICE_GROUP_INFO_KHR
    - VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_DEVICE_GROUP_INFO_KHR

If VK_KHR_surface is supported:

- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_CAPABILITIES_KHR

If VK_KHR_swapchain is supported:

- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_ACQUIRE_NEXT_IMAGE_INFO_KHR
    - VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_SWAPCHAIN_INFO_KHR
    - VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_INFO_KHR
    - VK_STRUCTURE_TYPE_DEVICE_GROUP_SWAPCHAIN_CREATE_INFO_KHR
    - VK_STRUCTURE_TYPE_IMAGE_SWAPCHAIN_CREATE_INFO_KHR
- Extending VkSwapchainCreateFlagBitsKHR:
    - VK_SWAPCHAIN_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR

New Built-in Variables

- `DeviceIndex`

New SPIR-V Capabilities

- `DeviceGroup`

Version History

- Revision 1, 2016-10-19 (Jeff Bolz)
    - Internal revisions
- Revision 2, 2017-05-19 (Tobias Hector)
    - Removed extended memory bind functions to VK_KHR_bind_memory2, added dependency on that extension, and device-group-specific structs for those functions.

- Revision 3, 2017-10-06 (Ian Elliott)
    - Corrected Vulkan 1.1 interactions with the WSI extensions. All Vulkan 1.1 WSI interactions are with the VK_KHR_swapchain extension.
- Revision 4, 2017-10-10 (Jeff Bolz)
    - Rename “SFR” bits and structure members to use the phrase “split instance bind regions”.

VK_KHR_device_group_creation

Name String

`VK_KHR_device_group_creation`

Extension Type

Instance extension

Registered Extension Number

71

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jeff Bolz [GitHub]jeffbolznv

Other Extension Metadata

Last Modified Date

2016-10-19

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA

Description

This extension provides instance-level commands to enumerate groups of physical devices, and to create a logical device from a subset of one of those groups. Such a logical device can then be used with new features in the `VK_KHR_device_group` extension.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- vkEnumeratePhysicalDeviceGroupsKHR

New Structures

- VkPhysicalDeviceGroupPropertiesKHR
- Extending VkDeviceCreateInfo:
  - VkDeviceGroupDeviceCreateInfoKHR

New Enum Constants

- VK_KHR_DEVICE_GROUP_CREATION_EXTENSION_NAME
- VK_KHR_DEVICE_GROUP_CREATION_SPEC_VERSION
- VK_MAX_DEVICE_GROUP_SIZE_KHR
- Extending VkMemoryHeapFlagBits:
  - VK_MEMORY_HEAP_MULTI_INSTANCE_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DEVICE_GROUP_DEVICE_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GROUP_PROPERTIES_KHR

Examples

```
    VkDeviceCreateInfo devCreateInfo = { VK_STRUCTURE_TYPE_DEVICE_CREATE_INFO };
    // (not shown) fill out devCreateInfo as usual.
    uint32_t deviceGroupCount = 0;
    VkPhysicalDeviceGroupPropertiesKHR *props = NULL;

    // Query the number of device groups
    vkEnumeratePhysicalDeviceGroupsKHR(g_vkInstance, &deviceGroupCount, NULL);

    // Allocate and initialize structures to query the device groups
    props = (VkPhysicalDeviceGroupPropertiesKHR *)malloc(deviceGroupCount*sizeof
(VkPhysicalDeviceGroupPropertiesKHR));
    for (i = 0; i < deviceGroupCount; ++i) {
        props[i].sType = VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GROUP_PROPERTIES_KHR;
        props[i].pNext = NULL;
    }
    vkEnumeratePhysicalDeviceGroupsKHR(g_vkInstance, &deviceGroupCount, props);

    // If the first device group has more than one physical device. create
    // a logical device using all of the physical devices.
    VkDeviceGroupDeviceCreateInfoKHR deviceGroupInfo = {
VK_STRUCTURE_TYPE_DEVICE_GROUP_DEVICE_CREATE_INFO_KHR };
    if (props[0].physicalDeviceCount > 1) {
        deviceGroupInfo.physicalDeviceCount = props[0].physicalDeviceCount;
        deviceGroupInfo.pPhysicalDevices = props[0].physicalDevices;
```

```
        devCreateInfo.pNext = &deviceGroupInfo;
    }

    vkCreateDevice(props[0].physicalDevices[0], &devCreateInfo, NULL, &g_vkDevice);
    free(props);
```

Version History

- Revision 1, 2016-10-19 (Jeff Bolz)
    - Internal revisions

VK_KHR_draw_indirect_count

Name String

`VK_KHR_draw_indirect_count`

Extension Type

Device extension

Registered Extension Number

170

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2017-08-25

IP Status

No known IP claims.

Contributors

- Matthaeus G. Chajdas, AMD
- Derrick Owens, AMD
- Graham Sellers, AMD
- Daniel Rakos, AMD
- Dominik Witczak, AMD
- Piers Daniell, NVIDIA

Description

This extension is based on the `VK_AMD_draw_indirect_count` extension. This extension allows an application to source the number of draws for indirect drawing calls from a buffer.

Applications might want to do culling on the GPU via a compute shader prior to drawing. This enables the application to generate an arbitrary number of drawing commands and execute them without host intervention.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. However, if Vulkan 1.2 is supported and this extension is not, the commands vkCmdDrawIndirectCount and vkCmdDrawIndexedIndirectCount are optional. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- vkCmdDrawIndexedIndirectCountKHR
- vkCmdDrawIndirectCountKHR

New Enum Constants

- `VK_KHR_DRAW_INDIRECT_COUNT_EXTENSION_NAME`
- `VK_KHR_DRAW_INDIRECT_COUNT_SPEC_VERSION`

Version History

- Revision 1, 2017-08-25 (Piers Daniell)
  - Initial draft based on VK_AMD_draw_indirect_count

VK_KHR_driver_properties

Name String

`VK_KHR_driver_properties`

Extension Type

Device extension

Registered Extension Number

197

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Daniel Rakos [GitHub]drakos-amd

Other Extension Metadata

Last Modified Date

2018-04-11

IP Status

No known IP claims.

Contributors

- Baldur Karlsson
- Matthaeus G. Chajdas, AMD
- Piers Daniell, NVIDIA
- Alexander Galazin, Arm
- Jesse Hall, Google
- Daniel Rakos, AMD

Description

This extension provides a new physical device query which allows retrieving information about the

driver implementation, allowing applications to determine which physical device corresponds to which particular vendor's driver, and which conformance test suite version the driver implementation is compliant with.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- VkConformanceVersionKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceDriverPropertiesKHR

New Enums

- VkDriverIdKHR

New Enum Constants

- `VK_KHR_DRIVER_PROPERTIES_EXTENSION_NAME`
- `VK_KHR_DRIVER_PROPERTIES_SPEC_VERSION`
- VK_MAX_DRIVER_INFO_SIZE_KHR
- VK_MAX_DRIVER_NAME_SIZE_KHR
- Extending VkDriverId:
  - VK_DRIVER_ID_AMD_OPEN_SOURCE_KHR
  - VK_DRIVER_ID_AMD_PROPRIETARY_KHR
  - VK_DRIVER_ID_ARM_PROPRIETARY_KHR
  - VK_DRIVER_ID_BROADCOM_PROPRIETARY_KHR
  - VK_DRIVER_ID_GGP_PROPRIETARY_KHR
  - VK_DRIVER_ID_GOOGLE_SWIFTSHADER_KHR
  - VK_DRIVER_ID_IMAGINATION_PROPRIETARY_KHR
  - VK_DRIVER_ID_INTEL_OPEN_SOURCE_MESA_KHR
  - VK_DRIVER_ID_INTEL_PROPRIETARY_WINDOWS_KHR
  - VK_DRIVER_ID_MESA_RADV_KHR
  - VK_DRIVER_ID_NVIDIA_PROPRIETARY_KHR
  - VK_DRIVER_ID_QUALCOMM_PROPRIETARY_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRIVER_PROPERTIES_KHR

Version History

- Revision 1, 2018-04-11 (Daniel Rakos)
  - Internal revisions

VK_KHR_dynamic_rendering

Name String

`VK_KHR_dynamic_rendering`

Extension Type

Device extension

Registered Extension Number

45

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1
and
VK_KHR_depth_stencil_resolve
or
Vulkan Version 1.2

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Tobias Hector [GitHub]tobski

Extension Proposal

VK_KHR_dynamic_rendering

Other Extension Metadata

Last Modified Date

2021-10-06

Contributors

- Tobias Hector, AMD
- Arseny Kapoulkine, Roblox
- François Duranleau, Gameloft
- Stuart Smith, AMD
- Hai Nguyen, Google
- Jean-François Roy, Google

- Jeff Leger, Qualcomm
- Jan-Harald Fredriksen, Arm
- Piers Daniell, Nvidia
- James Fitzpatrick, Imagination
- Piotr Byszewski, Mobica
- Jesse Hall, Google
- Mike Blumenkrantz, Valve

Description

This extension allows applications to create single-pass render pass instances without needing to create render pass objects or framebuffers. Dynamic render passes can also span across multiple primary command buffers, rather than relying on secondary command buffers.

This extension also incorporates VK_ATTACHMENT_STORE_OP_NONE_KHR from `VK_QCOM_render_pass_store_ops`, enabling applications to avoid unnecessary synchronization when an attachment is not written during a render pass.

New Commands

- vkCmdBeginRenderingKHR
- vkCmdEndRenderingKHR

New Structures

- VkRenderingAttachmentInfoKHR
- VkRenderingInfoKHR
- Extending VkCommandBufferInheritanceInfo:
  - VkCommandBufferInheritanceRenderingInfoKHR
- Extending VkGraphicsPipelineCreateInfo:
  - VkPipelineRenderingCreateInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDynamicRenderingFeaturesKHR

New Enums

- VkRenderingFlagBitsKHR

New Bitmasks

- VkRenderingFlagsKHR

New Enum Constants

- `VK_KHR_DYNAMIC_RENDERING_EXTENSION_NAME`
- `VK_KHR_DYNAMIC_RENDERING_SPEC_VERSION`
- Extending VkAttachmentStoreOp:
  - VK_ATTACHMENT_STORE_OP_NONE_KHR
- Extending VkRenderingFlagBits:
  - VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT_KHR
  - VK_RENDERING_RESUMING_BIT_KHR
  - VK_RENDERING_SUSPENDING_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_RENDERING_INFO_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PIPELINE_RENDERING_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_INFO_KHR
  - VK_STRUCTURE_TYPE_RENDERING_INFO_KHR

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Version History

- Revision 1, 2021-10-06 (Tobias Hector)
  - Initial revision

VK_KHR_dynamic_rendering_local_read

Name String

`VK_KHR_dynamic_rendering_local_read`

Extension Type

Device extension

Registered Extension Number

233

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_dynamic_rendering
or
Vulkan Version 1.3

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Tobias Hector [GitHub]tobski

Extension Proposal

VK_KHR_dynamic_rendering_local_read

Other Extension Metadata

Last Modified Date

2023-11-03

Contributors

- Tobias Hector, AMD
- Hans-Kristian Arntzen, Valve
- Connor Abbott, Valve
- Pan Gao, Huawei
- Lionel Landwerlin, Intel
- Shahbaz Youssefi, Google
- Alyssa Rosenzweig, Valve
- Jan-Harald Fredriksen, Arm
- Mike Blumenkrantz, Valve

- Graeme Leese, Broadcom
- Piers Daniell, Nvidia
- Stuart Smith, AMD
- Daniel Story, Nintendo
- James Fitzpatrick, Imagination
- Piotr Byszewski, Mobica
- Spencer Fricke, LunarG
- Tom Olson, Arm
- Michal Pietrasiuk, Intel
- Matthew Netsch, Qualcomm
- Marty Johnson, Khronos
- Wyvern Wang, Huawei
- Jeff Bolz, Nvidia
- Samuel (Sheng-Wen) Huang, MediaTek

Description

This extension enables reads from attachments and resources written by previous fragment shaders within a dynamic render pass.

New Commands

- vkCmdSetRenderingAttachmentLocationsKHR
- vkCmdSetRenderingInputAttachmentIndicesKHR

New Structures

- Extending VkGraphicsPipelineCreateInfo, VkCommandBufferInheritanceInfo:
  - VkRenderingAttachmentLocationInfoKHR
  - VkRenderingInputAttachmentIndexInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDynamicRenderingLocalReadFeaturesKHR

New Enum Constants

- `VK_KHR_DYNAMIC_RENDERING_LOCAL_READ_EXTENSION_NAME`
- `VK_KHR_DYNAMIC_RENDERING_LOCAL_READ_SPEC_VERSION`
- Extending VkImageLayout:
  - VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR
- Extending VkStructureType:

- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_LOCAL_READ_FEATURES_KHR
- VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_LOCATION_INFO_KHR
- VK_STRUCTURE_TYPE_RENDERING_INPUT_ATTACHMENT_INDEX_INFO_KHR

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4, with the KHR suffix omitted. However, Vulkan 1.4 implementations only have to support local read for storage resources and single sampled color attachments.

Support for reading depth/stencil attachments and multi-sampled attachments are respectively gated behind the new boolean `dynamicRenderingLocalReadDepthStencilAttachments` and `dynamicRenderingLocalReadMultisampledAttachments` properties, as described in the Version 1.4 appendix.

The original type, enum, and command names are still available as aliases of the core functionality.

Version History

- Revision 1, 2023-11-03 (Tobias Hector)
  - Initial revision

VK_KHR_external_fence

Name String

`VK_KHR_external_fence`

Extension Type

Device extension

Registered Extension Number

114

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_external_fence_capabilities

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jesse Hall [GitHub]critsec

Other Extension Metadata

Last Modified Date

2017-05-08

IP Status

No known IP claims.

Contributors

- Jesse Hall, Google
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Cass Everitt, Oculus
- Contributors to `VK_KHR_external_semaphore`

Description

An application using external memory may wish to synchronize access to that memory using fences. This extension enables an application to create fences from which non-Vulkan handles that reference the underlying synchronization primitive can be exported.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending VkFenceCreateInfo:
  - VkExportFenceCreateInfoKHR

New Enums

- VkFenceImportFlagBitsKHR

New Bitmasks

- VkFenceImportFlagsKHR

New Enum Constants

- `VK_KHR_EXTERNAL_FENCE_EXTENSION_NAME`
- `VK_KHR_EXTERNAL_FENCE_SPEC_VERSION`
- Extending VkFenceImportFlagBits:
  - VK_FENCE_IMPORT_TEMPORARY_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_EXPORT_FENCE_CREATE_INFO_KHR

Issues

This extension borrows concepts, semantics, and language from `VK_KHR_external_semaphore`. That extension's issues apply equally to this extension.

Version History

- Revision 1, 2017-05-08 (Jesse Hall)
  - Initial revision

VK_KHR_external_fence_capabilities

Name String

`VK_KHR_external_fence_capabilities`

Extension Type

Instance extension

Registered Extension Number

113

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jesse Hall [GitHub]critsec

Other Extension Metadata

Last Modified Date

2017-05-08

IP Status

No known IP claims.

Contributors

- Jesse Hall, Google
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Cass Everitt, Oculus
- Contributors to `VK_KHR_external_semaphore_capabilities`

Description

An application may wish to reference device fences in multiple Vulkan logical devices or instances, in multiple processes, and/or in multiple APIs. This extension provides a set of capability queries

and handle definitions that allow an application to determine what types of “external” fence handles an implementation supports for a given set of use cases.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- vkGetPhysicalDeviceExternalFencePropertiesKHR

New Structures

- VkExternalFencePropertiesKHR
- VkPhysicalDeviceExternalFenceInfoKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceIDPropertiesKHR

New Enums

- VkExternalFenceFeatureFlagBitsKHR
- VkExternalFenceHandleTypeFlagBitsKHR

New Bitmasks

- VkExternalFenceFeatureFlagsKHR
- VkExternalFenceHandleTypeFlagsKHR

New Enum Constants

- `VK_KHR_EXTERNAL_FENCE_CAPABILITIES_EXTENSION_NAME`
- `VK_KHR_EXTERNAL_FENCE_CAPABILITIES_SPEC_VERSION`
- VK_LUID_SIZE_KHR
- Extending VkExternalFenceFeatureFlagBits:
  - VK_EXTERNAL_FENCE_FEATURE_EXPORTABLE_BIT_KHR
  - VK_EXTERNAL_FENCE_FEATURE_IMPORTABLE_BIT_KHR
- Extending VkExternalFenceHandleTypeFlagBits:
  - VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_FD_BIT_KHR
  - VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR
  - VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT_KHR
  - VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT_KHR
- Extending VkStructureType:

    - VK_STRUCTURE_TYPE_EXTERNAL_FENCE_PROPERTIES_KHR
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_FENCE_INFO_KHR
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ID_PROPERTIES_KHR

Version History

- Revision 1, 2017-05-08 (Jesse Hall)
    - Initial version

VK_KHR_external_memory

Name String

`VK_KHR_external_memory`

Extension Type

Device extension

Registered Extension Number

73

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_external_memory_capabilities
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2016-10-20

IP Status

No known IP claims.

Interactions and External Dependencies

- Interacts with `VK_KHR_dedicated_allocation`.
- Interacts with `VK_NV_dedicated_allocation`.

Contributors

- Faith Ekstrand, Intel
- Ian Elliott, Google
- Jesse Hall, Google
- Tobias Hector, Imagination Technologies
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA

- Matthew Netsch, Qualcomm Technologies, Inc.
- Daniel Rakos, AMD
- Carsten Rohde, NVIDIA
- Ray Smith, ARM
- Lina Versace, Google

Description

An application may wish to reference device memory in multiple Vulkan logical devices or instances, in multiple processes, and/or in multiple APIs. This extension enables an application to export non-Vulkan handles from Vulkan memory objects such that the underlying resources can be referenced outside the scope of the Vulkan logical device that created them.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending VkBufferCreateInfo:
  - VkExternalMemoryBufferCreateInfoKHR
- Extending VkImageCreateInfo:
  - VkExternalMemoryImageCreateInfoKHR
- Extending VkMemoryAllocateInfo:
  - VkExportMemoryAllocateInfoKHR

New Enum Constants

- `VK_KHR_EXTERNAL_MEMORY_EXTENSION_NAME`
- `VK_KHR_EXTERNAL_MEMORY_SPEC_VERSION`
- VK_QUEUE_FAMILY_EXTERNAL_KHR
- Extending VkResult:
  - VK_ERROR_INVALID_EXTERNAL_HANDLE_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_EXPORT_MEMORY_ALLOCATE_INFO_KHR
  - VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_BUFFER_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_IMAGE_CREATE_INFO_KHR

Issues

1) How do applications correlate two physical devices across process or Vulkan instance boundaries?

RESOLVED: New device ID fields have been introduced by VK_KHR_external_memory_capabilities. These fields, combined with the existing VkPhysicalDeviceProperties::driverVersion field can be used to identify compatible devices across processes, drivers, and APIs. VkPhysicalDeviceProperties::pipelineCacheUUID is not sufficient for this purpose because despite its description in the specification, it need only identify a unique pipeline cache format in practice. Multiple devices may be able to use the same pipeline cache data, and hence it would be desirable for all of them to have the same pipeline cache UUID. However, only the same concrete physical device can be used when sharing memory, so an actual unique device ID was introduced. Further, the pipeline cache UUID was specific to Vulkan, but correlation with other, non-extensible APIs is required to enable interoperation with those APIs.

2) If memory objects are shared between processes and APIs, is this considered aliasing according to the rules outlined in the Memory Aliasing section?

RESOLVED: Yes. Applications must take care to obey all restrictions imposed on aliased resources when using memory across multiple Vulkan instances or other APIs.

3) Are new image layouts or metadata required to specify image layouts and layout transitions compatible with non-Vulkan APIs, or with other instances of the same Vulkan driver?

RESOLVED: Separate instances of the same Vulkan driver running on the same GPU should have identical internal layout semantics, so applications have the tools they need to ensure views of images are consistent between the two instances. Other APIs will fall into two categories: Those that are Vulkan- compatible, and those that are Vulkan-incompatible. Vulkan-incompatible APIs will require the image to be in the GENERAL layout whenever they are accessing them.

Note this does not attempt to address cross-device transitions, nor transitions to engines on the same device which are not visible within the Vulkan API. Both of these are beyond the scope of this extension.

4) Is a new barrier flag or operation of some type needed to prepare external memory for handoff to another Vulkan instance or API and/or receive it from another instance or API?

RESOLVED: Yes. Some implementations need to perform additional cache management when transitioning memory between address spaces and other APIs, instances, or processes which may operate in a separate address space. Options for defining this transition include:

- A new structure that can be added to the pNext list in VkMemoryBarrier, VkBufferMemoryBarrier, and VkImageMemoryBarrier.
- A new bit in VkAccessFlags that can be set to indicate an “external” access.
- A new bit in VkDependencyFlags
- A new special queue family that represents an “external” queue.

A new structure has the advantage that the type of external transition can be described in as much detail as necessary. However, there is not currently a known need for anything beyond differentiating between external and internal accesses, so this is likely an over-engineered solution. The access flag bit has the advantage that it can be applied at buffer, image, or global granularity, and semantically it maps pretty well to the operation being described. Additionally, the API already includes VK_ACCESS_MEMORY_READ_BIT and VK_ACCESS_MEMORY_WRITE_BIT which appear to

be intended for this purpose. However, there is no obvious pipeline stage that would correspond to an external access, and therefore no clear way to use VK_ACCESS_MEMORY_READ_BIT or VK_ACCESS_MEMORY_WRITE_BIT. VkDependencyFlags and VkPipelineStageFlags operate at command granularity rather than image or buffer granularity, which would make an entire pipeline barrier an internal→external or external→internal barrier. This may not be a problem in practice, but seems like the wrong scope. Another downside of VkDependencyFlags is that it lacks inherent directionality: there are no `src` and `dst` variants of it in the barrier or dependency description semantics, so two bits might need to be added to describe both internal→external and external→internal transitions. Transitioning a resource to a special queue family corresponds well with the operation of transitioning to a separate Vulkan instance, in that both operations ideally include scheduling a barrier on both sides of the transition: Both the releasing and the acquiring queue or process. Using a special queue family requires adding an additional reserved queue family index. Reusing VK_QUEUE_FAMILY_IGNORED would have left it unclear how to transition a concurrent usage resource from one process to another, since the semantics would have likely been equivalent to the currently-ignored transition of VK_QUEUE_FAMILY_IGNORED → VK_QUEUE_FAMILY_IGNORED. Fortunately, creating a new reserved queue family index is not invasive.

Based on the above analysis, the approach of transitioning to a special “external” queue family was chosen.

5) Do internal driver memory arrangements and/or other internal driver image properties need to be exported and imported when sharing images across processes or APIs.

RESOLVED: Some vendors claim this is necessary on their implementations, but it was determined that the security risks of allowing opaque metadata to be passed from applications to the driver were too high. Therefore, implementations which require metadata will need to associate it with the objects represented by the external handles, and rely on the dedicated allocation mechanism to associate the exported and imported memory objects with a single image or buffer.

6) Most prior interoperation and cross-process sharing APIs have been based on image-level sharing. Should Vulkan sharing be based on memory-object sharing or image sharing?

RESOLVED: These extensions have assumed memory-level sharing is the correct granularity. Vulkan is a lower-level API than most prior APIs, and as such attempts to closely align with to the underlying primitives of the hardware and system-level drivers it abstracts. In general, the resource that holds the backing store for both images and buffers of various types is memory. Images and buffers are merely metadata containing brief descriptions of the layout of bits within that memory.

Because memory object-based sharing is aligned with the overall Vulkan API design, it enables the full range of Vulkan capabilities with external objects. External memory can be used as backing for sparse images, for example, whereas such usage would be awkward at best with a sharing mechanism based on higher-level primitives such as images. Further, aligning the mechanism with the API in this way provides some hope of trivial compatibility with future API enhancements. If new objects backed by memory objects are added to the API, they too can be used across processes with minimal additions to the base external memory APIs.

Earlier APIs implemented interop at a higher level, and this necessitated entirely separate sharing

APIs for images and buffers. To co-exist and interoperate with those APIs, the Vulkan external sharing mechanism must accommodate their model. However, if it can be agreed that memory-based sharing is the more desirable and forward-looking design, legacy interoperation constraints can be considered another reason to favor memory-based sharing: while native and legacy driver primitives that may be used to implement sharing may not be as low-level as the API here suggests, raw memory is still the least common denominator among the types. Image-based sharing can be cleanly derived from a set of base memory- object sharing APIs with minimal effort, whereas image-based sharing does not generalize well to buffer or raw-memory sharing. Therefore, following the general Vulkan design principle of minimalism, it is better to expose interopability with image-based native and external primitives via the memory sharing API, and place sufficient limits on their usage to ensure they can be used only as backing for equivalent Vulkan images. This provides a consistent API for applications regardless of which platform or external API they are targeting, which makes development of multi-API and multi-platform applications simpler.

7) Should Vulkan define a common external handle type and provide Vulkan functions to facilitate cross-process sharing of such handles rather than relying on native handles to define the external objects?

RESOLVED: No. Cross-process sharing of resources is best left to native platforms. There are myriad security and extensibility issues with such a mechanism, and attempting to re-solve all those issues within Vulkan does not align with Vulkan's purpose as a graphics API. If desired, such a mechanism could be built as a layer or helper library on top of the opaque native handle defined in this family of extensions.

8) Must implementations provide additional guarantees about state implicitly included in memory objects for those memory objects that may be exported?

RESOLVED: Implementations must ensure that sharing memory objects does not transfer any information between the exporting and importing instances and APIs other than that required to share the data contained in the memory objects explicitly shared. As specific examples, data from previously freed memory objects that used the same underlying physical memory, and data from memory objects using adjacent physical memory must not be visible to applications importing an exported memory object.

9) Must implementations validate external handles the application provides as inputs to memory import operations?

RESOLVED: Implementations must return an error to the application if the provided memory handle cannot be used to complete the requested import operation. However, implementations need not validate handles are of the exact type specified by the application.

Version History

- Revision 1, 2016-10-20 (James Jones)
    - Initial version

VK_KHR_external_memory_capabilities

Name String

`VK_KHR_external_memory_capabilities`

Extension Type

Instance extension

Registered Extension Number

72

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2016-10-17

IP Status

No known IP claims.

Interactions and External Dependencies

- Interacts with `VK_KHR_dedicated_allocation`.
- Interacts with `VK_NV_dedicated_allocation`.

Contributors

- Ian Elliott, Google
- Jesse Hall, Google
- James Jones, NVIDIA

Description

An application may wish to reference device memory in multiple Vulkan logical devices or

instances, in multiple processes, and/or in multiple APIs. This extension provides a set of capability queries and handle definitions that allow an application to determine what types of “external” memory handles an implementation supports for a given set of use cases.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- vkGetPhysicalDeviceExternalBufferPropertiesKHR

New Structures

- VkExternalBufferPropertiesKHR
- VkExternalMemoryPropertiesKHR
- VkPhysicalDeviceExternalBufferInfoKHR
- Extending VkImageFormatProperties2:
  - VkExternalImageFormatPropertiesKHR
- Extending VkPhysicalDeviceImageFormatInfo2:
  - VkPhysicalDeviceExternalImageFormatInfoKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceIDPropertiesKHR

New Enums

- VkExternalMemoryFeatureFlagBitsKHR
- VkExternalMemoryHandleTypeFlagBitsKHR

New Bitmasks

- VkExternalMemoryFeatureFlagsKHR
- VkExternalMemoryHandleTypeFlagsKHR

New Enum Constants

- `VK_KHR_EXTERNAL_MEMORY_CAPABILITIES_EXTENSION_NAME`
- `VK_KHR_EXTERNAL_MEMORY_CAPABILITIES_SPEC_VERSION`
- VK_LUID_SIZE_KHR
- Extending VkExternalMemoryFeatureFlagBits:
  - VK_EXTERNAL_MEMORY_FEATURE_DEDICATED_ONLY_BIT_KHR
  - VK_EXTERNAL_MEMORY_FEATURE_EXPORTABLE_BIT_KHR
  - VK_EXTERNAL_MEMORY_FEATURE_IMPORTABLE_BIT_KHR

- Extending VkExternalMemoryHandleTypeFlagBits:
  - VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT_KHR
  - VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT_KHR
  - VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT_KHR
  - VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT_KHR
  - VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT_KHR
  - VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR
  - VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_EXTERNAL_BUFFER_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_EXTERNAL_IMAGE_FORMAT_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_BUFFER_INFO_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_IMAGE_FORMAT_INFO_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ID_PROPERTIES_KHR

Issues

1) Why do so many external memory capabilities need to be queried on a per-memory-handle-type basis?

This is because some handle types are based on OS-native objects that have far more limited capabilities than the very generic Vulkan memory objects. Not all memory handle types can name memory objects that support 3D images, for example. Some handle types cannot even support the deferred image and memory binding behavior of Vulkan and require specifying the image when allocating or importing the memory object.

2) Do the VkExternalImageFormatPropertiesKHR and VkExternalBufferPropertiesKHR structs need to include a list of memory type bits that support the given handle type?

No. The memory types that do not support the handle types will simply be filtered out of the results returned by vkGetImageMemoryRequirements and vkGetBufferMemoryRequirements when a set of handle types was specified at image or buffer creation time.

3) Should the non-opaque handle types be moved to their own extension?

Perhaps. However, defining the handle type bits does very little and does not require any platform-specific types on its own, and it is easier to maintain the bitfield values in a single extension for now. Presumably more handle types could be added by separate extensions though, and it would be midly weird to have some platform-specific ones defined in the core spec and some in extensions

4) Do we need a `D3D11_TILEPOOL` type?

No. This is technically possible, but the synchronization is awkward. D3D11 surfaces must be synchronized using shared mutexes, and these synchronization primitives are shared by the entire memory object, so D3D11 shared allocations divided among multiple buffer and image bindings

may be difficult to synchronize.

5) Should the Windows 7-compatible handle types be named “KMT” handles or “GLOBAL_SHARE” handles?

KMT, simply because it is more concise.

6) How do applications identify compatible devices and drivers across instance, process, and API boundaries when sharing memory?

New device properties are exposed that allow applications to correctly correlate devices and drivers. A device and driver UUID that must both match to ensure sharing compatibility between two Vulkan instances, or a Vulkan instance and an extensible external API are added. To allow correlating with Direct3D devices, a device LUID is added that corresponds to a DXGI adapter LUID. A driver ID is not needed for Direct3D because mismatched driver component versions are not currently supported on the Windows OS. Should support for such configurations be introduced at the OS level, further Vulkan extensions would be needed to correlate userspace component builds.

Version History

- Revision 1, 2016-10-17 (James Jones)
    - Initial version

VK_KHR_external_semaphore

Name String

`VK_KHR_external_semaphore`

Extension Type

Device extension

Registered Extension Number

78

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_external_semaphore_capabilities

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2016-10-21

IP Status

No known IP claims.

Contributors

- Faith Ekstrand, Intel
- Jesse Hall, Google
- Tobias Hector, Imagination Technologies
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Matthew Netsch, Qualcomm Technologies, Inc.
- Ray Smith, ARM
- Lina Versace, Google

Description

An application using external memory may wish to synchronize access to that memory using semaphores. This extension enables an application to create semaphores from which non-Vulkan handles that reference the underlying synchronization primitive can be exported.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending VkSemaphoreCreateInfo:
  - VkExportSemaphoreCreateInfoKHR

New Enums

- VkSemaphoreImportFlagBitsKHR

New Bitmasks

- VkSemaphoreImportFlagsKHR

New Enum Constants

- `VK_KHR_EXTERNAL_SEMAPHORE_EXTENSION_NAME`
- `VK_KHR_EXTERNAL_SEMAPHORE_SPEC_VERSION`
- Extending VkSemaphoreImportFlagBits:
  - VK_SEMAPHORE_IMPORT_TEMPORARY_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_CREATE_INFO_KHR

Issues

1) Should there be restrictions on what side effects can occur when waiting on imported semaphores that are in an invalid state?

RESOLVED: Yes. Normally, validating such state would be the responsibility of the application, and the implementation would be free to enter an undefined state if valid usage rules were violated. However, this could cause security concerns when using imported semaphores, as it would require the importing application to trust the exporting application to ensure the state is valid. Requiring this level of trust is undesirable for many potential use cases.

2) Must implementations validate external handles the application provides as input to semaphore state import operations?

RESOLVED: Implementations must return an error to the application if the provided semaphore

state handle cannot be used to complete the requested import operation. However, implementations need not validate handles are of the exact type specified by the application.

Version History

- Revision 1, 2016-10-21 (James Jones)
  - Initial revision

VK_KHR_external_semaphore_capabilities

Name String

`VK_KHR_external_semaphore_capabilities`

Extension Type

Instance extension

Registered Extension Number

77

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- James Jones [GitHub]cubanismo

Other Extension Metadata

Last Modified Date

2016-10-20

IP Status

No known IP claims.

Contributors

- Jesse Hall, Google
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA

Description

An application may wish to reference device semaphores in multiple Vulkan logical devices or instances, in multiple processes, and/or in multiple APIs. This extension provides a set of capability queries and handle definitions that allow an application to determine what types of “external” semaphore handles an implementation supports for a given set of use cases.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- vkGetPhysicalDeviceExternalSemaphorePropertiesKHR

New Structures

- VkExternalSemaphorePropertiesKHR
- VkPhysicalDeviceExternalSemaphoreInfoKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceIDPropertiesKHR

New Enums

- VkExternalSemaphoreFeatureFlagBitsKHR
- VkExternalSemaphoreHandleTypeFlagBitsKHR

New Bitmasks

- VkExternalSemaphoreFeatureFlagsKHR
- VkExternalSemaphoreHandleTypeFlagsKHR

New Enum Constants

- `VK_KHR_EXTERNAL_SEMAPHORE_CAPABILITIES_EXTENSION_NAME`
- `VK_KHR_EXTERNAL_SEMAPHORE_CAPABILITIES_SPEC_VERSION`
- VK_LUID_SIZE_KHR
- Extending VkExternalSemaphoreFeatureFlagBits:
  - VK_EXTERNAL_SEMAPHORE_FEATURE_EXPORTABLE_BIT_KHR
  - VK_EXTERNAL_SEMAPHORE_FEATURE_IMPORTABLE_BIT_KHR
- Extending VkExternalSemaphoreHandleTypeFlagBits:
  - VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT_KHR
  - VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT_KHR
  - VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR
  - VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT_KHR
  - VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_EXTERNAL_SEMAPHORE_PROPERTIES_KHR

- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_SEMAPHORE_INFO_KHR
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ID_PROPERTIES_KHR

Version History

- Revision 1, 2016-10-20 (James Jones)
  - Initial revision

VK_KHR_format_feature_flags2

Name String

`VK_KHR_format_feature_flags2`

Extension Type

Device extension

Registered Extension Number

361

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_VERSION_1_2
- Interacts with VK_EXT_filter_cubic
- Interacts with VK_EXT_sampler_filter_minmax
- Interacts with VK_IMG_filter_cubic

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Lionel Landwerlin [GitHub]llandwerlin

Other Extension Metadata

Last Modified Date

2021-07-01

IP Status

No known IP claims.

Contributors

- Lionel Landwerlin, Intel
- Faith Ekstrand, Intel
- Tobias Hector, AMD
- Spencer Fricke, Samsung Electronics

- Graeme Leese, Broadcom
- Jan-Harald Fredriksen, ARM

Description

This extension adds a new VkFormatFeatureFlagBits2KHR 64bits format feature flag type to extend the existing VkFormatFeatureFlagBits which is limited to 31 flags. At the time of this writing 29 bits of VkFormatFeatureFlagBits are already used.

Because VkFormatProperties2 is already defined to extend the Vulkan 1.0 vkGetPhysicalDeviceFormatProperties command, this extension defines a new VkFormatProperties3KHR to extend the VkFormatProperties.

On top of replicating all the bits from VkFormatFeatureFlagBits, VkFormatFeatureFlagBits2KHR adds the following bits :

- VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT_KHR and VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT_KHR specify that an implementation supports reading and writing, respectively, a given VkFormat through storage operations without specifying the format in the shader.
- VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT_KHR specifies that an implementation supports depth comparison performed by `OpImage*Dref*` instructions on a given VkFormat. Previously the result of executing a `OpImage*Dref*` instruction on an image view, where the `format` was not one of the depth/stencil formats with a depth component, was undefined. This bit clarifies on which formats such instructions can be used.

Prior to version 2 of this extension, implementations exposing the `shaderStorageImageReadWithoutFormat` and `shaderStorageImageWriteWithoutFormat` features may not report VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT_KHR and VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT_KHR in VkFormatProperties3KHR::`bufferFeatures`. Despite this, buffer reads/writes are supported as intended by the original features.

New Structures

- Extending VkFormatProperties2:
  - VkFormatProperties3KHR

New Enums

- VkFormatFeatureFlagBits2KHR

New Bitmasks

- VkFormatFeatureFlags2KHR

New Enum Constants

- `VK_KHR_FORMAT_FEATURE_FLAGS_2_EXTENSION_NAME`

- VK_KHR_FORMAT_FEATURE_FLAGS_2_SPEC_VERSION
- Extending VkFormatFeatureFlagBits2:
  - VK_FORMAT_FEATURE_2_BLIT_DST_BIT_KHR
  - VK_FORMAT_FEATURE_2_BLIT_SRC_BIT_KHR
  - VK_FORMAT_FEATURE_2_COLOR_ATTACHMENT_BIT_KHR
  - VK_FORMAT_FEATURE_2_COLOR_ATTACHMENT_BLEND_BIT_KHR
  - VK_FORMAT_FEATURE_2_COSITED_CHROMA_SAMPLES_BIT_KHR
  - VK_FORMAT_FEATURE_2_DEPTH_STENCIL_ATTACHMENT_BIT_KHR
  - VK_FORMAT_FEATURE_2_DISJOINT_BIT_KHR
  - VK_FORMAT_FEATURE_2_MIDPOINT_CHROMA_SAMPLES_BIT_KHR
  - VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT_KHR
  - VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT_KHR
  - VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_LINEAR_BIT_KHR
  - VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT_KHR
  - VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT_KHR
  - VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT_KHR
  - VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT_KHR
  - VK_FORMAT_FEATURE_2_STORAGE_IMAGE_ATOMIC_BIT_KHR
  - VK_FORMAT_FEATURE_2_STORAGE_IMAGE_BIT_KHR
  - VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT_KHR
  - VK_FORMAT_FEATURE_2_STORAGE_TEXEL_BUFFER_ATOMIC_BIT_KHR
  - VK_FORMAT_FEATURE_2_STORAGE_TEXEL_BUFFER_BIT_KHR
  - VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT_KHR
  - VK_FORMAT_FEATURE_2_TRANSFER_DST_BIT_KHR
  - VK_FORMAT_FEATURE_2_TRANSFER_SRC_BIT_KHR
  - VK_FORMAT_FEATURE_2_UNIFORM_TEXEL_BUFFER_BIT_KHR
  - VK_FORMAT_FEATURE_2_VERTEX_BUFFER_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_3_KHR

If VK_EXT_filter_cubic or VK_IMG_filter_cubic is supported:

- Extending VkFormatFeatureFlagBits2:

    - VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT

If Vulkan Version 1.2 or `VK_EXT_sampler_filter_minmax` is supported:

- Extending VkFormatFeatureFlagBits2:
    - VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_MINMAX_BIT_KHR

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Version History

- Revision 2, 2022-07-20 (Lionel Landwerlin)
    - Clarify that VK_FORMAT_FEATURE_2_STORAGE_(READ|WRITE)_WITHOUT_FORMAT_BIT also apply to buffer views.
- Revision 1, 2020-07-21 (Lionel Landwerlin)
    - Initial draft

VK_KHR_get_memory_requirements2

Name String

`VK_KHR_get_memory_requirements2`

Extension Type

Device extension

Registered Extension Number

147

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Faith Ekstrand [GitHub]gfxstrand

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Contributors

- Faith Ekstrand, Intel
- Jeff Bolz, NVIDIA
- Jesse Hall, Google

Description

This extension provides new queries for memory requirements of images and buffers that can be easily extended by other extensions, without introducing any additional commands. The Vulkan 1.0 VkMemoryRequirements and VkSparseImageMemoryRequirements structures do not include `sType` and `pNext` members. This extension wraps them in new structures with these members, so an application can query a chain of memory requirements structures by constructing the chain and letting the implementation fill them in. A new command is added for each `vkGet*MemoryRequrements` command in core Vulkan 1.0.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- vkGetBufferMemoryRequirements2KHR
- vkGetImageMemoryRequirements2KHR
- vkGetImageSparseMemoryRequirements2KHR

New Structures

- VkBufferMemoryRequirementsInfo2KHR
- VkImageMemoryRequirementsInfo2KHR
- VkImageSparseMemoryRequirementsInfo2KHR
- VkMemoryRequirements2KHR
- VkSparseImageMemoryRequirements2KHR

New Enum Constants

- `VK_KHR_GET_MEMORY_REQUIREMENTS_2_EXTENSION_NAME`
- `VK_KHR_GET_MEMORY_REQUIREMENTS_2_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_BUFFER_MEMORY_REQUIREMENTS_INFO_2_KHR
  - VK_STRUCTURE_TYPE_IMAGE_MEMORY_REQUIREMENTS_INFO_2_KHR
  - VK_STRUCTURE_TYPE_IMAGE_SPARSE_MEMORY_REQUIREMENTS_INFO_2_KHR
  - VK_STRUCTURE_TYPE_MEMORY_REQUIREMENTS_2_KHR
  - VK_STRUCTURE_TYPE_SPARSE_IMAGE_MEMORY_REQUIREMENTS_2_KHR

Version History

- Revision 1, 2017-03-23 (Faith Ekstrand)
  - Internal revisions

VK_KHR_get_physical_device_properties2

Name String

`VK_KHR_get_physical_device_properties2`

Extension Type

Instance extension

Registered Extension Number

60

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jeff Bolz [GitHub]jeffbolznv

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Ian Elliott, Google

Description

This extension provides new queries for device features, device properties, and format properties that can be easily extended by other extensions, without introducing any further queries. The Vulkan 1.0 feature/limit/formatproperty structures do not include `sType`/`pNext` members. This extension wraps them in new structures with `sType`/`pNext` members, so an application can query a chain of feature/limit/formatproperty structures by constructing the chain and letting the implementation fill them in. A new command is added for each `vkGetPhysicalDevice*` command in core Vulkan 1.0. The new feature structure (and a `pNext` chain of extending structures) can also be passed in to device creation to enable features.

This extension also allows applications to use the physical-device components of device extensions before vkCreateDevice is called.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- vkGetPhysicalDeviceFeatures2KHR
- vkGetPhysicalDeviceFormatProperties2KHR
- vkGetPhysicalDeviceImageFormatProperties2KHR
- vkGetPhysicalDeviceMemoryProperties2KHR
- vkGetPhysicalDeviceProperties2KHR
- vkGetPhysicalDeviceQueueFamilyProperties2KHR
- vkGetPhysicalDeviceSparseImageFormatProperties2KHR

New Structures

- VkFormatProperties2KHR
- VkImageFormatProperties2KHR
- VkPhysicalDeviceImageFormatInfo2KHR
- VkPhysicalDeviceMemoryProperties2KHR
- VkPhysicalDeviceProperties2KHR
- VkPhysicalDeviceSparseImageFormatInfo2KHR
- VkQueueFamilyProperties2KHR
- VkSparseImageFormatProperties2KHR
- Extending VkDeviceCreateInfo:
  - VkPhysicalDeviceFeatures2KHR

New Enum Constants

- `VK_KHR_GET_PHYSICAL_DEVICE_PROPERTIES_2_EXTENSION_NAME`
- `VK_KHR_GET_PHYSICAL_DEVICE_PROPERTIES_2_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_2_KHR
  - VK_STRUCTURE_TYPE_IMAGE_FORMAT_PROPERTIES_2_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FEATURES_2_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_FORMAT_INFO_2_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PROPERTIES_2_KHR

- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROPERTIES_2_KHR
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SPARSE_IMAGE_FORMAT_INFO_2_KHR
- VK_STRUCTURE_TYPE_QUEUE_FAMILY_PROPERTIES_2_KHR
- VK_STRUCTURE_TYPE_SPARSE_IMAGE_FORMAT_PROPERTIES_2_KHR

Examples

```
// Get features with a hypothetical future extension.
VkHypotheticalExtensionFeaturesKHR hypotheticalFeatures =
{
    .sType = VK_STRUCTURE_TYPE_HYPOTHETICAL_FEATURES_KHR,
    .pNext = NULL,
};

VkPhysicalDeviceFeatures2KHR features =
{
    .sType = VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FEATURES_2_KHR,
    .pNext = &hypotheticalFeatures,
};

// After this call, features and hypotheticalFeatures have been filled out.
vkGetPhysicalDeviceFeatures2KHR(physicalDevice, &features);

// Properties/limits can be chained and queried similarly.

// Enable some features:
VkHypotheticalExtensionFeaturesKHR enabledHypotheticalFeatures =
{
    .sType = VK_STRUCTURE_TYPE_HYPOTHETICAL_FEATURES_KHR,
    .pNext = NULL,
};

VkPhysicalDeviceFeatures2KHR enabledFeatures =
{
    .sType = VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FEATURES_2_KHR,
    .pNext = &enabledHypotheticalFeatures,
};

enabledFeatures.features.xyz = VK_TRUE;
enabledHypotheticalFeatures.abc = VK_TRUE;

VkDeviceCreateInfo deviceCreateInfo =
{
    .sType = VK_STRUCTURE_TYPE_DEVICE_CREATE_INFO,
    .pNext = &enabledFeatures,
    ...
    .pEnabledFeatures = NULL,
};
```

```
    VkDevice device;
    vkCreateDevice(physicalDevice, &deviceCreateInfo, NULL, &device);
```

Version History

- Revision 1, 2016-09-12 (Jeff Bolz)
  - Internal revisions
- Revision 2, 2016-11-02 (Ian Elliott)
  - Added ability for applications to use the physical-device components of device extensions before vkCreateDevice is called.

VK_KHR_global_priority

Name String

VK_KHR_global_priority

Extension Type

Device extension

Registered Extension Number

189

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Tobias Hector [GitHub]tobski

Other Extension Metadata

Last Modified Date

2021-10-22

Contributors

- Tobias Hector, AMD
- Contributors to VK_EXT_global_priority
- Contributors to VK_EXT_global_priority_query

Description

In Vulkan, users can specify device-scope queue priorities. In some cases it may be useful to extend this concept to a system-wide scope. This device extension allows applications to query the global queue priorities supported by a queue family, and then set a priority when creating queues. The default queue priority is VK_QUEUE_GLOBAL_PRIORITY_MEDIUM_EXT.

Implementations can report which global priority levels are treated differently by the implementation. It is intended primarily for use in system integration along with certain platform-specific priority enforcement rules.

The driver implementation will attempt to skew hardware resource allocation in favor of the higher-priority task. Therefore, higher-priority work may retain similar latency and throughput characteristics even if the system is congested with lower priority work.

The global priority level of a queue shall take precedence over the per-process queue priority (`VkDeviceQueueCreateInfo::pQueuePriorities`).

Abuse of this feature may result in starving the rest of the system from hardware resources. Therefore, the driver implementation may deny requests to acquire a priority above the default priority (VK_QUEUE_GLOBAL_PRIORITY_MEDIUM_EXT) if the caller does not have sufficient privileges. In this scenario VK_ERROR_NOT_PERMITTED_EXT is returned.

The driver implementation may fail the queue allocation request if resources required to complete the operation have been exhausted (either by the same process or a different process). In this scenario VK_ERROR_INITIALIZATION_FAILED is returned.

New Structures

- Extending VkDeviceQueueCreateInfo:
  - VkDeviceQueueGlobalPriorityCreateInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceGlobalPriorityQueryFeaturesKHR
- Extending VkQueueFamilyProperties2:
  - VkQueueFamilyGlobalPriorityPropertiesKHR

New Enums

- VkQueueGlobalPriorityKHR

New Enum Constants

- `VK_KHR_GLOBAL_PRIORITY_EXTENSION_NAME`
- `VK_KHR_GLOBAL_PRIORITY_SPEC_VERSION`
- VK_MAX_GLOBAL_PRIORITY_SIZE_KHR
- Extending VkQueueGlobalPriority:
  - VK_QUEUE_GLOBAL_PRIORITY_HIGH_KHR
  - VK_QUEUE_GLOBAL_PRIORITY_LOW_KHR
  - VK_QUEUE_GLOBAL_PRIORITY_MEDIUM_KHR
  - VK_QUEUE_GLOBAL_PRIORITY_REALTIME_KHR
- Extending VkResult:
  - VK_ERROR_NOT_PERMITTED_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DEVICE_QUEUE_GLOBAL_PRIORITY_CREATE_INFO_KHR

    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GLOBAL_PRIORITY_QUERY_FEATURES_KHR
    - VK_STRUCTURE_TYPE_QUEUE_FAMILY_GLOBAL_PRIORITY_PROPERTIES_KHR

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4 with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

Issues

1) Can we additionally query whether a caller is permitted to acquire a specific global queue priority in this extension?

RESOLVED: No. Whether a caller has enough privilege goes with the OS, and the Vulkan driver cannot really guarantee that the privilege will not change in between this query and the actual queue creation call.

2) If more than 1 queue using global priority is requested, is there a good way to know which queue is failing the device creation?

RESOLVED: No. There is not a good way at this moment, and it is also not quite actionable for the applications to know that because the information may not be accurate. Queue creation can fail because of runtime constraints like insufficient privilege or lack of resource, and the failure is not necessarily tied to that particular queue configuration requested.

Version History

- Revision 1, 2021-10-22 (Tobias Hector)
    - Initial draft

VK_KHR_image_format_list

Name String

`VK_KHR_image_format_list`

Extension Type

Device extension

Registered Extension Number

148

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Faith Ekstrand [GitHub]gfxstrand

Other Extension Metadata

Last Modified Date

2017-03-20

IP Status

No known IP claims.

Contributors

- Faith Ekstrand, Intel
- Jan-Harald Fredriksen, ARM
- Jeff Bolz, NVIDIA
- Jeff Leger, Qualcomm
- Neil Henning, Codeplay

Description

On some implementations, setting the VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT on image creation can cause access to that image to perform worse than an equivalent image created without VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT because the implementation does not know what view formats will be paired with the image.

This extension allows an application to provide the list of all formats that **can** be used with an image when it is created. The implementation may then be able to create a more efficient image that supports the subset of formats required by the application without having to support all formats in the format compatibility class of the image format.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending VkImageCreateInfo, VkSwapchainCreateInfoKHR, VkPhysicalDeviceImageFormatInfo2:
  - VkImageFormatListCreateInfoKHR

New Enum Constants

- `VK_KHR_IMAGE_FORMAT_LIST_EXTENSION_NAME`
- `VK_KHR_IMAGE_FORMAT_LIST_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_IMAGE_FORMAT_LIST_CREATE_INFO_KHR

Version History

- Revision 1, 2017-03-20 (Faith Ekstrand)
  - Initial revision

VK_KHR_imageless_framebuffer

Name String

`VK_KHR_imageless_framebuffer`

Extension Type

Device extension

Registered Extension Number

109

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
and
VK_KHR_maintenance2
or
Vulkan Version 1.1
and
VK_KHR_image_format_list
or
Vulkan Version 1.2

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Tobias Hector [GitHub]tobias

Other Extension Metadata

Last Modified Date

2018-12-14

Contributors

- Tobias Hector
- Graham Wihlidal

Description

This extension allows framebuffers to be created without the need for creating images first, allowing more flexibility in how they are used, and avoiding the need for many of the confusing compatibility rules.

Framebuffers are now created with a small amount of additional metadata about the image views that will be used in VkFramebufferAttachmentsCreateInfoKHR, and the actual image views are provided at render pass begin time via VkRenderPassAttachmentBeginInfoKHR.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- VkFramebufferAttachmentImageInfoKHR
- Extending VkFramebufferCreateInfo:
  - VkFramebufferAttachmentsCreateInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceImagelessFramebufferFeaturesKHR
- Extending VkRenderPassBeginInfo:
  - VkRenderPassAttachmentBeginInfoKHR

New Enum Constants

- `VK_KHR_IMAGELESS_FRAMEBUFFER_EXTENSION_NAME`
- `VK_KHR_IMAGELESS_FRAMEBUFFER_SPEC_VERSION`
- Extending VkFramebufferCreateFlagBits:
  - VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENTS_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENT_IMAGE_INFO_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGELESS_FRAMEBUFFER_FEATURES_KHR
  - VK_STRUCTURE_TYPE_RENDER_PASS_ATTACHMENT_BEGIN_INFO_KHR

Version History

- Revision 1, 2018-12-14 (Tobias Hector)
  - Internal revisions

VK_KHR_index_type_uint8

Name String

VK_KHR_index_type_uint8

Extension Type

Device extension

Registered Extension Number

534

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2023-06-06

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA

Description

This extension allows `uint8_t` indices to be used with vkCmdBindIndexBuffer.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceIndexTypeUint8FeaturesKHR

New Enum Constants

- `VK_KHR_INDEX_TYPE_UINT8_EXTENSION_NAME`
- `VK_KHR_INDEX_TYPE_UINT8_SPEC_VERSION`
- Extending VkIndexType:
  - VK_INDEX_TYPE_UINT8_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INDEX_TYPE_UINT8_FEATURES_KHR

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4 with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

Version History

- Revision 1, 2023-06-06 (Piers Daniell)
  - Internal revisions

VK_KHR_line_rasterization

Name String

`VK_KHR_line_rasterization`

Extension Type

Device extension

Registered Extension Number

535

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2023-06-08

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Allen Jensen, NVIDIA
- Faith Ekstrand, Intel

Description

This extension adds some line rasterization features that are commonly used in CAD applications and supported in other APIs like OpenGL. Bresenham-style line rasterization is supported, smooth rectangular lines (coverage to alpha) are supported, and stippled lines are supported for all three line rasterization modes.

New Commands

- vkCmdSetLineStippleKHR

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceLineRasterizationFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceLineRasterizationPropertiesKHR
- Extending VkPipelineRasterizationStateCreateInfo:
  - VkPipelineRasterizationLineStateCreateInfoKHR

New Enums

- VkLineRasterizationModeKHR

New Enum Constants

- `VK_KHR_LINE_RASTERIZATION_EXTENSION_NAME`
- `VK_KHR_LINE_RASTERIZATION_SPEC_VERSION`
- Extending VkDynamicState:
  - VK_DYNAMIC_STATE_LINE_STIPPLE_KHR
- Extending VkLineRasterizationMode:
  - VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR
  - VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR
  - VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR
  - VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_LINE_STATE_CREATE_INFO_KHR

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4 with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

When Version 1.4 is supported, the `bresenhamLines` feature must be supported.

Issues

1) Do we need to support Bresenham-style and smooth lines with more than one rasterization

sample? i.e. the equivalent of glDisable(GL_MULTISAMPLE) in OpenGL when the framebuffer has more than one sample?

RESOLVED: Yes. For simplicity, Bresenham line rasterization carries forward a few restrictions from OpenGL, such as not supporting per-sample shading, alpha to coverage, or alpha to one.

Version History

- Revision 1, 2019-05-09 (Jeff Bolz)
  - Initial draft

VK_KHR_load_store_op_none

Name String

VK_KHR_load_store_op_none

Extension Type

Device extension

Registered Extension Number

527

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Extension Proposal

VK_KHR_load_store_op_none

Other Extension Metadata

Last Modified Date

2023-05-16

Contributors

- Shahbaz Youssefi, Google
- Bill Licea-Kane, Qualcomm Technologies, Inc.
- Tobias Hector, AMD

Description

This extension provides VK_ATTACHMENT_LOAD_OP_NONE_KHR and VK_ATTACHMENT_STORE_OP_NONE_KHR, which are identically promoted from the VK_EXT_load_store_op_none extension.

New Enum Constants

- VK_KHR_LOAD_STORE_OP_NONE_EXTENSION_NAME

- VK_KHR_LOAD_STORE_OP_NONE_SPEC_VERSION
- Extending VkAttachmentLoadOp:
  - VK_ATTACHMENT_LOAD_OP_NONE_KHR
- Extending VkAttachmentStoreOp:
  - VK_ATTACHMENT_STORE_OP_NONE_KHR

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4 with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

NOTE	While VK_ATTACHMENT_STORE_OP_NONE is part of Vulkan 1.3, this extension was not promoted to core Vulkan 1.3 either in whole or in part. This functionality was promoted from VK_KHR_dynamic_rendering.

Version History

- Revision 1, 2023-05-16 (Shahbaz Youssefi)
  - Initial revision, based on VK_EXT_load_store_op_none.

VK_KHR_maintenance1

Name String

`VK_KHR_maintenance1`

Extension Type

Device extension

Registered Extension Number

70

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2018-03-13

Contributors

- Dan Ginsburg, Valve
- Daniel Koch, NVIDIA
- Daniel Rakos, AMD
- Jan-Harald Fredriksen, ARM
- Faith Ekstrand, Intel
- Jeff Bolz, NVIDIA
- Jesse Hall, Google
- John Kessenich, Google
- Michael Worcester, Imagination Technologies
- Neil Henning, Codeplay Software Ltd.
- Piers Daniell, NVIDIA
- Slawomir Grajewski, Intel

- Tobias Hector, Imagination Technologies
- Tom Olson, ARM

Description

`VK_KHR_maintenance1` adds a collection of minor features that were intentionally left out or overlooked from the original Vulkan 1.0 release.

The new features are as follows:

- Allow 2D and 2D array image views to be created from 3D images, which can then be used as color framebuffer attachments. This allows applications to render to slices of a 3D image.
- Support vkCmdCopyImage between 2D array layers and 3D slices. This extension allows copying from layers of a 2D array image to slices of a 3D image and vice versa.
- Allow negative height to be specified in the VkViewport::`height` field to perform y-inversion of the clip-space to framebuffer-space transform. This allows apps to avoid having to use `gl_Position.y = -gl_Position.y` in shaders also targeting other APIs.
- Allow implementations to express support for doing just transfers and clears of image formats that they otherwise support no other format features for. This is done by adding new format feature flags VK_FORMAT_FEATURE_TRANSFER_SRC_BIT_KHR and VK_FORMAT_FEATURE_TRANSFER_DST_BIT_KHR.
- Support vkCmdFillBuffer on transfer-only queues. Previously vkCmdFillBuffer was defined to only work on command buffers allocated from command pools which support graphics or compute queues. It is now allowed on queues that just support transfer operations.
- Fix the inconsistency of how error conditions are returned between the vkCreateGraphicsPipelines and vkCreateComputePipelines functions and the vkAllocateDescriptorSets and vkAllocateCommandBuffers functions.
- Add new VK_ERROR_OUT_OF_POOL_MEMORY_KHR error so implementations can give a more precise reason for vkAllocateDescriptorSets failures.
- Add a new command vkTrimCommandPoolKHR which gives the implementation an opportunity to release any unused command pool memory back to the system.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- vkTrimCommandPoolKHR

New Bitmasks

- VkCommandPoolTrimFlagsKHR

New Enum Constants

- `VK_KHR_MAINTENANCE1_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE1_SPEC_VERSION`
- `VK_KHR_MAINTENANCE_1_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE_1_SPEC_VERSION`
- Extending VkFormatFeatureFlagBits:
  - VK_FORMAT_FEATURE_TRANSFER_DST_BIT_KHR
  - VK_FORMAT_FEATURE_TRANSFER_SRC_BIT_KHR
- Extending VkImageCreateFlagBits:
  - VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT_KHR
- Extending VkResult:
  - VK_ERROR_OUT_OF_POOL_MEMORY_KHR

Issues

1. Are viewports with zero height allowed?

   RESOLVED: Yes, although they have low utility.

Version History

- Revision 1, 2016-10-26 (Piers Daniell)
  - Internal revisions
- Revision 2, 2018-03-13 (Jon Leech)
  - Add issue for zero-height viewports

VK_KHR_maintenance2

Name String

`VK_KHR_maintenance2`

Extension Type

Device extension

Registered Extension Number

118

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Michael Worcester [GitHub]michaelworcester

Other Extension Metadata

Last Modified Date

2017-09-05

Contributors

- Michael Worcester, Imagination Technologies
- Stuart Smith, Imagination Technologies
- Jeff Bolz, NVIDIA
- Daniel Koch, NVIDIA
- Jan-Harald Fredriksen, ARM
- Daniel Rakos, AMD
- Neil Henning, Codeplay
- Piers Daniell, NVIDIA

Description

`VK_KHR_maintenance2` adds a collection of minor features that were intentionally left out or overlooked from the original Vulkan 1.0 release.

The new features are as follows:

- Allow the application to specify which aspect of an input attachment might be read for a given subpass.
- Allow implementations to express the clipping behavior of points.
- Allow creating images with usage flags that may not be supported for the base image's format, but are supported for image views of the image that have a different but compatible format.
- Allow creating uncompressed image views of compressed images.
- Allow the application to select between an upper-left and lower-left origin for the tessellation domain space.
- Adds two new image layouts for depth stencil images to allow either the depth or stencil aspect to be read-only while the other aspect is writable.

Input Attachment Specification

Input attachment specification allows an application to specify which aspect of a multi-aspect image (e.g. a depth/stencil format) will be accessed via a `subpassLoad` operation.

On some implementations there **may** be a performance penalty if the implementation does not know (at vkCreateRenderPass time) which aspect(s) of multi-aspect images **can** be accessed as input attachments.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- VkInputAttachmentAspectReferenceKHR
- Extending VkImageViewCreateInfo:
  - VkImageViewUsageCreateInfoKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDevicePointClippingPropertiesKHR
- Extending VkPipelineTessellationStateCreateInfo:
  - VkPipelineTessellationDomainOriginStateCreateInfoKHR
- Extending VkRenderPassCreateInfo:
  - VkRenderPassInputAttachmentAspectCreateInfoKHR

New Enums

- VkPointClippingBehaviorKHR
- VkTessellationDomainOriginKHR

New Enum Constants

- VK_KHR_MAINTENANCE2_EXTENSION_NAME
- VK_KHR_MAINTENANCE2_SPEC_VERSION
- VK_KHR_MAINTENANCE_2_EXTENSION_NAME
- VK_KHR_MAINTENANCE_2_SPEC_VERSION
- Extending VkImageCreateFlagBits:
  - VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT_KHR
  - VK_IMAGE_CREATE_EXTENDED_USAGE_BIT_KHR
- Extending VkImageLayout:
  - VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL_KHR
  - VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL_KHR
- Extending VkPointClippingBehavior:
  - VK_POINT_CLIPPING_BEHAVIOR_ALL_CLIP_PLANES_KHR
  - VK_POINT_CLIPPING_BEHAVIOR_USER_CLIP_PLANES_ONLY_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_IMAGE_VIEW_USAGE_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_POINT_CLIPPING_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_DOMAIN_ORIGIN_STATE_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_RENDER_PASS_INPUT_ATTACHMENT_ASPECT_CREATE_INFO_KHR
- Extending VkTessellationDomainOrigin:
  - VK_TESSELLATION_DOMAIN_ORIGIN_LOWER_LEFT_KHR
  - VK_TESSELLATION_DOMAIN_ORIGIN_UPPER_LEFT_KHR

Input Attachment Specification Example

Consider the case where a render pass has two subpasses and two attachments.

Attachment 0 has the format VK_FORMAT_D24_UNORM_S8_UINT, attachment 1 has some color format.

Subpass 0 writes to attachment 0, subpass 1 reads only the depth information from attachment 0 (using inputAttachmentRead) and writes to attachment 1.

```
    VkInputAttachmentAspectReferenceKHR references[] = {
        {
            .subpass = 1,
            .inputAttachmentIndex = 0,
            .aspectMask = VK_IMAGE_ASPECT_DEPTH_BIT
        }
```

```
    };

    VkRenderPassInputAttachmentAspectCreateInfoKHR specifyAspects = {
        .sType =
VK_STRUCTURE_TYPE_RENDER_PASS_INPUT_ATTACHMENT_ASPECT_CREATE_INFO_KHR,
        .pNext = NULL,
        .aspectReferenceCount = 1,
        .pAspectReferences = references
    };


    VkRenderPassCreateInfo createInfo = {
        ...
        .pNext = &specifyAspects,
        ...
    };

    vkCreateRenderPass(...);
```

Issues

1) What is the default tessellation domain origin?

RESOLVED: Vulkan 1.0 originally inadvertently documented a lower-left origin, but the conformance tests and all implementations implemented an upper-left origin. This extension adds a control to select between lower-left (for compatibility with OpenGL) and upper-left, and we retroactively fix unextended Vulkan to have a default of an upper-left origin.

Version History

- Revision 1, 2017-04-28

VK_KHR_maintenance3

Name String

VK_KHR_maintenance3

Extension Type

Device extension

Registered Extension Number

169

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jeff Bolz [GitHub]jeffbolznv

Other Extension Metadata

Last Modified Date

2017-09-05

Contributors

- Jeff Bolz, NVIDIA

Description

VK_KHR_maintenance3 adds a collection of minor features that were intentionally left out or overlooked from the original Vulkan 1.0 release.

The new features are as follows:

- A limit on the maximum number of descriptors that are supported in a single descriptor set layout. Some implementations have a limit on the total size of descriptors in a set, which cannot be expressed in terms of the limits in Vulkan 1.0.
- A limit on the maximum size of a single memory allocation. Some platforms have kernel interfaces that limit the maximum size of an allocation.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- vkGetDescriptorSetLayoutSupportKHR

New Structures

- VkDescriptorSetLayoutSupportKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceMaintenance3PropertiesKHR

New Enum Constants

- `VK_KHR_MAINTENANCE3_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE3_SPEC_VERSION`
- `VK_KHR_MAINTENANCE_3_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE_3_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_SUPPORT_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_3_PROPERTIES_KHR

Version History

- Revision 1, 2017-08-22

VK_KHR_maintenance4

Name String

`VK_KHR_maintenance4`

Extension Type

Device extension

Registered Extension Number

414

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2021-10-25

Interactions and External Dependencies

- Requires SPIR-V 1.2 for `LocalSizeId`

Contributors

- Lionel Duc, NVIDIA
- Faith Ekstrand, Intel
- Spencer Fricke, Samsung
- Tobias Hector, AMD
- Lionel Landwerlin, Intel
- Graeme Leese, Broadcom
- Tom Olson, Arm
- Stu Smith, AMD
- Yiwei Zhang, Google

Description

VK_KHR_maintenance4 adds a collection of minor features, none of which would warrant an entire extension of their own.

The new features are as follows:

- Allow the application to destroy their VkPipelineLayout object immediately after it was used to create another object. It is no longer necessary to keep its handle valid while the created object is in use.
- Add a new maxBufferSize implementation-defined limit for the maximum size VkBuffer that **can** be created.
- Add support for the SPIR-V 1.2 LocalSizeId execution mode, which can be used as an alternative to LocalSize to specify the local workgroup size with specialization constants.
- Add a guarantee that images created with identical creation parameters will always have the same alignment requirements.
- Add new vkGetDeviceBufferMemoryRequirementsKHR, vkGetDeviceImageMemoryRequirementsKHR, and vkGetDeviceImageSparseMemoryRequirementsKHR to allow the application to query the image memory requirements without having to create an image object and query it.
- Relax the requirement that push constants must be initialized before they are dynamically accessed.
- Relax the interface matching rules to allow a larger output vector to match with a smaller input vector, with additional values being discarded.
- Add a guarantee for buffer memory requirement that the size memory requirement is never greater than the result of aligning create size with the alignment memory requirement.

New Commands

- vkGetDeviceBufferMemoryRequirementsKHR
- vkGetDeviceImageMemoryRequirementsKHR
- vkGetDeviceImageSparseMemoryRequirementsKHR

New Structures

- VkDeviceBufferMemoryRequirementsKHR
- VkDeviceImageMemoryRequirementsKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMaintenance4FeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceMaintenance4PropertiesKHR

New Enum Constants

- `VK_KHR_MAINTENANCE_4_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE_4_SPEC_VERSION`
- Extending VkImageAspectFlagBits:
  - VK_IMAGE_ASPECT_NONE_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_DEVICE_BUFFER_MEMORY_REQUIREMENTS_KHR
  - VK_STRUCTURE_TYPE_DEVICE_IMAGE_MEMORY_REQUIREMENTS_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_PROPERTIES_KHR

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Issues

None.

Version History

- Revision 1, 2021-08-18 (Piers Daniell)
  - Internal revisions
- Revision 2, 2021-10-25 (Yiwei Zhang)
  - More guarantees on buffer memory requirements

VK_KHR_maintenance5

Name String

`VK_KHR_maintenance5`

Extension Type

Device extension

Registered Extension Number

471

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.1
and
VK_KHR_dynamic_rendering
or
Vulkan Version 1.3

API Interactions

- Interacts with VK_VERSION_1_2
- Interacts with VK_VERSION_1_3
- Interacts with VK_VERSION_1_4
- Interacts with VK_ARM_pipeline_opacity_micromap
- Interacts with VK_ARM_shader_instrumentation
- Interacts with VK_EXT_attachment_feedback_loop_layout
- Interacts with VK_EXT_buffer_device_address
- Interacts with VK_EXT_conditional_rendering
- Interacts with VK_EXT_descriptor_buffer
- Interacts with VK_EXT_fragment_density_map
- Interacts with VK_EXT_graphics_pipeline_library
- Interacts with VK_EXT_pipeline_creation_cache_control
- Interacts with VK_EXT_pipeline_protected_access
- Interacts with VK_EXT_transform_feedback
- Interacts with VK_KHR_acceleration_structure
- Interacts with VK_KHR_buffer_device_address
- Interacts with VK_KHR_dynamic_rendering

- Interacts with VK_KHR_fragment_shading_rate
- Interacts with VK_KHR_pipeline_executable_properties
- Interacts with VK_KHR_pipeline_library
- Interacts with VK_KHR_ray_tracing_pipeline
- Interacts with VK_KHR_video_decode_queue
- Interacts with VK_KHR_video_encode_queue
- Interacts with VK_NV_device_generated_commands
- Interacts with VK_NV_displacement_micromap
- Interacts with VK_NV_ray_tracing
- Interacts with VK_NV_ray_tracing_motion_blur

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Stu Smith [GitHub]stu-s

Extension Proposal

VK_KHR_maintenance5

Other Extension Metadata

Last Modified Date

2023-05-02

Interactions and External Dependencies

Contributors

- Stu Smith, AMD
- Tobias Hector, AMD
- Shahbaz Youssefi, Google
- Slawomir Cygan, Intel
- Lionel Landwerlin, Intel
- James Fitzpatrick, Imagination Technologies
- Andrew Garrard, Imagination Technologies
- Ralph Potter, Samsung
- Pan Gao, Huawei
- Jan-Harald Fredriksen, ARM
- Jon Leech, Khronos
- Mike Blumenkrantz, Valve

Description

VK_KHR_maintenance5 adds a collection of minor features, none of which would warrant an entire extension of their own.

The new features are as follows:

- A new VK_FORMAT_A1B5G5R5_UNORM_PACK16_KHR format
- A new VK_FORMAT_A8_UNORM_KHR format
- A property to indicate that multisample coverage operations are performed after sample counting in EarlyFragmentTests mode
- Relax VkBufferView creation requirements by allowing subsets of the associated VkBuffer usage using VkBufferUsageFlags2CreateInfoKHR
- A new command vkCmdBindIndexBuffer2KHR, allowing a range of memory to be bound as an index buffer
- vkGetDeviceProcAddr must return NULL for supported core functions beyond the version requested by the application.
- A property to indicate that the sample mask test is performed after sample counting in EarlyFragmentTests mode
- vkCmdBindVertexBuffers2 now supports using VK_WHOLE_SIZE in the pSizes parameter.
- A default size of 1.0 is used if PointSize is not written
- Shader modules are deprecated - applications can now pass VkShaderModuleCreateInfo as a chained structure to pipeline creation via VkPipelineShaderStageCreateInfo
- A function vkGetRenderingAreaGranularityKHR to query the optimal render area for a dynamic rendering instance.
- A property to indicate that depth/stencil texturing operations with VK_COMPONENT_SWIZZLE_ONE have defined behavior
- Add vkGetImageSubresourceLayout2KHR and a new function vkGetDeviceImageSubresourceLayoutKHR to allow the application to query the image memory layout without having to create an image object and query it.
- Allow VK_REMAINING_ARRAY_LAYERS as the layerCount member of VkImageSubresourceLayers
- Adds stronger guarantees for propagation of VK_ERROR_DEVICE_LOST return values
- A property to indicate whether PointSize controls the final rasterization of polygons if polygon mode is VK_POLYGON_MODE_POINT
- Two properties to indicate the non-strict line rasterization algorithm used
- Two new flags words VkPipelineCreateFlagBits2KHR and VkBufferUsageFlagBits2KHR
- Physical-device-level functions can now be called with any value in the valid range for a type beyond the defined enumerants, such that applications can avoid checking individual features, extensions, or versions before querying supported properties of a particular enumerant.
- Clarification that copies between images of any type are allowed, treating 1D images as 2D

images with a height of 1.

New Commands

- vkCmdBindIndexBuffer2KHR
- vkGetDeviceImageSubresourceLayoutKHR
- vkGetImageSubresourceLayout2KHR
- vkGetRenderingAreaGranularityKHR

New Structures

- VkDeviceImageSubresourceInfoKHR
- VkImageSubresource2KHR
- VkRenderingAreaInfoKHR
- VkSubresourceLayout2KHR
- Extending VkBufferViewCreateInfo, VkBufferCreateInfo, VkPhysicalDeviceExternalBufferInfo, VkDescriptorBufferBindingInfoEXT:
  - VkBufferUsageFlags2CreateInfoKHR
- Extending VkComputePipelineCreateInfo, VkGraphicsPipelineCreateInfo, VkRayTracingPipelineCreateInfoKHR:
  - VkPipelineCreateFlags2CreateInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMaintenance5FeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceMaintenance5PropertiesKHR

New Enums

- VkBufferUsageFlagBits2KHR
- VkPipelineCreateFlagBits2KHR

New Bitmasks

- VkBufferUsageFlags2KHR
- VkPipelineCreateFlags2KHR

New Enum Constants

- `VK_KHR_MAINTENANCE_5_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE_5_SPEC_VERSION`
- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_INDEX_BUFFER_BIT_KHR

    - VK_BUFFER_USAGE_2_INDIRECT_BUFFER_BIT_KHR
    - VK_BUFFER_USAGE_2_STORAGE_BUFFER_BIT_KHR
    - VK_BUFFER_USAGE_2_STORAGE_TEXEL_BUFFER_BIT_KHR
    - VK_BUFFER_USAGE_2_TRANSFER_DST_BIT_KHR
    - VK_BUFFER_USAGE_2_TRANSFER_SRC_BIT_KHR
    - VK_BUFFER_USAGE_2_UNIFORM_BUFFER_BIT_KHR
    - VK_BUFFER_USAGE_2_UNIFORM_TEXEL_BUFFER_BIT_KHR
    - VK_BUFFER_USAGE_2_VERTEX_BUFFER_BIT_KHR
- Extending VkFormat:
    - VK_FORMAT_A1B5G5R5_UNORM_PACK16_KHR
    - VK_FORMAT_A8_UNORM_KHR
- Extending VkPipelineCreateFlagBits2:
    - VK_PIPELINE_CREATE_2_ALLOW_DERIVATIVES_BIT_KHR
    - VK_PIPELINE_CREATE_2_DERIVATIVE_BIT_KHR
    - VK_PIPELINE_CREATE_2_DISABLE_OPTIMIZATION_BIT_KHR
    - VK_PIPELINE_CREATE_2_DISPATCH_BASE_BIT_KHR
    - VK_PIPELINE_CREATE_2_VIEW_INDEX_FROM_DEVICE_INDEX_BIT_KHR
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_BUFFER_USAGE_FLAGS_2_CREATE_INFO_KHR
    - VK_STRUCTURE_TYPE_DEVICE_IMAGE_SUBRESOURCE_INFO_KHR
    - VK_STRUCTURE_TYPE_IMAGE_SUBRESOURCE_2_KHR
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_FEATURES_KHR
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_PROPERTIES_KHR
    - VK_STRUCTURE_TYPE_PIPELINE_CREATE_FLAGS_2_CREATE_INFO_KHR
    - VK_STRUCTURE_TYPE_RENDERING_AREA_INFO_KHR
    - VK_STRUCTURE_TYPE_SUBRESOURCE_LAYOUT_2_KHR

If VK_KHR_dynamic_rendering or Vulkan Version 1.3 and VK_EXT_fragment_density_map is supported:

- Extending VkPipelineCreateFlagBits2:
    - VK_PIPELINE_CREATE_2_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT

If VK_KHR_dynamic_rendering or Vulkan Version 1.3 and VK_KHR_fragment_shading_rate is supported:

- Extending VkPipelineCreateFlagBits2:
    - VK_PIPELINE_CREATE_2_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KH

R

If VK_EXT_attachment_feedback_loop_layout is supported:

- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT
  - VK_PIPELINE_CREATE_2_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT

If VK_EXT_conditional_rendering is supported:

- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_CONDITIONAL_RENDERING_BIT_EXT

If VK_EXT_descriptor_buffer is supported:

- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT
  - VK_BUFFER_USAGE_2_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT
  - VK_BUFFER_USAGE_2_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT
- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_DESCRIPTOR_BUFFER_BIT_EXT

If VK_EXT_graphics_pipeline_library is supported:

- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_LINK_TIME_OPTIMIZATION_BIT_EXT
  - VK_PIPELINE_CREATE_2_RETAIN_LINK_TIME_OPTIMIZATION_INFO_BIT_EXT

If VK_EXT_transform_feedback is supported:

- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT
  - VK_BUFFER_USAGE_2_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT

If VK_KHR_acceleration_structure is supported:

- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR
  - VK_BUFFER_USAGE_2_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR

If VK_KHR_pipeline_executable_properties is supported:

- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR
  - VK_PIPELINE_CREATE_2_CAPTURE_STATISTICS_BIT_KHR

If VK_KHR_pipeline_library is supported:

- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_LIBRARY_BIT_KHR

If VK_KHR_ray_tracing_pipeline is supported:

- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_SHADER_BINDING_TABLE_BIT_KHR
- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR
  - VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR
  - VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR
  - VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR
  - VK_PIPELINE_CREATE_2_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR
  - VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_AABBS_BIT_KHR
  - VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR

If VK_KHR_video_decode_queue is supported:

- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_VIDEO_DECODE_DST_BIT_KHR
  - VK_BUFFER_USAGE_2_VIDEO_DECODE_SRC_BIT_KHR

If VK_KHR_video_encode_queue is supported:

- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_VIDEO_ENCODE_DST_BIT_KHR
  - VK_BUFFER_USAGE_2_VIDEO_ENCODE_SRC_BIT_KHR

If Vulkan Version 1.2 or VK_KHR_buffer_device_address or `VK_EXT_buffer_device_address` is supported:

- Extending VkBufferUsageFlagBits2:
  - VK_BUFFER_USAGE_2_SHADER_DEVICE_ADDRESS_BIT_KHR

If Vulkan Version 1.3 or `VK_EXT_pipeline_creation_cache_control` is supported:

- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_EARLY_RETURN_ON_FAILURE_BIT_KHR
  - VK_PIPELINE_CREATE_2_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT_KHR

If Vulkan Version 1.4 or VK_EXT_pipeline_protected_access is supported:

- Extending VkPipelineCreateFlagBits2:
  - VK_PIPELINE_CREATE_2_NO_PROTECTED_ACCESS_BIT_EXT
  - VK_PIPELINE_CREATE_2_PROTECTED_ACCESS_ONLY_BIT_EXT

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4 with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

Version History

- Revision 1, 2022-12-12 (Stu Smith)
  - Initial revision

VK_KHR_maintenance6

Name String

`VK_KHR_maintenance6`

Extension Type

Device extension

Registered Extension Number

546

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.1

API Interactions

- Interacts with VK_EXT_descriptor_buffer
- Interacts with VK_KHR_push_descriptor

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Jon Leech [GitHub]oddhack

Extension Proposal

VK_KHR_maintenance6

Other Extension Metadata

Last Modified Date

2023-08-03

Interactions and External Dependencies

- Interacts with `VK_EXT_robustness2`

Contributors

- Jon Leech, Khronos
- Stu Smith, AMD
- Mike Blumenkrantz, Valve
- Ralph Potter, Samsung
- James Fitzpatrick, Imagination Technologies
- Piers Daniell, NVIDIA

- Daniel Story, Nintendo

Description

VK_KHR_maintenance6 adds a collection of minor features, none of which would warrant an entire extension of their own.

The new features are as follows:

- VkBindMemoryStatusKHR may be included in the `pNext` chain of VkBindBufferMemoryInfo and VkBindImageMemoryInfo, allowing applications to identify individual resources for which memory binding failed during calls to vkBindBufferMemory2 and vkBindImageMemory2.
- A new property `fragmentShadingRateClampCombinerInputs` to indicate if an implementation clamps the inputs to fragment shading rate combiner operations.
- VK_NULL_HANDLE is allowed to be used when binding an index buffer, instead of a valid VkBuffer handle. When the `nullDescriptor` feature is enabled, every index fetched results in a value of zero.
- A new property `maxCombinedImageSamplerDescriptorCount` to indicate the maximum number of descriptors needed for sampled images with $Y'C_BC_R$ conversion.
- A new property `blockTexelViewCompatibleMultipleLayers` indicating whether VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT is allowed to be used with `layerCount` > 1
- `pNext` extensible *2 versions of all descriptor binding commands.

New Commands

- vkCmdBindDescriptorSets2KHR
- vkCmdPushConstants2KHR

If VK_EXT_descriptor_buffer is supported:

- vkCmdBindDescriptorBufferEmbeddedSamplers2EXT
- vkCmdSetDescriptorBufferOffsets2EXT

If VK_KHR_push_descriptor is supported:

- vkCmdPushDescriptorSet2KHR
- vkCmdPushDescriptorSetWithTemplate2KHR

New Structures

- VkBindDescriptorSetsInfoKHR
- VkPushConstantsInfoKHR
- Extending VkBindBufferMemoryInfo, VkBindImageMemoryInfo:
  - VkBindMemoryStatusKHR

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMaintenance6FeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceMaintenance6PropertiesKHR

If VK_EXT_descriptor_buffer is supported:

- VkBindDescriptorBufferEmbeddedSamplersInfoEXT
- VkSetDescriptorBufferOffsetsInfoEXT

If VK_KHR_push_descriptor is supported:

- VkPushDescriptorSetInfoKHR
- VkPushDescriptorSetWithTemplateInfoKHR

New Enum Constants

- `VK_KHR_MAINTENANCE_6_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE_6_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_BIND_DESCRIPTOR_SETS_INFO_KHR
  - VK_STRUCTURE_TYPE_BIND_MEMORY_STATUS_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_PUSH_CONSTANTS_INFO_KHR

If VK_EXT_descriptor_buffer is supported:

- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_BIND_DESCRIPTOR_BUFFER_EMBEDDED_SAMPLERS_INFO_EXT
  - VK_STRUCTURE_TYPE_SET_DESCRIPTOR_BUFFER_OFFSETS_INFO_EXT

If VK_KHR_push_descriptor is supported:

- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_INFO_KHR
  - VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_WITH_TEMPLATE_INFO_KHR

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4 with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

Version History

- Revision 1, 2023-08-01 (Jon Leech)
  - Initial revision

VK_KHR_map_memory2

Name String

VK_KHR_map_memory2

Extension Type

Device extension

Registered Extension Number

272

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Faith Ekstrand [GitHub]gfxstrand

Extension Proposal

VK_KHR_map_memory2

Other Extension Metadata

Last Modified Date

2023-03-14

Interactions and External Dependencies

- None

Contributors

- Faith Ekstrand, Collabora
- Tobias Hector, AMD

Description

This extension provides extensible versions of the Vulkan memory map and unmap commands. The new commands are functionally identical to the core commands, except that their parameters are specified using extensible structures that can be used to pass extension-specific information.

New Commands

- vkMapMemory2KHR
- vkUnmapMemory2KHR

New Structures

- VkMemoryMapInfoKHR
- VkMemoryUnmapInfoKHR

New Enums

- VkMemoryUnmapFlagBitsKHR

New Bitmasks

- VkMemoryUnmapFlagsKHR

New Enum Constants

- `VK_KHR_MAP_MEMORY_2_EXTENSION_NAME`
- `VK_KHR_MAP_MEMORY_2_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_MEMORY_MAP_INFO_KHR
  - VK_STRUCTURE_TYPE_MEMORY_UNMAP_INFO_KHR

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4 with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

Version History

- Revision 0, 2022-08-03 (Faith Ekstrand)
  - Internal revisions
- Revision 1, 2023-03-14
  - Public release

VK_KHR_multiview

Name String

`VK_KHR_multiview`

Extension Type

Device extension

Registered Extension Number

54

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_multiview

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jeff Bolz [GitHub]jeffbolznv

Other Extension Metadata

Last Modified Date

2016-10-28

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for `GL_EXT_multiview`

Contributors

- Jeff Bolz, NVIDIA

Description

This extension has the same goal as the OpenGL ES `GL_OVR_multiview` extension. Multiview is a rendering technique originally designed for VR where it is more efficient to record a single set of commands to be executed with slightly different behavior for each “view”.

It includes a concise way to declare a render pass with multiple views, and gives implementations freedom to render the views in the most efficient way possible. This is done with a multiview configuration specified during render pass creation with the VkRenderPassMultiviewCreateInfo passed into VkRenderPassCreateInfo::pNext.

This extension enables the use of the SPV_KHR_multiview shader extension, which adds a new ViewIndex built-in type that allows shaders to control what to do for each view. If using GLSL there is also the GL_EXT_multiview extension that introduces a highp int gl_ViewIndex; built-in variable for vertex, tessellation, geometry, and fragment shaders.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceMultiviewFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceMultiviewPropertiesKHR
- Extending VkRenderPassCreateInfo:
  - VkRenderPassMultiviewCreateInfoKHR

New Enum Constants

- VK_KHR_MULTIVIEW_EXTENSION_NAME
- VK_KHR_MULTIVIEW_SPEC_VERSION
- Extending VkDependencyFlagBits:
  - VK_DEPENDENCY_VIEW_LOCAL_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_RENDER_PASS_MULTIVIEW_CREATE_INFO_KHR

New Built-In Variables

- ViewIndex

New SPIR-V Capabilities

- MultiView

Version History

- Revision 1, 2016-10-28 (Jeff Bolz)
  - Internal revisions

VK_KHR_push_descriptor

Name String

`VK_KHR_push_descriptor`

Extension Type

Device extension

Registered Extension Number

81

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_VERSION_1_1
- Interacts with VK_KHR_descriptor_update_template

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Jeff Bolz [GitHub]jeffbolznv

Other Extension Metadata

Last Modified Date

2017-09-12

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Michael Worcester, Imagination Technologies

Description

This extension allows descriptors to be written into the command buffer, while the implementation is responsible for managing their memory. Push descriptors may enable easier porting from older APIs and in some cases can be more efficient than writing descriptors into descriptor sets.

New Commands

- vkCmdPushDescriptorSetKHR

If Vulkan Version 1.1 or VK_KHR_descriptor_update_template is supported:

- vkCmdPushDescriptorSetWithTemplateKHR

New Structures

- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDevicePushDescriptorPropertiesKHR

New Enum Constants

- `VK_KHR_PUSH_DESCRIPTOR_EXTENSION_NAME`
- `VK_KHR_PUSH_DESCRIPTOR_SPEC_VERSION`
- Extending VkDescriptorSetLayoutCreateFlagBits:
  - VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PUSH_DESCRIPTOR_PROPERTIES_KHR

If Vulkan Version 1.1 or VK_KHR_descriptor_update_template is supported:

- Extending VkDescriptorUpdateTemplateType:
  - VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTORS_KHR

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4 with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

Version History

- Revision 1, 2016-10-15 (Jeff Bolz)
  - Internal revisions
- Revision 2, 2017-09-12 (Tobias Hector)
  - Added interactions with Vulkan 1.1

VK_KHR_relaxed_block_layout

Name String

`VK_KHR_relaxed_block_layout`

Extension Type

Device extension

Registered Extension Number

145

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- John Kessenich [GitHub]johnkslang

Other Extension Metadata

Last Modified Date

2017-03-26

IP Status

No known IP claims.

Contributors

- John Kessenich, Google

Description

The `VK_KHR_relaxed_block_layout` extension allows implementations to indicate they can support more variation in block `Offset` decorations. For example, placing a vector of three floats at an offset of 16×N + 4.

See Offset and Stride Assignment for details.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Enum Constants

- `VK_KHR_RELAXED_BLOCK_LAYOUT_EXTENSION_NAME`
- `VK_KHR_RELAXED_BLOCK_LAYOUT_SPEC_VERSION`

Version History

- Revision 1, 2017-03-26 (JohnK)

VK_KHR_sampler_mirror_clamp_to_edge

Name String

VK_KHR_sampler_mirror_clamp_to_edge

Extension Type

Device extension

Registered Extension Number

15

Revision

3

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Tobias Hector [GitHub]tobski

Other Extension Metadata

Last Modified Date

2019-08-17

Contributors

- Tobias Hector, Imagination Technologies
- Jon Leech, Khronos

Description

VK_KHR_sampler_mirror_clamp_to_edge extends the set of sampler address modes to include an additional mode (VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE) that effectively uses a texture map twice as large as the original image in which the additional half of the new image is a mirror image of the original image.

This new mode relaxes the need to generate images whose opposite edges match by using the original image to generate a matching “mirror image”. This mode allows the texture to be mirrored only once in the negative s, t, and r directions.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2. However, if Vulkan 1.2 is

supported and this extension is not, the VkSamplerAddressMode VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE is optional. Since the original extension did not use an author suffix on the enum VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE, it is used by both core and extension implementations.

New Enum Constants

- VK_KHR_SAMPLER_MIRROR_CLAMP_TO_EDGE_EXTENSION_NAME
- VK_KHR_SAMPLER_MIRROR_CLAMP_TO_EDGE_SPEC_VERSION
- Extending VkSamplerAddressMode:
  - VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE
  - VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE_KHR

Example

Creating a sampler with the new address mode in each dimension

```
VkSamplerCreateInfo createInfo =
{
    .sType = VK_STRUCTURE_TYPE_SAMPLER_CREATE_INFO,
    // Other members set to application-desired values
};

createInfo.addressModeU = VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE;
createInfo.addressModeV = VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE;
createInfo.addressModeW = VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE;

VkSampler sampler;
VkResult result = vkCreateSampler(
    device,
    &createInfo,
    &sampler);
```

Issues

1) Why are both KHR and core versions of the VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE token present?

RESOLVED: This functionality was intended to be required in Vulkan 1.0. We realized shortly before public release that not all implementations could support it, and moved the functionality into an optional extension, but did not apply the KHR extension suffix. Adding a KHR-suffixed alias of the non-suffixed enum has been done to comply with our own naming rules.

In a related change, before spec revision 1.1.121 this extension was hardwiring into the spec Makefile so it was always included with the Specification, even in the core-only versions. This has now been reverted, and it is treated as any other extension.

Version History

- Revision 1, 2016-02-16 (Tobias Hector)
  - Initial draft
- Revision 2, 2019-08-14 (Jon Leech)
  - Add KHR-suffixed alias of non-suffixed enum.
- Revision 3, 2019-08-17 (Jon Leech)
  - Add an issue explaining the reason for the extension API not being suffixed with KHR.

VK_KHR_sampler_ycbcr_conversion

Name String

VK_KHR_sampler_ycbcr_conversion

Extension Type

Device extension

Registered Extension Number

157

Revision

14

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_maintenance1
and
VK_KHR_bind_memory2
and
VK_KHR_get_memory_requirements2
and
VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_EXT_debug_report

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Andrew Garrard [GitHub]fluppeteer

Other Extension Metadata

Last Modified Date

2017-08-11

IP Status

No known IP claims.

Contributors

- Andrew Garrard, Samsung Electronics
- Tobias Hector, Imagination Technologies

- James Jones, NVIDIA
- Daniel Koch, NVIDIA
- Daniel Rakos, AMD
- Romain Guy, Google
- Jesse Hall, Google
- Tom Cooksey, ARM Ltd
- Jeff Leger, Qualcomm Technologies, Inc
- Jan-Harald Fredriksen, ARM Ltd
- Jan Outters, Samsung Electronics
- Alon Or-bach, Samsung Electronics
- Michael Worcester, Imagination Technologies
- Jeff Bolz, NVIDIA
- Tony Zlatinski, NVIDIA
- Matthew Netsch, Qualcomm Technologies, Inc

Description

The use of $Y'C_BC_R$ sampler conversion is an area in 3D graphics not used by most Vulkan developers. It is mainly used for processing inputs from video decoders and cameras. The use of the extension assumes basic knowledge of $Y'C_BC_R$ concepts.

This extension provides the ability to perform specified color space conversions during texture sampling operations for the $Y'C_BC_R$ color space natively. It also adds a selection of multi-planar formats, image aspect plane, and the ability to bind memory to the planes of an image collectively or separately.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. However, if Vulkan 1.1 is supported and this extension is not, the `samplerYcbcrConversion` capability is optional. The original type, enum, and command names are still available as aliases of the core functionality.

Promotion to Vulkan 1.4

If Vulkan 1.4 is supported, support for the `samplerYcbcrConversion` capability is required.

New Object Types

- VkSamplerYcbcrConversionKHR

New Commands

- vkCreateSamplerYcbcrConversionKHR

- vkDestroySamplerYcbcrConversionKHR

New Structures

- VkSamplerYcbcrConversionCreateInfoKHR
- Extending VkBindImageMemoryInfo:
  - VkBindImagePlaneMemoryInfoKHR
- Extending VkImageFormatProperties2:
  - VkSamplerYcbcrConversionImageFormatPropertiesKHR
- Extending VkImageMemoryRequirementsInfo2:
  - VkImagePlaneMemoryRequirementsInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceSamplerYcbcrConversionFeaturesKHR
- Extending VkSamplerCreateInfo, VkImageViewCreateInfo:
  - VkSamplerYcbcrConversionInfoKHR

New Enums

- VkChromaLocationKHR
- VkSamplerYcbcrModelConversionKHR
- VkSamplerYcbcrRangeKHR

New Enum Constants

- `VK_KHR_SAMPLER_YCBCR_CONVERSION_EXTENSION_NAME`
- `VK_KHR_SAMPLER_YCBCR_CONVERSION_SPEC_VERSION`
- Extending VkChromaLocation:
  - VK_CHROMA_LOCATION_COSITED_EVEN_KHR
  - VK_CHROMA_LOCATION_MIDPOINT_KHR
- Extending VkFormat:
  - VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16_KHR
  - VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16_KHR
  - VK_FORMAT_B16G16R16G16_422_UNORM_KHR
  - VK_FORMAT_B8G8R8G8_422_UNORM_KHR
  - VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16_KHR
  - VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16_KHR
  - VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16_KHR
  - VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16_KHR
  - VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16_KHR

    - VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16_KHR
    - VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16_KHR
    - VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16_KHR
    - VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16_KHR
    - VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16_KHR
    - VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16_KHR
    - VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16_KHR
    - VK_FORMAT_G16B16G16R16_422_UNORM_KHR
    - VK_FORMAT_G16_B16R16_2PLANE_420_UNORM_KHR
    - VK_FORMAT_G16_B16R16_2PLANE_422_UNORM_KHR
    - VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM_KHR
    - VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM_KHR
    - VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM_KHR
    - VK_FORMAT_G8B8G8R8_422_UNORM_KHR
    - VK_FORMAT_G8_B8R8_2PLANE_420_UNORM_KHR
    - VK_FORMAT_G8_B8R8_2PLANE_422_UNORM_KHR
    - VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM_KHR
    - VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM_KHR
    - VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM_KHR
    - VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16_KHR
    - VK_FORMAT_R10X6G10X6_UNORM_2PACK16_KHR
    - VK_FORMAT_R10X6_UNORM_PACK16_KHR
    - VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16_KHR
    - VK_FORMAT_R12X4G12X4_UNORM_2PACK16_KHR
    - VK_FORMAT_R12X4_UNORM_PACK16_KHR
- Extending VkFormatFeatureFlagBits:
    - VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT_KHR
    - VK_FORMAT_FEATURE_DISJOINT_BIT_KHR
    - VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT_KHR
    - VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT_KHR
    - VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT_KHR
    - VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT_KHR
    - VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCT

ION_FILTER_BIT_KHR

- Extending VkImageAspectFlagBits:
  - VK_IMAGE_ASPECT_PLANE_0_BIT_KHR
  - VK_IMAGE_ASPECT_PLANE_1_BIT_KHR
  - VK_IMAGE_ASPECT_PLANE_2_BIT_KHR
- Extending VkImageCreateFlagBits:
  - VK_IMAGE_CREATE_DISJOINT_BIT_KHR
- Extending VkObjectType:
  - VK_OBJECT_TYPE_SAMPLER_YCBCR_CONVERSION_KHR
- Extending VkSamplerYcbcrModelConversion:
  - VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY_KHR
  - VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_2020_KHR
  - VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_601_KHR
  - VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_709_KHR
  - VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_IDENTITY_KHR
- Extending VkSamplerYcbcrRange:
  - VK_SAMPLER_YCBCR_RANGE_ITU_FULL_KHR
  - VK_SAMPLER_YCBCR_RANGE_ITU_NARROW_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_BIND_IMAGE_PLANE_MEMORY_INFO_KHR
  - VK_STRUCTURE_TYPE_IMAGE_PLANE_MEMORY_REQUIREMENTS_INFO_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLER_YCBCR_CONVERSION_FEATURES_KHR
  - VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_IMAGE_FORMAT_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_INFO_KHR

Version History

- Revision 1, 2017-01-24 (Andrew Garrard)
  - Initial draft
- Revision 2, 2017-01-25 (Andrew Garrard)
  - After initial feedback
- Revision 3, 2017-01-27 (Andrew Garrard)
  - Higher bit depth formats, renaming, swizzle
- Revision 4, 2017-02-22 (Andrew Garrard)
  - Added query function, formats as RGB, clarifications

- Revision 5, 2017-04-?? (Andrew Garrard)
  - Simplified query and removed output conversions
- Revision 6, 2017-04-24 (Andrew Garrard)
  - Tidying, incorporated new image query, restored transfer functions
- Revision 7, 2017-04-25 (Andrew Garrard)
  - Added cosited option/midpoint requirement for formats, “bypassConversion”
- Revision 8, 2017-04-25 (Andrew Garrard)
  - Simplified further
- Revision 9, 2017-04-27 (Andrew Garrard)
  - Disjoint no more
- Revision 10, 2017-04-28 (Andrew Garrard)
  - Restored disjoint
- Revision 11, 2017-04-29 (Andrew Garrard)
  - Now Ycbcr conversion, and KHR
- Revision 12, 2017-06-06 (Andrew Garrard)
  - Added conversion to image view creation
- Revision 13, 2017-07-13 (Andrew Garrard)
  - Allowed cosited-only chroma samples for formats
- Revision 14, 2017-08-11 (Andrew Garrard)
  - Reflected quantization changes in BT.2100-1

VK_KHR_separate_depth_stencil_layouts

Name String

VK_KHR_separate_depth_stencil_layouts

Extension Type

Device extension

Registered Extension Number

242

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1
and
VK_KHR_create_renderpass2
or
Vulkan Version 1.2

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2019-06-25

Contributors

- Daniel Koch, NVIDIA
- Jeff Bolz, NVIDIA
- Jesse Barker, Unity
- Tobias Hector, AMD

Description

This extension allows image memory barriers for depth/stencil images to have just one of the VK_IMAGE_ASPECT_DEPTH_BIT or VK_IMAGE_ASPECT_STENCIL_BIT aspect bits set, rather than require both. This allows their layouts to be set independently. To support depth/stencil images with

different layouts for the depth and stencil aspects, the depth/stencil attachment interface has been updated to support a separate layout for stencil.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending VkAttachmentDescription2:
  - VkAttachmentDescriptionStencilLayoutKHR
- Extending VkAttachmentReference2:
  - VkAttachmentReferenceStencilLayoutKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceSeparateDepthStencilLayoutsFeaturesKHR

New Enum Constants

- `VK_KHR_SEPARATE_DEPTH_STENCIL_LAYOUTS_EXTENSION_NAME`
- `VK_KHR_SEPARATE_DEPTH_STENCIL_LAYOUTS_SPEC_VERSION`
- Extending VkImageLayout:
  - VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL_KHR
  - VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL_KHR
  - VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL_KHR
  - VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_STENCIL_LAYOUT_KHR
  - VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_STENCIL_LAYOUT_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SEPARATE_DEPTH_STENCIL_LAYOUTS_FEATURES_KHR

Version History

- Revision 1, 2019-06-25 (Piers Daniell)
  - Internal revisions

VK_KHR_shader_atomic_int64

Name String

`VK_KHR_shader_atomic_int64`

Extension Type

Device extension

Registered Extension Number

181

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Aaron Hagan [GitHub]ahagan

Other Extension Metadata

Last Modified Date

2018-07-05

Interactions and External Dependencies

- This extension provides API support for `GL_ARB_gpu_shader_int64` and `GL_EXT_shader_atomic_int64`

Contributors

- Aaron Hagan, AMD
- Daniel Rakos, AMD
- Jeff Bolz, NVIDIA
- Neil Henning, Codeplay

Description

This extension advertises the SPIR-V **Int64Atomics** capability for Vulkan, which allows a shader to contain 64-bit atomic operations on signed and unsigned integers. The supported operations include OpAtomicMin, OpAtomicMax, OpAtomicAnd, OpAtomicOr, OpAtomicXor, OpAtomicAdd,

OpAtomicExchange, and OpAtomicCompareExchange.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. However, if Vulkan 1.2 is supported and this extension is not, the `shaderBufferInt64Atomics` capability is optional. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderAtomicInt64FeaturesKHR

New Enum Constants

- `VK_KHR_SHADER_ATOMIC_INT64_EXTENSION_NAME`
- `VK_KHR_SHADER_ATOMIC_INT64_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_INT64_FEATURES_KHR

New SPIR-V Capabilities

- `Int64Atomics`

Version History

- Revision 1, 2018-07-05 (Aaron Hagan)
  - Internal revisions

VK_KHR_shader_draw_parameters

Name String

`VK_KHR_shader_draw_parameters`

Extension Type

Device extension

Registered Extension Number

64

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

SPIR-V Dependencies

- SPV_KHR_shader_draw_parameters

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Daniel Koch [GitHub]dgkoch

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for `GL_ARB_shader_draw_parameters`

Contributors

- Daniel Koch, NVIDIA Corporation
- Jeff Bolz, NVIDIA
- Daniel Rakos, AMD
- Jan-Harald Fredriksen, ARM
- John Kessenich, Google
- Stuart Smith, IMG

Description

This extension adds support for the following SPIR-V extension in Vulkan:

- `SPV_KHR_shader_draw_parameters`

The extension provides access to three additional built-in shader variables in Vulkan:

- `BaseInstance`, containing the `firstInstance` parameter passed to drawing commands,
- `BaseVertex`, containing the `firstVertex` or `vertexOffset` parameter passed to drawing commands, and
- `DrawIndex`, containing the index of the draw call currently being processed from an indirect drawing call.

When using GLSL source-based shader languages, the following variables from `GL_ARB_shader_draw_parameters` can map to these SPIR-V built-in decorations:

- `in int gl_BaseInstanceARB;` → `BaseInstance`,
- `in int gl_BaseVertexARB;` → `BaseVertex`, and
- `in int gl_DrawIDARB;` → `DrawIndex`.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1. However, the `shaderDrawParameters` feature bit was added to distinguish whether it is actually available or not.

New Enum Constants

- `VK_KHR_SHADER_DRAW_PARAMETERS_EXTENSION_NAME`
- `VK_KHR_SHADER_DRAW_PARAMETERS_SPEC_VERSION`

New Built-In Variables

- `BaseInstance`
- `BaseVertex`
- `DrawIndex`

New SPIR-V Capabilities

- `DrawParameters`

Issues

1) Is this the same functionality as `GL_ARB_shader_draw_parameters`?

RESOLVED: It is actually a superset, as it also adds in support for arrayed drawing commands.

In GL for `GL_ARB_shader_draw_parameters`, `gl_BaseVertexARB` holds the integer value passed to the parameter to the command that resulted in the current shader invocation. In the case where the

command has no `baseVertex` parameter, the value of `gl_BaseVertexARB` is zero. This means that `gl_BaseVertexARB` = `baseVertex` (for `glDrawElements` commands with `baseVertex`) or 0. In particular there are no `glDrawArrays` commands that take a `baseVertex` parameter.

Now in Vulkan, we have `BaseVertex` = `vertexOffset` (for indexed drawing commands) or `firstVertex` (for arrayed drawing commands), and so Vulkan's version is really a superset of GL functionality.

Version History

- Revision 1, 2016-10-05 (Daniel Koch)
  - Internal revisions

VK_KHR_shader_expect_assume

Name String

`VK_KHR_shader_expect_assume`

Extension Type

Device extension

Registered Extension Number

545

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_expect_assume

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Kevin Petit [GitHub]kpet

Extension Proposal

VK_KHR_shader_expect_assume

Other Extension Metadata

Last Modified Date

2023-12-06

IP Status

No known IP claims.

Contributors

- Kevin Petit, Arm
- Tobias Hector, AMD
- James Fitzpatrick, Imagination Technologies

Description

This extension allows the use of the `SPV_KHR_expect_assume` extension in SPIR-V shader modules which enables SPIR-V producers to provide optimization hints to the Vulkan implementation.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderExpectAssumeFeaturesKHR

New Enum Constants

- `VK_KHR_SHADER_EXPECT_ASSUME_EXTENSION_NAME`
- `VK_KHR_SHADER_EXPECT_ASSUME_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_EXPECT_ASSUME_FEATURES_KHR

New SPIR-V Capabilities

- ExpectAssumeKHR

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4 with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

Version History

- Revision 1, 2023-12-06 (Kevin Petit)
  - Initial revision

VK_KHR_shader_float16_int8

Name String

`VK_KHR_shader_float16_int8`

Extension Type

Device extension

Registered Extension Number

83

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Alexander Galazin [GitHub]alegal-arm

Other Extension Metadata

Last Modified Date

2018-03-07

Interactions and External Dependencies

- This extension interacts with `VK_KHR_8bit_storage`
- This extension interacts with `VK_KHR_16bit_storage`
- This extension interacts with `VK_KHR_shader_float_controls`
- This extension provides API support for `GL_EXT_shader_explicit_arithmetic_types`

IP Status

No known IP claims.

Contributors

- Alexander Galazin, Arm
- Jan-Harald Fredriksen, Arm
- Jeff Bolz, NVIDIA
- Graeme Leese, Broadcom

- Daniel Rakos, AMD

Description

The VK_KHR_shader_float16_int8 extension allows use of 16-bit floating-point types and 8-bit integer types in shaders for arithmetic operations.

It introduces two new optional features shaderFloat16 and shaderInt8 which directly map to the Float16 and the Int8 SPIR-V capabilities. The VK_KHR_shader_float16_int8 extension also specifies precision requirements for half-precision floating-point SPIR-V operations. This extension does not enable use of 8-bit integer types or 16-bit floating-point types in any shader input and output interfaces and therefore does not supersede the VK_KHR_8bit_storage or VK_KHR_16bit_storage extensions.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. However, if Vulkan 1.2 is supported and this extension is not, both the shaderFloat16 and shaderInt8 capabilities are optional. The original type, enum, and command names are still available as aliases of the core functionality.

Promotion to Vulkan 1.4

If Vulkan 1.4 is supported, support for the shaderInt8 capability is required.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceFloat16Int8FeaturesKHR
  - VkPhysicalDeviceShaderFloat16Int8FeaturesKHR

New Enum Constants

- VK_KHR_SHADER_FLOAT16_INT8_EXTENSION_NAME
- VK_KHR_SHADER_FLOAT16_INT8_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FLOAT16_INT8_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT16_INT8_FEATURES_KHR

Version History

- Revision 1, 2018-03-07 (Alexander Galazin)
  - Initial draft

VK_KHR_shader_float_controls

Name String

`VK_KHR_shader_float_controls`

Extension Type

Device extension

Registered Extension Number

198

Revision

4

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_float_controls

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Alexander Galazin [GitHub]alegal-arm

Other Extension Metadata

Last Modified Date

2018-09-11

IP Status

No known IP claims.

Contributors

- Alexander Galazin, Arm
- Jan-Harald Fredriksen, Arm
- Jeff Bolz, NVIDIA
- Graeme Leese, Broadcom
- Daniel Rakos, AMD

Description

The VK_KHR_shader_float_controls extension enables efficient use of floating-point computations through the ability to query and override the implementation's default behavior for rounding modes, denormals, signed zero, and infinity.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceFloatControlsPropertiesKHR

New Enums

- VkShaderFloatControlsIndependenceKHR

New Enum Constants

- VK_KHR_SHADER_FLOAT_CONTROLS_EXTENSION_NAME
- VK_KHR_SHADER_FLOAT_CONTROLS_SPEC_VERSION
- Extending VkShaderFloatControlsIndependence:
  - VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_32_BIT_ONLY_KHR
  - VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_ALL_KHR
  - VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_NONE_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FLOAT_CONTROLS_PROPERTIES_KHR

New SPIR-V Capabilities

- DenormPreserve
- DenormFlushToZero
- SignedZeroInfNanPreserve
- RoundingModeRTE
- RoundingModeRTZ

Issues

1) Which instructions must flush denorms?

RESOLVED: Only floating-point conversion, floating-point arithmetic, floating-point relational (except OpIsNaN, OpIsInf), and floating-point GLSL.std.450 extended instructions must flush denormals.

2) What is the denorm behavior for intermediate results?

RESOLVED: When a SPIR-V instruction is implemented as a sequence of other instructions:

- in the `DenormFlushToZero` execution mode, the intermediate instructions may flush denormals, the final result of the sequence **must** not be denormal.
- in the `DenormPreserve` execution mode, denormals must be preserved throughout the whole sequence.

3) Do denorm and rounding mode controls apply to `OpSpecConstantOp`?

RESOLVED: Yes, except when the opcode is `OpQuantizeToF16`.

4) The SPIR-V specification says that `OpConvertFToU` and `OpConvertFToS` unconditionally round towards zero. Do the rounding mode controls specified through the execution modes apply to them?

RESOLVED: No, these instructions unconditionally round towards zero.

5) Do any of the "Pack" GLSL.std.450 instructions count as conversion instructions and have the rounding mode applied?

RESOLVED: No, only instructions listed in "section 3.32.11. Conversion Instructions" of the SPIR-V specification count as conversion instructions.

6) When using inf/nan-ignore mode, what is expected of `OpIsNan` and `OpIsInf`?

RESOLVED: These instructions must always accurately detect inf/nan if it is passed to them.

Version 4 API Incompatibility

The original versions of `VK_KHR_shader_float_controls` shipped with booleans named "separateDenormSettings" and "separateRoundingModeSettings", which at first glance could have indicated "they can all be set independently, or not". However the spec language as written indicated that the 32-bit value could always be set independently, and only the 16- and 64-bit controls needed to be the same if these values were VK_FALSE.

As a result of this slight disparity, and lack of test coverage for this facet of the extension, we ended up with two different behaviors in the wild, where some implementations worked as written, and others worked based on the naming. As these are hard limits in hardware with reasons for exposure as written, it was not possible to standardize on a single way to make this work within the existing API.

No known users of this part of the extension exist in the wild, and as such the Vulkan WG took the unusual step of retroactively changing the once boolean value into a tri-state enum, breaking source compatibility. This was however done in such a way as to retain ABI compatibility, in case any code using this did exist; with the numerical values 0 and 1 retaining their original specified meaning, and a new value signifying the additional "all need to be set together" state. If any applications exist today, compiled binaries will continue to work as written in most cases, but will need changes before the code can be recompiled.

Version History

- Revision 4, 2019-06-18 (Tobias Hector)
  - Modified settings restrictions, see Version 4 API incompatibility
- Revision 3, 2018-09-11 (Alexander Galazin)
  - Minor restructuring
- Revision 2, 2018-04-17 (Alexander Galazin)
  - Added issues and resolutions
- Revision 1, 2018-04-11 (Alexander Galazin)
  - Initial draft

VK_KHR_shader_float_controls2

Name String

VK_KHR_shader_float_controls2

Extension Type

Device extension

Registered Extension Number

529

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.1
and
VK_KHR_shader_float_controls

SPIR-V Dependencies

- SPV_KHR_float_controls2

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Graeme Leese [GitHub]gnl21

Extension Proposal

VK_KHR_shader_float_controls2

Other Extension Metadata

Last Modified Date

2023-05-16

Interactions and External Dependencies

- This extension requires SPV_KHR_float_controls2.

Contributors

- Graeme Leese, Broadcom

Description

This extension enables use of the more expressive fast floating-point math flags in the SPV_KHR_float_controls2 extension. These flags give finer- grained control over which optimizations compilers may apply, potentially speeding up execution while retaining correct

results.

The extension also adds control over the fast-math modes to the GLSL extended instruction set, making these operations more consistent with SPIR-V and allowing their use in situations where floating-point conformance is important.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderFloatControls2FeaturesKHR

New Enum Constants

- `VK_KHR_SHADER_FLOAT_CONTROLS_2_EXTENSION_NAME`
- `VK_KHR_SHADER_FLOAT_CONTROLS_2_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT_CONTROLS_2_FEATURES_KHR

New SPIR-V Capabilities

- FloatControls2

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4 with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

Version History

- Revision 1, 2023-05-16 (Graeme Leese)
  - Initial draft

VK_KHR_shader_integer_dot_product

Name String

`VK_KHR_shader_integer_dot_product`

Extension Type

Device extension

Registered Extension Number

281

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_integer_dot_product

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Kevin Petit [GitHub]kpet

Extension Proposal

VK_KHR_shader_integer_dot_product

Other Extension Metadata

Last Modified Date

2021-06-16

Interactions and External Dependencies

- This extension interacts with `VK_KHR_shader_float16_int8`.

IP Status

No known IP claims.

Contributors

- Kévin Petit, Arm Ltd.
- Jeff Bolz, NVidia
- Spencer Fricke, Samsung

- Jesse Hall, Google
- John Kessenich, Google
- Graeme Leese, Broadcom
- Einar Hov, Arm Ltd.
- Stuart Brady, Arm Ltd.
- Pablo Cascon, Arm Ltd.
- Tobias Hector, AMD
- Jeff Leger, Qualcomm
- Ruihao Zhang, Qualcomm
- Pierre Boudier, NVidia
- Jon Leech, The Khronos Group
- Tom Olson, Arm Ltd.

Description

This extension adds support for the integer dot product SPIR-V instructions defined in SPV_KHR_integer_dot_product. These instructions are particularly useful for neural network inference and training but find uses in other general-purpose compute applications as well.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderIntegerDotProductFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceShaderIntegerDotProductPropertiesKHR

New Enum Constants

- `VK_KHR_SHADER_INTEGER_DOT_PRODUCT_EXTENSION_NAME`
- `VK_KHR_SHADER_INTEGER_DOT_PRODUCT_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_PROPERTIES_KHR

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

New SPIR-V Capabilities

- `DotProductInputAllKHR`
- `DotProductInput4x8BitKHR`
- `DotProductInput4x8BitPackedKHR`
- `DotProductKHR`

Version History

- Revision 1, 2021-06-16 (Kévin Petit)
  - Initial revision

VK_KHR_shader_non_semantic_info

Name String

`VK_KHR_shader_non_semantic_info`

Extension Type

Device extension

Registered Extension Number

294

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

SPIR-V Dependencies

- SPV_KHR_non_semantic_info

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Baldur Karlsson [GitHub]baldurk

Other Extension Metadata

Last Modified Date

2019-10-16

IP Status

No known IP claims.

Contributors

- Baldur Karlsson, Valve

Description

This extension allows the use of the `SPV_KHR_non_semantic_info` extension in SPIR-V shader modules.

New Enum Constants

- `VK_KHR_SHADER_NON_SEMANTIC_INFO_EXTENSION_NAME`
- `VK_KHR_SHADER_NON_SEMANTIC_INFO_SPEC_VERSION`

Promotion to Vulkan 1.3

Functionality in this extension is included in core Vulkan 1.3. Because the extension has no API controlling its functionality, this results only in a change to the SPIR-V Extensions table.

Version History

- Revision 1, 2019-10-16 (Baldur Karlsson)
  - Initial revision

VK_KHR_shader_subgroup_extended_types

Name String

`VK_KHR_shader_subgroup_extended_types`

Extension Type

Device extension

Registered Extension Number

176

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Neil Henning [GitHub]sheredom

Other Extension Metadata

Last Modified Date

2019-01-08

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for `GLSL_EXT_shader_subgroup_extended_types`

Contributors

- Jeff Bolz, NVIDIA
- Jan-Harald Fredriksen, Arm
- Neil Henning, AMD
- Daniel Koch, NVIDIA
- Jeff Leger, Qualcomm
- Graeme Leese, Broadcom
- David Neto, Google
- Daniel Rakos, AMD

Description

This extension enables the Non Uniform Group Operations in SPIR-V to support 8-bit integer, 16-bit integer, 64-bit integer, 16-bit floating-point, and vectors of these types.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderSubgroupExtendedTypesFeaturesKHR

New Enum Constants

- `VK_KHR_SHADER_SUBGROUP_EXTENDED_TYPES_EXTENSION_NAME`
- `VK_KHR_SHADER_SUBGROUP_EXTENDED_TYPES_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_EXTENDED_TYPES_FEATURES_KHR

Version History

- Revision 1, 2019-01-08 (Neil Henning)
  - Initial draft

VK_KHR_shader_subgroup_rotate

Name String

`VK_KHR_shader_subgroup_rotate`

Extension Type

Device extension

Registered Extension Number

417

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_subgroup_rotate

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Kevin Petit [GitHub]kpet

Extension Proposal

VK_KHR_shader_subgroup_rotate

Last Modified Date

2024-01-29

IP Status

No known IP claims.

Contributors

- Kévin Petit, Arm Ltd.
- Tobias Hector, AMD
- Jon Leech, Khronos
- Matthew Netsch, Qualcomm
- Jan-Harald Fredriksen, Arm Ltd.
- Graeme Leese, Broadcom

- Tom Olson, Arm Ltd.
- Spencer Fricke, LunarG Inc.

This extension adds support for the subgroup rotate instruction defined in SPV_KHR_subgroup_rotate.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderSubgroupRotateFeaturesKHR

New Enum Constants

- `VK_KHR_SHADER_SUBGROUP_ROTATE_EXTENSION_NAME`
- `VK_KHR_SHADER_SUBGROUP_ROTATE_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_ROTATE_FEATURES_KHR
- Extending VkSubgroupFeatureFlagBits:
  - VK_SUBGROUP_FEATURE_ROTATE_BIT_KHR
  - VK_SUBGROUP_FEATURE_ROTATE_CLUSTERED_BIT_KHR

New SPIR-V Capabilities

- GroupNonUniformRotateKHR

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4 with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

Version History

- Revision 2, 2024-01-29 (Kévin Petit)
  - Add VK_SUBGROUP_FEATURE_ROTATE_BIT_KHR and VK_SUBGROUP_FEATURE_ROTATE_CLUSTERED_BIT_KHR
- Revision 1, 2023-06-20 (Kévin Petit)
  - Initial revision

VK_KHR_shader_terminate_invocation

Name String

`VK_KHR_shader_terminate_invocation`

Extension Type

Device extension

Registered Extension Number

216

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_terminate_invocation

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Jesse Hall [GitHub]critsec

Other Extension Metadata

Last Modified Date

2020-08-11

IP Status

No known IP claims.

Contributors

- Alan Baker, Google
- Jeff Bolz, NVIDIA
- Jesse Hall, Google
- Ralph Potter, Samsung
- Tom Olson, Arm

Description

This extension adds Vulkan support for the SPV_KHR_terminate_invocation SPIR-V extension. That SPIR-V extension provides a new instruction, OpTerminateInvocation, which causes a shader invocation to immediately terminate and sets the coverage of shaded samples to 0; only previously executed instructions will have observable effects. The OpTerminateInvocation instruction, along with the OpDemoteToHelperInvocation instruction from the VK_EXT_shader_demote_to_helper_invocation extension, together replace the OpKill instruction, which could behave like either of these instructions. OpTerminateInvocation provides the behavior required by the GLSL discard statement, and should be used when available by GLSL compilers and applications that need the GLSL discard behavior.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceShaderTerminateInvocationFeaturesKHR

New Enum Constants

- VK_KHR_SHADER_TERMINATE_INVOCATION_EXTENSION_NAME
- VK_KHR_SHADER_TERMINATE_INVOCATION_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TERMINATE_INVOCATION_FEATURES_KHR

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Version History

- Revision 1, 2020-08-11 (Jesse Hall)

VK_KHR_spirv_1_4

Name String

`VK_KHR_spirv_1_4`

Extension Type

Device extension

Registered Extension Number

237

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

Vulkan Version 1.1
and
VK_KHR_shader_float_controls

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Jesse Hall [GitHub]critsec

Other Extension Metadata

Last Modified Date

2019-04-01

IP Status

No known IP claims.

Interactions and External Dependencies

- Requires SPIR-V 1.4.

Contributors

- Alexander Galazin, Arm
- David Neto, Google
- Jesse Hall, Google
- John Kessenich, Google
- Neil Henning, AMD
- Tom Olson, Arm

Description

This extension allows the use of SPIR-V 1.4 shader modules. SPIR-V 1.4's new features primarily make it an easier target for compilers from high-level languages, rather than exposing new hardware functionality.

SPIR-V 1.4 incorporates features that are also available separately as extensions. SPIR-V 1.4 shader modules do not need to enable those extensions with the `OpExtension` opcode, since they are integral parts of SPIR-V 1.4.

SPIR-V 1.4 introduces new floating-point execution mode capabilities, also available via `SPV_KHR_float_controls`. Implementations are not required to support all of these new capabilities; support can be queried using VkPhysicalDeviceFloatControlsPropertiesKHR from the `VK_KHR_shader_float_controls` extension.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Enum Constants

- `VK_KHR_SPIRV_1_4_EXTENSION_NAME`
- `VK_KHR_SPIRV_1_4_SPEC_VERSION`

Issues

1. Should we have an extension specific to this SPIR-V version, or add a version-generic query for SPIR-V version? SPIR-V 1.4 does not need any other API changes.

RESOLVED: Just expose SPIR-V 1.4.

Most new SPIR-V versions introduce optionally-required capabilities or have implementation-defined limits, and would need more API and specification changes specific to that version to make them available in Vulkan. For example, to support the subgroup capabilities added in SPIR-V 1.3 required introducing VkPhysicalDeviceSubgroupProperties to allow querying the supported group operation categories, maximum supported subgroup size, etc. While we could expose the parts of a new SPIR-V version that do not need accompanying changes generically, we will still end up writing extensions specific to each version for the remaining parts. Thus the generic mechanism will not reduce future spec-writing effort. In addition, making it clear which parts of a future version are supported by the generic mechanism and which cannot be used without specific support would be difficult to get right ahead of time.

2. Can different stages of the same pipeline use shaders with different SPIR-V versions?

RESOLVED: Yes.

Mixing SPIR-V versions 1.0-1.3 in the same pipeline has not been disallowed, so it would be inconsistent to disallow mixing 1.4 with previous versions. SPIR-V 1.4 does not introduce anything that should cause new difficulties here.

3. Must Vulkan extensions corresponding to SPIR-V extensions that were promoted to core in 1.4 be enabled in order to use that functionality in a SPIR-V 1.4 module?

RESOLVED: No, with caveats.

The SPIR-V 1.4 module does not need to declare the SPIR-V extensions, since the functionality is now part of core, so there is no need to enable the Vulkan extension that allows SPIR-V modules to declare the SPIR-V extension. However, when the functionality that is now core in SPIR-V 1.4 is optionally supported, the query for support is provided by a Vulkan extension, and that query can only be used if the extension is enabled.

This applies to any SPIR-V version; specifically for SPIR-V 1.4 this only applies to the functionality from `SPV_KHR_float_controls`, which was made available in Vulkan by `VK_KHR_shader_float_controls`. Even though the extension was promoted in SPIR-V 1.4, the capabilities are still optional in implementations that support `VK_KHR_spirv_1_4`.

A SPIR-V 1.4 module does not need to enable `SPV_KHR_float_controls` in order to use the capabilities, so if the application has *a priori* knowledge that the implementation supports the capabilities, it does not need to enable `VK_KHR_shader_float_controls`. However, if it does not have this knowledge and has to query for support at runtime, it must enable `VK_KHR_shader_float_controls` in order to use VkPhysicalDeviceFloatControlsPropertiesKHR.

Version History

- Revision 1, 2019-04-01 (Jesse Hall)
  - Internal draft versions

VK_KHR_storage_buffer_storage_class

Name String

`VK_KHR_storage_buffer_storage_class`

Extension Type

Device extension

Registered Extension Number

132

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

SPIR-V Dependencies

- SPV_KHR_storage_buffer_storage_class

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Alexander Galazin [GitHub]alegal-arm

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Contributors

- Alexander Galazin, ARM
- David Neto, Google

Description

This extension adds support for the following SPIR-V extension in Vulkan:

- `SPV_KHR_storage_buffer_storage_class`

This extension provides a new SPIR-V `StorageBuffer` storage class. A `Block`-decorated object in this class is equivalent to a `BufferBlock`-decorated object in the `Uniform` storage class.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1.

New Enum Constants

- `VK_KHR_STORAGE_BUFFER_STORAGE_CLASS_EXTENSION_NAME`
- `VK_KHR_STORAGE_BUFFER_STORAGE_CLASS_SPEC_VERSION`

Version History

- Revision 1, 2017-03-23 (Alexander Galazin)
  - Initial draft

VK_KHR_synchronization2

Name String

`VK_KHR_synchronization2`

Extension Type

Device extension

Registered Extension Number

315

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

API Interactions

- Interacts with VK_EXT_blend_operation_advanced
- Interacts with VK_EXT_conditional_rendering
- Interacts with VK_EXT_device_generated_commands
- Interacts with VK_EXT_fragment_density_map
- Interacts with VK_EXT_mesh_shader
- Interacts with VK_EXT_transform_feedback
- Interacts with VK_KHR_acceleration_structure
- Interacts with VK_KHR_fragment_shading_rate
- Interacts with VK_KHR_ray_tracing_pipeline
- Interacts with VK_NV_device_generated_commands
- Interacts with VK_NV_mesh_shader
- Interacts with VK_NV_ray_tracing
- Interacts with VK_NV_shading_rate_image

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Tobias Hector [GitHub]tobski

Other Extension Metadata

Last Modified Date

2020-12-03

Interactions and External Dependencies

- Interacts with VK_KHR_create_renderpass2

Contributors

- Tobias Hector

Description

This extension modifies the original core synchronization APIs to simplify the interface and improve usability of these APIs. It also adds new pipeline stage and access flag types that extend into the 64-bit range, as we have run out within the 32-bit range. The new flags are identical to the old values within the 32-bit range, with new stages and bits beyond that.

Pipeline stages and access flags are now specified together in memory barrier structures, making the connection between the two more obvious. Additionally, scoping the pipeline stages into the barrier structs allows the use of the MEMORY_READ and MEMORY_WRITE flags without sacrificing precision. The per-stage access flags should be used to disambiguate specific accesses in a given stage or set of stages - for instance, between uniform reads and sampling operations.

Layout transitions have been simplified as well; rather than requiring a different set of layouts for depth/stencil/color attachments, there are generic VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR and VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR layouts which are contextually applied based on the image format. For example, for a depth format image, VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR is equivalent to VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL_KHR. VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR also functionally replaces VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL.

Events are now more efficient, because they include memory dependency information when you set them on the device. Previously, this information was only known when waiting on an event, so the dependencies could not be satisfied until the wait occurred. That sometimes meant stalling the pipeline when the wait occurred. The new API provides enough information for implementations to satisfy these dependencies in parallel with other tasks.

Queue submission has been changed to wrap command buffers and semaphores in extensible structures, which incorporate changes from Vulkan 1.1, VK_KHR_device_group, and VK_KHR_timeline_semaphore. This also adds a pipeline stage to the semaphore signal operation, mirroring the existing pipeline stage specification for wait operations.

Other miscellaneous changes include:

- Events can now be specified as interacting only with the device, allowing more efficient access to the underlying object.
- Image memory barriers that do not perform an image layout transition can be specified by

setting `oldLayout` equal to `newLayout`.

    - E.g. the old and new layout can both be set to VK_IMAGE_LAYOUT_UNDEFINED, without discarding data in the image.

- Queue family ownership transfer parameters are simplified in some cases.
- Extensions with commands or functions with a VkPipelineStageFlags or VkPipelineStageFlagBits parameter have had those APIs replaced with equivalents using VkPipelineStageFlags2KHR.
- The new event and barrier interfaces are now more extensible for future changes.
- Relevant pipeline stage masks can now be specified as empty with the new VK_PIPELINE_STAGE_NONE_KHR and VK_PIPELINE_STAGE_2_NONE_KHR values.
- VkMemoryBarrier2KHR can be chained to VkSubpassDependency2, overriding the original 32-bit stage and access masks.

New Base Types

- VkFlags64

New Commands

- vkCmdPipelineBarrier2KHR
- vkCmdResetEvent2KHR
- vkCmdSetEvent2KHR
- vkCmdWaitEvents2KHR
- vkCmdWriteTimestamp2KHR
- vkQueueSubmit2KHR

New Structures

- VkBufferMemoryBarrier2KHR
- VkCommandBufferSubmitInfoKHR
- VkDependencyInfoKHR
- VkImageMemoryBarrier2KHR
- VkSemaphoreSubmitInfoKHR
- VkSubmitInfo2KHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
    - VkPhysicalDeviceSynchronization2FeaturesKHR
- Extending VkSubpassDependency2:
    - VkMemoryBarrier2KHR

New Enums

- VkAccessFlagBits2KHR

- VkPipelineStageFlagBits2KHR
- VkSubmitFlagBitsKHR

New Bitmasks

- VkAccessFlags2KHR
- VkPipelineStageFlags2KHR
- VkSubmitFlagsKHR

New Enum Constants

- `VK_KHR_SYNCHRONIZATION_2_EXTENSION_NAME`
- `VK_KHR_SYNCHRONIZATION_2_SPEC_VERSION`
- Extending VkAccessFlagBits:
  - VK_ACCESS_NONE_KHR
- Extending VkAccessFlagBits2:
  - VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT_KHR
  - VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT_KHR
  - VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT_KHR
  - VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT_KHR
  - VK_ACCESS_2_HOST_READ_BIT_KHR
  - VK_ACCESS_2_HOST_WRITE_BIT_KHR
  - VK_ACCESS_2_INDEX_READ_BIT_KHR
  - VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT_KHR
  - VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT_KHR
  - VK_ACCESS_2_MEMORY_READ_BIT_KHR
  - VK_ACCESS_2_MEMORY_WRITE_BIT_KHR
  - VK_ACCESS_2_NONE_KHR
  - VK_ACCESS_2_SHADER_READ_BIT_KHR
  - VK_ACCESS_2_SHADER_SAMPLED_READ_BIT_KHR
  - VK_ACCESS_2_SHADER_STORAGE_READ_BIT_KHR
  - VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT_KHR
  - VK_ACCESS_2_SHADER_WRITE_BIT_KHR
  - VK_ACCESS_2_TRANSFER_READ_BIT_KHR
  - VK_ACCESS_2_TRANSFER_WRITE_BIT_KHR
  - VK_ACCESS_2_UNIFORM_READ_BIT_KHR
  - VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT_KHR

- Extending VkEventCreateFlagBits:
  - VK_EVENT_CREATE_DEVICE_ONLY_BIT_KHR
- Extending VkImageLayout:
  - VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR
  - VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR
- Extending VkPipelineStageFlagBits:
  - VK_PIPELINE_STAGE_NONE_KHR
- Extending VkPipelineStageFlagBits2:
  - VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT_KHR
  - VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT_KHR
  - VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT_KHR
  - VK_PIPELINE_STAGE_2_BLIT_BIT_KHR
  - VK_PIPELINE_STAGE_2_BOTTOM_OF_PIPE_BIT_KHR
  - VK_PIPELINE_STAGE_2_CLEAR_BIT_KHR
  - VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT_KHR
  - VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT_KHR
  - VK_PIPELINE_STAGE_2_COPY_BIT_KHR
  - VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT_KHR
  - VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT_KHR
  - VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT_KHR
  - VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT_KHR
  - VK_PIPELINE_STAGE_2_HOST_BIT_KHR
  - VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT_KHR
  - VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT_KHR
  - VK_PIPELINE_STAGE_2_NONE_KHR
  - VK_PIPELINE_STAGE_2_PRE_RASTERIZATION_SHADERS_BIT_KHR
  - VK_PIPELINE_STAGE_2_RESOLVE_BIT_KHR
  - VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT_KHR
  - VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT_KHR
  - VK_PIPELINE_STAGE_2_TOP_OF_PIPE_BIT_KHR
  - VK_PIPELINE_STAGE_2_TRANSFER_BIT_KHR
  - VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT_KHR
  - VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT_KHR
  - VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT_KHR
- Extending VkStructureType:

    - VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER_2_KHR
    - VK_STRUCTURE_TYPE_COMMAND_BUFFER_SUBMIT_INFO_KHR
    - VK_STRUCTURE_TYPE_DEPENDENCY_INFO_KHR
    - VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER_2_KHR
    - VK_STRUCTURE_TYPE_MEMORY_BARRIER_2_KHR
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SYNCHRONIZATION_2_FEATURES_KHR
    - VK_STRUCTURE_TYPE_SEMAPHORE_SUBMIT_INFO_KHR
    - VK_STRUCTURE_TYPE_SUBMIT_INFO_2_KHR
- Extending VkSubmitFlagBits:
    - VK_SUBMIT_PROTECTED_BIT_KHR

If VK_EXT_blend_operation_advanced is supported:

- Extending VkAccessFlagBits2:
    - VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT

If VK_EXT_conditional_rendering is supported:

- Extending VkAccessFlagBits2:
    - VK_ACCESS_2_CONDITIONAL_RENDERING_READ_BIT_EXT
- Extending VkPipelineStageFlagBits2:
    - VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT

If VK_EXT_device_generated_commands is supported:

- Extending VkAccessFlagBits2:
    - VK_ACCESS_2_COMMAND_PREPROCESS_READ_BIT_EXT
    - VK_ACCESS_2_COMMAND_PREPROCESS_WRITE_BIT_EXT
- Extending VkPipelineStageFlagBits2:
    - VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT

If VK_EXT_fragment_density_map is supported:

- Extending VkAccessFlagBits2:
    - VK_ACCESS_2_FRAGMENT_DENSITY_MAP_READ_BIT_EXT
- Extending VkPipelineStageFlagBits2:
    - VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT

If VK_EXT_mesh_shader is supported:

- Extending VkPipelineStageFlagBits2:
    - VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT

    - VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT

If VK_EXT_transform_feedback is supported:

- Extending VkAccessFlagBits2:
    - VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT
    - VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT
    - VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT
- Extending VkPipelineStageFlagBits2:
    - VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT

If VK_KHR_acceleration_structure is supported:

- Extending VkAccessFlagBits2:
    - VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR
    - VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR
- Extending VkPipelineStageFlagBits2:
    - VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR

If VK_KHR_fragment_shading_rate is supported:

- Extending VkAccessFlagBits2:
    - VK_ACCESS_2_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR
- Extending VkPipelineStageFlagBits2:
    - VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR

If VK_KHR_ray_tracing_pipeline is supported:

- Extending VkPipelineStageFlagBits2:
    - VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Examples

See https://github.com/KhronosGroup/Vulkan-Docs/wiki/Synchronization-Examples

Version History

- Revision 1, 2020-12-03 (Tobias Hector)
    - Internal revisions

VK_KHR_timeline_semaphore

Name String

`VK_KHR_timeline_semaphore`

Extension Type

Device extension

Registered Extension Number

208

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Faith Ekstrand [GitHub]gfxstrand

Other Extension Metadata

Last Modified Date

2019-06-12

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension interacts with `VK_KHR_external_semaphore_capabilities`
- This extension interacts with `VK_KHR_external_semaphore`
- This extension interacts with `VK_KHR_external_semaphore_win32`

Contributors

- Jeff Bolz, NVIDIA
- Yuriy O'Donnell, Epic Games
- Faith Ekstrand, Intel
- Jesse Hall, Google
- James Jones, NVIDIA

- Jeff Juliano, NVIDIA
- Daniel Rakos, AMD
- Ray Smith, Arm

Description

This extension introduces a new type of semaphore that has an integer payload identifying a point in a timeline. Such timeline semaphores support the following operations:

- Host query - A host operation that allows querying the payload of the timeline semaphore.
- Host wait - A host operation that allows a blocking wait for a timeline semaphore to reach a specified value.
- Host signal - A host operation that allows advancing the timeline semaphore to a specified value.
- Device wait - A device operation that allows waiting for a timeline semaphore to reach a specified value.
- Device signal - A device operation that allows advancing the timeline semaphore to a specified value.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- vkGetSemaphoreCounterValueKHR
- vkSignalSemaphoreKHR
- vkWaitSemaphoresKHR

New Structures

- VkSemaphoreSignalInfoKHR
- VkSemaphoreWaitInfoKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceTimelineSemaphoreFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceTimelineSemaphorePropertiesKHR
- Extending VkSemaphoreCreateInfo, VkPhysicalDeviceExternalSemaphoreInfo:
  - VkSemaphoreTypeCreateInfoKHR
- Extending VkSubmitInfo, VkBindSparseInfo:
  - VkTimelineSemaphoreSubmitInfoKHR

New Enums

- VkSemaphoreTypeKHR
- VkSemaphoreWaitFlagBitsKHR

New Bitmasks

- VkSemaphoreWaitFlagsKHR

New Enum Constants

- `VK_KHR_TIMELINE_SEMAPHORE_EXTENSION_NAME`
- `VK_KHR_TIMELINE_SEMAPHORE_SPEC_VERSION`
- Extending VkSemaphoreType:
  - VK_SEMAPHORE_TYPE_BINARY_KHR
  - VK_SEMAPHORE_TYPE_TIMELINE_KHR
- Extending VkSemaphoreWaitFlagBits:
  - VK_SEMAPHORE_WAIT_ANY_BIT_KHR
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_SEMAPHORE_SIGNAL_INFO_KHR
  - VK_STRUCTURE_TYPE_SEMAPHORE_TYPE_CREATE_INFO_KHR
  - VK_STRUCTURE_TYPE_SEMAPHORE_WAIT_INFO_KHR
  - VK_STRUCTURE_TYPE_TIMELINE_SEMAPHORE_SUBMIT_INFO_KHR

Issues

1) Do we need a new object type for this?

RESOLVED: No, we just introduce a new type of semaphore object, as `VK_KHR_external_semaphore_win32` already uses semaphores as the destination for importing D3D12 fence objects, which are semantically close/identical to the proposed synchronization primitive.

2) What type of payload the new synchronization primitive has?

RESOLVED: A 64-bit unsigned integer that can only be set to strictly increasing values by signal operations and is not changed by wait operations.

3) Does the new synchronization primitive have the same signal-before-wait requirement as the existing semaphores do?

RESOLVED: No. Timeline semaphores support signaling and waiting entirely asynchronously. It is the responsibility of the application to avoid deadlock.

4) Does the new synchronization primitive allow resetting its payload?

RESOLVED: No, allowing the payload value to “go backwards” is problematic. Applications looking for reset behavior should create a new instance of the synchronization primitive instead.

5) How do we enable host waits on the synchronization primitive?

RESOLVED: Both a non-blocking query of the current payload value of the synchronization primitive, and a blocking wait operation are provided.

6) How do we enable device waits and signals on the synchronization primitive?

RESOLVED: Similar to `VK_KHR_external_semaphore_win32`, this extension introduces a new structure that can be chained to VkSubmitInfo to specify the values signaled semaphores should be set to, and the values waited semaphores need to reach.

7) Can the new synchronization primitive be used to synchronize presentation and swapchain image acquisition operations?

RESOLVED: Some implementations may have problems with supporting that directly, thus it is not allowed in this extension.

8) Do we want to support external sharing of the new synchronization primitive type?

RESOLVED: Yes. Timeline semaphore specific external sharing capabilities can be queried using vkGetPhysicalDeviceExternalSemaphoreProperties by chaining the new VkSemaphoreTypeCreateInfoKHR structure to its `pExternalSemaphoreInfo` structure. This allows having a different set of external semaphore handle types supported for timeline semaphores vs. binary semaphores.

9) Do we need to add a host signal operation for the new synchronization primitive type?

RESOLVED: Yes. This helps in situations where one host thread submits a workload but another host thread has the information on when the workload is ready to be executed.

10) How should the new synchronization primitive interact with the ordering requirements of the original `VkSemaphore`?

RESOLVED: Prior to calling any command which **may** cause a wait operation on a binary semaphore, the application **must** ensure that the semaphore signal operation that has been submitted for execution and any semaphore signal operations on which it depends (if any) **must** have also been submitted for execution.

11) Should we have separate feature bits for different sub-features of timeline semaphores?

RESOLVED: No. The only feature which cannot be supported universally is timeline semaphore import/export. For import/export, the application is already required to query available external handle types via vkGetPhysicalDeviceExternalSemaphoreProperties and provide the semaphore type by adding a VkSemaphoreTypeCreateInfoKHR structure to the `pNext` chain of VkPhysicalDeviceExternalSemaphoreInfo so no new feature bit is required.

Version History

- Revision 1, 2018-05-10 (Faith Ekstrand)
  - Initial version
- Revision 2, 2019-06-12 (Faith Ekstrand)
  - Added an initialValue parameter to timeline semaphore creation

VK_KHR_uniform_buffer_standard_layout

Name String

`VK_KHR_uniform_buffer_standard_layout`

Extension Type

Device extension

Registered Extension Number

254

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Graeme Leese [GitHub]gnl21

Other Extension Metadata

Last Modified Date

2019-01-25

Contributors

- Graeme Leese, Broadcom
- Jeff Bolz, NVIDIA
- Tobias Hector, AMD
- Faith Ekstrand, Intel
- Neil Henning, AMD

Description

This extension enables tighter array and structure packing to be used with uniform buffers.

It modifies the alignment rules for uniform buffers, allowing for tighter packing of arrays and structures. This allows, for example, the std430 layout, as defined in GLSL to be supported in uniform buffers.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceUniformBufferStandardLayoutFeaturesKHR

New Enum Constants

- `VK_KHR_UNIFORM_BUFFER_STANDARD_LAYOUT_EXTENSION_NAME`
- `VK_KHR_UNIFORM_BUFFER_STANDARD_LAYOUT_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_UNIFORM_BUFFER_STANDARD_LAYOUT_FEATURES_KHR

Version History

- Revision 1, 2019-01-25 (Graeme Leese)
  - Initial draft

VK_KHR_variable_pointers

Name String

`VK_KHR_variable_pointers`

Extension Type

Device extension

Registered Extension Number

121

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
and
VK_KHR_storage_buffer_storage_class
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_variable_pointers

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jesse Hall [GitHub]critsec

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Contributors

- John Kessenich, Google
- Neil Henning, Codeplay
- David Neto, Google
- Daniel Koch, Nvidia
- Graeme Leese, Broadcom

- Weifeng Zhang, Qualcomm
- Stephen Clarke, Imagination Technologies
- Faith Ekstrand, Intel
- Jesse Hall, Google

Description

The VK_KHR_variable_pointers extension allows implementations to indicate their level of support for the SPV_KHR_variable_pointers SPIR-V extension. The SPIR-V extension allows shader modules to use invocation-private pointers into uniform and/or storage buffers, where the pointer values can be dynamic and non-uniform.

The SPV_KHR_variable_pointers extension introduces two capabilities. The first, VariablePointersStorageBuffer, **must** be supported by all implementations of this extension. The second, VariablePointers, is optional.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted, however support for the variablePointersStorageBuffer feature is made optional. The original type, enum, and command names are still available as aliases of the core functionality.

Promotion to Vulkan 1.4

If Vulkan 1.4 is supported, support for the variablePointers and variablePointersStorageBuffer features is required.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceVariablePointerFeaturesKHR
  - VkPhysicalDeviceVariablePointersFeaturesKHR

New Enum Constants

- VK_KHR_VARIABLE_POINTERS_EXTENSION_NAME
- VK_KHR_VARIABLE_POINTERS_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTERS_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTER_FEATURES_KHR

New SPIR-V Capabilities

- VariablePointers
- VariablePointersStorageBuffer

Issues

1) Do we need an optional property for the SPIR-V `VariablePointersStorageBuffer` capability or should it be mandatory when this extension is advertised?

RESOLVED: Add it as a distinct feature, but make support mandatory. Adding it as a feature makes the extension easier to include in a future core API version. In the extension, the feature is mandatory, so that presence of the extension guarantees some functionality. When included in a core API version, the feature would be optional.

2) Can support for these capabilities vary between shader stages?

RESOLVED: No, if the capability is supported in any stage it must be supported in all stages.

3) Should the capabilities be features or limits?

RESOLVED: Features, primarily for consistency with other similar extensions.

Version History

- Revision 1, 2017-03-14 (Jesse Hall and John Kessenich)
  - Internal revisions

VK_KHR_vertex_attribute_divisor

Name String

`VK_KHR_vertex_attribute_divisor`

Extension Type

Device extension

Registered Extension Number

526

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Extension Proposal

VK_KHR_vertex_attribute_divisor

Other Extension Metadata

Last Modified Date

2023-09-20

IP Status

No known IP claims.

Contributors

- Shahbaz Youssefi, Google
- Contributors to `VK_EXT_vertex_attribute_divisor`

Description

This extension is based on the `VK_EXT_vertex_attribute_divisor` extension. The only difference is the new property `supportsNonZeroFirstInstance`, which indicates support for non-zero values in `firstInstance`. This allows the extension to be supported on implementations that have traditionally only supported OpenGL ES.

New Structures

- VkVertexInputBindingDivisorDescriptionKHR
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceVertexAttributeDivisorFeaturesKHR
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR
- Extending VkPipelineVertexInputStateCreateInfo:
  - VkPipelineVertexInputDivisorStateCreateInfoKHR

New Enum Constants

- `VK_KHR_VERTEX_ATTRIBUTE_DIVISOR_EXTENSION_NAME`
- `VK_KHR_VERTEX_ATTRIBUTE_DIVISOR_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_FEATURES_KHR
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_PROPERTIES_KHR
  - VK_STRUCTURE_TYPE_PIPELINE_VERTEX_INPUT_DIVISOR_STATE_CREATE_INFO_KHR

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4 with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

Version History

- Revision 1, 2023-09-20 (Shahbaz Youssefi)
  - First Version, based on `VK_EXT_vertex_attribute_divisor`

VK_KHR_vulkan_memory_model

Name String

`VK_KHR_vulkan_memory_model`

Extension Type

Device extension

Registered Extension Number

212

Revision

3

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_KHR_vulkan_memory_model

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Jeff Bolz [GitHub]jeffbolznv

Other Extension Metadata

Last Modified Date

2018-12-10

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Alan Baker, Google
- Tobias Hector, AMD
- David Neto, Google
- Robert Simpson, Qualcomm Technologies, Inc.
- Brian Sumner, AMD

Description

The VK_KHR_vulkan_memory_model extension allows use of the features guarded by the `VulkanMemoryModel`, `VulkanMemoryModelDeviceScope`, and `VulkanMemoryModelAvailabilityVisibilityChains` capabilities in shader modules. The Vulkan Memory Model formally defines how to synchronize memory accesses to the same memory locations performed by multiple shader invocations.

NOTE	Version 3 of the spec added a member (`vulkanMemoryModelAvailabilityVisibilityChains`) to VkPhysicalDeviceVulkanMemoryModelFeaturesKHR, which is an incompatible change from version 2.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. However, if Vulkan 1.2 is supported and this extension is not, the `vulkanMemoryModel` capability is optional. The original type, enum, and command names are still available as aliases of the core functionality.

Promotion to Vulkan 1.3

If Vulkan 1.3 is supported, support for the `vulkanMemoryModel` and `vulkanMemoryModelDeviceScope` capabilities is required.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceVulkanMemoryModelFeaturesKHR

New Enum Constants

- `VK_KHR_VULKAN_MEMORY_MODEL_EXTENSION_NAME`
- `VK_KHR_VULKAN_MEMORY_MODEL_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_MEMORY_MODEL_FEATURES_KHR

New SPIR-V Capabilities

- `VulkanMemoryModelKHR`

Version History

- Revision 1, 2018-06-24 (Jeff Bolz)
  - Initial draft
- Revision 3, 2018-12-10 (Jeff Bolz)
  - Add vulkanMemoryModelAvailabilityVisibilityChains member to the

VkPhysicalDeviceVulkanMemoryModelFeaturesKHR structure.

VK_KHR_zero_initialize_workgroup_memory

Name String

`VK_KHR_zero_initialize_workgroup_memory`

Extension Type

Device extension

Registered Extension Number

326

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Alan Baker [GitHub]alan-baker

Other Extension Metadata

Last Modified Date

2020-11-18

IP Status

No known IP claims.

Contributors

- Alan Baker, Google
- Jeff Bolz, Nvidia
- Faith Ekstrand, Intel

Description

This extension allows the use of a null constant initializer on shader Workgroup memory variables, allowing implementations to expose any special hardware or instructions they may have. Zero initialization is commonly used by applications running untrusted content (e.g. web browsers) as way of defeating memory-scraping attacks.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeaturesKHR

New Enum Constants

- `VK_KHR_ZERO_INITIALIZE_WORKGROUP_MEMORY_EXTENSION_NAME`
- `VK_KHR_ZERO_INITIALIZE_WORKGROUP_MEMORY_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ZERO_INITIALIZE_WORKGROUP_MEMORY_FEATURES_KHR

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Version History

- Revision 1, 2020-11-18 (Alan Baker)
  - Internal draft version

VK_EXT_depth_clamp_zero_one

Name String

VK_EXT_depth_clamp_zero_one

Extension Type

Device extension

Registered Extension Number

422

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to VK_KHR_depth_clamp_zero_one extension

Contact

- Graeme Leese [GitHub]gnl21

Other Extension Metadata

Last Modified Date

2021-07-29

Contributors

- Graeme Leese, Broadcom

Description

This extension gives defined behavior to fragment depth values which end up outside the conventional [0, 1] range. It can be used to ensure portability in edge cases of features like depthBias. The particular behavior is chosen to match OpenGL to aid porting or emulation.

Promotion to VK_KHR_depth_clamp_zero_one

All functionality in this extension is included in VK_KHR_depth_clamp_zero_one, with the suffix change to KHR. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDepthClampZeroOneFeaturesEXT

New Enum Constants

- VK_EXT_DEPTH_CLAMP_ZERO_ONE_EXTENSION_NAME
- VK_EXT_DEPTH_CLAMP_ZERO_ONE_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLAMP_ZERO_ONE_FEATURES_EXT

Version History

- Revision 1, 2021-07-29 (Graeme Leese)
  - Internal revisions

VK_EXT_descriptor_buffer

Name String

`VK_EXT_descriptor_buffer`

Extension Type

Device extension

Registered Extension Number

317

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1
and
VK_KHR_buffer_device_address
and
VK_EXT_descriptor_indexing
or
Vulkan Version 1.2
and
VK_KHR_synchronization2
or
Vulkan Version 1.3

API Interactions

- Interacts with VK_EXT_fragment_density_map
- Interacts with VK_KHR_acceleration_structure
- Interacts with VK_NV_ray_tracing

Deprecation State

- *Deprecated* by VK_EXT_descriptor_heap extension

Contact

- Tobias Hector [GitHub]tobski

Extension Proposal

VK_EXT_descriptor_buffer

Other Extension Metadata

Last Modified Date

2021-06-07

IP Status

No known IP claims.

Contributors

- Tobias Hector, AMD
- Stu Smith, AMD
- Maciej Jesionowski, AMD
- Boris Zanin, AMD
- Hans-Kristian Arntzen, Valve
- Connor Abbott, Valve
- Baldur Karlsson, Valve
- Mike Blumenkrantz, Valve
- Graeme Leese, Broadcom
- Jan-Harald Fredriksen, Arm
- Rodrigo Locatti, NVIDIA
- Jeff Bolz, NVIDIA
- Piers Daniell, NVIDIA
- Jeff Leger, QUALCOMM
- Lionel Landwerlin, Intel
- Slawomir Grajewski, Intel

Description

This extension introduces new commands to put shader-accessible descriptors directly in memory, making the management of descriptor data more explicit.

Deprecation by `VK_EXT_descriptor_heap`

Functionality in this extension is deprecated by the VK_EXT_descriptor_heap extension. See Descriptor Management: Replaced by Descriptor Heaps.

New Commands

- vkCmdBindDescriptorBufferEmbeddedSamplersEXT
- vkCmdBindDescriptorBuffersEXT
- vkCmdSetDescriptorBufferOffsetsEXT
- vkGetBufferOpaqueCaptureDescriptorDataEXT
- vkGetDescriptorEXT

- vkGetDescriptorSetLayoutBindingOffsetEXT
- vkGetDescriptorSetLayoutSizeEXT
- vkGetImageOpaqueCaptureDescriptorDataEXT
- vkGetImageViewOpaqueCaptureDescriptorDataEXT
- vkGetSamplerOpaqueCaptureDescriptorDataEXT

If VK_KHR_acceleration_structure or `VK_NV_ray_tracing` is supported:

- vkGetAccelerationStructureOpaqueCaptureDescriptorDataEXT

New Structures

- VkBufferCaptureDescriptorDataInfoEXT
- VkDescriptorAddressInfoEXT
- VkDescriptorBufferBindingInfoEXT
- VkDescriptorGetInfoEXT
- VkImageCaptureDescriptorDataInfoEXT
- VkImageViewCaptureDescriptorDataInfoEXT
- VkSamplerCaptureDescriptorDataInfoEXT
- Extending VkBufferCreateInfo, VkImageCreateInfo, VkImageViewCreateInfo, VkSamplerCreateInfo, VkAccelerationStructureCreateInfoKHR, VkTensorViewCreateInfoARM, VkAccelerationStructureCreateInfo2KHR:
  - VkOpaqueCaptureDescriptorDataCreateInfoEXT
- Extending VkDescriptorBufferBindingInfoEXT:
  - VkDescriptorBufferBindingPushDescriptorBufferHandleEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceDescriptorBufferFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceDescriptorBufferPropertiesEXT

If VK_EXT_fragment_density_map is supported:

- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceDescriptorBufferDensityMapPropertiesEXT

If VK_KHR_acceleration_structure or `VK_NV_ray_tracing` is supported:

- VkAccelerationStructureCaptureDescriptorDataInfoEXT

New Unions

- VkDescriptorDataEXT

New Enum Constants

- VK_EXT_DESCRIPTOR_BUFFER_EXTENSION_NAME
- VK_EXT_DESCRIPTOR_BUFFER_SPEC_VERSION
- Extending VkAccelerationStructureCreateFlagBitsKHR:
  - VK_ACCELERATION_STRUCTURE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT
- Extending VkAccessFlagBits2:
  - VK_ACCESS_2_DESCRIPTOR_BUFFER_READ_BIT_EXT
- Extending VkBufferCreateFlagBits:
  - VK_BUFFER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT
- Extending VkBufferUsageFlagBits:
  - VK_BUFFER_USAGE_PUSH_DESCRIPTORS_DESCRIPTOR_BUFFER_BIT_EXT
  - VK_BUFFER_USAGE_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT
  - VK_BUFFER_USAGE_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT
- Extending VkDescriptorSetLayoutCreateFlagBits:
  - VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
  - VK_DESCRIPTOR_SET_LAYOUT_CREATE_EMBEDDED_IMMUTABLE_SAMPLERS_BIT_EXT
- Extending VkImageCreateFlagBits:
  - VK_IMAGE_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT
- Extending VkImageViewCreateFlagBits:
  - VK_IMAGE_VIEW_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT
- Extending VkPipelineCreateFlagBits:
  - VK_PIPELINE_CREATE_DESCRIPTOR_BUFFER_BIT_EXT
- Extending VkSamplerCreateFlagBits:
  - VK_SAMPLER_CREATE_DESCRIPTOR_BUFFER_CAPTURE_REPLAY_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_BUFFER_CAPTURE_DESCRIPTOR_DATA_INFO_EXT
  - VK_STRUCTURE_TYPE_DESCRIPTOR_ADDRESS_INFO_EXT
  - VK_STRUCTURE_TYPE_DESCRIPTOR_BUFFER_BINDING_INFO_EXT
  - VK_STRUCTURE_TYPE_DESCRIPTOR_BUFFER_BINDING_PUSH_DESCRIPTOR_BUFFER_HANDLE_EXT
  - VK_STRUCTURE_TYPE_DESCRIPTOR_GET_INFO_EXT
  - VK_STRUCTURE_TYPE_IMAGE_CAPTURE_DESCRIPTOR_DATA_INFO_EXT
  - VK_STRUCTURE_TYPE_IMAGE_VIEW_CAPTURE_DESCRIPTOR_DATA_INFO_EXT
  - VK_STRUCTURE_TYPE_OPAQUE_CAPTURE_DESCRIPTOR_DATA_CREATE_INFO_EXT

    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_BUFFER_DENSITY_MAP_PROPERTIES_EXT
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_BUFFER_FEATURES_EXT
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_BUFFER_PROPERTIES_EXT
    - VK_STRUCTURE_TYPE_SAMPLER_CAPTURE_DESCRIPTOR_DATA_INFO_EXT

If VK_KHR_acceleration_structure or `VK_NV_ray_tracing` is supported:

- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CAPTURE_DESCRIPTOR_DATA_INFO_EXT

Version History

- Revision 1, 2021-06-07 (Stu Smith)
    - Initial revision

VK_EXT_descriptor_indexing

Name String

`VK_EXT_descriptor_indexing`

Extension Type

Device extension

Registered Extension Number

162

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
and
VK_KHR_maintenance3
or
Vulkan Version 1.1

SPIR-V Dependencies

- SPV_EXT_descriptor_indexing

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Jeff Bolz [GitHub]jeffbolznv

Other Extension Metadata

Last Modified Date

2017-10-02

Interactions and External Dependencies

- This extension provides API support for `GL_EXT_nonuniform_qualifier`

Contributors

- Jeff Bolz, NVIDIA
- Daniel Rakos, AMD
- Slawomir Grajewski, Intel
- Tobias Hector, Imagination Technologies

Description

This extension adds several small features which together enable applications to create large descriptor sets containing substantially all of their resources, and selecting amongst those resources with dynamic (non-uniform) indexes in the shader. There are feature enables and SPIR-V capabilities for non-uniform descriptor indexing in the shader, and non-uniform indexing in the shader requires use of a new `NonUniformEXT` decoration defined in the `SPV_EXT_descriptor_indexing` SPIR-V extension. There are descriptor set layout binding creation flags enabling several features:

- Descriptors can be updated after they are bound to a command buffer, such that the execution of the command buffer reflects the most recent update to the descriptors.
- Descriptors that are not used by any pending command buffers can be updated, which enables writing new descriptors for frame N+1 while frame N is executing.
- Relax the requirement that all descriptors in a binding that is “statically used” must be valid, such that descriptors that are not accessed by a submission need not be valid and can be updated while that submission is executing.
- The final binding in a descriptor set layout can have a variable size (and unsized arrays of resources are allowed in the `GL_EXT_nonuniform_qualifier` and `SPV_EXT_descriptor_indexing` extensions).

Note that it is valid for multiple descriptor arrays in a shader to use the same set and binding number, as long as they are all compatible with the descriptor type in the pipeline layout. This means a single array binding in the descriptor set can serve multiple texture dimensionalities, or an array of buffer descriptors can be used with multiple different block layouts.

There are new descriptor set layout and descriptor pool creation flags that are required to opt in to the update-after-bind functionality, and there are separate `maxPerStage*` and `maxDescriptorSet*` limits that apply to these descriptor set layouts which **may** be much higher than the pre-existing limits. The old limits only count descriptors in non-updateAfterBind descriptor set layouts, and the new limits count descriptors in all descriptor set layouts in the pipeline layout.

Promotion to Vulkan 1.2

Vulkan APIs in this extension are included in core Vulkan 1.2, with the EXT suffix omitted. However, if Vulkan 1.2 is supported and this extension is not, the `descriptorIndexing` capability is optional. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Promotion to Vulkan 1.4

If Vulkan 1.4 is supported, support for the `shaderUniformTexelBufferArrayDynamicIndexing` and `shaderStorageTexelBufferArrayDynamicIndexing` capabilities is required.

New Structures

- Extending VkDescriptorSetAllocateInfo:
  - VkDescriptorSetVariableDescriptorCountAllocateInfoEXT
- Extending VkDescriptorSetLayoutCreateInfo:

    - VkDescriptorSetLayoutBindingFlagsCreateInfoEXT
- Extending VkDescriptorSetLayoutSupport:
    - VkDescriptorSetVariableDescriptorCountLayoutSupportEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
    - VkPhysicalDeviceDescriptorIndexingFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
    - VkPhysicalDeviceDescriptorIndexingPropertiesEXT

New Enums

- VkDescriptorBindingFlagBitsEXT

New Bitmasks

- VkDescriptorBindingFlagsEXT

New Enum Constants

- `VK_EXT_DESCRIPTOR_INDEXING_EXTENSION_NAME`
- `VK_EXT_DESCRIPTOR_INDEXING_SPEC_VERSION`
- Extending VkDescriptorBindingFlagBits:
    - VK_DESCRIPTOR_BINDING_PARTIALLY_BOUND_BIT_EXT
    - VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT_EXT
    - VK_DESCRIPTOR_BINDING_UPDATE_UNUSED_WHILE_PENDING_BIT_EXT
    - VK_DESCRIPTOR_BINDING_VARIABLE_DESCRIPTOR_COUNT_BIT_EXT
- Extending VkDescriptorPoolCreateFlagBits:
    - VK_DESCRIPTOR_POOL_CREATE_UPDATE_AFTER_BIND_BIT_EXT
- Extending VkDescriptorSetLayoutCreateFlagBits:
    - VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT_EXT
- Extending VkResult:
    - VK_ERROR_FRAGMENTATION_EXT
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_BINDING_FLAGS_CREATE_INFO_EXT
    - VK_STRUCTURE_TYPE_DESCRIPTOR_SET_VARIABLE_DESCRIPTOR_COUNT_ALLOCATE_INFO_EXT
    - VK_STRUCTURE_TYPE_DESCRIPTOR_SET_VARIABLE_DESCRIPTOR_COUNT_LAYOUT_SUPPORT_EXT
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_INDEXING_FEATURES_EXT
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DESCRIPTOR_INDEXING_PROPERTIES_EXT

Version History

- Revision 1, 2017-07-26 (Jeff Bolz)
  - Internal revisions
- Revision 2, 2017-10-02 (Jeff Bolz)
  - ???

VK_EXT_device_fault

Name String

`VK_EXT_device_fault`

Extension Type

Device extension

Registered Extension Number

342

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to VK_KHR_device_fault extension

Contact

- Ralph Potter [GitLab]r_potter

Extension Proposal

VK_EXT_device_fault

Other Extension Metadata

Last Modified Date

2021-03-10

IP Status

No known IP claims.

Contributors

- Ralph Potter, Samsung
- Stuart Smith, AMD
- Jan-Harald Fredriksen, ARM
- Mark Bellamy, ARM
- Andrew Ellem, Google
- Alex Walters, IMG
- Jeff Bolz, NVIDIA

- Baldur Karlsson, Valve

Description

Device loss can be triggered by a variety of issues, including invalid API usage, implementation errors, or hardware failures.

This extension introduces a new command: vkGetDeviceFaultInfoEXT, which may be called subsequent to a VK_ERROR_DEVICE_LOST error code having been returned by the implementation. This command allows developers to query for additional information on GPU faults which may have caused device loss, and to generate binary crash dumps, which may be loaded into external tools for further diagnosis.

New Commands

- vkGetDeviceFaultInfoEXT

New Structures

- VkDeviceFaultAddressInfoEXT
- VkDeviceFaultCountsEXT
- VkDeviceFaultInfoEXT
- VkDeviceFaultVendorBinaryHeaderVersionOneEXT
- VkDeviceFaultVendorInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceFaultFeaturesEXT

New Enums

- VkDeviceFaultAddressTypeEXT
- VkDeviceFaultVendorBinaryHeaderVersionEXT

New Enum Constants

- `VK_EXT_DEVICE_FAULT_EXTENSION_NAME`
- `VK_EXT_DEVICE_FAULT_SPEC_VERSION`
- Extending VkDeviceFaultAddressTypeKHR:
  - VK_DEVICE_FAULT_ADDRESS_TYPE_EXECUTE_INVALID_EXT
  - VK_DEVICE_FAULT_ADDRESS_TYPE_INSTRUCTION_POINTER_FAULT_EXT
  - VK_DEVICE_FAULT_ADDRESS_TYPE_INSTRUCTION_POINTER_INVALID_EXT
  - VK_DEVICE_FAULT_ADDRESS_TYPE_INSTRUCTION_POINTER_UNKNOWN_EXT
  - VK_DEVICE_FAULT_ADDRESS_TYPE_NONE_EXT
  - VK_DEVICE_FAULT_ADDRESS_TYPE_READ_INVALID_EXT

    - VK_DEVICE_FAULT_ADDRESS_TYPE_WRITE_INVALID_EXT
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_DEVICE_FAULT_COUNTS_EXT
    - VK_STRUCTURE_TYPE_DEVICE_FAULT_INFO_EXT
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FAULT_FEATURES_EXT

Version History

- Revision 2, 2023-04-05 (Ralph Potter)
    - Restored two missing members to the XML definition of VkDeviceFaultVendorBinaryHeaderVersionOneEXT. No functional change to the specification.
- Revision 1, 2020-10-19 (Ralph Potter)
    - Initial revision

VK_EXT_extended_dynamic_state

Name String

`VK_EXT_extended_dynamic_state`

Extension Type

Device extension

Registered Extension Number

268

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Piers Daniell [GitHub]pdaniell-nv

Other Extension Metadata

Last Modified Date

2019-12-09

IP Status

No known IP claims.

Contributors

- Dan Ginsburg, Valve Corporation
- Graeme Leese, Broadcom
- Hans-Kristian Arntzen, Valve Corporation
- Jan-Harald Fredriksen, Arm Limited
- Faith Ekstrand, Intel
- Jeff Bolz, NVIDIA
- Jesse Hall, Google
- Philip Rebohle, Valve Corporation
- Stuart Smith, Imagination Technologies

- Tobias Hector, AMD

Description

This extension adds some more dynamic state to support applications that need to reduce the number of pipeline state objects they compile and bind.

New Commands

- vkCmdBindVertexBuffers2EXT
- vkCmdSetCullModeEXT
- vkCmdSetDepthBoundsTestEnableEXT
- vkCmdSetDepthCompareOpEXT
- vkCmdSetDepthTestEnableEXT
- vkCmdSetDepthWriteEnableEXT
- vkCmdSetFrontFaceEXT
- vkCmdSetPrimitiveTopologyEXT
- vkCmdSetScissorWithCountEXT
- vkCmdSetStencilOpEXT
- vkCmdSetStencilTestEnableEXT
- vkCmdSetViewportWithCountEXT

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceExtendedDynamicStateFeaturesEXT

New Enum Constants

- `VK_EXT_EXTENDED_DYNAMIC_STATE_EXTENSION_NAME`
- `VK_EXT_EXTENDED_DYNAMIC_STATE_SPEC_VERSION`
- Extending VkDynamicState:
  - VK_DYNAMIC_STATE_CULL_MODE_EXT
  - VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE_EXT
  - VK_DYNAMIC_STATE_DEPTH_COMPARE_OP_EXT
  - VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE_EXT
  - VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE_EXT
  - VK_DYNAMIC_STATE_FRONT_FACE_EXT
  - VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY_EXT
  - VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT_EXT

    - VK_DYNAMIC_STATE_STENCIL_OP_EXT
    - VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE_EXT
    - VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE_EXT
    - VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT_EXT
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_FEATURES_EXT

Promotion to Vulkan 1.3

All dynamic state enumerants and commands in this extension are included in core Vulkan 1.3, with the EXT suffix omitted. The feature structure is not promoted. Extension interfaces that were promoted remain available as aliases of the core functionality.

Issues

1) Why are the values of `pStrides` in vkCmdBindVertexBuffers2 limited to be between 0 and the maximum extent of the binding, when this restriction is not present for the same static state?

Implementing these edge cases adds overhead to some implementations that would require significant cost when calling this function, and the intention is that this state should be more or less free to change.

VK_EXT_vertex_input_dynamic_state allows the stride to be changed freely when supported via vkCmdSetVertexInputEXT.

Version History

- Revision 1, 2019-12-09 (Piers Daniell)
    - Internal revisions

VK_EXT_extended_dynamic_state2

Name String

`VK_EXT_extended_dynamic_state2`

Extension Type

Device extension

Registered Extension Number

378

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Vikram Kushwaha [GitHub]vkushwaha-nv

Other Extension Metadata

Last Modified Date

2021-04-12

IP Status

No known IP claims.

Contributors

- Vikram Kushwaha, NVIDIA
- Piers Daniell, NVIDIA
- Jeff Bolz, NVIDIA

Description

This extension adds some more dynamic state to support applications that need to reduce the number of pipeline state objects they compile and bind.

New Commands

- vkCmdSetDepthBiasEnableEXT
- vkCmdSetLogicOpEXT
- vkCmdSetPatchControlPointsEXT
- vkCmdSetPrimitiveRestartEnableEXT
- vkCmdSetRasterizerDiscardEnableEXT

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceExtendedDynamicState2FeaturesEXT

New Enum Constants

- `VK_EXT_EXTENDED_DYNAMIC_STATE_2_EXTENSION_NAME`
- `VK_EXT_EXTENDED_DYNAMIC_STATE_2_SPEC_VERSION`
- Extending VkDynamicState:
  - VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE_EXT
  - VK_DYNAMIC_STATE_LOGIC_OP_EXT
  - VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT
  - VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE_EXT
  - VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_2_FEATURES_EXT

Promotion to Vulkan 1.3

The dynamic state enumerants VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE_EXT, VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE_EXT, and VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE_EXT; and the corresponding commands in this extension are included in core Vulkan 1.3, with the EXT suffix omitted. The enumerants and commands for dynamic logic operation and patch control points are not promoted, nor is the feature structure. Extension interfaces that were promoted remain available as aliases of the core functionality.

Version History

- Revision 1, 2021-04-12 (Vikram Kushwaha)
  - Internal revisions

VK_EXT_host_image_copy

Name String

`VK_EXT_host_image_copy`

Extension Type

Device extension

Registered Extension Number

271

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1
and
VK_KHR_copy_commands2
and
VK_KHR_format_feature_flags2
or
Vulkan Version 1.3

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Extension Proposal

VK_EXT_host_image_copy

Other Extension Metadata

Last Modified Date

2023-04-26

Contributors

- Shahbaz Youssefi, Google
- Faith Ekstrand, Collabora
- Hans-Kristian Arntzen, Valve
- Piers Daniell, NVIDIA

- Jan-Harald Fredriksen, Arm
- James Fitzpatrick, Imagination
- Daniel Story, Nintendo

Description

This extension allows applications to copy data between host memory and images on the host processor, without staging the data through a GPU-accessible buffer. This removes the need to allocate and manage the buffer and its associated memory. On some architectures it may also eliminate an extra copy operation. This extension additionally allows applications to copy data between images on the host.

To support initializing a new image in preparation for a host copy, it is now possible to transition a new image to VK_IMAGE_LAYOUT_GENERAL or other host-copyable layouts via vkTransitionImageLayoutEXT. Additionally, it is possible to perform copies that preserve the swizzling layout of the image by using the VK_HOST_IMAGE_COPY_MEMCPY_BIT_EXT flag. In that case, the memory size needed for copies to or from a buffer can be retrieved by chaining VkSubresourceHostMemcpySizeEXT to `pLayout` in vkGetImageSubresourceLayout2EXT.

New Commands

- vkCopyImageToImageEXT
- vkCopyImageToMemoryEXT
- vkCopyMemoryToImageEXT
- vkGetImageSubresourceLayout2EXT
- vkTransitionImageLayoutEXT

New Structures

- VkCopyImageToImageInfoEXT
- VkCopyImageToMemoryInfoEXT
- VkCopyMemoryToImageInfoEXT
- VkHostImageLayoutTransitionInfoEXT
- VkImageSubresource2EXT
- VkImageToMemoryCopyEXT
- VkMemoryToImageCopyEXT
- VkSubresourceLayout2EXT
- Extending VkImageFormatProperties2:
  - VkHostImageCopyDevicePerformanceQueryEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceHostImageCopyFeaturesEXT
- Extending VkPhysicalDeviceProperties2:

    - VkPhysicalDeviceHostImageCopyPropertiesEXT
- Extending VkSubresourceLayout2:
    - VkSubresourceHostMemcpySizeEXT

New Enums

- VkHostImageCopyFlagBitsEXT

New Bitmasks

- VkHostImageCopyFlagsEXT

New Enum Constants

- VK_EXT_HOST_IMAGE_COPY_EXTENSION_NAME
- VK_EXT_HOST_IMAGE_COPY_SPEC_VERSION
- Extending VkFormatFeatureFlagBits2:
    - VK_FORMAT_FEATURE_2_HOST_IMAGE_TRANSFER_BIT_EXT
- Extending VkHostImageCopyFlagBits:
    - VK_HOST_IMAGE_COPY_MEMCPY_BIT_EXT
    - VK_HOST_IMAGE_COPY_MEMCPY_EXT
- Extending VkImageUsageFlagBits:
    - VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_COPY_IMAGE_TO_IMAGE_INFO_EXT
    - VK_STRUCTURE_TYPE_COPY_IMAGE_TO_MEMORY_INFO_EXT
    - VK_STRUCTURE_TYPE_COPY_MEMORY_TO_IMAGE_INFO_EXT
    - VK_STRUCTURE_TYPE_HOST_IMAGE_COPY_DEVICE_PERFORMANCE_QUERY_EXT
    - VK_STRUCTURE_TYPE_HOST_IMAGE_LAYOUT_TRANSITION_INFO_EXT
    - VK_STRUCTURE_TYPE_IMAGE_TO_MEMORY_COPY_EXT
    - VK_STRUCTURE_TYPE_MEMORY_TO_IMAGE_COPY_EXT
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_FEATURES_EXT
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_PROPERTIES_EXT
    - VK_STRUCTURE_TYPE_SUBRESOURCE_HOST_MEMCPY_SIZE_EXT

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4, with the EXT suffix omitted. However, the feature is made optional in Vulkan 1.4. The original type, enum, and command names are still available as aliases of the core functionality.

A Vulkan 1.4 implementation that has a VK_QUEUE_GRAPHICS_BIT queue must support either:

- the `hostImageCopy` feature; or
- an additional queue that supports VK_QUEUE_TRANSFER_BIT.

Additionally, all queues supporting VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT must also advertise VK_QUEUE_TRANSFER_BIT.

Issues

1) When uploading data to an image, the data is usually loaded from disk. Why not have the application load the data directly into a `VkDeviceMemory` bound to a buffer (instead of host memory), and use vkCmdCopyBufferToImage? The same could be done when downloading data from an image.

RESOLVED: This may not always be possible. Complicated Vulkan applications such as game engines often have decoupled subsystems for streaming data and rendering. It may be unreasonable to require the streaming subsystem to coordinate with the rendering subsystem to allocate memory on its behalf, especially as Vulkan may not be the only API supported by the engine. In emulation layers, the image data is necessarily provided by the application in host memory, so an optimization as suggested is not possible. Most importantly, the device memory may not be mappable by an application, but still accessible to the driver.

2) Are `optimalBufferCopyOffsetAlignment` and `optimalBufferCopyRowPitchAlignment` applicable to host memory as well with the functions introduced by this extension? Or should there be new limits?

RESOLVED: No alignment requirements for the host memory pointer.

3) Should there be granularity requirements for image offsets and extents?

RESOLVED: No granularity requirements, i.e. a granularity of 1 pixel (for non-compressed formats) and 1 texel block (for compressed formats) is assumed.

4) How should the application deal with layout transitions before or after copying to or from images?

RESOLVED: An existing issue with linear images is that when emulating other APIs, it is impossible to know when to transition them as they are written to by the host and then used bindlessly. The copy operations in this extension are affected by the same limitation. A new command is thus introduced by this extension to address this problem by allowing the host to perform an image layout transition between a handful of layouts.

Version History

- Revision 0, 2021-01-20 (Faith Ekstrand)
  - Initial idea and xml
- Revision 1, 2023-04-26 (Shahbaz Youssefi)
  - Initial revision

VK_EXT_load_store_op_none

Name String

`VK_EXT_load_store_op_none`

Extension Type

Device extension

Registered Extension Number

401

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to VK_KHR_load_store_op_none extension
  - Which in turn was *promoted* to Vulkan 1.4

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Other Extension Metadata

Last Modified Date

2021-06-06

Contributors

- Shahbaz Youssefi, Google
- Bill Licea-Kane, Qualcomm Technologies, Inc.
- Tobias Hector, AMD

Description

This extension incorporates VK_ATTACHMENT_STORE_OP_NONE_EXT from `VK_QCOM_render_pass_store_ops`, enabling applications to avoid unnecessary synchronization when an attachment is not written during a render pass.

Additionally, VK_ATTACHMENT_LOAD_OP_NONE_EXT is introduced to avoid unnecessary synchronization when an attachment is not used during a render pass at all. In combination with VK_ATTACHMENT_STORE_OP_NONE_EXT, this is useful as an alternative to preserve attachments in applications that cannot decide if an attachment will be used in a render pass until after the necessary pipelines have been created.

Promotion to VK_KHR_load_store_op_none

All functionality in this extension is included in VK_KHR_load_store_op_none, with the suffix changed to KHR. The original enum names are still available as aliases of the KHR functionality.

New Enum Constants

- VK_EXT_LOAD_STORE_OP_NONE_EXTENSION_NAME
- VK_EXT_LOAD_STORE_OP_NONE_SPEC_VERSION
- Extending VkAttachmentLoadOp:
  - VK_ATTACHMENT_LOAD_OP_NONE_EXT
- Extending VkAttachmentStoreOp:
  - VK_ATTACHMENT_STORE_OP_NONE_EXT

> **NOTE** While VK_ATTACHMENT_STORE_OP_NONE is part of Vulkan 1.3, this extension was not promoted to core either in whole or in part. This functionality was promoted from VK_KHR_dynamic_rendering.

Version History

- Revision 1, 2021-06-06 (Shahbaz Youssefi)
  - Initial revision, based on VK_QCOM_render_pass_store_ops.
  - Added VK_ATTACHMENT_LOAD_OP_NONE_EXT.

VK_EXT_opacity_micromap

Name String

VK_EXT_opacity_micromap

Extension Type

Device extension

Registered Extension Number

397

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_acceleration_structure
and
VK_KHR_synchronization2
or
Vulkan Version 1.3

API Interactions

- Interacts with VK_VERSION_1_4
- Interacts with VK_KHR_maintenance5

SPIR-V Dependencies

- SPV_EXT_opacity_micromap

Deprecation State

- *Promoted* to VK_KHR_opacity_micromap extension

Contact

- Christoph Kubisch [GitHub]pixeljetstream
- Eric Werness

Extension Proposal

VK_EXT_opacity_micromap

Other Extension Metadata

Last Modified Date

2022-08-24

Interactions and External Dependencies

- This extension provides API support for GLSL_EXT_opacity_micromap

Contributors

- Christoph Kubisch, NVIDIA
- Eric Werness, NVIDIA
- Josh Barczak, Intel
- Stu Smith, AMD

Description

When adding transparency to a ray traced scene, an application can choose between further tessellating the geometry or using an any-hit shader to allow the ray through specific parts of the geometry. These options have the downside of either significantly increasing memory consumption or adding runtime overhead to run shader code in the middle of traversal, respectively.

This extension adds the ability to add an *opacity micromap* to geometry when building an acceleration structure. The opacity micromap compactly encodes opacity information which can be read by the implementation to mark parts of triangles as opaque or transparent. The format is externally visible to allow the application to compress its internal geometry and surface representations into the compressed format ahead of time. The compressed format subdivides each triangle into a set of subtriangles, each of which can be assigned either two or four opacity values. These opacity values can control if a ray hitting that subtriangle is treated as an opaque hit, complete miss, or possible hit, depending on the controls described in Ray Opacity Micromap.

This extension provides:

- a VkMicromapEXT structure to store the micromap,
- functions similar to acceleration structure build functions to build the opacity micromap array, and
- a structure to extend VkAccelerationStructureGeometryTrianglesDataKHR to attach a micromap to the geometry of the acceleration structure.

New Object Types

- VkMicromapEXT

New Commands

- vkBuildMicromapsEXT
- vkCmdBuildMicromapsEXT
- vkCmdCopyMemoryToMicromapEXT
- vkCmdCopyMicromapEXT
- vkCmdCopyMicromapToMemoryEXT
- vkCmdWriteMicromapsPropertiesEXT
- vkCopyMemoryToMicromapEXT
- vkCopyMicromapEXT

- vkCopyMicromapToMemoryEXT
- vkCreateMicromapEXT
- vkDestroyMicromapEXT
- vkGetDeviceMicromapCompatibilityEXT
- vkGetMicromapBuildSizesEXT
- vkWriteMicromapsPropertiesEXT

New Structures

- VkCopyMemoryToMicromapInfoEXT
- VkCopyMicromapInfoEXT
- VkCopyMicromapToMemoryInfoEXT
- VkMicromapBuildInfoEXT
- VkMicromapBuildSizesInfoEXT
- VkMicromapCreateInfoEXT
- VkMicromapTriangleEXT
- VkMicromapUsageEXT
- VkMicromapVersionInfoEXT
- Extending VkAccelerationStructureGeometryTrianglesDataKHR:
  - VkAccelerationStructureTrianglesOpacityMicromapEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceOpacityMicromapFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceOpacityMicromapPropertiesEXT

New Enums

- VkBuildMicromapFlagBitsEXT
- VkBuildMicromapModeEXT
- VkCopyMicromapModeEXT
- VkMicromapCreateFlagBitsEXT
- VkMicromapTypeEXT
- VkOpacityMicromapFormatEXT
- VkOpacityMicromapSpecialIndexEXT

New Bitmasks

- VkBuildMicromapFlagsEXT
- VkMicromapCreateFlagsEXT

New Enum Constants

- VK_EXT_OPACITY_MICROMAP_EXTENSION_NAME
- VK_EXT_OPACITY_MICROMAP_SPEC_VERSION
- Extending VkAccessFlagBits2:
  - VK_ACCESS_2_MICROMAP_READ_BIT_EXT
  - VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT
- Extending VkBufferUsageFlagBits:
  - VK_BUFFER_USAGE_MICROMAP_BUILD_INPUT_READ_ONLY_BIT_EXT
  - VK_BUFFER_USAGE_MICROMAP_STORAGE_BIT_EXT
- Extending VkBuildAccelerationStructureFlagBitsKHR:
  - VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_BIT_EXT
  - VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_EXT
  - VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_BIT_EXT
  - VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_EXT
  - VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_BIT_EXT
  - VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_EXT
- Extending VkGeometryInstanceFlagBitsKHR:
  - VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_BIT_EXT
  - VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_EXT
  - VK_GEOMETRY_INSTANCE_FORCE_OPACITY_MICROMAP_2_STATE_BIT_EXT
  - VK_GEOMETRY_INSTANCE_FORCE_OPACITY_MICROMAP_2_STATE_EXT
- Extending VkObjectType:
  - VK_OBJECT_TYPE_MICROMAP_EXT
- Extending VkOpacityMicromapFormatKHR:
  - VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT
  - VK_OPACITY_MICROMAP_FORMAT_4_STATE_EXT
- Extending VkOpacityMicromapSpecialIndexKHR:
  - VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_OPAQUE_EXT
  - VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_TRANSPARENT_EXT
  - VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_OPAQUE_EXT
  - VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_TRANSPARENT_EXT
- Extending VkPipelineCreateFlagBits:
  - VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_EXT
- Extending VkPipelineStageFlagBits2:

    - VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
- Extending VkQueryType:
    - VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT
    - VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT
- Extending VkStructureType:
    - VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_TRIANGLES_OPACITY_MICROMAP_EXT
    - VK_STRUCTURE_TYPE_COPY_MEMORY_TO_MICROMAP_INFO_EXT
    - VK_STRUCTURE_TYPE_COPY_MICROMAP_INFO_EXT
    - VK_STRUCTURE_TYPE_COPY_MICROMAP_TO_MEMORY_INFO_EXT
    - VK_STRUCTURE_TYPE_MICROMAP_BUILD_INFO_EXT
    - VK_STRUCTURE_TYPE_MICROMAP_BUILD_SIZES_INFO_EXT
    - VK_STRUCTURE_TYPE_MICROMAP_CREATE_INFO_EXT
    - VK_STRUCTURE_TYPE_MICROMAP_VERSION_INFO_EXT
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_FEATURES_EXT
    - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_PROPERTIES_EXT

If VK_KHR_maintenance5 or Vulkan Version 1.4 is supported:

- Extending VkBufferUsageFlagBits2:
    - VK_BUFFER_USAGE_2_MICROMAP_BUILD_INPUT_READ_ONLY_BIT_EXT
    - VK_BUFFER_USAGE_2_MICROMAP_STORAGE_BIT_EXT
- Extending VkPipelineCreateFlagBits2:
    - VK_PIPELINE_CREATE_2_RAY_TRACING_OPACITY_MICROMAP_BIT_EXT

Promotion to VK_KHR_opacity_micromap

Functionality in this extension is included in VK_KHR_opacity_micromap. The type, enum and command names are still aliases of the KHR functionality with some exceptions.

While most application-visible functionality is available through both extensions, two notable differences to be aware of are:

- The `hostMicromapCommands` feature and its associated host-side build, copy, and query commands are only exposed by VK_EXT_opacity_micromap; those commands are legacy and are not included in VK_KHR_opacity_micromap.
- The two extensions differ in how opacity micromap data is consumed by ray query traversal. With VK_EXT_opacity_micromap enabled, opacity micromap data present in the acceleration structure is always consumed. With only VK_KHR_opacity_micromap enabled and its `micromap` feature enabled, opacity micromap data is consumed only when the shader declares the `OpacityMicromapIdKHR` execution mode with its specialization-constant operand equal to VK_TRUE. If both extensions are enabled, the VK_EXT_opacity_micromap behavior applies and the execution mode is not required. Traversal initiated by pipeline trace ray instructions is

unaffected by this difference; it requires VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_KHR under either extension.

See the VK_KHR_opacity_micromap proposal for details on the differences.

Reference Code

```
uint32_t BarycentricsToSpaceFillingCurveIndex(float u, float v, uint32_t level)
{
    u = clamp(u, 0.0f, 1.0f);
    v = clamp(v, 0.0f, 1.0f);

    uint32_t iu, iv, iw;

    // Quantize barycentric coordinates
    float fu = u * (1u << level);
    float fv = v * (1u << level);

    iu = (uint32_t)fu;
    iv = (uint32_t)fv;

    float uf = fu - float(iu);
    float vf = fv - float(iv);

    if (iu >= (1u << level)) iu = (1u << level) - 1u;
    if (iv >= (1u << level)) iv = (1u << level) - 1u;

    uint32_t iuv = iu + iv;

    if (iuv >= (1u << level))
        iu -= iuv - (1u << level) + 1u;

    iw = ~(iu + iv);

    if (uf + vf >= 1.0f && iuv < (1u << level) - 1u) --iw;

    uint32_t b0 = ~(iu ^ iw);
    b0 &= ((1u << level) - 1u);
    uint32_t t = (iu ^ iv) & b0;

    uint32_t f = t;
    f ^= f >> 1u;
    f ^= f >> 2u;
    f ^= f >> 4u;
    f ^= f >> 8u;
    uint32_t b1 = ((f ^ iu) & ~b0) | t;

    // Interleave bits
    b0 = (b0 | (b0 << 8u)) & 0x00ff00ffu;
    b0 = (b0 | (b0 << 4u)) & 0x0f0f0f0fu;
```

```
    b0 = (b0 | (b0 << 2u)) & 0x33333333u;
    b0 = (b0 | (b0 << 1u)) & 0x55555555u;
    b1 = (b1 | (b1 << 8u)) & 0x00ff00ffu;
    b1 = (b1 | (b1 << 4u)) & 0x0f0f0f0fu;
    b1 = (b1 | (b1 << 2u)) & 0x33333333u;
    b1 = (b1 | (b1 << 1u)) & 0x55555555u;

    return b0 | (b1 << 1u);
}
```

Issues

(1) Is the build actually similar to an acceleration structure build?

- Resolved: The build should be much lighter-weight than an acceleration structure build, but the infrastructure is similar enough that it makes sense to keep the concepts compatible.

(2) Why does VkMicromapUsageEXT not have type/pNext?

- Resolved: There can be a very large number of these structures, so doubling the size of these can be significant memory consumption. Also, an application may be loading these directly from a file which is more compatible with it being a flat structure. The including structures are extensible and are probably a more suitable place to add extensibility.

(3) Why is there a SPIR-V extension?

- Resolved: There is a ray flag. To be consistent with how the existing ray tracing extensions work that ray flag needs its own extension.

(4) Should there be indirect micromap build?

- Resolved: Not for now. There is more in-depth usage metadata required and it seems less likely that something like a GPU culling system would need to change the counts for a micromap.

(5) Should micromaps have a micromap device address?

- Resolved: There is no need right now (can just use the handle) but that is a bit different from acceleration structures, though the two are not completely parallel in their usage.

(6) Why are the alignment requirements defined as a mix of hardcoded values and caps?

- Resolved: This is most parallel with the definition of `VK_KHR_acceleration_structure` and maintaining commonality makes it easier for applications to share memory.

Version History

- Revision 2, 2022-06-22 (Eric Werness)
  - EXTify and clean up for discussion
- Revision 1, 2022-01-01 (Eric Werness)
  - Initial revision

VK_EXT_pipeline_library_group_handles

Name String

`VK_EXT_pipeline_library_group_handles`

Extension Type

Device extension

Registered Extension Number

499

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_ray_tracing_pipeline
and
VK_KHR_pipeline_library

Deprecation State

- *Promoted* to VK_KHR_pipeline_library_group_handles extension

Contact

- Hans-Kristian Arntzen [GitHub]HansKristian-Work

Extension Proposal

VK_EXT_pipeline_library_group_handles

Other Extension Metadata

Last Modified Date

2023-01-25

IP Status

No known IP claims.

Contributors

- Hans-Kristian Arntzen, Valve
- Stuart Smith, AMD
- Ricardo Garcia, Igalia
- Lionel Landwerlin, Intel
- Eric Werness, NVIDIA
- Daniel Koch, NVIDIA

Description

When using pipeline libraries in ray tracing pipelines, a library might get linked into different pipelines in an incremental way. An application can have a strategy where a ray tracing pipeline is comprised of N pipeline libraries and is later augmented by creating a new pipeline with N + 1 libraries. Without this extension, all group handles must be re-queried as the group handle is tied to the pipeline, not the library. This is problematic for applications which aim to decouple construction of record buffers and the linkage of ray tracing pipelines.

To aid in this, this extension enables support for querying group handles directly from pipeline libraries. Group handles obtained from a library **must** remain bitwise identical in any `VkPipeline` that links to the library.

With this feature, the extension also improves compatibility with DXR 1.1 AddToStateObject(), which guarantees that group handles returned remain bitwise identical between parent and child pipelines. In addition, querying group handles from COLLECTION objects is also supported with that API.

Promotion to `VK_KHR_pipeline_library_group_handles`

All functionality in this extension is included in `VK_KHR_pipeline_library_group_handles`, with the suffix change to KHR. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePipelineLibraryGroupHandlesFeaturesEXT

New Enum Constants

- `VK_EXT_PIPELINE_LIBRARY_GROUP_HANDLES_EXTENSION_NAME`
- `VK_EXT_PIPELINE_LIBRARY_GROUP_HANDLES_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_LIBRARY_GROUP_HANDLES_FEATURES_EXT

Version History

- Revision 1, 2023-01-25 (Hans-Kristian Arntzen)
  - Initial draft

VK_EXT_pipeline_protected_access

Name String

`VK_EXT_pipeline_protected_access`

Extension Type

Device extension

Registered Extension Number

467

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Extension Proposal

VK_EXT_pipeline_protected_access

Other Extension Metadata

Last Modified Date

2022-07-28

Contributors

- Shahbaz Youssefi, Google
- Jörg Wagner, Arm
- Ralph Potter, Samsung
- Daniel Koch, NVIDIA

Description

This extension allows protected memory access to be specified per pipeline as opposed to per device. Through the usage of this extension, any performance penalty paid due to access to protected memory will be limited to the specific pipelines that make such accesses.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePipelineProtectedAccessFeaturesEXT

New Enum Constants

- `VK_EXT_PIPELINE_PROTECTED_ACCESS_EXTENSION_NAME`
- `VK_EXT_PIPELINE_PROTECTED_ACCESS_SPEC_VERSION`
- Extending VkPipelineCreateFlagBits:
  - VK_PIPELINE_CREATE_NO_PROTECTED_ACCESS_BIT_EXT
  - VK_PIPELINE_CREATE_PROTECTED_ACCESS_ONLY_BIT_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_PROTECTED_ACCESS_FEATURES_EXT

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4 with the EXT suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

Version History

- Revision 1, 2022-07-28 (Shahbaz Youssefi)
  - Internal revisions

VK_EXT_pipeline_robustness

Name String

VK_EXT_pipeline_robustness

Extension Type

Device extension

Registered Extension Number

69

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to Vulkan 1.4

Contact

- Jarred Davies

Other Extension Metadata

Last Modified Date

2022-07-12

Interactions and External Dependencies

- Interacts with VK_EXT_robustness2
- Interacts with VK_EXT_image_robustness
- Interacts with VK_KHR_ray_tracing_pipeline

Contributors

- Jarred Davies, Imagination Technologies
- Alex Walters, Imagination Technologies
- Piers Daniell, NVIDIA
- Graeme Leese, Broadcom Corporation
- Jeff Leger, Qualcomm Technologies, Inc.
- Faith Ekstrand, Intel
- Lionel Landwerlin, Intel

- Shahbaz Youssefi, Google, Inc.

Description

This extension allows users to request robustness on a per-pipeline stage basis.

As `robustBufferAccess` and other robustness features may have an adverse effect on performance, this extension is designed to allow users to request robust behavior only where it may be needed.

New Structures

- Extending VkGraphicsPipelineCreateInfo, VkComputePipelineCreateInfo, VkPipelineShaderStageCreateInfo, VkRayTracingPipelineCreateInfoKHR:
  - VkPipelineRobustnessCreateInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePipelineRobustnessFeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDevicePipelineRobustnessPropertiesEXT

New Enums

- VkPipelineRobustnessBufferBehaviorEXT
- VkPipelineRobustnessImageBehaviorEXT

New Enum Constants

- `VK_EXT_PIPELINE_ROBUSTNESS_EXTENSION_NAME`
- `VK_EXT_PIPELINE_ROBUSTNESS_SPEC_VERSION`
- Extending VkPipelineRobustnessBufferBehavior:
  - VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_DEVICE_DEFAULT_EXT
  - VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_DISABLED_EXT
  - VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_2_EXT
  - VK_PIPELINE_ROBUSTNESS_BUFFER_BEHAVIOR_ROBUST_BUFFER_ACCESS_EXT
- Extending VkPipelineRobustnessImageBehavior:
  - VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_DEVICE_DEFAULT_EXT
  - VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_DISABLED_EXT
  - VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS_2_EXT
  - VK_PIPELINE_ROBUSTNESS_IMAGE_BEHAVIOR_ROBUST_IMAGE_ACCESS_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_ROBUSTNESS_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_ROBUSTNESS_PROPERTIES_EXT

    - VK_STRUCTURE_TYPE_PIPELINE_ROBUSTNESS_CREATE_INFO_EXT

Promotion to Vulkan 1.4

Functionality in this extension is included in core Vulkan 1.4 with the EXT suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

Version History

- Revision 1, 2022-07-12 (Jarred Davies)
    - Initial version

VK_EXT_present_mode_fifo_latest_ready

Name String

VK_EXT_present_mode_fifo_latest_ready

Extension Type

Device extension

Registered Extension Number

362

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_swapchain

Deprecation State

- *Promoted* to VK_KHR_present_mode_fifo_latest_ready extension

Contact

- Lionel Duc [GitHub]nvlduc

Extension Proposal

VK_EXT_present_mode_fifo_latest_ready

Other Extension Metadata

Last Modified Date

2024-05-28

IP Status

No known IP claims.

Contributors

- James Jones, NVIDIA
- Lionel Duc, NVIDIA

Description

This device extension adds a new present mode, VK_PRESENT_MODE_FIFO_LATEST_READY_EXT.

This tear-free present mode behaves much like VK_PRESENT_MODE_FIFO_KHR, except that each vertical blanking period dequeues consecutive present requests until the latest ready is found to update the current image.

While this seems similar in concept to VK_PRESENT_MODE_MAILBOX_KHR, the fundamental difference is that the processing of the present requests is done during vblank. From the application perspective, this means for example, that in a flip-based model, a single vblank **may** cause multiple swapchain images to be released at once, while VK_PRESENT_MODE_MAILBOX_KHR **may** continuously be releasing images as new requests become ready.

This additional present mode is useful when using a time-based present API.

Promotion to `VK_KHR_present_mode_fifo_latest_ready`

All functionality in this extension is included in `VK_KHR_present_mode_fifo_latest_ready`, with the suffix changed to KHR. The original type and enum names are still available as aliases of the KHR names.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDevicePresentModeFifoLatestReadyFeaturesEXT

New Enum Constants

- `VK_EXT_PRESENT_MODE_FIFO_LATEST_READY_EXTENSION_NAME`
- `VK_EXT_PRESENT_MODE_FIFO_LATEST_READY_SPEC_VERSION`
- Extending VkPresentModeKHR:
  - VK_PRESENT_MODE_FIFO_LATEST_READY_EXT
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_MODE_FIFO_LATEST_READY_FEATURES_EXT

Version History

- Revision 1, 2024-05-28 (Lionel Duc)
  - Internal revisions

VK_EXT_robustness2

Name String

`VK_EXT_robustness2`

Extension Type

Device extension

Registered Extension Number

287

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to VK_KHR_robustness2 extension

Contact

- Liam Middlebrook [GitHub]liam-middlebrook

Other Extension Metadata

Last Modified Date

2020-01-29

IP Status

No known IP claims.

Contributors

- Liam Middlebrook, NVIDIA
- Jeff Bolz, NVIDIA

Description

This extension adds stricter requirements for how out of bounds reads and writes are handled. Most accesses **must** be tightly bounds-checked, out of bounds writes **must** be discarded, out of bound reads **must** return zero. Rather than allowing multiple possible (0,0,0,x) vectors, the out of bounds values are treated as zero, and then missing components are inserted based on the format as described in Component Substitution and vertex input attribute extraction.

These additional requirements **may** be expensive on some implementations, and should only be

enabled when truly necessary.

This extension also adds support for “null descriptors”, where VK_NULL_HANDLE **can** be used instead of a valid handle. Accesses to null descriptors have well-defined behavior, and do not rely on robustness.

Promotion to VK_KHR_robustness2

All functionality in this extension is included in VK_KHR_robustness2, with the suffix changed to KHR. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceRobustness2FeaturesEXT
- Extending VkPhysicalDeviceProperties2:
  - VkPhysicalDeviceRobustness2PropertiesEXT

New Enum Constants

- VK_EXT_ROBUSTNESS_2_EXTENSION_NAME
- VK_EXT_ROBUSTNESS_2_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_FEATURES_EXT
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_PROPERTIES_EXT

Issues

1. Why do VkPhysicalDeviceRobustness2PropertiesEXT::robustUniformBufferAccessSizeAlignment and VkPhysicalDeviceRobustness2PropertiesEXT::robustStorageBufferAccessSizeAlignment exist?

RESOLVED: Some implementations cannot efficiently tightly bounds-check all buffer accesses. Rather, the size of the bound range is padded to some power of two multiple, up to 256 bytes for uniform buffers and up to 4 bytes for storage buffers, and that padded size is bounds-checked. This is sufficient to implement D3D-like behavior, because D3D only allows binding whole uniform buffers or ranges that are a multiple of 256 bytes, and D3D raw and structured buffers only support 32-bit accesses.

Examples

None.

Version History

- Revision 1, 2019-11-01 (Jeff Bolz, Liam Middlebrook)
  - Initial draft

VK_EXT_surface_maintenance1

Name String

`VK_EXT_surface_maintenance1`

Extension Type

Instance extension

Registered Extension Number

275

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_surface
and
VK_KHR_get_surface_capabilities2

Deprecation State

- *Promoted* to VK_KHR_surface_maintenance1 extension

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Extension Proposal

VK_EXT_surface_maintenance1

Other Extension Metadata

Last Modified Date

2022-12-16

Interactions and External Dependencies

- Promoted to `VK_KHR_surface_maintenance1`

Contributors

- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Lionel Landwerlin, Intel
- Shahbaz Youssefi, Google
- Chris Forbes, Google
- Ian Elliott, Google
- Hans-Kristian Arntzen, Valve

- Daniel Stone, Collabora

Description

VK_EXT_surface_maintenance1 adds a collection of window system integration features that were intentionally left out or overlooked in the original VK_KHR_surface extension.

The new features are as follows:

- Allow querying number of min/max images from a surface for a particular presentation mode.
- Allow querying a surface's scaled presentation capabilities.
- Allow querying a surface for the set of presentation modes which can be easily switched between without requiring swapchain recreation.

Promotion to VK_KHR_surface_maintenance1

All functionality in this extension is included in VK_KHR_surface_maintenance1, with the suffix changed to KHR. The original type, enum, and command names are still available as aliases of the KHR functionality.

New Structures

- Extending VkPhysicalDeviceSurfaceInfo2KHR:
  - VkSurfacePresentModeEXT
- Extending VkSurfaceCapabilities2KHR:
  - VkSurfacePresentModeCompatibilityEXT
  - VkSurfacePresentScalingCapabilitiesEXT

New Enums

- VkPresentGravityFlagBitsEXT
- VkPresentScalingFlagBitsEXT

New Bitmasks

- VkPresentGravityFlagsEXT
- VkPresentScalingFlagsEXT

New Enum Constants

- VK_EXT_SURFACE_MAINTENANCE_1_EXTENSION_NAME
- VK_EXT_SURFACE_MAINTENANCE_1_SPEC_VERSION
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_SURFACE_PRESENT_MODE_COMPATIBILITY_EXT
  - VK_STRUCTURE_TYPE_SURFACE_PRESENT_MODE_EXT

- VK_STRUCTURE_TYPE_SURFACE_PRESENT_SCALING_CAPABILITIES_EXT

Version History

- Revision 0, 2019-02-27 (Lionel Landwerlin)
  - Internal revisions
- Revision 0, 2020-06-15 (James Jones)
  - Internal revisions
- Revision 1, 2022-11-09 (Shahbaz Youssefi)
  - Add functionality and complete spec

VK_EXT_swapchain_maintenance1

Name String

`VK_EXT_swapchain_maintenance1`

Extension Type

Device extension

Registered Extension Number

276

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_swapchain
and
VK_EXT_surface_maintenance1
and
VK_KHR_get_physical_device_properties2
or
Vulkan Version 1.1

Deprecation State

- *Promoted* to VK_KHR_swapchain_maintenance1 extension

Contact

- Shahbaz Youssefi [GitHub]syoussefi

Extension Proposal

VK_EXT_swapchain_maintenance1

Other Extension Metadata

Last Modified Date

2022-12-16

Interactions and External Dependencies

- Promoted to `VK_KHR_swapchain_maintenance1`

Contributors

- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Shahbaz Youssefi, Google
- Chris Forbes, Google

- Ian Elliott, Google
- Yiwei Zhang, Google
- Charlie Lao, Google
- Lina Versace, Google
- Ralph Potter, Samsung
- Igor Nazarov, Samsung
- Hyunchang Kim, Samsung
- Suenghwan Lee, Samsung
- Munseong Kang, Samsung
- Joonyong Park, Samsung
- Hans-Kristian Arntzen, Valve
- Lisa Wu, Arm
- Daniel Stone, Collabora
- Pan Gao, Huawei

Description

VK_EXT_swapchain_maintenance1 adds a collection of window system integration features that were intentionally left out or overlooked in the original VK_KHR_swapchain extension.

The new features are as follows:

- Specify a fence that will be signaled when the resources associated with a present operation **can** be safely destroyed.
- Allow changing the present mode a swapchain is using at per-present granularity.
- Allow applications to define the behavior when presenting a swapchain image to a surface with different dimensions than the image. Using this feature **may** allow implementations to avoid returning VK_ERROR_OUT_OF_DATE_KHR in this situation.
- Allow applications to defer swapchain memory allocation for improved startup time and memory footprint.
- Allow applications to release previously acquired images without presenting them.

Promotion to VK_KHR_swapchain_maintenance1

All functionality in this extension is included in VK_KHR_swapchain_maintenance1, with the suffix changed to KHR. The original type, enum, and command names are still available as aliases of the KHR functionality.

New Commands

- vkReleaseSwapchainImagesEXT

New Structures

- VkReleaseSwapchainImagesInfoEXT
- Extending VkPhysicalDeviceFeatures2, VkDeviceCreateInfo:
  - VkPhysicalDeviceSwapchainMaintenance1FeaturesEXT
- Extending VkPresentInfoKHR:
  - VkSwapchainPresentFenceInfoEXT
  - VkSwapchainPresentModeInfoEXT
- Extending VkSwapchainCreateInfoKHR:
  - VkSwapchainPresentModesCreateInfoEXT
  - VkSwapchainPresentScalingCreateInfoEXT

New Enum Constants

- `VK_EXT_SWAPCHAIN_MAINTENANCE_1_EXTENSION_NAME`
- `VK_EXT_SWAPCHAIN_MAINTENANCE_1_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SWAPCHAIN_MAINTENANCE_1_FEATURES_EXT
  - VK_STRUCTURE_TYPE_RELEASE_SWAPCHAIN_IMAGES_INFO_EXT
  - VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_FENCE_INFO_EXT
  - VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_MODES_CREATE_INFO_EXT
  - VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_MODE_INFO_EXT
  - VK_STRUCTURE_TYPE_SWAPCHAIN_PRESENT_SCALING_CREATE_INFO_EXT
- Extending VkSwapchainCreateFlagBitsKHR:
  - VK_SWAPCHAIN_CREATE_DEFERRED_MEMORY_ALLOCATION_BIT_EXT

Version History

- Revision 0, 2019-05-28 (James Jones)
  - Initial revisions
- Revision 1, 2022-08-21 (Shahbaz Youssefi)
  - Add functionality and complete spec

VK_EXT_validation_features

Name String

`VK_EXT_validation_features`

Extension Type

Instance extension

Registered Extension Number

248

Revision

6

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Deprecated* by VK_EXT_layer_settings extension

Special Use

- Debugging tools

Contact

- Karl Schultz [GitHub]karl-lunarg

Other Extension Metadata

Last Modified Date

2018-11-14

IP Status

No known IP claims.

Contributors

- Karl Schultz, LunarG
- Dave Houlton, LunarG
- Mark Lobodzinski, LunarG
- Camden Stocker, LunarG
- Tony Barbour, LunarG
- John Zulauf, LunarG

Description

This extension provides the VkValidationFeaturesEXT structure that can be included in the `pNext` chain of the VkInstanceCreateInfo structure passed as the `pCreateInfo` parameter of vkCreateInstance. The structure contains an array of VkValidationFeatureEnableEXT enum values that enable specific validation features that are disabled by default. The structure also contains an array of VkValidationFeatureDisableEXT enum values that disable specific validation layer features that are enabled by default.

Deprecation by `VK_EXT_layer_settings`

Functionality in this extension is subsumed into the `VK_EXT_layer_settings` extension.

New Structures

- Extending VkInstanceCreateInfo, VkShaderModuleCreateInfo, VkShaderCreateInfoEXT, VkPipelineShaderStageCreateInfo, VkGraphicsPipelineCreateInfo, VkComputePipelineCreateInfo, VkRayTracingPipelineCreateInfoKHR:
  - VkValidationFeaturesEXT

New Enums

- VkValidationFeatureDisableEXT
- VkValidationFeatureEnableEXT

New Enum Constants

- `VK_EXT_VALIDATION_FEATURES_EXTENSION_NAME`
- `VK_EXT_VALIDATION_FEATURES_SPEC_VERSION`
- Extending VkStructureType:
  - VK_STRUCTURE_TYPE_VALIDATION_FEATURES_EXT

Version History

- Revision 1, 2018-11-14 (Karl Schultz)
  - Initial revision
- Revision 2, 2019-08-06 (Mark Lobodzinski)
  - Add Best Practices enable
- Revision 3, 2020-03-04 (Tony Barbour)
  - Add Debug Printf enable
- Revision 4, 2020-07-29 (John Zulauf)
  - Add Synchronization Validation enable
- Revision 5, 2021-05-18 (Tony Barbour)
  - Add Shader Validation Cache disable

- Revision 6, 2023-09-25 (Christophe Riccio)
    - Marked as deprecated by VK_EXT_layer_settings

Appendix F: Vulkan Roadmap Milestones

Roadmap milestones are intended to be supported by mid-to-high-end smartphones, tablets, laptops, consoles, and desktop devices.

Each milestone indicates support for a set of extensions, features, limits, and formats across these devices, and should be supported by all such new hardware shipping by the end of the target year or shortly thereafter.

Roadmap 2022

The Roadmap 2022 milestone is intended to be supported by newer mid-to-high-end devices shipping in 2022 or shortly thereafter across mainstream smartphone, tablet, laptop, console, and desktop devices.

Required API Versions

This milestone requires Vulkan 1.3.

Required Features

The following core optional features are required to be supported:

- Vulkan 1.0 Optional Features
  - `fullDrawIndexUint32`
  - `imageCubeArray`
  - `independentBlend`
  - `sampleRateShading`
  - `drawIndirectFirstInstance`
  - `depthClamp`
  - `depthBiasClamp`
  - `samplerAnisotropy`
  - `occlusionQueryPrecise`
  - `fragmentStoresAndAtomics`
  - `shaderStorageImageExtendedFormats`
  - `shaderUniformBufferArrayDynamicIndexing`
  - `shaderSampledImageArrayDynamicIndexing`
  - `shaderStorageBufferArrayDynamicIndexing`
  - `shaderStorageImageArrayDynamicIndexing`
- Vulkan 1.1 Optional Features
  - `samplerYcbcrConversion`

- Vulkan 1.2 Optional Features
  - samplerMirrorClampToEdge
  - descriptorIndexing
  - shaderUniformTexelBufferArrayDynamicIndexing
  - shaderStorageTexelBufferArrayDynamicIndexing
  - shaderUniformBufferArrayNonUniformIndexing
  - shaderSampledImageArrayNonUniformIndexing
  - shaderStorageBufferArrayNonUniformIndexing
  - shaderStorageImageArrayNonUniformIndexing
  - shaderUniformTexelBufferArrayNonUniformIndexing
  - shaderStorageTexelBufferArrayNonUniformIndexing
  - descriptorBindingSampledImageUpdateAfterBind
  - descriptorBindingStorageImageUpdateAfterBind
  - descriptorBindingStorageBufferUpdateAfterBind
  - descriptorBindingUniformTexelBufferUpdateAfterBind
  - descriptorBindingStorageTexelBufferUpdateAfterBind
  - descriptorBindingUpdateUnusedWhilePending
  - descriptorBindingPartiallyBound
  - descriptorBindingVariableDescriptorCount
  - runtimeDescriptorArray
  - scalarBlockLayout

Required Limits

The following core increased limits are **required**

Table 109. Vulkan 1.0 Limits

Limit Name	Unsupported Limit	Core Limit	Milestone Limit	Limit Type[1]
maxImageDimension1D	-	4096	8192	min
maxImageDimension2D	-	4096	8192	min
maxImageDimensionCube	-	4096	8192	min
maxImageArrayLayers	-	256	2048	min
maxUniformBufferRange	-	16384	65536	min
bufferImageGranularity	-	131072	4096	max
maxPerStageDescriptorSamplers	-	16	64	min

Limit Name	Unsupported Limit	Core Limit	Milestone Limit	Limit Type[1]
maxPerStageDescriptorUniformBuffers	-	12	15	min
maxPerStageDescriptorStorageBuffers	-	4	30	min
maxPerStageDescriptorSampledImages	-	16	200	min
maxPerStageDescriptorStorageImages	-	4	16	min
maxPerStageResources	-	128	200	min
maxDescriptorSetSamplers	-	96	576	min, *n* × PerStage
maxDescriptorSetUniformBuffers	-	72	90	min, *n* × PerStage
maxDescriptorSetStorageBuffers	-	24	96	min, *n* × PerStage
maxDescriptorSetSampledImages	-	96	1800	min, *n* × PerStage
maxDescriptorSetStorageImages	-	24	144	min, *n* × PerStage
maxFragmentCombinedOutputResources	-	4	16	min
maxComputeWorkGroupInvocations	-	128	256	min
maxComputeWorkGroupSize	-	(128,128,64)	(256,256,64)	min
subTexelPrecisionBits	-	4	8	min
mipmapPrecisionBits	-	4	6	min
maxSamplerLodBias	-	2	14	min
pointSizeGranularity	0.0	1.0	0.125	max, fixed point increment
lineWidthGranularity	0.0	1.0	0.5	max, fixed point increment
standardSampleLocations	-	-	VK_TRUE	implementation-dependent
maxColorAttachments	-	4	7	min

Table 110. Vulkan 1.1 Limits

Limit Name	Unsupported Limit	Core Limit	Milestone Limit	Limit Type[1]
subgroupSize	-	1/4	4	implementation-dependent
subgroupSupportedStages	-	VK_SHADER_STAGE_COMPUTE_BIT	VK_SHADER_STAGE_COMPUTE_BIT VK_SHADER_STAGE_FRAGMENT_BIT	implementation-dependent
subgroupSupportedOperations	-	VK_SUBGROUP_FEATURE_BASIC_BIT	VK_SUBGROUP_FEATURE_BASIC_BIT VK_SUBGROUP_FEATURE_VOTE_BIT VK_SUBGROUP_FEATURE_ARITHMETIC_BIT VK_SUBGROUP_FEATURE_BALLOT_BIT VK_SUBGROUP_FEATURE_SHUFFLE_BIT VK_SUBGROUP_FEATURE_SHUFFLE_RELATIVE_BIT VK_SUBGROUP_FEATURE_QUAD_BIT	implementation-dependent

Table 111. Vulkan 1.2 Limits

Limit Name	Unsupported Limit	Core Limit	Milestone Limit	Limit Type[1]
shaderSignedZeroInfNanPreserveFloat16	-	-	VK_TRUE	implementation-dependent
shaderSignedZeroInfNanPreserveFloat32	-	-	VK_TRUE	implementation-dependent

Limit Name	Unsupported Limit	Core Limit	Milestone Limit	Limit Type[1]
maxPerStageDescriptorUpdateAfterBindInputAttachments	0	4	7	min

Table 112. Vulkan 1.3 Limits

Limit Name	Unsupported Limit	Core Limit	Milestone Limit	Limit Type[1]
maxSubgroupSize	-	-	4	min

Required Extensions

The following extensions are **required**

VK_KHR_global_priority

Roadmap 2024

The Roadmap 2024 milestone is intended to be supported by newer mid-to-high-end devices shipping in 2024 or shortly thereafter across mainstream smartphone, tablet, laptop, console, and desktop devices.

Two of the core aims of this roadmap profile are to enable developers to rely on a number of important rasterization and shader features have been available for a long time, but until now have not enjoyed wide support.

Shader features required include smaller types (8/16-bit integers and 16-bit floats), reconvergence guarantees for subgroup ops (VK_KHR_shader_maximal_reconvergence and VK_KHR_shader_quad_control), and more consistent floating-point handling (VK_KHR_shader_float_controls2 and round-to-nearest-even for 32-/16-bit floats). Rasterization features include requiring support for multi-draw indirect, shader draw parameters, 8-bit indices, better line rasterization definitions, and local reads when using dynamic rendering. A few other features have been added opportunistically, in lieu of shipping a Vulkan 1.4 in the same time frame, such as push descriptors and the various minor improvements included in VK_KHR_maintenance5.

Required Milestones

This milestone requires the Roadmap 2022 profile.

Required Features

The following core optional features are required to be supported:

- Vulkan 1.0 Optional Features
  - multiDrawIndirect

    - shaderImageGatherExtended
    - shaderInt16
- Vulkan 1.1 Optional Features
    - shaderDrawParameters
    - storageBuffer16BitAccess
- Vulkan 1.2 Optional Features
    - shaderInt8
    - shaderFloat16
    - storageBuffer8BitAccess

Required Limits

The following core increased limits are **required**

Table 113. Vulkan 1.0 Limits

Limit Name	Unsupported Limit	Core Limit	Profile Limit	Limit Type[1]
maxBoundDescriptorSets	-	4	7	min
maxColorAttachments	-	4	8	min
timestampComputeAndGraphics	-	FALSE	TRUE	Boolean

Table 114. Vulkan 1.2 Limits

Limit Name	Unsupported Limit	Core Limit	Profile Limit	Limit Type[1]
shaderRoundingModeRTEFloat16	-	FALSE	TRUE	Boolean
shaderRoundingModeRTEFloat32	-	FALSE	TRUE	Boolean

Required Extensions

The following extensions are **required**

- VK_KHR_dynamic_rendering_local_read
- VK_KHR_load_store_op_none
- VK_KHR_shader_quad_control
- VK_KHR_shader_maximal_reconvergence
- VK_KHR_shader_subgroup_uniform_control_flow
- VK_KHR_shader_subgroup_rotate
- VK_KHR_shader_float_controls2

- VK_KHR_shader_expect_assume
- VK_KHR_line_rasterization
- VK_KHR_vertex_attribute_divisor
- VK_KHR_index_type_uint8
- VK_KHR_map_memory2
- VK_KHR_maintenance5
- VK_KHR_push_descriptor

Roadmap 2026

The Roadmap 2026 milestone is intended to be supported by newer mid-to-high-end devices shipping in 2026 or shortly thereafter across mainstream smartphone, tablet, laptop, console, and desktop devices.

This milestone raises the baseline feature support of Vulkan implementations as with prior milestones. In addition, this milestone makes guarantees about swapchain and presentation support for implementations.

Highlights of this roadmap include support for the following extensions:

- VK_KHR_fragment_shading_rate
- VK_KHR_shader_clock
- VK_KHR_compute_shader_derivatives
- VK_KHR_cooperative_matrix
- VK_KHR_surface and VK_KHR_swapchain
- VK_KHR_present_mode_fifo_latest_ready
- VK_KHR_present_id2 and VK_KHR_present_wait2

Several other extensions and limits are also required by this milestone as indicated below.

Required Milestones

This requires the Roadmap 2024 milestone.

Required Versions

Vulkan 1.4 is required by this milestone.

Required Features

The following core optional features are required to be supported:

- Vulkan 1.4 Optional Features
  - `hostImageCopy`

- VK_KHR_robustness2 Features
  - robustBufferAccess2
  - robustImageAccess2
  - nullDescriptor

Required Limits

The following core increased limits are **required**

Table 115. Vulkan 1.0 Limits

Limit Name	Core	2026	Limit Type[1]
maxPerStageDescriptorUniformBuffers	15	200	min
maxPerStageDescriptorStorageBuffers	4	200	min
maxPerStageDescriptorInputAttachments	4	8	min
maxDescriptorSetStorageBuffers	96	1800	min
maxDescriptorSetUniformBuffers	96	1800	min
maxDescriptorSetInputAttachments	4	8	min
maxVertexOutputComponents	64	124	min
maxTessellationControlPerVertexInputComponents	64	128	min
maxTessellationControlPerVertexOutputComponents	64	128	min
maxTessellationControlTotalOutputComponents	2048	4096	min
maxTessellationEvaluationInputComponents	64	128	min
maxTessellationEvaluationOutputComponents	64	128	min
maxGeometryOutputComponents	64	128	min
maxFragmentInputComponents	64	112	min
maxFragmentOutputAttachments	4	8	min
maxComputeSharedMemorySize	16384	32768	min
subPixelPrecisionBits	4	8	min
maxViewportDimensions.width	7680	8192	min
maxViewportDimensions.height	7680	8192	min
maxFramebufferWidth	7680	8192	min
maxFramebufferHeight	7680	8192	min

Required Extensions

The following extensions are **required**

- VK_KHR_robustness2
- VK_KHR_pipeline_binary
- VK_KHR_fragment_shading_rate
- VK_KHR_shader_clock
- VK_KHR_workgroup_memory_explicit_layout
- VK_KHR_compute_shader_derivatives
- VK_KHR_maintenance7
- VK_KHR_maintenance8
- VK_KHR_maintenance9
- VK_KHR_depth_clamp_zero_one
- VK_KHR_copy_memory_indirect
- VK_KHR_shader_untyped_pointers
- VK_KHR_surface
- VK_KHR_swapchain
- VK_KHR_present_mode_fifo_latest_ready
- VK_KHR_present_id2
- VK_KHR_present_wait2
- VK_KHR_surface_maintenance1
- VK_KHR_swapchain_maintenance1
- VK_KHR_cooperative_matrix

Appendix G: Legacy and Superseded Functionality

Functionality in the specification such as commands and structures **may** be marked as *legacy*. This label indicates that either functionality that supersedes it is available, or that the functionality has simply fallen out of favor for other reasons. Forward-looking applications **should** avoid using functionality marked as legacy where possible.

All legacy functionality is tagged in the XML registry, with information about what supersedes it. Any extension or version can tag functionality in any extension or version as legacy in the XML, however this is not always reflected in the specification. Functionality is only tagged as legacy in the specification if the superseding solution is added by a core version or extension at the same level or higher in the following hierarchy:

1. Core Version
2. KHR extension
3. EXT extension
4. Vendor extension

An explanatory section is provided in all cases to explain why functionality was marked as legacy, what functionality supersedes it (if any), and what applications **should** do instead.

NOTE	Legacy and superseded functionality is tagged in the xml registry via a deprecation tag, for both core versions and extensions. It is expected that tooling such as the validation layers will be able to warn users when they are making use of any functionality that is either legacy or superseded by the version and extensions the application has enabled.

Legacy functionality remains available for use in the specification, but it **may** be less capable than newer functionality. Interactions with legacy functionality will often be omitted when new extensions or features are developed, so legacy functionality **may** not work with the latest features.

NOTE	Another mechanism exists to indicate that an entire extension is considered legacy, using different terminology. See the Deprecation section of the Extensions chapter for more information.

In very rare instances, some functionality will be tagged as unused; this indicates when functionality was introduced and later discovered to be fundamentally flawed in some fashion. Application input in these cases **must** be well formed to handle implementations which tried to make use of this functionality, newer applications **must** not use the functionality, and newer implementations **must** do nothing with any input they receive. Unused parameters **must** be zeroed out, unused functions **must** not be called, unused enums **must** not be passed to APIs, and unused structure types **must** not be passed in `pNext` chains.

List of Legacy Functionality

When functionality is marked as legacy or if an extension supersedes it, an explanation will be added to the following sections.

NOTE	This list is a work in progress - we intend to add other items to this section over time. Examples include shader modules (superseded by VK_KHR_maintenance5) and anything superseded by a new version of the same functionality (e.g. VK_KHR_synchronization2).

Physical Device Queries: Superseded via version 2

VK_KHR_get_physical_device_properties2 was incorporated into Vulkan 1.1, which introduced new versions of several physical device query functions. These provide the same functionality as the Vulkan 1.0 functionality but with greater extensibility.

When querying device features, vkGetPhysicalDeviceFeatures2 **should** be used instead of vkGetPhysicalDeviceFeatures. When enabling device features, VkPhysicalDeviceFeatures2 **should** be provided in the `pNext` chain of VkDeviceCreateInfo instead of using VkDeviceCreateInfo ::`pEnabledFeatures`.

Physical Device Surface Queries: Superseded via version 2

Some parts of VK_KHR_surface were superseded by VK_KHR_get_surface_capabilities2, which introduced new versions of some physical device surface query functions. These provide the same functionality as the VK_KHR_surface functionality but with greater extensibility.

When querying surface capabilities, vkGetPhysicalDeviceSurfaceCapabilities2KHR **should** be used instead of vkGetPhysicalDeviceSurfaceCapabilitiesKHR.

When querying surface formats, vkGetPhysicalDeviceSurfaceFormats2KHR **should** be used instead of vkGetPhysicalDeviceSurfaceFormatsKHR.

Version Macros: Superseded via replacements including API variant

The version macros that do not take the API variant into account, such as VK_MAKE_VERSION or VK_VERSION_MAJOR, are superseded by those that do, such as VK_MAKE_API_VERSION or VK_API_VERSION_MAJOR.

Instead of VK_API_VERSION, specific version defines (e.g. VK_API_VERSION_1_0) or the VK_MAKE_API_VERSION macro should be used instead.

Device Layers: Superseded via instance layers

Previous versions of this specification distinguished between instance and device layers. Instance layers were only able to intercept commands that operate on `VkInstance` and `VkPhysicalDevice`, except they were not able to intercept vkCreateDevice. Device layers were enabled for individual devices when they were created, and could only intercept commands operating on that device or its child objects.

Device-only layers are now marked as legacy, and this specification no longer distinguishes between instance and device layers. Layers are enabled during instance creation, and are able to intercept all commands operating on that instance or any of its child objects. At the time this was marked as legacy, there were no known device-only layers and no compelling reason to create one.

The `enabledLayerCount` parameter of VkDeviceCreateInfo **must** be zero.

Render Pass Functions: Superseded via version 2

VK_KHR_create_renderpass2 and Vulkan 1.2 introduced new versions of several render pass functions. These provide the same functionality as the Vulkan 1.0 functionality but with greater extensibility.

NOTE	Render pass objects and all related commands are further superseded by dynamic rendering.

Render Pass Objects: Superseded via dynamic rendering

VK_KHR_dynamic_rendering and Vulkan 1.3 added a new way to specify render passes without needing to create VkFramebuffer and VkRenderPass objects. However, subpass functionality had no equivalent, meaning dynamic rendering was only suitable as a substitute for content not using subpasses.

VK_KHR_dynamic_rendering_local_read and Vulkan 1.4 later allowed the expression of most subpass functionality in core or extensions. Any subpass functionality which was not replicated is still expressible but requires applications to split work over multiple dynamic render pass instances. Functionality not covered with local reads would result in most or all vendors splitting the subpass internally.

Outside of vendor extensions, applications are advised to make use of vkCmdBeginRendering and vkCmdEndRendering to manage render passes from this API version onward.

Sampler and Buffer View Objects: Unnecessary with Descriptor Heaps

When using descriptor heaps, the creation of sampler and buffer view objects are wholly unnecessary. Instead, these objects are directly converted to descriptors via vkWriteSamplerDescriptorsEXT and vkWriteResourceDescriptorsEXT, skipping the need to create objects altogether. In the case of samplers, samplers can also be embedded directly into a shader via shader bindings, which only requires the VkSamplerCreateInfo, rather than a created object.

The creation of image views can also be skipped similarly for images used as descriptors, but image views themselves are still used elsewhere in the spec, such as for render passes.

Descriptor Management: Replaced by descriptor Heaps

Descriptor heaps provide a complete alternative for managing shader resources, and can be used where present instead of the resource management provided by Vulkan 1.0. Descriptor heaps can similarly be used instead of VK_EXT_descriptor_buffer.

While it is possible to use a mix of these in the same application, they cannot be used at the same time, and there are potential performance penalties for switching on some implementations.

Synchronization Commands: Deprecation via version 2

VK_KHR_synchronization2 was incorporated into Vulkan 1.3, which introduced new versions of synchronization functions. These provide the same functionality as the Vulkan 1.0 functionality but with greater extensibility.

Synchronization 2 commands **should** be used instead of synchronization 1 commands.

There is one piece of functionality in the original synchronization commands that cannot be mapped easily however. vkCmdSetEvent did not include an access scope; instead the access scope for an event dependency was fully specified by vkCmdWaitEvents. This allowed execution only dependencies to be expressed with cache access management handled at the end of the command. While the same dependencies can be expressed in VK_KHR_synchronization2, it is not a 1:1 mapping, requiring additional commands to be recorded, and so may not be as efficient.

VK_KHR_maintenance9 added the VK_DEPENDENCY_ASYMMETRIC_EVENT_BIT_KHR dependency flag, which enables this use case with vkCmdSetEvent2 and vkCmdWaitEvents2, making it efficient to express the same dependencies.

New flag values added by this extension are additionally extended to 64-bits.

32-bit Flags: Extended by 64-bit Flags

Initially Vulkan used 32-bit types to representing sets of flags (`Vk*Flags`) and individual flag bits (`Vk*FlagBits`). New functionality added many new flag bits. When the 32-bit types run out of available bits, new 64-bit flags (`Vk*Flags2`) and flag bits (`Vk*FlagBits2`) types are introduced.

These new types include all of the flag bits defined by the 32-bit types they correspond to, while allowing up to 32 additional bits. The new types are contained by corresponding extending structures. Such structures are used to specify or query flags by adding them to a `pNext` chain. When specifying 64-bit flags, those flags are used instead of the 32-bit flags in the base structure being extended. When querying 64-bit flags, both the 64-bit flags in the `pNext` chain and the 32-bit flags in the in the base structure being extended are returned.

64-bit flag and flag bit types are named the same as the 32-bit types they supersede, but with the addition of a “2” ahead of any vendor suffix. For example, VkPipelineCreateFlagBits and VkPipelineCreateFlags are superseded by VkPipelineCreateFlagBits2 and VkPipelineCreateFlags2, respectively.

Buffer Commands: Superseded by device address commands

Most commands that previously took buffer objects, including all command buffer commands, have now been superseded by equivalent commands or structures that use VkDeviceAddressRangeKHR structures to specify the address range directly. Some of these need a little additional information from the application in the form of VkAddressCommandFlagBitsKHR, which previously would have been retrieved from the buffer object itself.

Host-Side Build Commands: Superseded by Device Commands

VK_KHR_acceleration_structure and VK_EXT_opacity_micromap originally provided parallel host-side and device-side build, copy, and query commands for micromaps and acceleration structures. These host commands are enabled by the `micromapHostCommands` feature and the `accelerationStructureHostCommands` feature.

The following host commands are superseded by their command buffer equivalents:

- vkBuildAccelerationStructuresKHR is superseded by vkCmdBuildAccelerationStructuresKHR.
- vkCopyAccelerationStructureKHR is superseded by vkCmdCopyAccelerationStructureKHR.
- vkCopyAccelerationStructureToMemoryKHR is superseded by vkCmdCopyAccelerationStructureToMemoryKHR.
- vkCopyMemoryToAccelerationStructureKHR is superseded by vkCmdCopyMemoryToAccelerationStructureKHR.
- vkWriteAccelerationStructuresPropertiesKHR is superseded by vkCmdWriteAccelerationStructuresPropertiesKHR (used together with a query pool, and read back via vkGetQueryPoolResults or copied with vkCmdCopyQueryPoolResults).

The corresponding micromap host commands in VK_EXT_opacity_micromap are superseded similarly:

- vkBuildMicromapsEXT is superseded by vkCmdBuildMicromapsEXT.
- vkCopyMicromapEXT is superseded by vkCmdCopyMicromapEXT.
- vkCopyMicromapToMemoryEXT is superseded by vkCmdCopyMicromapToMemoryEXT.
- vkCopyMemoryToMicromapEXT is superseded by vkCmdCopyMemoryToMicromapEXT.
- vkWriteMicromapsPropertiesEXT is superseded by vkCmdWriteMicromapsPropertiesEXT.

Appendix H: API Boilerplate

This appendix defines Vulkan API features that are infrastructure required for a complete functional description of Vulkan, but do not logically belong elsewhere in the Specification.

Vulkan Header Files

Vulkan is defined as an API in the C99 language. Khronos provides a corresponding set of header files for applications using the API, which may be used in either C or C++ code. The interface descriptions in the specification are the same as the interfaces defined in these header files, and both are derived from the `vk.xml` XML API Registry, which is the canonical machine-readable description of the Vulkan API. The Registry, scripts used for processing it into various forms, and documentation of the registry schema are available as described at https://registry.khronos.org/vulkan/#apiregistry .

Language bindings for other languages can be defined using the information in the Specification and the Registry. Khronos does not provide any such bindings, but third-party developers have created some additional bindings.

Vulkan Combined API Header `vulkan.h` (Informative)

Applications normally will include the header `vulkan.h`. In turn, `vulkan.h` always includes the following headers:

- `vk_platform.h`, defining platform-specific macros and headers.
- `vulkan_core.h`, defining APIs for the Vulkan core and all registered extensions *other* than window system-specific and provisional extensions, which are included in separate header files.

In addition, specific preprocessor macros defined at the time `vulkan.h` is included cause header files for the corresponding window system-specific and provisional interfaces to be included, as described below.

Vulkan Platform-Specific Header `vk_platform.h` (Informative)

Platform-specific macros and interfaces are defined in `vk_platform.h`. These macros are used to control platform-dependent behavior, and their exact definitions are under the control of specific platforms and Vulkan implementations.

Platform-Specific Calling Conventions

On many platforms the following macros are empty strings, causing platform- and compiler-specific default calling conventions to be used.

`VKAPI_ATTR` is a macro placed before the return type in Vulkan API function declarations. This macro controls calling conventions for C++11 and GCC/Clang-style compilers.

`VKAPI_CALL` is a macro placed after the return type in Vulkan API function declarations. This macro controls calling conventions for MSVC-style compilers.

VKAPI_PTR is a macro placed between the '(' and '*' in Vulkan API function pointer declarations. This macro also controls calling conventions, and typically has the same definition as VKAPI_ATTR or VKAPI_CALL, depending on the compiler.

With these macros, a Vulkan function declaration takes the form of:

```
VKAPI_ATTR <return_type> VKAPI_CALL <command_name>(<command_parameters>);
```

Additionally, a Vulkan function pointer type declaration takes the form of:

```
typedef <return_type> (VKAPI_PTR *PFN_<command_name>)(<command_parameters>);
```

Platform-Specific Header Control

If the VK_NO_STDINT_H macro is defined by the application at compile time, extended integer types used by the Vulkan API, such as uint8_t, **must** also be defined by the application. Otherwise, the Vulkan headers will not compile. If VK_NO_STDINT_H is not defined, the system <stdint.h> is used to define these types. There is a fallback path when Microsoft Visual Studio version 2008 and earlier versions are detected at compile time.

If the VK_NO_STDDEF_H macro is defined by the application at compile time, size_t, **must** also be defined by the application. Otherwise, the Vulkan headers will not compile. If VK_NO_STDDEF_H is not defined, the system <stddef.h> is used to define this type.

Vulkan Core API Header vulkan_core.h

Applications that do not make use of window system-specific extensions may simply include vulkan_core.h instead of vulkan.h, although there is usually no reason to do so. In addition to the Vulkan API, vulkan_core.h also defines and / or uses a small number of C preprocessor macros that are described below.

Vulkan Header File Compile Time Controls

If the VK_NO_PROTOTYPES macro is defined by an application at compile time, prototypes for Vulkan APIs will not be included. Only typedefs for API function pointers will be defined.

This is intended for applications using their own function loader and dispatch mechanism.

If the macro is not defined by the application, prototypes for Vulkan APIs will be included.

If the VK_ONLY_EXPORTED_PROTOTYPES macro is defined by an application at compile time, only prototypes for Vulkan APIs tagged as `"exported"` in the API XML will be included. For non-tagged APIs, only typedefs for API function pointers will be defined.

This is intended to match APIs which are statically exported by the Vulkan loader. At present, the exported APIs are only those defined by Vulkan core versions.

If the macro is not defined by the application, prototypes for all Vulkan APIs will be included.

Vulkan Header File Version Number

VK_HEADER_VERSION is the version number of the vulkan_core.h header. This value is kept synchronized with the patch version of the released Specification.

```
// Provided by VK_VERSION_1_0
// Version of this file
#define VK_HEADER_VERSION 366
```

VK_HEADER_VERSION_COMPLETE is the complete version number of the vulkan_core.h header, comprising the major, minor, and patch versions. The major/minor values are kept synchronized with the complete version of the released Specification. This value is intended for use by automated tools to identify exactly which version of the header was used during their generation.

Applications should not use this value as their VkApplicationInfo::apiVersion. Instead applications should explicitly select a specific fixed major/minor API version using, for example, one of the VK_API_VERSION_*_* values.

```
// Provided by VK_VERSION_1_0
// Complete version of this file
#define VK_HEADER_VERSION_COMPLETE VK_MAKE_API_VERSION(0, 1, 4, VK_HEADER_VERSION)
```

VK_API_VERSION is now commented out of vulkan_core.h and **cannot** be used.

WARNING	This functionality is superseded by Vulkan Version 1.0. See Legacy Functionality for more information.

```
// Provided by VK_VERSION_1_0
//#define VK_API_VERSION VK_MAKE_API_VERSION(0, 1, 0, 0) // Patch version should
always be set to 0
```

Vulkan Handle Macros

VK_DEFINE_HANDLE defines a dispatchable handle type.

```
// Provided by VK_VERSION_1_0
#define VK_DEFINE_HANDLE(object) typedef struct object##_T* object;
```

- object is the name of the resulting C type.

The only dispatchable handle types are those related to device and instance management, such as VkDevice.

VK_DEFINE_NON_DISPATCHABLE_HANDLE defines a non-dispatchable handle type.

```
// Provided by VK_VERSION_1_0
#ifndef VK_DEFINE_NON_DISPATCHABLE_HANDLE
    #if (VK_USE_64_BIT_PTR_DEFINES==1)
        #define VK_DEFINE_NON_DISPATCHABLE_HANDLE(object) typedef struct object##_T
*object;
    #else
        #define VK_DEFINE_NON_DISPATCHABLE_HANDLE(object) typedef uint64_t object;
    #endif
#endif
```

- `object` is the name of the resulting C type.

Most Vulkan handle types, such as VkBuffer, are non-dispatchable.

NOTE

The `vulkan_core.h` header allows the VK_DEFINE_NON_DISPATCHABLE_HANDLE and VK_NULL_HANDLE definitions to be overridden by the application. If VK_DEFINE_NON_DISPATCHABLE_HANDLE is already defined when `vulkan_core.h` is compiled, the default definitions for VK_DEFINE_NON_DISPATCHABLE_HANDLE and VK_NULL_HANDLE are skipped. This allows the application to define a binary-compatible custom handle which **may** provide more type-safety or other features needed by the application. Applications **must** not define handles in a way that is not binary compatible - where binary compatibility is platform dependent.

`VK_NULL_HANDLE` is a reserved value representing a non-valid object handle. It may be passed to and returned from Vulkan commands only when specifically allowed.

```
// Provided by VK_VERSION_1_0
#ifndef VK_DEFINE_NON_DISPATCHABLE_HANDLE
    #if (VK_USE_64_BIT_PTR_DEFINES==1)
        #if (defined(__cplusplus) && (__cplusplus >= 201103L)) || (defined(_MSVC_LANG)
&& (_MSVC_LANG >= 201103L))
            #define VK_NULL_HANDLE nullptr
        #else
            #define VK_NULL_HANDLE ((void*)0)
        #endif
    #else
        #define VK_NULL_HANDLE 0ULL
    #endif
#endif
#ifndef VK_NULL_HANDLE
    #define VK_NULL_HANDLE 0
#endif
```

`VK_USE_64_BIT_PTR_DEFINES` defines whether the default non-dispatchable handles are declared using either a 64-bit pointer type or a 64-bit unsigned integer type.

`VK_USE_64_BIT_PTR_DEFINES` is set to '1' to use a 64-bit pointer type or any other value to use a 64-bit unsigned integer type.

```
// Provided by VK_VERSION_1_0
#ifndef VK_USE_64_BIT_PTR_DEFINES
    #if defined(__LP64__) || defined(_WIN64) || (defined(__x86_64__) &&
!defined(__ILP32__) ) || defined(_M_X64) || defined(__ia64) || defined (_M_IA64) ||
defined(__aarch64__) || defined(__powerpc64__) || (defined(__riscv) && __riscv_xlen ==
64)
        #define VK_USE_64_BIT_PTR_DEFINES 1
    #else
        #define VK_USE_64_BIT_PTR_DEFINES 0
    #endif
#endif
```

NOTE

The `vulkan_core.h` header allows the `VK_USE_64_BIT_PTR_DEFINES` definition to be overridden by the application. This allows the application to select either a 64-bit pointer type or a 64-bit unsigned integer type for non-dispatchable handles in the case where the predefined preprocessor check does not identify the desired configuration.

NOTE

This macro was introduced starting with the Vulkan 1.2.174 headers, and its availability can be checked at compile time by requiring `VK_HEADER_VERSION >= 174`.

It is not available if you are using older headers, such as may be shipped with an older Vulkan SDK. Developers requiring this functionality may wish to include a copy of the current Vulkan headers with their project in this case.

Window System-Specific Header Control (Informative)

To use a Vulkan extension supporting a platform-specific window system, header files for that window system **must** be included at compile time, or platform-specific types **must** be forward-declared. The Vulkan header files are unable to determine whether or not an external header is available at compile time, so platform-specific extensions are provided in separate headers from the core API and platform-independent extensions, allowing applications to decide which ones they need to be defined and how the external headers are included.

Extensions dependent on particular sets of platform headers, or that forward-declare platform-specific types, are declared in a header named for that platform. Before including these platform-specific Vulkan headers, applications **must** include both `vulkan_core.h` and any external native headers the platform extensions depend on.

As a convenience for applications that do not need the flexibility of separate platform-specific Vulkan headers, `vulkan.h` includes `vulkan_core.h`, and then conditionally includes platform-specific Vulkan headers and the external headers they depend on. Applications control which platform-specific headers are included by #defining macros before including `vulkan.h`.

The correspondence between platform-specific extensions, external headers they require, the platform-specific header which declares them, and the preprocessor macros which enable inclusion by `vulkan.h` are shown in the following table.

Table 116. Window System Extensions and Headers

Extension Name	Window System Name	Platform-specific Header	Required External Headers	Controlling vulkan.h Macro
VK_KHR_android_surface	Android	vulkan_android.h	None	VK_USE_PLATFORM_ANDROID_KHR
VK_KHR_wayland_surface	Wayland	vulkan_wayland.h	<wayland-client.h>	VK_USE_PLATFORM_WAYLAND_KHR
VK_KHR_win32_surface, VK_KHR_external_memory_win32, VK_KHR_win32_keyed_mutex, VK_KHR_external_semaphore_win32, VK_KHR_external_fence_win32, VK_NV_external_memory_win32, VK_NV_win32_keyed_mutex	Microsoft Windows	vulkan_win32.h	<windows.h>	VK_USE_PLATFORM_WIN32_KHR
VK_KHR_xcb_surface	X11 Xcb	vulkan_xcb.h	<xcb/xcb.h>	VK_USE_PLATFORM_XCB_KHR
VK_KHR_xlib_surface	X11 Xlib	vulkan_xlib.h	<X11/Xlib.h>	VK_USE_PLATFORM_XLIB_KHR
VK_EXT_directfb_surface	DirectFB	vulkan_directfb.h	<directfb/directfb.h>	VK_USE_PLATFORM_DIRECTFB_EXT
VK_EXT_acquire_xlib_display	X11 XRAndR	vulkan_xlib_xrandr.h	<X11/Xlib.h>, <X11/extensions/Xrandr.h>	VK_USE_PLATFORM_XLIB_XRANDR_EXT
VK_GGP_stream_descriptor_surface, VK_GGP_frame_token	Google Games Platform	vulkan_ggp.h	<ggp_c/vulkan_types.h>	VK_USE_PLATFORM_GGP
VK_MVK_ios_surface	iOS	vulkan_ios.h	None	VK_USE_PLATFORM_IOS_MVK
VK_MVK_macos_surface	macOS	vulkan_macos.h	None	VK_USE_PLATFORM_MACOS_MVK
VK_OHOS_surface	OHOS	vulkan_ohos.h	None	VK_USE_PLATFORM_OHOS
VK_NN_vi_surface	VI	vulkan_vi.h	None	VK_USE_PLATFORM_VI_NN
VK_FUCHSIA_imagepipe_surface	Fuchsia	vulkan_fuchsia.h	<zircon/types.h>	VK_USE_PLATFORM_FUCHSIA
VK_EXT_metal_surface	Metal on CoreAnimation	vulkan_metal.h	None	VK_USE_PLATFORM_METAL_EXT
VK_QNX_screen_surface	QNX Screen	vulkan_screen.h	<screen/screen.h>	VK_USE_PLATFORM_SCREEN_QNX

NOTE This section describes the purpose of the headers independently of the specific underlying functionality of the window system extensions themselves. Each extension name will only link to a description of that extension when viewing a specification built with that extension included.

Provisional Extension Header Control (Informative)

Provisional extensions **should** not be used in production applications. The functionality defined by such extensions **may** change in ways that break backwards compatibility between revisions, and before final release of a non-provisional version of that extension.

Provisional extensions are defined in a separate *provisional header*, `vulkan_beta.h`, allowing applications to decide whether or not to include them. The mechanism is similar to window system-specific headers: before including `vulkan_beta.h`, applications **must** include `vulkan_core.h`.

NOTE Sometimes a provisional extension will include a subset of its interfaces in `vulkan_core.h`. This may occur if the provisional extension is promoted from an existing vendor or EXT extension and some of the existing interfaces are defined as aliases of the provisional extension interfaces. All other interfaces of that provisional extension which are not aliased will be included in `vulkan_beta.h`.

As a convenience for applications, `vulkan.h` conditionally includes `vulkan_beta.h`. Applications **can** control inclusion of `vulkan_beta.h` by #defining the macro `VK_ENABLE_BETA_EXTENSIONS` before including `vulkan.h`.

NOTE Starting in version 1.2.171 of the Specification, all provisional enumerants are protected by the macro `VK_ENABLE_BETA_EXTENSIONS`. Applications needing to use provisional extensions must always define this macro, even if they are explicitly including `vulkan_beta.h`. This is a minor change to behavior, affecting only provisional extensions.

NOTE This section describes the purpose of the provisional header independently of the specific provisional extensions which are contained in that header at any given time. The extension appendices for provisional extensions note their provisional status, and link back to this section for more information. Provisional extensions are intended to provide early access for bleeding-edge developers, with the understanding that extension interfaces may change in response to developer feedback. Provisional extensions are very likely to eventually be updated and released as non-provisional extensions, but there is no guarantee this will happen, or how long it will take if it does happen.

Video Std Headers

Performing video coding operations usually involves the application having to provide various parameters, data structures, or other syntax elements specific to the particular video compression standard used, and the associated semantics are covered by the specification of those.

The interface descriptions of these are available in the header files derived from the `video.xml` XML file, which is the canonical machine-readable description of data structures and enumerations that are associated with the externally-provided video compression standards.

Table 117. Video Std Headers

Video Std Header Name	Description	Header File	Related Extensions
`vulkan_video_codecs_common`	Codec-independent common definitions	`<vk_video/vulkan_video_codecs_common.h>`	-
`vulkan_video_codec_h264std`	ITU-T H.264 common definitions	`<vk_video/vulkan_video_codec_h264std.h>`	VK_KHR_video_decode_h264, VK_KHR_video_encode_h264
`vulkan_video_codec_h264std_decode`	ITU-T H.264 decode-specific definitions	`<vk_video/vulkan_video_codec_h264std_decode.h>`	VK_KHR_video_decode_h264
`vulkan_video_codec_h264std_encode`	ITU-T H.264 encode-specific definitions	`<vk_video/vulkan_video_codec_h264std_encode.h>`	VK_KHR_video_encode_h264
`vulkan_video_codec_h265std`	ITU-T H.265 common definitions	`<vk_video/vulkan_video_codec_h265std.h>`	VK_KHR_video_decode_h265, VK_KHR_video_encode_h265
`vulkan_video_codec_h265std_decode`	ITU-T H.265 decode-specific definitions	`<vk_video/vulkan_video_codec_h265std_decode.h>`	VK_KHR_video_decode_h265
`vulkan_video_codec_h265std_encode`	ITU-T H.265 encode-specific definitions	`<vk_video/vulkan_video_codec_h265std_encode.h>`	VK_KHR_video_encode_h265
`vulkan_video_codec_vp9std`	VP9 common definitions	`<vk_video/vulkan_video_codec_vp9std.h>`	VK_KHR_video_decode_vp9
`vulkan_video_codec_vp9std_decode`	VP9 decode-specific definitions	`<vk_video/vulkan_video_codec_vp9std_decode.h>`	VK_KHR_video_decode_vp9
`vulkan_video_codec_av1std`	AV1 common definitions	`<vk_video/vulkan_video_codec_av1std.h>`	VK_KHR_video_decode_av1
`vulkan_video_codec_av1std_decode`	AV1 decode-specific definitions	`<vk_video/vulkan_video_codec_av1std_decode.h>`	VK_KHR_video_decode_av1
`vulkan_video_codec_av1std_encode`	AV1 encode-specific definitions	`<vk_video/vulkan_video_codec_av1std_encode.h>`	VK_KHR_video_encode_av1

Appendix I: Invariance

The Vulkan specification is not pixel exact. It therefore does not guarantee an exact match between images produced by different Vulkan implementations. However, the specification does specify exact matches, in some cases, for images produced by the same implementation. The purpose of this appendix is to identify and provide justification for those cases that require exact matches.

Repeatability

The obvious and most fundamental case is repeated issuance of a series of Vulkan commands. For any given Vulkan and framebuffer state vector, and for any Vulkan command, the resulting Vulkan and framebuffer state **must** be identical whenever the command is executed on that initial Vulkan and framebuffer state. This repeatability requirement does not apply when using shaders containing side effects (image and buffer variable stores and atomic operations), because these memory operations are not guaranteed to be processed in a defined order.

One purpose of repeatability is avoidance of visual artifacts when a double-buffered scene is redrawn. If rendering is not repeatable, swapping between two buffers rendered with the same command sequence **may** result in visible changes in the image. Such false motion is distracting to the viewer. Another reason for repeatability is testability.

Repeatability, while important, is a weak requirement. Given only repeatability as a requirement, two scenes rendered with one (small) polygon changed in position might differ at every pixel. Such a difference, while within the law of repeatability, is certainly not within its spirit. Additional invariance rules are desirable to ensure useful operation.

Multi-Pass Algorithms

Invariance is necessary for a whole set of useful multi-pass algorithms. Such algorithms render multiple times, each time with a different Vulkan mode vector, to eventually produce a result in the framebuffer. Examples of these algorithms include:

- “Erasing” a primitive from the framebuffer by redrawing it, either in a different color or using the XOR logical operation.
- Using stencil operations to compute capping planes.

Invariance Rules

For a given Vulkan device:

Rule 1 *For any given Vulkan and framebuffer state vector, and for any given Vulkan command, the resulting Vulkan and framebuffer state **must** be identical each time the command is executed on that initial Vulkan and framebuffer state.*

Rule 2 *Changes to the following state values have no side effects (the use of any other state value is not affected by the change):*

Required:

- *Color and depth/stencil attachment contents*
- *Scissor parameters (other than enable)*
- *Write masks (color, depth, stencil)*
- *Clear values (color, depth, stencil)*

Strongly suggested:

- *Stencil parameters (other than enable)*
- *Depth test parameters (other than enable)*
- *Blend parameters (other than enable)*
- *Logical operation parameters (other than enable)*

Corollary 1 *Fragment generation is invariant with respect to the state values listed in Rule 2.*

Rule 3 *The arithmetic of each per-fragment operation is invariant except with respect to parameters that directly control it.*

Corollary 2 *Images rendered into different color attachments of the same framebuffer, either simultaneously or separately using the same command sequence, are pixel identical.*

Rule 4 *Identical pipelines will produce the same result when run multiple times with the same input. The wording "Identical pipelines" means VkPipeline objects that have been created with identical SPIR-V binaries and identical state, which are then used by commands executed using the same Vulkan state vector. Invariance is relaxed for shaders with side effects, such as performing stores or atomics.*

Rule 5 *All fragment shaders that either conditionally or unconditionally assign* `FragCoord.z` *to* `FragDepth` *are depth-invariant with respect to each other, for those fragments where the assignment to* `FragDepth` *actually is done.*

If a sequence of Vulkan commands specifies primitives to be rendered with shaders containing side effects (image and buffer variable stores and atomic operations), invariance rules are relaxed. In particular, rule 1, corollary 2, and rule 4 do not apply in the presence of shader side effects.

The following weaker versions of rules 1 and 4 apply to Vulkan commands involving shader side effects:

Rule 6 *For any given Vulkan and framebuffer state vector, and for any given Vulkan command, the contents of any framebuffer state not directly or indirectly affected by results of shader image or buffer variable stores or atomic operations* ***must*** *be identical each time the command is executed on that initial Vulkan and framebuffer state.*

Rule 7 *Identical pipelines will produce the same result when run multiple times with the same input as long as:*

- *shader invocations do not use image atomic operations;*

- *no framebuffer memory is written to more than once by image stores, unless all such stores write the same value; and*
- *no shader invocation, or other operation performed to process the sequence of commands, reads memory written to by an image store.*

NOTE

The OpenGL specification has the following invariance rule: Consider a primitive p' obtained by translating a primitive p through an offset (x, y) in window coordinates, where x and y are integers. As long as neither p' nor p is clipped, it **must** be the case that each fragment f' produced from p' is identical to a corresponding fragment f from p except that the center of f' is offset by (x, y) from the center of f.

This rule does not apply to Vulkan and is an intentional difference from OpenGL.

When any sequence of Vulkan commands triggers shader invocations that perform image stores or atomic operations, and subsequent Vulkan commands read the memory written by those shader invocations, these operations **must** be explicitly synchronized.

Tessellation Invariance

When using a pipeline containing tessellation evaluation shaders, the fixed-function tessellation primitive generator consumes the input patch specified by an application and emits a new set of primitives. The following invariance rules are intended to provide repeatability guarantees. Additionally, they are intended to allow an application with a carefully crafted tessellation evaluation shader to ensure that the sets of triangles generated for two adjacent patches have identical vertices along shared patch edges, avoiding “cracks” caused by minor differences in the positions of vertices along shared edges.

Rule 1 *When processing two patches with identical outer and inner tessellation levels, the tessellation primitive generator will emit an identical set of point, line, or triangle primitives as long as the pipeline used to process the patch primitives has tessellation evaluation shaders specifying the same tessellation mode, spacing, vertex order, and point mode decorations. Two sets of primitives are considered identical if and only if they contain the same number and type of primitives and the generated tessellation coordinates for the vertex numbered m of the primitive numbered n are identical for all values of m and n.*

Rule 2 *The set of vertices generated along the outer edge of the subdivided primitive in triangle and quad tessellation, and the tessellation coordinates of each, depend only on the corresponding outer tessellation level and the spacing decorations in the tessellation shaders of the pipeline.*

Rule 3 *The set of vertices generated when subdividing any outer primitive edge is always symmetric. For triangle tessellation, if the subdivision generates a vertex with tessellation coordinates of the form (0, x, 1-x), (x, 0, 1-x), or (x, 1-x, 0), it will also generate a vertex with coordinates of exactly (0, 1-x, x), (1-x, 0, x), or (1-x, x, 0), respectively. For quad tessellation, if the subdivision generates a vertex with coordinates of (x, 0) or (0, x), it will also generate a vertex with coordinates of exactly (1-x, 0) or (0, 1-x), respectively. For isoline tessellation, if it generates vertices at (0, x) and (1, x) where x is not zero, it will also generate vertices at exactly (0, 1-x) and (1, 1-x), respectively.*

Rule 4 *The set of vertices generated when subdividing outer edges in triangular and quad tessellation*

must *be independent of the specific edge subdivided, given identical outer tessellation levels and spacing. For example, if vertices at (x, 1 - x, 0) and (1-x, x, 0) are generated when subdividing the w = 0 edge in triangular tessellation, vertices* ***must*** *be generated at (x, 0, 1-x) and (1-x, 0, x) when subdividing an otherwise identical v = 0 edge. For quad tessellation, if vertices at (x, 0) and (1-x, 0) are generated when subdividing the v = 0 edge, vertices* ***must*** *be generated at (0, x) and (0, 1-x) when subdividing an otherwise identical u = 0 edge.*

Rule 5 *When processing two patches that are identical in all respects enumerated in rule 1 except for vertex order, the set of triangles generated for triangle and quad tessellation* ***must*** *be identical except for vertex and triangle order. For each triangle n1 produced by processing the first patch, there* ***must*** *be a triangle n2 produced when processing the second patch each of whose vertices has the same tessellation coordinates as one of the vertices in n1.*

Rule 6 *When processing two patches that are identical in all respects enumerated in rule 1 other than matching outer tessellation levels and/or vertex order, the set of interior triangles generated for triangle and quad tessellation* ***must*** *be identical in all respects except for vertex and triangle order. For each interior triangle n1 produced by processing the first patch, there* ***must*** *be a triangle n2 produced when processing the second patch each of whose vertices has the same tessellation coordinates as one of the vertices in n1. A triangle produced by the tessellator is considered an interior triangle if none of its vertices lie on an outer edge of the subdivided primitive.*

Rule 7 *For quad and triangle tessellation, the set of triangles connecting an inner and outer edge depends only on the inner and outer tessellation levels corresponding to that edge and the spacing decorations.*

Rule 8 *The value of all defined components of* `TessCoord` *will be in the range [0, 1]. Additionally, for any defined component x of* `TessCoord`*, the results of computing 1.0-x in a tessellation evaluation shader will be exact. If any floating-point values in the range [0, 1] fail to satisfy this property, such values* ***must*** *not be used as tessellation coordinate components.*

Appendix J: Lexicon

This appendix defines terms, abbreviations, and API prefixes used in the Specification.

Glossary

The terms defined in this section are used consistently throughout the Specification and may be used with or without capitalization.

Accessible (Descriptor Binding)

A descriptor binding is accessible to a shader stage if that stage is included in the `stageFlags` of the descriptor binding. Descriptors using that binding **can** only be used by stages in which they are accessible.

Acquire Operation (Resource)

An operation that acquires ownership of an image subresource or buffer range.

Active (Descriptor Type)

When a descriptor with *mutable* type is updated with vkUpdateDescriptorSets, the active descriptor type changes. When the descriptor is consumed by shaders, it is the active descriptor type which determines validity, i.e. `VkDescriptorSetLayoutBinding`::`descriptorType` is replaced with the active descriptor type. A mismatch in active descriptor type and consumption by shader is considered an undefined descriptor.

Active (Transform Feedback)

Transform feedback is made active after vkCmdBeginTransformFeedbackEXT executes and remains active until vkCmdEndTransformFeedbackEXT executes. While transform feedback is active, data written to variables in the output interface of the last pre-rasterization shader stage of the graphics pipeline are captured to the bound transform feedback buffers if those variables are decorated for transform feedback.

Adjacent Vertex

A vertex in an adjacency primitive topology that is not part of a given primitive, but is accessible in geometry shaders.

Active Object (Ray Tracing)

A primitive or instance in a ray tracing acceleration structure which has a corresponding ID, and is not *inactive* (meaning that it is visible to rays).

Advanced Blend Operation

Blending performed using one of the blend operation enums introduced by the `VK_EXT_blend_operation_advanced` extension. See Advanced Blending Operations.

Alias (API type/command)

An identical definition of another API type/command with the same behavior but a different name.

Aliased Range (Memory)

A range of a device memory allocation that is bound to multiple resources simultaneously.

Allocation Scope

An association of a host memory allocation to a parent object or command, where the allocation's lifetime ends before or at the same time as the parent object is freed or destroyed, or during the parent command.

API command

Any command defined in the Vulkan specification. These entry points all have a vk prefix.

Aspect (Image)

Some image types contain multiple kinds (called "aspects") of data for each pixel, where each aspect is used in a particular way by the pipeline and **may** be stored differently or separately from other aspects. For example, the color components of an image format make up the color aspect of the image, and **can** be used as a framebuffer color attachment. Some operations, like depth testing, operate only on specific aspects of an image.

Attachment (Render Pass)

A zero-based integer index name used in render pass creation to refer to a framebuffer attachment that is accessed by one or more subpasses. The index also refers to an attachment description which includes information about the properties of the image view that will later be attached.

Availability Operation

An operation that causes the values generated by specified memory write accesses to become available for future access.

Available

A state of values written to memory that allows them to be made visible.

Axis-aligned Bounding Box

A box bounding a region in space defined by extents along each axis and thus representing a box where each edge is aligned to one of the major axes.

Back-Facing

See Facingness.

Backward Prediction

Sample prediction performed during video decode and encode operations based on reference pictures that occur temporally (in display order) after the current picture.

Backward Reference

A reference picture that occurs temporally (in display order) after the current picture.

Batch

A single structure submitted to a queue as part of a queue submission command, describing a set of queue operations to execute.

Backwards Compatibility

A given version of the API is backwards compatible with an earlier version if an application, relying only on valid behavior and functionality defined by the earlier specification, is able to correctly run against each version without any modification. This assumes no active attempt by that application to not run when it detects a different version.

Binary Semaphore

A semaphore with a boolean payload indicating whether the semaphore is signaled or unsignaled. Represented by a VkSemaphore object created with a semaphore type of VK_SEMAPHORE_TYPE_BINARY .

Binding (Memory)

An association established between a range of a resource object and a range of a memory object. These associations determine the memory locations affected by operations performed on elements of a resource object. Memory bindings are established using the vkBindBufferMemory command for non-sparse buffer objects, using the vkBindImageMemory command for non-sparse image objects , and using the vkQueueBindSparse command for sparse resources .

Blend Constant

Four floating-point (RGBA) values used as an input to blending.

Blending

Arithmetic operations between a fragment color value and a value in a color attachment that produce a final color value to be written to the attachment.

Buffer

A resource that represents a linear array of data in device memory. Represented by a VkBuffer object.

Buffer Device Address

A 64-bit value used in a shader to access buffer memory through the `PhysicalStorageBuffer` storage class.

Buffer View

An object that represents a range of a specific buffer, and state controlling how the contents are interpreted. Represented by a VkBufferView object.

Built-In Variable

A variable decorated in a shader, where the decoration makes the variable take values provided by the execution environment or values that are generated by fixed-function pipeline stages.

Built-In Interface Block

A block defined in a shader containing only variables decorated with built-in decorations, and is used to match against other shader stages.

Clip Coordinates

The homogeneous coordinate space in which vertex positions (`Position` decoration) are written by pre-rasterization shader stages.

Clip Distance

A built-in output from pre-rasterization shader stages defining a clip half-space against which the primitive is clipped.

Clip Volume

The intersection of the view volume with all clip half-spaces.

Coding Block (Video)

An element of a video picture, comprising a rectangular block of texel values that are coded as a single unit in video coding. Video coding blocks have a corresponding coding block size (width and height) defining the dimensions of these elements in units of texels. These dimensions are usually codec-specific and a single video compression standard **may** support more than one coding block type and size. Video compression standards **may** use multiple coding block types to organize coding blocks of different kinds and sizes into hierarchies.

Color Attachment

A subpass attachment point, or image view, that is the target of fragment color outputs and blending.

Color Renderable Format

A VkFormat where VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT is set in one of the following, depending on the image's tiling:

- VkFormatProperties::`linearTilingFeatures`
- VkFormatProperties::`optimalTilingFeatures`
- VkDrmFormatModifierPropertiesEXT::`drmFormatModifierTilingFeatures`

Combined Image Sampler

A descriptor type that includes both a sampled image and a sampler.

Command Buffer

An object that records commands to be submitted to a queue. Represented by a VkCommandBuffer object.

Command Buffer Nesting Level

The Command Buffer Nesting Level of a secondary command buffer is equal to the maximum nesting level of all secondary command buffers executed by that command buffer plus one, where a secondary command buffer that executes no other secondary command buffers has a nesting level of zero.

Command Pool

An object that command buffer memory is allocated from, and that owns that memory. Command pools aid multithreaded performance by enabling different threads to use different allocators, without internal synchronization on each use. Represented by a VkCommandPool object.

Compatible Allocator

When allocators are compatible, allocations from each allocator **can** be freed by the other

allocator.

Compatible Image Formats

When formats are compatible, images created with one of the formats **can** have image views created from it using any of the compatible formats. Also see *Size-Compatible Image Formats*.

Compatible Queues

Queues within a queue family. Compatible queues have identical properties.

Complete Mipmap Chain

The entire set of mip levels that can be provided for an image, from the largest application-specified mip level size down to the *minimum mip level size*. See Image Mip Level Sizing.

Completed Operation

A deferred operation whose corresponding command has been executed to completion. See Deferred Host Operations

Component (Format)

A distinct part of a format. Color components are represented with `R`, `G`, `B`, and `A`. Depth and stencil components are represented with `D` and `S`. Formats **can** have multiple instances of the same component. Some formats have other notations such as `E` or `X` which are not considered a component of the format.

Compressed Texel Block

An element of an image having a block-compressed format, comprising a rectangular block of texel values that are encoded as a single value in memory. Compressed texel blocks of a particular block-compressed format have a corresponding width, height, and depth defining the dimensions of these elements in units of texels, and a size in bytes of the encoding in memory.

Constant Integral Expressions

A SPIR-V constant instruction whose type is `OpTypeInt`. See *Constant Instruction* in section 2.2.1 “Instructions” of the Khronos SPIR-V Specification.

Cooperative Matrix

A SPIR-V type where the storage for and computations performed on the matrix are spread across a set of invocations such as a subgroup.

Coverage Index

The index of a sample in the coverage mask.

Coverage Mask

A bitfield associated with a fragment representing the samples that were determined to be covered based on the result of rasterization, and then subsequently modified by fragment operations or the fragment shader.

Cull Distance

A built-in output from pre-rasterization shader stages defining a cull half-space where the primitive is rejected if all vertices have a negative value for the same cull distance.

Cull Volume

The intersection of the view volume with all cull half-spaces.

Decode Output Picture

A video picture resource used to store the result of a video decode operation.

Decoded Picture Buffer

An indexed set of reference pictures used by a video session. Abbreviated as DPB.

Decoded Picture Buffer Slot

An entry within a DPB that **can** be associated with a particular reference picture.

Decoded Picture Buffer Slot Index

The index of a DPB slot within its encompassing DPB.

Decoration (SPIR-V)

Auxiliary information such as built-in variables, stream numbers, invariance, interpolation type, relaxed precision, etc., added to variables or structure-type members through decorations.

Deferrable Command

A command which allows deferred execution of host-side work. See Deferred Host Operations.

Deferrable Operation

A single logical item of host-side work which can be deferred. Represented by the VkDeferredOperationKHR object. See Deferred Host Operations.

Depth/Stencil Attachment

A subpass attachment point, or image view, that is the target of depth and/or stencil test operations and writes.

Depth/Stencil Format

A VkFormat that includes depth and/or stencil components.

Depth/Stencil Image (or ImageView)

A VkImage (or VkImageView) with a depth/stencil format.

Depth/Stencil Resolve Attachment

A subpass attachment point, or image view, that is the target of a multisample resolve operation from the corresponding depth/stencil attachment at the end of the subpass.

Derivative Group

A set of compute, mesh, task, or fragment shader invocations that cooperate to compute derivatives, including implicit derivatives for sampled image operations.

Descriptor

Information about a resource or resource view written into a descriptor set that is used to access the resource or view from a shader.

Descriptor Binding

An entry in a descriptor set layout corresponding to zero or more descriptors of a single descriptor type in a set. Defined by a VkDescriptorSetLayoutBinding structure.

Descriptor Pool

An object that descriptor sets are allocated from, and that owns the storage of those descriptor sets. Descriptor pools aid multithreaded performance by enabling different threads to use different allocators, without internal synchronization on each use. Represented by a VkDescriptorPool object.

Descriptor Set

An object that resource descriptors are written into via the API, and that **can** be bound to a command buffer such that the descriptors contained within it **can** be accessed from shaders. Represented by a VkDescriptorSet object.

Descriptor Set Layout

An object defining the set of resources (types and counts) and their relative arrangement (in the binding namespace) within a descriptor set. Used when allocating descriptor sets and when creating pipeline layouts. Represented by a VkDescriptorSetLayout object.

Device

The processor(s) and execution environment that perform tasks requested by the application via the Vulkan API.

Device Group

A set of physical devices that support accessing each other's memory and recording a single command buffer that **can** be executed on all the physical devices.

Device Index

A zero-based integer that identifies one physical device from a logical device. A device index is valid if it is less than the number of physical devices in the logical device.

Device Mask

A bitmask where each bit represents one device index. A device mask value is valid if every bit that is set in the mask is at a bit position that is less than the number of physical devices in the logical device.

Device Memory

Memory accessible to the device. Represented by a VkDeviceMemory object.

Device-Level Command

Any command that is dispatched from a logical device, or from a child object of a logical device.

Device-Level Functionality

All device-level commands and objects, and their structures, enumerated types, and enumerants. Additionally, physical-device-level functionality defined by a device extension is also considered device-level functionality.

Device-Level Object

Logical device objects and their child objects. For example, VkDevice, VkQueue, and VkCommandBuffer objects are device-level objects.

Device-Local Memory

Memory that is connected to the device, and **may** be more performant for device access than host-local memory.

Direct Drawing Commands

Drawing commands that take all their parameters as direct arguments to the command (and not sourced via structures in buffer memory as the *indirect drawing commands*). Includes vkCmdDrawMultiIndexedEXT, vkCmdDrawMultiEXT, vkCmdDrawMeshTasksEXT, vkCmdDraw, and vkCmdDrawIndexed.

Disjoint

Disjoint planes are *image planes* to which memory is bound independently.
A *disjoint image* consists of multiple *disjoint planes*, and is created with the VK_IMAGE_CREATE_DISJOINT_BIT bit set.

Dispatchable Command

A non-global command. The first argument to each dispatchable command is a dispatchable handle type.

Dispatchable Handle

A handle of a pointer handle type which **may** be used by layers as part of intercepting API commands.

Dispatching Commands

Commands that provoke work using a compute pipeline, such as vkCmdDispatch. See Dispatching Commands.

Drawing Commands

Commands that provoke work using a graphics pipeline. Includes vkCmdDraw, vkCmdDrawIndexed, vkCmdDrawIndirectCount, vkCmdDrawIndexedIndirectCount, vkCmdDrawIndirectCountKHR, vkCmdDrawIndexedIndirectCountKHR, vkCmdDrawMultiIndexedEXT, vkCmdDrawMultiEXT, vkCmdDrawMeshTasksEXT, vkCmdDrawMeshTasksIndirectEXT, vkCmdDrawMeshTasksIndirectCountEXT, vkCmdDrawIndirect, and vkCmdDrawIndexedIndirect.

Duration (Command)

The *duration* of a Vulkan command refers to the interval between calling the command and its return to the caller.

Dynamic Storage Buffer

A storage buffer whose offset is specified each time the storage buffer is bound to a command buffer via a descriptor set.

Dynamic Uniform Buffer

A uniform buffer whose offset is specified each time the uniform buffer is bound to a command buffer via a descriptor set.

Dynamically Uniform

See *Dynamically Uniform* in section 2.2 “Terms” of the Khronos SPIR-V Specification.

Encode Input Picture

A video picture resource used as the input of a video encode operation.

Element

Arrays are composed of multiple elements, where each element exists at a unique index within that array. Used primarily to describe data passed to or returned from the Vulkan API.

Explicitly-Enabled Layer

A layer enabled by the application by adding it to the enabled layer list in vkCreateInstance or vkCreateDevice.

Event

A synchronization primitive that is signaled when execution of previous commands completes through a specified set of pipeline stages. Events can be waited on by the device and polled by the host. Represented by a VkEvent object.

Executable State (Command Buffer)

A command buffer that has ended recording commands and **can** be executed. See also Initial State and Recording State.

Execution Dependency

A dependency that guarantees that certain pipeline stages’ work for a first set of commands has completed execution before certain pipeline stages’ work for a second set of commands begins execution. This is accomplished via pipeline barriers, subpass dependencies, events, or implicit ordering operations.

Execution Dependency Chain

A sequence of execution dependencies that transitively act as a single execution dependency.

Explicit chroma reconstruction

An implementation of sampler $Y'C_BC_R$ conversion which reconstructs reduced-resolution chroma samples to luma resolution and then separately performs texture sample interpolation. This is distinct from an implicit implementation, which incorporates chroma sample reconstruction into texture sample interpolation.

Extension Scope

The set of objects and commands that **can** be affected by an extension. Extensions are either device scope or instance scope.

Extending Structure

A structure type which may appear in the `pNext` *chain* of another structure, extending the

functionality of the other structure. Extending structures may be defined by either core API versions or extensions.

External Handle

A resource handle which has meaning outside of a specific Vulkan device or its parent instance. External handles **may** be used to share resources between multiple Vulkan devices in different instances, or between Vulkan and other APIs. Some external handle types correspond to platform-defined handles, in which case the resource **may** outlive any particular Vulkan device or instance and **may** be transferred between processes, or otherwise manipulated via functionality defined by the platform for that handle type.

External synchronization

A type of synchronization **required** of the application, where parameters defined to be externally synchronized **must** not be used simultaneously in multiple threads.

Facingness (Polygon)

A classification of a polygon as either front-facing or back-facing, depending on the orientation (winding order) of its vertices.

Facingness (Fragment)

A fragment is either front-facing or back-facing, depending on the primitive it was generated from. If the primitive was a polygon (regardless of polygon mode), the fragment inherits the facingness of the polygon. All other fragments are front-facing.

Fence

A synchronization primitive that is signaled when a set of batches or sparse binding operations complete execution on a queue. Fences **can** be waited on by the host. Represented by a VkFence object.

Field (Video)

Possibly discontinuous subregions of a frame. Frames **may** consist of two fields, a top field and a bottom field.

Flat Shading

A property of a vertex attribute that causes the value from a single vertex (the provoking vertex) to be used for all vertices in a primitive, and for interpolation of that attribute to return that single value unaltered.

Format Features

A set of features from VkFormatFeatureFlagBits that a VkFormat is capable of using for various commands. The list is determined by factors such as VkImageTiling.

Forward Prediction

Sample prediction performed during video decode and encode operations based on reference pictures that occur temporally (in display order) before the current picture.

Forward Reference

A reference picture that occurs temporally (in display order) before the current picture.

Fragment

A rectangular framebuffer region with associated data produced by rasterization and processed by fragment operations including the fragment shader.

Fragment Area

The width and height, in pixels, of a fragment.

Fragment Density

The ratio of fragments per framebuffer area in the x and y direction.

Fragment Density Texel Size

The (w,h) framebuffer region in pixels that each texel in a fragment density map applies to.

Fragment Input Attachment Interface

Variables with `UniformConstant` storage class and a decoration of `InputAttachmentIndex` that are statically used by a fragment shader's entry point, which receive values from input attachments.

Fragment Output Interface

A fragment shader entry point's variables with `Output` storage class, which output to color and/or depth/stencil attachments.

Frame (Video)

A multi-dimensional array of luma samples and an optional multi-dimensional array of chroma samples.

Fragment Tile Image Interface

A fragment shader entry point's variables with `TileImageEXT` storage class and a decoration of `Location`, which are used to read values from color attachments.

Framebuffer

A collection of image views and a set of dimensions that, in conjunction with a render pass, define the inputs and outputs used by drawing commands. Represented by a VkFramebuffer object.

Framebuffer Attachment

One of the image views used in a framebuffer.

Framebuffer Coordinates

A coordinate system in which adjacent pixels' coordinates differ by 1 in x and/or y, with (0,0) in the upper left corner and pixel centers at half-integers.

Framebuffer-Space

Operating with respect to framebuffer coordinates.

Framebuffer-Local

A framebuffer-local dependency guarantees that only for a single framebuffer region, the first set of operations happens-before the second set of operations.

Framebuffer-Global

A framebuffer-global dependency guarantees that for all framebuffer regions, the first set of operations happens-before the second set of operations.

Framebuffer Region

A framebuffer region is a set of sample (x, y, layer, view, sample) coordinates that is a subset of the entire framebuffer.

Frame Rate

A non-Vulkan term for Image Present Rate (IPR).

Front-Facing

See Facingness.

Full Compatibility

A given version of the API is fully compatible with another version if an application, relying only on valid behavior and functionality defined by either of those specifications, is able to correctly run against each version without any modification. This assumes no active attempt by that application to not run when it detects a different version.

Global Command

A Vulkan command for which the first argument is not a dispatchable handle type.

Global Workgroup

A collection of local workgroups dispatched by a single dispatching or single mesh task drawing command.

Handle

An opaque integer or pointer value used to refer to a Vulkan object. Each object type has a unique handle type.

Happen-after, happens-after

A transitive, irreflexive, and antisymmetric ordering relation between operations. An execution dependency with a source of **A** and a destination of **B** enforces that **B** happens-after **A**. The inverse relation of happens-before.

Happen-before, happens-before

A transitive, irreflexive, and antisymmetric ordering relation between operations. An execution dependency with a source of **A** and a destination of **B** enforces that **A** happens-before **B**. The inverse relation of happens-after.

Helper Invocation

A fragment shader invocation that is created solely for the purposes of evaluating derivatives for use in non-helper fragment shader invocations, and which does not have side effects.

Host

The processor(s) and execution environment that the application runs on, and that the Vulkan API is exposed on.

Host Mapped Device Memory

Device memory that is mapped for host access using vkMapMemory.

Host Mapped Foreign Memory

Memory owned by a foreign device that is mapped for host access.

Host Memory

Memory not accessible to the device, used to store implementation data structures.

Host-Accessible Subresource

A buffer, or a linear image subresource in either the VK_IMAGE_LAYOUT_PREINITIALIZED or VK_IMAGE_LAYOUT_GENERAL layout. Host-accessible subresources have a well-defined addressing scheme which can be used by the host.

Host-Local Memory

Memory that is not local to the device, and **may** be less performant for device access than device-local memory.

Host-Visible Memory

Device memory that **can** be mapped on the host and **can** be read and written by the host.

ICD

Installable Client Driver. An ICD is represented as a VkPhysicalDevice.

Identically Defined Objects

Objects of the same type where all arguments to their creation or allocation functions, with the exception of `pAllocator`, are

1. Vulkan handles which refer to the same object or
2. identical scalar or enumeration values or
3. Host pointers which point to an array of values or structures which also satisfy these three constraints.

Image

A resource that represents a multi-dimensional formatted interpretation of device memory. Represented by a VkImage object.

Image Present Duration

The amount of time the application intends for each newly-presented image to be visible to the user. This value **should** be a multiple of the refresh cycle duration.

Image Present Rate

The number of newly-presented images the application intends to present each second (a.k.a. frame rate). On fixed refresh rate displays, this value **should** be a multiple of the refresh rate.

Image Subresource

A specific mipmap level, layer, and set of aspects of an image.

Image Subresource Range

A set of image subresources that are contiguous mipmap levels and layers.

Image View

An object that represents an image subresource range of a specific image, and state controlling how the contents are interpreted. Represented by a VkImageView object.

Immutable Sampler

A sampler descriptor provided at descriptor set layout creation time for a specific binding. This sampler is then used for that binding in all descriptor sets allocated with the layout, and it **cannot** be changed.

Implicit chroma reconstruction

An implementation of sampler $Y'C_BC_R$ conversion which reconstructs the reduced-resolution chroma samples directly at the sample point, as part of the normal texture sampling operation. This is distinct from an *explicit chroma reconstruction* implementation, which reconstructs the reduced-resolution chroma samples to the resolution of the luma samples, then filters the result as part of texture sample interpolation.

Implicitly-Enabled Layer

A layer enabled by a loader-defined mechanism outside the Vulkan API, rather than explicitly by the application during instance or device creation.

Inactive Object (Ray Tracing)

A primitive or instance in a ray tracing acceleration structure which has a corresponding ID, but which will never report an intersection with any ray.

Index Buffer

A buffer bound via vkCmdBindIndexBuffer which is the source of index values used to fetch vertex attributes for a vkCmdDrawIndexed or vkCmdDrawIndexedIndirect command.

Indexed Drawing Commands

Drawing commands which use an *index buffer* as the source of index values used to fetch vertex attributes for a drawing command. Includes vkCmdDrawIndexed, vkCmdDrawIndexedIndirectCount, vkCmdDrawIndexedIndirectCountKHR, vkCmdDrawMultiIndexedEXT, and vkCmdDrawIndexedIndirect.

Indirect Commands

Drawing or dispatching commands that source some of their parameters from structures in buffer memory. Includes vkCmdDrawIndirect, vkCmdDrawIndexedIndirect, vkCmdDrawIndirectCount, vkCmdDrawIndexedIndirectCount, vkCmdDrawIndirectCountKHR, vkCmdDrawIndexedIndirectCountKHR, vkCmdDrawMeshTasksIndirectEXT, vkCmdDrawMeshTasksIndirectCountEXT, and vkCmdDispatchIndirect.

Indirect Drawing Commands

Drawing commands that source some of their parameters from structures in buffer memory. Includes vkCmdDrawIndirect, vkCmdDrawIndirectCount, vkCmdDrawIndexedIndirectCount, vkCmdDrawIndirectCountKHR, vkCmdDrawIndexedIndirectCountKHR,

vkCmdDrawMeshTasksIndirectEXT, vkCmdDrawMeshTasksIndirectCountEXT, and vkCmdDrawIndexedIndirect.

Initial State (Command Buffer)

A command buffer that has not begun recording commands. See also Recording State and Executable State.

Inline Uniform Block

A descriptor type that represents uniform data stored directly in descriptor sets, and supports read-only access in a shader.

Input Attachment

A descriptor type that represents an image view, and supports unfiltered read-only access in a shader, only at the fragment's location in the view.

Instance

The top-level Vulkan object, which represents the application's connection to the implementation. Represented by a VkInstance object.

Instance-Level Command

Any command that is dispatched from an instance, or from a child object of an instance, except for physical devices and their children.

Instance-Level Functionality

All instance-level commands and objects, and their structures, enumerated types, and enumerants.

Instance-Level Object

High-level Vulkan objects, which are not physical devices, nor children of physical devices. For example, VkInstance is an instance-level object.

Instance (Memory)

In a logical device representing more than one physical device, some device memory allocations have the requested amount of memory allocated multiple times, once for each physical device in a device mask. Each such replicated allocation is an instance of the device memory.

Instance (Resource)

In a logical device representing more than one physical device, buffer and image resources exist on all physical devices but **can** be bound to memory differently on each. Each such replicated resource is an instance of the resource.

Inter Block (Video)

A video coding block that uses inter prediction.

Inter Prediction (Video)

Sample prediction derived from reference picture samples or metadata.

Internal Synchronization

A type of synchronization **required** of the implementation, where parameters not defined to be externally synchronized **may** require internal mutexing to avoid multithreaded race conditions.

Intra Block (Video)

A video coding block that only uses intra prediction.

Intra Prediction (Video)

Sample prediction that is not derived from reference picture samples or metadata.

Invocation (Shader)

A single execution of an entry point in a SPIR-V module. For example, a single vertex's execution of a vertex shader or a single fragment's execution of a fragment shader.

Invocation Group

A set of shader invocations that are executed in parallel and that **must** execute the same control flow path in order for control flow to be considered dynamically uniform.

Invocation Repack Instruction

A ray tracing instruction where the implementation **may** change the set of invocations that are executing.

IPD

Image Present Duration.

Join (Deferred Host Operations)

The act of instructing a thread to participate in the execution of a deferred operation. See Deferred Host Operations.

Legacy (feature)

A feature which is no longer recommended as the correct or best way to achieve its intended purpose. Legacy features are typically superseded by new features which are the currently recommended way to achieve that purpose.

Linear Resource

A resource is *linear* if it is one of the following:

- a VkBuffer
- a VkImage created with VK_IMAGE_TILING_LINEAR
- a VkImage created with VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT and whose Linux DRM format modifier is `DRM_FORMAT_MOD_LINEAR` Because a VkAccelerationStructureKHR resource does not have memory bound to it directly, it is considered neither linear nor non-linear. However, the VkBuffer on which a VkAccelerationStructureKHR resource is placed is a linear resource.

A resource is *non-linear* if it is one of the following:

- a VkImage created with VK_IMAGE_TILING_OPTIMAL

- a VkImage created with VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT and whose Linux DRM format modifier is not DRM_FORMAT_MOD_LINEAR

Linux DRM Format Modifier

A 64-bit, vendor-prefixed, semi-opaque unsigned integer describing vendor-specific details of an image's memory layout. In Linux graphics APIs, *modifiers* are commonly used to specify the memory layout of externally shared images. An image has a *modifier* if and only if it is created with `tiling` equal to VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT. For more details, refer to the appendix for extension `VK_EXT_image_drm_format_modifier`.

Local Workgroup

A collection of compute shader invocations invoked by a single dispatching command, which share data via `WorkgroupLocal` variables and can synchronize with each other.

Logical Device

An object that represents the application's interface to the physical device. The logical device is the parent of most Vulkan objects. Represented by a VkDevice object.

Logical Operation

Bitwise operations between a fragment color value and a value in a color attachment, that produce a final color value to be written to the attachment.

Lost Device

A state that a logical device **may** be in as a result of unrecoverable implementation errors, or other exceptional conditions.

Mappable

See Host-Visible Memory.

Memory Dependency

A memory dependency is an execution dependency which includes availability and visibility operations such that:

- The first set of operations happens-before the availability operation
- The availability operation happens-before the visibility operation
- The visibility operation happens-before the second set of operations

Memory Domain

A memory domain is an abstract place to which memory writes are made available by availability operations and memory domain operations. The memory domains correspond to the set of agents that the write **can** then be made visible to. The memory domains are *host*, *device*, *shader*, *workgroup instance* (for workgroup instance there is a unique domain for each compute workgroup) and *subgroup instance* (for subgroup instance there is a unique domain for each subgroup).

Memory Domain Operation

An operation that makes the writes that are available to one memory domain available to another memory domain.

Memory Heap

A region of memory from which device memory allocations **can** be made.

Memory Type

An index used to select a set of memory properties (e.g. mappable, cached) for a device memory allocation.

Mesh Shading Pipeline

A graphics pipeline where the primitives are assembled explicitly in the shader stages. In contrast to the primitive shading pipeline where input primitives are assembled by fixed function processing.

Mesh Tasks Drawing Commands

Drawing commands which create shader invocations organized in workgroups for drawing mesh tasks. Includes vkCmdDrawMeshTasksEXT, vkCmdDrawMeshTasksIndirectEXT, and vkCmdDrawMeshTasksIndirectCountEXT .

Minimum Mip Level Size

The smallest size that is permitted for a mip level. For conventional images this is 1x1x1. See Image Mip Level Sizing.

Mip Tail Region

The set of mipmap levels of a sparse residency texture that are too small to fill a sparse block, and that **must** all be bound to memory collectively and opaquely.

Multi-planar

A *multi-planar format* (or "planar format") is an image format consisting of more than one *plane*, identifiable with a `_2PLANE` or `_3PLANE` component to the format name and listed in Formats Requiring Sampler $Y'C_BC_R$ Conversion for VK_IMAGE_ASPECT_COLOR_BIT Image Views. A *multi-planar image* (or "planar image") is an image of a multi-planar format.

Nested Command Buffers

A nested command buffer is a secondary command buffer that is executed by another secondary command buffer, which may itself execute other secondary command buffers.

Non-Dispatchable Handle

A handle of an integer handle type. Handle values **may** not be unique, even for two objects of the same type.

Non-Indexed Drawing Commands

Drawing commands for which the vertex attributes are sourced in linear order from the vertex input attributes for a drawing command (i.e. they do not use an *index buffer*). Includes vkCmdDraw, vkCmdDrawIndirectCount, vkCmdDrawIndirectCountKHR, vkCmdDrawMultiEXT, and vkCmdDrawIndirect.

Normalized

A value that is interpreted as being in the range [0,1] as a result of being implicitly divided by some other value.

Normalized Device Coordinates

A coordinate space after perspective division is applied to clip coordinates, and before the viewport transformation converts them to framebuffer coordinates.

Obsoleted (feature)

A feature is obsolete if it can no longer be used.

Opaque Capture Address

A 64-bit value representing the device address of a buffer or memory object that is expected to be used by trace capture/replay tools in combination with the `bufferDeviceAddress` feature.

Overdraw

Happens when more than one fragment invocation writes to the same (x, y, layer, view, sample) coordinate during a render pass instance. Usually refers to the performance impact of either rendering too much transparent geometry or redundantly rendering opaque geometry which is replaced by future fragments.

Overlapped Range (Aliased Range)

The aliased range of a device memory allocation that intersects a given image subresource of an image or range of a buffer.

Ownership (Resource)

If an entity (e.g. a queue family) has ownership of a resource, access to that resource is well-defined for access by that entity.

Packed Format

A format whose components are stored as a single texel block in memory, with their relative locations defined within that element.

Payload

Importable or exportable reference to the internal data of an object in Vulkan.

Peer Memory

An instance of memory corresponding to a different physical device than the physical device performing the memory access, in a logical device that represents multiple physical devices.

Physical Device

An object that represents a single device in the system. Represented by a VkPhysicalDevice object.

Physical-Device-Level Command

Any command that is dispatched from a physical device.

Physical-Device-Level Functionality

All physical-device-level commands and objects, and their structures, enumerated types, and enumerants.

Physical-Device-Level Object

Physical device objects. For example, VkPhysicalDevice is a physical-device-level object.

Picture Partition (Video)

Collective term that refers to a subregion of a video picture resource that **may** be decoded or encoded independently according to the partitioning scheme defined in the corresponding video compression standard. Picture partitions **may** be non-rectangular.

Pipeline

An object controlling how graphics or compute work is executed on the device. A pipeline includes one or more shaders, as well as state controlling any non-programmable stages of the pipeline. Represented by a VkPipeline object.

Pipeline Barrier

An execution and/or memory dependency recorded as an explicit command in a command buffer, that forms a dependency between the previous and subsequent commands.

Pipeline Cache

An object that **can** be used to collect and retrieve information from pipelines as they are created, and **can** be populated with previously retrieved information in order to accelerate pipeline creation. Represented by a VkPipelineCache object.

Pipeline Layout

An object defining the set of resources (via a collection of descriptor set layouts) and push constants used by pipelines that are created using the layout. Used when creating a pipeline and when binding descriptor sets and setting push constant values. Represented by a VkPipelineLayout object.

Pipeline Library

A pipeline that cannot be directly used, instead defining a set of shaders and shader groups which will be linked into other pipelines.

Pipeline Stage

A logically independent execution unit that performs some of the operations defined by an action command.

Pipeline Trace Ray Instruction

A ray tracing instruction which traces a ray into an acceleration structure when using ray tracing pipelines. One of:

- `OpTraceRayKHR`
- `OpHitObjectTraceRayEXT`
- `OpHitObjectTraceReorderExecuteEXT`

`pNext` Chain

A set of structures chained together through their `pNext` members.

Planar

See *multi-planar*.

Plane

An *image plane* is part of the representation of an image, containing a subset of the color components necessary to represent the texels in the image and with a contiguous mapping of coordinates to bound memory. Most images consist only of a single plane, but some formats spread the components across multiple image planes. The host-accessible properties of each image plane are accessible for a linear layout using vkGetImageSubresourceLayout. If a multi-planar image is created with the VK_IMAGE_CREATE_DISJOINT_BIT bit set, the image is described as *disjoint*, and its planes are therefore bound to memory independently.

Point Sampling (Rasterization)

A rule that determines whether a fragment sample location is covered by a polygon primitive by testing whether the sample location is in the interior of the polygon in framebuffer-space, or on the boundary of the polygon according to the tie-breaking rules.

Poison

Equivalent to poison in the SPIR-V specification. Can only be generated by shader operations. Poison values made available to the host or read by device operations outside of a SPIR-V instruction behave as undefined values for all such accesses.

Potential Format Features

The union of all VkFormatFeatureFlagBits that the implementation supports for a specified VkFormat, over all supported image tilings.

Pre-rasterization

Operations that execute before rasterization, and any state associated with those operations.

Presentable image

A `VkImage` object obtained from a `VkSwapchainKHR` used to present to a `VkSurfaceKHR` object.

Preserve Attachment

One of a list of attachments in a subpass description that is not read or written by the subpass, but that is read or written on earlier and later subpasses and whose contents **must** be preserved through this subpass.

Primary Command Buffer

A command buffer that **can** execute secondary command buffers, and **can** be submitted directly to a queue.

Primitive Shading Pipeline

A graphics pipeline where input primitives are assembled by fixed function processing. It is the counterpart to mesh shading.

Primitive Topology

State controlling how vertices are assembled into primitives, e.g. as lists of triangles, strips of lines, etc.

Promoted (feature)

A feature from an older extension is considered promoted if it is made available as part of a new core version or newer extension with wider support.

Protected Buffer

A buffer to which protected device memory **can** be bound.

Protected-capable Device Queue

A device queue to which protected command buffers **can** be submitted.

Protected Command Buffer

A command buffer which **can** be submitted to a protected-capable device queue.

Protected Device Memory

Device memory which **can** be visible to the device but **must** not be visible to the host.

Protected Image

An image to which protected device memory **can** be bound.

Provisional

A feature is released provisionally in order to get wider feedback on the functionality before it is finalized. Provisional features may change in ways that break backwards compatibility, and thus are not recommended for use in production applications.

Provoking Vertex

The vertex in a primitive from which flat shaded attribute values are taken. This is generally the “first” vertex in the primitive, and depends on the primitive topology.

PTLAS

Partitioned Top Level Acceleration Structures

Push Constants

A small bank of values writable via the API and accessible in shaders. Push constants allow the application to set values used in shaders without creating buffers or modifying and binding descriptor sets for each update.

Push Constant Interface

The set of variables with `PushConstant` storage class that are statically used by a shader entry point, and which receive values from push constant commands.

Push Descriptors

Descriptors that are written directly into a command buffer rather than into a descriptor set. Push descriptors allow the application to set descriptors used in shaders without allocating or modifying descriptor sets for each update.

Descriptor Update Template

An object specifying a mapping from descriptor update information in host memory to elements in a descriptor set, which helps enable more efficient descriptor set updates.

Query Pool

An object containing a number of query entries and their associated state and results. Represented by a VkQueryPool object.

Queue

An object that executes command buffers and sparse binding operations on a device. Represented by a VkQueue object.

Queue Family

A set of queues that have common properties and support the same functionality, as advertised in VkQueueFamilyProperties.

Queue Operation

A unit of work to be executed by a specific queue on a device, submitted via a queue submission command. Each queue submission command details the specific queue operations that occur as a result of calling that command. Queue operations typically include work that is specific to each command, and synchronization tasks.

Queue Submission

Zero or more batches and an optional fence to be signaled, passed to a command for execution on a queue. See the Devices and Queues chapter for more information.

Ray Tracing Command

Commands that provoke work using a ray tracing pipeline. Includes vkCmdTraceRaysKHR, and vkCmdTraceRaysIndirectKHR .

RC

Refresh Cycle.

Reconstructed Picture

A video picture resource reconstructed from a compressed bitstream using video decode or encode operations that **can** be used as a reference picture by future video decode or encode operations with the same video session.

Recording State (Command Buffer)

A command buffer that is ready to record commands. See also Initial State and Executable State.

Reference Picture

A video picture resource used by video decode and encode operations to provide predictions of the values of samples in the subsequently decoded or encoded pictures.

Reference Picture Metadata

Opaque state associated with a DPB slot, maintained by a video session.

Refresh Cycle

The periodic process for updating the contents of the Presentation Engine's display.

Refresh Cycle Duration

The amount of time from the start of one refresh cycle to the next.

Refresh Rate

The number of refresh cycles per second.

Release Operation (Resource)

An operation that releases ownership of an image subresource or buffer range.

Render Pass

An object that represents a set of framebuffer attachments and phases of rendering using those attachments. Represented by a VkRenderPass object.

Render Pass Instance

A use of a render pass in a command buffer.

Required Extensions

Extensions that **must** be enabled alongside extensions dependent on them (see Extension Dependencies).

Reset (Command Buffer)

Resetting a command buffer discards any previously recorded commands and puts a command buffer in the initial state.

Residency Code

An integer value returned by sparse image instructions, indicating whether any sparse unbound texels were accessed.

Resolve Attachment

A subpass attachment point, or image view, that is the target of a multisample resolve operation from the corresponding color attachment at the end of the subpass.

Retired Swapchain

A swapchain that has been used as the `oldSwapchain` parameter to vkCreateSwapchainKHR. Images cannot be acquired from a retired swapchain, however images that were acquired (but not presented) before the swapchain was retired **can** be presented.

Sample Index

The index of a sample within a single set of samples.

Sample Shading

Invoking the fragment shader multiple times per fragment, with the covered samples partitioned among the invocations.

Sampled Image

A descriptor type that represents an image view, and supports filtered (sampled) and unfiltered read-only access in a shader.

Sampler

An object containing state controlling how sampled image data is sampled (or filtered) when accessed in a shader. Also a descriptor type describing the object. Represented by a VkSampler object.

Secondary Command Buffer

A command buffer that **can** be executed by a primary command buffer, and **must** not be submitted directly to a queue.

Self-Dependency

A subpass dependency from a subpass to itself, i.e. with `srcSubpass` equal to `dstSubpass`. A self-dependency is not automatically performed during a render pass instance, rather a subset of it **can** be performed via vkCmdPipelineBarrier during the subpass.

Semaphore

A synchronization primitive that supports signal and wait operations, and **can** be used to synchronize operations within a queue or across queues. Represented by a VkSemaphore object.

Shader

Instructions selected (via an entry point) from a shader module, which are executed in a shader stage.

Shader Call

An instruction which **may** cause execution to continue in a different shader stage.

Shader Code

A stream of instructions used to describe the operation of a shader.

Shader Group

A set of Shader Stages that are part of a VkPipeline containing multiple of such sets. This allows the device to make use of all the shader groups from the bound pipeline independently.

Shader Module

A collection of shader code, potentially including several functions and entry points, that is used to create shaders in pipelines. Represented by a VkShaderModule object.

Shader Stage

A stage of the graphics or compute pipeline that executes shader code.

Shading Rate

The ratio of the number of fragment shader invocations generated in a fully covered framebuffer region to the size (in pixels) of that region.

Shared presentable image

A presentable image created from a swapchain with VkPresentModeKHR set to either VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR or VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR.

Side Effect

A store to memory or atomic operation on memory from a shader invocation.

Single-plane format

A format that is not *multi-planar*.

Size-Compatible Image Formats

When a compressed image format and an uncompressed image format are size-compatible, it means that the texel block size of the uncompressed format **must** equal the texel block size of the compressed format.

Sparse Block

An element of a sparse resource that can be independently bound to memory. Sparse blocks of a particular sparse resource have a corresponding size in bytes that they use in the bound memory.

Sparse Image Block

A sparse block in a sparse partially-resident image. In addition to the sparse block size in bytes, sparse image blocks have a corresponding width, height, and depth defining the dimensions of these elements in units of texels or compressed texel blocks, the latter being used in case of sparse images having a block-compressed format.

Sparse Unbound Texel

A texel read from a region of a sparse texture that does not have memory bound to it.

Static Use

An object in a shader is statically used by a shader entry point if any function in the entry point's call tree contains an instruction using the object. A reference in the entry point's interface list does not constitute a static use. Static use is used to constrain the set of descriptors used by a shader entry point.

Storage Buffer

A descriptor type that represents a buffer, and supports reads, writes, and atomics in a shader.

Storage Image

A descriptor type that represents an image view, and supports unfiltered loads, stores, and atomics in a shader.

Storage Texel Buffer

A descriptor type that represents a buffer view, and supports unfiltered, formatted reads, writes, and atomics in a shader.

Subgroup

A set of shader invocations that **can** synchronize and share data with each other efficiently. In compute shaders, the *local workgroup* is a superset of the subgroup.

Subgroup Mask

A bitmask for all invocations in the current subgroup with one bit per invocation, starting with

the least significant bit in the first vector component, continuing to the last bit (less than SubgroupSize) in the last required vector component.

Subpass

A phase of rendering within a render pass, that reads and writes a subset of the attachments.

Subpass Dependency

An execution and/or memory dependency between two subpasses described as part of render pass creation, and automatically performed between subpasses in a render pass instance. A subpass dependency limits the overlap of execution of the pair of subpasses, and **can** provide guarantees of memory coherence between accesses in the subpasses.

Subpass Description

Lists of attachment indices for input attachments, color attachments, depth/stencil attachment, resolve attachments, depth/stencil resolve, and preserve attachments used by the subpass in a render pass.

Subset (Self-Dependency)

A subset of a self-dependency is a pipeline barrier performed during the subpass of the self-dependency, and whose stage masks and access masks each contain a subset of the bits set in the identically named mask in the self-dependency.

Superseded (feature)

A legacy feature can be superseded by newer functionality that achieves the same goals through a newer style of API, being more efficient, or otherwise following current recommendations for use of Vulkan.

Texel Block

A single addressable element of an image with an uncompressed VkFormat, or a single compressed block of an image with a compressed VkFormat.

Texel Block Size

The size (in bytes) used to store a texel block of a compressed or uncompressed image.

Texel Coordinate System

One of three coordinate systems (normalized, unnormalized, integer) defining how texel coordinates are interpreted in an image or a specific mipmap level of an image.

Tile Image

A per-tile view of a framebuffer attachment. If the VK_EXT_shader_tile_image extension is enabled, the framebuffer is considered to be divided into tiles.

Timeline Semaphore

A semaphore with a strictly increasing 64-bit unsigned integer payload indicating whether the semaphore is signaled with respect to a particular reference value. Represented by a VkSemaphore object created with a semaphore type of VK_SEMAPHORE_TYPE_TIMELINE.

Undefined Value

A value that does not have a specific defined bit pattern. Undefined values are not invariant. When read by a shader operation, undefined values are read as *poison*.

Uniform Control Flow

See *Uniform Control Flow* in section 2.2 “Terms” of the Khronos SPIR-V Specification.

Uniform Texel Buffer

A descriptor type that represents a buffer view, and supports unfiltered, formatted, read-only access in a shader.

Uniform Buffer

A descriptor type that represents a buffer, and supports read-only access in a shader.

Units in the Last Place (ULP)

A measure of floating-point error loosely defined as the smallest representable step in a floating-point format near a given value. For the precise definition see Precision and Operation of SPIR-V instructions or Jean-Michel Muller, “On the definition of ulp(x)”, RR-5504, INRIA. Other sources may also use the term “unit of least precision”.

Unnormalized

A value that is interpreted according to its conventional interpretation, and is not normalized.

Unprotected Buffer

A buffer to which unprotected device memory **can** be bound.

Unprotected Command Buffer

A command buffer which **can** be submitted to an unprotected device queue or a protected-capable device queue.

Unprotected Device Memory

Device memory which **can** be visible to the device and **can** be visible to the host.

Unprotected Image

An image to which unprotected device memory **can** be bound.

User-Defined Variable Interface

A shader entry point’s variables with `Input` or `Output` storage class that are not built-in variables.

Vector Type

A SPIR-V `OpTypeVector` or `OpTypeVectorIdEXT` .

Vertex Input Attribute

A graphics pipeline resource that produces input values for the vertex shader by reading data from a vertex input binding and converting it to the attribute’s format.

Vertex Stream

A vertex stream is where the last pre-rasterization shader stages outputs vertex data, which then

goes to the rasterizer, is captured to a transform feedback buffer, or both. Geometry shaders **can** emit primitives to multiple independent vertex streams. Each vertex emitted by the geometry shader is directed at one of the vertex streams.

Validation Cache

An object that **can** be used to collect and retrieve validation results from the validation layers, and **can** be populated with previously retrieved results in order to accelerate the validation process. Represented by a VkValidationCacheEXT object.

Variable-Sized Descriptor Binding

A descriptor binding whose size will be specified when a descriptor set is allocated using this layout.

Vertex Input Binding

A graphics pipeline resource that is bound to a buffer and includes state that affects addressing calculations within that buffer.

Vertex Input Interface

A vertex shader entry point's variables with `Input` storage class, which receive values from vertex input attributes.

Video Bitstream Buffer

A resource that represents a linear array of data in device memory storing encoded video data. Represented by a VkBuffer object.

Video Coding Scope

A series of subsequent commands recorded into a command buffer starting with a vkCmdBeginVideoCodingKHR command and ending with a vkCmdEndVideoCodingKHR command that encompasses a set of video decode or encode operations.

Video Coding Operations

Any operations recorded into a command buffer within a video coding scope, including video decode and encode operations.

Video Decode Operation

An operation consuming data from a video bitstream buffer and zero or more reference pictures, and producing data to a decode output picture and an optional reconstructed picture.

Video Encode Operation

An operation consuming data from an encode input picture and zero or more reference pictures, and producing data to a video bitstream buffer and an optional reconstructed picture.

Video Picture Resource

A resource that represents a multi-dimensional formatted interpretation of device memory to be used with a video session as a decode output picture, encode input picture, reconstructed picture, and/or reference picture. It **may** contain metadata associated with a particular video session it is used with. Represented by a VkImage object and referred to using VkImageView objects created from it.

Video Session

A resource that represents and maintains the state needed to perform video decode or encode operations. Represented by a VkVideoSessionKHR object.

Video Session Parameters

A resource that stores preprocessed codec-specific parameters used with a compatible video session in video codec operations. Represented by a VkVideoSessionParametersKHR object.

Video Transcoding

The process of using the outputs of video decoding operations as inputs in video encoding operations.

View Mask

When multiview is enabled, a view mask is a property of a subpass controlling which views the rendering commands are broadcast to.

View Volume

A subspace in homogeneous coordinates, corresponding to post-projection x and y values between -1 and +1, and z values between 0 and +1.

Viewport Transformation

A transformation from normalized device coordinates to framebuffer coordinates, based on a viewport rectangle and depth range.

Visibility Operation

An operation that causes available values to become visible to specified memory accesses.

Visible

A state of values written to memory that allows them to be accessed by a set of operations.

Common Abbreviations

The abbreviations and acronyms defined in this section are sometimes used in the Specification and the API where they are considered clear and commonplace.

Src

Source

Dst

Destination

Min

Minimum

Max

Maximum

Rect
: Rectangle

Info
: Information

LOD
: Level of Detail

Log
: Logarithm

ID
: Identifier

UUID
: Universally Unique Identifier

Op
: Operation

R
: Red color component

G
: Green color component

B
: Blue color component

A
: Alpha color component

RTZ
: Round towards zero

RTE
: Round to nearest even

Video-Specific Abbreviations

The following abbreviations and acronyms are used in the context of video decode and encode operations to refer to commonly used video compression terms in their usual abbreviated form:

AVC
: Advanced Video Coding

Bipred
: Bidirectional Prediction

CABAC
: Context-Adaptive Binary Arithmetic Coding

CAVLC
: Context-Adaptive Variable-Length Coding

CBR
: Constant Bit Rate

CTB
: Coding Tree Block

Diff
: Difference

DPB
: Decoded Picture Buffer

GOP
: Group Of Pictures

HDR
: High Dynamic Range

HEVC
: High Efficiency Video Coding

HRD
: Hypothetical Reference Decoder

IDC
: Indicator

IDR
: Instantaneous Decoder Refresh

MB
: Macroblock

MV
: Motion Vector

NALU
: Network Abstraction Layer Unit

OBU
: Open Bitstream Unit

PCM
: Pulse-Code Modulation

Pic
: Picture

Pred
: Prediction

PPS
: Picture Parameter Set

QP
: Quantization Parameter

RC
: Rate Control

SPS
: Sequence Parameter Set

Std
: Standard

VBR
: Variable Bit Rate

VCL
: Video Coding Layer

VPS
: Video Parameter Set

Prefixes

Prefixes are used in the API to denote specific semantic meaning of Vulkan names, or as a label to avoid name clashes, and are explained here:

VK/Vk/vk
: Vulkan namespace
All types, commands, enumerants, and defines in this specification are prefixed with these two characters.

PFN/pfn
: Function Pointer

Denotes that a type is a function pointer, or that a variable is of a pointer type.

p

Pointer
Variable is a pointer.

vkCmd

Commands that record commands in command buffers
These API commands do not result in immediate processing on the device. Instead, they record the requested action in a command buffer for execution when the command buffer is submitted to a queue.

s

Structure
Used to denote the `VK_STRUCTURE_TYPE*` member of each structure in `sType`

Appendix K: Credits (Informative)

Vulkan 1.4 is the result of contributions from many people and companies participating in the Khronos Vulkan Working Group, as well as input from the Vulkan Advisory Panel.

Members of the Working Group, including the company that they represented at the time of their most recent contribution, are listed in the following section. Some specific contributions made by individuals are listed together with their name.

Working Group Contributors to Vulkan

- Aaron Greig, Codeplay Software Ltd. (versions 1.1, 1.4)
- Aaron Hagan, AMD (versions 1.1)
- Abhishek Sashi Nair, Huawei Technologies Co., Ltd. (version 1.4)
- Adam Jackson, Red Hat (versions 1.0, 1.1)
- Adam Śmigielski, Mobica (version 1.0)
- Aditi Verma, Qualcomm Technologies, Inc. (version 1.3)
- Ahmed Abdelkhalek, AMD (versions 1.3, 1.4) (video subgroup chair)
- Aidan Fabius, Lynx (version 1.2)
- Aitor Camacho, LunarG (version 1.4)
- Alan Baker, Google (versions 1.1, 1.2, 1.3, 1.4)
- Alan Vines, Imagination Technologies (version 1.4)
- Alan Ward, Google (versions 1.1, 1.2)
- Alastair Murray, Codeplay (version 1.4)
- Alejandro Piñeiro, Igalia (version 1.1)
- Aleks Todorov, Google (version 1.4)
- Aleksander Netzel, Epic Games (version 1.4)
- Alex Bourd, Qualcomm Technologies, Inc. (versions 1.0, 1.1, 1.4)
- Alex Crabb, Caster Communications (versions 1.2, 1.3, 1.4)
- Alex Dai, Oppo (version 1.4)
- Alex Krstic, Qualcomm Technologies, Inc. (version 1.4)
- Alex Todorov, Google (version 1.4)
- Alex Walters, Imagination Technologies (versions 1.2, 1.3, 1.4)
- Alexander Galazin, Imagination Technologies (versions 1.0, 1.1, 1.2, 1.3, 1.4) (conformance test subgroup chair)
- Alexey Knyazev, Independent, Khronos (version 1.4)
- Alexey Sachkov, Intel (version 1.3)

- Ali Asif, Huawei Technologies Co., Ltd. (version 1.4)
- Allan MacKinnon, Google (version 1.3)
- Allen Hux, Intel (version 1.0)
- Alon Or-bach, Arm (versions 1.0, 1.1, 1.2, 1.3, 1.4) (system integration subgroup chair)
- Alyssa Rosenzweig, Valve Software (version 1.4)
- Anastasia Stulova, Arm (versions 1.2, 1.3, 1.4)
- Andreas Vasilakis, Think Silicon (version 1.2)
- Andres Gomez, Igalia (version 1.1)
- Andrew Cox, Samsung Electronics (version 1.0)
- Andrew Ellem, Google (version 1.3)
- Andrew Garrard, Imagination Technologies (versions 1.0, 1.1, 1.2, 1.3, 1.4) (format wrangler)
- Andrew Poole, Samsung Electronics (version 1.0)
- Andrew Rafter, Samsung Electronics (version 1.0)
- Andrew Richards, Codeplay Software Ltd. (version 1.0)
- Andrew Woloszyn, Google (versions 1.0, 1.1)
- Andrew Zhi, AMD (version 1.4)
- Ann Thorsnes, Khronos (versions 1.2, 1.3)
- Anthony Salvi, Adobe (version 1.4)
- Antoine Labour, Google (versions 1.0, 1.1)
- Anton Berko, Mediatek (version 1.4)
- Anuj Phogat, Netflix (version 1.4)
- Aras Pranckevičius, Unity Technologies (version 1.0)
- Arpit Agarwal, Qualcomm Technologies, Inc. (version 1.4)
- Arseny Kapoulkine, Roblox (version 1.3)
- Ashley Smith, AMD (version 1.4)
- Ashwin Kolhe, NVIDIA (version 1.0)
- Aurelio Reis, NVIDIA (version 1.4)
- Avinash Baliga, NVIDIA (version 1.4)
- Balaji Calidas, Qualcomm Technologies, Inc. (version 1.4)
- Baldur Karlsson, Valve Software (versions 1.1, 1.2, 1.3, 1.4)
- Barthold Lichtenbelt, NVIDIA (version 1.1)
- Bas Nieuwenhuizen, Google (versions 1.1, 1.2, 1.4)
- Ben Ashbaugh, Intel (version 1.4)
- Ben Bowman, Imagination Technologies (version 1.0)
- Benj Lipchak, Unknown (version 1.0)

- Bill Hollings, Brenwill (versions 1.0, 1.1, 1.2, 1.3, 1.4)
- Bill Licea-Kane, Qualcomm Technologies, Inc. (versions 1.0, 1.1)
- Blaine Kohl, Khronos (versions 1.2, 1.3)
- Bob Fraser, Google (version 1.3)
- Boris Zanin, Mobica (versions 1.2, 1.3)
- Brad Grantham, LunarG (version 1.4)
- Brent E. Insko, Intel (version 1.0)
- Brian Ellis, Qualcomm Technologies, Inc. (version 1.0)
- Brian Paul, VMware (versions 1.2, 1.3)
- CAI Kangying, Huawei Technologies Co., Ltd. (version 1.4)
- Caio Marcelo de Oliveira Filho, Intel (versions 1.2, 1.3)
- Calle Ledjefors, Tencent (version 1.4)
- Cass Everitt, Oculus VR (versions 1.0, 1.1)
- Cemil Azizoglu, Canonical (version 1.0)
- Chang-Hyo Yu, Samsung Electronics (version 1.0)
- Charles Giessen, LunarG (versions 1.3, 1.4)
- Charlie Lao, Google (version 1.4)
- Charlie Turner, Igalia (version 1.4)
- Cheng Ge, Tencent (version 1.4)
- Chengping Luo, Mediatek (version 1.4)
- Chia-I Wu, LunarG (version 1.0)
- Chris Bieneman, Microsoft (version 1.4)
- Chris Frascati, Qualcomm Technologies, Inc. (version 1.0)
- Chris Glover, Google (versions 1.3, 1.4)
- Christian Forfang, Arm (version 1.3)
- Christoph Kubisch, NVIDIA (versions 1.3, 1.4)
- Christophe Riccio, LunarG (versions 1.0, 1.1, 1.4)
- Cody Northrop, LunarG (version 1.0)
- Colin Riley, AMD (version 1.1)
- Connor Abbott, Valve Software (version 1.4)
- Caterina Shablia, Collabora (version 1.4)
- Cort Stratton, Google (versions 1.1, 1.2)
- Courtney Goeltzenleuchter, Google (versions 1.0, 1.1, 1.3)
- Craig Davies, Huawei Technologies Co., Ltd. (version 1.2)
- Craig Graham, Samsung Electronics (version 1.4)

- Dae Kim, Imagination Technologies (versions 1.1, 1.4)
- Damien Leone, NVIDIA (version 1.0)
- Dan Baker, Oxide Games (versions 1.0, 1.1)
- Dan Ginsburg, Valve Software (versions 1.0, 1.1, 1.2, 1.3, 1.4)
- Dan Sinclair, Google (version 1.4)
- Daniel Johnston, Intel (versions 1.0, 1.1)
- Daniel Koch, NVIDIA (versions 1.0, 1.1, 1.2, 1.3, 1.4) (ray tracing subgroup chair)
- Daniel Rakos, RasterGrid (versions 1.0, 1.1, 1.2, 1.3, 1.4)
- Daniel Schürmann, Valve Software (version 1.4)
- Daniel Stone, Collabora (versions 1.1, 1.2, 1.4)
- Daniel Story, Nintendo (version 1.4)
- Daniel Vetter, Intel (version 1.2)
- Daniele Vettorel, Google (version 1.4)
- Danny Huynh, Samsung Electronics (version 1.4)
- Dave Aronson, Adobe (version 1.4)
- Dave Desormeaux, LunarG (version 1.4)
- Dave Evans, Samsung Electronics (version 1.4)
- David Airlie, Red Hat (versions 1.0, 1.1, 1.2, 1.3)
- David Mao, AMD (versions 1.0, 1.2)
- David McCloskey, Juice Labs (version 1.4)
- David Miller, Miller & Mattson (versions 1.0, 1.1) (Vulkan reference card)
- David Neto, Google (versions 1.0, 1.1, 1.2, 1.3, 1.4)
- David Pankratz, Huawei Technologies Co., Ltd. (version 1.3)
- David Wilkinson, AMD (version 1.2)
- David Yu, Pixar (version 1.0)
- Dean Sekulic, Croteam (version 1.4)
- Dejan Mircevski, Google (version 1.1)
- Dennis Adams, Sony (version 1.4)
- Derek Lamberti, Arm (version 1.4)
- Diego Novillo, Google (versions 1.3, 1.4)
- Dimitris Georgakakis, Think Silicon (version 1.3)
- Dominik Witczak, AMD (versions 1.0, 1.1, 1.3)
- Donald Scorgie, Imagination Technologies (version 1.2)
- Dzmitry Malyshau, Mozilla (versions 1.1, 1.2, 1.3)
- Ed Hutchins, Oculus (version 1.2)

- Eddie Hatfield, Google (version 1.4)
- Eduardo Lima, Igalia (version 1.4)
- Emily Stearns, Khronos (versions 1.2, 1.3, 1.4)
- Enrique de Lucas, Imagination Technologies (version 1.4)
- Eric Sullivan, NVIDIA (version 1.4)
- Faith Ekstrand, Collabora (versions 1.0, 1.1, 1.2, 1.3, 1.4)
- Florian Penzkofer, Epic (version 1.4)
- Frank (LingJun) Chen, Qualcomm Technologies, Inc. (version 1.0)
- Frank Yang, Google (version 1.4)
- François Duranleau, Gameloft (versions 1.3, 1.4)
- Fred Liao, Mediatek (version 1.0)
- Gabe Dagani, Samsung Electronics (versions 1.0, 1.4)
- Gabor Sines, AMD (version 1.2)
- Gang Chen, Intel (version 1.4)
- George Erfesoglou, OVR Tech (version 1.4)
- Graeme Leese, Broadcom (versions 1.0, 1.1, 1.2, 1.3, 1.4)
- Graham Connor, Imagination Technologies (version 1.0)
- Graham Sellers, Electronic Arts (versions 1.0, 1.1, 1.4)
- Graham Wihlidal, Electronic Arts (version 1.3)
- Greg Fischer, LunarG (versions 1.1, 1.4)
- Greg Roth, Microsoft (version 1.4)
- Gregory Grebe, AMD (versions 1.3, 1.4)
- Hai Nguyen, NVIDIA (versions 1.2, 1.3, 1.4)
- Hamish Todd, Imagination Technologies (version 1.4)
- Hans-Kristian Arntzen, Valve Software (versions 1.1, 1.2, 1.3, 1.4)
- Heman Gala, MAGIX (version 1.4)
- Henri Verbeet, Codeweavers (version 1.2)
- Hugo Devillers, Saarland University (version 1.4)
- Hwanyong Lee, Kyungpook National University (version 1.0)
- Iago Toral, Igalia (versions 1.1, 1.2)
- Ian Ameline, Huawei Technologies Co., Ltd. (version 1.4)
- Ian Elliott, Google (versions 1.0, 1.1, 1.2, 1.4)
- Ian Richardson, Intel (version 1.4)
- Ian Romanick, Intel (versions 1.0, 1.1, 1.3, 1.4)
- Ingrid Kelly, NVIDIA (version 1.4)

- J. D. Rouan, Google (version 1.4)
- Jaebaek Seo, Google (version 1.4)
- Jake Turner, Valve Software (version 1.4)
- Jakob Bornecrantz, Collabora (version 1.4)
- James Fitzpatrick, Imagination Technologies (versions 1.3, 1.4)
- James Hughes, Oculus VR (version 1.0)
- James Jones, NVIDIA (versions 1.0, 1.1, 1.2, 1.3, 1.4)
- James Riordon, Khronos (versions 1.2, 1.3)
- Jamie Madill, Google (versions 1.3, 1.4)
- Jan Hermes, Continental Corporation (versions 1.0, 1.1, 1.4)
- Jan-Harald Fredriksen, Arm (versions 1.0, 1.1, 1.2, 1.3, 1.4)
- Jarred Davies, Imagination Technologies (version 1.4)
- Jean Geffroy, id Software (version 1.4)
- Jean-François Roy, Google (versions 1.1, 1.2, 1.3)
- Jean-Noé Morrisette, Epic Games (version 1.4)
- Jeff Bolz, NVIDIA (versions 1.0, 1.1, 1.2, 1.3, 1.4)
- Jeff Juliano, NVIDIA (versions 1.0, 1.1, 1.2, 1.4)
- Jeff Kiel, NVIDIA (version 1.4)
- Jeff Leger, Qualcomm Technologies, Inc. (versions 1.1, 1.3, 1.4)
- Jeff Phillips, Khronos (versions 1.3, 1.4)
- Jeff Vigil, Samsung Electronics (versions 1.0, 1.1, 1.2, 1.3, 1.4)
- Jens Owen, Google (versions 1.0, 1.1)
- Jeremy Gebben, LunarG (version 1.4)
- Jeremy Hayes, LunarG (versions 1.0, 1.4)
- Jesse Barker, Unity Technologies (versions 1.0, 1.1, 1.2, 1.3, 1.4)
- Jesse Hall, Google (versions 1.0, 1.1, 1.2, 1.3, 1.4)
- Jessica Heerboth, NVIDIA (version 1.4)
- Jingye Wang, Huawei Technologies Co., Ltd. (version 1.4)
- Joe Davis, Samsung Electronics (version 1.1)
- Johannes van Waveren, Oculus VR (versions 1.0, 1.1)
- John Anthony, Arm (versions 1.2, 1.3, 1.4)
- John Kessenich, Google (versions 1.0, 1.1, 1.2, 1.3) (SPIR-V and GLSL for Vulkan spec author)
- John McDonald, Valve Software (versions 1.0, 1.1, 1.2, 1.3)
- John Stone, University of Illinois (version 1.4)
- John Zulauf, LunarG (versions 1.1, 1.2, 1.3)

- Jon Ashburn, LunarG (version 1.0)
- Jon Leech, Independent (versions 1.0, 1.1, 1.2, 1.3, 1.4) (XML toolchain, normative language, release wrangler)
- Jonas Gustavsson, Samsung Electronics (versions 1.0, 1.1)
- Jonas Meyer, Epic Games (versions 1.2, 1.3)
- Jonathan Hamilton, Imagination Technologies (version 1.0)
- Jordan Justen, Intel (version 1.1)
- Jordan Robinson, Samsung Electronics (version 1.4)
- Joshua Ashton, Valve Software (versions 1.3, 1.4)
- Jules Blok, Independent (version 1.4)
- Jungwoo Kim, Samsung Electronics (versions 1.0, 1.1)
- Jörg Wagner, Arm (version 1.1)
- Kalle Raita, Google (version 1.1)
- Karen Ghavam, LunarG (versions 1.1, 1.2, 1.3, 1.4)
- Karl Schultz, LunarG (versions 1.1, 1.2)
- Kathleen Mattson, Khronos (versions 1.0, 1.1, 1.2)
- Kaye Mason, Google (version 1.2)
- Keith Packard, Valve Software (version 1.2)
- Kenneth Benzie, Codeplay Software Ltd. (versions 1.0, 1.1)
- Kenneth Russell, Google (version 1.1)
- Kerch Holt, NVIDIA (versions 1.0, 1.1)
- Kevin O'Neil, AMD (version 1.1)
- Kevin Petit, Arm (versions 1.3, 1.4)
- Kris Rose, Khronos (versions 1.2, 1.3, 1.4)
- Kristian Kristensen, Intel (versions 1.0, 1.1)
- Krzysztof Iwanicki, Samsung Electronics (version 1.0)
- Larry Seiler, Intel (version 1.0)
- Laura Hermanns, Epic Games (version 1.4)
- Laura Shubel, Caster Communications (version 1.3)
- Lauri Ilola, Nokia (version 1.1)
- Lei Zhang, Google (version 1.2)
- Lenny Komow, LunarG (versions 1.1, 1.2)
- Leonid Fedorov, Huawei Technologies Co., Ltd. (version 1.4)
- Liam Middlebrook, NVIDIA (version 1.3)
- Lina Versace, Google (versions 1.0, 1.1, 1.2, 1.4) (system integration subgroup chair)

- Lionel Duc, NVIDIA (version 1.4)
- Lionel Landwerlin, Intel (versions 1.1, 1.2, 1.4)
- Lisie Aartsen, Khronos (versions 1.3, 1.4)
- Liz Maitral, Khronos (version 1.2)
- Lorenzo Dal Col, Khronos (version 1.4) (conformance test subgroup chair)
- Lou Kramer, AMD (version 1.3)
- Lutz Latta, Lucasfilm (version 1.0)
- Lynne Iribarren, Independent (version 1.4)
- Maciej Jesionowski, AMD (version 1.1)
- Mahmoud Kharsa, Samsung Electronics (version 1.4)
- Mais Alnasser, AMD (versions 1.1, 1.4)
- Marcin Kantoch, AMD (version 1.3)
- Marcin Rogucki, Mobica (version 1.1)
- Maria Rovatsou, Codeplay Software Ltd. (version 1.0)
- Mariusz Merecki, Intel (versions 1.3, 1.4)
- Marijn Suijten, Traverse Research (version 1.4)
- Mark Bellamy, Arm (versions 1.2, 1.3, 1.4)
- Mark Callow, Independent (versions 1.0, 1.1, 1.2, 1.3)
- Dhruv Mark Collins, Igalia (version 1.4)
- Mark Kilgard, NVIDIA (versions 1.1, 1.2)
- Mark Lobodzinski, LunarG (versions 1.0, 1.1, 1.2)
- Mark Young, LunarG (versions 1.1, 1.3, 1.4)
- Markus Tavenrath, NVIDIA (version 1.1)
- Marty Johnson, Khronos (versions 1.3, 1.4)
- Mateusz Przybylski, Intel (version 1.0)
- Mathias Heyer, NVIDIA (versions 1.0, 1.1)
- Mathias Schott, NVIDIA (versions 1.0, 1.1)
- Mathieu Robart, Arm (version 1.2)
- Matt Netsch, Qualcomm Technologies, Inc. (versions 1.1, 1.3, 1.4) (conformance test subgroup chair)
- Matthew Rusch, NVIDIA (version 1.3)
- Matthäus Chajdas, Intel (versions 1.1, 1.2, 1.3, 1.4)
- Maurice Ribble, Qualcomm Technologies, Inc. (versions 1.0, 1.1)
- Maxim Lukyanov, Samsung Electronics (version 1.0)
- Maximilien Dagois, Google (version 1.4)

- Michael Antilla, Google (version 1.4)
- Michael Blumenkrantz, Valve Software (versions 1.3, 1.4)
- Michael Lentine, Google (version 1.0)
- Michael O'Hara, AMD (version 1.1)
- Michael Phillip, Samsung Electronics (version 1.2)
- Michael Wong, Codeplay Software Ltd. (version 1.1)
- Michael Worcester, Imagination Technologies (versions 1.0, 1.1)
- Michal Pietrasiuk, Intel (versions 1.0, 1.3, 1.4)
- Mick Beaver, Electronic Arts (version 1.4)
- Mika Isojarvi, Google (versions 1.0, 1.1)
- Mike Schuchardt, LunarG (versions 1.1, 1.2, 1.4)
- Mike Stroyan, LunarG (version 1.0)
- Mike Weiblen, LunarG (versions 1.1, 1.2, 1.3)
- Minyoung Son, Samsung Electronics (version 1.0)
- Mitch Singer, AMD (versions 1.0, 1.1, 1.2, 1.3, 1.4)
- Mythri Venugopal, Samsung Electronics (version 1.0)
- Máté Ferenc Nagy-Egri, RasterGrid (version 1.4)
- Natalie Chouinard, Google (version 1.4)
- Nathan Gauër, Google (version 1.4)
- Nathaniel Cesario, LunarG (version 1.4)
- Naveen Leekha, Google (version 1.0)
- Neil Henning, AMD (versions 1.0, 1.1, 1.2, 1.3)
- Neil Hickey, Arm (version 1.2)
- Neil Trevett, NVIDIA (versions 1.0, 1.1, 1.2, 1.3, 1.4) (portability subgroup chair)
- Nick Penwarden, Epic Games (version 1.0)
- Nicola Zaghen, Innosilicon (version 1.4)
- Nicolai Hähnle, AMD (version 1.1)
- Nikhil Joshi, NVIDIA (version 1.4)
- Niklas Smedberg, Unity Technologies (version 1.0)
- Norbert Nopper, Independent (versions 1.0, 1.1)
- Nuno Subtil, NVIDIA (versions 1.1, 1.2, 1.3)
- Pan Gao, Huawei Technologies Co., Ltd. (version 1.4)
- Pat Brown, NVIDIA (version 1.0)
- Patrick Boyle, Qualcomm Technologies, Inc. (version 1.4)
- Patrick Cozzi, Independent (version 1.1)

- Patrick Doane, Activision (versions 1.0, 1.4)
- Patrick Han, Samsung Electronics (version 1.4)
- Patrick Mours, NVIDIA (version 1.4)
- Pau Baiget, Google (version 1.4)
- Peter Kutz, Adobe (version 1.4)
- Peter Lohrmann, AMD (versions 1.0, 1.2)
- Peter Quayle, Imagination Technologies (version 1.4)
- Petros Bantolas, Imagination Technologies (version 1.1)
- Philip Rebohle, Valve Software (version 1.3)
- Pierre Boudier, NVIDIA (versions 1.0, 1.1, 1.2, 1.3)
- Pierre-Loup Griffais, Valve Software (versions 1.0, 1.1, 1.2, 1.3)
- Piers Daniell, NVIDIA (versions 1.0, 1.1, 1.2, 1.3, 1.4)
- Ping Liu, Intel (versions 1.3, 1.4)
- Piotr Bialecki, Intel (version 1.0)
- Piotr Byszewski, Mobica (version 1.3)
- Prabindh Sundareson, Samsung Electronics (version 1.0)
- Pradyuman Singh, NVIDIA (version 1.4)
- Pyry Haulos, Google (versions 1.0, 1.1) (conformance test subgroup chair)
- Rachel Bradshaw, Caster Communications (version 1.3)
- Rajeev Rao, Qualcomm Technologies, Inc. (version 1.2)
- Ralph Potter, Samsung Electronics (versions 1.1, 1.2, 1.3, 1.4) (working group chair)
- Ran Wang, Google (version 1.4)
- Raun Krisch, Samsung Electronics (version 1.3)
- Ravi Patlegar, AMD (version 1.4)
- Ravikiran Pallapatula, Huawei Technologies Co., Ltd. (version 1.4)
- Ray Smith, Arm (versions 1.0, 1.1, 1.2)
- Ricardo Garcia, Igalia (versions 1.3, 1.4)
- Richard Huddy, Samsung Electronics (versions 1.2, 1.3)
- Richard Van Natta, Firaxis (version 1.4)
- Richard Wang, Huawei Technologies Co., Ltd. (version 1.4)
- Rick Hammerstone, Qualcomm Technologies, Inc. (version 1.4)
- Rob Barris, NVIDIA (version 1.1)
- Rob Stepinski, Transgaming (version 1.0)
- Robert Simpson, Qualcomm Technologies, Inc. (versions 1.0, 1.1, 1.3, 1.4)
- Rohan Garg, Intel (version 1.4)

- Rolando Caloca Olivares, Epic Games (versions 1.0, 1.1, 1.2, 1.3)
- Ronan Keryell, Xilinx (versions 1.3, 1.4)
- Roy Ju, Mediatek (version 1.0)
- Rufus Hamade, Imagination Technologies (version 1.0)
- Ruihao Zhang, Qualcomm Technologies, Inc. (versions 1.1, 1.2, 1.3, 1.4)
- SU Fujia, Huawei Technologies Co., Ltd. (version 1.4)
- Sam Martin, Arm (version 1.4)
- Samuel (Sheng-Wen) Huang, Mediatek (versions 1.3, 1.4)
- Samuel Iglesias Gonsalvez, Igalia (version 1.3)
- Samuel Pitoiset, Valve Software (version 1.4)
- Sandeep Kakarlapudi, Arm (version 1.4)
- Sascha Willems, Self (versions 1.3, 1.4)
- Sean Ellis, Arm (version 1.0)
- Sean Harmer, KDAB Group (versions 1.0, 1.1)
- Serdar Kocdemir, Google (version 1.4)
- Sergey Solovyev, Tencent (version 1.4)
- Seth Williams, NVIDIA (version 1.4)
- Shahbaz Youssefi, Google (version 1.4)
- Shannon Woods, NVIDIA (versions 1.0, 1.1, 1.2, 1.3, 1.4)
- Shaun Heald, Innosilicon (version 1.4)
- Sinuhe Hardegree, Samsung Electronics (version 1.4)
- Slawomir Cygan, Intel (versions 1.0, 1.1, 1.3, 1.4)
- Slawomir Grajewski, Intel (versions 1.0, 1.1, 1.3, 1.4)
- Sorel Bosan, AMD (version 1.1)
- Soroush Kashani, Imagination Technologies (version 1.4)
- Spencer Fricke, LunarG (versions 1.2, 1.3, 1.4)
- Stefanus Du Toit, Google (version 1.0)
- Stephane Cerveau, Igalia (version 1.4)
- Stephen Huang, Mediatek (version 1.1)
- Steve Hill, Broadcom (versions 1.0, 1.2)
- Steve Viggers, Core Avionics & Industrial Inc. (versions 1.0, 1.2)
- Steve Winston, Holochip (versions 1.3, 1.4)
- Steven Gu, Huawei Technologies Co., Ltd. (version 1.4)
- Steven Perron, Google (version 1.4)
- Stuart Smith, AMD (versions 1.0, 1.1, 1.2, 1.3, 1.4)

- Sujeevan Rajayogam, Google (version 1.3)
- Theresa Foley, Intel (version 1.0)
- Thomas Caissard, Adobe (version 1.4)
- Tiago Costa, Epic Games (version 1.4)
- Tilmann Scheller, Samsung Electronics (version 1.1)
- Tim Lewis, Khronos (versions 1.3, 1.4)
- Timo Suoranta, AMD (version 1.0)
- Timothy Lottes, AMD (versions 1.0, 1.1)
- Timur Kristof, Valve Software (version 1.4)
- Ting Wei, Arm (version 1.4)
- Tobias Hector, AMD (versions 1.0, 1.1, 1.2, 1.3, 1.4) (validity language and toolchain)
- Tobin Ehlis, LunarG (version 1.0)
- Tom Olson, Arm (versions 1.0, 1.1, 1.2, 1.3, 1.4) (working group chair)
- Tomasz Bednarz, Independent (version 1.1)
- Tomasz Kubale, Intel (version 1.0)
- Tony Barbour, LunarG (versions 1.0, 1.1, 1.2)
- Tony Zlatinski, NVIDIA (versions 1.3, 1.4)
- Trevor Black, Google (version 1.4)
- Tyler Nowicki, AMD (version 1.4)
- Vassili Nikolaev, NVIDIA (version 1.4)
- Vatsalya Prasad, Huawei Technologies Co., Ltd. (version 1.4)
- Victor Eruhimov, Unknown (version 1.1)
- Vikram Kushwaha, NVIDIA (versions 1.3, 1.4)
- Vikram Tarikere, Imagination Technologies (version 1.4)
- Vincent Hindriksen, Stream HPC (versions 1.2, 1.3)
- Víctor Jáquez, Igalia (version 1.4)
- Wang Jingye, Huawei Technologies Co., Ltd. (version 1.4)
- Wasim Abbas, Arm (version 1.3)
- Wayne Lister, Imagination Technologies (version 1.0)
- William Wheeler, Imagination Technologies (version 1.4)
- Wolfgang Engel, Unknown (version 1.1)
- Wooyoung Kim, Qualcomm Technologies, Inc. (version 1.4)
- Wyvern Wang, Huawei Technologies Co., Ltd. (version 1.3)
- Yajun Zhang, Tencent (version 1.4)
- Yanjun Zhang, VeriSilicon (versions 1.0, 1.1, 1.2, 1.3, 1.4)

- Youngwang Kim, Arm (version 1.4)
- Yu Chang Wang, Huawei Technologies Co., Ltd. (version 1.4)
- Yuehai Du, Qualcomm Technologies, Inc. (version 1.4)
- Yunxing Zhu, Huawei Technologies Co., Ltd. (version 1.3)
- Yury Uralsky, NVIDIA (version 1.4)
- Zehui Lin, Huawei Technologies Co., Ltd. (version 1.4)
- Žiga Markuš, LunarG (version 1.4)

Other Credits

The Vulkan Advisory Panel members provided important real-world usage information and advice that helped guide design decisions.

The wider Vulkan community have provided useful feedback, questions, and specification changes that have helped improve the quality of the Specification via GitHub.

Administrative support to the Working Group for Vulkan 1.1 and beyond was provided by Khronos staff including Ann Thorsnes, Blaine Kohl, Dennis Fu, Dominic Agoro-Ombaka, Emily Stearns, Jeff Phillips, Kendra Casper, Lisie Aartsen, Liz Maitral, Marty Johnson, Tim Lewis, and Xiao-Yu CHENG; and by Alex Crabb, Laura Shubel, and Rachel Bradshaw of Caster Communications.

Administrative support for Vulkan 1.0 was provided by Andrew Riegel, Elizabeth Riegel, Glenn Fredericks, Kathleen Mattson and Michelle Clark of Gold Standard Group.

Technical support was provided by James Riordon, site administration of Khronos.org and OpenGL.org.